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DESIGN QUALIFICATION
COLD ROOMS WITH REFRIGERATION SYSTEM

FG RELEASE

M/S. EGYPTIAN COMPANY FOR DRUGS AND VETERINARY VACCINES, EGYPT.

DESIGN QUALIFICATION FOR
COLD ROOM SYSTEM

SUBMITTED TO:

M/s.

SUBMITTED BY:

M/s. FABTECH TECHNOLOGIES INTERNATIONAL LTD.
717. JANKI CENTRE, OFF VEERA DESAI ROAD.
ANDHERI (W) MUMBAI-400053

Website: www.fabtechnologies.com

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1. Objective:

To establish the documented evidences for the qualification of machine design as per the user requirement specifications (URS).

2. Scope:

This protocol will define the responsibilities, acceptance criteria, basis of design, technical specifications, list of major bought out parts, utility Requirements, safety & cGMP features and documentation requirements.

The Protocol also defines the basis of Installation Qualification, (IQ) and Operational Qualification (OQ) and Performance Qualification (PQ).

The Design Qualification is an essential part of the qualification mechanism and will serve as a basic document for the validation responsibility.

3. Responsibilities:

M/s. EGYPTIAN COMPANY FOR DRUGS AND VETERINARY VACCINES, EGYPT.

To provide the URS for the Equipment.

To perform design qualification.

M/S. FABTECH TECHNOLOGIES INTERNATIONAL LTD.

1. To design, engineer and provide the complete technical details of the equipment pertaining to its design qualification viz.

- i. Machine overview
- ii. Equipment orientation with layout,
- iii. Specifications of the sub-components / bought out items, and their make, model & quantity, and backup records/ brochures,
- iv. Details of Utilities,
- v. Identification of components for calibration
- vi. Material of construction of major components
- vii. Brief operational description
- viii. Safety features and alarms
- ix. Pre-installation requirements

2. To confirm the safe delivery of the equipment to the user site.
3. To ensure that no un-authorized and / or unrecorded design modifications shall take place. If at any point in time, any change is desired in the mutually agreed design, Change control procedure shall be followed and documented.
4. To ensure the proper installation and commissioning of the equipment.

4. Description of Machine

Cold Room With Refrigeration System

A. DESCRIPTION:

Cold Rooms are made out of insulated Prefabricated PUF Panels. Pre-Fabricated type PUF Panels consist of two facing of relatively thin metal sheet of high strength enclosing a core, which is relatively thick and light but has adequate stiffness in a direction normal to face of the panel. The Core is made of Polyurethane which is the best insulation material in the World.

The Advantages of insulated panels with polyurethane are excellent and durable thermal insulation. High load bearing capacity at low weight absolute water and vapour barrier excellent air tightness and free of thermal bridges which results in considerable energy savings. Surface finish facings providing resistance to weather and aggressive environment.

B. PURPOSE:

The purpose of this document is to verify supply of equipment/system as per the user requirement. The reference of the document is our offer and/or Purchase Order. The user will go through the documents to verify requirement is satisfied.

5. Technical Specification of System / Equipment :

Description	ROOM 5
	FG Release
* Cold Room Size	13987 mm
	5297 mm
	4000 mm
* Room Temperature	2 to 8 Deg
* Refrigeration Capacity	4TR
* Cooling Capacity	12096 Kcal/Hr
* Compressor	Make - COPELAND / DANFOSS/EQUIVALENT Hermetic
	SCROLL
* Refrigerant	Freon R 404a
* Refrigerant ET	(-)5 Deg C
* Refrigerant CT	50 Deg C
* Condenser	Air Cooled
* Condenser Fan	Axial Type
* Evaporator	Fan Coil UNIT
* Electrical Power Supply	380V (+/- 10%) 50Hz
* Door Cold Room mm	1800 x 3000 – Sliding
* No of Doors	2 No
* PUF Panel THICK mm	100
* Cool Down Time	6 Hour
* PUF Panel Covering	GIPP / GIPP
* Columns	NIL
* Ante Room	NIL(Additional
	Panel 5mx4m)

|PRE FABRICATED PUF PANELS |
* Density	40 Kg/Mt3 CFC Free
* Panel Assembly Type	Tongue & Groove with CAMLOCK
* Panel Covering	Inner: SS Pre-Coated Sheet
	Outer: GI Pre-coated Sheet
* Locking Arrangement	CAMLOCK

|COLD ROOM PANELS |
|* Insulation Thickness |100mm |

6. SCOPE OF SUPPLY:

Condensing Unit	
Compressor	COPELAND / Danfoss / Emerson /Equivalent Make,
	Semi Hermetic
Compressor Accessories	CCH etc if required
Condenser	Air Cooled Type with Copper Tubes and Aluminium
	Fins
Evaporator Unit	Fan Coil UNIT manufactured with Aluminum Fins and
	Copper Tubes, of required Surface Area, and the
	same shall be provided with required size Axial
	Fans, and with Drain PAN, and required standard
	Accessories.
Controls	Switches, Thermostats, Thermostatic Expansion
	Valves,
	Solenoid Valve, Temperature Controller cum
	Indicators, etc
	Provided as required.
Electrical Cum Control	Complete Full Function Electrical Panel equipped
Panel	with Terminal Block, Contactors, Thermal Overload
Relays with Single Phasing Preventers, Off Delay	
Timer, Push Buttons, Indicating Lamps, with	
Interlocking System. Electrical Wiring from Panel	

|to Motors & Instruments. |
 ||
 |Refrigeration Piping |Interconnecting Refrigeration piping inside |
 |condensing unit and evaporator unit shall be |
 |providing with require valves and fittings. And |
 |External Refrigeration Piping from Condensing Unit|
 |to Evaporator Unit shall be done at site. |
 ||
 |Insulation |Suction Line Insulation |
 ||
 |Refrigerant and Oil |First Charge |
 ||
 |PUF Panels |Prefabricated PUF Panels, 100mm Thick, with GI / |
 |GI Pre |
 |coated Inner & Outer Surface (as per Tech |
 |Specifications), |
 |for walls and ceiling, average of 40Kg/M3 Density |
 |complete |
 |with Camlocks. . |
 ||
 |Panel Doors |80mm Thick & As per Technical Specifications Inner|
 |& |
 |Outer Surface with Hardware such as Lock, Hinges, |
 |Door |
 |Closer |
 ||
 |Accessories |Installation Accessories such as Silicon Sealant, |
 |Angles, Channels & Hardware etc. |

7. IDENTIFICATION OF COMPONENTS FOR CALIBRATION

Sr.	Component	Specification	Range	
No.				
01.	Temperature	Make: Castle/	(-)50 Deg.C to	Test & Calibration
	Controller-1	Equivalent	+99.9 Deg.C	Certificate submitted
		Accuracy: +/-1%		
01.	Temperature	Make:	(-)50 Deg.C to	Test & Calibration
	Datalogger	Welltronix/	+99.9 Deg.C	Certificate submitted
		Equivalent	Accuracy: +/-1%	
03	Pressure	Make: Akvalo/	0 to 500 PSI for	Test & Calibration
	Gauge	Equivalent GI.	High Pressure &	Certificate submitted
		Filled	(-)30 to 300 PSI	
		for Low Pressure		

8. COMPONENT FOR CERTIFICATION

Sr. No.	COMPONENT	CERTIFICATION
01.	Overall System	Test Certificate
02	Compressor	Test Certificate
03.	Condenser	Pressure Testing Report
04.	Cond. Fan	Test Certificate
05.	Evaporator	Test/Inspection Certificate
06.	Temperature Controller-1	Test & Calibration Certificate
07.	Temperature Data Logger	Test & Calibration Certificate
08.	Pressure Gauge	Test & Calibration Certificate

9. ABBREVIATIONS

ABBREVIATIONS	FULL FORM
P.O.	Purchase Order
MOC	Material of Construction
URS	User Requirement Specifications
RPM	Rotations per Minute
HP	Horse Power
SS	Stainless Steel
IQ	Installation Qualification
OQ	Operational Qualification
PQ	Performance Qualification
FAT	Factory Acceptance Test
Temp.	Temperature
HZ	Hertz
PSI	Pound Per Square Inch
G.I.	Galvanized Iron
TR	Ton of Refrigeration
GIPP	Galvanized Iron Pre-Painted

10. Design qualification summary:

Design parameters and functional specification (DQ) for Cold storage with Refrigeration Unit are studied for M/s. Egyptian company for Drugs and veterinary vaccines project. System functional specification & design specification was checked with the in-house requirement and cGMP requirements, and found satisfactory.

11. Change Control:

Any changes in the design and functional requirements mentioned above should be documented along with justifications for these.

The approval team should approve these changes.

	Name	Signature	Date
Written By			
Verified By			

12. Design qualification protocol approval sheet:

PROTOCOL AUTHORS: M/S. FABTECH TECHNOLOGIES INTERNATIONAL LTD.

Responsibility	Department	Name	Signature
		Date	
Prepared By	HVAC	Tauseef Uddin	
Checked By	HVAC	Fahad khan	

PROTOCOL REVIEWED AND APPROVED BY: M/S. EGYPTIAN COMPANY FOR DRUGS AND VETERINARY VACCINES, EGYPT.

Responsibility	Department	Name	Signature
		Date	
Reviewed By			
Reviewed By			
Reviewed By			
Approved By			

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“USER REQUIREMENT SPECIFICATION for DEEP FREEZER”

Name of the Equipment /	DEEP FREEZER (-80 deg)
System	
Project/ Facility	
URS No.	

Revision Index:

Revision	Effective	Reason for Revision
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	security requirements	
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1. OBJECTIVE:

The purpose of this document is to provide the appropriate design and performance requirements for Deep freezer, so as to meet the in-house requirements as well as compliance with current Good Manufacturing Practices.

The User Requirements Specification (URS) is written to facilitate to identify the company's needs and proposes a design suitable for Hilleman laboratories.

2.0 SCOPE

This document is applicable for the Deep freezer used for storage of Rota virus bulk antigen in different size of containers.

3.0 EQUIPMENT DESCRIPTION

The Deep freezer is to store the Rota virus bulk fluid. The deep freezers placed at a temperature of 25°C or ambient and the humidity in the area is 55 ± 5 %.

The shelves and inner chamber is made of stainless steel and the outer body is made of mild steel duly powder coated.

The Freezer is insulated with high density non-CFC polyurethane which helps in preventing the loss of temperature, thermal protection and structural strength. Enhanced acoustic insulation helps to reduce compressor sound. Double doors ensure that exposure to incoming air is kept to a minimum even when the outer door is opened. Both the doors are latched.

The down-feed evaporator for efficient refrigerant flow and washable condenser filter to maintain peak cooling efficiency. The filter also is washable for economy and do-it-yourself maintenance.

Deep freezer design hermetically sealed compressor for low temperature applications. A highly efficient vapour compression refrigeration cycle helps in attaining the required temperature within a short time period. The combination of efficient technology and construction brings the added benefit of considerable energy savings.

The temperature up to -86°C is controlled by a digital temperature indicator cum controller. The instrument is designed to work on 220V AC Supply (Model with 110V).

Salient Points

- Ecofriendly Non-CFC refrigerants.
- The whole unit is mounting on castors for free movement of the freezer.
- Key operated mechanical door lock is a standard feature.
- Specially designed air-cooled condenser ensures peak performance even at high ambient conditions.
- Microprocessor based control system with digital LED display of temperature.
- Main door with inner FRP Lining and durable magnetic gasket. Durable inner doors for each compartment.
- A hot line on the mouth of the chamber is provided to minimize cooling losses and to help improve the cooling efficiency.
- Should have Standby Cooling System to significantly minimize breakdowns. The switchover to Standby system is automatic & immediate upon detection of a fault in Primary. If desired, the switchovers can also be scheduled or manual ones.
- Automatic defrost by utilizing a heater to defrost the condenser coils, which cycles on and off, preventing any ice build-up. Their robust compressors and CFC-free insulation help ensure temperature uniformity

The following SS cans with Rota virus bulk will be stored. The provided slotted angle racks can be adjusted as per size of container

Can size -Liters	OD (inches)	Height (inches)
2	4	14.9
5	9	8.7
20	9	24

4.0 Recipe provision / data saving / data back-up / data security requirements

Recipe provision required

Yes ☐ No ☐

Data saving facility required Yes ☐ No ☐

(If yes mention the device for saving: _____ / Supplier to specify)

Data back-up required (during power failure) Yes ☐ No ☐

(If yes mention the device for saving: _____ / Supplier to specify)

Data back-up required (during life-cycle) Yes ☐ No ☐

(If yes mention the device for saving: _____ / Supplier to specify)

Data security required (during life-cycle) Yes ☐ No ☐

(If yes mention the device for saving: _____ / Supplier to specify)

5.0 Drawings

All drawings to be provided in hard copy prior to supply for approval.

6.0 Expected documentation

The Supplier shall provide the documentation for preliminary review.

The Supplier shall provide documentation reflecting "as-built" condition with final delivery.

Following documents shall be delivered by the suppliers during the procurement life cycle:

Sr. No	Document details	Required
1	Design Specification document (Hardware)	☐
2	Functional Specification (operational details)	☐
3	PLC Alarm/Interlock/Safety/ communication/power	☐
4	failure test procedures	☐
5	Process and Instrumentation Diagram (P&ID)	☐
6	Instrument Listing with tag numbers	☐
7	Control Schematics (in case of PLC logic)	☐
8	diagrams	☐
9	Control Panel Assembly Drawings	☐
10	Machine Assembly Drawings	☐
11	Bill of Materials	☐
12	Operator, Maintenance and Service Manuals	☐
13	Spare Parts List	☐
14	MOC certificates	☐
15	Calibration certificates of instruments	☐
16	Test certificates of components/test devices	☐
17	Air Flow Diagrams	☐
18	GA drawing	☐
19	Isometric drawing	☐

- |19 |Electrical drawings | |
- |20 |Component Cut Sheets | |
- |22 |PLC Program Printout and Disk File | |
- |23 |HMI Configuration Printout and Disk File | |
- |24 |List of consumables | |
- |25 |Other (Specify) | |

✓: Applicable & required
✗ : Not applicable

7.0 Training requirements

Vendor to provide formal operational training of equipment during commissioning

8.0 Other requirements

All documents must be provided in English.

9.0 Abbreviations

- |S. No |Abbreviation |Description |
- | |cGMP |Current Good Manufacturing Practices |
- | |GA |General arrangement |
- | |HMI |Human machine interface |
- | |MOC |Material of construction |
- | |NA |Not applicable |
- | |PLC |Programmable Logic control |
- | |SS |Stainless Steel |
- | |IPA |Iso propyl alcohol |
- | |UOM |Unit of measurement |

“USER REQUIREMENT SPECIFICATION for DISPENSING BOOTH”

Name of the Equipment /	Dispensing Booth
System	
Project/ Facility	
URS No.	

Revision Index:

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1. OBJECTIVE:

The purpose of this document is to provide the appropriate design and performance requirements for Dispensing booth, so as to meet the in-house requirements as well as compliance with current Good Manufacturing Practices.

The User Requirements Specification (URS) is written to facilitate to identify the company's needs and proposes a design suitable for Hilleman laboratories.

2.0 SCOPE

This document is applicable for the Dispensing booth used for ware house and process activities.

3.0 EQUIPMENT DESCRIPTION

Dispensing booth shall be operated as clean room work station. It is provided with HEPA filter, which prevents the airborne dusts by down draught technique. The unit eliminates powder contamination to protect the operator and the surrounding environment. Dispensing booth is also called sampling booth or weighing booth. It shall be designed to maintain the Reverse laminar air supply from the HEPA Filter through blower and the exhaust connects to the outside the work place. System shall be installed in grade C area. The air intake for the blower shall be from the surrounding grade C area and should maintain the required face velocity from the filter face. The mode of operation shall be of recirculation type. All the safety interlocks shall be considered with respect to operator and product contamination. There shall be pre-filter at least 10 µ grade provided above the HEPA to decrease the load on HEPA filter which is of 0.3 µ filter.

System shall contain the following main components:

1. Motor with blower
2. Fluorescent Lamp
3. Pre filter
4. HEPA filter
5. Magnehelic Gauge

Equipment design and principle of working

The Dispensing booth station is required to be designed to provide cleanliness under the area to meet grade A requirements.

1. HEPA filters shall be of having filtration efficiency of 0.3 µ

particles and the MOC of micro fiber-glass media.

2. Noise Level: Not more than 80 dB at one meter distance from equipment.

3. Integrity test port at upstream of HEPA filters.

4. Blower shall statically and dynamically balance to supply air flow with sufficient capacity and static pressure.

5. LAF surrounding glass length shall be up to the working height.

6. Local indication of power supply to be provided.

7. Non- viable particle/ viable particle should meet grade A requirement.

8. Design shall be made for ease of maintenance.

9. Monochromatic light, Florescent light and UV light fixtures shall be fixed .

10. Initial/final pressure drop limits to be provided for these LAF's from supplier end.

4.0 Capacity /Model

5.0 Drawings

All drawings to be provided in hard copy prior to supply for approval.

6.0 Expected documentation

The Supplier shall provide the documentation for preliminary review. The Supplier shall provide documentation reflecting "as-built" condition with final delivery.

Following documents shall be delivered by the suppliers during the life cycle.

|S.No. |Document details |

|1 |Design Specification document |

|2 |Functional Specification (operational details) |

|3 |PLC Alarm/Interlock/Safety/ communication/power |

| |failure test procedures |

|4 |Process and Instrumentation Diagram(P&ID) |

|5 |Instrument Listing with tag numbers |

|6 |Control Schematics (in case of PLC logic diagrams) |

|7 |Control Panel Assembly Drawings |

|8 |Operator, Maintenance and Service Manuals |

|9 |Spare Parts List |

|10 |MOC certificates |

|11 |Calibration certificates of instruments |

|12 |Test certificates of components/test devices |

|13 |Weld certificates |

|14 |GA drawing |

|15 |Isometric drawing |

|16 |Electrical drawings |

|17 |Component Cut Sheets |

|18 |Other (Specify) – IQ, OQ, PQ Documents |

7.0 Training requirements

Vendor to provide formal operational training of equipment during commissioning

8.0 Other requirements

All documents must be provided in English.

9.0 Abbreviations

|S. No |Abbreviation |Description |

| |cGMP |Current Good Manufacturing Practices |

| |GA |General arrangement |

| |HMI |Human machine interface |

| |MOC |Material of construction |

| |NA |Not applicable |

| |PLC |Programmable Logic control |

| |SS |Stainless Steel |

| |IPA |Iso propyl alcohol |

| |UOM |Unit of measurement |

“USER REQUIREMENT SPECIFICATION for FILTER INTEGRITY TESTER”

Name of the Equipment /	Filter Integrity Tester
System	
Project/ Facility	
URS No.	

Revision Index:

Revision	Effective	Reason for Revision
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1. OBJECTIVE:

The purpose of this document is to provide the appropriate design and performance requirements for Filter Integrity tester, so as to meet the in-house requirements as well as compliance with current Good Manufacturing Practices.

The User Requirements Specification (URS) is written to facilitate to identify the company's needs and proposes a design suitable for Hilleman laboratories.

2.0 SCOPE

This document is applicable for the Filter Integrity tester used for process activities.

3.0 EQUIPMENT DESCRIPTION

The integrity testing equipment is required to be verifying the integrity of the both hydrophilic and hydrophobic filters. The filters are using for filtration of gases like compressed air and nitrogen, liquids like aqueous products, detoxification agent (NaOCl), disinfectant solutions and water for injection at a temperature of 25°C or ambient and the humidity in the area is 55 ± 5 %.

The system shall be suitable for wetting of the filters with suitable wetting fluids like IPA, Water and methanol at supplier specified concentration.

The system shall be suitable for testing of integrity by bubble point method and the Maximum pressure applied will be 100 psi.

The system will be used for testing the integrity of disc filters, capsule filters and cartridge filters.

System shall have the feature for wetting the filters at a pressure NMT 1.0 Kg/Cm2.

OPERATIONAL AND DESIGN REQUIREMENTS

1. Details of input material (RM/Intermediate)

1. RM

Type of RM : Filters

Characteristics of RM : PTFE/ PVDF/ PES/N66
Membrane filters (Housing is cartridge or capsule)

2. Intermediate

Description : Compressed air/ Nitrogen

Characteristics : Gas

2. Information of output/utility (product) :

The system shall be tested the integrity of the filters by various test procedures

like bubble point, Water intrusion test etc and shall give the result, the result

shall be printed digitally and graphical representation.

The system shall also print the result whether pass/ fail on the print out along

with the operator details.

The data shall be stored in the system itself during paper outages/ out of stock/

printer malfunctioning.

5.0 PROCESS DESCRIPTION

The Hydrophobic filters are using for filtration of the gases like compressed air and nitrogen which are going to be used into the Lyo process, hence the environment or utility after filtration shall be sterile, to have assurance over the sterility due to the filtered gas, the filter used for filtration shall be confirmed for its integrity at specified pore size.

The Hydrophilic filters (0.2 μ Cartridge or Capsule filters) are using for filtration of the final product bulk solution into the sterile vessel. To have assurance over the sterility on filtered solution, the filters used in the process are to be tested for integrity for specified pore size.

Equipment design and principle of working

The equipment is required to be design to perform the integrity test of the 0.2/0.22 μ hydrophilic/ hydrophobic filters by bubble point method.

The filters are of different types such as capsule, cartridge and disc filters and required to be tested the bubble point after wetting the filter membrane using filter supplier recommendations.

The equipment shall be suitable to work by the means of nitrogen as well as compressed air and the result shall be given accurate.

The equipment shall be portable and easy to handle during idle conditions or as and when required.

6.0 FUNCTIONAL REQUIREMENTS

The system shall have the multi-level passwords and each level shall have its unique features.

System shall have provision to store multiple recipes for different type of filters and the number of recipes shall be at least 10.

System shall show the display during the time of test running for its stage and the pressure applied.

It should have the provision for making or modify entries for the variable details like filter serial number/ catalog number, operator name, Employee code, etc...

It should have periodic password expiry or facility to change the password at its own level as and when required.

The system shall have the feature to store the tested data for a period of time to have back up.

It shall also have the provision to transfer the data to other portable computers or data collection system.

The system shall be suitable for changing the UOM of the pressure as per requirement

The test report shall be customized as per requirement on site during installation.

The system shall also have facility to check the integrity of filters by other methods like Water intrusion test, diffusion method, hydro corr test, air flow measurement etc...

7.0 Data collection and reporting

The following parameters should be recorded and stored in the system

automatically while complying to the norms of 21CFR part 11 whichever is applicable.

Recipe provision / data saving / data back-up / data security requirements:

Recipe provision required Yes ☐ No ☐

Data saving facility required Yes ☐ No ☐

(If yes mention the device for saving: _____ / Supplier to specify)

Data back-up required (during power failure) Yes ☐ No ☐ No ☐

(If yes mention the device for saving: _____ / Supplier to specify)

Data back-up required (during life-cycle) Yes ☐ No ☐

(If yes mention the device for saving: _____ / Supplier to specify)

Data security required (during life-cycle) Yes ☐ No ☐

(If yes mention the device for saving: _____ / Supplier to specify)

Performance Features

The system shall be suitable for operating at the following parameters

Condition	Limit
Temperature	23 ± 2 °C
RH	NMT 60%
Wetting solution	Methanol/ IPA/ Water
Gas	Compressed air/ Nitrogen
Operating Pressure	100 psi
Data reporting	Data with graph
Report type	Customized as per user requirement
Electric supply	3.5 Amps

Capacity

One filter at a time and shall complete the cycle with minimum possible time

Data backup for recipe storage is at least 10 and test reports shall be stored for 500 No's

Automation and safety features

The recipe shall be designed and saved, can be retrieved for execution.

The system shall not be affected due to the wetting solution or gases being used and it shall have provision to protect against them.

The testing shall be done automatically for all steps once the set up done and commanded for start the test.

It shall be protected with the different level passwords.

The software of the system shall be protected for virus attacks

The Integrity testing equipment should have a PLC/HMI that includes all process instruments to control the critical parameters associated with the Integrity testing equipment and also enable the operator to run auto & manual operation as per process and instrumentation diagram flow.

The control system features considered for control system should comply with the 21 CFR part 11 norms

The System should have at least three different access levels.

Operator

View/ Load recipe

Entry of Filter details

Start or abort the test

System status (e.g. Ready, Running) shall be displayed

Print reports

Supervisor

View / Load recipe

Creation/ deletion of recipe

Review the backup data

Print the reports (Current and back up)

Administrator

Password management

Modification of Functional configuration

Recipe modification

Considerations for electronic compliance

The electrical system of the equipment shall be housed as per the cGMP and cGEP standards, with adequate safety.

8.0 Drawings

All drawings to be provided in AutoCAD and hard copy prior to supply for approval

Expected documentation

The Supplier shall provide the documentation for preliminary review.

The Supplier shall provide documentation reflecting “as-built” condition with final delivery.

Following documents shall be delivered by the suppliers during the procurement life cycle:

S.No	Document details	Required (✓/ ✗)
1	Design Specification document (Hardware)	
2	Functional Specification (operational details)	
3	PLC Alarm/Interlock/Safety/ communication/power	
	failure test procedures	
4	Process and Instrumentation Diagram (P&ID)	
5	Instrument Listing with tag numbers	
6	Control Schematics (in case of PLC logic diagrams)	
7	Control Panel Assembly Drawings	
8	Machine Assembly Drawings	
9	Bill of Materials	
10	Operator, Maintenance and Service Manuals	
11	Spare Parts List	
12	MOC certificates	
13	Calibration certificates of instruments	
14	Test certificates of components/test devices	
15	Air Flow Diagrams	
17	GA drawing	
18	Isometric drawing	
19	Electrical drawings	
20	Component Cut Sheets	
22	PLC Program Printout and Disk File	
23	HMI Configuration Printout and Disk File	
24	List of consumables	
25	Other (Specify)	

✓: Applicable & required

✗: Not applicable

9.0 Utilities available

Power, Compressed air and nitrogen is available at user point.

10.0 Training requirements

Vendor to provide formal operational training of equipment during commissioning

11.0 Other requirements

All documents must be provided in English

12.0 ABBREVIATIONS:

Abbreviation	Description
cGMP	Current Good Manufacturing Practices
GA	General arrangement
HMI	Human machine interface
MOC	Material of construction
NA	Not applicable
PLC	Programmable Logic control
SS	Stainless Steel
IPA	Iso propyl alcohol
UOM	Unit of measurement

- Counter () ♦
- Temperature () ♦
- Velocity/CFM () ♦
- LOD () ♦
- Dew point () ♦
- RPM () ♦

(Give details of printouts required)

- Time () ♦
- Pressure () ♦
- Cuts/min () ♦
- Alarms and Warnings. () ♦
- System Status (e.g. "off,"

[?] [?] [?]
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 [?] "on," ♦ "standby" states, etc.).
 Relative Humidity () ♦
 Other (Specify: _____ (_____) ♦

USER REQUIREMENT SPECIFICATION
(URS)

FOR
COLD ROOM

Approvals of the URS:

Department | Name | Signature | Date

Project: BioBridge Healthcare Solutions Pvt. Ltd., India | Project: BioBridge Healthcare Solutions Pvt. Ltd., India | Project: BioBridge Healthcare Solutions Pvt. Ltd., India | Project: BioBridge Healthcare Solutions Pvt. Ltd., India

Prepared by: | Manoj Kumar | | 18.07.2023

Checked/Reviewed by: | Rakesh Kumar | | 18.07.2023

INFARMA – Indústria Farmacêutica, Lda | INFARMA – Indústria Farmacêutica, Lda | INFARMA – Indústria Farmacêutica, Lda | INFARMA – Indústria Farmacêutica, Lda

Reviewed by: | | |

Reviewed by: | | |

Approved by: | | |

1.0 | Name of Equipment | Cold Room

2.0 | Purpose of Use | Cold room shall be used for storage of vaccine vials

3.0 | Justification | It's required for the storage of vaccine vials

4.0 | Equipment Details | Technical Specification of Room: -

Room Size: 7.50 (L) x 4.03(W) x 3.0(H) in meter

Room temp: 2 To 8°C

Surrounding building temp: 37±1°C

Specification of PUF panels: -

PUF Panels: -

60 mm Thick PUF Panel with cam lock 0.45 mm Thick PPGI Sheet on both side of the surface Inclusive

Flashing Silicon Sealant, Aluminum Pop Rivets and Cam Lock Plug, With Bubble wrapping

Manual Sliding door with lock

Manual Sliding Door W 1500 mm x H 2100 mm Single leaf Heavy duty 60 mm thickness for Positive temp.

(Density 42kg/m³) with both side 0.6 mm PPGI Sheet Heavy duty Aluminum Track Aluminum Door Profile, SS

304 Door Opener & Closure EPDM

Gasket Cladding for Installation on 60 mm Thick Panels SS 304 Integrated Pad Lock,

PUF Slab

60 mm Thick PUF Slab with One side Trafelt

Lighting: LED type light

Working Condition: - 2-8 °C

5.0 | Construction Material | Prefabricated Construction

Panel MOC : PPGI

Type of insulation : PUF

Density of panels : 40 +/- 2 Kg/m³

Thickness of insulation : 60 mm

Type of panel : free Standing

Wall lamination : Inside/Outside-PPGI

Ceiling lamination : Inside/Outside-PPGI

6.0 | Component Details | Refrigeration System:

Should designed for Summer / Monsoon seasons

Indoor design 2°C

Operation design for 24hrs

Control Logic: -

The evaporator fan is interconnected with the compressor. First the evaporator fan starts, if the HP/LP switch and thermostat are all found healthy then only compressor will start. This is to protect the compressor. In case, if any of these are not working, the compressor will not start.

The compressor is interconnected with the temperature controller. The sensor linked with temperature controller senses the return air temp in room, if the temp is below the set parameter the compressor will trip and vise – versa.

The thermostatic expansion valve is provided in between the condenser and near evaporator unit. The need of expansion valve is to regulate the flow of refrigerant as per load in room. The probe connected to valve is mounted just after the evaporator, it senses the refrigerant required as per the requirement the valve regulate.

For the safety of compressor, the High pressure and Low-pressure switch is provided in condenser unit. If the discharge pressure is high the compressor will trip and if the suction pressure is low, the compressor will trip automatically.

7.0 | Utility Requirements | 230 Volt AC / 1 Phase, 50 Hz

8.0 | Accessories | Inclusive of all accessories as required for normal operation of equipment

9.0 | Documents Required | Complete documentation Package with following documents should be delivered at least one month in advance.

Design Qualification (DQ)

Installation Qualification (IQ)

Operational Qualification (OQ)

Performance Qualification (PQ)

Operational manual (2 Sets)

Maintenance manual (2 Sets)

Spare part list

Other certificates should be delivered after performing relevant test/calibration before installation

10.0 | Training | The vendor shall impart successful training for operation and maintenance for at least two personnel from user department and engineering /projects department and shall provide evaluation report.

11.0 | Commissioning scope | The supplier/vendor shall provide complete commissioning of the cold room. The supplier/vendor shall ensure that the equipment is fully functional and meets all the specified requirements.

12.0 | Operational requirements | Equipment should be user-friendly

13.0 | Startup support | The supplier/vendor shall assist with the startup of the equipment

14.0 | Deviations | Any deviations or modifications to the standard machine specifications must be clearly communicated and documented.

Deviations should not negatively impact the overall performance and functionality

15.0 | Cleaning procedures | The supplier/vendor shall provide detailed instructions on cleaning and maintenance procedures.

16.0 | Packaging | The equipment and its components should be packaged securely to prevent any damage during transportation.

The supplier/vendor should use suitable packaging materials and techniques to ensure safe delivery.

17.0 | Delivery | The supplier/vendor shall arrange for the safe transportation and delivery to the specified location.

They should provide a clear delivery schedule and keep the user informed of any delays or changes.

USER REQUIREMENT SPECIFICATION

(URS)

FOR

DYNAMIC PASSBOX

Approvals of the URS:

Department | Name | Signature | Date

Project: BioBridge Healthcare Solutions Pvt. Ltd., India | Project: BioBridge Healthcare Solutions Pvt.

Ltd., India | Project: BioBridge Healthcare Solutions Pvt. Ltd., India | Project: BioBridge Healthcare

Solutions Pvt. Ltd., India

Prepared by: | Manoj Kumar | | 03.07.2023

Checked/Reviewed by: | Rakesh Kumar | | 03.07.2023

INFARMA – Indústria Farmacêutica, Lda | INFARMA – Indústria Farmacêutica, Lda | INFARMA – Indústria

Farmacêutica, Lda | INFARMA – Indústria Farmacêutica, Lda

Reviewed by: | | |

Reviewed by: | | |

Approved by: | | |

1.0 | Name of Equipment | Dynamic pass box

2.0 | Purpose of Use | Dynamic pass boxes used to maintain the cleanliness and integrity of controlled environments

3.0 | Justification | The primary function of a dynamic pass box is to maintain the cleanliness and integrity of controlled environments.

4.0 | Equipment Details | The dynamic Pass box is provided with electromagnetic interlocking arrangement for doors. The electromagnetic locking device is maintained with help of two Electric magnet fixed on each door. As soon as one door is opened the opposite side's magnet lock remain activated & door gets lock into pass box and white light will get "ON". And when both doors are in locked condition white light will get "OFF"

5.0 | Construction Material | All parts to be designed and manufactured in line with the current standards

6.0 | Component Details | Should be over all size of DPB 1170(W) x 1170 (D) x 1590(H) MM

Should have Fresh air pre-filters Box type, EU-7 rating, 3 µ

Should have Return air pre-filters Box type, EU-4 rating, 10 µ

Should have HEPA filters Box type, EU-13 rating, 0.3 µ

Should have centrifugal type blower

Should have pass box controller

Should have provision of LED type tube light

Should have electromagnetic lock

Should have door release button with status indication

Should have magnehelic gauges across HEPA filters

Should have DOP/PAO port

Should have MCB

Should be noise level of the dynamic pass box is less than 75dB at 1 meter from Dynamic pass box

Should have provision of Alarm warnings system

7.0 | Utility Requirements | Electrical Supply: 1 Phase + Neutral + Earthing

8.0 | Accessories | Inclusive of all accessories as required for normal operation of equipment

9.0 | Documents Required | Complete documentation Package with following documents should be delivered at least one month in advance.

Design Qualification (DQ)

Installation Qualification (IQ)

Operational Qualification (OQ)

Performance Qualification (PQ)

Operational manual (2 Sets)

Maintenance manual (2 Sets)

Spare part list

Other certificates should be delivered after performing relevant test/calibration before installation

10.0 | Training | The vendor shall impart successful training for operation and maintenance for at least two personnel from user department and engineering /projects department and shall provide evaluation report.

11. | Commissioning scope | The supplier/vendor shall provide complete commissioning of the dynamic pass box.

The supplier/vendor shall ensure that the equipment is fully functional and meets all the specified requirements.

12. | Operational requirements | Equipment should be user-friendly

13. | Startup support | The supplier/vendor shall assist with the startup of the equipment

14. | Deviations | Any deviations or modifications to the standard machine specifications must be clearly communicated and documented.

Deviations should not negatively impact the overall performance and functionality

15. | Cleaning procedures | The supplier/vendor shall provide detailed instructions on cleaning and maintenance procedures.

16. | Packaging | The equipment and its components should be packaged securely to prevent any damage during transportation.

The supplier/vendor should use suitable packaging materials and techniques to ensure safe delivery.

17. | Delivery | The supplier/vendor shall arrange for the safe transportation and delivery to the specified location.

They should provide a clear delivery schedule and keep the user informed of any delays or changes.

USER REQUIREMENT SPECIFICATION
(URS)

For

AIR HANDLING UNIT(AHU)- 01

Approvals of the URS:

Department | Name | Signature | Date

Project: BioBridge Healthcare Solutions Pvt. Ltd., India | Project: BioBridge Healthcare Solutions Pvt. Ltd., India | Project: BioBridge Healthcare Solutions Pvt. Ltd., India | Project: BioBridge Healthcare Solutions Pvt. Ltd., India

Prepared by: | Manoj Kumar | | 03.07.2023

Checked/Reviewed by: | Rakesh Kumar | | 03.07.2023

INFARMA – Indústria Farmacêutica, Lda | INFARMA – Indústria Farmacêutica, Lda | INFARMA – Indústria Farmacêutica, Lda | INFARMA – Indústria Farmacêutica, Lda

Reviewed by: | | |

Reviewed by: | | |

Approved by | | |

1.0 | Name of Equipment | Air Handling Unit (AHU) - 01

2.0 | Purpose of Use | Air Handling Units (AHU) use a critical role in maintaining controlled environments within biopharmaceutical industries. These industries require strict environmental conditions to ensure product quality, safety, and compliance with regulatory standards.

3.0 | Justification | A User Requirement Specification (URS) for an Air Handling Unit (AHU) is a critical document in the design and procurement process of HVAC (Heating, Ventilation, and Air Conditioning) systems. This document outlines the specific requirements and expectations of the AHU to ensure it meets the needs of the facility and its users.

4.0 | Equipment Details | The equipment shall be mounted on service floor, temperature range of 15 to 40 °C.

5.0 | Construction Material | All parts to be designed and manufactured in line with the current standards

6.0 | Component Details | Should be capacity of AHU-01 11000 CFM

Should be Double skin Floor Mounted Air Handling Unit

Should have outer skin 0.8 mm G.I pre coated with guard film

Should have inner skin 0.8 mm G.I plain sheet

Should have panel thickness 43 mm thick double skin panel with PUF

Should be frame structure aluminum extruded anodized profile with thermal break & internal coving

Should be color of AHU RAL-5012(Sky Blue)

Should have C' Type Glass filled nylon handle

Should have damper M.O.C aluminum

Should have cooling coil Fin & Tube Type Coil for Chilled Water Application

Should have drain pan provision with M.O.C S.S.304 With 19mm thick close cell Nitrile rubber Insulation

Should have pre-filters in mixing chamber with Efficiency 90% - 10µ (M-5)

Should have fresh air filter on fresh air opening with Efficiency 90% - 10µ (G-4)

Should have bleed air filter on top Bleed damper opening with Efficiency 90-95% - 5µ (F-8)

Should have fine filter on Supply Plenum in AHU with Efficiency 90-95% - 5µ (F-8)

Should have volume control dampers (Manual Operated)

Should have fire damper to be provided with limit switch and wiring up to control panel to have interlock with AHU.

Should have magnehelic gauge across pre-filters 0-25 MMWC range

Should have magnehelic gauge across fine-filters 0-50 MMWC range

Should be fan plug with FVD

| | Details of Duct

Should be fabricated type ducts

Should be thermal insulation of duct 25 mm thick on supply duct and 13 mm thick on return duct

Should be Closed cell nitrile rubber insulation with one side aluminum foil

| | Chilled Water Manifold Details

Should be piping of manifold size 65 NB

Should have Y-type strainer

Should have Butterfly valves

Should have 3-way valves

Should have Temperature Gauge 0 to 50 0C, 02 Nos

Should have pressure Gauge 0 to 7 Kg/cm², 02Nos

7.0 | Utility Requirements | Electrical Supply: 3 Phase + Neutral + Earthing, Included VFD

8.0 | Accessories | Inclusive of all accessories as required for normal operation of equipment

9.0 | Documents Required | Complete documentation Package with following documents should be delivered at least one month in advance.

Design Qualification (DQ)

Installation Qualification (IQ)

Operational Qualification (OQ)

Performance Qualification (PQ)

General Arrangement Drawing (GA) of AHU

Operational manual (2 Sets)

Maintenance manual (2 Sets)

Spare part list

Other certificates should be delivered after performing relevant test/calibration before installation

10.0 | Training | The vendor shall impart successful training for operation and maintenance for at least two personnel from user department and engineering /projects department and shall provide evaluation report.

11.0 | Commissioning scope | The supplier/vendor shall provide complete commissioning of the Air handling unit This includes installation, setup.

The supplier/vendor shall ensure that the machine is fully functional and meets all the specified requirements.

12.0 | Operational requirements | The machine should have user-friendly controls for operators to set and adjust various parameters such as In, out temperature, air balancing and other relevant settings. These controls should be easily accessible.

Post installation support

The supplier/vendor shall provide post-installation support for the AHU.

13.0 | Startup support | The supplier/vendor shall assist with the startup of the machine after installation.

They should provide guidance on initial facility runs and help optimize the machine settings.

14.0 | Deviations | Any deviations or modifications to the standard machine specifications must be clearly communicated and documented.

Deviations should not negatively impact the overall performance and functionality of the machine.

15.0 | Cleaning procedures | The supplier/vendor shall provide detailed instructions on cleaning and maintenance procedures.

16.0 | Packaging | The AHU and its components should be packaged securely to prevent any damage during transportation.

The supplier/vendor should use suitable packaging materials and techniques to ensure safe delivery.

17.0 | Delivery | The supplier/vendor shall arrange for the safe transportation and delivery of the AHU to the specified location.

They should provide a clear delivery schedule and keep the user informed of any delays or changes.

USER REQUIREMENT SPECIFICATION (URS)

For

AIR HANDLING UNIT(AHU)- 02

Approvals of the URS:

Department | Name | Signature | Date

Project: BioBridge Healthcare Solutions Pvt. Ltd., India | Project: BioBridge Healthcare Solutions Pvt.

Ltd., India | Project: BioBridge Healthcare Solutions Pvt. Ltd., India | Project: BioBridge Healthcare

Solutions Pvt. Ltd., India

Prepared by: | Manoj Kumar | | 03.07.2023

Checked/Reviewed by: | Rakesh Kumar | | 03.07.2023

INFARMA – Indústria Farmacêutica, Lda | INFARMA – Indústria Farmacêutica, Lda | INFARMA – Indústria

Farmacêutica, Lda | INFARMA – Indústria Farmacêutica, Lda

Reviewed by: | | |

Reviewed by: | | |

Approved by: | | |

1.0 | Name of Equipment | Air Handling Unit (AHU) - 02

2.0 | Purpose of Use | Air Handling Units (AHU) use a critical role in maintaining controlled environments within biopharmaceutical industries. These industries require strict environmental conditions to ensure product quality, safety, and compliance with regulatory standards.

3.0 | Justification | A User Requirement Specification (URS) for an Air Handling Unit (AHU) is a critical document in the design and procurement process of HVAC (Heating, Ventilation, and Air Conditioning) systems. This document outlines the specific requirements and expectations of the AHU to ensure it meets the needs of the facility and its users.

4.0 | Equipment Details | The equipment shall be mounted on service floor, temperature range of 15 to 40 °C.

5.0 | Construction Material | All parts to be designed and manufactured in line with the current standards

6.0 | Component Details | Should be capacity of AHU 4000 CFM

Should be Double skin Floor Mounted Air Handling Unit

Should have outer skin 0.8 mm G.I pre coated with guard film

Should have inner skin 0.8 mm G.I plain sheet

Should have panel thickness 43 mm thick double skin panel with PUF

Should be frame structure aluminum extruded anodized profile with thermal break & internal coving

Should be color of AHU RAL-5012(Sky Blue)

Should have C' Type Glass filled nylon handle

Should have damper M.O.C aluminum

Should have cooling coil Fin & Tube Type Coil for Chilled Water Application

Should have drain pan provision with M.O.C S.S.304 With 19mm thick close cell Nitrile rubber Insulation

Should have pre-filters in mixing chamber with Efficiency 90% - 10μ (M-5)

Should have fresh air filter on fresh air opening with Efficiency 90% - 10μ (G-4)

Should have bleed air filter on top Bleed damper opening with Efficiency 90-95% - 5μ (F-8)

Should have fine filter on Supply Plenum in AHU with Efficiency 90-95% - 5μ (F-8)

Should have volume control dampers (Manual Operated)

Should have fire damper to be provided with limit switch and wiring up to control panel to have interlock with AHU.

Should have magnehelic gauge across pre-filters 0-25 MMWC range

Should have magnehelic gauge across fine-filters 0-50 MMWC range

Should be plug fan type with FVD

| | Details of Duct

Should be fabricated type ducts

Should be thermal insulation of duct 25 mm thick on supply duct and 13 mm thick on return duct

Should be Closed cell nitrile rubber insulation with one side aluminum foil

| | Chilled Water Manifold Details

Should be piping of manifold size 40 NB

Should have Y-type strainer

Should have Butterfly valves

Should have 3-way valves

Should have Temperature Gauge 0 to 50 °C, 02 Nos

Should have pressure Gauge 0 to 7 Kg/cm², 02Nos

7.0 | Utility Requirements | Electrical Supply: 3 Phase + Neutral + Earthing, Included VFD controller.

8.0 | Accessories | Inclusive of all accessories as required for normal operation of equipment

9.0 | Documents Required | Complete documentation Package with following documents should be delivered at least one month in advance.

Design Qualification (DQ)

Installation Qualification (IQ)

Operational Qualification (OQ)

Performance Qualification (PQ)

General Arrangement Drawing (GA) of AHU

Operational manual (2 Sets)

Maintenance manual (2 Sets)

Spare part list

Other certificates should be delivered after performing relevant test/calibration before installation

10.0 | Training | The vendor shall impart successful training for operation and maintenance for at least two personnel from user department and engineering /projects department and shall provide evaluation report.

11.0 | Commissioning scope | The supplier/vendor shall provide complete commissioning of the Air handling unit This includes installation, setup.

The supplier/vendor shall ensure that the machine is fully functional and meets all the specified requirements.

12.0 | Operational requirements | The machine should have user-friendly controls for operators to set and adjust various parameters such as In, out temperature, air balancing and other relevant settings. These controls should be easily accessible.

Post installation support

The supplier/vendor shall provide post-installation support for the AHU.

13.0 | Startup support | The supplier/vendor shall assist with the startup of the machine after installation.

They should provide guidance on initial facility runs and help optimize the machine settings.

14.0 | Deviations | Any deviations or modifications to the standard machine specifications must be

clearly communicated and documented.

Deviations should not negatively impact the overall performance and functionality of the machine.

15.0 | Cleaning procedures | The supplier/vendor shall provide detailed instructions on cleaning and maintenance procedures.

16.0 | Packaging | The AHU and its components should be packaged securely to prevent any damage during transportation.

The supplier/vendor should use suitable packaging materials and techniques to ensure safe delivery.

17.0 | Delivery | The supplier/vendor shall arrange for the safe transportation and delivery of the AHU to the specified location.

They should provide a clear delivery schedule and keep the user informed of any delays or changes.

USER REQUIREMENT SPECIFICATION
(URS)

For

AIR HANDLING UNIT(AHU)- 03

Approvals of the URS:

Department | Name | Signature | Date

Project: BioBridge Healthcare Solutions Pvt. Ltd., India | Project: BioBridge Healthcare Solutions Pvt. Ltd., India | Project: BioBridge Healthcare Solutions Pvt. Ltd., India | Project: BioBridge Healthcare Solutions Pvt. Ltd., India

Prepared by: | Manoj Kumar | | 03.07.2023

Checked/Reviewed by: | Rakesh Kumar | | 03.07.2023

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Reviewed by: | | |

Reviewed by: | | |

Approved by | | |

1.0 | Name of Equipment | Air Handling Unit (AHU) - 03

2.0 | Purpose of Use | Air Handling Units (AHU) use a critical role in maintaining controlled environments within biopharmaceutical industries. These industries require strict environmental conditions to ensure product quality, safety, and compliance with regulatory standards.

3.0 | Justification | A User Requirement Specification (URS) for an Air Handling Unit (AHU) is a critical document in the design and procurement process of HVAC (Heating, Ventilation, and Air Conditioning) systems. This document outlines the specific requirements and expectations of the AHU to ensure it meets the needs of the facility and its users.

4.0 | Equipment Details | The equipment shall be mounted on service floor, temperature range of 15 to 40 °C.

5.0 | Construction Material | All parts to be designed and manufactured in line with the current standards

6.0 | Component Details | Should be capacity of AHU 2600 CFM

Should be Double skin Floor Mounted Air Handling Unit

Should have outer skin 0.8 mm G.I pre coated with guard film

Should have inner skin 0.8 mm G.I plain sheet

Should have panel thickness 43 mm thick double skin panel with PUF

Should be frame structure aluminum extruded anodized profile with thermal break & internal coving

Should be color of AHU RAL-5012(Sky Blue)

Should have C' Type Glass filled nylon handle

Should have damper M.O.C aluminum

Should have cooling coil Fin & Tube Type Coil for Chilled Water Application

Should have drain pan provision with M.O.C S.S.304 With 19mm thick close cell Nitrile rubber Insulation

Should have pre-filters in mixing chamber with Efficiency 90% - 10μ (M-5)

Should have fresh air filter on fresh air opening with Efficiency 90% - 10μ (G-4)

Should have bleed air filter on top Bleed damper opening with Efficiency 90-95% - 5μ (F-8)

Should have fine filter on Supply Plenum in AHU with Efficiency 90-95% - 5μ (F-8)

Should have volume control dampers (Manual Operated)

Should have fire damper to be provided with limit switch and wiring up to control panel to have interlock with AHU.

Should have magnehelic gauge across pre-filters 0-25 MMWC range

Should have magnehelic gauge across fine-filters 0-50 MMWC range

Should be plug fan type with FVD

| | Details of Duct

Should be fabricated type ducts

Should be thermal insulation of duct 25 mm thick on supply duct and 13 mm thick on return duct

Should be Closed cell nitrile rubber insulation with one side aluminum foil

| | Chilled Water Manifold Details

Should be piping of manifold size 40 NB

Should have Y-type strainer

Should have Butterfly valves

Should have 3-way valves

Should have Temperature Gauge 0 to 50 0C, 02 Nos

Should have pressure Gauge 0 to 7 Kg/cm2,02Nos

7.0 | Utility Requirements | Electrical Supply: 3 Phase + Neutral + Earthing, Included VFD controller.

8.0 | Accessories | Inclusive of all accessories as required for normal operation of equipment

9.0 | Documents Required | Complete documentation Package with following documents should be delivered at least one month in advance.

Design Qualification (DQ)

Installation Qualification (IQ)

Operational Qualification (OQ)

Performance Qualification (PQ)

General Arrangement Drawing (GA) of AHU

Operational manual (2 Sets)

Maintenance manual (2 Sets)

Spare part list

Other certificates should be delivered after performing relevant test/calibration before installation

10.0 | Training | The vendor shall impart successful training for operation and maintenance for at least two personnel from user department and engineering /projects department and shall provide evaluation report.

11.0 | Commissioning scope | The supplier/vendor shall provide complete commissioning of the Air handling unit This includes installation, setup.

The supplier/vendor shall ensure that the machine is fully functional and meets all the specified requirements.

12.0 | Operational requirements | The machine should have user-friendly controls for operators to set and adjust various parameters such as In, out temperature, air balancing and other relevant settings. These controls should be easily accessible.

Post installation support

The supplier/vendor shall provide post-installation support for the AHU.

13.0 | Startup support | The supplier/vendor shall assist with the startup of the machine after installation.

They should provide guidance on initial facility runs and help optimize the machine settings.

14.0 | Deviations | Any deviations or modifications to the standard machine specifications must be clearly communicated and documented.

Deviations should not negatively impact the overall performance and functionality of the machine.

15.0 | Cleaning procedures | The supplier/vendor shall provide detailed instructions on cleaning and maintenance procedures.

16.0 | Packaging | The AHU and its components should be packaged securely to prevent any damage during transportation.

The supplier/vendor should use suitable packaging materials and techniques to ensure safe delivery.

17.0 | Delivery | The supplier/vendor shall arrange for the safe transportation and delivery of the AHU to the specified location.

They should provide a clear delivery schedule and keep the user informed of any delays or changes.

USER REQUIREMENT SPECIFICATION
(URS)

For

AIR-COOLED SCREW CHILLER (70TR)

Approvals of the URS:

Department | Name | Signature | Date

Project: BioBridge Healthcare Solutions Pvt. Ltd., India | Project: BioBridge Healthcare Solutions Pvt.

Ltd., India | Project: BioBridge Healthcare Solutions Pvt. Ltd., India | Project: BioBridge Healthcare

Solutions Pvt. Ltd., India

Prepared by: | Manoj Kumar | | 06.07.2023

Checked/Reviewed by: | Rakesh Kumar | | 06.07.2023

INFARMA – Indústria Farmacêutica, Lda | INFARMA – Indústria Farmacêutica, Lda | INFARMA – Indústria

Farmacêutica, Lda | INFARMA – Indústria Farmacêutica, Lda

Reviewed by: | | |

Reviewed by: | | |

Approved by: | | |

1.0 | Name of Equipment | Air-cooled Screw chiller (70 TR)

2.0 | Purpose of Use | Air-cooled Screw chiller is use to temperature control or Heat Load handling for a facility.

3.0 | Justification | The justification for procuring 70 TR capacity Air Cooled Screw chiller units is based on their cooling capacity, energy efficiency, reliability, environmental considerations, scalability, and other factors that align with the specific requirements and goals of our project or facility

4.0 | Equipment Details | 70 TR capacity Air Cooled Screw chiller units comprising of sealed Screw compressor, Air Cooled condenser, evaporator, refrigerant piping and control device, micro-processor-based control panel etc. are to be installed at outside premised in provided space. Two nos. of Chilled water pumps (1 Working +1 Stand By) to be installed in same place.

5.0 | Construction Material | All parts to be designed and manufactured in line with the current standards

6.0 | Component Details | Air-cooled Screw chiller Capacity: 70 TR

Leaving Temperature : 7.00 °C

Entering Temperature. : 12 °C .

Compressor / Refrigerant : R410A

Condenser Type : Air cooled

Condenser Fan Quantity : 04

Air Flow Rate : 98542CMH

Compressor Type : Semi Hermetic Screw

Compressor Quantity : 01

Compressor Motor Speed : Vendor to specify

Evaporators Type : Tube type

Pump Type : Centrifugal

Noise Level : NMT 80 DB

Dia mention in mm(L*W*H) : 3198 * 2235 * 2432

Minimum Clearance Required : Vendor to specify

Expansion Valve : Thermostatic

7.0 | Utility Requirements | Electrical Supply: 3 Phase + Neutral + Earthing

8.0 | Accessories | Inclusive of all accessories as required for normal operation of equipment

9.0 | Documents Required | Complete documentation Package with following documents should be delivered at least one month in advance.

Design Qualification (DQ)

Installation Qualification (IQ)

Operational Qualification (OQ)

Performance Qualification (PQ)

Operational manual (2 Sets)

Maintenance manual (2 Sets)

Spare part list

Other certificates should be delivered after performing relevant test/calibration before installation

10.0 | Training | The vendor shall impart successful training for operation and maintenance for at least two personnel from user department and engineering /projects department and shall provide evaluation report.

11.0 | Commissioning scope | The supplier/vendor shall provide complete commissioning of the Air-cooled Screw chiller This includes installation, setup, and integration of the machine to be installed at outside premised in provided space.

The supplier/vendor shall ensure that the machine is fully functional and meets all the specified requirements.

12.0 | Operational requirements | The machine should have user-friendly controls for operators to set and adjust various parameters such as In, out temperature and other relevant settings. These controls should be easily accessible

Post installation support

The supplier/vendor shall provide post-installation support for the Air-cooled Screw chiller.

13.0 | Startup support | The supplier/vendor shall assist with the startup of the machine after installation.

They should provide guidance on initial production runs and help optimize the machine settings.

14.0 | Deviations | Any deviations or modifications to the standard machine specifications must be clearly communicated and documented.

Deviations should not negatively impact the overall performance and functionality of the machine.

15. | Cleaning procedures | The supplier/vendor shall provide detailed instructions on cleaning and maintenance procedures.

16.0 | Packaging | The Air-cooled Screw chiller should be packaged securely to prevent any damage during transportation.

The supplier/vendor should use suitable packaging materials and techniques to ensure safe delivery.

17.0 | Delivery | The supplier/vendor shall arrange for the safe transportation and delivery of the Air-cooled Screw chiller to the specified location.

They should provide a clear delivery schedule and keep the user informed of any delays or changes.

USER REQUIREMENT SPECIFICATION
(URS)

FOR

VIAL STICKER LABELLING MACHINE

Capacity – 150- 250 Vial/ minute

Approvals of the URS:

Department | Name | Signature | Date

Project: BioBridge Healthcare Solutions Pvt. Ltd., India | Project: BioBridge Healthcare Solutions Pvt.

Ltd., India | Project: BioBridge Healthcare Solutions Pvt. Ltd., India | Project: BioBridge Healthcare

Solutions Pvt. Ltd., India

Prepared by: | Manoj Kumar | | 21.07.2023

Checked/Reviewed by: | Rakesh Kumar | | 21.07.2023

INFARMA – Indústria Farmacêutica, Lda | INFARMA – Indústria Farmacêutica, Lda | INFARMA – Indústria

Farmacêutica, Lda | INFARMA – Indústria Farmacêutica, Lda

Reviewed by: | | |

Reviewed by: | | |

Approved by | | |

1.0 | Name of Equipment | Vial Sticker Labelling Machine

2.0 | Purpose of Use | The equipment being defined is intended to receive product Vial from an infeed conveyor, apply a label, overprint batch coding details and discharge the Vial onto an out feed conveyor. The equipment will provide label for Vial transport through the machine.

The labeler system will be operated adjacent to conveyor systems on an integrated packaging line and will be located in a designated operating area within a CGMP packaging operation.

3.0 | Justification | This URS defines the automatic labeling machine to place pressure sensitive labels on vaccine Vials.

Applying a pressure sensitive label to vials.

Printing product information on the label.

Inspecting pre-printed and printed characters on the label.

4.0 | Equipment Details | The equipment shall be mounted in Packing area with controlled environment, temperature range of 15 to 40 °C.

5.0 | Construction Material | All parts to be designed and manufactured in line with the Mild steel structure clad with stainless steel 304 sheet to meet GMP norms.

6.0 | Machine Details | Should have Output Speed: 150 to 250 Products/Minute (Speed depends on label length and products size)

Should have Dimensions: Approx. 2135 mm L x 1230 mm W x 875 + 50 mm H.

Should have Net Weight Approx. 450 Kg

Should have Label Length: 30 mm to 180 mm

Should have Label Width: 16 mm to 100 mm

Should have Label Roll Diameter: Max. 300 mm X ID 75 mm.

Should have Round Glass Vials: 2 mL to 10 mL.

Should have Label Roll Empty - Machine Stop System with Alarm.

Should have Missing Label in Roll - Machine Stop System with Alarm.

| | | Should have Special Label Sensor for sensing Transparent Film Labels which does not have any sensing mark on it.

Should have Option of Thermal Ink Jet Printer with various speed ranges available.

Toughened Glass or Acrylic Safety Cabinet for Machine.

Should have tower Light showing the machine status / operation.

Should have Color or Mono Touch Screen with PLC for machine operation and showing various error messages and information on display.

Should be Special Sensing system for checking missing label on product.

Should have be Low Compressed Air - Machine Stop with Alarm system for pneumatically operated coding system

Should have Vision System for checking Barcode, presence or absence of Coding or pharma code on labels.

Should have Online pneumatically operated product rejection system.

7.0 | 7.0 | Utility Requirements | Electrical supply: 1 Phase + Neutral + Earthing

Electrical load: 1 KW

8.0 | 8.0 | Accessories | All accessories should be included as required for normal operation of equipment

9.0 | 9.0 | Documents Required | Complete documentation Package with following documents should be delivered at least one month in advance.

Design Qualification (DQ)

Installation Qualification (IQ)

Operational Qualification (OQ)

Performance Qualification (PQ)

Operational manual (2 Sets)

Maintenance manual (2 Sets)

Spare part list

Other certificates should be delivered after performing relevant test/calibration before installation

10.0 | 10.0 | Training | The vendor shall impart successful training for operation and maintenance for at least two personnel from user department and engineering /projects department and shall provide evaluation report.

11.0 | 11.0 | Commissioning scope | The supplier/vendor shall provide complete commissioning of the Vial Sticker Labelling Machine.

This includes installation, setup, and integration of the machine into the existing production line.

The supplier/vendor shall ensure that the machine is fully functional and meets all the specified requirements.

12.0 | 12.0 | Operational requirements | The Vial Sticker Labelling Machine should have a capacity of 150-250 vials per minute.

It should be capable of accurately applying labels to vials of different sizes and shapes.

The machine should have a user-friendly interface for easy operation and control.

It should be compatible with the existing production line and integrate smoothly with other equipment

Post installation support

The supplier/vendor shall provide post-installation support for the Vial Sticker Labelling Machine.

13.0 | 13.0 | Startup support | The supplier/vendor shall assist with the startup of the machine after installation.

They should provide guidance on initial production runs and help optimize the machine settings.

14.0 | 14.0 | Deviations | Any deviations or modifications to the standard machine specifications must be clearly communicated and documented.

Deviations should not negatively impact the overall performance and functionality of the machine.

15.0 | 15.0 | Cleaning procedures | The supplier/vendor shall provide detailed instructions on cleaning

and maintenance procedures.

16.0 | 16.0 | Packaging | The Vial Sticker Labelling Machine should be packaged securely to prevent any damage during transportation.

The supplier/vendor should use suitable packaging materials and techniques to ensure safe delivery.

17.0 | 17.0 | Delivery | The supplier/vendor shall arrange for the safe transportation and delivery of the Vial Sticker Labelling Machine to the specified location.

They should provide a clear delivery schedule and keep the user informed of any delays or changes.

USER REQUIREMENT SPECIFICATION

(URS)

FOR

DE-DUSTING TUNNEL

Approvals of the URS:

Department | Name | Signature | Date

Project: BioBridge Healthcare Solutions Pvt. Ltd., India | Project: BioBridge Healthcare Solutions Pvt.

Ltd., India | Project: BioBridge Healthcare Solutions Pvt. Ltd., India | Project: BioBridge Healthcare

Solutions Pvt. Ltd., India

Prepared by: | Manoj Kumar | | 21.07.2023

Checked/Reviewed by: | Rakesh Kumar | | 21.07.2023

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Farmacêutica, Lda | INFARMA – Indústria Farmacêutica, Lda

Reviewed by: | | |

Reviewed by: | | |

Approved by: | | |

1.0 | Name of Equipment | De-Dusting tunnel

2.0 | Purpose of Use | The Equipment will be used for online removing of dust.

3.0 | Justification | This URS defines the de-dusting tunnel machine used for removing of dust.

4.0 | Equipment Details | The equipment shall be mounted in controlled environment, temperature range of 15 to 40 °C.

5.0 | Construction Material | All parts to be designed and manufactured in line with mild steel structure clad with stainless steel 304 sheet to meet GMP norms

6.0 | Component Details | Should have Combination Filter Efficiency 5 Micron & 10 Micron.

Should have Pre Filter:- EU 6, 95% Down to 5 Micron.

Should have Supply Motor Blower.

Should have Exhaust Motor Blower.

Should have SS 304 Dust Collector Tray.

Should have PVC Curtains Front & Back Side.

Should have Emergency Switch:- Entry & Exit Side.

Should have SS Air Nozzle.

Should have SS Perforation Grill.

Should have Exhaust Duct.

7.0 | Utility Requirements | Electrical supply: 1 Phase + Neutral + Earthing

8.0 | Accessories | All accessories should be included as required for normal operation of equipment

9.0 | Documents Required | Complete documentation Package with following documents should be delivered at least one month in advance.

Design Qualification (DQ)

Installation Qualification (IQ)

Operational Qualification (OQ)

Performance Qualification (PQ)

Operational manual (2 Sets)

Maintenance manual (2 Sets)

Spare part list

Other certificates should be delivered after performing relevant test/calibration before installation.

10.0 | Training | The vendor shall impart successful training for operation and maintenance for at least two personnel from user department and engineering /projects department and shall provide evaluation report.

11.0 | Commissioning scope | The supplier/vendor shall provide complete commissioning of the DE-Dusting tunnel.

The supplier/vendor shall ensure that the machine is fully functional and meets all the specified requirements.

12.0 | Operational requirements | The de-dusting tunnel should meet applicable regulatory standards and environmental requirements. The operational requirement would specify the need for compliance with relevant regulations to ensure the system's legality and adherence to environmental guidelines.

Post installation support

The supplier/vendor shall provide post-installation support for the DE-Dusting tunnel.

13.0 | Startup support | The supplier/vendor shall assist with the startup of the machine after installation.

14.0 | Deviations | Any deviations or modifications to the standard machine specifications must be clearly communicated and documented.

Deviations should not negatively impact the overall performance and functionality of the machine.

15.0 | Cleaning procedures | The supplier/vendor shall provide detailed instructions on cleaning and maintenance procedures

16.0 | Packaging | The DE-Dusting tunnel should be packaged securely to prevent any damage during transportation.

The supplier/vendor should use suitable packaging materials and techniques to ensure safe delivery.

17.0 | Delivery | The supplier/vendor shall arrange for the safe transportation and delivery of DE-Dusting tunnel to the specified location.

They should provide a clear delivery schedule and keep the user informed of any delays or changes.

USER REQUIREMENT SPECIFICATION

(URS)

FOR

ELECTRIC STACKER

Approvals of the URS:

Department | Name | Signature | Date

Project: BioBridge Healthcare Solutions Pvt. Ltd., India | Project: BioBridge Healthcare Solutions Pvt.

Ltd., India | Project: BioBridge Healthcare Solutions Pvt. Ltd., India | Project: BioBridge Healthcare

Solutions Pvt. Ltd., India

Prepared by: | Manoj Kumar | | 06.07.2023

Checked/Reviewed by: | Rakesh Kumar | | 06.07.2023

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Farmacêutica, Lda | INFARMA – Indústria Farmacêutica, Lda

Reviewed by: | | |

Reviewed by: | | |

Approved by: | | |

1.0 | Name of Equipment | Electric Stacker

2.0 | Purpose of Use | The electric stacker used to safe movement of materials and products. Its use can

contribute to improved productivity, reduced operational costs, and enhanced safety and compliance.

3.0 | Justification | An electric stacker can significantly enhance the efficiency of material handling and stacking tasks. It allows for quicker and more precise movements of goods, reducing downtime and increasing overall productivity.

4.0 | Equipment Details | It is used in warehouses and distribution centers to lift and move loads vertically and horizontally

5.0 | Construction Material | All parts to be designed and manufactured in line with the current standards

6.0 | Component Details | Should be fully Electric drive

Should be maximum capacity 2000 Kg

Should be lift height 4500 mm

Should be wheel type "polyurethane"

Should be Stacker travel speed (Laden / Unladen) 5/5.3 Km/h

Should be lift speed (Laden/ unladen) 58/148 mm/sec

Should have Electromagnetic Breaks

Should have AC traction motor

Should have 1.5 KW drive motor

Should have 3 KW lift motor

Should have safety belly button switch for operators' safety

Should have battery discharge indicator

Should have provision of quick battery change system

Should have 24 V and 210 AH capacity of battery

Should have provision of battery charging port

Should have Electric power steering system for smooth operation while maneuvering in confined area

Should have provision of control system

7.0 | Utility Requirements | NA

8.0 | Accessories | Inclusive all accessories as required for operation of Electric stacker

9.0 | Documents Required | Complete documentation Package with following documents should be delivered at least one month in advance.

Operational manual (2 Sets)

Maintenance manual (2 Sets)

10.0 | Training | The vendor shall impart training for operation and maintenance

11.0 | Commissioning scope | The supplier/vendor shall provide complete commissioning of the Electric stacker This includes installation, setup.

The supplier/vendor shall ensure that the stacker is fully functional and meets all the specified requirements.

12.0 | Operational requirements | The stacker should have user-friendly controls for operators to set relevant settings. These controls should be easily accessible.

13.0 | Startup support | The supplier/vendor shall assist with the startup of the stacker after setup.

They should provide guidance on initial facility runs and help optimize the machine settings

14.0 | Deviations | Any deviations or modifications to the standard machine specifications must be clearly communicated and documented.

Deviations should not negatively impact the overall performance and functionality of the stacker.

15.0 | Cleaning procedures | The supplier/vendor shall provide detailed instructions on cleaning and maintenance procedures.

16.0 | Packaging | The stacker and its components should be packaged securely to prevent any damage during transportation.

The supplier/vendor should use suitable packaging materials and techniques to ensure safe delivery.

17.0 | Delivery | The supplier/vendor shall arrange for the safe transportation and delivery of the stacker to the specified location.

They should provide a clear delivery schedule and keep the user informed of any delays or changes.

USER REQUIREMENT SPECIFICATION
(URS)

For

AIR COMPRESSOR

Approvals of the URS:

Department | Name | Signature | Date

Project: BioBridge Healthcare Solutions Pvt. Ltd., India | Project: BioBridge Healthcare Solutions Pvt.

Ltd., India | Project: BioBridge Healthcare Solutions Pvt. Ltd., India | Project: BioBridge Healthcare

Solutions Pvt. Ltd., India

Prepared by: | Manoj Kumar | | 06.07.2023

Checked/Reviewed by: | Rakesh Kumar | | 06.07.2023

INFARMA – Indústria Farmacêutica, Lda | INFARMA – Indústria Farmacêutica, Lda | INFARMA – Indústria

Farmacêutica, Lda | INFARMA – Indústria Farmacêutica, Lda

Reviewed by: | | |

Reviewed by: | | |

Approved by: | | |

1.0 | Name of Equipment | Air-cooled Screw Air Compressor unit

2.0 | Purpose of Use | Compressed air is commonly used for instrumentation and control systems within pharmaceutical manufacturing facilities. It provides a clean and reliable source of pneumatic power for various sensors, actuators, and control devices used in process automation, ensuring precise and consistent operations.

3.0 | Justification | In pharmaceutical manufacturing, maintaining aseptic conditions is paramount to prevent contamination and ensure product quality. Compressed air can be used for aseptic operations, such as sterilizing equipment, providing clean air for various sensors, actuators, and control devices used in process automation, ensuring precise and consistent operations.

4.0 | Equipment Details | 21 CFM capacity Air Cooled Screw air compressor units comprising of sealed Screw compressor, including refrigerant air dryer & receiver. are to be installed at outside premises in provided space.

5.0 | Construction Material | All parts to be designed and manufactured in line with the current standards

6.0 | Component Details | Should be screw type air compressor.

Should be actual capacity of Air compressor 21 CFM

Should be compressor receiver capacity 200 liters

Should be refrigerant type air dryer

Should be pressure of air compressor 10kg/cm2

Should have dry Paper Type Suction Air Filter with Silencer

Should have three-way solenoid valve required for load/unload regulation of the compressor

Should have air / oil temperature switch

Should have provision to shut down the compressor in case of too high element outlet air temperature

Should have sight glass for oil level indication and oil filling arrangement

Should have safety valve

Should have oil draining arrangement

Should have three stage Air Oil Separation system

Should have emergency stop button

7.0 | Utility Requirements | Electrical Supply: 3 Phase + Neutral + Earthing

8.0 | Accessories | Should be provided as required for operation of equipment

9.0 | Documents Required | Complete documentation Package with following documents should be delivered at least one month in advance.

Design Qualification (DQ)

Installation Qualification (IQ)

Operational Qualification (OQ)

Performance Qualification (PQ)

General Arrangement Drawing (GA)

Operational manual (2 Sets)

Maintenance manual (2 Sets)

Spare part list

Other certificates should be delivered after performing relevant test/calibration before installation

10.0 | Training | The vendor shall impart successful training for operation and maintenance for at least two personnel from user department and engineering /projects department and shall provide evaluation report.

11.0 | Commissioning scope | The supplier/vendor shall provide complete commissioning of the Air-cooled Screw compressor This includes installation, setup, and integration of the machine into the existing production line.

The supplier/vendor shall ensure that the machine is fully functional and meets all the specified requirements.

12.0 | Operational requirements | The machine should have user-friendly controls for operators to set and adjust various parameters such as In, out air pressure and other relevant settings. These controls should be easily accessible.

Post installation support

The supplier/vendor shall provide post-installation support for the Air-cooled Screw compressor.

13.0 | Startup support | The supplier/vendor shall assist with the startup of the machine after installation.

They should provide guidance on initial and help optimize the machine settings.

14.0 | Deviations | Any deviations or modifications to the standard machine specifications must be clearly communicated and documented.

Deviations should not negatively impact the overall performance and functionality of the machine.

15.0 | Cleaning procedures | The supplier/vendor shall provide detailed instructions on cleaning and maintenance procedures.

16.0 | Packaging | The Air-cooled Screw compressor should be packaged securely to prevent any damage during transportation.

The supplier/vendor should use suitable packaging materials and techniques to ensure safe delivery.

17.0 | Delivery | The supplier/vendor shall arrange for the safe transportation and delivery of the to the specified location.

They should provide a clear delivery schedule and keep the user informed of any delays or changes.

UG COLLECTION TANKS
ANNEXURE A - PLOT LAYOUT

GF - ETP / STP

UNDER GROUND 500 KL TANK -1 UNDER GROUND 500 KL TANK -2 UNDER GROUND 500 KL TANK -3
KILL TANK CHILLER ROOM BOILER ROOM BOREWELL

GF - WATER PRE-TREATMENT
UNDERGROUND SUMP
CONTAINER

CONTAINER
12.19x2.44x2.59

CONTAINER
12.19x2.44x2.59

12.19x2.44x2.59
ROAD ROAD
SPM
RECEIVING
BAY
FG PLASMA RM
DISCPATCH RECEIVING RECEIVING
BAY BAY
3D
D4

UP

ROAD

LIFT
FG
26.5X9.45 PLASMA & RM SAMPLING APPROVED RM
3.05X2.40 17.6X9.40 9.90X9.40

ROAD

FF - ENGINEERING OFFICE

GF - UTILITY BLOCK
FACILITY 02
FUTURE PROVISION-PLASMA STORAGE GF - QUARANTINE /
PLASMA STORAGE / FACILITY 01 BULK MANUFACTURING AREA
TESTING LABS /
28.6X45.0 WAREHOUSE
25.7X43.0 BULK MANUFACTURING AREA FUTURE PROVISION
50.7m x 74.2m 55.9m x 76.1m
40478.47 Sqft 45772.9 Sqft

WORKSHOP
LIFT

ROAD

ROAD ADMIN
46.9X17.1
7.21X10.3

UP
D4

UP

LIFT
LIFT

LIFT
UP

D8 D8 D8

+ 9.35 EGL
+ 9.35 EGL

ROAD
+ 10.08 FFL
+10.08 FFL

ROAD 810KVA, 3Ø, 415V
GENERATOR-3

DG
810KVA, 3Ø, 415V
GENERATOR-1

DG

m2.1
ELOH NAM
HTPED
TIP LIO TNRUB

2000x1800mm
WIDE GATE 91°0'17"

+9.47 EGL
HT
1800 x 2100mm
METAL GATE
FFL
+9.78 FGL + 9.93

X X
WELD MESH
TRANSFORMER VCB

FENCING
STR
BE-1 11KV TANGEDCO

X X X X
METERING
SOAK PIT RETAINER WALL

SLOPE SLOPE SLOPE
FFL
PANEL ROOM 2Nos - 1500x100x10mmTHK

FROM TANGEDCO
+ 10.08

11KV I/C SUPPLY
INSERT PLATE
F

SLOPE

SLOPE
HV SIDE

X X X
SOAK PIT RETAINER WALL

SOAK PIT RETAINER WALL
G (HT 5.3 M FROM FGL)

FIRE BARRIER WALL
ENTRY/EXIT 8M WIDE FFL FFL 11KV LAE
DP
EL + 14.77
BE
W + 9.93 + 10.08 HT

X X X
VCB

SLOPE

SLOPE

O/G
LV SIDE

WELD MESH
FENCING
X X X
11KV CONSUMER STR 11KV TANGEDCO
S SLOPE SLOPE

SOAK PIT RETAINER WALL
SLOPE
DP STRUCTURE BE-2
DP STRUCTURE

X
CLIENT :- LIFENITY WELLNESS INTERNATIONAL LTD.MUMBAI

GROUND FLOOR DESIGN &
PROJECT BY :-

BIOBRIDGE HEALTHCARE SOLUTION PVT.LTD.

PREPARED Y.T.J.
ADDRESS :-13, Rachna Blossom, Jagdishnagar,
Aundh, Pune -411007, Maharastra, India. G.K.
CHECKED

DATE 14.12.2023
APPROVED

SCALE 1:100 -
TITLE- PLOT LAYOUT GF BY CLIENT
SIZE -
(BULK MANUFATURING) DWG.NO 01
REVISION-R0

UG COLLECTION TANKS
GF - ETP / STP
Annexure B_Conceptual Layout-GF

UNDER GROUND 500 KL TANK -1 UNDER GROUND 500 KL TANK -2 UNDER GROUND 500 KL TANK -3
KILL TANK CHILLER ROOM BOILER ROOM BOREWELL

GF - WATER PRE-TREATMENT
24985

UNDERGROUND SUMP
CONTAINER

CONTAINER
12.19x2.44x2.59

CONTAINER
12.19x2.44x2.59

12.19x2.44x2.59
ROAD ROAD
8028

29249
SPM

PLASMA
RECEIVING
BAY 52821
FG RM
DISCPATCH RECEIVING RECEIVING
BAY BAY

CORRIDOR PLASMA TRANSFER

6D
3D

3D
D4

39.4X1.90
STORE CHANGE CORRIDOR
2.40X 49.89
11.6X2.90 PROVISION FOR FACILITY 2
ROOM

6D
UP

3.00X 2.40

ROAD
6D D6

350
500

500

500

500

500

500
1200
UTILITY & PROCESS PIPING 300
D6 D6 3D D3 D6 500

250 L
RM

1713
RLAF RLAF PIPING SKID PIPING SKID 500 PIPING SKID PIPING SKID PIPING SKID PIPING SKID PIPING SKID

1200
COLUMN

AKTA-1"
REJECT

1050
CONTROL PANEL

CONTROL PANEL
RECALLED RECEIPT 700 TFF CHROMO

HH

PLASMA RM
ROOM
HH HH

GOODS FG DISCPATCH FG IN SAMPLING 3.00X2.52 SAMPLING 1200 685

2275
500
2.40X 2.40 3.00X 2.10 3.00X2.10
REJECT

LIFT
1200
2.40X2.40 2275 CONTROL PANEL

PLASMA SEROLOGY

COLUMN
CHROMO
STORE 1.07X 1.90 1.07X1.90

300 L

300 L
700

ROOM

1200
1050

500
MH

1200
D3 D3
RECEIPT TESTING
COLUMN
OFFICE
D3
2.95X4.00
DE-DUSTING
STAGE 1 (F)/ CHROMO

BV 107
D3

500 L
2250
TUNNEL
APPROVED RM
UTILITY & PROCESS PIPING
520
16.9X4.35 2.03X4.35 2.40X 4.00 3.00X 4.00 AIR LOCK-2 AIR LOCK-2 600

STAGE 2 (F)
500
1.87X0.90
D3 1.87X0.90
D3

250 L

PV 107.1
PV 107.2
9.85X 5.80

1500 L
PLASMA DAY

1500 L

2000 L
2000 L
HH
4.90X1.90

4014
STAGE 12 (E)

COLUMN
CHROMO
4.80X 3.50
DPB DPB
STAGE 3 (F)

2140

3128
AIR LOCK-1 AIR LOCK-1
12.0X8.00

1920

1920
AKTA-1"
CONTROL PANEL

D6 1.87X0.90 D3 1.87X0.90

2175
D3 D3 D3

PV 107.3
STORE

1500 L
300

PIPING SKID
8.72X8.00
500
3.00X2.51

1500 L
2500

1500 L
1000 L
16.5X8.00

COLUMN
CHROMO
SD SD SD SD

500 L

500 L

500 L
3D
1.95X11.4
6.55X10.0

500
1050
6.50X5.25 D6 DPB ANTE ROOM 2-8 MH MH MH

CENTRIFUGE
CONT.
500
5.00X2.25

3D
2.30X7.67

2500
1500 L
750

PIPING SKID
450

AKTA-1"
600

2613
450
6.05X4.15 AIR LOCK-2 500

3D
COLD ROOM -3 COLD ROOM -4 3D D3 300

ICE PACK COLD ROOM -1 COLD ROOM -2 3.00X1.09 1.50X 1.67 PIPING SKID PIPING SKID 500 PIPING SKID PIPING SKID PIPING SKID PIPING SKID PIPING SKID

5.55X4.95 5.55X4.95 5.53X4.95
4.10 X 4.95 5.50X 4.95 DISPENCING ROOM
11803

600
D3 D6 D6 D6

500

500
WASHING BAG BAR 3.45X 3.45
PLASMA AREA CODING,

6D
QUARANTINE 2.00X2.95 MINI POOLS RM

ROAD
2.90X2.95 QUARANTINE AIR LOCK-1

CORRIDOR
1.50X 1.70
CORRIDOR

D3
RLAF 2.00X2.00
D6 DPB
35.3X2.00

D6

6D

6D
D3

4D D6 D6 D6

600
500

500
6D

D6

DPB
1200
CORRIDOR 1050

COLUMN
CHROMO
2.05X 25.5

AKTA-1"
PIPING SKID PIPING SKID PIPING SKID PIPING SKID

1200
AKTA-1"

1050
D6

600
2275

DPB

2275
1500 L
D3 D3 1000 L
MH
CHROMO
COLUMN

1200
500 L
BUFFER PLASMA BAG CUTTING & STAGE 8 N
1200

1500 L
1500 L
THAWING
PLASMA STORAGE PLASMA STORAGE PLASMA STORAGE PLASMA STORAGE
POOLING 9.00X7.00
3D

3D
~ 1000 L ~ 1000 L ~ 1000 L ~ 1000 L
PREPARATION 3.22X8.90
3D

3D

FF - ENGINEERING OFFICE
~ 70 KL GV ~ 70 KL GV ~ 70 KL GV ~ 70 KL GV
5.40X5.30 5.40X5.30 5.40X5.30 5.40X5.30
9.00X6.00 6.20X8.90

6D
D6
STAGE 4 (F)

GF - UTILITY BLOCK
D6

PRE - VIRUS CONTROLLED CORRIDOR
D6 14.5X13.8

DPB
PIPING SKID PIPING SKID PIPING SKID
CHROMO

600
COLUMN

1200
AKTA-1"
1200
3D

3D
PLASMA STORAGE PLASMA STORAGE PLASMA STORAGE PLASMA STORAGE
~ 1000 L ~ 1000 L ~ 1000 L ~ 1000 L
3D

3D

1753
TANK STORAGE
500
~ 70 KL GV ~ 70 KL GV ~ 70 KL GV ~ 70 KL GV

1000 L

1000 L
1000 L

693
500
5.40X5.32 5.40X5.32 5.40X5.32 5.40X5.32

2000 L

PIPING SKID
1500 L
1500 L
9.00X8.10

PIPING SKID
1500 L

500 L
1000 L

BUFFER
FACILITY 02
MH

TFF

2500
2.00X23.6

500
SYSTEMS

PIPING SKID
2000 L
PIPING SKID PIPING SKID PIPING SKID

PIPING SKID
1500 L
PIPING SKID

9.52X7.20

2.00X28.5
BULK MANUFACTURING AREA
10000 STAGE 5 (I)

566

566
600
PLASMA STORAGE PLASMA STORAGE
3D

3D

PLASMA STORAGE PLASMA STORAGE
9.00X16.5
1200

COLUMN
CHROMO
~ 1000 L ~ 1000 L PIPING SKID PIPING SKID PIPING SKID
PIPING SKID
~ 1000 L ~ 1000 L
3D

3D

1200
~ 70 KL GV ~ 70 KL GV

~ 70 KL GV ~ 70 KL GV

AKTA-1"
500
5.40X5.32 5.40X5.32 D6

TFF
5.40X5.32 5.40X5.32

6D

TFF

2500
CONTROL PANEL CONTROL PANEL

D4
MH 600

HH+SG

HH+SG
FUTURE PROVISION
600
2500

500 L
1000 L

1500 L
2000 L
CORRIDOR

492
CIP / SIP 2.00X48.4

500 L
PIPING SKID

500 L
ANTE ROOM

ANTE ROOM

500 L
COLUMN PACKING

MH
9.00X6.20

300
MH MH

AND SKIDS
55.9m x 76.1m

1500 L
1500 L
3D

3D

PLASMA STORAGE PLASMA STORAGE PLASMA STORAGE PLASMA STORAGE
~ 1000 L ~ 1000 L ~ 1000 L ~ 1000 L PIPING SKID PIPING SKID
3D

3D

500
~ 70 KL GV ~ 70 KL GV ~ 70 KL GV ~ 70 KL GV
9.52X7.30

6D
5.40X5.32 5.40X5.32 5.40X5.32 5.40X5.32

FUTURE PROVISION-PLASMA STORAGE

PIPING SKID
CORRIDOR 431

1500 L
1.80X42.9
GF - QUARANTINE / 1.80X42.9
2.00X52.8
COOL ZONE
500

D6
FILTER
DEPTH
STAGE 7A CORRIDOR

45772.9 Sqft

500
PLASMA STORAGE /
28.4X45.0 9.00X2.00
PIPING SKID PIPING SKID

1.80X62.6

4D

6D

PIPING SKID
1000 L
TESTING LABS / 14.5X14.6
WAREHOUSE AUTO-

500
1200

DHS
1050

COLUMN
CHROMO
PIPING SKID

FILTER
NANO
500

AKTA-1"
1200

PIPING SKID
CLAVE

500 L
2291

1500
600

MH
PLASMA STORAGE PLASMA STORAGE
4.45X3.00
3D

3D

PLASMA STORAGE PLASMA STORAGE

2275
1500
~ 1000 L ~ 1000 L ~ 1000 L ~ 1000 L
76307

3D

3D

4.45X3.00

500
~ 70 KL GV ~ 70 KL GV ~ 70 KL GV ~ 70 KL GV MH

600

250 L
5.40X5.32 5.40X5.32 5.40X5.32 5.40X5.32

250 L
150 L
150 L
D4

500 L
D4

PIPING SKID

PV 116
800 800

500 L

600
800 800

MH

1814

6D

6D
CONTROL PANEL

CONTROL PANEL
HH+SG HH+SG

D4

UTILITY & PROCESS PIPING

UTILITY & PROCESS PIPING
HH HH

CONTROL PANEL

CONTROL PANEL
1500 L

PIPING SKID
2.50X1.88

600

755

500
WASHING

500
TFF
1200

2500
1050

COLUMN
CHROMO
3070 1200
1050

COLUMN
CHROMO

AKTA-1"
9.00X4.50

1200
AKTA-1"
PIPING SKID

1200
1500 L
2500

2275
2275
3D

3D

PLASMA STORAGE PLASMA STORAGE PLASMA STORAGE PLASMA STORAGE
COLUMN PACKING

600

500
~ 1000 L ~ 1000 L ~ 1000 L ~ 1000 L
3D

3D

~ 70 KL GV ~ 70 KL GV ~ 70 KL GV ~ 70 KL GV
CONTROL PANEL

PIPING SKID
1000 L
5.40X5.32 5.40X5.32 5.40X5.32 5.40X5.32
AND SKIDS

150 L
HH+SG
D4

DE A/L STAGE 10 T/

WORKSHOP
JANITOR 11.6X15.5

500
GOWN 2.00X3.00
STAGE 6 (I) STAGE 9 N STAGE 13 E

PIPING SKID

500 L
2.00X3.00
4.80X4.00

MH
D4

D4

STAGE 11 F 8.00X10.5 8.50X10.5
4D
4D

A/L GOWN 9.00X15.5
JANITOR
3D

3D

PLASMA STORAGE PLASMA STORAGE PLASMA STORAGE PLASMA STORAGE
~ 1000 L ~ 1000 L ~ 1000 L ~ 1000 L 2.00X3.00 2.00X3.00
3D

3D

~ 70 KL GV ~ 70 KL GV ~ 70 KL GV ~ 70 KL GV
4.80X2.00
D4

D4

4D

5.40X5.32 5.40X5.32 5.40X5.32 5.40X5.32

LIFT
D4

D4

DE A/L 6D 6D 6D 6D
GOWN 2.00X3.00
2.00X3.00
4.80X3.70
ROAD

6D

6D
D4

D4 CORRIDOR 2.50X2.00
4D

34.9X2.00
4D
3D

3D

PLASMA STORAGE PLASMA STORAGE PLASMA STORAGE PLASMA STORAGE 4D D4 D4
A/L GOWN
D6 D6
~ 1000 L ~ 1000 L ~ 1000 L ~ 1000 L
3D

3D

CONTROL PANEL CONTROL PANEL

~ 70 KL GV ~ 70 KL GV ~ 70 KL GV ~ 70 KL GV
5.40X5.35 5.40X5.35 5.40X5.35 5.40X5.35
2.00X3.00 2.00X3.00 4.80X2.30 7556

HH+SG

HH+SG
DHS
D4

D4

4D

COOL ZONE
8000 5.75X4.45 WASHING

9.00X2.00
D6

BUFFER TANK STORAGE-2 CIP / SIP COLD ROOM 4.70X9.00
PREPARATION
1D/2S

1D/2S

1D/2S

1D/2S

1D/2S

1D/2S

1D/2S

1D/2S
500

500

500

500

500

500

500

500

4D
6.40X9.00 6.40X9.00 6.45X9.00
1D/2S

1D/2S

1D/2S

1D/2S

1D/2S

1D/2S

1D/2S

1D/2S
500

500

500

500

500

500

500

500

KBD/CM/BP KBD/CM/BP KBD/CM/BP KBD/CM/BP KBD/CM/BP KBD/CM/BP KBD/CM/BP KBD/CM/BP KBD/CM/BP

CHEMICAL DAY STORE 5.12X9.00
TEL

TEL

TEL

TEL

TEL

TEL

TEL

TEL

Leg Space/CPU Leg Space/CPU Leg Space/CPU Leg Space/CPU Leg Space/CPU Leg Space/CPU Leg Space/CPU Leg
Space/CPU
1D/2S 1D/2S 1D/2S 1D/2S 1D/2S 1D/2S 1D/2S 1D/2S

ADMIN
DOCUMENT STORAGE

48.7X2.00

D4
D6
COMPUTER COMPUTER COMPUTER COMPUTER COMPUTER
DESK DESK DESK DESK DESK DESK

[illegible]

CHR-2 CHR-2

4D
D1 CHR-2 CHR-2
2.20X2.10 2.20X2.10 2.20X2.10 2.20X2.10
F-AID ROOM LOCKER MANUFACTURING
TEA ROOM

UP
2.49X4.40
D1

D4 D4 D4 ROOM D4
(LADIES) OFFICE
COMPUTER COMPUTER COMPUTER COMPUTER
WAITING ROOM (LADIES) 2.50X4.40
DESK DESK DESK DESK DESK
3.00X3.95
3.45X4.40 9.30X4.40
13.7X6.20 CHR-1 CHR-1
D1

CHR-1 CHR-1
7.16X2.00 8D 2.20X2.10 2.20X2.10 2.20X2.10 2.20X2.10
D3 D3 D4 D4 D4 D4 D4 D4 D4 D4
CONFERENCE
14.7X5.05 CAFETERIA
D3

MEETING ROOM 5.00 X 5.00
5.00 X 5.00
D6

7.80X11.20 CORRIDOR CORRIDOR
D4

12.6X5.05
D4

ENTRANCE LOBBY 18.4X2.00 D4
RECEPTION

9.30X2.00
D1

±0.00

16AMPS 16AMPS
6M
5.00 X 6.00 6M COMPUTER COMPUTER COMPUTER COMPUTER COMPUTER COMPUTER
DESK DESK DESK DESK DESK DESK
UP

4.66X6.80 3.00X3.95 4D D4 4D D4
16AMPS 16AMPS
D1

6M 6M

REJECT ROOM
D4

LOCKER 4.75X1.90
D1 D1 D1
ROOM REST ROOM D1 D1 D1
D4
D1

8D DOCUMENT CABINETS LIFT 9.13X4.80 (GENTS) (LADIES) D4
OFFICE
TOILET (GENTS) 3.45X4.40 TOILET (LADIES) 5.30X4.40
4.60X4.40
4.89X4.40
LIFT

4.90X2.60 PPM RECEIPT

A
D4

CORRIDOR LIFT
46.8X2.00 4.00X2.45
UP

D8
MAIN ENTRY
D8
1.49X1.90
D8 A
+ 9.35 EGL
+ 9.35 EGL

ROAD
+ 10.08 FFL
+10.08 FFL
8000

ROAD 810KVA, 3Ø, 415V
GENERATOR-3

DG
810KVA, 3Ø, 415V
GENERATOR-1

DG
18215

m2.1
ELOH NAM
HTPED
TIP LIO TNRUB

2000x1800mm
WIDE GATE 91°0'17"

+9.47 EGL
HT
1800 x 2100mm
METAL GATE
FFL
+9.78 FGL + 9.93

X X
WELD MESH
TRANSFORMER VCB

FENCING
STR
BE-1 11KV TANGEDCO

X
METERING
SOAK PIT RETAINER WALL

X X
SLOPE SLOPE SLOPE
FFL
PANEL ROOM 2Nos - 1500x100x10mmTHK

FROM TANGEDCO
+ 10.08

11KV I/C SUPPLY
X
INSERT PLATE
F

SLOPE

SLOPE
HV SIDE

FIRE BARRIER WALL

X
SOAK PIT RETAINER WALL

SOAK PIT RETAINER WALL
G (HT 5.3 M FROM FGL)

X X
ENTRY/EXIT 8M WIDE FFL FFL 11KV LAE
DP
EL + 14.77
BE
W + 9.93 + 10.08 HT

X
VCB

SLOPE

X X
SLOPE

O/G
LV SIDE

WELD MESH
FENCING
X
11KV CONSUMER STR 11KV TANGEDCO
S SLOPE SLOPE SLOPE BE-2

X X
DP STRUCTURE DP STRUCTURE
SOAK PIT RETAINER WALL

X
DOOR
CLIENT :- LIFENITY WELLNESS INTERNATIONAL LTD.MUMBAI

GROUND FLOOR
SCHEDULE-GROUND
FLOOR
NAME SIZE DESIGN &
D1 PROJECT BY :-
750 x 2100
D2 800 x 2100 BIOBRIDGE HEALTHCARE SOLUTION PVT.LTD.
D3 900 x 2100
PREPARED Y.T.J.
D4 1000 x 2100 ADDRESS :-13, Rachna Blossom, Jagdishnagar,
D6 Aundh, Pune -411007, Maharastra, India.
1500 x 2100 CHECKED G.K.

D8 2000 x 2100 DATE 14.12.2023
APPROVED
SCALE 1:100 -
ED TITLE- CONCEPTUAL LAYOUT-GF BY CLIENT
1000 x 2100 SIZE -
(BULK MANUFATURING) DWG.NO 01
SD 1500 x 2100 REVISION-R0

D4
Annexure C_Conceptual Layout-FF

104.1X2.00
UP

D3
D8 D8 D8 D8

SPM RECEIPT REJECT

LIFT
2.00X4.80
3.30 X 4.00 3.00 X 3.30
FINISHED GOODS STAGING
13.15 X 7.00
D8 D8 LABELING STORE
8.0 X 7.00

4.50X2.00

D8
8D D4

WATER SYSTEM & PB PACKING OFFICE
3.00X2.55

4D
ENGINEERING STORE
LABELING &
PACKING
2.05X10.1
8.0 X 5.0

CANTEEN
PACKING
29.9X17.1 VISITOR

4D
3.00X2.00

4D
4D
23.8X10.3 SPM STORE,
21.3X6.80
D8

4D
PACKING CHANGE
27.5 X 17.1 MALE
3.0 X 3.0
8.05 X 10.1

4D
VISUAL &
INSPECTION

4D
5.00X5.05
3.05X3.05

4D
PACKING CHANGE
BUFFER ROOM FEMALE
3.00 X 2.40
D3

2.95X2.00

4D

4D
DPB

D8
CORRIDOR
26.40X2.00
DPB

D4
DPB

4D
1.00X1.00
CAPPING
5.83X4.00

VIAL COLLECTION
TRAY
LYO SERVICE AREA LYO LOAD AL

8.06X4.03 5.58X4.03 2.59X4.00
CONVEYOR
SWING

AUTOMATIC FOUR HEAD ALUMINUM CAP
SEALING MACHINE, MODEL-AEWCS-120
D4
4D

4D

4D

4D

4D
PAL 1 PAL 2 PAL 3 PAL 4
4.53X2.00 4.50X2.00 2.21X4.08 2.21X4.08
FILL FINISH
5.83X8.16

CORRIDOR

4D

4D
EXIT 2 EXIT 1 VIAL COLLECTION

2.00X21.9 & LOADING TRAY

4.53X2.00 4.50X2.00

36" TURNTABLE
NEW ADD
D4
D2

CONVEYOR
SWING
LAUNDRY AREA DOUBLE STERILE CORRIDOR
UNLOADING AREA
4.88X1.60 5.66X1.76
2.00X41.9 DOOR 3.07X4.76
3.57X3.16 2.59X15.71
D4 WASHER
1.50X2.00 1.80X60.7
MAL
1.23X1.53

D4
D4
TUMBLE PACKING AREA FORMULATION
DRYER 5.58X2.92

D4
PAL 1.50X2.00
1.23X1.50

STABLIZING ZONE
1500

D4
D4
D3 3D

COOLING ZONE
D8

MAL
2.53X1.50 MAL PAL

D4

D4
1.47X1.47 1.50X1.50

STERILIZING ZONE
3D D3 DISINFECTANT
5.58X3.08
PAL

4D
2.53X1.50

D4

D4

DRYING ZONE
DPB
1.00X1.00

D4
FACILITY 02
MATERIAL PREP WASH
6.98X3.55 VIAL WASHING
5.45X1.58
AC UNLOAD
1.00X2.50 5.83X10.55
2.35X1.55 5.58X3.55

D4
DPB

D4
4D
CORRIDOR
9.11X2.00
CLEAN JC

FUTURE PROVISION
D4 D4 D4 2.54X3.00
D4
DPB

PAL-2
1.70X1.80
PRIMARY CHANGE PRIMARY CHANGE PRIMARY CHANGE
MAL ENTRY
VISITOR FEMALE MALE
2.00X4.00 COLD ROOM
D4
4.50X4.00 4.50X4.00 4.50X6.08 DE-CARTONING
3.59X3.00 2.90X3.44
PAL-1
1.70X1.80
R&D OFFICE
5.15X2.70 R&D LAB -1 R&D LAB -2 D4 D4 D4 D4 D4 D4 D4
DPB

55.9X76.1
9.29X2.70 9.29X2.70
3D 3D 3D CORRIDOR
24.3X2.40
PASSAGE
1DE

R&D ENTRANCE
D3

18.6X1.50
5.15X2.20
D3 D3
SERVICE FLOOR

D4
3D
2.40X1.73 LIFT

54.4X28.2
R&D LAB-4 R&D LAB -3

7.20X2.00 9.29X2.70 9.29X2.70
D3

BULK STAGING
17.4 X 6.00

78.3X26.3 4.90 X4.26
D3

D3

DOCUMENT QA QC STAFF
STORAGE DOCUMENTATION
3.65X3.15
2.40X3.00 3.29X3.15 PM TESTING STABILITY ROOM
1.15X2.70 ROOM RETENTION
2.40X2.37 4.25X5.90 SAMPLES ROOM
2.40X5.90
D3

D6
QA MANGER QC MANEGER
CONFERENCE
2.40X3.57 2.40X2.70 2.09X2.70
APPROVED RUBBER
D3 D3 D3 D3 D6 BUNG STORE
D3
5.00 X 5.00
D3

CORRIDOR
WASH

CORRIDOR

EYE
1.20X9.34

15.0X1.20
TRANING HALL
GENETS
D3

5.15X14.3 D3
D3 D3
CHANGE ROOM
2.40X1.80
D3

D4
BALANCE ROOM
CHEMICAL /GLASS 2.40X2.40
2.00X14.4 WARE STORAGE
D3

2.40X3.00
D3

LADIES
CHANGE ROOM
HOT ZONE /WASHING
2.40X2.00 INSTRUMENT ROOM
D3

2.40X5.87
4.00X5.87

4.90X2.00
D3

WET LAB

TOC /LPC 5.99X3.42 2.00X24.3
SAMPLE
RECIPET 2.40X2.82 3.54X2.45
2.40X1.97 D4
D3

SAMPLE
D3

RECIPET
HATCH

LAF

4D
AIR LOCK 2

PPM STORE
D1
0.9X2.23

17.25X 6.78,
DPB

AUTOCLAVE
MLT LAF
2.56X1.92 COOLING 19.20 X 11.40

UP
ZONE AUTOCLAVE
D3 2.16X1.97 LAF MEDIA
LAF 3.38X1.97
DPB

STERILITY
D3

STERILE PREP./STOREGE
D1 ROOM D3
1.17X1.23 1.17X1.23

INCUBATION PASSAGE 1.97X4.00 2.40X11.4
AIR LOCK 2

1.77X2.94 1.20X2.99
1.34X2.52 Biosafety

4.25X4.00
BIOSEFTY
D1

7.20X2.15
AIR LOCK 1

0.9X2.23

AIR LOCK 3
D1
D1

AIR LOCK 2
D1

D1 D1
D3

1.65X0.935 AIR LOCK 1 AIR LOCK
1.60X0.935
1.00X1.95 1.17X1.95
AIR LOCK 1

DPB
Cabinet

3.05X1.00 AIR LOCK 4 AIR LOCK 5
D1

D1

1.65X0.935 1.60X0.935 3.30X4.90 3.30X5.05
6D 4.65X2.00
D4

4D

D3
D1 D1 D1 D1
D4

CORRIDOR
1DE

1D
1D

18.6X1.45
UP

D3 D3 D3 D3 D3 D1 D1
4.70X6.80
DPB

1D
1D

2.00X7.00
BET ROOM MICRO OFFICE LADIES CHANGE GENETS CHANGE
MEDIA FILL DECONTAMINATION
MEDIA FILL ROOM ROOM
INCUBATION-1 & WASHING ROOM 2.40X2.37 2.40X2.37 2.40X7.00
INCUBATION-2
2.40X2.37 2.00X2.37 2.00X2.37
4.25X2.37 2.90X2.37
1D

QUARANTINE SAMPLING UNDERTEST
D1 D1 D6 LIFT D6 D6
LIFT

A 54.9X2.00
UP 34.8X2.00
LIFT

A
DOOR
SCHEDULE-GROUND
CLIENT :- LIFENITY WELLNESS INTERNATIONAL LTD.MUMBAI
FLOOR
NAME SIZE
DESIGN &
D1 750 x 2100 PROJECT BY :-

FIRST FLOOR
D2 800 x 2100
D3 900 x 2100 BIOBRIDGE HEALTHCARE SOLUTION PVT.LTD.
D4 1000 x 2100 ADDRESS :-13, Rachna Blossom, Jagdishnagar, PREPARED Y.T.J.

D6 1500 x 2100 Aundh, Pune -411007, Maharastra, India.
CHECKED G.K.

D8 2000 x 2100 DATE 14.12.2023
APPROVED
SCALE 1:100 -
ED 1000 x 2100 TITLE- CONCEPTUAL LAYOUT-FF BY CLIENT
SIZE -
(FILL FINISH FACILITY) DWG.NO 02

UP
D4
Annexure D_Terrace Floor

LIFT
TERRACE FLOOR TERRACE FLOOR
FACILITY 01 FACILITY 02
FUTURE PROVISION FUTURE PROVISION
106.6X76.1 55.9X76.1
LIFT

D4

4D
UP
D4

UP

LIFT
LIFT

A UP
LIFT

A
DOOR
SCHEDULE-GROUND
CLIENT :- LIFENITY WELLNESS INTERNATIONAL LTD.MUMBAI
FLOOR

TERRACE FLOOR NAME
D1
D2
SIZE
750 x 2100
800 x 2100
DESIGN &
PROJECT BY :-

D3 900 x 2100 BIOBRIDGE HEALTHCARE SOLUTION PVT.LTD.
D4 1000 x 2100 ADDRESS :-13, Rachna Blossom, Jagdishnagar, PREPARED Y.T.J.

D6 1500 x 2100 Aundh, Pune -411007, Maharastra, India.
CHECKED G.K.

D8 2000 x 2100 DATE 14.12.2023
APPROVED
SCALE 1:100 TITLE- TERRACE FLOOR -
ED 1000 x 2100 SIZE - FUTURE PROVISION
BY CLIENT

DWG.NO 03
ED1 1200 x 2100 REVISION-R0

Annexure E_Section and Elevation Layout

1200

1200
TERRACE

TERRACE

6000

6000
FF

FF

6000

6000
PL

1200
PL
GL

GL
76500

EAST SIDE ELEVATION WEST SIDE ELEVATION

1200
TERRACE

6000
FF

6000
PL

1200
GL

NORTH SIDE ELEVATION

1200
TERRACE

6000
FF

6000
PL

1200
GL

SOUTH SIDE ELEVATION

1200

TERRACE TERRACE

MEDIA FILL
DESTRUCTION &
MEDIA FILL
INCUBATION-2
PASSAGE
SERVICE FLOOR LADIES CHANGE
FACILITY 2 PASSAGE PPM STORE PASSAGE TOILET
PASSAGE
PASSAGE TOILET ROOM
MICRO OFFICE BET ROOM WASHING ROOM
INCUBATION-1
6000

GENETS CHANGE
FUTURE PROVISION SERVICE FLOOR ROOM

FF FF

FACILITY 02
REST ROOM LOCKER MAIN ENTRY CAFETERIA TOILET ADMIN
PPM RECEIPT OFFICE TOILET (LADIES) TOILET (GENTS)
BULK MANUFACTURING AREA (LADIES)

6150
ROOM PASSAGE
6000

ADMIN
FUTURE PROVISION (GENTS) CONFERENCE

ENTERANCE MEETING ROOM
LOBBY

D D

PL PL

GL GL

SOUTH SIDE ELEVATION SECTION A-A
CLIENT :- LIFENITY WELLNESS INTERNATIONAL LTD.MUMBAI

DESIGN &
PROJECT BY :-

BIOBRIDGE HEALTHCARE SOLUTION PVT.LTD.

ADDRESS :-13, Rachna Blossom, Jagdishnagar, PREPARED Y.T.J.

Aundh, Pune -411007, Maharastra, India.
CHECKED G.K.

DATE 07.12.2023
SCALE 1:100 TITTLE-SECTION & ELEVATION LAYOUT APPROVED -
BY CLIENT
SIZE -
DWG.NO 04
REVISION-R0

[SHEET: Cover Sheet]
| BASIC INFORMATION |
| Name of the Customer |
| Constitution |
| Industry |
| Financials available / used from |
| Currency |
| Auditors |
| Last Day of Financial Year |
| Exchange Rate |
INSTRUCTIONS: ||
1. | Following colour codes are used in the template: |
| Input cells (Unlocked) |
| Cells containing formulae (Locked) |
| Cells containing text (Locked) |
[SHEET: BS Assets]
| (in lacs) |||||
BALANCE SHEET (ASSETS) INPUT | 2018 | 2019 | 2020 | 2021 | 2022 | 2023 | 2024 |||||
| Audited | Audited | Audited | Provisional | Projected | Projected | Projected |||||
CURRENT ASSETS |||||
Cash & Bank Balances | 364.61 | 643.9 | 884.24 | 1713.19 | 1155.2513380153653 | 997.996354593502 |
906.1614263418383 |||||
Investments |||||
Govt. and other trustee Securities |||||
Fixed Deposits with Banks |||||
Others |||||
Receivables |||||
Receivables Other than Deferred & exports (Domestic) | 489.17 | 608.06 | 1450.52 | 833.4 |
3110.978082191781 | 3266.5269863013696 | 3429.853335616438 |||||
Receivables under Bills discounting |||||
Export Receivables |||||
Note : |||||
1. All receivable upto 180 days only to be included. |||||
2. Sale bills negotiated under LC to be excluded. |||||
Deferred receivable (due within one year.) |||||
Inventory |||||
Raw Materials - Imported |||||
Raw Materials - Indigenous | 69.39 | 61.87 | 65.07 | 81.01 | 89.11100000000002 | 98.02210000000002 |
107.82431000000004 |||||
Work in process | 186.67 | 1265 | 839.46 | 520.5 | 546.525 | 573.85125 | 602.5438125000001 |||||
Finished Goods (incl Traded Goods) |||||
Other consumable spares - Imported |||||
Other consumable spares - Indigenous |||||
Sub total (Inventory) | 256.06 | 1326.87 | 904.53 | 601.51 | 635.636 | 671.8733500000001 |
710.3681225000001 |||||
Advances to suppliers |||||
Short Term loans and advances | 567.3 | 1675.6 | 1635.3 | 2263.83 | 2317.11494515625 | 2373.0691587473957
| 2431.863038018099 |||||
Investments in Mutual Funds |||||
Other Current Assets | 809.27 | 760.12 | 856.52 | 607.44 | 985.683159722222 | 1034.9673177083334 |
1086.71568359375 |||||
TOTAL CURRENT ASSETS | 2486.41 | 5014.549999999999 | 5731.109999999999 | 6019.369999999999 |
8204.663525085618 | 8344.4331673506 | 8564.961606070125 |||||
FIXED ASSETS |||||
Gross Block | 1223.7 | 1273.46 | 2309.91 | 2836.1 | 2554.3215 | 2947.091125 | 2841.51434375 |||||
Less: Accumulated Depreciation |||||
Net Block | 1223.7 | 1273.46 | 2309.91 | 2836.1 | 2554.3215 | 2947.091125 | 2841.51434375 |||||
Capital Work in progress |||||
NON-CURRENT ASSETS ||||| 4 | 4.229999999999997 | 8.469999999999999 ||
||| 0.5 | 0.5 | 4.5 |||
Investments / Book Debts / Advances / Deposits (which are not current assets): ||||| 3.5 |
3.729999999999997 | 3.969999999999999 |||
Investments in Group concerns ||||| 20.01 |||||
Loans to group concerns / Advances to subsidiaries |||||
Investments in others | 151.58 | 121.98 | 50.01 | 50.01 | 50.01 | 50.01 | 450 | 1.5158 | 1.2198 | 0.5001
| 0.5001 | 0.5001 | 0.5001 | 4.5
Security Deposits | 2662.85 | 2048.46 | 2008.2 | 2001.45 | 2350.5629 | 2374.068529 | 2397.80921429 |
26.6285 | 20.4846 | 20.082 | 20.0145 | 23.505629 | 23.740685290000002 | 23.9780921429
Deferred receivables (maturity exceeding one year) | 32.53 | 38.89 | 55.56 | 41.43 | 0 | 0 | 0 |
0.3253000000000003 | 0.3889 | 0.5556 | 0.4143 | 0 | 0 | 0
Debtors > 6 mths. | 661.21 | 816.5 | 157.36 | 610.18 | 0 | 0 | 0 | 6.612100000000001 | 8.165 | 1.5736 |
6.1018 | 0 | 0 | 0
Others (Loans & Advances non current in nature, ICD's | 0 | 0 | 362.43 | 325.84 | 0 | 0 | 0 | 0 | 0 |
3.6243 | 3.2584 | 0 | 0 | 0
Investments in Mutual Funds, FDS | 66.43 | 235.6 | 10 | 34.2 | 34.2 | 34.2 | 34.2 | 0.6643000000000001 |
2.356 | 0.1 | 0.342 | 0.342 | 0.342 | 0.342
Other Non-current Assets
(incl. Dues from directors) | 0 | 0 | 64.87 | 67.99 | 0 | 0 | 0 | 0 | 0 | 0.6487 | 0.6799 | 0 | 0 | 0
TOTAL NON CURRENT ASSETS | 3574.6 | 3261.43 | 2708.43 | 3131.0999999999995 | 2434.7729 | 2458.278529 |
2882.00921429 |||||
Intangible Assets:
Goodwill, Patents & trademarks | 9.52 | 9.21 | 4.73 | 7.39 | 5.542499999999995 | 4.156874999999999 |
3.1176562500000005 |||||
Accumulated Losses,Preliminary expenses,Miscellaneous expenditure not w/off, Other deferred revenue
expenses |||||
TOTAL ASSETS | 7294.23 | 9558.649999999998 | 10754.179999999998 | 11993.96 | 13199.300425085617 |
13753.959696350601 | 14291.602820360124 |||||
DIFFERENCE IN B/S | 0 | 0 | 0 | 0 | 0 | 0 | 0 |||||
[SHEET: BS Liabilities]
| (in lacs) |||||

BALANCE SHEET (LIABILITIES) INPUT | 2018 | 2019 | 2020 | 2021 | 2022 | 2023 | 2024 | ||||
 | Audited | Audited | Audited | Provisional | Projected | Projected | Projected | ||||
 CURRENT LIABILITIES |||||
 Short term borrowings from Banks (including bills purchased, discounted & excess borrowings placed on repayment basis) |||||
 Bank Borrowings - SBI | 656.81 | 1720.87 | -13.86 | 1126.65 | 3500 | 3500 | 3500 | ||||
 Bank Borrowings - HDFC ||| 0 | 45.19 | 400 | 400 | 400 | ||||
 Sub Total | 656.81 | 1720.87 | -13.86 | 1171.8400000000001 | 3900 | 3900 | 3900 | ||||
 Short term borrowings from Associates & Group Concerns |||||
 Short term borrowings from Others |||||
 Creditors for purchases | 1214.58 | 1510.45 | 1995.98 | 2682.6 | 175.18338133561656 | 183.9590583818493 | 193.2949455475171 | ||||
 Advances/ payments from customers/deposits from dealers. | 67.01 | 134.12 | 310.88 | 284.45 | 711.125 | 782.2375000000001 | 860.4612500000002 | ||||
 Provisions ||||| 601.51 | 601.51 |
 - Tax | 92.68 | 119.36 | 293.55 | 185.44 | 203.94261875000046 | 218.51642689236104 | 236.88771624913178 | 2682.6 | 2263 |
 -Others | 323.12 | 229.71 | 37.88 | 44.41 | 74.125 | 107.43249999999999 | 157.38374999999996 | | -2081.09 | 2864.51 |
 ||||| 833.4 | 2682 |
 Dividends Payable ||||| 182.51000000000022 | S
 Interest payable on CC ||| 8.02 ||||| 833 | D
 Instalments Machinery/ Vehicles AXIS/ HDFC Etc ||| 116.59 | 231.84 | 161.65 | 184.18 | 184.18 | 184.18 |
 ||| 1015.5100000000002 | T
 ||||| 545.51 | A
 Installments of Term Loans NSIC ||| 0 | 141.7 | 0 | 0 | 0 | ||| 470.0000000000002 | TOTAL S D
 ||||| 352.50000000000017 | NET DP
 COVID- SBI |||| 109.02 | 92.63 | 0 | 0 | ||||
 Other Payable | 264.89 | 762.6300000000001 | 542.72 | 545.51 | 150.77656874999934 | 198.93040748263883 | 54.2115395566257 | ||||
 Total Current Liabilities | 2619.0899999999997 | 4593.73 | 3407.0100000000007 | 5326.62 | 5491.962568835617 | 5575.255892756849 | 5586.4192013523125 | ||||
 TERM LIABILITIES |||||
 Debentures |||||
 Preference share capital |||||
 Dealer's Deposit |||||
 Deferred Tax Liability |||||
 Term Loans - From Banks | 352.64 | 196.41 | 247.85 | 92.63 | 552.5500000000001 | 368.3666666666667 | 184.1833333333328 | ||||
 Long Term Provisions | 14.68 | 16.7 | 18.96 | 31.74 | 0 | 0 | 0 | | 2821.1591875 | 2895.676834375 | 2897.5142558047946 |
 Term Deposits | 23.86 | ||||| 1404.39 | 1404.39 | 1404.39 |
 Unsecured loans others |||||
 Unsecured loans (Quasi Equity) | 19.09 | 19.09 | 19.09 | 0 | 0 | 0 | 0 | | 1416.7691874999998 | 1491.286834375 | 1493.1242558047945 |
 Sales tax deferred | 1066.56 | 1143.68 | 1592.42 | 1404.39 | 1404.39 | 1404.39 | 1404.39 | | 711.125 | 782.2375000000001 | 860.4612500000002 |
 Total Term Liabilities | 1476.83 | 1375.88 | 1878.3200000000002 | 1528.76 | 1956.94 | 1772.7566666666667 | 1588.5733333333333 | | 211.44261875000046 | 226.01642689236104 | 244.38771624913178 |
 ||||| 66.615 | 99.92249999999999 | 149.88374999999996 |
 TOTAL OUTSIDE LIABILITIES | 4095.9199999999996 | 5969.61 | 5285.3300000000001 | 6855.38 | 7448.902568835618 | 7348.012559423516 | 7174.992534685645 | | 184.18 | 184.18 |
 ||||| 92.63 | 0 | 0 |
 NET WORTH ||||| 150.77656874999934 | 198.93040748263883 | 54.2115395566257 |
 Equity Share Capital |||||
 Share Capital (Paid-up) | 179.4 | 190.4 | 183.2 | 183.2 | 183.2 | 183.2 | ||||
 Share Application (finalised for allotment) |||||
 Sub Total (Share Capital) | 179.4 | 179.4 | 190.4 | 183.2 | 183.2 | 183.2 | ||||
 General Reserve | 3018.91 | 3409.64 | 5278.45 | 4955.38 | 5567.197856250001 | 6222.747136927084 | 6933.410285674479 | ||||
 Capital Reserve |||||
 Capital Redemption Reserve |||||
 Statutory Reserve |||||
 Employees Stock Option Outstanding |||||
 Revaluation Reserve |||||
 Partners capital / Proprietor's capital |||||
 Other Reserves & Surplus: |||||
 Share Premium |||||
 Capital subsidy |||||
 Others |||||
 Balance in P&L Account (+ / -) |||||
 NET WORTH | 3198.31 | 3589.04 | 5468.849999999999 | 5138.58 | 5750.397856250001 | 6405.947136927084 | 7116.610285674479 | ||||
 ||||| 18320180 |||
 TOTAL LIABILITIES | 7294.23 | 9558.65 | 10754.18 | 11993.96 | 13199.300425085617 | 13753.959696350601 | 14291.602820360124 | | 512752619 | ||
 ||||| 9263771 |||
 DIFFERENCE IN B/S | 0 | 0 | 0 | 0 | 0 | 0 | 0 | | 140439400 |||
 ||||| 3173579 ||| | | |
 |||| 66.615 | 99.92249999999999 | 149.88374999999996 | | 142257332 |||
 |||| 7.51 | 7.51 | 7.5 | | 268260926 |||
 |||| 74.125 | 107.43249999999999 | 157.38374999999996 | | 99160541 |||
 ||||| 5769094 | 22985086 | 17215992 |
 ||||| 1199397442 |||
 ||||| 495536627 |||
 ||||| 512752619 |||
 [SHEET: P&L]
 | (in lacs) |||||
 PROFIT & LOSS ACCOUNT INPUT | 2018 | 2019 | 2020 | 2021 | 2022 | 2023 | 2024 | ||||
 | Audited | Audited | Audited | Provisional | Projected | |||||
 Gross Sales |||||
 - Domestic | 12698.95 | 10900.11 | 14783.52 | 11355.07 | 14193.8375 | 14903.529375 | 15648.70584375 | |||

|
 - Export |||||
 Sub Total (Gross Sales) | 12698.95 | 10900.11 | 14783.52 | 11355.07 | 14193.8375 | 14903.529375 |
 15648.70584375 ||||
 Less Excise Duty |||||
 Net Sales | 12698.95 | 10900.11 | 14783.52 | 11355.07 | 14193.8375 | 14903.529375 | 15648.70584375 ||
 |
 % wise rise/fall in net sales as compared to previous year || -14.165265632197938 | 35.62725513779218 |
 -23.19102622379515 | 25 | 5.00000000000003 | 5 ||||
 Other Operating Income |||||
 Export Incentives |||||
 Duty Drawback |||||
 Others | 205.54 | 119.17 | 190.42 | 117.94 | 129.734 | 142.70740000000004 | 156.97814000000005 ||||
 Total Operating Income | 12904.490000000002 | 11019.28 | 14973.94 | 11473.01 | 14323.5715 | 15046.236775
 | 15805.68398375 ||||
 Cost of Sales |||||
 Raw materials consumed |||||
 i) Imported |||||
 ii) Indigenous | 2044.45 | 2642.71 | 3399.11 | 2780.62 | 3491.684025 | 3606.65410875 | 3786.9868141875 |
 |||
 | 2035.2899999999997 | 2650.23 | 3399.1 | 2764.68 | 3483.5830250000004 | 3597.7430087499997 |
 3777.1846041875 ||||
 Other Spares consumed |||||
 i) Imported |||||
 ii) Indigenous |||||
 Power and fuel | 103.97 | 170.62 | 226.4 | 184.15 | 283.87675 | 298.0705875 | 312.974116875 ||||
 Direct labour and wages | 442.48 | 465 | 504.31 | 491.4799999999996 | 638.7226875 | 670.658821875 |
 704.19176296875 ||||
 Repair and Maintenance |||||
 Other manufacturing expenses | 7819.27 | 6654.86 | 8418.5444 | 6242.2079 | 8090.487374999999 |
 8495.01174375 | 8919.7623309375 ||||
 Depreciation | 232.87 | 235.88 | 310.85 | 282.47 | 283.616 | 358.616 | 386.616 ||||
 Sub Total | 10643.040000000003 | 10169.07 | 12859.214400000003 | 9980.9279 | 12788.386837499998 |
 13429.011261874999 | 14110.531024968748 ||||
 Add: Op. Stock of WIP | 873.99 | 186.67 | 1265 | 839.46 | 520.5 | 546.525 | 573.85125 ||||
 Less: Cl. Stock of WIP | 186.67 | 1265 | 839.46 | 520.5 | 546.525 | 573.85125 | 602.5438125000001 ||||
 Total Cost of Production | 11330.360000000002 | 9090.74 | 13284.754400000002 | 10299.887900000002 |
 12762.361837499999 | 13401.685011874999 | 14081.838462468748 ||||
 Add Opening Stock of Finished Goods | 60.23 | 69.39 | 65.06 | 65.07 | 81.01 | 89.11100000000002 |
 98.02210000000002 ||||
 Less: closing balance | 69.39 | 61.87 | 65.07 | 81.01 | 89.11100000000002 | 98.02210000000002 |
 107.82431000000004 ||||
 Total Cost of Sales | 11321.200000000003 | 9098.259999999998 | 13284.744400000001 | 10283.947900000001 |
 12754.260837499998 | 13392.773911875 | 14072.036252468748 ||||
 Selling, Gen. & Administration Exp | 655.05 | 1197.7 | 443.5056 | 340.6520999999996 | 425.8151249999997
 | 447.10588125 | 469.46117531249996 ||||
 Cost of Sales + SGA | 11976.250000000002 | 10295.96 | 13728.250000000002 | 10624.6 | 13180.075962499997 |
 13839.879793125 | 14541.497427781247 ||||
 Operating Profit before Interest | 928.2399999999998 | 723.3200000000015 | 1245.6899999999987 |
 848.4099999999999 | 1143.4955375000027 | 1206.3569818750002 | 1264.186559687521 ||||
 Interest payment to Banks |||||
 Interest - WC | 39.77 | 121.22 | 124.61 | 43.4 | 292.5 | 292.5 | 292.5 ||||
 Interest - Term Loans | 77.13 | 62.110000000000014 | 89.7 | 106.43 | 35.22506250000001 |
 39.79127430555557 | 24.135690972222243 ||||
 Interest payment to FIs |||||
 Interest - WC |||||
 Interest - Term Loans |||||
 Total Interest | 116.9 | 183.33 | 214.31 | 149.83 | 327.72506250000004 | 332.2912743055556 |
 316.6356909722222 ||||
 | 0.009205485492895082 | 0.016819096321046303 | 0.01449654750695369 | 0.01319498690893143 |
 0.02308925000022017 | 0.022296146499564008 | 0.020233985745133336 ||||
 Operating Profit after Interest | 811.3399999999998 | 539.9900000000015 | 1031.3799999999987 |
 698.5799999999998 | 815.7704750000026 | 874.0657075694446 | 947.55086499653 ||||
 Non Operating Items |||||
 Add Non Operating Income |||||
 Profit on sale of assets/ investments |||||
 Investments and Dividend/Interest Income |||||
 Forex gains |||||
 Non-op. income from subsidiaries |||||
 Tax Refund |||||
 Other Non Operating Income |||||
 Total non-operating Income | 0 | 0 | 0 | 0 | 0 | 0 | 0 ||||
 Deduct Non Operating Expenses |||||
 Loss on sale of assets |||||
 Prem. Expenses w/off |||||
 Other Non- operating exp. |||||
 Total Non-operating expenses | 0 | 0 | 0 | 0 | 0 | 0 | 0 ||||
 Net of Non-operating Income / Expenses | 0 | 0 | 0 | 0 | 0 | 0 | 0 ||||
 Profit Before Interest Depreciation & Taxes (PBITD) | 1161.11 | 959.2000000000015 | 1556.5399999999986 |
 1130.8799999999999 | 1427.1115375000027 | 1564.9729818750002 | 1650.8025559687521 ||||
 Profit Before Tax | 811.3399999999998 | 539.9900000000015 | 1031.3799999999987 | 698.5799999999998 |
 815.7704750000026 | 874.0657075694446 | 947.55086499653 ||||
 Provision for Taxation: |||||
 Current | 275.87 | 155.62 | 251.57999999999998 | 187.14000000000001 | 203.94261875000046 |
 218.51642689236104 | 236.88771624913178 ||||
 Deferred | -11.46 | -6.36 |||||
 Sub Total: Provision for Taxation | 264.41 | 149.26 | 251.57999999999998 | 187.14000000000001 |
 203.94261875000046 | 218.51642689236104 | 236.88771624913178 ||||
 Net Profit After Tax | 546.9299999999998 | 390.7300000000015 | 779.7999999999988 | 511.4399999999998 |
 611.8278562500022 | 655.5492806770835 | 710.6631487473982 ||||
 Extraordinary Items adjustments: |||||
 Extraordinary Income adjustments (+) |||||

Extraordinary Expenses adjustments (-) |||||

SubTotal Extraordinary items | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 556.0999999999999 | 383.21000000000106

||||| 546.93 | 390.73

Adjusted PAT (excl Extraordinary Items) | 546.9299999999998 | 390.7300000000015 | 779.7999999999988 |

511.4399999999998 | 611.8278562500022 | 655.5492806770835 | 710.6631487473982 || 9.169999999999959 |

-7.519999999998959

Dividend Paid ||||| 7810.1 | 6662.38

On Equity Capital ||||| 9.17 | 7.52

On Preference Sh. Capital ||||| 7819.27 | 6654.86

Dividend tax |||||

Partners' withdrawal |||||

Dividend (%) |||||

Retained Profit | 546.9299999999998 | 390.7300000000015 | 779.7999999999988 | 511.4399999999998 |

611.8278562500022 | 655.5492806770835 | 710.6631487473982 |||

Cash Accruals | 779.7999999999998 | 626.6100000000015 | 1090.6499999999987 | 793.9099999999999 |

895.4438562500022 | 1014.1652806770835 | 1097.2791487473983 |||

PBDIT/Sales | 0.09143354371818141 | 0.0879991193556775 | 0.10528886219249534 | 0.09959251682288175 |

0.10054444666567464 | 0.10500687068797086 | 0.1054913149018053 |||

Operating Profits/Sales | 0.06389032164076555 | 0.04953986702886498 | 0.06976552269013055 |

0.06152141730522135 | 0.05747356731398416 | 0.05864823596990727 | 0.06055138836768257 |||

PBT/Sales | 0.06389032164076555 | 0.04953986702886498 | 0.06976552269013055 | 0.06152141730522135 |

0.05747356731398416 | 0.05864823596990727 | 0.06055138836768257 |||

PAT/Sales | 0.043068915146527845 | 0.035846427237890395 | 0.052747924716170354 | 0.04504067346128204 |

0.04310517548548814 | 0.04398617697743046 | 0.04541354127576197 |||

Cash Accruals/ Sales | 0.06140665173104862 | 0.05748657582354687 | 0.07377471671158145 |

0.069916786070011 | 0.06308680483695846 | 0.06804866519593004 | 0.07011948206475138 |||

[SHEET: Assessment]

FBF Method | 2018 | 2019 | 2020 | 2021 | 2022 | 2023 | 2024 |||||

Total Current Assets | 2486.41 | 5014.5499999999999 | 5731.1099999999999 | 6019.3699999999999 |

8204.663525085618 | 8344.4331673506 | 8564.961606070125 |||||

Other Current Liabilities | 1962.2799999999999 | 2872.8599999999997 | 3420.8700000000001 | 4154.78 |

1591.9625688356173 | 1675.2558927568489 | 1686.4192013523125 |||||

Working Capital Gap | 524.1300000000001 | 2141.6899999999996 | 2310.2399999999998 | 1864.5899999999992 |

6612.700956250001 | 6669.1772745937515 | 6878.542404717812 |||||

25% of WCG | 131.03250000000003 | 535.4224999999999 | 577.5599999999999 | 466.1474999999998 |

1653.1752390625002 | 1667.2943186484379 | 1719.635601179453 |||||

25% of TCA | 621.6025 | 1253.6374999999998 | 1432.7774999999997 | 1504.8424999999997 | 2051.1658812714045

| 2086.10829183765 | 2141.240401517531 |||||

Actual/ Projected WC | -132.67999999999984 | 420.81999999999997 | 2324.0999999999998 | 834.4499999999991 |

2712.7009562500007 | 2769.1772745937515 | 2978.542404717812 |||||

Flexible Bank Finance WCG | 393.0975000000001 | 1606.2674999999997 | 1732.6799999999998 |

1398.4424999999994 | 4959.5257171875 | 5001.882955945313 | 5158.906803538359 |||||

Flexible Bank Finance TCA | -97.47249999999985 | 888.0524999999998 | 877.4624999999983 |

359.7474999999995 | 4561.535074978596 | 4583.068982756102 | 4737.3020032002805 |||||

Flexible Bank Finance NWC | 656.81 | 1720.87 | -13.860000000000127 | 1030.14 | 3900 | 3900 | 3900 |||

|||

Limit recommended | 656.81 | 1720.87 | -13.860000000000127 | 1030.14 | 3900 | 3900 | 3900 |||||

Net Sales | 12698.95 | 10900.11 | 14783.52 | 11355.07 | 14193.8375 | 14903.529375 | 15648.70584375 |||

||||

NWC to TCA% | -5.336207624647578 | 8.391979340120246 | 40.552353732522995 | 13.86274643359686 |

33.06291535242078 | 33.18592430494568 | 34.7758990840878 |||||

Flexible Bank Finance to TCA% | 26.415997361657972 | 34.317535970326354 | -0.24183796856106637 |

17.11375110684341 | 47.533941984041356 | 46.737746252910185 | 45.53435472770839 |||||

Other Current Liabilities to TCA% | 78.92021026298961 | 57.2904846895534 | 59.68948423603807 |

69.02350245955972 | 19.403142663537864 | 20.076329442144132 | 19.689746188203813 |||||

Sundry Creditors to TCA% | 48.84874176020849 | 30.121346880577526 | 34.827110280556475 |

44.56612569089457 | 2.1351683807629205 | 2.204572254250042 | 2.256810414777877 |||||

Turnover Method | (C in lacs) |||||

| 2018 | 2019 | 2020 | 2021 | 2022 | 2023 | 2024 |||||

Net Sales | 12698.95 | 10900.11 | 14783.52 | 11355.07 | 14193.8375 | 14903.529375 | 15648.70584375 |||

||||

25% of Net Sales | 3174.7375 | 2725.0275 | 3695.88 | 2838.7675 | 3548.459375 | 3725.88234375 |

3912.1764609375 |||||

Margin @ 5% of Sales | 634.9475000000001 | 545.0055000000001 | 739.176 | 567.7535 | 709.691875 |

745.17646875 | 782.4352921875001 |||||

Available Margin* | -132.67999999999984 | 420.8199999999997 | 2324.0999999999998 | 834.4499999999991 |

2712.7009562500007 | 2769.1772745937515 | 2978.542404717812 |||||

Eligible Limit as per 5% Margin | 2539.79 | 2180.022 | 2956.704 | 2271.014 | 2838.7675 | 2980.705875 |

3129.74116875 |||||

Eligible Limit as per Available Margin | 3307.4175 | 2304.2075000000004 | 1371.7800000000002 |

2004.3175000000008 | 835.7584187499992 | 956.7050691562486 | 933.6340562196879 |||||

Limit recommended | 2539.79 | 2180.022 | 1371.7800000000002 | 2004.3175000000008 | 835.7584187499992 |

956.7050691562486 | 933.6340562196879 |||||

Working Capital assessment |||||

| 2018 | 2019 | 2020 | 2021 | 2022 | 2023 | 2024 |||||

Raw Material | 186.67 | 1265 | 839.46 | 520.5 | 546.525 | 573.85125 | 602.5438125000001 |

1.8666999999999998 | 12.65 | 8.3946 | 5.205 | 5.46525 | 5.738512500000001 | 6.025438125000001

Cost of Production | 11330.3600000000002 | 9090.74 | 13284.7544000000002 | 10299.8879000000002 |

12762.361837499999 | 13401.685011874999 | 14081.838462468748 | 113.30360000000002 | 90.9074 |

132.84754400000003 | 102.99887900000002 | 127.62361837499999 | 134.01685011875 | 140.81838462468747

FG | 1096.28615 | 634.38144 | 534.92589 | 1008.28 | 1039.79 | #REF! | #REF! |||||

months cost of sales | 9.888626988183649 | 2.505876539308533 | 2.0087089534490747 | 1.818623995382555 |

1.8365440094200767 | #REF! | #REF! |||||

Raw Material (Days) | 6.013449705040261 | 50.79069470692155 | 23.064250250648215 | 18.445103659817498 |

15.630463039674806 | 15.629057544958338 | 15.617882007996236 |||||

Finished Goods | 69.39 | 61.87 | 65.07 | 81.01 | 89.11100000000002 | 98.02210000000002 |

107.82431000000004 | 0.6939 | 0.6187 | 0.6507 | 0.8101 | 0.8911100000000002 | 0.9802210000000002 |

1.0782431000000003

Cost of Sales | 11321.200000000003 | 9098.259999999998 | 13284.744400000001 | 10283.947900000001 |

12754.260837499998 | 13392.773911875 | 14072.036252468748 | 113.21200000000003 | 90.98259999999999 |

132.84744400000002 | 102.83947900000001 | 127.54260837499999 | 133.92773911875 | 140.7203625246875

Total Inventory/Sales (Days) | 2.2371612549906366 | 2.482073495371643 | 1.7878063201577288 |

2.875223628855607 | 2.550168560483622 | 2.6714455672455273 | 2.796743303094857 |||||

Domestic receivables (Days Gross dom.Sales) | 1150.38 | 1424.56 | 1607.88 | 1443.58 | 3110.978082191781 |

3266.5269863013696 | 3429.853335616438 | 11.503800000000002 | 14.2456 | 16.0788 | 14.435799999999999 |

31.109780821917813 | 32.6652698630137 | 34.29853335616438

Net Sales | 12698.95 | 10900.11 | 14783.52 | 11355.07 | 14193.8375 | 14903.529375 | 15648.70584375 |

126.9895 | 109.00110000000001 | 147.83520000000001 | 113.55069999999999 | 141.938375 | 149.03529375 |

156.4870584375

Receivables Turnover (Days) | 33.06483606912383 | 47.702674560164986 | 39.69800155849216 |

46.402769864034305 | 80 | 80 | 80 | |||||

Purchases | 2044.45 | 2642.71 | 3399.11 | 2780.62 | 3491.684025 | 3606.65410875 | 3786.9868141875 |

20.4445 | 26.4271 | 33.9911 | 27.8062 | 34.91684025 | 36.0665410875 | 37.869868141875

Creditors | 1214.58 | 1510.45 | 1995.98 | 2682.6 | 175.18338133561656 | 183.9590583818493 |

193.2949455475171 | 12.1458 | 15.1045 | 19.9598 | 26.826 | 1.7518338133561655 | 1.839590583818493 |

1.932949455475171

creditors/purchases (Days) | 216.84154662623197 | 208.61700678470206 | 214.33042767077265 |

352.13333716940826 | 18.312634742916075 | 18.61699355823346 | 18.630288032830357 | |||||

Working Capital Cycle (Days) | -175.52609959707723 | -107.64156402224387 | -149.78036954147456 |

-284.41024001670087 | 79.86799685724235 | 79.68350955397041 | 79.78433727826075 | |||||

Total Amount of Term Loan | 0 |||| | uuuiioioioioioioioioioio || 100 |||||

No. Total Installments | 84 ||||| |||||

DSCR Calculation ||||| |||||

Financial Year Ending/Year of Operation | I Year | II Year | III Year | IV Year | V Year | VI Year | VI

Year |||||

| 2018 | 2019 | 2020 | 2021 | 2022 | 2023 | 2024 |||||

Sources ||||| |||||

Net Profit after Tax | 546.9299999999998 | 390.73000000000015 | 779.7999999999988 | 511.4399999999998 |

611.8278562500022 | 655.5492806770835 | 710.6631487473982 | |||||

Interest on term loan | 77.13 | 62.110000000000014 | 89.7 | 106.43 | 35.22506250000001 |

39.79127430555557 | 24.135690972222243 | |||||

Deprecation | 232.87 | 235.88 | 310.85 | 282.47 | 283.616 | 358.616 | 386.616 | |||||

Preliminary Expenses W/o | 0 | 0 | 0 | 0 | 0 | 0 | 0 | |||||

Total(A) | 856.9299999999998 | 688.72000000000015 | 1180.3499999999999 | 900.3399999999999 |

930.6689187500023 | 1053.956554982639 | 1121.4148397196204 | |||||

Repayment Obligations ||||| |||||

Term loan installment | 0 | 0 | 0 | 141.7 | 0 | 0 | 0 | |||||

Interest on term loan | 77.13 | 62.110000000000014 | 89.7 | 106.43 | 35.22506250000001 |

39.79127430555557 | 24.135690972222243 | |||||

Total Repayment(B) | 77.13 | 62.110000000000014 | 89.7 | 248.13 | 35.22506250000001 | 39.79127430555557 |

24.135690972222243 | |||||

Debt Service Coverage Ratio(A/B) | 11.110203552443924 | 11.08871357269363 | 13.158862876254169 |

3.6285011888929186 | 26.420646343778724 | 26.487127476475113 | 46.46292666782386 | |||||

Average DSCR | 11.683656626970585 | ||||| |||||

[SHEET: Financial Indicators]

0 ||||| |||||

Financial Indicators | C in lacs) ||||| |||||

| 2018 | 2019 | 2020 | 2021 | 2022 | 2023 | 2024 |||||

Lead-up Capital | 179.4 | 179.4 | 190.4 | 183.2 | 183.2 | 183.2 | 183.2 | 1.794 | 1.794 |

1.9040000000000001 | 1.8319999999999999 | 1.8319999999999999 | 1.8319999999999999 | 1.8319999999999999 |

|

Reserves & Surplus | 3018.91 | 3409.64 | 5278.45 | 4955.38 | 5567.197856250001 | 6222.747136927084 |

6933.410285674479 | | 30.1891 | 34.096399999999996 | 52.7845 | 49.5538 | 55.67197856250001 |

62.227471369270845 | 69.33410285674479 | |

Intangible Assets | 9.52 | 9.21 | 4.73 | 7.39 | 5.542499999999995 | 4.156874999999999 |

3.1176562500000005 | | 0.09519999999999999 | 0.09210000000000002 | 0.0473 | 0.0739 | 0.05542499999999995

| 0.04156874999999995 | 0.031176562500000005 | | 100

Unsecured loans (Quasi equity) | 19.09 | 19.09 | 19.09 | 0 | 0 | 0 | 0 | | 0.1909 | 0.1909 | 0.1909 | 0 |

0 | 0 | 0 | |

Tangible Net Worth | 3188.79 | 3579.83 | 5464.12 | 5131.19 | 5744.855356250001 | 6401.790261927084 |

7113.492629424479 | | 31.8879 | 35.7983 | 54.6412 | 51.311899999999994 | 57.44855356250001 |

64.01790261927084 | 71.13492629424479 | |

Long Term Liabilities | 1476.83 | 1375.88 | 1878.3200000000002 | 1528.76 | 1956.94 | 1772.7566666666667 |

1588.5733333333333 | | 14.7683 | 13.7588 | 18.7832 | 15.2876 | 19.5694 | 17.727566666666668 |

15.885733333333333 | |

Capital Employed | 4665.62 | 4955.71 | 7342.4400000000005 | 6659.95 | 7701.795356250001 |

8174.546928593751 | 8702.065962757812 | | 46.6562 | 49.5571 | 73.4244 | 66.59949999999999 |

77.01795356250001 | 81.7454692859375 | 87.02065962757811 | |

Net Block | 1223.7 | 1273.46 | 2309.91 | 2836.1 | 2554.3215 | 2947.091125 | 2841.51434375 | | 12.237 |

12.7346 | 23.0991 | 28.361 | 25.543215 | 29.47091125 | 28.4151434375 | |

Non Current Assets | 3574.6 | 3261.43 | 2708.43 | 3131.0999999999995 | 2434.7729 | 2458.278529 |

2882.00921429 | | 35.746 | 32.6143 | 27.0843 | 31.310999999999993 | 24.347729 | 24.58278529 |

28.8200921429 | |

Net Working Capital | -132.67999999999984 | 420.8199999999997 | 2324.099999999998 | 834.4499999999991 |

2712.7009562500007 | 2769.1772745937515 | 2978.542404717812 | | -1.3267999999999984 | 4.208199999999997 |

23.240999999999982 | 8.344499999999991 | 27.127009562500007 | 27.691772745937513 | 29.78542404717812 | |

Current Assets | 2486.41 | 5014.549999999999 | 5731.109999999999 | 6019.369999999999 | 8204.663525085618

| 8344.4331673506 | 8564.961606070125 | | 24.864099999999997 | 50.14549999999999 | 57.31109999999999 |

60.19369999999999 | 82.04663525085618 | 83.444331673506 | 85.64961606070125 | |

Current liabilities | 2619.0899999999997 | 4593.73 | 3407.0100000000007 | 5326.62 | 5491.962568835617 |

5575.255892756849 | 5586.4192013523125 | | 26.190899999999996 | 45.93729999999999 | 34.070100000000004 |

53.2662 | 54.91962568835617 | 55.75255892756849 | 55.86419201352312 | |

Current Ratio | 0.9493411833881235 | 1.0916074736651915 | 1.6821523858163017 | 1.1300543308890063 |

1.4939401757112005 | 1.4966906143611016 | 1.533175599138137 | |||||

Debt Equity Ratio | 0.4631317835291756 | 0.38434227323643866 | 0.34375562159747594 | 0.2979347870571934 |

0.3406421708896441 | 0.2769157679547325 | 0.22331833546327265 | |||||

Debt Equity Ratio with USL as Quasi Capital | 0.45442472910458 | 0.3769992108743735 | 0.339076927566152 |

0.2979347870571934 | 0.3406421708896441 | 0.2769157679547325 | 0.22331833546327265 | |||||

DER (TOL/TNW) | 1.284474675347075 | 1.667568013006204 | 0.967279269122933 | 1.336021468704141 |

1.2966214302909702 | 1.1478058884752649 | 1.0086455287810066 | |||||

DER (TOL/TNW) with USL as Quasi Capital | 1.2708798334102271 | 1.6534182476965311 | 0.9604301130177397 |

1.336021468704141 | 1.2966214302909702 | 1.1478058884752649 | 1.0086455287810066 | 100 |||||

Net Sales | 12698.95 | 10900.11 | 14783.52 | 11355.07 | 14193.8375 | 14903.529375 | 15648.70584375 | |

126.9895 | 109.00110000000001 | 147.83520000000001 | 113.55069999999999 | 141.938375 | 149.03529375 |

BioBridge Master Knowledge — Page 62

BioBridge Master Knowledge — Page 63

	2937942															
	19	Chromatography AKTA-1 (30 L/hr pipe MOC PP)	1	14657140											5862856	
	8794284															
	20	Chromatography Column Dia 300mm x Height 750mm	1	4896570											1958628	
	2937942															
	21	Chromatography Column Dia 600mm x Height 500mm	1	14244570											5697828	
	8546742															
	22	Chromatography AKTA-1(160 L/hr pipe MOC PP)	1	14657140											5862856	
	8794284															
	23	Chromatography Column Dia 450mm x Height 500mm	1	12018850											4807540	
	7211310															
	24	TFF System 40 sqm	1	19350000											7740000	
	25	Cassette 40 sqm	1	5590000											2236000	
	26	TFF System 30 sqm	2	38700000											15480000	
	27	Cassette 30 sqm	2	8600000											3440000	
	28	TFF System 20 sqm	1	9460000											3784000	
	29	Cassette 20 sqm	1	3010000											1204000	
	30	TFF System 5 sqm	2	7310000											2924000	
	31	Cassette 5 sqm	2	688000											275200	
	32	Continuous Centrifuge 1000L/hr	2	30000000											12000000	
18000000																
	33	Process vessel 150L	3	3225000											1290000	
	34	Process vessel 250L	4	5160000											2064000	
	35	Process vessel (Mobile tank) 300L	2	2580000											1032000	
	258000															
	36	Process vessel 1500L	20	43000000											17200000	
4300000																
	37	Process vessel 2000L	6	15480000											6192000	
1548000																
	38	Buffer Vessel 500L	14	24080000											9632000	
2408000																
	39	Buffer Vessel 1000L	10	19350000											7740000	
1935000																
	40	Depth filtration system	2	2100000											840000	
	41	Nano filter holder	1	500000											200000	
	42	Cold room 2-8°C	2	1500000											600000	
	43	Rectangular tank	1	430000											172000	
	44	Buffer preparation tank	2	1720000											688000	
	45	Dispensing Booth	1	400000											160000	
	46	Analytical Balance	1	170000											68000	102000
	47	Precision Balance	1	150000											60000	90000
	48	Commercial Balance	1	120000											48000	72000
	49	PVC bag Sealer	1	5000											2000	3000
	50	Rotary Bag Sealer	1	300000											120000	180000
	51	pH/ Conductivity meter	2	300000											120000	180000
	52	Magnetic Stirrer	2	150000											60000	90000
	53	Peristaltic pump	2	200000											80000	120000
	54	Filter Integrity machine	1	2000000											800000	1200000
	55	Fume Hood	1	450000											180000	270000
	56	Autoclave 1000L	2	4000000											1600000	
	57	DHS 1000L	1	2000000											800000	
	58	Ceiling LAF	3	1200000											480000	
	59	Drying Oven	2	1000000											400000	600000
	60	Acid tank 1000L	2	3870000											1548000	
	61	Base tank 1000L	2	3870000											1548000	
	62	Central CIP system	2	2150000											860000	
	63	Refrigerator 200L	1	150000											60000	90000
	64	Laminar air flow cabinet	1	435000											174000	261000
	65	Dynamic pass box	4	480000											192000	
	66	Tumble Dryer 25 Kg	1	2000000											800000	1200000
	67	Barrier washer Extractor 25Kg	1	3000000											1200000	
1800000																
	68	SS Furniture (Table, chair, racks, Ladders etc)	NA	4000000												
	1600000	2400000														
	69	Lockers	NA	200000											80000	120000
	70	Laptop / Computers	10	500000											200000	300000
	71	Office furniture / Cabinets	NA	300000											120000	
180000																
	Total Amount in INR		720018520	0	0	0	0	0	0	0	0	115498348	30592000			
60513524	75231536	30307500	176875522	90770286	112847304	6061500	5891000	6172000	9258000							
0	0															
[SHEET: Fill finish]																
Plasma Project																
Annual Production Capacity 150,000L																
FILL FINISH EQUIPMENT			Monthly Cash Flow													
S/No	Name of Equipment	Quantity	Individual Cost (INR)	M1	M2	M3	M4	M5	M6	M7	M8					
M9	M10	M11	M12	M13	M14	M15										
1	Formulation Vessel 100L	1	2500000				750000					1500000		250000		
2	Ceiling LAF	2	800000				240000					480000		80000		

3	Laminar air flow cabinet	1	435000						130500							261000		43500
4	Continuous Particle Counter	1	3000000						900000							1800000		300000
5	Dynamic pass box	7	840000						252000							504000		84000
6	Pass Box static	1	38000						11400							22800		3800
7	Vial washing machine	1	4500000	1350000							2700000		450000					
8	Sterilizing tunnel	1	4500000	1350000							2700000		450000					
9	Vial filling & Sealing line 120 vials /min	1	7500000	2250000							4500000							
	750000																	
10	Lyophilizer (1sq meter)	1	5500000	1650000							3300000		550000					
11	Labelling machine	1	900000	270000							540000		90000					
12	Double door autoclave 1000L	1	2000000	600000							1200000		200000					
13	Cold room	1	800000						240000							480000		80000
14	SS furniture (Table, Chair, racks, cabinets etc)	1	2000000						60000									
140000																		
Total Amount in INR		33513000	7470000	0	0	0	2583900	0	0	0	0	0	19987800					
0	761300	2710000																
[SHEET: QC]																		
Plasma Project																		
Annual Production Capacity 150,000L																		
QC & R&D EQUIPMENT			Monthly Cash Flow															
S/No	Name of Equipment	Quantity	Individual Cost (INR)	M1	M2	M3	M4	M5	M6	M7	M8							
M9	M10	M11	M12	M13	M14	M15												
1	Akta pure -150 (For R&D)	2	4000000															
2	Columns XK 50/100 (For R & D)	1	700000															
3	Columns XK 50/60 (For R&D)	1	700000															
4	Column XK 50/30 (For R & D)	3	1400000															
5	HPLC-UV	1	3400000															
6	HPLC Columns and accessories	1	1000000															
7	Electrophoresis (Minigel) with trans blot	2	600000															
8	Gel Doc system densitometer	1	500000															
9	Power packs	2	200000															
10	Endotoxin Detection system (Tube heating block)	1	200000															
11	Micropipettes- variable volume range of 2-20µL, 20-200µL, 100-1000µL, 500-5000µL and multichannel																	
30-300µL	2 set each	300000																
12	Lab Refrigerator 2-8 ° C (SS) with data control	4	450000															
13	Flame photometer	1	450000															
14	Weighing Balance Analytical	2	350000															
15	UV-VIS spectrophotometer Double Beam with data acquisition.	1	900000															
16	Volumetric KF	1	700000															
17	Moisture Analyser	1	800000															
18	Stability Chambers (5±3° C, 25, 37 deg. C) 200L (1set)	3	1400000															
19	Muffle Furnance	1	80000															
20	pH and conductivity meter with printer	2	260000															
21	Vaccum Oven	1	120000															
22	Melting point apparatus	1	90000															
23	Incubator Sterility Test 500L	2	1440000															
24	Incubators for Normal Media plates Incubation 500L	1	300000															
25	Incubator for EMP and water samples 500L	2	600000															
26	Hot Air Oven	1	160000															
27	Thermostatic water bath (10-20 L)	2	200000															
28	Vaccum pump	2	120000															
29	Desiccator Glass	1	4000															
30	High Quality Laboratory water generation system (10 L /hr) Milli -Q water	1	1450000															
31	Sonicator	1	180000															
32	Vortex Shaker	6	210000															
33	Magnetic Stirrer with hot plate	4	400000															
34	LAF Vertical	2	600000															
35	Biosafety Cabinet	2	1400000															
36	Autoclave Double door	1	1250000															
37	Dry Heat Steriliser	1	750000															
38	Fume Hood (Table Top)	1	450000															
39	Deep freezer (-20 Degrees C) with Data acquisition	1	400000															
40	Air sampler	1	300000															
41	Particle counter	1	2000000															
42	Decontamination autoclave	1	1000000															
43	Microscope Binocular	1	100000															
44	Colony Counter	1	150000															
45	Microplate reader ELISA	1	600000															
46	Microplate washer	1	350000															
47	TOC analyser (Grab Samples)	1	2500000															
48	Timers/ Stop watch	5	10000															
49	Ice flaking machine	1	250000															
50	Standard Weight Box	1 Set	1000000															
51	Printer/scanner / xerox machine	1	500000															
52	Centrifuge table top	1	450000															
53	Size exclusion	L	150000															
54	Anion	L	150000															
55	Cation	L	150000															
56	Affinity	L	150000															
57	SS furniture racks	25	250000															
58	Computers / Laptops	10	500000															
59	Office Furniture	NA	500000															
60	Office Cabinets for Documents	10	120000							O	P							
	TOTAL	39694000							11908200		23816400	3969400						
[SHEET: QA]																		
Plasma Project																		
Annual Production Capacity 150,000L																		
QA EQUIPMENT			Monthly Cash Flow															
S/No	Name of Equipment	Quantity	Individual Cost (INR)	M1	M2	M3	M4	M5	M6	M7	M8							
M9 | M10 | M11 | M12 | M13 | M14 | M15 | M16

1	Anemometer	1	15000						
2	Lux meter	1	12000						
3	Datalogger 32 channel	1	400000						
4	Paper shredder	1	10000						
5	Laminator	1	10000						
6	Printer/scanner / xerox machine	1	500000						
7	Computers / Laptops	10	500000						
8	Numbering Machine	5	25000						
9	Spiral Binding Machine	1	15000						
10	A3 +A4 Lamination Machine	1	25000						
11	Bi-Focal Glass	1	80000						
12	Tachometer	2	6000						
13	Clamp Meter/ Multi meter	5	10000						
14	Level Indicator	2	8000						
15	Measuring TAPE	2	1500						
16	Stop Watch	5	50000						
17	Temperature recorders 24 Nos. (data Loggers)	20	200000						
18	Office Cabinets for Documents	20	240000						
19	Projector for conference / meeting room	1	300000						
20	Office Furniture	NA	700000						
Total Amount in INR			3107500				932250		
[SHEET: FG_Storage]									
Plasma Project									
Annual Production Capacity 150,000L									
FINISHED GOODS STORAGE			Monthly Cash Flow						
S/No	Name of Equipment	Quantity	Individual Cost (INR)	M1	M2	M3	M4	M5	M6
M9	M10	M11	M12	M13	M14	M15	M16	M17	
1	Air curtain	10	100000						
2	PVC pallets	50	300000						
3	Storage racks SS	50	1500000						
4	Hydraulic trolley	1	250000						
Total Amount in INR			2150000				645000		
[SHEET: STP]									
Plasma Project									
Annual Production Capacity 150,000L									
ETP / STP Plant			Monthly Cash Flow						
S/No	Name of Equipment	Quantity	Individual Cost (INR)	M1	M2	M3	M4	M5	M6
M9	M10	M11	M12	M13	M14	M15			
1	Biowaste reception tank 15000L	1	2500000						
2	Kill tank 500L	2							
3	Bar Screen	NA							
4	Collection tank	NA							
5	Equalization tank	NA			750000	1500000	250000		
6	Neutralization tank	NA							
7	Chemical coagulation	NA							
8	Biological oxidation tank	NA							
9	Filtration System	NA							
STP BUDGET ESTIMATE IN INR									
1	Primary treatment plant : Bar screening - 3 stages, pH correction, BOD & COD removal secondary clarifier		800000						
2	Secondary treatment : Electro coagulation process, Flash mixer, HRSC clarifier, Sand filter								
	240000			480000	80000				
3	Sludge Management: Sludge thickner, Screw pump, Filter press, Sludge storage								
Total Amount in INR			3300000	0	0	0	990000	0	1980000
0	0								
[SHEET: Admin]									
Plasma Project									
Annual Production Capacity 150,000L									
Admin Equipment			Monthly Cash Flow						
S/No	Name of Equipment	Quantity	Individual Cost (INR)	M1	M2	M3	M4	M5	M6
M9	M10	M11	M12	M13	M14	M15	M16	M17	
1	Office Cabinets for Documents	20	240000						
2	Computers / Laptops	50	2500000						
3	Office Furniture	NA	1200000						
4	Printer/scanner / xerox machine	1	500000						
5	Water Dispenser	2	40000						
6	Projector for conference / meeting room	2	600000						
Total Amount in INR			5080000				1524000		
[SHEET: Canteen]									
Plasma Project									
Annual Production Capacity 150,000L									
Canteen Equipment			Monthly Cash Flow						
S/No	Name of Equipment	Quantity	Individual Cost (INR)	M1	M2	M3	M4	M5	M6
M9	M10	M11	M12	M13	M14	M15	M16	M17	
1	Refrigerator	2	300000						
2	SS furniture (serving table etc)	NA	400000						
3	SS Water cooler dispenser	1	50000						
4	Microwaves	2	60000						
5	Kitchenware	na	500000						
6	Bain marie	1	50000						
Total Amount in INR			1360000				408000		

Techno-Commercial Feasibility cum Detailed Project Report for setting up a Plasma Fractionation Facility
150,000 Litre / Annum Capacity

Document number:

LWIL/F-DPR/2023-00 Dated.15.12.2023

Prepared for:

Lifenity Wellness International Ltd,
12th Floor, D-Wing, Trade World, Kamala Mills Compound, Lower Parel, Mumbai -400 013

Contact: Mr. T. Rajshekar

Prepared by:

BioBridge Healthcare Solutions Pvt. Ltd,
#13, Rachana Blossom, Jagdishnagar, Aundh, Pune 411007

www.biobridge.in

Contact: Dr. Ganesh Kumraj

Email: ganesh@biobridge.in

Mob: +91 9004349016

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CHAPTER 1 INTRODUCTION

1.1 Project Objective Statement:

The objective of this project is to establish and operate a state-of-the-art Human Plasma Derived Medicinal Products (HPDMP) manufacturing facility with an annual production capacity of 150,000 liters. This capacity will be subsequently increased to 500,000 liters per annum in the near future.

This facility aims to consistently and efficiently produce high-quality plasma derived products such as Human Serum Albumin (HSA), Human Normal Immunoglobulin (IgG), Hemophilia factors VIII, IX, Fibrin sealant etc. and Monoclonal antibodies (Mabs) while ensuring compliance with regulatory standards and best industry practices.

The primary goals of this project include:

Meeting Growing Demand: To address the increasing demand for plasma-derived products, ensuring a stable and reliable supply of essential treatments.

Quality Assurance: Implementing rigorous quality control and quality assurance processes to consistently produce safe and effective plasma derived products that meet or exceed regulatory requirements.

Operational Efficiency: Optimizing manufacturing processes to maximize productivity, minimize waste, and reduce operational costs to maintain competitiveness in the market.

Regulatory Compliance: Adhering to all relevant regulations and standards, including current Good Manufacturing Practices (cGMP), to ensure product safety, quality and efficacy.

Safety and Sustainability: Prioritizing the safety of both employees and the environment by implementing robust safety measures and eco-friendly practices throughout the facility.

Research and Development: Supporting ongoing research and development efforts to improve product quality, expand the product portfolio, and explore innovative techniques and newer technologies.

Supply Chain Reliability: Building a secure and efficient supply chain to source raw materials, transport finished products, and ensure timely deliveries to end users.

Stakeholder Satisfaction: Ensuring customer satisfaction by providing consistent and reliable plasma-derived products while maintaining a favorable relationship with employees, investors, and the local community.

By achieving these objectives, this Human Plasma Derived Medicinal Products manufacturing facility will play a pivotal role in meeting the critical healthcare needs of patients while contributing to advancements in the plasma industry.

Statutory Regulation

(A detailed list of provided in the References section at the end of the document)

In general, the following guidelines are referred to:

National Blood Policy, NBTC, Ministry of Health and Family Welfare.

National Policy for Access to Plasma Derived Medicinal Products from Human Plasma for

Clinical/Therapeutic Use: Addendum to National Blood Policy, NBTC, Ministry of Health and Family Welfare.

Technical manual, DGHS, MOHFW Govt. of India 2003.

Drugs and Cosmetic act 1940 and rules thereunder.

Schedule M Drugs and Cosmetic act 1940.

ISPE Baseline Guide Vol. 6, Biopharmaceutical Manufacturing Facilities ISPE Baseline Guide Vol. 3, Sterile Manufacturing Facilities.

WHO Good manufacturing practices for biological products.

Good manufacturing practice guideline for human and veterinary medicinal product.

PIC/s Guide to Good Manufacturing Practice for Medicinal Products.

USFDA Guidance for Industry – Sterile Drug Products produced by Aseptic Processing.

Executive Summary of The Project:

The Human Plasma Derived Medicinal Products manufacturing facility will be a green field project. The area earmarked for the project is approx. 6 acres in an industrial area on the outskirts of Nagpur city.

Different areas for the plasma manufacturing facility are identified and listed below:

Bulk Manufacturing Facility (for processing 150,000 L / annum)

Formulation, Fill Finish with lyophilization & Packaging

QC and R&D block

Administrative and QA Block

Ware House for Plasma, Packaging, Testing lab (for 32,000 L plasma)

Ware House (Finished Goods Storage)

Expansion area

Bulk Manufacturing Facility (for processing additional 350,000 L / annum)

Ware House for Plasma, Packaging, Testing lab (for additional 32,000 L plasma)

Supporting Blocks

Canteen

Engineering Office

Boiler House

Work Shop

Substation and Generator

Bore Well

Bio-Kill System

WTP/ETP/STP System

Fire Fighting System

Scrap Yard

Guard House and Reception Room

Security Cabin/Post

Boundary Wall and Parking etc.

Internal Roads and Drainage System

Rainwater Harvesting System

PROPOSED SITE PLAN

Annexure A : PLOT LAYOUT GF - (BULK MANUFACTURING)

Annexure B : Conceptual Layout – GF (Bulk manufacturing)

Figure 1: Proposed Site Plan

The company has identified different products to be manufactured in stagewise manner. List of products proposed to be manufactured are as follows:

Stage-1:

Albumin

Human Immunoglobulin

Factor VIII

Factor IX

Fibrin sealant

Fibrinogen

Stage-2:

Anti-D

Factor XI

Factor XIII

Hepatitis B IgG

Measles IgG

Protein C

Rabies IgG

Tetanus IgG

Thrombin

Varicella Zoster IgG

Table 1: Working Time Model Bulk Manufacturing

Table 2: Working Time Model Fill Finish

Environment Due Diligence:

As the proposed facility is required to handle biological products, proper segregation of type of wastes generated shall be adopted. Based on the categorization of the waste, appropriate treatment philosophy shall be adopted. Following broad categories are identified as wastes:

Process waste

General wash

Solid Process waste

General Solid wastes – Packaging materials etc.,

Gaseous waste from Process & Labs

Sewage

The facility shall be designed as zero waste discharge plant; meaning that every possible waste within the facility is categorized, treated with appropriate treatment step in line local pollution control board guidelines and then discharged safely. Soil contamination due to waste is also considered as a key factor and necessary steps & waste handling methods are adopted to avoid the same. As part of the design, all biologically waste will be treated in Primary treatment systems before letting it into a common wastewater treatment plant.

For the facility design, a detailed EHS document shall be prepared to identify risks and mitigation plans. During design, Construction & operation, EHS policy shall be made integral part of the sustainable operation.

CHAPTER 2

MARKET ANALYSIS

2.1 Human Plasma Derived Products and Its Share in The Total Components:

There are several medicinal products which are derived from human blood plasma by Plasma fractionation. The process for large scale purification was developed about 60 years ago by Cohn and co-workers. Therapeutically important proteins are Human serum albumin (HSA), Immunoglobulin G (IgG), Coagulation factor VIII, factor IX and Protease inhibitors.

HSA is one of the most abundant proteins in plasma and, together with IgGs, constitutes 80% of total plasma proteins. According to the data, HSA represented approximately 50% of total sales of therapeutic plasma protein products. The purified HSA is required for the treatment of hypo proteinemia, burn and heavy loss of blood due to injuries and other medical conditions. Plasma contains about 60 g/L of protein, out of which about 57 grams are used for many therapeutic products. This makes this human material unique as a source of multiple therapeutic products.

The concentration of different plasma proteins is variable. HSA is major protein component, followed by IgG. The wide concentration range of plasma proteins (less than 1 mg/mL to 40 g/L) is one of the potential challenges in designing separation and purification procedures.

2.2 Volume of Human Plasma Fractionated Worldwide (2022 to 2032):

In 2010, 78 fractionation plants were operating in the world, 25 of which were in China, 26 in Europe and 8 in the United States of America. The world plasma fractionation capacity was about 48.4 million liters. More than 500 tons of human serum albumin (HSA) and more than 40 tons of intravenous immunoglobulins (IVIg) are produced annually from more than 34 million liters of plasma. Total plasma marked of 2008 is depicted in pie chart (Figure 2).

Figure 2: Total plasma market of 2008 (Source: Marketing Research Bureau, Inc)

In 2015-16, the worldwide plasma market grew to a \$21,174 Million, almost doubled in terms of revenue from 2008. As compared to the 2008 statistics, in 2016 to IVIg and SCIG had a major share in plasma market, increasing from 29% to 47%. The market share for Albumin also grew from 9.7% to 15.7%. The market share for Factor VIII and Factor IX remained almost the same of 1 to 2 % from 2008 to 2016.

Figure 3: Plasma market in 2016 (Source: Marketing Research Bureau, Inc)

Global Plasma Fractionation Market Forecasts from 2022 to 2032

The Global Plasma Fractionation Market Size was valued at USD 27.1 Billion in 2022 (Fig 3).

The Market Size is growing at a CAGR of 6.7% from 2022 to 2032.

The Worldwide Plasma Fractionation Market Size is expected to reach USD 51.8 Billion by 2032.

Asia Pacific region is expected to grow the fastest during the forecasted period.

Figure 4: Global Fractionation Market

Companies in Domestic and International markets:

Key Market Players (Indian Manufacturers)

Plasma Gen Biosciences Pvt. Ltd. Bangalore

Reliance Life Sciences, Mumbai

Biocon Limited Bangalore

Intas Pharmaceuticals Ltd. Ahmedabad

Virchow Biotech, Hyderabad

Fusion Healthcare, Hyderabad

ICHOR Biologics Pvt. Ltd., Hyderabad

Bharat Serums and Vaccines Limited, Mumbai (Sales and marketing)

Table 3: Domestic companies' products

Key market players (International Manufacturers):

CSL Behring, USA

Bio plasma limited, USA
Pfizer, USA
Kedrion Biopharma, USA
ADMA Biologics, USA
Takeda Pharmaceutical Company Limited
Biotest AG, Germany
Octa pharma, Switzerland
Grifols, Spain
Kamada Pharmaceuticals, Israel
SK Plasma, South Korea

Table 4: International companies' products

Global plasma Market Overview:

Table 5: Plasma Derived Therapy Market Share, By Region, 2022 *

Source: Precedence Research, Plasma derived market size, A report 2032

Future roadmap for plasma fractionation industry in India:

Global plasma fractionation program, plasma protein therapies & newer clinical indications for use of these products, has led to a rapidly increasing demand.

Due to the limited volumes of plasma available for fractionation and fractionation capacities, has led to various challenges in availability of these products.

Unfortunately, plasma fractionation program did not get enough attention, focus and motivation in India because of challenges like availability of plasma, less components processing units, cold chain infrastructure, political will and availability of solid plasma ecosystem.

As per literature information, in 2020 India, wasted 6.5 Lakh units of blood and blood components on annual basis (Source: Print line Media Pvt Ltd). A report published by Ministry of Health and FW (2018-2019); the total collection of blood was 12.2 million units out of which 75.05% of blood was componentized by NACO supported blood component separating unit (BCSU). Thus, the wastage of 24.95% hampered the availability of blood components to the patients. Though, with the passage of time there has been a significant improvement in the overall blood banking systems and availability of plasma products in India, still much needs to be done considering clinical needs of the country.

There is a need to create a vision document for short-term, mid-term and long-term self-sufficiency of plasma products in India. This can be divided into following two categories:

Availability of safe plasma: How to overall increase the quantity and quality of plasma for fractionation:

Increase voluntary blood collection, with increased repeat donor base.

As a part of the licensing process blood component separation should be mandatory and a plan for current licensed blood banks to migrate to complete blood separation units in the next 3 years.

Making blood component therapy mandatory in clinical practice, so that blood therapy can be minimized, and patients can use plasma products, which are relatively safer and easily accessible.

Create a solid regulatory pathway for permitting source plasma collection.

Regulatory system reform: Create a facilitative regulatory landscape for plasma industry to grow and evolve which would involve office of DCGI, NIB, Department of AIDS Control (DAC), IPC, NPPA, NACO, clinicians, patient groups, industry representatives and NGOs by;

Forming a national level body for steering plasma protein initiative in India.

Making a robust regulatory system for collection.

Controlling pricing for plasma and increasing indigenous manufacturing capabilities.

More funding for products and patient care.

Fast track regulatory approval for importing different categories of plasma/fractions/intermediate.

2.4 Market Trend for The Next 10 Years and Shift in Products (2023-2032):

With the growth of IgG sales, some relatively niche plasma proteins have had a significant impact on the market in the 2010s.

C1 esterase inhibitor (C1-INH): In 2010, the sales of C1 esterase inhibitor (C1-INH) for the treatment of hereditary angioedema (HAE) amounted to about \$200 million globally. This was 1.5% of total plasma proteins sales because the product had just been launched in late 2008 in the United States, and European sales were limited by the product's approval confined to only a few countries. By 2016, the sales reached over \$1.1 billion (5.2% of global PDMPs sales), as the product gained rapid acceptance mainly in the United States where the awareness of treatment and diagnosis of HAE were boosted by Shire and CSL Behring. As a result, C1-INH became the fourth largest plasma protein in terms of dollar sales worldwide, behind IgG, albumin and plasma-derived factor VIII. The market for C1-INH is continuing to grow strongly, and it is likely to increase its share in the future. A new formulation for subcutaneous injection will further facilitate product adoption. In addition, C1-INH is under investigation for transplant where antibody mediated rejection (AMR) has occurred.

Alpha-1 Antitrypsin (AAT): Other specialty products which continue to grow at a stronger pace is prothrombin complex concentrate (PCC) as well as alpha-1 antitrypsin (AAT). Global sales of AAT reached \$1.0 billion in 2016, or 4.7% of the total market. This is up slightly from 4.1% of sales in 2010, as the market in the United States has sustained approximately a low two-digit annual sales growth for the past 6 years, driven mostly by new patient identification and higher dosing. Most of the global AAT sales are generated in the United States, though sales of this therapy in Europe are trending upward after approval through the central European Medicines Agency (EMA) in August 2015. AAT is also under investigation for new indications, such as graft vs. host disease after hematopoietic stem cell transplantation (HSCT) as well as type 1 diabetes. If either of these areas show clinical trial success, the market would be expected to experience higher demand for the therapy.

Prothrombin complex concentrate (PCC): PCC have nearly reached the \$500 million in sales mark in 2016, up from under \$300 million in 2010. This impressive growth was due to the introduction of the first four-factor (factors II, VII, IX and X) coagulation product in the United States by CSL Behring in 2013 for the urgent reversal of vitamin K antagonist (VKA) anticoagulation in adults with acute major bleeding. Sales of this product reached close to \$200 million within four years of product launch. Sales also increased in Europe for the same indication, while at present little PCC is used in other markets around the world.

In the United States, the market share of the plasma-derived factor VIII sales fell from 11% in 2010 to 9% in 2016, while the share of plasma-derived factor IX products fell from 20% to 6%. The latter's market share drop was due to the faster adoption of the extended half-life factor IX products than those of factor VIII, because they offer longer intervals between infusions than the standard recombinant and plasma-derived products.

Globally, the plasma-derived factor VIII market share went from 27% in 2010 to 21% in 2016. For, the plasma-derived factor IX' market share fell from 34% in 2010 to 21% in 2016. According to NPPA report of 2022, government has set the pricing for scheduled drugs and plasma products.

2.5 Government Market and the Rates:

A market survey and several documents and reports were studied to see the price of various plasma products in government and private sector respectively (Table 6 and Table 7).

Table 6: Pricing of Plasma Products (Government of India)

@Source: NPPA (National Pricing Pharmaceutical Authority 2022)

Figure 5: Stage 1 - Plasma products price

Figure 6: Stage 2 - Plasma products price

Table 7: Price List for Plasma Products (Private Manufacturers)

Figure 7: Price List for Plasma Products from Private Manufacturers

CHAPTER 3 TECHNOLOGY

3.1 Analysis of Various Technologies Available (Centrifugation and filtration for solid / Liquid

Separation, Thermal and Chemical Inactivation Procedure, Ultrafiltration / Diafiltration, Nanofiltration, Chromatography or Any Other Technology or Any Hybrid Model)

Several technologies are utilized for plasma purification at commercial scale, each has its advantages, disadvantages and applications in isolating specific proteins of human interest. Some of the key methods are as under:

Methods for Plasma purification

Plasma Fractionation method with ethyl alcohol

Cohn+ Liquid Chromatography

Heat Shock technology

Ammonium sulphate+ Liquid Chromatography

Ion Exchange Chromatography (IEC)

Plasma Fractionation Method with ethyl alcohol:

Cohn and colleagues developed a protein purification method by using plasma fractionation technique for the first time. This approach is based on albumin solubility difference with other plasma proteins. As compared to other plasma proteins, albumin sediments are at lower pH and higher ethanol concentration. In this method there are five important parameters for separation of plasma protein.

Ethyl alcohol (8- 40%)

Temperature (-3 to -10 °C)

pH (4.5-7.2)

Protein concentration (5.1- 8 %)

Ionic strength (0.14- 0.01).

Protein separation is based on the increase in ethyl alcohol concentration thus albumin will be seen in the last fraction.

The method is suitable for large-scale and industrial production.

Disadvantages of Method:

Purity of the product

Denaturation of protein

The Cohn method combined with liquid chromatography:

The original Cohn method was modified by Oncley et al. in 1949 and Kistler-Nitschman et al. in 1962 in order to improve albumin purity.

The method is used for the purification of plasma proteins at the industrial scale

In this method after ethanol precipitation liquid chromatography step is added. Albumin is a by- product and other important therapeutic proteins including factor VIII, intravenous immunoglobulin (IVIG) and some other proteins are isolated from plasma.

Advantages:

Increased purity of proteins

Higher yields

Process Flow Diagram -Modified Cohn Method

Cohn Modified

Supernatant A

Heat shock method:

Albumin is stable during thermal fluctuation comparing to the other proteins of the plasma; therefore, it can be purified by heat shock purification method.

Albumin protein resists to increasing temperature up to 60°C at which potential pathogen are inactivated. After inactivation 0.04 M caprylic acid is added to stabilize the protein at pH 5 at 60°C.

At this stage other proteins of the serum are denatured and precipitated in solution. The final product (albumin) is concentrated by precipitation and ultrafiltration with purity of app 98%

Ammonium Sulphate precipitation combined with liquid chromatography:

Another method for albumin purification is ammonium sulphate precipitation combined with liquid chromatography which yields approximately more than 90 percent purified albumin.

In this method albumin is separated from immunoglobulins by 50% ammonium sulphate. Then lipids are removed using ice- cold acetone at 2-8°C.

For further purification of albumin from other contaminating proteins (especially transferrin which is the main contaminant, DNases and immunoglobulins), ion exchange and size exclusion chromatography are used.

Ion exchange chromatography (IEC):

IEC is widely used for albumin and other protein purification. Among them the anion exchange method has the most widely used. Three major steps of anion exchange consisting pre-treatment, purification and polishing.

The purification consists of two anion and cation exchange chromatography followed by polishing step comprising of two ultrafiltration and one gel filtration steps.

Process Flow Diagram Purification and Polishing Methods for Human Serum

Albumin (HSA)

The limitations of method:

Long processing time

To overcome these limitations, the new method for ion exchange membrane chromatography (IEM) using ameliorated adsorbers was developed.

The membrane chromatography method is simple and has high separation efficiencies due to shorter diffusion times, reduction in buffer usage, small foot print requirements. No complex hardware or packing is required.

Disadvantage:

Expensive membrane

Low efficiency

Table No 8: Comparison of Different Methods for Industrial Application

3.2 Recommending a Cost Effective and High Productivity Technology:

On the basis of table 8, for purity analysis and percentage recovery there are individual methods like fractionation and combined methods like Cohn + LC and IEC which can be considered.

Lifenthy should accept the method which give high productivity and acceptable quality of desired protein (purity levels) as per the regulatory expectations.

Cohn method when combined with chromatographic techniques is the most well-known method. In this method other proteins of plasma such as coagulation factors, immunoglobulins etc. would be separated in upstream, and albumin will be separated in the last fraction. Due to the robustness and high purity of plasma in large volumes, this technique, is cost- effective for plasma therapeutic protein production. Hence, this method currently seems to be a major method for albumin production.

However, taking into consideration, the latest developments in plasma fractionation technologies the detailed method along with the process flow diagram is recommended in section 4 of this document.

3.3 Obligations for technology service provider:

The selected Technology service provider shall have the following obligations:

Technology Transfer Plan:

To provide a detailed plan outlining the steps and timeline for the technology transfer. This plan shall include milestones, responsibilities, and contingencies. There should be a Lifenity steering committee to monitor the progress. All the deviations/changes should be approved by the committee.

Documentation:

To provide detailed documentation that describes the technology (Methods), equipment URS, buffer compositions, process steps, product specifications, and relevant data etc., This documentation is crucial for Lifenity to understand and replicate the technology.

Checklist of the details to be provided to Lifenity by the technology service provider:

Manufacturing:

RMs PPM and SPM and excipient specifications and source.

Dispensing instructions.

Detailed process flow diagrams.

Process validation requirements.

Batch Manufacturing record (BMR), SOPs, Master formulas.

List of SOPs for operation of equipment and other relevant procedures.

List of specifications and In-process controls (IPC).

Process flow diagrams (PFD) along with description of chromatography and other critical process steps.

Details of Equipment (Capacities, designs, qualification and placement with utilities requirements.

Manufacturing implementation sequence/model with process time for each unit operation.

Control strategy based on PPQ lots defining CPPs, CQAs and IPCs.

Yield calculation.

Cleaning procedures and validation.

Quality Control and Assurance:

Requirements of different areas for QC activities.

List of general and product related SOPs and STPs with specifications.

Raw materials and Active starting materials control and release strategies.

List of documentation for QC and QA activities.

Procedures for batch release, recall and complaints handling by quality assurance department.

Details of ware housing activities (Receipts of incoming RMS and active materials, sampling, Quarantine, testing and, release activities.

Cold chain requirements in ware house storage areas for active materials and finished products.

QC tests, STPs and specifications.

Validation of analytical and microbiology techniques.

Stability protocols as per regulatory expectations.

List of SOPs for operation of equipment and other relevant procedures.

List of OOS, OOT and any other quality related events/incidents relevant to Lifenity.

Details of primary packaging and secondary packaging materials with sources.

EHS:

SOPs and STPs related to EHS

Training:

It is the responsibility of technology transfer partner to provide on job training and qualification of the staff for each unit operation. Lifenity is responsible to provide necessary resources for trainings.

Training will comprise of knowledge transfer sessions, hands-on training, and exchange of personnel.

Lab Scale Implementation:

Lifenity shall initiate the technology transfer at lab scale to test its feasibility and performance in a real-world setting before full-scale deployment. The technology service provider shall provide the lab scale technology.

Regulatory Compliance:

Technology provider to ensure that the provided technology and its implementation comply with relevant regulatory requirements and support necessary approvals from regulatory authorities.

Scale-Up and Commercialization:

Technology provider shall provide support for Scale up of the technology for commercial production, at least three batches of each product and deployment. This shall involve optimizing manufacturing processes, securing additional resources, and launching the manufactured product with the given technology into the market.

Monitoring and Evaluation:

Provide continuous monitoring support at the production floor to Lifenity scientists for the performance of the transferred technology and conducting evaluations to assess its impact, identify areas of improvement and address any technology related issues that may arise.

Post-Transfer Support:

Providing ongoing support to address any challenges, optimize processes, and explore further opportunities.

CHAPTER 4

MANUFACTURING

4.1 As per the project implementation model, the following products shall be manufactured from 150,000L/Year of plasma.

Albumin

Factor VIII

Factor IX

Human Immunoglobulin (IgG)

Fibrin sealant

Fibrinogen

4.2 Manufacturing Process for Deriving the Above Suggested Products Derived from Plasma

4.2.1 Process Flow Diagram- Common Fractionation Steps

PFD- 1

PROCEDURE FOR COMMON FRACTIONATION STEPS:

P.1.F CUTTING AND POOLING

Approved plasma bags issued by QA are collected and thawed in a thawing tank with suitable provision for placement, retrieval and subsequent removal of thawing medium from the tank. The bags are checked for leakages after thawing and discarded if leakage is found. The thawed bags are cut under a bench type laminar flow and plasma collected in to suitable containers and finally pooled in process vessel to be taken for the next step.

P.2.F FILTRATION

Collected plasma after thawing is filtered to remove any suspended particles which may clog chromatography resins and column bed supports. Filtration is done using a depth filter which doesn't bind any proteins of interest. Filtered plasma is collected in a suitable tank with provisions to connect to a chromatography system for following process steps.

P.3.F CHROMATOGRAPHY – 1F

Filtered plasma is subjected to chromatography 1F to separate coagulation factor– F-VIII from rest of the fractions of interest. This step is essential to both separate F-VIII and also further polish plasma for

other chromatography steps. Fraction of interest is collected using patterns on chromatograms, can be programmed for automatic collection.

P.4.F LOAD PREPARATION FOR 2F

pH of the collected eluate from Chromo – 1F, is adjusted to 7.0 and loaded on the next chromatography column.

P.5.F CHROMATOGRAPHY – 2F

pH adjusted eluate from Step 4 is subjected to chromatography -2F to separate F -IX from the remaining fractions. F-IX is collected separately for further process and remaining fractions are further processed.

P.6.F TANGENTIAL FLOW FILTRATION (TFF) FOR CONCENTRATION

F-IX depleted sample is collected and concentrated by TFF. Trans-membrane pressures (TMP) are adjusted to achieve maximum permeate flow while minimizing fouling of the membrane. Automated TFF skids are used to precisely control and maintain the TMP.

P.7.F CHROMATOGRAPHY – 3F

Concentrated solution from Step-6 is subjected to chromatography-3F to separate buffer components from earlier steps and the sample is condition for further process steps. The main purpose of this step is to maintain suitable conditions for removal of other abundant protein impurities which may interfere with polishing chromatography steps of individual fractions.

P.8.F IMPURITY REMOVAL

pH of the eluate from Step-7 is adjusted to precipitate impurities from products of interest.

P.9.F EUGLOBULIN PRECIPITATION

Impurities like Euglobulin are precipitated using precise temperature and pH conditions. Temperature is maintained at less than 10 °C and pH less than 6.

P.10.F PRECIPITATE SEPERATION USING CENTRIFUGATION

The sample if passed through a continuous centrifuge to separate the precipitate. Precipitates are separated as a slurry and discarded. Supernatant is used for the subsequent steps.

4.2.2 Process Flow Diagram IVIG

IVIG PURIFICATION PROCESS:

P.11.F LOAD PREPARATION FOR 4F

Conductivity of supernatant from Step # 10 is adjusted to facilitate favorable conditions for separation of IVIG and Albumin.

P.12.I CHROMATOGRAPHY – 4F

Conductivity adjusted sample is subjected to Chromatography 4F to separate Albumin from IVIG, this is the most critical chromatography step. Unlike in Cohn's process this gives a single step separation. The chromatography step is also operated in both flow through and elution modes to achieve the required separation. Albumin and IVIG are separated as different fractions after this step and collected in different process vessels. Individual fractions are processed separately to further purify and polished suitable for filling in glass vials.

P.13.I TANGENTIAL FLOW FILTRATION (TFF) FOR CONCENTRATION

IVIG fraction is collected and concentrated to about batch volume using TFF. Trans- membrane pressures (TMP) are adjusted to achieve maximum permeate flow while minimizing fouling of the membrane. Automated TFF skids can be used to precisely control and maintain the TMP.

P.14.I pH AND CONDUCTIVITY ADJUSTMENT

Both pH and conductivity of concentrated solution from step 13 is adjusted using NaOH and Acetic acid.

P.15.I FILTRATION

pH and conductivity adjusted, concentrated IVIG fraction is filtered using 1.2-micron filter to clarify, suitable for loading on next chromatography step.

P.16.I CHROMATOGRAPHY - 5I

Filtered IVIG sample is subjected to Chromatography 5I to further purify IVIG from other protein impurities. Impurities and IVIG are collected as separate fractions, impurities are discarded and IVIG is further processed.

P.17.I pH ADJUSTMENT

pH of the fraction collected from step 16 is adjusted with 1M Acetic acid for following viral inactivation step. This is ideally done in the same vessel in which the fraction is collected.

P.18.I CONDUCTIVITY ADJUSTMENT

Conductivity of collected fraction is adjusted. This is ideally done in the same vessel in which the fraction is collected and pH adjusted.

P.19.I LOW pH TREATMENT FOR VIRAL INACTIVATION

pH of the collected fraction is adjusted to 4.0 with 1M Acetic acid. This is held at pH 4.0 at 37 °C for 10 hours for viral inactivation. This step is most critical with regulatory and quality prospective.

P.20.I FILTRATION

Viral inactivated IVIG fraction is filtered using 1.2-micron filter to clarify, suitable for loading on next chromatography step.

P.21.I SOLVENT DETERGENT TREATMENT FOR VIRAL INACTIVATION - VERY CRITICAL STEP FOR REGULATORS

The solvent/detergent (S/D) pathogen inactivation method was developed for the inactivation of enveloped viruses in plasma-derived medicinal products. Currently, S/D treatment for pathogen reduction of plasma for transfusion and plasma-derived medicinal products is the most common, validated and robust industrial method for the virucidal treatment of blood products. Although many millions of therapeutic doses of S/D inactivated products have been used to date, there is no report on transmission of viral infections caused by enveloped viruses in recipients. The safety of the S/D treatment is also shown by the absence of reports of toxicity or antibodies against neoantigens after large- scale exposure of patients to several million transfused units.

1% TNBP and 1% Triton X are added calculated as final volume v/v. TNBP and Triton are prepared in a buffer before addition to the bulk of fraction. Stock solutions are prepared.

P.22.I CHROMATOGRAPHY - 6I – FINAL CHROMATOGRAPHY STEP IN IVIG PROCESS

Filtered IVIG sample is subjected to Chromatography 6I to effectively remove added solvents and detergents in P.21.1 This also works for removal of endotoxins in the solution. IVIG bound to resin at an Acidic pH is selectively eluted using elution buffers at pH 9.0 in a process called as pH gradient elution.

P.23.I pH ADJUSTMENT

pH of the eluate from Chromatography 6I step is adjusted to preserve structural stability of IVIG and provide more stable conditions.

P.24.I NANO FILTRATION – (OPTIONAL STEP)

Nano filtration further ascertains barrier mode removal of viral particles from IVIG solution.

Plano make filters are used in the process.

P.25.I TANGENTIAL FLOW FILTRATION (TFF) FOR CONCENTRATION AND DIA FILTRATION

IVIG fraction is concentrated using TFF about 5% IVIG w/v concentration followed by buffer exchange by dia-filtration at the same TFF running conditions.

P.26.I TANGENTIAL FLOW FILTRATION (TFF) FOR CONCENTRATION

Dia-Filtered IVIG is concentrated to 6% w/v to facilitate final formulation at 5% w/v using the same TFF conditions. TFF conditions are maintained as in previous steps.

P.27.I FORMULATION

Buffered IVIG solution is formulated by adding 10% w/v pharmacopeial grade Mannitol. pH of this final solution is maintained at 4.2.

P.28.I PRE-FINAL FILTRATION

Filtration is done to decrease turbidity of the solution and also decrease microbial load on final sterile filtration step. This filtration is done using sandwich filter rated 1.2/0.6 micron.

P.29.I STERILE FILTRATION

Sterile filtration is done to further reduce bio burden and the process is not claimed a sterile operation. Final terminal filtration is performed before fill-finish stage.

P.30.I FILLING IN TO VIALS

Washed, sterile and pyrogen free vials are used to fill bulk solution of IVIG.

4.2.3 Process Flow Diagram Albumin Purification:

ALBUMIN PURIFICATION PROCESS:

P.31.A CHROMATOGRAPHY – 7A

Albumin fraction from chromatography step 4F is used to load on Chromatography 7A and separation is achieved by binding and elution mode. This step is final chromatography step in Albumin process.

P.32.A TANGENTIAL FLOW FILTRATION (TFF) FOR CONCENTRATION

Elate collected from Chromo-7A is concentrated to 6-7% w/v and collected in a vessel suitable for next process step.

P.33.A HEAT TREATMENT

Concentrated albumin solution is collected in a process vessel having control for temperature and time (57 degree C/ 3 hours). End point of completion of heat treatment is taken as stable optical density@ 600 nm. Heating of material results in a slightly turbid solution with suspended impurities. This solution is cooled to room temperature or even up to 10 °C for better separation. Chromatography is performed at room temperature.

P.34.A CENTRIFUGATION AND FILTRATION

Albumin solution after heat treatment is centrifuged and filtered to remove suspended proteinaceous contaminant particles formed during heat treatment. Continuous centrifuge can be used for this step.

P.35.A CONCENTRATION AND DIA FILTRATION

Above heat treated sample is Concentrated to 12% and Dia filtered.

P.36.A CAPRYLATE ADDITION AND pH ADJUSTMENT

Caprylate is added to concentrated and dia. filtered solution. The pH of the solution is adjusted to 4.5. An IPQC check is done to check actual caprylate concentration and if required, adjusted the concentration.

P.37.A LOW pH INCUBATION

Solution is incubated at room temperature for about 7 hours followed by filtration to remove suspended particles.

P.38.A pH ADJUSTMENT

pH of the above solution is adjusted to 7.0

P.39.A CONCENTRATION AND DIA FILTRATION

Albumin solution is concentrated to formulation concentration and finally subjected to Pasteurization at 60 °C and finally filled.

4.2.4 Process Flow Diagram Factor IX

FACTOR IX PURIFICATION PROCESS:

P.40.N LOAD PREPARATION CHROMATOGRAPHY 8N

Elate from chromatography step 2F is suitably adjusted for pH and Conductivity followed by concentration using TFF. Solvent and detergent are added to achieve viral inactivation, followed by filtration.

P.41.N CHROMATOGRAPHY 8N

Chromatography 8N is performed in adsorption mode and eluted fraction is collected in a separate process tank

P.42.N LOAD PREPARATION CHROMATOGRAPHY 9N

Elate from chromatography step 8N is suitably adjusted for pH and conductivity followed by concentration using TFF.

P.43.N CHROMATOGRAPHY 9N

Chromatography is performed in adsorption mode and eluate fraction is collected in a separate process tank. This fraction is suitably concentrated and passed through a nano filter to achieve viral removal. This fraction is sterile filtered, formulated and filled in suitable vials.

4.2.5 Process Flow Diagram Thrombin

THROMBIN PURIFICATION STEPS:

P.44.T LOAD FOR CHROMATOGRAPHY 10T

Load from chromatography 10T is obtained from fraction of chromatography 8N.

P.45.T CHROMATOGRAPHY 10T

Chromatography 10T is performed in adsorption mode and eluate fraction is collected in a separate process tank. This fraction is suitably concentrated and passed through a nano filter to achieve viral removal. This fraction is sterile filtered, formulated and filled in suitable vials.

4.2.6 Process Flow Diagram Fibrinogen:

FIBRINOGEN PURIFICATION STEPS:

P.46.F LOAD FOR CHROMATOGRAPHY 11F

Load from chromatography 11F is obtained from fraction of chromatography 1F. The solution is precipitated and solubilized in suitable buffer solution.

P.47.F CHROMATOGRAPHY 11F Chromatography 11F is performed in adsorption mode and eluate fraction is collected in separate process tank. This fraction is suitably concentrated and passed through a nano filter to achieve viral removal. This fraction is sterile filtered, formulated and filled in suitable vials.

4.2.7 Process Flow Diagram Factor VIII

FACTOR VIII PURIFICATION PROCESS

P.48.E LOAD FOR CHROMATOGRAPHY 12E

Load from chromatography 12E is obtained from fraction of chromatography 1F, is filtered and checked for conductivity and pH, adjust pH and conductivity, if required before loading on the chromatography column.

P.49.E CHROMATOGRAPHY 12E

Chromatography 12E is performed in non-adsorption mode and flow through fraction is collected in a separate process tank.

P.50.E LOAD FOR CHROMATOGRAPHY 13E

Load for chromatography 13E is obtained from fraction of chromatography 12E, fraction is checked for conductivity and pH, adjust the conductivity and pH, if required before loading on the chromatography column.

P.51.E CHROMATOGRAPHY 13E

Chromatography 13E is performed in adsorption mode and eluted fraction is collected in a separate process tank. This fraction is suitably concentrated and passed through a nano filter to achieve viral removal. This fraction is sterile filtered, formulated and filled in vials.

4.2.8 Manufacturing for Stage-2 Products (PFD)

ANTI D IgG

Factor XI

ANTI MEASLES

TETANUS IMMUNOGLOBULIN

HEPATITIS-B IMMUNOGLOBULIN

RABIES IMMUNOGLOBULIN

4.3 Equipment Required to Manufacture for the Proposed Capacity of 150,000 L / annum

4.3.1 Equipment's for Plasma Bulk Manufacturing Facility

4.3.2 Equipment's for Fill Finish Production facility

4.3.3 Equipment's for QC / R&D

4.3.4 Equipment for QA

4.3.5 Equipment's for Plasma storage

4.3.6 Equipment in Finished Goods Storage

4.3.7 Equipment in Engg / Critical System

4.4 Detailed design

This document describes the strategic approach to the facility design, technology basis and products selected in stage 1 (150,000 L/ annum) and Stage 2 (500,000 L / annum)

The site is situated on the outskirts of Nagpur city spread over 6 acres.

The process design shall be based on Chromatography system-based separation.

It will be dedicated to manufacture of Human plasma fractions which include the following products in

Stage 1

Albumin

Human Immunoglobulin

Factor VIII

Factor IX

Fibrin sealant

Fibrinogen

The plasma processing capacity in Stage 1 will be 150,000 L / annum

The facility design shall have scope of increasing the plasma processing capacity to 500,000 L / annum.

Besides the products identified in stage 1 the following products have been identified in Stage 2. An

appropriate Technology partner shall be identified for providing the process know-how.

Anti-D

Factor XI

Factor XIII

Hepatitis B IgG

Measles IgG

Protein C

Rabies IgG

Tetanus IgG

Thrombin

Varicella Zoster IgG

The plant is designed as a monobloc facility which will be constructed as a two-floor facility

The ground floor shall contain the following operational units

Bulk plasma fractionation unit (150,000 L/ annum)

Warehouse Plasma storage (32000 L)

Admin / QA / Reception

Warehouse (RM/PPM/SPM/ Finished goods)

Electrical transformer area

Engineering workshop

Utility block

Raw water storage tank

Borewell

Boiler room

Chiller room

Kill tank system

Scrap yard

FUTURE EXPANSION AREA

Bulk plasma fractionation unit (350,000 L/ annum)

Warehouse Plasma storage (32000 L)

The first floor shall contain the following areas

Canteen area

Water system and engineering store

SPM store

Visual inspection

Label store

Change rooms

Fill finish with Lyo area

Quality Control area

Service area (for AHU's)

CHAPTER 5

PROPOSED SITE PLAN

Concept design including Structural, Architectural, Civil and Electrical:

FIG 1: CONCEPTUAL SITE MASTER PLAN

Conceptual Lay out Ground Floor (Bulk Manufacturing):

The bulk manufacturing area is designed in a manner to provide unidirectional flow of man, material and waste from the facility to avoid mix-ups and cross contamination.

All the ancillary functions are located on the ground floor and separated from bulk manufacturing area.

In the ground floor following facilities are present:

Bulk manufacturing facility -01

Bulk manufacturing facility -02 (Future provision)

Quarantine Plasma Storage/Testing lab/Ware house

Future provision for plasma storage

Finished goods storage & dispatch

Administrative Block

Utility Block

Workshop

DG Set and Transformer area

UG collection tank, ETP/STP & Kill Tank

Bore well, UG 500KL storage tanks

Chiller and Boiler rooms

UG SUMP

Bulk Manufacturing Block

Change Rooms :Bulk manufacturing facility-01, personnel entering the processing area will undergo a series of two dress changes applicable to clean rooms of different classifications. The separate change rooms

are provided for male and Females.

Locker Room: This shall be the first change where the personnel remove their street shoes and clothes and keep these belongings in the provided double door lockers.

For the exit from facilities, separate change rooms are provided.

In exit change room personnel shall remove the clothing (Overall, head gear, gown and pant and reach the Locker room and put on their street clothes from the lockers and will exit the facility.

Preparation and Processing areas:

To support the manufacturing operations Buffer Prep.(Support area) shall be responsible to prepare various materials, reagent, chemicals and sterile materials to be used in the manufacturing operations.

The provisions for dry and moist heat sterilisations are provided in this area.

The manufacturing facility is construed and designed in a manner to protect product from any mix-ups, cross contamination and making the operator's work path and activities less complex and enabling compliance.

The processing areas for thawing of bulk material (Plasma), fractionation and downstream processing (Chromatography) are separately provided in the lay out to avoid any mix ups and cross contamination.

Plasma storage/ Testing lab /ware house/ Future provision plasma storage:

All the incoming and released raw materials, packaging material required to support the manufacturing shall be stored in the ware house.

All the incoming plasma shall be sampled and tested in ware house testing lab.

There is a provision for the storage at -20degrees C for of plasma in the ware house.

For the envisaged future expansion of manufacturing capacities from 150000Lit.to.500000Lit / year an additional storage area is reserved in the ware house, this area shall be furnished during future expansion of the capacities.

Finished goods storage and despatch:

All the released material shall be stored in the released cold rooms. At the time of shipment, the packed material shall be transferred from finished goods dispatch area in the ground Floor.

Administrative Block:

All the administrative support functions shall be dealt in this block.

Holding tank /ETP/STP:

All the liquid waste generated in the bulk manufacturing area shall be properly segregated. The biological waste material shall be transferred to holding tank from there it will be transferred to kill tank for final processing and discharge to ETP through holding tank.

03 UG water storage tanks (5KL) each & Utility Block:

The ground water shall be transferred to UG storage 03 tanks having capacities of 5KL each. The raw water will be treated and will be used as feed water for the generation of PW, WFI and clean steam.

Engineering work shop:

All the maintenance and repair activities shall be undertaken in the work shop.

DG and Transformer room:

For the uninterrupted power supply separate area is designed in the lay out.

Chiller and Boiler room:

The facility of chilled water and plant steam required for various system shall be routed through chillers and boilers room.

First Floor: On the first-floor following facilities are housed:

Formulation / Fill Finish area

Finished goods quarantine

Labeling, packing & Visual Inspection

Water System and Eng. store

Bulk Store

SPM store

PPM store

R&D lab

QA/QC

Canteen

FIG 2: CONCEPTUAL LAYOUT FOR BULK MANUFACTURING FACILITY

The bulk manufacturing area is designed in a manner to provide unidirectional flow of man, material and waste from the facility to avoid mix-ups and cross contamination.

All the ancillary functions are located on the ground floor and separated from bulk manufacturing area.

In the ground floor of Bulk manufacturing facilities following key areas are present:

Bag Cutting and pooling area

Plasma Thawing Area

Downstream processing (Chromatography)

Columns packaging & skids

Washing area

DHS / Autoclave area

Cold room

Buffer Preparation

Plasma day store

Chemical Day store

Change rooms

Change Rooms: Bulk manufacturing facility -01, personnel entering the processing area will undergo a series of two dress changes applicable to clean rooms of different classifications. The separate change rooms are provided for male and Females.

Locker Room: This will be the first change where the personnel remove their street shoes and clothes and keep these belongings in the provided double door lockers.

For the exit from facilities separate change rooms are provided.

In exit change room personnel shall remove the clothing (Overall, head gear, gown and pant and reach the Locker room and put on their street clothes from the lockers and will exit the facility.

Preparation and Processing areas:

To support the manufacturing operations Buffer preparation (Support)area shall be responsible to prepare various materials, reagent, chemicals and sterile materials to be used in the manufacturing operations.

The provisions for dry and moist heat sterilisations are provided in the area.

The manufacturing facility is construed and designed in a manner to protect product from any mix-ups, cross contamination and making the operator's work path and activities less complex and enabling compliance.

The processing areas (Upstream) bag cutting , pooling and segregated plasma thawing areas are provided.

There are separate areas for Plasma fractionation and downstream processing in stagewise manner (Chromatography) provided in the lay out to avoid any mix ups and cross contamination.

For packing of chromatographic columns dedicated area is provided in the layout of bulk manufacturing facility

Cold Rooms:

Cold rooms are provided in the facilities for the storage of heat labile materials like chromatographic gels, in process samples and materials.

FIG 3: CONCEPTUAL LAYOUT FOR BULK PLASMA STORAGE FACILITY

Plasma storage facility is housed on the ground floor adjacent to the raw material storage area.

All the Raw materials shall take through the dedusting tunnel to clean and remove the dust from the external surface of RM containers. After sampling of RM, the goods shall be transferred to RM quarantine area pending the testing and release of materials from quality control lab.

The RMs which do not meet the specifications shall be transferred to rejection room. Rejection room has access control. Only authorised persons are allowed to enter the area.

In coming plasms is taken inside the ware house through plasma receipt corridor.

Incoming plasms is sampled in plasma sampling room, equipped with (RLAF), remaining plasma is transferred to plasma quarantine area having -20 degrees C cold facility.

After the completion of serological testing and other necessary test in the serology lab and QC lab

plasma is shifted to released plasma cold room at -20degrees C.

Barcoding for identification of each lot of bulk plasma is done in Barcoding room.

There is a provision for future expansion of cold facilities to cater to 500000 Lit Plasma.

FIG 4: CONCEPTUAL LAYOUT FOR FILL FINISH FACILITY

Fill finish facility is housed on first floor of the bulk manufacturing facilities.

For entry into the area separate entries for males, females and visitors are provided. After series of changes, personnel shall enter into the core areas of fill finish facility.

In the core area formulation area is present.

In the formulation area all the released bulks shall be formulated on campaign basis. The formulated bulks shall be filled, capped through automatic machine and inspected for any abnormalities (particulate matter) visually. Separate laundry area is provided for washing of soiled gowns.

Following are the key areas in Fill Finish facility

1. Formulation
2. Fill Finish area
3. Lyophilisation
4. Change Room
5. Disinfectant and Material Prep area
6. Cold room
7. Clean laundry
8. Packing area

FIG 5: QC AND R&D FACILITY

On the FF of Bulk manufacturing facilities R&D, Quality Assurance and quality control facilities are located.

Four labs are dedicated for research and development activities. The technology at small scale shall be transferred to R&D labs.

Adjacent to this, quality control labs are housed. Primarily quality control labs shall be responsible for testing

Raw materials, PPM and SPMs.

In-process testing of intermediate products

Bulk plasma -Physicochemical and microbiological test

Formulated Bulk plasma products

Fill finished plasma products

Microbial testing of critical utilities.

Microbial testing of environmental samples from clean rooms.

Following labs are envisaged in quality control

- 1 Sample receipt
2. Chemical / Glassware Storage
- 3 TOC / LPC
4. Instrument Room
5. Balance Room
- 6 Wet Lab(Chemical)
7. Hot Zone/ Washing
8. MLT
9. Incubator rooms
- 10 Sterility room
- 11 Washing & Sterilization area
- 12 Media prep and Storage
- 13 BET Room
- 14 Stability Testing

Quality Assurance:

Annexed to QC labs quality assurance department is located. In quality assurance following functions of site will be undertaken

Documentation and control

Trend analysis of critical utilities and Change management

Audits, Inspections, Pest control and Vendors management

Validation. Qualification and Calibration

Complaints and Recalls

Lot Release and Product review.

Trainings

FIG 6: ELEVATION DRAWING

5.3 Lay out Philosophy:

Plasma fractionation and processing is complex and critical manufacturing operation with respect to the receipt of plasma (Starting material) its downstream processing, formulation and fill finish. The facility requires strict cGMP and regulatory compliance to ensure the product quality and safety of the operator and environment.

The facility lay out adheres to GMP guidelines (Schedule M of Drugs and cosmetic act 1940 and Rules 1945).

The layout philosophy is based on the integral evaluation of processes, potential risks, man and material movement, generation and flow of solid and liquid waste etc.

The manufacturing facility is construed and designed in a separate block to protect product from any mix-ups, cross contamination and making the operator's work path and activities less complex and enabling compliance. Flow of personnel, clean and dirty materials are unidirectional with logical sequence of operations.

HVAC zoning of the facility (control areas) for the manufacturing operations is done based on the operation and its criticality (Plasma collection, fractionation, purification, and storage) to minimize the risk of contamination and to maintain physical conditions like temperature and humidity. The physical conditions should also be suitable to the comforts of the personnel working with protective clothing.

In the lay out provision for separate change rooms for entry and exit from clean areas is made. Change rooms design contains airlocks to provide physical separation of the different stages of gowning.

CHAPTER 6

SOURCING AND HANDLING OF RAW MATERIAL

6.1 Exchange Value of Plasma per Liter and Projected Value Over a Period of Seven Years:

As per the literature survey and discussion with different stake holders the exchange value of plasma in 2023-2024 is INR 2200/Liter and in next 5 years the estimated plasma price shall be expected to be ≈INR 2800/L.

Figure 8: Exchange value of plasma per liter and projected value in INR

6.2 Donors' Selection Criteria:

In India, the plasma industry is dependent on voluntary blood donation. Human blood is the source of plasma which is used to manufacture various therapeutic plasma proteins after fractionation.

In blood banks the cellular components (RBC, WBC, and platelets etc.), are separated out for different applications and the plasma is collected from blood banks by plasma fractionators for clinical / therapeutic use as per "National Plasma Policy" of Ministry of Health & Family Welfare.

The selection and management of blood donors is an essential part of the blood donation process. National health authorities and blood transfusion services are responsible for ensuring that the national system is in place for the selection of all blood donors through an assessment of their suitability to donate blood.

Following are the selection criteria for donors:

Health

Age (18-65 years)

Time between blood donations

Blood pressure

Body temperature

Hemoglobin content etc.

Test for HBsAg*

Test for anti-HIV*

Test for anti-HCV*

*All three tests should be negative. An additionally test for STD is also performed.

6.3 Plasma Donation Testing, Epidemiological Surveillance, Post Donation Information:

Plasma Donation testing:

The collected plasma is screened for the following:

Test for HBsAg

Test for anti-HIV

Test for anti-HCV

All three tests should be negative. Additionally test for STD is also performed.

Epidemiological surveillance, post donation information:

To ensure optimal long-term safety of plasma for fractionation, it is recommended to establish continuous epidemiological surveillance of the donor population.

The objective of this survey is to know, as precisely as possible, the prevalence and disease incidence, and trends of infectious markers that are relevant to the safety of medicinal plasma products so that appropriate counter-measures can be taken in a timely manner.

The system should not only be able to gather epidemiological data at the national and regional level but also among the donor populations that are providing blood or plasma for fractionation at individual blood establishments within a country or a region. The information from the epidemiological surveillance can furthermore be used to detect and assess.

Differences in disease patterns between donor population.

Trends of infectious marker.

The relevance of any preventive measures such as a strengthening of donor selection process, additional exclusion criteria, or implementation of any additional screening tests to avoid contamination of plasma products.

To detect any new or emerging pathogens in the community.

6.4 Handling of Blood, Plasma Freezing, Storage, Cold Chain and Complete Logistics Plan:

Handling of Blood:

A standardized and validated procedure for the bleeding shall be followed.

The presterilized container (Bags) shall be used to collect the blood.

Identity of the donor should be confirmed with relevant documents before venipuncture.

The collected blood used for plasma fractionation should be mixed well with the anticoagulant solution to avoid risks of activation of the coagulation cascade.

Presently, automated blood collectors are used in which plasma is separated from cells and the cells are resuspended and returned to donor in series of return cycles.

In most of the centers the centrifugation or filtration or combination of both process is followed to separate the cellular components from plasma.

The collected blood should be weighed and labelled properly in order to ensure full traceability at each stage of plasma production.

Information on the label of the blood storage containers should include:

Name of the blood product

Volume or weight

Unique donor identification number

Name of the blood collection center

Shelf life

Storage temperature

Name, content and volume of anticoagulant

Laboratory samples should also have the same information and must be stored securely at appropriate temperature

Blood Holding:

It has been shown that whole blood anticoagulated with CPD, transported and stored at 22°C for up to 8 hr prior to separation of plasma is suitable for the production of plasma for fractionation.

However, storage of blood beyond 24 hours the factor VIII activity is reduced by 15 to 20 percent. Rapid cooling of whole blood to 22°C +/- 2°C after collection protects factor VIII and may allow storage of blood for 24 hr.

Transportation/storage of collected blood at 4°C with different anticoagulants (ACD, ACD-adenine, or CPD) consistently appears to reduce the FVIII content, but not necessarily that of other proteins.

Blood holding time for the extraction of various components should be validated by the company.

Plasma Processing and Transportation:

After separation of plasma, freezing is an important processing step that has an impact on content of factor VIII. Several aspects of the freezing conditions of plasma for fractionation have been evaluated in this document based on the literature search.

Freezing of Plasma:

The requirements for the temperature at which plasma should be frozen depend upon the type of proteins to be fractionated.

As per pharmacopeial guidelines, recovered or apheresis plasma for fractionation to be used for the manufacturing of labile proteins (Factor VIII concentrate) should be frozen rapidly, within 24 hours of collection, at -20 °C or below.

However, freezing conditions are currently under debate. It is suggested that recovered plasma used to

manufacture stable plasma proteins (e.g., albumin and immunoglobulins) should be frozen within 72 hours of collection at –20 °C or below.

Plasma freezing process should be validated by life Nity to extract the desired plasma proteins.

Storage of Plasma:

Plasma for fractionation should be stored at –20 °C or below. When plasma is stored at -20°C it has the shelf life of 2 years. However, when plasma is stored below -20°C has shelf life of 3 years.

Stability/Shelf-life studies should be conducted on at least three Pooled Plasma bulks during initial phase as a minimum by the company and subsequently one batch of pooled plasma should be taken for stability on annually basis after approval from the quality assurance department. The stability studies should be well documented.

Transportation of plasma and its validation:

Transport of plasma pools (Bulk) and samples to the processing site should be done according to approved and validated procedures. Temperature monitoring device should be used to record the temperature during transportation. Any deviations in the temperature during transportation should be notified to QA department for impact assessment (compliance and quality).

Simulation studies should be conducted at site to validate the packaging and its transportation.

6.5 Sourcing of Plasma from Government and Non-Government Blood Banks:

There are different sources of plasma in India- Government, NGOs and some private facilities recognized by the government. The Govt. has the full responsibility for the blood program.

The surplus plasma from government hospitals and blood collection centers and blood banks are sold through tenders.

As a minimum there are certain conditions for plasma fractionator to fulfill before applying for plasma procurement as follows:

The plasma fractionator must be licensed by the Government of India for plasma fractionation and manufacturing clinically usable products in a cGMP facility.

The fractionator is accountable to update plasma procurement center about the nature and quantity of all the products manufactured from the plasma received from the center.

The fractionator must ensure that none of the products recovered from this plasma are exported before fulfilling domestic demand.

Buy Back policy:

In some cases, government buy plasma derived products from the fractionator for use.

6.6 Equipment for Proper Collection, Storage and Separation of Blood Components:

Following is the list of equipment required during the different stages of blood processing:

Blood Collection

Donor weighing balance

Haemoglobinometer

Sphygmomanometer

Stethoscope

Blood mixer and shaker

Tube stripper and Di-electric tube sealer

Needle destroyer

Blood Processing

Refrigerated Centrifuges

Double pan balance

Plasma Expresser- Manual/Automated

Blood Storage

Blood Bank Refrigerator

-20°C freezers

-80°C deep freezers

Platelet Agitator and Incubator

Cell Separators (Apheresis machine)

Blood Testing

pH Meter

Cell Counter (optional)

Coagulometer (optional)

Automated Cell Grouping system (optional)

Equipment for Gel technology (optional)

Table top centrifuge

Serological water bath

ELISA reader (plate reader / Strip reader)

VDRL shaker

Blood Distribution

Transportation vans (optional)

Insulated blood bags container with provision for storage between 2 °C to 8 °C

6.7 Importing of Plasma from Neighboring Countries and Indian Regulations:

India is highly import dependent for its plasma product needs, which are considered to be life-saving medicines. At this point in time, India produces majorly only two plasma products, albumin and intravenous immunoglobulin (IVIg) from local plasma.

Because of changing disease pattern, improved clinical diagnostic facilities, affordability, medical tourism and many other factors the demand of blood products is increasing. This creates a huge demand and supply gap and many patients lose battle of life because of non-availability of these products. Various hyper-immune plasma proteins (Anti-D, tetanus immunoglobulin, hepatitis-B immunoglobulin, and rabies immunoglobulin) are 100% imported or prepared from imported bulk.

6.7.1 Indian Regulations & Regulatory Bodies:

The Plasma Policy aims at making available, easily accessible and adequate supply of high quality of human plasma derived proteins for clinical/ therapeutic use. In India the plasma is separated from safe and quality blood from voluntary non-remunerated regular blood donor. Blood and plasma are regulated by different rules mention in drugs and cosmetic act 1940 and by different regulatory agencies as mentioned below.

Following regulations and rules are applicable:

The Drugs and Cosmetics Rules, 1945, framed under the Drugs and Cosmetics Act, 1940:

Includes human blood under the definition of "Drugs" under Section 2(b)

Regulates functions of blood banks, which provide recovered plasma for plasma fractionation (Part XII C specifically regulates the collection and storage of plasma for fractionation)

The National AIDS Control Organization (NACO), under the Ministry of Health and Family Welfare) to oversee operation of blood banking services in India.

National Blood Transfusion Council (NBTC) is the apex policy-making body, under NACO, making central licensing mandatory, increasing voluntary donation base and setting of quality standards.

State Blood Transfusion Councils (SBTCs) are responsible for the smooth functioning of blood transfusion services at a state level.

Drug Standard Control Organization (CDSCO) and State Drug Control Organizations' play a vital role in key aspects such as, approval of licenses, and enforcement of standard transfusion practices to ensure safe,

quality and efficacious blood and its components in clinical practices.

National Institute of Biologicals (NIB) Nodal governmental organization under the Ministry of Health and Family Welfare responsible for the testing and release of biologicals in India including blood plasma.

CHAPTER 7

PRODUCTION PROGRAM

7.1 Product and Product mix:

In phase 1 of the project the following products are proposed to be manufactured from normal human plasma.

Human Serum Albumin

Normal Immunoglobulin

Factor VIII

Factor IX

Fibrin Sealant

Depending upon the market demand and production capacities, product mix strategy can be formulated.

Various permutations and combinations are possible to manufacture different products. On the basis of PFD given in Chapter 4, different products shall be purified in different steps during the downstream processing of pooled plasma bulk (fractionation of plasma).

However, it is suggested that Lifinity shall take the first campaign of product which is at the highest demand followed by the products with moderate and low demand. As an alternate, the high yielding product shall be manufacture first to have sufficient stock followed by a campaign for low yield and moderately yield products.

Plasma requirements based on the manufacturing capacities:

Considering 1,50,000 Liters of plasma processing capacity, plasma bags requirement are presented in table below:

Table:9

Production Capacities and Yield Calculations:

HSA is major protein component, followed by IgG in plasma. The wide concentration range of plasma proteins (less than 1 mg/mL to 40 g/L is one of the potential challenges in designing production capacities.

Source: (J.D. Andrade et al. Plasma protein adsorption: the big twelve, Ann. N.Y. Acad. Sci., p. 158-172,1990. & Chapter -2 of DPR).

However, for calculation purpose below mentioned protein concentrations in normal plasma are considered.

It has been reported in the literature that depending upon the downstream processing and sterile/terminal filtration, recovery of the protein of interest is 50% at final stage.

Source: (Overview of Albumin and Its Purification Methods: Adv Pharm Bull, 2016, 6(4), 495-507 doi: 10.15171/apb.2016.063 <http://apb.tbzmed.ac.ir>) as shown in the table below.

Basis of Yield calculations:

Based on vessel size, the batch size assumed is 1000 liters, on every alternate date, a new batch can be initiated, therefore 15 batches shall be processed per month (one batch/ alternate day).

The yield for 1000 liters batch is as mentioned below.

Table 10: Yield for 1000L batch

In the EOI document, the phase 1 target is to process 1,50,000 Liters of plasma annually. The output obtained with 1.5 lac liters of plasma for the products mentioned for stage1 is as under.

Table 11: 150,000 Lit annual capacity

Therefore, to achieve the plasma processing target, 15,000 liters of plasma shall be processed on monthly basis at site.

Production Scaleup from 150,000 to 500,000 lit per annum:

Lifinity will increase its plasma processing capacity to 500,000 liters annually in phase 2 of the project implementation.

To achieve the plasma processing target, 50,000 liters of plasma shall be processed monthly. Two strategies, can be followed:

Strategy 1:

To keep number of batches equivalent to stage-1 (i.e., 15 batches per month), the individual batch size shall be 3400 liters. The yield calculations are given below.

Table 12: Yield for 3400 L batch

Strategy 2:

Alternatively, if the batch size is further increased to 5000 liters, the number batches shall be reduced to 10 batches per month as one batch can be initiated after every 3 days. The yield for a 5000 liters batch is as mentioned below.

Table 13: Yield for 5000 L batch

Total Production capacity:

The total production capacity with 500,000 Lit. of Plasma/year is given in the table below.

Table 14: Total production capacity with 500,000 Lit. of Plasma/year

Note:

For scale up to 500,000 Liters only 6 products (HSA, IgG, F-VIII, F-IX Fibrin Sealant, Fibrinogen) are considered.

Fibrin sealant is prepared by mixing Fibrinogen and Thrombin; therefore, production capacity of fibrin sealant is equal to the production capacity of Thrombin.

CHAPTER 8

PROJECT IMPLEMENTATION MODEL

Lifinity shall implement the PDMP manufacturing in a phased manner.

A suitable technology partner for the Plasma fractions shall be identified and contracted with for process know-how. The agreement will fulfil criteria mentioned in Section 3.3 of this document.

The project shall be initiated at small scale to establish the process for the products identified in stage 1.

Once the process, yields and quality are established as per the specifications, the project will proceed to the commercial scale stage

Commercial scale:

The company will initially procure 150,000 liters of plasma per year for fractionation to produce the following product in Phase-1.

Human Serum Albumin (HSA)

Human Immunoglobulin (IgG)

Factor VIII

Factor IX

Fibrin sealant

Fibrinogen

All the areas of the manufacturing block (Facility-1) will have sufficient space to cater the volumes and products without compromising on any of the cGMP and regulatory expectations.

The following systems installed during Phase-1 of the project will also be matching the requirements for Phase-2 (Expansion Project) capacities.

Clean utilities

Black utilities

Filling

Freeze drying and

Packaging lines for the final product

8.2 Facility Expansion:

For handling additional 350,000L of Plasma:

In Phase-2, the company shall procure additional 350,000 Liters of plasma to manufacture the following additional products.

Anti-D IgG

Factor XI & XIII

Fibrinogen

Hepatitis B IgG

Immunoglobulin G

Measles IgG

Protein C

Rabies IgG

Tetanus IgG

Thrombin

Varicella Zoster IgG

While implementing Phase-2 of the project, Lifenity shall extend the manufacturing operations to Facility-2 and shall procure the following additional systems to match the capacities of processing 350,000 L/year of plasma which are as follow:

Tank Systems

Chromatography Systems

Autoclaves

Dry heat sterilizers

Quality control and Quality assurance units developed during Phase-1 will be also cater to the Phase -2 products and other ancillary activities.

Central ware house space shall be adequate for future expansion. However, the additional cold Chain equipment (Freezers) and storage racks etc. for the storage of active material (plasma) and finished products shall be procured during Stage 2 expansion.

In Phase-1 of the project the bulk manufacturing operations will run in a single shift of 8 hours working /day, with effective working for 6 hours. The shifts can further be evaluated based on number of batches for processing/week. Likewise, the filling operations will run in scale down mode to meet the market requirement.

Monoclonal antibodies (Mabs) – Future considerations:

A monoclonal antibody are Y-shaped glycoproteins that detect and bind strongly to their antigen.

Monoclonal antibodies stems from a single white blood cell clone. Monoclonals not only bind to a single epitope of a target antigen but also activates immune responses, e.g. antibody-dependent cell-mediated cytotoxicity (ADCC) and recruits further immune cells such as phagocytes and are thus highly specific

Monoclonal antibody production:

Several technologies for monoclonal antibody production have been developed and the respective downstream processes are optimized. Some of the industrial scale technologies are:

Hybridoma cells production

Phage display technologies

Single B cell technologies

Hybridoma production:

In this technology fusion of antibody producing B cells of an immunized mouse with myeloma cells is done produce unlimited antibodies. Hybridoma cell line that can be maintained indefinitely

Disadvantages:

Use of lab animals for immunisation

Risk of mutations in the antibody with aging of hybridoma

Expression of additional variable domains by the cell line

Long lead times and labour-intensive operation

Phage display technology:

Phage display technology circumvent several disadvantages of hybridoma technologies and speed up the process of Mab generation.

The technology relies on molecular biology and cell culture methods to genetically engineer bacteriophages to display antibodies on their surface.

In this method binding of bacteriophages with desired antibodies onto immobilized antigens is done. The high affinity binders are then used to infect bacteria and collect their DNA for sequencing. The sequence help to design genes to be used in antibody production in cell cultures.

The overall process is much faster than hybridoma cell generation and doesn't need any laboratory animals.

Single B cell technologies:

Single B cell technologies are another milestone in Mab development. The technique involves the screening and sorting of ex vivo cultures of single B cells from blood samples of immunized animals or humans to select for promising antibody candidates.

Genetic information is then extracted from the candidate cells mRNA after reverse transcription into cDNA, amplification and subsequent sequencing and cloning into expression vectors for Mab production.

In this method there is no need to create a hybridoma cell line, thus process become efficient and timeline are shorted by several weeks.

Depending upon the process to be adopted for the manufacturing of monoclonal antibody a new production block shall be designed to meet the process and cGMP requirements.

Detailed working for the Mabs manufacturing unit will be proposed separately in future.

In general, the Mab production suite shall have cell culture, fermentation, harvesting, primary purification, secondary purification, sterile filtration.

However, for formulation and fill finish existing facilities shall be used areas.

Table 15: Following steps are performed in Mabs production (General)

For implementation of Mabs project in future, there shall be a requirement of cell culture and molecular biology labs in the quality control with some additional piece of equipment.

Present lay out of quality control unit will be adequate for plasma products (Phase-1 & 2).

However, for Mabs products testing, expansion in existing QC to cater the cell culture need shall be required.

CHAPTER 9

GMP AND REGULATORY CONSIDERATIONS

9.1 Premises, Layout & Process Engineering:

Plasma fractionation and processing is complex and critical manufacturing operation with respect to the receipt of plasma (Starting material) its downstream processing, formulation and fill finish.

Plasma fractionation facility require strict cGMP and regulatory compliance to ensure the product quality and safety of the products and operators. The facility's layout and process engineering are critical components for setting up of the green field project. Following are key considerations for setting up a plasma fractionation facility with respect to Premises, Layout and Process engineering:

Location and Surroundings:

The factory building(s) for manufacture of plasma products shall be situated and shall have measures to avoid risk of contamination from external environment including open sewage, drains, public lavatory or any factory which produce disagreeable or obnoxious odor or fumes, excessive soot, dust, smoke, chemical or biological emission.

Building:

The building(s) used for the purpose shall be designed, constructed, adapted and maintained to suit the manufacturing operations so as to permit production under hygienic conditions. They shall conform to the conditions laid down in the Factories Act, 1948 (63 of 1948).

Premises - General Requirements:

The premises used for manufacturing, processing, warehousing, packaging, labelling and testing purposes shall be:

Compatible with other products manufacturing operations that may be carried out in the same or adjacent area.

The manufacturing facility shall be designed / constructed / maintained to prevent entry of insects, pests, birds, vermin's and rodents. Interior surface (walls, floors and ceilings) shall be smooth and free from cracks, and permit easy cleaning, painting and disinfection.

There should be adequate space for orderly and logical placement of equipment, materials and movement of personnel in order to prevent:

(a) The risk of mix-up between different categories of products or with raw materials, intermediates and in-process material.

(b) Possibilities of contamination and cross- contamination by providing suitable segregation, zoning and access controls etc.

Layout Considerations:

During designing of facility, Lifenity must ensure the following points:

Plasma products will be produced in an independent block/building.

The facility is designed to prevent cross-contamination between process areas, easy cleaning and ensure the efficient flow of materials and personnel.

The facility design adheres to GMP guidelines (Schedule M of Drugs and cosmetic act 1940 and Rules 1945).

Ancillary areas like rest and refreshment rooms are separate from manufacturing areas.

Facilities for changing, storing clothes and for washing and toilet purposes are easily accessible and adequate for the number of users. Provision for separate toilets for males and females is made and toilet is not connected with production or storage areas.

Change rooms design contains airlocks to provide physical separation of the different stages of gowning. Flushed effectively with filtered air. Change rooms are of the same grade as the area into which it opens finally.

Provision of separate change rooms for entry and exit from clean areas.

Hand-washing facilities are provided in the first change room.

Flow of personnel, clean and dirty materials are unidirectional with logical sequence of operations.

Sinks and drains are excluded from Grade A and B areas where aseptic operations are carried out.

Drains are fitted with effective easily cleanable traps and with air brakes to prevent backflow.

The production and dispensing areas are well lighted, effectively ventilated.

HVAC Zoning:

Air supply for manufacturing area will be filtered through bacteria retaining filters (HEPA Filters) and will be at a pressure higher than in the adjacent areas unless otherwise required.

The filters to be checked for performance on installation and periodically thereafter and maintain written records.

HVAC zoning of the facility (control areas) for the manufacturing operations is done based on the operation and its criticality (Plasma collection, fractionation, purification, and storage) to minimize the risk of contamination and to maintain physical conditions like temperature and humidity. The physical conditions should also be suitable to the comforts of the personnel working with protective clothing.

Routine microbial and particulate counts of the manufacturing area will be carried out during manufacturing operations to assess the effective ness of HVAC.

The results of particulate matter and microbes to be continuously monitored against documented in-house standards and regulatory guidelines.

Based on the quality of air, manufacturing facility is divided into different grades/ Zones. In general, plasma processing facility is divided into four grades (Classes) based on the air cleanliness levels (Microbial and Particulate matter (Refer to table No 15 and 16).

Facility Grades:

Grade A: Clean rooms are used for high-risk operations, and require a unidirectional airflow system to maintain high air quality. Unidirectional airflow should provide a homogeneous air velocity of 0.36–0.54 m/s (guidance value) at a defined test position 15–30 cm below the terminal filter or air distributor system.

The velocity at working level should not be less than 0.36 m/s.

Grade B: In aseptic preparation and filling, this is the background environment for the Grade A zone.

Grades C and D: Clean areas for carrying out less critical stages of the manufacturing of plasma products -downstream processing, chromatographic purification and solution preparations.

Table 16: Recommended Maximum Allowable Particle in clean room

Table 17: Recommended limits for Microbial contamination in Clean Rooms

Clean Utility Systems:

Pure steam, purified water and water for injection are the clean utilities required to support the clean operations in plasmas manufacturing/processing facilities. Monitor the clean utilities for the quality and compliance regularly as per the laid procedures and the regulatory standards.

Equipment Placement:

Equipment and processing areas will be arranged to facilitate efficient workflow and minimize the risk of contamination. Ensure that equipment and materials are easily accessible for maintenance and cleaning. Processing areas must be equipped with dedicated equipment designed for specific purpose.

Waste Management:

Waste will be removed from the manufacturing area or sent to the bio-waste system in a safe manner. To ensure containment compliance with the environment, the facility must implement measures to prevent accidental spills or leaks. This includes the use of appropriate containment systems, spill response plans, and monitoring for any environmental impacts.

All sewage waste pipes will run below natural ground level (however this depends on the actual site conditions along the pipe routing) in proper gradient and connected to nearest available chamber.

Procedures must be in place for the safe removal of different types of waste such as chemical, biological, expired reagents, unused samples and solvents etc.

Design the drainage systems in such a way that effluents can be effectively neutralized or decontaminated to minimize the risk of cross-contamination.

Provisions should be there for the proper and safe storage of waste materials awaiting disposal.

Toxic substances and flammable materials must be stored in suitably designed, separate, enclosed cupboards, as required by national legislation.

Liquid effluent poses a safety or contamination risk (Biological liquid waste), should be treated in the

kill tank before its discharge into non biological /main drain.

Testing Facilities:

Testing facilities must be located in the separate building/floor.

Validation and Documentation:

Lifenity and PMC will validate the facility layout and design to ensure it meets GMP and regulatory requirements. Comprehensive documentation of the facility's layout and design, including as-built drawings shall be maintained at site and a robust document control system shall be established in quality Assurance department to maintain records.

Accessibility and Safety:

Ensure that the facility layout allows for easy access in the event of an emergency. The facilities must be equipped with safety features like fire hydrant systems, eyewash and body shower stations where ever necessary.

Quality Assurance & Regulatory Compliance:

The quality assurance system appropriate to the manufacture of plasma products shall ensure that the products are designed and developed in a way that takes account of the requirements of GMP and other associated codes such as those of GLP and GCP.

Adequate arrangement must be made at site for manufacture, supply chain and use of the correct starting and packaging materials.

Self-Inspection and Quality audit:

In the quality assurance department of Lifenity adequate procedures shall be designed to ensure self-inspections and quality audits periodically.

Plasma Process Considerations:

Plasma fractionation involves critical steps to separate plasma into its various components and produce high-quality plasma-derived products while adhering to Good Manufacturing Practices (GMP) and regulatory requirements.

Following are the key considerations for plasma processing:

Plasma Collection and Testing: A reliable and secure source of plasma, involving strict donor selection, screening and testing plan in place to screen for infectious agents, including HIV, Hepatitis B and C, and other potential contaminants in pooled plasma.

Fractionation Process: A well-defined fractionation process to separate plasma into its various protein components, such as albumin, immunoglobulins, and plasma factors. Robust filtration and purification processes will be followed by Lifenity to separate and purify these proteins.

Process Validation: The fractionation process must be validated to demonstrate its ability to consistently produce products. Process performance qualification (PPQ) shall be performed on first three initial batches to ensure that the process is in a state of control during commercial production.

Process Control and Monitoring: Robust process control systems to monitor critical process parameters (e.g., temperature, pH, pressure) in real-time. Written procedures must be in place for addressing deviations and out-of-specification results.

Raw Material Control: Documented procedures for receiving, testing, and release of raw materials and starting materials used in the fractionation process. Lifenity must ensure that all materials meet quality, safety standards and plasma source traceability.

Safety: Validated viral and pathogen inactivation and removal steps to ensure the safety of plasma-derived products. Implement measures to prevent contamination during processing.

Quality Control Testing: Comprehensive quality control testing plan, including in-process testing and final product testing, to ensure the safety, purity, and potency of the plasma products. Specifications for raw materials, intermediates, and final products shall be written and approved.

Equipment monitoring, Calibration and Validation: All the critical equipment used in the manufacturing and control of plasma are maintained, calibrated and qualified as per the written procedures.

Training and Qualification: Ensure that all personnel involved in the plasma fractionation process are well-trained with respect to the SOPs and STPs, understand GMP requirements, and are aware of their roles in ensuring product quality and safety.

Maintain records of personnel training and qualifications.

Writing and editing documentation: Documentation for all existing processes must be written and edited regularly, to ensure that all new hires and other personnel can understand the best practices and standards

Risk assessments of current equipment/processes: Assessment of current processes and equipment to determine potential risks and mitigation plan shall be in place.

Traceability: A robust documentation system must be in place for tracking and tracing individual plasma donations and the products derived from them.

Packaging and Labelling: Appropriate packaging and labelling procedures for the final plasma-derived products, ensuring they meet regulatory requirements.

Post-Market Surveillance: Post-market surveillance system to monitor the safety and efficacy of products in the market.

9.2 Regulatory Guidelines GMP, Drugs and Cosmetics act 1940 and Rules, CDSCO, WHO- GMP

Indian Guidelines:

The Drugs and Cosmetics Rules, 1945, framed under the Drugs and Cosmetics Act 1940:

Includes human blood under the definition of "Drugs" under Section 2 (b)

Regulates functions of blood banks, which provide recovered plasma for plasma fractionation

Part XII C specifically regulates the collection and storage of plasma for fractionation

In the year 1967, Central Govt. (Ministry of Health) enacted a separate provision in Schedule F Part XII B of Drugs & Cosmetics Rules.

Schedule F of the Drugs and Cosmetics Rules specifically pertains to the requirements for the manufacture, sale, and distribution of plasma-derived products, including immunoglobulins, clotting factor concentrates, and other blood

Drugs and Cosmetics Rules, Part X-B: Conditions for License (Rule 122EA to Rule 122P)

Destruction of infected blood and implementation of Bio-Medical Wastes (Management and Handling) Rules, 1996

Applicable forms for License:

The National AIDS Control Organization (NACO), under the Ministry of Health and Family Welfare) to oversee operation of blood banking services in India.

National Blood Transfusion Council (NBTC) is the apex policy-making body, under NACO, making central licensing mandatory, increasing voluntary donation base and setting of quality standards.

Similarly, State Blood Transfusion Councils (SBTCs) are responsible for the smooth functioning of blood transfusion services at a state level.

Drug Standard Control Organization (CDSCO) and State Drug Control Organizations play a vital role in approval of licenses, and enforcement of standard transfusion practices to ensure safe, quality and efficacious blood and its components in clinical practices.

National Institute of Biologicals (NIB) is the nodal governmental organization under the Ministry of Health and Family Welfare responsible for the testing and release of biologicals in India including blood plasma.

Policies and Guidelines for Fractionation:

The National Blood Policy 2002 regulates blood quality and the Guidelines of NBTC 2016 has been brought

out by NACO. NBTC has formulated guidelines for recovery of processing charges for blood and blood Components across blood banks in all sectors. It has also laid down uniform norms for issuance of No Objection Certificates (NOCs) for new blood banks, renewal of license, criteria for designation blood bank as RBTC, Guideline on bulk transfer of blood and Guidelines on exchange of surplus plasma with indigenous fractionators has been released by NBTC to support voluntary blood donation and rationalize blood utilization.

National Policy for Access to Plasma Derived Medicinal Products from Human Plasma for Clinical / Therapeutic Use covers the process of collecting standard plasma and transporting them under optimum conditions for fractionation, identifying critical parameters for safety, and ensuring compliance with regulatory requirements and training.

National Pharmaceutical Pricing Authority: Under the Ministry of Chemicals and Fertilizers, Department of Pharmaceuticals, the prices of medicines are controlled by the National Pharmaceutical Pricing Authority. Currently, plasma products under price control include albumin, AHF, anti-D immunoglobulin, anti-tetanus immunoglobulin and anti-rabies immunoglobulin however plasma itself is not under price control.

Pre-clinical safety and Clinical Evaluation of Plasma derived products:

European guidelines

Preclinical:

June 2011 EMA/CHMP/ICH/731268/1998 Committee for medicinal products for human use (CHMP)

Clinical Evaluation of Plasma derived products: Guideline on clinical investigation of recombinant and human plasma-derived factor IX products: 21 May 2015, EMA/CHMP/BPWP/144552/2009 Rev. 1, Corr. 1 Committee for medicinal products for human use (CHMP).

Guideline on clinical investigation of recombinant and human plasma-derived factor VIII products: 21 May

2015 EMA/CHMP/BPWP/144533/2009 rev Committee for medicinal products for human use (CHMP).

Guidance on the Clinical Investigation of Human Normal Immunoglobulin for Intravenous Administration (IVIg). CPMP/BPWP/388/95 rev.1.

Guidance on the Clinical Investigation of Human Normal Immunoglobulin for Subcutaneous and Intramuscular Use. EMEA/CPMP/BPWP/283/00.

Indian guidelines

As per Indian drugs and cosmetic act human blood plasma is included under the definition of "Drugs" under Section 2 (b)

As per the Indian regulations if the company which manufactures human plasma products similar to the composition and product profile currently marketed then the clinical trial waiver is given for marketing authorization. The company shall apply to DCGI &SLA for the waiver with relevant supporting documents. In case of hyper immune sera products appropriate clinical trial for safety and efficacy shall be conducted.

9.3 Containment Compliance

Blood and blood-derived products are highly sensitive and can be easily contaminated. Bloodborne pathogens, such as HIV, hepatitis B and C, can pose significant risks to both staff and patients if proper containment procedures are not followed. Maintaining a contained environment helps minimize the risk of spreading infectious diseases in the population.

Containment measures in the manufacturing premises shall be established such as isolation rooms, negative pressure environments, controlled airflow systems and proper handling procedures.

Plasma fractionation facilities must adhere to specific biosafety levels as defined by regulatory authorities, such as the Centers for Disease Control and Prevention (CDC). Plasma fractionation facilities operate at BSL-2 (Biosafety Level 2). It must include:

Access control, lab coats, gloves, eye protection (PPE) to be used by personnel working in the facility, and the use of biological safety cabinets during procedures.

Continuous monitoring of air pressure differentials, temperature, and humidity within the clean rooms.

Alarms should be triggered if any parameter falls outside the acceptable range.

Scheduled regular maintenance, testing, and certification of the HVAC system.

The exhaust air from the clean rooms potential for any contaminating microbes should be passed through the scrubbers to protect the environment.

9.4 Validation of Processes, Equipment and Services

Approaches to validation:

Lifinity will organize and plan qualification and validation in a manner that will ensure product quality, safety and efficacy throughout its life-cycle.

Qualification and validation shall be done in accordance with predetermined protocols, and the results to be documented, in reports.

An appropriate and effective quality management system shall be in place at Lifinity supporting the organization, planning, execution and management of qualification and validation activities.

Senior management will provide sufficient resources to perform validation in a timely manner. Management and persons responsible from quality assurance and manufacturing to be actively involved in the validation activities.

All relevant SOPs for operation, maintenance and calibration shall be prepared during qualification.

Training shall be provided to operators, and training records need to be updated and maintained.

Equipment Qualification/Validation:

Equipment qualification helps mitigate risks, prevent equipment failures, reduce downtime, enhance product quality, and ensures the overall operational excellence of an organization.

Equipment Validation comprises a series of steps which includes User Requirement Specification (URS), Design, Installation, Operational, and Performance Qualification.

User Requirement Specification (URS): Lifinity will prepare a document describing the requirements for the equipment to be sourced. The requirements will include specifications and must ensure that possible GMP risks are addressed; include technical requirements; and associated documentation.

Design qualification (DQ): DQ will provide the documented evidence that the design specifications were met and are in accordance with the URS. FAT and SAT needs to be performed where required, to verify the suitability of the system at site, prior to the subsequent stages of qualification and this must be appropriately documented.

Installation qualification (IQ): IQ will provide documented evidence that the installation was complete and satisfactory, including supporting utilities. The design specifications, including purchase specifications, drawings, manuals, lists of spare parts and vendor details, will be verified during IQ. Components installed must be verified, and documented evidence should be provided that components meet specifications, are traceable and are of the appropriate construction material. Applicable control and measuring devices, identified through impact or risk assessment, should be calibrated.

Operational Qualification (OQ): OQ should provide documented evidence that utilities, systems or equipment operate in accordance with operational specifications. Tests will be designed to demonstrate satisfactory operation over the normal operating range, as well as at the limits of its operating conditions. Worst-case conditions must be included in the testing. Operation controls, alarms, switches, displays and other operational components shall be tested.

Performance qualification (PQ): PQ will normally follow the successful completion of IQ and OQ. PQ will be performed under conditions simulating the intended use, to provide documented evidence that equipment/system can consistently perform.

Process validation:

Process validation is defined as the collection and evaluation of data, from the process design stage through commercial production, which establishes scientific evidence that a process is capable of consistently delivering quality product. It ensures that quality, safety and efficacy by design are built into the product.

Collect and evaluate the data throughout the life-cycle of a product from the process design stage through to commercial production and provides scientific evidence that a process is capable of consistently delivering a quality product.

Data need to be available for inspection at Lifinity manufacturing location.

A risk-assessment approach will be followed, to determine the scope and extent to which process(es) and starting material variability may affect product quality.

The critical process steps and critical process parameters (CPPs) will be identified, justified monitored and documented based on relevant studies carried out during the design stage and process knowledge.

Lifinity will ensure that the principles of process validation described in the guidelines are implemented.

Evaluate and compare the historical data when there is a change in the process.

When changes are planned to existing systems or processes need to be followed and documented. Sufficient data is collected to demonstrate that the revised process and product will meet desired quality, consistent and approved specifications.

CHAPTER 10

ALLIED EQUIPMENT & FACILITIES

10.1 Technology Based Allied Equipment Requirements

Refer section 13 Annexures

10.2 Facilities

Laboratories for allied testing should be separated from Production and Quality control labs. However, the materials required for testing such as media, filters, filtration devices etc. can be taken from QC labs. Dedicated glass ware should be used for any testing activities.

Example of allied test/activities are ETP samples, Black utilities, Ground water, testing of clean steam for physical parameters, compressed air and calibration activities etc.

Specialized temperature-controlled warehouse for storage of RMs, In-Process materials and finished goods:

Adequate areas shall be designed to allow sufficient and orderly warehousing of various categories of materials and products like starting and packaging materials, intermediates, bulks and finished products, products in quarantine, released, rejected, returned or recalled, machine and equipment spare parts and change items etc.

Warehousing areas need to be designed and adapted to ensure good storage conditions. Ware house shall be clean, dry and maintained with acceptable temperature and humidity limits where ever required.

Temperature and humidity will be monitored and recorded. A temperature monitoring and recording system that are independent from the temperature regulation system with alarm systems will be placed.

Storage areas must have program for pest control. The record shall be maintained.

Proper racks, bins and platforms must be provided for the storage of materials. The quarantine area in the warehousing will be clearly demarcated. Access to the areas must be restricted to authorized persons.

There shall be a separate sampling area in the warehousing for active and sterile raw materials/starting materials and excipients from the unsterile RMs.

Sampling and dispensing of sterile materials shall be conducted under aseptic condition conforming to Grade A(LAF). Sampling of sterile materials shall be performed in a dedicated area within the manufacturing facility in the absence of suitable areas in the ware house. Segregation shall be provided for the storage of rejected, recalled or returned materials or products.

Materials or products in this area should be suitably marked and secured. Access to these areas and materials must be restricted.

Highly hazardous, poisonous and explosive materials such as alcohols, narcotics and other substances presenting potential risks of abuse, fire or explosion shall be stored in explosion proof area, away from the central ware house in safe and secure manner.

Adequate fire protection measures shall be provided in conformity with the rules of the concerned civic authority in the ware house.

Printed packaging materials shall be stored in safe, separate and secure areas. Regular checks shall be made to ensure adequate steps are taken against spillage, breakage and leakage of containers.

Receiving and dispatch bays shall protect materials and products from adverse weather conditions.

The ware house shall have different areas for the receipt, quarantine, undertest, released, hazardous and rejected materials.

Primary packaging materials, labels, and secondary packaging material and finished products shall be stored in the separate area of the ware house with access controls systems in the critical areas of the ware house.

Incoming critical materials (such as sterile solutions, plasma bag and testing reagents) shall be physically quarantined immediately after receipt and until they are released for use by the authorized person.

Materials and reagents shall be stored under the conditions established by the manufacturer and in an orderly manner that permits segregation by batch or lot and stock rotation. Storage, issuance and usage should follow the "first-expiring first-out" principle (FEFO).

10.3 Materials Required Like Resins, Filters etc.

Ion exchange resins

Size exclusion resin – 1

Size exclusion resin - 2

Multi grade filters (MGF)

Softener

Filters 0.5 Micron

SMBS (Sodium Metabisulphite)

Chlorine

Common Salt

NaOH

Hydrochloric acid

10.4 SOP/STP and Biological Waste Management

The SOP's and STP for the allied testing facilities are as under:

Procedure for Decontamination of chemical and biological waste.

Procedure for estimation of BOD and COD in ETP samples.

pH determination of ETP samples.

Procedure for testing compressed air and pure steam.

Procedure for calibration of measuring devices of manufacturing, ware house, quality control and engineering.

Procedure for treatment of Biowaste

Procedure for collection segregation and disposal of Biowaste

Training on Biowaste management

Procedure for usage of treated water for horticulture purpose

Procedure for discharge of Biowaste in common drainage system

Procedure for management of spillage control

Biowaste management (BWM):

The disposal of sewage and effluents (solid, liquid and gas) from the manufactory should be in conformity with the requirements of Environment Pollution Control Board. All bio-medical waste must be destroyed as per the provisions of the Bio-Medical Waste (Management and Handling) Rules, 1996 and amendments thereof. Additional precautions to be taken for the storage and disposal of rejected waste material. Records shall be maintained for all disposal of waste. Provisions shall be made for the proper and safe storage of waste materials awaiting disposal. Hazardous, toxic substances and flammable materials shall be stored in suitably designed and segregated, enclosed areas in conformity with Central and State Legislations.

CHAPTER 11

SUBSIDY CONSIDERATIONS

11.1 Location / SEZ Considerations in Maharashtra

Special Economic Zone (SEZ) is a specifically delineated duty-free enclave with all required infrastructure provided under single administrative umbrella primarily meant for locating industries which manufacture and export goods and services.

These enclaves shall be deemed to be foreign territory for the purposes of trade operations and duties and tariffs.

Under section 3, a Special Economic Zone can be established either by Central Government, State Government or its agencies, private / public / joint sector or any person for manufacture of goods or rendering services or for both or as a free trade and warehousing zone.

As per the requirements of SEZ, the unit must be segregated into processing and non-processing areas. A processing area is that area in which all the manufacturing activities are performed. At least 50% of the total area shall be earmarked.

The non-processing area is the area which is intended to provide support facilities to the processing area activities and may include support areas like utilities block, power generation plant (DG), engineering services, ETP and other infrastructures as per the requirements of the current project.

In SEZ 100 percent Foreign Direct Investments are allowed (FDI) in some of the selected categories.

As per MIHAN-SEZ guidelines (Rules and Procedures) following class of incentives are available to SEZ enterprises.

11.2 Custom and Excise

SEZ units may import or procure from the domestic sources, duty free, all their requirements of capital goods, raw materials, consumables, spares, packing materials, office equipment, DG sets etc. for implementation of their project in the Zone without any license or specific approval.

11.3 Duty Free Import/Domestic Procurement of Goods for Setting up of SEZ Units

Goods imported/procured locally duty free could be utilized over the approval period of 5 years.

11.4 Sales Exemption

Domestic sales by SEZ units will be exempt from single administrative document (SAD). Domestic sale of finished products, by-products on payment of applicable custom duty. Domestic sale rejects, waste and scrap on payment of applicable Custom duty on the transaction value.

11.5 Income Tax:

Physical export benefit 100%.

IT exemption (10A) for first 5 years and 50% for 2 years thereafter.

Reinvestment allowance to the extent of 50% of ploughed back profits.

11.6 Carry forward of losses & Commercial borrowing:

Banking / Insurance/External Commercial Borrowing.

Setting up Off-shore Banking Units (OBUs) is allowed in SEZs. OBU's allowed 100% Income Tax exemption on profit for 3 years and 50 % for next two years.

External commercial borrowings by units up to \$ 500 million a year allowed without any maturity restrictions.

11.7 Central Sales Tax:

Exemption to sales made from Domestic Tariff Area to SEZ units.

11.8 Service Tax:

Exemption from Service Tax to SEZ units.

11.9 Other Exemptions:

Freedom to bring in export proceeds without any time limit.

Flexibility to keep 100% of export proceeds in exchange earners foreign currency account (EEFC). Freedom to make overseas investment from it.

Exemption from interest rate surcharge on import finance.

SEZ units allowed to 'write-off' unrealized export bills.

11.10 Additional Incentives:

The additional incentives are exemption of import license, custom examination of imported goods, exemption from public hearing under Environment Impact Assessment Notification etc.

CHAPTER 12

PROJECT TIMELINES

CHAPTER 13

BUDGET ESTIMATES

Budget Estimates

List of Annexures

Annexure 1: Administration Equipment and Office Furniture

Annexure 2: Canteen Equipment

Annexure 3: Utility Equipment

Annexure 4: Plasma Storage Equipment

Annexure 5: Bulk Plasma Manufacturing Equipment

Annexure 6: Fill Finish Equipment

Annexure 7: QC / R&D Equipment

Annexure 8: QA Equipment

Annexure 9: Finished Goods Storage

Annexure 10 : ETP / STP

CHAPTER 14

REVENUE AND COST ESTIMATION AND PAYBACK PERIOD

YIELD CALCULATIONS FOR PLASMA FRACTIONS

REVENUE FORECAST IN CRORE INR FROM PROCESSING 150,000 L / ANNUM

PROFIT AND LOSS STATEMENT FOR STAGE 1 – 150,000 L/ ANNUM

CHAPTER 15

RECOMMENDATION

The objective of this project is to establish and operate a state-of-the-art Human Plasma Derived Medicinal Products (HPDMP) manufacturing facility with an aim to address the increasing demand for plasma-derived products by Optimized manufacturing processes maximize productivity, minimize waste, and reduce operational costs in order to maintain competitiveness in the market.

This Human Plasma Derived Medicinal Products manufacturing facility will play a pivotal role in meeting the critical healthcare needs of patients while contributing to advancements in the plasma industry.

In phase-1 of the project annual production capacity will be 150,000 liters of plasma processing, which

will be subsequently increased to 500,000 liters per annum in future.

In this report several technologies utilized for plasma purification at commercial scale were evaluated exhaustively with its merits and demerits. Some of the key technologies evaluated are:

Plasma Fractionation method with ethyl alcohol

Cohn+ Liquid Chromatography

Heat Shock technology

Ammonium sulphate+ Liquid Chromatography

Chromatographic

It is concluded and recommended that Chromatographic purification with sterile filtration is best technology for the purification of plasma proteins for stage-1 products as it will give highly productivity and acceptable quality of desired proteins (purity levels) as per the regulatory expectations.

For stage-2 products, the technology evaluation and recommendations will be made in future.

CHAPTER 16

Layouts

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CHAPTER 17

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LIST OF ABBREVIATIONS: LIST OF ABBREVIATIONS: LIST OF ABBREVIATIONS:
1. AAT : Alpha-1 Antitrypsin
2. AIDS : Acquired Immunodeficiency Syndrome
3. BOD : Biological Oxygen Demand
4. CDSCO : Central Drug Standard Control Organization
5. cGMP : Current Good Manufacturing Practices
6. CO2 : Carbon Dioxide
7. COD : Chemical Oxygen Demand
8. CPPs : Critical Process Parameters
9. CQA : Critical Quality Attributes
10. DCGI : Drug Controller General of India
11. EHS : Environment, Health, and Safety
12. EMA : European Medicine Agency
13 EMP : Environmental Monitoring Program
14. ETP : Effluent Treatment Plant
15. FAT : Factory Acceptance Test
16. G : Gram
17. GCP : Good Clinical Practice
18. GLP : Good Laboratory Practice
19. GMP : Good Manufacturing Practice
20. HAE : Hereditary Angioedema
21. HbsAg : Hepatitis B surface antigen
22. HEPA : High Efficiency Particulate Air
23. HIV : Human Immunodeficiency Virus
24. HAS : Human Serum Albumin
25. HVAC : Heating, Ventilation, and Air Conditioning
26. IEC : Ion Exchange Chromatography
27. IgG : Immunoglobulin
28. IPC : In Process Control
29. IU : International Unit
30. Mabs : Monoclonal Antibodies

31. | mL | : Milliliter
 32. | NA | : Not Applicable
 33. | NACO | : National AIDS Control Organisation
 34. | NAT | : Nucleic acid Amplification testing
 35. | NBTC | : National Blood Transfusion Council
 36. | NGO | : Non-Governmental Organization
 37. | NIB | : National Institute of Biologicals
 38. | NPPA | : National Pharmaceutical Pricing Authority
 39. | OOS | : Out of Specification
 40. | OOT | : Out of Trend
 41. | PCC | : Prothrombin complex concentrate
 42. | PDMP | : Plasma Derived Medicinal Products
 43. | PFD | : Process Flow Diagram
 44. | PPQ | : Process Performance Qualification
 45. | QA | : Quality Assurance
 46. | QC | : Quality Control
 47. | RM | : Raw Material
 48. | SAT | : Site Acceptance Test
 49. | SEC | : Size exclusion chromatography
 50. | SBTC | : State Blood Transfusion Councils
 51. | SOP | : Standard Operating Procedure
 52. | SS | : Stainless Steel
 53. | STD | : Sexually Transmitted Disease
 54. | STP | : Sewage Treatment Plant
 55. | TFF | : Tangential Flow Filtration
 56. | TMP | : Trans-Membrane Pressure
 57. | UPS | Uninterrupted Power Supply
 58. | URS | : User Requirement Specification
 59. | USD | : United States Dollar
 60. | WHO | : World Health Organization
 S/No. | PARTICULARS | BULK MANUFACTURING FACILITY

1 | Days per year | 365 days
 2 | Effective working days per year (*) | 300 days
 3 | Working days per week | 7
 4 | Working hours per day | 24
 5 | Working shifts per day | 3
 6 | Working hours per shift | 8
 7 | Effective working hours per shift (**) | 6(**)
 8 | (-) Cleaning and change over | 2
 9 | Effective production hours per day | 18
 (*) Effective working days per year should exclude: Public holidays, Scheduled Maintenance, Media fill & Re-qualification of equipment.
 (**) For calculations, 6 hours/shift has been taken as effective hours / shift considering two hours for cleaning, changeover, machine setting, etc. | (*) Effective working days per year should exclude: Public holidays, Scheduled Maintenance, Media fill & Re-qualification of equipment.
 (**) For calculations, 6 hours/shift has been taken as effective hours / shift considering two hours for cleaning, changeover, machine setting, etc. | (*) Effective working days per year should exclude: Public holidays, Scheduled Maintenance, Media fill & Re-qualification of equipment.
 (**) For calculations, 6 hours/shift has been taken as effective hours / shift considering two hours for cleaning, changeover, machine setting, etc.

S/No. | PARTICULARS | BULK MANUFACTURING FACILITY
 1 | Days per year | 365 days
 2 | Effective working days per year (*) | 260 days – 300 days
 3 | Working days per week | 5 – 7
 4 | Working hours per day | 16 – 24
 5 | Working shifts per day | 2 - 3
 6 | Working hours per shift | 8
 7 | Effective working hours per shift (**) | 7-8(**)
 8 | (-) Machine setup | 1 - 2
 9 | (-) Cleaning and change over | 8
 10 | Effective hours for filling line per shift | 6
 (*) Effective working days per year should exclude: Public holidays, Scheduled Maintenance, Media fill & Re-qualification of equipment.
 (**) Effective working hours should exclude break, operator gowning. Considering staggered break for critical equipment e.g., filling lines. | (*) Effective working days per year should exclude: Public holidays, Scheduled Maintenance, Media fill & Re-qualification of equipment.
 (**) Effective working hours should exclude break, operator gowning. Considering staggered break for critical equipment e.g., filling lines. | (*) Effective working days per year should exclude: Public holidays, Scheduled Maintenance, Media fill & Re-qualification of equipment.
 (**) Effective working hours should exclude break, operator gowning. Considering staggered break for critical equipment e.g., filling lines.

Company Name | Products | Turnover
 Plasma Gen
 Biosciences Pvt.
 Ltd. Bangalore
www.plasmagen.com | PlasmaGlob, PlasmaGlob 10, PlasmaHep, PlasmaRAB, PlasmaRho-D I.V., AlbuMax-Neo, Verbumin | Annual Revenue \$14.1M as on Mar 31, 2022
 Reliance Life Sciences,
 Mumbai
<https://www.rellife.com/> | AlbuRel (Albumin), HemoRel-A (Factor VIII), HepaRel (Hepatitis B immunoglobulin - HBIG), ImmunoRel (Immunoglobulin G), ReliSeal (Fibrin Sealant) | Company revenue ■
 6,804.80 lakh as on March 31, 2023
 Biocon Limited, Bangalore
<https://www.biocon.com/> | Albubet, Invex | Annual Revenue Rs. 115,501 million as on March 31, 2023
 Intas Pharmaceuticals Ltd. Ahmedabad
www.intaspharma.com/ | Albucel 20%, Globucel 10%, Albucel 5%, Factocel IX, Celestera 1, Globucel-VF, Globucel 5%, Factocel VIII, Pro-throm, Albucel LS | Annual Revenue
 Rs. 6,753
 crore in FY 2022-2023 (Domestic market)
 Virchow Biotech, Hyderabad
<http://virchowbiotech.com/> | V- IMMUNE, V – ALB | Annual revenue \$500 million
 Fusion Healthcare, Hyderabad

www.fusionhealthc.com/ | Genexglob | Annual Revenue \$36.5M as on Mar 31, 2022
ICHOR Biologics Pvt. Ltd., Hyderabad
https://ichor.in/ | Hemalb (Human Albumin), Immuglo/Iggamma (Human Normal Immunoglobulin), Clotnine (Human Coagulation factor IX), Clotate (Human Coagulation factor VIII), Fibroseal (Fibrin Sealant kit), Sphthromb (Thrombin), Fibrina (Fibrinogen), Ichorab 150/300, Hepchor 100/200, Ichor 300 | Company current revenue \$1.06 B
Bharat Serums and Vaccines Limited, Mumbai
https://bsvsgroup.com/ | Gamma IV, Bharglob, Berirab-P, Tetglob, Histoglob | Company revenue Rs. 1,20,703.90 lakhs as on Mar 31, 2023
Company Name | Products
CSL Behring, USA
https://www.csl.com | Beriglobin P, Berinert, Berinert 2000/3000, Berirab P, Evogam, HAEGARDA, Hepatitis B Immunoglobulin P, Hizentra, Intragram P, Intragram 10, Privigen, Rhophylac, Tetagam P, Tetanus Immunoglobulin-VF IM, Zoster Immunoglobulin-VF, AFSTYLA, Beriate, Beriplast, Beriplex P/N, Biostate, Cluviat, Cluvot, Confidex, Corifact, Factor X P Behring, Fibrogammin P, Haemate P, Haemocomplettan P, Humate-P, IDELVION, KCENTRA, Kybernin, Kybernin P, MONOFIX-VF, Prothrombinex-VF, RiaSTAP, Tachosil, Thrombotrol VF, Voncento, Albapure, Albumeon, Albumex, Albuminar, Alunate, Alburex, AlbuRx, Human Albumin
Pfizer, USA
https://www.pfizer.com/ | BeneFIX, Cutaquig, Panzyga
Kedrion Biopharma, USA
https://www.kedrion.com/ | Gammaplex, Gammaked, Ig Vena/Humaglobin liquid/ Kedrigamma/Venital, Subgam, Optiglobin, Naxiglo/Keycute, Ryplazim, Coagadex, Emolcot/Humaclo/Plasamclot/Emowil/Klott/Koate, gy, Optivate, Nuwiq, Wilfactin, Aimafix/Kedrifix/Humafactor-9/Ixed, Replenine-VF, Rhogam/ Immunorho/Keyrho/Micrhogam Human anti D, D-gam, Immunohbs 180 IU, Uman albumin / Uman serum / Human albumin / Kedrialb / Albuplex / Albital / Kedbumin / Albuked / Albumina LFB, Zenalb / Albuminex, Kedrab, Venbig / Keyvenb / Vebiked, Immunohbs / Uman big / Kedhbs, Tetanus gamma / TETIG, Uman complex / Kedcom / Pronativ, AT III Kedrion / Atked, Plasmasafe / Plasmagrade
ADMA Biologics, USA
https://www.admabiologics.com | ASCENIV, BIVIGAM, Nabi-HB
Takeda Pharmaceutical Company Limited
https://www.takeda.com/ | Flexbumin, Gammagard Liquid, Gammagard IGA, Cuvitru, Ceprotin, Glassia, Aralast NP, Immunate, Prothrombin NF 600, Cinryze
Biotest AG, Germany
https://www.biotest.com/ | Haemoctin SDH, Haemonine
Varitect CP, Zutectra, Hepatect CP, Cytotect CP, Intratect, Yimmugo,
Pentaglobin, Albiomin, Biseko
Octa pharma, Switzerland
https://www.octapharma.com/ | Alburnorm, Atenativ, Cutaquig, Fibryga, Gammanorm, Nuwiq, Octagam, Octanate, Octanine-F, OctaplasLG, Octaplex, Panzyga, Rhesonativ, Wilate
Grifols, Spain
https://www.grifols.com/ | Albutein 5%, Albutein 20%, Anbinex, Factor IX Grifols, Fanhdi, Flebigamma DIF 100 mg/ml Solution, Flebogamma DIF 50 mg/ml solution, Gamma Anti-Tetanus Grifols, Plasbumin 20%, Igamad, Igamplia, Igantibe, Nivliva
Kamada Pharmaceuticals, Israel
https://www.kamada.com/ | Cytogam, Glassia, Hepagam B, KamRho D I.M., KamRho D I.V, KamRAB, Varizig, WinRho, Gammaplex, Hepatect CP, Intratect, Megalotect CP, Optivate 1000, Replenine-VF-1000, Varitect CP, Zenalb, Zutectra
SK Plasma, South Korea
https://www.skplasma.com/en | SK Albumin 20% inj., SK Albumin 5% Inj., Liv-Gamma SN Inj., IV Hepabulin SN Inj., Tetabulin SN Inj. Prefilled syringe, SK Antithrombin III Inj. 500 I.U., ETBI Inj. 250 IU
Region | Revenue Share 2022(%)
North America | 53
Asia Pacific | 17
Europe | 22
Latin America | 5
MEA | 3
S/No. | Name of the product | Dosage form | Unit/ pack size | Ceiling price
INR (Round of)
Stage 1 Products | Stage 1 Products | Stage 1 Products | Stage 1 Products | Stage 1 Products
1 | Albumin Injection | Injection 20% | 100 mL | 4000
2 | Coagulation factor IX | Powder | 600 IU | 14000.00
3 | Coagulation factor VIII | Powder | 250 IU | 4500.00
4 | Coagulation factor VIII | Powder | 500 IU | 10000.00
5 | Human normal immunoglobulin | Solution for infusion 5% | 1 mL | 200.00
6 | Human normal immunoglobulin | Solution for infusion 10% | 1 mL | 250.00
7 | Human normal immunoglobulin | Solution for infusion 16.5% | 1 mL | 550.00
8 | Fibrinogen | Powder | 1000 mg | 12,000
Stage 2 Products | Stage 2 Products | Stage 2 Products | Stage 2 Products | Stage 2 Products
8 | Diphtheria antitoxin | Injection | 10000 IU | 1500.00
9 | Hepatitis B immunoglobulin | Injection | Each Pack | 6200.00
10 | Anti-rabies immunoglobulin | Injection | 150 IU/mL /1 mL | 3100.00
11 | Anti-Tetanus immunoglobulin | Injection | 300 IU/mL / 1 mL | 200.00
12 | Anti-Tetanus immunoglobulin | Injection | 250 IU Each pack | 180.00
S/No. | Product Name | Manufacturer | Pack Size | Pack Size | Cost (Rs.)
S/No. | Product Name | Manufacturer | Conc. | Volume | Cost (Rs.)
Human Albumin | Human Albumin | Human Albumin | Human Albumin | Human Albumin | Human Albumin
1 | Albumin –
Albuprist - 5 | Pristyn Reviver | 5% | 100 mL | 2500
2 | Albumin – Vi- Alb | Virchow | 5% | 100 mL | 2500
3 | Albumin - Alburel | Reliance | 20% | 100 mL | 3800 to 4500
4 | Albumin - Alburel | Reliance | 20% | 50 mL | 2900
5 | Albumin - Albuprime | Intas | 20% | 100 mL | 3900
6 | Albumin - Albucel | Intas | 20% | 100 mL | 4500
7 | Albumin - Hetalbu | Hetero Healthcare | 20% | 100 mL | 4200
8 | Albumin -Flexalbumin | Baxter | 20% | 100 mL | 4000
9 | Albumin | Grifols | 20% | 100 mL | 4100
10 | Albumin – Uman Albumin | Kedroin | 20% | 100 mL | 3000
11 | Albumin – Hemalb | Ichor | 20% | 100 mL | 3300

BioBridge Master Knowledge — Page 91

46 | Filter Integrity machine
 47 | Fume Hood
 48 | Autoclave 1000L
 49 | DHS 1000L
 50 | Ceiling LAF
 51 | Drying Oven
 52 | Acid tank 1000L
 53 | Base tank 1000L
 54 | Central CIP system
 55 | Refrigerator 200L
 56 | Laminar air flow cabinet
 57 | Dynamic pass box
 58 | Tumble Dryer 25 Kg
 59 | Barrier washer Extractor 25Kg
 S/No. | Name of Equipment
 1 | Formulation Vessel 100L
 2 | Ceiling LAF
 3 | Laminar air flow cabinet
 4 | Continuous Particle Counter
 5 | Dynamic pass box
 6 | Pass Box static
 7 | Vial washing machine
 8 | Sterilizing tunnel
 9 | Vial filling & Sealing line 120 vials /min
 10 | Lyophilizer (1sq meter)
 11 | Labelling machine
 12 | Double door autoclave 1000L
 13 | Cold room
 S/No. | Name of Equipment
 1 | Akta pure -150
 2 | Columns XK 50/100
 3 | Columns XK 50/60
 4 | Column XK 50/30
 5 | HPLC-UV
 6 | HPLC Columns and accessories
 7 | Electrophoresis (Minigel) with trans blot
 8 | Gel Doc system densitometer
 9 | Power packs
 10 | Endotoxin Detection system (Tube heating block)
 11 | Micropipettes- variable volume range of 2-20µL, 20-200µL, 100-1000µL, 500-5000µL and multichannel 30-300µL
 12 | Lab Refrigerator 2-8 ° C (SS) with data control
 13 | Flame photometer
 14 | Weighing Balance Analytical
 15 | UV-VIS spectrophotometer Double Beam with data acquisition.
 16 | Volumetric KF
 17 | Moisture Analyser
 18 | Stability Chambers (5±3° C, 25, 37 deg. C) 200L (1set)
 19 | Muffle Furnance
 20 | pH and conductivity meter with printer
 21 | Vaccum Oven
 22 | Melting point apparatus
 23 | Incubator Sterility Test 500L
 24 | Incubators for Normal Media plates Incubation 500L
 25 | Incubator for EMP and water samples 500L
 26 | Hot Air Oven
 27 | Thermostatic water bath (10-20 L)
 28 | Vaccum pump
 29 | Desiccator Glass
 30 | High Quality Laboratory water generation system (10 L /hr) Milli -Q water
 31 | Sonicator
 32 | Vortex Shaker
 33 | Magnetic Stirrer with hot plate
 34 | LAF Vertical
 35 | Biosafety Cabinet
 36 | Autoclave Double door
 37 | Dry Heat Steriliser
 38 | Fume Hood (Table Top)
 39 | Deep freezer (-20 Degrees C) with Data acquisition
 40 | Air sampler
 41 | Particle counter
 42 | Decontamination autoclave
 43 | Microscope Binocular
 44 | Colony Counter
 45 | Microplate reader ELISA
 46 | Microplate washer
 47 | TOC analyser (Grab Samples)
 48 | Timers/ Stop watch
 49 | Ice flaking machine
 50 | Standard Weight Box
 51 | Printer/scanner / xerox machine
 52 | Centrifuge table top
 53 | Size exclusion
 54 | Anion
 55 | Cation
 56 | Affinity
 S/No. | Name of Equipment
 1 | Anemometer
 2 | Lux meter
 3 | Datalogger 32 channel
 4 | Paper shredder
 5 | Laminator

6 | Printer/scanner / xerox machine
 7 | Computers / Laptops
 8 | Numbering Machine
 9 | Spiral Binding Machine
 10 | A3 +A4 Lamination Machine
 11 | Bi-Focal Glass
 12 | Tachometer
 13 | Clamp Meter/ Multi meter
 14 | Level Indicator
 15 | Measuring TAPE
 16 | Stop Watch
 17 | Temperature recorders 24 Nos. (data Loggers)
 18 | Office Cabinets for Documents
 19 | Projector for conference / meeting room
 S/No. | Name of Equipment
 1 | Analytical weighing Balance
 2 | RLAF
 3 | Precision Balance
 4 | Commercial Balance
 5 | Refrigerator 200L
 6 | Deep freezer -20°C (1000L)
 7 | Air curtain
 8 | PVC pallets
 9 | Storage racks SS
 10 | Sampling rods
 11 | Liquid sampler
 12 | Dedusting tunnel
 13 | DPB
 14 | LAF cabinet
 S/No. | Name of Equipment
 1 | Air curtain
 2 | PVC pallets
 3 | Storage racks SS
 4 | Hydraulic trolley
 5 | Cold Rooms
 S/No. | Name of Equipment
 Clean Utilities | Clean Utilities
 1 | Soft water generation and storage system
 2 | Purified water generation storage and distribution system
 3 | Water for injection generation, storage and distribution system
 4 | Pure steam generator and distribution system
 Black Utilities | Black Utilities
 1 | Plant steam generation and distribution System
 2 | Chiller
 3 | Stand by Chiller
 4 | Compressed air generation and distribution system
 Effluent Treatment Plant (ETP) | Effluent Treatment Plant (ETP)
 1 | Biowaste reception tank 15000L
 2 | Kill tank 500L
 3 | Bar Screen
 4 | Collection tank
 5 | Equalization tank
 6 | Neutralization tank
 7 | Chemical coagulation
 8 | Biological oxidation tank
 9 | Filtration System
 Sewage Treatment Plant (STP) | Sewage Treatment Plant (STP)
 1 | Primary treatment plant: Bar screening - 3 stages, pH correction, BOD & COD removal secondary clarifier
 2 | Secondary treatment: Electro coagulation process, Flash mixer, HRSC clarifier, Sand filter
 3 | Sludge Management: Sludge thickener, Screw pump, Filter press, Sludge storage
 Item | Volume in Liters
 Plasma Storage capacity | 32000
 Plasma required per month | 30000
 Plasma required per 10 days | 10000
 Volume of plasma per bag | 0.2
 No. of bags required per 10 days | 50000
 No. of bags required per month | 150000
 Product | Protein availability (gm) | Protein Recovery | Conc. / vial | No. of vials /batch
 Albumin (gm) | 40,000 | 20000 | 20 GM | 900
 Human IgG (gm) | 12,500 | 6250 | 5 GM | 1000
 Factor VIII in IU | 10,00,000 | 12500 IU | 250 IU | 50
 Factor IX in IU | 10,00,000 | 90000 IU | 600 IU | 150
 Fibrinogen (gm) | 3,000 | 1500 | 70 mg | 80
 Thrombin in IU | 9,00,000 | 90000 | 1000 IU | 80
 Fibrinogen | 3000 | 1490 | 1000 mg | 1400
 Product | Protein availability | Protein Recovery | Conc. / Vial | No. of vials/year
 Albumin (gm) | 60,00,000 | 30,00,000 | 20 gm | 1,35,000
 Human IgG (gm) | 18,75,000 | 9,37,500 | 5 gm | 1,50,000
 Factor VIII in IU | 15,00,00,000 | 18,75,000 | 250 IU | 7,500
 Factor IX in IU | 15,00,00,000 | 13,500,000 | 600 IU | 22,500
 Fibrinogen (gm) | 4,50,000 | 2,25,000 | 70 mg | 12,000
 Thrombin in IU | 13,50,00,000 | 1,35,00,000 | 1000 IU | 12,000
 Fibrinogen | 450,000 | 225,000 | 1000 mg | 2,10,000
 Product | Protein availability (gm) | Protein Recovery (gm) | Conc. / Vial | No. of vials manufactured /batch
 Albumin | 1,36,000 | 68,000 | 20 gm | 3,060
 Human IgG | 42,500 | 21,250 | 5 gm | 3,400
 Factor VIII in IU | 34,00,000 | 8,16,000 | 250 IU | 170
 Factor IX in IU | 34,00,000 | 8,50,000 | 600 IU | 510
 Fibrinogen | 10,200 | 5,100 | 70 mg | 272
 Thrombin IU | 30,60,000 | 3,06,000 | 1000 IU |

Fibrinogen | 10,200 | 5100 | 1000 mg | 4,760
Product | Protein availability (gm) | Protein Recovery (gm) | Conc. / Vial | No. of vials manufactured /batch
Albumin | 2,00,000 | 100,000 | 20 gm | 4,500
Human IgG | 62,500 | 31,250 | 5 gm | 5,000
Factor VIII in IU | 50,00,000 | 1,200,000 | 250 IU | 250
Factor IX in IU | 50,00,000 | 1,250,000 | 600 IU | 750
Fibrinogen | 15,000 | 7,500 | 70 mg | 400
Thrombin IU | 45,00,000 | 450,000 | 1000 IU |
Fibrinogen | 15,000 | 7,500 | 1000 mg | 7,000
Product | Protein availability (gm) | Protein Recovery (gm) | Conc. / Vial | No. of vials /Year
Albumin | 2,00,00,000 | 1,00,00,000 | 20 gm | 450000
Human IgG | 62,50,000 | 31,25,000 | 5 gm | 500000
Factor VIII in IU | 50,00,00,000 | 12,00,00,000 | 250 IU | 25000
Factor IX in IU | 50,00,00,000 | 12,50,00,000 | 600 IU | 75000
Fibrinogen | 15,00,000 | 7,50,000 | 70 | 40000
Thrombin IU | 45,00,00,000 | 4,50,00,000 | 1000 IU | 40000
Fibrinogen | 15,00,000 | 7,50,000 | 1000 mg | 700,000
S/No. | Step | Purpose | Remark
1 | Cell culture and Fermentation | To generate the desired bulk of antibodies |
2 | Centrifugation | To remove cell debris, lipids, and clotted material |
3 | Filtration (0.45 µm) | To remove smaller contaminating material |
4 | Concentration (UF) | To concentrate the desired antibodies in the material |
5 | Ion exchange Chromatography (IEC) | To remove nucleic acids and endotoxins |
6 | Size exclusion chromatography (SEC) | SEC is preferred over IEC because size is relatively constant and charge on the protein may vary depending upon the pH as molecules are protonated and deprotonated
7 | Immunoaffinity Chromatography | Antibody will bind to specific antigen attached to the agarose matrix
| It will help to achieve highest purity
8 | Sterile filtration | To remove microbial contamination |
Grade | Maximum Allowable Particle/m3 | Maximum Allowable Particle/m3 | Maximum Allowable Particle/m3 |
Maximum Allowable Particle/m3
Grade | During Rest | During Rest | During Operation | During Operation
Grade | ≥ 0.5µm | ≥ 5µm | ≥ 0.5µm | ≥ 5µm
A | 3500 | 0 | 3500 | 0
B | 35,000 | 0 | 350000 | 2000
C | 350000 | 2000 | 3500000 | 20000
D | 3500000 | 20000 | Not Defined | Not Defined
Grade | Air sample (CFU/m3) | Settle plates (diameter 90 mm) (CFU/4 hours) | Contact plates (diameter 55 mm) (CFU/plate) | Glove print (5 fingers) (CFU/glove)
A | < 1 | < 1 | < 1 | < 1
B | 10 | 5 | 5 | 5
C | 100 | 50 | 25 | -
D | 200 | 100 | 50 | -
Rule 122 EA | Definitions
Rule 122F to 122N | License Application Procedure / Inspection / Reporting by Inspection Team / Grant or Rejection of License / Duration of license / Appeal provision
License granted by Licensing Authority
Approval by Central License Approving Authority
License is granted & delivered to Applicant
Rule 122 O | Procedure of Cancellation and Suspension of Licenses
Rule 122 P | Conditions of licenses:
Testing of whole blood / Component / Product Inspections
Reporting to SLA / CLAA about changes in staff / premise
Directions issued by SLA / CLAA / Recall Directions
Conditions for distribution of whole blood / component / products
To comply provisions of the Act and Rules
Product | License | Renewal
Blood and Components | 27-C | 28-C 26-G
Blood Products | 27-E | 28-E 26-I
1 | The Drugs and Cosmetics Rules, 1945, framed under the Drugs and Cosmetics Act 1940.
2 | Schedule F Part XII B of Drugs & Cosmetics Rules.
3 | Drugs and Cosmetics Rules, Part X-B: Conditions for License (Rule 122EA to Rule 122P).
4 | Destruction of infected blood and implementation of Bio-Medical Wastes (Management and Handling) Rules, 1996.
5 | The National Blood Policy 2002 regulates blood quality and the Guidelines of NBTC 2016.
6 | The National AIDS Control Organization (NACO), under the Ministry of Health and Family Welfare) to oversee operation of blood banking services in India.
7 | National Blood Transfusion Council (NBTC) is the apex policy-making body, under NACO, making central licensing mandatory, increasing voluntary donation base and setting of quality standards.
8 | State Blood Transfusion Councils (SBTCs) are responsible for the smooth functioning of blood transfusion services at a state level.
9 | Drug Standard Control Organization (CDSCO) and State Drug Control Organizations' play a vital role in key aspects such as, approval of licenses, and enforcement of standard transfusion practices to ensure safe, quality and efficacious blood and its components in clinical practices.
10 | National Institute of Biologicals (NIB) Nodal governmental organization under the Ministry of Health and Family Welfare responsible for the testing and release of biologicals in India including blood plasma.
11 | ISPE Baseline Guide Vol. 6, Biopharmaceutical Manufacturing Facilities ISPE Baseline Guide Vol. 3, Sterile Manufacturing Facilities.
12 | ISO EN 14644: Clean room and associated controlled environments.
13 | WHO Good manufacturing practices for biological products.
14 | Good manufacturing practice guideline for human and veterinary medicinal product.
15 | PIC/s Guide to Good Manufacturing Practice for Medicinal Products.
16 | US FDA Guidance for Industry – Sterile Drug Products produced by Aseptic Processing.
17 | WHO recommendations for the production, control and regulation of human plasma for fractionation, Annex 4, TRS No 941.
18 | WHO guidelines on good manufacturing practices for blood establishments WHO Technical Report Series, No. 961, 2011.
19 | June 2011 EMA/CHMP/ICH/731268/1998 Committee for medicinal products for human use (CHMP).
20 | Guideline on clinical investigation of recombinant and human plasma-derived factor IX products: May 2015, EMA/CHMP/BPWP/144552/2009 Rev. 1, Corr. 1 Committee for medicinal products for human use (CHMP).

21 | Guideline on clinical investigation of recombinant and human plasma-derived factor VIII products: 21 May 2015 EMA/CHMP/BPWP/144533/2009 rev Committee for medicinal products for human use (CHMP).

22 | Guidance on the Clinical Investigation of Human Normal Immunoglobulin for Intravenous Administration (IVIg). CPMP/BPWG/388/95 rev.1.

23 | Guidance on the Clinical Investigation of Human Normal Immunoglobulin for Subcutaneous and Intramuscular Use. EMEA/CPMP/BPWP/283/00.

24 | Precedence Research, Plasma derived market size, A report 2032.

25 | Marketing Research Bureau, Inc.

26 | Annex 2: WHO good manufacturing practices for biological products.

27 | Introduction to United States Environmental Protection Agency Containment Buildings (40 CFR Parts 264/265, Subpart DD).

28 | WHO good manufacturing practices for sterile pharmaceutical products; Annex 6, TRS No. 961, 2011.

29 | Biosafety in Microbiological and Biomedical Laboratories. 6th Edition U.S. Department of Health and Human Services Public Health Service Centers for Disease Control and Prevention National Institutes of Health Revised June 2020.

30 | WHO Guidelines on heating, ventilation and air-conditioning systems for non-sterile pharmaceutical product, Annex 8.

31 | WHO Good manufacturing practices: guidelines on validation, Annex 3.

32 | 01/2009:0918; Human normal immunoglobulin for intravenous administration, European Pharmacopoeia.

33 | Albumin / Official Monographs.

34 | 01/2008:1643; Human coagulation factor VIII (rDNA), European Pharmacopoeia.

35 | 1998:1223; Human coagulation factor IX, freeze-dried, European Pharmacopoeia.

36 | 07/2013:0275; Human coagulation factor VIII, European Pharmacopoeia.

37 | 01/2008:20711; Assay of Human coagulation factor IX.

38 | 01/2008:20712; Assay of Heparin in Coagulation factors.

39 | Ramin Raoufinia; Overview of Albumin and Its Purification Methods Adv Pharm Bull, 2016, 6(4), 495-507 doi: 10.15171/apb.2016.063.

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BUDGET ESTIMATE IN INR | BUDGET ESTIMATE IN INR | BUDGET ESTIMATE IN INR | BUDGET ESTIMATE IN INR

1 | Office Cabinets for Documents | 20 | 2,40,000.00

2 | Computers / Laptops | 50 | 25,00,000.00

3 | Office Furniture | NA | 12,00,000.00

4 | Printer/scanner / xerox machine | 1 | 5,00,000.00

5 | Water Dispenser | 2 | 40,000.00

6 | Projector for conference / meeting room | 2 | 6,00,000.00

Total Amount in INR | Total Amount in INR | Total Amount in INR | 50,80,000.00

BUDGET ESTIMATE IN INR | BUDGET ESTIMATE IN INR | BUDGET ESTIMATE IN INR | BUDGET ESTIMATE IN INR

S/No | Name of Equipment | Quantity | Approx Cost (INR)

1 | Refrigerator | 2 | 3,00,000.00

2 | SS furniture (serving table etc) | NA | 4,00,000.00

3 | SS Water cooler dispenser | 1 | 50,000.00

4 | Microwaves | 2 | 60,000.00

5 | Kitchenware | na | 5,00,000.00

6 | Bain marie | 1 | 50,000.00

Total Amount in INR | Total Amount in INR | Total Amount in INR | 13,60,000.00

BUDGET ESTIMATE IN INR | BUDGET ESTIMATE IN INR | BUDGET ESTIMATE IN INR | BUDGET ESTIMATE IN INR

S/No | Name of Equipment | Quantity | Approx Cost (INR)

| Clean Utilities | |

1 | Soft water generation 8000L/hour and storage system | 1 | 3,00,00,000.00

2 | Purified water generation 5000L/ hour storage tank 10KL and distribution system | 1 | 3,00,00,000.00

3 | Water for injection generation 1000L / hour, Storage tank 10KL and distribution system | 1 | 3,00,00,000.00

4 | Pure steam generator 1000Kg / hour and distribution system | 1 | 3,00,00,000.00

| Black Utilities | |

1 | Plant steam generation 2000Kg/Hour and distribution System | 2 | 70,00,000.00

2 | Compressed air generation and distribution system | 2 | 75,00,000.00

3 | PASSENGER LIFT X 2 NOS | 2 | 70,00,000

4 | MATERIAL LIFT X 3 NOS | 3 | 60,00,000

Total Amount in INR | Total Amount in INR | Total Amount in INR | 5,75,00,000.00

EQUIPMENT BUDGET ESTIMATE IN INR | EQUIPMENT BUDGET ESTIMATE IN INR | EQUIPMENT BUDGET ESTIMATE IN INR |

EQUIPMENT BUDGET ESTIMATE IN INR

S/No | Name of Equipment | Quantity | Approx Cost (INR)

1 | Analytical weighing Balance | 1 | 1,70,000.00

2 | RLAf | 3 | 12,00,000.00

3 | Precision Balance | 1 | 1,44,000.00

4 | Commercial Balance | 1 | 1,20,000.00

5 | Refrigerator 200L | 2 | 2,18,000.00

6 | Deep freezer -20°C (1000L) | 32 | 1,44,00,000.00

7 | Air curtain | 10 | 1,00,000.00

8 | PVC pallets | 50 | 3,00,000.00

9 | Storage racks SS | 25 | 7,00,000.00

10 | Sampling rods | 2 | 10,000.00

11 | Liquid sampler | 2 | 20,000.00

12 | Dedusting tunnel | 1 | 12,50,000.00

13 | DPB | 4 | 6,00,000.00

14 | LAF cabinet | 1 | 4,50,000.00

15 | Computer / Laptops | 4 | 1,20,000.00

16 | Office Furniture | NA | 2,00,000.00

17 | Office Cabinets | 10 | 1,20,000.00

Total Amount in INR | Total Amount in INR | Total Amount in INR | 2,01,22,000.00

EQUIPMENT BUDGET ESTIMATE IN INR | EQUIPMENT BUDGET ESTIMATE IN INR | EQUIPMENT BUDGET ESTIMATE IN INR |

EQUIPMENT BUDGET ESTIMATE IN INR

S/No | Name of Equipment | Quantity | Approx Cost (INR)

1 | Chromatography AKTA-1 (471 L/hr (Pipe MOC SS) | 1 | 1,71,00,000

2 | Chromatography Column Dia 1000mm x Height 500mm | 2 | 4,46,12,200

3 | Chromatography AKTA-1 (500 L/hr pipe MOC SS) | 1 | 1,71,00,000

4 | Chromatography Column Dia 1000mm x Height 500mm | 1 | 2,23,06,100

5 | Chromatography AKTA-1 (785 L/hr pipe MOC PP) | 1 | 2,00,31,420

6 | Chromatography Column Dia 1000mm x Height 500mm | 1 | 2,23,06,100

7 | Chromatography AKTA-1 (1178 L/hr pipe MOC PP) | 1 | 2,00,31,420

8 | Chromatography Column Dia 1000mm x Height 500mm | 2 | 4,46,12,210

9 | Chromatography AKTA-1 (864 L/hr pipe MOC PP) | 1 | 2,00,31,420

10 | Chromatography Column Dia 1000mm x Height 500mm | 1 | 2,23,06,100

11 | Chromatography AKTA-1 (864 L/hr pipe MOC PP) | 1 | 2,00,31,420

12 | Chromatography Column Dia 1000mm x Height 500mm | 1 | 2,23,06,100

13 | Chromatography AKTA-1 (667 L/hr pipe MOC PP) | 1 | 1,71,00,000

14 | Chromatography Column Dia 1000mm x Height 500mm | 1 | 2,23,06,100

15 | Chromatography Column Dia 300mm x Height 500mm | 1 | 48,96,570

16 | Chromatography AKTA-1 (785 L/hr pipe MOC PP) | 1 | 2,00,31,420

17 | Chromatography Column Dia 1000mm x Height 500mm | 1 | 2,23,06,100

18 | Chromatography Column Dia 300mm x Height 500mm | 1 | 48,96,570

19 | Chromatography AKTA-1 (30 L/hr pipe MOC PP) | 1 | 1,46,57,140

20 | Chromatography Column Dia 300mm x Height 750mm | 1 | 48,96,570

21 | Chromatography Column Dia 600mm x Height 500mm | 1 | 1,42,44,570

22 | Chromatography AKTA-1(160 L/hr pipe MOC PP) | 1 | 1,46,57,140

23 | Chromatography Column Dia 450mm x Height 500mm | 1 | 1,20,18,850

24 | TFF System 40 sqm | 1 | 1,93,50,000

25 | Cassette 40 sqm | 1 | 55,90,000

26 | TFF System 30 sqm | 2 | 3,87,00,000

27 | Cassette 30 sqm | 2 | 86,00,000

28 | TFF System 20 sqm | 1 | 94,60,000

29 | Cassette 20 sqm | 1 | 30,10,000

30 | TFF System 5 sqm | 2 | 73,10,000

31 | Cassette 5 sqm | 2 | 6,88,000

32 | Continuous Centrifuge 1000L/hr | 2 | 3,00,00,000

33 | Process vessel 150L | 3 | 32,25,000

34 | Process vessel 250L | 4 | 51,60,000

35 | Process vessel (Mobile tank) 300L | 2 | 25,80,000

36 | Process vessel 1500L | 20 | 4,30,00,000

37 | Process vessel 2000L | 6 | 1,54,80,000

38 | Buffer Vessel 500L | 14 | 2,40,80,000

39 | Buffer Vessel 1000L | 10 | 1,93,50,000

40 | Depth filtration system | 2 | 21,00,000

41 | Nano filter holder | 1 | 5,00,000

42 | Cold room 2-8°C | 2 | 15,00,000

43 | Rectangular tank | 1 | 4,30,000

44 | Buffer preparation tank | 2 | 17,20,000

45 | Dispensing Booth | 1 | 4,00,000

46 | Analytical Balance | 1 | 1,70,000

47 | Precision Balance | 1 | 1,50,000

48 | Commercial Balance | 1 | 1,20,000

49 | PVC bag Sealer | 1 | 5,000

50 | Rotary Bag Sealer | 1 | 3,00,000

51 | pH/ Conductivity meter | 2 | 3,00,000

52 | Magnetic Stirrer | 2 | 1,50,000

53 | Peristaltic pump | 2 | 2,00,000

54 | Filter Integrity machine | 1 | 20,00,000

55 | Fume Hood | 1 | 4,50,000

56 | Autoclave 1000L | 2 | 40,00,000

57 | DHS 1000L | 1 | 20,00,000

58 | Ceiling LAF | 3 | 12,00,000

59 | Drying Oven | 2 | 10,00,000

60 | Acid tank 1000L | 2 | 38,70,000

61 | Base tank 1000L | 2 | 38,70,000

62 | Central CIP system | 2 | 21,50,000

63 | Refrigerator 200L | 1 | 1,50,000
 64 | Laminar air flow cabinet | 1 | 4,35,000
 65 | Dynamic pass box | 4 | 4,80,000
 66 | Tumble Dryer 25 Kg | 1 | 20,00,000
 67 | Barrier washer Extractor 25Kg | 1 | 30,00,000
 68 | SS Furniture (Table, chair, racks, Ladders etc) | NA | 40,00,000
 69 | Lockers | NA | 2,00,000
 70 | Laptop / Computers | 10 | 5,00,000
 71 | Office furniture / Cabinets | NA | 3,00,000
 Total Amount in INR | Total Amount in INR | Total Amount in INR | 70,29,18,520
 EQUIPMENT BUDGET ESTIMATE IN INR | EQUIPMENT BUDGET ESTIMATE IN INR | EQUIPMENT BUDGET ESTIMATE IN INR |
 EQUIPMENT BUDGET ESTIMATE IN INR
 S/No | Name of Equipment | Quantity | Approx Cost (INR)
 1 | Formulation Vessel 100L | 1 | 25,00,000.00
 2 | Ceiling LAF | 2 | 8,00,000.00
 3 | Laminar air flow cabinet | 1 | 4,35,000.00
 4 | Continuous Particle Counter | 1 | 30,00,000.00
 5 | Dynamic pass box | 7 | 8,40,000.00
 6 | Pass Box static | 1 | 38,000.00
 7 | Vial washing machine | 1 | 45,00,000.00
 8 | Sterilizing tunnel | 1 | 45,00,000.00
 9 | Vial filling & Sealing line 120 vials /min | 1 | 75,00,000.00
 10 | Lyophilizer (1sq meter) | 1 | 55,00,000.00
 11 | Labelling machine | 1 | 9,00,000.00
 12 | Double door autoclave 1000L | 1 | 20,00,000.00
 13 | Cold room | 1 | 8,00,000.00
 14 | SS furniture (Table, Chair, racks, cabinets etc) | 1 | 2,00,000.00
 Total Amount in INR | Total Amount in INR | Total Amount in INR | 3,26,13,000.00
 EQUIPMENT BUDGET ESTIMATE IN INR | EQUIPMENT BUDGET ESTIMATE IN INR | EQUIPMENT BUDGET ESTIMATE IN INR |
 EQUIPMENT BUDGET ESTIMATE IN INR
 1 | Akta pure -150 | 2 | 40,00,000
 2 | Columns XK 50/100 | 1 | 7,00,000
 3 | Columns XK 50/60 | 1 | 7,00,000
 4 | Column XK 50/30 | 3 | 14,00,000
 5 | HPLC-UV | 1 | 34,00,000
 6 | HPLC Columns and accessories | 1 | 10,00,000
 7 | Electrophoresis (Minigel) with trans blot | 2 | 6,00,000
 8 | Gel Doc system densitometer | 1 | 5,00,000
 9 | Power packs | 2 | 2,00,000
 10 | Endotoxin Detection system (Tube heating block) | 1 | 2,00,000
 11 | Micropipettes- variable volume range of 2-20µL, 20-200µL, 100-1000µL, 500-5000µL and multichannel
 30-300µL | 2 set each | 3,00,000
 12 | Lab Refrigerator 2-8 ° C (SS) with data control | 4 | 4,50,000
 13 | Flame photometer | 1 | 4,50,000
 14 | Weighing Balance Analytical | 2 | 3,50,000
 15 | UV-VIS spectrophotometer Double Beam with data acquisition. | 1 | 9,00,000
 16 | Volumetric KF | 1 | 7,00,000
 17 | Moisture Analyser | 1 | 8,00,000
 18 | Stability Chambers (5±3° C, 25, 37 deg. C) 200L (1set) | 3 | 14,00,000
 19 | Muffle Furnance | 1 | 80,000
 20 | pH and conductivity meter with printer | 2 | 2,60,000
 21 | Vaccum Oven | 1 | 1,20,000
 22 | Melting point apparatus | 1 | 90,000
 23 | Incubator Sterility Test 500L | 2 | 14,40,000
 24 | Incubators for Normal Media plates Incubation 500L | 1 | 3,00,000
 25 | Incubator for EMP and water samples 500L | 2 | 6,00,000
 26 | Hot Air Oven | 1 | 1,60,000
 27 | Thermostatic water bath (10-20 L) | 2 | 2,00,000
 28 | Vaccum pump | 2 | 1,20,000
 29 | Desiccator Glass | 1 | 4,000
 30 | High Quality Laboratory water generation system (10 L /hr) Milli -Q water | 1 | 14,50,000
 31 | Sonicator | 1 | 1,80,000
 32 | Vortex Shaker | 6 | 2,10,000
 33 | Magnetic Stirrer with hot plate | 4 | 4,00,000
 34 | LAF Vertical | 2 | 6,00,000
 35 | Biosafety Cabinet | 2 | 14,00,000
 36 | Autoclave Double door | 1 | 12,50,000
 37 | Dry Heat Steriliser | 1 | 7,50,000
 38 | Fume Hood (Table Top) | 1 | 4,50,000
 39 | Deep freezer (-20 Degrees C) with Data acquisition | 1 | 4,00,000
 40 | Air sampler | 1 | 3,00,000
 41 | Particle counter | 1 | 20,00,000
 42 | Decontamination autoclave | 1 | 10,00,000
 43 | Microscope Binocular | 1 | 1,00,000
 44 | Colony Counter | 1 | 1,50,000
 45 | Microplate reader ELISA | 1 | 6,00,000
 46 | Microplate washer | 1 | 3,50,000
 47 | TOC analyser (Grab Samples) | 1 | 25,00,000
 48 | Timers/ Stop watch | 5 | 10,000
 49 | Ice flaking machine | 1 | 2,50,000
 50 | Standard Weight Box | 1 Set | 10,00,000
 51 | Printer/scanner / xerox machine | 1 | 5,00,000
 52 | Centrifuge table top | 1 | 4,50,000
 53 | Size exclusion | L | 1,50,000.00
 54 | Anion | L | 1,50,000.00
 55 | Cation | L | 1,50,000.00
 56 | Affinity | L | 1,50,000.00
 57 | SS furniture racks | 25 | 2,50,000
 58 | Computers / Laptops | 10 | 5,00,000
 59 | Office Furniture | NA | 5,00,000
 60 | Office Cabinets for Documents | 10 | 1,20,000
 Total Amount in INR | Total Amount in INR | Total Amount in INR | 3,96,94,000.00

BUDGET ESTIMATE IN INR | BUDGET ESTIMATE IN INR | BUDGET ESTIMATE IN INR | BUDGET ESTIMATE IN INR

- 1 | Anemometer | 1 | 15,000.00
- 2 | Lux meter | 1 | 12,000.00
- 3 | Datalogger 32 channel | 1 | 4,00,000.00
- 4 | Paper shredder | 1 | 10,000.00
- 5 | Laminator | 1 | 10,000.00
- 6 | Printer/scanner / xerox machine | 1 | 5,00,000.00
- 7 | Computers / Laptops | 10 | 5,00,000.00
- 8 | Numbering Machine | 5 | 25,000.00
- 9 | Spiral Binding Machine | 1 | 15,000.00
- 10 | A3 +A4 Lamination Machine | 1 | 25,000.00
- 11 | Bi-Focal Glass | 1 | 80,000.00
- 12 | Tachometer | 2 | 6,000.00
- 13 | Clamp Meter/ Multi meter | 5 | 10,000.00
- 14 | Level Indicator | 2 | 8,000.00
- 15 | Measuring TAPE | 2 | 1,500.00
- 16 | Stop Watch | 5 | 50,000.00
- 17 | Temperature recorders 24 Nos. (data Loggers) | 20 | 2,00,000.00
- 18 | Office Cabinets for Documents | 20 | 2,40,000.00
- 19 | Projector for conference / meeting room | 1 | 3,00,000.00
- 20 | Office Furniture | NA | 7,00,000.00

Total Amount in INR | Total Amount in INR | Total Amount in INR | 31,07,500.00

BUDGET ESTIMATE IN INR | BUDGET ESTIMATE IN INR | BUDGET ESTIMATE IN INR | BUDGET ESTIMATE IN INR

- 1 | Air curtain | 10 | 1,00,000.00
- 2 | PVC pallets | 50 | 3,00,000.00
- 3 | Storage racks SS | 50 | 15,00,000.00
- 4 | Hydraulic trolley | 1 | 2,50,000.00

Total Amount in INR | Total Amount in INR | Total Amount in INR | 21,50,000.00

BUDGET ESTIMATE IN INR | BUDGET ESTIMATE IN INR | BUDGET ESTIMATE IN INR | BUDGET ESTIMATE IN INR

S/No | Name of Equipment | Qty. | Approx Cost (INR)

- 1 | Biowaste reception tank 15000L | 1 | 25,00,000.00
- 2 | Kill tank 500L | 2 | 25,00,000.00
- 3 | Bar Screen | NA | 25,00,000.00
- 4 | Collection tank | NA | 25,00,000.00
- 5 | Equalization tank | NA | 25,00,000.00
- 6 | Neutralization tank | NA | 25,00,000.00
- 7 | Chemical coagulation | NA | 25,00,000.00
- 8 | Biological oxidation tank | NA | 25,00,000.00
- 9 | Filtration System | NA | 25,00,000.00

BUDGET ESTIMATE IN INR | BUDGET ESTIMATE IN INR | BUDGET ESTIMATE IN INR | BUDGET ESTIMATE IN INR

- 1 | Primary treatment plant : Bar screening - 3 stages, pH correction, BOD & COD removal secondary clarifier | | 8,00,000.00
 - 2 | Secondary treatment : Electro coagulation process, Flash mixer, HRSC clarifier, Sand filter | | 8,00,000.00
 - 3 | Sludge Management: Sludge thickner, Screw pump, Filter press, Sludge storage | | 8,00,000.00
- Total Amount in INR | Total Amount in INR | Total Amount in INR | 33,00,000.00

[SHEET: yield_CALCULATIONS]
| YIELD CALCULATIONS FOR PLASMA FRACTIONS ||||
| Product | PRICE REALISED PER VIAL IN INR (2023) | YIELD IN NUMBER OF VIALS / 1,000 L PLASMA | YIELD IN NUMBER OF VIALS / 150,000 L PLASMA | YIELD IN NUMBER OF VIALS / 500,000 L PLASMA
| 1. Human serums Albumin (HSA) - 20% | 3200 | 900 | 225000 | 450000
| 2. Human IVIG - 5% | 6600 | 1000 | 250000 | 500000
| 3. Clotting factor VIII | 3000 | 50 | 12500 | 25000
| 4. Clotting factor IX | 6200 | 150 | 37500 | 75000
| 5. Fibrin SEALANT | 3000 | 80 | 20000 | 40000
| 6. Fibrinogen | 8000 | 1400 | 375000 | 700000
| YIELD CALCULATIONS FOR PLASMA FRACTIONS Government Tender Rates ||||
| Product | PRICE REALISED PER VIAL IN INR (2023) | YIELD IN NUMBER OF VIALS / 1,000 L PLASMA | YIELD IN NUMBER OF VIALS / 250,000 L PLASMA | YIELD IN NUMBER OF VIALS / 500,000 L PLASMA
| 1. Human serums Albumin (HSA) - 20% | 3003 | 900 | 225000 | 450000
| 2. Human IVIG - 5% (Average) | 7000 | 1000 | 250000 | 500000
| 3. Clotting factor VIII | 1607 | 50 | 12500 | 25000
| 4. Clotting factor IX | 5000 | 150 | 37500 | 75000
| 5. Fibrin SEALANT | 2500 | 80 | 20000 | 40000
| 6. Fibrinogen | 5000 | 1400 | 375000 | 700000
[SHEET: REVENUE_FORECAST]
REVENUE FORECAST IN CRORE INR FROM PROCESSING
250,000 L / ANNUM (% CAPACITY UTILIZATION) |||||
| YEAR 1
(0%) | YEAR 2
(0%) | YEAR 3
(50%) | YEAR 4 (100%) | YEAR 5 (100%) | YEAR 6 (100%) | YEAR 7 (100%)
| NO PRODUCTION || Revenue – by product (150,000 LIT OF PLASMA / ANNUM) ||||
Human Serum Albumin - 20% | 0 | 0 | 36 | 72 | 72 | 72 | 72
IVIG | 0 | 0 | 82.5 | 165 | 165 | 165 | 165
Factor VIII | 0 | 0 | 1.875 | 3.75 | 3.75 | 3.75 | 3.75
Factor IX | 0 | 0 | 11.625 | 23.25 | 23.25 | 23.25 | 23.25
Fibrin sealant | 0 | 0 | 3 | 6 | 6 | 6 | 6
Fibrinogen | 0 | 0 | 150 | 300 | 300 | 300 | 300
TOTAL / ANNUM | 0 | 0 | 285 | 570 | 570 | 570 | 570
REVENUE FORECAST IN CRORE INR FROM PROCESSING
250,000 L / ANNUM (% CAPACITY UTILIZATION) Gov Tender |||||
| YEAR 1
(0%) | YEAR 2
(0%) | YEAR 3
(50%) | YEAR 4 (100%) | YEAR 5 (100%) | YEAR 6 (100%) | YEAR 7 (100%)
| NO PRODUCTION || Revenue – by product (150,000 LIT OF PLASMA / ANNUM) ||||
Human Serum Albumin - 20% | 0 | 0 | 33.78375 | 67.5675 | 67.5675 | 67.5675 | 67.5675
IVIG | 0 | 0 | 87.5 | 175 | 175 | 175 | 175
Factor VIII | 0 | 0 | 1.004375 | 2.00875 | 2.00875 | 2.00875 | 2.00875
Factor IX | 0 | 0 | 9.375 | 18.75 | 18.75 | 18.75 | 18.75
Fibrin sealant | 0 | 0 | 2.5 | 5 | 5 | 5 | 5
Fibrinogen | 0 | 0 | 93.75 | 187.5 | 187.5 | 187.5 | 187.5
TOTAL / ANNUM | 0 | 0 | 227.91312499999998 | 455.82624999999996 | 455.82624999999996 | 455.82624999999996 | 455.82624999999996
| 455.82624999999996
[SHEET: PROFIT_LOSS]
Projected Profit & Loss Statement@250,000 LIT PER ANNUM |||||
S. No. | Particulars | Year-1 | Year-2 | Year-3 | Year-4 | Year-5 | Year-6 | Year-7
1 | HUMAN SERUM ALBUMIN |||||
| No of HSA vials sold per annum | 0 | 0 | 112500 | 225000 | 225000 | 225000 | 225000
| Selling price of HAS/vial || | 3200 | 3440 | 3440 | 3520 | 3520
| Revenue from HSA | 0 | 0 | 360000000 | 774000000 | 774000000 | 792000000 | 792000000
2 | IVIG |||||
| No of IVIG vials sold per annum | 0 | 0 | 125000 | 250000 | 250000 | 250000 | 250000
| Selling price of IVIG/vial || | 6600 | 7095 | 7095 | 7260 | 7260
| Revenue from IVIG | 0 | 0 | 825000000 | 1773750000 | 1773750000 | 1815000000 | 1815000000
3 | FACTOR VIII |||||
| No of Factor VIII vials sold per annum | 0 | 0 | 6250 | 12500 | 12500 | 12500 | 12500
| Selling price of Factor VIII/vial || | 3000 | 3225 | 3225 | 3300 | 3300
| Revenue from Factor VIII | 0 | 0 | 18750000 | 40312500 | 40312500 | 41250000 | 41250000
4 | FACTOR IX |||||
| No of Factor IX vials sold per annum | 0 | 0 | 18750 | 37500 | 37500 | 37500 | 37500
| Selling price of Factor IX/vial || | 6200 | 6665 | 6665 | 6820 | 6820
| Revenue from Factor IX | 0 | 0 | 116250000 | 249937500 | 249937500 | 255750000 | 255750000
5 | FIBRIN SEALANT |||||
| No of Fibrin sealant vials sold per annum | 0 | 0 | 10000 | 20000 | 20000 | 20000 | 20000
| Selling price of Fibrin sealant/vial || | 3000 | 3225 | 3225 | 3300 | 3300
| Revenue from Fibrin sealant | 0 | 0 | 30000000 | 64500000 | 64500000 | 66000000 | 66000000
6 | FIBRINOGEN |||||
| No of Fibrinogen vials sold per annum | 0 | 0 | 187500 | 375000 | 375000 | 375000 | 375000
| Selling price of Fibrinogen/vial || | 8000 | 8600 | 8600 | 8800 | 8800
| Revenue from Fibrinogen | 0 | 0 | 1500000000 | 3225000000 | 3225000000 | 3300000000 | 3300000000
B | Gross Revenue from Operations INR | 0 | 0 | 2850000000 | 6127500000 | 6127500000 | 6270000000 | 6270000000
| Gross Revenue from Operations in INR cr | 0 | 0 | 285 | 612.75 | 612.75 | 627 | 627
C | Expenses |||||
a | Insurance @0.05% of project value | 0 | 0.8795000000000001 | 0.8795000000000001 | 0.8795000000000001
| 0.8795000000000001 | 0.8795000000000001 | 0.8795000000000001
b | Procurement of Fresh Frozen Plasma in cr INR | 0 | 2 | 16.5 | 36 | 36 | 39 | 39
c | Other Chemicals in cr INR | 0.5 | 2 | 8 | 8 | 17 | 17 | 5
d | Packaging in cr INR | 0 | 0 | 7 | 10 | 10 | 20 | 20
e | Labour/Employee expense in cr INR | 0.5 | 1.2 | 5 | 10 | 11 | 12 | 13
f | Annual Maintenance Cost | 0.25 | 0.25 | 1.5 | 2 | 2.5 | 3 | 3.5
g | Marketing and sales in cr INR | 0 | 4 | 6 | 6 | 6 | 9 | 9
h | Technology transfer fees @ lumpsum 2.5 cr | 0.5 | 1 | 1 | 0 | 0 | 0 | 0
i | Other Exp. In cr INR (includes statutory expenses, registration costs etc) | 1 | 4 | 2 | 2 | 2 | 2 |

2
| Sub-total | 2.75 | 15.3295 | 47.8795 | 74.87950000000001 | 85.37950000000001 | 102.87950000000001 | 92.37950000000001
D | Gross margin = (Revenue-COGS) / revenue % | NA | NA | 83.20017543859649 | 87.7797633618931 | 86.06617707058344 | 83.59178628389155 | 85.26642743221691
E | Breakeven point = fixed cost / gross profit margin | NA | NA | 27.7605341 | 54.77435425000001 | 59.39851815 | 63.4766515 | 60.637903800000004
F | EBIT | -2.75 | -15.3295 | 237.1205 | 537.8705 | 527.3705 | 524.1205 | 534.6205
| Depreciation and Amortization | 17.59 | 17.59 | 17.59 | 17.59 | 17.59 | 17.59 | 17.59
G | EBITDA | -20.34 | -32.9195 | 219.5305 | 520.2805 | 509.7805 | 506.5305 | 517.0305
Projected Profit & Loss Statement@150,000 LIT PER ANNUM Gov. Tender |||||
S. No. | Particulars | Year-1 | Year-2 | Year-3 | Year-4 | Year-5 | Year-6 | Year-7
1 | HUMAN SERUM ALBUMIN |||||
| No of HSA vials sold per annum | 0 | 0 | 112500 | 225000 | 225000 | 225000 | 225000
| Selling price of HAS/vial | | 3003 | 3228.225 | 3228.225 | 3303.3 | 3303.3
| Revenue from HSA | 0 | 0 | 337837500 | 726350625 | 726350625 | 743242500 | 743242500
2 | IVIG |||||
| No of IVIG vials sold per annum | 0 | 0 | 125000 | 250000 | 250000 | 250000 | 150000
| Selling price of IVIG/vial | | 7000 | 7525 | 7525 | 7700 | 7700
| Revenue from IVIG | 0 | 0 | 875000000 | 1881250000 | 1881250000 | 1925000000 | 1155000000
3 | FACTOR VIII |||||
| No of Factor VIII vials sold per annum | 0 | 0 | 6250 | 12500 | 12500 | 7500 | 7500
| Selling price of Factor VIII/vial | | 1607 | 1727.525 | 1727.525 | 1767.7 | 1767.7
| Revenue from Factor VIII | 0 | 0 | 10043750 | 21594062.5 | 21594062.5 | 13257750 | 13257750
4 | FACTOR IX |||||
| No of Factor IX vials sold per annum | 0 | 0 | 18750 | 37500 | 37500 | 22500 | 22500
| Selling price of Factor IX/vial | | 5000 | 5375 | 5375 | 5500 | 5500
| Revenue from Factor IX | 0 | 0 | 93750000 | 201562500 | 201562500 | 123750000 | 123750000
5 | FIBRIN SEALANT |||||
| No of Fibrin sealant vials sold per annum | 0 | 0 | 10000 | 20000 | 20000 | 12000 | 12000
| Selling price of Fibrin sealant/vial | | 2500 | 2687.5 | 2687.5 | 2750 | 2750
| Revenue from Fibrin sealant | 0 | 0 | 25000000 | 53750000 | 53750000 | 33000000 | 33000000
6 | FIBRINOGEN |||||
| No of Fibrinogen vials sold per annum | 0 | 0 | 187500 | 375000 | 375000 | 375000 | 375000
| Selling price of Fibrinogen/vial | | 5000 | 5375 | 5375 | 5500 | 5500
| Revenue from Fibrinogen | 0 | 0 | 937500000 | 2015625000 | 2015625000 | 2062500000 | 2062500000
B | Gross Revenue from Operations INR | 0 | 0 | 2279131250 | 4900132187.5 | 4900132187.5 | 4900750250 | 4130750250
| Gross Revenue from Operations in INR cr | 0 | 0 | 227.913125 | 490.01321875 | 490.01321875 | 490.075025 | 413.075025
C | Expenses |||||
a | Insurance @0.05% of project value | 0 | 0 | 0.8795000000000001 | 0.8795000000000001 | 0.8795000000000001 | 0.8795000000000001 | 0.8795000000000001
b | Procurement of Fresh Frozen Plasma in cr INR | 0 | 2 | 16.5 | 36 | 36 | 39 | 39
c | Other Chemicals in cr INR | 0.5 | 2 | 8 | 8 | 17 | 17 | 5
d | Packaging in cr INR | 0 | 0 | 7 | 10 | 10 | 20 | 20
e | Labour/Employee expense in cr INR | 0.5 | 1.2 | 5 | 10 | 11 | 12 | 13
f | Annual Maintenance Cost | 0.25 | 0.25 | 1.5 | 2 | 2.5 | 3 | 3.5
g | Marketing and sales in cr INR | 0 | 4 | 6 | 6 | 9 | 9 | 9
h | Technology transfer fees @ lumpsum 2.5 cr | 0.5 | 1 | 1 | 0 | 0 | 0 | 0
i | Other Exp. In cr INR (includes statutory expenses, registration costs etc) | 1 | 4 | 2 | 2 | 2 | 2 | 2
2
| Sub-total | 2.75 | 14.45 | 47.8795 | 74.87950000000001 | 85.37950000000001 | 102.87950000000001 | 92.37950000000001
D | Gross margin = (Revenue-COGS) / revenue % | NA | NA | 78.99221468706551 | 84.71888162710916 | 82.57608229063717 | 79.00739789790349 | 77.63614491096381
E | Breakeven point = fixed cost / gross profit margin | NA | NA | 27.7605341 | 54.77435425000001 | 59.39851815 | 63.4766515 | 60.637903800000004
F | EBIT | -2.75 | -14.45 | 180.033625 | 415.13371875 | 387.195525 | 320.695525 | 303.105525
| Depreciation and Amortization | 17.59 | 17.59 | 17.59 | 17.59 | 17.59 | 17.59 | 17.59
G | EBITDA | -20.34 | -32.04 | 162.443625 | 397.54371875000004 | 387.04371875000004 | 369.605525 | 303.105525
[SHEET: NPV IRR]
	COMPUTATION FOR NPV - IRR		
	Capital Expenditure (year 1)	-175.9	
	-20.34	-18.833333333333332	
	-32.9195	-28.223165294924552	
	219.5305	174.27038878727834	
	520.2805	382.42169932786504	
	509.7805	346.9480424754651	
	506.5305	319.2001361899124	
	517.0305	301.68233080757864	
	IRR	0.6520425109216972	1477.4660989598417
	NPV	1323.4660989598417	
	COMPUTATION FOR NPV - IRR		
	Capital Expenditure (year 1)	-175.9	
	-20.34	-18.833333333333332	
	-32.04	-27.469135802469133	
	162.443625	128.95298687319004	
	397.54371875000004	292.2065010729671	
	387.04371875000004	263.41545149870774	
	369.605525	232.91417578318396	
	303.105525	176.8591625883866	
	IRR	0.5394495419942553	1048.045808680633
	NPV	894.045808680633	

YIELD CALCULATIONS FOR PLASMA FRACTIONS

YIELD IN

PRICE REALISED YIELD IN NUMBER YIELD IN NUMBER

NUMBER OF

Product PER VIAL IN INR OF VIALS / OF VIALS / 500,000 L

VIALS / 1,000 L

(2023) 150,000 L PLASMA PLASMA

PLASMA

1. Human serums Albumin (HSA) - 20% 3,200 900 1,35,000 4,50,000

2. Human IVIG - 5% 6,600 1,000 1,50,000 5,00,000

3. Clotting factor VIII 3,000 50 7,500 25,000

4. Clotting factor IX 6,200 150 22,500 75,000

5. Fibrin SEALANT 3,000 80 12,000 40,000

6. Fibrinogen 8,000 1,400 2,10,000 7,00,000

REVENUE FORECAST IN CRORE INR FROM PROCESSING

150,000 L / ANNUM (% CAPACITY UTILIZATION)

YEAR 1 YEAR 2 YEAR 3 YEAR 4 YEAR 5 YEAR 6 YEAR 7

(0%) (0%) (50%) (100%) (100%) (100%) (100%)

NO PRODUCTION Revenue – by product (150,000 LIT OF PLASMA /

Human Serum

- - 21.60 43.20 43.20 43.20 43.20

Albumin - 20%

IVIG - - 49.50 99.00 99.00 99.00 99.00

Factor VIII - - 1.13 2.25 2.25 2.25 2.25

Factor IX - - 6.98 13.95 13.95 13.95 13.95

Fibrin sealant - - 1.80 3.60 3.60 3.60 3.60

Fibrinogen - - 84.00 168.00 168.00 168.00 168.00

TOTAL / ANNUM - - 165.00 330.00 330.00 330.00 330.00

Projected Profit & Loss Statement@150,000 LIT PER ANNUM

S. No. Particulars Year-1 Year-2 Year-3 Year-4 Year-5 Year-6 Year-7

1 HUMAN SERUM ALBUMIN

No of HSA vials sold per annum 0 0 67,500 1,35,000 1,35,000 1,35,000 1,35,000

Selling price of HAS/vial 3,200 3,440 3,440 3,520 3,520

Revenue from HSA 0 0 21,60,00,000 46,44,00,000 46,44,00,000 47,52,00,000 47,52,00,000

2 IVIG

No of IVIG vials sold per annum 0 0 75,000 1,50,000 1,50,000 1,50,000 1,50,000

Selling price of IVIG/vial 6,600 7,095 7,095 7,260 7,260

Revenue from IVIG 0 0 49,50,00,000 1,06,42,50,000 1,06,42,50,000 1,08,90,00,000 1,08,90,00,000

3 FACTOR VIII

No of Factor VIII vials sold per annum 0 0 3,750 7,500 7,500 7,500 7,500

Selling price of Factor VIII/vial 3,000 3,225 3,225 3,300 3,300

Revenue from Factor VIII 0 0 1,12,50,000 2,41,87,500 2,41,87,500 2,47,50,000 2,47,50,000

4 FACTOR IX

No of Factor IX vials sold per annum 0 0 11,250 22,500 22,500 22,500 22,500

Selling price of Factor IX/vial 6,200 6,665 6,665 6,820 6,820

Revenue from Factor IX 0 0 6,97,50,000 14,99,62,500 14,99,62,500 15,34,50,000 15,34,50,000

5 FIBRIN SEALANT

No of Fibrin sealant vials sold per annum 0 0 6,000 12,000 12,000 12,000 12,000

Selling price of Fibrin sealant/vial 3,000 3,225 3,225 3,300 3,300

Revenue from Fibrin sealant 0 0 1,80,00,000 3,87,00,000 3,87,00,000 3,96,00,000 3,96,00,000

6 FIBRINOGEN

No of Fibrinogen vials sold per annum 0 0 1,05,000 2,10,000 2,10,000 2,10,000 2,10,000

Selling price of Fibrinogen/vial 8,000 8,600 8,600 8,800 8,800

Revenue from Fibrinogen 0 0 84,00,00,000 1,80,60,00,000 1,80,60,00,000 1,84,80,00,000 1,84,80,00,000

B Gross Revenue from Operations INR 0 0 1,65,00,00,000 3,54,75,00,000 3,54,75,00,000 3,63,00,00,000

3,63,00,00,000

Gross Revenue from Operations in INR cr 0 0 165.00 354.75 354.75 363.00 363.00

C Expenses

a Insurance @0.05% of project value 0 0.88 0.88 0.88 0.88 0.88 0.88

Procurement of Fresh Frozen Plasma in cr

0 2 17 36 36 39 39

b INR

c Other Chemicals in cr INR 0.5 2 8 8 17 17 5

d Packaging in cr INR 0 0 7 10 10 20 20

e Labour/Employee expense in cr INR 0.5 1.2 5 10 11 12 13

f Annual Maintenance Cost 0.25 0.25 2 2 3 3 4

g Marketing and sales in cr INR 0 4 6 6 6 9 9

Technology transfer fees @ lumpsum 2.5

0.5 1 1 - - -

h cr

Other Exp. In cr INR (includes statutory

1 4 2 2 2 2 2

i expenses, registration costs etc)

Sub-total 2.75 15.33 47.88 74.88 85.38 102.88 92.38

Gross margin = (Revenue-COGS) / NA NA 70.98 78.89 75.93 71.66 74.55

D revenue %

Breakeven point = fixed cost / gross NA NA 27.76 54.77 59.40 63.48 60.64

E profit margin

F EBIT -2.75 -15 117.12 279.87 269.37 260.12 270.62

Depreciation and Amortization 17.59 17.59 17.59 17.59 17.59 17.59 17.59

G EBITDA -20.3 -33 99.53 262.28 251.78 242.53 253.03

COMPUTATION FOR NPV - IRR

Capital

Expenditure -175.90

(year 1)

-20.34 -18.83
-32.92 -28.22
99.53 79.01
262.28 192.78
251.78 171.36
242.53 152.84
253.03 147.64
IRR 40% 696.57
NPV 542.57

[SHEET: yield_REVENUE_FORECAST]
| YIELD CALCULATIONS FOR PLASMA FRACTIONS |||||
| Product | PRICE REALISED PER VIAL IN INR (2023) | YIELD IN NUMBER OF VIALS / 1,000 L PLASMA | YIELD IN
NUMBER OF VIALS / 150,000 L PLASMA | YIELD IN NUMBER OF VIALS / 500,000 L PLASMA ||
1. Human serums Albumin (HSA) - 20%	3200	900	135000	450000	
2. Human IVIG - 5%	6600	1000	150000	500000	
3. Clotting factor VIII	3000	50	7500	25000	
4. Clotting factor IX	6200	150	22500	75000	
5. Fibrin SEALANT	3000	80	12000	40000	
6. Fibrinogen	8000	1400	210000	700000	
REVENUE FORECAST IN CRORE INR FROM PROCESSING					
150,000 L / ANNUM					
	YEAR 1	YEAR 2	YEAR 3		
(50%)	YEAR 4 (100%)	YEAR 5	YEAR 6	YEAR 7	
	NO PRODUCTION		Revenue – by product (150,000 LIT OF PLASMA / ANNUM)		
Human Serum Albumin - 20%	0	0	21.6	43.2	43.2
IVIG	0	0	49.5	99	99
Factor VIII	0	0	1.125	2.25	2.25
Factor IX	0	0	6.975	13.95	13.95
Fibrin sealant	0	0	1.8	3.6	3.6
Fibrinogen	0	0	84	168	168
TOTAL / ANNUM	0	0	165	330	330
[SHEET: NPV_IRR]					
Projected Profit & Loss Statement@150,000 LIT PER ANNUM					
S. No.	Particulars	Year-1	Year-2	Year-3	Year-4
Expenditure (year 1)	-175.9				
1	HUMAN SERUM ALBUMIN				
	No of HSA vials sold per annum	0	0	67500	135000
-28.223165294924552					
	Selling price of HAS/vial		3200	3440	3440
79.01051986485798					
	Revenue from HSA	0	0	216000000	464400000
192.78399730638017					
2	IVIG				
	No of IVIG vials sold per annum	0	0	75000	150000
152.8353546927728					
	Selling price of IVIG/vial		6600	7095	7095
	Revenue from IVIG	0	0	495000000	1064250000
0.4005285957306175	696.5718173348853				
3	FACTOR VIII				
	No of Factor VIII vials sold per annum	0	0	3750	7500
	Selling price of Factor VIII/vial		3000	3225	3225
	Revenue from Factor VIII	0	0	11250000	24187500
4	FACTOR IX				
	No of Factor IX vials sold per annum	0	0	11250	22500
	Selling price of Factor IX/vial		6200	6665	6665
	Revenue from Factor IX	0	0	69750000	149962500
5	FIBRIN SEALANT				
	No of Fibrin sealant vials sold per annum	0	0	6000	12000
	Selling price of Fibrin sealant/vial		3000	3225	3225
	Revenue from Fibrin sealant	0	0	18000000	38700000
6	FIBRINOGEN				
	No of Fibrinogen vials sold per annum	0	0	105000	210000
	Selling price of Fibrinogen/vial		8000	8600	8600
	Revenue from Fibrinogen	0	0	840000000	1806000000
B	Gross Revenue from Operations INR	0	0	1650000000	3547500000
3630000000					
	Gross Revenue from Operations in INR cr	0	0	165	354.75
C	Expenses				
a	Insurance @0.05% of project value	0	0.8795000000000001	0.8795000000000001	
0.8795000000000001	0.8795000000000001	0.8795000000000001	0.8795000000000001		
b	Procurement of Fresh Frozen Plasma in cr INR	0	2	16.5	36
c	Other Chemicals in cr INR	0.5	2	8	8
d	Packaging in cr INR	0	0	7	10
e	Labour/Employee expense in cr INR	0.5	1.2	5	10
f	Annual Maintenance Cost	0.25	0.25	1.5	2
g	Marketing and sales in cr INR	0	4	6	6
h	Technology transfer fees @ lumpsum 2.5 cr	0.5	1	1	0
i	Other Exp. In cr INR (includes statutory expenses, registration costs etc)	1	4	2	2
2					
	Sub-total	2.75	15.3295	47.8795	74.87950000000001
92.37950000000001					
D	Gross margin = (Revenue-COGS) / revenue %	NA	NA	70.98212121212121	78.892318534179
75.93248766737139	71.65853994490358	74.55110192837465			
E	Breakeven point = fixed cost / gross profit margin	NA	NA	27.7605341	54.77435425000001
59.39851815	63.4766515	60.637903800000004			
F	EBIT	-2.75	-15.3295	117.12049999999999	279.8705
	Depreciation and Amortization	17.59	17.59	17.59	17.59
G	EBITDA	-20.34	-32.9195	99.53049999999999	262.2805

[SHEET: Lifenity Cash Flow 16.07.25]				
PLASMA PROJECT CASE FLOW FOR 150000 L PER ANNUM WITH PROJECT COST OF INR 175 CR				
Month	Monthly CAPEX (INR)	Cumulative CAPEX (INR)	1750000000	Percentage of total Items Items
Description				
1	19250000	57750000	3.3	Detailed Engineering, & land prep Filling line Equipment : Vial washing, sterilization, filling sealing and labelling Utilities : Plant Steam and compressed air sytems
2	19250000	38500000		
3	19250000	57750000		
4	48416666.66666666	106166666.6666667	145249999.99999997	8.299999999999999 Start of Procurement of long lead items ETP/STP equipment Clean Utilities : Soft water / PW/ WFI / PSG & distribution systems Fill finish area Equipment: Formulation vessel etc
5	48416666.66666666	154583333.3333333		
6	48416666.66666666	203000000		
7	126923611.1111111	329923611.111111	819875000.0000001	46.850000000000001 Civil works, large equipment procurement Bulk Plasma Production Equipment: Chromatography Systems, Columns, Buffer vessels, Process vessels & table top & clean room equipment QC Equipment
8	126923611.1111111	456847222.2222221		
9	126923611.1111111	583770833.3333333		
10	146368055.5555556	730138888.8888888		
11	146368055.5555556	876506944.4444444		
12	146368055.5555556	1022875000		
13	146368055.5555556	1169243055.555556	501125000	28.635714285714286 Installation, MEP systems, HVAC Plasma Storage Area: Deep freezers & other table top equipment QA / QC , Finished Goods storage area / Fill finish area items and intallations in respective area Canteen / Admin items
14	146368055.5555556	1315611111.111111		
15	69805555.55555555	1385416666.666667		
16	46194444.44444445	1431611111.111111		
17	46194444.44444445	1477805555.555556		
18	46194444.44444445	1524000000		
19	26750000	1550750000	185500000	10.6 Commissioning & Qualification, documentation, small tool/furnishing Plasma Storage Area: Deep freezers 2nd Lot order Commissioning and Qualification of Production equipment
20	41750000	1592500000		
21	41750000	1634250000		
22	41750000	1676000000		
23	16750000	1692750000		
24	16750000	1709500000		
25	16750000	1726250000	40500000	2.3142857142857145 Handover, minor punch list items
26	16750000	1743000000		
27	1750000	1744750000		
28	1750000	1746500000		
29	1750000	1748250000		
30	1750000	1750000000		
GRAND TOTAL 1750000000 100				
Our Schedule				
Extra Months				

[SHEET: Lifenity Cash Flow 16.07.25]				
PLASMA PROJECT CASE FLOW FOR 150000 L PER ANNUM WITH PROJECT COST OF INR 175 CR				
Month	Monthly CAPEX (INR)	Cumulative CAPEX (INR)	Section wise total	Percentage of total Items
1	19250000	19250000	57750000	3.3 Detailed Engineering, & land preparation
2	19250000	38500000		
3	19250000	57750000		
4	48416600	106166600	145249800	8.299988571428571 Start of Procurement of long lead items.
Clean Utilities : Soft water / PW/ WFI / PSG & distribution systems. Fill finish area Equipment:				
Formulation vessel etc				
5	48416600	154583200		
6	48416600	202999800		
7	126923600	329923400	819874800	46.84998857142857 Civil works. Large equipment procurement.
Bulk Plasma Production Equipment: Chromatography Systems, Columns, Buffer vessels, Process vessels & table top & clean room equipment. ETP STP equipment				
8	126923600	456847000		
9	126923600	583770600		
10	146368000	730138600		
11	146368000	876506600		
12	146368000	1022874600		
13	153118000	1175992600	541624700	30.949982857142857 Installation, MEP systems, HVAC. Plasma Storage Area: Deep freezers & other table top equipment. QA / QC , Finished Goods storage area / Fill finish area items and intallations in respective area. Canteen / Admin items. Plasma Storage Area: Deep freezers
14	153118000	1329110600		
15	76555500	1405666100		
16	52944400	1458610500		
17	52944400	1511554900		
18	52944400	1564499300		
19	26750000	1591249300	185500700	10.60004 Commissioning & Qualification, documentation, small tool/furnishing. Insurance, EPF/ ESI, Labour cess etc., Handover.
20	41750000	1632999300		
21	41750000	1674749300		
22	41750000	1716499300		
23	16750700	1733250000		
24	16750000	1750000000		
TOTAL	1750000000		100	

Techno-Commercial Feasibility cum Detailed Project Report for setting up a Plasma Fractionation Facility

150,000 Liter / Annum Capacity

Document number:

LWIL/F-DPR/2023-00 Dated.15.12.2023

Prepared for:

Lifenity Wellness International Ltd,

12th Floor, D-Wing, Trade World, Kamala Mills Compound, Lower Parel, Mumbai -400 013

Contact: Mr. T. Rajshekhar

Prepared by:

BioBridge Healthcare Solutions Pvt. Ltd,

#13, Rachana Blossom, Jagdishnagar, Aundh, Pune 411007

www.biobridge.in

Contact: Dr. Ganesh Kumraj Email: ganesh@biobridge.in Mob: +91 9004349016

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LIST OF ABBREVIATIONS:

NA : Not Applicable

NACO : National AIDS Control Organisation

NAT : Nucleic acid Amplification testing

NBTC : National Blood Transfusion Council

NGO : Non-Governmental Organization

NIB : National Institute of Biologicals

NPPA : National Pharmaceutical Pricing Authority

OOS : Out of Specification

OOT : Out of Trend

PCC : Prothrombin complex concentrate

PDMP : Plasma Derived Medicinal Products

PFD : Process Flow Diagram

PPQ : Process Performance Qualification

QA : Quality Assurance

QC : Quality Control

RM : Raw Material

SAT : Site Acceptance Test

SEC : Size exclusion chromatography

SBTC : State Blood Transfusion Councils

SOP : Standard Operating Procedure

SS : Stainless Steel

STD : Sexually Transmitted Disease

STP : Sewage Treatment Plant

TFF : Tangential Flow Filtration

TMP : Trans-Membrane Pressure

UPS Uninterrupted Power Supply

URS : User Requirement Specification

USD : United States Dollar

WHO : World Health Organization

CHAPTER 1 INTRODUCTION

Project Objective Statement:

The objective of this project is to establish and operate a state-of-the-art Human Plasma Derived Medicinal Products (HPDMP) manufacturing facility with an annual production capacity of 150,000 liters.

This capacity will be subsequently increased to 500,000 liters per annum in the near future.

This facility aims to consistently and efficiently produce high-quality plasma derived products such as Human Serum Albumin (HSA), Human Normal Immunoglobulin (IgG), Hemophilia factors VIII, IX, Fibrin sealant etc. and Monoclonal antibodies (Mabs) while ensuring compliance with regulatory standards and best industry practices.

The primary goals of this project include:

Meeting Growing Demand: To address the increasing demand for plasma-derived products, ensuring a stable and reliable supply of essential treatments.

Quality Assurance: Implementing rigorous quality control and quality assurance processes to consistently produce safe and effective plasma derived products that meet or exceed regulatory requirements.

Operational Efficiency: Optimizing manufacturing processes to maximize productivity, minimize waste, and reduce operational costs to maintain competitiveness in the market. **Regulatory Compliance:** Adhering to all relevant regulations and standards, including current Good Manufacturing Practices (cGMP), to ensure product safety, quality and efficacy.

Safety and Sustainability: Prioritizing the safety of both employees and the environment by implementing robust safety measures and eco-friendly practices throughout the facility.

Research and Development: Supporting ongoing research and development efforts to improve product quality, expand the product portfolio, and explore innovative techniques and newer technologies.

Supply Chain Reliability: Building a secure and efficient supply chain to source raw materials, transport finished products, and ensure timely deliveries to end users.

Stakeholder Satisfaction: Ensuring customer satisfaction by providing consistent and reliable plasma-derived products while maintaining a favorable relationship with employees, investors, and the local community.

By achieving these objectives, this Human Plasma Derived Medicinal Products manufacturing facility will play a pivotal role in meeting the critical healthcare needs of patients while contributing to advancements in the plasma industry.

Statutory Regulation

(A detailed list of provided in the References section at the end of the document) In general, the following guidelines are referred to:

National Blood Policy, NBTC, Ministry of Health and Family Welfare.

National Policy for Access to Plasma Derived Medicinal Products from Human Plasma for

Clinical/Therapeutic Use: Addendum to National Blood Policy, NBTC, Ministry of Health and Family Welfare.

Technical manual, DGHS, MOHFW Govt. of India 2003.

Drugs and Cosmetic act 1940 and rules thereunder.

Schedule M Drugs and Cosmetic act 1940.

ISPE Baseline Guide Vol. 6, Biopharmaceutical Manufacturing Facilities ISPE Baseline Guide Vol. 3, Sterile Manufacturing Facilities.

WHO Good manufacturing practices for biological products.

Good manufacturing practice guideline for human and veterinary medicinal product.

PIC/s Guide to Good Manufacturing Practice for Medicinal Products.

USFDA Guidance for Industry – Sterile Drug Products produced by Aseptic Processing.

Executive Summary of The Project:

The Human Plasma Derived Medicinal Products manufacturing facility will be a green field project. The area earmarked for the project is approx. 6 acres in an industrial area on the outskirts of Nagpur city. Different areas for the plasma manufacturing facility are identified and listed below:

Bulk Manufacturing Facility (for processing 150,000 L / annum)

Formulation, Fill Finish with lyophilization & Packaging

QC and R&D block

Administrative and QA Block

Ware House for Plasma, Packaging, Testing lab (for 32,000 L plasma)

Ware House (Finished Goods Storage)

Expansion area

Bulk Manufacturing Facility (for processing additional 350,000 L / annum)

Ware House for Plasma, Packaging, Testing lab (for additional 32,000 L plasma)

Supporting Blocks

Canteen

Engineering Office

Boiler House

Work Shop

Substation and Generator

Bore Well

Bio-Kill System

WTP/ETP/STP System

Fire Fighting System

Scrap Yard

Guard House and Reception Room

Security Cabin/Post

Boundary Wall and Parking etc.

Internal Roads and Drainage System

Rainwater Harvesting System

PROPOSED SITE PLAN

Annexure A

Figure 1: Plot Layout GROUND FLOOR

Annexure B

Figure 2: Conceptual Layout – GF

The company has identified different products to be manufactured in stagewise manner. List of products proposed to be manufactured are as follows:

Stage-1:

Albumin

Human Immunoglobulin

Factor VIII

Factor IX

Fibrin sealant

Fibrinogen

Stage-2:

Anti-D

Factor XI

Factor XIII

Hepatitis B IgG

Measles IgG

Protein C

Rabies IgG

Tetanus IgG

Thrombin

Varicella Zoster IgG

Table 1: Working Time Model Bulk Manufacturing

Table 2: Working Time Model Fill Finish

Environment Due Diligence:

As the proposed facility is required to handle biological products, proper segregation of type of wastes generated shall be adopted. Based on the categorization of the waste, appropriate treatment philosophy shall be adopted. Following broad categories are identified as wastes:

Process waste

General wash

Solid Process waste

General Solid wastes – Packaging materials etc.,

Gaseous waste from Process & Labs

Sewage

The facility shall be designed as zero waste discharge plant; meaning that every possible waste within the facility is categorized, treated with appropriate treatment step in line local pollution control board guidelines and then discharged safely. Soil contamination due to waste is also considered as a key factor and necessary steps & waste handling methods are adopted to avoid the same. As part of the design, all biologically waste will be treated in Primary treatment systems before letting it into a common wastewater treatment plant.

For the facility design, a detailed EHS document shall be prepared to identify risks and mitigation plans. During design, Construction & operation, EHS policy shall be made integral part of the sustainable operation.

CHAPTER 2 MARKET ANALYSIS

Human Plasma Derived Products and Its Share in The Total Components:

There are several medicinal products which are derived from human blood plasma by Plasma fractionation.

The process for large scale purification was developed about 60 years ago by Cohn and co-workers.

Therapeutically important proteins are Human serum albumin (HSA), Immunoglobulin G (IgG), Coagulation factor VIII, factor IX and Protease inhibitors.

HSA is one of the most abundant proteins in plasma and, together with IgGs, constitutes 80% of total plasma proteins. According to the data, HSA represented approximately 50% of total sales of therapeutic plasma protein products. The purified HSA is required for the treatment of hypo proteinemia, burn and heavy loss of blood due to injuries and other medical conditions. Plasma contains about 60 g/L of protein, out of which about 57 grams are used for many therapeutic products. This makes this human material unique as a source of multiple therapeutic products.

The concentration of different plasma proteins is variable. HSA is major protein component, followed by IgG. The wide concentration range of plasma proteins (less than 1 mg/mL to 40 g/L) is one of the potential challenges in designing separation and purification procedures.

Volume of Human Plasma Fractionated Worldwide (2022 to 2032):

In 2010, 78 fractionation plants were operating in the world, 25 of which were in China, 26 in Europe and

8 in the United States of America. The world plasma fractionation capacity was about 48.4 million liters. More than 500 tons of human serum albumin (HSA) and more than 40 tons of intravenous immunoglobulins (IVIg) are produced annually from more than 34 million liters of plasma. Total plasma marked of 2008 is depicted in pie chart (Figure 2).

Figure 3: Total plasma market of 2008 (Source: Marketing Research Bureau, Inc)

In 2015-16, the worldwide plasma market grew to a \$21,174 Million, almost doubled in terms of revenue from 2008. As compared to the 2008 statistics, in 2016 to IVIg and SCIG had a major share in plasma market, increasing from 29% to 47%. The market share for Albumin also grew from 9.7% to 15.7%. The market share for Factor VIII and Factor IX remained almost the same of 1 to 2 % from 2008 to 2016.

Figure 4: Plasma market in 2016 (Source: Marketing Research Bureau, Inc) Global Plasma Fractionation Market Forecasts from 2022 to 2032

The Global Plasma Fractionation Market Size was valued at USD 27.1 Billion in 2022 (Fig 3).

The Market Size is growing at a CAGR of 6.7% from 2022 to 2032.

The Worldwide Plasma Fractionation Market Size is expected to reach USD 51.8 Billion by 2032.

Asia Pacific region is expected to grow the fastest during the forecasted period.

Figure 5: Global Fractionation Market

Companies in Domestic and International markets: Key Market Players (Indian Manufacturers)

Plasma Gen Biosciences Pvt. Ltd. Bangalore

Reliance Life Sciences, Mumbai

Biocon Limited Bangalore

Intas Pharmaceuticals Ltd. Ahmedabad

Virchow Biotech, Hyderabad

Fusion Healthcare, Hyderabad

ICHOR Biologics Pvt. Ltd., Hyderabad

Bharat Serums and Vaccines Limited, Mumbai (Sales and marketing)

Table 3: Domestic companies' Plasma products

Key market players (International Manufacturers):

CSL Behring, USA

Bio plasma limited, USA

Pfizer, USA

Kedrion Biopharma, USA

ADMA Biologics, USA

Takeda Pharmaceutical Company Limited

Biotech AG, Germany

Octa pharma, Switzerland

Grifols, Spain

Kamada Pharmaceuticals, Israel

SK Plasma, South Korea

Table 4: International companies' Plasma products

Global plasma Market Overview:

Table 5: Plasma Derived Therapy Market Share, By Region, 2022 *

Source: Precedence Research, Plasma derived market size, A report 2032

Future roadmap for plasma fractionation industry in India:

Global plasma fractionation program, plasma protein therapies & newer clinical indications for use of these products, has led to a rapidly increasing demand.

Due to the limited volumes of plasma available for fractionation and fractionation capacities, has led to various challenges in availability of these products.

Unfortunately, plasma fractionation program did not get enough attention, focus and motivation in India because of challenges like availability of plasma, less components processing units, cold chain infrastructure, political will and availability of solid plasma ecosystem.

As per literature information, in 2020 India, wasted 6.5 Lakh units of blood and blood components on annual basis (Source: Print line Media Pvt Ltd). A report published by Ministry of Health and FW (2018-2019); the total collection of blood was 12.2 million units out of which 75.05% of blood was componentized by NACO supported blood component separating unit (BCSU). Thus, the wastage of 24.95% hampered the availability of blood components to the patients. Though, with the passage of time there has been a significant improvement in the overall blood banking systems and availability of plasma products in India, still much needs to be done considering clinical needs of the country.

There is a need to create a vision document for short-term, mid-term and long-term self-sufficiency of plasma products in India. This can be divided into following two categories:

Availability of safe plasma: How to overall increase the quantity and quality of plasma for fractionation:

Increase voluntary blood collection, with increased repeat donor base.

As a part of the licensing process blood component separation should be mandatory and a plan for current licensed blood banks to migrate to complete blood separation units in the next 3 years.

Making blood component therapy mandatory in clinical practice, so that blood therapy can be minimized, and patients can use plasma products, which are relatively safer and easily accessible.

Create a solid regulatory pathway for permitting source plasma collection.

Regulatory system reform: Create a facilitative regulatory landscape for plasma industry to grow and evolve which would involve office of DCGI, NIB, Department of AIDS Control (DAC), IPC, NPPA, NACO, clinicians, patient groups, industry representatives and NGOs by;

Forming a national level body for steering plasma protein initiative in India.

Making a robust regulatory system for collection.

Controlling pricing for plasma and increasing indigenous manufacturing capabilities.

More funding for products and patient care.

Fast track regulatory approval for importing different categories of plasma/fractions/intermediate.

Market Trend for The Next 10 Years and Shift in Products (2023-2032):

With the growth of IgG sales, some relatively niche plasma proteins have had a significant impact on the market in the 2010s.

C1 esterase inhibitor (C1-INH): In 2010, the sales of C1 esterase inhibitor (C1-INH) for the treatment of hereditary angioedema (HAE) amounted to about \$200 million globally. This was 1.5% of total plasma proteins sales because the product had just been launched in late 2008 in the United States, and European sales were limited by the product's approval confined to only a few countries. By 2016, the sales reached over \$1.1 billion (5.2% of global PDMPs sales), as the product gained rapid acceptance mainly in the United States where the awareness of treatment and diagnosis of HAE were boosted by Shire and CSL Behring. As a result, C1-INH became the fourth largest plasma protein in terms of dollar sales worldwide, behind IgG, albumin and plasma-derived factor VIII. The market for C1-INH is continuing to grow strongly, and it is likely to increase its share in the future. A new formulation for subcutaneous injection will further facilitate product adoption. In addition, C1-INH is under investigation for transplant where antibody mediated rejection (AMR) has occurred.

Alpha-1 Antitrypsin (AAT): Other specialty products which continue to grow at a stronger pace is prothrombin complex concentrate (PCC) as well as alpha-1 antitrypsin (AAT). Global sales of AAT reached \$1.0 billion in 2016, or 4.7% of the total market. This is up slightly from 4.1% of sales in 2010, as the

market in the United States has sustained approximately a low two-digit annual sales growth for the past 6 years, driven mostly by new patient identification and higher dosing. Most of the global AAT sales are generated in the United States, though sales of this therapy in Europe are trending upward after approval through the central European Medicines Agency (EMA) in August 2015. AAT is also under investigation for new indications, such as graft vs. host disease after hematopoietic stem cell transplantation (HSCT) as well as type 1 diabetes. If either of these areas show clinical trial success, the market would be expected to experience higher demand for the therapy.

Prothrombin complex concentrate (PCC): PCC have nearly reached the \$500 million in sales mark in 2016, up from under \$300 million in 2010. This impressive growth was due to the introduction of the first four-factor (factors II, VII, IX and X) coagulation product in the United States by CSL Behring in 2013 for the urgent reversal of vitamin K antagonist (VKA) anticoagulation in adults with acute major bleeding. Sales of this product reached close to \$200 million within four years of product launch. Sales also increased in Europe for the same indication, while at present little PCC is used in other markets around the world.

In the United States, the market share of the plasma-derived factor VIII sales fell from 11% in 2010 to 9% in 2016, while the share of plasma-derived factor IX products fell from 20% to 6%. The latter's market share drop was due to the faster adoption of the extended half-life factor IX products than those of factor VIII, because they offer longer intervals between infusions than the standard recombinant and plasma-derived products.

Globally, the plasma-derived factor VIII market share went from 27% in 2010 to 21% in 2016. For, the plasma-derived factor IX' market share fell from 34% in 2010 to 21% in 2016. According to NPPA report of 2022, government has set the pricing for scheduled drugs and plasma products.

Government Market and the Rates:

A market survey and several documents and reports were studied to see the price of various plasma products in government and private sector respectively (Table 6 and Table 7).

Table 6: Pricing of Plasma Products (Government of India)

@Source: NPPA (National Pricing Pharmaceutical Authority 2022)

Figure 6: Stage 1 - Plasma products price

Figure 7: Stage 2 - Plasma products price (Government)

Table 7: Price for Plasma Products (Private Manufacturers)

Figure 8: Prices for Plasma Products from Private Manufacturers

CHAPTER 3 TECHNOLOGY

Analysis of Various Technologies Available (Centrifugation and filtration for solid / Liquid Separation, Thermal and Chemical Inactivation Procedure, Ultrafiltration / Diafiltration, Nanofiltration, Chromatography or Any Other Technology or Any Hybrid Model)

Several technologies are utilized for plasma purification at commercial scale, each has its advantages, disadvantages and applications in isolating specific proteins of human interest. Some of the key methods are as under:

Methods for Plasma purification

Plasma Fractionation method with ethyl alcohol

Cohn+ Liquid Chromatography

Heat Shock technology

Ammonium sulphate+ Liquid Chromatography

Ion Exchange Chromatography (IEC)

Plasma Fractionation Method with ethyl alcohol:

Cohn and colleagues developed a protein purification method by using plasma fractionation technique for the first time. This approach is based on albumin solubility difference with other plasma proteins. As compared to other plasma proteins, albumin sediments are at lower pH and higher ethanol concentration. In this method there are five important parameters for separation of plasma protein.

Ethyl alcohol (8- 40%)

Temperature (-3 to -10 °C)

pH (4.5-7.2)

Protein concentration (5.1-8 %)

Ionic strength (0.14- 0.01).

Protein separation is based on the increase in ethyl alcohol concentration thus albumin will be seen in the last fraction.

The method is suitable for large-scale and industrial production.

Disadvantages of Method:

Purity of the product

Denaturation of protein

The Cohn method combined with liquid chromatography:

The original Cohn method was modified by Oncley et al. in 1949 and Kistler-Nitschmann et al. in 1962 in order to improve albumin purity.

The method is used for the purification of plasma proteins at the industrial scale

In this method after ethanol precipitation liquid chromatography step is added. Albumin is a by- product and other important therapeutic proteins including factor VIII, intravenous immunoglobulin (IVIG) and some other proteins are isolated from plasma.

Advantages:

Increased purity of proteins

Higher yields

Process Flow Diagram -Modified Cohn Method

Cohn Modified

Supernatant A

Heat shock method:

Albumin is stable during thermal fluctuation comparing to the other proteins of the plasma; therefore, it can be purified by heat shock purification method.

Albumin protein resists to increasing temperature up to 60°C at which potential pathogen are inactivated.

After inactivation 0.04 M caprylic acid is added to stabilize the protein at pH 5 at 60°C.

At this stage other proteins of the serum are denatured and precipitated in solution. The final product (albumin) is concentrated by precipitation and ultrafiltration with purity of app 98%

Ammonium Sulphate precipitation combined with liquid chromatography:

Another method for albumin purification is ammonium sulphate precipitation combined with liquid chromatography which yields approximately more than 90 percent purified albumin. In this method albumin is separated from immunoglobulins by 50% ammonium sulphate. Then lipids are removed using ice- cold acetone at 2-8°C.

For further purification of albumin from other contaminating proteins (especially transferrin which is the main contaminant, DNases and immunoglobulins), ion exchange and size exclusion chromatography are used.

Ion exchange chromatography (IEC):

IEC is widely used for albumin and other protein purification. Among them the anion exchange method has the most widely used. Three major steps of anion exchange consisting pre- treatment, purification and

polishing.

The purification consists of two anion and cation exchange chromatography followed by polishing step comprising of two ultrafiltration and one gel filtration steps.

Process Flow Diagram Purification and Polishing Methods for Human Serum Albumin (HSA)

The limitations of method:

Long processing time

To overcome these limitations, the new method for ion exchange membrane chromatography (IEM) using ameliorated adsorbers was developed.

The membrane chromatography method is simple and has high separation efficiencies due to shorter diffusion times, reduction in buffer usage, small foot print requirements. No complex hardware or packing is required.

Disadvantage:

Expensive membrane

Low efficiency

Table No 8: Comparison of Different Methods for Industrial Application

Recommending a Cost Effective and High Productivity Technology:

On the basis of table 8, for purity analysis and percentage recovery there are individual methods like fractionation and combined methods like Cohn + LC and IEC which can be considered.

Lifinity should accept the method which give high productivity and acceptable quality of desired protein (purity levels) as per the regulatory expectations.

Cohn method when combined with chromatographic techniques is the most well-known method. In this method other proteins of plasma such as coagulation factors, immunoglobulins etc. would be separated in upstream, and albumin will be separated in the last fraction. Due to the robustness and high purity of plasma in large volumes, this technique, is cost- effective for plasma therapeutic protein production. Hence, this method currently seems to be a major method for albumin production.

However, taking into consideration, the latest developments in plasma fractionation technologies the detailed method along with the process flow diagram is recommended in section 4 of this document.

Obligations for technology service provider:

The selected Technology service provider shall have the following obligations:

Technology Transfer Plan:

To provide a detailed plan outlining the steps and timeline for the technology transfer. This plan shall include milestones, responsibilities, and contingencies. There should be a Lifinity steering committee to monitor the progress. All the deviations/changes should be approved by the committee.

Documentation:

To provide detailed documentation that describes the technology (Methods), equipment URS, buffer compositions, process steps, product specifications, and relevant data etc., This documentation is crucial for Lifinity to understand and replicate the technology.

Checklist of the details to be provided to Lifinity by the technology service provider:

Manufacturing:

RMs PPM and SPM and excipient specifications and source.

Dispensing instructions.

Detailed process flow diagrams.

Process validation requirements.

Batch Manufacturing record (BMR), SOPs, Master formulas.

List of SOPs for operation of equipment and other relevant procedures.

List of specifications and In-process controls (IPC).

Process flow diagrams (PFD) along with description of chromatography and other critical process steps.

Details of Equipment (Capacities, designs, qualification and placement with utilities requirements.

Manufacturing implementation sequence/model with process time for each unit operation.

Control strategy based on PPQ lots defining CPPs, CQAs and IPCs.

Yield calculation.

Cleaning procedures and validation.

Quality Control and Assurance:

Requirements of different areas for QC activities.

List of general and product related SOPs and STPs with specifications.

Raw materials and Active starting materials control and release strategies.

List of documentation for QC and QA activities.

Procedures for batch release, recall and complaints handling by quality assurance department.

Details of ware housing activities (Receipts of incoming RMS and active materials, sampling, Quarantine, testing and, release activities.

Cold chain requirements in ware house storage areas for active materials and finished products.

QC tests, STPs and specifications.

Validation of analytical and microbiology techniques.

Stability protocols as per regulatory expectations.

List of SOPs for operation of equipment and other relevant procedures.

List of OOS, OOT and any other quality related events/incidents relevant to Lifinity.

Details of primary packaging and secondary packaging materials with sources.

EHS:

SOPs and STPs related to EHS

Training:

It is the responsibility of technology transfer partner to provide on job training and qualification of the staff for each unit operation. Lifinity is responsible to provide necessary resources for trainings.

Training will comprise of knowledge transfer sessions, hands-on training, and exchange of personnel.

Lab Scale Implementation:

Lifinity shall initiate the technology transfer at lab scale to test its feasibility and performance in a real-world setting before full-scale deployment. The technology service provider shall provide the lab scale technology.

Regulatory Compliance:

Technology provider to ensure that the provided technology and its implementation comply with relevant regulatory requirements and support necessary approvals from regulatory authorities.

Scale-Up and Commercialization:

Technology provider shall provide support for Scale up of the technology for commercial production, at least three batches of each product and deployment. This shall involve optimizing manufacturing processes, securing additional resources, and launching the manufactured product with the given technology into the market.

Monitoring and Evaluation:

Provide continuous monitoring support at the production floor to Lifinity scientists for the performance of the transferred technology and conducting evaluations to assess its impact, identify areas of improvement and address any technology related issues that may arise.

Post-Transfer Support:

Providing ongoing support to address any challenges, optimize processes, and explore further opportunities.

CHAPTER 4 MANUFACTURING

As per the project implementation model, the following products shall be manufactured from 150,000L/Year of plasma.

Albumin

Factor VIII

Factor IX

Human Immunoglobulin (IgG)

Fibrin sealant

Fibrinogen

Manufacturing Process for Deriving the Above Suggested Products Derived from Plasma

Process Flow Diagram- Common Fractionation Steps

PFD- 1

PROCEDURE FOR COMMON FRACTIONATION STEPS:

P.1.F CUTTING AND POOLING

Approved plasma bags issued by QA are collected and thawed in a thawing tank with suitable provision for placement, retrieval and subsequent removal of thawing medium from the tank. The bags are checked for leakages after thawing and discarded if leakage is found. The thawed bags are cut under a bench type laminar flow and plasma collected in to suitable containers and finally pooled in process vessel to be taken for the next step.

P.2.F FILTRATION

Collected plasma after thawing is filtered to remove any suspended particles which may clog chromatography resins and column bed supports. Filtration is done using a depth filter which doesn't bind any proteins of interest. Filtered plasma is collected in a suitable tank with provisions to connect to a chromatography system for following process steps.

P.3.F CHROMATOGRAPHY – 1F

Filtered plasma is subjected to chromatography 1F to separate coagulation factor– F-VIII from rest of the fractions of interest. This step is essential to both separate F-VIII and also further polish plasma for other chromatography steps. Fraction of interest is collected using patterns on chromatograms, can be programmed for automatic collection.

P.4.F LOAD PREPARATION FOR 2F

pH of the collected eluate from Chromo – 1F, is adjusted to 7.0 and loaded on the next chromatography column.

P.5.F CHROMATOGRAPHY – 2F

pH adjusted eluate from Step 4 is subjected to chromatography -2F to separate F -IX from the remaining fractions. F-IX is collected separately for further process and remaining fractions are further processed.

P.6.F TANGENTIAL FLOW FILTRATION (TFF) FOR CONCENTRATION

F-IX depleted sample is collected and concentrated by TFF. Trans-membrane pressures (TMP) are adjusted to achieve maximum permeate flow while minimizing fouling of the membrane. Automated TFF skids are used to precisely control and maintain the TMP.

P.7.F CHROMATOGRAPHY – 3F

Concentrated solution from Step-6 is subjected to chromatography-3F to separate buffer components from earlier steps and the sample is condition for further process steps. The main purpose of this step is to maintain suitable conditions for removal of other abundant protein impurities which may interfere with polishing chromatography steps of individual fractions.

P.8.F IMPURITY REMOVAL

pH of the eluate from Step-7 is adjusted to precipitate impurities from products of interest.

P.9.F EUGLOBULIN PRECIPITATION

Impurities like Euglobulin are precipitated using precise temperature and pH conditions. Temperature is maintained at less than 10 °C and pH less than 6.

P.10.F PRECIPITATE SEPERATION USING CENTRIFUGATION

The sample if passed through a continuous centrifuge to separate the precipitate. Precipitates are separated as a slurry and discarded. Supernatant is used for the subsequent steps.

Process Flow Diagram IVIG

IVIG PURIFICATION PROCESS:

P.11.F LOAD PREPARATION FOR 4F

Conductivity of supernatant from Step # 10 is adjusted to facilitate favorable conditions for separation of IVIG and Albumin.

P.12.I CHROMATOGRAPHY – 4F

Conductivity adjusted sample is subjected to Chromatography 4F to separate Albumin from IVIG, this is the most critical chromatography step. Unlike in Cohn's process this gives a single step separation. The chromatography step is also operated in both flow through and elution modes to achieve the required separation. Albumin and IVIG are separated as different fractions after this step and collected in different process vessels. Individual fractions are processed separately to further purify and polished suitable for filling in glass vials.

P.13.I TANGENTIAL FLOW FILTRATION (TFF) FOR CONCENTRATION

IVIG fraction is collected and concentrated to about batch volume using TFF. Trans- membrane pressures (TMP) are adjusted to achieve maximum permeate flow while minimizing fouling of the membrane. Automated TFF skids can be used to precisely control and maintain the TMP.

P.14.I pH AND CONDUCTIVITY ADJUSTMENT

Both pH and conductivity of concentrated solution from step 13 is adjusted using NaOH and Acetic acid.

P.15.I FILTRATION

pH and conductivity adjusted, concentrated IVIG fraction is filtered using 1.2-micron filter to clarify, suitable for loading on next chromatography step.

P.16.I CHROMATOGRAPHY - 5I

Filtered IVIG sample is subjected to Chromatography 5I to further purify IVIG from other protein impurities. Impurities and IVIG are collected as separate fractions, impurities are discarded and IVIG is further processed.

P.17.I pH ADJUSTMENT

pH of the fraction collected from step 16 is adjusted with 1M Acetic acid for following viral inactivation step. This is ideally done in the same vessel in which the fraction is collected.

P.18.I CONDUCTIVITY ADJUSTMENT

Conductivity of collected fraction is adjusted. This is ideally done in the same vessel in which the fraction is collected and pH adjusted.

P.19.I LOW pH TREATMENT FOR VIRAL INACTIVATION

pH of the collected fraction is adjusted to 4.0 with 1M Acetic acid. This is held at pH 4.0 at 37 °C for 10 hours for viral inactivation. This step is most critical with regulatory and quality prospective.

P.20.I FILTRATION

Viral inactivated IVIG fraction is filtered using 1.2-micron filter to clarify, suitable for loading on next chromatography step.

P.21.I SOLVENT DETERGENT TREATMENT FOR VIRAL INACTIVATION - VERY CRITICAL STEP FOR REGULATORS

The solvent/detergent (S/D) pathogen inactivation method was developed for the inactivation of enveloped

viruses in plasma-derived medicinal products. Currently, S/D treatment for pathogen reduction of plasma for transfusion and plasma-derived medicinal products is the most common, validated and robust industrial method for the virucidal treatment of blood products. Although many millions of therapeutic doses of S/D inactivated products have been used to date, there is no report on transmission of viral infections caused by enveloped viruses in recipients. The safety of the S/D treatment is also shown by the absence of reports of toxicity or antibodies against neoantigens after large-scale exposure of patients to several million transfused units.

1% TNBP and 1% Triton X are added calculated as final volume v/v. TNBP and Triton are prepared in a buffer before addition to the bulk of fraction. Stock solutions are prepared.

P.22.I CHROMATOGRAPHY - 6I – FINAL CHROMATOGRAPHY STEP IN IVIG PROCESS

Filtered IVIG sample is subjected to Chromatography 6I to effectively remove added solvents and detergents in P.21.1 This also works for removal of endotoxins in the solution. IVIG bound to resin at an Acidic pH is selectively eluted using elution buffers at pH 9.0 in a process called as pH gradient elution.

P.23.I pH ADJUSTMENT

pH of the eluate from Chromatography 6I step is adjusted to preserve structural stability of IVIG and provide more stable conditions.

P.24.I NANO FILTRATION – (OPTIONAL STEP)

Nano filtration further ascertains barrier mode removal of viral particles from IVIG solution. Plano make filters are used in the process.

P.25.I TANGENTIAL FLOW FILTRATION (TFF) FOR CONCENTRATION AND DIA FILTRATION IVIG fraction is concentrated using TFF about 5% IVIG w/v concentration followed by buffer exchange by dia-filtration at the same TFF running conditions.

P.26.I TANGENTIAL FLOW FILTRATION (TFF) FOR CONCENTRATION

Dia-Filtered IVIG is concentrated to 6% w/v to facilitate final formulation at 5% w/v using the same TFF conditions. TFF conditions are maintained as in previous steps.

P.27.I FORMULATION

Buffered IVIG solution is formulated by adding 10% w/v pharmacopeial grade Mannitol. pH of this final solution is maintained at 4.2.

P.28.I PRE-FINAL FILTRATION

Filtration is done to decrease turbidity of the solution and also decrease microbial load on final sterile filtration step. This filtration is done using sandwich filter rated 1.2/0.6 micron.

P.29.I STERILE FILTRATION

Sterile filtration is done to further reduce bio burden and the process is not claimed a sterile operation. Final terminal filtration is performed before fill-finish stage.

P.30.I FILLING IN TO VIALS

Washed, sterile and pyrogen free vials are used to fill bulk solution of IVIG.

Process Flow Diagram Albumin Purification:

ALBUMIN PURIFICATION PROCESS:

P.31.A CHROMATOGRAPHY – 7A

Albumin fraction from chromatography step 4F is used to load on Chromatography 7A and separation is achieved by binding and elution mode. This step is final chromatography step in Albumin process.

P.32.A TANGENTIAL FLOW FILTRATION (TFF) FOR CONCENTRATION

Eluate collected from Chromo-7A is concentrated to 6-7% w/v and collected in a vessel suitable for next process step.

P.33.A HEAT TREATMENT

Concentrated albumin solution is collected in a process vessel having control for temperature and time (57 degree C/ 3 hours). End point of completion of heat treatment is taken as stable optical density@ 600 nm. Heating of material results in a slightly turbid solution with suspended impurities. This solution is cooled to room temperature or even up to 10 °C for better separation. Chromatography is performed at room temperature.

P.34.A CENTRIFUGATION AND FILTRATION

Albumin solution after heat treatment is centrifuged and filtered to remove suspended proteinaceous contaminant particles formed during heat treatment. Continuous centrifuge can be used for this step.

P.35.A CONCENTRATION AND DIA FILTRATION

Above heat treated sample is Concentrated to 12% and Dia filtered.

P.36.A CAPRYLATE ADDITION AND pH ADJUSTMENT

Caprylate is added to concentrated and dia. filtered solution. The pH of the solution is adjusted to 4.5.

An IPQC check is done to check actual caprylate concentration and if required, adjusted the concentration.

P.37.A LOW pH INCUBATION

Solution is incubated at room temperature for about 7 hours followed by filtration to remove suspended particles.

P.38.A pH ADJUSTMENT

pH of the above solution is adjusted to 7.0

P.39.A CONCENTRATION AND DIA FILTRATION

Albumin solution is concentrated to formulation concentration and finally subjected to

Pasteurization at 60 °C and finally filled.

Process Flow Diagram Factor IX

FACTOR IX PURIFICATION PROCESS:

P.40.N LOAD PREPARATION CHROMATOGRAPHY 8N

Eluate from chromatography step 2F is suitably adjusted for pH and Conductivity followed by concentration using TFF. Solvent and detergent are added to achieve viral inactivation, followed by filtration.

P.41.N CHROMATOGRAPHY 8N

Chromatography 8N is performed in adsorption mode and eluted fraction is collected in a separate process tank

P.42.N LOAD PREPARATION CHROMATOGRAPHY 9N

Eluate from chromatography step 8N is suitably adjusted for pH and conductivity followed by concentration using TFF.

P.43.N CHROMATOGRAPHY 9N

Chromatography is performed in adsorption mode and eluate fraction is collected in a separate process tank. This fraction is suitably concentrated and passed through a nano filter to achieve viral removal.

This fraction is sterile filtered, formulated and filled in suitable vials.

Process Flow Diagram Thrombin

THROMBIN PURIFICATION STEPS:

P.44.T LOAD FOR CHROMATOGRAPHY 10T

Load from chromatography 10T is obtained from fraction of chromatography 8N.

P.45.T CHROMATOGRAPHY 10T

Chromatography 10T is performed in adsorption mode and eluate fraction is collected in a separate process tank. This fraction is suitably concentrated and passed through a nano filter to achieve viral removal.

This fraction is sterile filtered, formulated and filled in suitable vials.

Process Flow Diagram Fibrinogen:

FIBRINOGEN PURIFICATION STEPS:

P.46.F LOAD FOR CHROMATOGRAPHY 11F

Load from chromatography 11F is obtained from fraction of chromatography 1F. The solution is precipitated and solubilized in suitable buffer solution.

P.47.F CHROMATOGRAPHY 11F Chromatography 11F is performed in adsorption mode and eluate fraction is collected in separate process tank. This fraction is suitably concentrated and passed through a nano filter to achieve viral removal. This fraction is sterile filtered, formulated and filled in suitable vials.

Process Flow Diagram Factor VIII

FACTOR VIII PURIFICATION PROCESS

P.48.E LOAD FOR CHROMATOGRAPHY 12E

Load from chromatography 12E is obtained from fraction of chromatography 1F, is filtered and checked for conductivity and pH, adjust pH and conductivity, if required before loading on the chromatography column.

P.49.E CHROMATOGRAPHY 12E

Chromatography 12E is performed in non-adsorption mode and flow through fraction is collected in a separate process tank.

P.50.E LOAD FOR CHROMATOGRAPHY 13E

Load for chromatography 13E is obtained from fraction of chromatography 12E, fraction is checked for conductivity and pH, adjust the conductivity and pH, if required before loading on the chromatography column.

P.51.E CHROMATOGRAPHY 13E

Chromatography 13E is performed in adsorption mode and eluted fraction is collected in a separate process tank. This fraction is suitably concentrated and passed through a nano filter to achieve viral removal.

This fraction is sterile filtered, formulated and filled in vials.

Manufacturing for Stage-2 Products (PFD)

ANTI D IgG

Factor XI

ANTI MEASLES

TETANUS IMMUNOGLOBULIN

HEPATITIS-B IMMUNOGLOBULIN

RABIES IMMUNOGLOBULIN

Equipment Required to Manufacture for the Proposed Capacity of 150,000 L / annum

Equipment's for Plasma Bulk Manufacturing Facility

Equipment's for Fill Finish Production facility

Equipment's for QC / R&D

Equipment for QA

Equipment's for Plasma storage

Equipment in Finished Goods Storage

Equipment in Eng. / Critical System

Detailed design

This document describes the strategic approach to the facility design, technology basis and products selected in stage 1 (150,000 L / annum) and Stage 2 (500,000 L / annum)

The site is situated on the outskirts of Nagpur city spread over 6 acres.

The process design shall be based on Chromatography system-based separation.

It will be dedicated to manufacture of Human plasma fractions which include the following products in

Stage 1

Albumin

Human Immunoglobulin

Factor VIII

Factor IX

Fibrin sealant

Fibrinogen

The plasma processing capacity in Stage 1 will be 150,000 L / annum

The facility design shall have scope of increasing the plasma processing capacity to 500,000 L / annum.

Besides the products identified in stage 1 the following products have been identified in Stage

An appropriate Technology partner shall be identified for providing the process know-how.

Anti-D

Factor XI

Factor XIII

Hepatitis B IgG

Measles IgG

Protein C

Rabies IgG

Tetanus IgG

Thrombin

Varicella Zoster IgG

The plant is designed as a monobloc facility which will be constructed as a two-floor facility

The ground floor shall contain the following operational units

Bulk plasma fractionation unit (150,000 L / annum)

Warehouse Plasma storage (32000 L)

Admin / QA / Reception

Warehouse (RM/PPM/SPM/ Finished goods)

Electrical transformer area

Engineering workshop

Utility block

Raw water storage tank

Borewell

Boiler room

Chiller room

Kill tank system

Scrap yard

FUTURE EXPANSION AREA

Bulk plasma fractionation unit (350,000 L / annum)

Warehouse Plasma storage (32000 L)

The first floor shall contain the following areas

Canteen area

Water system and engineering store

SPM store

Visual inspection

Label store

Change rooms

Fill finish with Lyo area

Quality Control area

Service area (for AHU's)

CHAPTER 5 PROPOSED SITE PLAN

Concept design:

Figure 9: Conceptual Layout for Site Master Plan

Site Master Plan

Conceptual Layout Ground Floor (Bulk Manufacturing):

The bulk manufacturing area is designed in a manner to provide unidirectional flow of man, material and waste from the facility to avoid mix-ups and cross contamination.

All the ancillary functions are located on the ground floor and separated from bulk manufacturing area.

In the ground floor following facilities are present:

Bulk manufacturing facility -01

Bulk manufacturing facility -02 (Future provision)

Quarantine Plasma Storage/Testing lab/Ware house

Future provision for plasma storage

Finished goods storage & dispatch

Administrative Block

Utility Block

Workshop

DG Set and Transformer area

UG collection tank, ETP/STP & Kill Tank

Bore well, UG 500KL storage tanks

Chiller and Boiler rooms

UG SUMP

Bulk Manufacturing Block

Change Rooms: Bulk manufacturing facility-01, personnel entering the processing area will undergo a series of two dress changes applicable to clean rooms of different classifications. The separate change rooms are provided for male and Females.

Locker Room: This shall be the first change where the personnel remove their street shoes and clothes and keep these belongings in the provided double door lockers.

For the exit from facilities, separate change rooms are provided.

In exit change room personnel shall remove the clothing (Overall, head gear, gown and pant and reach the Locker room and put on their street clothes from the lockers and will exit the facility.

Preparation and Processing areas:

To support the manufacturing operations Buffer Prep.(Support area) shall be responsible to prepare various materials, reagent, chemicals and sterile materials to be used in the manufacturing operations. The provisions for dry and moist heat sterilisations are provided in this area.

The manufacturing facility is construed and designed in a manner to protect product from any mix-ups, cross contamination and making the operator's work path and activities less complex and enabling compliance.

The processing areas for thawing of bulk material (Plasma), fractionation and downstream processing (Chromatography) are separately provided in the Layout to avoid any mix ups and cross contamination.

Plasma storage/ Testing lab /ware house/ Future provision plasma storage:

All the incoming and released raw materials, packaging material required to support the manufacturing shall be stored in the ware house.

All the incoming plasma shall be sampled and tested in ware house testing lab. There is a provision for the storage at -20degrees C for of plasma in the ware house.

For the envisaged future expansion of manufacturing capacities from 150000Lit.to.500000Lit / year an additional storage area is reserved in the ware house, this area shall be furnished during future expansion of the capacities.

Finished goods storage and despatch:

All the released material shall be stored in the released cold rooms. At the time of shipment, the packed material shall be transferred from finished goods dispatch area in the ground Floor.

Administrative Block:

All the administrative support functions shall be dealt in this block.

Holding tank /ETP/STP:

All the liquid waste generated in the bulk manufacturing area shall be properly segregated. The biological waste material shall be transferred to holding tank from there it will be transferred to kill tank for final processing and discharge to ETP through holding tank.

03 UG water storage tanks (5KL) each & Utility Block:

The ground water shall be transferred to UG storage 03 tanks having capacities of 5KL each. The raw water will be treated and will be used as feed water for the generation of PW, WFI and clean steam.

Engineering work shop:

All the maintenance and repair activities shall be undertaken in the work shop.

DG and Transformer room:

For the uninterrupted power supply separate area is designed in the Layout.

Chiller and Boiler room:

The facility of chilled water and plant steam required for various system shall be routed through chillers and boilers room.

First Floor: On the first-floor following facilities are housed:

Formulation / Fill Finish area

Finished goods quarantine

Labeling, packing & Visual Inspection

Water System and Eng. store

Bulk Store

SPM store

PPM store

R&D lab

QA/QC

Canteen

Figure 10: Conceptual Layout for Bulk Manufacturing Facility

The bulk manufacturing area is designed in a manner to provide unidirectional flow of man, material and waste from the facility to avoid mix-ups and cross contamination.

All the ancillary functions are located on the ground floor and separated from bulk manufacturing area.

In the ground floor of Bulk manufacturing facilities following key areas are present:

Bag Cutting and pooling area

Plasma Thawing Area

Downstream processing (Chromatography)

Columns packaging & skids

Washing area

DHS / Autoclave area

Cold room

Buffer Preparation

Plasma day store

Chemical Day store

Change rooms

Change Rooms: Bulk manufacturing facility -01, personnel entering the processing area will undergo a series of two dress changes applicable to clean rooms of different classifications. The separate change rooms are provided for male and Females.

Locker Room: This will be the first change where the personnel remove their street shoes and clothes and keep these belongings in the provided double door lockers.

For the exit from facilities separate change rooms are provided.

In exit change room personnel shall remove the clothing (Overall, head gear, gown and pant and reach the Locker room and put on their street clothes from the lockers and will exit the facility.

Preparation and Processing areas:

To support the manufacturing operations Buffer preparation (Support) area shall be responsible to prepare various materials, reagent, chemicals and sterile materials to be used in the manufacturing operations.

The provisions for dry and moist heat sterilisations are provided in the area.

The manufacturing facility is construed and designed in a manner to protect product from any mix-ups, cross contamination and making the operator's work path and activities less complex and enabling compliance.

The processing areas (Upstream) bag cutting, pooling and segregated plasma thawing areas are provided.

There are separate areas for Plasma fractionation and downstream processing in stagewise manner (Chromatography) provided in the Layout to avoid any mix ups and cross contamination.

For packing of chromatographic columns dedicated area is provided in the layout of bulk manufacturing facility

Cold Rooms:

Cold rooms are provided in the facilities for the storage of heat labile materials like chromatographic gels, in process samples and materials.

Figure 11: Conceptual Layout for Bulk Plasma Storage Facility

Plasma storage facility is housed on the ground floor adjacent to the raw material storage area. All the Raw materials shall take through the dedusting tunnel to clean and remove the dust from the external surface of RM containers. After sampling of RM, the goods shall be transferred to RM quarantine area pending the testing and release of materials from quality control lab.

The RMs which do not meet the specifications shall be transferred to rejection room. Rejection room has access control. Only authorised persons are allowed to enter the area.

In coming plasmas is taken inside the ware house through plasma receipt corridor.

Incoming plasmas is sampled in plasma sampling room, equipped with (RLAF), remaining plasma is transferred to plasma quarantine area having -20 degrees C cold facility.

After the completion of serological testing and other necessary test in the serology lab and QC lab., plasma is shifted to released plasma cold room at -20degrees C.

Barcoding for identification of each lot of bulk plasma is done in Barcoding room. There is a provision for future expansion of cold facilities to cater to 500000 Lit Plasma.

Figure 12: Conceptual Layout for Fill Finish Facility

Fill finish facility is housed on first floor of the bulk manufacturing facilities.

For entry into the area separate entries for males, females and visitors are provided. After series of changes, personnel shall enter into the core areas of fill finish facility.

In the core area formulation area is present.

In the formulation area all the released bulks shall be formulated on campaign basis. The formulated bulks shall be filled, capped through automatic machine and inspected for any abnormalities (particulate matter) visually. Separate laundry area is provided for washing of soiled gowns.

Following are the key areas in Fill Finish facility

Formulation

Fill Finish area 3.Lyophilisation

Change Room

Disinfectant and Material Prep area

Cold room

Clean laundry

Packing area

Figure 13: Conceptual payout for QC and R&D Facility

On the FF of Bulk manufacturing facilities R&D, Quality Assurance and quality control facilities are located.

Four labs are dedicated for research and development activities. The technology at small scale shall be transferred to R&D labs.

Adjacent to this, quality control labs are housed. Primarily quality control labs shall be responsible for testing

Raw materials, PPM and SPMs.

In-process testing of intermediate products

Bulk plasma -Physicochemical and microbiological test

Formulated Bulk plasma products

Fill finished plasma products

Microbial testing of critical utilities.

Microbial testing of environmental samples from clean rooms.

Following labs are envisaged in quality control 1 Sample receipt

2.Chemical / Glassware Storage 3 TOC / LPC

Instrument Room

Balance Room

6 Wet Lab(Chemical)

Hot Zone/ Washing

MLT

Incubator rooms

Sterility room

Washing &Sterilization area

Media prep and Storage

BET Room

Stability Testing

Quality Assurance:

Annexed to QC labs quality assurance department is located. In quality assurance following functions of site will be undertaken

Documentation and control

Trend analysis of critical utilities and Change management

Audits, Inspections, Pest control and Vendors management

Validation. Qualification and Calibration

Complaints and Recalls

Lot Release and Product review.

Trainings

Figure 14: Elevation Drawing

Layout Philosophy:

Plasma fractionation and processing is complex and critical manufacturing operation with respect to the receipt of plasma (Starting material) its downstream processing, formulation and fill finish. The facility requires strict cGMP and regulatory compliance to ensure the product quality and safety of the operator and environment.

The facility Layout adheres to GMP guidelines (Schedule M of Drugs and cosmetic act 1940 and Rules 1945).

The layout philosophy is based on the integral evaluation of processes, potential risks, man and material movement, generation and flow of solid and liquid waste etc.

The manufacturing facility is construed and designed in a separate block to protect product from any mix-ups, cross contamination and making the operator's work path and activities less complex and enabling compliance. Flow of personnel, clean and dirty materials are unidirectional with logical sequence of operations.

HVAC zoning of the facility (control areas) for the manufacturing operations is done based on the operation and its criticality (Plasma collection, fractionation, purification, and storage) to minimize the risk of contamination and to maintain physical conditions like temperature and humidity. The physical conditions should also be suitable to the comforts of the personnel working with protective clothing.

In the Layout provision for separate change rooms for entry and exit from clean areas is made. Change rooms design contains airlocks to provide physical separation of the different stages of gowning.

CHAPTER 6 SOURCING AND HANDLING OF RAW MATERIAL

Exchange Value of Plasma per Liter and Projected Value Over a Period of Seven Years: As per the literature survey and discussion with different stake holders the exchange value of plasma in 2023-2024 is INR 2200/Liter and in next 5 years the estimated plasma price shall be expected to be ≈INR 2800/L.

Figure 15: Exchange value of plasma per liter and projected value in INR

Donors' Selection Criteria:

In India, the plasma industry is dependent on voluntary blood donation. Human blood is the source of plasma which is used to manufacture various therapeutic plasma proteins after fractionation.

In blood banks the cellular components (RBC, WBC, and platelets etc.), are separated out for different applications and the plasma is collected from blood banks by plasma fractionators for clinical / therapeutic use as per "National Plasma Policy" of Ministry of Health & Family Welfare.

The selection and management of blood donors is an essential part of the blood donation process. National health authorities and blood transfusion services are responsible for ensuring that the national system is in place for the selection of all blood donors through an assessment of their suitability to donate blood.

Following are the selection criteria for donors:

Health

Age (18-65 years)

Time between blood donations

Blood pressure

Body temperature

Hemoglobin content etc.

Test for HBsAg*

Test for anti-HIV*

Test for anti-HCV*

*All three tests should be negative. An additionally test for STD is also performed.

Plasma Donation Testing, Epidemiological Surveillance, Post Donation Information: Plasma Donation testing:

The collected plasma is screened for the following:

Test for HBsAg

Test for anti-HIV

Test for anti-HCV

All three tests should be negative. Additionally test for STD is also performed.

Epidemiological surveillance, post donation information:

To ensure optimal long-term safety of plasma for fractionation, it is recommended to establish continuous epidemiological surveillance of the donor population.

The objective of this survey is to know, as precisely as possible, the prevalence and disease incidence, and trends of infectious markers that are relevant to the safety of medicinal plasma products so that appropriate counter-measures can be taken in a timely manner.

The system should not only be able to gather epidemiological data at the national and regional level but also among the donor populations that are providing blood or plasma for fractionation at individual blood establishments within a country or a region. The information from the epidemiological surveillance can furthermore be used to detect and assess.

Differences in disease patterns between donor population.

Trends of infectious marker.

The relevance of any preventive measures such as a strengthening of donor selection process, additional exclusion criteria, or implementation of any additional screening tests to avoid contamination of plasma products.

To detect any new or emerging pathogens in the community.

Handling of Blood, Plasma Freezing, Storage, Cold Chain and Complete Logistics Plan: Handling of Blood:

A standardized and validated procedure for the bleeding shall be followed.

The presterilized container (Bags) shall be used to collect the blood.

Identity of the donor should be confirmed with relevant documents before venipuncture.

The collected blood used for plasma fractionation should be mixed well with the anticoagulant solution to avoid risks of activation of the coagulation cascade.

Presently, automated blood collectors are used in which plasma is separated from cells and the cells are resuspended and returned to donor in series of return cycles.

In most of the centers the centrifugation or filtration or combination of both process is followed to separate the cellular components from plasma.

The collected blood should be weighed and labelled properly in order to ensure full traceability at each stage of plasma production.

Information on the label of the blood storage containers should include:

Name of the blood product

Volume or weight

Unique donor identification number

Name of the blood collection center

Shelf life

Storage temperature

Name, content and volume of anticoagulant

Laboratory samples should also have the same information and must be stored securely at appropriate

temperature

Blood Holding:

It has been shown that whole blood anticoagulated with CPD, transported and stored at 22°C for up to 8 hr prior to separation of plasma is suitable for the production of plasma for fractionation.

However, storage of blood beyond 24 hours the factor VIII activity is reduced by 15 to 20 percent. Rapid cooling of whole blood to 22°C +/- 2°C after collection protects factor VIII and may allow storage of blood for 24 hr.

Transportation/storage of collected blood at 4°C with different anticoagulants (ACD, ACD- adenine, or CPD) consistently appears to reduce the FVIII content, but not necessarily that of other proteins.

Blood holding time for the extraction of various components should be validated by the company.

Plasma Processing and Transportation:

After separation of plasma, freezing is an important processing step that has an impact on content of factor VIII. Several aspects of the freezing conditions of plasma for fractionation have been evaluated in this document based on the literature search.

Freezing of Plasma:

The requirements for the temperature at which plasma should be frozen depend upon the type of proteins to be fractionated.

As per pharmacopeial guidelines, recovered or apheresis plasma for fractionation to be used for the manufacturing of labile proteins (Factor VIII concentrate) should be frozen rapidly, within 24 hours of collection, at -20 °C or below.

However, freezing conditions are currently under debate. It is suggested that recovered plasma used to manufacture stable plasma proteins (e.g., albumin and immunoglobulins) should be frozen within 72 hours of collection at -20 °C or below.

Plasma freezing process should be validated by life Nity to extract the desired plasma proteins.

Storage of Plasma:

Plasma for fractionation should be stored at -20 °C or below. When plasma is stored at -20°C it has the shelf life of 2 years. However, when plasma is stored below -20°C has shelf life of 3 years.

Stability/Shelf-life studies should be conducted on at least three Pooled Plasma bulks during initial phase as a minimum by the company and subsequently one batch of pooled plasma should be taken for stability on annually basis after approval from the quality assurance department. The stability studies should be well documented.

Transportation of plasma and its validation:

Transport of plasma pools (Bulk) and samples to the processing site should be done according to approved and validated procedures. Temperature monitoring device should be used to record the temperature during transportation. Any deviations in the temperature during

transportation should be notified to QA department for impact assessment (compliance and quality).

Simulation studies should be conducted at site to validate the packaging and its transportation.

Sourcing of Plasma from Government and Non-Government Blood Banks:

There are different sources of plasma in India- Government, NGOs and some private facilities recognized by the government. The Govt. has the full responsibility for the blood program.

The surplus plasma from government hospitals and blood collection centers and blood banks are sold through tenders.

As a minimum there are certain conditions for plasma fractionator to fulfill before applying for plasma procurement as follows:

The plasma fractionator must be licensed by the Government of India for plasma fractionation and manufacturing clinically usable products in a cGMP facility.

The fractionator is accountable to update plasma procurement center about the nature and quantity of all the products manufactured from the plasma received from the center.

The fractionator must ensure that none of the products recovered from this plasma are exported before fulfilling domestic demand.

Buy Back policy:

In some cases, government buy plasma derived products from the fractionator for use.

Equipment for Proper Collection, Storage and Separation of Blood Components: Following is the list of equipment required during the different stages of blood processing: Blood Collection

Donor weighing balance

Haemoglobinometer

Sphygmomanometer

Stethoscope

Blood mixer and shaker

Tube stripper and Di-electric tube sealer

Needle destroyer

Blood Processing

Refrigerated Centrifuges

Double pan balance

Plasma Expresser- Manual/Automated

Blood Storage

Blood Bank Refrigerator

-20°C freezers

-80°C deep freezers

Platelet Agitator and Incubator

Cell Separators (Apheresis machine)

Blood Testing

pH Meter

Cell Counter (optional)

Coagulometer (optional)

Automated Cell Grouping system (optional)

Equipment for Gel technology (optional)

Table top centrifuge

Serological water bath

ELISA reader (plate reader / Strip reader)

VDRL shaker

Blood Distribution

Transportation vans (optional)

Insulated blood bags container with provision for storage between 2 °C to 8 °C

Importing of Plasma from Neighboring Countries and Indian Regulations:

India is highly import dependent for its plasma product needs, which are considered to be life- saving medicines. At this point in time, India produces majorly only two plasma products, albumin and intravenous immunoglobulin (IVIG) from local plasma.

Because of changing disease pattern, improved clinical diagnostic facilities, affordability, medical tourism and many other factors the demand of blood products is increasing. This creates a huge demand and supply gap and many patients lose battle of life because of non- availability of these products. Various hyper-immune plasma proteins (Anti-D, tetanus

immunoglobulin, hepatitis-B immunoglobulin, and rabies immunoglobulin) are 100% imported or prepared from imported bulk.

Indian Regulations & Regulatory Bodies:

The Plasma Policy aims at making available, easily accessible and adequate supply of high quality of human plasma derived proteins for clinical/ therapeutic use. In India the plasma is separated from safe and quality blood from voluntary non-remunerated regular blood donor. Blood and plasma are regulated by different rules mention in drugs and cosmetic act 1940 and by different regulatory agencies as mentioned below.

Following regulations and rules are applicable:

The Drugs and Cosmetics Rules, 1945, framed under the Drugs and Cosmetics Act, 1940:

Includes human blood under the definition of "Drugs" under Section 2(b)

Regulates functions of blood banks, which provide recovered plasma for plasma fractionation (Part XII C specifically regulates the collection and storage of plasma for fractionation)

The National AIDS Control Organization (NACO), under the Ministry of Health and Family Welfare) to oversee operation of blood banking services in India.

National Blood Transfusion Council (NBTC) is the apex policy-making body, under NACO, making central licensing mandatory, increasing voluntary donation base and setting of quality standards.

State Blood Transfusion Councils (SBTCs) are responsible for the smooth functioning of blood transfusion services at a state level.

Drug Standard Control Organization (CDSCO) and State Drug Control Organizations' play a vital role in key aspects such as, approval of licenses, and enforcement of standard transfusion practices to ensure safe, quality and efficacious blood and its components in clinical practices. National Institute of Biologicals (NIB) Nodal governmental organization under the Ministry of Health and Family Welfare responsible for the testing and release of biologicals in India including blood plasma.

CHAPTER 7 PRODUCTION PROGRAM

Product and Product mix:

In phase 1 of the project the following products are proposed to be manufactured from normal human plasma.

Human Serum Albumin

Normal Immunoglobulin

Factor VIII

Factor IX

Fibrin Sealant

Fibrinogen

Depending upon the market demand and production capacities, product mix strategy can be formulated.

Various permutations and combinations are possible to manufacture different products. On the basis of PFD given in Chapter 4, different products shall be purified in different steps during the downstream processing of pooled plasma bulk (fractionation of plasma).

However, it is suggested that Lifenity shall take the first campaign of product which is at the highest demand followed by the products with moderate and low demand. As an alternate, the high yielding product shall be manufacture first to have sufficient stock followed by a campaign for low yield and moderately yield products.

Plasma requirements based on the manufacturing capacities:

Considering 1,50,000 Liters of plasma processing capacity, plasma bags requirement are presented in table below:

Table:9 Plasma requirements based on the manufacturing capacities:

Production Capacities and Yield Calculations:

HSA is major protein component, followed by IgG in plasma. The wide concentration range of plasma proteins (less than 1 mg/mL to 40 g/L is one of the potential challenges in designing production capacities.

Source: (J.D. Andrade et al. Plasma protein adsorption: the big twelve, Ann. N.Y. Acad. Sci., p. 158-172,1990. & Chapter -2 of DPR).

However, for calculation purpose below mentioned protein concentrations in normal plasma are considered.

It has been reported in the literature that depending upon the downstream processing and sterile/terminal filtration, recovery of the protein of interest is 50% at final stage.

Source: (Overview of Albumin and Its Purification Methods: Adv Pharm Bull, 2016, 6(4), 495- 507 doi: 10.15171/apb.2016.063 <http://apb.tbzmed.ac.ir>) as shown in the table below.

Basis of Yield calculations:

Based on vessel size, the batch size assumed is 1000 liters, on every alternate date, a new batch can be initiated, therefore 15 batches shall be processed per month (one batch/ alternate day).

The yield for 1000 liters batch is as mentioned below.

Table 10: Yields for 1000L batch

In the EOI document, the phase 1 target is to process 1,50,000 Liters of plasma annually. The output obtained with 1.5 lac liters of plasma for the products mentioned for stage1 is as under.

Table 11: Yields for 150,000 Lit annual capacity

Therefore, to achieve the plasma processing target, 15,000 liters of plasma shall be processed on monthly basis at site.

Production Scaleup from 150,000 to 500,000 lit per annum:

Lifenity will increase its plasma processing capacity to 500,000 liters annually in phase 2 of the project implementation.

To achieve the plasma processing target, 50,000 liters of plasma shall be processed monthly. Two strategies, can be followed:

Strategy 1: To keep number of batches equivalent to stage-1 (i.e., 15 batches per month), the individual batch size shall be 3400 liters. The yield calculations are given below.

Table 12: Yields for 3400 L batch

Strategy 2: Alternatively, if the batch size is further increased to 5000 liters, the number batches shall be reduced to 10 batches per month as one batch can be initiated after every 3 days. The yield for a 5000 liters batch is as mentioned below.

Table 13: Yields for 5000 L batch

Total Production capacity:

The total production capacity with 500,000 Lit. of Plasma/year is given in the table below.

Table 14: Total production capacity with 500,000 Lit. of Plasma/year

Note:

For scale up to 500,000 Liters only 6 products (HSA, IgG, F-VIII, F-IX Fibrin Sealant, Fibrinogen) are considered.

Fibrin sealant is prepared by mixing Fibrinogen and Thrombin; therefore, production capacity of fibrin sealant is equal to the production capacity of Thrombin.

CHAPTER 8 PROJECT IMPLEMENTATION MODEL

Lifenity shall implement the PDMP manufacturing in a phased manner.

A suitable technology partner for the Plasma fractions shall be identified and contracted with for process know-how. The agreement will fulfil criteria mentioned in Section 3.3 of this document.

The project shall be initiated at small scale to establish the process for the products identified in

stage 1.

Once the process, yields and quality are established as per the specifications, the project will proceed to the commercial scale stage

Commercial scale:

The company will initially procure 150,000 liters of plasma per year for fractionation to produce the following product in Phase-1.

Human Serum Albumin (HSA)

Human Immunoglobulin (IgG)

Factor VIII

Factor IX

Fibrin sealant

Fibrinogen

All the areas of the manufacturing block (Facility-1) will have sufficient space to cater the volumes and products without compromising on any of the cGMP and regulatory expectations.

The following systems installed during Phase-1 of the project will also be matching the requirements for Phase-2 (Expansion Project) capacities.

Clean utilities

Black utilities

Filling

Freeze drying and

Packaging lines for the final product

Facility Expansion:

For handling additional 350,000L of Plasma:

In Phase-2, the company shall procure additional 350,000 Liters of plasma to manufacture the following additional products.

Anti-D IgG

Factor XI & XIII

Fibrinogen

Hepatitis B IgG

Immunoglobulin G

Measles IgG

Protein C

Rabies IgG

Tetanus IgG

Thrombin

Varicella Zoster IgG

While implementing Phase-2 of the project, Lifenity shall extend the manufacturing operations to Facility-2 and shall procure the following additional systems to match the capacities of processing additional 350,000 L/year of plasma which are as follow:

Tank Systems

Chromatography Systems

Autoclaves

Dry heat sterilizers

Quality control and Quality assurance units developed during Phase-1 will be also cater to the Phase -2 products and other ancillary activities.

Central ware house space shall be adequate for future expansion. However, the additional cold Chain equipment (Freezers) and storage racks etc. for the storage of active material (plasma) and finished products shall be procured during Stage 2 expansion.

In Phase-1 of the project the bulk manufacturing operations will run in a single shift of 8 hours working /day, with effective working for 6 hours. The shifts can further be evaluated based on number of batches for processing/week. Likewise, the filling operations will run in scale down mode to meet the market requirement.

Monoclonal antibodies (Mabs) – Future considerations:

A monoclonal antibody are Y-shaped glycoproteins that detect and bind strongly to their antigen.

Monoclonal antibodies stems from a single white blood cell clone. Monoclonals not only bind to a single epitope of a target antigen but also activates immune responses, e.g. antibody-dependent cell-mediated cytotoxicity (ADCC) and recruits further immune cells such as phagocytes and are thus highly specific

Monoclonal antibody production:

Several technologies for monoclonal antibody production have been developed and the respective downstream processes are optimized. Some of the industrial scale technologies are:

Hybridoma cells production

Phage display technologies

Single B cell technologies

Hybridoma production:

In this technology fusion of antibody producing B cells of an immunized mouse with myeloma cells is done produce unlimited antibodies. Hybridoma cell line that can be maintained indefinitely

Disadvantages:

Use of lab animals for immunisation

Risk of mutations in the antibody with aging of hybridoma

Expression of additional variable domains by the cell line

Long lead times and labour-intensive operation

Phage display technology:

Phage display technology circumvent several disadvantages of hybridoma technologies and speed up the process of Mab generation.

The technology relies on molecular biology and cell culture methods to genetically engineer bacteriophages to display antibodies on their surface.

In this method binding of bacteriophages with desired antibodies onto immobilized antigens is done. The

high affinity binders are then used to infect bacteria and collect their DNA for

sequencing. The sequence help to design genes to be used in antibody production in cell cultures.

The overall process is much faster than hybridoma cell generation and doesn't need any laboratory animals.

Single B cell technologies:

Single B cell technologies are another milestone in Mab development. The technique involves the screening and sorting of ex vivo cultures of single B cells from blood samples of immunized animals or humans to select for promising antibody candidates.

Genetic information is then extracted from the candidate cells mRNA after reverse transcription into cDNA, amplification and subsequent sequencing and cloning into expression vectors for Mab production.

In this method there is no need to create a hybridoma cell line, thus process become efficient and timeline are shorted by several weeks.

Depending upon the process to be adopted for the manufacturing of monoclonal antibody a new production block shall be designed to meet the process and cGMP requirements.

Detailed working for the Mabs manufacturing unit will be proposed separately in future.

In general, the Mab production suite shall have cell culture, fermentation, harvesting, primary purification, secondary purification, sterile filtration.

However, for formulation and fill finish existing facilities shall be used areas.

Table 15: Steps performed in MABs production (General)

For implementation of Mabs project in future, there shall be a requirement of cell culture and molecular biology labs in the quality control with some additional piece of equipment.

Present Layout of quality control unit will be adequate for plasma products (Phase-1 & 2). However, for Mabs products testing, expansion in existing QC to cater the cell culture need shall be required.

CHAPTER 9 GMP AND REGULATORY

CONSIDERATIONS

Premises, Layout & Process Engineering:

Plasma fractionation and processing is complex and critical manufacturing operation with respect to the receipt of plasma (Starting material) its downstream processing, formulation and fill finish.

Plasma fractionation facility require strict cGMP and regulatory compliance to ensure the product quality and safety of the products and operators. The facility's layout and process engineering are critical components for setting up of the green field project. Following are key considerations for setting up a plasma fractionation facility with respect to Premises, Layout and Process engineering:

Location and Surroundings:

The factory building(s) for manufacture of plasma products shall be situated and shall have measures to avoid risk of contamination from external environment including open sewage, drains, public lavatory or any factory which produce disagreeable or obnoxious odor or fumes, excessive soot, dust, smoke, chemical or biological emission.

Building:

The building(s) used for the purpose shall be designed, constructed, adapted and maintained to suit the manufacturing operations so as to permit production under hygienic conditions. They shall conform to the conditions laid down in the Factories Act, 1948 (63 of 1948).

Premises - General Requirements:

The premises used for manufacturing, processing, warehousing, packaging, labelling and testing purposes shall be:

Compatible with other products manufacturing operations that may be carried out in the same or adjacent area.

The manufacturing facility shall be designed / constructed / maintained to prevent entry of insects, pests, birds, vermin's and rodents. Interior surface (walls, floors and ceilings) shall be smooth and free from cracks, and permit easy cleaning, painting and disinfection.

There should be adequate space for orderly and logical placement of equipment, materials and movement of personnel in order to prevent:

The risk of mix-up between different categories of products or with raw materials, intermediates and in-process material.

Possibilities of contamination and cross- contamination by providing suitable segregation, zoning and access controls etc.

Layout Considerations:

During designing of facility, Lifinity must ensure the following points:

Plasma products will be produced in an independent block/building.

The facility is designed to prevent cross-contamination between process areas, easy cleaning and ensure the efficient flow of materials and personnel.

The facility design adheres to GMP guidelines (Schedule M of Drugs and cosmetic act 1940 and Rules 1945).

Ancillary areas like rest and refreshment rooms are separate from manufacturing areas.

Facilities for changing, storing clothes and for washing and toilet purposes are easily accessible and adequate for the number of users. Provision for separate toilets for males and females is made and toilet is not connected with production or storage areas.

Change rooms design contains airlocks to provide physical separation of the different stages of gowning. Flushed effectively with filtered air. Change rooms are of the same grade as the area into which it opens finally.

Provision of separate change rooms for entry and exit from clean areas.

Hand-washing facilities are provided in the first change room.

Flow of personnel, clean and dirty materials are unidirectional with logical sequence of operations.

Sinks and drains are excluded from Grade A and B areas where aseptic operations are carried out.

Drains are fitted with effective easily cleanable traps and with air brakes to prevent backflow.

The production and dispensing areas are well lighted, effectively ventilated.

HVAC Zoning:

Air supply for manufacturing area will be filtered through bacteria retaining filters (HEPA Filters) and will be at a pressure higher than in the adjacent areas unless otherwise required.

The filters to be checked for performance on installation and periodically thereafter and maintain written records.

HVAC zoning of the facility (control areas) for the manufacturing operations is done based on the operation and its criticality (Plasma collection, fractionation, purification, and storage) to minimize the risk of contamination and to maintain physical conditions like temperature and humidity. The physical conditions should also be suitable to the comforts of the personnel working with protective clothing.

Routine microbial and particulate counts of the manufacturing area will be carried out during manufacturing operations to assess the effective ness of HVAC.

The results of particulate matter and microbes to be continuously monitored against documented in-house standards and regulatory guidelines.

Based on the quality of air, manufacturing facility is divided into different grades/ Zones. In general, plasma processing facility is divided into four grades (Classes) based on the air cleanliness levels (Microbial and Particulate matter (Refer to table No 15 and 16).

Facility Grades:

Grade A: Clean rooms are used for high-risk operations, and require a unidirectional airflow system to maintain high air quality. Unidirectional airflow should provide a homogeneous air velocity of 0.36–0.54 m/s (guidance value) at a defined test position 15–30 cm below the terminal filter or air distributor system.

The velocity at working level should not be less than 0.36 m/s.

Grade B: In aseptic preparation and filling, this is the background environment for the Grade A zone.

Grades C and D: Clean areas for carrying out less critical stages of the manufacturing of plasma products -downstream processing, chromatographic purification and solution preparations.

Table 16: Recommended Maximum Allowable Particles in Clean Rooms

Table 17: Recommended limits for Microbial contamination in Clean Rooms

Clean Utility Systems:

Pure steam, purified water and water for injection are the clean utilities required to support the clean operations in plasms manufacturing/processing facilities. Monitor the clean utilities for the quality and compliance regularly as per the laid procedures and the regulatory standards.

Equipment Placement:

Equipment and processing areas will be arranged to facilitate efficient workflow and minimize the risk of contamination. Ensure that equipment and materials are easily accessible for maintenance and cleaning. Processing areas must be equipped with dedicated equipment designed for specific purpose.

Waste Management:

Waste will be removed from the manufacturing area or sent to the bio-waste system in a safe manner. To ensure containment compliance with the environment, the facility must implement measures to prevent accidental spills or leaks. This includes the use of appropriate containment systems, spill response plans, and monitoring for any environmental impacts.

All sewage waste pipes will run below natural ground level (however this depends on the actual site conditions along the pipe routing) in proper gradient and connected to nearest available chamber.

Procedures must be in place for the safe removal of different types of waste such as chemical, biological, expired reagents, unused samples and solvents etc.

Design the drainage systems in such a way that effluents can be effectively neutralized or decontaminated to minimize the risk of cross-contamination.

Provisions should be there for the proper and safe storage of waste materials awaiting disposal.

Toxic substances and flammable materials must be stored in suitably designed, separate, enclosed cupboards, as required by national legislation.

Liquid effluent poses a safety or contamination risk (Biological liquid waste), should be treated in the kill tank before its discharge into non biological /main drain.

Testing Facilities:

Testing facilities must be located in the separate building/floor.

Validation and Documentation:

Lifentiy and PMC will validate the facility layout and design to ensure it meets GMP and regulatory requirements. Comprehensive documentation of the facility's layout and design, including as-built drawings shall be maintained at site and a robust document control system shall be established in quality Assurance department to maintain records.

Accessibility and Safety:

Ensure that the facility layout allows for easy access in the event of an emergency. The facilities must be equipped with safety features like fire hydrant systems, eyewash and body shower stations where ever necessary.

Quality Assurance & Regulatory Compliance:

The quality assurance system appropriate to the manufacture of plasma products shall ensure that the products are designed and developed in a way that takes account of the requirements of GMP and other associated codes such as those of GLP and GCP.

Adequate arrangement must be made at site for manufacture, supply chain and use of the correct starting and packaging materials.

Self-Inspection and Quality audit:

In the quality assurance department of Lifentiy adequate procedures shall be designed to ensure self-inspections and quality audits periodically.

Plasma Process Considerations:

Plasma fractionation involves critical steps to separate plasma into its various components and produce high-quality plasma-derived products while adhering to Good Manufacturing Practices (GMP) and regulatory requirements.

Following are the key considerations for plasma processing:

Plasma Collection and Testing: A reliable and secure source of plasma, involving strict donor selection, screening and testing plan in place to screen for infectious agents, including HIV, Hepatitis B and C, and other potential contaminants in pooled plasma.

Fractionation Process: A well-defined fractionation process to separate plasma into its various protein components, such as albumin, immunoglobulins, and plasma factors. Robust filtration and purification processes will be followed by Lifentiy to separate and purify these proteins.

Process Validation: The fractionation process must be validated to demonstrate its ability to consistently produce products. Process performance qualification (PPQ) shall be performed on first three initial batches to ensure that the process is in a state of control during commercial production.

Process Control and Monitoring: Robust process control systems to monitor critical process parameters (e.g., temperature, pH, pressure) in real-time. Written procedures must be in place for addressing deviations and out-of-specification results.

Raw Material Control: Documented procedures for receiving, testing, and release of raw materials and starting materials used in the fractionation process. Lifentiy must ensure that all materials meet quality, safety standards and plasma source traceability.

Safety: Validated viral and pathogen inactivation and removal steps to ensure the safety of plasma-derived products. Implement measures to prevent contamination during processing.

Quality Control Testing: Comprehensive quality control testing plan, including in-process testing and final product testing, to ensure the safety, purity, and potency of the plasma products. Specifications for raw materials, intermediates, and final products shall be written and approved.

Equipment monitoring, Calibration and Validation: All the critical equipment used in the manufacturing and control of plasma are maintained, calibrated and qualified as per the written procedures.

Training and Qualification: Ensure that all personnel involved in the plasma fractionation process are well-trained with respect to the SOPs and STPs, understand GMP requirements, and are aware of their roles in ensuring product quality and safety. Maintain records of personnel training and qualifications.

Writing and editing documentation: Documentation for all existing processes must be written and edited regularly, to ensure that all new hires and other personnel can understand the best practices and standards

Risk assessments of current equipment/processes: Assessment of current processes and equipment to determine potential risks and mitigation plan shall be in place.

Traceability: A robust documentation system must be in place for tracking and tracing individual plasma donations and the products derived from them.

Packaging and Labelling: Appropriate packaging and labelling procedures for the final plasma-derived products, ensuring they meet regulatory requirements.

Post-Market Surveillance: Post-market surveillance system to monitor the safety and efficacy of products in the market.

Regulatory Guidelines GMP, Drugs and Cosmetics act 1940 and Rules, CDSCO, WHO- GMP

Indian Guidelines:

The Drugs and Cosmetics Rules, 1945, framed under the Drugs and Cosmetics Act 1940:

Includes human blood under the definition of "Drugs" under Section 2 (b)

Regulates functions of blood banks, which provide recovered plasma for plasma fractionation

Part XII C specifically regulates the collection and storage of plasma for fractionation

In the year 1967, Central Govt. (Ministry of Health) enacted a separate provision in Schedule F Part XII B of Drugs & Cosmetics Rules.

Schedule F of the Drugs and Cosmetics Rules specifically pertains to the requirements for the manufacture, sale, and distribution of plasma-derived products, including immunoglobulins, clotting factor concentrates, and other blood

Drugs and Cosmetics Rules, Part X-B: Conditions for License (Rule 122EA to Rule 122P)

Destruction of infected blood and implementation of Bio-Medical Wastes (Management and Handling) Rules, 1996

Applicable forms for License:

The National AIDS Control Organization (NACO), under the Ministry of Health and Family Welfare) to oversee operation of blood banking services in India.

National Blood Transfusion Council (NBTC) is the apex policy-making body, under NACO, making central licensing mandatory, increasing voluntary donation base and setting of quality standards. Similarly, State Blood Transfusion Councils (SBTCs) are responsible for the smooth functioning of blood transfusion services at a state level.

Drug Standard Control Organization (CDSCO) and State Drug Control Organizations play a vital role in approval of licenses, and enforcement of standard transfusion practices to ensure safe, quality and efficacious blood and its components in clinical practices.

National Institute of Biologicals (NIB) is the nodal governmental organization under the Ministry of Health and Family Welfare responsible for the testing and release of biologicals in India including blood plasma.

Policies and Guidelines for Fractionation:

The National Blood Policy 2002 regulates blood quality and the Guidelines of NBTC 2016 has been brought out by NACO. NBTC has formulated guidelines for recovery of processing charges for blood and blood Components across blood banks in all sectors. It has also laid down uniform norms for issuance of No Objection Certificates (NOCs) for new blood banks, renewal of license, criteria for designation blood bank as RBTC, Guideline on bulk transfer of blood and Guidelines on exchange of surplus plasma with indigenous fractionators has been released by NBTC to support voluntary blood donation and rationalize blood utilization.

National Policy for Access to Plasma Derived Medicinal Products from Human Plasma for Clinical / Therapeutic Use covers the process of collecting standard plasma and transporting them under optimum conditions for fractionation, identifying critical parameters for safety, and ensuring compliance with regulatory requirements and training.

National Pharmaceutical Pricing Authority: Under the Ministry of Chemicals and Fertilizers, Department of Pharmaceuticals, the prices of medicines are controlled by the National Pharmaceutical Pricing Authority. Currently, plasma products under price control include albumin, AHF, anti-D immunoglobulin, anti-tetanus immunoglobulin and anti-rabies immunoglobulin however plasma itself is not under price control.

Pre-clinical safety and Clinical Evaluation of Plasma derived products: European guidelines

Preclinical:

June 2011 EMA/CHMP/ICH/731268/1998 Committee for medicinal products for human use (CHMP)

Clinical Evaluation of Plasma derived products: Guideline on clinical investigation of recombinant and human plasma-derived factor IX products: 21 May 2015, EMA/CHMP/BPWP/144552/2009 Rev. 1, Corr. 1 Committee for medicinal products for human use (CHMP).

Guideline on clinical investigation of recombinant and human plasma-derived factor VIII products: 21 May 2015 EMA/CHMP/BPWP/144533/2009 rev Committee for medicinal products for human use (CHMP).

Guidance on the Clinical Investigation of Human Normal Immunoglobulin for Intravenous Administration (IVIg). CPMP/BPWG/388/95 rev.1.

Guidance on the Clinical Investigation of Human Normal Immunoglobulin for Subcutaneous and Intramuscular Use. EMEA/CPMP/BPWG/283/00.

Indian guidelines

As per Indian drugs and cosmetic act human blood plasma is included under the definition of "Drugs" under Section 2 (b)

As per the Indian regulations if the company which manufactures human plasma products similar to the composition and product profile currently marketed then the clinical trial waiver is given for marketing authorization. The company shall apply to DCGI &SLA for the waiver with relevant supporting documents. In case of hyper immune sera products appropriate clinical trial for safety and efficacy shall be conducted.

Containment Compliance

Blood and blood-derived products are highly sensitive and can be easily contaminated. Bloodborne pathogens, such as HIV, hepatitis B and C, can pose significant risks to both staff and patients if proper containment procedures are not followed. Maintaining a contained environment helps minimize the risk of spreading infectious diseases in the population.

Containment measures in the manufacturing premises shall be established such as isolation rooms, negative pressure environments, controlled airflow systems and proper handling procedures.

Plasma fractionation facilities must adhere to specific biosafety levels as defined by regulatory authorities, such as the Centers for Disease Control and Prevention (CDC). Plasma fractionation facilities operate at BSL-2 (Biosafety Level 2). It must include:

Access control, lab coats, gloves, eye protection (PPE) to be used by personnel working in the facility, and the use of biological safety cabinets during procedures.

Continuous monitoring of air pressure differentials, temperature, and humidity within the clean rooms.

Alarms should be triggered if any parameter falls outside the acceptable range.

Scheduled regular maintenance, testing, and certification of the HVAC system.

The exhaust air from the clean rooms potential for any contaminating microbes should be passed through the scrubbers to protect the environment.

Validation of Processes, Equipment and Services

Approaches to validation:

Lifinity will organize and plan qualification and validation in a manner that will ensure product quality, safety and efficacy throughout its life-cycle.

Qualification and validation shall be done in accordance with predetermined protocols, and the results to be documented, in reports.

An appropriate and effective quality management system shall be in place at Lifinity supporting the organization, planning, execution and management of qualification and validation activities.

Senior management will provide sufficient resources to perform validation in a timely manner. Management and persons responsible from quality assurance and manufacturing to be actively involved in the validation activities.

All relevant SOPs for operation, maintenance and calibration shall be prepared during qualification.

Training shall be provided to operators, and training records need to be updated and maintained.

Equipment Qualification/Validation:

Equipment qualification helps mitigate risks, prevent equipment failures, reduce downtime, enhance product quality, and ensures the overall operational excellence of an organization.

Equipment Validation comprises a series of steps which includes User Requirement Specification (URS), Design, Installation, Operational, and Performance Qualification.

User Requirement Specification (URS): Lifinity will prepare a document describing the requirements for the equipment to be sourced. The requirements will include specifications and must ensure that possible GMP risks are addressed; include technical requirements; and associated documentation.

Design qualification (DQ): DQ will provide the documented evidence that the design specifications were met and are in accordance with the URS. FAT and SAT needs to be performed where required, to verify the suitability of the system at site, prior to the subsequent stages of qualification and this must be appropriately documented.

Installation qualification (IQ): IQ will provide documented evidence that the installation was complete and satisfactory, including supporting utilities. The design specifications, including purchase specifications, drawings, manuals, lists of spare parts and vendor details, will be verified during IQ. Components installed must be verified, and documented evidence should be provided that components meet specifications, are traceable and are of the appropriate construction material. Applicable control and measuring devices, identified through impact or risk assessment, should be calibrated.

Operational Qualification (OQ): OQ should provide documented evidence that utilities, systems or equipment operate in accordance with operational specifications. Tests will be designed to demonstrate satisfactory operation over the normal operating range, as well as at the limits of its operating conditions. Worst-case conditions must be included in the testing. Operation controls, alarms, switches, displays and other operational components shall be tested.

Performance qualification (PQ): PQ will normally follow the successful completion of IQ and OQ. PQ will be performed under conditions simulating the intended use, to provide documented evidence that equipment/system can consistently perform.

Process validation:

Process validation is defined as the collection and evaluation of data, from the process design stage through commercial production, which establishes scientific evidence that a process is capable of consistently delivering quality product. It ensures that quality, safety and efficacy by design are built into the product.

Collect and evaluate the data throughout the life-cycle of a product from the process design stage through to commercial production and provides scientific evidence that a process is capable of consistently delivering a quality product.

Data need to be available for inspection at Lifinity manufacturing location.

A risk-assessment approach will be followed, to determine the scope and extent to which process(es) and starting material variability may affect product quality.

The critical process steps and critical process parameters (CPPs) will be identified, justified monitored and documented based on relevant studies carried out during the design stage and process knowledge.

Lifinity will ensure that the principles of process validation described in the guidelines are implemented.

Evaluate and compare the historical data when there is a change in the process.

When changes are planned to existing systems or processes need to be followed and documented. Sufficient data is collected to demonstrate that the revised process and product will meet desired quality, consistent and approved specifications.

CHAPTER 10 ALLIED EQUIPMENT & FACILITIES

Technology Based Allied Equipment Requirements

Refer Annexures

Facilities

Laboratories for allied testing should be separated from Production and Quality control labs. However, the materials required for testing such as media, filters, filtration devices etc. can be taken from QC labs. Dedicated glass ware should be used for any testing activities.

Example of allied test/activities are ETP samples, Black utilities, Ground water, testing of clean steam for physical parameters, compressed air and calibration activities etc.

Specialized temperature-controlled warehouse for storage of RMs, In-Process materials and finished goods:

Adequate areas shall be designed to allow sufficient and orderly warehousing of various categories of materials and products like starting and packaging materials, intermediates, bulks and finished products, products in quarantine, released, rejected, returned or recalled, machine and equipment spare parts and change items etc.

Warehousing areas need to be designed and adapted to ensure good storage conditions. Ware house shall be clean, dry and maintained with acceptable temperature and humidity limits where ever required.

Temperature and humidity will be monitored and recorded. A temperature monitoring and recording system that are independent from the temperature regulation system with alarm systems will be placed.

Storage areas must have program for pest control. The record shall be maintained.

Proper racks, bins and platforms must be provided for the storage of materials. The quarantine area in the warehousing will be clearly demarcated. Access to the areas must be restricted to authorized persons.

There shall be a separate sampling area in the warehousing for active and sterile raw materials/starting materials and excipients from the unsterile RMs.

Sampling and dispensing of sterile materials shall be conducted under aseptic condition conforming to Grade A (LAF). Sampling of sterile materials shall be performed in a dedicated area within the manufacturing facility in the absence of suitable areas in the ware house. Segregation shall be provided for the storage of rejected, recalled or returned materials or products.

Materials or products in this area should be suitably marked and secured. Access to these areas and materials must be restricted.

Highly hazardous, poisonous and explosive materials such as alcohols, narcotics and other substances presenting potential risks of abuse, fire or explosion shall be stored in explosion proof area, away from the central ware house in safe and secure manner.

Adequate fire protection measures shall be provided in conformity with the rules of the concerned civic authority in the ware house.

Printed packaging materials shall be stored in safe, separate and secure areas. Regular checks shall be made to ensure adequate steps are taken against spillage, breakage and leakage of containers.

Receiving and dispatch bays shall protect materials and products from adverse weather conditions.

The ware house shall have different areas for the receipt, quarantine, undertest, released, hazardous and rejected materials.

Primary packaging materials, labels, and secondary packaging material and finished products shall be stored in the separate area of the ware house with access controls systems in the critical areas of the ware house.

Incoming critical materials (such as sterile solutions, plasma bag and testing reagents) shall be physically quarantined immediately after receipt and until they are released for use by the authorized person.

Materials and reagents shall be stored under the conditions established by the manufacturer and in an orderly manner that permits segregation by batch or lot and stock rotation. Storage, issuance and usage should follow the "first-expiring first-out" principle (FEFO).

Materials Required Like Resins, Filters etc.

Ion exchange resins

Size exclusion resin – 1

Size exclusion resin - 2

Multi grade filters (MGF)

Softener

Filters 0.5 Micron

SMBS (Sodium Metabisulphite)

Chlorine

Common Salt

NaOH

Hydrochloric acid

SOP/STP and Biological Waste Management

The SOP's and STP for the allied testing facilities are as under:

Procedure for Decontamination of chemical and biological waste.

Procedure for estimation of BOD and COD in ETP samples.

pH determination of ETP samples.

Procedure for testing compressed air and pure steam.

Procedure for calibration of measuring devices of manufacturing, ware house, quality control and engineering.

Procedure for treatment of Biowaste

Procedure for collection segregation and disposal of Biowaste

Training on Biowaste management

Procedure for usage of treated water for horticulture purpose

Procedure for discharge of Biowaste in common drainage system

Procedure for management of spillage control

Biowaste management (BWM):

The disposal of sewage and effluents (solid, liquid and gas) from the manufactory should be in conformity with the requirements of Environment Pollution Control Board. All bio-medical waste must be destroyed as per the provisions of the Bio-Medical Waste (Management and Handling) Rules, 1996 and amendments thereof.

Additional precautions to be taken for the storage and disposal of rejected waste material. Records shall be maintained for all disposal of waste. Provisions shall be made for the proper and safe storage of waste materials awaiting disposal. Hazardous, toxic substances and flammable materials shall be stored in suitably designed and segregated, enclosed areas in conformity with Central and State Legislations.

CHAPTER 11 SUBSIDY CONSIDERATIONS

Location / SEZ Considerations in Maharashtra

Special Economic Zone (SEZ) is a specifically delineated duty-free enclave with all required infrastructure provided under single administrative umbrella primarily meant for locating industries which manufacture and export goods and services.

These enclaves shall be deemed to be foreign territory for the purposes of trade operations and duties and tariffs.

Under section 3, a Special Economic Zone can be established either by Central Government, State Government or its agencies, private / public / joint sector or any person for manufacture of goods or rendering services or for both or as a free trade and warehousing zone.

As per the requirements of SEZ, the unit must be segregated into processing and non- processing areas. A processing area is that area in which all the manufacturing activities are performed. At least 50% of the total area shall be earmarked.

The non-processing area is the area which is intended to provide support facilities to the processing area activities and may include support areas like utilities block, power generation plant (DG), engineering services, ETP and other infrastructures as per the requirements of the current project.

In SEZ 100 percent Foreign Direct Investments are allowed (FDI) in some of the selected categories.

As per MIHAN-SEZ guidelines (Rules and Procedures) following class of incentives are available to SEZ enterprises.

Custom and Excise

SEZ units may import or procure from the domestic sources, duty free, all their requirements of capital goods, raw materials, consumables, spares, packing materials, office equipment, DG sets etc. for implementation of their project in the Zone without any license or specific approval.

Duty Free Import/Domestic Procurement of Goods for Setting up of SEZ Units

Goods imported/procured locally duty free could be utilized over the approval period of 5 years.

Sales Exemption

Domestic sales by SEZ units will be exempt from single administrative document (SAD). Domestic sale of finished products, by-products on payment of applicable custom duty.

Domestic sale rejects, waste and scrap on payment of applicable Custom duty on the transaction value.

Income Tax:

Physical export benefit 100%.

IT exemption (10A) for first 5 years and 50% for 2 years thereafter. Reinvestment allowance to the extent of 50% of ploughed back profits.

Carry forward of losses & Commercial borrowing:

Banking / Insurance/External Commercial Borrowing.

Setting up Off-shore Banking Units (OBUs) is allowed in SEZs. OBU's allowed 100% Income Tax exemption on profit for 3 years and 50 % for next two years.

External commercial borrowings by units up to \$ 500 million a year allowed without any maturity restrictions.

Central Sales Tax:

Exemption to sales made from Domestic Tariff Area to SEZ units.

Service Tax:

Exemption from Service Tax to SEZ units.

Other Exemptions:

Freedom to bring in export proceeds without any time limit.

Flexibility to keep 100% of export proceeds in exchange earners foreign currency account (EEFC). Freedom to make overseas investment from it.

Exemption from interest rate surcharge on import finance. SEZ units allowed to 'write-off' unrealized export bills.

Additional Incentives:

The additional incentives are exemption of import license, custom examination of imported goods, exemption from public hearing under Environment Impact Assessment Notification etc.

CHAPTER 12 PROJECT TIMELINES

CHAPTER 13 BUDGET ESTIMATES

Budget Estimates

Refer to Chapter 17

CHAPTER 14 REVENUE AND COST ESTIMATION AND PAYBACK PERIOD

YIELD CALCULATIONS FOR PLASMA FRACTIONS

REVENUE FORECAST IN CRORE INR FROM PROCESSING PLASMA 150,000 L / ANNUM

PROFIT AND LOSS STATEMENT FOR STAGE 1 – 150,000 L/ ANNUM

CHAPTER 15 RECOMMENDATION

The objective of this project is to establish and operate a state-of-the-art Human Plasma Derived Medicinal Products (HPDMP) manufacturing facility with an aim to address the increasing demand for plasma-derived products by Optimized manufacturing processes maximize productivity, minimize waste, and reduce operational costs in order to maintain competitiveness in the market.

This Human Plasma Derived Medicinal Products manufacturing facility will play a pivotal role in meeting the critical healthcare needs of patients while contributing to advancements in the plasma industry.

In phase-1 of the project annual production capacity will be 150,000 liters of plasma processing, which

will be subsequently increased to 500,000 liters per annum in future.

In this report several technologies utilized for plasma purification at commercial scale were evaluated exhaustively with its merits and demerits. Some of the key technologies evaluated are:

Plasma Fractionation method with ethyl alcohol

Cohn+ Liquid Chromatography

Heat Shock technology

Ammonium sulphate+ Liquid Chromatography

Chromatographic

It is concluded and recommended that Chromatographic purification with sterile filtration is best technology for the purification of plasma proteins for stage-1 products as it will give highly productivity and acceptable quality of desired proteins (purity levels) as per the regulatory expectations.

For stage-2 products, the technology evaluation and recommendations will be made in future.

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State Blood Transfusion Councils (SBTCs) are responsible for the smooth functioning of blood transfusion services at a state level.

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2. AIDS : Acquired Immunodeficiency Syndrome	
3. BOD : Biological Oxygen Demand	
4. CDSCO : Central Drug Standard Control Organization	
5. cGMP : Current Good Manufacturing Practices	
6. CO ₂ : Carbon Dioxide	
7. COD : Chemical Oxygen Demand	
8. CPPs : Critical Process Parameters	
9. CQA : Critical Quality Attributes	
10. DCGI : Drug Controller General of India	
11. EHS : Environment, Health, and Safety	
12. EMA : European Medicine Agency	
13 EMP : Environmental Monitoring Program	
14. ETP : Effluent Treatment Plant	
15. FAT : Factory Acceptance Test	
16. g : Gram	
17. GCP : Good Clinical Practice	
18. GLP : Good Laboratory Practice	
19. GMP : Good Manufacturing Practice	
20. HAE : Hereditary Angioedema	
21. HBsAg : Hepatitis B surface antigen	
22. HEPA : High Efficiency Particulate Air	
23. HIV : Human Immunodeficiency Virus	
24. HSA : Human Serum Albumin	
25. HVAC : Heating, Ventilation, and Air Conditioning	
26. IEC : Ion Exchange Chromatography	
27. IgG : Immunoglobulin	
28. IPC : In Process Control	
29. IU : International Unit	
30. Mabs : Monoclonal Antibodies	
31. mL : Milliliter	
S/No. PARTICULARS BULK MANUFACTURING FACILITY	
1 Days per year	365 days
2 Effective working days per year (*)	300 days
3 Working days per week	7
4 Working hours per day	24
5 Working shifts per day	3
6 Working hours per shift	8
7 Effective working hours per shift (**)	6(**)
8 (-) Cleaning and change over	2
9 Effective production hours per day	18
(*) Effective working days per year should exclude: Public holidays, Scheduled Maintenance, Media fill & Re-qualification of equipment.	
(**) For calculations, 6 hours/shift has been taken as effective hours / shift considering two hours for cleaning, changeover, machine setting, etc. (*) Effective working days per year should exclude: Public holidays, Scheduled Maintenance, Media fill & Re-qualification of equipment.	
(**) For calculations, 6 hours/shift has been taken as effective hours / shift considering two hours for cleaning, changeover, machine setting, etc. (*) Effective working days per year should exclude: Public holidays, Scheduled Maintenance, Media fill & Re-qualification of equipment.	
(**) For calculations, 6 hours/shift has been taken as effective hours / shift considering two hours for cleaning, changeover, machine setting, etc.	
S/No. PARTICULARS BULK MANUFACTURING FACILITY	
1 Days per year	365 days
2 Effective working days per year (*)	260 days – 300 days
3 Working days per week	5 – 7
4 Working hours per day	16 – 24

5 | Working shifts per day | 2 - 3
6 | Working hours per shift | 8
7 | Effective working hours per shift (**) | 7-8(**)
8 | (-) Machine setup | 1 - 2
9 | (-) Cleaning and change over | 8
10 | Effective hours for filling line per shift | 6
(*) Effective working days per year should exclude: Public holidays, Scheduled Maintenance, Media fill & Re-qualification of equipment.
(**) Effective working hours should exclude break, operator gowning. Considering staggered break for critical equipment e.g., filling lines. | (*) Effective working days per year should exclude: Public holidays, Scheduled Maintenance, Media fill & Re-qualification of equipment.
(**) Effective working hours should exclude break, operator gowning. Considering staggered break for critical equipment e.g., filling lines. | (*) Effective working days per year should exclude: Public holidays, Scheduled Maintenance, Media fill & Re-qualification of equipment.
(**) Effective working hours should exclude break, operator gowning. Considering staggered break for critical equipment e.g., filling lines.
Company Name | Products | Turnover
Plasma Gen Biosciences Pvt.
Ltd. Bangalore www.plasmagen.com | PlasmaGlob, PlasmaGlob 10, PlasmaHep, PlasmaRAB, PlasmaRho-D I.V., AlbuMax-Neo,
Verbunin | Annual Revenue \$14.1M as on Mar 31, 2022
Reliance Life Sciences, Mumbai <https://www.rellife.com/> | AlbuRel (Albumin), HemoRel-A (Factor VIII), HepaRel (Hepatitis B immunoglobulin - HBIG), ImmunoRel (Immunoglobulin G),
ReliSeal (Fibrin Sealant) | Company revenue INR 6,804.80 lakh as on March 31, 2023
Biocon Limited, Bangalore <https://www.biocon.com/> | Albubet, Invex | Annual Revenue INR 115,501 million as on March 31, 2023
Intas Pharmaceuticals Ltd. Ahmedabad www.intaspharma.com/ | Albucel 20%, Globucel 10%, Albucel 5%, Factocel IX, Celestase 1, Globucel- VF, Globucel 5%, Factocel VIII, Pro-throm, Albucel LS | Annual Revenue INR. 6,753 crore in FY 2022-2023 (Domestic market)
Virchow Biotech, Hyderabad <http://virchowbiotech.com/> | V- IMMUNE, V – ALB | Annual revenue \$500 million
Fusion Healthcare, Hyderabad | Genexglob | Annual Revenue \$36.5M as on Mar 31, 2022
www.fusionhealthc.com/ | |
ICHOR Biologics Pvt. Ltd., Hyderabad <https://ichor.in/> | Hemalb (Human Albumin), Immuglo/Igamm (Human Normal Immunoglobulin), Clotnine (Human Coagulation factor IX), Clotate (Human Coagulation factor VIII), Fibroseal (Fibrin Sealant kit), Sphthromb (Thrombin), Fibrina (Fibrinogen), Ichorab 150/300, Hepchor 100/200, Ichor 300 | Company current revenue \$1.06 B
Bharat Serums and Vaccines Limited, Mumbai
<https://bsvgroup.com/> | Gamma IV, Bharglob, Berirab-P, Tetglob, Histoglob | Company revenue INR. 1,20,703.90 lakhs as on Mar 31, 2023
Company Name | Products
CSL Behring, USA <https://www.csl.com> | Beriglobin P, Berinert, Berinert 2000/3000, Berirab P, Evogam, HAEGARDA, Hepatitis B Immunoglobulin P, Hizentra, Intragram P, Intragram 10, Privigen, Rhophylac, Tetagam P, Tetanus Immunoglobulin-VF IM, Zoster Immunoglobulin-VF, AFSTYLA, Beriate, Beriplast, Beriplex P/N, Biostate, Cluviat, Cluvot, Confidex, Corifact, Factor X P Behring, Fibrogammin P, Haemate P, Haemocompletan P, Humate-P, IDELVION, KCENTRA, Kyberin, Kyberin P, MONOFIX-VF, Prothrombinex-VF, RiaSTAP, Tachosil, Thrombotrol VF, Voncento, Albapure, Albumeon, Albumex, Albuminar, Albumate, Alburex,
AlbuRx, Human Albumin
Pfizer, USA <https://www.pfizer.com/> | BeneFIX, Cutaquig, Panzyga
Kedron Biopharma, USA <https://www.kedron.com/> | Gammaplex, Gammaked, Ig Vena/Humaglobin liquid/
Kedrigamma/Venital, Subgam, Optiglobin, Naxiglo/Keycute, Ryplazim, Coagadex, Emolcot/Humaclo/Plasamclot/Emowil/Klott/Koate, gy, Optivate, Nuwiq, Wilfactin, Aimagix/Kedrix/Humafactor-9/Ixred, Replenine-VF, Rhogam/
Immunorho/Keyrho/Micrhogam Human anti D, D- gam, Immunohbs 180 IU, Uman albumin / Uman serum / Human albumin / Kedrialb / Albuplex /
Albital / Kedbumin / Albuked / Albumina LFB, Zenalb / Albuminex, Kedrab, Venbig / Keyvenb / Vebiked, Immunohbs / Uman big / Kedhbs, Tetanus gamma / TETIG, Uman complex / Kedcom / Pronativ, AT III Kedron /
Atked, Plasmasafe / Plasmagrade
ADMA Biologics, USA
<https://www.admabiologics.com> | ASCENIV, BIVIGAM, Nabi-HB
Takeda Pharmaceutical Company Limited <https://www.takeda.com/> | Flexbumin, Gammagard Liquid, Gammagard IGA, Cuvitru, Ceprotin, Glassia, Aralast NP, Immunate, Prothrombin NF 600, Cinryze
Biotest AG, Germany <https://www.biotest.com/> | Haemoctin SDH, Haemonine Varitect CP, Zutectra, Hepatect CP, Cytotect CP, Intratect, Yimmugo,
Pentaglobin, Albiomin, Biseko
Octa pharma, Switzerland <https://www.octapharma.com/> | Albunorm, Atenativ, Cutaquig, Fibryga, Gammanorm, Nuwiq, Octagam, Octanate, Octanine-F, OctaplasLG, Octaplex, Panzyga, Rhesonativ, Wilate Grifols, Spain <https://www.grifols.com/> | Albutein 5%, Albutein 20%, Anbinex, Factor IX Grifols, Fanhdi, Flebigamma DIF 100 mg/ml solution, Flebogamma DIF 50 mg/ml solution, Gamma Anti- Tetanus Grifols, Plasbumin 20%, Igamad, Igamplia,
Igantibe, Nivliva
Kamada Pharmaceuticals, Israel <https://www.kamada.com/> | Cytogam, Glassia, Hepagam B, KamRho D I.M., KamRho D I.V, KamRAB, Varizig, WinRho, Gammaplex, Hepatect CP, Intratect, Megalotect CP, Optivate 1000, Replenine-VF-1000, Varitect CP,
Zenalb, Zutectra
SK Plasma, South Korea <https://www.skplasma.com/en> | SK Albumin 20% inj., SK Albumin 5% Inj., Liv-Gamma SN Inj., IV Hepabulin SN Inj., Tetabulin SN Inj.
Prefilled syringe, SK Antithrombin III Inj. 500 I.U.,
ETBI Inj. 250 IU
Region | Revenue Share 2022(%)
North America | 53
Asia Pacific | 17
Europe | 22
Latin America | 5
MEA | 3
S/No. | Name of the product | Dosage form | Unit/ pack size | Ceiling price INR (Round of)
Stage 1 Products | Stage 1 Products | Stage 1 Products | Stage 1 Products | Stage 1 Products
1 | Albumin Injection | Injection 20% | 100 mL | 4000
2 | Coagulation factor IX | Powder | 600 IU | 14000.00

3 | Coagulation factor VIII | Powder | 250 IU | 4500.00
4 | Coagulation factor VIII | Powder | 500 IU | 10000.00
5 | Human normal immunoglobulin | Solution for infusion 5% | 1 mL | 200.00
6 | Human normal immunoglobulin | Solution for infusion 10% | 1 mL | 250.00
7 | Human normal immunoglobulin | Solution for infusion 16.5% | 1 mL | 550.00
8 | Fibrinogen | Powder | 1000 mg | 12,000
Stage 2 Products | Stage 2 Products | Stage 2 Products | Stage 2 Products | Stage 2 Products
9 | Diphtheria antitoxin | Injection | 10000 IU | 1500.00
10 | Hepatitis B immunoglobulin | Injection | Each Pack | 6200.00
11 | Anti-rabies immunoglobulin | Injection | 150 IU/mL
/1 mL | 3100.00
12 | Anti-Tetanus immunoglobulin | Injection | 300 IU/mL
/ 1 mL | 200.00
13 | Anti-Tetanus immunoglobulin | Injection | 250 IU Each pack | 180.00
4000 | 4000 | 4000 | 4000 | 4000 | |||||
4000 | 4000 | 4000 | 4000 | 4000 | |||||
4000 | 4000 | 4000 | 4000 | 4000 | |||||
4000 | 4000 | 4000 | 4000 | 4000 | |||||
4000 | 4000 | 4000 | 4000 | 4000 | |||||
4000 | 4000 | 4000 | 4000 | 4000 | 4500 | 4500 | 4500 | 4500 | ||||
4000 | 4000 | 4000 | 4000 | 4000 | |||||
100 mL
Injection 20%

Albumin Injection | 100 mL
Injection 20%

Albumin Injection | 100 mL
Injection 20%

Albumin Injection | 600 IU
Powder

Coagulation factor IX | 600 IU
Powder

Coagulation factor IX | 600 IU
Powder

Coagulation factor IX | 250 IU
Powder

Coagulation factor VIII | 250 IU
Powder

Coagulation factor VIII | 500 IU
Powder

Coagulation factor VIII | 500 IU
Powder

Coagulation factor VIII | 500 IU
Powder

Coagulation factor VIII | 1 mL
Solution for infusion 5%
Human IgG | 1 mL
Solution for infusion 10%
Human IgG | 1 mL
Solution for infusion 16.5%
Human IgG
S/No. | Product Name | Manufacturer | Pack Size | Pack Size | Cost (Rs.)
S/No. | Product Name | Manufacturer | Conc. | Volume | Cost (Rs.)
Human Albumin | Human Albumin | Human Albumin | Human Albumin | Human Albumin | Human Albumin
1 | Albumin –Albuprist - 5 | Pristyn Reviver | 5% | 100 mL | 2500
2 | Albumin – Vi- Alb | Virchow | 5% | 100 mL | 2500
3 | Albumin - Alburel | Reliance | 20% | 100 mL | 3800 to
4500
4 | Albumin - Alburel | Reliance | 20% | 50 mL | 2900
5 | Albumin - Albuprime | Intas | 20% | 100 mL | 3900
6 | Albumin - Albucel | Intas | 20% | 100 mL | 4500
7 | Albumin - Hetalbu | Hetero Healthcare | 20% | 100 mL | 4200
8 | Albumin -Flexalbumin | Baxter | 20% | 100 mL | 4000
9 | Albumin | Grifols | 20% | 100 mL | 4100
10 | Albumin – Uman Albumin | Kedroin | 20% | 100 mL | 3000
11 | Albumin – Hemalb | Ichor | 20% | 100 mL | 3300
12 | Albumin - Abivclor | Emcure | 20% | 100 mL | 5500
13 | Albumin - Albuzest | Alniche | 20% | 100 mL | 4900
14 | Albumin - Albumax | Dev Pharma | 20% | 100 mL | 3300
15 | Albumin - Flexalbumin | Baxter | 25% | 100 mL | 5000
Immunoglobulin (IgG) | Immunoglobulin (IgG) | Immunoglobulin (IgG) | Immunoglobulin (IgG) |
Immunoglobulin (IgG) | Immunoglobulin (IgG)
1 | Human IgG | Aizen Life | 5% | 100 mL | 10500
2 | Human IgG Immuno Rel | Reliance | 5% | 100 mL | 11500
3 | Human IgG Immuglo | Ichor | 5% | 100 mL | 9500
4 | Human IgG | Virchow | 5% | 100 mL | 6500
5 | Human IgG Globucel | Intas | 5% | 100 mL | 8000
6 | Human IgG ibivig | India bulls | 5% | 100 mL | 8900
7 | Human IgG - Ivnex | Biocon | 5% | 100 mL | 8950

8 | Human IgG Immuglo | Ichor | 5% | 100 mL | 9500
9 | Human IgG ImmunoRel | Reliance | 10% | 100 mL | 18000
10 | Human IgG Glublocel | Intas | 10% | 100 mL | 9000
Factor VIII | Factor VIII | Factor VIII | Factor VIII | Factor VIII | Factor VIII
1 | F VIII- Novo eight | Intas | 250 IU | - | 3500
2 | F VIII- Immunate | Baxter | 250 IU | - | 3100
3 | F VIII- Alphanate | Grifols | 250 IU | - | 3500
Factor IX | Factor IX | Factor IX | Factor IX | Factor IX | Factor IX
1 | F IX – Factocel IX | Intas | 600 IU | - | 8000
2 | F IX – Immuline | Baxter | 600 IU | - | 10000
Method | Purity
% | Advantage | Disadvantage | Clinical Mfg. | Large Scale Mfg. | Rec.
(%)

Plasma Fractionation | 96 | Low cost, easily accessibility, Bacteriostatic nature of process, ethanol is accessible and inexpensive, Safety of therapeutic product | Protein denaturation, Need raw material with high quality, Need to refrigerated tanks,
Flame proof facility and warehouse design for ethanol handling and storage | $\sqrt{\quad} | \sqrt{\quad} | \approx 57$
Cohn+ LC | 99 | Low cost, Quality of manufactured product is superior of Cohn method, High purity, High yield, Safety of therapeutic product | Possibility of protein denaturation and/or aggregation during the addition of ethanol, Flame proof facility and warehouse design for ethanol handling and storage | $\sqrt{\quad} | \sqrt{\quad} | \approx 59$
Heat Shock | 98 | High purity | Protein denaturation | $\sqrt{\quad} | \sqrt{\quad} | \approx 57$
Ammonium sulfate+ LC | N/A | Sufficient purity, Suitable for major laboratory approach, Yield of albumin: ≤ 40 g/liter of serum | Not suitable for major clinical approach, Separate processing areas required for Ammonium sulfate salts | $X | \sqrt{\quad} | \approx 63$
IEC | ≤ 95 | High yield, Adsorption capacity is high, better reproducibility, High recovery, Convenient for large-scale production | High media, Capital costs and Ligand leakage, Interactions between albumin and the ion exchange sorbents,
Slow separation | $\sqrt{\quad} | \sqrt{\quad} | \approx 66$
S/No. | Name of Equipment

1	Chromatography AKTA-1 (471 L/hr (Pipe MOC SS)
2	Chromatography AKTA-1 (500 L/hr pipe MOC SS)
3	Chromatography AKTA-1 (785 L/hr pipe MOC PP)
4	Chromatography AKTA-1 (1178 L/hr pipe MOC PP)
5	Chromatography AKTA-1 (864 L/hr pipe MOC PP)
6	Chromatography AKTA-1 (864 L/hr pipe MOC PP)
7	Chromatography AKTA-1 (667 L/hr pipe MOC PP)
8	Chromatography AKTA-1 (785 L/hr pipe MOC PP)
9	Chromatography AKTA-1 (30 L/hr pipe MOC PP)
10	Chromatography AKTA-1(160 L/hr pipe MOC PP)
11	Chromatography Column Dia 300mm x Height 500mm
12	Chromatography Column Dia 300mm x Height 750mm
13	Chromatography Column Dia 600mm x Height 500mm
14	Chromatography Column Dia 1000mm x Height 500mm
15	Chromatography Column Dia 450mm x Height 500mm
16	TFF System 40 sqm
17	Cassette 40 sqm
18	TFF System 30 sqm
19	Cassette 30 sqm
20	TFF System 20 sqm
21	Cassette 20 sqm
22	TFF System 5 sqm
23	Cassette 5 sqm
24	Continuous Centrifuge 1000L/hr
25	Process vessel 150L
26	Process vessel 250L
27	Process vessel (Mobile tank) 300L
28	Process vessel 1500L
29	Process vessel 2000L
30	Buffer Vessel 500L
31	Buffer Vessel 1000L
32	Depth filtration system
33	Nano filter holder
34	Cold room 2-8°C
35	Rectangular tank
36	Buffer preparation tank
37	Dispensing Booth
38	Analytical Balance
39	Precision Balance
40	Commercial Balance
41	PVC bag Sealer
42	Rotary Bag Sealer
43	pH/ Conductivity meter
44	Magnetic Stirrer
45	Peristaltic pump
46	Filter Integrity machine
47	Fume Hood
48	Autoclave 1000L
49	DHS 1000L
50	Ceiling LAF
51	Drying Oven
52	Acid tank 1000L
53	Base tank 1000L
54	Central CIP system
55	Refrigerator 200L
56	Laminar air flow cabinet
57	Dynamic pass box
58	Tumble Dryer 25 Kg
59	Barrier washer Extractor 25Kg
S/No.	I Name of Equipment

1 | Formulation Vessel 100L
 2 | Ceiling LAF
 3 | Laminar air flow cabinet
 4 | Continuous Particle Counter
 5 | Dynamic pass box
 6 | Pass Box static
 7 | Vial washing machine
 8 | Sterilizing tunnel
 9 | Vial filling & Sealing line 120 vials /min
 10 | Lyophilizer (2 sq meter)
 11 | Labelling machine
 12 | Double door autoclave 1000L
 13 | Cold room
 S/No. | Name of Equipment
 1 | Akta pure -150
 2 | Columns XK 50/100
 3 | Columns XK 50/60
 4 | Column XK 50/30
 5 | HPLC-UV
 6 | HPLC Columns and accessories
 7 | Electrophoresis (Minigel) with trans blot
 8 | Gel Doc system densitometer
 9 | Power packs
 10 | Endotoxin Detection system (Tube heating block)
 11 | Micropipettes- variable volume range of 2-20µL, 20-200µL, 100-1000µL, 500-5000µL and multichannel 30-300µL
 12 | Lab Refrigerator 2-8 ° C (SS) with data control
 13 | Flame photometer
 14 | Weighing Balance Analytical
 15 | UV-VIS spectrophotometer Double Beam with data acquisition.
 16 | Volumetric KF
 17 | Moisture Analyser
 18 | Stability Chambers (5±3° C, 25, 37 deg. C) 200L (1set)
 19 | Muffle Furnace
 20 | pH and conductivity meter with printer
 21 | Vacuum Oven
 22 | Melting point apparatus
 23 | Incubator Sterility Test 500L
 24 | Incubators for Normal Media plates Incubation 500L
 25 | Incubator for EMP and water samples 500L
 26 | Hot Air Oven
 27 | Thermostatic water bath (10-20 L)
 28 | Vacuum pump
 29 | Desiccator Glass
 30 | High Quality Laboratory water generation system (10 L /hr) Milli -Q water
 31 | Sonicator
 32 | Vortex Shaker
 33 | Magnetic Stirrer with hot plate
 34 | LAF Vertical
 35 | Biosafety Cabinet
 36 | Autoclave Double door
 37 | Dry Heat Steriliser
 38 | Fume Hood (Table Top)
 39 | Deep freezer (-20 Degrees C) with Data acquisition
 40 | Air sampler
 41 | Particle counter
 42 | Decontamination autoclave
 43 | Microscope Binocular
 44 | Colony Counter
 45 | Microplate reader ELISA
 46 | Microplate washer
 47 | TOC analyser (Grab Samples)
 48 | Timers/ Stop watch
 49 | Ice flaking machine
 50 | Standard Weight Box
 51 | Printer/scanner / xerox machine
 52 | Centrifuge table top
 53 | Size exclusion
 54 | Anion
 55 | Cation
 56 | Affinity
 S/No. | Name of Equipment
 1 | Anemometer
 2 | Lux meter
 3 | Datalogger 32 channel
 4 | Paper shredder
 5 | Laminator
 6 | Printer/scanner / xerox machine
 7 | Computers / Laptops
 8 | Numbering Machine
 9 | Spiral Binding Machine
 10 | A3 +A4 Lamination Machine
 11 | Bi-Focal Glass
 12 | Tachometer
 13 | Clamp Meter/ Multi meter
 14 | Level Indicator
 15 | Measuring TAPE
 16 | Stop Watch
 17 | Temperature recorders 24 Nos. (data Loggers)
 18 | Office Cabinets for Documents
 19 | Projector for conference / meeting room
 S/No. | Name of Equipment

1 | Analytical weighing Balance
 2 | RLA
 3 | Precision Balance
 4 | Commercial Balance
 5 | Refrigerator 200L
 6 | Deep freezer -20°C (1000L)
 7 | Air curtain
 8 | PVC pallets
 9 | Storage racks SS
 10 | Sampling rods
 11 | Liquid sampler
 12 | Dedusting tunnel
 13 | DPB
 14 | LAF cabinet
 S/No. | Name of Equipment
 1 | Air curtain
 2 | PVC pallets
 3 | Storage racks SS
 4 | Hydraulic trolley
 5 | Cold Rooms
 S/No. | Name of Equipment
 Clean Utilities | Clean Utilities
 1 | Soft water generation and storage system
 2 | Purified water generation storage and distribution system
 3 | Water for injection generation, storage and distribution system
 4 | Pure steam generator and distribution system
 Black Utilities | Black Utilities
 1 | Plant steam generation and distribution System
 2 | Chiller
 3 | Stand by Chiller
 4 | Compressed air generation and distribution system
 Effluent Treatment Plant (ETP) | Effluent Treatment Plant (ETP)
 1 | Biowaste reception tank 15000L
 2 | Kill tank 500L
 3 | Bar Screen
 4 | Collection tank
 5 | Equalization tank
 6 | Neutralization tank
 7 | Chemical coagulation
 8 | Biological oxidation tank
 9 | Filtration System
 Sewage Treatment Plant (STP) | Sewage Treatment Plant (STP)
 1 | Primary treatment plant: Bar screening - 3 stages, pH correction, BOD & COD removal secondary clarifier
 2 | Secondary treatment: Electro coagulation process, Flash mixer, HRSC clarifier, Sand filter
 3 | Sludge Management: Sludge thickener, Screw pump, Filter press, Sludge storage
 2200 2250 |||||
 2200 2250 |||||
 2200 2250 |||||
 2200 2250 |||||
 2200 2250 |||||
 2200 2250 |||||
 Item | Volume in Liters
 Plasma Storage capacity | 32000
 Plasma required per month | 30000
 Plasma required per 10 days | 10000
 Volume of plasma per bag | 0.2
 No. of bags required per 10 days | 50000
 No. of bags required per month | 150000
 Product | Protein availability
 (gm) | Protein Recovery | Conc. / vial | No. of vials /batch
 Albumin (gm) | 40,000 | 20000 | 20 GM | 900
 Human IgG (gm) | 12,500 | 6250 | 5 GM | 1000
 Factor VIII in IU | 10,00,000 | 12500 IU | 250 IU | 50
 Factor IX in IU | 10,00,000 | 90000 IU | 600 IU | 150
 Fibrinogen (gm) | 3,000 | 1500 | 70 mg | 80
 Thrombin in IU | 9,00,000 | 90000 | 1000 IU | 80
 Fibrinogen | 3000 | 1490 | 1000 mg | 1400
 Product | Protein availability | Protein Recovery | Conc. / Vial | No. of vials/year
 Albumin (gm) | 60,00,000 | 30,00,000 | 20 gm | 1,35,000
 Human IgG (gm) | 18,75,000 | 9,37,500 | 5 gm | 1,50,000
 Factor VIII in IU | 15,00,00,000 | 18,75,000 | 250 IU | 7,500
 Factor IX in IU | 15,00,00,000 | 13,500,000 | 600 IU | 22,500
 Fibrinogen (gm) | 4,50,000 | 2,25,000 | 70 mg | 12,000
 Thrombin in IU | 13,50,00,000 | 1,35,00,000 | 1000 IU | 12,000
 Fibrinogen | 450,000 | 225,000 | 1000 mg | 2,10,000
 Product | Protein availability (gm) | Protein Recovery (gm) | Conc. / Vial | No. of vials manufactured /batch
 Albumin | 1,36,000 | 68,000 | 20 gm | 3,060
 Human IgG | 42,500 | 21,250 | 5 gm | 3,400
 Factor VIII in IU | 34,00,000 | 8,16,000 | 250 IU | 170
 Factor IX in IU | 34,00,000 | 8,50,000 | 600 IU | 510
 Fibrinogen | 10,200 | 5,100 | 70 mg | 272
 Thrombin IU | 30,60,000 | 3,06,000 | 1000 IU | 272
 Fibrinogen | 10,200 | 5100 | 1000 mg | 4,760
 Product | Protein availability (gm) | Protein Recovery (gm) | Conc. / Vial | No. of vials manufactured /batch
 Albumin | 2,00,000 | 100,000 | 20 gm | 4,500
 Human IgG | 62,500 | 31,250 | 5 gm | 5,000
 Factor VIII in IU | 50,00,000 | 1,200,000 | 250 IU | 250

Factor IX in IU | 50,00,000 | 1,250,000 | 600 IU | 750
Fibrinogen | 15,000 | 7,500 | 70 mg | 400
Thrombin IU | 45,00,000 | 450,000 | 1000 IU | 400
Fibrinogen | 15,000 | 7,500 | 1000 mg | 7,000
Product | Protein
availability (gm) | Protein
Recovery (gm) | Conc. /
Vial | No. of vials /Year
Albumin | 2,00,00,000 | 1,00,00,000 | 20 gm | 450000
Human IgG | 62,50,000 | 31,25,000 | 5 gm | 500000
Factor VIII in IU | 50,00,00,000 | 12,00,00,000 | 250 IU | 25000
Factor IX in IU | 50,00,00,000 | 12,50,00,000 | 600 IU | 75000
Fibrinogen | 15,00,000 | 7,50,000 | 70 | 40000
Thrombin IU | 45,00,00,000 | 4,50,00,000 | 1000 IU | 40000
Fibrinogen | 15,00,000 | 7,50,000 | 1000 mg | 700,000
S/No. | Step | Purpose | Remark
1 | Cell culture and
Fermentation | To generate the desired bulk of
antibodies |
2 | Centrifugation | To remove cell debris, lipids, and clotted material |
3 | Filtration (0.45 µm | To remove smaller contaminating material |
4 | Concentration (UF) | To concentrate the desired antibodies in the material |
5 | Ion exchange Chromatography
(IEC) | To remove nucleic acids and endotoxins |
6 | Size exclusion chromatography (SEC) | | SEC is preferred over IEC because size is relatively constant
and charge on the protein may vary depending upon the pH as molecules are protonated and
deprotonated
7 | Immunoaffinity Chromatography | Antibody will bind to specific antigen attached to the agarose matrix
| It will help to achieve highest purity
8 | Sterile filtration | To remove microbial contamination |
Grade | Maximum Allowable Particle/m3 | Maximum Allowable Particle/m3 | Maximum Allowable Particle/m3 |
Maximum Allowable Particle/m3
Grade | During Rest | During Rest | During Operation | During Operation
Grade | ≥ 0.5µm | ≥ 5µm | ≥ 0.5µm | ≥ 5µm
A | 3500 | 0 | 3500 | 0
B | 35,000 | 0 | 350000 | 2000
C | 350000 | 2000 | 3500000 | 20000
D | 3500000 | 20000 | Not Defined | Not Defined
Grade | Air sample (CFU/m3) | Settle plates (diameter 90 mm) (CFU/4
hours) | Contact plates (diameter 55 mm)
(CFU/plate) | Glove print (5 fingers) (CFU/glove)
A | < 1 | < 1 | < 1 | < 1
B | 10 | 5 | 5 | 5
C | 100 | 50 | 25 | -
D | 200 | 100 | 50 | -
Rule 122 EA | Definitions
Rule 122F to 122N | License Application Procedure / Inspection / Reporting by Inspection Team / Grant or
Rejection of License / Duration of license / Appeal provision
License granted by Licensing Authority
Approval by Central License Approving Authority License is granted & delivered to Applicant
Rule 122 O | Procedure of Cancellation and Suspension of Licenses
Rule 122 P | Conditions of licenses:
Testing of whole blood / Component / Product Inspections Reporting to SLA / CLAA about changes in staff /
premise Directions issued by SLA / CLAA / Recall Directions Conditions for distribution of whole blood /
component / products
To comply provisions of the Act and Rules
Product | License | Renewal
Blood and Components | 27-C | 28-C 26-G
Blood Products | 27-E | 28-E 26-I
Annexure No | Annexure No | Description | Description
NA | NA | Project Cost Summary | Project Cost Summary
1 | 1 | Administration equipment and office furniture | Administration equipment and office furniture
2 | 2 | Canteen equipment | Canteen equipment
3 | 3 | Utility equipment | Utility equipment
4 | 4 | Plasma storage equipment | Plasma storage equipment
5 | 5 | Bulk plasma manufacturing equipment | Bulk plasma manufacturing equipment
6 | 6 | Fill finish equipment | Fill finish equipment
7 | 7 | QC / R&D equipment | QC / R&D equipment
8 | 8 | QA equipment | QA equipment
9 | 9 | Finished goods storage | Finished goods storage
10 | 10 | ETP / STP | ETP / STP
| Layouts | Layouts |
A | A | Plot Layout-GF (Bulk Manufacturing) | Plot Layout-GF (Bulk Manufacturing)
B | B | Conceptual Layout-GF (Bulk Manufacturing) | Conceptual Layout-GF (Bulk Manufacturing)
C | C | Conceptual Layouts-FF (Fill Finish) | Conceptual Layouts-FF (Fill Finish)
D | D | Terrace Floor (Future Provision) | Terrace Floor (Future Provision)
E | E | Section and Elevation Layout | Section and Elevation Layout
PROJECT COST SUMMARY | PROJECT COST SUMMARY | PROJECT COST SUMMARY | PROJECT COST SUMMARY | PROJECT COST
SUMMARY
Sr. No. | ITEMS | DESCRIPTION | COST IN INR | REMARKS
1 | SUB Head 1 | CIVIL CONSTRUCTION | |
1A | Building work (Civil) | MAIN BUILDING (GROUND - FIRST - TERRACE FLOOR)
APPROX. 174,000 sqft | 22,50,00,000 |
1B | | UTILITY AND OTHER SUPPORT BUILDINGS | 2,60,00,000 |
1C | | ADMINISTRATION BUILDING | 1,00,00,000 |
1D | | RAIN WATER HARVESTING | 6,00,000 |
1E | | EPOXY FLOORING (PRODUCTION / QC / R&D) | 2,00,00,000 |
| | | |
1F | Development work civil | ROADS & LANDSCAPING | 1,52,00,000 |
| | | |
2 | SUB Head II | EQUIPMENT | |
2A | ADMINISTRATION | ADMINISTRATION EQUIPMENT AND OFFICE FURNITURE | 50,80,000 | ANNEX 1

2B | CANTEEN | CANTEEN EQUIPMENT | 13,60,000 | ANNEX 2
 2C | UTILITIES | UTILITIES (EXCLUDING HVAC) | 5,75,00,000 | ANNEX 3
 2D | WAREHOUSE (PLASMA STORAGE) | PLASMA STORAGE EQUIPMENT | 2,01,22,000 | ANNEX 4
 2E | BULK PLASMA | BULK PLASMA PRODUCTION EQUIPMENT | 72,00,18,520 | ANNEX 5
 2F | FILL FINISH FACILITY | FILL FINISH EQUIPMENT | 3,26,13,000 | ANNEX 6
 2G | QUALITY CONTROL / R&D | QC/R&D EQUIPMENT | 3,96,94,000 | ANNEX 7
 2H | QUALITY ASSURANCE | QA EQUIPMENT | 31,07,500 | ANNEX 8
 2I | WAREHOUSE (FG) | FINISHED GOODS STORAGE | 21,50,000 | ANNEX 9
 3 | SUB Head III | ELECTRICAL WORK | |
 3A | ELECTRICAL | ELECTRICAL COST FOR ALL BLOCKS (including UPS) | 9,00,00,000 |
 3B | | TRANSFORMER - 3000 KVA | 65,00,000 |
 3C | | DG SET 1000 KVA x 2 nos | 1,40,00,000 |
 3D | INSTRUMENTATION | FIRE ALARM, CCTV SYSTEM | 3,00,00,000 |
 3E | MECHANICAL / HVAC | HVAC FOR ALL BLOCKS INCLUDING DUCTING AND CHILLERS | 11,00,00,000 |
 3F | | CLEAN ROOM MODULAR PARTITION (Bulk/ QC/ Warehouse / Plasma storage) | 9,00,00,000 |
 3G | | COLD ROOM (4 NOS) | 80,00,000 |
 3H | | BOILER & PIPING x 1 nos | 75,00,000 |
 3I | | AIR COMPRESSOR | 75,00,000 |
 3J | | UTILITY PIPING | 1,25,00,000 |
 3K | | KILL TANK with storage | 25,00,000 |
 3L | | ETP /STP | 33,00,000 | ANNEX 10
 3M | | SOLAR POWER EQUIPMENT | 90,00,000 |
 ||||
 4 | SUB Head IV | LICENSING AND PROFESSIONAL FEES | |
 4A | | TECHNOLOGY TRANSFER FEE (6 products) | 2,20,00,000 |
 4B | | PROJECT CONSULTANCY CHARGES | 1,50,00,000 |
	Grand Total:	1,60,62,45,020
	Add for third party quality assurance @ 1%	1,60,62,450
	Total:	1,62,23,07,470
	Add 3% contingencies	4,86,69,224
	Total:	1,67,09,76,694
	Add EPF, ESI @ 4.25%	7,10,16,510
	Total:	1,74,19,93,204
	Add labour cess @1%	1,74,19,932
	Grand Total	1,75,94,13,136
	Say	1,759
PROJECT COST SUMMARY - ANNEX 1	PROJECT COST SUMMARY - ANNEX 1	PROJECT COST SUMMARY - ANNEX 1
PROJECT COST SUMMARY - ANNEX 1		
ADMIN EQUIPMENT BUDGET ESTIMATE IN INR	ADMIN EQUIPMENT BUDGET ESTIMATE IN INR	ADMIN EQUIPMENT BUDGET ESTIMATE IN INR
1	Office Cabinets for Documents	20
2	Computers / Laptops	50
3	Office Furniture	NA
4	Printer/scanner / xerox machine	1
5	Water Dispenser	2
6	Projector for conference / meeting room	2
Total Amount in INR	Total Amount in INR	Total Amount in INR
PROJECT COST SUMMARY - ANNEX 2	PROJECT COST SUMMARY - ANNEX 2	PROJECT COST SUMMARY - ANNEX 2
PROJECT COST SUMMARY - ANNEX 2		
CANTEEN EQUIPMENT BUDGET ESTIMATE IN INR	CANTEEN EQUIPMENT BUDGET ESTIMATE IN INR	CANTEEN EQUIPMENT BUDGET ESTIMATE IN INR
S/No	Name of Equipment	Quantity
1	Refrigerator	2
2	SS furniture (serving table etc)	NA
3	SS Water cooler dispenser	1
4	Microwaves	2
5	Kitchenware	na
6	Bain marie	1
Total Amount in INR	Total Amount in INR	Total Amount in INR
PROJECT COST SUMMARY - ANNEX 3	PROJECT COST SUMMARY - ANNEX 3	PROJECT COST SUMMARY - ANNEX 3
PROJECT COST SUMMARY - ANNEX 3		
UTILITY BUDGET ESTIMATE IN INR	UTILITY BUDGET ESTIMATE IN INR	UTILITY BUDGET ESTIMATE IN INR
UTILITY BUDGET ESTIMATE IN INR		
S/No	Name of Equipment	Quantity
Clean Utilities		
1	Soft water generation 8000L/hour and storage system	1
2	Purified water generation 5000L/ hour storage tank 10KL and distribution system	1
3	Water for injection generation 1000L / hour, Storage tank 10KL and distribution system	1
4	Pure steam generator 1000Kg / hour and distribution system	1
Black Utilities		
1	Plant steam generation 2000Kg/Hour and distribution System	2
2	Compressed air generation and distribution system	2
3	PASSENGER LIFT X 2 NOS	2
4	MATERIAL LIFT X 3 NOS	3
 Total Amount in INR | Total Amount in INR | Total Amount in INR | 5,75,00,000.00
 PROJECT COST SUMMARY - ANNEX 4 | PROJECT COST SUMMARY - ANNEX 4 | PROJECT COST SUMMARY - ANNEX 4 |
 PROJECT COST SUMMARY - ANNEX 4
 PLASMA STORAGE EQUIPMENT BUDGET ESTIMATE IN INR | PLASMA STORAGE EQUIPMENT BUDGET ESTIMATE IN INR |
 PLASMA STORAGE EQUIPMENT BUDGET ESTIMATE IN INR | PLASMA STORAGE EQUIPMENT BUDGET ESTIMATE IN INR
 S/No | Name of Equipment | Quantity | Approx Cost (INR)
 1 | Analytical weighing Balance | 1 | 1,70,000.00
 2 | RLAf | 3 | 12,00,000.00
 3 | Precision Balance | 1 | 1,44,000.00
 4 | Commercial Balance | 1 | 1,20,000.00
 5 | Refrigerator 200L | 2 | 2,18,000.00
 6 | Deep freezer -20°C (1000L) | 32 | 1,44,00,000.00

7 | Air curtain | 10 | 1,00,000.00
 8 | PVC pallets | 50 | 3,00,000.00
 9 | Storage racks SS | 25 | 7,00,000.00
 10 | Sampling rods | 2 | 10,000.00
 11 | Liquid sampler | 2 | 20,000.00
 12 | Dedusting tunnel | 1 | 12,50,000.00
 13 | DPB | 4 | 6,00,000.00
 14 | LAF cabinet | 1 | 4,50,000.00
 15 | Computer / Laptops | 4 | 1,20,000.00
 16 | Office Furniture | NA | 2,00,000.00
 17 | Office Cabinets | 10 | 1,20,000.00
 Total Amount in INR | Total Amount in INR | Total Amount in INR | 2,01,22,000.00
 PROJECT COST SUMMARY - ANNEX 5 | PROJECT COST SUMMARY - ANNEX 5 | PROJECT COST SUMMARY - ANNEX 5 |
 PROJECT COST SUMMARY - ANNEX 5
 BULK PLASMA MANUFACTURING EQUIPMENT BUDGET ESTIMATE IN INR | BULK PLASMA MANUFACTURING EQUIPMENT BUDGET
 ESTIMATE IN INR | BULK PLASMA MANUFACTURING EQUIPMENT BUDGET ESTIMATE IN INR | BULK PLASMA MANUFACTURING
 EQUIPMENT BUDGET ESTIMATE IN INR
 S/No | Name of Equipment | Quantity | Approx Cost (INR)
 1 | Chromatography AKTA-1 (471 L/hr (Pipe MOC SS) | 1 | 1,71,00,000
 2 | Chromatography Column Dia 1000mm x Height 500mm | 2 | 4,46,12,200
 3 | Chromatography AKTA-1 (500 L/hr pipe MOC SS) | 1 | 1,71,00,000
 4 | Chromatography Column Dia 1000mm x Height 500mm | 1 | 2,23,06,100
 5 | Chromatography AKTA-1 (785 L/hr pipe MOC PP) | 1 | 2,00,31,420
 6 | Chromatography Column Dia 1000mm x Height 500mm | 1 | 2,23,06,100
 7 | Chromatography AKTA-1 (1178 L/hr pipe MOC PP) | 1 | 2,00,31,420
 8 | Chromatography Column Dia 1000mm x Height 500mm | 2 | 4,46,12,210
 9 | Chromatography AKTA-1 (864 L/hr pipe MOC PP) | 1 | 2,00,31,420
 10 | Chromatography Column Dia 1000mm x Height 500mm | 1 | 2,23,06,100
 11 | Chromatography AKTA-1 (864 L/hr pipe MOC PP) | 1 | 2,00,31,420
 12 | Chromatography Column Dia 1000mm x Height 500mm | 1 | 2,23,06,100
 13 | Chromatography AKTA-1 (667 L/hr pipe MOC PP) | 1 | 1,71,00,000
 14 | Chromatography Column Dia 1000mm x Height 500mm | 1 | 2,23,06,100
 15 | Chromatography Column Dia 300mm x Height 500mm | 1 | 48,96,570
 16 | Chromatography AKTA-1 (785 L/hr pipe MOC PP) | 1 | 2,00,31,420
 17 | Chromatography Column Dia 1000mm x Height 500mm | 1 | 2,23,06,100
 18 | Chromatography Column Dia 300mm x Height 500mm | 1 | 48,96,570
 19 | Chromatography AKTA-1 (30 L/hr pipe MOC PP) | 1 | 1,46,57,140
 20 | Chromatography Column Dia 300mm x Height 750mm | 1 | 48,96,570
 21 | Chromatography Column Dia 600mm x Height 500mm | 1 | 1,42,44,570
 22 | Chromatography AKTA-1(160 L/hr pipe MOC PP) | 1 | 1,46,57,140
 23 | Chromatography Column Dia 450mm x Height 500mm | 1 | 1,20,18,850
 24 | TFF System 40 sqm | 1 | 1,93,50,000
 25 | Cassette 40 sqm | 1 | 55,90,000
 26 | TFF System 30 sqm | 2 | 3,87,00,000
 27 | Cassette 30 sqm | 2 | 86,00,000
 28 | TFF System 20 sqm | 1 | 94,60,000
 29 | Cassette 20 sqm | 1 | 30,10,000
 30 | TFF System 5 sqm | 2 | 73,10,000
 31 | Cassette 5 sqm | 2 | 6,88,000
 32 | Continuous Centrifuge 1000L/hr | 2 | 3,00,00,000
 33 | Process vessel 150L | 3 | 32,25,000
 34 | Process vessel 250L | 4 | 51,60,000
 35 | Process vessel (Mobile tank) 300L | 2 | 25,80,000
 36 | Process vessel 1500L | 20 | 4,30,00,000
 37 | Process vessel 2000L | 6 | 1,54,80,000
 38 | Buffer Vessel 500L | 14 | 2,40,80,000
 39 | Buffer Vessel 1000L | 10 | 1,93,50,000
 40 | Depth filtration system | 2 | 21,00,000
 41 | Nano filter holder | 1 | 5,00,000
 42 | Cold room 2-8°C | 2 | 15,00,000
 43 | Rectangular tank | 1 | 4,30,000
 44 | Buffer preparation tank | 2 | 17,20,000
 45 | Dispensing Booth | 1 | 4,00,000
 46 | Analytical Balance | 1 | 1,70,000
 47 | Precision Balance | 1 | 1,50,000
 48 | Commercial Balance | 1 | 1,20,000
 49 | PVC bag Sealer | 1 | 5,000
 50 | Rotary Bag Sealer | 1 | 3,00,000
 51 | pH/ Conductivity meter | 2 | 3,00,000
 52 | Magnetic Stirrer | 2 | 1,50,000
 53 | Peristaltic pump | 2 | 2,00,000
 54 | Filter Integrity machine | 1 | 20,00,000
 55 | Fume Hood | 1 | 4,50,000
 56 | Autoclave 1000L | 2 | 40,00,000
 57 | DHS 1000L | 1 | 20,00,000
 58 | Ceiling LAF | 3 | 12,00,000
 59 | Drying Oven | 2 | 10,00,000
 60 | Acid tank 1000L | 2 | 38,70,000
 61 | Base tank 1000L | 2 | 38,70,000
 62 | Central CIP system | 2 | 21,50,000
 63 | Refrigerator 200L | 1 | 1,50,000
 64 | Laminar air flow cabinet | 1 | 4,35,000
 65 | Dynamic pass box | 4 | 4,80,000
 66 | Tumble Dryer 25 Kg | 1 | 20,00,000
 67 | Barrier washer Extractor 25Kg | 1 | 30,00,000
 68 | SS Furniture (Table, chair, racks, Ladders etc) | NA | 40,00,000
 69 | Lockers | NA | 2,00,000
 70 | Laptop / Computers | 10 | 5,00,000
 71 | Office furniture / Cabinets | NA | 3,00,000
 Total Amount in INR | Total Amount in INR | Total Amount in INR | 72,00,18,520
 PROJECT COST SUMMARY - ANNEX 6 | PROJECT COST SUMMARY - ANNEX 6 | PROJECT COST SUMMARY - ANNEX 6 |
 PROJECT COST SUMMARY - ANNEX 6

FILL FINISH EQUIPMENT | FILL FINISH EQUIPMENT | FILL FINISH EQUIPMENT | FILL FINISH EQUIPMENT

S/No | Name of Equipment | Quantit

y | Approx Cost (INR)

1 | Formulation Vessel 100L | 1 | 25,00,000.00

2 | Ceiling LAF | 2 | 8,00,000.00

3 | Laminar air flow cabinet | 1 | 4,35,000.00

4 | Continuous Particle Counter | 1 | 30,00,000.00

5 | Dynamic pass box | 7 | 8,40,000.00

6 | Pass Box static | 1 | 38,000.00

7 | Vial washing machine | 1 | 45,00,000.00

8 | Sterilizing tunnel | 1 | 45,00,000.00

9 | Vial filling & Sealing line 120 vials /min | 1 | 75,00,000.00

10 | Lyophilizer (1sq meter) | 1 | 55,00,000.00

11 | Labelling machine | 1 | 9,00,000.00

12 | Double door autoclave 1000L | 1 | 20,00,000.00

13 | Cold room | 1 | 8,00,000.00

14 | SS furniture (Table, Chair, racks, cabinets etc) | 1 | 2,00,000.00

Total Amount in INR | Total Amount in INR | Total Amount in INR | 3,26,13,000.00

PROJECT COST SUMMARY - ANNEX 7 | PROJECT COST SUMMARY - ANNEX 7 | PROJECT COST SUMMARY - ANNEX 7 |

PROJECT COST SUMMARY - ANNEX 7

QC & R&D EQUIPMENT | QC & R&D EQUIPMENT | QC & R&D EQUIPMENT | QC & R&D EQUIPMENT

EQUIPMENT BUDGET ESTIMATE IN INR | EQUIPMENT BUDGET ESTIMATE IN INR | EQUIPMENT BUDGET ESTIMATE IN INR |

EQUIPMENT BUDGET ESTIMATE IN INR

1 | Akta pure -150 (For R&D) | 2 | 40,00,000

2 | Columns XK 50/100 (For R & D) | 1 | 7,00,000

3 | Columns XK 50/60 (For R&D) | 1 | 7,00,000

4 | Column XK 50/30 (For R & D) | 3 | 14,00,000

5 | HPLC-UV | 1 | 34,00,000

6 | HPLC Columns and accessories | 1 | 10,00,000

7 | Electrophoresis (Minigel) with trans blot | 2 | 6,00,000

8 | Gel Doc system densitometer | 1 | 5,00,000

9 | Power packs | 2 | 2,00,000

10 | Endotoxin Detection system (Tube heating block) | 1 | 2,00,000

11 | Micropipettes- variable volume range of 2-20µL, 20-200µL, 100-

1000µL, 500-5000µL and multichannel 30-300µL | 2 set each | 3,00,000

12 | Lab Refrigerator 2-8 ° C (SS) with data control | 4 | 4,50,000

13 | Flame photometer | 1 | 4,50,000

14 | Weighing Balance Analytical | 2 | 3,50,000

15 | UV-VIS spectrophotometer Double Beam with data acquisition. | 1 | 9,00,000

16 | Volumetric KF | 1 | 7,00,000

17 | Moisture Analyser | 1 | 8,00,000

18 | Stability Chambers (5±3° C, 25, 37 deg. C) 200L (1set) | 3 | 14,00,000

19 | Muffle Furnance | 1 | 80,000

20 | pH and conductivity meter with printer | 2 | 2,60,000

21 | Vaccum Oven | 1 | 1,20,000

22 | Melting point apparatus | 1 | 90,000

23 | Incubator Sterility Test 500L | 2 | 14,40,000

24 | Incubators for Normal Media plates Incubation 500L | 1 | 3,00,000

25 | Incubator for EMP and water samples 500L | 2 | 6,00,000

26 | Hot Air Oven | 1 | 1,60,000

27 | Thermostatic water bath (10-20 L) | 2 | 2,00,000

28 | Vaccum pump | 2 | 1,20,000

29 | Desiccator Glass | 1 | 4,000

30 | High Quality Laboratory water generation system (10 L /hr) Milli -Q

water | 1 | 14,50,000

31 | Sonicator | 1 | 1,80,000

32 | Vortex Shaker | 6 | 2,10,000

33 | Magnetic Stirrer with hot plate | 4 | 4,00,000

34 | LAF Vertical | 2 | 6,00,000

35 | Biosafety Cabinet | 2 | 14,00,000

36 | Autoclave Double door | 1 | 12,50,000

37 | Dry Heat Steriliser | 1 | 7,50,000

38 | Fume Hood (Table Top) | 1 | 4,50,000

39 | Deep freezer (-20 Degrees C) with Data acquisition | 1 | 4,00,000

40 | Air sampler | 1 | 3,00,000

41 | Particle counter | 1 | 20,00,000

42 | Decontamination autoclave | 1 | 10,00,000

43 | Microscope Binocular | 1 | 1,00,000

44 | Colony Counter | 1 | 1,50,000

45 | Microplate reader ELISA | 1 | 6,00,000

46 | Microplate washer | 1 | 3,50,000

47 | TOC analyser (Grab Samples) | 1 | 25,00,000

48 | Timers/ Stop watch | 5 | 10,000

49 | Ice flaking machine | 1 | 2,50,000

50 | Standard Weight Box | 1 Set | 10,00,000

51 | Printer/scanner / xerox machine | 1 | 5,00,000

52 | Centrifuge table top | 1 | 4,50,000

53 | Size exclusion | L | 1,50,000.00

54 | Anion | L | 1,50,000.00

55 | Cation | L | 1,50,000.00

56 | Affinity | L | 1,50,000.00

57 | SS furniture racks | 25 | 2,50,000

58 | Computers / Laptops | 10 | 5,00,000

59 | Office Furniture | NA | 5,00,000

60 | Office Cabinets for Documents | 10 | 1,20,000

|| TOTAL | 3,96,94,000

PROJECT COST SUMMARY - ANNEX 8 | PROJECT COST SUMMARY - ANNEX 8 | PROJECT COST SUMMARY - ANNEX 8 |

PROJECT COST SUMMARY - ANNEX 8

QA EQUIPMENT BUDGET ESTIMATE IN INR | QA EQUIPMENT BUDGET ESTIMATE IN INR | QA EQUIPMENT BUDGET ESTIMATE

IN INR | QA EQUIPMENT BUDGET ESTIMATE IN INR

1 | Anemometer | 1 | 15,000.00

2 | Lux meter | 1 | 12,000.00

3 | Datalogger 32 channel | 1 | 4,00,000.00
4 | Paper shredder | 1 | 10,000.00
5 | Laminator | 1 | 10,000.00
6 | Printer/scanner / xerox machine | 1 | 5,00,000.00
7 | Computers / Laptops | 10 | 5,00,000.00
8 | Numbering Machine | 5 | 25,000.00
9 | Spiral Binding Machine | 1 | 15,000.00
10 | A3 +A4 Lamination Machine | 1 | 25,000.00
11 | Bi-Focal Glass | 1 | 80,000.00
12 | Tachometer | 2 | 6,000.00
13 | Clamp Meter/ Multi meter | 5 | 10,000.00
14 | Level Indicator | 2 | 8,000.00
15 | Measuring TAPE | 2 | 1,500.00
16 | Stop Watch | 5 | 50,000.00
17 | Temperature recorders 24 Nos. (data Loggers) | 20 | 2,00,000.00
18 | Office Cabinets for Documents | 20 | 2,40,000.00
19 | Projector for conference / meeting room | 1 | 3,00,000.00
20 | Office Furniture | NA | 7,00,000.00
Total Amount in INR | Total Amount in INR | Total Amount in INR | 31,07,500.00
PROJECT COST SUMMARY - ANNEX 9 | PROJECT COST SUMMARY - ANNEX 9 | PROJECT COST SUMMARY - ANNEX 9 |
PROJECT COST SUMMARY - ANNEX 9
FINISHED GOODS STORAGE | FINISHED GOODS STORAGE | FINISHED GOODS STORAGE | FINISHED GOODS STORAGE
1 | Air curtain | 10 | 1,00,000.00
2 | PVC pallets | 50 | 3,00,000.00
3 | Storage racks SS | 50 | 15,00,000.00
4 | Hydraulic trolley | 1 | 2,50,000.00
Total Amount in INR | Total Amount in INR | Total Amount in INR | 21,50,000.00
PROJECT COST SUMMARY - ANNEX 10 | PROJECT COST SUMMARY - ANNEX 10 | PROJECT COST SUMMARY - ANNEX 10 |
PROJECT COST SUMMARY - ANNEX 10
ETP BUDGET ESTIMATE IN INR | ETP BUDGET ESTIMATE IN INR | ETP BUDGET ESTIMATE IN INR | ETP BUDGET
ESTIMATE IN INR
S/No | Name of Equipment | Qty. | Approx Cost
(INR)
1 | Biowaste reception tank 15000L | 1 | 25,00,000.00
2 | Kill tank 500L | 2 | 25,00,000.00
3 | Bar Screen | NA | 25,00,000.00
4 | Collection tank | NA | 25,00,000.00
5 | Equalization tank | NA | 25,00,000.00
6 | Neutralization tank | NA | 25,00,000.00
7 | Chemical coagulation | NA | 25,00,000.00
8 | Biological oxidation tank | NA | 25,00,000.00
9 | Filtration System | NA | 25,00,000.00
STP BUDGET ESTIMATE IN INR | STP BUDGET ESTIMATE IN INR | STP BUDGET ESTIMATE IN INR | STP BUDGET
ESTIMATE IN INR
1 | Primary treatment plant : Bar screening - 3 stages, pH
correction, BOD & COD removal secondary clarifier | | 8,00,000.00
2 | Secondary treatment : Electro coagulation process, Flash
mixer, HRSC clarifier, Sand filter | | 8,00,000.00
3 | Sludge Management: Sludge thickner, Screw pump, Filter
press, Sludge storage | | 8,00,000.00
Total Amount in INR | Total Amount in INR | Total Amount in INR | 33,00,000.00
Figure No | Description
1 | Plot Layout-GF Bulk manufacturing
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16 | Recommended maximum allowable particles in clean rooms
17 | Recommended limits for microbial contamination in clean rooms

FEASIBILITY AND DETAILED PROJECT REPORT

Techno-Commercial Feasibility cum
Detailed Project Report for setting
up a Plasma Fractionation Facility
150,000 Liter / Annum Capacity
Document number:
LWIL/F-DPR/2023-00 Dated.15.12.2023

Prepared for:

Lifenity Wellness International Ltd,
12th Floor, D-Wing, Trade World, Kamala Mills Compound, Lower
Parel, Mumbai -400 013
Contact: Mr. T. Rajshekhar
Prepared by:

BioBridge Healthcare Solutions Pvt. Ltd,
#13, Rachana Blossom, Jagdishnagar, Aundh, Pune 411007
www.biobridge.in
Contact: Dr. Ganesh Kumraj
Email: ganesh@biobridge.in
Mob: +91 9004349016

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LIST OF ABBREVIATIONS:

1. AAT : Alpha-1 Antitrypsin
2. AIDS : Acquired Immunodeficiency Syndrome
3. BOD : Biological Oxygen Demand
4. CDSCO : Central Drug Standard Control Organization
5. cGMP : Current Good Manufacturing Practices
6. CO₂ : Carbon Dioxide
7. COD : Chemical Oxygen Demand
8. CPPs : Critical Process Parameters
9. CQA : Critical Quality Attributes
10. DCGI : Drug Controller General of India
11. EHS : Environment, Health, and Safety
12. EMA : European Medicine Agency
13. EMP : Environmental Monitoring Program
14. ETP : Effluent Treatment Plant
15. FAT : Factory Acceptance Test
16. g : Gram
17. GCP : Good Clinical Practice
18. GLP : Good Laboratory Practice
19. GMP : Good Manufacturing Practice
20. HAE : Hereditary Angioedema
21. HBsAg : Hepatitis B surface antigen
22. HEPA : High Efficiency Particulate Air
23. HIV : Human Immunodeficiency Virus
24. HSA : Human Serum Albumin
25. HVAC : Heating, Ventilation, and Air Conditioning
26. IEC : Ion Exchange Chromatography
27. IgG : Immunoglobulin
28. IPC : In Process Control
29. IU : International Unit

30. Mabs : Monoclonal Antibodies
31. mL : Milliliter

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32. NA : Not Applicable
33. NACO : National AIDS Control Organisation
34. NAT : Nucleic acid Amplification testing
35. NBTC : National Blood Transfusion Council
36. NGO : Non-Governmental Organization
37. NIB : National Institute of Biologicals
38. NPPA : National Pharmaceutical Pricing Authority
39. OOS : Out of Specification
40. OOT : Out of Trend
41. PCC : Prothrombin complex concentrate
42. PDMP : Plasma Derived Medicinal Products
43. PFD : Process Flow Diagram
44. PPQ : Process Performance Qualification
45. QA : Quality Assurance
46. QC : Quality Control
47. RM : Raw Material
48. SAT : Site Acceptance Test
49. SEC : Size exclusion chromatography
50. SBTC : State Blood Transfusion Councils
51. SOP : Standard Operating Procedure
52. SS : Stainless Steel
53. STD : Sexually Transmitted Disease
54. STP : Sewage Treatment Plant
55. TFF : Tangential Flow Filtration
56. TMP : Trans-Membrane Pressure
57. UPS Uninterrupted Power Supply
58. URS : User Requirement Specification
59. USD : United States Dollar
60. WHO : World Health Organization

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CHAPTER 1

INTRODUCTION

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1.1 Project Objective Statement:

The objective of this project is to establish and operate a state-of-the-art Human Plasma Derived Medicinal Products (HPDMP) manufacturing facility with an annual production capacity of 150,000 liters. This capacity will be subsequently increased to 500,000 liters per annum in the near future.

This facility aims to consistently and efficiently produce high-quality plasma derived products such as Human Serum Albumin (HSA), Human Normal Immunoglobulin (IgG), Hemophilia factors VIII, IX, Fibrin sealant etc. and Monoclonal antibodies (Mabs) while ensuring compliance with regulatory standards and best industry practices.

The primary goals of this project include:

Meeting Growing Demand: To address the increasing demand for plasma-derived products, ensuring a stable and reliable supply of essential treatments.

Quality Assurance: Implementing rigorous quality control and quality assurance processes to consistently produce safe and effective plasma derived products that meet or exceed regulatory requirements.

Operational Efficiency: Optimizing manufacturing processes to maximize productivity, minimize waste, and reduce operational costs to maintain competitiveness in the market.

Regulatory Compliance: Adhering to all relevant regulations and standards, including current Good Manufacturing Practices (cGMP), to ensure product safety, quality and efficacy.

Safety and Sustainability: Prioritizing the safety of both employees and the environment by implementing robust safety measures and eco-friendly practices throughout the facility.

Research and Development: Supporting ongoing research and development efforts to improve product quality, expand the product portfolio, and explore innovative techniques and newer technologies.

Supply Chain Reliability: Building a secure and efficient supply chain to source raw materials, transport finished products, and ensure timely deliveries to end users.

Stakeholder Satisfaction: Ensuring customer satisfaction by providing consistent and reliable plasma-derived products while maintaining a favorable relationship with employees, investors, and the local community.

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By achieving these objectives, this Human Plasma Derived Medicinal Products manufacturing facility will play a pivotal role in meeting the critical healthcare needs of patients while contributing to advancements in the plasma industry.

Statutory Regulation

(A detailed list of provided in the References section at the end of the document)

In general, the following guidelines are referred to:

- National Blood Policy, NBTC, Ministry of Health and Family Welfare.
- National Policy for Access to Plasma Derived Medicinal Products from Human Plasma for Clinical/Therapeutic Use: Addendum to National Blood Policy, NBTC, Ministry of

Health and Family Welfare.

- Technical manual, DGHS, MOHFW Govt. of India 2003.
- Drugs and Cosmetic act 1940 and rules thereunder.
- Schedule M Drugs and Cosmetic act 1940.
- ISPE Baseline Guide Vol. 6, Biopharmaceutical Manufacturing Facilities ISPE Baseline Guide Vol. 3, Sterile Manufacturing Facilities.
- WHO Good manufacturing practices for biological products.
- Good manufacturing practice guideline for human and veterinary medicinal product.
- PIC/s Guide to Good Manufacturing Practice for Medicinal Products.
- USFDA Guidance for Industry – Sterile Drug Products produced by Aseptic Processing.

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1.2 Executive Summary of The Project:

The Human Plasma Derived Medicinal Products manufacturing facility will be a green field project. The area earmarked for the project is approx. 6 acres in an industrial area on the outskirts of Nagpur city.

Different areas for the plasma manufacturing facility are identified and listed below:

- Bulk Manufacturing Facility (for processing 150,000 L / annum)
- Formulation, Fill Finish with lyophilization & Packaging
- QC and R&D block
- Administrative and QA Block
- Ware House for Plasma, Packaging, Testing lab (for 32,000 L plasma)
- Ware House (Finished Goods Storage)

Expansion area

- Bulk Manufacturing Facility (for processing additional 350,000 L / annum)
- Ware House for Plasma, Packaging, Testing lab (for additional 32,000 L plasma)

Supporting Blocks

- Canteen
- Engineering Office
- Boiler House
- Work Shop
- Substation and Generator
- Bore Well
- Bio-Kill System
- WTP/ETP/STP System
- Fire Fighting System
- Scrap Yard
- Guard House and Reception Room
- Security Cabin/Post
- Boundary Wall and Parking etc.
- Internal Roads and Drainage System
- Rainwater Harvesting System

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PROPOSED SITE PLAN

Annexure A

Figure 1: Plot Layout GROUND FLOOR

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Annexure B

Figure 2: Conceptual Layout – GF

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The company has identified different products to be manufactured in stagewise manner. List of products proposed to be manufactured are as follows:

Stage-1:

- Albumin
- Human Immunoglobulin
- Factor VIII
- Factor IX
- Fibrin sealant
- Fibrinogen

Stage-2:

- Anti-D
- Factor XI
- Factor XIII
- Hepatitis B IgG
- Measles IgG
- Protein C
- Rabies IgG
- Tetanus IgG
- Thrombin
- Varicella Zoster IgG

Table 1: Working Time Model Bulk Manufacturing

BULK MANUFACTURING

S/No. PARTICULARS

FACILITY

- 1 Days per year 365 days
 - 2 Effective working days per year (*) 300 days
 - 3 Working days per week 7
 - 4 Working hours per day 24
 - 5 Working shifts per day 3
 - 6 Working hours per shift 8
 - 7 Effective working hours per shift (**) ()
 - 6 **
 - 8 (-) Cleaning and change over 2
 - 9 Effective production hours per day 18
- (*) Effective working days per year should exclude: Public holidays, Scheduled Maintenance, Media fill & Re-qualification of equipment.
- (**) For calculations, 6 hours/shift has been taken as effective hours / shift considering two hours for cleaning, changeover, machine setting, etc.

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Table 2: Working Time Model Fill Finish

BULK MANUFACTURING

S/No. PARTICULARS

FACILITY

- 1 Days per year 365 days
 - 2 Effective working days per year (*) 260 days – 300 days
 - 3 Working days per week 5–7
 - 4 Working hours per day 16 – 24
 - 5 Working shifts per day 2-3
 - 6 Working hours per shift 8
 - 7 Effective working hours per shift (**) 7-8(**)
 - 8 (-) Machine setup 1-2
 - 9 (-) Cleaning and change over 8
 - 10 Effective hours for filling line per shift 6
- (*) Effective working days per year should exclude: Public holidays, Scheduled Maintenance, Media fill & Re-qualification of equipment.
- (**) Effective working hours should exclude break, operator gowning. Considering staggered break for critical equipment e.g., filling lines.

Environment Due Diligence:

As the proposed facility is required to handle biological products, proper segregation of type of wastes generated shall be adopted. Based on the categorization of the waste, appropriate treatment philosophy shall be adopted. Following broad categories are identified as wastes:

- Process waste
- General wash
- Solid Process waste
- General Solid wastes – Packaging materials etc.,
- Gaseous waste from Process & Labs
- Sewage

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The facility shall be designed as zero waste discharge plant; meaning that every possible waste within the facility is categorized, treated with appropriate treatment step in line local pollution control board guidelines and then discharged safely. Soil contamination due to waste is also considered as a key factor and necessary steps & waste handling methods are adopted to avoid the same. As part of the design, all biologically waste will be treated in Primary treatment systems before letting it into a common wastewater treatment plant. For the facility design, a detailed EHS document shall be prepared to identify risks and mitigation plans. During design, Construction & operation, EHS policy shall be made integral part of the sustainable operation.

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CHAPTER 2

MARKET ANALYSIS

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2.1 Human Plasma Derived Products and Its Share in The Total Components:

There are several medicinal products which are derived from human blood plasma by Plasma fractionation. The process for large scale purification was developed about 60 years ago by Cohn and co-workers. Therapeutically important proteins are Human serum albumin (HSA), Immunoglobulin G (IgG), Coagulation factor VIII, factor IX and Protease inhibitors. HSA is one of the most abundant proteins in plasma and, together with IgGs, constitutes 80% of total plasma proteins. According to the data, HSA represented approximately 50% of total sales of therapeutic plasma protein products. The purified HSA is required for the treatment of hypo proteinemia, burn and heavy loss of blood due to injuries and other medical conditions. Plasma contains about 60 g/L of protein, out of which about 57 grams are used for many therapeutic products. This makes this human material unique as a source of multiple therapeutic products.

The concentration of different plasma proteins is variable. HSA is major protein component, followed by IgG. The wide concentration range of plasma proteins (less than 1 mg/mL to 40 g/L) is one of the potential challenges in designing separation and purification procedures.

2.2 Volume of Human Plasma Fractionated Worldwide (2022 to 2032):

In 2010, 78 fractionation plants were operating in the world, 25 of which were in China, 26 in Europe and 8 in the United States of America. The world plasma fractionation capacity was about 48.4 million liters. More than 500 tons of human serum albumin (HSA) and more than 40 tons of intravenous immunoglobulins (IVIg) are produced annually from more than 34 million liters of plasma. Total plasma market of 2008 is depicted in pie chart (Figure 2).

Figure 3: Total plasma market of 2008 (Source: Marketing Research Bureau, Inc)

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In 2015-16, the worldwide plasma market grew to a \$21,174 Million, almost doubled in terms of revenue from 2008. As compared to the 2008 statistics, in 2016 to IVIg and SCIg had a major share in plasma market, increasing from 29% to 47%. The market share for Albumin also grew from 9.7% to 15.7%. The market share for Factor VIII and Factor IX remained almost the same of 1 to 2 % from 2008 to 2016.

Figure 4: Plasma market in 2016 (Source: Marketing Research Bureau, Inc)

Global Plasma Fractionation Market Forecasts from 2022 to 2032

- The Global Plasma Fractionation Market Size was valued at USD 27.1 Billion in 2022 (Fig 3).
- The Market Size is growing at a CAGR of 6.7% from 2022 to 2032.
- The Worldwide Plasma Fractionation Market Size is expected to reach USD 51.8 Billion by 2032.
- Asia Pacific region is expected to grow the fastest during the forecasted period.

Figure 5: Global Fractionation Market

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2.3 Companies in Domestic and International markets:

Key Market Players (Indian Manufacturers)

- Plasma Gen Biosciences Pvt. Ltd. Bangalore
- Reliance Life Sciences, Mumbai
- Biocon Limited Bangalore
- Intas Pharmaceuticals Ltd. Ahmedabad
- Virchow Biotech, Hyderabad
- Fusion Healthcare, Hyderabad
- ICHOR Biologics Pvt. Ltd., Hyderabad
- Bharat Serums and Vaccines Limited, Mumbai (Sales and marketing)

Table 3: Domestic companies' Plasma products

Company Name Products Turnover

Plasma Gen PlasmaGlob, PlasmaGlob Annual Revenue \$14.1M
Biosciences Pvt. 10, PlasmaHep, as on Mar 31, 2022

Ltd. Bangalore PlasmaRAB, PlasmaRho-D

www.plasmagen.com I.V., AlbuMax-Neo,

Verbumin

Reliance Life Sciences, AlbuRel (Albumin), Company revenue INR

Mumbai HemoRel-A (Factor VIII), 6,804.80 lakh as on March

<https://www.rellife.com/> HepaRel (Hepatitis B 31, 2023

immunoglobulin - HBIG),

ImmunoRel

(Immunoglobulin G),

ReliSeal (Fibrin Sealant)

Biocon Limited, Bangalore Albubet, Invex Annual Revenue INR

<https://www.biocon.com/> 115,501 million as on

March 31, 2023

Intas Pharmaceuticals Ltd. AlbuGel 20%, GlobuGel Annual Revenue

Ahmedabad 10%, AlbuGel 5%, FactoGel INR. 6,753 crore in FY

www.intaspharma.com/ IX, Celesterase 1, GlobuGel- 2022-2023 (Domestic market)

VF, GlobuGel 5%, FactoGel

VIII, Pro-throm, AlbuGel LS

Virchow Biotech, Hyderabad V- IMMUNE, V – ALB Annual revenue \$500

<http://virchowbiotech.com/> million

Fusion Healthcare, Genexglob Annual Revenue \$36.5M

Hyderabad as on Mar 31, 2022

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www.fusionhealthc.com/

ICHOR Biologics Pvt. Ltd., Hemalb (Human Albumin), Company current revenue

Hyderabad Immuglo/Igamma (Human \$1.06 B

Normal Immunoglobulin),

<https://ichor.in/>

Clotnine (Human

Coagulation factor IX),

Clotate (Human

Coagulation factor VIII),

FibroSeal (Fibrin Sealant

kit), Sphthromb (Thrombin),

Fibrina (Fibrinogen),

Ichorab 150/300, Hepchor

100/200, Ichor 300

Bharat Serums and Gamma IV, Bharglob, Company revenue INR.

Key market players (International Manufacturers):

- CSL Behring, USA
- Bio plasma limited, USA
- Pfizer, USA
- Kedrion Biopharma, USA
- ADMA Biologics, USA
- Takeda Pharmaceutical Company Limited
- Biotest AG, Germany
- Octa pharma, Switzerland
- Grifols, Spain
- Kamada Pharmaceuticals, Iseral
- SK Plasma, South Korea

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Table 4: International companies' Plasma products
Company Name Products
CSL Behring, USA Beriglobin P, Berinert, Berinert 2000/3000, Berirab P,
<https://www.csl.com> Evogam, HAEGARDA, Hepatitis B Immunoglobulin P,
Hizentra, Intragram P, Intragram 10, Privigen,
Rhophylac, Tetagam P, Tetanus Immunoglobulin-VF
IM, Zoster Immunoglobulin-VF, AFSTYLA, Beriate,
Beriplast, Beriplex P/N, Biostate, Cluviat, Cluvot,
Confidex, Corifact, Factor X P Behring, Fibrogammin
P, Haemate P, Haemocomplettan P, Humate-P,
IDELVION, KCENTRA, Kybernin, Kybernin
P, MONOFIX-VF, Prothrombinex-VF, RiaSTAP,
Tachosil, Thrombotrol VF, Voncento, Albapure,
Albumeon, Albumex, Albuminar, Albunate, Alburex,
AlbuRx, Human Albumin
Pfizer, USA BeneFIX, Cutaquig, Panzyga
<https://www.pfizer.com/>

Kedrion Biopharma, USA Gammaplex, Gammaked, Ig Vena/Humaglobin liquid/
<https://www.kedrion.com/> Kedrigamma/Venital, Subgam, Optiglobin,
Naxiglo/Keycute, Ryplazim, Coagadex,
Emolcot/Humaclo/Plasamclot/Emowil/Klott/Koate,
gy, Optivate, Nuwiq, Wilfactin,
Aimafix/Kedrifix/Humafactor-9/Ixed, Replenine-VF,
Rhogam/
Immunorho/Keyrho/Micrhogam Human anti D, D-
gam, Immunohbs 180 IU, Uman albumin / Uman
serum /
Human albumin / Kedrialb / Albuplex /
Albital / Kedbumin / Albuked / Albumina LFB, Zenalb
/ Albuminex, Kedrab, Venbig / Keyvenb / Vebiked,
Immunohbs / Uman big / Kedhbs, Tetanus gamma /
TETIG,
Uman complex / Kedcom / Pronativ, AT III Kedrion /
Atked, Plasmasafe / Plasmagrade
ADMA Biologics, USA ASCENIV, BIVIGAM, Nabi-HB
<https://www.admabiologics.com>
Takeda Pharmaceutical Company Flexbumin, Gammagard Liquid, Gammagard IGA,
Limited Cuvitru, Ceprotin, Glassia, Aralast NP, Immunate,
<https://www.takeda.com/> Prothrombin NF 600, Cinryze

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Biotest AG, Germany Haemoctin SDH, Haemonine
<https://www.biotest.com/> Varitact CP, Zutectra, Hepatect CP, Cytotect CP,
Intratect, Yimmugo,
Pentaglobin, Albiomin, Biseko
Octa pharma, Switzerland Albunorm, Atenativ, Cutaquig, Fibryga, Gammanorm,
<https://www.octapharma.com/> Nuwiq, Octagam, Octanate, Octanine-F, OctaplasLG,
Octaplex, Panzyga, Rhesonativ, Wilate

Grifols, Spain Albutein 5%, Albutein 20%, Anbinex, Factor IX Grifols,
<https://www.grifols.com/> Fanhdi, Flebigamma DIF 100 mg/ml Solution,
Flebogamma DIF 50 mg/ml solution, Gamma Anti-
Tetanus Grifols, Plasbumin 20%, Igamad, Igampia,
Igantibe, Nivliva
Kamada Pharmaceuticals, Israel Cytogam, Glassia, Hepagam B, KamRho D I.M.,
<https://www.kamada.com/> KamRho D I.V., KamRAB, Varizig, WinRho,
Gammaplex, Hepatect CP, Intratect, Megalotect CP,
Optivate 1000, Replenine-VF-1000, Varitact CP,
Zenalb, Zutectra
SK Plasma, South Korea SK Albumin 20% inj., SK Albumin 5% Inj., Liv-Gamma
<https://www.skplasma.com/en> SN Inj., IV Hepabulin SN Inj., Tetabulin SN Inj.
Prefilled syringe, SK Antithrombin III Inj. 500 I.U.,
ETBI Inj. 250 IU

Global plasma Market Overview:

Table 5: Plasma Derived Therapy Market Share, By Region,2022 *

Source: Precedence Research, Plasma derived market size, A report 2032

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Future roadmap for plasma fractionation industry in India:

Global plasma fractionation program, plasma protein therapies & newer clinical indications for use of these products, has led to a rapidly increasing demand.

Due to the limited volumes of plasma available for fractionation and fractionation capacities, has led to various challenges in availability of these products.

Unfortunately, plasma fractionation program did not get enough attention, focus and motivation in India because of challenges like availability of plasma, less components processing units, cold chain infrastructure, political will and availability of solid plasma ecosystem.

As per literature information, in 2020 India, wasted 6.5 Lakh units of blood and blood components on annual basis (Source: Print line Media Pvt Ltd). A report published by Ministry of Health and FW (2018-2019); the total collection of blood was 12.2 million units out of which 75.05% of blood was componentized by NACO supported blood component separating unit (BCSU). Thus, the wastage of 24.95% hampered the availability of blood components to the patients. Though, with the passage of time there has been a significant improvement in the overall blood banking systems and availability of plasma products in India, still much needs to be done considering clinical needs of the country.

There is a need to create a vision document for short-term, mid-term and long-term self-sufficiency of plasma products in India. This can be divided into following two categories:

Availability of safe plasma: How to overall increase the quantity and quality of plasma for fractionation:

- Increase voluntary blood collection, with increased repeat donor base.
- As a part of the licensing process blood component separation should be mandatory and a plan for current licensed blood banks to migrate to complete blood separation units in the next 3 years.
- Making blood component therapy mandatory in clinical practice, so that blood therapy can be minimized, and patients can use plasma products, which are relatively safer and easily accessible.
- Create a solid regulatory pathway for permitting source plasma collection.

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Regulatory system reform: Create a facilitative regulatory landscape for plasma industry to grow and evolve which would involve office of DCGI, NIB, Department of AIDS Control (DAC), IPC, NPPA, NACO, clinicians, patient groups, industry representatives and NGOs by;

- Forming a national level body for steering plasma protein initiative in India.
- Making a robust regulatory system for collection.
- Controlling pricing for plasma and increasing indigenous manufacturing capabilities.
- More funding for products and patient care.
- Fast track regulatory approval for importing different categories of plasma/fractions/intermediate.

2.4 Market Trend for The Next 10 Years and Shift in Products (2023-2032):

With the growth of IgG sales, some relatively niche plasma proteins have had a significant impact on the market in the 2010s.

1. C1 esterase inhibitor (C1-INH): In 2010, the sales of C1 esterase inhibitor (C1-INH) for the treatment of hereditary angioedema (HAE) amounted to about \$200 million globally. This was 1.5% of total plasma proteins sales because the product had just been launched in late 2008 in the United States, and European sales were limited by the product's approval confined to only a few countries. By 2016, the sales reached over \$1.1 billion (5.2% of global PDMPs sales), as the product gained rapid acceptance mainly in the United States where the awareness of treatment and diagnosis of HAE were boosted by Shire and CSL Behring. As a result, C1-INH became the fourth largest plasma protein in terms of dollar sales worldwide, behind IgG, albumin and plasma-derived factor VIII. The market for C1-INH is continuing to grow strongly, and it is likely to increase its share in the future. A new formulation for subcutaneous injection will further facilitate product adoption. In addition, C1-INH is under investigation for transplant where antibody mediated rejection (AMR) has occurred.

2. Alpha-1 Antitrypsin (AAT): Other specialty products which continue to grow at a stronger pace is prothrombin complex concentrate (PCC) as well as alpha-1 antitrypsin (AAT). Global sales of AAT reached \$1.0 billion in 2016, or 4.7% of the total market. This is up slightly from 4.1% of sales in 2010, as the market in the United States has sustained approximately a low two-digit annual sales growth for the past 6 years, driven mostly by new patient

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identification and higher dosing. Most of the global AAT sales are generated in the United States, though sales of this therapy in Europe are trending upward after approval through the central European Medicines Agency (EMA) in August 2015. AAT is also under investigation for new indications, such as graft vs. host disease after hematopoietic stem cell transplantation (HSCT) as well as type 1 diabetes. If either of these areas show clinical trial success, the market would be expected to experience higher demand for the therapy.

3. Prothrombin complex concentrate (PCC): PCC have nearly reached the \$500 million in sales mark in 2016, up from under \$300 million in 2010. This impressive growth was due to the introduction of the first four-factor (factors II, VII, IX and X) coagulation product in the United States by CSL Behring in 2013 for the urgent reversal of vitamin K antagonist (VKA) anticoagulation in adults with acute major bleeding. Sales of this product reached close to \$200 million within four years of product launch. Sales also increased in Europe for the same indication, while at present little PCC is used in other markets around the world. In the United States, the market share of the plasma-derived factor VIII sales fell from 11% in 2010 to 9% in 2016, while the share of plasma-derived factor IX products fell from 20% to 6%. The latter's market share drop was due to the faster adoption of the extended half-life factor IX products than those of factor VIII, because they offer longer intervals between infusions than the standard recombinant and plasma-derived products. Globally, the plasma-derived factor VIII market share went from 27% in 2010 to 21% in 2016. For, the plasma-derived factor IX' market share fell from 34% in 2010 to 21% in 2016. According to NPPA report of 2022, government has set the pricing for scheduled drugs and plasma products.

2.5 Government Market and the Rates:
A market survey and several documents and reports were studied to see the price of various plasma products in government and private sector respectively (Table 6 and Table 7).

Table 6: Pricing of Plasma Products (Government of India)

Name of the Unit/ Ceiling price
S/No. Dosage form
product pack size INR (Round of)
Stage 1 Products
1 Albumin Injection Injection 20% 100 mL 4000
2 Coagulation factor IX Powder 600 IU 14000.00
3 Coagulation factor VIII Powder 250 IU 4500.00
4 Coagulation factor VIII Powder 500 IU 10000.00
Human normal Solution for
5 1 mL 200.00
immunoglobulin infusion 5%
Human normal Solution for
6 1 mL 250.00
immunoglobulin infusion 10%
Human normal Solution for
7 1 mL 550.00
immunoglobulin infusion 16.5%
8 Fibrinogen Powder 1000 mg 12,000
Stage 2 Products
9 Diphtheria antitoxin Injection 10000 IU 1500.00
Hepatitis B
10 Injection Each Pack 6200.00
immunoglobulin
Anti-rabies 150 IU/mL
11 Injection 3100.00
immunoglobulin /1 mL
Anti-Tetanus 300 IU/mL
12 Injection 200.00
immunoglobulin / 1 mL
Anti-Tetanus 250 IU Each
13 Injection 180.00
immunoglobulin pack

@Source: NPPA (National Pricing Pharmaceutical Authority 2022)

Stage 1- Pricing of Plasma Products
15100
14100 14000
13100
12100
11100
10000
10100
9100
Price (INR)
8100
7100
6100
5100 4500
4000
4100
3100
2100
1100 200 250 550
100
100 mL 600 IU 250 IU 500 IU 1 mL 1 mL 1 mL
Injection 20% Powder Powder Powder Solution for Solution for Solution for
infusion 5% infusion 10% infusion 16.5%
Albumin Injection Coagulation Coagulation Coagulation Human IgG Human IgG Human IgG
factor IX factor VIII factor VIII

Figure 6: Stage 1 - Plasma products price

Stage 2- Price of Plasma Products	
7000	
6200	
6000	
5000	
Price (INR)	
4000	
3100	
3000	
2000	1500
1000	
200	180
0	
10000 IU Each Pack 150 IU/mL -1 mL 300 IU/mL - 1 mL 250 IU Each pack	
Injection Injection Injection Injection Injection	
Diphtheria antitoxin Hepatitis B Anti-rabies Anti-Tetanus Anti-Tetanus	
immunoglobulin immunoglobulin immunoglobulin immunoglobulin	

Figure 7: Stage 2 - Plasma products price (Government)

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Table 7: Price for Plasma Products (Private Manufacturers)

Pack Size	Cost
S/No. Product Name Manufacturer	
Conc. Volume (Rs.)	
Human Albumin	
1 Albumin –Albuprist - 5 Pristyn Reviver 5% 100 mL	2500
2 Albumin – Vi- Alb Virchow 5% 100 mL	2500
Albumin - Alburel 3800 to	
3 Reliance 20% 100 mL	4500
4 Albumin - Alburel Reliance 20% 50 mL	2900
5 Albumin - Albuprime Intas 20% 100 mL	3900
6 Albumin - Albucl Intas 20% 100 mL	4500
Albumin - Hetalbu Hetero	
7 20% 100 mL	4200
Healthcare	
8 Albumin -Flexalbumin Baxter 20% 100 mL	4000
9 Albumin Grifols 20% 100 mL	4100
Albumin – Uman	
10 Kedroin 20% 100 mL	3000
Albumin	
11 Albumin – Hemalb Ichor 20% 100 mL	3300
12 Albumin - Abivclor Emcure 20% 100 mL	5500
13 Albumin - Albuzest Alniche 20% 100 mL	4900
14 Albumin - Albumax Dev Pharma 20% 100 mL	3300
15 Albumin - Flexalbumin Baxter 25% 100 mL	5000
Immunoglobulin (IgG)	
1 Human IgG Aizen Life 5% 100 mL	10500
Human IgG	
2 Reliance 5% 100 mL	11500
Immuno Rel	
3 Human IgG Immuglo Ichor 5% 100 mL	9500
4 Human IgG Virchow 5% 100 mL	6500
5 Human IgG Globucel Intas 5% 100 mL	8000
6 Human IgG ibivig India bulls 5% 100 mL	8900
7 Human IgG - Ivnex Biocon 5% 100 mL	8950
8 Human IgG Immuglo Ichor 5% 100 mL	9500
9 Human IgG ImmunoRel Reliance 10% 100 mL	18000
10 Human IgG Globucel Intas 10% 100 mL	9000
Factor VIII	
1 F VIII- Novo eight Intas 250 IU -	3500
2 F VIII- Immunate Baxter 250 IU -	3100
3 F VIII- Alphanate Grifols 250 IU -	3500
Factor IX	
1 F IX – Factocel IX Intas 600 IU -	8000
2 F IX – Immuneine Baxter 600 IU -	10000

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Price for Plasma Products from Private Manufacturers	
16000	
14000	13500
12000	
10000	9200 9000

Price (INR)

8000

6000 5000

4100

4000 3400

2500

2000

0

Conc. 5% 20% 25% 5% 10% 250 IU 600 IU

Albumin Albumin Albumin Human IgG Human IgG Factor VIII Factor IX

Pack size : 100 mL

Figure 8: Prices for Plasma Products from Private Manufacturers

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CHAPTER 3

TECHNOLOGY

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3.1 Analysis of Various Technologies Available (Centrifugation and filtration for solid / Liquid Separation, Thermal and Chemical Inactivation Procedure, Ultrafiltration / Diafiltration, Nanofiltration, Chromatography or Any Other Technology or Any Hybrid Model)

Several technologies are utilized for plasma purification at commercial scale, each has its advantages, disadvantages and applications in isolating specific proteins of human interest. Some of the key methods are as under:

Methods for Plasma purification

1. Plasma Fractionation method with ethyl alcohol
2. Cohn+ Liquid Chromatography
3. Heat Shock technology
4. Ammonium sulphate+ Liquid Chromatography
5. Ion Exchange Chromatography (IEC)

1. Plasma Fractionation Method with ethyl alcohol:

Cohn and colleagues developed a protein purification method by using plasma fractionation technique for the first time. This approach is based on albumin solubility difference with other plasma proteins. As compared to other plasma proteins, albumin sediments are at lower pH and higher ethanol concentration. In this method there are five important parameters for separation of plasma protein.

- Ethyl alcohol (8- 40%)
- Temperature (-3 to -10 °C)
- pH (4.5-7.2)
- Protein concentration (5.1-.8 %)
- Ionic strength (0.14- 0.01).

Protein separation is based on the increase in ethyl alcohol concentration thus albumin will be seen in the last fraction.

The method is suitable for large-scale and industrial production.

Disadvantages of Method:

- Purity of the product
- Denaturation of protein

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2. The Cohn method combined with liquid chromatography:

The original Cohn method was modified by Oncley et al. in 1949 and Kistler-Nitschman et al. in 1962 in order to improve albumin purity.

The method is used for the purification of plasma proteins at the industrial scale

In this method after ethanol precipitation liquid chromatography step is added. Albumin is a by- product and other important therapeutic proteins including factor VIII, intravenous immunoglobulin (IVIG) and some other proteins are isolated from plasma.

Advantages:

- Increased purity of proteins
- Higher yields

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Process Flow Diagram -Modified Cohn Method

Cohn Modified

Cohn / Oncley Kistler / Nitschman

Fresh Frozen Plasma Fresh Frozen Plasma

Cryo precipitation Cryo precipitation

Cryo supernatant

Cryosupernatant

Affinity IEX

Chromatography

AT III

Fraction 1 precipitation AT III Factor IX

Fraction 1 precipitation

(8% Ethanol, pH=7.2) Prothrombin

(8% Ethanol, pH=7.2)

Fraction 1

Supernatant Precipitation A

(19% Ethanol, pH=5.85)

Supernatant A

IEX

α 1 PI

Albumin

Fraction II + III

precipitation

(20-25% Ethanol, pH=6.9)

Fraction IV- 1 precipitation Fraction IV precipitation

(18% Ethanol, pH=5.2) (40% Ethanol, pH=5.85)

α 1 PI

Fraction IV- 4 precipitation

(40% Ethanol, pH=5.8) Fraction C precipitation

(40% Ethanol, pH=4.8)

Fraction V precipitation

(40% Ethanol, pH=4.8)

Fraction D precipitation

(10% Ethanol, pH=4.6)

Extraction

(10% Ethanol, pH=4.5)

(40% Ethanol, pH=5.2)

Purified

Albumin

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3. Heat shock method:

Albumin is stable during thermal fluctuation comparing to the other proteins of the plasma; therefore, it can be purified by heat shock purification method.

Albumin protein resists to increasing temperature up to 60°C at which potential pathogen are inactivated. After inactivation 0.04 M caprylic acid is added to stabilize the protein at pH 5 at 60°C.

At this stage other proteins of the serum are denatured and precipitated in solution. The final product (albumin) is concentrated by precipitation and ultrafiltration with purity of app 98%

4. Ammonium Sulphate precipitation combined with liquid chromatography:

Another method for albumin purification is ammonium sulphate precipitation combined with liquid chromatography which yields approximately more than 90 percent purified albumin.

In this method albumin is separated from immunoglobulins by 50% ammonium sulphate.

Then lipids are removed using ice- cold acetone at 2-8°C.

For further purification of albumin from other contaminating proteins (especially transferrin which is the main contaminant, DNases and immunoglobulins), ion exchange and size exclusion chromatography are used.

5. Ion exchange chromatography (IEC):

IEC is widely used for albumin and other protein purification. Among them the anion exchange method has the most widely used. Three major steps of anion exchange consisting pre-treatment, purification and polishing.

The purification consists of two anion and cation exchange chromatography followed by polishing step comprising of two ultrafiltration and one gel filtration steps.

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Process Flow Diagram Purification and Polishing Methods for Human Serum

Albumin (HSA)

Euglobulin free

Step 1:

Pretreatment plasma

Anion Exchange

Step 2:

Purification

Cation Exchange

Ultrafiltration

Step 3: Polishing

Gel Filtration

Ultrafiltration

Albumin

The limitations of method:

- Long processing time

To overcome these limitations, the new method for ion exchange membrane chromatography (IEM) using ameliorated adsorbers was developed.

The membrane chromatography method is simple and has high separation efficiencies due to shorter diffusion times, reduction in buffer usage, small foot print requirements. No complex hardware or packing is required.

Disadvantage:

- Expensive membrane
- Low efficiency

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Table No 8: Comparison of Different Methods for Industrial Application

Large
Purity Clinical Rec.
Method Advantage Disadvantage Scale
(%) Mfg. %
Mfg.
Low cost, easily Protein denaturation,
accessibility, Need raw material
Plasma Fractionation

Bacteriostatic with high quality,
nature of Need to refrigerated
process, tanks,
96 ethanol is Flame proof facility $\sqrt{\sqrt{\approx 57}}$
accessible and and warehouse
inexpensive, design for ethanol
Safety of handling and storage
therapeutic
product
Low cost, Possibility of protein
Quality of denaturation and/or
manufactured aggregation during
product is the addition of
Cohn+ LC

superior of ethanol, Flame proof
Cohn method, facility and $\sqrt{\sqrt{\approx 59}}$
99
High purity, warehouse design for
High yield, ethanol handling and
Safety of storage
therapeutic
product
High purity Protein denaturation
Shock
Heat

98 $\sqrt{\sqrt{\approx 57}}$

Sufficient Not suitable for major
Ammonium sulfate+ LC

purity, clinical approach,
Suitable for Separate processing
major areas required for
laboratory Ammonium sulfate X $\sqrt{\sqrt{\approx 63}}$
N/A
approach, Yield salts
of albumin: $\leq$
40 g/liter of
serum
High yield, High media, Capital
Adsorption costs and Ligand
capacity is high, leakage, Interactions
better between albumin and
IEC

≤ 95 reproducibility, the ion exchange $\sqrt{\sqrt{\approx 66}}$
High recovery, sorbents,
Convenient for Slow separation
large-scale
production

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3.2 Recommending a Cost Effective and High Productivity Technology:
On the basis of table 8, for purity analysis and percentage recovery there are individual methods like fractionation and combined methods like Cohn + LC and IEC which can be considered.
Lifenity should accept the method which give high productivity and acceptable quality of

desired protein (purity levels) as per the regulatory expectations. Cohn method when combined with chromatographic techniques is the most well-known method. In this method other proteins of plasma such as coagulation factors, immunoglobulins etc. would be separated in upstream, and albumin will be separated in the last fraction. Due to the robustness and high purity of plasma in large volumes, this technique, is cost-effective for plasma therapeutic protein production. Hence, this method currently seems to be a major method for albumin production. However, taking into consideration, the latest developments in plasma fractionation technologies the detailed method along with the process flow diagram is recommended in section 4 of this document.

3.3 Obligations for technology service provider:

The selected Technology service provider shall have the following obligations:

Technology Transfer Plan:

To provide a detailed plan outlining the steps and timeline for the technology transfer. This plan shall include milestones, responsibilities, and contingencies. There should be a Lifenity steering committee to monitor the progress. All the deviations/changes should be approved by the committee.

Documentation:

To provide detailed documentation that describes the technology (Methods), equipment URS, buffer compositions, process steps, product specifications, and relevant data etc., This documentation is crucial for Lifenity to understand and replicate the technology.

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Checklist of the details to be provided to Lifenity by the technology service provider:

1. Manufacturing:

- RMs PPM and SPM and excipient specifications and source.
- Dispensing instructions.
- Detailed process flow diagrams.
- Process validation requirements.
- Batch Manufacturing record (BMR), SOPs, Master formulas.
- List of SOPs for operation of equipment and other relevant procedures.
- List of specifications and In-process controls (IPC).
- Process flow diagrams (PFD) along with description of chromatography and other critical process steps.
- Details of Equipment (Capacities, designs, qualification and placement with utilities requirements).
- Manufacturing implementation sequence/model with process time for each unit operation.
- Control strategy based on PPQ lots defining CPPs, CQAs and IPCs.
- Yield calculation.
- Cleaning procedures and validation.

2. Quality Control and Assurance:

- Requirements of different areas for QC activities.
- List of general and product related SOPs and STPs with specifications.
- Raw materials and Active starting materials control and release strategies.
- List of documentation for QC and QA activities.
- Procedures for batch release, recall and complaints handling by quality assurance department.
- Details of warehousing activities (Receipts of incoming RMS and active materials, sampling, Quarantine, testing and, release activities).
- Cold chain requirements in ware house storage areas for active materials and finished products.
- QC tests, STPs and specifications.

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- Validation of analytical and microbiology techniques.
- Stability protocols as per regulatory expectations.
- List of SOPs for operation of equipment and other relevant procedures.
- List of OOS, OOT and any other quality related events/incidents relevant to Lifenity.
- Details of primary packaging and secondary packaging materials with sources.

3. EHS:

- SOPs and STPs related to EHS

4. Training:

It is the responsibility of technology transfer partner to provide on job training and qualification of the staff for each unit operation. Lifenity is responsible to provide necessary resources for trainings.

Training will comprise of knowledge transfer sessions, hands-on training, and exchange of personnel.

5. Lab Scale Implementation:

Lifenity shall initiate the technology transfer at lab scale to test its feasibility and performance in a real-world setting before full-scale deployment. The technology service provider shall provide the lab scale technology.

6. Regulatory Compliance:

Technology provider to ensure that the provided technology and its implementation comply with relevant regulatory requirements and support necessary approvals from regulatory authorities.

7. Scale-Up and Commercialization:

Technology provider shall provide support for Scale up of the technology for commercial production, at least three batches of each product and deployment. This shall involve optimizing manufacturing processes, securing additional resources, and launching the manufactured product with the given technology into the market.

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8. Monitoring and Evaluation:

Provide continuous monitoring support at the production floor to Lifenity scientists for the performance of the transferred technology and conducting evaluations to assess its impact, identify areas of improvement and address any technology related issues that may arise.

9. Post-Transfer Support:

Providing ongoing support to address any challenges, optimize processes, and explore further opportunities.

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CHAPTER 4

MANUFACTURING

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4.1 As per the project implementation model, the following products shall be manufactured from 150,000L/Year of plasma.

- Albumin
- Factor VIII
- Factor IX
- Human Immunoglobulin (IgG)
- Fibrin sealant
- Fibrinogen

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4.2 Manufacturing Process for Deriving the Above Suggested Products Derived from Plasma

4.2.1 Process Flow Diagram- Common Fractionation Steps

PFD- 1

COMMON FRACTIONATION STEPS

POOLED PLASMA

FILTRATION

FRACTION TO

CHROMATOGRAPHY 11F

CHROMATOGRAPHY 1F

FRACTION TO

LOAD PREPARATION

CHROMATOGRAPHY 12E

FOR 2F

CHROMATOGRAPHY 2F

FRACTION 2 SENT TO

TANGENTIAL FLOW CHROMATOGRAPHY

FILTRATION 8N

CHROMATOGRAPHY 3F

IMPURITY REMOVAL

EUGLOBULIN

PRECIPITATION

CENTRIFUGATION

TO IVIG PROCESS

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PROCEDURE FOR COMMON FRACTIONATION STEPS:

P.1.F CUTTING AND POOLING

Approved plasma bags issued by QA are collected and thawed in a thawing tank with suitable provision for placement, retrieval and subsequent removal of thawing medium from the tank. The bags are checked for leakages after thawing and discarded if leakage is found. The thawed bags are cut under a bench type laminar flow and plasma collected in to suitable containers and finally pooled in process vessel to be taken for the next step.

P.2.F FILTRATION

Collected plasma after thawing is filtered to remove any suspended particles which may clog chromatography resins and column bed supports. Filtration is done using a depth filter which doesn't bind any proteins of interest. Filtered plasma is collected in a suitable tank with provisions to connect to a chromatography system for following process steps.

P.3.F CHROMATOGRAPHY – 1F

Filtered plasma is subjected to chromatography 1F to separate coagulation factor– F-VIII from rest of the fractions of interest. This step is essential to both separate F-VIII and also further polish plasma for other chromatography steps. Fraction of interest is collected using patterns on chromatograms, can be programmed for automatic collection.

P.4.F LOAD PREPARATION FOR 2F

pH of the collected eluate from Chromo – 1F, is adjusted to 7.0 and loaded on the next chromatography column.

P.5.F CHROMATOGRAPHY – 2F

pH adjusted eluate from Step 4 is subjected to chromatography -2F to separate F -IX from the remaining fractions. F-IX is collected separately for further process and remaining fractions are further processed.

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P.6.F TANGENTIAL FLOW FILTRATION (TFF) FOR CONCENTRATION

F-IX depleted sample is collected and concentrated by TFF. Trans-membrane pressures (TMP) are adjusted to achieve maximum permeate flow while minimizing fouling of the membrane. Automated TFF skids are used to precisely control and maintain the TMP.

P.7.F CHROMATOGRAPHY – 3F

Concentrated solution from Step-6 is subjected to chromatography-3F to separate buffer components from earlier steps and the sample is condition for further process steps. The main purpose of this step is to maintain suitable conditions for removal of other abundant protein impurities which may interfere with polishing chromatography steps of individual fractions.

P.8.F IMPURITY REMOVAL

pH of the eluate from Step-7 is adjusted to precipitate impurities from products of interest.

P.9.F EUGLOBULIN PRECIPITATION

Impurities like Euglobulin are precipitated using precise temperature and pH conditions. Temperature is maintained at less than 10 °C and pH less than 6.

P.10.F PRECIPITATE SEPERATION USING CENTRIFUGATION

The sample is passed through a continuous centrifuge to separate the precipitate. Precipitates are separated as a slurry and discarded. Supernatant is used for the subsequent steps.

4.2.2 Process Flow Diagram IVIG

IVIG PURIFICATION STEP

LOAD PREPARATION
FOR 4F

CHROMATOGRAPHY
4F FRACTION 2 SENT TO
CHROMATOGRAPHY 7A

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TANGENTIAL
FLOW FILTRATION

pH & CONDUCTIVITY
ADJUSTMENT

FILTRATION

CHROMATOGRAPHY
5I

pH ADJUSTMENT

CONDUCTIVITY
ADJUSTMENT

VIRAL INACTIVATION

FILTRATION

VIRAL INACTIVATION

CHROMATOGRAPHY
6I

TANGENTIAL
FLOW FILTRATION

FILLING TO VIALS

IVIG PURIFICATION PROCESS:

P.11.F LOAD PREPARATION FOR 4F

Conductivity of supernatant from Step # 10 is adjusted to facilitate favorable conditions for separation of IVIG and Albumin.

P.12.I CHROMATOGRAPHY – 4F

Conductivity adjusted sample is subjected to Chromatography 4F to separate Albumin from IVIG, this is the most critical chromatography step. Unlike in Cohn's process this gives a single step separation. The chromatography step is also operated in both flow through and elution modes to achieve the required separation. Albumin and IVIG are separated as different fractions after this step and collected in different process vessels. Individual fractions are processed separately to further purify and polished suitable for filling in glass vials.

P.13.I TANGENTIAL FLOW FILTRATION (TFF) FOR CONCENTRATION

IVIG fraction is collected and concentrated to about batch volume using TFF. Trans-membrane pressures (TMP) are adjusted to achieve maximum permeate flow while minimizing fouling of the membrane. Automated TFF skids can be used to precisely control and maintain the TMP.

P.14.I pH AND CONDUCTIVITY ADJUSTMENT

Both pH and conductivity of concentrated solution from step 13 is adjusted using NaOH and Acetic acid.

P.15.I FILTRATION

pH and conductivity adjusted, concentrated IVIG fraction is filtered using 1.2-micron filter to clarify, suitable for loading on next chromatography step.

P.16.I CHROMATOGRAPHY - 5I

Filtered IVIG sample is subjected to Chromatography 5I to further purify IVIG from other protein impurities. Impurities and IVIG are collected as separate fractions, impurities are discarded and IVIG is further processed.

P.17.I pH ADJUSTMENT

pH of the fraction collected from step 16 is adjusted with 1M Acetic acid for following viral inactivation step. This is ideally done in the same vessel in which the fraction is collected.

P.18.I CONDUCTIVITY ADJUSTMENT

Conductivity of collected fraction is adjusted. This is ideally done in the same vessel in which the fraction is collected and pH adjusted.

P.19.I LOW pH TREATMENT FOR VIRAL INACTIVATION

pH of the collected fraction is adjusted to 4.0 with 1M Acetic acid. This is held at pH 4.0 at 37 °C for 10 hours for viral inactivation. This step is most critical with regulatory and quality prospective.

P.20.I FILTRATION

Viral inactivated IVIG fraction is filtered using 1.2-micron filter to clarify, suitable for loading on next chromatography step.

P.21.I SOLVENT DETERGENT TREATMENT FOR VIRAL INACTIVATION - VERY CRITICAL STEP FOR REGULATORS

The solvent/detergent (S/D) pathogen inactivation method was developed for the inactivation of enveloped viruses in plasma-derived medicinal products. Currently, S/D treatment for pathogen reduction of plasma for transfusion and plasma-derived medicinal products is the most common, validated and robust industrial method for the virucidal treatment of blood products. Although many millions of therapeutic doses of S/D inactivated products have been used to date, there is no report on transmission of viral infections caused by enveloped viruses in recipients. The safety of the S/D treatment is also shown by the absence of reports of toxicity or antibodies against neoantigens after large- scale exposure of patients to several million transfused units.

1% TNBP and 1% Triton X are added calculated as final volume v/v. TNBP and Triton are prepared in a buffer before addition to the bulk of fraction. Stock solutions are prepared.

P.22.I CHROMATOGRAPHY - 6I – FINAL CHROMATOGRAPHY STEP IN IVIG PROCESS

Filtered IVIG sample is subjected to Chromatography 6I to effectively remove added solvents and detergents in P.21.1 This also works for removal of endotoxins in the solution. IVIG bound to resin at an Acidic pH is selectively eluted using elution buffers at pH 9.0 in a process called as pH gradient elution.

P.23.I pH ADJUSTMENT

pH of the eluate from Chromatography 6I step is adjusted to preserve structural stability of IVIG and provide more stable conditions.

P.24.I NANO FILTRATION – (OPTIONAL STEP)

Nano filtration further ascertains barrier mode removal of viral particles from IVIG solution. Plano make filters are used in the process.

P.25.I TANGENTIAL FLOW FILTRATION (TFF) FOR CONCENTRATION AND DIA FILTRATION

IVIG fraction is concentrated using TFF about 5% IVIG w/v concentration followed by buffer exchange by dia-filtration at the same TFF running conditions.

P.26.I TANGENTIAL FLOW FILTRATION (TFF) FOR CONCENTRATION

Dia-Filtered IVIG is concentrated to 6% w/v to facilitate final formulation at 5% w/v using the same TFF conditions. TFF conditions are maintained as in previous steps.

P.27.I FORMULATION

Buffered IVIG solution is formulated by adding 10% w/v pharmacopeial grade Mannitol. pH of this final solution is maintained at 4.2.

P.28.I PRE-FINAL FILTRATION

Filtration is done to decrease turbidity of the solution and also decrease microbial load on final sterile filtration step. This filtration is done using sandwich filter rated 1.2/0.6 micron.

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P.29.I STERILE FILTRATION

Sterile filtration is done to further reduce bio burden and the process is not claimed a sterile operation. Final terminal filtration is performed before fill-finish stage.

P.30.I FILLING IN TO VIALS

Washed, sterile and pyrogen free vials are used to fill bulk solution of IVIG.

4.2.3 Process Flow Diagram Albumin Purification:

ALBUMIN PURIFICATION STEPS

FRACTION FROM 4F
CHROMATOGRAPHY

CHROMATOGRAPHY
7A

TANGENTIAL
FLOW FILTRATION

HEAT TREATMENT

CENTRIFUGATION
AND FILTRATION

pH ADJUSTMENT

LOW pH INCUBATION

CONCENTRATION
AND DIAFILTRATION

FILLING TO VIALS

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ALBUMIN PURIFICATION PROCESS:

P.31.A CHROMATOGRAPHY – 7A

Albumin fraction from chromatography step 4F is used to load on Chromatography 7A and separation is achieved by binding and elution mode. This step is final chromatography step in Albumin process.

P.32.A TANGENTIAL FLOW FILTRATION (TFF) FOR CONCENTRATION

Eluate collected from Chromo-7A is concentrated to 6-7% w/v and collected in a vessel suitable for next process step.

P.33.A HEAT TREATMENT

Concentrated albumin solution is collected in a process vessel having control for temperature and time (57 degree C/ 3 hours). End point of completion of heat treatment is taken as stable optical density@ 600 nm. Heating of material results in a slightly turbid solution with suspended impurities. This solution is cooled to room temperature or even up to 10 °C for better separation. Chromatography is performed at room temperature.

P.34.A CENTRIFUGATION AND FILTRATION

Albumin solution after heat treatment is centrifuged and filtered to remove suspended proteinaceous contaminant particles formed during heat treatment. Continuous centrifuge can be used for this step.

P.35.A CONCENTRATION AND DIA FILTRATION

Above heat treated sample is Concentrated to 12% and Dia filtered.

P.36.A CAPRYLATE ADDITION AND pH ADJUSTMENT

Caprylate is added to concentrated and dia. filtered solution. The pH of the solution is adjusted to 4.5. An IPQC check is done to check actual caprylate concentration and if required, adjusted the concentration.

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P.37.A LOW pH INCUBATION

Solution is incubated at room temperature for about 7 hours followed by filtration to remove suspended particles.

P.38.A pH ADJUSTMENT

pH of the above solution is adjusted to 7.0

P.39.A CONCENTRATION AND DIA FILTRATION

Albumin solution is concentrated to formulation concentration and finally subjected to Pasteurization at 60 °C and finally filled.

4.2.4 Process Flow Diagram Factor IX

FACTOR IX PURIFICATION STEPS

FRACTION FROM 2F
CHROMATOGRAPHY

LOAD PREPARATION
FOR 8N

CHROMATOGRAPHY FRACTION TO
8N CHROMATOGRAPHY
10 T
LOAD PREPARATION
FOR 9N

CHROMATOGRAPHY
9N

CONC. AND
DIAFILTRATION

FILLING TO VIALS

LYOPHILISATION

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FACTOR IX PURIFICATION PROCESS:

P.40.N LOAD PREPARATION CHROMATOGRAPHY 8N

Eluate from chromatography step 2F is suitably adjusted for pH and Conductivity followed by concentration using TFF. Solvent and detergent are added to achieve viral inactivation, followed by filtration.

P.41.N CHROMATOGRAPHY 8N

Chromatography 8N is performed in adsorption mode and eluted fraction is collected in a separate process tank

P.42.N LOAD PREPARATION CHROMATOGRAPHY 9N

Eluate from chromatography step 8N is suitably adjusted for pH and conductivity followed by concentration using TFF.

P.43.N CHROMATOGRAPHY 9N

Chromatography is performed in adsorption mode and eluate fraction is collected in a separate process tank. This fraction is suitably concentrated and passed through a nano filter to achieve viral removal. This fraction is sterile filtered, formulated and filled in suitable vials.

4.2.5 Process Flow Diagram Thrombin

THROMBIN PURIFICATION STEPS

FRACTION FROM 8N
CHROMATOGRAPHY

LOAD PREPARATION
FOR 10T

CHROMATOGRAPHY
10T

CONC. AND
DIAFILTRATION

FILLING TO VIAL AND
LYOPHILIZATION

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THROMBIN PURIFICATION STEPS:

P.44.T LOAD FOR CHROMATOGRAPHY 10T

Load from chromatography 10T is obtained from fraction of chromatography 8N.

P.45.T CHROMATOGRAPHY 10T

Chromatography 10T is performed in adsorption mode and eluate fraction is collected in a separate process tank. This fraction is suitably concentrated and passed through a nano filter to achieve viral removal. This fraction is sterile filtered, formulated and filled in suitable vials.

4.2.6 Process Flow Diagram Fibrinogen:

FIBRINOGEN PURIFICATION STEPS

FRACTION FROM 1F
CHROMATOGRAPHY

LOAD PREPARATION

FOR 11F

CHROMATOGRAPHY
11F

CONC. AND
DIAFILTRATION

FILLING TO VIAL

LYOPHILISATION

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FIBRINOGEN PURIFICATION STEPS:

P.46.F LOAD FOR CHROMATOGRAPHY 11F
Load from chromatography 11F is obtained from fraction of chromatography 1F. The solution is precipitated and solubilized in suitable buffer solution.
P.47.F CHROMATOGRAPHY 11F Chromatography 11F is performed in adsorption mode and eluate fraction is collected in separate process tank. This fraction is suitably concentrated and passed through a nano filter to achieve viral removal. This fraction is sterile filtered, formulated and filled in suitable vials.

4.2.7 Process Flow Diagram Factor VIII

FACTOR VIII PURIFICATION STEPS

FRACTION FROM 1F
CHROMATOGRAPHY

LOAD PREPARATION FOR
12E

CHROMATOGRAPHY 12E

LOAD PREPARATION FOR
13E

CHROMATOGRAPHY 13E

CONCENTRATION AND
DIAFILTRATION

FILLING TO VIALS

LYOPHILISATION

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FACTOR VIII PURIFICATION PROCESS

P.48.E LOAD FOR CHROMATOGRAPHY 12E
Load from chromatography 12E is obtained from fraction of chromatography 1F, is filtered and checked for conductivity and pH, adjust pH and conductivity, if required before loading on the chromatography column.

P.49.E CHROMATOGRAPHY 12E
Chromatography 12E is performed in non-adsorption mode and flow through fraction is collected in a separate process tank.

P.50.E LOAD FOR CHROMATOGRAPHY 13E
Load for chromatography 13E is obtained from fraction of chromatography 12E, fraction is checked for conductivity and pH, adjust the conductivity and pH, if required before loading on the chromatography column.

P.51.E CHROMATOGRAPHY 13E
Chromatography 13E is performed in adsorption mode and eluted fraction is collected in a separate process tank. This fraction is suitably concentrated and passed through a nano filter to achieve viral removal. This fraction is sterile filtered, formulated and filled in vials.

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4.2.8 Manufacturing for Stage-2 Products (PFD)

ANTI D IgG

ANTI D IgG PURIFICATION STEPS

POOLED PLASMA

COMMON PLASMA
FRACTIONATION

FILTRATION

LOAD PREPARATION FOR
CHROMATOGRAPHY
(pH & Cond Adjusted)

ION EXCHANGE
CHROMATOGRAPHY

MATRIX RINSING
(WASH SOLUTION)

ELUTION OF ANTI D IgG

CONCENTRATION OF
ANTI D IgG

FILLING TO VIALS AND
LYOPHILIZATION

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Factor XI

FACTOR XI PURIFICATION STEPS

POOLED PLASMA

COMMON PLASMA
FRACTIONATION

FILTRATION

ANION EXCHANGE
CHROMATOGRAPHY

DEAE-SEPHADEX
CHROMATOGRAPHY

MEP HYPERCEL COLUMN

ELUTION OF
COAGULATION FACTOR XI

FILLING TO VIALS

LYOPHILISATION

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ANTI MEASLES

ANTI MEASLES ANTIBODY

POOLED HYPER IMMUNE
PLASMA

COMMON PLASMA
FRACTIONATION

FILTRATION

ABSORPTION WITH MPA

INCUBATION AND
CENTRIFUGATION

WASHING AND BUFFER
TREATMENT

CENTRIFUGATION AND
NEUTRALIZATION

CHROMATOGRAPHIC
PURIFICATION

FILLING TO VIALS

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TETANUS IMMUNOGLOBULIN

TETANUS IMMUNOGLOBULIN

POOLED HYPER IMMUNE
PLASMA

COLD ETHANOL

PRECIPITATION

FRACTION II+III/CAPRYLATE
PPTN/DEPTH FILTRATION

CARPYLATE INC/DEPTH
FILTRATION

ION EXCHANGE
CHROMATOGRAPHY

ULTRAFILTRATION/
DIAFILTRATION

NANOFILTRATION

ULTRAFILTRATION/
FINAL FORMULATION

STERILE
FILTRATION/FILLING

LOW pH STORAGE

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HEPATITIS-B IMMUNOGLOBULIN

HEPATITIS-B IMMUNOGLOBULIN

POOLED HYPER IMMUNE PLASMA

PRECIPITATION I

PRECIPITATION II AND
FILTRATION

PRECIPITATION III AND FILTRATION

DISSOLUTION AND CLARIFYING
FILTRATION

DIAFILTRATION

ANION EXCHANGE CHROMATOGRAPHY

INACTIVATION

CATION EXCHANGE CHROMATOGRAPHY

DIAFILTRATION / NANOFILTRATION

ADDITION OF STABILIZERS

pH ADJUSTMENT

STERILIZATION/ FILLING

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RABIES IMMUNOGLOBULIN

RABIES IMMUNOGLOBULIN

POOLED HYPER IMMUNE
PLASMA

CHECK THE RABIES
ANTIBODY TITRE

SEPERATION OF
FRACTION I,II AND III

SEPERATION OF
COMPONENT I AND III

PREPARATION OF
COMPONENT II

REFINING OF
COMPONENT II

ION EXCHANGE
CHROMATOGRAPHY

DIALYSIS AND
INACTIVATION

LOW pH INCUBATION

STERILE

FILTRATION/FILLING

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4.3 Equipment Required to Manufacture for the Proposed Capacity of 150,000 L / annum

4.3.1 Equipment's for Plasma Bulk Manufacturing Facility

S/No. Name of Equipment

- 1 Chromatography AKTA-1 (471 L/hr (Pipe MOC SS)
- 2 Chromatography AKTA-1 (500 L/hr pipe MOC SS)
- 3 Chromatography AKTA-1 (785 L/hr pipe MOC PP)
- 4 Chromatography AKTA-1 (1178 L/hr pipe MOC PP)
- 5 Chromatography AKTA-1 (864 L/hr pipe MOC PP)
- 6 Chromatography AKTA-1 (864 L/hr pipe MOC PP)
- 7 Chromatography AKTA-1 (667 L/hr pipe MOC PP)
- 8 Chromatography AKTA-1 (785 L/hr pipe MOC PP)
- 9 Chromatography AKTA-1 (30 L/hr pipe MOC PP)
- 10 Chromatography AKTA-1(160 L/hr pipe MOC PP)
- 11 Chromatography Column Dia 300mm x Height 500mm
- 12 Chromatography Column Dia 300mm x Height 750mm
- 13 Chromatography Column Dia 600mm x Height 500mm
- 14 Chromatography Column Dia 1000mm x Height 500mm
- 15 Chromatography Column Dia 450mm x Height 500mm
- 16 TFF System 40 sqm
- 17 Cassette 40 sqm
- 18 TFF System 30 sqm
- 19 Cassette 30 sqm
- 20 TFF System 20 sqm
- 21 Cassette 20 sqm
- 22 TFF System 5 sqm
- 23 Cassette 5 sqm
- 24 Continuous Centrifuge 1000L/hr
- 25 Process vessel 150L
- 26 Process vessel 250L
- 27 Process vessel (Mobile tank) 300L
- 28 Process vessel 1500L

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- 29 Process vessel 2000L
- 30 Buffer Vessel 500L
- 31 Buffer Vessel 1000L
- 32 Depth filtration system
- 33 Nano filter holder
- 34 Cold room 2-8°C
- 35 Rectangular tank
- 36 Buffer preparation tank
- 37 Dispensing Booth
- 38 Analytical Balance
- 39 Precision Balance
- 40 Commercial Balance
- 41 PVC bag Sealer
- 42 Rotary Bag Sealer
- 43 pH/ Conductivity meter
- 44 Magnetic Stirrer
- 45 Peristaltic pump
- 46 Filter Integrity machine
- 47 Fume Hood
- 48 Autoclave 1000L
- 49 DHS 1000L
- 50 Ceiling LAF
- 51 Drying Oven
- 52 Acid tank 1000L
- 53 Base tank 1000L
- 54 Central CIP system
- 55 Refrigerator 200L
- 56 Laminar air flow cabinet
- 57 Dynamic pass box
- 58 Tumble Dryer 25 Kg
- 59 Barrier washer Extractor 25Kg

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4.3.2 Equipment's for Fill Finish Production facility

S/No. Name of Equipment

- 1 Formulation Vessel 100L
- 2 Ceiling LAF
- 3 Laminar air flow cabinet
- 4 Continuous Particle Counter
- 5 Dynamic pass box
- 6 Pass Box static
- 7 Vial washing machine
- 8 Sterilizing tunnel
- 9 Vial filling & Sealing line 120 vials /min

10 Lyophilizer (2 sq meter)
11 Labelling machine
12 Double door autoclave 1000L
13 Cold room

4.3.3 Equipment's for QC / R&D

S/No. Name of Equipment

1 Akta pure -150
2 Columns XK 50/100
3 Columns XK 50/60
4 Column XK 50/30
5 HPLC-UV
6 HPLC Columns and accessories
7 Electrophoresis (Minigel) with trans blot
8 Gel Doc system densitometer
9 Power packs
10 Endotoxin Detection system (Tube heating block)
Micropipettes- variable volume range of 2-20µL, 20-200µL, 100-1000µL, 500-
11
5000µL and multichannel 30-300µL
12 Lab Refrigerator 2-8 ° C (SS) with data control

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13 Flame photometer
14 Weighing Balance Analytical
15 UV-VIS spectrophotometer Double Beam with data acquisition.
16 Volumetric KF
17 Moisture Analyser
18 Stability Chambers (5±3° C, 25, 37 deg. C) 200L (1set)
19 Muffle Furnace
20 pH and conductivity meter with printer
21 Vacuum Oven
22 Melting point apparatus
23 Incubator Sterility Test 500L
24 Incubators for Normal Media plates Incubation 500L
25 Incubator for EMP and water samples 500L
26 Hot Air Oven
27 Thermostatic water bath (10-20 L)
28 Vacuum pump
29 Desiccator Glass
30 High Quality Laboratory water generation system (10 L /hr) Milli -Q water
31 Sonicator
32 Vortex Shaker
33 Magnetic Stirrer with hot plate
34 LAF Vertical
35 Biosafety Cabinet
36 Autoclave Double door
37 Dry Heat Steriliser
38 Fume Hood (Table Top)
39 Deep freezer (-20 Degrees C) with Data acquisition
40 Air sampler
41 Particle counter
42 Decontamination autoclave
43 Microscope Binocular

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44 Colony Counter
45 Microplate reader ELISA
46 Microplate washer
47 TOC analyser (Grab Samples)
48 Timers/ Stop watch
49 Ice flaking machine
50 Standard Weight Box
51 Printer/scanner / xerox machine
52 Centrifuge table top
53 Size exclusion
54 Anion
55 Cation
56 Affinity

4.3.4 Equipment for QA

S/No. Name of Equipment

1 Anemometer
2 Lux meter
3 Datalogger 32 channel
4 Paper shredder
5 Laminator
6 Printer/scanner / xerox machine
7 Computers / Laptops
8 Numbering Machine
9 Spiral Binding Machine
10 A3 +A4 Lamination Machine
11 Bi-Focal Glass
12 Tachometer
13 Clamp Meter/ Multi meter
14 Level Indicator

15 Measuring TAPE

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- 16 Stop Watch
- 17 Temperature recorders 24 Nos. (data Loggers)
- 18 Office Cabinets for Documents
- 19 Projector for conference / meeting room

4.3.5 Equipment's for Plasma storage

- S/No. Name of Equipment
- 1 Analytical weighing Balance
 - 2 RLAf
 - 3 Precision Balance
 - 4 Commercial Balance
 - 5 Refrigerator 200L
 - 6 Deep freezer -20°C (1000L)
 - 7 Air curtain
 - 8 PVC pallets
 - 9 Storage racks SS
 - 10 Sampling rods
 - 11 Liquid sampler
 - 12 Dedusting tunnel
 - 13 DPB
 - 14 LAF cabinet

4.3.6 Equipment in Finished Goods Storage

- S/No. Name of Equipment
- 1 Air curtain
 - 2 PVC pallets
 - 3 Storage racks SS
 - 4 Hydraulic trolley
 - 5 Cold Rooms

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4.3.7 Equipment in Eng. / Critical System

- S/No. Name of Equipment
- Clean Utilities
- 1 Soft water generation and storage system
 - 2 Purified water generation storage and distribution system
 - 3 Water for injection generation, storage and distribution system
 - 4 Pure steam generator and distribution system
- Black Utilities
- 1 Plant steam generation and distribution System
 - 2 Chiller
 - 3 Stand by Chiller
 - 4 Compressed air generation and distribution system

Effluent Treatment Plant (ETP)

- 1 Biowaste reception tank 15000L
 - 2 Kill tank 500L
 - 3 Bar Screen
 - 4 Collection tank
 - 5 Equalization tank
 - 6 Neutralization tank
 - 7 Chemical coagulation
 - 8 Biological oxidation tank
 - 9 Filtration System
- Sewage Treatment Plant (STP)
- Primary treatment plant: Bar screening - 3 stages, pH correction, BOD & COD
- 1 removal secondary clarifier
- Secondary treatment: Electro coagulation process, Flash mixer, HRSC clarifier,
- 2 Sand filter
 - 3 Sludge Management: Sludge thickener, Screw pump, Filter press, Sludge storage

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4.4 Detailed design

This document describes the strategic approach to the facility design, technology basis and products selected in stage 1 (150,000 L / annum) and Stage 2 (500,000 L / annum)

The site is situated on the outskirts of Nagpur city spread over 6 acres.

The process design shall be based on Chromatography system-based separation.

It will be dedicated to manufacture of Human plasma fractions which include the following products in Stage 1

- Albumin
- Human Immunoglobulin
- Factor VIII
- Factor IX
- Fibrin sealant
- Fibrinogen

The plasma processing capacity in Stage 1 will be 150,000 L / annum

The facility design shall have scope of increasing the plasma processing capacity to 500,000 L

/ annum.

Besides the products identified in stage 1 the following products have been identified in Stage

2. An appropriate Technology partner shall be identified for providing the process know-how.

- Anti-D
- Factor XI
- Factor XIII
- Hepatitis B IgG
- Measles IgG
- Protein C
- Rabies IgG
- Tetanus IgG
- Thrombin
- Varicella Zoster IgG

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The plant is designed as a monobloc facility which will be constructed as a two-floor facility

The ground floor shall contain the following operational units

- a. Bulk plasma fractionation unit (150,000 L/ annum)
- b. Warehouse Plasma storage (32000 L)
- c. Admin / QA / Reception
- d. Warehouse (RM/PPM/SPM/ Finished goods)
- e. Electrical transformer area
- f. Engineering workshop
- g. Utility block
- h. Raw water storage tank
- i. Borewell
- j. Boiler room
- k. Chiller room
- l. Kill tank system
- m. Scrap yard

FUTURE EXPANSION AREA

- i. Bulk plasma fractionation unit (350,000 L/ annum)
- ii. Warehouse Plasma storage (32000 L)

The first floor shall contain the following areas

1. Canteen area
2. Water system and engineering store
3. SPM store
4. Visual inspection
5. Label store
6. Change rooms
7. Fill finish with Lyo area
8. Quality Control area
9. Service area (for AHU's)

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CHAPTER 5

PROPOSED SITE PLAN

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5.1 Concept design:

Figure 9: Conceptual Layout for Site Master Plan

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5.2 Site Master Plan

Conceptual Layout Ground Floor (Bulk Manufacturing):

The bulk manufacturing area is designed in a manner to provide unflow of man, material and waste from the facility to avoid mix-ups and cross contamination.

All the ancillary functions are located on the ground floor and separated from bulk manufacturing area.

In the ground floor following facilities are present:

1. Bulk manufacturing facility -01
2. Bulk manufacturing facility -02 (Future provision)
3. Quarantine Plasma Storage/Testing lab/Ware house
4. Future provision for plasma storage
5. Finished goods storage & dispatch
6. Administrative Block
7. Utility Block
8. Workshop
9. DG Set and Transformer area
10. UG collection tank, ETP/STP & Kill Tank
11. Bore well, UG 500KL storage tanks
12. Chiller and Boiler rooms
13. UG SUMP

Bulk Manufacturing Block

Change Rooms: Bulk manufacturing facility-01, personnel entering the processing area will

undergo a series of two dress changes applicable to clean rooms of different classifications. The separate change rooms are provided for male and Females.
Locker Room: This shall be the first change where the personnel remove their street shoes and clothes and keep these belongings in the provided double door lockers.
For the exit from facilities, separate change rooms are provided.
In exit change room personnel shall remove the clothing (Overall, head gear, gown and pant and reach the Locker room and put on their street clothes from the lockers and will exit the facility.

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Preparation and Processing areas:
To support the manufacturing operations Buffer Prep.(Support area) shall be responsible to prepare various materials, reagent, chemicals and sterile materials to be used in the manufacturing operations. The provisions for dry and moist heat sterilisations are provided in this area.
The manufacturing facility is construed and designed in a manner to protect product from any mix-ups, cross contamination and making the operator's work path and activities less complex and enabling compliance.
The processing areas for thawing of bulk material (Plasma), fractionation and downstream processing (Chromatography) are separately provided in the Layout to avoid any mix ups and cross contamination.

Plasma storage/ Testing lab /ware house/ Future provision plasma storage:
All the incoming and released raw materials, packaging material required to support the manufacturing shall be stored in the ware house.
All the incoming plasma shall be sampled and tested in ware house testing lab.
There is a provision for the storage at -20degrees C for of plasma in the ware house.
For the envisaged future expansion of manufacturing capacities from 150000Lit.to.500000Lit / year an additional storage area is reserved in the ware house, this area shall be furnished during future expansion of the capacities.

Finished goods storage and despatch:
All the released material shall be stored in the released cold rooms. At the time of shipment, the packed material shall be transferred from finished goods dispatch area in the ground Floor.

Administrative Block:
All the administrative support functions shall be dealt in this block.

Holding tank /ETP/STP:

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All the liquid waste generated in the bulk manufacturing area shall be properly segregated. The biological waste material shall be transferred to holding tank from there it will be transferred to kill tank for final processing and discharge to ETP through holding tank.

03 UG water storage tanks (5KL) each & Utility Block:
The ground water shall be transferred to UG storage 03 tanks having capacities of 5KL each. The raw water will be treated and will be used as feed water for the generation of PW, WFI and clean steam.

Engineering work shop:
All the maintenance and repair activities shall be undertaken in the work shop.

DG and Transformer room:
For the uninterrupted power supply separate area is designed in the Layout.

Chiller and Boiler room:
The facility of chilled water and plant steam required for various system shall be routed through chillers and boilers room.

First Floor: On the first-floor following facilities are housed:

- 1) Formulation / Fill Finish area
- 2) Finished goods quarantine
- 3) Labeling, packing & Visual Inspection
- 4) Water System and Eng. store
- 5) Bulk Store
- 6) SPM store
- 7) PPM store
- 8) R&D lab
- 9) QA/QC
- 10) Canteen

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Figure 10: Conceptual Layout for Bulk Manufacturing Facility

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The bulk manufacturing area is designed in a manner to provide unidirectional flow of man, material and waste from the facility to avoid mix-ups and cross contamination.
All the ancillary functions are located on the ground floor and separated from bulk manufacturing area.

In the ground floor of Bulk manufacturing facilities following key areas are present:

1. Bag Cutting and pooling area
2. Plasma Thawing Area
3. Downstream processing (Chromatography)
4. Columns packaging & skids
5. Washing area
6. DHS / Autoclave area
7. Cold room
8. Buffer Preparation
9. Plasma day store
10. Chemical Day store
11. Change rooms

Change Rooms: Bulk manufacturing facility -01, personnel entering the processing area will undergo a series of two dress changes applicable to clean rooms of different classifications. The separate change rooms are provided for male and Females.

Locker Room: This will be the first change where the personnel remove their street shoes and clothes and keep these belongings in the provided double door lockers.

For the exit from facilities separate change rooms are provided.

In exit change room personnel shall remove the clothing (Overall, head gear, gown and pant and reach the Locker room and put on their street clothes from the lockers and will exit the facility.

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Preparation and Processing areas:

To support the manufacturing operations Buffer preparation (Support) area shall be responsible to prepare various materials, reagent, chemicals and sterile materials to be used in the manufacturing operations. The provisions for dry and moist heat sterilisations are provided in the area.

The manufacturing facility is construed and designed in a manner to protect product from any mix-ups, cross contamination and making the operator's work path and activities less complex and enabling compliance.

The processing areas (Upstream) bag cutting, pooling and segregated plasma thawing areas are provided. There are separate areas for Plasma fractionation and downstream processing in stagewise manner (Chromatography) provided in the Layout to avoid any mix ups and cross contamination.

For packing of chromatographic columns dedicated area is provided in the layout of bulk manufacturing facility

Cold Rooms:

Cold rooms are provided in the facilities for the storage of heat labile materials like chromatographic gels, in process samples and materials.

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Figure 11: Conceptual Layout for Bulk Plasma Storage Facility

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Plasma storage facility is housed on the ground floor adjacent to the raw material storage area.

All the Raw materials shall take through the dedusting tunnel to clean and remove the dust from the external surface of RM containers. After sampling of RM, the goods shall be transferred to RM quarantine area pending the testing and release of materials from quality control lab.

The RMs which do not meet the specifications shall be transferred to rejection room. Rejection room has access control. Only authorised persons are allowed to enter the area.

In coming plasms is taken inside the ware house through plasma receipt corridor.

Incoming plasms is sampled in plasma sampling room, equipped with (RLAF), remaining plasma is transferred to plasma quarantine area having -20 degrees C cold facility.

After the completion of serological testing and other necessary test in the serology lab and QC lab., plasma is shifted to released plasma cold room at -20degees C.

Barcoding for identification of each lot of bulk plasma is done in Barcoding room.

There is a provision for future expansion of cold facilities to cater to 500000 Lit Plasma.

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Figure 12: Conceptual Layout for Fill Finish Facility

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Fill finish facility is housed on first floor of the bulk manufacturing facilities.

For entry into the area separate entries for males, females and visitors are provided. After series of changes, personnel shall enter into the core areas of fill finish facility.

In the core area formulation area is present.

In the formulation area all the released bulks shall be formulated on campaign basis. The formulated bulks shall be filled, capped through automatic machine and inspected for any abnormalities (particulate matter) visually. Separate laundry area is provided for washing of soiled gowns.

Following are the key areas in Fill Finish facility

1. Formulation
2. Fill Finish area
3. Lyophilisation
4. Change Room
5. Disinfectant and Material Prep area
6. Cold room

7. Clean laundry
8. Packing area

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Figure 13: Conceptual payout for QC and R&D Facility

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On the FF of Bulk manufacturing facilities R&D, Quality Assurance and quality control facilities are located.

Four labs are dedicated for research and development activities. The technology at small scale shall be transferred to R&D labs.

Adjacent to this, quality control labs are housed. Primarily quality control labs shall be responsible for testing

- Raw materials, PPM and SPMs.
- In-process testing of intermediate products
- Bulk plasma -Physicochemical and microbiological test
- Formulated Bulk plasma products
- Fill finished plasma products
- Microbial testing of critical utilities.
- Microbial testing of environmental samples from clean rooms.

Following labs are envisaged in quality control

- 1 Sample receipt
2. Chemical / Glassware Storage
- 3 TOC / LPC
4. Instrument Room
5. Balance Room
- 6 Wet Lab(Chemical)
7. Hot Zone/ Washing
8. MLT
9. Incubator rooms
10. Sterility room
11. Washing & Sterilization area
12. Media prep and Storage
13. BET Room
14. Stability Testing

Quality Assurance:

Annexed to QC labs quality assurance department is located. In quality assurance following functions of site will be undertaken

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1. Documentation and control
2. Trend analysis of critical utilities and Change management
3. Audits, Inspections, Pest control and Vendors management
4. Validation. Qualification and Calibration
5. Complaints and Recalls
6. Lot Release and Product review.
7. Trainings

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Figure 14: Elevation Drawing

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5.3 Layout Philosophy:

Plasma fractionation and processing is complex and critical manufacturing operation with respect to the receipt of plasma (Starting material) its downstream processing, formulation and fill finish. The facility requires strict cGMP and regulatory compliance to ensure the product quality and safety of the operator and environment.

The facility Layout adheres to GMP guidelines (Schedule M of Drugs and cosmetic act 1940 and Rules 1945).

The layout philosophy is based on the integral evaluation of processes, potential risks, man and material movement, generation and flow of solid and liquid waste etc.

The manufacturing facility is construed and designed in a separate block to protect product from any mix-ups, cross contamination and making the operator's work path and activities less complex and enabling compliance. Flow of personnel, clean and dirty materials are unidirectional with logical sequence of operations.

HVAC zoning of the facility (control areas) for the manufacturing operations is done based on the operation and its criticality (Plasma collection, fractionation, purification, and storage) to minimize the risk of contamination and to maintain physical conditions like temperature and humidity. The physical conditions should also be suitable to the comforts of the personnel working with protective clothing.

In the Layout provision for separate change rooms for entry and exit from clean areas is made. Change rooms design contains airlocks to provide physical separation of the different stages of gowning.

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CHAPTER 6
SOURCING AND
HANDLING OF RAW
MATERIAL

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6.1 Exchange Value of Plasma per Liter and Projected Value Over a Period of Seven Years:
As per the literature survey and discussion with different stake holders the exchange value of plasma in 2023-2024 is INR 2200/Liter and in next 5 years the estimated plasma price shall be expected to be ≈INR 2800/L.

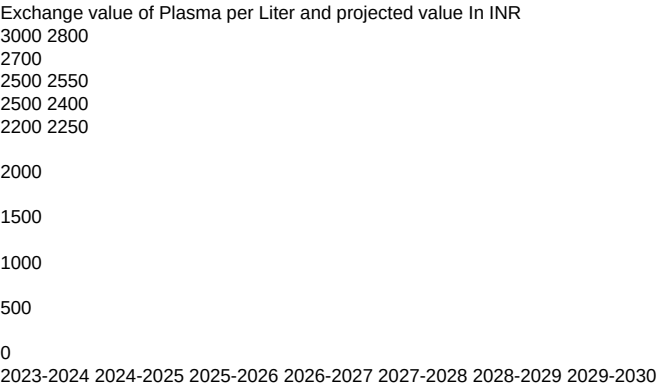


Figure 15: Exchange value of plasma per liter and projected value in INR

6.2 Donors' Selection Criteria:
In India, the plasma industry is dependent on voluntary blood donation. Human blood is the source of plasma which is used to manufacture various therapeutic plasma proteins after fractionation.
In blood banks the cellular components (RBC, WBC, and platelets etc.), are separated out for different applications and the plasma is collected from blood banks by plasma fractionators for clinical / therapeutic use as per "National Plasma Policy" of Ministry of Health & Family Welfare.
The selection and management of blood donors is an essential part of the blood donation process. National health authorities and blood transfusion services are responsible for ensuring that the national system is in place for the selection of all blood donors through an assessment of their suitability to donate blood.
Following are the selection criteria for donors:
1. Health
2. Age (18-65 years)
3. Time between blood donations

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- 4. Blood pressure
 - 5. Body temperature
 - 6. Hemoglobin content etc.
 - 7. Test for HBsAg*
 - 8. Test for anti-HIV*
 - 9. Test for anti-HCV*
- *All three tests should be negative. An additionally test for STD is also performed.

6.3 Plasma Donation Testing, Epidemiological Surveillance, Post Donation Information:
Plasma Donation testing:
The collected plasma is screened for the following:
1. Test for HBsAg
2. Test for anti-HIV
3. Test for anti-HCV
All three tests should be negative. Additionally test for STD is also performed.
Epidemiological surveillance, post donation information:
To ensure optimal long-term safety of plasma for fractionation, it is recommended to establish continuous epidemiological surveillance of the donor population.
The objective of this survey is to know, as precisely as possible, the prevalence and disease incidence, and trends of infectious markers that are relevant to the safety of medicinal plasma products so that appropriate counter-measures can be taken in a timely manner.
The system should not only be able to gather epidemiological data at the national and regional level but also among the donor populations that are providing blood or plasma for fractionation at individual blood establishments within a country or a region. The information from the epidemiological surveillance can furthermore be used to detect and assess.
● Differences in disease patterns between donor population.
● Trends of infectious marker.
● The relevance of any preventive measures such as a strengthening of donor selection process, additional exclusion criteria, or implementation of any additional screening tests to avoid contamination of plasma products.
● To detect any new or emerging pathogens in the community.

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6.4 Handling of Blood, Plasma Freezing, Storage, Cold Chain and Complete Logistics Plan:

Handling of Blood:

A standardized and validated procedure for the bleeding shall be followed.

- The presterilized container (Bags) shall be used to collect the blood.
- Identity of the donor should be confirmed with relevant documents before venipuncture.
- The collected blood used for plasma fractionation should be mixed well with the anticoagulant solution to avoid risks of activation of the coagulation cascade.
- Presently, automated blood collectors are used in which plasma is separated from cells

and the cells are resuspended and returned to donor in series of return cycles.

- In most of the centers the centrifugation or filtration or combination of both process is

followed to separate the cellular components from plasma.

- The collected blood should be weighed and labelled properly in order to ensure full

traceability at each stage of plasma production.

- Information on the label of the blood storage containers should include:

- o Name of the blood product
- o Volume or weight
- o Unique donor identification number
- o Name of the blood collection center
- o Shelf life
- o Storage temperature
- o Name, content and volume of anticoagulant
- o Laboratory samples should also have the same information and must be stored securely at appropriate temperature

Blood Holding:

It has been shown that whole blood anticoagulated with CPD, transported and stored at 22°C for up to 8 hr prior to separation of plasma is suitable for the production of plasma for fractionation.

However, storage of blood beyond 24 hours the factor VIII activity is reduced by 15 to 20 percent. Rapid cooling of whole blood to 22°C +/- 2°C after collection protects factor VIII and may allow storage of blood for 24 hr.

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Transportation/storage of collected blood at 4°C with different anticoagulants (ACD, ACD-adenine, or CPD) consistently appears to reduce the FVIII content, but not necessarily that of other proteins.

Blood holding time for the extraction of various components should be validated by the company.

Plasma Processing and Transportation:

After separation of plasma, freezing is an important processing step that has an impact on content of factor VIII. Several aspects of the freezing conditions of plasma for fractionation have been evaluated in this document based on the literature search.

Freezing of Plasma:

The requirements for the temperature at which plasma should be frozen depend upon the type of proteins to be fractionated.

As per pharmacopeial guidelines, recovered or apheresis plasma for fractionation to be used for the manufacturing of labile proteins (Factor VIII concentrate) should be frozen rapidly, within 24 hours of collection, at -20 °C or below.

However, freezing conditions are currently under debate. It is suggested that recovered plasma used to manufacture stable plasma proteins (e.g., albumin and immunoglobulins) should be frozen within 72 hours of collection at -20 °C or below.

Plasma freezing process should be validated by life Nity to extract the desired plasma proteins.

Storage of Plasma:

Plasma for fractionation should be stored at -20 °C or below. When plasma is stored at -20°C it has the shelf life of 2 years. However, when plasma is stored below -20°C has shelf life of 3 years.

Stability/Shelf-life studies should be conducted on at least three Pooled Plasma bulks during initial phase as a minimum by the company and subsequently one batch of pooled plasma should be taken for stability on annually basis after approval from the quality assurance department. The stability studies should be well documented.

Transportation of plasma and its validation:

Transport of plasma pools (Bulk) and samples to the processing site should be done according to approved and validated procedures. Temperature monitoring device should be used to record the temperature during transportation. Any deviations in the temperature during

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transportation should be notified to QA department for impact assessment (compliance and quality).

Simulation studies should be conducted at site to validate the packaging and its transportation.

6.5 Sourcing of Plasma from Government and Non-Government Blood Banks:

There are different sources of plasma in India- Government, NGOs and some private facilities recognized by the government. The Govt. has the full responsibility for the blood program. The surplus plasma from government hospitals and blood collection centers and blood banks are sold through tenders.

As a minimum there are certain conditions for plasma fractionator to fulfill before applying

for plasma procurement as follows:

1. The plasma fractionator must be licensed by the Government of India for plasma fractionation and manufacturing clinically usable products in a cGMP facility.
2. The fractionator is accountable to update plasma procurement center about the nature and quantity of all the products manufactured from the plasma received from the center.
3. The fractionator must ensure that none of the products recovered from this plasma are exported before fulfilling domestic demand.

Buy Back policy:

In some cases, government buy plasma derived products from the fractionator for use.

6.6 Equipment for Proper Collection, Storage and Separation of Blood Components:

Following is the list of equipment required during the different stages of blood processing:

Blood Collection

- Donor weighing balance
- Haemoglobinometer
- Sphygmomanometer
- Stethoscope
- Blood mixer and shaker
- Tube stripper and Di-electric tube sealer
- Needle destroyer

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Blood Processing

- Refrigerated Centrifuges
- Double pan balance
- Plasma Expresser- Manual/Automated

Blood Storage

- Blood Bank Refrigerator
- -20°C freezers
- -80°C deep freezers
- Platelet Agitator and Incubator
- Cell Separators (Apheresis machine)

Blood Testing

- pH Meter
- Cell Counter (optional)
- Coagulometer (optional)
- Automated Cell Grouping system (optional)
- Equipment for Gel technology (optional)
- Table top centrifuge
- Serological water bath
- ELISA reader (plate reader / Strip reader)
- VDRL shaker

Blood Distribution

- Transportation vans (optional)
- Insulated blood bags container with provision for storage between 2 °C to 8 °C

6.7 Importing of Plasma from Neighboring Countries and Indian Regulations:

India is highly import dependent for its plasma product needs, which are considered to be life-saving medicines. At this point in time, India produces majorly only two plasma products, albumin and intravenous immunoglobulin (IVIG) from local plasma.

Because of changing disease pattern, improved clinical diagnostic facilities, affordability, medical tourism and many other factors the demand of blood products is increasing. This creates a huge demand and supply gap and many patients lose battle of life because of non-availability of these products. Various hyper-immune plasma proteins (Anti-D, tetanus

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immunoglobulin, hepatitis-B immunoglobulin, and rabies immunoglobulin) are 100% imported or prepared from imported bulk.

6.7.1 Indian Regulations & Regulatory Bodies:

The Plasma Policy aims at making available, easily accessible and adequate supply of high quality of human plasma derived proteins for clinical/ therapeutic use. In India the plasma is separated from safe and quality blood from voluntary non-remunerated regular blood donor. Blood and plasma are regulated by different rules mention in drugs and cosmetic act 1940 and by different regulatory agencies as mentioned below.

Following regulations and rules are applicable:

The Drugs and Cosmetics Rules, 1945, framed under the Drugs and Cosmetics Act, 1940:

- Includes human blood under the definition of "Drugs" under Section 2(b)
- Regulates functions of blood banks, which provide recovered plasma for plasma fractionation (Part XII C specifically regulates the collection and storage of plasma for fractionation)

The National AIDS Control Organization (NACO), under the Ministry of Health and Family Welfare) to oversee operation of blood banking services in India.

National Blood Transfusion Council (NBTC) is the apex policy-making body, under NACO, making central licensing mandatory, increasing voluntary donation base and setting of quality standards.

State Blood Transfusion Councils (SBTCs) are responsible for the smooth functioning of blood transfusion services at a state level.

Drug Standard Control Organization (CDSCO) and State Drug Control Organizations' play a vital role in key aspects such as, approval of licenses, and enforcement of standard transfusion practices to ensure safe, quality and efficacious blood and its components in clinical practices.

National Institute of Biologicals (NIB) Nodal governmental organization under the Ministry of Health and Family Welfare responsible for the testing and release of biologicals in India including blood plasma.

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CHAPTER 7
PRODUCTION
PROGRAMPage 97
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7.1 Product and Product mix:

In phase 1 of the project the following products are proposed to be manufactured from normal human plasma.

- Human Serum Albumin
- Normal Immunoglobulin
- Factor VIII
- Factor IX
- Fibrin Sealant
- Fibrinogen

Depending upon the market demand and production capacities, product mix strategy can be formulated.

Various permutations and combinations are possible to manufacture different products. On the basis of PFD given in Chapter 4, different products shall be purified in different steps during the downstream processing of pooled plasma bulk (fractionation of plasma).

However, it is suggested that Lifinity shall take the first campaign of product which is at the highest demand followed by the products with moderate and low demand. As an alternate, the high yielding product shall be manufacture first to have sufficient stock followed by a campaign for low yield and moderately yield products.

Plasma requirements based on the manufacturing capacities:

Considering 1,50,000 Liters of plasma processing capacity, plasma bags requirement are presented in table below:

Table:9 Plasma requirements based on the manufacturing capacities:

Item	Volume in Liters
Plasma Storage capacity	32000
Plasma required per month	30000
Plasma required per 10 days	10000
Volume of plasma per bag	0.2
No. of bags required per 10 days	50000
No. of bags required per month	150000

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7.2 Production Capacities and Yield Calculations:

HSA is major protein component, followed by IgG in plasma. The wide concentration range of plasma proteins (less than 1 mg/mL to 40 g/L is one of the potential challenges in designing production capacities.

Source: (J.D. Andrade et al. Plasma protein adsorption: the big twelve, Ann. N.Y. Acad. Sci., p. 158-172,1990. & Chapter -2 of DPR).

However, for calculation purpose below mentioned protein concentrations in normal plasma are considered.

It has been reported in the literature that depending upon the downstream processing and sterile/terminal filtration, recovery of the protein of interest is 50% at final stage.

Source: (Overview of Albumin and Its Purification Methods: Adv Pharm Bull, 2016, 6(4), 495-507 doi: 10.15171/apb.2016.063 <http://apb.tbzmed.ac.ir>) as shown in the table below.

Basis of Yield calculations:

Based on vessel size, the batch size assumed is 1000 liters, on every alternate date, a new batch can be initiated, therefore 15 batches shall be processed per month (one batch/alternate day).

The yield for 1000 liters batch is as mentioned below.

Table 10: Yields for 1000L batch

Protein availability	No. of vials	Product Protein Recovery Conc. / vial (gm) /batch
Albumin (gm)	40,000	20000 20 GM 900
Human IgG (gm)	12,500	6250 5 GM 1000
Factor VIII in IU	10,00,000	12500 IU 250 IU 50
Factor IX in IU	10,00,000	90000 IU 600 IU 150
Fibrinogen (gm)	3,000	1500 70 mg 80
Thrombin in IU	9,00,000	90000 1000 IU
Fibrinogen	3000	1490 1000 mg 1400

In the EOI document, the phase 1 target is to process 1,50,000 Liters of plasma annually. The output obtained with 1.5 lac liters of plasma for the products mentioned for stage1 is as under.

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Table 11: Yields for 150,000 Lit annual capacity

No. of	Product Protein availability	Protein Recovery Conc. / Vial
vials/year		
Albumin (gm)	60,00,000	30,00,000 20 gm 1,35,000
Human IgG (gm)	18,75,000	9,37,500 5 gm 1,50,000
Factor VIII in IU	15,00,00,000	18,75,000 250 IU 7,500
Factor IX in IU	15,00,00,000	13,500,000 600 IU 22,500
Fibrinogen (gm)	4,50,000	2,25,000 70 mg 12,000

Thrombin in IU 13,50,00,000 1,35,00,000 1000 IU
Fibrinogen 450,000 225,000 1000 mg 2,10,000

Therefore, to achieve the plasma processing target, 15,000 liters of plasma shall be processed on monthly basis at site.

Production Scaleup from 150,000 to 500,000 lit per annum:
Lifentiy will increase its plasma processing capacity to 500,000 liters annually in phase 2 of the project implementation.

To achieve the plasma processing target, 50,000 liters of plasma shall be processed monthly. Two strategies, can be followed:

Strategy 1: To keep number of batches equivalent to stage-1 (i.e., 15 batches per month), the individual batch size shall be 3400 liters. The yield calculations are given below.

Table 12: Yields for 3400 L batch

No. of vials	Protein	Protein Conc. /	Product manufactured
availability (gm)	Recovery (gm)	Vial	/batch
Albumin 1,36,000	68,000	20 gm	3,060
Human IgG 42,500	21,250	5 gm	3,400
Factor VIII in IU 34,00,000	8,16,000	250 IU	170
Factor IX in IU 34,00,000	8,50,000	600 IU	510
Fibrinogen 10,200	5,100	70 mg	272
Thrombin IU 30,60,000	3,06,000	1000 IU	
Fibrinogen 10,200	5100	1000 mg	4,760

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Strategy 2: Alternatively, if the batch size is further increased to 5000 liters, the number batches shall be reduced to 10 batches per month as one batch can be initiated after every 3 days. The yield for a 5000 liters batch is as mentioned below.

Table 13: Yields for 5000 L batch

No. of vials	Protein	Protein Conc. /	Product manufactured
availability (gm)	Recovery (gm)	Vial	/batch
Albumin 2,00,000	100,000	20 gm	4,500
Human IgG 62,500	31,250	5 gm	5,000
Factor VIII in IU 50,00,000	1,200,000	250 IU	250
Factor IX in IU 50,00,000	1,250,000	600 IU	750
Fibrinogen 15,000	7,500	70 mg	400
Thrombin IU 45,00,000	450,000	1000 IU	
Fibrinogen 15,000	7,500	1000 mg	7,000

Total Production capacity:
The total production capacity with 500,000 Lit. of Plasma/year is given in the table below.
Table 14: Total production capacity with 500,000 Lit. of Plasma/year

Protein	Protein Conc. /	Product No. of vials /Year
availability (gm)	Recovery (gm)	Vial
Albumin 2,00,00,000	1,00,00,000	20 gm 450000
Human IgG 62,50,000	31,25,000	5 gm 500000
Factor VIII in IU 50,00,00,000	12,00,00,000	250 IU 25000
Factor IX in IU 50,00,00,000	12,50,00,000	600 IU 75000
Fibrinogen 15,00,000	7,50,000	70 40000
Thrombin IU 45,00,00,000	4,50,00,000	1000 IU
Fibrinogen 15,00,000	7,50,000	1000 mg 700,000

Note:
1. For scale up to 500,000 Liters only 6 products (HSA, IgG, F-VIII, F-IX Fibrin Sealant, Fibrinogen) are considered.
2. Fibrin sealant is prepared by mixing Fibrinogen and Thrombin; therefore, production capacity of fibrin sealant is equal to the production capacity of Thrombin.

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CHAPTER 8
PROJECT
IMPLEMENTATION
MODEL

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8.1 Lifentiy shall implement the PDMP manufacturing in a phased manner.
A suitable technology partner for the Plasma fractions shall be identified and contracted with for process know-how. The agreement will fulfil criteria mentioned in Section 3.3 of this document.
The project shall be initiated at small scale to establish the process for the products identified in stage 1.
Once the process, yields and quality are established as per the specifications, the project will proceed to the commercial scale stage
Commercial scale:
The company will initially procure 150,000 liters of plasma per year for fractionation to

produce the following product in Phase-1.

- Human Serum Albumin (HSA)
- Human Immunoglobulin (IgG)
- Factor VIII
- Factor IX
- Fibrin sealant
- Fibrinogen

All the areas of the manufacturing block (Facility-1) will have sufficient space to cater the volumes and products without compromising on any of the cGMP and regulatory expectations.

The following systems installed during Phase-1 of the project will also be matching the requirements for Phase-2 (Expansion Project) capacities.

- Clean utilities
- Black utilities
- Filling
- Freeze drying and
- Packaging lines for the final product

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8.2 Facility Expansion:

8.2.1 For handling additional 350,000L of Plasma:

In Phase-2, the company shall procure additional 350,000 Liters of plasma to manufacture the following additional products.

- Anti-D IgG
- Factor XI & XIII
- Fibrinogen
- Hepatitis B IgG
- Immunoglobulin G
- Measles IgG
- Protein C
- Rabies IgG
- Tetanus IgG
- Thrombin
- Varicella Zoster IgG

While implementing Phase-2 of the project, Lifenity shall extend the manufacturing operations to Facility-2 and shall procure the following additional systems to match the capacities of processing additional 350,000 L/year of plasma which are as follow:

- Tank Systems
- Chromatography Systems
- Autoclaves
- Dry heat sterilizers

Quality control and Quality assurance units developed during Phase-1 will be also cater to the Phase -2 products and other ancillary activities.

Central ware house space shall be adequate for future expansion. However, the additional cold Chain equipment (Freezers) and storage racks etc. for the storage of active material (plasma) and finished products shall be procured during Stage 2 expansion.

In Phase-1 of the project the bulk manufacturing operations will run in a single shift of 8 hours working /day, with effective working for 6 hours. The shifts can further be evaluated based on number of batches for processing/week. Likewise, the filling operations will run in scale down mode to meet the market requirement.

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8.2.2 Monoclonal antibodies (Mabs) – Future considerations:

A monoclonal antibody are Y-shaped glycoproteins that detect and bind strongly to their antigen. Monoclonal antibodies stems from a single white blood cell clone. Monoclonals not only bind to a single epitope of a target antigen but also activates immune responses, e.g. antibody-dependent cell-mediated cytotoxicity (ADCC) and recruits further immune cells such as phagocytes and are thus highly specific

Monoclonal antibody production:

Several technologies for monoclonal antibody production have been developed and the respective downstream processes are optimized. Some of the industrial scale technologies are:

- Hybridoma cells production
- Phage display technologies
- Single B cell technologies

Hybridoma production:

In this technology fusion of antibody producing B cells of an immunized mouse with myeloma cells is done produce unlimited antibodies. Hybridoma cell line that can be maintained indefinitely

Disadvantages:

- Use of lab animals for immunisation
- Risk of mutations in the antibody with aging of hybridoma
- Expression of additional variable domains by the cell line
- Long lead times and labour-intensive operation

Phage display technology:

Phage display technology circumvent several disadvantages of hybridoma technologies and speed up the process of Mab generation.

The technology relies on molecular biology and cell culture methods to genetically engineer bacteriophages to display antibodies on their surface.

In this method binding of bacteriophages with desired antibodies onto immobilized antigens is done. The high affinity binders are then used to infect bacteria and collect their DNA for

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sequencing. The sequence help to design genes to be used in antibody production in cell cultures.
The overall process is much faster than hybridoma cell generation and doesn't need any laboratory animals.

Single B cell technologies:

Single B cell technologies are another milestone in Mab development. The technique involves the screening and sorting of ex vivo cultures of single B cells from blood samples of immunized animals or humans to select for promising antibody candidates.

Genetic information is then extracted from the candidate cells mRNA after reverse transcription into cDNA, amplification and subsequent sequencing and cloning into expression vectors for Mab production.

In this method there is no need to create a hybridoma cell line, thus process become efficient and timeline are shorted by several weeks.

Depending upon the process to be adopted for the manufacturing of monoclonal antibody a new production block shall be designed to meet the process and cGMP requirements.

Detailed working for the Mabs manufacturing unit will be proposed separately in future.

In general, the Mab production suite shall have cell culture, fermentation, harvesting, primary purification, secondary purification, sterile filtration.

However, for formulation and fill finish existing facilities shall be used areas.

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Table 15: Steps performed in MABs production (General)

S/No. Step Purpose Remark

1 Cell culture and To generate the desired bulk of Fermentation antibodies

2 Centrifugation To remove cell debris, lipids, and clotted material

3 Filtration (0.45 µm To remove smaller contaminating material

4 Concentration (UF) To concentrate the desired antibodies in the material

5 Ion exchange To remove nucleic acids and Chromatography endotoxins (IEC)

6 Size exclusion SEC is preferred over IEC chromatography because size is relatively

(SEC) constant and charge on

the protein may vary depending upon the pH

as molecules are

protonated and

deprotonated

7 Immunoaffinity Antibody will bind to specific It will help to achieve Chromatography antigen attached to the highest purity

agarose matrix

8 Sterile filtration To remove microbial contamination

For implementation of Mabs project in future, there shall be a requirement of cell culture and molecular biology labs in the quality control with some additional piece of equipment.

Present Layout of quality control unit will be adequate for plasma products (Phase-1 & 2).

However, for Mabs products testing, expansion in existing QC to cater the cell culture need shall be required.

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CHAPTER 9 GMP AND REGULATORY CONSIDERATIONS

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9.1 Premises, Layout & Process Engineering:

Plasma fractionation and processing is complex and critical manufacturing operation with respect to the receipt of plasma (Starting material) its downstream processing, formulation and fill finish.

Plasma fractionation facility require strict cGMP and regulatory compliance to ensure the product quality and safety of the products and operators. The facility's layout and process engineering are critical components for setting up of the green field project. Following are key considerations for setting up a plasma fractionation facility with respect to Premises, Layout and Process engineering:

Location and Surroundings:

The factory building(s) for manufacture of plasma products shall be situated and shall have measures to avoid risk of contamination from external environment including open sewage, drains, public lavatory or any factory which produce disagreeable or obnoxious odor or fumes, excessive soot, dust, smoke, chemical or biological emission.

Building:

The building(s) used for the purpose shall be designed, constructed, adapted and maintained to suit the manufacturing operations so as to permit production under hygienic conditions. They shall conform to the conditions laid down in the Factories Act, 1948 (63 of 1948).

Premises - General Requirements:

The premises used for manufacturing, processing, warehousing, packaging, labelling and testing purposes shall be:

- Compatible with other products manufacturing operations that may be carried out in the same or adjacent area.
- The manufacturing facility shall be designed / constructed / maintained to prevent entry of insects, pests, birds, vermin's and rodents. Interior surface (walls, floors and ceilings) shall be smooth and free from cracks, and permit easy cleaning, painting and disinfection.
- There should be adequate space for orderly and logical placement of equipment, materials and movement of personnel in order to prevent:

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(a) The risk of mix-up between different categories of products or with raw materials, intermediates and in-process material.

(b) Possibilities of contamination and cross- contamination by providing suitable segregation, zoning and access controls etc.

Layout Considerations:

During designing of facility, Lifenity must ensure the following points:

1. Plasma products will be produced in an independent block/building.
2. The facility is designed to prevent cross-contamination between process areas, easy cleaning and ensure the efficient flow of materials and personnel.
3. The facility design adheres to GMP guidelines (Schedule M of Drugs and cosmetic act 1940 and Rules 1945).
4. Ancillary areas like rest and refreshment rooms are separate from manufacturing areas.
5. Facilities for changing, storing clothes and for washing and toilet purposes are easily accessible and adequate for the number of users. Provision for separate toilets for males and females is made and toilet is not connected with production or storage areas.
6. Change rooms design contains airlocks to provide physical separation of the different stages of gowning. Flushed effectively with filtered air. Change rooms are of the same grade as the area into which it opens finally.
7. Provision of separate change rooms for entry and exit from clean areas.
8. Hand-washing facilities are provided in the first change room.
9. Flow of personnel, clean and dirty materials are unidirectional with logical sequence of operations.
10. Sinks and drains are excluded from Grade A and B areas where aseptic operations are carried out.
11. Drains are fitted with effective easily cleanable traps and with air brakes to prevent backflow.
12. The production and dispensing areas are well lighted, effectively ventilated.

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HVAC Zoning:

Air supply for manufacturing area will be filtered through bacteria retaining filters (HEPA Filters) and will be at a pressure higher than in the adjacent areas unless otherwise required.

The filters to be checked for performance on installation and periodically thereafter and maintain written records.

HVAC zoning of the facility (control areas) for the manufacturing operations is done based on the operation and its criticality (Plasma collection, fractionation, purification, and storage) to minimize the risk of contamination and to maintain physical conditions like temperature and humidity. The physical conditions should also be suitable to the comforts of the personnel working with protective clothing.

Routine microbial and particulate counts of the manufacturing area will be carried out during manufacturing operations to assess the effective ness of HVAC.

The results of particulate matter and microbes to be continuously monitored against documented in-house standards and regulatory guidelines.

Based on the quality of air, manufacturing facility is divided into different grades/ Zones. In general, plasma processing facility is divided into four grades (Classes) based on the air cleanliness levels (Microbial and Particulate matter (Refer to table No 15 and 16).

Facility Grades:

Grade A: Clean rooms are used for high-risk operations, and require a unidirectional airflow system to maintain high air quality. Unidirectional airflow should provide a homogeneous air velocity of 0.36–0.54 m/s (guidance value) at a defined test position 15–30 cm below the terminal filter or air distributor system.

The velocity at working level should not be less than 0.36 m/s.

Grade B: In aseptic preparation and filling, this is the background environment for the Grade A zone.

Grades C and D: Clean areas for carrying out less critical stages of the manufacturing of plasma products -downstream processing, chromatographic purification and solution preparations.

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Table 16: Recommended Maximum Allowable Particles in Clean Rooms

Maximum Allowable Particle/m³

Grade During Rest During Operation

≥ 0.5µm ≥ 5µm ≥ 0.5µm ≥ 5µm

A 3500 0 3500 0

B 35,000 0 350000 2000

C 350000 2000 3500000 20000

D 3500000 20000 Not Defined Not Defined

Table 17: Recommended limits for Microbial contamination in Clean Rooms

Grade Air sample Settle plates Contact plates Glove print

(CFU/m³) (diameter 90 (diameter 55 (5 fingers)

mm) (CFU/4 mm) (CFU/glove)
hours) (CFU/plate)
A <1 <1 <1 <1
B 10 5 5 5
C 100 50 25 -
D 200 100 50 -

Clean Utility Systems:

Pure steam, purified water and water for injection are the clean utilities required to support the clean operations in plasma manufacturing/processing facilities. Monitor the clean utilities for the quality and compliance regularly as per the laid procedures and the regulatory standards.

Equipment Placement:

Equipment and processing areas will be arranged to facilitate efficient workflow and minimize the risk of contamination. Ensure that equipment and materials are easily accessible for maintenance and cleaning. Processing areas must be equipped with dedicated equipment designed for specific purpose.

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Waste Management:

1. Waste will be removed from the manufacturing area or sent to the bio-waste system in a safe manner. To ensure containment compliance with the environment, the facility must implement measures to prevent accidental spills or leaks. This includes the use of appropriate containment systems, spill response plans, and monitoring for any environmental impacts.
2. All sewage waste pipes will run below natural ground level (however this depends on the actual site conditions along the pipe routing) in proper gradient and connected to nearest available chamber.
3. Procedures must be in place for the safe removal of different types of waste such as chemical, biological, expired reagents, unused samples and solvents etc.
4. Design the drainage systems in such a way that effluents can be effectively neutralized or decontaminated to minimize the risk of cross-contamination.
5. Provisions should be there for the proper and safe storage of waste materials awaiting disposal.
6. Toxic substances and flammable materials must be stored in suitably designed, separate, enclosed cupboards, as required by national legislation.
7. Liquid effluent poses a safety or contamination risk (Biological liquid waste), should be treated in the kill tank before its discharge into non biological /main drain.

Testing Facilities:

Testing facilities must be located in the separate building/floor.

Validation and Documentation:

Lifinity and PMC will validate the facility layout and design to ensure it meets GMP and regulatory requirements. Comprehensive documentation of the facility's layout and design, including as-built drawings shall be maintained at site and a robust document control system shall be established in quality Assurance department to maintain records.

Accessibility and Safety:

Ensure that the facility layout allows for easy access in the event of an emergency. The facilities must be equipped with safety features like fire hydrant systems, eyewash and body shower stations where ever necessary.

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Quality Assurance & Regulatory Compliance:

The quality assurance system appropriate to the manufacture of plasma products shall ensure that the products are designed and developed in a way that takes account of the requirements of GMP and other associated codes such as those of GLP and GCP.

Adequate arrangement must be made at site for manufacture, supply chain and use of the correct starting and packaging materials.

Self-Inspection and Quality audit:

In the quality assurance department of Lifinity adequate procedures shall be designed to ensure self-inspections and quality audits periodically.

Plasma Process Considerations:

Plasma fractionation involves critical steps to separate plasma into its various components and produce high-quality plasma-derived products while adhering to Good Manufacturing Practices (GMP) and regulatory requirements.

Following are the key considerations for plasma processing:

1. Plasma Collection and Testing: A reliable and secure source of plasma, involving strict donor selection, screening and testing plan in place to screen for infectious agents, including HIV, Hepatitis B and C, and other potential contaminants in pooled plasma.
2. Fractionation Process: A well-defined fractionation process to separate plasma into its various protein components, such as albumin, immunoglobulins, and plasma factors. Robust filtration and purification processes will be followed by Lifinity to separate and purify these proteins.
3. Process Validation: The fractionation process must be validated to demonstrate its ability to consistently produce products. Process performance qualification (PPQ) shall be performed on first three initial batches to ensure that the process is in a state of control during commercial production.
4. Process Control and Monitoring: Robust process control systems to monitor critical process parameters (e.g., temperature, pH, pressure) in real-time. Written procedures must be in place for addressing deviations and out-of-specification results.
5. Raw Material Control: Documented procedures for receiving, testing, and release of raw materials and starting materials used in the fractionation process. Lifinity must ensure that all materials meet quality, safety standards and plasma source traceability.

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6. Safety: Validated viral and pathogen inactivation and removal steps to ensure the safety of plasma-derived products. Implement measures to prevent contamination during processing.
7. Quality Control Testing: Comprehensive quality control testing plan, including in-process testing and final product testing, to ensure the safety, purity, and potency of the plasma products. Specifications for raw materials, intermediates, and final products shall be written and approved.
8. Equipment monitoring, Calibration and Validation: All the critical equipment used in the manufacturing and control of plasma are maintained, calibrated and qualified as per the written procedures.
9. Training and Qualification: Ensure that all personnel involved in the plasma fractionation process are well-trained with respect to the SOPs and STPs, understand GMP requirements, and are aware of their roles in ensuring product quality and safety. Maintain records of personnel training and qualifications.
10. Writing and editing documentation: Documentation for all existing processes must be written and edited regularly, to ensure that all new hires and other personnel can understand the best practices and standards
11. Risk assessments of current equipment/processes: Assessment of current processes and equipment to determine potential risks and mitigation plan shall be in place.
12. Traceability: A robust documentation system must be in place for tracking and tracing individual plasma donations and the products derived from them.
13. Packaging and Labelling: Appropriate packaging and labelling procedures for the final plasma-derived products, ensuring they meet regulatory requirements.
14. Post-Market Surveillance: Post-market surveillance system to monitor the safety and efficacy of products in the market.

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9.2 Regulatory Guidelines GMP, Drugs and Cosmetics act 1940 and Rules, CDSCO, WHO-GMP

Indian Guidelines:

1. The Drugs and Cosmetics Rules, 1945, framed under the Drugs and Cosmetics Act 1940:
 - Includes human blood under the definition of "Drugs" under Section 2 (b)
 - Regulates functions of blood banks, which provide recovered plasma for plasma fractionation
 - Part XII C specifically regulates the collection and storage of plasma for fractionation
2. In the year 1967, Central Govt. (Ministry of Health) enacted a separate provision in Schedule F Part XII B of Drugs & Cosmetics Rules.
3. Schedule F of the Drugs and Cosmetics Rules specifically pertains to the requirements for the manufacture, sale, and distribution of plasma-derived products, including immunoglobulins, clotting factor concentrates, and other blood
4. Drugs and Cosmetics Rules, Part X-B: Conditions for License (Rule 122EA to Rule 122P)
 - Rule 122 EA Definitions
 - Rule 122F to 122N License Application Procedure / Inspection / Reporting by Inspection Team / Grant or Rejection of License / Duration of license / Appeal provision
 - License granted by Licensing Authority
 - Approval by Central License Approving Authority
 - License is granted & delivered to Applicant
 - Rule 122 O Procedure of Cancellation and Suspension of Licenses
 - Rule 122 P Conditions of licenses:
 - Testing of whole blood / Component / Product Inspections
 - Reporting to SLA / CLAA about changes in staff / premise
 - Directions issued by SLA / CLAA / Recall Directions
 - Conditions for distribution of whole blood / component / products
 - To comply provisions of the Act and Rules
5. Destruction of infected blood and implementation of Bio-Medical Wastes (Management and Handling) Rules, 1996

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6. Applicable forms for License:
 - Product License Renewal
 - Blood and Components 27-C 28-C 26-G
 - Blood Products 27-E 28-E 26-I
7. The National AIDS Control Organization (NACO), under the Ministry of Health and Family Welfare) to oversee operation of blood banking services in India.
8. National Blood Transfusion Council (NBTC) is the apex policy-making body, under NACO, making central licensing mandatory, increasing voluntary donation base and setting of quality standards. Similarly, State Blood Transfusion Councils (SBTCs) are responsible for the smooth functioning of blood transfusion services at a state level.
9. Drug Standard Control Organization (CDSCO) and State Drug Control Organizations play a vital role in approval of licenses, and enforcement of standard transfusion practices to ensure safe, quality and efficacious blood and its components in clinical practices.
10. National Institute of Biologicals (NIB) is the nodal governmental organization under the Ministry of Health and Family Welfare responsible for the testing and release of biologicals in India including blood plasma.

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Policies and Guidelines for Fractionation:

1. The National Blood Policy 2002 regulates blood quality and the Guidelines of NBTC

2016 has been brought out by NACO. NBTC has formulated guidelines for recovery of processing charges for blood and blood Components across blood banks in all sectors. It has also laid down uniform norms for issuance of No Objection Certificates (NOCs) for new blood banks, renewal of license, criteria for designation blood bank as RBTC. Guideline on bulk transfer of blood and Guidelines on exchange of surplus plasma with indigenous fractionators has been released by NBTC to support voluntary blood donation and rationalize blood utilization.

2. National Policy for Access to Plasma Derived Medicinal Products from Human Plasma for Clinical / Therapeutic Use covers the process of collecting standard plasma and transporting them under optimum conditions for fractionation, identifying critical parameters for safety, and ensuring compliance with regulatory requirements and training.

3. National Pharmaceutical Pricing Authority: Under the Ministry of Chemicals and Fertilizers, Department of Pharmaceuticals, the prices of medicines are controlled by the National Pharmaceutical Pricing Authority. Currently, plasma products under price control include albumin, AHF, anti-D immunoglobulin, anti-tetanus immunoglobulin and anti-rabies immunoglobulin however plasma itself is not under price control.

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Pre-clinical safety and Clinical Evaluation of Plasma derived products:

European guidelines

Preclinical:

1. June 2011 EMA/CHMP/ICH/731268/1998 Committee for medicinal products for human use (CHMP)

2. Clinical Evaluation of Plasma derived products: Guideline on clinical investigation of recombinant and human plasma-derived factor IX products: 21 May 2015, EMA/CHMP/BPWP/144552/2009 Rev. 1, Corr. 1 Committee for medicinal products for human use (CHMP).

3. Guideline on clinical investigation of recombinant and human plasma-derived factor VIII products: 21 May 2015 EMA/CHMP/BPWP/144533/2009 rev Committee for medicinal products for human use (CHMP).

4. Guidance on the Clinical Investigation of Human Normal Immunoglobulin for Intravenous Administration (IVIg). CPMP/BPWG/388/95 rev.1.

5. Guidance on the Clinical Investigation of Human Normal Immunoglobulin for Subcutaneous and Intramuscular Use. EMEA/CPMP/BPWG/283/00.

Indian guidelines

As per Indian drugs and cosmetic act human blood plasma is included under the definition of "Drugs" under Section 2 (b)

As per the Indian regulations if the company which manufactures human plasma products similar to the composition and product profile currently marketed then the clinical trial waiver is given for marketing authorization. The company shall apply to DCGI &SLA for the waiver with relevant supporting documents. In case of hyper immune sera products appropriate clinical trial for safety and efficacy shall be conducted.

9.3 Containment Compliance

Blood and blood-derived products are highly sensitive and can be easily contaminated. Bloodborne pathogens, such as HIV, hepatitis B and C, can pose significant risks to both staff and patients if proper containment procedures are not followed. Maintaining a contained environment helps minimize the risk of spreading infectious diseases in the population.

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Containment measures in the manufacturing premises shall be established such as isolation rooms, negative pressure environments, controlled airflow systems and proper handling procedures.

Plasma fractionation facilities must adhere to specific biosafety levels as defined by regulatory authorities, such as the Centers for Disease Control and Prevention (CDC). Plasma fractionation facilities operate at BSL-2 (Biosafety Level 2). It must include:

1. Access control, lab coats, gloves, eye protection (PPE) to be used by personnel working in the facility, and the use of biological safety cabinets during procedures.
2. Continuous monitoring of air pressure differentials, temperature, and humidity within the clean rooms. Alarms should be triggered if any parameter falls outside the acceptable range.
3. Scheduled regular maintenance, testing, and certification of the HVAC system.
4. The exhaust air from the clean rooms potential for any contaminating microbes should be passed through the scrubbers to protect the environment.

9.4 Validation of Processes, Equipment and Services

Approaches to validation:

1. Lifenity will organize and plan qualification and validation in a manner that will ensure product quality, safety and efficacy throughout its life-cycle.
2. Qualification and validation shall be done in accordance with predetermined protocols, and the results to be documented, in reports.
3. An appropriate and effective quality management system shall be in place at Lifenity supporting the organization, planning, execution and management of qualification and validation activities.
4. Senior management will provide sufficient resources to perform validation in a timely manner. Management and persons responsible from quality assurance and manufacturing to be actively involved in the validation activities.
5. All relevant SOPs for operation, maintenance and calibration shall be prepared during qualification.
6. Training shall be provided to operators, and training records need to be updated and maintained.

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Equipment Qualification/Validation:

Equipment qualification helps mitigate risks, prevent equipment failures, reduce downtime, enhance product quality, and ensures the overall operational excellence of an organization.

1. Equipment Validation comprises a series of steps which includes User Requirement Specification (URS), Design, Installation, Operational, and Performance Qualification.
2. User Requirement Specification (URS): Lifenity will prepare a document describing the requirements for the equipment to be sourced. The requirements will include specifications and must ensure that possible GMP risks are addressed; include technical requirements; and associated documentation.
3. Design qualification (DQ): DQ will provide the documented evidence that the design specifications were met and are in accordance with the URS. FAT and SAT needs to be performed where required, to verify the suitability of the system at site, prior to the subsequent stages of qualification and this must be appropriately documented.
4. Installation qualification (IQ): IQ will provide documented evidence that the installation was complete and satisfactory, including supporting utilities. The design specifications, including purchase specifications, drawings, manuals, lists of spare parts and vendor details, will be verified during IQ. Components installed must be verified, and documented evidence should be provided that components meet specifications, are traceable and are of the appropriate construction material. Applicable control and measuring devices, identified through impact or risk assessment, should be calibrated.
5. Operational Qualification (OQ): OQ should provide documented evidence that utilities, systems or equipment operate in accordance with operational specifications. Tests will be designed to demonstrate satisfactory operation over the normal operating range, as well as at the limits of its operating conditions. Worst-case conditions must be included in the testing. Operation controls, alarms, switches, displays and other operational components shall be tested.
6. Performance qualification (PQ): PQ will normally follow the successful completion of IQ and OQ. PQ will be performed under conditions simulating the intended use, to provide documented evidence that equipment/system can consistently perform.

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Process validation:

Process validation is defined as the collection and evaluation of data, from the process design stage through commercial production, which establishes scientific evidence that a process is capable of consistently delivering quality product. It ensures that quality, safety and efficacy by design are built into the product.

1. Collect and evaluate the data throughout the life-cycle of a product from the process design stage through to commercial production and provides scientific evidence that a process is capable of consistently delivering a quality product.
2. Data need to be available for inspection at Lifenity manufacturing location.
3. A risk-assessment approach will be followed, to determine the scope and extent to which process(es) and starting material variability may affect product quality.
4. The critical process steps and critical process parameters (CPPs) will be identified, justified monitored and documented based on relevant studies carried out during the design stage and process knowledge.
5. Lifenity will ensure that the principles of process validation described in the guidelines are implemented.
6. Evaluate and compare the historical data when there is a change in the process.
7. When changes are planned to existing systems or processes need to be followed and documented. Sufficient data is collected to demonstrate that the revised process and product will meet desired quality, consistent and approved specifications.

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CHAPTER 10**ALLIED EQUIPMENT & FACILITIES**

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10.1 Technology Based Allied Equipment Requirements

- Refer Annexures

10.2 Facilities

Laboratories for allied testing should be separated from Production and Quality control labs. However, the materials required for testing such as media, filters, filtration devices etc. can be taken from QC labs. Dedicated glass ware should be used for any testing activities. Example of allied test/activities are ETP samples, Black utilities, Ground water, testing of clean steam for physical parameters, compressed air and calibration activities etc. Specialized temperature-controlled warehouse for storage of RMs, In-Process materials and finished goods:

Adequate areas shall be designed to allow sufficient and orderly warehousing of various categories of materials and products like starting and packaging materials, intermediates, bulks and finished products, products in quarantine, released, rejected, returned or recalled, machine and equipment spare parts and change items etc. Warehousing areas need to be designed and adapted to ensure good storage conditions. Warehouse shall be clean, dry and maintained with acceptable temperature and humidity limits where ever required. Temperature and humidity will be monitored and recorded. A temperature monitoring and recording system that are independent from the temperature regulation system with alarm systems will be placed. Storage areas must have program for pest control. The record shall be maintained. Proper racks, bins and platforms must be provided for the storage of materials. The quarantine

area in the warehousing will be clearly demarcated. Access to the areas must be restricted to authorized persons.

There shall be a separate sampling area in the warehousing for active and sterile raw materials/starting materials and excipients from the unsterile RMs.

Sampling and dispensing of sterile materials shall be conducted under aseptic condition conforming to Grade A (LAF). Sampling of sterile materials shall be performed in a dedicated area within the manufacturing facility in the absence of suitable areas in the ware house. Segregation shall be provided for the storage of rejected, recalled or returned materials or products.

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Materials or products in this area should be suitably marked and secured. Access to these areas and materials must be restricted.

Highly hazardous, poisonous and explosive materials such as alcohols, narcotics and other substances presenting potential risks of abuse, fire or explosion shall be stored in explosion proof area, away from the central ware house in safe and secure manner.

Adequate fire protection measures shall be provided in conformity with the rules of the concerned civic authority in the ware house.

Printed packaging materials shall be stored in safe, separate and secure areas. Regular checks shall be made to ensure adequate steps are taken against spillage, breakage and leakage of containers.

Receiving and dispatch bays shall protect materials and products from adverse weather conditions.

The ware house shall have different areas for the receipt, quarantine, undertest, released, hazardous and rejected materials.

Primary packaging materials, labels, and secondary packaging material and finished products shall be stored in the separate area of the ware house with access controls systems in the critical areas of the ware house.

Incoming critical materials (such as sterile solutions, plasma bag and testing reagents) shall be physically quarantined immediately after receipt and until they are released for use by the authorized person.

Materials and reagents shall be stored under the conditions established by the manufacturer and in an orderly manner that permits segregation by batch or lot and stock rotation. Storage, issuance and usage should follow the "first-expiring first-out" principle (FEFO).

10.3 Materials Required Like Resins, Filters etc.

- Ion exchange resins
- Size exclusion resin – 1
- Size exclusion resin - 2
- Multi grade filters (MGF)
- Softener
- Filters 0.5 Micron
- SMBS (Sodium Metabisulphite)

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- Chlorine
- Common Salt
- NaOH
- Hydrochloric acid

10.4 SOP/STP and Biological Waste Management

The SOP's and STP for the allied testing facilities are as under:

- Procedure for Decontamination of chemical and biological waste.
- Procedure for estimation of BOD and COD in ETP samples.
- pH determination of ETP samples.
- Procedure for testing compressed air and pure steam.
- Procedure for calibration of measuring devices of manufacturing, ware house, quality control and engineering.
- Procedure for treatment of Biowaste
- Procedure for collection segregation and disposal of Biowaste
- Training on Biowaste management
- Procedure for usage of treated water for horticulture purpose
- Procedure for discharge of Biowaste in common drainage system
- Procedure for management of spillage control

Biowaste management (BWM):

The disposal of sewage and effluents (solid, liquid and gas) from the manufactory should be in conformity with the requirements of Environment Pollution Control Board. All bio-medical waste must be destroyed as per the provisions of the Bio-Medical Waste (Management and Handling) Rules, 1996 and amendments thereof.

Additional precautions to be taken for the storage and disposal of rejected waste material. Records shall be maintained for all disposal of waste. Provisions shall be made for the proper and safe storage of waste materials awaiting disposal. Hazardous, toxic substances and flammable materials shall be stored in suitably designed and segregated, enclosed areas in conformity with Central and State Legislations.

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CHAPTER 11

SUBSIDY

CONSIDERATIONS

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11.1 Location / SEZ Considerations in Maharashtra

Special Economic Zone (SEZ) is a specifically delineated duty-free enclave with all required infrastructure provided under single administrative umbrella primarily meant for locating industries which manufacture and export goods and services.

These enclaves shall be deemed to be foreign territory for the purposes of trade operations and duties and tariffs.

Under section 3, a Special Economic Zone can be established either by Central Government, State Government or its agencies, private / public / joint sector or any person for manufacture of goods or rendering services or for both or as a free trade and warehousing zone.

As per the requirements of SEZ, the unit must be segregated into processing and non-processing areas. A processing area is that area in which all the manufacturing activities are performed. At least 50% of the total area shall be earmarked.

The non-processing area is the area which is intended to provide support facilities to the processing area activities and may include support areas like utilities block, power generation plant (DG), engineering services, ETP and other infrastructures as per the requirements of the current project.

In SEZ 100 percent Foreign Direct Investments are allowed (FDI) in some of the selected categories.

As per MIHAN-SEZ guidelines (Rules and Procedures) following class of incentives are available to SEZ enterprises.

11.2 Custom and Excise

SEZ units may import or procure from the domestic sources, duty free, all their requirements of capital goods, raw materials, consumables, spares, packing materials, office equipment, DG sets etc. for implementation of their project in the Zone without any license or specific approval.

11.3 Duty Free Import/Domestic Procurement of Goods for Setting up of SEZ Units

Goods imported/procured locally duty free could be utilized over the approval period of 5 years.

11.4 Sales Exemption

Domestic sales by SEZ units will be exempt from single administrative document (SAD).

Domestic sale of finished products, by-products on payment of applicable custom duty.

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Domestic sale rejects, waste and scrap on payment of applicable Custom duty on the transaction value.

11.5 Income Tax:

Physical export benefit 100%.

IT exemption (10A) for first 5 years and 50% for 2 years thereafter.

Reinvestment allowance to the extent of 50% of ploughed back profits.

11.6 Carry forward of losses & Commercial borrowing:

Banking / Insurance/External Commercial Borrowing.

Setting up Off-shore Banking Units (OBUs) is allowed in SEZs. OBU's allowed 100% Income Tax exemption on profit for 3 years and 50 % for next two years.

External commercial borrowings by units up to \$ 500 million a year allowed without any maturity restrictions.

11.7 Central Sales Tax:

Exemption to sales made from Domestic Tariff Area to SEZ units.

11.8 Service Tax:

Exemption from Service Tax to SEZ units.

11.9 Other Exemptions:

Freedom to bring in export proceeds without any time limit.

Flexibility to keep 100% of export proceeds in exchange earners foreign currency account (EEFC). Freedom to make overseas investment from it.

Exemption from interest rate surcharge on import finance.

SEZ units allowed to 'write-off' unrealized export bills.

11.10 Additional Incentives:

The additional incentives are exemption of import license, custom examination of imported goods, exemption from public hearing under Environment Impact Assessment Notification etc.

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CHAPTER 12

PROJECT TIMELINES

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CHAPTER 13

BUDGET ESTIMATES

13.0 Budget Estimates
Refer to Chapter 17

CHAPTER 14
REVENUE AND COST
ESTIMATION AND
PAYBACK PERIOD

YIELD CALCULATIONS FOR PLASMA FRACTIONS

REVENUE FORECAST IN CRORE INR FROM PROCESSING
PLASMA 150,000 L / ANNUM

PROFIT AND LOSS STATEMENT FOR STAGE 1 – 150,000 L/ ANNUM

CHAPTER 15
RECOMMENDATION

The objective of this project is to establish and operate a state-of-the-art Human Plasma Derived Medicinal Products (HPDMP) manufacturing facility with an aim to address the increasing demand for plasma-derived products by Optimized manufacturing processes maximize productivity, minimize waste, and reduce operational costs in order to maintain competitiveness in the market.

This Human Plasma Derived Medicinal Products manufacturing facility will play a pivotal role in meeting the critical healthcare needs of patients while contributing to advancements in the plasma industry.

In phase-1 of the project annual production capacity will be 150,000 liters of plasma processing, which will be subsequently increased to 500,000 liters per annum in future.

In this report several technologies utilized for plasma purification at commercial scale were evaluated exhaustively with its merits and demerits. Some of the key technologies evaluated are:

1. Plasma Fractionation method with ethyl alcohol
2. Cohn+ Liquid Chromatography
3. Heat Shock technology
4. Ammonium sulphate+ Liquid Chromatography
5. Chromatographic

It is concluded and recommended that Chromatographic purification with sterile filtration is best technology for the purification of plasma proteins for stage-1 products as it will give highly productivity and acceptable quality of desired proteins (purity levels) as per the regulatory expectations.

For stage-2 products, the technology evaluation and recommendations will be made in future.

CHAPTER 16
REFERENCES

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- 2 Schedule F Part XII B of Drugs & Cosmetics Rules.
- 3 Drugs and Cosmetics Rules, Part X-B: Conditions for License (Rule 122EA to Rule 122P).
- 4 Destruction of infected blood and implementation of Bio-Medical Wastes (Management and Handling) Rules, 1996.
- 5 The National Blood Policy 2002 regulates blood quality and the Guidelines of NBTC 2016.
- 6 The National AIDS Control Organization (NACO), under the Ministry of Health and Family Welfare) to oversee operation of blood banking services in India.
- 7 National Blood Transfusion Council (NBTC) is the apex policy-making body, under

NACO, making central licensing mandatory, increasing voluntary donation base and setting of quality standards.

8 State Blood Transfusion Councils (SBTCs) are responsible for the smooth functioning of blood transfusion services at a state level.

9 Drug Standard Control Organization (CDSCO) and State Drug Control Organizations' play a vital role in key aspects such as, approval of licenses, and enforcement of standard transfusion practices to ensure safe, quality and efficacious blood and its components in clinical practices.

10 National Institute of Biologicals (NIB) Nodal governmental organization under the Ministry of Health and Family Welfare responsible for the testing and release of biologicals in India including blood plasma.

11 ISPE Baseline Guide Vol. 6, Biopharmaceutical Manufacturing Facilities ISPE Baseline Guide Vol. 3, Sterile Manufacturing Facilities.

12 ISO EN 14644: Clean room and associated controlled environments.

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15 PIC/s Guide to Good Manufacturing Practice for Medicinal Products.

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34 01/2008:1643; Human coagulation factor VIII (rDNA), European Pharmacopoeia.

35 1998:1223; Human coagulation factor IX, freeze-dried, European Pharmacopoeia.

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CHAPTER 17

List of Annexures

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List of Annexures

Annexure No Description

NA Project Cost Summary

1 Administration equipment and office furniture

2 Canteen equipment

3 Utility equipment

4 Plasma storage equipment

5 Bulk plasma manufacturing equipment

6 Fill finish equipment

7 QC / R&D equipment

8 QA equipment

9 Finished goods storage

10 ETP / STP

Layouts

A Plot Layout-GF (Bulk Manufacturing)

B Conceptual Layout-GF (Bulk Manufacturing)

C Conceptual Layouts-FF (Fill Finish)

D Terrace Floor (Future Provision)

E Section and Elevation Layout

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PROJECT COST SUMMARY

Sr. No. ITEMS DESCRIPTION COST IN INR REMARKS

1 SUB Head I CIVIL CONSTRUCTION

MAIN BUILDING (GROUND - FIRST - TERRACE FLOOR)

1A Building work (Civil) 22,50,00,000

APPROX. 174,000 sqft

1B UTILITY AND OTHER SUPPORT BUILDINGS 2,60,00,000

1C ADMINISTRATION BUILDING 1,00,00,000

1D RAIN WATER HARVESTING 6,00,000

1E EPOXY FLOORING (PRODUCTION / QC / R&D) 2,00,00,000

1F Development work civil ROADS & LANDSCAPING 1,52,00,000

2 SUB Head II EQUIPMENT

2A ADMINISTRATION ADMINISTRATION EQUIPMENT AND OFFICE FURNITURE 50,80,000 ANNEX 1

2B CANTEEN CANTEEN EQUIPMENT 13,60,000 ANNEX 2

2C UTILITIES UTILITIES (EXCLUDING HVAC) 5,75,00,000 ANNEX 3

2D WAREHOUSE (PLASMA STORAGE) PLASMA STORAGE EQUIPMENT 2,01,22,000 ANNEX 4

2E BULK PLASMA BULK PLASMA PRODUCTION EQUIPMENT 72,00,18,520 ANNEX 5

2F FILL FINISH FACILITY FILL FINISH EQUIPMENT 3,26,13,000 ANNEX 6

2G QUALITY CONTROL / R&D QC/R&D EQUIPMENT 3,96,94,000 ANNEX 7

2H QUALITY ASSURANCE QA EQUIPMENT 31,07,500 ANNEX 8

2I WAREHOUSE (FG) FINISHED GOODS STORAGE 21,50,000 ANNEX 9

3 SUB Head III ELECTRICAL WORK

3A ELECTRICAL ELECTRICAL COST FOR ALL BLOCKS (including UPS) 9,00,00,000

3B TRANSFORMER - 3000 KVA 65,00,000

3C DG SET 1000 KVA x 2 nos 1,40,00,000

3D INSTRUMENTATION FIRE ALARM, CCTV SYSTEM 3,00,00,000

HVAC FOR ALL BLOCKS INCLUDING DUCTING AND

3E MECHANICAL / HVAC 11,00,00,000
CHILLERS
CLEAN ROOM MODULAR PARTITION
3F 9,00,00,000
(Bulk/ QC/ Warehouse / Plasma storage)
3G COLD ROOM (4 NOS) 80,00,000
3H BOILER & PIPING x 1 nos 75,00,000
3I AIR COMPRESSOR 75,00,000
3J UTILITY PIPING 1,25,00,000
3K KILL TANK with storage 25,00,000
3L ETP /STP 33,00,000 ANNEX 10
3M SOLAR POWER EQUIPMENT 90,00,000

4 SUB Head IV LICENSING AND PROFESSIONAL FEES
4A TECHNOLOGY TRANSFER FEE (6 products) 2,20,00,000
4B PROJECT CONSULTANCY CHARGES 1,50,00,000

Grand Total: 1,60,62,45,020
Add for third party quality assurance @ 1% 1,60,62,450
Total: 1,62,23,07,470
Add 3% contingencies 4,86,69,224
Total: 1,67,09,76,694
Add EPF, ESI @ 4.25% 7,10,16,510
Total: 1,74,19,93,204
Add labour cess @1% 1,74,19,932
Grand Total 1,75,94,13,136
Say 1,759 LAKHS

1 of 13
PROJECT COST SUMMARY - ANNEX 1
ADMIN EQUIPMENT BUDGET ESTIMATE IN INR
1 Office Cabinets for Documents 20 2,40,000.00
2 Computers / Laptops 50 25,00,000.00
3 Office Furniture NA 12,00,000.00
4 Printer/scanner / xerox machine 1 5,00,000.00
5 Water Dispenser 2 40,000.00
6 Projector for conference / meeting room 2 6,00,000.00
Total Amount in INR 50,80,000.00

PROJECT COST SUMMARY - ANNEX 2
CANTEEN EQUIPMENT BUDGET ESTIMATE IN INR
S/No Name of Equipment Quantity Approx Cost (INR)
1 Refrigerator 2 3,00,000.00
2 SS furniture (serving table etc) NA 4,00,000.00
3 SS Water cooler dispenser 1 50,000.00
4 Microwaves 2 60,000.00
5 Kitchenware na 5,00,000.00
6 Bain marie 1 50,000.00
Total Amount in INR 13,60,000.00

PROJECT COST SUMMARY - ANNEX 3
UTILITY BUDGET ESTIMATE IN INR
S/No Name of Equipment Quantity Approx Cost (INR)
Clean Utilities
1 Soft water generation 8000L/hour and storage system 1
Purified water generation 5000L/ hour storage tank 10KL and
2 1
distribution system
3,00,00,000.00
Water for injection generation 1000L / hour, Storage tank 10KL
3 1
and distribution system
4 Pure steam generator 1000Kg / hour and distribution system 1
Black Utilities
1 Plant steam generation 2000Kg/Hour and distribution System 2 70,00,000.00
2 Compressed air generation and distribution system 2 75,00,000.00
3 PASSENGER LIFT X 2 NOS 2 70,00,000
4 MATERIAL LIFT X 3 NOS 3 60,00,000

Total Amount in INR 5,75,00,000.00
PROJECT COST SUMMARY - ANNEX 4
PLASMA STORAGE EQUIPMENT BUDGET ESTIMATE IN INR
S/No Name of Equipment Quantity Approx Cost (INR)
1 Analytical weighing Balance 1 1,70,000.00
2 RLAF 3 12,00,000.00
3 Precision Balance 1 1,44,000.00
4 Commercial Balance 1 1,20,000.00
5 Refrigerator 200L 2 2,18,000.00
6 Deep freezer -20°C (1000L) 32 1,44,00,000.00
7 Air curtain 10 1,00,000.00
8 PVC pallets 50 3,00,000.00
9 Storage racks SS 25 7,00,000.00
10 Sampling rods 2 10,000.00
11 Liquid sampler 2 20,000.00
12 Dedusting tunnel 1 12,50,000.00
13 DPB 4 6,00,000.00
14 LAF cabinet 1 4,50,000.00
15 Computer / Laptops 4 1,20,000.00
16 Office Furniture NA 2,00,000.00
17 Office Cabinets 10 1,20,000.00
Total Amount in INR 2,01,22,000.00
PROJECT COST SUMMARY - ANNEX 5
BULK PLASMA MANUFACTURING EQUIPMENT BUDGET ESTIMATE IN INR
S/No Name of Equipment Quantity Approx Cost (INR)

1 Chromatography AKTA-1 (471 L/hr (Pipe MOC SS) 1 1,71,00,000
 2 Chromatography Column Dia 1000mm x Height 500mm 2 4,46,12,200
 3 Chromatography AKTA-1 (500 L/hr pipe MOC SS) 1 1,71,00,000
 4 Chromatography Column Dia 1000mm x Height 500mm 1 2,23,06,100
 5 Chromatography AKTA-1 (785 L/hr pipe MOC PP) 1 2,00,31,420
 6 Chromatography Column Dia 1000mm x Height 500mm 1 2,23,06,100
 7 Chromatography AKTA-1 (1178 L/hr pipe MOC PP) 1 2,00,31,420
 8 Chromatography Column Dia 1000mm x Height 500mm 2 4,46,12,210
 9 Chromatography AKTA-1 (864 L/hr pipe MOC PP) 1 2,00,31,420
 10 Chromatography Column Dia 1000mm x Height 500mm 1 2,23,06,100
 11 Chromatography AKTA-1 (864 L/hr pipe MOC PP) 1 2,00,31,420
 12 Chromatography Column Dia 1000mm x Height 500mm 1 2,23,06,100
 13 Chromatography AKTA-1 (667 L/hr pipe MOC PP) 1 1,71,00,000
 14 Chromatography Column Dia 1000mm x Height 500mm 1 2,23,06,100
 15 Chromatography Column Dia 300mm x Height 500mm 1 48,96,570
 16 Chromatography AKTA-1 (785 L/hr pipe MOC PP) 1 2,00,31,420
 17 Chromatography Column Dia 1000mm x Height 500mm 1 2,23,06,100
 18 Chromatography Column Dia 300mm x Height 500mm 1 48,96,570
 19 Chromatography AKTA-1 (30 L/hr pipe MOC PP) 1 1,46,57,140
 20 Chromatography Column Dia 300mm x Height 750mm 1 48,96,570
 21 Chromatography Column Dia 600mm x Height 500mm 1 1,42,44,570
 22 Chromatography AKTA-1(160 L/hr pipe MOC PP) 1 1,46,57,140
 23 Chromatography Column Dia 450mm x Height 500mm 1 1,20,18,850
 24 TFF System 40 sqm 1 1,93,50,000
 25 Cassette 40 sqm 1 55,90,000
 26 TFF System 30 sqm 2 3,87,00,000
 27 Cassette 30 sqm 2 86,00,000
 28 TFF System 20 sqm 1 94,60,000
 29 Cassette 20 sqm 1 30,10,000
 30 TFF System 5 sqm 2 73,10,000
 31 Cassette 5 sqm 2 6,88,000
 32 Continuous Centrifuge 1000L/hr 2 3,00,00,000
 33 Process vessel 150L 3 32,25,000
 34 Process vessel 250L 4 51,60,000
 35 Process vessel (Mobile tank) 300L 2 25,80,000
 36 Process vessel 1500L 20 4,30,00,000
 37 Process vessel 2000L 6 1,54,80,000
 38 Buffer Vessel 500L 14 2,40,80,000
 39 Buffer Vessel 1000L 10 1,93,50,000
 40 Depth filtration system 2 21,00,000
 41 Nano filter holder 1 5,00,000
 42 Cold room 2-8°C 2 15,00,000
 43 Rectangular tank 1 4,30,000
 44 Buffer preparation tank 2 17,20,000
 45 Dispensing Booth 1 4,00,000
 46 Analytical Balance 1 1,70,000
 47 Precision Balance 1 1,50,000
 48 Commercial Balance 1 1,20,000
 49 PVC bag Sealer 1 5,000
 50 Rotary Bag Sealer 1 3,00,000
 51 pH/ Conductivity meter 2 3,00,000
 52 Magnetic Stirrer 2 1,50,000
 53 Peristaltic pump 2 2,00,000
 54 Filter Integrity machine 1 20,00,000
 55 Fume Hood 1 4,50,000
 56 Autoclave 1000L 2 40,00,000
 57 DHS 1000L 1 20,00,000
 58 Ceiling LAF 3 12,00,000
 59 Drying Oven 2 10,00,000
 60 Acid tank 1000L 2 38,70,000
 61 Base tank 1000L 2 38,70,000
 62 Central CIP system 2 21,50,000
 63 Refrigerator 200L 1 1,50,000
 64 Laminar air flow cabinet 1 4,35,000
 65 Dynamic pass box 4 4,80,000
 66 Tumble Dryer 25 Kg 1 20,00,000
 67 Barrier washer Extractor 25Kg 1 30,00,000
 68 SS Furniture (Table, chair, racks, Ladders etc) NA 40,00,000
 69 Lockers NA 2,00,000
 70 Laptop / Computers 10 5,00,000
 71 Office furniture / Cabinets NA 3,00,000
 Total Amount in INR 72,00,18,520
 PROJECT COST SUMMARY - ANNEX 6
 FILL FINISH EQUIPMENT
 Quantit
 S/No Name of Equipment Approx Cost (INR)
 y
 1 Formulation Vessel 100L 1 25,00,000.00
 2 Ceiling LAF 2 8,00,000.00
 3 Laminar air flow cabinet 1 4,35,000.00
 4 Continuous Particle Counter 1 30,00,000.00
 5 Dynamic pass box 7 8,40,000.00
 6 Pass Box static 1 38,000.00
 7 Vial washing machine 1 45,00,000.00
 8 Sterilizing tunnel 1 45,00,000.00
 9 Vial filling & Sealing line 120 vials /min 1 75,00,000.00
 10 Lyophilizer (1sq meter) 1 55,00,000.00
 11 Labelling machine 1 9,00,000.00
 12 Double door autoclave 1000L 1 20,00,000.00
 13 Cold room 1 8,00,000.00
 14 SS furniture (Table, Chair, racks, cabinets etc) 1 2,00,000.00
 Total Amount in INR 3,26,13,000.00

PROJECT COST SUMMARY - ANNEX 7

QC & R&D EQUIPMENT

EQUIPMENT BUDGET ESTIMATE IN INR

1 Akta pure -150 (For R&D) 2 40,00,000
2 Columns XK 50/100 (For R & D) 1 7,00,000
3 Columns XK 50/60 (For R&D) 1 7,00,000
4 Column XK 50/30 (For R & D) 3 14,00,000
5 HPLC-UV 1 34,00,000
6 HPLC Columns and accessories 1 10,00,000
7 Electrophoresis (Minigel) with trans blot 2 6,00,000
8 Gel Doc system densitometer 1 5,00,000
9 Power packs 2 2,00,000
10 Endotoxin Detection system (Tube heating block) 1 2,00,000
Micropipettes- variable volume range of 2-20µL, 20-200µL, 100-
11 2 set each 3,00,000
1000µL, 500-5000µL and multichannel 30-300µL
12 Lab Refrigerator 2-8 ° C (SS) with data control 4 4,50,000
13 Flame photometer 1 4,50,000
14 Weighing Balance Analytical 2 3,50,000
15 UV-VIS spectrophotometer Double Beam with data acquisition. 1 9,00,000
16 Volumetric KF 1 7,00,000
17 Moisture Analyser 1 8,00,000
18 Stability Chambers (5±3° C, 25, 37 deg. C) 200L (1set) 3 14,00,000
19 Muffle Furnance 1 80,000
20 pH and conductivity meter with printer 2 2,60,000
21 Vaccum Oven 1 1,20,000
22 Melting point apparatus 1 90,000
23 Incubator Sterility Test 500L 2 14,40,000
24 Incubators for Normal Media plates Incubation 500L 1 3,00,000
25 Incubator for EMP and water samples 500L 2 6,00,000
26 Hot Air Oven 1 1,60,000
27 Thermostatic water bath (10-20 L) 2 2,00,000
28 Vaccum pump 2 1,20,000
29 Desiccator Glass 1 4,000
High Quality Laboratory water generation system (10 L /hr) Milli -Q
30 1 14,50,000
water
31 Sonicator 1 1,80,000
32 Vortex Shaker 6 2,10,000
33 Magnetic Stirrer with hot plate 4 4,00,000
34 LAF Vertical 2 6,00,000
35 Biosafety Cabinet 2 14,00,000
36 Autoclave Double door 1 12,50,000
37 Dry Heat Steriliser 1 7,50,000
38 Fume Hood (Table Top) 1 4,50,000
39 Deep freezer (-20 Degrees C) with Data acquisition 1 4,00,000
40 Air sampler 1 3,00,000
41 Particle counter 1 20,00,000
42 Decontamination autoclave 1 10,00,000
43 Microscope Binocular 1 1,00,000
44 Colony Counter 1 1,50,000
45 Microplate reader ELISA 1 6,00,000
46 Microplate washer 1 3,50,000
47 TOC analyser (Grab Samples) 1 25,00,000
48 Timers/ Stop watch 5 10,000
49 Ice flaking machine 1 2,50,000
50 Standard Weight Box 1 Set 10,00,000
51 Printer/scanner / xerox machine 1 5,00,000
52 Centrifuge table top 1 4,50,000
53 Size exclusion L 1,50,000.00
54 Anion L 1,50,000.00
55 Cation L 1,50,000.00
56 Affinity L 1,50,000.00
57 SS furniture racks 25 2,50,000
58 Computers / Laptops 10 5,00,000
59 Office Furniture NA 5,00,000
60 Office Cabinets for Documents 10 1,20,000

TOTAL 3,96,94,000

PROJECT COST SUMMARY - ANNEX 8

QA EQUIPMENT BUDGET ESTIMATE IN INR

1 Anemometer 1 15,000.00
2 Lux meter 1 12,000.00
3 Datalogger 32 channel 1 4,00,000.00
4 Paper shredder 1 10,000.00
5 Laminator 1 10,000.00
6 Printer/scanner / xerox machine 1 5,00,000.00
7 Computers / Laptops 10 5,00,000.00
8 Numbering Machine 5 25,000.00
9 Spiral Binding Machine 1 15,000.00
10 A3 +A4 Lamination Machine 1 25,000.00
11 Bi-Focal Glass 1 80,000.00
12 Tachometer 2 6,000.00
13 Clamp Meter/ Multi meter 5 10,000.00
14 Level Indicator 2 8,000.00
15 Measuring TAPE 2 1,500.00
16 Stop Watch 5 50,000.00
17 Temperature recorders 24 Nos. (data Loggers) 20 2,00,000.00
18 Office Cabinets for Documents 20 2,40,000.00
19 Projector for conference / meeting room 1 3,00,000.00
20 Office Furniture NA 7,00,000.00
Total Amount in INR 31,07,500.00

PROJECT COST SUMMARY - ANNEX 9

FINISHED GOODS STORAGE

- 1 Air curtain 10 1,00,000.00
- 2 PVC pallets 50 3,00,000.00
- 3 Storage racks SS 50 15,00,000.00
- 4 Hydraulic trolley 1 2,50,000.00
- Total Amount in INR 21,50,000.00

PROJECT COST SUMMARY - ANNEX 10

ETP BUDGET ESTIMATE IN INR

Approx Cost

S/No Name of Equipment Qty.

(INR)

- 1 Biowaste reception tank 15000L 1
- 2 Kill tank 500L 2
- 3 Bar Screen NA
- 4 Collection tank NA
- 5 Equalization tank NA 25,00,000.00
- 6 Neutralization tank NA
- 7 Chemical coagulation NA
- 8 Biological oxidation tank NA
- 9 Filtration System NA

STP BUDGET ESTIMATE IN INR

Primary treatment plant : Bar screening - 3 stages, pH

1

correction, BOD & COD removal secondary clarifier

Secondary treatment : Electro coagulation process, Flash

2 8,00,000.00

mixer, HRSC clarifier, Sand filter

Sludge Management: Sludge thickner, Screw pump, Filter

3

press, Sludge storage

Total Amount in INR 33,00,000.00

FEASIBILITY AND DETAILED PROJECT REPORT

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FEASIBILITY AND DETAILED PROJECT REPORT

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BALANCE SHEET (LIABILITIES) INPUT | 2018 | 2019 | 2020 | 2021 | 2022 | 2023 | 2024 | ||||
 | Audited | Audited | Audited | Provisional | Projected | Projected | Projected | ||||
 CURRENT LIABILITIES |||||
 Short term borrowings from Banks (including bills purchased, discounted & excess borrowings placed on repayment basis) |||||
 Bank Borrowings - SBI | 656.81 | 1720.87 | -13.86 | 1126.65 | 3500 | 3500 | 3500 | ||||
 Bank Borrowings - HDFC || | 0 | 45.19 | 400 | 400 | 400 | ||||
 Sub Total | 656.81 | 1720.87 | -13.86 | 1171.8400000000001 | 3900 | 3900 | 3900 | ||||
 Short term borrowings from Associates & Group Concerns |||||
 Short term borrowings from Others |||||
 Creditors for purchases | 1214.58 | 1510.45 | 1995.98 | 2682.6 | 175.18338133561656 | 183.9590583818493 | 193.2949455475171 | ||||
 Advances/ payments from customers/deposits from dealers. | 67.01 | 134.12 | 310.88 | 284.45 | 711.125 | 782.2375000000001 | 860.4612500000002 | ||||
 Provisions ||||| | 601.51 | | 601.51 |
 - Tax | 92.68 | 119.36 | 293.55 | 185.44 | 203.94261875000046 | 218.51642689236104 | 236.88771624913178 | | 2682.6 | | 2263 |
 -Others | 323.12 | 229.71 | 37.88 | 44.41 | 74.125 | 107.43249999999999 | 157.38374999999996 | | -2081.09 | | 2864.51 |
 ||||| | 833.4 | | 2682 |
 Dividends Payable ||||| | 182.51000000000022 | S
 Interest payable on CC || | 8.02 | ||||| | 833 | D
 Instalments Machinery/ Vehicles AXIS/ HDFC Etc || | 116.59 | 231.84 | 161.65 | 184.18 | 184.18 | 184.18 |
 || | 1015.5100000000002 | T
 ||||| | 545.51 | A
 Installments of Term Loans NSIC || | 0 | 141.7 | 0 | 0 | 0 | || | 470.0000000000002 | TOTAL S D
 ||||| | 352.50000000000017 | NET DP
 COVID- SBI || | 109.02 | 92.63 | 0 | 0 | ||||
 Other Payable | 264.89 | 762.6300000000001 | 542.72 | 545.51 | 150.77656874999934 | 198.93040748263883 | 54.2115395566257 | ||||
 Total Current Liabilities | 2619.0899999999997 | 4593.73 | 3407.0100000000007 | 5326.62 | 5491.962568835617 | 5575.255892756849 | 5586.4192013523125 | ||||
 TERM LIABILITIES |||||
 Debentures |||||
 Preference share capital |||||
 Dealer's Deposit |||||
 Deferred Tax Liability |||||
 Term Loans - From Banks | 352.64 | 196.41 | 247.85 | 92.63 | 552.5500000000001 | 368.3666666666667 | 184.1833333333328 | ||||
 Long Term Provisions | 14.68 | 16.7 | 18.96 | 31.74 | 0 | 0 | 0 | | 2821.1591875 | 2895.676834375 | 2897.5142558047946 |
 Term Deposits | 23.86 | ||||| | 1404.39 | 1404.39 | 1404.39 |
 Unsecured loans others |||||
 Unsecured loans (Quasi Equity) | 19.09 | 19.09 | 19.09 | 0 | 0 | 0 | 0 | | 1416.7691874999998 | 1491.286834375 | 1493.1242558047945 |
 Sales tax deferred | 1066.56 | 1143.68 | 1592.42 | 1404.39 | 1404.39 | 1404.39 | 1404.39 | | 711.125 | 782.2375000000001 | 860.4612500000002 |
 Total Term Liabilities | 1476.83 | 1375.88 | 1878.3200000000002 | 1528.76 | 1956.94 | 1772.7566666666667 | 1588.5733333333333 | | 211.44261875000046 | 226.01642689236104 | 244.38771624913178 |
 ||||| | 66.615 | 99.92249999999999 | 149.88374999999996 |
 TOTAL OUTSIDE LIABILITIES | 4095.9199999999996 | 5969.61 | 5285.3300000000001 | 6855.38 | 7448.902568835618 | 7348.012559423516 | 7174.992534685645 | | 184.18 | 184.18 | 184.18 |
 ||||| | 92.63 | 0 | 0 |
 NET WORTH ||||| | 150.77656874999934 | 198.93040748263883 | 54.2115395566257 |
 Equity Share Capital |||||
 Share Capital (Paid-up) | 179.4 | 179.4 | 190.4 | 183.2 | 183.2 | 183.2 | 183.2 | ||||
 Share Application (finalised for allotment) |||||
 Sub Total (Share Capital) | 179.4 | 179.4 | 190.4 | 183.2 | 183.2 | 183.2 | 183.2 | ||||
 General Reserve | 3018.91 | 3409.64 | 5278.45 | 4955.38 | 5567.197856250001 | 6222.747136927084 | 6933.410285674479 | ||||
 Capital Reserve |||||
 Capital Redemption Reserve |||||
 Statutory Reserve |||||
 Employees Stock Option Outstanding |||||
 Revaluation Reserve |||||
 Partners capital / Proprietor's capital |||||
 Other Reserves & Surplus: |||||
 Share Premium |||||
 Capital subsidy |||||
 Others |||||
 Balance in P&L Account (+ / -) |||||
 NET WORTH | 3198.31 | 3589.04 | 5468.849999999999 | 5138.58 | 5750.397856250001 | 6405.947136927084 | 7116.610285674479 | ||||
 ||||| | 18320180 | ||
 TOTAL LIABILITIES | 7294.23 | 9558.65 | 10754.18 | 11993.96 | 13199.300425085617 | 13753.959696350601 | 14291.602820360124 | | 512752619 | ||
 ||||| | 9263771 | ||
 DIFFERENCE IN B/S | 0 | 0 | 0 | 0 | 0 | 0 | 0 | | 140439400 | ||
 ||||| | 3173579 | || | | |
 |||| | 66.615 | 99.92249999999999 | 149.88374999999996 | | 142257332 | ||
 |||| | 7.51 | 7.51 | 7.5 | | 268260926 | ||
 |||| | 74.125 | 107.43249999999999 | 157.38374999999996 | | 99160541 | ||
 ||||| | 5769094 | 22985086 | 17215992 |
 ||||| | 1199397442 | ||
 ||||| | 495536627 | ||
 ||||| | 512752619 | ||
 [SHEET: P&L]
 | (in lacs) |||||
 PROFIT & LOSS ACCOUNT INPUT | 2018 | 2019 | 2020 | 2021 | 2022 | 2023 | 2024 | ||||
 | Audited | Audited | Audited | Provisional | Projected | ||||
 Gross Sales |||||
 - Domestic | 12698.95 | 10900.11 | 14783.52 | 11355.07 | 14193.8375 | 14903.529375 | 15648.70584375 | |||

|
 - Export |||||
 Sub Total (Gross Sales) | 12698.95 | 10900.11 | 14783.52 | 11355.07 | 14193.8375 | 14903.529375 |
 15648.70584375 ||||
 Less Excise Duty |||||
 Net Sales | 12698.95 | 10900.11 | 14783.52 | 11355.07 | 14193.8375 | 14903.529375 | 15648.70584375 ||
 |
 % wise rise/fall in net sales as compared to previous year || -14.165265632197938 | 35.62725513779218 |
 -23.19102622379515 | 25 | 5.00000000000003 | 5 ||||
 Other Operating Income |||||
 Export Incentives |||||
 Duty Drawback |||||
 Others | 205.54 | 119.17 | 190.42 | 117.94 | 129.734 | 142.70740000000004 | 156.97814000000005 ||||
 Total Operating Income | 12904.490000000002 | 11019.28 | 14973.94 | 11473.01 | 14323.5715 | 15046.236775
 | 15805.68398375 ||||
 Cost of Sales |||||
 Raw materials consumed |||||
 i) Imported |||||
 ii) Indigenous | 2044.45 | 2642.71 | 3399.11 | 2780.62 | 3491.684025 | 3606.65410875 | 3786.9868141875 |
 |||
 | 2035.2899999999997 | 2650.23 | 3399.1 | 2764.68 | 3483.5830250000004 | 3597.7430087499997 |
 3777.1846041875 ||||
 Other Spares consumed |||||
 i) Imported |||||
 ii) Indigenous |||||
 Power and fuel | 103.97 | 170.62 | 226.4 | 184.15 | 283.87675 | 298.0705875 | 312.974116875 ||||
 Direct labour and wages | 442.48 | 465 | 504.31 | 491.47999999999996 | 638.7226875 | 670.658821875 |
 704.19176296875 ||||
 Repair and Maintanance |||||
 Other manufacturing expenses | 7819.27 | 6654.86 | 8418.5444 | 6242.2079 | 8090.487374999999 |
 8495.01174375 | 8919.7623309375 ||||
 Depreciation | 232.87 | 235.88 | 310.85 | 282.47 | 283.616 | 358.616 | 386.616 ||||
 Sub Total | 10643.040000000003 | 10169.07 | 12859.214400000003 | 9980.9279 | 12788.386837499998 |
 13429.011261874999 | 14110.531024968748 ||||
 Add: Op. Stock of WIP | 873.99 | 186.67 | 1265 | 839.46 | 520.5 | 546.525 | 573.85125 ||||
 Less: Cl. Stock of WIP | 186.67 | 1265 | 839.46 | 520.5 | 546.525 | 573.85125 | 602.5438125000001 ||||
 Total Cost of Production | 11330.360000000002 | 9090.74 | 13284.754400000002 | 10299.887900000002 |
 12762.361837499999 | 13401.685011874999 | 14081.838462468748 ||||
 Add Opening Stock of Finished Goods | 60.23 | 69.39 | 65.06 | 65.07 | 81.01 | 89.11100000000002 |
 98.02210000000002 ||||
 Less: closing balance | 69.39 | 61.87 | 65.07 | 81.01 | 89.11100000000002 | 98.02210000000002 |
 107.82431000000004 ||||
 Total Cost of Sales | 11321.200000000003 | 9098.259999999998 | 13284.744400000001 | 10283.947900000001 |
 12754.260837499998 | 13392.773911875 | 14072.036252468748 ||||
 Selling, Gen. & Administration Exp | 655.05 | 1197.7 | 443.5056 | 340.65209999999996 | 425.81512499999997
 | 447.10588125 | 469.46117531249996 ||||
 Cost of Sales + SGA | 11976.250000000002 | 10295.96 | 13728.250000000002 | 10624.6 | 13180.075962499997 |
 13839.879793125 | 14541.497427781247 ||||
 Operating Profit before Interest | 928.23999999999998 | 723.32000000000015 | 1245.6899999999987 |
 848.4099999999999 | 1143.4955375000027 | 1206.3569818750002 | 1264.186559687521 ||||
 Interest payment to Banks |||||
 Interest - WC | 39.77 | 121.22 | 124.61 | 43.4 | 292.5 | 292.5 | 292.5 ||||
 Interest - Term Loans | 77.13 | 62.110000000000014 | 89.7 | 106.43 | 35.22506250000001 |
 39.79127430555557 | 24.135690972222243 ||||
 Interest payment to FIs |||||
 Interest - WC |||||
 Interest - Term Loans |||||
 Total Interest | 116.9 | 183.33 | 214.31 | 149.83 | 327.72506250000004 | 332.2912743055556 |
 316.6356909722222 ||||
 | 0.009205485492895082 | 0.016819096321046303 | 0.01449654750695369 | 0.01319498690893143 |
 0.02308925000022017 | 0.022296146499564008 | 0.020233985745133336 ||||
 Operating Profit after Interest | 811.33999999999998 | 539.99000000000015 | 1031.3799999999987 |
 698.5799999999998 | 815.7704750000026 | 874.0657075694446 | 947.55086499653 ||||
 Non Operating Items |||||
 Add Non Operating Income |||||
 Profit on sale of assets/ investments |||||
 Investments and Dividend/Interest Income |||||
 Forex gains |||||
 Non-op. income from subsidiaries |||||
 Tax Refund |||||
 Other Non Operating Income |||||
 Total non-operating Income | 0 | 0 | 0 | 0 | 0 | 0 | 0 ||||
 Deduct Non Operating Expenses |||||
 Loss on sale of assets |||||
 Prem. Expenses w/off |||||
 Other Non- operating exp. |||||
 Total Non-operating expenses | 0 | 0 | 0 | 0 | 0 | 0 | 0 ||||
 Net of Non-operating Income / Expenses | 0 | 0 | 0 | 0 | 0 | 0 | 0 ||||
 Profit Before Interest Depreciation & Taxes (PBITD) | 1161.11 | 959.20000000000015 | 1556.5399999999986 |
 1130.8799999999999 | 1427.1115375000027 | 1564.9729818750002 | 1650.8025559687521 ||||
 Profit Before Tax | 811.3399999999998 | 539.99000000000015 | 1031.3799999999987 | 698.5799999999998 |
 815.7704750000026 | 874.0657075694446 | 947.55086499653 ||||
 Provision for Taxation: |||||
 Current | 275.87 | 155.62 | 251.57999999999998 | 187.14000000000001 | 203.94261875000046 |
 218.51642689236104 | 236.88771624913178 ||||
 Deferred | -11.46 | -6.36 |||||
 Sub Total: Provision for Taxation | 264.41 | 149.26 | 251.57999999999998 | 187.14000000000001 |
 203.94261875000046 | 218.51642689236104 | 236.88771624913178 ||||
 Net Profit After Tax | 546.92999999999998 | 390.73000000000015 | 779.7999999999988 | 511.4399999999998 |
 611.8278562500022 | 655.5492806770835 | 710.6631487473982 ||||
 Extraordinary Items adjustments: |||||
 Extraordinary Income adjustments (+) |||||

Extraordinary Expenses adjustments (-) |||||

SubTotal Extraordinary items | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 556.0999999999999 | 383.21000000000106

||||| 546.93 | 390.73

Adjusted PAT (excl Extraordinary Items) | 546.9299999999998 | 390.7300000000015 | 779.7999999999988 |

511.4399999999998 | 611.8278562500022 | 655.5492806770835 | 710.6631487473982 || 9.169999999999959 |

-7.519999999998959

Dividend Paid ||||| 7810.1 | 6662.38

On Equity Capital ||||| 9.17 | 7.52

On Preference Sh. Capital ||||| 7819.27 | 6654.86

Dividend tax |||||

Partners' withdrawal |||||

Dividend (%) |||||

Retained Profit | 546.9299999999998 | 390.7300000000015 | 779.7999999999988 | 511.4399999999998 |

611.8278562500022 | 655.5492806770835 | 710.6631487473982 |||

Cash Accruals | 779.7999999999998 | 626.6100000000015 | 1090.6499999999987 | 793.9099999999999 |

895.4438562500022 | 1014.1652806770835 | 1097.2791487473983 |||

PBDIT/Sales | 0.09143354371818141 | 0.0879991193556775 | 0.10528886219249534 | 0.09959251682288175 |

0.10054444666567464 | 0.10500687068797086 | 0.1054913149018053 |||

Operating Profits/Sales | 0.06389032164076555 | 0.04953986702886498 | 0.06976552269013055 |

0.06152141730522135 | 0.05747356731398416 | 0.05864823596990727 | 0.06055138836768257 |||

PBT/Sales | 0.06389032164076555 | 0.04953986702886498 | 0.06976552269013055 | 0.06152141730522135 |

0.05747356731398416 | 0.05864823596990727 | 0.06055138836768257 |||

PAT/Sales | 0.043068915146527845 | 0.035846427237890395 | 0.052747924716170354 | 0.04504067346128204 |

0.04310517548548814 | 0.04398617697743046 | 0.04541354127576197 |||

Cash Accruals/ Sales | 0.06140665173104862 | 0.05748657582354687 | 0.07377471671158145 |

0.069916786070011 | 0.06308680483695846 | 0.06804866519593004 | 0.07011948206475138 |||

[SHEET: Assessment]

FBF Method | 2018 | 2019 | 2020 | 2021 | 2022 | 2023 | 2024 |||||

Total Current Assets | 2486.41 | 5014.5499999999999 | 5731.1099999999999 | 6019.3699999999999 |

8204.663525085618 | 8344.4331673506 | 8564.961606070125 |||||

Other Current Liabilities | 1962.2799999999999 | 2872.8599999999997 | 3420.8700000000001 | 4154.78 |

1591.9625688356173 | 1675.2558927568489 | 1686.4192013523125 |||||

Working Capital Gap | 524.1300000000001 | 2141.6899999999996 | 2310.2399999999998 | 1864.5899999999992 |

6612.700956250001 | 6669.1772745937515 | 6878.542404717812 |||||

25% of WCG | 131.03250000000003 | 535.4224999999999 | 577.5599999999995 | 466.1474999999998 |

1653.1752390625002 | 1667.2943186484379 | 1719.635601179453 |||||

25% of TCA | 621.6025 | 1253.6374999999998 | 1432.7774999999997 | 1504.8424999999997 | 2051.1658812714045

| 2086.10829183765 | 2141.240401517531 |||||

Actual/ Projected WC | -132.67999999999984 | 420.81999999999997 | 2324.0999999999998 | 834.4499999999991 |

2712.7009562500007 | 2769.1772745937515 | 2978.542404717812 |||||

Flexible Bank Finance WCG | 393.0975000000001 | 1606.2674999999997 | 1732.6799999999998 |

1398.4424999999994 | 4959.5257171875 | 5001.882955945313 | 5158.906803538359 |||||

Flexible Bank Finance TCA | -97.47249999999985 | 888.0524999999998 | 877.4624999999983 |

359.7474999999995 | 4561.535074978596 | 4583.068982756102 | 4737.3020032002805 |||||

Flexible Bank Finance NWC | 656.81 | 1720.87 | -13.860000000000127 | 1030.14 | 3900 | 3900 | 3900 |||

|||

Limit recommended | 656.81 | 1720.87 | -13.860000000000127 | 1030.14 | 3900 | 3900 | 3900 |||||

Net Sales | 12698.95 | 10900.11 | 14783.52 | 11355.07 | 14193.8375 | 14903.529375 | 15648.70584375 |||

||||

NWC to TCA% | -5.336207624647578 | 8.391979340120246 | 40.552353732522995 | 13.86274643359686 |

33.06291535242078 | 33.18592430494568 | 34.7758990840878 |||||

Flexible Bank Finance to TCA% | 26.415997361657972 | 34.317535970326354 | -0.24183796856106637 |

17.11375110684341 | 47.533941984041356 | 46.737746252910185 | 45.53435472770839 |||||

Other Current Liabilities to TCA% | 78.92021026298961 | 57.2904846895534 | 59.68948423603807 |

69.02350245955972 | 19.403142663537864 | 20.076329442144132 | 19.689746188203813 |||||

Sundry Creditors to TCA% | 48.84874176020849 | 30.121346880577526 | 34.827110280556475 |

44.56612569089457 | 2.1351683807629205 | 2.204572254250042 | 2.256810414777877 |||||

Turnover Method | (C in lacs) |||||

| 2018 | 2019 | 2020 | 2021 | 2022 | 2023 | 2024 |||||

Net Sales | 12698.95 | 10900.11 | 14783.52 | 11355.07 | 14193.8375 | 14903.529375 | 15648.70584375 |||

||||

25% of Net Sales | 3174.7375 | 2725.0275 | 3695.88 | 2838.7675 | 3548.459375 | 3725.88234375 |

3912.1764609375 |||||

Margin @ 5% of Sales | 634.9475000000001 | 545.0055000000001 | 739.176 | 567.7535 | 709.691875 |

745.17646875 | 782.4352921875001 |||||

Available Margin* | -132.67999999999984 | 420.8199999999997 | 2324.0999999999998 | 834.4499999999991 |

2712.7009562500007 | 2769.1772745937515 | 2978.542404717812 |||||

Eligible Limit as per 5% Margin | 2539.79 | 2180.022 | 2956.704 | 2271.014 | 2838.7675 | 2980.705875 |

3129.74116875 |||||

Eligible Limit as per Available Margin | 3307.4175 | 2304.2075000000004 | 1371.7800000000002 |

2004.3175000000008 | 835.7584187499992 | 956.7050691562486 | 933.6340562196879 |||||

Limit recommended | 2539.79 | 2180.022 | 1371.7800000000002 | 2004.3175000000008 | 835.7584187499992 |

956.7050691562486 | 933.6340562196879 |||||

Working Capital assessment |||||

| 2018 | 2019 | 2020 | 2021 | 2022 | 2023 | 2024 |||||

Raw Material | 186.67 | 1265 | 839.46 | 520.5 | 546.525 | 573.85125 | 602.5438125000001 |

1.8666999999999998 | 12.65 | 8.3946 | 5.205 | 5.46525 | 5.738512500000001 | 6.025438125000001

Cost of Production | 11330.3600000000002 | 9090.74 | 13284.7544000000002 | 10299.8879000000002 |

12762.361837499999 | 13401.685011874999 | 14081.838462468748 | 113.30360000000002 | 90.9074 |

132.847544000000003 | 102.998879000000002 | 127.62361837499999 | 134.01685011875 | 140.81838462468747

FG | 1096.28615 | 634.38144 | 534.92589 | 1008.28 | 1039.79 | #REF! | #REF! |||||

months cost of sales | 9.888626988183649 | 2.505876539308533 | 2.0087089534490747 | 1.818623995382555 |

1.8365440094200767 | #REF! | #REF! |||||

Raw Material (Days) | 6.013449705040261 | 50.79069470692155 | 23.064250250648215 | 18.445103659817498 |

15.630463039674806 | 15.629057544958338 | 15.617882007996236 |||||

Finished Goods | 69.39 | 61.87 | 65.07 | 81.01 | 89.11100000000002 | 98.02210000000002 |

107.82431000000004 | 0.6939 | 0.6187 | 0.6507 | 0.8101 | 0.8911100000000002 | 0.9802210000000002 |

1.0782431000000003

Cost of Sales | 11321.200000000003 | 9098.259999999998 | 13284.744400000001 | 10283.947900000001 |

12754.260837499998 | 13392.773911875 | 14072.036252468748 | 113.21200000000003 | 90.98259999999999 |

132.84744400000002 | 102.83947900000001 | 127.54260837499999 | 133.92773911875 | 140.7203625246875

Total Inventory/Sales (Days) | 2.2371612549906366 | 2.482073495371643 | 1.7878063201577288 |

2.875223628855607 | 2.550168560483622 | 2.6714455672455273 | 2.796743303094857 ||||| |
Domestic receivables (Days Gross dom.Sales) | 1150.38 | 1424.56 | 1607.88 | 1443.58 | 3110.978082191781 |
3266.5269863013696 | 3429.853335616438 | 11.503800000000002 | 14.2456 | 16.0788 | 14.435799999999999 |
31.109780821917813 | 32.6652698630137 | 34.29853335616438
Net Sales | 12698.95 | 10900.11 | 14783.52 | 11355.07 | 14193.8375 | 14903.529375 | 15648.70584375 |
126.9895 | 109.00110000000001 | 147.83520000000001 | 113.55069999999999 | 141.938375 | 149.03529375 |
156.4870584375
Receivables Turnover (Days) | 33.06483606912383 | 47.702674560164986 | 39.69800155849216 |
46.402769864034305 | 80 | 80 | 80 | ||||| |
Purchases | 2044.45 | 2642.71 | 3399.11 | 2780.62 | 3491.684025 | 3606.65410875 | 3786.9868141875 |
20.4445 | 26.4271 | 33.9911 | 27.8062 | 34.91684025 | 36.0665410875 | 37.869868141875
Creditors | 1214.58 | 1510.45 | 1995.98 | 2682.6 | 175.18338133561656 | 183.9590583818493 |
193.2949455475171 | 12.1458 | 15.1045 | 19.9598 | 26.826 | 1.7518338133561655 | 1.839590583818493 |
1.932949455475171
creditors/purchases (Days) | 216.84154662623197 | 208.61700678470206 | 214.33042767077265 |
352.13333716940826 | 18.312634742916075 | 18.61699355823346 | 18.630288032830357 | ||||| |
Working Capital Cycle (Days) | -175.52609959707723 | -107.64156402224387 | -149.78036954147456 |
-284.41024001670087 | 79.86799685724235 | 79.68350955397041 | 79.78433727826075 | ||||| |
Total Amount of Term Loan | 0 |||| | uuuiioioioioioioioioioio || 100 |||| |
No. Total Installments | 84 ||||| ||||| |
DSCR Calculation ||||| ||||| |
Financial Year Ending/Year of Operation | I Year | II Year | III Year | IV Year | V Year | VI Year | VI
Year ||||| |
| 2018 | 2019 | 2020 | 2021 | 2022 | 2023 | 2024 ||||| |
Sources ||||| ||||| |
Net Profit after Tax | 546.9299999999998 | 390.73000000000015 | 779.7999999999998 | 511.4399999999998 |
611.8278562500022 | 655.5492806770835 | 710.6631487473982 | ||||| |
Interest on term loan | 77.13 | 62.110000000000014 | 89.7 | 106.43 | 35.22506250000001 |
39.79127430555557 | 24.135690972222243 | ||||| |
Deprecation | 232.87 | 235.88 | 310.85 | 282.47 | 283.616 | 358.616 | 386.616 | ||||| |
Preliminary Expenses W/o | 0 | 0 | 0 | 0 | 0 | 0 | 0 | ||||| |
Total(A) | 856.9299999999998 | 688.72000000000015 | 1180.3499999999999 | 900.3399999999999 |
930.6689187500023 | 1053.956554982639 | 1121.4148397196204 | ||||| |
Repayment Obligations ||||| ||||| |
Term loan installment | 0 | 0 | 0 | 141.7 | 0 | 0 | 0 | ||||| |
Interest on term loan | 77.13 | 62.110000000000014 | 89.7 | 106.43 | 35.22506250000001 |
39.79127430555557 | 24.135690972222243 | ||||| |
Total Repayment(B) | 77.13 | 62.110000000000014 | 89.7 | 248.13 | 35.22506250000001 | 39.79127430555557 |
24.135690972222243 | ||||| |
Debt Service Coverage Ratio(A/B) | 11.110203552443924 | 11.08871357269363 | 13.158862876254169 |
3.6285011888929186 | 26.420646343778724 | 26.487127476475113 | 46.46292666782386 | ||||| |
Average DSCR | 11.683656626970585 | ||||| ||||| |
[SHEET: Financial Indicators]
0 ||||| ||||| ||||| |
Financial Indicators | C in lacs) ||||| ||||| ||||| |
| 2018 | 2019 | 2020 | 2021 | 2022 | 2023 | 2024 ||||| ||||| |
Paid-up Capital | 179.4 | 179.4 | 190.4 | 183.2 | 183.2 | 183.2 | 183.2 | 1.794 | 1.794 |
1.9040000000000001 | 1.8319999999999999 | 1.8319999999999999 | 1.8319999999999999 | 1.8319999999999999 |
|
Reserves & Surplus | 3018.91 | 3409.64 | 5278.45 | 4955.38 | 5567.197856250001 | 6222.747136927084 |
6933.410285674479 | | 30.1891 | 34.096399999999996 | 52.7845 | 49.5538 | 55.67197856250001 |
62.227471369270845 | 69.33410285674479 | |
Intangible Assets | 9.52 | 9.21 | 4.73 | 7.39 | 5.5424999999999995 | 4.156874999999999 |
3.1176562500000005 | | 0.09519999999999999 | 0.09210000000000002 | 0.0473 | 0.0739 | 0.05542499999999995
| 0.041568749999999995 | 0.031176562500000005 | | 100
Unsecured loans (Quasi equity) | 19.09 | 19.09 | 19.09 | 0 | 0 | 0 | 0 | 0.1909 | 0.1909 | 0.1909 | 0 |
0 | 0 | 0 | |
Tangible Net Worth | 3188.79 | 3579.83 | 5464.12 | 5131.19 | 5744.855356250001 | 6401.790261927084 |
7113.492629424479 | | 31.8879 | 35.7983 | 54.6412 | 51.311899999999994 | 57.44855356250001 |
64.01790261927084 | 71.13492629424479 | |
Long Term Liabilities | 1476.83 | 1375.88 | 1878.3200000000002 | 1528.76 | 1956.94 | 1772.7566666666667 |
1588.5733333333333 | | 14.7683 | 13.7588 | 18.7832 | 15.2876 | 19.5694 | 17.727566666666668 |
15.885733333333333 | |
Capital Employed | 4665.62 | 4955.71 | 7342.4400000000005 | 6659.95 | 7701.795356250001 |
8174.546928593751 | 8702.065962757812 | | 46.6562 | 49.5571 | 73.4244 | 66.59949999999999 |
77.01795356250001 | 81.7454692859375 | 87.02065962757811 | |
Net Block | 1223.7 | 1273.46 | 2309.91 | 2836.1 | 2554.3215 | 2947.091125 | 2841.51434375 | | 12.237 |
12.7346 | 23.0991 | 28.361 | 25.543215 | 29.47091125 | 28.4151434375 | |
Non Current Assets | 3574.6 | 3261.43 | 2708.43 | 3131.0999999999995 | 2434.7729 | 2458.278529 |
2882.00921429 | | 35.746 | 32.6143 | 27.0843 | 31.310999999999993 | 24.347729 | 24.58278529 |
28.8200921429 | |
Net Working Capital | -132.67999999999984 | 420.8199999999997 | 2324.099999999998 | 834.4499999999991 |
2712.7009562500007 | 2769.1772745937515 | 2978.542404717812 | | -1.3267999999999984 | 4.208199999999997 |
23.240999999999982 | 8.344499999999991 | 27.127009562500007 | 27.691772745937513 | 29.78542404717812 | |
Current Assets | 2486.41 | 5014.549999999999 | 5731.109999999999 | 6019.369999999999 | 8204.663525085618
| 8344.4331673506 | 8564.961606070125 | | 24.864099999999997 | 50.14549999999999 | 57.31109999999999 |
60.19369999999999 | 82.04663525085618 | 83.444331673506 | 85.64961606070125 | |
Current liabilities | 2619.0899999999997 | 4593.73 | 3407.0100000000007 | 5326.62 | 5491.962568835617 |
5575.255892756849 | 5586.4192013523125 | | 26.190899999999996 | 45.93729999999999 | 34.070100000000004 |
53.2662 | 54.91962568835617 | 55.75255892756849 | 55.86419201352312 | |
Current Ratio | 0.9493411833881235 | 1.0916074736651915 | 1.6821523858163017 | 1.1300543308890063 |
1.4939401757112005 | 1.4966906143611016 | 1.533175599138137 | ||||| ||||| |
Debt Equity Ratio | 0.4631317835291756 | 0.38434227323643866 | 0.34375526159747594 | 0.2979347870571934 |
0.3406421708896441 | 0.2769157679547325 | 0.22331833546327265 | ||||| ||||| |
Debt Equity Ratio with USL as Quasi Capital | 0.45442472910458 | 0.3769992108743735 | 0.339076927566152 |
0.2979347870571934 | 0.3406421708896441 | 0.2769157679547325 | 0.22331833546327265 | ||||| ||||| |
DER (TOL/TNW) | 1.284474675347075 | 1.667568013006204 | 0.967279269122933 | 1.336021468704141 |
1.2966214302909702 | 1.1478058884752649 | 1.0086455287810066 | ||||| ||||| |
DER (TOL/TNW) with USL as Quasi Capital | 1.2708798334102271 | 1.6534182476965311 | 0.9604301130177397 |
1.336021468704141 | 1.2966214302909702 | 1.1478058884752649 | 1.0086455287810066 | 100 ||||| ||||| |
Net Sales | 12698.95 | 10900.11 | 14783.52 | 11355.07 | 14193.8375 | 14903.529375 | 15648.70584375 | |
126.9895 | 109.00110000000001 | 147.83520000000001 | 113.55069999999999 | 141.938375 | 149.03529375 |

BioBridge Master Knowledge — Page 193

2712.700956250001 | 2769.177274593751 | 2978.542404717812 | | 4.208200000000004 | 23.241000000000003 |
8.344500000000004 | 27.12700956250001 | 27.69177274593751 | 29.78542404717812 | | |
Difference | | 6.821210263296962e-13 | 0 | 1.1368683772161603e-12 | 0 | 0 | 0 | | | | | | | | | |
[SHEET: depn]
| | | tangible assets | % | as at 01-04-2011 | addition | deletion | as at 31-03-2012 | as at 01-04-2011
| depreciation for the year | as at 31-03-2012 | as at 31-03-2012 | as at 31-03-2011
| | | Eletrical instalation | 13.91 | 4802412 | 123087 | | 4925499 | 407389 | 645941 | 1053330 | 3872169
| 4395023
| | | furniture and fictures | 18.1 | 690267 | 170343 | | 860610 | 150348 | 116536 | 266884 | 593726 |
539919
| | | material handling equipment | 13.91 | 12900 | 0 | | 12900 | 5704 | 6075 | 11779 | 1121 | 7196
| | | Office equipment | 13.91 | 155291 | 217356 | | 372647 | 23667 | 40154 | 63821 | 308826 | 131624
| | | Factory equipment | 13.91 | 1474277 | 271698 | | 1745975 | 233729 | 172016 | 405745 | 1340230 |
1240548
| | | plant and machinary | 13.91 | 28562911 | 11636908 | | 40199819 | 2982981 | 4136166 | 7119147 |
33080672 | 25579930
| | | computers | 40 | 660786 | 117680 | | 778466 | 233564 | 204938 | 438502 | 339964 | 427222
| | | vehicles | 25.89 | 3102290 | 2472508 | | 5574798 | 1489460 | 887322 | 2376782 | 3198016 | 1612830
| | | factory building | 10 | 8023063 | 1175219 | | 9198282 | 311571 | 865153 | 1176724 | 8021558 |
7711492
| | | land | 0 | 1785570 | 0 | | 1785570 | 0 | 0 | 0 | 1785570 | 1785570
| | | leasehold land | | 0 | 8054410 | | 8054410 | 0 | 54820 | 54820 | 7999590 | 0
| | | intangible assets | | | | | | | | | |
| | | Total | | 49269767 | 24239209 | | 73508976 | 5838413 | 7129121 | 12967534 | 60541442 | 43431354
| | | previous year | | 6219636 | 43050132 | | 49269767 | 2146321 | 3692091 | 5838412 | 43431356 |
4073315
| | | computer software | | | 322600 | | 322600 | | 44639 | | 277961 |
| | | Capital WIP | | | 1940097 | | 1940097 | | | 1940097 |
[SHEET: GST Details]
				2021-04-01 00:00:00	3.2904									
				2021-05-01 00:00:00	3.3799									
				2021-06-01 00:00:00	7.465700000000001									
				2021-07-01 00:00:00	6.1107000000000005				100					
				2021-08-01 00:00:00	12.650699999999999									
				2021-09-01 00:00:00	7.4192									
				2021-10-01 00:00:00	4.7287									
				2021-11-01 00:00:00	9.916699999999999									
				2021-12-01 00:00:00	18.352899999999998									
				73.314900000000002										
				97.753200000000002										
[SHEET: Term Loan Calculation]														
				Type of Loan	2017	2018	2019	2020	2021					
				120										
				199										
				141										
				92										
				108										
				259										
				101										
				Total	0	0	0	0	0					
				Within Year	0	0	0	0						
[SHEET: ABSTRACT OF COST]														
PROJECT COST SUMMARY														
Sr. No.	ITEMS	DESCRIPTION	COST IN INR	REMARKS										
1	SUB Head 1	CIVIL CONSTRUCTION												
1A	Building work (Civil)	MAIN BUILDING (GROUND - FIRST - TERRACE FLOOR)												
APPROX. 174,000 sqft	225000000													
1B		UTILITY AND OTHER SUPPORT BUILDINGS	26000000											
1C		ADMINISTRATION BUILDING	10000000											
1D		RAIN WATER HARVESTING	600000											
1E		EPOXY FLOORING (PRODUCTION / QC / R&D)	20000000											
1F	Development work civil	ROADS & LANDSCAPING	15200000											
2	SUB Head II	EQUIPMENT												
2A	ADMINISTRATION	ADMINISTRATION EQUIPMENT AND OFFICE FURNITURE	5080000	ANNEX 1										
2B	CANTEEN	CANTEEN EQUIPMENT	1360000	ANNEX 2										
2C	UTILITIES	UTILITIES (EXCLUDING HVAC)	67500000	ANNEX 3										
2D	WAREHOUSE (PLASMA STORAGE)	PLASMA STORAGE EQUIPMENT	20122000	ANNEX 4										
2E	BULK PLASMA	BULK PLASMA PRODUCTION EQUIPMENT	720018520	ANNEX 5										
2F	FILL FINISH FACILITY	FILL FINISH EQUIPMENT	32613000	ANNEX 6										
2G	QUALITY CONTROL / R&D	QC/R&D EQUIPMENT	39694000	ANNEX 7										
2H	QUALITY ASSURANCE	QA EQUIPMENT	3107500	ANNEX 8										
2I	WAREHOUSE (FG)	FINISHED GOODS STORAGE	2150000	ANNEX 9										
3	SUB Head III	ELECTRICAL WORK												
3A	ELECTRICAL	ELECTRICAL COST FOR ALL BLOCKS (including UPS)	90000000											
3B		TRANSFORMER - 3000 KVA	6500000											
3C		DG SET 1000 KVA x 2 nos	14000000											
3D	INSTRUMENTATION	FIRE ALARM, CCTV SYSTEM	30000000											
3E	MECHANICAL / HVAC	HVAC FOR ALL BLOCKS INCLUDING DUCTING AND CHILLERS	110000000											
3F		CLEAN ROOM MODULAR PARTITION												
(Bulk/ QC/ Warehouse / Plasma storage)	90000000													
3G		COLD ROOM (4 NOS)	8000000											
3H		BOILER & PIPING x 1 nos	7500000											
3I		AIR COMPRESSOR	7500000											
3J		UTILITY PIPING	12500000											
3K		KILL TANK with storage	2500000											
3L		ETP /STP	5200000	ANNEX 10										
3M		SOLAR POWER EQUIPMENT	9000000											
4	SUB Head IV	LICENSING AND PROFESSIONAL FEES												
4A		TECHNOLOGY TRANSFER FEE (6 products)	22000000											
4B		PROJECT CONSULTANCY CHARGES	15000000											
		Grand Total:	1618145020											
		Add for third party quality assurance @ 1%	16181450.200000001											

		Total:	1634326470.2
		Add 3% contingencies	49029794.106
		Total:	1683356264.306
		Add EPF, ESI @ 4.25%	71542641.233005
		Total:	1754898905.539005
		Add labour cess @1%	17548989.055390052
		Grand Total	1772447894.5943952
		Say	1759
[SHEET: ANNEX1_Admin]			
ANNEX 1			
ADMIN EQUIPMENT BUDGET ESTIMATE IN INR			
1	Office Cabinets for Documents	20	240000
2	Computers / Laptops	50	2500000
3	Office Furniture	NA	1200000
4	Printer/scanner / xerox machine	1	500000
5	Water Dispenser	2	40000
6	Projector for conference / meeting room	2	600000
Total Amount in INR			5080000
[SHEET: ANNEX2_CANTEEN]			
ANNEX 2			
CANTEEN EQUIPMENT BUDGET ESTIMATE IN INR			
S/No	Name of Equipment	Quantity	Approx Cost (INR)
1	Refrigerator	2	300000
2	SS furniture (serving table etc)	NA	400000
3	SS Water cooler dispenser	1	50000
4	Microwaves	2	60000
5	Kitchenware	na	500000
6	Bain marie	1	50000
Total Amount in INR			1360000
[SHEET: ANNEX3_UTILITY]			
ANNEX 3			
UTILITY BUDGET ESTIMATE IN INR			
S/No	Name of Equipment	Quantity	Approx Cost (INR)
	Clean Utilities		
1	Soft water generation 10000/hour and storage system	1	40000000
2	Purified water generation 5000L/ hour storage tank 10KL and distribution system	1	
3	Water for injection generation 1000L / hour, Storage tank 10KL and distribution system	1	
4	Pure steam generator 1500Kg / hour and distribution system	1	
	Black Utilities		
1	Plant steam generation 2000Kg/Hour and distribution System	2	7000000
2	Compressed air generation and distribution system	2	7500000
3	PASSENGER LIFT X 2 NOS	2	7000000
4	MATERIAL LIFT X 3 NOS	3	6000000
Total Amount in INR			67500000
[SHEET: ANNEX4_Plasma_storage]			
ANNEX 4			
PLASMA STORAGE EQUIPMENT BUDGET ESTIMATE IN INR			
S/No	Name of Equipment	Quantity	Approx Cost (INR)
1	Analytical weighing Balance	1	170000
2	RLAF	3	1200000
3	Precision Balance	1	144000
4	Commercial Balance	1	120000
5	Refrigerator 200L	2	218000
6	Deep freezer -20°C (1000L)	32	14400000
7	Air curtain	10	100000
8	PVC pallets	50	300000
9	Storage racks SS	25	700000
10	Sampling rods	2	10000
11	Liquid sampler	2	20000
12	Dedusting tunnel	1	1250000
13	DPB	4	600000
14	LAF cabinet	1	450000
15	Computer / Laptops	4	120000
16	Office Furniture	NA	200000
17	Office Cabinets	10	120000
Total Amount in INR			20122000
[SHEET: ANNEX5_BULK_PLASMA]			
ANNEX 5			
BULK PLASMA MANUFACTURING EQUIPMENT BUDGET ESTIMATE IN INR			
S/No	Name of Equipment	Quantity	Approx Cost (INR)
1	Chromatography AKTA-1 (471 L/hr (Pipe MOC SS)	1	17100000
2	Chromatography Column Dia 1000mm x Height 500mm	2	44612200
3	Chromatography AKTA-1 (500 L/hr pipe MOC SS)	1	17100000
4	Chromatography Column Dia 1000mm x Height 500mm	1	22306100
5	Chromatography AKTA-1 (785 L/hr pipe MOC PP)	1	20031420
6	Chromatography Column Dia 1000mm x Height 500mm	1	22306100
7	Chromatography AKTA-1 (1178 L/hr pipe MOC PP)	1	20031420
8	Chromatography Column Dia 1000mm x Height 500mm	2	44612210
9	Chromatography AKTA-1 (864 L/hr pipe MOC PP)	1	20031420
10	Chromatography Column Dia 1000mm x Height 500mm	1	22306100
11	Chromatography AKTA-1 (864 L/hr pipe MOC PP)	1	20031420
12	Chromatography Column Dia 1000mm x Height 500mm	1	22306100
13	Chromatography AKTA-1 (667 L/hr pipe MOC PP)	1	17100000
14	Chromatography Column Dia 1000mm x Height 500mm	1	22306100
15	Chromatography Column Dia 300mm x Height 500mm	1	4896570
16	Chromatography AKTA-1 (785 L/hr pipe MOC PP)	1	20031420
17	Chromatography Column Dia 1000mm x Height 500mm	1	22306100
18	Chromatography Column Dia 300mm x Height 500mm	1	4896570
19	Chromatography AKTA-1 (30 L/hr pipe MOC PP)	1	14657140
20	Chromatography Column Dia 300mm x Height 750mm	1	4896570
21	Chromatography Column Dia 600mm x Height 500mm	1	14244570
22	Chromatography AKTA-1(160 L/hr pipe MOC PP)	1	14657140

| 23 | Chromatography Column Dia 450mm x Height 500mm | 1 | 12018850
 | 24 | TFF System 40 sqm | 1 | 19350000
 | 25 | Cassette 40 sqm | 1 | 5590000
 | 26 | TFF System 30 sqm | 2 | 38700000
 | 27 | Cassette 30 sqm | 2 | 8600000
 | 28 | TFF System 20 sqm | 1 | 9460000
 | 29 | Cassette 20 sqm | 1 | 3010000
 | 30 | TFF System 5 sqm | 2 | 7310000
 | 31 | Cassette 5 sqm | 2 | 688000
 | 32 | Continuous Centrifuge 1000L/hr | 2 | 30000000
 | 33 | Process vessel 150L | 3 | 3225000
 | 34 | Process vessel 250L | 4 | 5160000
 | 35 | Process vessel (Mobile tank) 300L | 2 | 2580000
 | 36 | Process vessel 1500L | 20 | 43000000
 | 37 | Process vessel 2000L | 6 | 15480000
 | 38 | Buffer Vessel 500L | 14 | 24080000
 | 39 | Buffer Vessel 1000L | 10 | 19350000
 | 40 | Depth filtration system | 2 | 2100000
 | 41 | Nano filter holder | 1 | 500000
 | 42 | Cold room 2-8°C | 2 | 1500000
 | 43 | Rectangular tank | 1 | 430000
 | 44 | Buffer preparation tank | 2 | 1720000
 | 45 | Dispensing Booth | 1 | 400000
 | 46 | Analytical Balance | 1 | 170000
 | 47 | Precision Balance | 1 | 150000
 | 48 | Commercial Balance | 1 | 120000
 | 49 | PVC bag Sealer | 1 | 5000
 | 50 | Rotary Bag Sealer | 1 | 300000
 | 51 | pH/ Conductivity meter | 2 | 300000
 | 52 | Magnetic Stirrer | 2 | 150000
 | 53 | Peristaltic pump | 2 | 200000
 | 54 | Filter Integrity machine | 1 | 2000000
 | 55 | Fume Hood | 1 | 450000
 | 56 | Autoclave 1000L | 2 | 4000000
 | 57 | DHS 1000L | 1 | 2000000
 | 58 | Ceiling LAF | 3 | 1200000
 | 59 | Drying Oven | 2 | 1000000
 | 60 | Acid tank 1000L | 2 | 3870000
 | 61 | Base tank 1000L | 2 | 3870000
 | 62 | Central CIP system | 2 | 2150000
 | 63 | Refrigerator 200L | 1 | 150000
 | 64 | Laminar air flow cabinet | 1 | 435000
 | 65 | Dynamic pass box | 4 | 480000
 | 66 | Tumble Dryer 25 Kg | 1 | 2000000
 | 67 | Barrier washer Extractor 25Kg | 1 | 3000000
 | 68 | SS Furniture (Table, chair, racks, Ladders etc) | NA | 4000000
 | 69 | Lockers | NA | 200000
 | 70 | Laptop / Computers | 10 | 500000
 | 71 | Office furniture / Cabinets | NA | 300000
 | Total Amount in INR | | 720018520
 [SHEET: ANNEX6_FILL FINISH]
 | ANNEX 6 | | |
 | FILL FINISH EQUIPMENT | | |
 | S/No | Name of Equipment | Quantity | Approx Cost (INR)
 | 1 | Formulation Vessel 100L | 1 | 2500000
 | 2 | Ceiling LAF | 2 | 800000
 | 3 | Laminar air flow cabinet | 1 | 435000
 | 4 | Continuous Particle Counter | 1 | 3000000
 | 5 | Dynamic pass box | 7 | 840000
 | 6 | Pass Box static | 1 | 38000
 | 7 | Vial washing machine | 1 | 4500000
 | 8 | Sterilizing tunnel | 1 | 4500000
 | 9 | Vial filling & Sealing line 120 vials /min | 1 | 7500000
 | 10 | Lyophilizer (1sq meter) | 1 | 5500000
 | 11 | Labelling machine | 1 | 9,00,000.00
 | 12 | Double door autoclave 1000L | 1 | 2000000
 | 13 | Cold room | 1 | 800000
 | 14 | SS furniture (Table, Chair, racks, cabinets etc) | 1 | 200000
 | Total Amount in INR | | 32613000
 [SHEET: ANNEX7_QC]
 11 | | | |
ANNEX 7		
QC & R&D EQUIPMENT		
EQUIPMENT BUDGET ESTIMATE IN INR		
1	Akta pure -150 (For R&D)	2
2	Columns XK 50/100 (For R & D)	1
3	Columns XK 50/60 (For R&D)	1
4	Column XK 50/30 (For R & D)	3
5	HPLC-UV	1
6	HPLC Columns and accessories	1
7	Electrophoresis (Minigel) with trans blot	2
8	Gel Doc system densitometer	1
9	Power packs	2
10	Endotoxin Detection system (Tube heating block)	1
11	Micropipettes- variable volume range of 2-20µL, 20-200µL, 100-1000µL, 500-5000µL and multichannel 30-300µL	2 set each
12	Lab Refrigerator 2-8 ° C (SS) with data control	4
13	Flame photometer	1
14	Weighing Balance Analytical	2
15	UV-VIS spectrophotometer Double Beam with data acquisition.	1
16	Volumetric KF	1
17	Moisture Analyser	1

| 18 | Stability Chambers (5±3° C, 25, 37 deg. C) 200L (1set) | 3 | 1400000
 | 19 | Muffle Furnance | 1 | 80000
 | 20 | pH and conductivity meter with printer | 2 | 260000
 | 21 | Vaccum Oven | 1 | 120000
 | 22 | Melting point apparatus | 1 | 90000
 | 23 | Incubator Sterility Test 500L | 2 | 1440000
 | 24 | Incubators for Normal Media plates Incubation 500L | 1 | 300000
 | 25 | Incubator for EMP and water samples 500L | 2 | 600000
 | 26 | Hot Air Oven | 1 | 160000
 | 27 | Thermostatic water bath (10-20 L) | 2 | 200000
 | 28 | Vaccum pump | 2 | 120000
 | 29 | Desiccator Glass | 1 | 4000
 | 30 | High Quality Laboratory water generation system (10 L /hr) Milli -Q water | 1 | 1450000
 | 31 | Sonicator | 1 | 180000
 | 32 | Vortex Shaker | 6 | 210000
 | 33 | Magnetic Stirrer with hot plate | 4 | 400000
 | 34 | LAF Vertical | 2 | 600000
 | 35 | Biosafety Cabinet | 2 | 1400000
 | 36 | Autoclave Double door | 1 | 1250000
 | 37 | Dry Heat Steriliser | 1 | 750000
 | 38 | Fume Hood (Table Top) | 1 | 450000
 | 39 | Deep freezer (-20 Degrees C) with Data acquisition | 1 | 400000
 | 40 | Air sampler | 1 | 300000
 | 41 | Particle counter | 1 | 2000000
 | 42 | Decontamination autoclave | 1 | 1000000
 | 43 | Microscope Binocular | 1 | 100000
 | 44 | Colony Counter | 1 | 150000
 | 45 | Microplate reader ELISA | 1 | 600000
 | 46 | Microplate washer | 1 | 350000
 | 47 | TOC analyser (Grab Samples) | 1 | 2500000
 | 48 | Timers/ Stop watch | 5 | 10000
 | 49 | Ice flaking machine | 1 | 250000
 | 50 | Standard Weight Box | 1 Set | 1000000
 | 51 | Printer/scanner / xerox machine | 1 | 500000
 | 52 | Centrifuge table top | 1 | 450000
 | 53 | Size exclusion | L | 150000
 | 54 | Anion | L | 150000
 | 55 | Cation | L | 150000
 | 56 | Affinity | L | 150000
 | 57 | SS furniture racks | 25 | 250000
 | 58 | Computers / Laptops | 10 | 500000
 | 59 | Office Furniture | NA | 500000
 | 60 | Office Cabinets for Documents | 10 | 120000
 || | TOTAL | 39694000
 [SHEET: ANNEX8_QA]
 | ANNEX 8 || |
 | QA EQUIPMENT BUDGET ESTIMATE IN INR || |
 | 1 | Anemometer | 1 | 15000
 | 2 | Lux meter | 1 | 12000
 | 3 | Datalogger 32 channel | 1 | 400000
 | 4 | Paper shredder | 1 | 10000
 | 5 | Laminator | 1 | 10000
 | 6 | Printer/scanner / xerox machine | 1 | 500000
 | 7 | Computers / Laptops | 10 | 500000
 | 8 | Numbering Machine | 5 | 25000
 | 9 | Spiral Binding Machine | 1 | 15000
 | 10 | A3 +A4 Lamination Machine | 1 | 25000
 | 11 | Bi-Focal Glass | 1 | 80000
 | 12 | Tachometer | 2 | 6000
 | 13 | Clamp Meter/ Multi meter | 5 | 10000
 | 14 | Level Indicator | 2 | 8000
 | 15 | Measuring TAPE | 2 | 1500
 | 16 | Stop Watch | 5 | 50000
 | 17 | Temperature recorders 24 Nos. (data Loggers) | 20 | 200000
 | 18 | Office Cabinets for Documents | 20 | 240000
 | 19 | Projector for conference / meeting room | 1 | 300000
 | 20 | Office Furniture | NA | 700000
 | Total Amount in INR || | 3107500
 [SHEET: ANNEX9_FG_STORAGE]
 | ANNEX 9 || |
 | FINISHED GOODS STORAGE || |
 | 1 | Air curtain | 10 | 100000
 | 2 | PVC pallets | 50 | 300000
 | 3 | Storage racks SS | 50 | 1500000
 | 4 | Hydraulic trolley | 1 | 250000
 | Total Amount in INR || | 2150000
 [SHEET: ANNEX10_ETP STP]
 | ANNEX 10 || |
 | ETP BUDGET ESTIMATE IN INR || |
 | S/No | Name of Equipment | Qty. | Approx Cost (INR)
 | 1 | Biowaste reception tank 15000L | 1 | 4000000
2	Kill tank 1000L	2
3	Bar Screen	NA
4	Collection tank	NA
5	Equalization tank	NA
6	Neutralization tank	NA
7	Chemical coagulation	NA
8	Biological oxidation tank	NA
9	Filtration System	NA
STP BUDGET ESTIMATE IN INR		
1	Primary treatment plant : Bar screening - 3 stages, pH correction, BOD & COD removal secondary clarifier	

| 2 | Secondary treatment : Electro coagulation process, Flash mixer, HRSC clarifier, Sand filter | |
| 3 | Sludge Management: Sludge thickner, Screw pump, Filter press, Sludge storage | |
| Total Amount in INR | | 5200000

PROJECT COST SUMMARY

Sr. No. ITEMS DESCRIPTION COST IN INR REMARKS

1 SUB Head 1 CIVIL CONSTRUCTION

MAIN BUILDING (GROUND - FIRST - TERRACE FLOOR)

1A Building work (Civil) 22,50,00,000

APPROX. 174,000 sqft

1B UTILITY AND OTHER SUPPORT BUILDINGS 2,60,00,000

1C ADMINISTRATION BUILDING 1,00,00,000

1D RAIN WATER HARVESTING 6,00,000

1E EPOXY FLOORING (PRODUCTION / QC / R&D) 2,00,00,000

1F Development work civil ROADS & LANDSCAPING 1,52,00,000

2 SUB Head II EQUIPMENT

2A ADMINISTRATION ADMINISTRATION EQUIPMENT AND OFFICE FURNITURE 50,80,000 ANNEX 1

2B CANTEEN CANTEEN EQUIPMENT 13,60,000 ANNEX 2

2C UTILITIES UTILITIES (EXCLUDING HVAC) 5,75,00,000 ANNEX 3

2D WAREHOUSE (PLASMA STORAGE) PLASMA STORAGE EQUIPMENT 2,01,22,000 ANNEX 4

2E BULK PLASMA BULK PLASMA PRODUCTION EQUIPMENT 72,00,18,520 ANNEX 5

2F FILL FINISH FACILITY FILL FINISH EQUIPMENT 3,26,13,000 ANNEX 6

2G QUALITY CONTROL / R&D QC/R&D EQUIPMENT 3,96,94,000 ANNEX 7

2H QUALITY ASSURANCE QA EQUIPMENT 31,07,500 ANNEX 8

2I WAREHOUSE (FG) FINISHED GOODS STORAGE 21,50,000 ANNEX 9

3 SUB Head III ELECTRICAL WORK

3A ELECTRICAL ELECTRICAL COST FOR ALL BLOCKS (including UPS) 9,00,00,000

3B TRANSFORMER - 3000 KVA 65,00,000

3C DG SET 1000 KVA x 2 nos 1,40,00,000

3D INSTRUMENTATION FIRE ALARM, CCTV SYSTEM 3,00,00,000

HVAC FOR ALL BLOCKS INCLUDING DUCTING AND

3E MECHANICAL / HVAC 11,00,00,000

CHILLERS

CLEAN ROOM MODULAR PARTITION

3F 9,00,00,000

(Bulk/ QC/ Warehouse / Plasma storage)

3G COLD ROOM (4 NOS) 80,00,000

3H BOILER & PIPING x 1 nos 75,00,000

3I AIR COMPRESSOR 75,00,000

3J UTILITY PIPING 1,25,00,000

3K KILL TANK with storage 25,00,000

3L ETP /STP 33,00,000 ANNEX 10

3M SOLAR POWER EQUIPMENT 90,00,000

4 SUB Head IV LICENSING AND PROFESSIONAL FEES

4A TECHNOLOGY TRANSFER FEE (6 products) 2,20,00,000

4B PROJECT CONSULTANCY CHARGES 1,50,00,000

Grand Total: 1,60,62,45,020

Add for third party quality assurance @ 1% 1,60,62,450

Total: 1,62,23,07,470

Add 3% contingencies 4,86,69,224

Total: 1,67,09,76,694

Add EPF, ESI @ 4.25% 7,10,16,510

Total: 1,74,19,93,204

Add labour cess @1% 1,74,19,932

Grand Total 1,75,94,13,136

Say 1,759 LAKHS

1 of 13

PROJECT COST SUMMARY - ANNEX 1

ADMIN EQUIPMENT BUDGET ESTIMATE IN INR

1 Office Cabinets for Documents 20 2,40,000.00

2 Computers / Laptops 50 25,00,000.00

3 Office Furniture NA 12,00,000.00

4 Printer/scanner / xerox machine 1 5,00,000.00

5 Water Dispenser 2 40,000.00

6 Projector for conference / meeting room 2 6,00,000.00

Total Amount in INR 50,80,000.00

PROJECT COST SUMMARY - ANNEX 2

CANTEEN EQUIPMENT BUDGET ESTIMATE IN INR

S/No Name of Equipment Quantity Approx Cost (INR)

1 Refrigerator 2 3,00,000.00

2 SS furniture (serving table etc) NA 4,00,000.00

3 SS Water cooler dispenser 1 50,000.00

4 Microwaves 2 60,000.00

5 Kitchenware na 5,00,000.00

6 Bain marie 1 50,000.00

Total Amount in INR 13,60,000.00

PROJECT COST SUMMARY - ANNEX 3

UTILITY BUDGET ESTIMATE IN INR

S/No Name of Equipment Quantity Approx Cost (INR)

Clean Utilities

1 Soft water generation 8000L/hour and storage system 1

Purified water generation 5000L/ hour storage tank 10KL and

2 1

distribution system

3,00,00,000.00

Water for injection generation 1000L / hour, Storage tank 10KL

3 1

and distribution system

4 Pure steam generator 1000Kg / hour and distribution system 1

Black Utilities

- 1 Plant steam generation 2000Kg/Hour and distribution System 2 70,00,000.00
- 2 Compressed air generation and distribution system 2 75,00,000.00
- 3 PASSENGER LIFT X 2 NOS 2 70,00,000
- 4 MATERIAL LIFT X 3 NOS 3 60,00,000

Total Amount in INR 5,75,00,000.00

PROJECT COST SUMMARY - ANNEX 4

PLASMA STORAGE EQUIPMENT BUDGET ESTIMATE IN INR

S/No Name of Equipment Quantity Approx Cost (INR)

- 1 Analytical weighing Balance 1 1,70,000.00
- 2 RLAf 3 12,00,000.00
- 3 Precision Balance 1 1,44,000.00
- 4 Commercial Balance 1 1,20,000.00
- 5 Refrigerator 200L 2 2,18,000.00
- 6 Deep freezer -20°C (1000L) 32 1,44,00,000.00
- 7 Air curtain 10 1,00,000.00
- 8 PVC pallets 50 3,00,000.00
- 9 Storage racks SS 25 7,00,000.00
- 10 Sampling rods 2 10,000.00
- 11 Liquid sampler 2 20,000.00
- 12 Dedusting tunnel 1 12,50,000.00
- 13 DPB 4 6,00,000.00
- 14 LAF cabinet 1 4,50,000.00
- 15 Computer / Laptops 4 1,20,000.00
- 16 Office Furniture NA 2,00,000.00
- 17 Office Cabinets 10 1,20,000.00

Total Amount in INR 2,01,22,000.00

PROJECT COST SUMMARY - ANNEX 5

BULK PLASMA MANUFACTURING EQUIPMENT BUDGET ESTIMATE IN INR

S/No Name of Equipment Quantity Approx Cost (INR)

- 1 Chromatography AKTA-1 (471 L/hr (Pipe MOC SS) 1 1,71,00,000
- 2 Chromatography Column Dia 1000mm x Height 500mm 2 4,46,12,200
- 3 Chromatography AKTA-1 (500 L/hr pipe MOC SS) 1 1,71,00,000
- 4 Chromatography Column Dia 1000mm x Height 500mm 1 2,23,06,100
- 5 Chromatography AKTA-1 (785 L/hr pipe MOC PP) 1 2,00,31,420
- 6 Chromatography Column Dia 1000mm x Height 500mm 1 2,23,06,100
- 7 Chromatography AKTA-1 (1178 L/hr pipe MOC PP) 1 2,00,31,420
- 8 Chromatography Column Dia 1000mm x Height 500mm 2 4,46,12,210
- 9 Chromatography AKTA-1 (864 L/hr pipe MOC PP) 1 2,00,31,420
- 10 Chromatography Column Dia 1000mm x Height 500mm 1 2,23,06,100
- 11 Chromatography AKTA-1 (864 L/hr pipe MOC PP) 1 2,00,31,420
- 12 Chromatography Column Dia 1000mm x Height 500mm 1 2,23,06,100
- 13 Chromatography AKTA-1 (667 L/hr pipe MOC PP) 1 1,71,00,000
- 14 Chromatography Column Dia 1000mm x Height 500mm 1 2,23,06,100
- 15 Chromatography Column Dia 300mm x Height 500mm 1 48,96,570
- 16 Chromatography AKTA-1 (785 L/hr pipe MOC PP) 1 2,00,31,420
- 17 Chromatography Column Dia 1000mm x Height 500mm 1 2,23,06,100
- 18 Chromatography Column Dia 300mm x Height 500mm 1 48,96,570
- 19 Chromatography AKTA-1 (30 L/hr pipe MOC PP) 1 1,46,57,140
- 20 Chromatography Column Dia 300mm x Height 750mm 1 48,96,570
- 21 Chromatography Column Dia 600mm x Height 500mm 1 1,42,44,570
- 22 Chromatography AKTA-1(160 L/hr pipe MOC PP) 1 1,46,57,140
- 23 Chromatography Column Dia 450mm x Height 500mm 1 1,20,18,850
- 24 TFF System 40 sqm 1 1,93,50,000
- 25 Cassette 40 sqm 1 55,90,000
- 26 TFF System 30 sqm 2 3,87,00,000
- 27 Cassette 30 sqm 2 86,00,000
- 28 TFF System 20 sqm 1 94,60,000
- 29 Cassette 20 sqm 1 30,10,000
- 30 TFF System 5 sqm 2 73,10,000
- 31 Cassette 5 sqm 2 6,88,000
- 32 Continuous Centrifuge 1000L/hr 2 3,00,00,000
- 33 Process vessel 150L 3 32,25,000
- 34 Process vessel 250L 4 51,60,000
- 35 Process vessel (Mobile tank) 300L 2 25,80,000
- 36 Process vessel 1500L 20 4,30,00,000
- 37 Process vessel 2000L 6 1,54,80,000
- 38 Buffer Vessel 500L 14 2,40,80,000
- 39 Buffer Vessel 1000L 10 1,93,50,000
- 40 Depth filtration system 2 21,00,000
- 41 Nano filter holder 1 5,00,000
- 42 Cold room 2-8°C 2 15,00,000
- 43 Rectangular tank 1 4,30,000
- 44 Buffer preparation tank 2 17,20,000
- 45 Dispensing Booth 1 4,00,000
- 46 Analytical Balance 1 1,70,000
- 47 Precision Balance 1 1,50,000
- 48 Commercial Balance 1 1,20,000
- 49 PVC bag Sealer 1 5,000
- 50 Rotary Bag Sealer 1 3,00,000
- 51 pH/ Conductivity meter 2 3,00,000
- 52 Magnetic Stirrer 2 1,50,000
- 53 Peristaltic pump 2 2,00,000
- 54 Filter Integrity machine 1 20,00,000
- 55 Fume Hood 1 4,50,000
- 56 Autoclave 1000L 2 40,00,000
- 57 DHS 1000L 1 20,00,000
- 58 Ceiling LAF 3 12,00,000
- 59 Drying Oven 2 10,00,000
- 60 Acid tank 1000L 2 38,70,000
- 61 Base tank 1000L 2 38,70,000

62 Central CIP system 2 21,50,000
 63 Refrigerator 200L 1 1,50,000
 64 Laminar air flow cabinet 1 4,35,000
 65 Dynamic pass box 4 4,80,000
 66 Tumble Dryer 25 Kg 1 20,00,000
 67 Barrier washer Extractor 25Kg 1 30,00,000
 68 SS Furniture (Table, chair, racks, Ladders etc) NA 40,00,000
 69 Lockers NA 2,00,000
 70 Laptop / Computers 10 5,00,000
 71 Office furniture / Cabinets NA 3,00,000
 Total Amount in INR 72,00,18,520
 PROJECT COST SUMMARY - ANNEX 6
 FILL FINISH EQUIPMENT

Quantit
 S/No Name of Equipment Approx Cost (INR)

y
 1 Formulation Vessel 100L 1 25,00,000.00
 2 Ceiling LAF 2 8,00,000.00
 3 Laminar air flow cabinet 1 4,35,000.00
 4 Continuous Particle Counter 1 30,00,000.00
 5 Dynamic pass box 7 8,40,000.00
 6 Pass Box static 1 38,000.00
 7 Vial washing machine 1 45,00,000.00
 8 Sterilizing tunnel 1 45,00,000.00
 9 Vial filling & Sealing line 120 vials /min 1 75,00,000.00
 10 Lyophilizer (1sq meter) 1 55,00,000.00
 11 Labelling machine 1 9,00,000.00
 12 Double door autoclave 1000L 1 20,00,000.00
 13 Cold room 1 8,00,000.00
 14 SS furniture (Table, Chair, racks, cabinets etc) 1 2,00,000.00
 Total Amount in INR 3,26,13,000.00
 PROJECT COST SUMMARY - ANNEX 7

QC & R&D EQUIPMENT

EQUIPMENT BUDGET ESTIMATE IN INR

1 Akta pure -150 (For R&D) 2 40,00,000
 2 Columns XK 50/100 (For R & D) 1 7,00,000
 3 Columns XK 50/60 (For R&D) 1 7,00,000
 4 Column XK 50/30 (For R & D) 3 14,00,000
 5 HPLC-UV 1 34,00,000
 6 HPLC Columns and accessories 1 10,00,000
 7 Electrophoresis (Minigel) with trans blot 2 6,00,000
 8 Gel Doc system densitometer 1 5,00,000
 9 Power packs 2 2,00,000
 10 Endotoxin Detection system (Tube heating block) 1 2,00,000
 Micropipettes- variable volume range of 2-20µL, 20-200µL, 100-
 11 2 set each 3,00,000
 1000µL, 500-5000µL and multichannel 30-300µL
 12 Lab Refrigerator 2-8 ° C (SS) with data control 4 4,50,000
 13 Flame photometer 1 4,50,000
 14 Weighing Balance Analytical 2 3,50,000
 15 UV-VIS spectrophotometer Double Beam with data acquisition. 1 9,00,000
 16 Volumetric KF 1 7,00,000
 17 Moisture Analyser 1 8,00,000
 18 Stability Chambers (5±3° C, 25, 37 deg. C) 200L (1set) 3 14,00,000
 19 Muffle Furnance 1 80,000
 20 pH and conductivity meter with printer 2 2,60,000
 21 Vaccum Oven 1 1,20,000
 22 Melting point apparatus 1 90,000
 23 Incubator Sterility Test 500L 2 14,40,000
 24 Incubators for Normal Media plates Incubation 500L 1 3,00,000
 25 Incubator for EMP and water samples 500L 2 6,00,000
 26 Hot Air Oven 1 1,60,000
 27 Thermostatic water bath (10-20 L) 2 2,00,000
 28 Vaccum pump 2 1,20,000
 29 Desiccator Glass 1 4,000
 High Quality Laboratory water generation system (10 L /hr) Milli -Q
 30 1 14,50,000
 water
 31 Sonicator 1 1,80,000
 32 Vortex Shaker 6 2,10,000
 33 Magnetic Stirrer with hot plate 4 4,00,000
 34 LAF Vertical 2 6,00,000
 35 Biosafety Cabinet 2 14,00,000
 36 Autoclave Double door 1 12,50,000
 37 Dry Heat Steriliser 1 7,50,000
 38 Fume Hood (Table Top) 1 4,50,000
 39 Deep freezer (-20 Degrees C) with Data acquisition 1 4,00,000
 40 Air sampler 1 3,00,000
 41 Particle counter 1 20,00,000
 42 Decontamination autoclave 1 10,00,000
 43 Microscope Binocular 1 1,00,000
 44 Colony Counter 1 1,50,000
 45 Microplate reader ELISA 1 6,00,000
 46 Microplate washer 1 3,50,000
 47 TOC analyser (Grab Samples) 1 25,00,000
 48 Timers/ Stop watch 5 10,000
 49 Ice flaking machine 1 2,50,000
 50 Standard Weight Box 1 Set 10,00,000
 51 Printer/scanner / xerox machine 1 5,00,000
 52 Centrifuge table top 1 4,50,000
 53 Size exclusion L 1,50,000.00
 54 Anion L 1,50,000.00

55 Cation L 1,50,000.00
 56 Affinity L 1,50,000.00
 57 SS furniture racks 25 2,50,000
 58 Computers / Laptops 10 5,00,000
 59 Office Furniture NA 5,00,000
 60 Office Cabinets for Documents 10 1,20,000
 TOTAL 3,96,94,000
 PROJECT COST SUMMARY - ANNEX 8
 QA EQUIPMENT BUDGET ESTIMATE IN INR
 1 Anemometer 1 15,000.00
 2 Lux meter 1 12,000.00
 3 Datalogger 32 channel 1 4,00,000.00
 4 Paper shredder 1 10,000.00
 5 Laminator 1 10,000.00
 6 Printer/scanner / xerox machine 1 5,00,000.00
 7 Computers / Laptops 10 5,00,000.00
 8 Numbering Machine 5 25,000.00
 9 Spiral Binding Machine 1 15,000.00
 10 A3 +A4 Lamination Machine 1 25,000.00
 11 Bi-Focal Glass 1 80,000.00
 12 Tachometer 2 6,000.00
 13 Clamp Meter/ Multi meter 5 10,000.00
 14 Level Indicator 2 8,000.00
 15 Measuring TAPE 2 1,500.00
 16 Stop Watch 5 50,000.00
 17 Temperature recorders 24 Nos. (data Loggers) 20 2,00,000.00
 18 Office Cabinets for Documents 20 2,40,000.00
 19 Projector for conference / meeting room 1 3,00,000.00
 20 Office Furniture NA 7,00,000.00
 Total Amount in INR 31,07,500.00
 PROJECT COST SUMMARY - ANNEX 9
 FINISHED GOODS STORAGE
 1 Air curtain 10 1,00,000.00
 2 PVC pallets 50 3,00,000.00
 3 Storage racks SS 50 15,00,000.00
 4 Hydraulic trolley 1 2,50,000.00
 Total Amount in INR 21,50,000.00
 PROJECT COST SUMMARY - ANNEX 10
 ETP BUDGET ESTIMATE IN INR
 Approx Cost
 S/No Name of Equipment Qty.
 (INR)
 1 Biowaste reception tank 15000L 1
 2 Kill tank 500L 2
 3 Bar Screen NA
 4 Collection tank NA
 5 Equalization tank NA 25,00,000.00
 6 Neutralization tank NA
 7 Chemical coagulation NA
 8 Biological oxidation tank NA
 9 Filtration System NA
 STP BUDGET ESTIMATE IN INR
 Primary treatment plant : Bar screening - 3 stages, pH
 1
 correction, BOD & COD removal secondary clarifier
 Secondary treatment : Electro coagulation process, Flash
 2 8,00,000.00
 mixer, HRSC clarifier, Sand filter
 Sludge Management: Sludge thickner, Screw pump, Filter
 3
 press, Sludge storage
 Total Amount in INR 33,00,000.00

4 | Pure steam generator 1000Kg / hour and distribution system | 1 | 3,00,00,000.00
 | Black Utilities | |
 1 | Plant steam generation 2000Kg/Hour and distribution System | 2 | 70,00,000.00
 2 | Compressed air generation and distribution system | 2 | 75,00,000.00
 3 | PASSENGER LIFT X 2 NOS | 2 | 70,00,000
 4 | MATERIAL LIFT X 3 NOS | 3 | 60,00,000
 Total Amount in INR | Total Amount in INR | Total Amount in INR | 5,75,00,000.00
 EQUIPMENT BUDGET ESTIMATE IN INR | EQUIPMENT BUDGET ESTIMATE IN INR | EQUIPMENT BUDGET ESTIMATE IN INR |
 EQUIPMENT BUDGET ESTIMATE IN INR
 S/No | Name of Equipment | Quantity | Approx Cost (INR)
 1 | Analytical weighing Balance | 1 | 1,70,000.00
 2 | RLAF | 3 | 12,00,000.00
 3 | Precision Balance | 1 | 1,44,000.00
 4 | Commercial Balance | 1 | 1,20,000.00
 5 | Refrigerator 200L | 2 | 2,18,000.00
 6 | Deep freezer -20°C (1000L) | 32 | 1,44,00,000.00
 7 | Air curtain | 10 | 1,00,000.00
 8 | PVC pallets | 50 | 3,00,000.00
 9 | Storage racks SS | 25 | 7,00,000.00
 10 | Sampling rods | 2 | 10,000.00
 11 | Liquid sampler | 2 | 20,000.00
 12 | Dedusting tunnel | 1 | 12,50,000.00
 13 | DPB | 4 | 6,00,000.00
 14 | LAF cabinet | 1 | 4,50,000.00
 15 | Computer / Laptops | 4 | 1,20,000.00
 16 | Office Furniture | NA | 2,00,000.00
 17 | Office Cabinets | 10 | 1,20,000.00
 Total Amount in INR | Total Amount in INR | Total Amount in INR | 2,01,22,000.00
 EQUIPMENT BUDGET ESTIMATE IN INR | EQUIPMENT BUDGET ESTIMATE IN INR | EQUIPMENT BUDGET ESTIMATE IN INR |
 EQUIPMENT BUDGET ESTIMATE IN INR
 S/No | Name of Equipment | Quantity | Approx Cost (INR)
 1 | Chromatography AKTA-1 (471 L/hr (Pipe MOC SS) | 1 | 1,71,00,000
 2 | Chromatography Column Dia 1000mm x Height 500mm | 2 | 4,46,12,200
 3 | Chromatography AKTA-1 (500 L/hr pipe MOC SS) | 1 | 1,71,00,000
 4 | Chromatography Column Dia 1000mm x Height 500mm | 1 | 2,23,06,100
 5 | Chromatography AKTA-1 (785 L/hr pipe MOC PP) | 1 | 2,00,31,420
 6 | Chromatography Column Dia 1000mm x Height 500mm | 1 | 2,23,06,100
 7 | Chromatography AKTA-1 (1178 L/hr pipe MOC PP) | 1 | 2,00,31,420
 8 | Chromatography Column Dia 1000mm x Height 500mm | 2 | 4,46,12,210
 9 | Chromatography AKTA-1 (864 L/hr pipe MOC PP) | 1 | 2,00,31,420
 10 | Chromatography Column Dia 1000mm x Height 500mm | 1 | 2,23,06,100
 11 | Chromatography AKTA-1 (864 L/hr pipe MOC PP) | 1 | 2,00,31,420
 12 | Chromatography Column Dia 1000mm x Height 500mm | 1 | 2,23,06,100
 13 | Chromatography AKTA-1 (667 L/hr pipe MOC PP) | 1 | 1,71,00,000
 14 | Chromatography Column Dia 1000mm x Height 500mm | 1 | 2,23,06,100
 15 | Chromatography Column Dia 300mm x Height 500mm | 1 | 48,96,570
 16 | Chromatography AKTA-1 (785 L/hr pipe MOC PP) | 1 | 2,00,31,420
 17 | Chromatography Column Dia 1000mm x Height 500mm | 1 | 2,23,06,100
 18 | Chromatography Column Dia 300mm x Height 500mm | 1 | 48,96,570
 19 | Chromatography AKTA-1 (30 L/hr pipe MOC PP) | 1 | 1,46,57,140
 20 | Chromatography Column Dia 300mm x Height 750mm | 1 | 48,96,570
 21 | Chromatography Column Dia 600mm x Height 500mm | 1 | 1,42,44,570
 22 | Chromatography AKTA-1(160 L/hr pipe MOC PP) | 1 | 1,46,57,140
 23 | Chromatography Column Dia 450mm x Height 500mm | 1 | 1,20,18,850
 24 | TFF System 40 sqm | 1 | 1,93,50,000
 25 | Cassette 40 sqm | 1 | 55,90,000
 26 | TFF System 30 sqm | 2 | 3,87,00,000
 27 | Cassette 30 sqm | 2 | 86,00,000
 28 | TFF System 20 sqm | 1 | 94,60,000
 29 | Cassette 20 sqm | 1 | 30,10,000
 30 | TFF System 5 sqm | 2 | 73,10,000
 31 | Cassette 5 sqm | 2 | 6,88,000
 32 | Continuous Centrifuge 1000L/hr | 2 | 3,00,00,000
 33 | Process vessel 150L | 3 | 32,25,000
 34 | Process vessel 250L | 4 | 51,60,000
 35 | Process vessel (Mobile tank) 300L | 2 | 25,80,000
 36 | Process vessel 1500L | 20 | 4,30,00,000
 37 | Process vessel 2000L | 6 | 1,54,80,000
 38 | Buffer Vessel 500L | 14 | 2,40,80,000
 39 | Buffer Vessel 1000L | 10 | 1,93,50,000
 40 | Depth filtration system | 2 | 21,00,000
 41 | Nano filter holder | 1 | 5,00,000
 42 | Cold room 2-8°C | 2 | 15,00,000
 43 | Rectangular tank | 1 | 4,30,000
 44 | Buffer preparation tank | 2 | 17,20,000
 45 | Dispensing Booth | 1 | 4,00,000
 46 | Analytical Balance | 1 | 1,70,000
 47 | Precision Balance | 1 | 1,50,000
 48 | Commercial Balance | 1 | 1,20,000
 49 | PVC bag Sealer | 1 | 5,000
 50 | Rotary Bag Sealer | 1 | 3,00,000
 51 | pH/ Conductivity meter | 2 | 3,00,000
 52 | Magnetic Stirrer | 2 | 1,50,000
 53 | Peristaltic pump | 2 | 2,00,000
 54 | Filter Integrity machine | 1 | 20,00,000
 55 | Fume Hood | 1 | 4,50,000
 56 | Autoclave 1000L | 2 | 40,00,000
 57 | DHS 1000L | 1 | 20,00,000
 58 | Ceiling LAF | 3 | 12,00,000
 59 | Drying Oven | 2 | 10,00,000
 60 | Acid tank 1000L | 2 | 38,70,000
 61 | Base tank 1000L | 2 | 38,70,000

62 | Central CIP system | 2 | 21,50,000
63 | Refrigerator 200L | 1 | 1,50,000
64 | Laminar air flow cabinet | 1 | 4,35,000
65 | Dynamic pass box | 4 | 4,80,000
66 | Tumble Dryer 25 Kg | 1 | 20,00,000
67 | Barrier washer Extractor 25Kg | 1 | 30,00,000
68 | SS Furniture (Table, chair, racks, Ladders etc) | NA | 40,00,000
69 | Lockers | NA | 2,00,000
70 | Laptop / Computers | 10 | 5,00,000
71 | Office furniture / Cabinets | NA | 3,00,000
Total Amount in INR | Total Amount in INR | Total Amount in INR | 70,29,18,520
EQUIPMENT BUDGET ESTIMATE IN INR | EQUIPMENT BUDGET ESTIMATE IN INR | EQUIPMENT BUDGET ESTIMATE IN INR |
EQUIPMENT BUDGET ESTIMATE IN INR
S/No | Name of Equipment | Quantity | Approx Cost (INR)
1 | Formulation Vessel 100L | 1 | 25,00,000.00
2 | Ceiling LAF | 2 | 8,00,000.00
3 | Laminar air flow cabinet | 1 | 4,35,000.00
4 | Continuous Particle Counter | 1 | 30,00,000.00
5 | Dynamic pass box | 7 | 8,40,000.00
6 | Pass Box static | 1 | 38,000.00
7 | Vial washing machine | 1 | 45,00,000.00
8 | Sterilizing tunnel | 1 | 45,00,000.00
9 | Vial filling & Sealing line 120 vials /min | 1 | 75,00,000.00
10 | Lyophilizer (1sq meter) | 1 | 55,00,000.00
11 | Labelling machine | 1 | 9,00,000.00
12 | Double door autoclave 1000L | 1 | 20,00,000.00
13 | Cold room | 1 | 8,00,000.00
14 | SS furniture (Table, Chair, racks, cabinets etc) | 1 | 2,00,000.00
Total Amount in INR | Total Amount in INR | Total Amount in INR | 3,26,13,000.00
EQUIPMENT BUDGET ESTIMATE IN INR | EQUIPMENT BUDGET ESTIMATE IN INR | EQUIPMENT BUDGET ESTIMATE IN INR |
EQUIPMENT BUDGET ESTIMATE IN INR
1 | Akta pure -150 | 2 | 40,00,000
2 | Columns XK 50/100 | 1 | 7,00,000
3 | Columns XK 50/60 | 1 | 7,00,000
4 | Column XK 50/30 | 3 | 14,00,000
5 | HPLC-UV | 1 | 34,00,000
6 | HPLC Columns and accessories | 1 | 10,00,000
7 | Electrophoresis (Minigel) with trans blot | 2 | 6,00,000
8 | Gel Doc system densitometer | 1 | 5,00,000
9 | Power packs | 2 | 2,00,000
10 | Endotoxin Detection system (Tube heating block) | 1 | 2,00,000
11 | Micropipettes- variable volume range of 2-20µL, 20-200µL, 100-1000µL, 500-5000µL and multichannel
30-300µL | 2 set each | 3,00,000
12 | Lab Refrigerator 2-8 ° C (SS) with data control | 4 | 4,50,000
13 | Flame photometer | 1 | 4,50,000
14 | Weighing Balance Analytical | 2 | 3,50,000
15 | UV-VIS spectrophotometer Double Beam with data acquisition. | 1 | 9,00,000
16 | Volumetric KF | 1 | 7,00,000
17 | Moisture Analyser | 1 | 8,00,000
18 | Stability Chambers (5±3° C, 25, 37 deg. C) 200L (1set) | 3 | 14,00,000
19 | Muffle Furnace | 1 | 80,000
20 | pH and conductivity meter with printer | 2 | 2,60,000
21 | Vaccum Oven | 1 | 1,20,000
22 | Melting point apparatus | 1 | 90,000
23 | Incubator Sterility Test 500L | 2 | 14,40,000
24 | Incubators for Normal Media plates Incubation 500L | 1 | 3,00,000
25 | Incubator for EMP and water samples 500L | 2 | 6,00,000
26 | Hot Air Oven | 1 | 1,60,000
27 | Thermostatic water bath (10-20 L) | 2 | 2,00,000
28 | Vaccum pump | 2 | 1,20,000
29 | Desiccator Glass | 1 | 4,000
30 | High Quality Laboratory water generation system (10 L /hr) Milli -Q water | 1 | 14,50,000
31 | Sonicator | 1 | 1,80,000
32 | Vortex Shaker | 6 | 2,10,000
33 | Magnetic Stirrer with hot plate | 4 | 4,00,000
34 | LAF Vertical | 2 | 6,00,000
35 | Biosafety Cabinet | 2 | 14,00,000
36 | Autoclave Double door | 1 | 12,50,000
37 | Dry Heat Steriliser | 1 | 7,50,000
38 | Fume Hood (Table Top) | 1 | 4,50,000
39 | Deep freezer (-20 Degrees C) with Data acquisition | 1 | 4,00,000
40 | Air sampler | 1 | 3,00,000
41 | Particle counter | 1 | 20,00,000
42 | Decontamination autoclave | 1 | 10,00,000
43 | Microscope Binocular | 1 | 1,00,000
44 | Colony Counter | 1 | 1,50,000
45 | Microplate reader ELISA | 1 | 6,00,000
46 | Microplate washer | 1 | 3,50,000
47 | TOC analyser (Grab Samples) | 1 | 25,00,000
48 | Timers/ Stop watch | 5 | 10,000
49 | Ice flaking machine | 1 | 2,50,000
50 | Standard Weight Box | 1 Set | 10,00,000
51 | Printer/scanner / xerox machine | 1 | 5,00,000
52 | Centrifuge table top | 1 | 4,50,000
53 | Size exclusion | L | 1,50,000.00
54 | Anion | L | 1,50,000.00
55 | Cation | L | 1,50,000.00
56 | Affinity | L | 1,50,000.00
57 | SS furniture racks | 25 | 2,50,000
58 | Computers / Laptops | 10 | 5,00,000
59 | Office Furniture | NA | 5,00,000
60 | Office Cabinets for Documents | 10 | 1,20,000

Total Amount in INR | Total Amount in INR | Total Amount in INR | 3,96,94,000.00
 BUDGET ESTIMATE IN INR | BUDGET ESTIMATE IN INR | BUDGET ESTIMATE IN INR | BUDGET ESTIMATE IN INR
 1 | Anemometer | 1 | 15,000.00
 2 | Lux meter | 1 | 12,000.00
 3 | Datalogger 32 channel | 1 | 4,00,000.00
 4 | Paper shredder | 1 | 10,000.00
 5 | Laminator | 1 | 10,000.00
 6 | Printer/scanner / xerox machine | 1 | 5,00,000.00
 7 | Computers / Laptops | 10 | 5,00,000.00
 8 | Numbering Machine | 5 | 25,000.00
 9 | Spiral Binding Machine | 1 | 15,000.00
 10 | A3 +A4 Lamination Machine | 1 | 25,000.00
 11 | Bi-Focal Glass | 1 | 80,000.00
 12 | Tachometer | 2 | 6,000.00
 13 | Clamp Meter/ Multi meter | 5 | 10,000.00
 14 | Level Indicator | 2 | 8,000.00
 15 | Measuring TAPE | 2 | 1,500.00
 16 | Stop Watch | 5 | 50,000.00
 17 | Temperature recorders 24 Nos. (data Loggers) | 20 | 2,00,000.00
 18 | Office Cabinets for Documents | 20 | 2,40,000.00
 19 | Projector for conference / meeting room | 1 | 3,00,000.00
 20 | Office Furniture | NA | 7,00,000.00
 Total Amount in INR | Total Amount in INR | Total Amount in INR | 31,07,500.00
 BUDGET ESTIMATE IN INR | BUDGET ESTIMATE IN INR | BUDGET ESTIMATE IN INR | BUDGET ESTIMATE IN INR
 1 | Air curtain | 10 | 1,00,000.00
 2 | PVC pallets | 50 | 3,00,000.00
 3 | Storage racks SS | 50 | 15,00,000.00
 4 | Hydraulic trolley | 1 | 2,50,000.00
 Total Amount in INR | Total Amount in INR | Total Amount in INR | 21,50,000.00
 BUDGET ESTIMATE IN INR | BUDGET ESTIMATE IN INR | BUDGET ESTIMATE IN INR | BUDGET ESTIMATE IN INR
 S/No | Name of Equipment | Qty. | Approx Cost (INR)
 1 | Biowaste reception tank 15000L | 1 | 25,00,000.00
 2 | Kill tank 500L | 2 | 25,00,000.00
 3 | Bar Screen | NA | 25,00,000.00
 4 | Collection tank | NA | 25,00,000.00
 5 | Equalization tank | NA | 25,00,000.00
 6 | Neutralization tank | NA | 25,00,000.00
 7 | Chemical coagulation | NA | 25,00,000.00
 8 | Biological oxidation tank | NA | 25,00,000.00
 9 | Filtration System | NA | 25,00,000.00
 BUDGET ESTIMATE IN INR | BUDGET ESTIMATE IN INR | BUDGET ESTIMATE IN INR | BUDGET ESTIMATE IN INR
 1 | Primary treatment plant : Bar screening - 3 stages, pH correction, BOD & COD removal secondary clarifier | | 8,00,000.00
 2 | Secondary treatment : Electro coagulation process, Flash mixer, HRSC clarifier, Sand filter | | 8,00,000.00
 3 | Sludge Management: Sludge thickner, Screw pump, Filter press, Sludge storage | | 8,00,000.00
 Total Amount in INR | Total Amount in INR | Total Amount in INR | 33,00,000.00

D4

104.1X2.00
UP

D3
D8 D8 D8 D8

SPM RECEIPT REJECT

LIFT
2.00X4.80
3.30 X 4.00 3.00 X 3.30
FINISHED GOODS STAGING
13.15 X 7.00
D8 D8 LABELING STORE
8.0 X 7.00

4.50X2.00

D8
8D D4

WATER SYSTEM & PB PACKING OFFICE
3.00X2.55

4D
ENGINEERING STORE
LABELING &
PACKING
2.05X10.1
8.0 X 5.0

CANTEEN
PACKING
29.9X17.1 VISITOR

4D
3.00X2.00

4D
4D
23.8X10.3 SPM STORE,
21.3X6.80
D8

4D
PACKING CHANGE
27.5 X 17.1 MALE
3.0 X 3.0
8.05 X 10.1

4D
VISUAL &
INSPECTION

4D
5.00X5.05
3.05X3.05

4D
PACKING CHANGE
BUFFER ROOM FEMALE
3.00 X 2.40
D3

2.95X2.00

4D

4D
DPB

D8
CORRIDOR
26.40X2.00
DPB

D4
DPB

4D
1.00X1.00
CAPPING
5.83X4.00

VIAL COLLECTION
TRAY
LYO SERVICE AREA LYO LOAD AL
8.06X4.03 5.58X4.03 2.59X4.00

CONVEYOR SWING	
AUTOMATIC FOUR HEAD ALUMINUM CAP SEALING MACHINE, MODEL-AEWCS-120	
D4	
4D	
4D	
4D	
4D	
PAL 1 PAL 2 PAL 3 PAL 4	
4.53X2.00 4.50X2.00 2.21X4.08 2.21X4.08	
FILL FINISH	
5.83X8.16	
CORRIDOR	
4D	
4D	
EXIT 2 EXIT 1 VIAL COLLECTION	
2.00X21.9 & LOADING TRAY	
4.53X2.00 4.50X2.00	
36" TURNTABLE	
NEW ADD	
D4	
D2	
CONVEYOR SWING	
LAUNDRY AREA DOUBLE STERILE CORRIDOR	
UNLOADING AREA	
4.88X1.60 5.66X1.76	
2.00X41.9 DOOR 3.07X4.76	
3.57X3.16 2.59X15.71	
D4 WASHER	
1.50X2.00 1.80X60.7	
MAL	
1.23X1.53	
D4	
D4	
TUMBLE PACKING AREA FORMULATION	
DRYER 5.58X2.92	
D4	
PAL 1.50X2.00	
1.23X1.50	
STABLIZING ZONE	
1500	
D4	
D4	
D3 3D	
COOLING ZONE	
D8	
MAL	
2.53X1.50 MAL PAL	
D4	
D4	
1.47X1.47 1.50X1.50	
STERILIZING ZONE	
3D D3 DISINFECTANT	
5.58X3.08	
PAL	
4D	
2.53X1.50	
D4	
D4	
DRYING ZONE	
DPB	
1.00X1.00	
D4	

FACILITY 02
MATERIAL PREP WASH
6.98X3.55 VIAL WASHING
5.45X1.58
AC UNLOAD
1.00X2.50 5.83X10.55
2.35X1.55 5.58X3.55

DPB

D4
D4
4D
CORRIDOR
9.11X2.00
CLEAN JC

FUTURE PROVISION
D4 D4 D4 2.54X3.00
D4
DPB

PAL-2
1.70X1.80
PRIMARY CHANGE PRIMARY CHANGE PRIMARY CHANGE
MAL ENTRY
VISITOR FEMALE MALE
2.00X4.00 COLD ROOM
D4
4.50X4.00 4.50X4.00 4.50X6.08 DE-CARTONING
3.59X3.00 2.90X3.44
PAL-1
1.70X1.80
R&D OFFICE
5.15X2.70 R&D LAB -1 R&D LAB -2 D4 D4 D4 D4 D4 D4
DPB

55.9X76.1
9.29X2.70 9.29X2.70
3D 3D 3D CORRIDOR
24.3X2.40
PASSAGE
1DE

R&D ENTRANCE
D3

18.6X1.50
5.15X2.20
D3 D3
SERVICE FLOOR

D4
3D
2.40X1.73 LIFT

54.4X28.2
R&D LAB-4 R&D LAB -3

7.20X2.00 9.29X2.70 9.29X2.70
D3

BULK STAGING
17.4 X 6.00

78.3X26.3 4.90 X4.26
D3

D3

DOCUMENT QA QC STAFF
STORAGE DOCUMENTATION
3.65X3.15
2.40X3.00 3.29X3.15 PM TESTING STABILITY ROOM
1.15X2.70 ROOM RETENTION
2.40X2.37 4.25X5.90 SAMPLES ROOM
2.40X5.90
D3

D6
QA MANGER QC MANEGER
CONFERENCE
2.40X3.57 2.40X2.70 2.09X2.70
APPROVED RUBBER
D3 D3 D3 D3 D6 BUNG STORE
D3
5.00 X 5.00
D3

CORRIDOR
WASH
CORRIDOR

EYE
1.20X9.34

15.0X1.20
TRAINING HALL
GENETS
D3

5.15X14.3 D3
D3 D3
CHANGE ROOM
2.40X1.80
D3

D4
BALANCE ROOM
CHEMICAL /GLASS 2.40X2.40
2.00X14.4 WARE STORAGE
D3

2.40X3.00
D3

LADIES
CHANGE ROOM
HOT ZONE /WASHING
2.40X2.00 INSTRUMENT ROOM
D3

2.40X5.87
4.00X5.87

4.90X2.00
D3

WET LAB

TOC /LPC 5.99X3.42 2.00X24.3
SAMPLE
RECIPET 2.40X2.82 3.54X2.45
2.40X1.97 D4
D3

SAMPLE
D3

RECIPET
HATCH

LAF

4D
AIR LOCK 2

PPM STORE
D1
0.9X2.23

17.25X 6.78,
DPB

AUTOCLAVE
MLT LAF
2.56X1.92 COOLING 19.20 X 11.40

UP
ZONE AUTOCLAVE
D3 2.16X1.97 LAF MEDIA
LAF 3.38X1.97
DPB

STERILITY
D3

STERILE PREP./STOREGE
D1 ROOM D3
1.17X1.23 1.17X1.23

INCUBATION PASSAGE 1.97X4.00 2.40X11.4
AIR LOCK 2

1.77X2.94 1.20X2.99
1.34X2.52 Biosafety

4.25X4.00
BIOSEFTY
D1

7.20X2.15
AIR LOCK 1
0.9X2.23

AIR LOCK 3
D1
D1

AIR LOCK 2
D1

D1 D1
D3

1.65X0.935 1.60X0.935 AIR LOCK 1 AIR LOCK
1.00X1.95 1.17X1.95
AIR LOCK 1

DPB
Cabinet

3.05X1.00 AIR LOCK 4 AIR LOCK 5
D1

D1

1.65X0.935 1.60X0.935 3.30X4.90 3.30X5.05
6D 4.65X2.00
D4

4D

D3
D1 D1 D1 D1
D4

CORRIDOR
1DE

1D
1D

18.6X1.45
UP

D3 D3 D3 D3 D3 D1 D1
4.70X6.80
DPB

1D
1D

2.00X7.00
BET ROOM MICRO OFFICE LADIES CHANGE GENETS CHANGE
MEDIA FILL DECONTAMINATION
MEDIA FILL ROOM ROOM
INCUBATION-1 & WASHING ROOM 2.40X2.37 2.40X2.37 2.40X7.00
INCUBATION-2
2.40X2.37 2.00X2.37 2.00X2.37
4.25X2.37 2.90X2.37
1D

QUARANTINE SAMPLING UNDERTEST
D1 D1 D6 LIFT D6 D6
LIFT

A 54.9X2.00
UP 34.8X2.00
LIFT

A
DOOR
SCHEDULE-GROUND
CLIENT :- LIFENITY WELLNESS INTERNATIONAL LTD.MUMBAI
FLOOR
NAME SIZE
DESIGN &
D1 750 x 2100 PROJECT BY :-

FIRST FLOOR
D2 800 x 2100
D3 900 x 2100 BIOBRIDGE HEALTHCARE SOLUTION PVT.LTD.
D4 1000 x 2100 ADDRESS :-13, Rachna Blossom, Jagdishnagar, PREPARED Y.T.J.

D6 1500 x 2100 Aundh, Pune -411007, Maharastra, India.
CHECKED G.K.

D8 2000 x 2100 DATE 14.12.2023
APPROVED
SCALE 1:100 -
ED 1000 x 2100 TITLE- CONCEPTUAL LAYOUT-FF BY CLIENT
SIZE -
(FILL FINISH FACILITY) DWG.NO 02
ED1 1200 x 2100 REVISION-R0

UG COLLECTION TANKS
GF - ETP / STP

UNDER GROUND 500 KL TANK -1 UNDER GROUND 500 KL TANK -2 UNDER GROUND 500 KL TANK -3
KILL TANK CHILLER ROOM BOILER ROOM BOREWELL

GF - WATER PRE-TREATMENT
24985

UNDERGROUND SUMP
CONTAINER

CONTAINER
12.19x2.44x2.59

CONTAINER
12.19x2.44x2.59

12.19x2.44x2.59
ROAD ROAD
8028

29249
SPM

PLASMA
RECEIVING
BAY 52821
FG RM
DISCPATCH RECEIVING RECEIVING
BAY BAY

CORRIDOR PLASMA TRANSFER

6D
3D

3D
D4

39.4X1.90
STORE CHANGE CORRIDOR
2.40X 49.89
11.6X2.90 PROVISION FOR FACILITY 2
ROOM

6D
UP

3.00X 2.40

ROAD
6D D6

350
500

500

500

500

500

500
1200
UTILITY & PROCESS PIPING 300
D6 D6 3D D3 D6 500

250 L
RM

1713
RLAF RLAF PIPING SKID PIPING SKID 500 PIPING SKID PIPING SKID PIPING SKID COLUMN PIPING SKID PIPING SKID

1200
AKTA-1"
REJECT

1050
CONTROL PANEL

CONTROL PANEL
RECALLED RECEIPT 700 TFF CHROMO

HH
PLASMA RM
ROOM FG IN
HH HH

GOODS FG DISCPATCH SAMPLING 3.00X2.52 SAMPLING 1200 685

2275
500
2.40X 2.40 3.00X 2.10 3.00X2.10
REJECT

LIFT
1200
2.40X2.40 2275 CONTROL PANEL

PLASMA SEROLOGY

COLUMN
CHROMO
STORE 1.07X1.90

300 L
1.07X 1.90

300 L
700

ROOM

1200
1050

500
MH

1200
D3 D3
RECEIPT TESTING
COLUMN
OFFICE
D3
2.95X4.00
DE-DUSTING
STAGE 1 (F)/ CHROMO

BV 107
D3

500 L
2250
TUNNEL
APPROVED RM
UTILITY & PROCESS PIPING
520
16.9X4.35 2.03X4.35 2.40X 4.00 3.00X 4.00 AIR LOCK-2 AIR LOCK-2 600

STAGE 2 (F)
500
1.87X0.90
D3 1.87X0.90
D3

250 L

PV 107.1
PV 107.2
9.85X 5.80

1500 L
PLASMA DAY

1500 L

2000 L
2000 L
HH
4.90X1.90

4014
STAGE 12 (E)

COLUMN
CHROMO
4.80X 3.50
DPB DPB
STAGE 3 (F)

2140

3128
AIR LOCK-1 AIR LOCK-1
12.0X8.00

1920

1920
AKTA-1"
CONTROL PANEL

D6 1.87X0.90 D3 1.87X0.90

2175
D3 D3 D3

PV 107.3
STORE

1500 L
300

PIPING SKID
8.72X8.00
500
3.00X2.51

1500 L
2500

1500 L
1000 L
16.5X8.00

COLUMN
CHROMO
SD SD SD SD

500 L

500 L

500 L
3D
1.95X11.4
6.55X10.0

500
1050
6.50X5.25 D6 DPB ANTE ROOM 2-8 MH MH MH

CENTRIFUGE
CONT.
500
5.00X2.25

3D
2.30X7.67

2500
1500 L
750

PIPING SKID
450

AKTA-1"
600

2613
450
6.05X4.15 500

3D
AIR LOCK-2
COLD ROOM -3 COLD ROOM -4 3D D3 300

ICE PACK COLD ROOM -1 COLD ROOM -2 3.00X1.09 1.50X 1.67 PIPING SKID PIPING SKID 500 PIPING SKID PIPING SKID PIPING SKID PIPING SKID PIPING SKID

5.55X4.95 5.55X4.95 5.53X4.95
4.10 X 4.95 5.50X 4.95 DISPENCING ROOM
11803

600
D3 D6 D6 D6

500

500
WASHING BAG BAR 3.45X 3.45
PLASMA AREA CODING,

6D
QUARANTINE 2.00X2.95 MINI POOLS RM

ROAD
2.90X2.95 QUARANTINE AIR LOCK-1
CORRIDOR

1.50X 1.70
CORRIDOR

D3
RLAF 2.00X2.00
D6 DPB
35.3X2.00

D6

6D

6D
D3

4D D6 D6 D6

600
500

500
6D

D6

DPB
1200
CORRIDOR 1050

COLUMN
CHROMO
2.05X 25.5

AKTA-1"
PIPING SKID PIPING SKID PIPING SKID PIPING SKID

1200
AKTA-1"

1050
D6

600
2275

DPB

2275
1500 L
D3 D3 1000 L
MH
CHROMO
COLUMN

1200
500 L
BUFFER PLASMA BAG CUTTING & STAGE 8 N
1200

1500 L
1500 L
THAWING
PLASMA STORAGE PLASMA STORAGE PLASMA STORAGE PLASMA STORAGE
POOLING 9.00X7.00
3D

3D
~ 1000 L ~ 1000 L ~ 1000 L ~ 1000 L
PREPARATION 3.22X8.90
3D

3D

FF - ENGINEERING OFFICE
~ 70 KL GV ~ 70 KL GV ~ 70 KL GV ~ 70 KL GV
5.40X5.30 5.40X5.30 5.40X5.30 5.40X5.30
9.00X6.00 6.20X8.90

6D
D6
STAGE 4 (F)

GF - UTILITY BLOCK
D6

PRE - VIRUS CONTROLLED CORRIDOR
D6 14.5X13.8

DPB
PIPING SKID PIPING SKID PIPING SKID
CHROMO
600

COLUMN

1200
AKTA-1"
1200
3D

3D
PLASMA STORAGE PLASMA STORAGE PLASMA STORAGE PLASMA STORAGE
~ 1000 L ~ 1000 L ~ 1000 L ~ 1000 L
3D

3D

1753
TANK STORAGE
500
~ 70 KL GV ~ 70 KL GV ~ 70 KL GV ~ 70 KL GV

1000 L
1000 L

1000 L

693
5.40X5.32 5.40X5.32 5.40X5.32 5.40X5.32 500

2000 L

PIPING SKID
1500 L
1500 L
9.00X8.10

PIPING SKID
1500 L

500 L
1000 L

BUFFER
FACILITY 02
MH

TFF

2500
2.00X23.6

500
SYSTEMS

PIPING SKID
2000 L
PIPING SKID PIPING SKID PIPING SKID

PIPING SKID
PIPING SKID

1500 L
9.52X7.20

2.00X28.5
10000 STAGE 5 (I)
BULK MANUFACTURING AREA

566

566
600
PLASMA STORAGE PLASMA STORAGE
3D

3D

PLASMA STORAGE PLASMA STORAGE
9.00X16.5
1200

COLUMN
CHROMO
~ 1000 L ~ 1000 L PIPING SKID PIPING SKID PIPING SKID
PIPING SKID
~ 1000 L ~ 1000 L
3D

3D

~ 70 KL GV ~ 70 KL GV

1200
~ 70 KL GV ~ 70 KL GV

AKTA-1"
500
5.40X5.32 5.40X5.32 D6

TFF
5.40X5.32 5.40X5.32

6D

TFF

2500
CONTROL PANEL CONTROL PANEL

D4
MH
600

HH+SG

HH+SG
FUTURE PROVISION
600
2500

500 L
1000 L

1500 L
2000 L
CORRIDOR

492
CIP / SIP 2.00X48.4

500 L
PIPING SKID

500 L
ANTE ROOM

ANTE ROOM

500 L
COLUMN PACKING

MH
9.00X6.20

300
MH MH

AND SKIDS
55.9m x 76.1m

1500 L
1500 L
3D

3D

PLASMA STORAGE PLASMA STORAGE PLASMA STORAGE PLASMA STORAGE
~ 1000 L ~ 1000 L ~ 1000 L ~ 1000 L PIPING SKID PIPING SKID
3D

3D

500
~ 70 KL GV ~ 70 KL GV ~ 70 KL GV ~ 70 KL GV
9.52X7.30

6D
5.40X5.32 5.40X5.32 5.40X5.32 5.40X5.32

FUTURE PROVISION-PLASMA STORAGE

PIPING SKID
CORRIDOR 431

1500 L
1.80X42.9
GF - QUARANTINE / 1.80X42.9
2.00X52.8
COOL ZONE
500

D6
FILTER
DEPTH
STAGE 7A CORRIDOR

45772.9 Sqft

500
PLASMA STORAGE /
28.4X45.0 9.00X2.00
PIPING SKID PIPING SKID

1.80X62.6

4D

6D

PIPING SKID
1000 L
TESTING LABS / 14.5X14.6
WAREHOUSE AUTO-

500
1200

DHS
1050

COLUMN
CHROMO
PIPING SKID

FILTER
NANO
500

AKTA-1"
1200

PIPING SKID
CLAVE

500 L
2291

1500
600

MH
PLASMA STORAGE PLASMA STORAGE
4.45X3.00
3D

3D

PLASMA STORAGE PLASMA STORAGE

2275
1500
~ 1000 L ~ 1000 L ~ 1000 L ~ 1000 L
76307

3D

3D

4.45X3.00

500
~ 70 KL GV ~ 70 KL GV ~ 70 KL GV ~ 70 KL GV MH

600

250 L
5.40X5.32 5.40X5.32 5.40X5.32 5.40X5.32

250 L
150 L
150 L
D4

500 L
D4

PIPING SKID

PV 116
800 800

500 L

600
800 800

MH

1814

6D

6D
CONTROL PANEL

CONTROL PANEL
HH+SG HH+SG

D4

UTILITY & PROCESS PIPING

UTILITY & PROCESS PIPING
HH HH

CONTROL PANEL

CONTROL PANEL
1500 L

PIPING SKID
2.50X1.88

600

755

500
WASHING

500
TFF
1200

2500
1050

COLUMN
CHROMO
3070 1200
1050

COLUMN
CHROMO

AKTA-1"
9.00X4.50

1200
AKTA-1"
PIPING SKID

1200
1500 L
2500

2275
2275
3D

3D

PLASMA STORAGE PLASMA STORAGE PLASMA STORAGE PLASMA STORAGE
COLUMN PACKING

600

500
~ 1000 L ~ 1000 L ~ 1000 L ~ 1000 L
3D

3D

~ 70 KL GV ~ 70 KL GV
CONTROL PANEL

~ 70 KL GV ~ 70 KL GV

PIPING SKID
1000 L
5.40X5.32 5.40X5.32 5.40X5.32 5.40X5.32
AND SKIDS

150 L
HH+SG
D4

DE A/L STAGE 10 T/

WORKSHOP

JANITOR 11.6X15.5

500
GOWN 2.00X3.00
STAGE 6 (I) STAGE 9 N STAGE 13 E

PIPING SKID
2.00X3.00

500 L
4.80X4.00

MH
D4

D4

STAGE 11 F 8.00X10.5 8.50X10.5
4D
4D

A/L GOWN
9.00X15.5
JANITOR
3D

3D

PLASMA STORAGE PLASMA STORAGE PLASMA STORAGE PLASMA STORAGE
~ 1000 L ~ 1000 L ~ 1000 L ~ 1000 L 2.00X3.00 2.00X3.00
3D

3D

~ 70 KL GV ~ 70 KL GV ~ 70 KL GV ~ 70 KL GV
4.80X2.00
D4

D4

4D

5.40X5.32 5.40X5.32 5.40X5.32 5.40X5.32

LIFT
D4

D4

DE A/L 6D 6D 6D 6D
GOWN 2.00X3.00
2.00X3.00
4.80X3.70
ROAD

6D

6D
D4

D4 CORRIDOR 2.50X2.00
4D

34.9X2.00
4D
3D

3D

PLASMA STORAGE PLASMA STORAGE PLASMA STORAGE PLASMA STORAGE 4D D4 D4
A/L GOWN
D6 D6
~ 1000 L ~ 1000 L ~ 1000 L ~ 1000 L
3D

3D

CONTROL PANEL CONTROL PANEL

~ 70 KL GV ~ 70 KL GV ~ 70 KL GV ~ 70 KL GV
5.40X5.35 5.40X5.35 5.40X5.35 5.40X5.35
2.00X3.00 2.00X3.00 4.80X2.30 7556

HH+SG

HH+SG
DHS
D4

D4

4D

COOL ZONE
8000 5.75X4.45 WASHING

9.00X2.00
D6

BUFFER TANK STORAGE-2 CIP / SIP COLD ROOM 4.70X9.00
PREPARATION
1D/2S

1D/2S

1D/2S

1D/2S

1D/2S

1D/2S

1D/2S

1D/2S
500

500

500

500

500

500

500

500

4D
6.40X9.00 6.40X9.00 6.45X9.00
1D/2S

1D/2S

1D/2S

1D/2S

1D/2S

1D/2S

1D/2S
500

500

500

500

500

500

500

500

KBD/CM/BP KBD/CM/BP KBD/CM/BP KBD/CM/BP KBD/CM/BP KBD/CM/BP KBD/CM/BP KBD/CM/BP

CHEMICAL DAY STORE 5.12X9.00
TEL

TEL

TEL

TEL

TEL

TEL

TEL

TEL

DE

4.66X6.80 3.00X3.95 4D D4 4D D4
16AMPS 16AMPS
D1

6M 6M

REJECT ROOM
D4

LOCKER 4.75X1.90
D1 D1 D1
ROOM REST ROOM D1 D1 D1
D4
D1

8D DOCUMENT CABINETS LIFT 9.13X4.80 (LADIES) D4
OFFICE
(GENTS)
TOILET (GENTS) 3.45X4.40 TOILET (LADIES) 5.30X4.40
4.60X4.40
4.89X4.40
LIFT

4.90X2.60 PPM RECEIPT

A
D4

CORRIDOR LIFT
46.8X2.00 4.00X2.45
UP

D8
MAIN ENTRY
D8
1.49X1.90
D8 A
+ 9.35 EGL
+ 9.35 EGL

ROAD
+ 10.08 FFL
+10.08 FFL
8000

ROAD 810KVA, 3Ø, 415V 810KVA, 3Ø, 415V
GENERATOR-3 GENERATOR-1

DG DG
18215

m2.1
ELOH NAM
HTPED
TIP LIO TNRUB

2000x1800mm
WIDE GATE

91°0'17"
+9.47 EGL
HT
1800 x 2100mm
METAL GATE
FFL
+9.78 FGL + 9.93

X
WELD MESH
TRANSFORMER VCB

FENCING
X X
STR
11KV TANGEDCO
BE-1
METERING
SOAK PIT RETAINER WALL

X
SLOPE SLOPE SLOPE
FFL
PANEL ROOM

X
2Nos - 1500x100x10mmTHK

FROM TANGEDCO
+ 10.08

11KV I/C SUPPLY
X X

INSERT PLATE
F

SLOPE
HV SIDE

SLOPE

FIRE BARRIER WALL
SOAK PIT RETAINER WALL

SOAK PIT RETAINER WALL
G (HT 5.3 M FROM FGL)

X
FFL FFL 11KV LAE
DP

X
ENTRY/EXIT 8M WIDE
EL + 14.77
BE
W + 9.93 + 10.08 HT

X
VCB

SLOPE

X X
SLOPE

O/G

WELD MESH
LV SIDE

FENCING
X
11KV CONSUMER STR 11KV TANGEDCO
S SLOPE SLOPE SLOPE BE-2

X
DP STRUCTURE DP STRUCTURE
SOAK PIT RETAINER WALL

X X
DOOR
CLIENT :- LIFENITY WELLNESS INTERNATIONAL LTD.MUMBAI

GROUND FLOOR
SCHEDULE-GROUND
FLOOR
NAME SIZE DESIGN &
D1 PROJECT BY :-
750 x 2100
D2 800 x 2100 BIOBRIDGE HEALTHCARE SOLUTION PVT.LTD.
D3 900 x 2100
PREPARED Y.T.J.
D4 1000 x 2100 ADDRESS :-13, Rachna Blossom, Jagdishnagar,
D6 Aundh, Pune -411007, Maharastra, India.
1500 x 2100 CHECKED G.K.

D8 2000 x 2100 DATE 14.12.2023
APPROVED
SCALE 1:100 -
ED TITLE- CONCEPTUAL LAYOUT-GF BY CLIENT
1000 x 2100 SIZE -
(BULK MANUFATURING) DWG.NO 01
SD 1500 x 2100 REVISION-R0

D4
UP

LIFT
TERRACE FLOOR TERRACE FLOOR
FACILITY 01 FACILITY 02
FUTURE PROVISION FUTURE PROVISION
106.6X76.1 55.9X76.1
LIFT

D4

4D
UP
D4

UP

LIFT
LIFT

A UP
LIFT

A
DOOR
SCHEDULE-GROUND
CLIENT :- LIFENITY WELLNESS INTERNATIONAL LTD.MUMBAI
FLOOR

TERRACE FLOOR NAME
D1
D2
SIZE
750 x 2100
800 x 2100
DESIGN &
PROJECT BY :-

D3 900 x 2100 BIOBRIDGE HEALTHCARE SOLUTION PVT.LTD.
D4 1000 x 2100 ADDRESS :-13, Rachna Blossom, Jagdishnagar, PREPARED Y.T.J.

D6 1500 x 2100 Aundh, Pune -411007, Maharastra, India.
CHECKED G.K.

D8 2000 x 2100 DATE 14.12.2023
APPROVED
SCALE 1:100 TITLE- TERRACE FLOOR -
ED 1000 x 2100 SIZE - FUTURE PROVISION
BY CLIENT

DWG.NO 03
ED1 1200 x 2100 REVISION-R0

D4
UP

23.1X9.10

LIFT
SPM STORE,
CANTEEN WATER SYSTEM & 27.5 X 19.2
25.9X19.2 ENGINEERING STORE 8.05 X 10.1
29.9X19.2

SERVICE FLOOR
54.4X56.8
FILLING AREA
26.2X56.8
FACILITY 02
15.0X10.1
25.9X26.3
FUTURE PROVISION
R & D LAB
25.9X7.05
LIFT
55.9X76.1
QC AREA
25.9X23.3

D4

4D
UP
D4

UP

LIFT
LIFT

LIFT
UP

CLIENT :- LIFENITY WELLNESS INTERNATIONAL LTD.MUMBAI

DESIGN &
PROJECT BY :-

FIRST FLOOR BIOBRIDGE HEALTHCARE SOLUTION PVT.LTD.

ADDRESS :-13, Rachna Blossom, Jagdishnagar,
Aundh, Pune -411007, Maharastra, India.
PREPARED Y.T.J.

CHECKED G.K.

DATE 14.12.2023
APPROVED
SCALE 1:100 -
TITLE- PLOT LAYOUT-FF BY CLIENT
SIZE -
(FILL FINISH FACILITY) DWG.NO 02
REVISION-R0

UG COLLECTION TANKS
GF - ETP / STP

UNDER GROUND 500 KL TANK -1 UNDER GROUND 500 KL TANK -2 UNDER GROUND 500 KL TANK -3
KILL TANK CHILLER ROOM BOILER ROOM BOREWELL

GF - WATER PRE-TREATMENT
UNDERGROUND SUMP
CONTAINER

CONTAINER
12.19x2.44x2.59

CONTAINER
12.19x2.44x2.59

12.19x2.44x2.59
ROAD ROAD
SPM
RECEIVING
BAY
FG PLASMA RM
DISCPATCH RECEIVING RECEIVING
BAY BAY
3D
D4

UP

ROAD

LIFT
FG
26.5X9.45 PLASMA & RM SAMPLING APPROVED RM
3.05X2.40 17.6X9.40 9.90X9.40

ROAD

FF - ENGINEERING OFFICE

GF - UTILITY BLOCK
FACILITY 02
FUTURE PROVISION-PLASMA STORAGE GF - QUARANTINE /
PLASMA STORAGE / FACILITY 01 BULK MANUFACTURING AREA
TESTING LABS /
28.6X45.0 WAREHOUSE
25.7X43.0 BULK MANUFACTURING AREA FUTURE PROVISION
50.7m x 74.2m 55.9m x 76.1m
40478.47 Sqft 45772.9 Sqft

WORKSHOP
LIFT

ROAD

ROAD ADMIN
46.9X17.1
7.21X10.3

UP
D4

UP

LIFT
LIFT

LIFT
UP

D8 D8 D8

+ 9.35 EGL
+ 9.35 EGL

ROAD
+ 10.08 FFL
+10.08 FFL

ROAD 810KVA, 3Ø, 415V
GENERATOR-3

DG
810KVA, 3Ø, 415V
GENERATOR-1

DG

m2.1
ELOH NAM

HTPED
TIP LIO TNRUB

2000x1800mm
WIDE GATE 91°0'17"

+9.47 EGL
HT
1800 x 2100mm
METAL GATE
FFL
+9.78 FGL + 9.93

X
WELD MESH
TRANSFORMER VCB

FENCING
X X
STR
11KV TANGEDCO
BE-1
METERING
SOAK PIT RETAINER WALL

X
SLOPE SLOPE SLOPE
FFL
PANEL ROOM

X
2Nos - 1500x100x10mmTHK

FROM TANGEDCO
+ 10.08

11KV I/C SUPPLY
X
INSERT PLATE
F

SLOPE
HV SIDE

SLOPE

X X
SOAK PIT RETAINER WALL

SOAK PIT RETAINER WALL
G (HT 5.3 M FROM FGL)

FIRE BARRIER WALL
FFL FFL 11KV LAE
DP

X
ENTRY/EXIT 8M WIDE
EL + 14.77
BE
W + 9.93 + 10.08 HT

X
VCB

SLOPE

X X
SLOPE

O/G

WELD MESH
LV SIDE

FENCING
X
11KV CONSUMER STR 11KV TANGEDCO
S SLOPE SLOPE SLOPE BE-2

X
DP STRUCTURE DP STRUCTURE
SOAK PIT RETAINER WALL

X
X
CLIENT :- LIFENITY WELLNESS INTERNATIONAL LTD.MUMBAI

GROUND FLOOR DESIGN &
PROJECT BY :-

BIOBRIDGE HEALTHCARE SOLUTION PVT.LTD.

PREPARED Y.T.J.
ADDRESS :-13, Rachna Blossom, Jagdishnagar,
Aundh, Pune -411007, Maharastra, India. G.K.
CHECKED

DATE 14.12.2023
APPROVED
SCALE 1:100 -
TITLE- PLOT LAYOUT GF BY CLIENT
SIZE -
(BULK MANUFATURING) DWG.NO 01
REVISION-R0

1200

1200
TERRACE

TERRACE

6000

6000
FF

FF

6000

6000
PL

1200
PL
GL

GL
76500

EAST SIDE ELEVATION WEST SIDE ELEVATION

1200
TERRACE

6000
FF

6000
PL

1200
GL

NORTH SIDE ELEVATION

1200
TERRACE

6000
FF

6000
PL

1200
GL
163185

SOUTH SIDE ELEVATION
1200

TERRACE TERRACE

MEDIA FILL
DESTRUCTION &
MEDIA FILL
INCUBATION-2
PASSAGE
SERVICE FLOOR LADIES CHANGE
FACILITY 2 PASSAGE PPM STORE PASSAGE TOILET
PASSAGE
PASSAGE TOILET ROOM
MICRO OFFICE BET ROOM WASHING ROOM
INCUBATION-1
6000

GENETS CHANGE
FUTURE PROVISION SERVICE FLOOR ROOM

FF FF

FACILITY 02
REST ROOM LOCKER MAIN ENTRY CAFETERIA TOILET ADMIN
PPM RECEIPT OFFICE TOILET (LADIES) TOILET (GENTS)
BULK MANUFACTURING AREA (LADIES)

6150
ROOM PASSAGE
6000

ADMIN
FUTURE PROVISION (GENTS) CONFERENCE
ENTERANCE MEETING ROOM

LOBBY

D D

PL PL

GL GL

SOUTH SIDE ELEVATION SECTION A-A
CLIENT :- LIFENITY WELLNESS INTERNATIONAL LTD.MUMBAI

DESIGN &
PROJECT BY :-

BIOBRIDGE HEALTHCARE SOLUTION PVT.LTD.

ADDRESS :-13, Rachna Blossom, Jagdishnagar, PREPARED Y.T.J.

Aundh, Pune -411007, Maharastra, India.
CHECKED G.K.

DATE 08.12.2023
APPROVED
SCALE 1:100 TITLE-SECTION & ELEVATION LAYOUT -
BY CLIENT
SIZE - OPTION 1
DWG.NO 04
REVISION-R0

TECHNOLOGY TRANSFER

DOSSIER

Human Plasma-Derived Medicinal Products

(Albumin / IVIG / Factor VIII / Factor IX / Coagulation Factors)

Scope of Document

This document defines the scope, methodology, responsibilities, technical requirements for the transfer of manufacturing technology for human plasma-derived medicinal products in compliance with WHO cGMP and applicable regulatory requirements.

Prepared for:

BIOHEALTH, BAGHDAD, IRAQ

Prepared by:

BIOBRIDGE HEALTH CARE SOLUTIONS PVT. LTD, PUNE, INDIA

www.biobridge.in

Document No.: TT-BB-HPDMP-001-2026

Version: 2.0

Effective Date: 12-02-2026

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1.0 Administrative Section

1.1 Document Control

Title: Plasma derived medicinal products such as Human Serum Albumin (HSA), Human Normal Immunoglobulin (IgG), and Hemophilia factors VIII, IX & Fibrin sealant.

Version history & change control Rev. 02

Effective date 12.02.2026

Confidentiality statement: This dossier contains proprietary information relating to the manufacturing process and formulation of the human plasma-derived medicinal products.

Approvals (TT giver / TT receiver / QA)

TT Giver TT Receiver

BioBridge Healthcare Solutions Pvt. Ltd, BioHealth, Baghdad, Iraq
Pune, India

1.2 Technology Transfer Overview

Scope of technology transfer

The objective of this project is to establish and operate a state-of-the-art Human Plasma Derived Medicinal Products (HPDMP) manufacturing facility with an annual plasma processing capacity of 150,000 liters. This facility aims to consistently and efficiently produce high-quality plasma-derived products such as Human Serum Albumin (HSA), Human Normal Immunoglobulin (IgG), Hemophilia factors VIII and IX, Thrombin and Fibrinogen while ensuring compliance with regulatory standards and best industry practices.

Product/Fraction Name:

- Albumin
- Human Immunoglobulin
- Factor VIII
- Factor IX
- Thrombin
- Fibrinogen

Transfer type: Full

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Site involved: BioHealth, Amman, Jordan

Transfer strategy (phased / parallel/staggered):

o The plasma fractionation process will be transferred using a staggered strategy in a phase wise manner i.e. transfer of high-risk unit operations (viral safety, Primary chromatography) etc., in the first phase. However, low-risk support operations will be transferred in Parallel, subject to readiness criteria and QA sign-off.

1.3 Roles & Responsibilities

Role Responsibilities

Overall responsibility for plasma fractionation

Head of

operation; ensures compliance with GMP and SOPs,

Manufacturing

approves batch records.

Supervises day-to-day fractionation steps

(precipitation, filtration, and chromatography);
 Production Manager manages staff and equipment readiness. Ensure the complete documentation of activities on time in full.
 Interface for QA activities and GMP batch release.
 Approves SOPs, batch records, deviations, and change controls; ensures regulatory compliance; oversees audits. Overall responsible for the integral evaluation
 Quality Assurance
 of environmental conditions during manufacturing
 (QA) Head
 (Clean rooms), trends of clean utilities and products batch release/rejection. To organize and maintain the training records of personnel.
 Conducts in-process and final product testing (purity, potency, viral safety); validates analytical methods;
 Quality Control (QC) releases test results to QA department for final batch Manager release. Responsible for release of Raw materials and packaging materials after testing for production department. To train the staff for different quality

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related procedures and GLP practices. To initiate product stability program.
 Prepares and submits product dossiers in CTD format; maintains communication with Drug Office; ensures Regulatory Affairs
 confidentiality and compliance with current GMP Manager
 guidelines. To ensure products are registered on time as per the business plan.
 Oversees process validation, PPQ runs, RTD studies, Validation Manager equipment qualification, cleaning validation and validation of clean utilities and analytical techniques.
 Ensures cold chain, HVAC, water systems, and Engineering/Utilities cleanroom compliance; preventive maintains Lead equipment and calibration of measuring devices in the facility.
 Implements biosafety, donor screening protocols, and Safety & Compliance
 occupational safety measures; monitors risk Officer
 assessments.
 Maintains master manufacturing formula, SOPs, Documentation/
 logbooks, and traceability of plasma pools; ensures Records Officer
 audit readiness.
 Manages plasma storage, raw material inventory, and Warehouse &
 distribution of final products; ensures cold chain Logistics Lead
 integrity.
 To oversee sterile filtration, final formulation and fill Formulation & Fill–
 finish activities; ensures container closure integrity and Finish Manager
 product labelling as per the requirements.

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Technology Transfer Team structure
 Role Responsibilities
 Overall coordination; Timeline management, Risk Tech Transfer Lead management, and Stakeholder alignment. Reports to Head of Manufacturing.
 Process Development Provides process knowledge; supports scale-up, Scientist comparability, and troubleshooting.
 To support preparation of equipment, SOPs, Master Manufacturing
 formulas, Batch records, and batch execution readiness Representative
 at receiving site.
 Quality Assurance Oversees GMP compliance, deviation handling, out of (QA) Lead specification and change control during transfer.
 To support preparation of QC STPs, Specifications and Quality Control (QC) Validation of analytical methods, IPCs strategy,
 Lead initiation of stability testing of PPQ batches and product release testing at new site.
 Leads PPQ planning to oversee equipment Validation Manager
 qualification and cleaning validation.
 Regulatory Affairs & To support dossier updating, variation strategy, and Liaison support in communication with drug office.
 Documentation To support management of BMRs and STPs, SOP Coordinator harmonization, and traceability of transfer records.
 Engineering/Utilities To support engineering team in facility readiness, cold Support chain, cleanroom and clean utilities compliance.
 To support training needs assessments for technical staff, to coordinate in arranging training (External and Training Coordinator
 Internal), competency assessments, and area specific SOP familiarization.

2.0 Product & Regulatory Background

2.1 Product Description

2.1.1 Therapeutic indication

2.1.1.1 Human Albumin 20% infusion

It is used to treat blood volume loss. It works by replacing the body fluids lost due to excessive trauma, bleeding, surgery, or kidney dialysis. Thus, it is helpful in treating low blood albumin levels.

2.1.1.2 Human Normal Immunoglobulin (IVIG):

Immunoglobulin preparations are indicated in several clinical conditions.

An approved list of clinical conditions where IVIG is indicated is as under:

Primary Immunodeficiency (PID) Syndromes

Replacement therapy to promote passive immunity: The following PID syndromes can be treated with intravenous replacement of IgG and are considered well established in-

- Congenital agammaglobulinaemia and hypogammaglobulinaemia
- Common variable immunodeficiency
- X-linked agammaglobulinaemia
- Wiskott-Aldrich syndrome
- Severe combined immunodeficiency

IVIG may be preferred in patients who require an immediate or large increase in intravascular immunoglobulin concentrations, in patients with small muscle mass, and in patients with bleeding tendencies in whom Intramuscular (IM) injections are contraindicated.

Secondary Immunodeficiency (SID) Syndromes

IgG can also be used as replacement therapy in:

- Secondary hypogammaglobulinaemia in patients with B-cell chronic lymphocytic leukemia (CLL) or multiple myeloma (MM) with recurrent infections.
- Pediatric HIV-I infection who have bacterial infections: HIV-infected infants and children with hypogammaglobulinaemia

(IgG <400 mg/dL) should receive IVIG (400 mg/kg once every 2–4 weeks) to prevent serious bacterial infections.

Kawasaki Syndrome

IVIG is used in conjunction with aspirin therapy for initial treatment of the acute phase of Kawasaki disease.

Approximately ≥10% of patients with Kawasaki disease fail to respond to initial treatment with IVIG and aspirin therapy and have persistent fever or recurrent fever after an initial afebrile period. Retreatment with IVIG (within 24–48 hours of persistent or recrudescence fever) and continued aspirin therapy usually is recommended for these patients.

Idiopathic Thrombocytopenic Purpura

IVIG is indicated for the treatment of acute or chronic (e.g., >6 months duration) idiopathic thrombocytopenic Purpura.

Allogeneic Bone marrow transplantation (BMT)

In adults and children undergoing BMT, IVIG can be used to decrease the risk of infections (e.g., septicemia), interstitial pneumonia of infectious or idiopathic etiologies, and acute graft-versus-host disease.

Guillain-Barre syndrome

IVIG initiated within 2 weeks of symptom onset appears to be as effective as plasma exchange.

Chronic Inflammatory Demyelinating Polyneuropathy (CIDP)

IVIG may be used for the treatment of chronic inflammatory demyelinating polyneuropathy to improve neuromuscular disability and impairment and for maintenance therapy to prevent relapse.

2.1.1.3 Human coagulation Factor IX

Is indicated for treatment and prophylaxis of bleeding episodes in patients with Hemophilia B (Congenital Factor IX deficiency or Christmas disease).

2.1.1.4 Human Coagulation Factor VIII

Is indicated for the treatment and prophylaxis of bleeding in patients with hemophilia A (Congenital Factor VIII deficiency).

2.1.1.5 Human Fibrinogen

Therapy and prophylaxis of hemorrhagic diatheses in:

- Congenital hypo-, dys- or fibrinogenemia
- Acquired hypo fibrinogenemia resulting from
 - Disorders of synthesis in cases of severe liver parenchyma damage.
 - Increased intravascular consumption e.g. as a result of disseminated intravascular coagulation, hyperfibrinolysis.
 - Increased loss.

The most important clinical pictures associated with a defibrination syndrome are: Obstetrical complications, acute leukemia especially promyelocytic leukemia, liver cirrhosis, intoxications, extensive injuries, hemolysis after transfusion errors, operative interventions, infections, sepsis, all forms of shock as well as tumors especially in the lung, pancreas, uterus, and prostate.

2.1.1.6 Thrombin

Thrombin is indicated to aid hemostasis whenever oozing blood and minor bleeding from capillaries and small venules is accessible and control of bleeding by standard surgical techniques (such as suture, ligation, or cautery) is ineffective or impractical. In various types of surgeries, solutions of Thrombin may be used in conjunction with an Absorbable Gelatin Sponge.

2.1.2 Dosage form & presentation

S/No. Product/ Fraction Name Dosage Form

- 1 Albumin Injection
- 2 Human normal immunoglobulin Solution for infusion
- 3 Human Coagulation Factor IX Freeze dried
- 4 Human coagulation Factor VIII Freeze dried
- 5 Human Fibrinogen Freeze dried
- 6 Thrombin Freeze dried

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2.1.3 Target markets

Plasma fractionation manufacturers are increasingly targeting export markets across Southeast Asia, Africa, Latin America, and select Middle Eastern countries due to rising demand for affordable plasma-derived therapies and limited local manufacturing capacity in those regions. These markets offer regulatory flexibility, growing healthcare infrastructure, and strategic opportunities for firms with WHO-GMP compliance and certification.

2.2 Regulatory Status

2.2.1 Reference product status (commercial)

Product Reference / Commercial Examples (Licensed

Category Status Marketed Products

Human Serum Licensed, marketed plasma Albumin 5%, 20% Albumin (HSA) biologic solutions (CSL, Grifols, etc.)

Human Normal Licensed IVIG/SCIG Privigen, Gamunex-C, Immunoglobulin biologics OCTAGAM, HYQVIA

(IgG) Octapharma, Takeda

Factor VIII Licensed plasma coagulation Alphanate, Humate-P, (Plasma-Derived) factor Koate-DVI

Factor IX Licensed plasma coagulation

AlphaNine SD, Mononine

(Plasma-Derived) factor

Fibrinogen Licensed concentrate for

Fibryga, RiaSTAP

(Plasma-Derived) deficiency

Thrombin Licensed haemostatic agent Human thrombin

(Plasma-Derived) formulations

2.2.2 Applicable guidelines:

- ISPE Baseline Guide Vol. 6, Biopharmaceutical Manufacturing Facilities
- ISPE Baseline Guide Vol. 3, Sterile Manufacturing Facilities.
- WHO Good manufacturing practices for biological products.

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- Good manufacturing practice guideline for human and veterinary medicinal product.
- PIC/s Guide to Good Manufacturing Practice for Medicinal Products.
- USFDA Guidance for Industry – Sterile Drug Products produced by Aseptic Processing.

2.3 Critical Quality Attributes (CQAs)

Product/Fraction Name Critical Quality Attributes (CQAs)

Human Serum Albumin (HSA) Description, Identification by SDS-PAGE, pH, Haem content by Absorption at 403 nm, Molecular Size Distribution, Prekallikrein Activator, Total Protein, Bacterial Endotoxin, Bioburden, N -acetyl tryptophan, Sodium Caprylate.

Human Normal Description, pH, Identification by SDS-Immunoglobulin (IgG) PAGE, Osmolality, Total protein,

Thrombogenic (procoagulant) activity, Test

for viral and bacterial antibodies (at an

Immunoglobulin Concentration of 50 g/L),

Distribution of Immunoglobulin G

subclasses, Test for Fc function of

Immunoglobulin, Molecular size by liquid

chromatography, Prekallikrein activator,

Bioburden, Bacterial Endotoxin.

Factor VIII (Plasma-Derived) Description, Identification, pH, Osmolality,

Assay of Human Coagulation Factor VIII,

Bacterial Endotoxins, Bioburden.

Factor IX (Plasma-Derived) Description, Identification, pH (Neat sample),

Osmolality (Neat sample), Assay of Human

Coagulation Factor IX, Specific Activity,

Bioburden, Bacterial Endotoxins, Activity of

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Factor II, Activity of Factor VII, Activity of

Factor X.

Fibrinogen (Plasma-Derived) Description, pH, Fibrinogen Assay (Clottable Protein), Bioburden.

Thrombin (Plasma-Derived) Description, pH, Thrombin Assay (Identification), Bioburden.

3.0 Plasma Source & Starting Material Control

3.1 Plasma Collection System

Donor selection criteria

The selection and management of blood donors is an essential part of the blood donation process. Health authorities and blood transfusion services are responsible for ensuring that the national system is in place for the selection of all blood donors through an assessment of their suitability to donate blood. In selecting individuals for blood donation, it is most important to determine whether the person is in good health, to protect the donor against damage to his or her own

health and to protect the recipient against exposure to diseases or to medicinal products from the blood or blood products.

The following are the selection criteria for donors:

1. Health: A licensed physician or a person under the direct supervision of a licensed physician shall determine the health of a donor, and the donor shall be free from any disease transmissible by blood transfusion.
2. Age: Donors shall be healthy persons of either sex between the ages of 18 and 65 years.
3. Time between blood donations
4. Blood pressure: Systolic blood pressure between 12 and 24 kPa (90 and 180 mmHg); diastolic blood pressure between 6.67 and 13.3 kPa (50 and 100 mmHg).
5. Blood Temperature: Temperature not exceeding 37.5 °C.
6. Hemoglobin content: The hemoglobin concentration or erythrocyte volume fraction of the donor's blood shall be within normal limits, as defined by the national control authority or the national blood transfusion authority.

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7. Test for HBsAg: Should be negative.
 8. Test for anti-HIV: Should be negative.
 9. Test for anti-HCV: Should be negative.
- An additional test for STD is also performed.

Plasma collection method:

Plasma must be collected only from licensed blood centres and only for fractionation purposes, not for direct transfusion.

Plasma by Plasmapheresis (Preferred & Dedicated Method): Plasma is collected directly from the donor using an automated plasmapheresis machine, and red blood cells are re-infused to the donor.

Regulatory Requirements

- Only approved plasmapheresis machines shall be used
- Equipment must be validated, calibrated, and maintained
- Procedure conducted by trained technical staff
- Donor consent is mandatory

Process Steps

1. Donor selection & screening
 - o Healthy donors meeting eligibility criteria
 - o Medical examination and history review
 2. Blood separation
 - o Blood withdrawn and centrifuged/filtered
 - o Plasma separated; RBCs returned to donor
 3. Anticoagulant
 - o Approved anticoagulant (e.g., ACD)
 4. Volume limits
 - o As prescribed by regulatory (based on donor weight)
 5. Frequency
 - o Controlled donation frequency to ensure donor safety
- Advantages
- High plasma yield
 - Lower protein degradation
 - Preferred for the fractionation industry

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Testing strategy (Serology & NAT):

To ensure the viral safety and quality of human plasma intended for fractionation, serological and Nucleic Acid Amplification Testing (NAT) is mandatory for transfusion-transmissible infections, in compliance with applicable regulatory requirements.

Plasma must be negative / non-reactive for:

1. Serology (Non-Reactive)
 - HBsAg
 - Anti-HIV-1 & HIV-2
 - Anti-HCV
 - Syphilis (VDRL/TPHA)
2. NAT Testing (Non-Reactive)
 - HIV-1 RNA
 - HCV RNA
 - HBV DNA

(Testing may be performed on individual donations or validated mini pools)

3.2 Plasma Acceptance Specifications

Acceptance/rejection criteria:

Plasma intended for fractionation shall be accepted only when it is collected from medically eligible and adequately screened donors at a licensed plasma or blood collection center, following informed consent and approved donor selection criteria. Each plasma unit must be collected aseptically using approved anticoagulants and validated collection procedures. The plasma shall be non-reactive for all mandatory serological tests, including HBsAg, Anti-HIV-1 & 2, Anti-HCV antibodies, and Syphilis, and shall be non-reactive for NAT testing for HIV-1 RNA, HBV DNA, and HCV RNA, performed either on individual donations or on validated mini pools.

Only plasma that meets all testing requirements, has complete donor traceability and documentation, and complies with regulatory requirements shall be considered acceptable.

In addition, plasma must meet defined quality attributes, being clear and pale yellow in appearance, free from hemolysis, turbidity, clots, and contained in

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intact, leak-proof containers with correct labelling. Plasma shall be frozen within validated timelines, stored and transported under controlled frozen conditions ($\leq -20^{\circ}\text{C}$ or as validated), with documented maintenance of the cold chain and no temperature excursions during storage and transport.

Plasma shall be rejected, if it fails to meet any of the above criteria.

Deviations handling:

There should be systems in place to ensure compliance, all types of quality defects (e.g. in blood bags or test kits), and adverse events or reactions are documented, carefully investigated for causative factors and, where necessary, followed by the implementation of corrective actions to prevent recurrence. The corrective and preventive action system should ensure that nonconformity of the product or quality problems are corrected, that recurrence of the problem is prevented, and that the plasma fractionator is notified according to the agreed procedure. The blood establishment should have methods and procedures in place to channel product or quality problems into the corrective and preventive action system.

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4.0 Manufacturing Process Technology

4.1 Overall Process Flow Diagram

4.1.1 Process block diagram Plasma Fractionation

Process block Flow Diagram (PFD) for Plasma Fractionation

Frozen Plasma

Thawing at a temperature $\leq 30^{\circ}\text{C}$

Plasma bags cutting

Plasma bags pooling

Pooled Plasma, filtration with 105 μ , 45 μ , 10 μ , 5 μ & DP Filters

Load for Chromatography-1F

Chromatography-1F

Plasma Loading 18 to 28% of the CV

Fraction -1 (Factor - VIII) Fraction -2 (Fibrinogen) Fraction -3 (Factor-FIX, Albumin, IVIG)

Chromatography 2F

Loading ≤ 9 mL/mL of the resin

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FT concentration & Diafiltration Chromatography 8N

Dilution and pH adjustment

Euglobulin ppt

Euglobulin ppt

Centrifugation supernatant is loaded for chromatography 4F

Chromatography 4F Load ≤ 1.12 mL of plasma /mL of resin

Flowthrough (Albumin, IVIG) Elution Albumin

Chromatography 5I Chromatography 7A

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Process block Flow Diagram (PFD) for IVIG

PF Chromatography 4F Flow through

Concentration

pH 5.7 & Cond 1.7 adjusted ms/cm

Filtration

Load

IG Chromatography 5I

Flow Through

Concentration

Low pH Incubation at 37°C for 12 hrs. at pH 4.0 Cond: 2ms/cm

Filtration

SD Incubation at 30°C for 4 hrs.

1.2 μm Filtration

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Elution

pH adjustment to pH 4.0

Nano filtration

Concentration & Diafiltration

Bulk

Formulation

Sterile Filtration

DS stored at 2 to 8 °C

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Process block Flow Diagram (PFD) for Albumin

PF Chromatography 4F Eluate

Load

AL Chromatography 7A

Eluate

Concentration

Heat treatment at 57 °C for 3 Hrs.

Centrifugation, Filtration & Concentration

Virus Inactivation- Low pH Incubation for 10 Hrs.

Filtration & Concentration

.

Pasteurization Incubation for 10 Hrs.

Filtration, Concentration & Diafiltration

.

Bulk

Formulation

Drug

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Process block Flow Diagram (PFD) for Thrombin

FN Chromatography 8N WASH-2

Incubation for 5 hours at RT

Filtration with 1.2 µm

Load on Chromatography 10T

Addition of Stabilizer to Elution

Elution Concentration

Bulk

Sterile Filtration

Drug

Freeze at -20°C

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Process block Flow Diagram (PFD) for Factor VIII

PF Chromatography 1F Fraction-1

Load on Chromatography 12E

SD Incubation for 4 Hrs

Filtration with 1.2 µm

Load on Chromatography 13E

Elution Concentration

Chromatography 14E

Collection of all Eluates

Loaded 12.5% on Chromatography 14E

Addition of Stabilizer and Concentration

Formulation

Filtration

Drug

Freeze at -20°C

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Thawing at RT

Sterile Filtration

Lyophilized

Stored at 2 to 8 °C

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Process block Flow Diagram (PFD) for Fibrinogen

PF Chromatography 1F Fraction-2

Precipitation for 3 hours at RT

Centrifugation

Suspended slurry

Incubation for 1Hour

Filtration with 1.2 µm

Concentration and Diafiltration

SD incubation for 4 hours at RT

Load on Chromatography 11FB

Concentration and Diafiltration

Bulk

Sterile Filtration

Drug Freeze at -20°C

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Thawing at RT

Sterile Filtration

Lyophilized

Stored at 2 to 8 °C

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Process block Flow Diagram (PFD) for Factor IX

PF Chromatography 2F Eluate

Dilution and Concentration

Virus inactivation, Incubation for 4 Hrs

Filtration with 1.2 µm

Load on Chromatography 8N

Chromatography 8N Elution

Concentration

Loaded 2% on Chromatography 9N

Collection of all injection Eluates

Concentration

Bulk

Filtration

Drug Freeze at -25°C

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Thawing at RT

Sterile Filtration

Lyophilized

Stored at 2 to 8 °C

4.1.2 Fraction-wise yield mapping
Yields for 150,000 Lit of Plasma- Annual capacity

Protein Protein No. of
 Product Conc. / Vial
 availability Recovery vials/year
 Albumin (gm) 60,00,000 30,00,000 20 gm 1,35,000
 Human IgG (gm) 18,75,000 9,37,500 5 gm 1,50,000
 Factor VIII in IU 15,00,00,000 18,75,000 250 IU 7,500
 Factor IX in IU 15,00,00,000 13,500,000 600 IU 22,500
 Fibrinogen (gm) 4,50,000 2,25,000 70 mg
 12,000
 Thrombin in IU 13,50,00,000 1,35,00,000 1000 IU
 Fibrinogen 450,000 225,000 1000 mg 2,10,000

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4.2 Detailed Unit Operations (per fraction)

4.2.1 Plasma fractionation

4.2.1.1 Objective

The objective of plasma fractionation operation is to manufacture/purify the plasma-derived product using various purification methods including chromatography and TFF.

4.2.1.2 Scientific principle

Plasma fractionation by chromatography is based on the principle of selective protein separation through controlled interactions with a stationary phase Chromatography exploits charge, size, hydrophobicity, and affinity differences to isolate plasma proteins with higher purity and reproducibility.

4.2.1.3 Process parameters (CPPs)

Step CPP

Thawing and Plasma PW Temperature, Pressure during

Pooling & filtration filtration, OD600, pH, conductivity

Chromatography 1F Flow rate, Peak 1 Collection based

on UV280, Fraction-2 collection,

Fraction-3 collection, Flow rate

after elution, pressure

Chromatography 2F Flow rate, pH drop, equilibration

pH, Loading flow rate, Flow

through Collection based on UV280,

Elution buffer conductivity,

Elution collection criteria based on

UV280

Concentration and TMP, NWP, Diafiltration: till

diafiltration of 2F flow conductivity permeate reaches

through 2000 – 2500 μ S/cm

Euglobulin precipitation Dilution conductivity, Dilution pH,

Incubation time, Incubation

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temperature, Separator speed,

Outlet flow

Chromatography 4F Flow rate, pH drop, Equilibration

pH, Conductivity of sample, Flow

through Collection based on UV280,

Elution criteria, Loading capacity

4.2.1.4 Operating ranges

Step Operating range

Thawing and Plasma PW Temperature, Pressure during

Pooling & plasma filtration: 1 ± 0.5 bar, OD600: 0.4 to

filtration 0.8, pH: 7.03 ± 0.1 , Conductivity:

15 ± 0.1 mS/cm

Chromatography 1F HETP: $<3-3.5$, Asymmetry: 0.8 to

1.8, Flow rate: 550 L/H, Peak 1

Collection based on UV280: 0.4 AU

till descending, Fraction-2

collection: Peak 2 starts till 50 L,

Fraction-3 collection: After 50 L

from fraction 2 end till descending

0.6 AU), Flow rate after elution:

600 LPH, pressure: ≤ 2.5 bar

Chromatography 2F HETP: <3 , Asymmetry: 0.8 to 1.8,

Flow rate: 942 L/H, pH drop: 4.50

± 0.50 , equilibration pH: $7.00 \pm$

0.20, Loading flow rate: 785 LPH,

Flow through Collection based on

UV280: 100 mAU to 100 mAU,

Elution buffer conductivity: $33 \pm$

1.5 mS/cm, Elution collection

criteria based on UV280: 100 mAU

to 100 mAU

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Concentration and TMP: 0.5 to 1.0 bar, NWP: ≥ 120

diafiltration of 2F flow LMH/Bar, Diafiltration: till

through conductivity reaches 2-2.5 mS/cm

EUGLOBULIN Dilution conductivity: NMT 1.1

precipitation mS/cm, Dilution pH: 5.10 - 5.30,

Incubation time: 6-16 hours,

Incubation temperature: 2-8°C,

Separator speed: 9100 rpm, Outlet

flow: NMT 250 LPH,

Chromatography 4F HETP: <3 , Asymmetry: 0.8 to 1.8,

Flow rate: 3454 L/H, pH drop: 4.25

± 0.25 , Equilibration pH: $5.20 \pm$

0.10, Conductivity of sample: $1.4 \pm$

0.1 mS/cm, OD600, OD280, Flow
through Collection based on
UV280: 100 mAU to 100 mAU,
Elution criteria: 200 mAU to 200
mAU, Loading capacity: ≤ 1.12 m:
of plasma/mL of resin
4.2.1.5 In-process controls
Step In process control
Thawing and Plasma
OD600, pH, conductivity
Pooling & filtration
Chromatography 1F NA
Chromatography 2F pH, conductivity, OD280, OD600
Concentration and
diafiltration of 2F flow pH, conductivity, OD280, OD600
through
EUGLOBULIN
Conductivity, Dilution pH, OD600
precipitation
Chromatography 4F OD280, OD600

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4.2.1.6 Hold times

Refer to Annexure-7

4.2.1.7 Scale-up considerations

Refer to Annexure-8

4.2.2 Albumin

4.2.2.1 Objective

The objective of this operation is to manufacture/purify the
Albumin by various purification methods including
chromatography and TFF.

4.2.2.2 Scientific principle

Albumin is a concentrated human plasma protein solution that
works on the principle of colloid osmotic pressure (oncotic
pressure). It draws water from the interstitial and intracellular
spaces into the vascular compartment, thereby expanding plasma
volume and stabilizing circulation in critically ill patients.

4.2.2.3 Process parameters (CPPs)

Various process parameters (CPPs) for Albumin operation are as
below:

Step CPP

Chromatography 7A Flow rate, pH and conductivity,

Volume of buffer passed, Amount of

loaded protein, Elution collection

based on UV280

Concentration pH, TMP, NWP

Heat Treatment pH, Incubation temperature and time

Separation of

Separator speed, Outlet flow

supernatant

Filtration Pressure

Virus inactivation Concentration of Sodium Caprylate,

pH, Temperature and time

Depth Filtration Pressure

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Diafiltration and pH, TMP, NWP, cycles of

concentration diafiltration

Pasteurization Temperature and time

Filtration and pH Pressure, pH

adjustment

4.2.2.4 Operating ranges

Step Operating Ranges

Chromatography 7A HETP: <3 , Asymmetry: 0.8 to 1.8,

Flow rate: 1356 L/H, Sample pH:

4.50 ± 0.10 , Loading capacity: ≤ 60

mg protein/mL of resin, Elution:

>100 mAU to <400 mAU

Concentration Sample pH: 5.20 ± 0.10 , TMP: 0.5 to

1.0 bar, NWP: ≥ 120 LMH/Bar

Heat Treatment Sample pH: 5.20 ± 0.10 , Incubation

temperature: 57 ± 0.5 °C, Incubation

time: 6 hours

Separation of Separator speed: 9100 rpm, Outlet

supernatant flow: 200 LPH

Filtration Pressure: 1 ± 0.5 bar

Virus inactivation Concentration of Sodium Caprylate:

32mM, Sample pH: 4.50 ± 0.10 ,

Final albumin concentration: $100 \pm$

05 mg/mL, Incubation Temperature:

30 ± 2.0 °C and time: 10 Hours,

Temperature after incubation: 5 ± 3

°C

Depth Filtration Pressure: 1 ± 0.5 bar

Diafiltration and Sample pH: 7.00 ± 0.30 , TMP: 0.5 to

concentration 1.0 bar, NWP: ≥ 120 LMH/bar, cycles

of diafiltration: 3

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Pasteurization Temperature: 60 ± 1.0 °C and time:

5 ± 1 hours

Filtration and pH Pressure: <1.0 bar, pH: 6.80 ± 0.10

adjustment
 4.2.2.5 In-process controls
 Step In process controls
 Chromatography 7A pH, conductivity, OD280, OD600
 Concentration OD280, OD600, pH
 Heat Treatment OD600
 Separation of
 pH, conductivity, OD280, OD600
 supernatant
 Filtration OD600
 Virus inactivation pH, conductivity, OD280, OD600
 Depth Filtration pH, conductivity, OD600
 Diafiltration and
 OD280
 concentration
 Pasteurization OD600
 Filtration and pH
 pH, conductivity, OD280, OD600
 adjustment
 4.2.2.6 Hold times
 Refer to Annexure-7
 4.2.2.7 Scale-up considerations
 Refer to Annexure-8
 4.2.3 IVIG
 4.2.3.1 Objective
 The objective of this operation is to manufacture/purify the IVIG by various purification methods including chromatography and TFF.
 4.2.3.2 Scientific principle
 Intravenous immunoglobulin (IVIG) is a sterile, pooled human IgG preparation that provides passive immunity and

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immunomodulation. Its scientific principle rests on restoring protective IgG in immunodeficiency and modulating dysregulated immune pathways in autoimmune/inflammatory diseases, with quality, safety, and clinical use governed by regulatory guidance.
 4.2.3.3 Process parameters (CPPs)
 Various process parameters (CPPs) for Albumin operation are as below:
 Step CPPs
 Concentration TMP, NWP
 Chromatography 5I Flow rate, pH drop, sample pH, sample conductivity, Volume of buffer passed, Amount of loaded protein, Collection during loading based on UV280, Collection during washing, Loading capacity
 pH adjustment of flow
 pH, conductivity
 through
 Concentration of chromatography 5I flow TMP, NWP
 through
 Conductivity adjustment Conductivity
 Virus inactivation by Incubation pH, Incubation
 Low pH treatment Temperature and time
 Virus inactivation by
 Incubation Temperature: and time
 S/D treatment
 Filtration Pressure 1 ± 0.5 bar
 Chromatography 6I Flow rate, pH drop, Volume of buffer passed, Amount of loaded protein, Elution Collection based on UV280, Loading capacity

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Protein concentration
 Protein concentration
 adjustment
 Filtration Pressure 1 ± 0.5 bar
 Concentration and
 TMP, NWP, Diafiltration end point
 diafiltration
 Formulation Concentration of maltose
 4.2.3.4 Operating ranges
 Step Operating ranges
 Concentration TMP: ≤ 1.0 bar, NWP: ≥ 120 LMH/bar
 Chromatography 5I HETP: <3 , Asymmetry: 0.8 to 1.8,
 Flow rate: 2211 L/H, pH drop: 3.50 ± 0.50 , sample pH: 5.20 ± 0.05 , sample conductivity: 1200 – 1400 $\mu\text{S}/\text{cm}$,
 Volume of buffer passed, Amount of loaded protein, Collection during loading based on UV280: > 50 mAU to till completion of loading,
 Collection during washing: From loading end to UV280 50-100 mAU,
 Loading capacity: ≤ 35 mg of protein/mL of resin

pH adjustment of flow
pH: 4.00 ± 0.10
through
Concentration of
TMP: 0.5 to 1.0 bar, NWP: ≥ 120
chromatography 5l flow
LMH/bar
through
Conductivity adjustment Conductivity: 2.0 ± 0.2 mS/cm
Virus inactivation by Incubation pH: 4.00 ± 0.10 ,
Low pH treatment Incubation Temperature: $37.0 \pm 0.5^\circ\text{C}$ and time: 12 Hours

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Virus inactivation by Incubation Temperature: $30.0 \pm 3^\circ\text{C}$
S/D treatment and time: 6 Hours
Filtration Pressure 1 ± 0.5 bar , OD600: 1.0 to 1.5
Chromatography 6l HETP: <3 , Asymmetry: 0.8 to 1.8,
Flow rate: 667 L/H, pH drop: 4.50 ± 0.10 , Volume of buffer passed,
Amount of loaded protein, Elution
Collection based on UV280: UV rises to 50-100 mAU, Loading capacity: ≤ 80 mg of protein/mL of resin
Protein concentration
Protein concentration: 8-11 mg/mL
adjustment
Filtration Pressure 1 ± 0.5 bar
Concentration and TMP: 0.5 to 1.0 bar, NWP: ≥ 120
diafiltration LMH/bar, Diafiltration end point:
Till permeate conductivity reaches 900-1300 $\mu\text{S/cm}$
Formulation Concentration of maltose: 10% w/v
4.2.3.5 In-process controls
Step In process controls
Concentration OD280, OD600
Chromatography 5l OD280, OD600, pH, conductivity
pH adjustment of flow
pH, conductivity
through
Concentration of
chromatography 5l flow OD280
through
Conductivity adjustment OD280, OD600, pH and conductivity
Virus inactivation by
OD280
Low pH treatment

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Virus inactivation by
OD600
S/D treatment
Filtration OD280, OD600
Chromatography 6l OD280, OD600, pH, conductivity
Protein concentration
Protein concentration
adjustment
Filtration OD280, OD600
Concentration and
OD280, OD600, pH, Conductivity
diafiltration
Formulation Concentration of maltose, Protein
concentration, pH, conductivity
4.2.3.6 Hold times
Refer to Annexure-7
4.2.3.7 Scale-up considerations
Refer to Annexure-8
4.2.4 Fibrinogen
4.2.4.1 Objective
The objective of Fibrinogen operation is to manufacture/purify the Fibrinogen by various purification methods including chromatography and TFF.
4.2.4.2 Scientific principle
The scientific principle of fibrinogen is its role as the soluble precursor of fibrin, enabling rapid conversion into a stable clot under thrombin's action, thereby ensuring hemostasis and wound repair.
4.2.4.3 Process parameters (CPPs)
Step CPP
Precipitation Conductivity, pH, Incubation time
Separation using high Separator speed, pH, Outlet flow rate, speed separator Incubation time
Filtration Pressure, 1 ± 0.5 bar

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Concentration and
TMP, NWP
diafiltration
Viral inactivation by S/D Incubation time: $23 \pm 1^\circ\text{C}$ for 4 treatment hours
Chromatography 11FB Flow rate, pH drop, Volume of buffer

passed, Elution Collection based on
 UV280, Loading capacity
 Concentration of 11FB
 TMP, NWP
 eluate
 Addition of stabilizer pH
 4.2.4.4 Operating ranges
 Step Operating ranges
 Precipitation Conductivity: 100 ± 10 mS/cm, pH:
 4.00 ± 0.2, Incubation time: 3 Hours
 Separation using high Separator speed: 9100 rpm, pH: 7.5 ±
 speed separator 0.1, Outlet flow rate: NMT 120 LPH,
 Incubation time: 1 Hour
 Filtration Pressure 1 ± 0.5 bar
 Concentration and TMP: 0.5 to 1.0 bar, NWP: ≥ 120
 diafiltration LMH/bar, Diafiltration volume: 5
 volumes
 Viral inactivation by S/D
 Incubation time: 4 Hours
 treatment
 Chromatography 11FB HETP: <3, Asymmetry: 0.8 to 1.8,
 Flow rate: 603 L/H, pH drop: 2.0,
 Elution Collection based on UV280:
 0.05 AU to 0.05 AU, Loading
 capacity: NMT 1 mg of protein/mL
 of resin
 Concentration of 11FB TMP: 0.5 to 1.0 bar, NWP: ≥ 120
 eluate LMH/bar

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Addition of stabilizer pH: 7.2
 4.2.4.5 In-process controls
 Step In process controls
 Precipitation OD280, OD600
 Separation using high
 NA
 speed separator
 Filtration pH, Conductivity, OD280, OD600
 Concentration and
 pH, conductivity, OD280, OD600
 diafiltration

Chromatography 11FB pH, conductivity, OD280, OD600
 Concentration of 11FB
 BET, pH, conductivity, OD280, OD600
 eluate
 Addition of stabilizer pH, Conductivity, OD280, OD600
 4.2.4.6 Hold times
 Refer to Annexure-7
 4.2.4.7 Scale up considerations
 Refer to Annexure-8
 4.2.5 Factor VIII
 4.2.5.1 Objective
 The objective of Factor VIII operation is to manufacture/purify
 the Factor VIII by various purification methods including
 chromatography and TFF
 4.2.5.2 Scientific principle
 • Structure: FVIII is a ~300 kDa glycoprotein composed of
 domains A1, A2, B, A3, C1, and C2.
 • Function: Acts as a cofactor for Factor IXa in the intrinsic
 pathway of coagulation, forming a complex that activates
 Factor X, leading to thrombin generation and fibrin clot
 formation.

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• Association: Circulates bound to von Wille brand Factor
 (vWF), which protects FVIII from premature degradation.
 4.2.5.3 Process parameters (CPPs)
 Step CPP
 Load preparation Pressure
 Chromatography 12E Flow rate, pH drop, Elution
 Collection based on UV280
 Viral inactivation by S/D
 Incubation time, conductivity
 treatment
 Chromatography 13E Flow rate, Elution Collection based
 on UV280, Loading capacity
 Concentration of
 Chromatography 13E TMP, NW
 eluate
 Stabilizer addition pH
 Concentration TMP, NWP
 Formulation Pressure during filtration 1 ± 0.5 bar
 4.2.5.4 Operating ranges
 Step Operating range
 Load preparation Pressure: 0.5 bar to 1.0 bar
 Chromatography 12E HETP: <2, Asymmetry: 0.8 to 1.8,
 Flow rate: 785 L/H, pH drop: 3.50 ±
 0.5, Elution Collection based on
 UV280: 100 mAU to 100 mAU
 Viral inactivation by S/D Incubation time: 4 Hours,
 treatment Conductivity: 14-17 mS/cm

Chromatography 13E HETP: <3, Asymmetry: 0.8 to 1.8,
Flow rate: 79 L/H, Elution Collection
based on UV280: 10 mAU to 10 mAU,
Loading capacity: NMT 176 mL of
plasma/mL of resin

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Concentration of TMP: 0.5 to 1.0 bar, NWP: ≥ 120

Chromatography 13E LMH/bar, Activity: >50 IU/mL
eluate

Stabilizer addition pH: 6.70 - 6.80

Concentration TMP: 0.5 to 1.0 bar, NWP: ≥ 120

LMH/bar, Activity: >40 IU/mL

Formulation Pressure during filtration 1 ± 0.5 bar

4.2.5.5 In-process controls

Step In process control

Load preparation pH, conductivity, OD280, OD600,
visual appearance

Chromatography 12E NA

Viral inactivation by S/D

Conductivity, pH

treatment

Chromatography 13E NA

Concentration of

Chromatography 13E pH, conductivity, OD280, OD600
eluate

Stabilizer addition OD280, OD600, pH

Concentration pH, conductivity, OD280, OD600

Formulation pH, conductivity, OD280, OD600

4.2.5.6 Hold times

Refer to Annexure-7

4.2.5.7 Scale up considerations

Refer to Annexure-8

4.2.6 Factor IX

4.2.6.1 Objective

The objective of Factor IX operation is to manufacture/purify the
factor IX by various purification methods including
chromatography and TFF.

4.2.6.2 Scientific principle

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Factor IX (FIX) is a vitamin K-dependent coagulation protein
essential for thrombin generation via the intrinsic pathway. The
product is manufactured using validated fractionation and
purification steps, with viral safety ensured through orthogonal
inactivation/removal methods.

Potency is expressed in IU based on WHO standards, and the
product is characterized for identity, purity, and activity using
validated analytical methods. FIX is indicated for the treatment
and prophylaxis of bleeding episodes in patients with
hemophilia B.

4.2.6.3 Process parameters (CPPs)

Step CPP

Concentration Conductivity after dilution, TMP,

Feed pressure, NWP

Filtration Pressure 1 ± 0.5 bar

Virus inactivation by

Incubation temperature

S/D treatment

Chromatography 8N Flow rate, pH drop, sample pH,

Elution Collection based on UV280,

Loading capacity

Concentration of 8N

TMP, NWP

eluate

Chromatography 9N Flow rate, Peak 3 Collection based on

UV280, Loading capacity

Concentration of 9N

TMP, NWP

eluate

Pre filtration and Nano

Pressure: 1 ± 0.5 bar

filtration

Sterile filtration Pressure: 1 ± 0.5 bar

4.2.6.4 Operating ranges

Step Operating range

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Concentration Conductivity after dilution: 10 - 11

mS/cm, TMP: 0.5 to 1.0 bar, Feed

pressure: Maximum 2.0 bar, NWP: $\geq$

120 LMH/Bar

Filtration Pressure 1 ± 0.5 bar

Virus inactivation by

Incubation temperature: 25 ± 1 °C

S/D treatment

Chromatography 8N HETP: <3, Asymmetry: 0.8 to 1.8,

Flow rate: 71 L/H, pH drop: $6.50 \pm$

0.20, sample pH: 7.40 ± 0.10 , Elution

Collection based on UV280: 10 mAU

to 10 mAU, Loading capacity: ≤ 90

mL of plasma/mL of resin

Concentration of 8N TMP: 0.5 to 1.0 bar, TMP during
eluate concentration: 0.2 bar, NWP: ≥ 120
LMH/Bar
Chromatography 9N HETP: <1.5 , Asymmetry: 1.2 to 1.5,
Flow rate: 18 L/H, Peak 3 Collection
based on UV280: 10 mAU to 10 mAU,
Loading capacity: $\leq 2.0\%$, pH of
eluate: 7.4 ± 0.2 , conductivity of
eluate: 17.0 ± 1 mS/cm
Concentration of 9N TMP: 0.5 to 1.0 bar, NWP: ≥ 120
eluate LMH/Bar, protein concentration,
activity: >125 IU/mL
Pre filtration and Nano
Pressure 1 ± 0.5 bar
filtration
Sterile filtration Pressure 1 ± 0.5 bar
4.2.6.5 In-process controls
Step In process controls

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Concentration Conductivity after dilution, protein
concentration
Filtration pH, conductivity, OD280, OD600
Chromatography 8N pH, conductivity, OD280, OD600
Concentration of 8N
OD280, OD600
eluate
Chromatography 9N pH of eluate, conductivity of eluate,
OD280
Concentration of 9N Protein concentration, activity,
eluate OD280, OD600
Pre filtration and Nano
OD280, OD600
filtration
Sterile filtration OD280, OD600
4.2.6.6 Hold times
Refer to Annexure-7
4.2.6.7 Scale-up considerations
Refer to Annexure-8
4.2.7 Thrombin
4.2.7.1 Objective
The objective of Thrombin operation is to manufacture/purify the
Thrombin by various purification methods including
chromatography and TFF.
4.2.7.2 Scientific principle
The scientific principle of thrombin is based on enzyme-
mediated proteolysis and substrate-specific cleavage, enabling
the transformation of soluble fibrinogen into insoluble fibrin,
thereby ensuring controlled blood clot formation and vascular
integrity.

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4.2.7.3 Process parameters (CPPs)
Step CPP
Activation pH, conductivity, incubation time,
Incubation temperature
Load preparation Conductivity of sample, pH
Flow rate, pH drop, Elution
Chromatography 10T Collection based on UV280, Loading
capacity
Concentration and
diafiltration of 10T TMP, NWP
eluate
Filtration Pressure 1 ± 0.5 bar
4.2.7.4 Operating ranges
Step Operating range
Activation incubation time: 5 hours $\pm$ 10 minutes
, Incubation temperature: 20-25 oC
Load preparation Conductivity of sample: 1.9 - 2.4
mS/cm
Chromatography 10T HETP: <3 , Asymmetry: 0.8 to 1.8,
Flow rate: 85 L/H, pH drop: 7.5 ± 0.5 ,
Elution Collection based on UV280:
100 mAU to 100 mAU, Loading
capacity: equivalent to 300 L plasma
(NMT 65 mL of plasma/mL of resin)
Concentration and TMP: 0.5 to 1.0 bar, NWP: ≥ 120
diafiltration of 10T LMH/Bar, Diafiltration: 5 cycles
eluate
Filtration Pressure 1 ± 0.5 bar

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4.2.7.5 In-process controls
Step In process controls
Activation pH, conductivity, visual appearance,
OD280, OD600
Load preparation Conductivity of sample, OD280,
OD600, pH
Chromatography 10T pH, conductivity, OD280, OD600
Concentration and
diafiltration of 10T BET, activity
eluate

Filtration OD280, OD600, pH, conductivity, activity

4.2.7.6 Hold times

Refer to Annexure-7

4.2.7.7 Scale-up considerations

Refer to Annexure-8

5.0 Viral Safety Management

5.1 Virus Inactivation / Removal Steps

Human blood is the source of a wide range of medicinal products used for the prevention and treatment of a variety of often life-threatening injuries and diseases. Despite measures such as donor selection, testing of donations and of plasma pools, the transmission of blood-borne viruses by plasma and purified plasma products is still considered to constitute a risk to patients. Viral inactivation methods should be applied to all blood plasma-derived protein solutions. Viral inactivation of plasma is a critical step in ensuring the safety of plasma-derived medicinal products. The objective of viral inactivation is to effectively inactivate or remove enveloped and non-enveloped viruses that may be present in plasma, without adversely affecting the quality or biological activity of plasma proteins.

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5.2 Methods for Inactivation

5.2.1 Solvent/Detergent (S/D) treatment

One commonly employed method is Solvent/Detergent (S/D) treatment, which is highly effective against enveloped viruses such as HIV, HBV, and HCV. In this method, plasma or plasma pools are treated with a validated combination of a solvent (e.g., tri-n-butyl phosphate) and a detergent (e.g., Triton X-100 or polysorbate), under controlled temperature and time conditions. The S/D treatment disrupts the lipid envelope of viruses, rendering them non-infectious. Following treatment, the solvent and detergent are removed by validated downstream purification steps. This method is widely accepted by WHO and regulatory authorities due to its robustness and reproducibility.

Organic solvent/detergent mixtures disrupt the lipid membrane of enveloped viruses. Once disrupted, the virus can no longer bind to and infect cells. Non-enveloped viruses are not inactivated.

5.2.2 Pasteurization (Heat treatment)

Another viral inactivation approach is pasteurization (heat treatment), typically involving heating plasma protein solutions at 60 °C for a defined duration, in the presence of stabilizers to protect protein integrity. Pasteurization is effective against a broad range of viruses, including some non-enveloped viruses, and is commonly used for products such as human serum albumin. Process parameters such as temperature, time, and stabilizer concentration are carefully validated to ensure consistent viral inactivation while maintaining product quality.

5.2.3 Low pH treatment

Most proteins are damaged by exposure to the acidic conditions needed to kill viruses. For example, few viruses are killed at pH 5.0 – 5.5, a condition known to inactivate factor VIII. Immune globulin solutions are an exception.

Low pH treatment often combined with incubation at elevated temperatures, is also used as a viral inactivation step, particularly during immunoglobulin manufacturing. Exposure of plasma protein

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intermediates to acidic conditions for a defined period contributes to the inactivation of enveloped viruses and is validated as part of the overall viral safety profile of the manufacturing process.

5.3 Viral removal techniques

In addition to inactivation methods, viral removal techniques such as Nano filtration are employed to physically remove viruses based on size exclusion. Nano filtration membranes with defined pore sizes can remove both enveloped and non-enveloped viruses and are considered a highly effective and complementary viral safety measure.

All viral inactivation and removal steps are validated through virus clearance studies, using relevant model viruses in accordance with WHO and ICH Q5A guidelines. These studies demonstrate the cumulative viral reduction capacity of the manufacturing process. The effectiveness of each step is monitored through defined critical process parameters, and changes to the process are subject to formal change control.

Overall, the application of validated viral inactivation and removal methods ensures that plasma-derived medicinal products meet stringent safety standards and comply with national and international regulatory expectations.

6.0 Equipment & Facility Requirements

6.1 Equipment List & Specifications

S/No. Name of Equipment

- 1 Chromatography AKTA-1 (471 L/hr (Pipe MOC SS)
- 2 Chromatography AKTA-1 (500 L/hr pipe MOC SS)
- 3 Chromatography AKTA-1 (785 L/hr pipe MOC PP)
- 4 Chromatography AKTA-1 (1178 L/hr pipe MOC PP)
- 5 Chromatography AKTA-1 (864 L/hr pipe MOC PP)
- 6 Chromatography AKTA-1 (667 L/hr pipe MOC PP)
- 7 Chromatography AKTA-1 (785 L/hr pipe MOC PP)
- 8 Chromatography AKTA-1 (30 L/hr pipe MOC PP)

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- 9 Chromatography AKTA-1(160 L/hr pipe MOC PP)
- 10 Chromatography Column Dia 300 mm x Height 500 mm
- 11 Chromatography Column Dia 300 mm x Height 750 mm
- 12 Chromatography Column Dia 600 mm x Height 500 mm

13 Chromatography Column Dia 1000 mm x Height 500 mm
 14 Chromatography Column Dia 450 mm x Height 500 mm
 15 TFF System 40 sq. m
 16 TFF System 30 sq. m.
 17 TFF System 20 sq. m
 18 TFF System 5 sq. m
 19 Continuous Centrifuge 1000 L/hr
 20 Process vessel 150 L
 21 Process vessel 250 L
 22 Process vessel (Mobile tank) 300 L
 23 Process vessel 1500 L
 24 Process vessel 2000 L
 25 Formulation Vessel 100 L
 26 Lyophilizer (2 sq. m)

6.2 Facility Zoning & HVAC Concept

Air supply for manufacturing area will be filtered through bacteria retaining filters (HEPA Filters) and will be at a pressure higher than in the adjacent areas unless otherwise required.

HVAC zoning of the facility (control areas) for the manufacturing operations is done based on the operation and its criticality (Plasma collection, fractionation, purification, and storage) to minimize the risk of contamination and to maintain physical conditions like temperature and humidity. Cleanroom classification and pressure cascade are critical design elements in a plasma fractionation facility to ensure protection of product quality, prevent cross-contamination, and maintain compliance with requirements for plasma-derived medicinal products. Routine microbial and particulate counts of the manufacturing area will be carried out during manufacturing operations to assess the effectiveness of HVAC.

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The results of particulate matter and microbes to be continuously monitored against documented in-house standards and regulatory guidelines. The classification of areas is based on the nature of operations, risk of contamination, and whether the process is open or closed.

6.2.1 Cleanroom Classification

Plasma fractionation facilities typically operate using a combination of controlled but unclassified areas and classified cleanrooms, as most fractionation steps are conducted in closed systems. However, classification is applied wherever there is potential product exposure.

- Plasma receiving, frozen plasma storage, and raw material quarantine areas are generally maintained as controlled, non-classified areas with defined environmental controls, as plasma is frozen and fully contained.
- Plasma thawing, pooling, and open handling areas are normally classified as Grade C (ISO Class 7) cleanrooms. These areas involve handling of plasma before viral inactivation and therefore require controlled particulate and microbiological conditions.
- Fractionation and purification areas operating in closed systems are typically maintained as Grade C, depending on risk assessment. Where open manipulations occur, a minimum of Grade D classification is maintained.
- Viral inactivation and viral removal processing areas, including solvent/detergent treatment and Nano filtration preparation zones, are generally maintained at Grade C or higher, based on process exposure.
- Final sterile filtration and aseptic filling operations (if applicable) are conducted in Grade A environments with a Grade B background.
- Support areas, such as equipment washing, preparation rooms, and utilities, are maintained as Grade D, as appropriate.

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6.2.2 Pressure Cascade

A well-defined pressure cascade is maintained to ensure air flows from cleaner areas to less clean areas, thereby preventing ingress of contaminants into critical processing zones.

- Highest positive pressure is maintained in Grade A areas, followed by Grade B, Grade C, and Grade D areas.
- Grade D areas are maintained at a higher pressure relative to adjacent unclassified or controlled areas.
- Raw material receipt, plasma receiving, and waste handling areas are maintained at lower pressure or neutral pressure compared to processing areas.
- Waste corridors and rejected material areas are often maintained at negative pressure relative to clean areas to prevent contamination spread.
- Typical pressure differentials of 10–15 Pa are maintained between adjacent areas of different classifications, with continuous monitoring and alarm systems in place.

6.2.3 Material and Personnel flow

In a GMP-compliant plasma fractionation facility, material flow and personnel flow are designed to be unidirectional, segregated, and controlled to prevent cross-contamination, mix-ups, and microbial contamination.

Material Flow

Material flow within the plasma fractionation facility follows a one-way movement from receipt to dispatch, with clear segregation between quarantine, approved, in-process, and rejected materials. Human plasma is received in a designated plasma receiving area, where it is inspected, logged, and placed under quarantine in controlled frozen storage. Only plasma that has been released by Quality Assurance is transferred to the approved plasma storage area.

From approved storage, plasma is moved in a controlled manner to the

thawing and pooling area, followed by sequential transfer through

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fractionation, purification, and viral inactivation/removal steps. Intermediate products move progressively through defined processing zones of increasing cleanliness, with no back-tracking permitted. Rejected materials, waste, and by-products are removed through dedicated waste corridors or exit points, ensuring they do not intersect with routes used for approved materials. Finished products are transferred to designated finished goods storage areas prior to dispatch, maintaining complete traceability throughout the process.

Personnel Flow

Personnel flow is designed to minimize contamination risks and is strictly segregated from material flow wherever possible. Entry into production areas is permitted only to authorized, trained personnel. Staff enter through controlled access points and follow a defined gowning sequence, progressing from general areas to classified processing zones through airlocks and change rooms.

Personnel movement is unidirectional, from lower to higher cleanliness areas, and reverse movement is restricted or requires full de-gowning and re-gowning. Separate entry and exit routes are provided for personnel working in plasma handling, fractionation, filling, and support areas to prevent cross-contamination. Movement between different production areas is controlled by SOPs and access restrictions.

Dedicated personnel flows are established for waste handling, maintenance, and cleaning activities, ensuring these functions do not compromise GMP areas. Training programs ensure that all personnel understand and comply with defined movement patterns and hygiene requirements.

6.3 Utilities Requirements

6.3.1 Water Supply

An adequate supply of suitable pyrogen-free water shall be provided for use during the fractionation process and for the reconstitution and/or dilution of the plasma fractions before filling and freeze-drying.

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The two most commonly used types of water are Purified water and Water for injection. Water preparation and delivery systems should be tested at regular intervals for microbes, endotoxin content and conductivity. The water should be a continuously in circulation mode and should have no dead legs.

Water for injections is used for the preparation of final formulations.

6.3.2 Clean steam

An adequate supply of clean steam shall be provided for the operation of sterilizing and cleaning equipment. The steam shall meet the quality of water for injections.

6.3.3 Refrigeration (–30 °C to +5 °C)

Refrigeration is required for:

- Storing various source materials and plasma fractions.
- Plasma for fractionation should be stored at –20 °C or below.
- Keeping the various fractionation areas at the correct temperature.
- Keeping the process equipment at the correct temperature.
- Storing final products and /or under test materials at appropriate storage conditions.
- Storing final products awaiting dispatch at recommended storage conditions.

7.0 Process Validation Strategy

7.1 Validation Master Plan Alignment

Process Validation is the most important under cGMP requirements. The process validation master plan is intended to assist manufacturers in understanding quality management system (QMS) requirements as applicable to process validation and has general applicability to manufacturing process.

Process validation can be defined as documented evidence that the process, operated within established parameters, can perform effectively and reproducibly to produce a medical product meeting its predetermined specifications and quality attributes. Process validation can include the collection and evaluation of data, from the process design stage throughout production, that establish scientifically that a process is capable of consistently delivering a quality product.

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Continuous process verification (CPV) has been introduced to cover an alternative approach to process validation based on a continuous monitoring of manufacturing performance. This approach is based on the knowledge from product and process development studies and / or previous manufacturing experience. CPV may be applicable to both a traditional and enhanced approach to pharmaceutical development. It may use extensive in-line, on-line or at-line monitoring and / or controls to evaluate process performance. Process validation should confirm that the control strategy is sufficient to support the process design and quality of the product. The validation should cover all manufactured strengths and all manufacturing sites used for production of the marketed product.

7.2 PPQ Batches (Process performance qualification)

- Number of batches: 03
- Acceptance criteria as per the laid specifications.

8.0 Analytical & Quality Control Transfer

8.1 Specifications (per fraction)

In Process Quality Testing (IPQC)

S/No Process Volume for

Step QC test Range

. stage test

FVIII

NA

Pooled plasma Quantification
30 mL
1 PF after filtration FIX
NA
from each lot Quantification
Protein NA
Manufacturing FVIII
2 PF NA 65 mL
pooled plasma Quantification
Before the Eu
globulin pH & pH:-5.2 ± 0.1 &
pH &
3 PF Conductivity Cond:-1.0 ± 0.1 20 mL
Conductivity
adjusted mS/cm
sample

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The pooled
4 FIX peak 3 sample Activity 10 to 20 IU/mL 7.5 mL
from 9N
Concentrated
Not less than 125
5 FIX sterile filtrate Activity 10 mL
IU/mL
sample
Concentrated > 180 IU/mL &
6 Thrombin Activity & BET 2 mL
sample EU/ mL (Limits)
Concentrated
> 180 IU/mL &
7 Thrombin sterile filtrate Activity & BET 10 mL
EU/ mL (Limits)
DS sample
11FB Eluate Protein Conc. > 30 mg/mL &
8 Fibrinogen 2 mL
sample & BET EU/mL (Limits)
12 E load PF pH:- 7.0 ± 0.1 &
pH &
9 Factor VIII Fraction 1 Cond:- 17 ± 0.1 10 mL
Conductivity
sample mS/cm
13 E Load
S/D treated pH:-7.0 ± 0.1 &
pH &
10 Factor VIII sample Cond:-15 ± 2.0 10 mL
Conductivity
Conductivity mS/cm
Adjusted
pH:- 6.7 ± 0.1 &
pH ,
14 E Pooled Cond:- 17.0 ± 0.1
11 Factor VIII Conductivity & 12 mL
elution mS/cm & < 0.03
Activity
IU /mL
5I FT
concentrated pH:- 4.0 ± 0.1 &
pH ,
before the low Cond:- 2.0 ± 0.1
12 IVIG Conductivity & 25 mL
pH sample; mS/cm & 50
Protein Conc.
Conductivity mg/mL
Adjusted

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Bulk-API pH & Protein pH:- 4.0 ± 0.1 &
13 IVIG 25 mL
analysis Conc. 50 mg/mL
pH:- 4.0 ± 0.1 &
As per the Cond:-0.9 ± 0.1
14 IVIG DS analysis 25 mL
guidelines mS/cm &
50 mg/mL
Before low pH
pH & Protein pH:- 4.5 ± 0.1 &
15 Albumin incubation pH 15 mL
Conc. 120 mg/mL
adjustment
Bulk-API
analysis pH:- 7.0 ± 0.3 &
pH & Protein
16 Albumin sample after 220 ± 5 mg/mL 25 mL
Conc.
sterile 7.0 mS/cm
filtration
Drug product
Human Albumin 20%
S/No Tests Acceptance limits
.
Clear slightly viscous and greenish – yellow or

1 Description

amber to green in colour.

The main component of the preparation to be

2 Identification by SDS-PAGE examined corresponds to the albumin component of normal human serum.

pH (10 g/L solution in 9 g/L

3 6.7 - 7.3

NaCl)

Haem content by Absorption at

4 A403: < 0.15

403 nm

5 Molecular Size Distribution < 5% (w/v) aggregate protein

6 Prekallikrein Activator < 35 IU/mL

7 Total Protein 19 - 21 g/100 mL

8 Bacterial Endotoxin < 1.3 IU of Endotoxin per millilitre of solution

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9 Bioburden Nil CFU/10 mL

10 N -acetyl tryptophan Not more than 0.08 mmol/g of Protein

11 Sodium Caprylate Not less than 0.08 mmol/g of Protein

Human Coagulation Factor IX

Clear or slightly opalescent and colourless or

1 Description

slightly yellow solution.

It complies with the limits of assay.

2 Identification by Assay

Not less than 125 IU/mL.

3 pH (Neat sample) 6.5 – 7.5

4 Osmolality (Neat sample) > 240 mOsmol/kg

5 Assay of Human Coagulation

Not less than 125 IU/mL

Factor IX

6 Specific Activity > 50 IU of F IX/mg of Total Protein

7 Bioburden Nil CFU / 2 mL

8 Bacterial Endotoxins < 0.03 IU of Endotoxin/ IU of factor IX

9 Activity of Factor II < 5 % of the activity of Factor IX

10 Activity of Factor VII < 5 % of the activity of Factor IX

11 Activity of Factor X < 5 % of the activity of Factor IX

Dried Human Ant Haemophilic Fraction 250 IU

(Human Coagulation Factor VIII)

Clear or slightly opalescent, colourless or slightly

1 Description

yellow solution.

2 Identification by Assay Complies with the limits of assay.

3 pH 6.5 – 7.5

4 Osmolality > 240 mOsmol/kg

5 Assay of Human Coagulation

> 30 IU/mL

Factor VIII

6 Bacterial Endotoxins < 0.03 IU of Endotoxin per IU of Factor VIII

7 Bioburden NIL CFU/10 mL

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Human Normal Immunoglobulin for Intravenous use 5%

The liquid preparation should be clear or Slightly

1 Description

opalescent and colourless or pale yellow.

4.0 to 7.4 with the reference to a dilution of the

pH (1% test solution in 0.9% w/v

2 preparation to be examined containing 1.0 per cent

NaCl solution)

immunoglobulin.

The main component of the preparation to be

3 Identification by SDS-PAGE examined corresponds to the IgG component of Normal Human Serum.

4 Osmolality Not less than 240 mosmol/kg

4.5 % - 5.5 %

5 Total protein by Biuret method Minimum 3% w/v and between 90 % to 110 % of the quantity of protein stated on the label.

For each of the dilution of the

1:10 dilution

Thrombogenic (procoagulant) test sample the coagulation

6

activity time should be not less than

1:100 dilution

150 Seconds.

Antibodies to Viral Antigens:

Test for one viral and one > 3 times to that of the initial plasma pool.

bacterial antibodies Antibodies to Hepatitis B surface antigen.

7

(At an Immunoglobulin Antibodies to Bacterial Antigens:

Concentration of 50 g/L) > 3 times to that of the initial plasma pool.

Antibodies to Diphtheria Toxoid antigen.

Distribution of Immunoglobulin The distribution of IgG subclasses is similar to that

8

G subclasses of Normal Human Plasma.

The Fc function for the preparation under the

Test for Fc function of

9 examination. Its value is not less than that stated in Immunoglobulin the leaflet accompanying the reference preparation.
a) Relative Retention (RRT): For monomer and dimer, the relative retention to the corresponding

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peak to the chromatogram obtained with the reference solution is 1 ± 0.02 .
b) The sum of peak areas of monomer and dimer represent not less than 90.0 per cent of the total area
Molecular size by liquid of the chromatogram and the sum of the peak area chromatography of polymers and aggregates represents not more than 3.0 per cent of the total area of the chromatogram. Monomer & Dimer: > 90 %
Polymers & Aggregates: < 3 %
Maximum 35 IU per mL calculated with the Prekallikrein activator
11 reference to a dilution of the preparation to be (3% Test solution) examined containing 3.0 per cent immunoglobulin.

12 Bioburden Nil CFU/10 mL

13 Bacterial Endotoxin Less than 0.5 IU of Endotoxin per mL of solution.

Component-1 (Human Fibrinogen Concentrate) of Fibrin Sealant Kit (1 mL)

1 Description Almost colourless, clear or slightly turbid solution.

2 pH 6.5 – 8.0

Fibrinogen Assay (Clottable
3 > 30 mg/mL of Clottable Protein.
Protein)
4 Bioburden Nil CFU/2.0 mL

Component 2 (Thrombin 500 IU) of Fibrin Sealant Kit 1.0 mL
Almost colourless to pale yellow, clear or slightly

1 Description turbid solution.

2 pH 5.0 – 8.0

3 Thrombin Assay (Identification) > 180 IU/mL

4 Bioburden NIL CFU/2.0 mL

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8.2 Analytical Methods

Analytical methods play a critical role in ensuring the quality, safety, and efficacy of plasma-derived medicinal products at all stages of manufacturing.

8.2.1 Identity assays

Identity assays are performed to confirm the identity of plasma protein or intermediate at various fractionation, purification, or formulation stages. These assays are primarily based on immunochemical and physicochemical principles.

Commonly used methods include:

Immunochemical Methods: which rely on specific antigen–antibody reactions to confirm protein identity.

- Immuno turbidimetry/ Nephelometry;
- ELISA;
- Radial immunodiffusion.

Electrophoretic techniques: applied to verify molecular weight, protein profile, and homogeneity.

- SDS-PAGE/ Native PAGE;
- Capillary electrophoresis.

Chromatographic techniques: provide confirmatory evidence of identity through characteristic retention times and spectral properties when compared with reference standards.

Size-exclusion or reverse-phase HPLC, together with UV absorbance at 280nm.

8.2.2 Potency assays

Potency assays are designed to demonstrate the biological activity of plasma-derived products. These assays are critical for dose determination and batch release. For immunoglobulin preparations, potency is typically assessed using antigen-specific ELISA or other immunological assays that quantify antibody activity against defined antigens, often calibrated against WHO International Standards. For coagulation factors such as Factor VIII, Factor IX, fibrinogen, and thrombin, functional clotting

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assays (one-stage clotting assays, chromogenic assays, or the Clauss method for fibrinogen) are employed to measure their ability to participate effectively in the coagulation cascade. Albumin potency is generally demonstrated through physicochemical characteristics and binding capacity studies, supported by compliance with specifications rather than a direct biological assay.

8.2.3 Purity & Impurity testing methods

Purity and impurity testing is done to ensure that the final product consists predominantly of the desired protein and that contaminants arising from plasma, processing steps, or degradation are adequately controlled.

Purity is commonly evaluated using:

- SDS-PAGE;
- Capillary electrophoresis;
- Chromatographic techniques such as ion-exchange and size-exclusion HPLC,

Product-related impurities:

- Aggregates & fragments – SEC-HPLC
- Residual solvents (e.g., ethanol): Gas chromatography (GC)
- Process-related impurities: Precipitating agents, Residual caprylate / PEG

- Host plasma impurities: Prekallikrein activator (PKA),

Complement activation (C3a, C5a)

- Endotoxin: Bacterial Endotoxin Test (LAL)

8.2.4 Viral markers

Viral marker testing forms the cornerstone of viral safety in plasma fractionation and is applied to source plasma, plasma pools, intermediates, and, where required, final products. Source plasma is screened using validated serological assays for markers such as HBsAg, anti-HIV-1/2, anti-HCV, and syphilis, with anti-HBc testing applied according to regulatory requirements. In parallel, nucleic acid testing (NAT) is performed for HIV-1 RNA, HCV RNA, and HBV DNA to

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significantly reduce the window period for viral detection. Additional NAT testing for Parvovirus B19 DNA and, where applicable, HAV RNA is carried out on plasma pools to ensure compliance with regulatory limits. These routine tests are complemented by viral clearance studies, in which dedicated viral inactivation and removal steps (such as solvent/detergent treatment, pasteurization, low pH incubation, or nano filtration) are validated using model viruses to demonstrate robust and reproducible viral safety of the plasma-derived medicinal products.

9.0 Stability & Shelf-Life Data

Stability studies should be performed, taking into account ICH guidelines, especially "Quality of Biotechnological Products: Stability Testing of Biotechnological/Biological products" (Q5C), WHO Technical Reports, EMA/FDA guidelines, and national regulations.

The purpose of stability testing is to provide evidence that the quality of a drug substance or drug product varies with time under the influence of a variety of environmental and physical factors such as temperature, humidity, light, or agitation. Stability testing establishes and confirms a re-test period or shelf life for the drug substance & drug product under the recommended storage conditions. Shelf life is also referred to as dating period or expiry period in some regions.

9.1 Stability Study Design

The plasma-derived products need precisely defined storage temperatures, the storage conditions for the real-time/real-temperature stability studies may be confined to the proposed storage temperature.

Accelerated and Stress Conditions

The expiration dating should be based on real-time/real- temperature data. However, it is strongly suggested that studies be conducted on the drug substance and drug product under accelerated and stress conditions. Studies under accelerated conditions may provide useful support data for establishing the expiration date, provide product stability information for future product development (e.g., preliminary assessment of proposed manufacturing changes such as change in formulation, scale-up), assist in validation of analytical methods for the stability program, or generate information which may help

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elucidate the degradation profile of the drug substance or drug product. Studies under stress conditions may be useful in determining whether accidental exposures to conditions other than those proposed (e.g., during transportation) are deleterious to the product and for evaluating which specific test parameters may be the best indicators of product stability. Studies of the exposure of the drug substance or drug product to extreme conditions may help to reveal patterns of degradation; if so, such changes should be monitored under proposed storage conditions. The data of accelerated and stress stability studies are important for regulatory filing of the product with commitment of submission of rolling stability data at real time and temperature While the tripartite guideline on stability describes the conditions of the accelerated and stress study, the applicant should note that those conditions may not be appropriate for biotechnological/biological products. Conditions should be carefully selected on a case-by-case basis.

9.2 Stability-Indicating Parameters

There is no single stability-indicating assay or parameter that profiles the stability characteristics of a biological product. Consequently, the manufacturer should propose a stability-indicating profile that provides assurance that changes in the identity, purity and potency of the product will be detected. At the time of submission, applicants should have validated the methods that comprise the stability-indicating profile and the data should be available for review. The determination of which tests should be included will be product-specific.

Potency

When the intended use of a product is linked to a definable and measurable biological activity, testing for potency should be part of the stability studies. For the purpose of stability testing of the products described in this guideline, potency is the specific ability or capacity of a product to achieve its intended effect. It is based on the measurement of some attribute of the product and is determined by a suitable quantitative method. In general, potencies of biotechnological/biological products tested by different laboratories can be compared in a meaningful way only if expressed in relation to that of an appropriate reference material. For that purpose, a reference material calibrated

directly or indirectly against the corresponding national or international reference material should be included in the assay. Potency studies should be performed at appropriate intervals as defined in the stability protocol and the results should be reported in units of biological activity calibrated, whenever possible, against nationally or internationally recognized standard. Where no national or international reference standards exist, the assay results may be reported in in-house derived units using a characterized reference material. In some biotechnological/biological products, potency is dependent upon the conjugation of the active ingredient(s) to a second moiety or binding to an adjuvant.

Dissociation of the active ingredient(s) from the carrier used in conjugates or adjuvants should be examined in real-time/real-temperature studies (including conditions encountered during shipment). The assessment of the stability of such products may be difficult since, in some cases, in vitro tests for biological activity and physicochemical characterization are impractical or provide inaccurate results. Appropriate strategies (e.g., testing the product prior to conjugation/binding, assessing the release of the active compound from the second moiety, in vivo assays) or the use of an appropriate surrogate test should be considered to overcome the inadequacies of in vitro testing.

Purity and Molecular Characterization

For the purpose of stability testing of the products described in this guideline, purity is a relative term. Due to the effect of glycosylation, deamidation, or other heterogeneities, the absolute purity of a biotechnological/biological product is extremely difficult to determine. Thus, the purity of a biotechnological/biological product should be typically assessed by more than one method and the purity value derived is method-dependent. For the purpose of stability testing, tests for purity should focus on methods for determination of degradation products. The degree of purity, as well as individual and total amounts of degradation products of the biotechnological/biological product entered into the stability studies, should be reported and documented whenever possible. Limits of acceptable degradation should be derived from the analytical profiles of batches of the drug substance and drug product used in the preclinical and clinical studies. The use of relevant physicochemical, biochemical and immunochemical analytical

methods should permit a comprehensive characterization of the drug substance and/or drug product (e.g., molecular size, charge, hydrophobicity) and the accurate detection of degradation changes that may result from deamidation, oxidation, sulfoxidation, aggregation or fragmentation during storage. As examples, methods that may contribute to this include electrophoresis (SDS-PAGE, immuno electrophoresis, Western blot, iso-electro focusing), high-resolution chromatography (e.g., reversed-phase chromatography, gel filtration, ion exchange, affinity chromatography), and peptide mapping. Wherever significant qualitative or quantitative changes indicative of degradation product formation is detected during long-term, accelerated and/or stress stability studies, consideration should be given to potential hazards and to the need for characterisation and quantification of degradation products within the long-term stability program. Acceptable limits should be proposed and justified, taking into account the levels observed in material used in preclinical and clinical studies. For substances that cannot be properly characterized or products for which an exact analysis of the purity cannot be determined through routine analytical methods, the company should propose and justify alternative testing procedures.

Other Product Characteristics

The following product characteristics, though not specifically relating to biotechnological/biological products, should be monitored and reported for the drug product in its final container: Visual appearance of the product (colour and opacity for solutions/suspensions; colour, texture and dissolution time for powders), visible particulates in solutions or after the reconstitution of powders or lyophilized cakes, pH, and moisture level of powders and lyophilized products. Sterility testing or alternatives (e.g., container/closure integrity testing) should be performed at a minimum initially and at the end of the proposed shelf-life.

Additives (e.g., stabilizers, preservatives) or excipients may degrade during the storage period of the drug product. If there is any indication during preliminary stability studies that reaction or degradation of such materials adversely affect the quality of the drug product, these items may need to be monitored during the stability program.

10.0 Packaging, Storage & Distribution

Packaging requirements

The packaging /specification should include the following information:

- How the plasma containers are to be packed to prevent damage during shipment.
- Plasma products of different types should be kept discrete and packaged into separate cartons.
- Each carton should have a unique identification number or a bar code which should be clearly displayed on the carton and recorded in the shipping documentation.

Transportation/Distribution Requirement

- The recommendations for cold chain maintenance, as mentioned for plasma storage, should also apply during the transportation of plasma. The arrangements for temperature control and monitoring during shipping should be clearly defined and documented.
- The requirements for the number and location of temperature logging devices during shipping of final product should be based on a documented assessment of risk throughout the process and during transportation.
- The temperature to be maintained during transportation should be defined. The responsibilities of organizations and individuals during shipping should be identified.

Recommendation for Storage and Transportation of Plasma :

10.1 Container closure systems

A Container Closure System (CCS) is the complete packaging (container + closure) that holds and protects a product, ensuring its safety, stability, and integrity throughout its shelf life.

The container closure integrity validation should take into consideration any transportation or shipping requirements that may negatively impact the integrity of the container. The container closure system may be designed to improve patient safety, minimize waste, and improve ease of administration.

Key Components

Container: Glass vials, plastic bottles, ampoules, syringes.

Closure: Rubber stoppers, aluminium caps, screw caps, snap caps, seals, inner liners.

Secondary Packaging: Outer cartons, shrink-wraps, labels, tamper proof seals.

Critical Functions

Protection: Shields the product from oxygen, light, moisture, physical damage, and microbial contamination.

Containment: Prevents leakage or evaporation of the product.

Compatibility: Ensures no harmful substances leach from the packaging into the drug product.

Performance: Allows for proper drug delivery (e.g., injection, oral).

Compliance: Meets strict regulatory standards (FDA, EMA) for quality and safety.

Importance in Pharma

Product Integrity: Directly impacts drug efficacy and patient safety.

Sterility: Maintains sterility, crucial for injectable (parenteral) drugs.

Stability: Ensures the product remains stable under various storage conditions.

Physical characteristic of container: The physical characteristics include dimensional criteria (e.g., shape, neck finish, wall thickness and design tolerances), physical parameters critical to the consistent manufacture of a packaging component (e.g., unit weight) and performance characteristics (e.g., metering valve delivery volume, or the ease of movement of syringe plungers).

Chemical composition: MOC shall be considered during container closure systems validation.

Secondary packaging components: A secondary packaging component generally serves one or more of the following additional functions:

- Provides protection from excessive transmission of moisture or solvents into or out of the packaging system.
- Provides protection from excessive transmission of reactive gases into or out of the packaging system.
- Provides light protection for the packaging system.
- Provides protection for a packaging system that is flexible or needs extra protection from rough handling.
- Provides an additional measure of microbiological protection.

11.0 Appendices

Plasma Manufacturing Process Details : Annexure 1

Solution Preparation records : Annexure 2

Analytical method SOPs, Specification of DP and QC equipment list : Annexure 3

RM specification, Manufacturing records Final Bulk and Formulations : Annexure 4

Bill Of Materials -1000Lit Batch : Annexure 5

Proposed hold time & Storage conditions : Annexure 6

Scale up parameters : Annexure 7

REGISTRATION OF PHARMACEUTICAL PRODUCTS FOR
HUMAN USE IN THE ECONOMIC COMMUNITY
OF WEST AFRICAN STATES
Guidance for the Preparation of Applications in the
Common Technical Document (CTD) Format

June 2018
REGISTRATION OF PHARMACEUTICAL PRODUCTS FOR HUMAN
USE IN THE ECONOMIC COMMUNITY OF WEST AFRICAN STATES

Guidance for the Preparation of Applications in the
Common Technical Document (CTD) Format

June 2018
FOREWORD

A Common Technical Document (CTD) is an internationally agreed upon format for the organization and preparation of application dossiers for marketing authorization. Developed through the ICH process, it is used by an increasing number of regulatory authorities, including the European Union, the US Food and Drug Administration, Japan, and Canada among others. The WHO has adopted the CTD as the format for applications submitted to its Prequalified Programme.

ECOWAS, in recognizing that all the fifteen member states have different requirements for submission of their dossiers for the granting of marketing authorization, initiated interventions and activities to harmonize medical product registration as part of overall process of harmonizing drug regulations in the region.

This initiative resulted in the development of registration requirements based on CTD format that were validated, and adopted by member states in 2010. Subsequent to that a number of countries including Nigeria, Ghana, Liberia, Sierra Leone and The Gambia have largely adopted the CTD format into their processes of granting of marketing authorization.

In the process the eight WAEMU member states within the region had also developed a set of CTDs that are being used in some of the countries¹.

The effect is that industry will have to contend with two different set of requirements in CTD formats in the ECOWAS region that may lead to delays in the registration of critical products for public health interventions, such as those for HIV/AIDS, Malaria, Tuberculosis, the recent Ebola outbreak and other neglected tropical diseases, leading to limited access to these medicines by the people who need them most.

It has become imperative that in order to improve access and to facilitate the interventions that are supported by development partners and governments in the region, these two documents for medicines registration are reviewed, harmonized in line with international standards, approved and implemented by member states as part of the overall harmonization of medicine regulation.

Based on the experience of WHO in providing technical assistance to various regions for the development and harmonization of medicines registration, ECOWAS requested WHO to review the two documents existing in the region and align them into a single registration document based of the CTD format.

Medicines Regulatory staff will be trained in the harmonized requirements in CTD format and processes to facilitate their use at country level. The same requirements will be used when the need arises to register some medicines at the regional level under the auspices of WAHO.

It is believed that the use of the CTD by all NMRAs in the 15 member states will facilitate easy submissions, reduce cost of both submission and assessment, improve collaboration and information exchange between regulators, reduce registration life cycles and ultimately improve access to critical and essential medicines in the region.

Prof. Stanley OKOLO
Director -General

1 Regulation No. 06/2010 / CM / 2010 / UEMOA on the Procedures for the Registration of Pharmaceutical Products for Human Use in UEMOA Member States.

i Registration of Medicines for Human Use in the ECOWAS – Applications in CTD Format
PREFACE

Established on May 28 1975 via the treaty of Lagos, the Economic Community of West African States (ECOWAS) is a 15-member regional group with a mandate of promoting economic integration in all fields of activity of the constituting countries. Member countries making up ECOWAS are Benin, Burkina Faso, Cape Verde, Cote d' Ivoire, The Gambia, Ghana, Guinea, Guinea Bissau, Liberia, Mali, Niger, Nigeria, Sierra Leone, Senegal and Togo.

The West African Health Organization (WAHO) was formed in 1987 when the Heads of State and Government from all fifteen (15) countries in the ECOWAS adopted the Protocol creating the organization. The Protocol, which was subsequently ratified by each government in the sub-region, grants WAHO status as a Specialized Agency of ECOWAS and describes the organization's mission as follows: «The objective of the West African Health Organization shall be the attainment of the highest possible standard and protection of health of the peoples in the region through the harmonisation of the policies of the Member States, pooling of resources,

and cooperation with one another and with others for a collective and strategic combat against the health problems of the sub-region.»

The WAHO has taken actions under its current medicines and vaccines programme to facilitate access to essential and quality medicines, vaccines and essential health products and reduce the use of illicit drugs and counterfeiting in the region.

In 2010, WAHO developed the Common Technical Documents on Medicines Registration Harmonization for the Economic Community of West African States (ECOWAS)² for the purpose of harmonizing medicine registration and regulation within the region. At the same time, the West African Economic and Monetary Union (known as UEMOA: Union Économique et Monétaire Ouest-Africaine) developed Annexes to regulation N°06/2010/CM/UEMOA describing the Approval Procedures for Pharmaceuticals for Human Use in the UEMOA Member States.

The UEMOA is an organization of eight West African states established, in 1994, to promote economic integration among countries that share the CFA franc as a common currency. In July 2005, the UEMOA Council of Ministers adopted a regulation aimed at establishing a harmonized regulatory framework for pharmaceuticals.

The project to harmonize the ECOWAS and the UEMOA documents was initiated in 2015 and supported by the Bill and Malinda Gates Foundation and facilitated by the Regulatory Systems Strengthening Group at the World Health Organization. In December 2016 a draft registration guideline in the Common Technical Document (CTD) format was created by merging the ECOWAS and the UEMOA documents. The draft was reviewed at a validation workshop held in Abidjan, Cote d'Ivoire on April 18-19, 2016. This second draft for the Registration of Generic Pharmaceuticals for Human Use in the CTD format for the Economic Community of West African States reflects the decision to narrow the scope of the document to Multisource (generic) pharmaceutical products. This draft also reflects the comments received during the Abidjan workshop.

2 Common Technical Documents on Medicines Registration Harmonization for The Economic Community Of West African States (ECOWAS), September 2010.

Registration of Medicines for Human Use in the ECOWAS – Applications in CTD Format ii
ACKNOWLEDGEMENTS

This document was produced by the West African Health Organization (WAHO)'s Essential Medicines and Vaccines Programme in the Public Health and Research (DPHR) Department under the frameworks of the Sahel Women's Empowerment and Demographic Dividends (SWEDD) and West African Medicines Regulatory Harmonization (WA-MRH) Projects, linked to the African Medicines Regulatory Harmonization and Global Medicines Regulatory Harmonization Initiatives, hugely supported by technical and financial partners, notably among which are the World Health Organization (WHO), African Union Development Agency (AUDA-NEPAD), World Bank, Bill and Melinda Gates Foundation (BMGF), Swissmedic, and USAID.

This document has been developed on the basis of valuable inputs made during the consultation, and developmental phases as well as regional stakeholders' validation meetings.

We would like to express our gratitude to all institutions, partners and persons who have in numerous ways supported WAHO to strengthen and improve the accessibility of quality, safe, efficacious and affordable essential medicines in the ECOWAS region. A special thanks goes to the Union Economique et Monétaire Ouest Africaine (UEMOA) that supported WAHO by agreeing in 2014, on a single integrated roadmap for better regional coordination of medicines regulatory harmonization in West Africa.

Special appreciation goes particularly to the partners who provided their expertise and supported WAHO to develop the ECOWAS harmonized Common Technical Document (CTD), the WHO Team: Samvel Azatyan, Gabriela Zenhausern and Consultant Brigitte Zirger, UEMOA Team: Mahamane Hamidine and Carmelle Hounnou; NEPAD Team: Aggrey Ambali, Margareth Ndomondo-Sigonda, Hudu Mogtari; World Bank Team: Andreas Seiter, Aissatou Diack, Christophe Lemiere, Benjamin Botwe and Cassandra De Souza; and David Mukanga of BMGF

Sincere gratitude also goes to the WAHO Team for their relentless effort to ensure that the harmonized CTD gets developed, validated and adopted by the Assembly of Health Ministers: Carlos Brito, the Director of the Department of Public Health and Research, Sybil Nana Ama Ossei- Agyeman-Yeboah, the Programme Officer responsible for Essential Medicines and Vaccines in WAHO, and the WA-MRH Team: Assi Bernard Assi, Oluwafunmi Sopein-Mann and Pierre Kpatcha Tchamdja.

We wish to appreciate and to thank also the Heads of the National Medicines Regulatory Authorities (NMRAs) and WA-MRH Focal Persons in the fifteen (15) ECOWAS Member States.

We also thank the national, regional, continental and global Pharmaceutical Industrial Associations: Pharmaceutical Manufacturers Group of Manufacturing Association of Nigeria (PMG-MAN), Pharmaceutical Manufacturers Association of Ghana (PMAG), West African Pharmaceutical Manufacturers Association (WAPMA), African Pharmaceutical Manufacturers Association (FAPMA), and International Federation Pharmaceutical Manufacturers Association (IFPMA) and all contributors from ECOWAS Member States, numerous other Partners and the Civil Society who diligently reviewed and validated the harmonized CTD and gave valuable suggestions.

It is our expectation that the population of the ECOWAS region and the Pharmaceutical sector will benefit greatly from this ECOWAS harmonized Common Technical Document for registration of pharmaceutical for human use.

Prof. Stanley OKOLO

Director -General

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LIST OF ABBREVIATIONS	
AIDS Acquired Immune Deficiency Syndrome	
API Active Pharmaceutical Ingredient	
APIMF Active Pharmaceutical Ingredient Master File	
ATC Anatomical, Therapeutic and Chemical Classification	
Certificate of Suitability issued by the European Directorate for the Quality of Medicines and CEP	
Healthcare (EDQM)	
CPP Certificate of Pharmaceutical Product	
CTD Common Technical Document	
DMF Drug Master File	
ECOWAS Economic Community for West African States	
FPP Finished Pharmaceutical Products	
GMP Good Manufacturing Practices	
HIV Human Immunodeficiency Virus	
International Council for Harmonisation of Technical Requirements for the Registration ICH	
of Medicines for human use	
INN International Non-Proprietary Name	
MA Market Authorization	
NCE New Chemical Entities	
NMRA National Medicines Regulatory Authority	
OTC Over the Counter Medicines	
PIL Patient Information Leaflet	
POM Prescription-only Medicines	
SmPC Summary of Product Characteristics	

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GLOSSARY / DEFINITIONS

Active pharmaceutical ingredient (API)

Any substance or mixture of substances intended to be used in the manufacture of a pharmaceutical dosage form and that, when so used, becomes an active ingredient of that pharmaceutical dosage form. Such substances are intended to furnish pharmacological activity or other direct effect in the diagnosis, cure, mitigation, treatment, or prevention of disease or to affect the structure and function of the body.

Authorized person

The person recognized by the National Regulatory Authority as having the responsibility for ensuring that each batch of finished product has been manufactured, tested and approved for release in compliance with the laws and regulations in force in that country.

Batch records

All documents associated with the manufacture of a batch of bulk product or finished product. They provide a history of each batch of product and of all circumstances pertinent to the quality of the final product.

Bio-equivalence:

Two pharmaceutical products are bioequivalent if they are pharmaceutically equivalent or pharmaceutical alternatives, and their bio-availabilities, in terms of peak (C_{max} and T_{max}) and total exposure (area under the curve (AUC)) after administration of the same molar dose under the same conditions, are similar to such a degree that their effects can be expected to be essentially the same³.

Bulk Product:

Any product that has completed all processing and steps up to, but not including final packaging.

Drug Master File

A drug master file (DMF) is a master file that provides a full set of data on an API. In some countries, the term may also comprise data on an excipient or a component of a product such as a container.

Drug Substance

Another term used for the Active Pharmaceutical Ingredient

Finished product

A finished dosage form that has undergone all stages of manufacture, including packaging in its final container and labelling. Related terms – Intermediate Product, Bulk Product.

Generic Products

The term generic product means a pharmaceutical product, usually intended to be interchangeable with the innovator product, which is usually manufactured without a licence from the innovator company and marketed after expiry of the patent or other exclusivity rights.

Intermediate product

Partly processed product that must undergo further manufacturing steps before it becomes a bulk product.

Manufacture

All operations of purchase of materials and products, production, quality control, release, storage and distribution of pharmaceutical products, and the related controls.

3 WHO, THE BLUE BOOK, Marketing Authorization of Pharmaceutical Products with Special Reference to Multisource (Generic) Products, A manual for National Medicines Regulatory Authorities (NMRAs). 2nd EDITION

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Marketing authorization (product license, registration certificate)

An official document issued by a competent medicines regulatory authority for the purpose of marketing or free distribution of the product after evaluations for safety, efficacy and quality.

Master formula

A document or set of documents specifying the starting materials with their quantities and the packaging materials, together with a description of the procedures and precautions required to produce a specified quantity of a finished product as well as the processing instructions, including the in-process controls.

Master record

A document or set of documents that serve as a basis for the batch documentation.

Multisource (Generic) Product

Multisource pharmaceutical products are pharmaceutically equivalent products that may or may not be therapeutically equivalent. Only multisource pharmaceutical products that are therapeutically equivalent are interchangeable.

Pharmaceutical product

Any material or product intended for human use presented in its finished dosage form or as a starting material for use in such a dosage form that is subject to control by pharmaceutical legislation.

Production

All operations involved in the preparation of a pharmaceutical product, from receipt of materials, through processing, packaging and repackaging, labelling and re-labelling, to completion of the finished product.

Specification

List of detailed requirements with which the products or materials used or obtained during manufacture have to conform. It serves as a basis for quality evaluation.

Standard Operating Procedure (SOP)

An authorized written procedure giving instructions for performing operations not necessarily specific to a given product or material (e.g. equipment operation, maintenance and cleaning; validation; cleaning of pre-mises and environmental control; sampling and inspection).

Certain SOPs may be used to supplement product-specific master and batch production documentation.

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INTRODUCTION

This document provides guidance for the preparation of a product dossier (PD)/application in the Common Technical Document (CTD) format for the Registration of Medicines for Human Use in the Economic Community of West African States (ECOWAS).

The document describes how to organize and format the product dossier; it does not describe what information, studies or data are required. Therefore, when preparing a regulatory dossier, it is necessary to consult relevant guidance documents on technical (data) requirements.

The use of the CTD allows applicants to prepare dossiers for the NMRAs of ECOWAS without unnecessarily reformatting information that may already have been submitted to other international regulatory authorities. It will reduce the cost of both submission and assessment, support collaboration and information exchange between regulators, and increase the access to critical and essential medicines in the ECOWAS region.

The document is based on the International Council for Harmonization guidelines for Technical Requirements for Registration of Pharmaceuticals for Human Use, namely ICH M4, ICH M4Q, ICH M4S, and ICH M4E. This guideline should be read in conjunction with other applicable WHO and ICH reference documents that provide further guidance and recommendations on the topic-specific content requirements in Modules 2 to 5. References to these will be provided throughout this document.

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GENERAL PRINCIPLES OF PRESENTATION

This section provide an overview of the physical presentation of the regulatory dossier in CTD format.

Table

1 outlines the modular structure and main headings that should be used.

Some headings and/or subheadings may not be applicable to certain dossiers, such as those for multisource (generic) drugs or variations. When no information is required in a specific section or subsection, that heading or subheading should be omitted. The numbering of an omitted section should not be reused for another section.

Insert the corresponding sentence of the document in French

LANGUAGE

There are three official languages in the ECOWAS region. These are English, French and Portuguese. Applications for products seeking a region-wide market authorization shall be submitted in English.

However, the summary product characteristics, labelling and packaging information shall be submitted in all the three official languages of the region (English, French and Portuguese).

Where applicants wish to register the product in a specific country, the official language of that country

SHALL be used for the application, the technical package and supporting documents.

In cases where there is the need to translate a document from its original language into the other languages

used in the region, the accuracy of the translations is the responsibility of the applicant.

DATA PRESENTATION

Dossiers should be submitted in separately bound volumes for the different parts but shall be numbered serially (e.g. Vol.1 of 2) for ease of reference.

Text and tables should be prepared using margins that allow the document to be printed on A4 and 80g/m² paper. The left-hand margin should be sufficiently large that information is not obscured by the method of

binding. Font sizes for text, tables, flow diagrams and floor maps should be of a style and size that are large

enough to be easily legible, even after photocopying. Times New Roman, 12-point font is recommended for narrative text.

All pages shall be numbered appropriately with the format 'page x of y' to facilitate easy reference by evaluators.

Each section of the dossier must have a table of content and must be accurately referenced.

Acronyms

and abbreviation should be defined the first time they are used in each part.

REFERENCES AND TEXTS

International standards for citing references in any parts of the dossier must be followed. The latest edition

of any reference source, specifying the year of publication must be used.

Literature references should be cited in accordance with the current edition of the Uniform Requirements for

Manuscripts Submitted to Biomedical Journals, International Committee of Medical Journals Editors

(ICMJE).

Acronyms and abbreviations should be defined the first time they are used in each module. Where necessary, especially for analytical methods, specifications and procedures, copies of the relevant portions of the reference source(s) must be included.

All in-house processes quoted in the documentation must have been validated and appropriate references cited.

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STRUCTURE OF THE CTD FORMAT

Information within the CTD is organized into a series of structured documents which are in turn organized into modules.

Table 1 : Main Section Headings in the Common Technical Document (CTD) Format

Number Title and Main Section Headings

Module 1: Administrative and Product Information

1.0 Table of Contents (Modules 1 to 5)

1.1 Correspondence

1.2 Administrative Information

1.3 Product Information

1.4 Regional Requirements

1.5 Clinical Trial Information

1.A Appendix

Module 2: Common Technical Document (CTD) Summaries

2.1 CTD Table of Contents (Modules 2 to 5)

2.2 CTD Introduction

2.3 Quality Overall Summary

2.4 Nonclinical Overview

2.5 Clinical Overview

2.6 Nonclinical Written and Tabulated Summaries

2.7 Clinical Summary

Module 3: Quality

3.1 Table of Contents of Module 3

3.2 Body of Data

3.3 Literature References

Module 4: Nonclinical Study Reports

4.1 Table of Contents of Module 4

4.2 Study Reports

4.3 Literature References

Module 5: Clinical Study Reports

5.1 Table of Contents of Module 5

5.2 Tabular Listing of All Clinical Studies

5.3 Clinical Study Reports

5.4 Literature References

Red text indicates sections that are not normally needed for a generic drug

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MODULE 1: ADMINISTRATIVE AND PRODUCT INFORMATION

Module 1 should contain all administrative documents (for example, application forms, certifications, labeling and general correspondence, etc. The documents should be organized in the order below.

1.0 TABLE OF CONTENTS (ToC)

The Table of Contents (ToC) for the entire regulatory dossier should be placed in this section. It should list all documents included in Modules 1-5. A module-specific ToC is included with each Module.

1.1 CORRESPONDENCE

All correspondence-related documents submitted to the regulatory authority are to be placed in Module 1.1 unless otherwise indicated. Scientific information is not to be included in this Module.

1.1.1 Cover Letter

A cover letter should accompany any data being submitted to the regulatory authority. The cover letter should clearly state what is being submitted, including reference to the request letter (if applicable) and a

brief description of the package.

The cover letter should not contain any scientific information.

Note that the Question and Answer (Q and A) responses related to correspondence issued by the regulatory authority and the Note to Reviewer are assigned a specific location (1.1.3 and 1.1.6) and should not be included in the cover letter.

Any cross-referenced regulatory document should be clearly stated in the cover letter, and the following information should be included:

- application type, specify whether new, renewal or variation
- NMRA application number (issued by the NMRA)
- date of regulatory authorization if applicable
- brand name, DCI, dosage, presentation, dosage form
- manufacturer's name
- Applicant's name
- Number of samples submitted

A sample cover letter is provided in Annex B: FORMS

1.1.2 Copy of Correspondence Issued by the Regulatory Authority

A copy of the correspondence issued by the regulatory authority, being responded to should be placed in this section. This includes (but is not limited to) the following:

- A request for additional information (during screening or evaluation);
- A negative decision (e.g., deferrals or rejections).

1.1.3 Information Solicited by the Regulatory Authority

Solicited information is defined as information requested by the regulatory authority (copy of the regulator's correspondence is placed in section 1.1.2). Responses to these requests are to be provided in Question and Answer format, and placed in this section. The answers should summarize the response and cross-reference the supporting data that is to be placed in the appropriate Module of the regulatory document. No supporting data in connection with the answers is to be provided in this section.

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1.1.4 Meeting Information

Any meeting related information and documentation, with the exception of an Appeal meeting, are to be placed in this section. This includes (but is not limited to) the following:

- meeting information package;
- proposed meeting agenda;
- Presentation slides;
- Meeting minutes.

1.1.5 Request for Appeal Documentation

Any person aggrieved by a decision in relation to any application for marketing authorization of a pharmaceutical product may within two (2) months from the date of notice of the decision, make representations in writing to WAHO or the appropriate NMRA and submit additional data to support the appeal. Any documentation required as part of a manufacturer's Request to Appeal a regulatory decision is to be placed in this section.

1.1.6 General Note to Reviewer

The Note to Reviewer should be used to facilitate the review. These comments are NOT to be included in the cover letter.

Notes relating to the entire regulatory dossier (e.g., advising that the product is referred to by a foreign trade name throughout the regulatory document) should be placed in this section.

Notes relating to a specific section of the regulatory document should be placed at the beginning of each pertinent section. For example, this note can be used to identify changes in a section and/or document.

1.2 ADMINISTRATIVE INFORMATION

1.2.1 Application Forms

Completed and signed application forms should be placed in this section.
(See Model Application Form in Annex B)

1.2.2 Fee Forms

Application fees shall be paid for each application submitted. This shall be in the form prescribed by the applicable NMRA or WAHO.

Completed fee forms (proof/evidence of payment) should be placed in this section.
(See Model Fee Form in Annex B)

1.2.3 Certification and Attestation Forms

Completed and signed forms are to be placed in this section. These include, but are not limited to, the following:

- Certification of Suitability to the Monographs of the European Pharmacopoeia (CEP) issued by the European Directorate for the Quality of Medicines and Healthcare (EDQM),
- World Health Organization Confirmation of Prequalification (WHO-CPQ) – where applicable.
- The export authorization if the product is not registered in the country of manufacture OR a Certificate of Pharmaceutical Product issued by the competent authority in the country of manufacture
- The Product Certificate issued by the NMRA;
- Any and all Certificates of analysis as required in the quality guidance. (In reference to the module 3)

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1.2.4 Compliance and Site Information

Submit any that apply.

1.2.4.1 Good Manufacturing Practices

Good Manufacturing Practices (GMP) compliance information should be placed in this section. Regulatory GMP compliance status issued by other jurisdictions, including Date of last GMP and/or pre-approval inspection, and any observation-related information should also be placed in this section.

1.2.4.2 Other Compliance and Site Information Documents

Any other regulatory compliance and site-related information which is not currently mentioned should be placed in this section.

1.2.5 Authorization for Sharing Information

Letters authorizing the regulatory authority to access Drug Master Files (DMFs)/ active pharmaceutical ingredient master files or Site Reference files (SRF) should be included here. See model letter of access in ANNEX B.

1.2.6 Regional and International Regulatory Status

Provide evidence on registration status of the proposed product(s) and, approved indications in the country

of origin, other parts of the ECOWAS region and in other countries/regions.

Provide evidence if the API or Finished Pharmaceutical Product is prequalified by the WHO.

Depending upon the status of the product this may include, but not be limited to, the following:

- International / regional registration, review and/or marketing status, including date of filing, approval of
- product or supplemental changes in other jurisdictions, information regarding the withdrawal, stop of sale and/or market recall;
- Foreign refusals;
- Foreign clinical trial status;
- Date of the first registration of the product for all approved indications;
- Confirmation of filing or the date(s) of approval or withdrawal.
- Certification issued by the competent authority for the country of product origin that the product is marketed in the country of origin
- Certification issued by the applicable competent authority for the wholesale price excluding tax in the country where the product is marketed.

1.2.7 Trademark & Intellectual Property Information

Application for the registration of medicines at the WAHO or country level may be made by the manufacturer

of the product, or the license/ patent holder. In situations where applications are submitted by other entities,

a duly authenticated power of attorney from the manufacturer or license holder shall be submitted.

1.2.8 Post-Authorization Information

Periodic Safety Update Reports (PSURs) are to be filed in this section as applicable.

1.2.9 Other Administrative Information

This section is for any administrative information that does not have a designated location in the CTD format.

This section should NOT contain any scientific information.

Requests for bio-waivers are submitted in this section. Justification for the request for bio-waivers shall be

submitted as per Guidance on Bio-waivers.

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1.3 PRODUCT INFORMATION

The content of the Summary of Product Characteristics (SmPC), the Patient Information Leaflet and all labels

in any of the 3 official languages should be the same.

See Model Product Information Templates provided in ANNEX A

1.3.1 Summary of Product Characteristics (Prescribing Information)

A copy of the Summary of Product Characteristics (SmPC) is to be placed in this section. This includes all

alternate language SmPCs.

When revisions are requested during the course of an evaluation, an annotated version of the revised SmPC is required. The annotations should identify all changes made, either in relation to the last approved

SmPC

or in response to a request made by the regulatory authority.

1.3.2 Patient Information Leaflet

A copy of the Patient Information Leaflet (PIL) is to be placed in this section.

1.3.3 Container Labels

All container labels, including the inner and outer labels, should be provided in this section.

This should include the labels for all strengths, dosage forms and reconstitution diluents.

When additional revisions are requested during the course of the review, an annotated version of the revised

label maybe requested, and should be placed in this section.

1.3.4 Foreign Labelling

If the drug product has been marketed outside the region, the applicant shall supply the SmPC approved for

WHO prequalified products and those marketed elsewhere in the ECOWAS or other African regions, must clearly translate the labelling into the three official languages of ECOWAS

1.3.5 Reference Product Labelling

For multisource (generic) products, the SmPC for the Reference Product(s) is to be placed in this section.

1.4 REGIONAL SUMMARIES

1.4.1 Bioequivalence Trial Information

The completed Bioequivalence Trial Information Form (BTIF) for all pivotal comparative bioavailability

(bioe-

quivalence) studies should be placed in this section.

See BTIF Template provided in ANNEX C

1.A APPENDIX

MODULE 1.A.1 ELECTRONIC DOCUMENTS

Electronic versions of applications are encouraged either in searchable Portable Document Format (PDF) or

Word. This electronic document should be saved to a CD-ROM or other acceptable storage medium.
All electronic media submitted to support the drug regulatory document should be placed in this section.

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MODULE 2: COMMON TECHNICAL DOCUMENT (CTD) SUMMARIES

Module 2 includes the following 7 sections. For multisource (generic) pharmaceutical products, Modules 2.4-2.7 are not usually needed.

2.1 CTD Table of Contents (Modules 2-5)

2.2 CTD Introduction

2.3 Quality Overall Summary

2.4 Nonclinical Overview

2.5 Clinical Overview

2.1 CTD TABLE OF CONTENTS (MODULE 2-5)

The table of contents for Module 2 to 5 should be provided.

2.2 CTD INTRODUCTION

The introduction should include proprietary name, non-proprietary name or common name of the drug substance, company name, dosage form(s), strength(s), route of administration, and proposed indication(s).

It should briefly describe the contents of the Modules 2 to 5 with appropriate cross-references to them.

2.3 QUALITY OVERALL SUMMARY

The Quality Overall Summary (QOS) is a summary that follows the scope and the outline of the Body of Data in Module 3. The QOS should not include information, data or justification that was not already included in

Module 3 or in other parts of the CTD.

Complete the QOS-PD Template in Annex C following the guidance in this section.

Refer to ICH M4Q (R1)

2.3.S DRUG SUBSTANCE

For a drug product containing more than one drug substance, the information in module 2.3.S.1 to 2.3.S.7 should be submitted for each drug substance, clearly identifying the substance name and manufacturer in the title of each module.

2.3.S.1 General Information (name, manufacturer)

Include information from Module 3.2.S.1

2.3.S.2 Manufacture (name, physical address, i.e., site)

Include information from Module 3.2.S.2

Information on the manufacturer,

- Provide the name, address and responsibility of each manufacturer, including contractors, and each proposed production site or facility involved in manufacturing and testing.
- A brief description of the manufacturing process (including, for example, reference to starting materials, critical steps, and reprocessing) and the controls that are intended to result in the routine and consistent production of material(s) of appropriate quality; this could be presented as a flow diagram.
- A flow diagram, as provided in 3.2.S.2.2;
- A description of the Source and Starting Material and raw materials of biological origin used in the manufacture of the API, as described in 3.2.S.2.3;
- Highlight critical process intermediates, as described in 3.2.S.2.4;
- A description of process validation and/or evaluation, as described in 3.2.S.2.5.

2.3.S.3 Characterisation (name, manufacturer)

A summary of the interpretation of evidence of structure and isomerism, as described in 3.2.S.3.1, should be included.

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A tabulated summary of the data provided in 3.2.S.3.2, with graphical representation, where appropriate should be included.

2.3.S.4 Control of Drug Substance (name, manufacturer)

A brief summary of the justification of the specification(s), the analytical procedures, and validation should be included.

Specification from 3.2.S.4.1 should be provided.

A tabulated summary of the batch analyses from 3.2.S.4.4, with graphical representation where appropriate, should be provided.

2.3.S.5 Reference Standards or Materials (name, manufacturer)

Information from 3.2.S.5 (tabulated presentation, where appropriate) should be included.

2.3.S.6 Container Closure System (name, manufacturer)

A brief description and discussion of the information, from 3.2.S.6 should be included.

2.3.S.7 Stability (name, manufacturer)

This section should include a summary of the studies undertaken (conditions, batches, analytical procedures) and a brief discussion of the results and conclusions, the proposed storage conditions, retest date or shelf-life, where relevant, as described in 3.2.S.7.1.

The post-approval stability protocol, as described in 3.2.S.7.2, should be included.

A tabulated summary of the stability results from 3.2.S.7.3, with graphical representation where appropriate, should be provided.

2.3.P FINISHED PHARMACEUTICAL PRODUCT

2.3.P.1 Description and Composition of the Drug Product (name, dosage form)

Information from 3.2.P.1 should be provided.

Composition from 3.2.P.1 should be provided.

2.3.P.2 Pharmaceutical Development (name, dosage form)

A discussion of the information and data from 3.2.P.2 should be presented.

A tabulated summary of the composition of the formulations used in clinical trials and a presentation of dissolution profiles should be provided, where relevant.

2.3.P.3 Manufacture (name, dosage form)

Information from 3.2.P.3 should include:

- Information on the manufacturer.
- A brief description of the manufacturing process and the controls that are intended to result in the routine and consistent production of product of appropriate quality.
- A flow diagram, as provided under 3.2.P.3.3.
- A brief description of the process validation and/or evaluation, as described in 3.2.P.3.5.

2.3.P.4 Control of Excipients (name, dosage form)

A brief summary on the quality of excipients, as described in 3.2.P.4, should be included.

2.3.P.5 Control of Drug Product (name, dosage form)

A brief summary of the justification of the specification(s), a summary of the analytical procedures and validation, and characterisation of impurities should be provided.

Specification(s) from 3.2.P.5.1 should be provided.

A tabulated summary of the batch analyses provided under 3.2.P.5.4, with graphical representation where appropriate should be included.

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2.3.P.6 Reference Standards or Materials (name, dosage form)

Information from 3.2.P.6 (tabulated presentation, where appropriate) should be included.

2.3.P.7 Container Closure System (name, dosage form)

A brief description and discussion of the information in 3.2.P.7 should be included.

2.3.P.8 Stability (name, dosage form)

A summary of the studies undertaken (conditions, batches, analytical procedures) and a brief discussion of the results and conclusions of the stability studies and analysis of data should be included. Conclusions with respect to storage conditions and shelf-life and, if applicable, in-use storage conditions and shelf-life should be given.

A tabulated summary of the stability results from 3.2.P.8.3, with graphical representation where appropriate, should be included.

The post-approval stability protocol, as described in 3.2.P.8.2, should be provided.

2.3.A Appendices

2.3.R Regional Information

2.4 NON-CLINICAL OVERVIEW

The Nonclinical Overview should provide an integrated overall analysis of the information in the Module 4.

In general, the Nonclinical Overview should not exceed 30 pages.

The Nonclinical Overview should be presented in the following sequence:

- Overview of the nonclinical testing strategy
- Pharmacology
- Pharmacokinetics
- Toxicology
- Integrated overview and conclusions
- List of literature references

The Integrated Overview and Conclusions should clearly define the characteristics of the human pharmaceutical as demonstrated by the nonclinical studies and arrive at logical, well-argued conclusions supporting

the safety of the product for the intended clinical use. Taking the pharmacology, pharmacokinetics, and toxicology results into account, the implications of the nonclinical findings for the safe human use of the

pharmaceutical should be discussed (i.e., as applicable to labelling).

ICH M4S (R2) Module 2.4 provides guidance for the contents of the Non-clinical Overview. The non-clinical information in Module 2.4 and Module 4 is not normally required for multisource (generic) drug products.

However in some cases such as changes in safety impurity profile, the safety assessment studies should be conducted.

2.5 CLINICAL OVERVIEW

The Clinical Overview is intended to provide a critical analysis of the clinical data in the Common Technical

Document. The Clinical Overview will necessarily refer to application data provided in the comprehensive Clinical Summary, the individual clinical study reports (ICH E3), and other relevant reports; but it should primarily present the conclusions and implications of those data, and should not recapitulate them.

Specifically, the Clinical Summary should provide a detailed factual summarization of the clinical information

in the CTD, and the Clinical Overview should provide a succinct discussion and interpretation of these findings together with any other relevant information (e.g., pertinent animal data or product quality issues that

may have clinical implications).

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The clinical overview should be presented in the following order:

Table of Contents

- 2.5.1 Product Development Rationale
- 2.5.2 Overview of Biopharmaceutics
- 2.5.3 Overview of Clinical Pharmacology
- 2.5.4 Overview of Efficacy
- 2.5.5 Overview of Safety
- 2.5.6 Benefits and Risks Conclusions
- 2.5.7 Literature References

ICH M4E (R1) Module 2.5 provides guidance for the contents of the Clinical Overview.

2.6. NONCLINICAL WRITTEN AND TABULATED SUMMARIES

Nonclinical Written Summaries

Introduction

This guideline is intended to assist authors in the preparation of nonclinical pharmacology, pharmacokinetics, and toxicology written summaries in an acceptable format. This guideline is not intended to indicate what studies are required. It merely indicates an appropriate format for the nonclinical data that have been acquired.

The sequence and content of the Nonclinical Written Summary sections are described below. It should be emphasised that no guideline can cover all eventualities, and common sense and clear focus on the needs of the regulatory authority assessor are the best guides to constructing an acceptable document.

Therefore, applicants can modify the format if needed to provide the best possible presentation of the information, in

order to facilitate the understanding and devaluation of the results.

Whenever appropriate, age- and gender-related effects should be discussed. Relevant findings with stereoisomers and/or metabolites should be included, as appropriate. Consistent use of units throughout the Summary

will facilitate their review. A table for converting units might also be useful.

In the Discussion and Conclusion Sections, information should be integrated across studies and across species,

and exposure in the test animals should be related to exposure in humans given the maximum intended doses.

General Presentation Issues

Order of Presentation of Information within Sections

When available, in vitro studies should precede in vivo studies.

Where multiple studies of the same type need to be summarised within the Pharmacokinetics and Toxicology sections, studies should be ordered by species, by route, and then by duration (shortest duration first).

Species should be ordered as follows:

- Mouse
- Rat
- Hamster
- Other rodent
- Rabbit
- Dog
- Non-human primate
- Other non-rodent mammal
- Non-mammals

Routes of administration should be ordered as follows :

- The intended route for human use
- Oral
- Intravenous
- Intramuscular

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- Intraperitoneal
- Subcutaneous
- Inhalation
- Topical
- Other

Use of Tables and Figures

Although the Nonclinical Written Summaries are envisaged to be composed mainly of text, some information contained within them might be more effectively and/or concisely communicated through the use of appropriate tables or figures.

To allow authors flexibility in defining the optimal structure for the Written Summaries, tables and figures

should preferably be included within the text. Alternatively, they could be grouped together at the end of

each of the Nonclinical Written Summaries. Throughout the text, reference citations to the Tabulated Summaries should be included, in the following format: (Table X.X, Study/Report Number).

Length of Nonclinical Written Summaries

Although there is no formal limit to the length of the Nonclinical Written Summaries, it is recommended that

the total length of the three Nonclinical Written Summaries in general not exceed 100-150 pages.

Sequence of Written Summaries and Tabulated Summaries

The following order is recommended:

- Introduction
- Written Summary of Pharmacology
- Tabulated Summary of Pharmacology
- Written Summary of Pharmacokinetics
- Tabulated Summary of Pharmacokinetics
- Written Summary of Toxicology
- Tabulated Summary of Toxicology

Refer ICH M4S_R2 for detailed discussion of content of the non-clinical written and tabulated summaries.

2.7. CLINICAL SUMMARY

Preamble

The Clinical Summary is intended to provide a detailed, factual summarisation of all of the clinical information in the Common Technical Document. This includes information provided in ICH E3 clinical study reports;

information obtained from any meta-analyses or other cross-study analyses for which full reports have been included in Module 5; and post-marketing data for products that have been marketed in other regions.

The comparisons and analyses of results across studies provided in this document should focus on factual observations. In contrast, the CTD Clinical Overview document should provide critical analysis of the clinical study program and its results, including discussion and interpretation of the clinical findings and discussion of the place of the test drug in the armamentarium.

The length of the Clinical Summary will vary substantially according to the information to be conveyed, but it is anticipated that (excluding attached tables) the Clinical Summary will usually be in the range of 50 to 400 pages.

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Table of Contents

2.7.1 Summary of Biopharmaceutical Studies and Associated Analytical

Methods

2.7.1.1 Background and Overview

2.7.1.2 Summary of Results of Individual Studies

2.7.1.3 Comparison and Analyses of Results Across Studies

2.7.1.4 Appendix

2.7.2 Summary of Clinical Pharmacology Studies

2.7.2.1 Background and Overview

2.7.2.2 Summary of Results of Individual Studies

2.7.2.3 Comparison and Analyses of Results Across Studies

2.7.2.4 Special Studies

2.7.2.5 Appendix

2.7.3 Summary of Clinical Efficacy

2.7.3.1 Background and Overview of Clinical Efficacy

2.7.3.2 Summary of Results of Individual Studies

2.7.3.3 Comparison and Analyses of Results Across Studies

2.7.3.4 Analysis of Clinical Information Relevant to Dosing Recommendations

2.7.3.5 Persistence of Efficacy and/or Tolerance Effects

2.7.3.6 Appendix

2.7.4 Summary of Clinical Safety

2.7.4.1 Exposure to the Drug

2.7.4.2 Adverse Events

2.7.4.3 Clinical Laboratory Evaluations

2.7.4.4 Vital Signs, Physical Findings, and Other Observations Related to Safety

2.7.4.5 Safety in Special Groups and Situations

2.7.4.6 Post-marketing Data

2.7.4.7 Appendix

2.7.5 Literature References

2.7.6 Synopses of Individual Studies

Refer ICH M4E_R1 for detailed discussion of content of the clinical summary.

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MODULE 3: QUALITY

The Quality module follows the structure and illustrative explanations that are outlined in ICH M4Q (R1).

3.1 TABLE OF CONTENTS (MODULE 3)

The table of contents should give the location of each study report in Module 3

3.2. S BODY OF DATA - DRUG SUBSTANCE

The following information may be submitted as information for the API as applicable:

- Option 1 - Full details in the product dossier
- Option 2 – Confirmation that Drug Substance (API) is pre-qualified by the WHO
- Option 3 - A Certificate of Suitability of European Pharmacopeia (CEP).

For a drug product containing more than one drug substance, the information should be submitted for each drug substance.

Where reference is made to a CEP, the applicant must provide a letter of access from the CEP holder. The letter of access should be provided in Module 1.2.5. Evidence of WHO Pre-qualification should be provided in Module 1.2.6.

Refer to WHO Technical Report Series, No. 970 Annex 4

Option 2 Option 3

Option 1

API-PQP by WHO CEP

3.2.S.1.3 General properties Yes Yes

3.2.S.1 Manufacture

3.2.S.2.1 Manufacturer

3.2.S.2.2 Description of manufacturing process and process controls Yes

if sterility of FPP

3.2.S.2.3 Control of materials

depends on sterile

3.2.S.2.4 Controls of critical steps and intermediates

API

3.2.S.2.5 Process validation and/or Evaluation

3.2.S.2.6 Manufacturing Process development

3.2.S.3.1 Elucidation of structure and other Yes Yes

3.2.S.4 Control of Drug Substance

3.2.S.4.1 Specification Yes Yes
3.2.S.4.2 Analytical procedures and validation Yes Yes
3.2.S.4.3 Validation of Analytical Procedures Yes Yes
3.2.S.4.4 Batch analysis Yes Yes
3.2.S.5 Reference standards or materials Yes Yes
3.2.S.6 Container-closure system Yes
3.2.S.7 Stability
3.2.S.7.1 Stability Summary and Conclusion Yes
Yes
3.2.S.7.2 Post-approval stability Protocol and stability If longer retest
If longer retest period
commitment period or higher
or higher storage
storage tempe-
temperatures than the
3.2.S.7.3 Stability Data ratures than the
prequalified API
prequalified API

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3.2.S.1 General Information (name, manufacturer)

3.2.S.1.1 Nomenclature (name, manufacturer)

Information on the nomenclature of the drug substance should be provided. For example:

- Recommended International Non-proprietary Name (INN);
- Compendial name if relevant;
- Chemical name(s);
- Company or laboratory code;
- Other approved non-proprietary name(s);
- Chemical Abstracts Service (CAS) registry number.

3.2.S.1.2 Structure (name, manufacturer)

The structural formula, including relative and absolute stereochemistry, the molecular formula, and the rela-
tive molecular mass should be provided.

3.2.S.1.3 General Properties (name, manufacturer)

The structure, molecular formula, molecular weight and structural formula are specified. The chiral centres
if any are identified.

Refer to ICH Guidelines: Q6A and Q6B

3.2.S.2 Manufacturer (name, manufacturer)

3.2.S.2.1 Manufacturer(s) (name, manufacturer)

State the name and street address of each facility where manufacture (synthesis, production) of API
occurs,
including contractors, and each proposed production site or facility involved in manufacturing and
testing
should be provided. Provide phone number(s) and E-mail addresses. Include any alternative manufacturers.
Provide a valid manufacturing Authorization for the production of APIs. If available, attach a
certificate of
GMP compliance. (Submitted in Module 1.2.4)

3.2.S.2.2 Description of Manufacturing Process and Process Controls (name, manufacturer)

Description of the manufacturing process and the summary diagram of the active substance.
The description of the drug substance manufacturing process represents the applicant's commitment for
the manufacture of the drug substance. Information should be provided to adequately describe the manu-
facturing process and process control. For example, a flow diagram of the synthetic process(es) should be
provided that includes molecular formulae, weights, yield ranges, chemical structures of starting
materials,
intermediates, reagents and drug substance reflecting stereochemistry, and identified operating
conditions
and solvents.
Reprocessing steps should be identified and justified and any data to support this justification should
be
either referenced.
Refer to ICH Guidelines: Q5A, Q5B, and Q6B

3.2.S.2.3 Control of Materials (name, manufacturer)

Materials used in the manufacture of the drug substance (e.g., raw materials, starting materials,
solvents,
reagents, catalysts) should be listed identifying where each material is used in the process. Information
on
the quality and control of these materials should be provided.
Refer ICH Guidelines: Q6A and Q6B

3.2.S.2.4 Controls of Critical Steps and Intermediates (name, manufacturer)

Critical Steps: Tests and acceptance criteria (with justification including experimental data) performed
at cri-
tical steps identified in 3.2.S.2.2 of the manufacturing process to ensure that the process is controlled
should
be provided.
Intermediates: Information on the quality and control of intermediates isolated during the process should
be provided.

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Refer to ICH Guidelines: Q6A and Q6B

3.2.S.2.5 Process Validation and/or Evaluation (name, manufacturer)

Description of the validation process and evaluation of the manufacturing method.

3.2.S.2.6 Manufacturing Process Development (name, manufacturer)

A description and discussion should be provided of the significant changes made to the manufacturing pro-

cess and/or manufacturing site of the drug substance used in producing non-clinical, stability, scale-up, pilot, and , if available, production scale batches.
Reference should be made to the drug substance data provided in section 3.2.S.4.4.
Refer to ICH Guideline: Q3A

3.2.S.3 Characterisation (name, manufacturer)

Describe the method of Characterization

3.2.S.3.1 Elucidation of Structure and other Characteristics (name, manufacturer) /
Confirmation of structure based on e.g. Synthetic route and spectral analyses should be provided.
Information such as the potential for isomerism, the identification of stereochemistry, or the potential for forming polymorphs should be included.
Refer to ICH Guideline: Q6A

3.2.S.3.2 Impurities (name, manufacturer)
Information on impurities should be provided.
Refer to ICH Guidelines: Q3A, Q3C, Q5C, Q6A, and Q6B

3.2.S.4 Control of Drug Substance (name, manufacturer)

3.2.S.4.1 Specification (name, manufacturer)
The specification for the drug substance should be provided.
Refer to ICH Guidelines Q2 (R1) and Q6B, and WHO Technical Report Series, No. 970-Annex 4

3.2.S.4.2 Analytical Procedures (name, manufacturer)
The analytical procedures used for testing the drug substance should be provided.
Refer to ICH Guidelines: Q2 (R1)

3.2.S.4.3 Validation of Analytical Procedures (name, manufacturer)
Analytical validation information, including experimental data for the analytical procedures used for testing the drug substance should be provided.
Refer to ICH Guidelines: Q2 (R1) and Q6B

3.2.S.4.4 Batch Analyses (name, manufacturer)
Description of batches and results of batch analyses should be provided.
Refer to ICH Guidelines: Q3A, Q3C, Q6A, and Q6B

3.2.S.4.5 Justification of Specification (name, manufacturer)
Justification for the drug substance specification should be provided.
Refer to ICH Guidelines: Q3A, Q3C, Q6A and Q6B

3.2.S.5 Reference Standards or Materials (name, manufacturer)

Information on the reference standards or reference materials used for testing of the drug substance should be provided.

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Refer to ICH Guidelines: Q6A and Q6B

3.2.S.6 Container Closure System (name, manufacturer)

A description of the container closure system(s) should be provided, including the identity of materials of construction of each primary packaging component, and their specifications. The specifications should include description and identification (and critical dimensions with drawings, where appropriate). Non-compendia methods (with validation) should be included, where appropriate.
For non-functional secondary packaging components (e.g. those that do not provide additional protection) only a brief description should be provided. For functional secondary packaging components, additional information should be provided.
The suitability should be discussed with respect to for example, choice of materials, protection from moisture and light, compatibility of the materials of construction with the drug substance, including sorption to container and leaching, and/or safety of materials of construction.
Provide a certificate of analysis for the container closure materials in module 1.2.3
Refer to WHO Technical Report Series, No. 902 Annex 9

3.2.S.7 Stability (name, manufacturer)

3.2.S.7.1 Stability Summary and Conclusions (name, manufacturer)
The types of studies conducted, protocols used, and the results of the studies should be summarized. The summary should include results, for example, from forced degradation studies and stress conditions, as well as conclusions with respect to storage conditions and retest date or shelf-life, as appropriate.
Refer to ICH Guidelines: Q1A, Q1B, and Q5C, and WHO Technical Report Series, No. 953 Annex 9

3.2.S.7.2 Post-approval Stability Protocol and Stability Commitment (name, manufacturer)
The post-approval stability protocol and stability commitment should be provided.

Refer to documents: ICH Q1A (20), Q1B (22), Q1D (24), Q1E (23),
WHO Technical Report Series, No. 953, Annex 2

3.2.S.7.3 Stability Data (name, manufacturer)
Results of the stability studies (e.g. forced degradation studies and stress conditions) should be presented in an appropriate format such as tabular, graphical narrative. Information on the analytical procedures used to

generate the data and validation of the procedures should be included.
Refer to ICH Q1A, Q1B, Q1D, Q1E, Q2 (R1), and
Stability testing of active pharmaceutical ingredients and finished pharmaceutical products, WHO
Technical
Report Series, No. 953, Annex 2, 2009

3.2.P Body of Data – Finished Drug Product (name, dosage form)

3.2.P.1 Description and Composition of the Drug Product (name, dosage form)

A description of the FPP or drug product and its composition should be provided.

The information provided should include, for example:

- A description of the dosage form;
- Composition i.e., list of all components of the dosage form, and their amount on a per-unit basis (including overages, if any), the function of the components, and a reference to their quality standards (e.g. Compendia monographs or manufacturers specifications)
- Description of accompanying reconstitution diluent(s); and
- Type of container and closure used for the dosage form and accompanying reconstitution diluent, if applicable.

Note: For a drug product supplied with reconstitution diluents(s) the information on the diluent(s) should be provided in a separate part "P" as appropriate.

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Refer to ICH Guidelines: Q6A and Q6B

3.2.P.2 Pharmaceutical Development (name, dosage form)

The Pharmaceutical development section should contain information on the development studies conducted to establish that the dosage form, the formulation, manufacturing process, container-closure system, microbiological attributes and usage instructions are appropriate for the purpose specified in the product dossier. The studies described here are distinguished from routine control tests conducted according to specifications. Additionally, this section should identify and describe the formulation and process attributes (critical parameters) that can influence batch reproducibility, product performance and FPP quality.

Supportive data and results from specific studies or published literature can be included within or attached to the Pharmaceutical development section. Additional supportive data can be referenced to the relevant nonclinical or clinical sections of the product dossier.

Refer to ICH Guidelines: Q6A and Q6B, and WHO Technical Report Series, No. 970, Annex 4

3.2.P.2.1 Components of the Drug Product (name, dosage form)

3.2.P.2.1.1 Drug Substance (name, dosage form)

The compatibility of the drug substance with excipients listed in 3.2.P.1 should be discussed.

Additionally, key physicochemical characteristic (e.g. water content, solubility, and particle size distribution, polymorphic or solid state form) of the drug substance that can influence the performance of the drug product should be discussed. For fixed dose combination products, the compatibility of drug substances with each other should be discussed.

3.2.P.2.1.2 Excipients (name, dosage form)

The choice of excipients, their concentration and their characteristics that can influence the drug product performance should be discussed relative to their respective functions.

3.2.P.2.2 Drug Product (name, dosage form)

3.2.P.2.2.1 Formulation Development (name, dosage form)

A brief summary describing the development of the drug product should be provided, taking into consideration the proposed route of administration and usage. The differences between clinical formulations and the formulation (i.e. composition) described should be discussed. Results from comparative in vitro studies (e.g., dissolution), or comparative in- vivo studies (e.g., Bioequivalence) should be discussed when appropriate.

3.2.P.2.2.2 Overages (name, dosage form)

Any overages in the formulations(s) should be justified

3.2.P.2.2.3 Physicochemical and Biological Properties (name, dosage form)

3.2.P.2.3 Manufacturing Process Development (name, dosage form)

The selection and optimization of the manufacturing process described in 3.2.P.3.3, in particular its critical aspects, should be explained. Where relevant, the method of sterilization should be explained and justified.

Differences between the manufacturing processes used to produce pivotal clinical batches and the process described that can influence the performance of product should be discussed.

3.2.P.2.4 Container Closure System (name, dosage form)

The suitability of the container closure system used for the storage, transportation (shipping) and use of the drug product should be discussed. This discussion should consider, e.g. choice of material, protection from moisture and light, compatibility of the materials of construction with the dosage form (including sorption of

container and leaching) safety of materials of construction, and performance (such as reproducibility of the dose delivery from the device when presented as part of the drug product).

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3.2.P.2.5 Microbiological Attributes (name, dosage form)

Where appropriate, the microbiological attributes of the dosage form should be discussed, including, for example, the rationale for not performing microbial limits testing for non-sterile products and the selection and effectiveness of preservative systems in products containing antimicrobial preservatives. For sterile products, the integrity of the container-closure system to prevent microbial contamination should be addressed.

3.2.P.2.6 Compatibility (name, dosage form)

The compatibility of the FPP with reconstitution diluent(s) or dosage devices (e.g., precipitation of API in solution, sorption on injection vessels) stability should be addressed to provide appropriate and supportive information for the labelling.

3.2.P.3 Manufacture (name, dosage form)

3.2.P.3.1 Manufacturer(s) (name, dosage form)

The name, address, and responsibility of each manufacturer, including contractors, and each proposed production site or facility involved in manufacturing and testing should be provided. Include, for example, production, sterilization, packaging and quality control. For each site where the major production step(s) is/are carried out, attach (in module 1.2.6) a valid manufacturing authorization for pharmaceutical production. Attach an original WHO-type certificate of GMP issued by the competent authority.

3.2.P.3.2 Master Formula (name, dosage form)

A recent master formula should be provided that includes a list of components of the dosage form to be used in the manufacturing process, their amounts per batch basis including overages, and a reference to their quality standards.

3.2.P.3.3 Description of Manufacturing Process and Process Controls (name, dosage form)

A flow diagram should be presented giving the steps of the process and showing where materials enter the process. The critical steps and points at which process controls, intermediate tests or final product controls are conducted should be identified.

A narrative description of the manufacturing process, including packaging that represents the sequence of steps undertaken and the scale of production should also be provided. Novel processes or technologies and packaging operations that directly affect product quality should be described with a greater level of detail.

Equipment should at least, be identified by type (e.g. tumble blend, in-line homogenizer) and working capacity, where relevant.

Steps in the process should have the appropriate process parameters identified, such as time, temperature, or pH. Associated numeric values can be presented as an expected range. Numeric ranges for critical steps should be justified in section 3.2.P.3.4. In certain case, environmental conditions (e.g. experimentally documented temperature and relative humidity for hygroscopic FPPs) should be stated.

Proposals for the reprocessing of materials should be justified. Any data to support this justification should be either referenced or filed in this section (3.2.P.3.3).

Refer to ICH Guideline: Q6B

3.2.P.3.4 Controls of Critical Steps and Intermediates (name, dosage form)

Critical steps: Test and acceptance criteria should be provided (with justification, including experimental data) performed at the critical steps of the manufacturing process, to ensure that the process is controlled.

Intermediates: Information on the quality and control of intermediates isolated during the process should be provided.

Refer to ICH Guidelines: Q2 (R1), Q6A, and Q6B

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3.2.P.3.5 Process Validation and/or Evaluation (name, dosage form)

Description, documentation, and results of the validation and/or evaluation studies should be provided for critical steps or critical assays used in the manufacturing process (e.g. validation of the sterilization process or aseptic processing or filling).

Refer to ICH Guideline: Q6B

3.2.P.4 Control of Excipients (name, dosage form)

3.2.P.4.1 Specifications (name, dosage form)

The specifications for excipients should be provided.

Refer to ICH Guideline: Q6A and Q6B

3.2.P.4.2 Analytical Procedures (name, dosage form)

The certificate of analysis must be provided.

The analytical procedures used for testing the excipients should be provided where appropriate.

Refer to ICH Guidelines: ICH Q2 (R1)

3.2.P.4.3 Validation of Analytical Procedures (name, dosage form)

Analytical validation information, including experimental data, for the analytical procedures used for testing the excipients should be provided, where appropriate.
Refer to ICH Guidelines: ICH Q6B and ICH Q2 (R1)

3.2.P.4.4 Justification of Specifications (name, dosage form)
Justification for the proposed excipient specifications should be provided, where appropriate.
Refer to ICH Guidelines: Q3C and Q6B

3.2.P.4.5 Excipients of Human or Animal Origin (name, dosage form)
For excipients of human or animal origin information should be provided regarding adventitious agents (e.g. sources, specifications, description of the testing performed and viral safety data. (Details in 3.2.A.2).
Refer to ICH Guidelines: Q5A, Q5D, and Q6B

3.2.P.4.6 Novel Excipients (name, dosage form)
For excipient(s) used for the first time in an FPP or by a new route of administration, full details of manufacture, characterization, and controls, with cross-references to supporting safety data (non-clinical and/or clinical) should be provided according to the API and/or FPP format. (Details in 3.2.A.3).

3.2.P.5 Control of Drug Product (name, dosage form)

3.2.P.5.1 Specification(s) (name, dosage form)
The specification(s) for the drug product should be provided.
Refer to ICH Guidelines: Q3B, Q6A and Q6B
A list of general characteristics, specific standards, tests and limits for results for the FPP must be provided.
Two separate sets of specifications may be set out; at manufacture (at release) and at the end of shelf life.
Justification for the proposed specification should be provided.

3.2.P.5.2 Analytical Procedures (name, dosage form)
The analytical procedures used for testing the FPP should be provided.
Reference ICH Q2 (R1) and Q6B

3.2.P.5.3 Validation of Analytical Procedures (name, dosage form)
Analytical validation information, including experimental data, for the analytical procedures used for testing the FPP, should be provided.
Reference ICH Q2 (R1) and Q6B

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3.2.P.5.4 Batch Analyses
Results of not less than three batch consecutive analyses (including the date of manufacture, place of manufacture, batch size and use of batch tested) must be presented. The batch analysis must include the results obtained for all specifications at release.
Reference ICH Guidelines: Q3B, Q3C, Q6A, and Q6B
3.2.P.5.5 Characterisation of Impurities (name, dosage form)
Information on the characterization of impurities should be provided, if not previously provided in "3.2.S.3.2 Impurities".

Reference ICH Guidelines: Q3B, Q5C, Q6A, and Q6B

3.2.P.5.6 Justification of Specification(s) (name, dosage form)
Justification for the proposed drug product specification(s) should be provided.
Reference ICH Guidelines: Q3B, Q6A, and Q6B

3.2. P.6 Reference Standards or Materials (name, dosage form)

Information on the reference standards or reference materials used for testing of the FPP should be provided, if not previously provided in "3.2.S.5 Reference standards or materials".
Refer to ICH Guidelines: Q6A and Q6B
3.2. P.7 Container Closure System (name, dosage form)

A description of the container closure system should be provided, including the identity of materials of construction of each primary packaging component and its specification. The specifications should include description and identification (and critical dimensions, with drawings, where appropriate)).
Non-compendial methods (with validation) should be included where appropriate.
For non-functional secondary packaging components (e.g., those that neither provide additional protection nor serve to deliver the product), only a brief description should be provided. For functional secondary packaging components, additional information should be provided.
The suitability of the container closure system used for the storage, transportation (shipping) and use of the FPP should be discussed and located in 3.2.P.2.
Refer to WHO Technical Report Series, No. 902 – Annex 9. Officially recognized pharmacopoeias should be consulted for recommendations on the packaging information for FPPs.

3.2. P.8 Stability (name, dosage form)

The purpose of stability testing is to provide evidence of how the quality of an API or FPP varies with time under the influence of a variety of environmental factors such as temperature, humidity and light. The stability

bility programme also includes the study of product related factors that influence its quality, for example, interaction of API with excipients, container-closure systems and packaging materials.

3.2. P.8.1 Stability Summary and Conclusions (name, dosage form)

The types of studies conducted, protocols used, and the results of the studies should be summarized. The summary should include, for example conclusions with respect to storage and shelf-life and if applicable, in –use storage conditions and shelf-life.

Refer to ICH Guidelines: Q1A, Q1B, Q3B, and Q5C, Q6A and WHO Technical Report Series, No. 953 – Annex 2

3.2. P.8.2 Post-approval Stability Protocol and Stability Commitment (name, dosage form)

The post-approval stability protocol and stability commitment should be provided

Reference document: ICH Q1A and Q5C

3.2. P.8.3 Stability Data (name, dosage form)

Results of the stability studies should be presented in an appropriate format (e.g., tabular, graphical and

30 Registration of Medicines for Human Use in the ECOWAS – Applications in CTD Format narrative). Information on the analytical procedures used to generate the data and validation of these procedures should be included.

Information on characterisation of impurities is located in 3.2.P.5.5.

Refer to ICH Guidelines: Q1A, Q1B, Q2(R1) and Q5C

3.2. A APPENDICES (NAME, DOSAGE FORM)

3.2. R REGIONAL INFORMATION (NAME, DOSAGE FORM)

3.3 LITERATURE REFERENCES (NAME, DOSAGE FORM)

Key literature referenced should be provided, if applicable.

MODULE 4: NON-CLINICAL SUMMARIES

This module is not normally needed for multisource (generic) pharmaceutical products. It deals with the toxicity testing intended to justify the stability and safety of the product. The module is included for completeness to indicate the appropriate format and placement of the nonclinical data.

Refer to ICH M4S (R2) for additional detail on the organization of Module 4 and for ICH references on study design and data content.

4.1 TABLE OF CONTENTS (MODULE 4)

4.2 STUDY REPORTS

The study reports should be presented in the following order:

4.2.1 Pharmacology

4.2.1.1 Primary Pharmacodynamics

4.2.1.2 Secondary Pharmacodynamics

4.2.1.3 Safety Pharmacology

4.2.1.4 Pharmacodynamic Drug Interactions

4.2.2 Pharmacokinetics

4.2.2.1 Analytical Methods and Validation Reports (if separate reports are available)

4.2.2.2 Absorption

4.2.2.3 Distribution

4.2.2.4 Metabolism

4.2.2.5 Excretion

4.2.2.6 Pharmacokinetic Drug Interactions (nonclinical)

4.2.2.7 Other Pharmacokinetic Studies

4.2.3 Toxicology

4.2.3.1 Single-Dose Toxicity (in order by species, by route)

4.2.3.2 Repeat-Dose Toxicity (in order by species, by route, by duration; including supportive toxicokinetics evaluations)

4.2.3.3 Genotoxicity

4.2.3.3.1 In vitro

4.2.3.3.2 In vivo (supportive toxicokinetic evaluations)

4.2.3.4 Carcinogenicity (including supportive toxicokinetic evaluations)

4.2.3.4.1 Long-term studies (in order by species; including range-finding studies that cannot appropriately be

Registration of Medicines for Human Use in the ECOWAS – Applications in CTD Format 31 included under repeat-dose toxicity or pharmacokinetics)

4.2.3.4.2 Short- or medium-term studies (including range-finding studies that cannot appropriately be included under repeat-dose toxicity or pharmacokinetics)

4.2.3.4.3 Other studies

4.2.3.5 Reproductive and Developmental Toxicity

4.2.3.5.1 Fertility and early embryonic development

4.2.3.5.2 Embryo-fetal development

4.2.3.5.3 Prenatal and postnatal development, including maternal function

4.2.3.5.4 Studies in which the offspring (juvenile animals) are dosed and/or further evaluated.

4.2.3.6 Local Tolerance

4.2.3.7 Other Toxicity Studies (if available)

4.2.3.7.1 Antigenicity

4.2.3.7.2 Immunotoxicity

4.2.3.7.3 Mechanistic studies (if not included elsewhere)

4.2.3.7.4 Dependence

4.2.3.7.5 Metabolites

4.2.3.7.6 Impurities

4.2.3.7.7 Other

4.3 LITERATURE REFERENCES

MODULE 5: CLINICAL SUMMARIES

For multisource (generic) pharmaceutical products, only Module 5.3.1 Reports of Biopharmaceutical Studies would normally be needed. However, all parts of the module are included for completeness to indicate the appropriate format and placement of the clinical data.

ICH E3 provides guidance on the organisation of clinical study reports, other clinical data, and references within a Common Technical Document (CTD).

Module 5 provides the recommended organization for the placement of clinical study reports and related information to simplify preparation and review of dossiers and to ensure completeness. The placement of a report should be determined by the primary objective of the study. Each study report should appear in only one section. Where there are multiple objectives, the study should be cross-referenced in the various sections. An explanation such as "not applicable" or "no study conducted" should be provided when no report or information is available for a section or subsection. Refer to ICH M4E (R2) for additional detail on the organization of Module 5 and for additional ICH references on study design and data content.

5.1 TABLE OF CONTENTS (MODULE 5)

A Table of Contents for study reports should be provided.

5.2 TABULAR LISTING OF CLINICAL STUDIES

5.3 CLINICAL STUDY REPORTS

5.3.1 Reports of Bio-pharmaceutic Studies

Bioavailability (BA) studies evaluate the rate and extent of release of the active substance from the medicinal product. Comparative BA or bioequivalence (BE) studies may use Pharmacokinetic (PK), Pharmacodynamic (PD), clinical or in vitro dissolution endpoints, and may be either single dose or multiple doses. When the primary purpose of a study is to assess the PK of a drug, but also includes BA information, the study report

32 Registration of Medicines for Human Use in the ECOWAS – Applications in CTD Format should be submitted in Section 5.3.1, and referenced in Sections 5.3.1.1 and/or 5.3.1.2. WHO guideline QAS/04.109 Rev1

5.3.1.1 Bioavailability (BA) Study Reports

BA studies in this section should include

- studies comparing the release and systemic availability of a drug substance from a solid oral dosage form to the systemic availability of the drug substance given intravenously or as an oral liquid dosage form
- dosage form proportionality studies, and
- food-effect studies.

5.3.1.2 Comparative Bioavailability (BA) and Bioequivalence (BE) Study Reports

- Studies in this section compare the rate and extent of release of the drug substance from similar drug products (e.g., tablet to tablet, tablet to capsule). Comparative BA or BE studies may include comparisons between
 - the drug product used in clinical studies supporting effectiveness and the to-be-marketed drug product,
 - the drug product used in clinical studies supporting effectiveness and the drug product used in stability batches, and
 - similar drug products from different manufacturers.

5.3.1.3 In vitro-In vivo Correlation Study Reports

In vitro dissolution studies that provide BA information, including studies used in seeking to correlate in vitro data with in vivo correlations, should be placed in this section. Reports of in vitro dissolution tests used for batch quality control and/or batch release should be placed in the Quality section (module 3) of the CTD.

5.3.1.4 Reports of Bioanalytical and Analytical Methods for Human Studies

Bioanalytical and/or analytical methods for biopharmaceutic studies or in vitro dissolution studies should ordinarily be provided in individual study reports. Where a method is used in multiple studies, the method and its validation should be included once in Section 5.3.1.4 and referenced in the appropriate individual study reports.

5.3.2 Reports of Studies Pertinent to Pharmacokinetics using Human Biomaterials

5.3.2.1 Plasma Protein Binding Study Reports

5.3.2.2 Reports of Hepatic Metabolism and Drug Interaction Studies

5.3.2.3 Reports of Studies Using Other Human Biomaterials

5.3.3 Reports of Human Pharmacokinetic Studies

5.3.3.1 Healthy Subject PK and Initial Tolerability Study Reports

5.3.3.2 Patient PK and Initial Tolerability Study Reports

5.3.3.3 Intrinsic Factor PK Study Reports

5.3.3.4 Extrinsic Factor PK Study Reports

5.3.3.5 Population PK Study Reports

5.3.4 Reports of Human Pharmacodynamic Studies

5.3.4.1 Healthy Subject PD and PK/PD Study Reports

5.3.4.2 Patient PD and PK/PD Study Reports

5.3.5 Reports of Efficacy and Safety Studies

5.3.5.1 Study Reports of Controlled Clinical Studies Pertinent to the Claimed Indication

5.3.5.2 Study Reports of Uncontrolled Clinical Studies References

5.3.5.3 Reports of Analyses of Data from more than one study, including any formal integrated analyses, meta-analyses, and bridging analyses

5.3.5.4 Other Clinical Study Reports

5.3.6 Reports of Post-marketing Experience

For products that are currently marketed, reports that summarize marketing experience (including all significant safety observations) should be included.

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5.3.7 Case Report Forms and Individual Patient Listings (when submitted)

Case report forms and individual patient data listings that are described as appendices in the ICH or WHO clinical study report guideline should be placed in this section when submitted in the same order as the clinical study reports and indexed by study.

5.4 LITERATURE REFERENCES

Copies of referenced documents, including important published articles, official meeting minutes, or other regulatory guidance or advice should be provided here. This includes copies of all references cited in the

Clinical Overview, and copies of important references cited in the Clinical Summary or in the individual technical reports that were provided in Module 5. Only one copy of each reference should be provided. Copies of

references that are not included here should be immediately available on request.

REFERENCES:

ICH Common Technical Document References (<http://www.ich.org>)

1. ICH M4 - Organization of the Common Technical Document for the Registration of Pharmaceuticals for Human Use (2016)

2. ICH M4E(R2) - Common Technical Document for the Registration of Pharmaceuticals for Human Use: Efficacy (2016)

3. ICH M4Q(R1) - Common Technical Document for the Registration of Pharmaceuticals for Human Use: Quality (2002)

4. ICH M4S(R2) - Common Technical Document for the Registration of Pharmaceuticals for Human Use: Safety (2002)

ICH Quality Guidelines

1. ICH Q1A(R2) - Stability Testing of New Drug Substances and Products (2003)

2. ICH Q1B Stability Testing: Photostability Testing of New Drug Substances and Products (1996)

3. ICH Q1D - Bracketing and Matrixing Designs for Stability Testing of New Drug Substances and Products (2002)

4. ICH Q1E - Evaluation for Stability Data (2003)

5. ICH Q2(R1) - Validation of Analytical Procedures: Text and Methodology (2005) [combines the previous Q2A and Q2B Guidelines]

6. ICH Q3A(R2) - Impurities in New Drug Substances (2006)

7. ICH Q3B(R2) - Impurities in New Drug Products (2006)

8. ICH Q3C(R6) - Impurities: Guideline For Residual Solvents Q3C(2016)

9. ICH Q5A, Q5B, Q5C, Q5D Quality of Biological Products [not needed for multisource (generic) pharmaceutical products]

10. ICH Q6A - Specifications: Test Procedures and Acceptance Criteria for New Drug Substances and New Drug Products: Chemical Substances (1999)

11. ICH Q6B Specifications: Test Procedures and Acceptance Criteria for Biotechnological/Biological Products (1999) [not needed for multisource (generic) pharmaceutical products]

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World Health Organization Guidelines

1. Guidelines on packaging for pharmaceutical products In: WHO Expert Committee on Specifications for Pharmaceutical Preparations. Forty-third report. Geneva, World Health Organization, 2002 (WHO Technical Report Series, No. 902), Annex 9

2. Stability testing of active pharmaceutical ingredients and finished pharmaceutical products In: WHO Expert Committee on Specifications for Pharmaceutical Preparations. Forty-third report. Geneva, World Health Organization, 2009 (WHO Technical Report Series, No. 953), Annex 2. [Together with 2015 update table Stability Conditions for WHO Member States by Region]

3. Guideline on submission of documentation for a multisource (generic) finished pharmaceutical product (FPP): quality part, In WHO Expert Committee on Specifications for Pharmaceutical Preparations. Forty-third report. Geneva, World Health Organization, 2012 (WHO Technical Report Series, No. 970), Annex 4

4. Multisource (generic) pharmaceutical products: guidelines on registration requirements to establish interchangeability, In WHO Expert Committee on Specifications for Pharmaceutical Preparations: Forty-ninth report. World Health Organization, 2015 (WHO Technical Report Series, No. 992), Annex 7.

5. Guidance on the selection of comparator pharmaceutical products for equivalence assessment of interchangeable multisource (generic) products In WHO Expert Committee on Specifications for Pharma-

ceutical Preparations: Forty-ninth report. World Health Organization, (WHO Technical Report Series, No. 992), Annex 8 2015

6. Guidance for organizations performing in vivo bioequivalence studies (revision), In WHO Expert Committee on Specifications for Pharmaceutical Preparations: Fiftieth report.

7. WHO Technical Report Series, No. 996, Annex 9, 2016

World Health Organization Templates

[<https://extranet.who.int/prequal/content/who-medicines-prequalification-guidance>]

Quality Templates

1. Quality overall summary - product dossier (QOS-PD) (2016)

2. Quality information summary (QIS) (2016)

3. Quality information summary (QIS) of the finished pharmaceutical product (FPP) approved by the reference SRA (QIS-SRA) (2013)

Bioequivalence Template

1. Presentation of bioequivalence trial information form (BTIF) (2017)

2. Make reference to guideline for bioavailability and bioequivalence studies in

3. ECOWAS region, (2012), 4.5b page 32 and the WHO Template on Bio waiver]

Labelling Templates

1. Patient information leaflet – Template

Annotated Patient information leaflet template (2016)

Section Guidance for Part 3 Patient information Leaflet for WHOPAR 2016

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2. Summary Product Characteristics (SmPC) Template (2016)

Annotated Summary Product Characteristics (SmPC) Template (2016)

Section Guidance for Part 4 Summary Product Characteristics (SmPC) for WHOPAR 2016

3. Annotated Labelling Template (2016)

Section Guidance for Part 5 Labelling for a WHOPAR 2016

ANNEX A: PRODUCT LABELLING GUIDANCE

There are three official languages in the ECOWAS region. These are English, French and Portuguese. Applications for products seeking a region-wide market authorization shall be submitted in English.

However,

the summary product characteristics, labelling and packaging information shall be submitted in all the three official languages of the region (English, French and Portuguese).

Where applicants wish to register the product in a specific country, the official language of that country

SHALL be used for the application, the technical package and the product labelling.

In cases where there is the need to translate a document from its original language into the other languages

used in the region, the accuracy of the translations is the responsibility of the applicant. For applications

originating outside the ECOWAS region the translations shall be authenticated by the nearest Embassy of an

ECOWAS member state.

The Guidance and templates for product labelling shall be based on the ECOWAS guidance or the current WHO Guidance documents for the Package Leaflet, Summary of Product Characteristics and Labelling which is available from the WHO website at <https://extranet.who.int/prequal/content/contents-and-structure-whopar>.

MODULE 1.3.1 SUMMARY OF PRODUCT CHARACTERISTICS (SMPC)

The format of the SmPC document is to be consistent with the WHO SmPC template. The information is given in the official language of the ECOWAS Member State giving market authorization.

Refer to WHO SmPC Guidance and Annotated Template

Use WHO SmPC Template

MODULE 1.3.2 PATIENT INFORMATION LEAFLET

The format of the PIL is to be consistent with the WHO PIL template. The information is given in the official

language of the ECOWAS Member State giving market authorization.

Refer to WHO PIL Guidance and Annotated Template

Use WHO PIL Template

MODULE 1.3.3 CONTAINER LABELLING (INNER AND OUTER LABELS)

The primary and secondary packaging must include the following information in a legible, understandable and indelible manner. The information is given in the official language of the ECOWAS Member State giving market authorization.

The Container Labelling is to be consistent with the WHO template.

Refer to WHO Guidance

Use WHO Annotated Template

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B.1 MODEL COVER LETTER LINK

Refer to Model Cover Letter in separate document

B.2 MODEL APPLICATION FORM FOR DRUG MARKET AUTHORIZATION

Refer to Model Application Form in separate document

B.3 MODEL LETTERS OF ACCESS

Include this in Module 1 – Administration and Product Information, submodule 1.2.5 of the application in CTD format

Refer to Model letter of Access to the CEP in separate document

Refer to Model letter of Access to the APIMF in separate document

Registration of Medicines for Human Use in the ECOWAS – Applications in CTD Format 37 B.4 MODEL FEE FORM

Application for Market Authorization Fee Form
Form to be inserted in Module 1.2.2 of the Common Technical Document

1. IDENTIFICATION ("refer to table 1.1 on page 45")

Product Name
as per Drug Submission Application form (item 6):
Name of manufacturer
(Full legal name - no abbreviations)
as per Drug Submission Application form (item 10):
Address of manufacturer
as per Drug Submission Application form (items 11-31):
Telephone Number: Fax Number:
Contact Person: E-mail:
Name/address of company (billing contact) to whom invoice
is to be sent (if different from manufacturer):
Telephone Number: Fax Number:
Contact Person: E-mail:

Official use only
Invoice Number _____

Amount paid
Date of payment
Name and address of Bank

Cheque Number _____
Application Number _____
Screened by _____

38 Registration of Medicines for Human Use in the ECOWAS – Applications in CTD Format 1. 2. FEES FOR THE REVIEW OF APPLICATIONS FOR MARKETING AUTHORIZATION Check the appropriate box to indicate the applicable submission, supplement or application fee. # Application Class Description Fee (\$USD) •Check one

1 Registration of a \$
Pharmaceutical Product
1a. Fully Manufactured in \$
ECOWAS
1b. Imported into ECOWAS \$
2 Renewal of Registration \$
2a. Fully manufactured in \$
ECOWAS
2b. Imported into ECOWAS \$
3. Variation of Registration \$
3a. Major variation \$
3b. Minor variation \$

* Other fees including those for appeal, replacement of certificates, laboratory tests among others may be charged by various country MRAs as their legislation requires.

3. METHOD OF PAYMENT

Applications, whether submitted to WAHO or to NMRAs shall be accompanied by the appropriate application fees.

Application fees shall be paid for each application submitted. This shall be in the form of a Bankers' draft or

by wire transfer to the account to be designated by the applicable NMRA or WAHO.

- International bank draft
- Wire (include bank confirmation)

ANNEX C: TEMPLATES

Refer to World Health Organization Templates found at
[<https://extranet.who.int/prequal/content/who-medicines-prequalification-guidance>]

C.1 Module 2 Quality Overall Summary – product dossier (QOS-PD)

C.2 Module 2 Quality information Summary (QIS)

Quality information summary (QIS) of the finished pharmaceutical product (FPP) approved by the reference SRA (QIS-SRA)

C.3 Module 1.4.1 Presentation of Bioequivalence Trail Information Form (BTIF)

Request for a Bio waiver template (to be determined)

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ANNEX D: GUIDANCE DOCUMENTS

ANNEX E: MANAGEMENT OF APPLICATIONS AND STANDARD OPERATING PROCEDURES

For pharmaceutical products that have been evaluated by a stringent drug regulatory authority, and also those that have been prequalified by the World Health Organization, an abridged file abstract will be required.

As for other products, including those for specific or neglected tropical diseases, a complete file will be required.

1.4 GENERAL POLICIES ON APPLICATIONS

A separate application is required for each product. For purposes of clarification, one application could be submitted for products containing the same active ingredients and the same strength made by the same manufacturer at the same manufacturing site, to the same specifications and dosage form, but differing only in packing or pack sizes. On the other hand, separate applications shall be submitted for products that contain the same active ingredient(s) but of different salts, different strength, dosage form and proprietary or brand name.

1.4.1 Classes of Applications

Applications shall be classified into three (3)

- New Applications
- Renewal of applications (i.e., registration)
- Variation of Applications (i.e., of a registered product)

1.4.2 New Applications

Applications for the registration of a pharmaceutical product either submitted to the Director General of WAHO for the granting of region wide market authorization or submitted to a particular NMRA in the region for the very first time shall be considered a new application. In addition to the dossier submitted, the applicant shall provide:

- i. Samples of the product in the commercial pack(s) from one batch with batch certificates of analysis per the samples schedule defined by WAHO (check www.wahooas.org) or the NMRA.
- ii. Certificate of Pharmaceutical Product issued in accordance with the format approved by the WHO and issued by the competent drug regulatory authority of the country of Origin/ Manufacture (submitted in Module 1 of the PD).
- iii. A site master file of the plant in which the product was manufactured. (submitted in Module 3)
- iv. For NCEs and innovator products the pharmacovigilance plan shall be submitted. (submitted in Module 1.2.8 (PSURs))

1.4.3 Applications for Renewal of Registration

Applications for renewal of registration shall be made at least 3 months before the expiry of existing registration by submitting the following:

- i. Dully filled application form for renewal of registration
- ii. Batch Manufacturing Record (BMR) of a real batch manufactured within at most six months before the submission of the application.
- iii. Periodic Safety Update Reports (PSUR)
- iv. Proof of interchangeability for multisource (generics) as explained in Part 5.
- v. Any other requirements that WAHO or the NMRA may determine
- vi. Samples of the product in the commercial pack(s) from one batch with batch certificates of analysis per the samples schedule defined by WAHO or the NMRA.
- vii. A site master file of the plant in which the product was manufactured.

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1.4.4 Application for Variation of a registered product

Applications for variation to a registered product shall be made according to requirements stipulated below:

- i. Dully filled application form for variation of registration
- ii. Samples of the product reflecting the variation
- iii. A site Master File of the manufacturer (if the variation is or includes a change in the name, site and/ or address of the manufacturer).
- iv. Other documents to support or justify the variation.

1.5 SUBMISSION OF APPLICATION

Applications for the registration of products for which a region wide market authorization is being sought shall be made to the Director General of WAHO through the identified Coordinating NMRA indicated in the Expression of Interest (EOI) in accordance with the approved format. For products meant for marketing authorization in a specific country, the application shall be sent to the Head of the NMRA in that country.

1.6 APPLICATION FEES

Application fees shall be paid for each application submitted.-This shall be in the form of a Bankers' draft or by wire transfer to the account to be designated by the applicable NMRA or WAHO. Refer to the fee form in Annex B.

1.6.1 Application fee for registration of a pharmaceutical product:
Imported in to ECOWAS US\$
Fully manufactured in ECOWAS US\$

1.6.2 Application fee for renewal of registration of a pharmaceutical product:
Imported into ECOWAS US\$
Fully manufactured in ECOWAS US\$
Application fees for variation of Registration US\$
Other fees including those for appeal, replacement of certificates, laboratory tests among others may be charged by various country MRAs as their legislation requires.

1.7 RECEIPT AND EVALUATION OF DOSSIERS

Applications, whether submitted to WAHO or to NMRAs shall be accompanied by the appropriate application fees.

1.7.1 Evaluation process
WAHO and NMRAs will assign application numbers serially to applications received and evaluate them on a first in first out (FIFO) basis. Preference and priority will be given to products for priority public health diseases such as Malaria, HIV/AIDS, Tuberculosis, and other neglected tropical diseases. Special consideration will be given to products manufactured by companies operating in member countries of the ECOWAS region. A committee of experts will be constituted at the WAHO secretariat made of regulatory officials and other scientists from member countries. Where necessary, specialists will be consulted for professional opinion on various sections of the dossiers. At country level evaluation will be done by in country evaluators using this guideline and Standard Operating Procedures for evaluation of dossiers. Additional information may be requested for during evaluation and if no response is received within six months of the request, the application will be discontinued. For applications made to WAHO, laboratory analysis based on the validated in-house or pharmacopoeia methods submitted by applicants shall be performed in accordance with approved SOP in accredited quality control laboratories. For purposes of verification of compliance to cGMP, all applications shall be accompanied by a Site Master

Registration of Medicines for Human Use in the ECOWAS – Applications in CTD Format 41
File. An inspection may be conducted by WAHO or the NMRA of the country in which marketing authorization is being sought. Report and recommendations of WAHO inspection teams shall be shared with NMRAs to avoid duplication. Decisions on registration shall be based on the dossier evaluation report, quality control report and inspection report on compliance to cGMP.

1.8 TIMELINES

Complete applications for expedited registration (Locally manufactured and Priority Medicines only), Post Approval Variation and Renewal of registration will be processed within 90 working days of receiving the applications. Complete new applications will be processed within 12 months of receipt of the application. The applicant will be required to provide any requested additional data within 6 months. In case additional time is required, a formal request must be submitted.

1.9 WITHDRAWAL OF AN APPLICATION

When the applicant fails to submit written responses to queries within 6 months from the date of their issuance, it will be deemed that the applicant has withdrawn the application or if the queries have been resubmitted for a second time and the applicant provides unsatisfactory responses, the product will be disqualified and the application will be rejected. The applicant will be required to apply afresh.

1.10 VALIDITY OF REGISTRATION

The registration of a pharmaceutical product at WAHO or by country NMRA shall be valid for five (5) years unless otherwise suspended or revoked by the NMRA, or withdrawn by applicant.

1.11 APPEALS

Any person aggrieved by a decision in relation to any application for marketing authorization of a pharmaceutical product may within two (2) months from the date of notice of the decision, make representations in writing to WAHO or the appropriate NMRA and submit additional data to support the appeal. Documentation in support of the manufacturer's request to appeal a regulatory decision is placed in Module 1.1.

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ANNEX XX – SAMPLE COVER LETTER

<Applicant>
<Address>
<Address>
<Post code> <Town>

<Country

<Applicant's reference> <Date>

<National Medicines Regulatory Authority>

<Address>

<Address>

<Post code> <Town>

<Country>

Dear Sir/Madam,

Subject: Submission of Application Dossier(s) for Marketing Authorization of <Product Name(s),
<strength(s) of active pharmaceutical ingredient(s) and dosage form(s)

We are pleased to submit our Application Dossier(s) for the registration of human medicines in <the
ECOWAS
Region> or <the following ECOWAS member countries> for the following product(s):

Name of the medicinal product(s):

Pharmaceutical form(s) and strength(s):

INN/active Pharmaceutical ingredient(s):

ATC Code(s):

<The application seeks market authorization for a new product not previously marketed in the ECOWAS or
any member country.>

<The application seeks to renew the following marketing authorization(s)>

<The application seeks a market authorization for a variation of <indicate the product and market
authoriza-
tion # or previous application identifiers, e.g., NMRA's control numbers>>

<This submission responds to correspondence from <name of regulatory authority, (e.g., ECOWAS or member
country), dated <DAY, MONTH, YEAR>, related to application number <#####>. The regulator's correspon-
dence is provided in module 1.1.2. The response to the correspondence is provided in module 1.1.3.>

You will find enclosed the submission dossier as specified hereafter:

UU CTD format, 2 copies (following the regulator requirements):

☐ Soft copy

☐ Hard copy

☐ Both

UU CD ROM; Summaries in word format and body data in PDF format

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UU We confirm that all future submissions for this specific product will be submitted in this same format

UU We confirm that the electronic submission has been checked with up-to-date and state-of-the-art anti-
virus software.

UU The electronic submission contains the following modules:

• Module 1: Administrative information and product information

• Module 2: Overview and summaries

• Module 3: Quality

• Module 4: Non clinical study reports

• Module 5: Clinical study reports>

The relevant fees have been paid.>

<xxx samples of the drug product have been submitted with this application.>

Yours sincerely,

.....

<Signature>

<Name>

<Title>

<Phone number(s)>

<Email address>

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MODEL APPLICATION FORM

FOR DRUG MARKET AUTHORIZATION (ENG)

Submit 2 months in advance of the application dossier. Include this form in Section 1.2.1 of the CTD.

MODULE 1.2.1 – APPLICATION FORM

NMRA use only

Application No. Date of Receipt (YYYY-MM-DD)

Number of Volumes/CDs Received Number of. Samples Received

Fees Paid – YES/NO

Name of NMRA person who received the document

PART 1: APPLICATION INFORMATION

New product

Renewal of a market authorization

Reintroduction of a deleted Market Authorization (MA)

Variation of a market authorized product

Proposed area of marketing

■ ECOWAS Region
■ Selected ECOWAS Member State(s) (Please identify)
Number of volumes sent
Number of CDs sent

CHARACTERISTICS OF SAMPLES ENCLOSED

Number of samples
Expiry date.....
Lot /batch Numbers
Lot/batch certificate of analysis number:.....

PART 2: APPLICANT INFORMATION

A. Name and address of Applicant

Name of Company
(Complete legal name – do not use abbreviations)
Head office address:
Web site:
Email:.....

B. Person in the country/ECOWAS authorized to represent the applicant for THIS application

Name Title.....
Name of Company if applicable
(Complete legal name – do not use abbreviations)
Address
Tel..... Fax.....
e-mail.....

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PART 3: PRODUCT INFORMATION: (applicant completes this)

1. Brand Name, Trade Name or Product Name
(the name as it appears on the product label)
2. International Non-proprietary Name (INN) of the Active Pharmaceutical Ingredient (API)
.....
3. Strength of Active Pharmaceutical Ingredient (API) per unit dosage form:

PART 4 – PRODUCT FORMULATION INFORMATION

Pharmaceutical Dosage Form
Route(s) of Administration
Pharmaceutical Group
ATC Code
Proposed Indication/Use

Proposed Distribution Category
Controlled Substance
Prescription only Medicine (POM)
Pharmacy Medicine
Over the Counter Medicine
*

Container Type
Package Size
Visual description of the medicine
Proposed Storage conditions
Proposed Storage conditions after first opening
o
Proposed Shelf Life months at C
o
Proposed Shelf Life after first opening Hours/days/Weeks at C
o
Proposed Shelf Life after reconstitution or dilution Hours/days/Weeks at C

Medicinal (Active Ingredient(s)) and Excipients

CAS No. (if Ingredient Name Standar Strenght Units Per Calculated as applicable) d* h Base?

■ Yes ■ No

■ Yes ■ No

■ Yes ■ No

Name Excipients

*State the reference/monograph standard such as British Pharmacopeia (BP), United States Pharmacopeia (USP), Ph. Eur, Japanese Pharmacopeia, In-house monograph (Mfr Std) etc. used for Finished Medicinal Product

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PART 5: NATIONAL, REGIONAL AND INTERNATIONAL STATUS OF THE PRODUCT

For each country category indicate the status of an application or marketing status of the product.

Include the Certificates of Pharmaceutical Product and other appropriate documentation in Module 1.2.7 of the CTD.

■ Country of Origin

■ Countries with Stringent Regulatory Authorities

■ Other African Regions

■ ECOWAS Member States

■ Other

Authorized

Country:

Date

Proprietary Name of Product:

Authorization Number:

Market Application in progress Withdrawn (by applicant after Authorization)

Country: Country:

Date Submitted: Date
Proprietary Name of Product: Proprietary Name of Product:
Reason:
Rejected Suspended/revoked/stop sale (by NMRA)
Country: Country:
Date Date
Proprietary Name of Product: Proprietary Name of Product:
Reason Reason:

WHO PREQUALIFIED APIs AND FINISHED PRODUCTS

Are the APIs prequalified by the WHO?

- ☐ NO
☐ YES (provide evidence in Module 1.2.7 of the CTD)
Is the finished product prequalified by the WHO?
☐ NO
☐ YES (provide evidence in Module 1.2.7 of the CTD)

API with a CEP

- ☐ NO
☐ YES (provide evidence in Module 1.2.7 of the CTD)

PART 6 : PRICE (where applicable)

Wholesale price (taxes excluded).....

Treatment price.....

Public price.....

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PART 7 : DECLARATION BY APPLICANT

I, the undersigned, certify that the information and material included in this drug application are accurate and complete.

Name of Authorized Official Signature Date

YYYY MM DD

Title Telephone No. Fax No.

Email address

Name of Company to which the Authorized Official Belongs

If the authorized official is a third party acting on behalf of the applicant identified in Part 2, a letter of authorization, signed by the applicant, must be filed with the completed application form.

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SAMPLE LETTER OF ACCESS TO CEP

Include the letter in Module 1 – Administration and Product Information, submodule 1.2.6 of the application in CTD format

<Applicant>
<Address>
<Address>
<Post code> <Town>
<Country>
<Applicant's reference> <Date>
<National Medicines Regulatory Authority>
<Address>
<Address>
<Post code> <Town>
<Country>

Dear Sir/Madam,

Subject: Authorization to access Certificate of Suitability (CEP)

Reference is made to the above subject matter.

Consent is hereby granted to <NMRA> to make reference to this company's Certificate(s) of Suitability (CEPs)

<number(s)> for <API(s) name(s)> in the evaluation of applications relating to the registration of <medicine

name(s)> submitted to <name of NMRA> by <applicant's name>.

This consent <includes / does not include> authorization to supply information or extracts from or the whole

of the data to:

(Name of company or individual)

The API is manufactured by:

<Names and addresses of all manufacturing sites and manufacturing steps carried out at site>

A formal agreement exists between the applicant of the medicine and the manufacturer of the API, which ensures that information will be communicated between them and to the <NMRA> before any significant change is made to the site of manufacture, manufacturing procedure or quality control specifications of the

API. Except as permitted by the WAHO guidelines relating to changes to medicines, such changes will not be made to the API to be used in manufacture of the medicine destined to be distributed in <ECOWAS or member country> before written approval is granted by the NMRA.

I understand that the consequences of failure to obtain approval for changes where approval is necessary may include de-registration and recall of batches of medicines.

Any questions arising from the < NMRA's> evaluation of this CEP should be forwarded to:

(Name and address)

Yours faithfully

{Signature of Company Representative}
{Name}
{Position in Company}
{Date}

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ANNEX V: LETTER OF ACCESS TO ECOWAS APIMF

<Applicant>
<Address>
<Address>
<Post code> <Town>
<Country

<Applicant's reference> <Date>

<National Medicines Regulatory Authority>
<Address>
<Address>
<Post code> <Town>
<Country>

Dear Sir/Madam,

Subject: Authorization to access Active Pharmaceutical Ingredient Master File

Reference is made to the above subject matter.

Consent is hereby granted to (Name ECOWAS-NMRA) to make reference to this company's Active Pharmaceutical Ingredient Master File(s) for [API(s) name] in the evaluation of applications relating to the registration of [medicine name(s)] submitted to (name of ECOWAS-NMRA) by the (applicant's name).

This consent does/does not** include authorization to supply information or extracts from or the whole of the data to:

(Name of company or individual)

The substance is manufactured by:
(Names and addresses of all manufacturing sites and manufacturing steps carried out at site)

A copy of the applicant's Part of the APIMF as specified in the ECOWAS Active Pharmaceutical Ingredient Master File Procedure has been supplied to the applicant.

A formal agreement exists between the applicant of the medicine and the manufacturer of the API, which ensures that information will be communicated between them and to ECOWAS-NMRA before any significant change is made to the site of manufacture, manufacturing procedure or quality control specifications of the API. Except as permitted by the ECOWAS guidelines relating to changes to medicines, such changes will not be made to the API to be used in manufacture of the medicine destined to be distributed in ECOWAS before written approval is granted by the NMRA.

I understand that the consequences of failure to obtain approval for changes where approval is necessary may include de-registration and recall of batches of medicines.

This APIMF (or data identical to that contained therein) has also been submitted to and approved by the regulatory authorities in (list of countries with stringent regulatory systems), and the [name(s) of ECOWAS

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NMR(s)] is authorized to request and refer to the evaluation reports of these agencies. The (ECOWAS-NMRA) is also authorized to exchange its own evaluation reports with these and other regulatory authorities.

Any questions arising from the ECOWAS-NMRA's evaluation of this APIMF should be forwarded to:

{Name and address}

Yours faithfully

{Signature of Company Representative}
{Name}
{Position in Company}
{Date}

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MODULE 2.3
QUALITY OVERALL SUMMARY: PRODUCT DOSSIER (QOS-PD)

See sections 1.5, 3 and 4 of "Guideline on submission of documentation for a multisource (generic) finished pharmaceutical product (FPP): quality part" for general and detailed instructions on the completion of this template.

INTRODUCTION

Summary of product information:

Non-proprietary name(s) of the finished
pharmaceutical product(s) (FPP)
Proprietary name(s) of the finished
pharmaceutical product(s) (FPP)
International non-proprietary name(s) of the
active pharmaceutical ingredient(s) (API(s)),

including form (salt, hydrate, polymorph)
Applicant name and address
Dosage form
Reference Number(s)
Strength(s)
Route of administration
Proposed indication(s)
Title:
Name:
Contact person responsible for this
Phone:
application
Fax:
Email:

If there are other contacts who should be routinely copied into correspondence for this application they should also be listed below.

1

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2.3.S DRUG SUBSTANCE (or ACTIVE PHARMACEUTICAL INGREDIENT (API)) (NAME, MANUFACTURER)
Complete the following table for the option that applies for the submission of API information:

Name of API:
Name of API manufacturer:
Confirmation of API Prequalification document:
■ a copy of the confirmation of API Prequalification document should be provided in
■ Module 1, and
■ summaries of the relevant information should be provided under the appropriate sections (e.g. S.1.3, S.2, S.3.1, S.4.1 through S.4.4, S.5 and S.7; see Quality guideline).
Certificate of suitability to the European Pharmacopoeia (CEP):
■ is a written commitment provided that the applicant will inform WHO in the event that the CEP is withdrawn and acknowledged that withdrawal of the CEP will require additional consideration of the API data requirements to support the dossier:
■ yes, ■ no;
■ ■ a copy of the most current CEP (with annexes) and written commitment should be provided in Module 1;
■ the declaration of access should be filled out by the CEP holder on behalf of the FPP manufacturer or applicant to PQTM who refers to the CEP; and
■ summaries of the relevant information should be provided under the appropriate sections (e.g. S.1.3, S.3.1, S.4.1 through S.4.4, S.5, S.6 and S.7; see Quality guideline).
Active pharmaceutical ingredient master file (APIMF):
■ a copy of the letter of access should be provided in Module 1; and
■
■ summaries of the relevant information from the Open part should be provided under the appropriate sections; see Section 3.2.S in the Quality guideline.
Active pharmaceutical ingredient pre-qualified by WHO
■
Provide evidence from WHO
Full details in the PD:
■ ■ Summaries of the full information should be provided under the appropriate sections; see Section 3.2.S in the quality guideline.

2.3.S.1 General Information (name, manufacturer)

2.3.S.1.1 Nomenclature (name, manufacturer)

(a) (Recommended) International Non-proprietary name (INN):

(b) Compendial name, if relevant:

(c) Chemical name(s):

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(d) Company or laboratory code:

(e) Other non-proprietary name(s) (e.g. national name, USAN, BAN):

(f) Chemical Abstracts Service (CAS) registry number:

2.3.S.1.2 Structure (name, manufacturer)

(a) Structural formula, including relative and absolute stereochemistry:

(b) Molecular formula:

(c) Relative molecular mass:

2.3.S.1.3 General Properties (name, manufacturer)

(a) Physical description (e.g. appearance, colour, physical state):

(b) Solubilities:

In common solvents:

Quantitative aqueous pH solubility profile (pH 1.2 to 6.8) at 37°C:

Medium (e.g. pH 4.5 buffer) Solubility (mg/ml)

Dose/solubility volume calculation:

(c) Physical form (e.g. polymorphic form(s), solvate, hydrate):

Polymorphic form:

Solvate:

Hydrate:

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(d) Other:

Property

pH

pK

Partition coefficients

Melting/boiling points

Specific optical rotation (specify solvent)

Refractive index (liquids)

Hygroscopicity

UV absorption maxima/molar absorptivity

Other

2.3.S.2 Manufacture (name, manufacturer)

2.3.S.2.1 Manufacturer(s) (name, manufacturer)

(a) Name, address and responsibility (e.g. fabrication, packaging, labelling, testing, storage) of each manufacturer, including contractors and each proposed production site or facility involved in these activities:

API-PQ

Name and address Responsibility number/APIMF/CEP

(including block(s)/unit(s)) number (if applicable)

(b) Manufacturing authorization for the production of API(s) and, where available, certificate of GMP compliance (GMP information should be provided in Module 1):

2.3.S.2.2 Description of Manufacturing Process and Process Controls (name, manufacturer)

(a) Flow diagram of the synthesis process(es):

(b) Brief narrative description of the manufacturing process(es):

(c) Alternate processes and explanation of their use:

(d) Reprocessing steps and justification:

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2.3.S.2.3 Control of Materials (name, manufacturer)

(a) Name of starting material:

(c) Name and manufacturing site address of starting material manufacturer(s):

(d) Summary of the quality and controls of the starting materials used in the manufacture of the API:

Step / Starting Material Test(s)/method(s) Acceptance criteria

(e) Where the API(s) and the starting materials and reagents used to manufacture the API(s) are without risk of transmitting agents of animal spongiform encephalopathies, a letter of attestation confirming this can be found in:

2.3.S.2.4 Controls of Critical Steps and Intermediates (name, manufacturer)

(a) Summary of the controls performed at critical steps of the manufacturing process and on intermediates:

Step/materials Test(s)/method(s) Acceptance criteria

2.3.S.2.5 Process Validation and/or Evaluation (name, manufacturer)

(a) Description of process validation and/or evaluation studies (e.g. for aseptic processing and sterilization):

2.3.S.2.6 Manufacturing Process Development (name, manufacturer)

(a) Description and discussion of the significant changes made to the manufacturing process and/or manufacturing site of the API used in producing comparative bioavailability or bioequivalence, stability, scale-up, pilot and, if available, production scale batches:

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2.3.S.3 Characterisation (name, manufacturer)

2.3.S.3.1 Elucidation of Structure and other Characteristics (name, manufacturer)

(a) List of studies performed (e.g. IR, UV, NMR, MS, elemental analysis) and conclusion from the studies (e.g. whether results support the proposed structure):

(b) Discussion on the potential for isomerism and identification of stereochemistry (e.g. geometric isomerism, number of chiral centres and configurations) of the API batch(es) used in comparative bioavailability or biowaiver studies:

(c) Summary of studies performed to identify potential polymorphic forms (including solvates): <including identification of and data on the API lot used in bioavailability studies>

(d) Summary of studies performed to identify the particle size distribution of the API: <including identification of and data on the API lot used in bioavailability studies>

(e) Other characteristics:

2.3.S.3.2 Impurities (name, manufacturer)

(a) Identification of potential and actual impurities arising from the synthesis, manufacture and/or degradation:

i. List of API-related impurities (e.g. starting materials, by-products, intermediates, chiral impurities, degradation products), including chemical name, structure and origin:

API-related impurity
(chemical name and Structure Origin descriptor)

ii. List of process-related impurities (e.g. residual solvents, reagents), including compound names and step used in synthesis:

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Process-related impurity (compound name) Step used in synthesis

(b) Basis for setting the acceptance criteria for impurities:

i. Maximum daily dose (i.e. the amount of API administered per day) for the API, corresponding to ICH Reporting/Identification/Qualification Thresholds for the API-related impurities and the concentration limits (ppm) for the process-related impurities (e.g. residual solvents):

Maximum daily dose for the API: <x mg/day>

Test Parameter ICH threshold or concentration limit

API-related impurities Reporting Threshold

Identification Threshold

Qualification Threshold

Process-related impurities <solvent 1>

<solvent 2>, etc.

ii. Data on observed impurities for relevant batches (e.g. comparative bioavailability or biowaiver, stability batches):

Impurity Acceptance Results (include batch number* and use**)

(API-related and Criteria process-related)

* include strength, if reporting impurity levels found in the FPP (e.g. for comparative studies)

** e.g. comparative bioavailability or biowaiver studies, stability

iii. Justification of proposed acceptance criteria for impurities:

2.3.S.4 Control of the API (name, manufacturer)

2.3.S.4.1 Specification (name, manufacturer)

(a) API specifications of the FPP manufacturer:

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Standard (e.g. Ph.Int., Ph.Eur., BP, USP, in-house)
Specification reference number and version

Analytical procedure

Test Acceptance criteria

(Type/Source/Version)

Description

Identification

Impurities

Assay

etc.

2.3.S.4.2 Analytical Procedures (name, manufacturer)

(a) Summary of the analytical procedures (e.g. key method parameters, conditions, system suitability testing):

2.3.S.4.3 Validation of Analytical Procedures (name, manufacturer)

(a) Summary of the validation information (e.g. validation parameters and results):

See 2.3.R Regional Information for summaries of the validation information (i.e. 2.3.R.2 Analytical Procedures and Validation Information).

Summarized tabulated methods and validation may be provided in a separate file <provide reference>.

2.3.S.4.4 Batch Analyses (name, manufacturer)

(a) Description of the batches:

Use (e.g. comparative

Date and

Batch number Batch size bioavailability or

site of production

biowaiver, stability)

(b) Summary of batch analyses release results of the FPP manufacturer for relevant batches

(e.g. comparative bioavailability or biowaiver, stability):

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Test Acceptance Results

Criteria <batch x> <batch y> etc.

Description

Identification

Impurities

Assay

etc.

(c) Summary of analytical procedures and validation information for those procedures

not previously summarized in 2.3.S.4.2 and 2.3.S.4.3 (e.g. historical analytical procedures):

2.3.S.4.5 Justification of Specification (name, manufacturer)

(a) Justification of the API specification (e.g. evolution of tests, analytical procedures and

acceptance criteria, differences from officially recognized compendial standard(s)):

2.3.S.5 Reference Standards or Materials (name, manufacturer)

(a) Source (including lot number) of primary reference standards or reference materials (e.g.

Ph.Int., Ph.Eur., BP, USP, in-house):

(b) Characterization and evaluation of non-official (e.g. not from an officially recognized

pharmacopoeia) primary reference standards or reference materials (e.g. elucidation of

structure, certificate of analysis):

(c) Description of the process controls of the secondary reference standard (comparative

certificate of analysis and IR spectra against primary standard):

2.3.S.6 Container Closure System (name, manufacturer)

(a) Description of the container closure system(s) for the shipment and storage of the API (including

the identity of materials of construction of each primary packaging component and a brief

summary of the specifications):

Specifications (list parameters e.g.

Packaging component Materials of construction

identification (IR))

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(b) Other information on the container closure system(s) (e.g. suitability studies):

2.3.S.7 Stability (name, manufacturer)

2.3.S.7.1 Stability Summary and Conclusions (name, manufacturer)

(a) Summary of stress testing (e.g. heat, humidity, oxidation, photolysis, acid/base): and results:

Results (e.g. including discussion whether mass

Stress condition Treatment

balance and peak purity are observed)

Heat

Humidity

Oxidation

Photolysis

Acid

Base

Other

(b) Summary of accelerated and long-term testing parameters (e.g. studies conducted):

Completed (and

Storage condition Batch Container closure

Batch size proposed) testing

(■C, % RH) number system

intervals

Summary of the stability results observed for the above accelerated and long-term studies:

Test (limits) Results

Description

Moisture

Impurities

Assay

etc.

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(c) Proposed storage statement and re-test period (or shelf-life, as appropriate):

Container closure system Storage statement Re-test period*

* indicate if a shelf-life is proposed in lieu of a re-test period (e.g. in the case of labile APIs)

2.3.S.7.2 Post-approval Stability Protocol and Stability Commitment (name, manufacturer)

(a) Stability protocol for Primary stability batches (e.g. storage conditions (including tolerances), batch numbers and batch sizes, tests and acceptance criteria, testing frequency, container closure system(s)):

Parameter Details

Storage condition(s) (■C, % RH)

Batch number(s) / batch size(s) <primary batches>

Tests and acceptance criteria Description

Moisture

Impurities

Assay

etc.

Testing frequency

Container closure system(s)

(b) Stability protocol for Commitment batches (e.g. storage conditions (including tolerances), batch numbers (if known) and batch sizes, tests and acceptance criteria, testing frequency, container closure system(s)):

Parameter Details

Storage condition(s) (■C, % RH)

Batch number(s) / batch size(s) <not less than three production batches>

Tests and acceptance criteria Description

Moisture

Impurities

Assay

etc.

Testing frequency

Container closure system(s)

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(c) Stability protocol for Ongoing batches (e.g. storage conditions (including tolerances), batch sizes and annual allocation, tests and acceptance criteria, testing frequency, container closure system(s)):

Parameter Details

Storage condition(s) (■C, % RH)

Annual allocation <at least one production batch per year (unless none is produced that year) in each container closure system >

Tests and acceptance criteria Description

Moisture

Impurities

Assay

etc.

Testing frequency

Container closure system(s)

2.3.S.7.3 Stability Data (name, manufacturer)

(a) The actual stability results should be provided in Module 3.

(b) Summary of analytical procedures and validation information for those procedures not previously summarized in 2.3.S.4 (e.g. analytical procedures used only for stability studies):

2.3.P DRUG PRODUCT (or FINISHED PHARMACEUTICAL PRODUCT (FPP))

2.3.P.1 Description and Composition of the FPP

(a) Description of the FPP (in signed specifications):

(b) Composition of the FPP:

i. Composition, i.e. list of all components of the FPP and their amounts on a per unit basis and percentage basis (including individual components of mixtures prepared in-house (e.g. coatings) and overages, if any):

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Component and quality Function Strength (label claim)

standard (and grade, if

applicable) Quant. % Quant. % Quantity %

per unit per unit per unit

or per mL or per mL or per mL

<complete with appropriate titles e.g. Core tablet (Layer 1, Layer 2, etc. as applicable), Contents of capsule,

Powder for injection>

Subtotal 1

<complete with appropriate title e.g. Film-coating >

Subtotal 2

Total

ii. Composition of all components purchased as mixtures (e.g. colourants, coatings, capsule shells, imprinting inks):

(c) Description of accompanying reconstitution diluent(s), if applicable:

(d) Type of container closure system used for the FPP and accompanying reconstitution diluent, if applicable:

2.3.P.2 Pharmaceutical Development

2.3.P.2.1 Components of the FPP

2.3.P.2.1.1 Active Pharmaceutical Ingredient

(a) Discussion of the:

i. compatibility of the API(s) with excipients listed in 2.3.P.1:

ii. key physicochemical characteristics (e.g. water content, solubility, particle size distribution, polymorphic or solid state form) of the API(s) that can influence the performance of the FPP:

iii. for fixed-dose combinations, compatibility of APIs with each other:

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2.3.P.2.1.2 Excipients

(a) Discussion of the choice of excipients listed in 2.3.P.1 (e.g. their concentrations, their characteristics that can influence the FPP performance):

2.3.P.2.2 Finished Pharmaceutical Product

2.3.P.2.2.1 Formulation Development

(a) Summary describing the development of the FPP (e.g. route of administration, usage, optimization of the formulation, etc.):

(b) Information on primary (submission, registration, exhibit) batches including comparative bioavailability or biowaiver, stability, commercial:

i. Summary of batch numbers:

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Batch number(s) of the FPPs used in

<e.g. bioequivalence batch A12345> <e.g.

Bioequivalence or biowaiver

biowaiver batch X12345>

For proportional strength biowaiver: the

bioequivalence batch of the reference strength

Dissolution profile studies

Stability studies (primary batches)

<packaging configuration I>

< packaging configuration II>

<Add/delete as many rows as necessary>

Stability studies (production batches)

< packaging configuration I>

< packaging configuration II>

(Add/delete as many rows as necessary)

Validation studies (primary batches) if available

< packaging configuration I>

< packaging configuration II>

(Add/delete as many rows as necessary)

Validation studies (at least the first three

consecutive production batches)

or code(s)/version(s) for process validation

protocol(s)

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ii. Summary of formulations and discussion of any differences:

Component and Relevant batches

quality standard Comparative Stability Process validation Commercial

(e.g. NF, BP, bioavailability or (2.3.P.1)

Ph.Eur, in-house) biowaiver

<Batch nos. and <Batch nos. and <Batch nos. and <Batch nos. and

sizes> sizes> sizes> sizes>

Theor. % Theor. % Theor. % Theor. %

quantity quantity quantity quantity

per batch per per per

batch batch batch

<complete with appropriate title e.g. Core tablet, Contents of capsule, Powder for injection>

Subtotal 1

<complete with appropriate title e.g. Film-coating >

Subtotal 2

Total

(c) Description of batches used in the comparative in vitro studies (e.g. dissolution) and in the in vivo studies (e.g. comparative bioavailability or biowaiver), including strength, batch number, type of study and reference to the data (volume, page):

(d) Summary of results for comparative in vitro studies (e.g. dissolution):

Summary of the multi-point dissolution profiles for the biobatch(es) in three BCS media across the physiological pH range and the proposed medium if different from the BCS media:

(e) Summary of any information on in vitro-in vivo correlation (IVIVC) studies (with cross-reference to the studies in Module 5):

(f) For scored tablets, provide the rationale/justification for scoring:

2.3.P.2.2.2 Overages

(a) Justification of overages in the formulation(s) described in 2.3.P.1:

2.3.P.2.2.3 Physicochemical and Biological Properties

(a) Discussion of the parameters relevant to the performance of the FPP (e.g. pH, ionic strength, dissolution, particle size distribution, polymorphism, rheological properties):

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2.3.P.2.3 Manufacturing Process Development

(a) Discussion of the development of the manufacturing process of the FPP (e.g. optimization of the process, selection of the method of sterilization):

(b) Discussion of the differences in the manufacturing process(es) for the batches used in the comparative bioavailability or biowaiver studies and the process described in 2.3.P.3.3:

2.3.P.2.4 Container Closure System

(a) Discussion of the suitability of the container closure system (described in 2.3.P.7) used for the storage, transportation (shipping) and use of the FPP (e.g. choice of materials, protection from moisture and light, compatibility of the materials with the FPP):

(b) For a device accompanying a multi-dose container, a summary of the study results demonstrating the reproducibility of the device (e.g. consistent delivery of the intended volume for the lowest intended dose):

2.3.P.2.5 Microbiological Attributes

(a) Discussion of microbiological attributes of the FPP (e.g. preservative effectiveness studies):

2.3.P.2.6 Compatibility

(a) Discussion of the compatibility of the FPP (e.g. with reconstitution diluent(s) or dosage devices, co-administered FPPs):

2.3.P.3 Manufacture

2.3.P.3.1 Manufacturer(s)

(a) Name, address and responsibility (e.g. fabrication, packaging, labelling, testing) of each manufacturer, including contractors and each proposed production site or facility involved in manufacturing and testing:

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Name and address Responsibility

(include block(s)/unit(s))

(b) Manufacturing authorization, marketing authorization and, where available, WHO-type certificate of GMP (GMP information should be provided in Module 1):

2.3.P.3.2 Batch Formula

Largest intended commercial batch size:

Other intended commercial batch sizes:

<information on all intended commercial batch sizes should be in the QOS-PD>

(a) List of all components of the FPP to be used in the manufacturing process and their amounts on a per batch basis (including individual components of mixtures prepared in-house (e.g. coatings) and overages, if any):

Strength (label claim)

Master production document

reference number and version

Proposed commercial batch size(s) (e.g.

number of dosage units)

Quantity per Quantity per Quantity per

Component and quality standard

batch (e.g. batch (e.g. batch (e.g.

(and grade, if applicable)

kg/batch) kg/batch) kg/batch)

<complete with appropriate titles e.g. Core tablet (Layer 1, Layer 2, etc. as applicable), Contents of

capsule,
Powder for injection>

Subtotal 1
<complete with appropriate title e.g. Film-coating >

Subtotal 2
Total

2.3.P.3.3 Description of Manufacturing Process and Process Controls

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(a) Flow diagram of the manufacturing process:

(b) Narrative description of the manufacturing process, including equipment type and working capacity, process parameters:

(c) Justification of reprocessing of materials:

2.3.P.3.4 Controls of Critical Steps and Intermediates

(a) Summary of controls performed at the critical steps of the manufacturing process and on isolated intermediates:

Step Controls (parameters/limits/frequency of testing)
(e.g. granulation, compression, coating)

Proposed/validated holding periods for intermediates (including bulk product):

2.3.P.3.5 Process Validation and/or Evaluation

(a) Summary of the process validation and/or evaluation studies conducted (including product quality review(s) where relevant) and/or a summary of the proposed process validation protocol for the critical steps or critical assays used in the manufacturing process (e.g. protocol number, parameters, results):

Document code(s) for the process validation protocol(s) and/or report(s) (including reference number/version/date):

2.3.P.4 Control of Excipients

2.3.P.4.1 Specifications

(a) Summary of the specifications for in-house standard specifications:

2.3.P.4.2 Analytical Procedures

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(a) Summary of the analytical procedures for supplementary tests:

2.3.P.4.3 Validation of Analytical Procedures

(a) Summary of the validation information for the analytical procedures for supplementary tests (where applicable):

2.3.P.4.4 Justification of Specifications

(a) Justification of the specifications (e.g. evolution of tests, analytical procedures and acceptance criteria, exclusion of certain tests, differences from officially recognized compendial standard(s)):

2.3.P.4.5 Excipients of Human or Animal Origin

(a) For FPPs using excipients without risk of transmitting agents of animal spongiform encephalopathies, a letter of attestation confirming this can be found in:

(b) CEP(s) demonstrating TSE-compliance can be found in:

2.3.P.4.6 Novel Excipients

Novel excipients are not accepted in PQTm. See quality guideline for definition.

2.3.P.5 Control of FPP

2.3.P.5.1 Specification(s)

(a) Specification(s) for the FPP:

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Standard (e.g. Ph.Int., BP, USP, in-house)

Specification reference number and version

Acceptance criteria Acceptance criteria Analytical procedure

Test

(release) (shelf-life) (type/source/version)

Description

Identification

Impurities
Assay
etc.

2.3.P.5.2 Analytical Procedures

(a) Summary of the analytical procedures (e.g. key method parameters, conditions, system suitability testing):

2.3.P.5.3 Validation of Analytical Procedures

(a) Summary of the validation information (e.g. validation parameters and results):

2.3.P.5.4 Batch Analyses

(a) Description of the batches:

Use (e.g. comparative
Strength and Date and
Batch size bioavailability or biowaiver,
batch number site of production
stability)

(b) Summary of batch analyses release results for relevant batches (e.g. comparative bioavailability or biowaiver, stability):

Acceptance Results

Test

criteria <batch x> <batch y> etc.

Description

Identification

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Acceptance Results

Test

criteria <batch x> <batch y> etc.

Impurities

Assay

etc.

(c) Summary of analytical procedures and validation information for those procedures not previously summarized in 2.3.P.5.2 and 2.3.P.5.3 (e.g. historical analytical procedures):

2.3.P.5.5 Characterisation of Impurities

(a) Identification of potential and actual impurities:

Degradation product

(code name, chemical name Structure Origin
and compendial name (e.g.

USP RC A) if relevant)

Process-related impurity (compound name) Step used in the FPP manufacturing process

(b) Basis for setting the acceptance criteria for impurities:

i. Maximum daily dose (i.e. the amount of API administered per day) for the API, corresponding ICH Reporting/Identification/Qualification Thresholds for the degradation products in the FPP and the concentration limits (ppm) for the process-related impurities (e.g. residual solvents):

Maximum daily dose for the API: <x mg/day>

Test Parameter ICH threshold or
concentration limit

Degradation product Reporting Threshold

Identification Threshold

Qualification Threshold

Process-related impurities <solvent 1>

<solvent 2>, etc.

ii. Data on observed impurities for relevant batches (e.g. comparative bioavailability or

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Acceptance Results

Impurity

criteria

(degradation product

<batch no.,

and process-related)

strength, use>

iii. Justification of proposed acceptance criteria for impurities:

2.3.P.5.6 Justification of Specification(s)

(a) Justification of the FPP specification(s) (e.g. evolution of tests, analytical procedures and acceptance criteria, differences from officially recognized compendial standard(s)):

2.3.P.6 Reference Standards or Materials

(a) Source (including lot number) of primary reference standards or reference materials (e.g. Ph.Int., Ph.Eur., BP, USP, in-house) not discussed in 3.2.S.5:

(b) Characterization and evaluation of non-official (e.g. not from an officially recognized pharmacopoeia) primary reference standards or reference materials (e.g. elucidation of structure, certificate of analysis) not discussed in 3.2.S.5:

(c) Description of the process controls of the secondary reference standard (comparative certificate of analysis and IR spectra against primary standard) not discussed in 3.2.S.5:

2.3.P.7 Container Closure System

(a) Description of the container closure systems, including unit count or fill size, container size or volume:

Description Strength Unit count or fill size Container size
(including materials of (e.g. 60s, 100s etc.) (e.g. 5 ml, 100 ml etc.)
construction)

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(b) Summary of specifications of each primary and functional secondary (e.g. foil pouches) packaging components:

Specifications

Packaging component (list parameters e.g. identification (IR))

HDPE bottle

PP cap

Induction sealed liners

Blister films (PVC, etc)

Aluminum foil backing

etc.

(c) Other information on the container closure system(s):

2.3.P.8 Stability

2.3.P.8.1 Stability Summary and Conclusions

(a) Summary of stress testing and results (e.g. photostability studies, cyclic studies, freeze-thaw studies, demonstration of stability-indication of purity/assay method(s)):

(b) Summary of accelerated and long-term testing parameters (e.g. studies conducted):

Completed (and

Storage conditions Strength and Container

Batch size proposed) test

(■C, % RH) batch number closure system

intervals

Summary of additional stability studies, if applicable (with reference to data location) <e.g. studies at intermediate conditions, holding period studies for intermediates and bulk product, transport studies, in-use studies>:

Summary of the stability results observed for the above accelerated and long-term studies:

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Test Results

Description

Moisture

Impurities

Assay

etc.

(c) Proposed storage statement and shelf-life (and in-use storage conditions and in-use period, if applicable):

Container closure system Storage statement Shelf-life

2.3.P.8.2 Post-approval Stability Protocol and Stability Commitment

(a) Stability protocol for Primary stability batches (e.g. storage conditions (including tolerances), batch numbers and batch sizes, tests and acceptance criteria, testing frequency, container closure system(s)):

Parameter Details

Storage condition(s) (■C, % RH)

Batch number(s) / batch size(s) <primary batches>

Tests and acceptance criteria Description

Moisture

Impurities

Assay

etc.

Testing frequency

Container closure system(s)

(b) Stability protocol for Commitment batches (e.g. storage conditions (including tolerances), batch numbers (if known) and batch sizes, tests and acceptance criteria, testing frequency, container closure system(s)):

Parameter Details

Storage condition(s) (■C, % RH)

Batch number(s) / batch size(s) <not less than three production batches in each container closure system>

Tests and acceptance criteria Description

Moisture

Impurities

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Parameter Details

Assay

etc.

Testing Frequency

Container Closure System(s)

(c) Stability protocol for Ongoing batches (e.g. storage conditions (including tolerances), number of batches per strength and batch sizes, tests and acceptance criteria, testing frequency, container closure system(s)):

Parameter Details

Storage condition(s) (■C, % RH)

Batch size(s), annual allocation <at least one production batch per year (unless none is produced that year) in each container closure system >

Tests and acceptance criteria Description

Moisture

Impurities

Assay

etc.

Testing frequency

Container closure system(s)

2.3.P.8.3 Stability Data

(a) The actual stability results should be provided in Module 3.

(b) Summary of analytical procedures and validation information for those procedures not previously summarized in 2.3.P.5 (e.g. analytical procedures used only for stability studies):

(c) Bracketing and matrixing design and justification for Commitment and/or Ongoing stability batches, if applicable:

2.3.A APPENDICES

2.3.A.1 Facilities and Equipment (name, manufacturer)

(a) Summary of information on facilities and equipment, in addition to the information provided in other sections of the submission: Not applicable.

2.3.A.2 Adventitious Agents Safety Evaluation (name, dosage form, manufacturer)

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(a) Summary of the information assessing the risk with respect to potential contamination with adventitious agents: Not applicable.

2.3.A.3 Excipients

(a) Summary of the details of manufacture, characterization and controls, with cross references to supporting safety data (nonclinical and/or clinical) for the novel excipients: Not applicable. Novel excipients are not accepted in PQTm. See quality guideline for definition.

2.3.R REGIONAL INFORMATION

2.3.R.1 Production Documentation

2.3.R.1.1 Executed Production Documents

(a) List of batches (including strengths) for which executed production documents have been provided (e.g. comparative bioavailability or biowaiver batches):

2.3.R.1.2 Master Production Documents

(a) The blank master production documents for each strength, proposed commercial batch size and manufacturing facility should be provided in Module 3.

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2.3.R.2 Analytical Procedures and Validation Information

ANALYTICAL PROCEDURES AND VALIDATION INFORMATION SUMMARIES

ATTACHMENT NUMBER:

HPLC Method Summary Volume/Page:

Method name:

Method code: Version and/or Date:

Column(s) / temperature (if other than ambient):

Mobile phase (specify gradient program, if applicable):

Detector (and wavelength, if applicable):
Flow rate:
Injection volume:
Sample solution preparation and concentration
(expressed as mg/ml, let this be termed "A"):
Reference solution preparation and concentration
(expressed as mg/ml and as % of "A"):
System suitability solution concentration
(expressed as mg/ml and as % of "A"):
System suitability tests (tests and acceptance criteria):
Method of quantification (e.g. against API or impurity
reference standard(s)):
Other information (specify):

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ATTACHMENT NUMBER:

Validation Summary Volume/Page:
Analytes:
Typical retention times (RT)
Relative retention times (RT_{Imp}/RT_{API} or Int. Std.):
Relative response factor (RF_{Imp}/RF_{API}):
Specificity:
Number of concentrations:
Range (expressed as %
"A"):
Linearity / Range:
Slope:
Y-intercept:
Correlation coefficient (r²) :
Conc.(s) (expressed as %
"A"):
Accuracy:
Number of replicates:
Percent recovery
(avg/RSD):
Conc.(s) (expressed as %
Precision /
"A"):
Repeatability:
Number of replicates:
(intra-assay precision)
Result (avg/RSD):
Precision /
Parameter(s) altered:
Intermediate Precision:
Result (avg/RSD):
(days/analysts/equipment)
Limit of Detection (LOD): (expressed as % "A")
Limit of Quantitation (LOQ): (expressed as % "A")
Stability of solutions:
Robustness:

Other variables/effects:
Typical chromatograms or spectra may be found in:
Company(s) responsible for method validation:
Other information (specify):

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QUALITY INFORMATION SUMMARY (QIS)
<Dossier reference number: e.g. HA999>

FOREWORD

The QIS template should be completed to provide a condensed summary of the key quality information for product dossiers (PDs) containing APIs of synthetic or semi-synthetic origin and their corresponding products that are filed with the Prequalification Programme.

The QIS constitutes part of the Prequalification PD. The QIS provides an accurate record of technical data in the PD at the time of prequalification and thereafter serves as an official reference document during the course of GMP inspections, variation assessments and requalification assessments as performed by WHO. The QIS is a condensed version of the Quality Overall Summary – Product Dossier (QOS-PD) and represents the final, agreed upon key information from the PD review (inter alia identification of the manufacturer(s), API/FPP specifications, stability conclusions and relevant commitments).

The QIS template is structured to permit the rapid assembly of the QIS by copying requisite information from the corresponding portions of the QOS-PD filed with the original PD. It is acknowledged that the numbering of the sections may not be entirely sequential. Those sections not considered necessary to be included in the QIS have been removed (e.g. 2.3.S.5 Reference Standards or Materials) and the remaining sections have

retained
their numbering to be consistent with the original PD.

For original PDs, the QIS should be provided in Word format at the time of PD submission. The QIS should be revised and submitted with the change history (see table at the end of the template) each time additional data is provided during the assessment process. If no revision is necessary due to no change in the information, a statement should be made to this effect in the covering letter. For variations and requalification dossiers, the QIS should be completed in its entirety (regardless of the proposed change), it should include information on all strengths, with any changes highlighted and it should be provided at the time of filing.

When completing the QIS template, this covering foreword should be deleted.

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QUALITY INFORMATION SUMMARY (QIS)

INTRODUCTION

(a) Summary of product information:
Non-proprietary name(s) of the finished
pharmaceutical product(s) (FPP)
Proprietary name(s) of the finished
pharmaceutical product(s) (FPP)
International non-proprietary name(s) of the
active pharmaceutical ingredient(s) (API(s)),
including form (salt, hydrate, polymorph)
Applicant name and address
Dosage form
Reference Number(s)
Strength(s)
Route of administration
Proposed indication(s)

Title:
Primary contact person responsible for this
First name:
application1
Family Name:
Contact person's job title
Contact person's postal address
Unit
Building/PO Box number
Road/Street
Plant/Zone
Village/suburb
Town/City
District and Mandal
Province/State
Postal code
Country
Contact person's email address
Contact person's phone number

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1 Please note that the contact listed in this form will be the primary contact for email and mail communication for this specific application.

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(b) Administrative Summary:
Applicant's date of preparation or revision of the
QIS
Internal version and/or date of acceptance (WHO use only)

Related dossiers (e.g. FPP(s) with the same API(s) submitted to the Prequalification Team: medicines (PQTm) by the applicant):
Reference API, strength, dosage form API manufacturer
Prequalified
number (eg. Abacavir (as sulphate) 300 (including address if same
(Y/N)
(e.g. HA998) mg tablets) supplier as current dossier)

2.3.S DRUG SUBSTANCE (or ACTIVE PHARMACEUTICAL INGREDIENT (API)) (NAME, MANUFACTURER)

Indicate which option applies for the submission of API information: <check one only>

Name of API:

Name of API manufacturer:

☒ Confirmation of API prequalification document

■ Certificate of suitability to the European Pharmacopoeia (CEP)

Active pharmaceutical ingredient master file (APIMF) procedure:

APIMF number assigned by WHO (if known): _____; version number(s) including amendments (and/or

■
date(s)) of the open part: _____; version number(s) including amendments (and/or date(s)) of the
restricted part: : _____.

Full details in the PD

■
Document version number/identifier of current module 3.2.S: _____

2.3.S.2 Manufacture (name, manufacturer)

2.3.S.2.1 Manufacturer(s) (name, manufacturer)

(a) Name, address and responsibility (e.g. fabrication, packaging, labelling, testing, storage) of
each manufacturer, including contractors and each proposed production site or facility involved in these
activities:

API-PQ number

Name and address Letter of access

Responsibility /APIMF/CEP number

(including block(s)/unit(s)) provided?

(if applicable)

2.3.S.2.3 Control of Materials (name, manufacturer) – for API option 4 only

(a) Name of starting material:

(b) Name and manufacturing site address of starting material manufacturer(s):

2.3.S.4 Control of the API (name, manufacturer)

2.3.S.4.1 Specification (name, manufacturer)

(a) API specifications of the FPP manufacturer:

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Standard (e.g. Ph.Int., Ph.Eur., BP, USP, in-house)

Specification reference number and version

Analytical procedure

Test Acceptance criteria

(Type/Source/Version)

Description

Identification

Impurities

Assay

etc.

2.3.S.6 Container Closure System (name, manufacturer)

(a) Description of the container closure system(s) for the storage and shipment of the API:

2.3.S.7 Stability (name, manufacturer)

2.3.S.7.1 Stability Summary and Conclusions (name, manufacturer)

(c) Proposed storage conditions and re-test period (or shelf-life, as appropriate):

Container closure system Storage statement Re-test period*

* indicate if a shelf-life is proposed in lieu of a re-test period (e.g. in the case of labile APIs)

2.3.P DRUG PRODUCT (or FINISHED PHARMACEUTICAL PRODUCT (FPP))

2.3.P.1 Description and Composition of the FPP

(a) Description of the FPP (in signed specifications):

(b) Composition of the FPP:

(i) Composition, i.e. list of all components of the FPP and their amounts on a per
unit basis and percentage basis (including individual components of mixtures prepared
in-house (e.g. coatings) and overages, if any):

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Component and Function Strength (label claim)

quality standard (and

grade, if applicable)

Quant. % Quant. % Quantity %

per unit per unit per unit

or per mL or per mL or per mL

<complete with appropriate titles e.g. Core tablet (Layer 1, Layer 2, etc. as applicable), Contents of

capsule,

Powder for injection>

Subtotal 1

<complete with appropriate title e.g. Film-coating >

Subtotal 2

Total

(ii) Composition of all components purchased as mixtures (e.g. colourants,
coatings, capsule shells, imprinting inks):

(c) Description of accompanying reconstitution diluent(s), if applicable:
2.3.P.2.2.1 Formulation Development
(b) Information on primary (submission, registration, exhibit) batches including comparative bioavailability or biowaiver, stability, commercial:
(i) Summary of batch numbers:

Batch number(s) of the FPPs used in
<e.g. bioequivalence batch A12345> <e.g. biowaiver
Bioequivalence or biowaiver
batch X12345>
For proportional strength biowaiver: the
bioequivalence batch of the reference strength
Dissolution profile studies
Stability studies (primary batches)
<packaging configuration I>
< packaging configuration II>
<Add/delete as many rows as necessary>
Stability studies (production batches)
< packaging configuration I>
< packaging configuration II>
(Add/delete as many rows as necessary)
Validation studies (primary batches)
< packaging configuration I>
< packaging configuration II>
(Add/delete as many rows as necessary)
Validation studies (at least the first three
consecutive production batches)
or code(s)/version(s) for process validation
protocol(s)

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Summary of formulations and discussion of any differences:
Component and Relevant batches
quality standard
Comparative Stability Process validation Commercial
(e.g. NF, BP,
bioavailability or (2.3.P.1)
Ph.Eur, in-house)
biowaiver
<Batch nos. and sizes> <Batch nos. and <Batch nos. and <Batch nos. and
sizes> sizes> sizes>
Theor. % Theor. % Theor. % Theor. %
quantity quantity quantity quantity
per batch per per per
batch batch batch
<complete with appropriate titles e.g. Core tablet (Layer 1, Layer 2, etc. as applicable), Contents of
capsule, Powder
for injection>

Subtotal 1
<complete with appropriate title e.g. Film-coating >

Subtotal 2
Total

2.3.P.3 Manufacture
2.3.P.3.1 Manufacturer(s)
(a) Name, address and responsibility (e.g. fabrication, packaging, labelling, testing) of each
manufacturer, including contractors and each proposed production site or facility involved in
manufacturing and testing:

Name and address
Responsibility
(include block(s)/unit(s))

2.3.P.3.2 Batch Formula
Largest intended commercial batch size:
Other intended commercial batch sizes:
<information on all intended commercial batch sizes should be in the QIS>
(a) List of all components of the FPP to be used in the manufacturing process and their
amounts on a per batch basis (including components of mixtures prepared in-house
(e.g. coatings) and overages, if any):

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Strength (label claim)

Master production document
reference number and/or version

Proposed commercial batch size(s) (e.g. number of dosage units)
Component and quality standard Quantity per batch Quantity per batch Quantity per batch
(and grade, if applicable) (e.g. kg/batch) (e.g. kg/batch) (e.g. kg/batch)
<complete with appropriate titles e.g. Core tablet (Layer 1, Layer 2, etc. as applicable), Contents of capsule, Powder for injection>

Subtotal 1
<complete with appropriate title e.g. Film-coating >

Subtotal 2
Total

2.3.P.3.3 Description of Manufacturing Process and Process Controls

(a) Flow diagram of the manufacturing process:
(b) Narrative description of the manufacturing process, including equipment type and working capacity, process parameters:

2.3.P.3.4 Controls of Critical Steps and Intermediates

(a) Summary of controls performed at the critical steps of the manufacturing process and on isolated intermediates:

Step
Controls (parameters/limits/frequency of testing)
(e.g. granulation, compression, coating)

Proposed/validated holding periods for intermediates (including bulk product):

2.3.P.3.5 Process Validation and/or Evaluation

(a) Summary of the process validation and/or evaluation studies conducted and/or a summary of the proposed validation protocol for the critical steps or critical assays used in the manufacturing process (e.g. protocol number, parameters, results):

Document code(s) for the process validation protocol(s) and/or report(s) (including reference number/version/date):

2.3.P.5 Control of FPP

2.3.P.5.1 Specification(s)

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(a) Specification(s) for the FPP:

Standard (e.g. Ph.Int., BP, USP, in-house)
Specification reference number and version
Acceptance criteria Acceptance criteria Analytical procedure
Test
(release) (shelf-life) (type/source/version)
Description
Identification
Impurities
Assay
etc.

2.3.P.7 Container Closure System

(a) Description of the container closure systems, including unit count or fill size, container size or volume:

Description
Unit count or fill size Container size
(including materials of Strength
(e.g. 60s, 100s etc.) (e.g. 5 ml, 100 ml etc.)
construction)

2.3.P.8 Stability

2.3.P.8.1 Stability Summary and Conclusions

(c) Proposed storage statement and shelf-life (and in-use storage conditions and in-use period, if applicable):

Container closure system Storage statement Shelf-life

2.3.P.8.2 Post-approval Stability Protocol and Stability Commitment

(a) Stability protocol for Primary stability batches (e.g. storage conditions (including tolerances), batch numbers and batch sizes, tests and acceptance criteria, testing frequency, container closure system(s)):

Parameter Details
Storage condition(s) (■C, % RH)
Batch number(s) / batch size(s) <primary batches>
Tests and acceptance criteria Description
Moisture
Impurities
Assay
etc.

Testing frequency
Container closure system(s)

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(b) Stability protocol for Commitment batches (e.g. storage conditions (including tolerances), batch numbers (if known) and batch sizes, tests and acceptance criteria, testing frequency, container closure system(s)):

Parameter Details

Storage condition(s) (■C, % RH)

Batch number(s) / batch size(s) <not less than three production batches in each container closure system>

Tests and acceptance criteria Description

Moisture

Impurities

Assay

etc.

Testing frequency

Container closure system(s)

(c) Stability protocol for Ongoing Batches (e.g. storage conditions (including tolerances), number of batches per strength and batch sizes, tests and acceptance criteria, testing frequency, container closure system(s)):

Parameter Details

Storage condition(s) (■C, % RH)

Batch size(s), annual allocation <at least one production batch per year (unless none is produced that year) in each container closure system >

Tests and acceptance criteria Description

Moisture

Impurities

Assay

etc.

Testing frequency

Container closure system(s)

2.3.P.8.3 Stability Data

(c) Bracketing and matrixing design for commitment and/or continuing (i.e. ongoing) batches, if applicable:

WRITTEN COMMITMENTS OF THE MANUFACTURER – FOR WHO USE

Important note: The product information is an essential part of the medicinal product. The SmPC and PIL published with the WHOPAR have been quality assured by WHO experts and reflect the situation at the time of publication of the WHOPAR. These texts, i.e. the SmPC and the PIL are prequalified and should be adhered to. Generally, a deviation from the prequalified product information (especially as to contents) means the product can no longer be considered to be prequalified.

API

If applicable (primary stability study commitment):

The Applicant (or API manufacturer) undertook in writing (date of letter of commitment) to continue long-term

testing of <INN of API> for a period of time sufficient to cover the whole provisional re-test period (period ending

month/year) and to report any significant changes or out-of-specification results immediately to WHO for the

following batches :

<Batch numbers, manufacturing dates, batch size, primary packing materials>

If applicable (commitment stability studies):

Since stability data on three production scale batches were not provided with the application, the remaining

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number of production scale batches should be put on long-term stability testing. Any significant changes or out-of-specification results should be reported immediately to WHO. The approved stability protocol should be used for commitment batches.

API option 2 – CEP

The Applicant provided a commitment in writing (date of letter of commitment) to inform WHO in the event that the CEP is revised or withdrawn, and that revisions to the CEP will be handled as per variation 5 (Annex

5, TRS

981). Note that revisions or withdrawal will require additional consideration of the API data requirements to

support the dossier.

API option 3 – full details in the PD (ongoing stability study commitment)

The Applicant undertook in writing (date of letter of commitment) a commitment regarding ongoing stability studies. Unless otherwise justified, at least one batch per year of the product will be included in the stability

programme (unless none is produced during that year). The stability protocol will be that which was approved for primary batches (or the protocol was submitted for assessment). Out-of-specification results or significant atypical trends will be investigated. Any confirmed significant change, out-of-specification result, or significant atypical trend will be reported immediately to WHO. The possible impact on batches on the market will be considered in consultation with WHO inspectors.

FPP

If applicable (primary stability study commitment):

The Applicant undertook in writing (date of letter of commitment) to continue long-term testing of < FPP reference number, trade name (INN of API), strength, pharmaceutical form> for a period of time sufficient to

cover the whole provisional shelf-life (period ending month/year) and to report any out-of-specification results

or significant changes immediately to WHO for the following batches :

<Batch numbers, manufacturing dates, batch size, primary packing materials >

If applicable (commitment stability studies):

Since stability data on three production scale batches was not provided with the application, the Applicant

undertook in writing, (date of letter of commitment) to put the remaining number <e.g. additional two (2)>

production scale batches of < FPP reference number, trade name (INN of API), strength, pharmaceutical form,

primary packing material> on long-term stability testing. Any out-of-specification results or significant changes

during the study will immediately be reported to WHO. The approved stability protocol will be used for commitment batches.

If applicable (when the proposed largest commercial batch size is 200 000 units (x units) or less)

The Applicant undertook in writing (date of letter of commitment) to place the first three batches of any production size larger than x units on stability. The stability protocol will be that which was approved for primary

batches (or the protocol was submitted for assessment). Out-of-specification results or significant atypical

trends will be investigated. Any confirmed significant change, out-of-specification result, or

significant atypical

trend will be reported immediately to WHO.

Ongoing stability study commitment

The Applicant undertook in writing (date of letter of commitment) a commitment regarding ongoing stability

studies. Unless otherwise justified, at least one batch per year of the product manufactured in every primary

packaging type will be included in the stability programme (unless none is produced during that year).

The stability protocol will be that which was approved for primary batches (or the protocol was submitted and found

acceptable). Out-of-specification results or significant atypical trends will be investigated. Any confirmed

significant change, out-of-specification result, or significant atypical trend will be reported immediately to WHO.

The possible impact on batches on the market will be considered in consultation with WHO inspectors.

If applicable (validation of production batches)

Validation data on production scale batches of not less than three (3) consecutive batches of <FPP reference

number, trade name (INN of API), strength, pharmaceutical form, primary packing material> was not provided

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with the application. Therefore, the Applicant submitted a written commitment (date of letter of commitment)

that three consecutive production batches would be prospectively validated and a validation report—in accordance with the details of the validation protocol provided in the dossier— would be made available

as

soon as possible for evaluation by assessors or for verification by the WHO inspection team.

Change History

Date of preparation of original QIS:

Date of revised Section (e.g.

Revision

version S.2.1)

QIS (2017-03-10)

<Dossier Reference Number(s) e.g. HA999>

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■

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PRESENTATION OF BIOEQUIVALENCE TRIAL INFORMATION

BIOEQUIVALENCE TRIAL INFORMATION

General Instructions:

Please review all the instructions thoroughly and carefully prior to completing the bioequivalence trial information form (BTIF). Neither the format nor the content of the document (text and tables) should be changed, except for setting horizontal page layout in subsections including wide tables.

Provide as much detailed, accurate and final information as possible. Note that the greyed areas are NOT to be completed in by the applicant but are for WHO use only.

Please state the exact location (Annex number) of appended documents in the relevant sections of the BTIF. For example, in section 3.4.3.1 under point b), indicate in which Annex (number) the Certificate of Analysis can be found. This procedure must be followed throughout the entire document where location of annexed documents is requested. Please ensure that the electronic submission has the same file structure and naming as the one employed to state the location of the documents and to include annexes of the BTIF as separate files.

Before submitting the completed BTIF, kindly check that you have provided all requested information and enclosed all requested documents.

Should you have any questions regarding this Form, please contact the WHO Prequalification Team - Medicines.

A properly filled out and signed original copy of the BTIF with all its annexes (including a copy on CD-ROM) must be submitted to the Prequalification of Medicines Programme together with the bioequivalence part of the dossier to the address below once the dossier has been accepted for assessment and the dossier has been allocated a WHO reference number. Note however a softcopy of the BTIF should be included already in the initial dossier submission to Geneva (please see Step 1 and Step 2 of the submission procedure http://www.who.int/prequal/info_applicants/info_for_applicants_dossier_SMF.htm).

CONFIDENTIAL

Attention: WHO Prequalification of Medicines Programme

Product Name:

UNICEF Supply Division

Oceanvej 10 - 12

2100 Copenhagen Ø

Denmark

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Application Form:

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Presentation of Bioequivalence Trial Information

■

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BIOEQUIVALENCE TRIAL INFORMATION

1 SUMMARY

1.1 Summary of bioequivalence studies performed

(Provide a brief description of each comparative bioavailability study included in the submission)

1.2 Tabulation of the composition of the formulation(s) proposed for marketing and those used for bioequivalence studies

(State the location of the master formulae in the quality part of the submission)

(Tabulate the composition of the biobatch using the table below. For solid oral dosage forms the table should contain only the ingredients in tablet core /contents of a capsule. A copy of the table should be filled in for the film coating / hard capsule, if any.

Important: If the formulation proposed for marketing and those used for bioequivalence studies are not identical, copies of this table should be filled in for each formulation with clear identification

in which bioequivalence study the respective formulation was used.)

Composition of the batches used for bioequivalence studies

Batch number

Batch size (number of unit doses)1

Comments, if any

Comparison of unit dose compositions and of clinical FPP batches

(duplicate this table for each strength, if compositions are different)

Unit dose Unit dose Biobatch Biobatch

Ingredients (and quality standard) Function

(mg) (%) (kg) (%)

Total

Equivalence of the compositions or justified differences

Maximum intended commercial batch size

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Bioequivalence batches should be at least of pilot scale (10% of production scale or 100,000 capsules/tablets whichever is the greater) and manufacturing method should be the same as for production scale.

Application Form:

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■

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2 CLINICAL STUDY REPORT

a) Study number:

b) Study title:

c) Location of study protocol:

d) Start and stop dates for each phase of the clinical study:

e) Dates of product administration:

2.1 ETHICS

a) State the name of review committee, date of approval of protocol and consent form and the location of approval letter in the submission

b) State location of a reference copy of the informed consent form

2.2 INVESTIGATORS AND STUDY ADMINISTRATIVE STRUCTURE

a) Name of principal investigator(s) (State location of c.v. in the submission)

b) Clinical Facility (Name and full mailing address)

c) Clinical Laboratories (Name and full mailing address)

d) Analytical Laboratories (Name and full mailing address)

e) Company performing pharmacokinetic/statistical analysis (Name and full mailing address)

2.3 STUDY OBJECTIVES

Briefly state the study objectives.

2.4 INVESTIGATIONAL PLAN

2.4.1 Overall study design and plan — description

(Describe the type of study design employed in 1-2 sentences)

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2.4.2 Selection of study population

2.4.2.1 Inclusion Criteria

(List the inclusion criteria applied to subjects)

2.4.2.2 Exclusion Criteria

(List the exclusion criteria applied to subjects)

2.4.2.3 Health Verification

(State location of the individual data included in the submission)

a) List criteria used and all tests performed in order to judge health status

b) Indicate when tests were performed

c) Study site normal values

(State location in submission of study site normal values for blood clinical chemistry, haematology, and urinalysis clinical screen)

d) Report any results that were outside of study site normal values

(State location in submission of the summary of anomalous values)

2.4.2.4 Removal of Trial subjects from Trial or Assessment

a) Number of subjects enrolled in the study

(All subjects including alternates, withdrawals, and dropouts)

b) Alternates

(Please note: Generally all subjects enrolled in the study should be included in the data set i.e., alternate subjects are strongly discouraged. However, in cases where there are alternate subjects, describe the procedure of including/excluding the alternates and whether alternates have been included in the study)

c) Withdrawals/dropouts

(Identify each withdrawal/dropout by subject and provide the reason for withdrawal/dropout and at what point in the study the withdrawal/dropout occurred)

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2.4.3 Products Administered

2.4.3.1 Test Product

a) Batch number, size, date of manufacture and expiry date for the test product

b) Potency (measured content) of test product as a percentage of label claim as per validated assay method

(This information should be cross-referenced to the location of the certificate of analysis in the submission)

2.4.3.2 Comparator (Reference) Product

(Append to this template a copy of product labelling (snap shot of the box, on which the name of the product, name and address of the manufacturer, batch number, and expiry date are clearly visible on the labelling))

a) Name and manufacturer of the comparator product and market where the comparator product was purchased

b) Batch number and expiry date for the comparator product

c) Purchase, shipment, storage of the comparator product
(Indicate from which company/pharmaceutical distributor the comparator product has been obtained. Clearly indicate in chronological order the steps and dates of shipment/transport from company of purchase to the study site. In addition, the storage conditions should be given. This information should be cross-referenced to location in submission of documents (e.g. receipts) proving conditions)

d) Potency (measured content) of the comparator product as a percentage of label claim, as measured by the same laboratory and under the same conditions as the test product
(This information should be cross-referenced to the location of the certificate of analysis in the submission)

e) Justification of choice of comparator product
(Provide short summary here and cross-reference to location of comprehensive justification in study protocol)

2.4.4 Selection of doses in the study

a) State dose administered

(Indicate the number of dosage units comprising a single dose, e.g., 400 mg as 1 x 400 mg or 2 x 200 mg tablets)

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2.4.5 Selection and Timing of Dose for Each Subject

a) State volume and type of fluid consumed with dose

b) Interval between doses (i.e., length of washout)

c) Protocol for the administration of food and fluid

d) Restrictions on posture and physical activity during the study

2.4.6 Blinding

2.4.6.1 Identify which of the following were blinded. If any of the groups were not blinded, provide a justification for not doing so

a) study monitors: Yes ☐ / No ☐ If No, justify:

b) subjects: Yes ☐ / No ☐ If No, justify:

c) analysts: Yes ☐ / No ☐ If No, justify:

2.4.6.2 Identify who held the study code and when the code was broken

2.4.7 Drug Concentration Measurements

2.4.7.1 Biological fluid(s) sampled

2.4.7.2 Sampling protocol

a) Number of samples collected per subject

b) Volume of fluid collected per sample

c) Total volume of fluid collected per subject per phase of the study

d) List the study sampling times

e) Identify any deviations from the sampling protocol

(State location of summary in the submission)

(Describe and explain reasons for deviations from sampling protocol. Comment on impact on study. Indicate whether the deviations were accounted for in the pharmacokinetic analysis)

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2.4.7.3 Sample Handling

a) Describe the method of sample collection

b) Describe sample handling and storage procedures

2.5 Comments from review of Section 2 – WHO use only

3 TRIAL SUBJECTS

3.1 Demographic and other baseline characteristics

a) Identify study population (i.e., normal, healthy adult volunteers or patients)

b) Summary of ethnic origin and gender of subjects

- c) Identify subjects noted to have special characteristics and state notable characteristics (e.g. fast acetylators of debrisoquine)
- d) Range and mean age ■ SD of subjects
- e) Range and mean height and weight ■ SD of subjects
- f) Identify subjects whose ratio is not within 15% of the values given on a standard height/weight table

3.2 Subjects who smoke

- a) Number of smokers included in the study
- b) Indicate how many cigarettes smoked per day per subject
- c) Comment on the impact on study

3.3 Comments from review of Section 3 – WHO use only

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4 PROTOCOL DEVIATIONS

- 4.1 Protocol deviations during the clinical study
(Describe any such deviations and discuss their implications with respect to bioequivalence)

4.2 Comments from review of Section 4 – WHO use only

5 SAFETY EVALUATION

- 5.1 Identify adverse events observed
(List any adverse events by subject number. State whether a reaction occurred following administration of the test or reference product, identify any causal relationships, and note any treatments required. State location of this summary in the submission.)
(Discuss the implications of the observed adverse events with respect to bioequivalence.)

5.2 Comments from review of Section 5 – WHO use only

6 EFFICACY EVALUATION

EFFICACY RESULTS AND TABULATIONS OF INDIVIDUAL TRIAL SUBJECTS
DATA

- 6.1 Presentation of data
 - a) State location in submission of tables of mean and individual subject concentrations
 - b) State location in submission of (mean and individual) linear and semi-logarithmic subject drug concentration vs. time plots

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- 6.2 Pharmacokinetic (PK) parameters
 - a) State how the pharmacokinetic parameters were calculated/obtained for AUC_{0-t}, AUC_{0-∞}, C_{max}, t_{max}, the elimination rate constant, and t_{1/2} (indicate location of description in protocol)
 - b) State whether actual sampling time points were used for estimation of the pharmacokinetic parameters
 - c) Complete the table below

Test	Reference
Interindividual	Interindividual
Arithmetic Standard	Arithmetic Standard
Parameter coefficient of variation (%)	coefficient of variation (%)
AUC _{0-t} (units)	
AUC _{0-∞} (units)	
C _{max} (units)	
t _{max} (units)	
t _{1/2} (units)	

- d) Ratio of AUC_{0-t} to AUC_{0-∞}
(State mean ratio for both test and reference, state location in submission where individual ratios can be found)

- 6.3 Statistical analysis
(State the method of calculation of the 90% confidence intervals for the ratio of test formulation over the reference formulation and indicate how treatment, period, sequence and subjects within

sequence were included as factors in the ANOVA. Provide the following results from the ANOVA (parametric) on the logarithmically transformed AUC0-t and CMAX and other relevant parameters. State software used for computing ANOVA.)

a) Geometric means, results from ANOVA, Degrees of Freedom (DF) and derived CV (intra-subject)

% Ratio of 90 %

Parameter Test Reference geometric Confidence DF CV (%)
means interval

AUC0-t (units)

AUC0-inf (units)

Cmax (units)

b) Comparison of the results

(Compare the results, including mean values, inter- and intra-individual variability, of this study with published results (literature, product information of reference product (innovator), WHOPARs), and copies of the references used should be appended to this document)

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6.4 Discussion of results

(State location of the discussion of the results in the submission)

6.5 Comments from review of Section 6 – WHO use only

7 ANALYTICAL VALIDATION REPORT

7.1 Analytical technique

7.1.1 Validation protocol

(State the location of the validation protocol)

7.1.2 Identify analyte(s) monitored

7.1.3 Comment on source and validity of reference standard

7.1.4 Identify internal standard

7.1.5 Comment on source and validity of internal standard

7.1.6 Identify method of extraction

7.1.7 Identify analytical technique or method of separation employed

7.1.8 Identify method of detection

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7.1.9 Identify anticoagulant used (if applicable)

7.1.10 If based on a published procedure, state reference citation

7.1.11 Identify any deviations from protocol

7.2 Selectivity

(Address the methods to verify selectivity against endogenous/exogenous compounds & results)

7.3 Sensitivity

(Address the methods to verify sensitivity & results)

7.4 Carry-over

(Summarize the method to verify carry-over & results)

7.5 Standard curves

(State location in submission of tabulated raw data and back calculated data with descriptive statistics)

a) List number and concentration of calibration standards used

b) Describe the regression model used including any weighting

c) List the back-calculated concentrations of the calibration standards of the validation runs (highlight the values outside of the acceptance range, e.g., 15%, except 20% for LLOQ)

7.6 Quality control samples

a) Identify the concentrations of the QC samples and the storage conditions employed prior to their analysis

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7.7 Precision and accuracy during validation

- a) Summarize inter-day/inter-run accuracy and precision of the calibration standards during assay validation
- b) Summarize inter-day/inter-run accuracy and precision of the calibration standards during assay re-validation
(If applicable)
- c) Summarize inter-day/inter-run and intra-day/intra-run accuracy and precision of the QC samples during assay validation
- d) Summarize inter-day/inter-run and intra-day/intra-run accuracy and precision of the QC samples during assay re-validation
(If applicable)

7.8 Dilution integrity

(Summarize the method to verify dilution integrity & results)

7.9 Matrix effect (in case of MS detection)

(Summarize methods to verify the matrix effect & results)

7.10 Stability

(For each section provide the location of the raw data, a description of the methodology employed and a summary of the data.)

- a) Summarize data on long-term storage stability
- b) Summarize data on freeze-thaw stability
- c) Summarize data on bench top stability
- d) Summarize data on auto-sampler storage stability

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- (e) Summarize data from any other stability studies conducted
(e.g. long-term stock solution and working solution stability, short-term stock solution and working solution stability, dry-extract stability, wet-extract stability, stability in blood before sample processing)

7.11 Re-injection reproducibility

(Summarize the method to verify re-injection reproducibility & results)

7.12 Comments from review of Section 7 – WHO use only

8 BIOANALYTICAL STUDY REPORT

(State the location of the bioanalytical report for the analysis of the study subject samples)

8.1 Analytical technique

(Confirm whether the method is the same as the validated method and whether the same equipment was employed. Identify any differences between the validated method described above in Section 7 and the method employed for subject sample analyses)

8.1.1 Analytical protocol

(State the location of the analytical protocol)

8.1.2 Identify any deviations from protocol

8.1.3 Dates of subject sample analysis

8.1.4 Longest period of subject sample storage

(Identify the time elapsed between the first day of sample collection and the last day of subject sample analysis)

8.1.5 State whether all samples for a given subject were analysed together in a single analysis run

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8.2 Standard curves

(State location in submission of tabulated raw data and back calculated data with descriptive statistics)

- a) List number and concentration of calibration standards used

- b) State number of curves run during the study (valid and failed runs, including reasons of failure).
- c) Summarize descriptive data including slope, intercept, correlation coefficients
- d) List the back-calculated concentrations of the calibration standards of the study runs (highlight the values outside of the acceptance range, e.g., 15%, except 20% for LLOQ)

8.3 Quality control samples

- a) Identify the concentrations of the QC samples, their date of preparation and the storage conditions employed prior to their analysis
- b) State the number of QC samples in each analytical run per concentration
- c) List the back-calculated concentrations of the QC samples of the study runs (highlight the values outside of the acceptance range, e.g., 15%)
- d) Discuss whether the concentrations of the QC sample concentrations are similar to the concentrations observed in the study samples
- e) State the percentage of QC samples per run with respect to the total number samples assayed in each run

8.4 Precision and accuracy

- a) Summarize inter-day precision of back-calculated standards and inter-day and intra-day precision and accuracy of QC samples analysed during subject sample analysis

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8.5 Repeat analysis (re-analysis, re-injection and re-integration)

- a) List re-analysed samples by sample identification and include the following information for each re-analysis: initial value; reason for re-analysis; re-analysed value(s); accepted value; and reason for acceptance
- b) Report the number of re-analysis as a percentage of the total number samples assayed
- c) List re-injected samples by sample identification and include the following information for each re-injection: initial value; reason for re-injection; re-injected value; accepted value; and reason for acceptance
- d) Report the number of re-injections as a percentage of the total number samples assayed
- e) List re-integrated chromatograms by sample identification and include the following information for each re-integration: initial value; reason for re-integration; re-integrated value(s); accepted value; and reason for acceptance
- f) Report the number of re-integrated chromatograms as a percentage of the total number of samples assayed

8.6 Incurred sample reanalysis

(State location in the submission and summarize the results of incurred sample reanalysis, including the number of subject samples included in ISR and the total number of samples analysed in the study)

8.7 Chromatograms

(State the location in the submission where the sample chromatograms can be found. The chromatograms should be obtained from a minimum of two analytical batches and include at least 20% of the subjects, up to a maximum of five. A complete set includes standards, QC samples, pre-dose and post-dose subject samples for both phases. Each chromatogram should be clearly labelled with respect to the following: date of analysis; subject ID number; study period; sampling time; analyte; standard or QC, with concentration; analyte and internal standard peaks; peak heights and/or areas)

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8.8 Comments from review of Section 9 – WHO use only

9 QUALITY ASSURANCE

9.1 Internal quality assurance methods

(State locations in the submission where internal quality assurance methods and results are described for each of study sites (see 3.2 b-d.)

9.2 Monitoring, auditing, inspections

(Provide a list of all monitoring and auditing reports of the study, and of recent inspections of study sites by regulatory agencies. State locations in the submission of the respective reports for each study site (see 3.2 b-d.)

9.3 Comments from review of Section 10 – WHO use only

10.0 CONCLUSIONS AND RECOMMENDATIONS – WHO use only

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BULK SUPPLY AGREEMENT

THIS AGREEMENT is made on this on ----- by and between the following entities collectively referred to as the parties and each referred to individually as party. Indian Immunologicals Limited, a Company incorporated under the Indian Companies Act, having its registered office at -----, India (Here in after referred to as 'Supplier' which expression shall, unless repugnant to the context or meaning thereof be deemed to include its successors and assignees).

AND

-----Pharmaceuticals Company, a company duly incorporated and in existence in accordance with the laws of -----, with registered office ----- register #----- (hereinafter referred to as 'Purchaser' which expression shall, unless repugnant to the context or meaning thereof be deemed to include its successors and assignees)

WHEREAS

a.----- is engaged in the research, manufacture, sale and distribution of biological / pharmaceutical products for animal and human health in India and other Countries.

b. Purchaser is engaged in Importing, exporting, sale, manufacturing, registering and distribution of pharmaceutical products and biologicals in the Appointed Territory.

c. Purchaser is desirous of importing Ready to Fill Bulk product and thereafter undertake activities for, filling bulk, formulating, packing, registering and selling the finished products in the Appointed Territory, and to this end has requested -----, for supplying the Ready to Fill Bulk along with related documentation support to register the Ready to Fill Bulk in the appointed territory. The agreement covers only supply of Ready to Fill Bulk as Product and do not involve supply of any technical knowhow for formulating the finished product therein, and if any it shall be maintained as "Technical knowhow" assets of -----and 2nd party "Purchaser" is not authorized to use without prior conditioned approval from -----, Formulating and technology transfer can be agreed on later if the 2 parties agree on this in a separate annex.

d. ----- has accepted the said offer and the Parties wish to reduce the terms and covenants of the said offer and acceptance in writing.

NOW THEREFORE THIS AGREEMENT WITNESSETH and it is hereby agreed by and between the parties hereto as follows:

DEFINITIONS

'Appointed Territory' shall mean only the territory of '-----'.

Ready to Fill Bulk' shall mean preparations in bulk of vaccines in unfinished or semi-finished forms, their derivatives, reconstitutions, supplements and replacements, supplied by -----more particularly described in the Schedule.

Confidential Information' shall mean any and all information relating to the products of including but not limited to, technology, specifications, samples, specimens, cultures, know-how, trade secrets, processes, procedures, innovations, ideas, concepts, inventions, developments, improvements, laboratory analysis, results of any test, protocols, research records, data, software programs, schematics, designs, business methods, pricing, cost, suppliers, actual products, product concepts, product formulations, methods for the manufacture and use of the materials, manufacturing and customer information and all other information that is indicated as confidential, whether of a written or oral or visual nature. For the purpose of this Agreement, Confidential Information shall include such of the results, data and information generated while performing any obligation pursuant to this Agreement.

"Finished Product(s)" shall mean the vaccines manufactured by the Purchaser out of the Bulk Ready to Fill Bulk supplied by -----.

APPOINTMENT

1. Supplier appoints the Purchaser for a period of 3 years to import the Ready to Fill Bulk to store, formulate, pack and make finished product, and to sell and distribute the finished product in the Appointed Territory. The agreement between the parties shall be on a Non Exclusive basis.

SUPPLY

2. Purchaser shall raise the order for supply of Ready to Fill Bulk, to be delivered by the Supplier to the Purchaser. Purchaser shall be responsible for importing the Ready to Fill Bulk into Territory directly. Supplier will execute all orders placed by Purchaser on the explicit understanding that these are supplied on non-returnable basis. Once the order is placed on Supplier and accepted by it, the same can only be changed / cancelled by the Supplier.

3. Purchaser shall store the Ready to Fill Bulk, get the same formulated, filled and packed as Finished Products, and in this regard it shall follow all the applicable laws of the Appointed Territory, guidelines issued by Supplier from time to time. The Trade Marks or any Brand names the Purchaser consider to adopt shall not be in any manner similar or infringing Supplier's brand names or Trade Marks.

4. Purchaser shall follow the procedure as laid down by ----- and other statutory requirements of Appointed Territory, for preparation, storage, filling and packing of the Finished Products. Purchaser shall be solely responsible for any loss or damages arising out of or as a result of any deviation or deficiency in this regard.

5. Purchaser shall intimate any deficiency in the Ready to Fill Bulk within thirty (30) days of receipt. The parties hereby acknowledge that ----- shall not be responsible for any deficiency identified after the said period of thirty (30) days.

6. Purchaser shall not sell the Ready to Fill Bulk or Finished Products directly or indirectly outside the Appointed Territory.

7. Purchaser shall not make any representations or claims with respect to the Ready to Fill Bulk which are false or misleading in any material particular; they shall store and handle the Ready to Fill Bulk with all due care and diligence appropriate for the circumstances and in accordance with the guidelines, standards, codes of practice and regulations governing the storage and handling of therapeutic / pharmaceutical goods within the Appointed Territory including without limitation, taking such steps as are necessary to ensure that the Ready to Fill Bulk are not stolen or tampered with.

8. ----- shall deliver the Ready to Fill Bulk as per the agreed schedule.

9. For each shipment ----- shall provide to the Purchaser a Certificate of Analysis from Its Quality Assurance Department and any other documents that may be requested by local authorities to facilitate importation and local manufacturing process.

10. Purchaser shall start local quality testing as per supplier specification and method of analysis Within 30 (Thirty) working days of shipment arrival and results to be shared with supplier. In case of any quality defect both parties agreed to define the liability & Support re-testing if required by neutral lab.

FINANCIALS

11. All orders to be executed as above shall be subject to final confirmation for availability and price with ----- . The prices will be FOB and shall be expressed in US Dollars. The prices valid for the duration of the agreement and any extension thereafter and the minimum order quantity are as specified in the Schedule

1. The cost of customs clearance storage and other allied operations, if applicable, will be on account of the Purchaser.

12. The Distributor shall pay 50% at the time of order confirmation and 50% on or before delivery. Further both parties agreed to proceed with LC payment mode upon successful completion of 12 months and after achieving 1, 00,000 USD annual business.

CONFIDENTIALITY

13. Purchaser recognize and acknowledge that they will have access to Confidential Information that are valuable assets of _____. Purchaser covenant that they, including their employees, representatives or agents, shall not use the Confidential Information for any other purpose or disclose Confidential Information to any third party. In the event of a breach or threatened breach _____ shall be entitled to pursue any available remedy, including equitable remedies, against the Purchaser.

14. Purchaser shall provide Confidential Information only to those employees, representatives or agents who reasonably require access to such information, provided that such employees, representatives or agents are bound by confidentiality obligations no less stringent than those set forth in the present Agreement. Purchaser shall advise such employees, representatives and agents of the terms and conditions of this Agreement including, without limitation, the obligations to not to disclose to any other party. Purchaser shall be responsible for any breach of those obligations by any such employee, representatives or agent.

15. Purchaser agree that if they are provided with custody of any Confidential Information belonging to _____ that shall be retained during the tenure of the Agreement and shall be returned to _____ immediately at the end of the Agreement or at any time as may be requested by _____.

INTELLECTUAL PROPERTY RIGHTS

16. All right, title and interest in the Ready to Fill Bulk and Intellectual Property (which term shall include Confidential Information) relating thereto, except for the Trade Marks / Brand Names of the Finished Products which shall be owned by the Purchaser, whether oral or tangible, shall be owned by _____.

17. Purchaser hereby acknowledge Ill's ownership of the Intellectual Property rights and covenants with _____ that it shall not use the said Intellectual Property rights in any manner whatsoever, which may jeopardize or dilute _____'s sole and exclusive right, title and interest in them or their significance, distinctiveness or validity. Purchaser shall not claim any ownership of, or right to or interests in the said Intellectual Property rights.

18. Purchaser agree to promptly and fully disclose in writing all inventions, whether or not patentable, results, technology, improvements, discoveries, developments, original works of authorship, trade secrets, or intellectual property rights therein, that are conceived, developed, or reduced to practice relating to the said Intellectual Property and shall be absolutely owned by _____. Purchaser shall provide necessary assistance to _____ for protecting them. Further, Purchaser shall ensure that all employees, representatives or agents assign all such inventions, improvements etc., to _____ and to this effect Purchaser shall obtain a written obligation from such employees, representatives or agents.

19. Purchaser shall use the best endeavors to discover any infringement of Intellectual Properties coming to their knowledge and shall assist _____ in any action to be taken in such circumstances.

REGISTRATION AND APPROVALS

20. Purchaser shall be responsible for obtaining the requisite registrations, approvals, licenses, permits, etc., which are required for importing, storing, preparing/formulating, packing and distributing the Ready to Fill Bulk or Finished Product, as per the local laws of the Appointed Territory. All expenses in this regard shall be borne by Purchaser only.

21. _____ shall provide such documentation support as may be necessary to assist the Purchaser obtain the said registrations, approvals etc.

22. All expenses for conducting of trials, analysis, tests and for registration of the Ready to Fill Bulk or Finished Product in the Appointed Territory shall be borne by Purchaser. All expenses for registration pertaining to preparation of files, certificates, documents and samples, which are carried out in India, will be borne by _____.

WARRANTY AND INDEMNIFICATION

23. This Agreement has been duly authorized, executed and delivered and constitutes its legal, valid and binding obligation, enforceable in accordance with its terms and do not conflict with or constitute a breach or default of, its Articles of Incorporation (or equivalent instrument), By-Laws or any resolution of its Board of Directors, or any contract or other instrument by which it or its assets is bound, or any law or regulation or any judgment, order, license or permit issued by any government authority.

24. Purchaser warrant and represent that they are independently competent to, and has fully satisfied themselves of, the worth, use, composition, efficiency, compliance with health and other laws, and such other characteristics and features of the Bulk Ready to Fill Bulk or Finished Product as are relevant and material to wide and safe use, distribution and storage and application of the Bulk Ready to Fill Bulk or Finished Product, and that in doing so, they have not relied on any representation or warranty of _____, but has conducted their own independent inspection, analysis and tests, consulted with such health experts and legal counsel, as they deem necessary or appropriate for purposes of purchase, formulate, fill and sale of the Ready to Fill Bulk or Finished Product in the Territory.

25. Purchaser hereby indemnifies and holds harmless from / against any and all damages, losses, costs and expenses arising out of any suit, action or proceeding relating in any manner to import, formulate, fill, sale etc. of the Ready to Fill Bulk or Finished Product or performances of obligations herein mentioned including those relating to infringement of intellectual property rights of third parties.

26. Purchaser at its own cost and expenses shall keep all of the ready to fill bulk or finished product products covered under appropriate product liability insurance.

27. _____ warrants that the Ready to Fill Bulk supplied by it shall meet the requirement referred to technical specifications and International quality standards. _____ shall indemnify and hold harmless the Purchaser against any loss and damage resulting from manufacturing deficiencies in the Ready to Fill Bulk, which are attributable to its inherent manufacturing defect / process. Purchaser shall conduct analysis and tests, as may be necessary and inform _____, within thirty (30) working days of receipt of the Bulk Ready to Fill Bulk, of any manufacturing deficiencies. It is hereby understood that indemnity from _____ shall only be valid till the final lab results issued by related local authority allowing use of bulk (Refer to point 10).

28. Subject to the above Clauses, all the parties mutually agree, hereafter to indemnify, defend, and hold the other party and its directors, employees, representatives, officers and agents and their respective successors, heirs and assigns fully free and harmless against any liability, damage loss, or expense including reasonable attorneys fees and expenses incurred by or imposed upon any of the parties hereto in respect of the Indemnities, which shall arise out of the Intellectual Property rights manifested or owned by the parties hereunder in connection with any claims, suits, actions, demands or judgements arising out of any and all liability (including without limitation actions in the form of tort, product liability) warranty, or strict liability or actions based on statements made by or assurances held out with respect to the Product thereto as contemplated hereunder concerning any product, process, or service that is made, used, sold, imported or performed pursuant to violations of any

contractual or statutory central, state and other local laws in territory by either party hereto.

RECORDS AND DOCUMENTS

29. Purchaser shall keep records of their business relating to Ready to Fill Bulk or Finished Product and will share the same with _____ to the extent as agreed by the both parties from time to time during regular business hours. _____ or its authorized representative may examine such records.

FORCE MAJEURE

30. All the parties agree that any delay, breach or failure on the part of either party in complying with the terms of this Agreement shall not be treated as default or give rise to any claim for damage to or in favour of other party, if and to the extent, such delay, breach or failure is caused by occurrences of any event beyond the control of the party including but not limited to acts of God, fires, floods, explosions, war, civil commotion, strikes, lockouts, shortages of materials and other similar causes.

TERMINATION

31. Subject to the other terms herein, any party hereto may terminate this Agreement as per clause 33. In the event of termination, the obligations of each party as on the date of termination shall be valid and binding.

32. Purchaser shall upon such termination return to _____ the complete documentation and other informative material (including those relating to scientific or technical data about the Bulk Ready to Fill Bulk or Finished Product) which they had received before or during the period of validity of this Agreement or which is maintained by it pursuant to this Agreement or otherwise, without exception and explicitly undertakes that whatever Confidential Information / secret is imparted to the Purchaser shall not be divulged.

33. Either Party may terminate this Agreement by sending a written notice to the other Party at least 3 (three) months prior to such intended date of termination if:

- a) The other Party is in breach of any of its material obligations under this Agreement which, in the case of a breach capable of remedy, shall not have been remedied within sixty (60) days of that other Party being served with written notice to do so; or
- b) The other Party ceases to carry out its business, becomes insolvent, applies for or agrees to the appointment of a trustee, receiver or liquidator of its assets or voluntarily seeks relief under any law for the aid of debtors; or
- c) Upon the change of control over the other Party.

34. Purchaser shall, within a period of six months from termination or expiry of the Agreement clear all the Ready to Fill Bulk or Finished Product lying with them and there after shall not use / deal with the Ready to Fill Bulk or Finished Product or the Intellectual Property of _____.

JURISDICTION & ARBITRATION

35. This Agreement is governed in all respects by the laws of Singapore without giving effect to its conflict-of-law provisions. The parties agree to submit to the jurisdiction of the courts located in Singapore.

36. Any dispute arising out of or in connection with this agreement, including any question regarding its existence, validity or termination, shall be referred to and finally resolved by arbitration administered by the Singapore International Arbitration Centre ("SIAC") for the time being force, which rules are deemed to be incorporated by reference in this clause. The seat of the arbitration shall be Singapore. The language of the arbitration shall be in English.

MISCELLANEOUS

37. Neither this Agreement nor any rights hereunder shall be assignable or transferable by the parties without the prior consent of the other parties.

38. No party shall by reason of the termination or non-renewal of this Agreement be liable to the other for compensation, reimbursement or damages on account of the loss of prospective profits, or anticipated sales or on account of expenditures, investments, Leases, property improvements or commitments in connection with the business or goodwill of the parties or otherwise unless there is a legal obligation that will be affected by termination.

39. Any notice to be given under this Agreement shall be in writing and shall be either personally delivered to each party or sent by telex/fax or similar method of Communication confirmed by pre-paid registered mail.

40. The relationship between the parties is that of independent contractor. No party shall by reason of this Agreement be deemed to be an agent of each other or member of partnership or joint venture with the other party.

41. The relationship between the parties is that of independent contractors. No party shall by reason of this Agreement be deemed to be an agent of each other or member of partnership or joint venture with the other party.

The Agreement sets forth the entire agreement and understanding between the parties on the subject matter hereof as of the effective date hereof, and supersedes all prior agreements, understandings, discussions and negotiates between the parties. This Agreement shall be changed only by agreement in writing signed by duly authorized representatives of the parties.

IN WITNESS WHEREOF the parties hereto have set their respective hands to these presents and the duplicates hereof on the date, month and year hereinabove written.

Signed and delivered by Indian Immunologicals)

Limited

By

AJAY SAINI Ajay Sigh

Signed and delivered by Purchaser

By ADAM HATEM AHMED ELDABA

Witness:

1.SCHEDULE - 1

A. Details of the Ready to Fill Bulk.

Ad

B. Price, Minimum Annual Business and Minimum Order Quantity.

Ready to Fill Unit of Minimum

Bulk

Measure

Annual Business in Unit

Minimum Annual

Business of Unit

Measure,

Doses

Measure, Litres

Order

Unit of

CIF Air Minimum Price, USD Quantity, per Unit of Measure, Litres of Measure

inenta Valent Litres 3,500,000 1925 1800 100 - Initial Stage

Vaccine doses 600 - Standard Qty

After the initial stage

Hepatitis B Liters 1,500,000 825 490 75

Vaccine Doses
(Pedia)
Hepatitis B Litres 250,000 275 490 75
Vaccine Adult Doses
Hepatitis A NA
Vaccine
ATTESTED
G. DAYANAND
Asst. Secretary
Andhra Chamber of Commerce
Secunderabad-500 003 (India)

2.0 METHODOLOGY EMPLOYED IN FEASIBILITY REPORT

1. Market Feasibility Analysis

Methodology:

Market research (primary and secondary)

Demand-supply gap analysis

Competitor landscape

Forecasting (demand for generics, biosimilars, and vaccines in Uganda and neighbouring markets)

2. Technical Feasibility Study

Methodology:

Process technology evaluation

Capacity planning

Equipment and utility specification

Site selection and master layout design

Technology transfer assessment (especially for biopharma and vaccine segments)

3. Financial Feasibility Analysis

Methodology:

Capital expenditure (CAPEX) estimation

Operating expenditure (OPEX) analysis

Break-even analysis

Return on Investment (ROI), IRR, NPV

Funding models (equity, debt, PPP)

4. Regulatory and Legal Feasibility

Methodology:

Mapping of Uganda National Drug Authority (NDA) regulations

WHO GMP and biosafety compliance requirements

International regulatory alignment (EU, WHO PQ, USFDA if export is intended)

Licensing and IP due diligence

5. Human Resource

Methodology:

Skills mapping of local workforce

Training needs assessment

Staffing plan based on function (R&D, QC, production, engineering, etc.)

6. Risk Assessment and Mitigation Planning

Methodology:

SWOT analysis

Risk matrix (technical, financial, geopolitical, supply chain, IP risks)

Scenario planning

Importance:

Anticipates challenges and vulnerabilities, allowing the project team to plan contingency strategies.

7. Supply Chain and Logistics Feasibility

Methodology:

Mapping of import/export routes

Availability of raw materials and packaging

Cold chain logistics (critical for vaccines and biopharma)

Infrastructure assessment (ports, roads, air)

8. Stakeholder and Partnership Mapping

Methodology:

Identification of potential partners

Government engagement

Joint venture and PPP models assessment

10. Strategic Fit and Policy Alignment

Methodology:

Mapping to Uganda's Vision 2040 and industrialization strategy

Donor alignment (Global Fund, GAVI, WHO, etc.)

Fit with regional goals (EAC, AU vaccine manufacturing initiatives)

Attachment Point 20

While the construction of all the facilities will be undertaken in concurrent manner starting from Q2 2025, the commissioning, qualification, validation and product roll out from the facilities will be done in phased manner

Phase 1 - Oral Solid dosage form

Phase 2 - General Injectable facility

Phase 3 - Biopharma facility

Phase 4 - Vaccine facility

Phase 5 - Oncology OSD facility

However, depending upon the detailed survey demand, phases for the roll out of products can be changed

Comments on the Sunshine Medicals feasibility study

1. The Executive summary lacks critical information - ideally the Executive summary should be a high level summary of the entire report; providing insights on each of the feasibility components. GK
 2. Provide details of the methodology employed in the feasibility study. AS
 3. The Executive summary is not one of the report chapters. It's a stand-alone section of report i.e. a high level summary of the entire report and should thus not be numbered. GK
 4. In the Acronym EAC, C stands for community and not Council. BS
 5. Chapter one of the report should comprise of background information. Such as an introduction with key statistical information on the sector and Justification for the proposed investment.
 6. The regulatory framework forms a stand-alone chapter. It shouldn't be merged with market feasibility. Also the sections under regulatory framework need to be well introduced in the report. MK
 7. Provide the sources of information for the tables and figures. e.g. Table no. 6 BS
 8. Site selection and infrastructure cannot be part of market feasibility. They are technical aspects and should thus be covered under technical feasibility. The same should apply to technology partners too. BS
 9. The technical feasibility is overly summarized. There's need to add some details to it. BS
 10. Also, provide details of where the technology will be sourced.
 11. The production of the different drugs has been justified due to the high dependence of imported medicines (70% imported), increase in prevalence of communicable and non-commonable diseases in the country. However, a comprehensive market trend describing the past and current state of the market for pharmaceutical and biopharmaceutical was over looked. This is key in informing the market size, future market outlook of the industry, factors driving market growth. (quantitative data is required in undertaking the analysis of the market trends) MK
 12. The target market for the proposed products are not clearly spelt out and this is essential in knowing who the potential customers are. MK
 13. Proposed technological partners for facilities under phase 1, 2 and 3 have been identified with the exception of Biopharma products facility and Oncology OSD facility that also require the right technological partners due to the nature of medicine to be manufactured to treat life threatening diseases such as cancer, diabetes among others.
 14. Estimated output capacities for Biopharma products facility and Vaccine manufacturing facilities were overlooked and these have a significant impact on the projected financials of the project such as inaccurate profit and loss due to un specified quantities produced and sold. This also leads to incorrect projected cash flows.
 15. The project intends to partners with several technological partners, however, it would be prudent if letters of intent or draft partnership agreements are included as attachments to the report as proof for the proposed partnership. AS
 16. The proposed facilities were to be set up in Tororo, however, the location has changed to Kampala. There's need to ascertain the actual location of the proposed facilities. BS
 17. Also, from previous discussions, it was indicated that the project site is known. Therefore, discussions on the project site should be more specific. BS
 18. The manufacturing process/ technology process is essential to enable one appreciate the process lay out in the manufacture of the required drugs and vaccines, however, this was not availed. Therefore, Indicative process flows for the different product lines should also be indicated in the technical feasibility. MK
 19. The area required to set up the integrated facility is known however; the proposed plant lay out was overlooked. Therefore, the proposed factory impressions need to be included.
- Layout are not the part of Feasibility
20. Are all the facilities going to be constructed at the same time or they will be phased. If phased, avail the phasing plan and this will aid in mobilizing the required resources to set up the integrated facility in a phased manner.

Response:

While the construction of all the facilities will be undertaken in concurrent manner starting from Q2 2025, the commissioning, qualification, validation and product roll out from the facilities will be done in phased manner

Phase 1 - Oral Solid dosage form

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Phase 3 - Biopharma facility

Phase 4 - Vaccine facility

Phase 5 - Oncology OSD facility

21. More clarity is needed on some blocks under common blocks e.g. Animal house blocks and Miscellaneous blocks e.g. training and Induction rooms and Horticulture. What is their significance? -AS

22. The projected financials focused on the profit and loss and these are incomplete. The projected cash flows were not availed hence, ascertaining viability was a challenge. A soft copy of the financial projection incorporating the assumptions is required for further analysis to determine the viability of the project.

23. Also, Provide details of key financial analysis ratios

IRR

Financial Net present value (NPV)

Average Rate of Return (ROI)

Discount Rate

Weighted Average cost of Capital (WACC) if applicable

Payback Back Period

Economic NPV

24. Although the financing requirement is well spelt out in the report, the financing structure that describes how the project is to be financed is not known. Therefore, provide details of the funding plan i.e. who the potential contributors are, and the mode of Investment i.e. equity or debt.

25. Clarity is needed on the regulatory costs which are appearing under both Capital expenditure and operation expenditure. How different are they, shouldn't they be under operation expenditure only?

26. Provide specific details of the projected impact e.g. foreign exchange to be earned/ saved, social economic impact other than jobs created;

Executive summary

Introduction:

The objective of this project is to establish and operate a state-of-the-art green field multi-Product pharmaceutical, Biopharmaceutical and vaccine manufacturing plant in the Republic of Uganda. This plant is designed to address and transform health care needs of Uganda and East African council (EAC) by addressing critical health care challenges and fostering local manufacturing.

Market Analysis

The most prevalent cases of mortality and morbidity in Uganda the most prevalent causes of mortality and morbidity in Uganda are Malaria, Pneumonia, Tuberculosis, Anaemia, Hypertension, Diabetes and Cancer. Uganda is also endemic for all of the Neglected Tropical diseases (NTDs) namely lymphatic filariasis, soil transmitted helminthiasis, schistosomiasis and onchocerciasis and trachoma. Other notable NTDs that remain endemic are Buruli ulcer, leishmaniasis (cutaneous and visceral), leprosy and rabies.

National vaccine coverage is more than 95% for the essential vaccines.

In Uganda, the revenue in the pharmaceuticals market is estimated to reach US \$329.32 m in 2025. The largest market within this industry is Oncology Drugs, which is projected to have a market volume of US\$50.03m in 2025. The revenue is expected to exhibit an annual growth rate (CAGR 2025-2029) of 3.06%, resulting in a market volume of US \$371.58m by 2029.

There are very few local manufacturers for Pharmaceutical products with the capability to cater to Uganda demands and most of the demand is met through imports. There are no local API, vaccine or Mabs manufacturers.

Hence there is large unmet need for essential medicines including vaccines, coupled with significant population growth.

Regulatory requirements

In Uganda, the regulation of human medicinal products is overseen by the National Drug Authority (NDA), which ensures the safety, efficacy, quality, availability and cost means of providing satisfactory healthcare and safeguarding the appropriate use of medicines used within the country.

There are more than 30 published guidelines for various aspects such as cGMP, licensing, import export, registration process, vaccines, clinical trials, distribution practices, recall/ withdrawal of products from the market.

The National Drug Authority (NDA) is yet to have the maturity level 3 like Nigeria, Egypt, South Africa, Rwanda, Senegal, Tanzania and Zimbabwe.

Site selection and infrastructure

The selected site shall have access to adequate power supply, water supply, waste management facilities, transport links, proximity to industrial zones, road connectivity and availability of qualified and experienced manpower.

Technical feasibility

The plant design shall incorporate

OSD and capsule manufacturing

Syrup and liquid manufacturing

Injectable and parenteral manufacturing

Mabs and vaccine manufacturing

Overall site plan shall be of 95,500 sqm. which will include all of the above blocks along with supporting blocks.

The supporting blocks will be dedicated to each manufacturing block to avoid cross contamination and comply with regulatory requirements

Financial feasibility

The overall capex investment for this mega project 571 million USD spread over a period of 3 years for plant commissioning and qualification.

Overall the total operational expenses at 100% capacity utilisation is 335 Million USD.

The overall revenue in year 7 will be 510 Million USD from OSD, general Injectables, biopharmaceutical products, vaccines, oncology OSD and contract manufacturing.

The payback period is 5 years. First and second year losses shall be due to ramp-up & market entry. Full profitability is expected in Year 4-5 with a Gross Profit Margin of ~50-65%. The project IRR is expected at 6%.

Supply chain and Raw material sourcing

The success and longevity of this project will depend on assured and consistent supply of raw materials, APIs, vaccine bulk, Mabs bulk.

Long term agreements with diversified suppliers for API, excipients, vaccine bulk will be important for the manufacturing operations.

Human resources

To operate the facility at 100% capacity it is envisaged that a total of 640 staff from various disciplines such as pharmacy, life sciences, engineering, quality control, quality assurance, administration, finance, accounts, human resource, pharmaceutical and vaccine formulation etc., Skills development, training and development of manpower is important due to the paucity of locally available talent.

Risk assessment

The project is prone to several risk factors such as regulatory uncertainty, government policy shifts, bureaucratic delays and geopolitical instability in EAC region.

Economic risks also exist such as currency fluctuations, inflation, supply chain disruptions, finance access and market demand.

Risk mitigation strategies such as early engagements with the regulatory agencies, public private partnerships, hedging currency risks, strategic supplier agreements along with diversifying revenue streams.

Sustainability

Complying with local environmental rules by segregation and recycling of wastes, handling of hazardous waste, minimising API and solvent wastes will be important along with implementation of energy efficient strategies such as use of solar power, biomass/ biogas utilization, water recycling, rain water harvesting etc.,

HR challenges and mitigation

Human resources have its own challenges such as availability of local skilled workforce, training and capability gaps, attrition etc.,

The challenges can be addressed by demand mapping, staggered hiring, establishment of training and development academy, collaboration with local academy, talent retention schemes etc.,

Strategy for accelerated ROI on the project

The ROI on the project can be accelerated through phased commissioning of production lines, prioritize products and high demand, partner with buyers early, leverage contract manufacturing activities, optimise operational costs, use of hybrid distribution model, avail of government tax incentives.

Project SWOT analysis

The project has its strengths such as financial backing, having a diversified product portfolio, Uganda location, modern infrastructure and state of the art technology.

The weakness for the project include lack of pharmaceutical domain experience, high dependence on imports, skill gaps and high operating costs.

However, the opportunities include growing demand in Uganda and EAC, support from government and international health agencies, regional regulatory harmonization, first mover advantage and green field facility with future ready compliance.

The threats to the project include regulatory and compliance challenges, geopolitical instability, currency fluctuation, import dependency, global competition, product specific challenges and delays in project implementation.

Summary and recommendations

The Sunshine Pharmaceutical Plant project in Uganda is deemed highly viable based on a multi-dimensional feasibility study and information given in this report.

Uganda's pharmaceutical market presents substantial opportunities, with projected revenue reaching US\$329 million by 2025 and strong CAGR, especially in generic drugs, oncology, vaccines, and biopharmaceuticals.

The plant's integrated model, with Oral Solid Dosage (OSD) forms and more complex formulations such as vaccines and biologics, aligns with the market demand and local regulatory evolution under WHO-GMP frameworks.

SAMPLE LETTER OF AUTHORISATION

To
Letter of Authorization
We----- manufacturer in -----is applying for the registration of -----
Product-----
We hereby appoint ----- incorporated under the laws of----- and
having corporate office----- to register, market, distribute, import, release and
sell the below mentioned products:
At----- drug authority, The----- ministry of health, the ----- authority for
unified procurement –medical supply and management of medical technology and other authorities linked to
importation, release, registration, distribution, marketing and sale in the territory of -----
This letter of Authorization (LOA) is valid for ----- years from the date of issuance unless extended
further by ----- and shall stand annulled upon expiry of such
period.
For -----
Address-----
Name-----
Designation -----
Authorized Signature-----
Date-----
Product Name | Brand Name | Generic Name | Presentation
| | |

Resposn Point 20

While the construction of all the facilities will be undertaken in concurrent manner starting from Q2 2025, the commissioning, qualification, validation and product roll out from the facilities will be done in phased manner

Phase 1 - Oral Solid dosage form

Phase 2 - General Injectable facility

Phase 3 - Biopharma facility

Phase 4 - Vaccine facility

Phase 5 - Oncology OSD facility

Point 25

Response: The regulatory Cost shown under capital Expenditure head is for Building, office equipment and other materials required.

The operational expenditure shall cover expenses related to Clinical trials, registration, Liaison and licensures etc.

Query:

11. The production of the different drugs has been justified due to the high dependence of imported medicines (70% imported), increase in prevalence of communicable and non-commonable diseases in the country. However, a comprehensive market trend describing the past and current state of the market for pharmaceutical and biopharmaceutical was over looked. This is key in informing the market size, future market outlook of the industry, factors driving market growth. (quantitative data is required in undertaking the analysis of the market trends)

RESPONSE

1. Market Size & Composition

Total Pharmaceutical Market

Estimated at US\$519million in 2022, up from US\$434mn in 2018—a 4.7% CAGR from 2017–2022

(<https://scapartner.com/2022/01/12/uganda-pharma-market>)

Per Capita Pharmaceutical Spend was around US\$10 in 2017, versus US\$10–US\$11 today

(<https://scapartner.com/2022/01/12/uganda-pharma-market>)

OTC Market Segment

OTC market projected at US\$221mn in 2023, expected to reach US\$240.6mn by 2025, growing at ~9.4% CAGR (2023–2028) .

In 2025, vitamin & mineral products (OTC) will be nearly US\$102mn

(<https://www.statista.com/outlook/hmo/otc-pharmaceuticals/uganda>)

Overall Pharma Revenue

Full-spectrum pharmaceutical market is projected to hit US\$325.9mn in 2025, from a lower base, with a modest 3.4% CAGR projection to reach US\$384.8mn by 2030.

(Note: Some sources vary in market definitions—OTC-only vs prescription + generic.)

2. Local Production & Biopharma Industry

Local manufacturing accounts for ~10% of drug consumption; 90% of pharmaceuticals are imported (mainly from India & China) <https://www.afyanahaki.org/arma-programme/uganda>

The number of domestic manufacturers has risen from 11 in 2009 to ~29 in 2021

<https://bpspubs.onlinelibrary.wiley.com/doi/full/10.1111/bcp.15324>

Key player includes:

Quality Chemical Industries Ltd (Qcil/Cipla): Generated UGX265bn (~US\$70.9mn) revenue by March2024; capacity ~1.2billion pills/year, distributing across 31 African nations <https://www.qcil.com/>

3. Growth Drivers

High Disease Burden — Malaria, HIV/AIDS, TB, and vaccine-preventable diseases remain top health concerns

<https://www.afyanahaki.org/arma-programme/uganda>

Health Spending Surge — Healthcare budget hitting ~7.3% of GDP; pharma expenditure growing ~8.4% per year; estimated at US\$520mn by 2021 (Uganda Investment Authority).

Policy Support — Government incentives include tax breaks, BUBU policy, and import tariffs to aid local production <https://bpspubs.onlinelibrary.wiley.com/doi/full/10.1111/bcp.15324>

Retail Expansion — OTC and self-medication on the rise, driven by convenience and rural demand

<https://www.slideserve.com/meenakshykr/the-uganda-pharmaceutical-market-challenges-trends-and-innovations>

Regional Export Potential — Access to East African Community (EAC) markets (~175mn people) and under-construction pharma industrial parks .

Biopharma Innovation — Investment in mRNA vaccine & biologics capacity via companies like Dei Pharma (Wikipedia).

4. Outlook & Forecast (2023–2030)

Total market to approach ~US\$520mn by 2022, projected to reach US\$384–520mn by 2025 depending on source.

OTC segment: set to grow to US\$240mn by 2025 (~+9% CAGR) .

Exports & regional share: Opportunity to reduce import dependence; EAC regional manufacturing plan targets ≥50% local procurement

<https://www.monitor.co.ug/uganda/special-reports/the-state-of-african-pharmaceuticals-3552908>

Biotech leap: mRNA & vaccine production rollout could shift market dynamics.

5. Constraints & Risks

Import dependence: 90% reliance on imports; raw materials from India/China .

Financing gap: High cost of capital and 18–24 month payback for greenfield production facilities

<https://www.monitor.co.ug/uganda/special-reports/the-state-of-african-pharmaceuticals-3552908>

Regulatory/Quality Risks: Prevalence of counterfeits & substandard drugs; regulatory enforcement still developing.

6. Summary Table

Conclusion & Implications

Strong growth trajectory in both prescription and OTC segments—especially with rising self-medication.

Growing domestic capacity, led by companies like Qcil and Dei Pharma, is transforming Uganda from importer to emerging exporter.

Policy backing and EAC integration bolster long-term volume and export potential.

Remaining challenges include heavy import reliance, financing hurdles, and quality control—but investments underway are addressing these.

Opportunity window for investors and local manufacturers lies in generic drugs, vaccines, mRNA tech, and regional distribution platforms.

Metric | Value / Forecast

Pharmaceutical Market (2022) | US\$519mn

OTC Market (2025) | US\$221–241mn, CAGR ~9.4%

Total Pharma Revenue (2025) | US\$326mn

Manufacturing Share | ~10% domestic, 90% imports

Local Manufacturers (2021) | 29 firms

Leading Local Producer | Quality Chemical (Rev. US\$70.9mn in 2024)

Market Growth Drivers | Disease burden, GDP health spend, policy support

Key Challenges | Import dependency, financing costs, quality risk

The Total Covered Area for the facility is 95,500 Sqm. However, facility Lay outs are out of scope of this report.

Layout may be included in the final agreed report in future.

Point No-21

Animal House Block(AH):

In case of Biologics and vaccine essentially some of the test shall be carried out on small animals' like- Mice, Guinea Pigs, Rabbits and Wistar rats as per regulatory guide lines.

For housing of these experimental animals this block is mentioned in the report.

However, if there will be any contract testing Organisation (CTO) for in vivo studies this block will not be considered.

Miscellaneous blocks:

The training and induction room indicated in the report is a part of quality assurance systems (Technical trainings) .This facility shall be used for the class room training of existing staff and new joiners as per the training schedule laid by quality Assurance department.

Trainings and learning is important part of any Bio manufacturing setup and lot of emphasis on training needs is laid by regulatory bodies like US FDA, EMA,WHO and ICH guide lines.

Horticulture: Horticulture, gardening, sewage treatment and Effluent treatment are essential for environment and health protection in the Biopharma industry.

Essential environmental clearance shall be required for the factory Licensure procedures before start of any manufacturing activities.

FEASIBILITY REPORT FOR INTEGRATED PHARMA AND BIOPHARMACEUTICAL PROJECT FOR UGANDA

Prepared for:

SUNSHINE MEDICAL PTE LTD
UNIT 06-10, UB ONE BUILDING,
UB AVENUE 4,
Singapore – 408 830

Prepared by:

BIOBRIDGE HEALTHCARE SOLUTIONS PVT. LTD
13, RACHANA BLOSSOM, AUNDH
PUNE 411007
INDIA

DOCUMENT NUMBER: BB//SS/FSR/11/2025 DATED 17.04.2025

TABLE OF CONTENTS

Executive Summary

Introduction

The objective of this project is to establish and operate a state-of-the-art green field multi-Product pharmaceutical, Biopharmaceutical and vaccine manufacturing plant in the Republic of Uganda. This plant is designed to address and transform health care needs of Uganda and East African council (EAC) by addressing critical health care challenges and fostering local manufacturing.

To realise the above said goals and to address the critical health care needs of Uganda, pharmaceutical, Biopharmaceutical and vaccine manufacturing facility shall be established in concurrent manner.

Oral Solid Dosage (OSD) form shall be established by integrating state-of-the-art manufacturing technologies to ensure efficiency, quality, and scalability of wide range of high-demand solid dosage forms, including tablets, capsules, syrups, and advanced technologies such as soft gel capsules.

General Injectable facility will focus on manufacturing sterile pharmaceutical injectable-grade products, targeting key therapeutic segments.

Biopharma Products facility focuses exclusively on the formulation and fill- finish of biopharmaceutical products. This facility shall address the increasing demand for biologics, including Monoclonal antibodies (mAbs), Hormones, Insulin, analogs, and other therapeutic proteins.

Formulation and fill-finish of vaccines facility shall be designed to cater to both routine immunization programs and outbreak vaccines.

Oncology OSD facility shall be designed for highly potent cancer drugs.

While the construction of all the facilities will be undertaken in concurrent manner starting from Q2 2025, the commissioning, qualification, validation and product roll out from the facilities will be done in phased manner

Phase 1 - Oral Solid dosage form

Phase 2 - General Injectable facility

Phase 3 - Biopharma facility

Phase 4 - Vaccine facility

Uganda's pharmaceutical industry is a key component of the country's healthcare system, with demand steadily increasing due to population growth, rising disease burdens, and government initiatives to improve healthcare access.

The disease burden in Uganda is similar to other less developed African countries, with communicable diseases accounting for roughly 67% of the total burden.

In 2021 preventive chemotherapy of 133 million population was done against the NTD.

Expanded Program on Immunisation the coverage for DTP group, Polio, Rota virus, Pneumococcal vaccines is more than 80%, taking into consideration the birth Cohort there is a great potential for vaccines and Biopharmaceuticals. The birth cohort is 2024 is reported to be 1,000 people in Uganda was 34.55 in 2024.

Malaria vaccine immunisation is also potential in the vaccine segment with reported cases (300/1000)

Objective:

The objective of this report is to comprehensively study the feasibility and to assess the potential for establishing a sustainable and efficient pharmaceutical, biopharmaceutical and vaccine manufacturing plant in Uganda.

BioBridge is requested to provide a proposal for feasibility study services, which include:

Market analysis

Regulatory requirements

Site selection and infrastructure

Technical feasibility

Financial feasibility

Supply chain and raw material sourcing

Human resources

Risk assessment

Sustainability

Key HR challenges

Strategies for accelerated ROI in the project

SWOT analysis

Summary and recommendations

Goals

Health Care Access: To enhance Healthcare access by manufacturing pharmaceuticals, Biologics and vaccines locally thereby reducing the dependency on imported medicines and biologics.

Health Care needs: To address regional health care needs by manufacturing medicinal products which are required to treat most prevalent diseases in the region such as Malaria, Tuberculosis and AIDS.

Economic growth: To foster Economic growth by providing employment opportunity to local population and attracting global investors and venture capitalists.

Sustainability: To establish a sustainable pharma base in Uganda which is reliable and self-sufficient.

Supply Chain Reliability: Building a secure and efficient supply chain to source raw materials, transport finished products, and ensure timely deliveries to end users.

Stakeholder Satisfaction: Ensuring customer satisfaction by providing consistent and reliable products while maintaining a favourable relationship with employees, investors, and the local community.

Solution:

To realise the above said goals and to address the critical health care needs of Uganda, pharmaceutical, Biopharmaceutical and vaccine manufacturing facility shall be established in concurrent manner.

Oral Solid Dosage (OSD) form shall be established by integrating state-of-the-art manufacturing technologies to ensure efficiency, quality, and scalability of wide range of high-demand solid dosage forms, including tablets, capsules, syrups, and advanced technologies such as soft gel capsules.

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Phase 3 - Biopharma facility

Phase 4 - Vaccine facility

Phase 5 - Oncology OSD facility

East Africa, like the rest of Africa, faces a growing demand for vaccines, particularly as it works to strengthen its vaccine supply chain and boost domestic capabilities to meet the needs of its population.

The most prevalent causes of mortality and morbidity in Uganda are

Malaria, Pneumonia, Tuberculosis, Anaemia, Hypertension, Diabetes and Cancer.

Uganda is endemic for all of the NTDs amenable to preventive chemotherapy through mass drug administration (MDA), namely lymphatic filariasis, soil-transmitted helminthiasis, schistosomiasis and onchocerciasis and trachoma. Other notable NTDs that remain endemic are Buruli ulcer, leishmaniasis (cutaneous and visceral), leprosy and rabies

2.0 DETAILED FEASIBILITY REPORT

a) Market Analysis:

i) Demand for pharmaceutical products in Uganda

Uganda's pharmaceutical industry is a key component of the country's healthcare system, with demand steadily increasing due to population growth, rising disease burdens, and government initiatives to improve healthcare access.

The disease burden in Uganda is similar to other less developed African countries, with communicable diseases accounting for roughly 67% of the total burden.

Uganda's communicable disease burden is dominated by infectious and parasitic diseases. Malaria is considered as the major cause of deaths in the country.

The most prevalent causes of mortality and morbidity in Uganda are

Malaria, Pneumonia, Tuberculosis, Anaemia, Hypertension, Diabetes and Cancer.

Fig No.: 1

Malaria:

Comparison of Malaria mortality rate (2021), compared to 2015 baseline

TUBERCULOSIS –

Estimated mortality in the African Region.

TB treatment coverage in African region

Human Immunodeficiency Virus (HIV)

Number of People living with HIV and number of cases prevented in the Africa region

Cardiovascular diseases (CVDs)

Estimated deaths related to cardiovascular diseases in the African region in 2019

Chronic respiratory diseases (CRD)

Estimated deaths related to chronic respiratory diseases in the African Region in 2019

Neglected tropical diseases

Uganda is endemic for all of the NTDs amenable to preventive chemotherapy through mass drug administration (MDA), namely lymphatic filariasis, soil-transmitted helminthiasis, schistosomiasis and onchocerciasis and trachoma. Other notable NTDs that remain endemic are Buruli ulcer, leishmaniasis (cutaneous and visceral), leprosy and rabies

EXPANDED PROGRAMME ON IMMUNIZATION IN CHILDREN

Estimates of national vaccination coverage of select antigens among countries in the Eastern and Southern African Sub Region in 2021

Measles Rubella –Coverage in African Region

Yellow Fever - Total confirmed cases on yellow fever since 2015.

CANCER

Trends of Cancer Patients in Uganda: Fig:2

Fig:3

The cancer burden by type of cancer diagnosed among patients

MARKET VALUE:

In Uganda, the revenue in the pharmaceuticals market is estimated to reach US\$329.32m in 2025.

The largest market within this industry is Oncology Drugs, which is projected to have a market volume of US\$50.03m in 2025.

The revenue is expected to exhibit an annual growth rate (CAGR 2025-2029) of 3.06%, resulting in a market volume of US\$371.58m by 2029.

Uganda's pharmaceutical market is experiencing a surge in demand for affordable generic medications.

The East Africa generic drug market reached USD 2.4 billion in 2024 and is expected to exhibit a CAGR of 6.72% during 2025-2033.

Competitive Landscape:

East Africa Pharma manufacturing- An Overview

Around 35 active manufacturers (members of FEAPM); 2 facilities with WHO cGMP approval and pre-qualified products

Primarily compounding and formulation of most dosage forms including infusions

Almost all raw materials imported

High dependency on imported finished pharmaceutical products

Focus on WHO list of essential medicines

No API manufacturing

Increasingly well-regulated (NDA, PPB, TFDA)

Trade is mostly intra-East Africa

Companies currently having manufacturing base in Uganda

1. CIPLA Quality Chemicals Industries Limited, joint venture between a local and international pharmaceutical company, is cGMP certified manufacturing Anti-retro viral, anti-malarial and hepatitis B drugs, exports to Sub-Saharan countries and supplies drugs to the Global Fund

2. Kampala Pharmaceutical Industries Limited, is a local manufacturer and an affiliate of the Aga Khan development network. Product range includes analgesics, anti-asthma, anti-diabetic, anti-malarial, anti-parasites, antibiotics, cardiovascular drugs, cold remedies & Nutraceuticals

3. Beta Health Care Uganda Limited, an affiliate of Aspen Group, exports its products to Tanzania, Rwanda, Burundi, Ethiopia, Sudan, Somalia, Mauritius and West Africa

Key players in the market are actively implementing various strategies to expand their reach and market share by:

Enhancing their production capabilities, both in terms of volume and the range of products.

Investments in advanced manufacturing facilities to increase output and meet the rising demand for generic drugs.

Creating more efficient supply chains and distribution networks to ensure wider availability of their

products. They are reaching out to remote areas and partnering with local distributors to improve access. Forming alliances with local manufacturers, international partners, and healthcare providers to extend their reach and enhance their market position.

Knowledge and technology transfer, through Joint ventures, Alliances thereby strengthening local capacities.

Key Pharma Sector Data Estimate (Market share, Generic, Branded pharmaceuticals)

Table No.: 1

Demand for Pharmaceutical Products in Uganda

The demand for pharmaceutical products in Uganda is driven by various factors, including an increasing prevalence of both communicable and non-communicable diseases (NCDs), an expanding healthcare infrastructure, and growing government support for local production.

Below is an analysis of key therapeutic areas:

a) Oral Solid Dosage Forms (Tablets and Capsules)

Oral solid dosage forms constitute the majority of prescription medications in Uganda, covering a range of treatments from antibiotics to anti-hypertensives.

Uganda has a few established pharmaceutical manufacturers producing these drugs.

Despite local manufacturing efforts, a large portion of Uganda's medicine supply is still imported due to insufficient production capacity and the need for diverse active pharmaceutical ingredients (APIs).

Expanding local production capabilities, especially for essential generics, could help meet growing demand and reduce reliance on imports.

b) Oncology Treatments

Cancer cases in Uganda have been rising significantly, with estimates from the Uganda Cancer Institute (UCI) indicating an annual increase in new cases (Refer to Fig Nos.2&3.).

Currently, oncology drugs are almost entirely imported, making treatment expensive and often inaccessible to many patients.

There is an opportunity for local pharmaceutical manufacturers to invest in cancer drug production, particularly generic chemotherapy drugs, which could lower costs and improve accessibility and health care.

The Ugandan government and international health organizations are increasingly investing in cancer treatment centres, further boosting the demand for oncology drugs

c) Vaccines

Uganda has a demand for vaccines (approx. 1.7 million doses per annum for essential vaccines) due to childhood immunization programs, seasonal disease outbreaks, and global initiatives like the Global Vaccine Alliance (GAVI).

The country currently relies on imports for nearly all its vaccines, which poses challenges such as supply chain disruptions and high procurement costs.

Government efforts to establish vaccine production facilities could significantly boost the local pharmaceutical industry and create self-sufficiency in vaccine supply.

d) Monoclonal Antibodies (mAbs)

Monoclonal antibodies are used in cancer treatment, autoimmune diseases, and infectious diseases like COVID-19.

Uganda currently lacks the infrastructure for manufacturing biologic drugs & mAbs, entirely dependent on imports.

Drivers of Pharma market:

Rapid expansion of local pharmaceutical manufacturing capacities.

East Africa is experiencing a significant boost in its pharmaceutical production driven by a combination of increased domestic and international investment and initiatives, improved technological capabilities, and a strategic push for self-sufficiency in pharmaceutical supply. Furthermore, there is a rising awareness among healthcare professionals and patients about the comparable efficacy of generic drugs to brand-name counterparts.

Generic drug manufacturers are significantly strengthening their distribution networks to improve product reach. This expanded distribution is further facilitated by technological advancements, such as digital platforms and delivery systems, enabling broader dissemination of generic drugs across the region, fostering overall the market growth.

Favourable Government Initiatives

Governments across East Africa are progressively acknowledging the role of generic drugs in achieving healthcare affordability and accessibility. They are implementing regulatory reforms to simplify the approval process, thereby encouraging their use by streamlining this process, governments are reducing barriers to entry, making it easier for generic manufacturers to bring their products to market.

Moreover, there is an increase in governmental efforts to boost public awareness about the efficacy and affordability of generic drugs. Despite their lower cost, generic drugs are equally effective as their branded counterparts.

EAC Pharma – Opportunities

Large unmet need for essential medicines

Significant population growth anticipated

Double disease burden (growth of chronic diseases):

NCDs are predicted to account for 40% of illnesses & for 42% of deaths (5.1 million) by 2030 (WHO Global Health Estimates, 2013)

Growing middle class willing to pay for better treatment

New models of care being designed (e.g. PPP, medical insurance, etc.)

Existing pharmaceutical production base & enabling regulatory environment

African traditional and herbal medicines as a market niche

No dominant players

Political will and action

EAC Pharma – Challenges

Negative perceptions of locally manufactured products

Lack of capacity to manufacture advanced formulations due to a skills gap on product development and formulation expertise

High cost of borrowing (20-30%)

High cost of operations e.g. finance, utilities, infrastructure

Lack of market data

Poor and unreliable infrastructure (roads, utilities, etc)

Under resourced and weak institutions (Civil society, Regulatory agencies, etc..)

Vaccines Market Analysis:

Immunization is a proven tool for controlling and eliminating life-threatening infectious diseases and is estimated to avert millions of deaths each year. Universal immunization of children against eight vaccine-preventable diseases [tuberculosis, diphtheria, whooping cough (pertussis), tetanus, hepatitis B, respiratory diseases caused by Haemophilus influenzae, polio, and measles] is crucial to reducing infant and child mortality. J Health Popul Nutr . 2013 Mar;31(1):118–129. Do

Immunisation Schedule in Uganda:

Routine immunization schedule for Children below the age of one year cover vaccines for following preventable diseases

Oral Polio vaccine, BCG, Pentavalent, Pneumococcal 10 valent, Rotavirus vaccine, Injectable Polio vaccine (14 weeks of age), Measles vaccine & HPV (Female Children) (Republic of Uganda, Ministry of Health/UNICEF Report)

Based on the birth cohort in Uganda (1.7 million per year) the requirements of different vaccines are as under based on the government Immunization program:

Ref: <https://www.gavi.org/programmes-impact/country-hub/africa/uganda>

Table No.:2

Upcoming vaccines which shall be targeted in the future

mRNA based vaccines for pandemic preparedness, RSV (Respiratory Syncytial virus) vaccine, Recombinant Tuberculosis vaccine (under development globally), Rabies vaccine etc.

Demand of vaccines products in Uganda & EAC region:

Eastern Africa is experiencing a surge in demand for vaccines, driven by increasing awareness and government initiatives to improve healthcare access.

The Vaccines market in Eastern Africa is anticipated to achieve a revenue of US\$463.65m in 2025.

Looking ahead, the market is expected to exhibit a compound annual growth rate (CAGR) of 4.98% from 2025 to 2029, leading to a market volume of US\$377.97m by 2029.

Key market trends, challenges, and opportunities:

One of the key trends in the Vaccines segment in Eastern Africa and Uganda is the increasing availability of vaccines.

Pharmaceutical companies are expanding their operations in the region, and governments are working to improve vaccine distribution networks. This has made vaccines more accessible to people living in remote areas.

Secondly, in the market there is growing demand for new and innovative vaccines. As more people become vaccinated against common diseases, there is a need for vaccines that can prevent emerging infectious diseases.

Pharmaceutical companies are investing in research and development to create new vaccines that can address these challenges.

East Africa, like the rest of Africa, faces a growing demand for vaccines, particularly as it works to strengthen its vaccine supply chain and boost domestic capabilities to meet the needs of its population.

Challenges and Opportunities:

Challenges include the provision of vaccines in rural locations hampered by inadequate cold chain infrastructure, and vaccination acceptance rates that are impacted by cultural and religious beliefs.

Besides, there are also market opportunities, such as increasing research and development (R&D) spending for novel vaccines and technological advancements.

Growing Support from Various Organizations:

The support from international bodies such as the World Health Organization (WHO), World Bank, the United Nations Children's Fund (UNICEF), and the Global Alliance for Vaccines and Immunization (GAVI) is indispensable for the East African vaccine market. These organizations provide not only financial aid but also logistical support, crucial for the procurement and distribution of vaccines in the region.

2. Competitive Analysis of the Existing Pharmaceutical Market

Uganda's pharmaceutical market is composed of both local manufacturers and international suppliers. The competitive landscape is shaped by several key factors:

a) Local Pharmaceutical Manufacturers

Uganda has a small but growing pharmaceutical manufacturing industry.

Local manufacturers face challenges such as high production costs, limited access to raw materials, and competition from imported generics.

b) Import Dependence and Foreign Suppliers

Uganda imports about 70% of its pharmaceuticals, with India and China being the dominant suppliers.

International pharmaceutical companies have a strong presence in Uganda through distributors.

The high reliance on imports creates challenges in affordability, availability, and quality control.

c) Policy and Market Protection

Uganda has introduced policies like the "Buy Uganda, Build Uganda" (BUBU) initiative, which promotes local manufacturing.

However, the local pharmaceutical industry struggles to compete with cheaper imports, and implementation of these policies remains inconsistent.

3. Key Market Trends, Challenges, and Opportunities

a) Market Trends

Regulatory Harmonization in the EAC

The East African Community (EAC) is working towards regulatory harmonization, allowing Ugandan manufacturers to access larger regional markets.

Harmonization simplifies drug registration processes and could enhance trade within the region.

Growing Non-Communicable Disease (NCD) Burden

There is an increasing prevalence of NCDs such as diabetes, hypertension, and cancer in Uganda.

This shift is driving demand for chronic disease medications, which presents opportunities for local manufacturers to diversify production.

Expansion of Health Infrastructure

The Ugandan government is investing in expanding hospitals, clinics, and specialized treatment centres.

These investments are increasing the demand for pharmaceuticals, particularly in specialized areas like oncology and vaccines.

b) Key Challenges

Limited Access to Finance

Pharmaceutical manufacturing requires large capital investments for facility upgrades, research, and compliance with regulatory standards.

Access to affordable financing remains a major barrier for Ugandan pharmaceutical firms.

Dependence on Imported Raw Materials

Uganda imports most of its Active Pharmaceutical Ingredients (APIs) from India and China.

This dependence makes local manufacturers vulnerable to global supply chain disruptions.

Regulatory and Quality Compliance

Meeting international Good Manufacturing Practices (GMP) standards is essential for market expansion but requires costly upgrades.

Non-compliance can limit Ugandan pharmaceutical companies' ability to export to international markets

Shortage of Skilled Workforce

There is a lack of specialized professionals in pharmaceutical research, manufacturing, and regulatory affairs.

Developing a skilled workforce through training and academic programs is critical for industry growth.

c) Opportunities for Growth

Investment in Local Production

Increasing local manufacturing, especially in essential medicines and specialty areas like oncology and vaccines, can enhance Uganda's pharmaceutical security.

Public-Private Partnerships (PPPs) could play a crucial role in driving investment.

Regional Expansion and Export Potential

The EAC harmonization efforts present an opportunity for Ugandan pharmaceutical companies to expand their

reach beyond the domestic market.

By meeting international regulatory standards, local manufacturers could increase exports to neighbouring countries.

Public-Private Collaborations

Government initiatives supporting pharmaceutical production can help develop infrastructure and provide financial incentives for local companies.

Partnerships with international pharmaceutical firms could help bring technology and expertise to Uganda.

Biopharmaceutical and Vaccine Development

Investing in biologics and vaccine production could position Uganda as a leader in high-value pharmaceutical production.

Establishing local capacity for monoclonal antibody production could reduce reliance on costly imports.

B. Overview of the Ugandan Pharmaceutical Regulatory Framework and EAC Regulatory Systems

Introduction:

In Uganda, the regulation of human medicinal products is overseen by the National Drug Authority (NDA), which ensures the safety, efficacy, quality, availability and cost means of providing satisfactory healthcare and safeguarding the appropriate use of medicines used within the country.

The legal framework governing these activities is established by the National Drug Policy and Authority Act, Cap 206.

The National Drug Authority (NDA) was established in 1993 by the National Drug Policy and Authority Statute which in 2000 206 of the Laws of Uganda (2000 Edition).

The EAC Guide on Good Manufacturing Practices (GMP) is based on the World Health Organization (WHO) Good Manufacturing Practices guidelines in the WHO Technical Report Series No. 961, 2011. This guide also refers to the Pharmaceutical Inspection Convention Scheme (PIC/S) guide PE 009-9.

Table No.:3

Guidelines: National Drug Authority, Uganda- Human medicinal guideline

■

Maturity level of NDA (Uganda) with respect to WHO Qualification

Maturity Level 3, which is a key benchmark for drug regulatory bodies, ensures that national authorities can effectively oversee drug safety, efficacy, and quality, aligning with international norms.

The National Drug Authority (NDA) is seeking to overcome legal and regulatory challenges that have hindered its progress toward achieving Maturity Level 3, a global standard for drug regulators.

Only eight African countries have currently attained this status, making it a significant goal for Uganda. These include, Egypt, Rwanda, Senegal, Ghana, Nigeria, South Africa, Tanzania and Zimbabwe.

Key Regulatory Requirements

In Uganda, the journey of a new drug from conception to market involves a systematic approval process regulated by several governing bodies, predominantly the National Drug Authority (NDA).

Preclinical and Clinical Trials:

The process begins with preclinical studies, where laboratory and animal testing evaluate the drug's safety and efficacy. This stage is crucial, as it provides preliminary data that informs the next phases of research and development. Pharmaceutical companies need to document their findings and prepare a comprehensive submission package for the NDA.

Conducting clinical trials in Uganda requires prior authorization from the NDA.

Researchers, organizations must adhere to the "Guidelines for the Conduct of Clinical Trials in Uganda," which detail the procedures for submission, review, and approval of clinical trial applications. ■

Compliance with Good Clinical Practice (GCP) is mandatory, as outlined in the "Guidelines on Good Clinical Practice in the Conduct of Clinical Trials Involving Human Participants." ■

After completing the clinical trials, the pharmaceutical company must submit a New Drug Application (NDA) to the NDA, including all research data, manufacturing details, and proposed labelling.

Once the NDA reviews the application and finds it satisfactory, it grants approval for the drug to be marketed.

Good Manufacturing Practice (GMP):

Manufacturers of medicinal products must comply with GMP standards to ensure product quality. The NDA provides detailed guidelines, including "Guidelines on Good Manufacturing Practice for Medicinal Products Part I – Basic Requirements for Medicinal Products." ■

All human medicinal products must be registered with the NDA before they can be marketed or distributed in Uganda. The registration process involves a scientific evaluation of the product's safety, efficacy, and quality.

GMP Inspection:

National Drug Authority (NDA) is the regulatory body responsible for overseeing the compliance of pharmaceutical companies with GMP standards.

Regular GMP inspections and audits are conducted as one of the requirements for registration of medicinal products in the EAC.

The NDA conducts comprehensive evaluations to verify that facilities meet the specified criteria for manufacturing processes, hygiene, and personnel training.

Inspection involves review of documents, records, facilities and any other resources to assess their conformity to the requirements of EAC GMP of medicinal products.

Any non-compliance can result in severe penalties, including the suspension or revocation of manufacturing licenses.

Importation and Exportation:

The NDA regulates the import and export of drugs and pharmaceutical raw materials.

Drugs imported must be registered unless given special clearance by the NDA under Sec 8(4) NDPA (National drug policy & Authority) by NDA.

The import permit may be:

Annual import permit -which is issued for regular imports like pharmaceutical manufacturers or any organization, retail pharmacies, wholesale premises, that regularly imports drugs and other related substances. Ministry of Health of Uganda and its affiliated units may also apply for an annual import permit.

Provisional import permit is issued to non-regular importers who may import drugs, for some reasons.

The provisional import permit is issued per consignment. This usually applies to the organizations, which receive donations, physicians or individuals who may for specified reasons need to bring limited quantities of specialized drugs that are not available in the country. The permit is valid for one month.

Licensing:

Entities involved in the manufacture of pharmaceuticals must obtain appropriate licenses from the NDA. ■

Import License: (Form-24)

Eligibility to apply for import license: A licensed person (a person holding license to operate pharmacy or for manufacture drugs).

Import license is valid for one year and is valid within the calendar year within which it is issued.

Verification certificate: A consignment of drugs to be imported into Uganda shall before importation, be issued with a verification certificate (Form-25) by the NDA basing on the application submitted in this regard in Form-26.

Stability Data:

In Uganda, the National Drug Authority (NDA) requires stability data as part of the regulatory

requirements for pharmaceutical and biopharmaceutical products, ensuring quality, safety, and efficacy, and mandates that all medicinal products meet acceptable standards.

The NDA requires that applicants for drug registration provide results of stability testing, which is crucial for determining the shelf-life and storage conditions of the product.

Applicants need to submit all relevant stability data as part of the registration application.

The stability studies will determine the appropriate storage conditions for the product, including temperature and humidity, and these conditions must be reflected on the product label.

The product label must include an expiry date and storage instructions based on the stability data.

Stability studies should include Storage conditions i.e temperature and relative humidity for accelerated and stress conditions.

Stability studies should be done in accordance or with reference to WHO TRS 987, WHO TRS 953 Annex 2, WHO TRS 962 Annex 3, ICH Q1A and ICH Q5C guidelines.

A minimum of twelve months data at the time of submission should be provided in cases where storage periods greater than six months are requested.

For storage periods of less than six months, the stability data should cover the whole proposed shelf life.

The stability studies should be submitted in controlled documentation.

Post-approval stability program should include the stability program or stability commitment to be carried out once the drug product is on the market, including the number of batches to be included in the study each year and the tests to be performed.

These results should be submitted periodically to update the information on the stability of the drug product.

The stability of each dosage form should be separately documented.

Stability data submitted should be for at least three consecutive batches and include the following: Information on stability of drug product, quality control methods and rationale for the choice of tests for determining stability.

Information on the dates of manufacture of the lots, the lot numbers, the vial and dose size, and the scale of production.

For lyophilized products, the data supporting the shelf-life of the product reconstitution should be included.

If the drug product is frozen, data supporting the stability of the product through a stated number of freeze-thaw cycles should be provided.

Drugs Registration in Uganda:

All drugs to be manufactured or imported into Uganda have to be registered with the National Drug Authority, Uganda. A person who intends to manufacture, import or export a product shall, prior to the manufacture, importation or exportation of the product, apply to the Authority for registration of the product.

The registration process involves:

NDA inspection & approval of the manufacturing facility.

Validity of registration:

The Authority shall issue for the first registration, a certificate of registration which shall be valid for five years.

The holder of a certificate of registration shall pay an annual retention fee prescribed by the Authority for maintaining the registered human or veterinary drug or preparation, vaccine or other immunological products or surgical instrument, on the Register.

A certificate for renewal of registration shall be valid for one year from the date of issue.

The certificate of registration issued under these Regulations shall be kept at the premises of the holder of the certificate of registration.

The holder of a certificate of registration shall monitor the safety and effectiveness of the registered human or veterinary drug or preparation, vaccine or other immunological products or surgical instrument while on the market in Uganda and shall submit periodic reports as may be prescribed by the Authority.

The registration of a human or veterinary drug or preparation, vaccine or other immunological products or surgical instrument under these Regulations shall be valid except where the registration is suspended or cancelled by the Authority or is terminated by the holder of the certificate of registration.

Table No.:4

Marketing Authorisation Time lines:

Registration Timeline: 12 to 18 months

Post-Marketing Surveillance and Pharmacovigilance:

Under the National Drug Policy and Authority (NPD &A) Act (1993) the Uganda National Pharmacovigilance Centre (NPC) is responsible for PV efforts in the country and mandates the National Drug Authority (NDA) to monitor the safety, efficacy and quality of medicines.

Risk Assessment

Risk assessment is a crucial regulatory requirement, ensuring product safety, efficacy, and quality, and safeguarding public health. Proactive risk identification and mitigation, through methodologies like Quality Risk Management (QRM), are essential for regulatory compliance and building trust with stakeholders.

Manufacturers would be required to conduct a risk assessment on critical production and quality control processes and thereafter develop a mitigation plan whose effectiveness would be monitored by the regulator.

Table No.:5

Regulation Framework Timelines:

ii. Environmental and Safety Regulations Applicable to Pharmaceutical Manufacturing in Uganda

Uganda's regulatory framework ensures that pharmaceutical production safeguards human health, worker safety, and environmental sustainability.

Environmental Compliance

Environmental oversight in Uganda is primarily the responsibility of the National Environment Management Authority (NEMA). Under the National Environment Act, 2019, all industrial and manufacturing projects, including pharmaceutical facilities, are required to undertake an Environmental and Social Impact Assessment (ESIA) as a prerequisite for licensing and construction. The ESIA must identify potential environmental risks and outline mitigation strategies, with particular attention to waste management, water usage, emissions, and impacts on surrounding communities.

Pharmaceutical facilities must install and maintain Effluent Treatment Plants (ETPs) to process wastewater before discharge. Compliance with the Environmental Management (Waste Management) Regulations ensures proper handling of both hazardous and non-hazardous solid and liquid waste, especially chemical residues, solvents, and expired or rejected pharmaceutical products. The facility must also adhere to Uganda's obligations under the Basel Convention on the control of transboundary movements of hazardous waste.

Safety and Occupational Health Standards

The National Drug Authority (NDA) regulates the pharmaceutical sector and enforces Good Manufacturing Practices (GMP), including protocols that safeguard worker and product safety throughout the production lifecycle. This includes facility design to minimize cross-contamination, controlled cleanroom environments, and safe material handling procedures.

In parallel, the Occupational Safety and Health Act, 2006, administered by the Ministry of Gender, Labour and Social Development, mandates comprehensive safety practices in the workplace. Employers must implement hazard identification and risk mitigation strategies, ensure employee training, and provide Personal Protective Equipment (PPE) and emergency response systems. All operations must include clearly marked emergency exits, fire suppression systems, first aid facilities, and regular safety drills. Regular inspections and audits are conducted by both NDA and the Ministry of Labour to verify compliance.

Strategic Considerations

Full regulatory compliance not only avoids legal penalties and operational delays but also enhances the facility's reputation among stakeholders and investors. Additionally, adherence to these regulations supports sustainable operations and positions the facility for future certification under global quality and safety systems, enabling access to regional and international markets.

Table No.: 6

iii. Overview of EAC Regulatory Systems for Key Therapeutic Areas

General Observations on the EAC Medicines Regulatory Framework

Harmonization Framework: The EAC Medicines Regulatory Harmonization (MRH) Programme supports joint product assessments, GMP inspections, and reliance pathways to reduce duplication across Partner States.

National Regulatory Authorities (NRAs): While each EAC country retains autonomy (e.g., Kenya PPB, Uganda NDA, Rwanda FDA), they collaborate extensively on shared regulatory functions.

Global Alignment: Reliance on WHO Prequalification, Global Fund procurement standards, AVAREF for clinical trials, and engagement with the African Medicines Agency (AMA) for future alignment.

C. Site Selection and Infrastructure

i. Site Selection Criteria

a) Proximity to Raw Materials & Suppliers

Most of the Active Pharmaceutical Ingredients (APIs) and excipients used in pharmaceutical production in Africa are imported (mainly from India and China).

Ideal site locations:

Close to international ports, airports, or dry ports to reduce import lead times.

Access to local packaging material manufacturers to lower supply chain dependency.

Industrial zones with chemical and biotech suppliers to ensure consistent raw material availability.

b) Logistics & Transportation Considerations

Proximity to major highways, rail networks, and ports to facilitate raw material import and product distribution.

Cold chain logistics is crucial for vaccines and biologics, requiring locations near temperature-controlled warehouses and transport hubs.

Close to key healthcare markets to reduce distribution delays, especially for essential medicines.

c) Utilities & Infrastructure

Electricity: Reliable power supply is critical, especially for Injectables, vaccine production, and sterile drug manufacturing. Backup generators, UPS systems, and solar energy solutions should be considered.

Water Supply & Treatment:

High-quality water sources for pharmaceutical processing, purification, and cleaning.

Reverse Osmosis (RO) & Water for Injection (WFI) systems for injectable production.

Effluent treatment plants (ETP) and bio-waste disposal facilities for environmental compliance.

Air Filtration & HVAC Systems:

ISO-classified cleanrooms (ISO 5, ISO 6, ISO 7, ISO 8) for sterile and injectable production.

High-Efficiency Particulate Air (HEPA) filters and humidity-controlled areas for tablet and capsule manufacturing.

d) Skilled Workforce Availability

Proximity to universities and research institutions that offer pharmaceutical, biotechnology, and engineering programs.

Training partnerships with international pharmaceutical firms to build local expertise.

Employee housing and transport accessibility to attract and retain skilled professionals.

2. Infrastructure Requirements

a) Facility Layout & Zoning

Segregated production zones for different dosage forms:

OSD (Oral Solid Dosage): Tablets, capsules.

Syrups & Liquids: Bottled formulations.

Injectables & Vaccines: Requires containment cleanrooms and sterile environments.

Dedicated cleanroom areas for sensitive processes such as sterile drug filling and vaccine formulation.

Material & personnel movement control to prevent cross-contamination.

b) Manufacturing Equipment & Technology Integration

Automated production lines for efficiency and compliance with GMP standards.

SCADA (Supervisory Control and Data Acquisition) for real-time monitoring.

c) Cold Chain & Storage Facilities

Temperature-controlled warehouses for raw materials and finished pharmaceutical products.

Cold chain logistics for vaccines and biologics, including deep freezers and transport containers.

Automated inventory management systems to ensure compliance with GDP (Good Distribution Practices).

3. Finding the Right Technology Partners

a) Pharmaceutical Machinery & Equipment Suppliers

To establish a world-class pharmaceutical facility, partnering with technology providers specializing in pharmaceutical manufacturing equipment is essential. Key areas include:

Oral Solid Dosage (OSD) Equipment

Granulation, blending, and tablet compression machines

Leading suppliers: GEA Group, Bosch, Syntegon (formerly Bosch Packaging), Fette Compacting ACG Worldwide,

SED Pharma, Syntegon Technology, IMA Group, Glatt, Fette Compacting, Romaco, iPharMachine

Sterile Injectable & Vaccine Manufacturing Equipment

Aseptic filling, lyophilization, and cleanroom solutions

Leading suppliers: Bausch+Ströbel, IMA Pharma, Optima Pharma, NKP Pharma machinery, Cozzoli Machine

Company, Groninger, ARxIUM, NJM Packaging, Intellitech

Water Purification & WFI Systems

Suppliers: Merck Millipore, Veolia Water Technologies, MECO, NGK Insulators Ltd, Syntegon, Ovivo, Fabtech

Technologies, Aqua-Chem, Herco Wasser technik GmbH, NEWater, Praj Industries, CN water

Cleanroom & Air Filtration Systems

Suppliers: Camfil, AES Clean Technology, Thermo Fisher Scientific, Clean Air Technology, Atmos-Tech

Industries, Abtech Inc., Microflow Devices India PVT Ltd. Zeco Aircon Ltd

b) Digital & Automation Partners

SCADA & MES Systems: Siemens, Rockwell Automation, Honeywell.

Automated quality control & batch monitoring: Emerson, Siemens, ABB.

c) Regulatory & Compliance Partners

GMP compliance consulting:

Local regulatory bodies (e.g., Uganda National Drug Authority - NDA).

d) Waste Management & Sustainability Partners

Hazardous waste disposal & ETP solutions: Veolia, SUEZ, Ecolab.
Renewable energy integration for sustainability: Tesla Energy, Schneider Electric.
4. Strategic Collaborations & Public-Private Partnerships
Government Incentives & Special Economic Zones (SEZs)
Tax incentives for pharmaceutical manufacturers in Uganda and the East African Community (EAC).
Preferential land allocation and infrastructure support for investors.
Public-Private Partnerships (PPPs)
Collaborations with global health organizations (WHO, UNICEF, GAVI, USAID) for vaccine and essential medicine production.

ii. Infrastructure Requirements for a Pharmaceutical Manufacturing Facility near Kampala, Uganda
The proposed pharmaceutical manufacturing facility, will be located near Kampala, Uganda.
Given the scale and scope of this state-of-the-art facility, robust infrastructure is vital to ensure uninterrupted operations, regulatory compliance, and product quality.
Power Supply: The facility will require a high-capacity, stable power source to support continuous manufacturing, HVAC systems, cleanrooms, cold storage, and quality control labs. It will be connected to the national electricity grid via UMEME, with dedicated transformers and high-voltage infrastructure. Backup power will be provided through diesel generators and solar power systems to ensure energy security and sustainability.
Water Supply: A reliable water supply is essential for production processes, utilities, cleaning, and sanitation. The facility will include on-site water purification systems to generate Purified Water (PW) and Water for Injection (WFI), meeting GMP standards. Water will be sourced from the national supply network and supplemented by boreholes with storage reservoirs.
Waste Management: The facility will house a centralized Effluent Treatment Plant (ETP) and systems for safe disposal of solid and hazardous waste, in full compliance with NEMA guidelines. Special provisions will be made for pharmaceutical and biomedical waste, including incineration and disposal via licensed operators.
Transport Links: Proximity to Kampala's industrial zones, the Northern Corridor, and Entebbe International Airport ensures efficient import/export logistics and access to regional markets. Excellent road connectivity will facilitate material movement and regulatory oversight.

D. TECHNICAL FEASIBILITY

1. Machinery and Equipment Requirements

Each dosage form requires specific equipment to facilitate high-quality and compliant production.

a) Oral Solid Dosage (OSD) & Capsules Manufacturing

Granulation Equipment – Fluidized bed dryers, high-shear mixers.
Tablet Compression Machines – Single rotary, double rotary presses.
Capsule Fillers – Semi-automatic and fully automatic encapsulation machines.
Coating Machines – Film and sugar coating systems.
Blending & Mixing Equipment – V-blenders, ribbon mixers.

b) Syrup and Liquid Manufacturing

Sugar Dissolution & Mixing Tanks – Stainless steel reactors with agitation.
Homogenizers – For uniform syrup consistency.
Liquid Filling & Capping Machines – For bottles and ampoules.
Sterile Filtration Systems – For microbial safety.

c) Injectables & Parenteral Manufacturing

Aseptic Filling Lines – For ampoules, vials, prefilled syringes.
Lyophilization Equipment – For freeze-dried injectables.
Autoclaves & Sterilizers – For equipment and component sterilization.
Isolators & Restricted Access Barrier Systems (RABS) – To maintain sterility.

d) Monoclonal Antibodies (mAbs) & Vaccine Manufacturing

Fill-Finish Equipment – Sterile filling, capping, lyophilization.
Cold Chain & Stability Chambers – For biological storage.

3. Technology and Automation

a) Digitalization and Smart Manufacturing

Manufacturing Execution Systems (MES) for batch tracking.
SCADA (Supervisory Control and Data Acquisition) for process automation.
Predictive Maintenance to prevent downtime.

b) Quality Control & Assurance Technologies

Analytical Testing Equipment – HPLC, GC, UV Spectrophotometers.
Microbial Testing – Endotoxin testing, sterility assurance.

4. Compliance & Regulatory Considerations

Adherence to GMP, WHO and local drug regulatory guidelines.
Regular validation & qualification of equipment and processes (IQ, OQ, PQ).
Environmental compliance with waste disposal and emission control.
Manufacturing Area Requirements - Overall Site Plan & Area Distribution

Table No.:7

COMMON AREAS

Note: All production plants viz., OSD, Injectables, Bio-pharma, Vaccines and Oncology OSD shall be designed for future expansion capability

Table No.: 8

PLANT 1 – OSD BLOCK (ASSUMED OUTPUT CAPACITY IS 15 MILLION UNITS PER DAY)

Table No.: 9

PLANT 2 – mAbs AND BIOSIMILARS (ASSUMED OUTPUT CAPACITY IS 3 MILLION UNITS PER ANNUM)

Table No.: 10

PLANT 3 - Filling Lines for Vials, Ampoules, and Pre-Filled Syringes

Table No.: 11

PLANT 4 - Oncology OSD Facility for Production of Hormonal and Cytotoxic Oral Formulations

Additional Considerations

High-Containment Design:

Closed systems with negative pressure, airlocks, isolators, and barrier technology.

Cross-Contamination Control:

Separate air handling, material flows, and personnel flows for hormonal and cytotoxic drugs.

Table No.: -12

PLANT 5 – VACCINE fill finish Facility (ASSUMED CAPACITY OUTPUT IS 50 MILLION DOSES PER ANNUM)

E. FINANCIAL FEASIBILITY

This review provides an overview of CAPEX, Operational Expenditure (OPEX), financial projections, break-even analysis, and potential Return on Investment (ROI).

Cost Breakdown: CAPEX

Table No.: 13

1.2 Capital Expenditure (CAPEX)

The total investment is \$571.62 M USD, allocated across different manufacturing units and operations as follows:

Table No.: 14

1.3 Operational Expenditure (OPEX) Breakdown

Annual OPEX includes raw materials, labour, utilities, maintenance, logistics, and regulatory compliance.

Estimated OPEX in full scale operations:

Table No.: 15

* As per calculations

NOTE: The actual project estimates are USD 529 MUSD (including Capex and Opex) as per the DPR.

However, an additional amount of 42.62 MUSD taking the total amount to 571.62 MUSD (capex and Opex) is considered for product registrations, clinical trials, licensing, technology transfer, manpower training including contingency.

Financial Projections and Revenue Estimates

Table 16:

2.1 Estimated Annual Revenue (First 7 Years)

Revenue is projected based on market demand, production capacity, and price realization

Table No.: 17

Revenue Projections

Revenue Contribution by Segment:

OSD - 20%

General Injectable facility – 15%

Biopharmaceutical products – 15%

Vaccines – 15%

Oncology OSD - 25%

Contract Manufacturing & Exports: 10%

Break-Even Analysis

Table No.: 18

3.2 Payback Period & Profitability Timeline

Payback period: ~5 years

First-year losses due to ramp-up & market entry.

Full profitability expected in Year 4-5 with a Gross Profit Margin of ~60-75%.

4. Return on Investment (ROI) Analysis

4.1 Projected ROI

$ROI = \text{Annual Profit} / \text{Total investment} \times 100$

Year 3 Expected Net Profit: ~\$188.18M USD

ROI by Year 3: $188.18 / 571.62 \times 100 = 32.92\%$

Key Financial Risks & Mitigation Strategies

Table No.: 19

6. Conclusion & Strategic Recommendations

CAPEX of \$442.90 M is strategically allocated across multiple high-growth segments.

OPEX of ~\$128.72 M annually ensures robust operations & regulatory compliance.

Break-even is expected within the third year, with profitability by Year 3-4.

Strategic Recommendations:

Focus on Oncology & Biologics: Higher-margin products provide better financial returns.

Optimize API Sourcing: Reduce costs by investing in local API production or strategic long-term supplier contracts.

Enhance Export Potential: Leverage Uganda's EAC trade advantages to increase regional market penetration.

Automate Supply Chain & Manufacturing

Seek Investment & Government Support: Public-private partnerships for incentives & subsidies to lower CAPEX burden.

F. SUPPLY CHAIN AND RAW MATERIAL SOURCING

The supply chain must ensure a steady supply of raw materials, efficient production, strict quality control, and seamless distribution, while mitigating risks associated with logistics, regulatory compliance, and market fluctuations.

1. Key Components of the Supply Chain

A robust supply chain for a SUNSHINE MEDICAL PTE. LTD., company shall include the following key components:

Raw Material Sourcing (APIs, Excipients, Packaging Materials)

Inventory & Warehousing

Production & Manufacturing

Quality Control & Regulatory Compliance

Cold Chain & Storage (For Vaccines & Biosimilars)

Distribution & Logistics

Risk Management & Supply Chain Resilience

2. Raw Material Sourcing Strategy

2.1 Active Pharmaceutical Ingredients (APIs) and Excipients

Uganda does not have a strong domestic API production base, so APIs must be imported.

Primary API sourcing regions:

India & China – Major global API suppliers.

Europe (Germany, Switzerland, Italy) – High-quality APIs for oncology & biosimilars.

Africa (South Africa, Egypt, Morocco) – Growing pharmaceutical hubs for APIs.

Procurement Strategy:

Develop multiple supplier relationships to reduce dependency on a single country.

Use long-term contracts & Just-in-Time (JIT) supply models for high-cost APIs.

Maintain buffer stock of critical APIs (especially for oncology & vaccines).

2.2 Packaging Materials

Primary packaging: Blister packs, vials, ampoules, pre-filled syringes, HDPE bottles.

Secondary packaging: Cartons, labels, package inserts.

Suppliers: Kenya, India, UAE, Turkey, and South Africa.

2.3 Supply Chain for Biologics (Vaccines & Biosimilars)

Requires specialized cold chain logistics (-80°C for some biologics).

Partnerships with international biotech firms for technology transfer.

Cold storage facilities at airports & major distribution hubs.

3. Inventory & Warehousing Management

3.1 Raw Material Storage

Separate storage for APIs, excipients, and packaging materials.

Cold storage for temperature-sensitive raw materials.

Real-time inventory tracking system (ERP-based).

3.2 Finished Goods Warehousing

GMP-compliant warehouses with segregated storage for different product categories.

First-Expiry-First-Out (FEFO) system for batch management.

Strategic warehouse locations in Uganda and EAC countries for faster distribution.

4. Production & Manufacturing Workflow

4.1 Manufacturing Planning & Scheduling

Lean Manufacturing to reduce waste and optimize production costs.

Batch Manufacturing Record (BMR) & Batch Production Record (BPR) compliance.

4.2 Facility Layout & Integration with Supply Chain

Segregated production lines for oncology, OSD, syrups, vaccines, and biosimilars.
Integrated Manufacturing Execution Systems (MES) for real-time monitoring.
Automated raw material dispensing & production tracking.

5. Quality Control & Regulatory Compliance

Stringent QC checks for APIs, excipients, and packaging materials.
Real-time product traceability using barcoding & serialization.
Compliance with WHO-GMP, EAC regulatory frameworks, US FDA, EMA.
Regular audits of suppliers & third-party logistics providers.

6. Cold Chain & Storage for Vaccines & Biosimilars

Cold storage at 2-8°C, -20°C, and -80°C.
Real-time temperature monitoring
Strategic cold chain hubs in Kampala & EAC export markets.

7. Distribution & Logistics Strategy

7.1 Domestic Distribution (Uganda)

Direct distribution to hospitals, pharmacies, and wholesalers.
Regional distribution centres in Kampala, Entebbe, Jinja, and Mbarara.
GPS-tracked delivery fleet with route optimization for last-mile delivery.

7.2 East African Community (EAC) Export Strategy

Key export markets: Kenya, Tanzania, Rwanda, Burundi, South Sudan, DRC.
Regulatory harmonization under the EAC-MRH program for seamless cross-border approvals.
Dedicated pharmaceutical logistics hubs in Nairobi (Kenya) & Dar es Salaam (Tanzania).

7.3 International Exports

Oncology & biosimilars exports to high-value markets in Europe & Middle East.
Collaboration with global pharmaceutical distributors.

8. Risk Management & Supply Chain Resilience

8.1 Key Risks in the Supply Chain

Dependency on imported APIs → Risk of supply chain disruptions.
Regulatory delays → Impact on product approvals & market entry.
Cold chain failures → Risk for vaccines & biologics.
Political & economic instability in Uganda & EAC → Disruptions in logistics.

8.2 Risk Mitigation Strategies

Dual sourcing strategy for APIs & critical raw materials.
Investment in API manufacturing in Uganda (long-term goal).
Use of digital supply chain solutions (AI-driven demand forecasting).
Collaboration with global logistics providers (DHL, UPS, local 3PLs).

9. Digital Transformation in Supply Chain Management

9.1 AI Based Monitoring

AI-based demand forecasting to optimize inventory levels.
Cost of Maintaining a Robust Pharmaceutical Supply Chain in Uganda

Key Cost Components of a Pharmaceutical Supply Chain

Table No.: 21

Note: The cost varies depending on production scale, market reach, automation levels, and regulatory requirements.

2. Breakdown of Major Cost Elements

2.1 Raw Material & API Procurement Costs

APIs account for 60%-80% of drug production costs, particularly for oncology & biologics.
Uganda imports most APIs from India, China, and Europe, incurring import duties & logistics expenses.
Cost fluctuations due to supply chain disruptions, geopolitical tensions, and regulatory constraints.
Mitigation: Multi-supplier agreements & local API production investments.

2.2 Packaging Material Costs

Primary packaging (blister packs, vials, ampoules, prefilled syringes) is imported or locally sourced.
Secondary packaging (cartons, labels, inserts) needs compliance with international standards (e.g., serialization, barcoding).
Sustainable & eco-friendly packaging costs 10%-15% more but enhances regulatory acceptance.

2.3 Warehousing & Cold Storage

Cold storage for vaccines & biosimilars (-80°C to 2-8°C) incurs high energy & maintenance costs.
Storage facilities in Kampala & regional hubs (e.g., Jinja, Mbarara) require advanced temperature monitoring.

2.4 Manufacturing & Production Logistics

Includes the cost of transporting raw materials, managing production scheduling, and handling batch-to-batch variations.
Biologics and vaccines require specialized production environments, increasing utility & maintenance costs.

2.5 Quality Control & Compliance

Regulatory costs include GMP audits and licensing fees.
Maintaining high-end analytical labs for testing APIs, excipients, and final products.
Oncology & biosimilar testing requires more stringent & expensive analytical processes.

2.6 Cold Chain Infrastructure for Vaccines & Biologics

Ultra-low temperature storage (-80°C) for certain biologics.
GPS-tracked refrigerated trucks for secure transportation.
Backup power systems to ensure uninterrupted temperature control.

2.7 Distribution & Last-Mile Delivery

Cross-border regulatory compliance costs (for EAC markets).
Investment in regional distribution hubs (e.g., Nairobi, Dar es Salaam).
High-security transportation for oncology drugs to prevent theft.

2.8 IT & Digital Supply Chain Systems

Track-and-trace systems to prevent counterfeit drugs.
Automated inventory management to reduce wastage & improve efficiency.

2.9 Risk Management & Contingency Planning

Emergency stockpiling of critical APIs & biologics.
Alternative supply routes & backup logistics partners.
Investment in local API production to reduce import dependency.

2.10 Regulatory Compliance & Audits

WHO-GMP certifications for global market access.
Annual site inspections & compliance renewals.
Investment in training & capacity building for regulatory teams.

3. Cost Optimization Strategies

A. Local API & Packaging Manufacturing Investment

Reduce import costs by establishing API & packaging material production in Uganda.
Collaboration with African pharmaceutical hubs (South Africa, Egypt, Morocco).

B. Multi-Supplier & Regional Sourcing Strategy

Reduce risk by sourcing APIs from multiple suppliers.

EAC regional integration to reduce cross-border tariffs & logistics delays.

C. Green Energy & Sustainability Investments

Solar-powered cold storage to cut energy costs by 30-40%.

Eco-friendly packaging to reduce regulatory risks & enhance marketability.

G. HUMAN RESOURCES

In this project below listed facilities/blocks are planned to be constructed:

Oral Solid Dosage (OSD) Facility (Capacity Production of 15 million units/day (tablets, capsules, and syrups).

Biopharma Drug Product Facility (Capacity: Specialized for monoclonal antibodies, hormones, and insulin analogs)

Vaccine Formulation and Fill- Finish Facility (50 Million doses per annum)

Oncology OSD Facility

Injectable Facility (Capacity: High-speed filling lines for vials, ampoules, and pre-filled syringes)

Number of employees with different skill set shall be employed for all above mentioned facilities.

Table No.: 22

Summary of estimated Manpower required for Operations

Hiring of employees with skill sets and expertise:

It is important to have healthy employees with aseptic skills especially in manufacturing units to avoid any contamination during manufacturing of drug products.

Separate organograms are prepared for engineering, corporate, HR & administration, warehouse, Quality (QC & QA) and manufacturing units to hire sufficient and effective workforce considering cost effectiveness.

Details of Organograms:

Corporate -Employees with managerial skills, project skills and leadership skills etc. will be hired in corporate for different levels like CEO, Head of engineering, Head of operations/manufacturing, project manager, Head of quality, Head of finance, Head of logistics and Head of HR. Managers and group leaders from each section will be managed/lead by corporate Heads.

Engineering -In engineering section, employees with skills like automation, instrumentation including complicated manufacturing equipment, electrical, mechanical, utilities (Water, steam, Air) and information technology will be hired at different levels.

HR & Administration -In HR & admin section, employees with skills to handle transport, canteen, legal challenges like pollution control, employee act and licensing application and renewal etc., recruitment and day to day management of contractual / fixed employees, employee wellness, medical facility and general management of BMS like furniture, rest rooms, and external vendors for admin support will be hired.

Warehouse -In warehouse section, skills to ensure the safe, efficient, and compliant movement of materials and products is required. It includes logistics management skills. Knowledge of inventory management using software like SAP or ERP, document and material management is essential in warehouse.

Quality Unit - (Quality control and Quality assurance) is critical to ensuring the safety, efficacy, and compliance of drug products. In QC, testing skills, knowledge of GLP, Data analysis and sample management are essential skills. In QA, Process validation and control, regulatory compliance, Risk Management, Documentation, Audits and inspection skills, Change management and trainings are essential skills.

Manufacturing -In manufacturing, knowledge of GMP, operation of complicated/automated equipment or system, knowledge of process and aseptic behaviour/environment, documentation and technical skills for problem solving in day-to-day process are crucial skills.

Health, Safety standards for Human Resource working in Pharmaceuticals:

For biopharmaceutical projects, regular medical check-up including tests like eyesight, audiometry, blood test, breathing test, X-Ray, ECG and basic tests; at-least annually for all employees is essential.

Key aspects in HR feasibility

Workforce Availability

Over the course of the last decade, Uganda's economic growth has ranked among sub-Saharan Africa's strongest; indeed, the country's annualized average growth rate was 5.4% between 2010 and 2019 (World Bank, 2020).

Uganda has a young and growing workforce, with over 75% of its population under the age of 30.

Total population: Approx. 51 million (2025 estimates)

Cost of Human Resources

In Uganda, employee salaries are relatively low, but there are also some differences. For example, in the capital city of Kampala, the monthly salary of an ordinary employee is usually between 100-200USD. In other cities, this number may be even lower. The salary level will be higher for senior management and technical talents.

Average monthly salaries (estimates):

Unskilled worker: \$100–\$200

Mid-level professional: \$500–\$1,500

High-skilled/managerial: \$2,000+

Legal and Compliance Factors

Key employment laws:

Employment Act (2006): Governs employment terms, termination, wages, and working conditions.

Occupational Safety and Health Act (2006): Ensures workplace safety.

Workers' Compensation Act (2000): Covers workplace injury compensation.

Uganda's labour laws are primarily governed by the Employment Act of 2006, which establishes the minimum standards of employment in the country.

Uganda is committed to upholding international labour standards and has ratified several key

International Labour Organization (ILO) Conventions. These include the Freedom of Association and Protection of the Right to Organize Convention, 1948 (No. 87), the Right to Organise and Collective Bargaining Convention, 1949 (No. 98), and the Collective Bargaining Convention, 1981 (No. 154).

The Uganda Employment Act applies to all employees, irrespective of nationality that the employers hire under the service contract.

According to the employment act 2006 and labour laws in Uganda, the standard working hours of an employee is a maximum of eight hours per day and 48 hours a week; excluding lunch and worship time.

Employers planning to hire or relocate international employees to Uganda must understand the processes surrounding work permits and visas. Ensuring compliance with Ugandan immigration laws helps streamline the hiring process and avoid potential legal hurdles.

Cultural and Language Considerations

Ugandan communication often leans towards indirectness, especially when delivering negative feedback or expressing disagreement.

Cultural Lens: The Ugandan concept of "Gwasi" (shame) discourages public criticism. This can influence communication, leading individuals to deliver critical messages indirectly to avoid causing shame to the recipient.

Uganda is a multilingual nation like many African countries. However, English is targeted as a medium of instruction and language of examination.

H. RISK ASSESSMENT

Risk Assessment and Mitigation Strategies for SUNSHINE MEDICAL PTE. LTD., Plant in Uganda

1. Political Risks and Mitigation Strategies

Uganda, like many developing economies, has political uncertainties that can affect business operations. These include regulatory changes, policy instability, and geopolitical factors.

1.1 Key General Risks

Table No.:23

1.2 Risk Mitigation Strategies

Early Engagement with Regulators:

Work closely with Uganda's National Drug Authority (NDA) and the Ministry of Health to ensure timely approvals.

Seek WHO-GMP certification for global market access.

Public-Private Partnerships (PPP):

Collaborate with government bodies to access tax benefits, incentives, and investment protection.

Diversification of Export Markets:

Establish trade links with EAC, COMESA, and beyond to reduce dependence on a single market.

Strengthening IP Protection:

Register patents and trademarks regionally through ARIPO (African Regional Intellectual Property Organization) to safeguard innovations.

2. Economic Risks and Mitigation Strategies

Economic instability can impact capital investment, supply chain costs, and market demand for pharmaceutical products.

2.1 Key Economic Risks

Table No.:24

2.2 Risk Mitigation Strategies

- Localizing API Sourcing:

Develop partnerships with regional API manufacturers to reduce reliance on imports and mitigate forex risks.

- Hedging Against Currency Risks:

Utilize foreign exchange hedging instruments to stabilize procurement costs.

Maintain a mix of local and foreign currency reserves for operational expenses.

- Operational Cost Optimization:

Implement energy-efficient manufacturing to reduce electricity costs.

Automate production processes to minimize labour costs.

- Strategic Supplier Agreements:

Enter long-term contracts with multiple suppliers for key raw materials to ensure supply chain resilience.

- Diversifying Revenue Streams:

Expand into high-margin segments (oncology, vaccines, biosimilar).

Target institutional buyers like the government, NGOs, and international health organizations (e.g., GAVI, WHO, Global Fund).

3. Environmental Risks and Mitigation Strategies

Uganda has stringent environmental regulations for pharmaceutical manufacturing, including waste management, water usage, and emissions control. Non-compliance can lead to fines, production shutdowns, or reputational damage.

3.1 Key Environmental Risks

Table No.:25

3.2 Risk Mitigation Strategies

- Sustainable Water Management:

Implement water recycling & rainwater harvesting systems to reduce dependence on municipal supplies.

Use high-efficiency Clean-in-Place (CIP) systems to minimize water wastage.

- Advanced Waste Treatment Solutions:

Install Zero Liquid Discharge (ZLD) systems for pharmaceutical effluent treatment.

Set up on-site hazardous waste incinerators to manage cytotoxic and biological waste safely.

- Renewable Energy Integration:

Solar power installations on-site to reduce dependency on the national grid.

Waste heat recovery systems to improve energy efficiency.

- Green Manufacturing Practices:

Obtain ISO 14001 Environmental Certification to align with international best practices.

Conduct regular environmental audits to ensure compliance with NEMA standards.

4. Summary of Risk Mitigation Strategies

Table No.:25

5. Conclusion

While setting up a pharmaceutical manufacturing plant in Uganda presents general, economic, and environmental risks, these can be effectively managed through strategic planning and proactive mitigation measures.

I. SUSTAINABILITY

1. Sustainable Waste Management Strategies

SUNSHINE MEDICAL PTE. LTD., facility shall generate solid, liquid, and hazardous waste that must be effectively managed to comply with environmental regulations and protect ecosystems.

1.1 Segregation and Recycling of Solid Waste

Implement waste segregation at the source to separate biodegradable, non-biodegradable, recyclable, and hazardous waste.

Recycle packaging materials like paper, plastics, and aluminium foils used in blister packs.

Partner with certified waste management companies to handle non-recyclable hazardous waste.

1.2 Efficient Handling of Hazardous Waste

Incinerate cytotoxic oncology waste and contaminated biological waste in high-temperature incinerators (above 850°C).

Use autoclaves and chemical disinfection for biological waste.

Adopt encapsulation techniques for disposing of highly toxic chemicals and APIs safely.

1.3 Effluent Treatment and Liquid Waste Management

Install an Effluent Treatment Plant (ETP) and Zero Liquid Discharge (ZLD) system to ensure wastewater treatment before discharge.

Reuse treated water for landscaping, cooling systems, and non-potable applications.

1.4 Minimization of API & Solvent Waste

Implement solvent recovery units to recycle and reuse solvents in oncology and biopharma production.

Adopt green chemistry techniques to minimize hazardous waste production.

Optimize batch sizes and manufacturing processes to reduce raw material waste.

2. Energy Efficiency Strategies

Energy consumption will be a significant operational cost in pharmaceutical manufacturing. A sustainable energy plan shall ensure cost reduction and reduced carbon footprint.

2.1 Integration of Renewable Energy Sources

Solar Power Installation:

Install solar panels on rooftops to supplement grid electricity and reduce dependence on fossil fuels.
 Target at least 30-50% energy from solar for non-critical operations.

Biomass & Biogas Utilization:
 Utilize biomass boilers for steam generation using agricultural waste.
 Install a biogas plant to convert organic waste into energy.

2.2 Energy-Efficient HVAC & Cooling Systems
 Use energy-efficient HVAC systems with variable frequency drives (VFDs) to optimize airflow and minimize energy wastage.
 Implement cold chain optimization for vaccines and biopharma products, including energy-efficient refrigeration and insulation systems.
 Utilize waste heat recovery systems to capture and reuse excess heat from industrial processes.

2.3 Smart Lighting and Equipment Optimization
 Install LED lighting with motion sensors to reduce power consumption in low-traffic areas.
 Use high-efficiency motors in production lines to reduce energy waste.
 Implement real-time energy monitoring systems to track and optimize energy use.

3. Water Conservation and Efficient Usage
 Pharmaceutical production requires significant water usage for cooling, formulation, and cleaning processes. Water scarcity in Uganda makes conservation and recycling essential.

3.1 Water Recycling & Reuse
 Install Water Treatment Plants (WTPs) with multi-stage filtration, RO, and UV disinfection to reuse wastewater.
 Recycle treated wastewater for boilers, chillers, and non-potable cleaning processes.

3.2 Rainwater Harvesting
 Set up rainwater harvesting tanks to collect and store rainwater for non-production use.
 Utilize rainwater for landscaping, cooling towers, and toilet flushing.

3.3 Optimized Process Water Usage
 Implement Clean-in-Place (CIP) and Sterilize-in-Place (SIP) systems to reduce water usage in equipment cleaning.
 Use dry cleaning methods where possible to minimize the need for extensive water washing.
 Train staff on water conservation best practices to minimize waste.

4. Regulatory Compliance & Green Certifications
WHO-GMP & EAC Environmental Regulations: Ensure compliance with local and international environmental guidelines.
ISO 14001 Certification: Implement environmental management systems to enhance sustainability.
LEED Certification: Design facilities with sustainable building materials to meet green building standards.

5. Conclusion
 SUNSHINE MEDICAL PTE. LTD., shall require integrated waste management, energy-efficient operations, and water conservation measures. By adopting renewable energy, recycling waste, and optimizing resource usage, this facility can achieve long-term environmental and economic benefits while maintaining regulatory compliance and corporate social responsibility.

J. Key HR Challenges and mitigation strategies

1. Skilled Workforce Shortage
 Limited local experience in advanced pharmaceutical manufacturing (especially GMP-compliant operations for vaccines, oncology, and monoclonal antibodies).
 Shortage of experienced quality assurance (QA), quality control (QC), production, validation, and biotech-specific personnel.

2. Training and Capability Gaps
 Need for training in WHO-GMP and global standards.
 Low exposure to advanced equipment and automation technologies common in global pharma plants.

3. High Attrition or Brain Drain
 Trained personnel may be poached by competitors or seek jobs abroad due to better salaries.

4. Cultural & Communication Barriers
 Integration challenges when blending expat professionals with local teams in managerial and technical roles.

Mitigation & Risk Management Strategies

1. Early & Strategic Workforce Planning
 Workforce demand mapping: Identify headcount, roles, and experience levels required for each production phase.
 Staggered hiring aligned with the commissioning of each production unit.

2. Establish a Technical Training Academy
 Set up an in-house Pharma Training Institute or Centre of Excellence.
 Partner with international experts to develop GMP, EHS, and regulatory curricula.
 Simulate manufacturing environments (cleanrooms, QC labs, aseptic suites).
 The training program shall constitute of the following aspects
 Induction of new employees
 cGMP training
 GLP training
 On the job training consisting of SOP training, equipment training etc.,
 Safety training
 Supervisor training
 Manager training
 Job change training or after leave of absence
 Remedial training
 Contractor employee training

3. International Partnerships
 Collaborate with Indian, European, or South African pharma companies or universities for knowledge transfer.
 Consider short-term expat secondments (2–5 years) to build and mentor local teams.

4. Collaboration with Local Academia
 Tie up with Makerere University or other local institutions for:
 Curricula alignment with pharma needs.
 Internship/apprenticeship programs.
 Fast-tracking diploma and B.Pharm graduates into technical roles.

5. Talent Retention Schemes
 Offer competitive packages, training sponsorships, and bonded contracts.
 Career development pathways for technicians to move into supervisory or QA roles.
 Stock options or profit-sharing models to retain senior staff.

6. Outsource Non-Core Activities Initially
 To reduce early HR load, consider outsourcing packaging, engineering maintenance, and logistics until local capacity matures.

7. Human Resources Rollout Roadmap

Phase-Wise HR Rollout Plan (2025–2030)

A. Planning & Initial Setup (2025)

Workforce Planning: Finalize detailed manpower requirement for each functional block (production, QA/QC, engineering, admin, etc.).

Key Roles: Project HR Head, Talent Acquisition Manager, Training & Compliance Manager

Action Points:

Begin local hiring process for support staff, documentation personnel

Initiate expat hiring for plant design, validation, and commissioning oversight

Establish HR policies, on-boarding modules, payroll systems

B. OSD & Syrup Commissioning

Estimated Staff Requirement: (As per Table No 19)

Key Functions: Production operators, QA/QC analysts, warehouse team

Action Points:

Launch training academy in partnership with local universities (Makerere, Mbarara, etc.)

Train local hires in GMP compliance, SOP writing, and safety protocols

Expat engineers and QA managers to lead operations and mentor local staff

C. Oncology & Injectable Line Setup

Estimated Staff Requirement: +/- (As per Table No 19)

Key Functions: Cytotoxic handling experts, sterile Injectables line operators, QC microbiology

Action Points:

Specialized cytotoxic/Hormonal handling training

Safety and containment procedure modules

Expand academy offerings to include aseptic processing and EHS protocols

D. Vaccine & Biopharma Commissioning

Estimated Staff Requirement: -(As per Table No 19)

Key Functions: Biotech fill-finish experts, cold chain QA/QC

Action Points:

Partner with international biotech training programs (India, EU)

Establish a biotechnology learning track in the internal academy

E. Operations Stabilization & Local Leadership Development

Target Workforce: 1,000+ including indirect employees

Focus Areas: Succession planning, reducing dependency on expats

Action Points:

Develop career growth ladders for technicians and supervisors

Launch Leadership Development Programs (LDP) for Ugandan managers

Retention strategy implementation (bonuses, ESOPs, continuous training)

Training Strategy Summary

In-House Pharma Academy to train employees per year

Modules in:

GMP & regulatory compliance

Pharmaceutical operations (OSD/liquid/injectable/Oncology)

Aseptic techniques

Equipment maintenance & calibration

Training Partners: Local academia + global consultants

Risk Mitigation Measures

Challenge: Limited experienced pharma talent pool

Solution: Layered expat hiring, internal academy, local university tie-ups

Challenge: Attrition post-training

Solution: Bonded employment, career development programs, competitive pay

Challenge: Cultural integration

Solution: Cross-cultural workshops, expat-local mentorship structure.

K. STRATEGIES FOR ACCELERATED ROI ON THE PROJECT

1. Phased Commissioning of Production Lines

Start with OSD and syrups, which have lower regulatory complexity and shorter setup timelines.

These can begin generating revenue within 12–18 months, while more complex lines (biopharma, oncology, vaccines) are still under validation.

Helps spread OPEX and initiates early cash flow.

2. Prioritize Products with High Local and Regional Demand

Focus on essential medicines, high-burden diseases (malaria, diabetes, antibiotics), and therapeutic areas like oncology and vaccines with unmet needs.

Customize the portfolio to include WHO-prequalified and procurement-friendly products for faster uptake by public health programs.

3. Partner with Global Health Buyers Early

Engage with donor agencies (Global Fund, GAVI, WHO, USAID) to position your facility as a local supplier.

Secure advance purchase agreements (APAs) or long-term supply contracts, even during construction phase. These contracts anchor demand and improve investor confidence.

4. Leverage Contract Manufacturing (CDMO) for Revenue Boost

Offer your excess capacity for contract manufacturing of OSD, fill-finish, or vaccine products for regional or global pharma firms.

Acts as a bridge revenue stream before full-scale launch of your branded portfolio.

5. Accelerate Regulatory Approvals through EAC Harmonization

Register products using the EAC Medicines Regulatory Harmonization (MRH) program to access multiple countries simultaneously.

Cuts down duplicated efforts and approval timelines, speeding up market entry.

6. Optimize Operational Costs from Day 1

Invest in energy-efficient infrastructure (solar, efficient HVAC) and automation to lower long-term utility and labour costs.

Design supply chain to minimize storage and transportation overhead.

7. Use a Hybrid Distribution Model

Combine direct-to-hospital/clinic distribution with national wholesalers and tenders.

Reduces reliance on a single sales channel and speeds up product uptake across multiple segments.

8. Government Incentives and Tax Planning

Take full advantage of government incentives, including VAT deferments, customs waivers, tax holidays, and utility subsidies.

Negotiate land lease concessions or reduced utility tariffs to reduce fixed overheads.

9. Continuous Financial Monitoring

Set up a Project Management Office (PMO) and cash burn tracking system to detect cost overruns early.

Implement strict working capital control (inventory turnover, receivables) during operations to stay lean.

Table No:- 26

Example ROI Acceleration Path (Indicative)

L. SWOT ANALYSIS

Strengths

Robust Financial Backing

The promoters' strong financial background ensures ample capital availability for high-tech manufacturing infrastructure, reducing financing risks.

USD 571.62 million CAPEX provides a solid foundation for a state-of-the-art facility with capacity for future expansion.

Strategic Product Portfolio

Manufacturing a diverse mix of high-demand therapeutics—from essential OSDs and syrups to complex oncology drugs, vaccines, and biosimilars—positions the company to address both local and regional health needs.

Biosimilars and oncology Injectables offer high-margin opportunities with increasing global and regional demand.

Location Advantage – Uganda & EAC Access

Uganda provides duty-free access to the East African Community (EAC), reducing export barriers and expanding market reach.

Strategic geographic location in Central-East Africa allows access to under-served pharmaceutical markets in South Sudan, Rwanda, Burundi, DRC, etc.

Modern Infrastructure and Technology Potential

Opportunity to design a greenfield facility aligned with WHO-GMP, EMA, and USFDA standards from the outset.

Flexibility to incorporate cutting-edge technologies (e.g., continuous manufacturing, single-use systems for biologics).

Weaknesses

1. Lack of local Pharmaceutical Domain Experience

Local experience of handling large scale integrated pharma-biopharma project experience

Mitigation: Early recruitment of experienced pharma professionals and regulatory consultants is critical, as is establishing a governance board with pharma advisors.

2. High Dependence on Imports for APIs, Equipment & Components

Uganda lacks a strong API and excipient manufacturing ecosystem, so nearly all critical raw materials, consumables, and equipment must be imported—mainly from India, China, or Europe.

This increases lead times, procurement costs, customs complexities, and exposure to global supply chain disruptions.

3. Skills and Talent Gap in Local Workforce

Uganda's pharmaceutical industry is still developing, with limited availability of skilled technicians, validation engineers, QC analysts, and GMP-trained production staff, especially in sterile manufacturing or biotech.

Over-dependence on expats may drive up costs and hinder the development of sustainable local capability.

Cultural integration, language, and training gaps can affect productivity and retention if not addressed proactively.

4. High Operating Costs for Complex Products

Oncology, vaccines, and bio-similars require sophisticated cleanroom environments, cold chain management, and batch-specific quality testing.

Maintaining aseptic conditions, HVAC, water systems (PW/WFI), and validation for multiple production lines significantly increases OPEX.

Opportunities

1. Growing Demand in Uganda and EAC for Quality Medicines

The East African Community (EAC) is home to over 280 million people with growing health burdens like cancer, diabetes, infectious diseases, and vaccine-preventable illnesses.

There's high demand for affordable, high-quality generics, bio-similars, and essential vaccines.

Current reliance on imports from India, Europe, and China creates a gap for trusted local manufacturers with faster market responsiveness and cost advantages.

2. Support from Government and International Health Agencies

Uganda and other EAC governments are actively promoting local manufacturing through:

Tax breaks, incentives, and infrastructure subsidies.

Potential fast-track approvals for local production of priority medicines.

Global health organizations like GAVI, WHO, UNICEF, Global Fund, and African CDC are eager to support African-based production, especially for vaccines and pandemic preparedness.

This creates opportunities for donor-funded procurement, technology transfer, and concessional financing.

3. Regional Regulatory Harmonization (EAC-MRH)

The EAC's Medicines Regulatory Harmonization initiative reduces the complexity and time for market entry across multiple countries.

A product registered under EAC-MRH can be fast-tracked for approval in all EAC member states, expanding your potential market and improving return on investment (ROI).

4. First-Mover Advantage in Advanced Therapies

There are very few players in the region manufacturing oncology products, bio-similars, or vaccines. A well-executed entry into these segments provides:

High-margin sales.

Competitive edge due to regulatory barriers that deter others from entering.

Leverage to license or co-develop products with international partners.

5. Greenfield Facility – Built for Future-Ready Compliance

Starting from scratch allows for:

State-of-the-art modular facility design that meets WHO-GMP and global requirements from day one.

Scalability—facilities can be expanded based on market growth or strategic partnerships.

Integration of renewable energy systems, automation, which can lower long-term operating costs and improve environmental credentials.

6. Technology Transfer and Global Collaborations

Many mid-sized Western and Asian pharma companies are seeking CDMO or joint venture partners in Africa to enter emerging markets.

This facility, once validated and operational, can be positioned as a fill-finish hub or contract manufacturing site for such companies—especially for vaccines and biologics.

Threats

Regulatory and Compliance Challenges

Ensuring continuous compliance with WHO-GMP, NEMA (Ugandan environmental laws), NDA, and other international standards requires highly skilled QA/QC teams and constant oversight.

Geopolitical and Trade Instability

Political tensions, conflicts in neighbouring countries, or disruptions in trade routes could hamper logistics, exports, or sourcing of raw materials.

Currency Fluctuations and Import Dependency

Heavy reliance on imports for APIs and machinery subjects the plant to foreign exchange risks, especially if the Ugandan Shilling depreciates.

Intensifying Global Competition

Global pharmaceutical companies and Indian manufacturers could enter the region with cheaper generic

drugs, impacting price competitiveness.

Product-Specific Challenges

Biosimilars and vaccines involve complex regulatory pathways, high R&D costs, and the risk of batch failures or cold-chain disruptions.

Construction & Commissioning Delays

Incomplete funding during the CAPEX phase can halt construction, delay procurement of equipment, and postpone validation activities.

Pharmaceutical projects are highly sequential: civil work must be completed before HVAC installation, which precedes equipment commissioning, which in turn precedes qualification. A break at any point stalls everything.

Regulatory Non-Compliance

If the facility is only partially built or validated, it cannot obtain GMP licenses or product approvals.

This puts the company in a no-revenue, high-burn position, severely affecting investor confidence and credibility.

Supply Chain Penalties

Equipment vendors and raw material suppliers often work with milestone-based payments. Funding gaps can lead to order cancellations, higher renegotiated prices, or contractual penalties.

Lead times in pharma are long, so any reordering due to lapsed payments causes multi-month delays.

Reputational Damage & Loss of Stakeholder Confidence

Financial instability affects relationships with regulators, investors, banks, and potential customers (like international health agencies).

Once perception of financial distress sets in, it is harder to secure bridge financing or partnerships.

Strategic Recommendations

Hire pharma domain experts in key leadership roles: manufacturing, regulatory, quality assurance, and R&D.

Form strategic alliances with experienced CDMOs or technology licensors for oncology and biosimilar lines.

Invest in workforce development through partnerships with local universities and international training programs.

Develop dual-source supply chain strategies and build regional sourcing capacity for packaging and excipients.

Design for modular expansion, allowing scaling based on product demand and market growth.

Pursue green certifications (LEED, ISO 14001) to attract ESG-aligned investors and NGOs.

M. SUMMARY AND RECOMMENDATIONS

SUNSHINE MEDICAL PTE. LTD., Pharmaceutical Plant project in Uganda is deemed highly viable based on a multi-dimensional feasibility study and information given in this report.

Uganda's pharmaceutical market presents substantial opportunities, with projected revenue reaching US\$329 million by 2025 and strong CAGR, especially in generic drugs, oncology, vaccines, and biopharmaceuticals.

The plant's integrated model, with Oral Solid Dosage (OSD) forms and more complex formulations such as vaccines and biologics, aligns with the market demand and local regulatory evolution under WHO-GMP frameworks.

Section-Wise Recommendations

1. Oral Solid Dosage (OSD)

Summary:

Focused on high-volume generics for common infectious and chronic diseases.

Recommendations:

Leverage economies of scale for paracetamol, amoxicillin, anti-hypertensives, antidiabetics HIV, anti-tuberculosis and anti-parasitic drugs (malaria, Soil-transmitted helminthiases (STH)).

Introduce low-cost fixed-dose combinations tailored to local disease patterns.

Invest early in automated production lines and MES systems to ensure compliance and efficiency.

Explore regional distribution partnerships to penetrate COMESA markets.

2. General Injectables

Summary:

Aim at supplying critical care and anti-infective Injectables, including Anaesthetics, antibiotics, analgesics, and small volume parenteral.

Recommendations:

Prioritize aseptic filling lines and technologies to meet NRA and WHO-GMP standards.

Build cold chain infrastructure for temperature-sensitive products.

Early vendor development for critical raw materials, APIs and packaging materials.

Innovation in container closure system and devices to reduce foot print of secondary packaging- PP tubes)

Secure API contracts and quality agreements with reputed Indian/Chinese suppliers.

Position Injectables as a high-demand, medium-margin product line with broad application in public and private health facilities.

3. OSD Oncology

Summary:

High-margin segment identified as underserved in Uganda and East Africa.

Recommendations:

Focus initially on oral chemotherapeutics such as capecitabine, imatinib, and tamoxifen.

Ensure high-containment facility design and strict handling protocols to manage hazardous APIs.

Partner with regional cancer centres and NGOs to support market uptake and pharmacovigilance.

Consider future vertical integration into hormonal therapies and oral targeted therapies.

4. Vaccines

Summary:

Critical for public health with rising demand due to government immunization programs and donor support.

Recommendations:

Focus on routine immunization vaccines.

Establish technology partnerships with Indian vaccine makers for niche vaccines- Meningococcal, Pneumococcal etc..)

Develop lyophilization and cold chain logistics capacity early.

Explore WHO prequalification pathways to qualify for GAVI and UNICEF tenders.

5. Biopharmaceuticals

Summary:

Long-term strategic segment with growing demand for biosimilars and monoclonal antibodies in East Africa.

Recommendations:

Prioritize monoclonal antibodies for oncology and autoimmune diseases

Develop in-house analytical capabilities for quality assurance.

Build capacity in fill-finish operations.

Collaborate with academic institutions and biopharma innovators to incubate local talent and adapt global technology.

Market Opportunity and Demand

The feasibility study confirms that Uganda's pharmaceutical market is experiencing robust growth driven by increasing demand for affordable, high-quality generics and specialized drugs, including oncology

treatments and vaccines.

With a projected market revenue reaching approximately US\$329 million in 2025 and a promising CAGR for both the domestic and broader East African markets, the opportunity to capture unmet demand is significant.

Government is encouraging and supporting local production through initiatives like “Buy Uganda, Build Uganda.”

Financial Viability

The financial analysis demonstrates a strong investment proposition:

CAPEX: The total investment is estimated at US\$571.62 million, strategically distributed across various manufacturing units.

OPEX and Break-Even: Annual operational expenditures are projected at approximately US\$241.86 million, with break-even achieved within the third year.

ROI and Profitability: Projected ROI stands at around 23.97% by Year 3.

Recommendations

1. Prioritize High-Margin Segments

Focus Areas:

Oncology Drugs: Given the rising burden of cancer in Uganda and East Africa, investing in the production of generic chemotherapy drugs can capture an underserved market.

Biologics and Vaccines: The increasing demand for advanced therapies such as vaccines and biologics—including monoclonal antibodies and bio-similars—presents an opportunity to tap into segments with relatively lower competition and high value.

2. Optimize API and Raw Material Sourcing

Short-Term Strategy:

Leverage International Suppliers: The project should secure long-term contracts with API suppliers in India and China. This will help stabilize costs and ensure a reliable supply chain during the early phases of operation.

Long-Term Strategy:

Diversification: Develop a multi-sourcing strategy to mitigate risks associated with geopolitical tensions, supply chain disruptions, or fluctuations in raw material prices.

3. Invest in Advanced Technologies

Technology Implementation:

Automation and Digitalization: Adoption of automated production lines, Manufacturing Execution Systems (MES), and real-time monitoring systems can streamline operations.

Quality Control Systems: Investment in advanced analytical testing equipment will support stringent quality assurance protocol

4. Enhance Strategic Collaborations Through

Public-Private Partnerships (PPPs): Engage with government agencies and international health organizations to secure financial incentives, regulatory support, and access to infrastructure.

Industry Alliances: Form strategic alliances with global pharmaceutical companies for technology transfer.

Local Partnerships: Collaborate with local distributors, healthcare providers, and academic institutions to strengthen market access.

5. Strengthen Supply Chain Resilience

Key Measures:

Robust Logistics Network: Develop an integrated logistics system that includes reliable transportation, warehousing, and cold chain infrastructure.

Digital Supply Chain Solutions: Implement demand forecasting and real-time inventory management systems to optimize supply chain operations and minimize waste.

Contingency Planning: Establish dual sourcing strategies and maintain buffer stocks to mitigate the impact of any supply disruptions.

To enable efficiency, reliability, cost management and risk mitigation

6. Focus on Workforce Development

Workforce Strategy:

Training Programs: Invest in specialized training and capacity-building initiatives to develop the technical, regulatory, and operational skills needed in manufacturing, quality control, and engineering.

Talent Acquisition: Attract skilled professionals by offering competitive remuneration packages and career development opportunities.

Collaborations with Educational Institutions: Forge partnerships with local universities and technical institutes to create a pipeline of skilled labour tailored to the project's needs.

Conclusion

THE COMPREHENSIVE FEASIBILITY STUDY STRONGLY SUPPORTS THE VIABILITY OF THE UGANDA PHARMA PLANT PROJECT. WITH FAVOURABLE MARKET CONDITIONS, A SUPPORTIVE REGULATORY FRAMEWORK, ROBUST FINANCIAL PROJECTIONS, AND EFFECTIVE RISK MITIGATION STRATEGIES, THE PROJECT IS COMPLETELY FEASIBLE AND POISED FOR SUCCESS.

STRATEGIC INVESTMENTS IN TECHNOLOGY, LOCAL PARTNERSHIPS, AND HUMAN CAPITAL WILL FURTHER ENHANCE THE PLANT'S COMPETITIVE EDGE AND CONTRIBUTE TO SUSTAINABLE ECONOMIC GROWTH IN UGANDA AND THE BROADER EAST AFRICAN REGION.

IN LIGHT OF THESE FINDINGS, IT IS RECOMMENDED THAT THE PROJECT PROCEED AS PLANNED, WITH CAREFUL ATTENTION TO THE OUTLINED RECOMMENDATIONS TO ENSURE OPERATIONAL EXCELLENCE AND LONG-TERM PROFITABILITY.

List of Abbreviations

ADE : Adverse Drug Event

ADR : Adverse Drug Reaction

AEFI : Adverse Events Following Immunization

AI : Artificial Intelligence

API : Active Pharmaceutical Ingredients

ARIPO : African Regional Intellectual Property Organization

BB : BioBridge

BCG : Bacillus Calmette-Guérin

BEP : Break Even Point

BMR : Batch Manufacturing Record

BOPV : Bivalent Oral Polio Vaccine

BPR : Batch Production Record

CAGR : Compound Annual Growth Rate

CAPA : Corrective and Preventive Action

CAPEX : Capital Expenditure

CEO : Chief Executive Officer

cGMP : Current Good Manufacturing Practices

CIP : Clean-in-Place

COMESA : Common Market for Eastern and Southern Africa

CT : Clinical Trial

EAC : East African Council

ECG : Electrocardiogram

EPI : Expanded program for immunization

ETP : Effluent treatment plants
 EU : European Union
 FEFO : First-Expiry-First-Out
 FR : Feasibility Report
 GAVI : Global Vaccine Alliance
 GC : Gas Chromatography
 GCP : Good Clinical Practice
 GDP : Good Distribution Practices
 GLP : Good Lab Practices
 GMP : Good Manufacturing Practices
 HDPE : High-Density Polyethylene
 HEPA : High-Efficiency Particulate Air
 HPLC : High Performance Liquid Chromatography
 HPV : Human papillomavirus
 HR : Human Resource
 HVAC : Heating, Ventilation, and Air Conditioning
 ICH : International Council for Harmonisation
 IPV : Inactivated Polio Vaccine
 IQ : Installation Qualification
 ISO : International Organisation for Standardisation
 JIT : Just In Time
 LED : Light-emitting diode
 MA : Marketing Authorization
 mAbs : Monoclonal Antibodies
 MES : Manufacturing Execution Systems
 mRNA : messenger Ribonucleic Acid
 NCD : Non-Communicable Diseases
 NDA : National Drug Authority
 NDPA : National drug policy & Authority
 NEMA : National Environment Management Authority
 NGO : Non-governmental organization
 NPC : National Pharmacovigilance Centre
 OPEX : Operational Expenditure
 OQ : Operational Qualification
 OSD : Oral Solid Dosage
 PCV : Pneumococcal Conjugate Vaccine
 PIC/S : Pharmaceutical Inspection Convention Scheme
 PoE : Point of entry
 PPB : Pharmacy and Poisons Board
 PPP : Public-Private Partnership
 PQ : Performance Qualification
 QA : Quality Assurance
 QC : Quality Control
 QPPV : Qualified Person for Pharmacovigilance
 R & D : Research and Development
 RABS : Restricted Access Barrier Systems
 REV : Revision
 RO : Reverse Osmosis
 ROI : Return on Investment
 RSV : Respiratory Syncytial virus
 SCADA : Supervisory Control and Data Acquisition
 SEZ : Special Economic Zones
 SIP : Sterilize-in-Place
 SQM : Square Meter
 TFDA : Tanzania Food and Drugs Authority
 TRS : Technical Report Series
 UAE : United Arab Emirates
 UCI : Uganda Cancer Institute
 UNICEF : United Nations Children's Fund
 UPS : Uninterruptible Power Supply
 USAID : United States Agency for International Development
 USD : United States Dollar
 UV : Ultra Violet
 VFD : Variable Frequency Drive
 WFI : Water for Injection
 WHO : World Health Organization
 WTP : Water Treatment Plant
 ZLD : Zero Liquid Discharge

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Countries | Burundi | Kenya | Rwanda | Tanzania | Uganda

Market size (USD Million) | 75 | 740 | 75 | 400 | 450

CAGR 2024-2029 | 6-8% | 6-8% | 6-8% | 6-8% | 6-8%

Market Share of Locally Produced Pharma Drugs(% of overall market) | 3% | 30% | < 1% | 12% | 20%

Market By Segment | 56% Generic,44% Branded | 62% Generic,38% Branded | 54% Generic,46% Branded | 54%

Generic, 46% Branded | 80% Generic, 20% Branded

Type | Vaccine | Uganda Requirement (MD)

Bacterial | Pentavalent | 6

Bacterial | PCV10 | 6

Bacterial | BCG | 2

Live Virus | Measles Vaccine | 6

Live Virus | Bivalent Oral Polio Vaccine (BOPV) | 7

Live Virus | Bivalent Oral Polio Vaccine BOPV (Type 1 and Type 3) | 7

Inactivated Viral | Rotavirus vaccine | 4

Inactivated Viral | Inctivated Polio Vaccibe | 4

Inactivated Viral | Human Papilloma Virus vaccine (1,2,3&4) | 2

Inactivated Viral | Hepatitis B (Birth &Adult) | 4

Inactivated Viral | Yellow fever | 2

Title | Document Number

Guidelines on the conduct of Clinical trials in Uganda. | DPS/GDL/024 REV.01

Guidelines on good clinical practice in the conduct of Clinical trials involving human participants. | DPS/GDL/025 REV.01

Guidelines on good manufacturing practice for medicinal products part I – Basic requirements for medicinal products. | INS/GDL/001 (Part I)

Guidelines on good manufacturing practice for medicinal products part II - Basic requirements for active pharmaceutical ingredients. | INS/GDL/001 (Part II)

Guidelines on good manufacturing practice for medicinal products. | -

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Guidelines on the verification of applications for the importation and exportation of drugs and pharmaceutical raw and packaging materials. | INS/GDL/038 Rev.01

Guidelines on competence profile of supervising pharmacists for a pharmaceutical manufacturing facility. | INS/GDL/048 Rev.00

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Guidelines on variations to registered pharmaceutical products. | PAR/GDL/005 Rev .03

Guidelines on registration of similar bio-therapeutic products. | PAR/GDL/017 Rev .01

Guidelines on registration of bio-therapeutic products. | PAR/GDL/016 Rev .01

Guidelines on registration of human vaccines. | PAR/GDL/021 Rev .01

Guidelines on variation of registered bio-therapeutic products. | PAR/GDL/033 Rev .01

Guidelines on submission of documentation for registration of a pharmaceutical product for human use. | PAR/GDL/004 Rev .03

Guidelines for the provision of insurance cover for research participants in clinical trials in Uganda. |

DPS/GDL/023 Rev.00

Guidelines for introducing a locally manufactured new pharmaceutical product on the Uganda market. |

INS/GDL/004 Rev.o1

Guidelines on detecting and reporting adverse drug reactions in Uganda. | DPS/GDL/013 FRev.01

Guidelines on submission of documentation for renewal of registration of a pharmaceutical product for human use. | PAR/GDL/020 Rev.00

Guidelines on Good Distribution Practice for Pharmaceutical Products. | INS/GDL/002 Rev.01

Guidelines on submitting periodic safety update report and any other reports that may be relevant to determine the safety, efficacy and quality of a drug. | DPS/GDL/034 Rev.00

Guidelines for the Recall or Withdrawal of a Medical Product. | INS/DGL/037 Rev.00

Guidelines on variation to registered Human vaccines. | PAR/GDL/018Rev .01

National AEFI surveillance guidelines Uganda. | -

Regulatory Area | Action | Timeline (Working days)

Regulatory Inspection | GMP Report assessment to Manufacture after inspection | 45 days

Regulatory Inspection | Feedback after inspection- Domestic | 28 days

Regulatory Inspection | Regulatory decision after CAPA – Domestic (Local Facilities) | 17 days

Documentation | Submission and approval of Dossier | 220 days

Total no. of days | Total no. of days | 310 days

Regulatory Area | Action | Timeline (Working days)

Clinical Trial (CT) Oversight | Receipt, Screening and acknowledgment of a Clinical trial application (new, renewals and amendment) | 10 days

Clinical Trial (CT) Oversight | Regulatory decision on a Clinical trial application | 50 days

Clinical Trial (CT) Oversight | Annual renewal of Ongoing trials | 20 days

Clinical Trial (CT) Oversight | Amendment of CT authorization | 20 days

Clinical Trial (CT) Oversight | Feedback report to a client following a GCP Inspection | 45 days

Marketing Authorization (MA) | Assessment of application and request for additional information on application for MA (Human Drugs) | 18 months (396 days)

Marketing Authorization (MA) | Assessment of application and request for additional information on application for MA (Vaccines, Anti-cancer medicines and other vital (medicines)). | 6 months (132 days)

Marketing Authorization (MA) | Regulatory decision on MA after additional information is received | 3 months (66 days)

Marketing Authorization (MA) | Special Categories |

Marketing Authorization (MA) | Regulatory decisions on products already authorized for marketing by stringent regulatory authorities, WHO prequalified products and those approved under article 58 of EU regulations | 6 months (132 days)

Marketing Authorization (MA) | Variation and Notification |

Marketing Authorization (MA) | Regulatory decision on minor variation | 4 months (88 days)

Marketing Authorization (MA) | Regulatory decision on Major variation | 6 months (132 days)

Marketing Authorization (MA) | Regulatory decision on annual or immediate notification change | 80 days

Marketing Authorization (MA) | Drug Register |

Marketing Authorization (MA) | Publication of drug register | Monthly

Regulatory Inspection | GMP Physical inspection after receipt of application- Foreign | 180 days

Regulatory Inspection | GMP Report assessment to Manufacture after inspection | 45 days

Regulatory Inspection | Feedback after inspection- Domestic | 28 days

Regulatory Inspection | Regulatory decision after CAPA – Foreign | 20 days

Regulatory Inspection | Regulatory decision after CAPA – Domestic (Local Facilities) | 17 days

Premise Licensing | Feedback to the applicant on the regulatory decision for pharmacy renewal | 40 days

Premise Licensing | Feedback to the applicant on the regulatory decision for licensing of new pharmacy applicants | 30 days

Market Surveillance and Control | Verification of imports (registered drugs) | 2 days

Market Surveillance and Control | Clearance of Imports at ports of entry | 2 days

Market Surveillance and Control | Verification of imports (un-registered drugs) | 10 days

Market Surveillance and Control | Regulatory decision on drug promotional material | 10 days

Market Surveillance and Control | Publication of the database of approved promotional materials/advert | Monthly (5th of the month)

Vigilance | Acknowledgement of receipt of an ADE report (mail/letter) | 5 days

Vigilance | Feedback on serious ADE reports | 18 days

Vigilance | Feedback on serious AEFI reports to the expanded program for immunization (EPI) | 14 days

Laboratory Testing | Test results from mandatory testing after sampling (Medicine)- PoE | 48 Days

Laboratory Testing | Test results from client requests after acceptance by the lab (All Products) | 45 Days

Laboratory Testing | Test results for pre-market (Domestics Manufactures) | 60 Days

Laboratory Testing | Test results for post-market surveillance (Medicines) | 48 Days

National Regulatory System | Issuance of recall or alert for a substandard or falsified medicine or health care product | 14 Days

National Regulatory System | Acknowledgement of receipt of a product complaint | 5 Days

National Regulatory System | Feedback on a market (process, product, other) complaint | 20 Days

National Regulatory System | Feedback on the effectiveness of recall after receipt of the recall report from the LTR (Local Technical Representative) | 10 Days

National Regulatory System | Feedback on test results from the Lab to the client | 5 Days

Support to Regulatory areas | QUALITY MANAGEMENT SYSTEM |

Support to Regulatory areas | Acknowledgement of receipt of a service-related complaint | 2 Days

Support to Regulatory areas | Feedback on service-related complaints | 21 Days

Therapeutic Area | Regulatory Focus | Key EAC Harmonization Mechanisms | Challenges | Progress & Opportunities

Diabetes | Insulin and oral hypoglycaemic Bioequivalence and cold chain Chronic disease management alignment | Common Technical Document (CTD)

Mutual Recognition Procedure (MRP) | Cold chain logistics

Limited local insulin production | Streamlined approvals

Regional demand driving local production

Malaria | Fast-tracking of ACTs Monitoring of resistance Alignment with WHO PQ | Joint assessments

Regional malaria drug guidelines | Counterfeits Dependency on donor programs | Strong harmonization Integrated procurement systems
Antibiotics | Quality and safety evaluations GMP compliance AMR mitigation | AMR frameworks under WHO/Africa CDC

Regional pharmacovigilance | Overuse and poor prescription practices - Limited surveillance | Regulatory updates in progress

Enhanced AMR monitoring
Vaccines | High scrutiny: quality, safety, cold chain - Emergency use approval mechanisms | AVAREF support

Reliance on WHO PQ and joint inspections | Heavy import reliance

Capacity gaps for biologics | Regional capability strengthening

Ongoing vaccine regulation training
Biologics | Advanced assessments for mAbs, biosimilars, insulin analogs - GMP and comparability requirements | International partnerships for capacity building - Involvement of WHO, UNICEF | Limited local evaluation experience

High assessment complexity | Capacity building underway

Future potential for local biologics production
HIV/AIDS | ARV quality and resistance monitoring - Integration with national HIV programs - Rapid access to WHO PQ ARVs | Fast-track review mechanisms - Reliance on global procurement systems (e.g., Global Fund) | Stockouts and supply chain challenges - Limited local production | Increasing local registration capabilities

Supportive policy for ARV production
Tuberculosis (TB) | MDR-TB drugs evaluation

Diagnostic technologies

Fixed-dose combination approvals | WHO TB drug guidelines

Regional efforts to reduce import timelines | Complexity of MDR/XDR TB treatment

Dependence on donor supply | Regional labs and regulatory harmonization improving Potential for local packaging and supply

S. No | DESIGNATED AREA | APPROX AREA SQM

1 | CENTRAL WAREHOUSE FOR RAW MATERIAL, PRIMARY, SECONDARY AND TERTIARY PACKAGING MATERIAL, SOLVENTS, ENGINEERING SPARES, COLD STORAGE, FINISHED GOODS STORAGE | 30,000

2 | ENGINEERING BLOCK – HVAC AND AHU, WATER PURIFICATION, PW AND WFI SYSTEM | 15,000

3 | CENTRAL UTILITY BLOCK | 6,000

4 | EFFLUENT TREATMENT PLANT | 5,000

5 | SCRAP YARD | 500

6 | PRIMARY WATER TREATMENT AREA | 1,000

7 | ADMINISTRATION BLOCK (INCLUDES FINANCE, ACCOUNT, HR, SUPPLY CHAIN ETC.), CORPORATE OFFICES, IT AND DATA SECURITY, MEETING ROOMS AND CONFERENCE HALLS, DOCUMENT CONTROL, RECEPTION AND VISITOR LOUNGE | 13,000

8 | SECURITY BLOCK – CONTROL ROOM, FIRE SAFETY AND EMERGENCY CONTROL ROOM, EMPLOYEE SCREENING AND ACCESS CONTROL AREA, | 3,500

9 | CAR PARK | 4,000

10 | CAFETERIA | 4,000

11 | TRAINING AND INDUCTION ROOMS | 1,500

12 | MEDICAL AND FIRST AID ROOMS | 500

13 | RECREATIONAL AREA FOR EMPLOYEES | 2,000

10 | QUALITY CONTROL LABORATORY | 4000

11 | RESEARCH AND DEVELOPMENT LABORATORY | 2000

12 | QUALITY ASSURANCE SECTION | 2000

13 | REGULATORY AFFAIRS DEPARTMENT | 1000

14 | CLINICAL RESEARCH DEPARTMENT | 500

| TOTAL AREA | 95,500 SQM

Section | Size Required (sq. m) | Function

Granulation Area | 3,500 | Wet/dry granulation for tablet and capsule production

Blending & Compression | 2,500 | Mixing active ingredients and excipients

Tablet & Capsule Filling | 3,000 | Tablet compression and capsule filling

Coating | 2,000 | Film/sugar coating for tablets

Liquid Manufacturing | 3,500 | Syrup preparation, mixing, and holding tanks

Filling & Packaging | 5,000 | High-speed blister packing, bottling, and labelling

Utilities & HVAC | 2,500 | Power, water, and air handling units

Support Areas (Offices, Change Rooms, Cafeteria, etc.) | 2,500 | Administrative functions, employee welfare, and cleanroom airlocks

TOTAL AREA | 24,500 SQM |

Section | Size required (sq. m) | Function

Raw Material & Component Storage | 2,500 | Storage of vials, stoppers, API, excipients, and packaging materials

Buffer preparation | 1,500 | Preparation of buffers

Formulation Suite | 2,000 | Mixing and formulation of mAbs bulk solution

Aseptic Filling & Stoppering | 2,500 | High-speed filling, capping, and stoppering under ISO 5 conditions

Lyophilization (if required) | 2,000 | Freeze-drying for extended shelf-life stability

Capping, Inspection & Labelling | 1,500 | Visual inspection, capping, and final labelling

Primary & Secondary Packaging | 2,000 | Packaging in cartons, serialization, and labelling

Cold Chain Storage & Stability Testing | 2,500 | Refrigerated and frozen storage at 2-8°C and -20°C for mAbs stability

Utilities & HVAC Systems | 2,000 | Power supply, purified water (WFI), HVAC, and compressed air

Support Areas (Offices, Cleanrooms, Cafeteria, Change Rooms, etc.) | 2,500 | Administrative offices, gowning areas, and personnel movement

TOTAL AREA | 21,000 SQM |

Section | Size required (sq. m) | Function

Vial Washing & Depyrogenation | 2,000 | Cleaning and depyrogenation of vials before filling
 Ampoule Washing & Sterilization | 1,500 | Automated washing and sterilization of ampoules
 Pre-Filled Syringe Preparation | 2,000 | Syringe preparation and plunger assembly
 Filling & Stoppering (Vials & Ampoules) | 3,000 | High-speed aseptic filling and stoppering under ISO 5 conditions
 Filling & Sealing (Pre-Filled Syringes) | 2,500 | Filling and sealing of syringes under sterile conditions
 Lyophilization (if required) | 2,500 | Freeze-drying for vials requiring stability enhancement
 Capping & Inspection | 1,500 | Automated visual inspection, capping, and rejection of defective units
 Labelling & Secondary Packaging | 2,500 | Labelling, serialization, and secondary packaging
 Utilities & HVAC Systems | 2,000 | Purified water (WFI), air handling, and compressed air systems
 Support Areas (Gowning, Change Rooms, Offices, etc.) | 2,500 | Personnel movement, gowning, and administrative offices
 TOTAL AREA | 22,000 SQM |
 Section | Size required (sq. m) | Function
 Raw Material Storage | 2,500 | Isolated storage for APIs, excipients, and packaging components
 Granulation Area | 3,500 | High-shear wet/dry granulation and fluid bed drying
 Blending & Compression | 2,500 | Blending of APIs and excipients, tablet compression
 Encapsulation Area | 2,000 | Capsule filling with containment systems
 Coating (if required) | 2,000 | Tablet film/sugar coating with dust extraction
 Hormonal & Cytotoxic Segregation | 3,500 | Dedicated, fully enclosed areas with negative pressure
 Filling & Packaging | 4,000 | High-speed blister packing, bottling, and labelling
 Waste Management & Decontamination | 2,000 | Safe disposal of hazardous cytotoxic waste
 Utilities & HVAC Systems | 2,500 | High-efficiency air handling, WFI, compressed air
 Support Areas (Change Rooms, Gowning, Admin, etc.) | 3,000 | GMP-compliant personnel movement and administrative areas
 TOTAL AREA | 27,500 |
 Section | Size required (sq. m) | Function
 Vial Washing & Depyrogenation | 2,000 | Cleaning and depyrogenation of vials before filling
 Filling & Stoppering (Vials) | 3,000 | High-speed aseptic filling and stoppering under ISO 5 conditions
 Lyophilization (if required) | 2,500 | Freeze-drying for vials requiring stability enhancement
 Capping & Inspection | 1,500 | Automated visual inspection, capping, and rejection of defective units
 Labelling & Secondary Packaging | 2,500 | Labelling, serialization, and secondary packaging
 Utilities & HVAC Systems | 2,000 | Purified water (WFI), air handling, and compressed air systems
 Support Areas (Gowning, Change Rooms, Offices, etc.) | 2,500 | Personnel movement, gowning, and administrative offices
 TOTAL AREA | 16,000 SQM |
 Manufacturing Block wise area and cost | Manufacturing Block wise area and cost | Manufacturing Block wise area and cost | Manufacturing Block wise area and cost
 S. No. | Block Name | Sub Unit | Cost in MUSD
 | Plant 1 - OSD | Granulation Area | 29.60
 | Plant 1 - OSD | Blending & Compression | 29.60
 | Plant 1 - OSD | Tablet & Capsule Filling | 29.60
 | Plant 1 - OSD | Coating | 29.60
 | Plant 1 - OSD | Liquid Manufacturing | 29.60
 | Plant 1 - OSD | Filling & Packaging | 29.60
 | Plant 1 - OSD | Utilities & HVAC | 29.60
 | Plant 1 - OSD | Support Areas (Offices, Change Rooms, Cafeteria, etc.) | 29.60
 | Plant 2-mAbs and Biosimilar | Raw Material & Component Storage | 68.20
 | Plant 2-mAbs and Biosimilar | Buffer preparation | 68.20
 | Plant 2-mAbs and Biosimilar | Formulation Suite | 68.20
 | Plant 2-mAbs and Biosimilar | Aseptic Filling & Stoppering | 68.20
 | Plant 2-mAbs and Biosimilar | Lyophilization (if required) | 68.20
 | Plant 2-mAbs and Biosimilar | Capping, Inspection & Labelling | 68.20
 | Plant 2-mAbs and Biosimilar | Primary & Secondary Packaging | 68.20
 | Plant 2-mAbs and Biosimilar | Cold Chain Storage & Stability Testing | 68.20
 | Plant 2-mAbs and Biosimilar | Utilities & HVAC Systems | 68.20
 | Plant 2-mAbs and Biosimilar | Support Areas (Offices, Cafeteria, Change Rooms, etc.) | 68.20
 | Plant 3-Filling Lines | Vial Washing & Depyrogenation | 61.70
 | Plant 3-Filling Lines | Ampoule Washing & Sterilization | 61.70
 | Plant 3-Filling Lines | Pre-Filled Syringe Preparation | 61.70
 | Plant 3-Filling Lines | Filling & Stoppering (Vials & Ampoules) | 61.70
 | Plant 3-Filling Lines | Filling & Sealing (Pre-Filled Syringes) | 61.70
 | Plant 3-Filling Lines | Lyophilization (if required) | 61.70
 | Plant 3-Filling Lines | Capping & Inspection | 61.70
 | Plant 3-Filling Lines | Labelling & Secondary Packaging | 61.70
 | Plant 3-Filling Lines | Utilities & HVAC Systems | 61.70
 | Plant 3-Filling Lines | Support Areas (Gowning, Change Rooms, Offices, etc.) | 61.70
 | Plant 4-Oncology | Raw Material Staging and component storage | 81.95
 | Plant 4-Oncology | Granulation Area | 81.95
 | Plant 4-Oncology | Blending & Compression | 81.95
 | Plant 4-Oncology | Encapsulation Area | 81.95
 | Plant 4-Oncology | Coating (if required) | 81.95
 | Plant 4-Oncology | Hormonal & Cytotoxic Segregation | 81.95
 | Plant 4-Oncology | Filling & Packaging | 81.95
 | Plant 4-Oncology | Waste Management & Decontamination | 81.95
 | Plant 4-Oncology | Utilities & HVAC Systems | 81.95
 | Plant 4-Oncology | Support Areas (Change Rooms, Gowning, Admin, etc.) | 81.95
 | Plant 5-Vaccine | Vial Washing & Depyrogenation | 51.70
 | Plant 5-Vaccine | Filling & Stoppering (Vials) | 51.70
 | Plant 5-Vaccine | Lyophilization (if required) | 51.70
 | Plant 5-Vaccine | Capping & Inspection | 51.70
 | Plant 5-Vaccine | Labelling & Secondary Packaging | 51.70
 | Plant 5-Vaccine | Utilities & HVAC Systems | 51.70
 | Plant 5-Vaccine | Support Areas (Gowning, Change Rooms, Offices, etc.) | 51.70
 | Common Blocks | Animal House | 85.16
 | Common Blocks | Central warehouse | 85.16
 | Common Blocks | Engineering block | 85.16
 | Common Blocks | Central utility block | 85.16
 | Common Blocks | Administration block | 85.16
 | Common Blocks | Quality control laboratory | 85.16
 | Common Blocks | Research and development laboratory | 85.16

| Common Blocks | Quality assurance section | 85.16
 | Miscellaneous blocks | Car park | 10.70
 | Miscellaneous blocks | Cafeteria | 10.70
 | Miscellaneous blocks | Training and induction rooms | 10.70
 | Miscellaneous blocks | Medical and first aid rooms | 10.70
 | Miscellaneous blocks | Recreational area for employees | 10.70
 | Miscellaneous blocks | Regulatory affairs department | 10.70
 | Miscellaneous blocks | Clinical research department | 10.70
 | Miscellaneous blocks | Landscaping and horticulture | 10.70
 | Quality Assurance | Facility Commissioning & validations. | 40.70
 | Regulatory Affairs | Licensing & Statutory Cost | 13.20
 || TOTAL BUDGET ESTIMATE IN USD | 442.90 MUSD
 Manufacturing Unit | Approx.
 Investment (MUSD) | Approx. % of Total CAPEX
 Oral Solid Dosage (OSD) | 29.6 | 6.68%
 Biopharma Fill-Finish | 68.2 | 15.40%
 Injectables (SVP) | 61.7 | 13.93%
 Oncology (Hormonal & Cytotoxic Products) | 81.95 | 18.50%
 Vaccine Fill-Finish | 51.70 | 11.67%
 Common Infrastructure (Utilities, Warehousing, R&D, QA/QC, Admin, etc.) | 85.16 | 19.23%
 Miscellaneous Blocks (Car park, cafeteria, Training room, Medical room, Regulatory affairs, Clinical research, Landscaping etc., | 10.70 | 2.42%
 Facility commissioning and qualification | 40.70 | 9.19%
 Registration and licensing | 13.20 | 2.98%
 Total Investment | 442.90 MUSD | 100%
 Cost Component | Annual Cost (MUSD) | % of Total OPEX
 Raw Materials & APIs | 44 | 34.18%
 Labour & Workforce | 12 | 9.32%
 Utilities (Power, Water, Waste Management) | 22 | 17.09%
 Maintenance costs | 10 | 7.77%
 Packaging | 11 | 8.55%
 Marketing and Sales | 9 | 6.99%
 Regulatory Compliance & Quality Assurance
 And technology transfer | 10 | 7.77%
 Logistics & Distribution | 8 | 6.22%
 Insurance | 2.72 | 2.11%
 Total OPEX (Annual) | 128.72 MUSD | 100%
 Year | Revenue (MUSD)
 Year 1 | 4.50
 Year 2 | 120.00
 Year 3 | 350.00
 Year 4 | 350.00
 Year 5 | 510.00
 Year 6 | 510.00
 Year 7 | 510.00
 Risk Factor | Potential Impact | Mitigation Strategy
 Regulatory Delays | Slower product approvals | Early engagement with NDA (Uganda), WHO-GMP
 API Price Volatility | Increased raw material costs | Multi-supplier sourcing, API localization
 Market Competition | Lower pricing power | Differentiation through quality & innovation
 Logistics & Distribution Challenges | Delayed market penetration | Strong partnerships with distributors
 Cost Component
 Raw Material & API Procurement
 Packaging Material Costs
 Warehousing & Cold Storage
 Manufacturing & Production Logistics
 Quality Control & Compliance
 Cold Chain Infrastructure (Vaccines & Biologics)
 Distribution & Last-Mile Delivery
 IT, AI, and Digital Supply Chain Systems
 Risk Management & Contingency Planning
 Regulatory Compliance & Audits
 Plant/Section | Number of Personnel
 Oral Solid Dosage (OSD) | 80
 Biopharma Drug Product | 80
 Vaccine Formulation | 70
 Oncology | 50
 Injectable | 70
 Warehouse | 40
 Quality Control | 62
 Quality Assurance | 43
 Engineering | 80
 HR and Admin | 65
 Total | 640
 Risk Factor | Potential Impact
 Regulatory Uncertainty | Changes in pharmaceutical laws, taxation, or licensing requirements could delay operations or increase compliance costs.
 Government Policy Shifts | Import/export restrictions, price controls, or changes in incentives may affect profitability.
 Bureaucratic Delays | Lengthy approvals for manufacturing licenses, product registrations, and environmental permits may slow down market entry.
 Geopolitical Instability in EAC Region | Conflicts or tensions in neighbouring East African Community (EAC) countries could disrupt supply chains and exports.
 Intellectual Property & Patent Protection | Weak enforcement of patents and trademarks may lead to counterfeiting risks.
 Risk Factor | Potential Impact
 Currency Fluctuations | Exchange rate volatility can increase the cost of imported raw materials (APIs, excipients, packaging materials).
 Inflation & Rising Operational Costs | High inflation may lead to increased costs for labor, utilities, and raw materials.
 Supply Chain Disruptions | Dependence on international suppliers for APIs and critical components could lead to shortages.
 Limited Access to Affordable Financing | High-interest rates and limited government incentives can make

project financing costly.

Market Demand & Price Sensitivity | Price-sensitive consumers and limited insurance coverage may affect product affordability.

Risk Factor | Potential Impact

Water Scarcity | High water demand for manufacturing and cooling may lead to shortages, increasing operational costs.

Waste Disposal Challenges | Improper disposal of hazardous waste (solvents, API residues) can lead to environmental penalties.

Energy Dependence on Non-Renewables | High electricity costs and unreliable grid supply may lead to production disruptions.

Climate Change Effects | Increased heat and erratic rainfall could impact cooling systems and energy consumption.

Regulatory Non-Compliance | Failure to meet Uganda's National Environment Management Authority (NEMA) regulations can lead to legal actions.

Risk Category | Key Risks | Mitigation Strategies

General Risks | Regulatory uncertainty, bureaucracy, policy shifts | Engage early with regulators, form public-private partnerships, diversify export markets

Economic Risks | Currency fluctuations, inflation, financing challenges | Hedge currency risks, optimize operational costs, diversify revenue streams

Environmental Risks | Water scarcity, waste disposal, high energy costs | Implement sustainable waste management, use renewable energy, optimize water usage

Phase | Focus | Timeframe | Revenue Stream | Impact on ROI

Phase 1 | OSD + Syrups | Year 1–2 | Tender sales, private hospitals | Early cash flow

Phase 2 | Oncology + Injectables | Year 2–3 | High-margin products | Boost margin

Phase 3 | Vaccines + Biosimilars | Year 3–5 | Institutional procurement | Long-term revenue

Cross-Cutting | CDMO + Export | Year 1–5 | Capacity monetization | Parallel revenue

FEASIBILITY REPORT FOR INTEGRATED PHARMA AND
BIOPHARMACEUTICAL
PROJECT FOR UGANDA

Prepared for:

SUNSHINE MEDICAL PTE LTD UNIT 06-10, UB ONE BUILDING,
UB AVENUE 4,
Singapore – 408 830

Prepared by:

BIOBRIDGE HEALTHCARE SOLUTIONS PVT. LTD
13, RACHANA BLOSSOM, AUNDH
PUNE 411007
INDIA

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EXECUTIVE SUMMARY

Introduction:

The objective of this project is to establish and operate a state-of-the-art green field multi-Products Biopharmaceutical manufacturing plant in Republic of Uganda and to address and transform health care needs of East African council (EAC) by addressing critical health care challenges and fostering local manufacturing.

This facility aims to consistently and efficiently produce high-quality pharmaceutical, Biopharmaceuticals & vaccines in a phased manner

Objective:

The objective of this report is to study comprehensively the feasibility and to assess the potential for establishing a sustainable and efficient pharmaceutical manufacturing plant in Uganda.

BioBridge is requested to provide a proposal for feasibility study services, which include:

Market analysis

Regulatory requirements

Site selection and infrastructure

Technical feasibility

Financial feasibility

Supply chain and raw material sourcing

Human resources

Risk assessment

Sustainability

Goals

Health Care Access: To enhance Healthcare access by manufacturing high quality pharmaceuticals, Biologics and vaccines locally thereby reducing the dependency on imported medicines and biologics, ensuring cost effective treatment to masses.

Health Care needs: To address regional health care needs by manufacturing medicinal products which are required to treat most prevalent diseases in the region-Malaria, Tuberculosis, AIDS.

Economic growth: To foster Economic growth by providing employment opportunity to local population and attracting global investors and venture capitalists.

Sustainability: To establish a sustainable pharma base in Uganda which is reliable and self-sufficient.

Supply Chain Reliability: Building a secure and efficient supply chain to source raw materials, transport finished products, and ensure timely deliveries to end users.

Stakeholder Satisfaction: Ensuring customer satisfaction by providing consistent and reliable products while maintaining a favourable relationship with employees, investors, and the local community.

Solution:

To realise the above said goals and to address the critical health care needs of Uganda, Biopharmaceutical manufacturing facility shall be established in the phased manner.

In the first phase Oral Solid Dosage (OSD) form shall be established by integrate state-of-the-art manufacturing technologies to ensure efficiency, quality, and scalability of wide range of high-demand solid dosage forms, including tablets, capsules, syrups, and advanced technologies such as soft gel capsules.

In the phase -2, The Phase-2 General Injectables facility will focus on manufacturing sterile pharmaceutical injectable-grade products, targeting key therapeutic segments. Injectables facility shall be designed to meet the highest global standards for sterile manufacturing while maintaining cost efficiency and scalability

In the Phase-3, Biopharma Products facility focuses exclusively on the formulation and fill- finish of advanced biopharmaceutical products. This facility shall be commissioned to address the increasing demand for biologics, including Monoclonal antibodies(mabs), Hormones, Insulin, analogs, and other therapeutic proteins.

This facility shall be designed to ensure contamination controls, precision, and compliance with global standards for biopharmaceutical fill-finish processes.

The Phase-4, facility for formulation and fill-finish of critical vaccines to address major global and regional health challenges. This facility shall be designed to cater to both routine immunization programs and outbreak vaccines.

East Africa, like the rest of Africa, faces a growing demand for vaccines, particularly as it works to strengthen its vaccine supply chain and boost domestic capabilities to meet the needs of its population.

In the 5th and the final phase, Oncology OSD facility shall be installed to address the needs of anticancer drugs in the region.

2.0 DETAILED FEASIBILITY

REPORT

Market Analysis:

Demand for pharmaceutical products in Uganda

Uganda's pharmaceutical industry is a key component of the country's healthcare system, with demand steadily increasing due to population growth, rising disease burdens, and government initiatives to improve healthcare access.

The disease burden in Uganda is similar to other less developed African countries, with communicable diseases accounting for roughly 67% of the total burden.

Uganda's communicable disease burden is dominated by infectious and parasitic diseases. Malaria is considered as the major cause of deaths in the country.

The most prevalent causes of mortality and morbidity in Uganda are

Malaria, Pneumonia, Tuberculosis, Anaemia, Hypertension, Diabetes and Cancer.

Fig No.: 1

Fig. No.:2

Trends of Cancer Patients in Uganda:

Fig.No.:3

The cancer burden by type of cancer diagnosed among patients

MARKET VALUE:

In Uganda, the revenue in the pharmaceuticals market is estimated to reach US\$329.32m in 2025.

The largest market within this industry is Oncology Drugs, which is projected to have a market volume of US\$50.03m in 2025.

The revenue is expected to exhibit an annual growth rate (CAGR 2025-2029) of 3.06%, resulting in a market volume of US\$371.58m by 2029.

Uganda's pharmaceutical market is experiencing a surge in demand for affordable generic medications.

The East Africa generic drug market reached USD 2.4 billion in 2024 and is expected to exhibit a CAGR of 6.72% during 2025-2033.

Competitive Landscape:

East Africa Pharma manufacturing- An Overview

Around 35 active manufacturers (members of FEAPM); 2 facilities with WHO cGMP approval and pre-qualified products

Primarily compounding and formulation of most dosage forms including infusions

Almost all raw materials imported

High dependency on imported finished pharmaceutical products

Focus on WHO list of essential medicines

No API manufacturing

Increasingly well-regulated (NDA, PPB, TFDA)

Trade is mostly intra-East Africa

Companies currently having manufacturing base in Uganda

CIPLA Quality Chemicals Industries Limited, joint venture between a local and international

pharmaceutical company, is certified cGMP facilities manufacturing Anti-retro viral, anti-malarial and

hepatitis B drugs, exports to Sub-Saharan countries and supplies drugs to the Global Fund

Kampala Pharmaceutical Industries (1996) Limited, local manufacturer and an affiliate of the Aga Khan

development network. Product range includes analgesics, anti-asthma, anti-diabetic, anti-malarial,

anti-parasites, antibiotics, cardiovascular drugs, cold remedies & Nutraceuticals

Beta Health Care Uganda Limited, an affiliate of Aspen Group. Exports its products to Tanzania, Rwanda,

Burundi, Ethiopia, Sudan, Somalia, Mauritius and West Africa

Key players in the market are actively implementing various strategies to expand their reach and market share by:

Enhancing their production capabilities, both in terms of volume and the range of products.

Investments in advanced manufacturing facilities to increase output and meet the rising demand for generic drugs.

Creating more efficient supply chains and distribution networks to ensure wider availability of their products. They are reaching out to remote areas and partnering with local distributors to improve access.

Forming alliances with local manufacturers, international partners, and healthcare providers to extend their reach and enhance their market position.

Technology transfers, knowledge and technology transfer, through Joint ventures, Alliances thereby strengthening local capacities

Key Pharma Sector Data Estimate (Market share, Generic, Branded pharmaceuticals)

Table No.: 1

Source 2014 Estimation (Industry and Association)

Demand for Pharmaceutical Products in Uganda

The demand for pharmaceutical products in Uganda is driven by various factors, including an increasing prevalence of both communicable and non-communicable diseases (NCDs), an expanding healthcare infrastructure, and growing government support for local production.

Below is an analysis of key therapeutic areas:

Oral Solid Dosage Forms (Tablets and Capsules)

Oral solid dosage forms constitute the majority of prescription medications in Uganda, covering a range of treatments from antibiotics to anti-hypertensives.

Uganda has a few established pharmaceutical manufacturers producing these drugs.

Despite local manufacturing efforts, a large portion of Uganda's medicine supply is still imported due to insufficient production capacity and the need for diverse active pharmaceutical ingredients (APIs).

Expanding local production capabilities, especially for essential generics, could help meet growing demand and reduce reliance on imports.

Oncology Treatments

Cancer cases in Uganda have been rising significantly, with estimates from the Uganda Cancer Institute (UCI) indicating an annual increase in new cases (Refer to Fig Nos.2&3.).

Currently, oncology drugs are almost entirely imported, making treatment expensive and often inaccessible to many patients.

There is an opportunity for local pharmaceutical manufacturers to invest in cancer drug production, particularly generic chemotherapy drugs, which could lower costs and improve accessibility and health care.

The Ugandan government and international health organizations are increasingly investing in cancer treatment centres, further boosting the demand for oncology drugs

Vaccines

Uganda has a demand for vaccines (approx. 1.7 million doses per annum for essential vaccines) due to childhood immunization programs, seasonal disease outbreaks, and global initiatives like the Global Vaccine Alliance (GAVI).

The country currently relies on imports for nearly all its vaccines, which poses challenges such as supply chain disruptions and high procurement costs.

Government efforts to establish vaccine production facilities could significantly boost the local pharmaceutical industry and create self-sufficiency in vaccine supply.

Monoclonal Antibodies (mAbs)

Monoclonal antibodies are used in cancer treatment, autoimmune diseases, and infectious diseases like COVID-19.

Uganda currently lacks the infrastructure for manufacturing biologic drugs & mAbs, entirely dependent on imports.

Drivers of Pharma market:

Rapid expansion of local pharmaceutical manufacturing capacities.

East Africa is experiencing a significant boost in its pharmaceutical production driven by a combination of increased domestic and international investment and initiatives, improved technological capabilities, and a strategic push for self-sufficiency in pharmaceutical supply. Furthermore, there is a rising awareness among healthcare professionals and patients about the comparable efficacy of generic drugs to brand-name counterparts.

Generic drug manufacturers are significantly strengthening their distribution networks to improve product reach. This involves the development of efficient supply chain strategies and logistics networks to ensure generic drugs are accessible even in remote regions. This expanded distribution is further facilitated by technological advancements, such as digital platforms and delivery systems, enabling broader dissemination of generic drugs across the region, fostering overall the market

growth.

Favourable Government Initiatives

Governments across East Africa are progressively acknowledging the role of generic drugs in achieving healthcare affordability and accessibility. They are implementing regulatory reforms to simplify the approval process, thereby encouraging their use by streamlining this process, governments are reducing barriers to entry, making it easier for generic manufacturers to bring their products to market.

Moreover, there is an increase in governmental efforts to boost public awareness about the efficacy and affordability of generic drugs. Despite their lower cost, generic drugs are equally effective as their branded counterparts.

EAC Pharma – Opportunities

Large unmet need for essential medicines

Significant population growth anticipated

Double disease burden (growth of chronic diseases):

NCDs are predicted to account for 40% of illnesses & for 42% of deaths (5.1 million) by 2030 (WHO Global Health Estimates, 2013)

Growing middle class willing to pay for better treatment

New models of care being designed (e.g. PPP, medical insurance, etc.)

Existing pharmaceutical production base & enabling regulatory environment

African traditional and herbal medicines as a market niche

No dominant players

Political will and action

EAC Pharma – Challenges

Negative perceptions of locally manufactured products

Lack of capacity to manufacture advanced formulations due to a skills gap on product development and formulation expertise

High cost of borrowing (20-30%)

High cost of operations e.g. finance, utilities, infrastructure

Lack of market data

Poor and unreliable infrastructure (roads, utilities, etc)

Under resourced and weak institutions (Civil society, Regulatory agencies, etc)

Vaccines Market Analysis:

Immunization is a proven tool for controlling and eliminating life-threatening infectious diseases and is estimated to avert millions of deaths each year.

Universal immunization of children against eight vaccine-preventable diseases [tuberculosis, diphtheria, whooping cough (pertussis), tetanus, hepatitis B, respiratory diseases caused by *Haemophilus influenzae*, polio, and measles] is crucial to reducing infant and child mortality. J Health Popul Nutr . 2013

Mar;31(1):118–129. doi

Immunisation Schedule in Uganda:

Routine immunization schedule for Children below the age of one year cover vaccines for following preventable diseases

Oral Polio vaccine, BCG, Pentavalent, Pneumococcal 10 valent, Rotavirus vaccine, Injectable Polio vaccine

(14 weeks of age), Measles vaccine & HPV (Female Children) (Republic of Uganda, Ministry of Health/UNICEF

Report) Based on the birth cohort in Uganda (1.7 million) the requirements of different vaccines are as

under based on the government Immunization program:

Table No.:2

Upcoming vaccines which shall be targeted in the future

mRNA based vaccines for pandemic preparedness, RSV (Respiratory Syncytial virus) vaccine, Recombinant

Tuberculosis vaccine (under development globally), Rabies vaccine etc.

Demand of vaccines products in Uganda & EAC region:

Eastern Africa is experiencing a surge in demand for vaccines, driven by increasing awareness and

government initiatives to improve healthcare access.

The Vaccines market in Eastern Africa is anticipated to achieve a revenue of US\$463.65m in 2025.

Looking ahead, the market is expected to exhibit a compound annual growth rate (CAGR) of -4.98% from 2025 to 2029, leading to a market volume of US\$377.97m by 2029.

Key market trends, challenges, and opportunities:

One of the key trends in the Vaccines segment in Eastern Africa and Uganda is the increasing availability of vaccines.

Pharmaceutical companies are expanding their operations in the region, and governments are working to improve vaccine distribution networks. This has made vaccines more accessible to people living in remote areas. Secondly, in the market there is growing demand for new and innovative vaccines. As more people become vaccinated against common diseases, there is a need for vaccines that can prevent emerging infectious diseases. Pharmaceutical companies are investing in research and development to create new vaccines that can address these challenges.

East Africa, like the rest of Africa, faces a growing demand for vaccines, particularly as it works to strengthen its vaccine supply chain and boost domestic capabilities to meet the needs of its population.

Challenges and Opportunities:

Challenges include the provision of vaccines in rural locations hampered by inadequate cold chain infrastructure, and vaccination acceptance rates that are impacted by cultural and religious beliefs.

Besides, there are also market opportunities, such as increasing research and development (R&D) spending for novel vaccines and technological advancements.

Growing Support from Various Organizations:

The support from international bodies such as the World Health Organization (WHO), World Bank, the United Nations Children's

Fund (UNICEF), and the Global Alliance for Vaccines and Immunization (GAVI) is indispensable for the East African vaccine market. These organizations provide not only financial aid but also logistical support,

crucial for the procurement and distribution of vaccines in the region.

Competitive Analysis of the Existing Pharmaceutical Market

Uganda's pharmaceutical market is composed of both local manufacturers and international suppliers. The competitive landscape is shaped by several key factors:

Local Pharmaceutical Manufacturers

Uganda has a small but growing pharmaceutical manufacturing industry.

Local manufacturers face challenges such as high production costs, limited access to raw materials, and competition from imported generics.

Import Dependence and Foreign Suppliers

Uganda imports about 70% of its pharmaceuticals, with India and China being the dominant suppliers.

International pharmaceutical companies have a strong presence in Uganda through distributors.

The high reliance on imports creates challenges in affordability, availability, and quality control.

Policy and Market Protection

Uganda has introduced policies like the "Buy Uganda, Build Uganda" (BUBU) initiative, which promotes local manufacturing.

However, the local pharmaceutical industry struggles to compete with cheaper imports, and implementation of these policies remains inconsistent.

Key Market Trends, Challenges, and Opportunities

Market Trends

Regulatory Harmonization in the EAC

The East African Community (EAC) is working towards regulatory harmonization, allowing Ugandan manufacturers to access larger regional markets.

Harmonization simplifies drug registration processes and could enhance trade within the region.

Growing Non-Communicable Disease (NCD) Burden

There is an increasing prevalence of NCDs such as diabetes, hypertension, and cancer in Uganda.

This shift is driving demand for chronic disease medications, which presents opportunities for local manufacturers to diversify production.

Expansion of Health Infrastructure

The Ugandan government is investing in expanding hospitals, clinics, and specialized treatment centres.

These investments are increasing the demand for pharmaceuticals, particularly in specialized areas like oncology and vaccines.

Key Challenges

Limited Access to Finance

Pharmaceutical manufacturing requires large capital investments for facility upgrades, research, and compliance with regulatory standards.

Access to affordable financing remains a major barrier for Ugandan pharmaceutical firms.

Dependence on Imported Raw Materials

Uganda imports most of its Active Pharmaceutical Ingredients (APIs) from India and China.

This dependence makes local manufacturers vulnerable to global supply chain disruptions.

Regulatory and Quality Compliance

Meeting international Good Manufacturing Practices (GMP) standards is essential for market expansion but requires costly upgrades.

Non-compliance can limit Ugandan pharmaceutical companies' ability to export to international markets.

Shortage of Skilled Workforce

There is a lack of specialized professionals in pharmaceutical research, manufacturing, and regulatory affairs.

Developing a skilled workforce through training and academic programs is critical for industry growth.

Opportunities for Growth

Investment in Local Production

Increasing local manufacturing, especially in essential medicines and specialty areas like oncology and vaccines, can enhance Uganda's pharmaceutical security.

Public-Private Partnerships (PPPs) could play a crucial role in driving investment.

Regional Expansion and Export Potential

The EAC harmonization efforts present an opportunity for Ugandan pharmaceutical companies to expand their reach beyond the domestic market.

By meeting international regulatory standards, local manufacturers could increase exports to neighbouring countries.

Public-Private Collaborations

Government initiatives supporting pharmaceutical production can help develop infrastructure and provide financial incentives for local companies.

Partnerships with international pharmaceutical firms could help bring technology and expertise to Uganda.

Biopharmaceutical and Vaccine Development

Investing in biologics and vaccine production could position Uganda as a leader in high-value pharmaceutical production.

Establishing local capacity for monoclonal antibody production could reduce reliance on costly imports.

Overview of the Ugandan Pharmaceutical Regulatory Framework and EAC Regulatory Systems

Introduction:

In Uganda, the regulation of human medicinal products is overseen by the National Drug Authority (NDA), which ensures the safety, efficacy, quality, availability and cost means of providing satisfactory healthcare and safeguarding the appropriate use of medicines used within the country.

The legal framework governing these activities is established by the National Drug Policy and Authority Act, Cap 206.

The National Drug Authority (NDA) was established in 1993 by the National Drug Policy and Authority Statute which in 2000 206 of the Laws of Uganda (2000 Edition).

The EAC Guide on Good Manufacturing Practices (GMP) is based on the World Health Organization (WHO) Good Manufacturing Practices guidelines in the WHO Technical Report Series No. 961, 2011. This guide also refers to the Pharmaceutical Inspection Convention Scheme (PIC/S) guide PE 009-9.

Table No.:3

Guidelines: National Drug Authority, Uganda- Human medicinal guideline

Maturity level of NDA (Uganda) with respect to WHO Qualification

Maturity Level 3, which is a key benchmark for drug regulatory bodies, ensures that national authorities can effectively oversee drug safety, efficacy, and quality, aligning with international norms.

The National Drug Authority (NDA) is seeking to overcome legal and regulatory challenges that have hindered its progress toward achieving Maturity Level 3, a global standard for drug regulators.

Only eight African countries have currently attained this status, making it a significant goal for Uganda. These include, Egypt, Rwanda, Senegal, Ghana, Nigeria, South Africa, Tanzania and Zimbabwe.

Key Regulatory Requirements

In Uganda, the journey of a new drug from conception to market involves a systematic approval process regulated by several governing bodies, predominantly the National Drug Authority (NDA).

Preclinical and Clinical Trials:

The process begins with preclinical studies, where laboratory and animal testing evaluate the drug's safety and efficacy. This stage is crucial, as it provides preliminary data that informs the next phases of research and development. Pharmaceutical companies need to document their findings and prepare a comprehensive submission package for the NDA.

Conducting clinical trials in Uganda requires prior authorization from the NDA.

Researchers, organizations must adhere to the "Guidelines for the Conduct of Clinical Trials in Uganda," which detail the procedures for submission, review, and approval of clinical trial applications.

Compliance with Good Clinical Practice (GCP) is mandatory, as outlined in the "Guidelines on Good Clinical Practice in the Conduct of Clinical Trials Involving Human Participants."

After completing the clinical trials, the pharmaceutical company must submit a New Drug Application (NDA) to the NDA, including all research data, manufacturing details, and proposed labelling.

Once the NDA reviews the application and finds it satisfactory, it grants approval for the drug to be marketed.

Good Manufacturing Practice (GMP):

Manufacturers of medicinal products must comply with GMP standards to ensure product quality. The NDA provides detailed guidelines, including "Guidelines on Good Manufacturing Practice for Medicinal Products Part I – Basic Requirements for Medicinal Products."

All human medicinal products must be registered with the NDA before they can be marketed or distributed in Uganda. The registration process involves a scientific evaluation of the product's safety, efficacy,

and quality.

GMP Inspection:

National Drug Authority (NDA) is the regulatory body responsible for overseeing the compliance of pharmaceutical companies with GMP standards.

Regular GMP inspections and audits are conducted as one of the requirements for registration of medicinal products in the EAC.

The NDA conducts comprehensive evaluations to verify that facilities meet the specified criteria for manufacturing processes, hygiene, and personnel training.

Inspection involves review of documents, records, facilities and any other resources to assess their conformity to the requirements of EAC GMP of medicinal products.

Any non-compliance can result in severe penalties, including the suspension or revocation of manufacturing licenses.

Importation and Exportation:

The NDA regulates the import and export of drugs and pharmaceutical raw materials.

Drugs imported must be registered unless given special clearance by the NDA under Sec 8(4) NDPA (National drug policy & Authority) by NDA.

The import permit may be:

Annual import permit -which is issued for regular imports like pharmaceutical manufacturers or any organization, retail pharmacies, wholesale premises, that regularly imports drugs and other related substances. Ministry of Health of Uganda and its affiliated units may also apply for an annual import permit.

Provisional import permit is issued to non-regular importers who may import drugs, for some reasons.

The provisional import permit is issued per consignment. This usually applies to the organizations, which receive donations, physicians or individuals who may for specified reasons need to bring limited quantities of specialized drugs that are not available in the country. The permit is valid for one month.

Licensing:

Entities involved in the manufacture of pharmaceuticals must obtain appropriate licenses from the NDA.

Import License: (Form-24)

Eligibility to apply for import license: A licensed person (a person holding license to operate pharmacy or for manufacture drugs).

Import license is valid for one year and is valid within the calendar year within which it is issued.

Verification certificate: A consignment of drugs to be imported into Uganda shall before importation, be issued with a verification certificate (Form-25) by the NDA basing on the application submitted in this regard in Form-26.

Stability Data:

In Uganda, the National Drug Authority (NDA) requires stability data as part of the regulatory requirements for pharmaceutical and biopharmaceutical products, ensuring quality, safety, and efficacy, and mandates that all medicinal products meet acceptable standards.

The NDA requires that applicants for drug registration provide results of stability testing, which is crucial for determining the shelf-life and storage conditions of the product.

Applicants need to submit all relevant stability data as part of the registration application.

The stability studies will determine the appropriate storage conditions for the product, including temperature and humidity, and these conditions must be reflected on the product label.

The product label must include an expiry date and storage instructions based on the stability data.

Stability studies should include Storage conditions i.e temperature and relative humidity for accelerated and stress conditions.

Stability studies should be done in accordance or with reference to WHO TRS 987, WHO TRS 953 Annex 2, WHO TRS 962

Annex 3, ICH Q1A and ICH Q5C guidelines.

A minimum of twelve months data at the time of submission should be provided in cases where storage periods greater than six months are requested.

For storage periods of less than six months, the stability data should cover the whole proposed shelf life.

The stability studies should be submitted in controlled documentation.

Post-approval stability program should include the stability program or stability commitment to be carried out once the drug product is on the market, including the number of batches to be included in the study each year and the tests to be performed.

These results should be submitted periodically to update the information on the stability of the drug product.

The stability of each dosage form should be separately documented.

Stability data submitted should be for at least three consecutive batches and include the following:

- a) Information on stability of drug product, quality control methods and rationale for the choice of tests for determining stability.
- b) Information on the dates of manufacture of the lots, the lot numbers, the vial and dose size, and the scale of production.

For lyophilized products the data supporting the shelf-life of the product reconstitution should be included.

If the drug product is frozen, data supporting the stability of the product through a stated number of freeze-thaw cycles should be provided.

Drugs Registration in Uganda:

All drugs to be manufactured or imported into Uganda have to be registered with the National Drug Authority, Uganda. A person who intends to manufacture, import or export a product shall, prior to the manufacture, importation or exportation of the product, apply to the Authority for registration of the product.

The registration process involves:

NDA inspection & approval of the manufacturing facility.

Validity of registration:

The Authority shall issue for the first registration, a certificate of registration which shall be valid for five years.

The holder of a certificate of registration shall pay an annual retention fee prescribed by the Authority for maintaining the registered human or veterinary drug or preparation, vaccine or other immunological products or surgical instrument, on the Register.

A certificate for renewal of registration shall be valid for one year from the date of issue.

The certificate of registration issued under these Regulations shall be kept at the premises of the holder of the certificate of registration.

The holder of a certificate of registration shall monitor the safety and effectiveness of the registered human or veterinary drug or preparation, vaccine or other immunological products or surgical instrument while on the market in Uganda and shall submit periodic reports as may be prescribed by the Authority.

The registration of a human or veterinary drug or preparation, vaccine or other immunological products or surgical instrument under these Regulations shall be valid except where the registration is suspended or

cancelled by the Authority or is terminated by the holder of the certificate of registration.

Table No.:4

Marketing Authorisation Time lines: Registration Timeline: 12 to 18 months

Post-Marketing Surveillance and Pharmacovigilance:

Under the National Drug Policy and Authority (NPD &A) Act (1993) the Uganda National Pharmacovigilance Centre (NPC) is responsible for PV efforts in the country and mandates the National Drug Authority (NDA) to monitor the safety, efficacy and quality of medicines.

Risk Assessment

Risk assessment is a crucial regulatory requirement, ensuring product safety, efficacy, and quality, and safeguarding public

health. Proactive risk identification and mitigation, through methodologies like Quality Risk Management (QRM), are essential for regulatory compliance and building trust with stakeholders.

Manufacturers would be required to conduct a risk assessment on critical production and quality control processes and thereafter develop a mitigation plan whose effectiveness would be monitored by the regulator.

Table No.:5 Regulation Framework Timelines:

Environmental and Safety Regulations Applicable to Pharmaceutical Manufacturing in Uganda

Uganda's regulatory framework ensures that pharmaceutical production safeguards human health, worker safety, and environmental sustainability.

Environmental Compliance

Environmental oversight in Uganda is primarily the responsibility of the National Environment Management Authority (NEMA). Under the National Environment Act, 2019, all industrial and manufacturing projects, including pharmaceutical facilities, are required to undertake an Environmental and Social Impact Assessment (ESIA) as a prerequisite for licensing and construction. The ESIA must identify potential environmental risks and outline mitigation strategies, with particular attention to waste management, water usage, emissions, and impacts on surrounding communities.

Pharmaceutical facilities must install and maintain Effluent Treatment Plants (ETPs) to process wastewater before discharge. Compliance with the Environmental Management (Waste Management) Regulations ensures proper handling of both hazardous and non-hazardous solid and liquid waste, especially chemical residues, solvents, and expired or rejected pharmaceutical products. The facility must also adhere to Uganda's obligations under the Basel Convention on the control of transboundary movements of hazardous waste.

Safety and Occupational Health Standards

The National Drug Authority (NDA) regulates the pharmaceutical sector and enforces Good Manufacturing Practices (GMP), including protocols that safeguard worker and product safety throughout the production lifecycle.

This includes facility design to minimize cross-contamination, controlled cleanroom environments, and safe material handling procedures.

In parallel, the Occupational Safety and Health Act, 2006, administered by the Ministry of Gender, Labour and Social Development, mandates comprehensive safety practices in the workplace. Employers must implement hazard identification and risk mitigation strategies, ensure employee training, and provide Personal Protective Equipment (PPE) and emergency response systems. All operations must include clearly marked emergency exits, fire suppression systems, first aid facilities, and regular safety drills.

Regular inspections and audits are conducted by both NDA and the Ministry of Labour to verify compliance.

Manufacturers are encouraged to adopt internationally recognized standards such as ISO 14001 (Environmental Management Systems) and ISO 45001 (Occupational Health and Safety Management Systems) to enhance performance and facilitate future export readiness.

Strategic Considerations

Full regulatory compliance not only avoids legal penalties and operational delays but also enhances the facility's reputation among stakeholders and investors. Additionally, adherence to these regulations supports sustainable operations and positions the facility for future certification under global quality and safety systems, enabling access to regional and international markets.

Overview of EAC Regulatory Systems for Key Therapeutic Areas

General Observations on the EAC Medicines Regulatory Framework

Harmonization Framework: The EAC Medicines Regulatory Harmonization (MRH) Programme supports joint product assessments, GMP inspections, and reliance pathways to reduce duplication across Partner States.

National Regulatory Authorities (NRAs): While each EAC country retains autonomy (e.g., Kenya PPB, Uganda NDA, Rwanda FDA), they collaborate extensively on shared regulatory functions.

Global Alignment: Reliance on WHO Prequalification, Global Fund procurement standards, AVAREF for clinical trials, and engagement with the African Medicines Agency (AMA) for future alignment.

Site Selection and Infrastructure

Site Selection Criteria

Proximity to Raw Materials & Suppliers

Most of the Active Pharmaceutical Ingredients (APIs) and excipients used in pharmaceutical production in Africa are imported (mainly from India and China).

Ideal site locations:

Close to international ports, airports, or dry ports to reduce import lead times.

Access to local packaging material manufacturers to lower supply chain dependency.

Industrial zones with chemical and biotech suppliers to ensure consistent raw material availability.

Logistics & Transportation Considerations

Proximity to major highways, rail networks, and ports to facilitate raw material import and product distribution.

Cold chain logistics is crucial for vaccines and biologics, requiring locations near temperature-controlled warehouses and transport hubs.

Close to key healthcare markets to reduce distribution delays, especially for essential medicines.

Utilities & Infrastructure

Electricity: Reliable power supply is critical, especially for injectables, vaccine production, and sterile drug manufacturing. Backup

generators, UPS systems, and solar energy solutions should be considered.

Water Supply & Treatment:

High-quality water sources for pharmaceutical processing, purification, and cleaning.

Reverse Osmosis (RO) & Water for Injection (WFI) systems for injectable production.

Effluent treatment plants (ETP) and bio-waste disposal facilities for environmental compliance.

Air Filtration & HVAC Systems:

ISO-classified cleanrooms (ISO 5, ISO 6, ISO 7, ISO 8) for sterile and injectable production.

High-Efficiency Particulate Air (HEPA) filters and humidity- controlled areas for tablet and capsule manufacturing.

Skilled Workforce Availability

Proximity to universities and research institutions that offer pharmaceutical, biotechnology, and engineering programs.

Training partnerships with international pharmaceutical firms to build local expertise.

Employee housing and transport accessibility to attract and retain skilled professionals.

Infrastructure Requirements

Facility Layout & Zoning

Segregated production zones for different dosage forms:

OSD (Oral Solid Dosage): Tablets, capsules.

Syrups & Liquids: Bottled formulations.

Injectables & Vaccines: Requires high-containment cleanrooms and sterile environments.

Dedicated cleanroom areas for sensitive processes such as sterile drug filling and vaccine formulation.

Material & personnel movement control to prevent cross-contamination.

Manufacturing Equipment & Technology Integration

Automated production lines for efficiency and compliance with GMP standards.

SCADA (Supervisory Control and Data Acquisition) and MES (Manufacturing Execution Systems) for real-time monitoring.

Cold Chain & Storage Facilities

Temperature-controlled warehouses for raw materials and finished pharmaceutical products.

Cold chain logistics for vaccines and biologics, including deep freezers and transport containers.

Automated inventory management systems to ensure compliance with GDP (Good Distribution Practices).

Finding the Right Technology Partners

Pharmaceutical Machinery & Equipment Suppliers

To establish a world-class pharmaceutical facility, partnering with technology providers specializing in pharmaceutical manufacturing equipment is essential. Key areas include:

Oral Solid Dosage (OSD) Equipment

Granulation, blending, and tablet compression machines

Leading suppliers: GEA Group, Bosch, Syntegon (formerly Bosch Packaging), Fette Compacting ACG Worldwide,

SED Pharma, Syntegon Technology, IMA Group, Glatt, Fette Compacting, Romaco, iPharMachine

Sterile Injectable & Vaccine Manufacturing Equipment

Aseptic filling, lyophilization, and cleanroom solutions

Leading suppliers: Bausch+Ströbel, IMA Pharma, Optima Pharma, NKP Pharma machinery, Sartorius, Cozzoli

Machine Company, Groninger, ARxIUM, NJM Packaging, Intellitech

Water Purification & WFI Systems

Suppliers: Merck Millipore, Veolia Water Technologies, Komal Industries, MECO, NGK Insulators Ltd,

Syntegon, Ovivo, Fabtech Technologies, Aqua-Chem, Herco Wasser technik GmbH, NEWater

Cleanroom & Air Filtration Systems

Suppliers: Camfil, AES Clean Technology, Thermo Fisher Scientific, Clean Air Technology, Atmos-Tech

Industries, Abtech Inc., Microflow Devices India PVT Ltd. Zeco Aircon Ltd

Digital & Automation Partners

SCADA & MES Systems: Siemens, Rockwell Automation, Honeywell.

Automated quality control & batch monitoring: Emerson, Siemens, ABB.

Regulatory & Compliance Partners

GMP compliance consulting:

Local regulatory bodies (e.g., Uganda National Drug Authority - NDA).

Waste Management & Sustainability Partners

Hazardous waste disposal & ETP solutions: Veolia, SUEZ, Ecolab.

Renewable energy integration for sustainability: Tesla Energy, Schneider Electric.

Strategic Collaborations & Public-Private Partnerships

Government Incentives & Special Economic Zones (SEZs)

Tax incentives for pharmaceutical manufacturers in Uganda and the East African Community (EAC).

Preferential land allocation and infrastructure support for investors.

Public-Private Partnerships (PPPs)

Collaborations with global health organizations (WHO, UNICEF, GAVI, USAID) for vaccine and essential medicine production.

Infrastructure Requirements for a Pharmaceutical Manufacturing Facility near Kampala, Uganda

The proposed pharmaceutical manufacturing facility, will be located near Kampala, Uganda.

Given the scale and scope of this state-of-the-art facility, robust infrastructure is vital to ensure uninterrupted operations, regulatory compliance, and product quality.

Power Supply: The facility will require a high-capacity, stable power source to support continuous manufacturing, HVAC systems, cleanrooms, cold storage, and quality control labs. It will be connected to the national electricity grid via UMEME, with dedicated transformers and high-voltage infrastructure. Backup power will be provided through diesel generators and solar power systems to ensure energy security and sustainability.

Water Supply: A reliable water supply is essential for production processes, utilities, cleaning, and sanitation. The facility will include on-site water purification systems to generate Purified Water (PW) and Water for Injection (WFI), meeting GMP standards. Water will be sourced from the national supply network and supplemented by boreholes with storage reservoirs.

Waste Management: The facility will house a centralized Effluent Treatment Plant (ETP) and systems for safe disposal of solid and hazardous waste, in full compliance with NEMA guidelines. Special provisions will be made for pharmaceutical and biomedical waste, including incineration and disposal via licensed operators.

Transport Links: Proximity to Kampala's industrial zones, the Northern Corridor, and Entebbe International Airport ensures efficient import/export logistics and access to regional markets. Excellent road connectivity will facilitate material movement and regulatory oversight.

TECHNICAL FEASIBILITY

Machinery and Equipment Requirements

Each dosage form requires specific equipment to facilitate high-quality and compliant production.

Oral Solid Dosage (OSD) & Capsules Manufacturing

Granulation Equipment – Fluidized bed dryers, high-shear mixers.

Tablet Compression Machines – Single rotary, double rotary presses.

Capsule Fillers – Semi-automatic and fully automatic encapsulation machines.

Coating Machines – Film and sugar coating systems.

Blending & Mixing Equipment – V-blenders, ribbon mixers.

Syrup and Liquid Manufacturing

Sugar Dissolution & Mixing Tanks – Stainless steel reactors with agitation.

Homogenizers – For uniform syrup consistency.

Liquid Filling & Capping Machines – For bottles and ampoules.

Sterile Filtration Systems – For microbial safety.

Injectables & Parenteral Manufacturing

Aseptic Filling Lines – For ampoules, vials, prefilled syringes.

Lyophilization Equipment – For freeze-dried injectables.

Autoclaves & Sterilizers – For equipment and component sterilization.

Isolators & Restricted Access Barrier Systems (RABS) – To maintain sterility.

Monoclonal Antibodies (mAbs) & Vaccine Manufacturing

Fill-Finish Equipment – Sterile filling, capping, lyophilization.

Cold Chain & Stability Chambers – For biological storage.

3. Technology and Automation

Digitalization and Smart Manufacturing

Manufacturing Execution Systems (MES) for batch tracking.

SCADA (Supervisory Control and Data Acquisition) for process automation.

Predictive Maintenance to prevent downtime.

Quality Control & Assurance Technologies

Analytical Testing Equipment – HPLC, GC, UV Spectrophotometers.

Microbial Testing – Endotoxin testing, sterility assurance.

4. Compliance & Regulatory Considerations

Adherence to GMP, WHO and local drug regulatory guidelines.

Regular validation & qualification of equipment and processes (IQ, OQ, PQ).

Environmental compliance with waste disposal and emission control.

Manufacturing Area Requirements - Overall Site Plan & Area Distribution

Table No.: 6 COMMON AREAS

Table No.: 7

PLANT 1 – OSD BLOCK (ASSUMED OUTPUT CAPACITY IS 15 MILLION UNITS PER DAY)

Table No.: 8

PLANT 2 – mAbs AND BIOSIMILARS (ASSUMED OUTPUT CAPACITY IS 3 MILLION UNITS PER ANNUM)

Expansion & Scalability: The facility should allow for a future increase in production capacity.

Table No.: 9

PLANT 3 - Filling Lines for Vials, Ampoules, and Pre-Filled Syringes

Expansion & Scalability: Design with future capacity increases in mind.

PLANT 4 - Oncology OSD Facility for Production of Hormonal and Cytotoxic Oral Formulations

Table No.: 10

Additional Considerations

High-Containment Design:

Closed systems with negative pressure, airlocks, isolators, and barrier technology.

Cross-Contamination Control:

Separate air handling, material flows, and personnel flows

for hormonal and cytotoxic drugs.

Scalability & Future Expansion:

Facility should accommodate increasing demand and new drug formulations.

Table No.: 11

PLANT 5 – VACCINE fill finish Facility (ASSUMED CAPACITY OUTPUT IS 50 MILLION DOSES PER ANNUM)

FINANCIAL FEASIBILITY

This review provides an overview of CAPEX, Operational Expenditure (OPEX), financial projections, break-even analysis, and potential Return on Investment (ROI).

Cost Breakdown: CAPEX

Table No.: 12

Capital Expenditure (CAPEX)

The total investment is \$571M USD, allocated across different manufacturing units and operations as follows:

Table No.: 13

Operational Expenditure (OPEX) Breakdown

Annual OPEX includes raw materials, labour, utilities, maintenance, logistics, and regulatory compliance.

Estimated OPEX in full scale operations:

Table No.: 14

* As per calculations

Financial Projections and Revenue Estimates

Estimated Annual Revenue (First 5 Years)

Revenue is projected based on market demand, production capacity, and price realization. The estimated year-wise revenue is:

Table No.: 14

Revenue Contribution by Segment:

OSD & Injectables: 35%

Oncology: 25%

Vaccines & Biopharma: 30%

Contract Manufacturing & Exports: 10%

Break-Even Analysis

Break-Even Point Calculation

The break-even point (BEP) is determined by comparing total fixed costs (CAPEX amortization) with contribution margins.

Fixed Costs: Depreciation on \$571M CAPEX (~10-year depreciation cycle) → ~\$57M/year

Variable Costs: 65% of revenue

Average Selling Price & Margins: 35% Gross Margin

Break-Even Sales Volume Calculation:

Break-even revenue=Fixed Costs / Contribution Margin = 57M / 0.35=162M USD

Thus, the break-even point is reached within the third/fourth year of operation.

Payback Period & Profitability Timeline

Payback period: ~6 years

First-year losses due to ramp-up & market entry.

Full profitability expected in Year 4-5 with a Net Profit Margin of ~15- 20%.

Return on Investment (ROI) Analysis

Projected ROI

ROI = Annual Profit / Total investment × 100

Year 3 Expected Net Profit: ~\$150M USD

ROI by Year 3/4: 150/ 571 ×100 = 26.0%

Cumulative ROI over 6 years: 125% (fully recovering investment & generating profit).

Internal Rate of Return (IRR)

Estimated IRR: 18-22%, exceeding the pharmaceutical industry benchmark of 15-18%.

Strong returns in high-value segments (Oncology, Biopharma, and Vaccines)

Key Financial Risks & Mitigation Strategies

Table No.: 15

Conclusion & Strategic Recommendations

CAPEX of \$571M is strategically allocated across multiple high- growth segments.

OPEX of ~\$335M annually ensures robust operations & regulatory compliance.

Break-even is expected within the third year, with profitability by Year 4-5.

Strong ROI (~28% by Year 4) and IRR (~18-22%) ensure high financial viability.

Strategic Recommendations:

Focus on Oncology & Biologics: Higher-margin products provide better financial returns.

Optimize API Sourcing: Reduce costs by investing in local API production or strategic long-term supplier

contracts.

Enhance Export Potential: Leverage Uganda's EAC trade advantages to increase regional market penetration.

Automate Supply Chain & Manufacturing

Seek Investment & Government Support: Public-private partnerships for incentives & subsidies to lower CAPEX burden.

SUPPLY CHAIN AND RAW MATERIAL SOURCING

The supply chain must ensure a steady supply of raw materials, efficient production, strict quality control, and seamless distribution, while mitigating risks associated with logistics, regulatory compliance, and market fluctuations.

Key Components of the Supply Chain

A robust supply chain for a Sunshine company shall include the following key components:

Raw Material Sourcing (APIs, Excipients, Packaging Materials)

Inventory & Warehousing

Production & Manufacturing

Quality Control & Regulatory Compliance

Cold Chain & Storage (For Vaccines & Biosimilars)

Distribution & Logistics

Risk Management & Supply Chain Resilience

Raw Material Sourcing Strategy

Active Pharmaceutical Ingredients (APIs) and Excipients

Uganda does not have a strong domestic API production base, so APIs must be imported.

Primary API sourcing regions:

India & China – Major global API suppliers.

Europe (Germany, Switzerland, Italy) – High-quality APIs for oncology & biosimilars.

Africa (South Africa, Egypt, Morocco) – Growing pharmaceutical hubs for APIs.

Procurement Strategy:

Develop multiple supplier relationships to reduce dependency on a single country.

Use long-term contracts & Just-in-Time (JIT) supply models for high-cost APIs.

Maintain buffer stock of critical APIs (especially for oncology & vaccines).

Packaging Materials

Primary packaging: Blister packs, vials, ampoules, pre-filled syringes, HDPE bottles.

Secondary packaging: Cartons, labels, package inserts.

Suppliers: Kenya, India, UAE, Turkey, and South Africa.

Supply Chain for Biologics (Vaccines & Biosimilars)

Requires specialized cold chain logistics (-80°C for some biologics).

Partnerships with international biotech firms for technology transfer.

Cold storage facilities at airports & major distribution hubs.

Inventory & Warehousing Management

Raw Material Storage

Separate storage for APIs, excipients, and packaging materials.

Cold storage for temperature-sensitive raw materials.

Real-time inventory tracking system (ERP-based).

Finished Goods Warehousing

GMP-compliant warehouses with segregated storage for different product categories.

First-Expiry-First-Out (FEFO) system for batch management.

Strategic warehouse locations in Uganda and EAC countries for faster distribution.

Production & Manufacturing Workflow

Manufacturing Planning & Scheduling

Lean Manufacturing to reduce waste and optimize production costs.

Batch Manufacturing Record (BMR) & Batch Production Record (BPR) compliance.

Facility Layout & Integration with Supply Chain

Segregated production lines for oncology, OSD, syrups, vaccines, and biosimilars.

Integrated Manufacturing Execution Systems (MES) for real-time monitoring.

Automated raw material dispensing & production tracking.

Quality Control & Regulatory Compliance

Stringent QC checks for APIs, excipients, and packaging materials.

Real-time product traceability using barcoding & serialization.

Compliance with WHO-GMP, EAC regulatory frameworks, US FDA, EMA.

Regular audits of suppliers & third-party logistics providers.

Cold Chain & Storage for Vaccines & Biosimilars

Cold storage at 2-8°C, -20°C, and -80°C.

Real-time temperature monitoring

Strategic cold chain hubs in Kampala & EAC export markets.

Distribution & Logistics Strategy

Domestic Distribution (Uganda)

Direct distribution to hospitals, pharmacies, and wholesalers.

Regional distribution centres in Kampala, Entebbe, Jinja, and Mbarara.

GPS-tracked delivery fleet with route optimization for last-mile delivery.

East African Community (EAC) Export Strategy

Key export markets: Kenya, Tanzania, Rwanda, Burundi, South Sudan, DRC.

Regulatory harmonization under the EAC-MRH program for seamless cross-border approvals.

Dedicated pharmaceutical logistics hubs in Nairobi (Kenya) & Dar es Salaam (Tanzania).

International Exports

Oncology & biosimilars exports to high-value markets in Europe & Middle East.

Collaboration with global pharmaceutical distributors.

Risk Management & Supply Chain Resilience

Key Risks in the Supply Chain

Dependency on imported APIs → Risk of supply chain disruptions.

Regulatory delays → Impact on product approvals & market entry.

Cold chain failures → Risk for vaccines & biologics.

Political & economic instability in Uganda & EAC → Disruptions in logistics.

Risk Mitigation Strategies

Dual sourcing strategy for APIs & critical raw materials.

Investment in API manufacturing in Uganda (long-term goal).

Use of digital supply chain solutions (AI-driven demand forecasting).

Collaboration with global logistics providers (DHL, UPS, local 3PLs).

Digital Transformation in Supply Chain Management

AI Based Monitoring

AI-based demand forecasting to optimize inventory levels.

Cost of Maintaining a Robust Pharmaceutical Supply Chain in Uganda

Key Cost Components of a Pharmaceutical Supply Chain

Note: The cost varies depending on production scale, market reach, automation levels, and regulatory

requirements.

Breakdown of Major Cost Elements

Raw Material & API Procurement Costs

APIs account for 60%-80% of drug production costs, particularly for oncology & biologics.

Uganda imports most APIs from India, China, and Europe, incurring import duties & logistics expenses.

Cost fluctuations due to supply chain disruptions, geopolitical tensions, and regulatory constraints.

Mitigation: Multi-supplier agreements & local API production investments.

Packaging Material Costs

Primary packaging (blister packs, vials, ampoules, prefilled syringes) is imported or locally sourced.

Secondary packaging (cartons, labels, inserts) needs compliance with international standards (e.g., serialization, barcoding).

Sustainable & eco-friendly packaging costs 10%-15% more but enhances regulatory acceptance.

Warehousing & Cold Storage

Cold storage for vaccines & biosimilars (-80°C to 2-8°C) incurs high energy & maintenance costs.

Storage facilities in Kampala & regional hubs (e.g., Jinja, Mbarara) require advanced temperature monitoring.

Manufacturing & Production Logistics

Includes the cost of transporting raw materials, managing production scheduling, and handling batch-to-batch variations.

Biologics and vaccines require specialized production environments, increasing utility & maintenance costs.

Quality Control & Compliance

Regulatory costs include GMP audits and licensing fees.

Maintaining high-end analytical labs for testing APIs, excipients, and final products.

Oncology & biosimilar testing requires more stringent & expensive analytical processes.

Cold Chain Infrastructure for Vaccines & Biologics

Ultra-low temperature storage (-80°C) for certain biologics.

GPS-tracked refrigerated trucks for secure transportation.

Backup power systems to ensure uninterrupted temperature control.

Distribution & Last-Mile Delivery

Cross-border regulatory compliance costs (for EAC markets).

Investment in regional distribution hubs (e.g., Nairobi, Dar es Salaam).

High-security transportation for oncology drugs to prevent theft.

IT & Digital Supply Chain Systems

Track-and-trace systems to prevent counterfeit drugs.

Automated inventory management to reduce wastage & improve efficiency.

Risk Management & Contingency Planning

Emergency stockpiling of critical APIs & biologics.

Alternative supply routes & backup logistics partners.

Investment in local API production to reduce import dependency.

Regulatory Compliance & Audits

WHO-GMP certifications for global market access.

Annual site inspections & compliance renewals.

Investment in training & capacity building for regulatory teams.

Cost Optimization Strategies

Local API & Packaging Manufacturing Investment

Reduce import costs by establishing API & packaging material production in Uganda.

Collaboration with African pharmaceutical hubs (South Africa, Egypt, Morocco).

Multi-Supplier & Regional Sourcing Strategy

Reduce risk by sourcing APIs from multiple suppliers.

EAC regional integration to reduce cross-border tariffs & logistics delays.

Green Energy & Sustainability Investments

Solar-powered cold storage to cut energy costs by 30-40%.

Eco-friendly packaging to reduce regulatory risks & enhance marketability.

HUMAN RESOURCES

In this project below listed facilities/blocks are planned to be constructed:

Oral Solid Dosage (OSD) Facility (Capacity Production of 15 million units/day (tablets, capsules, and syrups).

Biopharma Drug Product Facility (Capacity: Specialized for monoclonal antibodies, hormones, and insulin analogs)

Vaccine Formulation and Fill- Finish Facility (50 Million doses per annum)

Oncology OSD Facility

Injectable Facility (Capacity: High-speed filling lines for vials, ampoules, and pre-filled syringes)

Number of employees with different skill set shall be employed for all above mentioned facilities.

Summarized table for estimated number of man power required for Operations

Table No.: 16

Hiring of employees with skill sets and expertise:

It is important to have healthy employees with aseptic skills especially in manufacturing units to avoid any contamination during manufacturing of drug products.

Separate organograms are prepared for engineering, corporate, HR & administration, warehouse, Quality (QC & QA) and manufacturing units to hire sufficient and effective workforce considering cost effectiveness.

Details of Organograms:

Corporate -Employees with managerial skills, project skills and leadership skills etc. will be hired in corporate for different levels like CEO, Head of engineering, Head of operations/manufacturing, project manager, Head of quality, Head of finance, Head of logistics and Head of HR. Managers and group leaders from each section will be managed/lead by corporate Heads.

Engineering -In engineering section, employees with skills like automation, instrumentation including complicated manufacturing equipment, electrical, mechanical, utilities (Water, steam, Air) and information technology will be hired at different levels.

HR & Administration -In HR & admin section, employees with skills to handle transport, canteen, legal challenges like pollution control, employee act and licensing application and renewal etc., recruitment and day to day management of contractual / fixed employees, employee wellness, medical facility and general management of BMS like furniture, rest rooms, and external vendors for admin support will be hired.

Warehouse -In warehouse section, skills to ensure the safe, efficient, and compliant movement of materials and products is required. It includes logistics management skills.

Knowledge of inventory management using software like SAP or ERP, document and material management is essential in warehouse.

Quality Unit -Quality (Quality control and Quality assurance) is critical to ensuring the safety, efficacy, and compliance of drug products. In

QC, testing skills, knowledge of GLP, Data analysis and sample management are essential skills. In QA,

Process validation and control, regulatory compliance, Risk Management, Documentation, Audits and

inspection skills, Change management and trainings are essential skills.

Manufacturing -In manufacturing, knowledge of GMP, operation of complicated/automated equipment or system, knowledge of process and aseptic behaviour/environment, documentation and technical skills for problem solving in day-to-day process are crucial skills.

Health, Safety standards for Human Resource working in Pharmaceuticals:

For biopharmaceutical projects, regular medical check-up including tests like eyesight, audiometry, blood test, breathing test, X-Ray, ECG and basic tests; at-least annually for all employees is essential.

Key aspects in HR feasibility

Workforce Availability

Over the course of the last decade, Uganda's economic growth has ranked among sub-Saharan Africa's strongest; indeed, the country's annualized average growth rate was 5.4% between 2010 and 2019 (World Bank, 2020).

Uganda has a young and growing workforce, with over 75% of its population under the age of 30.

Total population: Approx. 51 million (2025 estimates)

Cost of Human Resources

In Uganda, employee salaries are relatively low, but there are also some differences. For example, in the capital city of Kampala, the monthly salary of an ordinary employee is usually between 100- 200USD. In other cities, this number may be even lower. The salary level will be higher for senior management and technical talents.

Average monthly salaries (estimates):

Unskilled worker: \$100–\$200

Mid-level professional: \$500–\$1,500

High-skilled/managerial: \$2,000+

Legal and Compliance Factors

Key employment laws:

Employment Act (2006): Governs employment terms, termination, wages, and working conditions.

Occupational Safety and Health Act (2006): Ensures workplace safety.

Workers' Compensation Act (2000): Covers workplace injury compensation.

Uganda's labour laws are primarily governed by the Employment Act of 2006, which establishes the minimum standards of employment in the country.

Uganda is committed to upholding international labour standards and has ratified several key

International Labour Organization (ILO) Conventions. These include the Freedom of Association and

Protection of the Right to Organize Convention, 1948 (No. 87), the

Right to Organise and Collective Bargaining Convention, 1949 (No. 98), and the Collective Bargaining Convention, 1981 (No. 154).

The Uganda Employment Act applies to all employees, irrespective of nationality that the employers hire under the service contract.

According to the employment act 2006 and labour laws in Uganda, the standard working hours of an employee is a maximum of eight hours per day and 48 hours a week; excluding lunch and worship time.

Employers planning to hire or relocate international employees to Uganda must understand the processes surrounding work permits and visas. Ensuring compliance with Ugandan immigration laws helps streamline the hiring process and avoid potential legal hurdles.

Cultural and Language Considerations

Ugandan communication often leans towards indirectness, especially when delivering negative feedback or expressing disagreement.

Cultural Lens: The Ugandan concept of "Gwasi" (shame) discourages public criticism. This can influence communication, leading individuals to deliver critical messages indirectly to avoid causing shame to the recipient.

Uganda is a multilingual nation like many African countries. However, English is targeted as a medium of instruction and language of examination.

RISK ASSESSMENT

Risk Assessment and Mitigation Strategies for Sunshine Plant in Uganda

Political Risks and Mitigation Strategies

Uganda, like many developing economies, has political uncertainties that can affect business operations.

These include regulatory changes, policy instability, and geopolitical factors.

Key Political Risks

Table No.:17

Risk Mitigation Strategies

Early Engagement with Regulators:

Work closely with Uganda's National Drug Authority (NDA) and the Ministry of Health to ensure timely approvals.

Seek WHO-GMP certification for global market access.

Public-Private Partnerships (PPP):

Collaborate with government bodies to access tax benefits, incentives, and investment protection.

Diversification of Export Markets:

Establish trade links with EAC, COMESA, and beyond to reduce dependence on a single market.

Strengthening IP Protection:

Register patents and trademarks regionally through ARIPO (African Regional Intellectual Property Organization) to safeguard innovations.

Economic Risks and Mitigation Strategies

Economic instability can impact capital investment, supply chain costs, and market demand for pharmaceutical products.

Key Economic Risks

Table No.:18

Risk Mitigation Strategies

Localizing API Sourcing:

Develop partnerships with regional API manufacturers to reduce reliance on imports and mitigate forex risks.

Hedging Against Currency Risks:

Utilize foreign exchange hedging instruments to stabilize procurement costs.

Maintain a mix of local and foreign currency reserves for operational expenses.

Operational Cost Optimization:

Implement energy-efficient manufacturing to reduce electricity costs.

Automate production processes to minimize labour costs.

Strategic Supplier Agreements:

Enter long-term contracts with multiple suppliers for key raw materials to ensure supply chain resilience.

Diversifying Revenue Streams:

Expand into high-margin segments (oncology, vaccines, biosimilar).

Target institutional buyers like the government, NGOs, and international health organizations (e.g., GAVI, WHO, Global Fund).

Environmental Risks and Mitigation Strategies

Uganda has stringent environmental regulations for pharmaceutical manufacturing, including waste management, water usage, and emissions control. Non-compliance can lead to fines, production shutdowns, or reputational damage.

Key Environmental Risks

Table No.:19

Risk Mitigation Strategies

Sustainable Water Management:

Implement water recycling & rainwater harvesting systems to reduce dependence on municipal supplies.

Use high-efficiency Clean-in-Place (CIP) systems to minimize water wastage.

Advanced Waste Treatment Solutions:

Install Zero Liquid Discharge (ZLD) systems for pharmaceutical effluent treatment.

Set up on-site hazardous waste incinerators to manage cytotoxic and biological waste safely.

Renewable Energy Integration:

Solar power installations on-site to reduce dependency on the national grid.

Waste heat recovery systems to improve energy efficiency.

Green Manufacturing Practices:

Obtain ISO 14001 Environmental Certification to align with international best practices.

Conduct regular environmental audits to ensure compliance with NEMA standards.

Summary of Risk Mitigation Strategies

Table No.:20

Conclusion

While setting up a pharmaceutical manufacturing plant in Uganda presents political, economic, and environmental risks, these can be effectively managed through strategic planning and proactive mitigation measures.

I. SUSTAINABILITY

Sustainable Waste Management Strategies

Sunshine facility shall generate solid, liquid, and hazardous waste that must be effectively managed to comply with environmental regulations and protect ecosystems.

Segregation and Recycling of Solid Waste

Implement waste segregation at the source to separate biodegradable, non-biodegradable, recyclable, and hazardous waste.

Recycle packaging materials like paper, plastics, and aluminium foils used in blister packs.

Partner with certified waste management companies to handle non- recyclable hazardous waste.

Efficient Handling of Hazardous Waste

Incinerate cytotoxic oncology waste and contaminated biological waste in high-temperature incinerators (above 850°C).

Use autoclaves and chemical disinfection for microbiological and vaccine-related waste.

Adopt encapsulation techniques for disposing of highly toxic chemicals and APIs safely.

Effluent Treatment and Liquid Waste Management

Install an Effluent Treatment Plant (ETP) and Zero Liquid Discharge (ZLD) system to ensure wastewater treatment before discharge.

Implement reverse osmosis (RO) and advanced oxidation processes to remove pharmaceutical residues.

Reuse treated water for landscaping, cooling systems, and non- potable applications.

Minimization of API & Solvent Waste

Implement solvent recovery units to recycle and reuse solvents in oncology and biopharma production.

Adopt green chemistry techniques to minimize hazardous waste production.

Optimize batch sizes and manufacturing processes to reduce raw material waste.

Energy Efficiency Strategies

Energy consumption will be a significant operational cost in pharmaceutical manufacturing. A sustainable energy plan shall ensure cost reduction and reduced carbon footprint.

Integration of Renewable Energy Sources

Solar Power Installation:

Install solar panels on rooftops to supplement grid electricity and reduce dependence on fossil fuels.

Target at least 30-50% energy from solar for non-critical operations.

Biomass & Biogas Utilization:

Utilize biomass boilers for steam generation using agricultural waste.

Install a biogas plant to convert organic waste into energy.

Energy-Efficient HVAC & Cooling Systems

Use energy-efficient HVAC systems with variable frequency drives (VFDs) to optimize airflow and minimize energy wastage.

Implement cold chain optimization for vaccines and biopharma products, including energy-efficient refrigeration and insulation systems.

Utilize waste heat recovery systems to capture and reuse excess heat from industrial processes.

Smart Lighting and Equipment Optimization

Install LED lighting with motion sensors to reduce power consumption in low-traffic areas.

Use high-efficiency motors and compressors in production lines to reduce energy waste.

Implement real-time energy monitoring systems to track and optimize energy use.

Water Conservation and Efficient Usage

Pharmaceutical production requires significant water usage for cooling, formulation, and cleaning processes. Water scarcity in Uganda makes conservation and recycling essential.

Water Recycling & Reuse

Install Water Treatment Plants (WTPs) with multi-stage filtration, RO, and UV disinfection to reuse wastewater.

Recycle treated wastewater for boilers, chillers, and non-potable cleaning processes.

Rainwater Harvesting

Set up rainwater harvesting tanks to collect and store rainwater for non-production use.

Utilize rainwater for landscaping, cooling towers, and toilet flushing.

Optimized Process Water Usage

Implement Clean-in-Place (CIP) and Sterilize-in-Place (SIP) systems to reduce water usage in equipment cleaning.

Use dry cleaning methods where possible to minimize the need for extensive water washing.

Train staff on water conservation best practices to minimize waste.

Regulatory Compliance & Green Certifications

WHO-GMP & EAC Environmental Regulations: Ensure compliance with local and international environmental guidelines.

ISO 14001 Certification: Implement environmental management systems to enhance sustainability.

LEED Certification: Design facilities with sustainable building materials to meet green building standards.

Conclusion

Sunshine shall require integrated waste management, energy-efficient operations, and water conservation measures. By adopting renewable energy, recycling waste, and optimizing resource usage, this facility can

achieve long-term environmental and economic benefits while maintaining regulatory compliance and corporate social responsibility.

J. Key HR Challenges and mitigation strategies

Skilled Workforce Shortage

Limited local experience in advanced pharmaceutical manufacturing (especially GMP-compliant operations for vaccines, oncology, and monoclonal antibodies).

Shortage of experienced quality assurance (QA), quality control (QC), production, validation, and biotech-specific personnel.

Training and Capability Gaps

Need for training in WHO-GMP and global standards.

Low exposure to advanced equipment and automation technologies common in global pharma plants.

High Attrition or Brain Drain

Trained personnel may be poached by competitors or seek jobs abroad due to better salaries.

Cultural & Communication Barriers

Integration challenges when blending expat professionals with local teams in managerial and technical roles.

Mitigation & Risk Management Strategies

Early & Strategic Workforce Planning

Workforce demand mapping: Identify headcount, roles, and experience levels required for each production phase.

Staggered hiring aligned with the commissioning of each production unit.

Establish a Technical Training Academy

Set up an in-house Pharma Training Institute or Centre of Excellence.

Partner with international experts to develop GMP, EHS, and regulatory curricula.

Simulate manufacturing environments (cleanrooms, QC labs, aseptic suites).

The training program shall constitute of the following aspects

Induction of new employees

cGMP training

GLP training

On the job training consisting of SOP training, equipment training etc.,

Safety training

Supervisor training

Manager training

Job change training or after leave of absence

Remedial training

Contractor employee training

International Partnerships

Collaborate with Indian, European, or South African pharma companies or universities for knowledge transfer.

Consider short-term expat secondments (2–5 years) to build and mentor local teams.

Collaboration with Local Academia

Tie up with Makerere University or other local institutions for:

Curricula alignment with pharma needs.

Internship/apprenticeship programs.

Fast-tracking diploma and B.Pharm graduates into technical roles.

Talent Retention Schemes

Offer competitive packages, training sponsorships, and bonded contracts.

Career development pathways for technicians to move into supervisory or QA roles.

Stock options or profit-sharing models to retain senior staff.

Outsource Non-Core Activities Initially

To reduce early HR load, consider outsourcing packaging, engineering maintenance, and logistics until local capacity matures.

Human Resources Rollout Roadmap Phase-Wise HR Rollout Plan (2025–2030) Phase 1: Planning & Initial Setup (2025)

Workforce Planning: Finalize detailed manpower requirement for each functional block (production, QA/QC, engineering, admin, etc.).

Key Roles: Project HR Head, Talent Acquisition Manager, Training & Compliance Manager

Action Points:

Begin local hiring process for support staff, documentation personnel

Initiate expat hiring for plant design, validation, and commissioning oversight

Establish HR policies, on-boarding modules, payroll systems

Phase 2: OSD & Syrup Commissioning (2026)

Estimated Staff Requirement: (As per Table No 16)

Key Functions: Production operators, QA/QC analysts, warehouse team

Action Points:

Launch training academy in partnership with local universities (Makerere, Mbarara, etc.)

Train local hires in GMP compliance, SOP writing, and safety protocols

Expat engineers and QA managers to lead operations and mentor local staff

Phase 3: Oncology & Injectable Line Setup (2027)

Estimated Staff Requirement: +- (As per Table No 16)

Key Functions: Cytotoxic handling experts, sterile Injectables line operators, QC microbiology

Action Points:

Specialized cytotoxic/Hormonal handling training

Safety and containment procedure modules

Expand academy offerings to include aseptic processing and EHS protocols

Phase 4: Vaccine & Biopharma Commissioning (2028)

Estimated Staff Requirement: -(As per Table No 16)

Key Functions: Biotech fill-finish experts, cold chain QA/QC

Action Points:

Partner with international biotech training programs (India, EU)

Establish a biotechnology learning track in the internal academy

Phase 5: Operations Stabilization & Local Leadership Development (2029–2030)

Target Workforce: 1,000+ including indirect employees

Focus Areas: Succession planning, reducing dependency on expats

Action Points:

Develop career growth ladders for technicians and supervisors

Launch Leadership Development Programs (LDP) for Ugandan managers

Retention strategy implementation (bonuses, ESOPs, continuous training)

Training Strategy Summary

In-House Pharma Academy to train employees per year

Modules in:

GMP & regulatory compliance

Pharmaceutical operations (OSD/liquid/injectable/Oncology)

Aseptic techniques

Equipment maintenance & calibration

Training Partners: Local academia + global consultants Risk Mitigation Measures

Challenge: Limited experienced pharma talent pool

Solution: Layered expat hiring, internal academy, local university tie-ups

Challenge: Attrition post-training

Solution: Bonded employment, career development programs, competitive pay

Challenge: Cultural integration

Solution: Cross-cultural workshops, expat-local mentorship structure

K. STRATEGIES FOR ACCELERATED ROI ON THE PROJECT

Phased Commissioning of Production Lines

Start with OSD and syrups, which have lower regulatory complexity and shorter setup timelines.

These can begin generating revenue within 12–18 months, while more complex lines (biopharma, oncology, vaccines) are still under validation.

Helps spread OPEX and initiates early cash flow.

Prioritize Products with High Local and Regional Demand

Focus on essential medicines, high-burden diseases (malaria, diabetes, antibiotics), and therapeutic areas like oncology and vaccines with unmet needs.

Customize the portfolio to include WHO-prequalified and procurement-friendly products for faster uptake by public health programs.

Partner with Global Health Buyers Early

Engage with donor agencies (Global Fund, GAVI, WHO, USAID) to position your facility as a local supplier.

Secure advance purchase agreements (APAs) or long-term supply contracts, even during construction phase.

These contracts anchor demand and improve investor confidence.

Leverage Contract Manufacturing (CDMO) for Revenue Boost

Offer your excess capacity for contract manufacturing of OSD, fill- finish, or vaccine products for regional or global pharma firms.

Acts as a bridge revenue stream before full-scale launch of your branded portfolio.

Accelerate Regulatory Approvals through EAC Harmonization

Register products using the EAC Medicines Regulatory Harmonization (MRH) program to access multiple countries simultaneously.

Cuts down duplicated efforts and approval timelines, speeding up market entry.

Optimize Operational Costs from Day 1

Invest in energy-efficient infrastructure (solar, efficient HVAC) and automation to lower long-term utility and labor costs.

Design supply chain to minimize storage and transportation overhead.

Use a Hybrid Distribution Model

Combine direct-to-hospital/clinic distribution with national wholesalers and tenders.

Reduces reliance on a single sales channel and speeds up product uptake across multiple segments.

Government Incentives and Tax Planning

Take full advantage of government incentives, including VAT deferments, customs waivers, tax holidays, and utility subsidies.

Negotiate land lease concessions or reduced utility tariffs to reduce fixed overheads.

Continuous Financial Monitoring

Set up a Project Management Office (PMO) and cash burn tracking system to detect cost overruns early.

Implement strict working capital control (inventory turnover, receivables) during operations to stay lean.

Table No.:21 Example ROI Acceleration Path (Indicative)

Assumptions

Table No.:22

Financial Projections (Years 1–10)

Table No.:23

Key Insights

Break-Even Year: Year 4 (Cumulative cash flow surpasses initial CAPEX of USD 529M)

Annual ROI Post-Break Even: ~39.7% (Profit of USD 210M / USD 529M)

Cumulative ROI by Year 10: ~3.4x return (USD 1795M cumulative cash flow)

L. SWOT ANALYSIS

Strengths

Robust Financial Backing

The promoters' strong financial background ensures ample capital availability for high-tech manufacturing infrastructure, reducing financing risks.

USD 530 million CAPEX provides a solid foundation for a state-of-the-art facility with capacity for future expansion.

Strategic Product Portfolio

Manufacturing a diverse mix of high-demand therapeutics—from essential OSDs and syrups to complex oncology drugs, vaccines, and biosimilars—positions the company to address both local and regional health needs.

Biosimilars and oncology injectables offer high-margin opportunities with increasing global and regional demand.

Location Advantage – Uganda & EAC Access

Uganda provides duty-free access to the East African Community (EAC), reducing export barriers and expanding market reach.

Strategic geographic location in Central-East Africa allows access to under-served pharmaceutical markets in South Sudan, Rwanda, Burundi, DRC, etc.

Modern Infrastructure and Technology Potential

Opportunity to design a greenfield facility aligned with WHO- GMP, EMA, and USFDA standards from the outset.

Flexibility to incorporate cutting-edge technologies (e.g., continuous manufacturing, single-use systems for biologics).

Weaknesses

Lack of Pharmaceutical Domain Experience

The promoters' background in infrastructure, while valuable in construction and logistics, does not directly translate to the complex and highly regulated pharmaceutical industry.

Manufacturing pharmaceuticals—especially oncology, vaccines, and biosimilars—requires deep expertise in GMP compliance, validation protocols, aseptic processing, and regulatory science.

Without experienced pharma leadership, there is a risk of non-compliance, production inefficiencies, or market entry delays.

Mitigation: Early recruitment of experienced pharma professionals and regulatory consultants is critical, as is establishing a governance board with pharma advisors.

High Dependence on Imports for APIs, Equipment & Components

Uganda lacks a strong API and excipient manufacturing ecosystem, so nearly all critical raw materials,

consumables, and equipment must be imported—mainly from India, China, or Europe. This increases lead times, procurement costs, customs complexities, and exposure to global supply chain disruptions.

Biopharma and vaccine components (e.g., single-use systems, cold-chain logistics, cell lines) are even more sensitive to supply delays.

Skills and Talent Gap in Local Workforce

Uganda's pharmaceutical industry is still developing, with limited availability of skilled technicians, validation engineers, QC analysts, and GMP-trained production staff, especially in sterile manufacturing or biotech.

Over-dependence on expats may drive up costs and hinder the development of sustainable local capability. Cultural integration, language, and training gaps can affect productivity and retention if not addressed proactively.

High Operating Costs for Complex Products

Oncology, vaccines, and bio-similars require sophisticated cleanroom environments, cold chain management, and batch-specific quality testing.

Maintaining aseptic conditions, HVAC, water systems (PW/WFI), and validation for multiple production lines significantly increases OPEX.

There is also an increased risk of batch failures, which can result in high rework or discard costs—especially in biologics.

Opportunities

Growing Demand in Uganda and EAC for Quality Medicines

The East African Community (EAC) is home to over 280 million people with growing health burdens like cancer, diabetes, infectious diseases, and vaccine-preventable illnesses.

There's high demand for affordable, high-quality generics, bio-similars, and essential vaccines.

Current reliance on imports from India, Europe, and China creates a gap for trusted local manufacturers with faster market responsiveness and cost advantages.

Support from Government and International Health Agencies

Uganda and other EAC governments are actively promoting local manufacturing through:

- Tax breaks, incentives, and infrastructure subsidies.
- Potential fast-track approvals for local production of priority medicines.

Global health organizations like Gavi, WHO, UNICEF, Global Fund, and African CDC are eager to support African-based production, especially for vaccines and pandemic preparedness.

This creates opportunities for donor-funded procurement, technology transfer, and concessional financing.

Regional Regulatory Harmonization (EAC-MRH)

The EAC's Medicines Regulatory Harmonization initiative reduces the complexity and time for market entry across multiple countries.

A product registered under EAC-MRH can be fast-tracked for approval in all EAC member states, expanding your potential market and improving return on investment (ROI).

First-Mover Advantage in Advanced Therapies

There are very few players in the region manufacturing oncology products, bio-similars, or vaccines. A well-executed entry into these segments provides:

- High-margin sales.
- Competitive edge due to regulatory barriers that deter others from entering.
- Leverage to license or co-develop products with international partners.

Greenfield Facility – Built for Future-Ready Compliance

Starting from scratch allows for:

- State-of-the-art modular facility design that meets WHO-GMP and global requirements from day one.
- Scalability—facilities can be expanded based on market growth or strategic partnerships.
- Integration of renewable energy systems, automation, which can lower long-term operating costs and improve environmental credentials.

Technology Transfer and Global Collaborations

Many mid-sized Western and Asian pharma companies are seeking CDMO or joint venture partners in Africa to enter emerging markets.

This facility, once validated and operational, can be positioned as a fill-finish hub or contract manufacturing site for such companies—especially for vaccines and biologics.

Threats

Regulatory and Compliance Challenges

Ensuring continuous compliance with WHO-GMP, NEMA (Ugandan environmental laws), NDA, and other international standards requires highly skilled QA/QC teams and constant oversight.

Geopolitical and Trade Instability

Political tensions, conflicts in neighbouring countries, or disruptions in trade routes could hamper logistics, exports, or sourcing of raw materials.

Currency Fluctuations and Import Dependency

Heavy reliance on imports for APIs and machinery subjects the plant to foreign exchange risks, especially if the Ugandan Shilling depreciates.

Intensifying Global Competition

Global pharmaceutical companies and Indian manufacturers could enter the region with cheaper generic drugs, impacting price competitiveness.

Product-Specific Challenges

Biosimilars and vaccines involve complex regulatory pathways, high R&D costs, and the risk of batch failures or cold-chain disruptions.

Construction & Commissioning Delays

Incomplete funding during the CAPEX phase can halt construction, delay procurement of equipment, and postpone validation activities.

Pharmaceutical projects are highly sequential: civil work must be completed before HVAC installation, which precedes equipment commissioning, which in turn precedes qualification. A break at any point stalls everything.

Regulatory Non-Compliance

If the facility is only partially built or validated, it cannot obtain GMP licenses or product approvals. This puts the company in a no-revenue, high-burn position, severely affecting investor confidence and credibility.

Supply Chain Penalties

Equipment vendors and raw material suppliers often work with milestone-based payments. Funding gaps can lead to order cancellations, higher renegotiated prices, or contractual penalties.

Lead times in pharma are long, so any reordering due to lapsed payments causes multi-month delays.

Reputational Damage & Loss of Stakeholder Confidence

Financial instability affects relationships with regulators, investors, banks, and potential customers (like international health agencies).

Once perception of financial distress sets in, it is harder to secure bridge financing or partnerships.

Strategic Recommendations

Hire pharma domain experts in key leadership roles: manufacturing, regulatory, quality assurance, and

R&D.

Form strategic alliances with experienced CDMOs or technology licensors for oncology and biosimilar lines.

Invest in workforce development through partnerships with local universities and international training programs.

Develop dual-source supply chain strategies and build regional sourcing capacity for packaging and excipients.

Design for modular expansion, allowing scaling based on product demand and market growth.

Pursue green certifications (LEED, ISO 14001) to attract ESG-aligned investors and NGOs.

M. SUMMARY AND RECOMMENDATIONS

This chapter synthesizes the key findings from the detailed feasibility study and outlines strategic recommendations to guide the implementation of the Sunshine Pharmaceutical Plant project in Uganda.

The analysis takes into account market dynamics, regulatory frameworks, technical and financial evaluations, supply chain logistics, and human resource considerations, all of which contribute to a comprehensive assessment of the project's viability.

Summary of Key Findings Market Opportunity and Demand

The study confirms that Uganda's pharmaceutical market is experiencing

robust growth driven by increasing demand for affordable, high-quality generics and specialized drugs, including oncology treatments and vaccines. With a projected market revenue reaching approximately US\$329 million in 2025 and a promising CAGR for both the domestic and broader East African markets, the opportunity to capture unmet demand is significant.

Regulatory and Technical Feasibility

Uganda's regulatory framework, led by the National Drug Authority (NDA), is increasingly aligned with international standards, including WHO-GMP guidelines. The regulatory environment supports local production through initiatives like "Buy Uganda, Build Uganda," while technical assessments underscore the availability of advanced manufacturing technologies. The proposed facility's phased approach—from Oral Solid Dosage forms to advanced biopharmaceutical and vaccine production—ensures scalability and adaptability to evolving market needs.

Financial Viability

The financial analysis demonstrates a strong investment proposition:

CAPEX: The total investment is estimated at US\$529 million, strategically distributed across various manufacturing units.

OPEX and Break-Even: Annual operational expenditures are projected at approximately US\$335 million, with break-even achieved within the first year due to efficient cost management and high contribution margins.

ROI and Profitability: Projected ROI stands at around 28% by Year 3, with an IRR estimated between 18-22%, indicating a healthy profit trajectory and the potential for cumulative returns exceeding 125% over five years.

Supply Chain and Human Resources

Robust supply chain strategies—including multi-sourcing of APIs, strategic partnerships for cold chain logistics, and digital monitoring systems—ensure resilience against market fluctuations and global disruptions.

Additionally, the availability of a young, growing workforce in Uganda supports the project's operational requirements, with competitive labour costs and targeted skills development across manufacturing, quality control, and regulatory compliance.

Recommendations

Prioritize High-Margin Segments Focus Areas:

Oncology Drugs: Given the rising burden of cancer in Uganda and East Africa, investing in the production of generic chemotherapy drugs can capture an underserved market. These drugs not only command higher margins but also improve patient access to critical treatments.

Biologics and Vaccines: The increasing demand for advanced therapies such as vaccines and biologics—including monoclonal antibodies and bio-similars—presents an opportunity to tap into segments with relatively lower competition and high value. The facility's phased approach, starting with Oral Solid Dosage forms and expanding into biopharmaceutical products, supports this focus.

Rationale:

Market Trends: The feasibility study highlights strong growth projections in these segments, driven by unmet healthcare needs and increasing government investments in healthcare infrastructure.

Financial Benefits: High-margin products contribute significantly to profitability and help accelerate the return on investment (ROI). By emphasizing these areas, the project can achieve robust financial returns and capture a larger share of the regional market.

Optimize API and Raw Material Sourcing Short-Term Strategy:

Leverage International Suppliers: Initially, the project should secure long-term contracts with reputable API suppliers in India and China. This will help stabilize costs and ensure a reliable supply chain during the early phases of operation.

Long-Term Strategy:

Local Production Initiatives: Explore opportunities for setting up local or regional API manufacturing facilities. This approach will not only reduce import dependency and associated logistics risks but also potentially lower production costs over time.

Diversification: Develop a multi-sourcing strategy to mitigate risks associated with geopolitical tensions, supply chain disruptions, or fluctuations in raw material prices.

Rationale:

Cost Efficiency: APIs represent a significant portion of production costs. Optimizing sourcing will directly impact overall operational efficiency and cost management.

Risk Mitigation: Diversification and local production can shield the project from external market shocks and ensure continuity of operations.

Invest in Advanced Technologies Technology Implementation:

Automation and Digitalization: Adoption of automated production lines, Manufacturing Execution Systems (MES), and IoT-based monitoring systems can streamline operations, reduce labour costs, and enhance precision.

Quality Control Systems: Investment in advanced analytical testing equipment will support stringent quality assurance protocols. Digital systems also facilitate real-time data monitoring, which is critical for maintaining compliance with WHO-GMP standards.

Rationale:

Operational Efficiency: Cutting-edge technology reduces the risk of human error, increases production speed, and improves overall product consistency.

Regulatory Compliance: High-tech systems ensure that manufacturing processes meet international standards, making it easier to secure regulatory approvals and expand into export markets.

Scalability: Advanced technologies support a phased production approach, allowing the facility to scale operations smoothly as market demand grows.

Enhance Strategic Collaborations Collaboration Models:

Public-Private Partnerships (PPPs): Engage with government agencies and international health organizations to secure financial incentives, regulatory support, and access to infrastructure such as Special Economic Zones (SEZs).

Industry Alliances: Form strategic alliances with global pharmaceutical companies for technology transfer, joint ventures, or co-investment. This can accelerate the learning curve and enhance competitive positioning.

Local Partnerships: Collaborate with local distributors, healthcare providers, and academic institutions to strengthen market access and foster innovation.

Rationale:

Regulatory and Financial Support: Strategic partnerships can reduce upfront investment risks and expedite regulatory approvals.

Market Penetration: Collaborations with established players can facilitate faster market entry, improve distribution networks, and expand customer reach.

Innovation: Joint ventures and alliances can bring in new technological insights and operational expertise, enhancing the project's long-term viability.

Strengthen Supply Chain Resilience Key Measures:

Robust Logistics Network: Develop an integrated logistics system that includes reliable transportation, warehousing, and cold chain infrastructure. This is crucial for managing both raw materials and finished products, especially for temperature-sensitive biologics and vaccines.

Digital Supply Chain Solutions: Implement demand forecasting and real-time inventory management systems to optimize supply chain operations and minimize waste.

Contingency Planning: Establish dual sourcing strategies and maintain buffer stocks to mitigate the impact of any supply disruptions.

Rationale:

Efficiency and Reliability: A resilient supply chain ensures that production processes are not interrupted by external factors, which is critical for meeting market demand consistently.

Cost Management: Effective supply chain management reduces logistics costs and improves overall operational margins.

Risk Mitigation: Proactive risk management through diversification and advanced monitoring systems will safeguard the project against unforeseen disruptions.

Focus on Workforce Development Workforce Strategy:

Training Programs: Invest in specialized training and capacity-building initiatives to develop the technical, regulatory, and operational skills needed across various departments such as manufacturing, quality control, and engineering.

Talent Acquisition: Attract skilled professionals by offering competitive remuneration packages and career development opportunities. This includes hiring experts with experience in high-containment manufacturing, digital automation, and international regulatory compliance.

Collaborations with Educational Institutions: Forge partnerships with local universities and technical institutes to create a pipeline of skilled labour tailored to the project's needs.

Rationale:

Operational Excellence: A well-trained workforce is essential to maintain high standards of product quality, regulatory compliance, and overall operational efficiency.

Sustainability: Continuous professional development not only improves immediate operational outcomes but also contributes to long-term sustainability by fostering innovation and adaptability.

Competitive Advantage: Investing in human capital builds organizational strength and ensures that the facility remains competitive in an evolving market landscape.

Conclusion

The comprehensive feasibility study strongly supports the viability of the Uganda Pharma Plant project. With favourable market conditions, a supportive regulatory framework, robust financial projections, and effective risk mitigation strategies, the project is not only feasible but poised for success. Strategic investments in technology, local partnerships, and human capital will further enhance the plant's competitive edge and contribute to sustainable economic growth in Uganda and the broader East African region.

In light of these findings, it is recommended that the project proceed as planned, with careful attention to the outlined recommendations to ensure operational excellence and long-term profitability.

List of Abbreviations

ADE : Adverse Drug Event ADR : Adverse Drug Reaction

AEFI : Adverse Events Following Immunization AI : Artificial Intelligence

API : Active Pharmaceutical Ingredients

ARIPO : African Regional Intellectual Property Organization BB : Biobridge

BCG : Bacillus Calmette-Guérin BEP : Break Even Point

BMR : Batch Manufacturing Record BOPV : Bivalent Oral Polio Vaccine BPR : Batch Production Record

CAGR : Compound Annual Growth Rate

CAPA : Corrective and Preventive Action CAPEX : Capital Expenditure

CEO : Chief Executive Officer

cGMP : Current Good Manufacturing Practices CIP : Clean-in-Place

COMESA : Common Market for Eastern and Southern Africa CT : Clinical Trial

EAC : East African Community EAC : East African Council

ECG : Electrocardiogram

EPI : Expanded program for immunization ETP : Effluent treatment plants

EU : European Union FEFO : First-Expiry-First-Out FR : Feasibility Report

GAVI : Global Vaccine Alliance GC : Gas Chromatography GCP : Good Clinical Practice

GDP : Good Distribution Practices GLP : Good Lab Practices

GMP : Good Manufacturing Practices HDPE : High-Density Polyethylene

HEPA : High-Efficiency Particulate Air

HPLC : High Performance Liquid Chromatography HPV : Human papillomavirus

HR : Human Resource

HVAC : Heating, Ventilation, and Air Conditioning ICH : International Council for Harmonisation IPV : Inactivated Polio Vaccine

IQ : Installation Qualification

ISO : International Organisation for Standardisation JIT : Just In Time

LED : Light-emitting diode

MA : Marketing Authorization mAbs : Monoclonal Antibodies

MES : Manufacturing Execution Systems mRNA : messenger Ribonucleic Acid

NCD : Non-Communicable Diseases NDA : National Drug Authority

NDPA : National drug policy & Authority

NEMA : National Environment Management Authority NGO : Non-governmental organization

NPC : National Pharmacovigilance Centre OPEX : Operational Expenditure

OQ : Operational Qualification

OSD : Oral Solid Dosage

PCV : Pneumococcal Conjugate Vaccine

PIC/S : Pharmaceutical Inspection Convention Scheme PoE : Point of entry

PPB : Pharmacy and Poisons Board PPP : Public-Private Partnership PQ : Performance Qualification

QA : Quality Assurance

QC : Quality Control

QPPV : Qualified Person for Pharmacovigilance R & D : Research and Development
RABS : Restricted Access Barrier Systems REV : Revision
RO : Reverse Osmosis
ROI : Return on Investment
RSV : Respiratory Syncytial virus
SCADA : Supervisory Control and Data Acquisition SEZ : Special Economic Zones
SIP : Sterilize-in-Place
SQM : Square Meter
TFDA : Tanzania Food and Drugs Authority TRS : Technical Report Series
UAE : United Arab Emirates
UCI : Uganda Cancer Institute UNICEF : United Nations Children's Fund UPS : Uninterruptible Power Supply
USAID : United States Agency for International Development USD : United States Dollar
UV : Ultra Violet
VFD : Variable Frequency Drive WFI : Water for Injection
WHO : World Health Organization WTP : Water Treatment Plant ZLD : Zero Liquid Discharge

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 Countries | Burundi | Kenya | Rwanda | Tanzania | Uganda
 Market size (USD Million) | 75 | 740 | 75 | 400 | 450
 CAGR20007- 2014 | 13% | 15% | 16% | 9% | 9%
 Market Share of Locally Produced Pharma Drugs(% of overall market) | 3% | 30% | < 1% | 12% | 20%
 Market By Segment | 56%
 Generic,44% Brander | 62%
 Generic,38% Branded | 54%
 Generic,46% Branded | 54%
 Generic, 46%
 Branded | 80%
 Generic, 20%
 Branded
 Type | Vaccine | Uganda Requirement (MD)
 Bacterial | Pentavalent | 6
 Bacterial | PCV10 | 6
 Bacterial | BCG | 2
 Live Virus | Measles Vaccine | 6
 Live Virus | BOPV | 7
 Live Virus | BOPV (Ty1and Ty3) | 7
 Inactivate d Viral | Rota | 4
 Inactivate d Viral | IPV | 4
 Inactivate d Viral | HPV (1,2,3&4) | 2
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 Title | Document Number
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 Guidelines on variation of registered bio- therapeutic products. | PAR/GDL/033 Rev .01
 Guidelines on submission of documentation for registration of a pharmaceutical product for human use. | PAR/GDL/004 Rev .03
 Guidelines for the provision of insurance cover for research participants in clinical trials in Uganda. | DPS/GDL/023 Rev.00
 Guidelines for introducing a locally manufactured new pharmaceutical product on | INS/GDL/004 Rev.01 the Uganda market. |
 Guidelines on detecting and reporting adverse drug reactions in Uganda. | DPS/GDL/013 FRev.01
 Guidelines on submission of documentation for renewal of registration of a pharmaceutical product for human use. | PAR/GDL/020 Rev.00
 Guidelines on Good Distribution Practice for Pharmaceutical Products. | INS/GDL/002 Rev.01
 Guidelines on submitting periodic safety update report and any other reports that may be relevant to determine the safety, efficacy and quality of a drug. | DPS/GDL/034 Rev.00
 Guidelines for the Recall or Withdrawal of a Medical Product. | INS/DGL/037 Rev.00
 Guidelines on variation to registered Human vaccines. | PAR/GDL/018Rev .01
 National AEFI surveillance guidelines Uganda. | -
 Regulatory
 Area | Action | Timeline (Working days)
 Regulatory Inspection | GMP Report assessment to Manufacture after inspection | 45 days
 Regulatory Inspection | Feedback after inspection- Domestic | 28 days
 Regulatory Inspection | Regulatory decision after CAPA – Domestic (Local Facilities) | 17 days
 Documentation | Submission and approval of Dossier | 220 days
 Total no. of days | Total no. of days | 310 days
 Regulatory Area | Action | Timeline (Working days)
 Clinical Trial (CT) Oversight | Receipt, Screening and acknowledgment of a Clinical trial application (new, renewals and amendment) | 10 days
 Clinical Trial (CT) Oversight | Regulatory decision on a Clinical trial application | 50 days

Clinical Trial (CT) Oversight | Annual renewal of Ongoing trials | 20 days
 Clinical Trial (CT) Oversight | Amendment of CT authorization | 20 days
 Clinical Trial (CT) Oversight | Feedback report to a client following a GCP Inspection | 45 days
 Marketing Authorization (MA) | Assessment of application and request for additional information on application for MA (Human Drugs) | 18 months (396 days)
 Marketing Authorization (MA) | Assessment of application and request for additional information on application for MA (Vaccines, Anti-cancer medicines and other vital (medicines)). | 6 months (132 days)
 Marketing Authorization (MA) | Regulatory decision on MA after additional information is received | 3 months (66 days)
 | Special Categories |
 | Regulatory decisions on products already authorized for marketing by stringent regulatory authorities, WHO prequalified products and those approved under article 58 of EU regulations | 6 months (132 days)
 | Variation and Notification |
 | Regulatory decision on minor variation | 4 months (88 days)
 | Regulatory decision on Major variation | 6 months (132 days)
 | Regulatory decision on annual or immediate notification change | 80 days
 | Drug Register |
 | Publication of drug register | Monthly
 Regulatory Inspection | GMP Physical inspection after receipt of application- Foreign | 180 days
 Regulatory Inspection | GMP Report assessment to Manufacture after inspection | 45 days
 Regulatory Inspection | Feedback after inspection- Domestic | 28 days
 Regulatory Inspection | Regulatory decision after CAPA – Foreign | 20 days
 Regulatory Inspection | Regulatory decision after CAPA – Domestic (Local Facilities) | 17 days
 Premise Licensing | Feedback to the applicant on the regulatory decision for pharmacy renewal | 40 days
 Premise Licensing | Feedback to the applicant on the regulatory decision for licensing of new pharmacy applicants | 30 days
 Market Surveillance and Control | Verification of imports (registered drugs) | 2 days
 Market Surveillance and Control | Clearance of Imports at ports of entry | 2 days
 | Verification of imports (un- registered drugs) | 10 days
 | Regulatory decision on drug promotional material | 10 days
 | Publication of the database of approved promotional materials/advert | Monthly (5th of the month)
 Vigilance | Acknowledgement of receipt of an ADE report (mail/letter) | 5 days
 Vigilance | Feedback on serious ADE reports | 18 days
 Vigilance | Feedback on serious AEFI reports to the expanded program for immunization (EPI) | 14 days
 Laboratory Testing | Test results from mandatory testing after sampling (Medicine)- PoE | 48 Days
 Laboratory Testing | Test results from client requests after acceptance by the lab (All Products) | 45 Days
 Laboratory Testing | Test results for pre-market (Domestics Manufactures) | 60 Days
 Laboratory Testing | Test results for post-market surveillance (Medicines) | 48 Days
 National Regulatory System | Issuance of recall or alert for a substandard or falsified medicine or health care product | 14 Days
 National Regulatory System | Acknowledgement of receipt of a product complaint | 5 Days
 National Regulatory System | Feedback on a market (process, product, other) complaint | 20 Days
 National Regulatory System | Feedback on the effectiveness of recall after receipt of the recall report from the LTR (Local Technical Representative) | 10 Days
 National Regulatory System | Feedback on test results from the Lab to the client | 5 Days
 Support to Regulatory areas | QUALITY MANAGEMENT SYSTEM |
 Support to Regulatory areas | Acknowledgement of receipt of a service-related complaint | 2 Days
 Support to Regulatory areas | Feedback on service-related complaints | 21 Days
 Therapeutic Area | Regulatory Focus | Key EAC Harmonization Mechanisms | Challenges | Progress & Opportunities
 Diabetes | Insulin and oral hypoglycaemic Bioequivalence and cold chain Chronic disease management alignment | Common Technical Document (CTD) Mutual Recognition Procedure (MRP) | Cold chain logistics

Limited local insulin production | Streamlined approvals

Regional demand driving local production
 Malaria | Fast-tracking of ACTs Monitoring of resistance Alignment with WHO PQ | Joint assessments

Regional malaria drug guidelines | Counterfeits Dependency on donor programs | Strong harmonization Integrated procurement systems
 Antibiotics | Quality and safety evaluations GMP compliance AMR mitigation | AMR frameworks under WHO/Africa CDC

Regional pharmacovigilance | Overuse and poor prescription practices - Limited surveillance | Regulatory updates in progress

Enhanced AMR monitoring
 Vaccines | High scrutiny: quality, safety, cold chain - Emergency use approval mechanisms | AVAREF support

Reliance on WHO PQ and joint inspections | Heavy import reliance

Capacity gaps for biologics | Regional capability strengthening

Ongoing vaccine regulation training
 Biologics | Advanced assessments for mAbs, biosimilars, insulin analogs - GMP and comparability requirements | International partnerships for capacity building - Involvement of WHO, UNICEF | Limited

local evaluation experience

High assessment complexity | Capacity building underway

Future potential for local

biologics production

HIV/AIDS | ARV quality and resistance

monitoring - | Fast-track review mechanisms -

Reliance on global | Stockouts and supply

chain | Increasing local registration

capabilities

Therapeutic Area | Regulatory Focus | Key EAC Harmonization Mechanisms | Challenges | Progress & Opportunities

| Integration with national HIV programs - Rapid access to WHO

PQ ARVs | procurement systems (e.g., Global Fund) | challenges - Limited local production | Supportive

policy for ARV production

Tuberculosis (TB) | MDR-TB drugs evaluation

Diagnostic technologies

Fixed-dose combination approvals | WHO TB drug guidelines

Regional efforts to reduce import timelines | Complexity of MDR/XDR TB treatment

Dependence on donor supply | Regional labs and regulatory harmonization improving Potential for local packaging and supply

S. No | DESIGNATED AREA | APPROX AREA SQM

1 | CENTRAL WAREHOUSE FOR RAW MATERIAL, PRIMARY, SECONDARY AND TERTIARY PACKAGING MATERIAL, SOLVENTS, ENGINEERING SPARES, COLD STORAGE, FINISHED GOODS

STORAGE | 30,000

2 | ENGINEERING BLOCK – HVAC AND

AHU, WATER PURIFICATION, PW AND WFI SYSTEM | 15,000

3 | CENTRAL UTILITY BLOCK | 6,000

4 | EFFLUENT TREATMENT PLANT | 5,000

5 | SCRAP YARD | 500

6 | PRIMARY WATER TREATMENT AREA | 1,000

7 | ADMINISTRATION BLOCK (INCLUDES FINANCE, ACCOUNT, HR, SUPPLY CHAIN ETC.), CORPORATE OFFICES, IT AND DATA SECURITY, MEETING ROOMS AND CONFERENCE HALLS, DOCUMENT CONTROL, RECEPTION AND VISITOR

LOUNGE | 13,000

8 | SECURITY BLOCK – CONTROL ROOM, FIRE SAFETY AND EMERGENCY CONTROL ROOM, EMPLOYEE SCREENING AND ACCESS CONTROL

AREA, | 3,500

9 | CAR PARK | 4,000

10 | CAFETERIA | 4,000

11 | TRAINING AND INDUCTION ROOMS | 1,500

12 | MEDICAL AND FIRST AID ROOMS | 500

13 | RECREATIONAL AREA FOR EMPLOYEES | 2,000

10 | QUALITY CONTROL LABORATORY | 4000

11 | RESEARCH AND DEVELOPMENT LABORATORY | 2000

12 | QUALITY ASSURANCE SECTION | 2000

13 | REGULATORY AFFAIRS DEPARTMENT | 1000

14 | CLINICAL RESEARCH DEPARTMENT | 500

| TOTAL AREA | 95,500 SQM

Section | Size

Required (sq. m) | Function

Granulation Area | 3,500 | Wet/dry granulation for tablet and capsule production

Blending & Compression | 2,500 | Mixing active ingredients and excipients

Tablet & Capsule Filling | 3,000 | Tablet compression and capsule filling

Coating | 2,000 | Film/sugar coating for tablets

Liquid Manufacturing | 3,500 | Syrup preparation, mixing, and holding tanks

Filling & Packaging | 5,000 | High-speed blister packing, bottling, and labelling

Utilities & HVAC | 2,500 | Power, water, and air handling units

Support Areas (Offices, Change Rooms, Cafeteria,

etc.) | 2,500 | Administrative functions, employee welfare, and

cleanroom airlocks

TOTAL AREA | 24,500 SQM |

Section | Estimated Area (sq. m) | Function

Raw Material & Component Storage | 2,500 | Storage of vials, stoppers, API, excipients, and

packaging materials

Buffer preparation | 1,500 | Preparation of buffers

Formulation Suite | 2,000 | Mixing and formulation of mAbs bulk solution

Aseptic Filling & Stoppering | 2,500 | High-speed filling, capping, and stoppering under ISO 5 conditions

Lyophilization (if required) | 2,000 | Freeze-drying for extended shelf-life stability

Capping, Inspection & Labelling | 1,500 | Visual inspection, capping, and final labelling

Primary & Secondary Packaging | 2,000 | Packaging in cartons, serialization, and labelling

Cold Chain Storage & Stability Testing | 2,500 | Refrigerated and frozen storage at 2-8°C and -20°C for

mAbs stability

Utilities & HVAC Systems | 2,000 | Power supply, purified water (WFI), HVAC, and compressed

air

Support Areas (Offices,

Cleanrooms, Cafeteria, Change Rooms, etc.) | 2,500 | Administrative offices,

gowning areas, and personnel movement

TOTAL AREA | 21,000 SQM |

Section | Estimated Area (sq. m) | Function

Vial Washing & Depyrogenation | 2,000 | Cleaning and depyrogenation of vials before filling

Ampoule Washing & Sterilization | 1,500 | Automated washing and sterilization of ampoules

Pre-Filled Syringe Preparation | 2,000 | Syringe preparation and plunger assembly

Filling & Stoppering (Vials & Ampoules) | 3,000 | High-speed aseptic filling and stoppering under ISO 5

conditions

Filling & Sealing (Pre-Filled Syringes) | 2,500 | Filling and sealing of syringes under sterile

conditions

Support Areas (Gowning, Change Rooms, Offices, etc.) | 2,500 | Personnel movement, gowning, and administrative offices
TOTAL AREA | 22,000 SQM |

Raw Material Storage | 2,500 | Isolated storage for APIs, excipients, and packaging components

Blending & Compression	2,500	Blending of APIs and excipients, tablet compression
------------------------	-------	---

Coating (if required)	2,000	Tablet film/sugar coating with dust extraction
-----------------------	-------	--

Filling & Packaging | 4,000 | High-speed blister packing, bottling, and labelling

Utilities & HVAC Systems | 2.500 | High-efficiency air handling, WFI, compressed air

etc.) | 3,000 | GMP-compliant personnel movement and administrative areas

Section | Estimated

Vial Washing & Depyrogenation	2.000	Cleaning and depyrogenation of vials before filling
-------------------------------	-------	---

Lyophilization (if required)	2,500	Freeze-drying for vials requiring stability enhancement
------------------------------	-------	---

defective units

Utilities & HVAC Systems	2,000	Purified water (WFI), air
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Support Areas (Gowning, Change Rooms, Offices, etc.) | 2,500 | Personnel movement, gowning, and administrative offices

SOM I

wise area and cost | Manufacturing Block wise area and cost

No.	Block Name	Sub	Unit	Cost in USD
-----	------------	-----	------	-------------

2 | Plant 1 - OSD | Blending & Compression | 4 39 00 000

4. | Plant 1 - OSD | Coating | 4 39 00 000

6. | | Filling & Packaging |

9. | | Support Areas (O

10. | Plant 2 mAbs and Biosimilar | Buffer preparation | 8,25,00,000

11.	Plant 2 mAbs and Biosimilar Formulation Suite 8,25,00,000
12.	Plant 2 mAbs and Biosimilar Aseptic Filling & Stoppering 8

13.	Plant 2 mAbs and Biosimilar	Lyophilization (if required)	8,25,00,000
14.	Plant 3 mAbs and Biosimilar	Capping, Inspection & Labelling	8,25,00,000

16. | | Cold Chain Storage & Stability Testing |

17. || Utilities & HVAC Systems |

20. | Plant 3 Filling Lines | Ampoule Washing & Sterilization | 3

21.	Plant 3-Filling Lines	Pre-Filled Syringe Preparation	7,60,00,000
22.	Plant 3 Filling Lines	Filling & Stoppering Vials & Ampoules	7,60,00,000

23.	Plant 3-Filling Lines	Filling & Sealing (Pre-Filled Syringes)	7,600,000,000
24.	Plant 3-Filling Lines	Lyophilization (if required)	7,600,000,000

25. | Plant 3-Filling Lines | Capping & Inspection | 7,60,00,000

27.	Plant 3-Filling Lines	Utilities & HVAC Systems	7,60,00,000
28.	Plant 3-Filling Lines	Support Areas (Canning, Storage, Packaging)	1,00,00,000

29. | Plant 4- Oncology | Raw Material Staging and

30. | Plant 4- Oncology | Granulation Area | 9,62,50,000

32.	Plant 4- Oncology	Encapsulation Area	9,62,50,000
33.	Plant 4- Oncology	Coating (freezing)	2,33,50,000

34. | Plant 4- Oncology | Hormonal & Cytotoxic Segregation | 9,62,50,000

36. | Plant 4- Oncology | Waste Management & | 9,62,50,000

37. || Utilities & HVAC

39. | Plant 5- Vaccine | Vial Washing & Depyrogenation | 6,60,00,000

41. | Plant 5- Vaccine | Lyophilization (if required) | 6,60,00,000

43. | Plant 5- Vaccine | Labelling & Secondary Packaging |

45. | Plant 5- Vaccine | Support Areas (Gowning, Change Rooms, Office

47. | Common Blocks | Central warehouse | 9,94,72,000

49. | Common Blocks | Central utility block | 9,94,72,000

50. | Common Blocks | Administration block | 9,94,72,000
 51. | Common Blocks | Quality control laboratory | 9,94,72,000
 52. | Common Blocks | Research and development laboratory | 9,94,72,000
 53. | Common Blocks | Quality assurance section | 9,94,72,000
 54. | Miscellaneous blocks | Car park | 2,50,00,000
 55. | Miscellaneous blocks | Cafeteria | 2,50,00,000
 56. | Miscellaneous blocks | Training and induction rooms | 2,50,00,000
 57. | Miscellaneous blocks | Medical and first aid rooms | 2,50,00,000
 58. | Miscellaneous blocks | Recreational area for employees | 2,50,00,000
 59. | | Regulatory affairs department |
 60. | | Clinical research department |
 61. | | Landscaping and horticulture |
 62. | QUALITY ASSURANCE | FACILITY COMMISSIONING AND VALIDATION COSTS | 5,50,00,000
 63. | REGULATORY AFFAIRS | LICENCING AND STATUTORY COSTS | 2,75,00,000

|| TOTAL BUDGET ESTIMATE IN

USD | 57,16,22,000

|| | SAY 571 MUSD

Manufacturing Unit | Approx

Investment (MUSD) | Approx % of Total CAPEX

Oral Solid Dosage (OSD) | 43.9 | 7.68%

Biopharma Fill-Finish | 82.5 | 14.44%

Injectables (SVP) | 76.0 | 13.31%

Oncology (Hormonal & Cytotoxic Products) | 96.25 | 16.85%

Vaccine Fill-Finish | 66.0 | 11.55%

Common Infrastructure (Utilities, Warehousing, R&D, QA/QC, Admin, etc.) | 99.47 | 17.42%

Miscellaneous Blocks (Car park, cafeteria, Training room, Medical room, Regulatory

affairs, Clinical research, Landscaping etc., | 25.00 | 4.37%

Facility commissioning and qualification | 5.50 | 0.96%

Registration and licensing | 2.75 | 0.48%

Total Investment | 571

MUSD(571.6) | 100%

Cost Component | Annual Cost (MUSD) | % of Total OPEX

Raw Materials & APIs | 150(150.75)* | 45%

Labour & Workforce | 50(50.25)* | 15%

Utilities (Power, Water, Waste Management) | 40(40.20)* | 12%

Maintenance & Equipment Servicing | 25(26.80)* | 8%

Regulatory Compliance & Quality Assurance | 20(20.18)* | 6%

Logistics & Distribution | 30(30.15)* | 9%

Total OPEX (Annual) | \$335M USD | 100%

Year | Revenue (MUSD)

Year 1 | 5

Year 2 | 90

Year 3 | 150

Year 4 | 250

Year 5 | 450

Risk Factor | Potential Impact | Mitigation Strategy

Regulatory Delays | Slower product approvals | Early engagement with NDA (Uganda), WHO-GMP

API Price Volatility | Increased raw material costs | Multi-supplier sourcing, API localization

Market Competition | Lower pricing power | Differentiation through quality & innovation

Logistics & Distribution Challenges | Delayed market penetration | Strong partnerships with distributors

Cost Component

Raw Material & API Procurement

Packaging Material Costs

Warehousing & Cold Storage

Manufacturing & Production Logistics

Quality Control & Compliance

Cold Chain Infrastructure (Vaccines & Biologics)

Distribution & Last-Mile Delivery

IT, AI, and Digital Supply Chain Systems

Risk Management & Contingency Planning

Regulatory Compliance & Audits

Plant/Section | Number of Personnel

Oral Solid Dosage (OSD) | 80

Biopharma Drug Product | 80

Vaccine Formulation | 70

Oncology | 50

Injectable | 70

Warehouse | 40

Quality Control | 62

Quality Assurance | 43

Engineering | 80

HR and Admin | 65

Total | 640

Risk Factor | Potential Impact

Regulatory Uncertainty | Changes in pharmaceutical laws, taxation, or

licensing requirements could delay operations or increase compliance costs.

Government Policy Shifts | Import/export restrictions, price controls, or changes in incentives may

affect profitability.

Bureaucratic Delays | Lengthy approvals for manufacturing licenses,

product registrations, and environmental permits may slow down market entry.

Geopolitical Instability in EAC Region | Conflicts or tensions in neighbouring East African Community

(EAC) countries could disrupt supply

chains and exports.

Intellectual Property & Patent Protection | Weak enforcement of patents and trademarks may lead to

counterfeiting risks.

Risk Factor | Potential Impact

Currency Fluctuations | Exchange rate volatility can increase the cost of imported raw materials (APIs,

excipients, packaging

materials).

Inflation & Rising Operational Costs | High inflation may lead to increased costs for labor, utilities,

and raw materials.

Supply Chain Disruptions | Dependence on international suppliers for APIs and critical components could

lead to shortages.

Limited Access to Affordable Financing | High-interest rates and limited government incentives can make project financing costly.

Market Demand & Price Sensitivity | Price-sensitive consumers and limited insurance coverage may affect product affordability.

Risk Factor | Potential Impact

Water Scarcity | High water demand for manufacturing and cooling may lead to shortages, increasing operational costs.

Waste Disposal Challenges | Improper disposal of hazardous waste (solvents, API residues) can lead to environmental penalties.

Energy Dependence on Non-Renewables | High electricity costs and unreliable grid supply may lead to production disruptions.

Climate Change Effects | Increased heat and erratic rainfall could impact cooling systems and energy consumption.

Regulatory Non- Compliance | Failure to meet Uganda’s National Environment Management Authority (NEMA) regulations can lead to legal actions.

Risk Category | Key Risks | Mitigation Strategies

Political Risks | Regulatory uncertainty, bureaucracy, policy shifts | Engage early with regulators, form public-private partnerships, diversify export markets

Economic Risks | Currency fluctuations, inflation, financing challenges | Hedge currency risks, optimize operational costs, diversify revenue streams

Environmental Risks | Water scarcity, waste disposal, high energy costs | Implement sustainable waste management, use renewable energy, optimize water usage

Phase | Focus | Timeframe | Revenue Stream | Impact on ROI

Phase 1 | OSD + Syrups | Year 1–2 | Tender sales, private hospitals | Early cash flow

Phase 2 | Oncology + Injectables | Year 2–3 | High-margin products | Boost margin

Phase 3 | Vaccines + Biosimilars | Year 3–5 | Institutional procurement | Long-term revenue

Cross- Cutting | CDMO + Export | Year 1–5 | Capacity monetization | Parallel revenue

Segment | CAPEX (USD M) | Revenue (Annual, USD M) | OPEX (Annual, USD M)

OSD & Syrups | 43.9 | 45 | 15

Oncology | 96.25 | 90 | 30

Injectables | 76 | 60 | 25

Vaccines | 66 | 80 | 35

Biopharma | 82.5 | 100 | 40

Common blocks | 112 | - | 20

Miscellaneous blocks | 25 | - | 12

Facility commissioning | 55 | - | -

Licensing costs | 27.5 | - | -

Total | 571 | 375 | 165

Year | Revenue (USD M) | OPEX (USD M) | Profit (USD M) | Cumulative Cash Flow (USD M)

1 | 5 | 20 | - | -

2 | 90 | 80 | 60 | 60

3 | 150 | 165 | 98 | 158

4 | 250 | 165 | 165 | 323

5 | 450 | 165 | 298 | 621

6 | 500 | 165 | 325 | 946

7 | 500 | 165 | 325 | 1271

8 | 500 | 165 | 325 | 1596

9 | 500 | 165 | 325 | 1921

10 | 500 | 165 | 325 | 2246

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BIOBRIDGE HEALTHCARE SOLUTIONS PVT. LTD

FEASIBILITY REPORT
FOR INTEGRATED
PHARMA AND
BIOPHARMACEUTICAL
PROJECT FOR UGANDA

Prepared for:
SUNSHINE MEDICAL PTE LTD
UNIT 06-10, UB ONE BUILDING,
UB AVENUE 4,
Singapore – 408 830

Prepared by:
BIOBRIDGE HEALTHCARE SOLUTIONS PVT. LTD
13, RACHANA BLOSSOM, AUNDH
PUNE 411007
INDIA

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BIOBRIDGE HEALTHCARE SOLUTIONS PVT. LTD

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BIOBRIDGE HEALTHCARE

HEALTHCARE SOLUTIONSPVT.

SOLUTIONS PVT. LTD

LTD

1. EXECUTIVE SUMMARY

Introduction:

The objective of this project is to establish and operate a state-of-the-art green field multi-Products Biopharmaceutical manufacturing plant in Republic of Uganda and to address and transform health care needs of East African council (EAC) by addressing critical health care challenges and fostering local manufacturing.

This facility aims to consistently and efficiently produce high-quality pharmaceutical, Biopharmaceuticals & vaccines in a phased manner

Objective:

The objective of this report is to study comprehensively the feasibility and to assess the potential for establishing a sustainable and efficient pharmaceutical manufacturing plant in Uganda. BioBridge is requested to provide a proposal for feasibility study services, which include:

- Market analysis
- Regulatory requirements
- Site selection and infrastructure
- Technical feasibility
- Financial feasibility
- Supply chain and raw material sourcing
- Human resources
- Risk assessment
- Sustainability

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Goals

Health Care Access: To enhance Healthcare access by manufacturing high quality pharmaceuticals, Biologics and vaccines locally thereby reducing the dependency on imported medicines and biologics, ensuring cost effective treatment to masses.

Health Care needs: To address regional health care needs by manufacturing medicinal products which are required to treat most prevalent diseases in the region-Malaria, Tuberculosis, AIDS.

Economic growth: To foster Economic growth by providing employment opportunity to local population and attracting global investors and venture capitalists.

Sustainability: To establish a sustainable pharma base in Uganda which is reliable and self-sufficient.

Supply Chain Reliability: Building a secure and efficient supply chain to source raw materials, transport finished products, and ensure timely deliveries to end users.

Stakeholder Satisfaction: Ensuring customer satisfaction by providing consistent and reliable products while maintaining a favourable relationship with employees, investors, and the local community.

Solution:

To realise the above said goals and to address the critical health care needs of Uganda, Biopharmaceutical manufacturing facility shall be established in the phased manner.

In the first phase Oral Solid Dosage (OSD) form shall be established by integrate state-of-the-art manufacturing technologies to ensure efficiency, quality, and scalability of wide range of high-demand solid dosage forms, including tablets, capsules, syrups, and advanced technologies such as soft gel capsules.

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In the phase -2, The Phase-2 General Injectables facility will focus on manufacturing sterile pharmaceutical injectable-grade products, targeting key therapeutic segments. Injectables facility shall be designed to meet the highest global standards for sterile manufacturing while maintaining cost efficiency and scalability

In the Phase-3, Biopharma Products facility focuses exclusively on the formulation and fill- finish of advanced biopharmaceutical products. This facility shall be commissioned to address the increasing demand for biologics, including Monoclonal antibodies(mabs), Hormones, Insulin, analogs, and other therapeutic proteins. This facility shall be designed to ensure contamination controls, precision, and compliance with global standards for biopharmaceutical fill-finish processes.

The Phase-4, facility for formulation and fill-finish of critical vaccines to address major global and regional health challenges. This facility shall be designed to cater to both routine immunization programs and outbreak vaccines.

East Africa, like the rest of Africa, faces a growing demand for vaccines, particularly as it works to strengthen its vaccine supply chain and boost domestic capabilities to meet the needs of its population.

In the 5th and the final phase, Oncology OSD facility shall be installed to address the needs of anticancer drugs in the region.

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2.0 DETAILED
FEASIBILITY
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a) Market Analysis:
i) Demand for pharmaceutical products in Uganda

Uganda's pharmaceutical industry is a key component of the country's healthcare system, with demand steadily increasing due to population growth, rising disease burdens, and government initiatives to improve healthcare access.

The disease burden in Uganda is similar to other less developed African countries, with communicable diseases accounting for roughly 67% of the total burden.

Uganda's communicable disease burden is dominated by infectious and parasitic diseases. Malaria is considered as the major cause of deaths in the country.

The most prevalent causes of mortality and morbidity in Uganda are

Malaria, Pneumonia, Tuberculosis, Anaemia, Hypertension, Diabetes and Cancer.

Fig No.: 1

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Fig. No.:2

Trends of Cancer Patients in Uganda:

Fig.No.:3

The cancer burden by type of cancer diagnosed among patients

MARKET VALUE:

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• In Uganda, the revenue in the pharmaceuticals market is estimated to reach US\$329.32m in 2025.

• The largest market within this industry is Oncology Drugs, which is projected to have a market volume of US\$50.03m in 2025.

• The revenue is expected to exhibit an annual growth rate (CAGR 2025-2029) of 3.06%, resulting in a market volume of US\$371.58m by 2029.

• Uganda's pharmaceutical market is experiencing a surge in demand for affordable generic medications.

• The East Africa generic drug market reached USD 2.4 billion in 2024 and is expected to exhibit a CAGR of 6.72% during 2025-2033.

Competitive Landscape:

East Africa Pharma manufacturing- An Overview

• Around 35 active manufacturers (members of FEAPM); 2 facilities with WHO cGMP approval and pre-qualified products

• Primarily compounding and formulation of most dosage forms including infusions

• Almost all raw materials imported

• High dependency on imported finished pharmaceutical products

• Focus on WHO list of essential medicines

• No API manufacturing

• Increasingly well-regulated (NDA, PPB, TFDA)

• Trade is mostly intra-East Africa

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Companies currently having manufacturing base in Uganda

1. CIPLA Quality Chemicals Industries Limited, joint venture between a local and international pharmaceutical company, is certified cGMP facilities manufacturing Anti-retro viral, anti-malarial and hepatitis B drugs, exports to Sub-Saharan countries and supplies drugs to the Global Fund

2. Kampala Pharmaceutical Industries (1996) Limited, local manufacturer and an affiliate of the Aga Khan development network. Product range includes analgesics, anti-asthma, anti-diabetic, anti-malarial, anti-parasites, antibiotics, cardiovascular drugs, cold remedies & Nutraceuticals

3. Beta Health Care Uganda Limited, an affiliate of Aspen Group. Exports its products to Tanzania, Rwanda, Burundi, Ethiopia, Sudan, Somalia, Mauritius and West Africa

Key players in the market are actively implementing various strategies to expand their reach and market share by:

• Enhancing their production capabilities, both in terms of volume and the range of products.

• Investments in advanced manufacturing facilities to increase output and meet the rising demand for generic drugs.

- Creating more efficient supply chains and distribution networks to ensure wider availability of their products. They are reaching out to remote areas and partnering with local distributors to improve access.
- Forming alliances with local manufacturers, international partners, and healthcare providers to extend their reach and enhance their market position.
- Technology transfers, knowledge and technology transfer, through Joint ventures, Alliances thereby strengthening local capacities

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Key Pharma Sector Data Estimate (Market share, Generic, Branded pharmaceuticals)

Table No.: 1

Countries Burundi Kenya Rwanda Tanzania Uganda

Market size (USD 75 740 75 400 450 Million

CAGR20007- 13% 15% 16% 9% 9% 2014

Market Share of 3% 30% < 1% 12% 20%
Locally Produced
Pharma Drugs(%
of overall market)

Market By 56% 62% 54% 54% 80%
Segment Generic,44% Generic,38% Generic,46% Generic, Generic,
Brander Branded Branded 46% 20%
Branded Branded

Source 2014 Estimation (Industry and Association)

Demand for Pharmaceutical Products in Uganda

The demand for pharmaceutical products in Uganda is driven by various factors, including an increasing prevalence of both communicable and non-communicable diseases (NCDs), an expanding healthcare infrastructure, and growing government support for local production.

Below is an analysis of key therapeutic areas:

a) Oral Solid Dosage Forms (Tablets and Capsules)

- Oral solid dosage forms constitute the majority of prescription medications in Uganda, covering a range of treatments from antibiotics to anti-hypertensives.

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- Uganda has a few established pharmaceutical manufacturers producing these drugs.
- Despite local manufacturing efforts, a large portion of Uganda's medicine supply is still imported due to insufficient production capacity and the need for diverse active pharmaceutical ingredients (APIs).
- Expanding local production capabilities, especially for essential generics, could help meet growing demand and reduce reliance on imports.

b) Oncology Treatments

- Cancer cases in Uganda have been rising significantly, with estimates from the Uganda Cancer Institute (UCI) indicating an annual increase in new cases (Refer to Fig Nos.2&3.).
- Currently, oncology drugs are almost entirely imported, making treatment expensive and often inaccessible to many patients.
- There is an opportunity for local pharmaceutical manufacturers to invest in cancer drug production, particularly generic chemotherapy drugs, which could lower costs and improve accessibility and health care.
- The Ugandan government and international health organizations are increasingly investing in cancer treatment centres, further boosting the demand for oncology drugs

c) Vaccines

- Uganda has a demand for vaccines (approx. 1.7 million doses per annum for essential vaccines) due to childhood immunization programs, seasonal disease outbreaks, and global initiatives like the Global Vaccine Alliance (GAVI).

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- The country currently relies on imports for nearly all its vaccines, which poses challenges such as supply chain disruptions and high procurement costs.
- Government efforts to establish vaccine production facilities could significantly boost the local pharmaceutical industry and create self-sufficiency in vaccine supply.

d) Monoclonal Antibodies (mAbs)

- Monoclonal antibodies are used in cancer treatment, autoimmune diseases, and infectious diseases like COVID-19.
- Uganda currently lacks the infrastructure for manufacturing biologic drugs & mAbs, entirely dependent on imports.

Drivers of Pharma market:

- Rapid expansion of local pharmaceutical manufacturing capacities.
- East Africa is experiencing a significant boost in its pharmaceutical

production driven by a combination of increased domestic and international investment and initiatives, improved technological capabilities, and a strategic push for self-sufficiency in pharmaceutical supply. Furthermore, there is a rising awareness among healthcare professionals and patients about the comparable efficacy of generic drugs to brand-name counterparts.

- Generic drug manufacturers are significantly strengthening their distribution networks to improve product reach. This involves the development of efficient supply chain strategies and logistics networks to ensure generic drugs are accessible even in remote regions. This expanded distribution is further facilitated by technological advancements, such as digital platforms and delivery

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systems, enabling broader dissemination of generic drugs across the region, fostering overall the market growth.

Favourable Government Initiatives

Governments across East Africa are progressively acknowledging the role of generic drugs in achieving healthcare affordability and accessibility. They are implementing regulatory reforms to simplify the approval process, thereby encouraging their use by streamlining this process, governments are reducing barriers to entry, making it easier for generic manufacturers to bring their products to market.

Moreover, there is an increase in governmental efforts to boost public awareness about the efficacy and affordability of generic drugs. Despite their lower cost, generic drugs are equally effective as their branded counterparts.

EAC Pharma – Opportunities

- Large unmet need for essential medicines
- Significant population growth anticipated
- Double disease burden (growth of chronic diseases):
- NCDs are predicted to account for 40% of illnesses & for 42% of deaths (5.1 million) by 2030 (WHO Global Health Estimates, 2013)
- Growing middle class willing to pay for better treatment
- New models of care being designed (e.g. PPP, medical insurance, etc.)
- Existing pharmaceutical production base & enabling regulatory environment
- African traditional and herbal medicines as a market niche
- No dominant players
- Political will and action

EAC Pharma – Challenges

- Negative perceptions of locally manufactured products

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- Lack of capacity to manufacture advanced formulations due to a skills gap on product development and formulation expertise
- High cost of borrowing (20-30%)

- High cost of operations e.g. finance, utilities, infrastructure
- Lack of market data
- Poor and unreliable infrastructure (roads, utilities, etc)
- Under resourced and weak institutions (Civil society, Regulatory agencies, etc)

Vaccines Market Analysis:

Immunization is a proven tool for controlling and eliminating life-threatening infectious diseases and is estimated to avert millions of deaths each year. Universal immunization of children against eight vaccine-preventable diseases [tuberculosis, diphtheria, whooping cough (pertussis), tetanus, hepatitis B, respiratory diseases caused by *Haemophilus influenzae*, polio, and measles] is crucial to reducing infant and child mortality. J Health Popul Nutr . 2013 Mar;31(1):118–129. doi

Immunisation Schedule in Uganda:

Routine immunization schedule for Children below the age of one year cover vaccines for following preventable diseases
Oral Polio vaccine, BCG, Pentavalent, Pneumococcal 10 valent, Rotavirus vaccine, Injectable Polio vaccine (14 weeks of age), Measles vaccine & HPV (Female Children) (Republic of Uganda, Ministry of Health/UNICEF Report)
Based on the birth cohort in Uganda (1.7 million) the requirements of different vaccines are as under based on the government Immunization program:

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Table No.:2

Type Vaccine Uganda Requirement (MD)

Pentavalent 6

Bacterial PCV10 6

BCG 2

Measles Vaccine 6

Live Virus BOPV

7

BOPV (Ty1and Ty3)

Rota 4

IPV 4

HPV (1,2,3&4) 2

Inactivate Hepatitis B (Birth &Adult) 4

d Viral Yellow fever 2

Upcoming vaccines which shall be targeted in the future

mRNA based vaccines for pandemic preparedness, RSV (Respiratory Syncytial virus) vaccine, Recombinant Tuberculosis vaccine (under development globally), Rabies vaccine etc.

Demand of vaccines products in Uganda & EAC region:

Eastern Africa is experiencing a surge in demand for vaccines, driven by increasing awareness and government initiatives to improve healthcare access.

The Vaccines market in Eastern Africa is anticipated to achieve a revenue of US\$463.65m in 2025.

Looking ahead, the market is expected to exhibit a compound annual growth rate (CAGR) of -4.98% from 2025 to 2029, leading to a market volume of US\$377.97m by 2029.

Key market trends, challenges, and opportunities:

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One of the key trends in the Vaccines segment in Eastern Africa and Uganda is the increasing availability of vaccines. Pharmaceutical companies are expanding their operations in the region, and governments are working to improve vaccine distribution networks. This has made vaccines more accessible to people living in remote areas. Secondly, in the market there is growing demand for new and innovative vaccines. As more people become vaccinated against common diseases, there is a need for vaccines that can prevent emerging infectious diseases. Pharmaceutical companies are investing in research and development to create new vaccines that can address these challenges. East Africa, like the rest of Africa, faces a growing demand for vaccines, particularly as it works to strengthen its vaccine supply chain and boost domestic capabilities to meet the needs of its population.

Challenges and Opportunities:

Challenges include the provision of vaccines in rural locations hampered by inadequate cold chain infrastructure, and vaccination acceptance rates that are impacted by cultural and religious beliefs. Besides, there are also market opportunities, such as increasing research and development (R&D)

spending for novel vaccines and technological advancements.

Growing Support from Various Organizations:

The support from international bodies such as the World Health Organization (WHO), World Bank, the United Nations Children's Fund (UNICEF), and the Global Alliance for Vaccines and Immunization (GAVI) is indispensable for the East African vaccine market. These organizations provide not only financial aid but also logistical support, crucial for the procurement and distribution of vaccines in the region.

2. Competitive Analysis of the Existing Pharmaceutical Market

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Uganda's pharmaceutical market is composed of both local manufacturers and international suppliers. The competitive landscape is shaped by several key factors:

a) Local Pharmaceutical Manufacturers

- Uganda has a small but growing pharmaceutical manufacturing industry.
- Local manufacturers face challenges such as high production costs, limited access to raw materials, and competition from imported generics.

b) Import Dependence and Foreign Suppliers

- Uganda imports about 70% of its pharmaceuticals, with India and China being the dominant suppliers.
- International pharmaceutical companies have a strong presence in Uganda through distributors.
- The high reliance on imports creates challenges in affordability, availability, and quality control.

c) Policy and Market Protection

- Uganda has introduced policies like the "Buy Uganda, Build Uganda" (BUBU) initiative, which promotes local manufacturing.
- However, the local pharmaceutical industry struggles to compete with cheaper imports, and implementation of these policies remains inconsistent.

3. Key Market Trends, Challenges, and Opportunities

a) Market Trends

1. Regulatory Harmonization in the EAC

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o The East African Community (EAC) is working towards regulatory harmonization, allowing Ugandan manufacturers to access larger regional markets.

o Harmonization simplifies drug registration processes and could enhance trade within the region.

2. Growing Non-Communicable Disease (NCD) Burden

o There is an increasing prevalence of NCDs such as diabetes, hypertension, and cancer in Uganda.

o This shift is driving demand for chronic disease medications, which presents opportunities for local manufacturers to diversify production.

3. Expansion of Health Infrastructure

o The Ugandan government is investing in expanding hospitals, clinics, and specialized treatment centres.

o These investments are increasing the demand for pharmaceuticals, particularly in specialized areas like oncology and vaccines.

b) Key Challenges

1. Limited Access to Finance

o Pharmaceutical manufacturing requires large capital investments for facility upgrades, research, and compliance with

regulatory standards.

- o Access to affordable financing remains a major barrier for Ugandan pharmaceutical firms.

2. Dependence on Imported Raw Materials

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- o Uganda imports most of its Active Pharmaceutical Ingredients (APIs) from India and China.

- o This dependence makes local manufacturers vulnerable to global supply chain disruptions.

3. Regulatory and Quality Compliance

- o Meeting international Good Manufacturing Practices (GMP) standards is essential for market expansion but requires costly upgrades.

- o Non-compliance can limit Ugandan pharmaceutical companies' ability to export to international markets

4. Shortage of Skilled Workforce

- o There is a lack of specialized professionals in pharmaceutical research, manufacturing, and regulatory affairs.

- o Developing a skilled workforce through training and academic programs is critical for industry growth.

c) Opportunities for Growth

1. Investment in Local Production

- o Increasing local manufacturing, especially in essential medicines and specialty areas like oncology and vaccines, can enhance Uganda's pharmaceutical security.

- o Public-Private Partnerships (PPPs) could play a crucial role in driving investment.

2. Regional Expansion and Export Potential

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- o The EAC harmonization efforts present an opportunity for Ugandan pharmaceutical companies to expand their reach beyond the domestic market.

- o By meeting international regulatory standards, local manufacturers could increase exports to neighbouring countries.

3. Public-Private Collaborations

- o Government initiatives supporting pharmaceutical production can help develop infrastructure and provide financial incentives for local companies.

- o Partnerships with international pharmaceutical firms could help bring technology and expertise to Uganda.

4. Biopharmaceutical and Vaccine Development

- o Investing in biologics and vaccine production could position Uganda as a leader in high-value pharmaceutical production.

- o Establishing local capacity for monoclonal antibody production could reduce reliance on costly imports.

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B. Overview of the Ugandan Pharmaceutical Regulatory Framework and EAC Regulatory Systems

Introduction:

- In Uganda, the regulation of human medicinal products is overseen by the National Drug Authority (NDA), which ensures the safety, efficacy, quality, availability and cost means of providing satisfactory healthcare and safeguarding the appropriate use of medicines used within the country.

- The legal framework governing these activities is established by the National Drug Policy and Authority Act, Cap 206.
- The National Drug Authority (NDA) was established in 1993 by the National Drug Policy and Authority Statute which in 2000 206 of the Laws of Uganda (2000 Edition).
- The EAC Guide on Good Manufacturing Practices (GMP) is based on the World Health Organization (WHO) Good Manufacturing Practices guidelines in the WHO Technical Report Series No. 961, 2011. This guide also refers to the Pharmaceutical Inspection Convention Scheme (PIC/S) guide PE 009-9.

Table No.:3

Guidelines: National Drug Authority, Uganda- Human medicinal guideline

Title Document Number

Guidelines on the conduct of Clinical trials in DPS/GDL/024

Uganda. REV.01

Guidelines on good clinical practice in the DPS/GDL/025

conduct of Clinical trials involving human

REV.01

participants.

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Guidelines on good manufacturing practice for medicinal products part I – Basic requirements INS/GDL/001 (Part I) for medicinal products.

Guidelines on good manufacturing practice for medicinal products part II - Basic requirements INS/GDL/001 (Part II) for active pharmaceutical ingredients.

Guidelines on good manufacturing practice for

-

medicinal products.

Guidelines on licensing of pharmacies and

INS/GDL/035

pharmaceutical manufacturers.

Guidelines on the verification of applications for the importation and exportation of drugs and INS/GDL/038 Rev.01

pharmaceutical raw and packaging materials.

Guidelines on competence profile of supervising pharmacists for a pharmaceutical INS/GDL/048 Rev.00 manufacturing facility.

Guidelines on the conduct of clinical trials in

DPS/GDL/049

children, pregnant and lactating women in

REV.00

Uganda.

Guidelines on variations to registered PAR/GDL/005 Rev pharmaceutical products. .03

Guidelines on registration of similar bio- PAR/GDL/017 Rev therapeutic products. .01

Guidelines on registration of bio-therapeutic PAR/GDL/016 Rev products. .01

Guidelines on registration of human vaccines. PAR/GDL/021 Rev .01

Guidelines on variation of registered bio- PAR/GDL/033 Rev therapeutic products. .01

Guidelines on submission of documentation for

PAR/GDL/004 Rev

registration of a pharmaceutical product for

.03

human use.

Guidelines for the provision of insurance cover

DPS/GDL/023

for research participants in clinical trials in

Rev.00

Uganda.

Guidelines for introducing a locally

INS/GDL/004 Rev.01

manufactured new pharmaceutical product on

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the Uganda market.

Guidelines on detecting and reporting adverse DPS/GDL/013

drug reactions in Uganda. FRev.01

Guidelines on submission of documentation for

PAR/GDL/020

renewal of registration of a pharmaceutical

Rev.00

product for human use.

Guidelines on Good Distribution Practice for
INS/GDL/002 Rev.01
Pharmaceutical Products.
Guidelines on submitting periodic safety update
report and any other reports that may be DPS/GDL/034
relevant to determine the safety, efficacy and Rev.00
quality of a drug.
Guidelines for the Recall or Withdrawal of a
INS/DGL/037 Rev.00
Medical Product.
Guidelines on variation to registered Human PAR/GDL/018Rev
vaccines. .01
National AEFI surveillance guidelines Uganda. -

Maturity level of NDA (Uganda) with respect to WHO Qualification

Maturity Level 3, which is a key benchmark for drug regulatory bodies, ensures that national authorities can effectively oversee drug safety, efficacy, and quality, aligning with international norms.

The National Drug Authority (NDA) is seeking to overcome legal and regulatory challenges that have hindered its progress toward achieving Maturity Level 3, a global standard for drug regulators.

Only eight African countries have currently attained this status, making it a significant goal for Uganda. These include, Egypt, Rwanda, Senegal, Ghana, Nigeria, South Africa, Tanzania and Zimbabwe.

Key Regulatory Requirements

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In Uganda, the journey of a new drug from conception to market involves a systematic approval process regulated by several governing bodies, predominantly the National Drug Authority (NDA).

Preclinical and Clinical Trials:

- The process begins with preclinical studies, where laboratory and animal testing evaluate the drug's safety and efficacy. This stage is crucial, as it provides preliminary data that informs the next phases of research and development. Pharmaceutical companies need to document their findings and prepare a comprehensive submission package for the NDA.

- Conducting clinical trials in Uganda requires prior authorization from the NDA.

- Researchers, organizations must adhere to the "Guidelines for the Conduct of Clinical Trials in Uganda," which detail the procedures for submission, review, and approval of clinical trial applications.

- Compliance with Good Clinical Practice (GCP) is mandatory, as outlined in the "Guidelines on Good Clinical Practice in the Conduct of Clinical Trials Involving Human Participants."

- After completing the clinical trials, the pharmaceutical company must submit a New Drug Application (NDA) to the NDA, including all research data, manufacturing details, and proposed labelling.

- Once the NDA reviews the application and finds it satisfactory, it grants approval for the drug to be marketed.

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Good Manufacturing Practice (GMP):

Manufacturers of medicinal products must comply with GMP standards to ensure product quality. The NDA provides detailed guidelines, including "Guidelines on Good Manufacturing Practice for Medicinal Products Part I – Basic Requirements for Medicinal Products."

All human medicinal products must be registered with the NDA before they can be marketed or distributed in Uganda. The registration process involves a scientific evaluation of the product's safety, efficacy, and quality.

GMP Inspection:

- National Drug Authority (NDA) is the regulatory body responsible for overseeing the compliance of pharmaceutical companies with GMP standards.

- Regular GMP inspections and audits are conducted as one of the requirements for registration of medicinal products in the EAC.

- The NDA conducts comprehensive evaluations to verify that facilities meet the specified criteria for manufacturing processes, hygiene, and personnel training.
- Inspection involves review of documents, records, facilities and any other resources to assess their conformity to the requirements of EAC GMP of medicinal products.
- Any non-compliance can result in severe penalties, including the suspension or revocation of manufacturing licenses.

Importation and Exportation:

- The NDA regulates the import and export of drugs and pharmaceutical raw materials.

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- Drugs imported must be registered unless given special clearance by the NDA under Sec 8(4) NDPA (National drug policy & Authority) by NDA.
- The import permit may be:
 - a. Annual import permit -which is issued for regular imports like pharmaceutical manufacturers or any organization, retail pharmacies, wholesale premises, that regularly imports drugs and other related substances. Ministry of Health of Uganda and its affiliated units may also apply for an annual import permit.
 - b. Provisional import permit is issued to non-regular importers who may import drugs, for some reasons. The provisional import permit is issued per consignment. This usually applies to the organizations, which receive donations, physicians or individuals who may for specified reasons need to bring limited quantities of specialized drugs that are not available in the country. The permit is valid for one month.

c. Licensing:

- Entities involved in the manufacture of pharmaceuticals must obtain appropriate licenses from the NDA.
- Import License: (Form-24)
 - Eligibility to apply for import license: A licensed person (a person holding license to operate pharmacy or for manufacture drugs).
 - Import license is valid for one year and is valid within the calendar year within which it is issued.

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- Verification certificate: A consignment of drugs to be imported into Uganda shall before importation, be issued with a verification certificate (Form-25) by the NDA basing on the application submitted in this regard in Form-26.
- Stability Data:
 - In Uganda, the National Drug Authority (NDA) requires stability data as part of the regulatory requirements for pharmaceutical and biopharmaceutical products, ensuring quality, safety, and efficacy, and mandates that all medicinal products meet acceptable standards.
 - The NDA requires that applicants for drug registration provide results of stability testing, which is crucial for determining the shelf-life and storage conditions of the product.
 - Applicants need to submit all relevant stability data as part of the registration application.
 - The stability studies will determine the appropriate storage conditions for the product, including temperature and humidity, and these conditions must be reflected on the product label.
 - The product label must include an expiry date and storage instructions based on the stability data.
 - Stability studies should include Storage conditions i.e temperature and relative humidity for accelerated and stress conditions.

- Stability studies should be done in accordance or with reference to WHO TRS 987, WHO TRS 953 Annex 2, WHO TRS 962 Annex 3, ICH Q1A and ICH Q5C guidelines.

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- A minimum of twelve months data at the time of submission should be provided in cases where storage periods greater than six months are requested.

- For storage periods of less than six months, the stability data should cover the whole proposed shelf life.

- The stability studies should be submitted in controlled documentation.

- Post-approval stability program should include the stability program or stability commitment to be carried out once the drug product is on the market, including the number of batches to be included in the study each year and the tests to be performed.

- These results should be submitted periodically to update the information on the stability of the drug product.

- The stability of each dosage form should be separately documented.

- Stability data submitted should be for at least three consecutive batches and include the following:

- a) Information on stability of drug product, quality control methods and rationale for the choice of tests for determining stability.

- b) Information on the dates of manufacture of the lots, the lot numbers, the vial and dose size, and the scale of production.

- For lyophilized products the data supporting the shelf-life of the product reconstitution should be included.

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- If the drug product is frozen, data supporting the stability of the product through a stated number of freeze-thaw cycles should be provided.

Drugs Registration in Uganda:

All drugs to be manufactured or imported into Uganda have to be registered with the National Drug Authority, Uganda. A person who intends to manufacture, import or export a product shall, prior to the manufacture, importation or exportation of the product, apply to the Authority for registration of the product.

The registration process involves:

- NDA inspection & approval of the manufacturing facility.

Validity of registration:

- The Authority shall issue for the first registration, a certificate of registration which shall be valid for five years.
- The holder of a certificate of registration shall pay an annual retention fee prescribed by the Authority for maintaining the registered human or veterinary drug or preparation, vaccine or other immunological products or surgical instrument, on the Register.
- A certificate for renewal of registration shall be valid for one year from the date of issue.
- The certificate of registration issued under these Regulations shall be kept at the premises of the holder of the certificate of registration.
- The holder of a certificate of registration shall monitor the safety and effectiveness of the registered human or veterinary drug or preparation, vaccine or other immunological products or surgical instrument while on the market in Uganda and shall submit periodic reports as may be prescribed by the Authority.

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- The registration of a human or veterinary drug or preparation, vaccine or other immunological products or surgical instrument under these Regulations shall be valid except where the registration is suspended or cancelled by the Authority or is terminated by the holder of the

certificate of registration.

Table No.:4

Marketing Authorisation Time lines:
Registration Timeline: 12 to 18 months
Regulatory Timeline (Working
Action
Area days)
GMP Report assessment to
45 days
Manufacture after inspection
Feedback after inspection-
Regulatory 28 days
Domestic
Inspection
Regulatory decision after
CAPA – Domestic (Local 17 days
Facilities)

Submission and approval of
Documentation 220 days
Dossier

Total no. of days 310 days

Post-Marketing Surveillance and Pharmacovigilance:

- Under the National Drug Policy and Authority (NPD &A) Act (1993) the Uganda National Pharmacovigilance Centre (NPC) is responsible for PV efforts in the country and mandates the National Drug Authority (NDA) to monitor the safety, efficacy and quality of medicines.

Risk Assessment

- Risk assessment is a crucial regulatory requirement, ensuring product safety, efficacy, and quality, and safeguarding public

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health. Proactive risk identification and mitigation, through methodologies like Quality Risk Management (QRM), are essential for regulatory compliance and building trust with stakeholders.

- Manufacturers would be required to conduct a risk assessment on critical production and quality control processes and thereafter develop a mitigation plan whose effectiveness would be monitored by the regulator.

Table No.:5

Regulation Framework Timelines:

Regulatory Action Timeline
Area (Working days)
Clinical Trial Receipt, Screening and 10 days
(CT) Oversight acknowledgment of a Clinical trial
application (new, renewals and
amendment)
Regulatory decision on a Clinical 50 days
trial application
Annual renewal of Ongoing trials 20 days
Amendment of CT authorization 20 days
Feedback report to a client 45 days
following a GCP Inspection
Marketing Assessment of application and 18 months (396
Authorization request for additional information days)
(MA) on application for MA (Human
Drugs)
Assessment of application and 6 months (132
request for additional information days)
on application for MA (Vaccines,
Anti-cancer medicines and other
vital (medicines)).
Regulatory decision on MA after 3 months (66
additional information is received days)

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Special Categories

Regulatory decisions on products 6 months (132
already authorized for marketing days)
by stringent regulatory
authorities, WHO prequalified
products and those approved
under article 58 of EU regulations
Variation and Notification

Regulatory decision on minor 4 months (88 variation days)
 Regulatory decision on Major 6 months (132 variation days)
 Regulatory decision on annual or 80 days immediate notification change
 Drug Register
 Publication of drug register Monthly
 Regulatory GMP Physical inspection after 180 days
 Inspection receipt of application- Foreign
 GMP Report assessment to 45 days
 Manufacture after inspection
 Feedback after inspection- 28 days
 Domestic
 Regulatory decision after CAPA – 20 days
 Foreign
 Regulatory decision after CAPA – 17 days
 Domestic (Local Facilities)
 Premise Feedback to the applicant on the 40 days
 Licensing regulatory decision for pharmacy renewal
 Feedback to the applicant on the 30 days regulatory decision for licensing of new pharmacy applicants
 Market Verification of imports (registered 2 days Surveillance drugs)
 and Control Clearance of Imports at ports of 2 days entry

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Verification of imports (un- 10 days registered drugs)
 Regulatory decision on drug 10 days promotional material
 Publication of the database of Monthly (5th of approved promotional the month) materials/advert
 Vigilance Acknowledgement of receipt of 5 days an ADE report (mail/letter)
 Feedback on serious ADE reports 18 days
 Feedback on serious AEFI 14 days reports to the expanded program for immunization (EPI)
 Laboratory Test results from mandatory 48 Days
 Testing testing after sampling (Medicine)- PoE
 Test results from client requests 45 Days after acceptance by the lab (All Products)
 Test results for pre-market 60 Days (Domestics Manufactures)
 Test results for post-market 48 Days surveillance (Medicines)
 National Issuance of recall or alert for a 14 Days
 Regulatory substandard or falsified medicine System or health care product
 Acknowledgement of receipt of a 5 Days product complaint
 Feedback on a market (process, 20 Days product, other) complaint
 Feedback on the effectiveness of 10 Days recall after receipt of the recall report from the LTR (Local Technical Representative)
 Feedback on test results from the 5 Days Lab to the client

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Support to QUALITY MANAGEMENT
 Regulatory SYSTEM
 areas Acknowledgement of receipt of a 2 Days service-related complaint
 Feedback on service-related 21 Days complaints

ii. Environmental and Safety Regulations Applicable to Pharmaceutical Manufacturing in Uganda

Uganda's regulatory framework ensures that pharmaceutical production safeguards human health, worker safety, and environmental sustainability.

Environmental Compliance

Environmental oversight in Uganda is primarily the responsibility of the

National Environment Management Authority (NEMA). Under the National Environment Act, 2019, all industrial and manufacturing projects, including pharmaceutical facilities, are required to undertake an Environmental and Social Impact Assessment (ESIA) as a prerequisite for licensing and construction. The ESIA must identify potential environmental risks and outline mitigation strategies, with particular attention to waste management, water usage, emissions, and impacts on surrounding communities.

Pharmaceutical facilities must install and maintain Effluent Treatment Plants (ETPs) to process wastewater before discharge. Compliance with the Environmental Management (Waste Management) Regulations ensures proper handling of both hazardous and non-hazardous solid and liquid waste, especially chemical residues, solvents, and expired or rejected pharmaceutical products. The facility must also adhere to Uganda's obligations under the Basel Convention on the control of transboundary movements of hazardous waste.

Safety and Occupational Health Standards

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The National Drug Authority (NDA) regulates the pharmaceutical sector and enforces Good Manufacturing Practices (GMP), including protocols that safeguard worker and product safety throughout the production lifecycle. This includes facility design to minimize cross-contamination, controlled cleanroom environments, and safe material handling procedures.

In parallel, the Occupational Safety and Health Act, 2006, administered by the Ministry of Gender, Labour and Social Development, mandates comprehensive safety practices in the workplace. Employers must implement hazard identification and risk mitigation strategies, ensure employee training, and provide Personal Protective Equipment (PPE) and emergency response systems. All operations must include clearly marked emergency exits, fire suppression systems, first aid facilities, and regular safety drills.

Regular inspections and audits are conducted by both NDA and the Ministry of Labour to verify compliance. Manufacturers are encouraged to adopt internationally recognized standards such as ISO 14001 (Environmental Management Systems) and ISO 45001 (Occupational Health and Safety Management Systems) to enhance performance and facilitate future export readiness.

Strategic Considerations

Full regulatory compliance not only avoids legal penalties and operational delays but also enhances the facility's reputation among stakeholders and investors. Additionally, adherence to these regulations supports sustainable operations and positions the facility for future certification under global quality and safety systems, enabling access to regional and international markets.

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iii. Overview of EAC Regulatory Systems for Key Therapeutic Areas

Key EAC

Therapeutic Regulatory Progress &

Harmonization Challenges

Area Focus Opportunities

Mechanisms

Insulin and oral

Cold chain Streamlined

hypoglycaemic

Common Technical logistics approvals

Bioequivalence

Document (CTD)

Diabetes and cold chain

Mutual Recognition Limited local Regional

Chronic disease

Procedure (MRP) insulin demand driving

management

production local production

alignment

Fast-tracking of Strong

Joint assessments Counterfeits

ACTs Monitoring harmonization

Dependency

Malaria of resistance Integrated

Regional malaria on donor

Alignment with procurement

drug guidelines programs

WHO PQ systems

AMR frameworks Overuse and Regulatory

Quality and safety under WHO/Africa poor updates in

evaluations GMP CDC prescription progress

Antibiotics
 compliance AMR practices -
 mitigation Regional Limited Enhanced AMR
 pharmacovigilance surveillance monitoring
 Regional
 High scrutiny:
 AVAREF support Heavy import capability
 quality, safety,
 reliance strengthening
 cold chain -
 Vaccines Reliance on WHO
 Emergency use
 PQ and joint Capacity gaps Ongoing vaccine
 approval
 inspections for biologics regulation
 mechanisms
 training
 Capacity
 Advanced Limited local
 building
 assessments for International evaluation
 underway
 mAbs, biosimilars, partnerships for experience
 Biologics insulin analogs - capacity building -
 Future potential
 GMP and Involvement of WHO, High
 for local
 comparability UNICEF assessment
 biologics
 requirements complexity
 production
 ARV quality and Fast-track review Stockouts Increasing local
 HIV/AIDS resistance mechanisms - and supply registration
 monitoring - Reliance on global chain capabilities

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Key EAC
 Therapeutic Regulatory Progress &
 Harmonization Challenges
 Area Focus Opportunities
 Mechanisms
 Integration with procurement challenges -
 national HIV systems (e.g., Global Limited local Supportive
 programs - Rapid Fund) production policy for ARV
 access to WHO production
 PQ ARVs
 MDR-TB drugs
 evaluation Complexity of Regional labs
 WHO TB drug
 MDR/XDR TB and regulatory
 guidelines
 Diagnostic treatment harmonization
 Tuberculosis
 technologies improving
 (TB) Regional efforts to
 Dependence Potential for
 reduce import
 Fixed-dose on donor local packaging
 timelines
 combination supply and supply
 approvals

General Observations on the EAC Medicines Regulatory Framework

- Harmonization Framework: The EAC Medicines Regulatory Harmonization (MRH) Programme supports joint product assessments, GMP inspections, and reliance pathways to reduce duplication across Partner States.
- National Regulatory Authorities (NRAs): While each EAC country retains autonomy (e.g., Kenya PPB, Uganda NDA, Rwanda FDA), they collaborate extensively on shared regulatory functions.
- Global Alignment: Reliance on WHO Prequalification, Global Fund procurement standards, AVAREF for clinical trials, and engagement with the African Medicines Agency (AMA) for future alignment.

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C. Site Selection and Infrastructure
 i. Site Selection Criteria

a) Proximity to Raw Materials & Suppliers

- Most of the Active Pharmaceutical Ingredients (APIs) and excipients used in pharmaceutical production in Africa are imported (mainly from India and China).

- Ideal site locations:

- o Close to international ports, airports, or dry ports to reduce import lead times.

- o Access to local packaging material manufacturers to lower supply chain dependency.

- o Industrial zones with chemical and biotech suppliers to ensure consistent raw material availability.

b) Logistics & Transportation Considerations

- Proximity to major highways, rail networks, and ports to facilitate raw material import and product distribution.

- Cold chain logistics is crucial for vaccines and biologics, requiring locations near temperature-controlled warehouses and transport hubs.

- Close to key healthcare markets to reduce distribution delays, especially for essential medicines.

c) Utilities & Infrastructure

- Electricity: Reliable power supply is critical, especially for injectables, vaccine production, and sterile drug manufacturing. Backup

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generators, UPS systems, and solar energy solutions should be considered.

- Water Supply & Treatment:

- o High-quality water sources for pharmaceutical processing, purification, and cleaning.

- o Reverse Osmosis (RO) & Water for Injection (WFI) systems for injectable production.

- o Effluent treatment plants (ETP) and bio-waste disposal facilities for environmental compliance.

- Air Filtration & HVAC Systems:

- o ISO-classified cleanrooms (ISO 5, ISO 6, ISO 7, ISO 8) for sterile and injectable production.

- o High-Efficiency Particulate Air (HEPA) filters and humidity-controlled areas for tablet and capsule manufacturing.

d) Skilled Workforce Availability

- Proximity to universities and research institutions that offer pharmaceutical, biotechnology, and engineering programs.

- Training partnerships with international pharmaceutical firms to build local expertise.

- Employee housing and transport accessibility to attract and retain skilled professionals.

2. Infrastructure Requirements

a) Facility Layout & Zoning

- Segregated production zones for different dosage forms:

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- o OSD (Oral Solid Dosage): Tablets, capsules.

- o Syrups & Liquids: Bottled formulations.

- o Injectables & Vaccines: Requires high-containment cleanrooms and sterile environments.

- Dedicated cleanroom areas for sensitive processes such as sterile drug filling and vaccine formulation.

- Material & personnel movement control to prevent cross-contamination.

b) Manufacturing Equipment & Technology Integration

- Automated production lines for efficiency and compliance with GMP standards.

- SCADA (Supervisory Control and Data Acquisition) and MES (Manufacturing Execution Systems) for real-time monitoring.

c) Cold Chain & Storage Facilities

- Temperature-controlled warehouses for raw materials and finished pharmaceutical products.

- Cold chain logistics for vaccines and biologics, including deep freezers and transport containers.

- Automated inventory management systems to ensure compliance with GDP (Good Distribution Practices).

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3. Finding the Right Technology Partners

a) Pharmaceutical Machinery & Equipment Suppliers

To establish a world-class pharmaceutical facility, partnering with technology providers specializing in pharmaceutical manufacturing equipment is essential. Key areas include:

1. Oral Solid Dosage (OSD) Equipment

- o Granulation, blending, and tablet compression machines

- o Leading suppliers: GEA Group, Bosch, Syntegon (formerly Bosch Packaging), Fette Compacting ACG Worldwide, SED Pharma, Syntegon Technology, IMA Group, Glatt, Fette Compacting, Romaco, iPharMachine

2. Sterile Injectable & Vaccine Manufacturing Equipment

- o Aseptic filling, lyophilization, and cleanroom solutions

- o Leading suppliers: Bausch+Ströbel, IMA Pharma, Optima Pharma, NKP Pharma machinery, Sartorius, Cozzoli Machine Company, Groninger, ARxIUM, NJM Packaging, Intellitech

3. Water Purification & WFI Systems

- o Suppliers: Merck Millipore, Veolia Water Technologies, Komal Industries, MECO, NGK Insulators Ltd, Syntegon, Ovivo, Fabtech Technologies, Aqua-Chem, Herco Wasser technik GmbH, NEWater

4. Cleanroom & Air Filtration Systems

- o Suppliers: Camfil, AES Clean Technology, Thermo Fisher Scientific, Clean Air Technology, Atmos-Tech Industries, Abtech Inc., Microflow Devices India PVT Ltd. Zeco Aircon Ltd

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b) Digital & Automation Partners

- SCADA & MES Systems: Siemens, Rockwell Automation, Honeywell.

- Automated quality control & batch monitoring: Emerson, Siemens, ABB.

c) Regulatory & Compliance Partners

- GMP compliance consulting:

- o Local regulatory bodies (e.g., Uganda National Drug Authority - NDA).

d) Waste Management & Sustainability Partners

- Hazardous waste disposal & ETP solutions: Veolia, SUEZ, Ecolab.

- Renewable energy integration for sustainability: Tesla Energy, Schneider Electric.

4. Strategic Collaborations & Public-Private Partnerships

- Government Incentives & Special Economic Zones (SEZs)

- o Tax incentives for pharmaceutical manufacturers in Uganda and

the East African Community (EAC).

o Preferential land allocation and infrastructure support for investors.

- Public-Private Partnerships (PPPs)

o Collaborations with global health organizations (WHO, UNICEF, GAVI, USAID) for vaccine and essential medicine production.

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ii. Infrastructure Requirements for a Pharmaceutical Manufacturing Facility near Kampala, Uganda

The proposed pharmaceutical manufacturing facility, will be located near Kampala, Uganda.

Given the scale and scope of this state-of-the-art facility, robust infrastructure is vital to ensure uninterrupted operations, regulatory compliance, and product quality.

Power Supply: The facility will require a high-capacity, stable power source to support continuous manufacturing, HVAC systems, cleanrooms, cold storage, and quality control labs. It will be connected to the national electricity grid via UMEME, with dedicated transformers and high-voltage infrastructure. Backup power will be provided through diesel generators and solar power systems to ensure energy security and sustainability.

Water Supply: A reliable water supply is essential for production processes, utilities, cleaning, and sanitation. The facility will include on-site water purification systems to generate Purified Water (PW) and Water for Injection (WFI), meeting GMP standards. Water will be sourced from the national supply network and supplemented by boreholes with storage reservoirs.

Waste Management: The facility will house a centralized Effluent Treatment Plant (ETP) and systems for safe disposal of solid and hazardous waste, in full compliance with NEMA guidelines. Special provisions will be made for pharmaceutical and biomedical waste, including incineration and disposal via licensed operators.

Transport Links: Proximity to Kampala's industrial zones, the Northern Corridor, and Entebbe International Airport ensures efficient import/export logistics and access to regional markets. Excellent road connectivity will facilitate material movement and regulatory oversight.

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D. TECHNICAL FEASIBILITY

1. Machinery and Equipment Requirements

Each dosage form requires specific equipment to facilitate high-quality and compliant production.

a) Oral Solid Dosage (OSD) & Capsules Manufacturing

- Granulation Equipment – Fluidized bed dryers, high-shear mixers.
- Tablet Compression Machines – Single rotary, double rotary presses.
- Capsule Fillers – Semi-automatic and fully automatic encapsulation machines.
- Coating Machines – Film and sugar coating systems.
- Blending & Mixing Equipment – V-blenders, ribbon mixers.

b) Syrup and Liquid Manufacturing

- Sugar Dissolution & Mixing Tanks – Stainless steel reactors with agitation.
- Homogenizers – For uniform syrup consistency.
- Liquid Filling & Capping Machines – For bottles and ampoules.
- Sterile Filtration Systems – For microbial safety.

c) Injectables & Parenteral Manufacturing

- Aseptic Filling Lines – For ampoules, vials, prefilled syringes.
- Lyophilization Equipment – For freeze-dried injectables.
- Autoclaves & Sterilizers – For equipment and component sterilization.

- Isolators & Restricted Access Barrier Systems (RABS) – To maintain sterility.

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d) Monoclonal Antibodies (mAbs) & Vaccine Manufacturing

- Fill-Finish Equipment – Sterile filling, capping, lyophilization.
- Cold Chain & Stability Chambers – For biological storage.

3. Technology and Automation

a) Digitalization and Smart Manufacturing

- Manufacturing Execution Systems (MES) for batch tracking.
- SCADA (Supervisory Control and Data Acquisition) for process automation.
- Predictive Maintenance to prevent downtime.

b) Quality Control & Assurance Technologies

- Analytical Testing Equipment – HPLC, GC, UV Spectrophotometers.
- Microbial Testing – Endotoxin testing, sterility assurance.

4. Compliance & Regulatory Considerations

- Adherence to GMP, WHO and local drug regulatory guidelines.
- Regular validation & qualification of equipment and processes (IQ, OQ, PQ).
- Environmental compliance with waste disposal and emission control.

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Manufacturing Area Requirements - Overall Site Plan & Area Distribution

Table No.:6

COMMON AREAS

S. No DESIGNATED AREA APPROX AREA SQM

- 1 CENTRAL WAREHOUSE FOR RAW 30,000 MATERIAL, PRIMARY, SECONDARY AND TERTIARY PACKAGING MATERIAL, SOLVENTS, ENGINEERING SPARES, COLD STORAGE, FINISHED GOODS STORAGE
- 2 ENGINEERING BLOCK – HVAC AND 15,000 AHU, WATER PURIFICATION, PW AND WFI SYSTEM
- 3 CENTRAL UTILITY BLOCK 6,000
- 4 EFFLUENT TREATMENT PLANT 5,000
- 5 SCRAP YARD 500
- 6 PRIMARY WATER TREATMENT AREA 1,000
- 7 ADMINISTRATION BLOCK (INCLUDES 13,000 FINANCE, ACCOUNT, HR, SUPPLY CHAIN ETC.), CORPORATE OFFICES, IT AND DATA SECURITY, MEETING ROOMS AND CONFERENCE HALLS, DOCUMENT CONTROL, RECEPTION AND VISITOR LOUNGE
- 8 SECURITY BLOCK – CONTROL ROOM, 3,500 FIRE SAFETY AND EMERGENCY CONTROL ROOM, EMPLOYEE SCREENING AND ACCESS CONTROL AREA,
- 9 CAR PARK 4,000
- 10 CAFETERIA 4,000
- 11 TRAINING AND INDUCTION ROOMS 1,500
- 12 MEDICAL AND FIRST AID ROOMS 500
- 13 RECREATIONAL AREA FOR EMPLOYEES 2,000

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- 10 QUALITY CONTROL LABORATORY 4000
- 11 RESEARCH AND DEVELOPMENT 2000 LABORATORY

12 QUALITY ASSURANCE SECTION 2000
13 REGULATORY AFFAIRS DEPARTMENT 1000
14 CLINICAL RESEARCH DEPARTMENT 500
TOTAL AREA 95,500 SQM

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Table No.: 7

PLANT 1 – OSD BLOCK (ASSUMED OUTPUT CAPACITY IS 15 MILLION
UNITS PER DAY)

Size
Section Required Function
(sq. m)
Wet/dry granulation for tablet
Granulation Area 3,500
and capsule production
Mixing active ingredients and
Blending & Compression 2,500
excipients
Tablet compression and
Tablet & Capsule Filling 3,000
capsule filling
Coating 2,000 Film/sugar coating for tablets
Syrup preparation, mixing, and
Liquid Manufacturing 3,500
holding tanks
High-speed blister packing,
Filling & Packaging 5,000
bottling, and labelling
Power, water, and air handling
Utilities & HVAC 2,500
units
Support Areas (Offices, Administrative functions,
Change Rooms, Cafeteria, 2,500 employee welfare, and
etc.) cleanroom airlocks
TOTAL AREA 24,500 SQM

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Table No.: 8

PLANT 2 – mAbs AND BIOSIMILARS (ASSUMED OUTPUT CAPACITY
IS 3 MILLION UNITS PER ANNUM)

Estimated
Section Function
Area (sq. m)
Storage of vials, stoppers,
Raw Material & Component
2,500 API, excipients, and
Storage
packaging materials
Buffer preparation 1,500 Preparation of buffers
Mixing and formulation of
Formulation Suite 2,000
mAbs bulk solution
High-speed filling, capping,
Aseptic Filling & Stoppering 2,500 and stoppering under ISO 5
conditions
Freeze-drying for extended
Lyophilization (if required) 2,000
shelf-life stability
Capping, Inspection & Visual inspection, capping,
1,500
Labelling and final labelling
Primary & Secondary Packaging in cartons,
2,000
Packaging serialization, and labelling
Refrigerated and frozen
Cold Chain Storage &
2,500 storage at 2-8°C and -20°C
Stability Testing
for mAbs stability
Power supply, purified water
Utilities & HVAC Systems 2,000 (WFI), HVAC, and compressed
air
Support Areas (Offices, Administrative offices,
Cleanrooms, Cafeteria, 2,500 gowning areas, and personnel
Change Rooms, etc.) movement
TOTAL AREA 21,000 SQM

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- Expansion & Scalability: The facility should allow for a future increase in production capacity.

Table No.: 9

PLANT 3 - Filling Lines for Vials, Ampoules, and Pre-Filled Syringes

Estimated
Section Function
Area (sq. m)
Vial Washing & Cleaning and depyrogenation of
2,000
Depyrogenation vials before filling
Ampoule Washing & Automated washing and
1,500
Sterilization sterilization of ampoules
Pre-Filled Syringe Syringe preparation and plunger
2,000
Preparation assembly
High-speed aseptic filling and
Filling & Stoppering (Vials &
3,000 stoppering under ISO 5
Ampoules)
conditions
Filling & Sealing (Pre-Filled Filling and sealing of syringes
2,500
Syringes) under sterile conditions
Freeze-drying for vials requiring
Lyophilization (if required) 2,500
stability enhancement
Automated visual inspection,
Capping & Inspection 1,500 capping, and rejection of
defective units
Labelling & Secondary Labelling, serialization, and
2,500
Packaging secondary packaging
Purified water (WFI), air
Utilities & HVAC Systems 2,000 handling, and compressed air
systems
Support Areas (Gowning,
Personnel movement, gowning,
Change Rooms, Offices, 2,500
and administrative offices
etc.)
TOTAL AREA 22,000 SQM

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- Expansion & Scalability: Design with future capacity increases in mind.

PLANT 4 - Oncology OSD Facility for Production of Hormonal and Cytotoxic Oral Formulations

Table No.: 10

Estimated
Section Function
Area (sq. m)
Isolated storage for APIs,
Raw Material Storage 2,500 excipients, and packaging
components
High-shear wet/dry granulation
Granulation Area 3,500
and fluid bed drying
Blending of APIs and
Blending & Compression 2,500
excipients, tablet compression
Capsule filling with
Encapsulation Area 2,000
containment systems
Tablet film/sugar coating with
Coating (if required) 2,000
dust extraction
Hormonal & Cytotoxic Dedicated, fully enclosed
3,500
Segregation areas with negative pressure
High-speed blister packing,
Filling & Packaging 4,000
bottling, and labelling
Waste Management & Safe disposal of hazardous
2,000
Decontamination cytotoxic waste
High-efficiency air handling,
Utilities & HVAC Systems 2,500
WFI, compressed air
Support Areas (Change GMP-compliant personnel

Rooms, Gowning, Admin, 3,000 movement and administrative etc.) areas
TOTAL AREA 27,500

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Additional Considerations

- High-Containment Design:
 - o Closed systems with negative pressure, airlocks, isolators, and barrier technology.
- Cross-Contamination Control:
 - o Separate air handling, material flows, and personnel flows for hormonal and cytotoxic drugs.
- Scalability & Future Expansion:
 - o Facility should accommodate increasing demand and new drug formulations.

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Table No.: 11

PLANT 5 – VACCINE fill finish Facility (ASSUMED CAPACITY OUTPUT IS 50 MILLION DOSES PER ANNUM)

Estimated
Section Area (sq. Function
m)
Vial Washing & Cleaning and depyrogenation of 2,000
Depyrogenation vials before filling
High-speed aseptic filling and
Filling & Stoppering (Vials) 3,000 stoppering under ISO 5 conditions
Freeze-drying for vials requiring
Lyophilization (if required) 2,500
stability enhancement
Automated visual inspection,
Capping & Inspection 1,500 capping, and rejection of defective units
Labelling & Secondary Labelling, serialization, and 2,500
Packaging secondary packaging
Purified water (WFI), air
Utilities & HVAC Systems 2,000 handling, and compressed air systems
Support Areas (Gowning, Personnel movement, gowning, 2,500
Change Rooms, Offices, etc.) and administrative offices 16,000
TOTAL AREA
SQM

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E. FINANCIAL FEASIBILITY
This review provides an overview of CAPEX, Operational Expenditure (OPEX), financial projections, break-even analysis, and potential Return on Investment (ROI).

1.1 Cost Breakdown: CAPEX

Table No.: 12

Manufacturing Block wise area and cost
S.
Block Name Sub Unit
No. Cost in USD
1. Granulation Area
2. Blending & Compression
Plant 1 - OSD 4,39,00,000
3. Tablet & Capsule Filling
4. Coating

5. Liquid Manufacturing
6. Filling & Packaging
7. Utilities & HVAC
8. Support Areas (Offices, Change

Rooms, Cafeteria, etc.)
9. Raw Material & Component Storage
10. Buffer preparation
11.
Plant 2-mAbs Formulation Suite
12. and Biosimilar Aseptic Filling & Stoppering 8,25,00,000
13. Lyophilization (if required)
14. Capping, Inspection & Labelling

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15. Primary & Secondary Packaging
16. Cold Chain Storage & Stability
Testing
17. Utilities & HVAC Systems
18. Support Areas (Offices, Cafeteria,
Change Rooms, etc.)
19. Vial Washing & Depyrogenation
20. Ampoule Washing & Sterilization
21. Pre-Filled Syringe Preparation
22. Filling & Stoppering (Vials &
Ampoules)
23. Filling & Sealing (Pre-Filled Syringes)
Plant 3-Filling
7,60,00,000
24. Lines Lyophilization (if required)
25. Capping & Inspection
26. Labelling & Secondary Packaging
27. Utilities & HVAC Systems
28. Support Areas (Gowning, Change
Rooms, Offices, etc.)
29. Raw Material Staging and
component storage
30. Granulation Area
31. Blending & Compression
32. Plant 4- Encapsulation Area
9,62,50,000
33. Oncology
Coating (if required)
34. Hormonal & Cytotoxic Segregation
35. Filling & Packaging
36. Waste Management &

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Decontamination
37. Utilities & HVAC Systems
38. Support Areas (Change Rooms,
Gowning, Admin, etc.)
39. Vial Washing & Depyrogenation
40. Filling & Stoppering (Vials)
41. Lyophilization (if required)
42. Plant 5- Capping & Inspection 6,60,00,000
Vaccine
43. Labelling & Secondary Packaging
44. Utilities & HVAC Systems
45. Support Areas (Gowning, Change
Rooms, Offices, etc.)
46. Animal House
47. Central warehouse
48. Engineering block
49. Central utility block
Common
50. 9,94,72,000
Blocks Administration block
51. Quality control laboratory
52. Research and development
laboratory
53. Quality assurance section
54. Car park
55. Cafeteria
56. Miscellaneous Training and induction rooms 2,50,00,000
blocks
57. Medical and first aid rooms
58. Recreational area for employees

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59. Regulatory affairs department
60. Clinical research department
61. Landscaping and horticulture
62. QUALITY FACILITY COMMISSIONING AND
5,50,00,000

ASSURANCE VALIDATION COSTS
63. REGULATORY LICENCING AND STATUATORY
2,75,00,000
AFFAIRS COSTS
TOTAL BUDGET ESTIMATE IN
57,16,22,000
USD
SAY 571
MUSD

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1.2 Capital Expenditure (CAPEX)

The total investment is \$571M USD, allocated across different manufacturing units and operations as follows:

Table No.: 13

Approx
Approx % of
Manufacturing Unit Investment
Total CAPEX
(MUSD)
Oral Solid Dosage (OSD) 43.9 7.68%
Biopharma Fill-Finish 82.5 14.44%
Injectables (SVP) 76.0 13.31%
Oncology (Hormonal & Cytotoxic Products) 96.25 16.85%
Vaccine Fill-Finish 66.0 11.55%
Common Infrastructure (Utilities,
99.47 17.42%
Warehousing, R&D, QA/QC, Admin, etc.)
Miscellaneous Blocks (Car park, cafeteria,
Training room, Medical room, Regulatory 25.00 4.37%
affairs, Clinical research, Landscaping etc.,
Facility commissioning and qualification 5.50 0.96%
Registration and licensing 2.75 0.48%
571
Total Investment 100%
MUSD(571.6)

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1.3 Operational Expenditure (OPEX) Breakdown

Annual OPEX includes raw materials, labour, utilities, maintenance, logistics, and regulatory compliance. Estimated OPEX in full scale operations:

Table No.: 14

Annual Cost % of Total
Cost Component
(MUSD) OPEX
Raw Materials & APIs 150(150.75)* 45%
Labour & Workforce 50(50.25)* 15%
Utilities (Power, Water, Waste
40(40.20)* 12%
Management)
Maintenance & Equipment Servicing 25(26.80)* 8%
Regulatory Compliance & Quality
20(20.18)* 6%
Assurance
Logistics & Distribution 30(30.15)* 9%
Total OPEX (Annual) \$335M USD 100%

* As per calculations

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2. Financial Projections and Revenue Estimates

2.1 Estimated Annual Revenue (First 5 Years)

Revenue is projected based on market demand, production capacity, and price realization. The estimated year-wise revenue is:

Table No.: 14

Revenue
Year
(MUSD)
Year 1 5

Year 2 90
Year 3 150
Year 4 250
Year 5 450

Revenue Contribution by Segment:

- OSD & Injectables: 35%
- Oncology: 25%
- Vaccines & Biopharma: 30%
- Contract Manufacturing & Exports: 10%

3. Break-Even Analysis

3.1 Break-Even Point Calculation

The break-even point (BEP) is determined by comparing total fixed costs (CAPEX amortization) with contribution margins.

- Fixed Costs: Depreciation on \$571M CAPEX (~10-year depreciation cycle) → ~\$57M/year
- Variable Costs: 65% of revenue
- Average Selling Price & Margins: 35% Gross Margin

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Break-Even Sales Volume Calculation:

Break-even revenue=Fixed Costs / Contribution Margin = 57M / 0.35=162M USD

Thus, the break-even point is reached within the third/fourth year of operation.

3.2 Payback Period & Profitability Timeline

- Payback period: ~6 years
- First-year losses due to ramp-up & market entry.
- Full profitability expected in Year 4-5 with a Net Profit Margin of ~15-20%.

4. Return on Investment (ROI) Analysis

4.1 Projected ROI

ROI = Annual Profit / Total investment × 100

- Year 3 Expected Net Profit: ~\$150M USD
- ROI by Year 3/4: 150/ 571 ×100 = 26.0%
- Cumulative ROI over 6 years: 125% (fully recovering investment & generating profit).

4.2 Internal Rate of Return (IRR)

- Estimated IRR: 18-22%, exceeding the pharmaceutical industry benchmark of 15-18%.
- Strong returns in high-value segments (Oncology, Biopharma, and Vaccines)

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5. Key Financial Risks & Mitigation Strategies

Table No.: 15

Risk Factor	Potential Impact	Mitigation Strategy
Slower product Regulatory Delays	Early engagement with NDA approvals (Uganda), WHO-GMP	
Increased raw API Price Volatility	Multi-supplier sourcing, API material costs localization	
Lower pricing Market Competition	Differentiation through quality & power innovation	
Logistics & Distribution Challenges	Delayed market penetration	Strong partnerships with distributors

6. Conclusion & Strategic Recommendations

- CAPEX of \$571M is strategically allocated across multiple high-growth segments.
- OPEX of ~\$335M annually ensures robust operations & regulatory compliance.
- Break-even is expected within the third year, with profitability by Year

4-5.

- Strong ROI (~28% by Year 4) and IRR (~18-22%) ensure high financial viability.

Strategic Recommendations:

1. Focus on Oncology & Biologics: Higher-margin products provide better financial returns.
2. Optimize API Sourcing: Reduce costs by investing in local API production or strategic long-term supplier contracts.
3. Enhance Export Potential: Leverage Uganda's EAC trade advantages to increase regional market penetration.
4. Automate Supply Chain & Manufacturing
5. Seek Investment & Government Support: Public-private partnerships for incentives & subsidies to lower CAPEX burden.

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F. SUPPLY CHAIN AND RAW MATERIAL SOURCING

The supply chain must ensure a steady supply of raw materials, efficient production, strict quality control, and seamless distribution, while mitigating risks associated with logistics, regulatory compliance, and market fluctuations.

1. Key Components of the Supply Chain

A robust supply chain for a Sunshine company shall include the following key components:

1. Raw Material Sourcing (APIs, Excipients, Packaging Materials)
2. Inventory & Warehousing
3. Production & Manufacturing
4. Quality Control & Regulatory Compliance
5. Cold Chain & Storage (For Vaccines & Biosimilars)
6. Distribution & Logistics
7. Risk Management & Supply Chain Resilience

2. Raw Material Sourcing Strategy

2.1 Active Pharmaceutical Ingredients (APIs) and Excipients

Uganda does not have a strong domestic API production base, so APIs must be imported.

- Primary API sourcing regions:

- o India & China – Major global API suppliers.

- o Europe (Germany, Switzerland, Italy) – High-quality APIs for oncology & biosimilars.

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- o Africa (South Africa, Egypt, Morocco) – Growing pharmaceutical hubs for APIs.

- Procurement Strategy:

- o Develop multiple supplier relationships to reduce dependency on a single country.

- o Use long-term contracts & Just-in-Time (JIT) supply models for high-cost APIs.

- o Maintain buffer stock of critical APIs (especially for oncology & vaccines).

2.2 Packaging Materials

- Primary packaging: Blister packs, vials, ampoules, pre-filled syringes, HDPE bottles.

- Secondary packaging: Cartons, labels, package inserts.

- Suppliers: Kenya, India, UAE, Turkey, and South Africa.

2.3 Supply Chain for Biologics (Vaccines & Biosimilars)

- Requires specialized cold chain logistics (-80°C for some biologics).

- Partnerships with international biotech firms for technology transfer.

- Cold storage facilities at airports & major distribution hubs.

3. Inventory & Warehousing Management

3.1 Raw Material Storage

- Separate storage for APIs, excipients, and packaging materials.
- Cold storage for temperature-sensitive raw materials.
- Real-time inventory tracking system (ERP-based).

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3.2 Finished Goods Warehousing

- GMP-compliant warehouses with segregated storage for different product categories.
- First-Expiry-First-Out (FEFO) system for batch management.
- Strategic warehouse locations in Uganda and EAC countries for faster distribution.

4. Production & Manufacturing Workflow

4.1 Manufacturing Planning & Scheduling

- Lean Manufacturing to reduce waste and optimize production costs.
- Batch Manufacturing Record (BMR) & Batch Production Record (BPR) compliance.

4.2 Facility Layout & Integration with Supply Chain

- Segregated production lines for oncology, OSD, syrups, vaccines, and biosimilars.
- Integrated Manufacturing Execution Systems (MES) for real-time monitoring.
- Automated raw material dispensing & production tracking.

5. Quality Control & Regulatory Compliance

- Stringent QC checks for APIs, excipients, and packaging materials.
- Real-time product traceability using barcoding & serialization.
- Compliance with WHO-GMP, EAC regulatory frameworks, US FDA, EMA.
- Regular audits of suppliers & third-party logistics providers.

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6. Cold Chain & Storage for Vaccines & Biosimilars

- Cold storage at 2-8°C, -20°C, and -80°C.
- Real-time temperature monitoring
- Strategic cold chain hubs in Kampala & EAC export markets.

7. Distribution & Logistics Strategy

7.1 Domestic Distribution (Uganda)

- Direct distribution to hospitals, pharmacies, and wholesalers.
- Regional distribution centres in Kampala, Entebbe, Jinja, and Mbarara.
- GPS-tracked delivery fleet with route optimization for last-mile delivery.

7.2 East African Community (EAC) Export Strategy

- Key export markets: Kenya, Tanzania, Rwanda, Burundi, South Sudan, DRC.
- Regulatory harmonization under the EAC-MRH program for seamless cross-border approvals.

- Dedicated pharmaceutical logistics hubs in Nairobi (Kenya) & Dar es Salaam (Tanzania).

7.3 International Exports

- Oncology & biosimilars exports to high-value markets in Europe & Middle East.
- Collaboration with global pharmaceutical distributors.

8. Risk Management & Supply Chain Resilience

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8.1 Key Risks in the Supply Chain

- Dependency on imported APIs → Risk of supply chain disruptions.
- Regulatory delays → Impact on product approvals & market entry.
- Cold chain failures → Risk for vaccines & biologics.
- Political & economic instability in Uganda & EAC → Disruptions in logistics.

8.2 Risk Mitigation Strategies

- Dual sourcing strategy for APIs & critical raw materials.
- Investment in API manufacturing in Uganda (long-term goal).
- Use of digital supply chain solutions (AI-driven demand forecasting).
- Collaboration with global logistics providers (DHL, UPS, local 3PLs).

9. Digital Transformation in Supply Chain Management

9.1 AI Based Monitoring

- AI-based demand forecasting to optimize inventory levels.

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Cost of Maintaining a Robust Pharmaceutical Supply Chain in Uganda
1. Key Cost Components of a Pharmaceutical Supply Chain

Cost Component
Raw Material & API Procurement
Packaging Material Costs
Warehousing & Cold Storage
Manufacturing & Production Logistics
Quality Control & Compliance
Cold Chain Infrastructure (Vaccines & Biologics)
Distribution & Last-Mile Delivery
IT, AI, and Digital Supply Chain Systems
Risk Management & Contingency Planning
Regulatory Compliance & Audits

Note: The cost varies depending on production scale, market reach, automation levels, and regulatory requirements.

2. Breakdown of Major Cost Elements

2.1 Raw Material & API Procurement Costs

- APIs account for 60%-80% of drug production costs, particularly for oncology & biologics.
- Uganda imports most APIs from India, China, and Europe, incurring import duties & logistics expenses.
- Cost fluctuations due to supply chain disruptions, geopolitical tensions, and regulatory constraints.
- Mitigation: Multi-supplier agreements & local API production investments.

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2.2 Packaging Material Costs

- Primary packaging (blister packs, vials, ampoules, prefilled syringes) is imported or locally sourced.

- Secondary packaging (cartons, labels, inserts) needs compliance with international standards (e.g., serialization, barcoding).

- Sustainable & eco-friendly packaging costs 10%-15% more but enhances regulatory acceptance.

2.3 Warehousing & Cold Storage

- Cold storage for vaccines & biosimilars (-80°C to 2-8°C) incurs high energy & maintenance costs.

- Storage facilities in Kampala & regional hubs (e.g., Jinja, Mbarara) require advanced temperature monitoring.

2.4 Manufacturing & Production Logistics

- Includes the cost of transporting raw materials, managing production scheduling, and handling batch-to-batch variations.

- Biologics and vaccines require specialized production environments, increasing utility & maintenance costs.

2.5 Quality Control & Compliance

- Regulatory costs include GMP audits and licensing fees.

- Maintaining high-end analytical labs for testing APIs, excipients, and final products.

- Oncology & biosimilar testing requires more stringent & expensive analytical processes.

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2.6 Cold Chain Infrastructure for Vaccines & Biologics

- Ultra-low temperature storage (-80°C) for certain biologics.

- GPS-tracked refrigerated trucks for secure transportation.

- Backup power systems to ensure uninterrupted temperature control.

2.7 Distribution & Last-Mile Delivery

- Cross-border regulatory compliance costs (for EAC markets).

- Investment in regional distribution hubs (e.g., Nairobi, Dar es Salaam).

- High-security transportation for oncology drugs to prevent theft.

2.8 IT & Digital Supply Chain Systems

- Track-and-trace systems to prevent counterfeit drugs.

- Automated inventory management to reduce wastage & improve efficiency.

2.9 Risk Management & Contingency Planning

- Emergency stockpiling of critical APIs & biologics.

- Alternative supply routes & backup logistics partners.

- Investment in local API production to reduce import dependency.

2.10 Regulatory Compliance & Audits

- WHO-GMP certifications for global market access.

- Annual site inspections & compliance renewals.

- Investment in training & capacity building for regulatory teams.

3. Cost Optimization Strategies

A. Local API & Packaging Manufacturing Investment

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- Reduce import costs by establishing API & packaging material production in Uganda.

- Collaboration with African pharmaceutical hubs (South Africa, Egypt, Morocco).

B. Multi-Supplier & Regional Sourcing Strategy

- Reduce risk by sourcing APIs from multiple suppliers.
- EAC regional integration to reduce cross-border tariffs & logistics delays.

C. Green Energy & Sustainability Investments

- Solar-powered cold storage to cut energy costs by 30-40%.
- Eco-friendly packaging to reduce regulatory risks & enhance marketability.

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G. HUMAN RESOURCES

In this project below listed facilities/blocks are planned to be constructed:

1. Oral Solid Dosage (OSD) Facility (Capacity Production of 15 million units/day (tablets, capsules, and syrups).

2. Biopharma Drug Product Facility (Capacity: Specialized for monoclonal antibodies, hormones, and insulin analogs)

3. Vaccine Formulation and Fill- Finish Facility (50 Million doses per annum)

4. Oncology OSD Facility

5. Injectable Facility (Capacity: High-speed filling lines for vials, ampoules, and pre-filled syringes)

Number of employees with different skill set shall be employed for all above mentioned facilities.

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Summarized table for estimated number of man power required for Operations

Table No.: 16

Plant/Section Number of Personnel

Oral Solid Dosage (OSD) 80

Biopharma Drug Product 80

Vaccine Formulation 70

Oncology 50

Injectable 70

Warehouse 40

Quality Control 62

Quality Assurance 43

Engineering 80

HR and Admin 65

Total 640

i) Hiring of employees with skill sets and expertise:

It is important to have healthy employees with aseptic skills especially in manufacturing units to avoid any contamination during manufacturing of drug products.

Separate organograms are prepared for engineering, corporate, HR & administration, warehouse, Quality (QC & QA) and manufacturing units to hire sufficient and effective workforce considering cost effectiveness.

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Details of Organograms:

- Corporate -Employees with managerial skills, project skills and leadership skills etc. will be hired in corporate for different levels like CEO, Head of engineering, Head of operations/manufacturing, project manager, Head of quality, Head of finance, Head of logistics and Head of HR. Managers and group leaders from each section will be managed/lead by corporate Heads.
- Engineering -In engineering section, employees with skills like automation, instrumentation including complicated manufacturing equipment, electrical, mechanical, utilities (Water, steam, Air) and information technology will be hired at different levels.
- HR & Administration -In HR & admin section, employees with skills to handle transport, canteen, legal challenges like pollution control, employee act and licensing application and renewal etc., recruitment and day to day management of contractual / fixed employees, employee wellness, medical facility and general management of BMS like furniture, rest rooms, and external vendors for admin support will be hired.
- Warehouse -In warehouse section, skills to ensure the safe, efficient, and compliant movement of materials and products is required. It includes logistics management skills. Knowledge of inventory management using software like SAP or ERP, document and material management is essential in warehouse.
- Quality Unit -Quality (Quality control and Quality assurance) is critical

to ensuring the safety, efficacy, and compliance of drug products. In

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QC, testing skills, knowledge of GLP, Data analysis and sample management are essential skills. In QA, Process validation and control, regulatory compliance, Risk Management, Documentation, Audits and inspection skills, Change management and trainings are essential skills.

- Manufacturing -In manufacturing, knowledge of GMP, operation of complicated/automated equipment or system, knowledge of process and aseptic behaviour/environment, documentation and technical skills for problem solving in day-to-day process are crucial skills.

- ii) Health, Safety standards for Human Resource working in Pharmaceuticals:

For biopharmaceutical projects, regular medical check-up including tests like eyesight, audiometry, blood test, breathing test, X-Ray, ECG and basic tests; at-least annually for all employees is essential.

Key aspects in HR feasibility

- Workforce Availability

Over the course of the last decade, Uganda's economic growth has ranked among sub-Saharan Africa's strongest; indeed, the country's annualized average growth rate was 5.4% between 2010 and 2019 (World Bank, 2020).

Uganda has a young and growing workforce, with over 75% of its population under the age of 30.

Total population: Approx. 51 million (2025 estimates)

- Cost of Human Resources

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In Uganda, employee salaries are relatively low, but there are also some differences. For example, in the capital city of Kampala, the monthly salary of an ordinary employee is usually between 100-200USD. In other cities, this number may be even lower. The salary level will be higher for senior management and technical talents.

Average monthly salaries (estimates):

- o Unskilled worker: \$100–\$200

- o Mid-level professional: \$500–\$1,500

- o High-skilled/managerial: \$2,000+

- Legal and Compliance Factors

Key employment laws:

- o Employment Act (2006): Governs employment terms, termination, wages, and working conditions.

- o Occupational Safety and Health Act (2006): Ensures workplace safety.

- o Workers' Compensation Act (2000): Covers workplace injury compensation.

Uganda's labour laws are primarily governed by the Employment Act of 2006, which establishes the minimum standards of employment in the country.

Uganda is committed to upholding international labour standards and has ratified several key International Labour Organization (ILO)

Conventions. These include the Freedom of Association and Protection of the Right to Organize Convention, 1948 (No. 87), the

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Right to Organise and Collective Bargaining Convention, 1949 (No. 98), and the Collective Bargaining Convention, 1981 (No. 154).

The Uganda Employment Act applies to all employees, irrespective of nationality that the employers hire under the service contract.

- o According to the employment act 2006 and labour laws in Uganda, the standard working hours of an employee is a maximum of eight hours per day and 48 hours a week; excluding lunch and worship time.

Employers planning to hire or relocate international employees to Uganda must understand the processes surrounding work permits and visas. Ensuring compliance with Ugandan immigration laws helps streamline the hiring process and avoid potential legal hurdles.

- Cultural and Language Considerations

Ugandan communication often leans towards indirectness, especially when delivering negative feedback or expressing disagreement.

Cultural Lens: The Ugandan concept of "Gwasi" (shame) discourages public criticism. This can influence communication, leading individuals to deliver critical messages indirectly to avoid causing shame to the recipient.

Uganda is a multilingual nation like many African countries. However, English is targeted as a medium of instruction and language of examination.

H. RISK ASSESSMENT
Risk Assessment and Mitigation Strategies for Sunshine Plant in Uganda

1. Political Risks and Mitigation Strategies

Uganda, like many developing economies, has political uncertainties that can affect business operations. These include regulatory changes, policy instability, and geopolitical factors.

1.1 Key Political Risks

Table No.:17

Risk Factor	Potential Impact
Changes in pharmaceutical laws, taxation, or Regulatory licensing requirements could delay operations or Uncertainty increase compliance costs.	
Government Policy Import/export restrictions, price controls, or changes Shifts in incentives may affect profitability.	
Lengthy approvals for manufacturing licenses, Bureaucratic Delays product registrations, and environmental permits may slow down market entry.	
Conflicts or tensions in neighbouring East African Geopolitical Instability Community (EAC) countries could disrupt supply in EAC Region chains and exports.	
Intellectual Property & Weak enforcement of patents and trademarks may Patent Protection lead to counterfeiting risks.	

1.2 Risk Mitigation Strategies

I. Early Engagement with Regulators:

- Work closely with Uganda’s National Drug Authority (NDA) and the Ministry of Health to ensure timely approvals.
- Seek WHO-GMP certification for global market access.

II. Public-Private Partnerships (PPP):

- Collaborate with government bodies to access tax benefits, incentives, and investment protection.

III. Diversification of Export Markets:

- Establish trade links with EAC, COMESA, and beyond to reduce dependence on a single market.

IV. Strengthening IP Protection:

- Register patents and trademarks regionally through ARIPO (African Regional Intellectual Property Organization) to safeguard innovations.

2. Economic Risks and Mitigation Strategies

Economic instability can impact capital investment, supply chain costs, and market demand for pharmaceutical products.

2.1 Key Economic Risks

Table No.:18

Risk Factor	Potential Impact
Exchange rate volatility can increase the cost of Currency Fluctuations imported raw materials (APIs, excipients, packaging materials).	
Inflation & Rising High inflation may lead to increased costs for labor, Operational Costs utilities, and raw materials.	
Supply Chain Dependence on international suppliers for APIs and Disruptions critical components could lead to shortages.	
Limited Access to High-interest rates and limited government Affordable Financing incentives can make project financing costly.	
Market Demand & Price-sensitive consumers and limited insurance Price Sensitivity coverage may affect product affordability.	

2.2 Risk Mitigation Strategies

- Localizing API Sourcing:
 - Develop partnerships with regional API manufacturers to reduce reliance on imports and mitigate forex risks.
- Hedging Against Currency Risks:
 - Utilize foreign exchange hedging instruments to stabilize procurement costs.
 - Maintain a mix of local and foreign currency reserves for operational expenses.
- Operational Cost Optimization:
 - Implement energy-efficient manufacturing to reduce electricity costs.
 - Automate production processes to minimize labour costs.

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- Strategic Supplier Agreements:
 - Enter long-term contracts with multiple suppliers for key raw materials to ensure supply chain resilience.
- Diversifying Revenue Streams:
 - Expand into high-margin segments (oncology, vaccines, biosimilar).
 - Target institutional buyers like the government, NGOs, and international health organizations (e.g., GAVI, WHO, Global Fund).

3. Environmental Risks and Mitigation Strategies

Uganda has stringent environmental regulations for pharmaceutical manufacturing, including waste management, water usage, and emissions control. Non-compliance can lead to fines, production shutdowns, or reputational damage.

3.1 Key Environmental Risks

Table No.:19

Risk Factor	Potential Impact
High water demand for manufacturing and cooling	
Water Scarcity	may lead to shortages, increasing operational costs.
Waste Disposal	Improper disposal of hazardous waste (solvents, Challenges API residues) can lead to environmental penalties.
Energy Dependence	High electricity costs and unreliable grid supply on Non-Renewables may lead to production disruptions.
Climate Change	Increased heat and erratic rainfall could impact Effects cooling systems and energy consumption.
Failure to meet Uganda's National Environment Regulatory Non-Management Authority (NEMA) regulations	can lead Compliance to legal actions.

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3.2 Risk Mitigation Strategies

- Sustainable Water Management:
 - Implement water recycling & rainwater harvesting systems to reduce dependence on municipal supplies.
 - Use high-efficiency Clean-in-Place (CIP) systems to minimize water wastage.
- Advanced Waste Treatment Solutions:
 - Install Zero Liquid Discharge (ZLD) systems for pharmaceutical effluent treatment.
 - Set up on-site hazardous waste incinerators to manage cytotoxic and biological waste safely.
- Renewable Energy Integration:
 - Solar power installations on-site to reduce dependency on the national grid.
 - Waste heat recovery systems to improve energy efficiency.
- Green Manufacturing Practices:

- Obtain ISO 14001 Environmental Certification to align with international best practices.
- Conduct regular environmental audits to ensure compliance with NEMA standards.

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4. Summary of Risk Mitigation Strategies

Table No.:20

Risk Category	Key Risks	Mitigation Strategies
Regulatory	Regulatory uncertainty	Engage early with regulators, form public-private partnerships, shifts diversify export markets
Political	Risks bureaucracy	
Currency	fluctuations	Hedge currency risks, optimize
Economic	inflation, financing	operational costs, diversify
Risks	challenges revenue streams	
Water	scarcity, waste	Implement sustainable waste
Environmental	disposal, high energy	management, use renewable
Risks	costs energy	optimize water usage

5. Conclusion

While setting up a pharmaceutical manufacturing plant in Uganda presents political, economic, and environmental risks, these can be effectively managed through strategic planning and proactive mitigation measures.

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I. SUSTAINABILITY

1. Sustainable Waste Management Strategies

Sunshine facility shall generate solid, liquid, and hazardous waste that must be effectively managed to comply with environmental regulations and protect ecosystems.

1.1 Segregation and Recycling of Solid Waste

- Implement waste segregation at the source to separate biodegradable, non-biodegradable, recyclable, and hazardous waste.
- Recycle packaging materials like paper, plastics, and aluminium foils used in blister packs.
- Partner with certified waste management companies to handle non-recyclable hazardous waste.

1.2 Efficient Handling of Hazardous Waste

- Incinerate cytotoxic oncology waste and contaminated biological waste in high-temperature incinerators (above 850°C).
- Use autoclaves and chemical disinfection for microbiological and vaccine-related waste.
- Adopt encapsulation techniques for disposing of highly toxic chemicals and APIs safely.

1.3 Effluent Treatment and Liquid Waste Management

- Install an Effluent Treatment Plant (ETP) and Zero Liquid Discharge (ZLD) system to ensure wastewater treatment before discharge.
- Implement reverse osmosis (RO) and advanced oxidation processes to remove pharmaceutical residues.
- Reuse treated water for landscaping, cooling systems, and non-potable applications.

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1.4 Minimization of API & Solvent Waste

- Implement solvent recovery units to recycle and reuse solvents in oncology and biopharma production.
- Adopt green chemistry techniques to minimize hazardous waste production.
- Optimize batch sizes and manufacturing processes to reduce raw material waste.

2. Energy Efficiency Strategies

Energy consumption will be a significant operational cost in pharmaceutical manufacturing. A sustainable energy plan shall ensure cost reduction and

reduced carbon footprint.

2.1 Integration of Renewable Energy Sources

- Solar Power Installation:
 - o Install solar panels on rooftops to supplement grid electricity and reduce dependence on fossil fuels.
 - o Target at least 30-50% energy from solar for non-critical operations.
- Biomass & Biogas Utilization:
 - o Utilize biomass boilers for steam generation using agricultural waste.
 - o Install a biogas plant to convert organic waste into energy.

2.2 Energy-Efficient HVAC & Cooling Systems

- Use energy-efficient HVAC systems with variable frequency drives (VFDs) to optimize airflow and minimize energy wastage.
- Implement cold chain optimization for vaccines and biopharma products, including energy-efficient refrigeration and insulation systems.
- Utilize waste heat recovery systems to capture and reuse excess heat from industrial processes.

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2.3 Smart Lighting and Equipment Optimization

- Install LED lighting with motion sensors to reduce power consumption in low-traffic areas.
- Use high-efficiency motors and compressors in production lines to reduce energy waste.
- Implement real-time energy monitoring systems to track and optimize energy use.

3. Water Conservation and Efficient Usage

Pharmaceutical production requires significant water usage for cooling, formulation, and cleaning processes. Water scarcity in Uganda makes conservation and recycling essential.

3.1 Water Recycling & Reuse

- Install Water Treatment Plants (WTPs) with multi-stage filtration, RO, and UV disinfection to reuse wastewater.
- Recycle treated wastewater for boilers, chillers, and non-potable cleaning processes.

3.2 Rainwater Harvesting

- Set up rainwater harvesting tanks to collect and store rainwater for non-production use.
- Utilize rainwater for landscaping, cooling towers, and toilet flushing.

3.3 Optimized Process Water Usage

- Implement Clean-in-Place (CIP) and Sterilize-in-Place (SIP) systems to reduce water usage in equipment cleaning.
- Use dry cleaning methods where possible to minimize the need for extensive water washing.
- Train staff on water conservation best practices to minimize waste.

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4. Regulatory Compliance & Green Certifications

- WHO-GMP & EAC Environmental Regulations: Ensure compliance with local and international environmental guidelines.
- ISO 14001 Certification: Implement environmental management systems to enhance sustainability.
- LEED Certification: Design facilities with sustainable building materials to meet green building standards.

5. Conclusion

Sunshine shall require integrated waste management, energy-efficient operations, and water conservation measures. By adopting renewable energy, recycling waste, and optimizing resource usage, this facility can achieve long-term environmental and economic benefits while maintaining regulatory compliance and corporate social responsibility.

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J. Key HR Challenges and mitigation strategies

1. Skilled Workforce Shortage

- Limited local experience in advanced pharmaceutical manufacturing (especially GMP-compliant operations for vaccines, oncology, and monoclonal antibodies).
- Shortage of experienced quality assurance (QA), quality control (QC), production, validation, and biotech-specific personnel.

2. Training and Capability Gaps

- Need for training in WHO-GMP and global standards.
- Low exposure to advanced equipment and automation technologies common in global pharma plants.

3. High Attrition or Brain Drain

- Trained personnel may be poached by competitors or seek jobs abroad due to better salaries.

4. Cultural & Communication Barriers

- Integration challenges when blending expat professionals with local teams in managerial and technical roles.

Mitigation & Risk Management Strategies

1. Early & Strategic Workforce Planning

- Workforce demand mapping: Identify headcount, roles, and experience levels required for each production phase.
- Staggered hiring aligned with the commissioning of each production unit.

2. Establish a Technical Training Academy

- Set up an in-house Pharma Training Institute or Centre of Excellence.

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- Partner with international experts to develop GMP, EHS, and regulatory curricula.
- Simulate manufacturing environments (cleanrooms, QC labs, aseptic suites).

The training program shall constitute of the following aspects

- a. Induction of new employees
- b. cGMP training
- c. GLP training
- d. On the job training consisting of SOP training, equipment training etc.,
- e. Safety training
- f. Supervisor training
- g. Manager training
- h. Job change training or after leave of absence
- i. Remedial training
- j. Contractor employee training

3. International Partnerships

- Collaborate with Indian, European, or South African pharma companies or universities for knowledge transfer.
- Consider short-term expat secondments (2–5 years) to build and mentor local teams.

4. Collaboration with Local Academia

- Tie up with Makerere University or other local institutions for:
 - o Curricula alignment with pharma needs.
 - o Internship/apprenticeship programs.
 - o Fast-tracking diploma and B.Pharm graduates into technical roles.

5. Talent Retention Schemes

- Offer competitive packages, training sponsorships, and bonded contracts.

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- Career development pathways for technicians to move into supervisory or QA roles.
- Stock options or profit-sharing models to retain senior staff.

6. Outsource Non-Core Activities Initially

- To reduce early HR load, consider outsourcing packaging, engineering maintenance, and logistics until local capacity matures.

7. Human Resources Rollout Roadmap

Phase-Wise HR Rollout Plan (2025–2030)

Phase 1: Planning & Initial Setup (2025)

- Workforce Planning: Finalize detailed manpower requirement for each functional block (production, QA/QC, engineering, admin, etc.).
- Key Roles: Project HR Head, Talent Acquisition Manager, Training & Compliance Manager
- Action Points:
 - o Begin local hiring process for support staff, documentation personnel
 - o Initiate expat hiring for plant design, validation, and commissioning oversight
 - o Establish HR policies, on-boarding modules, payroll systems

Phase 2: OSD & Syrup Commissioning (2026)

- Estimated Staff Requirement: (As per Table No 16)
- Key Functions: Production operators, QA/QC analysts, warehouse team
- Action Points:
 - o Launch training academy in partnership with local universities (Makerere, Mbarara, etc.)
 - o Train local hires in GMP compliance, SOP writing, and safety protocols

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- o Expat engineers and QA managers to lead operations and mentor local staff

Phase 3: Oncology & Injectable Line Setup (2027)

- Estimated Staff Requirement: +- (As per Table No 16)
- Key Functions: Cytotoxic handling experts, sterile Injectables line operators, QC microbiology
- Action Points:
 - o Specialized cytotoxic/Hormonal handling training
 - o Safety and containment procedure modules
 - o Expand academy offerings to include aseptic processing and EHS protocols

Phase 4: Vaccine & Biopharma Commissioning (2028)

- Estimated Staff Requirement: -(As per Table No 16)
- Key Functions: Biotech fill-finish experts, cold chain QA/QC
- Action Points:
 - o Partner with international biotech training programs (India, EU)
 - o Establish a biotechnology learning track in the internal academy

Phase 5: Operations Stabilization & Local Leadership Development (2029–2030)

- Target Workforce: 1,000+ including indirect employees
- Focus Areas: Succession planning, reducing dependency on expats
- Action Points:
 - o Develop career growth ladders for technicians and supervisors
 - o Launch Leadership Development Programs (LDP) for Ugandan managers
 - o Retention strategy implementation (bonuses, ESOPs, continuous training)

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Training Strategy Summary

- In-House Pharma Academy to train employees per year
- Modules in:
 - o GMP & regulatory compliance
 - o Pharmaceutical operations (OSD/liquid/injectable/Oncology)
 - o Aseptic techniques
 - o Equipment maintenance & calibration
- Training Partners: Local academia + global consultants

Risk Mitigation Measures

- Challenge: Limited experienced pharma talent pool

- o Solution: Layered expat hiring, internal academy, local university tie-ups
- Challenge: Attrition post-training
- o Solution: Bonded employment, career development programs, competitive pay
- Challenge: Cultural integration
- o Solution: Cross-cultural workshops, expat-local mentorship structure

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K. STRATEGIES FOR ACCELERATED ROI ON THE PROJECT

1. Phased Commissioning of Production Lines

- Start with OSD and syrups, which have lower regulatory complexity and shorter setup timelines.
- These can begin generating revenue within 12–18 months, while more complex lines (biopharma, oncology, vaccines) are still under validation.
- Helps spread OPEX and initiates early cash flow.

2. Prioritize Products with High Local and Regional Demand

- Focus on essential medicines, high-burden diseases (malaria, diabetes, antibiotics), and therapeutic areas like oncology and vaccines with unmet needs.
- Customize the portfolio to include WHO-prequalified and procurement-friendly products for faster uptake by public health programs.

3. Partner with Global Health Buyers Early

- Engage with donor agencies (Global Fund, GAVI, WHO, USAID) to position your facility as a local supplier.
- Secure advance purchase agreements (APAs) or long-term supply contracts, even during construction phase.
- These contracts anchor demand and improve investor confidence.

4. Leverage Contract Manufacturing (CDMO) for Revenue Boost

- Offer your excess capacity for contract manufacturing of OSD, fill-finish, or vaccine products for regional or global pharma firms.
- Acts as a bridge revenue stream before full-scale launch of your branded portfolio.

5. Accelerate Regulatory Approvals through EAC Harmonization

- Register products using the EAC Medicines Regulatory Harmonization (MRH) program to access multiple countries simultaneously.

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- Cuts down duplicated efforts and approval timelines, speeding up market entry.

6. Optimize Operational Costs from Day 1

- Invest in energy-efficient infrastructure (solar, efficient HVAC) and automation to lower long-term utility and labor costs.
- Design supply chain to minimize storage and transportation overhead.

7. Use a Hybrid Distribution Model

- Combine direct-to-hospital/clinic distribution with national wholesalers and tenders.
- Reduces reliance on a single sales channel and speeds up product uptake across multiple segments.

8. Government Incentives and Tax Planning

- Take full advantage of government incentives, including VAT deferments, customs waivers, tax holidays, and utility subsidies.
- Negotiate land lease concessions or reduced utility tariffs to reduce fixed overheads.

9. Continuous Financial Monitoring

- Set up a Project Management Office (PMO) and cash burn tracking system to detect cost overruns early.
- Implement strict working capital control (inventory turnover, receivables) during operations to stay lean.

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Table No.:21

Example ROI Acceleration Path (Indicative)

Impact on
Phase Focus Timeframe Revenue Stream
ROI
Tender sales, private Early cash
Phase 1 OSD + Syrups Year 1–2
hospitals flow
Oncology + High-margin
Phase 2 Year 2–3 Boost margin
Injectables products
Vaccines + Institutional Long-term
Phase 3 Year 3–5
Biosimilars procurement revenue
Cross- Capacity Parallel
CDMO + Export Year 1–5
Cutting monetization revenue

Assumptions

Table No.:22

CAPEX Revenue (Annual, OPEX (Annual,
Segment
(USD M) USD M) USD M)
OSD & Syrups 43.9 45 15
Oncology 96.25 90 30
Injectables 76 60 25
Vaccines 66 80 35
Biopharma 82.5 100 40
Common blocks 112 - 20
Miscellaneous blocks 25 - 12
Facility commissioning 55 - -
Licensing costs 27.5 - -
Total 571 375 165

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Financial Projections (Years 1–10)

Table No.:23

Revenue OPEX Profit Cumulative Cash
Year
(USD M) (USD M) (USD M) Flow (USD M)
1 5 20 - -
2 90 80 60 60
3 150 165 98 158
4 250 165 165 323
5 450 165 298 621
6 500 165 325 946
7 500 165 325 1271
8 500 165 325 1596
9 500 165 325 1921
10 500 165 325 2246

Key Insights

- Break-Even Year: Year 4 (Cumulative cash flow surpasses initial CAPEX of USD 529M)
- Annual ROI Post-Break Even: ~39.7% (Profit of USD 210M / USD 529M)
- Cumulative ROI by Year 10: ~3.4x return (USD 1795M cumulative cash flow)

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L. SWOT ANALYSIS

Strengths

1. Robust Financial Backing
 - o The promoters' strong financial background ensures ample capital availability for high-tech manufacturing infrastructure, reducing financing risks.
 - o USD 530 million CAPEX provides a solid foundation for a state-of-the-art facility with capacity for future expansion.
2. Strategic Product Portfolio
 - o Manufacturing a diverse mix of high-demand therapeutics—from essential OSDs and syrups to complex oncology drugs, vaccines, and biosimilars—positions the company to address both local and regional health needs.

- o Biosimilars and oncology injectables offer high-margin opportunities with increasing global and regional demand.
- 3. Location Advantage – Uganda & EAC Access
 - o Uganda provides duty-free access to the East African Community (EAC), reducing export barriers and expanding market reach.
 - o Strategic geographic location in Central-East Africa allows access to under-served pharmaceutical markets in South Sudan, Rwanda, Burundi, DRC, etc.
- 4. Modern Infrastructure and Technology Potential
 - o Opportunity to design a greenfield facility aligned with WHO-GMP, EMA, and USFDA standards from the outset.
 - o Flexibility to incorporate cutting-edge technologies (e.g., continuous manufacturing, single-use systems for biologics).

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Weaknesses

1. Lack of Pharmaceutical Domain Experience

- The promoters' background in infrastructure, while valuable in construction and logistics, does not directly translate to the complex and highly regulated pharmaceutical industry.
- Manufacturing pharmaceuticals—especially oncology, vaccines, and biosimilars—requires deep expertise in GMP compliance, validation protocols, aseptic processing, and regulatory science.
- Without experienced pharma leadership, there is a risk of non-compliance, production inefficiencies, or market entry delays.
- Mitigation: Early recruitment of experienced pharma professionals and regulatory consultants is critical, as is establishing a governance board with pharma advisors.

2. High Dependence on Imports for APIs, Equipment & Components

- Uganda lacks a strong API and excipient manufacturing ecosystem, so nearly all critical raw materials, consumables, and equipment must be imported—mainly from India, China, or Europe.
- This increases lead times, procurement costs, customs complexities, and exposure to global supply chain disruptions.
- Biopharma and vaccine components (e.g., single-use systems, cold-chain logistics, cell lines) are even more sensitive to supply delays.

3. Skills and Talent Gap in Local Workforce

- Uganda's pharmaceutical industry is still developing, with limited availability of skilled technicians, validation engineers, QC analysts, and GMP-trained production staff, especially in sterile manufacturing or biotech.
- Over-dependence on expats may drive up costs and hinder the development of sustainable local capability.
- Cultural integration, language, and training gaps can affect productivity and retention if not addressed proactively.

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4. High Operating Costs for Complex Products

- Oncology, vaccines, and bio-similars require sophisticated cleanroom environments, cold chain management, and batch-specific quality testing.
- Maintaining aseptic conditions, HVAC, water systems (PW/WFI), and validation for multiple production lines significantly increases OPEX.
- There is also an increased risk of batch failures, which can result in high rework or discard costs—especially in biologics.

Opportunities

1. Growing Demand in Uganda and EAC for Quality Medicines

- The East African Community (EAC) is home to over 280 million people with growing health burdens like cancer, diabetes, infectious diseases, and vaccine-preventable illnesses.
- There's high demand for affordable, high-quality generics, bio-similars, and essential vaccines.
- Current reliance on imports from India, Europe, and China creates a gap for trusted local manufacturers with faster market responsiveness and cost advantages.

2. Support from Government and International Health Agencies

- Uganda and other EAC governments are actively promoting local manufacturing through:
 - o Tax breaks, incentives, and infrastructure subsidies.
 - o Potential fast-track approvals for local production of priority medicines.

- Global health organizations like Gavi, WHO, UNICEF, Global Fund, and African CDC are eager to support African-based production, especially for vaccines and pandemic preparedness.
- This creates opportunities for donor-funded procurement, technology transfer, and concessional financing.

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3. Regional Regulatory Harmonization (EAC-MRH)

- The EAC's Medicines Regulatory Harmonization initiative reduces the complexity and time for market entry across multiple countries.
- A product registered under EAC-MRH can be fast-tracked for approval in all EAC member states, expanding your potential market and improving return on investment (ROI).

4. First-Mover Advantage in Advanced Therapies

- There are very few players in the region manufacturing oncology products, bio-similars, or vaccines. A well-executed entry into these segments provides:
 - o High-margin sales.
 - o Competitive edge due to regulatory barriers that deter others from entering.
 - o Leverage to license or co-develop products with international partners.

5. Greenfield Facility – Built for Future-Ready Compliance

- Starting from scratch allows for:
 - o State-of-the-art modular facility design that meets WHO-GMP and global requirements from day one.
 - o Scalability—facilities can be expanded based on market growth or strategic partnerships.
 - o Integration of renewable energy systems, automation, which can lower long-term operating costs and improve environmental credentials.

6. Technology Transfer and Global Collaborations

- Many mid-sized Western and Asian pharma companies are seeking CDMO or joint venture partners in Africa to enter emerging markets.
- This facility, once validated and operational, can be positioned as a fill-finish hub or contract manufacturing site for such companies—especially for vaccines and biologics.

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Threats

1. Regulatory and Compliance Challenges

- o Ensuring continuous compliance with WHO-GMP, NEMA (Ugandan environmental laws), NDA, and other international standards requires highly skilled QA/QC teams and constant oversight.

2. Geopolitical and Trade Instability

- o Political tensions, conflicts in neighbouring countries, or disruptions in trade routes could hamper logistics, exports, or sourcing of raw materials.

3. Currency Fluctuations and Import Dependency

- o Heavy reliance on imports for APIs and machinery subjects the plant to foreign exchange risks, especially if the Ugandan Shilling depreciates.

4. Intensifying Global Competition

- o Global pharmaceutical companies and Indian manufacturers could enter the region with cheaper generic drugs, impacting price competitiveness.

5. Product-Specific Challenges

- o Biosimilars and vaccines involve complex regulatory pathways, high R&D costs, and the risk of batch failures or cold-chain disruptions.

6. Construction & Commissioning Delays

- Incomplete funding during the CAPEX phase can halt construction, delay procurement of equipment, and postpone validation activities.
- Pharmaceutical projects are highly sequential: civil work must be completed before HVAC installation, which precedes equipment commissioning, which in turn precedes qualification. A break at any point stalls everything.

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7. Regulatory Non-Compliance

- If the facility is only partially built or validated, it cannot obtain GMP licenses or product approvals.
- This puts the company in a no-revenue, high-burn position, severely affecting investor confidence and credibility.

8. Supply Chain Penalties

- Equipment vendors and raw material suppliers often work with milestone-based payments. Funding gaps can lead to order cancellations, higher renegotiated prices, or contractual penalties.
- Lead times in pharma are long, so any reordering due to lapsed payments causes multi-month delays.

9. Reputational Damage & Loss of Stakeholder Confidence

- Financial instability affects relationships with regulators, investors, banks, and potential customers (like international health agencies).
- Once perception of financial distress sets in, it is harder to secure bridge financing or partnerships.

Strategic Recommendations

- Hire pharma domain experts in key leadership roles: manufacturing, regulatory, quality assurance, and R&D.
- Form strategic alliances with experienced CDMOs or technology licensors for oncology and biosimilar lines.
- Invest in workforce development through partnerships with local universities and international training programs.
- Develop dual-source supply chain strategies and build regional sourcing capacity for packaging and excipients.
- Design for modular expansion, allowing scaling based on product demand and market growth.
- Pursue green certifications (LEED, ISO 14001) to attract ESG-aligned investors and NGOs.

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M. SUMMARY AND RECOMMENDATIONS

This chapter synthesizes the key findings from the detailed feasibility study and outlines strategic recommendations to guide the implementation of the Sunshine Pharmaceutical Plant project in Uganda.

The analysis takes into account market dynamics, regulatory frameworks, technical and financial evaluations, supply chain logistics, and human resource considerations, all of which contribute to a comprehensive assessment of the project's viability.

Summary of Key Findings

Market Opportunity and Demand

The study confirms that Uganda's pharmaceutical market is experiencing robust growth driven by increasing demand for affordable, high-quality generics and specialized drugs, including oncology treatments and vaccines. With a projected market revenue reaching approximately US\$329 million in 2025 and a promising CAGR for both the domestic and broader East African markets, the opportunity to capture unmet demand is significant.

Regulatory and Technical Feasibility

Uganda's regulatory framework, led by the National Drug Authority (NDA), is increasingly aligned with international standards, including WHO-GMP guidelines. The regulatory environment supports local production through initiatives like "Buy Uganda, Build Uganda," while technical assessments underscore the availability of advanced manufacturing technologies. The proposed facility's phased approach—from Oral Solid Dosage forms to advanced biopharmaceutical and vaccine production—ensures scalability and adaptability to evolving market needs.

Financial Viability

The financial analysis demonstrates a strong investment proposition:

- CAPEX: The total investment is estimated at US\$529 million, strategically distributed across various manufacturing units.

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- OPEX and Break-Even: Annual operational expenditures are projected at approximately US\$335 million, with break-even achieved within the first year due to efficient cost management and high contribution margins.
- ROI and Profitability: Projected ROI stands at around 28% by Year 3, with an IRR estimated between 18-22%, indicating a healthy profit

trajectory and the potential for cumulative returns exceeding 125% over five years.

Supply Chain and Human Resources

Robust supply chain strategies—including multi-sourcing of APIs, strategic partnerships for cold chain logistics, and digital monitoring systems—ensure resilience against market fluctuations and global disruptions. Additionally, the availability of a young, growing workforce in Uganda supports the project's operational requirements, with competitive labour costs and targeted skills development across manufacturing, quality control, and regulatory compliance.

Recommendations

1. Prioritize High-Margin Segments

Focus Areas:

- **Oncology Drugs:** Given the rising burden of cancer in Uganda and East Africa, investing in the production of generic chemotherapy drugs can capture an underserved market. These drugs not only command higher margins but also improve patient access to critical treatments.
- **Biologics and Vaccines:** The increasing demand for advanced therapies such as vaccines and biologics—including monoclonal antibodies and bio-similars—presents an opportunity to tap into segments with relatively lower competition and high value. The facility's phased approach, starting with Oral Solid Dosage forms and expanding into biopharmaceutical products, supports this focus.

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Rationale:

- **Market Trends:** The feasibility study highlights strong growth projections in these segments, driven by unmet healthcare needs and increasing government investments in healthcare infrastructure.
- **Financial Benefits:** High-margin products contribute significantly to profitability and help accelerate the return on investment (ROI). By emphasizing these areas, the project can achieve robust financial returns and capture a larger share of the regional market.

2. Optimize API and Raw Material Sourcing

Short-Term Strategy:

- **Leverage International Suppliers:** Initially, the project should secure long-term contracts with reputable API suppliers in India and China. This will help stabilize costs and ensure a reliable supply chain during the early phases of operation.

Long-Term Strategy:

- **Local Production Initiatives:** Explore opportunities for setting up local or regional API manufacturing facilities. This approach will not only reduce import dependency and associated logistics risks but also potentially lower production costs over time.
- **Diversification:** Develop a multi-sourcing strategy to mitigate risks associated with geopolitical tensions, supply chain disruptions, or fluctuations in raw material prices.

Rationale:

- **Cost Efficiency:** APIs represent a significant portion of production costs. Optimizing sourcing will directly impact overall operational efficiency and cost management.
- **Risk Mitigation:** Diversification and local production can shield the project from external market shocks and ensure continuity of operations.

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3. Invest in Advanced Technologies

Technology Implementation:

- **Automation and Digitalization:** Adoption of automated production lines, Manufacturing Execution Systems (MES), and IoT-based monitoring systems can streamline operations, reduce labour costs, and enhance precision.
- **Quality Control Systems:** Investment in advanced analytical testing equipment will support stringent quality assurance protocols. Digital systems also facilitate real-time data monitoring, which is critical for maintaining compliance with WHO-GMP standards.

Rationale:

- **Operational Efficiency:** Cutting-edge technology reduces the risk of human error, increases production speed, and improves overall product consistency.
- **Regulatory Compliance:** High-tech systems ensure that manufacturing processes meet international standards, making it easier to secure regulatory approvals and expand into export markets.
- **Scalability:** Advanced technologies support a phased production approach, allowing the facility to scale operations smoothly as market demand grows.

4. Enhance Strategic Collaborations

Collaboration Models:

- **Public-Private Partnerships (PPPs):** Engage with government agencies and international health organizations to secure financial incentives, regulatory support, and access to infrastructure such as Special Economic Zones (SEZs).
- **Industry Alliances:** Form strategic alliances with global pharmaceutical companies for technology transfer, joint ventures, or co-investment. This can accelerate the learning curve and enhance competitive positioning.

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- **Local Partnerships:** Collaborate with local distributors, healthcare providers, and academic institutions to strengthen market access and foster innovation.

Rationale:

- **Regulatory and Financial Support:** Strategic partnerships can reduce upfront investment risks and expedite regulatory approvals.
- **Market Penetration:** Collaborations with established players can facilitate faster market entry, improve distribution networks, and expand customer reach.
- **Innovation:** Joint ventures and alliances can bring in new technological insights and operational expertise, enhancing the project's long-term viability.

5. Strengthen Supply Chain Resilience

Key Measures:

- **Robust Logistics Network:** Develop an integrated logistics system that includes reliable transportation, warehousing, and cold chain infrastructure. This is crucial for managing both raw materials and finished products, especially for temperature-sensitive biologics and vaccines.
- **Digital Supply Chain Solutions:** Implement demand forecasting and real-time inventory management systems to optimize supply chain operations and minimize waste.
- **Contingency Planning:** Establish dual sourcing strategies and maintain buffer stocks to mitigate the impact of any supply disruptions.

Rationale:

- **Efficiency and Reliability:** A resilient supply chain ensures that production processes are not interrupted by external factors, which is critical for meeting market demand consistently.
- **Cost Management:** Effective supply chain management reduces logistics costs and improves overall operational margins.

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- **Risk Mitigation:** Proactive risk management through diversification and advanced monitoring systems will safeguard the project against unforeseen disruptions.

6. Focus on Workforce Development

Workforce Strategy:

- **Training Programs:** Invest in specialized training and capacity-building initiatives to develop the technical, regulatory, and operational skills needed across various departments such as manufacturing, quality control, and engineering.
- **Talent Acquisition:** Attract skilled professionals by offering competitive remuneration packages and career development opportunities. This includes hiring experts with experience in high-containment manufacturing, digital automation, and international regulatory compliance.
- **Collaborations with Educational Institutions:** Forge partnerships with local universities and technical institutes to create a pipeline of

skilled labour tailored to the project's needs.

Rationale:

- **Operational Excellence:** A well-trained workforce is essential to maintain high standards of product quality, regulatory compliance, and overall operational efficiency.
- **Sustainability:** Continuous professional development not only improves immediate operational outcomes but also contributes to long-term sustainability by fostering innovation and adaptability.
- **Competitive Advantage:** Investing in human capital builds organizational strength and ensures that the facility remains competitive in an evolving market landscape.

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Conclusion

The comprehensive feasibility study strongly supports the viability of the Uganda Pharma Plant project. With favourable market conditions, a supportive regulatory framework, robust financial projections, and effective risk mitigation strategies, the project is not only feasible but poised for success. Strategic investments in technology, local partnerships, and human capital will further enhance the plant's competitive edge and contribute to sustainable economic growth in Uganda and the broader East African region.

In light of these findings, it is recommended that the project proceed as planned, with careful attention to the outlined recommendations to ensure operational excellence and long-term profitability.

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List of Abbreviations

ADE : Adverse Drug Event
ADR : Adverse Drug Reaction
AEFI : Adverse Events Following Immunization
AI : Artificial Intelligence
API : Active Pharmaceutical Ingredients
ARIPO : African Regional Intellectual Property Organization
BB : Biobridge
BCG : Bacillus Calmette-Guérin
BEP : Break Even Point
BMR : Batch Manufacturing Record
BOPV : Bivalent Oral Polio Vaccine
BPR : Batch Production Record
CAGR : Compound Annual Growth Rate
CAPA : Corrective and Preventive Action
CAPEX : Capital Expenditure
CEO : Chief Executive Officer
cGMP : Current Good Manufacturing Practices
CIP : Clean-in-Place
COMESA : Common Market for Eastern and Southern Africa
CT : Clinical Trial
EAC : East African Community
EAC : East African Council
ECG : Electrocardiogram
EPI : Expanded program for immunization
ETP : Effluent treatment plants
EU : European Union
FEFO : First-Expiry-First-Out
FR : Feasibility Report
GAVI : Global Vaccine Alliance
GC : Gas Chromatography
GCP : Good Clinical Practice
GDP : Good Distribution Practices
GLP : Good Lab Practices
GMP : Good Manufacturing Practices
HDPE : High-Density Polyethylene

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HEPA : High-Efficiency Particulate Air
HPLC : High Performance Liquid Chromatography
HPV : Human papillomavirus
HR : Human Resource
HVAC : Heating, Ventilation, and Air Conditioning
ICH : International Council for Harmonisation
IPV : Inactivated Polio Vaccine
IQ : Installation Qualification
ISO : International Organisation for Standardisation
JIT : Just In Time

LED : Light-emitting diode
MA : Marketing Authorization
mAbs : Monoclonal Antibodies
MES : Manufacturing Execution Systems
mRNA : messenger Ribonucleic Acid
NCD : Non-Communicable Diseases
NDA : National Drug Authority
NDPA : National drug policy & Authority
NEMA : National Environment Management Authority
NGO : Non-governmental organization
NPC : National Pharmacovigilance Centre
OPEX : Operational Expenditure
OQ : Operational Qualification
OSD : Oral Solid Dosage
PCV : Pneumococcal Conjugate Vaccine
PIC/S : Pharmaceutical Inspection Convention Scheme
PoE : Point of entry
PPB : Pharmacy and Poisons Board
PPP : Public-Private Partnership
PQ : Performance Qualification
QA : Quality Assurance
QC : Quality Control
QPPV : Qualified Person for Pharmacovigilance
R&D : Research and Development
RABS : Restricted Access Barrier Systems
REV : Revision
RO : Reverse Osmosis

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ROI : Return on Investment
RSV : Respiratory Syncytial virus
SCADA : Supervisory Control and Data Acquisition
SEZ : Special Economic Zones
SIP : Sterilize-in-Place
SQM : Square Meter
TFDA : Tanzania Food and Drugs Authority
TRS : Technical Report Series
UAE : United Arab Emirates
UCI : Uganda Cancer Institute
UNICEF : United Nations Children's Fund
UPS : Uninterruptible Power Supply
USAID : United States Agency for International Development
USD : United States Dollar
UV : Ultra Violet
VFD : Variable Frequency Drive
WFI : Water for Injection
WHO : World Health Organization
WTP : Water Treatment Plant
ZLD : Zero Liquid Discharge

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[SHEET: Compiled Sheet vaccines]												
Annual Requirement(M Doses)												
Filling line NATURE OF VACCINE Vaccine Uganda Rwanda Tanzania Kenya Ethiopia Total												
Country Annual Requirement												
Vaccine												
Hexavalent												
LINE 1 BACTERIAL Pentavalent 6 1.5 7.5 5 12.5 32.5												
PCV10 MIC 6 5 11												
PCV13 MIC 1.5 7.5 12.5 21.5												
BCG 2 0.5 2.5 2 4.5 11.5												
Td 0.5 2.5 1.75 4.5 9.25												
Hib 4.5 4.5												
Rota vaccine 4 1 5 5 8.5 23.5												
Tetanus Toxoid 0.5 4.5 5												
INACTIVATED VIRAL IPV 4 1 2.5 1.75 4.5 13.75												
HPV(1,2,3&4) 2 0.5 2.5 1.75 4.5 11.25												
Hepatitis B(Birth &Adult) 4 0.5 4.5												
Yellowfever 2 2 4												
Seasonal Flu(Adult) 2 2												
LINE 2 LIVE VIRAL Measles Rubella 6 1 5 3.5 15.5												
Measles Vaccine 8.5 8.5												
BOPV 7 2 7.5 6.5 16.5 39.5												
BOPV (Ty1and Ty3)												
Note :Dose calculation is based on Birth Cohort and Number of doses required for complete immunization												
Country Birth Cohort												
Ethiopia 4 million												
Kenya 1.5 million												
tanzania 2.4million												
Rwanda 0.4 million												
Uganda 1.7 million												

[SHEET: African Continent Vac Require.]
 | African Continent Vaccine Requirements |||||
 || Vaccination ||||| Annual birth Cohort |
 | SADC Countries | Hep B No of doses | Total Hep B doses | MMR Dooses | Total MMR | DPT Doses | Total DPT
HPV Doses	Total HPV	Yellow fever	YF Doses								
1) Angola	3	4200000	2	2800000	3	4200000	2	5400000	1	1400000	1400000
2) Botswana	3	105000	2	70000	3	105000	2	405000			35000
3) Eswatini (Formerly Swaziland)	3	36000	2	24000	3	36000	2	180000			12000
4) Lesotho	3	90000	2	60000	3	90000	2	375000			30000
5) Malawi	3	1800000	2	1200000	3	1800000	2	3074000			600000
6) DR Congo (Kinshasa)	3	15000000	2	10000000	3	15000000	2	13650000	1	5000000	
5000000											
7) Namibia	3	120000	2	80000	3	120000	2	405000			40000
8) Madagascar	3	1800000	2	1200000	3	1800000	2	3900000			600000
9) Mauritius	3	36000	2	24000	3	36000	2	234000			12000
10) Seychelles	3	6000	2	4000	3	6000	2	17000			2000
11) Comoros	3	90000	2	60000	3	90000	2	110000			30000
12) Tanzania	3	4800000	2	3200000	3	4800000	2	10050000	1	1600000	1600000
13) South Africa	3	3600000	2	2400000	3	3600000	2	11050000			1200000
14) Zambia	3	3000000	2	2000000	3	3000000	2	3150000	1	1000000	1000000
15) Zimbabwe	3	1050000	2	700000	3	1050000	2	2850000	1	350000	350000
16) Mozambique	3	3000000	2	2000000	3	3000000	2	5624000			1000000
East Africa Community											
17) Eritrea	3	150000	2	100000	3	150000	2	600000			50000
18) Djibouti	3	21000	2	14000	3	21000	2	165000			7000
19) Somalia	3	2400000	2	1600000	3	2400000	2	2036000			800000
20) Kenya	3	3900000	2	2600000	3	3900000	2	10200000	1	1300000	1300000
21) Burundi	3	1800000	2	1200000	3	1800000	2	2080000			600000
22) Rwanda	3	1350000	2	900000	3	1350000	2	2720000			450000
23) Uganda	3	4800000	2	3200000	3	4800000	2	8840000			1600000
24) South Sudan	3	1050000	2	700000	3	1050000	2	1485000			350000
25) Sudan	3	2700000	2	1800000	3	2700000	2	5500000			900000
26) Ethiopia	3	6600000	2	4400000	3	6600000	2	16900000			2200000
Middle Africa Community											
27) Gabon	3	105000	2	70000	3	105000	2	400000			35000
28) Congo (Brazzaville)	3	450000	2	300000	3	450000	2	850000			150000
29) Central Africa Republic	3	750000	2	500000	3	750000	2	660000			250000
30) Guinea Equatorial	3	75000	2	50000	3	75000	2	250000			25000
31) Cameroun	3	2400000	2	1600000	3	2400000	2	4030000			800000
32) Chad	3	2100000	2	1400000	3	2100000	2	2260000			700000
West Africa Community											
33) Burkina Faso	3	1950000	2	1300000	3	1950000	2	3870000			650000
34) Niger	3	2700000	2	1800000	3	2700000	2	3026000			900000
35) Nigeria	3	24000000	2	16000000	3	24000000	2	31200000			8000000
36) Togo	3	510000	2	340000	3	510000	2	1300000			170000
37) Guinea Bissau	3	105000	2	70000	3	105000	2	265000			35000
38) Guinea Republic	3	1950000	2	1300000	3	1950000	2	1760000			650000
39) Senegal	3	1350000	2	900000	3	1350000	2	3000000			450000
40) Sierra Leone	3	1050000	2	700000	3	1050000	2	1100000			350000
41) Liberia	3	900000	2	600000	3	900000	2	700000			300000
42) Cote D'Ivoire	3	2250000	2	1500000	3	2250000	2	4160000			750000
43) Ghana	3	2250000	2	1500000	3	2250000	2	5700000	1	750000	750000
44) Mali	3	2700000	2	1800000	3	2700000	2	2400000	1	900000	900000
45) Cape Verde	3	36000	2	24000	3	36000	2	115000	1	12000	12000
46) Mauritania	3	750000	2	500000	3	750000	2	600000			250000
Total Doses	3	111885000	74590000	3	111885000	178646000	12312000	37295000			
Yellow fever given to travelers in majority of countries											
MMR vaccine is administered in Infants					At Birth					37.295	
HPV is administtered only to girls of 9 to 14 age which represents around 15% of total population											
		Adults	Travellers								
Vaccine	Doses Required	20% additional									
Hepatitis B doses requirement	111885000	134262000									
DPT doses required	111885000	134262000									
Yellow fever doses requirement	12312000	14774400									
MMR Dose requirement	74590000	89508000									
HPV Dose requirement	178646000	214375200									
		587181600									

[SHEET: NPV_IRR]

| Projected Profit & Loss Statement ||||||| COMPUTATION FOR NPV - IRR ||

S. No. | Particulars | Year-1 | Year-2 | Year-3 | Year-4 | Year-5 | Year-6 | Year-7 | Capital Expenditure (year 1) | -571 |

| 1 | ORAL SOLID DOSAGE (OSD) FACILITY (20%) ||||||| -77.6 | -71.85185185185185

| Revenue from OSD | 1 | 24 | 70 | 70 | 102 | 102 | 102 ||| -24.955000000000005 | -21.394890260631005

| 2 | GENERAL PHARMACEUTICAL INJECTABLE FACILITY (15%) |||||||| 137.045000000000002 |

108.79073947060915

| Revenue from Injectable facility | 0.75 | 18 | 52.5 | 52.5 | 76.5 | 76.5 | 76.5 |||

123.045000000000002 | 90.4417482373396

| 3 | BIOPHARMA PRODUCTS FACILITY (15%) |||||||| 233.045 | 158.60651115273097

| Revenue from biopharmaceutical products | 0.75 | 18 | 52.5 | 52.5 | 76.5 | 76.5 | 76.5 ||| 220.045

| 138.66567554749273

| 4 | VACCINE FACILITY (15%) |||||||| 211.045 | 123.14273046809703

| Revenue from vaccines | 0.75 | 18 | 52.5 | 52.5 | 76.5 | 76.5 | 76.5 || IRR | 0.06498097673514391 |

526.4006627637866

| 5 | ONCOLOGY OSD FACILITY (25%) |||||||| NPV || 372.4006627637866

| Revenue from Oncology OSD | 1.25 | 30 | 87.5 | 87.5 | 127.5 | 127.5 | 127.5 ||||

| 6 | CONTRACT MANUFACTURING/ EXPORT (10%) |||||||||

| Revenue from CMO / exports | 0 | 12 | 35 | 35 | 51 | 51 | 51 ||||

B | Gross Revenue from Operations MUSD | 4.5 | 120 | 350 | 350 | 510 | 510 | 510 ||||

C | Expenses |||||||||

| a | Insurance @0.05% of project value | 0 | 2.855 | 2.855 | 2.855 | 2.855 | 2.855 | 2.855 ||||

| b | Chemical API, ready to fill bulk / Antigens for vaccines, Biopharma bulk API | 3 | 20 | 50 | 60 |

70 | 70 | 70 ||||

| c | Excipients | 0.5 | 8 | 10 | 11 | 30 | 30 | 30 ||||

| d | Packaging in MUSD | 0.5 | 7 | 8 | 11 | 18 | 20 | 20 ||||

| e | Labour/Employee expense in MUSD | 5 | 8 | 10 | 10 | 12 | 13 | 15 ||||

| f | Annual Maintenance Cost in MUSD | 2 | 5 | 10 | 10 | 10 | 12 | 12 ||||

| g | Utilities (Power, water, waste management) | 5 | 15 | 29 | 31 | 35 | 38 | 40 ||||

| h | Marketing and sales in MUSD | 0 | 4 | 8 | 8 | 12 | 13 | 15 ||||

| i | Technology transfer fees lumpsum in MUSD | 2 | 5 | 5 | 0 | 0 | 0 | 0 ||||

| j | Other Exp. In MUSD (includes statutory expenses, registration costs, quality assurance, etc.) | 5

| 5 | 8 | 9 | 10 | 12 | 15 ||||

| k | Logistics and Distribution | 2 | 8 | 15 | 17 | 20 | 22 | 22 ||||

| Sub-total | 25 | 87.855 | 155.855 | 169.855 | 219.85500000000002 | 232.855000000000002 |

241.855000000000002 ||||

| D | Gross margin = (Revenue-COGS) / revenue % | -82 | 36.58869728529964 | 55.470000000000006 |

51.4700000000000006 | 56.891176470588235 | 54.3421568627451 | 52.57745098039216 ||||

| E | Breakeven revenue = fixed cost / gross profit margin | -0.3048780487804878 | 2.4011513532431175 |

2.8097169641247515 | 3.300077715173887 | 3.864483275603578 | 4.284978982121272 | 4.5999757593839155 |||

|

| F | EBIT | -20.5 | 32.144999999999996 | 194.145 | 180.145 | 290.145 | 277.145 | 268.145 ||||

| Depreciation and Amortization | 57.1 | 57.1 | 57.1 | 57.1 | 57.1 | 57.1 | 57.1 ||||

| G | EBITDA | -77.6 | -24.9550000000000005 | 137.045000000000002 | 123.045000000000002 | 233.045 | 220.045

| 211.045 ||||

[SHEET: Sheet1]

Manufacturing Block wise area and cost |||

S. No. | Block Name | Sub Unit | Cost in USD

1. | Plant 1 - OSD | Granulation Area | 43,900,000

2. | | Blending & Compression |

3. | | Tablet & Capsule Filling |

4. | | Coating |

| Liquid Manufacturing |

6. | | Filling & Packaging |

7. | | Utilities & HVAC |

8. | | Support Areas (Offices, Change Rooms, Cafeteria, etc.) |

9. | Plant 2-mAbs and Biosimilar | Raw Material & Component Storage | 82,500,000

10. | | Buffer preparation |

11. | | Formulation Suite |

12. | | Aseptic Filling & Stoppering |

13. | | Lyophilization (if required) |

14. | | Capping, Inspection & Labelling |

15. | | Primary & Secondary Packaging |

16. | | Cold Chain Storage & Stability Testing |

17. | | Utilities & HVAC Systems |

18. | | Support Areas (Offices, Cafeteria, Change Rooms, etc.) |

19. | Plant 3-Filling Lines | Vial Washing & Depyrogenation | 76,000,000

20. | | Ampoule Washing & Sterilization |

21. | | Pre-Filled Syringe Preparation |

22. | | Filling & Stoppering (Vials & Ampoules) |

23. | | Filling & Sealing (Pre-Filled Syringes) |

24. | | Lyophilization (if required) |

25. | | Capping & Inspection |

26. | | Labelling & Secondary Packaging |

27. | | Utilities & HVAC Systems |

28. | | Support Areas (Gowning, Change Rooms, Offices, etc.) |

29. | Plant 4-Oncology | Raw Material Staging and component storage | 96,250,000

30. | | Granulation Area |

31. | | Blending & Compression |

32. | | Encapsulation Area |

33. | | Coating (if required) |

34. | | Hormonal & Cytotoxic Segregation |

35. | | Filling & Packaging |

36. | | Waste Management & Decontamination |

37. | | Utilities & HVAC Systems |

38. | | Support Areas (Change Rooms, Gowning, Admin, etc.) |

39. | Plant 5-Vaccine | Vial Washing & Depyrogenation | 66,000,000

40. | | Filling & Stoppering (Vials) |

41. | | Lyophilization (if required) |

42. | | Capping & Inspection |
 43. | | Labelling & Secondary Packaging |
 44. | | Utilities & HVAC Systems |
 45. | | Support Areas (Gowning, Change Rooms, Offices, etc.) |
 46. | Common Blocks | Animal House | 99,472,000
 47. | | Central warehouse |
 48. | | Engineering block |
 49. | | Central utility block |
 50. | | Administration block |
 51. | | Quality control laboratory |
 52. | | Research and development laboratory |
 53. | | Quality assurance section |
 54. | Miscellaneous blocks | Car park | 25,000,000
 55. | | Cafeteria |
 56. | | Training and induction rooms |
 57. | | Medical and first aid rooms |
 58. | | Recreational area for employees |
 59. | | Regulatory affairs department |
 60. | | Clinical research department |
 61. | | Landscaping and horticulture |
 62. | Quality Assurance | Facility Commissioning & validations. | 55,000,000
 63. | Regulatory Affairs | Licensing & Statutory Cost | 27,500,000
 | | TOTAL BUDGET ESTIMATE IN USD | 571,622,000
 | | | SAY 571.62 MUSD

Experience with the first wP based fully liquid
hexavalent vaccine.
EasySix™ Vaccine

R K Suri
Senior Advisor & Former Chief Executive- Biologicals
Panacea Biotech Ltd
New Delhi, INDIA

October 31, 2018
EasySix
FULLY LIQUID wP BASED HEXAVALENT COMBINATION VACCINE

■ What is EasySix ?

■ Clinical development plan

■ Safety profile

■ Immunogenicity

■ Conclusions

1
WHAT IS EasySix® : Composition and Présentation

Fully Liquid Hexavalent Vaccine, each dose of 0.5 ml contains:

Ingredients Quantity per dose of 0.5 ml

Active Component Unit /Dose
Diphtheria toxoid ≥ 30 IU

Tetanus toxoid ≥ 60 IU

Inactivated w-Bordetella pertussis ≥ 4 IU

r Hepatitis B surface antigen ≥ 10 μ g

Haemophilus influenzae type b conjugated (PRP-TT) 10 μ g

Inactivated Salk Polio Virus type 1 40 DU
Inactivated Salk Polio Virus type 2 8 DU
Inactivated Salk Polio Virus type 3 32 DU

Inactive Component
NMT 1.25 mg
Al 3+ as Aluminium phosphate gel
3.3 mg
2-phenoxyethanol

Physiological saline q.s to 0.5 ml

Fully Liquid Hexavalent Vaccine, each dose of 0.5 ml contains
2

EasySix■: Clinical Development Plan

Clinical Studies

Phase I Study Phase II Study

Safety in toddlers : India Efficacy & Safety in infants - India
EasySix® : n = 24 EasySix® : n = 136
15-18 months
VS

PENTAVAC™ + IMOVAX® : n = 136
6 - 10 weeks

Goals of Clinical Development

■ To assess the safety and tolerability of fully liquid hexavalent DTwP-HepB-Hib-IPV vaccine (EasySix)

■ To show that the hexavalent combination vaccine is safe and induces protective immune responses
equivalent (non-inferior) to
previously licensed combination vaccines PENTAVAC™ and IMOVAX®

■ To support the use of the vaccine with vaccination regimen : 6, 10 and 14 weeks

3
Clinical Study
Study Title: To assess the safety and tolerability of fully liquid hexavalent DTwP-Hep B-Hib-IPV vaccine
(EasySix®) of Panacea Biotech Ltd. in healthy subjects

Sr. No. Parameters Results

1 No. of Healthy Subjects 24
(Age 15 - 18 months)
2 No. of Study Centers 02
Dr. Sharad Agarkhedkar
Dr. D.Y. Patil Medical College, Pune.

Dr. Ram K. Dhongade
Sant Dnyaneshwar Medical Education and Research Centre, Pune.
3 Study Duration 2 Months

4 Study Vaccine EasySix® , Single Dose,
Deep Intramuscular
5 ADR Observation Period 6 hours post vaccination and
Upto 7 days post-vaccination
6 Laboratory Tests: CBC (Complete Blood Count)
Hematological Parameters LFT (Liver Functions Test)
KFT (Kidney Function Test)

4
Safety and Reactogenicity

Sr. No. Local Reaction Grade N (%)

1 All 14 (58.33)
Pain / Mild 10 (41.67)
Tenderness
Moderate 4 (16.67)

2 All 11 (45.83)

Swelling Mild 10 (41.67)

Moderate 1 (4.17)

3 All 7 (29.17)

Hardness / Induration Mild 7 (29.17)

Moderate 0 (0)

4 All 6 (25)
Redness / Mild 6 (25)
Erythema
Moderate 0 (0)

Reference: Data on File.: EasySix™ Phase I Clinical Study, Data on File, Protocol No: PBL/CR/2010/04/CT,
Version:2.0 Dated:05-08-10

5
Results & Conclusion

■ Majority of Local Reactions were Mild

■ EasySix® is found to be Safe and Well Tolerated in all 24 Infants in Phase I trial

9
Phase III Trial

Comparative Study
DTwP-Hep B-Hib-IPV vaccine (EasySix®, Panacea Biotec Ltd.)

Versus

Pentavalent (DTwP-Hep B-Hib) (Pentavac®, Serum Institute)
with IPV (Imovax®, Sanofi Pasteur)

10
Publication

11
Phases III Clinical Trial

Study Title
An Open Label, Randomized, Multicenter study to Evaluate and Compare the Immunogenicity and Reactogenicity of EasySix® (Fully Liquid Hexavalent DTwP-HepB-Hib-IPV Vaccine, Panacea Biotec Ltd.) with Pentavac SD® (Pentavalent DTwP-HepB/Hib Vaccine, Serum Institute of India Pvt. Ltd.) co-administered with Imovax Polio® (Salk Based Inactivated Polio Vaccine, Sanofi Pasteur India Pvt. Ltd.) in Healthy Infants.

Aim
To assess and compare the Immunogenicity (primary objective) and Reactogenicity (secondary objective) of Fully liquid Hexavalent EasySix® vaccine (DTwP-HepB-Hib-IPV, Panacea Biotec) with Pentavac SD® (Pentavalent DTwP-HepB/Hib Vaccine, Serum Institute of India Ltd.) co-administered with Imovax Polio® (Sanofi Pasteur India Pvt. Ltd.) in Healthy Infants

Primary Endpoints
Proportion of subjects achieving seroprotection against diphtheria, tetanus, Hepatitis B, Hib; seroresponsiveness against pertussis; seroconversion against Polio virus type 1, type 2 and type 3, 4 weeks after three dose vaccination series of DTwP-HepB-Hib-IPV vaccine in the two treatment groups

Secondary Endpoints

- Incidence of solicited local and systemic reactions during 4 days of each vaccine dose (day 0-3).
- Incidence of unsolicited adverse events up to 4 weeks after each vaccination.
- Incidence of SAE during the entire study period.

Immunogenicity: ability to produce immune response;
reactogenicity to produce common, "expected" adverse reactions, associated signs and symptoms—fever, pain, redness etc

9

Study Design

Design Open Label, Multi-centric, Randomized, Parallel group, Comparative study (active control).

No. of Study Centers Four

1. Dr. Ram Dhongade- Sant Dnyaneshwar Medical Education Research Centre, Pune
2. Dr. S. M. Prasad - Dr B. R. Ambedkar Medical College, Bengaluru
3. Dr. A. Swamy Naidu- King George Hospital, Visakhapatnam
4. Dr J. Venkateswara Rao - Gandhi Medical College & Hospital, Hyderabad

Indication Primary immunization against Diphtheria, Pertussis, Tetanus, Hepatitis B, Haemophilus Influenza and Polio

Study Arms Two

Test Fully liquid hexavalent DTwP-HepB-Hib-IPV vaccine
(EasySix™, Panacea Biotec Ltd.)

Comparator Pentavalent DTwP- HepB-Hib Vaccine (Pentavac) co-administered with IPV (Salk Based Inactivated Polio Vaccine)
(Imovax)

Age Group 6 - 10 weeks (infants)

Enrolled Subjects 284 (Each arm with 142 Infants).

Dosage and Administration 0.5 ml of each vaccine administered by deep IM injection in the anterolateral aspect of thigh.

Schedule 6, 10, and 14 weeks.

Reference: A randomized, open label trial to evaluate and compare the immunogenicity and safety of a novel liquid hexavalent DTwP-Hib/Hep B-IPV (EasySix™) to licensed combination vaccines in healthy infants. Vaccine. 2018

10

Inclusion Criteria

- Infants 6-10 weeks of age, whose parents/LAR willing to give written informed consent prior to the study entry.
- Infants with good health as determined by:
 - Medical history
 - Physical examination
 - Clinical judgment of the investigator
- Judged to be able to attend all scheduled study visits and to comply with trial procedures.

12

Exclusion Criteria

- Infants weighing ■ 3.3 Kg at the time of enrollment.
- Infants less than 6 weeks or more than 10 weeks of age.
- Known HBsAg positivity in mother.
- Infants having history of immunization with vaccine other than zero polio, BCG and birth dose of HepB
- Infants with history of infection potentially related to any of the agents targeted by the DPT-HepB-Hib-IPV vaccine
- Presence of evolving or changing neurological disorder or Infants with history of seizures before receiving the vaccine. Initiation or continuation of pertussis vaccination should be deferred until an evolving neurological disorder can be excluded.
- History of household contact and/or intimate exposure to an individual with suspected poliomyelitis.
- Fever >38o C in past 3 days
- Any evidence of acute illness or infection within past 7 days.

13

Exclusion Criteria

- Planned or elective surgery during the course of the study.
- Infants with a known or suspected impairment of the immune function (congenital or hereditary), or those receiving immunosuppressive therapy, or received immunosuppressive therapy prior to study entry (including systemic or high doses of inhaled corticosteroids) or those who have received a parenteral immunoglobulin preparation
- Infants who have received any blood products, cytotoxic agents or radiotherapy.
- Infants with history of anaphylaxis, or any serious vaccine reaction, or allergy to any vaccine component.
- Have any clinically significant chronic disease (for example, cardiac, pulmonary, renal, gastrointestinal, hepatic, endocrine, cancer, skin or psychiatric disease or disorder or autoimmune disease under treatment) or major congenital defects, such that it would endanger the volunteer's wellbeing or which, in the opinion of the investigator, might interfere with the evaluation of the study objectives.
- Any evidence of thrombocytopenia or a bleeding disorder.
- Infants who have participated in another trial or received an investigational agent within 30 days of enrolment.

14

EasySix:Comparative Immunogenicity

Comparator*

EasySix™

(DTwP-Hep B-Hib with IPV)

(N = 136)
(N = 136)
Antigen
N (%) N (%)

Anti-Diphtheria 129 (94.85) 130 (95.59)

Anti-Tetanus 136 (100) 136 (100)

Anti-PRP
136 (100) 136 (100)
(Short term)
Anti-PRP
126 (92.65) 121 (88.97)
(Long term)
Anti-HBs 133 (97.79) 132 (97.06)

Anti-PT 93 (68.38) 90 (66.18)

Pertussis IgG 103 (75.74) 99 (72.79)

Anti-Polio Type 1 122 (89.71) 125 (91.91)

Anti-Polio Type 2 127 (93.38) 128 (94.12)

Anti-Polio Type 3 120 (88.24) 123 (90.44)

Seroconversion rate of EasySix against all antigens is comparable with DTwP-Hep B-Hib with IPV
Reference: A randomized, open label trial to evaluate and compare the immunogenicity and safety of a novel liquid hexavalent DTwP-Hib/Hep B-IPV (EasySix™) to licensed combination vaccines in healthy infants. Vaccine. 2018
15
Sero-protection /Seroconversion Response rate
Easy Six

120
EasySix Pentavac with Imovax Polio
100 97.79
95.59 100
100 97.06 94.12
91.91 90.44
94.85 92.65 93.38
Sero-protection/Seroconversion rate (%)

88.97
89.71 88.24

75.74
80
72.79
68.38
66.18

60

40

20

0
D T Hib HBs PT Pertussis PV 1 PV 2 PV 3
IgG
Antibody

EasySix Vaccine Exhibited Comparable Immunogenicity
Reference: A randomized, open label trial to evaluate and compare the immunogenicity and safety of a novel liquid hexavalent DTwP-Hib/Hep B-IPV (EasySix™) to licensed combination vaccines in healthy infants. Vaccine. 2018
16
Antibody GMTs (Immunogenicity Cohorts)

GMT
Antibody Group Timing N 95% CI
Value LL UL
Post Vaccination 136 0.77 0.62 0.96
EasySix™
Anti-Diphtheria Pre Vaccination 136 0.04 0.03 0.05
(IU/ml) Pentavac SD® Post Vaccination 136 0.98 0.78 1.22
co-administered
with Imovax Polio® Pre Vaccination 136 0.03 0.02 0.04
Post Vaccination 136 1.16 1.02 1.31
EasySix™
Anti-Tetanus Pre Vaccination 136 1.18 1.01 1.38
Pentavac SD ®
(IU/ml) Post Vaccination 136 1.12 1.00 1.25
co-administered
with Imovax Polio® Pre Vaccination 136 1.02 0.85 1.23
Post Vaccination 136 12.33 9.29 16.36
EasySix™
Pre Vaccination 136 0.81 0.67 0.98
Anti-PRP (µg/ml) ®
Pentavac SD Post Vaccination 136 8.97 6.87 11.71
co-administered

with Imovax Polio® Pre Vaccination 136 0.69 0.58 0.82
Post Vaccination 136 328.57 214.01 504.44
EasySix™
Pre Vaccination 136 0.00 0.00 0.01
Anti-HBs (mIU/ml) ®
Pentavac SD Post Vaccination 136 244.97 150.53 398.68
co-administered
with Imovax Polio® Pre Vaccination 136 0.00 0.00 0.02

20

Antibody GMTs (Immunogenicity Cohorts) (cont.)

GMT
Antibody Group Timing N 95% CI
Value LL UL
Post Vaccination 136 45.97 36.29 58.24
EasySix™
Pre Vaccination 136 4.65 3.59 6.03
Anti-PT (µg/ml) Pentavac SD® Post Vaccination 136 47.33 35.82 62.55
co-administered
Pre Vaccination 136 3.14 2.02 4.91
with Imovax Polio®
Post Vaccination 136 25.64 23.33 28.18
EasySix™
Pre Vaccination 136 12.21 10.50 14.21
Pertussis IgG ®
Pentavac SD Post Vaccination 136 21.69 19.55 24.07
(IU/ml)
co-administered
Pre Vaccination 136 10.31 8.56 12.42
with Imovax Polio®
Post Vaccination 136 155.92 106.90 227.44
EasySix™
Pre Vaccination 136 1.00 0.78 1.29
Anti-Polio Type 1 ®
Pentavac SD Post Vaccination 136 221.05 152.69 320.01
≥8 (1/dilution)
co-administered
Pre Vaccination 136 0.79 0.64 0.97
with Imovax Polio®
Post Vaccination 136 243.52 174.07 340.68
EasySix™
Pre Vaccination 136 1.13 0.87 1.47
Anti-Polio Type 2 ®
Pentavac SD Post Vaccination 136 354.13 250.05 501.52
≥8 (1/dilution)
co-administered
Pre Vaccination 136 0.82 0.66 1.03
with Imovax Polio®
Post Vaccination 136 197.10 131.11 296.31
EasySix™
Pre Vaccination 136 1.03 0.80 1.33
Anti-Polio Type 3 ®
Pentavac SD Post Vaccination 136 260.18 173.62 389.91
≥8 (1/dilution)
co-administered
Pre Vaccination 136 0.79 0.64 0.99
with Imovax Polio®

21

EasySix®:Safety Results

■ No death reported in the trial.

■ One SAE (Pneumonia) was reported in the trial: NOT RELATED to the study vaccine.

■ Most Systemic solicited and local events were mild/moderate and resolved completely with symptomatic treatment.

22

EasySix®: Safe & Well tolerated

EasySix Comparator *
Adverse Events (AEs)
N (%) (DTwP - HepB-Hib with IPV) N (%)
100 106
Any AEs
(70.42) (74.64)
1 2
Any Unsolicited AEs
(0.70) (1.40)
99 104
Any Solicited AEs
(69.71) (73.23)

DTwP-Hep B-Hib IPV

79

Any Solicited Local AEs
(55.63) 84 53
(59.15) (38.02)

■ Exhibited Less Pain and Tenderness.

■ Comparable Local (redness and swelling) and Systemic (fever and irritability / restlessness / fuzziness) AE's.

Reference: A randomized, open label trial to evaluate and compare the immunogenicity and safety of a novel liquid hexavalent DTwP-Hib/Hep B-IPV (EasySix™) to licensed combination vaccines in healthy infants. Vaccine. 2018

23

Conclusion

■ World's first fully liquid wP-based hexavalent vaccine is safe and immunogenic

■ Indian NRA approved through phase I and phase III clinical trials.

■ Non-inferiority was demonstrated at the level of proportion of infants developing seroprotective titers or showing seroconversion following the primary series of vaccine compared to the same target-antigens included in licensed combination vaccines.

■ Has comparable immunogenicity versus DTwP-Hep B-Hib with IPV in 284 Indian infants study.

■ Exhibited less pain and tenderness, less local and systemic adverse events, hence well tolerated.

■ 6 in 1 wP-based hexavalent vaccine for broad and long term protection.

9

Thank You for your kind attention !

VIAL DETAILS

Vial size |3 ml |
 Vial details |3 ml clear glass vials |
 Fill Volume |3.0 ml |

1. VIAL WASHING DETAILS

|1. Glass vial washing details |
 |i |Date of washing | |
 |ii |Vials washing machine EQ. ID. No. | |
 |2. Parameters for vial washing machine |
i	Starting		End time	
	time			
ii	Total time	hrs.....min		
iii	Total vials			
	washed			

Washing done by (Sign/Name/Department):

2. DEPYROGENATION DETAILS

|1. Glass vial Depyrogenation details |
i	Date of Depyrogenation	
ii	EQ.ID of Depyrogenation tunnel	
iii	Total vials Depyrogenated	

Depyrogenation done by (Sign/Name/Department):

3. FILLING OF VACCINE IN VIALS

Date of filling	:		
Batch start		Batch End	
Time(Hours)		Time(Hours)	
Total Batch Time :			
Total No. of vials filled	:		
Rejection of vials during			
Filling			
Total No. of filled vials sent for Sealing:			
Rejection of vials during	:		
Sealing			
Total no. of sealed vials	:		
sent for visual inspection			

Filling done by (Sign/Name/Department):

4. VISUAL INSPECTION

Date:

4.1 Room Temp / R.H% during visual inspection

Process stage	Time (hrs)	Temperature	R.H%	Recorded by
		(range 22° ± 2° C)	(NMT 55 %)	
Starting				
Middle				
End				

4.2 Duration of Process

Starting Time (	Completion Time (	Total time taken	Recorded by
hrs)	hrs)		

No of vials rejected	
Total no. of good	
vials	

Checked by: Date:

5. LABELLING OF FILLED VIALS

5.1 Room Temp / R.H% during labeling area

Process stage	Time (hrs)	Temp.	R.H %	Recorded by
		(range 22° ± 2° C)	(NMT 55%)	
Starting				
Middle				
End				

2. Duration of labeling Process

Starting Time (	Completion Time (	Total time taken	Recorded by
hrs)	hrs)		

5.3 Labeling Details

a. Date of receipt of vials :
 b. No. of vials received for labeling :

- Labeled by: Date:

6.1 Packaging Duration

6.2 Temperature of room during Packaging

Checked by	Signature	Date

- i) After packaging, the vaccine vials are stored at $5^{\circ} \pm 3^{\circ} \text{C}$.
- ii) Transfer the batch to stores by packing dept.

BioBridge Master Knowledge — Page 434

VIAL DETAILS

Vial size |2 ml |
 Vial details |2 ml clear glass vials – Siliconized |
 Fill Volume |0.65 ml |

1. VIAL WASHING DETAILS

|1. Glass vial washing details |
 |i |Date of washing | |
 |ii |Vials washing machine EQ.ID. No. | |
 |2. Parameters for vial washing machine |
i	Starting		End time	
	time			
ii	Total time	hrs.....min		
iii	Total vials			
	washed			

Washing done by (Sign/Name/Department):

2. DEPYROGENATION DETAILS

|1. Glass vial Depyrogenation details |
i	Date of Depyrogenation	
ii	EQ.ID of Depyrogenation tunnel	
iii	Total vials Depyrogenated	

Depyrogenation done by (Sign/Name/Department):

3. FILLING OF VACCINE IN VIALS

Date of filling	:		
Batch start		Batch End	
Time(Hours)		Time(Hours)	
Total Batch Time :			
Total No. of vials filled	:		
Rejection of vials during			
Filling			
Total No. of filled vials sent for Sealing:			
Rejection of vials during	:		
Sealing			
Total no. of sealed vials	:		
sent for visual inspection			

Filling done by (Sign/Name/Department):

4. VISUAL INSPECTION

Date:

4.1 Room Temp / R.H% during visual inspection

Process stage	Time (hrs)	Temperature	R.H%	Recorded by
		(range 22° ± 2° C)	(NMT 55 %)	
Starting				
Middle				
End				

4.2 Duration of Process

Starting Time (	Completion Time (	Total time taken	Recorded by
hrs)	hrs)		

No of vials rejected	
Total no. of good	
vials	

Checked by: Date:

5. LABELLING OF FILLED VIALS

5.1 Room Temp / R.H% during labeling area

Process stage	Time (hrs)	Temp.	R.H %	Recorded by
		(range 22° ± 2° C)	(NMT 55%)	
Starting				
Middle				
End				

2. Duration of labeling Process

Starting Time (	Completion Time (	Total time taken	Recorded by
hrs)	hrs)		

5.3 Labeling Details

a. Date of receipt of vials :
 b. No. of vials received for labeling :

c. Date of labeling :

d. Rejection / breakage during labeling :

e. Percentage of rejection during labeling :

f. No. of vials sent for packing :

g. Specimen label

Labeled by: Date:

6. PACKAGING OF VIALS

6.1 Packaging Duration

Sl. No.	Starting Time(hrs)	Completion Time(hrs)	Total time taken	Recorded by
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				

6.2 Temperature of room during Packaging

Process Stage	Time	Temperature ($22^{\circ} \pm 2^{\circ}$ C)	Recorded by
Start			
Middle			
End			

Checked by	Signature	Date

7. TRANSFER OF BATCH FROM PRODUCTION DEPARTMENT TO STORES

i) After packaging, the vaccine vials are stored at $5^{\circ} \pm 3^{\circ} \text{C}$.

ii) Transfer the batch to stores by packing dept.

Checked by	Signature	Date

Information on Filling and Stoppering

|Filling and Stoppering |

1.	Date of filling	
2.	Total volume of blend received for filling	
3.	Capacity of the syringes	
4.	Volume set for filling per syringes	
5.	Filling Line ID No.	
6.	Start time of filling	
7.	Completion time of filling	
8.	Total time taken for filling	
9.	Temperature of room during filling	
	(Acceptance Range $22^{\circ} \pm 2^{\circ}\text{C}$)	
10.	Total No. of syringes filled and plugged	
11.	No. of filled syringes used for online	
	volume checking	
12.	No. of syringes rejected	
13.	No. of syringes taken for In-house testing	
14.	No. of syringes taken as Reference	
	standards (QC)	
15.	No. of filled syringes transfer to cold	
	room ($5^{\circ}\pm 3^{\circ}\text{C}$) in warehouse till visual	
	inspection	
16.	Name of person responsible for filling and	
	stoppering	

1. Information on Visual Inspection

|Visual Inspection |

1.	Date of visual inspection	
2.	No. of filled syringes received from cold	
	room	
3.	Visual inspection initiated at	
4.	Visual inspection completed at	
5.	Total time taken for visual inspection	
6.	Room temperature during visual inspection	
7.	No. of filled syringes rejected	
8.	No. of syringes visually passed (accepted)	
9.	Percentage of rejection	
10.	No. of syringes handed over to labeling	
11.	Name of person responsible for visual	
	inspection and plungering	
12.	Remarks, if any	

2. Information on Labeling, Plunger Rod & Back Stopper Insertion

|Labeling, Plunger Rod & Back Stopper Insertion |

1.	No. of Syringes received for labeling	
2.	Date of labeling	
3.	Labeling started at	
4.	Labeling completed at	
5.	Total time taken for labeling	
6.	Room temperature during labeling	
	(Acceptance Range $22^{\circ} \pm 2^{\circ}\text{C}$)	
7.	Rejection during labeling	
8.	Percentage of rejection	
9.	No. of pre-filled syringes received for	
	plunger rod & back stop insertion	
10.	Breakage during Plunger rod & Back stop	
	insertion	
11.	Percentage of rejection	
12.	No. of labeled Syringes sent for packaging	
13.	Specimen label	
14.	Name(s) of Person(s) responsible for	
	labeling	
15.	Remarks (if any)	

3. Information on Packaging Protocol

|Packaging Protocol |

1.	Blister packaging	
	a) Date of receipt	
	b) No. of pre-filled syringes received	
	for blister packaging	
	c) Date of blister packaging	
	d) Blister Packaging started at	
	e) Blister Packaging completed at	
	f) Total time taken for blister packaging	
	Room Temperature during blister packaging	
	(Acceptance range $22^{\circ} \pm 2^{\circ}\text{C}$)	
	i) Breakage / rejection during blister	
	packaging	
	Percentage of breakage / rejection	
	No. of syringes blister packaged	
2.	Packaging of blistered syringes in monocartons	

	Date of receipt	
	No. of blistered packed pre-filled syringes	
	received for monocartons packaging	
	Date of packaging	
	Packaging started at	
	Packaging completed at	
	Total time taken for packaging	
	Temperature during packaging	
	(Acceptance range $22^{\circ} \pm 2^{\circ} \text{C}$)	
	Breakage / rejection during monocartons	
	packaging	
	Percentage of breakage / rejection	
	No. of blistered syringes packaged in	
	monocarton	
3.	No. of syringes packaged	
4.	Retained samples	
5.	No. of syringes taken for national control	
	lab	
6.	No. of syringes transferred to warehouse	
7.	Storage temperature after packaging	
8.	Name(s) of Person(s) responsible for	
	packaging	
9.	Remarks (if any)	

2.0 Manufacturing Flow Chart

[pic]

Aluminium Phosphate Gel

Aluminium Phosphate Gel

Tetanus Toxoid

Diphtheria Toxoid

Saline Solution

wP

HeP B

Saline Solution

Adjust the pH at 6-6.5 & stir for 24 hr at 150 rpm/25°C

Mix Comp I and Comp II

Adjust the pH at 6.5-7.5 & stir for 24 hr at 150 rpm/25°C

Adjust the pH at 6-7 & stir for 24 hr at 150 rpm/25°C

Add HiB followed by IPV & Saline

Stir for 30 min after each addition then adjust the pH at 6.5-7.5 & stir for another 2hrs at 150 rpm/25°C

Final Formulation

Store at 2-8°C

Component-I

Component-II

2-Phenoxyethanol

2-Phenoxyethanol

10 -15 min stirring
at 150 rpm/25°C

10 -15 min stirring
at 150 rpm/25°C

10 -15 min stirring
at 150 rpm/25°C

10 -15 min stirring
at 150 rpm/25°C

10 -15 min stirring
at 150 rpm/25°C

10 -15 min stirring
at 150 rpm/25°C

10 -15 min stirring
at 150 rpm/25°C

Filling

Labeling

Dispatch

In process control

In process control

Stir for 30 min at 150 rpm/25°C

[[pic] |EASYSIX (16 L) |Page No. |
		1 of 3
	PROCESS VALIDATION SAMPLING PLAN	

S No.	Day	Sample ID	Sample stage	Sample details	Testing details	Rationale
	Day 1	E6-10-1				
	PROCESS VALIDATION SAMPLING PLAN					

S No.	Day	Sample ID	Sample stage	Sample details	Testing details	Rationale
		E6-10-11	Sample after mixing	1x5 ml, 1x40	pH, Adsorption	Base line study
		com I & II 30 min	ml,1x10 ml	(D, T & HBsAg)	after mixing Com I	
		sample		& II addition		
		E6-10-12	Sample after addition	1x5 ml, 1x40	pH, Adsorption	To establish
		of Com III in com I &	ml,1x10 ml	(D, T & HBsAg)	incubation effect	
		II mix and 30 min				
		stirring				

[[pic] |EASYSIX (16 L) |Page No. |
		3 of 3
	PROCESS VALIDATION SAMPLING PLAN	

		E6-10-13	Sample after mixing	1x5 ml, 1x40 ml,1x10	pH, Adsorption	To establish
		com I, II & III and	ml	(D, T & HBsAg)	incubation effect	
		after 2hr incubation				
		E6-10-14	Final bulk sample	3X20ml,1X5ml,	Complete analysis	IPQC verifications
		(after storage at	1X25ml, 1X50ml		of process	
		(2-8oC) for pending				
		filling				
		E6-10-15	Final bulk retained	1 X 140 ml	Retained sample	IPQC verifications
		sample			of process	
		E6-10-16	Final bulk retained	1 X 100 ml	R&D Retained	R&D Retained sample
		sample		sample		
	Day 5	E6-10-17	Before start filling	1X5ml, 1X25ml	pH and sterility	IPQC verifications
		(priming sample)			of process	
		E6-10-18	Extractable volume	1X 10 PFS	Extractable	
		check after setting		volume		
		E6-10-19	Extractable volume	1X 10 PFS	Extractable	
		check end filling		volume		
		E6-10-20	Solution left over	1X 25 ml	Sterility	
		E6-10-21	Final Lot sample	302 PFS	For complete	
			analysis			

	Prepared By	Verified By	Approved By	
	Quality Assurance	Production	Quality Control	Quality Assurance
Name				
Sign				
Date				

[pic]

Formulation Protocol

Title	:	Formulation Procedure of EasySix Vaccine (DTwPPEBL
	-HepBPEBL -HibPEBL -IPVNI)	
Protocol No.	:	VDP/MS/14-V028
Issue Date	:	
Supersedes	:	None
No. of Pages	:	28, Excluding Cover Page.

Panacea Biotec Limited
Research & Development
Lalru, Punjab-140501, INDIA

Approval Sheet

This protocol is Prepared, Checked and Approved by as under:

	Name	Designation/Department	Sign/ Date
Prepared By			
Checked By			
Approved By			

LIST OF ABBREVIATIONS

|Definitions : |
%	:	Percent
AlPO₄	:	Aluminium phosphate.
ARM	:	Active raw material
CRT	:	Controlled room temperature
DT	:	Diphtheria Toxoid
HCl	:	Hydrochloric acid
Hep B	:	Hepatitis B surface Antigen
Hib	:	Haemophilus influenza type b
IPA	:	Isopropyl alcohol
IPV	:	Salk Inactivated polio vaccine
LAF	:	Laminar air flow
NaOH	:	Sodium hydroxide
°C	:	Degree centigrade
PRP-TT	:	Polyribosylribitol phosphate conjugated to Tetanus
		toxoid
2-POE	:	2-Phenoxyethanol
QA	:	Quality Assurance
QC	:	Quality Control
R&D	:	Research and Development
rpm	:	rotations per minute
SOP	:	Standard Operating Procedure
TT	:	Tetanus Toxoid
wP	:	Inactivated Whole cell pertussis bulk
PEBL	:	PanEra Biotec Ltd., India
NVI	:	Netherlands Vaccine Institute, The Netherlands

1. Purpose

To lay down the procedure for formulation and filling of EasySix vaccine using Diphtheria toxoid, Tetanus toxoid, pooled bulk of inactivated whole cell Bordetella pertussis, Purified Capsular Polysaccharide (PRP) from Haemophilus influenzae type b conjugated to Tetanus Toxoid bulk, Hepatitis B surface antigen and Salk Inactivated polio vaccine (IPV).

2. SCOPE

This procedure is applicable to batches of EasySix vaccine to be formulated from bulks of Diphtheria toxoid, Tetanus toxoid, pooled bulk of inactivated whole cell Bordetella pertussis, Purified Capsular Polysaccharide (PRP) from Haemophilus influenzae type b conjugated to Tetanus Toxoid bulk, Hepatitis B surface antigen and Salk Inactivated polio vaccine (IPV) at R&D, Panacea Biotec Ltd., Lalru and Vaccine Formulation Plant, Panacea Biotec Ltd., Baddi.

3. RESPONSIBILITY

3.1 Formulation research and development shall be responsible for the Preparation of this SOP and perform the procedure.

3.2 Analytical research and development shall be responsible for the testing of this formulation.

3.3 Quality Assurance department shall be responsible to ensure the compliance of this protocol.

4. Materials & Equipments

- Aluminium phosphate gel (Adju-phos), Sterile
- 2-Phenoxyethanol
- Physiological Saline (0.85%), Sterile
- HCl (0.1 N) & NaOH (0.1 N), Sterile
- Laminar Air Flow (LAF) unit and filter sterilized 70 % Isopropyl alcohol (IPA)
- Sterilized glass bottles with magnet
- Sterile mops / wipes.
- Sterile "Aseptic area" dresses & gloves
- Magnetic stirrer
- Sterile Formulation vessels
- SIP unit
- Sterile calibrated measuring cylinders, pipettes, sampling vials

- ## 5. DESCRIPTION

EasySix is a uniform suspension of Diphtheria Toxoid (DT), Tetanus Toxoid (TT), inactivated whole cell pertussis (wP), Hepatitis-B surface antigen (HBsAg), Haemophilus influenzae type b (Hib-TT) and Inactivated Polio Virus (Salk). The Diphtheria Toxoid (DT), Tetanus Toxoid (TT) and Hepatitis B (HBsAg) are adsorbed on aluminium phosphate gel and combined with inactivated whole cell pertussis (wP), Inactivated Polio Virus (Salk) and Haemophilus influenzae type b (Hib-TT). The vaccine is suspended in isotonic saline solution to which 2-phenoxethanol is added as a preservative.

6. MASTER FORMULA

Batch No. :

S	Ingredients	Item Code	Quantity/unit
No.	(0.5 ml)		
	Diphtheria Toxoid	25 Lf	
	Tetanus Toxoid	7.5 LF	
	Inactivated w-Bordetella	16 IOU	
	pertussis		
	Hepatitis B Surface Antigen	10 µg	
	Purified Capsular	10 µg	
	Polysaccharide (PRP) from		
	Haemophilus influenzae type b		
	conjugated to Tetanus Toxoid		
	Inactivated Polio Virus (Salk)		
	Type 1	40 D units	
	Type 2	8 D units	
	Type 3	32 D units	
	2-Phenoxyethanol	3.3 mg	
	Aluminium Phosphate Gel	0.6 mg	
	(Al+3 Content)		
	Sodium Chloride Solution	q.s. to 0.5 ml	
	(0.85%w/v)		

7. CALCULATIONS FOR DISPENSING OF MATERIALS

1. Active Raw Material (ARM)

The per batch quantity of active raw materials required to be dispensed is calculated as follows

Final volume of the Vaccine (Batch size) : VFVmL

S	Ingredients	Batch No.	Concentrat Overla	Quantit Quantity
No.		on ges y Per Required		
	of arm (x) (%)age ml (Y) (Y x VFV /			
	) (X)#			
	Diphtheria Toxoid Lf/ml 5% 52.5 Lf VD ml			
	Tetanus Toxoid Lf/ml 5% 15.75 VT ml			
	Lf			
	Inactivated IOU/ml 5% 33.6 VW ml			
	w-Bordetella IOU			
	pertussis			
	Purified Capsular µg/ml 10% 22 µg VHi ml			
	Polysaccharide			
	(PRP) from			
	Haemophilus			
	influenzae type b			
	conjugated to			
	Tetanus Toxoid			
	Hepatitis B µg/ml 10% 22 µg VHB ml			
	Surface Antigen			

2. Calculations for dispensing of Bulk of Inactivated Polio Virus (Salk)

S.	Ingredients	Batch No.	Overages	Concentration	VIPV=
No.	(%)	ion	VFV	Final	
	of arm	YIPV	Quantity		
	(YIPV)	Required			
		MIPVF=			
		VIPV X (1			
		+ 5/100)			
	Bulk of	5% 5 X	ml	ml	
	Inactivated Polio	concentrat			
	Virus (Salk)	e			

VFV = Final volume of the vaccine (batch size), VD = Volume of the Diphtheria toxoid, VT = Volume of the Tetanus toxoid, VW = Volume of the w-B. Pertussis, VHi = Volume of PRP-TT, VHb = Volume of Hepatitis B Surface Antigen, V IPV = Bulk of Inactivated Polio Virus (Salk).

3. Calculations for dispensing of excipients:

1. Determination of volumes of individual components

S. No.	Particulars	Percentage	Formula	Volume in ml
1	Batch size (in ml) of final vaccine	NA	VFV	
2	Total volume (in ml) of Component-I to be prepared	60%	$VCI = VFV \times 60$	
3	Total volume (in ml) of Component-II to be prepared	15%	$VCI = VFV \times 15$	
4	Total volume (in ml) of Component-III to be prepared	25%	$VCI = VFV \times 25$	

2. For Aluminium phosphate gel:

Batch number :

Supplier/Manufacturer :

S. No.	Particulars	Calculation/Qty.
1	Batch size (in ml) of final vaccine	VFV
2	Required aluminium ion concentration (Al ³⁺) per dose	0.60 mg
3	of 0.5 ml vaccine	
4	Required aluminium ion concentration (Al ³⁺) in final vaccine	1.20 mg
5	vaccine (mg per ml) = F	
6	Concentration of Aluminium Phosphate Gel (as per COA)	
7	(in Al ³⁺ ion) (in mg/ml) = Al	
8	Quantity of Aluminium Phosphate Gel to be added in final vaccine batch	$VAL = F \times VFV / Al$

1. Determination of volume of Aluminium phosphate gel in individual components

S. No.	Particulars	Percentage	Formula	Volume in ml
1	Batch size (in ml) of final vaccine	NA	VFV	
2	Total quantity of Aluminium Phosphate Gel to be added in final vaccine batch		VAL	
3	Total volume Aluminium	80%	$VALCI = VAL$	
4	Phosphate Gel required for Component-I		$X 80$	
5	Total volume Aluminium	20%	$VALCII = VAL$	
6	Phosphate Gel required for Component-II		$X 20$	

3. For 2-Phenoxyethanol

Batch number :

Supplier/Manufacturer :

S. No.	Particulars	Calculation/Qty.
1	Batch size (in ml) of final vaccine	VFV
2	Required concentration of 2-phenoxyethanol per dose	3.3 mg
3	of 0.5 ml vaccine	
4	Required concentration of 2-phenoxyethanol in final vaccine	6.6 mg
5	(per ml) = G	
6	Concentration of 2-phenoxyethanol (as per COA)	
7	(mg/ml) = P	
8	Quantity of 2-phenoxyethanol to be added in final vaccine batch	Vp

1. 2-POE contribution from ARm (DS)

Particulars	Diphtheria toxoid	Tetanus toxoid bulk	Bordetella
Volume of different component - I	$VD = VT = VW$		
Concentration of component - I	$CPDT = CPTT = CPWP$		

2-POE present in bulk		
(mg/ml)		
Amount of 2-POE	[(CPDT X VD) + (CPTT X VT) + (CPWP X VW)]	
contributed from ARM	=	
(APARM) (in mg)		

2. Determination of volume of 2-Phenoxyethanol solution for individual components

S. No.	Particulars	Calculation/Qty.
	Batch size (in ml) of final vaccine =VFV	
	Required concentration of 2-phenoxyethanol per 3.3 mg	
	dose of 0.5 ml vaccine	
	Required concentration of 2-phenoxyethanol in 6.6 mg	
	final vaccine (per ml) = G	
	Concentration of 2-phenoxyethanol (as per COA)	
	(in mg/ml) = P	
.	Total volume of Component-I to be prepared (in	
	ml)= (VCI)	
	Total volume of Component-II to be prepared (in	
	ml)= (VCII)	
	Total volume (in ml) of component III (in ml) =	
	(VCIII)	
	Quantity of 2-phenoxyethanol to be added in	
	Component-I (in ml) = VpCI = (VCI x G)	
	– (APARM) + (VCIII x G)	
	P	
	Quantity of 2-phenoxyethanol to be added in	
	Component-II (in ml) = VpCII = (VCII X G)	
	P	
	Quantity of 2-phenoxyethanol to be added in final	
	vaccine batch (in ml) = Vp = VpCI + VpCII	

4. For Sodium Chloride

Batch number :

Supplier/Manufacturer :

Calculations for volume of Sodium Chloride solution (0.85%w/v)

S. No.	Particulars	Calculation/Qty.
No.		
	Batch size (in ml) of final vaccine = VFV	
	Quantity of Sodium Chloride Solution to be added in	
	final vaccine batch (in ml) = VS = VFV - (VD + VT + Vw	
	+ VHi + VHb + VIPV +Vp+VAL)	

1. Determination of volume of Sodium Chloride solution for individual components

S. No.	Particulars	Volume in ml
	Batch size (in ml) of final vaccine =VFV	
	Total volume of Component-I to be prepared (in	
	ml)= (VCI)	
	Total volume of Component-II to be prepared (in	
	ml)= (VCII)	
	Total volume of Component-III to be prepared	
	(in ml)= (VCIII)	
	Total volume of Sodium Chloride Solution be	
	added in Component-I (in ml) = VSI = VCI –(VD +	
	VT + Vw + Vp Com I + VALCI)	
	Total volume of Sodium Chloride Solution be	
	added in Component-II (in ml)= VSII = VCII –(	
	VHb + Vp Com II + VALCII)	
	Total volume of Sodium Chloride Solution be	
	added in Component-III (in ml) = VSIII = VCIII	
	–(VHi + V IPV)	

8. ACTIVE INGREDIENTS & EXCIPIENTS OF EASYSIX VACCINE

Quantity of ingredients in a batch

Depending on batch size of vaccine, the quantities of active ingredient, Aluminium phosphate gel, 2-POE and normal saline are worked out.

S.No	Name of ingredient	Batch No. of ARM &	Required quantity of
	Solutions	ingredient in present	
		batch of 10.0 lts.	
1	Diphtheria toxoid	ml	
2	Tetanus toxoid	ml	
3	Bordetella pertussis	ml	
4	Hepatitis B Surface	ml	
	Antigen		
5	Purified Capsular	ml	
	Polysaccharide (PRP)		
	from		
	Haemophilusinfluenzae		
	type b conjugated to		
	Tetanus Toxoid		

6	Inactivated Polio		ml
	Virus (Salk)		
7	Al 3+ as Aluminium		ml
	phosphate gel		
8	2-POE		ml
9	Physiological saline		
		ml	
9. MATERIAL REQUISITION AND ISSUING SLIP

Requesting Deptt. : Requisition Slip No. :

Product: EasySix Vaccine Date:

Batch size: Batch No. :

Sr.	Name of	Spec. No.	Item Code
No	Material		
	Sodium Chloride	8.5 g	
	WFI	q.s to 1000 mL	

Filtration details of Physiological Saline:

Date of filtration		
Type of Filter used		
Filter serial no.		
used		
EQ ID of integrity		
testing machine		
Test Performed for		
checking the		
integrity		
Results of Pre		
filtration		
integrity testing of		
filter*		
Filtration Start	Filtration End Time	Total =time taken
Time	:Hrs	
:.....hrs		
Volume taken for	Volume recovered after	Filtration loss
filtration	filtration	
	=.....ml	=.....ml
=.....ml		
Results of Post		
filtration integrity		
testing of filter*		

Distribution of Physiological Saline Solution for Sterilization

S.No	Bottle No.	Quantity (ml)	To be used	Distribute by	Checked by
.			for		

Pre-sterilization Distribution Details

S.No.	Bottle	Wt. of	Volume of	Wt. of bottle	Checked by
	No.	empty	Physiological	with	
		bottle	Saline Solution	Physiological	
			Saline Solution		

Sterilization Details for Physiological Saline Solution

Date of Sterilization: Sterilization hold time:
..... Hrs toHrs

Start Time: Sterilization hold temp.

Post Sterilization Details of Physiological Saline Solution

S.No.	Bottle No.	Wt. of	Wt. of bottle	Evaporation	Checked
		bottle with	with	loss	by
		Physiological	Physiological	W2-W1	
		Saline	Saline Solution		
		Solution	after		
		before	sterilization		
		sterilizatio			
		In			

Preparation and Sterilization Record

|S.No|Particulars |Remarks |
|. | | |

.....

.....

.....

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shall be decided based on the batch size of final formulated bulk vaccine.

Batch number :

Supplier/Manufacturer :

Preparation of 0.1 N Hydrochloric Acid Solution

[S.No|Particulars |Qty. required for |Qty required |
.		one liter solution	mL
	Hydrochloric Acid	8.5 mL	
	(Concentrated)		
	WFI	q.s to 1000 mL	

Sterilization Details for 0.1 N Hydrochloric Acid Solution

Date of Sterilization: Sterilization hold time:
..... Hrs toHrs

Start Time: Sterilization hold temp.

Prepared by (Sign/Name/Department):

Remarks:

.....

4. 0.1 N Sodium Hydroxide solution – Weigh 4.0 g of Sodium hydroxide and dissolve in 800 mL of sterile water for injection (WFI). Ensure complete dissolution and make up the volume to 1000 mL with sterile WFI. Mix properly and sterilize the solution by autoclaving. Total quantity of the solution to be prepared shall be decided based on the batch size of final formulated bulk vaccine.

Batch number :

Supplier/Manufacturer :

Preparation of 0.1 N Sodium Hydroxide Solution

[S.No|Particulars |Qty. required for |Qty required |
.		one liter solution	mL
	Sodium Hydroxide	4.0 g	
	WFI	q.s to 1000 mL	

Sterilization Details for 0.1 N Sodium Hydroxide Solution

Date of Sterilization: Sterilization hold time:
..... Hrs toHrs

Start Time: Sterilization hold temp.

Prepared by (Sign/Name/Department):

Remarks:

.....

5. 2-Phenoxyethanol – Take calculated quantity of 2-POE and sterilize by filtration using PTFE membrane solution (0.2 µm). Total quantity of the solution to be prepared shall be decided based on the batch size of final formulated bulk vaccine.

Batch number :

Supplier/Manufacturer :

Sterilization Details of Filter used

Details of Filter Used:

Date of Sterilization: EQ. ID of Autoclave:
Load no. : Cycle No.
.....
Description :

Filtration details of 2-POE

[Date of filtration | |
Type of Filter used	
Filter serial no.	
used	
EQ ID of integrity	
testing machine	
Test Performed for	
checking the	
integrity	
Results of Pre	
filtration	
integrity testing of	

filter* | |
Filtration Start |Filtration End Time |Total =time taken |
Time |:Hrs | |
|:.....hrs | | |
Volume taken for |Volume recovered after|Filtration loss |
filtration |filtration | |
| |=.....ml |=.....ml |
|=.....ml | | |
Results of Post | |
filtration integrity| |
testing of filter* | |

Remarks:
.....

10. PREPARATION OF FORMULATION VESSEL

1. CIP of Blending Vessel

Final rinse of the blending vessel is done. Data are recorded and attached with BPR.

- 1. Date of CIP :
- 2. I.D. no. of blending vessel :
- 3. Capacity of blending vessel :
- 4. Name of previous Batch :
- 5. Batch No. of previous product :
- 6. CIP Cycle No. :
- 7. CIP time : Start time End time.....

CIP Set Point Details:

S.No. |Type of cycle |Set Wt. |Temperature|RPM |Time |Done by |
1 |Alkaline cleaning agent| | | | |
2 |1st Water for injection| | | | |
| |wash | | | | |
3 |2nd Water for injection| | | | |
| |wash | | | | |
4 |3rd Water for injection| | | | |
| |wash* | | | | |
5 |4th Water for injection| | | | |
| |wash* | | | | |

Note: * Repeated water for injection wash if first sample fails.

Result of water rinse after CIP
Date of |I.D. No. of |Name of previous |Result |A. Report No. |
CIP |blending vessel |product / Batch. No| | |
			Pass/ Fails	
	Lt			
	s vessel			

*Attach test request form with BPR.

Sign: Date:

2. Integrity Testing of Air Vent of Vessel:

Integrity of the air vent must be carried out.

EQ. ID. of integrity testing machine:
Date

Type of |Filter serial |Test performed |Result |Remarks |
filter |no. | | | |
| | | |Pass/ Fails | |

Test performed by..... Checked by.....

3. SIP of the Blending Vessel

1. Transfer of filtered physiological saline:

Transfer the filtered physiological saline in to the vessel

EQ.ID. :

S.No |Name of |Total Quantity |Quantity |Transferred|Checked |
| |solution |of filtrate (ml)|transferred |by |by |
| | |(ml) | | |
1 |Physiological | | | |
| |saline | | | |

2. SIP of the Blending Vessel with Physiological Saline:

Record the data of SIP of the vessel.

EQ. ID. No. of blending vessel:

Date of SIP:

Run no.	Temperature	Sterilization	Done by	Checked by		
		Hold				
		From	To			

11. Manufacturing Procedure

1. Precautions

1. Check all the requirements as per the batch size.
2. Enter into the clean area as per the laid procedure.
3. Mop the work bench of LAF with swab containing 70 % IPA.
4. Take all the required material inside the LAF after mopping them with 70 % IPA.
5. Compensate the water loss in saline and aluminium phosphate gel during autoclave by addition of WFI before use.
6. Temperature of formulation bulk should be monitored continuously.

2. Formulation/ Blending

1. Component I (DTwP)

- a) Add the following ingredients in sequence mention below
- b) Aseptically transfer the calculated quantity of Aluminium phosphate gel into the blending vessel.
- c) Aseptically transfer the calculated quantity of Tetanus Toxoid slowly under constant stirring.
- d) Stir for 10 to 15 mins. for mixing (stirring at 150 rpm/CRT).
- e) Aseptically transfer the calculated quantity of Diphtheria Toxoid slowly under constant stirring.
- f) Stir for 10 to 15 mins. for mixing (stirring at 150 rpm/CRT).
- g) Aseptically transfer the calculated quantity of filter sterilized 2-phenoxyethanol under constant stirring.
- h) Stir for 10 to 15 mins for mixing (stirring at 150 rpm/CRT).
- i) Aseptically transfer the calculated quantity of sodium chloride solution under constant stirring.
- j) Stir for 10 to 15 mins for mixing (stirring at 150 rpm/CRT).
- k) Check the pH, if it is not in range of 6.0 to 6.5, adjust the pH of formulation using sterile 0.1 N HCl or 0.1 N NaOH solution and record as follows:

pH adjustment of Component-I (Before wP addition and before Incubation)

S. No.	Component Name			
Stirring speed				
Incubation period	Start		Start Time	
	Date			
	End Date		End Time	
Net incubation	Hours minutes			
period				
Remarks				

l) After 24 hr of incubation is completed (Day II) aseptically transfer the calculated quantity of Inactivated Bordetella pertussis bulk solution under constant stirring.

- m) Stir for 10 to 15 mins for mixing (stirring at 150 rpm/CRT).
- n) Check the pH, if it is not in range of 6.5 to 7.5, adjust the pH of formulation using sterile / 0.1 N HCl or 0.1 N NaOH solution and record as follows:

pH adjustment of Component-I (After wP addition and before Incubation)

S. No.	Component Name			
Stirring speed				
Incubation period	Start		Start Time	
	Date			
	End Date		End Time	
Net incubation	Hours minutes			
period				
Remarks				

2. Formulation procedure for Component II (HepB) (Day II):

- a) Add the following ingredients in sequence mention below
- b) Aseptically transfer the calculated quantity of Aluminium phosphate gel into the blending vessel.
- c) Aseptically transfer the calculated quantity of Hepatitis B slowly under constant stirring.
- d) Stir for 10 to 15 mins. for mixing (stirring at 150 rpm/CRT).
- e) Aseptically transfer the calculated quantity of filter sterilized 2-phenoxyethanol under constant stirring.
- f) Stir for 10 to 15 mins for mixing (stirring at 150 rpm/CRT).
- g) Aseptically transfer the calculated quantity of sodium chloride solution under constant stirring.

h) Stir for 10 to 15 mins for mixing (stirring at 150 rpm/CRT).
 i) Check the pH, if it is not in range of 6.0 to 7.0, adjust the pH of formulation using sterile 0.1 N HCl or 0.1 N NaOH solution and record as follows:
 pH adjustment of Component-II(Before Incubation)

S. No.	Component Name
Stirring speed	
Incubation period	Start Start Time
Date	
End Date	End Time
Net incubation	Hours minutes
period	
Remarks	

3. Mixing of component I and component II (Day III):

- On day III aseptically transfer the component II in the component I
- Stir for 30 min for mixing (stirring at 150 rpm/CRT)

4. Final Mixing of Hib TT and IPV bulk in DTwP component (Day III):

- On day III aseptically transfer the calculated quantity of Hib TT bulk under constant stirring to the bulk of step 12.2.3 (b).
- Stir for 30 min for mixing (stirring at 150 rpm/CRT).
- Aseptically transfer the calculated quantity of IPV bulk and saline solution under constant stirring to it.
- Stir for 30 min for mixing (stirring at 150 rpm/CRT).
- Check the pH, if it is not in range of 6.5 to 7.5, adjust the pH of formulation using sterile 0.1 N HCl or 0.1 N NaOH solution and record as follows:
 pH adjustment of final bulk

S. No.	Component Name	Vol. of component used	Initial pH	Volume of 0.1 N HCl required	Volume of 0.1 N NaOH required	Final pH
--------	----------------	------------------------	------------	------------------------------	-------------------------------	----------

 Stir for 2 hr for mixing (stirring at 150 rpm/CRT) and for further storage keep the formulated bulk at 2-8°C.
 Aseptically take the sample of bulk EasySix vaccine for QC testing.
 Filling of vaccine
 Aseptically fill the formulated bulk vaccine in suitable containers for shipment.
 Store the vaccine at 5°C±3°C.
 Maintain the records of formulation & filling. After obtaining the results of analysis, prepare a Certificate of analysis for the batch.
 AnnexureS
 Annexure I : Manufacturing Flow Chart for formulation of EasySix Vaccine.
 Annexure II : Details of the filling record for single-dose glass vials
 Annexure III : Details of the filling records for multi-dose glass vials.
 Annexure IV : Details of the filling records for Pre-Filled Syringes.

Immunogenicity and safety of a newly developed liquid DTwP-Hib/HepB-IPV hexavalent vaccine (EasySix™) was evaluated and compared with administration of commercially licensed Pentavac SD® (DTwP-HepB/Hib), SII and Imovax Polio® vaccine (Sanofi) in an open-label, randomized multi-centric trial. 284 participants, aged 6-10 weeks, randomized in a 1:1 allocation, received three doses of test or comparator vaccines, administered 4 weeks apart. Immunogenicity of the vaccines was determined by measuring the baseline and post-vaccination antibody responses and comparing the proportions of subjects achieving seroprotection against the vaccine antigens; safety was evaluated in terms of solicited (local and systemic) and unsolicited incidences in the follow up phase.

Post-vaccination, seroprotection was achieved against all six vaccine antigens in both vaccine groups. The seroresponse rate as well as geometric mean titers of antibody for all vaccine components were comparable between EasySix™ and Pentavac SD®-Imovax Polio® group. Both vaccines had similar reactogenicity profiles and were well tolerated; all adverse events resolved completely without any sequelae. Only one serious adverse event was reported that completely resolved; it was regarded unconnected to the vaccine administered. This study demonstrated that immunogenicity and safety profiles of EasySix™ vaccine, manufactured by Panacea Biotec Ltd, are non-inferior to the commercially available vaccines. Clinical trial registration: CTRI/2015/02/005578.

Open-label trial: a trial that is not completely blinded, typically in that subjects and physicians have knowledge of the assigned treatment. Importantly, it does not mean "uncontrolled" or "single-arm" which is often what the term "open-label" is used to describe. •

[1.0 |Purpose: |
	To lay down the procedure for the determination of CDAP
	(1-cyano-4-dimethylaminopyridinium tetrafluoroborate) by high performance
	liquid chromatography (HPLC).
[2.0	Applicable to:
	This Standard Operating Procedure is applicable for the analysis of the
	samples in which CDAP (1-cyano-4-dimethylaminopyridinium tetrafluoroborate)
	is to be determined.
[3.0	Responsibility
	3.1
	procedure correctly.
	3.2
	that the procedure is complied with and for checking the results.
[4.0	Abbreviations and Definitions
	4.1
	4.2
	4.3
	4.4
	4.5
	4.6
	4.7
	4.8
	4.9
	4.10
	4.11
	4.12
	4.13
	4.14
	4.15
	4.16
	4.17
	4.18
	4.19
	4.20
	4.21
	4.22
	4.23
	4.24
	4.25
	4.26
	4.27
	4.28
	Suitability
	Test
	chromatographic system are adequate for the analysis
	to be done.
	The tests are based on the concept that the equipment,
	electronics, analytical operations and samples to be
	analyzed constitute an integral system that can be
	evaluated as such.
[5.0	Procedure
	5.1
	5.1.1
	5.1.2
	be done in order to eliminate the entry of possible particles,
	if any, into the system as well as the column.
	5.1.3
	5.1.4
	5.1.5
	aluminum foil.
	5.1.6
	5.1.7
	5.2
	5.2.1
	5.2.1.1
	5.2.1.2
	5.2.1.3
	5.2.1.4
	5.2.1.5
	5.2.2
	5.2.2.1
	5.2.2.2
	5.2.2.3
	5.2.2.4
	5.2.3
	5.2.3.1
	5.2.3.2
	5.2.3.3
	5.2.3.4
	5.2.3.5
	filters.
	5.2.3.6
	5.2.3.7
	5.2.3.8
	5.2.4
	5.2.4.1
	5.2.4.2
	5.2.4.3
	5.2.4.4
	5.2.4.5

			tetrafluoroborate)			
			5.2.4.6	2-Propanol		
		5.2.4.7	Hydrochloric acid 37 %			
5.3	Preparation of solutions:					
		Prepare all the solutions and carry out the labelling of the prepared				
		solutions and assign the lot numbers to the solutions.				
		5.3.1	Solution A			
			(NaH₂PO₄. H₂O, 200mM)			
			For a volume of 1 liter, weigh			
			NaH₂PO₄. H₂O	27.6g		
			Adjust the pH to 3.0 with HCl 37%.			
			Make up the solution to 1litre with purified water. Prepare			
			fresh before use.			
		5.3.2	Mobile phase for CDAP:			
			Take 700mL of methanol in a measuring cylinder and 300mL of			
			solution A to make the volume to 1000mL. Use the mobile phase			
			within four days from the date of preparation if it remains			
			clear and no turbidity or fibers observed in it.			
		5.3.3	Isopropyl Alcohol 70% Solution			
			(For storing and purging the HPLC system)			
			Take 700 mL of 2-propanol using a measuring cylinder and make			
			up the volume to 1000 mL with Purified Water in a volumetric			
			flask. This solution can be stored at room temperature for 3			
			months.			
		5.3.4	1 mg/mL of CDAP calibration standard stock:			
			Weigh 5mg of CDAP and make up the volume to 5mL with purified			
			water in a volumetric flask. This solution can be stored as			
			aliquots at -20°C for 2months.			
		5.3.5	Preparation of working calibration standard of 10µg/mL or			
			10ng/µL:			
			Take 10µL of 1 mg/mL of CDAP calibration standard stock and			
			add 990µL of purified water to make 1mL of solution in an			
			eppendorf or HPLC vial. Mix thoroughly. Prepare fresh before			
			use.			
			Note: The working calibration standard is prepared fresh			
			before use. All the solutions prepared for use are vacuum			
			filtered using Nylon 6, 6 membrane 0.20/0.22/0.45 µm filters			
			to eliminate the entry of possible particles, if any into the			
			system as well as the column. For organic solvents 0.45µm			
			filters should be used (filtration not necessary if the			
			solution is prepared in 0.2 µ filtered purified water).			
5.4	Preparing the HPLC System for run:					
5.5	Run Conditions:					
		5.5.1	Load method CDAP.M.			
		5.5.2	Flow Rate: 1.0 mL/min			
		5.5.3	Detection wavelength: 280 nm			
		5.5.4	Run time: 4 minutes			
		5.5.5	Mobile Phase: mobile phase for CDAP			
5.6	Equilibration of the system and the column:					
		Equilibrate the system and column with mobile phase for CDAP until a				
		stable baseline at 280 nm is obtained which can be viewed with the				
		online plot window.				
5.7	Sample Preparation and Application:					
		Test sample and standards are loaded in screw cap vials and placed				
		against the appropriate no. on the sample tray by clearly marking				
		the screw cap vial.				
5.8	Running of sequence:					
		5.8.1	Load sequence CDAP.S.			
		5.8.2	Open the sequence parameter option by clicking Sequence menu.			
		5.8.3	Enter information about the data directory and operator name.			
		5.8.4	In the shutdown option, click on the post-sequence option and			
			select LAMPOFF/STANDBY.			
		5.8.5	In sequence table option, specify the sample names, methods			
			and vials to run in the sequence as per the following table:			
		S. No	Sample Code	Sample details	No. of	Injectio
			injections	n volume		
			per Vial			
		1	Blank	Eluent Blank	1	10 µL
		2	Standard 3	System	6	0.5 µL
			(5ng/µL)	Suitability		
			test standard			
		3	Blank	Eluent Blank	1	10 µL
		4	Standard 1	1ng/µL	1	0.1 µL
			Standard			
		5	Standard 2	3ng/µL	1	0.3 µL
			Standard			
		6	Standard 3	5ng/µL	1	0.5 µL
			Standard			
		7	Standard 4	10ng/µL	1	1.0 µL
			Standard			
		8	Standard 5	20ng/µL	1	2.0 µL
			Standard			
		9	Blank	Eluent Blank	1	10 µL
		10	Test Sample	Sample	1	1.0 µL
			(s)			
		5.8.6	Click on run sequence option in run control menu to start the			
			sequence.			
		5.8.7	Each sample run will be of 4 minutes and after running all			
			the samples the sequence will automatically stop.			
		5.8.8	Inject the eluent blank after every 20 injections.			
			Note: Here, eluent blank refers to mobile phase. One extra			
			vial with same SST sample can be used if 6 injections are not			

||| |possible from single vial. |

||5.9 |Calculations: |

||5.9.1 |Results of the sample are calculated from the calibration |

||| |standards using the chemstation software. |

||| |5.9.1.1 |Open the chromatogram corresponding to 1ng/μL |

||| |calibration standard in the offline copy of |

||| |chemstation software in the Data Analysis mode. |

||| |5.9.1.2 |From the calibration menu select calibration |

||| |settings and write ng/μL in the Amount unit's |

||| |option. |

||| |5.9.1.3 |Select New Calibration Table from Calibration |

||| |menu and enter value1 in default amount option |

||| |and value 1 in level option. |

||| |5.9.1.4 |Now open the chromatogram for 3.0 ng/μL. Select |

||| |add level option from calibration menu and enter |

||| |value 3.0 in default amount and value 2 in level |

||| |option. |

||| |5.9.1.5 |Similarly open chromatograms for 5.0ng/μL, |

||| |10.0ng/μL and 20.0ng/μL and keep adding default |

||| |amount and appropriate value of level. |

||| |5.9.1.6 |Select calibration task icon and the calibration |

||| |table with required concentrations and levels can|

||| |be viewed in the calibration table view. A |

||| |calibration curve is also displayed in the |

||| |calibration curve field with value of correlation|

||| |coefficient of the curve. |

||| |5.9.1.7 |Save the calibration method under appropriate |

||| |name. |

||| |5.9.1.8 |Open the chromatogram of test sample and select |

||| |the above created calibration method for |

||| |integration of the test sample. |

||| |5.9.1.9 |In the specify report option of Report menu, |

||| |select ESTD in the Calculate field. |

||| |5.9.1.10 |In the report layout view click on the show |

||| |report preview with current template, method and |

||| |data file icon and view the result of the test |

||| |sample in terms of concentration (ng/μL). |

||| |5.9.1.11 |Calculations for Standard curve and samples: |

||| |The calculations are made according to the |

||| |equation of minimum squares. |

||| | $Y = mX + b$. |

||| |Where: m = slope |

||| |b = intercept |

||| |Y = absorbance |

||| |X = concentration |

||5.9.2 |The calculations for determination of amount of CDAP per dose|

||| |(10μg) of ARM of Haemophilus Influenzae type b conjugate |

||| |vaccine is as follows: |

||| |Example: If a sample has a PRP concentration of 1.0 mg/mL and|

||| |the amount of CDAP present in the sample is 5ng/μL or 5μg/mL.|

||| |The amount of CDAP present per dose(10μg) is calculated as |

||| |follows: |

||| |1000μg -----5μg |

||| |10 μg(dose)----- X |

||| | $X = 5\mu\text{g} \times 10 \mu\text{g}$ |

||| |1000 μg |

||| |= 0.05μg or |

||| |X = 50 ng/dose |

||5.9.3 |Calculations for System Suitability Test: |

||| |5.9.3.1 |Note the values of area under the curve and |

||| |retention time for standard 3(SST standard) . |

||5.10 |Acceptance Criteria System Suitability Test: |

||5.10.1|RSD calculated for Retention time of standard 3(5ng/μL) |

||| |should be <5.0%. |

||5.10.2|RSD calculated for Area of standard 3(5ng/μL) should be |

||| |<5.0%. |

||5.11 |Result Reporting: |

||5.11.1|Open CDAP.FRP template and take print out of the report. |

||5.11.2|Print report: Click on print option to take a printout of a |

||| |ChemStation report using the current report template and data|

||| |from the currently loaded data file. |

||5.11.3|Report the results of quantity of CDAP present/dose (10μg). |

||5.11.4|The report generated can be viewed with report review icon. |

||5.11.5|The printed report is signed by the analyst who does the |

||| |analysis and also checked by the other analyst/supervisor. |

||6.0 |Forms and Records: |

||6.1 |Determination of CDAP by high performance liquid chromatography. |

||7.0 |References: |

||9.1 |European pharmacopoeia. |

1.0 |Objective: |
 | |To lay down the procedure for quantifying Total and unconjugated (free) |
 | |PRP concentration. |
 2.0 |Scope: |
	This Standard Operating Procedure is applicable for the analysis of all
	the samples of PRP and PRP-TT bulk conjugate of Haemophilus Influenzae
	type b in which Total and unconjugated (free) PRP concentration is to be
	determined, received in the Quality control department.
3.0	Responsibility
	3.1
	procedure correctly.
	3.2
	complied with and the results are checked.
4.0	Abbreviations and Definitions
	4.1
	4.1.1
	4.1.2
	4.1.3
	4.1.4
	4.1.5
	4.1.6
	4.1.7
	4.1.8
	4.1.9
	4.1.10
	4.1.11
	4.1.12
	4.1.13
	4.1.14
	4.2
5.0	Procedure:
	5.1
	5.1.1
	5.1.2
	5.1.3
	5.1.4
	5.1.5
	5.1.6
	5.1.7
	5.2
	5.2.1
	5.2.2
	5.2.3

	parameters of correlation coefficient i.e. ≥ 0.990 .	
	5.4	Total PRP:
	5.4.1	Take sample volume such that the sample volumes used in
		test contain 2.5 to 25 μg of ribose i.e. lies within the
		calibration curve.
	5.4.2	Make up the volume to 2 mL using purified water
		respectively.
		Note: The sample volume can vary depending upon the ribose
		concentration. If there is any sample outside the range of the
		calibration curve, prepare an appropriate dilution of the sample.
	5.5	Unconjugated (Free) PRP:
	5.5.1	Take 1 mL of sample (dilute if required) having about
		100-150 μg PRP in micro centrifuge tube in duplicate, add
		100 μL of 1% DOC and mix it.
	5.5.2	Put this tube in an ice bath for 30 minutes.
	5.5.3	Take out and add 50 μL 1M HCl and mix thoroughly.
	5.5.4	Centrifuge the mixture at 4°C for 15 minutes at 4000 RPM.
	5.5.5	Take 1 mL of supernatant in 20 mL labeled test tube.
	5.5.6	Make up the volume to 2 mL using purified water
		respectively.
	5.5.7	The test is performed in duplicate for each sample.
	5.6	Treat 2 mL purified water as blank in thermoresistant test tube.
	5.7	Add 2 ml of 0.5 g/l Ferric Chloride solution to tubes of
		calibration curve, Blank and samples.
	5.8	Vortex and add 0.2 mL of 100 g/l Orcinol Solution to all test
		tubes.
	5.9	Close the tubes with aluminium foil and place the test tubes in
		boiling water bath for 20 minutes.
	5.10	Take out the tubes and cool in iced water.
	5.11	Allow the tubes to come at room temperature and set the wavelength
		in spectrophotometer at 670 nm. Zero the absorbance of blank and
		then measure the absorbance of calibration curve and test samples.
	5.12	Calculations:
	5.12.1	Once the absorbance of the standards and the test samples
		have been taken, calculation shall be done using the excel
		sheet carrying formulas for the required calculation.
	5.12.2	Always copy the master excel sheet (sheet carrying formulas)
		for the required calculation), never use the previously
		used excel sheet for new calculations.
	5.12.3	Copy the master excel sheet, keep it for use in respective
		folder and then fill in the required sample details for
		total & free sample treatments, sample dilution, name of
		analyst, date, etc.
	5.12.4	Enter the absorbance for calculations starting from the
		Calibration curve prepared between 2.5 and 25 g of Ribose
		and test samples. The calculations are made according to
		the equation of minimum squares.
		$Y = mX + b$.
		Where: m = slope
		b = intercept
		Y = absorbance
		X = concentration
		(Multiply by dilution factor)
	5.12.5	Calculation of actual dilution for free PRP:
		Sample volume taken = 1000 μL
		Total volume = 1150 μL (1000 μL sample + 100 μL 1% DOC + 50 μL
		1M HCl)
		Correction Factor = 1150 μL / 1000 μL = 1.15
		Actual dilution = dilution factor $\times$ 1.15
	5.12.6	Calculate the content of ribose in the test solution for
		each volume tested and PRP content by the formula :
		PRP content = Ribose content / 0.434
	5.12.7	Record all the calculations in the data record form and
		also attach the hard copy of the excel sheet used for
		calculation along it.
	5.6	Acceptance criteria:
	5.6.1	The coefficient of variation (CV) between the replicates of
		the samples should be $\leq 5\%$.
	6.0	Forms and Records:
	6.1	Determination of Total and Free PRP Concentration
	7.0	References:
	9.1	Biologicals (1998) 26, 33-38 – A simple and rapid method for
		measuring unconjugated capsular polysaccharide (PRP) of Haemophilus
		influenzae type b in PRP-TT conjugate vaccine.
	9.2	European pharmacopoeia.

[1.0 |Objective: |
	To lay down the procedure for immunological identification of Purified
	Polysaccharide (PRP HIB) and Bulk Conjugate by immunodiffusion using
	polyclonal antibodies.
[2.0	Scope:
	This Standard Operating Procedure is applicable to the Purified
	Polysaccharide (PRP HIB) and Bulk Conjugate of Haemophilus influenzae type
	b vaccine.
[3.0	Responsibility:
	3.1
	for executing the described procedure correctly.
	3.2
	that the procedure is complied with and for checking the results.
[4.0	Abbreviations and Definitions:
	4.1
	4.2
	4.3
	4.4
	4.5
	4.6
	4.7
	4.8
	4.9
	4.10
	4.11
	4.12
[5.0	Procedure
	5.1
	5.2
	other. For Bulk Conjugate, load the sample of ARM HIB TT in the
	central well and PRP antisera in one of the surrounding wells and
	tetanus antitoxin in the other.
	5.3
	slides in a box with lid on top containing pre-wetted filter paper
	sheets and incubate in an incubator at 37± 2°C for 48 hours and look
	for the line of identity.
	5.4
	5.5
	passes identity test
[6.0	Reporting:
	Identification Of Purified Polysaccharide And Bulk Conjugate Of Haemophilus
	influenzae Type B Conjugate Vaccine By Immunodiffusion Method.
[7.0	References:
	7.1

[1.0]Purpose: |
|To lay down the procedure for the estimation of molecular size of PRP-TT |
|conjugate in bulk conjugate of Haemophilus Influenzae type b conjugate |
|vaccine. |
[2.0]Applicable to: |
|This Standard Operating Procedure is applicable for evaluating the |
|molecular size distribution and percentage elution of bulk conjugate of |
|Haemophilus Influenzae type b conjugate vaccine. |
[3.0]Responsibility: |
3.1	The analyst shall be responsible for carrying out the described
	procedure correctly.
3.2	The Supervisor of the laboratory shall be responsible for verifying
	that the procedure is complied with and for checking the results.
[4.0]Abbreviations and Definitions:	
4.1	CDAP
4.2	cm
4.3	g
4.4	HIBTT
4.5	HPLC
4.6	i.d.
4.7	l
4.8	M
4.9	m
4.10	mAu
4.11	mg
4.12	min
4.13	mL
4.14	mm
4.15	nm
4.16	No.
4.17	OE
4.18	PRP-TT
4.19	QA
4.20	QC
4.21	QCR
4.22	RSD
4.23	SOP
4.24	SST
4.25	°C
4.26	µg
4.27	µL
4.28	µm
4.29	System
	liquid chromatographic methods. They are used to verify
	the test that the resolution and reproducibility of the
	chromatographic system are adequate for the analysis to be
	done.
	The tests are based on the concept that the equipment,
	electronics, analytical operations and samples to be
	analyzed constitute an integral system that can be
	evaluated as such.
[5. Procedure:	
5.1	Precautions to be taken:
	5.1.1
	5.1.2
	5.1.3
	5.1.4
	5.1.5
	5.1.6
5.2	Requirement:
	5.2.1
	5.2.2
	5.2.3
	5.2.4

		5.2.4.3	Sodium dihydrogen phosphate monohydrate				
		5.2.4.4	Purified water				
		5.2.4.5	Sodium azide				
		5.2.4.6	Salmon DNA				
		5.2.4.7	2-Propanol				
5.3	Preparation of solutions:						
		Prepare all the solutions and Carry out the labelling of the prepared					
		solutions and assign the lot numbers to the solutions.					
	5.3.1	Isopropyl alcohol 70%					
			Take 700 mL of 2-propanol in a volumetric flask and make up				
			the volume upto 1000 mL with Purified Water. This solution can				
			be stored at room temperature for 3 months.				
	5.3.2	Solution I: [Na₂HPO₄ (0.1M), NaCl (0.1M)]					
			For a volume of 1000 mL, weigh:				
			Na₂HPO₄	14.19 g			
			NaCl	5.84 g			
			Dissolve in 700mL of purified water in a volumetric flask and				
			make up the volume upto 1000mL. Prepare fresh solution before				
			use.				
	5.3.3	Solution II: NaH₂PO₄.H₂O, 0.1 M					
			For a volume of 500 mL, weigh:				
			NaH₂PO₄.H₂O	6.85g			
			Dissolve in 400mL of purified water in a volumetric flask and				
			make up the volume upto 500mL. Prepare fresh solution before				
			use.				
	5.3.4	0.1M Sodium Phosphate buffer					
			(0.1 M Na₂HPO₄,0.1M NaCl +0.1M NaH₂PO₄.H₂O, pH 7.0)				
			For a volume of 1000mL, take approximately 600mL of Solution I				
			and 400mL of Solution II. The pH of the solution should be 7.0				
			± 0.2. Use the mobile phase within four days from the date of				
			preparation if it remains clear and no turbidity or fibers				
			observed in it.				
	5.3.5	Void Volume Marker (Salmon DNA):					
			For a volume of 5 mL, weigh:				
			Salmon DNA	5 mg			
			Dissolve in purified water and make up volume upto 5 mL with				
			purified water in a volumetric flask. Store at -20 °C ± 5°C				
			for 1 month.				
	5.3.6	Total volume marker (Sodium Azide):					
			For a volume of 10 mL, weigh:				
			Sodium Azide	10 mg			
			Dissolve in purified water and make up volume upto 10 mL with				
			purified water in a volumetric flask. Store at 2-8 °C for 1				
			month.				
	5.3.7	Column Preservation Solution:					
			Note: All the solutions prepared for use are vacuum filtered				
			using Nylon 6, 6 membrane 0.20 µm or 0.22 µm , depending upon				
			the nature of solution to be filtered. For organic solvents				
			0.45µm filters should be used.				
5.4	Preparing the HPLC System for run, Agilent Series:						
5.5	Run Conditions:						
	5.5.1	Load method HIBTT.M.					
	5.5.2	Flow Rate: 0.5 mL/min					
	5.5.3	Injection Quantity : 2.5µg					
	5.5.4	Detection Wavelength: 214nm					
	5.5.5	Run time: 30 minutes					
5.6	Equilibration of the system and the column:						
		Equilibrate the column with mobile 0.1 M sodium phosphate buffer at					
		214nm until a stable baseline is observed on the online plot window.					
5.7	Sample Preparation and Application:						
	5.7.1	Test sample and standards are loaded in screw cap vials and					
			placed against the appropriate no. on the sample tray by				
			clearly marking the screw cap vial.				
	5.7.2	Load the required volume of sample as per the PRP					
			concentration equivalent to 2.5 µg in the sample vial.				
			Example: If a sample has a PRP concentration of 0.5 mg/mL, to				
			calculate volume required for 2.5 µg of protein,				
			0.5 mg (500 µg) ----1 mL				
			2.5 µg-----X mL				
			X mL = 2.5 µg x 1 mL				
			500 µg				
			X mL = 5 µL				
			Therefore, 5 µL of volume is taken for sample having PRP				
			concentration of 0.5 mg/mL.				
5.8	Running of sequence:						
	5.8.1	Load sequence HIBTT.S.					
	5.8.2	Open the sequence parameter option by clicking Sequence menu.					
	5.8.3	Enter information about the data directory and operator name.					
	5.8.4	In the shutdown option, click on the post-sequence option and					
			select LAMPOFF/STANDBY.				
	5.8.5	In sequence table option, specify the sample names, methods					
			and vials to run in the sequence as per the following table:				
			S No	Sample Code	Sample details	No. of	Injection
				inject	volume (µL)		
				ions			
			1	Blank	Eluent Blank	1	10
			2	Sodium azide (Total	System	6	1
			volume marker)	Suitability			
				Test			
			3	Blank	Eluent Blank	1	10
			4	Salmon DNA	Void volume	1	10
				marker			

			5	Blank	Eluent Blank	1	10
			6	Test Sample (s)	Sample	1	2.5µg of PRP
			Click on run sequence option in run control menu to start				
			the sequence.				
			5.8.8	Each sample run will be of 30 min and after running all			
			the samples the sequence will automatically stop.				
			5.9	Calculations :			
			Area under curve for the standards/sample is analyzed by				
			Chemstation software.				
			5.9.1	The retention time of the void volume is determined with			
			Salmon DNA.				
			5.9.2	The Retention time of the component (total volume) which			
			has full access to the pores of stationary phase is				
			determined with sodium azide.				
			5.9.3	Elution time of PRP-TT (Te) at KD = 0.50			
			The calculations are made as per the following equation:				
			Te = KD (Tt - To) + To or Te = 0.50 (Tt -				
			To) + To				
			Where,				
			Te = Elution time of PRP-TT Bulk Conjugate at KD = 0.50				
			Tt = Retention time of sodium azide				
			To = Retention time of Salmon DNA				
			KD = Distribution Coefficient				
			Note: Take average RT of sodium azide as Tt from EXCEL				
			for calculation of SST.				
			5.9.4	Calculation of percentage of sample eluted before a given			
			KD:				
			The percentage of sample eluted before a given KD is				
			calculated using ChemStation software.				
			5.9.4.1	Open the acquired chromatogram in the data			
			analysis view in the offline copy of the				
			software.				
			5.9.4.2	Optimize integration with initial events in the			
			signal specific events table.				
			5.9.4.3	Enter the value of Elution time (Te) calculated			
			in step 5.9.3 in integration with timed events				
			option or with manual integration and select				
			split peak option.				
			5.9.4.4	The percentage of sample eluted at $KD \leq 0.5$ is			
			calculated by adding the values of percentage of				
			area (Area %) obtained for bulk conjugate curve				
			till Te.				
			5.9.5	Calculations for System Suitability Test:			
			5.9.5.1	Note the values of area under the curve and			
			retention time for sodium azide (SST standard)				
			in Annexure 1 and also in EXCEL format for				
			calculation of Relative standard deviation.				
			5.9.5.2	Every time a sample is run with system			
			suitability test, calculate the results for				
			system suitability.				
			5.9.5.3	Attach the hard print of SST calculation along			
			with chromatograms and the record form.				
			5.10	Acceptance Criteria :			
			5.10.1	For System Suitability Test :			
			1				
			5.10.1.1	RSD calculated for Retention time should be <			
			1	5.0%.			
			5.10.1.2	RSD calculated for Area should be < 5.0%.			
			2				
			5.10.2	For ARM of Haemophilus Influenzae type b conjugate			
			2	vaccine Samples: KD (Molecular Size distribution): $\geq 85\%$,			
			KD ≤ 0.50				
			6.0	Forms and Records:			
			6.1	Molecular sizing of polyribosyl ribitol phosphate (PRP) – Tetanus			
			toxoid conjugate by high performance size exclusion chromatography:				
			7.0	References:			
			United States Pharmacopeia 29.				

[1.0 |Purpose: |
 ||To lay down the procedure for quantifying nucleic acid by absorbance. |
 [2.0 |Applicable to: |
	This Standard Operating Procedure is applicable for the analysis of samples
	of PRP of Haemophilus Influenzae type b in which Nucleic acid concentration
	is to be determined
[3.0	Responsibility:
	The analyst shall be responsible for executing the described procedure
	correctly. Supervisor shall be responsible for verifying that the
	procedure is complied with and the results are checked.
[4.0	Abbreviations:
	4.1
	4.2
	4.3
	4.4
	4.5
	4.6
	4.7
	4.8
[5.0	Procedure:
	5.1
	5.1.1
	5.1.2
	5.1.3
	5.1.4
	5.1.5
	5.2
	5.2.1
	5.2.2
	5.2.3
	5.2.4
	5.2.5
	5.2.6
	5.2.7
	5.2.8
[6.0	Reporting:
	6.1
[7.0	Reference:
	Not applicable

1.0 |Purpose: |
|To lay down the procedure for the quantification of ribose concentration. |
2.0 |Applicable to: |
|This Standard Operating Procedure is applicable to all the samples of PRP |
|and PRP-TT bulk conjugate of Haemophilus Influenzae type b in which ribose |
|concentration is to be determined. |
3.0 |Responsibility: |
|The analyst shall be responsible for executing the described procedure |
|correctly. Supervisor shall be responsible for verifying that the procedure|
|is complied with and the results are checked. |
4.0 |Abbreviations |
4.1	Abbreviations:	
4.1.1	CV	:Coefficient of variation
4.1.2	g	:Gram
4.1.3	mg	:Milligram
4.1.4	mL	:Millilitre
4.1.5	µl	:Micro litre
4.1.6	µg	:Microgram
4.1.7	nm	:Nanometer
4.1.8	PRP	:Polyribosyl Ribitol Phosphate
4.1.9	TT	:Tetanus Toxoid
4.1.10	g/l	:Grams per Litre
4.1.11	w/v	:Weight by Volume
5.0	Procedure:	
5.1	Preparation of Solutions:	
Prepare all the solutions as described below and label the solutions.		
5.1.1	0.5 g/l Ferric chloride solution:	
5.1.1.1	Weigh 50 mg of Ferric Chloride hexahydrate and	
transfer it to 100 mL volumetric flask.		
5.1.1.2	Dissolve and make up the volume to 100 mL with	
concentrated hydrochloric acid.		
5.1.1.3	Store the solution in a screw cap amber bottle, for a	
maximum of 6 months at room temperature.		
5.1.2	100 g/l Orcinol solution:	
5.1.2.1	Weigh 1gm of Orcinol monohydrate and transfer it to 10	
mL volumetric flask.		
5.1.2.2	Dissolve and make up the volume to 10 mL with absolute	
ethanol.		
5.1.2.3	Store the solution in a screw cap amber bottle at 2 to	
8 °C, for a maximum of 15 days.		
5.1.3	1 mg/ml Ribose Solution:	
5.1.3.1	Weigh 25 mg of D-Ribose and transfer it to a 100 mL	
volumetric flask.		
5.1.3.2	Dissolve in purified water and make up the volume to	
100 mL with purified water.		
5.1.3.3	Prepare fresh prior to use.	
5.1.4	25 µg /mL Ribose Solution:	
5.1.4.1	Take 1 mL of 1 mg/ml of Ribose solution and transfer	
it to a 10 mL volumetric flask.		
5.1.4.2	Make up the volume to 10 mL with purified water.	
5.1.4.3	Prepare fresh prior to use.	
5.1.5	10% w/v Cetyltrimethylammonium bromide (CTAB) Solution:	
5.1.5.1	Weigh 1gm of Cetyltrimethylammonium bromide and	
transfer it to 10 mL volumetric flask.		
5.1.5.2	Dissolve and make up the volume to 10 mL with Purified	
water.		
5.1.5.3	Solution shall be prepared freshly.	
5.1.6	0.25 M Sodium hydroxide:	
5.1.6.1	Weigh 1gm of Sodium hydroxide and dissolve in	
sufficient amount of purified water and make up the		
volume to 10 mL with Purified water.		
5.1.6.2	Store the solution in a screw cap bottle, for a	
maximum of 6 months at room temperature.		
5.2	Preparation of Calibration Curve:	
5.2.1	Prepare calibration curve (2.5-25 µg) with freshly prepared	
ribose solution (25 µg /ml) in previously labelled test tubes		
of 20 mL.		
5.2.2	Calibration Curve (2.5 - 25µg) is prepared in duplicates as	
follow:		
Ribose content (µg)	2.5	5
25µg/ml ribose (µl)	100	200
Purified water (µl)	900	800
5.2.3	Make up the volume to 2 mL using purified water.	
Note: The Calibration curve is accepted if it is meeting the		
parameters of correlation coefficient ie. ≥ 0.990 .		
5.3	Test:	
5.3.1	Take sample volume such that the sample volumes used in test	
contain 2.5 to 25 µg of ribose i.e. lies within the		
calibration curve.		
In case of dry polysaccharide prepare a solution containing		
about 1-5 mg/mL of polysaccharide use it as test sample)		
5.3.2	Make up the volume to 2 mL using purified water respectively.	
Note: In case of fermentation samples, treat 1 mL of the sample with		
50µl 10 % CTAB, vortex and keep for 10 minutes, centrifuge at 7000		
RPM for 5 minutes. Discard the supernatant; add 1 mL purified water,		
vortex and again centrifuge for 5 minutes at 7000 RPM. Discard the		
supernatant and let the pellet air dry. Solublise pellet with 1 mL		
0.25 M NaOH. Treat this as sample for ribose estimation.		
5.3.3	Treat 2 ml purified water as blank in thermoresistant test	
tube.		

		5.3.4	Add 2 ml of 0.5 g/l Ferric Chloride solution to tubes of
			calibration curve, Blank and sample.
		5.3.5	Vortex and add 0.2 ml of 100 g/l Orcinol Solution to all test
			tubes.
		5.3.6	Close the tubes with aluminium foil and place the test
			tubes in boiling water bath for 20 minutes.
		5.3.7	Take out the tubes and cool in ice cold water.
		5.3.8	Allow the tubes to come at room temperature.
		5.3.9	Set the wavelength in spectrophotometer at 670 nm. Zero the
			absorbance of blank and then measure the absorbance of
			calibration curve and test samples.
		5.4	Calculations:
		5.4.1	Once the absorbance of the standards and the test samples
			have been taken, calculation shall be done using the excel
			sheet carrying formulas for the required calculation.
		5.4.2	Enter the absorbance for calculations starting from the
			Calibration curve prepared between 2.5 and 25 g of Ribose and
			test samples. The calculations are made according to the
			equation of minimum squares.
			$Y = mX + b$.
			Where: m = slope
			b = intercept
			Y = absorbance
			X = concentration
			(Multiply by dilution factor)
		5.4.3	Calculate the content of ribose in the test solution for each
			volume tested.
		5.4.4	Conversion factor for PRP:
			PRP content = Ribose content / 0.434
		5.4.5	In case of PRP.
			Ribose (%) in PRP on dry weight basis:
			Ribose content in PRP($\mu\text{g/mL}$)
			Ribose content in dry PRP ($\mu\text{g/ml}$) =
			$\frac{\text{Ribose content in PRP}(\mu\text{g/mL})}{1 - \{\text{Moisture (\%)} / 100\}}$
			Ribose content in dry PRP ($\mu\text{g/mL}$)
			Ribose (%) = $\frac{\text{Ribose content in dry PRP}(\mu\text{g/mL})}{\text{Concentration of PRP solution}(\mu\text{g/mL})} \times 100$
			Concentration of PRP solution ($\mu\text{g/mL}$)
		5.5	Acceptance criteria:
		5.5.1	The coefficient of variation (CV) between the replicates of
			the samples should be $\leq 5\%$.
		5.5.2	Ribose content in the sample should be within the acceptance
			limits.
		6.0	Forms and Records:
			Determination Of Ribose Concentration By Orcinol Method .
		7.0	References:
			European pharmacopoeia.

1.0 |Objective: |
 |To lay down the procedure for quantifying the phosphorus concentration. |
 2.0 |Scope: |
 |This Standard Operating Procedure is applicable to all the samples of PRP |
 |of Haemophilus Influenzae type b in which phosphorous concentration is to |
 |be determined. |
 3.0 |Responsibility |
 |3.1 |The analyst shall be responsible for executing the described |
 |procedure correctly. |
 |3.2 |Supervisor shall be responsible for verifying that the procedure is |
 |complied with and the results are checked. |
 4.0 |Abbreviations |
4.1	Abbreviations:		
	4.1.1	CV	:Coefficient of variation
	4.1.2	g	:Gram
	4.1.3	mg	:Milligram
	4.1.4	mL	:Millilitre
	4.1.5	µl	:Micro litre
	4.1.6	µg	:Microgram
	4.1.7	nm	:Nanometer
	4.1.8	PN	:Part Number
	4.1.9	SD	:Standard Deviation
	4.1.10	PRP	:Polyribosyl Ribitol Phosphate
5.0	Procedure:		
5.1	Preparation of solutions:		
	5.1.1	100 µg/mL phosphorus solution:	
		5.1.1.	Weigh 0.2194 g of potassium dihydrogen phosphate and
		1	transfer it to 500 mL volumetric flask.
		5.1.1.	Dissolve and make up the volume to 500 mL with purified
		2	water.
		5.1.1.	This solution is stored at room temprature and used
		3	within 6 months.
	5.1.2	5 µg/mL Phosphorus solution:	
		5.1.2.	Take 5 mL of 100 µg/mL phosphorus solution and make up
		1	the volume to 100 mL with purified water. This solution
		is freshly prepared.	
	5.1.3	25 mg/mL ammonium molybdate solution:	
		5.1.3.	Weigh 625 mg of ammonium molybdate and transfer it to
		1	25 mL volumetric flask.
		5.1.3.	Dissolve and make up the volume to 25 mL with purified
		2	water.
		5.1.3.	This solution is stored at room temperature and used
		3	within 6 months.
	5.1.4	100 mg/mL ascorbic acid solution:	
		5.1.4.	Weigh 2.5 g of ascorbic acid and transfer it to 25 mL
		1	volumetric flask.
		5.1.4.	Dissolve and make up the volume to 25 mL with purified
		2	water.
		5.1.4.	This solution is freshly prepared.
		3	
	5.1.5	3 M Sulphuric acid:	
		5.1.5.	Take approximately 500 mL of purified water in a 1
		1	litre volumetric flask. To this add 162 mL concentrated
		Sulphuric acid slowly.	
		5.1.5.	Make up the volume to 1 litre with purified water.
		2	
		5.1.5.	This solution is stored at room temprature and used
		3	within 6 months.
	5.1.6	Ammonium molybdate reagent:	
		5.1.6.	Take 1 volume of 25g/l solution of ammonium molybdate,
		1	1 volume of 100 g/l solution of ascorbic acid, 1 volume
		of 3 M Sulphuric acid and 2 volumes of purified	
		water. This solution is freshly prepared.	
5.2	Preparation of Calibration Curve:		
	5.2.1	Prepare calibration curve (2.5-10 µg) with freshly prepared	
	5µg/mL phosphorus solution in previously labelled micro		
	centrifuge tube. Calibration curve is prepared in duplicates		
	as follow:		
	Phosphorus content (µg)	2.5	5
	5µg/mL phosphorus solution (µl)	500	1000
	Note: The Calibration curve is accepted if it is meeting the		
	parameters of correlation coefficient i.e. ≥ 0.990		
5.3	Test:		
	5.3.1	Dilute the sample so that the sample volumes (1mL) used in	
	test contain about 6 µg of phosphorus i.e. lies within the		
	calibration curve.		
		(In case of dry polysaccharide prepare a solution containing	
	about 1-5 mg/mL of polysaccharide and use it as test sample)		
	5.3.2	Treat 2 mL purified water as blank in thermoresistant test	
	tube.		
	5.3.3	Add 0.2 mL of concentrated sulphuric acid to tubes of	
	calibration curve, Blank and sample.		
	5.3.4	Heat the tubes in oil bath or turbotherm at 120°C for 1 hour	
	and then at 160°C until white fumes appear (about 1 hour).		
	5.3.5	Add 0.1 mL of perchloric acid to all the tubes and heat the	
	tubes at 160°C until the solution is decolourised (about 90		
	minutes).		
	5.3.6	Cool at room temperature and add 4 mL of purified water and 4	
	mL of ammonium molybdate reagent.		
	5.3.7	Heat in water bath at 37°C for 90 minutes and cool.	

		5.3.8	Transfer the contents of each tube to 10 mL volumetric flasks
		and make up the volume of each tube to 10 mL with purified	
		water.	
		5.3.9	Set the wavelength in spectrophotometer at 820 nm. Zero the
		absorbance of blank and then measure the absorbance of	
		calibration curve and test samples. The blue colour is stable	
		for several hours.	
		5.4	Calculations:
		5.4.1	Once the absorbance of the standards and the test samples
		have been taken, calculation shall be done using the excel	
		sheet carrying formulas for the required calculation.	
		5.4.4	Enter the absorbance for calculations starting from the
		Calibration curve prepared between 2.5 and 10 g of phosphorus	
		and test samples. The calculations are made according to the	
		equation of minimum squares.	
		$Y = mX + b$.	
		Where: m = slope	
		b = intercept	
		Y = absorbance	
		X = concentration	
		(Multiply by dilution factor)	
		5.4.5	Calculate the content of phosphorus (per mL) in the test
		sample by multiplying with the dilution factor.	
		5.4.6	In case of PRP,
		Phosphorous (%) in PRP on dry weight basis:	
		Phosphorous content in PRP($\mu\text{g/ml}$)	
		Phosphorous content in dry PRP =	

		($\mu\text{g/mL}$) $1 - \{\text{Moisture}\}$	
		($\%$) / 100}	
		Phosphorous content in dry PRP ($\mu\text{g/mL}$)	
		Phosphorous (%) = _____ x	
		100	
		Concentration of PRP solution ($\mu\text{g/mL}$)	
		5.5	Acceptance criteria:
		The coefficient of variation (CV) between the replicates of the	
		samples should be $\leq 5\%$.	
		6.0	Forms and Records:
		Determination of Phosphorous Concentration	
		7.0	Reference:
		European pharmacopoeia.	

1.0	Purpose:			
	To lay down the procedure for the estimation of molecular size of PRP in			
	purified polysaccharide of Haemophilus Influenzae type b conjugate vaccine.			
2.0	Applicable to:			
	This Standard Operating Procedure is applicable for evaluating KD			
	(molecular size distribution) of purified polysaccharide of Haemophilus			
	Influenzae type b conjugate vaccine.			
3.0	Responsibility:			
	3.1	The analyst shall be responsible for carrying out the described		
	procedure correctly. The Supervisor of the laboratory shall be			
	responsible for verifying that the procedure is complied with and for			
	checking the results.			
4.0	Abbreviations and Definitions:			
	4.1	cm	: centimetre	
	4.2	DNA	: Deoxyribonucleic acid	
	4.3	g	: Gram	
	4.4	HIBTT	: Haemophilus Influenzae type b - Tetanus toxoid	
	4.5	HPLC	: High Performance Liquid Chromatography	
	4.6	id	: Internal diameter	
	4.7	l	: length	
	4.8	m	: meter	
	4.9	mg	: milligram	
	4.10	mL	: Millilitre	
	4.11	mm	: milli meter	
	4.12	nm	: Nano meter	
	4.13	No.	: number	
	4.14	PRP	: Polyribosyl Ribitol Phosphate	
	4.15	RID	: Refractive index detector	
	4.16	RSD	: Relative Standard Deviation	
	4.17	SST	: System Suitability Test	
	4.18	°C	: Degree centigrade	
	4.19	µg	: microgram	
	4.20	µL	: Micro litre	
	4.21	µm	: Micro meter	
	4.22	System	: System suitability tests are an integral part of gas	
	Suitability	and liquid chromatographic methods. They are used to		
	Test	verify that the resolution and reproducibility of the		
	chromatographic system are adequate for the analysis to			
	be done.			
	The tests are based on the concept that the equipment,			
	electronics, analytical operations and samples to be			
	analyzed constitute an integral system that can be			
	evaluated as such.			
5.0	Procedure:			
	5.1	Precautions to be taken:		
		5.1.1	Always wear laboratory coat.	
		5.1.2	Always use filtered solutions. Filtration of solution should	
		be done in order to eliminate the entry of possible particles.		
		if any, into the system as well as the column.		
		5.1.3	Degas all solutions.	
		5.1.4	Avoid entry of bubbles into the system as well as column.	
		5.1.5	All the solution bottle openings should be well covered with	
		aluminum foil.		
		5.1.6	Check the calibration status of all equipment and instrument.	
	5.2	Requirements		
		5.2.1	Equipment:	
			5.2.1.1	Calibrated HPLC System with ChemStation software.
			1	
			5.2.1.2	Refrigerator
			2	
			5.2.1.3	Ultrasonic Cleaner
			3	
			5.2.1.4	Vacuum Pump
			4	
		5.2.2	Instrument:	
			5.2.2.1	pH Meter
			1	
			5.2.2.2	Analytical balance
			2	
			5.2.2.3	Magnetic stirrer
			3	
			5.2.2.4	Precision balance
			4	
		5.2.3	Material:	
			5.2.3.1	Two TSK gel G5000 PWxl columns {30 cm (l), 7.8 mm
			1	(i.d)} connected in series.
			5.2.3.2	Laboratory Glassware
			2	
			5.2.3.3	Screw cap vials
			3	
			5.2.3.4	Solvac Filter Holder
			4	
			5.2.3.5	Nylon 6, 6 membrane 0.20/0.22/.045 µm membrane filters.
			5	
			5.2.3.6	Auto pipette (2-20 (L), (20-200 (L), (100-1000 (L)
			6	
			5.2.3.7	Tips for the Auto pipette
			7	
			5.2.3.8	Eppendorf tubes
			8	

			5.2.3.	Aluminium foil			
			9				
		5.2.4	Reagents:				
			5.2.4.	Sodium Chloride (NaCl)			
			1				
			5.2.4.	Sodium azide			
			2				
			5.2.4.	Salmon DNA			
			3				
			5.2.4.	Acetone			
			4				
			5.2.4.	Purified water			
			5				
		5.3	Preparation of solutions:				
		Prepare all the solutions as described below and label of the					
		prepared solutions.					
		5.3.1	0.4% Sodium Chloride solution:				
			For a volume of 2litre, weigh:				
			NaCl	8.0 g			
			Dissolve in 700 mL of purified water in a beaker. Make up the				
			volume to 2 litres with purified water in a volumetric flask.				
			Use the mobile phase within four days from the date of				
			preparation if it remains clear and no turbidity or fibers				
			observed in it.				
		5.3.2	Void volume marker (Salmon DNA):				
			For a volume of 5 mL, weigh:				
			Salmon DNA	10 mg			
			Dissolve in 3mL of 0.2µm filtered purified water and make up				
			the volume to 5mL in a volumetric flask. Store at -20 °C ± 5				
			°C for 1 month.				
		5.3.3	Total volume marker (Acetone):				
			Take 10µL of acetone and dissolve in 990µL of 0.2µm filtered				
			purified water in an eppendorf. Mix thoroughly. Prepare fresh				
			before use.				
		5.3.4	Column Preservation Solution:				
			Note: All the solutions prepared for use are vacuum filtered				
			using Nylon 6, 6 membrane 0.20/0.22/0.45 µm, depending upon				
			the nature of solution to be filtered (filtration not				
			necessary if the solution is prepared in 0.2µ filtered				
			purified water). For organic solvents filtration not				
			necessary if the solution is HPLC grade.				
		5.4	Preparing the HPLC System for run.				
		5.5	Run Conditions:				
		5.5.1	Load method HIBTT RI.M.				
		5.5.2	Flow Rate: 0.5 mL/min				
		5.5.3	Detector used: RID				
		5.5.4	Run time: 60 minutes				
		5.5.5	Mobile Phase: 0.4% sodium chloride solution				
		5.6	Equilibration of the system and the column:				
		Equilibrate the system and column with 0.4% sodium chloride solution					
		unless a stable baseline for RID is obtained which can be viewed with					
		the online plot window.					
		5.7	Sample Preparation and Application:				
		5.7.1	Test sample and standards are loaded in screw cap vials and				
			placed against the appropriate no. on the sample tray by				
			clearly marking the screw cap vial.				
		5.7.2	Take 3 mg/mL solution of Purified polysaccharide and obtain				
			the value of PRP concentration.				
		5.7.3	Load the required volume of sample as per the PRP				
			concentration equivalent to 200µg in the sample vial.				
			Example: If a sample has a PRP concentration of 2mg/mL, to				
			calculate volume required for 200µg of polysaccharide,				
			2mg (2000µg) -----1mL				
			200µg-----X mL				
			X mL = 200µg x 1mL				
			2000µg				
			X mL = 0.1mL				
			Therefore, 100µL of volume is injected for sample having PRP				
			concentration of 2mg/mL.				
			Note: The application of the Calibration standards is carried				
			out every time whenever a test sample is applied. If several				
			samples are applied in one sequence then only one application				
			of the Calibration standards is enough.				
		5.8	Running of sequence:				
		5.8.1	Load sequence PRP RI.S.				
		5.8.2	Open the sequence parameter option by clicking Sequence menu.				
		5.8.3	Enter information about the data directory and operator name.				
		5.8.4	In the shutdown option, click on the post-sequence option and				
			select LAMPOFF/STANDBY.				
		5.8.5	In sequence table option, specify the sample names, methods				
			and vials to run in the sequence as per the following table:				
			S	Sample Code	Sample details	No. of	Injection
			No			injectio	Volume
						ns	
			1	Blank	Eluent Blank	1	10 µL
			2	Acetone	System	6	1 µL
				(Total volume	Suitability Test		
				marker)			
			3	Blank	Eluent Blank	1	10 µL
			4	Salmon DNA	Void volume	1	50 µL
				marker			

			5	Blank	Eluent Blank	1	10 µL
			6	Test Sample	Sample	1	200 µg of
			(s)		PRP		
		5.8.6	Click on run sequence option in run control menu to start the				
			sequence.				
		5.8.7	Each sample run will be of 60 min and after running all the				
			samples the sequence will automatically stop.				
		5.8.8	Inject the eluent blank after every 20 injections.				
			Note: One extra vial with same SST sample can be used if 6				
			injections are not possible from single vial.				
		5.9	Calculations:				
		5.9.1	The retention time of the unretained compound is determined				
			with Salmon DNA.				
		5.9.2	The Retention time of the component which has full access to				
			the pores of stationary phase is determined with Acetone.				
		5.9.3	Proceed for calculating KD value.				
			The calculations are made as per the following equation:				
			$KD = T_e - T_o$				
			$T_t - T_o$				
			Where,				
			T_e = Retention time of peak maxima of purified polysaccharide				
			sample				
			T_t = Retention time of Acetone				
			T_o = Retention time of Salmon DNA				
			Note: Take average RT of acetone as T_t from EXCEL for				
			calculation of SST.				
		5.9.4	Calculation of percentage of sample eluted before a given KD:				
			The percentage of sample eluted before a given KD is				
			calculated using ChemStation software.				
			5.9.4.1	Open the acquired chromatogram in the data analysis			
			view in the offline copy of the software.				
			5.9.4.2	Optimize integration with initial events in the			
			signal specific events table.				
			5.9.4.3	Enter the value of Elution time (T_e) calculated in			
			step 5.9.3 in integration with timed events option				
			or with manual integration and select split peak				
			option.				
			5.9.4.4	The percentage of sample eluted before a given KD			
			is the value of percentage of area (Area %)				
			obtained for polysaccharide curve till the				
			retention time of peak maxima.				
		5.9.5	Calculations for System Suitability Test:				
			5.9.5.1	Note the values of area under the curve and			
			retention time for Acetone (SST standard).				
			5.9.5.2	Every time a sample is run with system suitability			
			test, calculate the results for system suitability				
			and record the same.				
			5.9.5.3	Attach the hard print of SST calculation along with			
			chromatograms and the record form.				
		5.10	Acceptance Criteria:				
		5.10.1	For System Suitability Test:				
			5.10.1.1	RSD calculated for RT of Acetone should be <5.0%.			
			5.10.1.2	RSD calculated for area of Acetone should be <5.0%.			
		5.10.2	For purified polysaccharide samples:				
			KD (Molecular Size distribution): $\geq 40\%$, $KD \leq 0.3$				
		6.0	Reporting:				
		Molecular sizing of polyribosyl ribitol phosphate (PRP) by high performance					
		size exclusion chromatography.					
		7.0	Reference:				
		European pharmacopoeia.					

1.0 |Purpose: |
 |To lay down a procedure for quantifying the protein concentration in HIB-TT|
 |conjugate samples received from the production process. |
 2.0 |Applicable to: |
 |This Standard Operating Procedure is applicable to all the samples of |
 |HIB-TT conjugate received from the production process. |
 3.0 |Responsibility: |
 |The analyst is responsible for correctly executing the procedure. Head / |
 |Supervisor of the laboratory is responsible for verifying that the |
 |procedure is complied with and results are checked. |
 4.0 |Equipment: |
4.1	HIB-TT	: Haemophilus influenzae type b with tetanus toxoid
4.2	SOP	: Standard Operating Procedure
4.3	AR	: As required
5.0	Procedure:	
5.1	Materials and reagents:	
		5.1.1
		5.1.2
		5.1.3
		5.1.4
		5.1.5
		5.1.6
		5.1.7
		Materials:
		5.1.9
		5.1.10
		5.1.11
		5.1.12
		5.1.13
		5.1.14
		5.1.15
		5.1.16
		5.2
		5.2.1
		5.2.2
		5.2.3
		5.3
		5.3.1

		high range (100-500 µg/ml)						
		Water (ml)	0.9	0.8	0.7	0.6	0.5	
		RM BSA 1 mg/ml (ml)	0.1	0.2	0.3	0.4	0.5	
		Concentration	100	200	300	400	500	
		(µg/ml)						
		Note:						
		Prepare the calibration curve whenever there is any change in the						
		test conditions (as when the new batches of reagents are						
		prepared, recalibration of the equipment, change in the analysts						
		for new batch of reference material etc.)						
		The test is accepted if the parameters of the intercept, the						
		slope and the correlation coefficient are found within the						
		acceptance limits.						
		Record the results of the calibration curve.						
		5.3.2	Treatment of samples with possible interference:					
		Each sample is treated according to the composition of elements						
		that may interfere in determination of protein concentration						
		(Table 1). If one or more interfering elements are present in the						
		sample, carry out the treatment, washing or acid precipitation as						
		described below.						
		Table 1. Substances that interfere with Lowry test.						
		The numbers in parenthesis indicate the concentration limit that						
		should be present (according to the reports of the literature)						
		(2).						
		Buffers						
		Tris (100 µmol/L), Phosphate (100 mmol/L), Tricine (100µmol/L),						
		MOPS (10 µmol/L), Hepes (1 µmol/L), Bicine (5 µmol/L), Citrate (1						
		mmol/L), Glycylglycine (100 µmol/L).						
		Amino acids						
		Histidin (150 mmol/L), Tyrosine (150 mmol/L), Tryptophan (150						
		mmol/L), Cysteine (40 µmol/L), Glycine (1 mmol/L), Glutamate						
		(value not described in the literature).						
		Carbohydrates						
		Saccharose (4 mmol/L), Monosaccharides (glucose, fructose,						
		mannose, xilose, ramnose) (150 µmol/L), Glucosamine (5 mg/L),						
		Glycerol (100 mL/L), Ethylenglycol (1 g/L), Ficoll,						
		Polyvinylpyrrolidone.						
		Reducers						
		Phenol (except the nitrophenol), Dithiothreitol (20 µmol/L),						
		Glutathion (40 µmol/L), 2-Mercaptoethanol (700 µmol/L), Hydrazine						
		(5 mg/L), Ascorbic acid (100 µg/L).						
		Salts, metals, etc.						
		(NH₄)₂SO₄ (1.5 g/L), K⁺ (12 mmol/L), Hg²⁺ (15 µmol/L), Mn²⁺ (15						
		µmol/L), Co²⁺ (1 µmol/L), NaCl (0.7 mol/L), NaI (15 mmol/L)						
		Miscellaneous						
		EDTA (70 µmol/L), Haematin (15 µg/L), Triton X-100 (10 mg/mL),						
		oxidated lipids (10-200 µmol/L), Fatty acids (1 mmol/L),						
		Ampholine, Salicilate (20 µg/L), Penicillin (0.5 units/mL),						
		Acetyl acetone (2 µmol/L).						
		5.3.3	Preparation of the samples and control:					
		5.3.3.1	Prepare the blank with water by substituting the volume					
		1	used for the sample.					
		5.3.3.2	For Preparing the control of HIB-TT conjugate samples,					
		2	dilute it 5 fold .					
		For example: Take 0.2 ml of HIB-TT conjugate samples and add 0.8						
		ml of water to it. Prepare appropriate dilutions of the samples						
		accordingly. The dilutions are shown in the following table:						
		Sample	Dilution	Volume of the	Volume of water			
		factor	sample (ml)	(ml)				
		HIB-TT conjugate	1/5	0.200	0.800			
		samples (ARM)						
		Note:						
		The dilutions mentioned can vary depending upon the protein						
		concentration. If there is any sample outside the range of						
		the calibration curve, carry out the calculation for more						
		appropriate dilution.						
		5.3.4	Coloured reaction:					
		5.3.4.1	For the development of the colored reaction the solutions					
		1	should be mixed in the following proportion:					
		Sample	1.0 volume					
		Copper alkaline solution	5.0 volume					
		1N Folin reagent	0.5 volume					
		5.3.5	Test:					
		5.3.5.1	Switch on the thermostat bath and maintain it at a					
		1	temperature of 37 °C.					
		5.3.5.2	Take 0.4 ml of the blank , control and sample (already					
		2	diluted) and pour them in test tubes. (Perform the test					
		twice).						
		5.3.5.3	Add 2 ml of alkaline copper solution in each test tube.					
		3						
		5.3.5.4	Stir gently and incubate at 37 °C for 30 minutes.					
		4						
		5.3.5.5	Add 0.2 ml of 1 N Folin - Ciocalteu reagent. Stir					
		5	immediately after the addition of Folin until a					
		homogeneous colour is obtained.						
		5.3.5.6	Incubate at 37 °C for 20 minutes and read the absorbance					
		6	at 730 nm in the spectrophotometer					
		5.3.6	Calculations:					
		For calculating the protein concentration use the equation Y = mX						
		+ b.						
		Where: m = slope						
		b = intercept						

	Y = absorbance	
	X = concentration	
	5.3.6.	Use EXCEL programme for the preparation of calibration
	1	curve and report the results (Multiply by the dilution
		factor. Divide with 1000 to change $\mu\text{g/ml}$ to mg/ml).
	5.3.7	Acceptance criteria:
		The variation coefficient for replicate of the samples, control
		of the HIB-TT conjugate vaccine and Bovine Serum Albumin is (5
		% .
		The value of the HIB-TT conjugate controls is within the
		acceptance limits according to the graph controlling it.
	6.0	Reporting:
	6.1	Calibration curve of Test method
	7.0	References:
	7.1	Lowry, O.H., J. Rosebrough, A.L.Farr and R.J. Randall, R.J. (1951)
		J. Biol.Chem. 193, 265 - 275.
	7.2	Text Book of General Chemistry, volume II. Collective of professors
		of the University of the Havana. Ability of Pharmacy and Foods.
		Editorial Town and Education, 1990.

1.0|Objective: |
 |To lay down the procedure for the determination of free carrier protein in |
 |PRP-TT conjugate (bulk conjugate) of Haemophilus Influenzae type b |
 |conjugate vaccine. |
 2.0|Scope: |
 |This Standard Operating Procedure is applicable for the determination of |
 |free carrier protein present in bulk conjugate of Haemophilus Influenzae |
 |type b conjugate vaccine. |
 3.0|Responsibility |
3.1	The analyst shall be responsible for carrying out the described
	procedure correctly.
3.2	The Supervisor of the laboratory shall be responsible for
	verifying that the procedure is complied with and for checking the
	results.
4.0	Abbreviations and Definitions:
4.1	cm
4.2	g
4.3	HPLC
4.4	id
4.5	l
4.6	L
4.7	M
4.8	mAu
4.9	mg
4.10	min
4.11	mL
4.12	mm
4.13	nm
4.14	No.
4.15	QA
4.16	QC
4.17	QCR
4.18	rpm
4.19	RSD
4.20	SOP
4.21	µL
4.22	µm
4.23	µg
4.24	SST
4.25	System
	Suitability
	test
5.0	Procedure
5.1	Precautions to be taken:
	5.1.1
	5.1.2
	5.1.3
	5.1.4
	5.1.5
	5.1.6
5.2	Requirement:
	5.2.1
	5.2.2
	5.2.3

			5.2.3.	Tips for the Auto pipettes		
			7			
		5.2.4	Reagents:			
			5.2.4.	Sodium Chloride (NaCl)		
			1			
			5.2.4.	Disodium hydrogen phosphate anhydrous (Na₂HPO₄)		
			2			
			5.2.4.	Sodium dihydrogen phosphate monohydrate		
			3			
		5.2.4.	Purified water			
			4			
		5.3	Preparation of solutions:			
		Prepare all the solutions as described below and Carry out the				
		labelling of the prepared solutions and assign the lot numbers.				
		5.3.1	Isopropyl alcohol 70%			
		Take 700 mL of 2-propanol in a volumetric flask and make up				
		the volume to 1000 mL with Purified Water. This solution				
		can be stored at room temperature for 3 months.				
		5.3.2	Solution I			
		Na₂HPO₄ (0.1M), NaCl (0.1M)				
		For a volume of 1000 mL, weigh:				
		Na₂HPO₄	14.19 g			
		NaCl	5.84 g			
		Dissolve in 700mL of purified water in a volumetric flask				
		and make up the volume to 1000mL. Prepare fresh before use.				
		5.3.3	Solution II			
		(NaH₂PO₄.H₂O, 0.1 M)				
		For a volume of 500 mL, weigh:				
		NaH₂PO₄.H₂O	6.85g			
		Dissolve in 400mL of purified water in a volumetric flask				
		and make up the volume to 500mL. Prepare fresh before use.				
		5.3.4	0.1M Sodium Phosphate buffer			
		(0.1 M Na₂HPO₄, 0.1M NaCl +0.1M NaH₂PO₄.H₂O, pH 7.0)				
		For a volume of 1000mL, take approximately 600mL of				
		Solution I and 400mL of Solution II. The pH of the solution				
		should be 7.0 ± 0.2. Use the mobile phase within four days				
		from the date of preparation if it remains clear and no				
		turbidity or fibers observed in it.				
		5.3.5	Column Preservation Solution: Prepare and use it.			
		5.3.6	1 mg/mL of p-Aminobenzoic Acid Stock			
		For a Volume of 5 mL, weigh:				
		5.3.7	p-Aminobenzoic acid (4-Aminobenzoic acid) 5 mg			
		5.3.8	Dissolve in 5mL of purified water and make up the volume to			
		5mL.				
		5.3.9	Working standard of p-Aminobenzoic Acid			
		Dilute 1 mg/mL of the stock solution to 1/10 with purified				
		water.				
		Note: All the solutions prepared for use are vacuum				
		filtered using Nylon 6, 6 membrane 0.20 µm or 0.22 µm ,				
		depending upon the nature of solution to be filtered. For				
		organic solvents 0.45µm filters should be used (filtration				
		not necessary if the solution is prepared in 0.2 µ filtered				
		purified water).				
		5.4	Preparing the HPLC System for run, Agilent Series:			
		5.5	Run Conditions:			
		5.5.1	Load method HIBTT FREE PROTEIN.M.			
		5.5.2	Flow Rate: 0.75 mL/min			
		5.5.3	Detection Wavelength: 280nm			
		5.5.4	Run time: 20 minutes			
		5.6	Equilibration of the system and the column:			
		Equilibrate the column with mobile 0.1 M sodium phosphate buffer				
		at 280nm until a stable baseline is observed on the online plot				
		window.				
		5.7	Sample Preparation and Application:			
		5.7.1	Test sample and standards are loaded in screw cap vials and			
		placed against the appropriate no. on the sample tray by				
		clearly marking the screw cap vial.				
		5.7.2	Load the required volume of sample as per the PRP			
		concentration equivalent to 2.5 µg in the sample vial.				
		Example: If a sample has a PRP concentration of 0.5 mg/mL,				
		to calculate volume required for 2.5 µg of protein,				
		0.5 mg (500 µg) ----1 mL				
		2.5 µg-----X mL				
		X mL = 2.5 µg x 1 mL				
		500 µg				
		X mL = 5 µL				
		Therefore, 5 µL of volume is taken for sample having PRP				
		concentration of 0.5 mg/mL.				
		5.8	Running of sequence:			
		5.8.1	Load sequence HIBTT FREE PROTEIN.S			
		5.8.2	Open the sequence parameter option by clicking Sequence			
		menu.				
		5.8.3	Enter information about the data directory and operator			
		name.				
		5.8.4	In the shutdown option, click on the post-sequence option			
		and select LAMPOFF/STANDBY.				
		5.8.5	In sequence table option, specify the sample names, methods			
		and vials to run in the sequence as per the following				
		table:				
		S	Sample Code	Sample details	No. of	Injection
		No		Injection	Volume(µL)	
				ns		

			1	Blank	Eluent Blank	1	10
			2	p-Aminobenzoic	System	6	1
			acid	Suitability Test			
			3	Blank	Eluent Blank	1	10
			4	Test Sample(s)	Sample	1	2.5µg of
					PRP		
		5.8.6	Click on run sequence option in run control menu to start				
		the sequence.					
		5.8.7	Each sample run will be of 20 min. and after running all				
		the samples the sequence will automatically stop.					
		5.8.8	Inject the eluent blank after every 20 injections.				
		One extra vial with same SST sample can be used if 6					
		injections are not possible from single vial.					
		5.9	Calculations :				
		Area under curve for the standards/sample is analyzed by					
		Chemstation software.					
		5.9.1	Open the acquired chromatogram in the data analysis view in				
		the offline copy of the software.					
		5.9.2	Optimize integration with initial events in the signal				
		specific events table.					
		5.9.3	Enter the value of retention time of peak maxima in				
		integration with timed events option.					
		5.9.4	The percentage of free carrier protein in sample is				
		calculated by taking the percentage area of the peak maxima					
		corresponding to the lower molecular weight compared to the					
		peak maxima of higher molecular weight.					
		5.9.5	Calculations for System Suitability Test:				
		5.9.5.	Note the values of area under the curve and				
		1	retention time for p-Aminobenzoic acid. (SST				
		standard)					
		5.9.5.	Every time a sample is run with system suitability				
		2	test, calculate the results for system suitability				
		by creating a copy of Master EXCEL in the desired					
		folder and record the same.					
		5.9.5.	Attach the hard print of SST calculation along with				
		3	chromatograms and the record form.				
		5.10	Acceptance Criteria :				
		5.10.1	For System Suitability Test :				
		5.10.1.	RSD calculated for Retention time should be <5.0%.				
		1					
		5.10.1.	RSD calculated for Area should be <5.0%.				
		2					
		5.10.2	Free protein for ARM of Haemophilus Influenzae type b				
		conjugate vaccine Samples:					
		5.10.2.	Free protein	< 1%			
		1					
		5.11	Result Reporting:				
		5.11.1	Report the results of percentage of area of peak towards				
		higher Molecular weights (Conjugated) and percentage of					
		area of peak towards lower Molecular weights					
		(Unconjugated).					
		Note: Percentage of free carrier protein is the percentage					
		of area of peak towards lower molecular weights					
		(Unconjugated).					
		6.0	Forms and Records:				
		Determination of free carrier protein in polyribosyl ribitol phosphate					
		(PRP) - tetanus toxoid conjugate by high performance size exclusion					
		chromatography					
		7.0	References:				
		9.1	United States Pharmacopeia 29.				
		9.2	European pharmacopeia				

CONTENTS

OBJECTIVE:

To document the installation of the Air Handling Unit (AHU-01), ascertain that the installation is as per design Specifications lay down.

SCOPE:

This protocol is applicable to the AHU installed in M/s. INFARMA- Indústria Farmacêutica, Lda

PERFORMING THE INSTALLATION QUALIFICATON:

The installation should conform to this checklist. Record all observations and deviations in the space provided in the checklist. In case of deviations prepare a deviation report along with justifications, which should be approved by the Engineering Department, Quality Assurance Manager and Validation In charge.

4.0 CHECKLIST FOR INSTALLATION OF AHU-01

AHU 01- 11000 CFM - RECIRCULATION

DUCTING:-

ELECTRICAL CONNECTIONS

CHILLED WATER MANIFOLD DETAILS

DETAILS OF UTILITIES

5.0 Test Certificates for the following components to be attached -

6.0 RESULTS:

7.0 CONCLUSION -

BASED UPON THE OBSERVATIONS RECORDED IN THE TABLE GIVEN ABOVE, THE INSTALLATION OF THE AHU IS FOUND TO BE SATISFACTORY / IS FOUND NOT TO BE SATISFACTORY AS PER THE DESIGN SPECIFICATIONS. THEREFORE, INSTALLATION OF THE AHU IS / IS NOT QUALIFIED AND THE OPERATION QUALIFICATION CAN / CANNOT BE PERFORMED

APPROVAL FOR INSTALLATION QUALIFICATION

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6.0 | Results | 14

7.0 | Conclusion | 14

8.0 | Approvals | 15

Constructional Details: | Constructional Details: | Observations | Sign & Date

Design & Project By | Bio Bridge Healthcare Solutions Pvt. Ltd. | |

Associate Manufacture | Citizen Industries Pvt.Ltd.(a member of DAIKN group) | |

Dimensions

(L x W x H) | 4836 mm X 2036 mm X 1536 mm | |

Outer skin | 0.8 mm G.I. Pre Coated with Guard Film | |

Inner skin | 0.8 mm G.I. Plain Sheet. | |

Panel thickness | 43 mm thick Double Skin Panels with PUF | |

Insulation | Polyurethane, Density (40±2 kg/m³) | |

Profile | Thermal break | |

Type of unit | Double skin Floor Mounted Air Handling Unit | |

Frame structure | Aluminium Extruded Anodized Profile with thermal break & internal coving | |

Colour | RAL-5012 (Sky blue) | |

Skid | 75 x 40 mm - 2 mm thick G.I. Fabricated | |

Handle | C' Type Glass filled nylon | |

Hinges | Nylon with G.I. Pivot | |

Corner Coupler | Glass filled nylon | |

Damper M.O.C. | Aluminium | |

Technical Specifications: | Technical Specifications: | Observation | Signature

Fan air quantity | 11000 CFM | |

Supply air quantity | 11000 CFM | |

De humidified. air quantity | 11000 CFM | |

Static Pressure | 100 mm of water Column | |

Fan | Fan | Fan | Fan

Fan Model & Type | BNC-P710 I & Plug Fan | |

Make | Kruger | |

Fan BKW | 7.41 | |

Fan Speed | 1394 RPM | |

Sr. No | 123S1138-M-00-00 | |

Motor | Motor | Observation | Signature

Make | CG (Crompton Greaves) | |

KW | 11.0 kW - 4 Pole (IE-3 - Non FLP) | |

Type | 415 V, 3 Ph. TEFC, Sq. Cage, Foot Mounted, Induction Motor | |

Machine No | 11KP4 | |

Efficiency | 91.4 IE3 | |

Voltage | 415 V ± 10 % | |

Current | 20 A | |

Connection | Delta | |

Frame Size | ND 160M | |

Speed | 1465 RPM | |

Cooling Coil | Cooling Coil | Cooling Coil | Cooling Coil

Type | Fin & Tube Type Coil for Chilled Water Application | |

Size (mm) | 1270 x 1609 (50.00"H x 63.36"L) - 6 RD | |

Coil TR | 32.0 TR | |

Quantity | 1 No | |

Fin Spacing | 11 to 12 FPI | |

Coil Face Velocity | 2.54 M/S | |

Tube Diameter & Material | 0.50 mm thick, 12.7 mm O.D, Plain Copper Tubes | |

Fins Thickness & Material | 0.13 mm thick, (blue Coated) Aluminium Fins | |

Drain pan | Drain pan | Drain pan | Drain pan

MOC | S. S.304 With 19mm thick close cell Nitrile rubber Insulation | |

Pre-filter | Pre-filter | Pre-filter | Pre-filter

Efficiency | 90% - 10µ (M-5) | |

Size, Qty and Type | 610 x 610 x 305 - 6 Nos. Flange Type | |

Location | In mixing chamber | |

Filter Make | AAF | |

Fresh Air filter | Fresh Air filter | Fresh Air filter | Fresh Air filter
Efficiency | 90% - 10µ (G-4) ||
Size, Qty and Type | 305 x 610 x 50 - 1 No. Flange Type ||
Location | Fresh Air Opening ||
Filter Make | AAF ||
Bleed Air filter | Bleed Air filter | Observation | Signature
Efficiency | 90-95% - 5µ (F-8) ||
Size, Qty and Type | 305 x 610 x 305 - 1 No. Flange Type ||
Location | On the Top of Filter ||
Filter Make | AAF ||
Fine filter | Fine filter | Fine filter | Fine filter
Efficiency | 90-95% - 5µ (F-8) ||
Size, Qty and Type | 610 x 610 x 305 - 6 Nos. Flange Type ||
Location | Supply Plenum in AHU ||
Filter Make | AAF ||
Electrical Heater ||
MOC | SS Finned Tubular Type ||
KW | 18 ||
Quantity | 2 KW X 9 No ||
Volume Control Dampers (Manually Operated) | Volume Control Dampers (Manually Operated) | Volume Control
Dampers (Manually Operated) | Volume Control Dampers (Manually Operated)
MOC | The dampers will be of Aerofoil Design made out of Aluminum extruded section multi-louvered type
with framework, operating link, etc. ||
Location ||
• Supply air outlet | 510X1200 mm-01 Nos ||
• Return air inlet | 410X1950 mm-01 Nos ||
• Bleed air outlet | 250X250 mm-01 Nos ||
• Fresh air inlet | 250X250 mm-01 Nos ||
Fire Damper | Fire Damper | Fire Damper | Fire Damper
MOC | Fire Dampers shall meet the fire resistance rating of 2 hours as per UL555 and shall be tested and
certified by CBRI. Fire Damper frame shall be galvanized steel of 1.6mm thickness, formed into a C
channel in 4-piece construction with flanges suitable for duct connection. Damper Blades shall be single
skin galvanized steel of 1.6mm thickness and must have 3 accurately roll formed longitudinal grooves for
reinforcement and proper closure and operation. The blades shall be arranged in parallel to provide
positive shut off. Each Blade shall have axles supported on both sides with sintered bronze
self-lubricating bearings turning in an extruded hole in the frame. All Blades shall be interconnected
using linkage in the frame section. Each Damper shall be operated through a UL stamped fusible link rated
at 74- or 100-Degree C. The closing of the fire damper shall be through a suitable rated spring which
will be capable of closing the blades in the event of fusible link disintegration. The option of
indication of Blade Position should be made available through a manual lever. Each Damper shall be
supplied with a factory mounted sleeve made from Minimum 1mm thick galvanized steel of minimum 150mm
length. Fire damper to be provided with limit switch and wiring up to control panel to have interlock
with AHU. | Fire Dampers shall meet the fire resistance rating of 2 hours as per UL555 and shall be
tested and certified by CBRI. Fire Damper frame shall be galvanized steel of 1.6mm thickness, formed into
a C channel in 4-piece construction with flanges suitable for duct connection. Damper Blades shall be
single skin galvanized steel of 1.6mm thickness and must have 3 accurately roll formed longitudinal
grooves for reinforcement and proper closure and operation. The blades shall be arranged in parallel to
provide positive shut off. Each Blade shall have axles supported on both sides with sintered bronze
self-lubricating bearings turning in an extruded hole in the frame. All Blades shall be interconnected
using linkage in the frame section. Each Damper shall be operated through a UL stamped fusible link rated
at 74- or 100-Degree C. The closing of the fire damper shall be through a suitable rated spring which
will be capable of closing the blades in the event of fusible link disintegration. The option of
indication of Blade Position should be made available through a manual lever. Each Damper shall be
supplied with a factory mounted sleeve made from Minimum 1mm thick galvanized steel of minimum 150mm
length. Fire damper to be provided with limit switch and wiring up to control panel to have interlock
with AHU.
| Fusible link rated 74 OC, UL stamped ||
Size & Quantity | 510X1200 mm-01 Nos
410X1950 mm-01 Nos ||
Make | Airflow Product ||
Magnehelic Gauge across Pre-Filter | Magnehelic Gauge across Pre-Filter | Observation | Signature
Location | Across Pre filter ||
Range | 0-25 MMWC ||
Quantity | 01 Nos ||
Make | Omicron ||
Model No | 2000-25MM ||
Magnehelic Gauge across Fine Filter | Magnehelic Gauge across Fine Filter | Magnehelic Gauge across Fine
Filter | Magnehelic Gauge across Fine Filter
Location | Across Fine filter ||
Range | 0-50 MMWC ||
Quantity | 01 Nos ||
Make | Omicron ||
Model No | 2000-50MM ||
SUPPLY & RETURN DIFFUSER | SUPPLY & RETURN DIFFUSER | SUPPLY & RETURN DIFFUSER | SUPPLY & RETURN DIFFUSER
MOC | Aluminum Powder Coated, Air Ceiling diffuser with VCD of powder coated aluminum construction. ||
Size & Quantity | SA 525 mm x 525 mm = 23 Nos
RA 525 mm x 525 mm = 19 Nos
SA 450 mm x 450 mm = 05 Nos
RA 450 mm x 450 mm = 10 Nos ||

Make | Airflow Product | |
 Type | Fabricated ducts. | |
 Standard | Ducts will be fabricated as per SMACNA Standard. Class "C" | |
 Manufacturer of Sheets | Jindal | |
 Type of Sealant | Silicon Sealant | |
 Duct Fabricator | Asawa | |
 Thermal Insulation of Ducts | Thermal Insulation of Ducts | Thermal Insulation of Ducts | Thermal Insulation of Ducts
 Make | Aero foam | |
 MOC | Closed cell nitrile rubber insulation with one side aluminium foil | |
 Supply Duct | 25 mm Thick | |
 Return Duct | 13 mm Thick | |
 Electrical Panel | Panel for AHU with MCB of Suitable Range as Isolator. The Panel will have Cooling Fan arrangement with Guard and Drawing Folder Provision. The Panel to be Fabricated from 16swg sheet duly powder coated, included VFD Controller | |
 Volts | 415 V | |
 Phase | 3 Phase | |
 Earthing | Connected with existing Earthing grid | |
 Leakage Voltage between Neutral & Earth | Not more than 5 volts | |
 Temperature sensor, HMI (Human Machine Interface) Controller & Limit switch | Temperature sensor, HMI (Human Machine Interface) Controller & Limit switch | Temperature sensor, HMI (Human Machine Interface) Controller & Limit switch | Temperature sensor, HMI (Human Machine Interface) Controller & Limit switch | Temperature sensor, HMI (Human Machine Interface) Controller & Limit switch | Temperature sensor, HMI (Human Machine Interface) Controller & Limit switch | Temperature & Relative Humidity Sensor | Temperature & Relative Humidity Sensor | |
 Type | TRH | |
 Make | Autonics | |
 Mode No | THD-D2-C | |
 Measurement range | For Temp.: -10 ~ 40°C
 0 ~ 70°C | |
 Output | 4 ~ 20 mA or 0 ~ 10 V DC | |
 HMI (Human Machine Interface) Controller | HMI (Human Machine Interface) Controller | HMI (Human Machine Interface) Controller | HMI (Human Machine Interface) Controller
 Make | Hiecon | |
 Display Size | 7 inch | |
 Model | PI 3070i | |
 Communication port | RS422TX+/RS485 | |
 Printer Port | USB/Serial Port | |
 Working Temperature | 0 to 50° C | |
 Working Humidity | 10 To 90 % | |
 Input Voltage Range | DC 10 volt | |
 Qty | 01Nos | |
 Limit Switch | Limit Switch | Limit Switch | Limit Switch
 Make | Jainson | |
 Type | JLSR | |
 Range | 10 A, 500 V- 1 NO + 1NC | |
 VFD | VFD | VFD | VFD
 Make | Danfoss | |
 Input | AC 3 PH, 380 - 480 V, 42 / 36 A, 50 / 60 Hz | |
 Output | AC 3 PH, 0 - Input V 31 /27A, 0 - 400 Hz | |
 Model
 Model | VLT Micro Drive FC 51 | |
 Capacity (KW) | 11.0 kW | |
 Piping | Piping | Observation | Signature
 Make | Jindal | |
 Size | 65 NB | |
 Y-type Strainer | Y-type Strainer | Y-type Strainer | Y-type Strainer
 Make | Sant | |
 Size | 65 NB | |
 Quantity | 01 Nos | |
 Butterfly Valves | Butterfly Valves | Butterfly Valves | Butterfly Valves
 Make | Kitz | |
 Size | 65 NB | |
 Quantity | 06 Nos | |
 3-way valve | 3-way valve | 3-way valve | 3-way valve
 Make | Belimo | |
 Valve size | 50 NB | |
 Qty | 01 Nos | |
 Model No | R3040 | |
 Power Supply | 24V AC/DC, | |
 Control Signal /, feedback signal | 0.5-10 V DC | |
 Temperature Gauge | Temperature Gauge | Temperature Gauge | Temperature Gauge
 Make | H-guru | |
 Range & Quantity | 0 to 50 0C, 02 Nos. | |
 Quantity | 02 Nos. | |
 Pressure Gauge | Pressure Gauge | Pressure Gauge | Pressure Gauge
 Make | H-guru | |
 Range | 0 to 7 Kg/cm2 | |
 Quantity | 02 Nos. | |
 Air Vent | Air Vent | Air Vent | Air Vent
 Specification | Automatic purge valve with ball valve. | |
 Make | Anergy | |
 Size & Quantity | 12 MM dia -01 Nos | |
 Description | Specifications | Observations | Sign & Date
 Electrical | Electrical | Electrical | Electrical
 Phase | Three Phase, Including VFD | |
 Voltage | 415V | |
 Frequency | 50 Hz +/- 3% variations | |
 Chilled Water | Inlet Temp to AHU is 7-8 °C and out let Temperature is 12 °C | |
 Sr. No | Item Name | Available (Yes/No)
 1) | AHU Test Certificate |
 2) | Cooling Coil Test Certificate |
 3) | Fan Test Certificate |
 4) | Induction Motor Test Certificate |

5) | Volume Control Damper / Fire Damper Test Certificate |
6) | 10-micron filter |
7) | 3-micron filter |
8) | 0.3-micron filter |
9) | Duct Test Certificate |
10) | Ducting Insulation Test Certificate |
11) | Magnehelic Gauge Test Certificate |
12) | Electrical Panel Test Certificate |
13) | Pressure / Temperature Guage Test Certificate |
14) | Y strainer, 3-way Valve, Butterfly Valve Test Certificate |
Department | Name | Signature | Date
Project: BioBridge Healthcare Solutions Pvt. Ltd., India | Project: BioBridge Healthcare Solutions Pvt.
Ltd., India | Project: BioBridge Healthcare Solutions Pvt. Ltd., India | Project: BioBridge Healthcare
Solutions Pvt. Ltd., India
Prepared by: | | |
Checked/Reviewed by: | | |
INFARMA – Indústria Farmacêutica, Lda | INFARMA – Indústria Farmacêutica, Lda | INFARMA – Indústria
Farmacêutica, Lda | INFARMA – Indústria Farmacêutica, Lda
Reviewed by: | | |
Reviewed by: | | |
Approved by | | |

CONTENTS

OBJECTIVE:

To document the installation of the Air Handling Unit (AHU-02), ascertain that the installation is as per design Specifications lay down.

SCOPE:

This protocol is applicable to the AHU installed in M/s. INFARMA- Indústria Farmacêutica, Lda

PERFORMING THE INSTALLATION QUALIFICATON:

The installation should conform to this checklist. Record all observations and deviations in the space provided in the checklist. In case of deviations prepare a deviation report along with justifications, which should be approved by the Engineering Department, Quality Assurance Manager and Validation In charge.

4.0 CHECKLIST FOR INSTALLATION OF AHU-02

AHU 02- 4000 CFM - RECIRCULATION

DUCTING:-

ELECTRICAL CONNECTIONS

CHILLED WATER MANIFOLD DETAILS

DETAILS OF UTILITIES

5.0 Test Certificates for the following components to be attached -

6.0 RESULTS:

7.0 CONCLUSION -

BASED UPON THE OBSERVATIONS RECORDED IN THE TABLE GIVEN ABOVE, THE INSTALLATION OF THE AHU IS FOUND TO BE SATISFACTORY / IS FOUND NOT TO BE SATISFACTORY AS PER THE DESIGN SPECIFICATIONS. THEREFORE, INSTALLATION OF THE AHU IS / IS NOT QUALIFIED AND THE OPERATION QUALIFICATION CAN / CANNOT BE PERFORMED

APPROVAL FOR INSTALLATION QUALIFICATION

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5.0 | List of Test Certificate | 13

6.0 | Results | 14

7.0 | Conclusion | 14

8.0 | Approvals | 15

Constructional Details: | Constructional Details: | Observations | Sign & Date

Associate Manufacture | Citizen Industries Pvt.Ltd.(a member of DAIKN group) | |

Dimensions (L x W x H) | 4086 mxm X 1356 mm X 936 mm | |

Outer skin | 0.8mm G.I. Pre Coated with Guard Film | |

Inner skin | 0.8mm G.I. Plain Sheet. | |

Panel thickness | 43 mm thick Double Skin Panels with PUF | |

Insulation | Polyurethane, Density (40±2 kg/m³) | |

Profile | Thermal break | |

Type of unit | Double skin Floor Mounted Air Handling Unit | |

Frame structure | Aluminium Extruded Anodized Profile with thermal break & internal coving | |

Colour | RAL-5012 (Sky blue) | |

Skid | 75 x 40 mm - 2 mm thick G.I. Fabricated | |

Handle | C' Type Glass filled nylon | |

Hinges | Nylon with G.I. Pivot | |

Corner Coupler | Glass filled nylon | |

Damper M.O.C. | Aluminium | |

Technical Specifications: | Technical Specifications: | Technical Specifications: | Technical Specifications:

Fan air quantity | 4000 CFM | |

Supply air quantity | 4000 CFM | |

Dehum. air quantity | 4000 CFM | |

Static Pressure | 100 mm of water Column | |

Fan | Fan | Fan | Fan

Fan Model and Type | BNC-P400 & Plug Fan | |

Make | Kruger | |

Fan BkW | 2.90 | |

Fan Speed | 2635 RPM | |

Sr. No | 123S1140-M-00-00 | |

Motor | Motor | Motor | Motor

Make | CG (Crompton Greaves) | |

KW | 5.5 kW - 2 Pole (IE-3 - Non FLP) | |

Type | 415 V, 3 Ph. TEFC, Sq. Cage, Foot Mounted, Induction Motor | |

Machine No | 5.50KP2 | |

Efficiency | 89.2 IE3 | |

Voltage | 415 V ± 10 % | |

Current | 9.8 A | |

Connection | Delta | |

Frame Size | NG132S | |

Speed | 2900 RPM | |

Cooling Coil | Cooling Coil | Cooling Coil | Cooling Coil

Type | Fin & Tube Type Coil for Chilled Water Application | |

Size (mm) | 762 x 975 (30.00"H x 38.40"L) - 6 RD | |

Coil TR | 8.09 TR | |

Quantity | 1 No | |

Fin Spacing | 11 to 12 FPI | |

Coil Face Velocity | 2.54 M/S | |

Tube Diameter & Material | 0.50 mm thick,12.7 mm O.D, Plain Copper Tubes | |

Fins Thickness & Material | 0.13 mm thick, (blue Coated) Aluminium Fins | |

Drain pan | Drain pan | Drain pan | Drain pan

MOC | S. S.304 With 19mm thick close cell Nitrile rubber Insulation | |

Pre-filter | Pre-filter | Pre-filter | Pre-filter

Efficiency | 90% - 10µ (M-5) | |

Size, Qty and Type | 610 x 610 x 305 - 2 Nos. Flange Type | |

Location | In mixing chamber | |

Filter Make | AAF | |

Fresh Air filter | Fresh Air filter | Fresh Air filter | Fresh Air filter

Efficiency | 90% - 10μ (G-4) ||
Size, Qty and Type | 305 x 305 x 50 - 1 No. Flange Type ||
Location | Fresh Air Opening ||
Filter Make | AAF ||
Bleed Air filter | Bleed Air filter | Bleed Air filter | Bleed Air filter
Efficiency | 90-95% - 5μ (F-8) ||
Size, Qty and Type | 305 x 305 x 305 - 1 No. Flange Type ||
Location | On the Top of Filter ||
Filter Make | AAF ||
Fine filter | Fine filter | Fine filter | Fine filter
Efficiency | 90-95% - 5μ (F-8) ||
Size, Qty and Type | 610 x 610 x 305 - 2 Nos. Flange Type ||
Location | Supply Plenum in AHU ||
Filter Make | AAF ||
Electrical Heater | Electrical Heater | Electrical Heater | Electrical Heater
MOC | SS Finned Tubular Type ||
KW | 12 ||
Quantity | 2 KW X 6 No ||
Volume Control Dampers (Manually Operated) | Volume Control Dampers (Manually Operated) | Volume Control Dampers (Manually Operated) | Volume Control Dampers (Manually Operated)
MOC | The dampers will be of Aerofoil Design made out of Aluminum extruded section multi-louvered type with framework, operating link, etc. ||
Location |||
• Supply air outlet | 410X700 mm-01 Nos ||
• Return air inlet | 350X 800mm-01 Nos ||
• Bleed air outlet | 250X250 mm-01 Nos ||
• Fresh air inlet | 250X250 mm-01 Nos ||
Fire Damper | Fire Damper | Fire Damper | Fire Damper
MOC | Fire Dampers shall meet the fire resistance rating of 2 hours as per UL555 and shall be tested and certified by CBRI. Fire Damper frame shall be galvanized steel of 1.6mm thickness, formed into a C channel in 4-piece construction with flanges suitable for duct connection. Damper Blades shall be single skin galvanized steel of 1.6mm thickness and must have 3 accurately roll formed longitudinal grooves for reinforcement and proper closure and operation. The blades shall be arranged in parallel to provide positive shut off. Each Blade shall have axles supported on both sides with sintered bronze self-lubricating bearings turning in an extruded hole in the frame. All Blades shall be interconnected using linkage in the frame section. Each Damper shall be operated through a UL stamped fusible link rated at 74- or 100-Degree C. The closing of the fire damper shall be through a suitable rated spring which will be capable of closing the blades in the event of fusible link disintegration. The option of indication of Blade Position should be made available through a manual lever. Each Damper shall be supplied with a factory mounted sleeve made from Minimum 1mm thick galvanized steel of minimum 150mm length. Fire damper to be provided with limit switch and wiring up to control panel to have interlock with AHU. | Fire Dampers shall meet the fire resistance rating of 2 hours as per UL555 and shall be tested and certified by CBRI. Fire Damper frame shall be galvanized steel of 1.6mm thickness, formed into a C channel in 4-piece construction with flanges suitable for duct connection. Damper Blades shall be single skin galvanized steel of 1.6mm thickness and must have 3 accurately roll formed longitudinal grooves for reinforcement and proper closure and operation. The blades shall be arranged in parallel to provide positive shut off. Each Blade shall have axles supported on both sides with sintered bronze self-lubricating bearings turning in an extruded hole in the frame. All Blades shall be interconnected using linkage in the frame section. Each Damper shall be operated through a UL stamped fusible link rated at 74- or 100-Degree C. The closing of the fire damper shall be through a suitable rated spring which will be capable of closing the blades in the event of fusible link disintegration. The option of indication of Blade Position should be made available through a manual lever. Each Damper shall be supplied with a factory mounted sleeve made from Minimum 1mm thick galvanized steel of minimum 150mm length. Fire damper to be provided with limit switch and wiring up to control panel to have interlock with AHU. | Fire Dampers shall meet the fire resistance rating of 2 hours as per UL555 and shall be tested and certified by CBRI. Fire Damper frame shall be galvanized steel of 1.6mm thickness, formed into a C channel in 4-piece construction with flanges suitable for duct connection. Damper Blades shall be single skin galvanized steel of 1.6mm thickness and must have 3 accurately roll formed longitudinal grooves for reinforcement and proper closure and operation. The blades shall be arranged in parallel to provide positive shut off. Each Blade shall have axles supported on both sides with sintered bronze self-lubricating bearings turning in an extruded hole in the frame. All Blades shall be interconnected using linkage in the frame section. Each Damper shall be operated through a UL stamped fusible link rated at 74- or 100-Degree C. The closing of the fire damper shall be through a suitable rated spring which will be capable of closing the blades in the event of fusible link disintegration. The option of indication of Blade Position should be made available through a manual lever. Each Damper shall be supplied with a factory mounted sleeve made from Minimum 1mm thick galvanized steel of minimum 150mm length. Fire damper to be provided with limit switch and wiring up to control panel to have interlock with AHU.
| Fusible link rated 74 OC, UL stamped ||
Size & Quantity | 350X 800 mm-01 Nos
410X700 mm-01 Nos ||
Make | Airflow Product ||
Magnehelic Gauge across Pre-Filter | Magnehelic Gauge across Pre-Filter | Magnehelic Gauge across Pre-Filter | Magnehelic Gauge across Pre-Filter
Location | Across Pre filter ||
Range | 0-25 MMWC ||
Quantity | 01 Nos ||
Make | Omicron ||
Model No | 2000-25MM ||
Magnehelic Gauge across Fine Filter | Magnehelic Gauge across Fine Filter | Magnehelic Gauge across Fine Filter | Magnehelic Gauge across Fine Filter
Location | Across Fine filter ||
Range | 0-50 MMWC ||
Quantity | 01 Nos ||
Make | Omicron ||
Model No | 2000-50MM ||
SUPPLY & RETURN DIFFUSER | SUPPLY & RETURN DIFFUSER | SUPPLY & RETURN DIFFUSER | SUPPLY & RETURN DIFFUSER
MOC | Aluminum Powder Coated, Air Ceiling diffuser with VCD of powder coated aluminum construction. ||
Size & Quantity | SA 525 mm x 525 mm = 07 Nos
RA 525 mm x 525 mm = 07 Nos
SA 450 mm x 450 mm = 01 Nos
RA 450 mm x 450 mm = 01 Nos ||

Make | Airflow Product | |
 Type | Fabricated ducts. | |
 Standard | Ducts will be fabricated as per SMACNA Standard. Class "C" | |
 Manufacturer of Sheets | Jindal | |
 Type of Sealant | Silicon Sealant | |
 Duct Fabricator | Asawa | |
 Thermal Insulation of Ducts | Thermal Insulation of Ducts | Thermal Insulation of Ducts | Thermal Insulation of Ducts
 Make | Aero foam | |
 MOC | Closed cell nitrile rubber insulation with one side aluminium foil | |
 Supply Duct | 25 mm Thick | |
 Return Duct | 13 mm Thick | |
 Electrical Panel | Panel for AHU with MCB of Suitable Range as Isolator. The Panel will have Cooling Fan arrangement with Guard and Drawing Folder Provision. The Panel to be Fabricated from 16swg sheet duly powder coated, included VFD Controller | |
 Volts | 415 V | |
 Phase | 3 Phase | |
 Earthing | Connected with existing Earthing grid | |
 Leakage Voltage between Neutral & Earth | Not more than 5 volts | |
 Temperature sensor, HMI (Human Machine Interface) Controller & Limit switch | Temperature sensor, HMI (Human Machine Interface) Controller & Limit switch | Temperature sensor, HMI (Human Machine Interface) Controller & Limit switch | Temperature sensor, HMI (Human Machine Interface) Controller & Limit switch
 Temperature & Relative Humidity Sensor | Temperature & Relative Humidity Sensor | Temperature & Relative Humidity Sensor | Temperature & Relative Humidity Sensor
 Type | TRH | |
 Make | Autonics | |
 Mode No | THD-D2-C | |
 Measurement range | For Temp.: -10 ~ 40°C
 0 ~ 70°C | |
 Output | 4 ~ 20 mA or 0 ~ 10 V DC | |
 HMI (Human Machine Interface) Controller | HMI (Human Machine Interface) Controller | HMI (Human Machine Interface) Controller | HMI (Human Machine Interface) Controller
 Make | Hiecon | |
 Display Size | 7 inch | |
 Model | PI 3070i | |
 Communication port | RS422TX+/RS485 | |
 Printer Port | USB/Serial Port | |
 Working Temperature | 0 to 50° C | |
 Working Humidity | 10 To 90 % | |
 Input Voltage Range | DC 10 volt | |
 Qty | 01Nos | |
 Limit Switch | Limit Switch | Limit Switch | Limit Switch
 Make | Jainson | |
 Type | JLSR | |
 Range | 10 A, 500 V- 1 NO + 1NC | |
 VFD | VFD | VFD | VFD
 Make | Danfoss | |
 Input | AC 3 PH, 380 - 480 V, 42 / 36 A, 50 / 60 Hz | |
 Output | AC 3 PH, 0 - Input V 31 /27A, 0 - 400 Hz | |
 Model
 Model | VLT Micro Drive FC 51 | |
 Capacity (KW) | 5.5 KW | |
 Piping | Piping | Piping | Piping
 Make | Jindal | |
 Size | 40 NB | |
 Y-type Strainer | Y-type Strainer | Y-type Strainer | Y-type Strainer
 Make | Sant | |
 Size | 40 NB | |
 Quantity | 01 Nos | |
 Butterfly Valves | Butterfly Valves | Butterfly Valves | Butterfly Valves
 Make | Kitz | |
 Size | 40 NB | |
 Quantity | 06 Nos | |
 3-way valve | 3-way valve | 3-way valve | 3-way valve
 Make | Belimo | |
 Valve size | 32 NB | |
 Qty | 01 Nos | |
 Model No | R3040 | |
 Power Supply | 24V AC/DC, | |
 Control Signal /, feedback signal | 0.5-10 V DC | |
 Temperature Gauge | Temperature Gauge | Temperature Gauge | Temperature Gauge
 Make | H-guru | |
 Range & Quantity | 0 to 50 0C, 02 Nos. | |
 Quantity | 02 Nos. | |
 Pressure Gauge | Pressure Gauge | Pressure Gauge | Pressure Gauge
 Make | H-guru | |
 Range | 0 to 7 Kg/cm2 | |
 Quantity | 02 Nos. | |
 Air Vent | Air Vent | Air Vent | Air Vent
 Specification | Automatic purge valve with ball valve. | |
 Make | Anergy | |
 Size & Quantity | 12 MM dia -01 Nos | |
 Description | Specifications | Observations | Sign & Date
 Electrical | Electrical | Electrical | Electrical
 Phase | Three Phase, Including VFD | |
 Voltage | 415V | |
 Frequency | 50 Hz +/- 3% variations | |
 Chilled Water | Inlet Temp to AHU is 5-6 °C and out let Temperature is 12 °C | |
 Sr. No | Item Name | Whether Available
 1) | AHU Test Certificate |
 2) | Cooling Coil Test Certificate |
 3) | Fan Test Certificate |

4) | Induction Motor Test Certificate |
5) | Volume Control Damper / Fire Damper Test Certificate |
6) | 10-micron filter |
7) | 3-micron filter |
8) | 0.3-micron filter |
9) | Duct Test Certificate |
10) | Ducting Insulation Test Certificate |
11) | Magnehelic Gauge Test Certificate |
12) | Electrical Panel Test Certificate |
13) | Pressure / Temperature Gauge Test Certificate |
14) | Y strainer, 3-way Valve, Butterfly Valve Test Certificate |
Department | Name | Signature | Date
Project: BioBridge Healthcare Solutions Pvt. Ltd., India | Project: BioBridge Healthcare Solutions Pvt.
Ltd., India | Project: BioBridge Healthcare Solutions Pvt. Ltd., India | Project: BioBridge Healthcare
Solutions Pvt. Ltd., India
Prepared by: | | |
Checked/Reviewed by: | | |
INFARMA – Indústria Farmacêutica, Lda | INFARMA – Indústria Farmacêutica, Lda | INFARMA – Indústria
Farmacêutica, Lda | INFARMA – Indústria Farmacêutica, Lda
Reviewed by: | | |
Reviewed by: | | |
Approved by | | |

CONTENTS

OBJECTIVE:

To document the installation of the Air Handling Unit (AHU – 03), ascertain that the installation is as per design Specifications lay down.

SCOPE:

This protocol is applicable to the AHU installed in M/s. INFARMA- Indústria Farmacêutica, Lda

PERFORMING THE INSTALLATION QUALIFICATON:

The installation should conform to this checklist. Record all observations and deviations in the space provided in the checklist. In case of deviations prepare a deviation report along with justifications, which should be approved by the Engineering Department, Quality Assurance Manager and Validation In charge.

4.0 CHECKLIST FOR INSTALLATION OF AHU-03

AHU 03- 2600 CFM - RECIRCULATION

DUCTING: -

ELECTRICAL CONNECTIONS

CHILLED WATER MANIFOLD DETAILS

DETAILS OF UTILITIES

5.0 Test Certificates for the following components to be attached -

6.0 RESULTS:

7.0 CONCLUSION –

BASED UPON THE OBSERVATIONS RECORDED IN THE TABLE GIVEN ABOVE, THE INSTALLATION OF THE AHU IS FOUND TO BE SATISFACTORY / IS FOUND NOT TO BE SATISFACTORY AS PER THE DESIGN SPECIFICATIONS. THEREFORE, INSTALLATION OF THE AHU IS / IS NOT QUALIFIED AND THE OPERATION QUALIFICATION CAN / CANNOT BE PERFORMED

APPROVAL FOR INSTALLATION QUALIFICATION

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7.0 | Conclusion | 14

8.0 | Approvals | 15

Constructional Details: | Constructional Details: | Constructional Details: | Observations | Sign & Date
Design & Project By | Bio Bridge Healthcare Solutions Pvt. Ltd. | Bio Bridge Healthcare Solutions Pvt.

Ltd. | |

Associate Manufacture | Citizen Industries Pvt.Ltd.(a member of DAIKN group) | Citizen Industries

Pvt.Ltd.(a member of DAIKN group) | |

Dimensions (L x W x H) | 3986 mm X 1086 mm X 936 mm | 3986 mm X 1086 mm X 936 mm | |

Outer skin | 0.8mm G.I. Pre Coated with Guard Film | 0.8mm G.I. Pre Coated with Guard Film | |

Inner skin | 0.8mm G.I. Plain Sheet. | 0.8mm G.I. Plain Sheet. | |

Panel thickness | 43 mm Thick CFC Free Puff | 43 mm Thick CFC Free Puff | |

Insulation | Polyurethane, Density (40±2 kg/m³) | Polyurethane, Density (40±2 kg/m³) | |

Profile | Thermal break | Thermal break | |

Type of unit | Double skin Floor Mounted Air Handling Unit | Double skin Floor Mounted Air Handling Unit

| |

Frame structure | Aluminium Extruded Anodized Profile with thermal break & internal coving | Aluminium

Extruded Anodized Profile with thermal break & internal coving | |

Colour | RAL-5012 (Sky blue) | RAL-5012 (Sky blue) | |

Skid | 75 x 40 mm - 2 mm thick G.I. Fabricated | 75 x 40 mm - 2 mm thick G.I. Fabricated | |

Handle | C' Type Glass filled nylon | C' Type Glass filled nylon | |

Hinges | Nylon with G.I. Pivot | Nylon with G.I. Pivot | |

Corner Coupler | Glass filled nylon | Glass filled nylon | |

Damper M.O.C. | Aluminium | Aluminium | |

Technical Specifications: | Technical Specifications: | Technical Specifications: | Technical

Specifications: | Technical Specifications:

Fan air quantity | Fan air quantity | 2600 CFM | |

Supply air quantity | Supply air quantity | 2600 CFM | |

Dehum. air quantity | Dehum. air quantity | 2600 CFM | |

Static Pressure | Static Pressure | 100 mm of wc | |

Fan | Fan | Fan | Fan | Fan

Fan Model & Type | Fan Model & Type | BNC-P355 I & Plug Type | |

Make | Make | Kruger | |

Fan BkW | Fan BkW | 1.88 | |

Fan Speed | Fan Speed | 2751 RPM | |

Sr. No | Sr. No | 123Q3598-M-00-00 | |

Motor | Motor | Motor | Motor | Motor

Make | CG (Crompton Greaves) | CG (Crompton Greaves) | |

KW | 3.7 kW - 2 Pole (IE-3 - Non FLP) | 3.7 kW - 2 Pole (IE-3 - Non FLP) | |

Type | 415 V, 3 Ph. TEFC, Sq. Cage, Foot Mounted, Induction Motor | 415 V, 3 Ph. TEFC, Sq. Cage, Foot

Mounted, Induction Motor | |

Machine No | 3.70PN2 | 3.70PN2 | |

Efficiency | 87.8 IE3 | 87.8 IE3 | |

Voltage | 415 V ± 10 % | 415 V ± 10 % | |

Current | 6.7 A | 6.7 A | |

Connection | Delta | Delta | |

Frame Size | NS100L | NS100L | |

Speed | 2875 RPM | 2875 RPM | |

Cooling Coil | Cooling Coil | Cooling Coil | Cooling Coil | Cooling Coil

Type | Fin & Tube Type Coil for Chilled Water Application | Fin & Tube Type Coil for Chilled Water

Application | |

Size (mm) | 699 x 692 (27.50"H x 27.23"L) - 6 RD | 699 x 692 (27.50"H x 27.23"L) - 6 RD | |

Coil TR | 6.00 TR | 6.00 TR | |

Quantity | 1 No | 1 No | |

Fin Spacing | 11 to 12 FPI | 11 to 12 FPI | |

Coil Face Velocity | 2.54 M/S | 2.54 M/S | |

Tube Diameter & Material | 0.50 mm thick,12.7 mm O.D, Plain Copper Tubes | 0.50 mm thick,12.7 mm O.D,

Plain Copper Tubes | |

Fins Thickness & Material | 0.13 mm thick, (blue Coated) Aluminium Fins | 0.13 mm thick, (blue Coated)

Aluminium Fins | |
Drain pan | Drain pan | Drain pan | Drain pan | Drain pan
MOC | S. S.304 With 19mm thick close cell Nitrile rubber Insulation | S. S.304 With 19mm thick close cell Nitrile rubber Insulation | |
Pre-filter | Pre-filter | Pre-filter | Pre-filter | Pre-filter
Efficiency | 90% - 10µ (M-5) | 90% - 10µ (M-5) | |
Size, Qty and Type | 610 x 610 x 305 - 1 No. Flange Type
610 x 305 x 305 - 1 No. Flange Type | 610 x 610 x 305 - 1 No. Flange Type
610 x 305 x 305 - 1 No. Flange Type | |
Location | In mixing chamber | In mixing chamber | |
Filter Make | AAF | AAF | |
Fresh Air filter | Fresh Air filter | Fresh Air filter | Fresh Air filter | Fresh Air filter
Efficiency | 90% - 10µ (G-4) | 90% - 10µ (G-4) | |
Size, Qty and Type | 305 x 305 x 50 - 1 No. Flange Type | 305 x 305 x 50 - 1 No. Flange Type | |
Location | Fresh Air Opening | Fresh Air Opening | |
Filter Make | AAF | AAF | |
Bleed Air filter | Bleed Air filter | Bleed Air filter | Bleed Air filter
Efficiency | 90-95% - 5µ (F-8) | |
Size, Qty and Type | 305 x 305 x 305 - 1 No. Flange Type | |
Location | On the Top of Filter | |
Filter Make | AAF | |
Fine filter | Fine filter | Fine filter | Fine filter
Efficiency | 90-95% - 5µ (F-8) | |
Size, Qty and Type | 610 x 610 x 305 - 1 No. Flange Type
610 x 305 x 305 - 1 No. Flange Type | |
Location | Supply Plenum in AHU | |
Filter Make | AAF | |
Electrical Heater | Electrical Heater | Electrical Heater | Electrical Heater
MOC | SS Finned Tubular Type | |
KW | 6 | |
Quantity | 2 KW X 3 No | |
Volume Control Dampers (Manually Operated) | Volume Control Dampers (Manually Operated) | Volume Control Dampers (Manually Operated) | Volume Control Dampers (Manually Operated)
MOC | The dampers will be of Aerofoil Design made out of Aluminum extruded section multi-louvered type with framework, operating link, etc. | |
Location | | |
• Supply air outlet | 350X550 mm-01 Nos | |
• Return air inlet | 350X550 mm-01 Nos | |
• Bleed air outlet | 250X250 mm-01 Nos | |
• Fresh air inlet | 250X250 mm-01 Nos | |
Fire Damper | Fire Damper | Fire Damper | Fire Damper
MOC | Fire Dampers shall meet the fire resistance rating of 2 hours as per UL555 and shall be tested and certified by CBRI. Fire Damper frame shall be galvanized steel of 1.6mm thickness, formed into a C channel in 4-piece construction with flanges suitable for duct connection. Damper Blades shall be single skin galvanized steel of 1.6mm thickness and must have 3 accurately roll formed longitudinal grooves for reinforcement and proper closure and operation. The blades shall be arranged in parallel to provide positive shut off. Each Blade shall have axles supported on both sides with sintered bronze self-lubricating bearings turning in an extruded hole in the frame. All Blades shall be interconnected using linkage in the frame section. Each Damper shall be operated through a UL stamped fusible link rated at 74- or 100-Degree C. The closing of the fire damper shall be through a suitable rated spring which will be capable of closing the blades in the event of fusible link disintegration. The option of indication of Blade Position should be made available through a manual lever. Each Damper shall be supplied with a factory mounted sleeve made from Minimum 1mm thick galvanized steel of minimum 150mm length. Fire damper to be provided with limit switch and wiring up to control panel to have interlock with AHU. | Fire Dampers shall meet the fire resistance rating of 2 hours as per UL555 and shall be tested and certified by CBRI. Fire Damper frame shall be galvanized steel of 1.6mm thickness, formed into a C channel in 4-piece construction with flanges suitable for duct connection. Damper Blades shall be single skin galvanized steel of 1.6mm thickness and must have 3 accurately roll formed longitudinal grooves for reinforcement and proper closure and operation. The blades shall be arranged in parallel to provide positive shut off. Each Blade shall have axles supported on both sides with sintered bronze self-lubricating bearings turning in an extruded hole in the frame. All Blades shall be interconnected using linkage in the frame section. Each Damper shall be operated through a UL stamped fusible link rated at 74- or 100-Degree C. The closing of the fire damper shall be through a suitable rated spring which will be capable of closing the blades in the event of fusible link disintegration. The option of indication of Blade Position should be made available through a manual lever. Each Damper shall be supplied with a factory mounted sleeve made from Minimum 1mm thick galvanized steel of minimum 150mm length. Fire damper to be provided with limit switch and wiring up to control panel to have interlock with AHU. | Fire Dampers shall meet the fire resistance rating of 2 hours as per UL555 and shall be tested and certified by CBRI. Fire Damper frame shall be galvanized steel of 1.6mm thickness, formed into a C channel in 4-piece construction with flanges suitable for duct connection. Damper Blades shall be single skin galvanized steel of 1.6mm thickness and must have 3 accurately roll formed longitudinal grooves for reinforcement and proper closure and operation. The blades shall be arranged in parallel to provide positive shut off. Each Blade shall have axles supported on both sides with sintered bronze self-lubricating bearings turning in an extruded hole in the frame. All Blades shall be interconnected using linkage in the frame section. Each Damper shall be operated through a UL stamped fusible link rated at 74- or 100-Degree C. The closing of the fire damper shall be through a suitable rated spring which will be capable of closing the blades in the event of fusible link disintegration. The option of indication of Blade Position should be made available through a manual lever. Each Damper shall be supplied with a factory mounted sleeve made from Minimum 1mm thick galvanized steel of minimum 150mm length. Fire damper to be provided with limit switch and wiring up to control panel to have interlock with AHU.
| Fusible link rated 74 OC, UL stamped | |
Size & Quantity | 350X550 mm-02 Nos | |
Make | Airflow Product | |
Magnehelic Gauge across Pre-Filter | Magnehelic Gauge across Pre-Filter | |
Location | Across Pre filter | |
Range | 0-25 MMWC | |
Quantity | 01 Nos | |
Make | Omicron | |
Model No | 2000-25MM | |
Magnehelic Gauge across Fine Filter | Magnehelic Gauge across Fine Filter | |
Location | Across Fine filter | |

Range | 0-50 MMWC ||
 Quantity | 01 Nos ||
 Make | Omicron ||
 Model No | 2000-50MM ||
 SUPPLY & RETURN DIFFUSER | SUPPLY & RETURN DIFFUSER ||
 MOC | Aluminum Powder Coated, Air Ceiling diffuser with VCD of powder coated aluminum construction. ||
 Size & Quantity | SA 525 mm x 525 mm = 04 Nos
 RA 525 mm x 525 mm = 04 Nos
 SA 450 mm x 450 mm = 01 Nos
 RA 450 mm x 450 mm = 02 Nos ||
 Make | Airflow Product ||
 Type | Fabricated ducts. ||
 Standard | Ducts will be fabricated as per SMACNA Standard. Class "C" ||
 Manufacturer of Sheets | Jindal ||
 Type of Sealant | Silicon Sealant ||
 Duct Fabricator | Asawa ||
 Thermal Insulation of Ducts | Thermal Insulation of Ducts | Thermal Insulation of Ducts | Thermal Insulation of Ducts
 Make | Aero foam ||
 MOC | Closed cell nitrile rubber insulation with one side aluminium foil ||
 Supply Duct | 25 mm Thick ||
 Return Duct | 13 mm Thick ||
 Electrical Panel | Panel for AHU with MCB of Suitable Range as Isolator. The Panel will have Cooling Fan arrangement with Guard and Drawing Folder Provision. The Panel to be Fabricated from 16swg sheet duly powder coated, included VFD Controller | Panel for AHU with MCB of Suitable Range as Isolator. The Panel will have Cooling Fan arrangement with Guard and Drawing Folder Provision. The Panel to be Fabricated from 16swg sheet duly powder coated, included VFD Controller | Panel for AHU with MCB of Suitable Range as Isolator. The Panel will have Cooling Fan arrangement with Guard and Drawing Folder Provision. The Panel to be Fabricated from 16swg sheet duly powder coated, included VFD Controller
 Volts | 415 V ||
 Phase | 3 Phase ||
 Earthing | Connected with existing Earthing grid ||
 Leakage Voltage between Neutral & Earth | Not more than 5 volts ||
 Temperature sensor, HMI (Human Machine Interface) Controller & Limit switch | Temperature sensor, HMI (Human Machine Interface) Controller & Limit switch | Temperature sensor, HMI (Human Machine Interface) Controller & Limit switch | Temperature sensor, HMI (Human Machine Interface) Controller & Limit switch | Temperature sensor, HMI (Human Machine Interface) Controller & Limit switch
 Temperature & Relative Humidity Sensor | Temperature & Relative Humidity Sensor | Temperature & Relative Humidity Sensor | Temperature & Relative Humidity Sensor
 Type | TRH ||
 Make | Autonics ||
 Mode No | THD-D2-C ||
 Measurement range | For Temp.: -10 ~ 40°C
 0 ~ 70°C ||
 Output | 4 ~ 20 mA or 0 ~ 10 V DC ||
 HMI (Human Machine Interface) Controller | HMI (Human Machine Interface) Controller | HMI (Human Machine Interface) Controller | HMI (Human Machine Interface) Controller
 Make | Hiecon ||
 Display Size | 7 inch ||
 Model | PI 3070i ||
 Communication port | RS422TX+/RS485 ||
 Printer Port | USB/Serial Port ||
 Working Temperature | 0 to 50° C ||
 Working Humidity | 10 To 90 % ||
 Input Voltage Range | DC 10 volt ||
 Qty | 01Nos ||
 Limit Switch | Limit Switch | Limit Switch | Limit Switch
 Make | Jainson ||
 Type | JLSR ||
 Range | 10 A, 500 V- 1 NO + 1NC ||
 VFD | VFD | VFD | VFD
 Make | Danfoss ||
 Input | AC 3 PH, 380 - 480 V, 42 / 36 A, 50 / 60 Hz ||
 Output | AC 3 PH, 0 - Input V 31 /27A, 0 - 400 Hz ||
 Model
 Model | VLT Micro Drive FC 51 ||
 Capacity (KW) | 3.7 KW ||
 Piping | Piping | Piping | Piping
 Make | Jindal ||
 Size | 40 NB ||
 Y-type Strainer | Y-type Strainer | Y-type Strainer | Y-type Strainer
 Make | Sant ||
 Size | 40 NB ||
 Quantity | 01 Nos ||
 Butterfly Valves | Butterfly Valves | Butterfly Valves | Butterfly Valves
 Make | Kitz ||
 Size | 40 NB ||
 Quantity | 06 Nos ||
 3-way valve | 3-way valve | 3-way valve | 3-way valve
 Make | Belimo ||
 Valve size | 32 NB ||
 Qty | 01 Nos ||
 Model No | R3040 ||
 Power Supply | 24V AC/DC, ||
 Control Signal /, feedback signal | 0.5-10 V DC ||
 Temperature Gauge | Temperature Gauge | Temperature Gauge | Temperature Gauge
 Make | H-guru ||
 Range & Quantity | 0 to 50 OC, 02 Nos. ||
 Quantity | 02 Nos. ||
 Pressure Gauge | Pressure Gauge | Pressure Gauge | Pressure Gauge
 Make | H-guru ||
 Range | 0 to 7 Kg/cm2 ||
 Quantity | 02 Nos. ||

Air Vent | Air Vent | Air Vent | Air Vent
Specification | Automatic purge valve with ball valve. | |
Make | Anergy | |
Size & Quantity | 12 MM dia -01 Nos | |
Description | Specifications | Observations | Sign & Date
Electrical | Electrical | Electrical | Electrical
Phase | Three Phase, Including VFD | |
Voltage | 415V | |
Frequency | 50 Hz +/- 3% variations | |
Chilled Water | Inlet Temp to AHU is 5-6 °C and out let Temperature is 12 °C | |
Sr. No | Item Name | Whether Available
1) | AHU Test Certificate |
2) | Cooling Coil Test Certificate |
3) | Fan Test Certificate |
4) | Induction Motor Test Certificate |
5) | Volume Control Damper / Fire Damper Test Certificate |
6) | 10-micron filter |
7) | 3-micron filter |
8) | 0.3-micron filter |
9) | Duct Test Certificate |
10) | Ducting Insulation Test Certificate |
11) | Magnehelic Gauge Test Certificate |
12) | Electrical Panel Test Certificate |
13) | Pressure / Temperature Guage Test Certificate |
14) | Y strainer, 3-way Valve, Butterfly Valve Test Certificate |
Department | Name | Signature | Date
Project: BioBridge Healthcare Solutions Pvt. Ltd., India | Project: BioBridge Healthcare Solutions Pvt.
Ltd., India | Project: BioBridge Healthcare Solutions Pvt. Ltd., India | Project: BioBridge Healthcare
Solutions Pvt. Ltd., India
Prepared by: | | |
Checked/Reviewed by: | | |
INFARMA – Indústria Farmacêutica, Lda | INFARMA – Indústria Farmacêutica, Lda | INFARMA – Indústria
Farmacêutica, Lda | INFARMA – Indústria Farmacêutica, Lda
Reviewed by: | | |
Reviewed by: | | |
Approved by | | |

CONTENTS

OBJECTIVE:

To document the installation of the Air Compressor Unit, ascertain that the installation is as per design Specifications lay down.

SCOPE:

This protocol is applicable to the Air Compressor Unit installed in M/s. INFARMA-
Indústria Farmacêutica, Lda

PERFORMING THE INSTALLATION QUALIFICATION:

The installation should conform to this checklist. Record all observations and deviations in the space provided in the checklist. In case of deviations prepare a deviation report along with justifications, which should be approved by the Engineering Department, Quality Assurance Manager and Validation In charge.

CHECKLIST FOR INSTALLATION OF AIR COMPRESSOR UNIT

TECHNICAL SPECIFICATIONS OF THE SUB-COMPONENTS/BOUGHT OUT ITEMS, AND THEIR MAKE, MODEL & QUANTITY

DETAILS OF UTILITIES

IDENTIFICATION OF COMPONENTS FOR CALIBRATION

MATERIAL OF CONSTRUCTION

SAFETY FEATURES

5.0 Test Certificates for the following components to be attached -

6.0 RESULTS:

7.0 Deviation (If Any)

8.0 Justification for Acceptance of Deviation

7.0 CONCLUSION -

Based upon the observations recorded in the table given above, the installation of the ahu is found to be satisfactory / is found not to be satisfactory as per the design specifications. therefore, installation of the Air Compressor is / is not qualified and the operation qualification can / cannot be performed

APPROVAL FOR INSTALLATION QUALIFICATION

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3.0 | Performing the Installation Qualification | 2

4.0 | Checklist for Installation | 3

5.0 | List of Test Certificate | 9

6.0 | Results | 9

7.0 | Conclusion | 9

8.0 | Approvals | 10

Description | Specifications | Observations | Sign & Date

Equipment | Air Compressor Unit | |

URS No | URS/MOZ/INF/019 | |

Functional Overview | Compressed air unit provides a clean and reliable source of pneumatic power for various sensors, actuators, and control devices used in process automation, ensuring precise and consistent operations. | |

Design Overview | It covers aspects such as power source, compression mechanisms, control mechanisms, storage tank, and safety features. | |

Material of construction | All parts to be designed and manufactured in line with the current standards |

|
Specification (Machine) & Dimensions | 21 CFM capacity air cooled screw air compressor units comprising of sealed screw compressor, including refrigerant air dryer & receiver. are to be installed at outside premises in the space provided.

Shall be screw type air compressor.

Shall have capacity of compressed air be 21 CFM

Shall have compressor air receiving tank of capacity 200 Liters

Shall be refrigerant type air dryer

Supply compressed air at pressure of 10kg/cm²

Machine has dry paper type suction air filter with silencer

Equipped with three-way solenoid valve which is used for loading/unload and regulation of the compressor

Shall have air / oil temperature switch

Shall have provision to shut down the compressor in case of too high outlet air temperature

Shall have view glass to check oil level and

provision for oil filling. | |

Alarms | NA | |

Operation of Equipment | The machine should have user-friendly controls for operators to set and adjust various parameters such as in, out air pressure and other relevant settings. These controls should be easily accessible | |

Name of component | Description | Observations

(Yes/No) | Sign & Date

Air Compressor Screw | Air Compressor Screw | Air Compressor Screw | Air Compressor Screw

Model No | CPM7TMD | |

Make | CHICAGO PNEUMATIC | |

Actual Capacity | 21 CFM | |

Compressor Type | Screw | |

Compressor Input Power | 5.50 KW | |

Air Receiver Capacity | 200 Liters | |

Air Dryer | Refrigerated Air dryer | |

Pressure | 10kg/cm² | |

Size | 1420(L)x550(W)x1280(H) mm | |

Qty | 01 Nos | |

Compressor | Compressor | Compressor | Compressor

One Single Stage, rotary Screw Element complete with:

Dry paper type suction air filter with silencer

Conveniently located for easy replacement of filter element

Unloader with Integrated Regulating Valve for Load/Unload Control system.

Simple design with only one moving part, needs no regular adjustments.

Three-way solenoid valve required for load/unload regulation of the compressor.

Air / Oil temperature switch

To shut down the compressor in case of too high outlet air temperature | One Single Stage, rotary Screw

Element complete with:

Dry paper type suction air filter with silencer

Conveniently located for easy replacement of filter element

Unloader with Integrated Regulating Valve for Load/Unload Control system.

Simple design with only one moving part, needs no regular adjustments.
Three-way solenoid valve required for load/unload regulation of the compressor.
Air / Oil temperature switch
To shut down the compressor in case of too high outlet air temperature | |
Air Oil Receiver Tank consisting of | Air Oil Receiver Tank consisting of | Air Oil Receiver Tank consisting of | Air Oil Receiver Tank consisting of
View glass for checking oil level and has provision for oil filling.
Minimum Pressure valve to close off the compressor from the air net when the unit is stopped or running unloaded and to maintain required air pressure in the system for proper oil lubrication- Yes.
Safety valve.
Oil draining arrangement.
Three stage air oil Separation system
-First by cyclonic action (Centrifugal), Second by --
-gravitational (heavier particles separate down),
Third by passing through coalescent filter. | View glass for checking oil level and has provision for oil filling.
Minimum Pressure valve to close off the compressor from the air net when the unit is stopped or running unloaded and to maintain required air pressure in the system for proper oil lubrication- Yes.
Safety valve.
Oil draining arrangement.
Three stage air oil Separation system
-First by cyclonic action (Centrifugal), Second by --
-gravitational (heavier particles separate down),
Third by passing through coalescent filter. | |
Oil Cooler Assembly | Oil Cooler Assembly | Oil Cooler Assembly |
Oil cooler is compact cooler of aluminum for optimum heat transfer. Lower pressure drops and lower weight-Yes.
Oil Filter (spin on) mounted on air oil receiver for filtration of lubricating oil.
Thermostatic valve to regulate oil temperature within the system. | Oil cooler is compact cooler of aluminum for optimum heat transfer. Lower pressure drops and lower weight-Yes.
Oil Filter (spin on) mounted on air oil receiver for filtration of lubricating oil.
Thermostatic valve to regulate oil temperature within the system. | |
Power and Control Panel | Power and Control Panel | Power and Control Panel | Power and Control Panel
Power Panel incorporating; - Suitable DOL starter with contactors, relays etc.
Emergency stop button
Power cable suitable for 3 phase 4-wire connection of 3 m length approx., factory fitted.
Control panel
Dial indicators mounted on facia for "Discharge Air Pressure."
Hour meter mounted on facia for indicating" Total Running hours."
Electro pneumatic Energy saving control with pressure switch, solenoid valve, timer etc. for load & unload regulation, and to shutdown compressor when no air is required
Safety device to shut down compressor in case of high outlet air temperature. | Power Panel incorporating; - Suitable DOL starter with contactors, relays etc.
Emergency stop button
Power cable suitable for 3 phase 4-wire connection of 3 m length approx., factory fitted.
Control panel
Dial indicators mounted on facia for "Discharge Air Pressure."
Hour meter mounted on facia for indicating" Total Running hours."
Electro pneumatic Energy saving control with pressure switch, solenoid valve, timer etc. for load & unload regulation, and to shutdown compressor when no air is required
Safety device to shut down compressor in case of high outlet air temperature. | |
Drive Arrangement | Drive Arrangement | Drive Arrangement | Drive Arrangement
V Belt drive with trapezoidal v belts for-Reduced belt slip
-Reduced risk of belt failure
-Long life
-Guided belt tension arrangement | V Belt drive with trapezoidal v belts for-Reduced belt slip
-Reduced risk of belt failure
-Long life
-Guided belt tension arrangement | |
Air Outlet | Air Outlet | Air Outlet |
Ball Valve with 1/2" female outlet connection | Ball Valve with 1/2" female outlet connection | |
Description | Specifications | Observations
(Yes/No) | Sign & Date
Electrical | Electrical | Electrical | Electrical
Phase | Three Phase | |
Voltage | 415V | |
Frequency | 50 Hz | |
Description | Observations
(Yes/No) | Sign & Date
Pressure gauge | |
Description | Observations
(Yes/No) | Sign & Date
M.S., Aluminium & S G Iron | |
Description | Observations
(Yes/No) | Sign & Date
Temp. Switch & Safety Valve | |
Sr. No | Item Name | Available (Yes/No)
1) | Pressure gauge |
Department | Name | Signature | Date
Project: BioBridge Healthcare Solutions Pvt. Ltd., India | Project: BioBridge Healthcare Solutions Pvt. Ltd., India | Project: BioBridge Healthcare Solutions Pvt. Ltd., India | Project: BioBridge Healthcare Solutions Pvt. Ltd., India
Prepared by: | | |
Checked/Reviewed by: | | |
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Reviewed by: | | |
Reviewed by: | | |
Approved by | | |

CONTENTS

OBJECTIVE:

To document the installation of the Chiller Unit, ascertain that the installation is as per design

Specifications lay down.

SCOPE:

This protocol is applicable to the Chiller Unit installed in M/s. INFARMA- Indústria

Farmacêutica, Lda

PERFORMING THE INSTALLATION QUALIFICATION:

The installation should conform to this checklist. Record all observations and deviations in the space

provided in the checklist. In case of deviations prepare a deviation report along with justifications,

which should be approved by the Engineering Department, Quality Assurance Manager and Validation In

charge.

4.0 CHECKLIST FOR INSTALLATION OF CHILLER

EQUIPMENT DESIGN, PRINCIPLE AND REVIEW OF USER REQUIREMENTS SPECIFICATION (URS).

DETAILS OF UTILITIES

5.0 Test Certificates/Documents for the following components to be attached –

6.0 RESULTS:

7.0 CONCLUSION –

Based upon the observations recorded in the table given above, the installation of the Chiller is found

to be satisfactory / is found not to be satisfactory as per the design specifications. therefore,

installation of the Chiller is / is not qualified and the operation qualification can / cannot be

performed

APPROVAL FOR INSTALLATION QUALIFICATION

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2.0 | Scope | 2

3.0 | Performing the Installation Qualification | 2

4.0 | Checklist for Installation | 3

5.0 | List of Test Certificate | 15

6.0 | Results | 16

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8.0 | Approvals | 17

Description | Specifications | Observations

(Yes/No) | Sign & Date

Equipment | Air cooled screw chiller (70 TR) | |

URS No | URS/MOZ/INF/015 | |

Functional Overview | Air-cooled screw chiller is use to temperature control or heat load handling for a

facility | |

Design Overview | 70 TR capacity Air Cooled Screw chiller units comprising of sealed screw compressor,

Air cooled condenser, evaporator, refrigerant piping and control device, micro-processor-based control

panel etc. designed to be installed at outside premised in provided space. Two nos. of Chilled water

pumps (1 Working +1 Stand By) to be installed at same place. | |

Material of construction | All parts to be designed and manufactured in line with the current standards |

|

Specification (Machine) & Dimensions | Air-cooled Screw chiller capacity: 70 TR

Leaving Temperature: 7.00 °C

Entering Temperature: 12 °C.

Compressor / Refrigerant: R410A

Condenser Type: Air cooled

Condenser Fan Quantity: 04

Air Flow Rate: 98542CMH

Compressor Type: Semi hermetic screw

Compressor Quantity: 01

Compressor Motor Speed: 25 to 100% Variable

Evaporators Type: Tube type

Pump Type: Centrifugal

Noise Level: NMT 80 DB

Dia mention in mm(L*W*H): 3198 * 2235 * 2432

Expansion Valve: Thermostatic | |

Electrical connections | Three phase, 415 V Frequency 50Hz | |

AIR COOLED FLOODED SCREW CHILLER | AIR COOLED FLOODED SCREW CHILLER | AIR COOLED FLOODED SCREW CHILLER |

AIR COOLED FLOODED SCREW CHILLER | AIR COOLED FLOODED SCREW CHILLER

Specification | 'Screw' type air-cooled liquid chillers for the following operating conditions. The

chillers shall be complete with Mono/twin screw compressors, air- cooled condensers, flooded/DX type

evaporator shell and tube type, drive motor, control panel, first charge of R-134 a refrigerant, Oil,

expansion device etc., The chiller shall have stepless capacity control, as per specifications complete

with starter panel.

The evaporator will be of shell and tube type. Shell will be fabricated from rolled carbon steel plate

with fusion welded seams have carbon steel tube sheets, drilled and reamed to accommodate the tubes and

intermediate tube supports spaced no more than four feet apart. The refrigerant side will be designed &

tested in accordance with applicable pressure vessel code. Tubes shall be high-efficiency, internally and

externally enhanced type having copper tubes. Each tube will be roller expanded into the tube sheets

providing a leak proof seal and be individually replaceable. The evaporator shall be provided with

Insulation of 32 mm thick nitrile rubber insulation finished with glass cloth. | 'Screw' type air-cooled

liquid chillers for the following operating conditions. The chillers shall be complete with Mono/twin

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| The chiller shall be properly painted to protect all the exposed components from weathering & include integral refrigerant piping with liquid isolation valves, and wiring, flow switches / sensors, accessories as required and called for, automatic and safety controls mounted in central control panel, and all mounted on a steel frame.

Chillers Capacity 70 TR is proposed for plant. The chiller should have Single Compressor. The compressor unloading shall be from 30% to 100%. The chiller output & input shall be at the following conditions:

-Chilled Water Out - 7.00 °C
Chilled Water In - 12.00 °C
Operating Voltage 440 V + 15%, 50 Hz, 4 Wire, 3 Phase with Earthing and Starter panel. | The chiller shall be properly painted to protect all the exposed components from weathering & include integral refrigerant piping with liquid isolation valves, and wiring, flow switches / sensors, accessories as required and called for, automatic and safety controls mounted in central control panel, and all mounted on a steel frame.

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Operating Voltage 440 V + 15%, 50 Hz, 4 Wire, 3 Phase with Earthing and Starter panel.

Model Specifications- | Model Specifications- | Model Specifications- | Observations(Yes/No) | Sign & Date

Model No | LCA1-070X52H | LCA1-070X52H | |
Make | Blue Star | Blue Star | |
Actual Capacity | 69.63 TR | 69.63 TR | |
Compressor Input Power | 83.16 KW | 83.16 KW | |
Refrigerant | R134a | R134a | |
Expansion Valve | Thermostatic | Thermostatic | |
Qty | 01 Nos | 01 Nos | |
Evaporator- | Evaporator- | Evaporator- | Evaporator- | Evaporator-
Leaving Temperature | Leaving Temperature | 7.00 °C | |
Entering Temperature | Entering Temperature | 12.00 °C | |
Delta T | Delta T | 5.00 °C | |
Flow Rate | Flow Rate | 11.71 L/s | |
Fouling Factor | Fouling Factor | 0.0000180 M². K/W | |
Max Operating Pressure | Max Operating Pressure | 150 psig (Water Side) | |
Condenser- | Condenser- | Condenser- | Condenser- | Condenser-
Ambient Temperature | Ambient Temperature | 35.0 °C | |
Total Heat Rejection | Total Heat Rejection | 90.44 TR | |
Fan Quantity | Fan Quantity | 4 | |
FAN motor Power (each) | FAN motor Power (each) | 2.500 KW | |
FAN motor Power (total) | FAN motor Power (total) | 10.00 KW | |
Air Flow Rate | Air Flow Rate | 98542 CMH | |
Electrical Data- | Electrical Data- | Electrical Data- | Electrical Data- | Electrical Data-
Volts/Phase/Hertz | Volts/Phase/Hertz | 415/3/50.0 | |
Full Load Current | Full Load Current | 136.30 A | |
Starting Current | Starting Current | 331.30 A | |
Voltage Range | Voltage Range | 360-440 V | |
Physical Data- | Physical Data- | Physical Data- | Physical Data- | Physical Data-
Operating Weight | Operating Weight | 3032 kg | |
Length | Length | 3198 mm | |
Width | Width | 2235 mm | |
Height | Height | 2432 mm | |
Compressor Type | Compressor Type | Semi hermetic screw | |
Compressor Quantity | Compressor Quantity | 1 | |
Compressor Motor Speed | Compressor Motor Speed | 2950 rpm | |
Chilled Water Primary Pump | Chilled Water Primary Pump | Chilled Water Primary Pump | Chilled Water Primary Pump
Specification | Type: Centrifugal Pump

MOC: Pump casing CI and the impeller shall be Bronze / SS.

Vertical long coupled single stage, single suction type inline centrifugal pumps complete with C.I.

Casing, suction and discharge connections shall be of same size and shall be supplied with matching flanges. Impellers shall be made of bronze and shall be fully closed type. Pumps shall be dynamically balanced, and two plane balancing is required for the pumps if impeller diameter is less than 6 times the width of the impeller.

Pump Shaft shall be made of stainless steel, coupling shall be rigid spacer type made of high tensile aluminium alloy, shall be designed for easy removal on site. The pump design shall meet the requirements of easy removal of mechanical seal components for servicing and replacement without disturbing the pump or motor. TEFC Sq. cage non overloading type motor with efficiency level of EFF 2, mechanical seal shall be stainless steel multi-spring with secondary seal, carbon rotating face and silicon carbide stationary seal. Gland plate shall be made of 316 stainless steel. | Type: Centrifugal Pump

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Vertical long coupled single stage, single suction type inline centrifugal pumps complete with C.I.

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Make | Armstrong | |

Qty | 02 Nos (1 Working+1 Stand by) | |

Model | Model: Conventional - Constant Speed - 4300 - 50- 200 - 2p - 7.5 kW - | |

Total System Flow | 38.15 m³/hr. | |

System Head | 35 m | |

Pump Cons | Bronze Fitted; | |

Pump Rating | PN-16 | |

Motor | 7.5 KW | |

Enclosure: | TEFC | |

Efficiency | IE2 | |

Power Supply | 415/3/50 | |

INSULATED CHILLED WATER PIPING, VALVES & ACCESSORIES | INSULATED CHILLED WATER PIPING, VALVES & ACCESSORIES | INSULATED CHILLED WATER PIPING, VALVES & ACCESSORIES | INSULATED CHILLED WATER PIPING, VALVES & ACCESSORIES

MS- C class condenser water piping | MS- C class condenser water piping | MS- C class condenser water piping | MS- C class condenser water piping

Specification | MS-C class condenser water piping conforming to IS1239/IS3589 specifications cut to required lengths & installed with welded joints including fittings such as bends, elbows, tees, reducers, rubber gaskets, Piping shall be supported from slab/walls/floor by MS supports and half round wooden/ puff support below pipes. Insulated CHW piping; using MS "C" class pipes are insulated with insulation and aluminium cladding. Brush & clean pipe surface before applying insulation. Stick insulation material using a thin film of manufacturer-recommended adhesive on both pipe and insulation. Seal all joints airtight. Provide minimum 25mm thick insulation up to 40 mm diameter piping, and minimum 32 mm thick insulation from 50 to 150mm diameter and above piping.

Piping shall be supported from slab/walls/floor with adequate vibration isolation & half round wooden/puff support below pipes | MS-C class condenser water piping conforming to IS1239/IS3589 specifications cut to required lengths & installed with welded joints including fittings such as bends, elbows, tees, reducers, rubber gaskets, Piping shall be supported from slab/walls/floor by MS supports and half round wooden/ puff support below pipes. Insulated CHW piping; using MS "C" class pipes are insulated with insulation and aluminium cladding. Brush & clean pipe surface before applying insulation. Stick insulation material using a thin film of manufacturer-recommended adhesive on both pipe and insulation. Seal all joints airtight. Provide minimum 25mm thick insulation up to 40 mm diameter piping, and minimum 32 mm thick insulation from 50 to 150mm diameter and above piping.

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Piping shall be supported from slab/walls/floor with adequate vibration isolation & half round wooden/puff support below pipes | MS-C class condenser water piping conforming to IS1239/IS3589 specifications cut to required lengths & installed with welded joints including fittings such as bends, elbows, tees, reducers, rubber gaskets, Piping shall be supported from slab/walls/floor by MS supports and half round wooden/ puff support below pipes. Insulated CHW piping; using MS "C" class pipes are insulated with insulation and aluminium cladding. Brush & clean pipe surface before applying insulation. Stick insulation material using a thin film of manufacturer-recommended adhesive on both pipe and insulation. Seal all joints airtight. Provide minimum 25mm thick insulation up to 40 mm diameter piping, and minimum 32 mm thick insulation from 50 to 150mm diameter and above piping.

Piping shall be supported from slab/walls/floor with adequate vibration isolation & half round wooden/puff support below pipes

| | | Observation

(Yes/No) | Sign & Date

Make | Jindal | Jindal | |

Size & Quantity | As per design sizes | As per design sizes | |

Valves | Valves | Valves | Valves | Valves

Butterfly Valves (Manual) | Butterfly Valves (Manual) | Butterfly Valves (Manual) | Butterfly Valves (Manual) | Butterfly Valves (Manual)

Specification | Wafer type dual plate check valves. Check valves shall conform to PN-10 rating as per the specifications. Check valves shall be complete with companion flanges, nuts, bolts, gaskets etc. as required. | Wafer type dual plate check valves. Check valves shall conform to PN-10 rating as per the specifications. Check valves shall be complete with companion flanges, nuts, bolts, gaskets etc. as required. | Wafer type dual plate check valves. Check valves shall conform to PN-10 rating as per the specifications. Check valves shall be complete with companion flanges, nuts, bolts, gaskets etc. as required. | Wafer type dual plate check valves. Check valves shall conform to PN-10 rating as per the specifications. Check valves shall be complete with companion flanges, nuts, bolts, gaskets etc. as required. | Wafer type dual plate check valves. Check valves shall conform to PN-10 rating as per the specifications. Check valves shall be complete with companion flanges, nuts, bolts, gaskets etc. as required.

Make | Kitz | Kitz | |

Size & Quantity | 100 mm -06 Nos | 100 mm -06 Nos | |

Non-Return Valve | Non-Return Valve | Non-Return Valve | Non-Return Valve | Non-Return Valve

Specification | Wafer type dual plate check valves. Check valves shall conform to PN-10 rating as per the specifications. Check valves shall be complete with companion flanges, nuts, bolts, gaskets etc. as required. | Wafer type dual plate check valves. Check valves shall conform to PN-10 rating as per the specifications. Check valves shall be complete with companion flanges, nuts, bolts, gaskets etc. as required. | Wafer type dual plate check valves. Check valves shall conform to PN-10 rating as per the specifications. Check valves shall be complete with companion flanges, nuts, bolts, gaskets etc. as required. | Wafer type dual plate check valves. Check valves shall conform to PN-10 rating as per the specifications. Check valves shall be complete with companion flanges, nuts, bolts, gaskets etc. as required. | Wafer type dual plate check valves. Check valves shall conform to PN-10 rating as per the specifications. Check valves shall be complete with companion flanges, nuts, bolts, gaskets etc. as required.

Make | Castle /Normex | Castle /Normex | |

Size & Quantity | 100 mm-02 Nos | 100 mm-02 Nos | |

Y-strainer | Y-strainer | Y-strainer | Y-strainer | Y-strainer

Specification | Y-strainers as per specifications (PN - 10). Y strainers shall be complete with companion flanges, nuts, bolts, gaskets etc. as required. | Y-strainers as per specifications (PN - 10). Y strainers shall be complete with companion flanges, nuts, bolts, gaskets etc. as required. | Y-strainers as per specifications (PN - 10). Y strainers shall be complete with companion flanges, nuts, bolts, gaskets etc. as required. | Y-strainers as per specifications (PN - 10). Y strainers shall be complete with companion flanges, nuts, bolts, gaskets etc. as required. | Y-strainers as per specifications (PN - 10). Y strainers shall be complete with companion flanges, nuts, bolts, gaskets etc. as required.

Make | Sant | Sant | |

Size & Quantity | 100 mm-03 Nos | 100 mm-03 Nos | |

Manual Balancing Valves | Manual Balancing Valves | Manual Balancing Valves | Manual Balancing Valves | Manual Balancing Valves

Specification | Specification | Manual balancing valves complete with companion flanges, nuts, bolts, gaskets etc. as required. Balancing valves shall conform to PN-10 rating as per the specifications. | Manual balancing valves complete with companion flanges, nuts, bolts, gaskets etc. as required. Balancing valves shall conform to PN-10 rating as per the specifications. | Manual balancing valves complete with companion flanges, nuts, bolts, gaskets etc. as required. Balancing valves shall conform to PN-10 rating as per the specifications.

Make | Make | Castle | |

Size & Quantity | Size & Quantity | 100 mm-01 Nos | |

Expansion Bellows | Expansion Bellows | Expansion Bellows | Expansion Bellows | Expansion Bellows

Specification | Specification | Neoprene single arch rubber expansion bellows with integral reinforced rubber flanges, guide rods on both sides at inlet and outlet of water chilling machines and circulation water pumps, of approved make. (Rating PN-10) | Neoprene single arch rubber expansion bellows with integral reinforced rubber flanges, guide rods on both sides at inlet and outlet of water chilling machines and circulation water pumps, of approved make. (Rating PN-10) | Neoprene single arch rubber expansion bellows with integral reinforced rubber flanges, guide rods on both sides at inlet and outlet of water chilling machines and circulation water pumps, of approved make. (Rating PN-10)

Make | Make | Cori | |

Size & Quantity | Size & Quantity | 100 mm-06 Nos | |

Flow switches | Flow switches | Flow switches | Flow switches | Flow switches

Specification | Specification | Pedal type flow switches complete with control wiring between inlet and outlet of each chiller. These shall be BMS compatible. | Pedal type flow switches complete with control wiring between inlet and outlet of each chiller. These shall be BMS compatible. | Pedal type flow switches complete with control wiring between inlet and outlet of each chiller. These shall be BMS compatible.

Make | Make | Honeywell | |

Size & Quantity | Size & Quantity | 01 Nos | |

Thermometer | Thermometer | Thermometer | Thermometer

Specification | Alcohol/red mercury filled 'V' form thermometer of approved make conforming to tender specifications. (For Floor Mounted AHU, Chiller & Condenser) | Alcohol/red mercury filled 'V' form thermometer of approved make conforming to tender specifications. (For Floor Mounted AHU, Chiller & Condenser) | Alcohol/red mercury filled 'V' form thermometer of approved make conforming to tender specifications. (For Floor Mounted AHU, Chiller & Condenser)

Make | H-guru | |

Size & Quantity | 02 Nos | |

Pressure Gauges | Pressure Gauges | Pressure Gauges | Pressure Gauges

Specification | Industrial type pressure gauge of approved make of diameter 100 mm constructed out of SS 304 material's siphon shall be factory insulated with closed cell nitrile rubber in tubing form. Pressure gauges shall be fitted with a ball valve at the tube. (For Floor Mounted AHU, Chiller, Condenser & All Pumps) | Industrial type pressure gauge of approved make of diameter 100 mm constructed out of SS 304 material's siphon shall be factory insulated with closed cell nitrile rubber in tubing form. Pressure gauges shall be fitted with a ball valve at the tube. (For Floor Mounted AHU, Chiller, Condenser & All Pumps) | Industrial type pressure gauge of approved make of diameter 100 mm constructed out of SS 304 material's siphon shall be factory insulated with closed cell nitrile rubber in tubing form. Pressure gauges shall be fitted with a ball valve at the tube. (For Floor Mounted AHU, Chiller, Condenser & All Pumps)

Make | H-guru | |

Quantity | 06 Nos | |

Ball Valves | Ball Valves | Ball Valves | Ball Valves

Specification | Ball Valves as per the specifications. Ball valves shall be complete with companion flanges, nuts, bolts, gaskets etc. as required. | Ball Valves as per the specifications. Ball valves shall be complete with companion flanges, nuts, bolts, gaskets etc. as required. | Ball Valves as per the specifications. Ball valves shall be complete with companion flanges, nuts, bolts, gaskets etc. as required.

Make | Kitz | |

Size & Quantity | 25 mm Dia-08 Nos | |
Auto air Purge | Auto air Purge | Auto air Purge | Auto air Purge
Specification | Automatic purge valve with ball valve. | |
Make | Anergy | |
Size & Quantity | 25 MM dia -04 Nos | |
HVAC Panel | HVAC Panel | HVAC Panel | HVAC Panel
Specification | Incoming shall have provision for main electrical power panel: (Main Power Panel for HVAC System)
Supply, design, shop testing / inspection & commissioning of main HVAC power panel compartmental cubicle type freestanding with appropriate cable entries, front operating, front maintained wherever required complete with base frame etc. The electrical panels to be fabricated out of 14 / 16 SWG CRCA sheet, and surface treated with phosphating seven tank process and duly powder coated, Cu busbars, LED indicating lamps, Multi-Function Meter with built in Energy meter for Incomer etc.
Incoming to the MCC panel shall be given by client / electrical contractor.
Outgoing shall consist of Incomer MCCB/MCB with rotary operating mechanism, control transformer phase indication lamps, relays, ON/E. OFF/RESET push buttons, ON/OFF/TRIP Indication Lamps, auto/manual selector switch & potential free contacts for ON/OFF/TRIP status and interlocks.
HVAC panel shall have provision for
2 Chiller 1Working + 1 Stand by feeder
Chilled water Pump 1Working + 1 Stand | Incoming shall have provision for main electrical power panel: (Main Power Panel for HVAC System)
Supply, design, shop testing / inspection & commissioning of main HVAC power panel compartmental cubicle type freestanding with appropriate cable entries, front operating, front maintained wherever required complete with base frame etc. The electrical panels to be fabricated out of 14 / 16 SWG CRCA sheet, and surface treated with phosphating seven tank process and duly powder coated, Cu busbars, LED indicating lamps, Multi-Function Meter with built in Energy meter for Incomer etc.
Incoming to the MCC panel shall be given by client / electrical contractor.
Outgoing shall consist of Incomer MCCB/MCB with rotary operating mechanism, control transformer phase indication lamps, relays, ON/E. OFF/RESET push buttons, ON/OFF/TRIP Indication Lamps, auto/manual selector switch & potential free contacts for ON/OFF/TRIP status and interlocks.
HVAC panel shall have provision for
2 Chiller 1Working + 1 Stand by feeder
Chilled water Pump 1Working + 1 Stand | Incoming shall have provision for main electrical power panel: (Main Power Panel for HVAC System)
Supply, design, shop testing / inspection & commissioning of main HVAC power panel compartmental cubicle type freestanding with appropriate cable entries, front operating, front maintained wherever required complete with base frame etc. The electrical panels to be fabricated out of 14 / 16 SWG CRCA sheet, and surface treated with phosphating seven tank process and duly powder coated, Cu busbars, LED indicating lamps, Multi-Function Meter with built in Energy meter for Incomer etc.
Incoming to the MCC panel shall be given by client / electrical contractor.
Outgoing shall consist of Incomer MCCB/MCB with rotary operating mechanism, control transformer phase indication lamps, relays, ON/E. OFF/RESET push buttons, ON/OFF/TRIP Indication Lamps, auto/manual selector switch & potential free contacts for ON/OFF/TRIP status and interlocks.
HVAC panel shall have provision for
2 Chiller 1Working + 1 Stand by feeder
Chilled water Pump 1Working + 1 Stand
Quantity | 01 Nos | 01 Nos | 01 Nos
Power Cabling | Power Cabling | Power Cabling | Power Cabling
Specification | 1.1KV Grade XLPE/PVC insulated aluminum/copper multi stranded conductor steel of armored/flexible braid power cables as per the specification, in between HVAC equipment, controls, controllers, HVAC panel boards; including terminations. laying of cables in horizontal/vertical formation on cable trays, supported along RCC/steel structures if required, pipe racks and cable supports provided, include unloading, shifting up to site at various location/levels, unpacking & inspection. The clamping of cables by readymade spacers, saddles or clamps fabricated out of Aluminum strip 3 mm thick along vertical runs and wherever specified, cutting of the cable as per actual measurement, teasing and clamping of cable-on-cable tray Velcro tap of minimum 20mm width or clamping materials in Aluminum or PVC cable tie as per site hygiene level and hardware etc., | 1.1KV Grade XLPE/PVC insulated aluminum/copper multi stranded conductor steel of armored/flexible braid power cables as per the specification, in between HVAC equipment, controls, controllers, HVAC panel boards; including terminations. laying of cables in horizontal/vertical formation on cable trays, supported along RCC/steel structures if required, pipe racks and cable supports provided, include unloading, shifting up to site at various location/levels, unpacking & inspection. The clamping of cables by readymade spacers, saddles or clamps fabricated out of Aluminum strip 3 mm thick along vertical runs and wherever specified, cutting of the cable as per actual measurement, teasing and clamping of cable-on-cable tray Velcro tap of minimum 20mm width or clamping materials in Aluminum or PVC cable tie as per site hygiene level and hardware etc., | 1.1KV Grade XLPE/PVC insulated aluminum/copper multi stranded conductor steel of armored/flexible braid power cables as per the specification, in between HVAC equipment, controls, controllers, HVAC panel boards; including terminations. laying of cables in horizontal/vertical formation on cable trays, supported along RCC/steel structures if required, pipe racks and cable supports provided, include unloading, shifting up to site at various location/levels, unpacking & inspection. The clamping of cables by readymade spacers, saddles or clamps fabricated out of Aluminum strip 3 mm thick along vertical runs and wherever specified, cutting of the cable as per actual measurement, teasing and clamping of cable-on-cable tray Velcro tap of minimum 20mm width or clamping materials in Aluminum or PVC cable tie as per site hygiene level and hardware etc.,
Make | Polycab | RR Cable | |
Size | 4C x 2.5 mm² LT XLPE (2XWY) insulated Copper conductor PVC outer sheathed Armored cable. | |
Size | 4C x 4 mm² LT XLPE (2XWY) insulated Copper conductor PVC outer sheathed Armored cable. | |
Size | 3.5C x 185 mm² LT XLPE (2XWY) insulated Aluminium conductor PVC outer sheathed Armored cable. | |
Earthing: | Earthing: | Earthing: | Earthing:
Specification | 4.0 sq.mm earthing wire for HVAC equipment & controls (HVAC Vender Panel to HVAC Equipment) Main Earthing - Electrical Vendor Scope) | 4.0 sq.mm earthing wire for HVAC equipment & controls (HVAC Vender Panel to HVAC Equipment) Main Earthing - Electrical Vendor Scope) | 4.0 sq.mm earthing wire for HVAC equipment & controls (HVAC Vender Panel to HVAC Equipment) Main Earthing - Electrical Vendor Scope)
Make | Polycab | |
Cable Trays: | Cable Trays: | Cable Trays: | Cable Trays:
Specification | GI perforated cable trays in utility room for chiller, pumps, AHUs, ventilation units etc., for laying power & control cables between HVAC panel boards & equipment; including hardware, accessories like bends, tee's, supports, etc. | GI perforated cable trays in utility room for chiller, pumps, AHUs, ventilation units etc., for laying power & control cables between HVAC panel boards & equipment; including hardware, accessories like bends, tee's, supports, etc. | GI perforated cable trays in utility room for chiller, pumps, AHUs, ventilation units etc., for laying power & control cables between HVAC panel boards & equipment; including hardware, accessories like bends, tee's, supports, etc. | GI perforated cable trays in utility room for chiller, pumps, AHUs, ventilation units etc., for laying power & control cables

between HVAC panel boards & equipment; including hardware, accessories like bends, tee's, supports, etc.
MS support | MS support | |
Specification | MS support work required for Chilled water and condenser water piping or for any other work | |
Make | Jindal | |
Description | Specifications | Observations | Sign & Date
Electrical | Electrical | Electrical | Electrical
Phase | Three Phase | |
Voltage | 415 V | |
Frequency | 50 Hz | |
Sr. No | Item Name | Available (Yes/No)
1) | Chiller Test Certificate |
2) | Pump Test Certificate |
3) | Pipe Test Certificate |
4) | Y-Strainer Test Certificate |
5) | Balancing Valve Test Certificate |
6) | Expansion Bellows Test Certificate |
7) | Ball Valve Test Certificate |
8) | Thermometer Test Certificate |
9) | Pressure Gauges |
10) | Auto air Purge |
11) | Chilled water coil Test Certificate |
Department | Name | Signature | Date
Project: BioBridge Healthcare Solutions Pvt. Ltd., India | Project: BioBridge Healthcare Solutions Pvt. Ltd., India | Project: BioBridge Healthcare Solutions Pvt. Ltd., India | Project: BioBridge Healthcare Solutions Pvt. Ltd., India
Prepared by: | | |
Checked/Reviewed by: | | |
INFARMA – Indústria Farmacêutica, Lda | INFARMA – Indústria Farmacêutica, Lda | INFARMA – Indústria Farmacêutica, Lda | INFARMA – Indústria Farmacêutica, Lda
Reviewed by: | | |
Reviewed by: | | |
Approved by | | |

CONTENTS

1.0 OBJECTIVE:

To document the installation of the Cold Room Unit, ascertain that the installation is as per design Specifications lay down.

2.0 SCOPE:

This protocol is applicable to the Cold Room Unit installed in M/s. INFARMA- Indústria Farmacêutica, Lda

3.0 PERFORMING THE INSTALLATION QUALIFICATION:

The installation should conform to this checklist. Record all observations and deviations in the space provided in the checklist. In case of deviations prepare a deviation report along with justifications, which should be approved by the Engineering Department, Quality Assurance Manager and Validation In charge.

4.0 CHECKLIST FOR INSTALLATION OF COLD ROOM

EQUIPMENT DESIGN, PRINCIPLE AND REVIEW OF USER REQUIREMENTS

SPECIFICATION (URS).

TECHNICAL SPECIFICATIONS OF THE SUB-COMPONENTS/BOUGHT OUT ITEMS, AND

THEIR MAKE, MODEL & QUANTITY

DETAILS OF UTILITIES

5.0 List of Document/Certificates for the following components to be attached –

6.0 RESULTS:

7.0 List of Deviations: (If any)

8.0. Justification of Acceptance

7.0 CONCLUSION:

Based upon the observations recorded in the table given above, the installation of the cold room is found to be satisfactory / is found not to be satisfactory as per the design specifications. therefore, installation of the cold room is / is not qualified and the operation qualification can / cannot be performed.

APPROVAL FOR DESIGN QUALIFICATION

S/No | Table of Content | Page No.

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3.0 | Performing the Installation Qualification | 2

4.0 | Checklist for Installation | 3

5.0 | List of Documents | 17

6.0 | Results | 18

7.0 | Conclusion | 18

8.0 | Approvals | 19

Description | Specifications | Observations | Sign & Date

Equipment | Walk-In Cold Room (Capacity 90000L) | |

URS No | URS/MOZ/INF/010 | |

Functional Overview | The Cold Room is capable of providing 2-8°C temperature uniformly to all the material stacked on pallets and SS racks/ plastic tubs. | |

Design Overview | Cold room design Overview:

A pharmaceutical cold room is a temperature- controlled space that is used to store temperature-sensitive pharmaceutical products, such as medications, vaccines etc.

some of the common factors consider for design are:

Temperature range: The temperature range for a pharmaceutical cold room is between 0°C and 8°C,

Insulation: The insulation of a cold room is designed to ensure good thermal performance and energy efficiency, as well as prevent heat transfer and air leakage.

Refrigeration system: The refrigeration system of a cold room is designed to provide consistent and reliable cooling, as well as comply with the environmental and safety regulations. The refrigeration system is designed to meet the | |

| required temperature range and capacity and should be equipped with sensors and alarms to monitor and control the temperature.

Shelving and racking: The shelving and racking of a cold room is designed to maximize the storage space and allow easy access to the stored items. The shelving and racking should be made of stainless-steel materials that are durable, corrosion-resistant, and easy to clean.

Doors: The doors of a cold room are designed to be well-insulated and sealed to prevent temperature fluctuations and air infiltration. The doors are sturdy and easy to open and close.

Lighting: The lighting of a cold room is designed to provide adequate visibility and safety, as well as minimize heat generation and energy consumption. Energy-efficient LED lights are provided, as they have low heat output and high performance at low temperatures.

Flooring: Cold rooms are designed to provide epoxy flooring for easy cleaning and to prevent any microbial proliferation.

Bollards: cold rooms are protected by providing bollards in front of the cold rooms to prevent any physical damage to the cold room panels while using pellet truck/stackers | |

| Vinyl curtains: Cold rooms are provided with the vinyl curtains to prevent warm air infiltration while doors are open

Noise level: it is designed to provide noise level below 85 dB | |

Material of construction | Prefabricated Construction

Panel MOC- PPGI

Type of insulation- PUF

Density of panels-40 +/- 2Kg/m3

Thickness of insulation-60 mm

Type of panel- Free Standing

Wall lamination- Inside/Outside-PPGI

Ceiling lamination- Inside/Outside-PPGI | |

Specification (Machine) & Dimensions | Room Size: 7.50 (L) x 4.03(W) x 3.0(H) in meter Room temp: 2 To 8°C

Surrounding building temp: 37±1°C

Specification of PUF panels: - PUF Panels: -

60 mm Thick PUF Panel with cam lock 0.45 mm thick PPGI Sheet on both side of the surface Including flashing silicon sealant, aluminum Pop rivets and cam lock plug, with bubble wrapping | |

| Manual Sliding door with lock

Manual Sliding Door W 1500 mm x H 2100 mm single leaf heavy duty 60 mm thick. (Density 42kg/m3) with both side 0.6 mm PPGI Sheet heavy duty aluminum track aluminum door profile, SS 304 door opener & closure EPDM

Gasket cladding for Installation on 60 mm thick panels, SS 304 Integrated pad lock,

PUF Slab
60 mm Thick PUF slab with one side Trafelt

Lighting: LED type light

Working Condition: - 2-8 °C

Refrigeration System:
Should designed for Summer / Monsoon seasons
Indoor design 2-8°C

Refrigerant Used: R407C | |

| Control Logic: -

The evaporator fan is interconnected with the compressor. First the evaporator fan starts, if the HP/LP switch and thermostat are all found healthy then only compressor will start. This is to protect the compressor. In case any of these are not working, the compressor will not start.

The compressor is interconnected with the temperature controller. The sensor linked with temperature controller senses the return air temp in room, if the temp is below the set parameter the compressor will trip and vice – versa.

The thermostatic expansion valve is provided in between the condenser and near evaporator unit. The need of expansion valve is to regulate the flow of refrigerant as per load in room. The probe connected to valve is mounted just after the evaporator, it senses the refrigerant required as per the requirement the valve regulate.

For the safety of compressor, the High pressure and Low-pressure switch is provided in condenser unit. discharge pressure is high

the compressor will trip and if the suction | |

| pressure is low, the compressor will trip automatically. | |

Alarms | Alarm shall be provided

High Temperature Alarm / Low Temperature alarm.

HP/ LP Trip Alarm

Condenser Fan Fault alarms.

Under voltage and /High voltage display

Alarm communication audible and visual. | |

Data Logging | Cloud based data logging model

Printout can be taken via connecting external PC

Cold room unit resume after a power

failure without any data losses. | |

Electrical connections | 230 Volt AC / 1 Phase, 50 Hz | |

Drainage pipe | Through UPVC piping to external drain | |

Technical Specification of Room | Technical Specification of Room | Technical Specification of Room |

Technical Specification of Room

Specification | Specification | Observation

(Yes/No) | Sign & Date

Location | Raw Material & Finished Goods Storage Area | |

Internal Room Size in Meter | 7.50 (L) x 4.03(W) x 3.0(H) in meter | |

Cold room Volume | 90000 Ltr | |

Desired Room Temperature | 2 To 80C | |

Maximum temperature of external
environment (Surroundings of Building) | 36.7 0C | |

Pull Down Time | 18 Hours. | |

Product to be Stored | Vaccine Products | |

Prefabricated Construction | Prefabricated Construction | Prefabricated Construction | Prefabricated
Construction

Specification | Specification | |

Panel MOC | PPGI | |

Type of insulation | PUF | |

Density of panels | 40 +/- 2Kg/m3 | |

Thickness of insulation | 60 mm | |

Type of panel | Free Standing | |

Wall lamination | Inside/Outside-PPGI | |

Ceiling lamination | Inside/Outside-PPGI | |

Door | Door | Door | Door

Specification | Specification | Observation

(Yes/No) | Sign & Date

Type of door | Sliding type with lock | |

Lamination for door | PP | |

Size of door clear opening | 1500 (W) X 2100 (H) in inch | |

Type of mounting | Panel Mounted | |

No. of doors | 1 (Sliding Door) | |

Lighting | Lighting | |

Lighting | LED Type light | |

Power Supply | Power Supply | |

Incoming Power Supply | 230 Volt AC / 1 Phase, 50 Hz | |

Technical Specification of Refrigeration System | Technical Specification of Refrigeration System |

Technical Specification of Refrigeration System | Technical Specification of Refrigeration System | |

Particulars | Particulars | Units | Specification | Observation

(Yes/No) | Sign & date

Outdoor design condition | Outdoor design condition | Outdoor design condition | Outdoor design condition

| |

1.1 | Summer DBT /WBT | Deg. C | 36.7 / 24.0 | |

1.2 | Monsoon DBT/WBT | Deg. C | 28.61 / 26.0 | |

Indoor design condition | Indoor design condition | Deg. C | 2-8°C | |

Type of duty operation | Type of duty operation | HRS | 24 | |

Units cooling capacity | Units cooling capacity | BTU/Hr. | 27297 (1W+1S) | |

Location of installation | Location of installation | NA | Inside of shade | |

Pull down time | Pull down time | HRS | 18 | |

Product to be stored | Product to be stored | NA | Vaccine product | |

Product incoming temperature | Product incoming temperature | Deg C | 25 | |

Door Lock | Door Lock | NA | Yes | |

Technical data of Compressor | Technical data of Compressor | Technical data of Compressor | Technical
data of Compressor | Technical data of Compressor

Compressor | Unit | Specification | Observation

(Yes/No) | Sign & Table
 Make | NA | Emerson | |
 Model | NA | Hermetic compressor.
 CR53KQME TFD 522 | |
 No. of compressor/unit | Nos. | 2 X 2 unit (1 Working +1 Standby)
 In the case of normal operations spare
 unit will start and the other stop on 12- hour cycle basis | |
 Refrigerant | NA | R407c | |
 Evaporator temp | Deg C | - 20C | |
 Condenser temp | Deg C | 500C | |
 Total cooling capacity of unit at
 indoor/outdoor condition | BTU/Hr | 27297 X 2 | |
 Power consumption at
 indoor/outdoor condition | KW | 0.1 / 3.42 KW / unit | |
 Compressor motor voltage | VAC | 410 +/-10% | |
 Compressor frequency | Hz | 50 | |
 Compressor motor phase | NA | 3 | |
 HP cut out / cut in pressure | Psig | 300 / 350 | |
 LP cut out / cut in pressure | Psig | 20/ 35 | |
 No. of Compressors each unit | Nos. | 1 | |
 Technical Data of Condenser | Technical Data of Condenser | Technical Data
 of Condenser | Technical Data of Condenser
 Condenser | Unit | Specification | Observation
 (Yes/No) | Sign & Table
 Make | NA | ENZO | |
 Model / Type | NA | ENZR8004MT3ACDU / Air Cooled | |
 Tube material | NA | Copper | |
 Fin material / size | NA | Aluminium / 0.18 mm | |
 No. of condensers. | Nos. | 2 (1 Working + 1 Standby) | |
 Technical Data for Evaporator Coil | Technical Data for Evaporator Coil | Technical Data for Evaporator
 Coil | Observation
 (Yes/No) | Sign & Table
 Evaporator coil | Unit | Specification | |
 Make | NA | ENZO | |
 No. of evaporator per unit | Nos. | 2 (1 Working + 1 Standby) | |
 Rows deep | Nos. | 3R | |
 Tube material | Mm | Copper | |
 Fin Material/size | Mm | Aluminium / 0.18 mm | |
 No. Fins | Per Inch | 4.5 mm | |
 No. of Evaporators in the room | Nos. | 2 (1 Working + 1 Standby) | |
 Technical Data of Evaporator Fan | Technical Data of Evaporator Fan | Technical Data of Evaporator Fan |
 Technical Data of Evaporator Fan
 Evaporator fan | Specification | Observation
 (Yes/No) | Sign & Table
 Make | AXIAL | |
 Discharge type | FREE | |
 No. Fans /unit | 3 | |
 Technical Data for Condenser Fan | Technical Data for Condenser Fan | Technical Data for Condenser Fan |
 Technical Data for Condenser Fan
 Condenser fan | Specification | Observation
 (Yes/No) | Sign & Table
 Make | AXIAL | |
 Discharge type | FREE | |
 No. Fans & Qty | 2 | |
 Electrical Data | Electrical Data | Electrical Data | Electrical Data
 Particulars | Specification | Observation
 (Yes/No) | Sign & Table
 Compressor | 3 Phase, 410 VAC | |
 Compressor OLR | Internal | |
 Evaporator fan | 1 Phase, 230VAC | |
 MCB main | 32 A | |
 Phase Failure relay | In built in PCB | |
 Temperature Controller Details | Temperature Controller Details | Temperature Controller Details |
 Temperature Controller Details
 Make | DIXELL XR60CX-GB | |
 Supply | 1 Phase, 230 VAC | |
 No of Sensor | 01 | |
 Phase Failure relay | In built in PCB | |
 Technical Data of Temperature Controller and Controls | Technical Data of Temperature Controller and
 Controls | Technical Data of Temperature Controller and Controls | Technical Data of Temperature
 Controller and Controls
 HP / LP Switch | HP / LP Switch | Observation
 (Yes/No) | Sign & Table
 HP | Electro mechanical | |
 LP | Adjustable | |
 Accuracy | 1-5 Psig | |
 Digital Temperature Indicator and Controller | Digital Temperature Indicator and Controller | Observation
 (Yes/No) | Sign & Table
 Quantity | 01 | |
 Purpose | For controlling cooling circuit | |
 Type | Microprocessor based | |
 Accuracy (ambient temp. 25°C) | ±0.7 °C ±1 digit | |
 Temperature Range | 99 to -55 0C | |
 Sensor | PTC | |
 Resolution | 1 0C | |
 Temperature Indicator | Temperature Indicator | Observation
 (Yes/No) | Sign & Table
 Quantity | 01 | 01 | |
 Purpose | Local Display of cold room inside temperature | Local Display of cold room inside temperature |
 |
 Range | 99 to -55 0C | 99 to -55 0C | |
 Resolution | 1 0C | 1 0C | |

Temperature sensor application | Temperature sensor application | Temperature sensor application | |
Type | Qty | Purpose | Observation
(Yes/No) | Sign & Table
PTC | 01 | Controlling temperature for cooling circuit | |
Safety Features | Safety Features | Safety Features | Safety Features | Safety Features
Description | Details | Details | |
Door lock | Yes | Yes | |
Access Control System: | Access Control System: | Access Control System:
Specification | Observation
(Yes/No) | Sign & Table
The access control will be through manual pad lock system | |
Cold rooms not designed for 21 CFR compliance. However, it can be upgraded for same in the future, if
required. | |
Temperature monitoring System & DATA archival: | Temperature monitoring System & DATA archival: |
Temperature monitoring System & DATA archival:
Specification | Observation
(Yes/No) | Sign & Table
System designed for Cloud based data logging model | |
Printout can be taken via connecting external PC | |
Designed to resume after a power failure without any data losses | |
Additional Features | Additional Features | Additional Features
Specification | Observation
(Yes/No) | Sign & Table
System designed for automatic change over during normal operation. Start and stop of spare unit shall be
designed for every 12-hour cycle basis. | |
Cold rooms shall be constructed using modular clean room panel with PUF insulation. Standard cleaning
regimen using disinfectants can be followed. | |
Epoxy flooring for cold rooms. | |
Smoke detection sensors shall be installed adjacent to the cold rooms. | |
Cold room designed to prevent the possibility of condensed moisture dripping on stored material. | |
Vinyl curtains at door entrance. | |
Designed not to exceed noise level 85db | |
Emergency button provision inside the cold room chamber. | |
Cold room door gasket designed to ease and facilitate maintenance. Also designed for accessibility for
service. | |
Designed for 2-8°C with circulation fan | |
Description | Specifications | Observation
(Yes/No) | Sign & Table
Electrical | Electrical | Electrical | Electrical
Phase | Three Phase | |
Voltage | 415 V | |
Frequency | 50 Hz | |
MATERIAL OF CONSTRUCTION | MATERIAL OF CONSTRUCTION | MATERIAL OF CONSTRUCTION
Specifications | Observation | Sign & Table
The Cold Room panels are fabricated with PUF PPGI sheet to meet GMP requirements. | |
Sr. No | Item Name | Available (Yes/No)
1 | Cold Room Layout |
2 | Indoor Unit GA drawing |
3 | Outdoor Unit GA drawing |
4 | Panel box drawing |
5 | Electrical SLD Drawing |
6 | Compressor Specification |
7 | Fan Specification |
8 | Digital Controller |
Department | Name | Signature | Date
Project: BioBridge Healthcare Solutions Pvt. Ltd., India | Project: BioBridge Healthcare Solutions Pvt.
Ltd., India | Project: BioBridge Healthcare Solutions Pvt. Ltd., India | Project: BioBridge Healthcare
Solutions Pvt. Ltd., India
Prepared by: | | |
Checked/Reviewed by: | | |
INFARMA – Indústria Farmacêutica, Lda | INFARMA – Indústria Farmacêutica, Lda | INFARMA – Indústria
Farmacêutica, Lda | INFARMA – Indústria Farmacêutica, Lda
Reviewed by: | | |
Reviewed by: | | |
Approved by | | |

CONTENTS

OBJECTIVE:

To document the installation of the De dusting tunnel Unit, ascertain that the installation is as per design Specifications lay down.

SCOPE:

This protocol is applicable to the De dusting tunnel Unit installed in M/s. INFARMA- Indústria Farmacêutica, Lda

PERFORMING THE INSTALLATION QUALIFICATION:

The installation should conform to this checklist. Record all observations and deviations in the space provided in the checklist. In case of deviations prepare a deviation report along with justifications, which should be approved by the Engineering Department, Quality Assurance Manager and Validation In charge.

4.0 CHECKLIST FOR INSTALLATION OF DEDUSTING TUNNEL

EQUIPMENT DESIGN, PRINCIPLE AND REVIEW OF USER REQUIREMENTS SPECIFICATION (URS).

TECHNICAL SPECIFICATIONS OF THE SUB-COMPONENTS/BOUGHT OUT ITEMS, AND THEIR MAKE, MODEL & QUANTITY

Filters in the De dusting tunnel

The series of filters in the de dusting tunnel are provided to ensure removal of contamination in the inlet & outlet air. The detailed specifications for the filter series in the de dusting unit are described below.

Combination Filters

Gaskets and sealing materials.

The purpose of the gaskets and the other sealing materials is to provide perfect sealing to the removable/ dismountable parts, such as Pre Filter & Blower.

Instrumentation

Detailed specifications for different instruments for safety and process control are described below.

Blower Motor

Electronic Controller with Operating Switch

MCB

Contactors

Tube Light

Differential Pressure Gauges

Emergency Switch

Air Nozzle

Manual Conveyor Bed

Outlet air velocity

Noise level

Light Intensity

Guidance: These specifications will become acceptance criteria for the subsequent qualification/ validation activities, especially during the installation qualification.

DETAILS OF UTILITIES

IDENTIFICATION OF COMPONENTS FOR CALIBRATION

MATERIAL OF CONSTRUCTION

The following parts of the system should be provided with the suggested MOC as provided below to ensure compatibility with the product / material coming in contact.

SAFETY FEATURES

5.0 Test Certificates/documents for the following components to be attached -

6.0 RESULTS:

7. LIST OF DEVIATION

8. JUSTIFICATION FOR ACCEPTANCES:

9.0 CONCLUSION –

Based upon the observations recorded in the table given above, the installation of the de-dusting tunnel is found to be satisfactory / is found not to be satisfactory as per the design specifications.

therefore, installation of the de-dusting tunnel is / is not qualified and the operation qualification

can / cannot be performed

APPROVAL FOR INSTALLATION QUALIFICATION

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3.0 | Performing the Installation Qualification | 2

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5.0 | List of Test Certificate | 12

6.0 | Results | 13

7.0 | Conclusion | 13

8.0 | Approvals | 14

Description | Specifications | Observations | Sign & Date

Equipment | De-Dusting Tunnel | |

URS No | URS/MOZ/INF/017 | |

Functional Overview | De dusting tunnel is a specialized enclosed antechambers which are incorporated as entryways for clean-room and other controlled environments to reduce particulate contamination on the material.

De dusting tunnel utilize high-pressure, filtered air to remove dust, fibrous lint and other contaminants from object surfaces. The forceful "cleansing" of surfaces prior to entering clean environments reduces the number of airborne particulates introduced in the clean room. | |

Design Overview | De dusting tunnel is fabricated from clean-room-compatible steel materials and feature electronically powered blowers, filters and high-pressure jet nozzles. Air, with a velocity of 5,000 to 6,000 feet-per-minute (FPM), continuously streams from the jet nozzles for 15-45 seconds or set time, effectively removing loose particulate matter from the surface of material entering clean rooms. | |

Material of construction | Body of the de-dusting tunnel is made of mild steel structure clad with 18G, S.S. 304 sheet.

Blower casing- MS Powder Coating | |

Specification (Machine) & Dimensions | De dusting Tunnel dimensions

W=2200, D=,3500 H= 2050, mm.

In supply air filter EU-4 grade

Magnahelic gauge for pressure monitoring

Supply motor blower.

Exhaust motor blower.

Dust collection tray SS 304

Exhaust Duct

SS perforation grill

PVC curtains on the front and at the back side of the tunnel.

Dedusting tunnel is provided with the emergency power shut down switch front and back end. | |

Operation of Equipment | Once the material is received in the ware house receiving bay the material is unloaded after physical cleaning the material is loaded on to the conveyor in the dedusting tunnel manually and tunnel is switch ON. The motor blower and conveyor will start automatically and material will start moving on the conveyor, with the help of prefiltered air blower the surface of the material is blown with air coming through the prefilters.

The dust and the particulate matter shall be collected in the tray or may be exhausted out through the exhaust duct having prefilters. The dedusted material is received in the material receiving area of the ware house | |

De dusting tunnel | De dusting tunnel | Observation(Yes/No) | Sign & Date

overall size | 2200 (W) x 3500 (D) x 2050 (H) mm. | |

working size | 1500 (W) x 1500 (D) x 1350 (H) mm. | |

Details | Specifications | Observations

(Yes/No) | Sign & Date

Make | Bio Bridge | |

Rating | EU 4 | |

Type | Box | |

Size | 550 x 350 x 150 mm | |

Qty | 2 Nos Each | |

Efficiency | 90% down to 10 μ | |

Construction | Anodized Aluminium frame | |

Media | Non-woven synthetic media. | |

Details | Specifications | Observations

(Yes/No) | Sign & Date

Make | Perfect Rubber Industries | |

Material | Neoprene Rubber / Food Grade | |

Details | Specifications | Observations

(Yes/No) | Sign & Date

Make | Eyyani Electric Machines Pvt Ltd | |

Type | EDB-220-125-T2 | |

Article No | 2990 | |

Input | 440V AC, 50 HZ | |

Qty | 2 No. | |

Details | Specifications | Observations

(Yes/No) | Sign & Date

Make | Singh's Automation | |

Power Input | 230V | |

Qty | 1 No. | |

Details | Specifications | Observations

(Yes/No) | Sign & Date

Make | Siemens | |

Model | 5SL6210-7, 4 P, 10A | |

Current Capacity | 25 A | |

Qty | 1 No. | |

Details | Specifications | Observations

(Yes/No) | Sign & Date

Make | Siemens | |

Current Capacity | 18 A | |

Qty | 1 No. | |

Details | Specifications | Observations

(Yes/No) | Sign & Date

Make | Philips | |

Type | LED Panel | |

Capacity | 22 W | |

Qty | 1 No. | |

Details | Specifications | Observations

(Yes/No) | Sign & Date

Make | Omicron Sensing Pvt Ltd | |

Model No | DPGT4010 | |

Power Input | 230 V | |

Range | 0-50 mm WC | |

Location | Across Primary Filters | |

Qty | 1 No. | |

Details | Specifications | Observations

(Yes/No) | Sign & Date

Make | TEKNIC | |

Type | 2AML4 | |

Dial Dia | 40 mm | |

Dial Type | Push – Twist release | |

Qty | 1 No. | |

Details | Specifications | Observations

(Yes/No) | Sign & Date

Make | Bio Bridge | |

Dial Dia | 40 mm | |

M.O.C | S.S. 304 | |

Qty | 23 Nos. | |

Details | Specifications | Observations

(Yes/No) | Sign & Date

Make | Bio Bridge | |

Type | Roller | |

Size | 1340 x 3000 mm | |

M.O.C | S.S. 304 | |

Qty | 1 No. | |

Specifications | Observations

(Yes/No) | Sign & Date

The outlet velocity of air should be NLT 5000 FPM. | |

Specifications | Observations

(Yes/No) | Sign & Date

The noise level of the De dusting tunnel should be less than 90dB at 1 meter from machine. | |

Specifications | Observations(Yes/No) | Sign & Date

The level of light intensity in the working area should be more than 400 lux. | |

Description | Specifications | Observations

(Yes/No) | Sign & Date
Electrical | Electrical | Electrical |
Phase | Three Phase | |
Voltage | 440V | |
Frequency | 50Hz | |
S/No. | Component | Observations
(Yes/No) | Sign & Date
1 | Pressure Gauge | |
Part | MOC | Observations
(Yes/No) | Sign & Date
Blower Casing | M.S. Powder Coating | |
Main Cabinet | 18G, S.S. 304 | |
Impeller | Aluminium | |
Working Area | 18G, S.S. 304 | |
Filter housing | 18G, S.S. 304 | |
Gasket | Neoprene Rubber / Food Grade | |
S/No. | Parameter | Specifications | Observations
(Yes/No) | Sign & Date
1. | Safety aspect | All safety feature and interlock should be incorporated with proper information sign board. | |
2 | Noise level | Blower noise level should be less than 90 db. | |
3 | Emergency Switch | Dedusting tunnel provided with the emergency power shut down switch front and back end | |
Sr. No | Item Name | Available (Yes / No)
1) | De-dusting tunnel test certificate |
Department | Name | Signature | Date
Project: BioBridge Healthcare Solutions Pvt. Ltd., India | Project: BioBridge Healthcare Solutions Pvt. Ltd., India | Project: BioBridge Healthcare Solutions Pvt. Ltd., India | Project: BioBridge Healthcare Solutions Pvt. Ltd., India
Prepared by: | | |
Checked/Reviewed by: | | |
INFARMA – Indústria Farmacêutica, Lda | INFARMA – Indústria Farmacêutica, Lda | INFARMA – Indústria Farmacêutica, Lda | INFARMA – Indústria Farmacêutica, Lda
Reviewed by: | | |
Reviewed by: | | |
Approved by | | |

CONTENTS

OBJECTIVE:

To document the installation of the Dynamic Pass Box Unit, ascertain that the installation is as per design Specifications lay down.

SCOPE:

This protocol is applicable to the Dynamic Pass Box Unit installed in M/s. INFARMA- Indústria Farmacêutica, Lda

PERFORMING THE INSTALLATION QUALIFICATION:

The installation should conform to this checklist. Record all observations and deviations in the space provided in the checklist. In case of deviations prepare a deviation report along with justifications, which should be approved by the Engineering Department, Quality Assurance Manager and Validation In charge.

4.0 CHECKLIST FOR INSTALLATION OF PASS BOX

EQUIPMENT DESIGN, PRINCIPLE AND REVIEW OF USER REQUIREMENTS SPECIFICATION (URS).

Dynamic Pass Box: -

Material of construction

The following parts of the system should be provided with the suggested MOC as provided below to ensure compatibility with the product / material coming in contact.

Filters in the dynamic pass box unit

The series of filters in the dynamic pass box unit are provided to ensure removal of contamination in the inlet air coming into the contact of product. The detailed specifications for the filter series in the dynamic pass box unit system are described below.

Return Air Pre-filters:-

Fresh Air Pre-filters

Absolute Filters (HEPA)

Gaskets and sealing materials

The purpose of the gaskets and the other sealing materials is to provide perfect sealing to the removable/ dismountable parts, such as Pre Filter, HEPA Filter & Blower.

Instrumentation

Detailed specifications for different instruments for safety and process control are described below.

Blower

Tube light

Pass Box Controller

Electromagnetic Lock

Door Release Button with Status Indication

Magnehelic gauges (Across HEPA Filter)

DOP / PAO Port

MCB

Outlet air velocity

Noise level

Light Intensity

Alarms & Interlocks:

DETAILS OF UTILITIES

IDENTIFICATION OF COMPONENTS FOR CALIBRATION

MATERIAL OF CONSTRUCTION

The following parts of the system shall be provided with the suggested MOC as provided below to ensure compatibility with the product / material coming in contact.

SAFETY FEATURES

5.0 Test Certificates/documents for the following components to be attached -

6.0 RESULTS:

7. Deviations (If any)

8. Acceptance of deviation

9.0 CONCLUSION –

Based upon the observations recorded in the table given above, the installation of the ahu is found to be satisfactory / is found not to be satisfactory as per the design specifications. therefore, installation of the ahu is / is not qualified and the operation qualification can / cannot be performed

APPROVAL FOR INSTALLATION QUALIFICATION

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Description | Specifications | Observations | Sign & Date

Equipment | Dynamic Pass Box | |

URS No | URS/MOZ/INF/011 | |

Functional Overview | Dynamic pass box is normally used to transfer materials and goods between cleanrooms of different class or between cleanroom and non-clean room. The primary function of the DPB is to maintain the classification and integrity of the clean environment during operations. Besides UV light and an interlock system, dynamic pass box is equipped with a suction HEPA filter of around 0.3microns. |

Design Overview | The dynamic Pass box is provided with electromagnetic interlocking arrangement for doors. The electromagnetic locking device is maintained with help of two electric magnet fixed on each door. As soon as one door is opened from one side, the opposite side's magnet lock remain activated & door gets lock into pass box and white light will get "ON". And when both doors are in locked condition white light will get "OFF" | |

Material of construction | All parts designed and manufactured in line with current standards. The body of Equipment is made of S.S. 304 | |

Specification (Machine) & Dimensions | DPB dimensions

W=1120, D=1120, H=1585 mm.

Filters

Fresh air pre-filters EU 7 is 3µ

Return air filters EU 4 is 10 µ

HEPA filters EU 13 is 0.3 µ

Blower is centrifugal Type

Locking system is electromagnetic type

Magnehelic gauge is installed to monitor the differential pressure across HEPA filters.

DOP/PAO port is provided
LED tube light is provided.
Door release button with status indication
Switch for ON/OFF is MCB type.
Noise level is < 75dB at the distance of 1m from DPB ||
Alarms | In case of any Mechanical /Technical/Operational errors there is a provision of alarm system. |
|
Operation of Equipment | After cleaning the DPB internally with 70% IPA switch on the MCB, motor blower will start automatically. Keep it ON for some time till the differential pressure across the HEPA filters is maintained. Place the material to be transferred by opening the door from one side, the door of other side will not open due to the interlocking system. Once the material is placed allow the material to stand in the DPB for the time as per validation studies. Unload the material from other side by opening the door. ||
Details | Details | Observations
(Yes/No) | Sign & Date
overall size | 1120(W) x 1120 (D) x 1585(H) MM. ||
working size | 900(W) x 900(D) x 900 (H) mm. ||
Part | MOC | Observations
(Yes/No) | Sign & Date
Blower Casing | 1.20 mm S.S. 304 ||
Top Chamber | 1.20 mm S.S. 304 ||
Impeller | Aluminium ||
Working Area | 1.20 mm S.S. 304 ||
Filter Housing | 1.20 mm S.S. 304 ||
Gasket | Neoprene Rubber / Food Grade ||
Return Air Grill | S.S. 304 ||
Details | Specifications | Observations
(Yes/No) | Sign & Date
Make | Bio Bridge ||
Type | Box ||
Size | 560 x 225 x 50 MM ||
Rating | EU-4 ||
Efficiency | 90% down to 10 µ. ||
Initial Pressure Differential | 4 mm WC ||
Final Pressure Differential | 12 mm WC ||
Qty | 1 No. ||
Construction | Anodized Aluminium frame ends sealed with special adhesive. ||
Filtration Media | Non-Woven synthetic ||
Details | Specifications | Observations
(Yes/No) | Sign & Date
Make | Bio Bridge ||
Type | Box ||
Size | 200 x 200 x 50 mm ||
Rating | EU-7 ||
Efficiency | 95% down to 3 µ. ||
Initial Pressure Differential | 8 mm WC ||
Final Pressure Differential | 24 mm WC ||
Qty | 1 No. ||
Construction | Anodized aluminium frame ends sealed with special adhesive. ||
Filtration Media | Non-Woven synthetic ||
Details | Specifications | Observations
(Yes/No) | Sign & Date
Make | Bio Bridge ||
Rating | EU-13 ||
Type | Box ||
Size | 610 x 610 x 69 mm ||
Qty | 1 No. ||
Efficiency | 99.997% down to 0.3 µ. ||
Initial Pressure Differential | 15 mm WC ||
Final Pressure Differential | 45 mm WC ||
Construction | Anodized aluminium frame with Mini pleat pack and ends sealed with special adhesive. ||
Media | Submicron glass Fiber. ||
Details | Specifications | Observations
(Yes/No) | Sign & Date
Material | Neoprene Rubber / Food Grade ||
Details | Specifications | Observations
(Yes/No) | Sign & Date
Make | Fanstech ||
Type | Centrifugal ||
Model | FH355A ||
Air Flow | 2450 CMH ||
RPM | 1400 (Approx) ||
Motor Capacity | 180 W ||
Power Input | 230V, 50Hz ||
Qty | 1 No. ||
Details | Specifications | Observations
(Yes/No) | Sign & Date
Make | Philips ||
Type | LED ||
Capacity | 20 Watt ||
Power Input | 230V ||
Qty | 1 No. ||
Details | Specifications | Observations
(Yes/No) | Sign & Date
Make | StyraX Instruments (I) Pvt Ltd ||
Model | SPB-1052D ||
Power Requirement | 230 V / 1 Ph ||
Qty | 1 No. ||
Details | Specifications | Observations
(Yes/No) | Sign & Date
Make | StyraX Instruments (I) Pvt Ltd ||
Capacity | 300 lbs ||

Power Requirement | 24V / 250 mA ||
 Qty | 3 Nos. ||
 Details | Specifications | Observations
 (Yes/No) | Sign & Date
 Make | Styrax Instruments (I) Pvt Ltd ||
 Power Requirement | 24V / 250 mA ||
 Type | Touch/Push ||
 Qty | 2 Nos. ||
 Details | Specifications | Observations
 (Yes/No) | Sign & Date
 Make | Dwyer ||
 Model | 2000-50mm ||
 Range | 0 – 50 mm WC ||
 Location | Across HEPA filter ||
 Working Range | 8 – 24 mm WC ||
 Qty | 1 No. ||
 Details | Specifications | Observations
 (Yes/No) | Sign & Date
 Make | Bio Bridge ||
 Size- | 10 mm ||
 MOC | SS 304 ||
 Qty | No ||
 Details | Specifications | Observations
 (Yes/No) | Sign & Date
 Make | Siemens ||
 Pole | 2 Pole ||
 Amp | 6A ||
 Qty | 1 No. ||
 Specifications | Observations
 (Yes/No) | Sign & Date
 The velocity of air at 150 mm from the filter should be 90 FPM $\pm$ 20% to get laminar flow. ||
 Specifications | Observations
 (Yes/No) | Sign & Date
 The noise level of the dynamic pass box is less than 75dB at 1 meter from DPB. ||
 Specifications | Observations(Yes/No) | Sign & Date
 The level of light intensity in the working area should be more than 300 lux. ||
 Specifications | Observations (Yes/No) | Sign & Date
 Buzzer audio & visual alarm provided for any Mechanical /Technical/Operational issue.
 Only one door shall open at a given time when ON. When door is opened, white light gets ON.
 Guidance: These specifications will become acceptance criteria for the subsequent qualification/
 validation activities, especially during the installation qualification. ||
 Description | Specifications | Observations
 (Yes/No) | Sign & Date
 Electrical | Electrical | Electrical |
 Phase | Single Phase ||
 Voltage | 230V ||
 Frequency | 50 Hz ||
 S/No. | Component | Observations(Yes/No) | Sign & Date
 1 | Magnehelic Gauge ||
 Part | MOC | Observations | Sign & Date
 Blower Casing | 1.20 mm S.S. 304 ||
 Top Chamber | 1.20 mm S.S. 304 ||
 Impeller | Aluminium ||
 Working Area | 1.20 mm S.S. 304 ||
 Filter housing | 1.20 mm S.S. 304 ||
 Gasket | Neoprene Rubber / Food Grade ||
 Return air grill | S.S. 304 ||
 Specifications | Observations
 (Yes/No) | Sign & Date
 All doors are interlock ||
 Sr. No | Item Name | Available (Yes/No)
 1) | Pass box Test Certificate |
 Department | Name | Signature | Date
 Project: BioBridge Healthcare Solutions Pvt. Ltd., India | Project: BioBridge Healthcare Solutions Pvt.
 Ltd., India | Project: BioBridge Healthcare Solutions Pvt. Ltd., India | Project: BioBridge Healthcare
 Solutions Pvt. Ltd., India
 Prepared by: |||
 Checked/Reviewed by: |||
 INFARMA – Indústria Farmacêutica, Lda | INFARMA – Indústria Farmacêutica, Lda | INFARMA – Indústria
 Farmacêutica, Lda | INFARMA – Indústria Farmacêutica, Lda
 Reviewed by: |||
 Reviewed by: |||
 Approved by |||

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Performance Qualification Procedure

The Standard Operating Procedure (SOP.) shall be prepared based on instruction given in the operating manual of machine.

Prior to the qualification test, the personnel shall be trained by the Engineer from the supplier (if reqd.) on the performance feature of equipment. This training shall be recorded in section 7.1.

The trained person shall carry out the performance Qualification following the procedure mentioned in 7.2 for Operation & Record the observation of qualification test in test data sheet of 7.2.1 and performance check as in 7.3.

Operate the machine as per the draft SOP. Record the change if any and confirm the SOP. Report the confirmation of SOP in section 7.4.

Report the deficiency from the specified function. If any in the section 7.5.

Training

Name of Trainer: _____

Signature of Trainer: _____

Date: _____

Operation and testing

Purpose:

The purpose of this procedure is to check the overall performance & function as specified in the operating manual of machine.

Testing & Programming:

For the parameters specified, In the Operational Qualification, the Performance of the equipment has to be checked for certain amount of operations continuously. e.g.

i) Consistency of the output speed.

ii) Quality of labelling & Printing and consistency of the same.

* Note: Program Data may be changed, it depends on Label Length & Bottle / Product Shape / Size / Dia.

Test DATA Sheets

PERFORMANCE CHECK

SOP VERIFICATION

Draft SOP No.: _____ Effective Date: _____

Title: _____

Operate the machine as per the draft SOP and record the details given below:

Operated By: _____

Checked By: _____

The operating person understands and follows the SOP description (Yes/No) _____

Change required in draft SOP (If any):

SOP to be revised (Yes/No): _____

If yes, Review No. _____

Remarks: SOP Confirmed / Not Confirmed _____

Verified by:

Name: _____, Sign: _____, Date: _____

DEFICIENCY (If Any) & CORRECTIVE ACTION REPORT

Conducted by: _____ Date: _____

Approved by: _____ Date: _____

Verified by:

Name: _____, Sign: _____, Date: _____

Deviations (if any)

Summary :-

All the above mention points are checked for correct technical data as per Purchase Order & found to be satisfactory.

All the drawing in operating manual have been checked and approved. The documentation required is checked and found correct.

The Machine / Equipment as built confirm to the design specification, operate as per the operating instruction manual & draft SOP. All over Machine operations and performance found satisfactory; hence, the Performance Qualification is complete.

Any other Comments (if any)

Conclusion :-

Abbreviations: -

List of Documents:-

APPROVAL FOR PERFORMANCE QUALIFICATION

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13.0 | Approvals for Operational Qualification | 14

1.0 | Objective | Objective | Objective | Objective | Objective

| The objective of this protocol is to carry out the Performance qualification of the Automatic Sticker

Labelling Machine, Model: VSC/VLC-150 (Servo with Automation) + 36" Dia. Infeed & Outfeed Turn | The

objective of this protocol is to carry out the Performance qualification of the Automatic Sticker

Labelling Machine, Model: VSC/VLC-150 (Servo with Automation) + 36" Dia. Infeed & Outfeed Turn | The

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objective of this protocol is to carry out the Performance qualification of the Automatic Sticker

Labelling Machine, Model: VSC/VLC-150 (Servo with Automation) + 36" Dia. Infeed & Outfeed Turn

2.0 | Scope | Scope | Scope | Scope | Scope

| This Protocol is applicable for Performance Qualification of the equipment. | This Protocol is

applicable for Performance Qualification of the equipment. | This Protocol is applicable for Performance Qualification of the equipment. | This Protocol is applicable for Performance Qualification of the equipment. | This Protocol is applicable for Performance Qualification of the equipment.

3.0 | Purpose | Purpose | Purpose | Purpose | Purpose

| The purpose of this protocol is to validate the above Specified Automatic Sticker Labelling Machine, Model: VSC/VLC-150 (Servo with Automation) + 36" Dia. Infeed & Outfeed Turn Table for the operational qualifications to:

Assure that the Automatic Sticker Labelling Machine performance is adequate to support the working for which the system is intended.

Establish the suitability of the draft SOP prepared for the operation of Automatic Sticker Labelling Machine.

Document the initial performance of the Automatic Sticker Labelling Machine for further reference.

Demonstrate the suitability of the SOP established for the Automatic Sticker Labelling Machine | The purpose of this protocol is to validate the above Specified Automatic Sticker Labelling Machine, Model: VSC/VLC-150 (Servo with Automation) + 36" Dia. Infeed & Outfeed Turn Table for the operational qualifications to:

Assure that the Automatic Sticker Labelling Machine performance is adequate to support the working for which the system is intended.

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Assure that the Automatic Sticker Labelling Machine performance is adequate to support the working for which the system is intended.

Establish the suitability of the draft SOP prepared for the operation of Automatic Sticker Labelling Machine.

Document the initial performance of the Automatic Sticker Labelling Machine for further reference.

Demonstrate the suitability of the SOP established for the Automatic Sticker Labelling Machine

4.0 | Responsibility | Responsibility | Responsibility | Responsibility | Responsibility

| The Officer - Engineering & Production should be responsible for the checking the operations and recording data as per the procedure outlined in this protocol.

Executive - Validation should collect all the test data and should compile the result to make the reports of Operational Qualification & write SOP.

The Reports should be checked by Manager - Engineering & Plant.

The Manager-QA; should be finally approving the Validation Report. | The Officer - Engineering & Production should be responsible for the checking the operations and recording data as per the procedure outlined in this protocol.

Executive - Validation should collect all the test data and should compile the result to make the reports of Operational Qualification & write SOP.

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Executive - Validation should collect all the test data and should compile the result to make the reports of Operational Qualification & write SOP.

The Reports should be checked by Manager - Engineering & Plant.

The Manager-QA; should be finally approving the Validation Report.

5.0 | Description of System / Equipment | Description of System / Equipment | Description of System / Equipment | Description of System / Equipment | Description of System / Equipment

| Automatic Sticker Labelling Machine, Model: VSC/VLC-150 (Servo with Automation) + 36" Dia. Infeed & Outfeed Turn Table

To apply Sticker Labels up to 95 mm Label Height on vertical Small / large Round Bottle / Vials / Product of different diameter.

Machine is designed for the Standard Rated Speed up to 150 Bottles / Min., Depending upon Label Length & Bottle / Product Size / Dia.

36" Dia. Turn Table is provided at In-Feed of Automatic Sticker Labelling Machine. which will accumulate and feed the Round Bottle / Vial in vertical position to Automatic Sticker Labelling Machine to further process. Operator has to load the sufficient bottles / vials / products manually on turn table before start the machine.

Even, 36" Dia. Turn Table is provided at Outfeed of Machine which will collect / accumulate labelled bottle / vial / container. | Automatic Sticker Labelling Machine, Model: VSC/VLC-150 (Servo with Automation) + 36" Dia. Infeed & Outfeed Turn Table
To apply Sticker Labels up to 95 mm Label Height on vertical Small / large Round Bottle / Vials / Product of different diameter.
Machine is designed for the Standard Rated Speed up to 150 Bottles / Min., Depending upon Label Length & Bottle / Product Size / Dia.

36" Dia. Turn Table is provided at In-Feed of Automatic Sticker Labelling Machine. which will accumulate and feed the Round Bottle / Vial in vertical position to Automatic Sticker Labelling Machine to further process. Operator has to load the sufficient bottles / vials / products manually on turn table before start the machine.

Even, 36" Dia. Turn Table is provided at Outfeed of Machine which will collect / accumulate labelled bottle / vial / container. | Automatic Sticker Labelling Machine, Model: VSC/VLC-150 (Servo with Automation) + 36" Dia. Infeed & Outfeed Turn Table
To apply Sticker Labels up to 95 mm Label Height on vertical Small / large Round Bottle / Vials / Product of different diameter.
Machine is designed for the Standard Rated Speed up to 150 Bottles / Min., Depending upon Label Length & Bottle / Product Size / Dia.

36" Dia. Turn Table is provided at In-Feed of Automatic Sticker Labelling Machine. which will accumulate and feed the Round Bottle / Vial in vertical position to Automatic Sticker Labelling Machine to further process. Operator has to load the sufficient bottles / vials / products manually on turn table before start the machine.

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Even, 36" Dia. Turn Table is provided at Outfeed of Machine which will collect / accumulate labelled bottle / vial / container.
| Additional Automation Attachments / Features | Additional Automation Attachments / Features | Additional Automation Attachments / Features | Additional Automation Attachments / Features | Additional Automation Attachments / Features
| 1 | 'Advantech' Make, 15" Touch Screen Industrial PC (IPC) with 'Fatek' Make PLC. | 'Advantech' Make, 15" Touch Screen Industrial PC (IPC) with 'Fatek' Make PLC. | 'Advantech' Make, 15" Touch Screen Industrial PC (IPC) with 'Fatek' Make PLC. | 'Advantech' Make, 15" Touch Screen Industrial PC (IPC) with 'Fatek' Make PLC.
| 2 | 'Videojet' Make, Thermal Transfer (TTO) Printer (Model: VJ6330) with touch pad controller | 'Videojet' Make, Thermal Transfer (TTO) Printer (Model: VJ6330) with touch pad controller | 'Videojet' Make, Thermal Transfer (TTO) Printer (Model: VJ6330) with touch pad controller | 'Videojet' Make, Thermal Transfer (TTO) Printer (Model: VJ6330) with touch pad controller
| 3 | Missing Label Detection Sensor ('Leuze' make, Luminescence Sensor suitable for Paper label only) | Missing Label Detection Sensor ('Leuze' make, Luminescence Sensor suitable for Paper label only) | Missing Label Detection Sensor ('Leuze' make, Luminescence Sensor suitable for Paper label only) | Missing Label Detection Sensor ('Leuze' make, Luminescence Sensor suitable for Paper label only)
| 4 | 'Baumer' Make, Vision System Camera for Inspection of Barcode, Batch Code, Correct Printing Matter, Smudge Printing, Line Missing and Proper Position of Printing on Label. | 'Baumer' Make, Vision System Camera for Inspection of Barcode, Batch Code, Correct Printing Matter, Smudge Printing, Line Missing and Proper Position of Printing on Label. | 'Baumer' Make, Vision System Camera for Inspection of Barcode, Batch Code, Correct Printing Matter, Smudge Printing, Line Missing and Proper Position of Printing on Label. | 'Baumer' Make, Vision System Camera for Inspection of Barcode, Batch Code, Correct Printing Matter, Smudge Printing, Line Missing and Proper Position of Printing on Label.
| 5 | Label Roll Ending Alarm System on Dancing Roll Assy. ('P+F' Make) | Label Roll Ending Alarm System on Dancing Roll Assy. ('P+F' Make) | Label Roll Ending Alarm System on Dancing Roll Assy. ('P+F' Make) | Label Roll Ending Alarm System on Dancing Roll Assy. ('P+F' Make)
| 6 | 'Janatics' Make, Pneumatic Cylinder for Pneumatic Rejection System with S.S. rejection box. | 'Janatics' Make, Pneumatic Cylinder for Pneumatic Rejection System with S.S. rejection box. | 'Janatics' Make, Pneumatic Cylinder for Pneumatic Rejection System with S.S. rejection box. | 'Janatics' Make, Pneumatic Cylinder for Pneumatic Rejection System with S.S. rejection box.
| 7 | 'Leuze' Make, Encoder for Rejection | 'Leuze' Make, Encoder for Rejection | 'Leuze' Make, Encoder for Rejection | 'Leuze' Make, Encoder for Rejection
| 8 | 2nd Product Sensor ('Leuze' make) | 2nd Product Sensor ('Leuze' make) | 2nd Product Sensor ('Leuze' make) | 2nd Product Sensor ('Leuze' make)
| 9 | Fallen Product Detection Sensor with Machine Stop Facility ('Leuze' Make) | Fallen Product Detection Sensor with Machine Stop Facility ('Leuze' Make) | Fallen Product Detection Sensor with Machine Stop Facility ('Leuze' Make) | Fallen Product Detection Sensor with Machine Stop Facility ('Leuze' Make)
| 10 | Low Air Pressure Alarm / Switch with machine stop facility ('Panasonic' make) | Low Air Pressure Alarm / Switch with machine stop facility ('Panasonic' make) | Low Air Pressure Alarm / Switch with machine stop facility ('Panasonic' make) | Low Air Pressure Alarm / Switch with machine stop facility ('Panasonic' make)
| 11 | Safety Cabinet of Aluminium extruded profile frame & Transparent acrylic Door with Interlocking | Safety Cabinet of Aluminium extruded profile frame & Transparent acrylic Door with Interlocking | Safety Cabinet of Aluminium extruded profile frame & Transparent acrylic Door with Interlocking | Safety Cabinet of Aluminium extruded profile frame & Transparent acrylic Door with Interlocking

|||||
|||||

|||||

6.0 | Qualification Team | Qualification Team | Qualification Team | Qualification Team | Qualification Team

| Name of Employee | Name of Employee | Company Name | Designation | Signature

|||||

|||||

||| 12 | Three Color Light Stack ('Leuze' Make, Tower Light with Buzzer) |

||| 13 | Emergency Stop Switch (In case of any emergency) |

|||||

Sr. No. | Name of Trainee | Employee

ID / No. | Comments | Signature of Trainee

|||||

|||||

|||||

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|||||

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|||||

Sr. No | Test Particulars | Specified Function | Observation | Comments | Checked By | Verified by

|||||||

|||||||

|||||||

|||||||

|||||||

|||||||

|||||||

|||||||

Sr. No. | Test / Task | Observation | Comments | Checked

by | Verified

by

1 | Operate the equipment at the speed mentioned in DQ / Purchase Order. All over Machine & other automation attachment's Operations and Performance found Satisfactory. ||||

Description of deficiency & date observed:

Person responsible for corrective action and date assigned:

Corrective actions taken and date conducted:

Sr.

No. | Description of Deviation | Criticality

(Major/ Minor) | Resolved

(Yes/ No) | Verified Date

|||||

|||||

|||||

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|||||

VPM / BPM | Vial / Bottle per Minute | URS | User Requirement Specification

PLC | Programmable Logical Controller | SS | Stainless Steel

HMI | Human Machine Interface | HP | Horse Power

MMI | Man, Machine Interface | dB | Decibel

IPC / Indl. PC | Industrial Personal Computer | V | Volt

GA | General Arrangement | Hz | Hertz

DQ | Design Qualification | CVT | Constant Voltage Transformer

IQ | Installation Qualification | FRL | Filter-Regulator-Lubricator

OQ | Operational Qualification | MI | Milliliter

PQ | Performance Qualification | Mm | Millimeter

Mtr | Meter | AC | Alternative Current

Dia / Ø | Diameter | DC | Direct Current

Lg. | Length / Long | kW | Kilowatt

Ht. | Height | BOM | Bill of Material

CGMP | Current Good Manufacturing Practices | SOP | Standard Operating Procedure

MOC | Material of construction | KVA | Kilovolt-ampere

Etc. & i.e., | et cetera, that is (for example) | P+F | Pepperl + Fuchs

& | And | UPS | Uninterruptible Power Supply

CFM | cubic feet per minute | PB | Push Button

S/No. | Document Name | Document Name | Available

Yes/No

1 | Documents: FAT, DQ, IQ, OQ, PQ. | Documents: FAT, DQ, IQ, OQ, PQ. |

2 | Operation Instruction Manual | Operation Instruction Manual |

| 2.1 | G.A. Drawing of Machine (In Manual) |

| 2.2 | Industrial PC Screen (IPC) / Programming Procedure (In Manual) |

| 2.3 | Electrical Wiring Diagram (In Manual) |

3 | Data Sheet of AC. Drive (VFD) | Data Sheet of AC. Drive (VFD) |

4 | Data Sheet of Sensors. | Data Sheet of Sensors. |

5 | Test Reports & Certificates. | Test Reports & Certificates. |

5 | 5.1 | S.S. 304 Sheet ■ Laboratory Test Reports |

5 | 5.2 | Motor & Gear box ■ Manufacturer / Supplier Test Certificate |

5 | 5.3 | Test Certificates of Supplier on Maharshi's Letter Pad. |

Department | Name | Signature | Date

Supplied by: M/s BioBridge Healthcare Solutions Pvt. Ltd., India | Supplied by: M/s BioBridge Healthcare Solutions Pvt. Ltd., India | Supplied by: M/s BioBridge Healthcare Solutions Pvt. Ltd., India | Supplied by: M/s BioBridge Healthcare Solutions Pvt. Ltd., India

Installed by |||

Installed by |||

Client : INFARMA – Indústria Farmacêutica, Lda | Client : INFARMA – Indústria Farmacêutica, Lda | Client

: INFARMA – Indústria Farmacêutica, Lda | Client : INFARMA – Indústria Farmacêutica, Lda
Approved by |||
Approved by |||

[SHEET: Product List]

NaCl TM
 Comp. NaCl sol TM
 Comp. Gluco Sol TM
 Ringer lactato TM
 Glucose (An.) TM
 gluco (5%) TM
 gluco (10%) TM
 gluco (20%) TM
 gluco (mix)TM
 KCl TM
 Ringer lactato TM
 CaCl TM
 Ringer lactato TM
 C3H5O3Na TM
 Ringer lactato TM

[SHEET: RM assays]

| Sodium Chloride | |

| S. No. | Test | Method / Equipment

| 1 | Identification | Centrifuge

| 2 | Appearance |

| 3 | Acidity and Alkalinity | Chemical method

| 4 | Loss on Drying | Drying oven 105 degree

| 5 | BET | To be conducted if claimed

| 6 | Sterility | To be conducted if claimed

| | Impurities |

| 7 | Aluminium | Flouresence Spectroscopy

| 8 | Arsenic | Arsenic test apparatus

| 9 | Barium | Chemical method

| 10 | Ferocyanides | Chemical method

| 11 | Iodides | Chemical method

| 12 | Iron | Chemical method

| 13 | Limits of Bromides | UV Spectroscopy

| 14 | Limits of Phosphates | AAS

| 15 | Magnesium | Titration

| 16 | Alkaline earth metals | Titration

| 17 | Nitrites | UV Spectroscopy

| 18 | Sulfates | Chemical method

| Dextrose / Glucose | |

| S. No. | Test | Method / Equipment

| 1 | Identification | Chemical / IR Spectroscopy

| 2 | Water content | KF Apparatus

| 3 | Color and clarity of solution | Spectroscopy

| 4 | Content assay | HPLC

| 5 | Impurities | HPLC

| 6 | Residue on Ignition | Muffle furnace

| 7 | Acidity | Chemical method

| 8 | Optical rotation | Polarimeter

| 9 | Conductivity | Conductivity meter

| 10 | Soluble starch | Chemical method

| 11 | Sulfites | Chemical method

| Potassium Chloride | |

| S. No. | Test | Method / Equipment

| 1 | Identification | Chemical / IR Spectroscopy

| 2 | Acidity and Alkalinity | Chemical method

| 3 | Loss on Drying | Drying oven 105 degree

| | Impurities |

| 4 | Aluminium | Flouresence Spectroscopy

| 5 | Sodium | Chemical method

| 6 | Iodine | Chemical method

| 7 | Bromide | Chemical method

| 8 | Calcium and Magnesium | Chemical method

| 9 | Heavy metals | Chemical method

| Calcium Chloride dihydrate | |

| S. No. | Test | Method / Equipment

| 1 | Identification | Chemical / IR Spectroscopy

| 2 | ph | ph meter

| | Impurities |

| 3 | Aluminium | Flouresence Spectroscopy

| 4 | Heavy metals | Chemical method

| 5 | Iron and phosphates | Chemical method

| 6 | Magnesium | Chemical method

| 7 | Alkaline salts | Chemical method

| Sodium Lactate | |

| S. No. | Test | Method / Equipment

| 1 | Identification | Chemical / IR Spectroscopy

| 2 | ph | ph meter

| | Impurities |

| 3 | Chlorides | Chemical method

| 4 | Heavy metals | Chemical method

| 5 | Methanol and methyl ester | Chemical method

| 6 | Residual solvents | Chemical method

[SHEET: FPQC assays]

| 1. Sodium chloride products assays | |

| S. No. | Assays | Type of Assay

| 1 | Appearance |

| 2 | BET |

| 3 | Sterility |

| 4 | pH |

| 5 | Acidity and Alkalinity |

6	Assay for Concentration	
7	Osmolality	
8	Particulate Matter	
1.	Dextrose products assays	
S. No.	Assays	Type of Analysis
1	Appearance	
2	BET	
3	Sterility	
4	pH	
5	Identification	
6	Impurity related substance	
7	Assay for Sodium Chloride	
8	Assay for Dextrose	
9	Osmolality	
10	Particulate Matter	
1.	Ringer Lactate products assays	
S. No.	Assays	Type of Analysis
1	Appearance	
2	BET	
3	Sterility	
4	Ph	
5	Osmolality	
6	Sodium content	
7	Chlorides	
8	Potassium	
9	Calcium	
10	Lactate	
11	Particulate Matter	
[SHEET: Equipments list]		
QC equipment list for all products		
Sr.no.	Equipment name	Quantity
1	Biological microscope	1
2	Colony counter	1
3	Air sampler	1
4	Incubator 20-30 °C	2
5	Incubator 30-45 °C	2
6	Incubator 50-60 °C	1
7	LAL incubator	1
8	Vacuum pump	2
9	Sterility test apparatus	1
10	UV Spectrometer	1
11	analitica Balance	2
12	Polarimeter	2
13	pH Meter	1
14	Filtration unit with vacuum pump	1
15	Flame photometer	1
16	Magnetic Stirrer with hot plate	1
17	Water bath	2
18	Precision balance	2
19	Fume hood chamber	1
20	Karl fischer apparatus	1
21	Stop watch	4
22	Autotitrator	1
23	Bunsen burner	2
24	Particle counter	1
25	TOC	1
26	Incubator (22-25C)	
27	magnetic stirrer	2
28	Refrigerator	5
29	Vortex mixer	4
30	Biosafety Cabinet	2
31	MilliQ water system	1
32	Double Door Autoclave	1
33	Sealing machine	1
34	Thin layer Chromatography	
35	Incubator 20-30 °C Stability	1
36	Incubator 30-45 °C Stability	1
37	Leak test apparatus	1
38	Osmometer	1
39	Conductivity meter	1
40	Autopipette	10
41	Anaerobic jar	2
42	UV cabinet	1
43	vertical autoclave	1
44	dry heat sterilizer	1
45	polarimetro	1
46	bubble point apparatus	1
47	weight box	1
48	Hot air oven	1
49	Colony counter	1
50	Electric Muffle furnace	1
51	Celing Laminar Air Flow	1
52	Peristaltic pump	1
53	Laminar Air Flow	1
54	Deep Freezer -20	1
55	Automatic Washer	1
56	Scuff resistance tester	1
57	Burst strength tester (cardboard)	1

JULY 2019

Ensuring Access to Medicines
in East Africa: Lessons from
India

SAKTHIVEL SELVARAJ
Ensuring Access to Medicines in
East Africa: Lessons from India

SAKTHIVEL SELVARAJ
ABOUT THE AUTHOR

Sakthivel Selvaraj is Director, Health Economics, Financing and Policy, at the Public Health Foundation of India (PHFI), New Delhi. He is currently engaged in research, teaching and advocacy in the area of health systems, and specifically, healthcare financing and access to medicines. He was a Takemi Fellow (Post-Doctoral) at the Harvard School of Public Health, Boston, and a Fulbright Scholar in 2006-07. He has a PhD in Health Economics (1996-2001) from Jawaharlal Nehru University, New Delhi.

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© 2019 Observer Research Foundation. All rights reserved. No part of this publication may be reproduced or transmitted in any form or by any means without permission in writing from ORF. Ensuring Access to Medicines in East Africa: Lessons from India

ABSTRACT

Africa faces various challenges in ensuring equitable access to life-saving drugs. Drug prices in the continent are largely unregulated, making locally produced drugs more expensive than the imported ones. The African countries thus rely heavily on imports to meet their pharmaceutical needs. Further, the medicine delivery systems in the countries are encumbered by the poor implementation of regulatory policies, if they are in place at all. East African countries, in particular, struggle to nurture domestic drug-manufacturing capacities and suffer from a shortage of both factories and trained human resource. India, for its part, is widely regarded as a "global pharmacy," meeting its own local needs as well as foreign demand for medicines. This paper discusses the challenges facing the pharmaceutical sectors of five East African countries, and explores the potential for learnings from the successes of the Indian pharmaceutical industry.

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ENSURING ACCESS TO MEDICINES IN EAST AFRICA: LESSONS FROM INDIA

INTRODUCTION

With an estimated population of 1.2 billion, Africa accounts for
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approximately 16 percent of the global population but almost 24
percent of the global disease burden. Seventy percent of all HIV patients
2
and 90 percent of malaria fatalities are recorded in African countries. The lack of access to essential life-saving medicines is thus a major challenge in the continent. Africa faces various barriers in ensuring equitable access to drugs, including inadequate public expenditure, insufficient financial-risk protection, ineffective public procurement systems, exorbitant drug prices, and low availability of essential medicines at public health facilities. These countries rely primarily on imported pharmaceutical products to meet local needs, due to the lack of a robust domestic pharmaceutical industry. According to the World Health Organization (WHO), more than 90 percent of the 54 national medicines regulatory authorities (MRAs) across Africa have minimal or no capacity³ to regulate various aspects of medicine manufacture and delivery.

This paper studies the pharmaceutical sector in five East African countries: Ethiopia, Kenya, Tanzania, Rwanda and Mozambique, discussing domestic pharmaceutical production, drug-procurement processes, medication availability and prices, and pharmaceutical regulations. The final section of the paper examines the successes of India's pharmaceutical industry in specific states and concludes with an exploration of what the East African countries can learn from the Indian experience.

I. PHARMACEUTICAL MARKET AND PRODUCTION

East African countries have a limited capacity for manufacturing pharmaceutical products. Thus, they rely mostly on imports—of

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finished products, raw ingredients and active pharmaceutical ingredients (APIs)—to meet the rapidly growing local demands. Much of the East African market is dominated by generics and medicines imported from India. Given the rapid growth in the market, there is immense scope for existing local manufacturers, with government support, to enhance their capacities and for new manufacturers to emerge. However, the sustainability of large-scale local production is predicated upon its efficacy and the extent of government subsidies provided. With populations ranging between 11 million and 100 million in the five countries discussed in this paper, domestic production can thrive if handled correctly.

Ethiopia

The Ethiopian healthcare market has an annual growth rate of 25 percent, valued at US\$400–500 million per annum. The country has approximately 200 importers and 22 local industries of pharmaceutical products and medical consumables. Of the 22, nine are involved directly in the manufacture of products, only two of which are Good Manufacturing Practices (GMP) certified. Local manufacturers supply only about 20 percent of the local demand and 90 of the 380 drugs on the national essential medicine list (EML).⁴ Moreover, no local company manufactures APIs in the country.

Despite government support, the Ethiopian industries face several challenges, such as limited access to foreign currency, difficulties in procuring raw materials, and constraints in human-resource capacity. A 2013 survey by Ewen M. et al. indicated that 55.4 percent of the products across all sectors were manufactured in Ethiopia and the remaining imported from India (17.6 percent), Cyprus (seven percent), Germany (5.1 percent), France (2.1 percent) and South Korea (1.9 percent). The availability of branded generics, INN generics and

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originator brands in Ethiopia was 74.6 percent, 20.7 percent and 4.7 percent, respectively.

Kenya

In 2014, Kenya was the largest and fastest-growing pharmaceutical market (US\$740 million) in East Africa, with an estimated compound annual growth rate (CAGR) of 15 percent from 2007.¹⁰ The market share of locally produced pharmaceutical products by value increased from 28 percent in 2010 to 30 percent in 2014. Around 70 percent of the pharmaceutical products were imported from other countries, particularly India and China, which together accounted for 50 percent of the total imports. The market was dominated by generics (62 percent), and almost 50 percent of the market by value was made up by anti-infective immunological and cardiovascular agents.¹² The pharmaceutical market in Kenya is significantly concentrated: about 80 percent of the total production comes from the 10 biggest manufacturers.¹³

According to the Kenya National Bureau of Statistics, the imports of pharmaceutical products rose to US\$604.50 million in 2016 from US\$396.50 million in 2011, mainly on account of generic drugs and raw materials. While the country's domestic manufacturers depend on imports for APIs,¹⁴ amongst East African countries, Kenya leads in the export of pharmaceutical products. In 2015, nearly 50 percent of its pharmaceutical exports went to Uganda and Tanzania.^{15,16} The export value increased from US\$74.46 million in 2011 to US\$131.90 million in 2016.¹⁷ The total value of pharmaceutical exports in 2015 was US\$71,200,000 (1.20 percent of the total exports), while the total value of pharmaceutical imports was US\$572,533,000.¹⁸ Further, data from the Kenya Pharmacy and Poisons Board (KPPB) show that pharmaceutical sales in the country increased from US\$826 million in 2016 to US\$919 million in 2017.

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Despite the increase in export and local sales, the overall contribution of Kenya's pharmaceutical industry to the GDP remains negligible (1.22 percent in 2017).¹⁹ Pharmaceutical companies have now started joint ventures or buyouts with leading Indian (Strides, Cipla) and South African (Aspen Pharma) companies, and are using their own funds (e.g. CDC Capital's investment in Abacus Parenteral Drug Ltd.) to expand operations and improve manufacturing standards. However, companies are yet to invest in research and development, which is crucial for the development of the sector.²⁰

Tanzania

Tanzania has a robust pharmacy system: 88.9 percent of its health facilities have a functional pharmacy department, and 92.9 percent of the essential medicines are generally available.²¹ Branded drugs constitute 46 percent and generics, 54 percent of the market in the

country. Eight of all Tanzanian pharma companies are registered manufacturers, and their local production meets 31 percent of the national demand. The remaining 69 percent is imported, mainly from

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India. Raw ingredients, too, are mostly imported from other countries. In 2013, local production decreased to 15 percent of the pharmaceutical market share (US\$48.7 million), from 35 percent in 2009 (from US\$49.2 million). Pharma exports declined as well, from US\$7.9 million to US\$1.7 million. Consequently, the import value of pharmaceutical products increased from US\$99.4 million in 2009 to US\$286 million in 2013.²³

Rwanda

Rwanda's pharmaceutical market was estimated to be worth US\$75 million in 2014, with a CAGR of 12.85 percent from 2007 to 2014.²⁴ The country has only one domestic pharmaceutical factory, which

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manufactures a few essential medicines.^{25,26,27} The total local production comprises three percent of the overall market, of which 56 percent are generic drugs and 44 percent are branded. More than 95 percent of the drugs are imported to meet local demands, along with most of the manufacturing ingredients. In 2015, the total import value of pharmaceuticals was US\$83.1 million.²⁸ Of Rwanda's total exports,²⁹ pharmaceutical products make up only 0.04 percent, at US\$242,000.

Mozambique

Mozambique has a single pharmaceutical producer, which started its operations in 2013, with support from the Brazilian government. Public expenditure on pharmaceuticals increased from US\$78 million to US\$122 million between 2004 and 2012. In 2012, Mozambique's pharmaceutical imports were valued at approximately US\$140 million, a per-capita drug expenditure of US\$5.55. The share of public-sector imports is 85 percent (US\$122 million) of the total market value, mainly supported by external funds and donations and managed by the Ministry of Health (MoH). Mozambique has 54 formally registered private importers, but more than 50 percent of the imported drugs in 2012 came from the four biggest importers. Private operators import 2,572 registered pharmaceutical products in Mozambique from India (55 percent), Portugal (12 percent) and Brazil and Germany (combined, five percent).³²

II. PUBLIC-SECTOR PROCUREMENT OF DRUGS

A reliable procurement system ensures access to medicines in sufficient quantities at the lowest possible rates, so as to secure maximum value for the most number of beneficiaries possible. This requires an Essential Medicines List, quantity and quality assessment, regular procurement of medicines, supply-chain management and timely payment to

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suppliers. An inefficient system can lead to exorbitant prices and medicine shortages at the point of delivery.

Ethiopia

The Ethiopian government procures almost 70 percent of the country's pharmacological requirement through the Pharmaceutical Fund and Supply Agency (PFSA), via centralised international tenders. Procurement through the PFSA increased from US\$27 million in 2007 to US\$310 million in 2014. The government offers a 25-percent price preference to local manufacturers and makes an advance payment of 30 percent. Additionally, tax-free loans are provided for new investment and upgrading local production, 70 percent and 60 percent, respectively. Procurement prices for locally produced products are higher than imported products, at 1.20 times and 0.84 times the international reference prices (IRPs), respectively. Locally produced products are also priced 45 percent higher than imported products.³⁷

Kenya

Centralised and decentralised procurement of pharmaceutical products happens through tender bids (not publicly available). The Kenya Medical Supplies Agency (KEMSA), a semi-autonomous primary public procurement agency, is responsible for centralised procurement.^{38,39} The KEMSA receives funding from the Government of Kenya and other development partners, to procure medical supplies for rural health facilities. Around 70 percent of Kenya's pharmaceutical products are imported from other countries, with India and China together

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accounting for 50 percent of the total imports. The Kenyan government gives 15-percent price preference for all medicines produced in Kenya and 10 percent for all imported medicines traded by companies with at least 51 percent Kenyan ownership. Under special

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circumstances, Kenyan law has provisions for the importation of unlicensed drugs.⁴³

Tanzania

All public facilities in Tanzania use either an allocated financial budget or an established budget ceiling for the procurement of medicines and medical supplies. The budget allocated for pharmaceuticals in the public sector was increased from US\$14.1 million to US\$28.5 million from 2000 to 2007, and the spending on essential drugs was increased by almost 50

percent from 2002 to 2005. However, this was not enough to meet the national need for medicines. In 2005, approximately 75 percent of the total spending on drugs was borne by the Ministry of Health and Social Welfare (MoH&SW) of Tanzania, and the remaining 25 percent by donors. Government allocation included donations and funding from other agencies, e.g. the Global TB Fund and Global Alliance for Vaccines and Immunisation. The MoH&SW obtains funding from donors through the Ministry of Finance (MoF), and the MoF allocates a quarterly share of resources to the MoH&SW, which then distributes resources for drugs to the Medical Stores Department (MSD). The MSD must follow the Public

Procurement Act no. 21, 2004, in the process of procurement.

For essential drugs, internal resources are the main sources of funding. In 2005–06, funds were reported to be underutilised due to delays in the allocation of money, stock-out of drugs at the MSD, and the inflexibility of the system to adapt to such delays.⁴⁷ The procurement of essential medicines is the responsibility of the MSD tender board at the central stores; the managers at the zonal level; and individuals such as supplies officers, nursing officers and pharmacists. Over 80 percent of the essential medicines and supplies are procured through international competitive bidding, 10 percent through selective bidding and five percent through national competitive bidding. Once procured, the

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essential drugs are stored at the MSD and then distributed to primary health facilities and hospitals via zonal stores. Public wholesalers give a 15-percent price preference to the local manufacturers. A 2017 survey by Ewen M. et al. indicated that the Tanzanian government pays 0.69 and 1.34 times the IRPs for local and imported products, respectively.

In addition to the drugs on the EML, the MSD also procures medicines requested by health institutions. Similarly, only about 52 percent of health facilities procure medicines within the EML. Three main reasons for non-compliance were recorded: a) the prescribers did not agree with the Standard Treatment Guidelines (STG); b) local needs were not addressed; and c) suppliers did not have stocks for the EML products.

Rwanda

In Rwanda, the Medical Production and Procurement Division (MPPD), referral hospitals (BUFMAR) and private pharmacy wholesalers are responsible for procuring pharma products. Public procurement of health commodities is done through international competitive biddings.⁵² To import drugs, the WHO prequalification and the approval of the Stringent Regulatory Authority is required.⁵³ There is a provision in the law to import unlicensed medicines in special circumstances.

Mozambique

Mozambique's pharmaceutical market is dominated by the public sector and a highly concentrated group of private wholesalers and importers. In the public sector, drugs are procured—local and imported—and distributed through the MoH's Central Drugs and Medical Supplies Agency (CMAM). The CMAM uses an open-tender procedure for the procurement of pharmaceuticals. Occasionally, it buys drugs from

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private importers/wholesalers.⁵⁵ APIs are exempt from import duties to promote local production in Mozambique. The Mozambique

government gives a 15-percent price preference to the local producers, over the cheapest competing international bid. In case of emergencies and drug stock-outs, procurement is done through local producers to shorten the regular drug-acquisition cycle.⁵⁷

III. AVAILABILITY OF ESSENTIAL MEDICINES

The success of a government procurement system can be measured by, amongst others, the availability of medicines at the point of delivery. Where patients rely on the private sector, drug availability in private pharmacies is crucial as well.

Ethiopia

A 2013 survey conducted in Ethiopia found that the mean availability of medicines in the public sector was 64 percent,⁵⁸ with locally produced drugs being more available than imported ones, at 48 and 19 percent, respectively. The mean availability of branded generics was 37 percent (local) and 10 percent (imported) and that of INN generics was 14 percent (local) and nine percent (imported). No originator brand was found in the public sector.^{59,60}

In the private sector, the mean availability of imported and local pharma products, together, was 73 percent. Local products had greater availability than imported products, at 54 percent and 35 percent, respectively. Branded generics (local, 42 percent; imported, 29 percent) were more easily available than INN generics (local, 13 percent; imported, three percent). The availability of imported originator brands was nine percent, and no local originator brand was found in the private sector.^{61,62}

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90% Mean Availability of Drugs in Ethiopia and Tanzania
82%
Overall availability
80%
73% Locally produced
70%
70% drugs
64%
Local branded
58%
60% generics
54%
52%
48%
Local INN generics
50%
42%
Local originator
40% 37%
35%
32% brands
29%
30% 27% Imported drugs
21% 21%
19%
20% 19%
14% 15% Imported branded
13% 12%
10% 9%
generics
10% 9% 6% 5%
7%
3% 2%
imported INN
0% 0% 0% 0% 0% 0% generics
0%
Public sector of Private Sector of Public Sector of Private Sector of Imported originator
Ethiopia Ethiopia Tanzania Tanzania brands

Kenya

The WHO/HAI pricing survey conducted in Kenya in 2004 reported that the mean availability of generics (37.7 percent) was lower in the public sector than in the private sector (72.4 percent), and the mean availability of originator medicines, too, was lower in the public sector (66 percent) than in the private sector (81 percent).⁶³

Availability of Drugs in Kenya
90% 81%
80% 72.40%
70% 66%

60% Median availability of
generics
50%
37.70%
40%
30% Median availability of

originator brands

20%

10%

0%

Public Sector Private Sector

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Tanzania

The availability of essential drugs across Tanzania's MSDs varied significantly in 2005–06. The central MSD had 97 percent availability of drugs, while at some zonal stores, it was 51 percent. The availability of essential drugs at district hospitals was very low and varied between 51.3 percent and 59.5 percent. The average availability of drugs was lower at primary health facilities than in hospitals, at 52.8 percent and 64

74.6 percent, respectively.

In 2008, the MoH&SW conducted a survey in public healthcare facilities to assess the medicines supply management system. The overall availability of stock at the facility level was 65 percent and 65

ranged from 52 to 96 percent per category of product. In 2007–13, the median availability of selected generics in the public and private healthcare sectors was 37.8 percent and 50 percent, respectively.⁶⁶

Ewen M. et al. reported that in the public sector, the availability of imported products was greater, at 32 percent, than that of local products, at 21 percent. The mean availability of both was 52 percent. Branded generics (local, 15 percent; imported, 27 percent) had higher availability than INN generics (local, six percent; imported, five percent). The availability of imported originator brands was only four percent in the public sector.

In the private sector, the mean availability of medicines (local and imported) was 82 percent, with the availability of local products at 21 percent and imported at 70 percent. Branded generics (local, 19 percent; imported, 58 percent) had greater availability than INN generics (imported, two percent; local 12 percent). The availability of imported originator brands was seven percent.⁶⁷

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Mozambique

Russo G. et al. (2010) reported that in Mozambique, the median availability of generic drugs was 60 percent in the public sector and 73.5 percent in the private sector. Originator brands were available only at 68 three percent of private pharmacies.

Availability of Drugs in Mozambique

80% 73.50%

70%

60%

60%

50%

40%

30%

20%

10% 3%

0%

Median availability of Median availability of Median availability of generic drugs in public generic drugs in private originator brands in private sector sector sector

IV. PRICING AND PRICE REGULATION

Affordable pricing is crucial for ensuring equitable access to pharmaceuticals. In East Africa, prices for domestic products are generally higher than for imported products across sectors, and higher for most drugs in the private than in the public sector. The lack of price regulation makes medicines unaffordable in these countries, often resulting in patients forgoing treatment.

Ethiopia

In the public sector, local branded generics are priced 23 percent higher than imported branded generics, while in the private sector imported

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branded generics are priced 153 percent higher.⁶⁹ The median price ratio (MPR) of branded generics was 1.41 for local products and 1.14 for imported products. For INN generics, too, the MPR for local products are slightly higher than for imported products.^{70,71} Neither drug prices nor supply-chain mark-ups are regulated in Ethiopia. The government does not levy value-added tax (VAT) or goods and services tax/general sales tax on medicine, but an import duty of five percent is applicable to

72,73

all imported products and APIs. The Ethiopian government applies a 53-percent mark-up on imported products, but only a 17-percent mark-up on local products, thereby supporting the local manufacturers.

74,75

In the private sector, the MPR is lower for local products than for imported products, at MPR 1.8 and 5.44, respectively. Consumers have to pay 193 percent more for imported products. Imported branded generics (MPR 4.33) were 153 percent more expensive than local products (MPR 1.71). For INN generics, the MPR of local products was slightly higher than for imported products, at MPR 2.17 and 2.08, respectively. Originator brands were available at higher prices than the generics.^{76,77}

For branded generics in all sectors of healthcare, consumers pay approximately 13 percent less for local products (MPR 1.44) than for imports from India (MPR 1.64), 60 percent less than for those from Cyprus, and over 90 percent less than those from Germany. INN generics imported from India (MPR 1.57) are only slightly more expensive than those produced in Ethiopia (MPR 1.45).

Kenya

The WHO/HAI pricing survey (2004) showed that the public procurement prices of generics (MPR 0.65) were below the International Reference Prices (IRPs) in Kenya, while prices paid by

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consumers were greater than the IRPs. In the government facilities, patients paid 1.99 times the IRPs for generics and 3.6 times the IRPs for originator medicines. In the private sector, consumers paid 3.33 and 18.1 times higher prices for generics and originator drugs, respectively.

79

Tanzania

Public-sector patients in Tanzania frequently need to pay OOP to buy drugs and supplies. About 44.4 percent patients pay for drugs either directly through OOP spending (64.2 percent) or indirectly through the payment of user charges or prepayment for health insurance plans, such as the NHIF (16.5 percent), the CHF (11 percent) or other schemes (8.3 percent).⁸⁰ According to Ewen M. et al., 2017 consumers pay seven percent more for local products (MPR 1.44) than for imported products (MPR 1.35). Additionally, consumers pay 135 percent and 65 percent more than the government procurement prices for local and imported products, respectively.

In the private sector, consumers pay similar prices for local and imported products, 2.27 and 2.29 times the IRPs, respectively.⁸¹

Rwanda

The prices of health commodities in Rwanda are steep and unaffordable to the majority of the population as the country has no pharmaceutical pricing policy in either the public or the private sector.

Mozambique

Russo G. et al. conducted a study to investigate medicine prices in urban Mozambique and the forces influencing them. They collected wholesale and retail prices for 25 originators/innovators and generic drugs from 10 public and 34 private pharmacies. Sixteen drugs were selected out of

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the 25 on the HAI/WHO list and adapted to the country's context, and the remaining nine drugs were those prescribed most frequently in Mozambique.⁸³

The study demonstrated that retail prices for generic medicines in Mozambique's government urban outlets are lower than the MSH reference prices,⁸⁴ while in the private sector, prices are four to five times that of the MSH reference prices. Originator brands available only in the private sector are 23 times the MSH reference prices. The study noted intercity price variations for both generics and originator brands. In the private sector, branded versions of the same drug are four to five times more expensive than the generic version. However, even the lowest priced generics are, on average, 10 times more expensive in the private sector than in the public sector. Treatments for acute conditions are more affordable than treatments for chronic conditions such as diabetes, hypertension and psychiatric illnesses in both sectors.⁸⁵

V. PHARMACEUTICAL REGULATION

Pharmaceutical regulation at different stages of production, importation, distribution and dispensing is essential to ensure the

availability of quality drugs.

Ethiopia

Ethiopia has a comprehensive law for the regulation of drugs, to safeguard public health from unsafe and substandard medicines. Based on Proclamation 661, 2009, the medical legislation of Ethiopia aims to ensure that all drugs (local or imported) available in the market are safe, effective and consistently meet the acceptable quality standards. The governing body of medicine regulation in Ethiopia—the Food, Medicine and Healthcare Administration and Control Authority—applies

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numerous controls at all levels of pharmaceutical activities to regulate the manufacturing, distribution, marketing, prescription, labelling and dispensing of pharmaceutical products.⁸⁶

Kenya

Kenya's national medicine policy document is the Kenya National Pharmaceutical Policy (KNPP), last updated in 2010. The KNPP addresses important issues related to the selection of essential medicines, medicine financing, pricing, procurement, distribution and regulations, as well as the rational use of medicines, pharmacovigilance^{87,88,89}

and research. The MRA, monitored by the Pharmacy and Poison Board under the MoH, is involved in the African Medicines Regulatory Harmonisation and the Harmonisation of Medicines Registration in the East African Community. All pharmaceutical products (domestic or imported) must be registered with the MRA before being sold in Kenya. Pharmaceutical manufacturers are required to be registered under the MRA and comply with GMP, while wholesalers and distributors must comply with Good Distributing Practices. The country has a WHO-prequalified medicines quality-control laboratory (QCL) for the testing of medicines.^{90,91,92}

For imported products, Kenya has legal provisions for their testing

93

and inspection at the authorised port of entry as well as for the

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promotion and/or advertisement of medicines. Custom duty and VAT are not levied on the import of pharmaceutical products or

95,96 97

ingredients, only on some packaging materials and locally procured products.⁹⁸ The country has national STGs for most common illnesses

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and a national EML comprising 359 drugs. The copy of EML and STG are available in 42 percent and 39 percent of public health facilities, respectively.¹⁰⁰

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In 2001, Kenya modified the Intellectual Property (IP) Act, its national patent law, to comply with the Trade-Related Aspects of

101

Intellectual Property Rights (TRIPS) Agreement. The country allows patent protection for products and processes for a 20-year period. The law incorporates all flexibilities and safeguards in line with the TRIPS Agreement, namely, compulsory licensing, Bolar exemptions and parallel importation. However, as of 2010, the government had not yet

102

utilised these flexibilities to improve access to drugs. There are legal provisions for data exclusivity for pharmaceuticals, but none for patent extension or linkage between patent status and marketing

103

authorisation.

Tanzania

Tanzania has national medicines policies (NMPs), based on the WHO's recommendations, such as evidence-based selection, proper supply-chain management, rational use of medicines, affordable medicines, appropriate financing, human resource development, monitoring and evaluation, research, technical cooperation amongst countries, and

104

legislative and regulatory frameworks for pharmaceuticals.

The country has several pharmaceutical legislations: the Tanzania Food, Drugs and Cosmetic Act, 2003 (amendments in 2004 and 2014); Pharmacy Act, 2011; Traditional Medicines Act, 2002; Procurement Act, 2011; and Industrial and Consumer Chemical Act, 2003. The Tanzania Food and Drug Authority—a semi-autonomous supervisory body—regulates food, medical devices, cosmetics and herbal drugs in addition to pharmaceutical products.^{105,106} Medicine registration requires mandatory compliance with the national GMP (based on the WHO's guidelines).¹⁰⁷ The country has a WHO-prequalified medicines QCL, part

108,109

of the national MRA. Under special circumstances, an external

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laboratory may be contracted for the quality check of medicines.

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In 1991, the government enforced a first national drug policy, an STG and a national EML for Tanzania (NEMLIT), revised in 1997. According to a survey, the 2007 revised draft of NEMLIT was available at the MSD, but treatment guidelines were available only for tuberculosis (in the version last reviewed in 2005–06). It was found that the EML was available in 80 percent of zonal stores and 38 percent of health facility pharmacies.¹¹¹

Rwanda

Rwanda started work on its national medicines policy in 2009, which is yet to be approved.¹¹² Currently, the medicine regulatory function is performed by the MoH,¹¹³ and the Pharmacy Task Force under the MoH acts as the legislative authority. The country also has a WHO pre-qualified functional QCL under the Rwanda Standards Board of the ^{114,115}

Ministry of Trade, Industry and EAC Affairs. Some challenges currently affecting the pharmaceutical sector include the lack of an ¹¹⁶

autonomous MRA, high production and transportation costs, inadequate financial support, the lack of an investment strategy, a shortage of skilled personnel, and a dearth of information on market ¹¹⁷

size. As of 2016, pharmaceutical products were expected from patentability to take advantage of the flexibilities in the TRIPS.¹¹⁸

Mozambique

In Mozambique, pharmaceutical functions are regulated as per the Medicines Act under Law 4/98 and are supported by other regulations and norms. The Pharmaceutical Department under the MoH acts as the regulatory authority and all pharmaceutical products must be registered under it for quality and price control.^{119,120} The Pharmaceutical Department is supported by the Inspection Directorate and the National Laboratory for Medicines Quality Control.¹²¹ The national EML

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of the country (last updated in 1999) comprises 415 medicine formulations.¹²² Mozambique has no autonomous MRA.¹²³

While some East African countries have comprehensive laws and policies for the regulation of medicines, others are still working towards the introduction of such policies. However, legal provisions alone are not sufficient; it is crucial to ensure their implementation at all levels to strengthen the public health institutions and ensure equitable access to quality pharmaceuticals.

VI. LESSONS FOR EAST AFRICA

India is often referred to as a “global pharmacy” due to its sprawling pharmaceutical industry, which not only meets local needs but also th exports medicines across the globe. During the first half of the 20 century, India was dependent on countries such as the UK, France and Germany for pharmaceutical products. With the establishment of the Hindustan Antibiotics Ltd. in 1945 and the Indian Drugs and Pharmaceuticals Ltd. in 1961, the Indian government took its first steps towards self-reliance in pharmaceuticals. However, the local industry did not develop substantially during this period, and the Indian pharmaceutical industry consisted ¹²⁴

almost exclusively of multinationals. Further, until the 1970s, multinationals were not keen on manufacturing in India, and the product patent regime prevented Indian firms from manufacturing. This resulted in an underdeveloped domestic industry, on the one hand, and exorbitant drug prices due to monopolies on the other.¹²⁵

In 1972, the Patents and Designs Act, 1911, which provided product patent protection for pharmaceuticals, was replaced by the Indian Patent Act, 1970.¹²⁶ The new Act recognised only process patents, abolishing patents on breakthrough products that could help in the development of the domestic industry. This allowed domestic firms to

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legally produce and sell medicines that were patented in other parts of the world. The local industry then established a reverse-engineering capacity to develop and market copies of patented medicines at lower ¹²⁷

prices. This encouraged the domestic production of pharmaceuticals, in addition to measures such as restrictions on the import of finished formulations, high tariff rates, the requirement for firms to match the import of bulk drugs to the purchase from domestic sources at a fixed ratio, and equity ceilings on foreign participation. By 1991, Indian firms ¹²⁸

were producing 80 percent of the total formulations in the country.

Today, India has over 10,500 manufacturing units and 3,000 pharmaceutical companies. The country exported pharmaceuticals (APIs, generics and alternative systems of medicine) worth US\$16.8 billion in 2016–17 and accounts for almost 42 percent of the generic medicines produced globally, which is the combined market size of Africa and the Middle East. North America is India's largest export market for pharmaceuticals and buys 34 percent of India's total exports, followed by Africa, which receives over 19 percent of India's exports.

East African countries, on the other hand, are struggling to establish in-house manufacturing capacity. Currently, there are nine local pharmaceutical manufacturers in Ethiopia, eight in Tanzania, 42 in Kenya, and one each in Rwanda and Mozambique. These countries rely primarily on imports to meet local needs. Building more factories, creating and training human resources, and fostering technical collaborations with countries that have well-established local pharma industries should be a top priority for East Africa. This will ensure the sustainability of the industry in the long run as well as improved and equitable access to essential and life-saving medicines.

Indian states follow different models of procurement and supply-chain management. Tamil Nadu, for example, pioneered the system of

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pooled procurement by establishing the autonomous Tamil Nadu Medical Services Corporation in 1994. The model is that of centralised procurement and free decentralised distribution of medicines at public health facilities, wherein medicines on the state's EML are sourced into every health facility through district warehouses, using passbooks with monetary entitlements. All district warehouses are linked to a central computer at the head office, and all issue and receipt of medicines are computerised under the Drug Distribution Management System. Rajasthan replicated Tamil Nadu's model in 2011 and established the Rajasthan Medical Services Corporation. In both states, a two-bid process is undertaken, comprising a technical bid and a financial bid.

Tamil Nadu and Rajasthan have thus managed to ensure efficient procurement and supply-chain management, while some states still struggle with the availability of medicines at the point of care. An assessment of the procurement systems in different Indian states

showed that for a common basket of medicines, the mean availability on the day of the survey was the highest in Tamil Nadu (81 percent), followed by Rajasthan (79 percent) and the lowest in Jammu and Kashmir (38 percent). Further, the availability was generally low at primary health centres and increased with the level of care.

Ethiopia, Tanzania, Kenya and Mozambique have established EMLs and systems of procurement. However, the outcomes have been substandard so far, with low mean availability of drugs in the public sector. The countries can benefit from studying and adopting Tamil Nadu's model of centralised procurement and decentralised distribution, with the help of autonomous procurement agencies. Smaller countries may benefit from setting up regional procurement agencies to exploit monopsony power in buying medicines from other countries, including India. East Africa can also learn from the Indian patent regime for pharmaceuticals, which successfully incorporates TRIPS flexibilities. India transitioned to

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the product patent system to comply with the TRIPS Agreement in 2005, which allows the signatory countries certain flexibilities in defining the scope of patentability to safeguard their public interest. The Indian Patents Act, 1970 makes good use of this provision by discouraging frivolous patenting. When Novartis, a Swiss multinational corporation, attempted evergreening for its blockbuster medicine "Imatinib" (marketed under the brand name Glivec), used for the treatment of Chronic Myeloid Leukaemia, its patent application was rejected by the Government of India. The government's decision was later challenged by the company in the Supreme Court of India, which upheld it.

Compulsory licensing (CL), another flexibility allowed under TRIPS, lets governments authorise the production of a medicine by a company other than the originator/patent holder, in the interest of public health. The Indian government utilised this flexibility by issuing a compulsory licence to the Indian generic manufacturer Natco, for producing German pharmaceutical company Bayer AG's cancer medicine Sorafenib tosylate (marketed as Nexavar) in March 2012, thereby considerably driving down the price of medicine.

South Africa, too, holds some valuable lessons for East Africa. In 2018, the former approved its new IP policy, which entails a substantive review of all patent applications and incorporates flexibilities, such as allowing patent decisions to be challenged and provisions for

compulsory licences.¹³³ The international community commended the new IP policy as an important step toward safeguarding public health in the country, through improved access to medicines.

CONCLUSION

East African countries can learn from India's successes in pharmaceuticals. They must focus on setting up a robust and

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sustainable domestic manufacturing industry, capable of meeting domestic needs and reducing the region's reliance on imports from other countries. Pooled public procurement of medicines via an autonomous agency established solely for this purpose can be an important step towards improving the access situation in specific countries, with the possibility of setting up a single regional procurement agency for smaller countries.

The region must also work on improving regulation policies and ensuring their proper implementation. Better utilisation of the flexibilities under the TRIPS agreement, such as compulsory licensing and provisions for patent oppositions, will go a long way in improving access to affordable essential medicines by driving down prices of patented products.

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20, Rouse Avenue Institutional Area, New Delhi - 110 002, INDIA
Ph. : +91-11-35332000 Fax : +91-11-35332005
E-mail: contactus@orfonline.org
Website: www.orfonline.org

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TANZANIA MEDICINES AND MEDICAL DEVICES AUTHORITY

COMPENDIUM GUIDELINES FOR MARKETING AUTHORIZATION OF
MEDICINAL PRODUCTS

(Made under Regulation 4 (1) of the Tanzania Medicines and Medical Devices (Registration of Medicinal
Products) Regulations, 2015

July, 2020

P. O. Box 77150, Mabibo, Off Mandela Road, Dar es Salaam, Tel:+255-22-
2450512/2450751/2452108, Fax: +255-22-2450793, Website: www.tmda.go.tz, Email:
info@tmda.go.tz, Toll Free 0800110084

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These guidelines were reviewed in order to tackle challenges faced by our Applicants who wish to apply for marketing authorization of human medicinal products. The review also took into consideration those challenges that arose from international best practice requirements not aligning with Tanzania Medicines and Medical Devices Authority (TMDA) requirements.

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Foreword

The Tanzania Medicines and Medical Devices Authority (TMDA) was established with the mission of protecting and promoting public health by ensuring quality, safety and effectiveness of drugs and medical devices including invitro diagnostic devices. The first step towards achieving this goal is to conduct pre-marketing evaluation of products so as to ensure that they meet standards of quality, safety and effectiveness before the products are allowed into the market. This is a fundamental requirement for authorization of medicinal products in Tanzania.

In fulfilment of its mission, TMDA is duty-bound to ensure that data submitted as evidence of quality, safety and efficacy is solid, credible and submitted in manner that is logical and consistent. It is also the TMDA's responsibility to ensure that the requirements to be fulfilled by our applicants are up to date and in line with best national and international practices. With this in mind, these guidelines were reviewed to update applicants on the content and format of application dossiers and to align our requirements with international best practices.

This edition combines all previous guidelines into one compendium of guidelines and introduces a new guideline on naming of medicinal products which provide guidance to applicants on how to choose appropriate brand names for their products.

This compendium of guidelines provides requirements on the quality, safety and efficacy information for active pharmaceutical ingredients (APIs) and finished pharmaceutical products (FPPs) that should be submitted to TMDA to support product dossier. Alternative approaches to the principles and practices described in this document may be acceptable provided that they are supported by adequate scientific justification. It is also important to note that the TMDA may request information or material, or define conditions not specifically described in these guidelines, in order to adequately assess the quality, safety and efficacy of a pharmaceutical product.

Therefore, adherence to all guidelines within the compendium will ensure that all relevant information is provided in registration dossiers submitted for marketing authorization. This will facilitate efficient and effective evaluation as well as approval process. It will also help to avoid queries which results in unnecessary delays in giving approvals to new medicines thereby improving access to medicines of proven quality, safety and efficacy in the shortest possible time.

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Scope

This compendium as a whole document is entirely applicable for human medicinal products that have been manufactured by chemical synthesis. However, each individual guideline may have a wider scope of applicability.

Depending on the scope of each individual guideline within this compendium, the principles maybe applicable to other medicinal products such as human vaccines, biotherapeutics, biosimilars and veterinary medicinal products including veterinary vaccines. (Example guidelines on labeling, naming of medicinal products and package inserts)

Specific requirements for human vaccines, biotherapeutics, biosimilars and veterinary medicinal products including veterinary vaccines can be found in the Guidelines on Submission of Documentation for Marketing Authorization of Human Vaccines, Guidelines on Submission of Documentation for Marketing Authorization of Biotherapeutic Products, Guidelines on Submission of Documentation for Marketing Authorization of Similar Biotherapeutic Products, Guidelines on Submission of Documentation for Registration of Veterinary Pharmaceutical Products and Guidelines on Submission of Documentation for Registration Immunological Veterinary Products respectively.

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PART I:

GUIDELINES ON SUBMISSION OF DOCUMENTATION FOR MARKETING AUTHORIZATION OF HUMAN PHARMACEUTICAL PRODUCTS

1

Introduction

The "Guidelines on Submission of Documentation for Registration of Human Pharmaceutical Products' Second Edition, April 2020" is the TMDA publication which sets out procedures and requirements for the implementation of Pharmaceutical Products Registration through Common Technical Document (CTD). The CTD has five Modules:

Module 1: Administrative Requirements;

Module 2: The Quality Overall Summaries (QOS),

Module 3: The Quality Requirements for the Active Pharmaceutical Ingredients (API) and Finished Pharmaceutical Products (FPP),

Module 4: Pre-Clinical data Requirements,

Module 5: Clinical data Requirements.

The general objective of the Common Technical Document (CTD) guidelines is to provide internationally harmonized pharmaceuticals registration procedures using CTD in order to improve access to essential pharmaceuticals for prevention and treatment of priority disease conditions in Tanzania.

Adherence to the guidelines by the manufacturers/applicants will facilitate timely assessments and approvals of medicinal product dossiers by the regulatory authorities for pre-marketing evaluation, marketing authorization/registration and post-marketing review.

I wish to express my gratitude to all individuals from EAC Partner States' NMRA, regional and international organizations and EAC Secretariat and members of the Technical Working Group (TWG) on Medicines Evaluation and Registration (MER) of the East African Community Medicine Regulatory Harmonization (EAC MRH) Project who actively participated in the development of the guidelines.

Special thanks are also extended to TMDA staff and esteemed stakeholders; the dealers in pharmaceutical industry and the academia in particular members of the Tanzania Pharmaceutical Manufacturers Association (TPMA) and the Tanzania Association of Pharmaceutical Industries (TAPI) who discussed the draft guidelines and gave commendable inputs for improving the guidelines.

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Abbreviations and acronyms

API Active Pharmaceutical Ingredient

APIMF Active Pharmaceutical Ingredient Master File

CEP Certificate of Suitability to the monograph of Ph Eur monograph

CTD Common Technical Document

EAC East Africa Community

EAC-MRH East Africa Medicines Registration Harmonization

EAC-NMRA East Africa Partner State National Medicines Regulatory Authority

EDQM European Directorate for the Quality of Medicines

EU European Union

FPP Finished Pharmaceutical Product

GCP- Good Clinical Practice

GMP- Good Manufacturing Practice

ICH International Conference on Harmonization (of Technical Requirements for Registration of Pharmaceuticals for Human Use)

PD Product Dossier

PHIS Pharmaceutical Health Information System

PI Product Information

SDRA Stringent Drug Regulatory Authority

SmPC Summary of Product Characteristics

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Glossary

The definitions provided below apply to the words and phrases used in these guidelines. The following definitions are provided to facilitate interpretation of the guidelines.

Active pharmaceutical ingredient (API)

An active ingredient is any component that provides pharmacological activity or other direct effect in the diagnosis, cure, mitigation, treatment, or prevention of disease, or to affect the

structure or any function of the body of man or animals.
(USFDA Glossary of terms, it can be found online at [Drugs@FDA Glossary of Terms](#)).

Active Pharmaceutical Ingredient (API) starting material

A raw material, intermediate, or an API that is used in the production of an API and that is incorporated as a significant structural fragment into the structure of the API. (WHO Glossary of Terms).

Market Authorization Holder (MAH)

Is a person who holds authorization to place a medicinal product in the Tanzanian market and is responsible for that product.

Commitment batches

Production batches of an API or FPP for which the stability studies are initiated or completed post-approval through a commitment made in a regulatory application.

Comparator product

A pharmaceutical product with which the generic product is intended to be interchangeable in clinical practice. The comparator product will normally be the innovator product for which efficacy, safety and quality have been established.

Generic product

Is a medicinal product which has the same qualitative and quantitative composition in active substances and the same pharmaceutical form as the reference medicinal product, and whose bioequivalence with the reference medicinal product has been demonstrated by appropriate bioavailability studies.

(PHIS Glossary 2009, can be found online at: <http://phis.goeg.at/index.aspx?alias=phisglossary>)

Existing API

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An API that is not considered a new active substance, which has been previously approved through a finished product by a stringent regulatory authority. (WHO Glossary of Terms).

Finished pharmaceutical product (FPP)

A finished dosage form of a pharmaceutical product which has undergone all stages of manufacture, including packaging in its final container and labelling. (WHO Glossary of Terms).

Innovator medicinal product

Generally the medicinal product that was first authorized for marketing (normally as a patented product) on the basis of documentation of efficacy, safety and quality. (WHO Glossary of Terms).

Manufacturer

A manufacturer is a natural or legal person with responsibility for manufacturing of a medicinal product or active pharmaceutical ingredient. It involves operations such as production, packaging, repackaging, labelling and relabeling of pharmaceuticals.

(PHIS Glossary 2009, can be found on line at: <http://phis.goeg.at/index.aspx?alias=phisglossary>)

Mock-up

A copy of the flat artwork design in full colour, providing a replica of both the outer and immediate packaging, providing a two-dimensional presentation of the packaging/ labelling of the medicine. It is also referred to as a paper copy or computer generated version.

Officially recognized pharmacopoeia (or compendium)

The official recognized pharmacopoeias are British Pharmacopoeia (BP), European Pharmacopoeia (Ph Eur.), The International Pharmacopoeia (Ph.Int), Japanese Pharmacopoeia (JP) and United States Pharmacopoeia (USP).

On-going stability study

The study carried out by the manufacturer on production batches according to a predetermined schedule in order to monitor, confirm and extend the projected retest period (or shelf-life) of the API, or confirm or extend the shelf-life of the FPP. (WHO Glossary of Terms).

Pilot-scale batch

A batch of an API or FPP manufactured by a procedure fully representative of and simulating that to be applied to a full production-scale batch. For example, for solid oral dosage forms a pilot scale is generally, at a minimum, one-tenth that of a full production scale or 100 000 tablets

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or capsules, whichever is the larger; unless otherwise adequately justified. (WHO Glossary of Terms).

Primary batch

A batch of an API or FPP used in a stability study, from which stability data are submitted in a registration application for the purpose of establishing a retest period or shelf-life. (WHO Glossary of Terms).

Production batch

A batch of an API or FPP manufactured at production scale by using production equipment in a production facility as specified in the application.

Specimen

A sample of the actual printed outer and inner packaging materials and package leaflet.

Stringent Drug Regulatory Authority (SDRA)

A National Medicines Regulatory Authority which is strict, precise, exact with effective and well-functioning systems.

Among others, it includes regulatory authorities which are:-

- Members or observers or associates of the International Conference on Harmonization of Technical Requirements for Registration of Pharmaceuticals for Human Use (ICH) as of October, 2015

MEMBERS:

- European Union member States (Austria, Belgium, Bulgaria, Cyprus, Czech Republic, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Ireland, Italy, Latvia, Lithuania, Luxembourg, Malta, Poland, Portugal, Romania, Slovak Republic, Slovenia, Spain, Sweden, The Netherlands, and United Kingdom
- Japan
- United States

OBSERVERS:

- European Free Trade Association (EFTA) represented by Swiss Medic of Switzerland, and Health Canada (as may be updated from time to time).

ASSOCIATES through mutual recognition agreements: Australia, Norway, Iceland and Liechtenstein (as may be updated from time to time).

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- For medicines used exclusively outside the ICH region, positive opinions or tentative approval under any of the following three special regulatory schemes are recognized as stringent approval:-

- Article 58 of European Union Regulation (EC) No. 726/2004
- Canada S.C. 2004, c. 23 (Bill C-9) procedure
- United States FDA tentative approval (for antiretrovirals under the PEPFAR programme)

- A regulatory Authority that has been agreed by TMDA to have an effective and well-functioning medicines regulation systems.

7

1. INTRODUCTION

1.1 Background

This guideline provides guidance for applicants while preparing a Common Technical Document for the Registration of Medicines for Human Use (CTD) for submission to the TMDA. The document describes how to organize applications based on the International Conference on Harmonization of Technical Requirements for Registration of Pharmaceuticals for Human Use (ICH) guidelines on the CTD.

According to the CTD format, each application is a collection of documents, grouped into 5 modules. Module 1 prescribes Administrative Information and Prescribing Information requirements which is region specific. The Summaries, Quality, Non-clinical, and Clinical modules have been described in Modules 2 to 5, respectively. Applicants should not modify the overall organization of the CTD.

If not contained in the bulk of the documentation, any additional data should be included as addenda to the relevant part, together with additional expert comment that may be provided as a supplement to, or incorporated into, the relevant summary, overall summary or overview.

Information in these Modules should be present in relevant sections.

For application procedures refer Guidelines on Procedural Aspects for Application for Market Authorization for Human Medicinal Products.

1.2 Scope

These guidelines will assist applicants to prepare applications to register medicinal products for human use in East Africa Partner States. The format for applications is the Common Technical Document (CTD).

These guidelines apply to MA applications for medicinal products containing APIs of synthetic or semi-synthetic origin. Biological, biotechnological and herbal products are not covered by these guidelines.

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MODULE 1: ADMINISTRATIVE INFORMATION AND PRODUCT INFORMATION

Module 1 should contain all administrative documents (for example, application forms and certifications), labelling, general correspondence and annexes (environmental assessments, antibiotic resistance and overseas evaluation reports), as needed. Documents should be organized in the order listed below. Generally, all of the documents in Module 1, other than the annexes, can be provided in a single volume. The annexes to the module should be submitted in

separate volumes. Official language is English as a mandatory language for all medicines.

Products shall be evaluated on a First in First out (FIFO) basis and the timeline for review and approval should be within 12 months.

1.1 Comprehensive table of contents for all modules

Module 1 should include a comprehensive table of contents for the entire application. The comprehensive table of contents should include a complete list of all documents provided in the application by module. In the table of contents, the location of each document should be identified by referring to the volume numbers that contain the relevant documents and any tab identifiers. In general, the name for the tab identifier should be the name of the document.

1.2 Cover letter

Applicants should include a cover letter with all applications. A copy of the letter should be placed at the beginning of Module 1. The cover letter shall be signed by the Market Authorization Holder (Refer Annex I).

1.3 Application form

An application to register a medicinal product for human use must be accompanied by a completed application form (Annex II). The application form should be duly filled with relevant information and attachments, dated signed and stamped appropriately.

1.4 Product Information

Provide copies of all package inserts, labels and any information intended for distribution with the product to the patient.

If the Summary Product Characteristics (SmPC), has not been approved from SDRA at the time the application is submitted, a draft document may be included. The approved SmPC from SDRA should then be supplied to the TMDA as they become available.

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1.4.1 Prescribing information (Summary of Product Characteristics)

All prescription medicines should be accompanied by SmPC. Refer to Guidelines on Format and Content of Summary for Pharmaceutical Products Characteristics

1.4.2 Container labelling

Product should be labeled as prescribed in the Guidelines on Content and Format of Labels for Pharmaceutical Products

1.4.3 Patient information leaflet (PIL)

All medicinal preparations with potential for long term use and self-administered injections and Over the Counter (OTC) must contain a patient information leaflet. Languages used for PIL and labeling should be clearly expressed in English and French.

Refer Guidelines on Content and Format of Package Inserts for Pharmaceutical Products.

1.4.4 Mock-ups and specimens

If the product applicant has a specimen or mock-up of the sample(s) presentation of the medicine available at the time of initial application, it should be included in Module 1.4.4.

If there are multiple strengths and/or pack sizes, one representative specimen or mock-up for each will be sufficient. If batch number and expiry date are to be printed on the label during packaging, a statement to this effect should accompany the labels. If mock-ups or specimens are not available at the time of initial application, a text version may be submitted, however, mock-ups or specimens must be submitted to the TMDA, during the evaluation process and prior to finalization of the application.

1.5 Information about the experts

Experts must provide detailed reports of the documents and particulars, which constitute Modules 3, 4 and 5.

The requirement for these signed Expert Reports may be met by providing:

- The Quality Information Summary
- The Quality Overall Summary, Non-clinical Overview / Summary and
■ Clinical Overview / Summary in Module 2,
- A declaration signed by the experts in Module 1.6.
- Brief information on the educational background, training and occupational experience of the experts in Module 1.6.

Experts should indicate in their declarations the extent, if any of their professional or other involvement with the applicant / dossier owner and confirm that the report has been prepared

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by them or if not, any assistance provided and by whom. Reports should be based on an independent assessment of the dossier and references must be provided for any additional claims not supported by the dossier. A sample declaration form is provided as Annex III.

Additionally, a filled in Quality Information Summary as provided under Annex IV should be submitted.

1.6 Certificates of Suitability of monographs of the European pharmacopoeia (CEP) or APIMF Approval

If a CEP is available, the finished product applicant should present copy of CEP in module 1.7.

Applicant should provide the Letter of Access to CEP or Letter of Access to TMDA as appropriate from API manufacturer. These letters should be included in Module 1.7. (Refer Annex V and Annex VI)

1.7 Good Manufacturing Practice (GMP)

For all medicines, irrespective of the country of origin, all key manufacturing and/or processing steps in the production of active pharmaceutical ingredient ingredients and finished pharmaceutical products must be performed in plants that comply with TMDA GMP guidelines. Attach a WHO-type certificate of GMP. For more information on GMP requirements and application for GMP inspection, refer Guidelines on Good Manufacturing Practice for more guidance.

If available at the time of submission of application, GMP certificates from TMDA and/or SDRAs or an evidence for application for GMP inspection should be submitted in module 1.8.

1.8 Good Clinical Practice (GCP) or Good Laboratory Practice (GLP)

Provide evidence such as accredited certificate for GCP or GLP for the sites participating in the clinical studies

1.9 Regulatory status

1.9.1 Registration status from countries with Stringent Drug Regulatory Authorities (SDRAs)

Provide registration status of the medicinal product applied for registration in the countries with SDRAs and attach evidence(s) for the same.

1.9.2 Registration status in other countries

Provide registration status of the medicinal product applied for registration in the EAC region as well as other countries and attach evidence(s) for the same.

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1.9.3 List of countries in which a similar application has been submitted

The applicant should provide, in Module 1.9.1 of the dossier, a list of countries in which a similar application has been submitted, dates of submission (if available) and the status of these applications. This should detail approvals (with indications) and deferrals, withdrawals and rejections with reasons in each case.

1.9.4 Statement on whether an application for the product has been previously rejected, withdrawn or repeatedly deferred in any jurisdiction

Applicant must declare whether a marketing application for the medicine has been rejected prior to submission of the application to TMDA. If the medicine has been rejected, repeatedly deferred, withdrawn or suspended then reasons must be stated. If rejection occurs during the evaluation process, the TMDA should be informed.

1.10 Evidence of API and/or FPP prequalified by WHO

If an evidence indicating that the active pharmaceutical ingredient and/or finished pharmaceutical product are prequalified by WHO is available, it should be presented in Module 1.

1.11 Manufacturing and Marketing authorization

Submit a Certificate of Pharmaceutical Product in format recommended by the World Health Organization together with a valid Manufacturing Authorization for pharmaceutical production. If available, evidence for prequalification of medicinal product by WHO should be submitted.

1.12 Product samples

Sufficient number of samples should be submitted together with the application.

Batch number, Manufacturing Date and Expiry Date should be dynamically printed on packages for all medicines except in situations where there is space is a restriction, the details can be on secondary packages with the primary pack having at least the batch number and expiry date. Pre-printing of the batch number, manufacturing date and Expiry Date will not be acceptable.

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MODULE 2: OVERVIEW & SUMMARIES

2.1 Table of contents of Module 2

A table of contents for module 2 should be provided.

2.2 CTD Introduction

2.3 Quality overall summary (QOS)

The quality overall summary (QOS) is a summary that follows the scope and the outline of the Body of Data in Module 3.

The QOS should not include information, data or justification that was not already included in Module 3 or in other parts of the common technical document (CTD).

Complete Annex VII following the guidance below.

The Quality Information Summary as provided under Annex IV should also be provided.

2.3.S Active pharmaceutical ingredient (name, manufacturer)

2.3.S.1 General Information (name, manufacturer)

Information from 3.2.S.1 should be included.

2.3.S.2 Manufacture (name, physical address)

Information from 3.2.S.2 should be included:

Information on the manufacturer;

■ A brief description of the manufacturing process and the controls

■ A flow diagram, as provided in 3.2.S.2.2;

■ A description of the Source and Starting Material and raw materials of biological origin used in the manufacture of the API, as described in 3.2.S.2.3;

■ Highlight critical process intermediates, as described in 3.2.S.2.4;

■ A description of process validation and/or evaluation, as described in 3.2.S.2.5.

2.3.S.3 Characterization

A summary of the interpretation of evidence of structure and isomerism, as described in 3.2.S.3.1.

A tabulated summary of the data provided in 3.2.S.3.2, with graphical representation, where appropriate should be included.

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2.3.S.4 Control of Drug Substance

A brief summary of the justification of the specification(s), the analytical procedures, and validation should be included.

Specification from 3.2.S.4.1 should be provided.

A tabulated summary of the batch analyses from 3.2.S.4.4, with graphical representation where appropriate, should be provided.

2.3. S.5 Reference Standards or Materials

Information from 3.2.S.5 (tabulated presentation, where appropriate) should be included.

2.3.S.6 Container Closure System

A brief description and discussion of the information, from 3.2.S.6 should be included.

2.3.S.7 Stability

This section should include a summary of the studies undertaken (conditions, batches, analytical procedures) and a brief discussion of the results and conclusions, the proposed storage conditions, retest date or shelf-life, where relevant, as described in 3.2.S.7.1.

The post-approval stability protocol, as described in 3.2.S.7.2, should be included.

A tabulated summary of the stability results from 3.2.S.7.3, with graphical representation where appropriate, should be provided.

2.3.P Finished Pharmaceutical Product (name, dosage form)

2.3. P.1 Description and Composition of the Drug Product (name, dosage form)

Information from 3.2.P.1 should be provided.

Composition from 3.2.P.1 should be provided.

2.3.P.2 Pharmaceutical Development

A discussion of the information and data from 3.2.P.2 should be presented.

A tabulated summary of the composition of the formulations used in clinical trials and a presentation of dissolution profiles should be provided, where relevant.

2.3.P.3 Manufacture (name, physical address)

Information from 3.2.P.3 should include:

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Information on the manufacturer

A brief description of the manufacturing process and the controls that are intended to result in the routine and consistent production of product of appropriate quality.

A flow diagram, as provided under 3.2.P.3.3.

A brief description of the process validation and/or evaluation, as described in 3.2.P.3.5.

2.3.P.4 Control of Excipients

A brief summary on the quality of excipients, as described in 3.2.P.4, should be included.

2.3.P.5 Control of Drug Product

A brief summary of the justification of the specification(s), a summary of the analytical procedures and validation, and characterization of impurities should be provided.

Specification(s) from 3.2.P.5.1 should be provided.

A tabulated summary of the batch analyses provided under 3.2.P.5.4, with graphical representation where appropriate should be included.

2.3.P.6 Reference Standards or Materials

Information from 3.2.P.6 (tabulated presentation, where appropriate) should be included.

2.3.P.7 Container Closure System

A brief description and discussion of the information in 3.2.P.7 should be included.

2.3.P.8 Stability

A summary of the studies undertaken (conditions, batches, analytical procedures) and a brief discussion of the results and conclusions of the stability studies and analysis of data should be included. Conclusions with respect to storage conditions and shelf-life and, if applicable, in-use storage conditions and shelf-life should be given.

Stability studies should be provided for each pack type applied for registration.

A tabulated summary of the stability results from 3.2.P.8.3, with graphical representation where appropriate, should be included.

The post-approval stability protocol, as described in 3.2.P.8.2, should be provided.

2.4 Non-Clinical overview

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The non-clinical overview should provide an integrated overall analysis of the information in the Common Technical Document. In general, the Nonclinical Overview should not exceed about 30 pages.

The non-clinical overview should be presented in the following sequence:

- Overview of the nonclinical testing strategy
- Pharmacology
- Pharmacokinetics
- Toxicology
- Integrated overview and conclusions
- List of literature references

Studies conducted to establish the pharmacodynamic effects, the mode of action, and potential side effects should be evaluated and consideration should be given to the significance of any issues that arise.

The Integrated Overview and Conclusions should clearly define the characteristics of the human pharmaceutical as demonstrated by the nonclinical studies and arrive at logical, well-argued conclusions supporting the safety of the product for the intended clinical use. Taking the pharmacology, pharmacokinetics, and toxicology results into account, the implications of the nonclinical findings for the safe human use of the pharmaceutical should be discussed (i.e., as applicable to labelling).

Refer ICH Guidance on the Common Technical Document for the registration of pharmaceuticals for human use: Safety for guidance on the format and the content of this part.

Generic products are generally exempted in this module; however, in some cases such as changes in safety impurity profile, the safety assessment studies should be conducted.

2.5. Clinical overview

The Clinical Overview is intended to provide a critical analysis of the clinical data in the Common Technical Document. The Clinical Overview will necessarily refer to application data provided in the comprehensive Clinical Summary, the individual clinical study reports (ICH E3), and other relevant reports; but it should primarily present the conclusions and implications of those data, and should not recapitulate them. Specifically, the Clinical Summary should provide a detailed factual summarization of the clinical information in the CTD, and the Clinical Overview should provide a succinct discussion and interpretation of these findings together with any other relevant information.

The clinical Overview should be presented in the following sequence

- Product Development Rationale
- Overview of Biopharmaceutics
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- Overview of Clinical Pharmacology
- Overview of Efficacy
- Overview of Safety
- Benefits and Risks Conclusions
- Literature References

Refer ICH Guidance on the Common Technical Document for the registration of pharmaceuticals for human use: Efficacy (M4E) for guidance on the format and the content of this part.

2.6 Nonclinical Written and Tabulated Summaries

The following order is recommended:

- Introduction
- Written Summary of Pharmacology
- Tabulated Summary of Pharmacology
- Written Summary of Pharmacokinetics
- Tabulated Summary of Pharmacokinetics
- Written Summary of Toxicology
- Tabulated Summary of Toxicology

Refer ICH Guidance on the Common Technical Document for the registration of pharmaceuticals for human use: Safety for guidance on the format and the content of this part.

For generic products are generally exempted in this module; however, in some cases such as changes in safety impurity profile, the safety assessment studies should be conducted.

2.7 Clinical Summary

The Clinical Summary is intended to provide a detailed, factual summarization of all of the clinical information in the Common Technical Document. This includes information provided in ICH E3 clinical study reports; information obtained from any meta-analyses or other cross-study analyses for which full reports have been included in Module 5; and post-marketing data for products that have been marketed in other regions.

Refer ICH Guidance on the Common Technical Document for the registration of pharmaceuticals for human use: Efficacy for guidance on the content of this section.

The following order is recommended:

2.7.1 Summary of Biopharmaceutical Studies and Associated Analytical Methods: Generic applications

The objective of CTD Module 2.7.1 is to summarize all relevant information in the product dossier with regard to bioequivalence studies and/or comparative dissolution and associated analytical methods.

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Annex I of the Guideline on Therapeutic Equivalence Requirements: Presentation of Biopharmaceutical and Bio-analytical Data contains a set of template tables to assist applicants in the preparation of Module 2.7.1 with regard to data to be presented. Furthermore, it is anticipated that a standardized presentation will facilitate the evaluation process.

Refer the Guideline on Therapeutic Equivalence Requirements: Presentation of Biopharmaceutical and Bio-analytical Data for more guidance. (Annex V).

2.7.2 Summary of Clinical Pharmacology Studies

Refer the Guideline on Therapeutic equivalence requirements: Presentation of Biopharmaceutical and Bio-analytical Data for more guidance. (Annex V).

2.7.3 Summary of Clinical Efficacy

Refer the Guideline on Therapeutic equivalence requirements: Presentation of Biopharmaceutical and Bio-analytical Data for more guidance. (Annex V).

2.7.4 Summary of Clinical Safety

Refer the Guideline on Therapeutic equivalence requirements: Presentation of Biopharmaceutical and Bio-analytical Data for more guidance. (Annex V).

2.7.5 Literature References

2.7.6 Synopses of Individual Studies

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MODULE 3: QUALITY

3.1 Table of contents of Module 3

A Table of Contents should be provided that lists all of the reports and gives the location of each study report in the Common Technical Document.

3.2 Body of data

3.2. S Active pharmaceutical ingredient (API))

The API information can be submitted the order of preference in one of the following four options:

- a) Option 1: Certificate of suitability of European Pharmacopeia (CEP);
- b) Option 2: Active pharmaceutical ingredient pre-qualified by WHO;
- c) Option 3: Active Pharmaceutical Ingredient Master File approved by TMDA;
- d) Option 4: Full details in the Product Dossier (PD);

The applicant should clearly indicate at the beginning of the API section in the Marketing Authorization (MA) application and in the QOS how the information on the API for each API manufacturer is being submitted.

Where reference is made to CEP, the finished product applicant must have written permission to access the CEP from the CEP holder. Applicant should provide the Letter of Access to CEP, as appropriate from API manufacturer (Refer Annex V). Letter of access should be included in Module 1.7.

Where reference is made to TMDA approved APIMF, the finished product applicant must have written permission to access the APIMF from the company that supplied the APIMF and must provide the APIMF file number to TMDA. Applicant should provide the Letter of Access to TMDA, as appropriate from API manufacturer (Refer Annex VI). Letter of access should be included in Module 1.7.

The applicant's open part of the APIMF should be included in Module 3.2.S of the Quality documentation presented in the CTD format. The API manufacturer's restricted (closed) part is supplied to TMDA directly by the API manufacturer when required.

The API information submitted by the applicant/FPP manufacturer should include the

following for each of the options used.

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a) Option 1: Certificate of suitability of European Pharmacopeia (CEP)

A complete copy of the CEP (including any annexes) should be provided in Module 1. The declaration of access for the CEP should be duly filled out by the CEP holder on behalf of the FPP manufacturer or applicant.

In addition, a written commitment should be included that the applicant will inform TMDA in the event that the CEP is withdrawn. It should also be acknowledged by the applicant that withdrawal of the CEP will require additional consideration of the API data requirements to support the PD. The written commitment should accompany the copy of the CEP in Module 1.

Along with the CEP the applicant should supply the following information in the dossier, with data summarized in the QOS-PD:-

a) 3.2.S.1.3 General properties – discussions on any additional applicable physicochemical and other relevant API properties that are not controlled by the CEP and Ph. Eur. monograph, e.g. solubilities and polymorphs as per guidance in this section.

b) 3.2.S.3.1 Elucidation of structure and other characteristics – studies to identify polymorphs (exception: where the CEP specifies a polymorphic form) and particle size distribution, where applicable, as per guidance in this section.

c) 3.2.S.4.1 Specification – the specifications of the FPP manufacturer including all tests and limits of the CEP and Ph. Eur. monograph and any additional tests and acceptance criteria that are not controlled in the CEP and Ph. Eur. monograph, such as polymorphs and/or particle size distribution.

d) 3.2.S.4.2/3.2.S.4.3 Analytical procedures and validation – for any tests in addition to those in the CEP and Ph. Eur. monograph.

e) 3.2.S.4.4 Batch analysis – results from two batches of at least pilot scale, demonstrating compliance with the FPP manufacturer's API specifications.

f) 3.2.S.5 Reference standards or materials – information on the FPP manufacturer's reference standards.

g) 3.2.S.6 Container-closure system – specifications including descriptions and identification of primary packaging components.

h) 3.2.S.7 Stability – exception: where the CEP specifies a re-test period that is the same as or of longer duration than the re-test period proposed by the applicant.

i) In the case of sterile APIs, data on the sterilization process of the API, including validation data, should be included in the PD.

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b) Option 2: Active pharmaceutical ingredient pre-qualified by WHO

A complete copy of the Confirmation of API prequalification document should be provided in Module 1, together with the duly filled out authorization box in the name of the FPP manufacturer or applicant.

The applicant should supply the following information in the dossier, with data summarized in the QOS-PD:-

a) 3.2.S.1.3 General properties – discussions on any additional applicable physicochemical and other relevant API properties that are not controlled by the API manufacturer's specifications, e.g. solubilities and polymorphs according to the guidance in this section.

b) 3.2.S.2 – if the sterility of the FPP is based upon the sterile manufacture of the API then data on the sterilization process together with full validation data should be provided.

c) 3.2.S.3.1 Elucidation of structure and other characteristics – studies to identify polymorphs and particle size distribution, where applicable, according to the guidance in this section.

d) 3.2.S.4.1 Specification – the specifications of the FPP manufacturer including all tests and limits of the API manufacturer's specifications and any additional tests and acceptance criteria that are not controlled by the API manufacturer's specifications such as polymorphs and/or particle size distribution.

e) 3.2.S.4.2/3.2.S.4.3 Analytical procedures and validation – any methods used by the FPP manufacturer in addition to those in the API manufacturer's specifications.

f) 3.2.S.4.4 Batch analysis – results from two batches of at least pilot scale, demonstrating compliance with the FPP manufacturer's API specifications.

g) 3.2.S.5 Reference standards or materials – information on the FPP manufacturer's reference standards.

h) 3.2.S.7 Stability – data to support the retest period if either the proposed retest period is longer or the proposed storage conditions are at a higher temperature or humidity to that of the prequalified API.

c) Option 3: Active Pharmaceutical Ingredient Master File Approved by TMDA

i. Option 3 (a): A copy of confirmation of approval of the API by TMDA provided in Module 1, together with the duly filled out authorization box in the name of the FPP manufacturer or applicant.

The applicant should supply the following information in the dossier, with data summarized in the QOS-PD:-

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■ 3.2.S.1.3 General properties – discussions on any additional applicable physicochemical and other relevant API properties that are not controlled by the API manufacturer's specifications, e.g. solubilities and polymorphs according to the guidance in this section.

■ 3.2.S.2 – if the sterility of the FPP is based upon the sterile manufacture of the API then data on the sterilization process together with full validation data should be provided.

■ 3.2.S.3.1 Elucidation of structure and other characteristics – studies to identify polymorphs and particle size distribution, where applicable, according to the guidance in this section.

■ 3.2.S.4.1 Specification – the specifications of the FPP manufacturer including all tests and limits of the API manufacturer's specifications and any additional tests and acceptance criteria that are not controlled by the API manufacturer's specifications such as polymorphs and/or particle size distribution.

■ 3.2.S.4.2/3.2.S.4.3 Analytical procedures and validation – any methods used by the FPP manufacturer in addition to those in the API manufacturer's specifications.

■ 3.2.S.4.4 Batch analysis – results from two batches of at least pilot scale, demonstrating compliance with the FPP manufacturer's API specifications.

■ 3.2.S.5 Reference standards or materials – information on the FPP manufacturer's reference standards.

■ 3.2.S.7 Stability – data to support the retest period if either the proposed retest period is longer or the proposed storage conditions are at a higher temperature or humidity to that of the API approved.

ii. Option 3 (b): Full details on the API information submitted by the API manufacturer, provided that the APIMF contains all information listed under Module 3.

It is the responsibility of the applicant to ensure that the API manufacturer's APIMF restricted part is supplied to TMDA directly by the API manufacturer when required. A copy of the letter of access should be provided in the product dossier in Module 1.

APIMF holders can use the guidance provided for the option "Full details in the" for preparation of the relevant sections of the Open and Restricted parts of their APIMFs.

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d) Option 4: Full details by completing Section 3.2.S.1 - 3.2.S.7 of these guidelines

Information on the 3.2.S Active pharmaceutical ingredient sections, including full details of chemistry, manufacturing process, quality controls during manufacturing and process validation for the API, should be submitted in the FPP dossier as outlined in the subsequent sections of this guideline.

3.2.S.1 General information

3.2.S.1.1 Nomenclature

Information on the nomenclature of the API should be provided. For example:

- International Non-proprietary Name (INN); (Recommended)
- Compendial name, if relevant;
- Chemical name(s);
- Company or laboratory code;
- Other non-proprietary name(s) (e.g., national name, United States Adopted Name (USAN), British Approved Name (BAN)); and
- Chemical Abstracts Service (CAS) registry number.

The listed chemical names should be consistent with those appearing in scientific literature and those appearing on the product labelling information (e.g. summary of product characteristics, package leaflet (also known as patient information leaflet or PIL), labelling). Where several names exist, the preferred name should be indicated.

3.2.S.1.2 Structure

The structural formula, including relative and absolute stereochemistry, the molecular formula and the relative molecular mass should be provided.

This information should be consistent with that provided in section 3.2.S.1.1. For APIs existing as salts, the molecular mass of the free base or acid should also be provided.

3.2.S.1.3 General properties

A list should be provided of physicochemical and other relevant properties of the API.

This information can be used in developing the specifications, in formulating FPPs and in the testing for release and stability purposes.

The physical and chemical properties of the API should be discussed including the physical description, solubilities in common solvents (e.g. water, alcohols, dichloromethane, acetone), quantitative aqueous pH solubility profile (e.g. pH 1.2 to 6.8, dose/solubility volume),

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polymorphism, pH and pKa values, UV absorption maxima and molar absorptivity, melting point, refractive index (for a liquid), hygroscopicity, partition coefficient, etc. (see table in the

QOS). This list is not intended to be exhaustive, but provides an indication as to the type of information that could be included.

Some of the more relevant properties to be considered for APIs are discussed below in greater detail.

Physical description

The description should include appearance, colour and physical state. Solid forms should be identified as being crystalline or amorphous (see 3.2.S.3.1 for further information on API solid forms).

Solubilities/quantitative aqueous pH solubility profile

The following should be provided for all options for the submission of API data.

The solubilities in a number of common solvents should be provided (e.g. water, alcohols, dichloromethane, acetone).

The solubilities over the physiological pH range (pH 1.2 to 6.8) in several buffered media should be provided in mg/ml. If this information is not readily available (e.g. literature references), it should be generated in-house.

For solid oral dosage forms, the dose/solubility volume should be provided as determined by:

Dose/solubility volume = largest dosage strength (mg)
the minimum concentration of the drug (mg/ml)*

* corresponding to the lowest solubility determined over the physiological pH range (pH 1.2 to 6.8) and temperature (37 ± 0.5 °C).

As per the Biopharmaceutics Classification System (BCS), highly soluble (or highly water-soluble) APIs are those with a dose/solubility volume of less than or equal to 250 ml.

For example, compound A has as its lowest solubility at 37 ± 0.5 °C, 1.0 mg/ml at pH 6.8 and is available in 100 mg, 200 mg and 400 mg strengths. This API would not be considered a BCS highly soluble API as its dose/solubility volume is greater than 250 ml ($400 \text{ mg}/1.0 \text{ mg/ml} = 400 \text{ ml}$).

Polymorphism

- a) The polymorphic form(s) present in the proposed API should be listed in section 3.2.S.1.3;
- b) The description of manufacturing process and process controls (3.2.S.2.2) should indicate which polymorphic form is manufactured, where relevant; the literature references or studies performed to identify the potential polymorphic forms of the API, including the study results, should be provided in section 3.2.S.3.1; and if a polymorphic form is to be defined or limited (e.g. for APIs that are not BCS highly soluble and/or where polymorphism has been identified as an issue), details should be included in 3.2.S.4.1 through 3.2.S.4.5.

Additional information is included in the referenced sections of this guideline.

Particle size distribution

Studies performed to identify the particle size distribution of the API should be provided in section 3.2.S.3.1 (refer to this section of this guideline for additional information).

Information from literature

Supportive data and results from specific studies or published literature can be included within or attached to this section.

3.2.S.2 Manufacture

3.2.S.2.1 Manufacturer(s) (name, physical address)

The name, address, and responsibility of each manufacturer, including contractors, and each proposed production site or facility involved in manufacturing and testing should be provided.

The facilities involved in the manufacturing, packaging, labelling, testing and storage of the API should be listed. If certain companies are responsible only for specific steps (e.g. milling of the API) it should be clearly indicated.

The list of manufacturers/companies should specify the actual addresses of production or manufacturing site(s) involved (including block(s) and units(s)), rather than the administrative offices. Telephone number(s), fax number(s) and e-mail address(es) should be provided.

A valid manufacturing authorization should be provided for the production of APIs. If available, a certificate of GMP compliance should be provided in the product dossier Module 1.

3.2.S.2.2 Description of manufacturing process and process controls

The description of the API manufacturing process represents the applicant's commitment for

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the manufacture of the API. Information should be provided to adequately describe the manufacturing process and process controls. For example, a flow diagram of the synthetic process (es) should be provided that includes molecular formulae, weights, yield ranges,

chemical structures of starting materials, intermediates, reagents and API reflecting stereochemistry, and identifies operating conditions and solvents.

A sequential procedural narrative of the manufacturing process should be submitted. The narrative should include, for example, quantities of raw materials, solvents, catalysts and reagents reflecting the representative batch scale for commercial manufacture, identification of critical steps, process controls, equipment and operating conditions (e.g. temperature, pressure, pH, time).

Alternate processes should be explained and described with the same level of detail as the primary process. Reprocessing steps should be identified and justified. Any data to support this justification should be either referenced or filed in 3.2.S.2.5.

The following requirements apply to the second option for submission of API information, where full details are provided in the dossier.

The API starting material should be fully characterized with respect to identity and purity. The starting material for synthesis defines the starting point in the manufacturing process for an API to be described in an application. The applicant should propose and justify which substances should be considered as starting materials for synthesis. See section 3.2.S.2.3 for further guidance.

The recovery of materials, if any, should be described in detail with the step in which they are introduced into the process. Recovery operations should be adequately controlled such that impurity levels do not increase over time. For recovery of solvents, any processing to improve the quality of the recovered solvent should be described. Regarding recycling of filtrates (mother liquors) to obtain second crops, information should be available on maximum holding times of mother liquors and maximum number of times the material can be recycled. Data on impurity levels should be provided to justify recycling of filtrates.

Where there are multiple manufacturing sites for one API manufacturer, a comprehensive list in tabular form should be provided comparing the processes at each site and highlighting any differences.

All solvents used in the manufacture (including purification and/or crystallization step(s)) should be clearly identified. Solvents used in the final steps should be of high purity. Use of recovered solvents in the final steps of purification and/or crystallization is not recommended.

Where particle size is considered a critical attribute (see 3.2.S.3.1 for details), the particle size reduction method(s) (milling, micronization) should be described.

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Justification should be provided for alternate manufacturing processes. Alternate processes should be explained with the same level of detail as the primary process. It should be demonstrated that batches obtained by the alternate processes have the same impurity profile as the principal process. If the obtained impurity profile is different it should be demonstrated to be acceptable according to the requirements described under S.3.2.

3.2.S.2.3 Control of materials

Materials used in the manufacture of the API (e.g. raw materials, starting materials, solvents, reagents, catalysts) should be listed, identifying where each material is used in the process. Information on the quality and control of these materials should be provided. Information demonstrating that materials meet standards appropriate for their intended use should be provided.

In general, the starting material for synthesis described in the marketing authorization dossier should:

- be a synthetic precursor of one or more synthesis steps prior to the final API intermediate. Acids, bases, salts, esters and similar derivatives of the API, as well as the racemate of a single enantiomer API, are not considered final intermediates;
- be a well characterized, isolated and purified substance with its structure fully elucidated including its stereochemistry (when applicable);
- have well-defined specifications that include among others one or more specific identity tests and tests and limits for assay and specified, unspecified and total impurities; and
- be incorporated as a significant structural fragment into the structure of the API.

Copies of the specifications for the materials used in the synthesis, extraction, isolation and purification steps should be provided in the PD, including starting materials, reagents, solvents, catalysts and recovered materials. Confirmation should be provided that the specifications apply to materials used at each manufacturing site. A certificate of analysis of the starting material for synthesis should be provided. A summary of the information on starting materials should be provided in the QOS-PD.

The carry-over of impurities of the starting materials for synthesis into the final API should be considered and discussed.

A letter of attestation should be provided confirming that the API and the starting materials and reagents used to manufacture the API are without risk of transmitting agents of animal spongiform encephalopathies.

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3.2.S.2.4 Controls of critical steps and intermediates

Critical steps: Tests and acceptance criteria (with justification including experimental data) performed at critical steps identified in 3.2.S.2.2 of the manufacturing process to ensure that the process is controlled should be provided.

Intermediates: Information on the quality and control of intermediates isolated during the process should be provided.

The critical steps should be identified. These can be among others: steps where significant impurities are removed or introduced, steps introducing an essential molecular structural element such as a chiral centre or resulting in a major chemical transformation, steps having an impact on solid-state properties and homogeneity of the API that may be relevant for use in solid dosage forms.

Specifications for isolated intermediates should be provided and should include tests and acceptance criteria for identity, purity and assay, where applicable.

3.2.S.2.5 Process validation and/or evaluation

Process validation and/or evaluation studies for aseptic processing and sterilization should be included.

It is expected that the manufacturing processes for all APIs are properly controlled. If the API is prepared as sterile, a complete description should be provided for aseptic processing and/or sterilization methods. The controls used to maintain the sterility of the API during storage and transportation should also be provided. Alternate processes should be justified and described.

3.2.S.3 Characterization

3.2.S.3.1 Elucidation of structure and other characteristics

Confirmation of structure based on e.g. synthetic route and spectral analyses should be provided. Information such as the potential for isomerism, the identification of stereochemistry or the potential for forming polymorphs should also be included.

Elucidation of structure

The MA application should include quality assurance (QA) certified copies of the spectra, peak assignments and a detailed interpretation of the data of the studies performed to elucidate and/or confirm the structure of the API. The QOS should include a list of the studies performed and a conclusion from the studies (e.g. if the results support the proposed structure).

For APIs that are not described in an officially recognized pharmacopoeia, the studies carried

out to elucidate and/or confirm the chemical structure normally include elemental analysis, infrared (IR), ultraviolet (UV), nuclear magnetic resonance (NMR) and mass spectra (MS) studies. Other tests could include X-ray powder diffraction and differential scanning calorimetry (DSC).

For APIs that are described in an officially recognized pharmacopoeia, it is generally sufficient to provide copies of the IR spectrum of the API from each of the proposed manufacturer(s) run concomitantly with a pharmacopoeial reference standard.

Isomerism/Stereochemistry

Where the potential for stereoisomerism exists, a discussion should be included of the possible isomers that can result from the manufacturing process and the steps where chirality was introduced. The identity of the isomeric composition of the API to that of the API in the comparator product should be established. Information on the physical and chemical properties of the isomeric mixture or single enantiomer should be provided, as appropriate. The API specification should include a test to ensure isomeric identity and purity.

The potential for inter-conversion of the isomers in the isomeric mixture, or racemization of the single enantiomer should be discussed.

When a single enantiomer of the API is claimed for non-pharmacopoeial APIs, unequivocal proof of absolute configuration of asymmetric centres should be provided such as determined by X-ray of a single crystal.

If, based on the structure of the API, there is not a potential for stereoisomerism, it is sufficient to include a statement to this effect.

Polymorphism

Many APIs can exist in different physical forms in the solid state. Polymorphism is characterized as the ability of an API to exist as two or more crystalline phases that have different arrangements and/or conformations of the molecules in the crystal lattice. Amorphous solids consist of disordered arrangements of molecules and do not possess a distinguishable crystal lattice. Solvates are crystal forms containing either stoichiometric or non-stoichiometric amounts of a solvent. If the incorporated solvent is water the solvates are also commonly known as hydrates.

Polymorphic forms of the same chemical compound differ in internal solid-state structure and, therefore, may possess different chemical and physical properties, including packing, thermodynamic, spectroscopic, kinetic, interfacial and mechanical properties. These properties can have a direct impact on API processability, pharmaceutical product manufacturability and product quality/performance, including stability, dissolution and bioavailability. Unexpected appearance or disappearance of a polymorphic form may lead to serious pharmaceutical consequences.

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Applicants and API manufacturers are expected to have adequate knowledge about the polymorphism of the APIs used and/or produced. Information on polymorphism can come from the scientific literature, patents, compendia or other references to determine if polymorphism is a concern, e.g. for APIs that are not BCS highly soluble. In the absence of published data for APIs that are not BSC highly soluble, polymorphic screening will be

necessary to determine if the API can exist in more than one crystalline form. Polymorphic screening is generally accomplished via crystallization studies using different solvents and conditions.

There are a number of methods that can be used to characterize the polymorphic forms of an API. Demonstration of a non-equivalent structure by single crystal X-ray diffraction is currently regarded as the definitive evidence of polymorphism. X-Ray diffraction can also be used to provide unequivocal proof of polymorphism. Other methods, including microscopy, thermal analysis (e.g. DSC, thermal gravimetric analysis and hot-stage microscopy) and spectroscopy (e.g. IR, Raman, solid-state nuclear magnetic resonance (ssNMR)) is helpful to further characterize polymorphic forms. Where polymorphism is a concern, the applicants/manufacturers of APIs should demonstrate that a suitable method, capable of distinguishing different polymorphs, is available to them.

Polymorphism can also include solvation or hydration products (also known as pseudopolymorphs). If the API is used in a solvated form, the following information should be provided:

- a) Specifications for the solvent-free API in 3.2.S.2.4, if that compound is a synthetic precursor;
 - b) Specifications for the solvated API including appropriate limits on the weight ratio API to solvent (with data to support the proposed limits);
 - c) A description of the method used to prepare the solvate in 3.2.S.2.2.
- Particle size distribution

For APIs whose particle size distribution will have influence on FPP processability, stability, content uniformity, dissolution and bioavailability, specifications should include controls on the particle size distribution.

3.2.S.3.2 Impurities

Information on impurities should be provided.

Details on the principles for the control of impurities (e.g. reporting, identification and qualification) are outlined in the ICH Q3A and Q3C impurity guidelines. Discussion should be provided of the potential and actual impurities arising from the synthesis, manufacture or degradation of the API. This should cover starting materials, by-products, intermediates, chiral impurities and degradation products and should include the chemical names, structures

and origins. The discussion of pharmacopoeial APIs should not be limited to the impurities specified in the API monograph.

Refer: ICH Q3A: Impurities in New Drug Substances and ICH Q3C Impurities: Guideline for Residual Solvents

3.2.S.4 Control of the API

3.2.S.4.1 Specification

The specification for the API should be provided. Copies of the API specifications, dated and signed by authorized personnel (e.g. the person in charge of the quality control or quality assurance department) should be provided in the marketing authorization dossier, including specifications from each API manufacturer as well as those of the FPP manufacturer.

The FPP manufacturer's API specification should be summarized according to the table in the QOS template under the headings tests, acceptance criteria and analytical procedures (including types, sources and versions for the methods).

- a) The standard declared by the applicant could be an officially recognized compendial standard (BP, JP, Ph.Eur, Ph.Int. and USP) or a house (manufacturer's) standard.
- b) The specification reference number and version (e.g. revision number and/or date) should be provided for version control purposes.
- c) For the analytical procedures, the type should indicate the kind of analytical procedure used (e.g. visual, IR, UV, HPLC, laser diffraction), the source refers to the origin of the analytical procedure (BP, JP, Ph.Eur, Ph.Int, USP, in-house) and the version (e.g. code number/version/date) should be provided for version control purposes.

In cases where there is more than one API manufacturer, the FPP manufacturer's API specifications should be one single compiled set of specifications that is identical for each manufacturer. It is acceptable to lay down in the specification more than one acceptance criterion and/or analytical method for a single parameter with the statement "for API from manufacturer A" (e.g. in the case of residual solvents).

Any non-routine testing should be clearly identified as such and justified along with the proposal on the frequency of non-routine testing.

3.2.S.4.2 Analytical procedures

The analytical procedures used for testing the API should be provided. Copies of the in-house analytical procedures used to generate testing results provided in the PD, as well as those proposed for routine testing of the API by the FPP manufacturer should be provided. Unless modified, it is not necessary to provide copies of officially recognized compendial analytical procedures.

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3.2.S.4.3 Validation of analytical procedures

Analytical validation information, including experimental data for the analytical procedures used for testing the API, should be provided.

Copies of the validation reports for the analytical procedures used to generate testing results

provided in the PD, as well as those proposed for routine testing of the API by the FPP manufacturer, should be provided.

Tables should be used to summarize the validation information of the analytical procedures of the FPP manufacturer for determination of residual solvents, assay and purity of the API, in section 2.3.S.4.3 of the QOS. The validation data for other methods used to generate assay and purity data in the PD can be summarized in 2.3.S.4.4 (c) or 2.3.S.7.3 (b) of the QOS.

The compendial methods as published are typically validated based on an API or an FPP originating from a specific manufacturer. Different sources of the same API or FPP can contain impurities and/or degradation products that were not considered during the development of the monograph. Therefore, the monograph and compendial method should be demonstrated suitable to control the impurity profile of the API from the intended source(s).

In general verification is not necessary for compendial API assay methods. However, specificity of a specific compendial assay method should be demonstrated if there are any potential impurities that are not specified in the compendial monograph. If an officially recognized compendial method is used to control API-related impurities that are not specified in the monograph, full validation of the method is expected with respect to those impurities.

If an officially recognized compendial standard is claimed and an in-house method is used in lieu of the compendial method (e.g. for assay or for specified impurities), equivalency of the in-house and compendial methods should be demonstrated. This could be accomplished by performing duplicate analyses of one sample by both methods and providing the results from the study. For impurity methods, the sample analyzed should be the API spiked with impurities at concentrations equivalent to their specification limits.

Refer ICHQ2: Validation of Analytical Procedures: Text and Methodology for more guidance

3.2.S.4.4 Batch analyses

Description of batches and results of batch analyses should be provided. The information provided should include batch number, batch size, date and production site of relevant API batches.

Copies of the certificates of analysis, both from the API manufacturer(s) and the FPP manufacturer, should be provided for the profiled batches and any company responsible for generating the test results should be identified. This data is used to evaluate consistency in API quality. The FPP manufacturer's test results should be summarized in the QOS.

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For quantitative tests (e.g. individual and total impurity tests and assay tests), it should be ensured that actual numerical results are provided rather than vague statements such as "within limits" or "conforms".

A discussion and justification should be provided for any incomplete analyses (e.g. results not tested according to the proposed specification).

3.2.S.4.5 Justification of specification

Justification for the API specification should be provided.

A discussion should be provided on the inclusion of certain tests, evolution of tests, analytical procedures and acceptance criteria, differences from the officially recognized compendial standard(s), etc. If the officially recognized compendial methods have been modified or replaced, a discussion should be included.

The justification for certain tests, analytical procedures and acceptance criteria may have been discussed in other sections of the PD (e.g. impurities, particle-size distribution) and does not need to be repeated here, although a cross-reference to their location should be provided.

Refer ICH Q6A Specifications: Test Procedures and Acceptance Criteria for New Drug Substances and New Drug Products: Chemical Substances, for more guidance

3.2.S.5 Reference standards or materials

Information on the reference standards or reference materials used for testing of the API should be provided. Information should be provided on the reference standard(s) used to generate data in the PD, as well as those to be used by the FPP manufacturer in routine API and FPP testing.

The source(s) of the reference standards or materials used in the testing of the API should be provided (e.g. those used for the identification, purity, assay tests). These could be classified as primary or secondary reference standards.

A suitable primary reference standard should be obtained from an officially recognized pharmacopoeial source (BP, JP, Ph.Eur, Ph.Int, USP) where one exists and the lot number should be provided. Primary reference standards from officially recognized pharmacopoeial sources do not need further structural elucidation.

Otherwise a primary standard may be a batch of the API that has been fully characterized (e.g. by IR, UV, NMR, MS analyses). Further purification techniques may be needed to render the material acceptable for use as a chemical reference standard. The purity requirements for a chemical reference substance depend upon its intended use. A chemical reference substance

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proposed for an identification test does not require meticulous purification, since the presence of a small percentage of impurities in the substance often has no noticeable effect on the test. On the other hand, chemical reference substances that are to be used in assays should possess a high degree of purity (such as 99.5% on the dried or water-/solvent-free basis). Absolute content of the primary reference standard must be declared and should follow the scheme:

100% minus organic impurities (quantitated by an assay procedure, e.g. HPLC, DSC, etc.)
minus inorganic impurities minus volatile impurities by loss on drying (or water content
minus residual solvents).

A secondary (or in-house) reference standard can be used by establishing it against a suitable primary reference standard, e.g. by providing legible copies of the IR of the primary and secondary reference standards run concomitantly and by providing its certificate of analysis, including assay determined against the primary reference standard. A secondary reference standard is often characterized and evaluated for its intended purpose with additional procedures other than those used in routine testing (e.g. if additional solvents are used during the additional purification process that are not used for routine purposes).

3.2.S.6 Container-closure system

A description of the container-closure system(s) should be provided, including the identity of materials of construction of each primary packaging component, and their specifications. The specifications should include description and identification (and critical dimensions with drawings, where appropriate). Non compendial methods (with validation) should be included, where appropriate.

For non-functional secondary packaging components (e.g. those that do not provide additional protection), only a brief description should be provided. For functional secondary packaging components, additional information should be provided.

The suitability should be discussed with respect to, for example, choice of materials, protection from moisture and light, compatibility of the materials of construction with the API, including sorption to container and leaching, and/or safety of materials of construction.

Primary packaging components are those that are in direct contact with the API or FPP. The specifications for the primary packaging components should be provided and should include a specific test for identification (e.g. IR).

Copies of the labels applied on the secondary packaging of the API should be provided and should include the conditions of storage. In addition, the name and address of the manufacturer of the API should be stated on the container, regardless of whether re-labelling is conducted at any stage during the API distribution process.

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3.2.S.7 Stability

Refer Guidelines on Stability Requirements for Testing Active Pharmaceutical Ingredients (APIs) and Finished Pharmaceutical Products (FPPs)

3.2.P Finished pharmaceutical product (FPP)

3.2. P.1 Description and Composition of the FPP

A description of the FPP and its composition should be provided. The information provided should include:

Description of the dosage form

The description of the FPP should include the physical description, available strengths, release mechanism (e.g. immediate, modified (delayed or extended)), as well as any other distinguishable characteristics.

Composition

This is a list of all components of the dosage form, and their amount on a per unit basis (including overages, if any), the function of the ingredients, and a reference to their quality standards [e.g. Compendial monographs (BP, USP, JP, Ph. Eur etc) or manufacturer's specifications (IH)].

The tables in the QOS template should be used to summarize the composition of the FPP and express the quantity of each component on a per unit basis (e.g. mg per tablet, mg per ml, mg per vial) and quantity per batch. The individual ingredient for mixtures prepared in-house (e.g. coatings) should be included in the tables, where applicable.

All ingredients used in the manufacturing process should be included, including those that may not be added to every batch (e.g. acid and alkali), those that may be removed during processing (e.g. solvents) and any others (e.g. nitrogen, silicon for stoppers). If the FPP is formulated using an active moiety, then the composition for the active ingredient should be clearly indicated (e.g. "1 mg of active ingredient base = 1.075 mg active ingredient hydrochloride"). All overages should be clearly indicated (e.g. "contains 2% overage of the API to compensate for manufacturing losses").

The ingredients should be declared by their proper or common names, quality standards (BP, JP, Ph.Eur, Ph.Int, USP, in-house) and, if applicable, their grades (e.g. "Microcrystalline Cellulose NF (PH 102)") and special technical characteristics (e.g. lyophilized, micronized, solubilized, emulsified).

The function of each component (e.g. diluent/filler, binder, disintegrant, lubricant, glidant, granulating solvent, coating agent, antimicrobial preservative) should be stated. If an excipient performs multiple functions, the predominant function should be indicated.

• Description of accompanying reconstitution diluent(s)

For FPPs supplied with reconstitution diluent(s) that have been assessed and considered acceptable (registered) in connection with another product dossier, a brief description of the reconstitution diluents(s) should be provided.

For FPPs supplied with reconstitution diluent(s) have not been assessed and considered acceptable in connection with another product dossier, the information on the diluent(s) should be provided in a separate FPP portion ("3.2.P"), as appropriate.

- Type of container and closure used for the dosage form and accompanying reconstitution diluent, if applicable

The container-closure used for the FPP (and accompanying reconstitution diluent, if applicable) should be briefly described, with further details provided under 3.2.P.7 Container-closure system, e.g. "The product is available in HDPE bottles with polypropylene caps (in sizes of 100s, 500s and 1000s) and in PVC/aluminium foil unit dose blisters (in packages of 100s) (cards of 5 × 2, 10 cards per package)."

3.2.P.2 Pharmaceutical development

The Pharmaceutical development section should contain information on the development studies conducted to establish that the dosage form, the formulation, manufacturing process, container-closure system, microbiological attributes and usage instructions are appropriate for the purpose specified in the product dossier. The studies described here are distinguished from routine control tests conducted according to specifications. Additionally, this section should identify and describe the formulation and process attributes (critical parameters) that can influence batch reproducibility, product performance and FPP quality. Supportive data and results from specific studies or published literature can be included within or attached to the Pharmaceutical development section. Additional supportive data can be referenced to the relevant nonclinical or clinical sections of the product dossier.

Pharmaceutical development information should include, at a minimum:

- a) the definition of the quality target product profile (QTPP) as it relates to quality, safety and efficacy, considering for example the route of administration, dosage form, bioavailability, strength and stability;
- b) identification of the potential critical quality attributes (CQAs) of the FPP so as to adequately control the product characteristics that could have an impact on quality;
- c) discussion of the potential CQAs of the API(s), excipients and container-closure system(s) including the selection of the type, grade and amount to deliver drug product of the desired quality; and
- d) discussion of the selection criteria for the manufacturing process and the control strategy required to manufacture commercial lots meeting the QTPP in a consistent manner.

These features should be discussed as part of the product development using the principles of risk management over the entire life-cycle of the product.

3.2.P.2.1 Components of the FPP

3.2.P.2.1.1 Active pharmaceutical ingredient

The compatibility of the API with excipients listed in 3.2.P.1 should be discussed. Additionally, key physicochemical characteristics (e.g. water content, solubility, particle size distribution, polymorphic or solid state form) of the API that can influence the performance of the FPP should be discussed. For fixed-dose combinations, the compatibility of APIs with each other should be discussed.

Physicochemical characteristics of the API may influence both the manufacturing capability and the performance of the FPP.

3.2.P.2.1.2 Excipients

The choice of excipients listed in 3.2.P.1, their concentration and their characteristics that can influence the FPP performance should be discussed relative to their respective functions.

3.2.P.2.2 Finished pharmaceutical product

3.2.P.2.2.1 Formulation development

A brief summary describing the development of the FPP should be provided, taking into consideration the proposed route of administration and usage. The differences between the comparative bioavailability or biowaiver formulations and the formulation (i.e. composition) described in 3.2.P.1 should be discussed. Results from comparative in vitro studies (e.g. dissolution) or comparative in vivo studies (e.g. bioequivalence) should be discussed when appropriate.

If the proposed FPP is a functionally scored tablet, a study should be undertaken to ensure the uniformity of dose in the tablet fragments. The data provided in the PD should include a description of the test method, individual values, mean and relative standard deviation (RSD) of the results. Uniformity testing (i.e. content uniformity or mass variation, depending on the requirement for the whole tablet) should be performed on each split portion from a minimum of 10 randomly selected whole tablets.

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In vitro dissolution or drug release

A discussion should be included as to how the development of the formulation relates to development of the dissolution method(s) and the generation of the dissolution profile.

The results of studies justifying the choice of in vitro dissolution or drug release conditions (e.g. apparatus, rotation speed, medium) should be provided.

Data should also be submitted to demonstrate whether the method is sensitive to changes in manufacturing processes and/or changes in grades and/or amounts of critical excipients and particle size where relevant. The dissolution method should be sensitive to any changes in the product that would result in a change in one or more of the pharmacokinetic parameters.

For slower dissolving immediate-release products (e.g. Q = 80% in 90 minutes), a second time point may be warranted (e.g. Q = 60% in 45 minutes).

Modified-release FPPs should have a meaningful in vitro release rate (dissolution) test that is used for routine quality control. Preferably this test should possess in vitro–in vivo correlation. Results demonstrating the effect of pH on the dissolution profile should be submitted if appropriate for the type of dosage form.

For extended-release FPPs, the testing conditions should be set to cover the entire time period of expected release (e.g. at least three test intervals chosen for a 12-hour release and additional test intervals for longer duration of release). One of the test points should be at the early stage of drug release (e.g. within the first hour) to demonstrate absence of dose dumping. At each test point, upper and lower limits should be set for individual units. Generally the acceptance range at each intermediate test point should not exceed 25% or 12.5% of the targeted value. Dissolution results should be submitted for several lots, including those lots used for pharmacokinetic and bioavailability or biowaiver studies. Recommendations for conducting and assessing comparative dissolution profiles can be found in the Guidelines on Therapeutic Equivalence Requirements.

3.2.P.2.2.2 Overages

Any overages in the formulation(s) described in 3.2.P.1 should be justified. Justification of an overage to compensate for loss during manufacture should be provided, including the step(s) where the loss occurs, the reasons for the loss and batch analysis release data (assay results).

Overages for the sole purpose of extending the shelf-life of the FPP are generally not acceptable.

3.2. P.2.2.3 Physicochemical and biological properties

Parameters relevant to the performance of the FPP, such as pH, ionic strength, dissolution, re-dispersion, reconstitution, particle size distribution, aggregation, polymorphism, rheological properties, biological activity or potency and/or immunological activity, should be addressed.

3.2.P.2.3 Manufacturing process development

The selection and optimization of the manufacturing process described in 3.2.P.3.3, in particular its critical aspects, should be explained. Where relevant, the method of sterilization should be explained and justified.

Where relevant, justification for the selection of aseptic processing or other sterilization methods over terminal sterilization should be provided.

Differences between the manufacturing process(es) used to produce comparative bioavailability or bio-waiver batches and the process described in 3.2.P.3.3 that can influence the performance of the product should be discussed.

The scientific rationale for the selection, optimization and scale-up of the manufacturing process described in 3.2.P.3.3 should be explained; in particular the critical aspects (e.g. rate of addition of granulating fluid, massing time, granulation end-point). A discussion of the critical process parameters (CPP), controls and robustness with respect to the QTPP and CQA of the product should be included.

3.2.P.2.4 Container-closure system

The suitability of the container-closure system (described in 3.2.P.7) used for the storage, transportation (shipping) and use of the FPP should be discussed. This discussion should consider, e.g. choice of materials, protection from moisture and light, compatibility of the materials of construction with the dosage form (including sorption to container and leaching) safety of materials of construction and performance (such as reproducibility of the dose delivery from the device when presented as part of the FPP).

The suitability of the container-closure system used for the storage, transportation (shipping) and use of any intermediate/in-process products (e.g. premixes, bulk FPP) should also be discussed.

3.2.P.2.5 Microbiological attributes

Where appropriate the microbiological attributes of the dosage form should be discussed, including, for example, the rationale for not performing microbial limits testing for non-sterile products and the selection and effectiveness of preservative systems in products containing antimicrobial preservatives. For sterile products the integrity of the container-closure system to prevent microbial contamination should be addressed.

Where an antimicrobial preservative is included in the formulation, the amount used should be justified by submission of results of the product formulated with different concentrations of the preservative(s) to demonstrate the least necessary but still effective concentration. The effectiveness of the agent should be justified and verified by appropriate studies (e.g. USP or Ph.Eur general chapters on antimicrobial preservatives) using a batch of the FPP. If the lower limit for the proposed acceptance criterion for the assay of the preservative is less than 90.0%, the effectiveness of the agent should be established with a batch of the FPP containing a concentration of the antimicrobial preservative corresponding to the lower proposed acceptance criteria.

3.2.P.2.6 Compatibility

The compatibility of the FPP with reconstitution diluent(s) or dosage devices (e.g. precipitation of API in solution, sorption on injection vessels, stability) should be addressed to provide appropriate and supportive information for the labelling.

Where a device is required for oral liquids or solids (e.g. solutions, emulsions, suspensions and powders/granules for such reconstitution) that are intended to be administered immediately after being added to the device, the compatibility studies mentioned in the following paragraphs are not required.

Where sterile, reconstituted products are to be further diluted, compatibility should be demonstrated with all diluents over the range of dilution proposed in the labelling. These studies should preferably be conducted on aged samples. Where the labelling does not specify the type of containers, compatibility (with respect to parameters such as appearance, pH, assay, levels of individual and total degradation products, sub-visible particulate matter and extractables from the packaging components) should be demonstrated in glass, PVC and polyolefin containers. However, if one or more containers are identified in the labelling, compatibility of admixtures needs to be demonstrated only in the specified containers.

Studies should cover the duration of storage reported in the labelling (e.g. 24 hours under controlled room temperature and 72 hours under refrigeration). Where the labelling specifies co-administration with other FPPs, compatibility should be demonstrated with respect to the principal FPP as well as the co-administered FPP (i.e. in addition to other aforementioned parameters for the mixture, the assay and degradation levels of each co-administered FPP should be reported).

Refer ICH Q8 guidelines: Pharmaceutical Development for more guidance

Note: For an established non sterile generic product, a product quality review may satisfy the requirements of sections 3.2.P.2.2.1 (a), 3.2.P.2.3 (a) of the PD and QOS (See Annex VIII)

3.2.P.3 Manufacture

3.2.P.3.1 Manufacturer(s) (name, physical address)

The name, address and responsibility of each manufacturer, including contractors, and each proposed production site or facility involved in manufacturing and testing should be provided.

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The facilities involved in the manufacturing, packaging, labelling and testing should be listed. If certain companies are responsible only for specific steps (e.g. manufacturing of an intermediate) it should be clearly indicated. The list of manufacturers/companies should specify the actual addresses of production or manufacturing site(s) involved (including block(s) and unit(s)), rather than the administrative offices.

A valid manufacturing authorization for pharmaceutical production, as well as a marketing authorization, should be submitted to demonstrate whether that the product is registered or licensed in accordance with national requirements. Attach a WHO-type certificate of GMP.

Regulatory situation in other countries

The countries should be listed in which this product has been granted a marketing authorization (attach evidence for marketing authorization), this product has been withdrawn from the market and/or this application for marketing has been rejected, deferred or withdrawn. (Module 1, 1.10 Regulatory Status).

3.2.P.3.2 Batch formula

A batch formula should be provided that includes a list of all components of the dosage form to be used in the manufacturing process, their amounts on a per batch basis, including overages, and a reference to their quality standards.

The tables in the QOS template should be used to summarize the batch formula of the FPP for each proposed commercial batch size and express the quantity of each component on a per batch basis, including a statement of the total weight or measure of the batch.

All ingredients used in the manufacturing process should be included, including those that may not be added to every batch (e.g. acid and alkali), those that may be removed during processing (e.g. solvents) and any others (e.g. nitrogen, silicon for stoppers). If the FPP is formulated using an active moiety, then the composition for the active ingredient should be clearly indicated (e.g. "1 kg of active ingredient base = 1.075 kg active ingredient hydrochloride"). All overages should be clearly indicated (e.g. "Contains 5 kg (corresponding to 2%) overage of the API to compensate for manufacturing losses").

The ingredients should be declared by their proper or common names, quality standards (e.g. BP, JP, Ph.Eur, Ph.Int, USP, house) and, if applicable, their grades (e.g. "Microcrystalline Cellulose NF (PH 102)") and special technical characteristics (e.g. lyophilized, micronized, solubilized, emulsified).

3.2.P.3.3 Description of manufacturing process and process controls

A flow diagram should be presented giving the steps of the process and showing where

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materials enter the process. The critical steps and points at which process controls, intermediate tests or final product controls are conducted should be identified.

A narrative description of the manufacturing process, including packaging that represents the sequence of steps undertaken and the scale of production should also be provided. Novel processes or technologies and packaging operations that directly affect product quality should be described with a greater level of detail. Equipment should, at least, be identified by type (e.g. tumble blender, in-line homogenizer) and working capacity, where relevant.

Steps in the process should have the appropriate process parameters identified, such as time, temperature or pH. Associated numeric values can be presented as an expected range. Numeric ranges for critical steps should be justified in section 3.2.P.3.4. In certain cases, environmental conditions (e.g. low humidity for an effervescent product) should be stated.

The maximum holding time for bulk FPP prior to final packaging should be stated. The holding time should be supported by the submission of stability data, if longer than 30 days. For an aseptic FPP, the holding time of the filtered product prior to filling should be supported by the submission of stability data, if longer than 24 hours.

Proposals for the reprocessing of materials should be justified. Any data to support this justification should be either referenced or filed in this section.

Provide a copy of the master formula and a copy of a manufacturing record for a real batch.

3.2.P.3.4 Controls of critical steps and intermediates

Critical steps: tests and acceptance criteria should be provided (with justification, including experimental data) performed at the critical steps identified in 3.2.P.3.3 of the manufacturing process, to ensure that the process is controlled.

Intermediates: information on the quality and control of intermediates isolated during the process should be provided.

Examples of applicable in-process controls include:

- (a) Granulations: moisture (limits expressed as a range), blend uniformity (e.g. low-dose tablets), bulk and tapped densities and particle size distribution;
- (b) Solid oral products: average weight, weight variation, hardness, thickness, friability, and disintegration checked periodically throughout compression, weight gain during coating;
- (c) Semi-solids: viscosity, homogeneity, pH;
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- (d) Transdermal dosage forms: assay of API–adhesive mixture, weight per area of coated patch without backing;
- (e) Metered dose inhalers: fill weight or volume, leak testing, valve delivery;
- (f) Dry powder inhalers: assay of API–excipient blend, moisture, weight variation of individually contained doses such as capsules or blisters;
- (g) Liquids: pH, specific gravity, clarity of solutions;
- (h) Parenterals: appearance, clarity, fill volume or weight, pH, filter integrity tests, particulate matter, leak testing of ampoules, pre-filtration and/or pre-sterilization bio-burden testing.

3.2.P.3.5 Process validation and/or evaluation

Description, documentation and results of the validation and/or evaluation studies should be provided for critical steps or critical assays used in the manufacturing process (e.g. validation of the sterilization process or aseptic processing or filling).

A product quality review may be submitted in lieu of the information below.

The following information should be provided:

- a) A copy of the process validation protocol, specific to this FPP, that identifies the critical equipment and process parameters that can affect the quality of the FPP and defines testing parameters, sampling plans, analytical procedures and acceptance criteria;
- b) A commitment that three consecutive, production-scale batches of this FPP will be subjected to prospective validation in accordance with the above protocol. The applicant should submit a written commitment that information from these studies will be available for verification.
- c) Validation information relating to the adequacy and efficacy of any sterilization process (e.g. pharmaceutical product, packaging component should be submitted).

The process validation protocol should include inter alia the following:

- a) A reference to the current master production document;
- b) A discussion of the critical equipment;
- c) The process parameters that can affect the quality of the FPP (critical process parameters (CPPs)) including challenge experiments and failure mode operation;

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- d) Details of the sampling: sampling points, stages of sampling, methods of sampling and the sampling plans (including schematics of blender/storage bins for uniformity testing of the final blend);
- e) The testing parameters/acceptance criteria including in-process and release specifications and including comparative dissolution profiles of validation batches against the batch(es) used in the bioavailability or biowaiver studies;
- f) The analytical procedures or a reference to appropriate section(s) of the dossier;
- g) The methods for recording/evaluating results; and
- h) The proposed timeframe for completion of the protocol.

The manufacture of sterile FPPs needs a well-controlled manufacturing area (e.g. a strictly controlled environment, highly reliable procedures and appropriate in-process controls). A detailed description of these conditions, procedures and controls should be provided.

The sterilization process should be described in detail and evidence should be provided to confirm that it will produce a sterile product with a high degree of reliability and that the physical and chemical properties as well as the safety of the FPP will not be affected. Details such as temperature range and peak dwell time for an FPP and the container-closure should be provided. Although standard autoclaving cycles of 121 °C for 15 minutes or more would not need a detailed rationale, such justifications should be provided for reduced temperature cycles or elevated temperature cycles with shortened exposure times. If ethylene oxide is used, studies and acceptance criteria should control the levels of residual ethylene oxide and related compounds.

Filters used should be validated with respect to pore size, compatibility with the product, absence of extractables and lack of adsorption of the API or any of the components.

For the validation of aseptic filling of parenteral products that cannot be terminally sterilized, simulation process trials should be conducted. This involves filling ampoules with culture media under normal conditions, followed by incubation and control of microbial growth. Results on microbial contamination levels should be provided.

Note: For an established generic product a product quality review may satisfy the requirements of sections 3.2.P.3.5 of the PD and QOS (Annex VIII).

Refer FDA Guidance for Industry Process Validation: General Principles and Practices for more guidance at: <http://www.fda.gov/downloads/Drugs/.../Guidances/UCM070336.pdf>

3.2.P.4 Control of excipients

3.2.P.4.1 Specifications

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The specifications for excipients should be provided.

The specifications from the FPP manufacturer should be provided for all excipients, including those that may not be added to every batch (e.g. acid and alkali), those that do not appear in the final FPP (e.g. solvents) and any others used in the manufacturing process (e.g. nitrogen, silicon for stoppers).

If the standard claimed for an excipient is an officially recognized compendial standard, it is sufficient to state that the excipient is tested according to the requirements of that standard, rather than reproducing the specifications found in the officially recognized compendial monograph.

If the standard claimed for an excipient is a non-compendial standard (e.g. house standard) or includes tests that are supplementary to those appearing in the officially recognized compendial monograph, a copy of the specification for the excipient should be provided.

For excipients of natural origin, microbial limit testing should be included in the specifications.

For oils of plant origin (e.g. soy bean, peanut) the absence of aflatoxins or biocides should be demonstrated.

The colours permitted for use are limited to those listed in the "Japanese pharmaceutical excipients", the EU "List of permitted food colours", and the FDA "Inactive ingredient guide". For proprietary mixtures, the supplier's product sheet with the qualitative formulation should be submitted, in addition to the FPP manufacturer's specifications for the product including identification testing.

For flavours the qualitative composition should be submitted, as well as a declaration that the excipients comply with foodstuff regulations (e.g. USA or EU).

Information that is considered confidential may be submitted directly to TMDA by the supplier with reference to the specific related product. If additional purification is undertaken on commercially available excipients details of the process of purification and modified specifications should be submitted.

3.2.P.4.2 Analytical procedures

The analytical procedures used for testing the excipients should be provided where appropriate. Copies of analytical procedures from officially recognized compendial monographs do not need to be submitted.

3.2.P.4.3 Validation of analytical procedures

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Analytical validation information, including experimental data, for the analytical procedures used for testing the excipients should be provided as in accordance to ICHQ6A.

Copies of analytical validation information are generally not submitted for the testing of excipients, with the exception of the validation of in-house methods where appropriate.

3.2.P.4.4 Justification of specifications

Justification for the proposed excipient specifications should be provided where appropriate.

A discussion of the tests that are supplementary to those appearing in the officially recognized compendial monograph should be provided.

Refer to ICHQ2A, ICHQ2B and ICHQ6A for more guidance

3.2.P.4.5 Excipients of human or animal origin

For excipients of human or animal origin, information should be provided regarding adventitious agents (e.g. sources, specifications, description of the testing performed and viral safety data).

The following excipients should be addressed in this section: gelatin, phosphates, stearic acid, magnesium stearate and other stearates. If from plant origin a declaration to this effect will suffice.

For these excipients from animal origin, a letter of attestation should be provided confirming that the excipients used to manufacture the FPP are without risk of transmitting agents of animal spongiform encephalopathies.

Refer:

■ ICH Q5A Viral safety Evaluation of Biotechnology Products derived from Cell Lines of Human or Animal Origin.

■ ICH Q5D Quality of Biotechnological Products: Derivation and Characterisation of Cell Substrates Used for Production of Biotechnological/Biological Products.

■ Q6B Test Procedures and Acceptance Criteria for Biotechnological/Biological Products.

3.2.P.4.6 Novel excipients

For excipient(s) used for the first time in an FPP or by a new route of administration, full details of manufacture, characterization and controls, with cross references to supporting safety data (nonclinical and/or clinical), should be provided according to the API and/or FPP format

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3.2.P.5 Control of FPP

3.2.P.5.1 Specification(s)

The specification(s) for the FPP should be provided. A copy of the FPP specification(s) from the company responsible for the batch release of the FPP should be provided. The specifications should be dated and signed by the authorized personnel (i.e. the person in charge of the quality control and quality assurance departments) should be provided in the PD. Two separate sets of specifications may be set out: after packaging of the FPP (release) and at the end of the shelf-life. Any differences between release and shelf-life tests and acceptance criteria should be clearly indicated and justified.

The specifications should be summarized according to the tables in the QOS template including the tests, acceptance criteria and analytical procedures (including types, sources and versions for the methods).

Skip testing is acceptable for parameters such as identification of colouring materials and microbial limits, when justified by the submission of acceptable supportive results for five production batches. When skip-testing justification has been accepted, the specifications should include a footnote, stating at minimum the following skip-testing requirements: at minimum every tenth batch and at least one batch annually is tested. In addition, for stability-indicating parameters such as microbial limits, testing will be performed at release and shelf-life during stability studies.

Refer ICHQ3B, ICHQ3C, ICHQ6A for more guidance.

3.2.P.5.2 Analytical procedures

The analytical procedures used for testing the FPP should be provided. Copies of the in-house analytical procedures used during pharmaceutical development (if used to generate testing results provided in the PD) as well as those proposed for routine testing should be provided. Unless modified, it is not necessary to provide copies of officially recognized compendial analytical procedures.

Refer to ICH Q2 for more guidance.

3.2.P.5.3 Validation of analytical procedures

Analytical validation information, including experimental data, for the analytical procedures used for testing the FPP should be provided.

Copies of the validation reports for the in-house analytical procedures used during pharmaceutical development (if used to support testing results provided in the MA application) as well as those proposed for routine testing should be provided.

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As recognized by regulatory authorities and pharmacopoeias themselves, verification of compendial methods can be necessary. The compendial methods, as published, are typically validated based on an API or an FPP originating from a specific manufacturer. Different sources of the same API or FPP can contain impurities and/or degradation products or excipients that were not considered during the development of the monograph. Therefore, the monograph and compendial method(s) should be demonstrated suitable for the control of the proposed FPP.

For officially recognized compendial FPP assay methods, verification should include a demonstration of specificity, accuracy and repeatability (method precision). If an officially recognized compendial method is used to control related substances that are not specified in the monograph, full validation of the method is expected with respect to those related substances.

If an officially recognized compendial standard is claimed and an in-house method is used in lieu of the compendial method (e.g. for assay or for related compounds), equivalency of the in-house and compendial methods should be demonstrated. This could be accomplished by

performing duplicate analyses of one sample by both methods and providing the results from the study. For related compound methods, the sample analysed should be the placebo spiked with related compounds at concentrations equivalent to their specification limits.

Refer to ICH Q2 for more guidance.

3.2.P.5.4 Batch analyses

A description of batches and results of batch analyses should be provided.

Information should include strength and batch number, batch size, date and site of production and use (e.g. used in comparative bioavailability or biowaiver studies, preclinical and clinical studies (if relevant), stability, pilot, scale-up and if available, production-scale batches) on relevant FPP batches used to establish the specification(s) and evaluate consistency in manufacturing.

Analytical results tested by the company responsible for the batch release of the FPP should be provided for not less than three batches of at least one commercial scale batch and two pilot scale batches. Copies of the certificates of analysis for these batches should be provided and the company responsible for generating the testing results should be identified.

The discussion of results should focus on observations noted for the various tests, rather than reporting comments such as "all tests meet specifications". This should include ranges of analytical results where relevant. For quantitative tests (e.g. individual and total impurity tests and assay tests), it should be ensured that actual numerical results are provided rather than vague statements such as "within limits" or "conforms" (e.g. "levels of degradation product A ranged from 0.2 to 0.4%"). Dissolution results should be expressed at minimum as both the average and range of individual results.

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A discussion and justification should be provided for any incomplete analyses (e.g. results not tested according to the proposed specification).

Refer ICH Q3B, Q3C and Q6A for more guidance.

3.2.P.5.5 Characterization of impurities

Information on the characterization of impurities should be provided, if not previously provided in "3.2.S.3.2 Impurities".

A discussion should be provided of all impurities that are potential degradation products (including those among the impurities identified in 3.2.S.3.2 as well as potential degradation products resulting from interaction of the API with other APIs (FDCs), excipients or the container-closure system) and FPP process-related impurities (e.g. residual solvents in the manufacturing process for the FPP).

Refer ICH Q3B, Q3C and Q6A for more guidance.

3.2.P.5.6 Justification of specification(s)

Justification for the proposed FPP specification(s) should be provided.

A discussion should be provided on the omission or inclusion of certain tests, evolution of tests, analytical procedures and acceptance criteria, differences from the officially recognized compendial standard(s), etc. If the officially recognized compendial methods have been modified or replaced, a discussion should be included.

The justification for certain tests, analytical procedures and acceptance criteria (e.g. degradation products, dissolution method development) may have been discussed in other sections of the marketing authorization dossier and does not need to be repeated here, although a cross-reference to their location should be provided.

3.2.P.6 Reference standards or materials

Information on the reference standards or reference materials used for testing of the FPP should be provided, if not previously provided in "3.2.S.5 Reference standards or materials".

See Section 3.2.S.5 for information that should be provided on reference standards or materials. Information should be provided on reference materials of FPP degradation products, where not included in 3.2.S.5.

3.2.P.7 Container-closure system

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A description of the container-closure systems should be provided, including the identity of materials of construction of each primary packaging component and its specification.

The specifications should include description and identification (and critical dimensions, with drawings where appropriate). Non-compendial methods (with validation) should be included, where appropriate.

For non-functional secondary packaging components (e.g. those that neither provide additional protection nor serve to deliver the product), only a brief description should be provided. For functional secondary packaging components, additional information should be provided.

Suitability information should be located in 3.2.P.2.

Descriptions, materials of construction and specifications should be provided for the packaging components that are:

- a) In direct contact with the dosage form (e.g. container, closure, liner, desiccant, filler);
- b) Used for drug delivery (including the device(s) for multi-dose solutions, emulsions,

- suspensions and powders/granules for such);
- c) Used as a protective barrier to help ensure stability or sterility; and
- d) Necessary to ensure FPP quality during storage and shipping.

Specifications for the primary packaging components should include a specific test for identification (e.g. IR). Specifications for film and foil materials should include limits for thickness or area weight.

Refer FDA Guidance for Industry Container Closure Systems for Packaging Human Drugs and Biologics for more guidance.

3.2.P.8 Stability

The purpose of stability testing is to provide evidence of how the quality of an API or FPP varies with time under the influence of a variety of environmental factors such as temperature, humidity and light. The stability programme also includes the study of product-related factors that influence its quality, for example, interaction of API with excipients, container-closure systems and packaging materials.

3.2.P.8.1 Stability Summary and Conclusion

The types of studies conducted, protocols used, and the results of the studies should be summarized. The summary should include, for example, conclusions with respect to storage conditions and shelf-life, and, if applicable, in-use storage conditions and shelf-life.

3.2.P.8.2 Post-approval Stability Protocol and Stability Commitment

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3.2.P.8.3 Stability Data

Results of the stability studies should be presented in an appropriate format (e.g. tabular, graphical and narrative). Information on the analytical procedures used to generate the data and validation of these procedures should be included.

Refer Guidelines on Stability Requirements for Testing Active Pharmaceutical Ingredients (APIs) and Finished Pharmaceutical Products (FPPs)

3.2. REGIONAL INFORMATION

3.2.R1 Production documentation

Submit Batch Manufacturing Record (BMR) of a real batch manufactured within at most six months before the submission of the application.

MODULE 4: NON CLINICAL STUDY REPORTS

This chapter presents an agreed format for the organization of the nonclinical reports in the Common Technical Document for applications that will be submitted.

This guidance is not intended to indicate what studies are required. It merely indicates an appropriate format for the nonclinical data that have been acquired and provide references to other guideline which may be used for populating this format.

4.1 Table of Contents of Module 4

A Table of Contents should be provided that lists all of the nonclinical study reports and gives the location of each study report in the Common Technical Document.

4.2 Study Reports

The study reports should be presented in the following order:

4.2.1 Pharmacology

Refer ICH Guideline on Nonclinical Safety Studies for the Conduct of Human Clinical Trials and marketing authorization for Pharmaceuticals (M3) for the nonclinical safety studies recommended to support human clinical trials of a given scope and duration as well as marketing authorization for pharmaceuticals.

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Refer ICH Guideline on Safety Pharmacology Studies for Human Pharmaceuticals (S7A) for the definition, objectives and scope of safety pharmacology studies. It also addresses which studies are needed before initiation of Phase 1 clinical studies as well as information needed for marketing.

Refer ICH Guideline on The Non-Clinical Evaluation of the Potential for Delayed Ventricular Repolarization (QT Interval Prolongation) by Human Pharmaceuticals (S7B) for a non-clinical testing strategy for assessing the potential of a test substance to delay ventricular repolarization. This Guideline includes information concerning non-clinical assays and integrated risk assessments.

4.2.1.1 Primary Pharmacodynamics

4.2.1.2 Secondary Pharmacodynamics

4.2.1.3 Safety Pharmacology

4.2.1.4 Pharmacodynamic Drug Interactions

4.2.2 Pharmacokinetics

4.2.2.1 Analytical Methods and Validation Reports (if separate reports are available)

4.2.2.2 Absorption

4.2.2.3 Distribution

4.2.2.4 Metabolism

4.2.2.5 Excretion

4.2.2.6 Pharmacokinetic Drug Interactions (nonclinical)

4.2.2.7 Other Pharmacokinetic Studies

Refer ICH Guideline on Pharmacokinetics: Guidance for Repeated Dose Tissue Distribution Studies (S3B) for guidance on circumstances when repeated dose tissue distribution studies should be considered (i.e., when appropriate data cannot be derived from other sources). It also gives recommendations on the conduct of such studies.

4.2.3 Toxicology

Refer ICH Note for Guidance on Toxicokinetics: The Assessment of Systemic Exposure in Toxicity Studies (S3A) for guidance on developing test strategies in toxicokinetics and the need to integrate pharmacokinetics into toxicity testing, in order to aid in the interpretation of the toxicology findings and promote rational study design development.

4.2.3.1 Single-Dose Toxicity (in order by species, by route)

4.2.3.2 Repeat-Dose Toxicity (in order by species, by route, by duration; including supportive toxicokinetics evaluations)

Refer The Committee for Human Medicinal Products (CHMP) Guideline on repeated dose toxicity for guidance on the conduct of repeated dose toxicity studies of active substances intended for human use.

Refer ICH Guideline on Duration of Chronic Toxicity Testing in Animals (Rodent and Non Rodent Toxicity Testing) (S4) for the considerations that apply to chronic toxicity testing in rodents and non-

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rodents as part of the safety evaluation of a medicinal product. The text incorporates the guidance for repeat-dose toxicity tests.

4.2.3.3 Genotoxicity

Refer ICH Guidance on Genotoxicity Testing and Data Interpretation for Pharmaceuticals Intended for Human Use (S2) for specific guidance and recommendations for in vitro and in vivo tests and on the evaluation of test results. This document addressed two fundamental areas of genotoxicity testing: the identification of a standard set of assays to be conducted for registration, and the extent of confirmatory experimentation in any particular genotoxicity assay in the standard battery.

Refer the committee for medicinal products for human use (CHMP) guideline on the limits of genotoxic impurities for a general framework and practical approaches on how to deal with genotoxic impurities in new active substances. It also relates to new applications for existing active substances, where assessment of the route of synthesis, process control and impurity profile does not provide reasonable assurance that no new or higher levels of genotoxic impurities are introduced as compared to products currently authorized in the EU containing the same active substance. The same also applies to variations to existing Marketing Authorizations pertaining to the synthesis.

4.2.3.3.1 In vitro

4.2.3.3.2 In vivo (including supportive toxicokinetics evaluations)

4.2.3.4 Carcinogenicity (including supportive toxicokinetics evaluations)

Refer ICH Guideline on Need for Carcinogenicity Studies of Pharmaceuticals (S1A) for a consistent definition of the circumstances under which it is necessary to undertake carcinogenicity studies on new drugs. These recommendations take into account the known risk factors as well as the intended indications and duration of exposure.

Refer ICH Guideline on Testing for Carcinogenicity of Pharmaceuticals (S1B) for guidance on the need to carry out carcinogenicity studies in both mice and rats, and guidance is also given on alternative testing procedures which may be applied without jeopardizing safety.

Refer ICH Guideline on Dose Selection for Carcinogenicity Studies of Pharmaceuticals (S1C) for the criteria for selection of the high dose for carcinogenicity studies of therapeutics. The use of other pharmacodynamic-, pharmacokinetic-, or toxicity-based endpoints in study design should be considered based on scientific rationale and individual merits.

4.2.3.4.1 Long-term studies (in order by species; including range-finding studies that cannot appropriately be included under repeat-dose toxicity or pharmacokinetics)

4.2.3.4.2 Short- or medium-term studies (including range-finding studies that cannot appropriately be included under repeat-dose toxicity or pharmacokinetics)

4.2.3.4.3 Other studies

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4.2.3.5 Reproductive and Developmental Toxicity (including range-finding studies and supportive toxicokinetics evaluations) (If modified study designs are used, the following sub-headings should be modified accordingly.)

Refer ICH Guidance on Detection of Toxicity to Reproduction for Medicinal Products & Toxicity to Male Fertility (S5) for guidance on tests for reproductive toxicity. It defines the periods of treatment to be used in animals to better reflect human exposure to medical products and allow more specific identification of stages at risk.

Refer committee for medicinal products for human use (CHMP) guideline on the need for non-clinical testing in juvenile animals of pharmaceuticals for paediatric indications for guidance on the need for, role and timing of studies in juvenile animals in the non-clinical safety evaluation of medicinal products for

paediatric use.

4.2.3.5.1 Fertility and early embryonic development

4.2.3.5.2 Embryo-foetal development

4.2.3.5.3 Prenatal and postnatal development, including maternal function

4.2.3.5.4 Studies in which the offspring (juvenile animals) are dosed and/or further evaluated.

4.2.3.6 Local Tolerance

Refer the Committee for medicinal products for human use (CHMP) guideline on Non-clinical local tolerance testing of medicinal products for recommendations on the evaluation of local tolerance to be performed prior to human exposure to the product. The purpose of these studies is to ascertain whether medicinal products are tolerated at sites in the body, which may come into contact with products as the result of its administration in clinical use.

4.2.3.7 Other Toxicity Studies (if available)

4.2.3.7.1 Antigenicity

4.2.3.7.2 Immunotoxicity

Refer ICH Guideline on Immunotoxicity Studies for Human Pharmaceuticals (S8) for the recommendations on nonclinical testing for immunosuppression induced by low molecular weight drugs (non-biologicals). It applies to new pharmaceuticals intended for use in humans, as well as to marketed drug products proposed for different indications or other variations on the current product label in which

the change could result in unaddressed and relevant toxicologic issues. In addition, the Guideline might also apply to drugs in which clinical signs of immunosuppression are observed during clinical trials and following approval to market.

4.2.3.7.3 Mechanistic studies (if not included elsewhere)

4.2.3.7.4 Dependence

4.2.3.7.5 Metabolites

4.2.3.7.6 Impurities

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4.2.3.7.7 Other toxicity studies

4.2.3.7.7.1 Photosafety evaluation

A harmonized guideline on photosafety evaluation of pharmaceuticals is to be published through the ICH process.

For generic products are generally exempted in this module; however, in some cases such as changes in safety impurity profile, the safety assessment studies should be conducted.

For specific products

Refer ICH Guideline on clinical Evaluation for Anticancer Pharmaceuticals (S9) for information for pharmaceuticals that are only intended to treat cancer in patients with late stage or advanced disease regardless of the route of administration, including both small molecule and biotechnology-derived pharmaceuticals. It describes the type and timing of nonclinical studies in relation to the development of anticancer pharmaceuticals and references other guidance as appropriate.

Refer ICH Guidance on Preclinical Safety Evaluation of Biotechnology-Derived Pharmaceuticals (S6) for the pre-clinical safety testing requirements for biotechnological products. It addresses the use of animal models of disease, determination of when genotoxicity assays and carcinogenicity studies should be performed, and the impact of antibody formation on duration of toxicology studies.

Refer committee for medicinal products for human use (CHMP) guideline on Non-clinical development of fixed combinations of medicinal products for guidance on the non-clinical strategies to be considered when developing a fixed combination based on the different data available in order to support the safe human use as well as avoid unnecessary repetition of animal studies.

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MODULE 5: CLINICAL STUDY REPORTS

5.1 Table of Contents of Module 5

A Table of Contents for study reports should be provided.

5.2 Tabular Listing of All Clinical Studies

5.3 Clinical Study Reports

Refer ICH Guidance on the Common Technical Document for the registration of pharmaceuticals for human use: Efficacy (M4E) for guidance on the content of this section.

Refer ICH guidelines for the structure and content of clinical study report (E3).

5.3.1 Reports of Biopharmaceutics Studies

5.3.1.1 Bioavailability (BA) Study Reports

5.3.1.2 Comparative BA and Bioequivalence (BE) Study Reports

For Generic product

Refer Guidelines on Therapeutic Equivalence Requirements. \\./././././User/AppData/Local/Temp/EAC Guidelines on Therapeutic Equivalence Requirements EAC-TF-MED-REG-N4R0.docx

Refer Guidelines on Therapeutic Equivalence Requirements. ../../../../User/AppData/Local/Temp/EAC
Guidelines on Therapeutic Equivalence Requirements EAC-TF-MED-REG-N4R0.docx

5.3.1.3 In vitro-In vivo Correlation Study Reports

For Generic product

Refer Guidelines on Therapeutic Equivalence Requirements.

5.3.1.4 Reports of Bioanalytical and Analytical Methods for Human Studies

For Generic product

Refer Guidelines on Therapeutic Equivalence Requirements.

5.3.2 Reports of Studies Pertinent to Pharmacokinetics using Human Biomaterials

5.3.2.1 Plasma Protein Binding Study Reports

5.3.2.2 Reports of Hepatic Metabolism and Drug Interaction Studies

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5.3.2.3 Reports of Studies Using Other Human Biomaterials

5.3.3 Reports of Human Pharmacokinetic (PK) Studies

5.3.3.1 Healthy Subject PK and Initial Tolerability Study Reports

5.3.3.2 Patient PK and Initial Tolerability Study Reports

5.3.3.3 Intrinsic Factor PK Study Reports

5.3.3.4 Extrinsic Factor PK Study Reports

5.3.3.5 Population PK Study Reports

5.3.4 Reports of Human Pharmacodynamic (PD) Studies

5.3.4.1 Healthy Subject PD and PK/PD Study Reports

5.3.4.2 Patient PD and PK/PD Study Reports

5.3.5 Reports of Efficacy and Safety Studies

5.3.5.1 Study Reports of Controlled Clinical Studies Pertinent to the Claimed Indication

5.3.5.2 Study Reports of Uncontrolled Clinical Studies

5.3.5.3 Reports of Analyses of Data from More Than One Study

5.3.5.4 Other Clinical Study Reports

5.3.6 Reports of Post-Marketing Experience if available

5.3.7 Case Report Forms and Individual Patient Listings

Refer Guidelines on Therapeutic Equivalence Requirements and bio-wavers

5.4 Literature References

Refer list of the ICH guidelines on clinical studies

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Annex I – Cover Letter

<Applicant>

<Address>

<Address>

<Post code><Town>

<Country>

<Applicant's reference> <Date>

<National Medicines Regulatory Authority>

<Address>

<Address>

<Post code><Town>

<Country>

Dear Sir/Madam,

Subject: Submission of Application Dossier(s) for Marketing Authorization of
<Product Name(s), [strength(s) of active pharmaceutical ingredient(s) and
dosage form(s)>

We are pleased to submit our Application Dossier(s) for a registration of human medicines
which details are as follows:

Name of the medicinal product(s):

Pharmaceutical form(s) and strength(s):

INN/active Pharmaceutical ingredient(s):

ATC Code(s):

You will find enclosed the submission dossier as specified hereafter:

CTD format, 2 soft copies documents format

CD rom; Summaries in word format and body data in PDF format

We confirm that all future submissions for this specific product will be submitted in this
same format

We confirm that the electronic submission has been checked with up-to-date and state-of-the-art antivirus software.

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The electronic submission contains the following modules:

- Module 1: Administrative information and product information
- Module 2: Overview and summaries
- Module 3: Quality
- Module 4: Non clinical study reports
- Module 5: Clinical study reports

<The relevant fees have been paid.>

Yours sincerely,

.....
<Signature>

<Name>

<Title>

<Phone number(s)>

<Email address>

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Annex II: Application Form

MODULE 1: ADMINISTRATIVE INFORMATION

1.0 PARTICULARS OF THE PRODUCT

1.1 Type of the medicinal product application

New

Generic

Extension application

Duplicate license

Renewal*

* If variation has been made, information supporting the changes should be submitted. See variation guidelines for registered medicinal products.

1.2 Proprietary Name

1.3 International Non-proprietary Name (INN) of the Active Pharmaceutical Ingredient (API)

1.4 Strength of Active Pharmaceutical Ingredient (API) per unit dosage form:

1.5 Name and address (physical and postal) of Applicant

(Company) Name:

Address:

Country:

Telephone:

Telefax:

E-Mail:

1.6 Name and address (physical and postal) of Local Technical Representative

(Company) Name:

Address:

Country:

Telephone:

Telefax:

E-Mail:

1.7 Pharmaceutical Dosage form* and route of administration*

* List of standard terms for dosage forms and routes of administration is available in the guidelines on submission of documentation for registration of human medicinal products

1.7.1 Dosage form:

1.7.2 Route(s) of administration (use current list of standard terms)

1.8 Packing/pack size:

1.9 Visual description

(Add as many rows as necessary)

1.10 Proposed shelf life (in months):

1.10.1 Proposed shelf life (after reconstitution or dilution):

1.10.2 Proposed shelf life (after first opening container):

1.10.3 Proposed storage conditions:

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1.10.4 Proposed storage conditions after first opening:

1.11 Other sister medicinal products registered or applied for registration

1.11.1 Do you hold Marketing Authorization (s) of other medicinal product (s) containing the same active substance (s) in other countries?

If yes state; ■ Product name (s), strength (s), pharmaceutical form (s):

■ Partner States where product is authorized:

■ Marketing authorization number(s):

■ Indication(s):

1.11.2 Have you applied for Marketing Authorization medicinal product (s) containing the same active substance (s) in other countries?

■ Product name (s), strength (s), pharmaceutical form (s):

■ Indication(s):

1.12 Pharmacotherapeutic group and ATC Code

1.12.1 Pharmacotherapeutic group:

1.12.2 ATC Code: (Please use current ATC code)

1.12.3 If no ATC code has been assigned, please indicate if an application for ATC code has been made:

1.13 Distribution category: Controlled Drug POM Pharmacy Only OTC

General sale

(Applicants are invited to indicate which categories they are requesting, however, TMDA reserves the right to change and/or apply only those categories provided for in their national legislation)

1.14 Country of origin:

1.15 Product Marketing Authorization in the country of origin (Attach Certificate of Pharmaceutical Product from National Medicines Regulatory Authority). If not registered, state reasons

Authorized Withdrawn (by applicant after authorization)

Country: Country:

Date of authorization (dd-mm-yyyy): Date of withdrawal (dd-mm-yyyy):

Proprietary name:

Proprietary name: Reason for withdrawal:

Authorization number: Suspended/revoked (by competent authority)

Refused Country:

Country: date of suspension/revocation (dd-mm-yyyy):

Date of refusal (dd-mm-yyyy): Reason for suspension/revocation:

Reason for Refusal: Proprietary name:

1.16 List ICH and Observers where the product is approved.

1.17 Name(s) and complete physical address(es) of the manufacturer(s)

1.17.1 Name(s) and physical address (es) of the manufacturing site of the finished pharmaceutical product (FPP), including the final product release if different from the manufacturer. Alternative sites should be also declared here.

All manufacturing sites involved in the manufacturing process of each step of the finished product, stating the role of each including quality control / in-process testing

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sites should be listed.

(Add as many rows as necessary)

Name:

Company name:

Address:

Country:

Telephone:

Telefax:

E-Mail:

1.17.2 Name(s) and physical address(es) of the manufacturer(s) of the active pharmaceutical ingredient(s) (API)

(Add as many rows as necessary)

All manufacturing sites involved in the manufacturing process of each source of active substance, including quality control / in-process testing sites should be listed.

Name:

Company name:

Address:

Country:

Telephone:

Telefax:

E-Mail:

1.18 Name and address (physical and postal) of the Brokers and Suppliers (if applicable)

Name:

Company name:

Address:

Country:

Telephone:

Telefax:

E-Mail:

1.19 Name and address (physical and postal) of the person or company responsible for Pharmacovigilance

Name:

Company name:

Address:

Country:

Telephone:

Telefax:

E-Mail:

1.20 State the reference/monograph standard such as British Pharmacopeia, United States Pharmacopeia, Ph. Eur, Japanese Pharmacopeia, In-house monograph e.t.c. used for Finished Medicinal Product.

Qualitative and Quantitative composition of the active substance(s) and excipient(s)

1.21 A note should be given as to which quantity the composition refers (e.g. 1 capsule).

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Name of active Quantity / Unit of measure Reference/
ingredient(s)* dosage unit monograph standard

1.

2.

3.

e.t.c

Name Excipient(s)

1.

2.

3

e.t.c

Note: * Only one name for each substance should be given in the following order of priority: INN**, Pharmacopoeia, common name, scientific name

** The active substance should be declared by its recommended INN, accompanied by its salt or hydrate form if relevant.

Details of averages should not be included in the formulation columns but should be stated below:

- Active substance(s):

- Excipient(s):

Name and address (physical and postal) of the Contract Research Organisation(s)

1.22 where the clinical studies of the product were conducted or name and address of laboratory where comparative dissolution studies in support of bio-waiver were conducted. (If applicable)

Name:

Company name:

Address:

Country:

Telephone:

Telefax:

E-Mail:

2.0 DECLARATION BY AN APPLICANT

I, the undersigned certify that all the information in this form and accompanying documentation is correct, complete and true to the best of my knowledge.

I further confirm that the information referred to in my application dossier is available for verification during GMP inspection.

I also agree that I shall carry out pharmacovigilance to monitor the safety of the product in the market and provide safety update reports to the Tanzania Medicines and Medical Devices Authority.

I further agree that I am obliged to follow the requirements of the Legislations and Regulations 63 which are applicable to medicinal products.

I also consent to the processing of information provided by TMDA.

It is hereby confirmed that fees will be paid/have been paid according to the fees and charges regulations

Name:

Position in the company:.....

Signature:

Date:.....

Official stamp:.....

* Note: If fees have been paid, attach proof of payment

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Annex III: Expert Declaration Form

The following is an example of a suitable declaration form:

Quality /Non-clinical / Clinical (delete those not appropriate)

I, the undersigned, declare that I have:

i. the suitable technical or professional qualifications to act in this capacity (for more information, refer to the enclosed curriculum vitae).

ii. fully examined the data provided by the applicant and have provided references to the literature to support statements made that are not supported by the applicant's original data. This report presents an objective assessment of the nature and extent of the data.

iii. provided a report based on my independent assessment of the data provided.

iv. based my recommendations, regarding suitability for registration, on the data provided herewith. I have considered the attached data and have recommended as to suitability for registration of the intended dose forms and presentations according to the proposed product information document.

I further declare that this expert report represents my own view.

Further, I declare the following to be the full extent of the professional relationship between myself and the applicant:

.....
.....
.....
.....

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Annex IV: Quality Information Summary (QIS)

INTRODUCTION

(a) Summary of product information:

Non-proprietary name(s) of the finished

pharmaceutical product(s) (FPP)

Proprietary name(s) of the finished

pharmaceutical product(s) (FPP)

International non-proprietary name(s) of the

active pharmaceutical ingredient(s) (API(s)),

including form (salt, hydrate, polymorph)

Applicant name and address

Dosage form

Application Number

Strength

Route of administration

Proposed indication(s)

Local Technical Representative (Agency)

LTR Contact person details

Local Technical Representative (LTR) contact Surname: First Name:

person

Physical address details

Town/City
Postal code
Contact person's email address
Contact person's phone number

FPP manufacturer Qualified Person Surname: First Name:
FPP manufacturer Qualified person's contact details (including Physical address)
Unit /block
Road/Street
Plant
Village/suburb
Town/City
Postal code
Country
Contact person's email address
Contact person's phone number

(b) Administrative Summary:

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Applicant's date of preparation or revision
of the QIS
Version and/or date of acceptance (official use only)

Related dossiers (e.g. FPP(s) with the same API(s) submitted to TMDA by the applicant):

Application Registration API, strength, dosage API manufacturer
number status form (including address if same
) (eg. Irinotecan (as manufacturer as current dossier
(Y/N)
chloride) 20mg per ml
Solution)

2.3.S DRUG SUBSTANCE (or ACTIVE PHARMACEUTICAL INGREDIENT (API))
(NAME, MANUFACTURER)
Indicate which option applies for the submission of API information: <check one only>

Name of API:

Name of API
manufacturer:
Certificate of suitability to the European Pharmacopoeia (CEP)

■
Option 1.
Confirmation of API prequalification document:

■
Option 2
API approval number _____.

■
Option 3a.
Active pharmaceutical ingredient master file (EAC APIMF) procedure:
APIMF number assigned by TMDA (if known): _____ ; version number(s) including
■ amendments (and/or date(s)) of the open part: _____ ; version number(s) including
amendments (and/or date(s)) of the restricted part: : _____.

Option 3b.
Full details in the PD
Open part DMF version number _____
■ Restricted part DMF version number _____
Identifier of current module 3.2.S: _____
Option 4.

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2.3.S.2 Manufacture (name, manufacturer)
2.3.S.2.1 Manufacturer(s) (name, manufacturer)
(a) Name, address and responsibility (e.g. fabrication, packaging, labelling, testing,
storage) of each manufacturer, including contractors and each proposed production site
or facility involved in these activities:
Name and address Responsibility CEP number/ Letter of access
(including WHOAPI-PQ provided?
block(s)/unit(s)) number /WHO
APIMF/ TMDA
registration
No./Approved
APIMF/ if
applicable)

2.3.S.2.3 Control of Materials (name, manufacturer) – for API option 4 only
(a) Name of starting material:
(b) Name and manufacturing site address of starting material manufacturer(s):
2.3.S.4 Control of the API (name, manufacturer)
2.3.S.4.1 Specification (name, manufacturer)
(a) API specifications of the FPP manufacturer:
Standard (e.g. Ph.Int., Ph.Eur., BP, USP, in-house)

Specification reference number & version effective date
Test Acceptance criteria Analytical procedure
(Type/Source/Version)
Description
Identification
Impurities
Assay
etc.

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2.3.S.6 Container Closure System (name, manufacturer)

(a) Description of the container closure system(s) for the storage and shipment of the API:

2.3.S.7 Stability (name, manufacturer)

2.3.S.7.1 Stability Summary and Conclusions (name, manufacturer)

(c) Proposed storage conditions and re-test period (or shelf-life, as appropriate):

Container closure system Storage statement Re-test period*

* indicate if a shelf-life is proposed in lieu of a re-test period (e.g. in the case of labile APIs)

2.3.P DRUG PRODUCT (or FINISHED PHARMACEUTICAL PRODUCT (FPP))

Indicate which option applies for the submission of FPP information: <check one only>

Name of API:

Name of API

manufacturer:

■ Full details

■ WHO collaborative procedure

■ SRA Abridged procedure

■ EAC Mutual Recognition

■ EU Article 58 procedure

2.3.P.1 Description and Composition of the FPP

(a) Description of the FPP (in signed specifications):

(b) Composition of the FPP:

(i) Composition, i.e. list of all components of the FPP and their amounts on a per unit basis and percentage basis (including individual components of mixtures prepared in-house (e.g. coatings) and overages, if any):

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Component and Function Strength (label claim)

quality standard

(and grade, if

applicable) Quant. % Quant. % Quantity %

per unit per unit per unit

or per or per or per

mL mL mL

<complete with appropriate titles e.g. Core tablet (Layer 1, Layer 2, etc. as applicable),

Contents of capsule, Powder for injection>

Subtotal 1

<complete with appropriate title e.g. Film-coating >

Subtotal 2

Total

(ii) Composition of all components purchased as mixtures (e.g. colourants, coatings, capsule shells, imprinting inks):

(c) Description of accompanying reconstitution diluent(s), if applicable:

2.3.P.2.2.1 Formulation Development

(b) Information on primary (submission, registration, exhibit) batches including comparative bioavailability or biowaiver, stability, commercial:

(i) Summary of batch numbers:

Batch number(s) of the FPPs used in

70

Bioequivalence <e.g. bioequivalence batch A12345>.

Biowaiver <e.g. biowaiver batch X12345>

For proportional strength biowaiver: the

bioequivalence batch of the reference

strength

Dissolution profile studies

Stability studies (primary batches)

<packaging configuration I>

< packaging configuration II>

<Add/delete as many rows as necessary>

Stability studies (production batches)

< packaging configuration I>

< packaging configuration II>

(Add/delete as many rows as necessary)

Validation studies (primary batches)

< packaging configuration I>

< packaging configuration II>

(Add/delete as many rows as necessary)

Validation studies (at least the first three

consecutive production batches)

version(s) for process validation

protocol(s)

Summary of formulations and discussion of any differences:

Component and Relevant batches

quality

Comparative Stability Process Commercial

standard (e.g.

bioavailability or validation (2.3.P.1)

NF, BP, Ph.Eur,

biowaiver

in-house)

<Batch nos. and <Batch nos. and <Batch nos. and <Batch nos. and

sizes> sizes> sizes> sizes>
Theor. % Theor. % Theor. % Theor. %
quantity quantit quantit quantit
per batch y per y per y per
batch batch batch
<complete with appropriate titles e.g. Core tablet (Layer 1, Layer 2, etc. as applicable), Contents
of capsule, Powder for injection>

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Component and Relevant batches
quality Comparative Process Commercial
Stability
standard (e.g.
bioavailability or validation (2.3.P.1)
NF, BP, Ph.Eur,
biowaiver
in-house)
<Batch nos. and <Batch nos. and <Batch nos. and
sizes> sizes> sizes> sizes>
Theor. % Theor. % Theor. % Theor. %
quantity quantit quantit quantit
per batch y per y per y per
batch batch batch

Subtotal 1
<complete with appropriate title e.g. Film-coating >

Subtotal 2
Total

2.3.P.3 Manufacture

2.3.P.3.1 Manufacturer(s)

(a) Name, address and responsibility (e.g. fabrication, packaging, labelling,
testing) of each manufacturer, including contractors and each proposed production
site or facility involved in manufacturing and testing:
Name and address Responsibility
(include block(s)/unit(s))

2.3.P.3.2 Batch Formula

Largest intended commercial batch size:

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Other intended commercial batch sizes:

<information on all intended commercial batch sizes should be in the QIS>
(a) List of all components of the FPP to be used in the manufacturing process and
their amounts on a per batch basis (including components of mixtures
prepared in-house (e.g. coatings) and overages, if any):
Strength (label claim)

Master production document
reference number and/or version
Proposed commercial batch size(s)
(e.g. number of dosage units)
Component and quality standard Quantity per Quantity per Quantity per
(and grade, if applicable) batch (e.g. batch (e.g. batch (e.g.
kg/batch) kg/batch) kg/batch)
<complete with appropriate titles e.g. Core tablet (Layer 1, Layer 2, etc. as applicable), Contents
of capsule, Powder for injection>

Subtotal 1
<complete with appropriate title e.g. Film-coating >

Subtotal 2
Total

2.3.P.3.3 Description of Manufacturing Process and Process Controls

(a) Flow diagram of the manufacturing process:
(b) Narrative description of the manufacturing process, including equipment
type and working capacity, process parameters:
2.3.P.3.4 Controls of Critical Steps and Intermediates
(a) Summary of controls performed at the critical steps of the manufacturing
process and on isolated intermediates:
Step Controls (parameters/limits/frequency of
(e.g. granulation, compression, coating) testing)

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Proposed/validated holding periods for intermediates (including bulk product):

2.3.P.3.5 Process Validation and/or Evaluation

(a) Summary of the process validation and/or evaluation studies conducted and/or
a summary of the proposed validation protocol for the critical steps or critical assays
used in the manufacturing process (e.g. protocol number, parameters, results):
Document code(s) for the process validation protocol(s) and/or report(s) (including
reference number/version/date):

2.3.P.5 Control of FPP

2.3.P.5.1 Specification(s)

(a) Specification(s) for the FPP:
Standard (e.g. Ph.Int., BP, USP, in-house)

Specification reference number and version
Test Acceptance Acceptance criteria Analytical
criteria (shelf-life) procedure

(release) (type/source/version)
)
Description
Identification
Impurities
Assay
etc.

2.3.P.7 Container Closure System

(a) Description of the container closure systems, including unit count or fill size, container size or volume:

Description
Unit count or fill size Container size
(including materials Strength
(e.g. 60s, 100s etc.) (e.g. 5 ml, 100 ml etc.)
of construction)

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2.3.P.8 Stability

2.3.P.8.1 Stability Summary and Conclusions

(c) Proposed storage statement and shelf-life (and in-use storage conditions and in-use period, if applicable):

Container closure system Storage statement Shelf-life

2.3.P.8.2 Post-approval Stability Protocol and Stability Commitment

(a) Stability protocol for Primary stability batches (e.g. storage conditions (including tolerances), batch numbers and batch sizes, tests and acceptance criteria, testing frequency, container closure system(s)):

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Parameter Details

Storage condition(s) (■C, % RH)

Batch number(s) / batch size(s) <primary batches>

Tests and acceptance criteria Description

Moisture

Impurities

Assay

etc.

Testing frequency

Container closure system(s)

(b) Stability protocol for Commitment batches (e.g. storage conditions (including tolerances), batch numbers (if known) and batch sizes, tests and acceptance criteria, testing frequency, container closure system(s)):

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Parameter Details

Storage condition(s) (■C, % RH)

Batch number(s) / batch size(s) <not less than three production batches in each container closure system>

Tests and acceptance criteria Description

Moisture

Impurities

Assay

etc.

Testing frequency

Container closure system(s)

(c) Stability protocol for Ongoing Batches (e.g. storage conditions (including tolerances), number of batches per strength and batch sizes, tests and acceptance criteria, testing frequency, container closure system(s)):

Parameter Details

Storage condition(s) (■C, % RH)

Batch size(s), annual allocation <at least one production batch per year (unless none is produced that year) in each container closure system >

Tests and acceptance criteria Description

Moisture

Impurities

Assay

etc.

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Parameter Details

Testing frequency

Container closure system(s)

2.3.P.8.3 Stability Data

(c) Bracketing and matrixing design for commitment and/or continuing (i.e. ongoing) batches, if applicable:

WRITTEN COMMITMENTS OF THE MANUFACTURER – for TMDA use

API

If applicable (primary stability study commitment):

The Applicant (or API manufacturer) undertook in writing (date of letter of commitment) to continue long-term testing of <INN of API> for a period of time sufficient to cover the whole provisional re-test period (period ending month/year) and to report any significant changes or out-of-specification results immediately to TMDA for the following batches :
<Batch numbers, manufacturing dates, batch size, primary packing materials>

If applicable (commitment stability studies):

Since stability data on three production scale batches were not provided with the application, the remaining number of production scale batches should be put on long-term stability testing. Any significant changes or out-of-specification results should be reported immediately to TMDA. The approved stability protocol should be used for commitment batches.

API option 1 – CEP

The Applicant provided a commitment in writing (date of letter of commitment) to inform TMDA in the event that the CEP is revised or withdrawn, and that revisions to the CEP will be handled as per variation guidelines. Note that revisions or withdrawal will require additional consideration of the API data requirements to support the dossier.

API option 2 – WHOAPI-CPQ

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The Applicant provided a commitment in writing (date of letter of commitment) to inform TMDA in the event that the WHOAPI-CPQ is revised or withdrawn, and that revisions to the WHOAPI-CPQ will be handled as per variation TMDA Variation guidelines. Note that revisions or withdrawal will require additional consideration of the API data requirements to support the dossier.

API option 4 – full details in the PD (ongoing stability study commitment)

The Applicant undertook in writing (date of letter of commitment) a commitment regarding ongoing stability studies. Unless otherwise justified, at least one batch per year of the product will be included in the stability programme (unless none is produced during that year). The stability protocol will be that which was approved for primary batches (or the protocol was submitted for assessment). Out-of-specification results or significant atypical trends will be investigated. Any confirmed significant change or out-of-specification result will be reported immediately to TMDA. The possible impact on batches on the market will be considered in consultation with GMP inspection.

FPP

If applicable (primary stability study commitment):

The Applicant undertook in writing (date of letter of commitment) to continue long-term testing of < FPP reference number, trade name (INN of API), strength, pharmaceutical form> for a period of time sufficient to cover the whole provisional shelf-life (period ending month/year) and to report any out-of-specification results or significant changes immediately for the following batches :
<Batch numbers, manufacturing dates, batch size, primary packing materials >

If applicable (commitment stability studies):

Since stability data on three production scale batches was not provided with the application, the Applicant undertook in writing, (date of letter of commitment) to put the remaining number <e.g. additional two (2)> production scale batches of < FPP reference number, trade name (INN of API), strength, pharmaceutical form, primary packing material> on long-term stability testing. Any out-of-specification results or significant changes during the study will immediately be reported to TMDA. The approved stability protocol will be used for commitment batches.

If applicable (when the proposed largest commercial batch size is 200 000 units (x units) or less)

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The Applicant undertook in writing (date of letter of commitment) to place the first three batches of any production size larger than x units on stability. The stability protocol will be that which was approved for primary batches (or the protocol was submitted for assessment). Out-of-specification results or significant atypical trends will be investigated. Any confirmed significant change or out-of-specification result will be reported immediately to TMDA.

Ongoing stability study commitment

The Applicant undertook in writing (date of letter of commitment) a commitment regarding ongoing stability studies. Unless otherwise justified, at least one batch per year of the product manufactured in every primary packaging type will be included in the stability programme (unless none is produced during that year). The stability protocol will be that which was

approved for primary batches (or the protocol was submitted and found acceptable). Out-of-specification results or significant atypical trends will be investigated. Any confirmed significant change or out-of-specification result will be reported immediately to TMDA. The possible impact on batches on the market will be considered in consultation with GMP inspection.

If applicable (validation of production batches)

Validation data on production scale batches of not less than three (3) consecutive batches of <FPF reference number, trade name (INN of API), strength, pharmaceutical form, primary packing material> was not provided with the application. Therefore, the Applicant submitted a written commitment (date of letter of commitment) that three consecutive production batches would be prospectively validated and a validation report—in accordance with the details of the validation protocol provided in the dossier— would be made available as soon as possible for evaluation by assessors or for verification by the GMP inspection.

Change History

Date of preparation of original QIS:

Date of revised Section (e.g. Revision version S.2.1)

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Annex V: Letter of Access to CEP

<Applicant>
<Address>
<Address>
<Post code><Town>
<Country>

<Applicant's reference> <Date>

<National Medicines Regulatory Authority>
<Address>
<Address>
<Post code><Town>
<Country>

Dear Sir/Madam,

Subject: Authorization to access Certificate of Suitability (CEP)

Reference is made to the above subject matter.

Consent is hereby granted to the Tanzania Medicines and Medical Devices (TMDA) to make reference to this company's Certificate(s) of Suitability (CEPs) [number(s)] for [API(s) name(s)] in the evaluation of applications relating to the registration of [medicine name(s)] submitted to TMDA by (applicant's name).

This consent does/does not** include authorization to supply information or extracts from or the whole of the data to:

(Name of company or individual)

The API is manufactured by:
(Names and addresses of all manufacturing sites and manufacturing steps carried out at site)

A formal agreement exists between the applicant of the medicine and the manufacturer of the API which ensures that information will be communicated between them and to TMDA before any significant change is made to the site of manufacture, manufacturing procedure or quality control specifications of the API. Except as permitted by the guidelines relating to changes to medicines, such changes will not be made to the API to be used in manufacture of the medicine destined to be distributed in Tanzania before written approval is granted by TMDA.

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I understand that the consequences of failure to obtain approval for changes where approval is necessary may include de-registration and recall of batches of medicines.

Any questions arising from the TMDA's evaluation of this CEP should be forwarded to:

(Name and address)

Yours faithfully

{Signature of Company Representative}
{Name}
{Position in Company}
{Date}

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Annex VI: Letter of Access to APIMF

<Applicant>
<Address>
<Address>
<Post code><Town>
<Country>

<Applicant's reference> <Date>

<National Medicines Regulatory Authority>
<Address>
<Address>

<Post code><Town>
<Country>

Dear Sir/Madam,

Subject: Authorization to access Active Pharmaceutical Ingredient Master File

Reference is made to the above subject matter.

Consent is hereby granted to the Tanzania Medicines and Medical Devices Authority (TMDA) to make reference to this company's Active Pharmaceutical Ingredient Master File(s) for [API(s) name] in the evaluation of applications relating to the registration of [medicine name(s)] submitted to TMDA by the (applicant's name).

This consent does/does not** include authorization to supply information or extracts from or the whole of the data to:

(Name of company or individual)

The substance is manufactured by:
(Names and addresses of all manufacturing sites and manufacturing steps carried out at site)

A copy of the applicant's Part of the APIMF as specified in the Guidelines on Procedures for Evaluation of Quality of API has been supplied to the applicant.

A formal agreement exists between the applicant of the medicine and the manufacturer of the API which ensures that information will be communicated between them and to TMDA before any significant change is made to the site of manufacture, manufacturing procedure or quality

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control specifications of the API. Except as permitted by the TMDA guidelines relating to changes to medicines, such changes will not be made to the API to be used in manufacture of the medicine destined to be distributed in Tanzania before written approval is granted by the TMDA.

I understand that the consequences of failure to obtain approval for changes where approval is necessary may include de-registration and recall of batches of medicines.

This APIMF (or data identical to that contained therein) has also been submitted to and approved by the regulatory authorities in (list of countries with stringent regulatory systems), and TMDA is authorized to request and refer to the evaluation reports of these agencies. TMDA is also authorized to exchange its own evaluation reports with these and other regulatory authorities.

Any questions arising from evaluation of this APIMF should be forwarded to:

{Name and address}

Yours faithfully

{Signature of Company Representative}
{Name}
{Position in Company}
{Date}

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Annex VII: Quality Overall Summary – Product Dossier (QOS- PD)

Summary of product information:

Non-proprietary name of the finished pharmaceutical product (FPP)
Proprietary name of the finished pharmaceutical product (FPP)
International non-proprietary name(s) of the active pharmaceutical ingredient(s) (API(s)), including form (salt, hydrate, polymorph)
Applicant name and address
Dosage form
Reference Number(s)
Strength(s)
Route of administration
Proposed indication(s)
Contact information Name:
Phone:
Fax:
Email:

2.3.S ACTIVE PHARMACEUTICAL INGREDIENT (API))

Complete the following table for the option that applies for the submission of API information:

Name of API:
Name of API manufacturer:
■ Certificate of suitability to the European Pharmacopoeia (CEP):
■ is a written commitment provided that the applicant will inform TMDA in the event that the CEP is withdrawn and has acknowledged that withdrawal
■ of the CEP will require additional consideration of the API data requirements to support the dossier:
o ■ yes, ■ no;
■ a copy of the most current CEP (with annexes) and written commitment should be provided in Module 1;

- the declaration of access should be filled out by the CEP holder on behalf of the FPP manufacturer or applicant to TMDA who refers to the CEP; and
- summaries of the relevant information should be provided under the appropriate sections (e.g. S.1.3, S.3.1, S.4.1 through S.4.4, S.6 and S.7; see Quality guideline).
- Active pharmaceutical ingredient master file (APIMF):
- A copy of the letter of access should be provided in Module 1; and summaries of the

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relevant information from the Open part should be provided under the appropriate sections; see Section 3.2.S in the Quality guideline.

- Active pharmaceutical ingredient pre-qualified by WHO

Provide evidence from WHO

- Full details in the PD:

Summaries of the full information should be provided under the appropriate sections; see Section 3.2.S in the Quality guideline.

2.3.S.1 General Information

2.3.S.1.1 Nomenclature

- (Recommended) International Non-proprietary name (INN):
- Compendial name, if relevant:
- Chemical name(s):
- Company or laboratory code:
- Other non-proprietary name(s) (e.g. national name, USAN, BAN):
- Chemical Abstracts Service (CAS) registry number:

2.3.S.1.2 Structure

- Structural formula, including relative and absolute stereochemistry:
- Molecular formula:
- Relative molecular mass:

2.3.S.1.3 General Properties

- Physical description (e.g. appearance, colour, physical state):
- Solubilities:
In common solvents:
Quantitative aqueous pH solubility profile (pH 1 to 6.8):

Medium (e.g. pH 4.5 buffer) Solubility (mg/ml)

Dose/solubility volume calculation:

- Physical form (e.g. polymorphic form(s), solvate, hydrate):

Polymorphic form:

Solvate:

Hydrate:

- Other:

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Property

pH

pK

Partition coefficients

Melting/boiling points

Specific optical rotation

(specify solvent)

Refractive index (liquids)

Hygroscopicity

UV absorption maxima/molar

absorptivity

Other

2.3.S.2 Manufacture

2.3.S.2.1 Manufacturer(s)

- Name, address and responsibility (e.g. fabrication, packaging, labelling, testing, storage) of each manufacturer, including contractors and each proposed production site or facility involved in these activities:

Name and address Responsibility APIMF/CEP number (if including block(s)/unit(s)) applicable)

- Manufacturing authorization for the production of API(s) and, where available, certificate of GMP compliance (GMP information should be provided in Module 1):

2.3.S.2.2 Description of Manufacturing Process and Process Controls

- Flow diagram of the synthesis process(es):
- Brief narrative description of the manufacturing process(es):
- Alternate processes and explanation of their use:
- Reprocessing steps and justification:

2.3.S.2.3 Control of Materials

- Summary of the quality and controls of the starting materials used in the manufacture of the API:

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Step/starting material Test(s)/method(s) Acceptance criteria

- Name and manufacturing site address of starting material manufacturer(s):
- Where the API(s) and the starting materials and reagents used to manufacture the API(s)

are without risk of transmitting agents of animal spongiform encephalopathies, a letter of attestation confirming this can be found in:

2.3.S.2.4 Controls of Critical Steps and Intermediates

(a) Summary of the controls performed at critical steps of the manufacturing process and on intermediates:

Step/materials Test(s)/method(s) Acceptance criteria

2.3.S.2.5 Process Validation and/or Evaluation

(a) Description of process validation and/or evaluation studies (e.g. for aseptic processing and sterilization):

2.3.S.2.6 Manufacturing Process Development

(a) Description and discussion of the significant changes made to the manufacturing process and/or manufacturing site of the API used in producing comparative bioavailability or bio-waiver, stability, scale-up, pilot and, if available, production scale batches:

2.3.S.3 Characterisation

2.3.S.3.1 Elucidation of Structure and other Characteristics

(a) List of studies performed (e.g. IR, UV, NMR, MS, elemental analysis) and conclusion from the studies (e.g. whether results support the proposed structure):

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(b) Discussion on the potential for isomerism and identification of stereochemistry (e.g. geometric isomerism, number of chiral centres and configurations) of the API batch(es) used in comparative bioavailability or bioavailability studies:

(c) Summary of studies performed to identify potential polymorphic forms (including solvates):

(d) Summary of studies performed to identify the particle size distribution of the API:

(e) Other characteristics:

2.3.S.3.2 Impurities

(a) Identification of potential and actual impurities arising from the synthesis, manufacture and/or degradation:

(i) List of API-related impurities (e.g. starting materials, by-products, intermediates, chiral impurities, degradation products), including chemical name, structure and origin:

API-related impurity Structure Origin
(chemical name or descriptor)

(ii) List of process-related impurities (e.g. residual solvents, reagents), including compound names and step used in synthesis:

Process-related impurity (compound name) Step used in synthesis

(b) Basis for setting the acceptance criteria for impurities:

(i) Maximum daily dose (i.e. the amount of API administered per day) for the API, corresponding to ICH Reporting/Identification/Qualification Thresholds for the API-related impurities and the concentration limits (ppm) for the process-related impurities (e.g. residual solvents):

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Maximum daily dose for the <x mg/day>

API:

Test Parameter ICH threshold or
concentration limit

API-related impurities Reporting Threshold

Identification Threshold

Qualification Threshold

Process-related impurities <solvent 1>

<solvent 2>, etc.

(ii) Data on observed impurities for relevant batches (e.g. comparative bioavailability or bioavailability, stability batches):

Impurity Acceptance Results (include batch number* and use**)
(API-related and Criteria
process-related)

* include strength, if reporting impurity levels found in the FPP (e.g. for comparative studies)

** e.g. comparative bioavailability or bio-waiver studies, stability

(iii) Justification of proposed acceptance criteria for impurities:

2.3.S.4 Control of the API

2.3.S.4.1 Specification

(a) API specifications of the FPP manufacturer:

Standard (e.g. Ph.Int., Ph.Eur., BP, USP, House)

Specification reference number and version

Test Acceptance criteria Analytical procedure

(Type/Source/Version)

Description

Identification

Impurities

Assay

etc.

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2.3.S.4.2 Analytical Procedures

(a) Summary of the analytical procedures (e.g. key method parameters, conditions, system suitability testing):

2.3.S.4.3 Validation of Analytical Procedures

(a) Summary of the validation information (e.g. validation parameters and results for non-compendia methods):

(b) Summary of verification information on compendia methods

2.3.S.4.4 Batch Analyses

(a) Description of the batches:

Batch number Batch size Date and Use (e.g. comparative site of production bioavailability or biowaiver, stability)

(b) Summary of batch analyses release results of the FPP manufacturer for relevant batches (e.g. comparative bioavailability or bio-waiver, stability):

Test Acceptance Results

Criteria <batch x> <batch y> etc.

Description

Identification

Impurities

Assay

etc.

(c) Summary of analytical procedures and validation information for those procedures not previously summarized in 2.3.S.4.2 and 2.3.S.4.3 (e.g. historical analytical procedures):

2.3.S.4.5 Justification of Specification

(a) Justification of the API specification (e.g. evolution of tests, analytical procedures and acceptance criteria, differences from officially recognized compendial standard(s)):

2.3.S.5 Reference Standards or Materials

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(a) Source (including lot number) of primary reference standards or reference materials (e.g. Ph.Int., Ph.Eur., BP, USP, in-house):

(b) Characterization and evaluation of non-official (e.g. not from an officially recognized pharmacopoeia) primary reference standards or reference materials (e.g. elucidation of structure, certificate of analysis):

(c) Description of the process controls of the secondary reference standard (comparative certificate of analysis and IR spectra against primary standard) :

2.3.S.6 Container Closure System

(a) Description of the container closure system(s) for the shipment and storage of the API (including the identity of materials of construction of each primary packaging component and a brief summary of the specifications):

Packaging component Materials of construction Specifications (list parameters e.g. identification (IR))

(b) Other information on the container closure system(s) (e.g. suitability studies):

2.3.S.7 Stability

2.3.S.7.1 Stability Summary and Conclusions

(a) Summary of stress testing (e.g. heat, humidity, oxidation, photolysis, and acid/base): and results:

Stress condition Treatment Results (e.g. including discussion whether mass balance is observed)

Heat

Humidity

Oxidation

Photolysis

Acid

Base

Other

(b) Summary of accelerated and long-term testing parameters (e.g. studies conducted):

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Storage condition Batch number Batch Container closure Completed (and (■C, % RH) size system proposed) testing intervals

Summary of the stability results observed for the above accelerated and long-term studies:

Test Results

Description

Moisture

Impurities

Assay

etc.

(c) Proposed storage statement and re-test period (or shelf-life, as appropriate):

Container closure system Storage statement Re-test period*

* indicate if a shelf-life is proposed in lieu of a re-test period (e.g. in the case of labile APIs)

2.3.S.7.2 Post-approval Stability Protocol and Stability Commitment

(a) Stability protocol for Primary stability batches (e.g. storage conditions (including tolerances), batch numbers and batch sizes, tests and acceptance criteria, testing frequency, container closure system(s)):

Parameter Details

Storage condition(s) (■C, % RH)

Batch number(s) / batch size(s)

Tests and acceptance criteria Description

Moisture

Impurities

Assay

etc.

Testing frequency

Container closure system(s)

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Parameter Details

(b) Stability protocol for Commitment batches (e.g. storage conditions (including tolerances), batch numbers (if known) and batch sizes, tests and acceptance criteria, testing frequency, container closure system(s)):

Parameter Details

Storage condition(s) (■C, % RH)

Batch number(s) / batch <not less than three production batches>
size(s)

Tests and acceptance criteria Description

Moisture

Impurities

Assay

etc.

Testing frequency

Container closure system(s)

(c) Stability protocol for Ongoing batches (e.g. storage conditions (including tolerances), batch sizes and annual allocation, tests and acceptance criteria, testing frequency, container closure system(s)):

Parameter Details

Storage condition(s) (■C, % RH)

Annual allocation <at least one production batch per year (unless none is produced
that year)in each container closure system >

Tests and acceptance criteria Description

Moisture

Impurities

Assay

etc.

Testing frequency

Container closure system(s)

2.3.S.7.3 Stability Data

(a) The actual stability results should be provided in Module 3.

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(b) Summary of analytical procedures and validation information for those procedures not previously summarized in 2.3.S.4 (e.g. analytical procedures used only for stability studies):

2.3.P FINISHED PHARMACEUTICAL PRODUCT (FPP)

2.3.P.1 Description and Composition of the FPP

(a) Description of the FPP:

(b) Composition of the FPP:

(i) Composition, i.e. list of all components of the FPP and their amounts on a per unit basis and percentage basis (including individual components of mixtures prepared in-house (e.g. coatings) and overages, if any):

Component and Function Strength (label claim)
quality standard (and
grade, if applicable) Quant. per % Quant. per % Quantity per%
unit unit unit
<complete with appropriate title e.g. Core tablet, Contents of capsule, Powder for injection>

Subtotal 1
<complete with appropriate title e.g. Film-coating>

Subtotal 2
Total

(ii) Composition of all components purchased as mixtures (e.g.
colourants, coatings, capsule shells, imprinting inks):

(c) Description of accompanying reconstitution diluent(s), if applicable:

(d) Type of container closure system used for the FPP and accompanying
reconstitution diluent, if applicable:

2.3.P.2 Pharmaceutical Development

2.3.P.2.1 Components of the FPP

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2.3.P.2.1.1 Active Pharmaceutical Ingredient

(a) Discussion of the:

(i) compatibility of the API(s) with excipients listed in 2.3.P.1:

(ii) key physicochemical characteristics (e.g. water content,
solubility, particle size distribution, polymorphic or solid state form) of the
API(s) that can influence the performance of the FPP:

(iii) for fixed-dose combinations, compatibility of APIs with each
other:

2.3.P.2.1.2 Excipients

(a) Discussion of the choice of excipients listed in 2.3.P.1 (e.g. their
concentrations, their characteristics that can influence the FPP performance):

2.3.P.2.2 Finished Pharmaceutical Product

2.3.P.2.2.1 Formulation Development

(a) Summary describing the development of the FPP (e.g. route of
administration, usage, optimization of the formulation, etc.):

(b) Information on primary (submission, registration, exhibit) batches including
comparative bioavailability or bio-waiver, stability, commercial:

(i) Summary of batch numbers:

Batch number(s) of the FPPs used in
Bioequivalence or biowaiver
Dissolution profile studies
Stability studies (primary batches)
<packaging configuration I>
< packaging configuration II>
<Add/delete as many rows as necessary>
Stability studies (production batches)
< packaging configuration I>
< packaging configuration II>
(Add/delete as many rows as necessary)
Validation studies (primary batches) if available
< packaging configuration I>
< packaging configuration II>

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(Add/delete as many rows as necessary)
Validation studies (at least the first three
consecutive production batches)
or code(s)/version(s) for process validation
protocol(s)

(ii) Summary of formulations and discussion of any differences:

Component and Relevant batches
quality standard Comparative Stability Process validation Commercial
(e.g. NF, BP, bioavailability (2.3.P.1)
Ph.Eur, in-house) or biowaiver
<Batch nos. and
<Batch nos. and <Batch nos. and
sizes> sizes> sizes> sizes>
Theor. % Theor. % Theor. % Theor. %
quantity quantity per quantity per quantity per
per batch batch batch batch
<complete with appropriate title e.g. Core tablet, Contents of capsule, Powder
for injection>

Subtotal 1
<complete with appropriate title e.g. Film-coating>

Subtotal 2
Total

(c) Description of batches used in the comparative in vitro studies (e.g. dissolution) and in the in vivo studies (e.g. comparative bioavailability or biowaiver), including strength, batch number, type of study and reference to the data (volume, page):

(d) Summary of results for comparative in vitro studies
(e.g. dissolution)

(e) Summary of any information on in vitro-in vivo correlation (IVIVC) studies (with cross-reference to the studies in Module 5):

(f) For scored tablets, provide the rationale/justification for scoring:

2.3.P.2.2.2 Overages

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(a) Justification of overages in the formulation(s) described in 2.3.P.1:

2.3.P.2.2.3 Physicochemical and Biological Properties

(a) Discussion of the parameters relevant to the performance of the FPP
(e.g. pH, ionic strength, dissolution, particle size distribution, polymorphism, rheological properties):

2.3.P.2.3 Manufacturing Process Development

(a) Discussion of the development of the manufacturing process of the FPP (e.g. optimization of the process, selection of the method of sterilization):

(b) Discussion of the differences in the manufacturing process(es) for the batches used in the comparative bioavailability or biowaiver studies and the process described in 2.3.P.3.3:

2.3.P.2.4 Container Closure System

(a) Discussion of the suitability of the container closure system
(described in 2.3.P.7) used for the storage, transportation (shipping) and use of the FPP (e.g. choice of materials, protection from moisture and light, compatibility of the materials with the FPP):

(b) For a device accompanying a multi-dose container, a summary of the study results demonstrating the reproducibility of the device (e.g. consistent delivery of the intended volume):

2.3.P.2.5 Microbiological Attributes

(a) Discussion of microbiological attributes of the FPP (e.g. preservative effectiveness studies):

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2.3.P.2.6 Compatibility

(a) Discussion of the compatibility of the FPP (e.g. with reconstitution diluent(s) or dosage devices, co-administered FPPs):

2.3.P.3 Manufacture

2.3.P.3.1 Manufacturer(s)

(a) Name, address and responsibility (e.g. fabrication, packaging, labelling, testing) of each manufacturer, including contractors and each proposed production site or facility involved in manufacturing and testing:

Name and address Responsibility
(include block(s)/unit(s))

2.3.P.3.2 Batch Formula

(a) List of all components of the FPP to be used in the manufacturing process and their amounts on a per batch basis (including individual components of mixtures prepared in-house (e.g. coatings) and overages, if any):

Strength (label claim)

Master production document

reference number and/or version

Proposed commercial batch size(s) (e.g. number of dosage units)

Component and quality Quantity per batch Quantity per batch Quantity per batch

Standard (and grade, if applicable) (e.g. kg/batch) (e.g. kg/batch) (e.g. kg/batch)

<complete with appropriate title e.g. Core tablet, Contents of capsule, Powder for injection>

Subtotal 1
<complete with appropriate title e.g. Film-coating>

Subtotal 2

Total

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2.3.P.3.3 Description of Manufacturing Process and Process Controls

(a) Flow diagram of the manufacturing process:

(b) Narrative description of the manufacturing process, including equipment type and working capacity, process parameters:

(c) Justification of reprocessing of materials:

2.3.P.3.4 Controls of Critical Steps and Intermediates

(a) Summary of controls performed at the critical steps of the manufacturing process and on isolated intermediates:

Step Controls

(e.g. granulation, compression, coating)

2.3.P.3.5 Process Validation and/or Evaluation

(a) Summary of the process validation and/or evaluation studies conducted (including product quality review(s) where relevant) and/or a summary of the proposed process validation protocol for the critical steps or critical assays used in the manufacturing process (e.g. protocol number, parameters, results):

2.3.P.4 Control of Excipients

2.3.P.4.1 Specifications

(a) Summary of the specifications for officially recognized compendial excipients which include supplementary tests not included in the officially recognized compendial monograph(s):

2.3.P.4.2 Analytical Procedures

(a) Summary of the analytical procedures for supplementary tests:

2.3.P.4.3 Validation of Analytical Procedures

(a) Summary of the validation information for the analytical procedures for supplementary tests (where applicable):

100

2.3.P.4.4 Justification of Specifications

(a) Justification of the specifications (e.g. evolution of tests, analytical procedures and acceptance criteria, exclusion of certain tests, differences from officially recognized compendial standard(s)):

2.3.P.4.5 Excipients of Human or Animal Origin

(a) For FPPs using excipients without risk of transmitting agents of animal spongiform encephalopathies, a letter of attestation confirming this can be found in: (page and volume)
(b) CEP(s) demonstrating TSE-compliance can be found in: (page and volume)

2.3.P.4.6 Novel Excipients

For excipient(s) used for the first time in an FPP or by a new route of administration, full details of manufacture, characterization and controls, with cross references to supporting safety data (nonclinical and/or clinical), should be provided according to the API and/or FPP format.

2.3.P.5 Control of FPP

2.3.P.5.1 Specification(s)

Specification(s) for the FPP:

Standard (e.g. Ph.Int., BP, USP, House)

Specification reference number and version

Test Acceptance criteria Acceptance criteria Analytical procedure
(release) (shelf-life) (type/source/version)

Description

Identification

Impurities

Assay

etc.

2.3.P.5.2 Analytical Procedures

(a) Summary of the analytical procedures (e.g. key method parameters, conditions, system suitability testing):

2.3.P.5.3 Validation of Analytical Procedures

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(a) Summary of the validation information (e.g. validation parameters and results):

2.3.P.5.4 Batch Analyses

(a) Description of the batches:

Strength and Batch size Date and Use (e.g. comparative batch number site of production bioavailability or biowaiver, stability)

(b) Summary of batch analyses release results for relevant batches (e.g. comparative bioavailability or biowaiver, stability):

Test Acceptance Results
criteria <batch x> <batch y> etc.
Description
Identification
Impurities
Assay
etc.

(c) Summary of analytical procedures and validation information for those procedures not previously summarized in 2.3.P.5.2 and 2.3.P.5.3 (e.g. historical analytical procedures):

2.3.P.5.5 Characterisation of Impurities

(a) Identification of potential and actual impurities:

Degradation product Structure Origin
(chemical name or descriptor)

Process-related impurity Step used in the FPP manufacturing process
(compound name)

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(b) Basis for setting the acceptance criteria for impurities:

(i) Maximum daily dose (i.e. the amount of API administered per day) for the API, corresponding ICH Reporting/Identification/Qualification Thresholds for the degradation products in the FPP and the concentration limits (ppm) for the process-related impurities (e.g. residual solvents):

Maximum daily dose for the <x mg/day>

API:

Test Parameter ICH threshold or
concentration
limit

Degradation product Reporting Threshold

Identification Threshold

Qualification Threshold

Process-related impurities <solvent 1>
<solvent 2>, etc.

(ii) Data on observed impurities for relevant batches (e.g. comparative bioavailability or biowaiver):

Impurity Acceptance Results
(degradation product criteria
and process-related) <batch no.,
strength, use>

(iii) Justification of proposed acceptance criteria for impurities:

2.3.P.5.6 Justification of Specification(s)

(a) Justification of the FPP specification(s) (e.g. evolution of tests, analytical procedures and acceptance criteria, differences from officially recognized compendial standard(s)):

2.3.P.6 Reference Standards or Materials

(a) Source (including lot number) of primary reference standards or reference materials (e.g. Ph.Int., Ph.Eur., BP, USP, in-house) not discussed in 3.2.S.5:
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(b) Characterization and evaluation of non-official (e.g. not from an officially recognized pharmacopoeia) primary reference standards or reference materials (e.g. elucidation of structure, certificate of analysis) not discussed in 3.2.S.5:

(c) Description of the process controls of the secondary reference standard (comparative certificate of analysis and IR spectra against primary standard) not discussed in 3.2.S.5:

2.3.P.7 Container Closure System

(a) Description of the container closure systems, including unit count or fill size, container size or volume:

Description Strength Unit count or fill size Container size
(including materials of
construction)

(b) Summary of specifications of each primary and functional secondary (e.g. foil pouches) packaging components:

Packaging component Specifications
(list parameters e.g. identification (IR))
HDPE bottle

PP cap
Induction sealed liners
Blister films (PVC, etc)
Aluminum foil backing
etc.

(c) Other information on the container closure system(s):

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2.3.P.8 Stability

2.3.P.8.1 Stability Summary and Conclusions

(a) Summary of stress testing and results (e.g. photostability studies, cyclic studies, freeze-thaw studies):

(b) Summary of accelerated and long-term testing parameters (e.g. studies conducted):

Storage conditions (■C,
Strength and Batch size Container closure Completed (and
% RH) batch system proposed) test
number intervals

Summary of the stability results observed for the above accelerated and long-term studies:

Test Results
Description
Moisture
Impurities
Assay
etc.

(c) Proposed storage statement and shelf-life (and in-use storage conditions and in-use period, if applicable):

Container closure system Storage statement Shelf-life

2.3.P.8.2 Post-approval Stability Protocol and Stability Commitment

(a) Stability protocol for Primary stability batches (e.g. storage conditions (including tolerances), batch numbers and batch sizes, tests and acceptance criteria, testing frequency, container closure system(s)):

Parameter Details
Storage condition(s) (■C, % RH)
Batch number(s) / batch size(s)
Tests and acceptance criteria Description
Moisture
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Parameter Details
Impurities
Assay
etc.

Testing frequency
Container closure system(s)

(b) Stability protocol for Commitment batches (e.g. storage conditions (including tolerances), batch numbers (if known) and batch sizes, tests and acceptance criteria, testing frequency, container closure system(s)):

Parameter Details
Storage condition(s) (■C, % RH)
Batch number(s) / batch size(s) <not less than three production batches in each container closure system>
Tests and acceptance Criteria Description
Moisture
Impurities
Assay
etc.
Testing Frequency
Container Closure System(s)

(c) Stability protocol for Ongoing batches (e.g. storage conditions (including tolerances), number of batches per strength and batch sizes, tests and acceptance criteria, testing frequency, container closure system(s)):

Parameter Details
Storage condition(s) (■C, % RH)
Batch size(s), annual allocation <at least one production batch per year (unless none is produced that year) in each container closure system >
Tests and acceptance Criteria Description
Moisture
Impurities
Assay
etc.
Testing frequency
Container closure system(s)

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2.3.P.8.3 Stability Data

(a) The actual stability results should be provided in Module 3.

- (b) Summary of analytical procedures and validation information for those procedures not previously summarized in 2.3.P.5 (e.g. analytical procedures used only for stability studies):
- (c) Bracketing and matrixing design and justification for Commitment and/or Ongoing stability batches, if applicable:

2.3.A APPENDICES

2.3.A.1 Facilities and Equipment

- (a) Summary of information on facilities and equipment, in addition to the information provided in other sections of the submission: Not applicable.

2.3.A.2 Adventitious Agents Safety Evaluation

- (a) Summary of the information assessing the risk with respect to potential contamination with adventitious agents: Not applicable.

2.3.A.3 Excipients

- (a) Summary of the details of manufacture, characterization and controls, with cross references to supporting safety data (nonclinical and/or clinical) for the novel excipients: Not applicable. Novel excipients are not accepted in the Prequalification Programme. See quality guideline for definition.

2.3.R REGIONAL INFORMATION

2.3.R.1 Production Documentation

2.3.R.1.1 Executed Production Documents

- (a) List of batches (including strengths) for which executed production documents have been provided (e.g. comparative bioavailability or biowaiver batches):

2.3.R.1.2 Master Production Documents

- (a) The blank master production documents for each strength, proposed commercial batch size and manufacturing facility should be provided in Module 3.

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2.3.R.2 Analytical Procedures and Validation Information

ANALYTICAL PROCEDURES AND VALIDATION INFORMATION SUMMARIES

ATTACHMENT NUMBER:

HPLC Method Summary Volume/Page:

Method name:

Method code: Version and/or

Date:

Column(s) / temperature (if other than ambient):

Mobile phase (specify gradient program, if applicable):

Detector (and wavelength, if applicable):

Flow rate:

Injection volume:

Sample solution concentration

(expressed as mg/ml, let this be termed "A"):

Reference solution concentration

(expressed as mg/ml and as % of "A"):

System suitability solution concentration

(expressed as mg/ml and as % of "A"):

System suitability tests (tests and acceptance criteria):

Method of quantification (e.g. against API or impurity reference standard(s)):

Other information (specify):

ATTACHMENT NUMBER:

Validation Summary Volume/Page:

Analytes:

Typical retention times (RT)

Relative retention times (RTImp./RTAPI or Int. Std.):

Relative response factor (RFImp./RFAPI):

Specificity:

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ATTACHMENT NUMBER:

Linearity / Range: Number of concentrations:

Range (expressed as %

"A"):

Slope:

Y-intercept:

Correlation coefficient

(r²) :

Accuracy: Conc.(s) (expressed as %

"A"):

Number of replicates:

Percent recovery

(avg/RSD):
Precision / Conc.(s) (expressed as %
Repeatability: "A"):
(intra-assay precision) Number of replicates:
Result (avg/RSD):
Precision / Parameter(s) altered:
Intermediate Precision: Result (avg/RSD):
(days/analysts/equipment)
Limit of Detection (LOD): (expressed as % "A")
Limit of Quantitation (LOQ): (expressed as %
"A")
Robustness: Stability of solutions:

Other variables/effects:
Typical chromatograms or spectra may be found
in:
Company(s) responsible for method validation:
Other information (specify):

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ANNEX VIII: Product Quality Review (PQR) requirements for generic
pharmaceutical products

For an established generic product a product quality review may satisfy the requirements of
sections 3.2.P.2.2.1 (a), 3.2.P.2.3 (a) and 3.2.P.3.5 of the PD and QOS-PD.

A product quality review should be submitted with the objective of verifying the consistency
of the quality of the FPP and its manufacturing process.

Rejected batches should not be included in the analysis but must be reported separately
together with the reports of failure investigations, as indicated below.

Reviews should be conducted with not less than 10 consecutive batches manufactured over
the period of the last 12 months, or, where 10 batches were not manufactured in the last 12
months, not less than 25 consecutive batches manufactured over the period of the last 36
months and should include at least:

1. A review of starting and primary packaging materials used in the FPP, especially those from
new sources.
2. A tabulated review and statistical analysis of quality control and in-process control results.
3. A review of all batches that failed to meet established specification(s).
4. A review of all critical deviations or non-conformances and related investigations.
5. A review of all changes carried out to the processes or analytical methods.
6. A review of the results of the stability-monitoring programme.
7. A review of all quality-related returns, complaints and recalls, including export- only
medicinal products.
8. A review of the adequacy of previous corrective actions.
9. A list of validated analytical and manufacturing procedures and their revalidation dates.

Notes
Reviews must include data from all batches manufactured during the review period. Data
should be presented in tabular or graphical form (i.e. charts or graphs), when applicable.

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PART II:

GUIDELINES ON STABILITY TESTING REQUIREMENTS FOR ACTIVE
PHARMACEUTICAL INGREDIENTS (APIs) AND FINISHED PHARMACEUTICAL
PRODUCTS (FPPs)

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Abbreviations and Acronyms

APIs: Active Pharmaceutical Ingredient

EAC: East Africa Community

FDC: Fixed Dose Combination

FPP: Finished Pharmaceutical Product

FPPs: Finished Medicinal Products

ICH: International Conference on Harmonization

LVPs: Large Volume Parenterals

NMT: Not More Than

RH: Relative Humidity

SVPs: Small Volume Parenterals

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1. INTRODUCTION

1.1 Objective

The guideline describes the core stability data package required for active pharmaceutical
ingredients (APIs) and finished medicinal products (FPPs). However, alternative approaches
can be used when they are scientifically justified. The guideline is adopted from WHO

Technical Report Series, No. 953, Annex II. Further guidance can be found in International Conference on Harmonisation (ICH) guidelines (3) and in the WHO guidelines on the active pharmaceutical ingredient master file procedure (4).

It is recommended that the guideline should also be applied to products that are already being marketed, with allowance for an appropriate transition period, e.g. upon re-registration or upon re-evaluation.

1.2 Scope

The guideline applies to new and existing APIs and addresses information to be submitted in original and subsequent applications for marketing authorization of their related FPP for human use. The guideline is not applicable to stability testing for biologicals (for details on vaccines please see WHO guidelines for stability evaluation of vaccines (5)).

1.3 General principles

The purpose of stability testing is to provide evidence of how the quality of an API or FPP varies with time under the influence of a variety of environmental factors such as temperature, humidity and light. The stability programme also includes the study of product-related factors that influence its quality, for example, interaction of API with excipients, container closure systems and packaging materials. In fixed-dose combination FPPs (FDCs) the interaction between two or more APIs also has to be considered.

As a result of stability testing a re-test period for the API (in exceptional cases, e.g. for unstable APIs, a shelf-life is given) or a shelf-life for the FPP can be established and storage conditions can be recommended.

2. GUIDELINE

2.1 Active pharmaceutical ingredient

2.1.1 General

Information on the stability of the API is an integral part of the systematic approach to stability evaluation. Potential attributes to be tested on an API during stability testing are listed in the examples of testing parameters (Annex I).

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The re-test period or shelf-life assigned to the API by the API manufacturer should be derived from stability testing data.

2.1.2 Stress testing

Stress testing of the API can help identify the likely degradation products, which, in turn, can help establish the degradation pathways and the intrinsic stability of the molecule and validate the stability-indicating power of the analytical procedures used. The nature of the stress testing will depend on the individual API and the type of FPP involved.

For an API the following approaches may be used:-

- when available, it is acceptable to provide the relevant data published in the scientific literature to support the identified degradation products and pathways;

- when no data are available, stress testing should be performed.

Stress testing may be carried out on a single batch of the API. It should include the effect of temperature (in 10 °C increments (e.g. 50 °C, 60 °C, e.t.c.) above the temperature used for accelerated testing), humidity (e.g. 75% relative humidity (RH) or greater) and, where appropriate, acid stress, alkaline stress, oxidation and photolysis on the API. The testing should also evaluate the susceptibility of the API to hydrolysis across a justified range of pH values when in solution or suspension (10).

Assessing the necessity for photostability testing should be an integral part of a stress testing strategy. More details can be found in other guidelines (3).

Results from these studies will form an integral part of the information provided to regulatory authorities.

2.1.3 Selection of batches

Data from stability studies on at least three primary batches of the API should normally be provided. The batches should be manufactured to a minimum of pilot scale by the same synthesis route as production batches, and using a method of manufacture and procedure that simulates the final process to be used for production batches. The overall quality of the batches of API placed on stability studies should be representative of the quality of the material to be made on a production scale.

For existing active substances that are known to be stable, data from at least two primary batches should be provided.

2.1.4 Container closure system

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The stability studies should be conducted on the API packaged in a container closure system that is the same as, or simulates, the packaging proposed for storage and distribution.

2.1.5 Specification

Stability studies should include testing of those attributes of the API that are susceptible to change during storage and are likely to influence quality, safety and/or efficacy. The testing should cover, as appropriate, the physical, chemical, biological and microbiological attributes. A guide as to the potential attributes to be tested in the stability studies is provided in Appendix 1.

Validated stability-indicating analytical procedures should be applied. Whether and to what extent replication should be performed will depend on the results from validation studies (11).

2.1.6 Testing frequency

For long-term studies, frequency of testing should be sufficient to establish the stability profile of the API.

For APIs with a proposed re-test period or shelf-life of at least 12 months, the frequency of testing at the long-term storage condition should normally be every three months over the first year, every six months over the second year, and annually thereafter throughout the proposed re-test period or shelf-life.

At the accelerated storage condition, a minimum of three time points, including the initial and final time points (e.g. 0, 3 and 6 months), from a six month study is recommended. Where it is expected (based on development experience) that results from accelerated studies are likely to approach significant change criteria, increased testing should be conducted either by adding samples at the final time point or by including a fourth time point in the study design.

2.1.7 Storage conditions

In general an API should be evaluated under storage conditions (with appropriate tolerances) that test its thermal stability and, if applicable, its sensitivity to moisture. The storage conditions and the lengths of studies chosen should be sufficient to cover storage and shipment.

Storage condition tolerances are defined as the acceptable variations in temperature and relative humidity of storage facilities for stability studies. The equipment used should be capable of controlling the storage conditions within the ranges defined in these guidelines. The storage conditions should be monitored and recorded. Short-term environmental changes due to opening the doors of the storage facility are accepted as unavoidable. The effect of excursions due to equipment failure should be assessed, addressed and reported if judged to affect stability results. Excursions that exceed the defined tolerances for more than 24 hours should be described in the study report and their effects assessed.

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The long-term testing should normally take place over a minimum of 12 months for the number of batches specified in section 2.1.3 at the time of submission, and should be continued for a period of time sufficient to cover the proposed re-test period or shelf-life. For existing substances that are known to be stable, data covering a minimum of six months may be submitted. Additional data accumulated during the assessment period of the registration application should be submitted to the authorities upon request.

Available information on the stability of the API under accelerated and long term storage conditions should be provided, including information in the public domain or obtained from scientific literature. The source of the information should be identified.

The required long-term storage conditions for APIs are either 30°C ± 2 °C/65% ± 5% RH or 30 ± 20°C/75% ± 5% RH. Alternative conditions should be supported with appropriate evidence, which may include literature references or in-house studies, demonstrating that storage at 300C is inappropriate for the API. For APIs intended for storage in a refrigerator and those intended for storage in a freezer, refer section 2.1.7.1.

APIs intended for storage below -20 °C should be treated on a case-by-case basis. To establish the retest period, data should be provided on not less than three batches of at least pilot scale. The batches should be manufactured by the same synthesis route as production batches and using a method of manufacture and a procedure that simulates the final process to be used for production batches.

2.1.7.1 General case

Study Storage condition Minimum time period covered by data at submission

Long-term

30 °C ± 2 °C/65% RH ± 5% RH or

30 °C ± 2 °C/75% RH ± 5% RH

12 months or 6 months as described in point 2.1.7

Accelerated

40 °C ± 2 °C/75% RH ± 5% RH 6 months

2.1.7.2 Active pharmaceutical ingredients intended for storage in a refrigerator

Study Storage condition Minimum time period covered by data at submission

Long-term

5 °C ± 3 °C 12 months

Accelerated

25 °C ± 2 °C/60% RH ± 5% RH or

30 °C ± 2 °C/65% RH ± 5% RH or

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30 °C ± 2 °C/75% RH ± 5% RH

6 months

Whether accelerated stability studies are performed at 25 ± 2 °C/60% RH ± 5% RH or 30 °C ± 2 °C/65% RH ± 5% RH or 30 °C ± 2 °C/75% RH ± 5% RH is based on a risk-based evaluation.

Testing at a more severe long term condition can be an alternative to storage testing at 25 °C/60%RH or 30 °C/65%RH. Data on refrigerated storage should be assessed according to the evaluation section of this guideline, except where explicitly noted below. If significant change occurs between three and six months' testing at the accelerated storage condition, the proposed re-test period should be based on the data available at the long-term storage condition. If significant change occurs within the first three months' testing at the accelerated storage

condition a discussion should be provided to address the effect of short-term excursions outside the label storage condition, e.g. during shipping or handling. This discussion can be supported, if appropriate, by further testing on a single batch of the API for a period shorter than three months but with more frequent testing than usual. It is considered unnecessary to continue to test an API for the whole six months when a significant change has occurred within the first three months.

2.1.7.3 Active pharmaceutical ingredients intended for storage in a freezer

Study Storage condition Minimum time period covered by data at submission

Long-term

-20 °C ± 5 °C 12 months

In the rare case of any API of non-biological origin being intended for storage in a freezer, the re-test period or shelf-life should be based on the long-term data obtained at the long-term storage condition. In the absence of an accelerated storage condition for APIs intended to be stored in a freezer, testing on a single batch at an elevated temperature (e.g. 5 °C ± 3 °C or 25 °C ± 2 °C or 30 °C ± 2 °C) for an appropriate time period should be conducted to address the effect of short-term excursions outside the proposed label storage condition, e.g. during shipping or handling.

2.1.7.4 Active pharmaceutical ingredients intended for storage below -20°C

APIs intended for storage below -20 °C should be treated on a case-by-case basis.

2.1.8 Stability commitment

When the available long-term stability data on primary batches do not cover the proposed re-test period granted at the time of approval, a commitment should be made to continue the stability studies post-approval in order to firmly establish the re-test period or shelf-life.

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Where the submission includes long-term stability data on the number of production batches specified in section 2.1.3 covering the proposed re-test period, a post-approval commitment is considered unnecessary. Otherwise one of the following commitments should be made:-

■ If the submission includes data from stability studies on the number of production batches specified in section 2.1.3, a commitment should be made to continue these studies through the proposed re-test period.

■ If the submission includes data from stability studies on fewer than the number of production batches specified in section 2.1.3, a commitment should be made to continue these studies through the proposed re-test period and to place additional production batches, to a total of at least three, in long-term stability studies through the proposed re-test period.

■ If the submission does not include stability data on production batches, a commitment should be made to place the first two or three production batches (see section 2.1.3) on long-term stability studies through the proposed re-test period.

The stability protocol used for long-term studies for the stability commitment should be the same as that for the primary batches, unless otherwise scientifically justified.

2.1.9 Evaluation

The purpose of the stability study is to establish, based on testing a minimum of the number of batches specified in section 2.1.3, unless otherwise justified and authorized, of the API and evaluating the stability information (including, as appropriate, results of the physical, chemical, biological and microbiological tests), a re-test period applicable to all future batches of the API manufactured under similar circumstances. The degree of variability of individual batches affects the confidence that a future production batch will remain within specification throughout the assigned re-test period.

The data may show so little degradation and so little variability that it is apparent from looking at them that the requested re-test period will be granted. Under these circumstances it is normally unnecessary to go through the statistical analysis; providing a justification for the omission should be sufficient.

An approach for analysing the data on a quantitative attribute that is expected to change with time is to determine the time at which the 95% one-sided confidence limit for the mean curve intersects the acceptance criterion. If analysis shows that the batch-to-batch variability is small, it is advantageous to combine the data into one overall estimate. This can be done by first applying appropriate statistical tests (e.g. p values for level of significance of rejection of more than 0.25) to the slopes of the regression lines and zero time intercepts for the individual batches. If it is inappropriate to combine data from several batches, the overall re-test period should be based on the minimum time a batch can be expected to remain within acceptance criteria.

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The nature of any degradation relationship will determine whether the data should be transformed for linear regression analysis. Usually the relationship can be represented by a linear, quadratic or cubic function on an arithmetic or logarithmic scale. As far as possible, the choice of model should be justified by a physical and/or chemical rationale and should also take into account the amount of available data (parsimony principle to ensure a robust prediction). Statistical methods should be employed to test the goodness of fit of the data on all batches and combined batches (where appropriate) to the assumed degradation line or curve.

Limited extrapolation of the long-term data from the long-term storage condition beyond the observed range to extend the re-test period can be undertaken if justified. This justification should be based on what is known about the mechanism of degradation, the results of testing under accelerated conditions, the goodness of fit of any mathematical model, batch size and existence of supporting stability data. However, this extrapolation assumes that the same

degradation relationship will continue to apply beyond the observed data.

Any evaluation should cover not only the assay but also the levels of degradation products and other appropriate attributes. Where appropriate, attention should be paid to reviewing the adequacy of evaluation linked to FPP stability and degradation "behaviour" during the testing.

2.1.10 Statements and labelling

A storage statement should be established for display on the label based on the stability evaluation of the API. Where applicable specific instructions should be provided, particularly for APIs that cannot tolerate freezing or excursions in temperature. Terms such as "ambient conditions" or "room temperature" should be avoided.

The recommended labelling statements for use if supported by the stability studies are provided in Appendix 2.

A re-test period should be derived from the stability information, and a retest date should be displayed on the container label if appropriate.

2.1.11 On-going stability studies

The stability of the API should be monitored according to a continuous and appropriate programme that will permit the detection of any stability issue (e.g. changes in levels of degradation products). The purpose of the on-going stability programme is to monitor the API and to determine that the API remains, and can be expected to remain, within specifications under the storage conditions indicated on the label, within the re-test period in all future batches.

The on-going stability programme should be described in a written protocol and the results presented in a formal report.

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The protocol for an on-going stability programme should extend to the end of the re-test period and shelf-life and should include, but not be limited to, the following parameters:-

- i. number of batch (es) and different batch sizes, if applicable;
- ii. relevant physical, chemical, microbiological and biological test methods;
- iii. acceptance criteria;
- iv. reference to test methods;
- v. description of the container closure system(s);
- vi. testing frequency;
- vii. description of the conditions of storage (standardized conditions for long-term testing as described in these guidelines, and consistent with the API labelling, should be used);
- and
- viii. other applicable parameters specific to the API.

At least one production batch per year of API (unless none is produced during that year) should be added to the stability monitoring programme and tested at least annually to confirm the stability (12). In certain situations additional batches should be included in the on-going stability programme. For example, an on-going stability study should be conducted after any significant change or significant deviation to the synthetic route, process or container closure system which may have an impact upon the stability of the API (13).

Out-of-specification results or significant atypical trends should be investigated. Any confirmed significant change, out-of-specification result, or significant atypical trend should be reported immediately to the relevant finished product manufacturer. The possible impact on batches on the market should be considered in consultation with the relevant finished product manufacturers and the competent authorities.

A summary of all the data generated, including any interim conclusions on the programme, should be written and maintained. This summary should be subjected to periodic review.

2.2 Finished medicinal product

2.2.1 General

The design of the stability studies for the FPP should be based on knowledge of the behaviour and properties of the API, information from stability studies on the API and on experience gained from pre-formulation studies and investigational FPPs.

2.2.2 Selection of batches

Data from stability studies should be provided on at least three primary batches of the FPP. The primary batches should be of the same formulation and packaged in the same container closure system as proposed for marketing. The manufacturing process used for primary batches should simulate that to be applied to production batches and should provide product of the same quality and meeting the same specification as that intended for marketing. In the case of

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conventional dosage forms with APIs that are known to be stable, data from at least two primary batches should be provided.

Two of the three batches should be at least pilot-scale batches and the third one can be smaller, if justified. Where possible, batches of the FPP should be manufactured using different batches of the API(s).

Stability studies should be performed on each individual strength, dosage form and container type and size of the FPP unless bracketing or matrixing is applied.

2.2.3 Container closure system

Stability testing should be conducted on the dosage form packaged in the container closure system proposed for marketing. Any available studies carried out on the FPP outside its

immediate container or in other packaging materials can form a useful part of the stress testing of the dosage form or can be considered as supporting information, respectively.

2.2.4 Specification

Stability studies should include testing of those attributes of the FPP that are susceptible to change during storage and are likely to influence quality, safety, and/or efficacy. The testing should cover, as appropriate, the physical, chemical, biological and microbiological attributes, preservative content (e.g. antioxidant or antimicrobial preservative) and functionality tests (e.g. for a dose delivery system). Examples of testing parameters in the stability studies are listed in Appendix 1. Analytical procedures should be fully validated and stability-indicating. Whether and to what extent replication should be performed will depend on the results of validation studies.

Shelf-life acceptance criteria should be derived from consideration of all available stability information. It may be appropriate to have justifiable differences between the shelf-life and release acceptance criteria based on the stability evaluation and the changes observed on storage. Any differences between the release and shelf-life acceptance criteria for antimicrobial preservative content should be supported by a validated correlation of chemical content and preservative effectiveness demonstrated during development of the pharmaceutical product with the product in its final formulation (except for preservative concentration) intended for marketing. A single primary stability batch of the FPP should be tested for effectiveness of the antimicrobial preservative (in addition to preservative content) at the proposed shelf-life for verification purposes, regardless of whether there is a difference between the release and shelf-life acceptance criteria for preservative content.

2.2.5 Testing frequency

For long-term studies, frequency of testing should be sufficient to establish the stability profile of the FPP.

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For products with a proposed shelf-life of at least 12 months, the frequency of testing at the long-term storage condition should normally be every three months over the first year, every six months over the second year and annually thereafter throughout the proposed shelf-life.

At the accelerated storage condition, a minimum of three time points, including the initial and final time points (e.g. 0, 3 and 6 months), from a six-month study is recommended. Where an expectation (based on development experience) exists that results from accelerated testing are likely to approach significant change criteria, testing should be increased either by adding samples at the final time point or by including a fourth time point in the study design.

Reduced designs, i.e. matrixing or bracketing, where the testing frequency is reduced or certain factor combinations are not tested at all, can be applied if justified (3).

2.2.6 Storage conditions

In general an FPP should be evaluated under storage conditions with specified tolerances that test its thermal stability and, if applicable, its sensitivity to moisture or potential for solvent loss. The storage conditions and the lengths of studies chosen should be sufficient to cover storage, shipment and subsequent use with due regard to the climatic conditions in which the product is intended to be marketed.

Photostability testing, which is an integral part of stress testing, should be conducted on at least one primary batch of the FPP if appropriate. More details can be found in other guidelines (3).

The orientation of the product during storage, i.e. upright versus inverted, may need to be included in a protocol where contact of the product with the closure system may be expected to affect the stability of the products contained, or where there has been a change in the container closure system.

Storage condition tolerances are usually defined as the acceptable variations in temperature and relative humidity of storage facilities for stability studies. The equipment used should be capable of controlling the storage conditions within the ranges defined in these guidelines. The storage conditions should be monitored and recorded. Short-term environmental changes due to opening of the doors of the storage facility are accepted as unavoidable. The effect of excursions due to equipment failure should be assessed, addressed and reported if judged to affect stability results.

Excursions that exceed the defined tolerances for more than 24 hours should be described in the study report and their effects assessed. The long-term testing should cover a minimum of six or 12 months at the time of submission and should be continued for a period of time sufficient to cover the proposed shelf-life. For an FPP containing an API that is known to be stable and where no significant change is observed in the FPP stability studies at accelerated and long-term conditions for at least 6 months data covering a minimum of six months should be submitted.

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Additional data accumulated during the assessment period of the registration application should be submitted to the authorities if requested. Data from the accelerated storage condition can be used to evaluate the effect of short-term excursions outside the label storage conditions (such as might occur during shipping). Long-term and accelerated storage conditions for FPPs are detailed in the sections below. The general case applies if the FPP is not specifically covered by a subsequent section (2.1.7.1). Alternative storage conditions can be used if justified.

2.2.6.1 General case

Study Storage condition Minimum time period covered by data at submission

Long-term

30 °C ± 2 °C/75% RH ± 5% RH

12 months or 6 months as referred to in section 2.2.6

Accelerated
40 °C ± 2 °C/75% RH ± 5% RH 6 months

In general “significant change” for an FPP is defined as:

- i. A change from the initial content of API(s) of 5% or more detected by assay, or failure to meet the acceptance criteria for potency when using biological or immunological procedures. (Note: Other values may be applied, if justified, to certain products, such as multivitamins and herbal preparations.)
- ii. Any degradation product exceeding its acceptance criterion.
- iii. Failure to meet the acceptance criteria for appearance, physical attribute and functionality test (e.g. colour, phase separation, resuspendability, caking, hardness, and dose delivery per actuation). However, some changes in physical attributes (e.g. softening of suppositories, melting of creams, and partial loss of adhesion for transdermal products) may be expected under accelerated conditions. Also, as appropriate for the dosage form:
- iv. failure to meet the acceptance criterion for pH; or
- v. failure to meet the acceptance criteria for dissolution for 12 dosage units.

2.2.6.2 FPPs packaged in impermeable containers

Parameters required to classify the packaging materials as permeable or impermeable depend on the characteristics of the packaging material, such as thickness and permeability coefficient. The suitability of the packaging material used for a particular product is determined by its product characteristics. Containers generally considered to be moisture impermeable include glass ampoules.

Sensitivity to moisture or potential for solvent loss is not a concern for FPPs packaged in impermeable containers that provide a permanent barrier to passage of moisture or solvent.

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Thus stability studies for products stored in impermeable containers can be conducted under any controlled or ambient relative humidity condition.

2.2.6.3 FPPs packaged in semi-permeable containers

Aqueous-based products packaged in semi-permeable containers should be evaluated for potential water loss in addition to physical, chemical, biological and microbiological stability. This evaluation can be carried out under conditions of low relative humidity, as discussed below. Ultimately it should be demonstrated that aqueous-based FPPs stored in semi-permeable containers could withstand environments with low relative humidity.

Other comparable approaches can be developed and reported for non-aqueous, solvent-based products.

Study Storage condition Minimum time period covered by data at submission

Long-term
30 °C ± 2 °C/35% RH ± 5% RH
12 months

Accelerated
40 °C ± 2 °C/not more than
(NMT) 25% RH
6 months

Products meeting the long-term storage conditions and the accelerated conditions, as specified in the table above, have demonstrated the integrity of the packaging in semi-permeable containers.

A 5% loss in water from its initial value is considered a significant change for a product packaged in a semi-permeable container after an equivalent of three months' storage at 40 °C not more than (NMT) 25% RH. However, for small containers (1 ml or less) or unit-dose products, a water loss of 5% or more after an equivalent of three months' storage at 40 °C/NMT 25% RH may be appropriate, if justified.

An alternative approach to studies at the low relative humidity as recommended in the table above (for either long-term or accelerated testing) is to perform the stability studies under higher relative humidity and deriving the water loss at the low relative humidity through calculation. This can be achieved by experimentally determining the permeation coefficient for the container closure system or, as shown in the example below, using the calculated ratio of water loss rates between the two humidity conditions at the same temperature. The permeation coefficient for a container closure system can be experimentally determined by using the worst-case scenario (e.g. the most diluted of a series of concentrations) for the proposed FPP.

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2.2.6.4 FPPs intended for storage in a refrigerator

Study Storage condition Minimum time period covered by data at submission

Long-term
5 °C ± 3 °C 12 months

Accelerated
25 °C ± 2 °C/60% RH ± 5% RH or
30 °C ± 2 °C/65% RH ± 5% RH or
30 °C ± 2 °C/75% RH ± 5% RH
6 months

Whether accelerated stability studies are performed at 25 ± 2 °C/60% RH ± 5% RH or 30 °C ± 2 °C/65% RH ± 5% RH or 30 °C ± 2 °C/75% RH ± 5% RH is based on a risk-based evaluation. Testing at a more severe accelerated condition can be an alternative to the storage condition at

25 °C/60% RH or 30 °C/65% RH.

If the FPP is packaged in a semi-permeable container, appropriate information should be provided to assess the extent of water loss.

Data from refrigerated storage should be assessed according to the evaluation section of these guidelines, except where explicitly noted below.

If significant change occurs between three and six months' testing at the accelerated storage condition, the proposed shelf-life should be based on the data available from the long-term storage condition.

If significant change occurs within the first three months' testing at the accelerated storage condition, a discussion should be provided to address the effect of short-term excursions outside the label storage condition, e.g. during shipment and handling. This discussion can be supported, if appropriate, by further testing on a single batch of the FPP for a period shorter than three months but with more frequent testing than usual. It is considered unnecessary to continue to test a product throughout six months when a significant change has occurred within the first three months of accelerated studies at the specific condition chosen in accordance with the risk analysis.

2.2.6.5 FPPs intended for storage in a freezer

Study Storage condition Minimum time period covered by data at submission

Long-term

-20 °C ± 5 °C 12 months

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For FPPs intended for storage in a freezer, the shelf-life should be based on the long-term data obtained at the long-term storage condition. In the absence of an accelerated storage condition for FPPs intended to be stored in a freezer, testing on a single batch at an elevated temperature (e.g. 5 °C ± 3 °C or 25 °C ± 2 °C or 30 °C ± 2 °C) for an appropriate time period should be conducted to address the effect of short-term excursions outside the proposed label storage condition.

2.2.6.6 FPPs intended for storage below -20 °C

FPPs intended for storage at temperatures below -20 °C should be treated on a case-by-case basis.

2.2.7 Stability commitment

When the available long-term stability data on primary batches do not cover the proposed shelf-life granted at the time of approval, a commitment should be made to continue the stability studies post-approval to firmly establish the shelf-life.

Where the submission includes long-term stability data from the production batches as specified in section 2.2.2 covering the proposed shelf-life, a post-approval commitment is considered unnecessary. Otherwise, one of the following commitments should be made:-

1. If the submission includes data from stability studies on at least the number of production batches specified in section 2.2.2, a commitment should be made to continue the long-term studies throughout the proposed shelf-life and the accelerated studies for six months.
2. If the submission includes data from stability studies on fewer than the number of production batches specified in section 2.2.2, a commitment should be made to continue the long-term studies throughout the proposed shelf-life and the accelerated studies for six months, and to place additional production batches, to a total of at least three, on long-term stability studies throughout the proposed shelf-life and on accelerated studies for six months.
3. If the submission does not include stability data on production batches, a commitment should be made to place the first two or three production batches (see section 2.2.2) on long-term stability studies throughout the proposed shelf-life and on accelerated studies for six months.

The stability protocol used for studies on commitment batches should be the same as that for the primary batches, unless otherwise scientifically justified.

2.2.8 Evaluation

A systematic approach should be adopted to the presentation and evaluation of the stability information, which should include, as appropriate, results from the physical, chemical,

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biological and microbiological tests, including particular attributes of the dosage form (for example, dissolution rate for solid oral dosage forms).

The purpose of the stability study is to establish, based on testing a minimum number of batches of the FPP as specified in section 2.2.2, a shelf-life and label storage instructions applicable to all future batches of the FPP manufactured under similar circumstances. The degree of variability of individual batches affects the confidence that a future production batch will remain within specification throughout its shelf-life.

Where the data show so little degradation and so little variability that it is apparent from looking at the data that the requested shelf-life will be granted, it is normally unnecessary to go through the statistical analysis. However, a provisional shelf-life of 24 months may be established provided the following conditions are satisfied:

1. The API is known to be stable (not easily degradable).
2. Stability studies, as outlined above in section 2.1.11, have been performed and no significant

changes have been observed.

3. Supporting data indicate that similar formulations have been assigned a shelf-life of 24 months or more.

4. The manufacturer will continue to conduct long-term studies until the proposed shelf-life has been covered, and the results obtained will be submitted to the national medicines regulatory authority.

An approach for analysing the data on a quantitative attribute that is expected to change with time is to determine the time at which the 95% one-sided confidence limit for the mean curve intersects the acceptance criterion. If analysis shows that the batch-to-batch variability is small, it is advantageous to combine the data into one overall estimate. This can be done by first applying appropriate statistical tests (e.g. p values for level of significance of rejection of more than 0.25) to the slopes of the regression lines and zero time intercepts for the individual batches. If it is inappropriate to combine data from several batches, the overall shelf-life should be based on the minimum time a batch can be expected to remain within acceptance criteria.

The nature of any degradation relationship will determine whether the data should be transformed for linear regression analysis. Usually the relationship can be represented by a linear, quadratic or cubic function on an arithmetic or logarithmic scale. As far as possible, the choice of model should be justified by a physical and/or chemical rationale and should also take into account the amount of available data (parsimony principle to ensure a robust prediction).

Statistical methods should be employed to test the goodness of fit of the data on all batches and combined batches (where appropriate) to the assumed degradation line or curve.

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Limited extrapolation of the long-term data from the long-term storage condition beyond the observed range to extend the shelf-life can be undertaken, if justified. This justification should be based on what is known about the mechanism of degradation, the results of testing under accelerated conditions, the goodness of fit of any mathematical model, batch size and the existence of supporting stability data. However, this extrapolation assumes that the same degradation relationship will continue to apply beyond the observed data.

Any evaluation should consider not only the assay but also the degradation products and other appropriate attributes. Where appropriate, attention should be paid to reviewing the adequacy of evaluation linked to FPP stability and degradation "behaviour" during the testing.

2.2.9 Statements and labelling

A storage statement should be established for the label based on the stability evaluation of the FPP. Where applicable, specific instructions should be provided particularly for FPPs that cannot tolerate freezing. Terms such as "ambient conditions" or "room temperature" must be avoided.

There should be a direct link between the storage statement on the label and the demonstrated stability of the FPP. An expiry date should be displayed on the container label. The recommended labelling statements for use, if supported by the stability studies, are provided in Appendix 2.

In principle, FPPs should be packed in containers that ensure stability and protect the FPP from deterioration. A storage statement should not be used to compensate for inadequate or inferior packaging. Additional labelling statements could be used in cases where the results of the stability testing demonstrate limiting factors (see also Appendix 2).

2.2.10 In-use stability

The purpose of in-use stability testing is to provide information for the labelling on the preparation, storage conditions and utilization period of multi-dose products after opening, reconstitution or dilution of a solution, e.g. an antibiotic injection supplied as a powder for reconstitution.

As far as possible the test should be designed to simulate the use of the FPP in practice, taking into consideration the filling volume of the container and any dilution or reconstitution before use. At intervals comparable to those which occur in practice appropriate quantities should be removed by the withdrawal methods normally used and described in the product literature.

The physical, chemical and microbial properties of the FPP susceptible to change during storage should be determined over the period of the proposed in-use shelf-life. If possible, testing should be performed at intermediate time points and at the end of the proposed in-use shelf-life on the final amount of the FPP remaining in the container. Specific parameters, e.g. for liquids and semi-solids, preservatives, per content and effectiveness, need to be studied.

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A minimum of two batches, at least pilot-scale batches, should be subjected to the test. At least one of these batches should be chosen towards the end of its shelf-life. If such results are not available, one batch should be tested at the final point of the submitted stability studies.

This testing should be performed on the reconstituted or diluted FPP throughout the proposed in-use period on primary batches as part of the stability studies at the initial and final time points and, if full shelf-life, long term data are not available before submission, at 12 months or the last time point at which data will be available.

In general this testing need not be repeated on commitment batches (see 2.2.10).

2.2.11 Variations

Once the FPP has been registered, additional stability studies are required whenever variations that may affect the stability of the API or FPP are made, such as major variations (13).

The following are examples of such changes:

- change in the manufacturing process;
- change in the composition of the FPP;
- change of the immediate packaging;
- change in the manufacturing process of an API.

In all cases of variations, the applicant should investigate whether the intended change will or will not have an impact on the quality characteristics of APIs and/or FPPs and consequently on their stability.

The scope and design of the stability studies for variations and changes are based on the knowledge and experience acquired on APIs and FPPs.

The results of these stability studies should be communicated to the regulatory authorities concerned (14).

2.2.12 On-going stability studies

After a marketing authorization has been granted, the stability of the FPP should be monitored according to a continuous appropriate programme that will permit the detection of any stability issue (e.g. changes in levels of impurities or dissolution profile) associated with the formulation in the container closure system in which it is marketed. The purpose of the On-going stability programme is to monitor the product over its shelf-life and to determine that the product remains, and can be expected to remain, within specifications under the storage conditions on the label.

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This mainly applies to the FPP in the container closure system in which it is supplied, but consideration should also be given to inclusion in the programme of bulk products. For example, when the bulk product is stored for a long period before being packaged and/or shipped from a manufacturing site to a packaging site, the impact on the stability of the packaged product should be evaluated and studied. Generally this would form part of development studies, but where this need has not been foreseen, inclusion of a one-off study in the on-going stability programme could provide the necessary data. Similar considerations could apply to intermediates that are stored and used over prolonged periods.

The on-going stability programme should be described in a written protocol and results formalized as a report.

The protocol for an on-going stability programme should extend to the end of the shelf-life period and should include, but not be limited to, the following parameters:

1. Number of batch(es) per strength and different batch sizes, if applicable;
2. The batch size should be recorded, if different batch sizes are employed;
3. Relevant physical, chemical, microbiological and biological test
4. Methods;
5. Acceptance criteria;
6. Reference to test methods;
7. Description of the container closure system(s);
8. Testing frequency;
9. Description of the conditions of storage (standardized conditions for long-term testing as described in these guidelines, and consistent with the product labelling, should be used); and
10. Other applicable parameters specific to the FPP.

The protocol for the on-going stability programme can be different from that of the initial long-term stability study as submitted in the marketing authorization dossier provided that this is justified and documented in the protocol (for example, the frequency of testing, or when updating to meet revised recommendations).

The number of batches and frequency of testing should provide sufficient data to allow for trend analysis. Unless otherwise justified, at least one batch per year of product manufactured in every strength and every primary packaging type, if relevant, should be included in the stability programme (unless none is produced during that year). The principle of bracketing and matrixing designs may be applied if scientifically justified in the protocol (15).

In certain situations additional batches should be included in the on-going stability programme. For example, an on-going stability study should be conducted after any significant change or significant deviation to the processor container closure system. Any reworking, reprocessing or recovery operation should also be considered for inclusion (13).

Out-of-specification results or significant atypical trends should be investigated. Any confirmed significant change, out-of-specification result, or significant atypical trend should be reported

130 immediately to the relevant competent authorities. The possible impact on batches on the market should be considered in consultation with the relevant competent authorities.

A summary of all the data generated, including any interim conclusions on the programme, should be written and maintained. This summary should be subjected to periodic review.

3. GLOSSARY

The definitions provided below apply to the words and phrases used in this guideline. Although an effort has been made to use standard definitions as far as possible, they may have different meanings in other contexts and documents. The following definitions are provided to facilitate interpretation of the guidelines. The definitions are consistent with those published in other WHO quality assurance guidelines.

Accelerated testing

Studies designed to increase the rate of chemical degradation and physical change of an API or FPP by using exaggerated storage conditions as part of the stability testing programme. The data thus obtained, in addition to those derived from long-term stability studies, may be used to

assess longer term chemical effects under non-accelerated conditions and to evaluate the impact of short-term excursions outside the label storage conditions, as might occur during shipping. The results of accelerated testing studies are not always predictive of physical changes.

Bracketing

The design of stability schedule such that only samples at the extremes of certain design factors, e.g. strength and package size, are tested at all-time points as in a full design. The design assumes that the stability of any intermediate levels is represented by the stability of the extremes tested.

Where a range of strengths is to be tested, bracketing is applicable if the strengths are identical or very closely related in composition (e.g. for a tablet range made with different compression weights of a similar basic granulation, or a capsule range made by filling different plug fill weights of the same basic composition into different size capsule shells). Bracketing can be applied to different container sizes or different fills in the same container closure system.

Commitment batches

Production batches of an API or FPP for which the stability studies are initiated or completed post-approval through a commitment made in a regulatory application.

Impermeable containers

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Containers that provide a permanent barrier to the passage of gases or solvents e.g. sealed aluminium tubes for semisolids, sealed glass ampoules for solutions and aluminium/aluminium blisters for solid dosage forms.

In use

See Utilization period

Long-term stability studies

Experiments on the physical, chemical, biological, biopharmaceutical and microbiological characteristics of an API or FPP, during and beyond the expected shelf-life and storage periods of samples under the storage conditions expected in the intended market. The results are used to establish the re-test period or the shelf-life, to confirm the projected re-test period and shelf-life, and to recommend storage conditions.

Matrixing

The design of a stability schedule such that a selected subset of the total number of possible samples for all factor combinations is tested at a specified time point. At a subsequent time point, another subset of samples for all factor combinations is tested. The design assumes that the stability of each subset of samples tested represents the stability of all samples at a given time point. The differences in the samples for the same FPP should be identified as, for example, covering different batches, different strengths, different sizes of the same container closure system, and, possibly in some cases, different container closure systems.

On-going stability study

The study carried out by the manufacturer on production batches according to a predetermined schedule in order to monitor, confirm and extend the projected re-test period (or shelf-life) of the API, or confirm or extend the shelf-life of the FPP.

Pilot-scale batch

A batch of an API or FPP manufactured by a procedure fully representative of and simulating that to be applied to a full production-scale batch. For example, for solid oral dosage forms, a pilot scale is generally, at a minimum, one-tenth that of a full production scale or 100 000 tablets or capsules, whichever is the larger; unless otherwise adequately justified.

Provisional shelf-life

A provisional expiry date which is based on acceptable accelerated and available long-term data for the FPP to be marketed in the proposed container closure system.

Re-test date

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The date after which an active API should be re-examined to ensure that the material is still in compliance with the specification and thus is still suitable for use in the manufacture of an FPP.

Re-test period

The period of time during which the API is expected to remain within its specification and, therefore, can be used in the manufacture of a given FPP, provided that the API has been stored under the defined conditions. After this period a batch of API destined for use in the manufacture of an FPP should be re-tested for compliance with the specification and then used immediately. A batch of API can be re-tested multiple times and a different portion of the batch used after each re-test, as long as it continues to comply with the specification. For most substances known to be labile, it is more appropriate to establish a shelf-life than a re-test period. The same may be true for certain antibiotics.

Significant change (See section 2.2.6.1.)

In general "significant change" for an FPP is defined as:-

1. A 5% or more change in assay from its initial content of API(s), or failure to meet the acceptance criteria for potency when using biological or immunological procedures. (Note: other values may be applied, if justified, to certain products, such as multivitamins and herbal preparations.)

2. Any degradation product exceeding its acceptance criterion.

1. Failure to meet the acceptance criteria for appearance, physical attribute and functionality test (e.g. colour, phase separation, re-suspendability, caking, hardness, dose delivery per actuation). However, some changes in physical attributes (e.g. softening of suppositories, melting of creams or partial loss of adhesion for transdermal products) may be expected under accelerated conditions. Also, as appropriate for the dosage form.

4. Failure to meet the acceptance criterion for pH.

Or

5. Failure to meet the acceptance criteria for dissolution for 12 dosage units.

Stability indicating methods

Validated analytical procedures that can detect the changes with time in the chemical, physical or microbiological properties of the API or FPP, and that are specific so that the content of the API, degradation products, and other components of interest can be accurately measured without interference.

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Stability studies (stability testing)

Long-term and accelerated (and intermediate) studies undertaken on primary and/or commitment batches according to a prescribed stability protocol to establish or confirm the re-test period (or shelf-life) of an API or the shelf-life of an FPP.

Stress testing (of the API)

Studies undertaken to elucidate the intrinsic stability of API(s). Such testing is part of the development strategy and is normally carried out under more severe conditions than those used for accelerated testing.

Stress testing (of the FPP)

Studies undertaken to assess the effect of severe conditions on the FPP. Such studies include photo stability testing and specific testing on certain products (e.g. metered dose inhalers, creams, emulsions, refrigerated aqueous liquid products).

Supporting stability data

Supplementary data, such as stability data on small-scale batches, related formulations, and products presented in containers not necessarily the same as those proposed for marketing, and scientific rationales that support the analytical procedures, the proposed re-test period or the shelf-life and storage conditions.

Utilization period

A period of time during which a reconstituted preparation of the finished dosage form in an unopened multi-dose container can be used.

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2. Regional Guidelines on stability testing of active substances and pharmaceutical products for the WHO Eastern Mediterranean Region. August 2006(<http://www.emro.who.int/edb/media/pdf/EMRC5312En.pdf>).

3. The following ICH Guidelines may be consulted in the context of stability testing:-

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- International Conference on Harmonisation. ICH Q1B: Photostability testing of new drug substances and products (<http://www.ich.org/LOB/media/MEDIA412.pdf>).
- International Conference on Harmonisation. ICH Q1C: Stability testing of new dosage forms (<http://www.ich.org/LOB/media/MEDIA413.pdf>).
- International Conference on Harmonisation. ICH Q1D: Bracketing and matrixing designs for stability testing of new drug substances and products (<http://www.ich.org/LOB/media/MEDIA414.pdf>).
- International Conference on Harmonisation. ICH Q1E: Evaluation for stability data (<http://www.ich.org/LOB/media/MEDIA415.pdf>).
- International Conference on Harmonisation. ICH Q2R1: Validation of analytical procedures: text and methodology (<http://www.ich.org/LOB/media/MEDIA417.pdf>).
- International Conference on Harmonisation. ICH Q3A: Impurities in new drug substances (<http://www.ich.org/LOB/media/MEDIA422.pdf>).
- International Conference on Harmonisation. ICH Q3B: Impurities in new drug products (<http://www.ich.org/LOB/media/MEDIA421.pdf>).
- International Conference on Harmonisation. ICH Q5C: Stability testing of biotechnological/biological products (<http://www.ich.org/LOB/media/MEDIA427.pdf>).
- International Conference on Harmonisation. ICH Q6A: Specifications: Test procedures and acceptance criteria for new drug substances and new drug products: Chemical substances (<http://www.ich.org/LOB/media/MEDIA430.pdf>).
- International Conference on Harmonisation. ICH Q6B: Specifications: Test procedures and acceptance criteria for biotechnological/biological products (<http://www.ich.org/LOB/media/MEDIA432.pdf>).
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PART III:

GUIDELINES ON THERAPEUTIC EQUIVALENCE REQUIREMENTS

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ABBREVIATIONS AND ACRONYMS

APIs - Active Pharmaceutical Ingredients
BCS - Biopharmaceutics Classification System
BMGF - Bill and Melinda Gates Foundation
BMR - Batch Manufacturing Record
CoA - Certificate of Analysis
EAC - East African Community
EAC-MRH - East African Community Medicines Regulatory
EMA - European Medicines Agency
f₂ - Similarity factor
FEAPM - Federation of East African Pharmaceutical Manufacturers
GCP - Good Clinical Practice
GMP - Good Manufacturing Practice
LTR - Local Technical Representative
MA - Marketing Authorization
MAH - Marketing Authorization Holder
MER - Medicines Evaluation and Registration
NEPAD - New Partnership for African Development
pK_a – Dissociation constant
SD - Standard deviation
TWG - Technical Working Group
USFDA – United States Food and Drug Administration
WHO - World Health Organization

Ae(0-t) Cumulative urinary excretion of unchanged drug from administration until time t;
AUC(0-t): Area under the plasma concentration curve from administration to last observed concentration at time t;

AUC(0-∞): Area under the plasma concentration curve extrapolated to infinite time;
AUC(0-t): AUC during a dosage interval at steady state;
AUC(0-72h) Area under the plasma concentration curve from administration to 72h;
Cmax: Maximum plasma concentration;
Cmax,ss: Maximum plasma concentration at steady state;
residual area Extrapolated area (AUC(0-∞) - AUC(0-t))/ AUC(0-∞);
Rmax Maximal rate of urinary excretion;
tmax: Time until Cmax is reached;
tmax,ss: Time until Cmax,ss is reached;
t1/2: Plasma concentration half-life;
λz: Terminal rate constant;
SmPC Summary of Product Characteristics

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DEFINITIONS

Absorption - the uptake of substance from a solution into or across tissues. As a time dependent process; absorption can include passive diffusion, facilitated passive diffusion (with a carrier molecule), and active transport. A Pharmaceutical product is considered to be highly absorbed when the measured extent of absorption of the highest therapeutic dose is greater or equal to (≥) 85%. High absorption: ≥ 85% of the administered dose absorbed.

Active moiety (Active): is the term used for the therapeutically active entity in the final formulation of a medicine, irrespective of the form of the API. The active is alternative terminology with the same meaning. For example, if the API is propranolol hydrochloride, the active moiety (and the active) is propranolol.

Active Pharmaceutical Ingredient (API): A substance or compound that is intended to be used in the manufacture of a pharmaceutical product as a therapeutically active ingredient.

Bioavailability: refers to the rate and extent to which the API, or its active moiety, is absorbed from a pharmaceutical product and becomes available at the site of action. It may be useful to distinguish between the "absolute bioavailability" of a given dosage form as compared with that (100 %) following intravenous administration (e.g. oral solution vs. intravenous), and the "relative bioavailability" as compared with another form administered by the same or another non-intravenous route (e.g. tablets vs. oral solution).

Bioequivalence: Two pharmaceutical products are bioequivalent if they are pharmaceutically equivalent or pharmaceutical alternatives and if their bioavailabilities in terms of peak (Cmax and Tmax) and total exposure (AUC) after administration of the same molar dose under the same conditions are similar to such a degree that their effects with respect to both efficacy and safety can be expected to be essentially the same. Bioequivalence focuses on the equivalence of release of the active pharmaceutical ingredient from the pharmaceutical product and its subsequent absorption into the systemic circulation. Comparative studies using clinical or pharmacodynamic end points may also be used to demonstrate bioequivalence.

Biopharmaceutics Classification System (BCS)-based biowaivers are meant to reduce the need for establishing in vivo bioequivalence in situations where in vitro data may be considered to provide a reasonable estimate of the relative in vivo performance of two products. The BCS is a scientific approach designed to predict medicinal absorption based on the aqueous solubility and intestinal absorptive characteristics of the Pharmaceutical product.

Biowaiver: The term biowaiver is applied to a regulatory drug approval process when the dossier (application) is approved based on evidence of equivalence other than through in vivo equivalence testing.

Comparator product: is a pharmaceutical product with which the generic product is intended to be interchangeable in clinical practice. The comparator product will normally be the innovator product for which efficacy, safety and quality have been established.

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Critical dose medicinal - Medicinal product where comparatively small differences in dose or concentration lead to dose- and concentration-dependent, serious therapeutic failures and/or serious adverse medicinal reactions which may be persistent, irreversible, slowly reversible, or life threatening, which could result in hospitalization or prolongation of existing hospitalization, persistent or significant disability or incapacity, or death. Adverse reactions that require significant medical intervention to prevent one of these outcomes are also considered to be serious.

Dose solubility volume (DSV) - the highest therapeutic dose [milligram (mg)] divided by the solubility of the substance [milligram/milliliter (mg/mL)] at a given pH and temperature. For example, if a Pharmaceutical product has a solubility of 31 mg/mL at pH 4.5 (37°C) and the highest dose is 500 mg, then DSV = 500 mg/31 mg/mL = 16 mL at pH 4.5 (37°C).

Fixed-dose combination (FDC): A combination of two or more active pharmaceutical ingredients in a fixed ratio of doses. This term is used generically to mean a particular combination of active pharmaceutical ingredients irrespective of the formulation or brand. It may be administered as single entity products given concurrently or as a finished pharmaceutical product.

Generic Pharmaceutical Product is a pharmaceutically equivalent product that may or may not be therapeutically equivalent or bioequivalent. Generic pharmaceutical products that are therapeutically equivalent are interchangeable.

High solubility: A Pharmaceutical product is classified as highly soluble if the highest therapeutic dose of the Pharmaceutical product is completely soluble in 250 mL or less of solvent over the pH range of 1.2-6.8 at 37 ± 1°C, that is (i.e.), DSV ≤ 250 mL over the pH range.

Highest dose - highest approved therapeutic dose for the Pharmaceutical product in Tanzania. If not currently approved in Tanzania, the highest proposed dose is applicable.

Low absorption: less than (<) 85% of the administered dose absorbed.

Low solubility: A Pharmaceutical product is classified as a low solubility compound if the highest therapeutic dose of the Pharmaceutical product is not completely soluble in 250 mL of solvent at any pH within the pH range of 1.2-6.8 at $37 \pm 1^\circ\text{C}$, i.e., DSV greater than ($>$) 250 mL at any pH within the range.

Metabonate - a substance which appears to be a metabolite but is actually an artefact formed during experimental conditions [for example (e.g.), isolation and storage].

Pharmaceutical alternatives: Pharmaceutical products are pharmaceutical alternatives if they contain the same active moiety but differ either in chemical form (e.g. salt, ester) of that moiety or in the dosage form or strength, administered by the same route of administration but are

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otherwise not pharmaceutically equivalent. Pharmaceutical alternatives do not necessarily imply bioequivalence.

Pharmaceutical Dosage Form: A pharmaceutical dosage form is the form of the completed pharmaceutical product e.g. tablet, capsule, injection, elixir, suppository.

Pharmaceutical Equivalence: Pharmaceutical products are pharmaceutically equivalent if they contain the same amount of the same API(s) in the same dosage form, if they meet the same or comparable standards and if they are intended to be administered by the same route. Pharmaceutical equivalence does not necessarily imply bioequivalence as differences in the excipients and/or the manufacturing process can lead to changes in dissolution and/or absorption.

Pharmaceutical Product: Any preparation for human (or animal) use, containing one or more APIs with or without pharmaceutical excipients or additives, that is intended to modify or explore physiological systems or pathological states for the benefit of the recipient.

Proportionally Similar Dosage Forms/Products: Pharmaceutical products are considered proportionally similar in the following cases:-

Rapidly dissolving product - a product in which not less than 85% of the labelled amount is released within 30 minutes or less during a product dissolution test under the conditions specified in these guidelines.

Solution - a homogenous mixture in a single phase with no precipitate.

Therapeutic Equivalence: Two pharmaceutical products are therapeutically equivalent if they are pharmaceutically equivalent or are pharmaceutical alternatives and, after administration in the same molar dose, their effects with respect to both efficacy and safety are essentially the same, as determined from appropriate bioequivalence, pharmacodynamic, clinical or in vitro studies.

Very rapidly dissolving product - not less than 85% of the labelled amount is released within 15 minutes or less during a product dissolution test under the conditions specified in this guidelines.

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1.0 INTRODUCTION

The objective of this guideline is to specify the requirements for the design, conduct, and evaluation of bioequivalence studies for immediate release and modified release dosage forms with systemic action.

Two medicinal products containing the same active substance are considered bioequivalent if they are pharmaceutically equivalent or Pharmaceutical alternatives and their bioavailabilities (rate and extent) after administration in the same molar dose lie within acceptable predefined limits. These limits are set to ensure comparable in vivo performance, i.e. similarity in terms of safety and efficacy.

In bioequivalence studies, the plasma concentration time curve is generally used to assess the rate and extent of absorption. Selected pharmacokinetic parameters and pre-set acceptance limits allow the final decision on bioequivalence of the tested products. The absorption rate of a drug is influenced by pharmacokinetic parameters like AUC, the area under the concentration time curve, reflects the extent of exposure, C_{max} , the maximum plasma concentration or peak exposure, and the time to maximum plasma concentration, t_{max} . In applications for generic medicinal products to TMDA, the concept of bioequivalence is fundamental.

The purpose of establishing bioequivalence is to demonstrate equivalence in biopharmaceutics quality between the generic medicinal product and a comparator medicinal product in order to allow bridging of preclinical tests and of clinical trials associated with the comparator medicinal product. The definition for generic medicinal products is a product that has the same qualitative and quantitative composition in active substances and the same pharmaceutical form as the comparator medicinal product, and whose bioequivalence with the comparator medicinal product has been demonstrated by appropriate bioavailability studies. The different salts, esters, ethers, isomers, mixtures of isomers, complexes or derivatives of an active substance are considered to be the same active substance, unless they differ significantly in properties with regard to safety and/or efficacy. Furthermore, the various immediate-release oral pharmaceutical forms shall be considered to be one and the same pharmaceutical form. Other types of applications may also require demonstration of bioequivalence, including variations, fixed combinations, extensions and generic applications.

The recommendations on design and conduct given for bioequivalence studies in this guideline may also be applied to comparative bioavailability studies evaluating different formulations used during the development of a new medicinal product containing a new chemical entity and to comparative bioavailability studies included in extension or generic applications that are not based exclusively on bioequivalence data.

Generally, results from comparative bioavailability studies should be provided in support of the safety and efficacy of each proposed product and of each proposed strength included in the

submission. In the absence of such studies, a justification supporting a waiver of this requirement should be provided in this section for each product and each strength. For

example, if there are several strengths of the proposed product, and comparative bioavailability data has not been submitted for all strengths, the applicant should provide a scientific justification for not conducting studies on each strength. This justification may address issues such as the nature of the kinetics of the drug (e.g., linear versus non-linear), and the proportionality of the strengths for which a waiver is sought to the strength on which a comparative bioavailability study was conducted.

The statement of justification for waiver will include supporting data (e.g. comparative dissolution data) which should be provided in the relevant module(s) of the CTD submission (i.e., Modules 2-5). For example, comparative dissolution profiles should be provided in Module 3, Section 3.2.P.2 of the main Guidelines on Submission of Documentation for Marketing Authorization of Human Pharmaceutical Products (Pharmaceutical Development).

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2.0 SCOPE

This guideline focuses on recommendations for bioequivalence studies for immediate release formulations and modified release with systemic action. The scope is limited to chemical entities. Biological products are not covered by these guidelines.

In case bioequivalence cannot be demonstrated using drug concentrations, in exceptional circumstances pharmacodynamic or clinical endpoints may be needed.

Exemptions for carrying out bioequivalence studies

Omission of BE studies must be justified except if a product fulfills one or more of the following conditions:-

- a) Solutions, complex or simple, which do not contain any ingredient which can be regarded as a pharmacologically active substance;
- b) Simple aqueous solutions intended for intravenous injection or infusion containing the same active substance(s) in the same concentration as currently registered products. Simple solutions do not include complex solution such as micellar or liposomal solutions;
- c) Solutions for injection that contain the same active ingredients and excipients in the same concentrations as currently registered products and which are administered by the same route(s);
- d) Products that are powder for reconstitution as a solution and the solution meets either criterion (b) or (c) above;
- e) Oral immediate release tablets, capsules and suspensions containing active pharmaceutical ingredients eligible for BCS based biowaivers.
- f) Oral solutions containing the same active ingredient(s) in the same concentration as a currently registered or innovator oral solution and not containing excipients that may significantly affect gastric passage or absorption of the active ingredient(s);
- g) Products for topical use provided the product is intended to act without systemic absorption when applied locally;
- h) Products containing therapeutic substances, which are not systemically or locally absorbed i.e. an oral dosage form which is not intended to be absorbed (e.g., barium sulphate enemas, Antacid, Radioopaque Contrast Media, or powders in which no ingredient is absorbed etc.). If there is doubt as to whether absorption occurs, a study or justification may be required;

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- i) Otic or ophthalmic products prepared as aqueous solutions and containing the same active pharmaceutical ingredient(s) in the same concentration;
- j) The product is an oral solution, syrup, or other similarly solubilized form;
- k) The product is oro-dispersable product is eligible for a biowaiver application only if there is no buccal or sublingual absorption and the product is labelled to be consumed with water;
- l) The product is an inhalant volatile anaesthetic solution, Inhalation and nasal preparations;
- m) The product is a reformulated product by the original manufacturer that is identical to the original product except for colouring agents, flavouring agents or preservatives, which are recognized as having no influence upon bioavailability;

- n) Gases;

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3.0 MAIN GUIDELINE TEXT

3.1 Design, conduct and evaluation of bioequivalence studies

The design, conduct and evaluation of the Bioequivalence study should comply with ICH GCP requirements (E6).

In the following sections, requirements for the design and conduct of comparative bioavailability studies are formulated. Investigator(s) should have appropriate expertise, qualifications and competence to undertake a proposed study and is familiar with pharmacokinetic theories underlying bioavailability studies. The design should be based on a reasonable knowledge of the pharmacodynamics and/or the pharmacokinetics of the active substance in question.

The number of studies and study design depend on the physico-chemical characteristics of the substance, its pharmacokinetic properties and proportionality in composition, and should be justified accordingly. In particular it may be necessary to address the linearity of pharmacokinetics, the need for studies both in fed and fasting state, the need for enantioselective analysis and the possibility of waiver for additional strengths (see Sections 3.1.4, 3.1.5 and 3.1.6).

Module 2.7.1 should list all relevant studies carried out with the product applied for, i.e. bioequivalence studies comparing the formulation applied for (i.e. same composition and manufacturing process) with a Comparator medicinal product approved in the EAC. Studies should be included in the list regardless of the study outcome. Full study reports should be provided for all studies, except pilot studies for which study report synopses (in accordance with ICH E3) are sufficient. Full study reports for pilot studies should be available upon request. Study report synopses for bioequivalence or comparative bioavailability studies conducted during formulation development should also be included in Module 2.7. Bioequivalence studies comparing the product applied for with non-EAC Comparator products should not be submitted and do not need to be included in the list of studies.

3.1.1 Study design

Standard design

If two formulations are compared, a randomized, two-period, two-sequence single dose crossover design is recommended. The treatment periods should be separated by a wash out period sufficient to ensure that drug concentrations are below the lower limit of bioanalytical quantification in all subjects at the beginning of the second period. Normally at least 5 elimination half-lives are necessary to achieve this. The study should be designed in such a way that the treatment effect (formulation effect) can be distinguished from other effects. In order to reduce variability a cross over design usually is the first choice.

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Alternative designs

Under certain circumstances, provided the study design and the statistical analyses are scientifically sound, alternative well-established designs could be considered such as parallel design for substances with very long half-life and replicate designs e.g. for substances with highly variable pharmacokinetic characteristics (see Section 3.1.10). The study should be designed in such a way that the formulation effect can be distinguished from other effects.

Other designs or methods may be chosen in specific situations, but should be fully justified in the protocol and final study report. The subjects should be allocated to treatment sequences in a randomized order. In general, single dose studies will suffice, but there are situations in which steady-state studies may be required:-

(a) If problems of sensitivity preclude sufficiently precise plasma concentration measurement after single dose;

(b) If the intra-individual variability in the plasma concentrations or disposition rate is inherently large;

(c) in the case of dose- or time-dependent pharmacokinetics;

(d) in the case of extended release products (in addition to single dose studies)

In such steady-state studies, the administration scheme should follow the usual dosage recommendations.

Conduct of a multiple dose study in patients is acceptable if a single dose study cannot be conducted in healthy volunteers due to tolerability reasons, and a single dose study is not feasible in patients.

In the rare situation where problems of sensitivity of the analytical method preclude sufficiently precise plasma concentration measurements after single dose administration and where the concentrations at steady state are sufficiently high to be reliably measured, a multiple dose study may be acceptable as an alternative to the single dose study. However, given that a multiple dose study is less sensitive in detecting differences in C_{max}, this will only be acceptable if the applicant can adequately justify that the sensitivity of the analytical method cannot be improved and that it is not possible to reliably measure the parent compound after single dose administration taking into account also the option of using a supra-therapeutic dose in the bioequivalence study (see also Section 3.1.6). Due to the recent development in the bioanalytical methodology, it is unusual that parent drug cannot be measured accurately and precisely. Hence, use of a multiple dose study instead of a single dose study, due to limited sensitivity of the analytical method, will only be accepted in exceptional cases.

In steady-state studies, the washout period of the previous treatment can overlap with the

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build-up of the second treatment, provided the build-up period is sufficiently long (at least 5 times the terminal half-life).

3.1.2 Comparator and test products

Comparator Product

Test products in an application for a generic product or an extension of a generic product are normally compared with the corresponding dosage form of a comparator medicinal product, if available on the market. The product used as comparator product in the bioequivalence study should meet the criteria stipulated in Annex IV.

In an application for extension of a medicinal product which has been initially approved by TMDA and when there are several dosage forms of this medicinal product on the market, it is recommended that the dosage form used for the initial approval of the concerned medicinal product (and which was used in clinical efficacy and safety studies) is used as comparator

product, if available on the market.

The selection of the Comparator product used in a bioequivalence study should be based on assay content and dissolution data and is the responsibility of the Applicant. Unless otherwise justified, the assayed content of the batch used as test product should not differ more than 5% from that of the batch used as comparator product determined with the test procedure proposed for routine quality testing of the test product. The Applicant should document how a representative batch of the comparator product with regards to dissolution and assay content has been selected. It is advisable to investigate more than one single batch of the Comparator product when selecting Comparator product batch for the bioequivalence study. (to be removed and moved to guideline on comparator).

Test product

The test product used in the study should be representative of the product to be marketed and this should be discussed and justified by the applicant. For example, for oral solid forms for systemic action:-

a) The test product should usually originate from a batch of at least 1/10 of production scale or 100,000 units, whichever is greater, unless otherwise justified.

b) The production of batches used should provide a high level of assurance that the product and process will be feasible on an industrial scale.

In case of a production batch smaller than 100,000 units, a full production batch will be required.

c) The characterization and specification of critical quality attributes of the finished pharmaceutical product, such as dissolution, should be established from the test batch, i.e. the clinical batch for which bioequivalence has been demonstrated.

d) Samples of the product from additional pilot and/or full-scale production batches, submitted to support the application, should be compared with those of the bioequivalence study test batch, and should show similar in vitro dissolution profiles when employing suitable dissolution test conditions.

Comparative dissolution profile testing should be undertaken on the first three production batches.

If full scale production batches are not available at the time of submission, the applicant should not market a batch until comparative dissolution profile testing has been completed.

The results should be provided at a Competent Authority's request or if the dissolution profiles are not similar together with proposed action to be taken.

For other immediate release pharmaceutical forms for systemic action, justification of the representative nature of the test batch should be similarly established.

Impact of excipients

Identify any excipients present in either product that are known to impact on in vivo absorption processes. Provide a literature-based summary of the mechanism by which these effects are known to occur should be included and relevant full discussion enclosed, if applicable.

Comparative qualitative and quantitative differences between the compositions of the test and comparator products

Identify all qualitative (and quantitative, if available) differences between the compositions of the test and comparator products. The data obtained and methods used for the determination of the quantitative composition of the comparator product as required by the guidance documents should be summarized here for assessment.

Impact of the differences between the compositions of the test and comparator products

Provide a detailed comment on the impact of any differences between the compositions of the test and comparator products with respect to drug release and in vivo absorption

Packaging of study products

The comparator and test products should be packed in an individual way for each subject and period, either before their shipment to the trial site, or at the trial site itself. Packaging (including labelling) should be performed in accordance with good manufacturing practice.

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It should be possible to identify unequivocally the identity of the product administered to each subject at each trial period. Packaging, labelling and administration of the products to the subjects should therefore be documented in detail. This documentation should include all precautions taken to avoid and identify potential dosing mistakes. The use of labels with a tear-off portion is recommended.

Subjects

Number of subjects

The number of subjects to be included in the study should be based on an appropriate sample size calculation. The number of evaluable subjects in a bioequivalence study should not be less than 12.

The number of subjects should be determined using appropriate methods taking into account the error variance associated with the primary parameters to be studied (as estimated for a pilot experiment, from previous studies or from published data), the significance level desired and

the deviation from the comparator product compatible with bioequivalence ($\pm 20\%$) and compatible with safety and efficacy. For a parallel design study a greater number of subjects may be required to achieve sufficient study power.

Applicants should enter a sufficient number of subjects in the study to allow for dropouts. Because replacement of subjects could complicate the statistical model and analysis, dropouts generally should not be replaced.

Selection of subjects

The subject population for bioequivalence studies should be selected with the aim of permitting detection of differences between pharmaceutical products. The subject population for bioequivalence studies should be selected with the aim to minimise variability and permit detection of differences between pharmaceutical products. In order to reduce variability not related to differences between products, the studies should normally be performed in healthy volunteers unless the drug carries safety concerns that make this unethical. This model, in vivo healthy volunteers, is regarded as adequate in most instances to detect formulation differences and to allow extrapolation of the results to populations for which the comparator medicinal product is approved (the elderly, children, patients with renal or liver impairment, etc.).

The inclusion/exclusion criteria should be clearly stated in the protocol. Subjects to be enrolled in a crossover bioequivalence study should be between 18-50 years in age, preferably have a Body Mass Index between 18.5 and 30 kg/m².

The subjects should be screened for suitability by means of clinical laboratory tests, a medical history, and a physical examination. Depending on the drug's therapeutic class and safety profile, special medical investigations and precautions may have to be carried out before, 151 during and after the completion of the study.

Subjects could belong to either sex; however, the risk to women of childbearing potential should be considered. Subjects should preferably be non-smokers and without a history of alcohol or drug abuse. Phenotyping and/or genotyping of subjects may be considered for safety or pharmacokinetic reasons.

In parallel design studies, the treatment groups should be comparable in all known variables that may affect the pharmacokinetics of the active substance (e.g. age, body weight, sex, ethnic origin, smoking status, extensive/poor metabolic status). This is an essential pre-requisite to give validity to the results from such studies.

Inclusion of patients

If the investigated active substance is known to have adverse effects and the pharmacological effects or risks are considered unacceptable for healthy volunteers, it may be necessary to include patients instead, under suitable precautions and supervision. In this case the applicant should justify the alternative.

3.1.4 Study conduct

Standardisation of the bioequivalence studies

The test conditions should be standardized in order to minimize the variability of all factors involved except that of the products being tested. Therefore, it is recommended to standardize diet, fluid intake and exercise.

The time of day for ingestion should be specified. Subjects should fast for at least 8 hours prior to administration of the products, unless otherwise justified. As fluid intake may influence gastric passage for oral administration forms, the test and comparator products should be administered with a standardized volume of fluid (at least 150 ml). It is recommended that water is allowed as desired except for one hour before and one hour after drug administration and no food is allowed for at least 4 hours post-dose. Meals taken after dosing should be standardized in regard to composition and time of administration during an adequate period of time (e.g. 12 hours).

In case the study is to be performed during fed conditions, the timing of administration of the finished pharmaceutical product in relation to food intake is recommended to be according to the SmPC of the originator product. If no specific recommendation is given in the originator SmPC, it is recommended that subjects should start the meal 30 minutes prior to administration of the finished pharmaceutical product and eat this meal within 30 minutes.

As the bioavailability of an active moiety from a dosage form could be dependent upon gastrointestinal transit times and regional blood flows, posture and physical activity may need to be standardized.

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The subjects should abstain from food and drinks, which may interact with circulatory, gastrointestinal, hepatic or renal function (e.g. alcoholic drinks or certain fruit juices such as grapefruit juice) during a suitable period before and during the study. Subjects should not take any other concomitant medication (including herbal remedies) for an appropriate interval before as well as during the study. Contraceptives are, however, allowed. In case concomitant medication is unavoidable and a subject is administered other drugs, for instance to treat adverse events like headache, the use must be reported (dose and time of administration) and possible effects on the study outcome must be addressed. In rare cases, the use of a concomitant medication is needed for all subjects for safety or tolerability reasons (e.g. opioid antagonists, anti-emetics). In that scenario, the risk for a potential interaction or bioanalytical interference affecting the results must be addressed.

Medicinal products that according to the originator SmPC are to be used explicitly in combination with another product (e.g. certain protease inhibitors in combination with ritonavir) may be studied either as the approved combination or without the product recommended to be administered concomitantly.

In bioequivalence studies of endogenous substances, factors that may influence the endogenous

baseline levels should be controlled if possible (e.g. strict control of dietary intake).

Sampling times

Several samples of appropriate biological matrix (blood, plasma/serum, urine) are collected at various time intervals post-dose. The sampling schedule depends on the pharmacokinetic characteristics of the drug being tested. In most cases, plasma or serum is the matrix of choice. However, if the parent drug is not metabolized and is largely excreted unchanged and can be suitably assayed in the urine, urinary drug levels may be used to assess bioequivalence, if plasma/serum concentrations of the drug cannot be reliably measured.

A sufficient number of samples are collected during the absorption phase to adequately describe the plasma concentration-time profile should be collected. The sampling schedule should include frequent sampling around predicted T_{max} to provide a reliable estimate of peak exposure. Intensive sampling is carried out around the time of the expected peak concentration. In particular, the sampling schedule should be planned to avoid C_{max} being the first point of a concentration time curve. The sampling schedule should also cover the plasma concentration time curve long enough to provide a reliable estimate of the extent of exposure which is achieved if $AUC(0-t)$ covers at least 80% of $AUC(0-\infty)$. At least three to four samples are needed during the terminal log-linear phase in order to reliably estimate the terminal rate constant (which is needed for a reliable estimate of $AUC(0-\infty)$). AUC truncated at 72 h [$AUC(0-72h)$] may be used as an alternative to $AUC(0-t)$ for comparison of extent of exposure as the absorption phase has been covered by 72 h for immediate release formulations. A sampling period longer than 72 h is therefore not considered necessary for any immediate release formulation irrespective of the half-life of the drug. Sufficient numbers of samples should also be collected in the log-linear elimination phase of the drug so that the terminal elimination rate constant and half-life of the drug can be accurately determined. A sampling period extending to at least five terminal elimination half-lives of the drug or five the longest half-life of the pertinent analyte (if more than one analyte) is usually sufficient. The samples are appropriately processed and stored carefully under conditions that preserve the integrity of the analyte(s).

In multiple -dose studies, the pre-dose sample should be taken immediately before (within 5 minutes) dosing and the last sample is recommended to be taken within 10 minutes of the nominal time for the dosage interval to ensure an accurate determination of $AUC(0-\tau)$.

If urine is used as the biological sampling fluid, urine should normally be collected over no less than three times the terminal elimination half-life. However, in line with the recommendations on plasma sampling, urine does not need to be collected for more than 72 h. If rate of excretion is to be determined, the collection intervals need to be as short as feasible during the absorption phase (see also Section 3.1.5).

For endogenous substances, the sampling schedule should allow characterization of the endogenous baseline profile for each subject in each period. Often, a baseline is determined from 2-3 samples taken before the finished pharmaceutical products are administered. In other cases, sampling at regular intervals throughout 1-2 day(s) prior to administration may be necessary in order to account for fluctuations in the endogenous baseline due to circadian rhythms (see Section 3.1.5).

Washout period

Subsequent treatments should be separated by periods long enough to eliminate the previous dose before the next one (wash-out period). In steady-state studies wash-out of the last dose of the previous treatment can overlap with the build-up of the second treatment, provided the build-up period is sufficiently long (at least five(5) times the dominating half-life).

Fasting or fed conditions

In general, a bioequivalence study should be conducted under fasting conditions as this is considered to be the most sensitive condition to detect a potential difference between formulations. For products where the SmPC recommends intake of the innovator medicinal product on an empty stomach or irrespective of food intake, the bioequivalence study should hence be conducted under fasting conditions. For products where the SmPC recommends intake of the innovator medicinal product only in fed state, the bioequivalence study should generally be conducted under fed conditions.

However, for products with specific formulation characteristics (e.g. microemulsions, prolonged modified release, solid dispersions), bioequivalence studies performed under both fasted and fed conditions are required unless the product must be taken only in the fasted state or only in the fed state.

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In cases where information is required in both the fed and fasted states, it is acceptable to conduct either two separate two-way cross-over studies or a four-way cross-over study.

In studies performed under fed conditions, the composition of the meal is recommended to be according to the SmPC of the originator product. If no specific recommendation is given in the originator SmPC, the meal should be a high-fat (approximately 50 percent of total caloric content of the meal) and high -calorie (approximately 800 to 1000 kcal) meal. This test meal should derive approximately 150, 250, and 500-600 kcal from protein, carbohydrate, and fat, respectively. The composition of the meal should be described with regard to protein, carbohydrate and fat content (specified in grams, calories and relative caloric content (%)).

Characteristics to be investigated

Pharmacokinetic parameters (Bioavailability Metrics)

In bioavailability studies, the shape and area under the plasma concentration versus time curves are mostly used to assess rate (C_{max} , t_{max}) and extent (AUC) of exposure. Sampling points or periods should be chosen such that the concentration versus time profile is sufficiently defined to allow calculation of relevant parameters.

For single-dose studies, the following parameters should be measured or calculated:

- a) Area under the plasma, serum or blood concentration–time curve from time zero to time t (AUC_{0-t}), where t is the last sampling time point with a measurable concentration of the API in the individual formulation tested. The method of calculating AUC values should be specified. Non-compartmental methods should be used for pharmacokinetic calculations in bioequivalence studies;
- b) C_{max} is the maximum or peak concentration observed representing peak exposure of API (or metabolite) in plasma, serum or whole blood.

Usually AUC_{0-t} and C_{max} are considered to be the most relevant parameters for assessment of bioequivalence. In addition it is recommended that the following parameters be estimated:

- c) area under the plasma, serum or blood concentration–time curve from time zero to time infinity ($AUC_{0-\infty}$) representing total exposure, where $AUC_{0-\infty} = AUC_{0-t} + C_{last} / K_e$; C_{last} is the last measurable analyte concentration and K_e is the terminal or elimination rate constant calculated according to an appropriate method;
- d) t_{max} is the time after administration of the FPP at which C_{max} is observed.

For additional information the elimination parameters can be calculated:

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- $T_{1/2}$ is the plasma (serum, whole blood) half-life.

For multiple-dose studies conducted with modified-release products, the following parameters should be calculated:

- AUC_{τ} is AUC over one dosing interval (τ) at steady state;
- C_{max} ;
- C_{min} (C_{τ}) is concentration at the end of a dosing interval;
- peak trough fluctuation is percentage difference between C_{max} and C_{min} .

As release mechanisms of pharmaceutical products become more complex, e.g. products with an immediate-release and a modified-release component, additional parameters such as partial AUC measures may be necessary to ensure the bioequivalence of two products. When urine samples are used, cumulative urinary recovery (A_e) and maximum urinary excretion rate are employed instead of AUC and C_{max} .

Parent compound or metabolites

In principle, evaluation of bioequivalence should be based upon measured concentrations of the parent compound. The reason for this is that C_{max} of a parent compound is usually more sensitive to detect differences between formulations in absorption rate than C_{max} of a metabolite.

Inactive pro-drugs

Also for inactive pro-drugs, demonstration of bioequivalence for parent compound is recommended. The active metabolite does not need to be measured. However, some pro-drugs may have low plasma concentrations and be quickly eliminated resulting in difficulties in demonstrating bioequivalence for parent compound. In this situation it is acceptable to demonstrate bioequivalence for the main active metabolite without measurement of parent compound. In the context of this guideline, a parent compound can be considered to be an inactive pro-drug if it has no or very low contribution to clinical efficacy.

Use of metabolite data as surrogate for active parent compound

The use of a metabolite as a surrogate for an active parent compound is not encouraged. This can only be considered if the applicant can adequately justify that the sensitivity of the analytical method for measurement of the parent compound cannot be improved and that it is not possible to reliably measure the parent compound after single dose administration taking into account also the option of using a higher single dose in the bioequivalence study. Due to recent developments in bioanalytical methodology it is unusual that parent drug cannot be measured accurately and precisely. Hence, the use of a metabolite as a surrogate for active parent compound is expected to be accepted only in exceptional cases. When using metabolite data as a substitute for active parent drug concentrations, the applicant should present any available data supporting the view that the metabolite exposure will reflect parent drug and that the metabolite formation is not saturated at therapeutic doses.

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Enantiomers

The use of achiral bioanalytical methods is generally acceptable. However, the individual enantiomers should be measured when all the following conditions are met:-

- a) the enantiomers exhibit different pharmacokinetics;
- b) the enantiomers exhibit pronounced difference in pharmacodynamics;
- c) the exposure (AUC) ratio of enantiomers is modified by a difference in the rate of absorption.

The individual enantiomers should also be measured if the above conditions are fulfilled or are unknown. If one enantiomer is pharmacologically active and the other is inactive or has a low contribution to activity, it is sufficient to demonstrate bioequivalence for the active enantiomer.

The use of urinary data

If drug/API concentrations in blood are too low to be detected and a substantial amount (> 40 %) of the drug/API is eliminated unchanged in the urine, then urine may serve as the biological fluid to be sampled.

If a reliable plasma C_{max} can be determined, this should be combined with urinary data on the extent of exposure for assessing bioequivalence. When using urinary data, the applicant should present any available data supporting that urinary excretion will reflect plasma exposure.

When urine is collected:-

- a) The volume of each sample should be measured immediately after collection and included in the report.
- b) Urine should be collected over an extended period and generally no less than seven times the terminal elimination half-life, so that the amount excreted to infinity (Ae_∞) can be estimated.
- c) Sufficient samples should be obtained to permit an estimate of the rate and extent of renal excretion. For a 24-hour study, sampling times of 0 to 2, 2 to 4, 4 to 8, 8 to 12, and 12 to 24 hours post-dose are usually appropriate.
- d) The actual clock time when samples are collected, as well as the elapsed time relative to API administration, should be recorded.

Urinary Excretion Profiles:-

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In the case of API's predominantly excreted renally, the use of urine excretion data may be advantageous in determining the extent of drug/API input. However, justification should also be given when this data is used to estimate the rate of absorption.

Sampling points should be chosen so that the cumulative urinary excretion profiles can be defined adequately so as to allow accurate estimation of relevant parameters.

The following bioavailability parameters are to be estimated:-

- a) A_{et}, Ae_∞ as appropriate for urinary excretion studies.
- b) Any other justifiable characteristics.
- c) The method of estimating AUC-values should be specified.

Endogenous substances

If the substance being studied is endogenous, the calculation of pharmacokinetic parameters should be performed using baseline correction so that the calculated pharmacokinetic parameters refer to the additional concentrations provided by the treatment. Administration of supra-therapeutic doses can be considered in bioequivalence studies of endogenous drugs, provided that the dose is well tolerated, so that the additional concentrations over baseline provided by the treatment may be reliably determined. If a separation in exposure following administration of different doses of a particular endogenous substance has not been previously established this should be demonstrated, either in a pilot study or as part of the pivotal bioequivalence study using different doses of the comparator formulation, in order to ensure that the dose used for the bioequivalence comparison is sensitive to detect potential differences between formulations.

The exact method for baseline correction should be pre-specified and justified in the study protocol. In general, the standard subtractive baseline correction method, meaning either subtraction of the mean of individual endogenous pre-dose concentrations or subtraction of the individual endogenous pre-dose AUC, is preferred. In rare cases where substantial increases over baseline endogenous levels are seen, baseline correction may not be needed.

In bioequivalence studies with endogenous substances, it cannot be directly assessed whether carry-over has occurred, so extra care should be taken to ensure that the washout period is of an adequate duration.

3.1.6 Strength to be investigated

If several strengths of a test product are applied for, it may be sufficient to establish bioequivalence at only one or two strengths, depending on the proportionality in composition between the different strengths and other product related issues described below. The strength(s) to evaluate depends on the linearity in pharmacokinetics of the active substance.

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In case of non-linear pharmacokinetics (i.e. not proportional increase in AUC with increased dose) there may be a difference between different strengths in the sensitivity to detect potential differences between formulations. In the context of this guideline, pharmacokinetics is considered to be linear if the difference in dose-adjusted mean AUCs is no more than 25% when comparing the studied strength (or strength in the planned bioequivalence study) and the strength(s) for which a waiver is considered. In order to assess linearity, the applicant should consider all data available in the public domain with regard to the dose proportionality and review the data critically. Assessment of linearity will consider whether differences in dose-adjusted AUC meet a criterion of $\pm 25\%$.

If bioequivalence has been demonstrated at the strength(s) that are most sensitive to detect a potential difference between products, in vivo bioequivalence studies for the other strength(s) can be waived.

General bioequivalence criteria

The following general requirements must be met where a waiver for additional strength(s) is claimed:-

- a) the pharmaceutical products are manufactured by the same manufacturing process,
- b) the qualitative composition of the different strengths is the same,
- c) the composition of the strengths are quantitatively proportional, i.e. the ratio between the amount of each excipient to the amount of active substance(s) is the same for all strengths

(for immediate release products coating components, capsule shell, colour agents and flavours are not required to follow this rule),

If there is some deviation from quantitatively proportional composition, condition c is still considered fulfilled if condition i) and ii) or i) and iii) below apply to the strength used in the bioequivalence study and the strength(s) for which a waiver is considered:-

i. the amount of the active substance(s) is less than 5 % of the tablet core weight, the weight of the capsule content.

ii. the amounts of the different core excipients or capsule content are the same for the concerned strengths and only the amount of active substance is changed.

iii. the amount of a filler is changed to account for the change in amount of active substance. The amounts of other core excipients or capsule content should be the same for the concerned strengths.

d) An appropriate in vitro dissolution data should confirm the adequacy of waiving additional in vivo bioequivalence testing (see Section 3.2).

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Linear pharmacokinetics

For products where all the above conditions a) to d) are fulfilled, it is sufficient to establish bioequivalence with only one strength.

The bioequivalence study should in general be conducted at the highest strength. For products with linear pharmacokinetics and where the active pharmaceutical ingredient is highly soluble based on BCS Biowaiver, selection of a lower strength than the highest is also acceptable. Selection of a lower strength may also be justified if the highest strength cannot be administered to healthy volunteers for safety/tolerability reasons. Further, if problems of sensitivity of the analytical method preclude sufficiently precise plasma concentration measurements after single dose administration of the highest strength, a higher dose may be selected (preferably using multiple tablets of the highest strength). The selected dose may be higher than the highest therapeutic dose provided that this single dose is well tolerated in healthy volunteers and that there are no absorption or solubility limitations at this dose.

Non-linear pharmacokinetics

For drugs with non-linear pharmacokinetics characterized by a more than proportional increase in AUC with increasing dose over the therapeutic dose range, the bioequivalence study should in general be conducted at the highest strength. As for drugs with linear pharmacokinetics a lower strength may be justified if the highest strength cannot be administered to healthy volunteers for safety/tolerability reasons. Likewise a higher dose may be used in case of sensitivity problems of the analytical method in line with the recommendations given for products with linear pharmacokinetics above.

For drugs with a less than proportional increase in AUC with increasing dose over the therapeutic dose range, bioequivalence should in most cases be established both at the highest strength and at the lowest strength (or strength in the linear range), i.e. in this situation two bioequivalence studies are needed. If the non-linearity is not caused by limited solubility but is due to e.g. saturation of uptake transporters and provided that conditions a) to d) above are fulfilled and the test and comparator products do not contain any excipients that may affect gastrointestinal motility or transport proteins, it is sufficient to demonstrate bioequivalence at the lowest strength (or a strength in the linear range).

Selection of other strengths may be justified if there are analytical sensitivity problems preventing a study at the lowest strength or if the highest strength cannot be administered to healthy volunteers for safety/tolerability reasons.

Bracketing approach

Where bioequivalence assessment at more than two strengths is needed, e.g. because of deviation from proportional composition, a bracketing approach may be used. In this situation it can be acceptable to conduct two bioequivalence studies, if the strengths selected represent the extremes, e.g. the highest and the lowest strength or the two strengths differing most in

160 composition, so that any differences in composition in the remaining strengths is covered by the two conducted studies.

Where bioequivalence assessment is needed both in fasting and in fed state and at two strengths due to nonlinear absorption or deviation from proportional composition, it may be sufficient to assess bioequivalence in both fasting and fed state at only one of the strengths. Waiver of either the fasting or the fed study at the other strength(s) may be justified based on previous knowledge and/or pharmacokinetic data from the study conducted at the strength tested in both fasted and fed state. The condition selected (fasting or fed) to test the other strength(s) should be the one which is most sensitive to detect a difference between products.

Fixed combinations

The conditions regarding proportional composition should be fulfilled for all active substances of fixed combinations. When considering the amount of each active substance in a fixed combination the other active substance(s) can be considered as excipients. In the case of bilayer tablets, each layer may be considered independently.

3.1.7 Bioanalytical methodology

The bioanalysis of bioequivalence samples should be performed in accordance with the principles of Good Laboratory Practice (GLP). However, as human bioanalytical studies fall outside the scope of GLP, the sites conducting the studies are not required to be monitored as part of a national GLP compliance programme.

The bioanalytical methods used to determine the active principle and/or its biotransformation

products in plasma, serum, blood or urine or any other suitable matrix must be well characterized, fully validated and documented to yield reliable results that can be satisfactorily interpreted. Within study validation should be performed using Quality control samples in each analytical run.

The main objective of method validation is to demonstrate the reliability of a particular method for the quantitative determination of analyte(s) concentration in a specific biological matrix. The main characteristics of a bioanalytical method that is essential to ensure the acceptability of the performance and the reliability of analytical results includes but not limited to: selectivity, sensitivity, lower limit of quantitation, the response function (calibration curve performance), accuracy, precision and stability of the analyte(s) in the biological matrix under processing conditions and during the entire period of storage.

The lower limit of quantitation should be 1/20 of C_{max} or lower, as pre-dose concentrations should be detectable at 5% of C_{max} or lower (see Section 3.1.8 Carry-over effects).

Reanalysis of study samples should be predefined in the study protocol (and/or SOP) before the actual start of the analysis of the samples. Normally reanalysis of subject samples because of a pharmacokinetic reason is not acceptable. This is especially important for bioequivalence studies, as this may bias the outcome of such a study.

Analysis of samples should be conducted without information on treatment.

The validation report of the bioanalytical method should be included in Module 5 of the application.

3.1.8 Evaluation

In bioequivalence studies, the pharmacokinetic parameters should in general not be adjusted for differences in assayed content of the test and comparator batch. However, in exceptional cases where a comparator batch with an assay content differing less than 5% from test product cannot be found (see Section 3.1.2 on Comparator and test product) content correction could be accepted. If content correction is to be used, this should be pre-specified in the protocol and justified by inclusion of the results from the assay of the test and comparator products in the protocol.

Subject accountability

Ideally, all treated subjects should be included in the statistical analysis. However, subjects in a crossover trial who do not provide evaluable data for both of the test and comparator products (or who fail to provide evaluable data for the single period in a parallel group trial) should not be included.

The data from all treated subjects should be treated equally. It is not acceptable to have a protocol which specifies that 'spare' subjects will be included in the analysis only if needed as replacements for other subjects who have been excluded. It should be planned that all treated subjects should be included in the analysis, even if there are no drop-outs.

In studies with more than two treatment arms (e.g. a three period study including two comparators, one from EU and another from USA, or a four period study including test and comparator in fed and fasted states), the analysis for each comparison should be conducted excluding the data from the treatments that are not relevant for the comparison in question.

Reasons for exclusion

Unbiased assessment of results from randomized studies requires that all subjects are observed and treated according to the same rules. These rules should be independent from treatment or outcome. In consequence, the decision to exclude a subject from the statistical analysis must be made before bioanalysis.

In principle any reason for exclusion is valid provided it is specified in the protocol and the decision to exclude is made before bioanalysis. However the exclusion of data should be avoided, as the power of the study will be reduced and a minimum of 12 evaluable subjects is required.

Examples of reasons to exclude the results from a subject in a particular period are events such as vomiting and diarrhoea which could render the plasma concentration-time profile unreliable. In exceptional cases, the use of concomitant medication could be a reason for excluding a subject.

The permitted reasons for exclusion must be pre-specified in the protocol. If one of these events occurs it should be noted in the CRF as the study is being conducted. Exclusion of subjects based on these pre-specified criteria should be clearly described and listed in the study report.

Exclusion of data cannot be accepted on the basis of statistical analysis or for pharmacokinetic reasons alone, because it is impossible to distinguish the formulation effects from other effects influencing the pharmacokinetics.

The exceptions to this are:-

1) A subject with lack of any measurable concentrations or only very low plasma concentrations for comparator medicinal product. A subject is considered to have very low plasma concentrations if its AUC is less than 5% of comparator medicinal product geometric mean AUC (which should be calculated without inclusion of data from the outlying subject). The exclusion of data due to this reason will only be accepted in exceptional cases and may question the validity of the trial.

2) Subjects with non-zero baseline concentrations > 5% of C_{max}. Such data should be excluded from bioequivalence calculation (see carry-over effects below).

The above can, for immediate release formulations, be the result of subject non-compliance and an insufficient wash-out period, respectively, and should as far as possible be avoided by

mouth check of subjects after intake of study medication to ensure the subjects have swallowed the study medication and by designing the study with a sufficient wash-out period. The samples from subjects excluded from the statistical analysis should still be assayed and the results listed (see Presentation of data below).

As stated in Section 3.1.4, AUC(0-t) should cover at least 80% of AUC(0-∞). Subjects should not be excluded from the statistical analysis if AUC(0-t) covers less than 80% of AUC(0-∞), but if the percentage is less than 80% in more than 20% of the observations then the validity of the study may need to be discussed. This does not apply if the sampling period is 72 h or more and AUC(0-72h) is used instead of AUC(0-t).

Parameters to be analysed and acceptance limits

In studies to determine bioequivalence after a single dose, the parameters to be analysed are AUC(0-t), or, when relevant, AUC(0-72h), and C_{max}. For these parameters the 90% confidence interval for the ratio of the test and comparator products should be contained within the acceptance interval of 80.00-125.00%. To be inside the acceptance interval the lower bound should be ≥ 80.00% when rounded to two decimal places and the upper bound should be ≤ 125.00% when rounded to two decimal places.

For studies to determine bioequivalence of immediate release formulations at steady state, AUC(0-t) and C_{max,ss} should be analysed using the same acceptance interval as stated above.

In the rare case where urinary data has been used, A_e(0-t) should be analysed using the same acceptance interval as stated above for AUC(0-t). R_{max} should be analysed using the same acceptance interval as for C_{max}.

A statistical evaluation of t_{max} is not required. However, if rapid release is claimed to be clinically relevant and of importance for onset of action or is related to adverse events, there should be no apparent difference in median T_{max} and its variability between test and comparator product.

In specific cases of products with a narrow therapeutic range, the acceptance interval may need to be tightened (see Section 3.1.9). Moreover, for highly variable finished pharmaceutical products the acceptance interval for C_{max} may in certain cases be widened (see Section 3.1.10).

Statistical analysis

The assessment of bioequivalence is based upon 90% confidence intervals for the ratio of the population geometric means (test/comparator) for the parameters under consideration. This method is equivalent to two one-sided tests with the null hypothesis of bioequivalence at the 5% significance level.

The pharmacokinetic parameters under consideration should be analysed using ANOVA. The data should be transformed prior to analysis using a logarithmic transformation. A confidence interval for the difference between formulations on the log-transformed scale is obtained from the ANOVA model. This confidence interval is then back-transformed to obtain the desired confidence interval for the ratio on the original scale. A non-parametric analysis is not acceptable.

The precise model to be used for the analysis should be pre-specified in the protocol. The statistical analysis should take into account sources of variation that can be reasonably assumed to have an effect on the response variable. The terms to be used in the ANOVA model are usually sequence, subject within sequence, period and formulation. Fixed effects, rather than random effects, should be used for all terms.

Carry-over effects

A test for carry-over is not considered relevant and no decisions regarding the analysis (e.g. analysis of the first period only) should be made on the basis of such a test. The potential for carry-over can be directly addressed by examination of the pre-treatment plasma concentrations in period 2 (and beyond if applicable).

If there are any subjects for whom the pre-dose concentration is greater than 5 percent of the C_{max} value for the subject in that period, the statistical analysis should be performed with the data from that subject for that period excluded. In a 2-period trial this will result in the subject being removed from the analysis. The trial will no longer be considered acceptable if these exclusions result in fewer than 12 subjects being evaluable. This approach does not apply to endogenous drugs.

Two-stage design

It is acceptable to use a two-stage approach when attempting to demonstrate bioequivalence. An initial group of subjects can be treated and their data analysed. If bioequivalence has not been demonstrated an additional group can be recruited and the results from both groups combined in a final analysis. If this approach is adopted appropriate steps must be taken to preserve the overall type I error of the experiment and the stopping criteria should be clearly defined prior to the study.

The analysis of the first stage data should be treated as an interim analysis and both analyses conducted at adjusted significance levels (with the confidence intervals accordingly using an adjusted coverage probability which will be higher than 90%). For example, using 94.12% confidence intervals for both the analysis of stage 1 and the combined data from stage 1 and stage 2 would be acceptable, but there are many acceptable alternatives and the choice of how much alpha to spend at the interim analysis is at the company's discretion. The plan to use a two-stage approach must be pre-specified in the protocol along with the adjusted significance levels to be used for each of the analyses.

When analyzing the combined data from the two stages, a term for stage should be included in the ANOVA model.

Presentation of data

All individual concentration data and pharmacokinetic parameters should be listed by formulation together with summary statistics such as geometric mean, median, arithmetic mean, standard deviation, coefficient of variation, minimum and maximum. Individual plasma concentration/time curves should be presented in linear/linear and log/linear scale. The method used to derive the pharmacokinetic parameters from the raw data should be specified. The number of points of the terminal log-linear phase used to estimate the terminal rate constant (which is needed for a reliable estimate of AUC_∞) should be specified.

For the pharmacokinetic parameters that were subject to statistical analysis, the point estimate and 90% confidence interval for the ratio of the test and comparator products should be presented.

The ANOVA tables, including the appropriate statistical tests of all effects in the model, should be submitted.

The report should be sufficiently detailed to enable the pharmacokinetics and the statistical analysis to be repeated, e.g. data on actual time of blood sampling after dose, drug concentrations, the values of the pharmacokinetic parameters for each subject in each period and the randomization scheme should be provided.

Drop-out and withdrawal of subjects should be fully documented. If available, concentration data and pharmacokinetic parameters from such subjects should be presented in the individual listings, but should not be included in the summary statistics.

The bioanalytical method should be documented in a pre-study validation report. A bioanalytical report should be provided as well. The bioanalytical report should include a brief description of the bioanalytical method used and the results for all calibration standards and quality control samples. A representative number of chromatograms or other raw data should be provided covering the whole concentration range for all standard and quality control samples as well as the specimens analysed. This should include all chromatograms from at least 20% of the subjects with QC samples and calibration standards of the runs including these subjects.

If for a particular formulation at a particular strength multiple studies have been performed some of which demonstrate bioequivalence and some of which do not, the body of evidence must be considered as a whole. Only relevant studies, as defined in Section 3.0, need be considered. The existence of a study which demonstrates bioequivalence does not mean that those which do not can be ignored. The applicant should thoroughly discuss the results and justify the claim that bioequivalence has been demonstrated. Alternatively, when relevant, a combined analysis of all studies can be provided in addition to the individual study analyses. It is not acceptable to pool together studies which fail to demonstrate bioequivalence in the absence of a study that does.

Narrow therapeutic index drugs

In specific cases of products with a narrow therapeutic index, the acceptance interval for AUC should be tightened to 90.00-111.11%. Where C_{max} is of particular importance for safety, efficacy or drug level monitoring the 90.00-111.11% acceptance interval should also be applied for this parameter. For a list of narrow therapeutic index drugs (NTIDs), refer to the table below:-

Aprindine Carbamazepine
Clindamycin Clonazepam
Clonidine Cyclosporine
Digitoxin Digoxin
Disopyramide Ethinyl Estradiol
Ethosuximide Guanethidine
Isoprenaline Lithium Carbonate

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Methotrexate Phenobarbital
Phenytoin Prazosin
Primidone Procainamide
Quinidine Sulfonylurea compounds
Tacrolimus Theophylline compounds
Valproic Acid Warfarin
Zonisamide Glybuzole

Highly variable drugs or finished pharmaceutical products

Highly variable finished pharmaceutical products (HVDP) are those whose intra-subject variability for a parameter is larger than 30%. If an applicant suspects that a finished pharmaceutical product can be considered as highly variable in its rate and/or extent of absorption, a replicate cross-over design study can be carried out.

Those HVDP for which a wider difference in C_{max} is considered clinically irrelevant based on a sound clinical justification can be assessed with a widened acceptance range. If this is the case the acceptance criteria for C_{max} can be widened to a maximum of 69.84 – 143.19%. For the acceptance interval to be widened the bioequivalence study must be of a replicate design where it has been demonstrated that the within-subject variability for C_{max} of the comparator compound in the study is >30%. The applicant should justify that the calculated intra-subject variability is a reliable estimate and that it is not the result of outliers. The request for widened interval must be prospectively specified in the protocol.

The extent of the widening is defined based upon the within-subject variability seen in the bioequivalence study using scaled-average-bioequivalence according to $[U, L] = \exp [\pm k \cdot \text{SWR}]$, where U is the upper limit of the acceptance range, L is the lower limit of the acceptance range, k is the regulatory constant set to 0.760 and SWR is the within-subject standard deviation of the log-transformed values of C_{max} of the comparator product. The table below gives examples of

how different levels of variability lead to different acceptance limits using this methodology.

Within-subject CV (%)* Lower Limit Upper Limit

30 80 125
35 77.23 129.48
40 74.62 134.02
45 72.15 138.59
≥50 69.84 143.19

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The geometric mean ratio (GMR) should lie within the conventional acceptance range 80.00-125.00%.

The possibility to widen the acceptance criteria based on high intra-subject variability does not apply to AUC where the acceptance range should remain at 80.00 – 125.00% regardless of variability.

It is acceptable to apply either a 3-period or a 4-period crossover scheme in the replicate design study.

3.2 In vitro dissolution tests

General aspects of in vitro dissolution experiments are briefly outlined in (annexe I) including basic requirements how to use the similarity factor (f_2 -test).

3.2.1 In vitro dissolution tests complementary to bioequivalence studies

The results of in vitro dissolution tests at three different buffers (normally pH 1.2, 4.5 and 6.8) and the media intended for finished pharmaceutical product release (QC media), obtained with the batches of test and comparator products that were used in the bioequivalence study should be reported. Particular dosage forms like ODT (oral dispersible tablets) may require investigations using different experimental conditions. The results should be reported as profiles of percent of labelled amount dissolved versus time displaying mean values and summary statistics.

Unless otherwise justified, the specifications for the in vitro dissolution to be used for quality control of the product should be derived from the dissolution profile of the test product batch that was found to be bioequivalent to the comparator product (see Annex I).

In the event that the results of comparative in vitro dissolution of the biobatches do not reflect bioequivalence as demonstrated in vivo the latter prevails. However, possible reasons for the discrepancy should be addressed and justified.

3.2.2 In vitro dissolution tests in support of biowaiver of strengths

Appropriate in vitro dissolution should confirm the adequacy of waiving additional in vivo bioequivalence testing. Accordingly, dissolution should be investigated at different pH values as outlined in the previous sections (normally pH 1.2, 4.5 and 6.8) unless otherwise justified. Similarity of in vitro dissolution (see Annex I) should be demonstrated at all conditions within the applied product series, i.e. between additional strengths and the strength(s) (i.e. batch(es)) used for bioequivalence testing.

At pH values where sink conditions may not be achievable for all strengths in vitro dissolution may differ between different strengths. However, the comparison with the respective strength of the comparator medicinal product should then confirm that this finding is active

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pharmaceutical ingredient rather than formulation related. In addition, the applicant could show similar profiles at the same dose (e.g. as a possibility two tablets of 5 mg versus one tablet of 10 mg could be compared).

3.3 Study report

3.3.1 Bioequivalence study report

The report of a bioavailability or bioequivalence study should follow the template format as provided in the Comprehensive Bioequivalence Information Summary (CBIS), Annex V) in order to submit the complete documentation of its conduct and evaluation complying with GCP-rules.

The report of the bioequivalence study should give the complete documentation of its protocol, conduct and evaluation. It should be written in accordance with the ICH E3 guideline and be signed by the investigator.

Names and affiliations of the responsible investigator(s), the site of the study and the period of its execution should be stated. Audits certificate(s), if available, should be included in the report.

The study report should include evidence that the choice of the comparator medicinal product is in accordance with the list of comparator products. This should include the comparator product name, strength, pharmaceutical form, batch number, manufacturer, expiry date and country of purchase.

The name and composition of the test product(s) used in the study should be provided. The batch size, batch number, manufacturing date and, if possible, the expiry date of the test product should be stated.

Certificates of analysis of comparator and test batches used in the study should be included in an Annex to the study report.

Concentrations and pharmacokinetic data and statistical analyses should be presented in the level of detail described above (Section 3.1.8 Presentation of data).

3.3.2 Other data to be included in an application

The applicant should submit a signed statement confirming that the test product has the same quantitative composition and is manufactured by the same process as the one submitted for authorization. A confirmation whether the test product is already scaled-up for production should be submitted. Comparative dissolution profiles (see Section 3.2) should be provided.

The validation report of the bioanalytical method should be included in Module 5 of the application.

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Data sufficiently detailed to enable the pharmacokinetics and the statistical analysis to be repeated, e.g. data on actual times of blood sampling, drug concentrations, the values of the pharmacokinetic parameters for each subject in each period and the randomization scheme, should be available in a suitable electronic format (e.g. as comma separated and space delimited text files or Excel format) to be provided upon request.

3.4 Variation applications

If a product has been reformulated from the formulation initially approved or the manufacturing method has been modified in ways that may impact on the bioavailability, an in vivo bioequivalence study is required, unless otherwise justified. Any justification presented should be based upon general considerations, e.g. as per Annex III.

In cases where the bioavailability of the product undergoing change has been investigated and an acceptable level A correlation between in vivo performance and in vitro dissolution has been established, the requirements for in vivo demonstration of bioequivalence can be waived if the dissolution profile in vitro of the new product is similar to that of the already approved medicinal product under the same test conditions as used to establish the correlation (see Annex I).

For variations of products approved on full documentation on quality, safety and efficacy, the comparative medicinal product for use in bioequivalence and dissolution studies is usually that authorized under the currently registered formulation, manufacturing process, packaging etc.

When variations to a generic product are made, the comparative medicinal product for the bioequivalence study should normally be a current batch of the reference medicinal product. If a valid reference medicinal product is not available on the market, comparison to the previous formulation (of the generic product) could be accepted, if justified. For variations that do not require a bioequivalence study, the advice and requirements stated in other published regulatory guidance should be followed.

4 OTHER APPROACHES TO ASSESS THERAPEUTIC EQUIVALENCE

4.1 Comparative pharmacodynamics studies

Studies in healthy volunteers or patients using pharmacodynamics measurements may be used for establishing equivalence between two pharmaceuticals products. These studies may become necessary if quantitative analysis of the drug and/or metabolite(s) in plasma or urine cannot be made with sufficient accuracy and sensitivity. Furthermore, pharmacodynamics studies in humans are required if measurements of drug concentrations cannot be used as surrogate end points for the demonstration of efficacy and safety of the particular pharmaceutical product e.g., for topical products without intended absorption of the drug into the systemic circulation.

4.2 Comparative clinical studies

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If a clinical study is considered as being undertaken to prove equivalence, the same statistical principles apply as for the bioequivalence studies. The number of patients to be included in the study will depend on the variability of the target parameters and the acceptance range, and is usually much higher than the number of subjects in bioequivalence studies.

4.3 Special considerations for modified – release finished pharmaceutical products

For the purpose of these guidelines modified release products include:-

- i. Delayed release
- ii. Sustained release
- iii. Mixed immediate and sustained release
- iv. Mixed delayed and sustained release
- v. Mixed immediate and delayed release

Generally, these products should:-

- i. Acts as modified –release formulations and meet the label claim.
- ii. Preclude the possibility of any dose dumping effects.
- iii. There must be a significant difference between the performance of modified release product and the conventional release product when used as reference product.
- iv. Provide a therapeutic performance comparable to the reference immediate – release formulation administered by the same route in multiple doses (of an equivalent daily amount) or to the reference modified – release formulation.
- v. Produce consistent Pharmacokinetic performance between individual dosage units and
- vi. Produce plasma levels which lie within the therapeutic range (where appropriate) for the proposed dosing intervals at steady state.

If all of the above conditions are not met but the applicant considers the formulation to be acceptable, justification to this effect should be provided.

i. Study Parameters

Bioavailability data should be obtained for all modified release finished pharmaceutical products although the type of studies required and the Pharmacokinetics parameters which should be evaluated may differ depending on the active ingredient involved. Factors to be considered include whether or not the formulation represents the first market entry of the active

pharmaceutical ingredients, and the extent of accumulation of the drug after repeated dosing.

If formulation is the first market entry of the APIs, the products pharmacokinetic parameters should be determined. If the formulation is a second or subsequent market entry then the comparative bioavailability studies using an appropriate reference product should be performed.

ii. Study design

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Study design will be single dose or single and multiple dose based on the modified release products that are likely to accumulate or unlikely to accumulate both in fasted and non-fasting state. If the effects of food on the reference product is not known (or it is known that food affects its absorption), two separate two-way cross-over studies, one in the fasted state and the other in the fed state, may be carried out.

iii. Requirement for modified release formulations unlikely to accumulate

This section outlines the requirements for modified release formulations which are used at a dose interval that is not likely to lead to accumulation in the body ($AUC_{0-v}/AUC_{0-\infty} \geq 0.8$)

When the modified release product is the first marketed entry type of dosage form, the reference product should normally be the innovator immediate-release formulation. The comparison should be between a single dose of the modified release formulation and doses of the immediate-release formulation which it is intended to replace. The latter must be administered according to the established dosing regimen.

When the release product is the second or subsequent entry on the market, comparison should be with the reference modified release product for which bioequivalence is claimed.

Studies should be performed with single dose administration in the fasting state as well as following an appropriate meal at a specified time.

The following pharmacokinetic parameters should be calculated from plasma (or relevant biological matrix) concentration of the drug and/or major metabolites(s) AUC_0-t , AUC_0-t , $AUC_0-\infty$, C_{max} (where the comparison is with an existing modified release product) and K_{el} .

The 90% confidence interval calculated using log transformed data for the ratios (Test vs Reference) of the geometric mean AUC (for both AUC_0-t and AUC_0-t) and C_{max} (Where the comparison is with an existing modified release product) should generally be within the range 80 to 125% both in the fasting state and following the administration of an appropriate meal at a specified time before taking the drug.

The Pharmacokinetic parameters should support the claimed dose delivery attributes of the modified release – dosage form.

iv. Requirement for modified release formulations likely to accumulate

This section outlines the requirement for modified release formulations that are used at dose intervals that are likely to lead to accumulation ($AUC/AUC \leq 0.8$).

When a modified release product is the first market entry of the modified release type, the reference formulation is normally the innovator's immediate-release formulation. Both a single dose and steady state doses of the modified release formulation should be compared with doses

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of the immediate-release formulation which it is intended to replace. The immediate-release product should be administered according to the conventional dosing regimen.

Studies should be performed with single dose administration in the fasting state as well as following an appropriate meal. In addition, studies are required at steady state. The following pharmacokinetic parameters should be calculated from single dose studies; AUC_0-t , AUC_0-t , $AUC_0-\infty$, C_{max} (where the comparison is with an existing modified release product) and K_{el} . The following parameters should be calculated from steady state studies; AUC_0-t , C_{max} , C_{min} , C_{pd} , and degree of fluctuation.

When the modified release product is the second or subsequent modified release entry, single dose and steady state comparisons should normally be made with the reference modified release product for which bioequivalence is claimed.

90% confidence interval for the ratio of geometric means (Test/Reference drug) for AUC, C_{max} and C_{min} determined using log-transformed data should generally be within the range 80 to 125% when the formulation are compared at steady state.

90% confidence interval for the ratio of geometric means (Test/Reference drug) for AUC_0-t , C_{max} , and C_{min} determined using log-transferred data should generally be within the range 80 to 125% when the formulation are compared at steady state.

The Pharmacokinetic parameters should support the claimed attributes of the modified-release dosage form.

The Pharmacokinetic data may reinforce or clarify interpretation of difference in the plasma concentration data.

Where these studies do not show bioequivalence, comparative efficacy and safety data may be required for the new product.

Pharmacodynamic studies;

Studies in healthy volunteers or patients using pharmacodynamics parameters may be used for establishing equivalence between two pharmaceutical products. These studies may become necessary if quantitative analysis of the drug and/or metabolites (s) in plasma or urine cannot be made with sufficient accuracy and sensitivity. Furthermore, pharmacodynamic studies in humans are required if measurement of drug concentrations cannot be used as surrogate endpoints for the demonstration of efficacy and safety of the particular pharmaceutical product

e.g. for topical products without an intended absorption of the drug into the systemic circulation.

In case, only pharmacodynamic data is collected and provided, the applicant should outline what other methods were tried and why they were found unsuitable.

The following requirements should be recognized when planning, conducting and assessing the results from a pharmacodynamic study;

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i. The response measured should be a pharmacological or therapeutically effects which is relevant to the claims of efficacy and /or safety of the drug.

ii. The methodology adopted for carrying out the study the study should be validated for precision, accuracy, reproducibility and specificity.

iii. Neither the test nor reference product should produce a maximal response in the course of the study, since it may be impossible to distinguish difference between formulations given in doses that produce such maximal responses. Investigation of dose – response relationship may become necessary.

iv. The response should be measured quantitatively under double – blind conditions and be recorded in an instrument – produced or instrument recorded fashion on a repetitive basis to provide a record of pharmacodynamic events which are suitable for plasma concentrations. If such measurement is not possible recording on visual – analog scales may be used. In instances where data are limited to quantitative (categorized) measurement, appropriate special statistical analysis will be required.

v. Non – responders should be excluded from the study by prior screening. The criteria by which responder –are versus non –responders are identified must be stated in the protocol.

vi. Where an important placebo effect occur comparison between products can only be made by a priori consideration of the placebo effect in the study design. This may be achieved by adding a third period/phase with placebo treatment, in the design of the study.

vii. A crossover or parallel study design should be used, appropriate.

viii. When pharmacodynamic studies are to be carried out on patients, the underlying pathology and natural history of the condition should be considered in the design.

ix. There should be knowledge of the reproducibility of the base – line conditions.

x. Statistical considerations for the assessments of the outcomes are in principle, the same as in Pharmacokinetic studies.

xi. A correction for the potential non – linearity of the relationship between dose and area under the effect – time curve should be made on the basis of the outcome of the dose ranging study.

The conventional acceptance range as applicable to Pharmacokinetic studies and bioequivalence is not appropriate (too large) in most cases. This range should therefore be defined in the protocol on a case – to – case basis.

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Comparative clinical studies

The plasma concentration time – profile data may not be suitable to assess equivalence between two formulations. Whereas in some of the cases pharmacodynamic studies can be an appropriate to for establishing equivalence , in other instances this type of study cannot be performed because of lack of meaningful pharmacodynamic parameters which can be measured and comparative clinical study has be performed in order to demonstrate equivalence between two formulations. Comparative clinical studies may also be required to be carried out for certain orally administered finished pharmaceutical products when pharmacokinetic and pharmacodynamic studies are no feasible. However, in such cases the applicant should outline what other methods were why they were found unsuitable.

If a clinical study is considered as being undertaken to prove equivalence, the appropriate statistical principles should be applied to demonstrate bioequivalence. The number of patients to be included in the study will depend on the variability of the target parameter and the acceptance range, and is usually much higher than the number of subjects in bioequivalence studies.

The following items are important and need to be defined in the protocol advance:-

a. The target parameters which usually represent relevant clinical end –points from which the intensity and the onset, if applicable and relevant, of the response are to be derived.

b. The size of the acceptance range has to be defined case taking into consideration the specific clinical conditions. These include, among others, the natural course of the disease, the efficacy of available treatment and the chosen target parameter. In contrast to bioequivalence studies (where a conventional acceptance range is applied) the size of the acceptance in clinical trials cannot be based on a general consensus on all the therapeutic clinical classes and indications.

c. The presently used statistical method is the confidence interval approach. The main concern is to rule out t Hence, a one – sided confidence interval (For efficacy and/or safety) may be appropriate. The confidence intervals can be derived from either parametric or nonparametric methods.

d. Where appropriate, a placebo leg should be included in the design.

e. In some cases, it is relevant to include safety end-points in the final comparative assessments.

General aspects of dissolution testing as related to bioavailability

During the development of a medicinal product a dissolution test is used as a tool to identify formulation factors that are influencing and may have a crucial effect on the bioavailability of the drug. As soon as the composition and the manufacturing process are defined a dissolution test is used in the quality control of scale-up and of production batches to ensure both batch-to-batch consistency and that the dissolution profiles remain similar to those of pivotal clinical trial batches. Furthermore, in certain instances a dissolution test can be used to waive a bioequivalence study. Therefore, dissolution studies can serve several purposes:-

(a) Testing on product quality:-

- To get information on the test batches used in bioavailability/bioequivalence studies and pivotal clinical studies to support specifications for quality control.
- To be used as a tool in quality control to demonstrate consistency in manufacture.
- To get information on the comparator product used in bioavailability/bioequivalence studies and pivotal clinical studies.

(b) Bioequivalence surrogate inference

- To demonstrate in certain cases similarity between different formulations of an active substance and the reference medicinal product (biowaivers e.g., variations, formulation changes during development and generic medicinal products; see Section 3.2 and Annex III)
- To investigate batch to batch consistency of the products (test and comparator) to be used as basis for the selection of appropriate batches for the in vivo study.

Test methods should be developed product related based on general and/or specific pharmacopoeial requirements. In case those requirements are shown to be unsatisfactory and/or do not reflect the in vivo dissolution (i.e. biorelevance) alternative methods can be considered when justified that these are discriminatory and able to differentiate between batches with acceptable and non-acceptable performance of the product in vivo. Current state-of-the-art information including the interplay of characteristics derived from the BCS classification and the dosage form must always be considered.

Sampling time points should be sufficient to obtain meaningful dissolution profiles, and at least every 15 minutes. More frequent sampling during the period of greatest change in the dissolution profile is recommended. For rapidly dissolving products, where complete dissolution is within 30 minutes, generation of an adequate profile by sampling at 5- or 10-minute intervals may be necessary.

If an active substance is considered highly soluble, it is reasonable to expect that it will not cause any bioavailability problems if, in addition, the dosage system is rapidly dissolved in the physiological pH-range and the excipients are known not to affect bioavailability. In contrast, if

an active substance is considered to have a limited or low solubility, the rate limiting step for absorption may be dosage form dissolution. This is also the case when excipients are controlling the release and subsequent dissolution of the active substance. In those cases a variety of test conditions is recommended and adequate sampling should be performed.

Similarity of dissolution profiles

Dissolution profile similarity testing and any conclusions drawn from the results (e.g. justification for a biowaiver) can be considered valid only if the dissolution profile has been satisfactorily characterised using a sufficient number of time points.

For immediate release formulations, further to the guidance above, comparison at 15 min is essential to know if complete dissolution is reached before gastric emptying.

Where more than 85% of the drug is dissolved within 15 minutes, dissolution profiles may be accepted as similar without further mathematical evaluation.

In case more than 85% is not dissolved at 15 minutes but within 30 minutes, at least three time points are required: the first time point before 15 minutes, the second one at 15 minutes and the third time point when the release is close to 85%.

For modified release products, the advice given in the relevant guidance should be followed.

Dissolution similarity may be determined using the f_2 statistic as follows:

In this equation f_2 is the similarity factor, n is the number of time points, $R(t)$ is the mean percent reference drug dissolved at time t after initiation of the study; $T(t)$ is the mean percent test drug dissolved at time t after initiation of the study. For both the reference and test formulations, percent dissolution should be determined.

The evaluation of the similarity factor is based on the following conditions:

- A minimum of three time points (zero excluded)
- The time points should be the same for the two formulations
- Twelve individual values for every time point for each formulation
- Not more than one mean value of > 85% dissolved for any of the formulations.
- The relative standard deviation or coefficient of variation of any product should be less than 20% for the first point and less than 10% from second to last time point.

An f_2 value between 50 and 100 suggests that the two dissolution profiles are similar.

When the f_2 statistic is not suitable, then the similarity may be compared using model-dependent or model-independent methods e.g. by statistical multivariate comparison of the parameters of the Weibull function or the percentage dissolved at different time points.

Alternative methods to the f_2 statistic to demonstrate dissolution similarity are considered acceptable, if statistically valid and satisfactorily justified.

The similarity acceptance limits should be pre-defined and justified and not be greater than a 10% difference. In addition, the dissolution variability of the test and reference product data should also be similar; however, a lower variability of the test product may be acceptable.

Evidence that the statistical software has been validated should also be provided.

A clear description and explanation of the steps taken in the application of the procedure should be provided, with appropriate summary tables.

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Annex II: Bioequivalence study requirements for different dosage forms

Although this guideline concerns immediate release formulations, Annex II provides some general guidance on the bioequivalence data requirements for other types of formulations and for specific types of immediate release formulations.

When the test product contains a different salt, ester, ether, isomer, mixture of isomers, complex or derivative of an active substance than the comparator medicinal product, bioequivalence should be demonstrated in in vivo bioequivalence studies. However, when the active substance in both test and comparator products is identical (or contain salts with similar properties as defined in Annex III, Section III), in vivo bioequivalence studies may in some situations not be required as described below and in Annex III.

Oral immediate release dosage forms with systemic action

For dosage forms such as tablets, capsules and oral suspensions, bioequivalence studies are required unless a biowaiver is applicable (see Annex III). For oral dispersible tablets and oral solutions specific recommendations apply, as detailed below.

Oral dispersible tablets

An oral dispersible tablet (ODT) is formulated to quickly disperse in the mouth. Placement in the mouth and time of contact may be critical in cases where the active substance also is dissolved in the mouth and can be absorbed directly via the buccal mucosa. Depending on the formulation, swallowing of the e.g. coated substance and subsequent absorption from the gastrointestinal tract also will occur. If it can be demonstrated that the active substance is not absorbed in the oral cavity, but rather must be swallowed and absorbed through the gastrointestinal tract, then the product might be considered for a BCS based biowaiver (see Annex III). If this cannot be demonstrated, bioequivalence must be evaluated in human studies.

If the ODT test product is an extension to another oral formulation, a 3-period study is recommended in order to evaluate administration of the orodispersible tablet both with and without concomitant fluid intake. However, if bioequivalence between ODT taken without water and comparator formulation with water is demonstrated in a 2-period study, bioequivalence of ODT taken with water can be assumed.

If the ODT is a generic to an approved ODT comparator medicinal product, the following recommendations regarding study design apply:-

■ if the comparator medicinal product can be taken with or without water, bioequivalence should be demonstrated without water as this condition best resembles the intended use of the formulation. This is especially important if the substance may be dissolved and partly absorbed in the oral cavity. If bioequivalence is demonstrated when taken without water, bioequivalence when taken with water can be assumed.

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■ if the comparator medicinal product is taken only in one way (e.g. only with water), bioequivalence should be shown in this condition (in a conventional two-way crossover design).

■ if the comparator medicinal product is taken only in one way (e.g. only with water), and the test product is intended for additional ways of administration (e.g. without water), the conventional and the new method should be compared with the comparator in the conventional way of administration (3 treatment, 3 period, 6 sequence design).

In studies evaluating ODTs without water, it is recommended to wet the mouth by swallowing 20 ml of water directly before applying the ODT on the tongue. It is recommended not to allow fluid intake earlier than 1 hour after administration.

Other oral formulations such as orodispersible films, buccal tablets or films, sublingual tablets and chewable tablets may be handled in a similar way as for ODTs. Bioequivalence studies should be conducted according to the recommended use of the product.

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Annex III: BCS-based Biowaiver

I. Introduction

The BCS (Biopharmaceutics Classification System)-based biowaiver approach is meant to reduce in vivo bioequivalence studies, i.e., it may represent a surrogate for in vivo bioequivalence. In vivo bioequivalence studies may be exempted if an assumption of equivalence in in vivo performance can be justified by satisfactory in vitro data.

Applying for a BCS-based biowaiver is restricted to highly soluble active pharmaceutical ingredients with known human absorption and considered not to have a narrow therapeutic index (see Section 3.1.9). The concept is applicable to immediate release, solid pharmaceutical products for oral administration and systemic action having the same pharmaceutical form. However, it is not applicable for sublingual, buccal, and modified release formulations. For oral dispersible formulations the BCS-based biowaiver approach may only be applicable when absorption in the oral cavity can be excluded.

BCS-based biowaivers are intended to address the question of bioequivalence between specific test and reference/comparator products. The principles may be used to establish bioequivalence in applications for generic medicinal products, extensions of innovator products, variations that require bioequivalence testing, and between early clinical trial products and to-be-marketed products.

In situations where multiples strength formulations have been submitted for BCS based biowaiver, comparative dissolution should be provided for all the strength.

II. Summary Requirements

BCS-based biowaiver are applicable for an immediate release finished pharmaceutical product if:-

- the active pharmaceutical ingredient has been proven to exhibit high solubility and complete absorption (BCS class I; for details see Section III) and

- either very rapid (> 85 % within 15 min) or similarly rapid (85 % within 30 min) in vitro dissolution characteristics of the test and reference product has been demonstrated considering specific requirements (see Section IV.1) and

- excipients that might affect bioavailability are qualitatively and quantitatively the same. In general, the use of the same excipients in similar amounts is preferred (see Section IV.2).

BCS-based biowaiver are also applicable for an immediate release finished pharmaceutical product if:-

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- the active pharmaceutical ingredient has been proven to exhibit high solubility and limited absorption (BCS class III; for details see Section III) and

- very rapid (> 85 % within 15 min) in vitro dissolution of the test and reference product has been demonstrated considering specific requirements (see Section IV.1) and

- excipients that might affect bioavailability are qualitatively and quantitatively the same and

- other excipients are qualitatively the same and quantitatively very similar (see Section IV.2).

Generally the risks of an inappropriate biowaiver decision should be more critically reviewed (e.g. site-specific absorption, risk for transport protein interactions at the absorption site, excipient composition and therapeutic risks) for products containing BCS class III than for BCS class I active pharmaceutical ingredient.

III. Active Pharmaceutical Ingredient

Generally, sound peer-reviewed literature may be acceptable for known compounds to describe the active pharmaceutical ingredient characteristics of importance for the biowaiver concept.

Biowaiver may be applicable when the active substance(s) in test and reference products are identical.

Biowaiver may also be applicable if test and reference contain different salts provided that both belong to BCS-class I (high solubility and complete absorption; see Sections III.1 and III.2). Biowaiver is not applicable when the test product contains a different ester, ether, isomer, mixture of isomers, complex or derivative of an active substance from that of the comparator product, since these differences may lead to different bioavailabilities not deducible by means of experiments used in the BCS-based biowaiver concept.

The active pharmaceutical ingredient should not belong to the group of 'narrow therapeutic index' drugs (see Section 4.1.9 on narrow therapeutic index drugs).

III.1 Solubility

The pH-solubility profile of the active pharmaceutical ingredient should be determined and discussed. An API is considered highly soluble when the highest single therapeutic dose as determined by the relevant regulatory authority, typically defined by the labeling for the innovator product, is completely dissolved in 250 ml of buffers within the range of pH 1 – 6.8 at 37±1 °C. This demonstration requires the investigation in at least three buffers within this range (preferably at pH 1.2, 4.5 and 6.8) and in addition at the pKa, if it is within the specified pH range. Replicate determinations at each pH condition may be necessary to achieve an unequivocal solubility classification (e.g. shake-flask method or other justified method). Solution pH should be verified prior and after addition of the active pharmaceutical ingredient to a buffer.

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III.2 Absorption

The demonstration of complete absorption in humans is preferred for BCS-based biowaiver applications. For this purpose complete absorption is considered to be established where measured extent of absorption is ≥ 85 %. Complete absorption is generally related to high permeability.

Complete drug absorption should be justified based on reliable investigations in human. Data from either:-

- absolute bioavailability or

- mass-balance

studies could be used to support this claim.

When data from mass balance studies are used to support complete absorption, it must be ensured that the metabolites taken into account in determination of fraction absorbed are formed after absorption. Hence, when referring to total radioactivity excreted in urine, it should be ensured that there is no degradation or metabolism of the unchanged active pharmaceutical ingredient in the gastric or intestinal fluid. Phase 1 oxidative and Phase 2 conjugative

metabolism can only occur after absorption (i.e. cannot occur in the gastric or intestinal fluid). Hence, data from mass balance studies support complete absorption if the sum of urinary recovery of parent compound and urinary and faecal recovery of Phase 1 oxidative and Phase 2 conjugative drug metabolites account for $\geq 85\%$ of the dose.

In addition highly soluble active pharmaceutical ingredients with incomplete absorption, i.e. BCS-class III compounds, could be eligible for a biowaiver provided certain prerequisites are fulfilled regarding product composition and in vitro dissolution (see also Section IV.2 Excipients). The more restrictive requirements will also apply for compounds proposed to be BCS class I but where complete absorption could not convincingly be demonstrated.

Reported bioequivalence between aqueous and solid formulations of a particular compound administered via the oral route may be supportive as it indicates that absorption limitations due to (immediate release) formulation characteristics may be considered negligible. Well performed in vitro permeability investigations including reference standards may also be considered supportive to in vivo data.

IV. Finished pharmaceutical product

IV.1 In vitro Dissolution

IV.1.1 General Aspects

Investigations related to the medicinal product should ensure immediate release properties and prove similarity between the investigative products, i.e. test and reference show similar in vitro

dissolution under physiologically relevant experimental pH conditions. However, this does not establish an in vitro/in vivo correlation. In vitro dissolution should be investigated within the range of pH 1 – 6.8 (at least pH 1.2, 4.5, and 6.8). Additional investigations may be required at pH values in which the drug substance has minimum solubility. The use of any surfactant is not acceptable.

Test and reference products should meet requirements as outlined in Section 3.1.2 of the main guideline text. In line with these requirements it is advisable to investigate more than one single batch of the test and reference products.

Comparative in vitro dissolution experiments should follow current compendial standards.

Hence, thorough description of experimental settings and analytical methods including validation data should be provided. It is recommended to use 12 units of the product for each experiment to enable statistical evaluation. Usual experimental conditions are e.g.:-

- Apparatus: paddle or basket
- Volume of dissolution medium: 900 ml or less
- Temperature of the dissolution medium: 37 ± 1 °C
- Agitation:
 - paddle apparatus - usually 50 rpm
 - basket apparatus - usually 100 rpm
- Sampling schedule: e.g. 10, 15, 20, 30 and 45 min
- Buffer: pH 1.0 – 1.2 (usually 0.1 N HCl or SGF without enzymes), pH 4.5, and pH 6.8 (or SIF without enzymes); (pH should be ensured throughout the experiment; Ph.Eur. buffers recommended)
- Other conditions: no surfactant; in case of gelatin capsules or tablets with gelatin coatings the use of enzymes may be acceptable.

Complete documentation of in vitro dissolution experiments is required including a study protocol, batch information on test and reference batches, detailed experimental conditions, validation of experimental methods, individual and mean results and respective summary statistics.

IV.1.2 Evaluation of in vitro dissolution results

Finished pharmaceutical products are considered 'very rapidly' dissolving when more than 85 % of the labelled amount is dissolved within 15 min. In cases where this is ensured for the test and reference product the similarity of dissolution profiles may be accepted as demonstrated without any mathematical calculation.

Absence of relevant differences (similarity) should be demonstrated in cases where it takes more than 15 min but not more than 30 min to achieve almost complete (at least 85 % of labelled amount) dissolution. F2-testing (see Annex I) or other suitable tests should be used to demonstrate profile similarity of test and reference. However, discussion of dissolution profile differences in terms of their clinical/therapeutic relevance is considered inappropriate since the investigations do not reflect any in vitro/in vivo correlation.

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IV.2 Excipients

Although the impact of excipients in immediate release dosage forms on bioavailability of highly soluble and completely absorbable active pharmaceutical ingredients (i.e., BCS-class I) is considered rather unlikely it cannot be completely excluded. Therefore, even in the case of class I drugs it is advisable to use similar amounts of the same excipients in the composition of test like in the comparator product.

If a biowaiver is applied for a BCS-class III active pharmaceutical ingredient excipients have to be qualitatively the same and quantitatively very similar in order to exclude different effects on membrane transporters.

As a general rule, for both BCS-class I and III active pharmaceutical ingredients well-established excipients in usual amounts should be employed and possible interactions affecting drug bioavailability and/or solubility characteristics should be considered and discussed. A description of the function of the excipients is required with a justification whether the amount of each excipient is within the normal range. Excipients that might affect bioavailability, like e.g. sorbitol, mannitol, sodium lauryl sulfate or other surfactants, should be identified as well as their possible impact on:-

- gastrointestinal motility
- susceptibility of interactions with the active pharmaceutical ingredient (e.g. complexation)
- drug permeability
- interaction with membrane transporters

Excipients that might affect bioavailability should be qualitatively and quantitatively the same in the test product and the comparator product.

V. Fixed Combinations (FCs)

BCS-based biowaiver are applicable for immediate release FC products if all active substances in the FC belong to BCS-class I or III and the excipients fulfil the requirements outlined in Section IV.2. Otherwise in vivo bioequivalence testing is required.

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Biowaiver Application Form: Biopharmaceutics Classification System (BCS)

This application form is designed to facilitate information exchange between the Applicant and the Tanzania Medicines and Medical Devices Authority (TMDA) if the Applicant seeks to waive bioequivalence studies based on the Biopharmaceutics Classification System (BCS). For further information, please study the respective biowaiver guidance documents. This form is not to be used if a biowaiver is requested for additional strength(s) of a submitted product(s), in which case a separate "Biowaiver Application Form: Additional Strengths" should be used.

General Instructions:

Please review all the instructions thoroughly and carefully prior to completing the current Application Form.

- Provide as much detailed, accurate, and final information as possible.
- Please enter the data and information directly following the greyed areas.
- Please enclose the required documentation in full and state in the relevant sections of the Application Form the exact location (Annex number) of the appended documents. For example, in section 2.5 indicate in which Annex the Certificate of Analysis can be found.
- Please provide the document as an MS Word file.
- Do not paste snap-shots into the document.
- The appended electronic documents should be clearly identified in their file names, which should include the product name and Annex number.
- Before submitting the completed Application Form, kindly check that you have provided all requested information and enclosed all requested documents.
- Should you have any questions regarding this procedure, please contact Tanzania Medicines and Medical Devices Authority (TMDA).

The signed paper version of this Biowaiver Application Form together with Annexes (and their electronic copies on CD-ROM) should be included to the bioequivalence part of the submitted dossier and sent by surface mail to the following address:

Director General,
Tanzania Food and Drugs Authority,
Off Mandela Road, Mabibo-External
P.O. Box 77150
Dar es Salaam
Tanzania

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Administrative data

1. INN of active ingredient(s)

<< Please enter information here >>

2. Dosage form and strength

<< Please enter information here >>

3. Product Reference number

<< Please enter information here >>

4. Name of applicant and official address

<< Please enter information here >>

5. Name of manufacturer of finished product and official address

<< Please enter information here >>

6. Name and address of the laboratory or Contract Research Organisation(s) where the BCS-based biowaiver solubility and dissolution studies were conducted

<< Please enter information here >>

I, the undersigned, certify, that the information provided in this application and the attached documents is correct and true

Signed on behalf of

<Company>

_____ (Date)

_____ (Name and title)

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1. Justification for a BCS Biowaiver

1.1.Active Pharmaceutical Ingredient (API)

Please confirm that the proposed product contains the same active substance (e.g. salt, ester, ether, isomer) as the comparator.

<< Please enter information here >>

1.2. Therapeutic Index of the API

Please enclose a copy of the comparator product labelling and literature references employed to support that the drug does not exhibit a narrow therapeutic index for all authorised indications

<< Please enter information here >>

1.3. Pharmacokinetic properties of the API

Please enclose a copy of the literature references employed to document the PK properties (PK linearity or reasons for non-linearity).

<< Please enter information here >>

1.4.Dosage form

Please confirm that:

- the dosage form is an immediate release product for systemic action
- the posology is limited to oral administration
- the administration without water is not included in the proposed posology

<< Please enter information here >>

2. Solubility

(Completion of this section is not necessary if the API(s) are included on the list of biowaiver-eligible APIs in the PQTM document General notes on Biopharmaceutics Classification System (BCS)-based biowaiver applications.)

2.1. Maximum therapeutic dose of the API

Please enclose a copy of the labelling of the comparator product to document the maximum single dose that can be administered in a single administration (e.g. two tablets together).

<< Please enter information here >>

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2.2.Stability of the drug in the physiological pH range

Please discuss stability of the API in the pH range from 1.2 to 6.8 and in the gastrointestinal tract.

Please discuss the ability of the analytical method to distinguish the API from its degradation products.

<< Please enter information here >>

2.3.Method of solubility determination

Please describe method and conditions (e.g. shake flask method at 37±1°C)

Please indicate also location of the solubility study protocol.

<< Please enter information here >>

2.4.Solubility study dates

Please indicate dates of study protocol, study conductance and study report

<< Please enter information here >>

2.5. Analytical method validation

Please summarise the results and indicate location in the documentation.

<< Please enter information here >>

2.6. Results

Please indicate location of the solubility study report.

Please fill in the following table for the necessary pH values. Add as many rows as necessary to create a solubility – pH profile

Theoretical Observed pH Adjusted pH Individual Cs (mean and Amount
pH concentration CV(%)) that can be
at saturation dissolved in
(Cs) values 250 mL
pH 1.2 Experiment 1 Experiment 1 Experiment 1
Experiment 2 Experiment 2 Experiment 2

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Experiment 3 Experiment 3 Experiment 3
Intermediate Experiment 1 Experiment 1 Experiment 1
Hs Experiment 2 Experiment 2 Experiment 2
Experiment 3 Experiment 3 Experiment 3
pH 4.5 Experiment 1 Experiment 1 Experiment 1
Experiment 2 Experiment 2 Experiment 2
Experiment 3 Experiment 3 Experiment 3
Intermediate Experiment 1 Experiment 1 Experiment 1
pHs Experiment 2 Experiment 2 Experiment 2
Experiment 3 Experiment 3 Experiment 3
pH 6.8 Experiment 1 Experiment 1 Experiment 1
Experiment 2 Experiment 2 Experiment 2
Experiment 3 Experiment 3 Experiment 3
Other Experiment 1 Experiment 1 Experiment 1
intermediate Experiment 2 Experiment 2 Experiment 2
pH values Experiment 3 Experiment 3 Experiment 3
(e.g. pKa,

pKa-1,
pKa+1)

2.7. Plot the Solubility – pH profile

Please attach the plot of the pH-solubility profile based on the above data

<< Please enter information here >>

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Annex IV: Selection of a comparator product to be used in establishing interchangeability

I Introduction

This annex is intended to provide applicants with guidance with respect to selecting an appropriate comparator product to be used to prove therapeutic equivalence (i.e. interchangeability) of their product to an existing medicinal product(s) in the context of applying for marketing authorization.

II Comparator product

A product with which a generic product is intended to be interchangeable in clinical practice.

III Guidance on selection of a comparator product

General principles for the selection of comparator products are described in the Guidelines on Therapeutic Equivalence Requirements.

The innovator pharmaceutical product, which was first authorized for marketing, is the most logical comparator product to establish interchangeability, because its quality, safety and efficacy has been fully assessed and documented in pre-marketing studies and post-marketing monitoring schemes.

A generic pharmaceutical product should not be used as a comparator as long as an innovator pharmaceutical product is available, because this could lead to progressively less reliable similarity of future multisource products and potentially to a lack of interchangeability with the innovator.

Comparator products should be purchased from a well regulated market with stringent regulatory authority, i.e. from countries participating in the International Conference on Harmonization (ICH)¹

The applicant has the following options which are listed in order of preference:-

1. To choose an innovator product;
2. To choose a product which is approved and has been on the market in any of the SRA countries. The definition of an SRA is rephrased as follows. A regulatory authority that is:
a) a member of ICH prior to 23 October 2015, namely: the US Food and Drug Administration, the European Commission and the Ministry of Health, Labour and Welfare of Japan also represented by the Pharmaceuticals and Medical Devices Agency;
or b) an ICH observer prior to 23 October 2015, namely: the European Free Trade Association, as represented by Swissmedic and Health Canada; or c) a regulatory authority

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associated with an ICH member through a legally-binding, mutual recognition agreement prior to 23 October 2015, namely: Australia, Iceland, Liechtenstein and Norway

3. To choose the WHO recommended comparator product (as presented in the list of international comparator pharmaceutical products);

In case no recommended comparator product is identified; or in case the recommended comparator product cannot be located in a well regulated market with stringent regulatory authority as noted above, the applicant should consult TMDA regarding the choice of comparator before starting any studies.

IV Origin of the comparator product

Comparator products should be purchased from a well regulated market with stringent regulatory authority, i.e. from countries participating in the International Conference on Harmonization (ICH)¹. Within the submitted dossier, the country of origin of the comparator product should be reported together with lot number and expiry date, as well as results of pharmaceutical analysis to prove pharmaceutical equivalence.

Further in order to prove the origin of the comparator product the applicant must present all of the following documents:-

1. Copy of the comparator product labelling. The name of the product, name and address of the manufacturer, batch number, and expiry date should be clearly visible on the labelling.
2. Copy of the invoice from the distributor or company from which the comparator product was purchased. The address of the distributor must be clearly visible on the invoice.
3. Documentation verifying the method of shipment and storage conditions of the comparator product from the time of purchase to the time of study initiation.
4. A signed statement certifying the authenticity of the above documents and that the comparator product was purchased from the specified national market. The certification should be signed by the company executive responsible for the application for registration of pharmaceutical product.

In case the invited product has a different dose compared to the available acceptable

comparator product, it is not always necessary to carry out a bioequivalence study at the same dose level; if the active substance shows linear pharmacokinetics, extrapolation may be applied by dose normalization.

The bioequivalence of fixed-dose combination (FDC) should be established following the same general principles. The submitted FDC product should be compared with the respective innovator FDC product. In cases when no innovator FDC product is available on the market, individual component products administered in loose combination should be used as a comparator.

11 Countries officially participating in ICH are the ICH members European Union, Japan and USA; and the ICH observers Canada and Switzerland.

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ANNEX V: Bioequivalence Trial Information (BTIF)

GENERAL INSTRUCTIONS:

Please review all the instructions thoroughly and carefully prior to completing the Bioequivalence Trial Information Form (BTIF).

Provide as much detailed, accurate and final information as possible.

Please state the exact location (Annex number) of appended documents in the relevant sections of the BTIF. For example, in section 3.4.3.1 under point b), indicate in which Annex (number) the Certificate of Analysis can be found. This procedure must be followed throughout the entire document where location of annexed documents is requested.

Before submitting the completed BTIF, kindly check that you have provided all requested information and enclosed all requested documents.
Should you have any questions regarding this Form, please contact Tanzania Medicines and Medical Devices Authority (TMDA)

A properly filled out and signed original copy of the BTIF with all its annexes (including a copy on CD-ROM) must be submitted to TMDA together with the bioequivalence part of the dossier to the address below

Director General,
Tanzania Medicines and Medical Devices Authority,
Off Mandela Road, Mabibo-External,
P.O. Box 77150,
Dar es Salaam

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ASSESSMENT REPORT FOR GENERIC FINISHED PHARMACEUTICAL PRODUCTS (FPPS) NOT REGISTERED IN ICH REGIONS OR RELATED COUNTRIES

BIOEQUIVALENCE PART OF A NEW DOSSIER

Reference of the session
Date
Type of product

Type of dossier EFFICACY
Type of submission NEW
First assessor Name Signature

Second assessor Name Signature

Quality assessor (e.g., when dissolution Name Signature
profiles are submitted for comparison of the
compositions of clinical, stability and
validation batches, or a biowaiver for
additional strengths is requested.)
Reference Number

Date of the submission
Number of binders
SPC, PIL submitted (state location in submission)
SPC, PIL, Package Labelling acceptable Yes: _____ // No: _____
Proprietary Product Name (if relevant) *..

International Non-proprietary Name (INN) *..
of the Active Pharmaceutical Ingredient
(API), strength, pharmaceutical form.

Conclusion of the assessment ACCEPTED (no outstanding issues)
ADDITIONAL DATA REQUESTED
REJECTED
(please delete the wrong entries)

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Name and complete address of the supplier *..
(Applicant of the dossier)
Name and address of the Contract Research *..
Organisation(s) where the clinical studies
proving efficacy and safety of the product
were conducted.
(Add as much rows as necessary)

This product assessment report should be written in clear unambiguous language referring to deficiencies or lack of data submitted, as communication with the manufacturer may result from the assessment.

The report should be completed by at least two evaluators from different countries excluding whenever possible evaluators from the country of manufacturer/applicant responsible for assessing product quality including pharmaceutical and analytical aspects.

The assessment report should be typed with "Bookman Old Style 12" fonts.

The format of tables must not be changed.

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BIOEQUIVALENCE TRIAL INFORMATION

1 SUMMARY

1.1 Summary of bioequivalence studies performed

(Provide a brief description of each comparative bioavailability study included in the submission)

1.2 Tabulation of the composition of the formulation(s) proposed for marketing and those used for bioequivalence studies

(State the location of the master formulae in the quality part of the submission)

(Tabulate the composition of the biobatch using the table below. For solid oral dosage forms the table should contain only the ingredients in tablet core /contents of a capsule. A copy of the table should be filled in for the film coating / hard capsule, if any.

Important: If the formulation proposed for marketing and those used for bioequivalence studies are not identical, copies of this table should be filled in for each formulation with clear identification in which

bioequivalence study the respective formulation was used.)

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Composition of the batches used for bioequivalence studies

Batch number

Batch size (number of unit doses)¹

Comments, if any

Comparison of unit dose compositions and of clinical FPP batches

(duplicate this table for each strength, if compositions are different)

Ingredients (and quality Function Unit Unit Biobatch Biobatch

standard) dose dose (%) (kg) (%)

(mg)

Total

Equivalence of the compositions or justified differences

Maximum intended commercial batch size

¹ Bioequivalence batches should be at least of pilot scale (10% of production scale or 100,000 capsules/tablets whichever is the greater) and manufacturing method should be the same as for production scale.

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2 CLINICAL STUDY REPORT

a) Study number:

b) Study title:

c) Location of study protocol:

d) Start and stop dates for each phase of the clinical study:

e) Dates of product administration:

2.1 ETHICS

a) State the name of review committee, date of approval of protocol and consent form and the location of approval letter in the submission

b) State location of a reference copy of the informed consent form

2.2 INVESTIGATORS AND STUDY ADMINISTRATIVE STRUCTURE

a) Name of principal investigator(s) (State location of c.v. in the submission)

b) Clinical Facility (Name and full mailing address)

c) Clinical Laboratories (Name and full mailing address)

d) Analytical Laboratories (Name and full mailing address)

e) Company performing pharmacokinetic/statistical analysis (Name and full mailing address)

2.3 STUDY OBJECTIVES

Briefly state the study objectives.

2.4 INVESTIGATIONAL PLAN

2.4.1 Overall study design and plan — description

(Describe the type of study design employed in 1-2 sentences)

2.4.2 Selection of study population

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2.4.2.1 Inclusion Criteria

(List the inclusion criteria applied to subjects)

2.4.2.2 Exclusion Criteria

(List the exclusion criteria applied to subjects)

2.4.2.3 Health Verification

(State location of the individual data included in the submission)

a) List criteria used and all tests performed in order to judge health status

b) Indicate when tests were performed

c) Study site normal values

(State location in submission of study site normal values for blood clinical chemistry, haematology, and urinalysis clinical screen)

d) Report any results that were outside of study site normal values

(State location in submission of the summary of anomalous values)

2.4.2.4 Removal of Trial subjects from Trial or Assessment

a) Number of subjects enrolled in the study

(All subjects including alternates, withdrawals, and dropouts)

b) Alternates

(Please note: Generally all subjects enrolled in the study should be included in the data set i.e., alternate

subjects are strongly discouraged. However, in cases where there are alternate subjects, describe the procedure of including/excluding the alternates and whether alternates have been included in the study)

c) Withdrawals/dropouts

(Identify each withdrawal/dropout by subject and provide the reason for withdrawal/dropout and at what point in the study the withdrawal/dropout occurred)

2.4.3 Products Administered

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2.4.3.1 Test Product

a) Batch number, size, date of manufacture and expiry date for the test product

b) Potency (measured content) of test product as a percentage of label claim as per validated assay method

(This information should be cross-referenced to the location of the certificate of analysis in the submission)

2.4.3.2 Comparator (Reference) Product

(Append to this template a copy of product labelling (snap shot of the box, on which the name of the product, name and address of the manufacturer, batch number, and expiry date are clearly visible on the labelling)

a) Name and manufacturer of the comparator product and market where the comparator product was purchased

b) Batch number and expiry date for the comparator product

c) Purchase, shipment, storage of the comparator product

(Indicate from which company/pharmaceutical distributor the comparator product has been obtained.

Clearly indicate in chronological order the steps and dates of shipment/transport from company of purchase to the study site. In addition, the storage conditions should be given. This information should be

cross-referenced to location in submission of documents (e.g. receipts) proving conditions)

d) Potency (measured content) of the comparator product as a percentage of label claim, as measured by the same laboratory and under the same conditions as the test product

(This information should be cross-referenced to the location of the certificate of analysis in the submission)

e) Justification of choice of comparator product

(Provide short summary here and cross-reference to location of comprehensive justification in study protocol)

2.4.4 Selection of doses in the study

a) State dose administered

(Indicate the number of dosage units comprising a single dose, e.g., 400 mg as 1 x 400 mg or 2 x 200 mg tablets)

2.4.5 Selection and Timing of Dose for Each Subject

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a) State volume and type of fluid consumed with dose

b) Interval between doses (i.e., length of washout)

c) Protocol for the administration of food and fluid

d) Restrictions on posture and physical activity during the study

2.4.6 Blinding

2.4.6.1 Identify which of the following were blinded. If any of the groups were not blinded, provide a justification for not doing so

a) study monitors: Yes ☐ / No ☐ If No, justify:

b) subjects: Yes ☐ / No ☐ If No, justify:

c) analysts: Yes ☐ / No ☐ If No, justify:

2.4.6.2 Identify who held the study code and when the code was broken

2.4.7 Drug Concentration Measurements

2.4.7.1 Biological fluid(s) sampled

2.4.7.2 Sampling protocol

a) Number of samples collected per subject

b) Volume of fluid collected per sample

c) Total volume of fluid collected per subject per phase of the study

d) List the study sampling times

e) Identify any deviations from the sampling protocol

(State location of summary in the submission)

(Describe and explain reasons for deviations from sampling protocol. Comment on impact on study.

Indicate whether the deviations were accounted for in the pharmacokinetic analysis)

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2.4.7.3 Sample Handling

a) Describe the method of sample collection

b) Describe sample handling and storage procedures

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3 TRIAL SUBJECTS

3.1 Demographic and other baseline characteristics

a) Identify study population (i.e., normal, healthy adult volunteers or patients)

b) Summary of ethnic origin and gender of subjects

c) Identify subjects noted to have special characteristics and state notable characteristics

(e.g. fast acetylators of debrisoquine)

d) Range and mean age ■ SD of subjects

e) Range and mean height and weight ■ SD of subjects

f) Identify subjects whose ratio is not within 15% of the values given on a standard height/weight table

3.2 Subjects who smoke

a) Number of smokers included in the study

b) Indicate how many cigarettes smoked per day per subject

c) Comment on the impact on study

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4 PROTOCOL DEVIATIONS

4.1 Protocol deviations during the clinical study

(Describe any such deviations and discuss their implications with respect to bioequivalence)

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5 SAFETY EVALUATION

5.1 Identify adverse events observed

(List any adverse events by subject number. State whether a reaction occurred following administration of the test or reference product, identify any causal relationships, and note any treatments required.

State

location of this summary in the submission.)

(Discuss the implications of the observed adverse events with respect to bioequivalence.)

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6 EFFICACY EVALUATION

Efficacy results and tabulations of individual trial subjects data

6.1 Presentation of data

a) State location in submission of tables of mean and individual subject concentrations

b) State location in submission of (mean and individual) linear and semi-logarithmic subject drug concentration vs. time plots

6.2 Pharmacokinetic (PK) parameters

a) State how the pharmacokinetic parameters were calculated/obtained for AUC_{0-t}, AUC_{0-∞}, C_{max}, t_{max}, the elimination rate constant, and t_{1/2} (indicate location of description in protocol)

b) State whether actual sampling time points were used for estimation of the pharmacokinetic parameters

c) Complete the table below

Test Reference

e

Parameter Arithmetic Standard Interindividual Arithmetic Standard Interindividual

c mean deviation coefficient of

n coefficient n variation (%)

of variation

(%)

AUC_{0-t}

(units)

AUC_{0-∞}

(units)

C_{max} (units)

t_{max} (units)

t_{1/2} (units)

d) Ratio of AUC_{0-t} to AUC_{0-∞}

(State mean ratio for both test and reference, state location in submission where individual ratios can

be

found)

6.3 Statistical analysis

(State the method of calculation of the 90% confidence intervals for the ratio of test formulation over

the
reference formulation and indicate how treatment, period, sequence and subjects within sequence were

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included as factors in the ANOVA. Provide the following results from the ANOVA (parametric) on the logarithmically transformed AUC0-t and CMAX and other relevant parameters. State software used for computing ANOVA.)

a) Geometric means, results from ANOVA, Degrees of Freedom (DF) and derived CV (intra-subject)

Parameter Test Reference % Ratio of 90 % DF CV (%)

geometric Confidence

means interval

AUC0-t (units)

AUC0-inf

(units)

Cmax (units)

b) Comparison of the results

(Compare the results, including mean values, inter- and intra-individual variability, of this study with published results (literature, product information of reference product (innovator), WHOPARs), and copies of the references used should be appended to this document)

6.4 Discussion of results

(State location of the discussion of the results in the submission)

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7 ANALYTICAL VALIDATION REPORT

7.1 Analytical technique

7.1.1 Validation protocol

(State the location of the validation protocol)

7.1.2 Identify analyte(s) monitored

7.1.3 Comment on source and validity of reference standard

7.1.4 Identify internal standard

7.1.5 Comment on source and validity of internal standard

7.1.6 Identify method of extraction

7.1.7 Identify analytical technique or method of separation employed

7.1.8 Identify method of detection

7.1.9 Identify anticoagulant used (if applicable)

7.1.10 If based on a published procedure, state reference citation

7.1.11 Identify any deviations from protocol

7.2 Selectivity

(Address the methods to verify selectivity against endogenous/exogenous compounds & results)

7.3 Sensitivity

(Address the methods to verify sensitivity & results)

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7.4 Carry-over

(Summarize the method to verify carry-over & results)

7.5 Standard curves

(State location in submission of tabulated raw data and back calculated data with descriptive statistics)

a) List number and concentration of calibration standards used

b) Describe the regression model used including any weighting

c) List the back-calculated concentrations of the calibration standards of the validation runs (highlight the values outside of the acceptance range, e.g., 15%, except 20% for LLOQ)

7.6 Quality control samples

a) Identify the concentrations of the QC samples and the storage conditions employed prior to their analysis

7.7 Precision and accuracy during validation

a) Summarize inter-day/inter-run accuracy and precision of the calibration standards during assay validation

b) Summarize inter-day/inter-run accuracy and precision of the calibration standards during assay re-validation (If applicable)

c) Summarize inter-day/inter-run and intra-day/intra-run accuracy and precision of the QC samples during assay validation

d) Summarize inter-day/inter-run and intra-day/intra-run accuracy and precision of the QC samples during assay re-validation (If applicable)

7.8 Dilution integrity

(Summarize the method to verify dilution integrity & results)

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7.9 Matrix effect (in case of MS detection)

(Summarize methods to verify the matrix effect & results)

7.10 Stability

(For each section provide the location of the raw data, a description of the methodology employed and a summary of the data.)

a) Summarize data on long-term storage stability

b) Summarize data on freeze-thaw stability

c) Summarize data on bench top stability

d) Summarize data on auto-sampler storage stability

(e) Summarize data from any other stability studies conducted

(e.g. long-term stock solution and working solution stability, short-term stock solution and working solution stability, dry-extract stability, wet-extract stability, stability in blood before sample processing)

7.11 Re-injection reproducibility

(Summarize the method to verify re-injection reproducibility & results)

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8 BIOANALYTICAL STUDY REPORT

(State the location of the bioanalytical report for the analysis of the study subject samples)

8.1 Analytical technique

(Confirm whether the method is the same as the validated method and whether the same equipment was employed. Identify any differences between the validated method described above in Section 7 and the method employed for subject sample analyses)

8.1.1 Analytical protocol

(State the location of the analytical protocol)

8.1.2 Identify any deviations from protocol

8.1.3 Dates of subject sample analysis

8.1.4 Longest period of subject sample storage

(Identify the time elapsed between the first day of sample collection and the last day of subject sample analysis)

8.1.5 State whether all samples for a given subject were analysed together in a single analysis run

8.2 Standard curves

(State location in submission of tabulated raw data and back calculated data with descriptive statistics)

a) List number and concentration of calibration standards used

b) State number of curves run during the study (valid and failed runs, including reasons of failure).

c) Summarize descriptive data including slope, intercept, correlation coefficients

d) List the back-calculated concentrations of the calibration standards of the study runs (highlight the values outside of the acceptance range, e.g., 15%, except 20% for LLOQ)

8.3 Quality control samples

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a) Identify the concentrations of the QC samples, their date of preparation and the storage conditions employed prior to their analysis

b) State the number of QC samples in each analytical run per concentration

c) List the back-calculated concentrations of the QC samples of the study runs (highlight the values outside of the acceptance range, e.g., 15%)

d) Discuss whether the concentrations of the QC sample concentrations are similar to the concentrations observed in the study samples

e) State the percentage of QC samples per run with respect to the total number samples assayed in each run

8.4 Precision and accuracy

a) Summarize inter-day precision of back-calculated standards and inter-day and intra-day precision and accuracy of QC samples analysed during subject sample analysis

8.5 Repeat analysis (re-analysis, re-injection and re-integration)

a) List re-analysed samples by sample identification and include the following information for each re-analysis: initial value; reason for re-analysis; re-analysed value(s); accepted value; and reason for acceptance

b) Report the number of re-analysis as a percentage of the total number samples assayed

c) List re-injected samples by sample identification and include the following information for each re-injection: initial value; reason for re-injection; re-injected value; accepted value; and reason for acceptance

d) Report the number of re-injections as a percentage of the total number samples assayed

e) List re-integrated chromatograms by sample identification and include the following information for each re-integration: initial value; reason for re-integration; re-integrated value(s); accepted value; and reason for acceptance

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f) Report the number of re-integrated chromatograms as a percentage of the total number of samples assayed

8.6 Incurred sample reanalysis

(State location in the submission and summarize the results of incurred sample reanalysis, including the number of subject samples included in ISR and the total number of samples analysed in the study)

8.7 Chromatograms

(State the location in the submission where the sample chromatograms can be found. The chromatograms should be obtained from a minimum of two analytical batches and include at least 20% of the subjects, up to a maximum of five. A complete set includes standards, QC samples, pre-dose and post-dose subject samples for both phases. Each chromatogram should be clearly labelled with respect to the following: date of analysis; subject ID number; study period; sampling time; analyte; standard or QC, with concentration; analyte and internal standard peaks; peak heights and/or areas)

9 QUALITY ASSURANCE

9.1 Internal quality assurance methods

(State locations in the submission where internal quality assurance methods and results are described for each of study sites (see 3.2 b-d.)

9.2 Monitoring, auditing, inspections

(Provide a list of all monitoring and auditing reports of the study, and of recent inspections of study sites by regulatory agencies. State locations in the submission of the respective reports for each study site (see 3.2 b-d.)

REFERENCES

1. WHO guideline on bioequivalence studies.
2. India draft guideline for bioavailability and bioequivalence studies.
3. EMA guideline on the investigation of bioequivalence.
4. JP NIHS guideline for bioequivalence studies for different strengths of oral solid dosage forms.

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PART IV:

GUIDELINES ON FORMAT AND CONTENT OF SUMMARY OF PRODUCT CHARACTERISTICS FOR PHARMACEUTICAL PRODUCTS

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A PRINCIPLES OF PRESENTING INFORMATION

I. The SmPC should be worded in clear and concise language.

II. Each section of the SmPC should first deal with those issues that apply to the core population for whom the medicine is indicated followed (when necessary) by specific information for any relevant special population (e.g. children or elderly).

III. Consistent medical terminology from the Medical Dictionary for Regulatory Activities (MedDRA) should be used throughout the SmPC.

IV. The SmPC provides information on a particular medicinal product; therefore, it should not include reference to other medicinal products (e.g. through statement such as "Like other medicines of the same class ...") except when it is a class warning recommended by a competent authority.

B SMPC FORMAT AND CONTENT

The SmPC will be structured and populated as outlined in 1-10 below.

1. NAME OF THE MEDICINAL PRODUCT

Both the strength and the pharmaceutical form should follow the proprietary name. However, when otherwise referring to the medicinal product throughout the SmPC text, the strength and the pharmaceutical form do not have to be mentioned in the name. The International Non-proprietary Name (INN) or the usual common name of the active substance should be used when referring to properties of the active substance(s) rather than those of the product. The use of pronouns (e.g. "it") is encouraged whenever possible.

Strength

The strength should be the relevant quantity for identification and use of the product and should be consistent with the quantity stated in the quantitative composition and in the posology. Different strengths of the same medicinal product should be stated in the same way, e.g. 250 mg, 500 mg, 750mg. The use of decimal points should be avoided where these can be easily removed (e.g. 250 microgram, not 0.25 mg). However, where a range of medicinal products of the same pharmaceutical form includes strengths of more than one unit (e.g. 250 microgram, 1 mg and 6 mg), it may be more appropriate in certain cases to state the strengths in the same unit for the purpose of comparability (e.g. 0.25 mg, 1 mg and 6 mg). For safety reasons, micrograms and millions (e.g. for units) should always be spelled out in full rather than be abbreviated.

The pharmaceutical form of a medicinal product should be described by a standard term (see Appendix 1). No reference should be made to the route of administration or container unless these elements are part of the standard term or where there is a particular safety reason for their inclusion or where there are identical products, which may be distinguished only by reference to the route of administration or to the container.

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

Full details of the qualitative and quantitative composition in terms of the active substance(s) and excipients, knowledge of which are essential for proper administration of the medicinal product, should be provided in section 2 of the SmPC and as appropriate in section 4.3 or 4.4. Excipients which are required to be declared on the labelling (see Guidelines on Format and Content of Labels for Medicinal Products) should be stated here under a separate subheading qualitatively, and, quantitatively. The following standard statement should be included at the end of the section, i.e. 'for full list of excipients, see section 6.1'.

If a diluent is part of the medicinal product, information should be included in the relevant sections (usually sections 3, 6.1, 6.5 and 6.6).

Qualitative declaration

The active substance should be declared by its recommended INN accompanied by its salt or hydrate form if applicable. References to the pharmacopoeial quality should not be included.

Quantitative declaration

The quantity of the active substance should be expressed per dosage unit (for metered dose inhalation products, per delivered dose and/or per metered dose), per unit volume, or per unit of weight and should be related to the declaration of strength in section 1.

Quantity should be expressed in internationally recognised standard term which could be complemented with another term if more meaningful to healthcare professionals.

Salts and hydrates

Where the active substance is present in the form of a salt or hydrate, the quantitative composition should be expressed in terms of the mass (or biological activity in International (or other) units where appropriate) of the active moiety (base, acid or anhydrous material), e.g. '60 mg toremifene (as citrate)' or toremifene citrate equivalent to 60 mg toremifene'.

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Where a salt is formed in situ during the preparation of the finished product (i.e. formed during the mixture of a solvent and powder), the quantity of the active moiety should be stated, with a reference to the in situ formation of the salt.

In the case of established active substances in medicinal products where the strength has traditionally been expressed in the form of a salt or hydrate, the quantitative composition may be declared in terms of the salt or hydrate, e.g. '60 mg diltiazem hydrochloride'. This may also apply when the salt is formed in situ.

Esters and pro-drugs

If the active substance is an ester or pro-drug, the quantitative composition should be stated in terms of the quantity of the ester or pro-drug. When the active moiety is an active substance of an already approved medicinal product, the quantitative composition should also be stated in terms of the quantity of this active moiety (e.g. 75 mg of fosphenytoin is equivalent to 50 mg of phenytoin).

Oral powders for solution or suspension

The quantity of active substance should be stated per unit dose if the product is a single-dose preparation or otherwise per unit dose volume after reconstitution; a reference to the molar concentration may also be appropriate in some cases.

Parenterals excluding powders for reconstitution

For single-dose parenterals, where the total contents of the container are given in a single dose ('total use'), the quantity of active substance(s) should be stated per presentation (e.g. 20 mg etc.) not including any overages or overfill. The quantity per ml and the total labelled volume should also be given.

For single-dose parenterals, where the amount to be given is calculated on the basis of the patient's weight or body surface or other variable ('partial use'), the quantity of active substance(s) should be stated per ml. The quantity per total labelled volume should also be given. Overages or overfills should not be included.

For multi-dose and large volume parenterals, the quantity of active substance(s) should be stated per ml, per 100 ml, per 1000 ml, etc. as appropriate, except for multidose vaccines containing 'n' doses of the same dose. In this case, the strength should be expressed per dose volume. Overages or overfills should not be included.

Where appropriate, e.g. for X-ray contrast media, and parenterals containing inorganic salts, the quantity of active substance(s) should also be indicated in millimoles. For X-ray contrast

media with iodine-containing active substances, the quantity of iodine per ml should be stated in addition to the quantity of the active substance.

Powders for reconstitution prior to parenteral administration

When the product is a powder to be reconstituted prior to administration, the total quantity of active substance in the container should be stated not including overages or overfills, as well as the quantity per ml when reconstituted, unless there are several means of reconstituting, or different quantities used, which result in different final concentrations.

Concentrates

The quantity should be stated as the content per ml in the concentrate and as the total content of the active substance. The content per ml when diluted as recommended should also be included unless the concentrate is to be diluted to within a range of different final concentrations.

Transdermal patches

The following quantitative details should be given: the content of active substance(s) per patch, the mean dose delivered per unit time, and the area of the releasing surface, e.g. 'Each patch contains 750 micrograms of estradiol in a patch size of 10 cm², releasing a nominal 25 micrograms of estradiol per 24 hours'.

Multidose solid or semi-solid products

Quantity of active substance should be stated, where possible, per unit dose, otherwise per gram, per 100 g or percentage, as appropriate.

Biological medicinal products

Expression of strength

The quantity of biological medicinal products should be expressed in terms of mass units, units of biological activity, or International Units as appropriate for the particular product.

The biological origin of the active substance

The origin of the active substance should be defined briefly. Thus, the nature of any cellular system(s) used for production and, if relevant, the use of recombinant DNA technology should be specified. The entry should take the form: "produced in XXX cells <by recombinant DNA technology>". The following are examples of the application of this principle:

"produced in human diploid (MRC-5) cells",

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"produced in Escherichia coli cells by recombinant DNA technology",

"produced in chick-embryo cells",

"produced from the plasma of human donors",
"produced from human urine",

"produced from <animal> blood",

"produced from porcine pancreatic tissue",

"produced from porcine intestinal mucosa".

Special provisions for normal immunoglobulins

In the case of normal immunoglobulins, the IgG subclass distribution should be stated in terms of percent of total IgG present. The upper limit of the IgA content should follow.

Special provisions for vaccines

In the case of vaccines, the content of active substance per dose unit (e.g. per 0.5 ml) should be stated.

Adjuvants, if present, should be stated qualitatively and quantitatively.

Residues that are of special relevance (e.g. ovalbumin in egg derived vaccines) should be specified.

Additional specific guidance is available in CHMP guidelines on biotechnological medicinal products, e.g. the CHMP Guideline on the Pharmaceutical Aspects of the Product Information for Human Vaccines.

Herbal medicinal products

The quantitative declaration should be in accordance with the existing quality guidelines on herbal medicinal products.

3. PHARMACEUTICAL FORM

The pharmaceutical form should be described by a full standard term using the singular form. The term used in this section should be the same as the term used in section 1. A visual description of the appearance of the product (colour, markings, etc.) should be given, in a separate paragraph to the standard term, including information on the actual size of a solid oral formulation, e.g. In case of tablets designed with a score line, information should be given on whether or not reproducible dividing of the tablets has been shown. e.g. 'the score line is only to facilitate breaking for ease of swallowing and not to divide into equal doses', 'the tablet can be 220

divided into equal halves'. The visual description should be as presented in the finished product specifications.

Information on pH and osmolality should be provided, as appropriate. In case of products to be reconstituted before use, the appearance before reconstitution should be stated in this section. Appearance of the product after reconstitution should be stated in sections 4.2 and 6.6.

4. CLINICAL PARTICULARS

4.1 Therapeutic indications

The indication(s) should be stated clearly and concisely and should define the target disease or condition distinguishing between treatment (symptomatic, curative or modifying the evolution or progression of the disease), prevention (primary or secondary) and diagnostic indication. When appropriate it should define the target population especially when restrictions to the patient populations apply.

Study endpoints should not normally be included. The objective of a prevention indication may be mentioned in general terms only. This should also be done for the target population.

Where results from subsequent studies provide further definition or information on an authorized indication, such information, provided it does not itself constitute a new indication, may be considered for inclusion in section 5.1.

Mandatory conditions of product usage not covered more appropriately in other parts of the SmPC may also be included when relevant, e.g. concomitant dietary measures, lifestyle changes, or other therapy.

It should be stated in which age groups the product is indicated, specifying the age limits, e.g. 'X is indicated in <adults><neonates><infants><children><adolescents><aged x to y>years, months>'.>

If the product's indication depends on a particular genotype or the expression of a gene or a particular phenotype, this should be stated in the indication.

4.2 Posology and method of administration

In case of restricted medical prescription, this section should be started by specifying the conditions.

In case of specific safety need, any recommended restriction to a particular setting should also be stated (e.g. "restricted to hospital use only" or "appropriate resuscitation equipment should be available").

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The dosage should be clearly specified for each method/route of administration and for each indication, as appropriate.

Where appropriate, a reference to official recommendations should be made (e.g. for primary vaccination and antibiotics as well as for booster dose). Dose recommendations (e.g. mg, mg/kg, mg/m²) should be specified per dose interval for each category where appropriate (specify age/weight/body surface area of subsets of the population as appropriate). Frequency of dosing should be expressed using time units (e.g. once or twice daily or every 6 hour) and, to avoid confusion, abbreviations e.g. OD or BID should not be used.

Where appropriate, the following points should be addressed:

- the maximum recommended single, daily and/or total dose,
- the need for dose titration,
- the normal duration of use and any restrictions on duration and, if relevant, the need for tapering off, or advice on discontinuation,
- advice on action to be taken if one or more dose(s) is (are) missed, or e.g. in case of vomiting (the advice should be as specific as possible, taking into consideration the recommended frequency of dosing and relevant pharmacokinetic data)
- advice on preventive measures to avoid certain adverse drug reactions (e.g. administration of antiemetic's) with cross-reference to section 4.4,
- the intake of the product in relation to drink and food intake, together with a cross-reference to section 4.5 in case of specific interaction e.g. with alcohol, grapefruit or milk,
- advice regarding repeat use, with any information on intervals to be observed between courses of treatment, as appropriate,
- Interactions requiring specific dose adjustments with cross-reference to other appropriate sections of the SmPC (e.g. 4.4, 4.5, 4.8, 5.1, 5.2), and
- it may also be relevant to recommend not to prematurely discontinue a treatment in case of specific non-serious adverse reaction(s) that are frequent but transient or manageable with dose titration.

Where relevant to the particular product, the following should appear 'The potency of this medicinal product is expressed in <proprietary name> units. These units are not

222 interchangeable with the units used to express the potency of other <active substance name> preparations'.

Special populations

Dosage adjustments or other posology related information in specific patient groups should be stated where necessary, in well-defined sub-sections ordered by importance, e.g. regarding:

- elderly population; it should be made clear whether or not any dosage adjustment is necessary in any subsets of the elderly population, with cross-reference to other sections providing information in elderly, e.g. 4.4, 4.5, 4.8 or 5.2.
- renal impairment; the dose recommendation should relate as precisely as possible to the cut-off values for biochemical markers of renal impairment in clinical studies and to the results of these studies;
- hepatic impairment, specified according to the patients included in studies, for instance 'alcohol-related cirrhosis' and the definitions used in the studies, for instance Child-Pugh score/grade of the patients;
- patients with a particular genotype; with cross-reference to other relevant sections for further detail as appropriate;
- other relevant special population (e.g. patients with other concomitant disease or overweight patients).

Advice relevant for dosage adjustment e.g. from monitoring of clinical symptoms and signs, and/or laboratory investigations, including blood concentrations of the medicinal product should be mentioned when appropriate with cross-reference to other sections where appropriate.

Paediatric population

The specific sub-section 'paediatric population' should always be included and the information given should cover all subsets of the paediatric population, using a combination of the possible situations presented below as appropriate.

If the product is indicated in the paediatric population, posology recommendations should be given for each of the relevant subsets. The age limits should reflect the benefit-risk assessment of the available documentation for each subset.

If the posology is the same in adults and children, then a statement to this effect is sufficient; the posology does not need to be repeated.

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Dose recommendations (e.g. mg, mg/kg, mg/m²) should be specified per dose interval for the paediatric subsets where the product is indicated. Different subsets may require different dosing information. If necessary, recommendations in preterm and new borns should be presented taking into account the more appropriate age e.g. gestational age or the post-menstrual age.

Depending on the subset, the clinical data and available formulations, the dose will be expressed according to weight or body surface area, e.g. "children aged 2-4 years, 1 mg/kg bodyweight twice a day".

When appropriate, information on timing of intake of the product should consider children's daily life, e.g. school or sleep.

Where a product is indicated in children and no adequate paediatric formulation can be developed, detailed instructions on how to obtain an extemporaneous preparation shall be included in section 6.6 with a cross-reference in section 4.2.

Doses and method of administration in the various subsets may be presented in a tabulated format.

If there is no indication for the product in some or all subsets of the paediatric population, no posology recommendation can be made, but available information should be summarized using the following standard statements (one or combination of several as appropriate):

- The <safety><and><efficacy> of X in children aged x to y<months, years><or any other relevant subsets e.g. weight, pubertal age, gender><has><have> not <yet> been established.

One of the following statements should be added:

– <No data are available>.

or

– <Currently available data are described in section <4.8><5.1><5.2>but no recommendation on a posology can be made >

- X should not be used in children aged x to y<years, months><or any other relevant subsets e.g. weight, pubertal age, gender> because of <safety><efficacy> concern(s) <concern(s) to be stated with cross-reference to sections detailing data (e.g. 4.8 or 5.1) >.

- There is no relevant use of X in <the paediatric population><in children aged x to y><years, months><or any other relevant subsets e.g. weight, pubertal age, gender> in the indication(s) <specify indication(s)>.

- X is contraindicated in children aged x to y<years, months><or any other relevant subsets e.g. weight, pubertal age, gender><in the indication ...> (cross-reference to section 4.3).

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If there are more appropriate strength(s) and/or pharmaceutical form(s) for administration in some or all subsets of the paediatric population (e.g. oral solution for infants), these can be mentioned in section 4.2 of the SmPC of the less appropriate one(s).

E.g.: Other pharmaceutical forms/strengths may be more appropriate for administration to this population.

Method of administration

Any special precautions related to the manipulation or administration of the product (e.g. cytotoxic products) by healthcare professionals (including pregnant healthcare professionals), the patient or carers should be mentioned here under a specific sub-heading (<Precaution to be taken before manipulating or administering the product>), with a cross-reference to section 6.6 (or 12).

The route of administration and concise relevant instruction for correct administration and use should be given here. Information on instructions for preparation or reconstitution should be placed in section 6.6 'Special precautions for disposal of a used medicinal product and other handling of the product' (or in section 12 if appropriate) and cross-referenced here.

When supportive data are available, information on alternative method(s) to facilitate administration or acceptability should be given as explicitly as possible (e.g. possibility of crushing tablet, cutting tablet or transdermal patch, pulverizing tablet, opening capsules, mixing with food, dissolution in drinks – specifying if a proportion of the dose can be given) particularly for administration via feeding tubes.

Any specific recommendation for use related to the pharmaceutical form should be explained, e.g.:

- "the coated tablet should not be chewed because of <bad taste>,"
- "the enteric-coated tablet should not be crushed because coating prevents <pH sensitive degradation><irritant effects> on the gut",
- "the coated tablet should not be broken because the coating is intended to ensure a prolonged release (see 5.2)".

For parenteral formulations, information on the rate or speed of injection or infusion should be provided.

For parenteral formulations - in children, especially newborns in whom quite often fluids have to be restricted - it would be useful to have information on maximal concentration that can be safely administered (e.g. "no more than X mg of Y/ml of solution").

4.3 Contraindications

Situations where the medicinal product must not be given for safety reasons, i.e. contraindications, are the subject of this section. Such circumstances could include a particular

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clinical diagnosis, concomitant diseases, demographic factors (e.g. gender, age) or predispositions (e.g. metabolic or immunological factors, a particular genotype and prior adverse reactions to the medicine or class of medicines). The situations should be unambiguously, comprehensively and clearly outlined.

Other medicines or classes of medicine, which must not be used concomitantly or consecutively should be stated, based on either data or strong theoretical reasons. If applicable a cross-reference to section 4.5 should be made.

In general, patient populations not studied in the clinical trial programme should be mentioned in section 4.4 and not in this section unless a safety issue can be predicted (e.g. use of renally eliminated substances with narrow therapeutic margin in renal failure patients). If, however, patients have been excluded from studies due to a contraindication on grounds of safety, they should be mentioned in this section. If applicable a cross-reference to section 4.4 should be made.

Only if pregnancy or breastfeeding is contraindicated, should it be mentioned here. In section 4.6, a cross-reference should be made and further background information provided. Hypersensitivity to the active substance or to any of the excipients or residues from the manufacturing process should be included, as well as any contraindication arising from the presence of certain excipients.

For herbal medicinal products, hypersensitivity extended to other plants of the same family or to other parts of the same plant should be labelled as a contraindication, where applicable.

Lack of data alone should not lead to a contraindication. Where for safety reasons, the product should be contraindicated in a specific population, e.g. paediatric or a subset of the paediatric population, it should appear in this section with a cross-reference to the section giving detailed information on the safety issue. A contraindication in the paediatric population should be listed without a sub-heading.

4.4 Special warnings and precautions for use

The order of warnings and precautions should in principle be determined by the importance of the safety information provided.

The exact content of this section will be different for each product and the therapeutic conditions it is intended to treat. It is however suggested that the following items should be included where relevant to the specific product.

Information on a specific risk should be given in section 4.4 only when the risk leads to a precaution for use or when healthcare professionals have to be warned of this risk. Patient groups in which use of the medicinal product is contraindicated should be mentioned in section 4.3 only and not to be repeated here.

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The following should be described:

- The conditions, in which the use of the medicinal product could be acceptable, provided

that special conditions for use are fulfilled. In particular, specific risk minimization measures requested as part of a Risk Management Plan to ensure safe and effective use should be described in this section. (For example; "Liver function should be monitored before initiation of treatment and monthly thereafter", "Patients should be advised to immediately report any symptoms of depression and/or suicidal ideation", "Women of childbearing potential should use contraception", ...)

- Special patient groups that are at increased risk or are the only groups at risk of experiencing product or product class-related adverse reactions (usually serious or common), e.g. elderly, children, patients with renal or hepatic impairment (including the degree of impairment, e.g. mild, moderate or severe), patients having an anaesthetic or patients with cardiac failure. Cross-reference to section 4.8 on the differential effects in terms of frequency and severity of the specified adverse reaction should be provided.
- Serious adverse reactions to which healthcare professionals need to be alerted, the situations in which these may occur and the action that may be required, e.g. emergency resuscitation.
- If there are particular risks associated with starting the medicinal product (e.g. first dose effects) or stopping it (e.g. rebound, withdrawal effects), these should be mentioned in this section, together with the action required for prevention.
- Any measures which can be taken to identify patients at risk and prevent the occurrence, or detect early the onset or worsening of noxious conditions. If there is a need for awareness of symptoms or signs representing early warning of a serious adverse reaction, a statement should be included.
- Any need for specific clinical or laboratory monitoring should be stated. Recommendation for monitoring should address why, when and how the monitoring should be conducted in clinical practice. If dose reduction or other posology is recommended in such circumstances or conditions, this should be included in section 4.2 and cross-referenced here.
- Any warnings necessary for excipients or residues from the manufacturing process.
- For herbal preparations containing alcohol, information about the ethanol content in the medicinal product should be included in accordance with the Guideline on excipients in the label and package leaflet of medicinal products for human use.
- Any warnings necessary with respect to transmissible agents.
- Subjects or patients with a specific genotype or phenotype might either not respond to the treatment or be at risk of a pronounced pharmacodynamics effect or adverse reaction. These may arise because of non-functioning enzyme alleles, alternative metabolic pathways (governed by specific alleles), or transporter deficiencies. Such situations should be clearly described if known.
- Any particular risk associated with an incorrect route of administration (e.g. necrosis risk with extravasation of intravenous formulation, or neurological consequences of intravenous use instead of intramuscular use), should be presented, with advice on management if possible.

In exceptional cases, especially important safety information may be included in bold type within a box.

Any adverse reactions described in this section or known to result from conditions mentioned here should also be included in section 4.8.

Specific interference with laboratory tests should be mentioned when appropriate, e.g. Coombs test and Beta-lactams. They should be clearly identified with a subheading, e.g. "Interference with serological testing".

In general, descriptions of warnings and precautions regarding pregnancy and breast-feeding, ability to drive and use machines, and other aspects of interactions should be dealt with in sections 4.6, 4.7 and 4.5, respectively. However in specific cases of major clinical importance it might be more appropriate to describe specific precautionary measures in this section, e.g. contraception measures, or when concomitant use of another medicine is not recommended, and with cross reference to section 4.5, 4.6, or 4.7.

Paediatric population

When the product is indicated in one or more subsets of the paediatric population and there are warnings and precautions for use that are specific to the paediatric population or any subset of the paediatric population, they should be identified under this subheading. Any necessary warning or precaution in relation to long-term safety (e.g. on growth, neuro-behavioural development or sexual maturation) or specific monitoring (e.g. growth) in the paediatric population should be described.

When long-term safety data are necessary but not yet available, it should be stated in this section. Warnings should be included in case of possible significant or long-lasting impact on children's daily activities, such as learning ability or physical activities, or in case of impact on appetite or sleep pattern.

If measures are requested that are specific to the paediatric population for which the product is indicated (e.g. as part of a Risk Management Plan), these measures should be described in this section.

4.5 Interaction with other medicinal products and other forms of interaction

This section should provide information on the potential for clinically relevant interactions based on the pharmacodynamics properties and in vivo pharmacokinetic studies of the medicinal product, with a particular emphasis on the interactions, which result in a recommendation regarding the use of this medicinal product. This includes in vivo interaction

results, which are important for extrapolating an effect on a marker ('probe') substance to other medicinal products having the same pharmacokinetic property as the marker.

Interactions affecting the use of this medicinal product should be given first, followed by those interactions resulting in clinically relevant changes on the use of others. Interactions referred to in other sections of the SmPC should be described here and cross-referenced from other sections.

The order of presentation should be contraindicated combinations, those where concomitant use is not recommended, followed by others.

The following information should be given for each clinically relevant interaction:

a) Recommendations: these might be

- Contraindications of concomitant use (cross-refer to section 4.3),
- Concomitant use not recommended (cross-refer to section 4.4), and
- Precautions including dose adjustment (cross-refer to sections 4.2 or 4.4, as appropriate), mentioning specific situations where these may be required.

b) Any clinical manifestations and effects on plasma levels and AUC of parent compounds or active metabolites and/or on laboratory parameters.

c) Mechanism, if known. For example, interaction due to inhibition or induction of cytochrome P450 should be presented as such in this section, with a cross-reference to 5.2 where in vitro results on inhibition or induction potential should be summarized.

Interactions not studied in vivo but predicted from in vitro studies or deducible from other situations or studies should be described if they result in a change in the use of the medicinal product, cross-referring to sections 4.2 or 4.4.

This section should mention the duration of interaction when a medicinal product with clinically important interaction (e.g., enzyme inhibitor or inducer) is discontinued. Adjustment of dosing may be required as a result. The implication for the need for a washout period when using medicines consecutively should also be mentioned.

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Information on other relevant interactions such as with herbal medicinal products, food, alcohol, smoking, or pharmacologically active substances not used for medical purpose, should also be given. With regard to pharmacodynamics effects where there is a possibility of a clinically relevant potentiation or a harmful additive effect, this should be stated.

In vivo results demonstrating an absence of interaction should only be mentioned here if this is of major importance to the prescriber (e.g. in therapeutic area where potentially problematic interactions have been identified such as with anti-retroviral medicines).

If no interaction studies have been performed, this should be clearly stated.

Additional information on special populations

If there are patient groups in which the impact of an interaction is more severe, or the magnitude of an interaction is expected to be larger e.g., patients with decreased renal function (in case the parallel pathway is renal excretion), paediatric patients, elderly e.t.c, this information should be given here.

If interactions with other medicinal products depend on polymorphisms of metabolizing enzymes or certain genotypes, this should be stated.

Paediatric population

Information specific to a subset of the paediatric population should be given here if there is an indication for the particular age group.

The resulting exposure and clinical consequences of a pharmacokinetic interaction can differ between adults and children, or between older and younger children. Therefore;

■ any identified treatment recommendations should be given in relation to concomitant use in the paediatric subset(s) (e.g. dose adjustment, extra-monitoring of clinical effect marker/adverse reactions, therapeutic drug monitoring),

■ If the interaction studies have been performed in adults, the statement 'Interaction studies have only been performed in adults' should be included.

■ If the extent of an interaction is known to be similar in a paediatric age group to that in adults, this should be stated.

■ If this is not known, this should also be stated.

The same applies to pharmacodynamics drug interactions.

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In cases of food interaction leading to a recommendation on co-administration with a meal or specific food, it should be specified whether this is relevant for paediatric use (especially newborns and infants) whose diet is different (100 % milk in newborns).

Overall, section 4.5 should be presented in the simplest possible way to highlight the interactions resulting in a practical recommendation regarding the use of the medicinal product. Presentation in a tabulated format may help where interactions are numerous and various, such as with anti-viral products.

4.6 Fertility, pregnancy and lactation

General principles

Efforts should be made by the Marketing Authorization Applicant or Holder to provide the reasons for the recommendations for use in pregnant or lactating women and in women of childbearing potential.

This information is important for the healthcare professionals informing the patient.

In the overall assessment, all available knowledge should be taken into account, including clinical studies and post-marketing surveillance, pharmacological activity, results from non-clinical studies, and knowledge about compounds within the same class.

Efforts should be made to update the recommendations for use during pregnancy and lactation on the basis of increasing human experience in exposed pregnancies, which eventually supersede the animal data.

In case of contraindication, this should be included in section 4.3.

The following should be mentioned:

Women of childbearing potential / Contraception in males and females

Recommendations on the use of the medicinal product in women of childbearing potential should be given when appropriate including the need for pregnancy test or contraceptive measures. Where an effective contraception is required for patients or partners of patients during treatment or for a defined period before starting or after ending treatment, the rationale should be included in this section. If contraceptive measures are recommended, there should also be a cross-reference to section 4.5 (and possibly 4.4) in case of interaction with oral contraceptives.

Pregnancy

In general, clinical and non-clinical data should be followed by recommendations.

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With respect to non-clinical data,

■ only conclusions of the reproductive toxicity studies should be included in this section. Further details should be provided in section 5.3.

With respect to clinical data,

■ the section should include comprehensive information on relevant adverse events reported in the embryo, the foetus, neonates and pregnant women, when appropriate. The frequency of such events (for example the frequency of birth defects) should be specified when available.

■ the section should specify the extent of the human experience if no adverse events have been reported in pregnancy.

With respect to the recommendations:

■ Recommendations on the use of the medicinal product during the different periods of gestation, including the reason(s) for these recommendations, should be given.

■ Recommendations for the management of exposure during pregnancy when appropriate (including relevant specific monitoring such as foetal ultrasound, specific biological or clinical surveillance of the foetus or the neonate) should be given.

Cross-references can be included in sections 4.3, 4.4 and 4.8, as appropriate.

Breastfeeding

If available, clinical data should be mentioned (exposed breastfed infants) as the conclusions of kinetic studies (plasma concentrations in breastfed infants, transfer of the active substance and/or its metabolite(s) into human milk...). Information on adverse reactions in nursing neonates should be included if available.

Conclusions from non-clinical studies on the transfer of the active substance and/or its metabolite(s) into milk should be given only if no human data are available.

Recommendations should be given to stop or continue breastfeeding and/or to stop or continue the treatment in cases where treatment or breastfeeding discontinuation is recommended, and the reason should be provided.

Fertility

The main information on the possible effects of the medicinal product on male and female fertility should be included in section 4.6.

This section should include:

a) Clinical data if available.

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b) Relevant conclusions from nonclinical toxicity studies, if available. Further details should be included in section 5.3.

c) Recommendations for the use of the medicinal product when pregnancy is planned but fertility might be affected by treatment.

Cross-references could be included in section 4.3, if appropriate.

If there are no fertility data at all, then this should be clearly stated.

4.7 Effects on ability to drive and use machines

On the basis of the pharmacodynamics and pharmacokinetics profile, reported adverse reactions and/or specific studies in a relevant target population addressing the performance related to driving and road safety or using machines, specify whether the medicinal product has a) no or negligible influence b) minor influence, c) moderate influence or d) major influence on these abilities. Other important factors that affect the ability to drive and use machines should be considered if known, e.g. duration of the impairing effect and the development of tolerance or adverse reactions with continued use.

For situations c and d, special warnings/precautions for use should be mentioned here (and also in section 4.4 for situation d).

4.8 Undesirable effects

This section should include all adverse reactions from clinical trials, post-authorization safety studies and spontaneous reporting for which, after thorough assessment, a causal relationship between the medicinal product and the adverse event is at least a reasonable possibility, based for example, on their comparative incidence in clinical trials, or on findings from epidemiological studies and/or on an evaluation of causality from individual case reports. Adverse events, without at least a suspected causal relationship, should not be listed in the SmPC.

The content of this section should be justified in the Clinical Overview of the marketing authorization application based upon a best-evidence assessment of all observed adverse events and all facts relevant to the assessment of causality, severity and frequency. This section should be regularly reviewed and, if necessary, updated with the aim to ensure appropriate information to health care professionals on the safety profile of the product.

It is important that the whole section is worded in concise and specific language and does not include information such as claims regarding the absence of specific adverse reactions, comparative frequency statements other than as described below, or statements of general good

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tolerability such as “well tolerated”, “adverse reactions are normally rare”, etc. Statements on lack of proof of causal association should not be included.

In order to provide clear and readily accessible information, section 4.8 should be structured according to the following recommendations:

- a. Summary of the safety profile
- b. Tabulated summary of adverse reactions
- c. Description of selected adverse reactions
- d. <Paediatric population>
- e. <Other special population(s)>

a. Summary of the safety profile

The summary of the safety profile should provide information about the most serious and/or most frequently occurring adverse reactions.

If known, it may be helpful to indicate the timing when adverse reactions occur. For example, in order to prevent early discontinuation of a treatment, it may be important to inform about non-serious adverse reactions that are frequent in the beginning of the treatment but may disappear with its continuation. Another example would be to inform about adverse reaction associated with long-term use. Frequencies of cited adverse reactions should be stated as accurately as possible. This summary of the safety profile should be consistent with the important identified risks mentioned in the Safety Specification of the Risk Management Plan. The information should be consistent with the Table of Adverse Reactions (see section b). Cross-reference should be made to section 4.4 if relevant risk minimization measures have been proposed in that section.

An example of an acceptable statement is given below:

‘At the beginning of the treatment, epigastric pain, nausea, diarrhoea, headache or vertigo may occur; these reactions usually disappear within a few days even if treatment is continued. The most commonly reported adverse reactions during treatment are dizziness and headache, both occurring in approximately 6% of patients. Serious acute liver injury and agranulocytosis may occur rarely (less than 1 case per 1,000 patients)’

b. Tabulated list of adverse reactions

A single table (or structured listing) should list all adverse reactions with their respective frequency category. In some cases for common or very common reactions, and when it is necessary for the clarity of the information, frequency figures may be presented in the table.

Separate tables are acceptable in exceptional cases where the adverse reaction profiles markedly differ depending on the use of the product. For example, it might be the case for a product used for different indications (e.g. an oncology and a non-oncology indication) or at different posologies.

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The table should be introduced with a short paragraph stating the source of the safety database (e.g. from clinical trials, post-authorization safety studies or spontaneous reporting).

The table should be presented according to the MedDRA system organ classification. The system organ class (SOC) should be presented in the order shown in the annex. Adverse reactions descriptions should be based on the most suitable representation within the MedDRA terminology. This will usually be at the Preferred Term (PT) Level, although there may be instances where the use of Lowest Term Level or exceptionally group terms, such as High Level Terms may be appropriate. As a general rule, any adverse reactions should be assigned to the most relevant SOC related to the target organ. For example, PT ‘Liver function test abnormal’ should be assigned to the SOC ‘Hepatobiliary disorders’ rather than to the SOC ‘Investigations’.

Within each system organ class, the adverse reactions should be ranked under headings of

frequency, most frequent reactions first. Within each frequency grouping, adverse reactions should be presented in the order of decreasing seriousness. The names used to describe each of the frequency groupings should follow standard terms established in each official language using the following convention: Very common ($\geq 1/10$); common ($\geq 1/100$ to $< 1/10$); uncommon ($\geq 1/1,000$ to $< 1/100$); rare ($\geq 1/10,000$ to $< 1/1,000$); very rare ($< 1/10,000$).

In exceptional cases, if a frequency cannot be estimated from the available data, an additional category frequency 'not known' may be used. In case the expression "Frequency not known" is used, the following text should be added in the list of terms explaining the frequency categories: "not known (cannot be estimated from the available data)". The expressions isolated/single cases/reports should not be used.

Where additional details about an adverse reaction are described in section c), the reaction concerned should be highlighted, for example with an asterisk, and, "see section c)" should be included as a footnote.

Guidance on how to estimate the frequency of an adverse reaction is provided at the end of this chapter of the guideline.

d) Description of selected adverse reactions

This section should include information characterizing specific adverse reaction which may be useful to prevent, assess or manage the occurrence of an adverse reaction in clinical practice.

This section should include information characterizing individual serious and/or frequently occurring adverse reactions, or those where there have been reports of particularly severe cases. The information should provide frequency and may describe for example reversibility, time of onset, severity, duration, mechanism of the reaction (if of clinical relevance), dose relationship, relationship with duration of exposure or risk factors. Measures to be taken to avoid specific adverse reactions or actions to be taken if specific reactions occur should be mentioned under section 4.4 and cross-referenced here.

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Information on the occurrence of withdrawal reactions may be mentioned here with cross-reference to section 4.2 in case of need for tapering off or advice on discontinuation of the product.

Mention should be made here of any differences between different dosage forms in respect of adverse reactions.

In the case of combination products, information should be included in this sub-section pointing out which particular adverse reactions are usually attributable to which active substance of the combination, where known.

Any adverse reactions resulting directly from an interaction should be mentioned here and cross-referenced to section 4.5.

This section should also inform on adverse reactions with very low frequency or with delayed onset of symptoms which may not have been observed in relation to the product, but which are considered to be related to the same therapeutic, chemical or pharmacological class. The fact that this is a class attribution should be mentioned.

Any adverse reaction specific to excipients or residues from the manufacturing process should be included.

e) <Paediatric population>

A paediatric sub-section should always be included (unless irrelevant).

The extent and age characteristics of the safety database in children should be described (e.g. from clinical trials or pharmacovigilance data). Uncertainties due to limited experience should be stated.

If the observed safety profile is similar in children and adults this could be stated: e.g. "Frequency, type and severity of adverse reactions in children are <expected> to be the same as in adults". Similarly, it is appropriate to state whether the safety profiles in the different paediatric subsets are similar or not.

Any clinically relevant differences (i.e. in nature, frequency, seriousness or reversibility of adverse reactions) between the safety profiles in adult and paediatric populations, or in any relevant age groups, should be described and presented by age group. If there is a need for specific monitoring, this should be highlighted by cross-referencing to section 4.4. For clinically relevant differences, a separate table listing such adverse reactions by frequency can be added and presented by relevant age groups if appropriate. If some paediatric adverse reactions are considered common ($\geq 1/100$ to $< 1/10$) or very common ($\geq 1/10$), the frequencies should be provided in parentheses.

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In case of major difference with the safety profile in adults, a summary of the safety profile in children could be presented to facilitate the presentation of the information. Available information, from any source scientifically validated, on long-term safety in children (e.g. on growth, mental development and sexual maturation) should also be summarized, whether positive or negative, with cross-reference to section 5.1 if appropriate. Any risk factors such as duration of treatment or period at risk should be specified.

If relevant, symptoms of neonatal withdrawal should be listed in a separate paragraph with cross-reference with 4.6

e. <Other special populations>

This section may include information on any clinically relevant differences (i.e. in nature, frequency, seriousness or reversibility of adverse reactions, or need for monitoring) specifically observed in other special populations such as elderly, patients with renal impairment, patients

with hepatic impairment, patients with other diseases or a specific genotype. Cross-reference to other sections such as 4.3, 4.4 or 4.5 may be added as appropriate.

Adverse reactions may also be related to genetically determined product metabolism. Subjects or patients deficient in the specific enzyme may experience a different rate or severity of adverse reactions. This should be mentioned and where relevant correlated with data from clinical trials.

Further guidance on the estimation of frequency of adverse reactions

The estimation of the frequency of an adverse reaction depends on the data source (i.e. clinical trial, post-authorization safety study or spontaneous reporting), the quality of data collection and causality evaluation. If the choice of the frequency category is based on different sources, the category representing the highest frequency should be chosen unless a more specific method has been applied and thus resulted in an estimate of clearly higher validity, e.g. a pooled analysis across suitable studies.

Sources of data should use population exposed to the doses and treatment duration as recommended in the SmPC.

Reactions that are reported under different terms but represent the same phenomenon (e.g., sedation, somnolence, drowsiness) should ordinarily be grouped together as a single adverse reaction to avoid diluting or obscuring the true effect. Similarly, reactions that represent a syndrome complex should ordinarily be grouped together under an appropriate heading to avoid obscuring the full range of respective symptoms.

Adverse reactions from clinical trials

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Safety data from several studies should be pooled to increase the precision of adverse reaction rates as appropriate without introducing bias (e.g. major difference in population characteristics or exposure to the product).

The frequency of adverse reactions should be derived from pooled placebo-controlled studies if these data are available and the databases are sufficiently large to be informative. If these data are unavailable or not sufficiently informative, active-controlled data or possibly single-arm or add-on trials databases could be used to estimate frequencies.

Frequency should represent crude incidence rates (and not differences or relative risks calculated against placebo or other comparator).

When a common, very common or serious adverse reaction (e.g. suicide) also occurs in the placebo group with a relevant frequency, both incidence rates can be stated to put the risk into perspective (e.g. in subsection c).

Adverse reactions from safety studies

The choice of the frequency category to which any adverse reaction will be assigned is based on the point estimate of the crude incidence rate derived from a study designed in such a way that specific adverse events occurring in patients within a defined observation period would have been detected and reasonably attributed to the medicinal product. In this situation, it is possible to calculate a point estimate of the crude incidence rate using standard statistical methods. In cases where the original information is expressed as an incidence density (denominator expressed as person-time), an appropriate transformation into an incidence proportion should be performed for choosing the frequency category. Normally, incidence proportions for the most representative exposure period (e.g. 1 week, 3 months, 1 year) should be used to derive the frequency category. However, this may not be appropriate if the hazard function increases over time; in this case, the adverse reaction and its frequency pattern, when clinically relevant, should be properly described in section c).

The frequency category to be chosen for each adverse reaction should not be based on differences calculated against a comparator. However, when data are derived from a study with a non-exposed group and the rate difference attributed to the medicinal product is smaller than the baseline or background incidence rate, and if the adverse reaction is considered important, the background incidence may be provided (e.g. in section c).

Adverse reactions from spontaneous reporting

The number of spontaneous reports should not be stated because the number can quickly become outdated. Frequencies based on reporting rates from a spontaneous reporting system should not be used to assign frequency category. In case of an unexpected adverse reaction detected from spontaneous reporting, each adequately designed study where this adverse reaction could have been detected should be reviewed to choose a frequency category. If the adverse reaction has never been observed in clinical trials, then the upper limit of the 95%

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confidence interval is not higher than $3/X$, with X representing the total sample size summed up across all relevant clinical trials and studies (e.g. those with a follow-up long enough to detect the adverse reaction).

For example, if a particular adverse reaction has not been observed among 3600 subjects exposed to the product in clinical trials and studies, then the upper limit of the 95% confidence interval for the point estimate is $1/1200$ or less and the frequency category should be "rare", based on worst value of the point estimate. The rationale for the frequency category for that particular reaction could be explained in sub-section c).

4.9 Overdose

Describe acute symptoms and signs and potential sequelae of different dose levels of the medicinal product based on all available information including accidental intake, mistakes and suicide attempts by patients.

Taking into account all relevant evidence, describe management of overdose in man, e.g. in

relation to monitoring or use of specific agonists/antagonists, antidotes or methods to increase elimination of the medicinal product such as dialysis. However, there should not be any dosage recommendation of other medicinal products (e.g. antidotes) as it could create conflict with the SmPCs of those other products. If applicable, counteractive measures based on genetic factors should be described.

Additional information on special populations

Information specifically observed in special populations such as elderly, patients with renal impairment, patients with hepatic impairment, other concomitant diseases etc.

Paediatric population

If there are specific paediatric considerations, there should be a sub-section entitled 'paediatric population'.

Special mention should be made of those medicinal products/strength of formulation for which ingestion of only one dose unit by children can cause fatal poisoning.

5. PHARMACOLOGICAL PROPERTIES

5.1 Pharmacodynamic properties

Sections 5.1 – 5.3 should normally mention information, which is relevant to the prescriber and to other health-care professionals, taking into account the approved therapeutic indication(s) and the potential adverse drug reactions. Statements should be brief and precise.

The sections should be updated regularly when new information becomes available, especially in relation to the paediatric population.

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5.1 Pharmacodynamic properties

Describe:

- Pharmacotherapeutic group and ATC code:

Inclusion of the therapeutic subgroup (2nd level of WHO classification) with the 3rd level (pharmacological subgroup) and the 4th level (chemical subgroup) is recommended. If an ATC code is not yet available, this should be mentioned as 'not yet assigned'.

In case of medicinal product Authorized as similar biological medicinal product, the following statement will be included:

<<(Proprietary) Name> is a biosimilar medicinal product.

- Mechanism of action (if known)

- Pharmacodynamic effects.

- Clinical efficacy and safety

It may be appropriate to provide limited information, relevant to the prescriber, such as the main results (statistically compelling and clinically relevant) regarding pre-specified end points or clinical outcomes in the major trials, and giving the main characteristics of the patient population. Such information on clinical trials should be concise, clear, relevant and balanced, and should summarise evidence from relevant studies supporting the indication. The magnitude of effects should be described using absolute figures. (Relative risks or odd ratio should not be presented without absolute figures).

In the exceptional cases when clinically relevant information from subgroup or post-hoc analyses is presented, it should be identified as such in a balanced manner reflecting the limited robustness of both positive and negative secondary observations.

Any relevant pharmacogenetic information from clinical studies may be mentioned here. This should include any data showing a difference in benefit or risk depending on a particular genotype or phenotype.

Paediatric population

The results of all pharmacodynamic (clinically relevant) or efficacy studies conducted in children should be presented under this sub-heading.

Information should be updated when new relevant information becomes available.

Results should be presented by age or relevant subsets.

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When there are data available, but there is no Authorized paediatric indication, data should be presented and a cross-reference should always be made to section 4.2 and, as appropriate to 4.3.

In presenting results of studies, particular attention should be given to include the relevant safety data. For exploratory studies, the results of the main endpoints should be given with the main characteristics of the population studied and the doses used. When they are available, information and results of confirmatory studies should usually supersede and replace those of exploratory studies. For confirmatory studies, the objectives, the study duration, the doses used (and the formulation used if different from the marketed one), the main characteristics of the patient population studied (including age and numbers of patient), and the main results regarding pre-specified endpoints should be provided, whether positive or negative. If data are considered inconclusive, this should be stated.

The objective and the main results or the conclusion of any specific clinical safety study should also be given.

5.2 Pharmacokinetic properties

Pharmacokinetic properties of the active substance(s) relevant for the advised dose, strength

and the pharmaceutical formulation marketed should be given in this section. If these are not available, results obtained with other administration routes, other pharmaceutical forms or doses can be given as alternative.

Basic primary pharmacokinetic parameters, for instance bioavailability, clearance and half-life, should be given as mean values with a measure of variability.

Pharmacokinetics items, which could be included in this section when relevant, are given below.

a. General introduction, information about whether the medicinal product is a pro-drug or whether there are active metabolites, chirality, solubility, information on the population in which general pharmacokinetic data were obtained, etc.

b. General characteristics of the active substance(s) after administration of the medicinal product formulation to be marketed.

Absorption: complete or incomplete absorption; absolute and/or relative bioavailability; first pass effect; T_{max}; the influence of food; in case of locally applied medicinal product the systemic bioavailability; involvement of transport proteins. If available, information on the site of absorption in the gastro-intestinal tract should be stated (as it may be important for administration by enteral feeding tubes).

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Distribution: plasma protein binding; apparent volume of distribution per kilogram bodyweight (l/kg); tissue and/or plasma concentrations; pronounced multi-compartment behaviour; involvement of transport proteins.

Biotransformation: degree of metabolism; which metabolites; activity of metabolites and contribution to effect and toxicity; enzymes involved in metabolism; site of metabolism; results from in vitro interaction studies that indicate whether the new compound can induce/inhibit metabolic enzymes.

Elimination: elimination half-lives, total clearance; inter and/or intra-subject variability in total clearance; excretion routes of the unchanged substance and the metabolites including the relative portion of the hepatic and renal eliminated fraction, involvement of transport proteins.

Linearity/non-linearity: linearity/non-linearity of the pharmacokinetics of the active substance with respect to dose and/or time; if the pharmacokinetics are nonlinear with respect to dose and/or time, the underlying reason for the non-linearity should be presented. Additional relevant information should be included here.

c. Characteristics in specific groups of subjects or patients

■ Variations with respect to factors such as age, weight, gender, smoking status, polymorphic metabolism and concomitant pathological situations such as renal failure, hepatic disease, including degree of impairment. If the influence on pharmacokinetics is considered to be clinically relevant, it should be described here in quantitative terms (cross-reference to section 4.2 when applicable).

d. Pharmacokinetic/pharmacodynamics relationship(s)

■ Relationship between dose/concentration/pharmacokinetic parameter and effect (either true endpoint, validated surrogate endpoint or side effect).

■ The population studied should be described.

Paediatric population

Results of pharmacokinetic studies in the different paediatric age groups should be summarized, with comparison to adults if available. If appropriate, the dose producing similar product exposure as adults could be given. The pharmaceutical form(s) used for pharmacokinetic studies in children should be stated. Uncertainties due to limited experience should be stated.

5.3 Preclinical safety data

Information should be given on any findings in the non-clinical testing which could be of relevance for the prescriber, in recognizing the safety profile of the medicinal product used for the authorized indication(s), and which is not already included in other relevant sections of the SmPC.

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If the results of the non-clinical studies do not add to the information needed by the prescriber, then the results (either positive or negative) need not be repeated in the SmPC.

The findings of the non-clinical testing should be described in brief with qualitative statements as outlined in the following example:

■ Non-clinical data reveal no special hazard for humans based on conventional studies of safety pharmacology, repeated dose toxicity, genotoxicity, carcinogenic potential and toxicity to reproduction and development.

■ Effects in non-clinical studies were observed only at exposures considered sufficiently in excess of the maximum human exposure indicating little relevance to clinical use.

■ Adverse reactions not observed in clinical studies, but seen in animals at exposure levels similar to clinical exposure levels and with possible relevance to clinical use were as follows.

Findings of non-clinical studies relevant for use in the paediatric population, including juvenile animals and peri- or post- natal studies, should be presented with a discussion of their clinical relevance, under a sub-heading if necessary.

<Environmental Risk Assessment (ERA)>

Where relevant, conclusions on the environmental risk assessment of the product should be included, with reference to section 6.6.

6. PHARMACEUTICAL PARTICULARS

6.1 List of excipients

A list should be given of the excipients, expressed qualitatively only. All excipients, which are present in the product, should be included, even those present in small amounts, such as printing inks. Further details on the excipients to be declared may be found in the section on definitions and examples in the Guidelines on format and content of labels for medicinal products

For transdermal patches, all ingredients of the patch (including the adhesive, release liner and backing film) should be mentioned.

The active substance itself, residues of substances used during manufacture of the finished product (for example, solvents, head-space gases or antibiotics in vaccine manufacture), lubricants for prefilled syringes and constituents of capsule shells for inhalation powders not intended to be taken should not be included.

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However, certain residues such as residues of antibiotic or other antimicrobial agents used in production that are known allergens with a potential for inducing undesirable effects should be mentioned in section 4.3 or 4.4 as appropriate.

Excipients should be referred to by their recommended INN if existing, accompanied by the salt or hydrate form if relevant or by their recognized pharmacopoeial name. If an excipient has neither an INN nor a pharmacopoeia name, it should be described by its usual common name. References to the pharmacopoeial quality should not be included. E numbers should be given along with the common name of the excipient where they exist and when necessary for proper use, e.g. when the excipient is listed in the Guideline on the excipients in the label and package leaflet of medicinal products for human use (as having recognised action or effect).

The ingredients in excipient mixtures should be listed individually. In cases where the full composition of a flavour or fragrance is not known to the applicant or is too complex, it may be declared in general terms (e.g. 'orange flavour', 'citrus perfume'). However, any of the components, which are known to have a recognised action or effect, should be included.

Ingredients that may or may not be added for the pH adjustment should be followed by the parenthesis '(for pH-adjustment)'

Proprietary names or general descriptive names such as 'printing ink' should not be used in place of the common name of an ingredient or of a mixture of ingredients but may be used in conjunction with the name(s) of the ingredient(s), so long as it is clear which ingredients are described by the name. Chemically modified excipients should be declared in such a way as to avoid confusion with the unmodified excipients, e.g. 'pregelatinised starch'.

In the case of a product containing a covert marker for the purpose of tracking, tracing and authentication, a general term such as "authentication factor" should be included in the list of excipients instead of the name of the excipient, unless the excipient is one that is known to have a recognised action or effect.

For clarity, it is recommended that each excipient be listed on a separate line. It can be useful to list excipients according to the different parts of the product, e.g. tablet core/coat, capsule contents/shells, etc. For products that are presented in more than one container or in dual-chamber containers, the excipients should be listed per container or per chamber.

Abbreviations for excipients should not be used. However, where justified for space considerations, abbreviations for excipient names may appear on the labelling, on condition that these abbreviations are designated in section 6.1.

6.2 Incompatibilities

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Information on physical and chemical incompatibilities of the medicinal product with other products with which it is likely to be mixed or co-administered should be stated. This is particularly important for medicinal products to be reconstituted and/or diluted before parenteral administration. Significant interaction problems, e.g. sorption of products or product components to syringes, large volume parenteral containers, tubing, in-line filters, administration sets, etc. should be stated.

Statements concerning compatibility of the product with other medicinal products or devices should not be included in this section but in section 6.6. Statements concerning pharmacological and chemical/physical incompatibilities with food should be included in section 4.5. If appropriate, the standard statement, 'Not applicable', should be included.

For certain pharmaceutical forms, e.g. parenterals, either of the following standard statements should be included as appropriate:

- 'In the absence of compatibility studies, this medicinal product must not be mixed with other medicinal products.'
- 'This medicinal product must not be mixed with other medicinal products except those mentioned in section 6.6.'

6.3 Shelf life

The shelf life should be given for the medicinal product as packaged for sale and, if appropriate, after dilution or reconstitution or after first opening.

A clear statement of the shelf life should be given, in an appropriate unit of time.

An in-use shelf life may need to be stated for other medicinal products if development studies have found it to be necessary.

Additionally, if different concentrations need to be prepared, e.g. for use in children, the physicochemical stability throughout the entire concentration range should be stated; e.g. "The stability has been demonstrated between x mg/ml and y mg/ml for t hours/days at 25 °C and 2-8 °C".

In case of a paediatric indication, if no age appropriate formulation is available for children but an extemporaneous formulation could be prepared from an existing formulation, relevant physicochemical data on storage and stability should be included here with a cross-reference in sections 6.4 and 6.6."

In case of specific temporary storage conditions need to be provided to healthcare professionals or patients, e.g. for the purpose of ambulatory use (e.g. shelf-life 24 months at 2-8°C of which 3 months could be below 25°C), specific additional guidance should be provided as appropriate. Such information should always be based on stability data. In particular, the recommended temperature range and maximum duration of temporary storage should be specified. This guidance may also include the action to be taken after the product has been stored under the

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temporary storage conditions (e.g. discard immediately). Statements such as "These data are not recommendations for storage" should not be used.

No reference should be made to the container unless there are different shelf lives for different containers. Storage conditions should not be included, except for the storage conditions after opening (see the corresponding guideline). Statements such as 'Do not use after the expiry date' should not be included.

When a device is supplied together with a medicinal product, the in-use shelf-life of the device should be given where applicable.

6.4 Special precautions for storage

Storage warnings should be stated.

For storage of sterile products that have been opened, diluted or reconstituted, a cross-reference should be made to section 6.3.

Note that if a specific storage warning is required, the warning should be consistent between the SmPC, label and PIL.

A warning to keep the product out of the reach and sight of children should not be included in the SmPC.

6.5 Nature and contents of container

Reference should be made to the immediate container using recognized pharmacopoeial standard term; the material of construction of the immediate container should be stated ('glass vials', 'PVC/Aluminium blisters', 'HDPE bottles'); and any other component of the product should be listed, e.g. needles, swabs, measuring spoons, syringes, inhaler devices, desiccant. The graduation on measuring devices should be explained. The container of any solvent provided with the medicinal product should also be described. Excessive detail, e.g., concerning the colour of the stopper, the nature of the heat-seal lacquer, should usually not be included. For parenteral preparations, when enclosure colour is used to differentiate between the presentations of a product, this should be stated here.

If appropriate, it should be indicated if the container closure is child-resistant.

Examples on the text in this section:

'<Volume> ml suspension in a pre-filled syringe (glass) with plunger stopper (chlorobutyl rubber) with or without needle in pack sizes of 5 or 10.'

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'HDPE bottle with a child-resistant closure and a silica gel desiccant. Pack-sizes of 30, 60 or 90 film-coated tablets.'

All pack sizes should be listed. Pack sizes mentioned should include the number of units, number of doses (for e.g. multi-dose vaccines, inhalers, etc.), total weight or volume of the immediate container, as appropriate, and the number of containers present in any outer carton. If appropriate, a standard statement, 'Not all pack sizes may be marketed', should be included, in order to alert health professionals to the fact that not all listed pack sizes may be available for prescribing or dispensing.

Multiple unit packs for distribution purposes only do not constitute new pack sizes for marketing of the product and should therefore not be included in this section.

6.6 Special precautions for disposal <and other handling>

Instructions for disposal should be included here, if appropriate for the product.

Where special precautions for the handling and disposal of certain products such as cytotoxics and some biological products or waste material derived from it are advised, e.g. in the case of products containing live organisms, these should be stated in this section, as should, where relevant, the disposal of items which come into contact with the product, such as nappies, or spoons used to administer oral vaccines. If relevant, a cross-reference to conclusions on the environmental risk assessment described in section 5.3 can be included.

If applicable, e.g. for cytotoxics, the following standard statement should be included, 'Any unused product or waste material should be disposed of in accordance with local requirements.'

If there are no special use or handling instructions for the pharmacist or other healthcare professionals, the standard statement, 'No special requirements.' should be included.

Any directions necessary for the accurate preparation of certain products such as cytotoxics and some biological products and/or necessary for the protection of persons including parents or carers preparing or handling the product should be stated.

In section 4.2, instructions on handling of the product by the doctor, other health personnel, or patient should be included, as well as general information concerning the administration of the product (whether administered by the patient or the health personnel). If instructions for use/handling are needed where the medicinal product has to be prepared before use, e.g. where it must be suspended or diluted, this information has to be given here. For clarity, a cross-reference in section 4.2 to the relevant information in section 6.6 could be included, e.g. 'For instructions on dilution of the product before administration, see section 6.6.'

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It is recommended that only information necessary for the pharmacist or other health personnel to prepare the product for administration to the patient should be included here.

Information on the preparation (e.g. the suspension of a powder for injection, or preparing a dilution) of the medicinal should be included in section 6.6, regardless of who prepares the product (e.g. pharmacist, doctor, other health personnel, patient, parents or carers). In the case of products for reconstitution, the appearance of the product after reconstitution should be stated.

Statements concerning compatibility of the product with other medicinal products or devices can be given here provided the data have been provided in the dossier.

In the exceptional cases where a product is indicated in children and where no adequate paediatric formulation can be developed (based on duly justified scientific grounds), information on extemporaneous formulation should appear under a sub-heading "Use in the paediatric population" and should cross-refer to the section 4.2. Detailed instructions for the preparation of the extemporaneous formulation from the appropriate "adult" or other "older children" dosage form and additional information on extemporaneous formulations for use in younger children shall be provided and, where appropriate, the maximum storage time during which such preparation will conform to its specifications. When necessary, the required packaging material and storage conditions should be stated here.

Any specific warnings for the handling of the product should be in section 4.4.

Information on risks due to occupational exposure should be included in this section, with reference to section 4.4 or 4.8 if there is information in that section.

7. MARKETING AUTHORIZATION HOLDER AND MANUFACTURING SITE ADDRESSES

Name and permanent address or registered place of business of the Marketing Authorization Holder and manufacturing site(s) physical address.

Telephone, fax numbers or e-mail addresses may be included (not websites or emails linking to websites).

8. MARKETING AUTHORIZATION NUMBER

Item to be completed by the Marketing Authorization Holder once the Marketing Authorization has been granted by TMDA.

9. DATE OF FIRST <REGISTRATION> / RENEWAL OF THE <REGISTRATION>

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Item to be completed by the Marketing Authorization Holder once the Marketing Authorization has been granted or renewed.

Both the date of first authorization and, if the authorization has been renewed, the date of the (last) renewal should be stated in the format given in the following example:

Date of first authorization: 3 April 1985
Date of latest renewal: 3 April 2000

10. DATE OF REVISION OF THE TEXT

Leave blank in case of a first Marketing Authorization.

11. DOSIMETRY (IF APPLICABLE)

Full details of internal radiation dosimetry should be included in this section for radiopharmaceuticals.
For all other products, this section should be excluded.

12 INSTRUCTIONS FOR PREPARATION OF RADIOPHARMACEUTICALS (IF APPLICABLE)

For radiopharmaceuticals, additional detailed instructions for extemporaneous preparation and quality control of such preparation and, where appropriate, maximum storage time during which any intermediate preparation such as an eluate or the ready-to-use pharmaceutical will conform to its specifications.

Special instructions relating to the disposal of containers and unused contents should also be included.

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PART V:

GUIDELINES ON FORMAT AND CONTENT OF LABELS FOR MEDICINAL PRODUCTS

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1. GENERAL REQUIREMENTS

(a) The label text

Particulars in the label shall be easily legible, clearly comprehensible and indelible.

(b) Conformity with the Summary of Product Characteristics

The label text should be in conformity with the summary of products characteristics.

(c) Language

The labelling must be presented at least in English and/or Swahili. If more than one language is used, then all of the text must be in each language and the overall readability should not be adversely affected. The content of all language versions must be identical. It is recommended to group different text elements for each language, where appropriate.

(d) Products with different strengths

Container labels may look similar across multiple strengths of the same product or across multiple products within a company's product line.

Product labels for medicinal products with multiple pharmaceutical strengths, within a manufacturer's product line, should be designed such that the products are identifiable and can be significantly differentiated from one another. Colour differentiation on product labels should be an effective tool that can differentiate products within a manufacturer's product line.

When applying differentiating colour, the applicant should ensure that the text highlighted by the differentiating colour has adequate colour contrast against the background colour on the container label.

(e) Label layout and artwork

Images, pictograms and other graphics may be used to aid comprehension of the information on the labels of non-prescription medicines only and not on the container labels of prescription medicines.

Images, pictograms and other graphics should exclude any element of a promotional nature and should only be used to aid navigation, clarify or highlight certain aspects of the text and should not replace the actual text.

The overall layout should not be misleading or have any inappropriate connotations in a way that no doubt about the meaning of a particular pictogram will be perceived.

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2. PARTICULARS TO BE INCLUDED ON THE LABEL

(a) Outer packaging or, where there is no outer packaging, on the immediate packaging

The label should include at least the following:

- i. Proprietary Name where applicable
- ii. International Non-Proprietary name(s) of the Active Pharmaceutical Ingredient(s)
- iii. Amount of each Active Pharmaceutical Ingredient present in a dosage unit
- iv. List of excipients known to be a safety concern for some patients, e.g. lactose, gluten, metabisulfites, parabens, ethanol, or tartrazine. For parenterals and topical preparations, all excipients should be listed.
- v. Pharmaceutical form and contents of the container, e.g. number of dosage units, weight or volume.
- vi. Method and route(s) of administration and the statement "Read the patient information leaflet before use."
- vii. Special warning that the medicinal product must be stored out of the reach and sight of children ("Keep out of the reach and sight of children").
- viii. Other special warnings and handling precautions, if necessary (e.g. in case of specific toxicity of the agents)
- ix. The word "sterile" if the product is sterile
- x. Batch number assigned by the manufacturer
- xi. The manufacturing date
- xii. The expiry date
- xiii. Special storage conditions, if applicable
- xiv. Special precautions for disposal of unused medicinal products or waste material derived from such medicinal products, if appropriate
- xv. The name and address of the Marketing Authorization Holder
- xvi. Physical address of the site responsible for release of the finished product
- xvii. Advice on general classification for distribution, e.g., Controlled Medicines, Prescription Only Medicines, Pharmacy Only Medicines, Over-the-Counter and General Sales List
- xviii. Instruction on use
- xix. The proprietary name, strength and expiry date in braille (Marburg Medium)
- xx. A unique identifier (example a product specific barcode, unique code or registration number issued by NMRA).

(b) Guidance for small containers

For containers of less than or equal to 10 ml capacity that are marketed in an outer pack such as

a carton, and the outer pack bears all the required information, the immediate container should contain at least these minimum information (added):-

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- i. Brand Name of the FPP, INN name, strength, pharmaceutical form, active substance(s) and route(s) of administration
- ii. Method of administration
- iii. Batch number assigned by the manufacturer
- iv. Expiry date
- v. Manufacturing date if space is enough
- vi. Contents by weight, by volume or by unit
- vii. The name and address of the manufacturing site— or a logo that unambiguously identifies the company.
- viii. Directions for use, and any warnings or precautions that may be necessary

(c) Guidance for Blisters and strips

Blisters and strips should include, as a minimum, the following information (printed directly):-

- i. Name, strength and pharmaceutical form of the FPP.
- ii. Name and physical address of the manufacturing site (the site responsible for release of the finished product)
- iii. The batch number assigned by the manufacturer
- iv. The expiry date [Note that for co-blistered products, the expiry date is that of the product which expires first.]
- v. The manufacturing date, if space is enough
- vi. The batch number assigned by the manufacturer
- vii. Directions for use, and any warnings or precautions that may be necessary.

(d) Additional labelling information required by some Partner States

Partner State NMRAs may require the use of certain forms of labelling making it possible to indicate:-

- i. Price of the medicinal product;
- ii. The reimbursement conditions of social security organisations;
- iii. Identification and authenticity;
- iv. A statement that the product is a property of government

The information specific to a Partner State NMRA should be accommodated on the label in a box, to appear on one side of the pack. Each box should only be presented in the official language or languages of the Partner State concerned and should state the name of that Partner State.

3. CONTROL OF THE CONFORMITY OF THE LABELLING

The labelling of the medicinal product forms part of the authorization and it must, therefore, be approved by the NMRA when the authorization is granted.

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4. CHANGES TO THE LABELLING

Any changes to the labelling, which are not connected with the Summary of Product Characteristics, shall be notified to the NMRA where the authorization is granted. Therefore, if a Marketing Authorization Holder wishes either to introduce any label text additional to that in the decision or to change any aspect of the labelling he must first notify this change to the relevant mentioned NMRA, who shall inform the Marketing Authorization Holder whether the proposed change is accepted or not.

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PART VI:

GUIDELINES ON FORMAT AND CONTENT OF PACKAGE INSERTS FOR PHARMACEUTICAL PRODUCTS

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1. GENERAL REQUIREMENTS

1.1 The Package insert

Particulars in the package insert shall be easily legible, clearly comprehensible and indelible.

a) Type size and font

The following should be considered while selecting type size and font:

- i. The font should be easy to read; stylized fonts which are difficult to read should not be used;
- ii. The font should be such that similar letters/numbers such as "l", "I" and "1" can be easily distinguished from each other;
- iii. A minimum type size of 9 points, as measured in font 'Times New Roman', not narrowed, with a space between lines of at least 3 mm, should be used;
- iv. Widespread use of capitals is discouraged; however, capitals may be used for emphasis.

b) Paper

The quality of insert paper should be taken into consideration in order to ensure proper readability of the insert. The following should be considered:

- i. The paper weight should be such that the paper is sufficiently thick to reduce transparency which makes reading difficult, particularly where the text size is small.
- ii. Uncoated paper is preferred as glossy paper reflects light thus making information difficult to read.
- iii. When the leaflet is folded the creases should not interfere with the readability of the information or lead to tearing of the insert.

1.2 Conformity with the Summary of Product Characteristics

The package insert should be in conformity with the summary of products characteristics.

1.3 Language

The labelling must be presented at least in English. An active style of writing should be used instead of passive.

2. PARTICULARS TO BE INCLUDED ON THE PACKAGE INSERT

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2.1 CONTENT AND FORMAT OF THE PRESCRIBING INFORMATION

For prescription only medicines, the package insert should include prescribing information. The content and format for the prescribing information should follow that of Summary of Product Characteristics (SmPC). Please refer to the Guidelines on Format and Content of Summary of Product Characteristics for Pharmaceutical Products.

2.2 CONTENT AND FORMAT OF THE PATIENT INFORMATION LEAFLET

The patient information leaflet shall include the particulars outlined in the template in the following section.

The applicant should complete the template and delete the parts which are not applicable.

2.2.1 TEMPLATE FOR PATIENT INFORMATION LEAFLET

{{(Proprietary) name strength pharmaceutical form}}

{Active substance(s)}

Read all of this leaflet carefully before you start <taking><using> this medicine.

■ Keep this leaflet. You may need to read it again.

■ If you have any further questions, ask your <doctor, health care provider><or><pharmacist>.

■ This medicine has been prescribed for you. Do not pass it on to others. It may harm them, even if their symptoms are the same as yours.

■ If any of the side effects gets serious, or if you notice any side effects not listed in this leaflet, please tell your <doctor, health care provider><or><pharmacist>.>

In this leaflet:

- a) What {product name} is and what it is used for
- b) Before you <take><use> {product name}
- c) How to <take><use> {product name}
- d) Possible side effects

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- e) How to store {product name}
- f) Further information

[Delete sections that are not applicable]

a) WHAT {PRODUCT NAME} IS AND WHAT IT IS USED FOR

b) BEFORE YOU <TAKE><USE> {PRODUCT NAME}

Do not <take><use> {product name}

- <if you are allergic (hypersensitive) to {active substance(s)} or any of the other ingredients of {product name}>.

- <if ...>

Take special care with {product name}

- <if you ...>

- <when ...>

- <Before treatment with {product name},...>

<Taking><Using> other medicines

<Please tell your <doctor, health care provider><or><pharmacist> if you are taking or have recently taken any other medicines, including medicines obtained without a prescription.>

<Taking><Using> {product name} with food and drink

Pregnancy and breast-feeding

<Ask your <doctor, health care provider><or><pharmacist> for advice before taking any medicine.>

Driving and using machines

<Do not drive <because...>.>

<Do not use any tools or machines.>

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Important information about some of the ingredients of {product name}

c) HOW TO <TAKE><USE> {PRODUCT NAME}

<Always <take><use> {product name} exactly as your doctor or health care provider has told you. You should check with your <doctor, health care provider><or><pharmacist> if you are not sure.><The usual dose is...>

<Use in children>

If you <take><use> more {product name} than you should

If you forget to <take><use> {product name}

<Do not take a double dose to make up for a forgotten <tablet><dose><...>.>

If you stop <taking><using> {product name}

<If you have any further questions on the use of this product, ask your <doctor, health care provider><or><pharmacist>.>

d) POSSIBLE SIDE EFFECTS

Like all medicines, {product name} can cause side effects, although not everybody gets them.

If any of the side effects gets serious, or if you notice any side effects not listed in this leaflet, please tell your <doctor, health care provider><or><pharmacist>.

e) HOW TO STORE {PRODUCT NAME}

Keep out of the reach and sight of children.

<Do not store above °C>, <Store in the original <container><carton>>

Do not use {product name} after the expiry date which is stated on the <label><carton><bottle><...><after {abbreviation used for expiry date}>.><The expiry date refers to the last day of that month.>

<Do not use {product name} if you notice {description of the visible signs of deterioration}>.

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<Medicines should not be disposed of via wastewater or household waste. Ask your pharmacist how to dispose of medicines no longer required. These measures will help to protect the environment.>

f) FURTHER INFORMATION

What {product name} contains

- The active substance(s) is (are)...
- The other ingredient(s) is (are)...

What {product name} looks like and contents of the pack

Name and full physical address of Marketing Authorization Holder and Manufacturing site:-

{Name and address}

<{tel}>

<{fax}>

<{e-mail}>

For any information about this medicinal product, please contact the <local representative of the> supplier:

{Country} {Country}

{Name} {Name}

<{Address} <{Address}

B-0000 {City}> B-0000 {City}>

tel: + {telephone number} tel: + {telephone number}

<{e-mail}> <{e-mail}>

<as appropriate, add additional local representatives to the above table>

This leaflet was last approved in {MM/YYYY}.

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PART VII:

GUIDELINES ON COMMON GLOSSARY OF TERMS USED IN MEDICINES REGISTRATION

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ABBREVIATIONS AND ACRONYMS

BMGF - Bill and Melinda Gates Foundation

EAC - East African Community

EAC-MRH - East African Community Medicines Regulatory

EMA - European Medicines Agency

GMP - Good Manufacturing Practice

ICH - The International Council for Harmonisation of Technical Requirements
for Pharmaceuticals for Human Use

MA - Marketing Authorization

MAH - Marketing Authorization Holder

MER - Medicines Evaluation and Registration

NEPAD - New Partnership for African Development

NMRA - National Medicines Regulatory Authority

Requirements for Registration of Pharmaceuticals for Human Use

TWG - Technical Working Group

US FDA - United States Food and Drug Administration

WHO - World Health Organization

SRA - Stringent Regulatory Authority

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1. INTRODUCTION

Glossary of terms in medicines registration have been developed to minimize misunderstanding of words used in medicines registration as this process is at the nexus of many key stakeholders. There is also an increasing proliferation and duplication of terms and definitions, as the medicines registration field itself is still evolving and adapting itself to new and changing contexts.

The glossary provides information on the range of terms and definitions encountered in medicines registration. It does not present new or different definitions of terms, but draws together definitions from many existing sources. Changes to definitions have been minimal, and only made to unify the style of the Glossary, e.g. some spelling has been standardised, and the plural form of terms has been replaced by the singular form.

2. SELECTION OF TERMS

The terms were selected from existing glossaries appended to guidelines on application for registration of medicines from respective EAC Partner States. Also, the terms were selected from international guidelines such as USFDA, WHO, EMA, Health Canada and other international publications.

Furthermore, definitions were selected using the criteria of widespread acceptance, wide spread use and consultation from EAC National Medicines Regulatory Authorities.

3. SCOPE

This glossary of terms primarily addresses terms that are used in various EAC harmonized guidelines on registration of human medicinal products. The terms are defined in context of EAC Human Medicines Registration Harmonization Project as they are confined to the respective regulations and guidelines that are intended to be used in EAC Region.

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4. GLOSSARY

In the context of EAC-Human Medicines Registration Harmonization the following words/phrases are defined as follows:

Active pharmaceutical An active ingredient is any component that provides ingredient (API) pharmacological activity or other direct effect in the diagnosis, cure, mitigation, treatment, or prevention of disease, or to affect the structure or any function of the body of man or animals.

(USFDA Glossary of terms, it can be found in line at
Drugs@FDA Glossary of Terms).

Acceptance criteria The product specifications and acceptance/rejection criteria, such as acceptable quality level and unacceptable quality level, with an associated sampling plan, that are necessary for making a decision to accept or reject a lot or batch (or any other convenient subgroups of manufactured units).

(WHO guide to good manufacturing practice (GMP) requirements, at <http://www.who.int/vaccines-documents/DocsPDF/www9651.pdf>).

Active Pharmaceutical See Drug Master File (DMF)
Ingredient Master File-
(APIMF)

Active Substance See Active pharmaceutical ingredient (API)

Adverse reaction (Adverse An adverse drug reaction is a response to a medicinal Drug Reaction, ADR) product which is noxious and unintended and which occurs at doses normally used in man for the prophylaxis, diagnosis or therapy of disease or for the restoration, correction or modification of physiological function.

At:<http://www.who.unc.org/DynPage.aspx?id=13111&mn=1513>]

Applicant See Marketing Authorization Holder

Batch (or lot) A defined quantity of material produced in a process or

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series of processes so that it is expected to be homogeneous within specified limits.

(European Federation of Pharmaceutical Industries and Association; GMP for API manufacturer, at <http://apic.cefic.org/pub/1gmp-api9604.pdf>).

Batch number (or lot A distinctive combination of numbers and/or letters which number) specifically identifies a batch or lot and from which the production history can be determined.

(European Federation of Pharmaceutical Industries and Association; GMP for API manufacturer, at <http://apic.cefic.org/pub/1gmp-api9604.pdf>).

Bio-equivalence The absence of a significant difference in the rate and extent to which the active ingredient or active moiety in pharmaceutical equivalents or pharmaceutical alternatives becomes available at the site of action when administered at the same molar dose under similar conditions in an appropriately designed study.

(Glossary (terms and abbreviations)/EMA).

Bulk product Any product that has completed all processing stages up to, but not including, final packaging.

(WHO guide to good manufacturing practice (GMP) requirements, at <http://www.who.int/vaccines-documents/DocsPDF/www9651.pdf>).

Calibration A set of operations which establish, under specified conditions, the relationship between values indicated by a measuring instrument or measuring system, or values represented by a material measure, and the corresponding known values of a reference standard.

(European Federation of Pharmaceutical Industries and Association; GMP for API manufacturer, at <http://apic.cefic.org/pub/1gmp-api9604.pdf>).

Certificate of Pharmaceutical WHO-type certificate as defined in the WHO Certification Product (CPP) Scheme on the quality of pharmaceutical products moving in international commerce.

(WHO Model Quality Assurance System for Procurement Agencies; it can be found at <http://www.myaccessrh.org/documents/10157/37547/ModelQualityAssurance.pdf>).

Clinical trial (clinical study) A clinical trial is any systematic study on pharmaceutical

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products in human subjects, whether in patients or other volunteers, in order to discover or verify the effects of, and/or identify any adverse reaction to, investigational products, and/or to study the absorption, distribution, metabolism and excretion of the products with the object of ascertaining their efficacy and safety.

(WHO Medicines Regulatory Package. A Collection of Tools for Medicines Regulatory Authorities - Regulatory Support Series No. 014 at: <http://infocollections.org/medregpack/interface/files/glossary.pdf>)

Comparator An investigational or marketed product (i.e., active control), or placebo, used as a reference in a clinical trial.

(<http://www.gcpelpdesk.com/index.php/glossary/10-c>)

Comparator product A pharmaceutical product with which the multisource product is intended to be interchangeable in clinical practice

(https://www.who.int/medicines/areas/quality_safety/quality_assurance/guidanceontheselectionofcomparatorpharmproducts-etc_qas14-596_18072014.pdf)

Composition Composition in relation to a medicinal product means the ingredients of which it consists, proportions, degree of strength, quality and purity in which those ingredients are contained.

Kenya and Guidelines on submission for Documentation for Registration of Human medicinal Product-TMDA).

Conflict of interest A conflict of interest is a situation in which a public official's decisions are influenced by the official's personal interests.

[At: <http://wordnet.princeton.edu/>]

Contamination The unintended, non-process related, introduction of impurities of a chemical or microbiological nature, or of foreign matter, into or onto a material during production,

sampling, packaging or repackaging, storage or transport.
(European Federation of Pharmaceutical Industries and Association; GMP for API manufacturer, at <http://apic.cefic.org/pub/1gmp-api9604.pdf>).

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Continuous production A process in which a material is continuously produced in a step or series of steps. In a continuous process the batches of raw materials and the process parameters can be statistically, but not absolutely, correlated to the material produced in a given window of time.

(European Federation of Pharmaceutical Industries and Association; GMP for API manufacturer, at <http://apic.cefic.org/pub/1gmp-api9604.pdf>).

Controlled Medicines Narcotic medicines and psychotropic substances regulated by provisions of national medicines laws.

(WHO Medicines Regulatory Package. A Collection of Tools for Medicines Regulatory Authorities - Regulatory Support Series No. 014 at: <http://infocollections.org/medregpack/interface/files/glossary.pdf>)

Cross contamination Contamination of a material or product with another material or product, thus cross contamination is a particular form of contamination.

(European Federation of Pharmaceutical Industries and Association; GMP for API manufacturer, at <http://apic.cefic.org/pub/1gmp-api9604.pdf>).

Data exclusivity Data exclusivity is the protection of an originator pharmaceutical company's data preventing other parties from using these data for a commercial purpose.

(OECD – Pharmaceutical Pricing Policies in a Global Market, at: http://www.oecd.org/document/36/0,3343,en_2649_33929_41000996_1_1_1_37407,00.html).

Design Space The multidimensional combination and interaction of input variables (e.g., material attributes) and process parameters that have been demonstrated to provide assurance of quality.

(ICHQ8- Glossary at http://www.ema.europa.eu/docs/en_GB/document_library/Scientific_guideline/2009/09/WC500002872.pdf)

Direct to consumer Direct-to-consumer advertising (DTC advertising) usually advertising refers to the marketing of medicines aimed directly toward the public, rather than healthcare professionals. Forms of DTC advertising include TV, print, and radio.

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(OECD – Pharmaceutical Pricing Policies in a Global Market, at: http://www.oecd.org/document/36/0,3343,en_2649_33929_41000996_1_1_1_37407,00.html).

Distribution category Distribution category indicates how a drug product is sold or dispensed. For example Prescription Only Medicines (POM), Over the Counter (OTC) refer to national guideline).(WHO glossary of terms)

Dosage form See pharmaceutical form

Drug Master File (DMF) Is a master file that provides a full set of data on an active pharmaceutical ingredient (API). In other circumstances the term may also comprise data on an excipient.

(Guidelines to Submission of Applications for Registration of Drug, Pharmacy and Poisons Board-Kenya).

Drug Product A finished dosage form, for example, a tablet, capsule or solution that contains an active pharmaceutical ingredient, generally, but not necessarily, in association with inactive ingredients. Reference: Manufacturing, Processing, or Holding Active Pharmaceutical Ingredients FDA Guidance (<https://www.registrarcorp.com/fdadugs/definitions/>)

Drug Substance See Active pharmaceutical ingredient (API)

Marketing authorization issued to a product that is Duplicate license identical in terms of qualitative and quantitative composition, manufacturing process and sites as well as quality controls to an already registered medicinal product. The only difference would be the brand name and product labels.

Efficacy The ability of a drug to produce the intended effect as determined by scientific methods, for example in pre-clinical research or clinical research studies.

(WHO Glossary of terms used in Pharmacovigilance, at <http://who-umc.org/Graphics/24729.pdf>).

Essential medicines Essential medicines satisfy the priority health care needs of the population. They are selected with due regard to public

health relevance, evidence on efficacy and safety, and comparative cost-effectiveness.

(At: http://www.who.int/topics/essential_medicines/en/)

Ethics Committee (EC)/ Ethics Committees (EC) ensure that biomedical research Institutional Review Board follows international guidelines, including the Declaration (IRB) of Helsinki, the WHO and ICH Guidelines for Good Clinical Practice. The purpose of an EC in reviewing biomedical research is to contribute to safeguarding the dignity, rights, safety, and well-being of all actual or potential research participants. (Operational Guidelines for Ethics Committees That Review Biomedical Research Geneva 2000, can be found online at: <http://apps.who.int/tdr/publications/training-guideline-publications/operational-guidelines-ethics-biomedical-research/pdf/ethics.pdf>)

Excipient Is any constituent of a pharmaceutical form that is not an active pharmaceutical ingredient

(Guideline on excipients in the dossier for application for marketing authorization of a medicinal product, it can be found on line at

http://www.ema.europa.eu/docs/en_GB/document_library/Scientific_guideline/2009/09/WC500003380.pdf).

Extension application A new application that is a modification/addition to an already registered medicinal product. The modification/addition shall be such that it does not fulfil criteria for minor or major variations but is similar enough to the original (already registered) product in terms of quality, safety and efficacy. A new marketing authorization will be issued for extension applications.

Finished Pharmaceutical Product (FPP) A finished dosage form of a pharmaceutical product which has undergone all stages of manufacture, including packaging in its final container and labelling (WHO Glossary)

Formulary A formulary is a manual containing clinically oriented summaries of pharmacological information about selected medicines.

(How to develop a national formulary based on the WHO model 269

formulary, a practical guide Geneva 2004, can be found online at: <http://apps.who.int/medicinedocs/en/d/Js6171e/2.3.html>)

General Sales Medicines Medicines which may be sold either by way of retail or (GSM) wholesale in an open shop such as supermarkets.

Generic product Is a medicinal product which has the same qualitative and quantitative composition in active substances and the same pharmaceutical form as the reference medicinal product, and whose bioequivalence with the reference medicinal product has been demonstrated by appropriate bioavailability studies.

(PHIS Glossary 2009, can be found on line at:

<http://phis.goeg.at/index.aspx?alias=phisglossary>)

Good Clinical Practice A standard for the design, conduct, performance, monitoring, auditing, recording, analyses, and reporting of clinical trials that provides assurance that the data and reported results are credible and accurate, and that the rights, integrity, and confidentiality of trial subjects are protected.

(PHIS Glossary 2009, can be found on line at:

<http://phis.goeg.at/index.aspx?alias=phisglossary>)

Good Manufacturing Practice (GMP) Part of quality assurance which ensures that products are consistently produced and controlled to the quality standards appropriate to their intended use and as required by the marketing authorization.

(WHO A model quality assurance system for procurement agencies Geneva 2007, can be found online at:

<http://apps.who.int/medicinedocs/documents/s14866e/s14866e.pdf>)

Impurity Any component present in the active pharmaceutical ingredient other than the substance defined as the active pharmaceutical ingredient

(European Federation of Pharmaceutical Industries and Association; GMP for API manufacturer,

At <http://apic.cefic.org/pub/1gmp-api9604.pdf>).

Innovator medicinal product Generally the medicinal product that was first authorized for marketing (normally as a patented product) on the basis of documentation of efficacy, safety and quality. (WHO glossary of terms)

In-process control Checks performed during production in order to monitor and, if necessary, to adjust the process, including repeating a process step, to ensure that the product conforms to its specification.

(European Federation of Pharmaceutical Industries and Association; GMP for API manufacturer, at <http://apic.cefic.org/pub/1gmp-api9604.pdf>).

Intermediate Partly processed material which must undergo further production steps before it becomes an Active Ingredient. (European Federation of Pharmaceutical Industries and Association; GMP for API manufacturer, at <http://apic.cefic.org/pub/1gmp-api9604.pdf>).

International Conference on The International Conference on Harmonization of harmonisation of Technical Technical Requirements for Registration of Requirements for Pharmaceuticals for Human Use (ICH) is a project that Registration of brings together the regulatory authorities of Europe, Japan Pharmaceuticals for Human and the United States and experts from the pharmaceutical Use (ICH) industry in the three regions to discuss scientific and technical aspects of product registration. [At: <http://www.ich.org/cache/compo/276-254-1.html>]

Reference product Pharmaceutical product with which the new product is intended to be interchangeable in clinical practice. The reference product will normally be the innovator product for which efficacy, safety and quality have been established. Where the innovator product is not available, the product which is the market leader may be used as a reference product, provided that it has been authorized for marketing and its efficacy, safety and quality have been established and documented.

(<http://apps.who.int/medicinedocs/en/d/Js5516e/19.2.htm#Js5516e.19.2>)

International Non- INN is a unique name that is globally recognized and is proprietary Name (INN) public property. [WHO Guidance on INN at: <http://www.who.int/medicines/services/inn/innguidance/en/index.html>]

Label Is a descriptive matter, written, printed, stenciled, marked, 271 embossed or impressed on or attached to a packaging of any medicinal product. (Adapted from USFDA Glossary of terms, can be found in line at Drugs@FDA Glossary of Terms).

Law Laws define the roles, rights and obligations of all parties involved in the subject matter in general terms. (WHO Medicines Regulatory Package. A Collection of Tools for Medicines Regulatory Authorities - Regulatory Support Series No. 014 at: <http://infocollections.org/medregpack/interface/files/glossary.pdf>)

Legal category See Distribution category

Legislation Legislation corresponds to the first stage of the legislative process, in which laws are passed by the legislative body of government with regard to a subject matter such as the control of pharmaceuticals. (WHO A model quality assurance system for procurement agencies Geneva 2007, can be found online at: <http://apps.who.int/medicinedocs/documents/s14866e/s14866e.pdf>)

License Holder A license holder is an individual or a corporate entity possessing a marketing authorization for a medicinal product. (WHO Medicines Regulatory Package. A Collection of Tools for Medicines Regulatory Authorities - Regulatory Support Series No. 014 at: <http://infocollections.org/medregpack/interface/files/glossary.pdf>) (Also see Market authorization Holder).

Licensing system National legal provisions on who should manufacture, import or supply medicinal products, what qualifications people in the supplying agency should have, and who should dispense and sell pharmaceutical products. (WHO glossary of terms)

Local Technical A person or company with sufficient pharmaceutical Representative expertise that is incorporated within the specific country and who will be responsible for facilitating communication with the Applicant and when the product is registered shall assume all legal responsibilities.

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Manufacture Manufacturing includes all operations of receipt of (manufacturing) materials, production, packaging, repackaging, labelling, relabeling, quality control, release, storage and distribution

of active pharmaceutical ingredients and/or medicinal product.

[PHIS Glossary 2009, can be found on line at:

<http://phis.goeg.at/index.aspx?alias=phisglossary>]

Manufacturer A manufacturer is a natural or legal person with responsibility for manufacturing of a medicinal product or active pharmaceutical ingredient.

(PHIS Glossary 2009, can be found on line at:

<http://phis.goeg.at/index.aspx?alias=phisglossary>)

Market Authorization Holder Is a person who holds authorization to place a medicinal product in the EAC Partner States and is responsible for that product.

Marketing Authorization Means approval to market a medicinal product in an EAC (MA) Partner States.

(Glossary of terms and abbreviations/EMA, it can be found at

http://www.ema.europa.eu/docs/en_GB/document_library/Other/2010/12/WC500099907.pdf).

Medical device Means an article which is intended to be used for human beings or animals for the purpose of diagnosis, prevention, monitoring, treatment or alleviation of disease, diagnosis, monitoring, treatment, alleviation of or compensation for an injury or handicap, investigation, replacement or modification of the anatomy or of a physiological process, or control of conception and does not achieve its purpose by pharmacological, immunological or metabolic means.

(Upholding standards and public trust in pharmacy, at

<http://www.pharmacyregulation.org/sites/default/files/Glossary%20of%20terms%20used%20in%20GPhC%20standards%20Feb%202012.pdf>)

Medicinal Product Any substance or combination of substances which may be administered to human beings or animals with a view to making a medical diagnosis or to restoring, prevention, correcting or modifying physiological functions in human beings or animals.

Medicinal Substance See Active pharmaceutical ingredient (API)

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Medicines Regulatory Authority A national body that has the legal mandate to set objectives and administer the full spectrum of medicines regulatory activities.

(WHO Medicines Regulatory Package. A Collection of Tools for Medicines Regulatory Authorities - Regulatory Support Series No. 014 at:

<http://infocollections.org/medregpack/interface/files/glossary.pdf>).

National essential medicines The list of essential medicines that has been defined, list adopted, and published at country level.

(WHO A model quality assurance system for procurement agencies Geneva 2007, can be found online at:

<http://apps.who.int/medicinedocs/documents/s14866e/s14866e.pdf>)

Originator medicinal product An originator brand is generally the product that was first product/originator brand authorized worldwide for marketing (normally as a patented product) on the basis of the documentation of its efficacy, safety and quality, according to requirements at the time of authorization.

(HAI/WHO Measuring medicine prices, availability, affordability and price components (2nd Edition) and at:

<http://www.haiweb.org/medicineprices/manual/documents.html>)

Over-The-Counter medicines Are medicines which are safe and effective for use by the general public without a doctor's prescription.

(WHO Medicines Regulatory Package. A Collection of Tools for Medicines Regulatory Authorities Regulatory Support Series No. 014 at:

<http://infocollections.org/medregpack/interface/files/glossary.pdf>)

Packaging materials Any material used to protect an Active Pharmaceutical Ingredient or finished pharmaceutical product during storage and transport but excluding labels.(European Federation of Pharmaceutical Industries and Association; GMP for API manufacturer, at <http://apic.cefic.org/pub/1gmp-api9604.pdf>).

Patient Information Leaflet Packages insert which contains information for patient's (PIL) understanding of how to safely use a medicinal product.

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(USFDA Glossary of terms, can be found in line at [Drugs@FDA](http://www.fda.gov/oc/ohrt/Drugs@FDA) Glossary of Terms).

Pharmaceutical alternatives Medicinal products are considered pharmaceutical alternatives if they contain the same therapeutic moiety, but are different salt, esters, or complexes of that moiety, or are different dosage forms or strengths.

(USFDA Orange book, it can be found on line at

http://www.fda.gov/ohrms/dockets/ac/05/briefing/2005-4137B1_07_Nomenclature.pdf)

Pharmaceutical equivalents Medicinal products are considered to be pharmaceutical equivalents if they contain the same active ingredient(s) same dosage form and route of administration and they are identical in strength or concentration. (USFDA Glossary of terms, can be found in line at [Drugs@FDA Glossary of Terms](#))

Pharmaceutical form The pharmaceutical form is the pharmaceutical-technological form in which an active substance is made available. Pharmaceutical may be administered in solid form (e.g. tablets, powers), in semi-liquid form (e.g. ointments, pastes), in liquid form (e.g. drops, injectables, infusions) or in gaseous form (inhalation). (WHO glossary of terms).

Pharmaceutical Product A pharmaceutical product is any substance for human or veterinary use that is intended to modify or explore physiological systems or pathological states for the benefit of the recipient.

[WHO A model quality assurance system for procurement agencies Geneva 2007, can be found online at:

<http://apps.who.int/medicinedocs/documents/s14866e/s14866e.pdf>]

Pharmacy Pharmacies are premises which in accordance to the local legal provisions and definitions may operate as a facility in the provision of pharmacy services in the community or health facility setting.

(In WHO Operational package for assessing, monitoring and evaluating country pharmaceutical situations at:

http://www.who.int/medicines/publications/WHO_TCM_2007.2/en/)

Post-marketing surveillance Post-marketing surveillance is testing medicine samples to assess the quality of medicines that have already been

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licensed for public use.

(In WHO Operational package for assessing, monitoring and evaluating country pharmaceutical situations at:

http://www.who.int/medicines/publications/WHO_TCM_2007.2/en/)

Post-marketing surveillance Studies performed after the pharmaceutical product has study been marketed.

(In WHO Operational package for assessing, monitoring and evaluating country pharmaceutical situations at:

http://www.who.int/medicines/publications/WHO_TCM_2007.2/en/)

Pre-marketing The stage before a drug is available for prescription or sale to the public.(WHO Glossary of terms used in

Pharmacovigilance,

At <http://who-umc.org/Graphics/24729.pdf>

Prescription-Only Medicines Prescription-only medicines are medicines supplied only in licensed pharmacies on the presentation of signed prescriptions issued by a licensed and registered medical practitioner, licensed and/or registered dentist (for dental treatment only), and/or licensed and/or registered veterinarian (for animal treatment only), and the supply and dispensing of these medicines must be carried out by a pharmacist or under the supervision of a pharmacist. (WHO glossary of terms)

Procedures Description of the operations to be carried out, the precautions to be taken, and measures to be applied directly or indirectly related to the manufacture of an Active Ingredient and Finished Pharmaceutical product.

(European Federation of Pharmaceutical Industries and Association; GMP for API manufacturer, at

<http://apic.cefic.org/pub/1gmp-api9604.pdf>).

Process aids Materials used as aids in the manufacture of an Active Ingredient and Finished Pharmaceutical Product which themselves do not participate in a chemical or biological reaction.

(Federation of Pharmaceutical Industries and Association; GMP for API manufacturer, at <http://apic.cefic.org/pub/1gmp-api9604.pdf>).

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Product Information Product information refers to the summary of product characteristics (SmPC), labelling and patient information leaflet.

(Glossary of terms and abbreviations/EMA it can be found at

http://www.ema.europa.eu/docs/en_GB/document_library/Other/2010/12/WC500099907.pdf).

Promotion Promotion refers to all informational and persuasive activities by manufacturers and distributors, the effect of

which is to induce the prescription, supply, purchase and/or use of medicinal drugs.
[C:\Documents and Settings\CVialle\Desktop\Country profile - Instructions and glossary 14 Sept 2010\WHO. A model quality assurance system for procurement agencies.pdf Criteria for Medicinal Drug Promotion can be found online at:
<http://apps.who.int/medicinedocs/documents/whozip08e/whozip08e.pdf>]

Proprietary name Name given for marketing purposes to any ready-prepared medicine placed on the market.
(PHIS Glossary 2009, it can be found at http://phis.goeg.at/downloads/glossary/PHIS%20Glossary_UpdatedApril2011.pdf).

Quality Information The QIS is a condensed version of Summary (QIS) the Quality Overall Summary – Product Dossier (QOS-PD) and represents the final, agreed upon key information from the PD review (inter alia identification of the manufacturer(s), API/FPP specifications, stability conclusions and relevant commitments)

Qualification The action of proving that any equipment is properly installed, works correctly, and consistently produces the expected results. Qualification is part of, but not limited to, the validation process. (Federation of Pharmaceutical Industries and Association; GMP for API manufacturer, at <http://apic.cefic.org/pub/1gmp-api9604.pdf>).

Quality assurance It is the sum total of the organized arrangements made with the object of ensuring that Active Ingredients and Finished Pharmaceutical products are of the quality required for their intended use.
(Federation of Pharmaceutical Industries and Association; GMP for API manufacturer, at <http://apic.cefic.org/pub/1gmp-api9604.pdf>).

Quality attribute Any product characteristic which may reflect quality, or may affect safety or efficacy of the product during its expected shelf life.
(Federation of Pharmaceutical Industries and Association; GMP for API manufacturer, at <http://apic.cefic.org/pub/1gmp-api9604.pdf>).

Quality Control Quality control is the part of Good Manufacturing Practices (GMP) concerned with sampling, specifications, and testing and with the organization, documentation, and release procedures which ensure that the necessary and relevant tests are actually carried out and that materials are not released for use or products released for sale or supply, until their quality has been judged to be satisfactory.
(WHO Medicines Regulatory Package. A Collection of Tools for Medicines Regulatory Authorities - Regulatory Support Series No. 014 at: <http://infocollections.org/medregpack/interface/files/glossary.pdf>).

Quarantine The status of materials isolated physically or by other effective means whilst awaiting a decision on their subsequent use.
(Federation of Pharmaceutical Industries and Association; GMP for API manufacturer, at <http://apic.cefic.org/pub/1gmp-api9604.pdf>).

Rational use of medicines Rational use of medicines requires that patients receive medications appropriate to their clinical needs, in doses that meet their own individual requirements, for an adequate period of time, and at the lowest cost to them and their community.
(Promoting rational use of medicines: Core components Geneva 2002, can be found online at: <http://apps.who.int/medicinedocs/pdf/h3011e/h3011e.pdf>)

Raw materials Any material of defined quality used in the manufacture of an Active Ingredient, but excluding packaging materials or labels.
(European Federation of Pharmaceutical Industries and Associations; April 1996, Good manufacturing practices for Active ingredient manufacturers).

Recovery Any treatment of materials by a process intended to make them suitable for further use.
(Federation of Pharmaceutical Industries and Association; GMP for API manufacturer, at <http://apic.cefic.org/pub/1gmp-api9604.pdf>).

Registration See Marketing Authorization

Regulations The second stage of the legislative process (the first stage being legislation). Regulations are specifically designed to

provide the legal machinery to achieve the administrative and technical goals of legislation.

(WHO A model quality assurance system for procurement agencies Geneva 2007, can be found online at:

<http://apps.who.int/medicinedocs/documents/s14866e/s14866e.pdf>
f)

Regulatory Inspection A regulatory inspection is an officially conducted examination (i.e. review of quality assurance processes, personnel involved, any delegation of authority and audit) by relevant authorities at sites where pharmaceutical activities take place (i.e. manufacturing, wholesale, testing, distribution, clinical trials) to verify adherence to Good Practices.

(WHO Medicines Regulatory Package. A Collection of Tools for Medicines Regulatory Authorities - Regulatory Support Series No. 014 at:

<http://infocollections.org/medregpack/interface/files/glossary.pdf>)

Reprocessing The treatment of a batch or sub-batch of materials of unacceptable quality by repeating the same process steps from a defined stage of production so that its quality may be made acceptable.

(Federation of Pharmaceutical Industries and Association; GMP for API manufacturer, at <http://apic.cefic.org/pub/1gmp-api9604.pdf>).

Reworking The treatment of a batch or sub-batch of materials of unacceptable quality by using a process other than that used to produce the original material so that its quality may be made acceptable.

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(Federation of Pharmaceutical Industries and Association; GMP for API manufacturer, at <http://apic.cefic.org/pub/1gmp-api9604.pdf>).

Route of administration Is a way of administering a medicinal product to a site in a patient.

(USFDA Glossary of terms, can be found on line at Drugs@FDA Glossary of Terms).

Sample A sample is a portion of a material collected according to a defined sampling procedure.

WHO Medicines Regulatory Package. A Collection of Tools for Medicines Regulatory Authorities - Regulatory Support Series No. 014 at:

<http://infocollections.org/medregpack/interface/files/glossary.pdf>

Sampling Operations designed to obtain a representative portion of a pharmaceutical product, based on an appropriate statistical procedure, for a defined purpose, e.g. acceptance of consignments, batch release.

(WHO Medicines Regulatory Package. A Collection of Tools for Medicines Regulatory Authorities - Regulatory Support Series No. 014 at:

<http://infocollections.org/medregpack/interface/files/glossary.pdf>)

Side effect Any unintended effect of a pharmaceutical product occurring at normal dosage which is related to the pharmacological properties of the drug.

(WHO Glossary of terms used in Pharmacovigilance, at <http://who-umc.org/Graphics/24729.pdf>)

Specifications A document describing in detail the requirements such as physical, chemical, biological and microbiological test requirements with which the products or materials used or obtained during manufacture have to conform.

(A WHO guide to good manufacturing practice (GMP) requirements; it can be found at

<http://www.who.int/vaccinesdocuments/DocsPDF/www9651.pdf>
).

Specifications Test Procedures and Acceptance Criteria for active pharmaceutical ingredients and medicinal products.

(ICHQ8- Glossary at

http://www.ema.europa.eu/docs/en_GB/document_library/Scientific_guideline/2009/09/WC500002872.pdf)

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Standard operating An authorized written procedure giving instructions for procedure performing operations not necessarily specific to a given product or material but of a more general nature.

(WHO guide to good manufacturing practice (GMP)

requirements, at <http://www.who.int/vaccines-documents/DocsPDF/www9651.pdf>).

Stringent Regulatory A regulatory authority that is:

Authority a) A member of ICH prior to 23rd October 2015, namely the USFDA, the European Commission and the Ministry of Health, Labour and Welfare of Japan also represented by the Pharmaceuticals and Medical

Devices Agency, or

b) An ICH observer prior to 23rd October 2015, namely the European Free Trade Association, as represented by Swissmedic and Health Canada, or

c) A regulatory authority associated with an ICH member through a legally-binding, mutual recognition agreement prior to 23rd October 2015, namely Australia, Iceland, Liechtenstein and Norway.

Summary of Product Characteristics (SmPC) Authority. The SmPC serves as the basis for production of information for health personnel as well as for consumer information on labels and leaflets of medicinal products and for control of advertising.

(WHO Medicines Regulatory Package. A Collection of Tools for Medicines Regulatory Authorities - Regulatory Support Series

No. 014 at:

<http://infocollections.org/medregpack/interface/files/glossary.pdf>)

Tentative Approval If a generic drug product is ready for approval before the expiration of any patents or exclusivities accorded to the reference listed drug product, FDA issues a tentative approval letter to the applicant. The tentative approval letter details the circumstances associated with the tentative approval. FDA delays final approval of the generic drug product until all patent or exclusivity issues have been resolved. A tentative approval does not allow the applicant to market the generic drug product.

(USFDA Glossary of terms, it can be found on line at Drugs@FDA Glossary of Terms).

Theoretical yield The quantity that would be produced at any appropriate phase of manufacture, processing, or packaging of a

particular drug product, based upon the quantity of components to be used, in the absence of any loss or an error in actual production.

(WHO guide to good manufacturing practice (GMP) requirements, at <http://www.who.int/vaccines-documents/DocsPDF/www9651.pdf>).

Therapeutic Equivalence Medicinal products are considered to be therapeutically (TE) equivalent only if they are pharmaceutical equivalents or pharmaceutical alternatives and their effect are essentially the same. This can be and have been scientifically demonstrated to be bioequivalent.

(Adapted from WHO glossary of terms)

Unique identifier Is a unique code that is added to the medicinal product label (primary and/or secondary pack) in order to specifically identify and capture particulars of the product for market surveillance purposes. It may be in form of a code, barcode or security number that is unique for a specific product. The product registration number issued by the NMRA may be considered as a unique identifier.

Validation Action of proving and documenting that any procedure, process, equipment, activity or system will, with a high degree of assurance, lead to the expected results.

(Federation of Pharmaceutical Industries and Association; GMP for API manufacturer, at <http://apic.cefic.org/pub/1gmp-api9604.pdf>).

Variation Variation is a change to a Marketing Authorization that is considered to fundamentally alter the terms of the MA for a medicinal product.

(Glossary of terms and abbreviations/EMA it can be found at http://www.ema.europa.eu/docs/en_GB/document_library/Other/2010/12/WC500099907.pdf).

Wholesale All activities consisting of procuring, holding, supplying or exporting bulk medicinal products, apart from supplying medicinal products to the public.

(PHIS Glossary 2009, can be found on line at: <http://phis.goeg.at/index.aspx?alias=phisglossary>).

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PART VIII:

GUIDELINES ON LIST OF STANDARD TERMS FOR PHARMACEUTICAL DOSAGE FORMS AND ROUTES OF ADMINISTRATION

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ABBREVIATIONS AND ACRONYMS

NMRA - National Medicine Regulatory Authority

EAC - East Africa Community

CTD - Common Technical Document

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1. INTRODUCTION AND GUIDANCE FOR USE

1.1 Scope

This list of standard terms for pharmaceutical dosage forms and routes of administration will assist in knowing all dosage forms used and routes used, accurate dose, protected dosage forms e.g. coated tablets, sealed ampoules, masked taste and odour, placement of drugs within body tissues, sustained release medication, controlled release medication, optimal drug action, insertion of drugs into body cavities (rectal, vaginal) and Use of desired vehicle for insoluble drugs. It has the double purpose to bring information to user (patient/prescriber) and distinguishing medicinal products having the same trade/generic name. Because of labelling purposes it is imperative that any Standard Term and combination of Standard Terms is constructed with a view to the patient

However, information on the container and the route of administration need not always be included in the Standard Term but may appear elsewhere in the labelling, package leaflet and SmPC.

1.2 Guidance for use

The EAC List of Standard Terms covers dosage forms and routes of administration for the use in the marketing authorization application, Summary of Product Characteristics (SmPC), Patient Information leaflet and labelling of medicinal product for human use.

1.2.1 Definitions

For the purposes of the Standard Terms, the following definitions apply.

1.2.1.1 Pharmaceutical form

The pharmaceutical form may be:

- a) a dosage form;
- b) a combination of dosage forms; or
- c) a combination of dosage form(s) and route(s)/method(s) of administration and/or container/administration device.

In the assessment of marketing authorization applications, pharmaceutical forms that differ only with respect to the containers/administration devices may not always be considered as different pharmaceutical forms.

1.2.1.2 Dosage form:

The dosage form is the physical manifestation of a medicinal product that contains the active ingredient(s) and/or excipient(s) that are intended to be delivered to the patient; it

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may refer to the form of presentation or the form of administration, which in some cases are identical.

a) Form of presentation:

The form of presentation is the dosage form of a medicinal product as manufactured and, where applicable, before reconstitution; where reconstitution is required before administration to the patient, the term includes the eventual form of administration.

Examples: Powder for solution for injection;
Tablet

b) Form of administration

The form of administration is the dosage form of a medicinal product as administered to the patient, after any necessary reconstitution has been carried out.

Examples: Solution for injection;
Tablet

1.2.1.3 Combined term

A combined term is a combination of existing Standard Terms or elements thereof that is constructed in order to properly characterize a medicinal product; a combined term may be a combination of dosage forms, or a combination of dosage form(s) and route(s)/method(s) of administration and/or container/administration device.

Examples: Powder and solution for solution for injection;
Eye drops, solution in single-dose container.

1.2.2 Pharmaceutical forms

The list of Standard Terms does not distinguish between medicinal products as presented by the manufacturer (form of presentation) and medicinal products as administered to the patient (form of administration). However, for a term representing a form of presentation such as 'Powder for solution for injection', the words 'for solution for injection' indicate that a reconstitution is required, and that the resulting form of administration is 'solution for injection'.

The label of the medicinal product may be too small to permit the inclusion of the Standard Term(s). In addition to the Standard Terms given in the Dosage forms section, a number of patient-friendly terms (generally shortened versions of existing terms), which may be used for labelling only, in case of space limitation, are also proposed. Where a term contains two or more dosage form elements, these elements are linked by 'and'; e.g. 'Powder and solvent for solution for injection'

If the same pharmaceutical form may be used in alternative ways, these ways are separated by '/', e.g. 'Gargle/mouthwash', 'Chewable/dispersible tablet'.

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In the case of a powder that is dissolved in a small amount of solvent before it is diluted in a larger volume to be infused and this dilution is mandatory for safety reasons, the

term 'concentrate' should appear in the pharmaceutical form (e.g. 'Powder for concentrate for solution for infusion'). If the powder that is dissolved in a small amount of solvent can either be administered as such or be further diluted before administration (i.e. no safety issue), there is no need to use the term 'concentrate' (e.g. 'Powder for solution for infusion').

The term 'modified-release' is not sufficiently precise for describing a particular product. A more specific term such as 'prolonged-release' or 'gastro-resistant' should be used, wherever applicable.

1.2.3 Routes of administration

The route of administration indicates the part of the body on which, through which or into which the medicinal product is to be introduced. The short terms proposed may be used for labelling only.

Where several routes of administration are intended for a medicinal product, the focus should be placed on the primary use for the creation of a standard term or a combination of standard terms, for example 'Oral use' is sufficient as the primary use for a request of 'Oral/gastric/gastroenteral use'.

1.2.4 Procedure for the addition or modification of terms in the list of standard terms

1.3 Pharmaceutical forms and short terms

Standard terms short terms

BATH ADDITIVE BATH ADDITIVE
BLADDER IRRIGATION BLADDER IRRIGATION
BAR CHEWABLE BAR CHEWABLE
BLOOD FRACTION MODIFIER BLOOD FRACTION MODIFIER
BUCCAL FILM BUCCAL FILM
BUCCAL TABLET BUCCAL TABLET
CACHET CACHET
CAPSULE CAPSULE
CAPSULE, HARD SHELL CAPSULE
CAPSULE, SOFT SHELL CAPSULE
CHEWABLE CAPSULE, SOFT CHEWABLE CAPSULE
CHEWABLE TABLET CHEWABLE TABLET
COATED GRANULES IN SACHET GRANULES
COATED TABLET TABLET
COLLODION COLLODION
COMPRESSED LOZENGE LOZENGE

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CONCENTRATE AND SOLVENT FOR CONCENTRATE AND SOLVENT FOR
CONCENTRATE FOR SOLUTION FOR CONCENTRATE FOR SOLUTION FOR
INFUSION INFUSION
CONCENTRATE AND SOLVENT FOR CONCENTRATE AND SOLVENT FOR
CUTANEOUS SOLUTION CUTANEOUS SOLUTION
CONCENTRATE AND SOLVENT FOR CONCENTRATE AND SOLVENT FOR
SOLUTION FOR INFUSION SOLUTION FOR INFUSION
CONCENTRATE AND SOLVENT FOR CONCENTRATE AND SOLVENT FOR
SOLUTION FOR INJECTION SOLUTION FOR INJECTION
CONCENTRATE AND SOLVENT FOR CONCENTRATE AND SOLVENT FOR
SOLUTION FOR INJECTION/INFUSION SOLUTION FOR INJECTION/INFUSION
CONCENTRATE AND SOLVENT FOR CONCENTRATE AND SOLVENT FOR
SUSPENSION FOR INJECTION SUSPENSION FOR INJECTION
CONCENTRATE FOR CUTANEOUS CONCENTRATE FOR CUTANEOUS
SOLUTION SOLUTION
CONCENTRATE FOR CUTANEOUS CONCENTRATE FOR CUTANEOUS
SPRAY, EMULSION SPRAY, EMULSION
CONCENTRATE FOR DISPERSION FOR CONCENTRATE FOR DISPERSION FOR
INFUSION INFUSION
CONCENTRATE FOR EMULSION FOR CONCENTRATE FOR EMULSION FOR
INFUSION INFUSION
CONCENTRATE FOR GARGLE CONCENTRATE FOR GARGLE
CONCENTRATE FOR HAEMODIALYSIS CONCENTRATE FOR HAEMODIALYSIS
SOLUTION SOLUTION
CONCENTRATE SOLUTION FOR CONCENTRATE SOLUTION FOR
INTRAVESICAL USE INTRAVESICAL USE
CONCENTRATE FOR ORAL SOLUTION CONCENTRATE FOR ORAL SOLUTION
CONCENTRATE FOR ORAL SUSPENSION CONCENTRATE FOR ORAL SUSPENSION
CONCENTRATE FOR ORAL/RECTAL CONCENTRATE FOR ORAL/RECTAL
SOLUTION SOLUTION
CONCENTRATE FOR RECTAL CONCENTRATE FOR RECTAL
SOLUTION SOLUTION
CONCENTRATE FOR SOLUTION FOR STERILE CONCENTRATE
INFUSION
CONCENTRATE FOR SOLUTION FOR STERILE CONCENTRATE
INJECTION
CONCENTRATE FOR SOLUTION FOR STERILE CONCENTRATE
INJECTION/INFUSION
CONCENTRATE FOR SOLUTION FOR CONCENTRATE FOR SOLUTION FOR
PERITONEAL DIALYSIS PERITONEAL DIALYSIS
CREAM CREAM
CUTANEOUS EMULSION CUTANEOUS LIQUID
CUTANEOUS FOAM CUTANEOUS FOAM
CUTANEOUS PASTE CUTANEOUS PASTE
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CUTANEOUS PATCH CUTANEOUS PATCH
CUTANEOUS POWDER CUTANEOUS POWDER
CUTANEOUS SOLUTION CUTANEOUS LIQUID

CUTANEOUS SPONGE CUTANEOUS SPONGE
CUTANEOUS CUTANEOUS
SOLUTION/CONCENTRATE FOR SOLUTION/CONCENTRATE FOR
OROMUCOSAL SOLUTION OROMUCOSAL SOLUTION
CUTANEOUS PATCH CUTANEOUS PATCH
CUTANEOUS SPRAY CUTANEOUS SPRAY
CUTANEOUS SPRAY, EMULSION CUTANEOUS SPRAY
CUTANEOUS SPRAY, OINTMENT CUTANEOUS SPRAY
CUTANEOUS SPRAY, POWDER CUTANEOUS SPRAY
CUTANEOUS SPRAY, SOLUTION CUTANEOUS SPRAY
CUTANEOUS SPRAY, SUSPENSION CUTANEOUS SPRAY
CUTANEOUS STICK CUTANEOUS STICK
CUTANEOUS SUSPENSION CUTANEOUS LIQUID
CUTANEOUS OINTMENT CUTANEOUS OINTMENT
DENTAL EMULSION DENTAL LIQUID
DENTAL GEL DENTAL GEL
DENTAL INSERT DENTAL INSERT
DENTAL LIQUID DENTAL LIQUID
DENTAL PASTE DENTAL PASTE
DENTAL POWDER DENTAL POWDER
DENTAL SOLUTION DENTAL LIQUID
DENTAL STICK DENTAL STICK
DENTAL SUSPENSION DENTAL LIQUID
DENTURE LACQUER DENTURE LACQUER
DISPERSIBLE TABLET DISPERSIBLE TABLET
DISPERSION DISPERSION
DISPERSION FOR INJECTION DISPERSION FOR INJECTION
EAR CREAM EAR CREAM
EAR DROPS EAR DROPS
EAR DROPS, EMULSION EAR DROPS
EAR DROPS, POWDER AND SOLVENT EAR DROPS, POWDER AND SOLVENT
FOR SUSPENSION FOR SUSPENSION
EAR DROPS, SOLUTION EAR DROPS
EAR DROPS, SUSPENSION EAR DROPS
EAR DROPS, SUSPENSION IN SINGLE- DOSE CONTAINER DOSE CONTAINER
EAR GEL EAR GEL
EAR OINTMENT EAR OINTMENT
EAR POWDER EAR POWDER
EAR SPRAY, EMULSION EAR SPRAY

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EAR SPRAY, SOLUTION EAR SPRAY
EAR SPRAY, SUSPENSION EAR SPRAY
EAR STICK EAR STICK
EAR TAMPON EAR TAMPON
EAR WASH, EMULSION EAR WASH
EAR WASH, SOLUTION EAR WASH
EAR/EYE DROPS, SOLUTION EAR/EYE DROPS, SOLUTION
EAR/EYE DROPS, SUSPENSION EAR/EYE DROPS, SUSPENSION
EAR/EYE OINTMENT EAR/EYE OINTMENT
EAR/EYE/NASAL DROPS, SOLUTION EAR/EYE/NASAL DROPS, SOLUTION
EFFERVESCENT GRANULES EFFERVESCENT GRANULES
EFFERVESCENT POWDER EFFERVESCENT POWDER
EFFERVESCENT TABLET EFFERVESCENT TABLET
EFFERVESCENT VAGINAL TABLET EFFERVESCENT VAGINAL TABLET
EMULSION FOR INFUSION INFUSION
EMULSION FOR INJECTION INJECTION
EMULSION FOR INJECTION/INFUSION INJECTION/INFUSION
ENDOCERVICAL GEL ENDOCERVICAL GEL
ENDOSINUSIAL WASH, SUSPENSION ENDOSINUSIAL WASH, SUSPENSION
ENDOTRACHEOPULMONARY ENDOTRACHEOPULMONARY
INSTILLATION INSTILLATION
ENDOTRACHEOPULMONARY ENDOTRACHEOPULMONARY
INSTILLATION, POWDER AND INSTILLATION
SOLVENT FOR SOLUTION
ENDOTRACHEOPULMONARY ENDOTRACHEOPULMONARY
INSTILLATION, POWDER FOR INSTILLATION
SOLUTION
ENDOTRACHEOPULMONARY ENDOTRACHEOPULMONARY
INSTILLATION, SOLUTION INSTILLATION
ENDOTRACHEOPULMONARY ENDOTRACHEOPULMONARY
INSTILLATION, SUSPENSION INSTILLATION
ENEMA ENEMA
EYE CREAM EYE CREAM
EYE DROPS EYE DROPS
EYE DROPS, EMULSION EYE DROPS, EMULSION
EYE DROPS, POWDER AND SOLVENT EYE DROPS, POWDER AND SOLVENT
FOR SOLUTION FOR SOLUTION
EYE DROPS, POWDER AND SOLVENT EYE DROPS, POWDER AND SOLVENT
FOR SUSPENSION FOR SUSPENSION
EYE DROPS, PROLONGED-RELEASE EYE DROPS, PROLONGED-RELEASE
EYE DROPS, PROLONGED-RELEASE EYE DROPS, PROLONGED-RELEASE
SOLUTION IN SINGLE-DOSE SOLUTION IN SINGLE-DOSE
CONTAINER CONTAINER

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EYE DROPS, SOLUTION EYE DROPS
EYE DROPS, SOLUTION IN SINGLE-DOSE EYE DROPS, SOLUTION IN SINGLE-DOSE
CONTAINER CONTAINER
EYE DROPS, SOLVENT FOR EYE DROPS, SOLVENT FOR
RECONSTITUTION RECONSTITUTION

EYE DROPS, SUSPENSION EYE DROPS
EYE DROPS, SUSPENSION IN SINGLE- DOSE CONTAINER DOSE CONTAINER
EYE GEL EYE GEL
EYE GEL IN SINGLE-DOSE CONTAINER EYE GEL IN SINGLE-DOSE CONTAINER
EYE LOTION EYE LOTION
EYE LOTION, SOLVENT FOR EYE LOTION, SOLVENT FOR
RECONSTITUTION RECONSTITUTION
EYE OINTMENT EYE OINTMENT
EYE OINTMENT IN SINGLE-DOSE EYE OINTMENT IN SINGLE-DOSE
CONTAINER CONTAINER
FILM-COATED TABLET TABLET
GARGLE GARGLE
GARGLE, POWDER FOR SOLUTION GARGLE, POWDER FOR SOLUTION
GARGLE, TABLET FOR SOLUTION GARGLE, TABLET FOR SOLUTION
GARGLE/MOUTHWASH GARGLE/MOUTHWASH
GARGLE/NASAL WASH GARGLE/NASAL WASH
GAS AND SOLVENT FOR DISPERSION GAS AND SOLVENT FOR DISPERSION
FOR INJECTION/INFUSION FOR INJECTION/INFUSION
GASTROENTERAL EMULSION GASTROENTERAL LIQUID
GASTROENTERAL LIQUID GASTROENTERAL LIQUID
GASTROENTERAL SOLUTION GASTROENTERAL LIQUID
GASTROENTERAL SUSPENSION GASTROENTERAL LIQUID
GASTRO-RESISTANT CAPSULE GASTRO-RESISTANT CAPSULE
GASTRO-RESISTANT CAPSULE, HARD GASTRO-RESISTANT CAPSULE
GASTRO-RESISTANT CAPSULE, SOFT GASTRO-RESISTANT CAPSULE
GASTRO-RESISTANT GRANULES GASTRO-RESISTANT GRANULES
GASTRO-RESISTANT GRANULES FOR GASTRO-RESISTANT GRANULES FOR
ORAL SUSPENSION ORAL SUSPENSION
GASTRO-RESISTANT TABLET GASTRO-RESISTANT TABLET
GEL GEL
GINGIVAL GEL GINGIVAL GEL
GINGIVAL PASTE GINGIVAL PASTE
GINGIVAL SOLUTION GINGIVAL SOLUTION
GRANULES GRANULES
GRANULES AND SOLVENT FOR ORAL GRANULES AND SOLVENT FOR ORAL
SUSPENSION SUSPENSION
GRANULES AND SOLVENT FOR GRANULES AND SOLVENT FOR

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SUSPENSION FOR INJECTION SUSPENSION FOR INJECTION
GRANULES FOR ORAL SOLUTION GRANULES FOR ORAL SOLUTION
GRANULES FOR ORAL SUSPENSION GRANULES FOR ORAL SUSPENSION
GRANULES FOR ORAL/RECTAL GRANULES FOR ORAL/RECTAL
SUSPENSION SUSPENSION
GRANULES FOR ORAL DROPS, GRANULES FOR ORAL DROPS,
SOLUTION SOLUTION
GRANULES FOR RECTAL SUSPENSION GRANULES FOR RECTAL SUSPENSION
GRANULES FOR SYRUP GRANULES FOR SYRUP
GRANULES FOR VAGINAL SOLUTION GRANULES FOR VAGINAL SOLUTION
IMPLANT IMPLANT
IMPLANT IN PRE-FILLED SYRINGE IMPLANT IN PRE-FILLED SYRINGE
IMPLANTATION CHAIN IMPLANTATION CHAIN
IMPLANTATION TABLET IMPLANTATION TABLET
IMPREGNATED DRESSING IMPREGNATED DRESSING
IMPREGNATED PAD IMPREGNATED PAD
IMPREGNATED PLUG IMPREGNATED PLUG
INFUSION INFUSION
INHALATION GAS INHALATION GAS
INHALATION POWDER INHALATION POWDER
INHALATION POWDER, HARD INHALATION POWDER
CAPSULE
INHALATION POWDER, PRE- INHALATION POWDER
DISPENSED
INHALATION POWDER, TABLET INHALATION POWDER
INHALATION SOLUTION INHALATION SOLUTION
INHALATION VAPOUR INHALATION VAPOUR
INHALATION VAPOUR, CAPSULE INHALATION VAPOUR
INHALATION VAPOUR, EFFERVESCENT INHALATION VAPOUR, EFFERVESCENT
TABLET TABLET
INHALATION VAPOUR, EMULSION INHALATION VAPOUR
INHALATION VAPOUR, IMPREGNATED INHALATION VAPOUR
PAD
INHALATION VAPOUR, LIQUID INHALATION VAPOUR
INHALATION VAPOUR, OINTMENT INHALATION VAPOUR
INHALATION VAPOUR, POWDER INHALATION VAPOUR, POWDER
INHALATION VAPOUR, SOLUTION INHALATION VAPOUR
INHALATION VAPOUR, TABLET INHALATION VAPOUR
INJECTION INJECTION
INTESTINAL GEL INTESTINAL GEL
INTRAPERITONEAL SOLUTION INTRAPERITONEAL SOLUTION
INTRAUTERINE DELIVERY SYSTEM INTRAUTERINE DELIVERY SYSTEM
INTRAUTERINE FOAM INTRAUTERINE FOAM

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INTRAUTERINE LIQUID INTRAUTERINE LIQUID
INTRAVESICAL SOLUTION INTRAVESICAL SOLUTION
IRRIGATION SOLUTION IRRIGATION SOLUTION
LOZENGE LOZENGE
LIQUEFIED GAS FOR DENTAL USE LIQUEFIED GAS FOR DENTAL USE
LYOPHILISATE FOR OCULONASAL LYOPHILISATE FOR OCULONASAL
SUSPENSION SUSPENSION

LYOPHILISATE FOR USE IN DRINKING LYOPHILISATE FOR USE IN DRINKING
 WATER WATER
 MEDICATED CHEWING-GUM MEDICATED CHEWING-GUM
 MEDICATED NAIL LACQUER MEDICATED NAIL LACQUER
 MEDICATED PLASTER MEDICATED PLASTER
 MEDICATED SPONGE MEDICATED SPONGE
 MEDICATED THREAD MEDICATED THREAD
 MEDICATED VAGINAL TAMPON MEDICATED VAGINAL TAMPON
 MEDICINAL GAS, COMPRESSED MEDICINAL GAS, COMPRESSED
 MEDICINAL GAS, CRYOGENIC MEDICINAL GAS, CRYOGENIC
 MEDICINAL GAS, LIQUEFIED MEDICINAL GAS, LIQUEFIED
 MODIFIED-RELEASE CAPSULE, HARD MODIFIED-RELEASE CAPSULE, HARD
 MODIFIED-RELEASE CAPSULE, SOFT MODIFIED-RELEASE CAPSULE, SOFT
 MODIFIED-RELEASE GRANULES MODIFIED-RELEASE GRANULES
 MODIFIED-RELEASE GRANULES FOR MODIFIED-RELEASE GRANULES FOR
 ORAL SUSPENSION ORAL SUSPENSION
 MODIFIED-RELEASE TABLET MODIFIED-RELEASE TABLET
 MOUTHWASH MOUTHWASH
 MOUTHWASH, POWDER FOR MOUTHWASH, POWDER FOR
 SOLUTION SOLUTION
 MOUTHWASH, TABLET FOR SOLUTION MOUTHWASH, TABLET FOR SOLUTION
 MUCO-ADHESIVE BUCCAL TABLET MUCO-ADHESIVE BUCCAL TABLET
 NASAL CREAM NASAL CREAM
 NASAL DROPS NASAL DROPS
 NASAL DROPS, EMULSION NASAL DROPS
 NASAL DROPS, SOLUTION NASAL DROPS
 NASAL DROPS, SOLUTION IN SINGLE- NASAL DROPS, SOLUTION IN SINGLE-
 DOSE CONTAINER DOSE CONTAINER
 NASAL DROPS, SUSPENSION NASAL DROPS
 NASAL GEL NASAL GEL
 NASAL OINTMENT NASAL OINTMENT
 NASAL POWDER NASAL POWDER
 NASAL SPRAY NASAL SPRAY
 NASAL SPRAY, EMULSION NASAL SPRAY
 NASAL SPRAY, POWDER FOR NASAL POWDER
 SOLUTION

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NASAL SPRAY, SOLUTION NASAL SPRAY,
 NASAL SPRAY, SOLUTION IN SINGLE- NASAL SPRAY, SOLUTION IN SINGLE-
 DOSE CONTAINER DOSE CONTAINER
 NASAL SPRAY, NASAL SPRAY,
 SOLUTION/ROMUCOSAL SOLUTION SOLUTION/ROMUCOSAL SOLUTION
 NASAL SPRAY, SUSPENSION NASAL SPRAY, SUSPENSION
 NASAL STICK NASAL STICK
 NASAL WASH NASAL WASH
 NASAL/ROMUCOSAL SOLUTION NASAL/ROMUCOSAL SOLUTION
 NASAL/ROMUCOSAL SPRAY, NASAL/ROMUCOSAL SPRAY,
 SOLUTION SOLUTION
 NEBULISER EMULSION NEBULISER LIQUID
 NEBULISER LIQUID NEBULISER LIQUID
 NEBULISER SOLUTION NEBULISER LIQUID
 NEBULISER SUSPENSION NEBULISER LIQUID
 OINTMENT OINTMENT
 OPHTHALMIC INSERT OPHTHALMIC INSERT
 OPHTHALMIC STRIP OPHTHALMIC STRIP
 ORAL DROPS ORAL DROPS
 ORAL DROPS, EMULSION ORAL DROPS
 ORAL DROPS, GRANULES FOR ORAL DROPS, GRANULES FOR
 SOLUTION SOLUTION
 ORAL DROPS, LIQUID ORAL DROPS
 ORAL DROPS, POWDER FOR ORAL DROPS, POWDER FOR
 SUSPENSION SUSPENSION
 ORAL DROPS, SOLUTION ORAL DROPS
 ORAL DROPS, SUSPENSION ORAL DROPS
 ORAL EMULSION ORAL LIQUID
 ORAL GEL ORAL GEL
 ORAL GUM ORAL GUM
 ORAL LIQUID ORAL LIQUID
 ORAL LYOPHILISATE ORAL LYOPHILISATE
 ORAL PASTE ORAL PASTE
 ORAL POWDER ORAL POWDER
 ORAL SOLUTION ORAL LIQUID
 ORAL SOLUTION IN SINGLE-DOSE ORAL SOLUTION IN SINGLE-DOSE
 CONTAINER CONTAINER
 ORAL SOLUTION/CONCENTRATE FOR ORAL SOLUTION/CONCENTRATE FOR
 NEBULISER SOLUTION NEBULISER SOLUTION
 ORAL SUSPENSION ORAL LIQUID
 ORAL/RECTAL SOLUTION ORAL/RECTAL LIQUID
 ORAL/RECTAL SUSPENSION ORAL/RECTAL LIQUID
 ORODISPERSIBLE FILM ORODISPERSIBLE FILM

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ORODISPERSIBLE TABLET ORODISPERSIBLE TABLET
 OROMUCOSAL CAPSULE OROMUCOSAL CAPSULE
 OROMUCOSAL CREAM OROMUCOSAL CREAM
 OROMUCOSAL DROPS OROMUCOSAL DROPS
 OROMUCOSAL GEL OROMUCOSAL GEL
 OROMUCOSAL LIQUID OROMUCOSAL LIQUID
 OROMUCOSAL OINTMENT OROMUCOSAL OINTMENT
 OROMUCOSAL PASTE OROMUCOSAL PASTE
 OROMUCOSAL PATCH OROMUCOSAL PATCH

OROMUCOSAL POWDER IN POUCH OROMUCOSAL POWDER IN POUCH
OROMUCOSAL SOLUTION OROMUCOSAL LIQUID
OROMUCOSAL SPRAY, EMULSION OROMUCOSAL SPRAY
OROMUCOSAL SPRAY, SOLUTION OROMUCOSAL SPRAY
OROMUCOSAL SPRAY, SUSPENSION OROMUCOSAL SPRAY
OROMUCOSAL SUSPENSION OROMUCOSAL LIQUID
OROMUCOSAL/LARYNGOPHARYNGEA OROMUCOSAL/LARYNGOPHARYNGEA
L SOLUTION L SOLUTION
PASTILLE PASTILLE
PERIODONTAL GEL PERIODONTAL GEL
PERIODONTAL INSERT PERIODONTAL INSERT
PERIODONTAL POWDER PERIODONTAL POWDER
PESSARY PESSARY
PILLULES PILLULES
PILLULES IN SINGLE-DOSE CONTAINER PILLULES IN SINGLE-DOSE CONTAINER
PLASTER FOR PROVOCATION TEST PLASTER FOR PROVOCATION TEST
POUCH POUCH
POULTICE POULTICE
POWDER AND SOLVENT FOR POWDER AND SOLVENT FOR
SOLUTION FOR INJECTION SOLUTION FOR INJECTION
POWDER AND SOLVENT FOR POWDER AND SOLVENT FOR
CONCENTRATE FOR SOLUTION FOR CONCENTRATE FOR SOLUTION FOR
INFUSION INFUSION
POWDER AND SOLVENT FOR POWDER AND SOLVENT FOR
CUTANEOUS SOLUTION CUTANEOUS SOLUTION
POWDER AND SOLVENT FOR POWDER AND SOLVENT FOR
DISPERSION FOR INJECTION DISPERSION FOR INJECTION
POWDER AND SOLVENT FOR POWDER AND SOLVENT FOR
EMULSION FOR INJECTION EMULSION FOR INJECTION
POWDER AND SOLVENT FOR POWDER AND SOLVENT FOR
ENDOCERVICAL GEL ENDOCERVICAL GEL
POWDER AND SOLVENT FOR POWDER AND SOLVENT FOR
ENDOSINUSIAL SOLUTION ENDOSINUSIAL SOLUTION
POWDER AND SOLVENT FOR POWDER AND SOLVENT FOR

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EPILESIONAL SOLUTION EPILESIONAL SOLUTION
POWDER AND SOLVENT FOR GINGIVAL POWDER AND SOLVENT FOR GINGIVAL
GEL GEL
POWDER AND SOLVENT FOR POWDER AND SOLVENT FOR
IMPLANTATION PASTE IMPLANTATION PASTE
POWDER AND SOLVENT FOR POWDER AND SOLVENT FOR
INSTILLATION SOLUTION FOR INSTILLATION SOLUTION FOR
INTRAOCULAR USE INTRAOCULAR USE
POWDER AND SOLVENT FOR POWDER AND SOLVENT FOR
INTRAVESICAL SOLUTION INTRAVESICAL SOLUTION
POWDER AND SOLVENT FOR POWDER AND SOLVENT FOR
INTRAVESICAL SUSPENSION INTRAVESICAL SUSPENSION
POWDER AND SOLVENT FOR POWDER AND SOLVENT FOR
NEBULISER SOLUTION NEBULISER SOLUTION
POWDER AND SOLVENT FOR ORAL POWDER AND SOLVENT FOR ORAL
SOLUTION SOLUTION
POWDER AND SOLVENT FOR ORAL POWDER AND SOLVENT FOR ORAL
SUSPENSION SUSPENSION
POWDER AND SOLVENT FOR POWDER AND SOLVENT FOR
PROLONGED-RELEASE SUSPENSION PROLONGED-RELEASE SUSPENSION
FOR INJECTION FOR INJECTION
POWDER AND SOLVENT FOR SEALANT POWDER AND SOLVENT FOR SEALANT
POWDER AND SOLVENT FOR POWDER AND SOLVENT FOR
SOLUTION FOR INFUSION SOLUTION FOR INFUSION
POWDER AND SOLVENT FOR POWDER AND SOLVENT FOR
SOLUTION FOR INJECTION SOLUTION FOR INJECTION
POWDER AND SOLVENT FOR POWDER AND SOLVENT FOR
SOLUTION FOR INJECTION IN SOLUTION FOR INJECTION IN
CARTRIDGE CARTRIDGE
POWDER AND SOLVENT FOR POWDER AND SOLVENT FOR
SOLUTION FOR INJECTION IN PRE- SOLUTION FOR INJECTION IN PRE-
FILLED PEN FILLED PEN
POWDER AND SOLVENT FOR POWDER AND SOLVENT FOR
SOLUTION FOR INJECTION IN PRE- SOLUTION FOR INJECTION IN PRE-
FILLED SYRINGE FILLED SYRINGE
POWDER AND SOLVENT FOR POWDER AND SOLVENT FOR
SOLUTION FOR INJECTION/ SKIN- SOLUTION FOR INJECTION/ SKIN-
PRICK TEST PRICK TEST
POWDER AND SOLVENT FOR POWDER AND SOLVENT FOR
SOLUTION FOR INJECTION/INFUSION SOLUTION FOR INJECTION/INFUSION
POWDER AND SOLVENT FOR POWDER AND SOLVENT FOR
SUSPENSION FOR INJECTION SUSPENSION FOR INJECTION
POWDER AND SOLVENT FOR POWDER AND SOLVENT FOR
SUSPENSION FOR INJECTION IN PRE- SUSPENSION FOR INJECTION IN PRE-
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FILLED SYRINGE FILLED SYRINGE
POWDER AND SUSPENSION FOR POWDER AND SUSPENSION FOR
SUSPENSION FOR INJECTION SUSPENSION FOR INJECTION
POWDER FOR BLADDER IRRIGATION POWDER FOR BLADDER IRRIGATION
POWDER FOR CONCENTRATE FOR POWDER FOR CONCENTRATE FOR
DISPERSION FOR INFUSION DISPERSION FOR INFUSION
POWDER FOR CONCENTRATE FOR POWDER FOR CONCENTRATE FOR
INTRAVESICAL SUSPENSION INTRAVESICAL SUSPENSION
POWDER FOR CONCENTRATE FOR POWDER FOR CONCENTRATE FOR
SOLUTION FOR HAEMODIALYSIS SOLUTION FOR HAEMODIALYSIS
POWDER FOR CONCENTRATE FOR POWDER FOR CONCENTRATE FOR

SOLUTION FOR INFUSION SOLUTION FOR INFUSION
 POWDER FOR CONCENTRATE FOR POWDER FOR CONCENTRATE FOR
 SOLUTION FOR INJECTION/INFUSION SOLUTION FOR INJECTION/INFUSION
 POWDER FOR CUTANEOUS SOLUTION POWDER FOR CUTANEOUS SOLUTION
 POWDER FOR DENTAL SOLUTION POWDER FOR DENTAL SOLUTION
 POWDER FOR DISPERSION FOR POWDER FOR DISPERSION FOR
 INFUSION INFUSION
 POWDER FOR EPILESIONAL SOLUTION POWDER FOR EPILESIONAL SOLUTION
 POWDER FOR IMPLANTATION POWDER FOR IMPLANTATION
 SUSPENSION SUSPENSION
 POWDER FOR INFUSION POWDER FOR INFUSION
 POWDER FOR INJECTION POWDER FOR INJECTION
 POWDER FOR INTRAVESICAL POWDER FOR INTRAVESICAL
 SOLUTION SOLUTION
 POWDER FOR INTRAVESICAL POWDER FOR INTRAVESICAL
 SOLUTION/SOLUTION FOR INJECTION SOLUTION/SOLUTION FOR INJECTION
 POWDER FOR INTRAVESICAL POWDER FOR INTRAVESICAL
 SUSPENSION SUSPENSION
 POWDER FOR NEBULISER SOLUTION POWDER FOR NEBULISER SOLUTION
 POWDER FOR NEBULISER SUSPENSION POWDER FOR NEBULISER SUSPENSION
 POWDER FOR ORAL SOLUTION POWDER FOR ORAL SOLUTION
 POWDER FOR ORAL SUSPENSION POWDER FOR ORAL SUSPENSION
 POWDER FOR ORAL/RECTAL POWDER FOR ORAL/RECTAL
 SUSPENSION SUSPENSION
 POWDER FOR RECTAL SOLUTION POWDER FOR RECTAL SOLUTION
 POWDER FOR RECTAL SUSPENSION POWDER FOR RECTAL SUSPENSION
 POWDER FOR SOLUTION FOR INFUSION POWDER FOR SOLUTION FOR INFUSION
 POWDER FOR SOLUTION FOR POWDER FOR SOLUTION FOR
 INJECTION INJECTION
 POWDER FOR SOLUTION FOR POWDER FOR SOLUTION FOR
 INJECTION/INFUSION INJECTION/INFUSION
 POWDER FOR SOLUTION FOR POWDER FOR SOLUTION FOR
 INTRAOCULAR IRRIGATION INTRAOCULAR IRRIGATION
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 POWDER FOR SOLUTION FOR POWDER FOR SOLUTION FOR
 IONTOPHORESIS IONTOPHORESIS
 POWDER FOR SUSPENSION FOR POWDER FOR SUSPENSION FOR
 INJECTION INJECTION
 POWDER FOR SYRUP POWDER FOR SYRUP
 POWDER, DISPERSION AND SOLVENT POWDER, DISPERSION AND SOLVENT
 FOR CONCENTRATE FOR DISPERSION FOR CONCENTRATE FOR DISPERSION
 FOR INFUSION FOR INFUSION
 PRESSURISED INHALATION PRESSURISED INHALATION
 PRESSURISED INHALATION, EMULSION PRESSURISED INHALATION, EMULSION
 PRESSURISED INHALATION, SOLUTION PRESSURISED INHALATION, SOLUTION
 PRESSURISED INHALATION, PRESSURISED INHALATION,
 SUSPENSION SUSPENSION
 PROLONGED-RELEASE CAPSULE PROLONGED-RELEASE CAPSULE
 PROLONGED-RELEASE CAPSULE, HARD PROLONGED-RELEASE CAPSULE
 PROLONGED-RELEASE CAPSULE, SOFT PROLONGED-RELEASE CAPSULE
 PROLONGED-RELEASE GRANULES PROLONGED-RELEASE GRANULES
 PROLONGED-RELEASE GRANULES FOR PROLONGED-RELEASE GRANULES FOR
 ORAL SUSPENSION ORAL SUSPENSION
 PROLONGED-RELEASE SUSPENSION PROLONGED-RELEASE SUSPENSION
 FOR INJECTION FOR INJECTION
 PROLONGED-RELEASE SUSPENSION PROLONGED-RELEASE SUSPENSION
 FOR INJECTION IN PRE-FILLED SYRINGE FOR INJECTION IN PRE-FILLED SYRINGE
 PROLONGED-RELEASE TABLET PROLONGED-RELEASE TABLET
 RADIONUCLIDE GENERATOR RADIONUCLIDE GENERATOR
 RADIOPHARMACEUTICAL PRECURSOR RADIOPHARMACEUTICAL PRECURSOR
 RADIOPHARMACEUTICAL PRECURSOR, RADIOPHARMACEUTICAL PRECURSOR,
 SOLUTION SOLUTION
 RECTAL CAPSULE RECTAL CAPSULE
 RECTAL CREAM RECTAL CREAM
 RECTAL EMULSION RECTAL EMULSION
 RECTAL FOAM RECTAL FOAM
 RECTAL GEL RECTAL GEL
 RECTAL OINTMENT RECTAL OINTMENT
 RECTAL SOLUTION ENEMA
 RECTAL SUSPENSION ENEMA
 RECTAL TAMPON RECTAL TAMPON
 SEALANT SEALANT
 SHAMPOO SHAMPOO
 SOLUBLE TABLET SOLUBLE TABLET
 SOLUTION AND SUSPENSION FOR SOLUTION AND SUSPENSION FOR
 SUSPENSION FOR INJECTION IN PRE- SUSPENSION FOR INJECTION IN PRE-
 FILLED SYRINGE FILLED SYRINGE

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SOLUTION FOR BLOOD FRACTION SOLUTION FOR BLOOD FRACTION
 MODIFICATION MODIFICATION
 SOLUTION FOR CARDIOPLEGIA SOLUTION FOR CARDIOPLEGIA
 SOLUTION FOR SOLUTION FOR
 HAEMODIAFILTRATION HAEMODIAFILTRATION
 SOLUTION FOR HAEMODIALYSIS SOLUTION FOR HAEMODIALYSIS
 SOLUTION FOR SOLUTION FOR
 HAEMODIALYSIS/HAEMOFILTRATION HAEMODIALYSIS/HAEMOFILTRATION
 SOLUTION FOR HAEMOFILTRATION SOLUTION FOR HAEMOFILTRATION
 SOLUTION FOR INFUSION INTRAVENOUS INFUSION
 SOLUTION FOR INFUSION IN SOLUTION FOR INFUSION IN
 ADMINISTRATION SYSTEM ADMINISTRATION SYSTEM
 SOLUTION FOR INFUSION IN PRE- SOLUTION FOR INFUSION IN PRE-

FILLED SYRINGE FILLED SYRINGE
SOLUTION FOR INJECTION INJECTION
SOLUTION FOR INJECTION IN SOLUTION FOR INJECTION IN
CARTRIDGE CARTRIDGE
SOLUTION FOR INJECTION IN NEEDLE- SOLUTION FOR INJECTION IN NEEDLE-
FREE INJECTOR FREE INJECTOR
SOLUTION FOR INJECTION IN PRE- SOLUTION FOR INJECTION IN PRE-
FILLED PEN FILLED PEN
SOLUTION FOR INJECTION IN PRE- SOLUTION FOR INJECTION IN PRE-
FILLED SYRINGE FILLED SYRINGE
SOLUTION FOR INJECTION/INFUSION SOLUTION FOR INJECTION/INFUSION
SOLUTION FOR INJECTION, SOLUTION FOR INJECTION,
LYOPHILISATE LYOPHILISATE
SOLUTION FOR INFUSION, SOLUTION FOR INFUSION,
LYOPHILISATE LYOPHILISATE
SOLUTION FOR INJECTION/INFUSION SOLUTION FOR INJECTION/INFUSION
IN PRE-FILLED SYRINGE IN PRE-FILLED SYRINGE
SOLUTION FOR IONTOPHORESIS SOLUTION FOR IONTOPHORESIS
SOLUTION FOR ORGAN PRESERVATION SOLUTION FOR ORGAN PRESERVATION
SOLUTION FOR PERITONEAL DIALYSIS SOLUTION FOR PERITONEAL DIALYSIS
SOLUTION FOR PROVOCATION TEST SOLUTION FOR PROVOCATION TEST
SOLUTION FOR SEALANT SOLUTION FOR SEALANT
SOLUTION FOR SKIN-PRICK TEST SOLUTION FOR SKIN-PRICK TEST
SOLUTION FOR SKIN-SCRATCH TEST SOLUTION FOR SKIN-SCRATCH TEST
SOLVENT FOR PARENTERAL USE SOLVENT FOR PARENTERAL USE
SOLVENT FOR SOLUTION FOR SOLVENT FOR SOLUTION FOR
INFUSION INFUSION
SOLVENT FOR SOLUTION FOR SOLVENT FOR SOLUTION FOR
INTRAOCULAR IRRIGATION INTRAOCULAR IRRIGATION
STERILE CONCENTRATE STERILE CONCENTRATE

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STOMACH IRRIGATION STOMACH IRRIGATION
SUBLINGUAL FILM SUBLINGUAL FILM
SUBLINGUAL SPRAY, EMULSION SUBLINGUAL SPRAY
SUBLINGUAL SPRAY, SOLUTION SUBLINGUAL SPRAY
SUBLINGUAL SPRAY, SUSPENSION SUBLINGUAL SPRAY
SUBLINGUAL TABLET SUBLINGUAL TABLET
SUPPOSITORY SUPPOSITORY
SUSPENSION AND EFFERVESCENT SUSPENSION AND EFFERVESCENT
GRANULES FOR ORAL SUSPENSION GRANULES FOR ORAL SUSPENSION
SUSPENSION AND SOLUTION FOR SUSPENSION AND SOLUTION FOR
SPRAY SPRAY
SUSPENSION AND SOLVENT FOR SUSPENSION AND SOLVENT FOR
SUSPENSION FOR INJECTION SUSPENSION FOR INJECTION
SUSPENSION FOR INFUSION SUSPENSION FOR INFUSION
SUSPENSION FOR INJECTION IN SUSPENSION FOR INJECTION IN
CARTRIDGE CARTRIDGE
SUSPENSION FOR INJECTION IN PRE- SUSPENSION FOR INJECTION IN PRE-
FILLED PEN FILLED PEN
SUSPENSION FOR INJECTION IN PRE- SUSPENSION FOR INJECTION IN PRE-
FILLED SYRINGE FILLED SYRINGE
SUSPENSION FOR INJECTION, SUSPENSION FOR INJECTION,
LYOPHILISATE LYOPHILISATE
SYRUP SYRUP
TABLET TABLET
TABLET AND POWDER FOR ORAL TABLET AND POWDER FOR ORAL
SOLUTION SOLUTION
TABLET AND SOLVENT FOR RECTAL TABLET AND SOLVENT FOR RECTAL
SUSPENSION SUSPENSION
TABLET FOR RECTAL SOLUTION TABLET FOR RECTAL SOLUTION
TABLET FOR ORAL SUSPENSION TABLET FOR ORAL SUSPENSION
TABLET FOR VAGINAL SOLUTION TABLET FOR VAGINAL SOLUTION
TOOTHPASTE TOOTHPASTE
TRANSDERMAL GEL TRANSDERMAL GEL
TRANSDERMAL PATCH TRANSDERMAL PATCH
TRANSDERMAL SOLUTION TRANSDERMAL SOLUTION
TRANSDERMAL SPRAY, SOLUTION TRANSDERMAL SPRAY
TRANSDERMAL PATCH TRANSDERMAL PATCH
URETHRAL GEL URETHRAL GEL
URETHRAL STICK URETHRAL STICK
VAGINAL CAPSULE VAGINAL CAPSULE
VAGINAL CAPSULE, HARD VAGINAL CAPSULE
VAGINAL CAPSULE, SOFT VAGINAL CAPSULE
VAGINAL CREAM VAGINAL CREAM

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VAGINAL DELIVERY SYSTEM VAGINAL DELIVERY SYSTEM
VAGINAL EMULSION VAGINAL LIQUID
VAGINAL FOAM VAGINAL FOAM
VAGINAL GEL VAGINAL GEL
VAGINAL OINTMENT VAGINAL OINTMENT
VAGINAL SOLUTION VAGINAL LIQUID
VAGINAL SUSPENSION VAGINAL LIQUID
VAGINAL TABLET VAGINAL TABLET
WOUND STICK WOUND STICK

1.4 Routes of Administration

NAMES SHORT TERM
AURICULAR OTIC
BUCCAL BUCCAL

CONJUNCTIVAL CONJUNC
CUTANEOUS CUTAN
DENTAL DENTAL
ENDOCERVICAL E-CERVIC
ENDOSINUSIAL E-SINUS
ENDOTRACHEAL E-TRACHE
ENDOTRACHEOPULMONARY
EPIDURAL EPIDUR
EPILESIONAL EPILESIONAL
EXTRA-AMNIOTIC X-AMNI
EXTRACORPOREAL X-CORPOR
GASTRIC
GASTROENTERAL
GINGIVAL
HEMODIALYSIS HEMO
IMPLANT
INFILTRATION INFIL
INHALATIONAL
INTERSTITIAL INTERSTIT
INTRA-ABDOMINAL I-ABDOM
INTRA-AMNIOTIC I-AMNI
INTRA-ARTERIAL I-ARTER
INTRA-ARTICULAR I-ARTIC
INTRABILIARY I-BILI
INTRABRONCHIAL I-BRONCHI
INTRABURSAL I-BURSAL
INTRACAMERAL
INTRACARDIAC I-CARDI

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INTRACARTILAGINOUS I-CARTIL
INTRACAUDAL I-CAUDAL
INTRACAVERNOUS I-CAVERN
INTRACAVITARY I-CAVIT
INTRACEREBRAL I-CERE
INTRACERVICAL
INTRACISTERNAL I-CISTERN
INTRACORNEAL I-CORNE
INTRACORONARY I-CORONARY
INTRACORPUS CAVERNOSUM I-CORPOR
INTRADERMAL I-DERMAL
INTRADISCAL I-DISCAL
INTRADUCTAL I-DUCTAL
INTRADUODENAL I-DUOD
INTRADURAL I-DURAL
INTRA-EPIDERMAL I-EPIDERM
INTRA-ESOPHAGEAL I-ESO
INTRAHEPATIC
INTRALESIONAL I-LESION
INTRALYMPHATIC I-LYMPHAT
INTRAMEDULLARY I-MEDUL
INTRAMENINGEAL I-MENIN
INTRAMUSCULAR IM
INTRAOCULAR I-OCUL
INTRAOSSEOUS
INTRAOVARIAN I-OVAR
INTRAPERICARDIAL I-PERICARD
INTRAPERITONEAL I-PERITON
INTRAPLEURAL I-PLEURAL
INTRAPROSTATIC I-PROSTAT
INTRAPULMONARY I-PULMON
INTRASINAL I-SINAL
INTRASYNOVIAL I-SYNOV
INTRASTERNAL
INTRATHECAL IT
INTRATUMORAL I-TUMOR
INTRAUTERINE I-UTER
INTRAVENOUS IV
INTRAVENOUS DRIP IV DRIP
INTRAVENOUS BOLUS IV BOLUS
INTRAVASCULAR I-VASC
INTRAVITREAL I-VITRE
IN VITRO

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IONTOPHORESIS ION
LARYNGOPHARYNGEAL LARYN
NASAL NASAL
NASOGASTRIC NG
OCCLUSIVE DRESSING TECHNIQUE OCCLUS
OPHTHALMIC OPHTHALM
ORAL ORAL
OROMUCOSAL
OROPHARYNGEAL ORO
OTHER OTHER
PARENTERAL PAREN
PERIARTICULAR P-ARTIC
PERICUTANEOUS PERCUT
PERINEURAL P-NEURAL
PERIODONTAL P-ODONT
PERIOSSEOUS
RECTAL RECTAL
RETROBULBAR RETRO

ROUTE OF ADMINISTRATION NOT NA
APPLICABLE
SUBCONJUNCTIVAL S-CONJUNC
SUBCUTANEOUS SC
SUBLINGUAL SL
SUBMUCOSAL S-MUCOS
TOPICAL TOPIC
TRANSDERMAL T-DERMAL
TRANSPLACENTAL T-PLACENT
URETERAL URETER
URETHRAL URETH
VAGINAL VAGIN

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2. REFERENCES

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2. XEVMPD Routes of Administration, Standard Term List, EMA/136146/2012, February 2012
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PART IX:

GUIDELINES ON REGISTRATION OF FIXED DOSE COMBINATION MEDICINAL PRODUCTS

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Abbreviations

AIHW Australian Institute of Health and Welfare
API Active pharmaceutical ingredient
BCS Biopharmaceutics Classification Scheme
BCS #1 Biopharmaceutics class number 1 (the most favourable)
CHMP Committee for Medicinal Products for Human Use; see also
CPMP Committee for Medicinal Products for Human Use (CHMP), formerly the
Committee for Proprietary Medicinal Products
CPP Certificate of pharmaceutical product
EMA European Medicines Agency, formerly the European Medicines Evaluation
Agency
EU European Union
FDA Food and Drug Administration of the USA
FDC Fixed-dose combination (see Glossary)
FDC-FPP Fixed-dose combination finished pharmaceutical product
FPP Finished Pharmaceutical Product
GCP Good Clinical Practice
GLP Good Laboratory Practice
GMP Good Manufacturing Practice
GSP Good Storage Practice
GTDP Good Trade and Distribution Practice
ICH International Conference on Harmonization
IUTLD International Union of Tuberculosis and Lung Disease
MIC Minimum Inhibitory Concentration
PP Per-Protocol (a form of clinical trial design and analysis)
SPC Summary of Product Characteristics (see Glossary)
TGA Therapeutic Goods Administration
WHO World Health Organization

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1. Scope

1.1 The scope of these guidelines is covers prescription and non-prescription medicines.

Similar principles and guidance provided in this document should apply to the registration of prescription and non-prescription products. Nevertheless, the risk–benefit considerations (and consequently data requirements) may be different.

FDCs are getting highly popular in the pharmaceutical markets of developing countries and have been particularly flourishing in the last few years. Unfortunately, scientific literature has provided evidence that many FDCs being introduced in certain countries are irrational. Regulatory authorities should take due care in implementing this guidance and can also take guidance of the World Health Organization's (WHO) Model List of Essential Medicines, which provides examples of some rational FDCs.

1.2 The principles in these guidelines would also apply to chemical combinations and complexes that comprise more than one active.

1.3 The scientific principles applicable to FDC products will also be applied in the assessment of co-packaged medicines.

2. General considerations

2.1 These guidelines are intended to be used in conjunction with the EAC guidelines on submission of documentation for registration of human medicinal products.

2.2 Appendices 2, 3 and 4 provide guidance on subjects that are not exclusive to FDCs, but

are nevertheless important in this context, and for which suitable guidance is not otherwise readily available.

2.3 It is important that access to useful, new FDCs should not be delayed by unnecessary constraints. These guidelines are not intended to define the only means of demonstrating the advantages and disadvantages of a new FDC. In some cases an alternative approach may be appropriate, for example when:

2.3.1 Scientific developments allow alternative means of achieving the same goals.

2.3.2 A circumstance unique to the product in question can be demonstrated.

2.3.3 An original but acceptable approach is devised.

2.3.4 Sufficient alternative studies have been conducted which, although not exactly what the guidelines seek, nevertheless satisfy the criteria of quality, safety and efficacy.

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When these guidelines (or others referred to herein) describe evidence that is required, applicants may either: provide the requested evidence, or provide an alternative form of evidence that addresses the same issues. In this case, the application should include an explanation and justification of the approach taken.

2.4 It is not always necessary to generate new (original) data. Evidence may be obtained from the scientific literature, subject to its being of adequate quality (see Appendix 2 entitled Principles for determining whether data from the scientific literature are acceptable).

An application for a marketing authorization may comprise:

2.4.1 Entirely original data.

2.4.2 Entirely data from the literature.

2.4.3 Both original data and data from the literature (a "generic" submission).

For FDC-FPPs, it is likely that generic submissions will be the most common type.

The scientific literature rarely contains enough adequately validated information on quality to allow the full quality data set to be based solely on data from the literature. In particular, the complete formulation and method of manufacture are rarely specified. Consequently the quality data set is almost always either totally original or generic.

2.5 When these guidelines request that an applicant explain and/or justify non-conformity with requirements, a suitable argument should be included in the section that discusses the advantages and disadvantages of the combination (see below), together with cross-references to data elsewhere in the submission.

2.6 When an applicant is unsure of registration requirements or wishes to deviate from these guidelines, prior consultation with the relevant regulatory authority may be advantageous. However, applicants should not request advice until they have read all relevant guidelines and WHO's Marketing authorization of pharmaceutical products with special reference to multisource (generic) products: a manual for a drug regulatory authority (1999) or updates thereof. Not all of the guidelines in Tables 1–5 are necessarily relevant to a particular enquiry; the particulars of each case should be considered.

2.7 Risk–benefit assessments for FDCs should take into consideration any differences in anticipated patient populations. Consequently, decisions on the same data set may vary between different national drug regulatory authorities.

3. Definitions

The definitions given below apply solely to the terms as used in these guidelines. They may have different meanings in other contexts.

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Active pharmaceutical ingredient (API)

Any substance or mixture of substances intended to be used in the manufacture of a pharmaceutical dosage form. When so used the API becomes the active moiety as defined below, often termed simply the active. The API may be a salt, hydrate or other form of the active moiety, or may be the active moiety itself. Active moieties are intended to furnish pharmacological activity or other direct effect in the diagnosis, cure, mitigation, treatment, or prevention of disease or to affect the structure and function of the body.

Active moiety

The term used for the therapeutically active entity in the final formulation of therapeutic goods, irrespective of the form of the API. The active is alternative terminology with the same meaning. For example, if the API is propranolol hydrochloride, the active moiety (the active) is propranolol.

Applicant

The person or company who submits an application for marketing authorization of a new pharmaceutical product, an update to an existing marketing authorization or a variation to an existing market authorization.

Certificate of pharmaceutical product

A WHO-type certificate of the form described in Guidelines for implementation of the WHO Certification Scheme on the quality of pharmaceutical products moving in international commerce. Geneva, World Health Organization, 1998.

Comparator

The finished pharmaceutical product with which an FDC-FPP is to be compared. The

comparison may be by means of bioequivalence studies or clinical studies of safety and/or effectiveness. A single study may use more than one comparator, for example several single entity FPPs. A comparator may be a placebo.

Co-packaged product

A product consisting of two or more separate pharmaceutical products in their final dosage forms that are packaged together for distribution to patients in the co-packaging.

Drug

Any substance or product for human or veterinary use that is intended to modify or explore physiological states for the benefit of the recipient.

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Finished pharmaceutical product (FPP)

A product that has undergone all stages of production, including packaging in its final container and labelling. An FPP may contain one or more actives.

Fixed-dose combination (FDC)

A combination of two or more actives in a fixed ratio of doses. This term is used generically to mean a particular combination of actives irrespective of the formulation or brand. It may be administered as single entity products given concurrently or as a finished pharmaceutical product.

Fixed-dose combination finished pharmaceutical product (FDC-FPP)

A finished pharmaceutical product that contains two or more actives.

Generic products

The term generic product has somewhat different meanings in different jurisdictions. Use of this term has therefore been avoided as far as possible, and the term multisource pharmaceutical product is used instead (see the definition below). Multisource products may be marketed either under the approved non-proprietary name or under a brand (proprietary) name. They may be marketed in dosage forms and/or strengths different to those of the innovator products.

Where the term generic product is used, it means a pharmaceutical product, usually intended to be interchangeable with the innovator product, which is usually manufactured without a licence from the innovator company and marketed after expiry of the patent or other exclusivity rights. The term should not be confused with generic names for APIs.

Microbiology

A branch of science that refers to microbes of all of types, including bacteria, viruses, rickettsia, protozoa, fungi and prions. Derived words (such as microbiological) have a similar meaning.

Multisource (generic) pharmaceutical product

Multisource pharmaceutical products are pharmaceutically equivalent products that may or may not be therapeutically equivalent. Multisource pharmaceutical products that are therapeutically equivalent are interchangeable.

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New chemical (or biological) entities

Actives that have not previously been authorized for marketing as a drug for use in humans in the country in question.

Pharmaceutical equivalents

Products are pharmaceutical equivalents if they contain the same amount of the same actives in the same dosage form, if they meet comparable standards, and if they are intended to be administered by the same route. Pharmaceutical equivalence does not necessarily imply therapeutic equivalence, as differences in the excipients and/or manufacturing process and some other variables can lead to differences in product performance.

Pivotal clinical trials

Those clinical studies that provide the significant evidence that is the basis for the decision as to the risk–benefit assessment for a particular FDC.

Product information

The information provided by the supplier of an FPP that allows prescribers and consumers to ensure the safe and effective use of drugs. If it is written especially for prescribers, it may be termed prescribing information.

Reference product

A pharmaceutical product with which the new product is intended to be interchangeable in clinical practice. The reference product will normally be the innovator product for which efficacy, safety and quality have been established. Where the innovator product is not available, the product that is the market leader may be used as a reference product, provided that it has been authorized for marketing and its

efficacy, safety and quality have been established and documented.

Summary of product characteristics (SPC)

A term used in the European Union. Product information or data sheets in the European Union should be based on the approved SPC.

Well-established drugs

Actives that:-

— have been marketed for at least 5 years in countries that undertake active post marketing monitoring;

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— have been widely used in a sufficiently large number of subjects to permit the assumption that safety and efficacy are well known; and

— Have the same route of administration and strength and the same or similar indications as in those countries.

4. Scenarios

An application to register an FDC-FPP may fall into any one of the following four scenarios. These guidelines are intended to address the different requirements for each scenario.

4.1 Scenario 1.

The new FDC-FPP contains the same actives in the same doses as an existing FDC-FPP; that is it is a "generic" of the existing FDC-FPP; they are "multisource" products. The quality, safety and efficacy of the existing product have been established.

4.2 Scenario 2.

The new FDC-FPP contains the same actives in the same doses as an established regime of single entity products, and the dosage regimen is the same. Alternatively the established regime may involve combinations of single entities and FDCs, for example, a single entity FPP combined with an FDC-FPP that contains two actives. In all cases, the established regime has a well-characterized safety and efficacy profile, and all of the FPPs used in obtaining clinical evidence have been shown to be of good quality.

4.3 Scenario 3

- The new FDC-FPP combines actives that are of established safety and efficacy but have not previously been used in combination for this indication.
- The new FDC-FPP comprises a combination for which safety and efficacy have been established, but that will be used in a different dosage regimen.

4.4 Scenario 4.

The new FDC-FPP contains one or more new chemical entities.

5. Balancing the advantages and disadvantages of a new fixed-dose combination

5.1 In determining whether it is rational to combine actives into a single product, there are medical, quality and bioavailability considerations.

5.1.1 Quality issues may be addressed by much the same criteria that apply to single-component products and it is difficult to imagine a case in which essentially the same standards would not apply.

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5.1.2 Medical considerations are more complex and sometimes contradictory, for example, when increased efficacy is accompanied by increased toxicity. The decision as to whether to give marketing approval for a new FDC-FPP in scenarios 3 and 4 is often based on a consideration of the balance of advantages and disadvantages from the medical perspective.

5.1.3 Interpretation of the results of bioavailability and bioequivalence tests involves both quality and medical considerations. For example it is not acceptable that bioavailability is reduced or variable, when compared with that of single entity products, because of poor formulation, but an interaction between two actives that leads to an increased bioavailability may be one of the advantages that is taken into account when balancing advantages and disadvantages.

Balancing the advantages and disadvantages of a new FDC-FPP should form a major component of submissions pursuant to this guideline.

5.2 Submissions for marketing approval of a new FDC in scenarios 2, 3 and 4 should include a section in which the advantages of the new combination are weighed against the disadvantages. All the possible advantages and disadvantages of the combination should be listed and discussed. The discussion should be based on the available data and on scientific and medical principles. In less well-developed nations, and particularly where there are difficulties with transport and the logistics of distribution, other matters may need to be taken into account, such as:

5.2.1 The cost of the combination as compared with the cost of individual components.

5.2.2 Evidence as to whether the new FDC will improve the reliability of supply as a result of simplified distribution procedures. Improved patient adherence may result from more reliable (continuing) availability of the FDC-FPP than of all of the components as loose combinations of single entity products.

However, issues of cost and procurement alone are not sufficient reason to approve an FDC if it has not been justified by appropriate data and on scientific and medical principles.

5.3 From a scientific or medical perspective, FDCs are more likely to be useful when several of the following factors apply:

5.3.1 There is a medical rationale for combining the actives.

5.3.2 There is an identifiable patient group for which this combination of actives and doses is suitable therapy. The larger the patient group in question, the more significant is this factor. It is not appropriate to combine actives that separately treat conditions that do not commonly coexist.

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5.3.3 The combination has a greater efficacy than any of the component actives given alone at the same dose.

5.3.4 The incidence of adverse reactions in response to treatment with the combination is lower than in that response to any of the component actives given alone, for example as a result of a lower dose of one component or a protective effect of one component, and particularly when the adverse reactions are serious.

5.3.5 For antimicrobials, the combination results in a reduced incidence of resistance.

5.3.6 One drug acts as a booster for another (for example in the case of some antiviral drugs).

5.3.7 The component actives have compatible pharmacokinetics and/or pharmacodynamics. See comments under Pharmacokinetics and pharmacodynamics below (section 6.6.2).

5.3.8 Therapy is simplified, particularly when the existing therapy is complex or onerous (e.g. because of a "high tablet load").

5.3.9 One of the ingredients is intended to minimize abuse of the other ingredient (e.g. the combination of diphenoxylate with atropine, or buprenorphine with naloxone).

5.3.10 The active pharmaceutical ingredients are chemically and physico-chemically compatible or special formulation techniques have been used that adequately address any incompatibility.

5.3.11 Other potential advantages of FDCs over single entity products given concurrently in the same dose may include:

5.3.11.1 Convenience for prescribers and patients.

5.3.11.2 Better patient adherence.

5.3.11.3 Simplified logistics of procurement and distribution.

5.3.11.4 Lower cost.

These factors are important, but there may not necessarily be evidence to support them; they may be more significant when there is specific evidence available to support a particular case.

5.4 From a scientific or medical perspective, FDCs are less likely to be useful when one or more of the following factors apply:

5.4.1 The component actives are normally separately titrated to meet the patient's needs. Consequently:

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5.4.1.1 Either the doses of the components, and/or the ratio of doses, typically differ from patient to patient, and/or

5.4.1.2 Patients are likely to be taking different doses at different stages of treatment (for example initial treatment compared with long-term treatment).

These two factors are particularly significant when one or more of the actives has a narrow therapeutic index and/or a steep dose–response curve in the therapeutic range.

5.4.2 There is a higher incidence or greater severity of adverse reactions to the combination than with any of the ingredients given alone, or there are adverse reactions not seen in response to treatment with any of the individual ingredients.

5.4.3 There are unfavourable pharmacokinetic interactions between the ingredients, for example when one drug alters the metabolism, absorption or excretion of another. However, see comments under Pharmacokinetics and pharmacodynamics below (section 6.6.2) concerning circumstances in which such interaction is intended.

5.4.4 Dose adjustment is necessary in special populations, such as in people with renal or hepatic impairment.

5.4.5 The product (tablets or capsules), is so large that patients find it difficult to swallow.

6. Data requirements for marketing authorization of fixed-dose combination finished pharmaceutical products

6.1 General

6.1.1 The framework for issuing a marketing authorization for an FDC-FPP is the same as that for single entity FPPs and is stipulated in the EAC guidelines on submission of documentation for registration of human pharmaceutical products.

6.1.2 Data requirements for marketing authorization of FDC-FPPs depend broadly on the scenario into which the application falls (see sections 4.1–4.4 above). Table 1 summarizes these differences.

However, each application should be considered on its own merits using scientific judgement and logical argument.

6.1.3 All applications to register an FDC-FPP should include a draft “product information” or “summary of product characteristics” for indicated diseases, and any package information leaflet or patient information. See the more detailed discussion below (section 7).

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Summary of requirements for the various scenarios

This table is a list of the most likely set of requirements for marketing authorization of an FDC-FPP in each scenario. However each application should be considered on its own merits in relation to data requirements, using scientific judgement and logical argument. Some of the data may be provided in the form of literature studies.

Table 1: Requirements for marketing authorization of an FDC-FPP in each scenario

Requirement Scenario 1 Scenario 2 Scenario 3 Scenario 4

Rationale for the Not usually Not usually ✓ ✓
combination
Balancing Not usually Not usually ✓ ✓
advantages and
disadvantages of the
combination

Marketing status in ✓ ✓ ✓ ✓
other countries

Analysis of literature Possibly for Possibly for ✓ ✓
data in the pharmaceutical pharmaceutical
submission development development
Pharmaceutical ✓ ✓ ✓ ✓
development studies
GMP certification of ✓ ✓ ✓ ✓
sites of manufacture
A full quality data ✓ ✓ ✓ ✓
set
Bioavailability dataa Not usually Not usually Sometimes ✓
Bioequivalence data ✓ ✓ Sometimes Sometimes
Preclinical Not usually Not usually Sometimes ✓
pharmacology and
safety
Clinical safety and Not usually Not usually ✓ ✓
efficacy
Product information ✓ ✓ ✓ ✓
Plan for passive ✓ ✓ ✓ ✓
post- marketing
surveillance

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Requirement Scenario 1 Scenario 2 Scenario 3 Scenario 4

Plan for active post- Not usually Not usually ✓ ✓
marketing
surveillance

6.1.6 A full quality data set is required in all scenarios (see 6.3 below).

6.1.7 In general, preclinical or clinical safety and efficacy data are not required in scenario 1. If the risk–benefit assessment has been found to be acceptable for an FDC, then new brands may be approved on the basis of bioequivalence with the brand(s) used in pivotal clinical trials.

The applicant may, however, be asked to establish that a risk–benefit assessment has been conducted and found acceptable if, for example the drug regulatory authority to which the application is submitted is not convinced that this is the case or does not have access to the data.

6.1.8 If the FDC directly substitutes for an established regimen of single entity products, in relation to both actives and doses and for the same indication(s), a bioequivalence study may provide adequate evidence of safety and efficacy. This is scenario 2. The established regimen should have well-characterized safety and efficacy.

6.2 Good manufacturing practice
Evidence of GMP compliance for all the API and FPP sites should be provided in Module 1 of the PD.

6.3 Quality

6.3.1 In relation to quality, very similar principles apply to FDC-FPPs as apply to single entity products. However there are additional complexities arising from the need to consider two or more actives instead of one. These complexities are principally, but not exclusively, related to assay, stability, physicochemical properties (for example dissolution rate) and bioavailability/ bioequivalence. Consequently the following considerations (and others) may be pertinent.

6.3.2 Pharmaceutical development studies are especially important for FDC-FPPs because they are technically more demanding than single-component products. Issues that are specific to the development of FDC-FPPs include:

6.3.2.1 Chemical and physicochemical compatibility of the APIs in an FDC with one another

as well as with possible excipients.

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6.3.2.2 The degradability of each API under stress conditions in the presence of the others.

6.3.2.3 Uniformity of content of each active prior to compression (tablets) or filling (for instance capsules, sachets and suspension dosage forms). This study determines whether mixing during manufacture is adequate.

6.3.2.4 Analytical procedures. These should be validated for each active in the presence of the others during development of analytical methods for quality control of the finished product, stability testing and dissolution testing.

Validation should be conducted for each active in the presence of the others and in the presence of related synthesis (process) impurities and potential degradation products. In the case of high-performance liquid chromatography (HPLC) (a common analytical technique), possible interference by degradation products in the assay of the active can usually be controlled by peak purity testing.

6.3.2.5 The dissolution rate of each active in pilot formulations. Multipoint limits should normally be established for routine quality control of each active. For some FDCFPs, different dissolution media may be acceptable for the different actives.

6.3.2.6 Different assay procedures may be necessary for the different actives in the finished product, and for different purposes (e.g. dissolution testing may be needed rather than stability testing).

6.3.3 For solid dosage forms a test and limit for content uniformity should be applied to any active that is present at a weight of ≤ 25 mg or when the API comprises 25% or less of a dosage unit.

Typically, when any one API is present at less than 25 mg or less than 25% of the weight of a dosage unit, all of the actives are subjected to content uniformity testing.

If a solid dosage form is not subject to content uniformity testing, for example because all of the actives are present at a weight of greater than 25 mg and greater than 25% of the weight of a dosage unit, there should be a test and limit for mass variation.

6.3.4 Acceptance criteria for impurities in FDC-FPPs should be expressed with reference to the parent API (and not with reference to the total content of APIs). If an impurity results from reaction between two APIs, its acceptance limits should be expressed in terms of the API that represents the worst case. If available, a reference standard should be used to quantify the degradation product in percentage mass/mass with respect to the parent API. Alternatively, and if justified, other quantitative techniques that are described in Impurities in new drug products (revised) ICH-Q3B(R) (2003), may be applied.

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Note: there should be an approximate mass balance. Together with the remaining active, degradants expressed with reference to the parent compound should sum to approximately 100% of initial strength.

6.3.5 The specifications and defining characteristics of the product should be based on the most vulnerable active. For example expiry dates should be based on the stability of the least stable active.

6.3.6 In setting specifications, relevant pharmacopoeial monographs, WHO guidelines, ICH guidelines and EAC guidelines should be taken into account.

6.3.7 Specifications in addition to those in pharmacopoeias may be necessary for APIs in some cases, for example for particle size, residual solvents and synthesis-related impurities that are not covered by relevant monographs.

6.4 Bioavailability and bioequivalence

6.4.1 Data on bioequivalence provide a bridge between two pharmaceutical equivalents (see Glossary) when safety and efficacy data are available for one of the FPPs, but not for the other. By demonstrating that the two products lead to the same profile for plasma concentration over time, available safety and efficacy data for one of the products can be extrapolated to the other. The two products being compared may be different brands, or different batches of the same brand, for example when manufactured by different methods, at different sites or according to different formulations.

6.4.2 Data on bioequivalence may also be important when the same FPP is administered under different circumstances, for example before or after food, in different patient populations (such as children versus adults), or by different routes of administration (such as subcutaneous versus intramuscular injection).

6.4.3 In the context of these guidelines, an additional application of bioequivalence studies is in scenario 2 in which safety and efficacy data on single entity products given concurrently may be extrapolated to an FDC-FPP, provided that all of the conditions described elsewhere in these guidelines are met.

6.4.4 Evidence as to bioequivalence is required for scenarios 1 and 2, and sometimes for scenarios 3 and 4, for example when there are major differences between the formulation and/or method of manufacture of the product to be registered and that used in pivotal clinical trials.

6.4.5 If a study of bioequivalence finds that the two treatments are bioequivalent, it may be assumed that any pharmacokinetic interactions between the actives were the same, even if one treatment comprised an FDC-FPP and the other comprised separate products.

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6.4.6 Data on absolute bioavailability are usually required in scenario 4, i.e. comparison of the

area under the curve for plasma concentration over time after an intravenous injection with that after administration of the dosage form to be marketed, for example a tablet given orally.

6.4.7 A decision as to whether it is necessary to conduct a study of the effect of food on the bioavailability of an FDC-FPP should be based on what is known of the effect of food on the individual actives, and any relevant recommendations in the product information for the single entity products.

The effect of food should normally be studied in scenario 4.

6.4.8 Recommendations as to the conduct and analysis of bioequivalence studies are provided in the EAC guidelines on submission of documentation for registration of human medicinal products.

6.5 Preclinical pharmacology and safety

6.5.1 Preclinical data are not normally required in scenarios 1 and 2. Data may, however, be required in some circumstances, for example if an unusual excipient is included in the formulation or if the impurity profile differs significantly from that of reference products.

6.5.2 Preclinical data will be required in scenario 4 as for any new chemical entity. The standard of evidence should be the same as for any new chemical entity.

6.5.3 In scenario 3, preclinical studies may not be required if all the actives have been extensively used in humans in the same combination for a long period and the safety of the combination has been well demonstrated. Bridging studies may be appropriate in some cases, for example for a new ratio of doses.

6.5.4 If the safety of the combination in humans has not already been demonstrated (i.e. in scenarios 3 and 4), preclinical studies should be conducted on the actives administered in combination in order to investigate possible additive or synergistic toxicological effects.

The preclinical data that are required in scenarios 3 and 4 will vary according to the data that are already available. For example, by definition in scenario 3, the safety and efficacy of each active will have already been established, but that of the combination will not. In scenario 4, the safety and efficacy of one or more of the actives may already have been established, but not those of all the actives or of the combination.

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6.5.5 When preclinical data are required, the studies should aim to determine both the pharmacological and the adverse effects that may be expected from the combination of actives during clinical use.

6.5.6 As a general rule, preclinical studies on the combination should be performed with the actives in same the ratio as in the FDC-FPP in question. If this is not the case, the applicant should explain and justify the proportions used. A comparison of the systemic exposures in animals and humans will be relevant.

6.5.7 In the absence of relevant WHO guidelines, the ICH preclinical guidelines in Table 4 may be used as source of guidance.

6.5.8 Preclinical studies should comply with a suitable code of good laboratory practice (GLP); see, for example Handbook: Good laboratory practice: Quality practices for regulated non-clinical research and development. World Health Organization (2001)

6.5.9 Microbiological preclinical studies

In general this section is applicable to scenarios 3 and 4, but not to scenarios 1 and 2. There may be some exceptions, for example microbiological data may be appropriate in scenarios 1 and 2 if a different pathogen or resistance pattern is encountered.

6.5.9.1 In scenarios 3 and 4, when a new combination is proposed for an antimicrobial indication, microbiological studies may be needed to determine the advantage of the FDC over the individual active moieties against relevant pathogen(s), and especially when clinical trials of monotherapy are inappropriate or unethical.

6.5.9.2 Data from microbiological preclinical studies of FDCs are particularly useful when clinical trials of monotherapy are inappropriate or unethical.

6.5.9.3 Data from the following types of study should normally be available for the combination:

6.5.9.3.1 Characterization of microbiological activity in vitro and in vivo against laboratory strains and clinical isolates of the targeted pathogen(s), including those strains in the relevant geographical regions.

6.5.9.3.2 Characterization of microbiological activity in appropriate animal models of infection with the targeted pathogen(s).

6.5.9.3.3 If possible, characterization of the mechanism by which the actives exhibit additive or synergistic microbiological activity against the targeted pathogen(s).

6.5.9.3.4 The potential for antagonistic effects between the actives.

6.5.9.3.5 The potential for development of resistance by target pathogens.

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6.6 Clinical efficacy and safety

This section is in general applicable to scenarios 3 and 4 but not to scenarios 1 and 2. Bridging studies may sometimes be appropriate in scenario 3, for example for a new

ratio of doses or a longer duration of treatment.

6.6.1 General principles

6.6.1.1 The risk–benefit assessment for a new combination may be based on data generated using either the components given as single entity products concurrently or the FDC as a single FPP.

6.6.1.2 Any theoretical advantages of a particular combination should be confirmed by means of efficacy studies. The risk–benefit assessment should not be based on theoretical considerations only, or on extrapolation from other data.

6.6.1.3 If the actives in an FDC are intended to relieve different symptoms of a disease state, it is a prerequisite that these symptoms commonly occur simultaneously at a clinically relevant intensity and for a period of time such that simultaneous treatment is appropriate. Occurrence of the individual symptoms in isolation should not be indications for the FDC.

6.6.1.4 Clinical studies should be designed to determine whether the combination has an advantage over the component actives given alone in a substantial patient population. The data should demonstrate that each active contributes to the therapeutic effect of the combination.

It may not be essential to show that all of the components have efficacy when administered as single entities; for example clavulanic acid has little or no antimicrobial activity when given alone, but it enhances the efficacy of beta-lactam antibiotics.

6.6.1.5 In situations where comparative clinical trials are not feasible, for example when monotherapy is inappropriate or is unethical, an aggregate of clinical and preclinical data may be substituted. Such data may include:

6.6.1.5.1 Historical clinical data, preferably at an exposure comparable to that for the proposed FDC.

6.6.1.5.2 Bridging pharmacokinetic data.

6.6.1.5.3 Preclinical pharmacology and/or toxicology data.

6.6.1.5.4 In vitro data (e.g. microbiological studies).

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6.6.1.6 If the FDC is available in more than one strength or ratio of doses, there should be a risk–benefit assessment for each combination.

6.6.1.7 The choice of comparators for the purposes of safety and efficacy studies should be justified. They should normally represent the recognized treatment for the indication in question. As far as possible, comparators should be licensed products with well-established safety and efficacy profiles and of established quality. Unapproved or novel combinations should be avoided as comparators as they may introduce new efficacy or toxicity characteristics and thus complicate assessment of the combination under test.

6.6.1.8 If the combination is intended for long-term use, data on safety in patients will normally be required for 6 months or longer.

6.6.1.9 If one or more of the component actives has an established use and dosage regimen in indications unrelated to the indications of the FDC, existing experience as to its safety may nevertheless be taken into account, bearing in mind the relative doses for the two sets of indications.

6.6.1.10 End-points in clinical trials should be such as to characterize the advantages and disadvantages of the combination.

For example, for a combination designed to reduce the development of drug resistance, end-points might include the frequency of new drug resistance as well as the overall clinical outcome.

6.6.1.11 Parallel group comparisons are one means of demonstrating a therapeutic effect. A parallel placebo group should be included if feasible and if consistent with the indications under treatment. Multi factorial designs are another means by which it may be possible to demonstrate that a combination is superior to the individual actives.

6.6.1.12 In some cases, studies have to be specifically designed to confirm the minimal effective dose and the usual effective dose of the combination. Multiple dose–effect studies may be necessary.

6.6.1.13 The design and analysis of studies of efficacy and safety should consider (among other things) whether the combination is indicated as first- or second-line therapy.

6.6.1.14 In general, all of the actives in a combination should have a similar duration of action. If this is not the case, the applicant should explain and justify the combination.

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6.6.1.15 In general, the actives in a combination should have similar pharmacokinetics. If this is not the case, the applicant should explain and justify the combination.

6.6.1.16 If there is an increase in the number or severity of adverse reactions to the FDC as compared with those in response to the individual actives given alone, evidence and argument should be presented showing that the advantages of the

combination outweigh the disadvantages. These should be included in the section of the submission entitled "Balancing the advantages and disadvantages of a new FDC".

6.6.1.17 Data generated in clinical safety and efficacy studies should comply with the WHO Guidelines for good clinical practice (GCP) for trials on pharmaceutical products (1995).

6.6.2 Pharmacokinetics and pharmacodynamics

This section is generally applicable to scenarios 3 and 4, but not to scenarios 1 and 2. In scenarios 1 and 2, the information described below will usually already be available.

6.6.2.1 In general, it is desirable that there be no pharmacokinetic or pharmacodynamic interactions between the components of a combination. However, there are circumstances in which such an interaction is intentional and may even contribute to the therapeutic outcome.

For example:-

6.6.2.1.1 Ritonavir boosts the activity of protease inhibitors.

6.6.2.1.2 Carbidopa and benserazide both reduce decarboxylation of levodopa in the gut wall, and consequently reduce the dose of levodopa that should be administered.

6.6.2.1.3 Clavulanic acid reduces bacterial hydrolysis of beta lactam antibiotics and consequently both increases the concentration and prolongs the duration of effectiveness.

6.6.2.2 Tests should be conducted to elucidate any pharmacokinetic or pharmacodynamic interaction between the actives in a combination. Some interactions may be predictable from pharmacokinetic and enzyme profiles, but should be confirmed by experiment. Any interaction should be quantified so that its effect on safety and efficacy is either predictable or (preferably) has been tested in a clinical study.

This includes competing metabolic effects and effects on gastrointestinal efflux mechanisms or on renal excretion or re-absorption. Interactions may be additive, synergistic or antagonistic.

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6.6.2.3 If there is an unintended pharmacokinetic interaction between the actives, it should be demonstrated that the therapeutic advantages of the combination outweigh any disadvantages resulting from the interaction. Relevant argument and cross-references to data should be included in the section that discusses the balance between the advantages and disadvantages of the combination.

6.6.3 Additional guidelines for scenario 3

6.6.3.1 The risk–benefit assessment for a new combination may be based (at least in part) on a demonstration of the clinical non-inferiority of the combination to another product licensed for the same indication. See Appendix 4, entitled Superiority, equivalence and non-inferiority clinical trials, for more information.

6.6.3.2 Pharmacodynamic studies for new combinations should normally be conducted at several dose ratios of the actives unless the applicant can provide justification for not doing so.

6.6.4 Additional guidelines for scenario 4

6.6.4.1 When an FDC-FPP contains an active that is a new chemical entity, data requirements are the same as for any new chemical entity. In some circumstances, some of the preclinical and clinical data on safety and/or efficacy may have been generated from studies on the combination rather than on single entities, for example when one active confers a protective effect in relation to adverse reactions or when the actives act synergistically.

6.6.4.2 Dose-finding monotherapy studies should normally be conducted for the new chemical entity before commencing studies of combination therapy, unless the new chemical entity is not intended to have activity when used alone (such as clavulanic acid). Alternative approaches may be acceptable if they can be justified.

6.6.4.3 The pharmacokinetics and enzyme profile of any new chemical entity should be fully characterized, including prediction of possible interactions and pharmacokinetics in children if the new chemical entity could be used in that population (see also section 7.6.6 on Paediatric dosage forms).

6.6.5 Superiority, equivalence and non-inferiority trials and fixed-dose combinations

Appendix 4 defines superiority, equivalence and non-inferiority trials and makes some general points concerning different types of study.

More information can be found in the Committee for Medicinal Products for Human Use (CHMP) guidelines in Table 3.

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6.6.5.1 In the context of FDCs, equivalence trials are largely confined to bioequivalence studies.

6.6.5.2 An FDC-FPP should be shown, directly or indirectly, to be superior to the component actives given as single entity treatments.

Only a superiority trial can give the necessary statistical confidence. Submissions should discuss both the statistical significance and clinical relevance of the

results. Any alternative form of evidence that purports to address the same issues, for example one that concerns a dose–response surface, must be explained and justified with appropriate statistical confidence.

6.6.5.3 In clinical trials that are intended to test for superiority and/or non-inferiority, the choice of comparator should be carefully considered and will depend in part on the medical and ethical circumstances. The comparator may be:

6.6.5.3.1 The treatment whose risk–benefit profile is best supported by evidence or is at least well established.

6.6.5.3.2 One or more of the actives in the FDC given as a single treatment.

6.6.5.3.3 A placebo.

6.6.5.4 Depending on the claim, superiority or non-inferiority should be demonstrated for each specified clinical outcome. For example if the claim is less bone marrow depression, but similar efficacy, a non-inferiority outcome should be demonstrated for efficacy and a superiority outcome for safety.

6.6.6 Paediatric dosage forms

6.6.6.1 Different FDC-FPPs may be needed in paediatric populations from those needed in adults because of differences in pharmacokinetic and pharmacodynamic profiles of the actives, and for reasons of palatability. The doses of each active may need to be lower or higher, and the appropriate dose ratio may be different.

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Scenarios 1 and 2

6.6.6.2 In scenarios 1 and 2, when the combination of actives and doses has already been shown to be safe and effective in the paediatric population, a bioequivalence study in adults may be extrapolated to the paediatric population provided that the pharmacokinetics of all actives are well-established in both populations and it is known that there are no differences that could affect the outcome of the bioequivalence study. Extrapolation of bioequivalence data between age groups should be justified in these terms.

Scenarios 3 and 4

6.6.6.3 If the FDC is indicated in a paediatric population, but the combination of actives and doses has not been shown to be safe and effective in this population, suitable doses of the actives given in combination should be established. In some cases, it may be necessary to do this in more than one age group (see table 2 below).

Table 2: Paediatric populations

Paediatric populations

Neonate Birth to under 1 month

Infants 1 month to under 2 years

Children 2 years to under 12 years

Adolescents 12 years to under 16 years

From the age of 16 years, individuals are considered to be adults in the context of these guidelines.

6.6.6.4 The pharmacokinetic profile of each active should be established in the age groups for which the FDC is indicated.

6.6.6.5 If it is possible to define target plasma concentrations in both adults and the paediatric population for an FDC that has established safety and efficacy in adults, then it may be possible to define suitable doses in the paediatric population on the basis of pharmacokinetics. The task is easier for actives that have the same target concentrations in adults and the paediatric population, such as antimicrobials that have established minimum inhibitory concentrations (MICs) and established safety at these concentrations.

6.6.6.6 When defining target plasma concentrations in the paediatric population, possible differences in the concentration–effect relationship should be taken into account.

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6.6.6.7 If safe and effective use of the FDC has not been established in any age group, and extrapolation between groups is not possible based on pharmacokinetic data, then new clinical, and possibly also preclinical, safety and efficacy data should be obtained.

7. Product information (or summary of product characteristics) for fixed-dose combination finished pharmaceutical products

7.1 This section of the guideline applies to all scenarios.

7.2 The product information should be an integrated evaluation of the FDC, and not a summation of the product information for each of the actives.

7.3 The rationale for use of the product should be presented in terms of the combination rather than in terms of the individual actives.

7.4 Only those indications for which each active in the FDC makes a useful contribution should be included in the product information. Each indication should be a well-recognized disease state, modification of a physiological state, dysfunctional state, syndrome or pathological entity.

7.5 For each indication there should be a statement as to whether the FDC is recommended for first- or second-line therapy.

7.6 Any pharmacokinetic and pharmacodynamic interactions between the actives should be described in qualitative and, as far as possible, in quantitative terms.

7.7 All clinically relevant interactions between the FDC and other drugs should be described, together with the resulting contraindications and precautions. Any deviations from expected interactions known for the single components should be highlighted.

7.8 When safety experience with the FDC is limited in comparison with that for the individual components, safety experience from clinical trials and post marketing experience should be presented for both the FDC and the individual components, and should be identified as such.

7.9 If the safety profile for the combination is different to that for the individual actives, this should be highlighted. For example a combination of a fibrate and a statin might carry a risk of more frequent or more severe rhabdomyolysis than for either individual active.

8. Post marketing studies and variations

8.1 Post market monitoring of safety is an important part of the role of both drug regulatory authorities and manufacturers. It is especially important when there are unresolved

328 concerns regarding safety, and when a new product is intended for wide community use, as for example a new antimicrobial FDC-FPP for use in the treatment of tuberculosis, malaria or HIV/AIDS. See WHO's importance of pharmacovigilance: safety monitoring of medicinal products (2002). Manufacturers should have (and use) written operating procedures for continuous assessment of the safety and utilization of their products following marketing authorization; SOPs can be examined during a GMP inspection.

For antimicrobials, monitoring of patterns of resistance is an important component of pharmacovigilance. Note also that pharmacovigilance outcomes can differ with diet, ethnicity, comorbidity and other factors.

8.2 For scenarios 1 and 2, passive surveillance (spontaneous reporting) would usually be acceptable. For scenarios 3 and 4, additional active (prospective) surveillance should be considered, especially when there is an outstanding safety concern. For more information, see the draft ICH guideline Pharmacovigilance planning (Table 5), or later updates thereof.

To ensure that drug regulatory authorities are aware of proposed changes to product information, it is recommended that marketing approval letters contain this statement: "The product information may not be altered without prior approval, except for safety updates that further restrict use of the product. Any such safety-related changes should be notified to [name of regulatory authority] within five days of making the change."

8.3 Variations to pharmaceutical aspects of registered FDC-FPPs are subject to the EAC guidelines on variations to registered products.

To ensure that drug regulatory authorities are aware of proposed variations, it is recommended that marketing approval letters contain this statement: "No changes may be made to the product without prior approval, except for changes of the type listed in [name of regulatory authority]'s policy on 'Changes to pharmaceutical aspects which may be made without prior approval'. Conditions in that policy apply."

9. Guidelines for co-packaged fixed-dose combinations

A co-packaged product consists of two or more separate pharmaceutical products in their final dosage form that are packaged together for distribution to patients in the co-packaging.

1. Co-packaged products may fall into any of scenarios 1 to 4. The data requirements for each scenario are the same as those listed in Table 6 of this Annex.

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2. A full quality data set is required for all components of co-packaged pharmaceutical products, except for any component that already has marketing authorization in which case more limited requirements apply (see below).

3. If one or more of the pharmaceutical products already has marketing authorization, then the additional quality information to support co-packaging of those pharmaceutical products will typically be limited to data on stability of the products in the co-packaging.

However the manufacturer of each component pharmaceutical product should provide an assurance that the product as used in co-packaging will be identical in formulation and method of manufacture to the one that already has marketing authorization. This is especially important when the manufacturer of a component is not the manufacturer of the co-packaged product.

4. Submissions concerning co-packaged pharmaceutical products should take into account the Guidelines on packaging for pharmaceutical products. In: WHO Expert Committee on Specifications for Pharmaceutical Preparations. Thirty-sixth report. Geneva, World Health Organization, 2002 (WHO Technical Report Series, No. 902), Annex 9.

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PART X:

GUIDELINES ON PROCEDURAL ASPECTS FOR APPLICATIONS FOR

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ABBREVIATIONS AND ACRONYMS

BMGF - Bill and Melinda Gates Foundation

BMR - Batch Manufacturing Record

EAC - East African Community

EAC-MRH - East African Community Medicines Regulatory Harmonization

EMA - European Pharmaceutical products Agency

FEAPM - Federation of East African Pharmaceutical Manufacturers Harmonization

GCP - Good Clinical Practice

GMP - Good Manufacturing Practice

ICH - International Conference on Harmonization of Technical Requirements for Registration of Pharmaceuticals for Human Use

MA - Marketing Authorization

MAH - Marketing Authorization Holder

MER - Medicines Evaluation and Registration

NEPAD - New Partnership for African Development

TWG - Technical Working Group

WHO - World Health Organization

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1. INTRODUCTION

All pharmaceutical products in Tanzania are regulated by the Tanzania Medicines and Medical Devices Authority (TMDA). As a preliminary step, TMDA issues marketing authorization of all pharmaceutical products prior to their use in the country. The objective of pharmaceutical product marketing authorization is to ensure that pharmaceutical products marketed in Tanzania are safe, efficacious and of good quality and are manufactured in facilities that complies with the requirements prescribed in the TMDA Good Manufacturing Practice guidelines. All local manufacturers, wholesalers, distributors and importers of pharmaceutical products must be licensed before they can conduct their businesses.

The guideline covers the steps that are followed from the submission of a dossier to the final outcome, the timeframe and procedure for competent authorities to amend, where necessary the conditions of marketing authorization of a particular product.

2. SCOPE

The guideline is applicable for all types of application submitted to TMDA that include new application, renewal of application and application for variation of a registered pharmaceutical product.

3. TYPES OF APPLICATIONS

3.1 For purposes of submission, applications are classified into new application, application for variation of a registered pharmaceutical product and renewal application.

3.2 A new application is an application for registration of a pharmaceutical product that is intended to be placed on the market for the first time. A new application may only be made by the applicant and he shall be the person who signs the application form.

3.3 A new application for registration shall include submission of relevant documentation as provided in the main guidelines for registration of pharmaceutical products in use.

4. GENERAL REQUIREMENTS AND APPLICATION PROCEDURES FOR PHARMACEUTICAL PRODUCT REGISTRATION

4.1 All applications and supporting documents shall be in English. All submitted documents which are in any language other than English must be accompanied by a certified or notarized English translation.

4.2 The responsibility of applying for product marketing authorization rests with the company responsible for the introduction of the product into the Tanzanian market i.e. the Marketing Authorization Holder (MAH).

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4.3 Applications must be duly completed and supported by all of the required documents i.e. Module I to Module V in accordance with the guidelines on submission of documentation for marketing authorization. The submitted application will be screened for completeness within 30 working days. Dossiers which are incomplete will not be accepted for evaluation.

4.4 A dossier is a file that contains detailed scientific information on the chemistry, formulation, manufacturing, quality control and non-clinical and clinical studies that

demonstrates quality, safety and efficacy of active pharmaceutical ingredient(s) and the corresponding finished pharmaceutical product.

Different sections of the dossier shall be distinctly marked and page numbered in the style: page x of y and have a table of contents indicating the sections and page numbers. Where information is required in the application forms its location shall be cross referenced in dossier. Information for each section shall be printed on both sides of an A-4 paper which will be arranged sequentially on a 1.00 mm or more diameter stainless spring and clamped with a stainless steel binder of not less than 1.0 mm thick in an A4 expandable spring file. The file shall be of cardboard or paper material of not less than 600gsm.

4.5 The covering letter shall be submitted in hard copy and the entire dossier on a CD-ROM or the entire application be electronically submitted to TMDA through the online system.

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4.6 Data shall be presented on A4 and 80g/m paper with readily readable letters of at least 12 font sizes. Every page shall be numbered sequentially. Extension sheets, tables, diagrams and other supporting documents shall as far as possible be of the same size, well annotated, numbered and appropriately cross-referenced.

4.7 Application must be accompanied by two samples of the finished product as packaged for sale. The Authority may request for additional samples when need arises.

4.8 The registration fee as prescribed in the fees and charges regulations must be paid at the point of submission of the application.

5. PROCESSING OF APPLICATIONS (MANAGEMENT OF APPLICATIONS)

5.1 Upon acceptance of an application, an acknowledgement for the receipt of the application will be issued within and a reference number will be generated. The reference number shown in this acknowledgement should be used in all subsequent correspondences relating to the application.

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5.2 The Authority shall complete screening of the dossier for completeness within 30 working days from receiving such application.

5.3 In the event that the dossier is incomplete, the applicant will be notified of the deficiencies and requested to resubmit the application.

5.4 In case of a positive outcome during screening, the application shall be placed in the evaluation queue.

5.5 Review of application for marketing authorization of a product will follow the appropriate evaluation queue. Priority review may be granted where the product is intended for treatment of a serious or life-threatening disease. Evaluation of priority product shall be carried out within 120 working days from receiving the application.

5.6 Evaluation of the application shall be carried out within 240 working days from receiving the application.

5.7 Abridged evaluation may be carried out to pharmaceutical products that are registered in any of the agreed benchmark regulatory agencies.

5.8 During product evaluation, the Authority may request for further information and additional supporting documents from the applicant. This shall be considered as the first round of evaluation.

5.9 Applicant should make available such information or documentation requested after the first round of evaluation within 180 calendar days from the date of receipt of the request.

5.10 Applicant should make available any information or documentation requested after subsequent rounds of evaluation within 120 calendar days from the date of receipt of the request.

5.11 If no response is received from applicant after the timelines described in 5.9 and 5.10 above, the clock stops and the application will be cancelled if no formal request for extension of deadline has been made. A new application will have to be submitted if the MAH wishes to pursue marketing authorization of the product.

5.12 Evaluation of the additional information shall be carried out within 3 months from receiving such information. This shall be considered as the second round of evaluation and subsequent submission of additional information shall be considered as third round of evaluation and so forth.

5.13 Evaluation of one application shall not exceed five rounds of evaluation with the exception of administrative queries.

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5.14 The MAH will be informed of the decision in writing as to whether the application has been approved or rejected.

5.15 A registration number will be given when a product is registered. The registration number is specific for the product registered as specified in the registration documents. A certificate of registration shall be issued for the registered product.

5.16 For a product to be issued MA, it must be manufactured in a GMP compliant facility and studies conducted following GCP.

6. MAINTENANCE OF MARKETING AUTHORIZATION

6.1 The conditions for marketing authorization of pharmaceutical products are as follows:-

6.1.1 The product registered with the marketing authorization number as stated in the marketing authorization certificate shall have the name, composition, characteristics, specifications and origin as specified in the marketing authorization documents.

6.1.2 The holder of the marketing authorization certificate must supply such documents, items, samples, particulars or information as the Authority may require in relation to the registered product.

6.1.3 Changes in name, composition, characteristics, origin, specifications, manufacturer, packing, indications, labelling, package insert, product literature or any other particulars of the registered product shall not be made without prior notification and/ or approval from the Authority.

6.1.4 The labels for the registered product must comply with all of the labelling requirements as specified by the guidelines for labelling.

6.1.5 The registered product must only be indicated for use as approved by the Authority.

6.1.6 The holder of the marketing authorization certificate must inform the Authority of any adverse reactions or complaints on quality, safety and efficacy of the registered product immediately after he/she becomes aware of such adverse reactions or complaints.

6.1.7 The holder of the registration certificate must notify in writing to the Authority of any decision to withdraw the marketing authorization of the product and shall state the reasons for the decision.

6.2 MAH shall be required to pay retention fees as specified by the fees and charges regulations.

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6.3 The registration of a product shall be valid for 5 years or such period as specified in the registration certificate (unless sooner suspended or cancelled by the Authority).

6.4 The renewal of product registration should be done not later than three months prior to expiry. Applications for renewal of registration shall be made by submitting documents as outlines in guidelines on submission of documentation for renewal of registration.

7. CANCELLATION OR SUSPENSION OF MARKETING AUTHORIZATION

7.1 The Authority may cancel or suspend the marketing authorization of any product if there are deficiencies in safety, quality or efficacy of the product or failure to comply with the marketing authorization requirements or due to changes in national policies.

7.2 Such products may not be imported and marketed in the country. The holder of the registration certificate shall immediately surrender to the Authority the marketing authorization certificate upon cancellation or suspension of marketing authorization of the product.

8. APPEALS AGAINST AUTHORITY'S DECISIONS ON PHARMACEUTICAL PRODUCT MARKETING AUTHORIZATION

8.1 For products that have been suspended and cancelled marketing authorization by the Authority, MAH may make a written appeal to the Director General of TMDA to review its decision.

8.2 All notice of appeals must be made within thirty (30) calendar days from the date of the notification.

8.3 MAH shall make appeal by giving grounds for review for each reason given for the rejection of his product. The grounds for the request shall be based on the information that was submitted in the product dossier. Any additional or new information that was not earlier submitted will not be accepted. The Authority may review or uphold its earlier decision.

9. VARIATIONS IN PARTICULARS OF REGISTERED PRODUCTS

All variations to a registered product shall be made according to requirements stipulated in the variation guidelines.

9.1 EXTENSION APPLICATIONS

9.1.1 An extension application is an application that is a modification of an already registered medicinal products. The modification shall be such that it does not fulfil criteria for

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minor or major variations but is similar enough to the original product in terms of quality, safety and efficacy.

9.1.2 A marketing authorization holder may apply for extension of marketing authorization of an already registered product as an extension application. Such an application should be submitted as a new application however an abridged evaluation will be carried out.

9.1.3 Extension applications shall be applicable in the following situations:-

a) Changes or addition of a pharmaceutical form from multi-dose to single-dose of the finished product or vice versa.

b) Change or addition of strength of the finished product.

c) Change or addition of a route of administration of the finished product for products of the same pharmaceutical form.

d) Inclusion of medical devices that result in change of strength, pharmaceutical form or route of administration of the finished product.

9.1.4 An extension application shall be accompanied by the following:-

a) A duly filled in applicant form with the extension application box clearly marked (ticked).

b) The applicable registration fees for applications for registration of new applications.

c) A full dossier submitted in accordance to the requirements stipulated in the guidelines for submission of documentation for registration of human medicinal products.

d) A cover letter declaring the following:

- i. The name and registration number of the relevant product from which the extension is applied.
- ii. The marketing authorization holder for both products shall remain the same.

e) An overview of the nature of the extension being made.

f) Supporting data related to the proposed extension.

9.1.5 The final decision on whether an application meets the criteria for extension applications will lie with the Authority. In case of any doubt the MAH may contact the the Director General TMDA before filling for an extension application.

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9.2 Duplicate Licensing

9.2.1 The Authority shall authorize the same applicant to submit more than one application for a finished product when there are objective verifiable reasons on public health grounds regarding the availability of finished products to health-care professionals and/or patients, or for co-marketing reasons and/or for Export purposes.

9.2.2 Additionally, the holder of a marketing authorization can grant the use of product information to another marketing authorization holder, whereby the original marketing authorization holder acts as a contract manufacturer.

9.2.3 The assessment on whether the conditions of a duplicate application are met shall be done on a case-by-case basis, having regard to the facts of each application. The overall objectives being preservation of public health.

9.2.4 To assess whether an application refers to a particular finished product that has already been granted a marketing authorization, and consequently, whose application for a marketing authorization qualifies for a duplicate license, the composition in active substance(s) and the pharmaceutical form shall be considered. Thus, any finished product with the same qualitative and quantitative composition in active pharmaceutical ingredient (i.e. the same strength) and the same pharmaceutical form are to be considered as the same relevant product.

9.2.5 A duplicate product shall be identical in all marketing authorization requirements with the exception of brand name and any other specific requirements on labelling. Additionally, any variation made to the original marketing authorization should be applied for the duplicate license.

9.2.6 Conditions for a Duplicate Marketing Authorization are outlined hereafter:-

a) That the duplicate application shall be submitted by the same applicant that submitted and/or holds the marketing authorization/application that is being duplicated (hereafter "original marketing authorization/application").

b) That the original marketing authorization is valid. This step does not apply in case of duplicate applications that are submitted in parallel with the original marketing authorization application (i.e. in cases where the application for the original marketing authorization is still pending).

c) In cases where the duplicate marketing authorization is submitted on the basis of an informed consent application, there should be a letter of consent from the marketing authorization holder that owns the dossier that is referred to.

d) The original marketing authorizations to which the duplicate application relates has to be valid at the time of the submission of the duplicate application.

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9.2.7 The applicant for a duplicate license may fall under the following categories:-

a) Applicant is the same entity that applied for the original marketing authorization.

b) Applicant belongs to the same group of companies as the applicant of the original marketing authorization.

c) Applicant is an independent entity that has agreed to placing on the market the product with the applicant of the original marketing authorization (evidence of license agreement or other agreement that can be identified are required).

d) Applicant is an independent entity whereby there are license agreements with the marketing authorization holder of the product in respect of which the duplicate is asked but not for the placing on the market of that product.

e) Applicant is an independent entity that has got an agreement to purchase and/or use data from the company that has applied for a marketing authorization for the product for the first time but there is not an agreement regarding the placing on the market of the product.

9.2.8 All documents in accordance to the guidelines on submission of documentation for new applications should be submitted however an abridged evaluation shall be applied. In addition, the following shall be submitted when making a duplicate license application:-

a) A dully filled in application form (Annex II) with the duplicate license box clearly marked (ticked).

b) A cover letter detailing the following:-

- i. The name of the marketing authorization holder relevant for the duplicate application.
- ii. The name of the product relevant for the duplicate application.
- iii. The proposed brand name for the duplicate license.
- iv. The proposed marketing authorization holder for the duplicate license.

c) The applicable registration fees for applications for registration of new applications.

d) For co-marketing reasons, the evidence co-marketing (contract or letter of agreement between the companies).

e) For duplicates asked on grounds of the existence of patents protecting certain therapeutic indications or pharmaceutical forms, the applicant shall provide a commitment undertaking to extend the therapeutic indication(s)/ pharmaceutical form(s) of the duplicate marketing authorization as soon as the patent restrictions no

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longer exist. Alternatively, the applicant may also commit to withdraw the marketing authorization with restricted indications/pharmaceutical forms after the relevant patents are no longer in force. The SmPC shall be harmonized. The commitment letter shall be submitted alongside the Marketing authorization Application Dossier.

f) Letter of consent in the case of an "informed consent application".

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PART XI:

GUIDELINES ON NAMING OF MEDICINAL PRODUCTS

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1. Introduction

Prescription and medication error may occur partly due to medicinal products having sound-alike or look-alike brand names, unclear labeling or poorly designed label artwork. Hence this guideline was developed in order to provide Applicants with clear guidance on how to choose brand names for their medicinal products.

Tanzania Medicines and Medical Devices Authority may request changes to a brand name if based on evaluation, it is deemed to potentially: -

- a) cause confusion with the name of an existing medicine;
- b) mislead as to the composition of the product or the use;
- c) or otherwise unsafe.

Approval of the name does not imply that the marketing authorization holder is absolved of any responsibility in the incidence that actual or potential adverse reactions occur due to the brand name.

State legislations in particular trade laws, should be taken into consideration while proposing a specific brand name.

2. Scope

This guideline is applicable to all prescription and non-prescription medicinal products as well as medicinal products of biological origin.

The principles outlined in this guideline are also applicable for applications for variation of names of registered medicinal products.

3. Acknowledgments

This guideline was developed based on the European Medicines Agency's Guideline on the acceptability of names for human medicinal products processed through the centralized procedure.

4. Criteria applied when reviewing proposed brand names

The criteria listed below should be seen as general principles. The Authority may develop additional guidance on specific topics based on experience and may apply additional requirements not listed in this document during review of proposed brand names.

The criteria for acceptability of proposed brand names shall be based on public health concerns and in particular with regard to safety.

Applicants should ensure that the proposed name complies with the criteria outlined in this guideline before submitting an application for marketing authorization. Requirements and

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considerations taken during review are outlined hereafter:-

4.1 Safety concerns and other public health concerns in brand names

4.1.1 The brand name of a medicinal product should not be liable to cause confusion in print, handwriting or speech with the brand name of another medicinal product.

When assessing the potential for such confusion, the following aspects are considered:

- The indication(s);
- The patient population(s);
- The pharmaceutical form(s);
- The route(s) of administration;
- The strength(s);
- The setting for prescription, dispensing and use;
- The legal status/classification for supply:
- Medicinal product subject to medical prescription;
- Medicinal product not subject to medical prescription;
- Medicinal product subject to special medical prescription;
- Medicinal product subject to restricted medical prescription;
- Medicinal product subject to special and restricted medical prescription;
- Orphan (designation) status;
- (Potential) New pharmaceutical forms, routes of administration and/or strengths for the medicinal product concerned, as appropriate.
- The degree of similarity versus the potential for harm to the patient in case of mix-up.

It should be noted that the Authority will consider potential for confusion of proposed brand names with the brand names of authorized, suspended and revoked/withdrawn medicinal.

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When considering potential for confusion with the name of a withdrawn/revoked marketing authorization, in principle, a period of 5 years should have elapsed after the official invalidity of the marketing authorization according to national legislation (e.g. publication in the national gazette notice, etc.). This period could be reduced (e.g. the product was not marketed in Tanzania for a period preceding this 5 year period) if it can reasonably be justified by the Applicant. The period may be extended (e.g. that the withdrawal of the marketing authorization was linked to serious safety concerns and this has an impact on the potential risk to public health associated with the name) at the discretion of the Authority. The Authority may also consider potential safety concerns and other public health concerns associated to the re-use of identical brand names.

4.1.2 The brand name of a medicinal product should not convey misleading therapeutic and/or pharmaceutical connotations.

This also includes brand names that are similar or allude to the name of pharmaceutical companies if they are thought to be misleading and cause confusion at the level of product information.

4.1.3 The brand name of a medicinal product should not be misleading with respect to the composition of the product.

4.1.4 Consideration should be given to the phonetics and the potential difficulties a proposed brand name may create in terms of pronunciation in the official or native languages of Tanzania.

Consideration should also be given to the fact that very short brand names composed of, for instance, a string of letters, may be inappropriate to identify medicinal products in certain settings.

4.1.5 The use of qualifiers/abbreviations by letters as part of the brand name should in principle be acceptable on conditions.

Qualifiers consisting of a single letter or number(s) (Arabic and Roman) are discouraged, because they may be confused with the strength and/or posology of the medicinal product. However, the use of numbers may in certain cases be acceptable, e.g. vaccines (see section 4.3.1). The applicant may provide a justification for their inclusion.

The potential added benefit versus its potential risk to public health in case of medication error shall be taken into consideration when considering the acceptability of a qualifier/abbreviation. The following shall be considered:-

a) Whether the qualifier/abbreviation provides further information on characteristics of the medicinal product (e.g. duration of action, devices, route of administration, composition, patient population) without being misleading or

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provides for a differentiation, which may help healthcare professionals and/or patients to prescribe/select the appropriate medicinal product.

b) The applicability and use of the qualifier across all official and native languages.

Qualifiers or abbreviations should not require translation to provide further information in the users.

c) The potential risk resulting from more complex names, adversely affecting memorability, pronunciation and/or prescription of the medicinal product.

4.1.6 The brand name should not convey a promotional message with respect to the therapeutic and/or pharmaceutical characteristics and/or the composition of the medicinal product.

4.1.7 The brand name should not be offensive or have an inappropriate connotation in any of the official or native languages.

4.1.8 For a medicinal product containing a prodrug, a different brand name from the brand name of the medicinal product containing the related active substance is required.

4.1.9 The brand name should not comprise wholly of initial letters (acronyms) or code numbers nor include punctuation marks.

4.1.10 The importance of other elements such as labelling and pack design should be taken into consideration as contributing factors for the safe use of a medicinal product. These aspects should be discussed at the time of the review of mock-ups.

The following are examples where labelling and pack design may play a role in the final decision of acceptability of brand names:-

- a) The actual display of a brand name in the printed material may increase the level of similarity between two brand names or may convey a misleading connotation.
- b) The labelling and pack design may support the meaning of a qualifier which otherwise would have been rejected.

4.2 Use of international non-proprietary names in proposed brand names

When proposing a brand name, Applicants are advised to take into consideration WHO resolution (WHA46.19), where appropriate, i.e. "It would therefore be appreciated if brand names were not derived from international non-proprietary names (INNs) and if INN stems were not used in brand names".

Two types of INN concerns could be considered i.e. a potential similarity with its own or different INN or the inclusion of an INN stem into the proposed brand name(s).

The Applicant is strongly advised to review INN similarity and/or INN stem inclusion before requesting that the proposed brand name(s) be considered for a medicinal product.

The Authority will review the above cases on the basis of WHO World Health Assembly 346 resolution (WHA46.19) on protection of INNs/INN stems to prevent any potential risk of confusion between brand names and common names.

4.3 Product specific concerns in proposed brand names

4.3.1 For vaccines composed of several serotypes, when adding a new serotype the original brand name may be kept, it is recommended that the name is then followed by the number of serotypes present. The description of serotypes present is then listed in the qualitative and quantitative composition. An example of the format of the proposed brand name follows:

Brand name + X [number of serotypes]

The same applies when different types of antigens are added. This is of particular importance in situations where both vaccines are simultaneously available on the market in order to allow differentiation of the products.

4.3.2 For radiopharmaceutical medicinal products, the inclusion of target organs in the brand name should be avoided in order to prevent misleading connotations should an extension of the indication include new target organs.

In principle, numbers should not be used in the name to avoid confusion with the strength. In cases where the numbers appear in the radionuclide, these should be displayed in superscript, i.e. mass numberElement + [brand name]

Numbers included as part of commonly known abbreviations will be assessed on a case by case basis.

4.3.3 When reviewing the acceptability of brand names for orphan medicinal products, the same approach as for non-orphan medicinal products shall be applied. It is of particular importance in these cases to provide detailed information on the specific setting in which the product is dispensed and used as well as on the target population.

4.3.4 For non-prescription medicinal products, the use of qualifiers/abbreviations within the brand name should aid selection/identification/differentiation of the product by the patient and should minimize the risk of inappropriate use.

In view of the above considerations, the specific criteria as described under sections 4.1.5, 4.1.8 and 4.3.7 may not apply here.

In order to help self-selection and compliance by patients/consumers, it is acceptable that brand names have a positive connotation and/or be informative; labelling and pack design could be considered as contributing factors to this end. Carton and container labels are particularly critical for non-prescription medicinal products (general sale and over the counter).

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In case of a switch from "prescription" to "non-prescription" status of an already authorized medicinal product it is up to the Applicant to choose whether to vary/extend the existing marketing authorization and consequently retain the same

brand name or to submit a separate marketing-authorization application under a different brand name (see section 5). In exceptional cases, depending on the therapeutic context, the acceptability of the maintenance of the existing brand name may be further considered by the Authority during the evaluation process.

4.3.5 For generic/hybrid/similar biological medicinal products the same criteria apply as for any other medicinal products in respect to the brand name.

Special consideration should be given to the proposed brand name of a hybrid medicinal product to allow for differentiation when the latter differs in pharmaceutical form, strength, expression of active substance and/or indication from the reference medicinal product or other generics in the market.

4.3.6 Where the Applicant wishes to use instead of the brand name the common name or scientific name, together with a trademark or the name of the marketing-authorization holder/applicant, they should take into account the following rules:-

- a) If an INN recommended by the World Health Organization exists for the active moiety it should be used within the name of the medicinal product exactly as published without omissions or abbreviations. All the linguistic versions of the INN, including translations officially recognized at the national level, shall be considered to be the same name. If one does not exist, the usual common name should be used.
- b) If a Modified INN (INNM) recommended by the World Health Organization exists for the active moiety, it should be used within the name of the medicinal product exactly as published without omissions or abbreviations.
- c) Where the active moiety is an unpublished INNM the name of the medicinal product should be that as agreed by users of INNs (pharmacopoeia, regulatory bodies, stakeholders), in accordance with the WHO INNM working document 05.167/3.
- d) The 'name of the MAH' within the name of the medicinal product should correspond to all or part of the official name of the MAH as presented in the proof of establishment of the applicant/MAH.
- e) For consistency reasons, ease in prescription by healthcare professionals and database entries, punctuation marks in between the INN and the name of the Company/trademark are not acceptable (with the exception of fixed combinations, where multiple INNs should be clearly separated by slash '/').
- f) The proposed brand name should either be an brand name or the common name accompanied by a trademark or the name of the MAH.

4.3.7 The brand name of a fixed combination medicinal product should be sufficiently different from those of the individual active substances and/or those of other fixed combinations containing the same active substance(s).

The whole brand name of individual active substance(s) should not be inserted into the proposed brand name for the fixed dose combination.

4.3.8 As multiple applications can have an independent life (e.g. may develop a different indication at a later stage), the proposed brand names of such applications should not lead to confusion.

5. Regulatory aspects related to the acceptability of proposed brand names

Brand names for variation/extension/duplicate applications should be the same as those of the existing medicinal product. The addition of a qualifier to an already approved brand name constitutes a different brand name, which would require submission as new marketing authorization application.

In case the applicant wants to submit a separate marketing-authorization application for, e.g., a new indication, a different brand name shall be used.

The Authority may request the MAH to change the brand name of an already approved medicinal product if the approved brand name is deemed inappropriate.

5.1 Change of the brand name

The brand name can also be changed at a post-authorizations stage through an application for variation if the MAH wishes to change the name. Post-authorizations procedural requirements are outlined in the Guidelines on Variations on Registered Medicinal Products.

5.2 Report of prescription errors/medication errors due to the brand names of medicinal products:

The marketing authorization holder is responsible for reporting any adverse drug reactions resulting from:-

- a) Prescription errors/medication errors due to the brand name of the medicinal product (example mix up with another medicinal product resulting into an ADR).
- b) Misuse and/or abuse of a medicinal product caused by misleading therapeutic connotations of the brand name.

The ADRs should be reported in accordance to the procedures and guidance stipulated in the Guidelines on Safety and Vigilance of Medical Products and Health technologies.

349 References and useful websites

1. Guideline on the acceptability of names for human medicinal products processed through centralized procedure

2. WHO website: <http://www.who.int/en/>

3. Information on INNs: <http://apps.who.int/medicinedocs/en/d/Jh1806e/5.html>

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PART X

EAC GUIDELINES ON PROCEDURAL ASPECTS FOR APPLICATIONS FOR
MARKETING AUTHORIZATION OF PHARMACEUTICAL PRODUCTS

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ABBREVIATIONS AND ACRONYMS

BMGF - Bill and Melinda Gates Foundation

BMR - Batch Manufacturing Record

EAC - East African Community

EAC-MRH - East African Community Medicines Regulatory
Harmonization

EMA - European Medicines Agency

FEAPM - Federation of East African Pharmaceutical Manufacturers
Harmonization

GCP - Good Clinical Practice

GMP - Good Manufacturing Practice

ICH - International Conference on Harmonization of Technical
Requirements for Registration of Pharmaceuticals for Human Use

MA - Marketing Authorization

MAH - Marketing Authorization Holder

MER - Medicines Evaluation and Registration

NEPAD - New Partnership for African Development

NMRA - National Medicines Regulatory Authority

TWG - Technical Working Group

WHO - World Health Organization

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1. INTRODUCTION

All pharmaceutical products in the EAC region are regulated by individual Partner States' NMRA. As a preliminary step, the NMRA issues marketing authorization of all pharmaceutical products for human use prior to their use in the region. The objective of pharmaceutical product marketing authorization is to ensure that pharmaceutical products marketed in the region are safe, efficacious and of good quality and are manufactured in facilities that complies with the requirements prescribed in the EAC Good Manufacturing Practice guidelines. All local manufacturers, wholesalers, distributors and importers of pharmaceutical products must be licensed before they can conduct their businesses.

The guideline covers the steps that are followed from the submission of a dossier to the final outcome, the timeframe and procedure for competent authorities to amend, where necessary the conditions of marketing authorization of a particular product.

2. SCOPE

The guideline is applicable for all types of application submitted to the EAC Partner States National Medicines Regulatory Authorities (NMRAs) that include new application, renewal of application and application for variation of a registered pharmaceutical product.

3. TYPES OF APPLICATIONS

3.1 For purposes of submission to the NMRA, applications are classified into new application, application for variation of a registered pharmaceutical product and renewal application.

3.2 A new application is an application for registration of a pharmaceutical product that is intended to be placed on the market for the first time. A new application may only be made by the applicant and he shall be the person who signs the application form.

3.3 A new application for registration shall include submission of relevant documentation as provided in the main guidelines for registration of pharmaceutical products in use.

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4. GENERAL REQUIREMENTS AND APPLICATION PROCEDURES FOR PHARMACEUTICAL PRODUCT REGISTRATION

4.1 All applications and supporting documents shall be in English. All submitted documents which are in any language other than English must be accompanied by a certified or notarized English translation.

4.2 The responsibility of applying for product marketing authorization rests with the company responsible for the introduction of the product into the EAC market, i.e.: the Marketing Authorization Holder (MAH).

4.3 Applications must be duly completed and supported by all of the required documents i.e. Module I to Module V in accordance with the EAC Common Technical Document (CTD) for registration of pharmaceutical products. The submitted application will be screened for completeness within 30 working days. Dossiers which are incomplete will not be accepted for evaluation.

4.4 A dossier is a file that contains detailed scientific information on the chemistry, formulation, manufacturing, quality control and non-clinical and clinical studies that demonstrates quality, safety and efficacy of active pharmaceutical ingredient(s) and the corresponding finished pharmaceutical product.

Different sections of the dossier shall be distinctly marked and page numbered in the style: page x of y and have a table of contents indicating the sections and page numbers. Where information is required in the application forms its location shall be cross referenced in dossier. Information for each section shall be printed on both sides of an A-4 paper which will be arranged sequentially on a 1.00 mm or more diameter stainless spring and clamped with a stainless steel binder of not less than 1.0 mm thick in an A4 expandable spring file. The file shall be of cardboard or paper material of not less than 600gsm.

4.5 The covering letter shall be submitted in hard copy and the entire dossier on a CD-ROM or the entire application be electronically submitted to the NMRA.

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4.6 Data shall be presented on A4 and 80g/m paper with readily readable letters of at least 12 font sizes. Every page shall be numbered sequentially. Extension sheets, tables, diagrams and other supporting documents shall as far as possible be of the same size, well annotated, numbered and appropriately cross-referenced.

4.7 Application must be accompanied by two samples of the finished product as packaged for sale. The NMRAs may request for additional samples when need arises.

4.8 The processing fee as prescribed in the respective NMRA's fees and charges schedule must be paid to the NMRA at the point of submission of the application.

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5. PROCESSING OF APPLICATIONS (MANAGEMENT OF APPLICATIONS)

5.1 Upon acceptance of an application, an acknowledgement for the receipt of the application will be issued within and a reference number will be generated. The reference number shown in this acknowledgement should be used in all subsequent correspondences relating to the application.

5.2 The NMRA shall complete screening of the dossier for completeness within 30 working days from receiving such application.

5.3 In the event that the dossier is incomplete, it will be rejected. The applicant will be notified of the rejection.

5.4 In case of a positive outcome during screening, the NMRA shall notify the MAH in writing that the screening has been successfully completed and place the dossier in the evaluation queue.

5.5 Review of application for marketing authorization of a product will follow the appropriate evaluation queue. Priority review may be granted where the product is intended for treatment of a serious or life-threatening disease. Evaluation of priority product shall be carried out within 6 months from receiving the application.

5.6 Evaluation of the application shall be carried out within 12 months from receiving the application.

5.7 Abridged evaluation will be carried out to pharmaceutical products that are registered in any of the agreed benchmark regulatory agencies.

5.8 During product evaluation, the NMRA may request for further information and additional supporting documents from the applicant. This shall be considered as the first round of evaluation.

5.9 Applicant should make available such information or documentation requested after the first round of evaluation within 180 calendar days from the date of receipt of the request.

5.10 Applicant should make available any information or documentation requested after subsequent rounds of evaluation within 120 calendar days from the date of receipt of the request.

5.11 If no response is received from applicant after the timelines described in 5.9 and 5.10 above, the clock stops and the application will be cancelled if no formal request for extension of deadline has been made to the NMRA. A new application

will have to be submitted if the MAH wishes to pursue marketing authorization of the product.

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5.12 Evaluation of the additional information shall be carried out within 3 months from receiving such information. This shall be considered as the second round of evaluation and subsequent submission of additional information shall be considered as third round of evaluation and so forth.

5.13 Evaluation of one application shall not exceed four rounds of evaluation with the exception of administrative queries.

5.14 The MAH will be informed of the decision of the NMRA in writing as to whether the application has been approved or rejected.

5.15 A registration number will be given when a product is registered. The registration number is specific for the product registered as specified in the registration documents. A certificate of registration shall be issued for the registered product.

5.16 For a product to be issued MA, it must be manufactured in a GMP compliant facility and studies conducted following GCP.

6. MAINTENANCE OF MARKETING AUTHORIZATION

6.1 The conditions for marketing authorization of pharmaceutical products are as follows:-

6.1.1 The product registered with the marketing authorization number as stated in the marketing authorization certificate shall have the name, composition, characteristics, specifications and origin as specified in the marketing authorization documents.

6.1.2 The holder of the marketing authorization certificate must supply such documents, items, samples, particulars or information as the NMRA may require in relation to the registered product.

6.1.3 Changes in name, composition, characteristics, origin, specifications, manufacturer, packing, indications, labelling, package insert, product literature or any other particulars of the registered product shall not be made without prior approval from the NMRA.

6.1.4 The labels for the registered product must comply with all of the labelling requirements as specified by the guidelines for labelling.

6.1.5 The registered product must only be indicated for use as approved by the NMRA.

6.1.6 The holder of the marketing authorization certificate must inform the NMRA of any adverse reactions or complaints on quality, safety and efficacy of the registered product immediately after he/she becomes aware of such adverse reactions or complaints.

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6.1.7 The holder of the registration certificate must notify in writing to the NMRA of any decision to withdraw the marketing authorization of the product and shall state the reasons for the decision.

6.2 MAH shall be required to pay retention fees as specified by the respective NMRA.

6.3 The registration of a product shall be valid for 5 years or such period as specified in the registration certificate (unless sooner suspended or cancelled by the NMRA).

6.4 The renewal of product registration should be done not later than three months prior to expiry. Applications for renewal of registration shall be made by submitting the following:

- a) Duly filled in application form for registration.
- b) Batch Manufacturing Record (BMR) of a real batch manufactured within at most six months before the submission of the application.
- c) Details of all changes during validity of the registration.
- d) Two samples of the finished product as packaged for sale.
- e) A site master file that describes the manufacturing facilities.
- f) Non-refundable evaluation fee for registration of pharmaceutical product and GMP and GCP inspection fees for facilities not inspected and approved by the NMRA within a period specified by NMRA.

7. CANCELLATION OR SUSPENSION OF MARKETING AUTHORIZATION

7.1 The NMRA may cancel or suspend the marketing authorization of any product if there are deficiencies in safety, quality or efficacy of the product or failure to comply with the marketing authorization requirements or due to changes in national policies.

7.2 Such products may not be imported and marketed in the country. The holder of the registration certificate shall immediately surrender to the NMRA the marketing authorization certificate upon cancellation or suspension of marketing authorization of the product.

7.3 NMRA shall notify all Partner States on the decision taken by the respective NMRA.

8. APPEALS AGAINST NMRA'S DECISIONS ON PHARMACEUTICAL PRODUCT MARKETING AUTHORIZATION

8.1 For products that have been suspended and cancelled marketing authorization by the NMRA, MAH may make a written appeal to the NMRA to review its decision.

8.2 All notice of appeals must be made within thirty (30) calendar days from the date of the NMRA's notification.

8.3 MAH shall make appeal by giving grounds for review for each reason given for the rejection of his product. The grounds for the request shall be based on the information that was submitted in the product dossier. Any additional or new information that was not earlier submitted will not be accepted. The NMRA may review or uphold its earlier decision.

9. VARIATIONS IN PARTICULARS OF REGISTERED PRODUCTS

All variations to a registered product shall be made according to requirements stipulated in the EAC Application Guidelines for Variation of Registered Pharmaceutical Products.

9.1 EXTENSION APPLICATIONS

9.1.1 An extension application is an application that is a modification of an already registered medicinal products. The modification shall be such that it does not fulfil criteria for minor or major variations but is similar enough to the original product in terms of quality, safety and efficacy.

9.1.2 A marketing authorization holder may apply for extension of marketing authorization of an already registered product as an extension application. Such an application should be submitted as a new application however an abridged evaluation will be carried out.

9.1.3 Extension applications shall be applicable in the following situations:-

a) Changes or addition of a pharmaceutical form from multi-dose to single-dose of the finished product or vice versa.

b) Change or addition of strength of the finished product.

c) Change or addition of a route of administration of the finished product for products of the same pharmaceutical form.

d) Inclusion of medical devices that result in change of strength, pharmaceutical form or route of administration of the finished product.

9.1.4 An extension application shall be accompanied by the following:-

a) A duly filled in applicant form with the extension application box clearly marked (ticked).

b) The applicable registration fees for applications for registration of new applications.

c) A full dossier submitted in accordance to the requirements stipulated in the guidelines for submission of documentation for registration of human medicinal products.

d) A cover letter declaring the following:

- i. The name and registration number of the relevant product from which the extension is applied.
- ii. The marketing authorization holder for both products shall remain the same.

e) An overview of the nature of the extension being made.

f) Supporting data related to the proposed extension.

9.1.5 The NMRA final decision on whether an application meets the criteria for extension applications will lie with the NMRA. In case of any doubt the MAH may contact the NMRAs before filling for an extension application.

9.2 Duplicate Licensing

9.2.1 The NMRA shall authorize the same applicant to submit more than one application for a finished product when there are objective verifiable reasons on public health grounds regarding the availability of finished products to health-care professionals and/or patients, or for co-marketing reasons and/or for Export purposes.

9.2.2 Additionally, the holder of a marketing authorization can grant the use of product information to another marketing authorization holder, whereby the original marketing authorization holder acts as a contract manufacturer.

9.2.3 The assessment on whether the conditions of a duplicate application are met shall be done on a case-by-case basis, having regard to the facts of each application. The overall objectives being preservation of public health.

9.2.4 To assess whether an application refers to a particular finished product that has

already been granted a marketing authorization, and consequently, whose application for a marketing authorization qualifies for a duplicate license, the composition in active substance(s) and the pharmaceutical form shall be considered. Thus, any finished product with the same qualitative and quantitative

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composition in active pharmaceutical ingredient (i.e. the same strength) and the same pharmaceutical form are to be considered as the same relevant product.

9.2.5 A duplicate product shall be identical in all marketing authorization requirements with the exception of brand name and any other specific requirements on labelling. Additionally, any variation made to the original marketing authorization should be applied for the duplicate license.

9.2.6 Conditions for a Duplicate Marketing Authorization are outlined hereafter:-

a) That the duplicate application shall be submitted by the same applicant that submitted and/or holds the marketing authorization/application that is being duplicated (hereafter "original marketing authorization/application").

b) That the original marketing authorization is valid. This step does not apply in case of duplicate applications that are submitted in parallel with the original marketing authorization application (i.e. in cases where the application for the original marketing authorization is still pending).

c) In cases where the duplicate marketing authorization is submitted on the basis of an informed consent application, there should be a letter of consent from the marketing authorization holder that owns the dossier that is referred to.

d) The original marketing authorizations to which the duplicate application relates has to be valid at the time of the submission of the duplicate application.

9.2.7 The applicant for a duplicate license may fall under the following categories:-

a) Applicant is the same entity that applied for the original marketing authorization.

b) Applicant belongs to the same group of companies as the applicant of the original marketing authorization.

c) Applicant is an independent entity that has agreed to placing on the market the product with the applicant of the original marketing authorization (evidence of license agreement or other agreement that can be identified are required).

d) Applicant is an independent entity whereby there are license agreements with the marketing authorization holder of the product in respect of which the duplicate is asked but not for the placing on the market of that product.

e) Applicant is an independent entity that has got an agreement to purchase and/or use data from the company that has applied for a marketing authorization for the product for the first time but there is not an agreement regarding the placing on the market of the product.

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9.2.8 All documents in accordance to the guidelines on submission of documentation for new applications should be submitted however an abridged evaluation shall be applied. In addition, the following shall be submitted when making a duplicate license application:

a) A duly filled in application form (Annex II) with the duplicate license box clearly marked (ticked).

b) A cover letter detailing the following:

- i. The name of the marketing authorization holder relevant for the duplicate application.
- ii. The name of the product relevant for the duplicate application.
- iii. The proposed brand name for the duplicate license.
- iv. The proposed marketing authorization holder for the duplicate license.

c) The applicable registration fees for applications for registration of new applications.

d) For co-marketing reasons, the evidence co-marketing (contract or letter of agreement between the companies).

e) For duplicates asked on grounds of the existence of patents protecting certain therapeutic indications or pharmaceutical forms, the applicant shall provide a commitment undertaking to extend the therapeutic indication(s)/ pharmaceutical form(s) of the duplicate marketing authorization as soon as the patent restrictions no longer exist. Alternatively, the applicant may also commit to withdraw the marketing authorization with restricted indications/pharmaceutical forms after the relevant patents are no longer in force. The SmPC shall be harmonized. The commitment letter shall be submitted alongside the Marketing authorization Application Dossier.

f) Letter of consent in the case of an "informed consent application".

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Medicinal Products in the MENA Region –

Differences and Similarities to European Procedures

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Zohaira El Malahi

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Erstprüfer: Dr. Rose Schraitle

Zweitprüfer Dr. Birka Lehmann

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Maintenance of Marketing Authorization in the MENA Region

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List of Abbreviation

ANPP National Agency of Pharmaceutical Product

ANSM Agence nationale de securité du médicament et des produits de santé

API Active Pharmaceutical Ingredient

CA Competent Authority

CAPA Central Administration of Pharmaceutical Affairs

CEP Certificate of Suitability of Monographs of the European Pharmacopoeia

CHMP Committee of Human Medicinal Product

CMS Concerned Member State

CP Centralised Procedure

CTD Common Technical Document

DCP De-Centralised Procedure

DGPA & DC Directorate General of Pharmaceutical Affairs & Drug Control

DHA - Dubai Health Authority

DMF Drug Master File

DMP Direction des Médicaments et de la Pharmacie

DTA Iraqian Directorate of Technical Affairs

DPM - Pharmacy and Medicine Directorate

eCTD electronical Common Technical Document

EDA Egyptian Drug Authority

EMA European Medicines Agency

EPAR European Public Assessment Report

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EU European Union

FOB Free On Board

GCC – DR Gulf Central Committee for Drug Registration

GMP Good Manufacturing Practices

HA Health Authority

HAAD Health Authority of Abu Dhabi

ICH International Conference of Harmonization

KDFC - Kuwait Drug and Food Control Administration

KMCA - Kurdistan Medical Control Agency

JFDA - Jordan Food and Drug Administration

LNCM Laboratoire National de Contrôle des Médicaments

MENA Middle East North Africa

MoH Ministry of Health

MOHME Ministry of Health and Medical Education

MoPH Ministry of Public Health

MRP Mutual Recognition Procedure

MSDS Material Safety Data Sheet

NHRA National Health Regulatory Authority of Bahrain

NODCAR - National Organization for Drug Control & Research

NORCB - National Organization for Research and Control of Biologics

PIL Product Information Leaflet

PSUR Periodic Safety Update Reports

QP Qualified Person

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RMS Reference Member State

SFDA Saudi Food and Drug Authority

SmPC Summary of Product Characteristics

TSE Transmissible Spongiforme Encephalopathie

USA United States of America

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Maintenance of Marketing Authorization in the MENA Region

Executive summary

The aim of this master thesis is to identify and define the regulatory requirements for maintenance procedures of marketing authorization for medicinal products in the countries of the Middle East and North Africa.

The importance of marketing authorizations of medicinal products is not only limited in the approval of new marketing authorization. It is more important to maintain existing marketing authorizations in these countries to ensure the prevention, diagnosis and treatment of diseases and for patient rehabilitation.

In the last 10 years the MENA region has become one of the most emerging markets in the world. This is due to the growth of the population, the increasing affluence and the increase of life expectancy. Inside the region, the tendency of the increased numbers of MAH and manufactures increased since several is still ongoing and the authorities are developing and modifying new regulations and laws in order to justify this development.

Even the developing process of adequate regulations for medicinal products is still ongoing, in most of the MENA countries there are a lot of challenges. Many of the regulations are only available in Arabic. For the Maghreb States the submission dossier has to be provided in French. Additionally the political situation in some MENA country is not stable and the processes are not transparent enough. These barriers complicate the authorization and maintenance for medicinal products in the MENA region. Finally the population has to face up with the consequence of these barriers.

This master thesis will provide an overview about the maintenance procedures for marketing authorizations in the MENA region and to compare them with the European procedure. This comparison conveys how far the development of the pharmaceutical sector in the MENA region is in opposite to Europe and how the trend will develop in the next years.

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Maintenance of Marketing Authorizations in the MENA Region

1. Introduction in Regulatory Affairs Department in MENA

The MENA region is one of the "emerging" markets in the pharmaceutical sector in the world. The importance of this region for many multinational pharmaceutical companies is very high. This section will provide some basic information to gain a better understanding of how the pharmaceutical market is organized and to maintain a marketing authorization in the MENA region.

1.1. Definition of MENA and geographical location

The term "MENA" is defined as "Middle East and North Africa" and includes in general 22 countries from Morocco at the eastern border until Iran in the north of the region. (1)

Due to the fact, that a firm definition does not exist, there are several opinions that Turkey belongs to the MENA region. In this case we will follow the definition of the World Bank and exclude Turkey in this analysis.

According to the World Bank and UNICEF the following countries belong to the MENA region: Algeria, Bahrain, Egypt, Iran, Iraq, Israel, Jordan, Kuwait, Libya, Morocco, Oman, Qatar, Saudi Arabia, Syria, Tunisia, United Arab Emirates (UAE), West bank and Yemen. (2)

Figure 1: Overview of countries of the MENA region

Source: https://www.researchgate.net/figure/s-a-map-showing-list-of-MENA-countries-1_fig1_318837933

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Inside the MENA region we have to divide three subgroups:

Middle East:

- GCC countries: Saudi-Arabia, UAE, Bahrain, Oman, Qatar, Kuwait, Yemen
- Levant countries: Iran, Iraq, Syria, Lebanon, West Bank, Jordan, Israel, Egypt, Libya: These countries are not part of any union. (3)

North-Africa:

- Maghreb-States: The countries Morocco, Algeria and Tunisia belong to the Maghreb countries.

The political situation in the MENA region is middling. Since the beginning of the Arab revolutions in 2011, the political situation in some countries is quite instable. In Syria, Yemen and Libya the current situation is not transparent and there are still civil wars in these countries. The medical supply is quite inadequate and the procedures to authorize import or maintain medicinal products into the affected markets are not up-to-date and not in use. Due to this fact these three countries will not be taken into account for the aim of this master thesis.

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1.2. Pharmaceutical & Healthcare sector in the MENA region

The pharmaceutical and healthcare sector in the MENA region grows up since the 1970's. The need of medicinal products and healthcare was growing up due to the growth of the population of 532 Million people. The increased life expectancy of the population and the decreased death mortality are one of the most important reasons. Additionally, the change of the life cycle during the last years leads to many widespread diseases, such as Diabetes, cancer or cardio-vascular diseases. (4)

The region belongs to one of the Emerging markets in the world after China. The amount of the MENA region in the world belongs to \$32 Billion dollar, which makes only a percentage of 2 percent. (5) The pharmaceutical sector is estimated to grow from 9 to 11 percent until 2020. (6)

Figure 2: Increase of Diabetes as widespread disease in the world

Source: https://www.huffingtonpost.com/riva-greenberg/diabetes-stats_b_4273505.html

The increased demand of medicinal products is shown in the figure. The amount of people with diabetes will be increased in the MENA region about 96.2 percent until 2035.

Inside the MENA region nearly all competent authorities and Institutions are in the process to develop easier procedures for granting authorization of medicinal products to counteract against this need. This challenge leads to the growth of the pharmaceutical sector in the region. Since 1970 more than 140 local pharmaceutical companies are located in countries of the MENA region.

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Figure 3: Development of MENA Health Expenditure in comparison with other markets

Source: http://www.gphaonline.org/media/cms/Osama_Nabulsi.pdf

Most of the domestic pharmaceutical companies are generic drug manufacturer. This is due to the fact that the development of new innovative drug substances is not well advanced in the MENA region. A large part of the countries are dependent on import of medicinal products from USA, Europe and China. Especially for the manufacturing, the import of active drug substances and ingredients are essential. This heavy dependence aggravated the development of the pharmaceutical sector in the MENA region.

Another point is the strong individuality and the status of development inside the MENA region. The Gulf States are well-advanced in the development of appropriate procedures and regulations, but the Maghreb States are backward in this case. In the Maghreb States a strong dependence on laws, regulations and procedures during and after the authorization of medicinal products to the French Authority ANPP. This can be judged from the fact that the price for medicinal products in Algeria, Morocco and Tunisia are similar to the actual prices of medicinal products in France.

In the last decade, local authorities and institutions are in the progress of developing appropriate procedures for authorization and maintenance of MA of medicinal products in regard to ICH-Guidelines. This development leads also to an increase of the amount of privatization of pharmaceutical companies.

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Figure 4: Number of pharmaceutical companies in the countries of the MENA region (private and non-private companies)

The figure shows, that the pharmaceutical market size in almost all countries is mainly generated by state-own enterprises, except for Iraq, where private pharmaceuticals companies makes more than 50 percent of the market size. It is

expected that the amount of private companies will increase in the next years. This development shows that with the privatization of companies in the countries leads to more independence on imports of medicinal products in the region.

All these points show that the development of the pharmaceutical industry in the MENA region is in progress, but many important milestones still need to be reached. In contrast to the industrial nation as the United States, Europe or Japan the development and manufacturing of new innovative drug substances and biologicals are not well developed in the countries of the MENA region. Additionally the manufacturing and authorization of medicinal products is difficult in many countries, due to the long approval timelines of the authorities. The missing resources inside the authorities complicate the access of pharmaceutical products to the local markets. In nearly all countries the Ministries of Health contain appropriate departments for authorization of medicinal products instead of European countries, where "own" competent authorities are responsible for marketing authorization and post-approval activities of medicinal products.

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1.3. Overview of Competent Authorities in the MENA region

To understand the procedure of the maintenance of marketing authorizations in the MENA region it is important to get an overview about the National Competent Authorities in each country.

1.3.1. Competent Authorities in the Gulf States

Kingdom of Saudi Arabia (KSA)

Since 2003 the Saudi Food and Drug Authority (SFDA) is the competent authority for registrations, maintenance, quality, pharmacovigilance and import of medicinal products. (7) It is an independent authority from the Ministry of Health.

United Arab Emirate (UAE)

Seven Emirates belongs to the United Arab Emirate. Abu Dhabi (the capital), Dubai, Sharjah, Ajman, Umm Al-Qaiwain, Ras Al-Khaimah and Fujairah.

In UAE there are two competent authorities who are responsible for all regulatory issues of medicinal products.

The Health Authority of Abu Dhabi (HAAD) is responsible for medicinal products in the capital emirate Abu Dhabi and for Dubai it is the Dubai Health Authority (DHA). For the other emirates, the Ministry of Health and Prevention is responsible for all drug related issues. For the management and regulation of health services the Health Authority of Abu Dhabi and the Dubai Health Authority merged together. (8)

Bahrain

For all pharmaceutical issues as licensing, healthcare professional's regulation, Healthcare facilities regulation and complaints the National Health Regulatory Authority of Bahrain (NHRA) takes the responsible. (9) The NHRA is an independent institution to the Ministry of Health.

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Oman

The Directorate of Pharmaceutical Affairs & Drug Control as part of the Ministry of Health is responsible for registration, import, maintenance and quality of medicinal products authorized in Oman. (10)

Qatar

In Qatar, the department of Pharmacy & Drug Control as part of the Ministry of Public Health takes the responsibility for all drug related issues such as registration, quality control, inspection and drug release. (11)

Kuwait

The Kuwait Drug and Food Control Administration (KDFA) is part of the Ministry of Health. This department is the competent body for all drug related issues as quality control, authorization and assessment of marketing authorizations for medicinal products. (12)

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1.3.2. Competent Authorities in the Levant countries

Egypt

The Egyptian Drug Authority (EDA) is a competent authority body within the MoH, which is responsible for issues regarding authorization and control of medicinal products. (13)

- The Central Administration of Pharmaceutical Affairs (CAPA) is an entity within the EDA, which takes the responsibility for the assessment and licensing for medicinal products. (14)
- The National Organization for Drug Control & Research (NODCAR) is responsible for all Quality related issues of nationally authorized products in Egypt. (15)
- For Biologicals the National Organization for Research and Control of

Biologics (NORCB) is the main department of authorization, import & quality related issues for Biologicals. (16)

Iraq

In Iraq there is a special situation. The Iraqi Directorate of Technical Affairs (DTA) as part of the MoH is responsible for all drug related issue as registration, inspection and quality control for medicinal products in the Iraqi market. (17)

Since 1992 the MoH in the Kurdish autonomous area establish the Kurdistan Medical Control Agency (KMCA). This competent body is responsible for the registration, quality control, distribution and quality control of medicinal products in Kurdistan. (18)

Iran

The Food and Drug Department of the Ministry of Health and Medical Education (MOHME) is responsible for all issues related to the authorization, maintenance and marketing of medicinal products in Iran. (19); (20)

Israel

For the authorization, control and marketing of medicinal products (human, herbal and veterinary products) the MoH in Israel is responsible. The Registration Department is responsible for the licensing processes while the Institute for

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Standardization and Control of Pharmaceuticals is responsible for the quality control of authorized medicinal products in Israel. (21); (22)

Jordan

The Jordan Food and Drug Administration (JFDA) belong to the MoH and take the responsibility for licensing processes for medicinal products and food products.

Inside the JFDA the Registration Division is the main directorate for the authorization for medicinal products in Jordan. (23)

Lebanon

The Ministry of Public Health (MoPH) is responsible for all drug related issues.

The "Service of Pharmacy" as part of the MoPH has the following activities:

- Issuing certificates related to pharmacies and pharmacists' practice
- Drug pricing
- Drug industry management and control
- Narcotic drugs' imports, distribution, and statistics
- Medicinal imports/exports
- Drug registration and control
- Registration of non-medicinal health-related items
- Pharmacies' and drugstores' inspection
- Controlling fraud in the pharmaceutical industry (24)

West Bank

The General Directorate of Pharmacy as part of the Ministry of Health take the responsibility of all licensing, quality control and price issues for medicinal products authorized and marketed in Palestine. Inside the Directorate there are separate departments for each section

- Registration Department
- Drug Quality Department
- Pharmaceutical Policy Department
- Dangerous Drug Department (narcotics and psychotropic drugs)
- Drug Information Department

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- Import and Export Department (25)

For the authorization of medicinal products, medical devices, food products and cosmetic products the Drug Registration Department of the General Directorate of Pharmacy is the main contact point. (26)

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1.3.3. Competent Authorities in the Maghreb States Morocco

In Morocco, the Ministry of Health is the main body for healthcare for public health. The department "Direction des Médicaments et de la Pharmacie"(DMP) is responsible for all drug related issue, as authorizations, maintenance, monitoring and distribution of pharmaceutical products. (27)

Algeria

The "Pharmacy and Medicine Directorate" (DPM) is an independent body of the Ministry of Health and Population, which is responsible for all regulatory topics for medicinal products. (28) Since 2017, the National Agency of Pharmaceutical Product

(ANPP) replaced the DPM and took the responsibility as independent and non-government agency for all drug related issues as drug registration, maintenance and distribution. (29) (30) (31)

Tunisia

The Ministry of Public Health is the main body related to regulatory issues for medicinal products. Since 1981 the Directorate of Pharmacy and Medicine (DPM) is a part of the Ministry of Public Health is responsible for the management of drug related issues in Tunisia. (32)

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1.3.4. Gulf Central Committee for Drug Registration

The Gulf Central Committee for Drug Registration is a committee founded by six of the Gulf States in 1999. The members of the committee are

- Saudi Arabia
- Qatar
- Oman
- Bahrain
- Kuwait
- United Arab Emirates
- Yemen

The committee is responsible for the assessment and authorization of medicinal products, which should be authorized via the "Central Drug registration". Marketing authorizations from the GCC-DR will get a marketing license for all members of the GCC. (33)

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2. Marketing Authorization in the MENA Region

In the last years, the MENA region became more important for international pharmaceutical companies. This is due to the fact that 32 Billion Dollar was made in pharma sales and the trend is still growing (34).

Inside the MENA region, besides the diversity of culture, language, and tradition, there are significant differences in the processes for obtaining marketing authorizations in the countries. Therefore, it is difficult to describe the processes for the whole region. In this part we will take the opportunity to summarize the most important parts of marketing authorizations in the MENA region.

A main reason for the complexity and diversity of the processes for marketing authorization of medicinal products are due to the development of the health- and pharmaceutical sector in the respective countries. In the last years significant progresses were done in the MENA region, mainly in the GCC-States and in some Levant States as Egypt and Israel. The numbers of local manufacturers increased in this region over the recent years. (34). This indicates that the region tries to get independent from medicinal products imported from other parts of the world. The pharmaceutical and health sector grows in these countries also due to the increased population and medical needs in this region.

Even there are positive trends for the whole region, inside the MENA region there are significant differences in the regulatory "know-how" and experience. While many GCC countries and several Levant States as indicate much progresses in the healthcare and pharmaceutical development and regulatory of medicinal products, the Maghreb States are not on the same level. There is a historical background behind the significant differences inside the MENA region. Until 1956 the Maghreb States were under French colonial power and this had a major impact on all administrative bodies, including the Ministries of Health and the Committee for all drug related issues. Especially the Algerian and Tunisian Authorities often refer to the decision of the French NCA ANSM (Agence nationale de sécurité du médicament et des produits de santé).

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Despite the numerous progresses in the recent/ past few years, there are still many aspects inside the pharmaceutical sector, which are not yet regulated adequate enough. The NCA of the MENA region have limited experience in the fields of biologicals and gene-therapy. Their experience with such kind of medicinal products cannot be compared with the European or American NCA's. There are only a few special laws and regulations for these kinds of medicinal products and this indicates that this field is still in progress. (35)

The timeline to get a marketing authorization depends on certain countries. A general statement is not possible. While the authorization procedures in the GCC States, Maghreb States and some Levant countries take 2 – 4 years, the same procedure can take up to 5 years in Iraq and Iran, even it is the same dossier for a medicinal product, which were submitted to other MENA countries. In comparison to this, a marketing authorization for a Centralized procedure in Europe takes 300 days, including the decision of the European Commission.

In the most countries of the MENA region, a local agent is required for the communication and the submission to the Health Authorities. For a MAH outside the respective country, it is not allowed to submit any application to the authorities, except for medicinal products, which were authorized centrally by the Drug Registration of the GCC (GCC-DR). Due to this fact, many multi-national companies have several establishments in these countries to facilitate the communication to the

authorities and to fulfill the local requirements. The local agent is the main contact point between the local authority and the MAH. This obligatory is a problem for small and medium enterprises, because due to financial aspects and the short range not every company can establish local agents in these countries. The result is that medicinal products of these companies cannot be marketed in the MENA region or only a small portfolio is available.

Another important role beside the local contact point in the certain country is the format of the dossier. Nearly all countries require the dossier as CTD format (agreed in line with the ICH-M8 Guideline), but till now it is not possible to submit the dossier electronically in all countries of the MENA. Especially in some Levant states and in the Maghreb states, an electronically submission of an eCTD is currently not possible. (36)

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Many authorities still require the dossier in hard and soft copies as CDs/DVDs. The reason for this is that many countries are not yet ready for eCTD submissions because the technical conditions are still missing. On the contrary, Saudi Arabia and Bahrain require eCTD submissions via electronically portals since 2015. (37) In Egypt and Lebanon, the CTD is still not implemented. The documents have to be provided electronically as simple pdf dossier via Email and paper-based. (38)

Figure 5: Definition of CTD dossier according to ICH

Source: http://www.ich.org/fileadmin/Public_Web_Site/ICH_Products/CTD/CTD_triangle.pdf

For the Maghreb States, the documents in the CTD should be provided in French. This leads to a serious problem for the applicant, because for the two biggest markets - the US and Europe and most of the countries in the world, the language requirements of submission dossiers are in English. Therefore, it is quite complicated if the whole submission package needs to be translated into French. This additional workload leads to a high workload and a massive loss of time for the applicant.

Another special feature for marketing authorizations is that all authorities required a legalized CPP from the country of origin (COO). This requirement is valid for all MAH outside the respective country. The country of origin is defined as the country where the batch release takes place. (39)

It is also required, that the medicinal product has a valid license in the COO. In many countries it is not possible to maintain a marketing authorization (MA) without a valid license in the country of origin. If the MA is expired in the COO, the MAH requires a justification in the application form.

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All these points have to be taken into account for pre- and post-marketing authorization activities especially for renewal procedures.

The assessment processes at the authorities' sites are not transparent and clear to the applicant. The timelines are not well communicated and this leads to the consequence that the applicant has to consider a delay for market access of the medicinal product due to the absence of information about the timelines and the assessment process. Due to the missing information about timelines, the applicant must always be prepared for further requests from the authorities at any time of the renewal procedure.

Another point of discussion for receiving a marketing authorization in the MENA region is the fact, that there are no mutual recognition procedures inside the region. The authorities do not accept any approvals from other authorities of the region for guarantying a MA in the respective country. In opposite to the countries of the European Union, where a MA can be guaranteed via the Centralized, De-centralized or Mutual Recognition procedures, only national procedure for MA is available. The only exception is the central authorization procedure in the Gulf States. Since 1999 it becomes possible to get a marketing license for a medicinal product with one central procedure for all Gulf States and Yemen. Therefore, the Applicant has to submit a dossier for marketing authorization to the Gulf Central Committee for Drug Registration (GCC-DR) and after the assessment and the positive approval, the medicinal product can be marketed in all Gulf countries.

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3. Maintenance of Approved Medicinal Products – Middle East and North Africa

This part contains information about relevant procedures to maintain a marketing authorization of a medicinal product in the MENA Region. The main focus is the description of "renewal procedures" in this region. The first part of this section will be an overview about the renewal processes in the MENA region as whole region and in each individual country. In the second part of this section, there will be a discussion about other procedures to maintain a marketing authorization in different countries. The last part of this section will expand the issue of availability of harmonization processes of renewal or other "maintenance" procedures in the MENA region.

3.1. Renewal as procedure to maintain marketing authorizations in the MENA Region

For all countries in the Middle East and North Africa the Marketing Authorization holder is obligated to submit a renewal application to maintain the marketing authorization in the relevant countries.

As mentioned in the previous part, the MAH should have a local agent within a

country, if the MAH is stated outside the affected country. This local agent is responsible for the renewal application and he is stated as main contact point for the authorities.

Due to the fact that the development of regulations affecting the marketing authorization of biological are still in development, the requirements for renewal applications do not differ between chemical entities and biological. Only in Israel and Egypt, the national competent authorities require other documents for the renewal applications for chemical entities and for biologicals. (40); (41)

In general, all marketing authorizations have to be renewed every 5 years from the beginning of the initial registration. Except in Iran the timelines for renewal of marketing authorization is 4 years. (42) In Israel there is the exemption that if the product is not changed in the manufacturing processes; the renewal has to be done 10 years after the initial registration.

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The renewal application has to be submitted within the last year of the marketing authorization. Most of the countries require that the applicant has to submit the dossier within 3 upon 9 months before the marketing authorization expires. In Jordan, it is usually that the renewal will be submitted at latest 6 months after expiry of the marketing authorization. (43) (44)

In many MENA countries it is required to submit the dossier of the initial marketing authorization to the competent authorities for renewal application. It is unusual to submit a shortened dossier containing only Module 1 and Module 2 for the initial renewal application. For further renewals, some authorities as the DMP in Morocco only require a cover letter containing a declaration that the license should be continued in the relevant country. (45)

In some countries, there are minor differences between the required documents from local manufactured where the MAH is stated in the country and for imported products, where the MAH is stated outside the country. In cases that the medicinal products where imported from foreign countries, the authorities always require a CPP of the country where the batch release takes place. For renewal applications, the applicant has to submit a list of all approved variations, which were submitted and approved during the time of the marketing authorization. This requirement is valid for all MENA countries.

In many MENA countries, a valid license of the current manufacturing site is a condition for submission of renewal applications. Especially in the Gulf and Levant states, the authorities require a "manufacturing site renewal" before the renewal of the marketing authorization. Without a valid manufacturing license, a renewal application for a MA is not possible.

Another important requirement for a renewal application in the MENA region is a valid marketing authorization in the country, where the MAH is stated (except the MAH is stated in one country of the MENA region). Some countries refer during the renewal assessment to the marketing authorization and the opinion of FDA, EMA or from other authorities. For the Maghreb States as Algeria, the opinion of the French authority ANPP will be take into consider for the assessment. (46)

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Nearly all countries require a submission of a renewal application dossier as CTD sequence. In the Gulf countries, a submission of an eCTD becomes mandatory. (36), (39) (47)

In other countries, a submission of hard and soft copies of the dossier to the NCA's is required. Only in Egypt and in Lebanon a submission of the renewal as CTD format is not possible. It is required to submit the dossier via Email or as hard copy to the different departments of the authorities, which are involved in the assessment of the renewal application. (48); (49); (50)

The timeline for renewal applications in the MENA countries are different. The assessment timelines can take from 15 days up to 24 months. In Tunisia the assessment of a renewal takes only 15 days, but in Egypt a renewal application can take up to 24 months. (51) In general, a renewal application takes 6 months. For a part of the countries, a procedure timeline is not applicable.

Another important point to consider is the fact, that in most cases the assessment procedure after submission of the renewal application is not well communicated to the applicant. Due to this fact, the applicant has to consider that the authorities may send questions or a request for more documents to any time of procedures. Additionally the approval of the renewal application cannot be foreseen and this should be involved in the planning of the submission

The countries of the MENA region are very individual and every authority refers to their own standards. In the next section, the renewal application for each country will be discussed.

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3.2. Renewal procedure in the countries of the MENA region

In this section, the renewal procedure in each country will be discussed. At the beginning, the procedure of renewal in the GCC States, the Levant States and the Maghreb States will be specified.

3.2.1. Renewal application in the Gulf States

3.2.1.1. Kingdom of Saudi Arabia

Legal Basis

In the "Law of Pharmaceutical Establishments and Preparations" from 2004 and the "Executive Rules of the Institutions and Pharmaceutical Products Law" the mandatory for renewal of a MA in KSA is stated. The renewal application has to be submitted within 6 months before the marketing authorization expires. (37) (47)

Required documents

The Applicant has to submit the renewal dossier in CTD format. The application form for renewals is the same as for the initial marketing application. (37) (47)

For the renewal application it is mandatory to submit the Module 1 and Module 3 within the CTD format to the SFDA. It is mandatory to submit the renewal dossier electronically via "Saudi Drug Registration System (SDR System)". Additionally, it is required to submit the dossier via paper.

The following documents are required for the submission (52)

Module 1

- Cover Letter
- Application form [Annex I - Application Form for Renewal application in KSA]
- Table of content
- Product Information
- Certificate of suitability for TSE, and the
- price list

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For MAHs, which were not stated in Saudi Arabia it is required to include a Certificate of Pharmaceutical Product (CPP). (52)

- The CPP from COO

Module 3

- specification of the drug substance (3.2.S.4)
- specification of the finished product (3.2.P.5.1)
- stability data of the finished drug product (3.2.P.8).

Assessment procedure and timelines

The renewal process can be divided into two main phases.

Figure 6: Schematic figure showing the renewal process of a marketing authorization

Source - Saudi Food & Drug Administration Regulatory Framework for Drug Approvals (version 5.0)

The validation phase (Phase II) takes 10 days for authority to ensure that all requirements for the renewal application were fulfilled by the applicant.

The next phase is divided into three main sub-processes:

- The related product manager will start the assessment by distributing the renewal application to three departments (Quality, Efficacy, and Safety). All

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three departments have to finish the assessment within 50 days and forward their assessment reports back to the project manager. (47)

- Within 30 days the price department has to check the rules for pricing of the affected medicinal product. (47)
- Lastly, the inspection department of the Authority checks if the manufacturing line is valid to the last inspection. If there are any changes, which were not included in the last valid inspection report, the manufacturing site has to be inspected again. This phase can take up to 50 days.

After this processes the assessment report of the renewal application will be forwarded to the "Renewal committee". Within 10 days the committee can decide about the renewal application for approval, rejection or ask for further information from the applicant.

After the positive approval of the renewal application, the marketing authorization is valid for 5 years. (47)

3.2.1.2. United Arab Emirates

Legal Basis

For medicinal products authorized in the Arab Emirates, the Circular 28 and 29 from 2007 describes the mandatory and requirements for renewal applications of medicinal products. (53)

A marketing authorization for a medicinal product in UAE is valid for 5 years. To maintain a marketing authorization for medicinal products in the UAE it is required from the Ministry of Health to submit a renewal application within 3 months before the marketing authorization expires. (53)

Required documents

The Ministry of Health differs between originator and generic medicinal products. Additionally it is required to use different application forms for the renewal of "commercial medicinal products", herbal medicinal products, and products for general sales. (54)

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The MAH or the local partner (if the MAH is stated outside the UAE) is obligated to submit a renewal application. It is required to submit the following documents: (54)

- Cover Letter
- Application form
 - o [Annex II - Application Form for Renewal of a conventional medicinal products UAE];
 - o [Annex III - Renewal Application Form for General sale - UAE];
 - o [Annex IV - Application form for Renewal for Herbal Medicinal Product - UAE]
- declaration for the intention of the renewal of a marketing authorization. This declaration is important; otherwise the Authority will withdraw the marketing license for the affected medicinal product without () notification. (54) for MAH located outside of UAE
- legalized CPP from COO
- approved SmPC, PIL and approved artworks

The applicant has to submit an appointment for the submission of the renewal and have to pay the fees before submitting.

Assessment procedure and timelines

After submission of the required documents, the Registration Department will send a confirmation about the receipt of the renewal dossier to the Applicant. The renewal procedure will take 3 months after submission of the dossier. (54)

3.2.1.3. Bahrain

Legal Basis

The Pharmacy Law 18 from 1997 clearly states, that a medicinal products authorized in Bahrain needs a valid license for manufacturing and sales purposes. In Bahrain, a renewal application should be submitted every five years. (39)

Required documents

The NHRA has published a Renewal Guideline which includes all relevant information for the applicant.

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For the renewal application, it is required that the manufacturing site registration is still valid at the time of the renewal. Any variation should be approved before the renewal application, while a common submission of a renewal and a variation application for the same medicinal product is not allowed. (39)

The renewal application should be submitted via eCTD. The dossier should include the following documents: (39)

- Cover letter
- application form for renewals in Bahrain (see Annex V - Renewal Application Form – Bahrain)
- checklist from the NHRA for renewal applications [Annex VI - Renewal

Checklist – Bahrain]

- approved SmPC, PIL and artworks
- CPP of the country where the batch release of the medicinal product takes place.
- A valid GMP certificate of the manufacturing site
- Certificate of Suitability for TSE
- Certificate of analysis for the active substance and the finished drug product and
- Manufacturing registration certificate in Bahrain.

Assessment procedure and timelines

The application should be submitted three months before the marketing authorization expires. Before the submission, the Applicant has to request an appointment of submitting the renewal application.

For the renewal application, it is mandatory to perform a laboratory analyze of the medicinal products upon assessment. Therefore, the NHRA requires samples of the product and several certificates of the product composition, method of analysis, product specification and safety data of the material used in the medicinal product. (39)

There is no timeline for the renewal assessment for medicinal products in Bahrain.

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3.2.1.4. Oman

Legal Basis

A special feature for marketing authorizations in Oman is that the process of "renewal" is called "re-registration" at the Ministry of Health (MoH) and the Directorate General of Pharmaceutical Affairs & Drug Control (DGPA&DC).

As in all Gulf countries, the renewal of a marketing authorization has to be done every 5 years. The renewal application should be submitted within 6 months before the expiry date of the marketing authorization of the affected medicinal products. (55)

Required documents

Before submission of the "re-registration" the applicant has to submit a "submit request" to the MoH before submitting the renewal application.

The applicant has to submit the renewal application via eCTD. The following documents have to be provided (55)

- Cover Letter
- Application form (see Annex VII - Application Form for Marketing and Renewal Application - Oman]
- Samples of the medicinal product
- Pharmacovigilance system
- Risk Management Plan
- CPP
- Certificate of analysis for drug substance & finished product
- Alcohol content declaration
- Pork content declaration
- diluents & color agents in the product formula
- Price list

Assessment procedure and timelines

The DGPA & DC does not publish any information about the timelines and the procedures for re-registration of marketing authorizations in Oman.

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3.2.1.5. Qatar

In the "Law No. 1 of 1986 on the Registration of Pharmaceutical Companies and their Products" it is not stated, that a renewal application for medicinal products authorized in Qatar is required. (56).

Moreover the MoH in Qatar does not publish any information about information about renewal procedures for national marketing authorizations. It can therefore be concluded that a renewal application for a MA of a medicinal product is not required in Qatar and a MA receive an unlimited validity after approval.

Only if the product was approved via the centralized GCC procedure, a renewal according the GCC is required. The renewal application of the centralized procedure in the GCC region will be discussed in a later section.

3.2.1.6. Kuwait

Legal basis

In the ministerial decree 302/80, it is stated, that a medicinal product has to be renewed. The decree is not available for the public. The MoH publish a guidance document for the submission of registration and renewal applications based on the ministerial decree. (57)

A renewal application has to be submitted within 3 months before the MA expires.

Required documents

The following documents are required for the renewal application: (57)

- checklist for renewal application [Annex VIII - Renewal Checklist – Kuwait]
- legalized CPP
- currently approved SmPC
- long term and accelerated stability studies from production batches
- Samples of the finished medicinal product

Assessment procedure and timelines

Information about the assessment procedure and the timelines are not available.

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3.2.2. Renewal application in the Levant States

In this section, the renewal process of marketing authorization of medicinal products in Levant states will be discussed.

3.2.2.1. Egypt

Legal Basis

The renewal of medicinal products in Egypt is regulated in two main ministry decrees. For medicinal products, the decree 425/2015 includes all requirements for registration in Egypt. (48) For Biologicals, the renewal process is described in the

Ministry decree 297/2009. (58)

For medicinal products according to Ministry Decree 425/2015 the renewal process should be done every 10 years. (48) For Biologicals, a renewal of the marketing authorization is required every 5 years. (58)

There is no exact timeline, when the Marketing Authorization Holder has to submit the renewal application to the authority. In the Ministry Decree, it is only stated that the submission should be done in the last year of the expiry date of the marketing authorization. (48), (58)

Required documents

For local and imported medicinal products, the renewal should be performed within 2 years after the previous registration certificate. (48), (58)

In Egypt, the submission dossier cannot be submitted as eCTD sequence. The applicant has to submit the dossier as simple PDF documents.

The renewal application should be submitted to the following email addresses:

- For human drugs: hf296@eda.mohealth.gov.eg
- For Biologicals: Biological_App@Eda.mohealth.gov.eg (59)

It is mandatory to send a request before submission to the EDA. For medicinal products, it is required to submit the following documents: (48), (58)

- request form the latest certificate of registration, and the latest price certificate.

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- for biologicals, the applicant should provide beside the certificate of analysis and the latest price certificate, a list of all variations, the last batch release certificate and the renewal fee receipt.

After reply of the authority on the request for the renewal, the applicant has to submit a "preliminary re-registration file" via Email. This dossier should contain: (48); (58); (59)

- Application form [Annex IX - Renewal Application Form – Egypt]
- Registration certificate
- legalized CPP
- Letter of Authorization between MAH and local partner
- registration fees receipt

Procedure and Timeline

The EDA will inform the applicant via "Renewal Status Letter" that the marketing authorization is under renewal registration.

At the same time, the applicant has to submit the "preliminary re-registration file" to the following department for approval:

- Scientific committee: If no scientific reference is available
- Egyptian Pharmacovigilance Center
- Technical committee communication section:
- Naming & labels committee: if the trade name was changed or for the approval of changes of the labeling for post-stability and pharmacology committee
- Variation committee
- Pricing committee
- Stability committee
- Pharmacology committee
- Bioequivalence committee:

Assessment procedure and timelines

The timeline for the renewal approval can be from 12 up-to 24 months. (48); (58); (59)

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3.2.2.2. Iraq

In Iraq there is a special situation, that two authorities are responsible for all drug related issues. The Iraqi Directorate of Technical Affairs (DTA) as part of the MoH is responsible for all medicinal products authorized and marketed in Iraq.

For the Kurdish autonomous region, the KMCA in Erbil is responsible for the assessment of renewal applications.

Legal Basis

The legal basis for the renewal of marketing authorizations in Iraq is two important guidelines, which describes the renewal process in Iraq.

A marketing authorization in Iraq has to be renewed every five years for imported products and every 10 years for medicinal products manufactured locally. (60)

Required documents

The renewal application must be submitted three months before the expiry date of

the marketing authorization. The product can be still placed on the market after submission of the renewal application. (61)

The documents have to be sending out to the Directorate of Technical Affairs (DTA) as hard and soft copies. For the submission, an application form is not required. Instead of this, the manufacturer has to insert the following document: (60)

- Declaration that the composition of the finished product has not been changed.
- checklist for renewal application [Annex X - Renewal Checklist – Iraq]
- CPP from COO or CPP from one reference country and a letter from the Iraqi National Laboratory - This letter should confirm that the finished product has pass all required testing for the last five batches. In this case, it is allowed to import the medicinal product upon one year until the requirements are fulfilled. (60)

The other requirements for renewal application are the same as for an initial marketing authorization. (60)

Assessment procedure and timelines

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The DTA will start the assessment of the renewal application after receipt of all required documents. A detailed description of the procedure is not available. For the timeline of renewal, there is no timeline available.

3.2.2.3. Kurdistan

The KMCA in Erbil is responsible for the assessment of renewal applications in the Kurdish autonomous area in Iraq it is required to submit two samples of the medicinal product.

Legal Basis

The authority published a guidance document, which includes all relevant information for renewal applications of medicinal products authorized in Kurdistan. (62)

According to the Guideline, a marketing authorization has to be renewed every 5 years. (62)

Required documents

Additionally the following documents are required: (62)

- Original CPP from the country of origin of the manufacturer legalized by the chamber of commerce and the Iraqi embassy in the country of origin.
- Product composition certificate signed and stamped by the manufacturer;
- Finished product specifications signed and stamped by the manufacturer;
- Specifications of active and inactive ingredients signed and stamped by the manufacturer;
- Letter from the manufacturer company declaring that there are no changes made on the formula, manufacturing method and the specifications of active and inactive ingredient and finished product;
- Proof of payment of the registration fees.

3.2.2.4. Iran

Legal basis

The renewal process for medicinal products authorized in Iran is stated in the law No. D/1243 dated 9th April 2009 by the Ministry of Health & Medical Education (63) and

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the "GUIDELINES ON REGISTRATION OF PHARMACEUTICAL PRODUCTS FOR IMPORTS" provide the legal basis for renewal applications for medicinal products authorized in Iran. (42)

A marketing authorization of a medicinal product in Iran is valid for 4 years. (42)

The renewal application has to be submitted within six months before the marketing authorization expires. (63); (42)

Required documents

For the renewal dossier it is required to include the following documents: (42)

- Letter of authorization (in copy)
- Valid Representation Registration Certificate issued by the Iranian Ministry of Commerce
- Certificate of a pharmaceutical Product (CPP)
- A photocopy of the Responsible Pharmacist's license
- The bank receipt for license renewing fee
- A photocopy of the existing Marketing Authorization License
- An approval issued by the Product License Holder (PLH) or its Marketing Authorization Holder (MAH) on non-variation in the formulation, manufacturing methods and manufacturing site, and etc. of the medicine in question
- A sample of the packaging (including the box, label, strip, cartridge or vial, carton labeling or shrink-wrap)
- A specimen of the product
- The latest approval of the Division of Quality Control Laboratory of Deputy for

Food and Drug on the imported consignment of the medicine in question

Assessment procedure and timelines

The Iranian MOHME does not publish any information about the assessment procedure and the timelines for approval.

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3.2.2.5. Israel

Legal basis

The basis for the requirement of renewal processes of medicinal product in Israel is stated in the "Pharmacist Regulations for Medicinal products" from 1986. (64)

Additionally, the Guideline EX 005/02 from 2016 includes all processes for renewal applications in Israel. (65)

Required documents

Medicinal products, which were authorized and marketed in Israel has to be renewed every 5 years for the first time. In the Guideline it is stated:

"The director is authorized to renew the validity of a registration of a medicinal product in the Drug Registry for additional period, each of which is not longer than ten years, after the expiration of the registration in the Registry." (65)

The renewal follows two steps

1. At first, the MAH has to request a renewal of the "quality certificate" to the Institute of Standardization and Control of Pharmaceuticals in Jerusalem.

The Institute of Standardization checks the quantitative and qualitative composition of the medicinal product against the approved specification. After a positive outcome, the Institute certify, that the medicinal product meets the required standards. (65)

Submission

The renewal dossier for the quality certificate should be submitted as CTD dossier according to the ICH-M5 Guideline (65). The following parts of the dossier are required:

Module 1:

- Checklist for submission of renewal application (for chemical entities and biological.
- o [Annex XI – Application Form for renewal of chemical entities - Israel]
- o [Annex XII - Renewal Checklist for Biologicals – Israel]
- Cover Letter

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- Proof of Payment
- Valid GMP certificate
- QP approval for the active ingredient
- TSE statement
- CEP
- Certificate of Analysis for the drug substance and the finished product
- Valid MSDS for the active substance and excipients
- Currently approved SmPC & PIL
- Summary of validation status and information about the medicinal product for chemical entities

Module 3:

For the first renewal, it is required to submit all documents from this part of the CTD. For the next renewal applications the Institute required a reduced Module 3 which should include the following documents: (65)

- specifications for drug substance and finished medicinal product
 - stability data from the last 5 years
 - significant Type II Variations - If significant variations (variations of type II that require the replacement of entire chapters in the registration file) occurred since the last renewal and the version of the file at the Institute does not correspond to the file in the hands of the MAH as a result of that, a complete updated file should be filed at the time of renewal.
 - For any submission of a complete file a signed statement by the Appointed Pharmacist stating that the submitted file does not contain any non-reported changes or changes not approved by the Ministry of Health should be attached.
- The file evaluator is authorized to request the submission of the complete file at his/her own discretion, even if it is not the first renewal.

Assessment procedure and timelines

Information about the assessment process and the timeline for the approval of a renewal application are not published.

3.2.2.6. Jordan

Legal Basis

In Jordan, the renewal application is regulated in the Jordan Law No. 12 of 2013, the Criteria for Drug Registration from 2015 and Circular No. 2/9/1/12448 of 2015. (43) (44)

A renewal application should be submitted every 5 years. (66) The applicant should submit the dossier within 3 months after expiry of the marketing authorization. Every year, the JFDA informs the local representatives with a list of all renewed products about the upcoming renewals.

Required documents

For the first renewal of a medicinal product, the JFDA requires the following documents, which have to be submitted in a CTD structure.

The following documents has to be included in the renewal application (43)

Module 1:

- Cover Letter from local representative, the Marketing Authorization holder or manufacturer
- Cover Letter from of the technical manager of the company or the responsible pharmacist.
- Application form - [Annex XIII - Renewal Checklist – Application form for Marketing and Renewal Application - Jordan]
- Comparison table between registration file for initial registration and re-registration dossier
- List of all ingredients (human or animal origins) including all relevant certificates
- Confirmation letter from manufacturer that the manufacturing processes does not change
- Approval letters from JFDA about all post approval changes
- Technical agreement
- Product Information and approved artworks

- Summary of Product Characteristics (legalized from the NCA) of the country of origin
- Price certificate

Module 3:

- Drug composition
- Shelf life specification of the drug substance
- Certificate of Analysis of the finished product
- Stability Studies

Module 5:

- Periodic Safety Update Reports (PSUR)

If it is not the first renewal, the applicant has only to submit a legalized CPP of the COO and a price certificate.

Assessment procedure and timelines

There are no specific timelines for the renewal. According to the information of the authority it can take around two years for the assessment of the renewal.

3.2.2.7. Lebanon

Legal basis

The Decision 528/2017 and 293/2017 provides the basis for the requirement for the renewal (re-registration) of marketing authorization of medicinal products in Lebanon. (49); (50)

Since 2017 it is required for MAH to submit a renewal application for all registered medicinal products to the MoPH (Ministry of Public Health). The Technical Committee of the MoPH is responsible for all issues related to registrations of medicinal products.

The Marketing authorization will be cancelled, if the MAH does not submit a re-registration dossier for all registered medicinal products to the MoPH. Additionally the applicant has to provide a renewal plan with information about the importance of the affected medicinal products for Lebanon, the date of the registration and the countries, where the manufacturing takes place.

For renewal applications where the submission dossier is not completed until the 31th December 2017, the marketing authorization will be on hold. This has the consequence that it's forbidden to bring these medicinal products to the market. They will be listed as "non-marketed drugs" (49); (50)

In a Memorandum from 2016, the Ministry stated that the new re-registration process is still ongoing. Therefore it is not possible to give a timeline and procedure about the renewal process.

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3.2.2.8. West Bank

Legal Basis

According to the Guidance document "GUIDANCE ON PHARMACEUTICAL PRODUCTS REGISTRATION IN PALESTINE" from 2007, a marketing authorization for a medicinal product has to be renewed every 5 years. (67)

The renewal application has to be submitted at least 4 months before the MA expires to the Drug Control and Registration" Department of the MoH. It is possible to submit the application later, but if the submission will be performed more than 2 months after expiration date of the MA, the MA will be suspended.

Required documents

The application shall include the following: (67)

- Application form for the renewal – [Annex XIV – Application Form for Renewal

- Palestine]

- Proof of Payment
- Approved method of analysis for the finished medicinal product
- Specification of the finished product
- Current master formula for the product
- Current stability study (Shelf life study)
- Sufficient samples for analysis accompanied by a reference standard material from the active constituent(s). Palestinian National Authority General Directorate of Pharmacy Ministry of Health Drug Control and Registration Dept DC 002-0 37 December 2007
- Samples from the latest secondary packaging materials and from the aluminum foil primary packaging material.

Assessment procedures and timelines

The Department of Drug Control and Registration reserves the right to ask for any additional documents with regard to the registered drug file.

A timeline for the procedure and an overview about the assessment report are not available.

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3.2.3. Renewal application in the Maghreb States

3.2.3.1. Morocco

Legal basis

In Morocco, the following laws mention the obligatory and requirement for the renewal of marketing authorizations of medicinal product

- Law No. 17-04: Code of Medicine and Pharmacy; (68)
- Circular No. 48 DMP/00: Procedure for the Marketing Authorization Application of Pharmaceutical Products; (45)

A marketing authorization of a medicinal product has to be renewed every 5 years. The local contact of the MAH has to submit a renewal dossier 180 days before the marketing authorization expires. For medicinal products imported from outside the marketing authorization has to be valid in the COO as well. (45)

Required documents

Concerning the submission of the renewal application, the DMP differs between the first renewal application and the following renewal applications. For the first review, the applicant has to submit a full CTD dossier of a medicinal product except of Module 4 (Non-clinical) and Module 5 (Clinical).

It is required to submit two application forms (45)

- "DMP" application form – [Annex XV – DMP Application Form for Renewal -

Morocco]

- "LNCM" application form – [Annex XVI – LNCM Application Form for

Renewal - Morocco]

For the following renewal applications, it is only required to submit a cover letter, which states that the marketing authorization will be renewed. (45)

Assessment procedure and timelines

The assessment of the renewal application will be submitted to the DMP. The review of the quality part of the dossier will be done by the LNCM and the administrative review by the DMP. Within 6 months, the marketing application will be renewed.

3.2.3.2. Algeria

Legal Basis

The legal basis for the renewal of marketing authorization is the Decree No. 92-284 of 1992. (46)

In Algeria a marketing authorization of a medicinal product has to be renewed every 5 years. The submission of the renewal has to be done at earliest in 6 months and at latest in 3 month by the local agent of the MAH before the marketing authorization expires to the ANPP. (46)

Required documents

The ANPP differs between two types of scenario: (46)

1. In case that the dossier of the initial registration does not changed or if all variations were submitted on-time (which means that the dossier is at the time of renewal up-to-date), the applicant has to submit the following documents
 - Statement of the MAH, that the content of the technical information of the medicinal product has not been changed since the first registration
 - Proof of payment of the registration fees (also from variations which were submitted in the last five years)
 - legalized CPP from the country of the MAH
2. In case, that the current dossier of the medicinal product does not correspond to the initial registration dossier or if variations were not submitted previous the following dossier requirements has to be fulfilled
 - Renewal letter (2 copies) and statement from MAH (3 copies) that the changes of the medicinal product were approved.
 - Submission receipts of variations
 - Checklist list of local application form
 - Form A (legalized) –
 - o Annex XVIII – Annex A Application Form for Renewal - Algeria
 - o Annex XVII – Application Form for Renewal - Algeria
 - o Annex XIX – Annex II Application Form for Renewal - Algeria
 - CPP and/or marketing authorization Approval letters with annexes issued by the country, where the MAH is stated

- Manufacturing license for all sites involved in drug product manufacturing.
- GMP certificates for all sites participating in the drug product manufacturing.
- Free On Board (FOB) price attestation
- Therapeutic information form
- 1 finished product sample along with the corresponding original certificate of analysis;
- Copies of certificate of analysis for the active ingredient and excipients;
- Updated soft copy of the Common Technical Document (CTD) Module 3 (no hard copy is needed; only a CD/DVD) in addition to the updated labelling and summary of product characteristics;
- Proof of payment of the registration fees;
- Authenticated certificate of suitability (CoS) or Drug Master File (DMF) with letter of access;
- Statement mentioning all the parties involved in the product manufacturing;
- Copy of the Active Pharmaceutical Ingredient (API) GMP certificate;
- Variation forms signed and stamped by the MAH.

Legalization of required documents has to be done by the Chamber of Commerce, by the Ministry of Foreign Affairs and by the Algerian embassy in the country of origin of the MAH)

Assessment procedure and timelines

The renewal will be assessed within three to six months by the ANPP. (46)

3.2.3.3. Tunisia

Legal basis

In the Tunisian Law No. 85-91 from 1985 the requirement for MAH to renew the marketing authorization of medicinal products marketed in Tunisia is included. (69)

Additionally the Order from the MoH from 1996 describes the obligation and requirements for a renewal application in Tunisia. (70)

A marketing authorization in Tunis is valid for 5 years. A renewal application has to be submitted by the local contact point of the MAH 6 month before expiry date (51).

Required documents

The Tunisian law divides the requirement of the renewal application into two subgroups.

For medicinal products manufactured in Tunisia the following documents has to be

included in the renewal application: (51).

- Cover letter
- Application sheet in French signed by the applicant responsible pharmacist
- A Certificate from the responsible pharmacist of the company attesting that there is no change in the medicinal product files that had been submitted for the initial registration
- GMP certificate of manufacturing sites for the finished product
- Updated SmPC (Word and PDF versions)
- Proof of payment for the renewal fee
- Two samples of the medicinal products (Tunisian market sale model) accompanied by their certificate of analysis.

For medicinal products imported into Tunisia, the above mentioned documents have to be submitted too. Additionally it is required to submit: (51)

- Copy of the updated Marketing authorization in the country of origin (COO)
- CPP from COO

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- Approved SmPC in the country of origin (COO)
- Two sale model samples (Country of origin market) and the corresponding certificates of analysis
- Certificate of wholesale price excluding issued by competent authorities in the country of origin.

Procedure and timelines

The timeline for the renewal assessment is due within 15 days. (51)

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3.2. Other procedure to maintain a marketing authorization of a medicinal product in the MENA region

In all countries of the MENA region, a marketing authorization of a medicinal product will be suspended, if the benefit-risk relation is negative and if there are high risks for the public health.

A MA for a medicinal product can be withdrawn, if the qualitative and quantitative composition of the finished drug product does not fit with the specification submitted within the application for marketing authorization or if the documents provided in the initial registration are not in line with the manufacturing and pharmaceutical profile of the medicinal product. In some countries, it is required that at least 2 of 3 batches have to be compliant with the approved drug specification; otherwise the MA will be withdrawn by the NCA.

For imported medicinal products, it is required that the MA in the COO is still valid. Otherwise this could be an entitled reason to withdrawn the marketing authorization in the affected country.

Additionally a large number of MENA countries implement regulations, which defined that if the medicinal product is not placed on the market over a certain period, the marketing authorization will be cancelled.

This procedure is well known in the European countries as "Sunset Clause". The EMA defines the term as follows:

"The so-called "sunset clause" is a provision leading to the cessation of the validity of the marketing authorization..." (71)

The EMA differs between two possible scenarios to withdrawn a marketing authorization according to Sunset Clause: (71)

- if the medicinal product was not placed in the market within three years after the initial marketing authorization.
- if the medicinal products is not marketed for three years

Many countries of the MENA region implement similar rules, but in the opposite to the Member States of the European Union, the marketing authorization will be cancelled, if the medicinal product is not marketed within a defined period of time after getting

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the approval for MA. Only in a few countries of the Levant states and in Morocco, there are also other defined requirements to obtain a MA in these countries.

For medicinal products registered centrally in the GCC States, a marketing authorization can be withdrawn if the medicinal product is not placed on the market within two years after approval. It is required that the medicinal product is marketed at least in two countries of the GCC otherwise the marketing license will be suspended. (72)

The following tables provide an overview about existing rules regarding sunset clause in each country.

3.2.1. Gulf States

Table 1: Overview about Sunset Clause in the Gulf States

Country Sunset available Timelines Other requirements

[Yes/No]

KSA No N/A N/A

UAE Yes 12 months - Medicinal product
has to be marketed
within 12 months
after registration

Bahrain No N/A N/A

Oman No N/A N/A

Qatar Yes 3 months - Sunset clause for
GCC

- Pharmaceutical
company does not
marketed registered
products for 3
months (the
registration of the
pharmaceutical
company will be
withdrawn) (40)

Kuwait Yes 2 years - Medicinal product is

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not imported within 2
years after
registration

3.2.2. Levant States

Table 2: Overview about Sunset Clause in the Levant States

Country Sunset available Timelines Other requirements

[Yes/No]

Egypt Yes 18 months - product has to be
2 years manufactured locally
within 18 months
(41)

- product has to be
manufactured or
imported within 2
years from expiry
date of the last batch
(41)

Iraq Yes 1 year Medicinal product
has to be marketed
within 1 year after
initial MAA (73)

Kurdistan Yes 1 year - Same requirement
as in Iraq (73)

Iran No N/A N/A

Israel No N/A N/A

Jordan No N/A N/A

Lebanon Yes 1 year - product has to be
marketed within 1
year after registration
(74)

- if the sales of a
medicinal product is
very low, the MAA
can be cancelled

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(74)

West Bank No N/A N/A

3.2.3. Maghreb States

Table 3: Overview about Sunset Clause in the Maghreb States

Country Sunset available Timelines Other requirements

[Yes/No]

Morocco Yes -12 months - medicinal product
- 6 months has to be marketed
within 1 year of
obtaining a license
- continuous supply
of goods over 6
months and
maintenance of
safety stock required
(75)

Algeria Yes 12 months - medicinal product
has to be placed on
marketed within 1
year after registration
(46)

Tunisia No N/A N/A

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3.3. Harmonization of renewal procedure in MENA region

Inside the MENA region, there are three subgroups, but only the Gulf States has developed a central renewal procedure for marketing authorizations.

Legal Basis

In Article 29A of the "Executive Board of the Health Ministers' Council For GCC", a marketing authorization for a centrally registered medicinal product in the GCC region is valid for 5 years. . (76)

In case that the medicinal product was centrally approved by the GCC-Drug Registration Committee (GCC-DR), the renewal application has to be submitted within 3 months before the marketing authorization expires. (76)

Required documents

For products, which were authorized centrally via GCC-DR, it is not obligatory, that a local contact has to submit the application to the authority. The marketing license will be cancelled, if the applicant fails to submit a renewal application within 6-months after the expiry of the MA.

The following documents should be submitted to the committee of the GCC-DR, which is responsible for renewal of medicinal products: (76)

- Application form
- CPP or certificate of free sale legalized by the NCA of the COO
- List of manufacturers including the countries, who handle the medicinal product
- List of countries, where the product is
 - o authorized
 - o not authorized and the reasons
- Currently Approved EN version of the SmPC or PIL from COO (approved by the NCA of the COO)
- List of all approved variations and re-registration applications
- Declaration form MAH, that there is no change of the medicinal products. In case of any changes of the medicinal products (quality, safety or efficacy)

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related), the applicant has to submit the changes together with the re-registration dossier

- Update of Stability data and stability study for two batches
- Certificate of Suitability
- Drug Master File from Manufacturer (in case, if CEP is not available)
- Methods of analysis for finished product according to the last approved Pharmacopeia
- Certificate of Analysis for finished medicinal products
- result of follow-up studies the product-marketing
- Commercial Samples
- Sample or Artwork for External Package

For biologicals or herbal medicinal products, the Committee required the following documentation additionally:

- Microbial Contamination Test for oral solutions (if one of the excipients are animal or plant origin)

Assessment procedure and timelines

The timelines for the approval of a renewal application and the assessment procedure were not communicated by the GCC-DR.

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4. Differences and Similarities to European Procedures

The previous part describes the renewal procedures and maintenance procedures of MA in the MENA countries.

For pharmaceutical companies located in Europe, it's more interesting to know the similarities and differences to known European Procedures.

4.1 Renewal procedure for medicinal products authorized via CP, DCP and MRP in the European Union

To get a better overview about the differences and similarities of maintenance procedures of MAA between the countries of the European Union and the MENA region, it is important to get an overview about renewal procedures for medicinal products authorized in the Member States of the European Union. In Europe there are three possible procedures to receive a MA for a medicinal product in the member states of the EU.

4.1.1 Centralized procedure

The centralized procedure allows medicinal products to receive MAA in all Member States of the EU. For drug substances against cancer, metabolic diseases, infectious diseases, and biologicals, it is required to submit an application for MA via the centralized procedure.

Legal framework

In the European regulation EC 726/2004 it is stated, that a MA for medicinal products authorized via CP is valid for 5 years. The EMA Guideline "Guideline on the processing of renewals in the centralised procedure" describes the renewal process for CP products. (77)

The applicant has to submit a renewal application to the EMA within 9 months, before the MAA expires. It is required, that the Applicant appoints a submission date to the EMA. This step is important, because the EMA has to agree the proposed submission date with the Rapporteur and Co-Rapporteurs. (77)

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Required documents

The Applicant has to submit Module 1 and Module 2 of the dossier. Module 2 contains the Quality Addendum, Non-clinical and Clinical Addendum. The Risk-Management Plan has to be included in the dossier too. (77)

All required documents are listed in Annex XX – Required documents for renewal of CP products

It is possible to include changes in the renewal application. This has to be stated in the application form. If new clinical studies are available, it is obligatory to submit the data within the renewal application. (77)

Assessment procedure and timelines

The CHMP and the PRAC are involved in the assessment of a renewal application for a medicinal product. If the Rapporteur and Co-Rapporteur agree, that the benefit risk evaluation is positive, the CHMP recommend a positive opinion. This means that the MAA will get either an unlimited validity or validity for five years.

If the CHMP comes to the output, that the benefit risk evaluation is not positive and that the medicinal product has a potential risk for the public health, the Committee will adopt an "unfavorable" opinion. The MAA will suspend and be withdrawn from the European Market. (77)

The EMA will prepare an update of the European Public Assessment Report (EPAR), reflecting the renewal assessment and CHMP opinion. After the Commission Decision on the renewal, the updated EPAR shall be published.

The renewal process takes 120 days. (77)

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4.1.2 Decentralized procedure/Mutual recognition procedure

The DCP and MRP procedures are two other options to authorize a medicinal product in countries of the European Union. In opposite of the centralized procedure, the MAH can select in which countries

Legal Basis

In Article 24 of the European Directive 2001/83/EC, it is stated that a MAA for a medicinal product is valid for five years. If the the MAA is renewed for once, the validity will be unlimited. The "Guideline on the Processing of the Renewals in the Mutual Recognition and Decentralized Procedure" describes the renewal process for medicinal products, which were authorized via DCP or MRP. There is no difference between the renewal application of medicinal products authorized via DCP or MRP. (78)

The Applicant has to submit the renewal application within six months before the MAA expires. (78) It is important that the Applicant agrees a "common" renewal date with all Member States, which were involved in the process. (78)

Required documents

As for the renewal application for MAA authorized via Centralized Procedure, the Applicant has to submit the whole documentation of Module 1 and Module 2 (see Annex XXI – Required documents for renewal of DCP or MRP products).

The dossier should be submitted in eCTD. (78)

If Applicant wants to implement new amendments in the Product Information, it has to be discussed and agreed with the RMS before submission.

Assessment procedure and timelines

The aim of the renewal application is to assess the benefit-risk evaluation of a medicinal product based on new information and on PSURs.

During the renewal procedure, variations should not be included in the renewal procedures. However it is permitted to include administrative changes of the Product Information together in this procedure. The changes need to be listed in the application form (see Annex). It is recommended to discuss proposed amendments

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of the SmPC together with the responsible RMS before submission of the renewal application.

The timeline for a renewal for medicinal products authorized via DCP or MRP is around 90 days. During this time, a clock-stop period is possible. The Applicant may send further documentation to the RMS & CMS upon request within 30 days after the start of the clock-stop period.

It is possible to submit a "shortened" renewal application. This will follow a 30 days procedure. In this case, a clock-stop is not possible and the RMS will lead the assessment of the renewal.

After the first renewal, the MA will have an unlimited validity. In some exceptions, it is possible that the RMS required a renewal after 5 years.

4.1.4. Other maintenance procedures in the European Union

To maintain a MA for a medicinal product in the EU, it is required to obtain the sunset clause for medicinal products authorized under regulation 2001/83/EC.

For centrally authorized medicinal products it is stated in Article 14 of the regulation 726/2004 and in Article 24 of the regulation 2001/83/EC, [which is also valid for medicinal products authorized via DCP or MRP], it is stated that a MA will be valid, if one presentation of the medicinal product is placed on the market. If a medicinal product is not marketed within 3 years in any Member State of the EU (including the EEA States as Norway, Island and Lichtenstein) the MA will be withdrawn from the pharmaceutical market. (79)

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4.2. Differences and Similarities between the procedures in the MENA Region and the European Union

In the previous parts of this master thesis, several procedures to maintain a MAA in the MENA region and the European Union were discussed. The differences and similarities will be shown in detail.

Validity

All MA in European countries and in the MENA countries have a validity of 5 years, except Iran, where a MA for medicinal products is valid for 4 years. (42)

The main difference is if the first renewal application for a MAA authorized via CP, DCP or MRP is positive, the MAA has an unlimited validity. But this comes with an exception, except for medicinal products, which were authorized under "conditional approval or exceptional circumstances". For these products, they have to be renewed again.

All NCA's of the MENA countries require a renewal application for every five years. That also applies, even the qualitative and quantitative composition of the medicinal products does not change or the benefit-risk evaluation remains the same as in the initial application.

Date of Renewal

To maintain a MAA via a renewal application, applicant has to submit the dossier within 9 months for medicinal products authorized via CP, and 6 months for medicinal products authorized via DCP or MRP. In the MENA region, the timelines for renewal are quite different.

A majority of the MENA countries have a renewal timeline of 6 months before expiration of the MA. In several countries the renewal application should be submitted within 3 months.

Required documents

The Applicant has to submit the Module 1 and Module 2 of the CTD to the Authorities for renewal application of European MAA.

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In most cases, the NCA of nearly all countries of the MENA region requires documents from Module 1 and Module 3. The overall summaries of the quality, non-clinical and clinical part are not part of the assessment of the benefit-risk evaluation. The European CA does not require documents for Module 3, only on request.

This is a main difference between MENA and Europe, because the assessment of a renewal application based on information provided in Module 2. For NCA of the MENA region, the assessors are focused on data based on the CMC part of a dossier, because most of variations affect the quality of a medicinal product. The assessment of the benefit-risk evaluation based on the summaries provided in Module 2 is not sufficient for the NCA of the MENA region.

For the GCC and Levant States another possible reason could be that the data in Module 3 should improve, that the current manufacturing processes, the analysis and the specifications are up-to-date and not been changed without a corresponding notification to the authority. This fact could be seen as "additional" control from the NCA to avoid medicinal products with content which were not approved by the authorities.

This "additional control" could also be a possible reason, why the applicant has to provide a list of all submitted variations including the approvals of the NCA. It can be seen as further measures against corruption, because the MAH's and manufacturers cannot implement changes in the quality and quantity of a medicinal product without

approval of the Authorities. This also indicates that the confidence of the Authorities to the manufacturers and MAH's is not stabilized enough and it can be disassembled.

For imported products or medicinal products, where the MAH is stated outside the affected country, the applicant has to indicate that a MAA is still valid in the COO. This can usually be done by submitting a CPP either from the origin country of the MAH (if outside the affected country) or of the country where the manufacturing takes place. In some countries it is also required to submit a list of all valid MAA in the world of the relevant medicinal product. For European renewal procedures, a CPP is not required for submission. Most of the medicinal products authorized in the MENA region are mostly imported from European countries and the USA. On the one side the initial applications for new drug substances will be done in these countries. Most of the MAH are located there, the development of potential new drug substances takes

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place there and both markets rank among the most important markets for pharmaceuticals in the world.

On the other side the development of innovative medicinal products in the MENA region is not on the same level as in USA, Europe or Japan. The missing innovations and regulations for potential new drugs are still under development and it will take decades to reach the same level regarding development and marketing of medicinal products as the EU, USA and Japan.

The format of the submission dossier is another point, which should be considered. This point indicates only exceedingly few differences. Most of the MENA countries require a CTD according to ICH-M4 as in the European countries. Only in Lebanon and Egypt, a submission of the dossier as CTD format is still not possible. The dossier language is English, except in the Maghreb States. In these countries, the applicant has to provide the documents in French.

Assessment procedure

The assessment of renewal applications for European MAA is quite simple. The benefit-risk evaluation will be assessed based on the overall summaries. For NCA from MENA, the assessors evaluate the benefit-risk evaluation based on the data mainly provided from Module 3.

The difference in the approach of an assessment of a renewal application could be, that most of the NCA in the MENA region are relative "young" in opposite to the European NCA and not so much experienced with renewal application. In several countries, such as Lebanon, the obligatory for MAH to renew the MAA of their products was introduced in 2017 and also the renewal procedure for medicinal products authorized via centralized procedure of the GCC-DR is not as long as in Europe.

Another important point to consider is, the NCA of the MENA countries included other activities (eg. inspection of the local manufacturing) in the assessment of the renewal process. For European procedures, an inspection during a renewal procedure is not required. This additional inspection is time consuming and cause longer periods for the assessment of renewal applications.

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In opposite to the renewal procedures for marketing authorizations granted via CP, DCP or MRP, the applicant cannot submit variations within the renewal applications. The authorities recommend that all important variations, which may have an impact on the safety, efficacy and quality of a medicinal product, should be submitted and approved prior to the renewal application. This is also a reason, why the applicant has to submit all variations, which were submitted and approved prior to the renewal application- due to the fact that the Authorities try to avoid additional workload during the renewal process. Another plausible reason could be, that the NCA does not have enough resources for the assessment of variations and renewal application on the same time.

Timelines

For renewal application for CP, DCP or MRP products the timeline is 120 days. This consists of 90 days of assessment for the authorities and 30 days as "clock-stop".

Sunset clause

To maintain a MA for a medicinal product via Sunset clause exist for all European procedures and for the MENA countries. The requirements are nearly the same, except in Lebanon. The special situation there is that if the sales numbers for a medicinal product are not high enough, the product will be withdrawn from the market.

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Fact Sheet

Table 4: Differences and Similarities between EU and MENA

Renewal MA – EU	Renewal MA – MENA
Validity 5 years for initial MAA, Every 5 years unlimited validity after renewal* (Iran 4 years)	
Renewal Date 9 months for CP products	3 – 6 months

6 months for DCP/MRP products
 Important requirement N/A Valid MAA in COO
 Required documents Module 1 Module 1
 Module 2 Module 3
 Module 4
 Module 5
 • Documents will be provided in English language (except Maghreb States**)
 Assessment of benefit-risk evaluation
 Submission EU- e-CTD CTD (hard copy and soft copy), (no paper submission) additional paper submission optional
 Timelines 120 days (90 days assessment 15 days – 24 months + 30 days clock-stop) - CP
 90 days (DCP/MRP)
 Inclusion of other applications within renewal? • variation can be included within renewal application (for CP) application
 Mock-ups required No Yes (not for all MENA countries)
 Shortened renewal Yes (for DCP/MRP) No applicable? (except Morocco)
 Sunset clause available Yes Yes except
 * except for medicinal products authorized under “conditional approval” or “Exceptional circumstances”

** documentation should be provided in French

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5. Summary, Discussion and Outlook

The detailed reflection of procedures in the MENA region to maintain a marketing authorization for a medicinal product shows, that the regulatory affairs for medicinal products is in an ongoing process. During the time, many authorities in the MENA try to improve the procedures and regulations for pharmaceuticals. Especially due to the increased need of medicinal products and the development of the pharmaceutical sectors of local and private manufacturers, the authorities and institutions develop and improve existing laws and regulation to facilitate the post-authorization procedures in the MENA region.

On the other side this master-thesis also shows, that many NCA in the MENA region are still not enough regulated. Many issues need to be improved in future as the transparency and timelines for renewal procedures in the relevant countries.

The renewal applications for medicinal products authorized in the MENA region are (dependent from the country) very demanding. All NCA's in the region requires further renewal applications of a medicinal products, even the quality and the composition of the medicinal product does not changed.

These further renewal applications are not necessary, due to the fact, that neither the composition nor the quality of a medicinal product changed. Additionally the assessment of a renewal application is not transparent; in some countries as in Tunisia, the renewal application takes only 15 days, in Egypt the assessment can take up to 2 years.

There are some similarities between the European procedures and the procedures in the MENA region. Nearly all MENA countries, except Egypt and Lebanon, require a CTD dossier, in the Gulf States a submission even as eCTD is mandatory.

The main difference between European procedures and MENA procedures for renewal applications is the required documents.

Whereas the European HA's require the submission of Module 2, which contains the overall summaries for the quality, non-clinical and clinical part of a dossier, none of the HA's in the MENA region requires documentation from Module 2. In some countries, it is obligatory to include documents from Module 4 and Module 5 into the renewal application.

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The most of the authorities in the MENA require the submission of documents from Module 3. These documents will not be provided to the European HA's at time of renewal.

Regarding sunset clause, there requirements for medicinal products authorized in Europe or in the MENA region are nearly the same. In this case it is interesting to see, that a medicinal product can be withdrawn from the Lebanese pharmaceutical market, if the sales numbers are not high enough, even the medicinal product is placed on the market.

The maintenance procedures in the MENA are very individual and none of the countries accept the approval of a renewal from another MENA country, especially for the Levant and Maghreb States. For these subgroups, there are no “recognition” procedures for renewal applications. In this case the Gulf States are further developed, because it is possible to maintain one MA for a medicinal product, centralized via GCC-DR. The renewal for a MA is valid for all member of the GCC.

The objective of this thesis shows, how demanding the MENA region is. Especially for small and medium enterprises it is very difficult to authorize and maintain a MA for a medicinal product, due to the fact that besides the high request of documents, a local agent is required for all countries of the MENA region. Only in case of a centrally MA via GCC-DR, a local agent is not necessary.

In the author's point of view, the pharmaceutical sector of the MENA countries and the maintenance procedures for MA of medicinal products are in development and it can be expected with great confidence, that the regulatory procedures will be much better as today.

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Annex I - Application Form for Renewal application in KSA

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Annex II - Application Form for Renewal of a conventional medicinal products UAE

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Annex III - Renewal Application Form for General sale - UAE

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Annex IV - Application form for Renewal for Herbal Medicinal Product - UAE

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Annex V - Renewal Application Form – Bahrain

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Annex VI - Renewal Checklist – Bahrain

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Annex VII - Application Form for Marketing and Renewal Application - Oman

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Annex VIII - Renewal Checklist – Kuwait

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Annex IX - Renewal Application Form – Egypt

NEW HUMAN MEDICINE PRODUCT
REQUEST INQUIRY FORM

- Please note that, the company profile should be submitted in advance of the submission of the request inquiry form, in order to proceed with your application.
- You should submit a scanned copy of the receipt attached with the form (1000 L.E.), the company should specify the generic name, and strength and dosage form inside the receipt.
- Each form should be submitted for every individual product.
- The fee is UN refundable and applicable only for this product.
- All the items should be fulfilled completely.
- The company will receive a notification mail within 15 working days; if the box is closed and an action Letter if the box is opened.
- The reply email or the action letter will state the decline of the request and the decline of the request and the inquiry will be stored according to the receiving time
- For more information: rdhd@eda.mohealth.gov.eg

Abbreviations:

- FToll: Manufacturing agreement between two manufacturing companies.
- COO: Country of origin

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Company Profile Code: Receipt amount:

Receipt Number: Date of stamping of the receipt:

Section 1: Application Type.

1.1 Type of Local F Toll Toll
registration:
Imported Under Imported Toll / Under
license Bulk License

1.2 This Innovator Product
Application
Concern:
Generic Products

Line Extension

Section 2: Applicant Company Information:

2.1. Applicant Company Name:

2.2. Contact Email:

Section 3: Product Information.

3.1. Active ingredient(s) &
strength (s):

3.2. Dosage form:

3.3. Route of administration:

3.4. Pharmacotherapeutic
group:

[e.g. Antibacterial, Diuretic, ect.]

3.5. Dose:

3.6. Indication:

3.7. Reference:

3.8. Pack:

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3.9. Additional information:

Section 4: Manufacturer Information:

4.1. Manufacturer Name:

Section 5: Additional information (For Imported & Under License Products)

5.1. Date of issue & expiration of CPP:

5.2. Marketing Authorization Holder:

5.3. Manufacturing site of the finished product:

5.4. Manufacturer of accessories (if present):

5.5. Packager:

5.6. Batch Release:

Section 6: Declaration.

5.1. Applicant declaration

In accordance with the ministerial decree 296/2009 . I certify that the information supplied is
complete and correct and that no relevant information has been omitted.

*Name head of regulatory affairs manager:

*Date:

*Contact mobile No.

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Annex X - Renewal Checklist – Iraq

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Annex XI – Application Form for renewal of chemical entities - Israel

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Annex XII - Renewal Checklist for Biologicals – Israel

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Annex XIV – Application Form for Renewal - Palestine

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Annex XV – DMP Application Form for Renewal - Morocco

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Annex XVI – LCNM Application Form for Renewal - Morocco

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Annex XVII – Application Form for Renewal - Algeria

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Annex XVIII – Annex A Application Form for Renewal - Algeria

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Annex XIX – Annex II Application Form for Renewal - Algeria

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Annex XX – Required documents for renewal of CP products

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Annex XXI – Required documents for renewal of DCP or MRP products

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Hiermit erkläre ich an Eides statt, die Arbeit selbständig verfasst und keine anderen als die angegebenen Hilfsmittel verwendet zu haben.

Ort, Datum Unterschrift

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Yuxi Walvax Biotechnology Co., Ltd

Appendix 3-2 Number of employees engaged in the quality management, production, quality control, storage and distribution

Department Staff number Department Staff number

Quality Management 3 Production 5
Dept. Management Dept.

Quality Operation 4 Bacterial Vaccine 76
Dept. Room I

Quality Assurance 13 Bacterial Vaccine 51
Room II
Dept.
(pneumococcal
production)

Quality Supervision 9 Bacterial Vaccine 60
Dept. Room III

Risk Validation 5 Filling Workshop 124
Management Dept.

Quality Control 57 Supply Workshop 40
Laboratory

Laboratory Animal 17 Storage and 13
Center Transportation Dept.

Purchase Dept. 5 Distribution Dept. 9

Human Resources 4 EngineeringOperations 27
Dept.
Detp.

Sales Management 2 Engineering Support 37
Dept.

Module 1
Version No. Walvax-202007-00 Yuxi Walvax Biotechnology Co., Ltd.

PRODUCT REGISTRATION DOSSIER

13-valent Pneumococcal Polysaccharide Conjugate Vaccine
(Tetanus Toxoid Protein)

MODULE 1: ADMINISTRATIVE AND PRODUCT
INFORMATION

Yuxi Walvax Biotechnology Co., Ltd.

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Module 1
Version No. Walvax-202007-00 Yuxi Walvax Biotechnology Co., Ltd.

Module 1: Administrative and Product information
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1.5.4. Template of Batch release certificate
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1.2 Application form
Please refer to Annex 1- Application form

1.3 Site Master file
Please refer to Annex 2-Site Master File

1.4. Compliance information
1.4.1. Manufacturing license
Please refer to Annex 3- Manufacturing license-Yuxi Walvax

1.4.2. Copy of GMP certificate
Please refer to Annex 4- Inspection Conclusion of Drug GMP Certification 20200415

1.4.3. Copy of marketing authorizations for all formulations and presentations in the
Country of Manufacture
Please refer to Annex 5- Registration Certificate of PCV13-TT in China

1.4.4. Policy for assignment of batch number
Please refer to Annex 6-Batch numbering management procedures

1.5. Vaccine composition, presentations and scheduling information
1.5.1. Summary of Product Characteristic
Please refer to Annex 7- Summary of Product Characteristic-PCV13-TT

1.5.2 Artworks of labels of primary containers and secondary packaging for the product
It will be submitted later

1.5.3. Samples of package inserts (in English) to be used
Please refer to Annex 8-Package insert_PCV13-TT_Walvax_2020.07.17

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1.5.4. Template of Batch release certificate
Please refer to Annex 9- Batch release certificate

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80° increases to 121° by 1-hour heat cycling, with 3T
water, at 10T/h flow rate
70° increases to 80° by transient heating at 10T/h flow rate
Heating medium 3 Bar Industrial Steam

Sampling

N.O.
Sampling

N.O.

Lyophilizer Tank

Reserved

(For 1F Sub-
packaging &
Lyophilizer
Point-of-Use)

WFI Tank

(WFI Point-of-Use for 2F/3F)

WFI Tank

Sampling Sampling Sampling Sampling Sampling

(For 3F
(For 2F Conjugation Area) (For 2F MEN Sterile Area) Pneumonia
(For 2F HIB Sterile Area) (For 3F Pneumonia Sterile Area)
Conjugation Area)

WFI Tank WFI Tank
WFI Tank WFI Tank
WFI Tank

Pneumatic FRL Pneumatic Valve Terminal
Power Supply
Conductivity Probe
Pressure Air
Breather
Temperature Sensor

Pneumatic Membrane Valve
Pneumatic Membrane Valve
Pneumatic Membrane Valve
Pressure Sensor WFI
Adjusting Angle Valve Non-Return Valve Pressure Gauge

Membrane Valve
Flowmeter

Pneumatic
Adjusting Angle Valve

Sampling Valve
Cooled Water
Pipeline Filter

Cooler
PW Sampling

Cooler
Valve
Pump
Cooled Water Regulating Valve
Cooled Water
Temperature Sensor
Sanitary Relief Valve
Temperature Sensor Temperature Sensor Temperature Sensor
Temperature Sensor Liquid Level Switch
Pressure Sensor
Temperature Sensor
Preheater

Preheater

Preheater

Preheater
Preheater
Liquid Level Sensor
Evaporation Tower Evaporation Tower Evaporation Tower Evaporation Tower Evaporation Tower Evaporation
Tower
Sight Glass Sight Glass Sight Glass Sight Glass Sight Glass Sight Glass Sight Glass
Flow Orifice Plate Flow Orifice Plate
Flow Orifice Plate Flow Orifice Plate Flow Orifice Plate
Flow Orifice Plate
Trap Valve
Flow Orifice Plate Flow Orifice Plate Flow Orifice Plate
Pressure Sensor

Manual Membrane Valve
Condensate

Manual Membrane Valve

Manual Membrane Valve

Regulating Angle Valve
Manual Membrane Valve

Manual Membrane Valve
Manual Membrane Valve

Manual Membrane Valve
Manual Membrane Valve

Manual Membrane Valve

Manual Membrane Valve
Pressure Gauge

Manual Membrane Valve
Manual Membrane Valve

Manual Membrane Valve
Manual Membrane Valve

Manual Membrane Valve
Outlet water
Industrial Steam Regulating Valve Outlet water
Industrial Steam
Outlet water

Power Supply
Pressure Air
Softener
Rotor Flowmeter

Ejector
Pressure Air
NaClO Dosing Device NaHSO3 Dosing Device NaOH Dosing Device
Active Carbon Filter
Rotor Flowmeter Pressure Gauge
Vent Valve
Forward Inlet Valve
Forward Inlet Valve
Drinking Water
Pressure Gauge

Reverse Discharge Valve Reverse Discharge Valve
Forward Outlet Valve Forward Outlet Valve
Rotor Flowmeter
Pressure Gauge

Reverse Inlet Valve Reverse Inlet Valve
Hardness Tester Pressure Gauge
Forward Discharge Valve Forward Discharge Valve
Raw Water Pump Pressure Gauge
Thermal Outlet Valve
Pressure Gauge
Bottom Discharge Valve
Liquid Level Sensor Bottom Discharge Valve
Pressure Sensor Salt Box
Raw Water Tank
Waste
Waste
Power Supply
EDI Device
Pressure Sensor Conductivity Meter
Conductivity Meter
EDI Power Supply
Pressure Sensor
Reverse Osmosis Conductivity Meter
Pressure Sensor
Flow Sensor
Flow Sensor
Flow Sensor
Pressure Sensor
HP Pump
Temperature Sensor
Temperature Sensor
Pressure Gauge
EDI Concentrated EDI Concentrated
Water Pump Water Tank
Plate-type Heat Exchanger
Pressure Sensor
5µm Precision Filter
ORP Meter
Pressure Gauge Pressure Gauge
Waste
Condensate

Sampling

80°C increases to 121°C by 1-hour heat cycling, with 1T water at 6.5T/h flow rate
70° increases to 80° by transient heating at 6.5T/h flow rate
Heating medium 3 Bar Industrial Steam

Sampling

Process PA 3Bar/ PA Inlet Process PA 3Bar/ PA Inlet Process PA 3Bar/ PA Inlet Process PA 3Bar/ PA Inlet

PS Inlet PS Inlet
PS Inlet PS Inlet
Thermal Insulation

(For 3F Pneumonia Sterile Area)

Process PA 3Bar/ PA Inlet Process PA 3Bar/ PA Inlet Process PA 3Bar/ PA Inlet Process PA 3Bar/ PA Inlet
PS Inlet PS Inlet

80°C decreases to 2~8°C, at 600L/h 80°C decreases to 2~8°C, at 600L/h Chilled Water Return 80°C decreases to 2~8°C, at 600L/h 80°C decreases to 2~8°C, at 600L/h
Chilled Water Return
WFI Tank

Refrigerant (25% Refrigerant (25% Refrigerant (25% Ethylene Glycol) Return Ethylene Glycol) Return Refrigerant (25% Ethylene Glycol) Return
Ethylene Glycol) Return
Ethylene Glycol) Return

80°C decreases to 40°C, at 1000L/h 80°C decreases to 40°C, at 1000L/h

Chilled Water Inlet Chilled Water Inlet

Refrigerant (25% Ethylene Glycol) Inlet Refrigerant (25% Ethylene Glycol) Inlet Refrigerant (25% Ethylene Glycol) Inlet Refrigerant (25% Ethylene Glycol) Inlet

Process PA 3Bar/ PA Inlet Process PA 3Bar/ PA Inlet

Sampling Inlet/Trap Port Sampling Inlet/Trap Port

Sampling Inlet/Trap Port Sampling Inlet/Trap Port Sampling Inlet/Trap Port Sampling Inlet/Trap Port

Sampling

1WS310 Washroom 2WS310 Washroom
Mobile Rapid Trap Valve
2PN1310 Ultrafiltration Room 3PN1311 Ultrafiltration Room 3PN2311 Ultrafiltration Room 2PN2310 Ultrafiltration Room

Q=15T/h, H=85m

20°C increases to 85°C by 1-hour heat cycling with
1.2T water at 7T/h flow rate
Heating medium 3 Bar Industrial Steam

Sampling

Sampling

(For 3F Pneumonia Non-sterile Area)

PW Tank

1PN1310 Fermentation Room 1PN2310 Fermentation Room 1PN2310 Fermentation Room 1PN2311 PW Tank for CIP
Station;
1PN1311 PW Tank for CIP Station; 1PN1310 Fermentation Room

Sampling

Q=15T/h, H=50m

Medical Products Administration of Yunnan Province
Inspection Conclusion of Drug GMP Certification
No.YYZ YN20190084
Applicant Yuxi Walvax Biotechnology Co., Ltd.
Address No. 83 South Dongfeng Road, High & New Technology Industries
Development Zone, Yuxi, Yunnan Province, China

Inspection 13-valent Pneumococcal Polysaccharide Conjugate Vaccine
Scope

Acceptance GMP2019070032 Acceptance 12th July,2019
No. Date

Date of 21st -23rd August,2019; Inspector Bin Fan, Guanghui Yang,
th th
Inspection 25 -27 September,2019 Chongyue Zhao
Inspection After inspection, the 13-valent Pneumococcal Polysaccharide
Conclusion Conjugate Vaccine of this company complies with the "Good
Manufacturing Practices"(2010 edition).

Send to Yuxi Walvax Biotechnology Co., Ltd.

Copy to Pharmaceutical Production Supervision Division of Yunnan Medical
Products Administration; Center for Food and Drug Inspection of
Yunnan Province
Remarks It was only the conclusion for this inspection.

Approved and stamped by: Medical Products Administration of Yunnan Province
Issued Date: January 20, 2020

People's Republic of China
Drug Manufacturing License

Name of Manufacturer : Yuxi Walvax Biotechnology Co., Ltd. License No.: Dian 20160408
Registered Address : No. 83, South Dongfeng Road, High & New Classification code: Sb
Manufacturing Address and Scope:
Technology Industries Development Zone,
No. 83, South Dongfeng Road, High & New Technology Industries
Yuxi, Yunnan Province, China Development Zone, Yuxi, Yunnan Province, China.
Social credit code Legal : 91530400770492152K Preventive biological products***
Representative : Yunchun Li
Responsible Person : Zhen Huang
Quality Responsible person : Guoliang Fang
Expiry Date : December 31, 2020
Daily Regulatory Agency : Food and Drug Administration of Yuxi City

Daily Supervisor :Farong Lu, Zixuan Zhao, Chenyu Yang Issued and Stamped by: Food and Drug Administration
of Yunnan Province

Report Phone No. :12331 Signed by: Zhu Yang Issued Date: May 12, 2016

China Food and Drug Administration

QC testing: refined Ps, Ps derivative, Bulk Conjugates, Final bulk, Finished product. 8

Yuxi Walvax Biotechnology Co., Ltd

Yuxi Walvax Biotechnology Co., Ltd

Site Master File

Address: No. 83, South Dongfeng Road,

High & New Technology Industries Development Zone,
Yuxi, Yunnan Province, China

Revised on 2019-12-30

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Yuxi Walvax Biotechnology Co., Ltd

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Yuxi Walvax Biotechnology Co., Ltd

Site Master File

1. General Information

1.1 Brief information of the firm

Name, address and contact of the company

Name: Yuxi Walvax Biotechnology Co., Ltd

Registration address: No. 83, South Dongfeng Road, High and New Technology Industrial Development Zone, Yuxi, China

Manufacturing address: No. 83, South Dongfeng Road, High and New Technology Industrial Development Zone, Yuxi, China

Post code: 653100

Contact person: Yin Guoqing

(Manager of International Cooperation Department)

Fax: +86-0877-2076918

Telephone: +86-0877-2077789, +86-18313116198

Email: yinguoqing@walvax.com

Information of other manufacturing and testing site

Currently, the firm has only one site for manufacturing and testing.

Manufacturing site: No. 83, South Dongfeng Road, High and New Technology Industrial Development Zone, Yuxi, China

Short description of site

Walvax Biotechnology Co., Ltd. (hereinafter referred to as Walvax) is a modern biological pharmaceutical enterprise engaged in R & D, production and sales of vaccine products for human use and biological products. Walvax is a leading enterprise in strategic emerging industries in Yunnan province. Based on independent innovation, Walvax aims to provide safe and effective, high quality and advanced biopharmaceuticals products to domestic and foreign markets. Walvax was founded in 2001, with a registered capital of 1.537 billion Yuan, a total asset of 6.938 billion Yuan, a net capital of 5.013 billion Yuan. In November 2010, Walvax Biotechnology Co., Ltd. (stock name: Walvax Biotech, and stock code: 300142) was listed on Shenzhen Stock Exchange. Walvax is the first enterprise listed in China Growth Enterprise Market in Yunnan Province, with a market capitalization of 31.902 billion Yuan. There are 26 national, provincial and municipal innovation platforms such as

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Yuxi Walvax Biotechnology Co., Ltd national recognized Enterprise Technology Centre, R&D and production international scientific and technological cooperation base. There are about 1200 staffs in Walvax Biotech including more than

100 staffs that have a doctorate or master's degree, 15 high-end returned scientists, 3 staffs in Walvax are in the Chinese 1,000 Talents Plan issued by the central organization department in China. Walvax has one of the leading R&D and industrialization teams for vaccine and therapeutic monotherapy in China.

1.2 Authorized Pharmaceutical manufacturing activities of the site.

a) The production scope of the company is prophylactic vaccines.

See in Appendix 1- Manufacturing license-Yuxi Walvax-CN&EN

b) The products manufactured by Yuxi Walvax are as follows:

Table 1 The products manufactured by Yuxi Walvax

Product name	Approval date
Haemophilus Influenzae Type b Conjugate Vaccine	April 2nd, 2007
Group A and C Meningococcal Polysaccharide Conjugate Vaccine	January 4th, 2009
Group ACYW135 Meningococcal Polysaccharide Vaccine	March 23th, 2012
Group A and C Meningococcal Polysaccharide Vaccine	March 23th, 2012
Diphtheria, Tetanus and Acellular Pertussis Combined Vaccine, adsorbed	July 27th, 2015
23-valent pneumococcal polysaccharide vaccine	March 22th, 2017
13-valent pneumococcal polysaccharide conjugate vaccine	Dec 30th, 2019

c) GMP Certificates
See in Appendix 2-GMP certificates

1.3 Any other manufacturing activities

None. Only preventive biological products manufacturing activity is carried out and there is no other manufacturing activities carried out on the site.

2.Quality Management System of Manufacturer

2.1 The Quality Management System of Manufacturer

According to requirements of the ICH Q10 and other relevant regulations and as well as the company's strategic development needs, Yuxi Walvax has established a Quality Manual, defining quality policies and quality objectives in the production quality management. In order to achieve the

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Yuxi Walvax Biotechnology Co., Ltd

quality policies and quality objectives, a quality management system based on risk management has been set up, covering all the elements of GMP management, including organization and personnel, facilities, equipment, materials and products, production management, quality control, file management, qualification and validation, product distribution and recall etc., ensuring that products meet quality standards approved by the registration. Quality management system elements are shown in the following figure:

QM System

Risk Assessment Approve & Monitoring

Product
Facilities Materials & Qualification
Personnel Equipment Process QC Marketing
Control Release & Validation
Management

Plan Approve

Result Assess
Qualification

Method Val.
Verification

Verification
Acceptance

Complaints
Monitoring

Monitoring
Warehouse

Implement
Validation

Validation

Validation

Deviation

Sampling
Cleaning

Cleaning

Shipping
Training

Supplier

Control
Health

ADRS

Recall
Test

Test

Figure 0 Quality Management System Elements - Structure Chart

Yuxi Walvax has established six departments: Quality Operation Department, Quality Assurance Department, Quality Supervision Department, Quality Control Laboratory, Laboratory Animal Centre and Risk Validation Management, directly led by manager of Quality Management Department, responsible for all quality management work. The Quality Operation Department continuously improves quality management system through six methods: deviation management, change management, product quality review, self-inspection and product OOS/OOT management. The Quality Assurance Department is responsible for law & regulation research, document management, training management, material control and product release, to guarantee operations conform to requirements. The Quality Supervision Department is responsible for the whole monitoring of product quality activities. The Quality Control Laboratory and the Laboratory Animal Centre are responsible for quality testing of materials, semi-finished products and finished products and daily environment monitoring, to evaluate whether product quality conforms to quality standards or not. The Risk Validation Management Department is responsible for organizing comprehensive confirmation and validation management work, to guarantee that the manufacturing processes and operations meet the requirements demanded by regulations and manufacturing techniques. See the organization chart of Quality Management Department in the following chart.

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Yuxi Walvax Biotechnology Co., Ltd

Manager of Quality
Management Department

Vice Manager of Quality
Control Department

Quality Operation Department

Risk Validation Management Department

Laboratory for Quality Control

Centre for Test Animals

Quality Assurance Department

Quality Supervision Department

Figure 2 Organization chart of Quality Management System

Yuxi Walvax operates Quality Management System based on risk management system. Maintenance and improvement of quality management system is based on risk management system. Walvax considers organizations as core, quality plan as basis, documents operation and management as support, quality information as tie of operation of quality management. All operation procedures form quality record. At the same time, Plan-Do-Check-Action (PDCA) circulation is adopted, applying three systems, deviation system, change system, rectification and prevention system for process management control, thus guaranteeing effective operation of quality management system.

General information and brief introduction of the production department

The production system establishes 10 departments such as Production Management Department, Bacterial Vaccine Room I, and Bacterial Vaccine Room II, and is in charge of assistant general manager.

According to the production plan and production management department organizing regulating and controlling of production schedule, the production orders are issued. In the production order, the product batch number, validity period, manufacture date and so on are stipulated. The production order is the basis for the production organization activities in each workshop. The production order includes strain preparation order, polysaccharide production order, conjugate bulk production order, filling order, labeling order and packaging order, and the issuance of the orders is through the whole process of the product manufacturing.

Each production department organizes production activities according to the production order.

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Yuxi Walvax Biotechnology Co., Ltd

2.2 Release Procedures of Finished Products

Manufacturer's release procedures

Yuxi Walvax established Standard Operation procedure for Sampling (Code: WSYX-QS-SOP-00-09-0003) and Standard Management Procedure for Review and Release of Finished Product (Code: WSYX-QA-SMP-00-09-0008) to standardize sampling and audit release of finished product. After the finished products are produced, the quality supervisor samples the finished products and transfers them to the Quality Control Laboratory (QC) for inspection. The Quality Assurance Department shall review the information on the batch production record (instruction information, batch record review status, production environment confirmation, material balance) and batch testing record (batch record review status, status of inspection (sampling) application, inspection record, test result) and issue a Certificate of Analysis. For products that meet the specification, apply for batch release, according to the batch release conclusion, the authorized quality personnel decide release or not.

Yuxi Walvax has established the "Standard Management Procedure for Reserve Samples" (code: WSYX-QA-SMP-00-09-0002) to stipulate that the reserve samples volume of the finished products is 2 times the total inspection volume.

Role of authorized person/qualified person

There is only one authorized person and role of authorized person/qualified person in quarantine and release of finished products and in assessment of compliance with the Marketing Authorization is below:

■To make sure such every released batch production and control test as is in compliance with regulations, registration requirements and quality standard.

- Responsible for approval of release of every batch of raw materials and final products
- Responsible for dealing with un-qualified products and approval of deviation processing
- Responsible for approval of AEFI reports and product recalls
- Responsible for handling consumer complaints

Product release procedures of NRA

Yuxi Walvax's products have passed the inspection and been packaged. The batch release application is submitted by the batch release administrator into the management system for the batch issuance of biological products. Subsequently, the NRA authorizes the staff at the provincial drug administration to sample each batch of products and store the sealed samples in the refrigerator (the storage temperature is 2-8 °C). Afterwards, the samples are sent to the National Institutes for Food and Drug Control for testing. If the product meets the requirements, the National Institutes for Food and Drug Control will issue a release certificate for the batch and mail it to the company.

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Yuxi Walvax Biotechnology Co., Ltd

2.3 Management of Suppliers and contractors

Management of Suppliers

Yuxi Walvax has established Supplier Management Procedures (Code WSYX-QA-SMP-00-09-0001), SOP for Supplier Audit (Code: WSYX-QA-SOP-00-09-0001) to guide and regulate supplier management. The Quality Assurance Department has special posts and staff responsible for supplier control. Based on the scheme of risk management, supplier management adopts grading management of Grade I, II and III according to the degree of influence of supplied materials on product quality. In process of management, methods of documents auditing, on-site auditing management, quality assurance agreement, quality commitment letter, quality evaluation, service quality evaluation, questionnaires, dynamic management (including complaint management, change management) are adopted to evaluate quality of suppliers, in order to guarantee the normalization and validity of supplier management, provide basis for materials quality, and provide security for product quality.

Management of contractors

Not applicable. All of the manufacturing procedures and analysis are carried out by Yuxi Walvax.

2.4 Quality Risk Management

Policies of quality risk management of the firm

Through control systemic processes of risk assessment, risk control, risk communication and risk review and evaluation etc., Keep risk management throughout the life cycle of product production. Mainly taking preventive measures, Yuxi Walvax recognizes potential risk factors in each production process. Practices risk control measures into the implemented document system, to reduce risk of product quality, to improve quality management level, ensure the safety and vital interests of users as the ultimate purpose.

Scope and emphasis of quality risk management activities, and the process of risk recognition, risk evaluation, risk control, risk communication and risk review under quality risk management system

Yuxi Walvax considers product quality as the core, connecting public project equipment and facility system, manufacturing technique control, process quality control and product distribution in series to manage risks, emphasizing risk communication of risk management in each part and improving risk management effect. Walvax has established Risk Management Control Procedures (Code: WSYX-P-09-0012), taking the concept of whole-staff risk, commanding methods, self-service risk control, and cooperation & communication as the target. At the same time, Walvax guides risk management through annual Main Plan of Risk Management.

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Yuxi Walvax Biotechnology Co., Ltd

Yuxi Walvax has established risk management committee, quality head as its director. The members of the committee include manufacturing head, managers and key cadres of each department. Each department has established a quality risk management group; each department head is the group leader. Group members are person for each position and core technical employees. The quality management system, based on risk management, follows out the concept of whole-staff risk management, implementing risk management into daily manufacturing activities. Walvax identifies risk sources by risk recognition and evaluation on each process of manufacturing activity and management activity, finally the control document is formed, the risk evaluation report corresponds to risk level. At the same time, Walvax regularly carries out two risk reviews every year, to make sure the control effects of risk source, to continuously improve quality management system based on risk management, and to decrease risks lying in vaccine manufacturing and circulation control, thus guaranteeing the quality of vaccines.

Quality Risk Management Committee

Public engineering system Control of vaccine

Risk management Control of production technical Control of quality system

Risk management committee distribution Risk

committee process Risk management committee management committee

preparation and distribution

Risk control group of water system

conditioning system preparation and

Risk control group of air

air system preparation and

Risk control group of compressed

steam preparation and distribution

Risk control group of industrial

preparation and distribution

Risk control group of pure steam

production

Risk control group of Hib vaccine

meningococcal vaccine

Risk

vaccine

Risk control group of pertussis

vaccine

Risk control

vaccine
Risk control group of diphtheria
production
Risk control group of sub-filling
production
Risk control
training
Risk control group of manpower
management
Risk control group of equipment
suppliers
Risk control group of material
releasing management
Risk control group
management
Risk control group of quality control
for validation management
Risk control group of management
storage
Risk control
dispatching
Risk control
complaints and recalls
Risk control group
reactions
Risk control
control

group

group

group
group

group
group

of

of

of

of

of
of

of

product

product

product

product

adverse
tetanus

supply
of

Figure 3 Organization Chart of Quality Risk Management System

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Yuxi Walvax Biotechnology Co., Ltd

Figure 4 Organization Chart of Quality Assurance Department

2.5 Review of Product Quality

Yuxi Walvax has designed Management Regulation for Product Quality Review (Code: WSYX-QO-SMP-00-09-0001) according to ICH Q7a, EU GMP Part I - Basic Requirements for Medicinal Products and Annex 6 Good manufacturing practices for sterile pharmaceutical products, aiming for guiding, regulating and controlling annual product quality review.

The Quality Operation Department of Yuxi Walvax appoints special persons to compile templates of product quality review protocol and report, clearly regulating responsibilities, review range and contents. Templates of product quality review protocol and report will come into implementation after joint hearing and obtaining approval by Quality Management Department, production management, workshops involved in production and other departments. Yuxi Walvax collects data of equipment and facilities such as air purification system, water system and compressed air system, material supplier, intermediate control of manufacturing process and stability examination, adopts line trend chart and tables to make statistic analysis, thus forming annual product quality review report, proposing manufacturing and quality improvement measures to guide continuous improvement of manufacturing and quality management system for the next year.

3. Personnel
3.1 Organization chart

In order to comply with Good Production Practice (2010), Yuxi Walvax has 27 departments to meet drug production demands. The organization chart is covering production, quality, storage, product

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Yuxi Walvax Biotechnology Co., Ltd
distribution, and etc. Yuxi Walvax has an independent quality management department to perform the duty of quality assurance and control. Job descriptions of each department and position have been made to ensure proper implementation of staff in each department.

Figure 5 Organizational Chart of Yuxi Walvax

3.2 Outline of arrangements for basic and in-service training and how records are maintained
Yuxi Walvax has established Training Management Control Procedure (Code: WSYX-P-02-0002) and Work License Standard Management Procedure (Code: WSYX-HR-SMP-00-02-0002), which are used for guiding and regulating training for persons relevant with production and quality, thus to enhance employees' consciousness, experiences and abilities, guarantee drug production and decrease risks. The Human Resources Department and Quality Assurance Department are jointly responsible for training management, selecting proper training content adapting job responsibilities, operating according to training objects, and forming training plan reviewed by each department head and jointly approved by person in charge of production management and person in charge of quality management. Each department shall organize training by lecturing, actual operation, group discussion and other ways, and evaluate actual training results by written examination, Q&A and other methods. All records involved in training shall be archived and stored in the archive room.
Training management of new employees

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Yuxi Walvax Biotechnology Co., Ltd
New employees shall attend pre-job training. The training includes three parts. The first part is basic GMP knowledge, company introduction and other basic information. The second part is training of public management knowledge of intended department. The third part is training of professional knowledge and operation skills. New employees shall attend the three parts training above and pass the examination during probationary period. The HR Department will issue work license for new employees, indicating the new employees obtain qualification of independent working ability. See detailed training contents in the following table.

Table 0 Contents of Training for New Employees
Classification Contents Detailed Information Time Trainer

Company history, Relevant person
Company
organizations, products and of HR
introduction
management regulations, etc. department
Drug Management Relevant person
Rules and Regulations and of Quality
regulations Implementation, GMP Management
knowledge Department
Relevant person
General sanitation of Quality
Sanitation
requirements Management
Once Department
Basic
every 3 Relevant person
Knowledge Definition, classification,
Microbial months of Production
growing environment of
knowledge Management
microorganism
Department
Relevant person
Management Definitions, classifications and
of Quality
of deviations processing program of
Management
and changes deviations and changes
Department
Relevant person
Safe production, fire safety, of Safety
Safety
etc. Management
Department
Designed the training
Public Public according to different
Management management department responsibilities,
Within
Knowledge of knowledge for example, production
6
Intended of workshop: clothes changing Relevant person
months
Department department requirements, safe production, of department
after
recording, etc.
entry
Designed the training
Position Professional
according to different
Knowledge knowledge
department responsibilities,

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Yuxi Walvax Biotechnology Co., Ltd
Classification Contents Detailed Information Time Trainer

and actual for example, production
operations workshop: production
technique, equipment
operations, cleaning and
disinfection; laboratory:
analysis methods, test
methods, operations of
laboratory apparatus, etc.
Employee continuing training management

Continuing training management for employees is designed by HR Department, Quality Assurance Department and responsible persons of each department according to national regulation requirements, position requirements, self-inspection and external audit evaluation etc., including basic training and targeted training. Basic training is general GMP requirements, rules and regulations, basic company information and safety knowledge, which shall be learned by employees, applicable for all staff. The HR Department and Quality Assurance Department shall organize the training once a year together. The continuing trainers are those who are familiar with GMP, rules and regulations, and company profile. Targeted training is designed for qualification training of special professional operations, professional knowledge and special work, applicable for relevant operation personnel. Each relevant department shall organize targeted training once a year at least. The trainer shall be a professional employee or person from qualified training agencies. See detailed basic knowledge training and targeted training contents in the following tables.

Table 3 Basic Training Contents
Detailed Training
Classification Contents Trainer
Information Objects
Company history,
Company organizations,
Relevant person of HR
Introductio products and
Department
n management
regulations, etc.
Drug Management
Basic
Regulations and
Knowledge All staff Relevant person of
Rules and Implementation,
Training Quality Management
Regulations basic GMP
Department
knowledge, drug data
management
Relevant person of
General sanitation
Sanitation Quality Management
requirements
Department

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Yuxi Walvax Biotechnology Co., Ltd
Fire safety, lift
management,
Relevant person of
management of
Safety Safety Management
inflammable and
Department
explosive dangerous
chemicals
Manageme Definitions,
nt of classifications, Relevant person of
changes processing program Quality Management
and of deviations and Department
deviations changes

Table 4 Targeted Training Contents
Detailed
Classification Contents Training Objects Trainer
Information
Technology Persons related to
Professionals
Production knowledge of each production operation
related to
Process position and actual and process
each position
operations supervision
Usage, cleaning,
Professionals
Equipment disinfection and Employees of
related to
Operation maintenance of production operation
each position
equipment
Persons related with
Sterile Professionals
sterile production,
production — related to
sterile test and
knowledge each position
process supervision

Definition,
 Microbial classification, Persons related with
 Targeted knowledge growing production operation, Professionals
 Training and bio- environment, bio- microbial laboratory related to
 safety safety and process each position
 knowledge containment, etc. supervision
 of microorganism
 Analysis
 method and Professionals
 operations of — Laboratory personnel related to
 analysis each position
 apparatus
 Definition,
 Persons related with
 classification and
 production Professionals
 Complaints management
 management, quality responsible
 and recalls procedures of
 management and for the work
 complaints and
 storage management
 recalls

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 Yuxi Walvax Biotechnology Co., Ltd
 Detailed
 Classification Contents Training Objects Trainer
 Information
 Good Supply Persons related with
 Practice and production Professionals
 other — management, quality related to
 circulation management and each position
 rules storage management
 Material
 Person related with
 Management management rules, Professionals
 quality management,
 of materials supplier responsible
 storage management,
 and suppliers management for the work
 purchasing, etc.
 procedures
 Pressure vessel,
 electrician, welder,
 Qualified
 fork lift truck,
 Persons responsible national
 Special work operations of
 for related operations training
 laboratory
 agency
 animals,
 calibration, etc.

Except for annual training plan, there are other trainings derived from deviations, changes, regulation revision, self-inspection, etc. At the end of each year, evaluate whether it shall be included in basic training or targeted training in the following year through training summary.
 Training implementation and evaluation management of training examination

During implementation of training, training consultants carry out training by selecting correspondent training textbooks and adopting classroom learning, actual operation, special training of special organizations, etc. according to training contents. To ensure better training results, the training consultants will adopt written examination, on-site Q&A or on-site operation to confirm training results. The department responsible for training shall arrange failed employees to attend training and examination again.

Table 5 Management of training examination

No. Method Management

Trainer provides examination paper, organizes the examination and
 Written

1 mark the papers. Failed employees shall attend training again and take
 Examination

the examination again.

Training Consultant proposes on-site questions and evaluates the

2 On-site Q&A results. Failed employees shall attend training again and take the
 examination again.

Training Consultant makes on-site check and evaluates the results.

3 On-site operation Failed employees shall attend training again and take the examination
 again.

Management of training archives

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Yuxi Walvax Biotechnology Co., Ltd

The training archives include training plan, training textbooks, learning record of training, examination materials, training review plan and report. Each department transfers training archives to HR Department every year. The HR Department reviews the archives and then transfers all the archives to achieve room.

3.3 Health requirements for personnel engaged in production

Yuxi Walvax has established Standard Management Procedure for Staff Health (Code: WSYX-HR-SMP-00-02-0003), used for guiding staff health management. The HR Department is responsible for

establishment and maintenance of staff health archives. New employees shall take health check before entry, and employees take health check once a year. Personnel engaged in production and process supervision should be vaccinated corresponding vaccine. Personnel directly contacting with product manufacturing and related with process monitoring shall not have skin disease and infectious disease, and corrected visual acuity of persons responsible for physical appearance inspection shall be above 5.0, thus guaranteeing the staff meet health requirements and decreasing contamination risk caused by personnel.

Table 6 Physical Examination Items of Employees

Evaluation and

No. Employee Physical Examination Items

Dealing

If the health check

Liver function, hepatitis B, HAV-IgM,

results show skin disease

New electrocardiogram, chest fluoroscopy, internal

1 or infectious disease, he

Employee medicine, surgery, ophthalmology and

or she shall not be

otorhinolaryngology

employed.

If the health check

results of production

person and process

Internal medicine, surgery, ophthalmology and supervising person

otorhinolaryngology, liver function, hepatitis B, directly contacting with

HAV-IgM, electrocardiogram, chest fluoroscopy, products show skin

abdominal ultrasonography disease or infectious

2 Employee

disease, he or she shall

be transferred to other

position.

If corrected visual acuity

Physical appearance inspection person: check less than 5.0, he or she

eyesight every 3 months shall be transferred to

other position.

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3.4 Personnel hygiene requirements, including clothing.

Yuxi Walvax has established Standard Management Procedure for Staff Hygiene (Code: WSYX-QS-SMP-00-02-0001) for guiding personnel hygiene. All the staffs should take the hygiene knowledge training. They are allowed to operate after pass examination of the training. As required, operators are not allowed to wear make-ups or jewellerys. Operators must follow gowning procedures to change clothes, wash and disinfect hands. Operators are not allowed to use bare hands to touch raw materials, excipients, final bulk, intermediate, packing materials and surface of the equipment, which directly contact with products. Smoking, eating, storing food, drinks, cigarettes, personal medicines are prohibited in production area and storage area. Visitors entering production area must be approved by production or quality responsible person and accompanied by specified person. Visitors must follow gowning procedures in accordance with SMP before entering the cleanliness area. Operators who work in clean area must operate appropriately. Putting hands down, akimbo, putting hands under the arms or holding hands high over the shoulder are not allowed. Operators must put hands in front of chest, and avoid to squat. Operators must keep a distance between each other to avoid contact with each other (even with sterile gloves). Operators must check whether gowns are proper when operating in high-risk areas. Hand disinfection must be carried out before touching any material or surface. Disinfect hands every 10-20mins when just in contact with air.

4. Premises and Equipments

4.1 Premises

Yuxi Walvax is the industrialization base of Walvax Biotech Co., Ltd., and is one of the focused supporting enterprises of Yuxi Government. The plant has been built since 2003 and covers an area of 85000 m². The phase I construction was completed in 2005 with the construction area as 1170 m², which is leased to the group companies for producing new variety vaccines and monoclonal antibody drug. The phase II construction was completed in 2010 with the construction area as 21720 m². Phase III was completed in 2012, the area of which is 63610 m².

The workshops are distant from urban arterial road. Surrounding environment has no pollutant source to affect product quality. Air and water quality are in good condition. Environment in the plant is clean and tidy, meeting the drug production hygiene criteria. The plant has two entries to effectively prevent possible cross contamination: west entry for personnel and north entry for logistics. The internal layout is reasonable. Personnel channel and logistic channel are separated. Production areas and administrative areas are independent to protect the vaccine production from possible pollution. The roads are cement concrete pavement to avoid fugitive dust. The roads are smooth and clear, good for safety and fire protection.

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Quality Control Laboratory is responsible for physical and chemical testing, material testing, preparation and management of standard test samples, and microbial, biochemical testing. The labs for microbial limit test and sterile test are separated in different rooms, and equipped with independent HVAC systems to prevent the pollution and cross contamination.

Production workshops are strictly managed and controlled according to production process approved by National Medical Products Administration (NMPA). All the production and quality operations meet the WHO cGMP requirements.

Joints of process pipelines, colored steel plates and high-efficiency supply air filters and the juncture of lamps and lanterns and so on entering the clean management areas are sealed with silicone glass glue. Buffer rooms and air blocks are set between the clean production areas and the control areas, ensuring closed clean production environment, not polluted by the outside clean air.

Warehouses and storage areas

Virulent material are checked upon receiving and stored in double-locked virulent safe storage. The storage is audited by Production Management Department and inspected by Quality Assurance department, with clear classification and numbering.

There are two keys for the double-locked storage with two warehouse administrators. Each administrator keeps one key. The code of safety box is kept by appointed staff from Transportation Department. The key of safety box is kept by appointed staff from Production Management Department. Rest eating and smoking in virulent storage room are prohibited.

The warehouse administrators check chemical-unstable virulent material once every month and stable nature material once a quarter. If find out any problem, the administrator must have proper measures, and report to QA manager immediately.

In case of break or spill virulent material, the administrator must report to QA immediately. Personnel in the storage shall put on gas masks clothing and gloves, and then clean and ventilate under the supervision of QA. Any materials with virulent splash shall be cleaned carefully. All the materials that cannot clean or wash shall be destroyed.

Description of specific storage conditions

The cold storage area of Walvax is about 5000 m², totally meeting the storage requirement of Walvax's products (bulks and final products). The Validation Management Department of Walvax carried out validation according to the approved master validation plan. The validation result shows all the design, installation and performance of cold storage meet the storage temperature for products. Not only verified mercurial thermometers are established in cold storage for monitoring, but also online monitoring platform are established for monitoring real-time temperature. The monitoring platform

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has alarm system. If the temperature exceeds the scheduled range, the responsible person in Walvax and the person in charge of cold storage will be informed by satellite network system, thus making timely emergency measures to guarantee the storage temperature safety of the products.

Layouts and flow charts of the production areas, as well as layouts of warehouses and storage areas for the storage and handling of highly toxic, hazardous and sensitising materials indicated are attached in Appendix 4.

4.1.1 HVAC system.

The 9 sets of HVAC system in Meningococcal Vaccine Workshop meet the national requirements for production plant. The schematic diagrams are as follows:

The air shall be filtrated out of dust by fresh air filter, and then mixed with returned air from clean rooms. The mixed air shall be filtered by G4 primary-efficiency air filter, and then be cooled/dehumidification by 7℃ cooling water in surface cooling section. After heating and stabilizing temperature by 0.3MPa of low-pressure saturate steam, filter the air by F6 medium efficiency filter. The filtered air shall enter wind inlet duct after humidification in humidification section. De-noised by muffler in air inlet duct, the air is sent into the room through HEPA filter (H14). Air in some rooms (such as air lock room and dressing room) shall exhaust through air outlet, exhaust duct and air filter (F5). The returned air shall return through hair return duct, mix with fresh air and then enter primary-efficiency air filter for circulation. JK-MGa HVAC system in non-sterile area is full fresh air system. The exhausted air is filtered by H13 high efficiency filter. The operation process is shown in Figure :

1. Fresh air 7. Medium efficiency filtration section
2. Fresh air filtration section 8. Humidification section
3. Primary efficiency filtration section 9. Air outlet fan section
4. Cooling section 10. Air outlet filtration section
5. Heating section 11. High efficiency filtration section
6. Air inlet fan section 12. High efficiency filtration section

Name Models and requirements

Fresh air filtration section Located at the mixing section, model of fresh air filter: G3

Primary efficiency filtration section Equipped with differential pressure gauge, model of primary efficiency filter: G3

Medium efficiency filtration section Equipped with differential pressure gauge, model of medium efficiency filter: F5

High efficiency filtration section Terminal filtration, model of medium efficiency filter: H13

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Figure 6 Schematic diagrams of HVAC System (Meningococcal Vaccine Workshop)

The HVAC system in Fill & Finish Workshop meets the national requirements for production plant. The schematic diagrams are as follows:

The air shall be filtrated out of dust by fresh air filter and then be mixed with returned air from clean room in air returning section. The mixed air shall be filtered by primary-efficiency air filter, cooled/dehumidified by 0°C or 7°C cooling water at the surface cooling section, then be supplied to 0.3MPa low pressure saturated steam heating section for heating. Humidify the air with necessary industrial steam at the humidification section, supply the humidified air to medium-efficiency air filter for filtration and then enter the main air inlet duct. Before entering each room through the branch ducts, filter the air by high efficiency filter. For areas of grade A/B, Fan Filter Units (FFU) is added before entering the room after filtration by high efficiency filter. The schematic diagrams are shown in Figure :

1. Fresh air 7. Medium efficiency filtration section
2. Fresh air filtration section 8. Humidification section
3. Primary efficiency filtration section 9. Air outlet fan section
4. Cooling section 10. Air outlet filtration section
5. Heating section 11. High efficiency filtration section
6. Air inlet fan section 12. High efficiency filtration section

Name Models and requirements

Fresh air filtration section Located at the mixing section, model of fresh air filter: G3
 Primary efficiency filtration section Equipped with differential pressure gauge, model of primary efficiency filter: G3
 Medium efficiency filtration section Equipped with differential pressure gauge, model of primary efficiency filter: F5
 High efficiency filtration section Model of medium efficiency filter: H13
 FFU filtration section Terminal filtration, model of medium efficiency filter: H14

Figure 7 Schematic diagram of HVAC System (Fill & Finish Workshop)

The HVAC system in International Vaccine Formulation Centre meets the national requirements for production plant. The schematic diagrams are as follows:

The air shall be filtrated out of dust by fresh air filter (G4) above the air returning section and then be mixed with returned air from clean room in air returning section. The mixed air shall be filtered by medium-efficiency air filter, cooled/dehumidified by 0°C or 7°C cooling water at the surface cooling section, then transfer to be heated with 60/50 hot water at heating section, and then humidify the

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air with necessary clean steam at the humidification section, supply the air to medium-efficiency air filter (F8) for filtration and then enter the main air inlet duct. Before entering each room through the branch ducts, filter the air by high efficiency filter (H14). Then enter into each clean room, the pressure sensor feedbacks the pressure in each room to Air volume regulating valve to adjust the pressure in room, a part of the air is out of the room through the outlet machine after filtered with medium-efficiency air filter (F6). The schematic diagrams are shown in Figure :

Figure 8 Schematic diagram of HVAC System (International Vaccine Formulation Centre)

The 4 sets of HVAC system in Quality Control Laboratory meet the national requirements for production plant. The HVAC system is composed of air inlet ducts, air outlet ducts, and air handling units. The air-handling units include filters of each grade, fans, temperature/humidity adjustment devices, etc.

KX4-2 and JK4-2 systems are full fresh air systems for non-sterile area of Quality Control Laboratory. The working principles are as follows: the air shall be filtrated by G3 primary efficiency filter. The filtered mixed air shall be cooled and dehumidified by 7°C chilled water at the surface cooling section, then be supplied to 0.3MPa low pressure saturated steam heating section for heating. The heated air shall be filtered in F6 medium-efficiency air filter, humidified in humidification section and then be supplied into air inlet duct. De-noised by muffler in air inlet duct, the air is supplied into the end of duct-HEPA filter (H13) and then the air enters the room. Air in rooms shall be filtered through air outlet, air outlet duct, and air outlet filter (F15) and then exhausted out of the room.

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JK4-1 and KX4-1 systems are for sterile area of Quality Control Laboratory, among which KX4-1 is full fresh air system with the same working principles as HVAC systems in non-sterile area. The working principles of KX4-1 are as follows: the air shall be filtrated by G3 primary efficiency filter. The filtered mixed air shall be cooled and dehumidified by 7°C chilled water at the surface cooling section, then be supplied to 0.3 MPa low pressure saturated steam heating section for heating. The heated air shall be filtered in F6 medium-efficiency air filter, humidified in humidification section and then be supplied into air inlet duct. De-noised by muffler in air inlet duct, the air is supplied into the very end of duct-HEPA filter (H13) and then the air enters the room. Before entering KX4-1 HVAC system, the air shall be filtered by FFU high efficiency filter (H14). Air in some rooms (Buffer Room and Sterilizing Air Valve) shall be filtered through air outlet, air outlet duct, and air outlets filter (F5) and then exhausted out of the rooms. Air in other rooms shall be mixed with fresh air in air returning duct and then enter primary efficiency filter for circulation.

The schematic diagrams of JK4-1 and KX4-1 systems are shown in Figure and Figure :

1. Fresh air section 7. Medium efficiency filtration section
2. Primary filtration section 8. High efficiency filtration section
3. Cooling section 9. FFU
4. Heating section 10. Clean Room
5. Humidification section 11. Air outlet filtration section
6. Fan section 12. Air outlet fan section

Figure 9 Schematic diagram of KX4-1 HVAC System (QC Lab)

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1. Fresh air section 2. Disinfection valve
3. Primary filtration section 4. Cooling section
5. Heating section 6. Humidification section
7. Fan section 8. Medium efficiency filtration
9. HEPA filtration section 10. Clean Room
11. Air outlet filtration section 12. Air outlet fan section

Figure 10 Schematic diagram of JK4-1 HVAC System (QC Lab)

Distributed Control System (DCS) is used in all the air conditioning control systems. This system meets requirements of three different conditions of production: normal operation, disinfection and stop. It is capable of automatically controlling the air return system and temperature, humidity and pressure of some rooms. Rooms with air exhaust equipment are also set up with monitoring and

automatic adjustment of pressure. Fresh air ducts and air outlet ducts are equipped with electric control valves, which have chain control with the fan. When the system shut down unexpectedly, the fresh air and air outlet electric valve will be shut down automatically to prevent outdoor airflow back into the clean room.

Design Standard of HVAC system

HVAC systems of Meningococcal Vaccine Workshop, Fill & Finish Workshop and Quality Control Laboratory were designed and constructed according to requirements of Good Manufacturing Practice (2010), Design Specifications of Heating Ventilation and Air Conditioning GB 50019-2003, Design Specifications of Clean Room GB 50073-2001, Design Specifications of Pharmaceutical Industry Clean Room GB 50457-2008 and the production process of products. Room cleanliness is classified as grade A, B, C, D and E. Grade E is controlled non- classified (CNC). Pressure cascade between clean areas and black areas, or between different grades are not less than 10 Pa.

C + A design is used in Meningococcal Vaccine Workshop and areas of high contamination risk in Quality Control Laboratory, such as sampling, strain operation, centrifugation, ultrafiltration and conjugation. B + A design is used in areas of high contamination risk in International Vaccine Formulation Centre involved in process such as formulation, filling and capping. Preparation of final bulk and filling process are carried out in an independent air conditioning partition, while capping

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process is in another air conditioning partition, to avoid cross-contamination. Air supply in Grade B areas is unidirectional laminar airflow supplied from ceiling and returned from sidewalls, Fan Filter Units (FFU) are designed for final bulk preparation rooms in grade A clean zone, the air is processed with HEPA filter again to ensure grade A environment. Equipment in capping room and sub-packing room are with its own isolation system in Grade A clean zone, effectively control contamination and to guarantee cleanliness grade. Online Particle Monitoring System is established for critical Grade A areas (such as filling area). The design specifications [according to China GMP (2010), Appendix 1 “Sterile Drugs”) are as follows:

Table 7 Maximum Permitted Number of Particles Concentrate in Clean Area
Maximum Permitted Number of Particles /m3

Clean Grade	At Rest	In Operation (3)
	≥0.5µm	≥5.0µm (2) ≥0.5µm ≥5.0µm
A	(1) 3520	20 3520 20
B	3520	29 352000 2900
C	352000	2900 3520000 29000
D	3520000	29000 Not stated Not stated

Note: (1) In order to confirm the Grade A clean area, the sampling volume of each sampling point shall be not less than 1 m3.

Table 8 Standard of Microbial Monitoring in Clean Area in Operation (1)

Surface Microorganism				
Floating Settling Microbe				
Clean Grade Bacteria (■90mm) Contact Plate Glove Print				
cfu/m3 cfu/4 hours (2) (■55mm) (5 fingers)				
cfu/plate cfu/glove				
A ■1 ■1 ■1 ■1				
B 10 5 5 5				
C 100 50 25 ■				
D 200 100 50 ■				

Note: (1) The numbers in the table are average values. (2) The exposure time of single settle plate may be less than 4 hours, and multiple settle plates can be used to continuously monitor and accumulate counts in the same location.
Environmental monitoring procedure

According to the technical standard of ISO14644-1:2015, the company has established the Standard Operation Procedure of Dynamic Environmental Monitoring in Clean Area (Code: WSYX-QS-SOP-00-09-0001) and Standard Management Procedure of Routine Environmental Monitoring in Clean Area (Code: WSYX-QS-SMP-00-09-0002) to monitor and manage the clean environment. Routine

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environmental monitoring includes monthly environmental monitoring and in-operational monitoring of production process.

Table 9 Monthly Environmental Monitoring

Items	Specification	Test Frequency
QC Laboratory conducts	Critical operating room/area≥300Lx, Auxiliary quarterly monitoring of critical	illumination
working room/area≥150Lx	operating rooms / areas in the clean area.	a) twice a day in each room equipped with Pressure Gauge
Pressure Between adjacent rooms of different grade	≥10Pa, in the clean area.	Cascade between clean room and the outdoor ≥10Pa b) the production workshop
monthly monitors all the systems in clean area.	a) Engineering Operation	Department records the temperature and humidity of each air conditioning system
Temperature	45%-65%RH	twice a day.
and Humidity	18%-26%	b) QC lab conducts monthly monitoring of critical operating rooms / areas in the clean area.
Static state	Dynamic state	QC Lab conducts a

Grade comprehensive monitoring of
 $\geq 0.5\mu\text{m}$ $\geq 5\mu\text{m}$ $\geq 0.5\mu\text{m}$ $\geq 5\mu\text{m}$ the system-related rooms
 A 3520 20 3520 20 pass-through boxes, clean
 Particles B 3520 29 352000 2900 bench, biosafety cabinets,
 (Number/m³) C 352000 2900 3520000 29000 laminar flow carts) according
 Not not to the monthly monitoring
 D 3520000 29000
 stated stated plan.
 In order to ensure grade A cleanness, the sampling volume of each
 Note
 sampling point shall be not less than 1 m³. The above is the upper limit.
 Sedimentation QC lab conducts a
 Plankton
 Grade bacteria comprehensive monitoring of
 cfu/m³
 ($\Phi 90\text{mm}$) cfu/4 hours the system-related rooms
 A ■1 ■1 (pass-through boxes, clean
 B 10 5 benches, biosafety cabinets,
 Microorganism laminar flow carts) according
 C 100 50
 m to the monthly monitoring
 D 200 100 plan.
 The exposure time of every single settle plate can be less than 4 hours, and
 Note multiple settle plates can be used to continuously monitor and accumulate
 counts in the same location. The above is the upper limit.

Grade A ■0.36-0.54m/s Once half a year for grade A
 Air speed and B (laminar hood) clean
 Grade B ■0.36-0.54m/s
 area.

Air Flow Grade A: The airflow pattern should be vertical Once half a year for grade A
 Pattern unidirectional flow, with no turbulence. clean area.

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Test on Leakage rate of H13 PAO ■0.06% Once half a year for grade A
 HEPA Filter and B clean area; once a year
 Integrity Leakage rate of H14 PAO ■0.01% for grade C and D.

Table 10 Area, Item and Frequency of Monitoring of Production Process in Operation
 Monitoring Monitoring
 Monitoring Item Judgment Standards
 Area Period
 Suspended particulates, plankton,
 Dilute dispensing sedimentation bacteria, personnel Dynamic standard for
 Every batch
 Room surface microorganisms, surface grade A and B clean area
 microorganisms
 Suspended particulates, plankton,
 Dynamic standard for
 Filling Room sedimentation bacteria, personnel Every batch
 grade A and B clean area
 surface microorganisms,
 Suspended particulates, plankton,
 Secondary Dynamic standard for
 sedimentation bacteria, personnel Every batch
 lyophilization grade A and B clean area
 surface microorganisms,
 Dynamic standard for
 Suspended particulates,
 Rolling cap Every batch grade A, B and C clean
 microorganisms, personnel
 area
 Sterilization and Suspended particulates, Dynamic standard for
 Filtration sedimentation bacteria, personnel Every batch grade A and B clean area
 Operating Area surface microorganisms,
 Layouts of ventilation system Please refer to Appendix 5 HVAC system schematics

4.1.2 Water system
 Operational Principle

The purified water preparation system produces purified water by urban drinking water with the
 process of "pre-treatment+ reverse osmosis + Electrode ionization (EDI)". Detailed production
 process is: city water is sequentially treated by multi-medium filter, softener, activated carbon filter
 and precision filter, then processed by reverse osmosis membrane and EDI device to generate purified
 water. Water for injection preparation system uses purified water as raw materials and factory steam
 as source of the first evaporation heat tower. Six-effect distiller generates WFI. Detailed production
 process is: raw water (purified water) is pumped into the pipes of distiller and heat exchange with
 industrial steam in the shell of distiller. After distillation, the raw water is separated by separator
 and
 pure steam is obtained. The pure steam produced in the first evaporation tower is used to heat the
 second evaporation tower, and the rest can be done in the same manner. The pure steam produced in
 the last evaporation tower enters the condenser for cooling down to generate water for injection.
 Water system is designed with a programmable logic controller (PLC) + touch screen automatic
 control with display and limit alert for liquid grade, electric conductivity, temperature, automatic
 pressure monitoring. At the same time, parameters of key process (conductivity of producing water,
 temperature of producing water) are continuously monitored and recorded to ensure the quality of

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process water.

Designing Standards

It is designed according to WHO GMP, EU GMP, Chinese GMP, United States pharmacopoeia, European pharmacopoeia and Chinese pharmacopoeia. It is set with the purified water preparation system, water for injection preparation system and pure steam preparation system. The production capacity of purified water is 20 t/hr, 10 t/hr for water for injection, and 4 t/hr for pure steam. Production capacity of purified water; water for injection and pure steam is 20t/hr, 10t/hr and 4t/hr, respectively. Distribution systems of purified water and water for injection strictly divide the purified water and water for injection in sterile area and non-sterile area in bulk workshop and Fill & Finish Workshop by different storage tanks to avoid risk of cross-contamination. Online conductivity analysis meter and Total Organic Carbon (TOC) analysis meter are set up at the water returning section for real-time monitoring of water quality.

The hardware of the system is equipped according to engineering design and construction guidance of ASME BPE of American Society of Mechanical Engineers to avoid the blindness of pipe tube and angle. At the same time, the pipe has a certain slope (not less than 5‰) to facilitate the cleaning, drainage, and sterilization of the system. The storage tank and clean pipeline adopts 316L stainless steel material. The valve adopts diaphragm valve from Gemu of German. The water for injection system uses the no hidden corners T valve type of Gemu. The circulation pump is drainable centrifugal pump of Sweden Alfa Laval in sanitary grade. The heat exchanger is adopted the shell and table heat exchanger of German Neumo in sanitary grade. The respirator with electric heating jacket has a separate pure steam pipeline to sterilization in place (SIP).

As for the software of the system, automatic control systems of storage and distribution systems meet the requirements of computer validation system guidance GAMP5 (Good Automation Manufactory Practice 5), recommended by ISPE (International Society for Pharmaceutical Engineering). The control system is programmable control system (PLC), established with user groups of different limit level, thus guaranteeing the stability, controllability and verifiability of automatic control system and data integrity.

Yuxi Walvax adopts the method of risk assessment to assess the whole water system, from the User Requirement Specification (URS), Design Qualification (DQ), Factory Acceptance Test (FAT), Installation Qualification (IQ), Site Acceptance Test (SAT), Operational Qualification (OQ) and Performance Qualification, throughout the whole system process. The distribution system was designed, manufactured, installed and debugged by the Morimatsu group of Japan, Shanghai Morimatsu Process Equipment Engineering Co., LTD. The water for injection system adopts above 70°C heat preservation cycle, and installed online TOC analyser (GE, USA) in the water outlet, in

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order to real-time monitor the quality condition of water.

Water system operation and verification

According to the validation main plan of our company, the Risk & Validation Management Department is responsible for organizing the validation of the water system, and the validation period is one year. According to the validation results, the Standard Management Procedure for Process Water Monitoring (Code: WSYX-QS-SMP-00-03-0002) has been formulated to stipulate the process water management plan. The Quality Supervision Department is responsible for sampling from the supply and outlet of the water system every 14 days, and should complete the sampling and monitoring of all used points in three months. The results of the validation and the daily monitoring data show that the company's purified water and water for injection system is stable and can continue to produce process water in line with quality standard of Chinese Pharmacopoeia, Pharmacopoeia of United States and European Pharmacopoeia.

Clean procedures and arrangements for water systems

According to the validated period of disinfection / sterilization, the purified water system and water for injection system will be disinfected / sterilized periodically. In meningococcal vaccine workshop and International Vaccine Formulation Centre, disinfection of purified water system is at 80°C for 120 minutes, and sterilization of water for injection system is overheated water at 121°C for 30 minutes. In Phase II Fill & Finish Workshop, the purified water system and the water for injection system both adopt hot water at 121°C for 30 minutes for sterilization. In order to ensure the complete set of water systems meets the requirements, cleaning validations of purified water system and water for injection system are carried out every five years.

4.1.3 Other System

Yuxi Walvax has established Standard Operation Procedures of Receiving, Storage and Distribution of Hazardous Chemicals (code: WSYX-ST-SMP-00-03-0004). Hazardous Chemicals are stored in specific hazardous chemicals warehouse, shall be isolated, separated and stored separately according to the storage method, storage quantity and safe distance specified in relevant technical standards. Hazardous chemicals warehouse conforms to the relevant technical standards for safety, fire protection requirements, set up with obvious signs, and be managed by special personnel; All hazardous chemicals in and out of the warehouse are registered and checked regularly. It is carried out the two-man rule for highly toxic chemicals, two men in charge of receiving and dispatching, two men in charge of keeping accounts, two men in charge of two locks, two men in charge of transportation, and two men in charge of using. The company has established Processing Procedures of the management of unqualified product (Code: WSYX-QA-SMP-00-09-0011) and Processing

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Procedures of destroying of unqualified product (Code: WSYX-ST-SOP-00-05-0012), to standardize the disposal of waste Hazardous Chemicals.

The company has established Processing Procedures of the Bacteria Carrier Items and Thallus in the Meningococcal Vaccine Workshop I (code ■ WSYX-BV1-SOP-00-08-0011), in order to conduct sterilization of the equipment after using in non-sterile area, and the sterilization condition is moist heat sterilization (122.1±1°C, 30 min), to reduce the contamination risk from non-sterile items in the production operation area.

4.1.4 Description of planned preventive maintenance program for premises and the recording system.

The company has established Procedures of Equipment Preventive Maintenance (code: WSYX-PM-SMP-00-04-0012), to guide the management of equipment preventive maintenance. According to the demands of equipment preventive maintenance, at the end of every year, each of the equipment

management department shall draft the equipment preventive maintenance plan for the coming year, (1) it must be carried out a review on the equipment operation and maintenance situation of the previous year and analyse the reason of the maintenance have not completed as planned; (2) According to the level assessment before the purchase of equipment, the critical evaluation of components during equipment verification, statistically analyse the breakdown, maintenance and maintenance of the equipment in each department last year, review and analyse whether the frequency and measures of key maintenance components last year are reasonable, and whether the preventive maintenance frequency and measures last year are reasonable or not. According to the equipment preventive maintenance plan, during daily production, the equipment using department / management department shall establish "Equipment Maintenance Table", record the anomalies, failures, and maintenance of each device during its use. The information should be including equipment code, type, manufacturer, place site, failure date, description of failure, Maintenance and maintenance measures and etc. We review the table at the end of every year as reference for the equipment preventive maintenance plan for the next year.

4.2 Equipment
4.2.1 Production and Quality Control laboratory equipment

Equipment and facilities for the vaccines and diluents are all purchased from the famous equipment manufacturers both domestic and abroad. Equipment lists are as follows:

Table 11 List of Critical Equipment of Meningococcal Workshop

30	Yuxi Walvax Biotechnology Co., Ltd
No. Room No. Room Name Main Equipment Grade	
Fermentation	
1 1MG210 BIO-DVT-3000F Bioreactor D	
Room	
2 1MG211 CIP Unit CIP Unit Control System D	
Centrifugation CR712 High Capacity	
3 2MG209 D	
Room 1 Refrigerated Centrifuge	
CR712 Large Capacity	
4 D	
Refrigerated Centrifuge	
Centrifugation LYNX400021 High Speed	
5 2MG212 D	
Room 2 Floor Centrifuge	
RC 12BP+8 High	
6 D	
Capacity Centrifuge	
Sterile Filtration Sartocheck 3plus Filter Test	
7 3E224/2 B	
Room Membrane Integrity Tester	
LYNX400021 High Speed	
8 Centrifugation Floor Centrifuge	
3MG210/1 C	
Room 3 RC 12BP+8 High	
9	
Capacity Centrifuge	
Centrifugation LYNX400021 High Speed	
10 C	
Room 4 Floor Centrifuge	
3MG210/2	
Centrifugation RC 12BP+8 High	
11 C	
Room 4 Capacity Centrifuge	
UF/Ultrafiltration UFC-10CC ultrafiltration	
12 3MG211/2 C	
Room system	
UF/Ultrafiltration UFC-10CC ultrafiltration	
13 3MG211/1 C	
Room system	
AKTA process purification	
14 3MG213 Tomographic room C	
system	

Table 12 Equipment List of Supply Workshop
No. Room No. Room Name Main Equipment Grade
1 2D201/2 Inactivation Room Pulsation Vacuum Sterilizer E
2 Pulsation Vacuum Sterilizer
Arranging Packing
1WS215 One-hundred-grade Purified D
3 Room
Dry Heat Sterilizer
Clean Material
4 3MG208 Pulsation Vacuum Sterilizer D
Storage
5 Pulsation Vacuum Sterilizer D
Moist Heat
3WS214 One-hundred-grade Purified
6 Sterilization Room D
Dry Heat Sterilizer
Liquor Preparation
7 1MP210 Pulsation Vacuum Sterilizer D
Room
Room Temperature
8 4MG208 Pulsation Vacuum Sterilizer D
Storage Room
9 3E224/2 Sterile Filtration Pulsation Vacuum Sterilizer D

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No. Room No. Room Name Main Equipment Grade
Room

10 Pulsation Vacuum Sterilizer
Cleaning & Sterilizing One-hundred-grade Purified
2CC06 D
11 Room Double (single) Door
Sterilizer
12 1T08 Disinfection Pulsation Vacuum Sterilizer D
13 2T08 Disinfection Pulsation Vacuum Sterilizer D
14 4T03 Disinfection Pulsation Vacuum Sterilizer D
15 LC03 Laundry Room Pulsation Vacuum Sterilizer D
One-hundred-grade Purified
Primary Washing
16 1FD09 Double (single) Door D
Room
Sterilizer
Grade C Washing &
17 2FD23 Pulsation Vacuum Sterilizer C
Sterilizing Room
Grade C Washing &
18 1FD23 Pulsation Vacuum Sterilizer C
Sterilizing Room
Washing & Sterilizing
19 2FD42 Pulsation Vacuum Sterilizer D
of Aluminium Caps
Washing & Sterilizing
20 1FD42 Pulsation Vacuum Sterilizer D
of Aluminium Caps
Preparation &
21 1MD03 Pulsation Vacuum Sterilizer D
Sterilization Room
22 Back out &
1MG212 Pulsation Vacuum Sterilizer D
Inactivation Room
Pulsation Vacuum
23 1T08 Disinfection D
Sterilization Cabinet
Pulsation Vacuum
24 2T08 Disinfection D
Sterilization Cabinet
Pulsation Vacuum
25 4T03 Buffer Room D
Sterilization Cabinet
26 Grade C Washing &
1FL23 Pulsation Vacuum Sterilizer C
Sterilizing Room
27 Grade C Washing &
2FL23 Pulsation Vacuum Sterilizer C
Sterilizing Room
28 Washing & Sterilizing
1FL42 Pulsation Vacuum Sterilizer D
of Aluminium Caps
29 Washing & Sterilizing
2FL42 Pulsation Vacuum Sterilizer D
of Aluminium Caps
30 Grade C Washing &
1FD23 Pulsation Vacuum Sterilizer C
Sterilizing Room
31 Grade C Washing &
2FD23 Pulsation Vacuum Sterilizer C
Sterilizing Room

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No. Room No. Room Name Main Equipment Grade

32 Washing & Sterilizing
1FD42 Pulsation Vacuum Sterilizer D
of Aluminium Caps
33 Washing & Sterilizing
2FD42 Pulsation Vacuum Sterilizer D
of Aluminium Caps
34 Primary Washing Purified Double (single)
1FL09 D
Room Door Hot Heat Sterilizer

Table 13 Equipment List of QC Lab
Room
No. Room Name Main Equipment Grade
No.
Ultralow Temperature Refrigerator
1 Refrigerator Control
1QC303 (-20■)
Room Area
2 Drug Refrigerator■2-8■
3 Emergency Shower Device for Eyes
4 High Speed Freezing Centrifuge
5 Hitachi CR21G Super Centrifuge
6 FOSS Kjeldahl Apparatus
Physicochemical

Control
7 2QC301 Experiment Advanced 3250 Osmometer
Area
8 Area Herchces Table Centrifuge
Horizontal Electrophoresis
9
Instrument
Electric-Heated Thermostatic Water
10
Bath
Shimadzu UV-2600 UV-vis
11 Instrument Control
2QC302 spectrophotometer
Room Area
12 FOSS Temperature Control Digester
13 Automatic Potentiometric Titrator
14 pH Meter
Precision desk type pH and
Control
2QC303 Titration Room Conductivity double-parameter
15 Area
measuring instrument for
pharmaceutical water
16 Multi-parameter tester
17 Automatic optical rotator
Analytical
18 Mettler melting point instrument Control
2QC304 Instrument
GE AKTA Purifier Purification Area
19 Room
System
20 Sartorius Electronic Balance
Control
2QC305 Balance Room Digital display electronic tearing
21 Area
and pulling instrument

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Room
No. Room Name Main Equipment Grade
No.
Medical packaging performance
22
tester
High
Temperature Control
23 2QC307 Electrothermal blowing dry box
Instrument Area
Room
24 High Chamber Electric Furnace
Temperature Control
25 2QC307 Electric vacuum drying box
Instrument Area
26 Room Air Dry Oven

Analytical GE Sievers900 Total Organic
27 Control
2QC308 Instrument Carbon Analyser
Area
28 Room 3 Particles Analysis Instruments
Cleaning & MJ-series High-Pressure Steam
29 Control
2QC310 Sterilization Sterilizer
Area
30 Centre ZGDA-1.0D dry heat Sterilizer
Shimadzu LC20A High
31 Analytical Performance Liquid
Control
2QC312 Instrument Chromatography
Area
Room 1 Shimadzu GC-2014C Gas
32
Chromatograph
Shimadzu IRAffinity-1s Infrared
33
Analytical Spectroscopy
Control
2QC313 Instrument Sartorius CP225D Electronic
34 Area
Room 2 Balance
35 AVQ-2100 Moisture Meter
Control
36 1QC403 Laboratory 1 CTHI-250B incubator
Area
37 Washing & XG1.D Pulsation Vacuum Sterilizer
2QC414 Sterilizing D
38 ZGDB-1.2D dry heat Sterilizer
Centre
BCM-1600A Biological Clean
39
Bench

3QC406 Sterile Room B
LIII301C Dust Particle Online
40
Monitoring System
BCM-1600ABiological Clean
41
Bench
3QC409 Sterile Room B
LIII301C Dust Particle Online
42
Monitoring System
XLS-III Temperature and Humidity
43 Control
1QC404 Laboratory 2 Test Chamber
Area
44 CST4001 Automatic Temperature

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Room
No. Room Name Main Equipment Grade
No.
Test System
Control
45 1QC405 Laboratory 3 Mettler AL104 Electronic Balance
Area
BCM-1600A/BLB-1600 Biological
46
Non-Sterile Clean Bench
4QC416 C
47 Room CTHI-250B incubator
48 CTHI-250B incubator
49 BSC-1600 II A2 Biosafety Cabinet
Non-Sterile
4QC418 Thermo Fisher 3111 Carbon Dioxide C
50 Room
Incubator
Olympus CX23 Biological
51
Microscope
Non-Sterile
4QC420 BioMerieux VITEK 2 COMPACT C
Room
52 Bacterial Identification And Drug
Sensitive Tester
53 BSC-1600 II A2 Biosafety Cabinet
54 CTHI-250B incubator
Non-Sterile
4QC422 Ultralow temperature storage box (- C
55 Room
20■)
56 HYC-610 drug storage box
Olympus CX23 Biological
57
Microscope
Ultralow Temperature Freeze■-
58 Non-Sterile
4QC424 80■■■ C
Room
Thermo Fisher 3111 Carbon Dioxide
59
Incubator
60 BSC-1600 II A2 Biosafety Cabinet
61 1D4203 Sampling Room BSC-1600 II A2 Biosafety Cabinet C

Table 14 Equipment List of QA Department
No. Room No. Room Name Main Equipment Grade
1 Ultralow Temperature Freezer Control Area
Seed Bank
2 1GH103 Ultralow Temperature Freezer Control Area
No.1
3 Ultralow Temperature Freezer Control Area
4 Virus Seed Ultralow Temperature Freezer Control Area
3A205
5 library Ultralow Temperature Freezer Control Area

Table 15 Equipment List of Public Engineering System

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Room
No. Room Name Main Equipment Grade
No.
1 Phase II Water Purified Water Machine
2 3B4104 Production Multi-Effect Water Distillator General Area
3 Room Pure Vapour Generator
Phase II
4 3B4103 Freezing Oil - free Rotary Air Compressor General Area
Station
5 Phase III Purified Water Machine
Water
6 3U101 Multi-Effect Water Distillator General Area

Production
7 Room Pure Vapour Generator
Phase III
8 — Freezing Oil - free Rotary Air Compressor General Area
Station
9 Air Handling Unit JK2-MGa
10 Air Handling Unit JK2-MGb
11 Air Handling Unit JK2-MGb■K■
12 Phase III Air- Air Handling Unit JK2-MGc
13 — Conditioner Air Handling Unit JK2-MGc (K) General Area
14 Room Air Handling Unit JK2-Ea
15 Air Handling Unit JK2-Ec
16 Air Handling Unit JK2-WSa
17 Air Handling Unit JK2-MGd
18 Air- Air Handling Unit JK4-1
Conditioner
19 Air Handling Unit JK4-2
Room of
20 Air Handling Unit KX4-1 Outdoor
— Scientific
Environment
Research &
21 Quality Air Handling Unit KX4-2
Control
22 Air Handling Unit JK-1FLa
23 Air Handling Unit JK-1FLb
Air-
24 Air Handling Unit JK-1FLc
Conditioner
25 Room of Air Handling Unit JK-1FLd
3B3201 General Area
26 Phase II Fill Air Handling Unit JK-2FLa
& Finish
27 Air Handling Unit JK-2FLb
Centre
28 Air Handling Unit JK-2FLc
29 Air Handling Unit JK-1FDa

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Yuxi Walvax Biotechnology Co., Ltd
Room
No. Room Name Main Equipment Grade
No.
30 Air Handling Unit JK-1FDb
31 Air Handling Unit JK-1FDc
32 Air Handling Unit JK-1FDd
33 Air Handling Unit JK-2FDa
34 Air Handling Unit JK-2FDb
35 Air Handling Unit JK-2FDc
36 Air Handling Unit JK-1MX
37 Air Handling Unit JK-2MX
38 Air Handling Unit JK-1MD
39 Air Handling Unit JK-2MD
40 Air Handling Unit JK-LC
41 Air Handling Unit JK-SY
42 Air Handling Unit EK-1
43 Air Handling Unit EK-2
44 Air Handling Unit AHU1-1
45 Air Handling Unit AHU1-6
46 Air Handling Unit AHU1-7 Lyophilisation
Sub-packing
47 Air Handling Unit AHU1-16-1 Line 1
48 Air Handling Unit AHU1-16-4
49 Air Handling Unit K1-1
50 Air Handling Unit AHU1-2
51 Air Handling Unit AHU1-8
52 Air Handling Unit AHU1-9 Lyophilisation
International
Sub-packing
53 Vaccine Air Handling Unit AHU1-16-2
- Line 2
Formulation
54 Air Handling Unit AHU1-16-5
Centre
55 Air Handling Unit K2-1
56 Air Handling Unit AHU1-4
57 Air Handling Unit AHU1-11
58 Air Handling Unit AHU1-12 Injection Line
59 Air Handling Unit AHU1-16-3
60 Air Handling Unit AHU1-16-6
61 Air Handling Unit AHU1-15 Washing area
Clean laundry
62 Air Handling Unit AHU1-14
area
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Yuxi Walvax Biotechnology Co., Ltd
Room
No. Room Name Main Equipment Grade
No.
63 Air Handling Unit AHU1-5 Excipient
preparation
64 Air Handling Unit AHU1-13

area
Diluent
65 Air Handling Unit AHU1-16-10 Sterilization & drying area
66 Air Handling Unit AHU1-16-7 Associated zone (Grade D
67 Air Handling Unit AHU1-16-8 zone for sub-
68 Air Handling Unit AHU1-16-9 packing)
69 Air Handling Unit AHU1-17
70 Air Handling Unit LK1-1
Packaging area
71 Air Handling Unit LK1-2
72 Air Handling Unit LK1-3
73 Air Handling Unit K3-1 Warehouse

Table 16 Equipment List of Fill & Finish Workshop Room

No.	Room Name	Main Equipment	Grade
1	QCL—400 Ultrasonic Vial Cleaner		
2	Washing & KJQS-5ES Automatic Wet Way Air-Blast		
3	1FL11 Sterilizing linkage Rubber Plug Cleaner		D
4	Line MGS-620 Laminar Flow Dry Heat		
5	Sterilization Tunnel		
6	4 Aseptic Filling and BKYG-400 Sterile Liquid Filling		
7	1FL34 Lyophilizing		B
8	Machine Room		
9	6 1FL52 Capping Room GKZX-400 Capping Machine		C+A
10	7 QCL—400 Ultrasonic Vial Cleaner		
11	Cleaning & KJQS-5ES Automatic Wet Way Air-Blast		
12	8 Sterilizing linkage Rubber Plug Cleaner		D
13	2FL11		
14	Line MGS-620 Laminar Flow Dry Heat		
15	Sterilization Tunnel		
16	Aseptic Filling BKYG-400 Sterile Liquid Filling		
17	10 2FL34		B
18	Room Machine		
19	11 2FL52 Capping Room GKZX-400 Capping Machine		C+A
20	12 QCL—400 Ultrasonic Vial Cleaner		
21	Cleaning & KJQS-5ES Automatic Wet Way Air-Blast		
22	13 1FD11 Sterilizing linkage Rubber Plug Cleaner		D
23	Line MGS-620 Laminar Flow Dry Heat		
24	14 Sterilization Tunnel		
25	38		
26	Yuxi Walvax Biotechnology Co., Ltd		
27	Room		
28	No. Room Name Main Equipment		Grade
29	No.		
30	BKYG-400 Sterile Liquid Filling		
31	15 Aseptic Filling		
32	1FD34 Machine		B
33	Room		
34	16 Laminar Flow Cart		
35	17 1FD52 Capping Room GKZX-400 Capping Machine		C+A
36	18 QCL—400 Ultrasonic Vial Cleaner		
37	Cleaning & KJQS-5ES Automatic Wet Way Air-Blast		
38	19 2FD12 Sterilizing linkage Rubber Plug Cleaner		D
39	Line MGS-620 Laminar Flow Dry Heat		
40	20 Sterilization Tunnel		
41	BKYG-400 Sterile Liquid Filling		
42	21 Aseptic Filling		
43	2FD34 Machine		B
44	Room		
45	22 Laminar Flow Cart		
46	23 2FL52 Capping Room GKZX-400Capping Machine		C+A
47	Lyophilizing		
48	24 3B3111 Vacuum Freeze Drier		E
49	Room		
50	25 Micro Character Ink-Jet Printer		
51	Visual Inspection		
52	26 SY07 Compound Transfer Labeller System		E
53	Room		4
54	27 Light inspector		
55	28 Micro Character Ink-Jet Printer		
56	Visual Inspection		
57	29 3B3115 Small Round Bottle Transfer System		E
58	Room		3
59	30 Light inspector		
60	31 KHV180T Automatic Boxing Machine		
61	32 3B3108 Packing Room 01 Automatic Sellotape Case Sealer		E

Table 17 Equipment List of International Vaccine Formulation Centre

Room
No. Room Name Main Equipment Grade
No.
washing & Ultrasonic bottle washing machine
1FD1111 sterilization Rubber plug cleaner D
linkage room Laminar dry heat sterilization tunnel

Aseptic liquid filling machine
Aseptic freeze-
2FD1110 B
dried filling room Freeze-drying machine feed-in and
discharge system
Freeze-drying General
3B110 Freeze-drying machine
machine room Area
4FD1109 Capping room Aluminum cover rolling machine B
Isolated conveying system
3B101 D
Fully automatic light inspector
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Receipt /light Automatic leak detector
inspection/
Automatic labeling machine
labelling room 1
washing & Ultrasonic bottle washing machine
1FD2111 sterilization Rubber plug cleaner D
linkage room Laminar dry heat sterilization tunnel

Aseptic liquid filling machine
Aseptic freeze-
2FD2110 B
dried filling room Freeze-drying machine feed-in and
discharge system
Freeze-drying General
3B112 Freeze-drying machine
machine room Area

4FD2109 Capping room Aluminum cover rolling machine B

Diluent
3B122 sterilization Pulsatile vacuum sterilizer D
chamber

Isolated conveying system

Receipt /light Fully automatic light inspector
3B104 inspection/ D
labelling room 2 Automatic leak detector

Automatic labeling machine

Outer package Composite packing line (2 vials
PK101 CNC
room package)

4.2.2 Cleaning and Sanitation

Yuxi Walvax has established Standard Management Procedure for Cleaning and Disinfection (No. WSYX-RVM-SMP-0009-0001). It is specified the cleaning and disinfection methods for production workshop, process equipment, clean areas and inspection equipment in quality control laboratory. The company has established SOP of Clean Air Conditioning System Maintenance (Phase III)(Code: WSYX-EO-SOP-00-03-0025), SOP of Use and Maintenance of OZS Ozone Generators (Phase III) (Code: WSYX-EO-SOP-00-04-0010), SOP of Use and Maintenance of Purified Water/Water for Injection Distribution System (Phase III) (Code: WSYX-EO-SOP-00-03-0028), which specify the cleaning and disinfection methods for Air Conditioning and Water System. The company has developed SOP of Cleaning, disinfection and maintenance of BIOQUELLZ-2 Hydrogen Peroxide Generator (Code: WSYX-FP-SOP-00-04-0043), which specifies the sterilization 40
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method and frequency of vaporized hydrogen peroxide in critical operating room of Fill & Finish Workshop.

4.2.3 Description of planned preventive maintenance programs for equipment and the recording system

Preventive Maintenance System

The company has established Procedures of Equipment Preventive Maintenance (code: WSYX-PM-SMP-00-04-0012), to guide the management of equipment preventive maintenance. According to the demands of equipment preventive maintenance, at the end of every year, each of the equipment management department shall draft the equipment preventive maintenance plan for the coming year, (1) it must be carried out a review on the equipment operation and maintenance situation of the previous year and analyse the reason of the maintenance have not completed as planned; (2) According to the level assessment before the purchase of equipment, the critical evaluation of components during equipment verification, statistically analyse the breakdown, maintenance and maintenance of the equipment in each department last year, review and analyse whether the frequency and measures of key maintenance components last year are reasonable, and whether the preventive maintenance frequency and measures last year are reasonable or not. According to the equipment

preventive maintenance plan, during daily production, the equipment using department / management department shall establish "Equipment Maintenance Table", record the anomalies, failures, and maintenance of each device during its use. The information should be including equipment code, type, manufacturer, place site, failure date, description of failure, Maintenance and maintenance measures and etc. We review the table at the end of every year as reference for the equipment preventive maintenance plan for the next year.

Maintenance of daily production operation

Equipment preventive maintenance is divided into primary level maintenance and secondary level maintenance. Primary maintenance is a routine maintenance conducted during the daily use of equipment, and secondary maintenance is a professional maintenance by the Engineers.

Primary equipment preventive maintenance is carried out during the routine operation. The equipment using departments organize and implement the equipment maintenance according to the annual preventive maintenance plan.

As for the secondary preventive maintenance of equipment, the equipment using departments should apply for maintenance to Engineering Security Department 1 week before the maintenance according to the production plan and maintenance plan, According to the maintenance content and maintenance

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requirements, the Engineering Security Department shall determine the maintenance personnel and feedback them to the Production Management Department, which shall coordinate the relevant public systems and sign opinions according to the maintenance contents and maintenance requirements, and then feedback to the device using department. The device using department organize and cooperate with the professional engineering staff to complete the maintenance. In the process of secondary level equipment maintenance, if there are multiple projects, which need to be carried out at the same time, equipment should be maintained according to equipment maintenance class and critical equipment parts assessment, in the order of class A→B→C.

After the equipment maintenance, fill the Equipment Preventive Maintenance Records, and fill the Equipment Maintenance Log Book by the using department at the same time, as reference for equipment preventive maintenance plan for the next year.

4.2.4 Qualification and calibration including the recording system arrangements for computerized systems validation

According to the Chinese GMP and WHO GMP requirements, as per the risk assessment, the company has formulated the Validation Master Plan (document code: WSYX – RVM – TS0006 - 0051), clarifying all the validation key elements like responsibilities, plant, facilities, equipment, analytical instruments, manufacturing process and so on to ensure the consistency and rationality of the validation method. Computer system validation strategy, as shown in the following table:

Validation Strategy for Computerized System

Items Content

Design qualification (DQ) for Computerized System should meet the Design

requirements of GMP and production process. Qualification should at least

Qualification contain URS and design documents.

The installation qualification should contain the following items: criteria

Installation operation files, general drawing of the control system, the manuals for Qualification hardware and software, the configuration lists of hardware and software, input/output (I/O) lists and continuity check, test of environment and utilities.

Operation qualification should contain the following items: system security

Operation test, interface test, alarm and interlock function test, functional test and

Qualification disaster recovery test.

Performance qualification (PQ) is to confirm the validity and stability during

Performance the system running. It should be tested under the normal working

Qualification environment. Test items are determined by the overall expected operation effect (for example, testing all the characteristics of product quality).

5. Documentation

Arrangements for the preparation, revision and distribution of documentation

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The Company has enacted the Document Management Control Procedures (document code: WSYX-P-07-0008) to regulate the document drafting, revision, approval, distribution, control and archiving, so as to ensure that the documents are under control in different life cycles and to guarantee that the document management is consistent with the corresponding regulations and procedure requirements.

Document preparation/deletion:

Preparation of documents (policies, guidance documents, regulations and records) includes drafting of new documents and revision of existing documents. In the case of document addition, alteration or deletion, the preparation/deletion personnel shall fill in the Approval Form for Quality Document ■Addition ■Alteration ■Deletion, stating the reason and basis for document addition, alteration or deletion and other necessary information descriptions.

Upon completion of document preparation, the preparation personnel shall attach the draft printing on both sides in black and white to the Approval Form for Quality Document to the related personnel for review and approval.

Record preparation and approval process:

Each activity related to the GMP management shall have records to ensure that the product production, quality control, quality assurance and other activities are traceable. When record is designed, sufficient blank shall be left for filling out data and information.

Document review:

Document review includes format review and content review: a. format review shall be carried out by Quality Assurance Department, which shall check the corresponding contents (e.g. document number, version number, font, font size and space between lines, etc.) according to the document standard format specified in the SOP for Document Preparation, and sign the review basis and comments in the column "Review Comments about Drug Administration Regulations from Quality Assurance Department" in the Approval Form for Quality Document; b. content review shall be completed according to the approval process specified in "document preparation", where the document contents shall be concise and complete, accurate and rigorous in wording, clear, comprehensible and logical in expression for the regulatory, technological and management purposes, so as to ensure the document operability and regulatory compliance; the different managers shall review from different points of view according to the review sequence, and fill out the review basis and comments in the corresponding columns of the Approval Form for Quality Document. If any document to be reviewed involves any legal requirement (Labor Law or Civil Law), the column for joint review comments shall include the legal counsel's opinions. If any document to be jointly reviewed or reviewed must be kept confidential, the SOP for Confidentiality of Technological

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Documents shall be followed.

Document approval:

Document approval shall be completed by the approver specified in "document preparation". Documents must be approved before use, and the approved documents can be used for training. The blank column for approval in the Approval Form for Quality Document shall be crossed out by the document submitter but signed and dated.

Document distribution:

Distribution of policies, guidance documents or regulations

Documents shall be released among which job description shall be jointly released by Quality Assurance Department and Human Resources Department, while the remaining documents shall be released by Quality Assurance Department. Document system documents shall be confidential, the original of which shall be affixed with the cross-page seal of "Quality Assurance Department" before color printing and distribution. Each page of documents distributed to each department shall be affixed with the "control seal" for reference and use, in the middle for documents or on the top left corner for records, and "special seal for quality document distribution" shall be affixed to the top right corner of the home page of documents. Documents shall be released after preparation of distribution number. Before document distribution, the distribution range and number of documents shall be specified by Quality Assurance Department according to the document's coverage range. Quality Assurance Department shall prepare the distribution number with unified printing and distribution, and the recipients shall sign on the Quality Document Distribution Record Form.

Distribution of documents under other circumstances:

After document distribution, if dissenting on the distribution range or the number of documents distributed, the user department shall complete the Quality Document Distribution Range Application Approval Form, which shall be printed and distributed uniformly by Quality Assurance Department after approval from leaders with specific duties (Director of Production Management or Director of Quality Management). Quality Assurance Department shall determine the distribution range according to the document's coverage range or the approved Quality Document Distribution Range Application Approval Form, and distribute the documents after preparation of document number. The recipients shall sign on the Quality Document Distribution Record.

Document training:

Upon receipt of the documents, each department head shall organize the related personnel to learn, and complete the Training & Learning Record. Document learning shall be completed before the

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documents become effective, and document training shall follow the Training Management Control Procedures.

Document effectiveness:

Documents shall become effective within no more than 30 working days after official release, so as to ensure that each use site can obtain the applicable documents in current edition, and that the implementers can have sufficient time to learn and understand the documents before document implementations. The documents shall be implemented as of the effective date.

6. Production

6.1 Type of Product

Table 18 Manufactured products

Registration

No. Product name Dosage forms
approval date

Haemophilus influenzae Type b
1 Liquid for injection Apr. 2, 2007
Conjugate Vaccine (vial)

Haemophilus influenzae Type b
2 Liquid for injection June. 17, 2009
Conjugate Vaccine (syringe)

Lyophilization powder
Meningococcal Group A,C Bivalent
3 accompany with diluent for Jan. 4, 2009

Polysaccharide Conjugate Vaccine
injection

Group A and C Meningococcal Lyophilization powder
4 accompany with diluent for Mar. 23, 2012
Polysaccharide Vaccine injection

Group ACYW135 Meningococcal Lyophilization powder
5 accompany with diluent for Mar. 23, 2012
Polysaccharide Vaccine
injection

Diphtheria, Tetanus and Acellular
6 Pertussis Combined Vaccine, Liquid for injection July. 27, 2015
Adsorbed

23-valent Pneumococcal
7 Liquid for injection Mar. 22, 2017
Polysaccharide Vaccine

13-valent Pneumococcal Suspension for IM
8 Dec.30, 2019
Polysaccharide Conjugate Vaccine injection

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6.2 Process Validation

6.2.1 Description of general policy for process validation

According to the requirements of China GMP, WHO GMP and risk evaluation, Yuxi Walvax has established Validation Master Plan (Code: WSYX-RVM-TS-0006-0051), which clarifies the validation responsibility and all the key elements of validation for the factory buildings, facilities, equipment, test instruments and production process, to guarantee consistency and rationality of validation method.

The company organizes the validation and production in strict accordance with the regulations issued by China Food and Drug Administration, and has a specialized Risk Validation Management department, which is under Quality Management Department, fully responsible for organizing the validation tasks of the company to ensure the continuously stable product quality. The Risk Validation Management Department provides the engineering equipment validation management and the process validation management according to the validation category. Process validation management mainly involves the process validation of the products manufactured by the company.

Process validation includes prospective validation, synchronous validation and retrospective validation by type, and the standards and contents for different types of process validation have been identified in the overall validation plan. To ensure that the products are always within the acceptable range and in the controllable state, and to effectively control the deviations during the production, the company shall use the method of risk assessment according to the product intended use and the registration standard to evaluate the effects of the process parameters during the entire life cycle of manufacturing of the products, so as to determine the process validation protocol and to organize the implementation in strict accordance with the validation protocol. Upon formation of the validation results, the company shall carry out the risk assessment and analysis of the validation results to judge whether the product process parameters and the product quality are in acceptable and controllable state, forming the validation report before implementation.

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Figure 11 Flow Chart of Carrying out Process Validation

6.2.2 Arrangements for reprocessing or reworking.

1)According to national approved drug standard, rejected intermediates and products which cannot be reprocessed, such as conjugate bulk and finish products should be clearly marked and stored separately in restricted areas. They should be destroyed under the supervision of local drug administration. Whatever action taken should be approved and recorded by authorized personnel.
2)According to national approved drug standard, the rejected intermediates, such as polysaccharide bulk could be reprocessed. It is only permitted after a series of evaluation and approving by workshop manager, QA manager, manufacture director and quality director. If the specifications are met and if it is done in accordance with a defined and authorized procedure, the intermediates could be reprocessed. Record should be kept of the reprocessing.

6.3 Material management and warehousing

6.3.1 Arrangements for the handling of starting materials, packaging materials, bulk and finished products, including sampling, quarantine, release and storage.

To guarantee intended use of products, the company has established a comprehensive and scientific material management and storage system and has established the Material Management Control Procedures (Document No.:WSYX-P-05-0005) to guide and control the material management and storage. All the purchased materials are from the qualified suppliers approved by the Director of Quality Management. Primary seeds are purchased from China Bacterial Culture Collection Center of National Institutes for Food and Drug Control, with clear record, history and source. The company has established Testing SOPs for each material to test whether the material quality indicators meet the

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production quality requirements. The company uses the means of risk assessment to conduct evaluation and hierarchical management of the materials.
The purchased materials shall be managed by zone, identification and quality state, and those with special storage requirements (e.g. products with special temperature & humidity requirements, highly toxic products and easily manufactured narcotic drugs) shall be stored in special storage zones. Quality Management Department shall take samples for testing by batch, review the materials whether acceptable, and issue the certificate of approval if approved, permitting the materials to be put into production.

Figure 12 Flow Chart of Material Management System

6.3.2 Arrangements for the handling of rejected materials and products.

In strict accordance with the GMP (revised edition in 2010), the company has enacted the Rejected Product Management Procedures (Document No.: WSYX-QA-SMP-0009-0031) to standardize the handling of rejected products, so as to prevent the rejected products from being used in production. After the material, intermediate product and finished product are rejected as nonconforming product

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by the deviation / OOS investigation, the quality assurance department shall issue a certification of unqualification according to the minimum packing /container quantity. The warehouse management personnel shall transfer them to the unqualified warehouse and store them in isolation, and hang the red "unqualified" status identification and storage card.

Figure 13 Rejected Product Management Procedures

7. Quality Control:

7.1 Description of the quality control system and of the activities of the Quality Control Department.

QC Laboratory is a department subordinate to Quality Management Department, responsible for providing objective and scientific testing data for the review and release of raw and auxiliary materials, packaging materials, intermediate products, semi-finished products and finished products involved in the quality management system.

QC Laboratory, with the total building area of about 3905m², has such functions as product physiochemical testing, material testing, standard test sample preparation and management as well as microbiological and biochemical testing. According to the GMP requirements, laboratories for microbial limit test and sterility test shall be separated and provided with independent air-conditioning purification system to prevent contamination and cross contamination. QC Laboratory is divided into Microbiological and Biochemical Laboratory, Product Physiochemical Laboratory, Material Physiochemical Laboratory, Environmental Monitoring Room, Validation Workshop, Standard Test Sample Management Room and Quality Standard Research Room based on the testing functional requirements.

The personnel at all levels of the quality control laboratory shall be trained for national relevant laws and regulations and professional knowledge according to the regulations, and shall be certified to work after passing examinations. Job description shall be formulated for personnel at all levels according to the position business requirements, and the number of personnel is appropriate to the

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volume of positions business, so as to ensure accurate and effective completion of the work within the prescribed time.

Quality control laboratory has formulated the Quality Control Laboratory Management Procedures (document coding: WSYX - QC - SMP - 0009 – 0021), for the guidance and control of the management of the entire experimental system.

Figure 14 Quality control laboratory inspection work flow

The quality control laboratory shall promptly start the investigation process and take corresponding corrective and preventive measures in case of OOS and OOT of the test results.

7.2 Procedures for the release of the finished products.

Product release procedures of Manufacturer

Yuxi Walvax established Standard Operation procedure for Sampling (Code: WSYX-QS-SOP-00-09-0003) and Standard Management Procedure for Review and Release of Finished Product (Code: WSYX-QA-SMP-00-09-0008) to standardize sampling and audit release of finished product. After the finished products are produced, the quality supervisor samples the finished products and transfers them to the Quality Control Laboratory (QC) for inspection. The Quality Assurance Department shall review the information on the batch production record (instruction information, batch record review status, production environment confirmation, material balance) and batch testing record (batch record review status, status of inspection (sampling) application, inspection record, test result) and issue a

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Certificate of Analysis. For products that meet the specification, apply for batch release, according to the batch release conclusion, the authorized quality personnel decide release or not.

Yuxi Walvax has established the " Standard Management Procedure for Reserve Samples" (code: WSYX-QA-SMP-00-09-0002) to stipulate that the reserve samples volume of the finished products is 2 times the total inspection volume.

Product release procedures of NRA

Yuxi Walvax's products have passed the inspection and been packaged. The batch release application is submitted by the batch release administrator into the management system for the batch issuance of biological products. Subsequently, the NRA authorizes the staff at the provincial drug administration to sample each batch of products and store the sealed samples in the refrigerator (the storage temperature is 2-8 °C). Afterwards, the samples are sent to the National Institutes for Food and Drug Control for testing. If the product meets the requirements, the National Institutes for Food and Drug Control will issue a release certificate for the batch and mail it to the company.

Figure 15 FPP release flow chart

8. Distribution, complaints, and product recall

8.1 Arrangements and recording system for distribution.

After obtaining the "Batch Release Certificate of Biological Products", Yuxi Walvax will fill in the "Inspection Record for the Release of Finished Product" in accordance with the "Procedures for the Inspection and Release of Finished Products" (document code: WSYX-QA-SMP-00-09-0008). After the "Release Order of Finished Products" is approved and signed by the authorized quality manager, the products can be sold on the market.

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After Yuxi Walvax receives the purchase order, the distribution center of finished products is responsible for the delivery of the finished products, which are managed according to the "Procedure for the Receiving, Storage and Distributing of Finished Products" (document code: WSYX-DC-SOP-00-05-0003). The product is placed in the cold storage tank made of a non-fluorinated polyurethane special material, which has good thermal insulation performance. The automatic temperature recorder or 3M temperature label is placed with the goods to ensure the traceability of the temperature during cold chain shipment. The mode of transport is by refrigerated trucks, and if necessary, the products kept in the cold storage tanks are then transferred by air.

Yuxi Walvax has established a shipping record for the delivery of finished products, so that the shipment status of each batch of products, including product name, dosage form, specification, batch number, quantity, receiving unit, receiving address, contact information, delivery date, transportation mode, sales invoice number, can be traced.

8.2 Arrangements for the handling of complaints and product recalls.

Yuxi Walvax has set up special posts and personnel to handle product complaints, to review and analyze the complaints as well as their investigation and handling, and to implement and evaluate relevant corrective and preventive measures. In order to standardize the procedures for handling product complaints and adverse reactions, Yuxi Walvax established the "Management Regulations for Quality Complaint" (code: WSYX-QA-SMP-00-09-0038).

A recalling group has been founded with specific responsibilities in Yuxi Walvax, in which the General Manager is group leader and all department heads are group members. Established the "Management and Control Procedure for Product Recall and Distribution" (code: WSYX-P-11-0013) to clarify that product recalls are initiated immediately after product quality incidents or after serious adverse events have been identified to be related to product quality. Relevant recall procedures shall be carried out according to recalling procedure of Class 1, 2 and 3 specified in Management and Control Procedure for Product Recall and Distribution.

After Yuxi Walvax recalls the product, it will be sealed and stored in isolation. During the recall, relevant records shall be filled, including client-informing record, client feedback record, recalled product receiving record, quantity balance relation of products to be recalled and real recalled products, etc. The recalled products should be isolated with obvious identity. Finally, the destruction process is carried out under the supervision of the drug administration.

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Figure 16 Recall control procedures

9. Self inspection

9.1 Short description of the self-inspection system

Yuxi Walvax has established Self-inspection Management Control Procedures (Code: WSYX-P-12-0014), to guide and regulate the carrying out of self-inspection. The Quality Operation Department appoints professional persons with correspondent knowledge and skills to carry out at least twice self-inspections every year, and is responsible for annual self-inspection plan and self-inspection scheme. Through inspecting persons, plant facilities, equipment, materials and products, confirmation and validation, document management, production management, quality control and quality assurance, products distribution and recalls, etc., the Quality Operation Department puts forward necessary rectifying and preventive measures, and finally self-inspection report and record are formed, jointly approved by manufacture head and quality head.

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Yuxi Walvax divides self-inspection into company-level inspection and special inspection.

Table 19 Company-level Self-inspection protocol

Items/Procedures Content

Frequency of self-

Once every six months

inspection

1 Objectives of self-inspection

2 Basis for self-inspection

3 Content of self-inspection: Requirements of GMP, product content and requirements involved in Chinese Pharmacopoeia, problems of product quality

Self-inspection

review of the previous year, defects of the previous self-inspection, and defects

protocol

of outer-inspection of current year.

4 Members and responsibilities of self-inspection subgroups

5 Qualification information of self-inspectors

6 Inspection content of each self-inspection group.

1 The first meeting of self-inspection (training of self-inspection plan)

2 Self-inspection groups design detailed self-inspection plan according to

Carrying out of content of self-inspection designing, and implement according to the plan.

self-inspection 3 The last meeting of self-inspection (defect in self-inspection, Corrective and

Preventive Action (CAPA) rectifying discussion)

4 Tracking and confirmation of defects (Rectifying information table)

1 Summary of outer quality of this self-inspection

2 Distribution of defect corresponding to departments

Report of self-

3 Distribution of defects corresponding to chapter content in GMP

inspection

4 Comparison and analysis of all self-inspections

5 Defects and rectifying of current self-inspection

Table 20 Procedures and Contents of Special Self-inspection

Items/Procedures Contents

Conduct special self-inspection for the following situations

- 1 Renewal of regulation/NRA Supervision policy
- Frequency of self-
- 2 Before qualification of NRA or other authorities inspection
- 3 Occurrence of severe quality events
- 4 Change of management system
- 1 Objective of Self-inspection
- 2 Basis of self-inspection
- Plan for self- 3 Content of self-inspection: list according to special content
- inspection 4 Subgroup members and responsibilities of self-inspection group
- 5 Qualification information of self-inspectors
- 6 Inspection content of each self-inspection group
- 1 The first meeting of self-inspection (Training of self-inspection plan, confirmation of self-inspection members)
- Carrying out of
- 2 Self-inspection groups design detailed self-inspection plan according to self-inspection
- content of self-inspection designing, and implement according to the plan.
- 3 The last meeting of self-inspection (defect in self-inspection and CAPA

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 Items/Procedures Contents
 rectifying discussion)

- 4 Tracking and confirmation of defects (Rectifying information table)
- 1 Summary of outer quality of current self-inspection
- 2 Distribution of defect corresponding to departments
- Report of self-
- 3 Distribution of defects corresponding to chapter content in GMP inspection
- 4 Comparison and analysis of all self-inspections
- 5 Defects and rectifying of current self-inspection

Appendixes

Appendix 1-Manufacturing license-Yuxi Walvax-CN&EN
 Appendix 2-GMP Certificates
 Appendix 3-1- Curricula vitae for Key personnel
 Appendix 3-2-Number of employees in departments
 Appendix 4 - Layout of site and manufacturing areas, manufacturing flow chart
 Appendix 5-HVAC system schematics
 Appendix 6-Water system schematics
 Appendix 7-Equipment list
 Appendix 8-SOP for Product Recall and Distribution

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20°C increases to 85°C by 1-hour heat cycling, with 2T water at 14T/h flow rate
Heating medium 3 Bar Industrial Steam

Sampling

Sampling

(For 3F Pneumonia Sterile Area & Conjugation Area)

PW Tank

3WS309 Rough Wash Room 1MP311 Liquid Preparation Room 4PN1311 Ultrafiltration Room
Sampling

Q=16T/h, H=60m 3WS309 Rough Wash Room 3WS310 Washroom 3WS310 Washroom 3WS318CI PW Tank in P Station
1MP311 Liquid Preparation Room 4PN2311 Ultrafiltration Room

3WS311 Laundry

Sampling

Q=16T/h, H=60m

3PN2311 Ultrafiltration Room 2PN2310 Ultrafiltration Room

1WS310 Washroom 1WS310 Washroom 1/2WS309 Rough Wash Room 1/2WS309 Rough Wash Room 2WS310 Washroom 2WS310 Washroom 3PN1311 Ultrafiltration Room 2PN1310 Ultrafiltration Room
2WS311 Laundry
1WS311 Laundry

303# N Q 305#

■■■■■■■■ ■■■■■■■■ 304# ■■■

Experimental Animal Centre ■■■■■■■■ Dining Hall

Hazardous waste depots ■■■ ■■■■■■■■■■

Finished Product Distribution Centre

■■■■ ■■■■■■

B1 Floor■Underground Parking lot 307#

1st Floor: Association area & Air Conditioning Room ■■■■ ■■■■■■

B1 Floor■Underground Parking lot 306# ■■■ ■■■■ 1st Floor: Administrative area

■■■ ■■■■■■■■ ■■ ■■ 2nd Floor: Dining Hall

■■■

R

■■■ ■■■■ 1st Floor: Distribution area ■■■ ■■■ 1st Floor: Kitchen ■■■ ■■■ 3rd Floor: Standby Area

2nd Floor: Rabbits Feeding & Experimental Area

■■■■

M

■■■ ■■■ 2nd Floor: Dining Hall ■■■ ■■■ 4th Floor: Standby Area

■■■■ ■■ ■■/■■■■■■■■■■■

3rd Floor: Mice/Guinea pigs Feeding & Experimental Area

■■■ ■■■■ 2nd Floor: Finished Product Warehouse

■■■ ■■■■ 3rd Floor: Finished Product Warehouse

■■■ ■■■■

3rd Floor: Administrative area Comprehensive

■■■ ■■■ 5th Floor: Standby Area

■■■ ■■■ 6th Floor: Standby Area

Assembly Hall

Boiler Room ■■■ ■■/■■■■■■■■■■■ ■■ ■■■■ 4th Floor: Finished Product Warehouse ■■■ ■■■ 4th Floor: Dining Hall

Building ■■■ ■■■ 7th Floor: Standby Area

■■■■ ■■■■■■

B1 Floor■Underground Parking lot

23 22 ■■■■■■ 4th Floor: Mice/Guinea pigs Feeding & Experimental Area ■■■ ■■■■ 5th Floor■Finished Product

Warehouse ■■■ ■■■ 5th Floor: Standby Area ■■■ ■■■ 8th Floor: Standby Area

■■■ ■■■ 9th Floor: Standby Area

■■■ ■■■ 1st Floor: Assembly Hall

Fire pump station

■■■■

■■■ ■■■■■■ 1st Floor: Power Centre

■■■ ■■■■ 2nd Floor: Administrative area

204# 302# ■■■ ■■■■■■■■■■: 3rd Floor: Packing material warehouse & Air Conditioning Room

202#

K ■■■■■■■■

Logistic and Power Centre

K ■■■■■■■■

Logistics and Power Centre

J ■■■ ■■■■ 4th Floor: Administrative area

■■■ ■■■■■■■■■■ 5th Floor: Packing material warehouse & Air Conditioning Room

■■■ ■■■■■■■■ 6th Floor: Unqualified product warehouse

■■■ ■■■ 7th Floor: Warehouse:

■■■ ■■■■■■ 1st Floor: Power Centre ■■■ ■■■ 8th Floor■Standby Area

■■■■■■■■■■■ ■■ ■■■■■■■■■■ 2nd Floor: Air Conditioning Room & Warehouse ■■■■

Tetanus Vaccine Bulk Workshop

■■■ ■■■■■■■■■■ 1st Floor: Tetanus Vaccine Bulk Workshop

■■■ ■■■■ 2nd Floor: Administrative area

■■■ ■■■ 3rd Floor: Warehouse 301#

■■■ ■■■■ 4th Floor: Administrative area

203#

K ■■■■

Bacterial Vaccine Building

J

■■■■■■/■■■■■■■■■■■ ■■ ■■■■■■■■■■ 1st Floor: International Centre for Vaccines

Fill and Finish Centre/Pertussis Vaccine Bulk Workshop ■■■■ ■■■■■■ Interlayer of 1st Floor: Central

Control Room

■■■ Hib■■■■■■■■■■■■■■■■■■■ 2nd Floor: Bulk workshop of Haemophilus Influenza type b & Meningococcal Vaccine

201#

■■■■■■ L ■■ ■■■■■■ 1st Floor: Fill and Finish Centre

■■■■■ ■■■■■■ Interlayer of 1st Floor: Central Control Room

■■■ ■■■■■■■■■■ 2nd Floor: Pertussis Vaccine Bulk Workshop

■■■■■ ■■■■■■ Interlayer of 2nd Floor: Air Conditioning Room

■■■ ■■■■■■■■■■ 3rd Floor: Pneumococcal Vaccine Bulk Workshop

■■■■■ ■■■■■■ Interlayer of 3rd Floor: Air Conditioning Room

■■■ ■■■■■■■■■■ 4th Floor: Manufacturing workshop of Yuxi Jiuzhou Biological Co.,Ltd

Personnel Logistic Centre ■■■■■■ ■■■■■■ Interlayer of 2nd Floor: Air Conditioning Room ■■■■■■ ■■■■■■■■■■■■

Interlayer of 4th Floor: Air Conditioning Room of Yuxi Jiuzhou Biological Co.,Ltd

■■■ ■■■■ 3rd Floor: Archive room

■■■ ■■■■■■ 1st Floor: Personnel Logistic Hall

■■■ ■■■■ 2nd Floor: Administrative area

■■■ ■■■■ 3rd Floor: Administrative area

■■■■

E F

■■■ ■■■■■■■■QC■■■■

1st Floor: Warehouse of Yuxi Walvax, QC water production room

■■■ ■■■■■■■■■■■■■■■■

H 105#

104# 110# 2nd Floor: Quality Control Laboratory of Yuxi Jiuzhou Biological Co.,Ltd

■■■■■■■■ ■■■■■■

■■ ■■■■■■■■■■

3rd Floor: Quality control laboratory of Yuxi Walvax

■■ ■■■■■■■■■■

308#

■■■■■

G 106# ■■

Warehouse

I

Diphtheria vaccine bulk workshop Animal Breeding Centre

4th Floor■■Quality control laboratory of Yuxi Walvax

■■ ■■■■■■■■ Research and Quality Control Building ■■■■

■■ ■■■■■■■■ ■■ ■■■■■■

5th Floor■■Administrative area of Yuxi Walvax Power Distribution Centre ■■ ■■■■■■ 1st Floor:Warehouse of Yuxi Jiahe

1st Floor: Diphtheria vaccine bulk workshop 1st Floor: Animal Breeding Centre ■■ ■■■■■■ ■■ ■■■■

6th Floor■■Administrative area of Yuxi Walvax 1st Floor: Power Distribution Centre

103#

■■■■■■■■

Quality Control Administrative Building■■North■

101#

■■■■■■■■■■

C ■■ ■■■■■■ 1st Floor : Administrative area of Yuxi Walvax

■■ ■■■■■■■■■■ 2nd Floor:Quality control laboratory of Yuxi Zerun

■■ ■■■■■■■■■■ 3rd Floor■■Quality control laboratory of Yuxi Zerun

102#

■■■■■■■■■■

D

Innovative Cervical Cancer Vaccine Manufacturing Plant ■■ ■■■■■■ 4th Floor■■Administrative area of Yuxi Walvax Therapeutic Monoclonal Antibody Production Workshop

A

■■ ■■■■■■ 1st Floor: Finished Product warehouse & Storage and Transportation dept. ■■ ■■■■■■■■■■■■ 1st

Floor: Therapeutic Monoclonal Antibody Production Workshop of Yuxi JiaHe

■■ ■■■■■■■■■■ 2nd Floor: Bulk workshop & drug production workshop

■■ ■■■■■■■■■■ 3rd Floor: Packing Centre & solution preparation Centre

■■ ■■■■■■ 1st Floor: Administrative area of Yuxi Walvax

■■ ■■■■■■■■■■ 2nd Floor: Quality control laboratory of Yuxi Jiahe

■■ ■■■■■■■■■■ 3rd Floor: Quality control laboratory of Yuxi Jiahe

■■ ■■■■■■ 4th Floor: Administrative area of Yuxi Walvax

■■■■■■■■

Quality Control Administrative Building■■South■

■■■ Parking lot 25

26

■■■ ■■■■

Explosive and Dangerous Sewage disposal station

■■■ Parking lot Goods Warehouse

■■■

Landscape pond t

■ C arpor

■

■G u ard

■

■■■■■■

Total Site Layout

Add: No. 83, South Dongfeng Road, High and New Technology Industrial
Development Zone, Yuxi, 653100, China
E-mail: yingguoqing@walvax.com Tel: 0877 - 2077789
Website: www.walvax.com Fax: 0877 - 2076918

Appendix 6 List of major equipments used in manufacturing and quality control laboratories
Equipment list of meningococcal vaccine production workshop

Equipment Place for
No. Equipment Name Equipment No. Manufacturer
Model Storage/Usage

WYMWF0388
WYMWF0389
Carbon dioxide WYMWF0390 Strain preparation
1 311 Thermo Fisher
incubator WYMWF0391 room
WYMWF0392
WYMWF0393
WYMWF0182 Suzhou Antai
Strain preparation
2 Biosafety cabinet BSC-1600IIA2 Airtech
WYMWF0183 room
Co., Ltd.
WYMWF0205
WYMWF0206
WYMWF0207
WYMWF0208 Shanghai Devotion
WYMWF0209 BIO-DVT-3000 Biological Fermentation
3 Bioreactor
WYMWF0210 F Engineering Co., room
WYMWF0211 Ltd.
WYMWF0212
WYMWF0213
WYMWF0214
WYMWF0184 Suzhou Antai
Fermentation
4 Biosafety cabinet WYMWF0185 BSC-1600IIA2 Airtech
room
Co., Ltd.
WYMWF0222
WYMWF0223
WYMWF0224
WYMWF0225
WYMWF0226
WYMWF0227
WYMWF0228
Large capacity Centrifugation
5 WYMWF0229 CR 7 Hitachi co.
centrifuge room 1
WYMWF0230
WYMWF0231
WYMWF0232
WYMWF0233
WYMWF0234
WYMWF0235
WYMWF0236
6 Large capacity WYMWF0237 CR 7 Hitachi co. Centrifugation
Add: No. 83, South Dongfeng Road, High and New Technology Industrial
Development Zone, Yuxi, 653100, China
E-mail: yingguoqing@walvax.com Tel: 0877 - 2077789
Website: www.walvax.com Fax: 0877 - 2076918

centrifuge WYMWF0238 room 2
WYMWF0239
WYMWF0240
WYMWF0241
WYMWF0242
WYMWF0243
High speed floor WYMWF0244 Centrifugation
7 lynx4000 Thermo Fisher
centrifuge WYMWF0245 room 2
WYMWF0246
WYMWF0247
WYMWF0282
Large capacity Centrifugation
8 WYMWF0283 RC 12BP+ Thermo Fisher
centrifuge room 2
WYMWF0284
Shanghai Naili Fluid
9 Depositing tank WYMWF0221 3000L Depositing room
Equipment Co, Ltd
WYMWF0470 Ultra-filtration
10 Ultrafiltration system UFC10CC PALL
WYMWF0471 room
WYMWF0248
WYMWF0249
WYMWF0250
WYMWF0251
WYMWF0252

WYMWF0253
WYMWF0254
WYMWF0255
WYMWF0256
WYMWF0257
WYMWF0258
High speed floor WYMWF0259 Centrifugation
11 lynx4000 Thermo Fisher
centrifuge WYMWF0260 room 3
WYMWF0261
WYMWF0262
WYMWF0263
WYMWF0264
WYMWF0265
WYMWF0266
WYMWF0267
WYMWF0268
WYMWF0269
WYMWF0270
WYMWF0271
Large capacity WYMWF0285 Centrifugation
12 RC 12BP+ Thermo Fisher
centrifuge WYMWF0286 room 3
Add: No. 83, South Dongfeng Road, High and New Technology Industrial
Development Zone, Yuxi, 653100, China
E-mail: yingguoqing@walvax.com Tel: 0877 - 2077789
Website: www.walvax.com Fax: 0877 - 2076918

WYMWF0272
WYMWF0273
WYMWF0274
WYMWF0275
High speed floor WYMWF0276 Centrifugation
13 lynx4000 Thermo Fisher
centrifuge WYMWF0277 room 4
WYMWF0278
WYMWF0279
WYMWF0280
WYMWF0281
Large capacity WYMWF0287 Centrifugation
14 RC 12BP+ Thermo Fisher
centrifuge WYMWF0288 room 4
WYMWF0303
WYMWF0304
WYMWF0305
WYMWF0306
WYMWF0307
WYMWF0308 Shanghai Morimatsu
Extract and
15 Depositing tank WYMWF0309 -- Pressure Vessel Co.
deposit room
WYMWF0310 Ltd
WYMWF0311
WYMWF0312
WYMWF0313
WYMWF0314
WYMWF0315
Shanghai Suparnin
Mechanical And
Lyophilization
16 Vacuum freeze drier WYMWF0191 LYOBETA-50 Electrical
room
Technology Co.,
Ltd.
AKTA process Chromatography
17 WYMWF0202 AKTA process GE
purification system room
Ultra-filtration
18 Ultrafiltration system WYMWF0472 FS013K05C PALL
room
Add: No. 83, South Dongfeng Road, High and New Technology Industrial
Development Zone, Yuxi, 653100, China
E-mail: yingguoqing@walvax.com Tel: 0877 - 2077789
Website: www.walvax.com Fax: 0877 - 2076918

Equipment list of pneumococcal vaccine production workshop
Equipment Place for Storage
No. Name Equipment No. Equipment Model Manufacturer / Usage
1 Biosafety WYPVF0082 BSC-1600IIAZ Suzhou Antai Airtech Strain
cabinet WYPVF0083 Co., Ltd. of Jiangsu preparation
■1PN1324■
WYPVF0084 Suzhou Purification
Group
WYPVF0085
2 Carbon dioxide WYPVF0049 3111 Thermo Fisher Scientific Strain
WYPVF0050
incubator (China) preparation
WYPVF0051
■1PN1324■
WYPVF0052
3 Refrigerated WYPVF0030 QB-YC-618 Zhejiang Iceshare Strain
medicine cabinet WYPVF0031 Refrigerator Co., Ltd. freeze-drying

4 Biosafety ■1PN1325■
 Fermentaion
 WYPVF0086 BSC-1600IIAZ Suzhou Antai Airtech
 cabinet WYPVF0087 Co., Ltd. of Jiangsu workshop
 ■1PN1310■
 WYPVF0088 Suzhou Purification
 Group
 WYPVF0089
 5 Large capacity WYPVF0100 RC12BP+ Thermo Fisher Centrifugation
 WYPVF0101
 centrifuge workshop2
 WYPVF0102
 WYPVF0103 ■2PN1313■
 WYPVF0104
 WYPVF0105
 WYPVF0106
 6 High speed floor WYPVF0115 LYNX-4000 Thermo Centrifugation
 WYPVF0116
 centrifuge workshop2
 WYPVF0117
 WYPVF0118 ■2PN1313■
 WYPVF0119
 WYPVF0120
 WYPVF0121
 WYPVF0122
 WYPVF0123
 WYPVF0124
 WYPVF0125
 WYPVF0126
 WYPVF0127
 WYPVF0128
 WYPVF0129
 WYPVF0130
 WYPVF0131
 Add: No. 83, South Dongfeng Road, High and New Technology Industrial
 Development Zone, Yuxi, 653100, China
 E-mail: yingguoqing@walvax.com Tel: 0877 - 2077789
 Website: www.walvax.com Fax: 0877 - 2076918

WYPVF0132
 WYPVF0133
 WYPVF0134
 WYPVF0135
 WYPVF0136
 WYPVF0137
 WYPVF0138
 7 Centrifuge WYPVF0205 AvantiJ-26SXP Beckman Coulter (China) Centrifugation
 WYPVF0206 workshop1(2PN1
 WYPVF0207
 WYPVF0208 309)
 WYPVF0209
 WYPVF0210
 WYPVF0211
 WYPVF0212
 WYPVF0213
 WYPVF0214
 WYPVF0215
 WYPVF0216
 WYPVF0217
 WYPVF0218
 WYPVF0219
 WYPVF0220
 WYPVF0221
 WYPVF0222
 WYPVF0223
 WYPVF0224
 WYPVF0225
 WYPVF0226
 WYPVF0227
 WYPVF0228
 WYPVF0229
 WYPVF0230
 WYPVF0231
 WYPVF0232
 WYPVF0233
 WYPVF0234
 WYPVF0235
 WYPVF0236
 WYPVF0237
 WYPVF0238
 WYPVF0239
 WYPVF0240
 WYPVF0241
 WYPVF0242
 WYPVF0243
 WYPVF0244
 WYPVF0245
 WYPVF0246
 Add: No. 83, South Dongfeng Road, High and New Technology Industrial
 Development Zone, Yuxi, 653100, China
 E-mail: yingguoqing@walvax.com Tel: 0877 - 2077789
 Website: www.walvax.com Fax: 0877 - 2076918

WYPVF0247
 WYPVF0248
 WYPVF0249
 WYPVF0250
 WYPVF0251
 WYPVF0252
 WYPVF0253
 WYPVF0254
 WYPVF0255
 WYPVF0256
 WYPVF0257
 WYPVF0258
 WYPVF0259
 WYPVF0260
 8 A10L bioreactor WYPVF0261 BIO-DVT-3000F Shanghai Devotion Fermentaion
 9 A100L WYPVF0262 Biological Engineering workshop
 bioreact Co., Ltd. ■1PN1310■
 10 A900L
 or WYPVF0263
 bioreactor
 11 B10L bioreactor WYPVF0264

 12 B100L WYPVF0265
 bioreactor
 13 B900L WYPVF0266
 bioreactor
 14 C10L bioreactor WYPVF0267

 15 C100L WYPVF0268
 bioreactor
 16 C900L WYPVF0269
 bioreactor
 17 D10L bioreactor WYPVF0270
 18 D100L WYPVF0271
 bioreactor
 19 D900L WYPVF0272
 bioreactor
 20 E10L bioreactor WYPVF0273 BIO-DVT-3000F Shanghai Devotion Fermentaion
 21 E100L WYPVF0274 Biological Engineering workshop
 bioreactor Co., Ltd. ■1PN2310■
 22 E900L WYPVF0275
 bioreactor
 23 F10L bioreactor WYPVF0276

 24 F100L WYPVF0277
 bioreactor
 Add: No. 83, South Dongfeng Road, High and New Technology Industrial
 Development Zone, Yuxi, 653100, China
 E-mail: yingguoqing@walvax.com Tel: 0877 - 2077789
 Website: www.walvax.com Fax: 0877 - 2076918

 25 F900L WYPVF0278
 bioreactor
 26 H10L bioreactor WYPVF0279

 27 H100L WYPVF0280
 bioreactor
 28 H900L WYPVF0281
 bioreactor
 29 G10L bioreactor WYPVF0282

 30 G100L WYPVF0283
 bioreactor
 31 G900L WYPVF0284
 bioreactor
 32 FFU WYPVF0441 CU03-1200`600`2 MayAir (China) Centrifugation
 WYPVF0442 400-S-P workshop1
 WYPVF0443 ■2PN1309■

 WYPVF0444
 WYPVF0445 CU03-2100`1325` Centrifugation
 WYPVF0446 2400-S-P workshop2
 WYPVF0447 CU03-2400`2400` ■2PN1313■
 Ultra-filtration
 WYPVF0448 2400-S-P workshop
 CU03-2400`1120` ■2PN1310■
 WYPVF0449 Purification and
 2400-S-P precipitation
 WYPVF0450
 workshop
 33 WYPVF0295 UFC10CC PALL ■2PN1311■
 Ultrafiltration Ultra-filtration
 system WYPVF0296 workshop
 ■2PN1310■
 34 Small freeze WYPVF0026 FD5-3 SIM Freeze-drying
 dryer workshop(3PN13
 WYPVF0027
 12)
 35 Ultra-low WYPVF0034 DW-40L358 Zhejiang Iceshare Freeze-drying
 temperature WYPVF0035 Refrigerator Co., Ltd. workshop(3PN23
 freezer WYPVF0036 12)

36 Large capacity WYPVF0107 RC12BP+ Thermo Fisher Centrifugation
centrifuge WYPVF0108 workshop4
WYPVF0109 (3PN1310/2)
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Development Zone, Yuxi, 653100, China
E-mail: yingguoqing@walvax.com Tel: 0877 - 2077789
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WYPVF0110

WYPVF0111
WYPVF0112
WYPVF0113

WYPVF0114
37 High speed floor WYPVF0139 LYNX-4000 Thermo Fisher Centrifugation
centrifuge workshop3
(3PN1310/1)

WYPVF0140

WYPVF0141
WYPVF0142

WYPVF0143
WYPVF0144

WYPVF0145
WYPVF0146

WYPVF0147

WYPVF0148
WYPVF0149

WYPVF0150
WYPVF0151

WYPVF0152
WYPVF0153

WYPVF0154
WYPVF0155

WYPVF0156
WYPVF0157

WYPVF0158
WYPVF0159

WYPVF0160
WYPVF0161

WYPVF0162

WYPVF0163
WYPVF0164
Add: No. 83, South Dongfeng Road, High and New Technology Industrial
Development Zone, Yuxi, 653100, China
E-mail: yingguoqing@walvax.com Tel: 0877 - 2077789
Website: www.walvax.com Fax: 0877 - 2076918

High speed floor WYPVF0165 LYNX-4000 Thermo Fisher Centrifugation
centrifuge WYPVF0166 workshop3
WYPVF0167 (3PN2310/1)

WYPVF0168
WYPVF0169

WYPVF0170

WYPVF0171
WYPVF0172

WYPVF0173
WYPVF0174

WYPVF0175
WYPVF0176

WYPVF0177
WYPVF0178

WYPVF0179
WYPVF0180

WYPVF0181
WYPVF0182

WYPVF0183
WYPVF0184

WYPVF0185
WYPVF0186

WYPVF0187

WYPVF0188
WYPVF0189

WYPVF0190

High speed floor WYPVF0191 LYNX-4000 Thermo Fisher Centrifugation
centrifuge WYPVF0192 workshop4
WYPVF0193 (3PN1310/2)

WYPVF0194
WYPVF0195

WYPVF0196
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Development Zone, Yuxi, 653100, China
E-mail: yingguoqing@walvax.com Tel: 0877 - 2077789
Website: www.walvax.com Fax: 0877 - 2076918

WYPVF0197

High speed floor WYPVF0198 LYNX-4000 Thermo Fisher Centrifugation
centrifuge WYPVF0199 workshop4
WYPVF0200 (3PN2310/2)

WYPVF0201

WYPVF0202

WYPVF0203

WYPVF0204

38 Ultrafiltration WYPVF0297 UFC10CC PALL Ultra-filtration
system workshop(3PN13
WYPVF0298
11)
WYPVF0299 Ultra-filtration

WYPVF0300 workshop(3PN23
11)
39 FFU WYPVF0451 CU03-4000`1325` MayAir (China) Centrifugation
2400-S-P workshop3(3PN1
310/1)
WYPVF0452 Centrifugation
workshop3(3PN2
310/1)
WYPVF0453 Centrifugation
workshop4(3PN1

WYPVF0454 310/2)
Centrifugation
workshop4(3PN2

CU03-2400`1120` 310/2)
WYPVF0455 Extraction
2400-S-P precipitation
workshop(3PN13
09)

Extraction
WYPVF0456
precipitation
workshop(3PN23
WYPVF0457 09)
Ultra-filtration
workshop(3PN13
11)

WYPVF0458 Ultra-filtration
workshop(3PN23
11)

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40 WYPVF0070 Suzhou Antai Airtech Collection
Clean bench BLB-1600
Co., Ltd. of Jiangsu workshop(3PN23

Suzhou Purification 13)

Group Collection
WYPVF0071
workshop(3PN13
13)

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Development Zone, Yuxi, 653100, China
E-mail: yingguoqing@walvax.com Tel: 0877 - 2077789

Website: www.walvax.com Fax: 0877 - 2076918

Equipment list of QC lab
Equipment Equipment Equipment Place for Storage /
No. Name Manufacturer
No. Model Usage

1 pH meter WYQCF0040 Φ390 BECKMAN Titration room

Amersham Analytical
2 Constant flow pump WYQCF0080 P-1 Biosciences instrument room

Amersham Analytical
3 Fraction collector WYQCF0081 Frac-920 Biosciences instrument room

Chromatographic Amersham Analytical
4 column WYQCF0082 XK16/100 Biosciences instrument room

Amersham Analytical
5 Constant flow pump WYQCF0083 P-1 Biosciences instrument room

Amersham Analytical
6 Fraction collector WYQCF0084 Frac-920 Biosciences instrument room

Chromatographic Amersham Analytical
7 column WYQCF0085 XK16/100 Biosciences instrument room

Amersham Analytical
8 Constant flow pump WYQCF0086 P-1 Biosciences instrument room

Amersham Analytical
9 Fraction collector WYQCF0087 Frac-920 Biosciences instrument room

Chromatographic Amersham Analytical
10 column WYQCF0088 XK16/100 Biosciences instrument room

Amersham Analytical
11 Constant flow pump WYQCF0089 P-1 Biosciences instrument room

Amersham Analytical
12 Fraction collector WYQCF0090 Frac-920 Biosciences instrument room

Chromatographic Amersham Analytical
13 column WYQCF0091 XK16/100 Biosciences instrument room
Add: No. 83, South Dongfeng Road, High and New Technology Industrial
Development Zone, Yuxi, 653100, China
E-mail: yingguoqing@walvax.com Tel: 0877 - 2077789
Website: www.walvax.com Fax: 0877 - 2076918

Ultraviolet-visible Physicochemical
14 spectrophotometer WYQCF0243 UV-2600 Shimadzu experimental area

Ultraviolet-visible Physicochemical
15 spectrophotometer WYQCF0244 UV-2600 Shimadzu experimental area

Ultraviolet-visible UitraSpec Analytical
16 spectrophotometer WYQCF0364 2100pro GE instrument room

High performance
liquid Agilent Analytical
WYQCF0159
17 chromatograph 1260Infinity Technologies instrument room

Shanghai
Precision
Microcomputer melting Analytical
18 WYQCF0160 WRS-2 Scientific
point meter instrument room
Instrument Co.,
Ltd.

SLOAIR3350 Lighthouse Physicochemical
19 Particle counter WYQCF0189
5350 (USA) experimental area

Analytical
20 Automatic
Automatic polarimeter WYQCF0174 Autopol instrument room
polarimeter

Tianhe Medical
Analytical instrument
21 Particle analyzer WYQCF0186 GWF8JC Instrument Co.,
room
Ltd
Lighthouse Physicochemical
22 Particle counter WYQCF0190 SLOAIR33505350 (USA) experimental area
Shinva medical
Pulsation vacuum Wash and
23 WYQCF0320 XG1.D instrument Co.,
sterilizer sterilization center
Ltd.

Shinva medical
Pulsation vacuum Wash and
WYQCF0321 XG1.D instrument Co.,
sterilizer sterilization center
24 Ltd.
Laboratory Toc Analytical instrument
WYQCF0290 Sievers900 GE
25 Analyzer room
Microbiological air Physicochemical
26 WYQCF0300 Minicapt100 PMS
sampler(MC13F047) experimental area
Add: No. 83, South Dongfeng Road, High and New Technology Industrial
Development Zone, Yuxi, 653100, China
E-mail: yingguoqing@walvax.com Tel: 0877 - 2077789
Website: www.walvax.com Fax: 0877 - 2076918

Kunming
27 30-35 ■ incubator 2QC407
Shenghao
Kunming
28 20-25 ■ incubator 2QC408
Shenghao
Formall water jacket
29 carbon dioxide WYQCF0284 Formal Thermo Fisher 4QC424
incubator
BCM-1600A
BIOLOGICAL
30 WYQCF0304 BCM-1600A Suzhou Antai 3QC409
CLEAN
BENCH
Shanghai
Electro-thermostatic Yiheng
31 water WYQCF0281 HWS-24 Scientific 2QC301
bath Instruments
Ltd.
CTHI-250B
CONSTANT
32 TEMPERATURE WYQCF0317 CTHI-250B STIK 4QC422
HUMIDITY
CHAMBER
Ultra-low temperature Zhejiang
33 WYQCF0327 DW-3DL318 1QC303
freezer Iceshare
2-8■ refrigerated Zhejiang
34 WYQCF0309 QB-YC-308 1QC303
medicine cabinet Iceshare
Shanghai
DHG-9075A
Yiheng
ELECTRO-THERMO
35 WYQCF0294 DHG-9075A Scientific 2QC307
ST
Instruments
ATIC BLAST OVEN
Ltd.
36 8200 kjeldahl apparatus WYQCF0313 8200 FOSS 2QC302

Yuxi Walvax Biotechnology Co., Ltd.

Summary of product characteristics

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Yuxi Walvax Biotechnology Co., Ltd.

1. Name of the medicinal product

Generic Name: 13-valent Pneumococcal Polysaccharide Conjugate Vaccine

2. Qualitative and quantitative composition

1 dose (0.5 ml) contains:

Pneumococcal polysaccharide serotype 1 2.6 µg

Pneumococcal polysaccharide serotype 3 2.5 µg

Pneumococcal polysaccharide serotype 4 3.0 µg

Pneumococcal polysaccharide serotype 5 2.5 µg

Pneumococcal polysaccharide serotype 6A 2.5 µg

Pneumococcal polysaccharide serotype 6B 6.0 µg

Pneumococcal polysaccharide serotype 7F 2.85 µg

Pneumococcal polysaccharide serotype 9V 2.5 µg

Pneumococcal polysaccharide serotype 14 2.75 µg

Pneumococcal polysaccharide serotype 18C 3.25 µg

Pneumococcal polysaccharide serotype 19A 2.6 µg

Pneumococcal polysaccharide serotype 19F 2.75 µg

Pneumococcal polysaccharide serotype 23F 3.0 µg

Conjugated to Tetanus Toxoid carrier protein, adsorbed on aluminium phosphate.

Excipients with known effect

For the full list of excipients, see section 6.1.

3. Pharmaceutical form

Suspension for injection.

The vaccine is a homogeneous white suspension.

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Yuxi Walvax Biotechnology Co., Ltd.

4. Clinical particulars

4.1 Therapeutic indications

The vaccine elicits immune response in recipients following immunization, and is indicated for the prevention of invasive disease caused by 13 serotypes (1, 3, 4, 5, 6A, 6B, 7F, 9V, 14, 18C, 19A, 19F and 23F) of *Streptococcus pneumoniae*.

See sections 4.4 and 5.1 for information on protection against specific pneumococcal serotypes.

The use of PCV13-TT should be determined on the basis of official recommendations taking into consideration the risk of invasive disease and pneumonia in different age groups, underlying comorbidities as well as the variability of serotype epidemiology in different geographical areas.

4.2 Posology and method of administration

The immunisation schedules for PCV13-TT should be based on official recommendations.

Posology

Infants and children aged 6 weeks through 5 years of age (before 6th birthday).

The vaccine should be shaken well immediately prior to use, and be given intramuscularly. The preferred sites for injection are the anterolateral aspect of the thigh in infants and the deltoid muscle of the upper arm in toddlers and children. Each dose (0.5 mL) is to be injected. The vaccine should not be injected in and/or near the areas where there may be a nerve trunk and/or blood vessel.

Vaccination Schedule for Infants and Toddlers: 4-dose series in total.

The primary infant series consists of three doses, with the first dose recommended to be given at 2 months of age (minimum age: 6 weeks) with an interval of 2 months between doses; the booster dose (4th dose) is given between 12 and 15 months of age.

Alternatively, the primary infant series consists of three doses, with the first dose recommended to be given at 3 months of age with an interval of 1 months between doses; the booster dose (4th dose) is given between 12 and 15 months of age.

To sum up, the product is to be administered as a four-dose series at 2, 4, 6, and 12-15 months of age or 3, 4, 5, and 12-15 months of age, respectively, as described in Table 1 and Table 2.

Table 1 Vaccination Schedule for Infants 2 months of age

Dose Dose 1*† Dose 2† Dose 3† Dose 4‡

Age at Dose 2 months 4 months 6 months 12-15 months

*Dose 1 may be given as early as 6 weeks of age.

†The recommended dosing interval is 8 weeks.

‡The fourth dose should be administered at approximately 12–15 months of age, and at least 2 months after the

third dose.

Table 2 Vaccination Schedule for Infants 3 months of age

Dose Dose 1† Dose 2† Dose 3† Dose 4‡

Age at Dose 3 months 4 months 5 months 12-15 months

†The recommended dosing interval is 4 weeks.

‡The fourth dose should be administered at approximately 12–15 months of age, and at least 2 months after the

third dose.

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Infants 7 through 11 months of age: The primary series consists of 2 doses with an interval of at least 2 months between doses; the booster dose (3rd dose) is given at 12+ months of age, with an interval of at least 2 months apart from the 2nd dose.

Toddlers 12 through 23 months of age: 2 doses in total, with an interval of at least 2 months between doses.

Children 2 through 5 years of age: a single dose.

For children 7 months through 5 years of age who have not received the product, the catch-up schedule in Table 3 applies:

Table 3 Vaccination Schedule for Infants and Toddlers 7 months through 5 years of age

Age at First Dose Total Number of 0.5 mL Doses

7-11 months of age 3*

12-23 months of age 2†

24 months through 5 years of age (prior to the 6th birthday) 1

*The first 2 doses at least 4 weeks apart; third dose after the one-year birthday, separated from the second dose

by at least 2 months.

†Two doses at least 2 months apart.

4.3 Contraindications

Hypersensitivity to any component of the product, including active antigens, excipients, or tetanus toxoid, etc.

4.4 Special warnings and precautions for use

1. Do not vaccinate via intravenous route or by gluteal intramuscular injection, ensure the syringe needle is not puncturing blood vessel during inoculation.
2. Check if the package, container, label, appearance and expiration date of the vaccine are in compliance with corresponding requirements before administration. Do not use the vaccine in case that there is any crack in the container, loosened stopper, detached label, foreign particle(s) or discoloring inside the container, etc. Do not use the vaccine after the expiration date.
3. Use immediately after unsealing. A single human dose shall be used up each time according to prescribing information.
4. The vaccination should be postponed in case of fever, acute disease, and acute attack of chronic disease.
5. Appropriate monitoring and medical care and rescue measures should be readily available in case of occurrence of rare hypersensitivity reactions during vaccination. If allergic reactions occur after vaccination, please go to the vaccination site or hospital in time.
6. Cautions should be taken for vaccination in recipients with thrombocytopenia, any coagulopathy or those who are receiving anticoagulant treatment.
7. Preterm infants should be monitored for the potential risk of apnea during primary series. For preterm infants under hospitalization (gestational age ≤ 30 weeks at birth) vaccinated with PCV13-TT according to the recommended immunization schedule, at least 48 hours of monitoring should be considered. Given the benefit of vaccination in preterm infants, discontinued or deferred vaccination of the product is not recommended.

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8. Given there is still no available safety and immunogenicity data for PCV3-TT in immuno-compromised individuals (e.g., malignancy or nephrotic syndrome), vaccination in this special group should be considered on an individual basis.
9. The use of PCV13-TT does not replace the use of 23-valent pneumococcal polysaccharide vaccines in children ≥ 24 months of age with conditions such as sickle cell disease, asplenia, HIV infection, chronic illness, or those who are immuno-compromised.
10. Under no circumstances shall the tetanus toxoid contained in the vaccine replace the routine immunization of tetanus vaccine or tetanus containing vaccine.
11. This product cannot guarantee all recipients can be protected from any diseases caused by *Streptococcus pneumoniae*.

■PREGNANCY AND LACTATION■

4.5 Interaction with other medicinal products and other forms of interaction

There is no concomitant immunization data available either inside or outside of China for the product. During the phase III clinical trial of the product, the interval between vaccination of the product and any other diphtheria, tetanus and acellular pertussis (DTaP) containing vaccine was ≥ 10 days. No adverse impact of DTaP-containing vaccine was observed on the immunogenicity of PCV13-TT under the indicated immunization interval.

4.6 Fertility, pregnancy and lactation

Pregnancy

There is no data for using the product in pregnant women.

Breast-feeding

It is unknown whether this product is excreted in human milk.

Animal studies do not indicate direct or indirect harmful effects with respect to reproductive toxicity (see section 5.3).

4.7 Effects on ability to drive and use machines

PCV13-TT has no or negligible influence on the ability to drive and use machines. However, some of the effects mentioned under section 4.8 "Undesirable effects" may temporarily affect the ability to drive or use machines.

4.8 Adverse Reactions

4.8.1. Clinical trials of the vaccine

Two clinical trials (phase I and III) of the vaccine (PCV13-TT) were conducted within China, including 120 and 2760 subjects, respectively; among all subjects, 1754 have been vaccinated with at least one dose of PCV13-TT. The systematic safety observation period of PCV13-TT started from the first vaccination to 30 days after each subsequent dose, and the long-term safety observation for the majority of subjects was performed through 180 days after the last dose (after full immunization).

4.8.1.1. Summary description

The incidence of adverse reactions reported in clinical trials, according to the guidance on classifications of adverse events recommended by The Council for International Organizations of

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Medical Sciences (CIOMS), are classified as: very common ($\geq 10\%$), common ($\geq 1\%$ to $<10\%$), uncommon ($\geq 0.1\%$ to $<1\%$), rare ($\geq 0.01\%$ to $<0.1\%$), very rare ($<0.01\%$), the safety data post primary immunizations collected from phase I and III clinical trials of all populations included are summarized as follows:

① Systemic adverse reactions

Very common: fever, diarrhea

reactions N*=488~519 N*=493~519 N*=487~517 N*=186~198 N*=191~200 N*= 200 N*=200 N*=200 N*=200
%b %b %b %b %b % % % %
Fever
Any 28.07~58.00 27.38~66.28 21.56~61.12 20.43~53.54 20.94~61.50 41.50 44.00 20.50 23.50
Grade 1# 17.01~48.17 18.05~57.61 15.40~52.22 11.83~35.86 13.09~41.00 30.00 29.00 8.50 15.50
Grade 2# 10.86~17.73 9.53~22.35 6.78~18.18 9.14~23.23 7.85~30.50 14.50 17.00 10.00 8.00
Grade 3# 1.23~0.39 0.61~0.39 0.00~0.39 0.00~3.03 0.00~1.50 0.50 1.00 2.00 1.00
Grade 4# 0.00~0.19 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00
Fatigue / somnolence
Any 0.20~6.17 0.20~5.20 0.00~3.09 0.00~1.52 0.00~2.50 1.00 1.00 2.50 1.50
Grade 1c 0.20~5.01 0.20~4.24 0.00~2.71 0.00~1.52 0.00~2.00 0.50 0.50 2.50 1.50
Grade 2c 0.00~1.73 0.00~0.96 0.00~0.39 0.00 0.00~0.50 0.50 0.50 0.00 0.00
Grade 3c 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00
Nausea / vomiting
Any 1.64~5.78 0.81~3.66 0.21~6.00 0.00~5.56 1.05~4.00 1.50 2.50 1.00 2.50
Grade 1d 0.82~4.62 0.61~2.31 0.21~4.26 0.00~4.55 1.05~3.50 1.00 2.50 0.50 2.50
Grade 2d 0.61~1.35 0.20~1.54 0.00~2.13 0.00~1.52 0.00~0.50 0.50 0.00 0.50 0.00
Grade 3d 0.00~0.20 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00
Diarrhea
Any 2.46~12.91 2.23~11.95 0.41~11.03 1.61~7.07 1.05~9.00 4.00 1.50 1.00 2.00
Grade 1e 1.23~7.32 1.01~6.17 0.21~4.64 0.54~4.55 0.00~7.00 3.00 1.50 1.00 1.50
Grade 2e 1.23~5.78 1.01~6.17 0.21~7.74 1.08~2.53 1.05~2.50 1.00 0.00 0.00 0.50
Grade 3e 0.00~0.58 0.20~0.19 0.00 0.00 0.00 0.00 0.00 0.00 0.00
Myalgia
Any 0.00~0.19 0.00~0.39 0.00~0.19 0.00 0.00~0.50 0.00 0.00 0.50 0.50
Grade 1f 0.00~0.19 0.00~0.39 0.00~0.19 0.00 0.00~0.50 0.00 0.00 0.50 0.50
Grade 2f 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00
Grade 3f 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00
Cough
Any 0.82~7.32 1.22~6.55 0.21~5.61 0.54~5.05 1.05~4.50 5.00 5.50 4.00 5.50
Grade 1g 0.20~5.01 0.61~4.05 0.21~4.64 0.00~2.53 0.52~2.50 1.00 2.00 3.00 2.00
Grade 2g 0.61~2.31 0.61~2.70 0.00~0.77 0.54~2.53 0.52~2.00 4.00 3.50 1.00 3.00
Grade 3g 0.00 0.00~0.19 0.00 0.00 0.00 0.00 0.00 0.00 0.50
Grade 4g 0.00 0.00 0.00~0.19 0.00 0.00 0.00 0.00 0.00 0.00
Abnormal crying
Any 0.82~10.40 0.61~9.44 0.21~10.44 1.08~6.57 0.00~6.00 2.50 1.50 0.00 1.50
Grade 1h 0.61~6.94 0.20~6.74 0.00~6.77 0.00~5.05 0.00~4.50 0.50 0.50 0.00 1.00
Grade 2h 0.20~3.85 0.20~3.66 0.21~4.64 1.08~1.52 0.00~1.50 2.00 1.00 0.00 0.50
Grade 3h 0.00~0.19 0.19~0.20 0.00 0.00 0.00 0.00 0.00 0.00 0.00
Allergic reaction
Any 0.00~0.58 0.00~0.58 0.00~0.58 0.00 0.52~3.00 1.50 3.50 1.50 1.00
Grade 1i 0.00~0.19 0.00~0.19 0.00~0.39 0.00 0.00~0.50 0.50 1.50 0.00 0.00
Grade 2i 0.00~0.19 0.00~0.39 0.00 0.00 0.52~1.00 0.50 1.00 1.50 0.50
Grade 3i 0.00~0.19 0.00 0.00~0.19 0.00 0.00~1.50 0.50 1.00 0.00 0.50

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Any systemic reactionsj 30.33~67.82 29.21~71.48 22.38~67.12 22.04~58.59 22.51~66.00 46.50 50.00 24.00
28.50
Grade 1 19.06~57.61 19.27~62.81 15.81~57.64 12.37~42.93 14.14~49.00 33.50 34.00 13.00 19.50
Grade 2 12.91~25.82 10.75~30.83 7.19~27.08 10.22~27.27 9.42~34.50 20.50 20.50 12.00 11.50
Grade 3 1.43~1.35 0.81~0.96 0.00~0.58 0.00~3.03 0.00~3.00 1.00 2.00 2.00 2.00
Grade 4 0.00~0.19 0.00 0.00~0.19 0.00 0.00 0.00 0.00 0.00 0.00
*

N=number of subjects reporting "yes" for at least 1 day, or "no" for all 30 days.
a. Minimum age: 6 weeks.
b. The minimum and maximum incidence rates for systemic adverse reactions after primary and booster immunizations.
Grade 1, 37.1-37.5 ■; grade 2, 37.6-39.0 ■; grade 3, >39.0 ■.
c.
Grade 1, slight reduction in normal activities for■48 h, not affecting daily activities; grade 2, 20%-50% reduction in normal activities for >48 h, affecting daily activities mildly; grade 3, >50% reduction in normal activities, affecting daily activities severely and is unable to work.
d. Grade 1, once or twice/24 h, with basically normal intake and will not affect activities; grade 2, 2-5 times/24 h, with significantly reduced intake or restricted activities; grade 3, > 6 times/24 h, with no significant intake, and in need of intravenous infusion.
e. Grade 1, mild or transient with 2-3 times of loose stools/day, or mild diarrhea lasting■1 week; grade 2, moderate or persistent diarrhea, 4-5 times/day, or diarrhea lasting >1 week; >6 times of watery stool/day, or bloody diarrhea, orthostatic hypotension, electrolyte imbalance, in need of intravenous infusion of > 2 L.
f. Grade 1, not affecting daily activities; grade 2, tender muscles in non-vaccination sites, affecting daily activities mildly; grade 3, severe tender muscles, affecting daily activities severely.
g. Grade 1, transient, no treatment required; grade 2, persistent and controllable cough; grade 3, uncontrollable paroxysmal cough; grade 4, emergency care required.
h.
Grade 1, slight increase in frequency than usual; grade 2, doubled frequency than usual; grade 3, doubled frequency than usual, and lasts > 3 days.
i. Grade 1, itching with no rash; grade 2, local urticaria; grade 3, extensive urticaria, angioedema.
j. All systemic reactions, including body temperature ≥37.1 °C, fatigue/somnolence, nausea/vomiting, diarrhea, myalgia, cough, abnormal crying and allergic reactions.

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②Incidence and severity of unsolicited adverse reactions

The incidence of unsolicited adverse reactions for PCV13-TT was 0.06%-0.73%, and most episodes were grade 1 in severity. The documented symptoms include: nasal obstruction, rhinorrhea, nasopharyngitis, upper respiratory tract infection, oral ulcer, abdominal pain, abdominal distension, decreased appetite, hyperhidrosis, increased tearing and eye discharge and

red eyelids.

4.8.2. Safety monitoring of similar products inside and outside China

Despite the adverse reactions reported in clinical trials of PCV13-TT, the additional adverse reactions captured from a similar vaccine, the marketed 13-valent Pneumococcal Conjugate Vaccine (Prevenar 13®) which has also been investigated in clinical trials within China, were reported among 1182 infants aged 6 through 11 weeks vaccinated with at least one dose of Prevenar 13®, as irritability and decreased sleep; besides, in another 15 clinical trials conducted outside China enrolling 5075 toddlers and children 6 weeks through 5 years of age, additional adverse reactions were identified after the vaccination of Prevenar 13®, including insomnia, convulsion (including febrile convulsion), anaphylactic reaction (including facial edema, dyspnea and bronchial spasm) and hypotonia events.

In addition, the reported additional adverse reactions since the marketing of Prevenar 13® include dermatitis/urticaria in injection site, lymphadenopathy limited to vaccination site, hypersensitivity/hypersensitivity-like reactions (including shock), angioedema, erythema multiforme and Kawasaki syndrome, all are with indeterminate incidence rates due to the spontaneous reporting nature.

4.9 Overdose

Overdose with PCV13-TT is unlikely due to its presentation as a vial or a pre-filled syringe. There is no overdose data available for the product in recipients.

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5. Pharmacological properties

5.1 Pharmacodynamic properties

Pharmacotherapeutic group: vaccines, pneumococcal vaccines; ATC code: J07AL02

Prevenar 13 contains the 7 pneumococcal capsular polysaccharides that are in Prevenar 7 (4, 6B, 9V, 14, 18C, 19F, 23F) plus 6 additional polysaccharides (1, 3, 5, 6A, 7F, 19A) all conjugated to Tetanus Toxoid carrier protein.

Burden of disease

Infants and children aged 6 weeks to 5 years

Based on serotype surveillance in Europe performed before the introduction of Prevenar, Prevenar 13 is estimated to cover 73-100 % (depending on the country) of serotypes causing invasive pneumococcal disease (IPD) in children less than 5 years of age. In this age group, serotypes 1, 3, 5, 6A, 7F, and 19A account for 15.6 % to 59.7 % of invasive disease, depending on the country, the time period studied, and the use of Prevenar.

Acute otitis media (AOM) is a common childhood disease with different aetiologies. Bacteria can be responsible for 60-70 % of clinical episodes of AOM. *S. pneumoniae* is one of the most common causes of bacterial AOM worldwide.

PCV13-TT is estimated to cover over 90% of serotypes causing antimicrobial-resistant IPD.

Children and adolescents aged 6 to 17 years

In children and adolescents aged 6 to 17 years, the incidence of pneumococcal disease is low, however, there is an increased risk of morbidity and mortality in those with underlying comorbidities.

Adults ≥18 years and the elderly

Pneumonia is the most common clinical presentation of pneumococcal disease in adults.

The reported incidence of community-acquired pneumonia (CAP) and IPD in Europe varies by country, increases with age from 50 years and is highest in individuals aged ≥ 65 years. *S. pneumoniae* is the most frequent cause of CAP, and is estimated to be responsible for approximately 30% of all CAP cases requiring hospitalisation in adults in developed countries. Bacteraemic pneumonia (approximately 80% of IPD in adults), bacteraemia without a focus, and meningitis are the most common manifestations of IPD in adults. Based on surveillance data following the introduction of Prevenar but before the introduction of PCV13-TT in childhood vaccination programmes, the pneumococcal serotypes in PCV13-TT may be responsible for at least 50 – 76% (depending on country) of IPD in adults.

The risk for CAP and IPD in adults also increases with chronic underlying medical conditions, specifically, anatomical or functional asplenia, diabetes mellitus, asthma, chronic cardiovascular, pulmonary, kidney or liver disease, and it is highest in those who are immune-suppressed such as those with malignant haematological diseases or HIV infection.

PCV13-TT immunogenicity clinical studies in infants, children and adolescents

The protective efficacy of PCV13-TT against IPD has not been studied. As recommended by the World Health Organization (WHO) the assessment of potential efficacy against IPD in infants and young children has been based on a comparison of immune responses to the seven common serotypes shared between PCV13-TT and Prevenar, for which protective efficacy has been proven

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(for Prevenar (7-valent) efficacy in infants and children, see below). Immune responses to the additional 6 serotypes were also measured.

Immune responses following a three-dose primary infant series

In China, the pivotal phase III clinical trial of PCV13-TT was a randomized, blinded, positive control of imported similar vaccine (Prevnar), and non-inferiority study designed to evaluate the immunogenicity and safety of PCV13-TT and Prevnar in 2760 subjects 2 through 71 months of age (at least 6 weeks of age). A total of 1040 infants aged 3 months were randomized in a 1:1 ratio as study group and control group, with PCV13-TT or Prevnar given at 3, 4, 5 and 12-15 months of age, respectively; for 520 infants aged 2 months (at least 6 weeks of age), PCV13-TT were vaccinated at 2, 4, 6 and 12-15 months of age; the other three age groups of 7 through 11 months (vaccinated at 0, 2 months and 12-15 months of age, with an interval of at least 2 months between the third and second dose), 12 through 23 months (vaccinated at 0, 2 months), and 24 through 71 months (vaccinated at 0 month), each was with 400 subjects randomized in a 1:1 ratio to receive PCV13-TT or Prevnar, respectively. Enzyme-linked immunosorbent assay (ELISA) was utilized to detect the IgG antibody concentration in all serum samples collected from each group; additionally, 100 subjects in PCV13-TT and Prevnar groups in each age stratification, totaling 900, were included as a subgroup for opsonophagocytic activity (OPA) antibody assay, which provides the measurement of the ability of serum antibodies to eliminate pneumococci by promoting complement-mediated phagocytosis.

The main target population of the trial was infants 2 and 3 months of age (at least 6 weeks of age). For immunogenicity, the PCV13-TT (3 months) group was compared with Prevnar (3 months) group in the first place, followed by comparison between PCV13-TT (2 months) group versus Prevnar (3 months) group and then versus PCV13-TT (3 months) group in a sequential

manner. The immunogenicity measures included the proportion of subjects with vaccine type-specific pneumococcal IgG antibody concentration $\geq 0.35 \mu\text{g/mL}$ (positive rate) and IgG geometric mean concentration (GMC) one month post primary immunization(s) and one month post booster dose, as well as the proportion of subjects with vaccine type-specific pneumococcal OPA antibody titer $\geq 1:8$ (positive rate) and OPA geometric mean titer (GMT) one month post primary immunization(s) and one month post booster dose.

The noninferiority comparison was performed for the immune responses of 7 common serotypes between PCV13-TT and Pevnar groups. Once the lower limits of 2-sided 95% CI for IgG antibody positive rate of 6 additional serotypes in study group achieved $\geq 70\%$, superiority test was then performed to compare the immune responses to 6 additional serotypes and corresponding serotypes in control group; meanwhile, the 6 additional serotypes in study group were compared with the serotype with the lowest response among all the 7 common serotypes in control group to determine the non-inferiority. The non-inferiority was demonstrated as the lower limit of the 2-sided 97.5% CI for the difference in percentage of responders at $0.35 \mu\text{g/mL}$ between groups (study group - control group) $> -10\%$, or the lower limit of the 2-sided 97.5% CI for IgG geometric mean concentration (GMC) ratio (study group/control group) > 0.5 ; the superiority was demonstrated as the lower limit of 2-sided 95% CI for the difference in percentage of responders at $0.35 \mu\text{g/mL}$ between groups > 0 , and the lower limit of the 2-sided 95% CI for the IgG GMC ratio > 1 . The results demonstrate that:

In subjects aged 3 months, after primary immunizations (3 doses), the IgG antibody were compared between PCV13-TT group and Pevnar group, and 6 out of 7 common serotypes met the non-inferiority criterion with the exception of 6B; the 6 additional serotypes reached superiority threshold; besides, the 6 additional serotypes in PCV13-TT group were non-inferior to the serotype with the lowest response among all the 7 common serotypes in Pevnar group.

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In subjects aged 3 months, after full immunizations (4 doses), the IgG antibody were compared between PCV13-TT group and Pevnar group, and all 7 common serotypes met the non-inferiority criterion; the 6 additional serotypes reached superiority threshold; besides, the 6 additional serotypes in PCV13-TT group were non-inferior to the serotype with the lowest response among all the 7 common serotypes in Pevnar group.

In subjects aged 2 months (at least 6 weeks of age), after primary immunizations (3 doses), the IgG antibody were first compared between PCV13-TT (2 months) group and Pevnar (3 months) group, and all 7 common serotypes met the non-inferiority criterion; the 6 additional serotypes reached superiority threshold; besides, the 6 additional serotypes in PCV13-TT (2 months) group were non-inferior to the serotype with the lowest IgG response among all 7 common serotypes in Pevnar (3 months) group. The IgG antibody were then compared between PCV13-TT (2 months) group and PCV13-TT (3 months) group, and 12 out of the 13 serotypes met the non-inferiority criterion with the exception of serotype 5.

In subjects aged 2 months (at least 6 weeks of age), after full immunizations (4 doses), the IgG antibody were first compared between PCV13-TT (2 months) group with Pevnar (3 months) group, and all 7 common serotypes met the non-inferiority criterion; the 6 additional serotypes reached superiority threshold; besides, the 6 additional serotypes in PCV13-TT (2 months) group were non-inferior to the serotype with the lowest IgG response among all 7 common serotypes in Pevnar (3 months) group. The IgG antibody were then compared between PCV13-TT (2 months) group and PCV13-TT (3 months) group, and all the 13 serotypes met the non-inferiority criterion. The testing results of OPA antibody demonstrated that, all serotypes were capable of inducing functional OPA antibody reaction in all age groups included in the study.

The immunogenicity data from subjects 2 months (at least 6 weeks of age) through 71 months of age are shown in Table 6–Table 11

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Table 6 Proportion of subjects with IgG antibody concentration $\geq 0.35 \mu\text{g/mL}$ in subjects aged 2 months and 3 months (PPS)

Post primary immunizations Post booster immunization

PCV13-TT PCV13-TT % PCV13-TT PCV13-TT % % %

Pevnar (3m) % % Pevnar (3m)

(3m) (2m) (PCV13-TT (3m) (2m) (PCV13-TT (PCV13-TT (PCV13-TT (PCV13-TT (PCV13-TT

Serotypes N=471 N=487 N=482 [2m] – N=475 N=487 N=476 [3m] - [2m] – TT [2m] –

[3m] - Pevnar [2m] – Pevnar

PCV13-TT Pevnar Pevnar PCV13-TT

% % % [3m]] [3m]] % % %

[3m] [3m]] [3m]] [3m]]

(95%CI) (95%CI) (95%CI) (CI*) (CI*) (95%CI) (95%CI) (95%CI)

(CI*) (CI*) (CI*) (CI*)

100.0 99.18 100.0 0.82 0.82 0.00 99.79 99.79 99.37 -0.01 -0.42 -0.42

4

(99.22-100.0) (97.91-99.78) (99.24-100.0) (-0.10,1.74) (-0.10,1.74) (-0.93,0.91) (98.83-99.99)

(98.86-99.99) (98.17-99.87) (-0.66,0.65) (-1.36,0.51) (-1.36,0.52)

90.45 98.97 97.72 -8.53 -1.26 7.27 100.0 99.79 100.0 0.21 0.21 0.00

6B

(87.42-92.95) (97.62-99.67) (95.95-98.86) (-11.73,-5.32) (-3.09,0.58) (3.87,10.67) (99.23-100.0)

(98.86-99.99) (99.23-100.0) (-0.25,0.67) (-0.25,0.67) (-0.92,0.92)

99.15 99.79 99.79 -0.64 -0.00 0.64 100.0 100.0 100.0 0.00 0.00 0.00

9V

(97.84-99.77) (98.86-99.99) (98.85-99.99) (-1.70,0.41) (-0.66,0.65) (-0.41,1.70) (99.23-100.0)

(99.25-100.0) (99.23-100.0) (-0.90,0.92) (-0.90,0.92) (-0.92,0.92)

100.0 99.79 99.79 0.21 -0.00 -0.21 99.79 100.0 100.0 -0.21 0.00 0.21

Common 14

(99.22-100.0) (98.86-99.99) (98.85-99.99) (-0.25,0.67) (-0.66,0.65) (-0.67,0.26) (98.83-99.99)

(99.25-100.0) (99.23-100.0) (-0.68,0.26) (-0.90,0.92) (-0.26,0.68)

98.30 97.74 98.96 0.56 1.22 0.66 99.79 99.59 99.58 0.20 -0.01 -0.21

18C

(96.68-99.26) (95.99-98.87) (97.60-99.66) (-1.45,2.57) (-0.61,3.05) (-1.03,2.35) (98.83-99.99)

(98.52-99.95) (98.49-99.95) (-0.60,1.00) (-0.94,0.92) (-1.02,0.61)

100.0 100.0 100.0 0.00 0.00 0.00 100.0 100.0 100.0 0.00 0.00 0.00

19F

(99.22-100.0) (99.25-100.0) (99.24-100.0) (-0.90,0.93) (-0.90,0.91) (-0.93,0.91) (99.23-100.0)

(99.25-100.0) (99.23-100.0) (-0.90,0.92) (-0.90,0.92) (-0.92,0.92)
97.24 98.56 99.79 -1.32 1.23 2.55 100.0 99.79 99.79 0.21 -0.00 -0.21
23F
(95.33-98.52) (97.06-99.42) (98.85-99.99) (-3.40,0.76) (-0.07,2.52) (0.80,4.31) (99.23-100.0)
(98.86-99.99) (98.84-99.99) (-0.25,0.67) (-0.66,0.65) (-0.68,0.26)
99.58 57.49 100.0 42.08 42.51 0.42 99.79 91.38 99.79 8.41 8.41 0.00
1
(98.47-99.95) (52.97-61.93) (99.24-100.0) (37.65,46.51) (38.11,46.90) (-0.25,1.10) (98.83-99.99)
(88.52-93.71) (98.84-99.99) (5.89,10.94) (5.89,10.94) (-0.67,0.67)
98.09 45.17 96.06 52.91 50.88 -2.03 99.16 92.20 98.95 6.96 6.75 -0.21
3
Additiona (96.40-99.12) (40.69-49.72) (93.91-97.61) (48.32,57.50) (46.13,55.63) (-4.47,0.41)
(97.86-99.77) (89.45-94.42) (97.57-99.66) (4.44,9.48) (4.20,9.30) (-1.62,1.20)
I 98.51 16.63 90.87 81.88 74.24 -7.64 97.89 40.04 97.27 57.85 57.23 -0.63
5
(vs. (96.96-99.40) (13.43-20.24) (87.94-93.29) (78.40,85.36) (70.05,78.43) (-10.84,-4.45) (96.16-98.99)
(35.66-44.55) (95.38-98.54) (53.31,62.39) (52.64,61.82) (-2.86,1.61)
correspon 96.82 78.44 99.17 18.38 20.73 2.35 99.58 97.74 99.58 1.84 1.84 0.00
6A
ding (94.80-98.21) (74.52-82.01) (97.89-99.77) (14.39,22.36) (16.99,24.47) (0.32,4.39) (98.49-99.95)
(95.99-98.87) (98.49-99.95) (0.40,3.28) (0.40,3.28) (-0.94,0.94)
serotypes) 99.58 24.23 99.59 75.35 75.36 0.01 100.0 66.74 100.0 33.26 33.26 0.00
7F
(98.47-99.95) (20.49-28.29) (98.51-99.95) (71.49,79.20) (71.51,79.20) (-0.93,0.95) (99.23-100.0)
(62.36-70.91) (99.23-100.0) (29.08,37.45) (29.08,37.45) (-0.92,0.92)
100.0 97.13 100.0 2.87 2.87 0.00 100.0 99.79 100.0 0.21 0.21 0.00
19A
(99.22-100.0) (95.22-98.42) (99.24-100.0) (1.39,4.36) (1.39,4.36) (-0.93,0.91) (99.23-100.0)
(98.86-99.99) (99.23-100.0) (-0.20,0.61) (-0.20,0.61) (-0.92,0.92)
99.58 97.74 1.83 2.26 99.79 99.59 0.20 0.20
Additiona 1 \ \ \ \ \
(98.47-99.95) (95.99-98.87) (0.18,3.49) (0.94,3.58) (98.83-99.99) (98.52-99.95) (-0.60,1.00) (-0.50,0.90)
I 98.09 97.74 0.35 -1.68 99.16 99.59 -0.43 -0.64
(vs. 3 \ \ \ \ \
(96.40-99.12) (95.99-98.87) (-1.72,2.42) (-3.86,0.50) (97.86-99.77) (98.52-99.95) (-1.57,0.71)
(-1.72,0.44)
serotype
98.51 97.74 0.77 -6.87 97.89 99.59 -1.69 -2.32
with the 5 \ \ \ \ \
(96.96-99.40) (95.99-98.87) (-1.19,2.73) (-9.76,-3.98) (96.16-98.99) (98.52-99.95) (-3.31,-0.08)
(-3.89,-0.75)
lowest
96.82 97.74 -0.93 1.43 99.58 99.59 -0.01 -0.01
response 6A \ \ \ \ \
(94.80-98.21) (95.99-98.87) (-3.29,1.43) (-0.12,2.98) (98.49-99.95) (98.52-99.95) (-0.94,0.92)
(-0.82,0.80)
among all
7 99.58 97.74 1.83 1.84 100.0 99.59 0.41 0.41
7F \ \ \ \ \
common (98.47-99.95) (95.99-98.87) (0.18,3.49) (0.40,3.28) (99.23-100.0) (98.52-99.95) (-0.24,1.06)
(-0.16,0.98)
serotypes) 100.0 97.74 2.26 2.26 100.0 99.59 0.41 0.41
19A \ \ \ \ \
(99.22-100.0) (95.99-98.87) (0.75,3.77) (0.94,3.58) (99.23-100.0) (98.52-99.95) (-0.24,1.06) (-0.16,0.98)
Note: PCV13-TT (2m): PCV13-TT recipients 2 months of age; Pevnar (3m): Pevnar recipients 3 months of
age; PCV13-TT (3m): PCV13-TT recipients 3 months of age; N: subjects included in PPS; %: IgG antibody
positive rate; CI*: 2-sided
97.5% CI for common serotypes, 2-sided 97.5% CI for additional serotypes (vs. serotype with the lowest
response among all 7 common serotypes), 2-sided 95% CI for additional serotypes (vs. corresponding
serotypes).

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Table 7 IgG antibody GMC in subjects aged 2 months and 3 months (PPS)
Post primary immunizations Post booster immunization
PCV13-TT PCV13-TT Ratio Ratio Ratio PCV13-TT PCV13-TT Ratio Ratio Ratio
Pevnar (3m) Pevnar (3m)
(3m) (2m) (PCV13-TT (PCV13-TT (PCV13-TT (3m) (2m) (PCV13-TT (PCV13-TT (PCV13-TT
Serotypes
N=471 N=487 N=482 [3m]/Pevnar [2m]/Pevnar [2m]/PCV13 N=475 N=487 N=476 [3m]/Pevnar [2m]/Pevnar
[2m]/PCV13
GMC GMC GMC [3m]) [3m]) -TT [3m]) GMC GMC GMC [3m]) [3m]) -TT [3m])
(95%CI) (95%CI) (95%CI) (CI*) (CI*) (CI*) (95%CI) (95%CI) (95%CI) (CI*) (CI*) (CI*)
4.51 7.79 2.43 0.58 0.31 0.54 3.19 4.27 2.07 0.75 0.48 0.65
4
(4.20-4.84) (7.22-8.41) (2.28-2.60) (0.51,0.65) (0.28,0.35) (0.48,0.60) (2.96-3.44) (3.98-4.59)
(1.92-2.22) (0.66,0.84) (0.43,0.54) (0.58,0.73)
1.25 3.52 3.72 0.36 1.06 2.97 10.01 11.70 8.60 0.86 0.73 0.86
6B
(1.11-1.42) (3.25-3.81) (3.37-4.10) (0.30,0.42) (0.91,1.22) (2.47,3.56) (9.30-10.79) (10.78-12.70)
(7.93-9.32) (0.75,0.97) (0.64,0.84) (0.76,0.97)
3.72 4.47 3.66 0.83 0.82 0.98 4.22 3.87 3.83 1.09 0.99 0.91
9V
(3.45-4.02) (4.19-4.78) (3.40-3.94) (0.74,0.93) (0.73,0.92) (0.87,1.11) (3.90-4.56) (3.62-4.14)
(3.56-4.12) (0.97,1.23) (0.88,1.11) (0.80,1.03)
10.38 16.53 10.25 0.63 0.62 0.99 13.97 17.89 11.61 0.78 0.65 0.83
Common 14
(9.57-11.27) (15.09-18.11) (9.46-11.11) (0.55,0.72) (0.54,0.71) (0.87,1.13) (12.94-15.09) (16.77-19.09)
(10.85-12.44) (0.70,0.88) (0.58,0.72) (0.74,0.93)
2.91 4.80 2.04 0.61 0.43 0.70 2.44 3.90 2.04 0.63 0.52 0.83
18C

19F
(98.03-100.0) (98.07-100.0) (98.07-100.0) (98.07-100.0) (98.15-100.0) (98.16-100.0) (98.04-100.0)
(98.07-100.0)
97.30 96.83 99.47 99.47 98.99 97.99 100.0 100.0
23F
(93.81-99.12) (93.22-98.83) (97.09-99.99) (97.09-99.99) (96.40-99.88) (94.93-99.45) (98.04-100.0)
(98.07-100.0)
98.92 75.66 100.0 70.37 98.48 66.33 100.0 82.54
1
(96.15-99.87) (68.90-81.60) (98.07-100.0) (63.31-76.78) (95.64-99.69) (59.31-72.86) (98.04-100.0)
(76.36-87.67)
97.84 61.90 98.41 47.62 88.89 44.72 98.92 68.78
3
(94.56-99.41) (54.57-68.86) (95.43-99.67) (40.32-54.99) (83.66-92.90) (37.69-51.92) (96.17-99.87)
(61.65-75.31)
91.35 44.44 96.83 53.44 98.48 63.82 93.01 40.74
Additional 5
(86.34-94.98) (37.23-51.83) (93.22-98.83) (46.06-60.71) (95.64-99.69) (56.72-70.50) (88.34-96.23)
(33.67-48.11)
(vs. corresponding
95.68 77.78 98.41 93.65 96.46 91.96 98.39 93.12
serotypes) 6A
(91.66-98.11) (71.17-83.49) (95.43-99.67) (89.17-96.68) (92.85-98.57) (87.27-95.33) (95.36-99.67)
(88.53-96.29)
99.46 46.03 100.0 71.43 98.99 71.36 100.0 73.02
7F
(97.03-99.99) (38.77-53.42) (98.07-100.0) (64.42-77.75) (96.40-99.88) (64.54-77.53) (98.04-100.0)
(66.09-79.20)
100.0 97.88 100.0 99.47 100.0 99.50 100.0 100.0
19A
(98.03-100.0) (94.67-99.42) (98.07-100.0) (97.09-99.99) (98.15-100.0) (97.23-99.99) (98.04-100.0)
(98.07-100.0)
98.92 96.83 100.0 98.41 98.48 97.99 100.0 99.47
1
(96.15-99.87) (93.22-98.83) (98.07-100.0) (95.43-99.67) (95.64-99.69) (94.93-99.45) (98.04-100.0)
(97.09-99.99)
97.84 96.83 98.41 98.41 88.89 97.99 98.92 99.47
3
(94.56-99.41) (93.22-98.83) (95.43-99.67) (95.43-99.67) (83.66-92.90) (94.93-99.45) (96.17-99.87)
(97.09-99.99)
Additional
91.35 96.83 96.83 98.41 98.48 97.99 93.01 99.47
(vs. serotype with the 5
(86.34-94.98) (93.22-98.83) (93.22-98.83) (95.43-99.67) (95.64-99.69) (94.93-99.45) (88.34-96.23)
(97.09-99.99)
lowest response
95.68 96.83 98.41 98.41 96.46 97.99 98.39 99.47
among all 7 common 6A
(91.66-98.11) (93.22-98.83) (95.43-99.67) (95.43-99.67) (92.85-98.57) (94.93-99.45) (95.36-99.67)
(97.09-99.99)
serotypes)
99.46 96.83 100.0 98.41 98.99 97.99 100.0 99.47
7F
(97.03-99.99) (93.22-98.83) (98.07-100.0) (95.43-99.67) (96.40-99.88) (94.93-99.45) (98.04-100.0)
(97.09-99.99)
100.0 96.83 100.0 98.41 100.0 97.99 100.0 99.47
19A
(98.03-100.0) (93.22-98.83) (98.07-100.0) (95.43-99.67) (98.15-100.0) (94.93-99.45) (98.04-100.0)
(97.09-99.99)
Note: N: subjects included in PPS; CI*: 2-sided 95% CI for IgG antibody positive rate.

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Table 9 IgG antibody GMC in subjects aged 7 through 71 months (PPS)
Post primary immunizations Post booster immunization
7-11 months 12-23 months 24-71 months 7-11 months
PCV13-TT Prevnar PCV13-TT Prevnar PCV13-TT Prevnar PCV13-TT Prevnar
Serotypes
N=185 N=189 N=189 N=189 N=198 N=199 N=186 N=189
GMC GMC GMC GMC GMC GMC
GMC(CI*) GMC(CI*)
(CI*) (CI*) (CI*) (CI*) (CI*) (CI*)
4.26 4.23 5.01 5.08 8.90 8.58 3.06 3.12
4
(3.86-4.71) (3.88-4.62) (4.60-5.45) (4.63-5.58) (7.95-9.96) (7.63-9.66) (2.76-3.38) (2.85-3.41)
2.90 2.20 5.42 2.71 7.75 3.63 5.62 4.57
6B
(2.46-3.42) (1.92-2.52) (4.86-6.04) (2.35-3.11) (6.58-9.12) (3.04-4.32) (5.01-6.32) (4.08-5.11)
2.38 2.76 2.97 2.95 3.84 3.94 2.91 3.67
9V
(2.09-2.71) (2.43-3.12) (2.68-3.30) (2.64-3.30) (3.35-4.41) (3.36-4.62) (2.58-3.29) (3.31-4.07)
8.54 16.17 9.20 14.98 10.20 15.75 13.94 19.49
Common 14
(7.60-9.60) (14.53-17.99) (8.28-10.22) (13.51-16.61) (8.40-12.38) (12.82-19.34) (12.54-15.51)
(17.94-21.17)
3.78 3.44 5.03 3.92 8.82 7.47 2.97 3.32
18C
(3.30-4.34) (3.08-3.84) (4.63-5.46) (3.56-4.31) (7.56-10.29) (6.45-8.66) (2.62-3.36) (3.00-3.67)
7.21 4.09 8.80 4.20 11.45 5.71 7.54 3.92
19F
(6.43-8.08) (3.70-4.52) (8.10-9.55) (3.79-4.65) (10.05-13.04) (4.98-6.54) (6.84-8.31) (3.56-4.31)

3.56 2.70 4.32 2.96 4.42 4.29 4.53 4.21
23F
(3.10-4.10) (2.36-3.09) (3.89-4.79) (2.61-3.36) (3.80-5.14) (3.56-5.17) (4.01-5.12) (3.77-4.71)
2.23 0.47 2.25 0.42 2.06 0.31 1.98 0.58
1
(2.03-2.45) (0.40-0.56) (2.07-2.44) (0.35-0.50) (1.84-2.30) (0.26-0.39) (1.82-2.15) (0.53-0.65)
1.32 0.46 1.06 0.39 0.84 0.38 1.51 0.53
3
(1.20-1.45) (0.41-0.51) (0.98-1.14) (0.34-0.45) (0.75-0.94) (0.33-0.45) (1.38-1.66) (0.48-0.59)
0.82 0.19 1.01 0.27 1.60 0.31 0.86 0.24
Additional 5
(0.73-0.91) (0.15-0.25) (0.91-1.12) (0.22-0.32) (1.43-1.79) (0.25-0.38) (0.78-0.95) (0.20-0.29)
(vs. corresponding
1.79 0.54 2.46 1.17 2.35 1.29 2.99 1.63
serotypes) 6A
(1.51-2.12) (0.41-0.72) (2.08-2.92) (1.01-1.35) (2.00-2.76) (1.05-1.60) (2.64-3.37) (1.43-1.87)
5.73 0.14 5.47 0.41 4.60 0.43 4.91 0.46
7F
(5.04-6.51) (0.10-0.20) (5.04-5.94) (0.33-0.52) (4.00-5.28) (0.33-0.56) (4.46-5.39) (0.39-0.55)
5.06 1.50 7.81 2.67 9.56 3.32 6.28 2.17
19A
(4.54-5.64) (1.29-1.74) (7.08-8.61) (2.28-3.12) (8.47-10.79) (2.85-3.88) (5.71-6.91) (1.96-2.40)
2.23 2.20 2.25 2.71 2.06 3.63 1.98 3.12
1
(2.03-2.45) (1.92-2.52) (2.07-2.44) (2.35-3.11) (1.84-2.30) (3.04-4.32) (1.82-2.15) (2.85-3.41)
1.32 2.20 1.06 2.71 0.84 3.63 1.51 3.12
3
(1.20-1.45) (1.92-2.52) (0.98-1.14) (2.35-3.11) (0.75-0.94) (3.04-4.32) (1.38-1.66) (2.85-3.41)
Additional
0.82 2.20 1.01 2.71 1.60 3.63 0.86 3.12
(vs. serotype with the 5
(0.73-0.91) (1.92-2.52) (0.91-1.12) (2.35-3.11) (1.43-1.79) (3.04-4.32) (0.78-0.95) (2.85-3.41)
lowest response
1.79 2.20 2.46 2.71 2.35 3.63 2.99 3.12
among all 7 common 6A
(1.51-2.12) (1.92-2.52) (2.08-2.92) (2.35-3.11) (2.00-2.76) (3.04-4.32) (2.64-3.37) (2.85-3.41)
serotypes)
5.73 2.20 5.47 2.71 4.60 3.63 4.91 3.12
7F
(5.04-6.51) (1.92-2.52) (5.04-5.94) (2.35-3.11) (4.00-5.28) (3.04-4.32) (4.46-5.39) (2.85-3.41)
5.06 2.20 7.81 2.71 9.56 3.63 6.28 3.12
19A
(4.54-5.64) (1.92-2.52) (7.08-8.61) (2.35-3.11) (8.47-10.79) (3.04-4.32) (5.71-6.91) (2.85-3.41)
Note: N: subjects included in PPS; CI*: 2-sided 95% CI for IgG GMC.

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Table 10 Proportion of subjects with OPA antibody titer ≥1:8 in all age stratification (PPS)
Post primary immunizations Post booster immunization
3 months 2 months 7-11 months 12-23 months 24-71 months 3 months 2 months 7-11 months
Serotypes PCV13-TT Prevnar PCV13-TT PCV13-TT Prevnar PCV13-TT Prevnar PCV13-TT Prevnar PCV13-TT Prevnar
PCV13-TT PCV13-TT Prevnar
N=93 N=96 N=97 N=96 N=95 N=92 N=93 N=99 N=100 N=96 N=97 N=92 N=93 N=95
%(CI*) %(CI*) %(CI*) %(CI*) %(CI*) %(CI*) %(CI*) %(CI*) %(CI*) %(CI*) %(CI*) %(CI*) %(CI*)
100.0 98.97 100.0 100.0 100.0 100.0 100.0 96.00 100.0 100.0 100.0 100.0 100.0
100.0 100.0
4 (96.23- (94.39- (96.23- (96.19- (96.07- (96.11- (96.34- (90.07- (96.23- (96.27- (96.07- (96.11-
(96.11-100.0) (96.19-100.0)
100.0) 99.97) 100.0) 100.0) 100.0) 100.0) 100.0) 98.90) 100.0) 100.0) 100.0) 100.0)
98.96 92.78 81.25 89.47 94.57 91.40 97.98 98.00 98.96 97.94 100.0 93.55
89.25 95.79
6B (94.33- (85.70- (72.00- (81.49- (87.77- (83.75- (92.89- (92.96- (94.33- (92.75- (96.07- (86.48-
(81.11-94.72) (89.57-98.84)
99.97) 97.05) 88.49) 94.84) 98.21) 96.21) 99.75) 99.76) 99.97) 99.75) 100.0) 97.60)
100.0 100.0 100.0 100.0 100.0 100.0 100.0 100.0 100.0 100.0 100.0 100.0
100.0 100.0
9V (96.23- (96.27- (96.23- (96.19- (96.07- (96.11- (96.34- (96.38- (96.23- (96.27- (96.07- (96.11-
(96.11-100.0) (96.19-100.0)
100.0) 100.0) 100.0) 100.0) 100.0) 100.0) 100.0) 100.0) 100.0) 100.0) 100.0) 100.0)
100.0 100.0 98.96 97.89 100.0 100.0 92.93 95.00 100.0 100.0 100.0 100.0
100.0 100.0
Common 14 (96.23- (96.27- (94.33- (92.60- (96.07- (96.11- (85.97- (88.72- (96.23- (96.27- (96.07- (96.11-
(96.11-100.0) (96.19-100.0)
100.0) 100.0) 99.97) 99.74) 100.0) 100.0) 97.11) 98.36) 100.0) 100.0) 100.0) 100.0)
98.96 100.0 98.96 98.95 100.0 100.0 100.0 99.00 100.0 98.97 100.0 100.0
100.0 98.95
18C (94.33- (96.27- (94.33- (94.27- (96.07- (96.11- (96.34- (94.55- (96.23- (94.39- (96.07- (96.11-
(96.11-100.0) (94.27-99.97)
99.97) 100.0) 99.97) 99.97) 100.0) 100.0) 100.0) 99.97) 100.0) 99.97) 100.0) 100.0)
97.92 98.97 98.96 91.58 98.91 92.47 98.99 94.00 100.0 96.91 100.0 100.0
100.0 93.68
19F (92.68- (94.39- (94.33- (84.08- (94.09- (85.10- (94.50- (87.40- (96.23- (91.23- (96.07- (96.11-
(96.11-100.0) (86.76-97.65)
99.75) 99.97) 99.97) 96.29) 99.97) 96.92) 99.97) 97.77) 100.0) 99.36) 100.0) 100.0)
98.96 95.88 91.67 94.74 98.91 96.77 96.97 99.00 98.96 98.97 100.0 97.85
95.70 95.79
23F (94.33- (89.78- (84.24- (88.14- (94.09- (90.86- (91.40- (94.55- (94.33- (94.39- (96.07- (92.45-
(89.35-98.82) (89.57-98.84)
99.97) 98.87) 96.33) 98.27) 99.97) 99.33) 99.37) 99.97) 99.97) 99.97) 100.0) 99.74)
0.00 91.75 92.71 5.26 88.04 43.43 97.92 4.12 95.65 94.62
83.87 3.23 1.00 6.32

1 (0.00- (84.39- (85.55- (1.73- (79.61- (33.50- (92.68- (1.13- (89.24- (87.90- (74.80-90.68) (0.67-9.14) (0.03-5.45) (2.35-13.24) 3.77) 96.37) 97.02) 11.86) 93.88) 53.77) 99.75) 10.22) 98.80) 98.23) 15.63 100.0 100.0 46.32 100.0 75.27 100.0 48.00 97.92 37.11 100.0 100.0 96.77 56.84 3 (9.02- (96.27- (96.23- (36.02- (96.07- (65.24- (96.34- (37.90- (92.68- (27.52- (96.07- (96.11- (90.86-99.33) (46.28-66.97) 24.46) 100.0) 100.0) 56.85) 100.0) 83.63) 100.0) 58.22) 99.75) 47.52) 100.0) 100.0) Additiona 4.17 100.0 97.92 7.37 98.91 95.96 100.0 5.15 100.0 98.92 98.92 7.53 9.00 5.26 I 5 (1.15- (96.27- (92.68- (3.01- (94.09- (89.98- (96.23- (1.69- (96.07- (94.15- (94.15-99.97) (3.08-14.90) (4.20-16.40) (1.73-11.86) (vs. 10.33) 100.0) 99.75) 14.59) 99.97) 98.89) 100.0) 11.62) 100.0) 99.97) correspon 80.21 97.94 89.58 65.26 96.74 78.49 94.95 95.00 95.83 92.78 97.83 95.70 97.85 81.05 ding 6A (70.83- (92.75- (81.68- (54.80- (90.77- (68.76- (88.61- (88.72- (89.67- (85.70- (92.37- (89.35- (92.45-99.74) (71.72-88.37) serotypes) 87.64) 99.75) 94.89) 74.74) 99.32) 86.34) 98.34) 98.36) 98.85) 97.05) 99.74) 98.82) 37.50 100.0 100.0 35.79 100.0 49.46 100.0 75.00 100.0 52.58 98.91 100.0 100.0 42.11 7F (27.82- (96.27- (96.23- (26.21- (96.07- (38.93- (96.34- (65.34- (96.23- (42.18- (94.09- (96.11- (96.11-100.0) (32.04-52.67) 47.97) 100.0) 100.0) 46.28) 100.0) 60.03) 100.0) 83.12) 100.0) 62.81) 99.97) 100.0) 44.79 100.0 100.0 78.95 100.0 88.17 100.0 92.00 100.0 79.38 100.0 100.0 100.0 78.95 19A (34.63- (96.27- (96.23- (69.38- (96.07- (79.82- (96.34- (84.84- (96.23- (69.97- (96.07- (96.11- (96.11-100.0) (69.38-86.64) 55.29) 100.0) 100.0) 86.64) 100.0) 93.95) 100.0) 96.48) 100.0) 86.93) 100.0) 100.0) Note: N: subjects included in PPS; CI*: 2-sided 95% CI for OPA antibody positive rate.

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Table 11 OPA antibody GMT in subjects of all age stratification (PPS)											
Post primary immunizations Post booster immunization											
3 months	2 months	7-11 months	12-23 months	24-71 months	3 months	2 months	7-11 months				
Serotypes	PCV13-TT	Prevnar	PCV13-TT	Prevnar	PCV13-TT	Prevnar	PCV13-TT	Prevnar	PCV13-TT	Prevnar	PCV13-TT
PCV13-TT	PCV13-TT	Prevnar									
N=93	N=96	N=97	N=96	N=95	N=92	N=93	N=99	N=100	N=96	N=97	N=92
N=95	GMT(CI*)	GMT(CI*)	GMT(CI*)	GMT(CI*)	GMT(CI*)	GMT(CI*)	GMT(CI*)	GMT(CI*)	GMT(CI*)	GMT(CI*)	GMT(CI*)
GMT(CI*)	GMT(CI*)	GMT(CI*)									
1227.93	2291.32	1214.42	1014.62	1214.37	1756.58	1993.15	2710.14	1924.49	6182.37	6208.15	3440.10
1549.33	1465.72	4 (992.5-	(1947.8-	(940.3-	(866.1-	(1022.4-	(1426.1-	(1667.7-	(2176.9-	(1331.2-	(4931.4-
(4950.8-	(2690.0-	(1288.7-	(1253.1-	1519.2)	2695.4)	1568.5)	1188.7)	1442.4)	2163.6)	2382.2)	3374.1)
2782.2)	7750.7)	7784.8)	4399.3)	1862.7)	1714.4)	528.67	2928.36	811.17	237.34	599.57	923.67
1530.14	2511.69	2511.16	2661.93	7564.42	1948.21	890.24	1843.95	6B (329.58-	(2233.9-	(542.0-	(142.61-
(373.97-	(611.9-	(921.8-	(1829.2-	(1760.5-	(1948.9-	(5345-	(1460.3-	(599.4-	(1265.7-	848.02)	3838.7)
1214.1)	395.00)	961.29)	1394.4)	2539.9)	3448.8)	3581.9)	3635.9)	10705)	2599.2)	1322.2)	2686.4)
1922.48	1974.62	2192.74	1425.78	1692.48	3219.64	2824.34	2174.68	1980.18	6184.75	5484.79	3719.77
2075.15	2264.80	9V (1518.9-	(1612.0-	(1728.3-	(1153.3-	(1371.4-	(2417.8-	(2057.1-	(1710.1-	(1547.4-	(4870.3-
(4453.1-	(2889.7-	(1697.0-	(1886.3-	2433.2)	2418.9)	2781.9)	1762.6)	2088.7)	4287.4)	3877.8)	2765.4)
2534.1)	7854.0)	6755.5)	4788.3)	2537.6)	2719.2)	4511.21	5594.48	3571.78	2538.79	3060.32	6373.82
7303.79	2534.19	3395.54	7686.96	6221.93	6173.16	3761.98	2883.34	Common 14	(3494.9-	(4188.4-	(2833.6-
(1929.1-	(2260.7-	(5041.8-	(5760.0-	(1586.7-	(2193.4-	(6160.6-	(4855.7-	(4922.7-	(3002.8-	(2359.5-	5823.1)
7472.5)	4502.3)	3341.2)	4142.7)	8057.8)	9261.3)	4047.4)	5256.5)	9591.5)	7972.5)	7741.3)	4713.1)
3523.5)	1356.26	3129.49	518.67	952.30	1157.93	3133.89	3971.64	3572.76	3330.72	(4173.73	5507.66
1117.88	1491.46	1677.92	18C (1094.5-	(2353.5-	(417.13-	(748.0-	(908.2-	(2557.9-	(3197.8-	(2952.0-	2548.4-
(3269.4-	(4094.6-	(899.5-	(1230.2-	(1331.9-	1680.7)	4161.3)	644.92)	1212.4)	1476.4)	3839.5)	4932.8)
4324.1)	4353.1)	5328.2)	7408.5)	1389.3)	1808.2)	2146.1)	652.57	540.56	769.17	518.08	240.27
816.50	331.18	1133.78	333.59	2359.26	815.68	1966.07	751.55	258.60	19F (547.97-	(423.02-	(617.10-
(412.37-	(170.50-	(653.5-	(230.91-	(876.5-	(233.90-	(1898.8-	(600.1-	(1496.8-	(635.68-	(190.65-	777.14)
690.76)	958.72)	650.89)	338.60)	1020.2)	474.99)	1466.6)	475.75)	2931.3)	1108.8)	2582.5)	888.54)
350.77)	2090.78	5939.64	1072.95	669.00	1699.27	2069.47	3784.23	2025.10	3468.13	4943.66	16131.84
2529.55	1577.42	3607.17	23F (1452.4-	(4465.8-	(738.0-	(444.5-	(1116.1-	(1616.3-	(2584.5-	(1465.3-	(2544.3-
(3817.5-	(11742-	(1959.0-	(1197.5-	(2428.0-	3009.7)	7899.8)	1560.0)	1006.8)	2587.2)	2649.7)	5540.9)
2798.7)	4727.4)	6402.1)	22163)	3266.3)	2077.9)	5359.1)	62.13	92.51	56.40	42.36	321.14
158.59	87.89	2.00	2.24	2.23	8.16	2.07	2.28	2.34	1 (42.64-	(66.48-	(43.54-
(31.65-	(239.43-	(115.59-	(67.11-	(2.00-2.00)	(2.01-2.50)	(1.97-2.52)	(5.78-11.52)	(1.94-2.21)	(2.00-2.61)	(2.06-2.67)	90.51)
128.75)	73.05)	56.70)	430.74)	217.59)	115.11)	511.70	575.94	316.46	361.20	14.09	121.39
678.98	586.01	431.54	2.80	6.62	9.67	5.03	8.12	3 (376.29-	(444.43-	(264.28-	(304.70-
(10.43-	(101.55-	(523.20-	(440.25-	(359.88-	(2.35-3.33)	(4.88-8.97)	(6.60-14.16)	(3.77-6.72)	(6.04-10.91)	695.83)	746.38)
378.94)	428.17)	19.04)	145.11)	881.14)	780.04)	517.48)	450.24	411.73	115.18	192.91	84.80
1375.13	1027.83	222.49									

Additional 2.24 2.43 2.80 2.51 2.61 2.41
5 (358.12- (317.12- (90.52- (155.85- (64.41- (1061.9- (796.4- (173.71-
(vs. (1.99-2.51) (2.10-2.82) (2.17-3.62) (2.16-2.93) (2.07-3.29) (2.02-2.87)
566.05) 534.56) 146.55) 238.79) 111.65) 1780.8) 1326.6) 284.98)
correspondi
1079.96 275.54 1202.17 442.24 95.37 1434.20 487.96 1737.99 970.59 3254.75 1562.59 2546.12 1234.03
ng 416.90(234.
6A (792.9- (159.69- (895.0- (286.46- (51.43- (1009.8- (251.04- (1185.4- (655.6- (2221.0- (972.9- (1780.8-
(866.0-
serotypes) 51-741.16)
1471.0) 475.44) 1614.8) 682.71) 176.86) 2037.0) 948.50) 2548.1) 1437.0) 4769.7) 2509.6) 3640.3) 1758.5)
7741.10 12.76 2414.63 3589.60 5524.63 32.24 5886.86 171.55 7244.42 41.47 2616.75 4276.87 21.71
16.04
7F (6383.8- (7.68- (1872.4- (2955.9- (4317.5- (17.72- (4802.3- (98.79- (6006.0- (22.27- (1938.6- (3488.6-
(12.04-
(8.94-28.80)
9387.0) 21.19) 3113.9) 4359.2) 7069.2) 58.68) 7216.3) 297.89) 8738.3) 77.21) 3532.1) 5243.3) 39.14)
777.70 11.50 659.70 692.99 57.76 1368.75 166.43 2070.66 303.83 5503.11 72.99 2546.03 1141.93 71.60
19A (638.69- (7.55- (532.25- (565.53- (38.19- (1110.8- (110.60- (1638.8- (200.97- (4501.2- (46.46-
(2030.4- (928.0- (45.99-
946.97) 17.51) 817.66) 849.18) 87.34) 1686.6) 250.45) 2616.3) 459.33) 6728.1) 114.65) 3192.6) 1405.2)
111.47)
Note: N: subjects included in PPS; CI*: 2-sided 95% CI for OPA antibody GMT.

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5.2 Pharmacokinetic properties
Not applicable.

5.3 Preclinical safety data
The evaluation results showed that, the dosage of the vaccine is within a safety margin; minor irritation at injection site is observed after use in animals, while may resolve after discontinuation of treatment; in addition, there are no apparent abnormalities regarding the general conditions, body temperature, body weight, food consumption, blood biochemistry, hematology, urine, bone marrow, ophthalmology, electrocardiogram, organ weight, pathological findings of tissues and other measures, no obviously increased or inhibited immune function is noted; specific antibody to vaccine components can be detected; risk of anaphylaxis is present due to the use of carrier protein; agglutination effect is observed in rabbit red blood cells, while no hemolytic reaction is seen.
The pharmacological and toxicological tests studies of the vaccine products demonstrate excellent safety and immunogenicity, and thus can be safely administered clinically. The results of animal tests were shown in Table 12 to Table 15

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Yuxi Walvax Biotechnology Co., Ltd.

Table 12. Summary of Acute Toxicity Test and Abnormal Toxicity Test.
Observation Results Animal
Test Study Undertaker Duration
measurement Species
New
Yunnan Walvax 12
Immunogenicity Zealand
R&D Center months
white rabbits
During the 14-day observation after administration, the mice in each group had normal physical appearance, mental and motor state. The natural pores General conditions were clean. No obvious abnormal reactions were observed. No irritant reactions such as redness, swelling, ecchymosis, induration, suppuration and fester were observed.
Body weight and No abnormal changes were observed in body weight and food intake for National Chengdu Kunming
food intake mice in each group 14 days after administration.
Acute Toxicity Center for Safety 4 months species small
hematological and
On the 15th day of the experiment, no abnormal results were observed in Evaluation of Drugs mice biochemical
the hematological and biochemical measurements for mice in each group.
examination
On the 15th day of the experiment, no abnormal changes in the size, shape, Gross anatomy color and texture of the heart, liver, spleen, lung and kidney were observed results for mice in each group. No local irritant reactions such as swelling and induration in the muscles at the injection site.
All mice had normal physical appearance, mental and motor state. The natural pores were clean. No obvious abnormal reactions were observed
General Conditions
from the time the vaccine was administered until the end of the Mice, observation. Guinea pigs
On the 7th day after administration, the weight of each mouse increased
Weight National Chengdu
Abnormal compared with that before administration.
Center for Safety 2 months
Toxicity All guinea pigs had normal physical appearance, mental and motor state.
Evaluation of Drugs
The natural pores were clean. No obvious abnormal reactions were
General Conditions Guinea pigs
observed from the time the vaccine was administered until the end of the observation.
On the 7th day after administration, the weight of each guinea pig
Weight
increased compared with that before administration.

Table 13 Summary Table of Toxicity Test Results for Carrier Proteins (Tetanus Toxoid)
Specific toxicity test Toxicity irreversibility test
Animals should not have tetanus symptoms; at the end of the
Animals should not have tetanus symptoms; at the end of the experiment, the weight of
Standard experiment, the weight of each animal should increase compared
each animal should increase compared with that before administration.
with that before administration.

Pn1ps-TT 20090501 Up to standard Up to standard
Pn1ps-TT 20090502 Up to standard Up to standard
Pn1ps-TT 20090503 Up to standard Up to standard
Pn3ps-TT20100903
Up to standard Up to standard
Pn4ps-TT 20090201
Pn4ps-TT 20090202 Up to standard Up to standard
Pn4ps-TT 20090203 Up to standard Up to standard
Pn5ps-TT 20090401 Up to standard Up to standard
Pn5ps-TT 20090502 Up to standard Up to standard
Pn5ps-TT 20090503 Up to standard Up to standard
Pn6Aps-TT20101103
Up to standard Up to standard
Pn6Bps-TT20090301
Pn6Bps-TT20090302 Up to standard Up to standard
Pn6Bps-TT20090303 Up to standard Up to standard
Pn7Fps-TT20101003
Up to standard Up to standard
Pn9Vps-TT20090301
Pn9Vps-TT20090302 Up to standard Up to standard
Pn9Vps-TT20090303 Up to standard Up to standard
Pn14ps-TT20090301 Up to standard Up to standard
Pn14ps-TT20090302 Up to standard Up to standard
Pn14ps-TT20090303 Up to standard Up to standard
Pn18Cps-TT20090301 Up to standard Up to standard
Pn18Cps-TT20090402 Up to standard Up to standard
Pn18Cps-TT20090403 Up to standard Up to standard
Pn19Aps-TT20101103
Up to standard Up to standard
Pn19Fps-TT20090401
Pn19Fps-TT20090402 Up to standard Up to standard
Pn19Fps-TT20090403 Up to standard Up to standard
Pn23Fps-TT20090401 Up to standard Up to standard
Pn23Fps-TT20090402 Up to standard Up to standard
Pn23Fps-TT20090403 Up to standard Up to standard

Table 14 Summary Table of Administration Methods for Each Group in Systemic Active
Anaphylaxis Experiment

6 Guinea Pigs/Group (Half Male & Half Female)
Negative Control Group 0.5mL/kg (0.9% Sodium Chloride)
Positive Control Group 10mg/mL (Ovalbumin Solution)
7-Valent PCV Low-dose Group 0.05mL/kg (Bulk)
Group High-dose Group 0.5mL/kg (Bulk)
Low-dose Group 0.05mL/kg (Bulk)
20110304
High-dose Group 0.5mL/kg (Bulk)
Low-dose Group 0.05mL/kg (Bulk)
20110305
High-dose Group 0.5mL/kg (Bulk)
Low-dose Group 0.05mL/kg (Bulk)
20110306
High-dose Group 0.5mL/kg (Bulk)

Table 15 Summary Table of Administration Methods for Each Group in Passive Anaphylaxis
Experiment

2 Guinea Pigs/Group
Negative Control Group 0.5mL/kg (0.9% Sodium Chloride)
Positive Control Group 10mg/mL (Ovalbumin Solution)
7-Valent PCV Low-dose Group 0.05mL/kg (Bulk)
Group High-dose Group 0.5mL/kg (Bulk)
Low-dose Group 0.05mL/kg (Bulk)
20110304
High-dose Group 0.5mL/kg (Bulk)
Low-dose Group 0.05mL/kg (Bulk)
20110305
High-dose Group 0.5mL/kg (Bulk)
Low-dose Group 0.05mL/kg (Bulk)
20110306
High-dose Group 0.5mL/kg (Bulk)

6. Pharmaceutical particulars

6.1 List of excipients

Aluminum phosphate 0.5 mg Adjuvant
Sodium chloride 4.25 mg Excipient
Sodium dihydrogen phosphate 44.35 µg Excipient
Disodium hydrogen phosphate 19.0 µg Excipient
Water for injection Qs to 0.5 mL Excipient

6.2 Incompatibilities

In the absence of compatibility studies, this medicinal product must not be mixed with other

medicinal products.

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Yuxi Walvax Biotechnology Co., Ltd.

6.3 Shelf life

The shelf life of the vaccine is 24 months. Please use before the expiration date printed on the label on vial or small box.

6.4 Special precautions for storage

Store and transport at 2-8°C away from light.

Do not freeze. If the product is frozen, it cannot be used again.

Keep it out of the reach of children.

6.5 Nature and contents of container

0.5 ml suspension for injection in pre-filled syringe, 1 syringe /box.

1 ml vials, 1 vial/box.

6.6 Special precautions for disposal and other handling

During storage, a white deposit and clear supernatant can be observed. This does not constitute a sign of deterioration.

The vaccine should be shaken well to obtain a homogeneous white suspension prior to expelling air from the syringe, and should be inspected visually for any particulate matter and/or variation of physical aspect prior to administration. Do not use if the content appears otherwise.

No special requirements for disposal.

Any unused medicinal product or waste material should be disposed of in accordance with local requirements.

7. Marketing authorisation holder

Yuxi Walvax Biotechnology Co., Ltd.

8. Marketing authorisation number(s)

GuoYaoZhunZi S20190046

9. Date of first authorisation/renewal of the authorisation

Date of first authorisation: 30 December 2019

10. Date of revision of the text

15 June 2020

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Sampling
80°C increases to 121°C by 1-hour heat
cycling, with 1T water at 6.5T/h flow rate
70° increases to 80° by transient heating at
6.5T/h

Sampling

Process PA 3Bar/ PA Inlet
Process PA 3Bar/ PA Inlet

PS Inlet PS Inlet
Thermal Insulation

Sampling Inlet/
Trap Port
(For 3F Conjugation Area) Process PA 3Bar/ PA Inlet Process PA 3Bar/ PA Inlet
PS Inlet PS Inlet
PS Inlet

80°C decreases to 2~8°C, at 600L/h 80°C decreases to 2~8°C, at 600L/h Chilled Water Return Chilled Water
Return
WFI Tank
Chilled Water Return
Refrigerant (25% Refrigerant (25%
Ethylene Glycol) Return Ethylene Glycol) Return

80°C decreases to 40°C, at 1000L/h
80°C decreases to 40°C, at 1000L/h
80°C decreases to 40°C, at 3600 L/h
Sampling

Chilled Water Inlet Chilled Water Inlet
Chilled Water Inlet
Refrigerant (25% Ethylene Glycol) Inlet Refrigerant (25% Ethylene Glycol) Inlet Sampling Inlet/ Sampling
Inlet/ 3WS318 CIP Room
Trap Port Trap Port

Process PA 3Bar/ PA Inlet Process PA 3Bar/ PA Inlet

Process PA 3Bar/ PA Inlet

Sampling Inlet/Trap Port Sampling Inlet/Trap Port

Sampling
2MP305 Sterile Liquor Preparation Room Sampling Inlet/Trap Port

3WS310 Washroom 3WS310/ Washroom Tank

4PN2311 Ultrafiltration Room 4PN1311 Ultrafiltration Room
Mobile Rapid Trap Valve

1MP311 Liquid Preparation Room

Q=15T/h, H=58m

20°C increases to 85°C by 1-hour heat
cycling, with 5T water, at 21T/h flow rate
Heating medium 3 Bar Industrial Steam

Sampling

Reserved

Sampling

(For 3F Pneumonia Sterile Area & Bond Area)

PW Tank

(2F/3F Point-of-Use, Water Distiller, Pure Steam Generator)

PW Tank

Sampling Sampling
Sampling Sampling

Multi-effect Water Distillator Multi-effect Water Distillator Pure Steam Generator Pure Steam Generator

Discharge Discharge Electric Boiler Electric Boiler

Sampling

Sampling

Discharge

Sampling

Sampling Sampling Sampling

(For 3F Pneumonia Non-sterile Area)

(For 2F MEN Non-sterile Area) (For 2F HIB Non-sterile Area)
(For 2F Sterile Area & Conjugation Area)

PW Tank PW Tank
PW Tank
PW Tank

Add: No. 83, South Dongfeng Road, High and New Technology
Industrial Development Zone, Yuxi, 653100, China
E-mail: yingguoqing@walvax.com Tel: 0877 - 2077789
Website: www.walvax.com Fax: 0877 - 2076918

Manufacturing Process Flow Chart and Quality Control of 23-Valent Pneumococcal Polysaccharide Vaccine Bulk

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Add: No. 83, South Dongfeng Road, High and New Technology
Industrial Development Zone, Yuxi, 653100, China
E-mail: yingguoqing@walvax.com Tel: 0877 - 2077789
Website: www.walvax.com Fax: 0877 - 2076918

Filling and packaging process flow chart of 23-valent Pneumococcal Polysaccharide Vaccine

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Manufacturing Process Flow Chart and Quality Control of Group A and C
Meningococcal Polysaccharide vaccine bulk

Strain Amplification Dringking
Water

Fermentation and cultivation

Harvesting fermentation broth Reverse Osmosis

Critical quality
control points and Preparation of raw
items:
Identity test, total solid, polysaccharide
protein content, nucleic
acid content, O-acetyl
content, phosphorus
content (Group A), Preparation of refined Purified
v content,
sialic acid Water
(Group C), molecular polysaccharide
size, residual phenol
content, bacterial
endotoxin
Polysaccharide dissolution

Distillation
Critical quality control Sterilization and
points and items: filtration
Identity ■test, total solid,
protein content, nucleic acid
content, O-acetyl content,
phosphorus content (Group
A), sialic acid content, Polysaccharide
(Group C), molecular size, bulk Water for
residual phenol content,
bacterial endotoxin, sterility injection
test

Grade A Grade B Grade C Grade D Sealed
container

1
Filling and packing process flow chart of Group A and C Meningococcal
Polysaccharide vaccine

Group A Group C
conjugate bulk conjugate bulk
Main quality control points
and item: sterility test

Water Steriliza
Sterili Dilution and tion and Sodium chloride
for
zation preparation filtratio solution, lactose
injection n solution

Filling Sterili Bottle and rubber
zation stopper washing

Lyophilization

Sterili Aluminum cap
Capping zation washing

Light
inspection

Packing
material Packaging
Main quality control points and
items:
Finished Identity, Appearance, Weight
product
variation, Moisture content, pH,
Osmolality, Polysaccharide content,
Warehousing Molecular size distribution of
polysaccharide, Sterility, Abnormal
toxicity, Pyrogen test, Bacterial
endotoxin

Grade A Grade B Grade C Grade D Grade E Sealed
container

2

Curricula vitae for Key personnel

2.2.1.1 Person in charge of enterprise: Shi Jing

Table 2-1 Curricula Vitae for Person in Charge of Enterprise

Name Shi Jing Gender Male Birth Date 1979.01

Biophar
Master's Technical Associate
Education Major maceutic
Degree Title researcher
al

Current
Working
Position General Manager 16 years
Seniority
and Duty

1 Education Background

1997.09-2001.07 Majored in biotechnology, South China Agricultural University

2009.09-2012.07 Majored in biopharmaceutical technology, Kunming University of Science and Technology

2 Working Experience

2003.05-2006.04: Director of Fermentation Center, R & D dept. Walvax

2007.04-2009.04: Production workshop director, Yuxi Walvax

2009.04-2012.01: Deputy Manager, production Management Department, Yuxi Walvax

2012.01-2013.12: Manager of R&D dept. Yuxi Walvax

2012.12-2016.10: General Manager Assistant, Yuxi Walvax

2016.10-2018.01: Vice-General Manager, Yuxi Walvax

2018.02 until now: General Manager, Yuxi Walvax
2.2.1.2 Quality person, head of Quality Management - Fang Guoliang

Table 2-2 Curricula Vitae for Head of Quality Management

Name Fang Guoliang Gender Male Birth Date 1977.02

Senior
Education Master's Degree Major Bioengineering Technical Title
Engineer

Current
Vice General Manager Working
Position 19 years
Head of Quality Management Seniority
and Duty

1 Education Background

1997.09-2001.06: Bachelor's degree from Nanjing Forestry University

2013.09-2016.07: Master's degree from Life Science Academy of Kunming University of Science and Technology

2 Working Experience

2001-2003: Quality testing supervisor and quality management manager in Kunming White Horse Pharmaceutical Co., Ltd of Sanjiu Enterprise Group.

2005.05-2008.07: QA deputy manager of Yuxi Walvax Biotechnology Co., Ltd.

2008.08-2013.05: QA manager of Yuxi Walvax Biotechnology Co., Ltd.

2010.05-2014.01: Quality management manager of Yuxi Walvax Biotechnology Co., Ltd.

2011.12-Now: Quality responsible person of Yuxi Walvax Biotechnology Co., Ltd.

2013.02-Now: Quality person of Yuxi Walvax Biotechnology Co., Ltd.

2014.06-Now: Vice General manager of Yuxi Walvax Biotechnology Co., Ltd.
2.2.1.3 Head of Production Management – Lin Jianxiang

Table 2-3 Curricula Vitae for Head of Production Management

1982
Full name Lin Jianxiang Gender Male Date of birth

Education master major Biotechnology Technical title Engineer

Deputy General
Working
Present positions Manager, Production 16 years
Seniority
Management Personnel

1.Educational Background

1998.09- 2002.07, Honghe Prefecture Health School, Yunnan Province

2006.01-2009.01, China Pharmaceutical University

2011.03-2004 .01, Kunming Medical University

2. Working experience

2005.05- 2008.08, Director of Production Workshop, Yuxi Walvax Biotechnology Co., Ltd.

2008.08-2015.03, Director of sub-Packing Workshop, Yuxi Walvax Biotechnology Co., Ltd.

2015.03-2016.10, Production Management Manager, Yuxi Walvax Biotechnology Co., Ltd.

2016.10 -2018.01, General Manager Assistant, Yuxi Walvax Biotechnology Co., Ltd.

2018.01-now, Deputy General Manager of Yuxi Walvax Biotechnology Co., Ltd.

2.2.1.4 Head of Quality Assurance Department - Wu Xiuli

Table 2-4 Curricula Vitae for Head of Quality Assurance Department

Name Wu Xiuli Gender Female Birth Date 1981.11

Bachelor Assistant
Education Major Bioscience Technical Title
Degree Engineer

Current
QM Deputy Manager Working
Position 16 years
QA Manager Seniority
and Duty

1 Education Background

2010.09-2013.07: Yunnan Normal University, major in bioscience

2 Working Experience

2004.12-2006.03: Yunnan Walvax Biotechnology Co. Ltd., strain storage group of R&D department, work on Haemophilus Influenzae Type b Conjugate Vaccine and Group ACYW135 Meningococcal Polysaccharide Vaccine strain storage management

2006.03-2008.10: Yuxi Walvax Biotechnology Co. Ltd., supervisor of Hib conjugate vaccine stain amplification process

2008.10--2014.02: Yuxi Walvax Biotechnology Co. Ltd., workshop director of Hib conjugate vaccine, led a team work on production of Hib conjugate vaccine

2015.03-Now: Yuxi Walvax Biotechnology Co. Ltd., QA manager

2016.04-Now: Yuxi Walvax Biotechnology Co. Ltd., QM deputy manager
2.2.1.5 Head of Quality Operation Department – Shen Feng

Table 2-5 Curricula Vitae for Head of Quality Operation Department

Name Shen Feng Gender Male Birth Date 1983.3

Education Bachelor Degree Major Bioscience Technical Title Assistant Engineer

Quality Management Manager Working
Position 11 years
Quality Operation Manager Seniority

1 Education Background

2003.09-2007.07: Heilongjiang Jiamusi University, major in bioscience

2 Working Experience

2009.11-2012.06: Yuxi Walvax Biotechnology Co. Ltd., QA department

2012.06-2014.04: Yuxi Walvax Biotechnology Co. Ltd., site supervisor of QA department, conducted on establishment of quality supervision system and GMP certificate of products

2014.04-2015.04: Yuxi Walvax Biotechnology Co. Ltd., quality operation supervisor of quality management (QM) department, worked on perfect and improvement deviation and change progress

2015.04-2016.01: Yuxi Walvax Biotechnology Co. Ltd., quality operations deputy manager, work on establishment of quality operation system

2016.03-Now: Yuxi Walvax Biotechnology Co. Ltd., QM deputy manager and quality operations manager, worked on improvement of quality operation system and Halal Assurance System (HAS)

2.2.1.6 Head of Quality Supervision Department - Li Ding

Table 2-6 Curricula Vitae for Head of Quality Supervision Department

Name Li Ding Gender Male Birth Date 1986.12

Education Bachelor Degree Major Biotechnology Technical Title Engineer

Working

Position Deputy Manager of Quality Supervision 8 years

Seniority

1 Education Background

2005.09-2009.07: Yunnan Nationalities University, Biotechnology

2 Working Experience

2012.04-2014.03: Yuxi Walvax Biotechnology Co. Ltd., QA department, participated in GMP certificate and unannounced inspection of sub-packaging centre phrase II

2014.04-2016.04: Yuxi Walvax Biotechnology Co. Ltd., on-site supervisor of QA department, participated in GMP certificate and unannounced inspection of products. Meanwhile, published an article, Use of VVM label on Meningococcal Group A,C Bivalent Polysaccharide Conjugate Vaccine, in Chinese Journal of Biologicals

2016.05-Now: Yuxi Walvax Biotechnology Co. Ltd., Quality supervision deputy manager, lead the team to annual supervision and monitoring task (production process supervision, routine inspections, environmental dynamism monitoring and so on) to ensure the completion of KPI.

2.2.1.7 Head of Quality Control – Li Pengyang

Table 2-7 Curricula Vitae for Head of Quality Control

Name Li Pengyang Gender Male Birth Date 1989.01

Bachelor's pharmaceutical

Education Major Technical Title Engineer

Degree engineering

Working

Position Deputy Manager of QC Lab 10 years

Seniority

1 Education Background

2007.09-2011.06: Sichuan University, majored in pharmaceutical engineering

2 Working Experience

2011.10-2013.05, worked in Quality Supervision dept. was responsible for the quality supervision of meningococcal vaccine workshop and quality control laboratory. participated in the GMP certification of group A and C meningococcal polysaccharide vaccine and ACYW135 meningococcal polysaccharide vaccine.

2013.05-2014.04, supervisor of the Microbiology and Biochemistry Laboratory of the Quality Control Laboratory, is responsible for organizing the aseptic test, microbial limit test and production strain test for all samples of Yuxi Walvax. Familiar with the requirements of sterility test and microbial limit test in Chinese Pharmacopoeia.

2014.04-2015.04, the OOS/OOT supervisor of the Quality Operations Department. formulated the OOS and OOT survey management documents, and was responsible for organizing the tracking and investigation of the OOS and OOT results generated during the operation. Master failure mode risk management tools, responsible for the drafting and training of company risk management master plan.

2015.04-now, the deputy manager of the quality control laboratory, in charge of the quality control laboratory inspection management and quality management, and organizes the product inspection and stability sample inspection. At the same time, as the project manager of the Laboratory Information Management System (LIMS), responsible for system function design, auditing, validation and so on.

2.2.1.8 Head of Laboratory Animal Centre - Yan Yongqing

Table 2-8 Curricula Vitae for Head of Laboratory Animal Centre

Name Yan Yongqing Gender Male Birth Date 1986.02

Bachelor's Chemical Engineering

Education Major Technical Title Engineer

Degree and Technology

Working

Position Director of Laboratory Animal Centre 10 years

Seniority

1 Education Background

2004.09-2008.07: Nanjing Forestry University, Chemical Engineering and Technology (fine chemical engineering)

2 Working Experience

2010.05-2012.04: Employee of Quality Assurance Department of Yuxi Walvax Biotechnology Co., Ltd, responsible for supervision management of manufacturing sites, to guarantee the drug product, laying pavement for passing GMP examination;

2012.04-2015.03: Director of Quality Control Laboratory of Yuxi Walvax Biotechnology Co., Ltd, responsible for organizing and completing compilation and review of related documents of Laboratory Animal Centre, cooperatively completing application of usage approval of Laboratory Animal Centre, and improving examination methods for potency and toxicity of Diphtheria, Tetanus and Acellular Pertussis Combined Vaccine, Adsorbed;

2015.03-Now: Laboratory Animal Centre Manager of Yuxi Walvax Biotechnology Co., Ltd, established management model and team of Laboratory Animal Centre, possessed good practical ability, experimental R&D ability, and abilities of management, organization, coordination, and solving problems.

2.2.1.9 Head of Risk Validation Management Department – Xin Guangcheng

Table 2-9 Curricula Vitae for Head of Risk &Validation Management Department

Name Xin Guangcheng Gender Male Birth Date 1985.04

Education College Degree Major Chemistry Technical Title Technician

Current
Manager of Risk &Validation Management Working
Position 14 years
Department Seniority
and Duty

1 Education Background

2003.09-2006.07: Yuxi Normal University, Chemistry

2 Working Experience

2006.09-2009.09: Fill & finish employee of Yuxi Walvax Biotechnology Co., Ltd, responsible for sterile operation on filling position

2009.09-2011.10: Director of fill & finish of Yuxi Walvax Biotechnology Co., Ltd, responsible for operation management of filling procedures. Participated in a lot of FAT tests of sub-filling line, rubber bung washing machine and pulsation vacuum sterilizer

2011.10-2013.03: Director of Risk& Validation Management Department of Yuxi Walvax Biotechnology Co., Ltd, responsible for validation management of technology equipment, attended trainings of quality risk control of sterile manufacturing technology through ISPE member platform, and training of sterile manufacturing technology, etc.

2013.05-Now: Manager of the of Risk &Validation Management Department of Yuxi Walvax Biotechnology Co., Ltd, responsible for establishment and maintenance of comprehensive validation management systems.

2.2.1.10 Head of Production Management Department – Zhang Yan

Table 2-10 Curricula Vitae for Head of Production Management Department

Name Zhang Yan Gender Female Birth Date 1984.08

Education Master Major Bioengineering Technical Title Engineer

Current Position and Duty Manager of Engineering Operation Department Working Seniority 14 years

1 Education Background

2002.09-2006.07: Kunming University of Science and Technology, Pharmaceutical Engineering

2011.09-2014.07: Kunming University of Science and Technology, Bioengineering

2 Working Experiences

2006-2008: Employee of Quality Assurance Department of Yuxi Walvax Biotechnology Co., Ltd, responsible for quality supervision

2008-2010: On-site supervision director of Quality Assurance Department of Yuxi Walvax Biotechnology Co., Ltd. led department employees to complete on-site monitoring of all workshops

2010-2012: Office director of Quality Assurance Department of Yuxi Walvax Biotechnology Co., Ltd, responsible for organizing to complete revision and upgrading of company document system, and to complete GMP application document compiling

2012-2015.02: Deputy Manager of Quality Assurance Department of Yuxi Walvax Biotechnology Co., Ltd, participated GMP certification many times, and obtained GMP certifications.

2015.03-2016.09: Deputy Manager of Power Engineering Equipment Department of Yuxi Walvax Biotechnology Co., Ltd, responsible for cooperating with manager for department management work.

2016.10-2019.02: Manager of Engineering Operation Department of Yuxi Walvax Biotechnology Co., Ltd, responsible operation management of water system, air-conditioning system and automatic system, establishing and improving engineering management procedures.

2019.02-now, Manager of Production Management dept.of Yuxi Walvax.
2.2.1.11 Head of Bacterial Vaccine Workshop I (responsible for production of group ACYW135 meningococcal polysaccharide) - Wang Jing

Table 2-11 Curricula Vitae for Head of Meningococcal Vaccine Workshop

Name Wang Jing Gender Male Birth Date 1984.03

Food
Assistant
Education Master Degree Major Processing and Technical Title
Engineer
Security

Current Position Manager of Meningococcal Vaccine
Working Seniority 14 years
and Duty Workshop I

1 Education Background

2003.09-2006.07: Kunming University, Development and utilization of biological resources

2010.09-2013.07:Yunnan Agricultural University, Food processing and security

2 Working Experiences

2006.02-2006.07: Responsible for R&D of preparation of tetanus toxoid bulk in Yunnan Walvax Biotechnology Co., Ltd.

2006.07-2009.06: Strain process director of tetanus workshop in Yuxi Walvax Biotechnology Co., Ltd, mainly responsible for establishment and optimization, expansion of industrialized technology of preparation technology of tetanus toxoid strain, management of strain procedure persons and production, participated in GMP certification work of Haemophilus Influenzae Type b Conjugate Vaccine and Meningococcal Group A, C Bivalent Polysaccharide Conjugate Vaccine.

2009.07-2012.10: Vice director of Diphtheria-Pertussis-Tetanus (DTP) vaccine workshop of Yuxi Walvax Biotechnology Co., Ltd, mainly responsible for technology exploration, optimization and industrialized expansion of three kinds of DTP bulks, namely tetanus toxoid bulk, pertussis bulk and diphtheria toxoid bulk.

2012.11-2015.12: Diphtheria vaccine workshop director of Yuxi Walvax Biotechnology Co., Ltd, responsible for technology optimization and industrialization process of diphtheria toxoid bulk. In July, 2014, Yuxi Walvax had passed on-site inspection of Diphtheria, Tetanus and Acellular pertussis Combined Vaccine, Adsorbed, and got GMP certification.

2016.01-Now: Workshop director of Bacterial Vaccine Room 1 of Yuxi Walvax Biotechnology Co., Ltd, responsible for bulk production of four listed products in Hib vaccine workshop and meningococcal vaccine workshop

2.2.1.12 Head of Fill &Finish Workshop - Wang Yong

Table 2-12 Curricula Vitae for Head of Sub-packing Workshop

Name Wang Yong Gender Male Birth Date 1987.01

Technical Secondary Pharmaceutical
Education Major Technical Title Technician
Education Preparation

Current
Working
Position Manager of Sub-packing Workshop 14 years
Seniority
and Duty

1 Education Background

2004.09-2006.07: Chuxiong Hygienic School pharmaceuticals

2 Working Experience

2006.07-2008.06: Yuxi Walvax Biotechnology Co. Ltd., fill& finish workshop.

2008.07-2013.03: Yuxi Walvax Biotechnology Co. Ltd., freeze-drying process supervisor of sub-packaging workshop, combined production with quality requirements, developed and improved the freeze-drying technique, and optimized technological process and equipment management.

2013.04-2016.09: Yuxi Walvax Biotechnology Co. Ltd., deputy manager of fill & finish workshop, improved and optimized filling process, ensure the products pass the GMP certificate.

2016.10-Now: Yuxi Walvax Biotechnology Co. Ltd., fill & finish workshop director, in charge of filling process of listed products and department management.

2.2.1.13 Head of Engineering Security Department – Zhang Junxiang

Table 2-13 Curricula Vitae for Head of Engineering Security Department

Zhang
Name Gender Male Birth Date 1964.11
Junxiang

Bachelor's Mechanical manufacturing Technical
Education Major Technician
Degree technology and equipment. Title

Working
Position Manager of Engineering Security Department 14years
Seniority

1 Education Background

Graduated from Yunnan Institute of Technology in July, 1985, majoring in mechanical manufacturing technology and equipment.

2 Working Experience

1985.07-1994.04 Xinping Gas Sugar Factory
1994.04-1995.05 Xinping Food Factory
1995.05-1999.04 Xinping Glycerol Factory
1999.04-2005.11, worked in Yuxi Hongta Plastic Co., Ltd. and Yunnan Seed Coating Co., Ltd.
2005.11-2006.06, engineering dept. of Yuxi Walvax
2006.06-2015.08, Deputy Manager of Engineering Department, Yuxi Walvax
2015.08-2018.07, Manager of Safety Management Department, Yuxi Walvax
2018.07-now, Manager of Engineering Security Department, Yuxi Walvax
2.2.1.14 Head of Engineering Operation Department - Zhang Yan

Table 2-14 Curricula Vitae for Head of Engineering Operation Department

Name Zhang Yan Gender Female Birth Date 1984.08

Education Master Major Bioengineering Technical Title Engineer

Current Manager of Engineering Operation
Working Seniority 14 years
Position Department

1 Education Background

2002.09-2006.07: Kunming University of Science and Technology, Pharmaceutical Engineering
2011.09-2014.07: Kunming University of Science and Technology, Bioengineering

2 Working Experiences

2006-2008: Employee of Quality Assurance Department of Yuxi Walvax responsible for quality supervision
2008-2010: On-site supervision director of Quality Assurance Department of Yuxi Walvax Biotechnology Co., Ltd. led department employees to complete on-site monitoring of all workshops
2010-2012: Office director of Quality Assurance Department of Yuxi Walvax, responsible for organizing to complete revision and upgrading of company document system, and to complete GMP application document compiling
2012-2015.02: Deputy Manager of Quality Assurance Department of Yuxi Walvax, participated GMP certification many times, and obtained GMP certifications.
2015.03-2016.09: Deputy Manager of Power Engineering Equipment Department of Yuxi Walvax, responsible for cooperating with manager for department management work.
2016.10-2019.02: Manager of Engineering Operation Department of Yuxi Walvax Biotechnology Co., Ltd, responsible operation management of water system, air-conditioning system and automatic system, establishing and improving engineering management procedures.
2019.02-now, Manager of Production Management dept. of Yuxi Walvax Biotechnology Co., Ltd.
2.2.1.15 Head of Storage & Transportation Department: Wei Hongxu

Table 2-15 Curricula Vitae for Head of Storage & Transportation Department

Name Wei Hongxu Gender Male Birth Date 1987.05

Education College degree Major Biopharmaceutical Technical Title N/A

Current Manager of Storage & Transportation Working
14 years
Position Department Seniority

1 Education Background

2003.09-2006.07: Chuxiong Hygienic School, Pharmaceutical Preparations

2012.09-2015.07: Yuxi Normal University, Bioengineering

2 Working Experiences

2006.05-2013.12: Store man of Storage & Transportation Department of Yuxi Walvax Biotechnology Co., Ltd, responsible for storage management of materials.

2013.12-Now: Deputy Manager of Storage & Transportation Department of Yuxi Walvax Biotechnology Co., Ltd, responsible for organizing and completing of storage materials management, guaranteeing KPI, participating planning and designing of storage system and continuously improving storage control procedures.

2.2.1.16 Head of Product Distribution Centre: Huang Qiong

Table 2-16 Curricula Vitae for Head of Product Distribution Centre

Name Huang Qiong Gender Female Birth Date 1982.09

pharmaceutical
Education Master Major Technical Title Engineer
engineering

Current
Position and Manager of Product Distribution Centre Working Seniority 12 years
Duty

1 Education Background

2011.11-2014.12: Pharmaceutical Engineering, Kunming University of Technology

2003.09-2007.06 Chemistry, Yuxi Normal University

2 Working Experiences

2008.02-2010.12: Employee of production management dept. of Yuxi Walvax
2011.01-2013.06: Deputy manager of Storage and Transportation Department of Yuxi Walvax.

2013.07-2017.02: Deputy Manager of Production management Department of Yuxi Walvax

2017.03-2017.10: Deputy Manager of Marketing Department of Yuxi Walvax

2017.10-Now: Manager of Product Distribution Centre of Yuxi Walvax, responsible for managing final product storage and distribution, improving and perfecting final product distribution procedures.

2.2.1.17 Head of Purchasing Department: Liu Meixun

Table 2-17 Curricula Vitae for Head of Purchasing Department

Name Liu Meixun Gender Female Birth Date 1988.12

College
Education Major Accounting Technical Title N/A
Degree

Current
Manager of Purchasing Department Working Seniority 14 years
Position

1 Education Background

2003.09-2006.07: Chuxiong Medical College, Pharmacy

2011.01-2014.03; Yuxi Normal University, Accounting

2 Working Experience

2006.07-2006.12: Employee of supply workshop of Yuxi Walvax Biotechnology Co., Ltd, responsible for cleaning, sterilization, sorting cleanroom suits, and other basic preparation work.

2007.01-2008.10: Warehouseman of Logistics Department of Yuxi Walvax Biotechnology Co., Ltd, responsible for cooperating with department head for management of materials entering, check, in-warehouse and EX-warehouse.

2008.11-2009.06: Purchaser of Purchasing Department of Yuxi Walvax Biotechnology Co., Ltd, responsible for daily purchasing.

2009.07-2013.03: Plan and contract director of Purchasing Department of Yuxi Walvax Biotechnology Co., Ltd, responsible for daily management of purchasing contracts, established standing book template of daily purchasing management, laying foundation for digital management of department information.

2013.04-Now: Purchasing manager of Yuxi Walvax Biotechnology Co., Ltd, improved purchasing management, established purchasing team, and possessed good working ability and abilities of management, organization, coordination and problem solving.

Approval Date: 30th December, 2019
Revision Date: 17th July, 2020

Package Insert for 13-valent Pneumococcal Polysaccharide Conjugate Vaccine
Please read the package insert carefully and follow physician's guidance to use

■PRODUCT NAME■

Generic Name: 13-valent Pneumococcal Polysaccharide Conjugate Vaccine

■PRODUCT DESCRIPTION■

The vaccine is formulated by compounding the glycoconjugates of the capsular polysaccharide antigen of *Streptococcus pneumoniae* serotypes 1, 3, 4, 5, 6A, 6B, 7F, 9V, 14, 18C, 19A, 19F and 23F, individually conjugated to tetanus toxoid carrier protein. The individual polysaccharides are extracted from the cultures of *Streptococcus pneumoniae*, and purified through centrifugation, precipitation, ultrafiltration and column chromatography. The polysaccharides are chemically activated and derivatized, and then conjugated to tetanus toxoid carrier protein to form the glycoconjugate, with aluminum phosphate as adjuvant. The vaccine should be shaken well to obtain a homogeneous milky white suspension. During storage, a white deposit and clear supernatant might be observed due to adjuvant precipitation.

Active components: capsular polysaccharides of *Streptococcus pneumoniae* serotypes 1, 3, 4, 5, 6A, 6B, 7F, 9V, 14, 18C, 19A, 19F and 23F conjugated to tetanus toxoid carrier protein.

Excipients: sodium chloride, sodium dihydrogen phosphate, disodium hydrogen phosphate, aluminum phosphate.

■ELIGIBLES■

The vaccine is indicated for use in infants and children 6 weeks through 5 years of age (before 6th birthday).

■INDICATION■

The vaccine elicits immune response in recipients following immunization, and is indicated for the prevention of invasive disease caused by 13 serotypes (1, 3, 4, 5, 6A, 6B, 7F, 9V, 14, 18C, 19A, 19F and 23F) of *Streptococcus pneumoniae*.

The vaccine does not protect against diseases caused by *Streptococcus pneumoniae* serotypes that are not contained in the vaccine.

■NAME AND STRENGTH OF ACTIVE INGREDIENT(S)■

The vaccine is supplied as 0.5 mL/vial or in a single-dose prefilled syringe with 0.5 mL suspension for intramuscular injection. Each dose (0.5 mL) of the vaccine is formulated to contain polysaccharides of:

Streptococcus pneumoniae serotype 1 2.6 µg *Streptococcus pneumoniae* serotype 14 2.75 µg
Streptococcus pneumoniae serotype 3 2.5 µg *Streptococcus pneumoniae* serotype 18C 3.25 µg
Streptococcus pneumoniae serotype 4 3.0 µg *Streptococcus pneumoniae* serotype 19A 2.6 µg
Streptococcus pneumoniae serotype 5 2.5 µg *Streptococcus pneumoniae* serotype 19F 2.75 µg
Streptococcus pneumoniae serotype 6A 2.5 µg *Streptococcus pneumoniae* serotype 23F 3.0 µg
Streptococcus pneumoniae serotype 7F 2.85 µg *Streptococcus pneumoniae* serotype 6B 6.0 µg
Streptococcus pneumoniae serotype 9V 2.5 µg

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■DOSAGE AND ADMINISTRATION■

Since this product is a suspension containing an adjuvant, shake vigorously immediately prior to use to obtain a homogenous, white suspension in the vaccine container. This product is for intramuscular injection only. The preferred sites for injection are the anterolateral aspect of the thigh in infants and the deltoid muscle of the upper arm in toddlers and children. Each dose (0.5 mL) is to be injected. The vaccine should not be injected in and/or near the areas where there may be a nerve trunk and/or blood vessel.

Vaccination Schedule for Infants and Toddlers

The product is to be administered as a four-dose series at 2, 4, 6, and 12-15 months of age or 3, 4, 5, and 12-15 months of age, respectively, as described in Table 1 and Table 2.

Table 1 Vaccination Schedule for Infants 2 months of age

Dose Dose 1*† Dose 2† Dose 3† Dose 4‡

Age at Dose 2 months 4 months 6 months 12-15 months

*Dose 1 may be given as early as 6 weeks of age.

†The recommended dosing interval is 8 weeks.

‡The fourth dose should be administered at approximately 12–15 months of age, and at least 2 months after the

third dose.

Table 2 Vaccination Schedule for Infants 3 months of age

Dose Dose 1† Dose 2† Dose 3† Dose 4‡

Age at Dose 3 months 4 months 5 months 12-15 months

†The recommended dosing interval is 4 weeks.

‡The fourth dose should be administered at approximately 12–15 months of age, and at least 2 months after the

third dose.

For children 7 months through 5 years of age who have not received the product, the catch-up schedule in Table 3 applies:

Table 3 Vaccination Schedule for Infants and Toddlers 7 months through 5 years of age

Age at First Dose Total Number of 0.5 mL Doses

7-11 months of age 3*

12-23 months of age 2†

24 months through 5 years of age (prior to the 6th birthday) 1

*The first 2 doses at least 4 weeks apart; third dose after the one-year birthday, separated from the second dose

by at least 2 months.

†Two doses at least 2 months apart.

Children 2 through 5 years of age: a single dose.

■ADVERSE REACTIONS■

1. Clinical trials for PCV13-TT

Two clinical trials (phase I and III) of the vaccine (PCV13-TT) were conducted in China, including 120 and 2760 subjects, respectively; among all subjects (2880), 1754 have been vaccinated with

at least one dose of PCV13-TT. The safety observations were conducted in a systematic fashion for all subjects, starting from the first dose of vaccination, the safety observation will last for 30 days post each dose. For a majority of subjects excluding the dropout, the long-term safety observation period will start from the first vaccination till 180 days post the last dose of vaccination (after complete series).

(1) Summary

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The incidence rate of adverse reactions reported in clinical trials, according to the guidance on classifications of adverse events recommended by The Council for International Organizations of Medical Sciences (CIOMS), are classified as: very common ($\geq 10\%$), common ($\geq 1\%$ to $<10\%$), uncommon ($\geq 0.1\%$ to $<1\%$), rare ($\geq 0.01\%$ to $<0.1\%$) and very rare ($<0.01\%$). The safety data for the primary series of PCV-13 collected from both phase I and phase III clinical trials of all subjects are summarized as following:

① Systemic adverse reactions

Very common: fever, diarrhea

Common: crying, cough, nausea/vomiting, fatigue/somnolence, allergic reaction

Uncommon: myalgia

② Local adverse reactions

Very common: redness

Common: swelling, pain, induration

Uncommon: itching

Rare: rash (vaccination site)

③ Serious adverse reactions

There was one case of serious adverse event (SAE) reported during the primary series in the 3-months age group. This SAE was reported to be fever and considered to be possibly related to PCV13-TT. Other serious adverse events were judged to be irrelevant to PCV13-TT.

(2) Adverse reactions in phase III clinical trial

① The incidence and severity of solicited adverse reactions

The incidence and severity of solicited adverse reactions post primary and booster immunization in the phase III clinical trial of PCV13-TT comparing to the comparator vaccine (7-valent Prevnar, referred to as Prevnar) are summarized in Table 4 and Table 5.

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Table 4 The solicited local adverse reactions in phase III clinical trial in China

3, 4, 5 and 12 months 2a, 4, 6 and 12 months 7-11 months 12-23 months 2-5 years

Symptoms and grading of local adverse PCV13-TT Prevnar PCV13-TT Prevnar PCV13-TT Prevnar PCV13-TT Prevnar

reactions N*=488-519 N*=493-519 N*= 487-517 N*=186-198 N*=191-200 N*= 200 N*=200 N*= 200 N*=200

%b %b %b %b %b % % % % %

Pain

Any 0.20~2.31 0.00~1.93 0.21~1.93 0.54~7.07 0.52~3.50 5.00 1.50 9.50 10.00

Grade 1d 0.20~2.31 0.00~1.93 0.21~1.55 0.54~6.57 0.52~3.50 5.00 1.00 9.50 9.50

Grade 2d 0.00 0.00 0.00~0.39 0.00~0.51 0.00 0.00 0.50 0.00 0.50

Grade 3d 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00

Redness

Any 1.02~10.60 0.61~13.68 0.41~7.93 0.54~16.67 1.05~12.50 8.00 7.00 9.50 8.00

Grade 1c 0.20~6.74 0.20~8.67 0.21~5.42 0.00~9.09 1.05~11.50 4.50 3.50 4.50 5.00

Grade 2c 0.61~3.08 0.41~5.78 0.21~2.32 0.54~5.56 0.00~1.00 1.50 2.50 3.00 1.50

Grade 3c 0.20~1.16 0.00~0.19 0.00~0.19 0.00~2.02 0.00 2.00 1.00 2.00 1.50

Swelling

Any 0.82~1.93 0.41~3.47 0.62~3.29 1.61~7.58 1.05~3.50 6.50 3.00 12.50 8.00

Grade 1c 0.20~0.77 0.20~2.31 0.62~1.93 1.08~1.52 1.05~3.00 1.50 1.50 4.50 4.00

Grade 2c 0.41~0.96 0.20~1.16 0.00~1.35 0.54~4.55 0.00~0.50 3.00 1.00 4.50 2.50

Grade 3c 0.20~0.39 0.00~0.19 0.00~0.19 0.00~1.52 0.00 2.00 0.50 3.50 1.50

Induration

Any 0.20~2.89 0.00~3.47 0.00~2.71 0.00~3.03 0.00~0.50 0.50 0.50 2.50 0.50

Grade 1c 0.00~0.58 0.00~1.73 0.00~1.55 0.00~1.52 0.00~0.50 0.50 0.00 1.00 0.50

Grade 2c 0.20~1.93 0.00~1.35 0.00~1.35 0.00~0.51 0.00 0.00 0.50 1.50 0.00

Grade 3c 0.00~0.58 0.00~0.39 0.00 0.00~1.01 0.00 0.00 0.00 0.00 0.00

Itching

Any 0.00~0.39 0.00~0.39 0.00 0.00~0.51 0.00~0.50 1.00 0.00 5.00 1.00

Grade 1e 0.00~0.39 0.00~0.39 0.00 0.00~0.51 0.00~0.50 1.00 0.00 5.00 1.00

Grade 2e 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00

Grade 3e 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00

Rash (vaccination site)

Any 0.00 0.00 0.00 0.00 0.00 0.50 0.00 0.00 0.00

Grade 1c 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00

Grade 2c 0.00 0.00 0.00 0.00 0.00 0.50 0.00 0.00 0.00

Grade 3c 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00

Any local reactions 1.64~13.49 0.61~15.61 1.03~10.44 2.15~21.21 2.09~14.50 10.50 9.00 21.00 18.00

Grade 1 0.41~8.48 0.20~11.18 0.82~7.54 1.61~16.67 2.09~13.50 9.50 5.50 17.00 15.00

Grade 2 1.23~4.82 0.41~6.17 0.21~4.06 0.54~8.59 0.00~1.50 3.50 3.50 7.00 4.00

Grade 3 0.20~1.54 0.00~0.58 0.00~0.39 0.00~3.54 0.00 2.50 1.00 4.00 1.50

*

N=number of subjects reporting "yes" for at least 1 day, or "no" for all 30 days.

a. Minimum age: 6 weeks.

b. The minimum and maximum incidence rates for local adverse reactions after primary and booster immunizations.

c. Grade 1, <15 mm; grade 2, 15-30 mm; grade 3, >30 mm.

d.

Grade 1, not affecting activities; grade 2, affecting activities or multiple use of non-narcotic analgesics; grade 3, affecting daily activities or multiple use of narcotic analgesics.

e. Grade 1, mild itching in vaccination site; grade 2, moderate itching in limbs with vaccination; grade 3, systemic itching.

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Table 5 Incidence of solicited systemic adverse reactions in phase III clinical trial in China

3, 4, 5 and 12 months 2a, 4, 6 and 12 months 7-11 months 12-23 months 2-5 years

Symptoms and grading of systemic adverse PCV13-TT Prevnar PCV13-TT Prevnar PCV13-TT Prevnar PCV13-TT Prevnar

reactions N*=488-519 N*=493-519 N*=487-517 N*=186-198 N*=191-200 N*= 200 N*=200 N*=200 N*=200

%b %b %b %b %b % % % % %

Fever

Any 28.07~58.00 27.38~66.28 21.56~61.12 20.43~53.54 20.94~61.50 41.50 44.00 20.50 23.50
Grade 1# 17.01~48.17 18.05~57.61 15.40~52.22 11.83~35.86 13.09~41.00 30.00 29.00 8.50 15.50
Grade 2# 10.86~17.73 9.53~22.35 6.78~18.18 9.14~23.23 7.85~30.50 14.50 17.00 10.00 8.00
Grade 3# 1.23~0.39 0.61~0.39 0.00~0.39 0.00~3.03 0.00~1.50 0.50 1.00 2.00 1.00
Grade 4# 0.00~0.19 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00

Fatigue / somnolence

Any 0.20~6.17 0.20~5.20 0.00~3.09 0.00~1.52 0.00~2.50 1.00 1.00 2.50 1.50
Grade 1c 0.20~5.01 0.20~4.24 0.00~2.71 0.00~1.52 0.00~2.00 0.50 0.50 2.50 1.50
Grade 2c 0.00~1.73 0.00~0.96 0.00~0.39 0.00 0.00~0.50 0.50 0.50 0.00 0.00
Grade 3c 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00

Nausea / vomiting

Any 1.64~5.78 0.81~3.66 0.21~6.00 0.00~5.56 1.05~4.00 1.50 2.50 1.00 2.50
Grade 1d 0.82~4.62 0.61~2.31 0.21~4.26 0.00~4.55 1.05~3.50 1.00 2.50 0.50 2.50
Grade 2d 0.61~1.35 0.20~1.54 0.00~2.13 0.00~1.52 0.00~0.50 0.50 0.00 0.50 0.00
Grade 3d 0.00~0.20 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00

Diarrhea

Any 2.46~12.91 2.23~11.95 0.41~11.03 1.61~7.07 1.05~9.00 4.00 1.50 1.00 2.00
Grade 1e 1.23~7.32 1.01~6.17 0.21~4.64 0.54~4.55 0.00~7.00 3.00 1.50 1.00 1.50
Grade 2e 1.23~5.78 1.01~6.17 0.21~7.74 1.08~2.53 1.05~2.50 1.00 0.00 0.00 0.50
Grade 3e 0.00~0.58 0.20~0.19 0.00 0.00 0.00 0.00 0.00 0.00 0.00

Myalgia

Any 0.00~0.19 0.00~0.39 0.00~0.19 0.00 0.00~0.50 0.00 0.00 0.50 0.50
Grade 1f 0.00~0.19 0.00~0.39 0.00~0.19 0.00 0.00~0.50 0.00 0.00 0.50 0.50
Grade 2f 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00
Grade 3f 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00

Cough

Any 0.82~7.32 1.22~6.55 0.21~5.61 0.54~5.05 1.05~4.50 5.00 5.50 4.00 5.50
Grade 1g 0.20~5.01 0.61~4.05 0.21~4.64 0.00~2.53 0.52~2.50 1.00 2.00 3.00 2.00
Grade 2g 0.61~2.31 0.61~2.70 0.00~0.77 0.54~2.53 0.52~2.00 4.00 3.50 1.00 3.00
Grade 3g 0.00 0.00~0.19 0.00 0.00 0.00 0.00 0.00 0.00 0.50
Grade 4g 0.00 0.00 0.00~0.19 0.00 0.00 0.00 0.00 0.00 0.00

Abnormal crying

Any 0.82~10.40 0.61~9.44 0.21~10.44 1.08~6.57 0.00~6.00 2.50 1.50 0.00 1.50
Grade 1h 0.61~6.94 0.20~6.74 0.00~6.77 0.00~5.05 0.00~4.50 0.50 0.50 0.00 1.00
Grade 2h 0.20~3.85 0.20~3.66 0.21~4.64 1.08~1.52 0.00~1.50 2.00 1.00 0.00 0.50
Grade 3h 0.00~0.19 0.19~0.20 0.00 0.00 0.00 0.00 0.00 0.00 0.00

Allergic reaction

Any 0.00~0.58 0.00~0.58 0.00~0.58 0.00 0.52~3.00 1.50 3.50 1.50 1.00
Grade 1i 0.00~0.19 0.00~0.19 0.00~0.39 0.00 0.00~0.50 0.50 1.50 0.00 0.00
Grade 2i 0.00~0.19 0.00~0.39 0.00 0.00 0.52~1.00 0.50 1.00 1.50 0.50
Grade 3i 0.00~0.19 0.00 0.00~0.19 0.00 0.00~1.50 0.50 1.00 0.00 0.50
Any systemic reactionsj 30.33~67.82 29.21~71.48 22.38~67.12 22.04~58.59 22.51~66.00 46.50 50.00 24.00 28.50

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Grade 1 19.06~57.61 19.27~62.81 15.81~57.64 12.37~42.93 14.14~49.00 33.50 34.00 13.00 19.50
Grade 2 12.91~25.82 10.75~30.83 7.19~27.08 10.22~27.27 9.42~34.50 20.50 20.50 12.00 11.50
Grade 3 1.43~1.35 0.81~0.96 0.00~0.58 0.00~3.03 0.00~3.00 1.00 2.00 2.00 2.00
Grade 4 0.00~0.19 0.00 0.00~0.19 0.00 0.00 0.00 0.00 0.00 0.00

* N=number of subjects reporting "yes" for at least 1 day, or "no" for all 30 days.

a.

Minimum age: 6 weeks.

b. The minimum and maximum incidence rates for systemic adverse reactions after primary and booster immunizations.

Grade 1, 37.1-37.5 ■; grade 2, 37.6-39.0 ■; grade 3, >39.0 ■.

c. Grade 1, slight reduction in normal activities for ■48 h, not affecting daily activities; grade 2, 20%-50% reduction in normal activities for >48 h, affecting daily activities mildly; grade 3, >50% reduction in normal activities, affecting daily activities severely and is unable to work.

d. Grade 1, once or twice/24 h, with basically normal intake and will not affect activities; grade 2, 2-5 times/24 h, with significantly reduced intake or restricted activities; grade 3, > 6 times/24 h, with no significant intake, and in need of intravenous infusion.

e.

Grade 1, mild or transient with 2-3 times of loose stools/day, or mild diarrhea lasting ■1 week; grade 2, moderate or persistent diarrhea, 4-5 times/day, or diarrhea lasting >1 week; >6 times of watery stool/day, or bloody diarrhea, orthostatic hypotension, electrolyte imbalance, in need of intravenous infusion of > 2 L.

f.

Grade 1, not affecting daily activities; grade 2, tender muscles in non-vaccination sites, affecting daily activities mildly; grade 3, severe tender muscles, affecting daily activities severely.

g. Grade 1, transient, no treatment required; grade 2, persistent and controllable cough; grade 3, uncontrollable paroxysmal cough; grade 4, emergency care required.

h. Grade 1, slight increase in frequency than usual; grade 2, doubled frequency than usual; grade 3, doubled frequency than usual, and lasts > 3 days.

i. Grade 1, itching with no rash; grade 2, local urticaria; grade 3, extensive urticaria, angioedema.

j.

All systemic reactions, including body temperature ≥37.1 °C, fatigue/somnolence, nausea/vomiting, diarrhea, myalgia, cough, abnormal crying and allergic reactions.

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②Incidence and severity of unsolicited adverse reactions

The incidence of unsolicited adverse reactions for PCV13-TT was 0.06%-0.73% with most episodes were grade 1 in severity. The documented symptoms include: nasal obstruction, rhinorrhea, nasopharyngitis, upper respiratory tract infection, oral ulcer, abdominal pain, abdominal distension, decreased appetite, hyperhidrosis, increased tearing and eye discharge and red eyelids.

■CONTRAINDICATIONS■

Hypersensitivity to any component of the product, including active antigens, excipients, or tetanus toxoid, etc.

■PRECAUTIONS■

1. Do not vaccinate via intravenous route or by gluteal intramuscular injection, ensure the syringe needle is not puncturing blood vessel during inoculation.
2. Check if the package, container, label, appearance and expiration date of the vaccine are in compliance with corresponding requirements before administration. Do not use the vaccine in case that there is any crack in the container, loosened stopper, detached label, foreign particle(s) or discoloring inside the container, etc. Do not use the vaccine after the expiration date.
3. Use immediately after unsealing. A single human dose shall be used up each time according to prescribing information.
4. The vaccination should be postponed in case of fever, acute disease, and acute attack of chronic disease.
5. Appropriate monitoring and medical care and rescue measures should be readily available in case of occurrence of rare hypersensitivity reactions during vaccination. If allergic reactions occur after vaccination, please go to the vaccination site or hospital in time.
6. Cautions should be taken for vaccination in recipients with thrombocytopenia, any coagulopathy or those who are receiving anticoagulant treatment.
7. Preterm infants should be monitored for the potential risk of apnea during primary series. For preterm infants under hospitalization (gestational age ≤ 30 weeks at birth) vaccinated with PCV13-TT according to the recommended immunization schedule, at least 48 hours of monitoring should be considered. Given the benefit of vaccination in preterm infants, discontinued or deferred vaccination of the product is not recommended.
8. Given there is still no available safety and immunogenicity data for PCV13-TT in immuno-compromised individuals (e.g., malignancy or nephrotic syndrome), vaccination in this special group should be considered on an individual basis.
9. The use of PCV13-TT does not replace the use of 23-valent pneumococcal polysaccharide vaccines in children ≥ 24 months of age with conditions such as sickle cell disease, asplenia, HIV infection, chronic illness, or those who are immuno-compromised.
10. Under no circumstances shall the tetanus toxoid contained in the vaccine replace the routine immunization of tetanus vaccine or tetanus containing vaccine.
11. This product cannot guarantee all recipients can be protected from any diseases caused by *Streptococcus pneumoniae*.

■PREGNANCY AND LACTATION■

Pregnancy

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There is no data for using the product in pregnant women.

Breast-feeding

It is unknown whether this product is excreted in human milk.

■INTERACTIONS WITH OTHER MEDICAMENTS■

There is no concomitant immunization data available either inside or outside of China for the product. During the phase III clinical trial of the product, the interval between vaccination of the product and any other diphtheria, tetanus and acellular pertussis (DTaP) containing vaccine was ≥ 10 days. No adverse impact of DTaP-containing vaccine was observed on the immunogenicity of PCV13-TT under the indicated immunization interval.

■OVERDOSE AND TREATMENT■

Overdose with PCV13-TT is unlikely due to its presentation as a vial or a pre-filled syringe. There is no overdose data available for the product in recipients.

■PHARMACODYNAMIC/PHARMACOKINETICS■

Not applicable.

■CLINICAL TRIALS■

In China, the pivotal phase III clinical trial of PCV13-TT was a randomized and blinded non-inferiority study comparing to Prevnar. This study evaluated the immunogenicity and safety of PCV13-TT and Prevnar in 2760 subjects 2 through 71 months of age (at least 6 weeks of age). A total of 1040 infants aged 3 months were randomized in a 1:1 ratio as study group and control group, with PCV13-TT or Prevnar given at 3, 4, 5 and 12-15 months of age, respectively; for 520 infants aged 2 months (at least 6 weeks of age), PCV13-TT were vaccinated at 2, 4, 6 and 12-15 months of age. 1200 Children of 7 through 71 months of age, including children 7 months through 11 months of age, 12 months through 23 months of age and 24 months through 5 years of age (prior to the 6th birthday) who were naïve to pneumococcal conjugate vaccine, were split into three age groups with each containing 400 subjects randomized in a 1:1 ratio given 3, 2 or 1 dose of PCV13-TT or Prevnar, respectively, according to the age-appropriate schedules in Table 1 and Table 2. Enzyme-linked immunosorbent assay (ELISA) was utilized to detect the IgG antibody concentration in all serum samples collected from each group; additionally, a subsets of sera randomly obtained from 100 subjects in each age group for PCV13-TT and Prevnar, totaling 900, were tested by opsonophagocytic assay (OPA) to measure the ability of serotype-specific functional antibodies to eliminate corresponding pneumococci by promoting complement-mediated phagocytosis.

The main target population of the pivotal trial were infants of 2 and 3 months of age (at least 6 weeks of age). For immunogenicity, the PCV13-TT (3 months) group was compared with Prevnar (3 months) group in the first place, followed by the comparison between PCV13-TT (2 months) group versus Prevnar (3 months) group and then the comparison between PCV13-TT (2 months) versus PCV13-TT (3 months) group. The immunogenicity endpoints include the percentage of subjects with IgG antibody concentration ≥ 0.35 $\mu\text{g/mL}$ (positive rate) and the geometric mean concentration (GMC) of serotype-specific IgG antibody one month post the final dose of primary series and booster dose, as well as the percentage of subjects with vaccine serotype-specific OPA titer $\geq 1:8$ (positive rate) and OPA geometric mean titer (GMT) one month post the final dose of primary series and booster dose.

The non-inferiority comparison was performed for the immune responses of 7 common serotypes between PCV13-TT and Prevnar. For the 6 additional serotypes, if the lower limits of 95% CI for IgG antibody positive rate in study group achieved $\geq 70\%$, superiority assessment was then

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performed to compare the immune responses to 6 additional serotypes and corresponding serotypes in control group; meanwhile, the 6 additional serotypes in study group were compared with the serotype with the lowest response among all the 7 common serotypes in control group to determine the non-inferiority. The non-inferiority can be established either through demonstrating the lower limit of 97.5% CI for the difference of positive rate between groups (study group - control group) $> -10\%$, or the lower limit of 97.5% CI for IgG geometric mean concentration (GMC) ratio (study group/control group) > 0.5 ; Whereas the superiority can only be demonstrated when both lower limit of 95% CI for the difference of positive rates between groups (study group - control group) > 0 , and the lower limit of 95% CI for the IgG GMC ratio (study group/control group) > 1 .

The results are shown as below:

In subjects aged 3 months, after primary series (3 doses), the IgG antibody were compared between PCV13-TT group and Prevnar group. 6 out of 7 common serotypes met the non-inferiority criteria with the exception of 6B; the 6 additional serotypes reached superiority threshold; besides, the 6

additional serotypes in PCV13-TT group were non-inferior to the serotype with the lowest response among all the 7 common serotypes in Pevnar group.

In subjects aged 3 months, after complete series (4 doses), the IgG antibody were compared between PCV13-TT group and Pevnar group. All 7 common serotypes met the non-inferiority criteria; the 6 additional serotypes reached superiority threshold; besides, the 6 additional serotypes in PCV13-TT group were non-inferior to the serotype with the lowest response among all the 7 common serotypes in Pevnar group.

In subjects aged 2 months (at least 6 weeks of age), after primary series (3 doses), the IgG antibody were firstly compared between PCV13-TT (2 months) group and Pevnar (3 months) group, showing that all 7 common serotypes met the non-inferiority criteria and the 6 additional serotypes reached superiority threshold; besides, the 6 additional serotypes in PCV13-TT (2 months) group were non-inferior to the serotype with the lowest IgG response among all 7 common serotypes in Pevnar (3 months) group. The IgG antibody were then compared between PCV13-TT (2 months) group and PCV13-TT (3 months) group. 12 out of the 13 serotypes met the non-inferiority criterion with the exception of serotype 5.

In subjects aged 2 months (at least 6 weeks of age), after complete series (4 doses), the IgG antibody were firstly compared between PCV13-TT (2 months) group and Pevnar (3 months) group. All 7 common serotypes met the non-inferiority criteria and the 6 additional serotypes reached superiority threshold; besides, the 6 additional serotypes in PCV13-TT (2 months) group were non-inferior to the serotype with the lowest IgG response among all 7 common serotypes in Pevnar (3 months) group. The IgG antibody were then compared between PCV13-TT (2 months) group and PCV13-TT (3 months) group. All the 13 serotypes met the non-inferiority criteria.

The testing results of OPA antibody demonstrated that, all 13 serotypes of PCV13-TT were capable of inducing anti-pneumococcal OPA antibody in all age groups in the study group.

The immunogenicity data from subjects 2 months (at least 6 weeks of age) through 71 months of age are shown in Table 6 to Table 11.

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Table 6 Proportion of subjects with IgG antibody concentration $\geq 0.35 \mu\text{g/mL}$ in subjects aged 2 months and 3 months (PPS)											
Post primary immunizations Post booster immunization											
PCV13-TT PCV13-TT % PCV13-TT PCV13-TT % % %											
Pevnar (3m) % % Pevnar (3m)											
(3m) (2m) (PCV13-TT (3m) (2m) (PCV13-TT (PCV13-TT (PCV13-											
(PCV13-TT (PCV13-TT											
Serotypes N=471 N=487 N=482 [2m] – N=475 N=487 N=476 [3m] - [2m] – TT [2m] –											
[3m] - Pevnar [2m] – Pevnar											
PCV13-TT Pevnar Pevnar PCV13-TT											
% % % [3m]] [3m]] % % %											
[3m] [3m]] [3m]] [3m]											
(95%CI) (95%CI) (95%CI) (CI*) (CI*) (95%CI) (95%CI) (95%CI)											
(CI*) (CI*) (CI*) (CI*)											
100.0 99.18 100.0 0.82 0.82 0.00 99.79 99.79 99.37 -0.01 -0.42 -0.42											
4											
(99.22-100.0) (97.91-99.78) (99.24-100.0) (-0.10,1.74) (-0.10,1.74) (-0.93,0.91) (98.83-99.99)											
(98.86-99.99) (98.17-99.87) (-0.66,0.65) (-1.36,0.51) (-1.36,0.52)											
90.45 98.97 97.72 -8.53 -1.26 7.27 100.0 99.79 100.0 0.21 0.21 0.00											
6B											
(87.42-92.95) (97.62-99.67) (95.95-98.86) (-11.73,-5.32) (-3.09,0.58) (3.87,10.67) (99.23-100.0)											
(98.86-99.99) (99.23-100.0) (-0.25,0.67) (-0.25,0.67) (-0.92,0.92)											
99.15 99.79 99.79 -0.64 -0.00 0.64 100.0 100.0 100.0 0.00 0.00 0.00											
9V											
(97.84-99.77) (98.86-99.99) (98.85-99.99) (-1.70,0.41) (-0.66,0.65) (-0.41,1.70) (99.23-100.0)											
(99.25-100.0) (99.23-100.0) (-0.90,0.92) (-0.90,0.92) (-0.92,0.92)											
100.0 99.79 99.79 0.21 -0.00 -0.21 99.79 100.0 100.0 -0.21 0.00 0.21											
Common 14											
(99.22-100.0) (98.86-99.99) (98.85-99.99) (-0.25,0.67) (-0.66,0.65) (-0.67,0.26) (98.83-99.99)											
(99.25-100.0) (99.23-100.0) (-0.68,0.26) (-0.90,0.92) (-0.26,0.68)											
98.30 97.74 98.96 0.56 1.22 0.66 99.79 99.59 99.58 0.20 -0.01 -0.21											
18C											
(96.68-99.26) (95.99-98.87) (97.60-99.66) (-1.45,2.57) (-0.61,3.05) (-1.03,2.35) (98.83-99.99)											
(98.52-99.95) (98.49-99.95) (-0.60,1.00) (-0.94,0.92) (-1.02,0.61)											
100.0 100.0 100.0 0.00 0.00 0.00 100.0 100.0 100.0 0.00 0.00 0.00											
19F											
(99.22-100.0) (99.25-100.0) (99.24-100.0) (-0.90,0.93) (-0.90,0.91) (-0.93,0.91) (99.23-100.0)											
(99.25-100.0) (99.23-100.0) (-0.90,0.92) (-0.90,0.92) (-0.92,0.92)											
97.24 98.56 99.79 -1.32 1.23 2.55 100.0 99.79 99.79 0.21 -0.00 -0.21											
23F											
(95.33-98.52) (97.06-99.42) (98.85-99.99) (-3.40,0.76) (-0.07,2.52) (0.80,4.31) (99.23-100.0)											
(98.86-99.99) (98.84-99.99) (-0.25,0.67) (-0.66,0.65) (-0.68,0.26)											
99.58 57.49 100.0 42.08 42.51 0.42 99.79 91.38 99.79 8.41 8.41 0.00											
1											
(98.47-99.95) (52.97-61.93) (99.24-100.0) (37.65,46.51) (38.11,46.90) (-0.25,1.10) (98.83-99.99)											
(88.52-93.71) (98.84-99.99) (5.89,10.94) (5.89,10.94) (-0.67,0.67)											
98.09 45.17 96.06 52.91 50.88 -2.03 99.16 92.20 98.95 6.96 6.75 -0.21											
3											
(96.40-99.12) (40.69-49.72) (93.91-97.61) (48.32,57.50) (46.13,55.63) (-4.47,0.41) (97.86-99.77)											
(89.45-94.42) (97.57-99.66) (4.44,9.48) (4.20,9.30) (-1.62,1.20)											
Additional											
98.51 16.63 90.87 81.88 74.24 -7.64 97.89 40.04 97.27 57.85 57.23 -0.63											
(vs. 5											
(96.96-99.40) (13.43-20.24) (87.94-93.29) (78.40,85.36) (70.05,78.43) (-10.84,-4.45) (96.16-98.99)											
(35.66-44.55) (95.38-98.54) (53.31,62.39) (52.64,61.82) (-2.86,1.61)											
correspondi											
96.82 78.44 99.17 18.38 20.73 2.35 99.58 97.74 99.58 1.84 1.84 0.00											
ng 6A											
(94.80-98.21) (74.52-82.01) (97.89-99.77) (14.39,22.36) (16.99,24.47) (0.32,4.39) (98.49-99.95)											
(95.99-98.87) (98.49-99.95) (0.40,3.28) (0.40,3.28) (-0.94,0.94)											
serotypes)											
99.58 24.23 99.59 75.35 75.36 0.01 100.0 66.74 100.0 33.26 33.26 0.00											
7F											
(98.47-99.95) (20.49-28.29) (98.51-99.95) (71.49,79.20) (71.51,79.20) (-0.93,0.95) (99.23-100.0)											

7F
(97.03-99.99) (38.77-53.42) (98.07-100.0) (64.42-77.75) (96.40-99.88) (64.54-77.53) (98.04-100.0)
(66.09-79.20)
100.0 97.88 100.0 99.47 100.0 99.50 100.0 100.0
19A
(98.03-100.0) (94.67-99.42) (98.07-100.0) (97.09-99.99) (98.15-100.0) (97.23-99.99) (98.04-100.0)
(98.07-100.0)
98.92 96.83 100.0 98.41 98.48 97.99 100.0 99.47
1
(96.15-99.87) (93.22-98.83) (98.07-100.0) (95.43-99.67) (95.64-99.69) (94.93-99.45) (98.04-100.0)
(97.09-99.99)
97.84 96.83 98.41 98.41 88.89 97.99 98.92 99.47
3
(94.56-99.41) (93.22-98.83) (95.43-99.67) (95.43-99.67) (83.66-92.90) (94.93-99.45) (96.17-99.87)
(97.09-99.99)
Additional
91.35 96.83 96.83 98.41 98.48 97.99 93.01 99.47
(vs. serotype with the 5
(86.34-94.98) (93.22-98.83) (93.22-98.83) (95.43-99.67) (95.64-99.69) (94.93-99.45) (88.34-96.23)
(97.09-99.99)
lowest response
95.68 96.83 98.41 98.41 96.46 97.99 98.39 99.47
among all 7 common 6A
(91.66-98.11) (93.22-98.83) (95.43-99.67) (95.43-99.67) (92.85-98.57) (94.93-99.45) (95.36-99.67)
(97.09-99.99)
serotypes)
99.46 96.83 100.0 98.41 98.99 97.99 100.0 99.47
7F
(97.03-99.99) (93.22-98.83) (98.07-100.0) (95.43-99.67) (96.40-99.88) (94.93-99.45) (98.04-100.0)
(97.09-99.99)
100.0 96.83 100.0 98.41 100.0 97.99 100.0 99.47
19A
(98.03-100.0) (93.22-98.83) (98.07-100.0) (95.43-99.67) (98.15-100.0) (94.93-99.45) (98.04-100.0)
(97.09-99.99)
Note: N: subjects included in PPS; CI*: 2-sided 95% CI for IgG antibody positive rate.

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Table 9 IgG antibody GMCs in subjects aged 7 through 71 months (PPS)
Post primary immunizations Post booster immunization
7-11 months 12-23 months 24-71 months 7-11 months
PCV13-TT Prevnar PCV13-TT Prevnar PCV13-TT Prevnar PCV13-TT Prevnar
Serotypes
N=185 N=189 N=189 N=189 N=198 N=199 N=186 N=189
GMC GMC GMC GMC GMC GMC
GMC(CI*) GMC(CI*)
(CI*) (CI*) (CI*) (CI*) (CI*) (CI*)
4.26 4.23 5.01 5.08 8.90 8.58 3.06 3.12
4
(3.86-4.71) (3.88-4.62) (4.60-5.45) (4.63-5.58) (7.95-9.96) (7.63-9.66) (2.76-3.38) (2.85-3.41)
2.90 2.20 5.42 2.71 7.75 3.63 5.62 4.57
6B
(2.46-3.42) (1.92-2.52) (4.86-6.04) (2.35-3.11) (6.58-9.12) (3.04-4.32) (5.01-6.32) (4.08-5.11)
2.38 2.76 2.97 2.95 3.84 3.94 2.91 3.67
9V
(2.09-2.71) (2.43-3.12) (2.68-3.30) (2.64-3.30) (3.35-4.41) (3.36-4.62) (2.58-3.29) (3.31-4.07)
8.54 16.17 9.20 14.98 10.20 15.75 13.94 19.49
Common 14
(7.60-9.60) (14.53-17.99) (8.28-10.22) (13.51-16.61) (8.40-12.38) (12.82-19.34) (12.54-15.51)
(17.94-21.17)
3.78 3.44 5.03 3.92 8.82 7.47 2.97 3.32
18C
(3.30-4.34) (3.08-3.84) (4.63-5.46) (3.56-4.31) (7.56-10.29) (6.45-8.66) (2.62-3.36) (3.00-3.67)
7.21 4.09 8.80 4.20 11.45 5.71 7.54 3.92
19F
(6.43-8.08) (3.70-4.52) (8.10-9.55) (3.79-4.65) (10.05-13.04) (4.98-6.54) (6.84-8.31) (3.56-4.31)
3.56 2.70 4.32 2.96 4.42 4.29 4.53 4.21
23F
(3.10-4.10) (2.36-3.09) (3.89-4.79) (2.61-3.36) (3.80-5.14) (3.56-5.17) (4.01-5.12) (3.77-4.71)
2.23 0.47 2.25 0.42 2.06 0.31 1.98 0.58
1
(2.03-2.45) (0.40-0.56) (2.07-2.44) (0.35-0.50) (1.84-2.30) (0.26-0.39) (1.82-2.15) (0.53-0.65)
1.32 0.46 1.06 0.39 0.84 0.38 1.51 0.53
3
(1.20-1.45) (0.41-0.51) (0.98-1.14) (0.34-0.45) (0.75-0.94) (0.33-0.45) (1.38-1.66) (0.48-0.59)
0.82 0.19 1.01 0.27 1.60 0.31 0.86 0.24
Additional 5
(0.73-0.91) (0.15-0.25) (0.91-1.12) (0.22-0.32) (1.43-1.79) (0.25-0.38) (0.78-0.95) (0.20-0.29)
(vs. corresponding
1.79 0.54 2.46 1.17 2.35 1.29 2.99 1.63
serotypes) 6A
(1.51-2.12) (0.41-0.72) (2.08-2.92) (1.01-1.35) (2.00-2.76) (1.05-1.60) (2.64-3.37) (1.43-1.87)
5.73 0.14 5.47 0.41 4.60 0.43 4.91 0.46
7F
(5.04-6.51) (0.10-0.20) (5.04-5.94) (0.33-0.52) (4.00-5.28) (0.33-0.56) (4.46-5.39) (0.39-0.55)
5.06 1.50 7.81 2.67 9.56 3.32 6.28 2.17
19A
(4.54-5.64) (1.29-1.74) (7.08-8.61) (2.28-3.12) (8.47-10.79) (2.85-3.88) (5.71-6.91) (1.96-2.40)
2.23 2.20 2.25 2.71 2.06 3.63 1.98 3.12
1
(2.03-2.45) (1.92-2.52) (2.07-2.44) (2.35-3.11) (1.84-2.30) (3.04-4.32) (1.82-2.15) (2.85-3.41)
1.32 2.20 1.06 2.71 0.84 3.63 1.51 3.12
3

(1.20-1.45) (1.92-2.52) (0.98-1.14) (2.35-3.11) (0.75-0.94) (3.04-4.32) (1.38-1.66) (2.85-3.41)
Additional
0.82 2.20 1.01 2.71 1.60 3.63 0.86 3.12
(vs. serotype with the 5
(0.73-0.91) (1.92-2.52) (0.91-1.12) (2.35-3.11) (1.43-1.79) (3.04-4.32) (0.78-0.95) (2.85-3.41)
lowest response
1.79 2.20 2.46 2.71 2.35 3.63 2.99 3.12
among all 7 common 6A
(1.51-2.12) (1.92-2.52) (2.08-2.92) (2.35-3.11) (2.00-2.76) (3.04-4.32) (2.64-3.37) (2.85-3.41)
serotypes)
5.73 2.20 5.47 2.71 4.60 3.63 4.91 3.12
7F
(5.04-6.51) (1.92-2.52) (5.04-5.94) (2.35-3.11) (4.00-5.28) (3.04-4.32) (4.46-5.39) (2.85-3.41)
5.06 2.20 7.81 2.71 9.56 3.63 6.28 3.12
19A
(4.54-5.64) (1.92-2.52) (7.08-8.61) (2.35-3.11) (8.47-10.79) (3.04-4.32) (5.71-6.91) (2.85-3.41)
Note: N: subjects included in PPS; CI*: 2-sided 95% CI for IgG GMC.

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Table 10 Proportion of subjects with OPA antibody titer ≥1:8 in each age group (PPS)
Post primary immunizations Post booster immunization
3 months 2 months 7-11 months 12-23 months 24-71 months 3 months 2 months 7-11 months
Serotypes PCV13-TT Prevnar PCV13-TT PCV13-TT Prevnar PCV13-TT Prevnar PCV13-TT Prevnar PCV13-TT Prevnar PCV13-TT Prevnar
PCV13-TT PCV13-TT Prevnar
N=93 N=96 N=97 N=96 N=95 N=92 N=93 N=99 N=100 N=96 N=97 N=92 N=93 N=95
%(CI*) %(CI*) %(CI*) %(CI*) %(CI*) %(CI*) %(CI*) %(CI*) %(CI*) %(CI*) %(CI*) %(CI*) %(CI*)
100.0 98.97 100.0 100.0 100.0 100.0 100.0 96.00 100.0 100.0 100.0 100.0
100.0 100.0
4 (96.23- (94.39- (96.23- (96.19- (96.07- (96.11- (96.34- (90.07- (96.23- (96.27- (96.07- (96.11-
(96.11-100.0) (96.19-100.0)
100.0) 99.97) 100.0) 100.0) 100.0) 100.0) 98.90) 100.0) 100.0) 100.0) 100.0)
98.96 92.78 81.25 89.47 94.57 91.40 97.98 98.00 98.96 97.94 100.0 93.55
89.25 95.79
6B (94.33- (85.70- (72.00- (81.49- (87.77- (83.75- (92.89- (92.96- (94.33- (92.75- (96.07- (86.48-
(81.11-94.72) (89.57-98.84)
99.97) 97.05) 88.49) 94.84) 98.21) 96.21) 99.75) 99.76) 99.97) 99.75) 100.0) 97.60)
100.0 100.0 100.0 100.0 100.0 100.0 100.0 100.0 100.0 100.0 100.0 100.0
100.0 100.0
9V (96.23- (96.27- (96.23- (96.19- (96.07- (96.11- (96.34- (96.38- (96.23- (96.27- (96.07- (96.11-
(96.11-100.0) (96.19-100.0)
100.0) 100.0) 100.0) 100.0) 100.0) 100.0) 100.0) 100.0) 100.0) 100.0) 100.0)
100.0 100.0 98.96 97.89 100.0 100.0 92.93 95.00 100.0 100.0 100.0 100.0
100.0 100.0
Common 14 (96.23- (96.27- (94.33- (92.60- (96.07- (96.11- (85.97- (88.72- (96.23- (96.27- (96.07- (96.11-
(96.11-100.0) (96.19-100.0)
100.0) 100.0) 99.97) 99.74) 100.0) 100.0) 97.11) 98.36) 100.0) 100.0) 100.0) 100.0)
98.96 100.0 98.96 98.95 100.0 100.0 100.0 99.00 100.0 98.97 100.0 100.0
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18C (94.33- (96.27- (94.33- (94.27- (96.07- (96.11- (96.34- (94.55- (96.23- (94.39- (96.07- (96.11-
(96.11-100.0) (94.27-99.97)
99.97) 100.0) 99.97) 99.97) 100.0) 100.0) 100.0) 99.97) 100.0) 99.97) 100.0) 100.0)
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19F (92.68- (94.39- (94.33- (84.08- (94.09- (85.10- (94.50- (87.40- (96.23- (91.23- (96.07- (96.11-
(96.11-100.0) (86.76-97.65)
99.75) 99.97) 99.97) 96.29) 99.97) 96.92) 99.97) 97.77) 100.0) 99.36) 100.0) 100.0)
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(89.35-98.82) (89.57-98.84)
99.97) 98.87) 96.33) 98.27) 99.97) 99.33) 99.37) 99.97) 99.97) 99.97) 100.0) 99.74)
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(74.80-90.68) (0.67-9.14) (0.03-5.45) (2.35-13.24)
3.77) 96.37) 97.02) 11.86) 93.88) 53.77) 99.75) 10.22) 98.80) 98.23)
15.63 100.0 100.0 46.32 100.0 75.27 100.0 48.00 97.92 37.11 100.0 100.0
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3 (9.02- (96.27- (96.23- (36.02- (96.07- (65.24- (96.34- (37.90- (92.68- (27.52- (96.07- (96.11-
(90.86-99.33) (46.28-66.97)
24.46) 100.0) 100.0) 56.85) 100.0) 83.63) 100.0) 58.22) 99.75) 47.52) 100.0) 100.0)
4.17 100.0 97.92 7.37 98.91 95.96 100.0 5.15 100.0 98.92
Additional 98.92 7.53 9.00 5.26
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(vs. (94.15-99.97) (3.08-14.90) (4.20-16.40) (1.73-11.86)
10.33) 100.0) 99.75) 14.59) 99.97) 98.89) 100.0) 11.62) 100.0) 99.97)
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80.21 97.94 89.58 65.26 96.74 78.49 94.95 95.00 95.83 92.78 97.83 95.70
ding 97.85 81.05
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serotypes) (92.45-99.74) (71.72-88.37)
87.64) 99.75) 94.89) 74.74) 99.32) 86.34) 98.34) 98.36) 98.85) 97.05) 99.74) 98.82)
37.50 100.0 100.0 35.79 100.0 49.46 100.0 75.00 100.0 52.58 98.91 100.0
100.0 42.11
7F (27.82- (96.27- (96.23- (26.21- (96.07- (38.93- (96.34- (65.34- (96.23- (42.18- (94.09- (96.11-
(96.11-100.0) (32.04-52.67)
47.97) 100.0) 100.0) 46.28) 100.0) 60.03) 100.0) 83.12) 100.0) 62.81) 99.97) 100.0)
44.79 100.0 100.0 78.95 100.0 88.17 100.0 92.00 100.0 79.38 100.0 100.0
100.0 78.95
19A (34.63- (96.27- (96.23- (69.38- (96.07- (79.82- (96.34- (84.84- (96.23- (69.97- (96.07- (96.11-
(96.11-100.0) (69.38-86.64)
55.29) 100.0) 100.0) 86.64) 100.0) 93.95) 100.0) 96.48) 100.0) 86.93) 100.0) 100.0)

Note: N: subjects included in PPS; CI*: 2-sided 95% CI for OPA antibody positive rate.

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Table 11 OPA antibody GMT in subjects of each age group (PPS)
Post primary immunizations Post booster immunization
3 months 2 months 7-11 months 12-23 months 24-71 months 3 months 2 months 7-11 months
Serotypes PCV13-TT Prevnar PCV13-TT PCV13-TT Prevnar PCV13-TT Prevnar PCV13-TT Prevnar PCV13-TT Prevnar PCV13-TT Prevnar
PCV13-TT PCV13-TT Prevnar
N=93 N=96 N=97 N=96 N=95 N=92 N=93 N=99 N=100 N=96 N=97 N=92 N=93 N=95
GMT(CI*) GMT(CI*) GMT(CI*) GMT(CI*) GMT(CI*) GMT(CI*) GMT(CI*) GMT(CI*) GMT(CI*) GMT(CI*) GMT(CI*) GMT(CI*)
GMT(CI*) GMT(CI*) GMT(CI*)
1227.93 2291.32 1214.42 1014.62 1214.37 1756.58 1993.15 2710.14 1924.49 6182.37 6208.15 3440.10 1549.33
1465.72
4 (992.5- (1947.8- (940.3- (866.1- (1022.4- (1426.1- (1667.7- (2176.9- (1331.2- (4931.4- (4950.8-
(2690.0- (1288.7- (1253.1-
1519.2) 2695.4) 1568.5) 1188.7) 1442.4) 2163.6) 2382.2) 3374.1) 2782.2) 7750.7) 7784.8) 4399.3) 1862.7)
1714.4)
528.67 2928.36 811.17 237.34 599.57 923.67 1530.14 2511.69 2511.16 2661.93 7564.42 1948.21 890.24 1843.95
6B (329.58- (2233.9- (542.0- (142.61- (373.97- (611.9- (921.8- (1829.2- (1760.5- (1948.9- (5345- (1460.3-
(599.4- (1265.7-
848.02) 3838.7) 1214.1) 395.00) 961.29) 1394.4) 2539.9) 3448.8) 3581.9) 3635.9) 10705) 2599.2) 1322.2)
2686.4)
1922.48 1974.62 2192.74 1425.78 1692.48 3219.64 2824.34 2174.68 1980.18 6184.75 5484.79 3719.77 2075.15
2264.80
9V (1518.9- (1612.0- (1728.3- (1153.3- (1371.4- (2417.8- (2057.1- (1710.1- (1547.4- (4870.3- (4453.1-
(2889.7- (1697.0- (1886.3-
2433.2) 2418.9) 2781.9) 1762.6) 2088.7) 4287.4) 3877.8) 2765.4) 2534.1) 7854.0) 6755.5) 4788.3) 2537.6)
2719.2)
4511.21 5594.48 3571.78 2538.79 3060.32 6373.82 7303.79 2534.19 3395.54 7686.96 6221.93 6173.16 3761.98
2883.34
Common 14 (3494.9- (4188.4- (2833.6- (1929.1- (2260.7- (5041.8- (5760.0- (1586.7- (2193.4- (6160.6-
(4855.7- (4922.7- (3002.8- (2359.5-
5823.1) 7472.5) 4502.3) 3341.2) 4142.7) 8057.8) 9261.3) 4047.4) 5256.5) 9591.5) 7972.5) 7741.3) 4713.1)
3523.5)
1356.26 3129.49 518.67 952.30 1157.93 3133.89 3971.64 3572.76 3330.72(4173.73 5507.66 1117.88 1491.46
1677.92
18C (1094.5- (2353.5- (417.13- (748.0- (908.2- (2557.9- (3197.8- (2952.0- 2548.4- (3269.4- (4094.6-
(899.5- (1230.2- (1331.9-
1680.7) 4161.3) 644.92) 1212.4) 1476.4) 3839.5) 4932.8) 4324.1) 4353.1) 5328.2) 7408.5) 1389.3) 1808.2)
2146.1)
652.57 540.56 769.17 518.08 240.27 816.50 331.18 1133.78 333.59 2359.26 815.68 1966.07 751.55 258.60
19F (547.97- (423.02- (617.10- (412.37- (170.50- (653.5- (230.91- (876.5- (233.90- (1898.8- (600.1-
(1496.8- (635.68- (190.65-
777.14) 690.76) 958.72) 650.89) 338.60) 1020.2) 474.99) 1466.6) 475.75) 2931.3) 1108.8) 2582.5) 888.54)
350.77)
2090.78 5939.64 1072.95 669.00 1699.27 2069.47 3784.23 2025.10 3468.13 4943.66 16131.84 2529.55 1577.42
3607.17
23F (1452.4- (4465.8- (738.0- (444.5- (1116.1- (1616.3- (2584.5- (1465.3- (2544.3- (3817.5- (11742-
(1959.0- (1197.5- (2428.0-
3009.7) 7899.8) 1560.0) 1006.8) 2587.2) 2649.7) 5540.9) 2798.7) 4727.4) 6402.1) 22163) 3266.3) 2077.9)
5359.1)
62.13 92.51 56.40 42.36 321.14 158.59 87.89
2.00 2.24 2.23 8.16 2.07 2.28 2.34
1 (42.64- (66.48- (43.54- (31.65- (239.43- (115.59- (67.11-
(2.00-2.00) (2.01-2.50) (1.97-2.52) (5.78-11.52) (1.94-2.21) (2.00-2.61) (2.06-2.67)
90.51) 128.75) 73.05) 56.70) 430.74) 217.59) 115.11)
511.70 575.94 316.46 361.20 14.09 121.39 678.98 586.01 431.54
2.80 6.62 9.67 5.03 8.12
3 (376.29- (444.43- (264.28- (304.70- (10.43- (101.55- (523.20- (440.25- (359.88-
(2.35-3.33) (4.88-8.97) (6.60-14.16) (3.77-6.72) (6.04-10.91)
695.83) 746.38) 378.94) 428.17) 19.04) 145.11) 881.14) 780.04) 517.48)
450.24 411.73 115.18 192.91 84.80 1375.13 1027.83 222.49
Additional 2.24 2.43 2.80 2.51 2.61 2.41
5 (358.12- (317.12- (90.52- (155.85- (64.41- (1061.9- (796.4- (173.71-
(vs. (1.99-2.51) (2.10-2.82) (2.17-3.62) (2.16-2.93) (2.07-3.29) (2.02-2.87)
566.05) 534.56) 146.55) 238.79) 111.65) 1780.8) 1326.6) 284.98)
correspondi
1079.96 275.54 1202.17 442.24 95.37 1434.20 487.96 1737.99 970.59 3254.75 1562.59 2546.12 1234.03
ng 416.90(234.
6A (792.9- (159.69- (895.0- (286.46- (51.43- (1009.8- (251.04- (1185.4- (655.6- (2221.0- (972.9- (1780.8-
(866.0-
serotypes) 51-741.16)
1471.0) 475.44) 1614.8) 682.71) 176.86) 2037.0) 948.50) 2548.1) 1437.0) 4769.7) 2509.6) 3640.3) 1758.5)
7741.10 2414.63 3589.60 5524.63 32.24 5886.86 171.55 7244.42 41.47 2616.75 4276.87 21.71
12.76 16.04
7F (6383.8- (1872.4- (2955.9- (4317.5- (17.72- (4802.3- (98.79- (6006.0- (22.27- (1938.6- (3488.6-
(12.04-
(7.68-21.19) (8.94-28.80)
9387.0) 3113.9) 4359.2) 7069.2) 58.68) 7216.3) 297.89) 8738.3) 77.21) 3532.1) 5243.3) 39.14)
777.70 659.70 692.99 57.76 1368.75 166.43 2070.66 303.83 5503.11 72.99 2546.03 1141.93 71.60
11.50
19A (638.69- (532.25- (565.53- (38.19- (1110.8- (110.60- (1638.8- (200.97- (4501.2- (46.46- (2030.4-
(928.0- (45.99-
(7.55-17.51)
946.97) 817.66) 849.18) 87.34) 1686.6) 250.45) 2616.3) 459.33) 6728.1) 114.65) 3192.6) 1405.2) 111.47)
Note: N: subjects included in PPS; CI*: 2-sided 95% CI for OPA antibody GMT.

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■STORAGE CONDITION■

Transport and store refrigerated at 2°C to 8°C, protected from light.

DO NOT FREEZE. If the vaccine has been frozen. Please discard and keep the product in places out of children's reach.

■DOSAGE FORMS AND PACKAGING AVAILABLE■

1 mL vial, 0.5 mL/one single human dose/vial, 20 vials per package;
prefilled syringe, 0.5mL/one single human dose/prefilled syringe, 1 dose per package.

■SHELF LIFE■

The shelf life of the vaccine is 24 months. Please use before the expiration date printed on the label or small box.

■STANDARD■YBS00392019

■APPROVAL NUMBER■S20190046

■NAME AND ADDRESS OF MANUFACTURER■

Name: Yuxi Walvax Biotechnology Co., Ltd.

Address: No. 83 Dongfeng South Road, High-Tech Zone, Yuxi, Yunnan, China

Zip code: 653100

Tel: (+86) 0877-2076210

Fax: (+86) 0877-2076918

Website: www.walvax.com

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1 Objective

To standardize the management of products distribution, to ensure that all the products meet the requirements of the quality standard for temperature during storage and transport, eliminate defects during distribution; to establish the recall management system and improve company's recall system in order to timely and effectively recall products, to avoid or minimize drug quality incidents.

2 Definitions/Abbreviations

2.1 Finished products: products have completed all the production procedure and with final packaging.

2.2 Distribution: refers to a series of operations in the procedure that enterprise delivers products to agents or users, including allocate cargo and transport etc.

2.3 The first class logistics company (the mainline logistics distribution enterprise): the company that delivers the finished products directly from our company, as well as the national mainline logistics distribution enterprise which has been registered in our company and meet the requirement of distribution.

2.4 The secondary logistics company (the storage and distribution enterprise): the company connected with the first class logistics company, storage and distribution enterprise outside Yunnan Province, which has been registered in our company and meet the requirement of distribution.

2.5 Recall: in accordance with the prescribed procedures, pharmaceutical production enterprises take back the drugs which have been sold in the listed market but exist potential safety risk, That is to say, due to products violate regulations and/or registration information; the products existing defects; the product is reported that has serious adverse reactions etc., it should be taken back from the market, clinical trial, a batch or more.

2.6 Potential safety risk: it means, due to research or development or other reasons, the drug

may lead to an unreasonable harm for human health and life safety.

2.7 Voluntarily recall: depending on severity of case, pharmaceutical production enterprises
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start unsafe products recall on the premise that there's no official mandate, use the method of collection of information analysis, investigation and assessment.

2.8 Mandatory recall: the Drug Supervision and Administration Department considers that in some kind of product exists potential safety risk by investigation and assessment, and the enterprise should recall the medicine, but didn't, and order the enterprise to recall the medicine.

2.9 World Health Organization: WHO.

2.10 World Health Organization Pre-qualification: WHOPQ.

2.11 United Nations International Children's Emergency Fund: UNICEF.

3 Scope of Application

This procedure applies to management of entire procedure of finished products distribution and entire recall procedure for unqualified products or products unsuitable for clinic trial from market.

4 Responsibilities

4.1 Products Distribution Center

- Responsible for implementing the delivery of products;
- Responsible for acceptance and storage of recall products.

4.2 Marketing Center

- Responsible for connecting logistics company and monitoring products' temperature during transport;
- Responsible for supplying batch sales record in which is involved the recall;
- Responsible for delivering recall notice to domestic customers and storage & distribution enterprise outside Yunnan Province;
- Responsible for communicating with domestic customers and storage & distribution enterprise during the process of recall, as well as follow-up and feedback.

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4.3 International Cooperation Department

- Responsible for connecting Logistics Company with Market Center;
- Responsible for supplying recall products' foreign sales record with experimental or clinical sample;
- Responsible for delivering recall notice to the supervision organization of exporting country, customers or agents, if recall products is involved WHO PQ products, this department should report recall information to WHO and related procurement agencies such as: UNICEF etc.;
- Responsible for communicating with foreign customers or agents during the process of recall, as well as follow-up and feedback.

4.4 Quality Assurance Department

- Responsible for implementing and continuing the improvement of this procedure;
- Responsible for making recall notice and delivering to International Cooperation Department, Marketing Center;
- Responsible for making recall implementation plan and reporting to supervision institution according to regulations;
- Responsible for making recall information and reporting to national supervision institution, if involved WHO PQ products, the recall information should be submitted to International Cooperation Department in order to report to WHO and related procurement agencies;
- Responsible for following-up recall schedule and reporting to supervision institution as

requested;

——Responsible for evaluating recall situation and forming the final report of recall, submit it to national supervision institution, if involved WHOPQ products the final recall report should submit to WHO to make an announcement.

4.5 Quality Operation Department
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——Responsible for organizing investigation for products defects;

——Responsible for compiling Products Potential Security Risks Investigation and Assessment Report for company in order to take actions involve recall.

4.6 Quality Supervision Department

——Responsible for supervising implementation of products delivery;

——Responsible for implementing investigation for products defects;

——Responsible for following-up recall schedule and reporting it to Quality Assurance Department;

——Responsible for supervising recall products storage management.

4.7 Quality Management Department

——Responsible for helping to organize, coordinate and take part in products recall.

4.8 Quality Control Laboratory

—— Quality Control Lab responsible for testing recall products.

4.9 Finance Department

——Finance Department responsible for recalling products and financial processing of compensation.

4.10 Production Management Department

——Responsible for making an alternative supply approaches;

——Responsible for supplying the using situation of raw material, excipients, packing material in recall products.

4.11 Quality Authorized Person

——The Quality Authorized Person responsible for organizing related work for recall.

4.12 (Vice) General Manager

——(Vice) general manager responsible for guiding and supervising products recall,
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examination and approval for important decisions, communicating with media, public, and internal staff.

4.13 Other relevant departments responsible for supplying necessary information and helping recall task group for complete products recall.

5 Working Procedures

5.1 Products Distribution

5.1.1 Once Yuxi Walvax Biotechnology Co., Ltd Products Distribution Center receives "Letter of Advice", connect with first level logistics company in time, confirm products delivery date and related flight information.

5.1.2 Products distribution center confirms delivery date and flight information ■ and distributes products in time according to "Letter of Advice" and prepares relevant information (batch release certificate, products warehousing temperature records, quality report, certification, shipper letter of instruction etc.)

5.1.2.1 Pre-freeze ice pack in – 20°C freezer, continuously for more than 24 hours, until the cool storage liquid in ice pack completely frozen solid, ice-water mixture is not allowed in the ice pack, keep balance for ice pack at room temperature for 15 minutes or 30 minutes in 2 ~ 8°C refrigerator, the Marketing Center put ice pack or water pack according to the temperature situation in shipping destination.

5.1.3 In the process of products distribution, according to the principle "First in First Out", and "Short Term First Out".

5.1.4 Once products are ready, the products must be collected electronic supervision code or traceable code and uploading in time on the same day.

5.1.5 Once products are loaded in the ice storage tank, it must be packaged. After packing, the Products will be delivered to the first level logistics company, and connect immediately with the Local office of delivery destination.

5.1.6 First level logistics company delivers products to airport, and informs flight information to secondary logistics enterprise and Marketing Center in time, Marketing Center informs to Standard Management Procedure of Yuxi Walvax Biotechnology Co., Ltd

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local office in time.

5.1.7 Secondary logistics enterprise and Marketing Center local office will take products in airport after receiving flight information. delivery method will be decided according to local temperature situation. to ensure products temperature meet the requirement.

5.1.8 Please send back confirmation letter to our company by fax within 24 hours after receiving the products, if not to send back The Receipt Confirmation in time, this situation is defined as the products has been received and has had no doubt to the above content, if there is any objection, please explain it in attached page.

5.1.9 Products delivery center is responsible for supervising the entire process that company delivery products to first level logistics company, and airport. Marketing Center is responsible for supervising entire process from secondary logistics enterprise transport to CDC.

5.2 Products recall

5.2.1 Classification of recall: according to products' potential risk and severity of damage, the drug recall is divided into three classes:

5.2.1.1 Recall Class I: The use of this product may pose serious health hazards or in this product exists defects, which will pose a potential threat to patients' lives;

5.2.1.2 Recall Class II: The use of this product may pose a temporary or reversible health hazards or in this product defects are able to cause or discomfort, but not as severe as Class I;

5.2.1.3 Recall Class III: The use of this product generally does not causes health hazards, but need to be recalled due to other reasons or the defect may not have a significant impact on the patient's health.

5.2.2 Recall Classification Example Class Example

- 1) Wrong product (label and contents are different products)
 - 2) Correct product but wrong strength, with serious medical consequences
 - 3) Microbial contamination of the vaccine
- Recall Class I
- 4) Chemical contamination with serious medical consequences
 - 5) Mix up of some products with more than one container involved
 - 6) Wrong active ingredient in a multi-component product with serious medical

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consequences

- 1) Mislabelling e.g. wrong or missing text or figures
 - 2) Missing or incorrect information (leaflets or inserts)
 - 3) Chemical/physical contamination (significant impurities, cross contamination, Recall Class II particulates)
 - 4) Mix up of products in containers
 - 5) Non-compliance with specification (e.g. content, stability, with serious medical consequences, fill/ weight)
- 1) Faulty packaging (wrong or missing batch number or expiry date)
- Recall Class III
- 2) Faulty closure

5.2.3 Evaluation and Decision for Products Recall

5.2.3.1 In case of quality incident or severe adverse event which may be caused by products quality, Quality Operations Department organizes, investigates and fills in "Products Potential Security Risks Investigation and Assessment Report" to organize evaluation; it should be immediately reported to the Quality Authorized Person, according to the seriousness of the quality incident, Quality Authorized Person will organize relevant leader to determine the final measures including the recall.

5.2.3.2 Decision for recall should be based on investigation and evaluation of products

potential safety hazards.

Investigation content should include details as below (according to actual situation):

- Type, scope and reason of medicine adverse event that has happened;
- The use of medicine whether it meets the requirement of medicine specification and usage of the drug instructions stipulated by labels;
- Whether medicine quality meets the requirement national standards, whether the process of medicine production, meets the requirement GMP etc., whether medicine production coincide with approved process;
- Products storage, transport whether it meets the requirements;
- The component proportion ratio of product's main users;
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- Products batch, amount, circulation area or scope in which may exist potential safety hazards;
- Other factors which may affect the safety of medicine.

Main evaluation content should include:

- The possibility of the product causing harm, and whether it has been harmful to human health;
- Harmful impact on main users;
- Harmful impact on special groups, especially on high-risk groups;
- The severity and emergency degree of the hazard;
- The consequences caused by the hazard (short term and long term).

5.2.4 After evaluation and approval, company determines the recall, Quality Assurance Department shall fill in Products Recall Approval Form for approval.

5.2.5 Preparation for the recall

5.2.5.1 Set up a recall task group

Once decided the products recall, company should immediately set up a recall task group, prepare detail recall plan and recall action. If necessary any department can be requested for offering assistance to the recall task group.

5.2.5.2 Recall group member and responsibility (can be modified according to the practical situation)

Role Position Responsibility

1. Organize and supervise implementation of recall;

Group 2. Approval for important decisions during the process of recall;

General (vice) manager

Leader 3. Responsible for communicating with media, public, and internal staff.

1. Taking part in making recall implementation plan;

Group Quality Authorized 2. Responsible for preparing recall products list (name, batch Member Person number, amount etc.);

3. Responsible for reviewing products recall situation (amount,

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material balance).

Person in Charge of

Quality Management (or 1. Organize making recall implementation plan;

Group

Person in Charge of 2. Responsible for communicate with supervision institution Member

Quality Management during the process of recall.

Department)

Person in Charge of

Group

Quality Operations 1. Taking part in making recall implementation plan.

Member

Department

1. Responsible for making recall implementation plan;

2. Responsible for making recall information announcement and reporting to national supervision institution, if it is involved

World Health Organization-pre-qualification products,

International Business Department should report the recall information to WHO and related procurement agencies as well;

3. Responsible for making recall notice and delivering to International Business Department, Marketing Center;

Person in Charge of 4. Responsible for starting recall and submitting recall Group

Quality Assurance investigation & evaluation report and recall implementation Member

Department plan to supervision institution;

5. Responsible for following-up recall schedule and report to supervision institution as requested;

6. Responsible for evaluating recall situation and forming the final report of recall, submit it to national supervision institution, if it is involved WHOPQ products the final recall report should be submitted to WHO to make an announcement;

7. Organize periodically report recall decision group to recall situation (in case of emergency situation, report).

1. Taking part in making recall implementation plan;

Person in Charge of 2. Responsible for follow-up recall schedule and report to Group

Quality Supervision Quality Assurance Department; Member

Department 3. Responsible for supervising recall products storage management.

1. Take part in making recall implementation plan;

Group Person in Charge of 2. Help to complete recall products list (customers connect Member Marketing Center information, etc.);

3. Responsible for connecting with secondary logistics enterprise

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and monitoring temperature of product during transport;

4. Responsible for supplying batch sales records which involve the recall;

5. Responsible for delivering recall notice to domestic customers and storage & distribution enterprise (Secondary logistics enterprise);

6. Responsible during the process of recall with domestic customers and storage & distribution enterprise to communicate, follow-up and feedback;

7. Responsible with customers to negotiate an alternative supply approaches and/ or compensation plan.

1. Taking part in making recall implementation plan;

2. Responsible for connecting Marketing Center with logistics company;

3. Responsible for supplying recall products foreign sales record include clinical sample and sampling;

Person in Charge of 4. Responsible for delivering recall notice to the supervision Group

International Business organization of exporting country, customers or agents■if Member

Department recall products are involved with WHOPQ products ■ it should be reported as well recall information to WHO and related procurement agencies such as: UNICEF etc.;

5. Responsible for communicating with foreign customers or agents during the process of recall, as well as follow-up and feedback.

Person in Charge of 1. Taking part in making recall implementation plan;

Group

Products Distribution 2. Responsible for accepting and isolating storage recall Member

Center products.

Group Related Person in Charge 1. Taking part in making recall implementation plan;

Member of Quality Control Labs 2. Responsible for testing recall products (if necessary).

Related Person in Charge 1. Taking part in making recall implementation plan;

Group

of Production 2. Responsible for an alternative supply approaches production Member

Management Department (if necessary).

1. Taking part in making recall implementation plan;

Group Related Person in Charge

2. Responsible for recall products and financial processing of Member of Finance Department

compensation.

Group 1. Taking part in making recall implementation plan;

Lawyer

Member 2. To deal with legal actions. (When necessary).

5.2.5.3 Once the recall group is established, it should make the recall plan, organize and Standard Management Procedure of Yuxi Walvax Biotechnology Co., Ltd

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implement the recall immediately. The recall plan should identify every step, relevant person in charge, participants, the corresponding responsibilities and deadlines.

5.2.5.4 Preparation of Recall Plan

■ Technique preparation:

■ List recall involved products and batch number, amount, quantity of sale, inventory etc.;

■ Freeze materials and products related to recall products;

■ List agents or customers that need to be notified of the recall, this list should include specific contact information and specific address where the recall product sales to;

■ Prepare customers, agents, storage & distribution enterprise recall notification table (should include specific address to send back products and acceptance contact), recall notification table, see appendix 3 for details;

■ Prepare supervision institution "Vaccine Recall Incident Report Table" (supervision institution: the State Food and Drug Administration, products the supervision organization of exporting country, WHO), see appendix 4 for recall incident report;

■ Prepare an alternative supply approaches;

■ Primarily confirm how to return, collect, coordinate and final destruction.

■ Communicate preparation includes but not limited to following content:

■ Prepare different forms of reports, announces and text notifications for different communication objects (employees, cooperation partners, public media, etc.);

■ Solutions for the foreseeable external problems;

■ Different person in charge of communication and their contact information for different communication objects;

■ Choose the way to communication/media: such as meetings, telephone, fax, E-mail, the public media (TV announcement, broadcast);

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■ The frequency and manner of communication between recall task group and recall decision group.

■ Financial preparation include but not limited to following content:

■ Customers compensation plan;

■ Relevant funds and other financial preparation.

■ Legal preparation include but not limited to following content:

■ In some cases, companies need to full prepare for possible legal proceedings, including assignment or delegation of lawyer, and the preparation of relevant texts;

■ The preparatory works shall be accurately completed in time, so as to start the recall work as soon as possible.

■ Recall plan should include the following contents:

■ Product sales situation and the quantity of recalled;

■ The specific contents of the recall, including the organization, scope and timelines etc. for implementation;

■ The way to publish recall information and its scope;

■ The expected effect of the recall;

■ Handling measures after product recall;

■ Name and contact information of contact person.

5.2.4 Start Recall

5.2.4.1 Measures and time limit:

Recall Class Measures Start Time Limit

It must quickly pass the recall notice to agents, Quickly take action for the recall

Recall Class I customers (include clinical sample user or minimize latency as possible as the sampling person), reporting to national company can do

Recall Class supervision institution include our National

Within 24 hours or less
II Regulatory Authority, the supervision
Recall Class organization of exporting country■if involved If it's possible to start the recall within
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III WHOPQ products it should report to WHO as 48 hours, no more than 72 hours well.

5.2.4.3 During the process of recall, related stock products in company should be sealed immediately, stored in isolation, and clearly marked. During the process of recall, related records, including records of customer notification, customers feedback■recall products arrival records, the balance between the amount of products should be recalled and actual amount of recall products.

5.2.4.4 After starting products recall, the recall group should submit evaluation report and recall plans to National Regulatory Authority for the record Recall, in 1 day of Class I, within 3 days of Class II, within 7 days of recall Class III, in case of modification of recall plan, the group should inform National Regulatory Authority immediately.

- Investigation and evaluation report should include the following contents:
- Specific situation of recall medicine, including name, batch and other basis information;
- Reason of recall;
- Result of investigation and evaluation;
- Recall classification.

5.2.4.5 During the process of recall, recall group should follow-up involved recall customers or agents, to understand their recall situation and ask them to provide the "Recall Response List"■sees the detail in Annex 6■to our company, at the same time, recall group should report the medicine recall progress situation to National Regulatory Authority, the frequency is: Recall Class I, every day; Recall Class II, each 3 days; Recall Class III each 7 days, sees the" recall schedule report in Annex7.

5.2.5 Acceptance of recall products

5.2.5.1 Marketing Center is responsible for accepting recall products, verify amount, batch number of recall products, then store them uniformly, and fill in" recall products acceptance records".

5.2.5.2 The recall products should be store in isolation, should be marked clearly.
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5.2.6 Treatment of recall products

5.2.6.1 Recall task group is responsible for summarizing recall situation in time, evaluating recall schedule, effect and influence of recall products quality, puts forward recall products specific processing scheme and submit it to general manager for approval, at same time, submit it to National Regulatory Authority for the record or approval

5.2.6.2 According to the approved handling decision, the recalled products should be processed as soon as possible and record information in detail. The products that must be destroyed should be destroyed under the supervision of the National Regulatory Authority.

5.2.7 Corrective & preventive measures during the process of recall

5.2.7.1 During the process of recall, if the company finds that recall system was defective or needed to be further improved, it's necessary to put forward the corrective & preventive measures for the recall system itself.

5.2.7.2 Every group member of recall task group put forward relevant corrective & preventive measure for the defects and insufficiencies in the recall process, submits to specific leader for audit. Once approved, Quality Control Department will organize related department to make rectification and implementation.

5.2.7.3 Quality Assurance Department is responsible for supervising and tracking implementation situation of approved corrective & preventive measures until ?????

5.2.7.4 The recall system should be evaluated regularly to ensure its effectiveness in order to effectively start the recall in any time if necessary

5.2.8 Close recall

5.2.8.1 Once the recall is completed, it should be evaluated the effect of recall, and submit Medicine Recall Summary Report to Drug Supervision and Administration Department of the province, If the recall is not complete or more effective measures are needed to be taken after

investigate of Drug Supervision and Administration Department, company will be required to expand the scope of the recall, the recall shall be closed after completing all requests.

5.2.8.2 Recall is turned off, Quality Management Department is responsible for all related Standard Management Procedure of Yuxi Walvax Biotechnology Co., Ltd
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documents, records, data archiving, and long-term preservation.

5.2.8.2 Once the recall is closed, Quality Control Department is responsible for filing all related document, records and data for long term preservation.

6 Flow Chart
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Recall Management Control Flow Chart
Quality
Person Recall Task Group Specific Leader
Complaint
Information
No
Yes
Recall
Decisions Set Up a Recall Task Group
No
Make Recall Plan, Review and
Prepare for Recall Approval

Start Recall Yes

Evaluate Recall Situation
and Recall Products

Treatment for Recall
the Product

Recall Summary Report NO

Put Forward Corrective &
Preventive Measures Audit Measures

Execute Corrective &
Preventive Measures Yes

No
Track Implement
Situation

Yes

Evaluate the
Effectiveness of the
Measures
Yes

Close Recall

Related Description Non
7
Notices

7.1 In the process of transportation, it should be handled with care, and forbidden to Standard Management Procedure of Yuxi Walvax Biotechnology Co., Ltd
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inversion.

8 Relevant Documents

8.1 "Products Management Procedure" document code: WSYX-ST-SMP-0005-0004

8.2 "Warehouse Security Management Procedure" document code:
WSYX-ST-SMP-0003-0002

8.3 "Complaint with Adverse Reaction Management Procedure" document code:
WSYX-QA-SMP-0009-0034

8.4 "Medical Complaint Management Procedure" document code:
WSYX-QA-SMP-0009-0035

8.5 "Quality Complaint Management Procedure" document code:
WSYX-QA-SMP-0009-0036

9 Related records

9.1 "Customers Complaint Register Table" records document code:
WSYX-QA-SMP-0009-0034-R01

9.2 "Active Monitor Table" records document code: WSYX-QA-SMP-0009-0034-R02

9.3 "Quality Problem Active Monitor Statistics (Season) Report" records document code:
WSYX-QA-SMP-0009-0034-R03

9.4 "Vaccine Inoculation Adverse Reaction Active Monitor Statistics (Season) Report" records
document code: WSYX-QA-SMP-0009-0034-R04

9.5 "Adverse Reaction Complaint Feedback Table" Records Document Code:
WSYX-QA-SMP-0009-0035-R01

9.6 "Medicinal Complaint Investigation Table (Production and Test System)" records
document code: WSYX-QA-SMP-0009-0035-R02

9.7 "Adverse Reaction Complaint Investigation Report" records document code:
WSYX-QA-SMP-0009-0035-R03

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9.8 "Quality Problem Preliminary Investigation and Approval Form" records document code:
WSYX-QA-SMP-0009-0036-R01

9.9 "Quality Complaint Feedback Table" records document code:
WSYX-QA-SMP-0009-0036-R02

9.10 "Medicine Quality Complaint Investigation Table" records document code:
WSYX-QA-SMP-0009-0036-R03

9.11 "Quality Complaint Corrective & Preventive Measures Approval Form" records
document code: WSYX-QA-SMP-0009-0036-R04

9.12 "Quality Complaint Corrective & Preventive Measures Table" records document code:
WSYX-QA-SMP-0009-0036-R05

9.13 "Corrective & Preventive Measures Rectification/ Implementation Notification" records
document code: WSYX-QA-SMP-0009-0036-R06

9.14 "Quality Complaint Investigation Report" records document code:
WSYX-QA-SMP-0009-0036-R07

10 References or Standards

10.1 Page.58 "Material System GMP Implementation Guidance 2010 Edition" Order (No.79)
of the Ministry of health of the People's Republic of China 2011

10.2 Chapter 4 workshop and Implementation Section 3: warehousing area "Good
Manufacture Practice for Drugs (2010 Revision)" Order (No.79) of the Ministry of health of
the People's Republic of China 2011

10.3 Chapter 12 Articles 293 to 305 in "Good Manufacture Practice for Drugs (2010
Revision)" Order (No.79) of the Ministry of health of the People's Republic of China 2011

10.4 "Medicine Recall Management Method" State Food and Drug Administration (No. 29)
2007

10.5 WHO Technique Series Report Code N0.986.2014 GMP Annex2

10.6 PIC/Emergency Alert Handling and Recall Procedures for Quality Defects"01, 2011
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11 Appendixes

11.1 "Products Potential Security Risks Investigation and Assessment Report" records
document code: WSYX-P-11-0013-R01

11.2 "Products Recall Approval Form" records document code: WSYX-P-11-0013-R02

11.3 "Recall Notification Table" records document code: WSYX-P-11-0013-R03

11.4 "Vaccine Recall Incident Report Table" records document code: WSYX-P-11-0013-R04

11.5 "Recall Investigation and Evaluation Report" records document code: WSYX-P-11-0013-R05

11.6 "Recall Response Table" records document code: WSYX-P-11-0013-R06

11.7 "Recall Schedule Report Table" records document code: WSYX-P-11-0013-R07

11.8 "Recall Products Acceptance Records" records document code: WSYX-P-11-0013-R08

11.9 "Recall System Corrective & Preventive Measures Approval Form" records document code: WSYX-P-11-0013-R09

11.10 "Recall System Effectiveness Evaluation Table" records document code: WSYX-P-11-0013-R10

11.11 "Recall System Corrective & Preventive Measures Rectification/ Implementation Notification" records document code: WSYX-P-11-0013-R11

11.12 "Medicine Recall Summary Report" records document code: WSYX-P-11-0013-R12

12 Revision Histories

Revision basis, cause and revision content(Note: If the Version Effective Revision Issuance time bland is not enough to write, Remark number time department another page can be attached for description)

Clarify responsibilities, Quality increase responsibility of November 1, December 03 Control Production Management Non 2011 1, 2011

Department Department, clarify the responsibilities of quality Standard Management Procedure of Yuxi Walvax Biotechnology Co., Ltd Code WSYX-P-11-0013 Management and Control Version

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management department, and change the Quality Verification Department to Quality Control Labs

Delivery, transport, storage Quality management's responsibility

March 27, 04 March 24, 2014 Control change from Storage and 2014

Department Transportation Department to Marketing Center

Due to the company's international business needs, Quality the recall management process

June 17, 05 June 16, 2014 Assurance is not sufficient to meet the Non 2014

Department current operation needs, so company conducts revision and upgrade of this procedure

■■■■/CODING: WSYX-WHO- MenP(ACYW135)-0015

Certificate for the Release of Biological Products
LRA20201144

Generic Name: 13-valent Pneumococcal Polysaccharide Conjugate Vaccine

Manufacturer: Yuxi Walvax Biotechnology Co., Ltd.

Address: No. 83 South Dongfeng Road, High & New Technology Industries
Development Zone, Yuxi, Yunnan Province, China

Regis. No.: SP0408202001421 Lot No.: K202002001

Dosage Form: Suspension for
IM Injection Strength: 0.5ml/syringe. Each dose (0.5 mL) of
the vaccine contains:
Streptococcus pneumoniae serotype 1 2.6 µg
Streptococcus pneumoniae serotype 3 2.5 µg
Streptococcus pneumoniae serotype 4 3.0 µg
Streptococcus pneumoniae serotype 5 2.5 µg
Streptococcus pneumoniae serotype 6A 2.5 µg
Streptococcus pneumoniae serotype 7F 2.85 µg
Streptococcus pneumoniae serotype 9V 2.5 µg
Streptococcus pneumoniae serotype 14 2.75 µg
Streptococcus pneumoniae serotype 18C 3.25 µg
Streptococcus pneumoniae serotype 19A 2.6 µg
Streptococcus pneumoniae serotype 19F 2.75 µg
Streptococcus pneumoniae serotype 23F 3.0 µg
Streptococcus pneumoniae serotype 6B 6.0 µg

Expiry Date: Feb 04, 2022 Quantity: 141119 vials

The product mentioned above complies with the relevant provisions for lot release of
biological products and is approved for release.

As a minimum, this certificate is based on review of the summary protocol of
manufacturing and control.

Issued by: Miao Xu
(Official Stamp for State Food and Drug Administration)
30, March 2020

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1 Objective

Standardize batch number formulation principles and management requirements to ensure traceability of product information, avoid confusion and errors in samples.

2 Defined/Abbreviations

2.1 Batch: In the same production cycle, a certain number of uniform products produced with the same batch of raw materials and the same method.

2.2 Batch quantity: quantity of output in a single batch.

2.3 Batch number: A set of numbers or letters plus numbers used to identify a "batch". The batch number is used to trace and review the production history of a certain batch of drugs.

2.4 Sub-batch: A homogeneous group of bulk or final bulk is filled through several filling machine for filling process or the use of different freeze-drying machine for freeze-drying, that is, the formation of different sub batches. Sub-batches are part of the batch.

2.5 Intermediate Products: refers to the product that has completed part of the processing steps. Intermediate products still need further processing to become pre-packaged products.

2.6 Finished product: The product acquired by inspection, labeling and packaging of final bulk that has been filled (or freeze-dried) and sealed in the final container in an appropriate manner.

2.7 Bulk: A homogeneous substance used in the manufacture of a final preparation or final bulk. For multi-valent products, the final bulk is prepared from monovalent bulk solution.

3 Applicable scope

Applicable to all company product batch usage and management, including production, research and development, validation, etc.

4 Duty

4.1 Production Management Department

--Responsible for the preparation, determination, usage and issuance of batch number;
--Responsible for the management of batch numbers.

4.2 The Production Workshop/Department

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--Responsible for the implementation of the content of the management discipline.

4.3 Quality Supervision Department

--Responsible for the validation of compilation of batch numbers;
--Responsible for the supervision and management of the usage of batch numbers.

5 Working procedure

5.1 General principles for batch number preparation

5.1.1 The same batch number of products, should be from the same consistent source, with uniform quality, in accordance with the requirements of sampling test, based on which the whole batch of products can be assessed.

5.1.2 The batch number of the same product shall not be duplicated, and different batch number shall be used for different specifications of the same product.

5.1.3 In the compilation principle, the year should be written in 4 digits, month written in 2 digits, year serial number written in 3 digits.

5.1.4 Product Batch and sub-batch preparation should be able to make the whole process clear and easy to trace, in order to ensure that each batch of products processed by the process is consistent.

5.1.5 A single instalment of sub-batches should be limited to one of the following conditions permitting the development of sub-batches.

5.1.5.1 When the bulk or final bulk is prepared, if it is filtered by different filters, it shall be divided into different batches or sub-batches.

5.1.5.2 When the bulk or final bulk is directly filled into the final container, if different filling machines are used, it should be divided into different batches or sub-batches according to different filling machines.

5.1.5.3 If the bulk or final bulk use the same filling machine to be filled into the final container, then different freeze-drying machines are used for lyophilize, they should be divided into different batches or sub-batch according to different freeze-drying machines.

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5.1.5.4 When the bulk or final bulk is prepared, if it is purified with column chromatography twice or more, it should be divided into different batches or sub-batches according to the number of times it is purified with column chromatography.

5.1.6 The batch number for finished product shall be determined by the preparation time of final bulk, and the formulation date is the production date. Non-homogeneous preparation, mixing, dilution, filtration, filling of final bulk shall not be used as the same batch.

5.2 Definition of Batch Size:

Product

Batch Size Definition

Name

The amount after filling and lyophilization (annual inspection notes remaining

Strain

amount, freeze dry date, bacteria number).

Polysacch The amount of polysaccharide before sampling and after freeze drying when the

aride process is finished.

The amount of bulk in each bottle before sampling and after separation. If multiple bottles are showed on the Batch Record Audit List, batch size is the batch number Bulk

and total batch size on the production instructions (equals the total of batch size of bottles).

Finished Quantity of qualified products by light inspection (excluding production process products sterile test qualified quantity).

The actual amount of preparation for the bulk (if it goes through sterile filtration, Final bulk

the preparation amount before sterile filtration is the batch size).

If not separated by the bottle, the quantity before sampling and after completion of production process is the batch size.

Excipient Aluminum hydroxide Solution: After the completion of the process and separation, the amount before sampling is the batch size.

If multiple bottles are shown on the Batch Record Audit List, batch size is the batch SOP of Yuxi Walvax Biotechnology Co., Ltd.

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number and total batch size on the production instructions (total quantity of bottles).

Bulk Workshop: The amount after preparation before autoclave is the batch size.

Filling-packaging workshop: Before filling, the amount after preparation before Medium autoclave is the batch size. Finished product after the filling: vial: the remaining

amount after capping and removal of defect and empty vials. Pre-filling syringe: the remaining amount after removal of empty vial after filling.

5.3 Principles for compiling batch number of production strains

5.3.1 Compilation of batch number of strains is made up of the bacterium name code+ bacteria serogroup number (serotype) + seed lot category + year + month + annual serial number.

5.3.2 The Strain name code is abbreviated with the corresponding English names, and the seed lot class uses the abbreviation code corresponding to the English name.

Strain name Code table:

Species name Abbreviation

Haemophilus influenzae HI

Neisseria meningitidis NM

Bordetella pertussis BP

Corynebacterium diphtheriae CD

Clostridium tetanus CT

Streptococcus pneumoniae SP

Diphtheria Mutant strain DM

Seed Lot Category Code table

Seed Batch Category English abbreviation Code

Original seed lot O

Master seed lot M

Working seed lot W

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For example, HIBW201801001 means the 1st batch of Haemophilus influenzae type b of working seed lot produced in Jan. 2018.

5.4 Production of intermediate products Batch number Preparation principle

5.4.1 The batch number of the intermediate product is compiled with bacterium name code + bacterium serogroup number (model) + intermediate product category + year + month + annual serial number + Bottle number.

Intermediate Product Category Code table

Intermediate Product Category English name English abbreviation Code

Crude polysaccharide (HIB,

Crude polysaccharide CP

meningitis, pneumococcal)

Polysaccharide (HIB, meningitis,

Polysaccharide P

pneumococcal)

Derivative bulk (HIB, meningitis,

Derivative bulk D

pneumococcal)

Conjugate Bulk (HIB, meningitis,

Conjugate Bulk C

pneumococcal)

Bulk (diphtheria, pertussis, tetanus) Bulk B

Purified liquid (tetanus,

Purified liquid L

CRM197■

For example, NMAP201801002 refers to the 2nd batch of group A meningococcal polysaccharide produced in Jan. 2018.

SP1D201908001 refers to the 1st batch of serotype 1 pneumococcal polysaccharide produced

in Aug. 2019.

5.4.2 The batch number of bulk polysaccharide follows the batch number of the polysaccharide used. Batch number of the first dissolution of polysaccharide is followed by "-1", the second is followed by "-2". For example, NMCP201907001-1 refers to the bulk of polysaccharide of the SOP of Yuxi Walvax Biotechnology Co., Ltd.

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first dissolution of polysaccharide NMCP201907001.

5.4.3 Bulk produced with the same process and same equipment that is divided into bottles is distinguished by bottle numbers. If a single bottle is used, then the batch number is followed by "No.1 Bottle". For example, if NMCP201907001-1 bulk was divided into two bottles, then batch numbers were "NMCP201907001-1 (No.1 Bottle)" and "NMCP201907001-1 (No.2 Bottle)". If BPB201907001 acellular pertussis bulk was filled into a single bottle, then batch number was "BPB201907001 (No.1 Bottle)".

5.4.4 The tetanus toxoid bulk used in acellular DTaP vaccine production was divided into bottles after qualified by inspection, bottles used to divide a single mother bottle was numbered: batch number of mother bottle - bottle No. For example, if CTB201901001 (No.1 Bottle) was divided into 15 bottles, then the batch numbers were CTB201901001 (No.1 Bottle-1) to CTB201901001 (No.1 Bottle-15); if CTB201901001 (No.2 Bottle) was divided into 18 bottles, then the batch number were CTB201901001 (No.2 Bottle-1) to CTB201901001 (No.2 Bottle-18)

5.4.5 Bottle number is composed of "Bottle No. + numeric" in brackets.

5.4.6 Process number is composed of operation code + batch number of products. For example, ■NMAP201801003 refers to the 3rd batch produced in upstream process of group A meningitis polysaccharide. ■HibC201803015 (No. 1 Bottle) refers to the 15th batch produced in conjugation process of Hib polysaccharide.

Operation Codes:

Bacterial Vaccine Workshop 3
Bacterial Vaccine Workshop Diphtheria
Pertussis Vaccine Tetanus Vaccine
1 and 2 Vaccine
Workshop Workshop
Workshop
Strain amplification Strain Strain Strain

■ ■ ■ ■

process process process process

Upstream process of Fermentatio Extraction Extraction

■ ■ ■ ■

polysaccharides n process Process Process

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Downstream process Extraction Purification

■ ■ — — — ■

of polysaccharide Process n process

Polysaccharide Purification

■ ■ — — — — —

conjugation process process

5.4.7 The batch number of sub-batch is composed by the original batch number + numeric serial number.

For example, ■NMY201801003-1 refers to the 1st sub-batch of the 3rd batch of group Y meningococcal polysaccharide produced by downstream process in Jan. 2018.

5.4.8 Intermediate products in the production, if the use of strains, intermediate category has not changed, only the process changes, according to the process change point to add the appropriate code.

Process Change Point Code table

English

Process

Middle Product Name abbreviati

Change Point

on Code

Adsorption of acellular pertussis

Component C

component bulk

Tetanus toxin bulk Novel medium N
Serotype 5 pneumococcal
Conjugate C
polysaccharide
Group A and C meningococcal
polysaccharide conjugate bulk (used
for production of ACYW135 Carrier protein P
meningococcal polysaccharide
conjugate vaccine)
For example, CBPB201805001 refers to the 1st batch of acellular pertussis component bulk produced in May 2018.
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NCTB201805001 refers to the 1st batch of tetanus bulk (neopeptone) produced in May 2018.
CSP5P201907001 refers to the 1st batch of serotype 5 polysaccharide of 13-valent pneumococcal polysaccharide conjugate vaccine produced in Jul. 2019.
SP5P201907001 refers to the 1st batch of serotype 5 polysaccharide of 23-valent pneumococcal polysaccharide vaccine produced in Jul. 2019.
PNMCC201909001 refers to the 1st batch of group A meningococcal conjugate bulk of group ACYW135 meningococcal polysaccharide conjugate vaccine produced in Sep. 2019.
5.5 Principles of batch numbers of finished product
5.5.1 The production batch number is compiled with the finished product code + year + month + annual serial number.
5.5.2 Considering the practicality of marking of packages, finished product code is in capital letters.

Production Code table

Product Name Product Code Notes

Haemophilus Influenzae Type b Conjugate Vaccine (Vial) A Vial

Haemophilus Influenzae Type b Conjugate Vaccine (Pre-

B Pre-filled syringe

filled syringe)

Group A, C Meningococcal Bivalent Polysaccharide

C -

Conjugate Vaccine

Group A and C Meningococcal Polysaccharide Vaccine D -

Group ACYW135 Meningococcal Polysaccharide

E -

Vaccine

Diphtheria, Tetanus and Acellular Pertussis Combined

F -

Vaccine, adsorbed

23-valent Pneumococcal Polysaccharide Vaccine (Pre-

G Pre-filled syringe

filled syringe

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23-valent Pneumococcal Polysaccharide Vaccine (Vial) H Vial

13-valent Pneumococcal Polysaccharide Conjugate

I Vial

Vaccine (Vial)

Group ACYW135 Meningococcal Polysaccharide

J Halal product

Vaccine (halal)

13-valent Pneumococcal Polysaccharide Conjugate

K Pre-filled syringe

Vaccine (pre-filled syringe)

Group ACYW135 Meningococcal Polysaccharide

L -

Conjugate Vaccine

Tetanus vaccine M -

Acellular DTaP absorbed vaccine N -

Group A and C meningitis polysaccharide vaccine P For export

Acellular DTaP absorbed/Hib vaccine Q -

For example, D201805016 refers to the 16th batch of group A and C meningococcal

polysaccharide vaccine produced in May 2018.

5.6 Principles of batch numbers of the 3 batches inspected during site inspection is the same as

principles of production batch numbers.

5.7 Principles of compiling batch numbers of digestive fluid, medium and excipients

Category Code Table

Name English abbreviation Code

Digestive Fluid D

Medium M

Excipients E

5.7.1 The batch number of digestive fluids is compiled with digestive fluid Code + Digestive

Fluid Name Code + Year + Month + annual serial number.

Digestive Fluid Name Code table

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Name of digestive Fluid English abbreviation Code

Beef Gastric Enzyme Digestive Fluid G

Cheese Digestive Fluid C

Beef Pancreatic Enzyme Digestive Fluid P

36% Maltose Solution S

Thick's Digestive Fluid T

10% Yeast Extract Y

Martin Digestive Fluid M

Beef Infusion Broth I

For example, DC201801003 refers to the 3rd batch of cheese digestive fluid produced in Jan. 2018.

5.7.2 Medium batch numbers are composed of medium code + media name code + year + month + annual serial number.

Culture Medium Name Code table

Name of Culture medium English abbreviation Code

Pancreatic dairy soy peptone

P

agar medium

R2A Medium R

For example, MP201803015 refers to the 15th batch of pancreatic dairy soy peptone agar medium produced in Mar. 2018.

5.7.3 Excipients batch numbers are composed of excipient code + excipient name code + year + month + annual serial number.

Accessory Name Code Table

Excipient name English abbreviation Code

10% sodium chloride solution S

10% lactose solution L

10% lactose solution (Halal) H

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0.01mol/l phosphate buffer solution P

Aluminum hydroxide solution A

30% sucrose solution U

Troetamol solution M

For example, EP201803009 refers to the 9th batch of phosphate buffer solution produced in Mar. 2018.

5.8 Products, including strains, polysaccharides, bulk sold only to foreign markets, the letter E (Exit) is added before "5.3 Production of bacteria batch number preparation principle", "5.4 Production of intermediate product batch number preparation principle". For example, ESP6BP201805002 refers to the 2nd batch of serotype 6B pneumococcal polysaccharide produced in May 2018 which was sold abroad.

5.9 Principles of batch numbers of products for Non-listed sales (mainly in the compatibility of packaging materials compatibility, stability, etc.) is: G + finished product code (5.5.2 Production Code table) + year + month + annual serial number. For example, G1201805001 refers to the 1st batch of non-listed 13-valent pneumococcal polysaccharide conjugate vaccine produced in May 2018.

5.10 Principles for R&D project batch number.

5.10.1 Principle of compilation of R&D project batch number: add letter R (Research) in front of "5.3 Production of bacteria batch number preparation principle", "5.4 Production of intermediate product batch number preparation principle", "5.5 Principle of compiling batch number of finished products". For example RSP1P201801001 refers to the 1st batch in R&D of serotype 1 pneumococcal polysaccharide produced in Jan 2018.

5.10.2 When a new project is being developed, the project manager/team leader is responsible for compilation of easy to identify, non-repetitive bacterial name code/bacteria serotype number, intermediate product type, process code, finished product code, etc.

5.10.3 The same batch of R & D products using different groping methods is distinguished by SOP of Yuxi Walvax Biotechnology Co., Ltd.

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sub-batch, and sub-batch is used from the groping stage. For example, the batch number of group A meningococcal polysaccharide which is under groping methods are RNMAP201907001-1 and RNMAP201907001-2. If there are two methods used in formulation process of DTaP vaccine, the sub-batch of finished products are RM201907001-1 and RM201907001-2.

5.11 Validation of principles of batch number compilation.

5.11.1 The validation project batch number is mainly used in process validation projects, including product process validation and sterilization process simulation validation.

5.11.2 The principle of batch coding for process validation, to "5.3 Production of bacteria batch number preparation principle", "5.4 Production of intermediate product batch number preparation principle", "5.5 Production Principle of compiling batch number of finished products" letter V (Verify) is added in the front. For example, VCSP5P201907001 refers to the 1st batch of process validation of serotype 5 polysaccharide of 13-valent pneumococcal polysaccharide conjugate vaccine produced in Jul. 2019.

5.11.3 The principle of compiling the batch of sterilization process simulation validation: V + department code + year + month + annual serial number. For example, VFP201801001 refers to the 1st batch of sterilization process validation in filling and packaging workshop in Jan 2018. VPV201801003 refers to the 1st batch of sterilization process validation in pneumococcal vaccine workshop produced in Jan 2018.

5.11.4 When the sterilization process simulation verification involves the packaging workshop and the bulk workshop, the validation batch number is uniformly executed according to the filling and packing workshop validation batch number.

5.11.5 The department codes are detailed in the following table

Department Code

Bacterial Vaccines Haemophilus Influenza type b conjugate

HW

Workshop 1 vaccine Workshop
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Meningococcal vaccine Workshop MW
Bacterial Vaccines
Pneumococcal Vaccine Workshop PV
Workshop 2
Diphtheria Vaccine Workshop DW
Bacterial Vaccines
Tetanus Vaccine Workshop TW
Workshop 3
Pertussis Vaccine Workshop PW

Filling and Packing Workshop FP
Supply Workshop SW
5.12 If a non-list product validated qualified for continuous 3 batches, they call all be considered process validation. R&D batch number is the process validation batch number
6 Flow charts
N/A
7 Attention matters
N/A
8 Related documents
N/A
9 Related Records
N/A
10 Reference or standard basis
10.1 Good Manufacturing Practice (current edition)
10.2 Pharmacopoeia of the People's Republic of China (2015 Edition Part Three) procedure of biological product batching
10.3 Production and testing procedures of Hib conjugate vaccine
10.4 Production and testing procedures of groups A and C meningococcal polysaccharide conjugate vaccine
10.5 Production and testing procedures of groups A and C meningococcal polysaccharide
SOP of Yuxi Walvax Biotechnology Co., Ltd.

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vaccine
10.6 Production and testing procedures of group ACYW135 meningococcal polysaccharide vaccine
10.7 Production and testing procedures of absorbed acellular DTaP vaccine
10.8 Production and testing procedures of 23-valent pneumococcal polysaccharide vaccine
10.9 Production and testing procedures of 13-valent pneumococcal polysaccharide conjugate vaccine
11 Appendix
N/A
12 Revise history

Time Revision
Version Revision basis, reason and content Note
in force Department
Production Revise and improve the development and
1 Dec.
03 Management management of intermediate product and finished None
2011
Department product batch number.
Production
20 Feb.
04 Management Improve Batch Number Management Procedure. None
2012
Department
To prevent confusion in serial numbers of
meningococcal vaccine intermediate and finished
Production
1 Nov. product, considering advice from NIFDC, we
05 Management None
2012 improved the batch numbers of meningococcal
Department
vaccine intermediate and finished product as a risk
control strategy.
Add batch numbering principle of acellular pertussis
vaccine bulk, adsorption of diphtheria vaccine bulk
Production and adsorption acellular DTaP combined vaccine
1 Jan.
06 Management finished product; add batch management principle of None
2013
Department acellular pertussis vaccine bulk and diphtheria toxoid;
add pertussis fermentation process and diphtheria
extraction process.
SOP of Yuxi Walvax Biotechnology Co., Ltd.

Code WSYX-PM-SMP-00-08-0001

Batch numbering Version
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Add serial number establishment principles of pertussis extraction process; add month section of batch number for acellular pertussis vaccine bulk, diphtheria toxoid bulk and finished products.
1. Add the batch numbering principle of three batches
Production inspected in registration production site;
7 July
07 Management 2. Add the listed products batch number compiling None
2013
Department principle of streptococcus pneumoniae and its polysaccharides, conjugate.
During meningococcal vaccine GMP certification, the Production compiling principle of batch number can not
10 June
08 Management effectively distinguish varieties, and there is risk of None
2014
Department confusion. Therefore, the batch number compilation principle is improved.
Production In order to differentiate the products sold in China and
1 Jan.
09 Management halal products, the document was revised. None
2016
Department
In order to differentiate the products sold only in
Production
1 Nov. foreign markets, and emphasize the batch number
10 Management None
2016 formulation principle of industrialization and process
Department
validation products, this file was revised.
Add the batch number formulation principles of non-
Production
15 May listed products; add product code for 13-valent
11 Management None
2017 pneumococcal polysaccharide conjugate vaccine (Pre-
Department
filling syringe).
Production Add the product code for Group ACYW135
17 Aug.
12 Management meningococcal polysaccharide conjugate vaccine and None
2017
Department code for aluminum hydroxide gel.
Production Add the formulation principles of R&D batch number
01 Oct.
13 Management and validation item batch number, revised documents. None
2017
Department
Production Add the aluminum phosphate adjuvant code.
13 June
14 Management None
2018
Department
11 Mar. Production 1. Improve the compiling principle of the bulk
15 None
2019 Management polysaccharide batch number, process serial number,
SOP of Yuxi Walvax Biotechnology Co., Ltd.

Code WSYX-PM-SMP-00-08-0001
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management procedures number
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Department research and develop products bulk bottle number.
2. Establish the batch number principle for serotype 5 pneumococcal polysaccharide of 13-valent pneumococcal polysaccharide conjugate vaccine.
3. Normalize the name of DTaP vaccine.
4. Add the finished product code for DTaP combined vaccine.
Production Add products code of exported Group A&C
17 July
16 Management meningococcal polysaccharide vaccine. None
2019
Department
1. Add the code of crude polysaccharide and bulk polysaccharide derivate;
2. Add the principle of batch number of bottles of
Production tetanus toxoid bulk used in the production of acellular
06 Aug
17 Management DTaP combined vaccine None
2019
Department 3. Add the process change point code of group A and C meningococcal polysaccharide bulk conjugate (used to manufacture the Group ACYW135 meningococcal conjugate vaccine)
1. Add the products code of DTaP/Hib combined

Production
06 Jan. vaccine.
18 Management None
2020 2. Add the English abbreviation code for 30% sucrose
Department
solution and troetamol solution.

Layout Plan for Differential Pressure of Fill & Sub-packaging Center

■■■■/CODING: WSYX-WHO- MenP(ACYW135)-0017

China Food and Drug Administration

Drug Registration Certificate

Original No.:53110019

Acceptance No.: CXSS1800003 GUO Ratification No.: 2019S00755

Drug Name Generic Name: 13-valent Pneumococcal conjugate vaccine

English Name: 13-valent Pneumococcal conjugate vaccine

Active ingredient Capsular polysaccharides of 13 pneumococcal serotypes (1, 3, 4, 5, 6A, 6B, 7F, 9V, 14, 18C,

19A, 19F, and 23F)

Dosage form Injection Application Domestic drug registration

Strength 0.5ml per syringe. Each 0.5-mL dose Classification Biological products for prevention of vaccine contains:

2.6µg of polysaccharide type 1,

2.5µg of polysaccharide type 3,

3.0µg of polysaccharide type 4,

2.5µg of polysaccharide type 5,

2.5µg of polysaccharide type 6A,

6.0µg of polysaccharide type 6B,

2.85µg of polysaccharide type 7F,

2.5µg of polysaccharide type 9V,

2.75µg of polysaccharide type 14,

3.25µg of polysaccharide type 18C,

2.6µg of polysaccharide type 19A,

2.75µg of polysaccharide type 19F,

3.0µg of polysaccharide type 23F,

Standard YBS00392019 Shelf life 24 months

Conclusion This product complies with the relative requirements of "The Drug Registration Regulations of China", the registration is approved.

Manufacturer Yuxi Walvax Biotechnology Co., Ltd.

Add: No. 83 South Dongfeng Road, High-tech Zone, Yuxi, Yunnan Province, China

Ratification No. GYZZ S20190046 Valid date December 29th, 2024

Attachment Specification, instruction, and labeling

Main delivery to Yuxi Walvax Biotechnology Co., Ltd. Walvax Biotechnology Co., Ltd.

Report to Medical Products Administration of Yunnan Province, Yunnan Institute for Food and Drug

Control, National Institute for Food and Drug Control, Chinese Pharmacopoeia Commission,

Center for Drug Evaluation of NMPA, Center for Information of NMPA, Center for Food and

Drug Inspection of NMPA, Department of Drug Regulation of NMPA. The section of

manufacturing and control procedures of the registration standard is only sent to the applicant

(the main sending unit).

Note Applicant shall submit the research data of standard material and relevant material to the

NIFDC according to the "Application and filing method for pharmaceutical raw material of

standard substances"

China Food and Drug Administration

December 30nd, 2019

Manufacturing Process Flow Chart and Quality Control of Group ACYW135
Meningococcal Polysaccharide vaccine bulk

Strain Amplification Dringking
Water

Fermentation and cultivation

Harvesting fermentation broth Reverse Osmosis

Critical quality
control points and Preparation of raw
items:
Identity test, total solid, polysaccharide
protein content, nucleic
acid content, O-acetyl
content, phosphorus
content (Group A), Preparation of refined Purified
v content,
sialic acid Water
(Group C/Y/W135), polysaccharide
molecular size, residual
phenol content,
bacterial endotoxin
Polysaccharide dissolution

Distillation
Critical quality control Sterilization and
points and items: filtration
Identity ■test, total solid,
protein content, nucleic acid
content, O-acetyl content,
phosphorus content (Group
A), sialic acid content, Polysaccharide
(Group C/Y/W135), molecular bulk Water for
size, residual phenol content,
bacterial endotoxin, sterility injection
test

Grade A Grade B Grade C Grade D Sealed
container

1
Filling and packing process flow chart of Group ACYW135 Meningococcal
Polysaccharide vaccine

Group A Group C
polysaccharide polysaccharide
bulk bulk
Group Y Group W135
polysaccharide polysaccharide
bulk bulk Main quality control points
and item: sterility test

Water Steriliza
Sterili Dilution and tion and Sodium chloride
for
zation preparation filtratio solution, lactose
injection n solution

Filling Sterili Bottle and rubber
zation stopper washing

Lyophilization

Sterili Aluminum cap
Capping zation washing

Light
inspection

Packing
material Packaging
Main quality control points and
items:
Finished Identity, Appearance, Weight
product
variation, Moisture content, pH,
Osmolality, Polysaccharide content,

Warehousing
Molecular size distribution of
polysaccharide, Sterility, Abnormal
toxicity, Pyrogen test, Bacterial
endotoxin

Grade A Grade B Grade C Grade D Grade E Sealed
container

■■■■/CODING: WSYX-WHO- MenP(ACYW135)-0016

MODULE 2 COMMON TECHNICAL DOCUMENT SUMMARIES

Version No. Walvax-202007-00 Yuxi Walvax Biotechnology Co., Ltd.

PRODUCT REGISTRATION DOSSIER

13-valent Pneumococcal Polysaccharide Conjugate Vaccine
(Tetanus Toxoid Protein)

2.4 NONCLINICAL OVERVIEW

Yuxi Walvax Biotechnology Co., Ltd.

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MODULE 2 COMMON TECHNICAL DOCUMENT SUMMARIES

Version No. Walvax-202007-00 Yuxi Walvax Biotechnology Co., Ltd.

2.4 NONCLINICAL OVERVIEW

2.4.1 Overview of non-clinical trial strategies

Vaccines are preparations which contain antigens that induce the human body to produce specific active immunity, protecting the body from the damage caused by the original infection, toxins, and also the injury due to original antigen. Non-clinical research on vaccines includes, but is not limited to pharmacological (pharmacodynamics), toxicology (safety) evaluations. The main purpose of the preclinical animal safety evaluation of vaccines is to examine the safety of the vaccine through related animals, including the effects on immune organs and other toxic target organs, the reversibility of toxicity, and clinically relevant parameters, to predict possible adverse reactions in large population, and to reduce the risk faced with clinical trial subjects and clinical users, and to provide a basis for the development of clinical trial protocols. The possible toxic reactions of the vaccine include: direct damage to the body by the component itself as a toxic substance, immunodeficiency-related toxicity induced by the immune system, and toxicity caused by contaminants and residual impurities. Since the vaccine exerts the effects through inducing the immune system to produce antibody and / or by effector T cells, the most important potential toxicity of vaccine is derived from immune system-related toxicities.[1]

Vaccines are different from general chemicals, the pharmacological and toxicological experiments of vaccines are relatively special, and the pharmacological test mainly includes the rationale of action, the relationship between biological valence and dosage, the relationship between immune schedules and administration route and effectiveness, etc.; For toxicological test, major considerations include reactions at vaccination site and systemic pathological reactions, unexpected immune response of body to the vaccine product and the duration of the specific response. Since the above aspects are interrelated, the factors of pharmacology, toxicology and immunogenicity or biological valence should be taken into account comprehensively.[2]

For the toxicity test of the carrier protein in the conjugates of conjugate vaccines, current Chinese Pharmacopeia regulations should be followed if tetanus toxoid and diphtheria toxin are utilized as the carrier protein, to conduct the toxicity test of tetanus toxoid and diphtheria toxin for polysaccharide-protein conjugates.[3]

The vaccine is formulated by compounding the glycoconjugates of the capsular polysaccharide antigen of *Streptococcus pneumoniae* serotypes 1, 3, 4, 5, 6A, 6B, 7F, 9V, 14, 18C, 19A, 19F and 23F, individually conjugated to tetanus toxoid carrier protein. The individual polysaccharides are extracted from the cultures of *Streptococcus pneumoniae*, and purified through centrifugation, precipitation, ultrafiltration and column chromatography. The polysaccharides are chemically activated and derivatized, and then conjugated to tetanus toxoid carrier protein to form the glycoconjugate, with aluminum phosphate as adjuvant. This vaccine is used to prevent pneumonia, meningitis, otitis media and other diseases caused by *Streptococcus pneumoniae* serotypes 1, 3, 4, 5, 6A, 6B, 7F, 9V, 14, 18C, 19A, 19F and 23F. The following pharmacological tests were conducted based on product properties and expected function:

- 1) Immunogenicity test in New Zealand White Rabbit with intramuscular injection of various serotypes of antigens and 13 serotypes of antigen mixtures to examine the immunogenicity of monovalent conjugates and the compatibility of conjugates of 13 serotypes.
- 2) The study to determine the relationship between biological valence and immunization doses, aiming to investigate the relationship between antigen content in conjugates, vaccination doses and immunogenicity efficacy, so as to determine the content of various conjugates in finished products, and to provide reference for clinical vaccination procedures.

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MODULE 2 COMMON TECHNICAL DOCUMENT SUMMARIES

Version No. Walvax-202007-00 Yuxi Walvax Biotechnology Co., Ltd.

- 3) Immunogenicity test of vaccine products to determine the immunogenicity of preparation formulated based on the results of "the study to determine the relationship between biological valence and immunization doses", to validate the feasibility and rationality of the prescription of the preparation.

Vaccine antigens could produce the expected immunogenicity effects after vaccination in animals, but at the same time may also induce or cause potentially unexpected safety risks, in accordance with the requirements of relevant regulatory principles, the GLP qualified National Chengdu Center for Safety Evaluation of Drugs was commissioned to perform the following tests to evaluate the vaccine product:

- 1) Acute toxicity test in Kunming mice with intramuscular injection of the vaccine, to investigate the nature, degree, time effect and reversibility of the possible toxic reactions induced by vaccine, to evaluate target organ or target tissue toxicity, so as to define the reference range for clinical study.
- 2) Toxicity test in New Zealand rabbit and rhesus monkey with multiple intramuscular injection of the vaccine, to investigate the nature, degree, dose-response and time effect as well as reversibility of the potential toxic reactions induced by vaccine to immune organs and other target organ toxicity, so as to provide reference information for clinical studies.
- 3) Abnormal toxicity test in mice and guinea pigs with intraperitoneal injection of the vaccine, to examine the presence of exogenous contaminants and potential unexpected unsafe factors, so as to provide reference information for evaluating the safety of clinical use.
- 4) Specific toxicity test and toxicity reversibility test in guinea pigs with subcutaneous injection of the vaccine, to observe the occurrence of tetanus symptoms, presence of local symptoms of purulence, necrosis and other local reactions, so as to provide reference information for evaluating the safety of clinical use.

- 5) Active anaphylaxis test and passive cutaneous anaphylaxis tests in guinea pigs with injection of the vaccine, to observe the occurrence of systemic anaphylactic reactions and skin anaphylactic reactions, so as to provide reference information for evaluating the safety of clinical use.
- 6) Irritation test in rabbits with single-dose injection of the vaccine, to observe the irritations in rabbit's quadriceps induced by the product, so as to provide reference information for evaluating the safety of clinical use.
- 7) In vitro hemolysis test using red blood cells of rabbit with exposure to the vaccine, to observe the effect on the status of rabbit red blood cells, so as to provide reference information for evaluating the safety of clinical use.

The results of the above seven tests strongly underpinned the use of the vaccine in clinical studies. The evaluation results showed that, the dosage of the vaccine is within a safety margin; minor irritation at injection site is observed after use in animals, while may resolve after discontinuation of treatment; in addition, there are no apparent abnormalities regarding the general conditions, body temperature, body weight, food consumption, blood biochemistry, hematology, urine, bone marrow, ophthalmology, electrocardiogram, organ weight, pathological findings of tissues and other measures, no obviously increased or inhibited immune function is noted; specific antibody to vaccine components can be detected; risk of anaphylaxis is present due to the use of carrier protein; agglutination effect is observed in rabbit red blood cells, while no hemolytic reaction is seen.

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MODULE 2 COMMON TECHNICAL DOCUMENT SUMMARIES

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2.4.2 Pharmacology

1) Immunogenicity Test

The New Zealand white rabbit model was established, and the corresponding derivatives of conjugates were included as control, under the same system conditions, 0.5 mL of test articles were injected on Day 0, 14, 28 (except for serotype 6B as 3.2 µg/mL, other serotypes were 1.6 µg/mL), blood was collected on Day 21 and 35, ELISA was utilized to determine animal antigen-specific antibody titers with baseline data as control, to compare the pre- and post-immunization specific antibody titer, in order to determine the animal immunogenicity of polysaccharide conjugates. The results showed that 13 polysaccharide conjugates demonstrated significant immunogenicity relative to the corresponding polysaccharides.

Conjugates were diluted with 1 mg/mL aluminum phosphate solution (obtained by diluting aluminum phosphate using sodium chloride injection) to obtain monotype conjugate solution at 4 µg/mL (8 µg/mL for serotype 6B); Take a certain proportion from each of 13 conjugate dilutions to mix with 1 mg/mL aluminum phosphate solution, to form a uniform suspension containing 4 µg/mL of each serotype (8 µg/mL for serotype 6B), which was formulated to be a compound conjugate solution; 1 mg/mL aluminum phosphate was again used to create 2.5 fold dilution of the monotype conjugate solution, which was then compounded to form conjugates solution, 1.6 µg/mL (3.2 µg/mL for serotype 6B) conjugates solution was applied as test solution, with 0.5 mL injected in New Zealand rabbits on Day 1 and 14 intramuscularly, blood was collected on Day 21, using baseline values as control, to detect antibody level, so as to determine the antibody positive rate. The results indicated that, the 13 polysaccharide conjugates mixture generally imposed no effects on the immunogenicity in animals.

2) Study on the relationship between biological valence and dosage and immune doses

With reference to Wyeth's "7-valent pneumococcal conjugate vaccine" (in each dose of finished product, the polysaccharide content of each serotype is: 4 2 µg, 6B 4 µg, 9V 2 µg, 14 2 µg, 18C 2 µg, 19F 2 µg, 23F 2 µg), 13-valent pneumococcal polysaccharide conjugate vaccine (in each dose of finished product, the polysaccharide content of each serotype is: 1 2.2 µg, 3 2.2 µg, 4 2.2 µg, 5 2.2 µg, 6A 2.2 µg, 6B 4.4 µg, 7F 2.2 µg, 9V 2.2 µg, 14 2.2 µg, 18C 2.2 µg, 19A 2.2 µg, 19F 2.2 µg, 23F 2.2 µg, the following formulations are proposed:

Formulation 1: the polysaccharide content of each serotype in each dose: 1 1.8 µg, 3 1.8 µg, 4 1.8 µg, 5 1.8 µg, 6A 1.8 µg, 6B 3.6 µg, 7F 1.8 µg, 9V 1.8 µg, 14 1.8 µg, 18C 1.8 µg, 19A 1.8 µg, 19F 1.8 µg, 23F 1.8 µg;

Formulation 2: the polysaccharide content of each serotype in each dose: 1 2.0 µg, 3 2.0 µg, 4 2.0 µg, 5 2.0 µg, 6A 2.0 µg, 6B 4.0 µg, 7F 2.0 µg, 9V 2.0 µg, 14 2.0 µg, 18C 2.0 µg, 19A 2.0 µg, 19F 2.0 µg, 23F 2.0 µg;

Formulation 3: the polysaccharide content of each serotype in each dose: 1 2.2 µg, 3 2.2 µg, 4 2.2 µg, 5 2.2 µg, 6A 2.2 µg, 6B 4.4 µg, 7F 2.2 µg, 9V 2.2 µg, 14 2.2 µg, 18C 2.2 µg, 19A 2.2 µg, 19F 2.2 µg, 23F 2.2 µg;

With the same testing system and conditions, animal immunogenicity test was conducted in parallel with the commercially available Wyeth's "7-valent pneumococcal conjugate vaccine":

New Zealand white rabbit model was used, 0.5 mL test vaccines were intramuscularly injected on Day 0, 14 and 28 (samples were obtained by 2.5 fold dilution with 1 mg/mL aluminum phosphate), blood was collected on Day 21 and 35, and were compared with baseline to detect the antibody titer, and GMT was calculated; antigen-specific antibody titers of each serotype were pooled to assess the antigen-specific antibody titer levels of each serotype, and the antigen content of each

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serotype in each final bulk of each dose was determined as: 1 2.2 µg, 3 2.2 µg, 4 2.2 µg, 5 2.2 µg, 6A 2.2 µg, 6B 4.4 µg, 7F 2.2 µg, 9V 2.2 µg, 14 2.2 µg, 18C 2.2 µg, 19A 2.2 µg, 19F 2.2 µg, 23F 2.2 µg. The specific antibody titer levels after 2-dose and 3-dose series in animals were compared, showing higher GMT values post 3-dose series than 2-dose series.

3) Immunogenicity test of vaccine product

Three consecutive batches of vaccines formulated as per the formulation of "polysaccharide content of each serotype in each dose (0.5 mL): 1 2.2 µg, 3 2.2 µg, 4 2.2 µg, 5 2.2 µg, 6A 2.2 µg, 6B 4.4 µg, 7F 2.2 µg, 9V 2.2 µg, 14 2.2 µg, 18C 2.2 µg, 19A 2.2 µg, 19F 2.2 µg, 23F 2.2 µg" were diluted for 2.5 fold with 1 mg/mL aluminum phosphate to obtain test articles, respectively; New Zealand white rabbit model was used, and 0.5 mL of test articles were injected intramuscularly on Day 0, 14 and 28, blood was collected on Day 21 and 35, with baseline values as control, to detect antibody levels, the seroconversion rates and GMT were assessed. The results demonstrated good immunogenicity and acceptable lot-to-lot reproducibility and consistency of the vaccine products.

2.4.3 Pharmacokinetics

Not applicable for prophylactic biological products (vaccines).

2.4.4 Toxicology

1) Acute Toxicity Test

Kunming mice were intramuscularly injected with a single dose of the vaccine with a maximum injectable volume of 10 mL/kg, with overall good general conditions during 14 days post

vaccination, without obvious toxic reactions observed.

2) Repeated Dose Toxicity Tests

A New Zealand rabbit model

7-valent pneumococcal conjugate vaccine, adjuvant of aluminum phosphate and 13-JHYM administered to rabbits once every 2 weeks for 5 consecutive intramuscular injections might cause some irritation, while partially recovered after discontinuation. At 21 days after the first injection, the specific antibodies associated with vaccine components could be detected in rabbits. The peak antibody titer was observed 21~35 days after the first administration, and then decreased gradually, while the antibody level remained detectable 28 days after the last injection. In addition, 7-valent pneumococcal conjugate vaccine, adjuvant of aluminum phosphate and 13-JHYM imposed little impact on the rabbit's general condition, body temperature, weight, food consumption, blood biochemistry, hematology, bone marrow, organ weight and other indicators, is no obviously increased or inhibited immune function was noted, and there were no apparent pathological findings in organs or tissues.

B Rhesus Monkey Model

The vaccine product of 0.11 mL/animal and 0.5 mL/animal, and adjuvant of aluminum phosphate of 0.5 mL/animal were administered intramuscularly to monkeys once every 2 weeks for 5 consecutive injections, and the animals were recovered and observed for 6 weeks. The adjuvant of aluminum phosphate was irritant to the muscle at injection site, while might recover after discontinuation. In addition, no obvious abnormalities were observed for monkeys in each group with respect to general condition, body temperature, weight, food consumption, blood biochemistry, hematology, urine, bone marrow, ophthalmology, electrogram, organ weight, histopathology and other indicators, no obviously increased or inhibited immune function was noted, and there were no apparent pathological findings in organs or tissues. On 21 days after administration of the vaccine product, specific antibody associated with vaccine components could

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be detected in rhesus monkeys. The peak antibody titer was observed 21~35 days after the first administration, and then decreased gradually, while the antibody level remained detectable 28 days after the last injection.

3) Abnormal Toxicity Test

A Mice

Kunming mice were administered with a single intraperitoneal dose of this vaccine of 0.5 mL/animal (1 dose); from administration to the end of the observation, visual sign, mental and physical states were normal in all mice, with clean natural holes, without obvious abnormal reactions; at Day 7 after administration, the weight of each mouse increased as compared with baseline, demonstrating acceptable findings for abnormal toxicity test.

B Guinea Pigs

Guinea pigs were administered with a single intraperitoneal dose of this vaccine of 5 mL/animal (10 dose); from administration to the end of the observation, visual sign, mental and physical states were normal in all guinea pigs, with clean natural holes, without obvious abnormal reactions; at Day 7 after administration, the weight of each guinea pig increased as compared with baseline, demonstrating acceptable findings for abnormal toxicity test.

4) Test for carrier protein toxicity

A Specific Toxicity Test

Guinea pigs were administered with a single subcutaneous dose of this vaccine of 2 mL/animal; from administration to the end of the observation, visual sign, mental and physical states were normal in all guinea pigs, with clean natural holes, without tetanus symptoms; no abscess or necrosis at administration site were observed; at Day 21 after administration, the weight of each guinea pig increased as compared with baseline, demonstrating acceptable findings for specific toxicity test of carrier protein (tetanus toxoid) in guinea pigs.

B Toxicity Reversibility Test

Guinea pigs were administered with a single subcutaneous dose of this vaccine of 5 mL/animal; from administration to the end of the observation, visual sign, mental and physical states were normal in all guinea pigs, with clean natural holes, without tetanus symptoms; no abscess or necrosis at administration site were observed; at Day 21 after administration, the weight of each guinea pig increased as compared with baseline, demonstrating acceptable findings for the toxicity reversibility test of carrier protein (tetanus toxoid) in guinea pigs.

5) Anaphylaxis tests

A Systemic Active Anaphylaxis Test

Guinea pigs were intramuscularly administered with vaccine bulk at 0.05 mL bulk/kg, 0.5 mL bulk/kg from 3 batches of the product (lot numbers: 20110304, 20110305, 20110406), at the dose of 0.5 mL bulk/kg, positive active anaphylactic reactions were observed, while at the dose of 0.05 mL bulk/kg, negative results were noted for negative anaphylactic reactions in guinea pigs.

B Passive Cutaneous Anaphylaxis Test

Guinea pigs were intramuscularly administered with the vaccine products at 0.05 mL/kg, 0.5 mL/kg from 3 batches of the product (lot numbers: 20110304, 20110305, 20110406), and the anti-serum dilution was 1:2, 1:8, and 1:32, respectively, all were with negative findings in the passive cutaneous anaphylaxis test in the guinea pigs.

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6) Irritation Test of Single Intramuscular Injection

Rabbit quadriceps were administered with single intramuscular injection of low- and high-dose (0.5 mL/site, 1.5 mL/site, clinical preparation concentration) aluminum phosphate, 7-valent pneumococcal conjugate vaccine, the vaccine product (lot number 2010304, 20110 305, 2010406), then the low- and high-dose aluminum phosphate, high-dose 7-valent pneumococcal conjugate vaccine, high-dose vaccine products of 3 batches, and low-dose vaccine products of lot number 2010406 showed obvious irritation to rabbit quadriceps. The above muscle irritations partially resolved during a 16-day observation period.

7) In Vitro Hemolysis Test

Aluminum phosphate adjuvant (concentration of 1 mg/mL, the same concentration as aluminum phosphate in the vaccine products), the vaccine products (lot number 2010304, 2010305, 20110406) at different doses (0.5, 0.4, 0.3, 0.2, 0.1 mL/tube) induced significant agglutination effect in red rabbit cells, without hemolytic reaction.

2.4.5 Summary of evaluations and conclusion

The evaluation results showed that, the dosage of the vaccine is within a safety margin; minor irritation at injection site is observed after use in animals, while may resolve after discontinuation

of treatment; in addition, there are no apparent abnormalities regarding the general conditions, body temperature, body weight, food consumption, blood biochemistry, hematology, urine, bone marrow, ophthalmology, electrocardiogram, organ weight, pathological findings of tissues and other measures, no obviously increased or inhibited immune function is noted; specific antibody to vaccine components can be detected; risk of anaphylaxis is present due to the use of carrier protein; agglutination effect is observed in rabbit red blood cells, while no hemolytic reaction is seen.

The pharmacological and toxicological tests studies of the vaccine products demonstrate excellent safety and immunogenicity, and thus can be safely administered clinically.

2.4.6 Reference

[1] General principles for technical review of preclinical safety evaluation of biological products for prevention

[2] Technical guidelines for preclinical studies of preventive vaccines

[3] Technical guidelines for quality control and clinical study of conjugate vaccines

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PRODUCT REGISTRATION DOSSIER

13-valent Pneumococcal Polysaccharide Conjugate Vaccine
(Tetanus Toxoid Protein)

2.6 NONCLINICAL WRITTEN AND TABULATED SUMMARIES

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2.6 NONCLINICAL WRITTEN AND TABULATED SUMMARIES

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2.6.1 Introduction

The study product is a prophylactic biological product – vaccine, and is consisted of type 1, type 3, type 4, type 5, type 6a, type 6B, type 7F, type 9V, type 14, Type 18C, type 19A, type 19F and type 23F of pneumococcal capsular polysaccharide antigens covalent binding with tetanus toxoid carrier protein and adsorbed by aluminum phosphate adjuvant. This vaccine is intended to be used in the prevention of invasive infection caused by the above serotypes pneumococcus in infants and children aged 2 months to 2 years. The proposed dosage is 0.5mL per person, and the proposed route is intramuscular injection.

According to the requirements of regulatory authorities, necessary pharmacological and toxicological tests were carried out in preclinical non-clinical studies. Since the vaccine is different from ordinary chemicals, the experimental requirements of pharmacology and toxicology are special. The pharmacological tests of the study product mainly include immunogenicity, antigen compatibility, dose-response relationship, relationship between dose effect, order of administration and the overall effect, etc. In terms of toxicology, we mainly considered the pathological reactions at the injection site and systemic pathological reactions, as well as the unexpected immune response and the duration of these reactions. We carried out local muscle irritation test, hemolysis test, passive cutaneous anaphylaxis test, systemic acute toxicity test, repeat-dose toxicity test, abnormal toxicity test and active anaphylaxis test to evaluate toxicology. The results showed that the study product had good safety and immunogenicity, and could be safely used in clinical settings.

2.6.2 Pharmacology Written Summary

2.6.2.1 Brief Summary

According to the literature, New Zealand white rabbits, a sensitive species, were selected to be used in pharmacological studies and evaluation. The immunogenicity test, antigen compatibility test, dose-response relationship test and Experiment on the relationship between dose effect, order of administration and the overall effect were carried out. The results showed that the 13 types of polysaccharide conjugates had significant animal immunogenicity compared with the corresponding polysaccharides. No interference between the 13 antigens was observed. The contents of each type in each dose were as follows: serotype 1 2.2 µg, serotype 3 2.2 µg, serotype 4 2.2 µg, serotype 5 2.2 µg, serotype 6A 2.2 µg, serotype 6B 4.4 µg, serotype 7F 2.2 µg, serotype 9V 2.2 µg, serotype 14 2.2 µg, serotype 18C 2.2 µg, serotype 19A 2.2 µg, serotype 19F 2.2 µg and serotype 23F 2.2 µg. The antibody positive conversion rate and GMT level of the vaccine were similar to those of the same kind of vaccine sold in the market. The GMT value of 3-dose procedure was higher than that of 2-dose procedure.

2.6.2.2 Primary Pharmacodynamics

2.6.2.2.1 Immunogenicity

2.6.2.2.1.1 Monotype Immunogenicity of 13-type Conjugate Bulk

In order to prove the strong immunogenicity of the self-made polysaccharide conjugate bulk, the titer of specific antibody against animal antigen was detected by Elisa with corresponding derivatives of the conjugates immunizing animals under the same system conditions as control. The titers of specific antibodies before and after immunization were compared to determine whether the polysaccharide conjugates were immunogenic in animals.

2.6.2.2.1.1.1. Materials

Conjugate bulk: Pn1ps-TT 20100701, Pn3ps-TT 20100902, Pn4ps-TT 20100701, Pn5ps-TT 20100701, Pn6Aps-TT 20101102, Pn6Bps-TT 20100701, Pn7Fps-TT20101002, Pn9Vps-TT20100801, Pn14ps-TT20100801, Pn18Cps-TT20100801, Pn19Aps-TT 20101102, Pn19Fps-TT20100801, Pn23Fps-TT20100801; Polysaccharide derivatives Pn1ps-ADH 20100701, Pn3ps-ADH 20100902, Pn4ps-ADH 20100701, Pn5ps-ADH20100701, Pn6Aps-ADH 20101102, Pn6Bps-ADH20100701, Pn7Fps-ADH20101002, Pn9Vps-ADH20100801, Pn14ps-ADH20100801, Pn18Cps-ADH20100801, Pn19Aps-ADH 20101102, Pn19Fps-ADH20100801, Pn23Fps-ADH20100801. Sodium chloride injection A10011661 (manufactured by Kunming Nanjiang Pharmaceutical Co., Ltd). New Zealand white rabbits (1.8-2.2 kg, half male and half female, purchased from Animal Laboratory Center of Kunming Medical University).

2.6.2.2.1.1.2. Methods

(1) Test samples preparation: All types of antigen content referred to Prevnar Wyeth's Pneumococcal 7-Valent Conjugate Vaccine. Type 1, type 3, type 4, type 5, type 6A, type 7F, type 9V, type 14, type 18C, type 19A, type 19F and type 23F conjugates were diluted to 1.6 µg/mL respectively with sodium chloride injection, and type 6B conjugate was diluted to 3.2 µg/mL, serving as the test sample for conjugate bulks; Type 1, type 3, type 4, type 5, type 6A, type 7F, type 9V, type 14, type 18C, type 19A, type 19F and type 23F derivatives were diluted to 1.6 µg/mL respectively with sodium chloride injection, and type 6B derivative was diluted to 3.2 µg/mL, serving as the test sample for derivatives.

(2) Animal immunization: the New Zealand White Rabbit Model was adopted. The test samples were injected on Days 0, 14, 28 and blood samples were collected on Day 21 and Day 35. The antibody levels were detected to evaluate the antibody conversion rate and GMT with self-control.

2.6.2.2.1.1.3. Results

Table 1 Table of Results in the Test of Monotype Immunogenicity of 13-type Conjugate Bulk

2-dose procedure 3-dose procedure

Name and batch No.

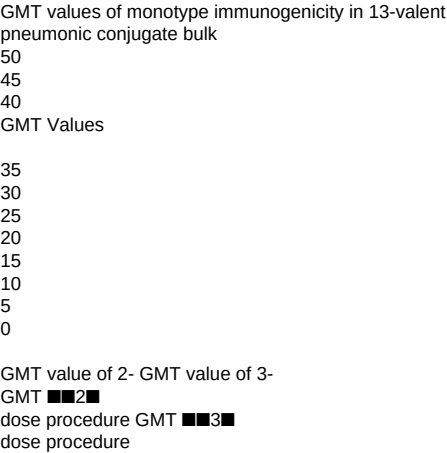
Positive conversion rate GMT Positive conversion rate GMT

Pn1ps-TT20100701	100%	18	100%	35
Pn1ps-ADH20100701	10%	1	10%	1
Pn3ps-TT 20100902	100%	23	100%	45
Pn3ps-ADH20100902	10%	1	10%	1
Pn4ps-TT 20100701	100%	26	100%	26
Pn4ps-ADH 20100701	10%	1	10%	1
Pn5ps-TT 20100701	100%	27	100%	42
Pn5ps-ADH 20100701	10%	1	10%	1
Pn6Aps-TT 20101102	100%	22	100%	30
Pn6Aps-ADH20101102	10%	1	10%	1
Pn6Bps-TT 20100701	100%	16	100%	22
Pn6Bps-ADH20100701	10%	1	10%	1
Pn7Fps-TT20101002	100%	21	100%	43
Pn7Fps-ADH20101002	10%	1	10%	1
Pn9Vps-TT20100801	100%	21	100%	32
Pn9Vps-ADH20100801	10%	1	20%	1
Pn14ps-TT20100801	100%	20	100%	45
Pn14ps-ADH20100801	10%	1	20%	1
Pn18Cps-TT20100801	100%	25	100%	42
Pn18Cps-ADH20100801	10%	1	10%	1
Pn19Aps-TT 20101102	100%	23	100%	45
Pn19Aps-ADH20101102	10%	1	10%	1
Pn19Fps-TT20100801	100%	24	100%	44
Pn19Fps-ADH20100801	10%	1	20%	1
Pn23Fps-TT20100801	100%	17	100%	28
Pn23Fps-ADH20100801	10%	1	10%	1

Figure 1 Positive Conversion Rates of Monotype Immunogenicity in 13-valent Pneumonic



Figure 2 GMT values of monotype immunogenicity in 13-valent pneumonic conjugate bulk



2.6.2.2.1.1.4. Conclusions
For the polysaccharide conjugates, the positive conversion rates of all serotypes were 100%, and the GMT values increased significantly. The results showed that the 13-type polysaccharide conjugates had significant immunogenicity in animals compared to the corresponding polysaccharides.

2.6.2.2.1.2. Immunogenicity Study of Finished Products
In order to evaluate the immunogenicity of the finished products in animals, referring to the methods published in the literatures, New Zealand female white rabbits were immunized with

different dosages of finished products with aluminium phosphate as control. Antiserum was collected after immunization, and the positive conversion rate as well as antibody GMT were detected using Elisa so as to evaluate the immunogenicity of the finished products in animals.

2.6.2.2.1.2.1. Materials

13-valent pneumococcal polysaccharide conjugate vaccine 20110707, aluminium phosphate 8169402 (manufactured by REHEIS General Chemical, US), New Zealand White rabbits (1.8-2.2 kg, half male and half female, purchased from Chengdu Dashuo Biotechnology Co., Ltd).

2.6.2.2.1.2.2. Methods

(1) Test samples preparation: The 13-valent pneumococcal polysaccharide conjugate vaccine was diluted 2.5 times with 1 mg/mL aluminum phosphate solution to 1.6 µg/mL (3.2 µg/mL for type 6B), serving as the test sample in this experiment.

(2) Immunization: The New Zealand female white rabbits were randomly divided into three groups. Rabbits in study group 1 were injected with 0.25 mL test sample through the hind limb (unilateral, 0.25 mL each side), rabbits in study group 2 were injected with 0.5 mL test sample through the hind limb (bilateral, 0.25 mL each side), and rabbits in the control group were injected with 0.5 mL aluminum phosphate (1 mg/mL, bilateral, 0.25 mL each side). The rabbits were immunized once respectively on Day 0, 14 and 28. Blood samples were collected before immunization, 7 days after the second dose and 7 days after the third dose. The serum was separated and the positive conversion rate as well as the antibody GMT were detected using ELISA.

2.6.2.2.1.2.3. Results

Table 2 Summary Table of the Positive Conversion Rates of Immunizing Rabbits Using 13-valent Pneumococcal Polysaccharide Conjugate Vaccine 20110707

Serotype

Group 1 3 4 5 6A 6B 7F 9V 14 18C 19A 19F 23F

Control

0 0 0 0 0 0 0 0 0 0 0 0 0

group

Study group

50% 60% 60% 50% 60% 40% 50% 60% 40% 50% 60% 50% 50%

1

Study group

70% 90% 80% 80% 90% 60% 80% 90% 60% 80% 90% 70% 80%

2

Table 3 Summary Table of the GMT Increasing Folds of Immunizing Rabbits Using 13-valent Pneumococcal Polysaccharide Conjugate Vaccine 20110707

Serotype

Group 1 3 4 5 6A 6B 7F 9V 14 18C 19A 19F 23F

Control

0 0 0 0 0 0 0 0 0 0 0 0 0

group

Study group

50% 60% 60% 50% 60% 40% 50% 60% 40% 50% 60% 50% 50%

1

Study group

70% 90% 80% 80% 90% 60% 80% 90% 60% 80% 90% 70% 80%

2

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2.6.2.2.1.2.4. Conclusions

The results showed that more than half of the serotypes achieved positive conversion after immunizing rabbits with the study vaccine, therefore the antigens in the vaccine had comparative immunogenicity. The post-immunization antibody titer of each type doubled when compared with the pre-immunization titer. The positive conversion rate was positively correlated with antibody titer and dosage. The body obviously gained resistance and protection against pneumococcus after immunization.

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2.6.2.2.1.3. Comparison of Immunogenicity with Similar Licensed Product

In order to compare the immunogenicity of the study vaccine with the similar licensed products, New Zealand white rabbits were immunized with the same dose of the study vaccine and Plevnar Wyeth's 7-Valent PCV. The antiserum was collected after immunization. The positive conversion rate and antibody GMT were detected using ELISA so as to compare the immunogenicity of each serotype between two products.

2.6.2.2.1.3.1. Materials

13-valent pneumococcal polysaccharide conjugate vaccine 20110707, Plevnar Wyeth's Pneumococcal 7-Valent Conjugate Vaccine E61516E52444 (2010.4-2013.3), New Zealand female white rabbits (1.5-2.5 kg, purchased from Chengdu Dashuo Biotechnology Co., Ltd).

2.6.2.2.1.3.2. Methods

(1) Test sample preparation: The Walvax's Pneumococcal 13-Valent Conjugate Vaccine (Lot No. 20110707) and Plevnar Wyeth's Pneumococcal 7-Valent Conjugate Vaccine (Lot No. E61516E52444) were diluted 2.5 times in 1 mg/mL aluminum phosphate solution respectively, each with the monotype concentration of 1.6 µg/mL (3.2 µg/mL for type 6B), serving as the test samples in this experiment.

(2) Immunization: The New Zealand female white rabbits were randomly divided into three groups. Rabbits in study group 1 were injected with 0.25 mL test sample of Walvax's Pneumococcal 13-Valent Conjugate Vaccine through the hind limb, rabbits in study group 2 were injected with 0.5 mL test sample of Plevnar Wyeth's Pneumococcal 7-Valent Conjugate Vaccine through the hind limb, and rabbits in the control group were injected with 0.5 mL aluminum phosphate (1 mg/mL). The rabbits were immunized once respectively on Day 0, 14 and 28. Blood samples were collected before immunization, 7 days after the second dose (Day 21) and 7 days after the third dose (Day 35). The serum was separated and the positive conversion rate as well as the antibody GMT were detected using ELISA.

2.6.2.2.1.3.3. Results

Table 4 Summary Table of the Positive Conversion Rates of Immunizing Rabbits Using the Study Vaccine and the Plevnar Wyeth's

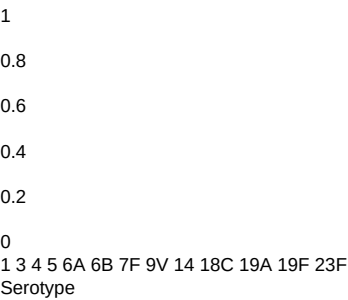
Serotype

1 3 4 5 6A 6B 7F 9V 14 18C 19A 19F 23F
Group
Control
0 0 0 0 0 0 0 0 0 0 0 0
group
Study
group 1
70% 90% 80% 80% 90% 60% 80% 90% 60% 80% 90% 70% 80%
(study
vaccine)
Study
group 2
(the / / 90% / / 60% / 80% 70% 80% / 80% 80%
Pevnar
Wyeth's)

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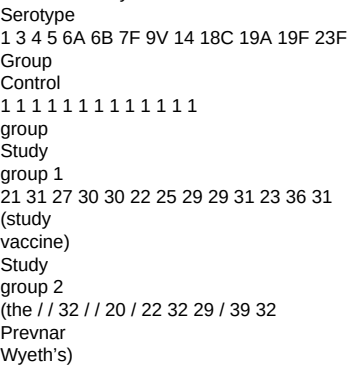
Figure 3 Comparison of the Positive Conversion Rates between the Study Vaccine and the Pevnar Wyeth's

Comparison of the Positive Conversion Rates between
the Study Vaccine and the Pevnar Wyeth's
Positive Concersion Rate



the Pevnar Wyeth's Study vaccine

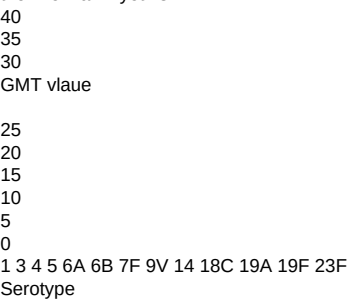
The positive conversion rates of the two study groups were compared using paired t-test. No significant difference was observed ($P=0.36 > 0.05$).
Table 5 Summary Table of the GMTs of Immunizing Rabbits Using the Study Vaccine and the Pevnar Wyeth's



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Figure 4 Comparison of the Positive Conversion Rates between the Study Vaccine and the Pevnar Wyeth's

Comparison of GMTs between the Study Vaccine and
the Pevnar Wyeth's



the Pevnar Wyeth's Study vaccine

The GMT values of the two study groups were compared using paired t-test. No significant difference was observed ($P=0.93 > 0.05$).
2.6.2.2.1.3.4. Conclusions

The results showed that the study vaccine had comparative immunogenicity as the Pevnar Wyeth's Pneumococcal 7-Valent Conjugate Vaccine.

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2.6.2.2.2 Antigen Compatibility of the 13-type Conjugates
In order to determine the antigen compatibility of the 13-type conjugates, the polysaccharide conjugates were diluted to 4 ug/mL (type 6b was diluted to 8 ug/mL) in aluminium phosphate solution referring to the dosage of Pevnar Wyeth's Pneumococcal 7-Valent Conjugate Vaccine. The monotype immunogenicity was compared with that of 13-type conjugates to determine whether the 13-type conjugates were compatible.

2.6.2.2.2.1. Materials
Conjugate bulk: Pn1ps-TT 20100701, Pn3ps-TT 20100902, Pn4ps-TT 20100701, Pn5ps-TT 20100701, Pn6Aps-TT 20101102, Pn6Bps-TT 20100701, Pn7Fps-TT20101002, Pn9Vps-TT20100801, Pn14ps-TT20100801, Pn18Cps-TT20100801, Pn19Aps-TT 20101102Pn19Fps-TT20100801, Pn23Fps-TT20100801; Pevnar Wyeth's Pneumococcal 7-Valent Conjugate Vaccine E61516E52444 (2010.4-2013.3). Aluminum phosphate (Batch No. 8169402, manufactured by REHEIS General Chemical, US). Sodium chloride injection A10011661 (manufactured by Kunming Nanjiang Pharmaceutical Co., Ltd). New Zealand white rabbits (1.8-2.2 kg, half male and half female, purchased from Chengdu Dashuo Biotechnology Co., Ltd).
2.6.2.2.2.2. Methods
(1) Test samples preparation: The conjugates were diluted to 4 ug/mL (6b type 8 ug/mL) in 1 mg/mL aluminum phosphate solution (obtained by diluting aluminum phosphate with sodium chloride injection) to obtain test samples for the monotype conjugate. The 13-type conjugates were mixed with 1 mg/mL aluminum phosphate solution to form a homogeneous suspension containing 4 ug/mL of each type conjugate (8 ug/mL for type 6b), serving as the compounding conjugate in this experiment.
(2) Immunogenicity tests: Test samples mentioned above were diluted 2.5 times to 1.6 ug/mL (3.2 ug/mL for type 6B) with 1 mg/mL aluminum phosphate solution to serve as samples in this test. Based on the New Zealand White Rabbit Model, 0.5 mL samples were intramuscularly injected on Day 0, 14 and blood samples were collected on Day 21. The antibody levels were detected to evaluate the antibody conversion rate with self-control.
2.6.2.2.2.3. Results
No significant difference was observed between the positive conversion rates in the monotype group and the compounding group based on the results obtained from paired data t-test ($p = 0.444 > 0.05$).

Table 6 Summary Table of the Positive Conversion Rates in Immunogenicity tests

Serotype 1 3 4 5 6A 6B 7F 9V 14 18C 19A 19F 23F

Conversion rates

in the

70 60 100 60 70 60 100 100 100 100 100 100 30

compounding

group

Conversion rates

in the monotype 100 100 100 100 100 60 100 100 80 100 80 70 60

group

2.6.2.2.2.4. Conclusions

The results showed that the monotype immunogenicity of conjugates was basically the same as that of the 13-valent mixture, indicating that the 13-type polysaccharide conjugates were well compatible with each other as expected.

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2.6.2.3 Secondary Pharmacodynamics
2.6.2.3.1. Experiment on the Relationship between Dose Effect, Order of Administration and the Overall Effect
To be added.

2.6.2.3.2. Validation Test of Formulation Preparation Process
To be added.

2.6.2.4 Safety Pharmacology
For the study product, potential unexpected adverse effects on physiological function, which refers to the effect of vaccine on central nervous system, cardiovascular system and respiratory system, were observed in the repeat-dose toxicity test. Please refer to the data of repeat-dose toxicity test for details

At the end of administration and recovery period, heart rate, no obvious abnormalities were observed in R-R interval, P-R interval, QRS wave time, Q-T interval, corrected Q-T interval and ECG shape of lead II of each monkey.

2.6.2.5 Pharmacodynamic Drug Interactions
Not applicable for vaccines.

2.6.2.6 Discussion and Conclusions
The study product is a multivalent combined vaccine made of 13 types of antigens. After immunization, the animals were able to produce specific immune response to 13 types of antigens. With the increase of the number of immune injections, the antibody level of immune response increased. The results of three consecutive batches of process validation showed that the study product had good immunogenicity and the formulation preparation process was feasible and stable.

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2.6.3 Pharmacology Tabulated Summary
Test samples: 13 valent pneumococcal polysaccharide conjugate vaccine (bulk of 13 serotypes of pneumococcal polysaccharide conjugates).

Experiment type Test sample Experiment Administration Experiment Experiment
type system route Institute No.
Immunogenicity bulk New Zealand Intramuscular Walvax R & /
test white rabbits injection D Center

model
Antigen bulk New Zealand Intramuscular Walvax R & /
compatibility test white rabbits injection D Center
model
Mice and Intraperitoneal Walvax R & /
guinea pigs injection D Center
Experiment on the Formulation New Zealand Intramuscular Walvax R & /
relationship white rabbits injection D Center
between dose model
effect, order of
administration and
the overall effect
Validation test of Formulation New Zealand Intramuscular Walvax R & /
formulation white rabbits injection D Center
preparation process model

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2.6.4 Pharmacokinetic Written Summary
Not applicable to biological products.

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2.6.5 Pharmacokinetic Tabulated Summary
Not applicable to biological products.

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2.6.6 Toxicology Written Summary

2.6.6.1 Brief Summary

In terms of toxicology, the pathological reactions at the injection site and the systemic pathological reactions as well as the unexpected immune responses and the duration of the reactions were mainly considered. The local muscle irritation test, hemolysis test, passive cutaneous anaphylaxis test, systemic acute toxicity test, repeat-dose toxicity test, abnormal toxicity test and active anaphylaxis test were carried out to evaluate toxicology.

The results showed that the dosage of the study product was within the safe range. The muscles at the injection site were irritated to some extent after the study product was given to the animals, but the irritation could be recovered after the administration ended. Apart from the local irritation, no obvious abnormalities were observed in the general conditions, body temperature, body weight, food intake, blood biochemistry, hematology, urine, bone marrow, ophthalmology, electrocardiogram, organ weight, histopathological examination and other measurements of the animals, and no obvious enhancement or inhibition of immune function was observed. Specific antibodies related to vaccine components can be detected. Considering the potential risk of allergy induced by the carrier protein, special attention should be paid in clinical use. The study product was able to coagulate rabbit red blood cells, but no hemolytic reaction was observed.

2.6.6.2 Single-Dose Toxicity

To be added.

2.6.6.3 Repeat-Dose Toxicity

To be added.

2.6.6.4 Genotoxicity

Not applicable to this product.

2.6.6.5 Carcinogenicity

Not applicable to the study product.

2.6.6.6 Reproductive and Development Toxicity

Not applicable to the study product.

2.6.6.7 Local Tolerance

2.6.6.7.1. Muscle Irritation Test

In accordance with the relevant requirements predefined by the Provisions for Drug Registration on the declaration of new drugs, the single intramuscular injection irritation test of 13-valent pneumococcal conjugate vaccine manufactured by Yunnan Walvax Biotechnology Co., Ltd. was performed by National Chengdu New Drug Safety Evaluation Center in rabbits.

2.6.6.7.1.1. Objectives

To observe the irritant reaction in quadriceps femoris of rabbits induced by 13-Valent pneumococcal conjugate vaccine and to provide a reference for evaluating the safety of clinical use.

2.6.6.7.1.2. Methods

A total of 11 groups were set in this experiment, including a negative control group, a low aluminum phosphate dosage group and a high aluminum phosphate dosage group, a low-dose group and a high-dose group for the 7-valent PCV, as well as a low-dose and a high-dose group

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for each of the 3 batches of 13-valent pneumococcal conjugate vaccine (batch no. 20110304, 20110305 and 20110406). Each group included 4 rabbits, half male and half female. Rabbits in the low-dose groups were given one dose of test sample at the corresponding concentration injected through quadriceps femoris in the volume of 0.5 mL per side, and rabbits in the high-dose groups were given one dose of test sample at the corresponding concentration injected through quadriceps femoris in the volume of 1.5 mL per side. Rabbits in the negative control group was given 0.9% sodium chloride injection in the volume of 1.5 mL per side. Two rabbits of each group were euthanized (CO₂ inhalation) and anatomized respectively 48 hours after administration and 16 days after administration. The injected quadriceps femoris was observed and graded, meanwhile histopathology was performed.

2.6.6.7.1.3. Results

(1) General conditions and gross anatomy results observed by naked eye: No local inflammation and congestion was observed at the injection site during the administration or recovery period.

About 48 hours after administration, in the high-dose group of 7-valent PCV, slight hyperemia was observed at the right injection site of the quadriceps femoris in 1 rabbit and moderate hyperemia was observed at the left injection site of the quadriceps femoris in 1 rabbit; in the high-dose group of 13-valent PCV (batch No. 20110304), moderate hyperemia was observed at the right injection site of the quadriceps femoris in 1 rabbit; in the high-dose group of 13-valent PCV (batch No. 20110406), moderate hyperemia was observed at the right injection site of the quadriceps femoris in 1 rabbit. No significant abnormalities were observed at the injection site of the quadriceps femoris in the rest rabbits. No obvious abnormal changes were observed in the muscle at the injection site in all rabbits 16 days after administration.

(2) Histopathology results: About 48 hours after administration, changes such as mild to severe degenerative necrosis of muscle fiber or muscle interstitial, bleeding, inflammatory cell infiltration were observed in the high-dose group of 7-valent PVC, the high-dose groups of 13-valent PCV of 3 different batch numbers and the low-dose group of 13-valent PCV with batch No. 20110305, indicating muscle irritation at the injection site. After 16 days of administration, mild interstitial inflammatory cell infiltration was still observed in the muscles at the injection site in the low-dose and high-dose group of aluminum phosphate, as well as the high-dose group of 13-valent PCV with batch No.20110305. No obvious abnormalities were observed in the muscles at the injection site in the rest groups, indicating that the above irritation could be partially recovered after the 16-day observation.

2.6.6.7.1.4. Conclusions

Obvious irritation to quadriceps femoris of rabbits was induced by the injection of low-dose and high-dose aluminum phosphate, high-dose 7-valent PCV, high-dose 13-valent PCV of 3 different batch numbers, low-dose 13-valent PCV with batch No.20110406. The muscle irritation can be recovered to some extent after 16-day observation.

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2.6.6.7.2. Passive Cutaneous Anaphylaxis Test

In accordance with the relevant requirements predefined by the Provisions for Drug Registration on the declaration of new drugs, the passive cutaneous anaphylaxis test of 13-valent pneumococcal conjugate vaccine manufactured by Yunnan Walvax Biotechnology Co., Ltd. was performed by National Chengdu New Drug Safety Evaluation Center through intramuscular injection in Guinea pigs.

2.6.6.7.2.1. Objectives

To observe whether the passive cutaneous anaphylaxis was induced by 13-valent pneumococcal conjugate vaccine in guinea pigs, and to provide a reference for evaluating the safety of clinical use.

2.6.6.7.2.2. Methods

(1) Antiserum preparation: Twenty guinea pigs (half male and half female) were randomly divide into 10 groups including a negative control group, a positive control group, a low-dose group (sensitizing dosage of 0.05 mL bulk/kg, which was double as the clinical proposed dosage) and a high-dose group (sensitizing dosage of 0.5 mL bulk/kg, which was 10 times as the clinical proposed dosage) for the 7-Valent PCV and for each of the 3 batches of 13-Valent PCV (batch no. 20110304, 20110305 and 20110406) respectively. When sensitized, guinea pigs in the 13-valent pneumococcal conjugate vaccine group and the 7-valent pneumococcal conjugate vaccine group were given samples or control through intramuscular injection at the corresponding concentration according to the above prescribed dosage. Guinea pigs in the positive control group were given 10mg/mL ovalbumin solution in a volume of 0.5 mL per guinea pig through intraperitoneal injection, while guinea pigs in the negative control group were given 0.9% sodium chloride injection (normal saline) through intramuscular injection in a volume of 0.5 mL/kg. Sensitization was carried out 5 times continuously (once every other day). On the 10th day after the last sensitization, blood samples were collected through aorta abdominalis for antiserum preparation. The serums of two animals in the same group were mixed and kept in reserve.

Table 7 Summary Table of Administration Methods for Each Group in Passive Anaphylaxis Experiment

Group Administration method (2 Guinea Pigs/Group)

Negative Control Group 0.5mL/kg (0.9% Sodium Chloride)

Positive Control Group 10mg/mL (Ovalbumin Solution)

7-Valent PCV Low-dose Group 0.05mL/kg (Bulk)

Group High-dose Group 0.5mL/kg (Bulk)

Low-dose Group 0.05mL/kg (Bulk)

20110304

High-dose Group 0.5mL/kg (Bulk)

Low-dose Group 0.05mL/kg (Bulk)

20110305

High-dose Group 0.5mL/kg (Bulk)

Low-dose Group 0.05mL/kg (Bulk)

20110306

High-dose Group 0.5mL/kg (Bulk)

(2) Locus coeruleus test: Sixty guinea pigs (half male and half female) were randomly divided into 10 groups at the same setting as the antiserum preparation stage. The antiserum of each group was diluted at the degree of 1:2, 1:8 and 1:32, and serum of each dilution degree was injected intracutaneously to each animal through 3 pre-depilated points (3×4cm²) on the back in the volume of 0.1 mL per point. Four hours after the intradermal injection, 1 mL mixture consisted of the same

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dosage of provocative antigens as in step (1) and the same amount of 1% evensland dye was injected intravenously to the guinea pigs. Thirty minutes after the injection, guinea pigs were anesthetized (pentobarbital sodium, about 60 mg/kg, intraperitoneally injected) and euthanized by abdominal aortic bleeding. The length and diameter of the locus coeruleus were measured and the locus coeruleus were photographed

2.6.6.7.2.3. Conclusions

For the negative control group, the diameters of intradermal locus coeruleus at the injection site were 0 mm at all dilution degrees; for the positive control group, the average diameter of locus coeruleus was 15.23 mm when the dilution degree was 1:32, 19.17 mm when the dilution degree was 1:8, and 22.6 mm when the dilution degree was 1:2. The diameters of locus coeruleus for all vaccine groups were 0mm at all dilution degrees.

2.6.6.7.3. Hemolysis Test

In accordance with the relevant requirements predefined by the Provisions for Drug Registration on the declaration of new drugs, the hemolysis test of the study product was performed by National Chengdu New Drug Safety Evaluation Center through intramuscular injection in Guinea pigs, with a purpose to observe the effect of the study product on the state of rabbit red blood cells and to provide reference for the safety evaluation of clinical use.

The groups in this experiment included the negative control group (0.9% sodium chloride injection), the positive control group (sterile water for injection), adjuvant control group (1 mg/mL aluminum phosphate concentration) and the study product of 3 batches (batch No. 20110304, 20110305, 20110406). Five groups of different dosages (0.5, 0.4, 0.3, 0.2, 0.1 mL/tube) were set for aluminum phosphate and the study product of each batch respectively. Rabbit red blood cell suspension (2%), aluminum phosphate adjuvant, the study product, 0.9% sodium chloride injection and sterilized water for injection were mixed in a certain proportion and placed in a 37 ± 0.5 °C incubator. The compound was observed at 0, 15, 30, 45, 60 minutes and 2, 3 hours respectively. Three parallel tubes were made for each dosage.

The results showed that no hemolytic and coagulation reaction were observed in the negative control group and no hemolytic reaction was observed in the positive control group after preserved at 37 ± 0.5 °C for 3 hours. In the aluminum phosphate adjuvant group, red blood cell aggregation was observed in each tube (0.5, 0.4, 0.3, 0.2, 0.1 mL/tube), and the degree of aggregation was dose-dependent, but no hemolytic reaction was observed. Red blood cell aggregation was observed in each tube (0.5, 0.4, 0.3, 0.2, 0.1 mL/tube) of three batches of the study product, and the degree of aggregation was lower than that of the aluminum phosphate group, but no significant dose-dependent relationship and no hemolytic reaction were observed.

In conclusion, under conditions of this experiment, aluminum phosphate adjuvant (concentration of 1 mg/mL, the same as the concentration of aluminum phosphate in the formulation of the study product) and the study product (batch No. 20110304, 20110305, 20110406) at different dosages (0.5, 0.4, 0.3, 0.2, 0.1 mL/tube) had aggregation effect on rabbit red blood cells, but no hemolytic reaction was observed.

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2.6.6.8 Other Toxicity Studies

2.6.6.8.1 Antigenicity

N/A

2.6.6.8.2 Immunotoxicity

N/A

2.6.6.8.3 Mechanistic studies (if not reported elsewhere)

N/A

2.6.6.8.4 Dependence

N/A

2.6.6.8.5 Studies on metabolites

N/A

2.6.6.8.6 Studies on impurities

N/A

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2.6.6.8.7 Other studies

2.6.6.8.7.1 Acute Toxicity Study

In accordance with the relevant requirements predefined by the Provisions for Drug Registration on the declaration of new drugs, the acute toxicity test of 13-valent pneumococcal conjugate vaccine manufactured by Yunnan Walvax Biotechnology Co., Ltd. was performed by National Chengdu New Drug Safety Evaluation Center through intramuscular injection in Kunming mice.

2.6.6.8.7.1.1 Objectives

To evaluate the nature, degree, time-effect relationship and reversibility of the possible toxic reaction; to determine the toxic target organs or target tissues; to provide a reference range of safety for clinical use.

2.6.6.8.7.1.2 Methods

The Walvax's 13-Valent PCV samples were divided into 3 groups by batch numbers of 20110304, 20110305, and 20110406, respectively. Mice in each group were administered with 13-Valent PCV of corresponding batch number at a concentration of 64.4 µg/mL (calculated based on the total amount of pneumococcal polysaccharide) in a volume of 10 mL/kg. Another negative control group was set and mice in the control group was injected with 0.9% sodium chloride injection at the volume of 10 mL/kg. Each group included 10 mice, half male and half female. The maximum administration volume for each mouse was a single intramuscular injection with corresponding test sample or sodium chloride injection (intramuscularly injected on bilateral hind legs simultaneously) in a volume of 10 mL/kg. The day of administration was defined as Day 1 in this experiment.

The general conditions of mice in each group was observed daily after administration; the body weight was measured twice and food intake was measured once each week; on the 15th day of this experiment, abdominal aortic blood was collected from all surviving mice after anesthetization (about 60 mg/kg of sodium pentobarbital, intraperitoneal injection) for hematological and biochemical examination. The mice were also systematically autopsied for comprehensive observation of animal appearances and internal organs.

2.6.6.8.7.1.3 Conclusions

The mice were in good condition during the 14-day observation after administration of Pneumococcal 13-Valent Conjugate Vaccine (batch no. 20110304, 20110305, 20110406) in the maximum dosage volume of 10 mL/kg. No obvious toxicity reaction was observed.

Table 8 Summary Table of Observation Results in Acute Toxicity Study

Observation

Results

measurement

During the 14-day observation after administration, the mice in each group had normal physical appearance, mental and motor state. The natural pores were clean.

General conditions

No obvious abnormal reactions were observed. No irritant reactions such as redness, swelling, ecchymosis, induration, suppuration and fester were observed.

Body weight and No abnormal changes were observed in body weight and food intake for mice in food intake each group 14 days after administration.

hematological and

On the 15th day of the experiment, no abnormal results were observed in the biochemical

hematological and biochemical measurements for mice in each group. examination

On the 15th day of the experiment, no abnormal changes in the size, shape, color

Gross anatomy and texture of the heart, liver, spleen, lung and kidney were observed for mice in results each group. No local irritant reactions such as swelling and induration in the muscles at the injection site.

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2.6.6.8.7.2 Abnormal Toxicity

2.6.6.8.7.2.1 Abnormal Toxicity Test in Mice

In accordance with the relevant requirements predefined by the Provisions for Drug Registration on the declaration of new drugs, the abnormal toxicity test of 13-valent pneumococcal conjugate vaccine manufactured by Yunnan Walvax Biotechnology Co., Ltd. was performed by National Chengdu New Drug Safety Evaluation Center through intraperitoneal injection in mice.

2.6.6.8.7.2.1.1 Objectives

To determine whether the samples were contaminated with foreign substances and whether there were unexpected unsafe factors; to provide a reference for evaluating the safety of clinical use.

2.6.6.8.7.2.1.2 Methods

One dosage group of 13-Valent PCV sample at the clinical proposed dosage (0.5 mL per mouse) was set in this experiment and a total of 5 male mice were included. Each mouse was injected with 0.5 mL of the sample once intraperitoneally. Measurements such as the animals' appearance, general behavior, mental state, inhalation, genitalia, crissum, fecal urine, articulo mortis or death were observed after administration and recorded once a day, for a total of 7 days. The animals were weighed after the 7-day observation ended. Specific results are showed in the table below.

Table 9 Summary Table of Observation Results in Abnormal Toxicity Test with mice

Observation

Results

measurements

All mice had normal physical appearance, mental and motor state. The natural

General

pores were clean. No obvious abnormal reactions were observed from the time the

Conditions

vaccine was administered until the end of the observation.

On the 7th day after administration, the weight of each mouse increased compared

Weight

with that before administration.

2.6.6.8.7.2.1.3 Conclusions

The abnormal toxicity test of 13-Valent pneumococcal conjugate vaccine in mice was up to standard.

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2.6.6.8.7.2.2 Abnormal Toxicity Test in Guinea Pigs

In accordance with the relevant requirements predefined by the Provisions for Drug Registration on the declaration of new drugs, the abnormal toxicity test of 13-valent pneumococcal conjugate vaccine manufactured by Yunnan Walvax Biotechnology Co., Ltd. was performed by National Chengdu New Drug Safety Evaluation Center through intraperitoneal injection in guinea pigs.

2.6.6.8.7.2.2.1 Methods

One dosage group of 13-Valent PCV sample at the clinical proposed dosage (5 mL per guinea pig, 10 doses) was set in this experiment and a total of 6 guinea pigs (half male and half female) were included. Each guinea pig was injected with 5 mL of the sample once intraperitoneally.

Measurements such as the animals' appearance, general behavior, mental state, inhalation, genitalia, crissum, fecal urine, articulo mortis or death were observed after administration and recorded once a day, for a total of 7 days. The animals were weighed after the 7-day observation ended. Specific results are showed in the table below.

Table 10 Summary Table of Observation Results in Abnormal Toxicity Test with guinea pigs

Observation

Results

measurements

All guinea pigs had normal physical appearance, mental and motor state. The

General

natural pores were clean. No obvious abnormal reactions were observed from the

Conditions

time the vaccine was administered until the end of the observation.

On the 7th day after administration, the weight of each guinea pig increased

Weight

compared with that before administration.

2.6.6.8.7.2.2.2 Conclusions

The abnormal toxicity test of 13-Valent pneumococcal conjugate vaccine in guinea pigs was up to standard.

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2.6.6.8.7.3 Specific Toxicity Test and Test for Reversion to Toxicity - Tetanus Toxoid Toxicity Tests

In accordance with the relevant requirements of the Technical Guidelines for Pre-clinical Research of Prophylactic Vaccines and the Technical Guidelines for Conjugate Vaccine Quality Control and Clinical Research, the specific toxicity test and toxicity irreversibility test targeting the carrier protein (tetanus toxoid) were performed for each type of conjugate bulk. Based on the methods of specific toxicity test and toxicity irreversibility test for tetanus toxoid specified in the "adsorptive

tetanus vaccine" part of Pharmacopoeia of the People's Republic of China (2010 edition, Volume III), the specific toxicity test and test for reversion to toxicity were performed for 13 types of conjugates, and the safety of each type of conjugate bulk was evaluated.

2.6.6.8.7.2.3.1 Materials

Conjugate bulks: Pn1ps-TT 20090501, 20090502, 20090503■Pn4ps-TT 20090201, 20090202, 20090203 ■ Pn5ps-TT 20090401, 20090502, 20090503 ■ Pn6Bps-TT 20090301, 20090302, 20090303 ■ Pn9Vps-TT20090301, 20090302, 20090303 ■ Pn14ps-TT20090301, 20090302, 20090303■Pn18Cps-TT20090301, 20090402, 20090403 ■ Pn19Fps-TT20090401, 20090402, 20090403 ■ Pn23Fps-TT20090401, 20090402, 20090403 ■ Pn3ps-TT20100903 ■ Pn6Aps-TT20101103■Pn7Fps-TT20101003■Pn19Aps-TT20101103. Healthy guinea pigs (250-350 g, purchased from Animal Laboratory Center of Kunming Medical University).

2.6.6.8.7.2.3.2 Methods

(1) Specific Toxicity Test

Each guinea pig (4 guinea pigs in total) was subcutaneously injected with 2 mL conjugate bulk and observed for 21 days.

(2) Test for Reversion to Toxicity

The TT flocculating unit was assigned as A Lf/mL, protein concentration as B µg/mL, and protein concentration in conjugate bulk as C µg/mL. One Lf corresponded to B/A µg protein. Conjugate bulk in a volume of 300 *(b/a)/C ml was diluted to 30ml (sterile operation) with PBS, and set aside at 37■for 42 days. Each guinea pig was subcutaneously injected with 5 mL sample (4 guinea pigs in total) and observed for 21 days.

2.6.6.8.7.2.3.3 Results

The results of specific toxicity test and test for reversion to toxicity were summarized in the table below.

Table 11 Summary Table of Toxicity Test Results for Carrier Proteins (Tetanus Toxoid)

Specific toxicity test Toxicity irreversibility test

Animals should not have tetanus Animals should not have tetanus

symptoms; at the end of the experiment, the symptoms; at the end of the experiment, the Standard

weight of each animal should increase weight of each animal should increase

compared with that before administration. compared with that before administration.

Pn1ps-TT

Up to standard Up to standard

20090501

Pn1ps-TT

Up to standard Up to standard

20090502

Pn1ps-TT

Up to standard Up to standard

20090503

Pn3ps-

Up to standard Up to standard

TT20100903

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Pn4ps-TT

20090201

Pn4ps-TT

Up to standard Up to standard

20090202

Pn4ps-TT

Up to standard Up to standard

20090203

Pn5ps-TT

Up to standard Up to standard

20090401

Pn5ps-TT

Up to standard Up to standard

20090502

Pn5ps-TT

Up to standard Up to standard

20090503

Pn6Aps-

TT20101103

Up to standard Up to standard

Pn6Bps-

TT20090301

Pn6Bps-

Up to standard Up to standard

TT20090302

Pn6Bps-

Up to standard Up to standard

TT20090303

Pn7Fps-

TT20101003

Up to standard Up to standard

Pn9Vps-

TT20090301

Pn9Vps-

Up to standard Up to standard

TT20090302

Pn9Vps-

Up to standard Up to standard

TT20090303

Pn14ps-

Up to standard Up to standard

TT20090301

Pn14ps-

Up to standard Up to standard

TT20090302

Pn14ps-
Up to standard Up to standard
TT20090303
Pn18Cps-
Up to standard Up to standard
TT20090301
Pn18Cps-
Up to standard Up to standard
TT20090402
Pn18Cps-
Up to standard Up to standard
TT20090403
Pn19Aps-
TT20101103
Up to standard Up to standard
Pn19Fps-
TT20090401
Pn19Fps-
Up to standard Up to standard
TT20090402
Pn19Fps-
Up to standard Up to standard
TT20090403
Pn23Fps-
Up to standard Up to standard
TT20090401
Pn23Fps-
Up to standard Up to standard
TT20090402
Pn23Fps-
Up to standard Up to standard
TT20090403

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2.6.6.8.7.2.3.4.2. Conclusions
Toxicity tests of the tetanus toxoid as the carrier protein were in compliance with regulations.

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2.6.6.8.7.4. Systemic Active Anaphylaxis Experiment
In accordance with the relevant requirements predefined by the Provisions for Drug Registration on the declaration of new drugs, the systemic active anaphylaxis test of 13-valent pneumococcal conjugate vaccine manufactured by Yunnan Walvax Biotechnology Co., Ltd. was performed by National Chengdu New Drug Safety Evaluation Center through intramuscular injection in Guinea pigs.

2.6.6.8.7.4.1. Objectives
To observe whether the systemic anaphylaxis was induced by 13-valent pneumococcal conjugate vaccine in guinea pigs; to provide a reference for evaluating the safety of clinical use.

2.6.6.8.7.4.2. Methods
A total of 10 groups were set in this experiment, including a negative control group, a positive control group, a low-dose group and a high-dose group for the 7-valent PCV, as well as a low-dose and a high-dose group for each of the 3 batches of 13-valent pneumococcal conjugate vaccine (batch no. 20110304, 20110305 and 20110406). Each group included six guinea pigs, half male and half female. The sensitizing dosage for the low-dose groups was 0.05 mL bulk per kg, which was twice as the proposed clinical dosage; the sensitizing dosage for the high-dose groups was 0.5 mL bulk per kg, which was 10 times as the proposed clinical dosage. When sensitized, guinea pigs in the 13-valent pneumococcal conjugate vaccine group and the 7-valent pneumococcal conjugate vaccine group were given samples or control through intramuscular injection at the corresponding concentration according to the above prescribed dosage. Guinea pigs in the positive control group were given 10mg/mL ovalbumin solution in a volume of 0.5 mL per guinea pig through intraperitoneal injection, while guinea pigs in the negative control group were given 0.9% sodium chloride injection (normal saline) through intramuscular injection in a volume of 0.5 mL/kg. Sensitization was carried out 5 times continuously (once every other day). On the 10th day after the last sensitization, the guinea pigs in each group were stimulated by a single injection through the dorsalis pedis vein at twice the sensitizing dosage. Systemic reactions and death of guinea pigs were observed for at least 30 minutes after irritation injection.

Table 12 Summary Table of Administration Methods for Each Group in Systemic Active Anaphylaxis Experiment

Administration method (6 Guinea Pigs/Group, half male
Group
& half female)
Negative Control Group 0.5mL/kg (0.9% Sodium Chloride)
Positive Control Group 10mg/mL (Ovalbumin Solution)
7-Valent PCV Low-dose Group 0.05mL/kg (Bulk)
Group High-dose Group 0.5mL/kg (Bulk)
Low-dose Group 0.05mL/kg (Bulk)
20110304
High-dose Group 0.5mL/kg (Bulk)
Low-dose Group 0.05mL/kg (Bulk)
20110305
High-dose Group 0.5mL/kg (Bulk)
Low-dose Group 0.05mL/kg (Bulk)
20110306
High-dose Group 0.5mL/kg (Bulk)

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2.6.6.8.7.4.3 Results

No anaphylaxis was observed for guinea pigs in the negative control group, 7-valent pneumococcal conjugate vaccine control group and the low-dose groups of 13-valent pneumococcal conjugate vaccine in three different batch numbers within 30 minutes after injection. All guinea pigs in the positive control group experienced strong positive reaction and all died within 4 minutes. Three guinea pigs in the 13-valent pneumococcal conjugate vaccine (batch No. 20110304, 0.5 mL/kg) group experienced strong positive allergic reaction with a duration of 6-21 minutes; one guinea pig experienced extremely positive reaction and died about 7 minutes after being stimulated. One guinea pig in the 13-valent pneumococcal conjugate vaccine (batch No. 20110305, 0.5 mL/kg) group experienced positive allergic reaction and one experienced strong positive reaction with a duration of 4-6 minutes; one guinea pig experienced extremely positive reaction and died about 7 minutes after being stimulated. Three guinea pigs in the 13-valent pneumococcal conjugate vaccine (batch No. 20110306, 0.5 mL/kg) group experienced positive allergic reaction and one experienced strong positive reaction with a duration of 23-30 minutes.

2.6.6.8.7.4.4. Conclusions

Under a dosage of 0.5 mL bulk/kg, the systemic active anaphylaxis tests on guinea pigs were all positive. Under a dosage of 0.05 mL bulk/kg, the systemic active anaphylaxis tests on guinea pigs were all negative.

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2.6.6.8.7.5. Long-Term Toxicity Test

2.6.6.8.7.5.1. Long-Term Toxicity Test in New Zealand Rabbits

In accordance with the relevant requirements predefined by the Provisions for Drug Registration on the researches of new drugs, toxicity test of repeated intramuscular injection (5 times in total) for 13-valent pneumococcal conjugate vaccine manufactured by Yunnan Walvax Biotechnology Co., Ltd. was performed by National Chengdu New Drug Safety Evaluation Center in New Zealand rabbits.

2.6.6.8.7.5.1.1. Objectives

To evaluate the preclinical safety of 13-valent pneumococcal conjugate vaccine, including the nature, extent, dose-response and time-effect relationship, as well as the reversibility of probable toxicities to the immune organs and other toxic target organs induced by the vaccine; to provide a reference for clinical researches.

2.6.6.8.7.5.1.2. Methods

A total of 6 groups were set in this experiment, including a low-dose group (0.13 mL per animal) and a high-dose group (0.5 mL per animal) for the 13-valent PCV, a low-dose group (0.13 mL per animal) and a high-dose group (0.5 mL per animal) for aluminum phosphate adjuvant at a concentration of 1 mg/mL, a 7-valent PCV group (0.5 mL per animal) as well as a negative control group (0.5 mL of 0.9% NaCl solution per animal). A total of 20 New Zealand rabbits were included in each group (half male and half female). The rabbits in each group were injected with corresponding sample or control through quadriceps femoris according to the given dosage. The administration was carried out once respectively on Week 0, 2, 4, 6 and 8 (5 times in total). The animals were observed for 28 days after the last administration. The day of the first injection was defined as Day 1 of this experiment.

The general conditions of the rabbits were observed daily during the administration and recovery period. The body weight and food intake were measured once each week. The rectal temperature was measured about 60 minutes and 24 hours after each administration. Antibodies were detected in the negative control group and the 13-valent PCV group before administration, 21 and 35 days after the first injection, 14 days after the last injection as well as at the end of the recovery period. T lymphocyte (CD4+, CD8+), cytokines (IL-2, IL-6, IFN- γ) and immunoglobulin (IgG) were detected 21 and 35 days after the first injection, 72 hours after the last injection as well as at the end of recovery period. Hematology, blood biochemistry and complements (C3, C4) of each group were detected 72 hours, 21 days and 35 days after the first injection, 72 hours after the last injection, as well as at the end of the recovery period. In each group, 12 rabbits (half male and half female) were euthanized 72 hours after the last injection and 8 rabbits (half male and half female) were euthanized at the end of the recovery period. Urine and bone marrow were examined and systematic autopsy as well as histopathological examination were performed.

2.6.6.8.7.5.1.3 Results

(1) General conditions: On the 12th day after the last administration, one male rabbit (No. 4M09) in the high-dose group of aluminum phosphate adjuvant died. The animal developed severe loose stool for three days before death. The death of this animal was an individual accident and should not be related to the injection of aluminum phosphate adjuvant. Other than that, no death was observed in other groups of rabbits during the period of administration and recovery. The rabbits had normal physical appearance, mental and motor state. The natural pores were clean. No obvious abnormal reactions were observed. No toxic reactions were observed. No local irritant reactions

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such as inflammation, ecchymosis, induration, suppuration and fester were observed at the injection site.

(2) Body weight and food intake: No significant abnormalities were observed in the body weight and food intake of the rabbits in each group at the detection time points in the administration and recovery period.

(3) Body temperature: No significant abnormalities were observed in the body temperature of the rabbits in each group at the detection time points after each administration.

(4) Hematology: No significant abnormalities were observed in the WBC count and classification, RBC, HGB, HCT, RET, PLT, PT and APTT at the detection time points in the administration and recovery period.

(5) Blood biochemical measurements: No significant abnormalities were observed in ALP, ALT, AST, GGT, TP, ALB, GLU, TBIL, GHOL, TG, CK, Urea, Crea and the electrolyte concentration (K⁺■Na⁺■Cl⁻) at the detection time points in the administration and recovery period.

(6) Immunological measurements: No significant abnormalities were observed in the complements (C3, C4), T lymphocyte (CD 4+, CD 8+), cytokines (IL-2, IL-6, IFN- γ) and Immunoglobulin (IgG) at the detection time points in the administration and recovery period. All serotypes of polysaccharide antibodies induced by 13-valent pneumococcal conjugate vaccine could be detected in the serum of rabbits in both low-dose and high-dose group 21 days after the

first administration. In the low-dose group, the levels of serotype 1, 3, 4, 5, 7F and 9V polysaccharide antibody peaked 21 days after the first administration, while serotype 6A, 6B, 14, 18C, 19A, 19F and 23F peaked 35 days after the first administration and then decreased gradually. Antibodies of certain titers could still be detected 28 days after the last administration. In the high-dose group, the levels of serotype 1, 3, 4, 5, 6A, 7F 9V 18C and 19A polysaccharide antibody peaked 21 days after the first administration, while serotype 6B, 14, 19F and 23F peaked 35 days after the first administration and then decreased gradually. Antibodies of certain titers could still be detected 28 days after the last administration. Except for serotype 5, 6A, 6B, 19F and 23F, the antibody titers of other polysaccharide serotypes were dose-related, and the serum antibody level of high-dose group was slightly higher than that of low-dose group.

(7) Urine measurements: No obvious abnormalities were observed in the grade distribution or positive rate of urine glucose, bilirubin, acetone bodies, specific gravity, occult blood, PH value, urocholinogen, nitrite and white blood cells in the urine of rabbits in each group 72 hours after the last administration and at the end of the recovery period.

(8) Bone marrow measurements: No obvious abnormalities were observed in granulocyte system, erythroid system, lymphocyte percentage, monocytes percentage, percentage of other cells, granulocyte-erythroid ratio, and megakaryocyte count in the bone marrow of rabbits in each group 72 hours after the last administration and at the end of the recovery period.

(9) Organ weight and coefficients: No obvious abnormalities were observed in the organ weight and coefficient of brain, heart, liver, spleen, lung, kidney, adrenal gland, thymus and reproductive system of the rabbits in each group 72 hours after the last administration and at the end of the recovery period.

(10) Gross anatomy and histopathology examination: Dark red areas of different sizes were observed in the muscles at the injection site of 1 rabbit (1 / 10) respectively in the high-dose group of 7-valent PCV, the high-dose group of aluminum phosphate adjuvant, as well as both the high-dose group and the low-dose group of 13-valent PCV 72 hours after the last administration. Gray areas (0.5×0.5cm) were observed in the muscles at the injection site of 1 Rabbit (1 / 10) in the

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high-dose group of 13-valent PCV. Other than the above changes, no abnormal changes associated with the study vaccine were observed with naked eye in the location, size, shape, color and texture of organs or tissues such as brain, heart, liver, spleen, lung, kidney, gastrointestinal tract and reproductive system. Mild to moderate degeneration of muscle fibers and inflammatory cell infiltration in different proportions were observed in the muscles of rabbits injected with 7-valent PCV, aluminum phosphate adjuvant and 13-valent PCV 72 hours after the last administration. Other than that, no histopathological changes were found in the brain, heart, liver, spleen, lung, kidney, gastrointestinal tract and reproductive system of the rabbits in each group.

At the end of the recovery period, pathological changes in muscles at the injection site could be recovered to a certain extent. The incidence and degree of the pathological changes were lower than that obtained 72 hours after the last administration except some individual animals in the high-dose group of aluminum phosphate adjuvant.

2.6.6.8.7.5.1.4 Conclusions

Certain irritation to the muscles at the injection sites of rabbits were observed for 7-Valent PCV, aluminum phosphate adjuvant and 13-Valent PCV, which partially recovered after the last administration. Relevant specific antibodies could be detected 21 days after the first injection of 13-Valent PCV. The antibody titer peaked 21-35 days after the first injection, and then reduced gradually. The antibody level remained at a certain level 28 days after the last injection. In addition, 7-valent PCV, aluminum phosphate adjuvant, and 13-valent PCV did not have any obvious impacts on measurements including the general conditions, body temperature, body weight, food intake, blood biochemical results, hematology, bone marrow and organ weight. No obvious enhancement or inhibition of the immune function was observed. No obvious abnormal reactions were observed in histopathological examinations of all organs.

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2.6.6.8.7.5.2 Long-Term Toxicity Test in Macaca mulatta

In accordance with the relevant requirements predefined by the Provisions for Drug Registration on the researches of new drugs, toxicity test of repeated intramuscular injection for 13-valent pneumococcal conjugate vaccine manufactured by Yunnan Walvax Biotechnology Co., Ltd. was performed by National Chengdu New Drug Safety Evaluation Center in Macaca mulatta.

2.6.6.8.7.5.2.1 Objectives

To evaluate the preclinical safety of 13-valent pneumococcal conjugate vaccine, including the nature, extent, dose-response and time-effect relationship, as well as the reversibility of probable toxicities to the immune organs and other toxic target organs induced by the vaccine; to provide a reference for clinical researches.

2.6.6.8.7.5.2.2 Methods

A total of 4 groups were set in this experiment including a negative control group (0.5 ml of 0.9% NaCl solution per animal), an aluminum phosphate group (0.5 mL of 1 mg/mL of aluminum phosphate per animal), as well as a low-dose group and a high-dose group (0.11 mL and 0.5 mL per animal, respectively) of 13-Valent PCV. A total of 8 monkeys were included in each group (half male and half female). The monkeys in each group were injected with corresponding sample or control at the given dose through lateral thigh muscle once every 2 weeks (5 times in total). The monkeys were observed for 6 weeks after the last administration. The day of first injection was defined as Day 1 in this experiment.

The general conditions of the monkeys was observed daily during administration period and recovery period. The body weight and food intake were measured once a week. The body temperature was measured before administration, 60 minutes after, and 3 hours after administration. Hematology, blood biochemical examination, complements (C3 and C4) detection, immunoglobulin (IgG) detection, circulating immune complex (CIC) detection, T lymphocyte subsets (CD3+, CD4+, CD8+) detection, and cytokines (IL-2, IL-6, IFN- γ , TNF- α) were performed 24 hours after the first, the third and the last administration as well as at the end of the recovery period. Urine and bone marrow smears were examined 24 hours after the last administration and at the end of the recovery period on the day of autopsy. II lead ECG and ophthalmologic examination were performed after the last administration and 1 day before the recovery period ended. Antibodies were detected 21 and 35 days before the first administration as well as 24 hours, 15 days and 29 days after the last administration. Four monkeys in each group (half male and half female) were anatomized 24 hours after the last administration and at the end of the recovery period, for systematic autopsy and histopathology examination.

2.6.6.8.7.5.2.3 Results

- (1) General conditions: During the administration and recovering observation period, the monkeys in each group had normal physical appearance, mental and motor state. The natural pores were clean. No obvious abnormal reactions were observed. No abnormal reactions were observed. No local irritant reactions such as inflammation, ecchymosis, induration, suppuration and fester were observed at the injection site.
- (2) Body weight and food intake: The weight gain of monkeys in each group was normal and no significant abnormalities were observed in the food intake at the detection time points in the administration and recovery period.
- (3) Body temperature: No obvious abnormalities were observed in the body temperature of the monkeys in each group at the detection time points in the administration period.

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- (4) Ophthalmologic examination: In each group, the dioptric media of the eyes were clear. The fundus veins of the eyes were clear without bleeding and exudation. No edema was observed in the optic papilla. The macula was clear, and the structure of the macula arch was normal. The diameter of the retinal artery was homogeneous, and no obvious dilation or tortuosity of the veins as well as other abnormal symptoms of the retina were observed.
- (5) Hematology: No significant abnormalities were observed in the WBC count and classification, RBC, HGB, HCT, RET, PLT, PT and APTT at the detection time points in the administration and recovery period.
- (6) Blood biochemical measurements: No significant abnormalities were observed in ALP, ALT, AST, GGT, TP, ALB, GLU, TBIL, GHOL, TG, CK, Urea, Crea and the electrolyte concentration (K^{+} ■ Na^{+} ■ Cl^{-}) at the detection time points in the administration and recovery period.
- (7) Immunological measurements: No significant abnormalities were observed in the complements (C3, C4), T lymphocyte (CD 4+, CD 8+), cytokines (IL-2, IL-6, IFN- γ) and Immunoglobulin (IgG) at the detection time points in the administration and recovery period. Serotype 1, 3, 4, 5, 6A, 7F, 9V, 14, 18C, 19A, and 23F polysaccharide antibodies induced by 13-valent pneumococcal conjugate vaccine could be detected in the serum of monkeys in each group 21 days after the first administration. Polysaccharide antibody of all serotypes peaked 21 to 35 days after the first administration and then decreased gradually. Antibodies of certain titers could still be detected 29 days after the last administration. Except for serotype 6B, the antibody titers of other polysaccharide serotypes were dose-related, and the serum antibody level of high-dose group was slightly higher than that of low-dose group.
- (8) Urine measurements: No obvious abnormalities were observed in the grade distribution or positive rate of urine glucose, bilirubin, acetone bodies, specific gravity, occult blood, PH value, urocholinogen, nitrite and white blood cells in the urine of monkeys in each group at the end of the administration and recovery period.
- (9) ECG measurements: No obvious abnormalities were observed in heart rate, R-R interval, P-R interval, QRS wave time, Q-T interval, corrected Q-T interval and the shape of II lead ECG of monkeys in each group at the end of the administration and recovery period.
- (10) Bone marrow measurements: No obvious abnormalities were observed in erythroid cells, granular cells, lymphocytes, monocytes and megakaryocyte percentage in the bone marrow of monkeys in each group at the end of the administration and recovery period.
- (11) Organ weight and coefficients: No obvious abnormalities were observed in the organ weight and coefficient of brain, heart, liver, spleen, lung, kidney, adrenal gland, thymus and reproductive system of the rabbits in each group 72 hours after the last administration and at the end of the recovery period.
- (12) Gross anatomy and histopathology examination: Dark red areas in a size of 0.9 * 0.4 cm – 1.0 * 0.7 cm were observed in the muscles at the injection site of 1 male monkey (1 / 4) in the high-dose group of 13-valent PCV at the end of administration. Other than the above changes, no abnormal changes associated with the study vaccine were observed with naked eye in the location, size, shape, color and texture of organs or tissues such as brain, heart, liver, spleen, lung, kidney, gastrointestinal tract and reproductive system. Mild inflammation was observed at the injection site for 2 monkeys (2/4) in the aluminum phosphate adjuvant group, and mild inflammation was observed at the injection site for 2 monkeys (2/4) in the high-dose group of 13-valent PCV. Other than the changes above, no histopathological changes were found in the rest organs and tissues of the monkeys in each group. The pathological changes in muscles at the injection site of monkeys in each group recovered obviously at the end of the recovery period.

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2.6.6.8.7.5.2.4 Conclusions

Certain irritation to the muscles at the injection sites of monkeys were observed for aluminum phosphate adjuvant and 13-Valent PCV, which partially recovered after the last administration. Other than that, no obvious impacts on measurements including the general conditions, body temperature, body weight, food intake, blood biochemical results, hematology, bone marrow, organ weight, ophthalmology and ECG were observed. No obvious enhancement or inhibition of the immune function was observed. Relevant specific antibodies could be detected 21 days after the first injection of 13-Valent PCV. The antibody titer peaked 21-35 days after the first injection, and then reduced gradually. The antibody level remained at a certain level 29 days after the last injection.

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2.6.6.9. Discussion and Conclusions

The results of toxicological tests showed that the dosage of the study product was within the safe range; the study product had certain irritation to the muscles at the injection site after the product was given to the animals, but the irritation could be recovered after the administration ended; apart from the irritation, no obvious abnormalities were observed in the general conditions, body temperature, body weight, food intake, blood biochemistry, hematology, urine, bone marrow, ophthalmology, electrocardiogram, organ weight, histopathological examination and other measurements of the animals. The specific antibody related to vaccine components could be detected; the carrier protein had a certain risk of allergy, suggesting that special attention should be paid to clinical use; the study product had aggregation effect on rabbit red blood cells, but no hemolytic reaction was observed.

After multiple intramuscular injections of the study product were given to the rabbits in the volume of 0.13 mL and 0.5 mL per rabbit, 13 serotypes of pneumococcal polysaccharide antibodies were detected in 21 days after the first administration. The peak time of antibody titer of different subtypes was slightly different, and the antibody with certain titer level could still be detected 28 days after the last administration. Except for serotypes 5, 6a, 6b, 19F and 23F, the antibody levels of other types of polysaccharides were dose-dependent. The above results showed that the product had good immunogenicity to New Zealand rabbits and could induce the production of specific antibodies. The C3 value of 0.5 mL/mouse group was slightly higher than that of the control group 72 hours after the last administration, which might be related to the activation of complement system by bacterial polysaccharide. However, 13-JHYM had no significant effect on immunological measurements such as T lymphocyte subsets (CD4+, CD8+, CD4+/CD8+) and cytokines (IL-2, IL-6, IFN- γ). After repeated injections, the licensed 7-valent pneumococcal conjugate vaccine, aluminum phosphate adjuvant and the study product could cause slight to moderate degeneration and necrosis of muscle fiber and inflammatory cell infiltration to the muscles at the injection site of rabbits, which indicated that the 7-valent pneumococcal conjugate vaccine, aluminum phosphate adjuvant and the study product had muscle irritation to the animals. The degree of pathological changes in the study product group was similar to that of the 7-valent pneumococcal conjugate vaccine group. After the end of the recovery period, the above pathological changes could be recovered to a certain extent.

After repeated intramuscular injections of the study product were given to the Rhesus macaques in the volume of 0.11 mL and 0.5 mL per animal, 13 serotypes of pneumococcal polysaccharide antibodies were detected in 21 days after the first administration. The peak time of antibody titers of different subtypes was slightly different, and antibodies with certain titer could still be detected 29 days after the last administration. Except for serotype 6B, a certain dose-dependent relationship between the antibody levels was observed for other types of polysaccharides. The above results showed that the study product had immunogenicity to Rhesus macaques and could induce specific antibody production. During the period of administration and recovery observation, no abnormal changes of immune measurements such as complements (C3, C4), immune complex (CIC), T lymphocyte subsets (CD3+, CD4+, CD8+, CD4+/CD8+), cytokines (IL-2, IL-6, IFN- γ , TNF- α), immunoglobulin (IgG) and other immunological measurements were observed for monkeys in each group, indicating that the product had no obvious interference effect on the immune function of the body. After repeated administration, slight to mild inflammation was observed at the injection site of 0.5 mL/monkey group. In the 0.5 mL/animal aluminum phosphate group, the same pathological changes were observed in the muscles at the injection site, and the incidence and degree of the pathological changes were basically the same as that of the 0.5 mL/animal study product group. These results suggested that the high dose of the study product had muscle irritation to the animals, which might be related to the aluminum phosphate adjuvant contained in the

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vaccine. The above pathological changes could be recovered 6 weeks after the administration ended.

The study product is made by covalent binding of pneumococcal capsular polysaccharide antigens with tetanus toxoid carrier protein and adsorbed by aluminum phosphate adjuvant. In the experiment, the red blood cells were observed to aggregate and adhere to the test tube wall in varying degrees in the aluminum phosphate adjuvant group and the study product group of 3 batches, showing a certain dose-dependent relationship. Aluminum phosphate had a stronger adsorption effect, which was able to adsorb viruses, bacteria and toxins produced by bacteria. Therefore, the above phenomenon might be caused by the adsorption effect of aluminum phosphate.

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2.6.7 Toxicology Tabulated Summary

Experiment Experiment Administration

Administration frequency Species Test sample

type period route

Acute toxicity Intramuscular Prototype

Once 15 days Mice

test injection drug

Repeat-dose Twice a week, 5 times in Intramuscular Prototype

12 weeks Rabbits

toxicity test 1 total injection drug

Repeat-dose Twice a week, 5 times in Intramuscular Rhesus Prototype

16 weeks

toxicity test 2 total injection macaques drug

Muscle Intramuscular Prototype

Once 16 days Rabbits

irritation test injection drug

Sensitization: once every

other day, 5 times in

Acute systemic

total ■ Intravenous Guinea Prototype

anaphylaxis 10 days

Provocation: once on the injection pigs drug

test

10th day after the last

sensitization

Passive

cutaneous Once every other day, 5 Intramuscular Guinea Prototype

20 days

anaphylaxis times in total injection pigs drug

test

Prototype

Hemolysis test / 1 day / Rabbits

drug

Mice,

Abnormal Intraperitoneal Prototype

Once 7 days guinea

toxicity test injection drug
pigs
Specific Subcutaneous Guinea
Once 21 days Bulk
toxicity test injection pigs
Test for
Intramuscular Guinea
reversion to Once 63 days Bulk
injection pigs
toxicity
New
Zealand
Long-term Once every 2 weeks, 5 Intramuscular Prototype
6 weeks rabbits,
toxicity test times in total injection drug
Rhesus
macaques

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PRODUCT REGIDTRATION DOSSIER

13-valent Pneumococcal Polysaccharide Conjugate Vaccine
(Tetanus Toxoid Protein)

2.5 CLINICAL OVERVIEW

Yuxi Walvax Biotechnology Co., Ltd.

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2.5. CLINICAL OVERVIEW

2.5.1. Product Development Rational

The vaccine elicits immune response in recipients following immunization, and is indicated for the prevention of invasive disease caused by 13 serotypes (1, 3, 4, 5, 6A, 6B, 7F, 9V, 14, 18C, 19A, 19F and 23F) of Streptococcus pneumoniae.

The vaccine does not protect against diseases caused by Streptococcus pneumoniae serotypes that are not contained in the vaccine.

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2.5.2. Overview of Biopharmaceutics

N/A.

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2.5.3. Overview of Clinical Pharmacology

N/A.

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2.5.4. Overview of Efficacy

2.5.4.1. Study Design

The 13-valent pneumococcal polysaccharide conjugate vaccine (PCV13-TT) manufactured by Yuxi Walx Biotechnology Co., Ltd. Is proposed for the active immunization of infants and children 6 weeks through 5 years of age (before 6th birthday) for the prevention of invasive disease and otitis media caused by Streptococcus pneumoniae serotypes 1,3, 4, 5, 6A, 6B, 7F, 9V, 14, 18C, 19A, 19F, and 23F.

Two clinical trials (phase I and III) of the vaccine (PCV13-TT) were conducted within China, including 120 and 2760 subjects, respectively; among all subjects, 1754 have been vaccinated with at least one dose of PCV13-TT. In China, the pivotal phase III clinical trial of PCV13-TT was a randomized, blinded, positive control of imported similar vaccine (Prenvar), and non-inferiority study designed to evaluate the immunogenicity and safety of PCV13-TT and Prenvar in 2760 subjects 2 through 71 months of age (at least 6 weeks of age). Table 1 presents an overview of PCV13-TT clinical trials.

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Table 1. Overview of Clinical Trials for PCV13-TT

Preliminary trial

Phase/Study No. Phase I PCV13-TT 001

Country China

Status Completed

Objective Primary To assess the safety of 13-Valent Pneumococcal Conjugate Vaccine in infants aged 2

months (≥6 weeks) and 3 months, children aged 2-5 years and adults aged 18 years old.

Secondary To explore the IgG immune response of 13-Valent Pneumococcal Polysaccharide Conjugate Vaccine in infants aged 2 months and 3 months, children aged 2-5 years and adults aged 18 years old.

Group PCV13-TT Group PCV7 Group

Vaccine Name PCV13-TT /

Manufacture Yuxi Walvax Biotechnology Co., Ltd /

Specification Pneumococcal polysaccharide serotype 1: 2.6 µg /

Pneumococcal polysaccharide serotype 3: 2.5 µg

Pneumococcal polysaccharide serotype 4:3.0 µg

Pneumococcal polysaccharide serotype 5:2.5 µg

Pneumococcal polysaccharide serotype 6A:2.5 µg

Pneumococcal polysaccharide serotype 6B:6.0 µg

Pneumococcal polysaccharide serotype 7F :2.85 µg

Pneumococcal polysaccharide serotype 9V:2.5 µg

Pneumococcal polysaccharide serotype 14:2.75 µg

Pneumococcal polysaccharide serotype 18C:3.25 µg

Pneumococcal polysaccharide serotype 19A:2.6 µg

Pneumococcal polysaccharide serotype 19F:2.75 µg

Pneumococcal polysaccharide serotype 23F:3.0 µg

Dosage

Items	Age	Primary Series	Boost Dose	Number	Primary Series	Boost Dose	Number
Schedule dose	Schedule dose	Schedule dose	Schedule dose	Schedule dose	Schedule dose	Schedule dose	Schedule dose
Group1	Adults aged 18 years and above	NA	1	NA	NA	30	NA
Group2	Children aged 2–5 years	NA	1	NA	NA	30	NA

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Group3 Infants aged 3 months month

Group4 Infants aged 2 months month

Pivotal trial

Phase/Study No. Phase III PCV13-TT 002

Country China

Status Completed

Objective Primary To assess the immunogenicity of Walvax PCV13-TT in healthy children aged 3 months and above. The proposed regimen consists of four doses, with a single intramuscular injection at 0, 1, 2 month for primary immunisation schedule and a single intramuscular injection at 12 to 15 months for booster immunisation schedule. To assess the non-inferiority of 7 common pneumococcal serotypes compared to

Prevfromnar 7® and the efficacy of 6 additional pneumococcal serotypes with same immunization schedule.

Secondary (1) To assess the immunogenicity of Walvax PCV13-TT in healthy children aged 2 months (≥6 weeks). The proposed regimen consists of four doses, with a single intramuscular injection at 0, 1 and 2 month for primary immunisation schedule and a single intramuscular injection at 12 to 15 months for booster immunisation schedule.

(2) To assess the non-inferiority of 7 common pneumococcal serotypes compared to Prevfromnar 7® and the efficacy of 6 additional pneumococcal serotypes with same immunization schedule.

(3) To assess antibody opsono-phagocytosis activity (OPA) Walvax PCV13-TT in healthy children aged 2 months (≥6 weeks) to 71months by different immunization procedures.

(4) To assess the safety of Walvax PCV13-TT in healthy children aged 2 months (≥6 weeks) to 71months. Evaluate safety of PCV13 in 6 weeks - 5 years old children

Group Test Group Control Group

Vaccine Name Walvax PCV13-TT Prevfromnar 7® (Pfizer)

Manufacture Yuxi Walvax Biotechnology Co., Ltd Wyeth Pharmaceuticals Inc.

Specification Pneumococcal polysaccharide serotype 1: 2.6 µg Each 0.5 ml dose contains 2 µg of polysaccharide from pneumococcal

Pneumococcal polysaccharide serotype 3: 2.5 µg serotypes 4, 9V, 14, 18C,19F,23F and 4 µg polysaccharide for 6B.

Pneumococcal polysaccharide serotype 4:3.0 µg

Pneumococcal polysaccharide serotype 5:2.5 µg

Pneumococcal polysaccharide serotype 6A:2.5 µg

Pneumococcal polysaccharide serotype 6B:6.0 µg

Pneumococcal polysaccharide serotype 7F :2.85 µg

Pneumococcal polysaccharide serotype 9V:2.5 µg

Pneumococcal polysaccharide serotype 14:2.75 µg

Pneumococcal polysaccharide serotype 18C:3.25 µg

Pneumococcal polysaccharide serotype 19A:2.6 µg

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Pneumococcal polysaccharide serotype 19F:2.75 µg

Pneumococcal polysaccharide serotype 23F:3.0 µg

Batch number 201511003 H29135H13880

Expiry Date 2017.11 2017.05

Items	Age	Primary Series	Boost Dose	Number	Primary Series	Boost Dose	Number
Schedule dose	Schedule dose	Schedule dose	Schedule dose	Schedule dose	Schedule dose	Schedule dose	Schedule dose
Group1	Infants aged 3 months	3	0, 1, 2	month	3	12-15month	1 520
	Infants aged 2 months	0, 2, 4	month	3	12-15month	1 520	N/A

Group2 Infants aged 7- 0, 2 month 2 12-15month(at 1 200 0, 2 month 2 12-15month(at least 2 1 200 11 least 2 months months apart from the months apart from the second dose) second dose)
Group3 Infants aged 0, 2 month 2 N/A N/A 200 0, 2 month 2 N/A N/A 200 12-23 months
Group4 Children aged 0 month 1 N/A N/A 200 0 month 1 N/A N/A 200 24-71 months

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2.5.4.2. Summary of Efficacy Results 2.5.4.2.1. Phase III Clinical Trial

In the main target population (3 months of age)
The common serotypes achieved noninferiority criteria for comparison between PCV13-TT and PCV7 group except for serotype 6B during primary series, meanwhile all common serotypes in PCV13-TT group were noninferior to that in PCV7 group after the booster dose; for additional serotypes, superiority criteria were met for PCV13-TT group vs PCV7 group after primary series; the common serotypes in PCV13-TT group aged 3-71 months were all noninferior to those in PCV7 group during primary series, besides, the additional serotypes in PCV13-TT group were all superior as compared to PCV7 group.

For PCV13-TT (2 months) group vs PCV7 (3 months) group
Both common and additional serotypes had achieved noninferiority or superiority criteria post primary series. For PCV13-TT (2 months) group vs PCV13-TT (3 months) group, 12 serotypes reached noninferiority criteria during primary series except for additional serotype 5; and all 13 serotypes were noninferior to those in PCV13-TT (3 months) group post booster dose.

In PCV13-TT recipients aged 7-11 months, 12-23 months, and 24-71 months
The proportions of subjects with IgG antibody concentration $\geq 0.35 \mu\text{g/mL}$ for common serotypes were all noninferior to those in PCV7 group. The additional serotypes in PCV13-TT group of 7-11 months had reached superiority criteria after primary series; for PCV13-TT groups of 12-23 months and 24-71 months, superiority criteria were generally met after immunization. The results before and after booster dose indicated decrease in the antibody levels of several serotypes in the 13 serotypes either in PCV13-TT group or PCV7 group to some extent, while the percentage of subjects with IgG antibody concentration $\geq 3.5 \mu\text{g/mL}$ and antibody GMC after 1 dose of booster immunization both reached or exceeded the levels observed after primary series, demonstrating significant effect of the booster dose, reflecting the immunological memory effect of the conjugate vaccine adequately. The antibody GMC post booster dose had more than 3-fold increase in the 2-6 months olds, and greater than 1-fold increase in subjects aged 7-11 months as compared with the GMC before booster dose.

To sum up, the immunogenicity of 7 common serotypes in main target population aged 3 months met the non-inferiority criteria after booster dose, and that of 6 new serotypes had already achieved the standard of superiority after primary immunizations in this clinical trial. The common and additional serotypes had fulfilled the non-inferiority or superiority standard in subjects 3-71 months of age after primary immunizations. The common and additional serotypes in PCV13-TT group (2 months) were noninferior and/or superior to that in PCV7 (3 months) group. In PCV13-TT (2 months) group, all 13 serotypes were noninferior to those in the PCV13-TT (3 months) group post booster dose. The results of immunogenicity evaluation in this clinical trial had achieved the study objectives predefined in the protocol.

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2.5.5. Overview of Safety 2.5.5.1. Phase I Clinical Trial

The incidence of vaccine-related adverse events was 80.00%, 53.33%, 66.67% and 70.00% in adults, children, 2-month-old and PCV13-TT (3 months) group, respectively. A total of 6 cases in the adult group experienced grade 3 local adverse reactions, corresponding to an incidence rate of 20% (6 / 30), including 1 case of local redness, 1 case of induration, 2 cases of swelling, 2 cases of redness combined with swelling. The incidence rate of adverse reactions with grade 3 severity in other age groups was less than 15%.

The most common symptoms of systemic adverse reactions were fever in adults and children aged 2-5 years, and were fever and diarrhea in 2-month-old and 3-month-old infants. Common local adverse reactions in adults included redness, pain and swelling at the vaccination site, and were pain and redness in children aged 2-5 years, as compared to redness in infants 2 and 3 months of age.

Adverse events mainly occurred on Day 0-7. No adverse events related to the study vaccine were collected on Day 8-30, and no serious adverse events were collected during the observation period.

2.5.5.2. Phase III Clinical Trial

The overall incidence of adverse events in PCV13-TT group was lower than that in PCV7 group. No significant intergroup difference was observed in the incidence of vaccine-related adverse events. The main symptom of systemic adverse reaction was fever. The incidence of systemic reactions in PCV13-TT group was lower than that in PCV7 group. Local adverse reactions were primarily redness, swelling and pain at the injection site. No significant difference was observed in the incidence of local reactions between PCV13-TT group and PCV7 group.

The incidence of vaccine-related grade 3 adverse events in PCV13-TT group and PCV7 group was 4.21% and 2.41%, respectively. The incidence of local adverse reactions with severity reaching grade 3 in PCV13-TT group was higher than that in PCV7 group, while no significant intergroup difference was observed in subjects aged 3 months.

A total of 80 cases (105 events) of SAEs were observed during primary series and 8 cases (10 events) of SAEs were noted during booster dose. No significant intergroup difference was observed in the incidence of SAEs. SAEs were not related to the study vaccine except for one case in PCV13-TT (3 months) group with fever determined to be probably related to the study vaccine.

The incidence of adverse events/reactions in PCV13-TT group was not higher than that in PCV7 group.

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2.5.6. Benefits and Risks Conclusions

N/A.

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2.5.7. Literature References

[1] Good Clinical Practice (GCP), National Medical Products Administration, 2003.6.4

[2] Technical guidelines for clinical trials of vaccines National Medical Products Administration 2004.12.3

[3] Guidelines for quality management of vaccine clinical trials (Trial), National Medical Products Administration 2013.10.31

[4] Provisions for Drug Registration National Medical Products Administration 2016.7.22

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PRODUCT REGISTRATION DOSSIER

13-valent Pneumococcal Polysaccharide Conjugate Vaccine
(Tetanus Toxoid Protein)

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2.2 CTD Introduction

INTRODUCTION

The Applicant has developed the 13-valent Pneumococcal Polysaccharide Conjugate Vaccine (Weuphoria), for use in infants and young children to prevent pneumococcal disease, caused by the 13 pneumococcal serotypes contained in the vaccine.

Weuphoria is a sterile liquid formulation of pneumococcal capsular polysaccharides of serotypes 1, 3, 4, 5, 6A, 6B, 7F, 9V, 14, 18C, 19A, 19F, and 23F, individually conjugated to tetanus toxoid. Phosphate-buffered Saline is also included in the Weuphoria formulation to improve process control and to provide further pH control following the addition of aluminum phosphate.

Manufacturer: Yuxi Walvax Biotechnology Co., Ltd.

Common name of drug product: 13-valent Pneumococcal Polysaccharide Conjugate Vaccine.

This vaccine is produced by culture, extraction and purification of polysaccharide antigens of 13 epidemic and infectious Streptococcus pneumoniae of serotypes of 1, 3, 4, 5, 6A, 6B, 7F, 9V, 14, 18C, 19A, 19F and 23F. After being activated and derived, the polysaccharide conjugates with tetanus toxoid carrier protein to form polysaccharide protein conjugate, which is prepared by mixing with a certain proportion and adding aluminum phosphate adjuvant. The vaccine is milky even suspension after shaking. After storage, white precipitates and supernatant can be seen due to the subsidence of adjuvant. No preservatives contained.

Table 1 Product Composition (each dose/0.5mL contains)

Active Substances	Nominal Composition	Function
Pneumococcal polysaccharide serotype 1	2.6 µg	Antigen
Pneumococcal polysaccharide serotype 3	2.5 µg	Antigen
Pneumococcal polysaccharide serotype 4	3.0 µg	Antigen
Pneumococcal polysaccharide serotype 5	2.5 µg	Antigen
Pneumococcal polysaccharide serotype 6A	2.5 µg	Antigen
Pneumococcal polysaccharide serotype 6B	6.0 µg	Antigen
Pneumococcal polysaccharide serotype 7F	2.85 µg	Antigen
Pneumococcal polysaccharide serotype 9V	2.5 µg	Antigen
Pneumococcal polysaccharide serotype 14	2.75 µg	Antigen
Pneumococcal polysaccharide serotype 18C	3.25 µg	Antigen
Pneumococcal polysaccharide serotype 19A	2.6 µg	Antigen
Pneumococcal polysaccharide serotype 19F	2.75 µg	Antigen
Pneumococcal polysaccharide serotype 23F	3.0 µg	Antigen
Aluminum phosphate	0.5 mg	Adjuvant
Sodium chloride	4.25 mg	Excipient
Sodium dihydrogen phosphate	44.35 µg	Excipient
Disodium hydrogen phosphate	19.0 µg	Excipient
Water for injection	Qs to 0.5 mL	Excipient

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Proprietary name:

Route of administration: The vaccine should be shaken well immediately prior to use, and be given intramuscularly. The preferred sites for injection are the anterolateral aspect of the thigh in infants and the deltoid muscle of the upper arm in toddlers and children. Each dose (0.5 mL) is to be injected. The vaccine should not be injected in and/or near the areas where there may be a nerve trunk and/or blood vessel.

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PRODUCT REGIDTRATION DOSSIER

13-valent Pneumococcal Polysaccharide Conjugate Vaccine
(Tetanus Toxoid Protein)

2.7 CLINICAL SUMMARY

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2.7 CLINICAL SUMMARY

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2.7.1. Biopharmaceutic Studies and Associated Analytical Methods
Pharmacokinetics
Pharmacokinetic studies were not performed in accordance with the note for guidance on clinical evaluation of new vaccines (CPMP/EWP/463/97).
Pharmacodynamics
As generally done with vaccines, pharmacodynamic studies are essentially comprised by the immunogenicity studies that characterise the immune response to vaccines. The detailed characterisation of the immunological response is discussed below.

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2.7.2. Clinical Pharmacology Studies
N/A

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2.7.3. Clinical Efficacy
2.7.3.1. Background
Streptococcus pneumoniae (S. pneumoniae) is a gram-positive encapsulated diplococcus and a major cause of mortality and morbidity worldwide with the highest incidence in infants under 2 years of age and in the elderly over 60 years of age. S. pneumoniae has been estimated to cause ~16 million deaths every year, including up to one million in children below 5 years of age. The highest morbidity and mortality rates have been reported from developing countries. The spectrum of disease encompasses invasive pneumococcal disease (IPD), such as sepsis and meningitis, lower respiratory infections, such as bacterial pneumonia and upper respiratory infections, such as acute otitis media (AOM). The relative incidences of the various disease entities are estimated to be that

for 1 case of IPD there are 100 cases of pneumonia and 1000 cases of otitis media. Prevnar, the first pneumococcal conjugate vaccine licensed in the United States (U.S.), was approved by the FDA for active immunization to prevent invasive pneumococcal disease (IPD) in February 2000. The serotypes in the vaccine were originally selected because, at that time, they accounted for approximately 80% of IPD in young children in North America. A high level of efficacy (aggregated for the 7 serotypes) in preventing vaccine serotype IPD was demonstrated in a clinical endpoint vaccine efficacy trial in infants. In October, 2002, an additional Prevnar indication was approved for active immunization of infants and toddlers against otitis media caused by vaccine serotypes. Efficacy against otitis media was supported by data from an acute otitis media efficacy trial, which involved tympanocentesis, as well as information on health care utilization for otitis media.

There are no efficacy studies with the 13-valent pneumococcal polysaccharide conjugate vaccine (Prevenar 13). The overall objective of the clinical development program was to evaluate the immunogenicity and safety of PCV13 in the target population. Non-inferiority of the immune response compared with licensed 7-valent Prevenar was specifically evaluated. Studies were designed to evaluate the immunogenicity and safety of the vaccine when used for primary, catch-up and booster vaccination; and to evaluate the presence of immune memory through administration of a booster dose.

Licensure criteria for pneumococcal conjugate vaccines

Over the past years, regulatory agencies and experts in the field have reflected upon the serological criteria for the evaluation and licensure of new pneumococcal conjugate vaccines. It was agreed to follow the same pathway as used for licensure of Hib and MenC conjugate vaccines, and after the demonstration of a high level of invasive pneumococcal disease efficacy in Northern California with Prevenar, to license future new pneumococcal conjugate vaccines for IPD purely on the basis of immunological data in comparison with the licensed vaccine. A consensus recommendation on criteria for licensure of new pneumococcal conjugate vaccines against IPD was reached at the WHO Expert Committee meeting in 2003 (WHO 2005, Jodar 2004, Lee 2003). The WHO has issued a technical report series (TRS 927, annex 2) with recommendations for the evaluation of new pneumococcal conjugate vaccines that reflect these principles.

The following criteria are recommended for use as the primary end-point for demonstration of non-inferiority against a registered vaccine:

- IgG antibody concentration, as measured by ELISA, in sera collected 4 weeks after a three-dose primary series is considered to be the optimal primary end-point and main licensing parameter.

- A single threshold or reference antibody concentration is recommended for use for all pneumococcal serotypes. A reference antibody concentration of 0.35 µg/mL, that has been

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determined through a pooled analysis of data from the efficacy trials with invasive disease end-points that have been completed to date, is recommended. This threshold does not necessarily predict protection in an individual subject.

- The reference value is defined on the basis of data obtained using ELISA without pre-adsorption with serotype 22F. Antibody concentrations determined using an alternative method will need to be bridged to this method to derive an equivalent threshold concentration. It is recommended that the assay used be calibrated against a reference assay.

- Direct clinical comparison of the registered (established) vaccine with the new one is the preferred method for evaluating new vaccine formulations.

- The percentage of responders (those in whom post-immunization antibody concentration is above the threshold) should be used as the criterion to determine non-inferiority.

- For the serotypes present in a registered vaccine, the percentage of responders to each serotype in the new formulation or combination should be compared with the percentage of responders to the same serotype in the registered vaccine in the same population.

Non-inferiority to antibody response for each of the serotypes in the registered vaccine is desirable, but not an absolute requirement. Registration of products in which one or more serotypes do not meet non-inferiority criteria would have to be decided on an individual basis.

- Serotypes not contained in a registered formulation may be evaluated for non-inferiority to the aggregate response to the serotypes in the registered vaccine. Failure of one or more new serotypes to meet this criterion may be considered on an individual basis.

- Additional criteria that must be met to support registration:

- In addition to showing non-inferiority with respect to the primary end-point, additional data to demonstrate the functional capacity of the antibody and induction of immunological memory in a subset of the sera are required for registration.

Functional antibodies

- Opsonophagocytic activity (OPA) as measured by opsonophagocytic assay after a three-dose priming series is required to demonstrate the functionality of antibodies.

- The method used to demonstrate OPA should be comparable to the reference assay.

Immunological memory

- Evidence of memory should be demonstrated. One possible method is to administer a booster dose of pneumococcal polysaccharide vaccine and to compare concentrations between age-matched unprimed and primed individuals; data from non-concurrent controls may be sufficient for the purposes of comparison.

- A full dose of polysaccharide vaccine should be used at this stage because the use of a reduced dose of the polysaccharide vaccine as a booster has not been sufficiently tested.

- Avidity of antibodies is also a useful marker for immunological memory.

- In follow-up of WHO 2003 meeting, a WHO workshop took place in January 2007 with the objective to formulate a plan for the standardisation of the pneumococcal OPA assay (WHO 2007).

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2.7.3.1.1. Clinical Trial Overview

2.7.3.1.1.1. Phase I clinical trials

Design

This clinical trial was designed as a single center, open, non-controlled trial.

Healthy subjects aged 2 months and above were included. The planned sample size was 120, and subjects were recruited in three stages in the order of: adults (18 years and above), children (2-5 years) and infant and toddlers (2-3 months, minimum 6 weeks).

A total of 30 subjects were enrolled in the first stage. Each subject was administered with 1 dose

of 13-valent pneumococcal polysaccharide conjugate vaccine (PCV13-TT). Blood samples were collected 3 times from all subjects before vaccination, on Day 3 post vaccination (interval 2 days, same below) and Day 30 post vaccination, respectively. The blood samples collected before vaccination and on Day 3 post vaccination were assessed by hematology, liver and kidney function tests, meanwhile, the serum IgG antibody concentration was measured using the blood sample taken before vaccination and on Day 30 post vaccination. Subjects who had no abnormal results of hematology, liver and kidney function tests before vaccination and met other inclusion criteria could proceed to get vaccinated. If no SAE related to study vaccine was identified by the investigators after 7 days' observation and no grade 4 vaccine-related abnormalities determined by the investigators as measured by hematology, liver and kidney function tests were detected on Day 3 after vaccination, then the study proceeded to the next stage.

A total of 30 subjects were enrolled in the second stage. Each subject was administered with 1 dose of PCV13-TT. Blood samples were collected 3 times from all subjects respectively before the vaccination, on Day 3 post vaccination (interval 2 days, same below) and Day 30 days post vaccination. The blood samples collected before vaccination and on Day 3 post vaccination were assessed by hematology, liver and kidney function tests, meanwhile, the serum IgG antibody concentration was measured using the blood sample taken before vaccination and on Day 30 post vaccination. Subjects with testing results within the normal range predefined in the protocol or not within the normal range but were determined by site investigators to be normal and in compliance with other inclusion criteria could be administered with the vaccine. If no SAE related to study vaccine was identified by the investigators after 7 days' observation and no grade 4 vaccine-related abnormalities determined by the investigators as measured by hematology, liver and kidney function tests were detected on Day 3 after vaccination, then the study proceeded to the next stage.

A total of 60 subjects were enrolled in the third stage and subjects were divided into 2 months (minimum age: 6 weeks) group and 3 months group with 30 subjects in each arm. Subjects in the 2 months group were administered with 3 doses of PCV13-TT according to the procedure of one dose each vaccinated at 0, 2 and 4 months and subjects in the 3 months group were administered with 3 doses of PCV13-TT following the procedure of one dose each at 0, 1 and 2 months. Blood samples were collected from all subjects twice respectively before the first dose and 30 days after the third dose. The serum was separated and the IgG antibody concentration was measured.

All subjects were observed for 30 minutes after each dose of PCV13-TT, and were under active observation for adverse events (AEs) within 0-7 days post vaccination, while for 8-30 days post vaccination, AEs were spontaneously reported and collected by subjects. Serious adverse events (SAEs) were monitored for 6 months after full immunization of PCV13-TT which was administered in accordance with the immunization procedure prescribed in the protocol.

Objective

Primary objective:

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To evaluate the safety of 13-valent pneumococcal polysaccharide conjugate vaccine (PCV13-TT) in healthy population aged 2-3 months (minimum age: 6 weeks), 2-5 years and 18 years and above.

Population

Healthy subjects aged 2 months and above.

Results

The incidence of adverse events related to the study vaccine in adults, children, 2 months and 3 months group was 80.00%, 53.33%, 66.67% and 70.00%, respectively. A total of 6 subjects in the adults group experienced local adverse reactions with severity reaching grade 3, including 1 case of local redness, 1 case of induration, 2 cases of swelling and 2 cases of redness with swelling, corresponding to an incidence of 20% (6/30) in the adults group. The incidence of adverse reactions with grade 3 severity in other age groups was less than 15%.

The most common symptom of systemic adverse reactions was fever in adults and children aged 2-5 years, while fever and diarrhea were more common in 2 months and 3 months group. Local adverse reactions in adults were mainly redness, pain and swelling at the vaccination site; pain and redness were common local reactions in children aged 2-5 years, and redness was the most common local symptom in 2 months and 3 months group.

Adverse events mainly occurred within 0-7 days post vaccination and no adverse events related to the study vaccine were collected during 8-30 days post vaccination. No serious adverse events were reported during the extended observation period.

Since the primary objective of the Phase I clinical trial was to evaluate the safety of the investigational vaccine, only the safety results of the Phase I clinical trial are presented in this section, while the immunogenicity results, as exploratory measures, are listed in Module 5.

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2.7.3.1.1.2. Phase III clinical trials

Design

This clinical trial was designed as a randomized, blinded, comparable vaccine-controlled, multicenter, non-inferiority clinical study.

This clinical trial was carried out in the healthy population, enrolling healthy children aged 2-71 months (minimum age: 6 weeks). The planned sample size was 2760.

In view of absence of serious adverse reactions 7 days after the first dose vaccination in the third stage of Phase I clinical trial, and the incidence of local and systemic reactions with severity of grade 3 or above was lower than 15%, phase III clinical trial was conducted to further evaluate the immunogenicity and safety of 13-valent pneumococcal polysaccharide conjugate vaccine (PCV13-TT) in the target population with expanded sample size.

Objective

To observe and evaluate the immunogenicity and safety of PCV13-TT in healthy children aged 2-71 months (minimum age: 6 weeks).

Population

This clinical trial was conducted in healthy population and the subjects were healthy children aged 2-71 months (minimum age: 6 weeks).

Results

This clinical trial, with a licensed 7-valent pneumococcal conjugate vaccine as control, was designed to evaluate the immunogenicity and safety of the 13-valent pneumococcal polysaccharide conjugate vaccine, jointly developed by Yunnan Walvax Biotechnology Co., Ltd. and Yuxi Walvax Biotechnology Co., Ltd, in subjects aged 3 months (vaccinated at month 0, 1, 2), 7-11 months (vaccinated at month 0, 2), 12-23 months (2 doses) and 24-71 months (1 dose), respectively, the 13-valent pneumococcal polysaccharide conjugate vaccine was also administered in subjects

2 month of age (minimum age: 6 weeks) enrolled to compare with subjects aged 3 months in terms of immunization procedure.

In the main target population (3 months of age), the common serotypes achieved noninferiority criteria for comparison between PCV13-TT and PCV7 group except for serotype 6B during primary series, meanwhile all common serotypes in PCV13-TT group were noninferior to that in PCV7 group after the booster dose; for additional serotypes, superiority criteria were met for PCV13-TT group vs PCV7 group after primary series; the common serotypes in PCV13-TT group aged 3-71 months were all noninferior to those in PCV7 group during primary series, besides, the additional serotypes in PCV13-TT group were all superior as compared to PCV7 group.

For PCV13-TT (2 months) group vs PCV7 (3 months) group, both common and additional serotypes had achieved noninferiority or superiority criteria post primary series. For PCV13-TT (2 months) group vs PCV13-TT (3 months) group, 12 serotypes reached noninferiority criteria during primary series except for additional serotype 5; and all 13 serotypes were noninferior to those in PCV13-TT (3 months) group post booster dose.

In PCV13-TT recipients aged 7-11 months, 12-23 months, and 24-71 months, the proportions of subjects with IgG antibody concentration $\geq 0.35 \mu\text{g/mL}$ for common serotypes were all noninferior to those in PCV7 group. The additional serotypes in PCV13-TT group of 7-11 months had reached superiority criteria after primary series; for PCV13-TT groups of 12-23 months and 24-71 months, superiority criteria were basically met after immunization.

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The results before and after booster dose indicated decrease in the antibody levels of several serotypes in the 13 serotypes either in PCV13-TT group or PCV7 group to some extent, while the percentage of subjects with IgG antibody concentration $\geq 3.5 \mu\text{g/mL}$ and antibody GMC after 1 dose of booster immunization both reached or exceeded the levels observed after primary series, demonstrating significant effect of the booster dose, reflecting the immunological memory effect of the conjugate vaccine adequately. The antibody GMC post booster dose had more than 3-fold increase in the 2-6 months olds, and greater than 1-fold increase in subjects aged 7-11 months as compared with the GMC before booster dose.

The overall incidence of adverse events in PCV13-TT group was lower than that in PCV7 group.

No significant intergroup difference was observed in the incidence of vaccine-related adverse events. The main symptom of systemic adverse reaction was fever. The incidence of systemic reactions in PCV13-TT group was lower than that in PCV7 group. Local adverse reactions were primarily redness, swelling and pain at the injection site. No significant difference was observed in the incidence of local reactions between PCV13-TT group and PCV7 group.

The incidence of vaccine-related grade 3 adverse events in PCV13-TT group and PCV7 group was 4.21% and 2.41%, respectively. The incidence of local adverse reactions with severity reaching grade 3 in PCV13-TT group was higher than that in PCV7 group, while no significant intergroup difference was observed in subjects aged 3 months.

A total of 80 cases (105 events) of SAEs were observed during primary series and 8 cases (10 events) of SAEs were noted during booster dose. No significant intergroup difference was observed in the incidence of SAEs. SAEs were not related to the study vaccine except for one case in PCV13-TT (3 months) group with fever determined to be probably related to the study vaccine. To sum up, the immunogenicity of 7 common serotypes in main target population aged 3 months met the non-inferiority criteria after booster dose, and that of 6 new serotypes had already achieved the standard of superiority after primary immunizations in this clinical trial. The common and additional serotypes had fulfilled the non-inferiority or superiority standard in subjects 3-71 months of age after primary immunizations. The common and additional serotypes in PCV13-TT group (2 months) were noninferior and/or superior to that in PCV7 (3 months) group. In PCV13-TT (2 months) group, all 13 serotypes were noninferior to those in the PCV13-TT (3 months) group post booster dose. The incidence of adverse events/reactions in PCV13-TT group was not higher than that in PCV7 group. The results of immunogenicity and safety evaluation in this clinical trial had achieved the study objectives predefined in the protocol.

The phase III clinical trial of 13-valent pneumococcal polysaccharide conjugate vaccine was intended to evaluate the immunogenicity and safety of 13-valent pneumococcal polysaccharide conjugate vaccine in healthy children aged 2-71 months (minimum age: 6 weeks) in a relatively large population (2760 subjects enrolled). Therefore, phase III clinical findings of safety and immunogenicity are exhibited in this section.

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2.7.3.1.2. Immunogenicity assessment

2.7.3.1.2.1. Immunogenicity observations

Immunogenicity observation in Phase III clinical trial:

Venous blood was collected before the first dose of vaccination and 30 days after the primary series; besides, venous blood was also collected before and 30 days after the booster dose in subjects indicated for booster immunization. All subjects were tested for 13 serotypes specific IgG antibodies before the first dose of vaccination, 30 days after primary immunization, 30 days before and 30 days after booster immunization, with the purpose to compare the proportion of subjects whose specific IgG antibody concentration $\geq 0.35 \mu\text{g/mL}$ and the geometric mean concentration of specific IgG antibody after vaccination between the investigational vaccine and the comparator vaccine.

Serum specific antibody opsonophagocytic activity (OPA) was detected in totally 900 subjects (100 subjects in each PCV13-TT and PCV7 group of different age strata). A total of 300 subjects aged 2-6 months (minimum age: 6 weeks) were tested for OPA 30 days after primary immunization, 30 days before and 30 days after booster immunization; a total of 200 subjects aged 7-11 months were tested for OPA before the first dose of vaccination, 30 days after primary immunization, before booster immunization and 30 days after booster immunization; and a total of 400 subjects aged 12-71 months were tested for OPA before the first dose of vaccination and 30 days after the full immunizations.

Blood collection performed before the first dose of vaccination, 30 days after primary immunizations and/or before booster immunization, 30 days after booster immunization was determined as valid blood collection. The valid blood collection rate should be not less than 85%.

2.7.3.1.2.2. Immunogenicity endpoints

2.7.3.1.2.2.1. Primary immunogenicity endpoints

(1) Percentage of subjects with pneumococcal serotype-specific IgG antibody concentration $\geq 0.35 \mu\text{g/mL}$ 30 days post-dose 3 in PCV13-TT group and PCV7 group

(2) Pneumococcal serotype-specific IgG geometric mean concentration (GMC) 30 days post-dose 3 in PCV13-TT group and PCV7 group

2.7.3.1.2.2.2. Secondary immunogenicity endpoints

- (1) Percentage of subjects with pneumococcal serotype-specific OPA antibody titer $\geq 1:8$ one month post-dose 3 in PCV13-TT group and PCV7 group
- (2) Percentage of subjects with pneumococcal serotype-specific OPA GMT one month post-dose 3 in PCV13-TT group and PCV7 group
- (3) Percentage of subjects with pneumococcal serotype-specific IgG antibody concentration ≥ 1.0 $\mu\text{g/mL}$ 30 days post-dose 3 in PCV13-TT group and PCV7 group
- (4) Percentage of subjects with pneumococcal serotype-specific IgG antibody concentration ≥ 0.35 $\mu\text{g/mL}$ 30 days post-dose 4 in PCV13-TT group and PCV7 group
- (5) Pneumococcal serotype-specific IgG GMC 30 days post-dose 4 in PCV13-TT group and PCV7 group
- (6) Percentage of subjects with pneumococcal serotype-specific OPA antibody titer $\geq 1:8$ one month post-dose 4 in PCV13-TT group and PCV7 group

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(7) Percentage of subjects with pneumococcal serotype-specific OPA GMT one month post-dose 4 in PCV13-TT group and PCV7 group

(8) Percentage of subjects with pneumococcal serotype-specific IgG antibody concentration ≥ 1.0 $\mu\text{g/mL}$ 30 days post-dose 4 in PCV13-TT group and PCV7 group

2.7.3.1.2.3. Immunogenicity Assessment Method

2.7.3.1.2.3.1. Sample size allocation and immunogenicity test subgroup

Table 1. Sample size allocation

Age (Month) PCV13-TT group PCV13-TT group

PCV7 group Total

(3-71M) (2M)

2-6

520 520 520 1560

(Minimum age of 6 weeks)

7-11 200 / 200 400

12-23 200 / 200 400

24-71 200 / 200 400

Total 1120 520 1120 2760

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2.7.3.1.2.3.2. Criterion for immunogenicity hypothesis testing

Common 7 serotypes: non-inferiority test

The evaluation of non-inferiority criterion for comparison of common 7 serotypes in PCV13-TT group and PCV7 group includes:

- (1) The lower limit of the 2-sided 97.5% CI for the rate difference of percentage of subjects with IgG antibody concentration ≥ 0.35 $\mu\text{g/mL}$ (PCV13-TT – PCV7) was greater than -10%
- (2) The lower limit of the 2-sided 97.5% CI for IgG antibody GMC ratio (GMR, PCV13-TT relative to PCV7) is greater than 0.5.

Any common serotype that meets the evaluation criteria 1) or 2) will be judged to be immunogenically non-inferior.

Additional 6 serotypes: superiority test

The evaluation of superiority criterion for comparison of additional 6 serotypes in PCV13-TT group and PCV7 group includes:

- (1) The lower limit of the 2-sided 95% CI for the rate difference of percentage of subjects with IgG antibody concentration ≥ 0.35 $\mu\text{g/mL}$ (PCV13-TT – PCV7 seroresponse or PCV7 reference) was greater than 0.
- (2) The lower limit of the 2-sided 95% CI for IgG antibody GMC ratio (GMR, PCV13-TT relative to PCV7 seroresponse or PCV7 reference) is greater than 1.
- (3) The lower limit of the 2-sided 95% CI for percentage of subjects with IgG antibody concentration ≥ 0.35 $\mu\text{g/mL}$ was greater than 70% in PCV13-TT group.

Any additional serotype, which simultaneously meets the evaluation criterion (1), (2) and (3), is judged to be superior.

PCV7 reference is referred to as the lowest response among the 7 common serotypes contained in PCV7 achieved by subjects who received PCV7.

The evaluation of non-inferiority criterion for secondary endpoints, including percentage of subjects with IgG antibody concentration ≥ 1.0 $\mu\text{g/mL}$, OPA titer $\geq 1:8$, GMT and GMT ratio were same as the above described evaluation criterion.

Sequential tests

In the hypothesis test, if the non-inferiority of PCV13-TT (3 months) group vs PCV7 group was established and additional 6 serotypes met the pre-specified superiority criterion, the non-inferiority test of PCV13-TT group (2 months) vs PCV7 group was performed. Then, if all the serotypes met the pre-specified evaluation criterion in PCV13-TT (2 months) group, the non-inferiority test of PCV13-TT group (2 months) vs PCV13-TT group (3 months) was performed. Strict statistical analysis was performed for the comparison of PCV13-TT (3 months) vs PCV7 (3 months), PCV13-TT (2 months) vs PCV7 (3 months), and PCV13-TT (2 months) vs PCV13-TT (3 months) in a sequential fashion, while other comparisons for non-primary populations (including age groups of 7-11, 12-23, 24-71 months) were statistically analyzed while without strict statistical inference.

The immunogenicity analysis would be performed in all eligible subjects with evaluable serum for testing as per the study protocol. Immunogenicity analysis was performed with the PPS data set as the primary analysis set.

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2.7.3.2. Summary of Results of Individual Studies

2.7.3.2.1. Study Populations for phase III clinical trials

Inclusion Criteria:

- Healthy infants (aged 2-71 months) as evidenced by medical history, physical examination, and investigators' judgement
- Subjects are in compliance with age requirements for this clinical trial (children from 6 weeks to 5 years old or adults over 18 years old)
- Parents / legal guardians of subjects must sign and date the informed consent in person
- Parents / legal guardians of subjects have been informed of all information related to the trial
- Subjects / legal guardians of subjects are able to follow requirements of the trial
- Subjects who have not been vaccinated with pneumococcal vaccines, and have not been administered any other preventive products within 7 days
- Axillary temperature $\leq 37^{\circ}\text{C}$

Exclusion Criteria:

Exclusion Criteria for First Vaccination

- Previously vaccination listed or vaccinated with trial pneumococcal vaccine
- History of invasive disease caused by *Streptococcus pneumoniae*, which was confirmed by bacterial culture
- Prior history of allergic reactions to any drug, vaccine, or vaccine-related serotype
- Vaccination-related fever of 39°C or more has occurred with biological products used for vaccination
- Birth weight $< 2.5\text{ kg}$
- There is a history or family history of allergies, convulsions, seizures, encephalopathy, and psychosis
- Abnormal births of infants (difficult labors, aided by instruments) or a history of asphyxia, neurological organ damage
- History of diagnosed thrombocytopenia or other coagulation disorders
- Infants with diagnosed pathological jaundice
- Known or suspected to suffer from diseases, including: respiratory system disease, acute infection or chronic disease, serious cardiovascular disease, kidney disease, malignant tumor, skin diseases, mothers with HIV infection (as evidenced by test report)
- Known serious congenital malformations or severe chronic disease; congenital deformities, developmental disorder, or a clinical diagnosis of severe chronic diseases (such as Down's syndrome, diabetes, sickle cell anemia, or neurological disorders, Guillain-Barre Syndrome)
- Known or suspected co-morbidities including: respiratory disorders, acute infections or active attack of chronic disease, serious cardiovascular disorders, liver and kidney disorders, malignant neoplasms, skin diseases.
- Received blood products or globulin. Hepatitis B hyper-immune globulin can be used
- Participating in other clinical trials
- Any condition that the investigator considered to have the potential to affect the evaluation of the study

Exclusion Criteria for 2nd, 3rd Vaccination or Booster Dose

- Subjects who had severe allergic reactions during the first or second vaccination
- Subjects who experienced serious adverse effects from the previous vaccination with causal relationship demonstrated
- After the first vaccination, the newly discovered or newly occurred conditions not conforming to the first dose of the inclusion criteria or to the exclusion criteria for the first dose
- Other exclusion criteria affirmed by investigators

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2.7.3.2.2. Pneumococcal immunogenicity results of Phase III Clinical Trial

2.7.3.2.2.1. Pneumococcal immunogenicity results for main population

2.7.3.2.2.1.1. Pneumococcal immunogenicity results for subjects aged 3 months in PCV13-TT group and PCV7 group

2.7.3.2.2.1.1.1. Comparison of percentage of subjects achieving IgG antibody concentration $\geq 0.35\text{ }\mu\text{g/mL}$ and the IgG antibody geometric mean concentrations (GMCs)

Post dose 3 IgG pneumococcal immunogenicity results with comparison to the PCV7 seroresponse for the 7 common serotypes

With the exception of serotype 6B, all common serotypes met the noninferiority criterion after primary series. The proportion of responders $\geq 0.35\text{ }\mu\text{g/mL}$ in the PCV13-TT group were all higher than 97% (except 6B). For serotype 6B, the GMC was $1.25\text{ }\mu\text{g/mL}$, with a lower limit of the 2-sided 97.5% CI of 0.30% (Table 2). For serotype 6B, the proportion of responders was 90.45%. The rate difference of proportions of responders to 6B was -8.53%, with a lower limit of the 2-sided 97.5% CI of -11.73%. All 7 common (except 6B) serotypes met the non-inferiority criterion.

Post dose 3 IgG pneumococcal immunogenicity results with comparison to the PCV7 seroresponse for the 6 additional serotypes

For the 6 additional serotypes, the proportions of responders in the PCV13-TT group were all higher than 96% and all additional serotypes met the pre-specified superiority criteria.

Post dose 3 IgG pneumococcal immunogenicity results with comparison to the PCV7 reference for the 6 additional serotypes

The PCV7 reference value for the 6 additional serotypes was based on serotype 23F from the PCV7 group. For serotype 18C, the proportion of responders was 97.74%. All additional serotypes met the non-inferiority criterion.

The PCV7 reference value for the 6 additional serotypes was serotype 6B from the PCV7 group. For serotype 6B, the GMC was 3.52. Five (5) of the 6 additional serotypes met the non-inferiority criterion for GMR (lower limit of the 2-sided 95% CI for the GMR >0.5). The exception was serotype 3 with a GMR of 0.48 (95% CI, 0.42 to 0.58).

Among the 6 additional serotypes, the serotypes 3, 5 and 6A did not meet the prespecified noninferiority criterion. The lower limits of the 2-sided 97.5% CI for serotypes 3, 5 and 6A were 0.42, 0.47 and 0.48, respectively. Other additional serotypes met the pre-specified noninferiority criterion.

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Table 2. Comparison of percentage of subjects aged 3 months achieving a pneumococcal IgG antibody concentration $\geq 0.35\text{ }\mu\text{g/mL}$ and GMCs post dose 3 of the primary series (PPS)

IgG antibody concentration $\geq 0.35\text{ }\mu\text{g/mL}$ IgG antibody GMC

PCV13-TT (N=471) PCV7 (N=487) Rate difference c PCV13-TT (N=471) PCV7 (N=487) GMC Ratio

95% CI GMC
Serotype na % 95% CI nb % 95% CI % GMC 95% CI 95% CI Ratio d 95%/97.5%CI e
/97.5%CI (N=1043)
PCV7
4 471 100.0 99.22-100.0 483 99.18 97.91-99.78 0.82 -0.10,1.74 4.51 4.20-4.84 7.79 7.22-8.41 0.58
0.51,0.65
6B 426 90.45 87.42-92.95 482 98.97 97.62-99.67 -8.53 -11.73, -5.32 1.25 1.11-1.42 3.52 3.25-3.81 0.36
0.30,0.42
9V 467 99.15 97.84-99.77 486 99.79 98.86-99.99 -0.64 -1.70,0.41 3.72 3.45-4.02 4.47 4.19-4.78 0.83
0.74,0.93
14 471 100.0 99.22-100.0 486 99.79 98.86-99.99 0.21 -0.25,0.67 10.38 9.57-11.27 16.53 15.09-18.11 0.63
0.55,0.72
18C 463 98.30 96.68-99.26 476 97.74 95.99-98.87 0.56 -1.45,2.57 2.91 2.68-3.15 4.80 4.37-5.28 0.61
0.52,0.70
19F 471 100.0 99.22-100.0 487 100.0 99.25-100.0 0.00 -0.90,0.93 6.95 6.49-7.45 9.00 8.43-9.60 0.77
0.69,0.86
23F 458 97.24 95.33-98.52 480 98.56 97.06-99.42 -1.32 -3.40,0.76 2.93 2.65-3.23 4.44 4.08-4.82 0.66
0.57,0.76
Additional
vs PCV7 seroresponse f
1 469 99.58 98.47-99.95 280 57.49 52.97-61.93 42.08 37.65,46.51 2.15 2.01-2.30 0.30 0.26-0.34 7.17
6.14,8.37
3 462 98.09 96.40-99.12 220 45.17 40.69-49.72 52.91 48.32,57.50 1.67 1.56-1.79 0.33 0.31-0.36 5.02
4.55,5.53
5 464 98.51 96.96-99.40 81 16.63 13.43-20.24 81.88 78.40,85.36 1.88 1.75-2.02 0.05 0.04-0.06 40.29
32.02,50.69
6A 456 96.82 94.80-98.21 382 78.44 74.52-82.01 18.38 14.39,22.36 1.96 1.77-2.16 0.58 0.50-0.67 3.38
2.83,4.03
7F 469 99.58 98.47-99.95 118 24.23 20.49-28.29 75.35 71.49,79.20 5.47 5.06-5.92 0.05 0.04-0.07 105.38
82.43,134.72
19A 471 100.0 99.22-100.0 473 97.13 95.22-98.42 2.87 1.38,4.36 3.85 3.62-4.09 1.06 0.97-1.16 3.63
3.25,4.06
vs PCV7 reference g
1 469 99.58 98.47-99.95 476 97.74 95.99-98.87 1.83 0.18,3.49 2.15 2.01-2.30 3.52 3.25-3.81 0.61 0.54,0.69
3 462 98.09 96.40-99.12 476 97.74 95.99-98.87 0.35 -1.72,2.42 1.67 1.56-1.79 3.52 3.25-3.81 0.48
0.42,0.54
5 464 98.51 96.96-99.40 476 97.74 95.99-98.87 0.77 -1.19,2.73 1.88 1.75-2.02 3.52 3.25-3.81 0.54
0.47,0.61
6A 456 96.82 94.80-98.21 476 97.74 95.99-98.87 -0.93 -3.29,1.43 1.96 1.77-2.16 3.52 3.25-3.81 0.56
0.48,0.64
7F 469 99.58 98.47-99.95 476 97.74 95.99-98.87 1.83 0.18,3.49 5.47 5.06-5.92 3.52 3.25-3.81 1.56
1.37,1.77
19A 471 100.0 99.22-100.0 476 97.74 95.99-98.87 2.26 0.75,3.77 3.85 3.62-4.09 3.52 3.25-3.81 1.09
0.97,1.23
N=number of subjects included in PPS for PCV13-TT and PCV7.
a. number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT group.
b. number of subjects with a determinate IgG antibody concentration to the given serotype for PCV7 group.
c. The rate difference of PCV13-TT – PCV7.
d. GMC ratio: PCV13-TT to PCV7 reference.
e. Two-sided 95% CI is for additional serotypes, and two-sided 97.5% CI is for common serotypes.
f. Difference in proportions: (PCV13-TT – PCV7 seroresponse) expressed as a percentage.
GMC ratio: PCV13-TT to PCV7 seroresponse. IgG antibody concentration ≥ 0.35 $\mu\text{g/mL}$ was calculated respectively, with 95% CI given.
g. Difference in proportions: (PCV13-TT – PCV7 reference) expressed as a percentage. For the additional serotypes, the reference value is derived from serotype 23F from the PCV7 group.
GMC ratio: PCV13-TT to PCV7 reference. For the additional serotypes, the reference value is derived from serotype 6B from the PCV7 group.

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Table 3. Comparison of percentage of subjects aged 3 months achieving a pneumococcal IgG antibody concentration ≥ 0.35 $\mu\text{g/mL}$ and GMCs post dose 3 of the primary series (FAS)

IgG antibody concentration ≥ 0.35 $\mu\text{g/mL}$ IgG antibody GMC
PCV13-TT (N=519) PCV7 (N=519) Rate difference c PCV13-TT (N=519) PCV7 (N=519) GMC Ratio
95% CI
Serotype na % 95% CI nb % 95% CI % GMC 95% CI GMC 95% CI Ratio d 95%/97.5%CI e
/97.5%CI
PCV7
4 497 95.76 93.65-97.32 499 96.15 94.11-97.63 -0.39 -3.13,2.36 3.46 3.06-3.92 6.40 5.64-7.26 0.54
0.44,0.66
6B 463 89.21 86.22-91.75 504 97.11 95.28-98.37 -7.90 -11.37,-4.43 1.23 1.09-1.39 3.26 2.97-3.58 0.38
0.32,0.45
9V 502 96.72 94.81-98.08 505 97.30 95.52-98.52 -0.58 -2.95,1.79 3.24 2.95-3.56 3.91 3.54-4.32 0.83
0.71,0.97
14 516 99.42 98.32-99.88 517 99.61 98.61-99.95 -0.19 -1.16,0.77 9.32 8.49-10.22 15.45 14.01-17.03 0.60
0.52,0.70
18C 493 94.99 92.75-96.70 496 95.57 93.42-97.17 -0.58 -3.53,2.37 2.42 2.18-2.70 4.29 3.84-4.79 0.57
0.47,0.67
19F 516 99.42 98.32-99.88 513 98.84 97.50-99.57 0.58 -0.71,1.87 6.26 5.80-6.76 8.12 7.48-8.81 0.77
0.68,0.88
23F 489 94.22 91.85-96.07 501 96.53 94.57-97.93 -2.31 -5.23,0.61 2.59 2.32-2.90 3.95 3.54-4.41 0.65
0.55,0.78
Additional
vs PCV7 seroresponse f
1 498 95.95 93.88-97.48 294 56.65 52.26-60.96 39.31 34.72,43.89 1.79 1.60-1.99 0.29 0.25-0.33 6.27
5.26,7.47
3 489 94.22 91.85-96.07 234 45.09 40.75-49.48 49.13 44.40,53.86 1.45 1.33-1.58 0.32 0.29-0.34 4.58
4.06,5.16

5 489 94.22 91.85-96.07 86 16.57 13.47-20.05 77.65 73.87,81.43 1.50 1.34-1.68 0.05 0.04-0.06 32.80 25.92,41.52
6A 488 94.03 91.63-95.91 403 77.65 73.82-81.16 16.38 12.25,20.50 1.80 1.62-2.00 0.57 0.49-0.65 3.18 2.66,3.81
7F 504 97.11 95.28-98.37 126 24.28 20.65-28.20 72.83 68.87,76.79 4.66 4.19-5.19 0.05 0.04-0.07 87.39 68.38,111.69
19A 517 99.61 98.61-99.95 499 96.15 94.11-97.63 3.47 1.73,5.21 3.57 3.35-3.81 1.03 0.94-1.13 3.46 3.09,3.88
vs PCV7 reference g
1 498 95.95 93.88-97.48 496 95.57 93.42-97.17 0.39 -2.42,3.19 1.79 1.60-1.99 3.26 2.97-3.58 0.55 0.47,0.65
3 489 94.22 91.85-96.07 496 95.57 93.42-97.17 -1.35 -4.41,1.71 1.45 1.33-1.58 3.26 2.97-3.58 0.45 0.39,0.52
5 489 94.22 91.85-96.07 496 95.57 93.42-97.17 -1.35 -4.41,1.71 1.50 1.34-1.68 3.26 2.97-3.58 0.46 0.39,0.54
6A 488 94.03 91.63-95.91 496 95.57 93.42-97.17 -1.54 -4.63,1.55 1.80 1.62-2.00 3.26 2.97-3.58 0.55 0.47,0.65
7F 504 97.11 95.28-98.37 496 95.57 93.42-97.17 1.54 -1.07,4.15 4.66 4.19-5.19 3.26 2.97-3.58 1.43 1.22,1.68
19A 517 99.61 98.61-99.95 496 95.57 93.42-97.17 4.05 1.93,6.16 3.57 3.35-3.81 3.26 2.97-3.58 1.10 0.96,1.25

N=number of subjects included in FAS for PCV13-TT and PCV7.

- number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT group.
 - number of subjects with a determinate IgG antibody concentration to the given serotype for PCV7 group.
 - The rate difference of PCV13-TT – PCV7.
 - GMC ratio: PCV13-TT to PCV7.
 - Two-sided 95% CI is for additional serotypes, and two-sided 97.5% CI is for common serotypes.
 - Difference in proportions: (PCV13-TT – PCV7 seroresponse) expressed as a percentage.
- GMC ratio: PCV13-TT to PCV7 seroresponse. IgG antibody concentration ≥ 0.35 $\mu\text{g/mL}$ was calculated respectively, with 95% CI given.
- Difference in proportions: (PCV13-TT – PCV7 reference) expressed as a percentage. For the additional serotypes, the reference value is derived from serotype 18C from the PCV7 group.
- GMC ratio: PCV13-TT to PCV7 reference. For the additional serotypes, the reference value is derived from serotype 6B from the PCV7 group.

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Booster Dose

Post dose 4 IgG pneumococcal immunogenicity results with comparison to the PCV7 seroresponse for the 7 common serotypes

The non-inferiority criterion at the 0.35 $\mu\text{g/mL}$ level was met for all the common serotypes after the booster dose. The lower limit of the 2-sided 97.5% CI for the rate difference of percentage of subjects with IgG antibody concentration ≥ 0.35 $\mu\text{g/mL}$ (PCV13-TT – PCV7) was greater than -10% for all common serotypes. (Table 4)

The non-inferiority criterion with a lower limit of the 2-sided 97.5% CI for the GMR (PCV13-TT relative to PCV7) greater than 0.5 was met for all the common serotypes after the booster dose.

Post dose 4 IgG pneumococcal immunogenicity results with comparison to the PCV7 seroresponse for the 6 additional serotypes

With the exception of serotype 19A, all common serotypes met the superiority criterion after booster dose. For serotype 9A, the rate difference was 0.21% with a lower limit of the 2-sided 95% CI of -0.20%.

All additional serotypes met the pre-specified superiority criteria for GMC.

Post dose 4 IgG pneumococcal immunogenicity results with comparison to the PCV7 reference for the 6 additional serotypes

The PCV7 reference value for the 6 additional serotypes was based on serotype 9V from the PCV7 group (the lowest response among the 7 serotypes contained in PCV7 achieved by subjects who received PCV7). For serotype 9V, the proportion of responders was 99.59%. All additional serotypes met the non-inferiority criterion for IgG antibody concentration ≥ 0.35 $\mu\text{g/mL}$.

The PCV7 reference value for the 6 additional serotypes was serotype 9V from the PCV7 group (the lowest response among the 7 serotypes contained in PCV7 achieved by subjects who received PCV7). For serotype 9V, the GMC was 3.87. Four (4) of the 6 additional serotypes met the non-inferiority criterion for GMR (lower limit of the 2-sided 95% CI for the GMR > 0.5). The exception was serotype 3 and 5 with a GMR of 0.49 (95% CI, 0.44 to 0.54) and 0.41 (95% CI, 0.37 to 0.46), respectively.

Among the 6 additional serotypes, serotypes 3, and 5 did not meet the prespecified noninferiority criterion. The lower limits of the 2-sided 97.5% CI for serotypes 3 and 5 were 0.42 and 0.37, respectively. Other additional serotypes meet the prespecified noninferiority criterion.

The geometric mean fold rises (GMFRs) of antibody concentrations from pre- to post-booster for the 7 common serotypes ranged from 4.13 (serotype 14) to 6.38 (18C) in the PCV13-TT group.

For the 6 additional serotypes, the fold rises ranged from 3.57 (serotype 7F) to 7.08 (serotype 19A) in the PCV13-TT group.

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Table 4. Comparison of percentage of subjects aged 3 months achieving a pneumococcal IgG antibody concentration ≥ 0.35 $\mu\text{g/mL}$ and GMCs post booster dose (PPS)

IgG antibody concentration ≥ 0.35 $\mu\text{g/mL}$ IgG antibody GMC

PCV13-TT (N=475) PCV7 (N=487) Rate difference c PCV13-TT (N=475) PCV7 (N=487) GMC Ratio

95% CI

Serotype na % 95% CI nb % 95% CI % GMC 95% CI GMC 95% CI Ratio d 95%/97.5%CI e

/97.5%CI

PCV7

4 474 99.79 98.83-99.99 486 99.79 98.86-99.99 -0.01 -0.66,0.65 3.19 2.96-3.44 4.27 3.98-4.59 0.75

0.66,0.84

6B 475 100.0 99.23-100.0 486 99.79 98.86-99.99 0.21 -0.25,0.67 10.01 9.30-10.79 11.70 10.78-12.70 0.86

0.75,0.97

9V 475 100.0 99.23-100.0 487 100.0 99.25-100.0 0.00 -0.90,0.92 4.22 3.90-4.56 3.87 3.62-4.14 1.09
0.97,1.23
14 474 99.79 98.83-99.99 487 100.0 99.25-100.0 -0.21 -0.68,0.26 13.97 12.94-15.09 17.89 16.77-19.09 0.78
0.70,0.88
18C 474 99.79 98.83-99.99 485 99.59 98.52-99.95 0.20 -0.60,1.00 2.44 2.27-2.63 3.90 3.59-4.23 0.63
0.55,0.71
19F 475 100.0 99.23-100.0 487 100.0 99.25-100.0 0.00 -0.90,0.92 12.18 11.25-13.20 7.84 7.33-8.38 1.56
1.38,1.75
23F 475 100.0 99.23-100.0 486 99.79 98.86-99.99 0.21 -0.25,0.67 6.9 6.39-7.45 10.81 9.91-11.80 0.64
0.56,0.73
Additional
vs PCV7 seroresponsef
1 474 99.79 98.83-99.99 445 91.38 88.52-93.71 8.41 5.89,10.94 2.79 2.61-2.99 0.67 0.62-0.72 4.18
3.76,4.65
3 471 99.16 97.86-99.77 449 92.20 89.45-94.42 6.96 4.44,9.48 1.89 1.77-2.01 1.02 0.95-1.09 1.86 1.69,2.04
5 465 97.89 96.16-98.99 195 40.04 35.66-44.55 57.85 53.31,62.39 1.58 1.47-1.70 0.16 0.14-0.19 9.63
8.06,11.51
6A 473 99.58 98.49-99.95 476 97.74 95.99-98.87 1.84 0.40,3.28 5.82 5.32-6.37 3.80 3.44-4.21 1.53
1.34,1.75
7F 475 100.0 99.23-100.0 325 66.74 62.36-70.91 33.26 29.08,37.45 4.59 4.29-4.91 0.31 0.26-0.36 15.01
12.58,17.91
19A 475 100.0 99.23-100.0 486 99.79 98.86-99.99 0.21 -0.20,0.61 13.34 12.37-14.39 3.00 2.78-3.24 4.45
4.00,4.95
vs PCV7 reference g
1 474 99.79 98.83-99.99 485 99.59 98.52-99.95 0.20 -0.60,1.00 2.79 2.61-2.99 3.87 3.62-4.14 0.72
0.65,0.80
3 471 99.16 97.86-99.77 485 99.59 98.52-99.95 -0.43 -1.57,0.71 1.89 1.77-2.01 3.87 3.62-4.14 0.49
0.44,0.54
5 465 97.89 96.16-98.99 485 99.59 98.52-99.95 -1.69 -3.31,-0.08 1.58 1.47-1.70 3.87 3.62-4.14 0.41
0.37,0.46
6A 473 99.58 98.49-99.95 485 99.59 98.52-99.95 -0.01 -0.94,0.92 5.82 5.32-6.37 3.87 3.62-4.14 1.50
1.32,1.71
7F 475 100.0 99.23-100.0 485 99.59 98.52-99.95 0.41 -0.24,1.06 4.59 4.29-4.91 3.87 3.62-4.14 1.19
1.06,1.32
19A 475 100.0 99.23-100.0 485 99.59 98.52-99.95 0.41 -0.24,1.06 13.34 12.37-14.39 3.87 3.62-4.14 3.45
3.07,3.87
N=number of subjects included in PPS for PCV13-TT and PCV7.
a. number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT group.
b. number of subjects with a determinate IgG antibody concentration to the given serotype for PCV7 group.
c. The rate difference of PCV13-TT – PCV7.
d. GMC ratio: PCV13-TT to PCV7 reference.
e. Two-sided 95% CI is for additional serotypes, and two-sided 97.5% CI is for common serotypes.
f. Difference in proportions: (PCV13-TT – PCV7 seroresponse) expressed as a percentage.
GMC ratio: PCV13-TT to PCV7 seroresponse. IgG antibody concentration ≥ 0.35 $\mu\text{g/mL}$ was calculated respectively, with 95% CI given.
g. Difference in proportions: (PCV13-TT – PCV7 reference) expressed as a percentage. For the additional serotypes, the reference value is derived from serotype 18C from the PCV7 group.
GMC ratio: PCV13-TT to PCV7 reference. For the additional serotypes, the reference value is derived from serotype 9V from the PCV7 group.

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Table 5. Comparison of percentage of subjects aged 3 months for the geometric mean fold rises (GMFRs) of antibody concentrations (PPS)
PCV13-TT (N=475) PCV7 (N=487)
Serotype P
GMFRs 95%CI GMFRs 95%CI
4 5.62 5.23-6.03 5.15 4.78-5.55 0.0980
6B 4.83 4.50-5.17 8.66 8.05-9.31 <0.0001
9V 7.11 6.57-7.69 6.06 5.63-6.52 0.0036
Common 14 4.13 3.82-4.46 3.70 3.46-3.96 0.0356
18C 6.38 5.77-7.04 7.63 6.96-8.36 0.0090
19F 5.43 4.93-5.99 4.40 4.08-4.76 0.0009
23F 5.35 5.01-5.71 9.71 8.99-10.49 <0.0001
1 3.96 3.73-4.20 1.83 1.70-1.99 <0.0001
3 4.80 4.51-5.11 5.19 4.79-5.62 0.1335
5 3.67 3.45-3.92 0.95 0.82-1.09 <0.0001
Additional
6A 4.79 4.42-5.19 5.66 5.16-6.20 0.0073
7F 3.57 3.32-3.84 1.02 0.88-1.19 <0.0001
19A 7.08 6.51-7.70 2.61 2.42-2.82 <0.0001

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2.7.3.2.2.1.1.2. Pneumococcal Serotype antibody concentrations in the subjects aged 3 months in PCV13-TT group and PCV7 group
Please refer to the section 2.2.1.3.2. in this clinical summary.

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2.7.3.2.2.1.1.3. Comparison of percentage of subjects achieving IgG antibody concentration ≥ 1.0 $\mu\text{g/mL}$ with comparison to the PCV7 seroresponse
Primary series
Data for the 7 common serotypes
With the exception of serotypes 6B and 23F, all common serotypes met the noninferiority criterion

after primary series. For serotype 6B and 23F, the lower limit of the 2-sided 95% CI was -35.86 and -12.81, respectively. (Table 6)
 Data for the 6 additional serotypes
 All common serotypes met the noninferiority criterion after primary series. The lower limit of the 2-sided 95% CI was all greater than 0.

Table 6. Comparison of percentage of subjects aged 3 months achieving a pneumococcal IgG antibody concentration ≥ 1.0 $\mu\text{g/mL}$ post dose 3 of the primary series (PPS)

IgG antibody concentration ≥ 1.0 $\mu\text{g/mL}$									
PCV13-TT (N=471) PCV7 (N=487) Rate difference c									
Serotype	na	%	95% CI	nb	%	95% CI	%	95% CI	/97.5%CI
PCV7									
4	456	96.82	94.80-98.21	480	98.56	97.06-99.42	-1.75	-3.65,0.16	
6B	288	61.15	56.58-65.57	448	91.99	89.21-94.24	-30.85	-35.86,-25.83	
9V	452	95.97	93.77-97.55	472	96.92	94.97-98.27	-0.95	-3.30,1.39	
14	466	98.94	97.54-99.65	484	99.38	98.21-99.87	-0.45	-1.60,0.71	
18C	431	91.51	88.62-93.86	465	95.48	93.24-97.15	-3.98	-7.10,-0.85	
19F	466	98.94	97.54-99.65	479	98.36	96.79-99.29	0.58	-0.88,2.04	
23F	412	87.47	84.14-90.33	472	96.92	94.97-98.27	-9.45	-12.81,-6.09	
Additional									
1	404	85.77	82.29-88.80	33	6.78	4.71-9.38	68.60	64.11,73.08	
3	355	75.37	71.22-79.20	11	2.26	1.13-4.01	78.21	74.39,82.02	
5	379	80.47	76.59-83.95	180	36.96	32.66-41.42	40.32	34.60,46.04	
6A	364	77.28	73.23-80.99	27	5.54	3.69-7.96	91.91	89.43,94.39	
7F	459	97.45	95.59-98.68	260	53.39	48.85-57.89	44.91	40.33,49.50	
19A	463	98.30	96.68-99.26	33	6.78	4.71-9.38	68.60	64.11,73.08	

N=number of subjects included in PPS for PCV13-TT and PCV7.

- a. number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT group.
- b. number of subjects with a determinate IgG antibody concentration to the given serotype for PCV7 group.
- c. The rate difference of PCV13-TT – PCV7.

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Table 7. Comparison of percentage of subjects aged 3 months achieving a pneumococcal IgG antibody concentration ≥ 1.0 $\mu\text{g/mL}$ post dose 3 of the primary series (FAS)

IgG antibody concentration ≥ 1.0 $\mu\text{g/mL}$									
PCV13-TT (N=519) PCV7 (N=519) Rate difference c									
Serotype	na	%	95% CI	nb	%	95% CI	%	95% CI	/97.5%CI
PCV7									
4	482	92.87	90.31-94.93	496	95.57	93.42-97.17	-2.70	-5.53,0.14	
6B	309	59.54	55.17-63.79	464	89.40	86.43-91.92	-29.87	-34.85,-24.88	
9V	477	91.91	89.22-94.11	488	94.03	91.63-95.91	-2.12	-5.23,0.99	
14	500	96.34	94.34-97.78	508	97.88	96.24-98.94	-1.54	-3.58,0.49	
18C	454	87.48	84.32-90.20	481	92.68	90.09-94.77	-5.20	-8.83,-1.58	
19F	497	95.76	93.65-97.32	499	96.15	94.11-97.63	-0.39	-2.78,2.01	
23F	436	84.01	80.57-87.06	487	93.83	91.41-95.74	-9.83	-13.60,-6.05	
Additional									
1	427	82.27	78.71-85.46	44	8.48	6.23-11.21	73.80	69.73,77.86	
3	370	71.29	67.19-75.15	38	7.32	5.23-9.91	63.97	59.48,68.46	
5	392	75.53	71.60-79.17	11	2.12	1.06-3.76	73.41	69.51,77.31	
6A	390	75.14	71.19-78.81	193	37.19	33.02-41.50	37.96	32.38,43.54	
7F	488	94.03	91.63-95.91	28	5.39	3.61-7.70	88.63	85.82,91.45	
19A	494	95.18	92.97-96.86	278	53.56	49.17-57.92	41.62	36.95,46.29	

N=number of subjects included in PPS for PCV13-TT and PCV7.

- a. number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT group.
- b. number of subjects with a determinate IgG antibody concentration to the given serotype for PCV7 group.
- c. The rate difference of PCV13-TT – PCV7.

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Booster Dose

Data for the 7 common serotypes

With the exception of serotypes 4 and 18C, the proportion of responders ≥ 1.0 $\mu\text{g/mL}$ in the PCV13-TT group were all higher than 95%. For serotype 4 and 18C, the proportion of responders ≥ 1.0 $\mu\text{g/mL}$ was 93.26% and 88.84%. All common serotypes met the noninferiority criterion after booster dose. The rate difference of percentage of subjects with IgG antibody concentration ≥ 1.0 $\mu\text{g/mL}$ (PCV13-TT – PCV7) was all greater than -10% (Table 8).

Data for the 6 additional serotypes

The proportion of responders ≥ 1.0 $\mu\text{g/mL}$ was higher than 95% except serotypes 1, 3, and 5A. The lower limit of the 2-sided 95% CI for the rate difference of percentage of subjects with IgG antibody concentration ≥ 1.0 $\mu\text{g/mL}$ (PCV13-TT – PCV7) was greater than 0 for all additional serotypes.

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Table 8 Comparison of percentage of subjects aged 3 months achieving a pneumococcal IgG antibody concentration ≥ 1.0 $\mu\text{g/mL}$ post booster dose (PPS)

IgG antibody concentration ≥ 1.0 $\mu\text{g/mL}$									
PCV13-TT (N=475) PCV7 (N=487) Rate difference c									
Serotype	na	%	95% CI	nb	%	95% CI	%	95% CI	/97.5%CI
PCV7									
4	443	93.26	90.62-95.35	470	96.51	94.47-97.95	-3.25	-6.03,-0.46	
6B	475	100.0	99.23-100.0	478	98.15	96.52-99.15	1.85	0.65,3.04	
9V	455	95.79	93.57-97.41	471	96.71	94.72-98.11	-0.93	-3.33,1.48	

14 474 99.79 98.83-99.99 487 100.0 99.25-100.0 -0.21 -0.62,0.20
18C 422 88.84 85.66-91.53 460 94.46 92.04-96.31 -5.61 -9.10,-2.13
19F 475 100.0 99.23-100.0 484 99.38 98.21-99.87 0.62 -0.08,1.31
23F 469 98.74 97.27-99.54 484 99.38 98.21-99.87 -0.65 -1.87,0.57
Additional
1 431 90.74 87.76-93.19 125 25.67 21.84-29.79 65.07 60.40,69.74
3 391 82.32 78.58-85.64 243 49.90 45.37-54.43 32.42 26.81,38.03
5 342 72.00 67.73-76.00 13 2.67 1.43- 4.52 69.33 65.05,73.61
6A 457 96.21 94.08-97.74 431 88.50 85.33-91.20 7.71 4.40,11.02
7F 466 98.11 96.43-99.13 69 14.17 11.19-17.59 83.94 80.61,87.27
19A 473 99.58 98.49-99.92 459 94.25 91.80-96.15 5.33 3.18,7.48
N=number of subjects included in PPS for PCV13-TT and PCV7.
a. number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT group.
b. number of subjects with a determinate IgG antibody concentration to the given serotype for PCV7 group.
c. The rate difference of PCV13-TT – PCV7.

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2.7.3.2.2.1.1.4. Comparison of percentage of subjects achieving an OPA antibody $\geq 1:8$ and the IgG antibody geometric mean titers (GMTs)
Primary series
Data for 7 common serotypes
For the 7 common serotypes, the percentage of subjects achieving an OPA antibody titer $\geq 1:8$ was 100% except serotypes 6B (89.25%) and 23F (95.70%) in PCV13-TT group. With the exception of serotypes 6B, the lower limit of the 2-sided 95% CI of rate difference of the percentage of subjects achieving an OPA antibody titer $\geq 1:8$ was greater than -10% for the common serotypes (Table 9). The lower limit of the 2-sided 95% CI of GMT ratio was greater than 0.5 for serotypes 9V, 14 and 19F. For other serotypes including 4, 6B, 18C and 19F, the lower limit of the 2-sided 95% CI of GMT ratio was ≥ 0.5 .
Data for 6 additional serotypes
For the 6 additional serotypes, the percentage of subjects achieving an OPA antibody titer $\geq 1:8$ was higher than 96% except serotype 1 (83.87%) in PCV13-TT group. The lower limit of the 2-sided 95% CI of rate difference of the percentage of subjects achieving an OPA antibody titer $\geq 1:8$ was greater than 0 for all the additional serotypes. The lower limit of the 2-sided 95% CI of GMT ratio for the additional serotypes was all greater than 1.

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Table 9. Comparison of percentage of subjects aged 3 months achieving a pneumococcal IgG antibody titer $\geq 1:8$ and GMTs post-dose 3 of the primary series (PPS)
OPA antibody titer $\geq 1:8$ OPA GMT
PCV13-TT (N=93) PCV7 (N=96) Rate difference c PCV13-TT (N=93) PCV7 (N=96) GMT Ratio
Serotype na % 95% CI nb % 95% CI % 95% CI /97.5%CI GMT 95% CI GMT 95% CI Ratio d 95%/97.5%CI e
Common
4 93 100.0 96.11-100.0 96 100.0 96.23-100.0 0.00 -3.90,4.05 1227.93 992.5-1519.2 2291.32 1947.8-2695.4 0.54 0.41-0.70
6B 83 89.25 81.11-94.72 95 98.96 94.33-99.97 -9.71 -16.33,-3.10 528.67 329.58-848.02 2928.36 2233.9-3838.7 0.18 0.11-0.31
9V 93 100.0 96.11-100.0 96 100.0 96.23-100.0 0.00 -3.90,4.05 1922.48 1518.9-2433.2 1974.62 1612.0-2418.9 0.97 0.72-1.33
14 93 100.0 96.11-100.0 96 100.0 96.23-100.0 0.00 -3.90,4.05 4511.21 3494.9-5823.1 5594.48 4188.4-7472.5 0.81 0.55-1.18
18C 93 100.0 96.11-100.0 95 98.96 94.33-99.97 1.04 -0.99,3.07 1356.26 1094.5-1680.7 3129.49 2353.5-4161.3 0.43 0.30-0.62
19F 93 100.0 96.11-100.0 94 97.92 92.68-99.75 2.08 -0.77,4.94 652.57 547.97-777.14 540.56 423.02-690.76 1.21 0.89-1.63
23F 89 95.70 89.35-98.82 95 98.96 94.33-99.97 -3.26 -7.86,1.34 2090.78 1452.4-3009.7 5939.64 4465.8-7899.8 0.35 0.22-0.56
Additional
1 78 83.87 74.80-90.68 0 0.00 0.00-3.77 83.87 76.40,91.35 62.13 42.64-90.51 2.00 2.00-2.00 31.06 21.50-44.88
3 90 96.77 90.86-99.33 15 15.63 9.02-24.46 81.15 73.05,89.25 511.70 376.29-695.83 2.80 2.35-3.33 182.81 129.07-258.94
5 92 98.92 94.15-99.97 4 4.17 1.15-10.33 94.76 90.24,99.27 450.24 358.12-566.05 2.24 1.99-2.51 201.42 156.52-259.20
6A 91 97.85 92.45-99.74 77 80.21 70.83-87.64 17.64 9.14,26.14 1079.96 792.9-1471.0 275.54 159.69-475.44 3.92 2.09-7.35
7F 93 100.0 96.11-100.0 36 37.50 27.82-47.97 62.50 52.82,72.18 7741.10 6383.8-9387.0 12.76 7.68-21.19 606.70 351.58-1046.93
19A 93 100.0 96.11-100.0 43 44.79 34.63-55.29 55.21 45.26,65.16 777.70 638.69-946.97 11.50 7.55-17.51 67.63 42.42-107.83
N=number of subjects included in PPS for PCV13-TT and PCV7.
a. number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT group.
b. number of subjects with a determinate IgG antibody concentration to the given serotype for PCV7 group.
c. The rate difference of PCV13-TT – PCV7.
d. GMT ratio: PCV13-TT to PCV7.

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Table 10. Comparison of percentage of subjects aged 3 months achieving a pneumococcal IgG antibody titer $\geq 1:8$ and GMTs post-dose 3 of the primary series (FAS)
OPA antibody titer $\geq 1:8$ OPA GMT
PCV13-TT (N=98) PCV7 (N=97) Rate difference c PCV13-TT (N=93) PCV7 (N=96) GMT Ratio

Serotype na % 95% CI nb % 95% CI % 95% CI /97.5%CI GMT 95% CI GMT 95% CI Ratio d 95%/97.5%CI e

Common

4 98 100.0 96.31-100.0 97 100.0 96.27-100.0 0.00 -3.88,3.81 1246.22 1011.4-1535.5 2307.23 1963.5-2711.1
0.54 0.42-0.70

6B 88 89.80 82.03-95.00 96 98.97 94.39-99.97 -9.17 -15.49,-2.85 553.85 352.92-869.16 2940.48
2249.2-3844.1 0.19 0.11-0.32

9V 98 100.0 96.31-100.0 97 100.0 96.27-100.0 0.00 -3.88,3.81 1966.80 1569.2-2465.1 1993.08 1629.1-2438.4
0.99 0.73-1.33

14 98 100.0 96.31-100.0 97 100.0 96.27-100.0 0.00 -3.88,3.81 4387.49 3416.4-5634.6 5604.71 4208.8-7463.6
0.78 0.54-1.14

18C 98 100.0 96.31-100.0 96 98.97 94.39-99.97 1.03 -0.98,3.04 1347.71 1096.1-1657.2 3142.30 2370.0-4166.3
0.43 0.30-0.61

19F 98 100.0 96.31-100.0 95 97.94 92.75-99.75 2.06 -0.77,4.89 655.55 551.84-778.74 542.42 425.53-691.42
1.21 0.90-1.62

23F 94 95.92 89.88-98.88 96 98.97 94.39-99.97 -3.05 -7.45,1.35 2108.20 1487.1-2988.8 5943.19
4481.9-7880.9 0.35 0.23-0.55

Additional

1 82 83.67 74.84-90.37 0 0.00 0.00-3.73 83.67 76.36,90.99 64.30 44.31-93.29 2.00 2.00-2.00 32.15
22.17-46.63

3 95 96.94 91.31-99.36 15 15.46 8.92-24.22 81.47 73.51,89.44 509.18 378.71-684.60 2.79 2.35-3.31 182.55
129.76-256.80

5 97 98.98 94.45-99.97 4 4.12 1.13-10.22 94.86 90.43,99.28 453.05 363.07-565.34 2.23 1.99-2.50 202.91
158.31-260.08

6A 96 97.96 92.82-99.75 78 80.41 71.11-87.78 17.55 9.17,25.93 1120.90 831.4-1511.2 282.21 164.15-485.19
3.97 2.15-7.33

7F 98 100.0 96.31-100.0 37 38.14 28.47-48.57 61.86 52.19,71.52 7730.89 6423.4-9304.5 13.20 7.96-21.90
585.75 343.55-998.70

19A 98 100.0 96.31-100.0 44 45.36 35.22-55.79 54.64 44.73,64.55 783.76 648.55-947.14 12.00 7.84-18.34
65.34 41.23-103.55

N=number of subjects included in FAS for PCV13-TT and PCV7.

a. number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT group.

b. number of subjects with a determinate IgG antibody concentration to the given serotype for PCV7 group.

c.

The rate difference of PCV13-TT – PCV7.

d. GMT ratio: PCV13-TT to PCV7.

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Booster Dose

Data for 7 common serotypes

For the 7 common serotypes, the percentage of subjects achieving an OPA antibody titer $\geq 1:8$ was 100% except serotypes 6B (98.96%) and 23F (98.96%) in PCV13-TT group. The lower limit of the 2-sided 95% CI of rate difference of the percentage of subjects achieving an OPA antibody titer $\geq 1:8$ was greater than -10% for all the common serotypes (Table 11). With the exception of serotypes 6B and 23F, the lower limit of the 2-sided 95% CI of rate difference of the percentage of subjects achieving an OPA antibody titer $\geq 1:8$ was greater than -10% for the common serotypes.

Data for 6 additional serotypes

For the 6 additional serotypes, the percentage of subjects achieving an OPA antibody titer $\geq 1:8$ was higher than 95% in PCV13-TT group. With the exception of serotypes 6A, the lower limit of the 2-sided 95% CI of rate difference of the percentage of subjects achieving an OPA antibody titer $\geq 1:8$ was greater than 0 for the additional serotypes. For the serotype 6A, the lower limit of the 2-sided 95% CI of rate difference of the percentage of subjects achieving an OPA antibody titer $\geq 1:8$ was -3.47%. The lower limit of the 2-sided 95% CI of rate difference of the percentage of subjects achieving an OPA antibody titer $\geq 1:8$ was greater than 1 for all the additional serotypes.

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Table 11. Comparison of percentage of subjects aged 3 months achieving a pneumococcal IgG antibody titer $\geq 1:8$ and GMTs post booster dose (PPS)

OPA antibody titer $\geq 1:8$ OPA GMT

PCV13-TT (N=96) PCV7 (N=97) Rate difference c PCV13-TT (N=96) PCV7 (N=97) GMT Ratio

Serotype na % 95% CI nb % 95% CI % 95% CI /97.5%CI GMT 95% CI GMT 95% CI Ratio d 95%/97.5%CI e

Common

4 96 100.0 96.23-100.0 97 100.0 96.27-100.0 0.00 -3.88,3.89 6182.37 4931.4-7750.7 6208.15 4950.8-7784.8
1.00 0.72-1.37

6B 95 98.96 94.33-99.97 95 97.94 92.75-99.75 1.02 -2.46,4.50 2661.93 1948.9-3635.9 7564.42 5345-10705
0.35 0.22-0.56

9V 96 100.0 96.23-100.0 97 100.0 96.27-100.0 0.00 -3.88,3.89 6184.75 4870.3-7854.0 5484.79 4453.1-6755.5
1.13 0.82-1.54

14 96 100.0 96.23-100.0 97 100.0 96.27-100.0 0.00 -3.88,3.89 7686.96 6160.6-9591.5 6221.93 4855.7-7972.5
1.24 0.89-1.72

18C 96 100.0 96.23-100.0 96 98.97 94.39-99.97 1.03 -0.98,3.04 4173.73 3269.4-5328.2 5507.66 4094.6-7408.5
0.76 0.52-1.11

19F 96 100.0 96.23-100.0 94 96.91 91.23-99.36 3.09 -0.35,6.54 2359.26 1898.8-2931.3 815.68 600.1-1108.8
2.89 1.99-4.21

23F 95 98.96 94.33-99.97 96 98.97 94.39-99.97 -0.01 -2.87,2.85 4943.66 3817.5-6402.1 16131.84 11742-22163
0.31 0.20-0.46

Additional

1 1 94 97.92 4 4.12 1.13-10.22 93.79 88.91,98.67 321.14 239.43-430.74 2.28 2.00-2.61 140.82 102.34-193.77
3 3 94 97.92 36 37.11 27.52-47.52 60.80 50.77,70.83 678.98 523.20-881.14 5.03 3.77-6.72 134.86
91.64-198.48

5 5 96 100.0 5 5.15 1.69-11.62 94.85 90.45,99.25 1375.13 1061.9-1780.8 2.61 2.07-3.29 527.57
373.44-745.31

6A 6A 92 95.83 90 92.78 85.70-97.05 3.05 -3.47,9.57 3254.75 2221.0-4769.7 1562.59 972.9-2509.6 2.08
1.14-3.82

7F 7F 96 100.0 51 52.58 42.18-62.81 47.42 37.49,57.36 7244.42 6006.0-8738.3 41.47 22.27-77.21 174.70
91.38-333.97

19A 19A 96 100.0 77 79.38 69.97-86.93 20.62 12.57,28.67 5503.11 4501.2-6728.1 72.99 46.46-114.65 75.40 46.06-123.44

N=number of subjects included in PPS for PCV13-TT and PCV7.

e. number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT group.

f. number of subjects with a determinate IgG antibody concentration to the given serotype for PCV7 group.

g.

The rate difference of PCV13-TT – PCV7.

h.

GMt ratio: PCV13-TT to PCV7.

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Table 12. Comparison of percentage of subjects aged 3 months achieving a pneumococcal IgG antibody titer $\geq 1:8$ and GMTs post booster dose (FAS)

OPA antibody titer $\geq 1:8$ OPA GMT

PCV13-TT (N=96) PCV7 (N=97) Rate difference c PCV13-TT (N=96) PCV7 (N=97) GMT Ratio

Serotype na % 95% CI nb % 95% CI % 95% CI GMT 95% CI GMT 95% CI Ratio d 95%CI e

Common

4 96 100.0 96.23-100.0 97 100.0 96.27-100.0 0.00 -3.88,3.89 6182.37 4931.4-7750.7 6208.15 4950.8-7784.8

1.00 0.72-1.37

6B 95 98.96 94.33-99.97 95 97.94 92.75-99.75 1.02 -2.46,4.50 2661.93 1948.9-3635.9 7564.42 5345-10705

0.35 0.22-0.56

9V 96 100.0 96.23-100.0 97 100.0 96.27-100.0 0.00 -3.88,3.89 6184.75 4870.3-7854.0 5484.79 4453.1-6755.5

1.13 0.82-1.54

14 96 100.0 96.23-100.0 97 100.0 96.27-100.0 0.00 -3.88,3.89 7686.96 6160.6-9591.5 6221.93 4855.7-7972.5

1.24 0.89-1.72

18C 96 100.0 96.23-100.0 96 98.97 94.39-99.97 1.03 -0.98,3.04 4173.73 3269.4-5328.2 5507.66 4094.6-7408.5

0.76 0.52-1.11

19F 96 100.0 96.23-100.0 94 96.91 91.23-99.36 3.09 -0.35,6.54 2359.26 1898.8-2931.3 815.68 600.1-1108.8

2.89 1.99-4.21

23F 95 98.96 94.33-99.97 96 98.97 94.39-99.97 -0.01 -2.87,2.85 4943.66 3817.5-6402.1 16131.84 11742-22163

0.31 0.20-0.46

Additional

1 94 97.92 92.68-99.75 4 4.12 1.13-10.22 93.79 88.91,98.67 321.14 239.43-430.74 2.28 2.00-2.61 140.82

102.34-193.77

3 94 97.92 92.68-99.75 36 37.11 27.52-47.52 60.80 50.77,70.83 678.98 523.20-881.14 5.03 3.77-6.72 134.86

91.64-198.48

5 96 100.0 96.23-100.0 5 5.15 1.69-11.62 94.85 90.45,99.25 1375.13 1061.9-1780.8 2.61 2.07-3.29 527.57

373.44-745.31

6A 92 95.83 89.67-98.85 90 92.78 85.70-97.05 3.05 -3.47,9.57 3254.75 2221.0-4769.7 1562.59 972.9-2509.6

2.08 1.14-3.82

7F 96 100.0 96.23-100.0 51 52.58 42.18-62.81 47.42 37.49,57.36 7244.42 6006.0-8738.3 41.47 22.27-77.21

174.70 91.38-333.97

19A 96 100.0 96.23-100.0 77 79.38 69.97-86.93 20.62 12.57,28.67 5503.11 4501.2-6728.1 72.99 46.46-114.65

75.40 46.06-123.44

N=number of subjects included in FAS for PCV13-TT and PCV7.

a. number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT group.

b. number of subjects with a determinate IgG antibody concentration to the given serotype for PCV7 group.

c.

The rate difference of PCV13-TT – PCV7.

d. GMT ratio: PCV13-TT to PCV7.

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2.7.3.2.2.1.2. Pneumococcal immunogenicity results for subjects in PCV13-TT (2 months) group and PCV7 (3 months) group

2.7.3.2.2.1.2.1. Comparison of percentage of subjects achieving IgG antibody concentration $\geq$

0.35 $\mu\text{g/mL}$ $\mu\text{g/mL}$ and the IgG antibody geometric mean concentrations (GMCs)

Post dose 3 IgG pneumococcal immunogenicity results with comparison to the PCV7

seroresponse for the 7 common serotypes

The IgG seroresponse rates $\geq 0.35 \mu\text{g/mL}$ were all higher than 97% for subjects using 0th, 2nd, and

4th month dosing schedule in the PCV13-TT (2 months) group. The IgG seroresponse rates ≥ 0.35

$\mu\text{g/mL}$ of all common serotypes met the noninferiority criterion after primary series for subjects

in PCV13-TT group compared with subjects using 0th, 1st, and 2nd month dosing schedule in PCV7

group (Table 13). With the exception of serotypes 4 and 18C, all common serotypes met the

noninferiority criterion after primary series. For serotypes 4 and 18C, the GMC was 2.43 $\mu\text{g/mL}$

and 2.04 $\mu\text{g/mL}$, respectively.

Post dose 3 IgG pneumococcal immunogenicity results with comparison to the PCV7

seroresponse for the 6 additional serotypes

With the exception of serotype 5, the IgG seroresponse rates $\geq 0.35 \mu\text{g/mL}$ were all higher than

96% for subjects using 0th, 2nd, and 4th month dosing schedule in the PCV13-TT (2 months) group.

For serotype 5, the IgG seroresponse rates $\geq 0.35 \mu\text{g/mL}$ was 90.87%. The lower limit of the 2-

sided 95% CI for the IgG seroresponse rate was greater than 70% for all additional serotypes. The

lower limit of the 2-sided 95% CI for rate difference (PCV13-TT group - PCV7) was greater than

0. The lower limit of the 2-sided 95% CI for GMR (PCV13-TT relative to PCV7) was greater than

0.5 and all additional serotypes met the superiority criterion.

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Table 13. Comparison of percentage of subjects achieving a pneumococcal IgG antibody concentration ≥ 0.35

$\mu\text{g/mL}$ and GMCs in PCV13-TT (2 months) group

and PCV7 (3 months) group post dose 3 of the primary series (PPS)

IgG antibody concentration $\geq 0.35 \mu\text{g/mL}$ IgG antibody GMC

PCV13-TT (2M)

PCV13-TT (2M) (N=482) PCV7 (3M) (N=487) Rate difference c PCV7 (3M) (N=487) GMC Ratio (N=482)

95% CI Ratio 95%/97.5%CI

Serotype na % 95% CI nb % 95% CI % GMC 95% CI GMC 95% CI d e /97.5%CI

PCV7

4 482 100.0 99.24-100.0 483 99.18 97.91-99.78 0.82 -0.10,1.74 2.43 2.28-2.60 7.79 7.22-8.41 0.31 0.28,0.35

6B 471 97.72 95.95-98.86 482 98.97 97.62-99.67 -1.26 -3.09,0.58 3.72 3.37-4.10 3.52 3.25-3.81 1.06 0.91,1.22

9V 481 99.79 98.85-99.99 486 99.79 98.86-99.99 -0.00 -0.66,0.65 3.66 3.40-3.94 4.47 4.19-4.78 0.82 0.73,0.92

14 481 99.79 98.85-99.99 486 99.79 98.86-99.99 -0.00 -0.66,0.65 10.25 9.46-11.11 16.53 15.09-18.11 0.62 0.54,0.71

18C 477 98.96 97.60-99.66 476 97.74 95.99-98.87 1.22 -0.61,3.05 2.04 1.87-2.23 4.80 4.37-5.28 0.43 0.37,0.49

19F 482 100.0 99.24-100.0 487 100.0 99.25-100.0 0.00 -0.90,0.91 5.46 5.14-5.80 9.00 8.43-9.60 0.61 0.55,0.67

23F 481 99.79 98.85-99.99 480 98.56 97.06-99.42 1.23 -0.07,2.52 3.86 3.53-4.21 4.44 4.08-4.82 0.87 0.76,1.00

Additional

1 482 100.0 99.24-100.0 280 57.49 52.97-61.93 42.51 38.11,46.90 2.09 1.95-2.23 0.30 0.26-0.34 6.96 5.97,8.12

3 463 96.06 93.91-97.61 220 45.17 40.69-49.72 50.88 46.13,55.63 1.41 1.32-1.50 0.33 0.31-0.36 4.22 3.83,4.65

5 438 90.87 87.94-93.29 81 16.63 13.43-20.24 74.24 70.05,78.43 0.79 0.73-0.85 0.05 0.04-0.06 16.88 13.42,21.24

6A 478 99.17 97.89-99.77 382 78.44 74.52-82.01 20.73 16.99,24.47 3.28 3.01-3.57 0.58 0.50-0.67 5.65 4.78-6.69

7F 480 99.59 98.51-99.95 118 24.23 20.49-28.29 75.36 71.51,79.20 4.76 4.42-5.13 0.05 0.04-0.07 91.73 72.03,116.82

19A 482 100.0 99.24-100.0 473 97.13 95.22-98.42 2.87 1.39,4.36 3.12 2.95-3.30 1.06 0.97-1.16 2.94 2.64,3.28

N=number of subjects included in PPS for PCV13-TT (2M) and PCV7 (3M).

a. number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT (2M) group.

b. number of subjects with a determinate IgG antibody concentration to the given serotype for PCV7 (3M) group.

c. The rate difference of PCV13-TT (2M) – PCV7 (3M).

d. GMC ratio: PCV13-TT (2M) to PCV7 (3M).

e. Two-sided 95% CI is for additional serotypes, and two-sided 97.5% CI is for common serotypes.

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Table 14. Comparison of percentage of subjects achieving a pneumococcal IgG antibody concentration ≥ 0.35 $\mu\text{g/mL}$ and GMCs in PCV13-TT (2 months) group and PCV7 (3 months) group post dose 3 of the primary series (FAS)

IgG antibody concentration ≥ 0.35 $\mu\text{g/mL}$ IgG antibody GMC

PCV13-TT (2M)

PCV13-TT (2M) (N=517) PCV7 (3M) (N=519) Rate difference c PCV7 (3M) (N=519) GMC Ratio (N=517)

95% CI

Serotype na % 95% CI nb % 95% CI % GMC 95% CI GMC 95% CI Ratio d 95%/97.5%CI e /97.5%CI

PCV7

4 492 95.16 92.94-96.85 499 96.15 94.11-97.63 -0.98 -3.82,1.86 1.94 1.74-2.17 6.40 5.64-7.26 0.54 0.44,0.66

6B 499 96.52 94.55-97.92 504 97.11 95.28-98.37 -0.59 -3.04,1.85 3.38 3.05-3.75 3.26 2.97-3.58 0.38 0.32,0.45

9V 503 97.29 95.50-98.51 505 97.30 95.52-98.52 -0.01 -2.27,2.25 3.11 2.82-3.43 3.91 3.54-4.32 0.83 0.71,0.97

14 511 98.84 97.49-99.57 517 99.61 98.61-99.95 -0.78 -1.99,0.44 8.73 7.87-9.68 15.45 14.01-17.03 0.60 0.52,0.70

18C 499 96.52 94.55-97.92 496 95.57 93.42-97.17 0.95 -1.76,3.66 1.82 1.65-2.00 4.29 3.84-4.79 0.57 0.47,0.67

19F 512 99.03 97.76-99.69 513 98.84 97.50-99.57 0.19 -1.24,1.62 4.92 4.60-5.27 8.12 7.48-8.81 0.77 0.68,0.88

23F 505 97.68 95.98-98.80 501 96.53 94.57-97.93 1.15 -1.19,3.48 3.40 3.10-3.74 3.95 3.54-4.41 0.65 0.55,0.78

Additional

1 501 96.91 95.02-98.22 294 56.65 52.26-60.96 40.26 35.74,44.78 1.87 1.73-2.02 0.29 0.25-0.33 6.55 5.59,7.67

3 469 90.72 87.88-93.08 234 45.09 40.75-49.48 45.63 40.67,50.59 1.21 1.11-1.32 0.32 0.29-0.34 3.82 3.39,4.30

5 452 87.43 84.26-90.16 86 16.57 13.47-20.05 70.86 66.57,75.15 0.69 0.63-0.76 0.05 0.04-0.06 15.13 12.03,19.03

6A 501 96.91 95.02-98.22 403 77.65 73.82-81.16 19.26 15.37,23.14 2.85 2.57-3.16 0.57 0.49-0.65 5.04 4.22,6.03

7F 503 97.29 95.50-98.51 126 24.28 20.65-28.20 73.01 69.07,76.96 4.04 3.67-4.43 0.05 0.04-0.07 75.71 59.54,96.28

19A 514 99.42 98.31-99.88 499 96.15 94.11-97.63 3.27 1.49,5.05 2.93 2.76-3.11 1.03 0.94-1.13 2.84 2.54,3.17

N=number of subjects included in FAS for PCV13-TT (2M) and PCV7 (3M).

a. number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT (2M) group.

b. number of subjects with a determinate IgG antibody concentration to the given serotype for PCV7 (3M) group.

c. The rate difference of PCV13-TT (2M) – PCV7 (3M).

d. GMC ratio: PCV13-TT (2M) to PCV7 (3M).

e. Two-sided 95% CI is for additional serotypes, and two-sided 97.5% CI is for common serotypes.

Booster dose
Post dose 4 IgG pneumococcal immunogenicity results with comparison to the PCV7 seroresponse for the 7 common serotypes
The non-inferiority criterion at the 0.35 µg/mL level was met for all the common serotypes after the booster dose (Table 15). The lower limit of the 2-sided 97.5% CI for the rate difference of percentage of subjects with IgG antibody concentration ≥0.35 µg/mL (PCV13-TT – PCV7) was greater than -10% for all common serotypes
Post dose 4 IgG pneumococcal immunogenicity results with comparison to the PCV7 reference for the 6 additional serotypes
The superiority criterion was met for all the additional serotypes (except serotype 19A) after the booster dose. For serotype 19A, the lower limit of the 2-sided 95% CI for the rate difference of percentage of subjects with IgG antibody concentration ≥0.35 µg/mL (PCV13-TT – PCV7) was -0.20%
The geometric mean fold rises (GMFRs) of antibody concentrations from pre- to post-boost for the 7 common serotypes ranged from 3.42 (serotype 14) to 5.69 (9V) in the PCV13-TT (2 months) group. For the 6 additional serotypes, the fold rises ranged from 3.00 (serotype 1) to 5.67 (serotype 19A) in the PCV13-TT (2 months) group.

Table 15. Comparison of percentage of subjects achieving a pneumococcal IgG antibody concentration ≥ 0.35 µg/mL and GMCs in PCV13-TT (2 months) group and PCV7 (3 months) group post dose 3 of the primary series (PPS)
IgG antibody concentration ≥ 0.35 µg/mL IgG antibody GMC
PCV13-TT (2M)
PCV13-TT (2M) (N=476) PCV7 (3M) (N=487) Rate difference c PCV7 (3M) (N=487) GMC Ratio (N=475)
95% CI
Serotype na % 95% CI nb % 95% CI % GMC 95% CI GMC 95% CI Ratio d 95%/97.5%CI e
/97.5%CI
PCV7
4 473 99.37 98.17-99.87 486 99.79 98.86-99.99 -0.42 -1.36,0.51 2.07 1.92-2.22 4.27 3.98-4.59 0.48 0.43-0.54
6B 476 100.0 99.23-100.0 486 99.79 98.86-99.99 0.21 -0.25,0.67 8.60 7.93-9.32 11.70 10.78-12.70 0.73 0.64-0.84
9V 476 100.0 99.23-100.0 487 100.0 99.25-100.0 0.00 -0.90,0.92 3.83 3.56-4.12 3.87 3.62-4.14 0.99 0.88-1.11
14 476 100.0 99.23-100.0 487 100.0 99.25-100.0 0.00 -0.90,0.92 11.61 10.85-12.44 17.89 16.77-19.09 0.65 0.58-0.72
18C 474 99.58 98.49-99.95 485 99.59 98.52-99.95 -0.01 -0.94,0.92 2.04 1.90-2.18 3.90 3.59-4.23 0.52 0.46-0.59
19F 476 100.0 99.23-100.0 487 100.0 99.25-100.0 0.00 -0.90,0.92 8.76 8.16-9.39 7.84 7.33-8.38 1.12 1.00-1.25
23F 475 99.79 98.84-99.99 486 99.79 98.86-99.99 -0.00 -0.66,0.65 5.70 5.25-6.18 10.81 9.91-11.80 0.53 0.46-0.60
Additional
1 475 99.79 98.84-99.99 445 91.38 88.52-93.71 8.41 5.89,10.94 1.98 1.86-2.12 0.67 0.62-0.72 2.97 2.68-3.30
3 471 98.95 97.57-99.66 449 92.20 89.45-94.42 6.75 4.20,9.30 1.53 1.44-1.64 1.02 0.95-1.09 1.51 1.37-1.66
5 463 97.27 95.38-98.54 195 40.04 35.66-44.55 57.23 52.64,61.82 1.13 1.06-1.21 0.16 0.14-0.19 6.92 5.80-8.25
6A 474 99.58 98.49-99.95 476 97.74 95.99-98.87 1.84 0.40,3.28 4.79 4.39-5.22 3.80 3.44-4.21 1.26 1.10-1.44
7F 476 100.0 99.23-100.0 325 66.74 62.36-70.91 33.26 29.08,37.45 4.05 3.79-4.32 0.31 0.26-0.36 13.22 11.09-15.76
19A 476 100.0 99.23-100.0 486 99.79 98.86-99.99 0.21 -0.20,0.61 9.32 8.66-10.03 3.00 2.78-3.24 3.11 2.80-3.45
N=number of subjects included in PPS for PCV13-TT (2M) and PCV7 (3M).
a. number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT (2M) group.
b. number of subjects with a determinate IgG antibody concentration to the given serotype for PCV7 (3M) group.
c. The rate difference of PCV13-TT (2M) – PCV7 (3M).
d. GMC ratio: PCV13-TT (2M) to PCV7 (3M).
e. Two-sided 95% CI is for additional serotypes, and two-sided 97.5% CI is for common serotypes.

Table 16 Comparison of Subjects for the geometric mean fold rises (GMFRs) of antibody concentrations in PCV13-TT (2 months) group and PCV7 (3 months) group post dose 3 of the primary series (PPS)
PCV13-TT (2M) (N=476) PCV7 (3M) (N=487)
Serotype P
GMFRs 95%CI GMFRs 95%CI
4 4.17 3.91-4.44 5.15 4.78-5.55 <0.0001
6B 4.26 3.98-4.55 8.66 8.05-9.31 <0.0001
9V 5.69 5.30-6.10 6.06 5.63-6.52 0.2176
Common 14 3.42 3.21-3.65 3.70 3.46-3.96 0.0959
18C 4.93 4.53-5.37 7.63 6.96-8.36 <0.0001
19F 4.20 3.92-4.51 4.40 4.08-4.76 0.3816
23F 4.35 4.08-4.64 9.71 8.99-10.49 <0.0001
1 3.00 2.83-3.17 1.83 1.70-1.99 <0.0001
3 3.91 3.70-4.14 5.19 4.79-5.62 <0.0001
5 3.50 3.26-3.75 0.95 0.82-1.09 <0.0001

Additional

6A 4.00 3.71-4.30 5.66 5.16-6.20 <0.0001

7F 3.17 3.00-3.35 1.02 0.88-1.19 <0.0001

19A 5.67 5.25-6.11 2.61 2.42-2.82 <0.0001

N=number of subjects included in PPS for PCV13-TT (2M) and PCV7 (3M).

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2.7.3.2.2.1.2.2. Pneumococcal Serotype Antibody Concentrations in the Subjects in PCV13-TT (2 months) group and PCV7 (3 months) group

Please refer to the section 2.7.3.2.2.1.3.2. in this clinical summary.

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2.7.3.2.2.1.2.3. Comparison of percentage of subjects achieving IgG antibody concentration ≥ 1.0 $\mu\text{g/mL}$ with comparison to the PCV7 seroresponse

Booster Dose

Data for the 7 common serotypes

With the exception of serotypes 4 and 18C, the proportion of responders ≥ 1.0 $\mu\text{g/mL}$ in the PCV13-TT (2M) group were all higher than 94% (Table 17). For serotype 4 and 18C, the proportion of responders ≥ 1.0 $\mu\text{g/mL}$ was 81.51% and 80.67%. The rate difference of percentage of subjects with IgG antibody concentration ≥ 1.0 $\mu\text{g/mL}$ (PCV13-TT – PCV7) was all greater than -10% except the serotypes 4 (-15%) and 18C (-13.68%).

Data for the 6 additional serotypes

The proportion of responders ≥ 1.0 $\mu\text{g/mL}$ was higher than 95% except serotypes 1, 3, and 5A. The lower limit of the 2-sided 95% CI for the rate difference of percentage of subjects with IgG antibody concentration ≥ 1.0 $\mu\text{g/mL}$ (PCV13-TT – PCV7) was greater than 0 for all additional serotypes.

Table 17 Comparison of percentage of subjects achieving a pneumococcal IgG antibody concentration ≥ 1.0 $\mu\text{g/mL}$ in PCV13-TT (2 months) group and PCV7 (3 months) group post booster dose (PPS)

IgG antibody concentration ≥ 1.0 $\mu\text{g/mL}$

PCV13-TT (2M) (N=476) PCV7 (3M) (N=487) Rate difference c

a b

Serotype n % 95% CI n % 95% CI % 95% CI

4 388 81.51 77.73-84.90 470 96.51 94.47-97.95 -15.00 -18.85,-11.15

6B 472 99.16 97.86-99.77 478 98.15 96.52-99.15 1.01 -0.44,2.46

9V 451 94.75 92.34-96.57 471 96.71 94.72-98.11 -1.97 -4.52,0.59

14 476 100.0 99.23-100.0 487 100.0 99.25-100.0 0.00 -0.75,0.77

18C 384 80.67 76.83-84.12 460 94.46 92.04-96.31 -13.78 -17.87,-9.70

19F 476 100.0 99.23-100.0 484 99.38 98.21-99.87 0.62 -0.08,1.31

23F 465 97.69 95.90-98.84 484 99.38 98.21-99.87 -1.69 -3.21,-0.18

1 387 81.30 77.50-84.71 125 25.67 21.84-29.79 55.64 50.41,60.86

3 338 71.01 66.71-75.05 243 49.90 45.37-54.43 21.11 15.08,27.14

5 256 53.78 49.19-58.33 13 2.67 1.43- 4.52 51.11 46.41,55.81

6A 454 95.38 93.09-97.08 431 88.50 85.33-91.20 6.88 3.47,10.28

7F 472 99.16 97.86-99.77 69 14.17 11.19-17.59 84.99 81.79,88.20

19A 476 100.0 99.23-100.0 459 94.25 91.80-96.15 5.75 3.68,7.82

N=number of subjects included in PPS for PCV13-TT (2M) and PCV7 (3M).

a. number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT (2M) group.

b. number of subjects with a determinate IgG antibody concentration to the given serotype for PCV7 (3M) group.

c. The rate difference of PCV13-TT (2M) – PCV7 (3M).

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2.7.3.2.2.1.2.4. Comparison of percentage of subjects achieving an OPA antibody $\geq 1:8$ and the IgG antibody geometric mean titers (GMTs)

Primary series

Data for 7 common serotypes

For the 7 common serotypes, the percentage of subjects achieving an OPA antibody titer $\geq 1:8$ was all higher than 92% in PCV13-TT (2 months) group. With the exception of serotypes 6B, the lower limit of the 2-sided 95% CI of rate difference of the percentage of subjects achieving an OPA antibody titer $\geq 1:8$ was greater than -10% for the common serotypes (Table 18). The lower limit of the 2-sided 95% CI of GMT ratio was greater than 0.5 for serotypes 9V and 19F. For other serotypes including 4, 6B, 14 18C and 23F, the lower limit of the 2-sided 95% CI of GMT ratio was $\blacksquare$ 0.5.

Data for 6 additional serotypes

For the 6 additional serotypes, the percentage of subjects achieving an OPA antibody titer $\geq 1:8$ was 100% except serotypes 1 (91.75%) and 6 (97.94%) in PCV13-TT (2 months) group. The lower limit of the 2-sided 95% CI of rate difference of the percentage of subjects achieving an OPA antibody titer $\geq 1:8$ was greater than 0 for all the additional serotypes. The lower limit of the 2-sided 95% CI of GMR (PCV13-TT relative to PCV7) for the additional serotypes was all greater than 1.

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Table 18. Comparison of percentage of subjects achieving a pneumococcal IgG antibody titer $\geq 1:8$ and GMTs in PCV13-TT (2 months) group and PCV7 (3 months)

group post-dose 3 of the primary series (PPS)

OPA antibody titer $\geq 1:8$ OPA GMT

PCV13-TT (2M) (N=97) PCV7 (3M) (N=96) Rate difference c PCV13-TT (2M) (N=97) PCV7 (3M) (N=96) GMT Ratio

Serotype na % 95% CI nb % 95% CI % 95% CI /97.5%CI GMT 95% CI GMT 95% CI Ratio d 95%/97.5%CI e

Common
4 96 98.97 94.39-99.97 96 100.0 96.23-100.0 -1.03 -3.04,0.98 1214.42 940.3-1568.5 2291.32 1947.8-2695.4
0.53 0.39-0.72
6B 90 92.78 85.70-97.05 95 98.96 94.33-99.97 -6.17 -11.71, -0.64 811.17 542.0-1214.1 2928.36
2233.9-3838.7 0.28 0.17-0.45
9V 97 100.0 96.27-100.0 96 100.0 96.23-100.0 0.00 -3.89,3.88 2192.74 1728.3-2781.9 1974.62 1612.0-2418.9
1.11 0.81-1.52
14 97 100.0 96.27-100.0 96 100.0 96.23-100.0 0.00 -3.89,3.88 3571.78 2833.6-4502.3 5594.48 4188.4-7472.5
0.64 0.44-0.92
18C 97 100.0 96.27-100.0 95 98.96 94.33-99.97 1.04 -0.99,3.07 518.67 417.13-644.92 3129.49 2353.5-4161.3
0.17 0.12-0.24
19F 96 98.97 94.39-99.97 94 97.92 92.68-99.75 1.05 -2.44,4.55 769.17 617.10-958.72 540.56 423.02-690.76
1.42 1.03-1.97
23F 93 95.88 89.78-98.87 95 98.96 94.33-99.97 -3.08 -7.53,1.37 1072.95 738.0-1560.0 5939.64 4465.8-7899.8
0.18 0.11-0.29
Additional
1 89 91.75 84.39-96.37 0 0.00 0.00-3.77 91.75 86.28,97.23 92.51 66.48-128.75 2.00 2.00-2.00 46.26
33.25-64.35
3 97 100.0 96.27-100.0 15 15.63 9.02-24.46 84.38 77.11,91.64 575.94 444.43-746.38 2.80 2.35-3.33 205.77
150.78-280.80
5 97 100.0 96.27-100.0 4 4.17 1.15-10.33 95.83 91.84,99.83 411.73 317.12-534.56 2.24 1.99-2.51 184.19
138.57-244.82
6A 95 97.94 92.75-99.75 77 80.21 70.83-87.64 17.73 9.27,26.19 1202.17 895.0-1614.8 275.54 159.69-475.44
4.36 2.36-8.07
7F 97 100.0 96.27-100.0 36 37.50 27.82-47.97 62.50 52.82,72.18 2414.63 1872.4-3113.9 12.76 7.68-21.19
189.24 107.89-331.94
19A 97 100.0 96.27-100.0 43 44.79 34.63-55.29 55.21 45.26,65.16 659.70 532.25-817.66 11.50 7.55-17.51
57.37 35.93-91.60
N=number of subjects included in PPS for PCV13-TT (2M) and PCV7 (3M).
a. number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT (2M) group.
b. number of subjects with a determinate IgG antibody concentration to the given serotype for PCV7 (3M) group.
c. The rate difference of PCV13-TT (2M) – PCV7 (3M).
d. GMC ratio: PCV13-TT (2M) to PCV7 (3M).
e. Two-sided 95% CI is for additional serotypes, and two-sided 97.5% CI is for common serotypes.

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Table 19. Comparison of percentage of subjects achieving a pneumococcal IgG antibody titer $\geq 1:8$ and GMTs in PCV13-TT (2 months) group and PCV7 (3 months) group post-dose 3 of the primary series (FAS)
OPA antibody titer $\geq 1:8$ OPA GMT
PCV13-TT (N=98) PCV7 (N=97) Rate difference c PCV13-TT (N=93) PCV7 (N=96) GMT Ratio
Serotype na % 95% CI nb % 95% CI % 95% CI /97.5%CI GMT 95% CI GMT 95% CI Ratio d 95%/97.5%CI e
Common
4 97 98.98 94.45-99.97 97 100.0 96.27-100.0 -1.02 -3.01,0.97 1231.00 954.3-1588.0 2307.23 1963.5-2711.1
0.53 0.40-0.72
6B 91 92.86 85.84-97.08 96 98.97 94.39-99.97 -6.11 -11.59,-0.63 835.16 558.0-1250.0 2940.48 2249.2-3844.1
0.28 0.18-0.46
9V 98 100.0 96.31-100.0 97 100.0 96.27-100.0 0.00 -3.88,3.81 2245.67 1766.1-2855.5 1993.08 1629.1-2438.4
1.13 0.82-1.54
14 98 100.0 96.31-100.0 97 100.0 96.27-100.0 0.00 -3.88,3.81 3651.57 2891.8-4611.0 5604.71 4208.8-7463.6
0.65 0.45-0.94
18C 98 100.0 96.31-100.0 96 98.97 94.39-99.97 1.03 -0.98,3.04 522.68 421.08-648.79 3142.30 2370.0-4166.3
0.17 0.12-0.24
19F 97 98.98 94.45-99.97 95 97.94 92.75-99.75 1.04 -2.42,4.50 786.74 629.77-982.85 542.42 425.53-691.42
1.45 1.05-2.01
23F 94 95.92 89.88-98.88 96 98.97 94.39-99.97 -3.05 -7.45,1.35 1098.20 756.1-1595.1 5943.19 4481.9-7880.9
0.18 0.12-0.29
Additional
1 90 91.84 84.55-96.41 0 0.00 0.00-3.73 91.84 86.42,97.26 94.23 67.80-130.94 2.00 2.00-2.00 47.11
33.91-65.45
3 98 100.0 96.31-100.0 15 15.46 8.92-24.22 84.54 77.34,91.73 591.13 455.03-767.94 2.79 2.35-3.31 211.92
155.12-289.54
5 98 100.0 96.31-100.0 4 4.12 1.13-10.22 95.88 91.92,99.83 412.18 318.33-533.71 2.23 1.99-2.50 184.61
139.30-244.66
6A 96 97.96 92.82-99.75 78 80.41 71.11-87.78 17.55 9.17,25.93 1225.83 913.1-1645.7 282.21 164.15-485.19
4.34 2.36-8.00
7F 98 100.0 96.31-100.0 37 38.14 28.47-48.57 61.86 52.19,71.52 2463.59 1909.4-3178.6 13.20 7.96-21.90
186.66 106.47-327.25
19A 98 100.0 96.31-100.0 44 45.36 35.22-55.79 54.64 44.73,64.55 674.88 543.13-838.59 12.00 7.84-18.34
56.26 35.07-90.25
N=number of subjects included in PPS for PCV13-TT and PCV7.
a. number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT group.
b. number of subjects with a determinate IgG antibody concentration to the given serotype for PCV7 group.
c. The rate difference of PCV13-TT – PCV7 reference.
d. GMC ratio: PCV13-TT to PCV7 reference.
e. Two-sided 95% CI is for additional serotypes, and two-sided 97.5% CI is for common serotypes.

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Booster dose
Data for 7 common serotypes
For the 7 common serotypes, the percentage of subjects achieving an OPA antibody titer $\geq 1:8$ was all 100% in PCV13-TT (2 months) group. The lower limit of the 2-sided 95% CI of rate difference of the percentage of subjects achieving an OPA antibody titer $\geq 1:8$ was greater than -10% for all

the common serotypes (Table 20.).The lower limit of the 2-sided 95% CI of GMT ratio was greater than 0.5 for serotypes 14 and 19F. For other serotypes including 4, 6B, 9V 18C and 23F, the lower limit of the 2-sided 95% CI of GMT ratio was ■ 0.5. The lower limit of the 2-sided 95% CI of GMT ratio for serotypes 4 and 9V was 0.40 and 0.49, respectively.

Data for 6 additional serotypes

For the 6 additional serotypes, the percentage of subjects achieving an OPA antibody titer ≥1:8 was all higher than 95% in PCV13-TT (2 months) group. For serotypes 3, 5 and 19A, the percentage of subjects achieving an OPA antibody titer ≥1:8 was all 100%.

With the exception of serotype 6A (-0.91%), the lower limit of the 2-sided 95% CI of rate difference of the percentage of subjects achieving an OPA antibody titer ≥1:8 was greater than 0 for all the additional serotypes. With the exception of serotype 6A (0.90%), the lower limit of the 2-sided 95% CI of GMR (PCV13-TT relative to PCV7) for the additional serotypes was all greater than 1.

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Table 20 Comparison of percentage of subjects achieving a pneumococcal IgG antibody titer ≥ 1:8 and GMTs in PCV13-TT (2 months) group and PCV7 (3 months)																	
group post booster dose (PPS)																	
OPA antibody titer ≥ 1:8 OPA GMT																	
PCV13-TT (2M) (N=92) PCV7 (3M) (N=97) Rate difference c PCV13-TT (2M) (N=92) PCV7 (3M) (N=97) GMT Ratio																	
Serotype	na	%	95% CI	nb	%	95% CI	%	95% CI	/97.5%CI	GMT	95% CI	GMT	95% CI	Ratio	d	95%/97.5%CI	e
Common																	
4	92	100.0	96.07-100.0	97	100.0	96.27-100.0	0.00	-3.91,4.07	3440.10	2690.0-4399.3	6208.15	4950.8-7784.8	0.55	0.40-0.77			
6B	92	100.0	96.07-100.0	95	97.94	92.75-99.75	2.06	-0.77,4.89	1948.21	1460.3-2599.2	7564.42	5345-10705	0.26	0.16-0.40			
9V	92	100.0	96.07-100.0	97	100.0	96.27-100.0	0.00	-3.91,4.07	3719.77	2889.7-4788.3	5484.79	4453.1-6755.5	0.68	0.49-0.94			
14	92	100.0	96.07-100.0	97	100.0	96.27-100.0	0.00	-3.91,4.07	6173.16	4922.7-7741.3	6221.93	4855.7-7972.5	0.99	0.71-1.39			
18C	92	100.0	96.07-100.0	96	98.97	94.39-99.97	1.03	-0.98,3.04	1117.88	899.5-1389.3	5507.66	4094.6-7408.5	0.20	0.14-0.29			
19F	92	100.0	96.07-100.0	94	96.91	91.23-99.36	3.09	-0.35,6.54	1966.07	1496.8-2582.5	815.68	600.1-1108.8	2.41	1.60-3.63			
23F	92	100.0	96.07-100.0	96	98.97	94.39-99.97	1.03	-0.98,3.04	2529.55	1959.0-3266.3	16131.84	11742-22163	0.16	0.10-0.24			
Additional																	
1	88	95.65	89.24-98.80	4	4.12	1.13-10.22	91.53	85.78,97.27	158.59	115.59-217.59	2.28	2.00-2.61	69.54	49.76-97.18			
3	92	100.0	96.07-100.0	36	37.11	27.52-47.52	62.89	53.27,72.50	586.01	440.25-780.04	5.03	3.77-6.72	116.40	77.72-174.32			
5	92	100.0	96.07-100.0	5	5.15	1.69-11.62	94.85	90.45,99.25	1027.83	796.4-1326.6	2.61	2.07-3.29	394.33	279.95-555.44			
6A	90	97.83	92.37-99.74	90	92.78	85.70-97.05	5.04	-0.91,10.99	2546.12	1780.8-3640.3	1562.59	972.9-2509.6	1.63	0.90-2.95			
7F	91	98.91	94.09-99.97	51	52.58	42.18-62.81	46.34	36.18,56.50	2616.75	1938.6-3532.1	41.47	22.27-77.21	63.10	31.41-126.77			
19A	92	100.0	96.07-100.0	77	79.38	69.97-86.93	20.62	12.57,28.67	158.59	115.59-217.59	72.99	46.46-114.65	69.54	49.76-97.18			

N=number of subjects included in PPS for PCV13-TT (2M) and PCV7 (3M).

a. number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT (2M) group.

b. number of subjects with a determinate IgG antibody concentration to the given serotype for PCV7 (3M) group.

c. The rate difference of PCV13-TT (2M) – PCV7 (3M).

d. GMT ratio: PCV13-TT (2M) to PCV7 (3M).

e. Two-sided 95% CI is for additional serotypes, and two-sided 97.5% CI is for common serotypes.

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Table 21. Comparison of percentage of subjects achieving a pneumococcal IgG antibody titer ≥ 1:8 and GMTs in PCV13-TT (2 months) group and PCV7 (3 months)															
group post booster dose (FAS)															
OPA antibody titer ≥ 1:8 OPA GMT															
PCV13-TT (2M) (N=96) PCV7 (3M) (N=97) Rate difference c PCV13-TT (2M) (N=96) PCV7 (3M) (N=97) GMT Ratio															
Serotype na % 95% CI nb % 95% CI % 95% CI GMT 95% CI GMT 95% CI Ratio d 95%CI e															
Common															
4	96	100.0	96.23-100.0	97	100.0	96.27-100.0	0.00	-3.88,3.89	3058.68	2339.9-3998.2	6208.15	4950.8-7784.8			
0.49	0.35-0.70														
6B	95	98.96	94.33-99.97	95	97.94	92.75-99.75	1.02	-2.46,4.50	1734.63	1266.1-2376.5	7564.42	5345-10705			
0.23	0.14-0.37														
9V	96	100.0	96.23-100.0	97	100.0	96.27-100.0	0.00	-3.88,3.89	3400.35	2623.6-4407.0	5484.79	4453.1-6755.5			
0.62	0.45-0.86														
14	95	98.96	94.33-99.97	97	100.0	96.27-100.0	0.00	-3.88,3.89	5364.66	4051.2-7104.0	6221.93	4855.7-7972.5			
0.86	0.59-1.25														
18C	95	98.96	94.33-99.97	96	98.97	94.39-99.97	1.03	-0.98,3.04	972.48	748.8-1263.0	5507.66	4094.6-7408.5			
0.18	0.12-0.26														
19F	96	100.0	96.23-100.0	94	96.91	91.23-99.36	3.09	-0.35,6.54	1758.11	1321.1-2339.8	815.68	600.1-1108.8			
2.16	1.42-3.27														
23F	95	98.96	94.33-99.97	96	98.97	94.39-99.97	-0.01	-2.87,2.85	2221.21	1658.4-2975.0	16131.84	11742-22163			
0.14	0.09-0.21														
Additional															
1	91	94.79	88.26-98.29	4	4.12	1.13-10.22	90.67	84.72,96.62	144.44	104.59-199.46	2.28	2.00-2.61	63.33		
44.83-89.48															
3	96	100.0	96.23-100.0	36	37.11	27.52-47.52	62.89	53.27,72.50	557.39	419.95-739.82	5.03	3.77-6.72	110.71		
74.09-165.43															
5	95	98.96	94.33-99.97	5	5.15	1.69-11.62	93.80	88.96,98.65	917.73	686.2-1227.4	2.61	2.07-3.29	352.09		

243.27-509.57
 6A 93 96.88 91.14-99.35 90 92.78 85.70-97.05 4.09 -2.12,10.31 2221.94 1520.6-3246.8 1562.59 972.9-2509.6
 1.42 0.78-2.60
 7F 94 97.92 92.68-99.75 51 52.58 42.18-62.81 45.34 35.00,55.68 2390.26 1728.2-3305.9 41.47 22.27-77.21
 57.64 28.66-115.94
 19A 95 98.96 94.33-99.97 77 79.38 69.97-86.93 19.58 11.27,27.88 2126.65 1594.0-2837.4 72.99 46.46-114.65
 29.14 17.09-49.68
 N=number of subjects included in FAS for PCV13-TT (2M) and PCV7 (3M).
 f. number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT (2M) group.
 g. number of subjects with a determinate IgG antibody concentration to the given serotype for PCV7 (3M) group.
 h. The rate difference of PCV13-TT (2M) – PCV7 (3M) reference.
 i. GMT ratio: PCV13-TT (2M) to PCV7 (3M) reference.
 j. Two-sided 95% CI is for additional serotypes, and two-sided 97.5% CI is for common serotypes.

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2.7.3.2.2.1.3. Pneumococcal immunogenicity results for subjects in PCV13-TT (2 months) group and PCV13-TT (3 months) group
 2.7.3.2.2.1.3.1. Comparison of percentage of subjects achieving IgG antibody concentration $\geq$ 0.35 μ g/mL and the IgG antibody geometric mean concentrations (GMCs)
 Post dose 3 IgG pneumococcal immunogenicity results for the 13 serotypes in subjects in PCV13-TT (2M) and PCV13-TT (3M) groups
 After primary series, except for serotype 5 with the lower limit of 2-sided 97.5% CI for rate difference (PCV13-TT [2 months] – PCV13-TT [3 months]) being lower than -10% (-10.84%), and serotypes 4 and 5 with the lower limits of 2-sided 97.5% CI for GMR (PCV13-TT [2 months] relative to PCV13-TT [3 months]) being lower than 0.5 (0.48 and 0.37, respectively), the noninferiority by either criterion of IgG antibody concentration or GMR was met for 12 out of the 13 serotypes (Table 22.)

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Table 22. Comparison of percentage of subjects achieving a pneumococcal IgG antibody concentration $\geq$ 0.35 μ g/mL and GMCs in PCV13-TT (2 months) group and PCV13-TT (3 months) group post dose 3 of the primary series (PPS)
 IgG antibody concentration $\geq$ 0.35 μ g/mL IgG antibody GMC
 PCV13-TT (2M) (N=482) PCV13-TT (3M) (N=471) Rate difference c PCV13-TT(2M) (N=471) PCV13-TT (3M) (N=487)
 GMC Ratio

Serotype	na	%	95% CI	nb	%	95% CI	%	97.5%CI	GMC	95% CI	GMC	95% CI	Ratio	d	97.5%CI
PCV7	4 482	100.0	99.24-100.0	471	100.0	99.22-100.0	0.00	-0.93,0.91	2.43	2.28-2.60	4.51	4.20-4.84	0.54	0.48,0.60	
6B	471	97.72	95.95-98.86	426	90.45	87.42-92.95	7.27	3.87,10.67	3.72	3.37-4.10	1.25	1.11-1.42	2.97	2.47,3.56	
9V	481	99.79	98.85-99.99	467	99.15	97.84-99.77	0.64	-0.41,1.70	3.66	3.40-3.94	3.72	3.45-4.02	0.98	0.87,1.11	
14	481	99.79	98.85-99.99	471	100.0	99.22-100.0	-0.21	-0.67,0.26	10.25	9.46-11.11	10.38	9.57-11.27	0.99	0.87,1.13	
18C	477	98.96	97.60-99.66	463	98.30	96.68-99.26	0.66	-1.03,2.35	2.04	1.87-2.23	2.91	2.68-3.15	0.70	0.61,0.81	
19F	482	100.0	99.24-100.0	471	100.0	99.22-100.0	0.00	-0.93,0.91	5.46	5.14-5.80	6.95	6.49-7.45	0.79	0.71,0.87	
23F	481	99.79	98.85-99.99	458	97.24	95.33-98.52	2.55	0.80,4.31	3.86	3.53-4.21	2.93	2.65-3.23	1.32	1.13,1.53	

1	482	100.0	99.24-100.0	469	99.58	98.47-99.95	0.42	-0.25,1.10	2.09	1.95-2.23	2.15	2.01-2.30	0.97	0.87,1.09	
3	463	96.06	93.91-97.61	462	98.09	96.40-99.12	-2.03	-4.47,0.41	1.41	1.32-1.50	1.67	1.56-1.79	0.84	0.76,0.94	
5	438	90.87	87.94-93.29	464	98.51	96.96-99.40	-7.64	-10.84, -4.45	0.79	0.73-0.85	1.88	1.75-2.02	0.42	0.37,0.47	
6A	478	99.17	97.89-99.77	456	96.82	94.80-98.21	2.35	0.32,4.39	3.28	3.01-3.57	1.96	1.77-2.16	1.67	1.44,1.95	
7F	480	99.59	98.51-99.95	469	99.58	98.47-99.95	0.01	-0.93,0.95	4.76	4.42-5.13	5.47	5.06-5.92	0.87	0.77,0.98	
19A	482	100.0	99.24-100.0	471	100.0	99.22-100.0	0.00	-0.93,0.91	3.12	2.95-3.30	3.85	3.62-4.09	0.81	0.74,0.89	

N=number of subjects included in PPS for PCV13-TT (2M) and PCV13-TT (3M)
 a. number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT (2M) group.
 b. number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT (3M) group.
 c. The rate difference of PCV13-TT (2M) – PCV13-TT (3M).
 d. GMC ratio: PCV13-TT (2M) to PCV13-TT (3M).

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Table 23. Comparison of percentage of subjects achieving a pneumococcal IgG antibody concentration $\geq$ 0.35 μ g/mL and GMCs in PCV13-TT (2 months) group and PCV13-TT (3 months) group post dose 3 of the primary series (FAS)
 IgG antibody concentration $\geq$ 0.35 μ g/mL IgG antibody GMC
 PCV13-TT (2M) (N=517) PCV13-TT (3M) (N=519) Rate difference c PCV13-TT(2M) (N=517) PCV13-TT (3M) (N=519)
 GMC Ratio
 Serotype na % 95% CI nb % 95% CI na 97.5% CI na 95% CI nb % 95% CI na 97.5%CI e
 PCV7

4 492 95.16 92.94-96.85 497 95.76 93.65-97.32 -0.60 -3.50,2.30 1.94 1.74-2.17 3.46 3.06-3.92 0.56
0.46,0.68
6B 499 96.52 94.55-97.92 463 89.21 86.22-91.75 7.31 3.76,10.86 3.38 3.05-3.75 1.23 1.09-1.39 2.75
2.29,3.30
9V 503 97.29 95.50-98.51 502 96.72 94.81-98.08 0.57 -1.80,2.94 3.11 2.82-3.43 3.24 2.95-3.56 0.96
0.82,1.12
14 511 98.84 97.49-99.57 516 99.42 98.32-99.88 -0.58 -1.88,0.71 8.73 7.87-9.68 9.32 8.49-10.22 0.94
0.80,1.10
18C 499 96.52 94.55-97.92 493 94.99 92.75-96.70 1.53 -1.28,4.33 1.82 1.65-2.00 2.42 2.18-2.70 0.75
0.63,0.89
19F 512 99.03 97.76-99.69 516 99.42 98.32-99.88 -0.39 -1.61,0.83 4.92 4.60-5.27 6.26 5.80-6.76 0.79
0.70,0.88
23F 505 97.68 95.98-98.80 489 94.22 91.85-96.07 3.46 0.73,6.19 3.40 3.10-3.74 2.59 2.32-2.90 1.31
1.11,1.55

1 501 96.91 95.02-98.22 498 95.95 93.88-97.48 0.95 -1.63,3.53 1.87 1.73-2.02 1.79 1.60-1.99 1.04
0.90,1.22
3 469 90.72 87.88-93.08 489 94.22 91.85-96.07 -3.50 -7.17,0.16 1.21 1.11-1.32 1.45 1.33-1.58 0.83
0.73,0.96
5 452 87.43 84.26-90.16 489 94.22 91.85-96.07 -6.79 -10.79,-2.80 0.69 0.63-0.76 1.50 1.34-1.68 0.46
0.39,0.55
6A 501 96.91 95.02-98.22 488 94.03 91.63-95.91 2.88 -0.01,5.77 2.85 2.57-3.16 1.80 1.62-2.00 1.59
1.34,1.88
7F 503 97.29 95.50-98.51 504 97.11 95.28-98.37 0.18 -2.11,2.48 4.04 3.67-4.43 4.66 4.19-5.19 0.87
0.74,1.02
19A 514 99.42 98.31-99.88 517 99.61 98.61-99.95 -0.19 -1.16,0.77 2.93 2.76-3.11 3.57 3.35-3.81 0.82
0.74,0.91

N=number of subjects included in FAS for PCV13-TT (2M) and PCV13-TT (3M)

- number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT (2M) group.
- number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT (3M) group.
- The rate difference of PCV13-TT (2M) – PCV13-TT (3M).
- GMC ratio: PCV13-TT (2M) to PCV13-TT (3M).

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Booster Dose

Post dose 4 IgG pneumococcal immunogenicity results for the 13 serotypes in subjects in PCV13-TT (2 months) and PCV13-TT (3 months) groups

After booster dose, all the 13 serotypes met the noninferiority criterion with respect to rate difference of proportion of responders (PCV13-TT [2 months] – PCV13-TT [3 months]) or GMR (PCV13-TT [2 months] relative to PCV13-TT [3 months]) for subjects in PCV13-TT groups 2 months of age and 3 months of age (Table 24).

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Table 24 Comparison of percentage of subjects achieving a pneumococcal IgG antibody concentration ≥ 0.35 $\mu\text{g/mL}$ and GMCs in PCV13-TT (2 months) group and PCV13-TT (3 months) group post booster dose (PPS)

IgG antibody concentration ≥ 0.35 $\mu\text{g/mL}$ IgG antibody GMC
PCV13-TT (2M) (N=476) PCV13-TT (3M) (N=475) Rate difference c PCV13-TT(2M) (N=476) PCV13-TT (3M) (N=475)
GMC Ratio

Serotype na % 95% CI nb % 95% CI % 97.5%CI GMC 95% CI GMC 95% CI Ratio d 97.5%CI

4 473 99.37 98.17-99.87 474 99.79 98.83-99.99 -0.42 -1.36,0.52 2.07 1.92-2.22 3.19 2.96-3.44 0.65
0.58,0.73
6B 476 100.0 99.23-100.0 475 100.0 99.23-100.0 0.00 -0.92,0.92 8.6 7.93-9.32 10.01 9.30-10.79 0.86
0.76,0.97
9V 476 100.0 99.23-100.0 475 100.0 99.23-100.0 0.00 -0.92,0.92 3.83 3.56-4.12 4.22 3.90-4.56 0.91
0.80,1.03
14 476 100.0 99.23-100.0 474 99.79 98.83-99.99 0.21 -0.26,0.68 11.61 10.85-12.44 13.97 12.94-15.09 0.83
0.74,0.93
18C 474 99.58 98.49-99.95 474 99.79 98.83-99.99 -0.21 -1.02,0.61 2.04 1.90-2.18 2.44 2.27-2.63 0.83
0.74,0.94
19F 476 100.0 99.23-100.0 475 100.0 99.23-100.0 0.00 -0.92,0.92 8.76 8.16-9.39 12.18 11.25-13.20 0.72
0.64,0.81
23F 475 99.79 98.84-99.99 475 100.0 99.23-100.0 -0.21 -0.68,0.26 5.7 5.25-6.18 6.9 6.39-7.45 0.83
0.73,0.94

1 475 99.79 98.84-99.99 474 99.79 98.83-99.99 0.00 -0.67,0.67 1.98 1.86-2.12 2.79 2.61-2.99 0.71
0.64,0.79
3 471 98.95 97.57-99.66 471 99.16 97.86-99.77 -0.21 -1.62,1.20 1.53 1.44-1.64 1.89 1.77-2.01 0.81
0.73,0.90
5 463 97.27 95.38-98.54 465 97.89 96.16-98.99 -0.63 -2.86,1.61 1.13 1.06-1.21 1.58 1.47-1.70 0.72
0.64,0.80
6A 474 99.58 98.49-99.95 473 99.58 98.49-99.95 0.00 -0.94,0.94 4.79 4.39-5.22 5.82 5.32-6.37 0.82
0.71,0.95
7F 476 100.0 99.23-100.0 475 100.0 99.23-100.0 0.00 -0.92,0.92 4.05 3.79-4.32 4.59 4.29-4.91 0.88
0.79,0.98
19A 476 100.0 99.23-100.0 475 100.0 99.23-100.0 0.00 -0.92,0.92 9.32 8.66-10.03 13.34 12.37-14.39 0.70
0.62,0.79

N=number of subjects included in PPS for PCV13-TT (2M) and PCV13-TT (3M)

- number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT (2M) group.
- number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT (3M) group.
- The rate difference of PCV13-TT (2M) – PCV13-TT (3M).

d. GMC ratio: PCV13-TT (2M) to PCV13-TT (3M).

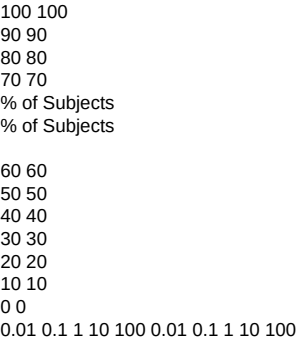
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2.7.3.2.2.1.3.2. Pneumococcal Serotype Antibody Concentrations in the Subjects in PCV13-TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group
(1) Primary Series
The results of pneumococcal serotype antibody concentrations for subjects in PCV13-TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group post primary series were shown in Table 25 to Table 37. The reverse cumulative distribution curves were shown in Figure 1 to 13.

Table 25. Distribution of the pneumococcal serotype 4 IgG antibody concentrations for subjects in PCV13-TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group post primary series

FAS N (%)	PPS N (%)	PCV13-TT (2M) (N=519)	PCV13-TT (3M) (N=519)	PCV13-TT (2M) (N=487)	PCV13-TT (3M) (N=487)	PCV7 (3M) (N=471)
≥0.01	506	97.87	506	97.50	507	97.69
≥0.1	505	97.68	506	97.50	507	97.69
≥1	429	82.98	482	92.87	496	95.57
≥10	15	2.90	77	14.84	194	37.38
≥100	0	0.00	0	0.00	0	0.00

N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT group (2M), PCV13-TT group (3M) and PCV7 group (3M).
Serotype 4, After Primary Series (FAS) Serotype 4, After Primary Series (PPS)



Antibody Concentration (µg/mL)

PCV13-TT (2M) PCV13-TT (3M) PCV7 (3M) PCV13-TT (2M) PCV13-TT (3M) PCV7 (3M)

Figure 1 Reverse cumulative distribution curves, pneumococcal serotype 4, IgG antibody concentrations for subjects in PCV13-TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group - post primary series.

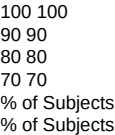
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Table 26 Distribution of the pneumococcal serotype 6B IgG antibody concentrations for subjects in PCV13-TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group post primary series

FAS N (%)	PPS N (%)	PCV13-TT (2M) (N=519)	PCV13-TT (3M) (N=519)	PCV13-TT (2M) (N=487)	PCV13-TT (3M) (N=487)	PCV7 (3M) (N=471)
≥0.01	514	99.42	508	97.88	517	99.61
≥0.1	514	99.42	508	97.88	517	99.61
≥1	446	86.27	309	59.54	464	89.40
≥10	77	14.89	18	3.47	65	12.52
≥100	0	0.00	0	0.00	0	0.00

N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT group (2M), PCV13-TT group (3M) and PCV7 group (3M).
Serotype 6B, After Primary Series (FAS) Serotype 6B, After Primary Series (PPS)

Serotype 6B, After Primary Series (FAS) Serotype 6B, After Primary Series (PPS)



60 60
50 50
40 40
30 30
20 20
10 10
0 0
0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration ■µg/mL
Antibody Concentration (µg/mL)

PCV13-TT (2M) PCV13-TT (3M) PCV7 (3M) PCV13-TT (2M) PCV13-TT (3M) PCV7 (3M)

Figure 2 Reverse cumulative distribution curves, pneumococcal serotype 6B, IgG antibody concentrations for subjects in PCV13-TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group - post primary series.

Table 27 Distribution of the pneumococcal serotype 9V IgG antibody concentrations for subjects in PCV13-TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group post primary series

FAS N (%)	PPS N (%)	PCV13-TT (2M)	PCV13-TT (3M)	PCV7 (3M)
(N=519)	(N=471)	(N=487)	(N=517)	(N=519)
n %	n %	n %	n %	n %
≥0.01	512 99.03	515 99.23	512 98.65	482 100.0
≥0.1	512 99.03	515 99.23	512 98.65	482 100.0
≥1	462 89.36	477 91.91	488 94.03	451 93.57
≥10	48 9.28	50 9.63	62 11.95	48 9.96
≥100	0 0.00	0 0.00	0 0.00	0 0.00

N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT group (2M), PCV13-TT group (3M) and PCV7 group (3M).

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Serotype 9V, After Primary Series (FAS) Serotype 9V, After Primary Series (PPS)

100 100
90 90
80 80
70 70

% of Subjects
% of Subjects

60 60
50 50
40 40
30 30
20 20
10 10
0 0
0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration ■µg/mL
Antibody Concentration (µg/mL)

PCV13-TT (2M) PCV13-TT (3M) PCV7 (3M) PCV13-TT (2M) PCV13-TT (3M) PCV7 (3M)

Figure 3 Reverse cumulative distribution curves, pneumococcal serotype 9V, IgG antibody concentrations for subjects in PCV13-TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group - post primary series.

Table 28 Distribution of the pneumococcal serotype 14 IgG antibody concentrations for subjects in PCV13-TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group post primary series

FAS N (%)	PPS N (%)	PCV13-TT (2M)	PCV13-TT (3M)	PCV7 (3M)
(N=519)	(N=471)	(N=487)	(N=517)	(N=519)
n %	n %	n %	n %	n %
≥0.01	514 99.42	518 99.81	518 99.81	482 100.0
≥0.1	514 99.42	518 99.81	518 99.81	482 100.0
≥1	497 96.13	500 96.34	508 97.88	475 98.55
≥10	261 50.48	268 51.64	384 73.99	260 53.94
≥100	0 0.00	0 0.00	0 0.00	0 0.00

N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT group (2M), PCV13-TT group (3M) and PCV7 group (3M).

Serotype 14, After Primary Series (FAS) Serotype 14, After Primary Series (PPS)

100 100
90 90
80 80
70 70
% of Subjects
% of Subjects

60 60
50 50
40 40
30 30
20 20
10 10
0 0
0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT (2M) PCV13-TT (3M) PCV7 (3M) PCV13-TT (2M) PCV13-TT (3M) PCV7 (3M)

Figure 4 Reverse cumulative distribution curves, pneumococcal serotype 14, IgG antibody concentrations for subjects in PCV13-TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group - post primary series.

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Table 29. Distribution of the pneumococcal serotype 18C IgG antibody concentrations for subjects in PCV13-TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group post primary series

FAS N (%)	PPS N (%)	PCV13-TT (2M) (N=519)	PCV13-TT (3M) (N=517)	PCV7 (3M) (N=487)
≥0.01	509 98.45	510 98.27	512 98.65	477 98.96
≥0.1	509 98.45	510 98.27	512 98.65	477 98.96
≥1	409 79.11	454 87.48	481 92.68	403 83.61
≥10	14 2.71	34 6.55	102 19.65	13 2.70
≥100	0 0.00	0 0.00	0 0.00	0 0.00

N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT group (2M), PCV13-TT group (3M) and PCV7 group (3M).

Serotype 18C, After Primary Series (FAS) Serotype 18C, After Primary Series (PPS)

100 100
90 90
80 80
70 70
% of Subjects
% of Subjects

60 60
50 50
40 40
30 30
20 20
10 10
0 0
0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT (2M) PCV13-TT (3M) PCV7 (3M) PCV13-TT (2M) PCV13-TT (3M) PCV7 (3M)

Figure 5 Reverse cumulative distribution curves, pneumococcal serotype 18C, IgG antibody concentrations for subjects in PCV13-TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group - post primary series.

Table 30. Distribution of the pneumococcal serotype 19F IgG antibody concentrations for subjects in PCV13-TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group post primary series

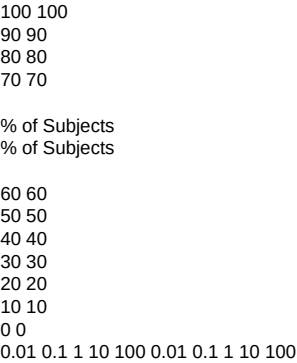
FAS N (%)	PPS N (%)	PCV13-TT (2M) (N=519)	PCV13-TT (3M) (N=517)	PCV7 (3M) (N=487)
≥0.01	517 100.0	519 100.0	518 99.81	482 100.0
≥0.1	517 100.0	519 100.0	518 99.81	482 100.0
≥1	497 96.13	497 95.76	499 96.15	478 99.17
≥10	85 16.44	149 28.71	224 43.16	85 17.63
≥100	0 0.00	0 0.00	0 0.00	0 0.00

N=number of subjects included in FAS and PPS for vaccine group.

n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT group (2M), PCV13-TT group (3M) and PCV7 group (3M).

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Serotype 19F, After Primary Series (FAS) Serotype 19F, After Primary Series (PPS)



Antibody Concentration ■µg/mL
Antibody Concentration (µg/mL)

PCV13-TT (2M) PCV13-TT (3M) PCV7 (3M) PCV13-TT (2M) PCV13-TT (3M) PCV7 (3M)

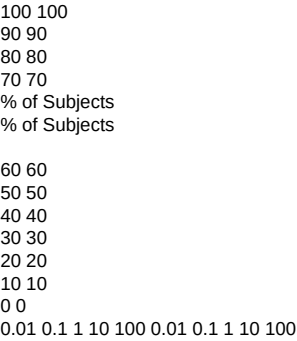
Figure 6 Reverse cumulative distribution curves, pneumococcal serotype 19F, IgG antibody concentrations for subjects in PCV13-TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group - post primary series.

Table 31. Distribution of the pneumococcal serotype 23F IgG antibody concentrations for subjects in PCV13-TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group post primary series

FAS N (%)	PPS N (%)	PCV13-TT (2M)	PCV13-TT (3M)	PCV7 (3M)
517	519	482	471	
≥0.01	516	99.81	514	99.04
≥0.1	516	99.81	514	99.04
≥1	458	88.59	436	84.01
≥10	79	15.28	53	10.21
≥100	0	0.00	0	0.00

N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT group (2M), PCV13-TT group (3M) and PCV7 group (3M).

Serotype 23F, After Primary Series (FAS) Serotype 23F, After Primary Series (PPS)



Antibody Concentration ■µg/mL
Antibody Concentration (µg/mL)

PCV13-TT (2M) PCV13-TT (3M) PCV7 (3M) PCV13-TT (2M) PCV13-TT (3M) PCV7 (3M)

Figure 7 Reverse cumulative distribution curves, pneumococcal serotype 23F, IgG antibody concentrations for subjects in PCV13-TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group - post primary series.

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Table 32. Distribution of the pneumococcal serotype 1 IgG antibody concentrations for subjects in PCV13-TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group post primary series

FAS N (%)	PPS N (%)	PCV13-TT (2M)	PCV13-TT (3M)	PCV7 (3M)
517	519	482	471	
≥0.01	516	99.81	514	99.04
≥0.1	516	99.81	514	99.04
≥1	458	88.59	436	84.01
≥10	79	15.28	53	10.21
≥100	0	0.00	0	0.00

N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT group (2M), PCV13-TT group (3M) and PCV7 group (3M).

(2M) (3M) (2M) (3M)
tration (N=519) (N=487)
(N=517) (N=519) (N=482) (N=471)
n % n % n % n % n % n %
≥0.01 516 99.81 509 98.07 476 91.71 482 100.0 471 100.0 450 92.40
≥0.1 516 99.81 509 98.07 475 91.52 482 100.0 471 100.0 450 92.40
≥1 404 78.14 427 82.27 44 8.48 396 82.16 404 85.77 41 8.42
≥10 10 1.93 12 2.31 1 0.19 10 2.07 10 2.12 1 0.21
≥100 0 0.00 0 0.00 0 0.00 0 0.00 0 0.00 0 0.00
N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT
group (2M), PCV13-
TT group (3M) and PCV7 group (3M).

Serotype 1, After Primary Series (FAS) Serotype 1, After Primary Series (PPS)

100 100
90 90
80 80
70 70

% of Subjects
% of Subjects

60 60
50 50
40 40
30 30
20 20
10 10
0 0
0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT (2M) PCV13-TT (3M) PCV7 (3M) PCV13-TT (2M) PCV13-TT (3M) PCV7 (3M)

Figure 8 Reverse cumulative distribution curves, pneumococcal serotype 1, IgG antibody concentrations for subjects in PCV13-TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group - post primary series.

Table 33 Distribution of the pneumococcal serotype 3 IgG antibody concentrations for subjects in PCV13-TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group post primary series
FAS N (%) PPS N (%)
PCV13-TT PCV13-TT PCV13-TT PCV13-TT
PCV7 (3M) PCV7 (3M)
Concentration (2M) (3M) (2M) (3M)
(N=519) (N=487)
(N=517) (N=519) (N=482) (N=471)
n % n % n % n % n % n %
≥0.01 515 99.61 518 99.81 514 99.04 482 100.0 471 100.0 486 99.79
≥0.1 511 98.84 505 97.30 493 94.99 482 100.0 471 100.0 470 96.51
≥1 336 64.99 370 71.29 38 7.32 333 69.09 355 75.37 33 6.78
≥10 0 0.00 6 1.16 1 0.19 0 0.00 5 1.06 1 0.21
≥100 0 0.00 0 0.00 0 0.00 0 0.00 0 0.00 0 0.00
N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT
group (2M), PCV13-
TT group (3M) and PCV7 group (3M).

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Serotype 3, After Primary Series (FAS) Serotype 3, After Primary Series (PPS)

100 100
90 90
80 80
70 70

% of Subjects
% of Subjects

60 60
50 50
40 40
30 30
20 20
10 10
0 0
0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT (2M) PCV13-TT (3M) PCV7 (3M) PCV13-TT (2M) PCV13-TT (3M) PCV7 (3M)

Figure 9 Reverse cumulative distribution curves, pneumococcal serotype 3, IgG antibody concentrations for subjects in PCV13-TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group - post primary series.

Table 34. Distribution of the pneumococcal serotype 5 IgG antibody concentrations for subjects in PCV13-TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group post primary series

FAS N (%) PPS N (%)

PCV13-TT PCV13-TT PCV13-TT PCV13-TT

PCV7 (3M) PCV7 (3M)

Concentration (2M) (3M) (2M) (3M)

(N=519) (N=487)

(N=517) (N=519) (N=482) (N=471)

n % n % n % n % n % n %

≥0.01 503 97.29 505 97.30 325 62.62 477 98.96 470 99.79 308 63.24

≥0.1 503 97.29 505 97.30 323 62.24 477 98.96 470 99.79 306 62.83

≥1 188 36.36 392 75.53 11 2.12 186 38.59 379 80.47 11 2.26

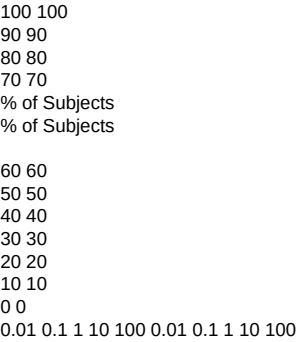
≥10 0 0.00 6 1.16 2 0.39 0 0.00 6 1.27 2 0.41

≥100 0 0.00 0 0.00 0 0.00 0 0.00 0 0.00 0 0.00

N=number of subjects included in FAS and PPS for vaccine group.

n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT group (2M), PCV13-TT group (3M) and PCV7 group (3M).

Serotype 5, After Primary Series (FAS) Serotype 5, After Primary Series (PPS)



Antibody Concentration ■µg/mL)

Antibody Concentration (µg/mL)

PCV13-TT (2M) PCV13-TT (3M) PCV7 (3M) PCV13-TT (2M) PCV13-TT (3M) PCV7 (3M)

Figure 10 Reverse cumulative distribution curves, pneumococcal serotype 5, IgG antibody concentrations for subjects in PCV13-TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group - post primary series.

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Table 35. Distribution of the pneumococcal serotype 6A IgG antibody concentrations for subjects in PCV13-TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group post primary series

FAS N (%) PPS N (%)

PCV13-TT PCV13-TT PCV13-TT PCV13-TT

PCV7 (3M) PCV7 (3M)

Concentration (2M) (3M) (2M) (3M)

(N=519) (N=487)

(N=517) (N=519) (N=482) (N=471)

n % n % n % n % n % n %

≥0.01 512 99.03 513 98.84 483 93.06 481 99.79 467 99.15 455 93.43

≥0.1 512 99.03 512 98.65 482 92.87 481 99.79 467 99.15 455 93.43

≥1 446 86.27 390 75.14 193 37.19 435 90.25 364 77.28 180 36.96

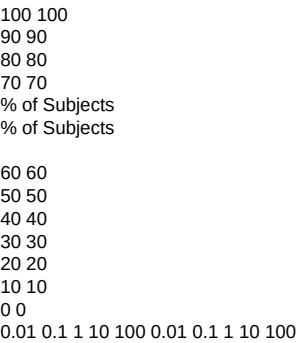
≥10 53 10.25 22 4.24 2 0.39 52 10.79 21 4.46 1 0.21

≥100 0 0.00 0 0.00 0 0.00 0 0.00 0 0.00 0 0.00

N=number of subjects included in FAS and PPS for vaccine group.

n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT group (2M), PCV13-TT group (3M) and PCV7 group (3M).

Serotype 6A, After Primary Series (FAS) Serotype 6A, After Primary Series (PPS)



Antibody Concentration ■µg/mL)

Antibody Concentration (µg/mL)

Figure 11 Reverse cumulative distribution curves, pneumococcal serotype 6A, IgG antibody concentrations for subjects in PCV13-TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group - post primary series.

TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group post booster dose
FAS N (%) PPS N (%)
PCV13-TT PCV13-TT
Concen PCV13-TT PCV7 (3M) PCV13-TT PCV7 (3M)
(3M) (N=486) (2M)
tration (2M) (N=486) (N=492) (3M) (N=475) (N=487)
(N=476)
n % n % n % n % n % n %
≥0.01 485 99.79 486 100.0 492 100.0 476 100.0 475 100.0 487 100.0
≥0.1 485 99.79 486 100.0 492 100.0 476 100.0 475 100.0 487 100.0
≥1 390 80.25 451 92.80 472 95.93 388 81.51 443 93.26 470 96.51
≥10 14 2.88 47 9.67 71 14.43 14 2.94 47 9.89 71 14.58
≥100 0 0.00 0 0.00 0 0.00 0 0.00 0 0.00 0 0.00
N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT
group (2M), PCV13-
TT group (3M) and PCV7 group (3M).

Serotype 4, After Booster Dose (FAS) Serotype 4, After Booster Dose (PPS)

100 100
90 90
80 80
70 70
% of Subjects
% of Subjects

60 60
50 50
40 40
30 30
20 20
10 10
0 0
0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT (2M) PCV13-TT (3M) PCV7 (3M) PCV13-TT (2M) PCV13-TT (3M) PCV7 (3M)

Figure 14 Reverse cumulative distribution curves, pneumococcal serotype 4, IgG antibody concentrations for subjects in PCV13-TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group-post booster dose.

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Table 39. Distribution of the pneumococcal serotype 6B IgG antibody concentrations for subjects in PCV13-TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group post booster dose
FAS N (%) PPS N (%)
PCV13-TT PCV13-TT
Concen PCV13-TT PCV7 (3M) PCV13-TT PCV7 (3M)
(3M) (N=486) (2M)
tration (2M) (N=486) (N=492) (3M) (N=475) (N=487)
(N=476)
n % n % n % n % n % n %
≥0.01 486 100.0 486 100.0 492 100.0 476 100.0 475 100.0 487 100.0
≥0.1 486 100.0 486 100.0 492 100.0 476 100.0 475 100.0 487 100.0
≥1 481 98.97 486 100.0 480 97.56 472 99.16 475 100.0 478 98.15
≥10 208 42.80 251 51.65 279 56.71 208 43.70 248 52.21 279 57.29
≥100 4 0.82 2 0.41 6 1.22 4 0.84 2 0.42 6 1.23
N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT
group (2M), PCV13-
TT group (3M) and PCV7 group (3M).

Serotype 6B, After Booster Dose (FAS) Serotype 6B, After Booster Dose (PPS)

100 100
90 90
80 80
70 70

% of Subjects
% of Subjects

60 60
50 50
40 40
30 30
20 20
10 10
0 0
0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT (2M) PCV13-TT (3M) PCV7 (3M) PCV13-TT (2M) PCV13-TT (3M) PCV7 (3M)

PCV13-TT (2M) PCV13-TT (3M) PCV7 (3M) PCV13-TT (2M) PCV13-TT (3M) PCV7 (3M)

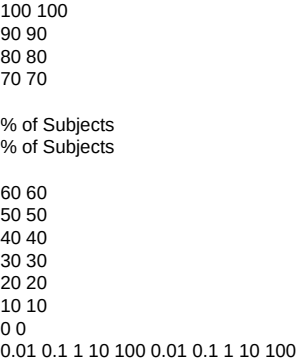
Figure 17 Reverse cumulative distribution curves, pneumococcal serotype 14, IgG antibody concentrations for subjects in PCV13-TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group - post booster dose.

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Table 42. Distribution of the pneumococcal serotype 18C IgG antibody concentrations for subjects in PCV13-TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group post booster dose

PCV13-TT PCV13-TT												
Concen PCV13-TT PCV7 (3M) PCV13-TT PCV7 (3M)												
(3M) (N=486) (2M)												
Concentration (2M) (N=486) (N=492) (3M) (N=475) (N=487)												
(N=476)												
n % n % n % n % n % n %												
≥0.01	484	99.59	485	99.79	490	99.59	476	100.0	474	99.79	486	99.79
≥0.1	484	99.59	485	99.79	490	99.59	476	100.0	474	99.79	486	99.79
≥1	385	79.22	430	88.48	462	93.90	384	80.67	422	88.84	460	94.46
≥10	9	1.85	20	4.12	71	14.43	9	1.89	18	3.79	71	14.58
≥100	0	0.00	0	0.00	0	0.00	0	0.00	0	0.00	0	0.00
N=number of subjects included in FAS and PPS for vaccine group.												
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT group (2M), PCV13-TT group (3M) and PCV7 group (3M).												

Serotype 18C, After Booster Dose (FAS) Serotype 18C, After Booster Dose (PPS)



Antibody Concentration ■µg/mL
Antibody Concentration (µg/mL)

PCV13-TT (2M) PCV13-TT (3M) PCV7 (3M) PCV13-TT (2M) PCV13-TT (3M) PCV7 (3M)

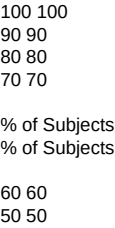
Figure 18 Reverse cumulative distribution curves, pneumococcal serotype 18C, IgG antibody concentrations for subjects in PCV13-TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group - post booster dose.

Table 43. Distribution of the pneumococcal serotype 19F IgG antibody concentrations for subjects in PCV13-TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group post booster dose

PCV13-TT PCV13-TT												
Concen PCV13-TT PCV7 (3M) PCV13-TT PCV7 (3M)												
(3M) (N=486) (2M)												
Concentration (2M) (N=486) (N=492) (3M) (N=475) (N=487)												
(N=476)												
n % n % n % n % n % n % n %												
≥0.01	486	100.0	486	100.0	492	100.0	476	100.0	475	100.0	487	100.0
≥0.1	486	100.0	486	100.0	492	100.0	476	100.0	475	100.0	487	100.0
≥1	484	99.59	486	100.0	487	98.98	476	100.0	475	100.0	484	99.38
≥10	185	38.07	282	58.02	182	36.99	185	38.87	281	59.16	182	37.37
≥100	1	0.21	5	1.03	0	0.00	1	0.21	5	1.05	0	0.00
N=number of subjects included in FAS and PPS for vaccine group.												
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT group (2M), PCV13-TT group (3M) and PCV7 group (3M).												

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Serotype 19F, After Booster Dose (FAS) Serotype 19F, After Booster Dose (PPS)



60 60
50 50
40 40
30 30
20 20
10 10
0 0
0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT (2M) PCV13-TT (3M) PCV7 (3M) PCV13-TT (2M) PCV13-TT (3M) PCV7 (3M)

Figure 21 Reverse cumulative distribution curves, pneumococcal serotype 1, IgG antibody concentrations for subjects in PCV13-TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group - post booster dose.

Table 46. Distribution of the pneumococcal serotype 3 IgG antibody concentrations for subjects in PCV13-TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group post booster dose

FAS N (%)	PPS N (%)	PCV13-TT	PCV13-TT	Concen	PCV13-TT	PCV7 (3M)	PCV13-TT	PCV7 (3M)
(3M)	(N=486)	(2M)	tration (2M)	(N=486)	(N=492)	(3M)	(N=475)	(N=487)
(N=476)	n %	n %	n %	n %	n %	n %	n %	n %
≥0.01	486	100.0	486	100.0	492	100.0	476	100.0
≥0.1	485	99.79	486	100.0	487	98.98	476	100.0
≥1	340	69.96	393	80.86	243	49.39	338	71.01
≥10	5	1.03	3	0.62	5	1.02	3	0.63
≥100	0	0.00	0	0.00	0	0.00	0	0.00

N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT group (2M), PCV13-TT group (3M) and PCV7 group (3M).

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Serotype 3, After Booster Dose (FAS) Serotype 3, After Booster Dose (PPS)

100 100
90 90
80 80
70 70

% of Subjects
% of Subjects

60 60
50 50
40 40
30 30
20 20
10 10
0 0
0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT (2M) PCV13-TT (3M) PCV7 (3M) PCV13-TT (2M) PCV13-TT (3M) PCV7 (3M)

Figure 22 Reverse cumulative distribution curves, pneumococcal serotype 3, IgG antibody concentrations for subjects in PCV13-TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group - post booster dose.

Table 47 Distribution of the pneumococcal serotype 5 IgG antibody concentrations for subjects in PCV13-TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group post booster dose

FAS N (%)	PPS N (%)	PCV13-TT	PCV13-TT	Concen	PCV13-TT	PCV7 (3M)	PCV13-TT	PCV7 (3M)
(3M)	(N=486)	(2M)	tration (2M)	(N=486)	(N=492)	(3M)	(N=475)	(N=487)
(N=476)	n %	n %	n %	n %	n %	n %	n %	n %
≥0.01	483	99.38	486	100.0	414	84.15	476	100.0
≥0.1	483	99.38	486	100.0	410	83.33	476	100.0
≥1	256	52.67	345	70.99	13	2.64	256	53.78
≥10	1	0.21	7	1.44	0	0.00	1	0.21
≥100	0	0.00	0	0.00	0	0.00	0	0.00

N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT group (2M), PCV13-TT group (3M) and PCV7 group (3M).

Serotype 5, After Booster Dose (FAS) Serotype 5, After Booster Dose (PPS)

100 100
90 90

80 80
70 70
% of Subjects
% of Subjects

60 60
50 50
40 40
30 30
20 20
10 10
0 0
0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT (2M) PCV13-TT (3M) PCV7 (3M) PCV13-TT (2M) PCV13-TT (3M) PCV7 (3M)

Figure 23 Reverse cumulative distribution curves, pneumococcal serotype 5, IgG antibody concentrations for subjects in PCV13-TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group - post booster dose.

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Table 48. Distribution of the pneumococcal serotype 6A IgG antibody concentrations for subjects in PCV13-TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group post booster dose

FAS N (%)	PPS N (%)	PCV13-TT	Concen	PCV13-TT	PCV7 (3M)	PCV13-TT	PCV7 (3M)
(3M)	(N=486)	(2M)	tration	(2M)	(N=486)	(N=492)	(3M) (N=475) (N=487)
(N=476)							
n %	n %	n %	n %	n %	n %	n %	n %
≥0.01	486	100.0	486	100.0	492	100.0	476 100.0 475 100.0 487 100.0
≥0.1	486	100.0	484	99.59	491	99.80	476 100.0 473 99.58 486 99.79
≥1	460	94.65	465	95.68	434	88.21	454 95.38 457 96.21 431 88.50
≥10	98	20.16	135	27.78	93	18.90	98 20.59 133 28.00 93 19.10
≥100	1	0.21	1	0.21	1	0.20	1 0.21 1 0.21 1 0.21
N=number of subjects included in FAS and PPS for vaccine group.							
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT group (2M), PCV13-TT group (3M) and PCV7 group (3M).							

Serotype 6A, After Booster Dose (FAS) Serotype 6A, After Booster Dose (PPS)

100 100
90 90
80 80
70 70

% of Subjects
% of Subjects

60 60
50 50
40 40
30 30
20 20
10 10
0 0
0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT (2M) PCV13-TT (3M) PCV7 (3M) PCV13-TT (2M) PCV13-TT (3M) PCV7 (3M)

Figure 24 Reverse cumulative distribution curves, pneumococcal serotype 6A, IgG antibody concentrations for subjects in PCV13-TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group - post booster dose.

Table 49. Distribution of the pneumococcal serotype 7F IgG antibody concentrations for subjects in PCV13-TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group post booster dose

FAS N (%)	PPS N (%)	PCV13-TT	PCV13-TT	Concen	PCV13-TT	PCV7 (3M)	PCV13-TT	PCV7 (3M)				
(3M)	(N=486)	(2M)	tration (2M)	(N=486)	(N=492)	(3M)	(N=475)	(N=487)				
(N=476)	n %	n %	n %	n %	n %	n %	n %	n %				
≥0.01	486	100.0	486	100.0	436	88.62	476	100.0	475	100.0	434	89.12
≥0.1	486	100.0	486	100.0	434	88.21	476	100.0	475	100.0	432	88.71
≥1	479	98.56	476	97.94	69	14.02	472	99.16	466	98.11	69	14.17
≥10	51	10.49	67	13.79	0	0.00	51	10.71	67	14.11	0	0.00
N=number of subjects included in FAS and PPS for vaccine group.												
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT group (2M), PCV13-TT group (3M) and PCV7 group (3M).												

Serotype 7F, After Booster Dose (FAS) Serotype 7F, After Booster Dose (PPS)

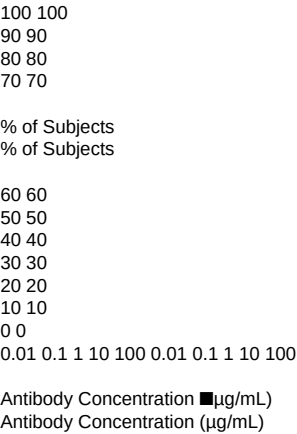


Figure 25 Reverse cumulative distribution curves, pneumococcal serotype 7F, IgG antibody concentrations for subjects in PCV13-TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group - post booster dose.

Table 50. Distribution of the pneumococcal serotype 19A IgG antibody concentrations for subjects in PCV13-TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group post booster dose

FAS N (%)	PPS N (%)	PCV13-TT (2M) (N=486)	PCV13-TT (3M) (N=492)	PCV7 (3M) (N=475)
≥0.01	100.0	100.0	100.0	100.0
≥0.1	100.0	100.0	100.0	100.0
≥1	100.0	100.0	100.0	100.0
≥10	100.0	100.0	100.0	100.0
≥100	100.0	100.0	100.0	100.0

Serotype 19A, After Booster Dose (FAS) Serotype 19A, After Booster Dose (PPS)

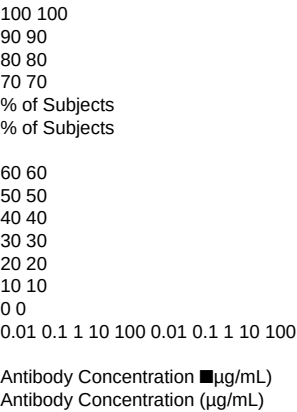


Figure 26 Reverse cumulative distribution curves, pneumococcal serotype 19A, IgG antibody concentrations for subjects in PCV13-TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group - post booster dose.

2.7.3.2.2.1.3.3. Comparison of percentage of subjects achieving IgG antibody concentration ≥ 1.0 µg/mL with comparison to the PCV7 seroresponse
Booster Dose
The proportion of responders (IgG antibody concentration ≥ 1.0 µg/mL) were relatively lower in terms of serotypes 4 (81.51%), 18C (80.67%), 1 (81.30%), and 3 (71.01%), with nadir value observed for serotype 5 (53.78%), while all other serotypes were with ≥ 94% of subjects with IgG antibody concentration ≥ 1.0 µg/mL. With the exception of serotype 4, 18C, 1, and 5 with the lower limits of 2-sided 97.5% CI for rate difference of proportion of responders (PCV13-TT [2 months] – PCV13-TT [3 months]) being < -10% (-15.90%, -12.71%, -13.80%, and -24.25%, respectively), all other serotypes met the non-inferiority criterion (Table 51.).

Table 51. Comparison of percentage of subjects achieving a pneumococcal IgG antibody concentration $\geq$ 1.0 μ g/mL in PCV13-TT (2 months) group and PCV13-TT (3 months) group post booster dose (PPS)
IgG antibody concentration $\geq$ 1.0 μ g/mL

PCV13-TT (2M) (N=476) PCV13-TT (3M) (N=487) Rate difference c
a b

Serotype	N	%	95% CI	N	%	95% CI	%	97.5%CI
4	388	81.51	77.73-84.90	443	93.26	90.62-95.35	-11.75	-15.90,-7.60
6B	472	99.16	97.86-99.77	475	100.0	99.23-100.0	-0.84	-1.66,-0.02
9V	451	94.75	92.34-96.57	455	95.79	93.57-97.41	-1.04	-3.74,1.66
14	476	100.0	99.23-100.0	474	99.79	98.83-99.99	0.21	-0.20,0.62
18C	384	80.67	76.83-84.12	422	88.84	85.66-91.53	-8.17	-12.71,-3.63
19F	476	100.0	99.23-100.0	475	100.0	99.23-100.0	0.00	-0.77,0.77
23F	465	97.69	95.90-98.84	469	98.74	97.27-99.54	-1.05	-2.73,0.63
1	387	81.30	77.50-84.71	431	90.74	87.76-93.19	-9.43	-13.80,-5.07
3	338	71.01	66.71-75.05	391	82.32	78.58-85.64	-11.31	-16.64,-5.98
5	256	53.78	49.19-58.33	342	72.00	67.73-76.00	-18.22	-24.25,-12.19
6A	454	95.38	93.09-97.08	457	96.21	94.08-97.74	-0.83	-3.38,1.72
7F	472	99.16	97.86-99.77	466	98.11	96.43-99.13	1.05	-0.42,2.53
19A	476	100.0	99.23-100.0	473	99.58	98.49-99.95	0.42	-0.16,1.00

N=number of subjects included in PPS for PCV13-TT (2M) and PCV13-TT (3M)

a. number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT (2M) group.

b. number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT (3M) group.

c. The rate difference of PCV13-TT (2M) – PCV13-TT (3M).

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2.7.3.2.2.1.3.4. Comparison of percentage of subjects achieving an OPA antibody $\geq$ 1:8 and the IgG antibody geometric mean titers (GMTs)

Primary series

The proportion of subjects achieving an OPA antibody titer $\geq$ 1:8 was mostly $\geq$ 95% except for serotype 1 (91.75%) and 6B (92.78%), however, the lower limits of 2-sided 97.5% CI for rate difference of percentage of subjects achieving an OPA titer $\geq$ 1:8 (PCV13-TT [2 months] – PCV13-TT [3 months]) after primary series were greater than -10% of all serotypes (Table 52.).The lower limit of 2-sided 97.5% CI for GMR (PCV13-TT [2 months] relative to PCV13-TT [3 months]) was lower than 0.5 for serotypes 18C, 23F, and 7F, and all other serotypes were compliant with the non-inferiority criterion.

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Table 52. Comparison of percentage of subjects achieving a pneumococcal IgG antibody titer $\geq$ 1:8 and GMTs in PCV13-TT (2 months) group and PCV13-TT (3 months) group post-dose 3 of the primary series (PPS)

OPA antibody titer $\geq$ 1:8 OPA GMT

PCV13-TT (2M) (N=92) PCV13-TT (3M) (N=96) Rate difference c PCV13-TT (2M) (N=92) PCV13-TT (3M) (N=96) GMT Ratio

Serotype	na	%	95% CI	nb	%	95% CI	%	97.5%CI	GMT	95% CI	GMT	95% CI	Ratio	d	97.5%CI
Common															
4	96	98.97	94.39-99.97	93	100.0	96.11-100.0	-1.03	-3.04,0.98	1214.42	940.3-1568.5	1227.93	992.5-1519.2			
0.99	0.71	-1.38													
6B	90	92.78	85.70-97.05	83	89.25	81.11-94.72	3.54	-4.60,11.67	811.17	542.0-1214.1	528.67	329.58-848.02			
1.53	0.83	-2.84													
9V	97	100.0	96.27-100.0	93	100.0	96.11-100.0	0.00	-4.15,3.93	2192.74	1728.3-2781.9	1922.48	1518.9-2433.2			
1.14	0.82	-1.59													
14	97	100.0	96.27-100.0	93	100.0	96.11-100.0	0.00	-4.15,3.93	3571.78	2833.6-4502.3	4511.21	3494.9-5823.1			
0.79	0.56	-1.11													
18C	97	100.0	96.27-100.0	93	100.0	96.11-100.0	0.00	-4.15,3.93	518.67	417.13-644.92	1356.26	1094.5-1680.7			
0.38	0.28	-0.52													
19F	96	98.97	94.39-99.97	93	100.0	96.11-100.0	-1.03	-3.04,0.98	769.17	617.10-958.72	652.57	547.97-777.14			
1.18	0.89	-1.56													
23F	93	95.88	89.78-98.87	89	95.70	89.35-98.82	0.18	-5.54,5.89	1072.95	738.0-1560.0	2090.78	1452.4-3009.7			
0.51	0.31	-0.86													
Additional															
1	89	91.75	84.39-96.37	78	83.87	74.80-90.68	7.88	-1.38,17.15	92.51	66.48-128.75	62.13	42.64-90.51	1.49		
0.91	-2.45														
3	97	100.0	96.27-100.0	90	96.77	90.86-99.33	3.23	-0.37,6.82	575.94	444.43-746.38	511.70	376.29-695.83			
1.13	0.76	-1.68													
5	97	100.0	96.27-100.0	92	98.92	94.15-99.97	1.08	-1.02,3.17	411.73	317.12-534.56	450.24	358.12-566.05			
0.91	0.65	-1.29													
6A	95	97.94	92.75-99.75	91	97.85	92.45-99.74	0.09	-4.00,4.17	1202.17	895.0-1614.8	1079.96	792.9-1471.0			
1.11	0.73	-1.70													
7F	97	100.0	96.27-100.0	93	100.0	96.11-100.0	0.00	-4.15,3.93	2414.63	1872.4-3113.9	7741.10	6383.8-9387.0			
0.31	0.23	-0.43													
19A	97	100.0	96.27-100.0	93	100.0	96.11-100.0	0.00	-4.15,3.93	659.70	532.25-817.66	777.70	638.69-946.97			
0.85	0.63	-1.13													

N=number of subjects included in PPS for PCV13-TT (2M) and PCV13-TT (3M)

a. number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT (2M) group.

b. number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT (3M) group.

c. The rate difference of PCV13-TT (2M) – PCV13-TT (3M).

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Table 53. Comparison of percentage of subjects achieving a pneumococcal IgG antibody titer $\geq 1:8$ and GMTs in PCV13-TT (2 months) group and PCV13-TT (3 months) group post-dose 3 of the primary series (FAS)

OPA antibody titer $\geq 1:8$ OPA GMT

PCV13-TT (2M) (N=92) PCV13-TT (3M) (N=96) Rate difference c PCV13-TT (2M) (N=92) PCV13-TT (3M) (N=96) GMT

Serotype	na %	95% CI	nb %	95% CI	% 97.5%CI	GMT	95% CI	GMT	95% CI	Ratio d	97.5%CI	e
Common												
4	97.98.98	94.45-99.97	98.100.0	96.31-100.0	-1.02	-3.01,0.97	1231.00	954.3-1588.0	1246.22	1011.4-1535.5		
0.99	0.71-1.37											
6B	91.92.86	85.84-97.08	88.89.80	82.03-95.00	3.06	-4.81,10.93	835.16	558.0-1250.0	553.85	352.92-869.16		
1.51	0.83-2.75											
9V	98.100.0	96.31-100.0	98.100.0	96.31-100.0	0.00	-3.87,3.87	2245.67	1766.1-2855.5	1966.80	1569.2-2465.1		
1.14	0.82-1.58											
14	98.100.0	96.31-100.0	98.100.0	96.31-100.0	0.00	-3.87,3.87	3651.57	2891.8-4611.0	4387.49	3416.4-5634.6		
0.83	0.59-1.17											
18C	98.100.0	96.31-100.0	98.100.0	96.31-100.0	0.00	-3.87,3.87	522.68	421.08-648.79	1347.71	1096.1-1657.2		
0.39	0.29-0.52											
19F	97.98.98	94.45-99.97	98.100.0	96.31-100.0	-1.02	-3.01,0.97	786.74	629.77-982.85	655.55	551.84-778.74		
1.20	0.91-1.59											
23F	94.95.92	89.88-98.88	94.95.92	89.88-98.88	0.00	-5.54,5.54	1098.20	756.1-1595.1	2108.20	1487.1-2988.8		
0.52	0.31-0.87											
Additional												
1	90.91.84	84.55-96.41	82.83.67	74.84-90.37	8.16	-0.94,17.27	94.23	67.80-130.94	64.30	44.31-93.29	1.47	
0.89	0.2-2.40											
3	98.100.0	96.31-100.0	95.96.94	91.31-99.36	3.06	-0.35,6.47	591.13	455.03-767.94	509.18	378.71-684.60		
1.16	0.78-1.72											
5	98.100.0	96.31-100.0	97.98.98	94.45-99.97	1.02	-0.97,3.01	412.18	318.33-533.71	453.05	363.07-565.34		
0.91	0.65-1.28											
6A	96.97.96	92.82-99.75	96.97.96	92.82-99.75	0.00	-3.96,3.96	1225.83	913.1-1645.7	1120.90	831.4-1511.2		
1.09	0.72-1.66											
7F	98.100.0	96.31-100.0	98.100.0	96.31-100.0	0.00	-3.87,3.87	2463.59	1909.4-3178.6	7730.89	6423.4-9304.5		
0.32	0.23-0.44											
19A	98.100.0	96.31-100.0	98.100.0	96.31-100.0	0.00	-3.87,3.87	674.88	543.13-838.59	783.76	648.55-947.14		
0.86	0.65-1.15											

N=number of subjects included in FAS for PCV13-TT (2M) and PCV13-TT (3M)

a. number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT (2M) group.

b. number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT (3M) group.

c. The rate difference of PCV13-TT (2M) – PCV13-TT (3M).

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Booster dose

For the 13 serotypes, the percentage of subjects achieving an OPA antibody titer $\geq 1:8$ was all higher than 95% in PCV13-TT (2 months) group. The lower limits of 2-sided 97.5% CI of rate difference for the percentage of subjects achieving an OPA antibody titer $\geq 1:8$ was greater than - 10% to all serotypes after booster dose (Table 54). The lower limits of 2-sided 97.5% CI for GMR was greater than 0.5 for serotypes 14, 19F, 3, and 5. All other serotypes had lower limits of 2-sided 97.5% CI for GMR of < 0.5.

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Table 54 Comparison of percentage of subjects achieving a pneumococcal IgG antibody titer $\geq 1:8$ and GMTs in PCV13-TT (2 months) group and PCV13-TT (3 months) group post booster dose (PPS)

OPA antibody titer $\geq 1:8$ OPA GMT

PCV13-TT (2M) (N=92) PCV13-TT (3M) (N=96) Rate difference c PCV13-TT (2M) (N=92) PCV13-TT (3M) (N=96) GMT

Serotype	na %	95% CI	nb %	95% CI	% 97.5%CI	GMT	95% CI	GMT	95% CI	Ratio d	97.5%CI
Common											
4	92.100.0	96.07-100.0	96.100.0	96.23-100.0	0.00	-3.99,4.27	3440.10	2690.0-4399.3	6182.37	4931.4-7750.7	
0.56	0.40-0.78										
6B	92.100.0	96.07-100.0	95.98.96	94.33-99.97	1.04	-0.99,3.07	1948.21	1460.3-2599.2	2661.93	1948.9-3635.9	
0.73	0.48-1.12										
9V	92.100.0	96.07-100.0	96.100.0	96.23-100.0	0.00	-3.99,4.27	3719.77	2889.7-4788.3	6184.75	4870.3-7854.0	
0.60	0.43-0.85										
14	92.100.0	96.07-100.0	96.100.0	96.23-100.0	0.00	-3.99,4.27	6173.16	4922.7-7741.3	7686.96	6160.6-9591.5	
0.80	0.59-1.10										
18C	92.100.0	96.07-100.0	96.100.0	96.23-100.0	0.00	-3.99,4.27	1117.88	899.5-1389.3	4173.73	3269.4-5328.2	
0.27	0.19-0.37										
19F	92.100.0	96.07-100.0	96.100.0	96.23-100.0	0.00	-3.99,4.27	1966.07	1496.8-2582.5	2359.26	1898.8-2931.3	
0.83	0.59-1.18										
23F	92.100.0	96.07-100.0	95.98.96	94.33-99.97	1.04	-0.99,3.07	2529.55	1959.0-3266.3	4943.66	3817.5-6402.1	
0.51	0.36-0.73										
Additional											
1	88.95.65	89.24-98.80	94.97.92	92.68-99.75	-2.26	-7.32,2.79	158.59	115.59-217.59	321.14	239.43-430.74	
0.49	0.32-0.76										
3	92.100.0	96.07-100.0	94.97.92	92.68-99.75	2.08	-0.77,4.94	586.01	440.25-780.04	678.98	523.20-881.14	
0.86	0.59-1.27										
5	92.100.0	96.07-100.0	96.100.0	96.23-100.0	0.00	-3.99,4.27	1027.83	796.4-1326.6	1375.13	1061.9-1780.8	
0.75	0.52-1.07										
6A	90.97.83	92.37-99.74	92.95.83	89.67-98.85	1.99	-2.99,6.98	2546.12	1780.8-3640.3	3254.75	2221.0-4769.7	
0.78	0.46-1.32										
7F	91.98.91	94.09-99.97	96.100.0	96.23-100.0	-1.09	-3.21,1.03	2616.75	1938.6-3532.1	7244.42	6006.0-8738.3	
0.36	0.25-0.51										
19A	92.100.0	96.07-100.0	96.100.0	96.23-100.0	0.00	-3.99,4.27	158.59	115.59-217.59	321.14	239.43-430.74	

0.49 0.32-0.76

N=number of subjects included in PPS for PCV13-TT (2M) and PCV13-TT (3M)

a. number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT (2M) group.

b. number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT (3M) group.

c. The rate difference of PCV13-TT (2M) – PCV13-TT (3M).

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Table 55 Comparison of percentage of subjects achieving a pneumococcal IgG antibody titer $\geq 1:8$ and GMTs

in PCV13-TT (2 months) group and PCV13-TT (3

months) group post booster dose (FAS)

OPA antibody titer $\geq 1:8$ OPA GMT

PCV13-TT (2M) (N=96 PCV13-TT (3M) (N=96) Rate difference c PCV13-TT (2M) (N=96) PCV13-TT (3M) (N=96) GMT Ratio

Serotype na % 95% CI nb % 95% CI 97.5% CI GMT 95% CI GMT 95% CI Ratio d 97.5%CI

Common

4 96 100.0 96.23-100.0 96 100.0 96.23-100.0 0.00 -4.01,4.01 3058.68 2339.9-3998.2 6182.37 4931.4-7750.7

0.49 0.35-0.70

6B 95 98.96 94.33-99.97 95 98.96 94.33-99.97 0.00 -2.87,2.87 1734.63 1266.1-2376.5 2661.93 1948.9-3635.9

0.65 0.42-1.01

9V 96 100.0 96.23-100.0 96 100.0 96.23-100.0 0.00 -4.01,4.01 3400.35 2623.6-4407.0 6184.75 4870.3-7854.0

0.55 0.39-0.78

14 95 98.96 94.33-99.97 96 100.0 96.23-100.0 -1.04 -3.07,0.99 5364.66 4051.2-7104.0 7686.96 6160.6-9591.5

0.70 0.49-1.00

18C 95 98.96 94.33-99.97 96 100.0 96.23-100.0 -1.04 -3.07,0.99 972.48 748.8-1263.0 4173.73 3269.4-5328.2

0.23 0.16-0.33

19F 96 100.0 96.23-100.0 96 100.0 96.23-100.0 0.00 -4.01,4.01 1758.11 1321.1-2339.8 2359.26 1898.8-2931.3

0.75 0.52-1.06

23F 95 98.96 94.33-99.97 95 98.96 94.33-99.97 0.00 -2.87,2.87 2221.21 1658.4-2975.0 4943.66 3817.5-6402.1

0.45 0.30-0.66

Additional

1 91 94.79 88.26-98.29 94 97.92 92.68-99.75 -3.1 -8.41,2.16 144.44 104.59-199.46 321.14 239.43-430.74

0.45 0.29-0.69

3 96 100.0 96.23-100.0 94 97.92 92.68-99.75 2.08 -0.77,4.94 557.39 419.95-739.82 678.98 523.20-881.14

0.82 0.56-1.20

5 95 98.96 94.33-99.97 96 100.0 96.23-100.0 -1.04 -3.07,0.99 917.73 686.2-1227.4 1375.13 1061.9-1780.8

0.67 0.45-0.98

6A 93 96.88 91.14-99.35 92 95.83 89.67-98.85 1.04 -4.26,6.34 2221.94 1520.6-3246.8 3254.75 2221.0-4769.7

0.68 0.40-1.17

7F 94 97.92 92.68-99.75 96 100.0 96.23-100.0 -2.08 -4.94,0.77 2390.26 1728.2-3305.9 7244.42 6006.0-8738.3

0.33 0.23-0.48

19A 95 98.96 94.33-99.97 96 100.0 96.23-100.0 -1.04 -3.07,0.99 2126.65 1594.0-2837.4 5503.11

4501.2-6728.1 0.39 0.27-0.55

N=number of subjects included in FAS for PCV13-TT (2M) and PCV13-TT (3M)

a. number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT (2M) group.

b. number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT (3M) group.

c. The rate difference of PCV13-TT (2M) – PCV13-TT (3M).

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2.7.3.2.2.1.4. Pneumococcal immunogenicity results for subjects aged 3-71 months in PCV13-TT group and PCV7 group

2.7.3.2.2.1.4.1. Comparison of percentage of subjects achieving IgG antibody concentration $\geq$

0.35 $\mu\text{g/mL}$ and the IgG antibody geometric mean concentrations (GMCs)

Post dose 3 IgG pneumococcal immunogenicity results with comparison to the PCV7

seroresponse for the 7 common serotypes

The IgG antibody concentration $\geq 0.35 \mu\text{g/mL}$ in the PCV13-TT group were all higher than 95%.

The lower limit of rate difference for the proportion of responders $\geq 0.35 \mu\text{g/mL}$ was all greater than -10%. With the exception of serotype 6B, the GMCs for all common serotypes were greater than 3.0 $\mu\text{g/mL}$. For serotype 6B, the GMC was 2.68 $\mu\text{g/mL}$. All of the 7 common serotypes met the non-inferiority criterion for the IgG antibody concentration $\geq 0.35 \mu\text{g/mL}$ and GMC after primary series (Table 56).

Post dose 3 IgG pneumococcal immunogenicity results with comparison to the PCV7 seroresponse for the 6 additional serotypes

For the 6 additional serotypes, the proportions of responders in the PCV13-TT group were all higher than 96% and all additional serotypes met the pre-specified superiority criteria.

All additional serotypes met the pre-specified superiority criteria for the IgG antibody concentration $\geq 0.35 \mu\text{g/mL}$ and GMC.

Post dose 3 IgG pneumococcal immunogenicity results with comparison to the PCV7 reference for the 6 additional serotypes

The PCV7 reference value for the 6 additional serotypes was based on serotype 23F from the PCV7 group. For serotype 23F, the proportion of responders was 98.31%. All additional serotypes met the non-inferiority criterion. The PCV7 reference value for the 6 additional serotypes was serotype 6B from the PCV7 group. For serotype 6B, the GMC was 3.11. Four (4) of the 6 additional serotypes met the non-inferiority criterion for GMR (lower limit of the 2-sided 97.5% CI for the GMR >0.5). The exceptions were serotypes 3 and 5 with a GMR of 0.42 (97.5% CI, 0.38 to 0.45).and 0.45 (97.5% CI, 0.41 to 0.50)., respectively.

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Table 56. Comparison of percentage of subjects aged 3-71 months achieving a pneumococcal IgG antibody concentration ≥ 0.35 $\mu\text{g/mL}$ and GMCs post dose 3 of the primary series (PPS)

IgG antibody concentration ≥ 0.35 $\mu\text{g/mL}$ IgG antibody GMC

PCV13-TT (N=1043) PCV7 (N=1064) Rate difference c PCV13-TT (N=1043) PCV7 (N=1064) GMC Ratio

Serotype na % 95% CI nb % 95% CI % 95% CI /97.5%CI GMC 95% CI GMC 95% CI Ratio d 95%/97.5%CI e

PCV7

4	1043	100.0	99.65-100.0	1059	99.53	98.91-99.85	0.47	-0.00,0.94	5.18	4.94-5.43	6.60	6.28-6.93	0.78
0.73,0.85													
6B	992	95.11	93.62-96.34	1053	98.97	98.16-99.48	-3.86	-5.11,-2.21	2.68	2.46-2.92	3.11	2.92-3.30	0.86
0.77,0.97													
9V	1031	98.85	98.00-99.40	1055	99.15	98.40-99.61	-0.30	-1.28,0.67	3.32	3.15-3.50	3.72	3.53-3.93	0.89
0.82,0.97													
14	1042	99.90	99.47-100.0	1062	99.81	99.32-99.98	0.09	-0.27,0.46	9.78	9.22-10.37	16.04	15.07-17.07	0.61
0.55,0.67													
18C	1032	98.95	98.12-99.47	1052	98.87	98.04-99.42	0.07	-0.94,1.09	4.15	3.91-4.41	4.74	4.47-5.03	0.88
0.80,0.96													
19F	1043	100.0	99.65-100.0	1064	100.0	99.65-100.0	0.00	-0.41,0.42	8.03	7.65-8.42	6.27	5.96-6.60	1.28
1.18,1.39													
23F	1022	97.99	96.94-98.75	1046	98.31	97.34-98.99	-0.32	-1.64,1.00	3.52	3.30-3.74	3.76	3.53-4.00	0.94
0.85,1.03													
Additional													
vs PCV7 seroresponsef													
1	1036	99.33	98.62-99.73	688	64.66	61.70-67.54	34.67	31.75,37.58	2.16	2.07-2.26	0.35	0.32-0.38	6.20
5.63,6.83													
3	1005	96.36	95.03-97.41	516	48.50	45.45-51.55	47.86	44.65,51.07	1.29	1.24-1.36	0.37	0.35-0.39	3.48
3.24,3.73													
5	1011	96.93	95.70-97.89	393	36.94	34.03-39.92	60.00	56.91,63.08	1.41	1.34-1.48	0.12	0.10-0.13	12.07
10.48,13.90													
6A	1010	96.84	95.59-97.81	889	83.55	81.19-85.73	13.28	10.82,15.75	2.08	1.94-2.23	0.75	0.68-0.83	2.76
2.45,3.12													
7F	1038	99.52	98.88-99.84	482	45.30	42.28-48.35	54.22	51.20,57.24	5.34	5.07-5.62	0.13	0.11-0.15	40.06
34.21,46.91													
19A	1043	100.0	99.65-100.0	1044	98.12	97.11-98.85	1.88	1.06,2.70	5.46	5.19-5.73	1.64	1.53-1.76	3.32
3.05,3.62													
vs PCV7 reference g													
1	1036	99.33	98.62-99.73	1046	98.31	97.34-98.99	1.02	-0.03,2.07	2.16	2.07-2.26	3.11	2.92-3.30	0.70
0.64,0.76													
3	1005	96.36	95.03-97.41	1046	98.31	97.34-98.99	-1.95	-3.53,-0.38	1.29	1.24-1.36	3.11	2.92-3.30	0.42
0.38,0.45													
5	1011	96.93	95.70-97.89	1046	98.31	97.34-98.99	-1.38	-2.87,0.11	1.41	1.34-1.48	3.11	2.92-3.30	0.45
0.41,0.50													
6A	1010	96.84	95.59-97.81	1046	98.31	97.34-98.99	-1.47	-2.98,0.03	2.08	1.94-2.23	3.11	2.92-3.30	0.67
0.60,0.74													
7F	1038	99.52	98.88-99.84	1046	98.31	97.34-98.99	1.21	0.20,2.22	5.34	5.07-5.62	3.11	2.92-3.30	1.72
1.57,1.88													
19A	1043	100.0	99.65-100.0	1046	98.31	97.34-98.99	1.69	0.81,2.58	5.46	5.19-5.73	3.11	2.92-3.30	1.76
1.61,1.92													

N=number of subjects included in PPS for PCV13-TT and PCV7.

a. number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT group.

b. number of subjects with a determinate IgG antibody concentration to the given serotype for PCV7 group.

c. The rate difference of PCV13-TT – PCV7.

d. GMC ratio: PCV13-TT to PCV7 reference.

e. Two-sided 95% CI is for additional serotypes, and two-sided 97.5% CI is for common serotypes.

f. Difference in proportions: (PCV13-TT – PCV7 seroresponse) expressed as a percentage. GMC ratio: PCV13-TT to PCV7 seroresponse.

g. Difference in proportions: (PCV13-TT – PCV7 reference) expressed as a percentage. For the additional serotypes, the reference value is derived from serotype 23F from the PCV7 group.

GMC ratio: PCV13-TT to PCV7 reference. For the additional serotypes, the reference value is derived from serotype 6B from the PCV7 group.

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Table 57. Comparison of percentage of subjects aged 3-71 months achieving a pneumococcal IgG antibody concentration ≥ 0.35 $\mu\text{g/mL}$ and GMCs post dose 3 of the primary series (FAS)

IgG antibody concentration ≥ 0.35 $\mu\text{g/mL}$ IgG antibody GMC

PCV13-TT (N=1043) PCV7 (N=1064) Rate difference c PCV13-TT (N=1117) PCV7 (N=1119) GMC Ratio

Serotype Na % 95% CI Nb % 95% CI % 95% CI /97.5%CI GMC 95% CI GMC 95% CI Ratio d 95%/97.5%CI e

PCV7

4	1080	96.69	95.46-97.66	1080	96.51	95.27-97.51	0.17	-1.54,1.89	4.14	3.81-4.50	5.42	4.99-5.90	0.76
0.67,0.87													
6B	1047	93.73	92.15-95.08	1091	97.50	96.40-98.33	-3.76	-5.70,1.83	2.52	2.31-2.74	2.91	2.73-3.11	0.87
0.77,0.98													
9V**	1079	96.68	95.46-97.66	1080	96.51	95.27-97.51	0.17	-1.55,1.89	2.90	2.71-3.11	3.20	2.96-3.45	0.91
0.81,1.02													
14	1111	99.46	98.83-99.80	1111	99.29	98.60-99.69	0.18	-0.57,0.93	8.97	8.41-9.56	14.66	13.68-15.70	0.61
0.55,0.68													
18C	1075	96.24	94.95-97.28	1078	96.34	95.06-97.36	-0.10	-1.89,1.70	3.47	3.20-3.77	4.04	3.72-4.38	0.86
0.75,0.98													
19F	1114	99.73	99.22-99.94	1111	99.29	98.60-99.69	0.45	-0.22,1.11	7.34	6.97-7.74	5.82	5.50-6.16	1.26
1.16,1.38													
23F	1066	95.43	94.04-96.58	1076	96.16	94.86-97.21	-0.72	-2.63,1.18	3.17	2.96-3.40	3.40	3.17-3.66	0.93
0.83,1.05													
Additional													
vs PCV7 seroresponsef													
1	1077	96.42	95.16-97.43	709	63.36	60.46-66.19	33.06	30.03,36.08	1.91	1.79-2.03	0.33	0.31-0.36	5.71
5.13,6.34													
3	1042	93.29	91.66-94.68	536	47.90	44.94-50.87	45.39	42.11,48.66	1.17	1.11-1.24	0.35	0.33-0.38	3.31

3.05,3.59
5 1046 93.64 92.05-95.00 401 35.84 33.02-38.72 57.81 54.66,60.96 1.22 1.13-1.30 0.11 0.10-0.13 10.96
9.48,12.67
6A 1053 94.27 92.74-95.56 917 81.95 79.57-84.16 12.32 9.69,14.96 1.92 1.78-2.06 0.72 0.65-0.79 2.67
2.36,3.02
7F 1088 97.40 96.29-98.25 496 44.33 41.39-47.29 53.08 50.02,56.13 4.72 4.42-5.04 0.13 0.11-0.15 35.80
30.58,41.89
19A 1115 99.82 99.35-99.98 1093 97.68 96.61-98.48 2.14 1.23,3.06 5.06 4.80-5.33 1.59 1.48-1.70 3.19
2.92,3.47
vs PCV7 reference g
1 1077 96.42 95.16-97.43 1076 96.16 94.86-97.21 0.26 -1.53,2.05 1.91 1.79-2.03 2.91 2.73-3.11 0.66
0.59,0.73
3 1042 93.29 91.66-94.68 1076 96.16 94.86-97.21 -2.87 -4.99,-0.76 1.17 1.11-1.24 2.91 2.73-3.11 0.40
0.36,0.44
5 1046 93.64 92.05-95.00 1076 96.16 94.86-97.21 -2.51 -4.60,-0.43 1.22 1.13-1.30 2.91 2.73-3.11 0.42
0.37,0.46
6A 1053 94.27 92.74-95.56 1076 96.16 94.86-97.21 -1.89 -3.91,0.14 1.92 1.78-2.06 2.91 2.73-3.11 0.66
0.59,0.74
7F 1088 97.40 96.29-98.25 1076 96.16 94.86-97.21 1.25 -0.43,2.92 4.72 4.42-5.04 2.91 2.73-3.11 1.62
1.46,1.80
19A 1115 99.82 99.35-99.98 1076 96.16 94.86-97.21 3.66 2.34,4.98 5.06 4.80-5.33 2.91 2.73-3.11 1.74
1.58,1.91
N=number of subjects included in FAS for PCV13-TT and PCV7. **The subjects for PCV13-TT in FAS were 1116
for IgG antibody GMC
a. number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT
group.
b.
number of subjects with a determinate IgG antibody concentration to the given serotype for PCV7 group.
c. The rate difference of PCV13-TT – PCV7.
d. GMC ratio: PCV13-TT to PCV7.
e. Two-sided 95% CI is for additional serotypes, and two-sided 97.5% CI is for common serotypes.
f. Difference in proportions: (PCV13-TT – PCV7 seroresponse) expressed as a percentage.
GMC ratio: PCV13-TT to PCV7 seroresponse. IgG antibody concentration ≥ 0.35 $\mu\text{g/mL}$ was calculated
respectively, with 95% CI given.
g. Difference in proportions: (PCV13-TT – PCV7 reference) expressed as a percentage. For the additional
serotypes, the reference value is derived from serotype 23F from the PCV7 group.
GMC ratio: PCV13-TT to PCV7 reference. For the additional serotypes, the reference value is derived from
serotype 6B from the PCV7 group

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2.7.3.2.2.1.4.2. Comparison of percentage of subjects achieving IgG antibody concentration $\geq$
1.0 $\mu\text{g/mL}$ with comparison to the PCV7 seroresponse
Primary series
Data for the 7 common serotypes
With the exception of serotypes 6B, all common serotypes met the noninferiority criterion after
primary series. For serotype 6B, the lower limit of the 2-sided 95% CI for rate difference of
subjects achieving antibody concentration ≥ 1.0 $\mu\text{g/mL}$ was -11.51(Table 58)
Data for the 6 additional serotypes
All common serotypes met the noninferiority criterion after primary series. The lower limit of the
2-sided 95% CI was all greater than 0.

Table 58. Comparison of percentage of subjects aged 3-71 months achieving a pneumococcal IgG antibody
concentration ≥ 0.35 $\mu\text{g/mL}$ and GMCs post booster dose(PPS)
IgG antibody concentration ≥ 1.0 $\mu\text{g/mL}$

PCV13-TT (N=1043) PCV7 (N=1064) Rate difference c
Serotype Na % 95% CI Nb % 95% CI % 95% CI /97.5%CI d
Common
4 1024 98.18 97.17-98.90 1050 98.68 97.80-99.28 -0.51 -1.57,0.56
6B 822 78.81 76.20-81.25 927 87.12 84.96-89.08 -8.31 -11.51,-5.12
9V 959 91.95 90.13-93.53 1002 94.17 92.59-95.50 -2.23 -4.40,-0.06
14 1036 99.33 98.62-99.73 1059 99.53 98.91-99.85 -0.20 -0.84,0.44
18C 991 95.01 93.51-96.25 1020 95.86 94.49-96.98 -0.85 -2.63,0.93
19F 1030 98.75 97.88-99.33 1041 97.84 96.77-98.62 0.92 -0.19,2.02
23F 949 90.99 89.08-92.66 983 92.39 90.63-93.91 -1.40 -3.76,0.96
Additional serotypes (vs PCV7 seroresponse)
1 915 87.73 85.58-89.66 104 9.77 8.06-11.72 77.95 75.28,80.63
3 650 62.32 59.30-65.27 109 10.24 8.49-12.22 52.08 48.62,55.54
5 682 65.39 62.41-68.28 37 3.48 2.46-4.76 61.91 58.82,65.00
6A 825 79.10 76.50-81.53 460 43.23 40.23-46.27 35.87 32.00,39.73
7F 1021 97.89 96.82-98.67 135 12.69 10.75-14.84 85.20 83.02,87.38
19A 1029 98.66 97.76-99.26 741 69.64 66.78-72.40 29.01 26.17,31.86
N=number of subjects included in PPS for PCV13-TT and PCV7.

a. number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT
group.
b. number of subjects with a determinate IgG antibody concentration to the given serotype for PCV7 group.
c. The rate difference of PCV13-TT – PCV7.
d. Two-sided 95% CI is for additional serotypes, and two-sided 97.5% CI is for common serotypes.

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Table 59 Comparison of percentage of subjects aged 3-71 months achieving a pneumococcal IgG antibody
concentration ≥ 0.35 $\mu\text{g/mL}$ and GMCs post booster dose(FAS)
IgG antibody concentration ≥ 1.0 $\mu\text{g/mL}$

PCV13-TT (N=1117) PCV7 (N=1119) Rate difference c
Serotype Na % 95% CI Nb % 95% CI % 95% CI /97.5%CI d
Common
4 1060 94.90 93.44-96.11 1071 95.71 94.35-96.82 -0.81 -2.57,0.94

6B 852 76.28 73.67-78.74 948 84.72 82.47-86.78 -8.44 -11.71,-5.18
 9V 991 88.80 86.80-90.59 1023 91.42 89.62-93.00 -2.62 -5.09,-0.15
 14 1082 96.87 95.67-97.81 1093 97.68 96.61-98.48 -0.81 -2.16,0.54
 18C 1023 91.58 89.80-93.15 1041 93.03 91.38-94.45 -1.44 -3.65,0.76
 19F 1075 96.24 94.95-97.28 1072 95.80 94.45-96.90 0.44 -1.18,2.06
 23F 980 87.74 85.67-89.60 1002 89.54 87.60-91.28 -1.81 -4.44,0.82
 Additional serotypes (vs PCV7 seroresponse)f
 1 947 84.78 82.54-86.84 107 9.56 7.90-11.44 75.22 72.50,77.94
 3 671 60.07 57.13-62.96 118 10.55 8.81-12.49 49.53 46.14,52.92
 5 699 62.58 59.67-65.42 38 3.40 2.41-4.63 59.18 56.15,62.21
 6A 859 76.90 74.32-79.35 476 42.54 39.62-45.50 34.36 30.56,38.17
 7F 1060 94.90 93.44-96.11 136 12.15 10.30-14.21 82.74 80.43,85.05
 19A 1076 96.33 95.05-97.35 769 68.72 65.91-71.43 27.61 24.68,30.54
 N=number of subjects included in FAS for PCV13-TT and PCV7.

- number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT group.
 - number of subjects with a determinate IgG antibody concentration to the given serotype for PCV7 group.
- The number of subjects for 9V was 1116.
- The rate difference of PCV13-TT – PCV7 reference.
 - Two-sided 95% CI is for additional serotypes, and two-sided 97.5% CI is for common serotypes.

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2.7.3.2.2.2. Pneumococcal immunogenicity results for non-primary population
 2.7.3.2.2.2.1. Pneumococcal immunogenicity results for subjects aged 7-11 months in PCV13-TT group and PCV7 group
 2.7.3.2.2.2.1.1. Comparison of percentage of subjects achieving IgG antibody concentration $\geq 0.35 \mu\text{g/mL}$ and the IgG antibody geometric mean concentrations (GMCs)
 Post dose 3 IgG pneumococcal immunogenicity results with comparison to the PCV7 seroresponse for the 7 common serotypes
 The IgG antibody concentration $\geq 0.35 \mu\text{g/mL}$ in the PCV13-TT group were all higher than 96%. The lower limit of rate difference for the proportion of responders $\geq 0.35 \mu\text{g/mL}$ was all greater than -10%. The GMCs for all common serotypes were greater than $2.0 \mu\text{g/mL}$. With the exception of serotype 14, 6 of the 7 common serotypes met the non-inferiority criterion for GMR after primary series. For serotype 14, the GMR was 0.53, with a lower limit of the 2-sided 97.5% CI of 0.45 (Table 60).

Post dose 3 IgG pneumococcal immunogenicity results with comparison to the PCV7 seroresponse for the 6 additional serotypes
 All additional serotypes met the pre-specified superiority criteria for the IgG antibody concentration $\geq 0.35 \mu\text{g/mL}$ and GMC.

Post dose 3 IgG pneumococcal immunogenicity results with comparison to the PCV7 reference for the 6 additional serotypes
 The PCV7 reference value for the 6 additional serotypes was based on serotype 23F from the PCV7 group. For serotype 23F, the proportion of responders was 96.83%. All additional serotypes met the non-inferiority criterion. The PCV7 reference value for the 6 additional serotypes was serotype 6B from the PCV7 group. For serotype 6B, the GMC was 2.20. All the 6 additional serotypes met the non-inferiority criterion for GMR (lower limit of the 2-sided 95% CI for the GMR >0.5).

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Table 60. Comparison of percentage of subjects aged 7-11 months achieving a pneumococcal IgG antibody concentration $\geq 0.35 \mu\text{g/mL}$ and GMCs post dose 3 of the primary series (PPS)

	IgG antibody concentration $\geq 0.35 \mu\text{g/mL}$	IgG antibody GMC	PCV13-TT (N=185)	PCV7 (N=189)	Rate difference c	PCV13-TT (N=185)	PCV7 (N=189)	GMC Ratio	Serotype Na	% 95% CI	Nb	% 95% CI	% 95% CI	/97.5%CI	GMC 95% CI	GMC 95% CI	Ratio d	95%CI e
Common																		
4	185	100.0	98.03-100.0	189	100.0	98.07-100.0	0.00	-1.93,1.97	4.26	3.86-4.71	4.23	3.88-4.62	1.01					
	0.88,1.15																	
6B	179	96.76	93.07-98.80	188	99.47	97.09-99.99	-2.71	-5.47,0.04	2.90	2.46-3.42	2.20	1.92-2.52	1.32					
	1.07,1.63																	
9V	180	97.30	93.81-99.12	185	97.88	94.67-99.42	-0.59	-3.70,2.52	2.38	2.09-2.71	2.76	2.43-3.12	0.86					
	0.72,1.03																	
14	185	100.0	98.03-100.0	189	100.0	98.07-100.0	0.00	-1.93,1.97	8.54	7.60-9.60	16.17	14.53-17.99	0.53					
	0.45,0.62																	
18C	184	99.46	97.03-99.99	188	99.47	97.09-99.99	-0.01	-1.49,1.47	3.78	3.30-4.34	3.44	3.08-3.84	1.10					
	0.92,1.31																	
19F	185	100.0	98.03-100.0	189	100.0	98.07-100.0	0.00	-1.93,1.97	7.21	6.43-8.08	4.09	3.70-4.52	1.76					
	1.52,2.05																	
23F	180	97.30	93.81-99.12	183	96.83	93.22-98.83	0.47	-2.95,3.89	3.56	3.10-4.10	2.70	2.36-3.09	1.32					
	1.09,1.60																	
Additional vs PCV7 seroresponse f																		
1	183	98.92	96.15-99.87	143	75.66	68.90-81.60	23.26	16.96,29.55	2.23	2.03-2.45	0.47	0.40-0.56	4.69					
	3.89,5.65																	
3	181	97.84	94.56-99.41	117	61.90	54.57-68.86	35.93	28.70,43.17	1.32	1.20-1.45	0.46	0.41-0.51	2.89					
	2.51,3.32																	
5	169	91.35	86.34-94.98	84	44.44	37.23-51.83	46.91	38.75,55.07	0.82	0.73-0.91	0.19	0.15-0.25	4.30					
	3.23,5.72																	
6A	177	95.68	91.66-98.11	147	77.78	71.17-83.49	17.90	11.29,24.51	1.79	1.51-2.12	0.54	0.41-0.72	3.31					
	2.38,4.62																	
7F	184	99.46	97.03-99.99	87	46.03	38.77-53.42	53.43	46.24,60.61	5.73	5.04-6.51	0.14	0.10-0.20	39.88					
	27.69,57.43																	
19A	185	100.0	98.03-100.0	185	97.88	94.67-99.42	2.12	0.06,4.17	5.06	4.54-5.64	1.50	1.29-1.74	3.37					

2.81,4.05
vs PCV7 reference g
1 183 98.92 96.15-99.87 183 96.83 93.22-98.83 2.09 -0.82,5.00 2.23 2.03-2.45 2.20 1.92-2.52 1.01
0.86,1.20
3 181 97.84 94.56-99.41 183 96.83 93.22-98.83 1.01 -2.25,4.27 1.32 1.20-1.45 2.20 1.92-2.52 0.60
0.51,0.71
5 169 91.35 86.34-94.98 183 96.83 93.22-98.83 -5.47 -10.23,-0.71 0.82 0.73-0.91 2.20 1.92-2.52 0.37
0.31,0.44
6A 177 95.68 91.66-98.11 183 96.83 93.22-98.83 -1.15 -5.00,2.70 1.79 1.51-2.12 2.20 1.92-2.52 0.81
0.65,1.01
7F 184 99.46 97.03-99.99 183 96.83 93.22-98.83 2.63 -0.08,5.35 5.73 5.04-6.51 2.20 1.92-2.52 2.61
2.16,3.14
19A 185 100.0 98.03-100.0 183 96.83 93.22-98.83 3.17 0.68,5.67 5.06 4.54-5.64 2.20 1.92-2.52 2.30
1.93,2.74
N=number of subjects included in PPS for PCV13-TT and PCV7.
a. number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT group.
b. number of subjects with a determinate IgG antibody concentration to the given serotype for PCV7 group.
c. The rate difference of PCV13-TT – PCV7.
d. GMC ratio: PCV13-TT to PCV7.
e. Two-sided 95% CI is for additional serotypes, and two-sided 97.5% CI is for common serotypes.
f. Difference in proportions: (PCV13-TT – PCV7 seroresponse) expressed as a percentage. GMC ratio: PCV13-TT to PCV7 seroresponse.
g. Difference in proportions: (PCV13-TT – PCV7 reference) expressed as a percentage. For the additional serotypes, the reference value is derived from serotype 23F from the PCV7 group. GMC ratio: PCV13-TT to PCV7 reference. For the additional serotypes, the reference value is derived from serotype 6B from the PCV7 group.

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Table 61. Comparison of percentage of subjects aged 7-11 months achieving a pneumococcal IgG antibody concentration $\geq 0.35 \mu\text{g/mL}$ and GMCs post dose 3 of the primary series (FAS)
IgG antibody concentration $\geq 0.35 \mu\text{g/mL}$ IgG antibody GMC
PCV13-TT (N=198) PCV7 (N=200) Rate difference c PCV13-TT (N=198) PCV7 (N=200) GMC Ratio
Serotype Na % 95% CI Nb % 95% CI % 95% CI /97.5%CI GMC 95% CI GMC 95% CI Ratio d 95%CI e
PCV7
4 191 96.46 92.85-98.57 191 95.50 91.63-97.92 0.96 -2.89,4.82 3.34 2.73-4.10 3.08 2.47-3.85 1.09
0.80,1.47
6B 187 94.44 90.28-97.19 195 97.50 94.26-99.18 -3.06 -6.91,0.80 2.75 2.33-3.25 2.02 1.76-2.33 1.36
1.09,1.69
9V** 187 94.92 90.86-97.54 187 93.50 89.14-96.49 1.42 -3.17,6.01 2.04 1.70-2.45 2.11 1.70-2.61 0.97
0.73,1.28
14 197 99.49 97.22-99.99 197 98.50 95.68-99.69 0.99 -0.96,2.95 7.82 6.85-8.93 13.54 11.52-15.91 0.58
0.47,0.71
18C 190 95.96 92.19-98.24 190 95.00 91.00-97.58 0.96 -3.12,5.04 3.01 2.43-3.73 2.53 2.02-3.17 1.19
0.87,1.62
19F 198 100.0 98.15-100.0 198 99.00 96.43-99.88 1.00 -0.38,2.38 6.59 5.80-7.48 3.75 3.36-4.18 1.76
1.49,2.08
23F 187 94.44 90.28-97.19 187 93.50 89.14-96.49 0.94 -3.73,5.62 3.19 2.73-3.72 2.36 2.01-2.77 1.35
1.08,1.69
Additional
vs PCV7 seroresponse f
1 190 95.96 92.19-98.24 147 73.50 66.81-79.48 22.46 15.76,29.16 2.07 1.87-2.30 0.46 0.39-0.54 4.51
3.74,5.43
3 186 93.94 89.65-96.83 120 60.00 52.85-66.85 33.94 26.38,41.50 1.19 1.06-1.34 0.42 0.37-0.48 2.80
2.36,3.33
5 175 88.38 83.08-92.49 85 42.50 35.56-49.67 45.88 37.71,54.06 0.77 0.68-0.87 0.18 0.14-0.23 4.29
3.21,5.72
6A 184 92.93 88.42-96.08 149 74.50 67.87-80.39 18.43 11.41,25.45 1.68 1.41-2.00 0.51 0.39-0.67 3.31
2.39,4.57
7F 192 96.97 93.52-98.88 90 45.00 37.98-52.18 51.97 44.67,59.27 5.11 4.41-5.92 0.14 0.10-0.20 35.48
24.84,50.67
19A 198 100.0 98.15-100.0 196 98.00 94.96-99.45 2.00 0.06,3.94 4.75 4.24-5.33 1.46 1.27-1.68 3.26
2.72,3.91
vs PCV7 reference g
1 190 95.96 92.19-98.24 187 93.50 89.14-96.49 2.46 -1.92,6.84 2.07 1.87-2.30 2.02 1.76-2.33 1.02
0.86,1.22
3 186 93.94 89.65-96.83 187 93.50 89.14-96.49 0.44 -4.33,5.21 1.19 1.06-1.34 2.02 1.76-2.33 0.59
0.49,0.71
5 175 88.38 83.08-92.49 187 93.50 89.14-96.49 -5.12 -10.74,0.50 0.77 0.68-0.87 2.02 1.76-2.33 0.38
0.31,0.46
6A 184 92.93 88.42-96.08 187 93.50 89.14-96.49 -0.57 -5.51,4.37 1.68 1.41-2.00 2.02 1.76-2.33 0.83
0.66,1.04
7F 192 96.97 93.52-98.88 187 93.50 89.14-96.49 3.47 -0.70,7.64 5.11 4.41-5.92 2.02 1.76-2.33 2.52
2.06,3.09
19A 198 100.0 98.15-100.0 187 93.50 89.14-96.49 6.50 3.08,9.92 4.75 4.24-5.33 2.02 1.76-2.33 2.35
1.96,2.81
N=number of subjects included in PPS for PCV13-TT and PCV7.
a. number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT group.
b. number of subjects with a determinate IgG antibody concentration to the given serotype for PCV7 group.
c. The rate difference of PCV13-TT – PCV7.
d. GMC ratio: PCV13-TT to PCV7.
e. Two-sided 95% CI is for additional serotypes, and two-sided 97.5% CI is for common serotypes.
f. Difference in proportions: (PCV13-TT – PCV7 seroresponse) expressed as a percentage. GMC ratio: PCV13-TT to PCV7 seroresponse.
g. Difference in proportions: (PCV13-TT – PCV7 reference) expressed as a percentage. For the additional serotypes, the reference value is derived from serotype 23F from the PCV7 group. GMC ratio: PCV13-TT to PCV7 reference. For the additional serotypes, the reference value is derived from serotype 6B

from the PCV7 group.

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Booster Dose

Post dose 4 IgG pneumococcal immunogenicity results with comparison to the PCV7 seroresponse for the 7 common serotypes
All common serotypes met the noninferiority criterion after Booster dose for the IgG antibody concentration $\geq 0.35 \mu\text{g/mL}$, the IgG antibody concentration ≥ 1.0 and GMC (Table 62).

Post dose 4 IgG pneumococcal immunogenicity results with comparison to the PCV7 seroresponse for the 6 additional serotypes
With the exception of serotype 19A, all common serotypes met the superiority criterion after booster dose for the IgG antibody concentration $\geq 0.35 \mu\text{g/mL}$. For serotype 19A, the rate difference was 0.00% with a lower limit of the 2-sided 95% CI of -1.93%. all common serotypes met the superiority criterion after booster dose for the IgG antibody concentration $\geq 1.0 \mu\text{g/mL}$. and GMC. The geometric mean fold rises (GMFRs) of antibody concentrations from pre- to post-boost for the 7 common serotypes ranged from 1.27 (serotype 4) to 1.96 (6B) in the PCV13-TT group. For the 6 additional serotypes, the fold rises ranged from 1.28 (serotype 7F) to 1.86 (serotype 6A) in the PCV13-TT (Table 63).

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Table 62. Comparison of percentage of subjects aged 7-11 months achieving a pneumococcal IgG antibody concentration $\geq 0.35 \mu\text{g/mL}$ and GMCs post booster dose

(PPS)															
IgG antibody concentration $\geq 0.35 \mu\text{g/mL}$ IgG antibody GMC															
PCV13-TT (N=186) PCV7 (N=189) Rate difference c PCV13-TT (N=186) PCV7 (N=189) GMC Ratio															
Serotype	Na	%	95% CI	Nb	%	95% CI	%	95% CI	/97.5%CI	GMC	95% CI	GMC	95% CI	Ratio	d 95% CI /97.5%CIe
Common															
4	186	100.0	98.04-100.0	189	100.0	98.07-100.0	0.00	-1.93,1.96	3.06	2.76-3.38	3.12	2.85-3.41	0.98	0.86,1.12	
6B	186	100.0	98.04-100.0	189	100.0	98.07-100.0	0.00	-1.93,1.96	5.62	5.01-6.32	4.57	4.08-5.11	1.23	1.05,1.45	
9V	185	99.46	97.04-99.99	189	100.0	98.07-100.0	-0.54	-1.59,0.51	2.91	2.58-3.29	3.67	3.31-4.07	0.79	0.68,0.93	
14	186	100.0	98.04-100.0	189	100.0	98.07-100.0	0.00	-1.93,1.96	13.94	12.54-15.51	19.49	17.94-21.17	0.72	0.63,0.82	
18C	185	99.46	97.04-99.99	188	99.47	97.09-99.99	-0.01	-1.48,1.47	2.97	2.62-3.36	3.32	3.00-3.67	0.89	0.76,1.05	
19F	186	100.0	98.04-100.0	189	100.0	98.07-100.0	0.00	-1.93,1.96	7.54	6.84-8.31	3.92	3.56-4.31	1.92	1.68,2.20	
23F	186	100.0	98.04-100.0	189	100.0	98.07-100.0	0.00	-1.93,1.96	4.53	4.01-5.12	4.21	3.77-4.71	1.08	0.91,1.27	
Additional															
vs PCV7 seroresponse f															
1	186	100.0	98.04-100.0	156	82.54	76.36-87.67	17.46	12.05,22.87	1.98	1.82-2.15	0.58	0.53-0.65	3.38	2.96,3.86	
3	184	98.92	96.17-99.87	130	68.78	61.65-75.31	30.14	23.37,36.91	1.51	1.38-1.66	0.53	0.48-0.59	2.85	2.48,3.28	
5	173	93.01	88.34-96.23	77	40.74	33.67-48.11	52.27	44.36,60.18	0.86	0.78-0.95	0.24	0.20-0.29	3.61	2.90,4.50	
6A	183	98.39	95.36-99.67	176	93.12	88.53-96.29	5.27	1.23,9.30	2.99	2.64-3.37	1.63	1.43-1.87	1.83	1.53,2.19	
7F	186	100.0	98.04-100.0	138	73.02	66.09-79.20	26.98	20.66,33.31	4.91	4.46-5.39	0.46	0.39-0.55	10.61	8.71,12.93	
19A	186	100.0	98.04-100.0	189	100.0	98.07-100.0	0.00	-1.93,1.96	6.28	5.71-6.91	2.17	1.96-2.40	2.90	2.52,3.33	

N=number of subjects included in PPS for PCV13-TT and PCV7.

- number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT group.
- number of subjects with a determinate IgG antibody concentration to the given serotype for PCV7 group.
- The rate difference of PCV13-TT – PCV7 seroresponse.
- GMC ratio: PCV13-TT to PCV7 seroresponse.
- Two-sided 95% CI is for additional serotypes, and two-sided 97.5% CI is for common serotypes.
- Difference in proportions: (PCV13-TT – PCV7) expressed as a percentage. GMC ratio: PCV13-TT to PCV7 .

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Table 63. Comparison of percentage of subjects aged 7-11 months for the geometric mean fold rises (GMFRs) of antibody concentrations (PPS)

Serotype PCV13-TT (N=185) PCV7 (N=188)						
P						
GMFRs 95%CI			GMFRs 95%CI			
4	1.27	1.20-1.34	1.31	1.24-1.38	0.4037	
6B	1.96	1.82-2.11	2.70	2.44-2.98	<0.0001	
9V	1.63	1.53-1.74	1.73	1.62-1.86	0.2039	
Common	14	1.64	1.51-1.78	1.45	1.32-1.59	0.0445
18C	1.31	1.22-1.41	1.54	1.44-1.65	0.0010	
19F	1.41	1.30-1.54	1.38	1.28-1.49	0.7036	
23F	1.66	1.54-1.78	2.02	1.84-2.21	0.0010	
1	1.31	1.24-1.39	1.15	1.07-1.25	0.0091	
3	1.61	1.52-1.71	1.64	1.51-1.78	0.7019	
5	1.51	1.41-1.61	0.87	0.73-1.04	<0.0001	
Additional						
6A	1.86	1.68-2.06	2.09	1.82-2.40	0.1793	

7F 1.28 1.18-1.39 1.04 0.89-1.22 0.0239
19A 1.59 1.42-1.78 1.13 1.01-1.26 <0.0001

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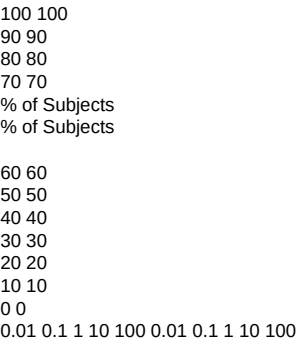
2.7.3.2.2.1.2. Pneumococcal Serotype antibody concentrations in the subjects aged 7-11 months in PCV13-TT group and PCV7 group

(1) Primary Series
The results of pneumococcal serotype antibody concentrations for subjects aged 7-11 months after primary series in PCV13-TT group and PCV7 group were shown in Table 64. to Table 76. The reverse cumulative distribution curves were shown in Figure 27 to 39.

Table 64. Distribution of the pneumococcal serotype 4 IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group post primary series

FAS N (%)	PPS N (%)								
PCV13-TT	PCV13-TT								
Concentration	PCV7 (N=200)	PCV7 (N=189)							
(N=198)	(N=185)								
n	%	n	%	n	%	n	%	n	%
≥0.01	192	96.97	192	96.00	185	100.0	189	100.0	
≥0.1	192	96.97	192	96.00	185	100.0	189	100.0	
≥1	188	94.95	187	93.50	182	98.38	185	97.88	
≥10	20	10.10	12	6.00	19	10.27	12	6.35	
≥100	0	0.00	0	0.00	0	0.00	0	0.00	
N=number of subjects included in FAS and PPS for vaccine group.									
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.									

Serotype 4, After Primary Series (FAS) Serotype 4, After Primary Series (PPS)



Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

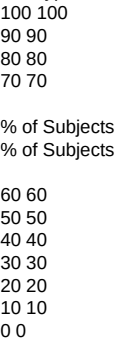
Figure 27 Reverse cumulative distribution curves, pneumococcal serotype 4, IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group - post primary series.

Table 65. Distribution of the pneumococcal serotype 6B IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group post primary series

FAS N (%)	PPS N (%)								
PCV13-TT	PCV13-TT								
Concentration	PCV7 (N=200)	PCV7 (N=189)							
(N=198)	(N=185)								
n	%	n	%	n	%	n	%	n	%
≥0.01	197	99.49	200	100.0	184	99.46	189	100.0	
≥0.1	197	99.49	200	100.0	184	99.46	189	100.0	
≥1	160	80.81	143	71.50	155	83.78	142	75.13	
≥10	19	9.60	11	5.50	17	9.19	11	5.82	
≥100	0	0.00	0	0.00	0	0.00	0	0.00	
N=number of subjects included in FAS and PPS for vaccine group.									
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.									

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Serotype 6B, After Primary Series (FAS) Serotype 6B, After Primary Series (PPS)



PCV13-TT PCV7 PCV13-TT PCV7

Figure 30 Reverse cumulative distribution curves, pneumococcal serotype 14, IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group - post primary series.

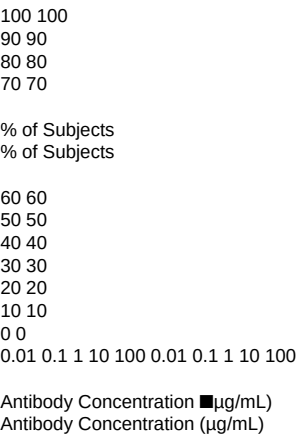
Table 68. Distribution of the pneumococcal serotype 18C IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group post primary series

FAS N (%)	PPS N (%)
PCV13-TT	PCV13-TT
Concentration PCV7 (N=200)	PCV7 (N=189)
(N=198)	(N=185)
n %	n %
≥0.01	191 96.46
≥0.1	191 96.46
≥1	181 91.41
≥10	19 9.60
≥100	0 0.00

N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

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Serotype 18C, After Primary Series (FAS) Serotype 18C, After Primary Series (PPS)



PCV13-TT PCV7 PCV13-TT PCV7

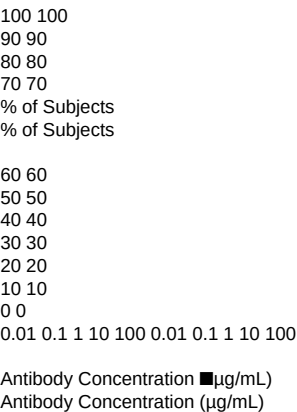
Figure 31 Reverse cumulative distribution curves, pneumococcal serotype 18C, IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group - post primary series.

Table 69. Distribution of the pneumococcal serotype 19F IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group post primary series

FAS N (%)	PPS N (%)
PCV13-TT	PCV13-TT
Concentration PCV7 (N=200)	PCV7 (N=189)
(N=198)	(N=185)
n %	n %
≥0.01	198 100.0
≥0.1	198 100.0
≥1	187 94.44
≥10	66 33.33
≥100	0 0.00

N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

Serotype 19F, After Primary Series (FAS) Serotype 19F, After Primary Series (PPS)



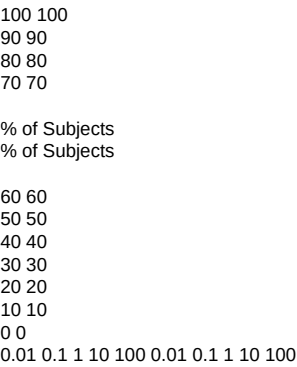
PCV13-TT PCV7 PCV13-TT PCV7

Figure 32 Reverse cumulative distribution curves, pneumococcal serotype 19F, IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group - post primary series.

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Table 70. Distribution of the pneumococcal serotype 23F IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group post primary series
FAS N (%) PPS N (%)
PCV13-TT PCV13-TT
Concentration PCV7 (N=200) PCV7 (N=189)
(N=198) (N=185)
n % n % n % n %
≥0.01 198 100.0 199 99.50 185 100.0 189 100.0
≥0.1 198 100.0 199 99.50 185 100.0 189 100.0
≥1 173 87.37 166 83.00 169 91.35 164 86.77
≥10 26 13.13 14 7.00 25 13.51 14 7.41
≥100 0 0.00 0 0.00 0 0.00 0 0.00
N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

Serotype 23F, After Primary Series (FAS) Serotype 23F, After Primary Series (PPS)



Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

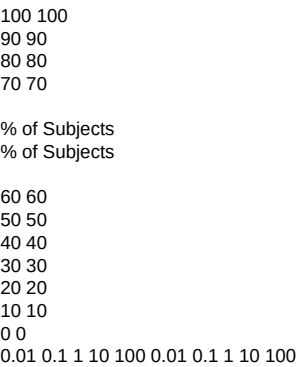
PCV13-TT PCV7 PCV13-TT PCV7

Figure 33 Reverse cumulative distribution curves, pneumococcal serotype 23F, IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group - post primary series.

Table 71. Distribution of the pneumococcal serotype 1 IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group post primary series
FAS N (%) PPS N (%)
PCV13-TT PCV13-TT
Concentration PCV7 (N=200) PCV7 (N=189)
(N=198) (N=185)
n % n % n % n %
≥0.01 198 100.0 195 97.50 185 100.0 184 97.35
≥0.1 198 100.0 193 96.50 185 100.0 183 96.83
≥1 176 88.89 25 12.50 170 91.89 25 13.23
≥10 3 1.52 0 0.00 3 1.62 0 0.00
≥100 0 0.00 0 0.00 0 0.00 0 0.00
N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

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Serotype 1, After Primary Series (FAS) Serotype 1, After Primary Series (PPS)



Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

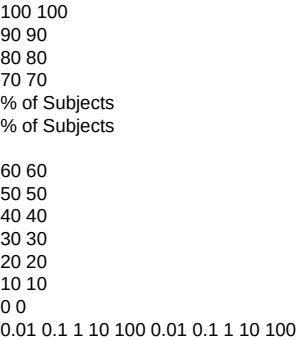
PCV13-TT PCV7 PCV13-TT PCV7

Figure 34 Reverse cumulative distribution curves, pneumococcal serotype 1, IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group - post primary series.

Table 72 Distribution of the pneumococcal serotype 3 IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group post primary series

FAS N (%)	PPS N (%)
PCV13-TT	PCV13-TT
Concentration PCV7 (N=200)	PCV7 (N=189)
(N=198)	(N=185)
n %	n %
n %	n %
≥0.01	198 100.0
199 99.50	185 100.0
189 100.0	
≥0.1	192 96.97
191 95.50	184 99.46
187 98.94	
≥1	127 64.14
26 13.00	124 67.03
24 12.70	
≥10	0 0.00
0 0.00	0 0.00
0 0.00	
N=number of subjects included in FAS and PPS for vaccine group.	
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.	

Serotype 3, After Primary Series (FAS) Serotype 3, After Primary Series (PPS)



Antibody Concentration (µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

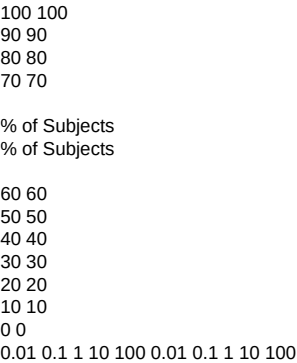
Figure 35. Reverse cumulative distribution curves, pneumococcal serotype 3, IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group - post primary series.

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Table 73. Distribution of the pneumococcal serotype 5 IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group post primary series

FAS N (%)	PPS N (%)
PCV13-TT	PCV13-TT
Concentration PCV7 (N=200)	PCV7 (N=189)
(N=198)	(N=185)
n %	n %
n %	n %
≥0.01	196 98.99
172 86.00	184 99.46
164 86.77	
≥0.1	196 98.99
172 86.00	184 99.46
164 86.77	
≥1	72 36.36
6 3.00	69 37.30
6 3.17	
≥10	0 0.00
0 0.00	0 0.00
0 0.00	
N=number of subjects included in FAS and PPS for vaccine group.	
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.	

Serotype 5, After Primary Series (FAS) Serotype 5, After Primary Series (PPS)



Antibody Concentration (µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

Figure 36 Reverse cumulative distribution curves, pneumococcal serotype 5, IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group - post primary series.

Table 74. Distribution of the pneumococcal serotype 6A IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group post primary series

FAS N (%) PPS N (%)

PCV13-TT PCV13-TT

Concentration PCV7 (N=200) PCV7 (N=189)

(N=198) (N=185)

n % n % n % n %

≥0.01 195 98.48 180 90.00 182 98.38 170 89.95

≥0.1 195 98.48 179 89.50 182 98.38 169 89.42

≥1 143 72.22 74 37.00 139 75.14 73 38.62

≥10 8 4.04 4 2.00 6 3.24 4 2.12

≥100 0 0.00 0 0.00 0 0.00 0 0.00

N=number of subjects included in FAS and PPS for vaccine group.

n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

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Serotype 6A, After Primary Series (FAS) Serotype 6A, After Primary Series (PPS)

100 100

90 90

80 80

70 70

% of Subjects

% of Subjects

60 60

50 50

40 40

30 30

20 20

10 10

0 0

0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration ■µg/mL)

Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

Figure 37 Reverse cumulative distribution curves, pneumococcal serotype 6A, IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group - post primary series.

Table 75 Distribution of the pneumococcal serotype 7F IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group post primary series

FAS N (%) PPS N (%)

PCV13-TT PCV13-TT

Concentration PCV7 (N=200) PCV7 (N=189)

(N=198) (N=185)

n % n % n % n %

≥0.01 197 99.49 159 79.50 184 99.46 149 78.84

≥0.1 197 99.49 158 79.00 184 99.46 149 78.84

≥1 189 95.45 19 9.50 183 98.92 19 10.05

≥10 34 17.17 1 0.50 33 17.84 1 0.53

≥100 0 0.00 0 0.00 0 0.00 0 0.00

N=number of subjects included in FAS and PPS for vaccine group.

n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

Serotype 7F, After Primary Series (FAS) Serotype 7F, After Primary Series (PPS)

100 100

90 90

80 80

70 70

% of Subjects

% of Subjects

60 60

50 50

40 40

30 30

20 20

10 10

0 0

0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration ■µg/mL)

Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

Figure 38 Reverse cumulative distribution curves, pneumococcal serotype 7F, IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group - post primary series.

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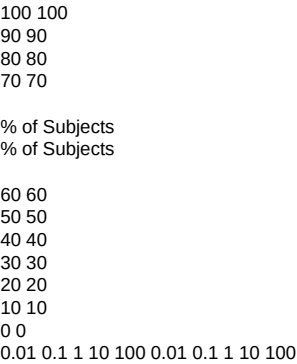
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Table 76. Distribution of the pneumococcal serotype 23F IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group post primary series

FAS N (%)	PPS N (%)
PCV13-TT	PCV13-TT
Concentration PCV7 (N=200)	PCV7 (N=189)
(N=198)	(N=185)
n %	n %
n %	n %
n %	n %
≥0.01	198 100.0
198 99.00	185 100.0
187 98.94	
≥0.1	198 100.0
198 99.00	185 100.0
187 98.94	
≥1	190 95.96
135 67.50	181 97.84
132 69.84	
≥10	32 16.16
4 2.00	31 16.76
4 2.12	
≥100	0 0.00
0 0.00	0 0.00
0 0.00	

N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

Serotype 19A, After Primary Series (FAS) Serotype 19A, After Primary Series (PPS)



Antibody Concentration (µg/mL) Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

Figure 39 Reverse cumulative distribution curves, pneumococcal serotype 19A, IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group - post primary series.

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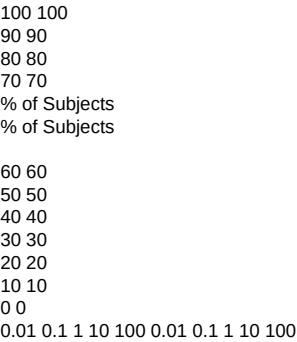
(2) Booster dose
The results of pneumococcal serotype antibody concentrations for subjects aged 7-11 months after booster dose in PCV13-TT group and PCV7 group were shown in Table 77. to Table 89. The reverse cumulative distribution curves were shown in Figure 40 to 52.

Table 77. Distribution of the pneumococcal serotype 4 IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group post booster dose

FAS N (%)	PPS N (%)
PCV13-TT	PCV13-TT
Concentration PCV7 (N=190)	PCV7 (N=189)
(N=186)	(N=186)
n %	n %
n %	n %
n %	n %
≥0.01	186 100.0
190 100.0	186 100.0
189 100.0	
≥0.1	186 100.0
190 100.0	186 100.0
189 100.0	
≥1	178 95.70
187 98.42	178 95.70
186 98.41	
≥10	9 4.84
9 4.74	9 4.84
9 4.76	
≥100	0 0.00
0 0.00	0 0.00
0 0.00	

N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

Serotype 4, After Booster Dose (FAS) Serotype 4, After Booster Dose (PPS)



Antibody Concentration (µg/mL) Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

Figure 40 Reverse cumulative distribution curves, pneumococcal serotype 4, IgG antibody concentrations

for subjects aged 7-11 months in PCV13-TT group and PCV7 group - post booster dose

Table 78. Distribution of the pneumococcal serotype 6B IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group post booster dose

FAS N (%) PPS N (%)

PCV13-TT PCV13-TT

Concentration PCV7 (N=190) PCV7 (N=189)

(N=186) (N=186)

n % n % n % n %

≥0.01 186 100.0 190 100.0 186 100.0 189 100.0

≥0.1 186 100.0 190 100.0 186 100.0 189 100.0

≥1 180 96.77 183 96.32 180 96.77 182 96.30

≥10 37 19.89 29 15.26 37 19.89 29 15.34

≥100 0 0.00 0 0.00 0 0.00 0 0.00

N=number of subjects included in FAS and PPS for vaccine group.

n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

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Serotype 6B, After After Booster Dose (FAS) Serotype 6B, After Booster Dose (PPS)

100 100

90 90

80 80

70 70

% of Subjects

% of Subjects

60 60

50 50

40 40

30 30

20 20

10 10

0 0

0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration ■μg/mL)

Antibody Concentration (μg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

Figure 41 Reverse cumulative distribution curves, pneumococcal serotype 6B, IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group - post booster dose

Table 79. Distribution of the pneumococcal serotype 9V IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group post booster dose

FAS N (%) PPS N (%)

PCV13-TT PCV13-TT

Concentration PCV7 (N=190) PCV7 (N=189)

(N=186) (N=186)

n % n % n % n %

≥0.01 186 100.0 190 100.0 186 100.0 189 100.0

≥0.1 186 100.0 190 100.0 186 100.0 189 100.0

≥1 171 91.94 182 95.79 171 91.94 181 95.77

≥10 10 5.38 13 6.84 10 5.38 13 6.88

≥100 0 0.00 0 0.00 0 0.00 0 0.00

N=number of subjects included in FAS and PPS for vaccine group.

n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

Serotype 9V, After Booster Dose (FAS) Serotype 9V, After Booster Dose (PPS)

100 100

90 90

80 80

70 70

% of Subjects

% of Subjects

60 60

50 50

40 40

30 30

20 20

10 10

0 0

0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration ■μg/mL)

Antibody Concentration (μg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

Figure 42 Reverse cumulative distribution curves, pneumococcal serotype 9V, IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group - post booster dose

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Table 80. Distribution of the pneumococcal serotype 14 IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group post booster dose

FAS N (%) PPS N (%)

PCV13-TT PCV13-TT

Concentration PCV7 (N=190) PCV7 (N=189)

(N=186) (N=186)

n % n % n % n %

≥0.01 186 100.0 190 100.0 186 100.0 189 100.0

≥0.1 186 100.0 190 100.0 186 100.0 189 100.0

≥1 186 100.0 190 100.0 186 100.0 189 100.0

≥10 121 65.05 163 85.79 121 65.05 162 85.71

≥100 1 0.54 0 0.00 1 0.54 0 0.00

N=number of subjects included in FAS and PPS for vaccine group.

n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

Serotype 14, After Booster Dose (FAS) Serotype 14, After Booster Dose (PPS)

100 100

90 90

80 80

70 70

% of Subjects

% of Subjects

60 60

50 50

40 40

30 30

20 20

10 10

0 0

0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration ■μg/mL)

Antibody Concentration (μg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

Figure 43 Reverse cumulative distribution curves, pneumococcal serotype 14, IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group - post booster dose

Table 81. Distribution of the pneumococcal serotype 18C IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group post booster dose

FAS N (%) PPS N (%)

PCV13-TT PCV13-TT

Concentration PCV7 (N=190) PCV7 (N=189)

(N=186) (N=186)

n % n % n % n %

≥0.01 185 99.46 190 100.0 185 99.46 189 100.0

≥0.1 185 99.46 190 100.0 185 99.46 189 100.0

≥1 174 93.55 184 96.84 174 93.55 183 96.83

≥10 8 4.30 8 4.21 8 4.30 8 4.23

≥100 0 0.00 0 0.00 0 0.00 0 0.00

N=number of subjects included in FAS and PPS for vaccine group.

n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

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Serotype 18C, After Booster Dose (FAS) Serotype 18C, After Booster Dose (PPS)

100 100

90 90

80 80

70 70

% of Subjects

% of Subjects

60 60

50 50

40 40

30 30

20 20

10 10

0 0

0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration ■μg/mL)

Antibody Concentration (μg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

Figure 44 Reverse cumulative distribution curves, pneumococcal serotype 18C, IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group - post booster dose

Table 82. Distribution of the pneumococcal serotype 19F IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group post booster dose

FAS N (%) PPS N (%)

PCV13-TT PCV13-TT

Concentration PCV7 (N=190) PCV7 (N=189)

(N=186) (N=186)

n % n % n % n %

≥0.01 186 100.0 190 100.0 186 100.0 189 100.0

≥0.1 186 100.0 190 100.0 186 100.0 189 100.0

≥1 186 100.0 187 98.42 186 100.0 186 98.41

≥10 55 29.57 15 7.89 55 29.57 15 7.94

≥100 0 0.00 0 0.00 0 0.00 0 0.00

N=number of subjects included in FAS and PPS for vaccine group.

n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

Serotype 19F, After Booster Dose (FAS) Serotype 19F, After Booster Dose (PPS)

100 100

90 90

80 80

70 70

% of Subjects

% of Subjects

60 60

50 50

40 40

30 30

20 20

10 10

0 0

0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration ■µg/mL)

Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

Figure 45 Reverse cumulative distribution curves, pneumococcal serotype 19F, IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group - post booster dose

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Table 83. Distribution of the pneumococcal serotype 23F IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group post booster dose

FAS N (%) PPS N (%)

PCV13-TT PCV13-TT

Concentration PCV7 (N=190) PCV7 (N=189)

(N=186) (N=186)

n % n % n % n %

≥0.01 186 100.0 190 100.0 186 100.0 189 100.0

≥0.1 186 100.0 190 100.0 186 100.0 189 100.0

≥1 180 96.77 181 95.26 180 96.77 181 95.77

≥10 28 15.05 19 10.00 28 15.05 19 10.05

≥100 0 0.00 0 0.00 0 0.00 0 0.00

N=number of subjects included in FAS and PPS for vaccine group.

n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

Serotype 23F, After Booster Dose (FAS) Serotype 23F, After Booster Dose (PPS)

100 100

90 90

80 80

70 70

% of Subjects

% of Subjects

60 60

50 50

40 40

30 30

20 20

10 10

0 0

0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration ■µg/mL)

Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

Figure 46 Reverse cumulative distribution curves, pneumococcal serotype 23F, IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group - post booster dose

Table 84. Distribution of the pneumococcal serotype 1 IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group post booster dose

FAS N (%) PPS N (%)

PCV13-TT PCV13-TT
Concentration PCV7 (N=190) PCV7 (N=189)
(N=186) (N=186)
n % n % n % n %
≥0.01 186 100.0 189 99.47 186 100.0 188 99.47
≥0.1 186 100.0 189 99.47 186 100.0 188 99.47
≥1 168 90.32 35 18.42 168 90.32 35 18.52
≥10 1 0.54 0 0.00 1 0.54 0 0.00
≥100 0 0.00 0 0.00 0 0.00 0 0.00
N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

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Serotype 1, After Booster Dose (FAS) Serotype 1, After Booster Dose (PPS)

100 100
90 90
80 80
70 70

% of Subjects
% of Subjects

60 60
50 50
40 40
30 30
20 20
10 10
0 0
0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

Figure 47 Reverse cumulative distribution curves, pneumococcal serotype 1, IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group - post booster dose

Table 85. Distribution of the pneumococcal serotype 3 IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group post booster dose
FAS N (%) PPS N (%)
PCV13-TT PCV13-TT
Concentration PCV7 (N=190) PCV7 (N=189)
(N=186) (N=186)
n % n % n % n %
≥0.01 186 100.0 190 100.0 186 100.0 189 100.0
≥0.1 186 100.0 190 100.0 186 100.0 189 100.0
≥1 136 73.12 38 20.00 136 73.12 38 20.11
≥10 0 0.00 1 0.53 0 0.00 1 0.53
≥100 0 0.00 0 0.00 0 0.00 0 0.00
N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

Serotype 3, After Booster Dose (FAS) Serotype 3, After Booster Dose (PPS)

100 100
90 90
80 80
70 70
% of Subjects
% of Subjects

60 60
50 50
40 40
30 30
20 20
10 10
0 0
0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

Figure 48 Reverse cumulative distribution curves, pneumococcal serotype 3, IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group - post booster dose

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Table 86. Distribution of the pneumococcal serotype 5 IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group post booster dose

FAS N (%) PPS N (%)
PCV13-TT PCV13-TT
Concentration PCV7 (N=190) PCV7 (N=189)
(N=186) (N=186)
n % n % n % n %
≥0.01 186 100.0 176 92.63 186 100.0 175 92.59
≥0.1 186 100.0 175 92.11 186 100.0 174 92.06
≥1 75 40.32 3 1.58 75 40.32 3 1.59
≥10 1 0.54 0 0.00 1 0.54 0 0.00
≥100 0 0.00 0 0.00 0 0.00 0 0.00
N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

Serotype 5, After Booster Dose (FAS) Serotype 5, After Booster Dose (PPS)

100 100
90 90
80 80
70 70

% of Subjects
% of Subjects

60 60
50 50
40 40
30 30
20 20
10 10
0 0
0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

Figure 49 Reverse cumulative distribution curves, pneumococcal serotype 5, IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group - post booster dose

Table 87. Distribution of the pneumococcal serotype 6A IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group post booster dose
FAS N (%) PPS N (%)
PCV13-TT PCV13-TT
Concentration PCV7 (N=190) PCV7 (N=189)
(N=186) (N=186)
n % n % n % n %
≥0.01 186 100.0 190 100.0 186 100.0 189 100.0
≥0.1 185 99.46 190 100.0 185 99.46 189 100.0
≥1 174 93.55 133 70.00 174 93.55 133 70.37
≥10 10 5.38 8 4.21 10 5.38 8 4.23
≥100 0 0.00 0 0.00 0 0.00 0 0.00
N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

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Serotype 6A, After Booster Dose (FAS) Serotype 6A, After Booster Dose (PPS)

100 100
90 90
80 80
70 70

% of Subjects
% of Subjects

60 60
50 50
40 40
30 30
20 20
10 10
0 0
0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

Figure 50 Reverse cumulative distribution curves, pneumococcal serotype 6A, IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group - post booster dose

Table 88. Distribution of the pneumococcal serotype 7F IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group post booster dose
FAS N (%) PPS N (%)
PCV13-TT PCV13-TT

Concentration PCV7 (N=190) PCV7 (N=189)
(N=186) (N=186)
n % n % n % n %
≥0.01 186 100.0 185 97.37 186 100.0 184 97.35
≥0.1 186 100.0 183 96.32 186 100.0 182 96.30
≥1 185 99.46 33 17.37 185 99.46 33 17.46
≥10 21 11.29 0 0.00 21 11.29 0 0.00
≥100 0 0.00 0 0.00 0 0.00 0 0.00
N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

Serotype 7F, After Booster Dose (FAS) Serotype 7F, After Booster Dose (PPS)

100 100
90 90
80 80
70 70
% of Subjects
% of Subjects

60 60
50 50
40 40
30 30
20 20
10 10
0 0
0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

Figure 51 Reverse cumulative distribution curves, pneumococcal serotype 7F, IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group - post booster dose

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Table 89. Distribution of the pneumococcal serotype 19A IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group post booster dose
FAS N (%) PPS N (%)
PCV13-TT PCV13-TT
Concentration PCV7 (N=190) PCV7 (N=189)
(N=186) (N=186)
n % n % n % n %
≥0.01 186 100.0 190 100.0 186 100.0 189 100.0
≥0.1 186 100.0 190 100.0 186 100.0 189 100.0
≥1 184 98.92 164 86.32 184 98.92 163 86.24
≥10 38 20.43 5 2.63 38 20.43 5 2.65
≥100 0 0.00 0 0.00 0 0.00 0 0.00
N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

Serotype 19A, After Booster Dose (FAS) Serotype 19A, After Booster Dose (PPS)

100 100
90 90
80 80
70 70
% of Subjects
% of Subjects

60 60
50 50
40 40
30 30
20 20
10 10
0 0
0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

Figure 52 Reverse cumulative distribution curves, pneumococcal serotype 19A, IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group - post booster dose

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2.7.3.2.2.1.3. Comparison of percentage of subjects achieving IgG antibody concentration ≥ 1.0 µg/mL with comparison to the PCV7 seroresponse
Primary series

Data for the 7 common serotypes

With the exception of serotype 9V, all common serotypes met the noninferiority criterion after primary series. For serotype 9V, the lower limit of the 2-sided 95% CI was -13.12. (Table 90)
Data for the 6 additional serotypes

All common serotypes met the noninferiority criterion after primary series. The lower limit of the 2-sided 95% CI was all greater than 0.

Table 90 Comparison of percentage of subjects aged 7-11 months achieving a pneumococcal IgG antibody concentration $\geq 1.0 \mu\text{g/mL}$ post dose 3 of the primary series (PPS)

IgG antibody concentration $\geq 1.0 \mu\text{g/mL}$

PCV13-TT (N=185) PCV7 (N=189) Rate difference c

Serotype na % 95% CI n b % 95% CI % 95 %CI

PCV7

4 182 98.38 95.33-99.66 185 97.88 94.67-99.42 0.49 -2.25,3.24
6B 155 83.78 77.67-88.78 142 75.13 68.34-81.12 8.65 0.52,16.79
9V 154 83.24 77.07-88.32 169 89.42 84.13-93.42 -6.17 -13.12,0.77
14 184 99.46 97.03-99.99 189 100.0 98.07-100.0 -0.54 -1.60,0.52
18C 176 95.14 90.97-97.75 177 93.65 89.17-96.68 1.48 -3.17,6.14
19F 179 96.76 93.07-98.80 182 96.30 92.52-98.50 0.46 -3.25,4.17
23F 169 91.35 86.34-94.98 164 86.77 81.10-91.25 4.58 -1.72,10.88

Additional

1 170 91.89 86.98-95.39 25 13.23 8.75-18.90 78.66 72.44,84.89
3 124 67.03 59.75-73.75 24 12.70 8.31-18.30 54.33 46.06,62.60
5 69 37.30 30.31-44.70 6 3.17 1.17- 6.78 34.12 26.72,41.53
6A 139 75.14 68.26-81.18 73 38.62 31.65-45.96 36.51 27.18,45.84
7F 183 98.92 96.15-99.87 19 10.05 6.16-15.25 88.87 84.33,93.40
19A 181 97.84 94.56-99.41 132 69.84 62.76-76.29 28.00 21.13,34.87

N=number of subjects included in PPS for PCV13-TT and PCV7.

a. number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT group.

b. number of subjects with a determinate IgG antibody concentration to the given serotype for PCV7 group.

c. The rate difference of PCV13-TT – PCV7.

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Table 91. Comparison of percentage of subjects aged 7-11 months achieving a pneumococcal IgG antibody concentration $\geq 1.0 \mu\text{g/mL}$ post dose 3 of the primary series (FAS)

IgG antibody concentration $\geq 1.0 \mu\text{g/mL}$

PCV13-TT (N=185) PCV7 (N=189) Rate difference c

Serotype na % 95% CI nb % 95% CI % 95 %CI

PCV7

4 188 94.95 90.91-97.55 187 93.50 89.14-96.49 1.45 -3.13,6.03
6B 160 80.81 74.62-86.05 143 71.50 64.71-77.64 9.31 0.99,17.63
9V** 158 80.20 73.95-85.53 171 85.50 79.84-90.07 -5.30 -12.70,2.10
14 191 96.46 92.85-98.57 193 96.50 92.92-98.58 -0.04 -3.66,3.58
18C 181 91.41 86.61-94.92 179 89.50 84.40-93.38 1.91 -3.85,7.68
19F 187 94.44 90.28-97.19 186 93.00 88.53-96.12 1.44 -3.32,6.21
23F 173 87.37 81.93-91.66 166 83.00 77.06-87.93 4.37 -2.59,11.34

Additional

1 176 88.89 83.66-92.90 25 12.50 8.26-17.90 76.39 70.05,82.73
3 127 64.14 57.04-70.82 26 13.00 8.67-18.47 51.14 43.00,59.29
5 72 36.36 29.66-43.48 6 3.00 1.11-6.42 33.36 26.26,40.47
6A 143 72.22 65.43-78.34 74 37.00 30.30-44.09 35.22 26.07,44.37
7F 189 95.45 91.55-97.90 19 9.50 5.82-14.44 85.95 80.96,90.95
19A 190 95.96 92.19-98.24 135 67.50 60.53-73.94 28.46 21.41,35.51

N=number of subjects included in FAS for PCV13-TT and PCV7.

a. number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT group.

b. number of subjects with a determinate IgG antibody concentration to the given serotype for PCV7 group.

c. The rate difference of PCV13-TT – PCV7.

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Booster Dose

Data for the 7 common serotypes

All the common serotypes met the pre-specified noninferiority criteria for the IgG antibody concentration $\geq 1.0 \mu\text{g/mL}$ and GMC (Table 92).

Data for the 6 additional serotypes

The superiority criterion was met for all the additional serotypes (except serotype 19A) for the proportion of responders $\geq 0.35 \mu\text{g/mL}$ after the booster dose. For serotype 19A, the lower limit of the 2-sided 95% CI for the rate difference of percentage of subjects with IgG antibody concentration $\geq 0.35 \mu\text{g/mL}$ (PCV13-TT – PCV7) was -1.93%. The superiority criterion was met for all the additional serotypes for the proportion of responders $\geq 1.0 \mu\text{g/mL}$ and GMC after the booster dose.

The geometric mean fold rises (GMFRs) of antibody concentrations from pre- to post-boost for the 7 common serotypes ranged from 1.27 (serotype 4) to 1.96 (6B) in the PCV13-TT group. For the 6 additional serotypes, the fold rises ranged from 1.28 (serotype 7F) to 1.86 (serotype 6A) in the PCV13-TT group (Table 93).

Table 92. Comparison of percentage of subjects aged 7-11 months achieving a pneumococcal IgG antibody concentration $\geq 1.0 \mu\text{g/mL}$ post dose 3 of the primary series (FAS)

IgG antibody concentration $\geq 1.0 \mu\text{g/mL}$

PCV13-TT (N=186) PCV7 (N=189) Rate difference c

Serotype na % 95% CI n b % 95% CI

% 95% CI % 95 %CI

PCV7

4 178 95.70 91.70-98.13 186 98.41 95.43-99.67 -2.71 -6.13,0.70

6B 180 96.77 87.05-95.42 182 96.30 92.52-98.50 0.48 -3.22,4.18
 9V** 171 91.94 91.83-98.16 181 95.77 91.83-98.16 -3.83 -8.68,1.02
 14 186 100.0 98.04-100.0 189 100.0 98.07-100.0 0.00 -1.93,1.96
 18C 174 93.55 89.00-96.62 183 96.83 93.22-98.83 -3.28 -7.60,1.05
 19F 186 100.0 98.04-100.0 186 98.41 95.43-99.67 1.59 -0.19,3.37
 23F 180 96.77 93.11-98.81 181 95.77 91.83-98.16 1.01 -2.83,4.84
 Additional
 1 168 90.32 85.14-94.16 35 18.52 13.25-24.80 71.80 64.82,78.78
 3 136 73.12 66.14-79.34 38 20.11 14.64-26.54 53.01 44.45,61.57
 5 75 40.32 33.21-47.75 3 1.59 0.33-4.57 38.74 31.46,46.01
 6A 174 93.55 89.00-96.62 133 70.37 63.31-76.78 23.18 15.77,30.58
 7F 185 99.46 97.04-99.99 33 17.46 12.33-23.64 82.00 76.49,87.52
 19A 184 98.92 96.17-99.87 163 86.24 80.50-90.81 12.68 7.55,17.81
 N=number of subjects included in FAS for PCV13-TT and PCV7.
 a. number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT group.
 b. number of subjects with a determinate IgG antibody concentration to the given serotype for PCV7 group.
 c. The rate difference of PCV13-TT – PCV7.

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Table 93. Comparison of percentage of subjects aged 7-11 months for the geometric mean fold rises (GMFRs) of antibody concentrations in PCV13-TT group and PCV7 group post booster dose (PPS)
 PCV13-TT (N=185) PCV7 (N=188)

Serotype P
 GMFRs 95%CI GMFRs 95%CI
 4 1.27 1.20-1.34 1.31 1.24-1.38 0.4037
 6B 1.96 1.82-2.11 2.70 2.44-2.98 <0.0001
 9V 1.63 1.53-1.74 1.73 1.62-1.86 0.2039
 Common 14 1.64 1.51-1.78 1.45 1.32-1.59 0.0445
 18C 1.31 1.22-1.41 1.54 1.44-1.65 0.0010
 19F 1.41 1.30-1.54 1.38 1.28-1.49 0.7036
 23F 1.66 1.54-1.78 2.02 1.84-2.21 0.0010
 1 1.31 1.24-1.39 1.15 1.07-1.25 0.0091
 3 1.61 1.52-1.71 1.64 1.51-1.78 0.7019
 5 1.51 1.41-1.61 0.87 0.73-1.04 <0.0001
 Additional
 6A 1.86 1.68-2.06 2.09 1.82-2.40 0.1793
 7F 1.28 1.18-1.39 1.04 0.89-1.22 0.0239
 19A 1.59 1.42-1.78 1.13 1.01-1.26 <0.0001

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2.7.3.2.2.2.1.4. Comparison of percentage of subjects achieving an OPA antibody $\geq 1:8$ and the IgG antibody geometric mean titers (GMTs)

Primary series

Data for 7 common serotypes

For the 6 of 7 common serotypes, the percentage of subjects achieving an OPA antibody titer $\geq 1:8$ was all higher than 90% in PCV13-TT group (Table 94). The exceptions were serotypes 6B, the percentage of subjects achieving an OPA antibody titer $\geq 1:8$ was 81.25%. With the exception of serotypes 6B and 23F, the lower limit of the 2-sided 95% CI of rate difference of the percentage of subjects achieving an OPA antibody titer $\geq 1:8$ was greater than -10% for the common serotypes. And the lower limit of the 2-sided 95% CI of GMR was greater than 0.5 except serotypes 6B and 23F.

Data for 6 additional serotypes

For the 5 of 6 additional serotypes, the percentage of subjects achieving an OPA antibody titer $\geq 1:8$ was all higher than 90% in PCV13-TT group. The exceptions were serotypes 6A, the percentage of subjects achieving an OPA antibody titer $\geq 1:8$ was 89.58%. The lower limit of the 2-sided 95% CI of rate difference of the percentage of subjects achieving an OPA antibody titer $\geq 1:8$ was all greater than 0 for the additional serotypes. The lower limit of the 2-sided 95% CI of GMR was all greater than 1 for all the additional serotypes.

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Table 94. Comparison of percentage of subjects achieving a pneumococcal IgG antibody titer $\geq 1:8$ and GMTs in PCV13-TT group and PCV7 group aged 7-11 months post-dose 3 of the primary series (PPS)

OPA antibody titer $\geq 1:8$ OPA GMT
 PCV13-TT (N=96) PCV7 (N=95) Rate difference a PCV13-TT (N=96) PCV7 (N=95)

Serotype 95%CI Rate diff 95%CI
 % % GMT GMT
 Common
 4 100.0 100.0 0.00 -3.81,3.77 1014.62 1214.37 0.84 0.66,1.05
 6B 81.25 89.47 -8.22 -18.18,1.73 237.34 599.57 0.40 0.20,0.79
 9V 100.0 100.0 0.00 -3.81,3.77 1425.78 1692.48 0.84 0.63,1.13
 14 98.96 97.89 1.06 -2.47,4.59 2538.79 3060.32 0.83 0.55,1.25
 18C 98.96 98.95 0.01 -2.88,2.90 952.30 1157.93 0.82 0.59,1.16
 19F 98.96 91.58 7.38 1.44,13.32 518.08 240.27 2.16 1.43,3.25
 23F 91.67 94.74 -3.07 -10.19,4.05 669.00 1699.27 0.39 0.22,0.70
 Additional
 1 92.71 5.26 87.45 80.57,94.32 56.40 2.24 25.18 19.03,33.30
 3 100.0 46.32 53.68 43.66,63.71 316.46 6.62 47.82 33.63,68.01
 5 97.92 7.37 90.55 84.57,96.53 115.18 2.43 47.33 35.72,62.72
 6A 89.58 65.26 24.32 12.96,35.68 442.24 95.37 4.64 2.19,9.82
 7F 100.0 35.79 64.21 54.57,73.85 3589.60 16.04 223.76 120.98,413.89

19A 100.0 78.95 21.05 12.85,29.25 692.99 57.76 12.00 7.58,18.99
a. The rate difference of PCV13-TT – PCV7.
b. GMC ratio: PCV13-TT to PCV7.

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Table 95 Comparison of percentage of subjects achieving a pneumococcal IgG antibody titer $\geq 1:8$ and GMTs in PCV13-TT group and PCV7 group aged 7-11 months post-dose 3 of the primary series (FAS)
OPA antibody titer $\geq 1:8$ OPA GMT
PCV13-TT (N=96) PCV7 (N=95) Rate difference a PCV13-TT (N=96) PCV7 (N=95)
Serotype 95%CI Ratiob 95%CI
% % % GMT GMT
Common
4 97.98 96.00 1.98 -2.76,6.72 895.43 933.75 0.96 0.66,1.40
6B 79.80 87.00 -7.20 -17.50,3.09 219.38 504.22 0.44 0.22,0.88
9V 97.98 96.00 1.98 -2.76,6.72 1229.12 1277.36 0.96 0.63,1.48
14 96.97 95.00 1.97 -3.48,7.41 2217.86 2401.55 0.92 0.56,1.53
18C 96.97 95.00 1.97 -3.48,7.41 834.16 889.27 0.94 0.60,1.46
19F 96.97 88.00 8.97 1.76,16.18 463.37 196.71 2.36 1.49,3.73
23F 89.90 91.00 -1.10 -9.27,7.07 600.51 1301.23 0.46 0.24,0.87
Additional
1 90.91 5.00 85.91 78.82,93.00 53.17 2.23 23.87 17.92,31.80
3 97.98 45.00 52.98 42.84,63.12 286.32 6.51 44.01 30.32,63.89
5 95.96 7.00 88.96 82.63,95.29 107.04 2.41 44.42 33.07,59.67
6A 87.88 63.00 24.88 13.44,36.32 396.28 83.54 4.74 2.24,10.04
7F 98.99 37.00 61.99 52.32,71.66 3247.79 17.07 190.22 102.28,353.74
19A 97.98 76.00 21.98 13.16,30.80 617.87 52.02 11.88 7.28,19.36
a. The rate difference of PCV13-TT – PCV7.
b. GMC ratio: PCV13-TT to PCV7.

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Booster dose
Data for 7 common serotypes
For the 7 common serotypes, the percentage of subjects achieving an OPA antibody titer $\geq 1:8$ was all higher than 90% in PCV13-TT-TT group (Table 96). The lower limit of the 2-sided 95% CI of rate difference of the percentage of subjects achieving an OPA antibody titer $\geq 1:8$ was greater than -10% for all the common serotypes. And the lower limit of the 2-sided 95% CI of GMR was greater than 0.5 except serotypes 6B and 23F.

Data for 6 additional serotypes
For the 6 additional serotypes, the percentage of subjects achieving an OPA antibody titer $\geq 1:8$ was all higher than 90% in PCV13-TT-TT group. The lower limit of the 2-sided 95% CI of rate difference of the percentage of subjects achieving an OPA antibody titer $\geq 1:8$ was all greater than 0 for the additional serotypes. The lower limit of the 2-sided 95% CI of GMR was all greater than 1 for all the additional serotypes.

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Table 96. Comparison of percentage of subjects achieving a pneumococcal IgG antibody titer $\geq 1:8$ and GMTs in PCV13-TT group and PCV7 group aged 7-11 months post booster dose (PPS)
OPA antibody titer $\geq 1:8$ OPA GMT
PCV13-TT (N=93) PCV7 (N=95) Rate difference a PCV13-TT (N=93) PCV7 (N=95)
Serotype 95%CI Ratiob 95%CI
% % % GMT GMT
Common
4 100.0 100.0 0.00 -3.81,3.89 1549.33 1465.72 1.06 0.83,1.34
6B 93.55 95.79 -2.24 -8.66,4.18 890.24 1843.95 0.48 0.28,0.83
9V 100.0 100.0 0.00 -3.81,3.89 2075.15 2264.80 0.92 0.70,1.20
14 100.0 100.0 0.00 -3.81,3.89 3761.98 2883.34 1.30 0.97,1.76
18C 100.0 98.95 1.05 -1.00,3.10 1491.46 1677.92 0.89 0.65,1.21
19F 100.0 93.68 6.32 1.42,11.21 751.55 258.60 2.91 2.06,4.11
23F 97.85 95.79 2.06 -2.94,7.06 1577.42 3607.17 0.44 0.27,0.71
Additional
1 94.62 6.32 88.31 81.60,95.01 87.89 2.34 37.49 27.82,50.54
3 100.0 56.84 43.16 33.20,53.12 431.54 8.12 53.15 37.65,75.03
5 98.92 5.26 93.66 88.71,98.62 222.49 2.41 92.37 68.35,124.83
6A 95.70 81.05 14.65 5.75,23.54 1234.03 416.90 2.96 1.51,5.80
7F 100.0 42.11 57.89 47.97,67.82 4276.87 21.71 197.01 105.77,366.96
19A 100.0 78.95 21.05 12.85,29.25 1141.93 71.60 15.95 9.80,25.96
a. The rate difference of PCV13-TT – PCV7.
b. GMC ratio: PCV13-TT to PCV7.

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Table 97. Comparison of percentage of subjects achieving a pneumococcal IgG antibody titer $\geq 1:8$ and GMTs in PCV13-TT group and PCV7 group aged 7-11 months post booster dose (FAS)
OPA antibody titer $\geq 1:8$ OPA GMT
PCV13-TT (N=96) PCV7 (N=95) Rate difference a PCV13-TT (N=96) PCV7 (N=95)
Serotype 95%CI Ratiob 95%CI
% % % GMT GMT

Common

4 100.0 100.0 0.00 -3.81,3.89 1549.33 1465.72 1.06 0.83,1.34
6B 93.55 95.79 -2.24 -8.66,4.18 890.24 1843.95 0.48 0.28,0.83
9V 100.0 100.0 0.00 -3.81,3.89 2075.15 2264.80 0.92 0.70,1.20
14 100.0 100.0 0.00 -3.81,3.89 3761.98 2883.34 1.30 0.97,1.76
18C 100.0 98.95 1.05 -1.00,3.10 1491.46 1677.92 0.89 0.65,1.21
19F 100.0 93.68 6.32 1.42,11.21 751.55 258.60 2.91 2.06,4.11
23F 97.85 95.79 2.06 -2.94,7.06 1577.42 3607.17 0.44 0.27,0.71
Additional
1 94.62 6.32 88.31 81.60,95.01 87.89 2.34 37.49 27.82,50.54
3 100.0 56.84 43.16 33.20,53.12 431.54 8.12 53.15 37.65,75.03
5 98.92 5.26 93.66 88.71,98.62 222.49 2.41 92.37 68.35,124.83
6A 95.70 81.05 14.65 5.75,23.54 1234.03 416.90 2.96 1.51,5.80
7F 100.0 42.11 57.89 47.97,67.82 4276.87 21.71 197.01 105.77,366.96
19A 100.0 78.95 21.05 12.85,29.25 1141.93 71.60 15.95 9.80,25.96
a. The rate difference of PCV13-TT – PCV7.
b. GMC ratio: PCV13-TT to PCV7.

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2.7.3.2.2.2.2. Pneumococcal immunogenicity results for subjects aged 12-23 months in PCV13-TT group and PCV7 group

2.7.3.2.2.2.2.1. Comparison of percentage of subjects achieving IgG antibody concentration $\geq$ 0.35 μ g/mL and the IgG antibody geometric mean concentrations (GMCs)

Data for 7 common serotypes

The proportion of responders $\geq$ 0.35 μ g/mL in the PCV13-TT-TT group were all higher than 99% and all common serotypes met the noninferiority criterion for the proportion of responders $\geq$ 0.35 μ g/mL (Table 98). With the exception of serotype 9V (2.97 μ g/mL), the GMCs for common serotypes were all higher than 4 μ g/mL. All common serotypes met the noninferiority criterion for the GMC.

Data for 6 additional serotypes with comparison to the PCV7 seroresponse for the 6 additional serotypes

The proportion of responders $\geq$ 0.35 μ g/mL in the PCV13-TT group were all higher than 96%, and the lower limit of the 2-sided 95% CI of proportion of responders $\geq$ 0.35 μ g/mL was all greater than 70%. With the exception of serotype 19A, all the additional serotypes were all met superiority criterion. For serotype 19A, the rate difference for the proportion of responders $\geq$ 0.35 μ g/mL was 0.53, with a lower limit of the 2-sided 95% CI of -0.51.

Data for 6 additional serotypes with comparison to the PCV7 reference for the 6 additional serotypes

The PCV7 reference value for the 6 additional serotypes was based on serotype 6B from the PCV7 group (the lowest response among the 7 serotypes contained in PCV7 achieved by subjects who received PCV7). For serotype 6B, the proportion of responders was 98.41%. All additional serotypes met the non-inferiority criterion.

The PCV7 reference value for the 6 additional serotypes was serotype 6B from the PCV7 group (the lowest response among the 7 serotypes contained in PCV7 achieved by subjects who received PCV7). For serotype 6B, the GMC was 2.71. All the additional serotypes met the non-inferiority criterion for GMC. Four (4) of the 6 additional serotypes met the non-inferiority criterion for GMR (lower limit of the 2-sided 95% CI for the GMR $>$ 0.5). The exceptions were serotype 3 and 5 with a GMR of 0.39 (95% CI, 0.33 to 0.46) and 0.37 (95% CI, 0.31 to 0.44), respectively.

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Table 98. Comparison of percentage of subjects aged 12-23 months achieving a pneumococcal IgG antibody concentration $\geq$ 0.35 μ g/mL and GMCs

post full immunizations (PPS)

IgG antibody concentration $\geq$ 0.35 μ g/mL IgG antibody GMC

PCV13-TT (N=189) PCV7 (N=189) Rate difference c PCV13-TT (N=198) PCV7 (N=189) GMC Ratio

Serotype Na % 95% CI Nb % 95% CI % 95 %CI GMC 95% CI GMC 95% CI Ratio d 95%CI e

PCV7

4 189 100.0 98.07-100.0 188 99.47 97.09-99.99 0.53 -0.51,1.56 5.01 4.60-5.45 5.08 4.63-5.58 0.99
0.87,1.12
6B 189 100.0 98.07-100.0 186 98.41 95.43-99.67 1.59 -0.19,3.37 5.42 4.86-6.04 2.71 2.35-3.11 2.00
1.68,2.39
9V** 189 100.0 98.07-100.0 187 98.94 96.23-99.87 1.06 -0.40,2.52 2.97 2.68-3.30 2.95 2.64-3.30 1.01
0.87,1.17
14 189 100.0 98.07-100.0 189 100.0 98.07-100.0 0.00 -1.93,1.93 9.20 8.28-10.22 14.98 13.51-16.61 0.61
0.53,0.71
18C 189 100.0 98.07-100.0 189 100.0 98.07-100.0 0.00 -1.93,1.93 5.03 4.63-5.46 3.92 3.56-4.31 1.28
1.13,1.45
19F 189 100.0 98.07-100.0 189 100.0 98.07-100.0 0.00 -1.93,1.93 8.80 8.10-9.55 4.20 3.79-4.65 2.09
1.84,2.39
23F 188 99.47 97.09-99.99 188 99.47 97.09-99.99 0.00 -1.46,1.46 4.32 3.89-4.79 2.96 2.61-3.36 1.46
1.24,1.72
Additional
vs PCV7 seroresponse f
1 189 100.0 98.07-100.0 133 70.37 63.31-76.78 29.63 23.12,36.14 2.25 2.07-2.44 0.42 0.35-0.50 5.33
4.39,6.49
3 186 98.41 95.43-99.67 90 47.62 40.32-54.99 50.79 43.45,58.13 1.06 0.98-1.14 0.39 0.34-0.45 2.71
2.31,3.17
5 183 96.83 93.22-98.83 101 53.44 46.06-60.71 43.39 35.85,50.92 1.01 0.91-1.12 0.27 0.22-0.32 3.79
3.05,4.71
6A 186 98.41 95.43-99.67 177 93.65 89.17-96.68 4.76 0.86,8.67 2.46 2.08-2.92 1.17 1.01-1.35 2.11
1.69,2.64
7F 189 100.0 98.07-100.0 135 71.43 64.42-77.75 28.57 22.13,35.01 5.47 5.04-5.94 0.41 0.33-0.52 13.28
10.44,16.88
19A 189 100.0 98.07-100.0 188 99.47 97.09-99.99 0.53 -0.51,1.56 7.81 7.08-8.61 2.67 2.28-3.12 2.93

2.43,3.52
vs PCV7 reference g
1 189 100.0 98.07-100.0 186 98.41 95.43-99.67 1.59 -0.19,3.37 2.25 2.07-2.44 2.71 2.35-3.11 0.83
0.71,0.98
3 186 98.41 95.43-99.67 186 98.41 95.43-99.67 0.00 -2.52,2.52 1.06 0.98-1.14 2.71 2.35-3.11 0.39
0.33,0.46
5 183 96.83 93.22-98.83 186 98.41 95.43-99.67 -1.59 -4.66,1.48 1.01 0.91-1.12 2.71 2.35-3.11 0.37
0.31,0.44
6A 186 98.41 95.43-99.67 186 98.41 95.43-99.67 0.00 -2.52,2.52 2.46 2.08-2.92 2.71 2.35-3.11 0.91
0.73,1.13
7F 189 100.0 98.07-100.0 186 98.41 95.43-99.67 1.59 -0.19,3.37 5.47 5.04-5.94 2.71 2.35-3.11 2.02
1.72,2.38
19A 189 100.0 98.07-100.0 186 98.41 95.43-99.67 1.59 -0.19,3.37 7.81 7.08-8.61 2.71 2.35-3.11 2.89
2.43,3.42

N=number of subjects included in PPS for PCV13-TT and PCV7.

- number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT group.
- number of subjects with a determinate IgG antibody concentration to the given serotype for PCV7 group.
- The rate difference of PCV13-TT – PCV7.
- GMC ratio: PCV13-TT to PCV7.
- Two-sided 95% CI is for additional serotypes, and two-sided 97.5% CI is for common serotypes.
- Difference in proportions: (PCV13-TT – PCV7 seroresponse) expressed as a percentage. GMC ratio: PCV13-TT to PCV7 seroresponse.
- Difference in proportions: (PCV13-TT – PCV7 reference) expressed as a percentage. For the additional serotypes, the reference value is derived from serotype 6B from the PCV7 group. GMC ratio: PCV13-TT to PCV7 reference. For the additional serotypes, the reference value is derived from serotype 6B from the PCV7 group.

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Table 99. Comparison of percentage of subjects aged 12-23 months achieving a pneumococcal IgG antibody concentration ≥ 0.35 $\mu\text{g/mL}$ and GMCs post full immunizations (FAS)

IgG antibody concentration ≥ 0.35 $\mu\text{g/mL}$ IgG antibody GMC

PCV13-TT (N=200) PCV7 (N=200) Rate difference c PCV13-TT (N=200) PCV7 (N=200) GMC Ratio

95% CI

Serotype Na % 95% CI Nb % 95% CI % GMC 95% CI GMC 95% CI Ratio d 95%CI e

/97.5%CI

PCV7

4 193 96.50 92.92-98.58 191 95.50 91.63-97.92 1.00 -2.84,4.84 3.87 3.17-4.72 3.99 3.30-4.83 0.97

0.74,1.28

6B 197 98.50 95.68-99.69 194 97.00 93.58-98.89 1.50 -1.40,4.40 4.92 4.33-5.59 2.52 2.19-2.91 1.95

1.61,2.36

9V** 193 96.50 92.92-98.58 190 95.00 91.00-97.58 1.50 -2.45,5.45 2.35 1.94-2.85 2.36 1.94-2.87 1.00

0.76,1.31

14 199 99.50 97.25-99.99 198 99.00 96.43-99.88 0.50 -1.19,2.19 8.32 7.34-9.43 13.01 11.38-14.87 0.64

0.53,0.77

18C 194 97.00 93.58-98.89 192 96.00 92.27-98.26 1.00 -2.60,4.60 3.99 3.32-4.79 3.02 2.47-3.69 1.32

1.01,1.73

19F 200 100.0 98.17-100.0 200 100.0 98.17-100.0 0.00 -1.83,1.83 7.98 7.19-8.85 3.90 3.50-4.35 2.04

1.76,2.37

23F 193 96.50 92.92-98.58 192 96.00 92.27-98.26 0.50 -3.22,4.22 3.90 3.43-4.44 2.66 2.32-3.06 1.46

1.21,1.77

Additional

vs PCV7 seroresponse f

1 193 96.50 92.92-98.58 135 67.50 60.53-73.94 29.00 22.03,35.97 1.98 1.73-2.26 0.39 0.33-0.47 5.03

4.01,6.31

3 190 95.00 91.00-97.58 92 46.00 38.95-53.17 49.00 41.46,56.54 0.93 0.82-1.06 0.36 0.31-0.42 2.58

2.12,3.15

5 186 93.00 88.53-96.12 102 51.00 43.85-58.12 42.00 34.22,49.78 0.85 0.73-1.00 0.24 0.20-0.29 3.54

2.74,4.58

6A 189 94.50 90.37-97.22 181 90.50 85.56-94.18 4.00 -1.15, 9.15 2.13 1.73-2.62 1.05 0.89-1.24 2.02

1.55,2.63

7F 195 97.50 94.26-99.18 137 68.50 61.57-74.87 29.00 22.21,35.79 4.73 4.15-5.39 0.39 0.31-0.49 12.10

9.37,15.62

19A 200 100.0 98.17-100.0 199 99.50 97.25-99.99 0.50 -0.48,1.48 7.07 6.32-7.91 2.53 2.17-2.94 2.80

2.32,3.38

vs PCV7 reference g

1 193 96.50 92.92-98.58 194 97.00 93.58-98.89 -0.50 -3.98,2.98 1.98 1.73-2.26 2.52 2.19-2.91 0.78

0.64,0.95

3 190 95.00 91.00-97.58 194 97.00 93.58-98.89 -2.00 -5.84,1.84 0.93 0.82-1.06 2.52 2.19-2.91 0.37

0.30,0.45

5 186 93.00 88.53-96.12 194 97.00 93.58-98.89 -4.00 -8.25,0.25 0.85 0.73-1.00 2.52 2.19-2.91 0.34

0.27,0.42

6A 189 94.50 90.37-97.22 194 97.00 93.58-98.89 -2.50 -6.45,1.45 2.13 1.73-2.62 2.52 2.19-2.91 0.84

0.66,1.08

7F 195 97.50 94.26-99.18 194 97.00 93.58-98.89 0.50 -2.70,3.70 4.73 4.15-5.39 2.52 2.19-2.91 1.87

1.54,2.27

19A 200 100.0 98.17-100.0 194 97.00 93.58-98.89 3.00 0.64,5.36 7.07 6.32-7.91 2.52 2.19-2.91 2.80

2.34,3.36

N=number of subjects included in PPS for PCV13-TT and PCV7.

- number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT group.
- number of subjects with a determinate IgG antibody concentration to the given serotype for PCV7 group.
- The rate difference of PCV13-TT – PCV7.
- GMC ratio: PCV13-TT to PCV7.
- Two-sided 95% CI is for additional serotypes, and two-sided 97.5% CI is for common serotypes.
- Difference in proportions: (PCV13-TT – PCV7 seroresponse) expressed as a percentage. GMC ratio: PCV13-TT to PCV7 seroresponse.
- Difference in proportions: (PCV13-TT – PCV7 reference) expressed as a percentage. For the additional

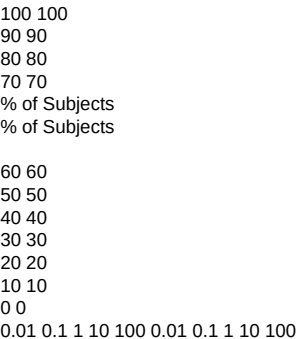
serotypes, the reference value is derived from serotype 6B from the PCV7 group. GMC ratio: PCV13-TT to PCV7 reference. For the additional serotypes, the reference value is derived from serotype 6B from the PCV7 group.

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2.7.3.2.2.2.2.2. Pneumococcal Serotype antibody concentrations in the subjects aged 12-23 months in PCV13-TT group and PCV7 group
The results of pneumococcal serotype antibody concentrations for subjects aged 12-23 months after full immunizations in PCV13-TT group and PCV7 group were shown in Table 100. to Table 112 The reverse cumulative distribution curves were shown in Figure 53 to 65.

Table 100. Distribution of the pneumococcal serotype 4 IgG antibody concentrations for subjects in PCV13-TT group and PCV7 group post full immunizations
FAS N (%) PPS N (%)
Concentration PCV13-TT (N=200) PCV7 (N=200) PCV13-TT (N=189) PCV7 (N=189)
n % n % n % n %
≥0.01 194 97.00 195 97.50 189 100.0 189 100.0
≥0.1 194 97.00 195 97.50 189 100.0 189 100.0
≥1 192 96.00 190 95.00 189 100.0 187 98.94
≥10 22 11.00 27 13.50 22 11.64 27 14.29
≥100 0 0.00 0 0.00 0 0.00 0 0.00
N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

Serotype 4, After Full Immunizations (FAS) Serotype 4, full immunizations (PPS)



Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

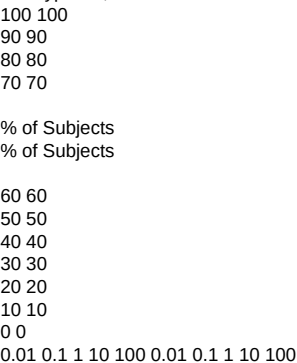
PCV13-TT PCV7 PCV13-TT PCV7

Figure 53 Reverse cumulative distribution curves, pneumococcal serotype 4, IgG antibody concentrations for subjects aged 12-23 months in PCV13-TT group and PCV7 group – post full immunizations

Table 101. Distribution of the pneumococcal serotype 6B IgG antibody concentrations for subjects in PCV13-TT group and PCV7 group post full immunizations
FAS N (%) PPS N (%)
Concentration PCV13-TT (N=200) PCV7 (N=200) PCV13-TT (N=189) PCV7 (N=189)
n % n % n % n %
≥0.01 200 100.0 200 100.0 189 100.0 189 100.0
≥0.1 200 100.0 200 100.0 189 100.0 189 100.0
≥1 191 95.50 167 83.50 188 99.47 164 86.77
≥10 38 19.00 19 9.50 37 19.58 19 10.05
≥100 0 0.00 0 0.00 0 0.00 0 0.00
N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

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Serotype 6B, After Full Immunizations (FAS) Serotype 6B, After Full Immunizations (PPS)



Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

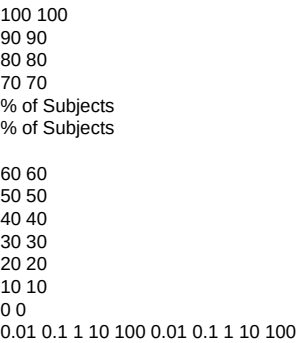
PCV13-TT PCV7 PCV13-TT PCV7

Figure 54 Reverse cumulative distribution curves, pneumococcal serotype 6B, IgG antibody concentrations for subjects aged 12-23 months in PCV13-TT group and PCV7 group post full immunizations

Table 102. Distribution of the pneumococcal serotype 9V IgG antibody concentrations for subjects in PCV13-TT group and PCV7 group post full immunizations

FAS N (%)	PPS N (%)						
Concentration PCV13-TT (N=200) PCV7 (N=200) PCV13-TT (N=189) PCV7 (N=189)							
n %	n %	n %	n %	n %	n %	n %	n %
≥0.01	194 97.00	194 97.00	189 100.0	189 100.0	189 100.0	189 100.0	189 100.0
≥0.1	194 97.00	194 97.00	189 100.0	189 100.0	189 100.0	189 100.0	189 100.0
≥1	175 87.50	176 88.00	173 91.53	173 91.53	173 91.53	173 91.53	173 91.53
≥10	9 4.50	14 7.00	9 4.76	14 7.41	9 4.76	14 7.41	14 7.41
≥100	0 0.00	0 0.00	0 0.00	0 0.00	0 0.00	0 0.00	0 0.00
N=number of subjects included in FAS and PPS for vaccine group.							
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.							

Serotype 9V, After Full Immunizations (FAS) Serotype 9V, After Full Immunizations (PPS)



Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

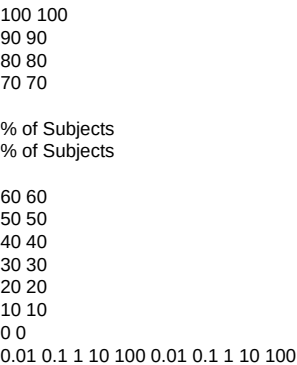
Figure 55 Reverse cumulative distribution curves, pneumococcal serotype 9V, IgG antibody concentrations for subjects aged 12-23 months in PCV13-TT group and PCV7 group post full immunizations

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Table 103. Distribution of the pneumococcal serotype 14 IgG antibody concentrations for subjects in PCV13-TT group and PCV7 group post full immunizations

FAS N (%)	PPS N (%)						
Concentration PCV13-TT (N=200) PCV7 (N=200) PCV13-TT (N=189) PCV7 (N=189)							
n %	n %	n %	n %	n %	n %	n %	n %
≥0.01	200 100.0	200 100.0	189 100.0	189 100.0	189 100.0	189 100.0	189 100.0
≥0.1	200 100.0	200 100.0	189 100.0	189 100.0	189 100.0	189 100.0	189 100.0
≥1	193 96.50	194 97.00	189 100.0	189 100.0	189 100.0	189 100.0	189 100.0
≥10	91 45.50	139 69.50	89 47.09	139 73.54	89 47.09	139 73.54	139 73.54
≥100	0 0.00	2 1.00	0 0.00	2 1.06	0 0.00	2 1.06	2 1.06
N=number of subjects included in FAS and PPS for vaccine group.							
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.							

Serotype 14, After Full Immunizations (FAS) Serotype 14, After Full Immunizations (PPS)



Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

Figure 56 Reverse cumulative distribution curves, pneumococcal serotype 14, IgG antibody concentrations for subjects aged 12-23 months in PCV13-TT group and PCV7 group post full immunizations

immunizations

Table 104. Distribution of the pneumococcal serotype 18C IgG antibody concentrations for subjects in PCV13-TT group and PCV7 group post full immunizations

FAS N (%) PPS N (%)

Concentration PCV13-TT (N=200) PCV7 (N=200) PCV13-TT (N=189) PCV7 (N=189)

n	%	n	%	n	%	n	%	
≥0.01	195	97.50	193	96.50	189	100.0	189	100.0
≥0.1	195	97.50	193	96.50	189	100.0	189	100.0
≥1	191	95.50	186	93.00	189	100.0	183	96.83
≥10	21	10.50	17	8.50	21	11.11	17	8.99
≥100	0	0.00	0	0.00	0	0.00	0	0.00

N=number of subjects included in FAS and PPS for vaccine group.

n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

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Serotype 18C, After Full Immunizations (FAS) Serotype 18C, After Full Immunizations (PPS)

100	100
90	90
80	80
70	70
% of Subjects	
% of Subjects	
60	60
50	50
40	40
30	30
20	20
10	10
0	0
0.01	0.1
1	1
10	100
0.01	0.1
0.1	1
10	100

Antibody Concentration ■(μg/mL)

Antibody Concentration (μg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

Figure 57 Reverse cumulative distribution curves, pneumococcal serotype 18C IgG antibody concentrations for subjects aged 12-23 months in PCV13-TT group and PCV7 group - post full immunizations

Table 105. Distribution of the pneumococcal serotype 19F IgG antibody concentrations for subjects in PCV13-TT group and PCV7 group post full immunizations

FAS N (%) PPS N (%)

Concentration PCV13-TT (N=200) PCV7 (N=200) PCV13-TT (N=189) PCV7 (N=189)

n	%	n	%	n	%	n	%	
≥0.01	200	100.0	200	100.0	189	100.0	189	100.0
≥0.1	200	100.0	200	100.0	189	100.0	189	100.0
≥1	193	96.50	191	95.50	189	100.0	185	97.88
≥10	85	42.50	23	11.50	84	44.44	23	12.17
≥100	0	0.00	0	0.00	0	0.00	0	0.00

N=number of subjects included in FAS and PPS for vaccine group.

n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

Serotype 19F, After Full Immunizations (FAS) Serotype 19F, After Full Immunizations (PPS)

100	100
90	90
80	80
70	70
% of Subjects	
% of Subjects	
60	60
50	50
40	40
30	30
20	20
10	10
0	0
0.01	0.1
0.1	1
1	10
100	0.01
0.1	0.1
1	10
100	100

Antibody Concentration ■(μg/mL)

Antibody Concentration (μg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

Figure 58 Reverse cumulative distribution curves, pneumococcal serotype 19F, IgG antibody concentrations for subjects aged 12-23 months in PCV13-TT group and PCV7 group - post full immunizations

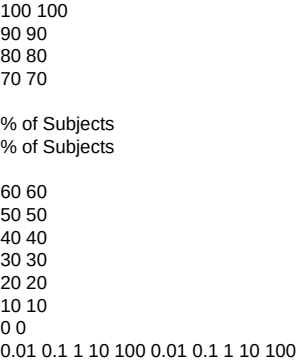
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Table 106. Distribution of the pneumococcal serotype 23F IgG antibody concentrations for subjects in PCV13-TT group and PCV7 group post full immunizations
FAS N (%) PPS N (%)
Concentration PCV13-TT (N=200) PCV7 (N=200) PCV13-TT (N=189) PCV7 (N=189)
n % n % n % n %
≥0.01 200 100.0 200 100.0 189 100.0 189 100.0
≥0.1 200 100.0 200 100.0 189 100.0 189 100.0
≥1 186 93.00 171 85.50 184 97.35 169 89.42
≥10 22 11.00 16 8.00 20 10.58 16 8.47
≥100 0 0.00 0 0.00 0 0.00 0 0.00
N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

Serotype 23F, After Full Immunizations (FAS) Serotype 23F, After Full Immunizations (PPS)



Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

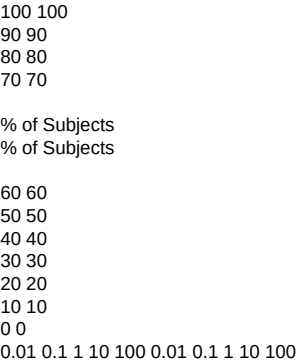
PCV13-TT PCV7 PCV13-TT PCV7

Figure 59 Reverse cumulative distribution curves, pneumococcal serotype 23F, IgG antibody concentrations for subjects aged 12-23 months in PCV13-TT group and PCV7 group post full immunizations

Table 107. Distribution of the pneumococcal serotype 1 IgG antibody concentrations for subjects in PCV13-TT group and PCV7 group post full immunizations
FAS N (%) PPS N (%)
Concentration PCV13-TT (N=200) PCV7 (N=200) PCV13-TT (N=189) PCV7 (N=189)
n % n % n % n %
≥0.01 198 99.00 190 95.00 189 100.0 181 95.77
≥0.1 198 99.00 190 95.00 189 100.0 181 95.77
≥1 179 89.50 25 12.50 176 93.12 25 13.23
≥10 1 0.50 0 0.00 1 0.53 0 0.00
≥100 0 0.00 0 0.00 0 0.00 0 0.00
N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

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Serotype 1, After Full Immunizations (FAS) Serotype 1, After Full Immunizations (PPS)



Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

Figure 60 Reverse cumulative distribution curves, pneumococcal serotype 1, IgG antibody concentrations for subjects aged 12-23 months in PCV13-TT group and PCV7 group - post full immunizations

Table 108. Distribution of the pneumococcal serotype 3 IgG antibody concentrations for subjects in PCV13-TT group and PCV7 group post full immunizations
FAS N (%) PPS N (%)
Concentration PCV13-TT (N=200) PCV7 (N=200) PCV13-TT (N=189) PCV7 (N=189)

n % n % n % n %
≥0.01 198 99.00 198 99.00 189 100.0 188 99.47
≥0.1 194 97.00 190 95.00 189 100.0 186 98.41
≥1 105 52.50 19 9.50 103 54.50 18 9.52
≥10 0 0.00 1 0.50 0 0.00 1 0.53
≥100 0 0.00 0 0.00 0 0.00 0 0.00
N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

Serotype 3, After Full Immunizations (FAS) Serotype 3, After Full Immunizations (PPS)

100 100
90 90
80 80
70 70
% of Subjects
% of Subjects

60 60
50 50
40 40
30 30
20 20
10 10
0 0
0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

Figure 61 Reverse cumulative distribution curves, pneumococcal serotype 3, IgG antibody concentrations for subjects aged 12-23 months in PCV13-TT group and PCV7 group - post full immunizations

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Table 109 Distribution of the pneumococcal serotype 5 IgG antibody concentrations for subjects in PCV13-TT group and PCV7 group post full immunizations
FAS N (%) PPS N (%)
Concentration PCV13-TT (N=200) PCV7 (N=200) PCV13-TT (N=189) PCV7 (N=189)
n % n % n % n %
≥0.01 194 97.00 181 90.50 188 99.47 175 92.59
≥0.1 194 97.00 181 90.50 188 99.47 175 92.59
≥1 93 46.50 5 2.50 93 49.21 5 2.65
≥10 0 0.00 0 0.00 0 0.00 0 0.00
N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

Serotype 5, After Full Immunizations (FAS) Serotype 5, After Full Immunizations (PPS)

100 100
90 90
80 80
70 70
% of Subjects
% of Subjects

60 60
50 50
40 40
30 30
20 20
10 10
0 0
0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

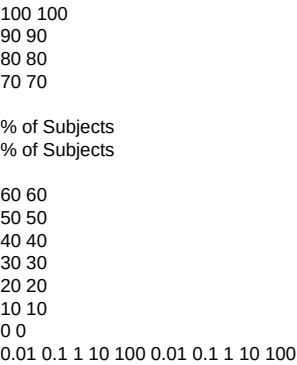
Figure 62 Reverse cumulative distribution curves, pneumococcal serotype 5, IgG antibody concentrations for subjects aged 12-23 months in PCV13-TT group and PCV7 group - post full immunizations

Table 110. Distribution of the pneumococcal serotype 6A IgG antibody concentrations for subjects in PCV13-TT group and PCV7 group post full immunizations
FAS N (%) PPS N (%)
Concentration PCV13-TT (N=200) PCV7 (N=200) PCV13-TT (N=189) PCV7 (N=189)
n % n % n % n %
≥0.01 194 97.00 197 98.50 186 98.41 188 99.47
≥0.1 194 97.00 197 98.50 186 98.41 188 99.47
≥1 167 83.50 99 49.50 164 86.77 97 51.32
≥10 13 6.50 3 1.50 12 6.35 3 1.59
≥100 0 0.00 0 0.00 0 0.00 0 0.00
N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

PCV7 group.

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Serotype 6A, After Full Immunizations (FAS) Serotype 6A, After Full Immunizations (PPS)



Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

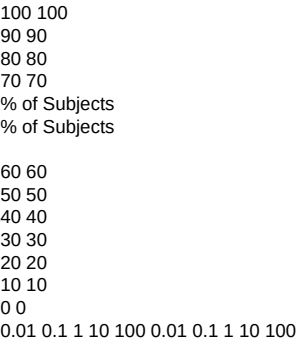
Figure 63 Reverse cumulative distribution curves, pneumococcal serotype 6A IgG antibody concentrations for subjects aged 12-23 months in PCV13-TT group and PCV7 group post full immunizations

Table 111. Distribution of the pneumococcal serotype 7F IgG antibody concentrations for subjects in PCV13-TT group and PCV7 group post full immunizations

FAS N (%)	PPS N (%)	Concentration PCV13-TT (N=200)	PCV7 (N=200)	PCV13-TT (N=189)	PCV7 (N=189)
n %	n %	n %	n %	n %	n %
≥0.01	199 99.50	186 93.00	189 100.0	176 93.12	
≥0.1	199 99.50	186 93.00	189 100.0	176 93.12	
≥1	191 95.50	36 18.00	188 99.47	36 19.05	
≥10	25 12.50	1 0.50	25 13.23	1 0.53	
≥100	0 0.00	0 0.00	0 0.00	0 0.00	

N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

Serotype 7F, After Full Immunizations (FAS) Serotype 7F, After Full Immunizations (PPS)



Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

Figure 64 Reverse cumulative distribution curves, pneumococcal serotype 7F, IgG antibody concentrations for subjects aged 12-23 months in PCV13-TT group and PCV7 group - post full immunizations

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Table 112. Distribution of the pneumococcal serotype 19A IgG antibody concentrations for subjects in PCV13-TT group and PCV7 group post full immunizations

FAS N (%)	PPS N (%)	Concentration PCV13-TT (N=200)	PCV7 (N=200)	PCV13-TT (N=189)	PCV7 (N=189)
n %	n %	n %	n %	n %	n %
≥0.01	200 100.0	199 99.50	189 100.0	188 99.47	
≥0.1	200 100.0	199 99.50	189 100.0	188 99.47	
≥1	194 97.00	174 87.00	189 100.0	168 88.89	
≥10	64 32.00	20 10.00	64 33.86	20 10.58	
≥100	1 0.50	2 1.00	1 0.53	2 1.06	

N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

Serotype 19A, After Full Immunizations (FAS) Serotype 19A, After Full Immunizations (PPS)

100 100
90 90
80 80
70 70

% of Subjects
% of Subjects

60 60
50 50
40 40
30 30
20 20
10 10
0 0
0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

Figure 65 Reverse cumulative distribution curves, pneumococcal serotype 19A, IgG antibody concentrations for subjects aged 12-23 months in PCV13-TT group and PCV7 group - post full immunizations

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2.7.3.2.2.2.3. Comparison of percentage of subjects achieving IgG antibody concentration $\geq 1.0 \mu\text{g/mL}$ with comparison to the PCV7 seroresponse
Data for the 7 common serotypes
All common serotypes met the noninferiority criterion after immunization. The lower limit of the 2-sided 95% CI for rate difference of IgG seroresponse rates $\geq 1.0 \mu\text{g/mL}$ was all greater than -10% (Table 113).
Data for the 6 additional serotypes
All additional serotypes met the noninferiority criterion after primary series. The lower limit of the 2-sided 95% CI rate difference of IgG seroresponse rates $\geq 1.0 \mu\text{g/mL}$ was all greater than 0.

Table 113. Comparison of percentage of subjects aged 12-23 months achieving a pneumococcal IgG antibody concentration $\geq 1.0 \mu\text{g/mL}$ post full immunizations (PPS)

IgG antibody concentration $\geq 1.0 \mu\text{g/mL}$									
PCV13-TT (N=189) PCV7 (N=189) Rate difference c									
Serotype	na	%	95% CI	nb	%	95% CI	%	95% CI	/97.5%CI
Common									
4	189	100.0	98.07-100.0	187	98.94	96.23-99.87	1.06	-0.40,2.52	
6B	188	99.47	97.09-99.99	164	86.77	81.10-91.25	12.70	7.76,17.64	
9V**	173	91.53	86.62-95.08	173	91.53	86.62-95.08	0.00	-5.61,5.61	
14	189	100.0	98.07-100.0	189	100.0	98.07-100.0	0.00	-1.93,1.93	
18C	189	100.0	98.07-100.0	183	96.83	93.22-98.83	3.17	0.68,5.67	
19F	189	100.0	98.07-100.0	185	97.88	94.67-99.42	2.12	0.06,4.17	
23F	184	97.35	93.93-99.14	169	89.42	84.13-93.42	7.94	2.99,12.88	
Additional									
1	176	93.12	88.53-96.29	25	13.23	8.75-18.90	79.89	73.87,85.92	
3	103	54.50	47.11-61.74	18	9.52	5.74-14.63	44.97	36.73,53.21	
5	93	49.21	41.88-56.56	5	2.65	0.86-6.07	46.56	39.08,54.05	
6A	164	86.77	81.10-91.25	97	51.32	43.96-58.64	35.45	26.84,44.06	
7F	188	99.47	97.09-99.99	36	19.05	13.71-25.38	80.42	74.73,86.12	
19A	189	100.0	98.07-100.0	168	88.89	83.52-92.99	11.11	6.63,15.59	

N=number of subjects included in PPS for PCV13-TT and PCV7.
a. number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT group.
b. number of subjects with a determinate IgG antibody concentration to the given serotype for PCV7 group.
c. The rate difference of PCV13-TT – PCV7

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Table 114. Comparison of percentage of subjects aged 12-23 months achieving a pneumococcal IgG antibody concentration $\geq 1.0 \mu\text{g/mL}$ post full immunizations (FAS)

IgG antibody concentration $\geq 1.0 \mu\text{g/mL}$									
PCV13-TT (N=200) PCV7 (N=200) Rate difference c									
Serotype	na	%	95% CI	nb	%	95% CI	%	95% CI	/97.5%CI
Common									
4	192	96.00	92.27-98.26	190	95.00	91.00-97.58	1.00	-3.06,5.06	
6B	191	95.50	91.63-97.92	167	83.50	77.62-88.36	12.00	6.11,17.89	
9V**	175	87.50	82.10-91.74	176	88.00	82.67-92.16	-0.50	-6.93,5.93	
14	193	96.50	92.92-98.58	194	97.00	93.58-98.89	-0.50	-3.98,2.98	
18C	191	95.50	91.63-97.92	186	93.00	88.53-96.12	2.50	-2.06,7.06	
19F	193	96.50	92.92-98.58	191	95.50	91.63-97.92	1.00	-2.84,4.84	
23F	186	93.00	88.53-96.12	171	85.50	79.84-90.07	7.50	1.47,13.53	
Additional									
1	179	89.50	84.40-93.38	25	12.50	8.26-17.90	77.00	70.75,83.25	
3	105	52.50	45.34-59.59	19	9.50	5.82-14.44	43.00	34.97,51.03	
5	93	46.50	39.44-53.67	5	2.50	0.82-5.74	44.00	36.76,51.24	
6A	167	83.50	77.62-88.36	99	49.50	42.37-56.64	34.00	25.37,42.63	
7F	191	95.50	91.63-97.92	36	18.00	12.94-24.04	77.50	71.45,83.55	

19A 194 97.00 93.58-98.89 174 87.00 81.53-91.33 10.00 4.77,15.23
N=number of subjects included in PPS for PCV13-TT and PCV7.
a. number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT group.
b. number of subjects with a determinate IgG antibody concentration to the given serotype for PCV7 group.
c. The rate difference of PCV13-TT – PCV7.

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2.7.3.2.2.2.4. Comparison of percentage of subjects achieving an OPA antibody $\geq 1:8$ and the IgG antibody geometric mean titers (GMTs)
Data for 7 common serotypes
For the 7 common serotypes, the percentage of subjects achieving an OPA antibody titer $\geq 1:8$ was all higher than 90% in PCV13-TT group (Table 115). The lower limit of the 2-sided 95% CI of rate difference of the percentage of subjects achieving an OPA antibody titer $\geq 1:8$ was greater than -10% for all the common serotypes. With the exception of serotypes 6B and 23F, the lower limit of the 2-sided 95% CI of GMR was greater than 0 % for the common serotypes.
Data for 6 additional serotypes
With the exception of serotype 1 (88.04%), the percentage of subjects achieving an OPA antibody titer $\geq 1:8$ was all higher than 90% in PCV13-TT group. The lower limit of the 2-sided 95% CI of rate difference of the percentage of subjects achieving an OPA antibody titer $\geq 1:8$ was greater than 0 for all the additional serotypes. The lower limit of the 2-sided 95% CI of GMR was greater than 1 for all the additional serotypes.

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Table 115. Comparison of percentage of subjects achieving a pneumococcal IgG antibody titer $\geq 1:8$ and GMTs in PCV13-TT group and PCV7 group aged 7-11 months post full immunizations (PPS)
OPA antibody titer $\geq 1:8$ OPA GMT
PCV13-TT (N=92) PCV7 (N=93) Rate difference a PCV13-TT (N=92) PCV7 (N=93)
Serotype 95%CI Ratiob 95%CI
% % % GMT GMT
Common
4 100.0 100.0 0.00 -3.89,3.93 1756.58 1993.15 0.88 0.67,1.16
6B 94.57 91.40 3.17 -4.18,10.51 923.67 1530.14 0.60 0.32,1.15
9V 100.0 100.0 0.00 -3.89,3.93 3219.64 2824.34 1.14 0.75,1.74
14 100.0 100.0 0.00 -3.89,3.93 6373.82 7303.79 0.87 0.63,1.22
18C 100.0 100.0 0.00 -3.89,3.93 3133.89 3971.64 0.79 0.59,1.06
19F 98.91 92.47 6.44 0.67,12.21 816.50 331.18 2.47 1.62,3.76
23F 98.91 96.77 2.14 -2.03,6.31 2069.47 3784.23 0.55 0.35,0.86
Additional
1 88.04 3.23 84.82 77.28,92.36 42.36 2.23 19.01 13.86,26.07
3 100.0 75.27 24.73 15.96,33.50 361.20 14.09 25.63 18.17,36.15
5 98.91 7.53 91.39 85.62,97.15 192.91 2.80 68.79 49.39,95.81
6A 96.74 78.49 18.24 9.14,27.35 1434.20 487.96 2.94 1.39,6.21
7F 100.0 49.46 50.54 40.38,60.70 5524.63 32.24 171.35 89.87,326.73
19A 100.0 88.17 11.83 5.26,18.39 1368.75 166.43 8.22 5.21,12.99
a. The rate difference of PCV13-TT – PCV7.
b. GMC ratio: PCV13-TT to PCV7.

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Table 116. Comparison of percentage of subjects achieving a pneumococcal IgG antibody titer $\geq 1:8$ and GMTs in PCV13-TT group and PCV7 group aged 7-11 months post full immunizations (FAS)
OPA antibody titer $\geq 1:8$ OPA GMT
PCV13-TT (N=100) PCV7 (N=100) Rate difference a PCV13-TT (N=100) PCV7 (N=100)
Serotype 95%CI Ratiob 95%CI
% % % GMT GMT
Common
4 94.00 95.00 -1.00 -7.32,5.32 1172.19 1391.19 0.84 0.51,1.40
6B 89.00 87.00 2.00 -7.00,11.00 655.69 1125.35 0.58 0.28,1.21
9V 97.00 96.00 1.00 -4.09,6.09 2331.78 2098.68 1.11 0.63,1.94
14 96.00 98.00 -2.00 -6.72,2.72 4424.86 5589.37 0.79 0.47,1.33
18C 94.00 95.00 -1.00 -7.32,5.32 2047.64 2719.64 0.75 0.43,1.31
19F 93.00 88.00 5.00 -3.10,13.10 574.96 258.02 2.23 1.31,3.79
23F 94.00 92.00 2.00 -5.07,9.07 1437.95 2625.25 0.55 0.30,1.01
Additional
1 83.00 3.00 80.00 71.91,88.09 35.78 2.21 16.18 11.68,22.41
3 98.00 77.00 21.00 12.31,29.69 310.59 15.08 20.59 14.22,29.81
5 93.00 7.00 86.00 78.93,93.07 146.14 2.74 53.36 36.59,77.81
6A 91.00 76.00 15.00 4.92,25.08 993.87 399.94 2.49 1.12,5.51
7F 99.00 52.00 47.00 37.02,56.98 4870.68 37.56 129.67 68.15,246.71
19A 94.00 85.00 9.00 0.59,17.41 922.97 136.75 6.75 3.87,11.76
a. The rate difference of PCV13-TT – PCV7.
b. GMC ratio: PCV13-TT to PCV7.

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2.7.3.2.2.2.3. Pneumococcal immunogenicity results for subjects aged 24-71 months in PCV13-TT group and PCV7 group
2.7.3.2.2.2.3.1. Comparison of percentage of subjects achieving IgG antibody concentration $\geq 0.35 \mu\text{g/mL}$ $\mu\text{g/mL}$ and the IgG antibody geometric mean concentrations (GMCs)

Data for the 7 common serotypes

The proportion of responders ≥ 0.35 $\mu\text{g/mL}$ in the PCV13-TT group were all higher than 98% and all common serotypes met the noninferiority criterion for the proportion of responders ≥ 0.35 $\mu\text{g/mL}$ (Table 117). The GMCs were all greater than 3.84 $\mu\text{g/mL}$. With the serotype 14, all common serotypes met the noninferiority criterion for GMC. The GMR to serotype 14 was 0.65, with a lower limit of the 2-sided 95% CI of -0.49.

Data for the 6 additional serotypes with comparison to the PCV7 seroresponse

With the exception of serotype 3 (88.89%), the proportion of responders ≥ 0.35 $\mu\text{g/mL}$ in the PCV13-TT group were all higher than 96%. The lower limit of the 2-sided 95% CI for the proportion of responders ≥ 0.35 $\mu\text{g/mL}$ was all greater than 70%. The lower limit of the 2-sided 95% CI for the rate difference for the proportion of responders ≥ 0.35 $\mu\text{g/mL}$ was all greater than 0 for all additional serotypes except serotypes 6A (-0.07) and 19A (-0.48). The GMR was all greater than 1 for all the additional serotypes.

Data for the 6 additional serotypes with comparison to the PCV7 reference

The PCV7 reference value for the 6 additional serotypes was based on serotype 23F from the PCV7 group (the lowest response among the 7 serotypes contained in PCV7 achieved by subjects who received PCV7). For serotype 23F, the proportion of responders was 97.99%. All additional serotypes met the non-inferiority criterion except serotype 3. For serotype 3, the lower limit of the 2-sided 95% CI for the proportion of responders ≥ 0.35 $\mu\text{g/mL}$ was -13.89.

The PCV7 reference value for the 6 additional serotypes was serotype 6B from the PCV7 group (the lowest response among the 7 serotypes contained in PCV7 achieved by subjects who received PCV7). For serotype 6B, the GMC was 3.63. The GMR >0.5 was met for 4 of 6 additional serotypes. The exceptions were serotype 3 and serotype 5. The lower limit of the 2-sided 95% CI for the GMR was met for 3 of 6 additional serotypes. The exceptions were serotype 1, 3 and 5.

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Table 117. Comparison of percentage of subjects aged 24-71 months achieving a pneumococcal IgG antibody concentration ≥ 0.35 $\mu\text{g/mL}$ and GMCs post full immunizations (PPS)

IgG antibody concentration ≥ 0.35 $\mu\text{g/mL}$ IgG antibody GMC

PCV13-TT (N=198) PCV7 (N=199) Rate difference c PCV13-TT (N=198) PCV7 (N=199) GMC Ratio

Serotype Na % 95% CI Nb % 95% CI % 95% CI GMC 95% CI GMC 95% CI Ratio d 95% CI e

PCV7

4 198 100.0 98.15-100.0 199 100.0 98.16-100.0 0.00 -1.84,1.85 8.90 7.95-9.96 8.58 7.63-9.66 1.04 0.88,1.22

6B 198 100.0 98.15-100.0 197 98.99 96.42-99.88 1.01 -0.38,2.39 7.75 6.58-9.12 3.63 3.04-4.32 2.14 1.68,2.71

9V 195 98.48 95.64-99.69 197 98.99 96.42-99.88 -0.51 -2.70,1.68 3.84 3.35-4.41 3.94 3.36-4.62 0.97 0.79,1.20

14 197 99.49 97.22-99.99 198 99.50 97.23-99.99 -0.00 -1.40,1.39 10.20 8.40-12.38 15.75 12.82-19.34 0.65 0.49,0.86

18C 196 98.99 96.40-99.88 199 100.0 98.16-100.0 -1.01 -2.40,0.38 8.82 7.56-10.29 7.47 6.45-8.66 1.18 0.95,1.46

19F 198 100.0 98.15-100.0 199 100.0 98.16-100.0 0.00 -1.84,1.85 11.45 10.05-13.04 5.71 4.98-6.54 2.01 1.66,2.42

23F 196 98.99 96.40-99.88 195 97.99 94.93-99.45 1.00 -1.40,3.40 4.42 3.80-5.14 4.29 3.56-5.17 1.03 0.81,1.31

Additional

vs PCV7 seroresponse f

1 195 98.48 95.64-99.69 132 66.33 59.31-72.86 32.15 25.37,38.94 2.06 1.84-2.30 0.31 0.26-0.39 6.54 5.17,8.27

3 176 88.89 83.66-92.90 89 44.72 37.69-51.92 44.17 35.99,52.34 0.84 0.75-0.94 0.38 0.33-0.45 2.19 1.80,2.66

5 195 98.48 95.64-99.69 127 63.82 56.72-70.50 34.67 27.78,41.56 1.60 1.43-1.79 0.31 0.25-0.38 5.13 4.06,6.48

6A 191 96.46 92.85-98.57 183 91.96 87.27-95.33 4.50 -0.07,9.08 2.35 2.00-2.76 1.29 1.05-1.60 1.82 1.39,2.37

7F 196 98.99 96.40-99.88 142 71.36 64.54-77.53 27.63 21.20,34.07 4.60 4.00-5.28 0.43 0.33-0.56 10.80 8.00,14.58

19A 198 100.0 98.15-100.0 198 99.50 97.23-99.99 0.50 -0.48,1.48 9.56 8.47-10.79 3.32 2.85-3.88 2.87 2.36,3.49

vs PCV7 reference g

1 195 98.48 95.64-99.69 195 97.99 94.93-99.45 0.49 -2.09,3.08 2.06 1.84-2.30 3.63 3.04-4.32 0.57 0.46,0.70

3 176 88.89 83.66-92.90 195 97.99 94.93-99.45 -9.10 -13.89,-4.31 0.84 0.75-0.94 3.63 3.04-4.32 0.23 0.19,0.29

5 195 98.48 95.64-99.69 195 97.99 94.93-99.45 0.49 -2.09,3.08 1.60 1.43-1.79 3.63 3.04-4.32 0.44 0.36,0.54

6A 191 96.46 92.85-98.57 195 97.99 94.93-99.45 -1.53 -4.75,1.70 2.35 2.00-2.76 3.63 3.04-4.32 0.65 0.51,0.82

7F 196 98.99 96.40-99.88 195 97.99 94.93-99.45 1.00 -1.40,3.40 4.60 4.00-5.28 3.63 3.04-4.32 1.27 1.01,1.58

19A 198 100.0 98.15-100.0 195 97.99 94.93-99.45 2.01 0.06,3.96 9.56 8.47-10.79 3.63 3.04-4.32 2.64 2.13,3.26

N=number of subjects included in PPS for PCV13-TT and PCV7.

a. number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT group.

b. number of subjects with a determinate IgG antibody concentration to the given serotype for PCV7 group.

c. The rate difference of PCV13-TT – PCV7.

d. GMC ratio: PCV13-TT to PCV7.

e. Two-sided 95% CI is for additional serotypes, and two-sided 97.5% CI is for common serotypes.

f. Difference in proportions: (PCV13-TT – PCV7 seroresponse) expressed as a percentage. GMC ratio: PCV13-TT to PCV7 seroresponse.

g. Difference in proportions: (PCV13-TT – PCV7 reference) expressed as a percentage. For the additional serotypes, the reference value is derived from serotype 23F from the PCV7 group. GMC ratio:

PCV13-TT to PCV7 reference. For the additional serotypes, the reference value is derived from serotype 6B from the PCV7 group.

Table 118. Comparison of percentage of subjects aged 24-71 months achieving a pneumococcal IgG antibody concentration $\geq 0.35 \mu\text{g/mL}$ and GMCs post full immunizations (FAS)

IgG antibody concentration $\geq 0.35 \mu\text{g/mL}$ IgG antibody GMC														
PCV13-TT (N=200) PCV7 (N=200) Rate difference c PCV13-TT (N=200) PCV7 (N=200) GMC Ratio														
95% CI														
Serotype Na % 95% CI Nb % 95% CI % GMC 95% CI GMC 95% CI Ratio d 95%CI e														
/97.5%CI														
PCV7														
4	199	99.50	97.25-99.99	199	99.50	97.25-99.99	0.00	-1.38,1.38	8.71	7.73-9.81	8.43	7.46-9.53	1.03	
0.87,1.22														
6B	200	100.0	98.17-100.0	198	99.00	96.43-99.88	1.00	-0.38,2.38	7.63	6.48-9.00	3.62	3.04-4.30	2.11	
1.66,2.68														
9V**	197	98.50	95.68-99.69	198	99.00	96.43-99.88	-0.50	-2.68,1.68	3.82	3.33-4.38	3.91	3.34-4.59	0.98	
0.79,1.20														
14	199	99.50	97.25-99.99	199	99.50	97.25-99.99	0.00	-1.38,1.38	10.01	8.24-12.16	15.60	12.70-19.15	0.64	
0.48,0.85														
18C	198	99.00	96.43-99.88	200	100.0	98.17-100.0	-1.00	-2.38,0.38	8.86	7.61-10.33	7.37	6.34-8.55	1.20	
0.97,1.49														
19F	200	100.0	98.17-100.0	200	100.0	98.17-100.0	0.00	-1.83,1.83	11.39	10.01-12.97	5.69	4.97-6.51	2.00	
1.66,2.42														
23F	197	98.50	95.68-99.69	196	98.00	94.96-99.45	0.50	-2.07,3.07	4.34	3.73-5.06	4.26	3.53-5.13	1.02	
0.80,1.30														
Additional														
vs PCV7 seroresponse f														
1	196	98.00	94.96-99.45	133	66.50	59.50-73.00	31.50	24.68,38.32	2.03	1.81-2.27	0.32	0.26-0.39	6.42	
5.08,8.11														
3	177	88.50	83.25-92.57	90	45.00	37.98-52.18	43.50	35.31,51.69	0.83	0.74-0.93	0.39	0.33-0.45	2.15	
1.77,2.61														
5	196	98.00	94.96-99.45	128	64.00	56.93-70.65	34.00	27.07,40.93	1.59	1.42-1.77	0.32	0.26-0.39	5.02	
3.97,6.34														
6A	192	96.00	92.27-98.26	184	92.00	87.33-95.36	4.00	-0.64,8.64	2.31	1.97-2.72	1.29	1.04-1.60	1.79	
1.37,2.34														
7F	197	98.50	95.68-99.69	143	71.50	64.71-77.64	27.00	20.52,33.48	4.51	3.91-5.19	0.43	0.33-0.56	10.55	
7.82,14.23														
19A	200	100.0	98.17-100.0	199	99.50	97.25-99.99	0.50	-0.48,1.48	9.47	8.40-10.68	3.31	2.84-3.86	2.86	
2.36,3.48														
vs PCV7 reference g														
1	196	98.00	94.96-99.45	196	98.00	94.96-99.45	0.00	-2.74,2.74	2.03	1.81-2.27	3.62	3.04-4.30	0.56	
0.46,0.69														
3	177	88.50	83.25-92.57	196	98.00	94.96-99.45	-9.50	-14.33,-4.67	0.83	0.74-0.93	3.62	3.04-4.30	0.23	
0.19,0.28														
5	196	98.00	94.96-99.45	196	98.00	94.96-99.45	0.00	-2.74,2.74	1.59	1.42-1.77	3.62	3.04-4.30	0.44	
0.36,0.54														
6A	192	96.00	92.27-98.26	196	98.00	94.96-99.45	-2.00	-5.34,1.34	2.31	1.97-2.72	3.62	3.04-4.30	0.64	
0.50,0.81														
7F	197	98.50	95.68-99.69	196	98.00	94.96-99.45	0.50	-2.07,3.07	4.51	3.91-5.19	3.62	3.04-4.30	1.25	
1.00,1.56														
19A	200	100.0	98.17-100.0	196	98.00	94.96-99.45	2.00	0.06,3.94	9.47	8.40-10.68	3.62	3.04-4.30	2.62	
2.12,3.23														

2.7.3.2.2.3.2. Pneumococcal Serotype antibody concentrations in the subjects aged 24-71 months in PCV13-TT group and PCV7 group
The results of pneumococcal serotype antibody concentrations for subjects aged 24-71 months after full immunizations in PCV13-TT group and PCV7 group were shown in Table 119. to Table 131. The reverse cumulative distribution curves were shown in Figure 66 to 78.

Table 119. Distribution of the pneumococcal serotype 4 IgG antibody concentrations for subjects aged 24-71 months in PCV13-TT group and PCV7 group post full immunization

FAS N (%) PPS N (%)														
PCV13-TT PCV13-TT														
Concentration PCV7 (N=200) PCV7 (N=199)														
(N=200) (N=198)														
n % n % n % n %														
≥ 0.01	200	100.0	200	100.0	198	100.0	199	100.0						
≥ 0.1	200	100.0	200	100.0	198	100.0	199	100.0						
≥ 1	198	99.00	198	99.00	197	99.49	198	99.50						
≥ 10	85	42.50	82	41.00	85	42.93	82	41.21						
≥ 100	0	0.00	0	0.00	0	0.00	0	0.00						

N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and

PCV7 group.

Serotype 4, After Full Immunization (FAS) Serotype 4, full immunization (PPS)
100 100
90 90
80 80
70 70
% of Subjects
% of Subjects

60 60
50 50
40 40
30 30
20 20
10 10
0 0
0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration (µg/mL) Antibody Concentration ■µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

Figure 66 Reverse cumulative distribution curves, pneumococcal serotype 4, IgG antibody concentrations for subjects aged 24-71 months in PCV13-TT group and PCV7 group - post full immunization

Table 120. Distribution of the pneumococcal serotype 6B IgG antibody concentrations for subjects aged 24-71 months in PCV13-TT group and PCV7 group post full immunization

FAS N (%)	PPS N (%)
PCV13-TT	PCV13-TT
Concentration PCV7 (N=200)	PCV7 (N=199)
(N=200)	(N=198)
n %	n %
≥0.01	200 100.0
≥0.1	200 100.0
≥1	192 96.00
≥10	81 40.50
≥100	4 2.00

N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

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Serotype 6B, After Full Immunization (FAS) Serotype 6B, After Full Immunization (PPS)
100 100
90 90
80 80
70 70
% of Subjects
% of Subjects

60 60
50 50
40 40
30 30
20 20
10 10
0 0
0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

Figure 67 Reverse cumulative distribution curves, pneumococcal serotype 6B, IgG antibody concentrations for subjects aged 24-71 months in PCV13-TT group and PCV7 group - post full immunization

Table 121. Distribution of the pneumococcal serotype 9V IgG antibody concentrations for subjects aged 24-71 months in PCV13-TT group and PCV7 group post full immunization

FAS N (%)	PPS N (%)
PCV13-TT	PCV13-TT
Concentration PCV7 (N=200)	PCV7 (N=199)
(N=200)	(N=198)
n %	n %
≥0.01	200 100.0
≥0.1	200 100.0
≥1	181 90.50
≥10	32 16.00
≥100	0 0.00

N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

Serotype 9V, After Full Immunization (FAS) Serotype 9V, After Full Immunization (PPS)

100 100

90 90
80 80
70 70
% of Subjects
% of Subjects

60 60
50 50
40 40
30 30
20 20
10 10
0 0
0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration ■µg/mL
Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

Figure 68 Reverse cumulative distribution curves, pneumococcal serotype 9V, IgG antibody concentrations for subjects aged 24-71 months in PCV13-TT group and PCV7 group - post full immunization

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Table 122. Distribution of the pneumococcal serotype 14 IgG antibody concentrations for subjects aged 24-71 months in PCV13-TT group and PCV7 group post full immunization

FAS N (%)	PPS N (%)	PCV13-TT	PCV13-TT
Concentration		PCV7 (N=200)	PCV7 (N=199)
(N=200)		(N=198)	
n	%	n	%
≥0.01	200	100.0	199 99.50 198 100.0 198 99.50
≥0.1	199	99.50	199 99.50 197 99.49 198 99.50
≥1	198	99.00	198 99.00 197 99.49 197 98.99
≥10	99	49.50	118 59.00 99 50.00 118 59.30
≥100	9	4.50	26 13.00 9 4.55 26 13.07

N=number of subjects included in FAS and PPS for vaccine group.

n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

Serotype 14, After Full Immunization (FAS) Serotype 14, After Full Immunization (PPS)

100 100
90 90
80 80
70 70

% of Subjects
% of Subjects

60 60
50 50
40 40
30 30
20 20
10 10
0 0
0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration ■µg/mL
Antibody Concentration (µg/mL)

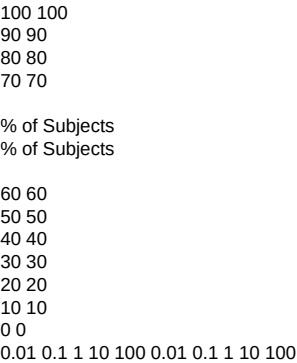
PCV13-TT PCV7 PCV13-TT PCV7

Figure 69 Reverse cumulative distribution curves, pneumococcal serotype 14, IgG antibody concentrations for subjects aged 24-71 months in PCV13-TT group and PCV7 group - post full immunization

Table 123. Distribution of the pneumococcal serotype 18C IgG antibody concentrations for subjects aged 24-71 months in PCV13-TT group and PCV7 group post full immunization									
FAS N (%)		PPS N (%)		PCV13-TT		PCV13-TT			
Concentration		PCV7 (N=200)		PCV7 (N=199)		(N=200)		(N=198)	
n	%	n	%	n	%	n	%	n	%
≥0.01	199	99.50	200	100.0	197	99.49	199	100.0	
≥0.1	199	99.50	200	100.0	197	99.49	199	100.0	
≥1	197	98.50	195	97.50	195	98.48	195	97.99	
≥10	98	49.00	73	36.50	97	48.99	73	36.68	
≥100	1	0.50	5	2.50	1	0.51	5	2.51	
N=number of subjects included in FAS and PPS for vaccine group.									
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.									

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Serotype 18C, After Full Immunization (FAS) Serotype 18C, After Full Immunization (PPS)



Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

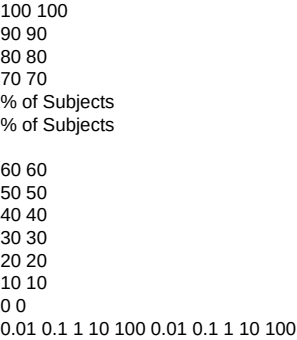
Figure 70 Reverse cumulative distribution curves, pneumococcal serotype 18C, IgG antibody concentrations for subjects aged 24-71 months in PCV13-TT group and PCV7 group - post full immunization

Table 124. Distribution of the pneumococcal serotype 19F IgG antibody concentrations for subjects aged 24-71 months in PCV13-TT group and PCV7 group post full immunization

FAS N (%)	PPS N (%)	PCV13-TT N (%)	PCV13-TT N (%)
Concentration	PCV7 (N=200)	PCV7 (N=199)	(N=200) (N=198)
n %	n %	n %	n %
≥0.01	200 100.0	200 100.0	198 100.0
≥0.1	200 100.0	200 100.0	198 100.0
≥1	198 99.00	196 98.00	196 98.99
≥10	110 55.00	54 27.00	109 55.05
≥100	4 2.00	1 0.50	4 2.02

N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

Serotype 19F, After Full Immunization (FAS) Serotype 19F, After Full Immunization (PPS)



Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

Figure 71 Reverse cumulative distribution curves, pneumococcal serotype 19F, IgG antibody concentrations for subjects aged 24-71 months in PCV13-TT group and PCV7 group - post full immunization

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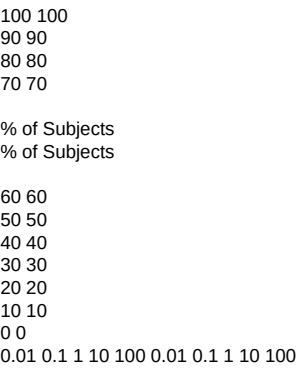
Table 125. Distribution of the pneumococcal serotype 23F IgG antibody concentrations for subjects aged 24-71 months in PCV13-TT group and PCV7 group post full immunization

FAS N (%)	PPS N (%)	PCV13-TT N (%)	PCV13-TT N (%)
Concentration	PCV7 (N=200)	PCV7 (N=199)	(N=200) (N=198)
n %	n %	n %	n %
≥0.01	200 100.0	199 99.50	198 100.0
≥0.1	200 100.0	199 99.50	198 100.0
≥1	185 92.50	178 89.00	184 92.93
≥10	41 20.50	43 21.50	41 20.71
≥100	0 0.00	4 2.00	0 0.00

N=number of subjects included in FAS and PPS for vaccine group.

n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

Serotype 23F, After Full Immunization (FAS) Serotype 23F, After Full Immunization (PPS)



Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

Figure 72 Reverse cumulative distribution curves, pneumococcal serotype 23F, IgG antibody concentrations for subjects aged 24-71 months in PCV13-TT group and PCV7 group - post full immunization

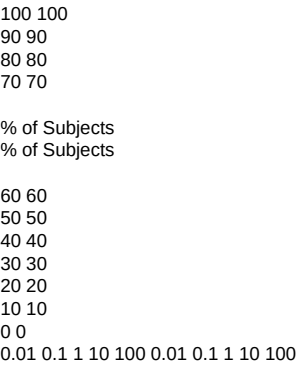
Table 126. Distribution of the pneumococcal serotype 1 IgG antibody concentrations for subjects aged 24-71 months in PCV13-TT group and PCV7 group post full immunization

FAS N (%)	PPS N (%)
PCV13-TT	PCV13-TT
Concentration PCV7 (N=200)	PCV7 (N=199)
(N=200)	(N=198)
n %	n %
n %	n %
n %	n %
≥0.01	200 100.0 186 93.00 198 100.0 185 92.96
≥0.1	200 100.0 186 93.00 198 100.0 185 92.96
≥1	165 82.50 13 6.50 165 83.33 13 6.53
≥10	5 2.50 0 0.00 5 2.53 0 0.00
≥100	0 0.00 0 0.00 0 0.00 0 0.00

N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

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Serotype 1, After Full Immunization (FAS) Serotype 1, After Full Immunization (PPS)



Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

Figure 73 Reverse cumulative distribution curves, pneumococcal serotype 1, IgG antibody concentrations for subjects aged 24-71 months in PCV13-TT group and PCV7 group - post full immunization

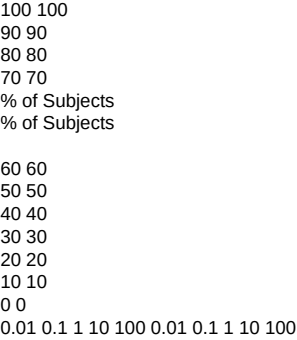
Table 127. Distribution of the pneumococcal serotype 3 IgG antibody concentrations for subjects aged 24-71 months in PCV13-TT group and PCV7 group post full immunization

FAS N (%)	PPS N (%)
PCV13-TT	PCV13-TT
Concentration PCV7 (N=200)	PCV7 (N=199)
(N=200)	(N=198)
n %	n %
n %	n %
n %	n %
≥0.01	200 100.0 199 99.50 198 100.0 198 99.50
≥0.1	199 99.50 189 94.50 198 100.0 188 94.47
≥1	69 34.50 35 17.50 68 34.34 34 17.09
≥10	2 1.00 1 0.50 2 1.01 1 0.50
≥100	0 0.00 0 0.00 0 0.00 0 0.00

N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

PCV7 group.

Serotype 3, After Full Immunization (FAS) Serotype 3, After Full Immunization (PPS)



Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

Figure 74 Reverse cumulative distribution curves, pneumococcal serotype 3, IgG antibody concentrations for subjects aged 24-71 months in PCV13-TT group and PCV7 group - post full immunization

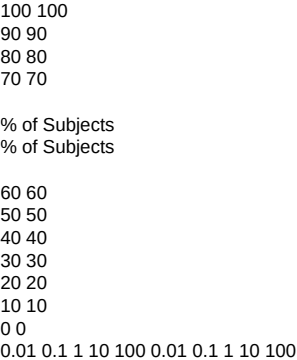
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Table 128. Distribution of the pneumococcal serotype 5 IgG antibody concentrations for subjects aged 24-71 months in PCV13-TT group and PCV7 group post full immunization

FAS N (%)	PPS N (%)	PCV13-TT	PCV13-TT
Concentration	PCV7 (N=200)	PCV7 (N=199)	(N=200) (N=198)
n %	n %	n %	n %
≥0.01	200	100.0	184 92.00 198 100.0 183 91.96
≥0.1	200	100.0	184 92.00 198 100.0 183 91.96
≥1	142	71.00	16 8.00 141 71.21 15 7.54
≥10	4	2.00	0 0.00 4 2.02 0 0.00
≥100	0	0.00	0 0.00 0 0.00 0 0.00

N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

Serotype 5, After Full Immunization (FAS) Serotype 5, After Full Immunization (PPS)



Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

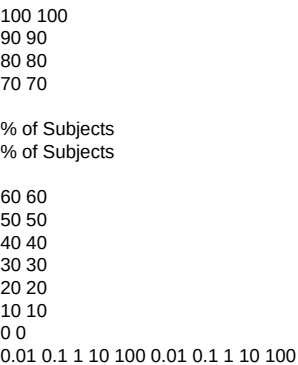
Figure 75 Reverse cumulative distribution curves, pneumococcal serotype 5, IgG antibody concentrations for subjects aged 24-71 months in PCV13-TT group and PCV7 group - post full immunization

FAS N (%)	PPS N (%)	PCV13-TT	PCV13-TT
Concentration	PCV7 (N=200)	PCV7 (N=199)	(N=200) (N=198)
n %	n %	n %	n %
≥0.01	199	99.50	196 98.00 197 99.49 195 97.99
≥0.1	199	99.50	196 98.00 197 99.49 195 97.99
≥1	159	79.50	110 55.00 158 79.80 110 55.28
≥10	19	9.50	12 6.00 19 9.60 12 6.03
≥100	0	0.00	2 1.00 0 0.00 2 1.01

N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

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Serotype 6A, After Full Immunization (FAS) Serotype 6A, After Full Immunization (PPS)



Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

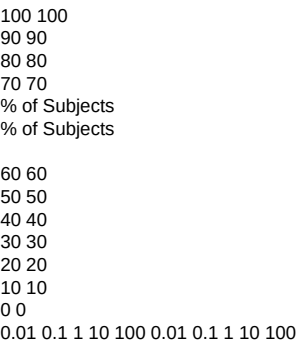
Figure 76 Reverse cumulative distribution curves, pneumococcal serotype 6A, IgG antibody concentrations for subjects aged 24-71 months in PCV13-TT group and PCV7 group - post full immunization

Table 130. Distribution of the pneumococcal serotype 7F IgG antibody concentrations for subjects aged 24-71 months in PCV13-TT group and PCV7 group post full immunizations

FAS N (%)	PPS N (%)	PCV13-TT PCV13-TT
Concentration	PCV7 (N=200)	PCV7 (N=199)
(N=200)	(N=198)	
n %	n %	n %
≥0.01	199 99.50	181 90.50
≥0.1	199 99.50	197 99.49
≥1	192 96.00	191 96.46
≥10	33 16.50	3 1.51
≥100	0 0.00	0 0.00

N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

Serotype 7F, After Full Immunization (FAS) Serotype 7F, After Full Immunization (PPS)



Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

Figure 77 Reverse cumulative distribution curves, pneumococcal serotype 7F, IgG antibody concentrations for subjects aged 24-71 months in PCV13-TT group and PCV7 group - post full immunization

Table 131. Distribution of the pneumococcal serotype 19A IgG antibody concentrations for subjects aged 24-71 months in PCV13-TT group and PCV7 group post full immunization

FAS N (%)	PPS N (%)	PCV13-TT PCV13-TT
Concentration	PCV7 (N=200)	PCV7 (N=199)
(N=200)	(N=198)	
n %	n %	n %
≥0.01	200 100.0	198 100.0
≥0.1	200 100.0	198 100.0
≥1	198 99.00	181 90.95
≥10	104 52.00	28 14.07
≥100	0 0.00	0 0.00

N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

PCV7 group.

Serotype 19A, After Full Immunization (FAS) Serotype 19A, After Full Immunization (PPS)

100 100
90 90
80 80
70 70

% of Subjects
% of Subjects

60 60
50 50
40 40
30 30
20 20
10 10
0 0
0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

Figure 78 Reverse cumulative distribution curves, pneumococcal serotype 19A, IgG antibody concentrations for subjects aged 24-71 months in PCV13-TT group and PCV7 group - post full immunization.

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2.7.3.2.2.3.3. Comparison of percentage of subjects achieving IgG antibody concentration ≥ 1.0 µg/mL with comparison to the PCV7 seroresponse
Data for the 7 common serotypes
The lower limit of the 2-sided 95% CI for rate difference of percentage of subjects with IgG antibody concentration ≥1.0 µg/mL (PCV13-TT – PCV7) was all greater than -10% for all common serotypes (Table 132).

Data for the 6 additional serotypes
The lower limit of the 2-sided 95% CI for rate difference of percentage of subjects with IgG antibody concentration ≥1.0 µg/mL (PCV13-TT – PCV7) was all greater than 0 for all additional serotypes.

Table 132. Comparison of percentage of subjects aged 24-71 months achieving a pneumococcal IgG antibody concentration ≥ 1.0 µg/mL post full immunizations (PPS)
IgG antibody concentration ≥ 1.0 µg/mL

PCV13-TT (N=198)	PCV7 (N=199)	Rate difference c
Serotype na	% 95% CI nb	% 95% CI % 95% CI /97.5%CI
Common		
4	197 99.49 97.22-99.99	198 99.50 97.23-99.99 -0.00 -1.40,1.39
6B	191 96.46 92.85-98.57	173 86.93 81.44-91.28 9.53 4.19,14.87
9V	180 90.91 86.01-94.52	188 94.47 90.32-97.21 -3.56 -8.67,1.55
14	197 99.49 97.22-99.99	197 98.99 96.42-99.88 0.50 -1.20,2.20
18C	195 98.48 95.64-99.69	195 97.99 94.93-99.45 0.49 -2.09,3.08
19F	196 98.99 96.40-99.88	195 97.99 94.93-99.45 1.00 -1.40,3.40
23F	184 92.93 88.42-96.08	178 89.45 84.32-93.35 3.48 -2.08,9.05
Additional		
1	165 83.33 77.40-88.24	13 6.53 3.52-10.91 76.80 70.58,83.02
3	68 34.34 27.76-41.41	34 17.09 12.13-23.05 17.26 8.83,25.69
5	141 71.21 64.37-77.41	15 7.54 4.28-12.13 63.67 56.38,70.97
6A	158 79.80 73.52-85.16	110 55.28 48.08-62.31 24.52 15.63,33.41
7F	191 96.46 92.85-98.57	53 26.63 20.63-33.35 69.83 63.17,76.49
19A	196 98.99 96.40-99.88	181 90.95 86.08-94.55 8.04 3.81,12.26
N=number of subjects included in PPS for PCV13-TT and PCV7.		
a. number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT group.		
b. number of subjects with a determinate IgG antibody concentration to the given serotype for PCV7 group.		
c. The rate difference of PCV13-TT – PCV7		

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Table 133. Comparison of percentage of subjects aged 24-71 months achieving a pneumococcal IgG antibody concentration ≥ 1.0 µg/mL post full immunizations (FAS)
IgG antibody concentration ≥ 1.0 µg/mL

PCV13-TT (N=200)	PCV7 (N=200)	Rate difference c
Serotype na	% 95% CI nb	% 95% CI % 95% CI /97.5%CI
Common		
4	198 99.00 96.43-99.88	198 99.00 96.43-99.88 0.00 -1.95,1.95
6B	192 96.00 92.27-98.26	174 87.00 81.53-91.33 9.00 3.61,14.39
9V**	181 90.50 85.56-94.18	188 94.00 89.75-96.86 -3.50 -8.73,1.73
14	198 99.00 96.43-99.88	198 99.00 96.43-99.88 0.00 -1.95,1.95
18C	197 98.50 95.68-99.69	195 97.50 94.26-99.18 1.00 -1.74,3.74
19F	198 99.00 96.43-99.88	196 98.00 94.96-99.45 1.00 -1.38,3.38
23F	185 92.50 87.93-95.74	178 89.00 83.82-92.98 3.50 -2.17,9.17
Additional		
1	165 82.50 76.51-87.50	13 6.50 3.51-10.86 76.00 69.72,82.28
3	69 34.50 27.94-41.53	35 17.50 12.50-23.49 17.00 8.57,25.43

5 142 71.00 64.18-77.18 16 8.00 4.64-12.67 63.00 55.67,70.33
 6A 159 79.50 73.23-84.87 110 55.00 47.82-62.02 24.50 15.62,33.38
 7F 192 96.00 92.27-98.26 53 26.50 20.52-33.19 69.50 62.81,76.19
 19A 198 99.00 96.43-99.88 182 91.00 86.15-94.58 8.00 3.80,12.20

N=number of subjects included in FAS for PCV13-TT and PCV7.

- number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT group.
- number of subjects with a determinate IgG antibody concentration to the given serotype for PCV7 group.
- The rate difference of PCV13-TT – PCV7.

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2.7.3.2.2.3.4. Comparison of percentage of subjects achieving an OPA antibody $\geq 1:8$ and the IgG antibody geometric mean titers (GMTs)

Data for 7 common serotypes

For the 7 common serotypes, the percentage of subjects achieving an OPA antibody titer $\geq 1:8$ was higher than 96% except serotypes 14 (92.93%) in PCV13-TT group (Table 134). The lower limit of the 2-sided 95% CI of rate difference of the percentage of subjects achieving an OPA antibody titer $\geq 1:8$ was greater than -10% for the common serotypes. With the exceptions of serotypes 14 and 23F, the lower limit of the 2-sided 95% CI of GMR was greater than 0.5. For serotypes 14 and 23F, the lower limit of the 2-sided 95% CI of GMR was 0.75 (95% CI, 0.39 to 1.41) and 0.58 (95% CI, 0.37 to 0.97), respectively.

Data the 6 additional serotypes

For the 6 additional serotypes, the percentage of subjects achieving an OPA antibody titer $\geq 1:8$ was higher than 94% except serotypes 1 (43.43%) in PCV13-TT group. With the exceptions of serotype 6A, the lower limit of the 2-sided 95% CI of rate difference of the percentage of subjects achieving an OPA antibody titer $\geq 1:8$ was greater than 0 for all the common serotypes. For serotype 6A, the rate difference was -0.05(95% CI, -6.12 to 6.02). The lower limit of the 2-sided 95% CI of GMR was greater than 1 for all the additional serotypes.

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Table 134. Comparison of percentage of subjects achieving a pneumococcal IgG antibody titer $\geq 1:8$ and GMTs in PCV13-TT group and PCV7 group aged 24-71 months post full immunizations (PPS)

OPA antibody titer $\geq 1:8$ OPA GMT

PCV13-TT (N=99) PCV7 (N=100) Rate difference a PCV13-TT (N=99) PCV7 (N=100)

Serotype 95%CI Ratiob 95%CI

% % % GMT GMT

Common

4 100.0 96.00 4.00 0.16,7.84 2710.14 1924.49 1.41 0.92,2.16
 6B 97.98 98.00 -0.02 -3.92,3.88 2511.69 2511.16 1.00 0.62,1.61
 9V 100.0 100.0 0.00 -3.62,3.66 2174.68 1980.18 1.10 0.78,1.55
 14 92.93 95.00 -2.07 -8.68,4.54 2534.19 3395.54 0.75 0.39,1.41
 18C 100.0 99.00 1.00 -0.95,2.95 3572.76 3330.72 1.07 0.77,1.49
 19F 98.99 94.00 4.99 -0.06,10.04 1133.78 333.59 3.40 2.20,5.26
 23F 96.97 99.00 -2.03 -5.93,1.87 2025.10 3468.13 0.58 0.37,0.91

Additional

1 43.43 1.00 42.43 32.48,52.39 8.16 2.07 3.95 2.78,5.61
 3 100.0 48.00 52.00 42.21,61.79 121.39 9.67 12.56 8.25,19.11
 5 95.96 9.00 86.96 80.14,93.78 84.80 2.51 33.74 24.67,46.16
 6A 94.95 95.00 -0.05 -6.12,6.02 1737.99 970.59 1.79 1.04,3.09
 7F 100.0 75.00 25.00 16.51,33.49 5886.86 171.55 34.32 19.09,61.70
 19A 100.0 92.00 8.00 2.68,13.23 2070.66 303.83 6.82 4.25,10.93

a. The rate difference of PCV13-TT – PCV7.

b. GMC ratio: PCV13-TT to PCV7.

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Table 135. Comparison of percentage of subjects achieving a pneumococcal IgG antibody titer $\geq 1:8$ and GMTs in PCV13-TT group and PCV7 group aged 24-71 months post full immunizations (FAS)

OPA antibody titer $\geq 1:8$ OPA GMT

PCV13-TT (N=100) PCV7 (N=100) Rate difference a PCV13-TT (N=99) PCV7 (N=100)

Serotype 95%CI Ratiob 95%CI

% % % GMT GMT

Common

4 99.00 96.00 3.00 -1.31,7.31 2521.58 1924.49 1.31 0.84,2.05
 6B 97.00 98.00 -1.00 -5.33,3.33 2338.71 2511.16 0.93 0.57,1.52
 9V 100.0 100.0 0.00 -3.62,3.62 2062.70 1980.18 1.04 0.73,1.49
 14 93.00 95.00 -2.00 -8.58,4.58 2534.75 3395.54 0.75 0.40,1.41
 18C 100.0 99.00 1.00 -0.95,2.95 3453.55 3330.72 1.04 0.74,1.45
 19F 99.00 94.00 5.00 -0.05,10.05 1120.94 333.59 3.36 2.17,5.19
 23F 96.00 99.00 -3.00 -7.31,1.31 1889.70 3468.13 0.54 0.34,0.87

Additional

1 43.00 1.00 42.00 32.10,51.90 8.05 2.07 3.89 2.75,5.51
 3 99.00 48.00 51.00 41.02,60.98 116.51 9.67 12.05 7.87,18.47
 5 95.00 9.00 86.00 78.95,93.05 81.69 2.51 32.50 23.61,44.75
 6A 94.00 95.00 -1.00 -7.32,5.32 1624.27 970.59 1.67 0.96,2.92
 7F 100.0 75.00 25.00 16.51,33.49 5723.35 171.55 33.36 18.52,60.10
 19A 100.0 92.00 8.00 2.68,13.23 2012.51 303.83 6.62 4.12,10.65

a. The rate difference of PCV13-TT – PCV7.

b. GMC ratio: PCV13-TT to PCV7.

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2.7.3.2.3. Discussion and conclusion

2.7.3.2.3.1. Discussion of immunogenicity

2.7.3.2.3.1.1. Discussion of subjects achieving IgG antibody concentration $\geq 0.35 \mu\text{g/mL}$ and the IgG antibody geometric mean concentrations (GMCs).

2.7.3.2.3.1.1.1. Primary series

Analysis of common serotypes

After primary immunization series, the IgG antibody concentration $\geq 0.35 \mu\text{g/mL}$ was high, though in common serotypes, the positive rate of 6B was 90.45% PCV13-TT (3 months) group, besides this, the subjects achieving IgG antibody concentration $\geq 0.35 \mu\text{g/mL}$ of all other serotypes in remaining age groups was above 95%, with antibody GMC consistently over $2.0 \mu\text{g/mL}$. Apart from the serotype 6B in subjects aged 3-71 months, 6B and 23F in subjects aged 3 months, and 9V in subjects aged 7-11 months, the lower limit of two-sided 95% CI for percentage difference (PCV13-TT – PCV7) of other IgG antibody with concentration $\geq 1.0 \mu\text{g/mL}$ was greater than -10%. The percentage of IgG antibody concentration $\geq 1.0 \mu\text{g/mL}$ in subjects of PCV13-TT group aged 12-23 months and 24-71 months were higher than that in PCV7 group. The lower limit of 97.5% CI was $> -10\%$ for the rate difference of antibody positive rate of 7 common serotypes (PCV13-TT – PCV7) in subjects aged 3-71 months after primary immunization series, the lower limit of 97.5% CI for antibody GMC ratio (PCV13-TT / PCV7) was > 0.5 , demonstrating the noninferiority between PCV13-TT group and PCV7 group for common serotypes.

In the 3-month main target population, 6 out of 7 common serotypes (except serotype 6B) met the noninferiority criterion after primary series, and serotype 6B met the noninferiority criterion after booster dose.

For PCV13-TT group subjects aged 2 months vs PCV7 group subjects aged 3 months, the lower limit of 97.5% CI was $> -10\%$ for the rate difference for the subjects achieving IgG antibody concentration $\geq 0.35 \mu\text{g/mL}$ of 7 common serotypes; the lower limit of 2-sided 97.5% CI was > 0.5 for the antibody GMC ratio of serotypes 6B, 9V, 14, 19F, and 23F. The 7 common serotypes in PCV13-TT group subjects aged 2 months were noninferior to PCV7 group subjects aged 3 months.

For PCV13-TT (2 months) group vs PCV13-TT (3 months) group, the additional serotype 5 was observed with a lower limit of 2-sided 97.5% CI $> -10\%$ (-10.84%) for the rate difference of the subjects achieving IgG antibody concentration $\geq 0.35 \mu\text{g/mL}$, and a lower limit of 2-sided 97.5% CI < 0.5 for antibody GMC ratio, besides, the common serotype 4 was with a lower limit of 2-sided 97.5% CI < 0.5 for antibody GMC, apart from the exceptions above, all other serotypes achieved noninferiority criteria in this comparison.

For 7-11 months age group, the lower limit of 95% CI was $> -10\%$ for the rate difference of the subjects achieving IgG antibody concentration $\geq 0.35 \mu\text{g/mL}$ (PCV13-TT – PCV7), the lower limit of 95% CI was > 0.5 for GMC ratio (PCV13-TT / PCV7) of 6 common serotypes except serotype 14. In 12-23 months age group and 24-71 months age group, the lower limit of 95% CI was $> -10\%$ for rate difference of the subjects achieving IgG antibody concentration $\geq 0.35 \mu\text{g/mL}$ (PCV13-TT – PCV7), and the lower limit of 95% CI was > 0.5 for GMC ratio (PCV13-TT / PCV7) for 7 common serotypes.

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Analysis of additional serotypes

In 3-71 months age groups, the subjects achieving IgG antibody concentration $\geq 0.35 \mu\text{g/mL}$ of all additional serotypes were $> 90\%$, and the lower limit of 95% CI was > 0 for the rate difference of the subjects achieving IgG antibody concentration $\geq 0.35 \mu\text{g/mL}$ (PCV13-TT – PCV7), besides, the lower limit of 95% CI was > 1 for GMC ratio (PCV13-TT / PCV7) of each additional serotype, demonstrating the achievement of superiority of each additional serotype in PCV13-TT group. The serotypes with lower subjects achieving IgG antibody concentration $\geq 0.35 \mu\text{g/mL}$ were 23F in 3-71 months age group, 18C in 3 months age group, 23F in 7-11 months age group, 6B in 12-23 months age group, and 23F in 24-71 months age group; serotype 6B was confirmed to be with the lowest antibody GMC in all PCV7 groups.

Of all comparisons, apart from the serotype 5 in 7-11 months and serotype 3 in 24-71 months age group, the lower limit of 97.5% / 95% CI was $> -10\%$ for the rate difference of subjects achieving IgG antibody concentration $\geq 0.35 \mu\text{g/mL}$ of additional serotypes with lowest subjects achieving IgG antibody concentration $\geq 0.35 \mu\text{g/mL}$, and thus were considered to be noninferior to common serotypes with lowest subjects achieving IgG antibody concentration $\geq 0.35 \mu\text{g/mL}$ according to noninferiority criteria.

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2.7.3.2.3.1.1.2. Discussion of booster immunizations

Analysis of common serotypes

After booster dose, in PCV13-TT (3 months) group, the subjects achieving IgG antibody concentration $\geq 0.35 \mu\text{g/mL}$ and antibody GMC of 7 common serotypes were noninferior to those in PCV7 group, both the t PCV13-TT group and PCV7 group had more than 3.5-fold increase in antibody GMC compared to that before booster dose. The subjects achieving IgG antibody concentration $\geq 0.35 \mu\text{g/mL}$ of 7 common serotypes in PCV13-TT (2 months) group were noninferior to PCV7 (3 months) group. The subjects achieving IgG antibody concentration $\geq 0.35 \mu\text{g/mL}$ and antibody GMC of 7 common serotypes in PCV13-TT group aged 7-11 months were noninferior to those in PCV7 group.

Analysis of additional serotypes

As to the 6 additional serotypes, the lower limit of 95% CI for subjects achieving IgG antibody concentration $\geq 0.35 \mu\text{g/mL}$ of 19A was -0.2%, though other serotypes were superior to PCV7 group; all the 6 additional serotypes attained superiority criteria with respect to antibody GMC, with > 3.5 -fold increase compared to that before booster dose in PCV13-TT (3 months) group. As to the 6 additional serotypes, despite the lower limit of 95% CI for antibody positive rate of 19A was -0.2%, other serotypes were superior to PCV7 group; all the 6 additional serotypes attained superiority criteria with respect to antibody GMC in PCV13-TT (2 months) group. The antibody positive rates and antibody GMC of 13 serotypes in PCV13-TT (2 months) group were noninferior to those in PCV13-TT (3 months) group, with > 3.5 -fold increase in GMC after

booster dose compared to that before.

For the 6 additional serotypes in PCV13-TT group aged 7-11 months, the lower limit of 95% CI for the rate difference of serotype 19A was -1.93%, though all other serotypes were superior to PCV7 group; all the 6 additional serotypes attained superiority criteria with respect to antibody GMC. The antibody GMC of 13 serotypes in PCV13-TT group aged 7-11 months and 7 serotypes in PCV7 group were with more than 1-fold increase after booster dose.

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2.7.3.2.3.1.2. Proportion of subjects with OPA antibody titer $\geq 1:8$ (OPA positive rate) and OPA geometric mean titer

2.7.3.2.3.1.2.1. Primary series

In subjects aged 3 months, after primary series of PCV13-TT, the OPA positive rates of 6 out of 7 common serotypes all exceeded 95% except for serotype 6B (89.25%), achieving the lower limit defined in non-inferiority test compared to PCV7 group. The OPA positive rates exceeded 95% in 5 additional serotypes except serotype 1 (83.87%). All of the 6 additional serotypes met the lower limit defined in superiority test compared to PCV7 group.

In subjects aged 2 months after primary series of PCV13-TT, the OPA positive rates of 13 serotypes all exceeded 90%, and 11 serotypes were with positive rates exceeding 95%. Except for serotype 6B, the remaining 6 of the 7 common serotypes met the lower limit defined in non-inferiority test compared to PCV7 (3 months) group. All of the 6 additional serotypes met the lower limit defined in superiority test.

In subjects aged 7-11 months, and in PCV13-TT group in PPS after immunization, the OPA positive rates of 11 serotypes among 13 serotypes exceeded 90%. The OPA positive rate of serotype 6B and 6A was respectively 81.25% and 89.58%. For OPA positive rate, the lower limit of 95% CI for the intergroup difference (PCV13-TT – PCV7) was greater than -10% (lower reference limit) in 5 of the 7 common serotypes for PCV13-TT group vs PCV7 group in PPS. The only two serotypes in PCV13-TT group with lower limit of 95% CI for the intergroup difference of OPA positive rate lower than -10% (lower reference limit) were serotype 6B and 23F. The lower limit of 95% CI for the intergroup difference of OPA positive rate all exceeded 0% in all 6 additional serotypes.

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2.7.3.2.3.1.2.2. Booster Dose

In subjects aged 3 months, after booster dose of PCV13-TT, the OPA positive rates of all common serotypes exceeded 98%, reaching the lower limit defined in non-inferiority test compared to PCV7 group. The OPA positive rates of additional serotypes all exceeded 95% and reached the lower limit defined in superiority test except for serotype 6A as compared with PCV7 group. The positive rate of serotype 6A was 95.83% in PCV13-TT group and 92.78% in PCV7 group, which were both at a relatively high level. The lower limit of 95% CI for difference of positive rates was -3.47%, which was consistent with the reported evidence of possible cross-protection between pneumococcal serotype 6B and 6A in prior literature. The GMT of serotype 7F in PCV13-TT (3 months) group post booster dose rose to a similar level as that after primary series, while GMTs of other serotypes after booster dose were higher than that after primary series to varied degrees. In subjects aged 2 months, after booster dose of PCV13-TT, the OPA positive rates of 10 serotypes reached 100%, meanwhile the positive rates of other 3 serotypes exceeded 95%. The OPA positive rates of 7 common serotypes all reached the lower limit defined in non-inferiority test when compared with PCV7 (3 months) group. Five of the 6 additional serotypes reached the lower limit defined in superiority test except for 6A. The positive rate for serotype 6A was 97.83% after booster dose in PCV13-TT (2 months) group. The lower limit of 95% CI for the difference between the PCV13-TT (2 months) group and PCV7 (3 months) group was -0.91%, which was similar to the result obtained in PCV13-TT (3 months) group for serotype 6A, consistent with the reported evidence of possible cross-protection between pneumococcal serotype 6B and 6A. The overall trend of the OPA positive rates for all serotypes was similar in PCV13-TT (2 months) group vs PCV7 (3 months) and PCV13-TT (3 months). Through booster dose, the GMT of serotype 3 rose to a similar level as that after primary series, while GMTs of other serotypes after booster dose were higher than that after primary series to varied degrees.

In subjects aged 7-11 months, after booster dose of PCV13-TT, the OPA positive rates of 13 serotypes all exceeded 90%, among which the positive rates of 8 serotypes reached 100%. For OPA positive rates, the lower limit of 95% CI for the difference of OPA positive rates (PCV13-TT - PCV7) was greater than -10% (lower reference limit) for the 7 common serotypes in PCV13-TT group vs PCV7 group. The lower limit of 95% CI for the difference of OPA positive rates all exceeded 0% in all 6 additional serotypes. For the GMT of OPA, the lower limit of 95% CI for GMT ratio (PCV13-TT / PCV7) was greater than 0.5 (lower reference limit) for 5 of the 7 common serotypes in PCV13-TT group vs PCV7 group. The only two serotypes with lower limit of 95% CI for GMT ratio lower than 0.5 (lower reference limit) in PCV13-TT group were serotype 6B and 23F, while GMTs (PPS) of serotype 6B and 23F in PCV13-TT group had achieved 110- and 197-fold increase as compared with the positive threshold (1:8), respectively. The lower limits of 95% CI for intergroup GMT ratio in 6 additional serotypes were all higher than 1.

In subjects aged 12-23 months, after full immunizations, the OPA positive rates of 12 serotypes among 13 serotypes in PCV13-TT group (PPS) all exceeded 90%. The OPA positive rate for serotype 1 was 88.04%. For OPA positive rate, the lower limit of 95% CI for the difference of OPA positive rates (PCV13-TT - PCV7) was greater than -10% (lower reference limit) for the 7 common serotypes in PCV13-TT group vs PCV7 group. The lower limit of 95% CI for the intergroup difference of OPA positive rates all exceeded 0% in all 6 additional serotypes. Considering the GMT of OPA, the lower limit of 95% CI for the intergroup GMT ratio was greater than 0.5 (lower reference limit) for 5 of the 7 common serotypes in PCV13-TT group vs PCV7 group. The only two serotypes in PCV13-TT group with lower limit of 95% CI for the intergroup GMT ratio lower than 0.5 were serotype 6B and 23F. The lower limit of 95% CI for intergroup GMT ratio in 6 additional serotypes was all higher than 1.

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In subjects aged 24-71 months, after full immunizations, the OPA positive rates of 12 serotypes among 13 serotypes in PCV13-TT group (PPS) all exceeded 90%. The OPA positive rate for

serotype 1 was 43.43%. For OPA positive rate, the lower limit of 95% CI for the intergroup difference of the OPA positive rates (PCV13-TT - PCV7) was greater than -10% (lower reference limit) for the 7 common serotypes in PCV13-TT group vs PCV7 group. The lower limit of 95% CI for the intergroup difference of OPA positive rate all exceeded 0% in 5 serotypes among the 6 additional serotypes. The OPA antibody positive rates of 6A in PCV13-TT group and PCV7 group in PPS were 94.95% and 95.00%, respectively, for the GMT of OPA, the lower limit of 95% CI for the GMT ratio was greater than 0.5 (lower reference limit) for 5 of the 7 common serotypes in PCV13-TT group vs PCV7 group. The only two serotypes in PCV13-TT group with lower limit of 95% CI for GMT ratio lower than 0.5 were serotype 14 and 23F. The lower limit of 95% CI for GMT ratio in 6 additional serotypes was all higher than 1.

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2.7.3.2.3.2. Immunogenicity Results

This clinical trial, with a licensed 7-valent pneumococcal conjugate vaccine as control, was designed to evaluate the immunogenicity and safety of the 13-valent pneumococcal polysaccharide conjugate vaccine, jointly developed by Yunnan Walvax Biotechnology Co., Ltd. and Yuxi Walvax Biotechnology Co., Ltd, in subjects aged 3 months (schedule of vaccination at 0, 1, 2 months), 7-11 months (month 0, 2 schedule), 12-23 months (2 doses) and 24-71 months (1 dose), respectively, the 13-valent pneumococcal polysaccharide conjugate vaccine was also administered in subjects 2 month of age (minimum age: 6 weeks) enrolled to compare with subjects aged 3 months in terms of immunization procedure.

In the main target population (3 months of age), the common serotypes achieved noninferiority criteria for comparison between PCV13-TT and PCV7 group except for serotype 6B during primary series, meanwhile all common serotypes in PCV13-TT group were noninferior to that in PCV7 group after the booster dose; for additional serotypes, superiority criteria were met for PCV13-TT group vs PCV7 group after primary series; the common serotypes in PCV13-TT group aged 3-71 months were all noninferior to those in PCV7 group during primary series, besides, the additional serotypes in PCV13-TT group were all superior as compared to PCV7 group.

For PCV13-TT (2 months) group vs PCV7 (3 months) group, both common and additional serotypes had achieved noninferiority or superiority criteria post primary series. For PCV13-TT (2 months) group vs PCV13-TT (3 months) group, 12 serotypes reached noninferiority criteria during primary series except for additional serotype 5; and all 13 serotypes were noninferior to those in PCV13-TT (3 months) group post booster dose.

In PCV13-TT recipients aged 7-11 months, 12-23 months, and 24-71 months, the proportions of subjects with IgG antibody concentration $\geq 0.35 \mu\text{g/mL}$ for common serotypes were all noninferior to those in PCV7 group. The additional serotypes in PCV13-TT group of 7-11 months had reached superiority criteria after primary series; for PCV13-TT groups of 12-23 months and 24-71 months, superiority criteria were generally met after immunization.

The results before and after booster dose indicated decrease in the antibody levels of several serotypes in the 13 serotypes either in PCV13-TT group or PCV7 group to some extent, while the percentage of subjects with IgG antibody concentration $\geq 3.5 \mu\text{g/mL}$ and antibody GMC after 1 dose of booster immunization both reached or exceeded the levels observed after primary series, demonstrating significant effect of the booster dose, reflecting the immunological memory effect of the conjugate vaccine adequately. The antibody GMC post booster dose had more than 3-fold increase in the 2-6 months olds, and greater than 1-fold increase in subjects aged 7-11 months as compared with the GMC before booster dose.

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2.7.4. Summary of Clinical Safety

2.7.4.1. Drug Exposure

N/A

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2.7.4.1.1. Overall safety assessment plan and safety study description

2.7.4.1.1.1. Phase I clinical trial

All subjects were observed for 30 minutes after each dose of 13-valent pneumococcal polysaccharide conjugate vaccine (PCV13-TT), and were under active observation for adverse events (AEs) within 0-7 days post vaccination, while for 8-30 days post vaccination, AEs were spontaneously reported and collected by subjects. Serious adverse events (SAEs) were collected for 6 months after full immunization of PCV13-TT which was administered in accordance with the immunization procedure prescribed in the protocol. Venous blood was collected from subjects in stage 1 and 2 before receiving the first dose and in 3 days after the first dose of PCV13-TT, so as to assess the changes in measures of hematology, liver and kidney function test.

Stage 1 (adults aged 18 and above): Adult subjects aged 18 years and above were recruited and the venous blood samples (no less than 5.0 ml) were collected after informed consenting to assess related measures of hematology, liver and kidney function test. One dose of PCV13-TT was administered to subjects with normal testing results and demonstrating eligibility by other inclusion criteria, and 30 subjects were enrolled in the stage. Thirty-minute medical observation was carried out after vaccination. Venous blood (no less than 3.5 ml) was collected on Day 3 after vaccination to assess related measures of hematology, liver and kidney function test, as well as to observe the changes in these measures, all subjects were actively observed to evaluate safety until Day 7 after vaccination. If no vaccine-related SAE was identified by the investigators, and no grade 4 vaccine-related abnormalities determined by the investigators as measured by hematology, liver and kidney function test on Day 3 after vaccination were detected, then the study proceeded to stage 2. The serum samples collected before vaccination were retained, and another round of venous blood (no less than 3.0 ml) collection was performed 30 days after one dose of vaccine in these subjects. The serum was separated, and the serum IgG antibody concentration was measured before and 30 days post vaccination.

Stage 2 (children aged 2-5 years old): Children subjects aged 2-5 years were recruited, and the venous blood samples (no less than 5.0 ml) were collected after the informed consent form was signed by their legal guardians to detect related measures of hematology, liver and kidney function tests. Subjects with testing results within the normal range predefined in the protocol or not within the normal range but were determined by site investigators to be normal and in compliance with

other inclusion criteria were given 1 dose of PCV13-TT, a total of 30 subjects were enrolled in this stage. If the results of hematology, liver and kidney function tests before vaccination were not within the normal range defined in study protocol while not exceeding 50% of the normal range (below the lower limit or above the upper limit), the results were to be evaluated by site investigators to determine whether the subjects were eligible for enrollment and vaccination. Thirty-minute medical observation was carried out after vaccination. Venous blood (no less than 3.5 ml) was collected on Day 3 after vaccination to assess related measures of hematology, liver and kidney function tests, as well as to observe the changes in these measures, all subjects were actively observed for safety until Day 7 after vaccination. If no SAE related to the study vaccine was identified by the investigators, and no grade 4 vaccine-related abnormalities determined by the investigators as measured by hematology, liver and kidney function tests were detected on Day 3 after vaccination, then the study proceeded to stage 3. The serum samples collected before vaccination were retained, and another round of venous blood (no less than 3.0 ml) collection was performed 30 days after one dose of vaccine administered in these subjects. The serum was separated, and the serum IgG antibody concentration was measured before and 30 days post vaccination.

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Stage 3 (2 months old group): Infant subjects aged 2 months (minimum age: 6 weeks) were recruited, and venous blood samples (no less than 3.0 ml) were collected after the informed consent form was signed by their legal guardians. The serum was separated and the serum IgG antibody concentration before immunization was detected. Three doses of PCV13-TT were administered according to the procedure of one dose each in 0, 2 and 4 months. A total of 30 subjects were enrolled. Thirty-minute medical observation was carried out after each vaccination, and all subjects were actively observed until Day 7 day after vaccination. Venous blood (no less than 3.0 ml) was collected 30 days after the 3rd dose of vaccine. The serum was separated and the serum IgG antibody concentration 30 days post immunization was detected.

Stage 3 (3-month group): Infant subjects aged 3 months (minimum age: 6 weeks) were recruited, and venous blood samples (no less than 3.0 ml) were collected after the informed consent form was signed by their legal guardians. The serum was separated and the serum IgG antibody concentration before immunization was detected. Three doses of PCV13-TT were administered according to the procedure of one dose each in 0, 1 and 2 months. A total of 30 subjects were enrolled. Thirty-minute medical observation was carried out after each vaccination, and all subjects were actively observed until Day 7 day after vaccination. Venous blood (no less than 3.0 ml) was collected 30 days after the 3rd dose of vaccine. The serum was separated and the serum IgG antibody concentration 30 days post immunization was detected.

All subjects were observed for 30 minutes after each dose of PCV13-TT, and were under active observation for AEs within 0-7 days post vaccination, while for 8-30 days post vaccination, AEs were spontaneously reported and collected by subjects. SAEs were collected for 6 months after full immunization PCV13-TT which was administered in accordance with the immunization procedure prescribed in the protocol.

In the third stage of phase I clinical trial, after the first dose of PCV13-TT was administered to the 2-month and 3-month group followed by 7-day active observation, the safety of phase I clinical trial was preliminarily evaluated. If no serious adverse events related to study vaccine were determined by the investigators, or the incidence of local or systemic adverse events with a severity of grade 3 or above was not higher than 15%, and no vaccine-related grade 4 abnormalities in hematology, liver and kidney function tests were observed on Day 3 after the first dose of vaccine administered to subjects in the first and second stage, then the program proceeded to phase III clinical trial. In the meantime, the follow-up immunization procedure and corresponding safety observation for the subjects in stage 3 of phase I clinical trial were continued and the data were collected to be included in the final safety analysis.

Data collation and clarification were finished within two months after the completion of stage 3 full immunization at each site. Phase I clinical study report was issued within one month of receiving immunogenicity test results after serious adverse events were collected 6 months after completing all stages of vaccination.

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2.7.4.1.1.2 Phase III clinical trial

All subjects were actively observed for 30 minutes and 0-7 days post vaccination as part of systematic safety monitoring, and the AEs were spontaneously reported by the subjects within 8-30 days post vaccination. SAEs occurred during 6 months after the full primary immunizations and / or 6 months after the full booster immunization were collected.

Safety observation was carried out after the administration of each dose of vaccine.

Immunogenicity was observed before the first dose of vaccine and 30 days after the primary immunizations. Immunogenicity observation was also performed before and 30 days after booster immunization for subjects in age group requiring booster immunization. Data collation and clarification were completed after finishing the on-site sampling. Statistical analysis and clinical study reports were completed after the blood test results returned.

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2.7.4.1.2. Overall exposure

N/A

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2.7.4.1.3. Demographics and other characteristics of the study population

2.7.4.1.3.1. Phase I clinical trial

The age and sex distribution of subjects included in phase I clinical trial are shown in Table 136.

Table 136 Age and gender distribution of subjects in phase I clinical trial

Age (months) Male Female

Groups N

Mean±SD Median Min, Max N % N %
≥18 years 30 35.79±12.53 30.67 18.04,64.02 16 53.33 14 46.67
2-5 years 30 4.04±1.26 4.06 2.06,5.94 14 46.67 16 53.33
2 months 30 0.19±0.03 0.19 0.13,0.24 14 46.67 16 53.33
3 months 30 0.28±0.02 0.28 0.25,0.33 16 53.33 14 46.67

2.7.4.1.3.2. Phase III clinical trial
The proportions of age and sex of subjects aged 2 months (minimum age: 6 weeks) and above are shown in Table 137 to

Table 141. Considering the age at which the comparator vaccine was approved for use in China, no PCV7 group was set up for 2 months old group (minimum age: 6 weeks). No statistical intergroup difference was observed in the mean age and sex distribution of SS and FAS data sets between the PCV13-TT and PCV7 groups in subjects aged 3-71 months. The baseline demographics were balanced and comparable.

Table 137 Baseline demographics of subjects aged 2 months (at least 6 weeks of age)

Data Age (months) Gender

Observations

set Mean±SD Median (Min, Max) Male (n%) Female (n%)

SS 517 2.27±0.40 2.23 (1.41,2.92) 236(45.65) 281(54.35)

FAS 517 2.27±0.40 2.23 (1.41,2.92) 236(45.65) 281(54.35)

*A total of 108 subjects in the age group of 2 months were 42-60 days of age, accounting for 20.89% of this age group.

Table 138 Baseline demographics of subjects aged 3 months

Obse Age (months) Gender

Data

Groups rvati Mediann P P

set Mean±SD Malen% Femalen%

ons (Min,Max)

PCV13-

519 3.33±0.30 3.25(2.73,3.91) 270(52.02) 249(47.98)

SS TT 0.4351 0.4939

PCV7 519 3.34±0.30 3.29(2.63,3.91) 281(54.14) 238(45.86)

PCV13- 519

3.33±0.30 3.25(2.73,3.91) 270(52.02) 249(47.98)

FAS TT 0.4351 0.4939

PCV7 519 3.34±0.30 3.29(2.63,3.91) 281(54.14) 238(45.86)

Table 139 Baseline demographics of subjects aged 7-11 months

Obse Age (months) Gender

Data

Groups rvati P P

set Mean±SD Median (Min,Max) Malen% Femalen%

ons

PCV13

198 9.34±1.37 9.30(6.90,11.79) 97(48.99) 101(51.01)

SS -TT 0.9616 0.4236

PCV7 200 9.33±1.33 9.23(6.93,11.76) 106(53.00) 94(47.00)

PCV13

198 9.34±1.37 9.30(6.90,11.79) 97(48.99) 101(51.01)

FAS -TT 0.9616 0.4236

PCV7 200 9.33±1.33 9.23(6.93,11.76) 106(53.00) 94(47.00)

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Table 140 Baseline demographics of subjects aged 12-23 months

Obse Age (months) Gender

Data

Groups rvati P P

set Mean±SD Median (Min,Max) Malen% Femalen%

ons

PCV13

200 17.36±3.25 17.50(11.63,23.59) 112(56.00) 88(44.00)

SS -TT 0.3604 0.6862

PCV7 200 17.66±3.27 17.38(11.89,23.62) 116(58.00) 84(42.00)

PCV13

200 17.36±3.25 17.50(11.63,23.59) 112(56.00) 88(44.00)

FAS -TT 0.3604 0.6862

PCV7 200 17.66±3.27 17.38(11.89,23.62) 116(58.00) 84(42.00)

Table 141 Baseline demographics of subjects aged 24-71 months

Obse Age (months) Gender

Data

Groups rvati P P

set Mean±SD Median (Min,Max) Malen% Femalen%

ons

PCV13

200 42.82±13.49 41.74(23.75,70.80) 99(49.50) 101(50.50)

SS -TT 0.1110 0.2708

PCV7 200 40.76±12.30 40.41(23.66,69.29) 110(55.00) 90(45.00)

PCV13

200 42.82±13.49 41.74(23.75,70.80) 99(49.50) 101(50.50)

FAS -TT 0.1110 0.2708

PCV7 200 40.76±12.30 40.41(23.66,69.29) 110(55.00) 90(45.00)

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2.7.4.2. Adverse events

2.7.4.2.1 Incidence of adverse event

2.7.4.2.1.1. Phase I clinical trial

Solicited adverse events in phase I clinical trials are by default associated with the study vaccine, and the relatedness of unsolicited adverse reactions was adjudicated by investigators. The overall adverse events in each stage is summarized in Table 142.

A total of 24 (80.00%) subjects in adults group experienced adverse events related to the study vaccine, of which 10 cases (33.33%) experienced systemic adverse reactions, but no grade 3 systemic adverse reaction was reported; 23 cases (76.67%) experienced local reactions, of which the severity of the reaction for 6 cases (20.00%) reached grade 3.

A total of 16 (53.33%) subjects in children group experienced adverse events related to the study vaccine, of which 7 cases (23.33%) experienced systemic adverse reactions, and the severity of 2 cases (6.67%) reached grade 3; 8 cases (26.67%) experienced local reactions, of which the severity of the reaction for 1 case (3.33%) reached grade 3.

A total of 20 (66.67%) subjects in the 2-month age group experienced adverse events related to the study vaccine, of which 18 cases (60.00%) experienced systemic adverse reactions, and the severity of 2 cases (6.67%) reached grade 3; 12 cases (40.00%) experienced local reactions, of which the severity of the reaction for 2 cases (6.67%) reached grade 3. Another case of adverse event which was unrelated to the study vaccine was reported as herpangina and the severity reached grade 3.

A total of 21 (70.00%) subjects in the 3-month age group experienced adverse events related to the study vaccine, of which 17 cases (56.67%) experienced systemic adverse reactions, but no grade 3 systemic adverse reaction was reported; 11 cases (36.67%) experienced local reactions, of which the severity of the reaction for 1 case (3.33%) reached grade 3. Another case of adverse event which was unrelated to the study vaccine was reported as upper respiratory tract infection and the severity reached grade 3.

No serious adverse event was collected during the observation period, and no drop-outs due to adverse events were noted.

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Table 142 Incidence of adverse reactions in phase I clinical trial												
≥18 years (N=30) 2-5 years (N=30) 2 months (N=30) 3 months (N=30)												
Events Cases % Events Events Cases % Events Events Cases % %												
Any AE 81 24 80.00 41 19 63.33 105 21 70.00 79 22 73.33												
Grade 1 49 23 76.67 33 18 60.00 67 21 70.00 58 20 66.67												
Grade 2 24 13 43.33 4 4 13.33 32 13 43.33 19 15 50.00												
Grade 3 8 6 20.00 4 2 6.67 6 4 13.33 2 2 6.67												
Related 35 16 53.33 66 21 70.00												
69 24 80.00 82 20 66.67												
AE												
Grade 1 43 23 76.67 28 15 50.00 59 20 66.67 52 19 63.33												
Grade 2 18 10 33.33 3 3 10.00 18 10 33.33 13 10 33.33												
Grade 3 8 6 20.00 4 2 6.67 5 4 13.33 1 1 3.33												
Solicited 81 20 66.67												
63 23 76.67 28 13 43.33 64 21 70.00												
AE												
Grade 1 37 21 70.00 21 12 40.00 58 20 66.67 50 19 63.33												
Grade 2 18 10 33.33 3 3 10.00 18 10 33.33 13 10 33.33												
Grade 3 8 6 20.00 4 2 6.67 5 4 13.33 1 1 3.33												
Systemic 13 7 23.33 40 17 56.67												
15 10 33.33 54 18 60.00												
reactions												
Grade 1 10 7 23.33 8 6 20.00 38 17 56.67 34 16 53.33												
Grade 2 5 4 13.33 3 3 10.00 14 8 26.67 6 6 20.00												
Grade 3 0 0 0.00 2 2 6.67 2 2 6.67 0 0 0.00												
Local 15 8 26.67 24 11 36.67												
48 23 76.67 27 12 40.00												
reactions												
Grade 1 27 21 70.00 13 7 23.33 20 11 36.67 16 10 33.33												
Grade 2 13 10 33.33 0 0 0.00 4 3 10.00 7 5 16.67												
Grade 3 8 6 20.00 2 1 3.33 3 2 6.67 1 1 3.33												
Unsolicited												
6 5 16.67 7 5 16.67 1 1 3.33 2 2 6.67												
AE												
Grade 1 6 5 16.67 7 5 16.67 1 1 3.33 2 2 6.67												
Grade 2 0 0 0.00 0 0 0.00 0 0 0.00 0 0 0.00												
Grade 3 0 0 0.00 0 0 0.00 0 0 0.00 0 0 0.00												
Unrelated												
12 8 26.67 6 6 20.00 23 11 36.67 13 10 33.33												
AE												
Grade 1 6 4 13.33 5 5 16.67 8 7 23.33 6 5 16.67												
Grade 2 6 6 20.00 1 1 3.33 14 8 26.67 6 6 20.00												
Grade 3 0 0 0.00 0 0 0.00 1 1 3.33 1 1 3.33												

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2.7.4.2.1.2. Phase III clinical trial

2.7.4.2.2.1 Primary series

In Phase III clinical trial, 790 (2282 events) cases of adverse events were observed in the 3-71 months group, and 851 (2440 events) cases of adverse events were observed in the PCV7 group. The overall incidence of adverse events was 70.73% and 76.05% in the PCV13-TT group and the PCV7 group, respectively. There was significant difference in the incidence of adverse events between groups, and the overall incidence of adverse events in the PCV13-TT group was lower than that in the PCV7 group.

The incidence of adverse events related to the study vaccine was 60.79% in the PCV13-TT group and 64.79% in the PCV7 group. The PCV13-TT group was slightly lower than the PCV7 group but no significant difference was observed.

The incidence of vaccine-independent adverse events was 35.00% in the PCV13-TT group and

Grade 1 24 19 3.66 23 21 4.05 0.8721
Grade 2 29 26 5.01 33 25 4.82 1.0000
Grade 3 // // // //
Grade 4 1 1 0.19 0 0 0.00 1.0000
AEs unrelated to
367 197 37.96 379 211 40.66 0.4088
vaccine
Grade 1 99 78 15.03 119 90 17.34 0.3540
Grade 2 231 132 25.43 227 144 27.75 0.4397
Grade 3 28 21 4.05 23 20 3.85 1.0000
Grade 4 9 7 1.35 10 8 1.54 1.0000
Solicited AE 72 55 10.60 81 56 10.79 1.0000
Grade 1 27 24 4.62 32 30 5.78 0.4850
Grade 2 42 34 6.55 45 31 5.97 0.7980
Grade 3 3 3 0.58 4 4 0.77 1.0000
Grade 4 // // // //
Unsolicited AE 295 181 34.87 298 191 36.80 0.5602
Grade 1 72 65 12.52 87 72 13.87 0.5823
Grade 2 189 124 23.89 182 135 26.01 0.4733
Grade 3 25 19 3.66 19 17 3.28 0.8656
Grade 4 9 7 1.35 10 8 1.54 1.0000

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Table 145 Overall incidence of adverse reactions in PCV13-TT and PCV7 groups of subjects aged 7-11 months

PCV13-TT (N=198) PCV7 (N=200)											
P											
Events			Cases			% Events			Cases		
Any AE											
Grade 1	224	114	57.58	252	128	64.00	0.0455				
Grade 2	194	102	51.52	232	115	57.50	0.2682				
Grade 3	23	19	9.60	12	12	6.00	0.1953				
Grade 4	2	2	1.01	9	5	2.50	0.4490				
AEs related to vaccine											
Grade 1	173	99	50.00	198	131	55.50	0.3154				
Grade 2	99	70	35.35	99	74	37.00	0.7551				
Grade 3	15	13	6.57	6	6	3.00	0.1053				
Grade 4	0	0	0.00	0	0	0.00	1.0000				
Solicited AE											
Grade 1	169	98	49.49	197	111	55.50	0.2695				
Grade 2	93	67	33.84	90	71	35.50	0.7527				
Grade 3	15	13	6.57	6	6	3.00	0.1053				
Grade 4	0	0	0.00	0	0	0.00	1.0000				
Unsolicited AE											
Grade 1	4	4	2.02	1	1	0.50	0.2140				
Grade 2	6	5	2.53	9	9	4.50	0.4156				
Grade 3	0	0	0.00	0	0	0.00	1.0000				
Grade 4	0	0	0.00	0	0	0.00	1.0000				
AEs unrelated to											
156 87 43.94 202 95 47.50 0.4831											
vaccine											
Grade 1	51	37	18.69	54	39	19.50	0.8987				
Grade 2	95	59	29.80	133	70	35.00	0.2853				
Grade 3	8	8	4.04	6	6	3.00	0.5993				
Grade 4	2	2	1.01	9	5	2.50	0.4490				
Solicited AE											
Grade 1	16	13	6.57	10	9	4.50	0.3896				
Grade 2	17	15	7.58	29	23	11.50	0.2324				
Grade 3	2	2	1.01	5	5	2.50	0.4490				
Grade 4	0	0	0.00	0	0	0.00	1.0000				
Unsolicited AE											
Grade 1	35	29	14.65	44	33	16.50	0.6789				
Grade 2	78	49	24.75	104	60	30.00	0.2619				
Grade 3	6	6	3.03	1	1	0.50	0.0669				
Grade 4	2	2	1.01	9	5	2.50	0.4490				

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Table 146 Overall incidence of adverse reactions in PCV13-TT and PCV7 groups of subjects aged 12-23 months

PCV13-TT (N=200) PCV7 (N=200)											
P											
Events			Cases			% Events			% Cases		
Any AE											
344	136	68.00	295	138	69.00	0.9143					
Grade 1											
169	95	47.50	146	90	45.00	0.6884					
Grade 2											
150	84	42.00	128	82	41.00	0.9192					
Grade 3											
17	13	6.50	18	14	7.00	1.0000					
Grade 4											
8	4	2.00	3	3	1.50	1.0000					
AEs related to vaccine											
197	106	53.00	178	114	57.00	0.4818					
Grade 1											
121	79	39.50	108	77	38.50	0.9184					
Grade 2											
65	47	23.50	62	49	24.50	0.9069					
Grade 3											
11	7	3.50	8	6	3.00	1.0000					
Grade 4											
0	0	0.00	0	0	0.00	1.0000					
Solicited AE											
188	105	52.50	168	111	55.50	0.6160					
Grade 1											
118	78	39.00	102	76	38.00	0.9182					
Grade 2											
59	47	23.50	58	47	23.50	1.0000					
Grade 3											
11	7	3.50	8	6	3.00	1.0000					

Grade 4 // // // //
 Unsolicited AE 9 8 4.00 10 8 4.00 1.0000
 Grade 1 3 3 1.50 6 4 2.00 1.0000
 Grade 2 6 5 2.50 4 4 2.00 1.0000
 Grade 3 // // // //
 Grade 4 // // // //
 AEs unrelated to
 147 72 36.00 117 62 31.00 0.3404
 vaccine
 Grade 1 48 32 16.00 38 28 14.00 0.6747
 Grade 2 85 54 27.00 66 45 22.50 0.3540
 Grade 3 6 6 3.00 10 9 4.50 0.6002
 Grade 4 8 4 2.00 3 3 1.50 1.0000
 Solicited AE 31 21 10.50 28 23 11.50 0.8732
 Grade 1 10 9 4.50 11 9 4.50 1.0000
 Grade 2 20 17 8.50 13 12 6.00 0.4411
 Grade 3 1 1 0.50 4 4 2.00 0.3719
 Grade 4 // // // //
 Unsolicited AE 116 67 33.50 89 52 26.00 0.1255
 Grade 1 38 28 14.00 27 33 11.50 0.5491
 Grade 2 65 46 23.00 53 37 18.50 0.3240
 Grade 3 5 5 2.50 6 5 2.50 1.0000
 Grade 4 8 4 2.00 3 3 1.50 1.0000

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Table 147 Overall incidence of adverse reactions in PCV13-TT and PCV7 groups of subjects aged 24-71 months

PCV13-TT (N=200) PCV7 (N=200)
 P
 Events Cases % Events Cases %
 Any AE 189 89 44.50 193 105 52.50 0.1333
 Grade 1 98 56 28.00 105 67 33.50 0.2786
 Grade 2 75 57 28.50 78 62 31.00 0.6619
 Grade 3 15 12 6.00 10 7 3.50 0.3475
 Grade 4 1 1 0.50 0 0 0.00 1.0000
 AEs related to vaccine 143 72 36.00 139 77 38.50 0.6792
 Grade 1 83 51 25.50 90 58 29.00 0.5006
 Grade 2 45 35 17.50 39 33 16.50 0.8942
 Grade 3 15 12 6.00 10 7 3.50 0.3475
 Grade 4 0 0 0.00 0 0 0.00 1.0000
 Solicited AE 141 72 36.00 134 75 37.50 0.8357
 Grade 1 82 51 25.50 90 58 29.00 0.5006
 Grade 2 44 34 17.00 34 31 15.50 0.7865
 Grade 3 15 12 6.00 10 7 3.50 0.3475
 Grade 4 // // // //
 Unsolicited AE 2 2 1.00 5 5 2.50 0.4490
 Grade 1 1 1 0.50 0 0 0.00 1.0000
 Grade 2 1 1 0.50 5 5 2.50 0.2152
 Grade 3 // // // //
 Grade 4 // // // //
 AEs unrelated to
 46 35 17.50 54 44 22.00 0.3150
 vaccine
 Grade 1 15 12 6.00 15 13 6.50 1.0000
 Grade 2 30 25 12.50 39 33 16.50 0.3202
 Grade 3 0 0 0.00 0 0 0.00 1.0000
 Grade 4 1 1 0.50 0 0 0.00 1.0000
 Solicited AE 12 8 4.00 10 9 4.50 1.0000
 Grade 1 4 3 1.50 3 3 1.50 1.0000
 Grade 2 8 6 3.00 7 7 3.50 1.0000
 Grade 3 // // // //
 Grade 4 // // // //
 Unsolicited AE 34 29 14.50 44 40 20.00 0.1854
 Grade 1 11 9 4.50 12 11 5.50 0.8192
 Grade 2 22 20 10.00 32 30 15.00 0.1732
 Grade 3 // // // //
 Grade 4 1 1 0.50 0 0 0.00 1.0000

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2.7.4.2.2.2. Booster immunization stage

2.7.4.2.2.2.1. Overall incidence of adverse events

The subjects with booster immunization included subjects aged 2 months and 3 months receiving 3-dose series, and those aged 7-11 months who received 2 doses of vaccination during Primary series, and booster dose was given as 1 dose.
 The incidence of adverse events was 35.46% in the PCV13-TT group and 34.50% in the PCV7 group for 3-11 month of age who received booster immunization. The severity was mainly grade 1 and 2. Thirteen and seven grade 3 adverse events were reported in the PCV13-TT group and the PCV7 group, and the incidence rate was respectively 1.93% and 1.02%. No significant difference was observed between the groups. One case of grade 4 adverse event was observed in each group (reported as SAE), which were both not related to the study vaccine.
 There was no significant difference in the incidence of adverse events among PCV13-TT and PCV7 groups in subjects 3 months and 7-11 month of age.
 Adverse events after booster immunization in the PCV13-TT and PCV7 group were shown in Table 148 to Table 150.

Table 148 Overall incidence in PCV13-TT and PCV7 groups post booster dose in subjects aged 3-11 months

PCV13-TT (N=674) PCV7 (N=684)											
P											
Events			Cases			% Events			Cases		
Any AE			328	239	35.46	346	236	34.50	0.7329		
Grade 1			158	135	20.03	172	141	20.61	0.8397		
Grade 2			153	127	18.84	164	120	17.54	0.5737		
Grade 3			15	13	1.93	9	7	1.02	0.1831		
Grade 4			2	1	0.15	1	1	0.15	1.0000		
AEs related to vaccine			242	197	29.23	241	193	28.22	0.7190		
Grade 1			132	119	17.66	151	126	18.42	0.7247		
Grade 2			101	90	13.35	85	74	10.82	0.1576		
Grade 3			9	8	1.19	5	4	0.58	0.2614		
Grade 4			//////								
Solicited AE			237	196	29.08	232	190	27.78	0.6303		
Grade 1			131	119	17.66	145	124	18.13	0.8321		
Grade 2			97	88	13.06	82	73	10.67	0.1801		
Grade 3			9	8	1.19	5	4	0.58	0.2614		
Grade 4			//////								
Unsolicited AE			5	5	0.74	9	9	1.32	0.4217		
Grade 1			1	1	0.15	6	6	0.88	0.1241		
Grade 2			4	4	0.59	3	3	0.44	0.7240		
Grade 3			//////								
Grade 4			//////								
AEs unrelated to											
vaccine			86	65	9.64	105	71	10.38	0.7178		
Grade 1			26	22	3.26	21	19	2.78	0.6368		
Grade 2			52	44	6.53	79	56	8.19	0.2544		
Grade 3			6	5	0.74	4	3	0.44	0.5035		
Grade 4			2	1	0.15	1	1	0.15	1.0000		
Solicited AE			18	12	1.78	28	18	2.63	0.3566		
Grade 1			7	5	0.74	4	3	0.44	0.5035		
Grade 2			9	8	1.19	23	17	2.49	0.1048		
Grade 3			2	2	0.30	1	1	0.15	0.6220		
Grade 4			//////								
Unsolicited AE			68	64	9.50	77	63	9.21	0.9258		
Grade 1			19	19	2.82	17	16	2.34	0.6108		
Grade 2			43	42	6.23	56	48	7.02	0.5867		
Grade 3			4	4	0.59	3	2	0.29	0.4489		
Grade 4			2	1	0.15	1	1	0.15	1.0000		

Table 149 Overall incidence in PCV13-TT and PCV7 groups post booster dose in subjects aged 3 months

PCV13-TT (N=488) PCV7 (N=493)											
P											
Events			Cases			% Events			Cases		
Any AE			260	186	38.11	270	179	36.31	0.5972		
AEs related to vaccine			190	153	31.35	187	149	30.22	0.7296		
Solicited AE			186	152	31.15	179	146	29.61	0.6272		
Unsolicited AE			4	4	0.82	8	8	1.62	0.3848		
AEs unrelated to											
vaccine			70	51	10.45	83	51	10.34	1.0000		
Solicited AE			17	11	2.25	28	18	3.65	0.2579		
Unsolicited AE			53	50	10.25	55	43	8.72	0.4464		

Table 150 Overall incidence in PCV13-TT and PCV7 groups post booster dose in subjects aged 7-11 months

PCV13-TT (N=186) PCV7 (N=191)											
P											
Events			Cases			% Events			Cases		
Any AE			68	53	28.49	76	57	29.84	0.8210		
AEs related to vaccine			52	44	23.66	54	44	23.04	0.9036		
Solicited AE			51	44	23.66	53	44	23.04	0.9036		
Unsolicited AE			1	1	0.54	1	1	0.52	1.0000		
AEs unrelated to											
vaccine			16	14	7.53	22	20	10.47	0.3705		
Solicited AE			1	1	0.54	0	0	0.00	0.4934		
Unsolicited AE			15	14	7.53	22	20	10.47	0.3705		

2.7.4.2.2.2. Adverse events occurred 30 minutes and 7 days post vaccination
The incidence of adverse events occurred 30 minutes and 7 days post vaccination is shown in tables Table 151 to Table 162.
No significant difference was observed in the incidence of adverse events within 30 minutes after vaccination between the PCV13-TT group and the PCV7 group of 3 month of age and 7-11 month of age.
Table 151 Incidence of adverse reactions 30 min post booster dose in subjects aged 3-11 months
PCV13-TT (N=674) PCV7 (N=684)

P

Events Cases % Events Cases %

Any AE 30 30 4.45 34 34 4.97 0.7017

AEs related to vaccine 30 30 4.45 34 34 4.97 0.7017

Solicited AE 30 30 4.45 34 34 4.97 0.7017

AEs unrelated to 1.0000

0 0 0.00 0 0 0.00

vaccine

Table 152 Incidence of adverse reactions 30 min post booster dose in subjects aged 3 months

PCV13-TT (N=488) PCV7 (N=493)

P

Events Cases % Events Cases %

Any AE 25 25 5.12 24 24 4.87 0.8843

AEs related to vaccine 25 25 5.12 24 24 4.87 0.8843

Solicited AE 25 25 5.12 24 24 4.87 0.8843

AEs unrelated to 1.0000

0 0 0.00 0 0 0.00

vaccine

Table 153 Incidence of adverse reactions 30 min post booster dose in subjects aged 7-11 months

PCV13-TT (N=186) PCV7 (N=191)

P

Events Cases % Events Cases %

Any AE 5 5 2.69 10 10 5.24 0.2925

AEs related to vaccine 5 5 2.69 10 10 5.24 0.2925

Solicited AE 5 5 2.69 10 10 5.24 0.2925

AEs unrelated to 1.0000

0 0 0.00 0 0 0.00

vaccine

Table 154 Incidence of adverse reactions 0-7 days post booster dose in subjects aged 3-11 months

PCV13-TT (N=674) PCV7 (N=684)

P

Events Cases % Events Cases %

Any AE 273 208 30.86 281 203 29.68 0.6371

AEs related to vaccine 240 197 29.23 238 191 27.92 0.6309

Solicited AE 237 196 29.08 232 190 27.78 0.6303

Unsolicited AE 3 3 0.45 6 6 0.88 0.5065

AEs unrelated to 0.3302

33 16 2.37 43 23 3.36

vaccine

Solicited AE 18 12 1.78 28 18 2.63 0.3566

Unsolicited AE 15 15 2.23 15 14 2.05 0.8530

Table 155 Incidence of adverse reactions 0-7 days post booster dose in subjects aged 3 months

PCV13-TT (N=488) PCV7 (N=493)

P

Events Cases % Events Cases %

Any AE 219 163 33.40 225 158 32.05 0.6832

AEs related to vaccine 188 153 31.35 184 147 29.82 0.6278

Solicited AE 186 152 31.15 179 146 29.61 0.6272

Unsolicited AE 2 2 0.41 5 5 1.01 0.4515

AEs unrelated to 31 15 3.07 0.3964

41 21 4.26

vaccine

Solicited AE 17 11 2.25 28 18 3.65 0.2579

Unsolicited AE 14 14 2.87 13 12 2.43 0.6959

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Table 156 Incidence of adverse reactions 0-7 days post booster dose in subjects aged 7-11 months

PCV13-TT (N=186) PCV7 (N=191)

P

Events Cases % Events Cases %

Any AE 54 45 24.19 56 45 23.56 0.9043

AEs related to vaccine 52 44 23.66 54 44 23.04 0.9036

Solicited AE 51 44 23.66 53 44 23.04 0.9036

Unsolicited AE 1 1 0.54 1 1 0.52 1.0000

AEs unrelated to

2 1 0.54 2 2 1.05 1.0000

vaccine

Solicited AE 1 1 0.54 0 0 0.00 0.4934

Unsolicited AE 1 1 0.54 2 2 1.05 1.0000

Table 157 Incidence of adverse reactions 8-30 days post booster dose in subjects aged 3-11 months

PCV13-TT (N=674) PCV7 (N=684)

P

Events Cases % Events Cases %

Any AE 52 52 7.57 61 51 7.46 1.0000

AEs related to vaccine 2 2 0.30 3 3 0.44 1.0000

Unsolicited AE 2 2 0.30 3 3 0.44 1.0000

AEs unrelated to 0.9162

50 49 7.27 58 48 7.02

vaccine

Unsolicited AE 50 49 7.27 58 48 7.02 0.9162

Table 158 Incidence of adverse reactions 8-30 days post booster dose in subjects aged 3 months

PCV13-TT (N=488) PCV7 (N=493)

P

Events Cases % Events Cases %

Any AE 39 39 7.99 42 34 6.90 0.5444

AEs related to vaccine 2 2 0.41 3 3 0.61 1.0000

Unsolicited AE 2 2 0.41 3 3 0.61 1.0000

AEs unrelated to 0.4524

37 37 7.58 39 31 6.29

vaccine
Unsolicited AE 37 37 7.58 39 31 6.29 0.4524
Table 159 Incidence of adverse reactions 8-30 days post booster dose in subjects aged 7-11 months
PCV13-TT (N=186) PCV7 (N=191)
P
Events Cases % Events Cases %
Any AE 13 12 6.45 19 17 8.90 0.4412
AEs related to vaccine 0 0 0.00 0 0 0.00 1.0000
AEs unrelated to 0.4412
13 12 6.45 19 17 8.90
vaccine
Unsolicited AE 13 12 6.45 19 17 8.90 0.4412

Table 160 Incidence of adverse reactions 30 days post booster dose in subjects aged 3-11 months
PCV13-TT (N=674) PCV7 (N=684)
P
Events Cases % Events Cases %
Any AE 3 3 0.45 4 4 0.58 1.0000
AEs related to vaccine 0 0 0.00 0 0 0.00 1.0000
AEs unrelated to
vaccine 3 3 0.45 4 4 0.58 1.0000
Unsolicited AE 3 3 0.45 4 4 0.58 1.0000

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Table 161 Incidence of adverse reactions 30 days post booster dose in subjects aged 3 months
PCV13-TT (N=488) PCV7 (N=493)
P
Events Cases % Events Cases %
Any AE 2 2 0.41 3 3 0.61 1.0000
AEs related to vaccine 0 0 0.00 0 0 0.00 1.0000
AEs unrelated to
vaccine 2 2 0.41 3 3 0.61 1.0000
Unsolicited AE 2 2 0.41 3 3 0.61 1.0000
Table 162 Incidence of adverse reactions 30 days post booster dose in subjects aged 7-11 months
PCV13-TT (N=186) PCV7 (N=191)
P
Events Cases % Events Cases %
Any AE 1 1 0.54 1 1 0.52 1.0000
AEs related to vaccine 0 0 0.00 0 0 0.00 1.0000
AEs unrelated to
1 1 0.54 1 1 0.52 1.0000
vaccine
Unsolicited AE 1 1 0.54 1 1 0.52 1.0000

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2.7.4.2.2.3. Occurrence of adverse events with different procedures in the PCV13-TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group
The incidence of vaccine-related adverse events after the booster immunization in the PCV13-TT (2 months) group was significantly lower than that in the PCV13-TT (3 months) group and PCV7 (3 months) group (p=0.0180). The incidence of fever and diarrhea in the PCV13-TT (2 months) group was lower than that in the PCV13-TT (3 months) and PCV7 (3 months) group (fever: p=0.0360, diarrhea: p=0.0156). No significant difference was observed among other symptom groups, please refer to
Table 163 to

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Table 164.

Table 163 Incidence of adverse reactions 30 days post booster dose in PCV13-TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group
PCV13-TT (2M) PCV13-TT (3M) PCV7 (3M)
(N=487) (N=488) (N=493) P
Events Cases % Events Cases % Events Cases %
Any AE 202 151 31.01 260 186 38.11 270 179 36.31 0.0523
Grade 1 115 95 19.51 124 105 21.52 128 107 21.70 0.6516
Grade 2 86 74 15.20 122 104 21.31 133 92 18.66 0.0461
Grade 3 1 1 0.21 14 12 2.46 9 7 1.42 0.0051
AEs related to
vaccine 130 116 23.82 190 153 31.35 187 149 30.22 0.0180
Grade 1 90 83 17.04 105 93 19.06 116 98 19.88 0.4929
Grade 2 40 39 8.01 76 70 14.34 66 56 11.36 0.0070
Grade 3 0 0 0.00 9 8 1.64 5 4 0.81 0.0075
Solicited AE 125 113 23.20 186 152 31.15 179 146 29.61 0.0126
Grade 1 88 81 16.63 104 93 19.06 111 96 19.47 0.4657
Grade 2 37 36 7.39 73 68 13.93 63 55 11.16 0.0039
Grade 3 0 0 0.00 9 8 1.64 5 4 0.81 0.0075
Unsolicited AE 5 5 1.03 4 4 0.82 8 8 1.62 0.5354
Grade 1 2 2 0.41 1 1 0.20 5 5 1.01 0.2677
Grade 2 3 3 0.62 3 3 0.61 3 3 0.61 1.0000
Grade 3 / / / / / / / / / /
AEs unrelated to
vaccine 72 48 9.86 70 51 10.45 83 51 10.34 0.9523
Grade 1 25 20 4.11 19 15 3.07 12 11 2.23 0.2535

Grade 2 46 37 7.60 46 39 7.99 67 45 9.13 0.6700
 Grade 3 1 1 0.21 5 4 0.82 4 3 0.61 0.4635
 Solicited AE 24 14 2.87 17 11 2.25 28 18 3.65 0.4287
 Grade 1 11 8 1.64 7 5 1.02 4 3 0.61 0.2567
 Grade 2 12 8 1.64 8 7 1.43 23 17 3.45 0.0797
 Grade 3 1 1 0.21 2 2 0.41 1 1 0.20 0.8504
 Unsolicited AE 48 42 8.62 53 50 10.25 55 43 8.72 0.6248
 Grade 1 14 14 2.87 12 12 2.46 8 8 1.62 0.3935
 Grade 2 34 31 6.37 38 37 7.58 44 37 7.51 0.7195
 Grade 3 0 0 0.00 3 3 0.61 3 2 0.41 0.2981

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Table 164 Incidence of systemic and local reactions post booster dose in subjects aged 7-11 months
 PCV13-TT (2M) PCV13-TT (3M) PCV7 (3M)
 (N=487) (N=488) (N=493) P
 Events Cases % Events Cases % Events Cases %
 Fever 114 105 21.56 146 137 28.07 149 135 27.38 0.0360
 Fatigue / 0 0 0.00
 somnolence 1 1 0.20 1 1 0.20 1.0000
 Nausea /
 vomiting 1 1 0.21 8 8 1.64 4 4 0.81 0.0523
 Diarrhea 2 2 0.41 12 12 2.46 11 11 2.23 0.0156
 Cough 1 1 0.21 4 4 0.82 6 6 1.22 0.2075
 Abnormal
 crying 1 1 0.21 4 4 0.82 3 3 0.61 0.4635
 Local pain 1 1 0.21 1 1 0.20 0 0 0.00 0.5536
 Local
 redness 2 2 0.41 5 5 1.02 3 3 0.61 0.5375
 Local
 swelling 3 3 0.62 4 4 0.82 2 2 0.41 0.6556
 Local 0 0 0.00
 induration 1 1 0.20 0 0 0.00 0.6642

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2.7.4.2.2. Analysis of adverse events
 2.7.4.2.2.1. Phase I clinical trial
 2.7.4.2.2.1.1. Adverse reactions after each vaccination in 2 months old group and 3 months old group
 The incidence of adverse events in the 2 months old group (vaccinations were given at 0, 2, 4 months) and the 3 months old group (vaccinations were given at month 0, 1, 2) are shown in Table 165 to Table 166.
 No growing tendency was observed for adverse reaction as the doses of vaccination increased in subjects administered with either immunization procedure. No significant difference was observed in the incidences of adverse events among each dose in the schedule of vaccination at 0, 2, 4 months, and a downward trend was noted as the doses increased in the schedule of vaccination at 0, 1, 2 months.

Table 165. Adverse reactions after each dose in subjects aged 2 months (schedule of vaccination at 0, 2, 4 months)
 1st Dose (N=30) 2nd Dose (N=30) 3rd Dose (N=30)
 Events Cases % Events Cases % Events Cases %
 Any AE 34 16 53.33 34 16 53.33 37 14 46.67
 Related AE 26 14 46.67 24 14 46.67 32 14 46.67
 Solicited AE 26 14 46.67 24 14 46.67 31 14 46.67
 Systemic reactions 24 12 40.00 15 12 40.00 15 10 33.33
 Local reactions 2 2 6.67 9 7 23.33 16 10 33.33
 Unsolicited AE 0 0 0.00 0 0 0.00 1 1 3.33
 AEs unrelated to
 8 5 16.67 10 6 20.00 5 3 10.00
 vaccine

Table 166. Adverse reactions after each dose in subjects aged 3 months (schedule of vaccination at 0, 1, 2 months)
 1st Dose (N=30) 2nd Dose (N=29) 3rd Dose (N=28)
 Events Cases % Events Cases % Events Cases %
 Any AE 39 17 56.67 27 15 51.72 13 11 39.29
 Related AE 34 16 53.33 20 13 44.83 12 10 35.71
 Solicited AE 33 16 53.33 19 13 44.83 12 10 35.71
 Systemic reactions 22 11 36.67 11 10 34.48 7 7 25.00
 Local reactions 11 7 23.33 8 7 24.14 5 4 14.29
 Unsolicited AE 1 1 3.33 1 1 3.45 0 0 0.00
 AEs unrelated to
 5 5 16.67 7 5 17.24 1 1 3.57
 vaccine

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2.7.4.2.3.1.2. Solicited adverse events – systemic and local reactions
 See Table 167-Table 170 for the incidences of systemic and local reactions.
 The most common systemic reaction was fever in all age groups, the incidence of fever in infants and toddlers was higher than that in adults and children aged 2-5 years. The severity of fever was

mainly grade 1, while 2 cases in 2-5 years old group and 1 case in 2 months old group reach grade 3. The incidences of vomiting, diarrhea, abnormal crying and other symptoms were also higher in subjects aged 2 and 3 months. Except for fever, one case in the 2 months old group had grade 3 fatigue / somnolence. The severity of other systemic reactions was mainly grade 1, and no grade 3 reaction was reported.

Common local reactions in adults and children aged 2-5 years were pain, redness and swelling. The severity of local pain for 1 case in the children group reached grade 3; the severity of local redness for 3 adults and 1 child reached grade 3; the severity of local swelling for 4 cases in the adults group reached grade 3; the severity of local induration for 1 case in the adults group reached grade 3. Other local adverse reactions were mainly grade 1. The most common symptoms were local tenderness, redness and induration in 2 months old group; tenderness, redness and swelling were more common in 3 months old group. The severity of local redness and local induration for 1 case in 2 months old group reached grade 3, and the severity of swelling for 1 case in 3-month group reached grade 3.

Within 30 minutes after vaccination, the primarily observed adverse reactions were fever, fatigue (mainly observed among adults) and local redness, including 1 case of fever, one case of fatigue and 7 cases of local redness in the adults group who received 1 dose of vaccine; one case of fever was observed in the children group who received 1 dose of vaccine; eight cases of fever and 7 cases of local redness were observed in the 2 months old group who received three-dose series given at 0, 2, 4 months; four cases of fever and 4 cases of local redness were observed in the 3 months old group who received three-dose series at 0, 1, 2 months.

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Table 167. Analysis of systemic reactions												
≥18 years (N=30)			2-5 years (N=30)			2 months (N=30)			3 months (N=30)			
Events	Cases	%	Events	Cases	%	Events	Cases	%	Events	Cases	%	
Fever												
8	7	23.33	9	7	23.33	27	16	53.33	20	14	46.67	
Grade 1												
5	4	13.33	5	5	16.67	21	16	53.33	19	13	43.33	
Grade 2												
3	3	10.00	2	2	6.67	5	5	16.67	1	1	3.33	
Grade 3												
0	0	0.00	2	2	6.67	1	1	3.33	0	0	0.00	
Fatigue / somnolence												
4	4	13.33	1	1	3.33	2	2	6.67	1	1	3.33	
Grade 1												
3	3	10.00	0	0	0.00	1	1	3.33	1	1	3.33	
Grade 2												
1	1	3.33	1	1	3.33	0	0	0.00	0	0	0.00	
Grade 3												
0	0	0.00	0	0	0.00	1	1	3.33	0	0	0.00	
Nausea / vomiting												
0	0	0.00	0	0	0.00	1	1	3.33	3	3	10.00	
Grade 1												
0	0	0.00	0	0	0.00	1	1	3.33	2	2	6.67	
Grade 2												
0	0	0.00	0	0	0.00	0	0	0.00	1	1	3.33	
Grade 3												
0	0	0.00	0	0	0.00	0	0	0.00	0	0	0.00	
Diarrhea												
0	0	0.00	1	1	3.33	6	5	16.67	5	5	16.67	
Grade 1												
0	0	0.00	1	1	3.33	4	4	13.33	3	3	10.00	
Grade 2												
0	0	0.00	0	0	0.00	2	2	6.67	2	2	6.67	
Grade 3												
0	0	0.00	0	0	0.00	0	0	0.00	0	0	0.00	
Cough												
3	3	10.00	1	1	3.33	6	3	10.00	4	3	10.00	
Grade 1												
2	2	6.67	1	1	3.33	5	2	6.67	4	3	10.00	
Grade 2												
1	1	3.33	0	0	0.00	1	1	3.33	0	0	0.00	
Grade 3												
0	0	0.00	0	0	0.00	0	0	0.00	0	0	0.00	
Crying												
0	0	0.00	1	1	3.33	11	7	23.33	7	7	23.33	
Grade 1												
0	0	0.00	1	1	3.33	5	4	13.33	5	5	16.67	
Grade 2												
0	0	0.00	0	0	0.00	6	5	16.67	2	2	6.67	
Grade 3												
0	0	0.00	0	0	0.00	0	0	0.00	0	0	0.00	
Allergic reactions												
0	0	0.00	0	0	0.00	1	1	3.33	0	0	0.00	
Grade 1												
0	0	0.00	0	0	0.00	1	1	3.33	0	0	0.00	
Grade 2												
0	0	0.00	0	0	0.00	0	0	0.00	0	0	0.00	
Grade 3												
0	0	0.00	0	0	0.00	0	0	0.00	0	0	0.00	

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Table 168. Analysis of local reactions											
≥18 years (N=30)			2-5 years (N=30)			2 months (N=30)			3 months (N=30)		
Events	Cases	%	Events	Cases	%	Events	Cases	%	Events	Cases	%
Local											
19	18	60.00	6	6	20.00	5	4	13.33	4	4	13.33
pain											
Grade 1			14	13	43.33	5	5	16.67	4	4	13.33
Grade 2			5	5	16.67	0	0	0.00	1	1	3.33
Grade 3			0	0	0.00	1	1	3.33	0	0	0.00
Local											
13	13	43.33	5	5	16.67	16	11	36.67	11	7	23.33
redness											
Grade 1			7	7	23.33	4	4	13.33	11	8	26.67
Grade 2			3	3	10.00	0	0	0.00	3	3	10.00
Grade 3			3	3	10.00	1	1	3.33	2	1	3.33
Local											
9	9	30.00	1	1	3.33	2	2	6.67	7	5	16.67
swelling											
Grade 1			1	1	3.33	1	1	3.33	2	2	6.67
Grade 2			4	4	13.33	0	0	0.00	0	0	0.00
Grade 3			4	4	13.33	0	0	0.00	0	0	0.00
Local											
4	4	13.33	0	0	0.00	4	3	10.00	2	2	6.67
induration											
Grade 1			2	2	6.67	0	0	0.00	3	2	6.67
Grade 2			1	1	3.33	0	0	0.00	0	0	0.00
Grade 3			1	1	3.33	0	0	0.00	1	1	3.33

Local
3 3 10.00 3 3 10.00 0 0 0.00 0 0 0.00
pruritis
Grade 1 3 3 10.00 3 3 10.00 0 0 0.00 0 0 0.00
Grade 2 0 0 0.00 0 0 0.00 0 0 0.00 0 0 0.00
Grade 3 0 0 0.00 0 0 0.00 0 0 0.00 0 0 0.00

Table 169. Analysis of systemic and local reactions 30 mins post vaccination
≥18 years (N=30) 2-5 years (N=30) 2 months (N=30) 3 months (N=30)
Events Cases % Events Cases % Events Cases % Events Cases %
Fever 1 1 3.33 1 1 3.33 11 8 26.67 5 4 13.33
Fatigue /
1 1 3.33 0 0 0.00 0 0 0.00 0 0 0.00
somnolence
Local
7 7 23.33 0 0 0.00 10 7 23.33 6 4 13.33
redness

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Table 170. Analysis of systemic and local reactions 0-7 days post vaccination
≥18 years (N=30) 2-5 years (N=30) 2 months (N=30) 3 months (N=30)
Events Cases % Events Cases % Events Cases % Events Cases %
Fever 8 7 23.33 9 7 23.33 27 16 53.33 20 14 46.67
Fatigue /
4 4 13.33 1 1 3.33 2 2 6.67 1 1 3.33
somnolence
Nausea
0 0 0.00 0 0 0.00 1 1 3.33 3 3 10.00
vomiting
Diarrhea 0 0 0.00 1 1 3.33 6 5 16.67 5 5 16.67
Cough 3 3 10.00 1 1 3.33 6 3 10.00 4 3 10.00
Crying 0 0 0.00 1 1 3.33 11 7 23.33 7 7 23.33
Allergic
0 0 0.00 0 0 0.00 1 1 3.33 0 0 0.00
reactions
Local pain 19 18 60.00 6 6 20.00 5 4 13.33 4 4 13.33
Local
13 13 43.33 5 5 16.67 16 11 36.67 11 7 23.33
redness
Local
9 9 30.00 1 1 3.33 2 2 6.67 7 5 16.67
swelling
Local
4 4 13.33 0 0 0.00 4 3 10.00 2 2 6.67
induration
Local
3 3 10.00 3 3 10.00 0 0 0.00 0 0 0.00
pruritis

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2.7.4.2.3.1.3. Unsolicited adverse events
A total of 70 unsolicited adverse events including laboratory tests were observed in 44 cases, including 13 vaccine-related cases (16 events) and 35 vaccine-independent cases (54 events). In the adult and 2-5-year-old children group, 9 cases (11 events) were observed with abnormal laboratory findings related to vaccination. The severity of abnormal laboratory results related to vaccine was grade 1, and were determined as clinically irrelevant events by investigators. Five other unsolicited adverse events, other than laboratory tests, were determined as "probably related to study vaccine" by the investigators, including one case of lymphadenopathy in adult group, one case of rhinorrhea (runny nose) in children group, 1 case of decreased appetite in 2 months old group, 1 case of dyspepsia and 1 case of vomiting in 3-month group. See Table 171-Table 174 for analysis of unsolicited events.

Table 171. Unsolicited adverse events
≥18 years (N=30) 2-5 years (N=30) 2 months (N=30) 3 months (N=30)
Event Case Event Case Event Case Event Case
% % % %
s s s s s s s
Any
43.3 30.0 36.6 36.6
unsolicited 18 13 13 9 24 11 15 11
3 0 7 7
AE
Study
16.6 16.6
vaccine- 6 5 7 5 1 1 3.33 2 2 6.67
7 7
related
Investigatio 13.3 16.6
5 4 6 5 // // //
ns 3 7
Other 1 1 3.33 1 1 3.33 1 1 3.33 2 2 6.67
Unrelated to
26.6 20.0 36.6 33.3
study 12 8 6 6 23 11 13 10
7 0 7 3
vaccine

Table 172. Abnormal laboratory findings

≥18 years (N=30)	2-5 years (N=30)
Events	Cases %
Leukocytopenia	1 1 3.33 0 0 0.00
Elevated white blood cell count	0 0 0.00 6 6 20.00
Increased alanine	1 1 3.33 0 0 0.00
aminotransferase	
Increased aspartate	1 1 3.33 0 0 0.00
aminotransferase	
Increased bilirubin	2 2 6.67 0 0 0.00
Increased hemoglobin	1 1 3.33 0 0 0.00
Elevated serum creatinine	1 1 3.33 0 0 0.00
Decreased blood urea	1 1 3.33 1 1 3.33
Elevated platelet count	1 1 3.33 1 1 3.33

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Table 173. Unsolicited events related to study vaccine

≥18 years (N=30)	2-5 years (N=30)	2 months (N=30)	3 months (N=30)
Events	Cases %	Events	Cases %
Decreased appetite	1 1 3.33		
Rhinorrhea	0 0 0.00 1 1 3.33	0 0 0.00 0 0 0.00	0 0 0.00
Dyspepsia	0 0 0.00 0 0 0.00	0 0 0.00 1 1 3.33	
Infant vomiting	0 0 0.00 0 0 0.00	0 0 0.00 1 1 3.33	
Lymphadenopathy	1 1 3.33	0 0 0.00 0 0 0.00	0 0 0.00

Table 174. Unsolicited events unrelated to study vaccine

≥18 years (N=30)	2-5 years (N=30)	2 months (N=30)	3 months (N=30)
Events	Cases %	Events	Cases %
Nasopharyngitis	2 2 6.67	1 1 3.33	3 3 10.00
Viral infection	0 0 0.00	0 0 0.00	0 0 0.00
Infectious diarrhea	0 0 0.00	0 0 0.00	1 1 3.33
Respiratory tract infection	0 0 0.00	1 1 3.33	0 0 0.00
Candida infection	0 0 0.00	0 0 0.00	1 1 3.33
Tracheitis	0 0 0.00	0 0 0.00	1 1 3.33
Upper respiratory tract infection	3 3 10.00	0 0 0.00	3 3 10.00
Rhinorrhea	1 1 3.33	1 1 3.33	0 0 0.00
Bronchitis	0 0 0.00	0 0 0.00	0 0 0.00
Herpangina	0 0 0.00	0 0 0.00	1 1 3.33
Cough	0 0 0.00	0 0 0.00	0 0 0.00
Rash	0 0 0.00	0 0 0.00	1 1 3.33
Papular rash	1 1 3.33	0 0 0.00	0 0 0.00
Eczema	0 0 0.00	0 0 0.00	1 1 3.33
Urticaria	0 0 0.00	0 0 0.00	1 1 3.33
Itching	0 0 0.00	1 1 3.33	0 0 0.00
Fever	0 0 0.00	0 0 0.00	4 4 13.33
Limb pain	1 1 3.33	0 0 0.00	0 0 0.00
Diarrhea	0 0 0.00	0 0 0.00	4 4 13.33
Decreased appetite	0 0 0.00	0 0 0.00	1 1 3.33
Dyspepsia	0 0 0.00	0 0 0.00	2 2 6.67
Infant vomiting	0 0 0.00	0 0 0.00	0 0 0.00

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2.7.4.2.3.1.4. Discussions and conclusions of safety

The incidence rates of vaccine-related adverse events in adults, children, 2 months old and 3 months old group were 80.00%, 53.33%, 66.67% and 70.00%, respectively. Over all 6 cases reported grade 3 local reactions in the adult group and the incidence rate was 20% (6 / 30), including 1 case of local redness, 1 case of induration, 2 cases of swelling and 2 cases of redness combined with swelling. The incidence rate of grade 3 adverse reactions in other age groups was all less than 15%.

Common symptoms of systemic adverse reactions were fever in adults and children aged 2-5 years, and fever and diarrhea in 2 months old and 3 months old infants; common local reactions were primarily redness, pain and swelling at the vaccination site in adults; pain and redness were the most common local reactions in children aged 2-5 years, and redness was common in infants 2 months and 3 months of age.

Adverse events mainly occurred within 0-7 days. No adverse events related to study vaccine were collected during 8-30 days, and no serious adverse events were collected during the observation period.

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2.7.4.2.2.2. Phase III clinical trial

2.7.4.2.2.2.1. Primary series

2.7.4.2.2.2.1.1. Solicited events

(1) Systemic reactions in PCV13-TT (3-71 months) group and PCV7 group
The incidence of systemic adverse reactions in the PCV13-TT and the PCV7 group was 54.52% and 58.98%, respectively, and the incidence of systemic adverse reactions in the PCV13-TT group was lower than that in the PCV7 group ($p=0.0364$). The main symptom of systemic adverse reaction was fever, which was less in PCV13-TT group than in PCV7 group ($p=0.0035$). No significant difference was observed for the incidence of other systemic reactions between groups. The systemic reactions were primarily grade 1 or 2, similar incidence rates of grade 3 systemic reactions were noted between PCV13-TT group and PCV7 group.
Systemic reactions and their incidence at different time observed in PCV13-TT and PCV7 group of the 3-71 months olds are shown in Table 175.

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Table 175. Analysis of systemic reactions in PCV13-TT and PCV7 groups of subjects aged 3-71 months
PCV13-TT (N=1117) PCV7 (N=1119)
P

Events	Cases	%	Events	Cases	%		
Systemic reactions	1190	609	54.52	1327	660	58.98	0.0364
Grade 1	829	477	42.70	914	531	47.45	0.0244
Grade 2	341	253	22.65	394	293	26.18	0.0549
Grade 3	19	19	1.70	19	19	1.70	1.0000
Grade 4	1	1	0.09	0	0	0.00	0.4996
Fever	816	531	47.54	972	602	53.80	0.0035
Grade 1	593	398	35.63	695	470	42.00	0.0021
Grade 2	209	187	16.74	268	227	20.29	0.0337
Grade 3	13	13	1.16	9	9	0.80	0.4022
Grade 4	1	1	0.09	0	0	0.00	0.4996
Fatigue / somnolence	47	42	3.76	39	37	3.31	0.5692
Grade 1	37	35	3.13	32	30	2.68	0.5323
Grade 2	10	10	0.90	7	7	0.63	0.4787
Grade 3	0	0	0.00	0	0	0.00	/
Nausea / vomiting	55	46	4.12	40	37	3.31	0.3166
Grade 1	42	36	3.22	30	29	2.59	0.3817
Grade 2	13	12	1.07	10	9	0.80	0.5212
Grade 3	0	0	0.00	0	0	0.00	/
Diarrhea	103	91	8.15	100	87	7.77	0.7552
Grade 1	56	55	4.92	57	52	4.65	0.7673
Grade 2	44	37	3.31	42	38	3.40	1.0000
Grade 3	3	3	0.27	1	1	0.09	0.3742
Myalgia	2	2	0.18	4	4	0.36	0.6871
Grade 1	2	2	0.18	4	4	0.36	0.6871
Grade 2	0	0	0.00	0	0	0.00	/
Grade 3	0	0	0.00	0	0	0.00	/
Cough	69	66	5.91	72	65	5.81	0.9285
Grade 1	41	39	3.49	38	34	3.04	0.5546
Grade 2	28	27	2.42	32	31	2.77	0.6902
Grade 3	0	0	0.00	2	2	0.18	0.4998
Crying	89	72	6.45	81	67	5.99	0.6625
Grade 1	56	47	4.21	53	47	4.20	1.0000
Grade 2	32	27	2.42	27	25	2.23	0.7812
Grade 3	1	1	0.09	1	1	0.09	1.0000
Allergic reactions	9	9	0.81	19	18	1.61	0.1198
Grade 1	2	2	0.18	5	5	0.45	0.4524
Grade 2	5	5	0.45	8	7	0.63	0.7738
Grade 3	2	2	0.18	6	6	0.54	0.2882

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(2) Local reactions in PCV13-TT (3-71 months) and PCV7 group
The incidence of local adverse reaction was 15.67% in the PCV13-TT group and 14.66% in the PCV7 group, indicating no significant difference. The most common symptom of local adverse reaction was redness at injection site, followed by swelling and pain. The severity of local reactions was mainly grade 1. The incidence of local reactions with severity of grade 3 in the PCV13-TT group was higher than that in the PCV7 group ($p=0.0007$). The symptoms of local reactions achieving grade 3 in the PCV13-TT group were mainly redness, swelling and induration, however, no intergroup difference was observed with respect to redness, swelling and induration in 3-month-old population. The symptoms mainly occurred in the population over 7-month-old.
Local reactions and their incidence at different time observed in PCV13-TT and PCV7 group of the 3-71 months olds are shown in Table 176.

Table 176. Analysis of local reactions in PCV13-TT and PCV7 groups of subjects aged 3-71 months
PCV13-TT (N=1117) PCV7 (N=1119)
P

Events	Cases	%	Events	Cases	%		
Local reactions	301	175	15.67	280	164	14.66	0.5170
Grade 1	177	130	11.64	200	126	11.26	0.7908
Grade 2	84	63	5.64	66	50	4.47	0.2110
Grade 3	40	28	2.51	14	8	0.71	0.0007
Local pain	59	55	4.92	43	40	3.57	0.1170
Grade 1	58	54	4.83	41	38	3.40	0.0895
Grade 2	1	1	0.09	2	2	0.18	1.0000
Grade 3	0	0	0.00	0	0	0.00	/
Local redness	128	123	11.01	155	126	11.26	0.8931
Grade 1	73	71	6.36	106	85	7.60	0.2805
Grade 2	37	36	3.22	42	40	3.57	0.7265
Grade 3	18	18	1.61	7	6	0.54	0.0141
Local swelling	68	63	5.64	55	47	4.20	0.1188
Grade 1	22	19	1.70	36	29	2.59	0.1886

Grade 2 29 29 2.60 14 14 1.25 0.0213
 Grade 3 17 16 1.43 5 5 0.45 0.0163
 Local rash 1 1 0.09 0 0 0.00 0.4996
 Grade 1 0 0 0.00 0 0 0.00 /
 Grade 2 1 1 0.09 0 0 0.00 0.4996
 Grade 3 0 0 0.00 0 0 0.00 /
 Local induration 30 27 2.42 22 21 1.88 0.3861
 Grade 1 9 9 0.81 12 11 0.98 0.8230
 Grade 2 16 14 1.25 8 8 0.71 0.2075
 Grade 3 5 5 0.45 2 2 0.18 0.2880
 Local pruritis 15 15 1.34 5 5 0.45 0.0258
 Grade 1 15 15 1.34 5 5 0.45 0.0258
 Grade 2 0 0 0.00 0 0 0.00 /
 Grade 3 0 0 0.00 0 0 0.00 /

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(3) The onset time of systemic and local reactions in PCV13-TT (3-71 months) and PCV7 group
 The occurrence of systemic and local adverse reactions in the PCV13-TT group and the PCV7 group 30 minutes and 7 days post vaccination are shown in Table 177 and Table 178. Within 30 minutes, only fever was observed as systemic reaction. The incidence of fever in the PCV7 group was higher than that in the PCV13-TT group. Redness was the main local reaction, and was similar between groups in terms of incidence rate.

Table 177. Systemic and local reactions 30 min after vaccination in subjects aged 3-71 months
 PCV13-TT (N=1117) PCV7 (N=1119)

P	Events	Cases	%	Events	Cases	%
Fever	147	127	11.37	175	162	14.48
Local pain	1	1	0.09	0	0	0.00
Local redness	46	44	3.94	37	32	2.86
Local swelling	1	1	0.09	0	0	0.00

Table 178. Systemic and local reactions 7 days min after vaccination in subjects aged 3-71 months
 PCV13-TT (N=1117) PCV7 (N=1119)

P	Events	Cases	%	Events	Cases	%
Fever	816	531	47.54	972	602	53.80
Fatigue / somnolence	47	42	3.76	39	37	3.31
Nausea / vomiting	55	46	4.12	40	37	3.31
Diarrhea	103	91	8.15	100	87	7.77
Myalgia	2	2	0.18	4	4	0.36
Cough	69	66	5.91	72	65	5.81
Abnormal crying	89	72	6.45	81	67	5.99
Allergic reactions	9	9	0.81	19	18	1.61
Local pain	59	55	4.92	43	40	3.57
Local redness	128	123	11.01	155	126	11.26
Local swelling	68	63	5.64	55	47	4.20
Local rash	1	1	0.09	0	0	0.00
Local induration	30	27	2.42	22	21	1.88
Local pruritis	15	15	1.34	5	5	0.45

(4) Systemic reactions and local reactions observed in PCV13-TT and PCV7 group of different age groups

No significant difference was observed in the incidence of systemic and local adverse reactions between PCV13-TT (3 months) and PCV7 (3 months) groups, the 7-11 months olds, the 12-23 months olds and the 24-71 months olds.

Fever was the most common systemic reaction in all age groups. The incidence of fever in 3-month age PCV13-TT group was lower than that in the PCV7 group ($p=0.0072$), and the incidence of allergies in 7-11 month age PCV13-TT group was lower than that in the PCV7 group ($p=0.0301$), the incidence of local itching in 24-71 months old PCV13-TT group was higher than that in the PCV7 group ($p=0.0358$). No significant difference was observed between PCV13-TT and PCV7 group of other ages in fever and other systemic or local symptoms.

The incidence of systemic and local reactions in the PCV13-TT and PCV7 group of each age stratification is shown in Table 179 to Table 182.

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Table 179. Systemic and local reactions after vaccination in subjects aged 3 months
 PCV13-TTN=519 PCV7N=519

P	Events	Cases	%	Events	Cases	%
Systemic reactions	778	352	67.82	862	371	71.48
Fever	515	301	58.00	628	344	66.28
Fatigue / somnolence	37	32	6.17	29	27	5.20
Nausea / vomiting	38	30	5.78	22	19	3.66
Diarrhea	76	67	12.91	74	62	11.95
Myalgia	1	1	0.19	2	2	0.39
Cough	40	38	7.32	41	34	6.55
Abnormal crying	68	54	10.40	62	49	9.44
Allergic reactions	3	3	0.58	4	3	0.58
Local reactions	107	70	13.49	150	81	15.61
Local pain	16	12	2.31	12	10	1.93
Local redness	59	55	10.60	92	71	13.68
Local swelling	12	10	1.93	25	18	3.47
Local induration	18	15	2.89	19	18	3.47
Local pruritis	2	2	0.39	2	2	0.39

Table 180. Systemic and local reactions after vaccination in subjects aged 7-11 months

PCV13-TT (N=198) PCV7 (N=200)									
P									
Events Cases % Events Cases %									
Systemic reactions 206 116 58.59 245 132 66.00 0.1476									
Fever 149 106 53.54 184 123 61.50 0.1281									
Fatigue / somnolence 3 3 1.52 5 5 2.50 0.7238									
Nausea vomiting 12 11 5.56 8 8 4.00 0.4912									
Diarrhea 15 14 7.07 19 18 9.00 0.5810									
Myalgia 0 0 0.00 1 1 0.50 1.0000									
Cough 11 10 5.05 9 9 4.50 0.8188									
Crying 16 13 6.57 13 12 6.00 0.8393									
Allergic reactions 0 0 0.00 6 6 3.00 0.0301									
Local reactions 71 42 21.21 48 29 14.50 0.0894									
Local pain 14 14 7.07 8 7 3.50 0.1221									
Local redness 34 33 16.67 30 25 12.50 0.2580									
Local swelling 16 15 7.58 8 7 3.50 0.0830									
Local induration 6 6 3.03 1 1 0.50 0.0669									
Local pruritis 1 1 0.51 1 1 0.50 1.0000									

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Table 181. Systemic and local reactions after vaccination in subjects aged 12-23 months

PCV13-TT (N=200) PCV7 (N=200)									
P									
Events Cases % Events Cases %									
Systemic reactions 143 93 46.50 141 100 50.00 0.5483									
Fever 110 83 41.50 110 88 44.00 0.6861									
Fatigue / somnolence 2 2 1.00 2 2 1.00 1.0000									
Nausea vomiting 3 3 1.50 5 5 2.50 0.7238									
Diarrhea 10 8 4.00 3 3 1.50 0.2201									
Cough 10 10 5.00 11 11 5.50 1.0000									
Crying 5 5 2.50 3 3 1.50 0.7238									
Allergic reactions 3 3 1.50 7 7 3.50 0.3375									
Local reactions 45 21 10.50 27 18 9.00 0.7364									
Local pain 10 10 5.00 3 3 1.50 0.0870									
Local redness 16 16 8.00 17 14 7.00 0.8498									
Local swelling 15 13 6.50 6 6 3.00 0.1569									
Local rash 1 1 0.50 0 0 0.00 1.0000									
Local induration 1 1 0.50 1 1 0.50 1.0000									
Local pruritis 2 2 1.00 0 0 0.00 0.4987									

Table 182. Systemic reactions after vaccination in subjects aged 24-71 months

PCV13-TT (N=200) PCV7 (N=200)									
P									
Events Cases % Events Cases %									
Systemic reactions 63 48 24.00 79 57 28.50 0.3633									
Fever 42 41 20.50 50 47 23.50 0.5463									
Fatigue / 5 5 2.50 3 3 1.50 0.7238									
somnolence									
Nausea vomiting 2 2 1.00 5 5 2.50 0.4490									
Diarrhea 2 2 1.00 4 4 2.00 0.6851									
Myalgia 1 1 0.50 1 1 0.50 1.0000									
Cough 8 8 4.00 11 11 5.50 0.6394									
Crying 0 0 0.00 3 3 1.50 0.2481									
Allergic reactions 3 3 1.50 2 2 1.00 1.0000									
Local reactions 78 42 21.00 55 36 18.00 0.5282									
Local pain 19 19 9.50 20 20 10.00 1.0000									
Local redness 19 19 9.50 16 16 8.00 0.7239									
Local swelling 25 25 12.50 16 16 8.00 0.1867									
Local induration 5 5 2.50 1 1 0.50 0.2152									
Local pruritis 10 10 5.00 2 2 1.00 0.0358									

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2.7.4.2.3.2.1.2. Vaccine-related unsolicited adverse events

Unsolicited adverse events (coincidental) that occurred 0-30 days post vaccination collected from the PCV13-TT and PCV7 groups during the observation period are shown in Table 183 to Table 189.

Vaccine-related unsolicited events in the PCV13-TT and PCV7 groups were mainly respiratory, thoracic and mediastinal disorders, gastrointestinal disorders, infections and infestations, general disorders and administration site conditions. General disorders and administration site conditions occurred mainly in 8-30 days post vaccination, and the most common symptom was fever.

Considering from the time of occurrence, unsolicited adverse events related to the study vaccine occurred mainly within 7 days post vaccination. In view of age groups, more types and higher incidence rates of unsolicited adverse events related to the vaccine were reported in the 3-month age group, and a decreasing trend was observed in 7-11-month age group, 12-23 months age group and 24-71-month age group sequentially.

No significant difference was observed between the PCV13-TT and PCV7 group in the incidence of any vaccine-related unsolicited adverse events.

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Table 183. Vaccine-related unsolicited events in PCV13-TT and PCV7 groups of subjects aged 3-71 months

PCV13-TT (N=1117) PCV7 (N=1119)

Events	Cases	Incidence rate	Events	Cases	Incidence rate	P
Unsolicited adverse reactions	75	63	5.64	81	66	5.90 0.8561
Metabolism and nutrition disorders	1	1	0.09	0	0	0.00 0.4996
Decreased appetite	1	1	0.09	0	0	0.00 0.4996
Infections and infestations	16	16	1.43	12	12	1.07 0.4551
Nasopharyngitis	3	3	0.27	1	1	0.09 0.3742
Tonsillitis	1	1	0.09	0	0	0.00 0.4996
Infectious pneumonia	0	0	0.00	1	1	0.09 1.0000
Upper respiratory tract infection	10	10	0.90	10	10	0.89 1.0000
Bronchitis	1	1	0.09	0	0	0.00 0.4996
Herpes virus infection	1	1	0.09	0	0	0.00 0.4996
Respiratory, thoracic and mediastinal disorders	28	25	2.24	34	29	2.59 0.6798
Rhinorrhea	22	19	1.70	30	27	2.41 0.2970
Nasal obstruction	3	3	0.27	1	1	0.09 0.3742
Cough	2	2	0.18	0	0	0.00 0.2494
Sneezing	1	1	0.09	2	2	0.18 1.0000
Increased production of sputum	0	0	0.00	1	1	0.09 1.0000
Psychiatric disorders	1	1	0.09	0	0	0.00 0.4996
Irritability	1	1	0.09	0	0	0.00 0.4996
Skin and subcutaneous tissue disorders	2	2	0.18	0	0	0.00 0.2494
Hyperhidrosis	1	1	0.09	0	0	0.00 0.4996
Eczema	1	1	0.09	0	0	0.00 0.4996
General disorders and administration site conditions	13	12	1.07	16	15	1.34 0.6994
Fever	13	12	1.07	16	15	1.34 0.6994
Gastrointestinal disorders	13	13	1.16	18	17	1.52 0.5821
Aphthous ulcer	1	1	0.09	0	0	0.00 0.4996
White stool	0	0	0.00	1	1	0.09 1.0000
Abdominal pain	0	0	0.00	1	1	0.09 1.0000
Diarrhea	5	5	0.45	10	10	0.89 0.3001
Abdominal distension	1	1	0.09	1	1	0.09 1.0000
Oral ulcer	2	2	0.18	1	1	0.09 0.6245
Increased frequency of defecation	1	1	0.09	0	0	0.00 0.4996
Gastrointestinal pain	1	1	0.09	0	0	0.00 0.4996
Dyspepsia	1	1	0.09	3	3	0.27 0.6247
Infant vomiting	1	1	0.09	0	0	0.00 0.4996
Hard stool	0	0	0.00	1	1	0.09 1.0000
Blood and lymphatic system disorders	1	1	0.09	0	0	0.00 0.4996
Cold extremities	1	1	0.09	0	0	0.00 0.4996
Eye disorders	0	0	0.00	1	1	0.09 1.0000
Swollen eyelids	0	0	0.00	1	1	0.09 1.0000

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Table 184. Unsolicited events possibly related to study vaccine 0-7 days post vaccination in subjects aged 3-71 months

Events	Cases	Incidence rate	Events	Cases	Incidence rate	P
PCV13-TT (N=1117) PCV7 (N=1119)						
Unsolicited adverse reactions	51	46	4.12	51	43	3.84 0.7469
Metabolism and nutrition disorders	1	1	0.09	0	0	0.00 0.4996
Decreased appetite	1	1	0.09	0	0	0.00 0.4996
Infections and infestations	15	15	1.34	11	11	0.98 0.4391
Nasopharyngitis	2	2	0.18	1	1	0.09 0.6245
Tonsillitis	1	1	0.09	0	0	0.00 0.4996
Infectious pneumonia	0	0	0.00	1	1	0.09 1.0000
Upper respiratory tract infection	10	10	0.90	9	9	0.80 0.8229
Bronchitis	1	1	0.09	0	0	0.00 0.4996
Herpes virus infection	1	1	0.09	0	0	0.00 0.4996
Respiratory, thoracic and mediastinal disorders	24	22	1.97	31	28	2.50 0.4749
Rhinorrhea	20	18	1.61	27	26	2.32 0.2864
Nasal obstruction	3	3	0.27	1	1	0.09 0.3742
Sneezing	1	1	0.09	2	2	0.18 1.0000
Increased production of sputum	0	0	0.00	1	1	0.09 1.0000
Psychiatric disorders	1	1	0.09	0	0	0.00 0.4996
Irritability	1	1	0.09	0	0	0.00 0.4996
Skin and subcutaneous tissue disorders	1	1	0.09	0	0	0.00 0.4996
Hyperhidrosis	1	1	0.09	0	0	0.00 0.4996
Gastrointestinal disorders	8	8	0.72	8	7	0.63 0.8028
Aphthous ulcer	1	1	0.09	0	0	0.00 0.4996
White stool	0	0	0.00	1	1	0.09 1.0000
Abdominal pain	0	0	0.00	1	1	0.09 1.0000
Abdominal distension	1	1	0.09	1	1	0.09 1.0000
Oral ulcer	2	2	0.18	1	1	0.09 0.6245
Increased frequency of defecation	1	1	0.09	0	0	0.00 0.4996
Gastrointestinal pain	1	1	0.09	0	0	0.00 0.4996
Dyspepsia	1	1	0.09	3	3	0.27 0.6247
Infant vomiting	1	1	0.09	0	0	0.00 0.4996
Hard stool	0	0	0.00	1	1	0.09 1.0000
Blood and lymphatic system disorders	1	1	0.09	0	0	0.00 0.4996
Cold extremities	1	1	0.09	0	0	0.00 0.4996
Eye disorders	0	0	0.00	1	1	0.09 1.0000
Swollen eyelids	0	0	0.00	1	1	0.09 1.0000

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Table 185. Unsolicited events possibly related to study vaccine 8-30 days post vaccination in subjects aged 3-71 months

PCV13-TT (N=1117) PCV7 (N=1119)

P

Events	Cases	Incidence rate	Events	Cases	Incidence rate
Unsolicited adverse reactions	24 18	1.61 30 26	2.32 0.2864		
Infections and infestations	1 1	0.09 1 1	0.09 1.0000		
Nasopharyngitis	1 1	0.09 0 0	0.00 0.4996		
Upper respiratory tract infection	0 0	0.00 1 1	0.09 1.0000		
Respiratory, thoracic and mediastinal disorders	4 4	0.36 3 2	0.18 0.4522		
Rhinorrhea	2 2	0.18 3 2	0.18 1.0000		
Cough	2 2	0.18 0 0	0.00 0.2494		
Skin and subcutaneous tissue disorders	1 1	0.09 0 0	0.00 0.4996		
Eczema	1 1	0.09 0 0	0.00 0.4996		
General disorders and administration site conditions	13 12	1.07 16 15	1.34 0.6994		
Fever	13 12	1.07 16 15	1.34 0.6994		
Gastrointestinal disorders	5 5	0.45 10 10	0.89 0.3001		
Diarrhea	5 5	0.45 10 10	0.89 0.3001		

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Table 186. Unsolicited events related to study vaccine in subjects aged 3 months
PCV13-TT (N=519) PCV7 (N=519)

Events	Cases	Incidence rate	Events	Cases	Incidence rate
Unsolicited adverse reactions	54 44	8.48 56 44	8.48 1.0000		
Metabolism and nutrition disorders	1 1	0.19 0 0	0.00 1.0000		
Decreased appetite	1 1	0.19 0 0	0.00 1.0000		
Infections and infestations	9 9	1.73 7 7	1.35 0.8021		
Nasopharyngitis	1 1	0.19 0 0	0.00 1.0000		
Tonsillitis	1 1	0.19 0 0	0.00 1.0000		
Infectious pneumonia	0 0	0.00 1 1	0.19 1.0000		
Upper respiratory tract infection	7 7	1.35 6 6	1.16 1.0000		
Respiratory, thoracic and mediastinal disorders	20 17	3.28 27 22	4.24 0.5143		
Rhinorrhea	14 11	2.12 23 20	3.85 0.1435		
Nasal obstruction	3 3	0.58 1 1	0.19 0.6243		
Cough	2 2	0.39 0 0	0.00 0.4995		
Sneezing	1 1	0.19 2 2	0.39 1.0000		
Increased production of sputum	0 0	0.00 1 1	0.19 1.0000		
Psychiatric disorders	1 1	0.19 0 0	0.00 1.0000		
Irritability	1 1	0.19 0 0	0.00 1.0000		
Skin and subcutaneous tissue disorders	1 1	0.19 0 0	0.00 1.0000		
Hyperhidrosis	1 1	0.19 0 0	0.00 1.0000		
General disorders and administration site conditions	11 10	1.93 10 9	1.73 1.0000		
Fever	11 10	1.93 10 9	1.73 1.0000		
Gastrointestinal disorders	10 10	1.93 12 12	2.31 0.8300		
Aphthous ulcer	1 1	0.19 0 0	0.00 1.0000		
Diarrhea	5 5	0.96 9 9	1.73 0.4207		
Abdominal distension	1 1	0.19 1 1	0.19 1.0000		
Increased frequency of defecation	1 1	0.19 0 0	0.00 1.0000		
Dyspepsia	1 1	0.19 2 2	0.39 1.0000		
Infant vomiting	1 1	0.19 0 0	0.00 1.0000		
Blood and lymphatic system disorders	1 1	0.19 0 0	0.00 1.0000		
Cold extremities	1 1	0.19 0 0	0.00 1.0000		

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Table 187. Unsolicited events related to study vaccine in subjects aged 7-11 months
PCV13-TT (N=198) PCV7 (N=200)

Events	Cases	Incidence rate	Events	Cases	Incidence rate
Unsolicited adverse reactions	10 9	4.55 10 9	4.50 1.0000		
Infections and infestations	3 3	1.52 2 2	1.00 0.6843		
Nasopharyngitis	0 0	0.00 1 1	0.50 1.0000		
Upper respiratory tract infection	2 2	1.01 1 1	0.50 0.6222		
Bronchitis	1 1	0.51 0 0	0.00 0.4975		
Respiratory, thoracic and mediastinal disorders	3 3	1.52 3 3	1.50 1.0000		
Rhinorrhea	3 3	1.52 3 3	1.50 1.0000		
Skin and subcutaneous tissue disorders	1 1	0.51 0 0	0.00 0.4975		
Eczema	1 1	0.51 0 0	0.00 0.4975		
General disorders and administration site conditions	2 2	1.01 3 3	1.50 1.0000		
Fever	2 2	1.01 3 3	1.50 1.0000		
Gastrointestinal disorders	1 1	0.51 2 2	1.00 1.0000		
Abdominal pain	0 0	0.00 1 1	0.50 1.0000		
Gastrointestinal pain	1 1	0.51 0 0	0.00 0.4975		
Dyspepsia	0 0	0.00 1 1	0.50 1.0000		

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Table 188. Unsolicited events related to study vaccine in subjects aged 12-23 months
PCV13-TT (N=200) PCV7 (N=200)

Events	Cases	Incidence rate	Events	Cases	Incidence rate
Unsolicited adverse reactions	9 8	4.00 10 8	4.00 1.0000		
Infections and infestations	3 3	1.50 1 1	0.50 0.6231		
Nasopharyngitis	2 2	1.00 0 0	0.00 0.4987		
Upper respiratory tract infection	1 1	0.50 1 1	0.50 1.0000		
Respiratory, thoracic and mediastinal disorders	4 4	2.00 3 3	1.50 1.0000		
Rhinorrhea	4 4	2.00 3 3	1.50 1.0000		
General disorders and administration site conditions	0 0	0.00 2 2	1.00 0.4987		

Fever 0 0 0.00 2 2 1.00 0.4987
 Gastrointestinal disorders 2 2 1.00 3 2 1.00 1.0000
 White stool 0 0 0.00 1 1 0.50 1.0000
 Diarrhea 0 0 0.00 1 1 0.50 1.0000
 Oral ulcer 2 2 1.00 0 0 0.00 0.4987
 Hard stool 0 0 0.00 1 1 0.50 1.0000
 Eye disorders 0 0 0.00 1 1 0.50 1.0000
 Swollen eyelids 0 0 0.00 1 1 0.50 1.0000
 Table 189. Unsolicited events related to study vaccine in subjects aged 24-71 months
 PCV13-TT (N=200) PCV7 (N=200)

P
 Events Cases Incidence rate Events Cases Incidence rate
 Unsolicited adverse reactions 2 2 1.00 5 5 2.50 0.4490
 Infections and infestations 1 1 0.50 2 2 1.00 1.0000
 Upper respiratory tract infection 0 0 0.00 2 2 1.00 0.4987
 Herpes virus infection 1 1 0.50 0 0 0.00 1.0000
 Respiratory, thoracic and mediastinal disorders 1 1 0.50 1 1 0.50 1.0000
 Rhinorrhea 1 1 0.50 1 1 0.50 1.0000
 General disorders and administration site conditions 0 0 0.00 1 1 0.50 1.0000
 Fever 0 0 0.00 1 1 0.50 1.0000
 Gastrointestinal disorders 0 0 0.00 1 1 0.50 1.0000
 Oral ulcer 0 0 0.00 1 1 0.50 1.0000

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2.7.4.2.3.2.1.3. Discussions and conclusions of safety

A total of 790 cases (2282 events) were observed in the PCV13-TT group (3-71 months), and 851 cases (2440 events) were observed in the PCV7 group. The overall incidence rate of adverse events in the PCV13-TT and PCV7 group was 70.73% and 76.05%, respectively. Significant difference was observed in the incidence of adverse events between groups, and the overall incidence of adverse events in the PCV13-TT group was lower than that in the PCV7 group. The incidence of adverse events related to the study vaccine was 60.79% in the PCV13-TT group and 64.79% in the PCV7 group, with merely numerical difference noted.

The incidence rates of adverse events were higher in 2-6 months group and 7-11 months group. The intergroup difference of adverse events was mainly observed in 7-11 months group. The incidence of systemic adverse reactions was 54.52% in the PCV13-TT group and 58.98% in the PCV7 group (p=0.0364). The incidence of local adverse reactions was 15.67% in the PCV13-TT group and 14.66% in the PCV7 group, with no significant difference observed between groups. Fever was the most common symptom of systemic adverse reaction. The incidence of fever was 47.54% in the PCV13-TT group, and was significantly lower compared to 53.80% in the PCV7 group (p=0.0035). The most common symptom of local adverse reaction was redness at injection site, followed by swelling and pain.

The severity of adverse events in the 3-71 months group was mainly grade 1 and 2, while grade 3 systemic reactions related to the study vaccine occurred in 4.21% and 2.41% in PCV13-TT and PCV7 group subjects, respectively. No significant difference was observed in the incidence of systemic reaction with severity of grade 3 between the PCV13-TT group and the PCV7 group. The incidence of local reactions assessed as grade 3 in the PCV13-TT group was higher than that in the PCV7 group (p=0.0007). The symptoms mainly included redness, swelling and induration observed in subjects aged over 7 months. No intergroup difference was observed in the incidence of grade 3 adverse events in infants 3 months of age.

Totally, 80 cases (105 events) of SAE were collected in primary series, including 12 cases (13 events) in 2 months group, and 34 cases (49 events) and 34 cases (43 events) in the PCV13-TT group and PCV7 group of the 3-71 months olds, respectively. No significant difference was observed in SAE incidence between the PCV13-TT and PCV7 group. With the exception of 1 case of SAE in the PCV13-TT group of the 3 months olds had fever that was possibly related to the study vaccine, other SAEs were not related to the study vaccine.

No intergroup difference was observed with regards to the incidence of adverse events resulting in drop-out or vaccine-independent solicited /unsolicited events.

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2.7.4.2.2.2.2. Booster dose

2.7.4.2.2.2.2.1. Systemic reactions and local reactions

No significant difference was observed in the incidence of systemic and local reactions among all 3-11 months old subjects and between the PCV13-TT and PCV7 group of different age groups, as shown in

Table 190 to
 Table 192.

Table 190. Systemic and local reactions post booster dose in subjects aged 3-11 months
 PCV13-TT (N=674) PCV7 (N=684)

P
 Events Cases % Events Cases %
 Fever 186 175 25.96 190 175 25.58 0.9013
 Fatigue / somnolence 1 1 0.15 1 1 0.15 1.0000
 Nausea / vomiting 8 8 1.19 6 6 0.88 0.6029
 Diarrhea 15 15 2.23 13 13 1.90 0.7064
 Cough 5 5 0.74 8 8 1.17 0.5793
 Crying 6 6 0.89 3 3 0.44 0.3388
 Allergic reactions 0 0 0.00 1 1 0.15 1.0000
 Local pain 2 2 0.30 1 1 0.15 0.6220
 Local redness 6 6 0.89 5 5 0.73 0.7720
 Local swelling 7 7 1.04 4 4 0.58 0.3825
 Local induration 1 1 0.15 0 0 0.00 0.4963

Table 191. Systemic and local reactions post booster dose in subjects aged 3 months
 PCV13-TT (N=488) PCV7 (N=493)

P

Events	Cases	%	Events	Cases	%		
Fever	146	137	28.07	149	135	27.38	0.8308
Fatigue / somnolence	1	1	0.20	1	1	0.20	1.0000
Nausea / vomiting	8	8	1.64	4	4	0.81	0.2617
Diarrhea	12	12	2.46	11	11	2.23	0.8362
Cough	4	4	0.82	6	6	1.22	0.7527
Crying	4	4	0.82	3	3	0.61	0.7243
Local pain	1	1	0.20	0	0	0.00	0.4975
Local redness	5	5	1.02	3	3	0.61	0.5040
Local swelling	4	4	0.82	2	2	0.41	0.4496
Local induration	1	1	0.20	0	0	0.00	0.4975

Table 192. Systemic and local reactions post booster dose in subjects aged 7-11 months

PCV13-TT (N=186) PCV7 (N=191)

Events	Cases	%	Events	Cases	%		
Fever	40	38	20.43	41	40	20.94	1.0000
Nausea / vomiting	0	0	0.00	2	2	1.05	0.4988
Diarrhea	3	3	1.61	2	2	1.05	0.6817
Cough	1	1	0.54	2	2	1.05	1.0000
Crying	2	2	1.08	0	0	0.00	0.2427
Allergic reactions	0	0	0.00	1	1	0.52	1.0000
Local pain	1	1	0.54	1	1	0.52	1.0000
Local redness	1	1	0.54	2	2	1.05	1.0000
Local swelling	3	3	1.61	2	2	1.05	0.6817

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2.7.4.2.2.2.2.2. Unsolicited adverse events related to the study vaccine

Unsolicited adverse events related to the study vaccine as determined by the investigators are listed in Table 193 to Table 195.

Five and nine cases of unsolicited adverse events in the PCV13-TT group and PCV7 group were judged to be related to the study vaccine. No significant difference was observed in the intergroup incidences.

Table 193. Unsolicited events related to study vaccine post booster dose in subjects aged 3-11 months

PCV13-TT (N=674) PCV7 (N=684)

Events	Cases	Incidence rate	Events	Cases	Incidence rate		
Unsolicited adverse reactions	5	5	0.74	9	9	1.32	0.4217
Infections and infestations	2	2	0.30	6	6	0.88	0.2877
Nasopharyngitis	0	0	0.00	1	1	0.15	1.0000
Upper respiratory tract infection	2	2	0.30	5	5	0.73	0.4520
Respiratory, thoracic and mediastinal disorders	2	2	0.30	2	2	0.29	1.0000
Rhinorrhea	0	0	0.00	2	2	0.29	0.4997
Nasal obstruction	1	1	0.15	0	0	0.00	0.4963
Expectoration	1	1	0.15	0	0	0.00	0.4963
General disorders and administration site conditions	1	1	0.15	0	0	0.00	0.4963
Fever	1	1	0.15	0	0	0.00	0.4963
Gastrointestinal disorders	0	0	0.00	1	1	0.15	1.0000
Dyspepsia	0	0	0.00	1	1	0.15	1.0000

Table 194. Unsolicited events related to study vaccine post booster dose in subjects aged 3 months

PCV13-TT (N=488) PCV7 (N=493)

Events	Cases	Incidence rate	Events	Cases	Incidence rate		
Unsolicited adverse reactions	4	4	0.82	8	8	1.62	0.3848
Infections and infestations	1	1	0.20	5	5	1.01	0.2174
Nasopharyngitis	0	0	0.00	1	1	0.20	1.0000
Upper respiratory tract infection	1	1	0.20	4	4	0.81	0.3738
Respiratory, thoracic and mediastinal disorders	2	2	0.41	2	2	0.41	1.0000
Rhinorrhea	0	0	0.00	2	2	0.41	0.4995
Nasal obstruction	1	1	0.20	0	0	0.00	0.4975
Sputum	1	1	0.20	0	0	0.00	0.4975
General disorders and administration site conditions	1	1	0.20	0	0	0.00	0.4975
Fever	1	1	0.20	0	0	0.00	0.4975
Gastrointestinal disorders	0	0	0.00	1	1	0.20	1.0000
Dyspepsia	0	0	0.00	1	1	0.20	1.0000

Table 195. Unsolicited events related to study vaccine post booster dose in subjects aged 7-11 months

PCV13-TT (N=186) PCV7 (N=191)

Event Case Incidence Event Case Incidence P

s s rate s s rate

1.000

Unsolicited adverse reactions 1 1 0.54 1 1 0.52

0

1.000

Infections and infestations 1 1 0.54 1 1 0.52

0

Upper respiratory tract 1.000

1 1 0.54 1 1 0.52

infection 0

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2.7.4.2.2.2.2.3. Discussions and conclusions of safety

The incidence of adverse events was respectively 35.46% and 34.50% in the PCV13-TT (3 months) and PCV7 (3 months) group. The incidence of adverse events with grade 3 severity was 1.93% and 1.02%, respectively, with no significant difference observed between groups. No significant difference was observed regarding the incidence of adverse events between the PCV13-TT and

PCV7 group in subjects aged 3 months and 7-11 months. The incidence of vaccine-related adverse events in the PCV13-TT (2 months) group was lower than that in the PCV13-TT (3 months) and PCV7 (3 months) groups.

As with primary immunization, the adverse reactions occurred in the stage of booster immunization were fever and local swelling. No significant difference was observed for the incidence of symptoms between the age groups of 3 months and 7-11 months. The incidence rates of fever and diarrhea in the subjects 2 months of age were significantly lower than that in the PCV13-TT (3 months) and PCV7 (3 months) groups.

The severity of adverse events was mainly grade 1 and 2. Thirteen cases of grade 3 adverse events were observed in the PCV13-TT group and seven cases were observed in the PCV7 group, corresponding to an incidence rate of respectively 1.93% and 1.02%. with no significant difference was observed between groups.

A total of 8 cases (10 events) of SAE were observed after booster immunization, including 3 cases in the PCV13-TT group of the 3-11 months olds, 4 cases in the PCV7 group of the 3-11 months olds and 1 case among the 2 months olds. No significant difference was observed between groups. All SAEs were not related to the study vaccine.

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2.7.4.2.3. General adverse events
2.7.4.2.3.1. Phase I clinical trial
N/A

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2.7.4.2.3.2. Phase III clinical trial
2.7.4.2.3.2.1. Primary series
2.7.4.2.3.2.1.1. Solicited events unrelated to study vaccine

A total of 231 cases (313 events) were determined by the investigators as solicited adverse events unrelated to study vaccine, including 109 cases in the PCV13-TT group and 122 cases in the PCV7 group, and the incidence rate was 9.76% and 10.90% respectively. No significant intergroup difference was observed for the incidence of vaccine-independent solicited adverse events.

Symptoms of solicited adverse events unrelated to study vaccine were mainly fever, cough, diarrhea, and nausea / vomiting.

The severity of vaccine-independent solicited adverse events was concentrated on grade 1 and 2. The severity of 6 cases (0.54%) in the PCV13-TT group and 13 cases (1.16%) in the PCV7 group were determined to be grade 3. No significant intergroup difference was observed in the incidence. Fever, cough, and diarrhea were the most common vaccine-independent solicited adverse events in all age groups.

Incidences of solicited adverse events unrelated to study vaccine are shown in Table 196 to Table 200.

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Table 196. Solicited adverse events unrelated to study vaccine in PCV13-TT and PCV7 groups of subjects aged 3-71 months
PCV13-TT (N=1117) PCV7 (N=1119)
P

Events	Cases	%	Events	Cases	%
Solicited AE unrelated					
150	109	9.76	163	122	10.90
0.4045					
to study					
Grade 1	57	49	4.39	56	51
4.56 0.9186					
Grade 2	87	72	6.45	94	73
6.52 1.0000					
Grade 3	6	6	0.54	13	13
1.16 0.1653					
Systemic reactions					
150	109	9.76	161	120	10.72
0.4856					
Grade 1	57	49	4.39	54	49
4.38 1.0000					
Grade 2	87	72	6.45	94	73
6.52 1.0000					
Grade 3	6	6	0.54	13	13
1.16 0.1653					
Fever	56	54	4.83	67	62
5.54 0.5046					
Grade 1	23	23	2.06	21	20
1.79 0.6482					
Grade 2	29	28	2.51	37	36
3.22 0.3748					
Grade 3	4	4	0.36	9	9
0.80 0.2654					
Fatigue / somnolence					
1	1	0.09	1	1	0.09
1.0000					
Grade 1	0	0	0.00	1	1
0.00 1.0000					
Grade 2	1	1	0.09	0	0
0.00 0.4996					
Grade 3	0	0	0.00	0	0
0.00 /					
Nausea / vomiting					
11	10	0.90	11	11	0.98
1.0000					
Grade 1	5	5	0.45	10	10
0.89 0.3001					
Grade 2	6	5	0.45	1	1
0.09 0.1242					
Grade 3	0	0	0.00	0	0
0.00 /					
Diarrhea					
23	22	1.97	24	23	2.06
1.0000					
Grade 1	7	6	0.54	7	7
0.63 1.0000					
Grade 2	15	15	1.34	13	12
1.07 0.5690					
Grade 3	1	1	0.09	4	4
0.36 0.3744					
Cough					
53	52	4.66	54	53	4.74
1.0000					
Grade 1	20	20	1.79	15	15
1.34 0.4010					
Grade 2	32	31	2.78	39	38
3.40 0.4634					
Grade 3	1	1	0.09	0	0
0.00 0.4996					
Abnormal crying					
1	1	0.09	1	1	0.09
1.0000					
Grade 1	0	0	0.00	0	0
0.00 /					
Grade 2	1	1	0.09	1	1
0.09 1.0000					
Grade 3	0	0	0.00	0	0
0.00 /					
Allergic reactions					
5	5	0.45	3	3	0.27
0.5067					
Grade 1	2	2	0.18	0	0
0.00 0.2494					
Grade 2	3	3	0.27	3	3
0.27 1.0000					

Grade 3 0 0 0.00 0 0 0.00 /
 Vaccination reactions /
 0 0 0.00 2 2 0.18 0.4998
 site redness
 Grade 1 0 0 0.00 2 2 0.18 0.4998
 Grade 2 0 0 0.00 0 0 0.00 /
 Grade 3 0 0 0.00 0 0 0.00 /

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Table 197. Solicited adverse events unrelated to study vaccine in subjects aged 3 months

PCV13-TT (N=519) PCV7 (N=519)

P

Events	Cases	%	Events	Cases	%
Fever	27	5.01	34	30	5.78
Fatigue / somnolence	1	0.19	0	0	0.00
Nausea / vomiting	3	0.39	5	5	0.96
Diarrhea	11	2.12	17	16	3.08
Cough	27	5.01	23	22	4.24
Abnormal crying	1	0.19	0	0	0.00
Allergic reactions	2	0.39	2	2	0.39

Table 198. Solicited adverse events unrelated to study vaccine in subjects aged 7-11 months

PCV13-TT (N=198) PCV7 (N=200)

P

Events	Cases	%	Events	Cases	%
Fever	14	7.07	15	14	7.00
Nausea / vomiting	3	1.52	4	4	2.00
Diarrhea	7	3.03	5	5	2.50
Cough	10	5.05	17	17	8.50
Abnormal crying	0	0.00	1	1	0.50
Allergic reactions	1	0.51	1	1	0.50

Vaccination site redness 0 0 0.00 1 1 0.50 1.0000

Table 199. Solicited adverse events unrelated to study vaccine in subjects aged 12-23 months

PCV13-TT (N=200) PCV7 (N=200)

P

Events	Cases	%	Events	Cases	%
Fever	12	5.50	14	14	7.00
Fatigue / somnolence	0	0.00	1	1	0.50
Nausea / vomiting	3	1.50	2	2	1.00
Diarrhea	3	1.50	2	2	1.00
Cough	13	6.50	8	8	4.00
Vaccination site redness	0	0.00	1	1	0.50

Table 200. Solicited adverse events unrelated to study vaccine in subjects aged 24-71 months

PCV13-TT (N=200) PCV7 (N=200)

P

Events	Cases	%	Events	Cases	%
Fever	3	1.50	4	4	2.00
Nausea / vomiting	2	1.00	0	0	0.00
Diarrhea	2	1.00	0	0	0.00
Cough	3	1.50	6	6	3.00
Allergic reactions	2	1.00	0	0	0.00

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2.7.4.2.3.2.1.2. Unsolicited events unrelated to study vaccine

A total of 717 cases (1155 events) of unsolicited adverse events unrelated to study vaccine were collected during the observation period, including 325 cases in the PCV13-TT group and 365 cases in the PCV7 group, with incidence rates being 31.51% and 32.62%, respectively. No significant intergroup difference was observed for the incidence of vaccine-independent unsolicited adverse events.

Unsolicited adverse events unrelated to study vaccine were primarily infections and infestations, followed by gastrointestinal disorders, respiratory, thoracic and mediastinal disorders, general disorders and administration site conditions. Fever was observed with the highest proportion within the class of general disorders and administration site conditions, and was of relatively higher incidence rates among subjects aged 3 months, 7-11 months, and 12-23 months that were within 2 years old. No significant difference was observed for unsolicited adverse events between the PCV13-TT group and the PCV7 group.

Incidences of vaccine-independent unsolicited adverse events in the PCV13-TT and PCV7 group of each age group are shown in Table 201 to Table 209

Table 201 Unsolicited events unrelated to study vaccine in subjects aged 3-71 months

PCV13-TT (N=1117) PCV7 (N=1119)

Incidence Incidence P

Events	Cases	Events	Cases
rate	rate		

Unsolicited adverse reactions	566	352	31.51	589	365	32.62	0.5868
Hepatobiliary disorders	0	0	0.00	1	1	0.09	1.0000
Abnormal liver function	0	0	0.00	1	1	0.09	1.0000
Infections and infestations	274	213	19.07	293	223	19.93	0.6311
Upper respiratory tract infection	163	135	12.09	182	150	13.40	0.3747
Nasopharyngitis	28	26	2.33	24	20	1.79	0.3764
Rhinitis	0	0	0.00	1	1	0.09	1.0000
Nasosinusitis	1	1	0.09	0	0	0.00	0.4996
Tonsillitis	3	3	0.27	4	4	0.36	1.0000
Viral encephalitis	1	1	0.09	1	1	0.09	1.0000
Viral rash	0	0	0.00	2	2	0.18	0.4998
Viral gastroenteritis	0	0	0.00	3	3	0.27	0.2497
Angular cheilitis	1	1	0.09	2	2	0.18	1.0000
Rubella	0	0	0.00	1	1	0.09	1.0000

Infectious enteritis 1 1 0.09 0 0 0.00 0.4996
Infectious pneumonia 16 16 1.43 10 10 0.89 0.2447
Infectious diarrhea 0 0 0.00 1 1 0.09 1.0000
Anal abscess 0 0 0.00 1 1 0.09 1.0000
Laryngitis 2 2 0.18 1 1 0.09 0.6245
Respiratory tract infection 1 1 0.09 0 0 0.00 0.4996
Conjunctivitis 0 0 0.00 1 1 0.09 1.0000
Oral herpes 2 2 0.18 3 3 0.27 1.0000
Measles 1 1 0.09 0 0 0.00 0.4996
Urethritis 0 0 0.00 1 1 0.09 1.0000
Sepsis 1 1 0.09 2 2 0.18 1.0000
Abscess 1 1 0.09 0 0 0.00 0.4996
Bacterial skin infections 1 1 0.09 0 0 0.00 0.4996
Tracheitis 2 2 0.18 2 2 0.18 1.0000
Upper respiratory tract viral infection 0 0 0.00 2 2 0.18 0.4998

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Table 202 Unsolicited events unrelated to study vaccine in subjects aged 3-71 months
PCV13-TT (N=1117) PCV7 (N=1119)

Incidence Incidence P
Events Cases Events Cases
rate rate
Upper respiratory and digestive
1 1 0.09 0 0 0.00 0.4996
tract infections
Hand-foot-mouth disease 9 9 0.81 12 12 1.07 0.6621
Gastroenteritis 1 1 0.09 2 2 0.18 1.0000
Bacterial tonsillitis 1 1 0.09 0 0 0.00 0.4996
Laryngeal tonsillitis 2 2 0.18 1 1 0.09 0.6245
Laryngitis 1 1 0.09 2 2 0.18 1.0000
Eye infections 1 1 0.09 0 0 0.00 0.4996
Roseola infantum 3 3 0.27 8 8 0.71 0.2254
Bronchitis 16 15 1.34 11 11 0.98 0.4391
Mycoplasma pneumonia 1 1 0.09 0 0 0.00 0.4996
Otitis media 2 2 0.18 2 1 0.09 0.6245
Herpes virus infection 1 1 0.09 2 2 0.18 1.0000
Herpangina 10 10 0.90 8 8 0.71 0.6460
Herpes laryngitis 0 0 0.00 1 1 0.09 1.0000
Investigations 1 1 0.09 7 6 0.54 0.1244
Abnormal electromyographic
0 0 0.00 1 1 0.09 1.0000
findings
Increased biological markers of
1 1 0.09 6 6 0.54 0.1244
myocardial necrosis
Nervous system disorders 2 2 0.18 2 2 0.18 1.0000
Febrile convulsion 1 1 0.09 1 1 0.09 1.0000
Psychomotor retardation 0 0 0.00 1 1 0.09 1.0000
Epilepsy 1 1 0.09 0 0 0.00 0.4996
Injury, poisoning and procedural
1 1 0.09 0 0 0.00 0.4996
complications
Animal bite 1 1 0.09 0 0 0.00 0.4996
Respiratory, thoracic and
94 80 7.16 86 72 6.43 0.5027
mediastinal disorders
Rhinorrhea 54 51 4.57 50 45 4.02 0.5334
Nasal obstruction 2 2 0.18 0 0 0.00 0.2494
Epistaxis 1 1 0.09 1 1 0.09 1.0000
Tonsillar hypertrophy 1 1 0.09 0 0 0.00 0.4996
Dysphonia 1 1 0.09 0 0 0.00 0.4996
Pneumonia 2 2 0.18 7 6 0.54 0.2882
Cough 31 30 2.69 23 21 1.88 0.2060
Sneezing 1 1 0.09 2 2 0.18 1.0000
Increased production of sputum 1 1 0.09 1 1 0.09 1.0000
Pharyngeal erythema 0 0 0.00 1 1 0.09 1.0000
Pharyngeal edema 0 0 0.00 1 1 0.09 1.0000
Immune system disorders 1 1 0.09 0 0 0.00 0.4996
Drug-induced hypersensitivity
1 1 0.09 0 0 0.00 0.4996
reactions
Skin and subcutaneous tissue
7 6 0.54 10 10 0.89 0.4529
disorders
Maculopapular rash 0 0 0.00 1 1 0.09 1.0000
Allergic dermatitis 0 0 0.00 1 1 0.09 1.0000
Erythema multiforme 1 1 0.09 0 0 0.00 0.4996

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Table 203 Unsolicited events unrelated to study vaccine in subjects aged 3-71 months
PCV13-TT (N=1117) PCV7 (N=1119)

Incidence Incidence P
Events Cases Events Cases
rate rate
Rash 0 0 0.00 1 1 0.09 1.0000
Eczema 0 0 0.00 5 5 0.45 0.0622

Blister rash 2 1 0.09 0 0 0.00 0.4996
 Urticaria 4 4 0.36 2 2 0.18 0.4522
 General disorders and administration site
 72 65 5.82 85 77 6.88 0.3402
 conditions
 Fever 71 64 5.73 84 76 6.79 0.3370
 Crying 1 1 0.09 0 0 0.00 0.4996
 Inflammation 0 0 0.00 1 1 0.09 1.0000
 Reproductive system and breast disorders 0 0 0.00 1 1 0.09 1.0000
 Vulvar erythema 0 0 0.00 1 1 0.09 1.0000
 Gastrointestinal disorders 109 97 8.68 99 86 7.69 0.3971
 Aphthous ulcer 3 3 0.27 1 1 0.09 0.3742
 Constipation 1 1 0.09 0 0 0.00 0.4996
 Enteritis 4 4 0.36 2 2 0.18 0.4522
 Intestinal obstruction 1 1 0.09 0 0 0.00 0.4996
 Inguinal hernia 1 1 0.09 0 0 0.00 0.4996
 Abdominal pain 2 2 0.18 1 1 0.09 0.6245
 Diarrhea 69 66 5.91 70 65 5.81 0.9285
 Abdominal distension 3 3 0.27 7 6 0.54 0.5069
 Functional gastrointestinal disorders 1 1 0.09 1 1 0.09 1.0000
 Oral ulcer 3 3 0.27 5 5 0.45 0.7261
 Gastritis 1 1 0.09 0 0 0.00 0.4996
 Dyspepsia 19 18 1.61 11 9 0.80 0.0851
 Infant vomiting 1 1 0.09 1 1 0.09 1.0000
 Blood and lymphatic system disorders 4 4 0.36 4 3 0.27 0.7260
 Granulopenia 2 2 0.18 0 0 0.00 0.2494
 Lymphadenitis 1 1 0.09 0 0 0.00 0.4996
 Anemia 1 1 0.09 3 3 0.27 0.6247
 Neutropenia 0 0 0.00 1 1 0.09 1.0000
 Eye disorders 1 1 0.09 1 1 0.09 1.0000
 Eye congestion 1 1 0.09 0 0 0.00 0.4996
 Blepharitis 0 0 0.00 1 1 0.09 1.0000

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Table 204 Unsolicited events unrelated to study vaccine in subjects aged 3 months

PCV13-TT (N=519) PCV7 (N=519)
 Incidence Incidence P
 Events Cases Events Cases
 rate rate
 Unsolicited adverse reactions 295 181 34.87 298 191 36.80 0.5602
 Infections and infestations 140 109 21.00 143 113 21.77 0.8204
 Nasopharyngitis 14 14 2.70 9 7 1.35 0.1847
 Rhinitis 0 0 0.00 1 1 0.19 1.0000
 Tonsillitis 0 0 0.00 1 1 0.19 1.0000
 Viral rash 0 0 0.00 2 2 0.39 0.4995
 Viral gastroenteritis 0 0 0.00 1 1 0.19 1.0000
 Angular cheilitis 0 0 0.00 1 1 0.19 1.0000
 Rubella 0 0 0.00 1 1 0.19 1.0000
 Infectious enteritis 1 1 0.19 0 0 0.00 1.0000
 Infectious pneumonia 8 8 1.54 9 9 1.73 1.0000
 Anal abscess 0 0 0.00 1 1 0.19 1.0000
 Laryngitis 0 0 0.00 1 1 0.19 1.0000
 Respiratory tract infection 1 1 0.19 0 0 0.00 1.0000
 Measles 1 1 0.19 0 0 0.00 1.0000
 Urethritis 0 0 0.00 1 1 0.19 1.0000
 Sepsis 1 1 0.19 0 0 0.00 1.0000
 Abscess 1 1 0.19 0 0 0.00 1.0000
 Tracheitis 2 2 0.39 2 2 0.39 1.0000
 Upper respiratory tract viral
 0 0 0.00 1 1 0.19 1.0000
 infection
 Upper respiratory tract infection 91 74 14.26 92 78 15.03 0.7923
 Hand-foot-mouth disease 1 1 0.19 1 1 0.19 1.0000
 Laryngitis 1 1 0.19 1 1 0.19 1.0000
 Eye infections 1 1 0.19 0 0 0.00 1.0000
 Roseola infantum 3 3 0.58 7 7 1.35 0.3414
 Bronchitis 10 9 1.73 6 6 1.16 0.6046
 Otitis media 1 1 0.19 2 1 0.19 1.0000
 Herpes virus infection 0 0 0.00 1 1 0.19 1.0000
 Herpangina 3 3 0.58 1 1 0.19 0.6243
 Herpes laryngitis 0 0 0.00 1 1 0.19 1.0000
 Investigations 1 1 0.19 2 2 0.39 1.0000
 Increased biological markers of
 1 1 0.19 2 2 0.39 1.0000
 myocardial necrosis
 Nervous system disorders 2 2 0.39 1 1 0.19 1.0000
 Febrile convulsion 1 1 0.19 0 0 0.00 1.0000
 Psychomotor retardation 0 0 0.00 1 1 0.19 1.0000
 Epilepsy 1 1 0.19 0 0 0.00 1.0000
 Respiratory, thoracic and
 45 38 7.32 40 33 6.36 0.6231
 mediastinal disorders
 Rhinorrhea 24 23 4.43 22 20 3.85 0.7558
 Dysphonia 1 1 0.19 0 0 0.00 1.0000
 Pneumonia 1 1 0.19 6 5 0.96 0.2174
 Cough 17 16 3.08 9 8 1.54 0.1468

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Table 205 Unsolicited events unrelated to study vaccine in subjects aged 3 months
PCV13-TT (N=519) PCV7 (N=519)

Incidence	Incidence	P	Events	Cases	Events	Cases
rate	rate					
Sneezing	1	1	0.19	1	1	0.19
Increased production of sputum	1	1	0.19	0	0	0.00
Pharyngeal erythema	0	0	0.00	1	1	0.19
Pharyngeal edema	0	0	0.00	1	1	0.19
Skin and subcutaneous tissue disorders	2	2	0.39	5	5	0.96
Maculopapular rash	0	0	0.00	1	1	0.19
Eczema	0	0	0.00	4	4	0.77
Urticaria	2	2	0.39	0	0	0.00
General disorders and administration site conditions	33	31	5.97	37	35	6.74
Fever	32	30	5.78	36	34	6.55
Crying	1	1	0.19	0	0	0.00
Inflammation	0	0	0.00	1	1	0.19
Reproductive system and breast disorders	0	0	0.00	1	1	0.19
Vulvar erythema	0	0	0.00	1	1	0.19
Gastrointestinal disorders	68	60	11.56	66	57	10.98
Aphthous ulcer	2	2	0.39	0	0	0.00
Enteritis	4	4	0.77	2	2	0.39
Abdominal pain	1	1	0.19	1	1	0.19
Diarrhea	43	41	7.90	48	45	8.67
Abdominal distension	3	3	0.58	5	4	0.77
Functional gastrointestinal disorders	1	1	0.19	0	0	0.00
Oral ulcer	1	1	0.19	1	1	0.19
Gastritis	1	1	0.19	0	0	0.00
Dyspepsia	11	10	1.93	8	6	1.16
Infant vomiting	1	1	0.19	1	1	0.19
Blood and lymphatic system disorders	3	3	0.58	3	2	0.39
Granulopenia	2	2	0.39	0	0	0.00
Anemia	1	1	0.19	2	2	0.39
Neutropenia	0	0	0.00	1	1	0.19
Eye disorders	1	1	0.19	0	0	0.00
Eye congestion	1	1	0.19	0	0	0.00

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Table 206 Unsolicited events unrelated to study vaccine in subjects aged 7-11 months
PCV13-TT (N=198) PCV7 (N=200)

Incidence	Incidence	P	Events	Cases	Events	Cases
rate	rate					
Unsolicited adverse reactions	121	75	37.88	158	82	41.00
Infections and infestations	56	44	22.22	75	49	24.50
Nasopharyngitis	5	5	2.53	7	6	3.00
Tonsillitis	2	2	1.01	1	1	0.50
Viral gastroenteritis	0	0	0.00	1	1	0.50
Angular cheilitis	1	1	0.51	1	1	0.50
Infectious pneumonia	3	3	1.52	0	0	0.00
Infectious diarrhea	0	0	0.00	1	1	0.50
Laryngitis	2	2	1.01	0	0	0.00
Oral herpes	0	0	0.00	2	2	1.00
Sepsis	0	0	0.00	2	2	1.00
Upper respiratory tract infection	32	26	13.13	47	36	18.00
Hand-foot-mouth disease	2	2	1.01	2	2	1.00
Gastroenteritis	0	0	0.00	1	1	0.50
Bacterial tonsillitis	1	1	0.51	0	0	0.00
Laryngeal tonsillitis	1	1	0.51	1	1	0.50
Roseola infantum	0	0	0.00	1	1	0.50
Bronchitis	2	2	1.01	2	2	1.00
Herpes virus infection	0	0	0.00	1	1	0.50
Herpangina	5	5	2.53	5	5	2.50
Investigations	0	0	0.00	1	1	0.50
Increased biological markers of myocardial necrosis	0	0	0.00	1	1	0.50
Respiratory, thoracic and mediastinal disorders	21	18	9.09	25	20	10.00
Rhinorrhea	15	14	7.07	18	15	7.50
Nasal obstruction	2	2	1.01	0	0	0.00
Cough	4	4	2.02	5	5	2.50
Sneezing	0	0	0.00	1	1	0.50
Increased production of sputum	0	0	0.00	1	1	0.50
Skin and subcutaneous tissue disorders	4	3	1.52	3	3	1.50
Allergic dermatitis	0	0	0.00	1	1	0.50
Erythema multiforme	1	1	0.51	0	0	0.00
Blister rash	2	1	0.51	0	0	0.00

Urticaria 1 1 0.51 2 2 1.00 1.0000
 General disorders and administration
 17 15 7.58 30 24 12.00 0.1770
 site conditions
 Fever 17 15 7.58 30 24 12.00 0.1770
 Gastrointestinal disorders 22 20 10.10 23 19 9.50 0.8675
 Aphthous ulcer 1 1 0.51 1 1 0.50 1.0000
 Diarrhea 17 16 8.08 16 14 7.00 0.7083
 Abdominal distension 0 0 0.00 1 1 0.50 1.0000
 Oral ulcer 0 0 0.00 2 2 1.00 0.4988
 Dyspepsia 4 4 2.02 3 3 1.50 0.7231
 Blood and lymphatic system disorders 1 1 0.51 1 1 0.50 1.0000
 Lymphadenitis 1 1 0.51 0 0 0.00 0.4975
 Anemia 0 0 0.00 1 1 0.50 1.0000

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Table 207 Unsolicited events unrelated to study vaccine in subjects aged 12-23 months
 PCV13-TT (N=200) PCV7 (N=200)

Incidence	Incidence P	Events	Cases	Events	Cases	rate	rate
Unsolicited adverse reactions	116 67 33.50 89 52 26.00 0.1255						
Hepatobiliary disorders	0 0 0.00 1 1 0.50 1.0000						
Abnormal liver function	0 0 0.00 1 1 0.50 1.0000						
Infections and infestations	56 40 20.00 52 38 19.00 0.8996						
Nasopharyngitis	4 3 1.50 5 4 2.00 1.0000						
Tonsillitis	1 1 0.50 2 2 1.00 1.0000						
Viral encephalitis	1 1 0.50 1 1 0.50 1.0000						
Infectious pneumonia	5 5 2.50 1 1 0.50 0.2152						
Conjunctivitis	0 0 0.00 1 1 0.50 1.0000						
Oral herpes	2 2 1.00 1 1 0.50 1.0000						
Bacterial skin infections	1 1 0.50 0 0 0.00 1.0000						
Upper respiratory tract viral infection	0 0 0.00 1 1 0.50 1.0000						
Upper respiratory tract infection	27 22 11.00 26 19 9.50 0.7420						
Upper respiratory and digestive tract infections	1 1 0.50 0 0 0.00 1.0000						
Hand-foot-mouth disease	6 6 3.00 8 8 4.00 0.7868						
Gastroenteritis	0 0 0.00 1 1 0.50 1.0000						
Laryngeal tonsillitis	1 1 0.50 0 0 0.00 1.0000						
Laryngitis	0 0 0.00 1 1 0.50 1.0000						
Bronchitis	3 3 1.50 2 2 1.00 1.0000						
Mycoplasma pneumonia	1 1 0.50 0 0 0.00 1.0000						
Herpes virus infection	1 1 0.50 0 0 0.00 1.0000						
Herpangina	2 2 1.00 2 2 1.00 1.0000						
Investigations	0 0 0.00 4 3 1.50 0.2481						
Abnormal electromyographic findings	0 0 0.00 1 1 0.50 1.0000						
Increased biological markers of myocardial necrosis	0 0 0.00 3 3 1.50 0.2481						
Nervous system disorders	0 0 0.00 1 1 0.50 1.0000						
Febrile convulsion	0 0 0.00 1 1 0.50 1.0000						
Injury, poisoning and procedural complications	1 1 0.50 0 0 0.00 1.0000						
Animal bite	1 1 0.50 0 0 0.00 1.0000						
Respiratory, thoracic and mediastinal disorders	21 18 9.00 10 10 5.00 0.1692						
Rhinorrhea	14 13 6.50 7 7 3.50 0.2509						
Epistaxis	0 0 0.00 1 1 0.50 1.0000						
Tonsillar hypertrophy	1 1 0.50 0 0 0.00 1.0000						
Pneumonia	1 1 0.50 1 1 0.50 1.0000						
Cough	5 5 2.50 1 1 0.50 0.2152						
Immune system disorders	1 1 0.50 0 0 0.00 1.0000						
Drug-induced hypersensitivity reactions	1 1 0.50 0 0 0.00 1.0000						
Skin and subcutaneous tissue disorders	1 1 0.50 0 0 0.00 1.0000						
Urticaria	1 1 0.50 0 0 0.00 1.0000						
General disorders and administration site conditions	19 16 8.00 13 13 6.50 0.7004						
Fever	19 16 8.00 13 13 6.50 0.7004						

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Table 208 Unsolicited events unrelated to study vaccine in subjects aged 12-23 months
 PCV13-TT (N=200) PCV7 (N=200)

Incidence	Incidence P	Events	Cases	Events	Cases	rate	rate
Gastrointestinal disorders	17 15 7.50 7 7 3.50 0.1230						
Constipation	1 1 0.50 0 0 0.00 1.0000						
Intestinal obstruction	1 1 0.50 0 0 0.00 1.0000						
Inguinal hernia	1 1 0.50 0 0 0.00 1.0000						
Diarrhea	9 9 4.50 4 4 2.00 0.2589						
Abdominal distension	0 0 0.00 1 1 0.50 1.0000						
Functional gastrointestinal disorders	0 0 0.00 1 1 0.50 1.0000						
Oral ulcer	1 1 0.50 1 1 0.50 1.0000						
Dyspepsia	4 4 2.00 0 0 0.00 0.1231						
Eye disorders	0 0 0.00 1 1 0.50 1.0000						
Blepharitis	0 0 0.00 1 1 0.50 1.0000						

Table 209 Unsolicited events unrelated to study vaccine in subjects aged 24-71 months

PCV13-TT (N=200) PCV7 (N=200)						
Incidence Incidence P						
Events Cases Events Cases						
rate rate						
Unsolicited adverse reactions 34 29 14.50 44 40 20.00 0.1854						
Infections and infestations 22 20 10.00 23 23 11.50 0.7472						
Nasopharyngitis 5 4 2.00 3 3 1.50 1.0000						
Nasosinusitis 1 1 0.50 0 0 0.00 1.0000						
Viral gastroenteritis 0 0 0.00 1 1 0.50 1.0000						
Upper respiratory tract infection 13 13 6.50 17 17 8.50 0.5697						
Hand-foot-mouth disease 0 0 0.00 1 1 0.50 1.0000						
Gastroenteritis 1 1 0.50 0 0 0.00 1.0000						
Bronchitis 1 1 0.50 1 1 0.50 1.0000						
Otitis media 1 1 0.50 0 0 0.00 1.0000						
Respiratory, thoracic and						
7 6 3.00 11 9 4.50 0.6002						
mediastinal disorders						
Rhinoirrhoea 1 1 0.50 3 3 1.50 0.6231						
Epistaxis 1 1 0.50 0 0 0.00 1.0000						
Cough 5 5 2.50 8 7 3.50 0.7710						
Skin and subcutaneous tissue						
0 0 0.00 2 2 1.00 0.4987						
disorders						
Rash 0 0 0.00 1 1 0.50 1.0000						
Eczema 0 0 0.00 1 1 0.50 1.0000						
General disorders and						
3 3 1.50 5 5 2.50 0.7238						
administration site conditions						
Fever 3 3 1.50 5 5 2.50 0.7238						
Gastrointestinal disorders 2 2 1.00 3 3 1.50 1.0000						
Abdominal pain 1 1 0.50 0 0 0.00 1.0000						
Diarrhea 0 0 0.00 2 2 1.00 0.4987						
Oral ulcer 1 1 0.50 1 1 0.50 1.0000						

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2.7.4.2.3.2.2. Booster dose
2.7.4.2.3.2.2.1. Solicited events unrelated to study vaccine
Solicited adverse events unrelated to study vaccine were all systemic reactions. Twelve solicited cases (occurred within 7 days post vaccination) were observed in the PCV13-TT group of the 3-11 months olds and 18 cases in the PCV7 group, which mainly occurred in subjects 3 months of age, and there was only 1 case observed in the PCV13-TT group of 7-11 months olds. No significant difference was observed for vaccine-independent solicited adverse events between groups (Table 210 to Table 211).

Table 210 Solicited events unrelated to study vaccine post booster dose in subjects aged 3-11 months
PCV13-TT (N=674) PCV7 (N=684)

P												
Events Cases % Events Cases %												
Solicited adverse events												
unrelated to study												
18	12	1.78	28	18	2.63	0.3566						
vaccine												
Fever												
6	6	0.89	11	11	1.61	0.3295						
Nausea / vomiting												
1	1	0.15	3	3	0.44	0.6245						
Diarrhea												
0	0	0.00	3	3	0.44	0.2495						
Cough												
11	11	1.63	10	10	1.46	0.8294						
Allergic reactions												
0	0	0.00	1	1	0.15	1.0000						

Table 211 Solicited events unrelated to study vaccine post booster dose in subjects aged 3 months

PCV13-TT (N=488) PCV7 (N=493)									
P									
Events		Cases	%	Events		Cases	%		
Solicited adverse events									
unrelated to study									
17	11	2.25	28	18	3.65	0.2579			
vaccine									
Fever	5	5	1.02	11	11	2.23	0.2065		
Nausea / vomiting	1	1	0.20	3	3	0.61	0.6243		
Diarrhea	0	0	0.00	3	3	0.61	0.2493		
Cough	11	11	2.25	10	10	2.03	0.8292		
Allergic reactions	0	0	0.00	1	1	0.20	1.0000		

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2.7.4.2.3.2.2.2. Unsolicited events unrelated to study vaccine
The incidence rate of unsolicited adverse events was respectively 9.5% and 9.21% in the PCV13-TT and PCV7 group during the observation period. No intergroup difference was observed for the incidence of unsolicited adverse events in each age group, which were mainly infections and infestations (Table 212 to Table 214).

Table 212 Unsolicited events unrelated to study vaccine post booster dose in subjects aged 3-11 months
PCV13-TT (N=674) PCV7 (N=684)

P									
Events Cases Incidence rate Events Cases Incidence rate									
Unsolicited adverse reactions									
68	64	9.50	77	63	9.21	0.9258			
Infections and infestations									
38	36	5.34	34	32	4.68	0.6196			
Nasopharyngitis									
1	1	0.15	0	0	0.00	0.4963			

Tonsillitis 1 1 0.15 0 0 0.00 0.4963
 Infectious pneumonia 1 1 0.15 1 1 0.15 1.0000
 Rotavirus gastroenteritis 1 1 0.15 0 0 0.00 0.4963
 Upper respiratory tract viral infection 1 1 0.15 0 0 0.00 0.4963
 Upper respiratory tract infection 28 27 4.01 32 30 4.39 0.7873
 Hand-foot-mouth disease 1 1 0.15 0 0 0.00 0.4963
 Gastroenteritis 1 1 0.15 0 0 0.00 0.4963
 Mycoplasma pneumonia 2 2 0.30 0 0 0.00 0.2461
 Herpangina 1 1 0.15 1 1 0.15 1.0000
 Injury, poisoning and procedural complications 0 0 0.00 3 2 0.29 0.4997
 Skull base fractures 0 0 0.00 1 1 0.15 1.0000
 Subcutaneous hematoma 0 0 0.00 1 1 0.15 1.0000
 Injury 0 0 0.00 1 1 0.15 1.0000
 Respiratory, thoracic and mediastinal disorders 19 19 2.82 16 16 2.34 0.6108
 Rhinorrhea 13 13 1.93 11 11 1.61 0.6854
 Cough 6 6 0.89 5 5 0.73 0.7720
 Skin and subcutaneous tissue disorders 0 0 0.00 1 1 0.15 1.0000
 Erythema 0 0 0.00 1 1 0.15 1.0000
 General disorders and administration site conditions 5 5 0.74 8 7 1.02 0.7735
 Fever 5 5 0.74 8 7 1.02 0.7735
 Gastrointestinal disorders 6 6 0.89 15 14 2.05 0.1128
 Enteritis 1 1 0.15 0 0 0.00 0.4963
 Diarrhea 3 3 0.45 9 9 1.32 0.1443
 Abdominal distension 0 0 0.00 1 1 0.15 1.0000
 Oral ulcer 0 0 0.00 1 1 0.15 1.0000
 Vomiting 0 0 0.00 1 1 0.15 1.0000
 Dyspepsia 2 2 0.30 3 3 0.44 1.0000

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Table 213 Unsolicited events unrelated to study vaccine post booster dose in subjects aged 3 months

PCV13-TT (N=488) PCV7 (N=493)

P

Events	Cases	Incidence rate	Events	Cases	Incidence rate
Unsolicited adverse reactions	53	50	10.25	55	43
Infections and infestations	27	26	5.33	22	21
Nasopharyngitis	1	1	0.20	0	0
Infectious pneumonia	1	1	0.20	1	1
Rotavirus gastroenteritis	1	1	0.20	0	0
Upper respiratory tract infection	21	20	4.10	21	20
Gastroenteritis	1	1	0.20	0	0
Mycoplasma pneumonia	1	1	0.20	0	0
Herpangina	1	1	0.20	0	0
Injury, poisoning and procedural complications	0	0	0.00	3	2
Skull base fractures	0	0	0.00	1	1
Subcutaneous hematoma	0	0	0.00	1	1
Injury	0	0	0.00	1	1
Respiratory, thoracic and mediastinal disorders	16	16	3.28	12	12
Rhinorrhea	12	12	2.46	8	8
Cough	4	4	0.82	4	4
Skin and subcutaneous tissue disorders	0	0	0.00	1	1
Erythema	0	0	0.00	1	1
General disorders and administration site conditions	4	4	0.82	6	5
Fever	4	4	0.82	6	5
Gastrointestinal disorders	6	6	1.23	11	10
Enteritis	1	1	0.20	0	0
Diarrhea	3	3	0.61	6	6
Oral ulcer	0	0	0.00	1	1
Vomiting	0	0	0.00	1	1
Dyspepsia	2	2	0.41	3	3

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Table 214 Unsolicited events unrelated to study vaccine post booster dose in subjects aged 7-11 months

PCV13-TT (N=186) PCV7 (N=191)

Incidence Incidence P

Events Cases Events Cases

rate rate

Unsolicited adverse reactions	15	14	7.53	22	20	10.47
Infections and infestations	11	10	5.38	12	11	5.76
Tonsillitis	1	1	0.54	0	0	0.4934
Upper respiratory tract viral infection	1	1	0.54	0	0	0.4934
Upper respiratory tract infection	7	7	3.76	11	10	5.24
Hand-foot-mouth disease	1	1	0.54	0	0	0.4934
Mycoplasma pneumonia	1	1	0.54	0	0	0.4934
Herpangina	0	0	0.00	1	1	0.52
Respiratory, thoracic and mediastinal disorders	3	3	1.61	4	4	2.09
Rhinorrhea	1	1	0.54	3	3	1.57
Cough	2	2	1.08	1	1	0.52
General disorders and administration site conditions	1	1	0.54	2	2	1.05
Fever	1	1	0.54	2	2	1.05
Gastrointestinal disorders	0	0	0.00	4	4	2.09

Diarrhea 0 0 0.00 3 3 1.57 0.2481
Abdominal distension 0 0 0.00 1 1 0.52 1.0000

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2.7.4.2.4. Death
N/A

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2.7.4.2.5. Severe adverse events
2.7.4.2.5.1. Phase I clinical trial
Fourteen subjects experienced grade 3 adverse events, including 1 case of local redness, 1 case of induration, 2 cases of swelling, 2 cases of redness with swelling after one dose of vaccine in adults ≥ 18 years old; 1 case of fever and 1 case of fever with local reaction in children group aged 2-5 years; 1 case of local redness (2 episodes), 1 case of local induration, 1 case of somnolence, 1 case of herpangina with fever (herpangina was not related to vaccination, while fever was determined to be probably related to vaccination) in subjects of 2 months old; 1 case of local reaction and 1 case of upper respiratory tract infection in the 3 months olds. Except for 3 cases receiving outpatient treatment, most of the patients recovered spontaneously. No serious adverse events were observed in this trial.
Adverse events with grade 3 severity were summarized in Table 215.

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Table 215 Adverse events with severity of grade 3

Days since	No. of Duration	Gender	Age	Dose	Type	Term of adverse events last	Severity	Treatment	Relevance
subjects (days)									
vaccination									
4	Female	27.4 yrs	1	Local reactions	Local redness	1 3	Grade 3	Untreated	Possibly related
5	Male	19.2 yrs	1	Local reactions	Local induration	1 2	Grade 3	Untreated	Possibly related
14	Female	29.3 yrs	1	Local reactions	Local redness	0 4	Grade 3	Untreated	Possibly related
14	Female	29.3 yrs	1	Local reactions	Local swelling	1 3	Grade 3	Untreated	Possibly related
20	Female	24 yrs	1	Local reactions	Local swelling	0 5	Grade 3	Untreated	Possibly related
24	Female	42.7 yrs	1	Local reactions	Local redness	5 9	Grade 3	Untreated	Possibly related
24	Female	42.7 yrs	1	Local reactions	Local swelling	5 13	Grade 3	Untreated	Possibly related
29	Male	25.2 yrs	1	Local reactions	Local swelling	0 8	Grade 3	Untreated	Possibly related
33	Female	4.1 yrs	1	Systemic reactions	Fever	0 6	Grade 3	Untreated	Possibly related
33	Female	4.1 yrs	1	Local reactions	Local pain	1 1	Grade 3	Untreated	Possibly related
33	Female	4.1 yrs	1	Local reactions	Local redness	1 1	Grade 3	Untreated	Possibly related
42	Male	5.8 yrs	1	Systemic reactions	Fever	3 5	Grade 3	Clinic visit	Possibly related
66	Female	3.7 mos	1	Local reactions	Local swelling	0 6	Grade 3	Untreated	Possibly related
infections and									
86	Male	3 mos	3	Upper respiratory tract infection		12 4	Grade 3	Clinic visit	Possibly unrelated
infestations									
91	Female	2.5 mos	3	Systemic reactions	Fever	5 6	Grade 3	Clinic visit	Possibly related
infections and									
91	Female	2.5 mos	3	Herpangina		5 6	Grade 3	Clinic visit	Definitely unrelated
infestations									
94	Male	2.2 mos	3	Local reactions	Local induration	1 5	Grade 3	Untreated	Possibly related
100	Male	1.7 mos	1	Systemic reactions	Fatigue / somnolence	0 7	Grade 3	Untreated	Possibly related
115	Female	2.6 mos	2	Local reactions	Local redness	1 1	Grade 3	Untreated	Possibly related
115	Female	2.6 mos	3	Local reactions	Local redness	1 2	Grade 3	Untreated	Possibly related

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2.7.4.2.5.2. Phase III clinical trial
2.7.4.2.5.2.1. Primary series
A total of 80 cases (105 events) of SAE were collected in the Primary series, including 34 cases (49 events) in the PCV13-TT group and 34 cases (43 events) in the PCV7 group of subjects aged 3-71 months. No significant difference was observed with respect to SAE incidence between the PCV13-TT and PCV7 group; a total of 12 cases (13 events) of SAE were reported in the subjects 2 months of age.
With the exception of 1 case of SAE in 3-month-old PCV13-TT group whose fever symptom was determined to be possibly related to the study vaccine, other SAEs were all independent of study vaccine. The SAE incidence in subjects aged 2 months and 3 months subjects was higher than that of other age groups. The major SAEs were "infections and infestations".
The severity of solicited symptoms or the most severe symptoms in SAEs was assessed in accordance with Guidelines for The Classification of Adverse Reactions in Clinical Trials of Prophylactic Vaccines, with a purpose to timely understand and analyze the possible risks of SAE. Results showed that the severity was grade 4 in 32 cases, grade 3 in 34 cases, and grade 2 in 2 cases, notwithstanding no intergroup difference was observed for SAEs with different severity.
See Table
Table 216 to Table 223for analysis of SAE incidence.

Table 216 Incidence of serious adverse events in PCV13-TT and PCV7 groups of subjects aged 3-71 months

PCV13-TT (N=1117)	PCV7 (N=1119)
P	
Events	Cases %
Any SAE	49 34 3.04 43 34 3.04 1.0000
SAE related to study vaccine	1 1 0.09 0 0 0.00 0.4996

SAE unrelated to study
48 33 2.95 43 34 3.04 1.0000
vaccine

Table 217 Incidence rate of serious adverse events in subjects aged 2 months

PCV13-TT (N=517)
Events Cases %
Any SAE 13 12 2.32
SAE related to study vaccine 0 0 0.00
SAE unrelated to study vaccine 13 12 2.32
Infections and infestations 10 10 1.93
Infectious pneumonia 5 5 0.97
Upper respiratory tract infection 1 1 0.19
Roseola infantum 1 1 0.19
Bronchitis 3 3 0.58
Injury, poisoning and procedural
1 1 0.19
complications
Contusion 1 1 0.19
Gastrointestinal disorders 2 2 0.39
Enteritis 1 1 0.19
Diarrhea 1 1 0.19

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Table 218 Incidence of serious adverse events in subjects aged 3 months

PCV13-TT (N=519) PCV7 (N=519)
P
Events Cases % Events Cases %
Any SAE 28 18 3.47 25 21 4.05 0.7445
SAE related to study
1 1 0.19 0 0 0.00 1.0000
vaccine
SAE unrelated to study
27 17 3.28 25 21 4.05 0.6206
vaccine
Infections and infestations 22 16 3.08 20 17 3.28 1.0000
Viral rash 0 0 0.00 1 1 0.19 1.0000
Infectious enteritis 1 1 0.19 0 0 0.00 1.0000
Infectious pneumonia 9 8 1.54 9 8 1.54 1.0000
Laryngitis 0 0 0.00 1 1 0.19 1.0000
Respiratory tract infection* 2 2 0.39 0 0 0.00 0.4995
Measles 1 1 0.19 0 0 0.00 1.0000
Sepsis 1 1 0.19 0 0 0.00 1.0000
Abscess 1 1 0.19 0 0 0.00 1.0000
Upper respiratory tract
5 3 0.58 5 5 0.96 0.7255
infection
Bronchitis 2 2 0.39 4 4 0.77 0.6866
Investigations 1 1 0.19 0 0 0.00 1.0000
Increased biological
markers of myocardial 1 1 0.19 0 0 0.00 1.0000
necrosis
Nervous system disorders 2 2 0.39 1 1 0.19 1.0000
Febrile convulsion 1 1 0.19 0 0 0.00 1.0000
Psychomotor retardation 0 0 0.00 1 1 0.19 1.0000
Epilepsy 1 1 0.19 0 0 0.00 1.0000
Respiratory, thoracic and
1 1 0.19 4 3 0.58 0.6243
mediastinal disorders
Pneumonia 1 1 0.19 4 3 0.58 0.6243
Gastrointestinal disorders 1 1 0.19 0 0 0.00 1.0000
Enteritis 1 1 0.19 0 0 0.00 1.0000
Blood and lymphatic
1 1 0.19 0 0 0.00 1.0000
system disorders
Granulopenia 1 1 0.19 0 0 0.00 1.0000
*1 case of respiratory tract infection is "possibly related" to study vaccine.

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Table 219 Incidence of serious adverse events in subjects aged 7-11 months

PCV13-TT (N=198) PCV7 (N=200)
P
Events Cases % Events Cases %
Any SAE 5 5 2.53 10 6 3.00 1.0000
SAE related to study 1.0000
0 0 0.00 0 0 0.00
vaccine
SAE unrelated to study 1.0000
5 5 2.53 10 6 3.00
vaccine
Infections and infestations 5 5 2.53 7 5 2.50 1.0000
Infectious pneumonia 3 3 1.52 0 0 0.00 0.1222
Sepsis 0 0 0.00 2 2 1.00 0.4988
Upper respiratory tract 0 0 0.00 3 3 1.50 0.2481
infection
Hand-foot-mouth disease 1 1 0.51 0 0 0.00 0.4975

Bacterial tonsillitis 1 1 0.51 0 0 0.00 0.4975
Bronchitis 0 0 0.00 2 2 1.00 0.4988
Investigations 0 0 0.00 1 1 0.50 1.0000
Increased biological markers of myocardial necrosis 0 0 0.00 1 1 0.50 1.0000
Skin and subcutaneous tissue disorders 0 0 0.00 1 1 0.50 1.0000
Urticaria 0 0 0.00 1 1 0.50 1.0000
Gastrointestinal disorders 0 0 0.00 1 1 0.50 1.0000
Oral ulcer 0 0 0.00 1 1 0.50 1.0000

Table 220 Incidence of serious adverse events in subjects aged 12-23 months
PCV13-TT (N=200) PCV7 (N=200)
P

Events Cases % Events Cases %
Any SAE 14 9 4.50 5 4 2.00 0.2589
SAE related to study 1.0000
0 0 0.00 0 0 0.00
vaccine
SAE unrelated to study 0.2589
14 9 4.50 5 4 2.00
vaccine
Infections and infestations 10 7 3.50 5 4 2.00 0.5431
Tonsillitis 1 1 0.50 0 0 0.00 1.0000
Infectious pneumonia 4 4 2.00 1 1 0.50 0.3719
Bacterial skin infections 1 1 0.50 0 0 0.00 1.0000
Upper respiratory and digestive tract infections 1 1 0.50 0 0 0.00 1.0000
Hand-foot-mouth disease 1 1 0.50 3 3 1.50 0.6231
Bronchitis 0 0 0.00 1 1 0.50 1.0000
Mycoplasma pneumoniae 1 1 0.50 0 0 0.00 1.0000
Herpangina 1 1 0.50 0 0 0.00 1.0000
Immune system disorders 1 1 0.50 0 0 0.00 1.0000
Drug-induced hypersensitivity reactions 1 1 0.50 0 0 0.00 1.0000
Gastrointestinal disorders 3 3 1.50 0 0 0.00 0.2481
Intestinal obstruction 1 1 0.50 0 0 0.00 1.0000
Inguinal hernia 1 1 0.50 0 0 0.00 1.0000
Oral ulcer 1 1 0.50 0 0 0.00 1.0000

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Table 221 Incidence of serious adverse events in subjects aged 24-71 months
PCV13-TT (N=200) PCV7 (N=200)
P

Events Cases % Events Cases %
Any SAE 2 2 1.00 3 3 1.50 1.0000
SAE related to study 1.0000
0 0 0.00 0 0 0.00
vaccine
SAE unrelated to study 1.0000
2 2 1.00 3 3 1.50
vaccine
Infections and infestations 2 2 1.00 3 3 1.50 1.0000
Viral encephalitis 0 0 0.00 1 1 0.50 1.0000
Infectious pneumonia 1 1 0.50 0 0 0.00 1.0000
Bacterial tonsillitis 0 0 0.00 1 1 0.50 1.0000
Bronchitis 1 1 0.50 1 1 0.50 1.0000

Table 222 Proportion of serious adverse events in PCV13-TT and PCV7 groups of subjects aged 3-71 months by severity
PCV13-TT (N=1117) PCV7 (N=1119)
Severity P

Cases % Cases %
Grade 2 1 0.09 1 0.09
0.9940
Grade 3 17 1.52 17 1.52
Note: Severity assessment is merely applicable for solicited symptoms or most severe symptoms of SAE.

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Table 223 Listing of serious adverse events in subjects during primary series
Term of Days since

Study Vaccine Age System organ Duration
Gender Site Dose adverse Description last Severity SAE leads to Relevance
No. groups (mos) classification (days)
events vaccination
The subject had diarrhea on 20160903, 5-6 times / day, all were yellow watery stool. On 20160906, the was taken to Puyang County People's Hospital Hospitalizatio
Third Gastrointestin for hospitalization and examination, revealing n or prolonged Possibly
69 PCV13-TT Male 1.5 Henan Enteritis 4 7 Grade 3
dose al disorders active bowel sounds. Vomiting persisted from length of unrelated
20160908 to 20160909, 2 times / day, and was hospital stay
gastric contents, not projectile vomiting. Diarrhea ended on September 12.
The subject had fever on 20160905 at 39.6 ■. The

subject was hospitalized in Puyang County People's Hospital on September 6 for treatment. The subject was examined on admission to hospital

Hospitalization

Roseola with heavy breathing sounds of both lungs. On Third Infections and n or prolonged Possibly 69 PCV13-TT Male 1.5 Henan infantu September 6, the subject was transferred to Puyang 4 7 Grade 3 dose infestations length of unrelated m City People's Hospital, physical examination hospital stay showed hyperemia of pharynx and heavy breathing sounds. On September 9, body temperature recovered to normal. On 20160910, the subject had new systemic rash.

The subject had fever on October 14, with a maximum body temperature of 39.2 ■, accompanied by rhinorrhea. The subject received 1 dose of intramuscular treatment in Qingzu hospital, and treated with Ankahuangnamin granules, cefixime granules and other drugs. The symptom of high body temperature still repeated. On October Upper 15, the subject was treated in the outpatient Hospitalization respirato Third Infections and department of Qingzu town hospital. The n or prolonged Possibly 98 PCV13-TT Female 2.8 Henan ry tract 55 5 Grade 4 dose infestations hematology test showed increased WBC. On length of unrelated infectio October 16, the subject was hospitalized for "acute hospital stay n upper respiratory infection" was admitted to the hospital. Physical examination on admission to hospital showed: T: 37.2 ■, P: 100 times / min, R: 100 times / min. WBC: $11.5 \times 10^9/L$, symptomatic and supportive treatment such as anti-inflammatory, heat clearing and detoxification after admission, and was discharged on October 20.

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On April 14, the subject had cough, gurgling in throat, more obvious after suckling, after treatment of oral medication in the department of outpatient in Maternal and Child Health Hospital, poor effects were observed, accompanied with foaming at

Infectio Hospitalization

mouth, so the subject was taken to Wenxian First Infections and us n or prolonged Possibly 115 PCV13-TT Male 2.1 Henan People's Hospital for treatment on April 19, and 6 7 Grade 4 dose infestations pneumo length of unrelated was hospitalized for "bronchopneumonia", physical nia hospital stay examination on admission findings: poor mental state, paleness, dry lips, hyperemia of pharynx, heavy breathing sounds in both lungs, crackles and stridor in lungs, chest X-ray indicated increased pulmonary texture.

Infectio Cough with sputum, occasional wheezing, slight Hospitalization

Third Infections and us shortness of breath, hyperemia of pharynx, heavy n or prolonged Definitely 142 PCV13-TT Male 2.2 Henan 100 6 Grade 4 dose infestations pneumo breathing sounds of both lungs, and rhonchi and length of unrelated nia crackles in the lung hospital stay Cough and rhinorrhea, slight shortness of breath, Hospitalization heavy breathing sounds in both lungs, wheezing First Infections and Bronchit n or prolonged Possibly 150 PCV13-TT Female 2.6 Henan and fine crackles were noted. Chest X-ray showed: 66 6 Grade 4 dose infestations is length of unrelated increased lung increased, and Acinetobacter hospital stay baumannii infection is found in throat swab culture. Cough and fever at 39.3 ■. Physical examination on admission findings: heavy breathing sounds of Infectio both lungs, with crackles, and poor mental state. Hospitalization First Infections and us Chest X-ray showed: both lungs with increased n or prolonged Possibly 155 PCV13-TT Male 2 Henan 37 8 Grade 4 dose infestations pneumo markings, thickening and disorder, and widened length of unrelated nia mediastinal shadow. Hematologic findings: WBC: hospital stay $7.76 \times 10^9/L$, RBC: $3.25 \times 10^{12}/L$, N: 43.4%, L: 44.1%.

Infectio Rhinorrhea, cough, asthmatic suffocation, fever, Hospitalization

Secon Infections and us heavy breathing sounds in both lungs, with fine n or prolonged Definitely 190 PCV13-TT Male 1.9 Henan 46 6 Grade 4 d dose infestations pneumo crackles and stridor. Chest X-ray: length of unrelated nia bronchopneumonia hospital stay The subject had fever on 20160625, with the highest temperature being 37.5 ■, accompanied by paroxysmal continuous cough and gurgling in Hospitalization First Infections and Bronchit throat; 2 episodes of vomiting were recorded. In the n or prolonged Possibly

228 PCV13-TT Female 2.7 Hebei 38 9 Grade 4
dose infestations is outpatient department of Dingxing county hospital, length of unrelated
the subject was treated with oral medication, while hospital stay
the cough worsened. The subject was examined in
Dingxing county hospital on 20160630 for a second

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time. Physical examination revealed: heavy
breathing sound of both lungs, with gurgling in
throat; hematologic findings included elevated
markers of infection, which primarily were lymph;
the chest X-ray showed increased and enhanced
markings of both lungs, with visible spots, which
traveled along the bronchi.

The subject developed cough without obvious
causes on 20170224, the cough was mild initially
and gradually aggravating, presenting as non-
spasmodic, nonproductive cough with
expectoration, wheezing appeared on 20170227,
exacerbating after physical activities, with fever
Hospitalizatio

(highest temperature of 38.9 ■), after oral

Third Infections and Bronchit n or prolonged Possibly

301 PCV13-TT Male 2.8 Hebei medication treatment (not specified), the conditions 160 7 Grade 3
dose infestations is length of unrelated
were not resolved. On 20170228, the subject was
hospital stay

taken to Zhengding County People's Hospital, and
underwent hematology test in outpatient

department, with the following findings: WBC

$8.87 \times 10^9/L$, N% 34.7%, L% 52.3%, HGB 107g/L,

PLT $429 \times 10^9/L$, CRP 7.5 mg/L; chest X-ray:

thickening markings in both lungs

Nearly one month ago (estimated date: 20170123),

the subject developed cough without obvious
causes, presenting as non-spasmodic cough, with
sputum, not easy to cough up, not treated in the
beginning, on 20170218, the subject was treated
with unspecified medication for about 3 days,

Infectio without poor response, vomiting occurred on Hospitalizatio

Third Infections and us 20170221, with non-projectile gastric contents, n or prolonged Possibly

323 PCV13-TT Female 2.2 Hebei 155 12 Grade 3

dose infestations pneumo with high frequency, loose stools, aggravated cough length of unrelated
nia with wheezing, and poor mental diet appeared in hospital stay

the past two days, on 20170223, the subject was
hospitalized in the People's Hospital of Zhengding
County for "bronchopneumonia and gastrointestinal
dysfunction". Physical examination findings:

hyperemia of pharynx, heavy breathing sounds in
both lungs, with wheezing

Injury, 20161004 Fall from bed, had fever at 40.0 ■ two Hospitalizatio

Third poisoning and Contusi days later, without vomiting, rhinorrhea or cough. n or prolonged Definitely

350 PCV13-TT Female 2.8 Hebei 12 8 Grade 3

dose procedural on On 1007, swelling and protrusion on the right top length of unrelated
complications head without rupture. On 1008, in the provincial hospital stay

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children's hospital, head CT showed hematoma of
right parietal scalp and fracture of right parietal
bone. Physical examination on admission to
hospital findings: body temperature 36.7 ■, heart
rate 120 / min, 34 breaths / min, conscious, in good
mental state, swelling and protrusion of the top
right scalp of 10 cm×8 cm, with palpable
fluctuation, no rupture or tenderness, red
maculopapules scattered in torso and both lower
limbs. Hyperemia of pharynx, heavy breathing
sounds in both lungs, positive Babinski sign on
both sides. Puncture of scalp hematoma was
performed on 20161015

20160922 Fever without causes, with peak
Hospitalizatio

frequency of 5-6 times / day, the highest

Shanx Secon Gastrointestin n or prolonged Possibly

424 PCV13-TT Female 1.8 Diarrhea temperature was 39.4 ■, accompanied with 8-9 62 7 Grade 3
i d dose al disorders length of unrelated

times / day of diarrhea as yellow green and watery
hospital stay

stool

Respirat Hospitalizatio

Fever, body temperature 38.2 ■, hyperemia of

First Infections and ory tract n or prolonged Definitely

545 PCV13-TT Male 3.9 Henan pharynx, heavy breathing sounds in both lungs, 41 4 Grade 3

dose infestations infectio length of unrelated

cough, sneezing, paroxysmal crying

n hospital stay

Increase
d
biologic
Hospitalizatio
al Creatine, kinase, isoenzyme 48 M/L↑, lactate
First n or prolonged Definitely
545 PCV13-TT Male 3.9 Henan Investigations markers dehydrogenase 379↑, A hydroxybutyrate 41 4 Grade 3
dose length of unrelated
of dehydrogenase 330↑
hospital stay
myocard
ial
necrosis
Blood and Hospitalizatio
First lymphatic Granulo Neutrophil count was $0.7 \times 10^9/L$, hematology test n or prolonged Definitely
545 PCV13-TT Male 3.9 Henan 41 4 Grade 3
dose system penia showed a lymphocyte ratio of 66.9%. length of unrelated
disorders hospital stay
The subject had fever on June 8, with the highest
temperature of 38.0 ■, coughing with sputum. On
Infectio Hospitalizatio
June 11, the subject was treated in Puyang people's
Secon Infections and us n or prolonged Possibly
572 PCV13-TT Male 3.5 Henan hospital and was admitted. Physical examination 19 7 Grade 3
d dose infestations pneumo length of unrelated
showed: mild hyperemia of pharynx, heavy
nia hospital stay
breathing sounds in both lungs, with wheezing and
crackles.

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Three hours before admission to hospital on June Hospitalizatio
Nervous Febrile
Secon 11, the subject had another episode of fever, with n or prolonged Possibly
572 PCV13-TT Male 3.5 Henan system convulsi 19 7 Grade 3
d dose body temperature of 37.8 ■, and eyes turning up, length of unrelated
disorders on
lasted for nearly 10 minutes, and relieved then. hospital stay
The subject had fever on July 3, with a maximum
Upper
body temperature of 39.2 ■, cough and rhinorrhea, Hospitalizatio
respirato
Third Infections and and was hospitalized in Puyang County maternal n or prolonged Definitely
589 PCV7 Male 3 Henan ry tract 11 7 Grade 3
dose infestations and child health care hospital on July 4. Physical length of unrelated
infectio
examination findings: slight hyperemia of pharynx hospital stay
n
and heavy breathing sounds of both lungs.
Upper
Hospitalizatio
respirato Fever, heavy breathing sounds in both lungs, red
Secon Infections and n or prolonged Definitely
603 PCV13-TT Female 3.5 Henan ry tract and swollen pharynx, the highest fever temperature 14 4 Grade 3
d dose infestations length of unrelated
infectio was 39.0 ■.
hospital stay
n
The subject had fever on 20160508 after getting a
cold, in the afternoon of the same day, the was
taken to the outpatient department of Xuzhen
Health Center for treatment, receiving ibuprofen
suspension orally 2 ML for one dose, and another
dose each on 0509 and 0510, respectively, with
Upper poor response. On 20160510 the subject was
Hospitalizatio
respirato admitted to Xuzhen health center, with body
First Infections and n or prolonged Possibly
603 PCV13-TT Female 3.5 Henan ry tract temperature on admission being 37.4 ■, and red 16 4 Grade 2
dose infestations length of unrelated
infectio pharynx. WBC $16.11 \times 10^9/L$, diagnosis on
hospital stay
n admission: acute upper respiratory tract infection.
20160511 Diarrhea 3-4 times/day, with yellow
loose stool. After anti-infective antipyretic
treatment and symptomatic treatment, the subject
was discharged from the hospital on 20160513.
Discharge diagnosis: acute upper respiratory tract
infection
Upper
Fever at 39.0 ■, combined with diarrhea, Hospitalizatio
respirato
First Infections and pharyngeal redness and swelling, heavy breathing n or prolonged Possibly
612 PCV13-TT Female 3.6 Henan ry tract 32 5 Grade 4
dose infestations sounds of both lungs, without rhonchi or crackles, length of unrelated
infectio
WBC count $16.33 \times 10^9/L$ hospital stay
n
Upper

Pharyngeal redness and swelling, clear breath Hospitalization
respiratory
First Infections and sounds in both lungs, no crackles, fever at 40.0 °C, n or prolonged Possibly
612 PCV13-TT Female 3.6 Henan ry tract 16 5 Grade 4
dose infestations WBC count 14.63×10⁹/L, hsCRP > 10 mg/L, length of unrelated
infection
CRP 48.9 mg/L, I9.62×10⁹/L, L 47.3% hospital stay
n

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The subject on 20160610 had cough with sputum, vomiting of gastric contents. On June 13, the Infectio subject developed dyspnea. On June 16, the was Hospitalization
Secon Infections and us taken to Puyang County People's Hospital for n or prolonged Possibly
624 PCV7 Male 3.8 Henan 21 7 Grade 3
d dose infestations pneumo treatment. Physical examination on admission length of unrelated
nia findings: heavy breathing sounds, with rhonchi and hospital stay
crackles. The highest temperature was 37.8 °C, and was discharged on June 22.
On 20160623, the subject had cough, fever, and asthma with a maximum body temperature of 39.5 °C. n 20160627, the subject was hospitalized
Respiratory, Hospitalization
in Xuzhen health hospital. Physical examination on Secon thoracic and Pneumo n or prolonged Possibly
637 PCV13-TT Male 3.1 Henan admission findings: red and swelling pharynx, 32 8 Grade 3
d dose mediastinal nia length of unrelated
heavy breathing sounds of both lungs, with rhonchi disorders hospital stay
and stridor. The subject was discharged on 20160704, recovered on July 9, and the cough symptoms disappeared.
Hospitalization
First Infections and Fever > 38.5 °C, WBC count 26.20×10⁹/L, CRP n or prolonged Definitely
646 PCV13-TT Female 3.9 Henan Sepsis 42 7 Grade 3
dose infestations 82 mg/L, increased respiratory rate and heart rate length of unrelated
hospital stay
Upper
Hospitalization
respiratory Fever 38.5 °C, hyperemia of pharynx, heavy
First Infections and n or prolonged Definitely
646 PCV13-TT Female 3.9 Henan ry tract breathing sounds in both lungs, increased WBC 42 7 Grade 4
dose infestations length of unrelated
infectio count and CRP hospital stay
n
The subject had fever of 38.7 °C on 20160624, with rhinorrhea and cough with sputum, on June 25 was
Hospitalization
treated in Puyang Hospital of traditional Chinese
First Infections and Bronchit n or prolonged Definitely
646 PCV13-TT Female 3.9 Henan Medicine. Physical examination on admission to 58 6 Grade 3
dose infestations is length of unrelated
hospital: red pharynx: T 38.1 °C with heavy
hospital stay
breathing sounds in both lungs, with gurgling in throat.
The subject had fever and cough on 20160702, with the maximum body temperature of 38.4 °C. The
Respiratory, Hospitalization
subject was hospitalized in Xuzhen health center on Secon thoracic and Pneumo n or prolonged Possibly
657 PCV7 Male 3.3 Henan July 21, physical examination on admission 43 7 Grade 3
d dose mediastinal nia length of unrelated
indicated heavy breathing sounds with few gurgling disorders hospital stay
in throat. The subject was discharged on July 27, and the cough symptoms disappeared on July 30.
Secon Infections and Infectio The main symptoms and signs were cough, Hospitalization Possibly
667 PCV7 Female 3.7 Henan 8 9 Grade 3
d dose infestations us gurgling in throat, heavy breathing sounds, with n or prolonged unrelated

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pneumo fine crackles, diagnosed as bronchopneumonia, and length of
nia was hospitalized during 20160611-0619. hospital stay
The subject had fever of 38.0 °C on 20160713,
Hospitalization
Infectio diarrhea of 5 times / day on July 15, with a small
Secon Infections and n or prolonged Possibly
686 PCV13-TT Female 3.5 Henan us amount of yellow mucus stool. The subject was 40 7 Grade 3
d dose infestations length of unrelated
enteritis hospitalized in Xuzhen hospital on July 20 and
hospital stay
discharged from the hospital and cured on July 26.
The subject had fever of 38.0 °C on 20160713. On the same day, the subject was given intramuscular
injection once in Xuzhenji clinic. On July 16, an

Hospitalization

Induration was observed on the injection site of Secon Infections and n or prolonged Possibly 686 PCV13-TT Female 3.5 Henan Abscess right hip. On July 20, the subject was hospitalized 40 7 Grade 3 d dose infestations length of unrelated in Xuzhen health hospital. On July 22, the subject hospital stay underwent abscess incision and drainage and was discharged on July 26. On August 5, the surgical wound healed.

The subject developed paroxysmal cough with gurgling in throat on 20160624. The subject was Infectio Hospitalization hospitalized in Puyang maternal and child health Secon Infections and us n or prolonged Possibly 698 PCV7 Male 3.3 Henan care hospital on 20160704. Physical examination 27 15 Grade 3 d dose infestations pneumo length of unrelated on admission indicated heavy breathing sounds in nia hospital stay both lungs and fixed crackles. The subject was discharged on 20160718. Rhinorrhea, cough, physical examination on admission indicated: poor mental state, heavy Infectio Hospitalization breathing sounds in both lungs during auscultation, First Infections and us n or prolonged Possibly 752 PCV13-TT Male 3.2 Henan with crackles. Chest X-ray showed: increased and 23 7 Grade 4 dose infestations pneumo length of unrelated thickened markings in both lungs, and flocculent nia hospital stay shadow was seen in the middle of the right lung. T: 38.2■

The subject had fever on April 15, with body temperature of 38.3 ■. After symptomatic treatment (unspecified), the symptoms combined Hospitalization First Infections and Bronchit with fever and cough still persisted intermittently. n or prolonged Possibly 766 PCV7 Male 3 Henan 39 7 Grade 4 dose infestations is The subject was hospitalized in the health care length of unrelated hospital on May 21. Physical examination on hospital stay admission indicated: hyperemia of pharynx, heavy breathing sounds in both lungs. Infectio Cough occurred on June 6, aggravated on June 21, Hospitalization Secon Infections and us with sputum, occasional asthma, hyperemia of n or prolonged Possibly 780 PCV7 Female 3 Henan 41 7 Grade 4 d dose infestations pneumo pharynx, heavy breathing sounds in both lungs, length of unrelated nia with rhonchi and crackles. hospital stay

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Fever on the evening of April 19, with the Infectio maximum temperature of 38.7 ■, accompanied by Hospitalization First Infections and us cough and vomiting (3 times / day, milk curds); on n or prolonged Possibly 801 PCV7 Male 3.9 Henan 2 7 Grade 4 dose infestations pneumo April 20, dyspnea, poor mental state, dry lips, length of unrelated nia hyperemia of pharynx, heavy breathing sounds in hospital stay both lungs, fine crackles and wheezing. Fever occurred on June 24. The village health center gave compound zinc ibuprofen and other drugs (unspecified details) with poor response. Fever persisted intermittently. On June 27, the body Hospitalization Secon Infections and Bronchit temperature was 38.0 ■. The subject was n or prolonged Possibly 801 PCV7 Male 3.9 Henan 30 6 Grade 4 d dose infestations is hospitalized in Wenxian County People's Hospital length of unrelated on the same day. Physical examination on hospital stay admission indicated: poor mental state, hyperemia of pharynx, heavy breathing sounds in both lungs, and occasional gurgling in throat. Infectio Hospitalization Third Infections and us Cough occurred on October 10, fever occurred on n or prolonged Definitely 801 PCV7 Male 3.9 Henan 98 5 Grade 4 dose infestations pneumo October 12, the highest temperature was 39.0 ■. length of unrelated nia hospital stay On April 24, the subject with wheezing developed cough, treated on April 25 in outpatient clinic with poor response, resulting in aggravating symptoms, accompanied with wheezing and occasionally choking cough. On April 29, the subject was diagnosed as asthmatic bronchopneumonia in the Infectio outpatient clinic, and was admitted to the hospital Hospitalization First Infections and us for intermittent cough. Physical examination on n or prolonged Possibly 809 PCV7 Female 3.9 Henan 10 9 Grade 4 dose infestations pneumo admission indicated: T: 36.5 ■, poor mental state, length of unrelated nia slight shortness of breath, hyperemia of pharynx, hospital stay heavy breathing sounds in both lungs during auscultation, with wheezing and crackles, chest X-ray showed: increased lung markings, blood RT: WBC:17.17×10⁹/L, L: 42.5%, N: 50.90%, HGB: 118 g/L indicative of bacterial infection, CRP 34 mg/L

The subject had cough on May 1st, and fever with Infectio 39.0 ■ on May 25th, the coughing persisted, and Hospitalizatio First Infections and us in poor mental state, slight shortness of breath, dry n or prolonged Possibly 828 PCV13-TT Female 3.8 Henan 36 6 Grade 4 dose infestations pneumo lips, hyperemia of pharynx, heavy breathing sounds length of unrelated nia in both lungs, with rhonchi and fine crackles. hospital stay Hematology test indicated: WBC: $5.03 \times 10^9/L$, N:

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7.72%, L: 86.91%. Chest X-ray showed increased lung markings.

Hospitalizatio

First Gastrointestin Loose and watery stool, 5-6 times / day, stool n or prolonged Possibly

828 PCV13-TT Female 3.8 Henan Enteritis 40 2 Grade 4

dose al disorders routine test: 6-7 leukocytes / HP length of unrelated

hospital stay

On December 8, 2016, the subject developed

paroxysmal cough, with sputum, and combined

with asthma on December 9, 2016. The subject was

Infectio given symptomatic treatment in the village health Hospitalizatio

Third Infections and us center (unspecified details) without remission. On n or prolonged Definitely

844 PCV13-TT Male 3.1 Henan 167 8 Grade 4

dose infestations pneumo December 11, 2016, the subject was hospitalized in length of unrelated

nia Wenxian maternal and child health care hospital for hospital stay

"bronchopneumonia". The chest X-ray showed

increased and thickened lung markings in both

lungs, with slightly blurred and flocculent sign

The body temperature was 37.3 ■ in 30 minutes

after immunization and 39.5 ■ on May 3. After

treatment with oral medicine and outpatient clinical

therapy, the body temperature decreased to about

37.5 ■, and reached 38.6 ■ on July 30. On July

Respirat 31, the subject was hospitalized in Zhengzhou Hospitalizatio

Secon Infections and dry tract children's Hospital. Physical examination on n or prolonged Possibly

877 PCV13-TT Female 3 Henan 62 10 Grade 4

d dose infestations infectio admission indicated: T: 38.0 ■, slight hyperemia length of related

n of pharynx, heavy breathing sounds in both lungs. hospital stay

T cells (CD3) were decreased to 48%. CD4 / CD8

was increased to 2.74, indicative of cellular

immune dysfunction. Vitamin B12 117 pg/mL,

decreased. cytomegalovirus IgG positive.

Complement C3 0.75 g/L, decreased.

On May 19, subjects had developed cough with

sputum due to cold, and occasionally had shortness

of breath. The subject was given oral medication in

Infectio the local clinic while with poor response. On May Hospitalizatio

First Infections and us 27, the cough worsened, and on May 29, the subject n or prolonged Possibly

904 PCV13-TT Female 3 Henan 20 8 Grade 4

dose infestations pneumo was hospitalized in Wenxian people's hospital for length of unrelated

nia "bronchopneumonia". Physical examination on hospital stay

admission indicated poor mental state, dry lips,

hyperemia of pharynx, heavy breathing sounds of

both lungs, with rhonchi and crackles.

Third Infections and Infectio On December 9, the subject had paroxysmal cough Hospitalizatio Definitely

904 PCV13-TT Female 3 Henan 152 11 Grade 4

dose infestations us induced by cold, with sputum and shortness of n or prolonged unrelated

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pneumo breath. Oral medication in the street clinic showed length of

nia poor response. On December 10, the subject was hospital stay

hospitalized in Wenxian people's hospital was for

"bronchopneumonia". Physical examination on

admission indicated hyperemia of pharynx, heavy

breathing sounds of both lungs, with crackles and

wheezing.

Infectio Hospitalizatio

First Infections and us Fever 38.2 ■, cough, heavy breathing sounds in n or prolonged Possibly

907 PCV13-TT Female 3.3 Henan 38 7 Grade 4

dose infestations pneumo both lungs, with rhonchi and crackles. length of unrelated

nia hospital stay

During 20160715-0724, the subject suffered from

upper respiratory tract infection, fever (39.0 ■),

cough with expectoration, and rhinorrhea. All

symptoms were relieved after treatment. Fever

occurred again on 0728, the highest temperature

was 39.5 ■, and cough symptoms were

aggravated. On 0728 the subject was diagnosed as

Infectio Hospitalizatio

"pneumonia" in the outpatient department of the

Third Infections and us n or prolonged Possibly

984 PCV7 Male 3.2 Hebei county hospital while was not treated. On 0729 the 37 4 Grade 4

dose infestations pneumo length of unrelated

subject was taken to Baoding children's Hospital

nia hospital stay

for further treatment. At 0729 13:50, the subject

was hospitalized for "bronchopneumonia". Physical

examination showed: heavy breathing sounds of both lungs, with fine crackles and gurgling in the throat, hyperemia of pharynx, herpes was present in the pharyngeal isthmus. After treatment, the subject recovered and was discharged at 0801 16:00. The subject had cough and fever on 20170113, with the highest temperature of 38.4 ■. No response Hospitalization was observed after home treatment (unspecified Third Infections and Bronchitis or prolonged Possibly 1048 PCV7 Female 3.1 Hebei medication). On 20170114, the subject was 183 5 Grade 4 dose infestations is length of unrelated hospitalized and treated in the outpatient hospital stay department of Dingxing Yunfeng maternity hospital and was discharged on 20170121. The subject was admitted to hospital on 20161122 for cough with wheezing lasting for 10 days, and Respiratory, Hospitalization aggravation with suffocation for 3 days. On Third thoracic and Pneumonia or prolonged Possibly 1067 PCV7 Male 3.2 Hebei 20161112, the subject had cough with wheezing 123 11 Grade 4 dose mediastinal nia length of unrelated with no obvious cause, and was treated with disorders hospital stay medications (unspecified) in the local outpatient clinic for 10 days, and the symptoms were relieved.

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On 20161119, the subject had cough with wheezing and mild suffocation due to cold, and was admitted to hospital on 20161122. After physical examination and auxiliary examination, the subject was diagnosed as asthmatic suffocating pneumonia, and was then treated with medications. After admission, examinations indicated elevated myocardial enzymes and myocardial cell damage, supplementary diagnosis showed myocardial injury, all symptoms were resolved and the subject was discharged after hospitalization on 20161202, with medicines prescribed after discharge. Since 20160901, the subject had gurgling in throat without obvious cause, no evident cough or fever, and the parents did not pay attention to it. On 20161015 the subject was taken by parents to the village clinic for examination while without definite diagnosis. After 3 days of treatment with Respiratory, oral medicines (unspecified type and dosage), there Hospitalization Third thoracic and Pneumonia was little improvement observed. For further n or prolonged Possibly 1067 PCV7 Male 3.2 Hebei 105 12 Grade 4 dose mediastinal nia diagnosis and treatment, the subject was taken to length of unrelated disorders Dingxing county hospital on 20161104. At hospital stay 20161104 12:00, the subject was hospitalized for "pneumonia". Physical examination on hospital admission indicated hyperemia of pharynx, heavy breathing sounds of both lungs, with crackles. Small patchy opacities were observed in the right upper lung field. On 20160721 the subject had cough and fever with no obvious cause. Antipyretic treatment (unspecified) was administered by village health room, followed by oral medication and enema therapy which resulted in poor response. On 0726, Infectio another episode of fever occurred (as high as Hospitalization Third Infections and us 40.0 ■), with frequent coughing and evident n or prolonged Possibly 1107 PCV13-TT Female 3.6 Hebei 30 6 Grade 3 dose infestations pneumo wheezing, and in poor mental state. On 20160727 length of unrelated nia the subject was examined in Zhengding County hospital stay People's Hospital, with body temperature being 40.0 ■, poor mental state, slight shortness of breath, small bubbling sound in both lungs, a small amount of wheezing sound, CRP 19.5 mg/L, chest X-ray showed thickened markings and disorder in

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both lungs, with scattered high-density patchy opacity in both lung field, diagnosed as bronchopneumonia, and was hospitalized. On 20160801, the symptoms resolved and the subject was discharged. On July 25, the subject's body temperature was 39.6 ■, breath was 40 times / min, heart rate was 120 beats / min, blood pressure was 85 / 50 mmHg. Hospitalization Third Infections and During the course of the disease, there were Kolik n or prolonged Possibly 1146 PCV13-TT Male 3.3 Hebei Measles 27 14 Grade 3 dose infestations spots in buccal mucosa, full body rash, with normal length of unrelated

skin during the occurrence of rashes, positive IgM hospital stay
antibody of measles virus. The subject was
discharged on August 7.
The subject had cough from June 26 to July 7, and
was not recovered after treatment with outpatient
medication. On July 9, the subject was hospitalized
in the Department of Pediatrics of Zhengding
County People's hospital. The subject was found by
Infectio Hospitalizatio
physical examination to have cough with sputum
Secon Infections and us n or prolonged Possibly
1153 PCV7 Female 3.3 Hebei and hyperemia of pharynx, heavy breathing sounds 31 8 Grade 4
d dose infestations pneumo length of unrelated
in both lungs, with gurgling in throat, the chest X-
nia hospital stay
ray DR indicated thickened and increased marking
in both lungs, with blurred presentation, and the
diagnosis was bronchopneumonia. The subject
recovered on July 16, and the event was considered
to be possibly unrelated to vaccination.
From 0520 to 0526, the highest temperature of
fever in the subject was 39.1 ■, with cough, and
was not resolved after outpatient treatment, the
subject was hospitalized in Zhengding County
People's Hospital on May 27. The body
Hospitalizatio
temperature was 38.4 ■, with hyperemia of
First Infections and Bronchit n or prolonged Possibly
1162 PCV13-TT Male 3.2 Hebei pharynx, heavy breathing sounds in both lungs, 18 5 Grade 3
dose infestations is length of unrelated
with some wheezing and gurgling in throat, WBC
hospital stay
7.36×10⁹/L, L% 25.4%, CRP 7.8 mg/L, chest X-
ray indicated thickened lung markings in both
lungs, and it was diagnosed as bronchitis, anti-
infectives were given, and the subject was
discharged on May 31.
Cough for 4-5 days, asthma for 1 day, hospitalized
Secon Infections and Bronchit Hospitalizatio Possibly
1175 PCV7 Female 3.3 Hebei on June 21, 2016, physical examination indicated: 13 10 Grade 3
d dose infestations is n or prolonged unrelated
body temperature 37.3 ■, slightly fast breathing,

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hyperemia of pharynx, heavy breathing sounds in length of
both lungs, with rhonchi and crackles, WBC hospital stay
5.0×10⁹/L, L% 70.8%, N% 21.3%, chest X-ray
indicated bronchitis, recovered and discharged on
June 30, 2016 after anti-infection treatment.
On 20160719 There was cough without obvious
cause, presenting as paroxysmal continuous cough
with sputum, which was not easy to cough up,
accompanied by rhinorrhea. On 20160721, the
subject was hospitalized, physical examination
Infectio Hospitalizatio
indicated: hyperemia of pharynx, heavy breathing
Secon Infections and us n or prolonged Possibly
1188 Group A Male 3.2 Hebei sounds, gurgling in throat both lungs can hear 30 6 Grade 3
d dose infestations pneumo length of unrelated
sputum sound, hematology test: WBC 8.6×10⁹ ^
nia hospital stay
9/L, L% 75.1%, GR% 14.0%, HGB 122 g/L, PLT
440×10⁹/L, CRP 1.3 mg/L, chest X-ray showed
bronchopneumonia. After symptomatic treatment,
the subject recovered and was discharged on
20160726.
On 20160815, the subject developed fever without
obvious cause, the highest temperature was
40.2 ■, with mild cough, and did not have enough
response to the treatment administered in local
Upper
clinic. The subject was hospitalized on 20160816. Hospitalizatio
respirato
Third Infections and Physical examination showed slightly swollen n or prolonged Possibly
1200 PCV7 Male 3.8 Hebei ry tract 25 4 Grade 3
dose infestations tonsils on both sides and congestion. After 3 days, length of unrelated
infectio
the fever subsided and rash appeared. WBC hospital stay
n
5.88×10⁹/L, W 31.5%, with the diagnosis of upper
respiratory tract infection, symptomatic anti-
infection treatment was administered, and then the
subject was cured and discharged on 20160819.
The subject's eyeballs did not follow the movement
of object, and may not remain stable when raising
Psycho Hospitalizatio
Nervous head, unresponsive, and cannot crawl, the subject
Shanx Secon motor n or prolonged Possibly
1260 PCV7 Male 3 system had to be supported and might not make steps while 10 7 Grade 3

i d dose retardati length of unrelated disorders walking, and would not actively grasp objects by on hospital stay either hand, bilateral Babinski sign, and could not roll over.
Upper
Hospitalizatio
respirato Fever at 39.0 ■, cough, poor mental state, heavy Shanx Third Infections and n or prolonged Possibly 1405 PCV7 Male 3.8 ry tract breathing sounds in both lungs, with hyperemia of 26 7 Grade 3
i dose infestations length of unrelated infectio pharynx hospital stay
n

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Fever at 38.3 ■, cough, groan, dyspnea, hyperemia
Respiratory, of pharynx, positive inspiratory three concave sign, Hospitalizatio
Shanx First thoracic and Pneumo heavy breathing sounds in both lungs, with n or prolonged Possibly 1485 PCV7 Female 3.1 4 7 Grade 3
i dose mediastinal nia crackles, chest X-ray in posteroanterior position length of unrelated disorders revealed increased and disordered lung markings, hospital stay especially in the right lower lung.
Convulsion, double eyes turned up, askew mouth, Hospitalizatio
Nervous
Shanx Third limb rigidity, hyperemia of pharynx, heavy n or prolonged Possibly 1499 PCV13-TT Female 3.4 system Epilepsy 24 6 Grade 3
i dose breathing sounds of both lungs, bilateral positive length of unrelated disorders
Babbitt sign hospital stay
Fever at 38.7 ■, accompanied by cough, Hospitalizatio
hoarseness, barking cough, inspiratory laryngeal
Shanx Third Infections and Laryngit n or prolonged Possibly 1537 PCV7 Male 3.5 stridor, circumoral cyanosis, hyperemia of pharynx, 215 5 Grade 3
i dose infestations is length of unrelated positive inspiratory three concave sign, heavy hospital stay
breathing sounds in both lungs.
Upper
Fever at 38.0 ■, poor sleep at night, irritable, Hospitalizatio
respirato
Shanx Third Infections and reduced milk intake. Poor mental state, red face, n or prolonged Possibly 1559 PCV7 Male 3.1 ry tract 80 7 Grade 3
i dose infestations red lips, throat congestion, heavy breathing sounds length of unrelated infectio
in both lungs. hospital stay
n
Hospitalizatio
Shanx Third Infections and Viral Many dark red herpes and maculopapular rashes n or prolonged Possibly 1560 PCV7 Male 3.2 216 5 Grade 3
i dose infestations rash present on the whole body length of unrelated hospital stay
Upper
Hospitalizatio
respirato Body temperature was 37.2 ■, hyperemia of
Shanx Third Infections and n or prolonged Possibly 1560 PCV7 Male 3.2 ry tract pharynx, swollen tonsils on both sides, heavy 216 5 Grade 3
i dose infestations length of unrelated infectio breathing sounds of both lungs.
hospital stay
n
Fever at T 39.1 ■, vomiting several times per day, Hospitalizatio
First Infections and Bronchit hyperemia of pharynx, heavy breathing sounds of n or prolonged Possibly 1580 PCV7 Male 8 Henan 16 7 Grade 4
dose infestations is both lungs, bronchitis as indicated by chest X-ray length of unrelated in posteroanterior view. hospital stay
Hospitalizatio
First Infections and Fever with shiver, WBC 12.22×10⁹/L, C-reactive n or prolonged Possibly 1580 PCV7 Male 8 Henan Sepsis 16 7 Grade 4
dose infestations protein 31.47 mg/L, T 38.5 ■ length of unrelated hospital stay
Skin and Hospitalizatio
First subcutaneous Urticari Rash mainly on chest and back, and reddish n or prolonged Possibly 1580 PCV7 Male 8 Henan 16 7 Grade 4
dose tissue a maculopapular rash. length of unrelated disorders hospital stay

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Increase
d
biologic
Hospitalizatio
al
First Abnormal myocardial zymogram CKMB-M 26 n or prolonged Possibly 1580 PCV7 Male 8 Henan Investigations markers 16 7 Grade 4

dose μL , LDH 301 μL , HBDH 270 μL length of unrelated
of
hospital stay
myocard
ial
necrosis

The subject had fever on 20160602, the highest
body temperature was 39.2 $^{\circ}\text{C}$, accompanied by
cough, wheezing on June 7, the clinical symptoms
were relieved after medication in local clinic,
coughing on June 15, with wheezing aggravated,
accompanied by fever of 38.0 $^{\circ}\text{C}$, after clinical
Infectio Hospitalizatio
medication the body temperature was reduced to
Secon Infections and us n or prolonged Possibly
1582 PCV13-TT Female 8.3 Henan normal, the subject was hospitalized in Puyang 2 8 Grade 3
d dose infestations pneumo length of unrelated
people's Hospital on June 17 for treatment,
nia hospital stay
physical examination on admission indicated:
hyperemia of pharynx, heavy breathing sounds in
both lungs, with apparent wheezing sounds,
gurgling in throat and scattered coarse crackles,
indicating positive serum mycoplasma pneumoniae
antibody, and was discharged on 20160624
Hand- Hospitalizatio

Fever, maximum temperature of 38.9 $^{\circ}\text{C}$, rash on
First Infections and foot- n or prolonged Possibly
1590 PCV13-TT Male 10.8 Henan hip, hands and feet, multiple herpes and ulcer in 3 6 Grade 4
dose infestations mouth length of unrelated
oral mucosa
disease hospital stay
Hospitalizatio

Fever at 39.8 $^{\circ}\text{C}$, C-reactive protein 22 mg/L , WBC
First Infections and n or prolonged Definitely
1606 PCV7 Female 9.9 Henan Sepsis $15.50 \times 10^9/\text{L}$, P: 136 times / min, R 32 times / 19 6 Grade 4
dose infestations length of unrelated
min
hospital stay
Increased respiratory rate, congestion of pharynx,
heavy breathing sounds of both lungs during the
Upper auscultation, on May 5, the subject had paroxysmal
Hospitalizatio

respirato cough accompanied with minor gurgling in the
First Infections and n or prolonged Definitely
1606 PCV7 Female 9.9 Henan ry tract throat, the on May 6, the maximum temperature 19 6 Grade 4
dose infestations length of unrelated
infectio was 39.8 $^{\circ}\text{C}$, the fever type was irregular, the
hospital stay
n laboratory test results showed positive mycoplasma
pneumonia, WBC count $15.50 \times 10^9/\text{L}$, neutrophil
count $9.99 \times 10^9/\text{L}$, lymphocyte count

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$4.28 \times 10^9/\text{L}$, and the C-reactive protein was 22
 mg/L .
Fever occurred on 20160630, with a high body
temperature of 38.9 $^{\circ}\text{C}$, the subject and was Hospitalizatio
Secon Gastrointestin Oral hospitalized in Qingzu town hospital on 20160703. n or prolonged Possibly
1615 PCV7 Male 10.4 Henan 3 7 Grade 3
d dose al disorders ulcer Physical examination indicated hyperemia of length of unrelated
pharynx, and two ulcer sites were found in oral hospital stay
cavity, the subject was discharged on 20160709
Upper

Fever, body temperature 39.2 $^{\circ}\text{C}$, hyperemia of Hospitalizatio
respirato
First Infections and pharynx, heavy breathing sounds in both lungs n or prolonged Possibly
1689 PCV7 Male 8 Henan ry tract 5 5 Grade 4
dose infestations during auscultation. WBC: $3.42 \times 10^9/\text{L}$, N: length of unrelated
infectio
 46.81% , L: 38.04% , HB: 94 g/L , PLT: $150 \times 10^9/\text{L}$ hospital stay
n

Upper Fever (39.5 $^{\circ}\text{C}$), poor mental state, hyperemia of
Hospitalizatio
respirato pharynx, heavy breathing sounds of both lungs
Secon Infections and n or prolonged Definitely
1702 PCV7 Female 8 Henan ry tract during auscultation, flushing face, dry lips, 28 5 Grade 4
d dose infestations length of unrelated
infectio hematology test: WBC: $12.29 \times 10^9/\text{L}$, N: 70.14% ,
hospital stay
n L: 22.44% , HGB: 90.0 g/L .

On 20160419 the subject had sneezing, and on
20160421 paroxysmal cough, with sputum that was
difficult to cough up. Wheezing symptoms
occurred on 20160422, with the highest severity
observed in the morning and during the night, the
highest fever was 38.1 $^{\circ}\text{C}$. On 20160423, the
subject was taken to the outpatient department of
Yunfeng maternity hospital in Dingxing for

Hospitalization
treatment, intravenous dripping was administered in
First Infections and Bronchitis or prolonged Possibly
1775 PCV7 Male 8.8 Hebei the outpatient department for one day, while no 8 8 Grade 4
disease infestations is length of unrelated
response was noted. On 20160424, the subject was
hospital stay
hospitalized in Dingxing Yunfeng maternity
hospital and was diagnosed as "bronchitis".
Physical examination indicated: hyperemia of
pharynx, heavy breathing sounds in both lungs,
with moderate gurgling in throat and wheezing.
After treatment of anti-infection, expectorant and
relieving asthma, the subject was discharged on
20160501.
On 20160622, the subject developed cough without
Infection Hospitalization
evident cause, accompanied by rhinorrhea,
First Infections and us or prolonged Possibly
1806 PCV13-TT Female 10.2 Hebei untreated; on 20160627, the subject had fever (the 78 7 Grade 3
disease infestations pneumo length of unrelated
maximum temperature was 38.8 ■), one episode of
infection hospital stay
spiking fever per day, cough was more severe than

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before, presenting as paroxysmal continuous cough,
with sputum that was difficult to cough up, and
symptoms worsened at night. The subject was
treated in personal clinic, with normal body
temperature, and the cough did not improve
significantly. For further diagnosis and treatment,
the subject was treated in Hebei children's Hospital
on 20160702 and was examined with findings of
heart rate 126 beats / min, 46 breaths / min,
hyperemia of pharynx, slight shortness of breath,
heavy breathing sounds in both lungs, with crackles
and wheezing, lymphocyte percentage 48.4%, chest
X-ray: bronchitis.
On July 20, 2016, the subject developed fever
without obvious cause, with body temperature of
38.5 ■ and rhinorrhea. The subject was treated on
20160721 in outpatient department with poor
response, and was hospitalized on 20160722.
Bacteria Physical examination indicated: body temperature Hospitalization
Second Infections and I 36.7 ■, heart rate 100 beats / min, 32 breaths / min or prolonged Possibly
1819 PCV13-TT Female 8 Hebei 32 5 Grade 3
disease infestations tonsillitis, hyperemia of pharynx, swollen tonsil, with length of unrelated
purulent coating, clear breath sounds in both lungs, hospital stay
rhonchi and crackles were absent, hemocyte count
5.02×10⁹/L, C-reactive protein 1.90 mg/L,
diagnosed as acute suppurative tonsillitis, and was
discharged on 20160726 after symptomatic
treatment.
Parents reported that the subject had fever,
rhinorrhea and other symptoms on 20160518,
Infection Hospitalization
which were not resolved after medication in
First Infections and us or prolonged Possibly
1837 PCV13-TT Female 7.9 Hebei outpatient department. On 20160605, the subject 27 7 .
disease infestations pneumo length of unrelated
was treated in the children's Hospital of Hebei
infection hospital stay
Province, and was admitted to the hospital for
"bronchopneumonia".
Several red herpes with the size of rice grains
distributed in bilateral palms and feet soles as well
Hand- Hospitalization
as in the hip, with the highest temperature of
First Infections and foot- or prolonged Possibly
1966 PCV7 Female 21 Henan 38.0 ■. The throat was congested and swollen, 50 5 Grade 4
disease infestations mouth length of unrelated
with grade 1 swollen tonsils of both sides, the red
disease hospital stay
spots were present in oral palate of the oral mucosa,
and soft neck without nuchal rigidity. The lab tests

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showed positive enterovirus EV71 infection and
positive EV infection.
The subject had fever on June 24, with the
maximum temperature of 38.5 ■. On June 27, the
Upper
subject vomited. The subject vomited immediately
respiratory
after eating, accompanied by poor mental state. On Hospitalization

ry and
 Secon Infections and June 27, the subject was treated in and admitted to n or prolonged Possibly
 2004 PCV13-TT Male 19.4 Henan digestiv 11 4 Grade 3
 d dose infestations Puyang Oilfield General Hospital, physical length of unrelated
 e tract
 examination indicated hyperemia of pharynx, tonsil hospital stay
 infectio
 swelling of grade 1, hyperemia of pharynx, heavy
 ns
 breathing sounds in both lungs, and active bowel
 sounds.
 Hospitalizatio
 When abdominal pressure was increased, a 3×4 cm
 First Gastrointestin Inguinal n or prolonged Definitely
 2019 PCV13-TT Female 23.2 Henan mass was noted in the right superficial inguinal 12 4 Grade 2
 dose al disorders hernia length of unrelated
 ring.
 hospital stay
 Hospitalizatio
 Fever at T 39.0 ■, buccal mucosa and upper palate
 First Gastrointestin Oral n or prolonged Possibly
 2026 PCV13-TT Male 17.9 Henan were with multiple ulcer sites, coated with 5 6 Grade 4
 dose al disorders ulcer length of unrelated
 purulence and surrounded by erythema
 hospital stay
 The subject had intermittent fever on 20160712,
 cough with sputum, the highest temperature was
 Hospitalizatio
 39.6 ■. The subject was treated in Qingzu town
 Secon Infections and Bronchit n or prolonged Possibly
 2028 PCV7 Male 17.3 Henan hospital on July 19, and discharged from hospital 23 6 Grade 3
 d dose infestations is length of unrelated
 on July 24, still coughing, with sputum, coughing
 hospital stay
 persisted on July 25, and the presence of sputum
 resolved.
 Hand- Hospitalizatio
 First Infections and foot- The body temperature was as high as 39.3 ■, red n or prolonged Possibly
 2028 PCV7 Male 17.3 Henan 6 9 Grade 4
 dose infestations mouth maculopapular rash was present in both feet and hip length of unrelated
 disease hospital stay
 Hand- Red brown maculopapular rash scattered in both Hospitalizatio
 First Infections and foot- hands and feet as well as hip, fever at 38.3 ■, n or prolonged Possibly
 2034 PCV7 Male 16.5 Henan 21 14 Grade 4
 dose infestations mouth herpes and ulcer scattered in buccal mucosa, length of unrelated
 disease positive EV71 hospital stay
 On June 15, the subject had rhinorrhea, with fever
 Infectio Hospitalizatio
 at 38.5 ■, and had cough on 18th, wheezing on
 First Infections and us n or prolonged Possibly
 2052 PCV13-TT Female 20.8 Henan 21st, in poor mental state, with slight shortness of 72 7 Grade 4
 dose infestations pneumo length of unrelated
 breath, heavy breathing sounds in both lungs, with
 nia hospital stay
 rhonchi and fine crackles.

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Infectio Hospitalizatio
 Fever at 39.0 ■, heavy breathing sounds in both
 Secon Infections and us n or prolonged Definitely
 2065 PCV13-TT Male 22.5 Henan lungs, with no obvious crackles, lung DR showed 14 6 Grade 4
 d dose infestations pneumo length of unrelated
 bronchopneumonia.
 nia hospital stay
 Hospitalizatio
 Fever at 39.0 ■, hyperemia of pharynx, red herpes
 Secon Infections and Herpang n or prolonged Definitely
 2065 PCV13-TT Male 22.5 Henan scattered in soft palate and pharynx, surrounded by 14 6 Grade 4
 d dose infestations ina length of unrelated
 erythema.
 hospital stay
 Hospitalizatio
 Secon Infections and Tonsillit Fever 39.0 ■, hyperemia of pharynx, swollen n or prolonged Definitely
 2065 PCV13-TT Male 22.5 Henan 14 6 Grade 4
 d dose infestations is tonsils of grade 3 on both sides. length of unrelated
 hospital stay
 Drug-
 induced Hospitalizatio
 Immune After pharmacotherapy in outpatient department,
 Secon hyperse n or prolonged Definitely
 2065 PCV13-TT Male 22.5 Henan system the fever symptoms were recurrent, and red rash 14 6 Grade 4
 d dose nsitivity length of unrelated
 disorders was scattered all over the body.
 reaction hospital stay
 s
 Bacteria Hospitalizatio
 There are two masses on the inner side of the right
 Secon Infections and l skin n or prolonged Definitely
 2065 PCV13-TT Male 22.5 Henan arm, one of which is white at the top. It is painful to 14 6 Grade 4

dose infestations infectio length of unrelated touch.
 ns hospital stay
 Cough, fever, body temperature of 40 ■,
 Infectio Hospitalizatio
 hyperemia of pharynx, heavy breathing sounds of
 First Infections and us n or prolonged Possibly
 2097 PCV13-TT Female 19.6 Henan both lungs during auscultation, with crackles, chest 17 6 Grade 4
 dose infestations pneumo length of unrelated
 X-ray showed increased and thickened markings in
 nia hospital stay
 both lungs, with blurred structure.
 Mycopl Hospitalizatio
 Xerocheilia, hyperemia of pharyngeal cavity, heavy
 First Infections and asma n or prolonged Definitely
 2114 PCV13-TT Male 12 Henan breathing sounds of both lungs during auscultation, 66 3 Grade 2
 dose infestations pneumo length of unrelated
 mycoplasma pneumoniae: weak positive
 nia hospital stay
 Vomiting, 5-6 times / day, with gastric content, the
 Intestina subject was hospitalized and treated, physical Hospitalizatio
 First Gastrointestin I examination on admission: poor mental state, dry n or prolonged Definitely
 2114 PCV13-TT Male 12 Henan 66 3 Grade 3
 dose al disorders obstructi lips. The abdominal plain film showed partial length of unrelated
 on intestinal gas in the abdomen, and several liquid hospital stay
 gas plane shadows of different sizes.
 On 20160516 the subject had rash with a maximum
 Hand- Hospitalizatio
 body temperature of 39.0 ■ and no vomiting. On
 First Infections and foot- n or prolonged Possibly
 2197 PCV13-TT Female 13.8 Hebei 20160519, the subject was treated in Hebei 36 9 Grade 3
 dose infestations mouth length of unrelated
 children's Hospital. Physical examination indicated
 disease hospital stay
 small herpes in both hands and feet, hyperemia of

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pharynx, and a few herpes were present in oral cavity, hands and feet, and around anus. Heart rate 112 beats / min. Enterovirus showed positive EV71 and EVM PCR, indicative of HFMD. After treatment, the subject was discharged on May 27.
 On May 18, the subject had fever for 4 hours, the highest temperature was 40.5 ■, the sudden twitching lasted for about 1 minute, and then resolved spontaneously, with intermittent cough. Hematology test indicated WBC $14.3 \times 10^9/L$, CPR 4.8 mg/L, ALT 86 μ/L , AST 63 μ/L , LDH 472 μ/L ; myocardial enzyme indicated HBDH 379 μ/L ,
 First Infections and us n or prolonged Possibly
 2209 PCV7 Male 16.6 Hebei CK 1286 μ/L , CKMB 53 μ/L ; hyperemia of 35 20 Grade 3
 dose infestations pneumo length of unrelated
 pharynx, which was treated in Zhengding County
 nia hospital stay
 People's Hospital from May 18 to May 23 during hospitalization. After symptomatic treatment, hyperemia of pharynx, with herpes, liver function and high myocardial enzyme remained high, and the subject was then transferred to the provincial children's hospital for treatment.
 On June 2, 2016, the subject had cough and fever, and was hospitalized on June 12, 2016. The main symptoms were cough for 10 days and aggravating for 1 day, with slight congestion of pharynx, with Infectio wheezing and some small or medium bubbling Hospitalizatio
 First Infections and us sounds in the lung. After admission, the chest X-ray n or prolonged Possibly
 2233 PCV13-TT Male 11.6 Hebei 52 6 Grade 3
 dose infestations pneumo showed thickened markings in both lungs, the length of unrelated
 nia subjects were given antitussive treatment as well as hospital stay
 therapies for thinning sputum and relieving asthma. The condition was improved and discharged.
 Discharge diagnosis: bronchopneumonia, June 17, 2016.
 Hospitalizatio
 Heavy breathing sounds WBC: $8.6 \times 10^9/L$, fever
 First Infections and Bronchit n or prolonged Possibly
 2411 PCV13-TT Male 33.7 Henan 39.4 ■, L: 0.138, N: 0.709, MP-IgM positive 1 5 Grade 4
 dose infestations is length of unrelated
 cough with sputum
 hospital stay
 1-6 On September 9, the subject had cough, and was month Infectio treated in the village health center with enema and Hospitalizatio
 s after Infections and us oral medicine (unspecified), while with poor n or prolonged Possibly
 2506 PCV13-TT Female 23.8 Henan 71 7 Grade 4
 last infestations pneumo response. On June 21, cough aggravated with length of unrelated
 vaccin nia sputum and occasional shortness of breath, hospital stay
 ation intramuscular "cefoperazone sulbactam" and

atomization (unspecified) were administered in Wenquan Town Hospital, while with poor response, on June 24, "bronchopneumonia" was hospitalized in county hospital. Physical examination indicated poor mental state, hyperemia of pharynx, heavy breathing sounds in both lungs, with rhonchi and crackles. WBC: $11.28 \times 10^9/L$, N: 23%, L: 66.2%, HGB: 93 g/L, indicating viral infection and mild anemia. Weakly positive MP-IgM.

On 20160624 the subject had cough, fever at $37.3^\circ C$, with sputum difficult to cough up, combined with low-grade fever at $37.3^\circ C$, the subject was treated in the individual outpatient 1-6 treatment of the village, while poor response was month Hospitalization achieved, and then the subject was admitted to s after Infections and Bronchitis or prolonged Possibly 2581 PCV7 Male 27.6 Hebei hospital on 20160709, physical examination 89 9 Grade 3 last infestations is length of unrelated indicated slight pharyngeal congestion, heavy vaccin hospital stay breathing sounds in both lungs, with mild gurgling ation in throat, chest X-ray showed thickened markings in both lungs, which was diagnosed as acute bronchitis, the subject recovered on 20160717 after symptomatic treatment and was discharged. Hospitalization On 20160722 the subject developed fever of n or prolonged $38.6^\circ C$ without obvious cause, combined with length of headache which was most significant in the hospital stay forehead, while might be relieved in standing position, the subject had projectile vomiting, and was gastric content. The subject was treated in the 1-6 neighboring village health center, but still had month recurrent fever and headache, and vomiting was not Viral s after Infections and obviously relieved, the subject was further treated Possibly 2589 PCV7 Female 63.5 Hebei encephalitis 103 17 Grade 3 last infestations in hospital on 20160723. Physical examination unrelated litis vaccin indicated hyperemia of pharynx, WBC ation $16.2 \times 10^9/L$, cerebrospinal fluid findings showed WBC $150 \times 10^6/L$, biochemical pressure 130 mmH₂O, diagnosed as viral encephalitis. After antiviral, anti-infection, reducing cranial pressure, anti-inflammatory, nutritional support for brain and organs, the subject recovered on 20160808 and was discharged.

1-6 Fever at $38.9^\circ C$, throat swelling and pain, mental month Bacteria malaise, poor appetite, yellow tongue with greasy Hospitalization Shanx s after Infections and I yellow tongue coating, floating pulse, hyperemia of n or prolonged Possibly 2752 PCV7 Male 32.1 176 10 Grade 2 i last infestations tonsillitis oral and pharyngeal cavity, hyperemia of bilateral length of unrelated vaccin s tonsils in bright red, swollen, with pus on the hospital stay ation surface.

2.7.4.2.5.2.2. Booster dose
A total of 8 cases (10 events) of SAE were observed after the booster immunization, including 3 cases in the PCV13-TT group and 4 cases in the PCV7 group of subjects aged 3-11 months (no significant intergroup difference was observed); and 1 case in PCV13-TT (2 months) group. SAEs after booster immunization were not related to the study vaccine (Table 224 to Table 229).

Table 224 Incidence of adverse events post booster dose in each PCV13-TT group

Observations	Any SAE	Related SAE	Unrelated SAE
Groups	Events	Cases	%
Events	Cases	%	Events
Cases	%	Events	Cases
%	Events	Cases	%
3-11 PCV13- 674	3-11 PCV13- 674	3-11 PCV13- 674	3-11 PCV13- 674
4 3 4 3	4 3 4 3	4 3 4 3	4 3 4 3
months TT	months TT	months TT	months TT
PCV7 684 5 4	PCV7 684 5 4	PCV7 684 5 4	PCV7 684 5 4
P 1.0000	1.0000	1.0000	1.0000
3 PCV13- 3	PCV13- 3	PCV13- 3	PCV13- 3

488 0 0 488 0 0
months TT months TT months TT
PCV7 493 3 2 PCV7 493 3 2 PCV7
P 0.4995 1.0000 0.4995
7-11 PCV13- 7-11 PCV13- 7-11 PCV13-
186 4 3 186 4 3
months TT months TT months TT
PCV7 191 2 2 PCV7 191 2 2 PCV7
P 0.6817 1.0000 0.6817
2 PCV13- 2 PCV13- 2 PCV13-
487 1 1 487 1 1
months TT months TT months TT

Table 225 Incidence of adverse events post booster dose in subjects aged 2 months
PCV13-TT (N=487)
Events Cases Incidence rate
Infections and infestations 1 1 0.21
Infectious pneumonia 1 1 0.21

Table 226 Incidence of adverse events post booster dose in subjects aged 3-11 months
PCV13-TT (N=674) PCV7 (N=684)
Incidence Incidence P
Events Cases Events Cases
rate rate
Infections and infestations 4 3 0.45 3 3 0.44 1.0000
Infectious pneumonia 2 2 0.30 2 2 0.29 1.0000
Hand-foot-mouth disease 1 1 0.15 0 0 0.00 0.4963
Mycoplasma pneumonia 1 1 0.15 0 0 0.00 0.4963
Herpangina 0 0 0.00 1 1 0.15 1.0000
Injury, poisoning and procedural
0 0 0.00 2 1 0.15 1.0000
complications
Skull base fractures 0 0 0.00 1 1 0.15 1.0000
Subcutaneous hematoma 0 0 0.00 1 1 0.15 1.0000

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Table 227 Incidence of adverse events post booster dose in subjects aged 3 months
PCV13-TT (N=488) PCV7 (N=493)
Incidence Incidence P
Events Cases Events Cases
rate rate
Infections and infestations 0 0 0.00 1 1 0.20 1.0000
Infectious pneumonia 0 0 0.00 1 1 0.20 1.0000
Injury, poisoning and procedural
0 0 0.00 2 1 0.20 1.0000
complications
Skull base fractures 0 0 0.00 1 1 0.20 1.0000
Subcutaneous hematoma 0 0 0.00 1 1 0.20 1.0000

Table 228 Incidence of adverse events post booster dose in subjects aged 7-11 months
PCV13-TT (N=186) PCV7 (N=191)
Incidence Incidence P
Events Cases Events Cases
rate rate
Infections and
4 3 1.61 2 2 1.05 0.6817
infestations
Infectious
2 2 1.08 1 1 0.52 0.6191
pneumonia
Hand-foot-mouth
1 1 0.54 0 0 0.00 0.4934
disease
Mycoplasma
1 1 0.54 0 0 0.00 0.4934
pneumonia
Herpangina 0 0 0.00 1 1 0.52 1.0000

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Table 229 Listing of serious adverse events in subjects post booster dose
Term of Days since
Study Vaccine Age System organ Duration
Gender Site adverse Description last Severity SAE leads to Relevance
No. groups (mos) classification (days)
events vaccination
The subject coughed without apparent cause
on 20170522. The cough was non-
spasmodic and paroxysmal, with sputum
difficult to cough up. The cough alternated
between severe and mild and became
apparent during night time. The subject
experienced fever on 20170603. The highest
body temperature was 39.2 and the
patterns of fever was irregular. The subject Hospitalization
Infections
Group Infectious was given oral drugs and atomization or prolonged Possibly

325 Male 2.7 Hebei and 78 6 Grade 3
 A pneumonia treatment (unspecified) while with poor length of unrelated
 infestations
 response, therefore was treated in outpatient hospital stay
 department of People's Hospital of
 Zhengding County on 20170605.
 Hematology results: WBC $10.40 \times 10^9/L$,
 CRP 2.90 mg/L. Chest X-ray results:
 thickened lung markings in both lungs, with
 scattered patchy opacities in both lung
 fields. The subject was admitted for
 "bronchopneumonia" for further treatment.
 The subject experienced headache and
 Injury,
 dizziness, with interruption of bone Hospitalization
 poisoning
 Skull base continuity in the right occipital. No or prolonged Definitely
 1497 PCV7 Male 3.2 Shanxi and 3 6 Grade 3
 fractures abnormality was observed in the residual length of unrelated
 procedural
 brain parenchyma. The brain middle was hospital stay
 complications
 structurally centered.
 Injury,
 Hospitalization
 poisoning A 4 cm×5 cm ecchymoma was observed in
 Subcutaneous or prolonged Definitely
 1497 PCV7 Male 3.2 Shanxi and the right nuchal occipital region, with 3 6 Grade 3
 hematoma length of unrelated
 procedural tenderness.
 hospital stay
 complications
 The subject experienced cough and
 expectoration on April 12th. The cough
 Hospitalization
 Infections worsened on April 15th. Physical
 Infectious or prolonged Possibly
 1536 PCV7 Female 3.3 Shanxi and examination on admission results: heavy 28 5 Grade 3
 pneumonia length of unrelated
 infestations breathing sounds in both lungs with gurgling
 hospital stay
 in chest and bubbling sound. X-ray results
 showed thickened, increased and blurred
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lung markings.
 The subject developed a fever of 38.5■ on
 November 27th due to a cold, with
 paroxysmal and productive cough, and
 occasional dyspnea. The subject was given
 Hospitalization
 Infections oral medicine treatment (unspecified) at the
 Infectious or prolonged Definitely
 1659 PCV7 Female 10.6 Henan and village clinic and the People's hospital in the 113 9 Grade 4
 pneumonia length of unrelated
 infestations county but with poor response. The cough
 hospital stay
 aggravated on November 30th and the
 subject was admitted into the People's
 Hospital of Wen County on December 2nd
 for bronchopneumonia.
 Herpes was identified on hands, feet, and hip
 with a body temperature of 39.2■. Physical
 examination on admission: The subject was
 in poor spirits. Scattered red papules were
 identified on hands, feet, and hip while
 individual herpes was identified. The
 Hospitalization
 Infections Hand-foot- diameter of the papules was approximately
 Group or prolonged Definitely
 1663 Male 11.3 Henan and mouth 2-4 mm, with thick wall. The papules were 25 5 Grade 4
 A length of unrelated
 infestations disease opaque, with turbid papular liquid.
 hospital stay
 Xerocheilia and pharyngeal cavity
 congestion were observed. A tiny herpes
 was identified in the palatopharyngeal
 arches with periphery redness. WBC:
 $8.76 \times 10^9/L$, L: 59.40%, N: 25.20%, HB:
 $113G/L$, indicative of viral infection.
 Hospitalization
 Infections Heavy breathing sounds in both lungs. Fever
 Group Mycoplasma or prolonged Definitely
 1663 Male 11.3 Henan and of 39.2■. Mycoplasma pneumonia weakly 25 5 Grade 4
 A pneumonia length of unrelated
 infestations positive.
 hospital stay
 The subject experienced fever of 38.5■
 with salivation in the mouth and crying
 while eating. Physical examination results

Hospitalization
Infections on admission: poor spirit and xerocheilia
or prolonged Definitely
1714 PCV7 Male 10.9 Henan and Herpangina and pharyngeal cavity congestion were 15 6 Grade 4
length of unrelated
infestations observed. Several tiny herpes was identified
hospital stay
in both palatopharyngeal arches with
periphery redness. Heavy breathing sounds
were identified in both lungs.
Group Infections Infectious The subject coughed without apparent cause Hospitalization Possibly
1817 Male 11.4 Hebei 85 9 Grade 3
A and pneumonia on around 20161120 (more than 10 days or prolonged unrelated

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infestations before admission on 20161201). The cough length of
was non-spasmodic and paroxysmal, with hospital stay
sputum difficult to cough up and wheezing
aggravated after exercise. The subject was
treated in the village clinic (unspecified) for
more than 10 days but with poor response.
The subject was treated in People's Hospital
of Zhengding County on 20161128.
Hematology results: WBC $16.77 \times 10^9/L$,
HGB 87G/L, CRP 8.5 MG/L, chest X-ray
results: thickened lung markings in both
lungs, with scattered patchy opacities in
both lung fields. The subject received
intravenous drip treatment at the outpatient
clinic, and the symptoms had slightly
improved.
The subject coughed without apparent cause
on 20161114. The cough was non-
spasmodic and paroxysmal, with sputum
difficult to cough up and rhinorrhea. The
highest body temperature was 39.2■ and
the patterns of fever was irregular, with
wheezing aggravated after exercise. The
subject received An'erning, cephalosporin,
pholcodine and enema at the local village Hospitalization
Infections
Group Infectious clinic for around 3 days but with poor or prolonged Possibly
1823 Male 8.8 Hebei and 75 5 Grade 3
A pneumonia response, therefore was taken to the length of unrelated
infestations
outpatient department of People's Hospital hospital stay
of Zhengding County treatment on
20161120. Hematology results: WBC
 $10.13 \times 10^9/L$, CRP 1.76 MG/L, chest X-
ray results: thickened and disordered lung
markings in both lungs. The outpatient
diagnosis was bronchopneumonia, which
was treated using cefoxitin, Reduning and
doxofylline through intravenous drip.

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2.7.4.2.6. Other important adverse events
N/A

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2.7.4.2.7. Analysis of adverse reactions by organ system or syndromes
N/A

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2.7.4.2.8. Adverse events narratives
SAE1: Bronchopneumonia (No. 190)
A 1.9-month boy, 46 days after PCV13-TT dose 2, was hospitalized for rhinorrhea, cough,
asthmatic suffocation, fever, heavy breathing sounds in both lungs, with fine crackles and stridor.
Chest X-ray: bronchopneumonia. Significant physical exam findings on hospital admission
included 37 0■. Treatment was discharged from 08 to 16 2016. This SAE is bronchial pneumonia
(asthma type).

SAE2: Bronchitis (No. 1107)
An 7-month girl, 16 days after PCV13-TT dose 1, had cough and fever with no obvious cause.
Antipyretic treatment (unspecified) was administered by village health room, followed by oral
medication and enema therapy which resulted in poor response. On 20160726, another episode of
fever occurred (as high as 40.0 ■), with frequent coughing and evident wheezing, and in poor
mental state. On 20160727 the subject was examined in Zhengding County People's Hospital,
with body temperature being 40.0 ■, poor mental state, slight shortness of breath, small bubbling
sound in both lungs, a small amount of wheezing sound, CRP 19.5 mg/L, chest X-ray showed

thickened markings and disorder in both lungs, with scattered high-density patchy opacity in both lung field, diagnosed as bronchopneumonia, and was hospitalized. On 20160801, the symptoms resolved and the subject was discharged.

SAE3: Upper respiratory and digestive tract infection (No. 2004)

A 19.4-month boy, 11 days after PCV13-TT dose 2, was hospitalized for fever with the maximum temperature of 38.5 °C. On June 27, the subject vomited. The subject vomited immediately after eating, accompanied by poor mental state. The subject hospitalized 3 days and recovered 5 days after discharge.

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2.7.4.3 Clinical laboratory evaluations

2.7.4.3.1. Phase I clinical trial

2.7.4.3.1.1. Laboratory tests

The hematology, liver and kidney function detection in this clinical trial were entrusted to the Laboratory of Shuangqiao Hospital in Chaoyang District, Beijing. See the tables below for methods used and the normal value reference range for each measurement.

Table 230 Detection method and reference range for hematology

Measurement Method Reference range

Male: 110-175 g/L

Hemoglobin (HGB) Automatic Blood Cell Analyzer

Female: 110-150 g/L

White blood cell count (WBC) Automatic Blood Cell Analyzer (3.5-9.5)×10⁹/L

Platelet count (PLT) Automatic Blood Cell Analyzer (125-350)×10⁹/L

Table 231 Detection method and reference range for liver function

Measurement Method Reference range

Male: 9.0-50.0 U/L

Alanine aminotransferase (ALT) Velocity method

Female: 7.0-40.0 U/L

Male: 15.0-40.0 U/L

Aspartate aminotransferase (AST) Velocity method

Female: 13.0-35.0 U/L

Total bilirubin (TBIL) Modified diazonium method 3.44-25.40 μmol/L

Table 232 Detection method and reference range for kidney function

Measurement Method Reference range

Male: 44.0-97.0 μmol/L

Creatinine (cr) Jaffe reaction

Female: 35.0-80.0 μmol/L

Blood urea nitrogen (BUN) Enzymatic method 2.9-8.2 mmol/L

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2.7.4.3.1.2. Serum antibody detection

All serum samples detection were entrusted to National Institutes for Food and Drug Control.

The antibody concentration of IgG in serum samples was detected using enzyme-linked immunosorbent assay (ELISA), the subjects achieving IgG antibody concentration ≥0.35 μg/mL and the geometric mean concentration (GMCs) were calculated.

Opsonophagocytic Assay (OPA) was used to detect the antibody opsonophagocytic activity of serum samples, meanwhile the subjects achieving an OPA antibody ≥1:8 and the geometric mean titers (GMTs) were calculated.

Testing standard samples and quality control standards were provided by the laboratory. The method and quality control of ELISA and OPA can be found in the standard procedure of laboratory.

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2.7.4.3.2. Phase III clinical trial

2.7.4.3.2.1. Processing, sorting, storage and transportation of blood samples

Venous blood should be left at room temperature for at least 2 hours or other measures should be taken to allow the blood samples to fully coagulate after collection. The serum should be separated within 24 hours. If the collected blood sample could not be centrifuged on the same day, it should be stored in a refrigerator at 2-8 °C. The separated serum was divided into two tubes for samples that required detection of only IgG: the testing tube 1 and the spare tube, and the sorting of serums gave priority to meet the needs of detection. For samples that needed to be tested for both IgG and OPA, the separated serum was divided into three tubes: testing tube 1, testing tube 2 and the spare tube, and the serum was collected according to the priority order of testing tube 1, testing tube 2 and the spare tube.

All sorted serums were stored at -20 °C or below and were kept frozen during the transportation.

2.7.4.3.2.2. Coding of serum samples

In principle, study numbers were used for the serum samples collected from the subjects and were labeled with different sampling time. Specific numbering standards referred to the SOP of Biological Sample Numbering in Clinical Trials.

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2.7.4.3.2.3. Blood sample detection

All serum samples detection was trusted to National Institutes for Food and Drug Control.

The antibody concentration of IgG in serum samples was detected using enzyme-linked immunosorbent assay (ELISA).

Opsonophagocytic Assay (OPA) was used to detect the antibody opsonophagocytic activity of serum samples.

Testing standard samples and quality control standards were provided by the laboratory.

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2.7.4.4. Vital Signs, physical examination, and other safety-related observations

N/A

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2.7.4.5. Safety of special population

N/A

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2.7.4.6. Post-marketing data

N/A

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2.7.5. Literature References

[1] Good Clinical Practice (GCP), National Medical Products Administration, 2003.6.4.

[2] Technical guidelines for clinical trials of vaccines (NMPA [2004] No.575) National Medical Products Administration, 2004.12.3.

[3] Guidelines for quality management of vaccine clinical trials (Trial), National Medical Products Administration, 2013.10.31.

[4] Provisions for Drug Registration, National Medical Products Administration, 2016.7.22.

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2.7.6. Synopses of Individual Studies

2.7.6.1. Phase I clinical trial

Table 233 Synopses of phase I clinical trial

Vaccine name 13-valent pneumococcal polysaccharide conjugate vaccine

Yunnan Walvax Biotechnology Co., Ltd.

Sponsor

Yuxi Walvax Biotechnology Co., Ltd

Study title Phase I clinical study of 13-valent pneumococcal polysaccharide conjugate vaccine

Clinical trial approval Approval No.: 2015L00591 Approved on April, 2015

Principal investigator

Nianming Shi, Beijing Chaoyang District Center for Disease Control and Prevention

and responsible units

Duration of clinical

From: June, 2015 To: December, 2016

trial

Number of subjects 120

Appearance inspection:

appears as milky white homogeneous suspension after shaking, comply with the regulation.

Extractable volume:

not less than 0.5 mL/vial, comply with the regulation.

pH value:

6.3 (specification: 5.5-7.5), comply with the regulation.

Aluminum phosphate content (mg/mL):

1.0 (specification:0.8-1.2), comply with the regulation.

Sodium chloride content (g/L):

Results of 8.5 (specification: 7.5-9.5), comply with the regulation.

investigational Polysaccharide content (µg /dose):

vaccine Serotype 1: 2.6; serotype 3: 2.5; serotype 4: 2.7; serotype 5: 2.6; serotype 6A: 2.8; serotype

6B: 5.5; serotype 7F: 3.0; serotype 9V: 2.7; serotype

14: 2.3; serotype 18C: 2.8; serotype 19A: 2.9; serotype 19F: 2.9; serotype 23F: 2.5.

(standard range: 2.2-3.3 for serotype 1,3, 4, 5, 6A, 7F, 9V, 14, 18C, 19A, 19F, 23F; 4.4-6.6 for 6B),

comply with the regulation.

Abnormal toxicity test:

All mice survived during the observation period, and no abnormal reaction was observed; weight gain was

observed in each mouse at the end of

observation, comply with the regulation.

All guinea pigs survived during the observation period, and no abnormal reaction was observed; weight

gain was observed in each guinea pig at

the end of observation, comply with the regulation.

Bacterial endotoxin content (EU/dose): not higher than 12.5, comply with the regulation.

Sterility test: No bacterial growth, comply with the regulation.

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Usage and dosage: 0.5 mL for

intramuscular injection

Adults aged

18 years and 1 dose

above

Batch No.: Children aged
13-valent pneumococcal polysaccharide conjugate vaccine, manufactured by Yuxi 1 dose
201511003 2-5 years
Investigational Walvax Biotechnology Co., Ltd, for injection, 0.5 mL/vial, each dose contains 2.2 µg
Expiration 2 One dose each at
vaccine of polysaccharides from pneumococcal serotype 1, 3, 4, 5, 6A, 7F, 9V, 14, 18C, 19A,
date:
19F, and 23F, and 4.4 µg polysaccharide for 6B. Infants aged 2 month 0, 2 and 4, 3
23/11/2017 months doses in total

One dose each at
Infants aged 3 month 0, 1 and 2
months months, 3 doses in
total

Safety evaluation measures:

Local reactions: administration site pain, itching, redness, swelling, induration, rash (injection site)
size and severity, and body movement
disorders.

Systemic reactions: observation of fever, allergy, headache, tiredness/fatigue, nausea/vomiting,
diarrhea, myalgia, cough and other discomfort or
Efficacy measures clinical adverse reactions.

Laboratory tests: hematology and urinalysis, hepatic and renal function correlates in blood biochemistry.

Immunogenicity evaluation measures:

Proportion of subjects with specific pneumococcal IgG antibody concentration ≥ 0.35 µg/mL (positive rate)

Specific pneumococcal IgG geometric mean concentration (GMC).

Safety:

The incidence of vaccine-related adverse events was 80.00%, 53.33%, 66.67% and 70.00% in adults,
children, 2-month-old and PCV13-TT (3
months) group, respectively. A total of 6 cases in the adult group experienced grade 3 local adverse
reactions, corresponding to an incidence rate
of 20% (6 / 30), including 1 case of local redness, 1 case of induration, 2 cases of swelling, 2 cases of
redness combined with swelling. The
incidence rate of adverse reactions with grade 3 severity in other age groups was less than 15%.

Clinical trial

The most common symptoms of systemic adverse reactions were fever in adults and children aged 2-5 years,
and were fever and diarrhea in 2-
evaluation

month-old and 3-month-old infants. Common local adverse reactions in adults included redness, pain and
swelling at the vaccination site, and
were pain and redness in children aged 2-5 years, as compared to redness in infants 2 and 3 months of
age.

Adverse events mainly occurred on Day 0-7. No adverse events related to the study vaccine were collected
on Day 8-30, and no serious adverse
events were collected during the observation period.

Immunogenicity:

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High IgG antibody positive rates and antibody levels were observed in adults and children aged 2-5 years
before immunization for all 13

serotypes. The IgG antibody positive rates of all serotypes reached 100% or over 80% after immunization,
the antibody level increased more than

2 folds after immunization. In 2-month-old and PCV13-TT (3 months) group, all the 13 serotypes were with
lower IgG antibody positive rates

and antibody levels than those in the adult and children groups before immunization, while after
immunization, the antibody positive rates all

reached 100% except for serotype 3, 5, 6A and 6B, and the fold increase of antibody was significantly
higher in the two infants groups compared
to adult group and children group.

The positive rates and antibody levels of OPA were lower than that of IgG in each age PCV7efore
immunization, while the level of OPA

antibody after immunization significantly increased with a higher fold increase compared with IgG. The
OPA positive rate of each serotype was $\geq$

90% in subjects aged 3 months after immunization.

The correlation analysis between IgG and OPA indicated high consistency between post-immunization
antibody positive rates of the two
measures, while some bias might exist due to the small sample size.

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2.7.6.2. Phase III clinical trial

Table 234 Synopses of phase III clinical trial

Vaccine name 13-valent pneumococcal polysaccharide conjugate vaccine

Sponsor Yunnan Walvax Biotechnology Co., Ltd.

Yuxi Walvax Biotechnology Co., Ltd

Study title Phase III Clinical Trial of 13-valent pneumococcal polysaccharide conjugate vaccine

Clinical trial Approval No.: 2015L00591 Approved on: April, 2015

approval

Principal Nianming Shi Beijing Chaoyang District Center for Disease Control and Prevention

investigator Yuliang Zhao Hebei Province Center for Disease Control and Prevention

and Shengli Xia Henan Province Center for Disease Control and Prevention

responsible Guohua Li Shanxi Province Center for Disease Control and Prevention

units

Duration of From: June, 2015 To: January, 2018

clinical trial

Number of 2760

subjects

Result of the National Institutes for Food and Drug Control certified that this vaccine's quality
complied with the Enterprise Registration Standard of 13-valent

investigational pneumococcal polysaccharide conjugate vaccine. Report No.: SZ201511716
vaccine
Investigational 13-valent pneumococcal polysaccharide conjugate vaccine, manufactured by Yuxi Walvax
Biotechnology Co., Ltd, Batch No.: 201511003
vaccine for injection, 0.5 mL/vial, each dose contains 2.2 µg of polysaccharide from pneumococcal
serotype 1, 3, 4, 5, 6A, Expiration date: November,
7F, 9V, 14, 18C, 19A, 19F, and 23F, and 4.4 µg polysaccharide for 6B. 2017

Comparator 7-valent pneumococcal conjugate vaccine, manufactured by Pfizer (Wyeth) Pharmaceutical Inc.,
for injection, 0.5 Batch No.: H29315H13880
vaccine mL/vial, each dose contains 2 µg of polysaccharide from pneumococcal serotypes 4, 9V, 14, 18C,
19F, 23F and 4 µg Expiration date: May, 2017
polysaccharide for 6B.
Usage and
dosage of Number of Primary series Booster dose
Age strata Groups
investigational subjects Schedule Dose(s) Schedule Dose
vaccine and PCV13-TT (3M) 520 Month 0, 1, 2 3 doses
comparator 2-6 months 12-15
PCV13-TT (2M) 520 Month 0, 2, 4 3 doses 1 dose
vaccine (≥ 6 weeks) months of age
PCV7 (3M) 520 Month 0, 1, 2 3 doses
7-11 months PCV13-TT 200 Month 0, 2 2 doses 1 dose

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12-15 months of age (≥
PCV7 200 Month 0, 2 2 doses 2 months apart from
the second dose)
PCV13-TT 200 Month 0, 2 2 doses //
12-23 months
PCV7 200 Month 0, 2 2 doses //
PCV13-TT 200 Month 0 1 dose //
24-71 months
PCV7 200 Month 0 1 dose //
Efficacy Immunogenicity:
measures ■ The percentage of subjects with vaccine-type specific IgG antibody concentration ≥0.35 µg/mL
and ≥ 1.0 µg/mL as well as the IgG geometric mean
concentration (GMC) measured 30 days after primary series.
■ The percentage of subjects with IgG vaccine-type specific IgG antibody concentration ≥0.35 µg/mL and ≥
1.0 µg/mL, as well as the IgG geometric
mean concentration (GMC) measured before and 30 days after booster dose.
■ The percentage of subjects with vaccine-type specific anti-pneumococcal OPA titer ≥1:8 and the
geometric mean titer (GMT) measured 30 days
after primary series and 30 days after booster dose.

Safety:
■ Adverse events within 30 days after full primary immunizations and within 30 days after booster dose.
Clinical trial In the main target population (3 months of age)
evaluation The common serotypes achieved noninferiority criteria for comparison between PCV13-TT and PCV7
group except for serotype 6B during primary
series, meanwhile all common serotypes in PCV13-TT group were noninferior to that in PCV7 group after the
booster dose; for additional serotypes,
superiority criteria were met for PCV13-TT group vs PCV7 group after primary series; the common serotypes
in PCV13-TT group aged 3-71 months
were all noninferior to those in PCV7 group during primary series, besides, the additional serotypes in
PCV13-TT group were all superior as compared
to PCV7 group.
For PCV13-TT (2 months) group vs PCV7 (3 months) group
Both common and additional serotypes had achieved noninferiority or superiority criteria post primary
series. For PCV13-TT (2 months) group vs
PCV13-TT (3 months) group, 12 serotypes reached noninferiority criteria during primary series except for
additional serotype 5; and all 13 serotypes
were noninferior to those in PCV13-TT (3 months) group post booster dose.
In PCV13-TT recipients aged 7-11 months, 12-23 months, and 24-71 months
The proportions of subjects with IgG antibody concentration ≥ 0.35 µg/mL for common serotypes were all
noninferior to those in PCV7 group. The
additional serotypes in PCV13-TT group of 7-11 months had reached superiority criteria after primary
series; for PCV13-TT groups of 12-23 months
and 24-71 months, superiority criteria were generally met after immunization.

The results before and after booster dose indicated decrease in the antibody levels of several serotypes
in the 13 serotypes either in PCV13-TT group or
PCV7 group to some extent, while the percentage of subjects with IgG antibody concentration ≥ 3.5 µg/mL
and antibody GMC after 1 dose of booster
immunization both reached or exceeded the levels observed after primary series, demonstrating significant
effect of the booster dose, reflecting the
immunological memory effect of the conjugate vaccine adequately. The antibody GMC post booster dose had
more than 3-fold increase in the 2-6
months olds, and greater than 1-fold increase in subjects aged 7-11 months as compared with the GMC
before booster dose.

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The overall incidence of adverse events in PCV13-TT group was lower than that in PCV7 group. No
significant intergroup difference was observed in
the incidence of vaccine-related adverse events. The main symptom of systemic adverse reaction was fever.
The incidence of systemic reactions in
PCV13-TT group was lower than that in PCV7 group. Local adverse reactions were primarily redness,
swelling and pain at the injection site. No

significant difference was observed in the incidence of local reactions between PCV13-TT group and PCV7 group.

The incidence of vaccine-related grade 3 adverse events in PCV13-TT group and PCV7 group was 4.21% and 2.41%, respectively. The incidence of local adverse reactions with severity reaching grade 3 in PCV13-TT group was higher than that in PCV7 group, while no significant intergroup difference was observed in subjects aged 3 months.

A total of 80 cases (105 events) of SAEs were observed during primary series and 8 cases (10 events) of SAEs were noted during booster dose. No significant intergroup difference was observed in the incidence of SAEs. SAEs were not related to the study vaccine except for one case in PCV13-TT (3 months) group with fever determined to be probably related to the study vaccine.

To sum up, the immunogenicity of 7 common serotypes in main target population aged 3 months met the non-inferiority criteria after booster dose, and that of 6 new serotypes had already achieved the standard of superiority after primary immunizations in this clinical trial. The common and additional serotypes had fulfilled the non-inferiority or superiority standard in subjects 3-71 months of age after primary immunizations. The common and additional serotypes in PCV13-TT group (2 months) were noninferior and/or superior to that in PCV7 (3 months) group. In PCV13-TT (2 months) group, all 13 serotypes were noninferior to those in the PCV13-TT (3 months) group post booster dose. The incidence of adverse events/reactions in PCV13-TT group was not higher than that in PCV7 group. The results of immunogenicity and safety evaluation in this clinical trial had achieved the study objectives predefined in the protocol.

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PRODUCT REGIDTRATION DOSSIER

13-valent Pneumococcal Polysaccharide Conjugate Vaccine
(Tetanus Toxoid Protein)

2.3 QUALITY OVERALL SUMMARY

Yuxi Walvax Biotechnology Co., Ltd.

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2.3 QUALITY OVERALL SUMMARY

2.3.S DRUG SUBSTANCE

2.3.S.1 General Information

1. Streptococcus Pneumoniae serotype 1 was obtained from CMCC (No. 31404)
For pneumococcal polysaccharide serotype 1:
Chemical Name: Serotype 1 Polysaccharide consists of a trisaccharide repeating subunit of -4-(D-galacturonopyranosyl-1-α)-3-(D-galacturonopyranosyl-1-α)-3-(2-acetamido-4-amino-2,4,6-trideoxy-D-galactopyranosyl-1-α) residues. The galacturonopyranosyl residues are partially O-acetylated. Each mole of the saccharide repeat unit carries a non-stoichiometric amount of O-acetyl groups per repeating unit.

* O
4 6 COOH
O HO HOOC
6 O
5 4 H2N H3C 6
HO 2 5
2 O
4 2
1 O 5 1
3 OH 3 OH 1 O 3 NHAC
*

n
Molecular size: determined by gel filtration through Sepharose CL-4B, release specification is
KD ≤ 0.15, recovery rate ≥ 75% before KD value reached 0.3.

2. Streptococcus Pneumoniae serotype 3 was obtained from CMCC (No. 31441)

For pneumococcal polysaccharide serotype 3:

Chemical Name: Serotype 3 Polysaccharide consists of a disaccharide repeating subunit of -3-(D-glucuronopyranosyl-1-β)-4-(D-glucopyranosyl-1-β) residues.

OH
4 HOOC 6
6 O 4
O
HO 2 2
* 5
O 5 *
O 3 1 HO
OH 3 1
OH
n

Molecular size: determined by gel filtration through Sepharose CL-4B, release specification is
KD ≤ 0.15, recovery rate ≥ 78% before KD value reached 0.3.

3. Streptococcus Pneumoniae serotype 4 was obtained from CMCC (No. 31448)

For pneumococcal polysaccharide serotype 4:

Chemical Name: Serotype 4 Polysaccharide consists of a repeating subunit of -3-(D-2-acetamido-2-deoxy-D-mannopyranosyl-1-β)-3-(L-2-acetamido-2-deoxy-D-fucopyranosyl-1-α)-3-(D-2-acetamido-2-deoxy-D-galactopyranosyl-1-α)-4-(D-2,3-S-pyruvyl-galactopyranosyl-1-α) residues.

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HO
4 6 OH 5
H3C
4 OH
HO O 4 3 1
O 6 O
* 5 NHAc O
HO 5
O O NHAc 2 1
3 2 1 2 3 NHAC
O OH
6
4
O
2
5
O 1
3
O
*
H3C n
COOH

Molecular size: determined by gel filtration through Sepharose CL-4B, release specification is
KD ≤ 0.15, recovery rate ≥ 73% before KD value reached 0.3.

4. Streptococcus Pneumoniae serotype 5 was obtained from CMCC (No. 31457)

For pneumococcal polysaccharide serotype 5:

Chemical Name: Serotype 5 Polysaccharide consists of a penta-saccharide repeating subunit of -4-(D-glucopyranosyl-1-β)-4-[L-2-acetamido-2-deoxy-3-(2-acetamido-2,6-dideoxy-L-talopyranosyl-1-α-2-D-glucuronopyranosyl-1-β)-fucopyranosyl-1-α]-3-(D-2-acetamido-2,6-dideoxy-D-xylo-hexopyranos-4-ulosyl-1-β) residues.

6 4
H3C 5
OH 6 CH3
4 1 O
4 6
O O O
2 3
* O 5 O O
HO O 2 NHAc 2 1
3 OH 1 3
NHAc
O
HOOC 6 2
1
*
4 5 n
3
O
CH3
HO 1 3
2

NHAC 5 OH
O
6 4
HO
H3C

Molecular size: determined by gel filtration through Sepharose CL-2B, release specification is
KD ≤ 0.60, recovery rate ≥ 70% before KD value reached 0.6.

5. Streptococcus Pneumoniae serotype 6A was obtained from CMCC (No. 31476)
For pneumococcal polysaccharide serotype 6A:
Chemical Name: Serotype 6A Polysaccharide consists of a repeating subunit of -2-(D-galactopyranosyl-1- α)-3-(D-glucopyranosyl-1- α)-3-(L-rhamnopyranosyl-1- α)-3-(D-ribitol-1-phosphate) residues.

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HO
4 6
OH

HO O
5 2
3
4 OH
HO 6
1
5 O
2 OH
O O
* 3 H3C
4 6
OH 1
5 O P O *
O 3
O OH HO
OH 1
2 O
n
OH

OH

Molecular size: determined by gel filtration through Sepharose CL-2B, release specification is
KD $\leq$ 0.50, recovery rate $\geq$ 50% before KD value reached 0.6.

6. Streptococcus Pneumoniae serotype 6B was obtained from CMCC (No. 31492)
For pneumococcal polysaccharide serotype 6B:
Chemical Name: Serotype 6B Polysaccharide consists of a repeating subunit of -2-(D-galactopyranosyl-1- α)-3-(D-glucopyranosyl-1- α)-3-(L-rhamnopyranosyl-1- α)-4-(D-ribitol-5-phosphate) residues.
HO

4
OH
6
HO O
5 2
3
4 OH
6
1 HO
5 O
2 OH
O
* 3 H3C
4 6
OH 1
5 O
O 3
O O P O *
OH 1
2
O HO
n
OH
OH

OH

Molecular size: determined by gel filtration through Sepharose CL-2B, release specification is
KD $\leq$ 0.50, recovery rate $\geq$ 85% before KD value reached 0.6.

7. Streptococcus Pneumoniae serotype 7F was obtained from CMCC (No. 31510)
For pneumococcal polysaccharide serotype 7F:
Chemical Name: Serotype 7F Polysaccharide consists of a branched repeating subunit of -6-[D-2-(galactopyranosyl-1- β)-galactopyranosyl-1- α]-3-(L-2-O-acetyl-rhamnopyranosyl-1- β)-4-(D-glucopyranosyl-1- β)-3-[D-2-acetamido-2-deoxy-4-(2-acetamido-2-deoxyglucopyranosyl-1- α)-2-(L-rhamnopyranosyl-1- β)-galactopyranosyl-1- β] residues.

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OH
OH
HO H3C
O
HO HO
HO
O H3C NHAc O

O
* HO O
O OH
O O O
O OH *
HO
O CH4
AcO
O NHAc
O O
HO OH HO OH
HO

HO

n

Molecular size: determined by gel filtration through Sepharose CL-4B, release specification is
KD ≤ 0.20, recovery rate ≥ 70% before KD value reached 0.3.

8. Streptococcus Pneumoniae serotype 9V was obtained from CMCC (No. 31547)

For pneumococcal polysaccharide serotype 9V:

Chemical Name: Serotype 9V Polysaccharide consists of a repeating subunit of -4-(D-glucopyranosyl-1-β)-4-(D-glucuronopyranosyl-1-α)-3-(D-galactopyranosyl-1-α)-3-(D-2-acetamido-2-deoxy-mannopyranosyl-1-β)-4-(D-glucopyranosyl-1-β) residues.

OH
O
* O
AcO HOOC
OH
O HO
O OH
O OAc
AcO O OH
OH *
HO NHAC O
O O
OH
O
HO OH
n

Molecular size: determined by gel filtration through Sepharose CL-2B, release specification is
KD ≤ 0.45, recovery rate ≥ 80% before KD value reached 0.6.

9. Streptococcus Pneumoniae serotype 14 was obtained from CMCC (No. 31605)

For pneumococcal polysaccharide serotype 14:

Chemical Name: Serotype 14 Polysaccharide consists of a repeating subunit of -4-(D-glucopyranosyl-1-β)-6-[D-2-acetamido-2-deoxy-4-(D-galactopyranosyl-1-β)-glucopyranosyl-1-β]-3-(D-galactopyranosyl-1-β) residues.

HO
OH
OH
O
O O *

* O
O
HO O OH
HO OH
OH
O O NHAc
HO
HO OH

n

Molecular size: determined by gel filtration through Sepharose CL-2B, release specification is
KD ≤ 0.30, recovery rate ≥ 55% before KD value reached 0.3.

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10. Streptococcus Pneumoniae serotype 18C was obtained from CMCC (No. 31689)

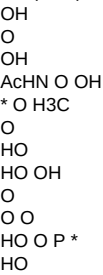
For pneumococcal polysaccharide serotype 18C:

Chemical Name: Serotype 18C Polysaccharide consists of a repeating subunit of -4-(D-glucopyranosyl-1-β)-4-[(2-glucopyranosyl-1-α-3-glycerophospho)-D-galactopyranosyl-1-β]-4-(D-glucopyranosyl-1-α)-3-(L-rhamnopyranosyl-1-α) residues.

O OH
O
O OH
* HO OH OH
O O O H3C
OH
O HO
HO
O OH OH O
P
OAc O
O O OH
O
HO OH
*
OH

HO
n
Molecular size: determined by gel filtration through Sepharose CL-4B, release specification is
KD ≤ 0.15, recovery rate ≥ 70% before KD value reached 0.3.

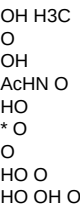
11. Streptococcus Pneumoniae serotype 19A was obtained from CMCC (No. 31708)
For pneumococcal polysaccharide serotype 19A:
Chemical Name: Serotype 19A Polysaccharide consists of a repeating subunit of -4-(D-2-acetamido-2-deoxy-D-mannopyranosyl-1-β)-4-(D-glucopyranosyl-1-α)-3-(L-rhamnopyranosyl-1-α-phosphate) residues.



n
Molecular size: determined by gel filtration through Sepharose CL-4B, release specification is
KD ≤ 0.45, recovery rate ≥ 80% before KD value reached 0.5.

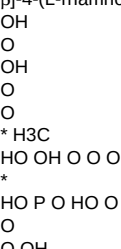
12. Streptococcus Pneumoniae serotype 19F was obtained from CMCC (No. 31696)
For pneumococcal polysaccharide serotype 19F:
Chemical Name: Serotype 19F Polysaccharide consists of a repeating subunit of -4-(D-2-acetamido-2-deoxy-D-mannopyranosyl-1-β)-4-(D-glucopyranosyl-1-α)-2-(L-rhamnopyranosyl-1-α-phosphate) residues.

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OH



n
Molecular size: determined by gel filtration through Sepharose CL-4B, release specification is
KD ≤ 0.20, recovery rate ≥ 75% before KD value reached 0.3.

13. Streptococcus Pneumoniae serotype 23F was obtained from CMCC (No. 31761)
For pneumococcal polysaccharide serotype 23F:
Chemical Name: Serotype 23F Polysaccharide consists of a repeating subunit of -4-(D-glucopyranosyl-1-β)-4-[D-2-(L-rhamnopyranosyl-1-α)-3-(glycerophospho)-galactopyranosyl-1-β]-4-(L-rhamnopyranosyl-1-β) residues.



HO n
Molecular size: determined by gel filtration through Sepharose CL-4B, release specification is
KD ≤ 0.15, recovery rate ≥ 85% before KD value reached 0.3.

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2.3.S.2 Manufacture
2.3.S.2.1 Manufacturer(s)

Name: Yuxi Walvax Biotechnology Co., Ltd.
Address: No. 83 South Dongfeng Road, High ■ New Technology Industries Development Zone, Yuxi, Yunnan Province, China
Function: Vaccine manufacture (All manufacturing procedures)
Tel: 0877-2076210
Fax: 0877-2076918
Manufacturing site: 4th floor, pneumococcal vaccine workshop, J building 301# (vaccine building); 1st floor, Tetanus vaccine workshop, K building 201#, No. 83 South Dongfeng Road, High ■ New Technology Industries Development Zone, Yuxi, Yunnan Province, China
Tel: 0877-2076210
Fax: 0877-2076918
QC site: 3rd and 4th floor, QC laboratory, G building 308# (Research & QC building), No. 83 South Dongfeng Road, High ■ New Technology Industries Development Zone, Yuxi, Yunnan Province, China
Tel: 0877-2076210
Fax: 0877-2076918

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2.3.S.2.2 Description of Manufacturing Process and Process Controls
2.3.S.2.2.1 Flow Chart of Manufacturing Process and Process Control

Lyophilized WB thawed with media

Cultivate for 20±4 h

Inoculate to solid medium for cultivating the 1st generation (1 Kolle flask)

Cultivate for 20±4 h

Inoculate to solid medium for cultivating the 2nd generation (5 Kolle flasks)

Bacterial colony is washed and collected for fermentation

Go into fermentation

Figure 1 Manufacturing Process Flow Chart for Pneumococcal Strain Recovery

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In-process control items:
Inoculate collected liquid to OD694
10 L cell tank (3-6 L) Purity test
Gram stain
Cultivate for 3-10 h

In-process control items:
Inoculate to 100 L OD694
amplification tank (30-60 L) Purity test
Gram stain
Cultivate for ≤ 6 h

In-process control items:
Inoculate to 900L OD694
fermentation tank (300-600 L) Purity test
Gram stain
Cultivate for 2-8 h

Killing bacteria with 10%
DOC for 30 min

Harvesting fermentation broth

Centrifuge for 15 min in 7000 rpm & collect supernatant

UF concentration of supernatant
10±2 fold, 100 kDa

1st Ethanol precipitation
(Concentration: 17-50%)
Centrifuge for 60 min in 10000 rpm & collect supernatant

2nd Ethanol precipitation
(Concentration: 30-78%)
Centrifuge for 15 min in 4000 rpm & collect precipitate

Precipitate washed with ethanol / acetone / ether

Go into

Pump dry & collect crude polysaccharide
polysaccharide
(Store at -20 ■) purification
Figure 2 Manufacturing Process Flow Chart for Polysaccharide Preparation Process I:
Crude Polysaccharide Preparation

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Crude polysaccharide dissolution

Cold phenol extraction
(3-7 times)

Centrifuge for 40 min in 10000
rpm & collect supernatant

UF/DF to remove phenol
MWCO: 50/100 kDa

1st Ethanol precipitation
(Concentration: 15-30%)

Centrifuge for 60 min in 10000
rpm & collect supernatant

UF/DF to remove nucleic acid
MWCO: 30/50/100 kDa

2nd Ethanol precipitation
(Concentration: 40-80%)

Centrifuge for 15 min in 4000 In-process control items:
rpm & collect precipitate Identification, Total solid, Protein
content, Nucleic acid content, Total
Precipitate washed with ethanol / nitrogen content, Phosphorus
acetone / ether content, Molecular weight, Residual
ethanol content, Residual ether
content, Residual acetone content,
Bacterial endotoxin test, Uronic acid
content, O-Acetyl content,
Pump dry & collect refined
Hexosamine content, Methyl
polysaccharide pentose content, Sodium
(Store at -20 ■) deoxycholate

Go into conjugation

Figure 3 Manufacturing Process Flow Chart for Polysaccharide Preparation Process II:
Refined Polysaccharide Preparation (Serotype 1, 2, 4, 5, 6A, 6B, 7F, 9N, 9V, 10A, 12F, 14,
17F, 18C, 19A, 19F, 22F, 23F, 33F)

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Crude polysaccharide dissolution

Cold phenol extraction
(3-7 times)

Centrifuge for 40 min in 10000
rpm & collect supernatant

UF/DF to remove phenol
MWCO: 50/100 kDa

CTAB precipitation
(Concentration: 2%)
Centrifuge for 40 min in 10000
rpm & collect precipitate

Re-dissolving with NaCl

Centrifuge for 30 min in 10000
rpm & collect supernatant

Ethanol precipitation
(Concentration: 18-33.3%)
In-process control items:
Centrifuge for 15 min in 4000 Identification, Total solid,
rpm & collect precipitate Protein content, Nucleic acid
content, Total nitrogen content,
Precipitate washed with ethanol / Phosphorus content, Molecular
acetone / ether weight, Residual ethanol
content, Residual ether content,
Residual acetone content,
Pump dry & collect refined Bacterial endotoxin test, Uronic
acid content, Sodium
polysaccharide
deoxycholate
(Store at -20 ■)

Go into conjugation

Figure 4 Manufacturing Process Flow Chart for Polysaccharide Preparation Process II:
Refined Polysaccharide Preparation (Serotype 3)

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Strains open

In-process control items:

Cultivation of the 1st generation Microscopy

Cultivate for 48±2 h

at 34.5±1 ■

In-process control items:

Cultivation of the 2nd generation Microscopy

Cultivate for 48±2 h

at 34.5±1 ■

In-process control items:

Cultivation of the 3rd generation Microscopy

Cultivate for 48±2 h

at 34.5±1 ■

In-process control items:

Fermentation & Cultivation Microscopy,

Flocculent unit ≥ 40 Lf/mL

Cultivate for 60-72 h

at 34.5±1 ■

Killing bacteria with

formaldehyde solution for 30 min

Go into purification

Figure 5 Manufacturing Process Flow Chart for Clostridium Tetani Fermentation

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Harvesting Add formaldehyde

fermentation broth solution

Centrifuge for 15

min in 7000 rpm & Sterile filtration

collect supernatant

(0.45+0.22 μm)

Filtration

(0.45 μm)

Detoxification

(37±1 ■)

UF concentration

(1/8-1/10 fold)

Filtration (0.45 μm)

Centrifuge for 15 & UF to remove

min in 10000 rpm & formaldehyde

collect supernatant

Filtration

(0.22 μm) Add NaCl & adjust

pH with NaHCO₃

Adjust pH with Na₂CO₃

1st (NH₄)₂SO₄ Sterile filtration

fractionation (0.45+0.22 μm)

(46-48 h) In-process control items:

Sterility test,

Centrifuge for 20

Bulk tetanus toxoid detoxification test,

min in 10000rpm &

stored at 2-8 ■ abnormal toxicity, toxicity

collect supernatant

reversion, flocculent unit,

Filtration protein content, purity,

(0.22 μm) Filtration (0.45 μm) & appearance, pH

UF to final concentration

2nd (NH₄)₂SO₄ at 50-100 mg/mL

fractionation

(18-24 h)

Filtration (0.45 μm)

Centrifuge for 15

min in 10000 rpm &

collect precipitate Column chromatography

Precipitate dissolution

& filtration

Concentration & sterile

filtration (0.45+0.22 μm)

UF to remove

ammonium In-process control items:

(15 times) Purified liquid of tetanus Protein content, sterility,

bacterial endotoxin, purity
Centrifuge for 15 toxoid stored at 2-8 ■
min in 10000 rpm &
collect precipitate
Add NaCl & adjust Go into conjugation
pH with NaHCO₃
Figure 6 Manufacturing Process Flow Chart for Purification of Tetanus Toxoid Carrier
Protein

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Refined polysaccharide dissolution
Concentration: 3-8 mg/mL

Polysaccharide activation with CNBR or
CDAP & derivation with ADH

Polysaccharide derivatives filtration
In-process control
items:
Total solid
Derivation rate
Derivatives UF to remove cyanide and ADH
Residual cyanide
MWCO: 10 kDa Molecular weight

Conjugation with tetanus toxoid
(TT: 0.1-1.3 mg/mg polysaccharide
EDAC: 1.5-10.1 mg/mL)

Reaction condition: 2-6
h, 2-8 ■, pH at 5-6.

Polysaccharide conjugate
(Store at 2-8 ■)

Conjugate dialysis
NMT 8 h

In-process control items:
Identification, Protein content,
Conjugate purified with column chromatography Polysaccharide content,
Molecular weight,
Appearance, Bacterial
endotoxin test, Free
Final sterile filtration polysaccharide content, Free
protein content,
Polysaccharide/protein ratio,
Residual ADH content,
Residual EDAC content,
Monovalent bulk conjugate Sterility test

Figure 7 Manufacturing Process Flow Chart and Quality Control for Monovalent Bulk
Conjugate

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2.3.S.2.2.2 Brief Description of Manufacturing Process
Walvax carries out production in accordance with the Drug Registration Standard for 13-valent
Pneumococcal Conjugate Vaccine (Standard No.: YBS00392019) approved by the National
Medical Products Administration.
The manufacturing process includes the following process: establishment and passage of the seed
batches, amplification of strain and large tank cultivation, sterilization of bacteria and harvest of
fermentation liquor, preparation of the crude polysaccharide and refined polysaccharide,
polysaccharide derivatization, polysaccharide conjugation with tetanus toxoid carrier protein.
After purification, sterilization and filtration, the conjugate bulk was obtained and stored at 2-8■.
The preparation of tetanus toxoid protein carrier includes the establishment and passage of seed
batches, amplification of bacteria and large tank fermentation, sterilization and harvest of
fermentation liquid after culture, ultrafiltration, two-stage salting out and formaldehyde
detoxification to obtain tetanus toxoid original liquid, and further purification to obtain the purified
tetanus toxoid liquid.

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2.3.S.2.3 Quality Control of Materials
2.3.S.2.3.1 Strain for Production
(1) Source of Strain
The strains used in the manufacturing are pneumococcus serotype 1, 3, 4, 5, 6A, 6B, 7F, 9V, 14,
18C, 19A, 19F and 23F strains, which are purchased from National Centre for Medical Culture
Collections of NIFDC.

Table 1 The name and CMCC code of pneumococcal strains
Serotypes Strain Name CMCC Code
1 Streptococcus pneumoniae serotype 1 31404
3 Streptococcus pneumoniae serotype 3 31441
4 Streptococcus pneumoniae serotype 4 31448

5 Streptococcus pneumoniae serotype 5 31457
6A Streptococcus pneumoniae serotype 6A 31476
6B Streptococcus pneumoniae serotype 6B 31492
7F Streptococcus pneumoniae serotype 7F 31510
9V Streptococcus pneumoniae serotype 9V 31547
14 Streptococcus pneumoniae serotype 14 31605
18C Streptococcus pneumoniae serotype 18C 31689
19A Streptococcus pneumoniae serotype 19A 31708
19F Streptococcus pneumoniae serotype 19F 31696
23F Streptococcus pneumoniae serotype 23F 31761

Clostridium tetani (CMCC 64008), which is used for producing the tetanus toxoid carrier protein, was purchased from National Centre for Medical Culture Collections of NIFDC.

(2) Establishment of Seed Lot
According to the Chinese Pharmacopoeia, Walvax established the Bacteria Management procedure (WSYX-QA-SMP-00-05-0001) for the bacteria bank establishment and the three-level seed bank management. The three-level seed bank includes original seed lot, master seed lot and working seed lot. The original seed is lyophilized after culturing two generations, as well as the working seed lot. The working seed lot is used for vaccine production. The biological characteristics of the working seed lot is consistent with the original seed lot, and each seed lot is stored below -60 ■.

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Original Master seed Working
seed (the 0th (the 2nd seed (the 4th
generation) generation) generation)

Strain open Strain open
Inoculated on solid Inoculated on solid
medium medium
Culture & Culture &
amplification at amplification at
37 ■ 37 ■
Sub-cultivate on solid Sub-cultivate on solid
medium medium
Culture &
amplification at Culture &
37 ■ amplification at 37 ■

Collection of Collection of
the bacteria the bacteria

Packaging Packaging

Lyophilization Lyophilization

Seal Seal

Test Test

Figure 8 Establishment Flow Chart of Pneumococcal Three-level Seed Bank

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Table 2 Three-level Seed Bank Information of 13 Serotypes of Pneumococcus												
Serotype	Generatio											
1 3 4 5 6A 6B 7F 9V 14 18C 19A 19F 23F												
Strain Info.	n											
CMCC												
31404 31441 31448 31457 31476 31492 31510 31547 31605 31689 31708 31696 31761												
Code												
Date of												
Original Purchas	2008.10 2009.04 2003.09 2008.10 2013.06 2003.09 2009.06 2003.09 2003.09 2003.09 2009.06											
2003.09 2003.09 0												
e												
Quantit												
1 1 1 1 1 1 1 1 1 1 1 1 1												
y (vial)												
2008102 2009051 2004040 2008101 2013060 2004042 2009062 2004050 2004051 2004052 2009072 2004060 2004061												
Lot No.												
8 0 1 5 8 3 1 3 4 4 6 2 5												
Master 2nd												
Quantit												
30 40 30 30 40 30 40 30 30 40 30 30												
y (vial)												
2008110 2009051 2004040 2008102 2013061 2004043 2009062 2004051 2004052 2004053 2009072 2004060 2004062												
Lot No.												
Workin 2 3 8 0 3 0 4 0 1 1 9 2												
4th												
g Quantit												
100 50 100 100 50 100 50 100 100 100 50 100 100												
y (vial)												

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Original seed (the Master seed (the Working seed (the 0th generation) 2nd generation) 4th generation)

Strain open Strain open
Inoculated on semi-solid Inoculated on semi-solid medium medium
Culture & Culture & amplification at amplification at 34.5±1 ■ 34.5±1 ■
Inoculated on liquid Inoculated on liquid medium medium
Culture & Culture & amplification at amplification at 34.5±1 ■ 34.5±1 ■

Collection of Collection of the bacteria the bacteria

Packaging Packaging

Lyophilization Lyophilization

Seal Seal

Test Test

Figure 9 Establishment Flow Chart of Clostridium Tetani Three-level Seed Bank

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Table 3 Three-level Seed Bank Information of Clostridium Tetani Strain
Clostridium tetani strain info. Generation
CMCC Code 64008
Original Date of Purchase 2003.06.27 0th
Quantity (Vial) 3
Lot No. 200403001
Master 2nd
Quantity (Vial) 16
Lot No. 20040320
Working 4th
Quantity (Vial) 38

(3) Strain detection
Based on Bergey's Manual of Determinative Bacteriology (the 8th Edition), Testing Methods for Streptococcus Pneumoniae on Experimental Animals (GB/T 14926.15-2001), and Vaccines (2004 Edition), Walvax established testing items and standards for the Streptococcus Pneumoniae strains and Clostridium tetani strains.
According to the test items and specifications in the Table 4, the three-level seed lot of 13 types Streptococcus Pneumoniae were tested. The results showed from the Table 5 to Table 17, which indicate that the master seed lot and working seed lot have the same biological, biochemical and serological characteristics as the original seed lot.

Table 4 Testing Items and Standards of Pneumococcal Strains
Test items Specifications
No growth on the common nutrient agar slant and good growth on the 10% Culture and Staining sheep blood agar slant. Colonies isolated and cultured on blood agar appear Microscopy grey and translucent, with neat edges and smooth, moist surfaces. Alpha hemolytic, G+ aureus, a double ball or short chain arrangement.
Glucose Positive
Raffinose Positive
Biochemical Melibiose Positive
Reactions
Inulin Positive
Sorbitol Negative
Serum Agglutination Agglutination with corresponding serum
Bile Salt Dissolution Test Positive
Aupu Tosin Test Positive
Capsule Swelling Test Capsule observable

Table 5 Test Results for Three-level Seed Lot of Serotype 1 Pneumococcal Strain
Original seed Master seed Working seed
Test items Standards
lot lot lot
No growth on the common nutrient agar
Culture characteristic slant and good growth Comply Comply Comply
on the 10% sheep blood agar slant.
Colonies isolated and cultured on blood agar
Colonial morphology Comply Comply Comply
appear grey and translucent, with neat

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edges and smooth, moist

surfaces, α hemolytic
G+ aureus, a double ball
Gram staining or short chain
arrangement
Capsule staining Capsule observable
Bile Salt Dissolution Test + + + +
Apu Tosin Test + + + +
Glucose + + + +
Raffinose + + + +
Biochemical
Melibiose + + + +
reaction
Inulin + + + +
Sorbitol - - - -
Agglutination Agglutination Agglutination
Agglutination with
Serum Agglutination with serotype 1 with serotype 1 with serotype 1
corresponding serum
serum serum serum
Meet the Meet the Meet the
Conclusion
requirement requirement requirement

Table 6 Test Results for Three-level Seed Lot of Serotype 3 Pneumococcal Strain
Original seed Master seed Working seed
Test items Standards
lot lot lot
No growth on the
common nutrient agar
Culture characteristic slant and good growth
on the 10% sheep blood
agar slant.
Colonies isolated and
cultured on blood agar
appear grey and
Colonial morphology
translucent, with neat
edges and smooth, moist
surfaces, α hemolytic
G+ aureus, a double ball
Gram staining or short chain
arrangement
Capsule staining Capsule observable
Bile Salt Dissolution Test + + + +
Apu Tosin Test + + + +
Glucose + + + +
Raffinose + + + +
Biochemical
Melibiose + + + +
reaction
Inulin + + + +
Sorbitol - - - -
Agglutination Agglutination Agglutination
Agglutination with
Serum Agglutination with serotype 3 with serotype 3 with serotype 3
corresponding serum
serum serum serum
Meet the Meet the Meet the
Conclusion
requirement requirement requirement

Table 7 Test Results for Three-level Seed Lot of Serotype 4 Pneumococcal Strain
Original seed Master seed Working seed
Test items Standards
lot lot lot
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No growth on the
common nutrient agar
Culture characteristic slant and good growth
on the 10% sheep blood
agar slant.
Colonies isolated and
cultured on blood agar
appear grey and
Colonial morphology
translucent, with neat
edges and smooth, moist
surfaces, α hemolytic
G+ aureus, a double ball
Gram staining or short chain
arrangement
Capsule staining Capsule observable
Bile Salt Dissolution Test + + + +
Apu Tosin Test + + + +
Glucose + + + +
Raffinose + + + +
Biochemical
Melibiose + + + +
reaction

Inulin + + + +
Sorbitol - - - -
Agglutination Agglutination Agglutination
Agglutination with
Serum Agglutination with serotype 4 with serotype 4 with serotype 4
corresponding serum
serum serum serum
Meet the Meet the Meet the
Conclusion
requirement requirement requirement

Table 8 Test Results for Three-level Seed Lot of Serotype 5 Pneumococcal Strain
Original seed Master seed Working seed
Test items Standards
lot lot lot
No growth on the
common nutrient agar
Culture characteristic slant and good growth Comply Comply Comply
on the 10% sheep blood
agar slant.
Colonies isolated and
cultured on blood agar
appear grey and
Colonial morphology Comply Comply Comply
translucent, with neat
edges and smooth, moist
surfaces, α hemolytic
G+ aureus, a double ball
Gram staining or short chain Comply Comply Comply
arrangement
Capsule staining Capsule observable Comply Comply Comply
Bile Salt Dissolution Test + + + +
Aupu Tosin Test + + + +
Glucose + + + +
Raffinose + + + +
Biochemical
Melibiose + + + +
reaction
Inulin + + + +
Sorbitol - - - -

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Agglutination Agglutination Agglutination
Agglutination with
Serum Agglutination with serotype 5 with serotype 5 with serotype 5
corresponding serum
serum serum serum
Meet the Meet the Meet the
Conclusion
requirement requirement requirement

Table 9 Test Results for Three-level Seed Lot of Serotype 6A Pneumococcal Strain
Original seed Master seed Working seed
Test items Standards
lot lot lot
No growth on the
common nutrient agar
Culture characteristic slant and good growth Comply Comply Comply
on the 10% sheep blood
agar slant.
Colonies isolated and
cultured on blood agar
appear grey and
Colonial morphology translucent, with neat Comply Comply Comply
edges and smooth,
moist surfaces, α
hemolytic
G+ aureus, a double
Gram staining ball or short chain Comply Comply Comply
arrangement
Capsule staining Capsule observable Comply Comply Comply
Bile Salt Dissolution Test + + + +
Aupu Tosin Test + + + +
Glucose + + + +
Raffinose + + + +
Biochemical
Melibiose + + + +
reaction
Inulin + + + +
Sorbitol - - - -
Agglutination Agglutination Agglutination
Agglutination with
Serum Agglutination with serotype with serotype with serotype
corresponding serum
6A serum 6A serum 6A serum
Meet the Meet the Meet the
Conclusion
requirement requirement requirement

Table 10 Test Results for Three-level Seed Lot of Serotype 6B Pneumococcal Strain
Original seed Master seed Working seed
Test items Standards
lot lot lot
No growth on the
common nutrient agar
Culture characteristic slant and good growth Comply Comply Comply
on the 10% sheep blood
agar slant.
Colonies isolated and
cultured on blood agar
appear grey and
Colonial morphology translucent, with neat Comply Comply Comply
edges and smooth,
moist surfaces, α
hemolytic
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G+ aureus, a double
Gram staining ball or short chain Comply Comply Comply
arrangement
Capsule staining Capsule observable Comply Comply Comply
Bile Salt Dissolution Test + + + +
Aupu Tosin Test + + + +
Glucose + + + +
Raffinose + + + +
Biochemical
Melibiose + + + +
reaction
Inulin + + + +
Sorbitol - - - -
Agglutination Agglutination Agglutination
Agglutination with
Serum Agglutination with serotype with serotype with serotype
corresponding serum
6B serum 6B serum 6B serum
Meet the Meet the Meet the
Conclusion
requirement requirement requirement

Table 11 Test Results for Three-level Seed Lot of Serotype 7F Pneumococcal Strain
Original seed Master seed Working seed
Test items Standards
lot lot lot
No growth on the
common nutrient agar
Culture characteristic slant and good growth Comply Comply Comply
on the 10% sheep blood
agar slant.
Colonies isolated and
cultured on blood agar
appear grey and
Colonial morphology translucent, with neat Comply Comply Comply
edges and smooth,
moist surfaces, α
hemolytic
G+ aureus, a double
Gram staining ball or short chain Comply Comply Comply
arrangement
Capsule staining Capsule observable Comply Comply Comply
Bile Salt Dissolution Test + + + +
Aupu Tosin Test + + + +
Glucose + + + +
Raffinose + + + +
Biochemical
Melibiose + + + +
reaction
Inulin + + + +
Sorbitol - - - -
Agglutination Agglutination Agglutination
Agglutination with
Serum Agglutination with serotype with serotype with serotype
corresponding serum
7F serum 7F serum 7F serum
Meet the Meet the Meet the
Conclusion
requirement requirement requirement

Table 12 Test Results for Three-level Seed Lot of Serotype 9V Pneumococcal Strain
Original seed Master seed Working seed
Test items Standards
lot lot lot

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No growth on the
common nutrient agar
Culture characteristic slant and good growth Comply Comply Comply

on the 10% sheep blood agar slant.
Colonies isolated and cultured on blood agar appear grey and
Colonial morphology Comply Comply Comply
translucent, with neat edges and smooth, moist surfaces, α hemolytic
G+ aureus, a double ball
Gram staining or short chain Comply Comply Comply
arrangement
Capsule staining Capsule observable Comply Comply Comply
Bile Salt Dissolution Test + + + +
Aupu Tosin Test + + + +
Glucose + + + +
Raffinose + + + +
Biochemical
Melibiose + + + +
reaction
Inulin + + + +
Sorbitol - - -
Agglutination Agglutination Agglutination
Agglutination with
Serum Agglutination with serotype with serotype with serotype
corresponding serum
9V serum 9V serum 9V serum
Meet the Meet the Meet the
Conclusion
requirement requirement requirement

Table 13 Test Results for Three-level Seed Lot of Serotype 14 Pneumococcal Strain

Original seed Master seed Working seed
Test items Standards
lot lot lot
No growth on the common nutrient agar
Culture characteristic slant and good growth Comply Comply Comply
on the 10% sheep blood agar slant.
Colonies isolated and cultured on blood agar appear grey and
Colonial morphology Comply Comply Comply
translucent, with neat edges and smooth, moist surfaces, α hemolytic
G+ aureus, a double ball
Gram staining or short chain Comply Comply Comply
arrangement
Capsule staining Capsule observable Comply Comply Comply
Bile Salt Dissolution Test + + + +
Aupu Tosin Test + + + +
Glucose + + + +
Raffinose + + + +
Biochemical
Melibiose + + + +
reaction
Inulin + + + +
Sorbitol - - -

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Agglutination Agglutination Agglutination
Agglutination with
Serum Agglutination with serotype with serotype with serotype
corresponding serum
14 serum 14 serum 14 serum
Meet the Meet the Meet the
Conclusion
requirement requirement requirement

Table 14 Test Results for Three-level Seed Lot of Serotype 18C Pneumococcal Strain

Original seed Master seed Working seed
Test items Standards
lot lot lot
No growth on the common nutrient agar
Culture characteristic slant and good growth Comply Comply Comply
on the 10% sheep blood agar slant.
Colonies isolated and cultured on blood agar appear grey and
Colonial morphology Comply Comply Comply
translucent, with neat edges and smooth, moist surfaces, α hemolytic
G+ aureus, a double ball
Gram staining or short chain Comply Comply Comply

arrangement
Capsule staining Capsule observable Comply Comply Comply
Bile Salt Dissolution Test + + + +
Aupu Tosin Test + + + +
Glucose + + + +
Raffinose + + + +
Biochemical
Melibiose + + + +
reaction
Inulin + + + +
Sorbitol - - - -
Agglutination Agglutination Agglutination
Agglutination with
Serum Agglutination with serotype with serotype with serotype
corresponding serum
18C serum 18C serum 18C serum
Meet the Meet the Meet the
Conclusion
requirement requirement requirement

Table 15 Test Results for Three-level Seed Lot of Serotype 19A Pneumococcal Strain

Original seed Master seed Working seed
Test items Standards
lot lot lot
No growth on the
common nutrient agar
Culture characteristic slant and good growth Comply Comply Comply
on the 10% sheep blood
agar slant.
Colonies isolated and
cultured on blood agar
appear grey and
Colonial morphology Comply Comply Comply
translucent, with neat
edges and smooth, moist
surfaces, α hemolytic

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G+ aureus, a double ball
Gram staining or short chain Comply Comply Comply
arrangement
Capsule staining Capsule observable Comply Comply Comply
Bile Salt Dissolution Test + + + +
Aupu Tosin Test + + + +
Glucose + + + +
Raffinose + + + +
Biochemical
Melibiose + + + +
reaction
Inulin + + + +
Sorbitol - - - -
Agglutination Agglutination Agglutination
Agglutination with
Serum Agglutination with serotype with serotype with serotype
corresponding serum
19A serum 19A serum 19A serum
Meet the Meet the Meet the
Conclusion
requirement requirement requirement

Table 16 Test Results for Three-level Seed Lot of Serotype 19F Pneumococcal Strain

Original seed Master seed Working seed
Test items Standards
lot lot lot
No growth on the
common nutrient agar
Culture characteristic slant and good growth Comply Comply Comply
on the 10% sheep blood
agar slant.
Colonies isolated and
cultured on blood agar
appear grey and
Colonial morphology Comply Comply Comply
translucent, with neat
edges and smooth, moist
surfaces, α hemolytic
G+ aureus, a double ball
Gram staining or short chain Comply Comply Comply
arrangement
Capsule staining Capsule observable Comply Comply Comply
Bile Salt Dissolution Test + + + +
Aupu Tosin Test + + + +
Glucose + + + +
Raffinose + + + +
Biochemical
Melibiose + + + +
reaction
Inulin + + + +
Sorbitol - - - -

Agglutination Agglutination Agglutination
Agglutination with
Serum Agglutination with serotype with serotype with serotype
corresponding serum
19F serum 19F serum 19F serum
Meet the Meet the Meet the
Conclusion
requirement requirement requirement

Table 17 Test Results for Three-level Seed Lot of Serotype 23F Pneumococcal Strain
Original seed Master seed Working seed
Test items Standards
lot lot lot
No growth on the
Culture characteristic Comply Comply Comply
common nutrient agar
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slant and good growth
on the 10% sheep blood
agar slant.
Colonies isolated and
cultured on blood agar
appear grey and
Colonial morphology Comply Comply Comply
translucent, with neat
edges and smooth, moist
surfaces, α hemolytic
G+ aureus, a double ball
Gram staining or short chain Comply Comply Comply
arrangement
Capsule staining Capsule observable Comply Comply Comply
Bile Salt Dissolution Test + + + +
Apu Tosin Test + + + +
Glucose + + + +
Raffinose + + + +
Biochemical
Melibiose + + + +
reaction
Inulin + + + +
Sorbitol - - - -
Agglutination Agglutination Agglutination
Agglutination with
Serum Agglutination with serotype with serotype with serotype
corresponding serum
23F serum 23F serum 23F serum
Meet the Meet the Meet the
Conclusion
requirement requirement requirement

According to the test items and specifications in the Table 18, the three-level seed lot of Clostridium Tetani were tested. The results showed in the Table 19, which indicate that the master seed lot and working seed lot have the same biological, biochemical and serological characteristics as the original seed lot.

Table 18 Testing Items and Standards of Clostridium Tetani Strain
Test items Specifications
Obligate anaerobic, preferential temperature at 37 ■,
turbid with gas produced and rotten stench in cooked
Growth Characteristic meat culture, permeated bacteria colony in blood agar
medium, flagellar motility in semi-solid puncture
Cultural
culture.
Characteristic
Gram negative, rod-shaped bacteria with little amount
of spores at the beginning of cultivation; transformed to
Staining Microscopy
Gram positive, drumstick-shaped bacteria with round
spores on the top after 48 h of cultivation.
Glucose Negative
Lactose Negative
Xylose Negative
Saccharose Negative
Biochemical Galactose Negative
Reactions Fructose Negative
Nitrate Negative
Gelatin liquefaction Positive
Hydrogen sulfide Positive
Dynamic experiment Positive
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Symptoms such as stiffened tail, cramp of hind legs or
Toxin production test muscle spasm of the whole body, or even death, should
Zoology appear.
experiment Mice in the positive control group should appear
Specific neutralization test obvious tetanus symptoms and die, while mice in the
test group survive.

Table 19 Test Results for Three-level Seed Lot of Clostridium Tetani Strain

Original seed	Master seed	Working seed
Test items Standards		
lot	lot	lot
Obligate anaerobic, preferential temperature at 37 ■, turbid with gas produced and rotten stench in cooked meat		
Growth Characteristic culture, permeated	Comply	Comply Comply
bacteria colony in blood agar medium, flagellar motility in semi-solid puncture culture.		
Gram negative, rod-shaped bacteria with little amount of spores at the beginning of cultivation; transformed		
Staining Microscopy	Comply	Comply Comply
to Gram positive, drumstick-shaped bacteria with round spores on the top after 48 h of cultivation.		
Glucose	----	
Lactose	----	
Xylose	----	
Saccharose	----	
Galactose	----	
Fructose	----	
Biochemical reaction Nitrate	----	
Gelatin	++++	
liquefaction		
Hydrogen	++++	
sulfide		
Dynamic	++++	
experiment		
Symptoms such as stiffened tail, cramp of		
Toxin		
hind legs or muscle production	Comply	Comply Comply
spasm of the whole test		
Zoology body, or even death, experiment should appear.		
Mice in the positive		
Spcific		
control group should neutralization	Comply	Comply Comply
appear obvious tetanus test		
symptoms and die,		

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while mice in the test group survive.
Meet the Meet the Meet the
Conclusion
requirement requirement requirement

(4) Study on the heredity stability of pneumococcal strains
[Pneumococcal strain]
In order to ensure the stability of both process and product quality, Walvax carried out heredity stability studies for strains of all 13 serotypes. Each strain is passed continuously to the 20th generation. During the process, bacteria from the 10th, 15th, and 20th generations are comprehensively tested. The process diagram is as follows:

Open a vial of lyophilized Working Seed Lot

Culture on solid medium for 20±4 h, then
↓ transfer the bacteria
2nd Generation

Culture on solid medium for 20±4 h, then
↓ transfer the bacteria

3rd Generation

Culture on solid medium for 20±4 h, then
↓ transfer the bacteria
4th Generation

Culture on liquid medium for 6 h, then
↓ transfer the bacteria

5th Generation
Culture on liquid medium for 6 h, then
transfer the bacteria
↓
6th Generation

Culture on liquid medium for 6 h/generation, then
↓ transfer the bacteria
10th Generation, sampling for tests
Culture on liquid medium for 6 h/generation,
↓ then transfer the bacteria

15th Generation, sampling for tests
Culture on liquid medium for 6 h/generation,
then transfer the bacteria
↓

20th Generation, sampling for tests
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↓
Terminate culturing, commence sterilization
Figure 10 The Process of Heredity Stability for Pneumococcal Strains

Testing results of strains from all generations have been constant with the Working Seed Lot.
There has been no change in biological, biochemical, or serological characteristics of the strains.
The stability of the strains used in production is proven.

[Clostridium tetani strain]
In order to ensure the stability of vaccine production process and product quality, Walvax carried out heredity stability studies of Clostridium tetani strain. Each strain is passed and cultured in the cooked meat semi-solid medium every year. The storage strain has been passed three times, and the production strain has been passed five times, without any decline in virulence and morphological variation. The test results of the strain show that its culture characteristics, staining microscopy, biochemical reaction and specific neutralization tests are same as the seed lot, which confirmed that the production strains of Clostridium tetani had good stability.

(5) Study on the virulence stability of pneumococcal strains
[Study Objective]
To investigate the virulence and virulence stability of pneumococcal serotype 1, 3, 4, 5, 6A, 6B, 7F, 9V, 14, 18C, 19A, 19F, and 23F production strain.
[Reference]
Manual of medical biological products, 235-241
[Experiment material]
The 3rd generation and the 20th generation of pneumococcal serotype 1, 3, 4, 5, 6A, 6B, 7F, 9V, 14, 18C, 19A, 19F, and 23F production strain.
[Experiment method]
The strain was diluted with normal saline and inoculated on solid medium containing 10% sheep blood. After incubation at 37±1 °C and CO₂ for 20±4 hours, the coating of pneumococcus was washed with normal saline and gradient diluted. The dilution was 1:10, 1:100, 1:103 until 1:108, the last three dilutions (1:106, 1:107, 1:108) were used to inject 18-22 g healthy mice (3 mice for each dilutions) into abdominal cavity at 0.5 ml/mouse.
[Criteriaions]
In 72 hours, if 2 of the 3 mice injected with 1:107 and 1:108 dilutions died, the strain is a strong virulent strain, otherwise it is a weak virulent strain.
[Experiment result]

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Table 20 The Virulence Test Result of the 3rd and the 20th Generation of all 13 Serotypes of Pneumococcal Strain

Generation	Serotypes the 3rd	the 20th
Items		
Dilutability	10-6 10-7 10-8 10-6 10-7 10-8	
1		
Mortality/totality	0/3 0/3 0/3 0/3 0/3 0/3	
Dilutability	10-6 10-7 10-8 10-6 10-7 10-8	
3		
Mortality/totality	1/3 0/3 0/3 1/3 0/3 0/3	
Dilutability	10-6 10-7 10-8 10-6 10-7 10-8	
4		
Mortality/totality	1/3 0/3 0/3 1/3 0/3 0/3	
Dilutability	10-6 10-7 10-8 10-6 10-7 10-8	
5		
Mortality/totality	0/3 0/3 0/3 0/3 0/3 0/3	
Dilutability	10-6 10-7 10-8 10-6 10-7 10-8	
6A		
Mortality/totality	1/3 0/3 0/3 0/3 0/3 0/3	
Dilutability	10-6 10-7 10-8 10-6 10-7 10-8	
6B		
Mortality/totality	0/3 0/3 0/3 1/3 0/3 0/3	
Dilutability	10-6 10-7 10-8 10-6 10-7 10-8	
7F		
Mortality/totality	1/3 1/3 0/3 1/3 1/3 0/3	
Dilutability	10-6 10-7 10-8 10-6 10-7 10-8	

9V
Mortality/totality 1/3 0/3 0/3 0/3 0/3 0/3
Dilutability 10-6 10-7 10-8 10-6 10-7 10-8
14
Mortality/totality 1/3 0/3 0/3 1/3 0/3 0/3
Dilutability 10-6 10-7 10-8 10-6 10-7 10-8
18C
Mortality/totality 1/3 0/3 0/3 1/3 0/3 0/3
Dilutability 10-6 10-7 10-8 10-6 10-7 10-8
19A
Mortality/totality 1/3 0/3 0/3 1/3 0/3 0/3
Dilutability 10-6 10-7 10-8 10-6 10-7 10-8
19F
Mortality/totality 1/3 1/3 0/3 1/3 0/3 0/3
Dilutability 10-6 10-7 10-8 10-6 10-7 10-8
23F
Mortality/totality 1/3 0/3 0/3 1/3 0/3 0/3

From Table 20, it can be seen that the 13 strains (pneumococcus serotypes 1, 3, 4, 5, 6A, 6B, 7F, 9V, 14, 18C, 19A, 19F, and 23F) used for production are attenuated vaccine strain. After 20 generations of continuous transplanting, the strain virulence did not change and the strain was still attenuated vaccine strain. It showed that the virulence of the strain is stable and the strains are suitable for vaccine development and production.

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2.3.S.2.3.2 Other Raw Materials
The company has formulated the Material Management Control Procedure (WSYX-P-05-0005) to guide and control the management and storage of the company's materials. All purchased materials are from qualified suppliers approved by the quality management director. For each kind of material, the company has established the inspection standard operating procedures according to the legal standards of the material to check whether the quality indicators of the material meet the quality requirements of the product production. At the same time, collect documents to prove whether the raw materials used in production contain ingredients of animal derived materials. If the materials contain ingredients of animal derived materials, further collect animal safety certification materials of the materials to ensure the safety of materials used in production and further ensure the safety of products. See the Table 21 and Table 22 for the composition of culture medium.

Table 21 Culture Medium Components for Production of 13 Serotypes of Pneumococcal

Polysaccharides
Medium name No. Material name
1 Tryptone
2 Soybean peptone
3 Dipotassium hydrogen phosphate
Composition of pneumococcal comprehensive solid 4 L-cysteine hydrochloride
medium 5 Tryptophan
6 Tyrosine
7 Agar
8 Sheep blood
9 Tryptone
10 Soybean peptone
11 L-cysteine hydrochloride
12 L-tryptophan
13 L-tyrosine
Dipotassium hydrogen phosphate
14
trihydrate
15 Sodium chloride
16 Catalase
17 Sodium bicarbonate
18 Mercaptoacetic acid
19 Nicotinic acid
20 Biotin
Composition of pneumococcal liquid medium 21 Vitamin B1
22 Vitamin B6
23 D-calcium Pantothenate
24 Uracil
25 Adenine sulfate
26 L-glutamine
27 L-asparagine
28 Magnesium sulfate anhydrous
Ferrous sulfate heptahydrate
29
(FeSO₄·7H₂O)
30 Riboflavin
31 Zinc sulfate (ZnSO₄·7H₂O)
32 Manganese Sulfate (MnSO₄·H₂O)
33 Choline chloride

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Table 22 Toxin Production Medium Components of Clostridium Tetani

Medium name No. Material name
1 Vitamin B1
2 Vitamin B2
3 Vitamin B6
Factor ■

4 Calcium pantothenate
5 Biotin
6 Absolute ethanol
7 Nicotinic acid
8 Uracil
9 Cystine
Factor ■
10 Zinc sulphate
11 Magnesium sulphate anhydrous
12 Hydrochloric acid
13 Beef stomach enzyme digestive fluid
14 Casein digestive fluid
15 Sodium acetate
16 Disodium hydrogen phosphate
17 Potassium dihydrogen phosphate
Production medium 18 Glycerine
19 Glucose
20 Factor ■
21 Factor ■
22 Yeast extract
23 Ferric chloride

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2.3.S.2.4 Controls of Critical Steps and Intermediates

Table 23 Critical Manufacture Steps of Serotypes 1, 4, 5, 6A, 6B, 7F, 9V, 14, 18C, 19A, 19F, 23F Pneumococcal Polysaccharide and Bulk Conjugate

Process Process variable

Medium volume used for dissolving strains (mL)

Number of opened Kirschner bottles

Temperature of the first-generation culture (■)

Carbon dioxide concentration of the first-generation culture

Time of the 1st generation culture (h)

Strain Amplification Process The number of the Kirschner flask which is bacteria passing into

Temperature of the 2nd generation culture (■)

Carbon dioxide concentration of the 2nd generation culture

Time of the 2nd generation culture (h)

Bacterial liquid volume collected (mL)

OD value

Seed tank culture volume (L)

Seed tank culture temperature (■)

Seed tank culture pH

Seed tank culture time (h)

Amplification tank culture volume (L)

Amplification tank culture temperature (■)

Amplification tank culture pH

Amplification tank culture time (h)

Fermentation tank culture volume (L)

Fermentation tank culture temperature (■)

Fermentation tank culture pH

Fermentation tank culture time (h)

Polysaccharide preparation process

Sterilant concentration

I:

Sterilization time (min)

Crude polysaccharide preparation

Centrifugal speed of fermentation liquid (rpm)

Centrifugal time of fermentation liquid (min)

Centrifugal temperature of fermentation liquid (min)

Ultrafiltration concentration fold

Final concentration of sodium acetate in the 1st ethanol precipitation

Concentration of ethanol in the 1st ethanol precipitation

Final concentration of sodium acetate in the 2nd ethanol precipitation

Concentration of ethanol in the 2nd ethanol precipitation

Times of crude polysaccharide washed by absolute ethanol

Times of crude polysaccharide washed by acetone

Times of crude polysaccharide washed by ether

Crude polysaccharide dried time (h)

Concentration of crude polysaccharide after dissolving (mg/mL)

Times of cold phenol extraction

Final concentration of ethanol for precipitating nucleic acid

Polysaccharide preparation process

Final concentration of ethanol for precipitating polysaccharide

II:

Times of refined polysaccharide washed by absolute ethanol

Refined polysaccharide preparation

Times of refined polysaccharide washed by acetone

Times of refined polysaccharide washed by ether

Time of refined polysaccharide dried (h)

Polysaccharide quantity used for conjugation (g)

Polysaccharide conjugate process Polysaccharide concentration after dissolving (mg/mL)

CNBR or CDAP quantity for per mg of polysaccharide (mg)

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pH value of activation

Activation temperature (■)

Activation time (min)

ADH quantity for per mg of polysaccharide (mg)
 pH value of derivatization
 Derivatization temperature (■)
 Derivatization reaction time (min)
 Polysaccharide derivative standing time (h)
 Standing temperature of polysaccharide derivative (■)
 Volume ratio of ultrafiltrate and derivative reaction fluid
 Ultrafiltration reaction times
 Storage temperature (■)
 Storage time (month)
 Carrier protein quantity for per mg of polysaccharide derivative (mg)
 Sodium chloride concentration (mol/L)
 pH value of conjugation
 EDAC quantity for per ml of mixed liquid (mg)
 Conjugation reaction time (h)
 Final pH of conjugation reaction
 Conjugate standing time (h)
 Conjugate standing temperature (■)
 Chromatographic column filler
 Collection KD value
 Pore size of sterilization & filtration membrane

Table 24 Critical Manufacture Steps of Serotypes 3 Pneumococcal Polysaccharide and Bulk

Conjugate
 Process Process variable
 Medium volume used for dissolving strains (mL)
 Number of opened Kirschner bottles
 Temperature of the 1st generation culture (■)
 Carbon dioxide concentration of the 1st generation culture
 Time of the 1st generation culture (h)
 Strain Amplification Process Number of the Kirschner flask which is bacteria passing into
 Temperature of the 2nd generation culture (■)
 Carbon dioxide concentration of the 2nd generation culture
 Time of the 2nd generation culture (h)
 Volume of bacterial liquid collected (mL)
 OD value
 Seed tank culture volume (L)
 Seed tank culture temperature (■)
 Seed tank culture pH
 Seed tank culture time (h)
 Amplification tank culture volume (L)
 Polysaccharide preparation process Amplification tank culture temperature (■)
 I: Amplification tank culture pH
 Crude polysaccharide preparation Amplification tank culture time (■)
 Fermentation tank culture volume (L)
 Fermentation tank culture temperature (■)
 Fermentation tank culture pH
 Fermentation tank culture time (h)
 Sterilant concentration

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Sterilization time (min)
 Centrifugal speed of fermentation liquid (rpm)
 Centrifugal time of fermentation liquid (min)
 Centrifugal temperature of fermentation liquid (■)
 Ultrafiltration concentration fold
 Final concentration of sodium acetate in the 1st ethanol precipitation
 Final concentration of ethanol in the 1st ethanol precipitation
 Final concentration of sodium acetate in the 2nd ethanol precipitation
 Final concentration of ethanol in the 2nd ethanol precipitation
 Times of crude polysaccharide washed by absolute ethanol
 Times of crude polysaccharide washed by acetone
 Times of crude polysaccharide washed by ether
 Crude polysaccharide dried time (h)
 Crude polysaccharide concentration after dissolving (mg/mL)
 Times of cold phenol extraction
 Final concentration of CTAB for precipitating nucleic acid
 Polysaccharide preparation process Final concentration of sodium acetate for precipitating polysaccharide
 II: Final concentration of ethanol for precipitating polysaccharide
 Refined polysaccharide preparation Times of refined polysaccharide washed by absolute ethanol
 Times of refined polysaccharide washed by acetone
 Times of refined polysaccharide washed by ether
 Refined polysaccharide dried time (h)
 Polysaccharide quantity used for conjugation (g)
 Polysaccharide concentration after dissolving (mg/mL)
 CNBR quantity for per mg of polysaccharide (mg)
 pH value of activation
 Activation temperature (■)
 Activation reaction time (min)
 ADH quantity for per mg of polysaccharide (mg)
 pH value of derivatization
 Derivatization temperature (■)
 Derivatization reaction time (min)
 Polysaccharide derivative standing time (h)
 Standing temperature of polysaccharide derivative (■)
 Volume ratio of ultrafiltrate and derivative reaction fluid
 Polysaccharide conjugate process Ultrafiltration times
 Storage temperature (■)
 Storage time (month)

Carrier protein quantity for per mg of polysaccharide derivative (mg)
Sodium chloride concentration (mol/L)
pH value of conjugation
EDAC quantity for per ml of mixed liquid (mg)
Conjugation reaction time (h)
Final pH of conjugation reaction
Conjugate standing time (h)
Conjugate standing temperature (■)
Chromatographic column filler
Collection KD value
Pore size of sterilization & filtration membrane

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2.3.S.2.5 Process Validation and/or Evaluation

The manufacturing process of 13 serotypes of Pneumococcal Conjugate Polysaccharide can be divided mainly into establishment of seed lot, preparation of crude polysaccharide, purification of polysaccharide, activation and derivatization of polysaccharide, conjugation with tetanus toxoid. The critical process parameters and the control range of each step were determined by the evaluation, and the key process parameters were strictly controlled within the predetermined range in the production process verification of the three batches of 13 serotypes of Pneumococcal Polysaccharide, Polysaccharide Derivate, Pneumococcal Conjugate Polysaccharide, each and all test item complies with the requirements as set forth in Manufacturing and Control Tests on 13-valent Pneumococcal Conjugate Vaccine. Through 13 types of Pneumococcal Polysaccharide Bulk Conjugate manufacturing validation, it proved that: 13 types of Pneumococcal Polysaccharide manufacturing process can sustainably produce desired products in accordance with Manufacturing and Control Tests on 13-valent Pneumococcal Polysaccharide Conjugate Vaccine, critical process parameters and quality attributes in the process of the entire process are under control.

Table 25 List of Process Validation Protocol and Report for Intermediate Products

No. Process validation protocol and report Code

Process validation protocol and report of serotype 1 pneumococcal

1 WSYX-BPVW-TS-0006-0190

polysaccharide

Process validation protocol and report of serotype 3 pneumococcal

2 WSYX-BPVW-TS-0006-0197

polysaccharide

Process validation protocol and report of serotype 4 pneumococcal

3 WSYX-BPVW-TS-0006-0198

polysaccharide

Process validation protocol and report of serotype 5 pneumococcal

4 WSYX-BPVW-TS-0006-0199

polysaccharide

Process validation protocol and report of serotype 6A pneumococcal

5 WSYX-BPVW-TS-0006-0240

polysaccharide

Process validation protocol and report of serotype 6B pneumococcal

6 WSYX-BPVW-TS-0006-0200

polysaccharide

Process validation protocol and report of serotype 7F pneumococcal

7 WSYX-BPVW-TS-0006-0201

polysaccharide

Process validation protocol and report of serotype 9V pneumococcal

8 WSYX-BPVW-TS-0006-0204

polysaccharide

Process validation protocol and report of serotype 14 pneumococcal

9 WSYX-BPVW-TS-0006-0208

polysaccharide

Process validation protocol and report of serotype 18C

10 WSYX-BPVW-TS-0006-0211

pneumococcal polysaccharide

Process validation protocol and report of serotype 19A

11 WSYX-BPVW-TS-0006-0212

pneumococcal polysaccharide

Process validation protocol and report of serotype 19F

12 WSYX-BPVW-TS-0006-0213

pneumococcal polysaccharide

Process validation protocol and report of serotype 23F

13 WSYX-BPVW-TS-0006-0216

pneumococcal polysaccharide

Process validation protocol and report of serotype 1 pneumococcal

14 WSYX-BV2-TS-0006-0110

polysaccharide bulk conjugate

Process validation protocol and report of serotype 3 pneumococcal

15 WSYX-BV2-TS-0006-0111

polysaccharide bulk conjugate

Process validation protocol and report of serotype 4 pneumococcal

16 WSYX-BV2-TS-0006-0112

polysaccharide bulk conjugate

Process validation protocol and report of serotype 5 pneumococcal

17 WSYX-BV2-TS-0006-0113

polysaccharide bulk conjugate

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Process validation protocol and report of serotype 6A pneumococcal

18 WSYX-BV2-TS-0006-0114

polysaccharide bulk conjugate

Process validation protocol and report of serotype 6B pneumococcal
 19 WSYX-BV2-TS-0006-0115
 polysaccharide bulk conjugate
 Process validation protocol and report of serotype 7F pneumococcal
 20 WSYX-BV2-TS-0006-0116
 polysaccharide bulk conjugate
 Process validation protocol and report of serotype 9V pneumococcal
 21 WSYX-BV2-TS-0006-0117
 polysaccharide bulk conjugate
 Process validation protocol and report of serotype 14 pneumococcal
 22 WSYX-BV2-TS-0006-0118
 polysaccharide bulk conjugate
 Process validation protocol and report of serotype 18C
 23 WSYX-BV2-TS-0006-0119
 pneumococcal polysaccharide bulk conjugate
 Process validation protocol and report of serotype 19A
 24 WSYX-BV2-TS-0006-0120
 pneumococcal polysaccharide bulk conjugate
 Process validation protocol and report of serotype 19F
 25 WSYX-BV2-TS-0006-0121
 pneumococcal polysaccharide bulk conjugate
 Process validation protocol and report of serotype 23F
 26 WSYX-BV2-TS-0006-0122
 pneumococcal polysaccharide bulk conjugate

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2.3.S.3 Characterization

2.3.S.3.1 Elucidation of Structure and other Characteristics

Walvax carried out 1H-NMR tests of the 13 serotypes of pneumococcal polysaccharides, and analyzed the corresponding spectra by comparison with ATCC standard polysaccharides. The results showed that the structures of Walvax's polysaccharides are in full accord with ATCC standard polysaccharides.

Table 26 General Information of 1H-NMR Tests of the 13 Serotypes of Pneumococcal Polysaccharides

Test information Walvax ATCC
 NMR machine Varian Inova 500 Varian Inova 500
 Frequency 500 MHz 500 MHz
 For serotypes 1, 3, 4, 5, 6A, 6B, 7F, 9V, 18C,
 For all 13 serotypes: D2O + 19A, 19F, 23F: D2O + 0.01%DMSO + 0.01%
 Solution system 0.05%wt TSP + 0.5 mg/mL DSS-d6
 maleic acid For serotype 14: D2O + 0.05%wt TSP +
 0.5mg/ml maleic acid
 Polysaccharide
 5 mg/mL 5 mg/mL
 concentration

Given the information in Table 26, test instrument and test frequency of Walvax's polysaccharides tests match those of ATCC standard polysaccharides tests. Walvax's polysaccharides used a solution system of D2O + 0.05% wt TSP + 0.5 mg/mL maleic acid, while ATCC serotypes 1, 3, 4, 5, 6A, 7F, 9V, 18C, 19A, 19F, 23F polysaccharides used a solution system of D2O + 0.01% DMSO + 0.01% DSS-d6, and serotype 14 used a solution system of D2O + 0.05% wt TSP + 0.5 mg/mL maleic acid. The two systems used maleic acid and DMSO respectively as quantitative and TSP and DSS-d6 respectively as internal standard, which has no influence on test results.

Table 27 Abbreviations for Polysaccharide 1H-NMR Spectra

Abbreviations	Name	Abbreviations	Name
D2O	Heavy water	Gal	Galactose
HDO	half-heavy water	GlcA	Glucuronic acid
Sodium 3 - (trimethylsilicon)			
TSP	GlcNAc	N-acetyl glucose	
deuteriopropionate			
DMSO	Dimethyl sulfoxide	ManNAc	N-acetyl mannose
Sodium 3,3,4,4,5,5-hexadeuteriu-2,2-			
DSS-d6	FucNAc	N-acetyl fucose	
dimethyl-2-cyclopentane-5-sulfonate			
2-acetylamino-2,6-dideoxy-D-			
C-poly	C-polysaccharide	Sug	
glutaraldehyde-hexose-4-ketone			
GalA	Galacturonic acid	Pyr	Pyruvic acid
GalNAc	N-acetyl galactose	Rha	Rhamnolipid
Glc	Glucose	-PO3	Phosphate radical

1

H-NMR spectra and corresponding comparisons with ATCC standard polysaccharides are showed as follows.

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(1) Structure of Serotype 1 Pneumococcal Polysaccharide:

→ 4) – α – GalpA(1 → 3) – α – GalpA(1 → 3) – α – 6dGalpNAc 4 N (1 →

* O

4 6 COOH

O HO HOOC

6 O

5 4 H2N H3C 6

HO 2 5

n

1

1

M

v

•

1

1

M

Y

Q

1

1

M

v

•

H

4

H

4

H

9

*

H

0

3

00

6

4

0

2

5
O 1
3
O
*
H3C n
COOH

Molecular size: determined by gel filtration through Sepharose CL-4B, release specification is
KD ≤ 0.15, recovery rate ≥ 73% before KD value reached 0.3.

(a) Walvax

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(b) ATCC

Figure 14 1H-NMR Spectra of Native Serotype 4 Pneumococcal Polysaccharide from (a)
Walvax and (b) ATCC

Comparison of 1H-NMR spectra showed Walvax's polysaccharide is the same as ATCC
polysaccharide, apart from quantitative DMSO peak and residual ethanol peak. Both spectra
showed structural peaks of Gal, GalNAc, FucNAc, ManNAc, N-Acetate, Pyr and residual C-
polyaccharide, which indicated the structure of Walvax's serotype 4 polysaccharide is in full
accord with ATCC standard polysaccharide.

(4) Structure of Serotype 5 Pneumococcal Polysaccharide:

→ 4 - β - D - Glcp - 1 → 4 - α - L - FucpNAc - 1 → 3 - β - D - Sugp - 1 →

3
↑
1
α - L - PnepNAc - 1 → 2 - β - D - GlcpA
6 4
H3C 5
OH 6 CH3
4 1 O
4 6
O O O
2 3
* O 5 O O
HO O 2 NHAc 2 1
3 OH 1 3
NHAc
O
HOOC 6 2
1
*
4 5 n
3
O
CH3
HO 1 3
2

NHAC 5 OH
O
6 4
HO
H3C

Molecular size: determined by gel filtration through Sepharose CL-2B, release specification is
KD ≤ 0.60, recovery rate ≥ 70% before KD value reached 0.6.

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(a) Walvax

(b) ATCC

Figure 15 1H-NMR Spectra of Native Serotype 5 Pneumococcal Polysaccharide from (a)
Walvax and (b) ATCC

Comparison of 1H-NMR spectra showed Walvax's polysaccharide is the same as ATCC
polysaccharide, apart from quantitative DMSO peak and residual ethanol peak. Both spectra
showed structural peaks of N-Acetate and residual C-polysaccharide, which indicated the structure
of Walvax's serotype 5 polysaccharide is in full accord with ATCC standard polysaccharide.

(5) Structure of Serotype 6A Pneumococcal Polysaccharide:

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→ 2)α-D-Galp(1 → 3)α-D-Glcp(1 → 3)α-L-Rhap(1 → 3)-D-Ribitol(5- PO3 →

HO
4 6
OH

HO O

5 2
3
4 OH
HO 6
1
5 O
2 OH
O O
* 3 H3C
4 6
OH 1
5 O P O *
O 3
O OH HO
OH 1
2 O
n
OH

OH

Molecular size: determined by gel filtration through Sepharose CL-2B, release specification is KD ≤ 0.50, recovery rate ≥ 50% before KD value reached 0.6.

(a) Walvax

(b) ATCC

Figure 16 1H-NMR Spectra of Native Serotype 6A Pneumococcal Polysaccharide from (a) Walvax and (b) ATCC

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Comparison of 1H-NMR spectra showed Walvax's polysaccharide is the same as ATCC polysaccharide, apart from quantitative DMSO peak and residual ethanol peak. Both spectra showed structural peaks of Gal, Glc, Rha and residual C-polysaccharide, which indicated the structure of Walvax's serotype 6A polysaccharide is in full accord with ATCC standard polysaccharide.

(6) Structure of Serotype 6B Pneumococcal Polysaccharide:

→ 2)α-D-Galp(1 → 3)α-D-Glcp(1 → 3)α-L-Rhap(1 → 4)-D-Ribitol(5- PO3 → HO

4
OH
6
HO O
5 2
3
4 OH
6
1 HO
5 O
2 OH
O
* 3 H3C
4 6
OH 1
5 O
O 3
O O P O *
OH 1
2
O HO
n
OH
OH

OH

Molecular size: determined by gel filtration through Sepharose CL-2B, release specification is KD ≤ 0.50, recovery rate ≥ 85% before KD value reached 0.6.

(a) Walvax

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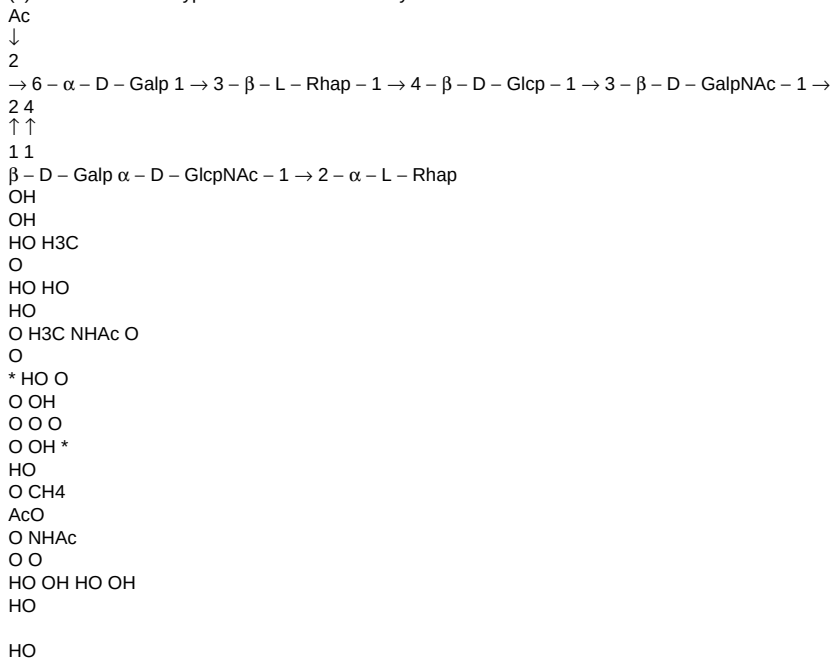
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(b) ATCC

Figure 17 1H-NMR Spectra of Native Serotype 6B Pneumococcal Polysaccharide from (a) Walvax and (b) ATCC

Comparison of 1H-NMR spectra showed Walvax's polysaccharide is the same as ATCC polysaccharide, apart from quantitative DMSO peak and residual ethanol peak. Both spectra showed structural peaks of Gal, Glc, Rha and residual C-polysaccharide, which indicated the structure of Walvax's serotype 6B polysaccharide is in full accord with ATCC standard polysaccharide.

(7) Structure of Serotype 7F Pneumococcal Polysaccharide:



n
Molecular size: determined by gel filtration through Sepharose CL-4B, release specification is
KD ≤ 0.20, recovery rate ≥ 70% before KD value reached 0.3.

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(a) Walvax

(b) ATCC

Figure 18 1H-NMR Spectra of Native Serotype 7F Pneumococcal Polysaccharide from (a) Walvax and (b) ATCC

Comparison of 1H-NMR spectra showed Walvax's polysaccharide is the same as ATCC polysaccharide, apart from quantitative DMSO peak. Both spectra showed broad and overlapped signals at the anomeric carbon region, which indicated the structure of Walvax's serotype 7F polysaccharide is in full accord with ATCC standard polysaccharide.

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(a) Walvax

(b) ATCC

Figure 19 1H-NMR Spectra of De-O-acetylated Serotype 7F Pneumococcal Polysaccharide from (a) Walvax and (b) ATCC

Comparison of 1H-NMR spectra showed the structural peaks (including Gal, Rha, GlcNAc, Acetate and residual C-polysaccharide) of Walvax's de-O-acetylated serotype 7F polysaccharide completely matched the peaks of the de-O-acetylated serotype 7F polysaccharide of ATCC.

(8) Structure of Serotype 9V Pneumococcal Polysaccharide:

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Molecular size: determined by gel filtration through Sepharose CL-2B, release specification is $KD \leq 0.45$, recovery rate $\geq 80\%$ before KD value reached 0.6.

(a) Walvax

(b) ATCC

Figure 20 1H-NMR Spectra of Native Serotype 9V Pneumococcal Polysaccharide from (a) Walvax and (b) ATCC

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Comparison of 1H-NMR spectra showed Walvax's polysaccharide is the same as ATCC polysaccharide, apart from quantitative DMSO peak. Both spectra showed broad and overlapped signals at the anomeric carbon region, which indicated the structure of Walvax's serotype 9V polysaccharide is in full accord with ATCC standard polysaccharide.

(a) Walvax

(b) ATCC

Figure 21 1H-NMR Spectra of De-O-acetylated Serotype 9V Pneumococcal Polysaccharide from (a) Walvax and (b) ATCC

Comparison of 1H-NMR spectra showed the structural peaks (including Glc, Gal, GlcA, ManNAc, Acetate) of Walvax's de-O-acetylated serotype 9V polysaccharide completely matched the peaks of the de-O-acetylated serotype 9V polysaccharide of ATCC.

(9) Structure of Serotype 14 Pneumococcal Polysaccharide:

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$\rightarrow 4 - \beta - D - Glcp - 1 \rightarrow 6 - \beta - D - GlcpNAc - 1 \rightarrow 3 - \beta - D - Galp - 1 \rightarrow$

4

↑

1

$\beta - D - Galp$

HO

OH

OH

O

O O *

* O

O

HO O OH

HO OH

OH

O O NHAc

HO

HO OH

n

Molecular size: determined by gel filtration through Sepharose CL-4B, release specification is $KD \leq 0.30$, recovery rate $\geq 55\%$ before KD value reached 0.3.

(a) Walvax

(b) ATCC

Figure 22 1H-NMR Spectra of Native Serotype 14 Pneumococcal Polysaccharide from (a) Walvax and (b) ATCC

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Comparison of 1H-NMR spectra showed Walvax's polysaccharide is the same as ATCC polysaccharide, apart from residual ethanol peak. Both spectra showed structural peaks of GlcNAc, Gal and residual C-polysaccharide, which indicated the structure of Walvax's serotype 14 polysaccharide is in full accord with ATCC standard polysaccharide.

(10) Structure of Serotype 18C Pneumococcal Polysaccharide:

$AcO - 3 - \alpha - D - Glcp$

1

↓

2

$\rightarrow 4 - \beta - D - Glcp - 1 \rightarrow 4 - \beta - D - Galp - 1 \rightarrow 4 - \alpha - D - Glcp - 1 \rightarrow 3 - \alpha - L - Rhap - 1 \rightarrow$

3

↑

Glycerol - 1 - OP O-2 O

O OH

O

O OH

* HO OH OH

O O O H3C

OH

O HO

HO
O OH OH O
P
OAc O
O O OH
O
HO OH
*
OH
HO
n

Molecular size: determined by gel filtration through Sepharose CL-4B, release specification is $KD \leq 0.15$, recovery rate $\geq 70\%$ before KD value reached 0.3.

(a) Walvax

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(b) ATCC

Figure 23 1H-NMR Spectra of Native Serotype 18C Pneumococcal Polysaccharide from (a) Walvax and (b) ATCC

Comparison of 1H-NMR spectra showed Walvax's polysaccharide is the same as ATCC polysaccharide, apart from quantitative DMSO peak. Both spectra showed broad and overlapped signals at the anomeric carbon region, which indicated the structure of Walvax's serotype 18C polysaccharide is in full accord with ATCC standard polysaccharide.

(a) Walvax

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(b) ATCC

Figure 24 1H-NMR Spectra of De-O-acetylated Serotype 18C Pneumococcal Polysaccharide from (a) Walvax and (b) ATCC

Comparison of 1H-NMR spectra showed the structural peaks (including Glc, Rha, Gal, Acetate) and residual C-polysaccharide peak of Walvax's de-O-acetylated serotype 18C polysaccharide completely matched the peaks of the de-O-acetylated serotype 18C polysaccharide of ATCC.

(11) Structure of Serotype 19A Pneumococcal Polysaccharide:
→ 4 - β - D - ManpNAc - 1 → 4 - α - D - Glcp - 1 → 3 - α - L - Rhap - 1 - ■■4 →

OH
O
OH
AcHN O OH
* O H3C
O
HO
HO OH
O
O O
HO O P *
HO

n
Molecular size: determined by gel filtration through Sepharose CL-4B, release specification is $KD \leq 0.45$, recovery rate $\geq 80\%$ before KD value reached 0.5.

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(a) Walvax

(b) ATCC

Figure 25 1H-NMR Spectra of Native Serotype 19A Pneumococcal Polysaccharide from (a) Walvax and (b) ATCC

Comparison of 1H-NMR spectra showed Walvax's polysaccharide is the same as ATCC polysaccharide, apart from quantitative DMSO peak and residual ethanol peak. Both spectra showed structural peaks of Rha, Glc, ManNAc, N-Acetate and residual C-polysaccharide, which indicated the structure of Walvax's serotype 19A polysaccharide is in full accord with ATCC standard polysaccharide.

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(12) Structure of Serotype 19F Pneumococcal Polysaccharide:
→ 4) - ■ - D - ManpNAc(1 → 4) - ■ - D - Glcp(1 → 2) - α - L - Rhap(1 - PO3 →

OH
OH H3C
O
OH

AcHN O
HO
* O
O
HO O
HO OH O

O
O
P *

OH

n

Molecular size: determined by gel filtration through Sepharose CL-4B, release specification is $KD \leq 0.20$, recovery rate $\geq 75\%$ before KD value reached 0.3.

(a) Walvax

(b) ATCC

Figure 26 1H-NMR Spectra of Native Serotype 19F Pneumococcal Polysaccharide from (a) Walvax and (b) ATCC

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Comparison of 1H-NMR spectra showed Walvax's polysaccharide is the same as ATCC polysaccharide, apart from quantitative DMSO peak and residual ethanol peak. Both spectra showed structural peaks of Rha, Glc, ManNAc, N-Acetate and residual C-polysaccharide, which indicated the structure of Walvax's serotype 19F polysaccharide is in full accord with ATCC standard polysaccharide.

(13) Structure of Serotype 23F Pneumococcal Polysaccharide:

α - L - Rhap

1

↓

2

$\rightarrow 4 - \beta - D - Glcp - 1 \rightarrow 4 - \beta - D - Galp - 1 \rightarrow 4 - \beta - L - Rhap - 1 \rightarrow$

3

PO2 OH

↑

HOCH2 - CH - CH2 OH

OH

O

OH

O

O

* H3C

HO OH O O O

*

HO P O HO O

O

O OH

HO OH O

OH

OH

H3C

HO n

Molecular size: determined by gel filtration through Sepharose CL-4B, release specification is $KD \leq 0.15$, recovery rate $\geq 85\%$ before KD value reached 0.3.

(a) Walvax

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(b) ATCC

Figure 27 1H-NMR Spectra of Native Serotype 23F Pneumococcal Polysaccharide from (a) Walvax and (b) ATCC

Comparison of 1H-NMR spectra showed Walvax's polysaccharide is the same as ATCC polysaccharide, apart from quantitative DMSO peak and residual ethanol peak. Both spectra showed structural peaks of Rha, Gal, Glc and residual C-polysaccharide, which indicated the structure of Walvax's serotype 23F polysaccharide is in full accord with ATCC standard polysaccharide.

2.3.S.3.2 Impurities

2.3.S.3.2.1 Impurities and their Limits in Polysaccharide Purification Process

(1) C-polysaccharide, nucleic acid and protein are the main impurities in refined polysaccharide production.

The quality of pneumococcal polysaccharides manufactured by Walvax are controlled according to the European Pharmacopeia. Control of C-polysaccharide are implemented via various purification steps. Table x shows residual C-polysaccharide contents in 12 serotypes of polysaccharides derived from NMR spectrum analysis, which are all below 20%, indicating the purification process can effectively remove C-polysaccharide impurities.

Table 28 Residual C-polysaccharide Content in 13 Serotypes of Pneumococcal Polysaccharide

Serotypes Batch No. Residual C-polysaccharide content % (W/W)

1 PN1201609001 20
 3 PN3201507003-1 None detected
 4 4-PVPV16011103 11.4
 5* PN5201610005 -
 6A PN6A201503001 1.4
 6B PN6B201408001 0.5
 7F 7F-PVPV15100701 None detected
 9V 9V-PVPV15090301 0.7
 14 PN14201403001 7.5

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18C 18C-PVPV15100801 0.9
 19A PN19A201408001 1.6
 19F PN19F201508002 1.2
 23F 23F-PVPV15090601 2.7

*: Since signature peak(s) and content of C-polysaccharide cannot be analyzed in NMR spectrum of serotype 5 polysaccharide due to its structural features, no result for serotype 5 is available.

During approval process of new drug of PPSV23, Walvax completed process optimization and validation studies, including impurities removal process validation. According to the established purification process of serotypes 1, 3, 4, 5, 6B, 7F, 9V, 14, 18C, 19A, 19F and 23F, 3 continuous batches of the above serotypes are purified, sampled in each step, and impurities removal rate was calculated with nucleic acid and protein content derived from polysaccharide content, and viability, reproducibility and stability was evaluated.

Crude polysaccharide solution, concentrated phenol extraction ultrafiltration solution and concentrated solution after precipitation of nucleic acid of serotypes 1, 4, 5, 6B, 7F, 9V, 14, 18C, 19A, 19F and 21F were sampled and analyzed for nucleic acid, protein and polysaccharide content, percentages of nucleic acid and protein were calculated, and impurities removal rate of each step was evaluated by comparison of data from before/after each steps. Table X shows the removal rate of nucleic acid and protein.

Table 29 Impurities Removal Rate of Purification Process of Serotypes 1, 4, 5, 6B, 7F, 9V, 14, 18C, 19A, 19F and 23F

After phenol After nucleic acid
 extraction and precipitation and

Serotypes Batch No. Index
 ultrafiltration ultrafiltration
 concentration concentration

Nucleic acid (%) 33.2 90.5
 RPN1201310001

Protein (%) 85.4 60.4

Nucleic acid (%) 29.4 92.5

1 RPN1201311002

Protein (%) 88.2 57.2

Nucleic acid (%) 35.6 95.1

RPN1201312003

Protein (%) 90.3 56.8

Nucleic acid (%) 36.1 93.4

RPN4201311001

Protein (%) 87.2 61.5

Nucleic acid (%) 34.5 91.7

4 RPN4201311002

Protein (%) 83.6 59.3

Nucleic acid (%) 39.3 89.1

RPN4201312003

Protein (%) 88.2 63.6

Nucleic acid (%) 36.2 92.2

RPN5 201310001

Protein (%) 87.3 52.3

Nucleic acid (%) 40.6 90.4

5 RPN5 201311002

Protein (%) 89.1 49.7

Nucleic acid (%) 35.4 87.4

RPN5 201312003

Protein (%) 84.4 58.5

Nucleic acid (%) 38.5 91.6

RPN6B201311001

Protein (%) 88.4 53.4

Nucleic acid (%) 35.2 87.8

6B RPN6B201312002

Protein (%) 93.5 58.8

Nucleic acid (%) 33.6 94.6

RPN6B201312003

Protein (%) 95.2 62.7

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Nucleic acid (%) 41.3 90.7

RPN7F201311001

Protein (%) 87.1 52.7

Nucleic acid (%) 39.5 93.5

7F RPN7F201312002

Protein (%) 90.6 57.7

Nucleic acid (%) 35.7 89.6
RPN7F201312003
Protein (%) 88.5 61.4
Nucleic acid (%) 34.1 90.9
RPN9V201311001
Protein (%) 89.5 56.6
Nucleic acid (%) 34.1 90.9
9V RPN9V201311001
Protein (%) 89.5 56.6
Nucleic acid (%) 34.1 90.9
RPN9V201311001
Protein (%) 89.5 56.6
Nucleic acid (%) 34.5 89.8
RPN14 201311001
Protein (%) 87.3 54.9
Nucleic acid (%) 32.6 91.4
14 RPN14 201311002
Protein (%) 85.3 63.5
Nucleic acid (%) 30.7 90.3
RPN14 201312003
Protein (%) 89.2 58.5
Nucleic acid (%) 38.4 89.4
RPN18C201311001
Protein (%) 84.7 58.8
Nucleic acid (%) 41.8 86.3
18C RPN18C201312002
Protein (%) 88.3 60.6
Nucleic acid (%) 36.6 93.2
RPN18C201312003
Protein (%) 86.8 61.7
Nucleic acid (%) 40.5 89.6
RPN19A201311001
Protein (%) 87.3 58.3
Nucleic acid (%) 85.7 63.4
19A RPN19A201312002
Protein (%) 36.3 91.3
Nucleic acid (%) 87.3 60.7
RPN19A201312003
Protein (%) 34.5 93.8
Nucleic acid (%) 40.6 91.4
RPN19F201311001
Protein (%) 88.4 61.7
Nucleic acid (%) 37.7 89.4
19F RPN19F201312002
Protein (%) 90.4 59.8
Nucleic acid (%) 35.6 92.3
RPN19F201312003
Protein (%) 92.7 63.2
Nucleic acid (%) 36.6 93.5
RPN23F201311001
Protein (%) 85.4 54.4
Nucleic acid (%) 35.2 91.2
23F RPN23F201312002
Protein (%) 87.7 58.8
Nucleic acid (%) 33.1 89.6
RPN23F201312003
Protein (%) 84.1 57.2
Note: Removal rate (%) = (impurities content before processing - impurities content after processing)/impurities content before processing ×100%

Table 29 shows that the nucleic acid and protein removal rate of 3 batches of 11 serotypes of polysaccharide was relatively stable, which indicates the purification process is stable.

Table 30 Impurities Removal Rate of 3 Batches of Serotype 3 Polysaccharide
After CTAB
After phenol
Serotype Batch No. Index precipitation and NaCl extraction and dissociation
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ultrafiltration
concentration
Nucleic acid (%) 30.2 95.4
RPN3 201310001
Protein (%) 85.7 40.3
Nucleic acid (%) 29.8 95.8
3 RPN3 201311002
Protein (%) 87.5 39.8
Nucleic acid (%) 32.4 96.4
RPN3 201312003
Protein (%) 88.7 41.2
Note: Removal rate (%) = (impurities content before processing - impurities content after processing)/impurities content before processing ×100%

Table 30 shows that the nucleic acid and protein removal rate of 3 batches of serotype 3 polysaccharide was relatively stable, which indicates the purification process is stable. Nucleic acid and protein impurity contents were monitored in the established purification process

of serotypes 1, 3, 4, 5, 6B, 7F, 9V, 14, 18C, 19A, 19F and 23F, and removal rates of each step was calculated. The test results show the removal rates were high and stable, content homogeneity was well controlled, which indicates good reproducibility and stability for the purification process in pneumococcal polysaccharide vaccine manufacture.

(2) Studies on organic solvents (reagents) removal in polysaccharide production
I. Types and purposes of organic reagents used in polysaccharide manufacture
In the production process of pneumococcal polysaccharide, organic solvents (reagents) are used. According to the Chinese Pharmacopeia (2010 version), similar reagents are used in pneumococcal polysaccharide vaccine, Vi capsular polysaccharide vaccine and Hib conjugate vaccine productions, which indicates this as a feature related to bacterium capsular polysaccharide vaccines.
Table 31 shows organic reagents used in 13-Valent Pneumococcal Polysaccharide Conjugate Vaccine.

Table 31 Usage of Organic Reagents and their Purposes
Organic solvent
Steps Used Objective
(reagent)
DOC Sterilization after fermentation Neutralization of pneumococcus
Purification of pneumococcal Removal of protein impurities by extraction with
Phenol
polysaccharide cold phenol
Crude polysaccharide production Ethanol is used as precipitation agent to separate
95% Ethanol
and purification polysaccharides and impurities
Washing of crude polysaccharide Anhydrous ethanol is used as dehydration agent,
Anhydrous ethanol
and refined polysaccharide and to remove alcohol-soluble impurities
Washing of crude polysaccharide Acetone is used as dehydration agent, and to
Acetone
and refined polysaccharide remove acetone-soluble impurities
Washing of crude polysaccharide Ether is used as dehydration agent, and to
Ether
and refined polysaccharide remove ether-soluble impurities
Due to its ability of precipitating capsular
Purification of serotype 3
CTAB polysaccharide and nucleic acid, CTAB is used
polysaccharide
to remove nucleic acid.

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II. Organic solvents (reagents) removal process
Table 32 Description of Organic Solvents (Reagents) Removal Process
Organic solvent
Removal process
(reagent)
After fermentation, DOC is added for sterilization, after which several steps are done, including ultrafiltration of crude polysaccharide, ethanol precipitation, ultrafiltration of DOC refined polysaccharide and ethanol precipitation. Due to low molecular size and good water solubility, DOC can be effectively removed from capsular polysaccharide.
Phenol is used in the removal of protein, after extraction with phenol, the mixture goes through ultrafiltration to remove phenol, after which several stages of ethanol precipitation and ultrafiltration were conducted. In the ultrafiltration and ethanol Phenol precipitation process, phenol can be effectively removed due to its low molecular size and good solubility in both water and organic solution. Moreover, in the process of washing with ethanol, acetone and ether, phenol was further separated and removed.
95% ethanol is used in precipitation of crude and refined polysaccharide and can be 95% ethanol effectively removed with ultrafiltration. Vacuum drying and washing with acetone and ether of refined polysaccharide can further remove ethanol.
Anhydrous Anhydrous ethanol is used to wash refined polysaccharides, it can be removed with ethanol acetone, ether and vacuum drying.
Acetone is used to wash refined polysaccharide; it can be removed by washing with Acetone ether and vacuum drying.
Ether Ether is used to wash refined polysaccharide; it can be removed by vacuum drying.
CTAB is used for removal of nucleic acid in serotype 3 polysaccharide; freezing, CTAB centrifugation and ethanol precipitation can remove most of the CTAB used, residual CTAB can be removed with washing using anhydrous ethanol.

III. Residual organic solvents (reagents) content
Table 33 shows the test results of residual organic reagents content (DOC, phenol, CTAB, ethanol, ether and acetone) of continuous 3 batches of polysaccharide.

Table 33 Residual Organic Reagents Content of Continuous 3 Batches of 13 serotypes of Polysaccharide
DOC Phenol CTAB Ethanol Ether Acetone
Serotypes Batch No.
µg/mL µg/mL µg/mL (%) (%) (%)
20090401 0 0 - 0.0050 0.0016 0.0015
Serotype 1
20090402 0 0 - 0.0035 0.0017 0.0015
polysaccharide
20090503 0 0 - 0.0033 0.0017 0.0015
20090602 0.09 0 0 0.0045 0.0003 0.0019
Serotype 3
20090603 0 0 0 0.0040 0.0003 0.0017

polysaccharide
 20090604 0 0 0 0.0040 0.0003 0.0017
 20090101 0 0 - 0.0089 0.0013 0.0059
 Serotype 4
 20090102 0 0 - 0.0088 0.0013 0.0060
 polysaccharide
 20090103 0 0 - 0.0089 0.0012 0.0059
 20090401 1.1 0 - 0.0098 0.0003 0.0059
 Serotype 5
 20090402 2.5 0 - 0.0102 0.0003 0.0060
 polysaccharide
 20090403 9.3 0 - 0.0103 0.0003 0.0058
 None
 6A-PVPV15090101 0 0 - 0.0565 0.0025
 Serotype 6A detected
 polysaccharide* None None
 6A-PVPV15110102 0.02 0 - 0.061
 detected detected

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None
 6A-PVPV15120103 0 0 - 0.026 0.002
 detected
 20090201 0 0 - 0.0182 0.0007 0.0125
 Serotype 6B
 20090202 0.5 0 - 0.0178 0.0007 0.0125
 polysaccharide
 20090203 2.3 0 - 0.0175 0.0006 0.0121
 20090802 0.1 0 - 0.0045 0.0011 0.0116
 Serotype 7F
 20090803 1.5 0 - 0.0045 0.0011 0.0118
 polysaccharide
 20090804 2.3 0 - 0.0046 0.0011 0.0117
 20090201 0 0 - 0.0069 0.0042 0.0064
 Serotype 9V
 20090302 0 0 - 0.0070 0.0041 0.0064
 polysaccharide
 20090303 0 0 - 0.0066 0.0068 0.0079
 20090301 0 0 - 0.0021 0.0009 0.0043
 Serotype 14
 20090302 0 0 - 0.0020 0.0009 0.0043
 polysaccharide
 20090303 0 0 - 0.0020 0.0010 0.0043
 20090301 0.09 0 - 0.0046 0.0018 0.0062
 Serotype 18C
 20090302 0. 0 - 0.0046 0.0018 0.0062
 polysaccharide
 20090303 2.64 0 - 0.0046 0.0018 0.0063
 20090902 0 0 - 0.0535 0.0004 0.0035
 Serotype 19A
 20090903 0 0 - 0.0534 0.0004 0.0035
 polysaccharide
 20090904 0 0 - 0.0544 0.0004 0.0035
 20090301 0 0 - 0.0571 0.0002 0.0031
 Serotype 19F
 20090302 0 0 - 0.0559 0.0002 0.0031
 polysaccharide
 20090403 0 0 - 0.0555 0.0002 0.0032
 20090401 0 0 - 0.0018 0.0025 0.0140
 Serotype 23F
 20090402 0 0 - 0.0019 0.0024 0.0137
 polysaccharide
 20090403 0 0 - 0.0019 0.0026 0.0140
 Notes: "-" indicates this item was not tested for this sample, "*" indicates the 3 continuous batches of serotype
 6A polysaccharide produced in process validation approval.

Those test results show good reproducibility of ethanol, ether and acetone concentration between continuous 3 batches. There's difference between different serotypes, but residual contents remain low, the average residual ethanol is 119.22 µg/mL (polysaccharide concentration at 3.0 mg/mL), the average residual ether is 8.87 µg/mL (polysaccharide concentration at 3.0 mg/mL), the average residual acetone is 47.09 µg/mL (polysaccharide concentration at 3.0 mg/mL). According to the chemical handbook, the boiling points of ethanol, ether and acetone at 1 atm is 78.37 °C, 34.6 °C and 56.48 °C respectively, the lower the boiling point, the easier it is to be removed with vacuum drying, which accounted for the lowest average residual of ether and highest average residual of ethanol.

The results of residual DOC, phenol and CTAB show that no residual phenol was present in 13 serotypes of polysaccharide. No CTAB was detected in the only serotype that used CTAB in the process, serotype 3. The average residual DOC concentration was low at 0.575 µg/mL (polysaccharide concentration at 3 mg/mL). The production process can effectively remove DOC, phenol and CTAB.

According to ICH and Chinese Pharmacopeia (2010 version), ethanol, ether and acetone are low toxicity solvents (type 3 solvents) and have limited impact on human health. Acute toxicity and short-term toxicity tests show they're almost non-toxic and non-genotoxic. No discussion is needed if content intake is 50 mg/day or lower.

Walvax conducted further validation of removal of product-related and process-related impurities through planning to be listed commercial-scale process validation during clinical studies. The

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results showed the removal of the above impurities is effective. The results are shown in Table 34 to Table 46.

Table 34 Test Result of Serotype 1 Pneumococcal Polysaccharide Impurities

Nucleic Sodium					
Sample Item	Protein	Ethanol	Acetone	Ether	
acid deoxycholate					
types Batch No.	≤ 2%	< 0.08%	< 0.03%	< 0.01%	
≤ 2%	< 0.2%				
Clinical					
PN1201504002	0.67	1.5	0.02	0.007	- -
sample					
Process					
PN1201609001	1.1	1.3	0.09	- 0.001	-
validation					
Process					
PN1201610002	0.5	1.1	0.1	- - -	
validation					
Process					
PN1201611004	0.1	0.5	- 0.017	0.0006	-
validation					
Process					
PN1201702001	0.8	1.3	0.07	- - -	
validation					

Table 35 Test Result of Serotype 3 Pneumococcal Polysaccharide Impurities

Nucleic Sodium					
Sample Item	Protein	Ethanol	Acetone	Ether	
acid deoxycholate					
types Batch No.	≤ 2%	< 0.08%	< 0.03%	< 0.01%	
≤ 2%	< 0.2%				
Clinical					
PN3201504001	0.10	0.1	0.14	0.002	- -
sample					
Process					
3-PVPV15100901	0.04	0.1	0.03	0.002	- -
validation					
Process					
3-PVPV15110902	0.09	0.4	0.16	- - -	
validation					
Process					
3-PVPV15120903	1.99	0.5	0.17	- - -	
validation					
Process					
SP3P201606002	0.10	- 0.02	0.002	- -	
validation					
Process					
SP3P201608003	0.10	0.03	0.07	0.002	- -
validation					

Table 36 Test Result of Serotype 4 Pneumococcal Polysaccharide Impurities

Nucleic Sodium					
Sample Item	Protein	Ethanol	Acetone	Ether	
acid deoxycholate					
types Batch No.	≤ 2%	< 0.08%	< 0.03%	< 0.01%	
≤ 2%	< 0.2%				
Clinical					
PN4 201503001	0.17	1.5	0.13	0.009	0.001 -
sample					
Process					
4-PVPV15121102	0.3	1.3	0.16	0.006	- -
validation					
Process					
4-PVPV16011103	0.6	2.7	0.11	0.001	- -
validation					
Process					
SP4P201605001	0.4	0.9	0.04	0.016	0.004 -
validation					

Table 37 Test Result of Serotype 5 Pneumococcal Polysaccharide Impurities

Nucleic Sodium					
Sample Item	Protein	Ethanol	Acetone	Ether	
acid deoxycholate					
types Batch No.	≤ 2%	< 0.08%	< 0.03%	< 0.01%	
≤ 2%	< 0.2%				
Clinical					
PN5201503001	0.2	5.89	0.13	0.02	0.0045 -
sample					
Process					
PN5201701001	0.7	7.30	0.17	0.005	- -
validation					
Process					
PN5201701002	0.8	6.42	0.12	0.0045	- -
validation					
Process					

PN5201701003 0.5 5.72 0.12 0.005 - -
validation

Table 38 Test Result of Serotype 6A Pneumococcal Polysaccharide Impurities
Nucleic Protei Sodium
Sample Item Ethanol Acetone Ether
acid n deoxycholate
types Batch No. < 0.08% < 0.03% < 0.01%
≤ 2% ≤ 2% < 0.2%
Clinical
PN6A201503001 0.17 0.95 0.15 0.062 0.006 0.004
sample
Process
6A-PVPV15090101 0.082 0.36 0 0.0565 0.0025 -
validation
Process
6A-PVPV15110102 0.08 1.16 0.02 0.061 - -
validation
Process
6A-PVPV15120103 0.10 1.30 0 0.026 0.002 -
validation

Table 39 Test Result of Serotype 6B Pneumococcal Polysaccharide Impurities
Nucleic Sodium
Sample Item Protein Ethanol Acetone Ether
acid deoxycholate
types Batch No. ≤ 2% < 0.08% < 0.03% < 0.01%
≤ 2% < 0.2%
Clinical
PN6B201503001 0.2 0.86 0.05 0.066 0.002 -
sample
Process
6B-PVPV15110202 0.1 0.4 0.11 0.064 0.004 -
validation
Process
6B-PVPV15120203 0.1 1.4 0.18 0.0275 0.002 -
validation
Process
SP6BP201605001 0.1 0.2 0.05 0.044 - -
validation

Table 40 Test Result of Serotype 7F Pneumococcal Polysaccharide Impurities
Nuclei Sodium
Sample Item Protein Ethanol Acetone Ether
c acid deoxycholate
types Batch No. ≤ 2% < 0.08% < 0.03% < 0.01%
≤ 2% < 0.2%
Clinical
PN7F201504001 0.44 2.5 0.05 0.001 0.002 -
sample
Process
7F-PVPV15100701 0.4 0.2 0.04 0.022 0.001 -
validation
Process
7F-PVPV16010703 0.2 1.1 0.05 0.005 - -
validation

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Process
SP7FP201606001 0.2 0.6 0.08 0.031 0.003 -
validation
Process
SP7FP201608003 0.4 0.9 0.04 0.016 0.007 0.003
validation

Table 41 Test Result of Serotype 9V Pneumococcal Polysaccharide Impurities
Nucleic Sodium
Sample Item Protein Ethanol Acetone Ether
acid deoxycholate
types Batch No. ≤ 2% < 0.08% < 0.03% < 0.01%
≤ 2% < 0.2%
Clinical
PN9V201503001 0.04 0.2 0.04 0.007 - -
sample
Process
9V-PVPV15120303 0.06 0.3 0.18 - 0.001 -
validation
Process
SP9VP201605001 0.10 0.1 0.03 0.001 - -
validation
Process
SP9VP201607002 0.10 0.2 0.05 0.013 0.0006 -
validation

Table 42 Test Result of Serotype 14 Pneumococcal Polysaccharide Impurities
Nucleic Sodium
Sample Item Protein Ethanol Acetone Ether
acid deoxycholate
types Batch No. ≤ 2% < 0.08% < 0.03% < 0.01%

≤ 2% < 0.2%
Clinical
PN14201503001 0.22 1.6 0.06 0.009 - -
sample
Process
PN14201610001 0.2 3.1 0.18 0.0095 - -
validation
Process
PN14201610002 0.3 2.7 0.09 0.0045 - -
validation
Process
PN14201703001 0.4 1.1 0.04 - - -
validation

Table 43 Test Result of Serotype 18C Pneumococcal Polysaccharide Impurities

Nucleic Sodium
Sample Item Protein Ethanol Acetone Ether
acid deoxycholate
types Batch No. ≤ 2% < 0.08% < 0.03% < 0.01%
≤ 2% < 0.2%
Clinical
PN18C201504001 0.14 0.96 0.09 0.02 - -
sample
Process
18C-PVPV15110802 0.2 0.8 0.06 0.034 0.002 -
validation
Process
18C-PVPV16010803 0.2 1.3 0.11 0.046 0.002 -
validation
Process
SP18CP201608003 0.7 1.2 0.003 0.006 - -
validation

Table 44 Test Result of Serotype 19A Pneumococcal Polysaccharide impurities

Nucleic Sodium
Sample Item Protein Ethanol Acetone Ether
acid deoxycholate
types Batch No. ≤ 2% < 0.08% < 0.03% < 0.01%
≤ 2% < 0.2%

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Clinical
PN19A201504001 0.05 0.5 0.07 0.015 0.002 -
sample
Process
19A-PVPV15110402 0.02 0.7 0.004 0.0145 0.002 -
validation
Process
19A-PVPV15120403 0.05 0.5 0.15 0.005 0.002 0.0001
validation
Process
SP19AP201606001 0.10 0.3 0.03 0.011 0.001 -
validation

Table 45 Test Result of Serotype 19F Pneumococcal Polysaccharide Impurities

Nucleic Sodium
Sample Item Protein Ethanol Acetone Ether
acid deoxycholate
types Batch No. ≤ 2% < 0.08% < 0.03% < 0.01%
≤ 2% < 0.2%
Clinical
PN19F201504001 0.09 0.6 0.1 0.013 0.002 -
sample
Process
19F-PVPV15110502 0.10 0.2 0 0.0145 - -
validation
Process
19F-PVPV15120503 0.10 0.3 0.13 0.0035 0.0005 -
validation
Process
SP19FP201606001 0.2 0.1 0.1 0.01 0.002 -
validation

Table 46 Test Result of Serotype 23F Pneumococcal Polysaccharide Impurities

Nucleic Sodium
Sample Item Protein Ethanol Acetone Ether
acid deoxycholate
types Batch No. ≤ 2% < 0.08% < 0.03% < 0.01%
≤ 2% < 0.2%
Clinical
PN23F201504001 0.18 0.6 0.04 0.017 0.002 -
sample
Process
23F-PVPV15120602 0.1 0.2 0.17 0.01 0.003 -
validation
Process
23F-PVPV16010603 0.1 0.3 0.08 0.0015 - -
validation
Process

Table 50 Test Results of 5 Batches of Serotype 5 Polysaccharide Derivative and Bulk Conjugate

Polysaccharide

Bulk conjugate derivative

Items Residual ADH Residual Derivation Free- Free-

Batch No. cyanide content EDAC rate polysaccharide protein

(≤ 10 (< 0.25 (≤ 5 (5.4-18.0%) (≤ 20%) (≤ 4.5%) ng/mg) μg/mL) μmol/L)

PN5201505002 3.6 5.31 0.240 1.9575 19.5 3.3

PN5201703002 0.7 13.19 0.012 0.3270 8.2 2.6

RSP5C201705001 2.1 17.89 0.139 0.7000 18.0 3.2

None

RSP5C201707002 2.5 16.08 0.102 0.8000 17.4 detected

RSP5C201708003 1.1 10.01 0.102 3.7000 15.4 0.5

Table 51 Test Results of 5 Batches of Serotype 6A Polysaccharide Derivative and Bulk Conjugate

Polysaccharide

Bulk conjugate derivative

Items Residual ADH Residual Derivation Free- Free-

Batch No. cyanide content EDAC rate polysaccharide protein

(≤ 10 (< 1.2 (≤ 5 (4.0-6.4%) (≤ 20%) (≤ 4.5%) ng/mg) μg/mL) μmol/L)

PN6A201504001 0.61 3.90 0.932 0.4355 19.7 1.97

PN6A201611001 0.80 4.90 0.215 3.213 17.4 3.3

None

RSP6AC201705001 4..27 0.163 3.6 13.7 3.6 detected

RSP6AC201707002 2.10 5.09 0.081 2.6 19.4 0.9

RSP6AC201708003 2.60 4.28 0.227 2.3 12.8 0.5

Table 52 Test Results of 5 Batches of Serotype 6B Polysaccharide Derivative and Bulk Conjugate

Polysaccharide

Bulk conjugate derivative

Items derivative

Batch No. Residual Derivation ADH Residual Free- Free- cyanide rate content EDAC polysaccharide protein

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(≤ 10 (1.5-3.4%) (< 0.25 (≤ 5 (≤ 25%) (≤ 4.0%) ng/mg) μg/mL) μmol/L)

PN6B 201504001 0.63 2.92 0.15050 0.637 17.3 0.13

PN6B 201611001 1.90 2.71 0.1740 2.409 17.9 1.4

None

RSP6BC201705001 2.33 0.0120 2.800 21.6 4.0 detected

RSP6BC201707002 3.10 2.57 0.0940 0.500 21.1 0.9

RSP6BC201708003 6.70 2.42 0.1412 0.100 9.2 2.1

Table 53 Test Results of 5 Batches of Serotype 7F Polysaccharide Derivative and Bulk Conjugate

Polysaccharide

Bulk conjugate derivative

Items Residual ADH Residual Derivation Free- Free-

Batch No. cyanide content EDAC rate polysaccharide protein

(≤ 10 (< 0.2 (≤ 5 (2.9-5.3%) (≤ 20%) (≤ 4.5%) ng/mg) μg/mL) μmol/L)

PN7F201504002 1.4 4.28 0.150 0.149 16.7 1.69

None

PN7F201704003 3.35 0.137 2.362 13.6 3.80 detected

None

RSP7FC201705001 4.76 0.066 3.200 18.4 4.00 detected

RSP7FC201706002 3.8 3.85 0.121 1.200 18.5 0.30

RSP7FC201708003 6.10 4.44 0.088 3.700 18.9 1.30

Table 54 Test Results of 5 Batches of Serotype 9V Polysaccharide Derivative and Bulk Conjugate

Polysaccharide

Bulk conjugate derivative

Items Residual ADH Residual Derivation Free- Free-

Batch No. cyanide content EDAC

rate polysaccharide protein
 (≤ 10 (< 0.7 (≤ 5
 (3.3-7.4%) (≤ 25%) (≤ 3.5%)
 ng/mg) µg/mL) µmol/L)
 PN9V201506002 0.6 3.94 0.582 3.2035 2.0 0.7
 None
 PN9V201612001 6.41 0.177 4.6130 1.1 2.2
 detected
 RSP9VC201705001 0.1 5.30 0.176 2.5000 6.8 2.9
 RSP9VC201707002 1.5 6.09 0.153 2.2000 12.0 1.6
 RSP9VC201709003 1.8 4.98 0.203 4.1000 11.8 0.7

Table 55 Test Results of 5 Batches of Serotype 14 Polysaccharide Derivative and Bulk Conjugate

Polysaccharide
 Bulk conjugate
 derivative
 Items Residual ADH Residual
 Derivation Free-
 Batch No. cyanide content EDAC Free-protein
 rate polysaccharide
 (≤ 10 (< 1.15 (≤ 5 (≤ 4.5%)
 (2.5-5.9%) (≤ 25%)
 ng/mg) µg/mL) µmol/L)
 PN14201505003 4.20 4.28 0.832 1.9098 8.1 0.71
 PN14201704003 0.05 3.93 0.208 3.6940 12.7 3.80

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None
 RSP14C201706001 5.11 0.016 2.4000 14.6 3.10
 detected
 None
 RSP14C201707002 4.86 0.120 3.8000 12.6 1.00
 detected
 RSP14C201709003 1.60 3.52 0.114 1.0000 23.9 1.00

Table 56 Test Results of 5 Batches of Serotype 18C Polysaccharide Derivative and Bulk Conjugate

Polysaccharide
 Bulk conjugate
 derivative
 Items Residual ADH Residual
 Derivation Free- Free-
 Batch No. cyanide content EDAC
 rate polysaccharide protein
 (≤ 10 (< 0.2 (≤ 5
 (2.0-5.4%) (≤ 30%) (≤ 3.5%)
 ng/mg) µg/mL) µmol/L)
 PN18C201505002 3.3 3.61 0.170 0.9155 10.5 2.01
 PN18C201612001 0.9 2.58 0.122 2.1130 12.0 0.60
 RSP18C201706001 3.5 3.39 0.097 0.9000 7.7 2.70
 None
 RSP18CC201707002 3.84 0.108 1.5000 26.2 0.9
 detected
 RSP18CC201708003 1.0 3.38 0.015 0.7000 22.6 0.7

Table 57 Test Results of 5 Batches of Serotype 19A Polysaccharide Derivative and Bulk Conjugate

Polysaccharide
 Bulk conjugate
 derivative
 Items Residual ADH Residual
 Derivation Free- Free-
 Batch No. cyanide content EDAC
 rate polysaccharide protein
 (≤ 10 (< 0.2 (≤ 5
 (3.5-8.0%) (≤ 30%) (≤ 3.5%)
 ng/mg) µg/mL) µmol/L)
 PN19A201504002 2.6 6.21 0.1700 0.3630 21.6 0.30
 PN19A201611001 2.8 5.63 0.1575 2.8845 10.2 2.00
 None
 RSP19AC201705001 3.97 0.1080 1.2000 27.4 2.90
 detected
 RSP19AC201707002 4.7 5.58 0.1300 1.3000 26.0 2.80
 RSP19AC201708003 7.0 6.36 0.1030 2.300 28.0 0.13

Table 58 Test Results of 5 Batches of Serotype 19F Polysaccharide Derivative and Bulk Conjugate

Polysaccharide
 Bulk conjugate
 derivative
 Items Residual ADH Residual
 Derivation Free- Free-
 Batch No. cyanide content EDAC
 rate polysaccharide protein
 (≤ 10 (< 0.2 (≤ 5
 (2.9-6.2%) (≤ 35%) (≤ 3.5%)
 ng/mg) µg/mL) µmol/L)
 PN19F201504001 6.5 5.09 0.174 1.3340 14.6 1.39

PN19F201611001 0.5 5.07 0.089 2.5945 6.1 2.00
RSP19FC201706001 3.8 3.92 0.073 3.9000 24.4 3.20
RSP19FC201707002 1.3 5.26 0.126 1.4000 30.1 2.40
RSP19FC201708003 5.9 5.00 0.066 3.1000 25.2 1.10

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Table 59 Test Results of 5 Batches of Serotype 23F Polysaccharide Derivative and Bulk Conjugate

Polysaccharide

Bulk conjugate

derivative

Items Residual ADH Residual

Derivation Free- Free-

Batch No. cyanide content EDAC

rate polysaccharide protein

(≤ 10 (< 0.45 (≤ 5

(0.9-3.7%) ($\leq 25\%$) ($\leq 4.0\%$)

ng/mg) $\mu\text{g/mL}$) $\mu\text{mol/L}$)

PN23F201505002 2.6 2.95 0.4065 2.2840 7.4 0.15

PN23F201611001 2.1 2.29 0.2600 2.4775 4.4 3.00

None

RSP23FC201705001 3.12 0.1620 0.7000 13.0 1.60

detected

RSP23FC201707002 3.7 2.35 0.1740 2.5000 24.9 2.10

RSP23FC201708003 2.3 2.00 0.1600 2.1000 18.7 0.60

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2.3.S.4 Control of Drug Substance

2.3.S.4.1 Justification of Specification for Drug Substance

2.3.S.4.1.1 Justification of Specification for Pneumococcal Polysaccharide

Table 60 Justification of Specification for Serotype 1 Pneumococcal Polysaccharide

Test items Test method Specifications Ph. Eur. WHO Rationale for specifications

Double immunodiffusion test is an

Double Appear positive By an immunochemical A test should be performed on

immunochemical method which

Identity test immunodiffusion response to specific method or other suitable the purified polysaccharide

to

confirms the presence of serotype 1

test antiserum type 1 methods verify its identity.

polysaccharide.

Not more than the limit

Each lot of purified

approved for the

polysaccharide should typically The specification is required by Ph.

Protein content Lowry method $\leq 2\%$ product, calculated with

contain no more Eur. (2150).

reference to the dried

than 3% by weight of protein.

substance

Sufficient polysaccharide shall

UV-vis be assayed to detect 2% nucleic The specification is required by Ph.

Nucleic acid content $\leq 2\%$ $\leq 2\%$

spectrometry acid Eur. (2150).

contamination accurately.

The specification is required by Ph.

Total nitrogen content Kjeldahl method 3.5-6.0% 3.5-6.0% 3.5-6.0%

Eur. (2150).

Chromogenic The specification is required by Ph.

Phosphorus content 0-1.5% 0-1.5% 0-1.5%

method Eur. (2150).

Chromogenic The specification is required by Ph.

Uronic acid content $\geq 45\%$ $\geq 45\%$ $\geq 45\%$

method Eur. (2150).

Chromogenic The specification is required by Ph.

O-acetyl group ≥ 1.8 ≥ 1.8 ≥ 1.8

method Eur. (2150).

Recovery rate is calculated as per

KD ≤ 0.15 , recovery Appropriate detection An acceptable level of

Polysaccharide Molecular Size

Gel rate at KD ≤ 0.30 system, an acceptable consistency should be agreed

Molecular size (KD) Distribution (WSYX-QC-TS-0006-

chromatography should be no less than value is established for with the NRA and can be

0597), which is approved by

75% each Ps. established either by process

NMPA.

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validation or by measurement

on each lot

Bacteria endotoxin is tested to limit

pyrogenic effect of the product. It is

Bacterial endotoxin Gel-clot method ≤ 0.5 EU/ μg // an additional quality control

according to in-house standard (WSYX-QC-TS-0609-0004). Gas Ethanol, acetone and ether are used Ethanol residual < 0.08% // chromatography to control quality of bulk Gas polysaccharide. These test items are Acetone residual < 0.03% // chromatography additional quality control according Gas to in-house standard (WSYX-QC-Ether residual < 0.01% // chromatography TS-0609-0004). The limit is set according to Quality Standard for Pneumococcal UV-vis Sodium deoxycholate < 0.20 // Polysaccharide Conjugate Vaccine spectrometry (WSYX-QC-TS-0006-0721), which is approved by NMPA. The pyrogen content of the purified polysaccharide should Pyrogen test required by WHO can Pyrogen // be determined and shown to be be represented by endotoxin test within acceptable limits agreed by the NRA.

Table 61 Justification of Specification for Serotype 3 Pneumococcal Polysaccharide Test items Test method Specifications Ph. Eur. WHO Rationale for specifications Double immunodiffusion test is a Double Appear positive By an immunochemical A test should be performed on immunochemical method which Identity test immunodiffusion response to specific method or other suitable the purified polysaccharide to confirms the presence of serotype 3 test antiserum type 3 methods verify its identity. polysaccharide. Each lot of purified Not more than the limit polysaccharide should typically The specification is required by Ph. Protein content Lowry method $\leq 5\%$ approved for the contain not more Eur. (2150). product, calculated with than 3% by weight of protein. Confidential 85 MODULE 2 COMMON TECHNICAL DOCUMENT SUMMARIES Version No. Walvax-202007-00 Yuxi Walvax Biotechnology Co., Ltd.

reference to the dried substance Sufficient polysaccharide shall UV-vis The specification is required by Ph. Nucleic acid content $\leq 2\%$ $\leq 2\%$ be assayed to detect 2% nucleic spectrometry Eur. (2150). acid contamination accurately. The specification is required by Ph. Total nitrogen content Kjeldahl method 0-1.0% 0-1.0% 0-1% Eur. (2150). Chromogenic The specification is required by Ph. Phosphorus content 0-1.0% 0-1.0% 0-1% method Eur. (2150). Chromogenic The specification is required by Ph. Uronic acid content $\geq 40\%$ $\geq 40\%$ $\geq 40\%$ method Eur. (2150). An acceptable level of Recovery rate is calculated as per $KD \leq 0.15$, recovery Appropriate detection consistency should be agreed Polysaccharide Molecular Size Gel rate at $KD \leq 0.30$ system, an acceptable with the NRA and can be Molecular size (KD) Distribution (WSYX-QC-TS-0006- chromatography should be no less than value is established for established either by process 0597), which is approved by 78% each Ps. validation or by measurement NMPA. on each lot. Bacteria endotoxin is tested to limit pyrogenic effect of the product. It is Bacterial endotoxin Gel-clot method ≤ 0.5 EU/ μg // an additional quality control according to in-house standard (WSYX-QC-TS-0609-0004). Gas Ethanol, acetone and ether are used Ethanol residual < 0.08% // chromatography to control quality of bulk Gas polysaccharide. These test items are Acetone residual < 0.03% // chromatography additional quality control according Gas to in-house standard (WSYX-QC-Ether residual < 0.01% // chromatography TS-0609-0004). The limit is set according to Quality Standard for Pneumococcal UV-vis Sodium deoxycholate < 0.2 // Polysaccharide Conjugate Vaccine

spectrometry
(WSYX-QC-TS-0006-0721), which
is approved by NMPA.

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The pyrogen content of the
purified polysaccharide should Pyrogen test result required by
Pyrogen / / be determined and shown to be WHO can be represented by
within acceptable limits agreed endotoxin test.
by the NRA.

Table 62 Justification of Specification for Serotype 4 Pneumococcal Polysaccharide
Test items Test method Specifications Ph. Eur. WHO Rationale for specifications
Double immunodiffusion test is an
Double Appear positive By an immunochemical A test should be performed on
immunochemical method which
Identity test immunodiffusion response to specific method or other suitable the purified polysaccharide
to
confirms the presence of serotype 4
test antiserum type 4 methods verify its identity.
polysaccharide.
Each lot of purified
polysaccharide should typically The specification is required by Ph.
Protein content Lowry method $\leq 3\% \leq 3\%$
contain no more than 3% by Eur. (2150).
weight of protein.
Sufficient polysaccharide shall
UV-vis The specification is required by Ph.
Nucleic acid content $\leq 2\% \leq 2\%$ be assayed to detect 2% nucleic
spectrometry Eur. (2150).
acid contamination accurately.
The specification is required by Ph.
Total nitrogen content Kjeldahl method 4.0-6.0% 4.0-6.0% 4-6%
Eur. (2150).
Chromogenic The specification is required by Ph.
Phosphorus content 0-1.5% 0-1.5% 0-1.5%
method Eur. (2150).
Chromogenic The specification is required by Ph.
Hexosamine content $\geq 40\% \geq 40\% \geq 40\%$
method Eur. (2150).
Chromogenic The specification is required by Ph.
Methyl pentose / / $\geq 10\%$
method Eur. (2150).
An acceptable level of
Recovery rate is calculated as per
KD ≤ 0.15 , recovery Appropriate detection consistency should be agreed
Polysaccharide Molecular Size
Gel rate at KD ≤ 0.30 system, an acceptable with the NRA and can be
Molecular size (KD) Distribution (WSYX-QC-TS-0006-
chromatography should be no less than value is established for established either by process
0597), which is approved by
75% each Ps. validation or by measurement
NMPA.
on each lot

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Bacteria endotoxin is tested to limit
pyrogenic effect of the product. It is
Bacterial endotoxin Gel-clot method $\leq 0.5 \text{ EU}/\mu\text{g}$ / / an additional quality control
according to in-house standard
(WSYX-QC-TS-0609-0004).
Gas Ethanol, acetone and ether are used
Ethanol residual $< 0.08\%$ / /
chromatography to control quality of bulk
Gas polysaccharide. These test items are
Acetone residual $< 0.03\%$ / /
chromatography additional quality control according
Gas to in-house standard (WSYX-QC-
Ether residual $< 0.01\%$ / /
chromatography TS-0609-0004).
The limit is set according to Quality
Standard for Pneumococcal
UV-vis
Sodium deoxycholate < 0.20 / / Polysaccharide Conjugate Vaccine
spectrometry
(WSYX-QC-TS-0006-0721), which
is approved by NMPA.
The pyrogen content of the
purified polysaccharide should
Pyrogen test required by WHO can
Pyrogen / / be determined and shown to be
be represented by endotoxin test
within acceptable limits agreed
by the NRA.

Table 63 Justification of Specification for Serotype 5 Pneumococcal Polysaccharide

Test items Test method Specifications Ph. Eur. WHO Rationale for specifications
 Double immunodiffusion test is an
 Double Appear positive By an immunochemical A test should be performed on
 immunochemical method which
 Identity test immunodiffusion response to specific method or other suitable the purified polysaccharide
 to
 confirms the presence of serotype 5
 test antiserum serotype 5 methods verify its identity.
 polysaccharide.
 Each lot of purified
 polysaccharide should typically The specification is required by Ph.
 Protein content Lowry method $\leq 7.5\% \leq 7.5\%$
 contain no more than 3% by Eur. (2150).
 weight of protein.

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Sufficient polysaccharide shall
 UV-vis The specification is required by Ph.
 Nucleic acid content $\leq 2\% \leq 2\%$ be assayed to detect 2% nucleic
 spectrometry Eur. (2150).
 acid contamination accurately.
 The specification is required by Ph.
 Total nitrogen content Kjeldahl method 2.5-6.0% 2.5-6.0% 2.5-6%
 Eur. (2150).
 Chromogenic The specification is required by Ph.
 Phosphorus content $\leq 2.0\% \leq 2.0\% < 2\%$
 method Eur. (2150).
 Chromogenic The specification is required by Ph.
 Uronic acid content $\geq 12\% \geq 12\% \geq 12\%$
 method Eur. (2150).
 Chromogenic The specification is required by Ph.
 Hexosamine content $\geq 20\% \geq 20\% \geq 20\%$
 method Eur. (2150).
 An acceptable level of
 Recovery rate is calculated as per
 $KD \leq 0.60$, recovery Appropriate detection consistency should be agreed
 Polysaccharide Molecular Size
 Gel rate at $KD \leq 0.60$ system, an acceptable with the NRA and can be
 Molecular size (KD) Distribution (WSYX-QC-TS-0006-
 chromatography should be no less than value is established for established either by process
 0597), which is approved by
 70% each Ps. validation or by measurement
 NMPA.
 on each lot
 Bacteria endotoxin is tested to limit
 pyrogenic effect of the product. It is
 Bacterial endotoxin Gel-clot method $\leq 0.5EU/\mu g$ // an additional quality control
 according to in-house standard
 (WSYX-QC-TS-0609-0004).
 Gas Ethanol, acetone and ether are used
 Ethanol residual $< 0.08\%$ //
 chromatography to control quality of bulk
 Gas polysaccharide. These test items are
 Acetone residual $< 0.03\%$ //
 chromatography additional quality control according
 Gas to in-house standard (WSYX-QC-
 Ether residual $< 0.01\%$ //
 chromatography TS-0609-0004).
 The limit is set according to Quality
 Standard for Pneumococcal
 UV-vis
 Sodium deoxycholate < 0.20 // Polysaccharide Conjugate Vaccine
 spectrometry
 (WSYX-QC-TS-0006-0721), which
 is approved by NMPA.
 The pyrogen content of the Pyrogen test required by WHO can
 Pyrogen // //
 purified polysaccharide should be represented by endotoxin test

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be determined and shown to be
 within acceptable limits agreed
 by the NRA.

Table 64 Justification of Specification for Serotype 6A Pneumococcal Polysaccharide
 Test items Test method Specifications Ph. Eur. WHO Rationale for specifications
 Double immunodiffusion test is an
 Double Appear positive By an immunochemical A test should be performed on
 immunochemical method which
 Identity test immunodiffusion response to specific method or other suitable the purified polysaccharide
 to
 confirms the presence of serotype
 test antiserum type 6A methods verify its identity.
 6A polysaccharide.
 Each lot of purified
 polysaccharide should typically The specification is required by Ph.

Protein content Lowry method $\leq 2\% \leq 7.5\%$
 contain no more than 3% by Eur. (2150).
 weight of protein.
 Sufficient polysaccharide shall
 UV-vis The specification is required by Ph.
 Nucleic acid content $\leq 1\% \leq 2\%$ be assayed to detect 2% nucleic
 spectrometry Eur. (2150).
 acid contamination accurately.
 The specification is required by Ph.
 Total nitrogen content Kjeldahl method 0-2.0% 2.5-6.0% /
 Eur. (2150).
 Chromogenic The specification is required by Ph.
 Phosphorus content 2.4-5.0% $\leq 2.0\%$ /
 method Eur. (2150).
 Chromogenic The specification is required by Ph.
 Methyl pentose content $\geq 15\% \geq 12\%$ /
 method Eur. (2150).
 An acceptable level of
 Recovery rate is calculated as per
 $KD \leq 0.50$, recovery Appropriate detection consistency should be agreed
 Polysaccharide Molecular Size
 Gel rate at $KD \leq 0.60$ system, an acceptable with the NRA and can be
 Molecular size (KD) Distribution (WSYX-QC-TS-0006-
 chromatography should be no less than value is established for established either by process
 0597), which is approved by
 50% each Ps. validation or by measurement
 NMPA.
 on each lot.
 Bacteria endotoxin is tested to limit
 pyrogenic effect of the product. It is
 Bacterial endotoxin Gel-clot method $\leq 0.5EU/\mu g$ / / an additional quality control
 according to in-house standard
 (WSYX-QC-TS-0609-0004).
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Gas Ethanol, acetone and ether are used
 Ethanol residual $< 0.08\%$ / /
 chromatography to control quality of bulk
 Gas polysaccharide. These test items are
 Acetone residual $< 0.03\%$ / /
 chromatography additional quality control according
 Gas to in-house standard (WSYX-QC-
 Ether residual $< 0.01\%$ / /
 chromatography TS-0609-0004).
 The limit is set according to Quality
 Standard for Pneumococcal
 UV-vis
 Sodium deoxycholate < 0.20 / / Polysaccharide Conjugate Vaccine
 spectrometry
 (WSYX-QC-TS-0006-0721), which
 is approved by NMPA.
 The pyrogen content of the
 purified polysaccharide should
 Pyrogen test required by WHO can
 Pyrogen / / / be determined and shown to be
 be represented by endotoxin test.
 within acceptable limits agreed
 by the NRA.

Table 65 Justification of Specification for Serotype 6B Pneumococcal Polysaccharide
 Test items Test method Specifications Ph. Eur. WHO Rationale for specifications
 Double immunodiffusion test is an
 Double Appear positive By an immunochemical A test should be performed on
 immunochemical method which
 Identity test immunodiffusion response to specific method or other suitable the purified polysaccharide
 to
 confirms the presence of serotype
 test antiserum type 6B methods verify its identity.
 6B polysaccharide.
 Each lot of purified
 polysaccharide should typically The specification is required by Ph.
 Protein content Lowry method $\leq 2\% \leq 7.5\%$
 contain no more than 3% by Eur. (2150).
 weight of protein.
 Sufficient polysaccharide shall
 UV-vis The specification is required by Ph.
 Nucleic acid content $\leq 2\% \leq 2\%$ be assayed to detect 2% nucleic
 spectrometry Eur. (2150).
 acid contamination accurately.
 The specification is required by Ph.
 Total nitrogen content Kjeldahl method 0-2.0% 2.5-6.0% 0-2.0%
 Eur. (2150).
 Chromogenic The specification is required by Ph.
 Phosphorus content 2.5-5.0% $\leq 2.0\%$ 2.5-5.0%
 method Eur. (2150).
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Chromogenic The specification is required by Ph.

Methyl pentose content $\geq 15\% \geq 12\% \geq 15\%$
method Eur. (2150).
An acceptable level of
Recovery rate is calculated as per
 $KD \leq 0.50$, recovery Appropriate detection consistency should be agreed
Polysaccharide Molecular Size
Gel rate at $KD \leq 0.60$ system, an acceptable with the NRA and can be
Molecular size (KD) Distribution (WSYX-QC-TS-0006-
chromatography should be no less than value is established for established either by process
0597), which is approved by
85% each Ps. validation or by measurement
NMPA.
on each lot.
Bacteria endotoxin is tested to limit
pyrogenic effect of the product. It is
Bacterial endotoxin Gel-clot method $\leq 0.5\text{EU}/\mu\text{g}$ / / an additional quality control
according to in-house standard
(WSYX-QC-TS-0609-0004).
Gas Ethanol, acetone and ether are used
Ethanol residual $< 0.08\%$ / /
chromatography to control quality of bulk
Gas polysaccharide. These test items are
Acetone residual $< 0.03\%$ / /
chromatography additional quality control according
Gas to in-house standard (WSYX-QC-
Ether residual $< 0.01\%$ / /
chromatography TS-0609-0004).
The limit is set according to Quality
Standard for Pneumococcal
UV-vis
Sodium deoxycholate < 0.20 / / Polysaccharide Conjugate Vaccine
spectrometry
(WSYX-QC-TS-0006-0721), which
is approved by NMPA.
The pyrogen content of the
purified polysaccharide should
Pyrogen test required by WHO can
Pyrogen / / / be determined and shown to be
be represented by endotoxin test
within acceptable limits agreed
by the NRA.

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Table 66 Justification of Specification for Serotype 7F Pneumococcal Polysaccharide
Test items Test method Specifications Ph. Eur. WHO Rationale for specifications
Double immunodiffusion test is an
Double Appear positive By an immunochemical A test should be performed on
immunochemical method which
Identity test immunodiffusion response to specific method or other suitable the purified polysaccharide
to
confirms the presence of serotype
test antiserum type 7F methods verify its identity.
7F polysaccharide.
Each lot of purified
polysaccharide should typically The specification is required by Ph.
Protein content Lowry method $\leq 5\% \leq 5\%$
contain no more than 3% by Eur. (2150).
weight of protein.
Sufficient polysaccharide shall
UV-vis The specification is required by Ph.
Nucleic acid content $\leq 2\% \leq 2\%$ be assayed to detect 2% nucleic
spectrometry Eur. (2150).
acid contamination accurately.
The specification is required by Ph.
Total nitrogen content Kjeldahl method 1.5-4.0% 1.5-4.0% 1.5-4.0%
Eur. (2150).
Chromogenic The specification is required by Ph.
Phosphorus content 0-1.0% 0-1.0% 0-1.0%
method Eur. (2150).
Chromogenic The specification is required by Ph.
Methyl pentose content $\geq 13\% \geq 13\% \geq 13\%$
method Eur. (2150).
Chromogenic The specification is required by Ph.
Hexosamine content / / 33.09%
method Eur. (2150).
Chromogenic The specification is required by Ph.
O-acetyl group content / / 3.5%
method Eur. (2150).
An acceptable level of
Recovery rate is calculated as per
 $KD \leq 0.20$, recovery Appropriate detection consistency should be agreed
Polysaccharide Molecular Size
Gel rate at $KD \leq 0.30$ system, an acceptable with the NRA and can be
Molecular size (KD) Distribution (WSYX-QC-TS-0006-
chromatography should be no less than value is established for established either by process
0597), which is approved by
70% each Ps. validation or by measurement
NMPA.
on each lot.

Bacteria endotoxin is tested to limit pyrogenic effect of the product. It is Bacterial endotoxin Gel-clot method $\leq 0.5 \text{ EU}/\mu\text{g}$ // an additional quality control according to in-house standard (WSYX-QC-TS-0609-0004).

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Gas Ethanol, acetone and ether are used
Ethanol residual $< 0.08\%$ // chromatography to control quality of bulk
Gas polysaccharide. These test items are
Acetone residual $< 0.03\%$ // chromatography additional quality control according
Gas to in-house standard (WSYX-QC-Ether residual $< 0.01\%$ // chromatography TS-0609-0004).
The limit is set according to Quality Standard for Pneumococcal UV-vis
Sodium deoxycholate < 0.20 // Polysaccharide Conjugate Vaccine spectrometry
(WSYX-QC-TS-0006-0721), which is approved by NMPA.
The pyrogen content of the purified polysaccharide should
Pyrogen test required by WHO can
Pyrogen // be determined and shown to be
be represented by endotoxin test.
within acceptable limits agreed
by the NRA.

Table 67 Justification of Specification for Serotype 9V Pneumococcal Polysaccharide
Test items Test method Specifications Ph. Eur. WHO Rationale for specifications
Double immunodiffusion test is an
Double Appear positive By an immunochemical A test should be performed on
immunochemical method which
Identity test immunodiffusion response to specific method or other suitable the purified polysaccharide
to
confirms the presence of serotype
test antiserum type 9V methods verify its identity.
9V polysaccharide.
Each lot of purified
polysaccharide should typically The specification is required by Ph.
Protein content Lowry method $\leq 2\% \leq 2\%$
contain not more than 3% by Eur. (2150).
weight of protein.
Sufficient polysaccharide shall
UV-vis The specification is required by Ph.
Nucleic acid content $\leq 2\% \leq 2\%$ be assayed to detect 2% nucleic
spectrometry Eur. (2150).
acid contamination accurately.
The specification is required by Ph.
Total nitrogen content Kjeldahl method 0.5-3.0% 0.5-3.0% 0.5-3.0%
Eur. (2150).
Chromogenic The specification is required by Ph.
Phosphorus content 0-1.0% 0-1.0% 0-1.0%
method Eur. (2150).

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Chromogenic The specification is required by Ph.
Uronic acid content $\geq 15\% \geq 15\% \geq 15\%$
method Eur. (2150).
Chromogenic The specification is required by Ph.
Hexosamine content $\geq 13\% \geq 13\% \geq 13\%$
method Eur. (2150).
An acceptable level of
Recovery rate is calculated as per
 $KD \leq 0.45$, recovery Appropriate detection consistency should be agreed
Polysaccharide Molecular Size
Gel rate at $KD \leq 0.60$ system, an acceptable with the NRA and can be
Molecular size (KD) Distribution (WSYX-QC-TS-0006-
chromatography should be no less than value is established for established either by process
0597), which is approved by
80% each Ps. validation or by measurement
NMPA.
on each lot.
Bacteria endotoxin is tested to limit
pyrogenic effect of the product. It is
Bacterial endotoxin Gel-clot method $\leq 0.5 \text{ EU}/\mu\text{g}$ // an additional quality control
according to in-house standard
(WSYX-QC-TS-0609-0004).
Gas Ethanol, acetone and ether are used
Ethanol residual $< 0.08\%$ // chromatography to control quality of bulk
Gas polysaccharide. These test items are
Acetone residual $< 0.03\%$ // chromatography additional quality control according

Gas to in-house standard (WSYX-QC-Ether residual < 0.01% // chromatography TS-0609-0004). The limit is set according to Quality Standard for Pneumococcal UV-vis Sodium deoxycholate < 0.20 // Polysaccharide Conjugate Vaccine spectrometry (WSYX-QC-TS-0006-0721), which is approved by NMPA. The pyrogen content of the purified polysaccharide should Pyrogen test required by WHO can Pyrogen /// be determined and shown to be be represented by endotoxin test within acceptable limits agreed by the NRA.

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Table 68 Justification of Specification for Serotype 14 Pneumococcal Polysaccharide
Test items Test method Specifications Ph. Eur. WHO Rationale for specifications
Double immunodiffusion test is an Double Appear positive By an immunochemical A test should be performed on immunochemical method which Identity test immunodiffusion response to specific method or other suitable the purified polysaccharide to confirms the presence of serotype test antiserum type 14 methods verify its identity. 14 polysaccharide. Each lot of purified polysaccharide should typically The specification is required by Ph. Protein content Lowry method $\leq 5\% \leq 5\%$ contain no more than 3% by Eur. (2150). weight of protein. Sufficient polysaccharide shall UV-vis The specification is required by Ph. Nucleic acid content $\leq 2\% \leq 2\%$ be assayed to detect 2% nucleic spectrometry Eur. (2150). acid contamination accurately. The specification is required by Ph. Total nitrogen content Kjeldahl method 1.5-4.0% 1.5-4.0% 1.5-4.0% Eur. (2150). Chromogenic The specification is required by Ph. Phosphorus content 0-1.0% 0-1.0% 0-1.0% method Eur. (2150). Chromogenic The specification is required by Ph. Hexosamine content $\geq 20\% \geq 20\% \geq 20\%$ method Eur. (2150). An acceptable level of Recovery rate is calculated as per $KD \leq 0.30$, recovery Appropriate detection consistency should be agreed Polysaccharide Molecular Size Gel rate at $KD \leq 0.30$ system, an acceptable with the NRA and can be Molecular size (KD) Distribution (WSYX-QC-TS-0006- chromatography should be no less than value is established for established either by process 0597), which is approved by 55% each Ps. validation or by measurement NMPA. on each lot. Bacteria endotoxin is tested to limit pyrogenic effect of the product. It is Bacterial endotoxin Gel-clot method $\leq 0.5 \text{ EU}/\mu\text{g}$ // an additional quality control according to in-house standard (WSYX-QC-TS-0609-0004). Gas Ethanol, acetone and ether are used Ethanol residual < 0.08% // chromatography to control quality of bulk Gas polysaccharide. These test items are Acetone residual < 0.03% // chromatography additional quality control according

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Gas to in-house standard (WSYX-QC-Ether residual < 0.01% // chromatography TS-0609-0004). The limit is set according to Quality Standard for Pneumococcal UV-vis Sodium deoxycholate < 0.20 // Polysaccharide Conjugate Vaccine spectrometry (WSYX-QC-TS-0006-0721), which is approved by NMPA. The pyrogen content of the purified polysaccharide should Pyrogen test required by WHO can Pyrogen /// be determined and shown to be

be represented by endotoxin test
within acceptable limits agreed
by the NRA.

Table 69 Justification of Specification for Serotype 18C Pneumococcal Polysaccharide

Test items Test method Specifications Ph. Eur. WHO Rationale for specifications

Double immunodiffusion test is an

Double Appear positive By an immunochemical A test should be performed on

immunochemical method which

Identity test immunodiffusion response to specific method or other suitable the purified polysaccharide
to

confirms the presence of serotype

test antiserum type 18C methods verify its identity.

18C polysaccharide.

Each lot of purified

polysaccharide should typically The specification is required by Ph.

Protein content Lowry method $\leq 3\% \leq 3\%$

contain not more than 3% by Eur. (2150).

weight of protein.

Sufficient polysaccharide shall

UV-vis The specification is required by Ph.

Nucleic acid content $\leq 2\% \leq 2\%$ be assayed to detect 2% nucleic

spectrometry Eur. (2150).

acid contamination accurately.

The specification is required by Ph.

Total nitrogen content Kjeldahl method 0-1.0% 0-1.0% 0-1.0%

Eur. (2150).

Chromogenic The specification is required by Ph.

Phosphorus content 2.4-4.9% 2.4-4.9% 2.4-4.9%

method Eur. (2150).

Chromogenic The specification is required by Ph.

Methyl pentose content $\geq 14\% \geq 14\% \geq 14\%$

method Eur. (2150).

Chromogenic Complies the requirement of Ph.

O-acetyl group content / / 4.24%

method Eur. (2150).

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An acceptable level of

Recovery rate is calculated as per

$KD \leq 0.15$, recovery Appropriate detection consistency should be agreed

Polysaccharide Molecular Size

Gel rate at $KD \leq 0.30$ system, an acceptable with the NRA and can be

Molecular size (KD) Distribution (WSYX-QC-TS-0006-

chromatography should be no less than value is established for established either by process

0597), which is approved by

70% each Ps. validation or by measurement

NMPA.

on each lot.

Bacteria endotoxin is tested to limit

pyrogenic effect of the product. It is

Bacterial endotoxin Gel-clot method $\leq 0.5 \text{ EU}/\mu\text{g}$ / / an additional quality control

according to in-house standard

(WSYX-QC-TS-0609-0004).

Gas Ethanol, acetone and ether are used

Ethanol residual $< 0.08\%$ / /

chromatography to control quality of bulk

Gas polysaccharide. These test items are

Acetone residual $< 0.03\%$ / /

chromatography additional quality control according

Gas to in-house standard (WSYX-QC-

Ether residual $< 0.01\%$ / /

chromatography TS-0609-0004).

The limit is set according to Quality

Standard for Pneumococcal

UV-vis

Sodium deoxycholate < 0.20 / / Polysaccharide Conjugate Vaccine

spectrometry

(WSYX-QC-TS-0006-0721), which

is approved by NMPA.

The pyrogen content of the

purified polysaccharide should

Pyrogen test required by WHO can

Pyrogen / / / be determined and shown to be

be represented by endotoxin test

within acceptable limits agreed

by the NRA.

Table 70 Justification of Specification for Serotype 19A Pneumococcal Polysaccharide

Test items Test method Specifications Ph. Eur. WHO Rationale for specifications

Double immunodiffusion test is an

Double Appear positive By an immunochemical A test should be performed on

immunochemical method which

Identity test immunodiffusion response to specific method or other suitable the purified polysaccharide

to

confirms the presence of serotype

test antiserum type 19A methods verify its identity.

19A polysaccharide.

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Each lot of purified polysaccharide should typically The specification is required by Ph. Protein content Lowry method $\leq 2\% \leq 2\%$ contain no more than 3% by Eur. (2150). weight of protein. Sufficient polysaccharide shall UV-vis The specification is required by Ph. Nucleic acid content $\leq 2\% \leq 2\%$ be assayed to detect 2% nucleic spectrometry Eur. (2150). acid contamination accurately. The specification is required by Ph. Total nitrogen content Kjeldahl method 0.6-3.5% 0.6-3.5% 0.6-3.5% Eur. (2150). Chromogenic The specification is required by Ph. Phosphorus content 3.0-7.0% 3.0-7.0% 3.0-7.0% method Eur. (2150). Chromogenic The specification is required by Ph. Hexosamine content $\geq 12\% \geq 12\% \geq 12\%$ method Eur. (2150). Chromogenic The specification is required by Ph. Methyl pentose content $\geq 20\% \geq 20\% \geq 20\%$ method Eur. (2150). An acceptable level of Recovery rate is calculated as per $KD \leq 0.15$, recovery Appropriate detection consistency should be agreed Polysaccharide Molecular Size Gel rate at $KD \leq 0.30$ system, an acceptable with the NRA and can be Molecular size (KD) Distribution (WSYX-QC-TS-0006-chromatography should be no less than value is established for established either by process 0597), which is approved by 75% each Ps. validation or by measurement NMPA. on each lot. Bacteria endotoxin is tested to limit pyrogenic effect of the product. It is Bacterial endotoxin Gel-clot method $\leq 0.5 \text{ EU}/\mu\text{g}$ // an additional quality control according to in-house standard (WSYX-QC-TS-0609-0004). Gas Ethanol, acetone and ether are used Ethanol residual $< 0.08\%$ // chromatography to control quality of bulk Gas polysaccharide. These test items are Acetone residual $< 0.03\%$ // chromatography additional quality control according Gas to in-house standard (WSYX-QC-Ether residual $< 0.01\%$ // chromatography TS-0609-0004). The limit is set according to Quality UV-vis Sodium deoxycholate < 0.20 // Standard for Pneumococcal spectrometry Polysaccharide Conjugate Vaccine

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(WSYX-QC-TS-0006-0721), which is approved by NMPA. The pyrogen content of the purified polysaccharide should Pyrogen test required by WHO can Pyrogen / / / be determined and shown to be be represented by endotoxin test within acceptable limits agreed by the NRA.

Table 71 Justification of Specification for Serotype 19F Pneumococcal Polysaccharide Test items Test method Specifications Ph. Eur. WHO Rationale for specifications Double immunodiffusion test is an Double Appear positive By an immunochemical A test should be performed on immunochemical method which Identity test immunodiffusion response to specific method or other suitable the purified polysaccharide to confirms the presence of serotype test antiserum type 19F methods verify its identity. 19F polysaccharide.

Each lot of purified polysaccharide should typically The specification is required by Ph. Protein content Lowry method $\leq 3\% \leq 3\%$ contain not more than 3% by Eur. (2150). weight of protein. Sufficient polysaccharide shall UV-vis The specification is required by Ph. Nucleic acid content $\leq 2\% \leq 2\%$ be assayed to detect 2% nucleic spectrometry Eur. (2150). acid contamination accurately. The specification is required by Ph. Total nitrogen content Kjeldahl method 1.4-3.5% 1.4-3.5% 1.4-3.5% Eur. (2150).

Chromogenic The specification is required by Ph.
 Phosphorus content 3.0-5.5% 3.0-5.5% 3.0-5.5%
 method Eur. (2150).
 Chromogenic The specification is required by Ph.
 Hexosamine content $\geq 12.5\%$ $\geq 12.5\%$ $\geq 12.5\%$
 method Eur. (2150).
 Chromogenic The specification is required by Ph.
 Methyl pentose content $\geq 20\%$ $\geq 20\%$ $\geq 20\%$
 method Eur. (2150).
 Recovery rate is calculated as per
 $KD \leq 0.15$, recovery Appropriate detection An acceptable level of
 Polysaccharide Molecular Size
 Gel rate at $KD \leq 0.30$ system, an acceptable consistency should be agreed
 Molecular size (KD) Distribution (WSYX-QC-TS-0006-
 chromatography should be no less than value is established for with the NRA and can be
 0597), which is approved by
 75% each Ps. established either by process
 NMPA.
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validation or by measurement
 on each lot
 Bacteria endotoxin is tested to limit
 pyrogenic effect of the product. It is
 Bacterial endotoxin Gel-clot method ≤ 0.5 EU/ μ g // an additional quality control
 according to in-house standard
 (WSYX-QC-TS-0609-0004).
 Gas Ethanol, acetone and ether are used
 Ethanol residual $< 0.08\%$ //
 chromatography to control quality of bulk
 Gas polysaccharide. These Test items
 Acetone residual $< 0.03\%$ //
 chromatography are additional quality control
 Gas according to in-house standard
 Ether residual $< 0.01\%$ //
 chromatography (WSYX-QC-TS-0609-0004).
 The limit is set according to Quality
 Standard for Pneumococcal
 UV-vis
 Sodium deoxycholate < 0.20 // Polysaccharide Conjugate Vaccine
 spectrometry
 (WSYX-QC-TS-0006-0721), which
 is approved by NMPA.
 The pyrogen content of the
 purified polysaccharide should
 Pyrogen test required by WHO can
 Pyrogen // be determined and shown to be
 be represented by endotoxin test
 within acceptable limits agreed
 by the NRA.

Table 72 Justification of Specification for Serotype 23F Pneumococcal Polysaccharide
 Test items Test method Specifications Ph. Eur. WHO Rationale for specifications
 Double immunodiffusion test is an
 Double Appear positive By an immunochemical A test should be performed on
 immunochemical method which
 Identity test immunodiffusion response to specific method or other suitable the purified polysaccharide
 to
 confirms the presence of serotype 1
 test antiserum type 23F methods verify its identity.
 polysaccharide.
 Each lot of purified
 polysaccharide should typically The specification is required by Ph.
 Protein content Lowry method $\leq 2\%$ $\leq 2\%$
 contain not more than 3% by Eur. (2150).
 weight of protein.
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Sufficient polysaccharide shall
 UV-vis The specification is required by Ph.
 Nucleic acid content $\leq 2\%$ $\leq 2\%$ be assayed to detect 2% nucleic
 spectrometry Eur. (2150).
 acid contamination accurately.
 The specification is required by Ph.
 Total nitrogen content Kjeldahl method 0-1.0% 0-1.0% 0-1%
 Eur. (2150).
 Chromogenic The specification is required by Ph.
 Phosphorus content 3.0-4.5% 3.0-4.5% 3.0-4.5%
 method Eur. (2150).
 Chromogenic The specification is required by Ph.
 Methyl pentose content $\geq 37\%$ $\geq 37\%$ $\geq 37\%$
 method Eur. (2150).
 Chromogenic The specification is required by Ph.
 O-acetyl group content // 4.24%
 method Eur. (2150).
 An acceptable level of
 Recovery rate is calculated as per
 $KD \leq 0.15$, recovery Appropriate detection consistency should be agreed

Polysaccharide Molecular Size
Gel rate at $KD \leq 0.30$ system, an acceptable with the NRA and can be
Molecular size (KD) Distribution (WSYX-QC-TS-0006-
chromatography should be no less than value is established either by process
0597), which is approved by
75% each Ps. validation or by measurement
NMPA.
on each lot
Bacteria endotoxin is tested to limit
pyrogenic effect of the product. It is
Bacterial endotoxin Gel-clot method ≤ 0.5 EU/ μ g // an additional quality control
according to in-house standard
(WSYX-QC-TS-0609-0004).
Gas Ethanol, acetone and ether are used
Ethanol residual $< 0.08\%$ //
chromatography to control quality of bulk
Gas polysaccharide. These test items are
Acetone residual $< 0.03\%$ //
chromatography additional quality control according
Gas to in-house standard (WSYX-QC-
Ether residual $< 0.01\%$ //
chromatography TS-0609-0004).
The limit is set according to Quality
Standard for Pneumococcal
UV-vis
Sodium deoxycholate < 0.20 // Polysaccharide Conjugate Vaccine
spectrometry
(WSYX-QC-TS-0006-0721), which
is approved by NMPA.
The pyrogen content of the Pyrogen test required by WHO can
Pyrogen // //
purified polysaccharide should be represented by endotoxin test

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be determined and shown to be
within acceptable limits agreed
by the NRA.

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2.3.S.4.1.2 Justification of Specification for Pneumococcal Polysaccharide Derivatives
Table 73 Justification of Specification for Pneumococcal Polysaccharide Derivatives
Test items Test method Specifications Ph. Eur. WHO Rationale for specifications
The manufacturer should
demonstrate consistency
of the degree of
modification of the
Serotype Rate (%) polysaccharide, either by
1 2.6 - 4.4 an assay of each batch of
3 5.0 - 14.0 the polysaccharide or by
4 3.0 - 11.0 validation of the
5 3.5 - 18.0 manufacturing process. The limit is set according to
6A 3.0 - 6.0 Depending on the Quality Standard for
Chromogenic 6B 1.5 - 3.4 conjugation chemicals Pneumococcal Polysaccharide
Derivation rate /
method 7F 2.9 - 5.3 used, consistency in Conjugate Vaccine (WSYX-
9V 3.3 - 7.4 degree of polysaccharide QC-TS-0006-0721), which is
14 2.5 - 5.9 activation may be approved by NMPA.
18C 2.0 - 4.8 determined as part of
19A 3.5 - 8.0 process validation or
19F 3.2 - 6.2 reflected by
23F 1.1 - 3.7 characteristics of vaccine
lots shown to have
adequate safety and
immunogenicity in
clinical trials.
Several methods for the
The limit is set according to
chemical modification of
Quality Standard for
polysaccharides prior to
Residual Pneumococcal Polysaccharide
Lowry method No more than 10 ng/mg / conjugation may be
cyanide Conjugate Vaccine (WSYX-
satisfactory. The chosen
QC-TS-0006-0721), which is
method should be
approved by NMPA.
approved by the NRA.
 $KD = 0.2$ - In the case of a The degree of size The limit is set according to
Gel $KD > 0.6$
Molecular size Serotype $KD 0.6$ size-reduced reduction of the Quality Standard for
chromatography recovery
recovery modified polysaccharide will Pneumococcal Polysaccharide
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$1 \leq 0.37 \leq 45\%$ - pneumococcal depend upon the Conjugate Vaccine (WSYX-
 $3 \leq 0.45 \leq 60\%$ - polysaccharide, manufacturing process. QC-TS-0006-0721), which is
 $4 \leq 0.42 \leq 65\%$ - the molecular- The average size approved by NMPA.
 $5 \leq 0.90 \leq 37\%$ size or distribution (degree of
 $6A \leq 0.62 \leq 60\%$ - molecular-mass polymerization) of the
 $6B \leq 0.33 \leq 50\%$ - distributions modified polysaccharide
 $7F \leq 0.43 \leq 60\%$ - determined by should be determined by a
 $9V \leq 0.46 \leq 70\%$ - size-exclusion suitable method and
 $14 \leq 0.40 \leq 65\%$ - chromatography shown to be consistent.
 $18C \leq 0.41 \leq 65\%$ - combined with The molecular size
 an appropriate distribution should be
 $19A \leq 0.48 \leq 65\%$ -
 detection specified for each
 $19F \leq 0.56 \leq 70\%$ -
 system. An serotype, with appropriate
 $23F \leq 0.46 \leq 75\%$ -
 acceptable limits for consistency, as
 value is the size may affect the
 established for reproducibility of the
 each modified conjugation process.
 pneumococcal
 polysaccharide.

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2.3.S.4.1.3 Justification of Specification for Bulk Tetanus Toxoid

Table 74 Justification of Specification for Bulk Tetanus Toxoid

Test items Test method Specifications Ph. Eur. WHO Rationale for specifications

Specification is set according to

Quality Standard for Pneumococcal

Polysaccharide Conjugate Vaccine

(WSYX-QC-TS-0006-0721), which

pH pH meter 6.6 - 7.4 / /

is approved by NMPA, and

supported by process capability and

is well within range of safety

concern.

Limit of The antigenic purity is

flocculation No less than 1500 no less than 1500Lf per The specification is required by Ph.

Purity /

Lf/mg milligram of protein Eur. (2150).

Kjedahl method nitrogen.

Specification is set according to

Quality Standard for Pneumococcal

Polysaccharide Conjugate Vaccine

Should be in

(WSYX-QC-TS-0006-0721), which

Sterility Film filtration accordance with the / /

is approved by NMPA, and

regulations

supported by process capability and

is well within range of safety

concern.

Observation on the

Specification is set according to

7th, 14th and 21st day

Quality Standard for Pneumococcal

showing no local

Polysaccharide Conjugate Vaccine

suppuration and

(WSYX-QC-TS-0006-0721), which

Specific toxicity Animal test necrosis, no tetanus / /

is approved by NMPA, and

symptoms. Weight

supported by process capability and

gain for each guinea

is well within range of safety

pig at the end of the

concern.

period.

Observation on the Specification is set according to

Toxicity reversal test Animal test / /

7th, 14th and 21st day Quality Standard for Pneumococcal

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showing no tetanus Polysaccharide Conjugate Vaccine

symptoms, weight (WSYX-QC-TS-0006-0721), which

gain for each guinea is approved by NMPA, and

pig at the end of the supported by process capability and

period. is well within range of safety

concern.

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2.3.S.4.1.4 Justification of Specification for Purified Liquid of Tetanus Toxoid
 Table 75 Justification of Specification for Purified Liquid of Tetanus Toxoid
 Test items Test method Specifications Ph. Eur. WHO Rationale for specifications
 The identity of tetanus toxoid is
 The identity may be determined tested in tetanus toxoid bulk
 Identity / / /
 serologically. production with flocculant test,
 which is a serological method.
 The limit is set according to Quality
 Standard for Pneumococcal
 No less than 16 Should be approved by the
 Protein content Lowry method / Polysaccharide Conjugate Vaccine
 mg/mL NRA.
 (WSYX-QC-TS-0006-0721), which
 is approved by NMPA.
 The limit is set according to Quality
 Standard for Pneumococcal
 No more than 10
 Bacterial endotoxin Gel-clot method / / Polysaccharide Conjugate Vaccine
 EU/mg
 (WSYX-QC-TS-0006-0721), which
 is approved by NMPA.
 The limit is set according to Quality
 Liquid
 Standard for Pneumococcal
 chromatography
 Purity No less than 90% / / Polysaccharide Conjugate Vaccine
 (UV absorption
 (WSYX-QC-TS-0006-0721), which
 at 280 nm)
 is approved by NMPA.

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2.3.S.4.1.5 Justification of Specification for Monovalent Bulk Conjugate
 Table 76 Justification of Specification for Monovalent Bulk Conjugate
 Test items Test method Specifications Ph. Eur. WHO Rationale for specifications
 Significant color change may be
 indicative of contamination or
 Clear, transparent liquid or
 Visual degradation. Levels of color are
 Appearance slightly white floccule after / /
 determination set to ensure consistency with
 shaking
 clinical, production and stability
 experience.
 A test should be performed on
 the monovalent bulk to verify
 A clear precipitation line
 its identity. The method Double immunodiffusion test is an
 should appear with the
 Double should be validated to show immunochemical method which
 corresponding
 Identity immunodiffusion / that it distinguishes the confirms the presence of a specific
 pneumococcal specific
 test desired monovalent material serotype of polysaccharide-tetanus
 antiserum and tetanus
 from all other polysaccharides toxoid conjugate.
 toxoid immune serum.
 and conjugates produced on
 that manufacturing site.
 No less than the limit
 below.
 Serotype Limit
 (µg/mL) The polysaccharide
 1 40 content is determined
 3 20 by a suitable physical The limit is set according to
 4 40 or chemical method or Quality Standard for
 Polysaccharide 5 40 by an Pneumococcal Polysaccharide
 Turbidimetry /
 content 6A 40 immunochemical Conjugate Vaccine (WSYX-QC-
 6B 40 method. The value TS-0006-0721), which is approved
 7F 40 complies with the by NMPA.
 9V 40 requirement approved
 14 40 for each serotype.
 18C 40
 19A 40
 19F 40

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23F 40
 The protein content is The protein content of the
 determined by a conjugate should be The limit is set according to
 suitable physical or determined by means of an Quality Standard for
 chemical method. The appropriate validated assay Pneumococcal Polysaccharide
 Protein content Lowry method No less than 70 µg/mL
 value complies with and comply with limits for the Conjugate Vaccine (WSYX-QC-
 the requirement product. Each batch should be TS-0006-0721), which is approved

approved for each tested for conjugated and by NMPA.
serotype. unbound protein.
Serotype Range
1 0.20 - 1.80
3 0.10 - 0.73
4 0.30 - 0.90
5 0.10 - 0.60 Determine the For each conjugate, the ratio The specification is required by
6A 0.10 - 0.50 polysaccharide to should be within the range Ph. Eur. (2150). The limit is set
protein ratio by approved for that particular according to Quality Standard for
6B 0.30 - 0.60
PS: protein ratio Calculation calculation. The value conjugate by the NRA and Pneumococcal Polysaccharide
7F 0.20 - 0.90
compiles with the should be consistent with Conjugate Vaccine (WSYX-QC-
9V 0.10 - 0.35
requirement approved vaccine shown to be effective TS-0006-0721), which is approved
14 0.20 - 0.60 for each serotype. in clinical trials. by NMPA.
18C 0.40 - 0.80
19A 0.20 - 0.70
19F 0.20 - 0.70
23F 0.25 - 0.60
KD < Molecular-size or The relative molecular size of
Serotype KD 0.2 molecular-mass the polysaccharide-protein
recovery distribution is conjugate should be
The specification is required by
1 ≥ 85% determined by size- determined for each bulk,
Ph. Eur. (2150). The limit is set
3 ≥ 95% exclusion using a gel matrix appropriate
according to Quality Standard for
Gel chromatography to the size of the conjugate.
Molecular size (KD) 4 ≥ 80% Pneumococcal Polysaccharide
chromatography combined with an The method should be
5 ≤ 80% Conjugate Vaccine (WSYX-QC-
appropriate detection validated with an emphasis on
6A 0.2 ≥ 80% TS-0006-0721), which is approved
system. An acceptable specificity to distinguish the
6B ≥ 90% value is established for polysaccharide-protein
by NMPA.
7F ≥ 90% the bulk conjugate of conjugate from other
9V ≥ 95% each polysaccharide. components that may be
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14 ≥ 90% Each batch must be present, e.g. unbound protein
18C ≥ 80% shown to comply with or polysaccharide. The size
19A ≥ 85% this limit. distribution specifications will
19F ≥ 80% be vaccine-specific and
23F ≥ 90% should be consistent with lots
shown to be immunogenic in
clinical trials.
Unbound
No more than the limit polysaccharide is
below. determined after
Serotype Limit (%) removal of the
1 25.0 conjugate, for example
Each batch of conjugate
3 20.0 by anion-exchange
should be tested for unbound
4 25.0 size-exclusion or The specification is required by
or free polysaccharide in
5 20.0 hydrophobic Ph. Eur. (2150). The limit is set
order to establish consistency
6A 20.0 chromatography, according to Quality Standard for
Free polysaccharide Anthrone-sulfate of production and to ensure
ultrafiltration, or other Pneumococcal Polysaccharide
content method 6B 25.0 that the amount present in the
validated methods. A Conjugate Vaccine (WSYX-QC-
7F 20.0 purified bulk is within the
value consistent with TS-0006-0721), which is approved
9V 25.0 limits agreed by the NRA
adequate by NMPA.
14 25.0 based on lots shown to be
immunogenicity as
18C 30.0 clinically safe and efficacious.
shown in clinical trials
19A 30.0 is established foreach
19F 35.0 serotype and each lot
23F 25.0 must be shown to
comply with this limit
Determine the content
by a suitable method,
either directly or by The limit is set according to
deriving the content Quality Standard for
Free carrier protein Gel by calculation from Pneumococcal Polysaccharide
No more than 5.0% /
content chromatography the results of other Conjugate Vaccine (WSYX-QC-
tests. The value TS-0006-0721), which is approved
complies with the by NMPA.
requirement approved
for each serotype.
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Where applicable, suitable tests are carried out to determine residues of The conjugate purification reagents used during procedures should remove inactivation and residual reagents used for purification. An conjugation and capping. The acceptable value for removal of reagents and The limit is set according to each reagent is reaction by products such as Quality Standard for established for the Chromogenic cyanide, 1-ethyl-3,3-(3- Pneumococcal Polysaccharide Residual EDAC No more than 5.0 µmol/L product and each method dimethylaminopropyl)- Conjugate Vaccine (WSYX-QC- batch of conjugate carbodiimide (EDAC) and TS-0006-0721), which is approved must be shown to others, depending on the by NMPA. comply with this limit conjugation chemistry, should Where validation be confirmed by suitable tests studies have or by validation of the demonstrated removal purification process. of residual reagents, the test on conjugate polysaccharides maybe omitted. No more than the limit below.

Limit
Serotype
(µg/mL)
1 0.15 The limit is set according to
3 0.25 Quality Standard for
Chromogenic 4 0.15 Pneumococcal Polysaccharide
ADH content / /
method 5 0.25 Conjugate Vaccine (WSYX-QC-
6A 1.20 TS-0006-0721), which is approved
6B 0.25 by NMPA.
7F 0.20
9V 0.70
14 1.15
18C 0.20

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19A 0.20
19F 0.20
23F 0.45
The bulk purified conjugate should be tested for bacterial and mycotic sterility in It complies with the accordance with the The specification is required by test for sterility, requirements of the revised Ph. Eur. (2150). The limit is set carried out using 10 General requirements for the according to Quality Standard for Meets the requirement of ml for each medium or sterility of biological Sterility Film filtration Pneumococcal Polysaccharide the test. the equivalent of 100 substances or by a method Conjugate Vaccine (WSYX-QC- doses for each approved by the NRA. If a TS-0006-0721), which is approved medium, whichever is preservative has been added by NMPA. less to the product, appropriate measures should be taken to prevent it from interfering with the test. To ensure an acceptable level of endotoxin in the final Should less than 0.75 product, the endotoxin content The specification is required by Bacterial endotoxin Gel-clot method No more than 0.75 EU/µg IU per microgram of of the monovalent bulk may Ph. Eur. (2150). polysaccharide. be determined and shown to be within acceptable limits agreed by the NRA.

The bulk conjugate should be
Observation on the 7th, 14th
tested for the absence of The limit is set according to
and 21st day showing no
specific toxicity of the carrier Quality Standard for
local suppuration and
Specific toxin of protein where appropriate. Pneumococcal Polysaccharide
Animal test necrosis, no tetanus /
carrier protein Absence of specific toxicity Conjugate Vaccine (WSYX-QC-
symptoms. Weight gain for
of the carrier protein may also TS-0006-0721), which is approved
each guinea pig at the end
be assessed through validation by NMPA.
of the period.
of the production process.

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2.3.S.4.2 Analytical Procedures 2.3.S.4.2.1 SOP for Polysaccharide Test

Table 77 SOP List of Pneumococcal Polysaccharides Test
Tests SOP Document code Test method
WSYX-QC-SOP-00-09- Double
Identity SOP for Identity Test
0201 immunodiffusion
SOP for Bacterial WSYX-QC-SOP-00-09-
Bacterial endotoxin Gel-clot method
Endotoxin 0203
SOP for Total Solid WSYX-QC-SOP-00-09- Heat-dry and weighing
Total solid
Content 0111 method
SOP for Protein Content WSYX-QC-SOP-00-09-
Protein content Lowry method
Determination 0102
SOP for Nucleic Acid WSYX-QC-SOP-00-09- UV-vis
Nucleic acid content
Content Determination 0103 spectrophotometry
SOP for Phosphorus WSYX-QC-SOP-00-09-
Phosphorus content Chromogenic method
Content Determination 0101
SOP for Hexosamine WSYX-QC-SOP-00-09-
Hexosamine content Chromogenic method
Content Determination 0127
SOP for Uronic Acid WSYX-QC-SOP-00-09-
Uronic acid content Chromogenic method
Content Determination 0129
SOP for O-acetyl group WSYX-QC-SOP-00-09-
O-acetyl group Chromogenic method
Content Determination 0112
SOP for Methyl pentose WSYX-QC-SOP-00-09-
Methyl pentose content Chromogenic method
Content Determination 0128
SOP for Molecular Size WSYX-QC-SOP-00-09-
Molecular size Gel chromatography
Determination 0104
SOP for Phenol Content WSYX-QC-SOP-00-09-
Phenol content Titration
Determination 0115
Residual ethanol SOP for Residual Gas chromatography
WSYX-QC-SOP-00-09-
Residual acetone Ethanol, Ether, Acetone Gas chromatography
0138
Residual ether Content Determination Gas chromatography
SOP for Total Nitrogen WSYX-QC-SOP-00-09-
Total nitrogen Kjeldahl method
Content Determination 0751
SOP for Residual
WSYX-QC-SOP-00-09- UV-vis
Sodium deoxycholate Sodium Deoxycholate
0137 spectrophotometry
Content Determination

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2.3.S.4.2.2 SOP for Polysaccharide Derivative Test

Table 78 SOP List of Pneumococcal Polysaccharide Derivatives test
Tests SOP Document code Test method
WSYX-QC-SOP-00-09- Heat-dry and weighing
Total solid SOP for Total Solid Content
0111 method
SOP for Adipic Dihydrazide WSYX-QC-SOP-00-09-
Chromogenic method
Content Determination 0105
Derivation rate
SOP for Glucose Analysis WSYX-QC-SOP-00-09-
Anthrone method

with Anthrone 0126
SOP for Residual Cyanide WSYX-QC-SOP-00-09-
Residual cyanide Chromogenic method
Content Determination 0106
SOP for Molecular Size WSYX-QC-SOP-00-09-
Molecular size Gel chromatography
Determination 0104

2.3.S.4.2.3 SOP for Carrier Protein

Table 79 SOP List of Bulk Tetanus Toxoid Test
Tests SOP Document code Test method
Specification of
Diphtheria, tetanus, and
WSYX-QC-TS-00-09-
Appearance acellular pertussis Visual determination
0001
combined vaccine,
adsorbed
SOP for pH Value WSYX-QC-SOP-00-09-
pH pH meter
Determination 0110
SOP for Residual WSYX-QC-SOP-00-09-
Residual ammonium Kjeldahl method
Ammonium Determination 0754
Protein nitrogen SOP for Protein Nitrogen WSYX-QC-SOP-00-09-
Kjeldahl method
content Content Determination 0774
SOP for Toxoid
WSYX-QC-SOP-00-09-
Flocculation units Flocculation Units Serological method
0011
Determination
Calculation (flocculation
WSYX-QC-SOP-00-09-
Purity SOP for Purity Test units/protein nitrogen
0209
content)
WSYX-QC-SOP-00-09- Membrane filtration
Sterility SOP for Sterility Test
0202 method
SOP for Protein Content - WSYX-QC-SOP-00-09-
Protein content Lowry method
Lowry Method 0134
SOP for Specific Toxicity
WSYX-QC-SOP-02-09-
Specific toxicity Test of Tetanus Toxoid Animal Test
0003
Bulk
SOP for Toxicity Reverse
WSYX-QC-SOP-02-09-
Toxicity Reversal Test of Tetanus Toxoid Animal Test
0002
Bulk
SOP for Detoxification
WSYX-QC-SOP-02-09-
Detoxification test Test of Tetanus Toxoid Animal Test
0004
Bulk

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Table 80 SOP List of Purified Liquid of Tetanus Toxoid Test
Tests SOP Document code Test method
SOP for Protein Content - WSYX-QC-SOP-00-09-
Protein content Lowry method
Lowry Method 0134
WSYX-QC-SOP-00-09- Membrane filtration
Sterility SOP for Sterility Test
0202 method
SOP for Bacterial Endotoxin WSYX-QC-SOP-00-09-
Bacterial endotoxin Gel-clot method
Test 0203
SOP for Purity Test of
WSYX-QC-SOP-00-09-
Purity Tetanus Toxoid Purified HPLC
0143
Liquid

2.3.S.4.2.4 SOP for Monovalent Bulk Conjugate

Table 81 SOP List of Monovalent Bulk Conjugate Test
Tests SOP Document code Test method
SOP for Appearance WSYX-QC-SOP-00-09-
Appearance Visual determination
Determination 0108
WSYX-QC-SOP-00-09- Double
Identity SOP for Identity Test
0201 immunodiffusion
SOP for Bacterial WSYX-QC-SOP-00-09-

Bacterial endotoxin Gel-clot method
Endotoxin 0203
WSYX-QC-SOP-00-09-
Sterility SOP for Sterility Test Membrane filtration
0202
SOP for Polysaccharide
Content Determination WSYX-QC-SOP-00-09- Rate Nephelometry
Polysaccharide content
with Rate Nephelometry 0141 Method
Method
SOP for Protein Content WSYX-QC-SOP-00-09-
Protein content Lowry method
Determination 0102
SOP for
Polysaccharide/protein WSYX-QC-SOP-00-09-
Polysaccharide/Protein Calculation
ratio 0125
Ratio Determination
SOP for Free
Polysaccharide
Free polysaccharide Determination of WSYX-QC-SOP-00-09-
Chromogenic method
content Pneumococcal 0123
Polysaccharide bulk
Conjugate
SOP for Free Protein WSYX-QC-SOP-00-09-
Free protein content Gel chromatography
Determination 0114
SOP for Molecular Size WSYX-QC-SOP-00-09-
Molecular size Gel chromatography
Determination 0104
SOP for Residual EDAC WSYX-QC-SOP-00-09-
Residual EDAC Chromogenic method
Determination 0107
SOP for ADH Content WSYX-QC-SOP-00-09-
ADH content Chromogenic method
Determination 0105

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2.3.S.4.3 Validation or Qualification of Analytical Procedures

Table 82 List of Test Method Validation/Qualification Protocol for Intermediate Product

Validation/Qualification

Test items Document ode Test Method

protocol

Validation Protocol of

Immuno-double

Identity Identification test for WSYX-QC-TS-0006-0697

diffusion

Pneumococcal Polysaccharide

Qualification Protocol of Protein

Protein content WSYX-QC-TS-0006-0592 Lowry method

Content Determination

Qualification Protocol of Nucleic

Nucleic Acid UV visible

Acid Content Determination for WSYX-QC-TS-0006-0719

Content spectrophotometry

Pneumococcal Polysaccharide

Qualification Protocol of Total

Total Nitrogen Nitrogen Content Determination WSYX-QC-TS-0006-0587 Kjeldahl method

for Pneumococcal Polysaccharide

Phosphorus Qualification Protocol of Total Chromogenic

WSYX-QC-TS-0006-0598

Content Solid content Determination method

Qualification Protocol of Uronic

Uronic Acid Chromogenic

Acid Determination for WSYX-QC-TS-0006-0589

Content method

Pneumococcal Polysaccharide

Qualification Protocol of

Hexosamine Chromogenic

Hexosamine Determination for WSYX-QC-TS-0006-0591

Content method

Pneumococcal Polysaccharide

Methyl Qualification Protocol of Methyl

Chromogenic

pentoses pentoses Determination for WSYX-QC-TS-0006-0590

method

Content Pneumococcal Polysaccharide

Qualification Protocol of O-acetyl

O-acetyl Chromogenic

Determination for Pneumococcal WSYX-QC-TS-0006-0599

Content method

Polysaccharide

Qualification Protocol of Gel

Molecular Size WSYX-QC-TS-0006-0597

Molecular Size Determination chromatography

Organic

Validation Protocol of Ethanol,

Solvent
Ether, Acetone Residues Gas
(ethanol, ether, WSYX-QC-TS-0006-0694
Determination for Pneumococcal chromatography
acetone)
Polysaccharide
residues
Qualification Protocol of Bacterial
Bacterial
Endotoxin Test Method for
Endotoxin WSYX-QC-TS-0006-07785 Gel-clot method
Pneumococcal Polysaccharide and
Content
Bulk Polysaccharide
Qualification Protocol of Sodium
Sodium Deoxycholate Content UV-vis
WSYX-QC-TS-0006-0721
Deoxycholate Determination for Pneumococcal spectrophotometry
Polysaccharide
Qualification Protocol of Phenol
Phenol content Content Determination for WSYX-QC-TS-0006-0717 Titration
Pneumococcal Polysaccharide
Validation Protocol of Free
Free
Polysaccharide Content Chromogenic
Polysaccharide WSYX-QC-TS-0006-0668
Determination for Pneumococcal method
Content
Bulk Polysaccharide
Free Protein Validation Protocol of Free Gel
WSYX-QC-TS-0006-0679
Content Protein Content Determination for chromatography
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Pneumococcal Bulk
Polysaccharide
Bacterial
Interference Test / /
Endotoxin Test
Method Applicability Test for
Microbial Limit Microbial Limit Test of 13-valent WSYX-QC-TS-0006-0553 /
Pneumococcal Bulk Conjugate
Qualification Protocol of
Polysaccharide Rate Nephelometry
Polysaccharide Content WSYX-QC-TS-0006-0647
Content Method
Determination
Method Applicability Test for
Membrane
Sterility Test Microbial Limit Test of 13-valent WSYX-QC-TS-0006-0543
filtration
Pneumococcal Conjugate Vaccine
Validation Protocol of Purity Test
Purity WSYX-QC-TS-0006-0740 HPLC
for purified Tetanus Toxoid
Protein Content Determination
Protein WSYX-QC-TS-0006-0564 Lowry Method
Method
Method Applicability Test for Membrane
Sterility Test WSYX-QC-TS-006-0551
Sterility Test of Excipients filtration

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2.3.S.4.4 Batch Analyses
Different batches of 13 serotypes of pneumococcal polysaccharide, 13 serotypes of polysaccharide
derivatives, bulk tetanus toxoid, purified liquid of
tetanus toxoid, 13 serotypes of bulk conjugate were analyzed based on the analytical procedures. The data
is showing in table below.
Table 83 Batch Analyses Data Statistics for Pneumococcal Polysaccharide
Test O-
Total Bacteria Sodi
items Identific Total Nucleic Phosph Uronic acetyl Hexosa Methyl Phenol
Protein nitroge Molecular size I Ether Acetone Ethanol m
ation solid acid orus acid group mine pentose residue
content n (KD) endotox residual residual residual deoxyc
Batch test mg/mL content content content conten content content g/L
content in holate
No. t
Appear
While
positive ≤ 0.5
KD <0.3,
response EU/μg
Specific 3.5- KD ≤ the < < < <
to - ≤ 2% ≤ 2% 0-1.5% ≥ 45% ≥ 1.8% / / /
ation 6.0% 0.15 recovery 0.01% 0.03% 0.08% 0.2%
specific ≤ 50

rate
antiseru EU/mL
≥75%
m type 1
1-
PVPV15 3.245 1.9% 0.8% 5.0% 0.3% 0 / ND ND 0.007% 0.02% 55.9% 5.8% / / /
101201
Comply Comply
1-
with the with the
PVPV15 2.93 1.5% 1.1% 3.7% 0.3% 0 / ND 0.001% ND 0.09% 60.8% 6.1% / / /
requirem requirem
121202
ents ents
1-
PVPV16 3.080 1.6% 0.9% 5.6% 0.6% 0.04 / ND ND ND 0.10% 57.4% 5.7% / / /
011203
Appear
While
positive ≤ 0.5
KD <
response EU/μg
Specific 0- KD≤0.1 0.3, the < < < <
to - ≤5% ≤2% ≤5% ≥ 40% / / / /
ation 1.0% 5 recovery 0.01% 0.03% 0.08% 0.2%
specific ≤ 50
rate
antiseru EU/mL
≥78%
m type 3

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Test O-
Total Bacteria Sodium
items Identific Total Nucleic Phosph Uronic acetyl Hexosa Methyl Phenol
Protein nitroge Molecular size I Ether Acetone Ethanol m
ation solid acid orus acid group mine pentose residue
content n (KD) endotox residual residual residual deoxyc
Batch test mg/mL content content content conten content content g/L
content in holate
No. t
3-
PVPV15 3.105 0.1% 0.04% 0.68% 0.00% 0 / ND ND 0.002% 0.030% 50.5% / / / /
100901
Comply Comply
3-
with the with the
PVPV15 3.065 0.4% 0.09% 0.65% 0.00% 0 / ND ND ND 0.16% 52.90% / / / /
requirem requirem
110902
ents ents
3-
PVPV15 2.325 0.5% 1.99% 0.86% 0.10% 0.04 / ND ND ND 0.170% 47.8% / / / /
120903
Appear
While
positive ≤ 0.5
KD <
response EU/μg
Specific 4.0- KD ≤ 0.3, the < < < <
to - ≤ 3% ≤ 2% 0-1.5% / / ≥ 40% / /
ation 6.0% 0.15 recovery 0.01% 0.03% 0.08% 0.2%
specific ≤ 50
rate
antiseru EU/mL
≥75%
m type 4
4-
PVPV15 2.600 1.30% 0.20% 5.73% 0.20% 0 / ND 0.002% 0.018% 0.110% / / 45.2% / /
101101
Comply Comply
4-
with the with the
PVPV15 3.140 1.3% 0.3% 5.73% 0.29% 0.04 / ND ND 0.006% 0.16% / / 46.4% / /
requirem requirem
121102
ents ents
4-
PVPV16 2.710 2.7% 0.6% 5.83% 0.28% 0 / ND ND 0.001% 0.11% / / 42.0% / /
011103
Appear ≤ 0.5
While
positive EU/μg
Specific 2.5- KD ≤ KD < < < < <
response - ≤ 7.5% ≤ 2% ≤ 2.0% / ≥ 20% / /
ation 6.0% 0.60 0.6, the 0.01% 0.03% 0.08% 0.2% ≥ 12%
to ≤ 50
recovery
specific EU/mL

Test O-
Total Bacteria Sodium
items Identification Total Nucleic Phosph Uronic acetyl Hexosa Methyl Phenol
Protein nitrogen Molecular size I Ether Acetone Ethanol m
ation solid acid orus acid group mine pentose residue
content n (KD) endotoxin residual residual residual deoxyc
Batch test mg/mL content content content content content content g/L
content in holoate
No. t
antiserum rate
m type 5 ≥70%
5-
PVPV15 3.080 3.90% 1.70% 3.64% 0.20% 0.35 / ND 0.0045% 0.020% 0.13% 13.60% / 21.70% / /
101301
Comply Comply
5-
with the with the
PVPV15 3.185 6.40% 1.50% 5.90% 0.30% 0.39 / ND ND ND 0.17% 13.3% / 23.5% / /
requirement requirement
121302
ents ents
5-
PVPV16 4.5% 1.1% 3.82% 0.2% 4.5% 0.35 / ND ND 0.0045% 0.12% 14.8% / 20.9% / /
011303
Appear
positive While
≤ 0.5
response KD <
EU/μg
Specific to 0- 2.5- KD ≤ 0.6, the < < < <
- ≤ 1.5% ≤ 1% / / ≥ 15% /
ation specific 2.0% 5.0% 0.50 recovery 0.01% 0.03% 0.08% 0.2% /
≤ 50
antiserum rate ≥
EU/mL
m type 50%
6A
6A-
PVPV15 2.020 0.36% 0.082% 1.3% 4.5% 0.39 / ND 0.0025% 0.0565% 0% / / / 23.7% /
090101
Comply Comply
6A-
with the with the
PVPV15 2.165 1.16% 0.08% 1.5% 4.17% 0.39 / ND ND 0.061% 0.02% / / / 22.6% /
requirement requirement
110102
ents ents
6A-
PVPV15 2.825 1.3% 0.1% 1.4% 4.17% 0.39 / ND 0.002% 0.026% 0% / / / 23.6% /
120103
Specific Appear 0- 2.5- KD ≤ While ≤ 0.5 < < < <
- ≤ 2% ≤ 2% / / ≥15% /
ation positive 2.0% 5.0% 0.50 KD<0.6, EU/μg 0.01% 0.03% 0.08% 0.2% /

Test O-
Total Bacteria Sodium
items Identification Total Nucleic Phosph Uronic acetyl Hexosa Methyl Phenol
Protein nitrogen Molecular size I Ether Acetone Ethanol m
ation solid acid orus acid group mine pentose residue
content n (KD) endotoxin residual residual residual deoxyc
Batch test mg/mL content content content content content content g/L
content in holoate
No. t
response the
to recovery ≤ 50
specific rate ≥ EU/mL
antiserum 85%
m type
6B
6B-
PVPV15 2.505 0.10% 0.10% 0.96% 4.40% 0.35 / ND 0.002% 0.066% 0.050% / / / 23.40% /
090201
Comply Comply
6B-
with the with the
PVPV15 2.105 0.40% 0.10% 1.09% 4.40% 0.3 / ND 0.004% 0.064% 0.110% / / / 23.80% /
requirement requirement
110202
ents ents
6B-
PVPV15 2.87 1.40% 0.10% 0.73% 4.10% 0.3 / ND 0.002% 0.0275% 0.180% / / / 23.50% /
120203
Appear

positive While
≤ 0.5
response KD <
EU/μg
Specific to 1.5- 0- KD ≤ 0.3, the < < < <
- ≤ 5% ≤ 2% / / / ≥ 13%
ation specific 4.0% 1.0% 0.20 recovery 0.01% 0.03% 0.08% 0.2%
≤ 50
antiseru rate ≥
EU/mL
m type 70%
7F-
7F-
PVPV15 Comply 2.830 0.2% 0.4% 3.07% 0.1% 0.15 / Comply ND 0.001% 0.022% 0.04% / / / 19.4% /
100701 with the with the
7F- requirem requirem
PVPV15 ents 3.430 1.2% 0.2% 3.38% 0.5% 0.12 / ents ND 0.002% 0.014% 0.12% / / / 19.0% /
110702

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Test O-
Total Bacteria Sodi
items Identific Total Nucleic Phosph Uronic acetyl Hexosa Methyl Phenol
Protein nitroge Molecular size I Ether Acetone Ethanol m
ation solid acid orus acid group mine pentose residue
content n (KD) endotox residual residual residual deoxyc
Batch test mg/mL content content content conten content content g/L
content in holate
No. t
7F-
<
PVPV16 2.585 1.1% 0.2% 3.87% 0.40% 0 / ND ND 0.005% / / 17.3% /
0.2% /
010703
Appear
positive While
≤ 0.5
response KD <
EU/μg
Specific to 0.5- KD ≤ 0.6, the < < < <
- ≤ 2% ≤ 2% 0-1.0% ≥ 15% / ≥ 13% / /
ation specific 3.0% 0.45 recovery 0.01% 0.03% 0.08% 0.2%
≤ 50
antiseru rate ≥
EU/mL
m type 80%
9V
9V-
PVPV15 3.075 0.4% 0.2% 1.95% 0.09% 0.35 / ND ND 0.007% 0.04% 22.2% / 13.0% / /
090301
Comply Comply
9V-
with the with the
PVPV15 2.920 0.3% 0.05% 2.36% 0.03% 0.30 / ND ND 0.020% 0.14% 26.1% / 14.6% / /
requirem requirem
110302
ents ents
9V-
PVPV15 2.625 0.3% 0.06% 2.48% 0.07% 0.30 / ND 0.001% ND 0.180% 24.3% / 15.7% / /
120303
Appear
positive While
≤ 0.5
response KD <
EU/μg
Specific to 1.5- KD ≤ 0.3, the < < < <
- ≤ 5% ≤ 2% 0-1.0% / / ≥ 20% / /
ation specific 4.0% 0.30 recovery 0.01% 0.03% 0.08% 0.2%
≤ 50
antiseru rate ≥
EU/mL
m type 55%
14
14-
Comply Comply
PVPV15 3.085 0.8% 1.2% 2.82% 0.20% 0.19 / ND ND 0.009% 0.06% / / 22.2% / /
with the with the
101001
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Test O-
Total Bacteria Sodi
items Identific Total Nucleic Phosph Uronic acetyl Hexosa Methyl Phenol
Protein nitroge Molecular size I Ether Acetone Ethanol m
ation solid acid orus acid group mine pentose residue
content n (KD) endotox residual residual residual deoxyc
Batch test mg/mL content content content conten content content g/L

content in holate
No. t
14- requirem requirem
0.1800
PVPV15 ents 2.385 1.9% 0.1% 3.35% 0.30% 0.08 / ents ND ND 0.0095% // 23.6% // %
121002
14-
0.0900
PVPV16 2.750 1.5% 0.7% 3.02% 0.30% 0 / ND ND 0.0045% // 22.9% // %
011003
Appear
positive While
≤ 0.5
response KD <
EU/μg
Specific to 0- 2.4- KD ≤ 0.3, the < < < <
- ≤ 3% ≤ 2% // ≥ 14% /
ation specific 1.0% 4.9% 0.15 recovery 0.01% 0.03% 0.08% 0.2%
≤ 50
antiseru rate ≥
EU/mL
m type 70%
18C
18C-
PVPV15 2.725 1.0% 0.2% 0.92% 3.00% 0.07 / ND ND 0.020% 0.090% // / 14.5% /
100801
Comply Comply
18C-
with the with the
PVPV15 2.130 0.8% 0.2% 0.89% 2.80% 0.08 / ND 0.002% 0.034% 0.060% // / 17.5% /
requirem requirem
110802
ents ents
18C-
PVPV16 1.990 1.3% 0.2% 0.95% 2.70% 0.08 / ND 0.002% 0.046% 0.110% // / 14.6% /
010803
Appear
positive While
≤ 0.5
response KD <
EU/μg
Specific to 0.6- 3.0- KD ≤ 0.5, the < < < <
- ≤ 2% ≤ 2% // ≥ 12% ≥ 20% /
ation specific 3.5% 7.0% 0.45 recovery 0.01% 0.03% 0.08% 0.2%
≤ 50
antiseru rate ≥
EU/mL
m type 80%
19A
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Test O-
Total Bacteria Sodi
items Identific Total Nucleic Phosph Uronic acetyl Hexosa Methyl Phenol
Protein nitroge Molecular size I Ether Acetone Ethanol m
ation solid acid orus acid group mine pentose residue
content n (KD) endotox residual residual residual deoxyc
Batch test mg/mL content content content conten content content g/L
content in holate
No. t
19A-
PVPV15 2.210 0.4% 0.3% 3.26% 4.90% 0.18 / ND 0.002% 0.015% 0.07% // 12.6% 29.0% /
090401
Comply Comply
19A-
with the with the
PVPV15 2.445 0.7% 0.02% 3.31% 4.7% 0.04 / ND 0.002% 0.0145 0.004% // 13.7% 31.3% /
requirem requirem
110402
ents ents
19A-
PVPV15 2.850 0.5% 0.05% 3.44% 4.9% 0.04 / 0.001% 0.002% 0.005% 0.15% // 15.0% 33.8% /
120403
Appear
positive While
≤ 0.5
response KD <
EU/μg
Specific to 1.4- 3.0- KD ≤ 0.3, the < < < < ≥
- ≤ 3% ≤ 2% // ≥ 20%
ation specific 3.5% 5.5% 0.20 recovery 0.01% 0.03% 0.08% 0.2% 12.5%
≤ 50
antiseru rate ≥
EU/mL
m type 75%
19F
19F-
PVPV15 2.750 0.4% 0.1% 3.27% 4.60% 0.04 / ND 0.002% 0.013% 0.10% // 19.10% 28.1% /

090501
Comply Comply
19F-
with the with the
PVPV15 2.535 0.2% 0.1% 3.00% 4.80% 0 / ND 0.0030% 0.0145% 0.00% / / 20.3% 27.9% /
requirem requirem
110502
ents ents
19F-
PVPV15 3.130 0.3% 0.1% 3.42% 4.80% 0 / ND 0.0005% 0.0035% 0.13% / / 18.30% 24.2% /
120503
Appear While
≤ 0.5
Specific positive 0- 3.0- KD ≤ KD < < < < <
- ≤ 2% ≤ 2% EU/μg / / / ≥ 37%
ation response 1.0% 4.5% 0.15 0.3, the 0.01% 0.03% 0.08% 0.2%
to recovery

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Test O-
Total Bacteria Sodi
items Identific Total Nucleic Phosph Uronic acetyl Hexosa Methyl Phenol
Protein nitroge Molecular size I Ether Acetone Ethanol m
ation solid acid orus acid group mine pentose residue
content n (KD) endotox residual residual residual deoxyc
Batch test mg/mL content content content conten content content g/L
content in holate
No. t
specific rate ≥ ≤ 50
antiseru 85% EU/mL
m type
23F
23F-
PVPV15 2.585 0.3% 0.2% 0.89% 3.80% 0.04 / ND 0.002% 0.017% 0.04% / / / 41.5% /
090601
Comply Comply
23F-
with the with the
PVPV15 3.415 0.2% 0.1% 0.88% 3.90% 0.08 / ND 0.003% 0.010% 0.17% / / / 41.8% /
requirem requirem
120602
ents ents
23F-
0.0015
PVPV16 3.180 0.3% 0.1% 0.97% 3.30% 0.00 / ND ND 0.0015% ND ND / 38.8% /
%
010603

Table 84 Batch Analyses Data Statistics for Polysaccharide Derivatives
Residual
Total solid Derivation Molecular Size Determination by SEC-MALS
cyanide Molecular size (KD)
Test item and mg/mL rate (%) Method
content
Serotypes specification
While KD = 0.2- Mn (Number- Mw (Mass
Batch No. Mw/Mn
- 2.6-4.4% ≤ 10 ng/mg KD ≤ 0.37 0.6, the recovery average Molecular average molar
(polydispersity)
rate ≥ 45% Weight) mass)
SP1C201908001 7.900 3.3% 6 0.32 88% 258.8 334.6 1.292
1 SP1C201909002 7.180 3.9% 2 0.26 68% 237.8 298.6 1.255
SP1C201909003 8.500 3.6% 7 0.33 81% 463.3 695.8 1.502
SP3C201908001 14.700 10.6% 1 0.14 64% 296.4 590.0 1.991
3 SP3C201909002 14.440 13.0% 1 0.18 67% 412.55 1207 2.9255
SP3C201909003 15.530 12.9% 1 0.15 65% 362.9 988.05 2.7235
SP4D202001001 13.055 5.8% 3 0.33 76% 163.15 210.4 1.2895
4
SP4D 202002002 10.855 6.0% 0 0.37 77% 155.5 191.35 1.2305
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SP4D 202005003 9.750 6.4% 1 0.29 65.2% 193.0 255.4 1.3235
SP5C201909001 8.365 12.6% 1 0.58 51% 27.01 34.45 1.275
5 SP5C201909002 9.065 11.9% 0 0.63 44% 24.86 31.95 1.285
SP5C201909003 9.945 12.6% 1 0.68 44% 27.61 40.76 1.477
SP6AC201909001 11.875 5.2% 0 0.39 83% 217.7 251.4 1.155
6A SP6AC201909002 10.460 5.2% 0 0.35 79% 256.7 303.0 1.181
SP6AC201909003 10.020 5.5% 0 0.41 85% 214.4 247.6 1.155
SP6BD 202001001 11.150 2.7% 3 0.30 74% 278.05 350.95 1.263
6B SP6BD 202001002 10.760 2.7% 0 0.30 78% 218.9 256.15 1.170
SP6BD 202002003 8.315 2.8% 0 0.30 74% 224.65 267.3 1.1895
SP7FD 202001001 9.925 4.3% 1 0.37 78% 404.6 607.2 1.5015
7F SP7FD 202002002 9.760 4.0% 0 0.32 76% 367.55 571.45 1.5545
SP7FD 202002003 7.220 4.3% 1 0.37 79% 357.75 472.4 1.3205
SP9VC201908001 8.750 4.7% 0 0.30 81% 636.4 1059 1.664
9V SP9VC201908002 10.085 5.0% 1 0.28 92% 959.3 2067.5 2.155
SP9VC201908003 8.850 4.6% 1 0.35 82% 685.2 1266 1.848

SP14D 202001001 11.775 3.9% 0 0.29 79% 923.35 2168.5 2.3485
14C SP14D 202002002 10.470 4.3% 0 0.33 72% 1048 2007.5 1.916
SP14D 202002003 11.695 4.3% 0 0.30 70% 1272.5 2185 1.717
SP18CC201908001 10.540 3.7% 1 0.32 74% 202.5 245.2 1.211
18C SP18CC201908002 11.670 4.0% 0 0.36 74% 209.0 250.4 1.198
SP18CC201909003 10.710 4.0% 0 0.35 73% 206.4 246.4 1.194
SP19AC201908001 14.170 6.4% 1 0.44 79% 161.3 191.4 1.186
19A SP19AC201908002 13.335 6.9% 0 0.44 82% 163.0 191.4 1.175
SP19AC201909003 14.025 7.1% 0 0.40 82% 179.6 215.2 1.199
SP19FC201909001 11.095 4.5% 0 0.37 76% 247.2 272.4 1.102
19F SP19FC201909002 11.220 4.8% 2 0.37 81% 195.7 227.6 1.163
SP19FC201909003 11.580 5.7% 2 0.37 76% 223.8 264.8 1.182
SP23FC201908002 11.790 2.5% 1 0.37 79% 194.6 223.8 1.150
23F SP23FC201909003 12.975 2.5% 0 0.36 78% 223.15 254.5 1.1405
SP23FD 202001001 10.425 2.6% 2 0.41 90% 224.75 275.8 1.227

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Table 85 Batch Analyses Data Statistics for Bulk Tetanus Toxoid
Residual
Test items Flocculent Detoxification Protein
Appearance pH Purity Sterility test Abnormal toxicity Toxicity reverse test ammoniu
Batch No. unit test content
m
Observed on the 7th,
14th, and 21st days
Animals should
after injection. There The guinea pigs must
not have
The antigenic was no local not have symptoms
Should be symptoms of
purity is not suppuration or of tetanus. Each
yellow or tetanus. Each
less than Should comply necrosis. Animals guinea pig whose
brownish - animal should not ≤ 0.25 -
Specification 6.6-7.4 1500Lf per with the should not have weight was more
yellow (Lf/mL) lose more weight mg/mL ($\mu\text{g/mL}$)
milligram of requirements. symptoms of tetanus. than that before the
transparent than before
protein Those who expired injection by the end
liquid injection, and the
nitrogen. and the weight of of the period was
survivors are
each animal increased qualified.
eligible.
than before injection
were eligible.
Comply with
201511004 Comply with the Comply with the Comply with the Comply with the
the 7.30 1540 1672.1 0.033 5689.1
(Bottle No.1) requirements. requirements. requirements. requirements.
requirements.
Comply with
CTB201606005 Comply with the Comply with the Comply with the Comply with the
the 7.27 800 1720.4 ND 2983.1
(Bottle No.1) requirements. requirements. requirements. requirements.
requirements.
Comply with
CTB201610006 Comply with the Comply with the Comply with the Comply with the
the 6.88 800 1698.5 0.007 3057.3
(Bottle No.1) requirements. requirements. requirements. requirements.
requirements.

Table 86 Batch Analyses Data Statistics for Purified Liquid of Tetanus Toxoid
Test items
Protein content Sterility Bacterial endotoxin Purity
Batch No.
Specification ≥ 16 mg/mL Should comply with requirements Should be not more than 10 EU/mg protein Should
not be less than 90%
CTL201909018 (Bottle No.1) 26 Comply with the requirements. Comply with the requirements. 100.0%
CTL201909020 (Bottle No.1) 24 Comply with the requirements. Comply with the requirements. 99.9%
CTL201909019 (Bottle No.1) 26 Comply with the requirements. Comply with the requirements. 100.0%
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Table 87 Batch Analyses Data Statistics for Monovalent Bulk Conjugate
Free Free
Polysaccharide Protein Polysaccharide- Molecular EDAC Bacterial ADH
polysaccharide protein Identification Sterility
content content protein ratio size (KD) residue endotoxin content
Test item and content content
specification Appear
Serotypes Should
positive
 $\geq 70 \leq 5.0 \leq 0.75$ EU/ μg comply with < 0.15
Batch No. ≥ 40 $\mu\text{g/mL}$ 0.20-1.80 $\geq 80\% \leq 25.0\% \leq 5.0\%$ response to
 $\mu\text{g/mL}$ $\mu\text{mol/L}$ polysaccharide the $\mu\text{g/mL}$
specific
requirements.

antisera

Comply with Comply with

RSP1C201706001 245.5 239.1 1.03 91.4 2.8 10.8 4.1 the < 0.75 EU/μg the 0.05 requirements. requirements.

Comply with Comply with

1 RSP1C201707002 258.0 170.5 1.51 99.4 2.9 15.8 ND the < 0.75 EU/μg the 0.06 requirements. requirements.

Comply with Comply with

RSP1C201709003 423.8 462.2 0.92 96.2 ND 14.0 1.3 the < 0.75 EU/μg the 0.086 requirements. requirements.

Comply with Comply with

RSP3C201706001 61.6 141.6 0.44 98.5 1.1 15.7 3.8 the < 0.75 EU/μg the 0.016 requirements. requirements.

Comply with Comply with

3 RSP3C201707002 77.2 355.8 0.22 97.8 4.4 6.2 0.48 the < 0.75 EU/μg the 0.160 requirements. requirements.

Comply with Comply with

RSP3C201708003 63.6 285.3 0.22 97.6 3.8 12.2 1.20 the < 0.75 EU/μg the 0.128 requirements. requirements.

Comply with Comply with

RSP4C201706001 154.3 254.1 0.61 98.9 0.9 19.6 4.7 the < 0.75 EU/μg the 0.082 requirements. requirements.

4

Comply with Comply with

RSP4C201707002 196.0 276.2 0.71 98.0 3.6 23.6 2.2 the < 0.75 EU/μg the 0.110 requirements. requirements.

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Comply with Comply with

RSP4C201709003 190.8 288.0 0.66 99.4 3.5 14.3 0.9 the < 0.75 EU/μg the 0.082 requirements. requirements.

Comply with Comply with

RSP5C201705001 91.1 431.0 0.21 94.3 0.7 18.0 3.2 the < 0.75 EU/μg the 0.139 requirements. requirements.

Comply with Comply with

5 RSP5C201707002 57.8 172.1 0.34 98.9 0.8 17.4 ND the < 0.75 EU/μg the 0.102 requirements. requirements.

Comply with Comply with

RSP5C201708003 99.8 224.6 0.44 96.8 3.7 15.4 0.5 the < 0.75 EU/μg the 0.102 requirements. requirements.

Comply with Comply with

RSP6AC201705001 139.1 547.4 0.25 95.6 3.6 13.7 3.6 the < 0.75 EU/μg the 0.163 requirements. requirements.

Comply with Comply with

6A RSP6AC201707002 154.6 493.0 0.31 88.1 2.6 19.4 0.9 the < 0.75 EU/μg the 0.081 requirements. requirements.

Comply with Comply with

RSP6AC201708003 86.8 191.5 0.45 97.7 2.3 12.8 0.5 the < 0.75 EU/μg the 0.227 requirements. requirements.

Comply with Comply with

RSP6BC201705001 152.9 327.4 0.47 98.6 2.8 21.6 4.0 the < 0.75 EU/μg the 0.012 requirements. requirements.

Comply with Comply with

6B RSP6BC201707002 170.2 349.9 0.49 93.5 0.5 21.1 0.9 the < 0.75 EU/μg the 0.094 requirements. requirements.

Comply with Comply with

RSP6BC201708003 124.8 296.5 0.42 96.1 0.1 9.2 2.1 the < 0.75 EU/μg the 0.142 requirements. requirements.

Comply with Comply with

7F RSP7FC201705001 284.1 643.5 0.44 97.2 3.2 18.4 4.0 the < 0.75 EU/μg the 0.066 requirements. requirements.

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Comply with Comply with

RSP7FC201706002 303.0 362.8 0.84 96.6 1.2 18.5 0.3 the < 0.75 EU/μg the 0.121 requirements. requirements.

Comply with Comply with

RSP7FC201708003 310.8 494.3 0.63 96.6 3.7 18.9 1.3 the < 0.75 EU/μg the 0.088 requirements. requirements.

Comply with Comply with

RSP9VC201705001 148.6 513.5 0.29 100.0 2.5 6.8 4.3 the < 0.75 EU/μg the 0.176 requirements. requirements.

Comply with Comply with

9V RSP9VC201707002 159.3 614.1 0.26 99.7 2.2 12.0 1.6 the < 0.75 EU/μg the 0.153 requirements. requirements.

Comply with Comply with

RSP9VC201709003 157.5 722.3 0.22 98.8 4.1 11.8 0.7 the < 0.75 EU/μg the 0.203 requirements. requirements.

Comply with Comply with

RSP14C201706001 143.2 480.7 0.30 95.9 2.4 14.6 3.1 the < 0.75 EU/μg the 0.016 requirements. requirements.

Comply with Comply with

14C RSP14C201707002 134.9 340.4 0.4 99.0 3.8 12.6 1.0 the < 0.75 EU/μg the 0.120 requirements. requirements.

Comply with Comply with

RSP14C201709003 125.0 305.8 0.41 97.2 1.0 23.9 1.0 the < 0.75 EU/μg the 0.114

requirements. requirements.
Comply with Comply with
RSP18CC201706001 106.6 195.8 0.54 97.7 0.9 7.7 2.7 the < 0.75 EU/μg the 0.097
requirements. requirements.
Comply with Comply with
18C RSP18CC201707002 153.3 263.1 0.58 86.9 1.5 26.2 0.9 the < 0.75 EU/μg the 0.108
requirements. requirements.
Comply with Comply with
RSP18CC201708003 144.4 199.8 0.72 96.9 0.7 22.6 0.7 the < 0.75 EU/μg the 0.015
requirements. requirements.

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Comply with Comply with
RSP19AC201705001 131.7 409.8 0.32 95.4 1.2 27.4 2.9 the < 0.75 EU/μg the 0.108
requirements. requirements.
Comply with Comply with
19A RSP19AC201707002 163.6 435.2 0.38 93.3 1.3 26.0 2.8 the < 0.75 EU/μg the 0.130
requirements. requirements.
Comply with Comply with
RSP19AC201708003 202.0 444.9 0.45 99.4 2.3 28.0 0.7 the < 0.75 EU/μg the 0.103
requirements. requirements.
Comply with Comply with
RSP19FC201706001-
195.7 500.3 0.39 88.8 3.9 24.4 3.2 the < 0.75 EU/μg the 0.073
1
requirements. requirements.
Comply with Comply with
19F RSP19FC201707002 145.6 237.8 0.61 98.1 1.4 30.1 2.4 the < 0.75 EU/μg the 0.126
requirements. requirements.
Comply with Comply with
RSP19FC201708003 134.0 300.3 0.45 96.4 3.1 25.2 1.1 the < 0.75 EU/μg the 0.066
requirements. requirements.
Comply with Comply with
RSP23FC201705001 85.4 232.8 0.37 98.7 0.7 13.0 1.6 the < 0.75 EU/μg the 0.162
requirements. requirements.
Comply with Comply with
23F RSP23FC201707002 91.3 215.0 0.42 100.0 2.5 24.9 2.1 the < 0.75 EU/μg the 0.174
requirements. requirements.
Comply with Comply with
RSP23FC201708003 75.8 212.3 0.36 97.5 2.1 18.7 0.6 the < 0.75 EU/μg the 0.160
requirements. requirements.

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2.3.S.5 Reference Standards or Materials
Table 88 List of Reference Standards or Materials used
Storage
Test Items Reference name Manufacturer Unit
Condition
National standard sample 19.73
Protein content NIFDC Below -20 ■
for protein content test mg/tube
Standard Phosphorus
Room
Phosphorus content (potassium NIFDC 100 mg/vial
temperature
dihydrophosphate)
Room
Uronic acid content Sodium glucuronate SIGMA 10 g/vial
temperature
D-glucosamine Room
Hexosamine content NIFDC 200 mg/vial
hydrochloride temperature
Methyl pentose Room
Rhamnose SIGMA 10 g/vial
content temperature
O-acetyl content Acetylcholine chloride SIGMA 25 g/vial Below -20 ■
Room
Dextran GE 10 g/vial
temperature
Molecule size Sinopharm
Room
Vitamin B12 Chemical Reagent 1 g/vial
temperature
Co., Ltd.
Residual sodium Room
Sodium deoxycholate SIGMA 500 g/vial
deoxycholate content temperature
Yunnan Yanglin
Shandian 500
Ethanol Store in shade
Pharmaceutical Co., mL/vial
Ltd.
Yunnan Yanglin
Residual organic Shandian 500
Ether Store in shade
solvent content Pharmaceutical Co., mL/vial

Ltd.
Yunnan Yanglin
Shandian 500
Acetone Store in shade
Pharmaceutical Co., mL/vial
Ltd.
Standard bacterial Zhanjiang Andusi
Bacterial endotoxin 10 EU/tube 2-8 ■
endotoxin sample Biology Co., Ltd.
ADH Adipic dihydrazide SIGMA 100 g/vial Below -20 ■
Sinopharm
Room
Glucose content Anhydrous glucose Chemical Reagent 500 g/vial
temperature
Co., Ltd.
Residual Cyanide Cyanogen bromide J&K Scientific Ltd. 100 g/vial 2-8 ■
Pneumococcal specific
antiserum (serotypes 1, 3, Danish National
1 ml/vial 2-8 ■
4, 5, 6A, 6B, 7F, 9V, 14, Serum Institute
Identity / 18C, 19A, 19F, 23F)
Polysaccharide Pneumococcal
content polysaccharide reference
10 mg/vial
(serotypes 1, 3, 4, 5, 6A, ATCC Below -20 ■
2 mg/vial
6B, 7F, 9V, 14, 18C, 19A,
19F, 23F)
EDAC Carbodiimide SIGMA 10 g/vial Below -20 ■
Sterility Test Staphylococcus aureus NIFDC / Below -20 ■

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2.3.S.6 Container Closure System

The 13 types polysaccharides are stored in centrifuge bottles, and the information of the centrifuge bottles is shown in Table 89.

Table 89 Information Regarding the Container Closure System for Pneumococcal Polysaccharide

Item Information
Container name Centrifuge bottle
Manufacturer Backman & Thermo Fisher Scientific
Container material PP
Container specification 250ml/bottle and 50ml/bottle
Container structure Cap + body

The 13 types pneumococcal polysaccharide bulk conjugates are stored in 10 L glass bottles, and the tetanus toxoid carrier protein bulk was stored in 500 mL glass bottles, the information of the glass bottles is shown in Table 90

Table 90 Information Regarding the Container Closure System for Bulk Glass bottle for polysaccharide bulk Glass bottle for bulk carrier

Item
conjugate protein
Container name Serum bottle Saline bottle
Chongzhou Shubo Scientific Instruments Sichuan Jianwei Hesheng Glass Co.,
Manufacturer
Co., Ltd Ltd
Container material Glass Soda-lime glass
Container
10L/bottle 500mL/bottle
specification
Container structure Body + rubber plug Body + rubber plug

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2.3.S.7 Stability

According to the requirements of preventive vaccine 13-valent Pneumococcal Conjugate Vaccine. Walvax has performed the stability studies of intermediate products, including 13 types of pneumococcal polysaccharide, pneumococcal polysaccharide bulk conjugate and carrier protein.

2.3.S.7.1 Stability Summary and Conclusion on Polysaccharide

Table 91 Batch information of Pneumococcal Polysaccharide used in Stability Study

Batch Storage
Serotype Batch No. Packaging Mfg.Date
quantity condition
Pneumococcal 1-PVPV15101201 11.12 g 2015.10.21
polysaccharide 1-PVPV15121202 13.74 g 2015.10.20
serotype 1 1-PVPV16011203 14.51 g 2016.01.11
Pneumococcal 3-PVPV15100901 35.24 g 2015.10.09
polysaccharide 3-PVPV15110902 24.06 g 2015.11.28
serotype 3 3-PVPV15120903 29.26 g 2015.12.24
Pneumococcal 4-PVPV15101101 28.21 g 2015.10.17
polysaccharide 4-PVPV16011102 24.64 g 2015.12.26
serotype 4 4-PVPV15121103 29.82 g 2016.01.07
Pneumococcal 5-PVPV15101301 11.81 g 2015.10.25

deoxycholate
Ether content ≤ 0.01%
Acetone
≤ 0.03%
content
Ethanol content ≤ 0.08%

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Conclusion

The pneumococcal polysaccharides of serotypes 1, 3, 4, 5, 6A, 6B, 7F, 9V, 14, 18C, 19A, 19F and 23F produced by Walvax were stored at -20 °C for 0 months, 1 year, 2 years, 3 years and 4 years respectively. The detection values of each test item at each test time point were in accordance with the Quality Standard for 13-Valent Pneumococcal Polysaccharide Conjugate Vaccine (WSYX-QC-TS-06-09-0005).

This stability study was conducted by the quality control laboratory in strict accordance with the stability study protocol, which can provide the basis for the safety, immunogenicity and validity of pneumococcal polysaccharide stored at -20 °C. To sum up, the pneumococcal polysaccharide produced by Walvax can meet the requirements of 4 years storage at -20 °C.

2.3.S.7.2 Stability Summary and Conclusion on Polysaccharide Derivatives

Table 93 Batch information of Polysaccharide Derivatives used in Stability Study

Batch Storage

Serotypes Batch No. Packaging Mfg.Date

quantity condition

Pneumococcal SP1C201908001 1476 mL 2019.08.23
polysaccharide SP1C201909002 1480 mL 2019.09.13
derivative serotype 1 SP1C201909003 1459 mL 2019.09.16
Pneumococcal SP3C201908001 1429 mL 2019.08.26
polysaccharide SP3C201909002 1519 mL 2019.09.18
derivative serotype 3 SP3C201909003 1469 mL 2019.09.20
Pneumococcal SP4D202001001 1800 mL 2020.01.16
polysaccharide SP4D202002002 2480 mL 2020.02.24
derivative serotype 4 SP4D202005003 1970 mL 2020.05.12
Pneumococcal SP5C201909001 1650 mL 2019.09.06
polysaccharide SP5C201909002 1690 mL 2019.09.09
derivative serotype 5 SP5C201909003 1470 mL 2019.09.11
Pneumococcal SP6AC201909001 1430 mL 2019.09.13
polysaccharide SP6AC201909002 1440 mL 2019.09.16
derivative serotype 6A SP6AC201909003 1490 mL 2019.09.18
Pneumococcal SP6BD202001001 2500 mL 2020.01.14
polysaccharide SP6BD202001002 Glass bottle 2491 mL 2-8 °C 2020.01.24
derivative serotype 6B SP6BD202002003 3290 mL 2020.02.24
Pneumococcal SP7FD202001001 1070 mL 2020.01.14
polysaccharide SP7FD202002002 2460 mL 2020.02.06
derivative serotype 7F SP7FD202002003 2620 mL 2020.02.15
Pneumococcal SP9VC201908001 1790 mL 2019.08.23
polysaccharide SP9VC201908002 1810 mL 2019.08.26
derivative serotype 9V SP9VC201908003 1790 mL 2019.08.28
Pneumococcal SP14D202001001 2000 mL 2020.01.16
polysaccharide SP14D202002002 3630 mL 2020.02.09
derivative serotype 14 SP14D202002003 3391 mL 2020.02.27
Pneumococcal SP18CC201908001 1469 mL 2019.08.30
polysaccharide SP18CC201908002 1469 mL 2019.09.02
derivative serotype 18C SP18CC201908003 1489 mL 2019.09.04
Pneumococcal SP19AC201908001 1500 mL 2019.08.30
polysaccharide SP19AC201908002 1530 mL 2019.09.02
derivative serotype 19A SP19AC201909003 1549 mL 2019.09.04

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Pneumococcal SP19FC201909001 1389 mL 2019.09.06
polysaccharide SP19FC201909002 1369 mL 2019.09.09
derivative serotype 19F SP19FC201909003 1389 mL 2019.09.11
Pneumococcal SP23FC201908002 1389 mL 2019.08.20
polysaccharide SP23FC201908003 1410 mL 2019.08.28
derivative serotype 23F SP23FD202001001 1611 mL 2019.09.20

Basis of Study

1. GMP
2. Chinese Pharmacopeia 2015, volume IV 9000, Guiding principles for stability test of bulk drugs and pharmaceutical product.
3. Management Procedures for Continuous Stability Investigation of Products (Document code: WSYX-QA-SMP-0009-0030)
4. Quality Standard of 13-Valent Pneumococcal Polysaccharide Conjugate Vaccine (Document code: WSYX-QC-TS-0609-0005).
5. Operation Procedure for Determination of Molecular Size with SEC-MALS Method (Document code: WSYX-QC-SOP-00-09-0143).

Test protocol

1. storage condition: 2-8 °C;
2. test intervals: initial, 10th day, 20th day, 30th day, 35th day;
3. test items and corresponding time:
0th day, 35th day: full test;
10th day, 20th day, 30th day: molecular size (SEC-MALS method);
35th day: add test items of 3 batches of serotypes 4, 6B, 7F, 14, and 1 batch of serotype 23F in the 35th day, more detail please see the stability report of polysaccharide derivatives.

Test items and standard

See the Table 94.

Table 94 Test Items and Specifications of Polysaccharide Derivatives

Test items Specification

Total solid /

5.0-

Serotype 1 2.6-4.4 Serotype 3 Serotype 4 3.0-11.0

Derivation rate 14.0

(%) Serotype 5 3.5-18.0 Serotype 6A 3.0-6.0 Serotype 6B 1.5-3.4

(ADH Serotype 7F 2.9-5.3 Serotype 9V 3.3-7.4 Serotype 14 2.5-5.9

/polysaccharide) Serotype 18C 2.0-4.8 Serotype 19A 3.5-8.0 Serotype 19F 3.2-6.2

Serotype 23F 1.1-3.7

Residual cyanide Should not more than 10 ng/mg

Serotype KD value Recovery rate at KD of 0.2-0.6

$1 \leq 0.37 \leq 45\%$

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$3 \leq 0.45 \leq 60\%$

$4 \leq 0.42 \leq 65\%$

Recovery rate before KD reached 0.6

$5 \leq 0.90$

$\geq 37\%$

$6A \leq 0.62 \leq 60\%$

Molecular size $6B \leq 0.33 \leq 50\%$

(Sephacrose CL- 7F $\leq 0.43 \leq 60\%$

4B filter) $9V \leq 0.46 \leq 70\%$

$14 \leq 0.40 \leq 65\%$

$18C \leq 0.41 \leq 65\%$

$19A \leq 0.48 \leq 65\%$

$19F \leq 0.56 \leq 70\%$

$23F \leq 0.46 \leq 75\%$

Molecular size

(SEC-MALS -

method)

Conclusion

According to Stability Protocol of Pneumococcal Polysaccharide Derivatives (Document code: WSYX-QC-SP-0609-0099, version 00, 01), 13 serotypes of Pneumococcal Polysaccharide Derivatives were kept at 2-8 ■ for 35 days, except batch SP4D202005003 at 35th day, the tests were not totally finished, all the test results of the other batches at each time point were in accordance with the Quality Standard of 13-Valent Pneumococcal Polysaccharide Conjugate Vaccine (Document code: WSYX-QC-TS-06-09-0002).

2.3.S.7.3 Stability Summary and Conclusion on Bulk Tetanus Toxoid

Table 95 Batch Information of Bulk Tetanus Toxoid used in Stability Study

Batch Storage Mfg Date

Sample Batch No. Packaging

quantity condition (Strain opening)

201511004 10000 mL 2015.11.17

Bulk tetanus toxoid CTB201606005 Glass bottle 59900 mL 2-8 ■ 2016.06.15

CTB201610006 59800 mL 2016.10.25

Basis of Study

1. GMP

2. Chinese Pharmacopeia 2015, volume IV, Guiding principles for stability test of bulk drugs and pharmaceutical product.

3. Management Procedure for Continuous Stability Inspection of Products (Document Code: WSYX-QA-SMP-0009-0030)

4. Quality Standard of Diphtheria, Tetanus and Acellular Pertussis Combined Vaccine, Adsorbed (Document Code: WSYX-QC-TS-0609-0003).

Test protocol

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1. storage condition: 6 ± 2 ■

2. test intervals: initial, 3rd month, 6th month, 9th month, 12th month, 18th month, 24th month, 36th month, 42nd month, 48th month.

3. test items and corresponding time:

Initial: detoxification test, appearance, pH, flocculation unit, purity, sterility test, specific toxicity

test, toxicity reversal test, protein content, residual ammonium;

6th month, 12th month, 18th month, 24th month, 36th month, 42nd month, 48th month: appearance, pH, limit of flocculation, purity, sterility test, specific toxicity test, toxicity reversal test,

protein

content, residual ammonium;

3rd month, 9th month: appearance, pH, limit of flocculation, purity, protein content, residual ammonium.

Test items and specification

See the Table 96

Table 96 Test Items and Specifications of Bulk Tetanus Toxoid

Test items Specifications

Appearance Should be yellow or brown yellow transparent liquid.

pH 6.6 - 7.4

Limit of flocculation

-
(Lf/mL)

Purity No less than 1500 Lf/mg

Sterility Should be in accordance with the regulations

Observation on the 7th, 14th and 21st day showing no local suppuration and

Specific toxicity necrosis, no tetanus symptoms. Weight gain for each guinea pig at the end of the period.

Observation on the 7th, 14th and 21st day showing no tetanus symptoms, weight

Toxicity reversal test

gain for each guinea pig at the end of the period.

Observation on the 7th, 14th and 21st day showing no tetanus symptoms, weight

Detoxification test

gain for each guinea pig at the end of the period.

Residual ammonium Not more than 0.25 mg/mL

Protein content (µg/mL) -

Conclusion

3 batches of Bulk Tetanus Toxoid, which are 201511004 (Bottle No.1), CTB201606005 (Bottle No.1), CTB201610006 (Bottle No.1) were kept at 2-8 ■ for 48 months, 42 months and 36 months respectively, test results were all complied with specification of Diphtheria, Tetanus and Acellular Pertussis Combined Vaccine, Adsorbed (Document code: WSYX-QC-TS-00-09-0001), the test results show that the bulk tetanus toxoid produced by Walvax can meet the requirement of 36 months stored at 2-8 ■.

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2.3.S.7.4 Stability Summary and Conclusion on Purified Liquid of Tetanus Toxoid

Table 97 Batch Information of Purified Liquid of Tetanus Toxoid used in Stability Study

Batch Storage

Sample Batch No. Packaging Mfg Date

quantity condition

CTL201909018 3000 mL 2019.09.20

Purified liquid of

CTL201909019 Glass bottle 3000 mL 2-8 ■ 2019.09.22

tetanus toxoid

CTL201909020 2500 mL 2019.09.30

Basis of Study

1. GMP

2. Chinese Pharmacopeia 2015, volume IV, Guiding principles for stability test of bulk drugs and pharmaceutical product.

3. Management Procedure for Continuous Stability Inspection of Products (Document Code: WSYX-QA-SMP-00-09-0010)

4. Quality Standard of Purified Liquid of Tetanus Toxoid (Document Code: WSYX-QC-TS-02-09-0002).

Test protocol

1. storage condition: 2-8 ■

2. test intervals: initial, 1st month, 2nd month, 3rd month, 6th month, 9th month, 12th month, 15th month, 18th month, 24th month, 30th month.

3. test items and corresponding time:

Initial, 12th month, 30th month: full test (purity, sterility test, protein content, bacterial endotoxin);

1st month, 2nd month, 3rd month, 6th month, 9th month, 15th month, 18th month, 24th month: purity, protein content;

Test items and specification

See the Table 98.

Table 98 Test Items and Specifications of Purified Liquid of Tetanus Toxoid

Test items Specifications

Protein content No less than 16 mg/mL

Bacterial endotoxin No more than 10 EU/mg

Purity No less than 90%

Sterility test Should comply with the requirement

Conclusion

3 batches of Purified Liquid of Tetanus Toxoid, which are CTL201909018 (Bottle No.1),

CTL201909019 (Bottle No.1), CTL201909020 (Bottle No.1) were kept at 2-8 ■ for 6 months,

tested at 0th month, 1st month, 2nd month, 3rd month, 6th month. Test results were all complied with Quality standard of Diphtheria, Tetanus and Acellular Pertussis Combined Vaccine, Adsorbed

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(Code: WSYX-QC-TS-00-09-0001). Test results show that the purified liquid of tetanus toxoid produced by Walvax can meet the requirement of 6 months stored at 2-8 ■.

2.3.S.7.5 Stability Summary and Conclusion on Monovalent Bulk Conjugate

Table 99 Batch Information of Monovalent Bulk Conjugate used in Stability Study

Batch Storage

Serotypes Batch No. Packaging Mfg. Date

quantity condition

Serotype 1 RSP1C201706001 9286 mL 2017.06.21

pneumococcal bulk RSP1C201707002 12302 mL 2017.07.21

conjugate RSP1C201709003 12660 mL 2017.09.06

Serotype 3 RSP3C201706001 22700 mL 2017.06.24
pneumococcal bulk RSP3C201707002 22700 mL 2017.07.26
conjugate RSP3C201708003 20600 mL 2017.08.19
Serotype 4 RSP4C201706001 20800 mL 2017.06.03
pneumococcal bulk RSP4C201707002 12300 mL 2017.07.26
conjugate RSP4C201709003 11400 mL 2017.09.03
Serotype 5 RSP5C201705001 11702 mL 2017.05.25
pneumococcal bulk RSP5C201707002 12200 mL 2017.07.13
conjugate RSP5C201708003 10700 mL 2017.08.16
Serotype 6A RSP6AC201705001 11496 mL 2017.05.19
pneumococcal bulk RSP6AC201707002 11491 mL 2017.07.01
conjugate RSP6AC201708003 10000 mL 2017.08.22
Serotype 6B RSP6BC201705001 21800 mL 2017.05.19
pneumococcal bulk RSP6BC201707002 22600 mL 2017.07.01
conjugate RSP6BC201708003 22900 mL 2017.08.25
Serotype 7F RSP7FC201705001 5255 mL 2017.05.30
pneumococcal bulk RSP7FC201706002 Glass bottle 11500 mL 2-8 ■ 2017.06.27
conjugate RSP7FC201708003 10500 mL 2017.08.28
Serotype 9V RSP9VC201705001 11400 mL 2017.05.22
pneumococcal bulk RSP9VC201707002 11000 mL 2017.07.05
conjugate RSP9VC201709003 11900 mL 2017.09.03
Serotype 14 RSP14C201706001 12156 mL 2017.06.24
pneumococcal bulk RSP14C201707002 11400 mL 2017.07.21
conjugate RSP14C201709003 11546 mL 2017.09.06
Serotype 18C RSP18CC201706001 20800 mL 2017.06.21
pneumococcal bulk RSP18CC201707002 22000 mL 2017.07.17
conjugate RSP18CC201708003 21800 mL 2017.08.13
Serotype 19A RSP19AC201705001 11686 mL 2017.05.22
pneumococcal bulk RSP19AC201707002 12131 mL 2017.07.05
conjugate RSP19AC201708003 11700 mL 2017.08.28
Serotype 19F RSP19FC201706001 12031 mL 2017.06.27
pneumococcal bulk RSP19FC201707002 12311 mL 2017.07.17
conjugate RSP19FC201708003 12000 mL 2017.08.31
Serotype 23F RSP23FC201705001 22162 mL 2017.05.28
pneumococcal bulk RSP23FC201707002 22600 mL 2017.07.13
conjugate RSP23FC201708003 11000 mL 2017.08.31

Basis of Study

1. GMP
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2. Chinese Pharmacopeia 2015, volume IV, Guiding principles for stability test of bulk drugs and pharmaceutical product.
3. Management Procedure for Continuous Stability Inspection of Products (Document Code: WSYX-QA-SMP-0009-0030)
4. Quality Standard of 13-valent Pneumococcal Polysaccharide Conjugate Vaccine (Document Code: WSYX-QC-TS-0609-0005).

Test protocol

1. storage condition: 2-8 ■
2. test intervals: initial, 1st month, 2nd month, 3rd month, 6th month, 9th month, 12th month.
3. test items and corresponding time:
Initial, 12th month: full test (sterility test, bacterial endotoxin, polysaccharide content, protein content, residual EDAC, polysaccharide/protein ratio, free-polysaccharide content, free-protein content, molecular size, identity test, ADH content);
1st month, 2nd month, 3rd month, 6th month, 9th month: polysaccharide content, protein content, polysaccharide/protein ratio, free-polysaccharide content, free-protein content, molecular size.

Test items and specification

See the Table 100.

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Table 100 Test Items and Specifications of Monovalent Bulk Conjugate
Polysaccharide Protein Free- Free- Residual Bacterial
Polysaccharide Molecular
Serotype content content polysaccharide protein EDAC endotoxin Identity test Sterility test
/ protein ratio size
μg/mL μg/mL content content μmol/L EU/μg
KD ≤ 0.2, Should comply
Should be positive
1 ≥ 45 ≥ 70 0.15-0.60 ≤ 25% ≤ 4.5% ≤ 5 ≤ 0.75 recovery rate with the
with antiserum type 1
≥ 80% requirement
KD ≤ 0.2, Should comply
Should be positive
3 ≥ 50 ≥ 60 0.40-0.90 ≤ 20% ≤ 4.0% ≤ 5 ≤ 0.75 recovery rate with the
with antiserum type 3
≥ 80% requirement
KD ≤ 0.2, Should comply
Should be positive
4 ≥ 45 ≥ 40 0.30-1.10 ≤ 25% ≤ 3.5% ≤ 5 ≤ 0.75 recovery rate with the
with antiserum type 4
≥ 80% requirement
KD ≤ 0.2, Should comply
Should be positive
5 ≥ 30 ≥ 100 0.10-0.30 ≤ 20% ≤ 4.5% ≤ 5 ≤ 0.75 recovery rate with the
with antiserum type 5

$\geq 80\%$ requirement
 $KD \leq 0.2$, Should be positive Should comply
 $6A \geq 40 \geq 80$ $0.10-0.50 \leq 20\% \leq 4.5\% \leq 5 \leq 0.75$ recovery rate with antiserum type with the
 $\geq 80\%$ 6A requirement
 $KD \leq 0.2$, Should be positive Should comply
 $6B \geq 40 \geq 60$ $0.20-0.65 \leq 25\% \leq 4.0\% \leq 5 \leq 0.75$ recovery rate with antiserum type with the
 $\geq 80\%$ 6B requirement
 $KD \leq 0.2$, Should be positive Should comply
 $7F \geq 80 \geq 180$ $0.15-0.45 \leq 20\% \leq 4.5\% \leq 5 \leq 0.75$ recovery rate with antiserum type with the
 $\geq 80\%$ 7F requirement
 $KD \leq 0.2$, Should be positive Should comply
 $9V \geq 30 \geq 50$ $0.10-0.50 \leq 25\% \leq 3.5\% \leq 5 \leq 0.75$ recovery rate with antiserum type with the
 $\geq 80\%$ 9V requirement

$KD \leq 0.2$, Should be positive Should comply
 $14 \geq 40 \geq 60$ $0.15-0.70 \leq 25\% \leq 4.5\% \leq 5 \leq 0.75$ recovery rate with antiserum type with the
 $\geq 80\%$ 14 requirement

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$KD \leq 0.2$, Should be positive Should comply
 $18C \geq 30 \geq 50$ $0.30-0.60 \leq 30\% \leq 3.5\% \leq 5 \leq 0.75$ recovery rate with antiserum type with the
 $\geq 80\%$ 18C requirement
 $KD \leq 0.2$, Should be positive Should comply
 $19A \geq 80 \geq 120$ $0.10-0.65 \leq 30\% \leq 3.5\% \leq 5 \leq 0.75$ recovery rate with antiserum type with the
 $\geq 80\%$ 19A requirement
 $KD \leq 0.2$, Should be positive Should comply
 $19F \geq 40 \geq 100$ $0.10-0.40 \leq 35\% \leq 3.5\% \leq 5 \leq 0.75$ recovery rate with antiserum type with the
 $\geq 80\%$ 19F requirement
 $KD \leq 0.2$, Should be positive Should comply
 $23F \geq 40 \geq 60$ $0.25-0.65 \leq 25\% \leq 4.0\% \leq 5 \leq 0.75$ recovery rate with antiserum type with the
 $\geq 80\%$ 23F requirement

Conclusion
 Serotypes 1, 3, 4, 5, 6A, 6B, 7F, 9V, 14, 18C, 19A, 19F, 23F Pneumococcal Bulk Conjugate were stored at 2-8 ■, tested at 0th month, 1st month, 2nd month, 3rd month, 6th month, 9th month, 12th month, test results were all complied with Quality Standard of 13-valent Pneumococcal Polysaccharide Conjugate Vaccine (Document code: WSYX-QC-TS-06-09-0005).
 This stability study was conducted by the quality control laboratory strictly sampled at each test point, could provide evidence for safety, effectiveness and validity of pneumococcal bulk conjugate stored at 2-8 ■. In conclusion, shelf life of 13 serotypes of pneumococcal bulk conjugate manufactured Walvax stored at 2-8 ■ is 12 months.

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2.3.P DRUG PRODUCT

2.3.P.1 Description and Composition of the Drug Product
 13-Valent Pneumococcal Polysaccharide Conjugate Vaccine is a sterile liquid suspension of capsular polysaccharide antigens of *Streptococcus pneumoniae* serotypes 1, 3, 4, 5, 6A, 6B, 7F, 9V, 14, 18C, 19A, 19F, and 23F, with each saccharide individually conjugated to Tetanus Toxoid protein. The vaccine contains 2.6 µg/dose of serotype 1 and 19A, 2.5 µg/dose of serotype 3, 5, 6A, 9V, 3.0 µg/dose of serotype 4 and 23F, 6 µg/dose of serotype 6B, 2.85 µg/dose of serotype 7F, 2.75 µg/dose of serotype 14 and 19F, 3.25 µg/dose of serotype 18C. The vaccine is formulated in Phosphate-buffered Saline containing 4.25 mg/dose NaCl, 44.35 µg/dose NaH₂PO₄, 19.0 µg/dose Na₂HPO₄, and contains 0.5 mg/dose of aluminum phosphate as an adjuvant.
 Each 1 mL syringe or vial contains a single 0.5 mL dose of vaccine for intramuscular administration, with no preservative.

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2.3.P.2 Pharmaceutical Development

2.3.P.2.1 Components of the Drug Product

2.3.P.2.1.1 Drug Substances

Tetanus toxoid was chosen as the carrier protein for the 13-valent Pneumococcal Polysaccharide Vaccine based on the previous Walvax experience with tetanus toxoid in the *Haemophilus influenzae* type b conjugate vaccine.
 Activation and derivation are two key steps in polysaccharide protein conjugation. After a thorough investigation on the 13 polysaccharide structures, different activation agents are used according to different structures of capsular polysaccharides, and polysaccharide-protein conjugates are prepared with indirect conjugation method. The method uses cyanide as activation agent to activate capsular polysaccharides, the activated polysaccharides are bonded with the linker, ADH, through covalent bonding to form polysaccharide-ADH derivatives. The derivatives are banded with carrier protein through covalent bonding with condensation agent EDAC to form polysaccharide-ADH-carrier protein conjugates.
 The immunogenicity of the combination containing 13 polysaccharide-protein conjugates and aluminum phosphate was compared to single antigen during compatibility test in pre-clinical research. Test results showed no difference between the mitogenicity of the combination and single antigen, which proves good immunogenicity of the combination.

2.3.P.2.1.2 Excipients

The excipients in the 13-valent Pneumococcal Polysaccharide Vaccine are NaCl at 8.5 mg/mL, sodium dihydrogen at 88.7 µg/mL, disodium hydrogen phosphate at 38 µg/mL and aluminum

phosphate at 0.5 mg/mL.
The phosphate/saline buffer was added to the formulation to provide pH control and suitable osmolality. Aluminium phosphate was added as an adjuvant to enhance immunogenicity.

2.3.P.2.2 Drug Product

2.3.P.2.2.1 Formulation Development

(1) Adjuvant selection

Aluminium adjuvant has been used in clinic for more than 80 years, which is the first classical adjuvant approved by FDA for human vaccine. In the development of 9-valent pneumococcal conjugate vaccine preparation process, Walvax referred to the successful experience of Wyeth 7-valent pneumococcal conjugate vaccine containing aluminium phosphate adjuvant, and the requirements of European Pharmacopoeia Adsorbed Pneumococcal Conjugate Vaccine (Annex 3-6) for aluminium adjuvant, and determined to add aluminium adjuvant in the preparation of 9-valent Pneumococcal Polysaccharide Conjugate Vaccine final bulk.

Walvax used aluminium hydroxide and aluminium phosphate as adjuvants to formulate 9-valent Pneumococcal Polysaccharide Conjugate Vaccine final bulk. After visual inspection, the

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appearance of final bulk with aluminium hydroxide adjuvant is milky uneven suspension with coagulum after shaking, and the appearance of final bulk with aluminium phosphate adjuvant is milky homogeneous suspension after shaking. Therefore, in the preparation of 13-valent Pneumococcal Polysaccharide Conjugate Vaccine final bulk, we choose aluminium phosphate as the adjuvant of the vaccine.

In addition, we checked the instructions of PCV10 (GSK) and PCV13 (Pfizer) which were launched in 2009 and 2010 respectively, and both of them also adopted the formula containing aluminium phosphate adjuvant. Up to now, the vaccine composition of pneumococcal conjugate vaccine approved for marketing in the world is shown in the Table 101.

Table 101 Vaccine Composition of PCV7, PCV13 and PCV10

Product

PCV7 (Pfizer) PCV13 (Pfizer) PCV10 (GSK)

Composition

18C: TT, 19F: DT, other

Carrier protein CRM197 CRM197

serotypes: PD

6B: 4.4 µg/vial

6B: 4 µg/vial 4, 18C, 19F: 3 µg/vial

Polysaccharide 1, 3, 4, 5, 6A, 7F, 9V, 14,

4, 9V, 14, 18C, 19F 23F: 1, 5, 6B, 7F, 9V, 14, 23F:

content 18C, 19A, 19F, 23F: 2.2

4µg/vial 1 µg/vial

µg/vial

0.5 mg of aluminium Aluminium phosphate, Aluminium phosphate, 0.5

Adjuvant

phosphate 0.125 mg of aluminium mg of aluminium

100 µg of Tween-80

Other excipients Sodium chloride Sodium chloride

295 µg of succinate

It can be seen from the Table 101 that all the three approved pneumococcal conjugate vaccines in the world are adsorbed with aluminium adjuvant without exception, and all of them use aluminium phosphate adjuvant, which can effectively induce the production of high-efficient protective antibodies.

(2) Specification and dose study

In the pre-clinical study of 13-valent Pneumococcal Polysaccharide Conjugate Vaccine, Walvax referred to the actual situation that the content of each antigen in all kinds of listed conjugate vaccines was 1-10 µg, and referred to the specification of PCV7 (Pfizer) and PCV13 (Pfizer), our company designed and carried out the specification and dose research comparing to PCV7 (Pfizer), and determined the product specification and dose through the immunogenicity test.

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2.3.P.2.3 Manufacturing Process Development

2.3.P.2.3.1 Determination of Adjuvants

According to the European Pharmacopoeia (5.0) Pneumococcal polysaccharide conjugate vaccine (adsorption) and other related literature, the characteristics (structure, charge, molecular size, etc.) of pneumococcal polysaccharide conjugate antigen, aluminium adjuvant is proposed. Aluminium hydroxide and aluminium phosphate commonly used in vaccines as adjuvant. After compounding with aluminium hydroxide and aluminium phosphate, respectively, the appropriate adjuvant can be determined through observing the appearance of complex.

Table 102 Appearance after Compounding with Adjuvant

Sample Appearance

Aluminium hydroxide milky uneven suspension with coagulum after shaking

Aluminium phosphate milky homogeneous suspension after shaking

The results in the Table 102 shows that aluminium phosphate has good compatibility with polysaccharides, while aluminium hydroxide is incompatible. Aluminium phosphate is selected as the adjuvant for this vaccine.

2.3.P.2.3.2 Determination of Buffer System

Antigenicity of an antigen is affected by its primary (molecular formula) and higher order (secondary and higher) structures. Higher order structures of antigen are dramatically affected by the solution system it's in, factors include pH, osmotic pressure, ionic strength, and so on, among which pH more drastically affects its higher order structures. Apart from controlling the osmotic pressure the same to that of body fluid and ionic strength within bearing range of human body, a suitable buffer system should be developed as well to keep constant pH.

In reference to Chinese Pharmacopeia 2010 Version, Volume III, Manufacturing and Testing Procedure of Diphtheria, Tetanus and Acellular Pertussis Combined Vaccine, Adsorbed, and Walvax's Production Process of Final Bulk Diphtheria, Tetanus and Acellular Pertussis Combined Vaccine, adsorbed, final bulk was added to a phosphate buffer solution of pH 6.5, then filled to vials and conducted accelerated experiment at 37 °C. Appearance and pH of the sample were constantly tested to ensure a constant pH, thus evaluate the phosphate buffer system. The test results showed that while formulating the final bulk, the NaH₂PO₄-Na₂HPO₄ buffer can stabilize the pH value effectively and ensure that the final bulk maintains a good physical appearance. The buffer system for final bulk is as follow:

NaH₂PO₄ 88.7 µg/mL
 Na₂HPO₄ 38.0 µg/mL
 NaCl 8.5 mg/mL

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2.3.P.2.3.3 Establishment of Final Bulk Preparation Process

In reference to Chinese Pharmacopeia 2010 Version, Volume III, Manufacturing and Testing Procedure of Diphtheria, Tetanus and Acellular Pertussis Combined Vaccine, Adsorbed, and Walvax's Manufacturing Process of Final Bulk Diphtheria, Tetanus and Acellular Pertussis Combined Vaccine, adsorbed, and Walvax's Production and Testing Procedure of 9-valent Pneumococcal Conjugate Vaccine, Walvax established the operation procedure of final bulk. Through the detection of appearance, pH, antigen content, aluminium phosphate content, sodium chloride content, sterility, pyrogen, abnormal toxicity index, to investigate whether it meets the predetermined standard.

The preparation process of final bulk is determined as follows:

(1) Preparation of the excipients stock solution

Weigh the medicinal grade sodium chloride in the container according to the formula amount, dry heat sterilization and remove the pyrogen; weigh the medicinal grade phosphate in another container according to the formula amount, dry heat sterilization and remove the pyrogen; add the formula amount of WFI into the container, mix well, autoclave and obtain the excipients stock solution, place at 2-8 °C for use;

(2) Weigh the formula amount of each serotype polysaccharide bulk conjugate, mix well to obtain the mixture.

(3) Measure the formula amount of aluminium phosphate adjuvant, phosphate stock solution (pH 6.5), sodium chloride stock solution and WFI, mix well to obtain the excipients mixture.

(4) Pump the polysaccharide bulk conjugate mixture into the excipients mixture while stirring to obtain the final bulk.

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2.3.P.2.4 Container Closure System

2.3.P.2.4.1 Primary Packaging (Vial)

Table 103 Brief Profile of Packaging Materials

Packaging material Vendor Specification Standard

Shuangfeng Gerresheimer Pharmaceutical

1 mL/vial

Neutral borosilicate Glass (Danyang)Co., Ltd. YBB00292005-2-

glass vial Ningbo Zhengli Pharmaceutical Packing 2015

1 mL/vial

Co.ltd.

Halogenated butyl Shengzhou Pharmaceutical Packing 11G002-A17-

Q/SRPYBB10-2019

rubber plugs for Material (China) Co., Ltd. D2T-S

injection Jiangsu Hualan New Pharmaceutical YBB00042005-2015

11-B

(chlorinated) Material Co., Ltd. YBB00052005-2015

Aluminium-plastics Hebei Jinhuan Packaging Co., Ltd 11.3×6.4 mm

YBB00372003-2015

combined caps Jingjiang Huaxin Capping Co., Ltd. 11×6.3 mm

2.3.P.2.4.2 Vial Compatibility Research

The compatibility research of vial packaging was conducted on neutral borosilicate glass vial (produced by Ningbo Zhengli Pharmaceutical Packaging Co., LTD) and halogenated butyl rubber plugs for injection (chlorinated) (produced by Jiangsu Hualan Pharmaceutical New Materials Co., Ltd).

Compatibility summary and conclusion on neutral borosilicate glass vial

Objective:

This test is to investigate the compatibility of PCV13 and middle-boro glass injection vials (hereinafter referred to as vials) manufactured by Ningbo Zhengli as inner packing material, to evaluate the migration of the packaging material and its safety, the influence of vaccine on packing systems, and then to provide basis of selection of packaging materials and safety of drug usage.

Basis:

1 The compatibility trial guidelines between pharmaceutical packaging materials and drug substances (YBB00142002-2015)

2 The compatibility research technology guidelines of chemical pharmaceutical injection and glass pharmaceutical packaging container (trial implementation)

3 The criterion of neutral borosilicate glass vial (YBB00292005-2-2015)

4 Chinese pharmacopoeia, 4th edition, 2015

5 ICH Q3D, Introduction of elements and impurities, Step 4, 16 December 2014

6 United States Pharmacopeia <1660>

Experimental design:

1.The influence of glass container on drug substances

1.1 Migration tests: 1 batch of neutral borosilicate glass vial were used to package 3 batches of vaccine to conduct compatibility test and placed upright or upside down. At the test points of

accelerated test and long-term test, elements of extraction were studied with validated method, estimating possibly migration of elements to drug substance.

1.2 Migration of organics of rubber plugs: testing small molecules antioxidants content such as 2, 6-tert-butyl-4-methyl phenol (BHT) in every point. (Test results are used to evaluate compatibility and whether to continue the test.)

2 The test items of influences of pharmaceuticals on inner surface of glass containers.

2.1 The trend of packaging materials damage and erosion: 1 batch of vaccine were filled to 3 batches of neutral borosilicate glass vial placed upright or upside down to test compatibility. At the test points of accelerated test and long-term, samples were taken to study inner surface erosion and silicon, boron elements in the drug by staining, atomic absorption spectrum, etc. The influence of drug substances on vials was evaluated by analysis of trends of silicon and silicon/boron rate.

Table 104 Experimental Design for Compatibility Study of Neutral Borosilicate Glass Vial

Packaging material batch
Content of No. Storage
Product name Batch No. Categories Sampling time Test items
research Batch No. Batch No. of condition
of vial rubber plug
0 month, 1st month, 2nd
month, 3rd month, 6th
month, 9th month, Cadmium, Lead,
3 batches of PCV13 2-8 ■
GI201709004 12th month, Arsenic, Silicon,
and corresponding 1 Migration 18th month,
GI201709009 201705006 X201709009 Boron, Antimony,
batch of packaging substances
GI201709010 24th month, 36th month Calcium,
material
25±2 ■, 0 month, 1st month, 2nd Aluminum
13-valent
humidity month, 3rd month, 6th
pneumococcal
60%±10% month
polysaccharide
conjugate 0 month, 1st month, 2nd
Glass peeling
month, 3rd month, 6th
vaccine (staining method),
month, 9th month,
1 batch of PCV13 Stability 2-8 ■ visible foreign
GI201709004 201705006 X201709009 12th month,
and corresponding 3 of packa bodies, changes of
GI201709006 201706001 X201709010 18th month,
batches of ging silicon Si
GI201709008 201706044 X201709011 24 month, 36th month
th
packaging material material concentration and
25±2 ■, 0 month, 1st month, 2nd
silicon / boron
humidity month, 3rd month, 6th
ratio
60%±10% month

Compatibility summary and conclusion on halogenated butyl rubber plugs

Objectives:

The test is conducted to evaluate the compatibility of PCV13 with halogenated butyl rubber plugs which used to injection were made by Jiangsu Hualan Pharmaceutical New Materials Co., Ltd, and to evaluate possible migration substances and its safety, test the influence of vaccine to package system, and then provide bases of selecting package materials and using drug safely.

Basis:

- 1 Compatibility test guideline for pharmaceutical packaging material and drug substance (YBB00142002-2015)
- 2 Compatibility research technology guidelines for chemical pharmaceutical and elastomeric seal (trial implementation)
- 3 Standard of halogenated rubber plugs (YBB00042005-2015)
- 4 Chinese pharmacopoeia, 4th edition, 2015
- 5 ICH Q3D, Introduction of elements and impurities, Step 4, dated 16 December 2014
- 6 United States Pharmacopeia <1663>
- 7 United States Pharmacopeia <1664>

Experimental design

1 Extraction test

1.1 Elements Extraction: Elements are the easiest extractable substances in rubber plugs.

Referencing Compatibility research technology guideline for chemical pharmaceutical and elastomeric seal (trial implementation), and ICH Q3D, we focus on detection of the following elements: magnesium, silicon, titanium, cadmium, lead, arsenic, mercury, cobalt, vanadium, nickel, antimony, lithium and copper. If high content of other elements, they shall be listed as test items for compatibility test.

1.2 Organics extraction: organics are mainly used as assistant agents during rubber plugs manufacturing process. Referencing Compatibility research technology guideline for chemical pharmaceutical and elastomeric seal (trial implementation), major assistant agents are

antioxidants (BHT and its degradants), sulfides (disulfuric alkyl phenol and its degradants), volatile organic compounds (C13, C21 rubber oligomers), lubricants (dimethyl silicone oil).
2 Rubber plugs will age as time goes by, physical properties will change, and it may influence fragments for the rubber plugs, thus it was investigated.

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Table 105 Experimental Design for Compatibility Study of Halogenated Butyl Rubber Plugs
Batch No. of package
Test Storage
Product name Batch No. Categories Sampling time Test items
content Batch No. Batch No. of condition
of vials rubber plugs

0 month, 3rd month, 6th
3 batches of month, 9th month, 12th month,
PCV13 and 2-8 ■ metal elements: Zn, Ca, Fe,
GI201709004 18th month, 24th month, 36th
Si, Mg, Cu, Li, Sb, Ni, V, Co,
corresponding Migration month
GI201709009 201705006 X201709009 Hg, As, Pb, Cd, Ti■
1 batch of substances
GI201709010 organic: C13 oligomers, C13
packaging
13-valent material chlorine oligomer, C21
pneumococcal oligomers, stearic acid,
polysaccharide 25±2 ■ palmitic acid, BHT,
conjugate 0 month, 1st month, 2nd antioxidant 1076, antioxidant
humidity
vaccine month, 3rd month, 6th month 1010, silicone oil, Vulcanizer
60%±10%
fragments (sulfur, p-tert-amyl
0 month, 3rd month, 6th phenol), 2-Chloro-4-(tert-
1 batch of month, 9th month, 12th month,
2-8 ■ pentyl)phenol■■■
PCV13 and Stability 18th month, 24th month, 36th
GI201709004 201705006 X201709009 Puncture exfoliation
corresponding of packa month
GI201709006 201706001 X201709010
3 batches of ging
GI201709008 201706044 X201709011 25±2 ■
packaging material 0 month, 1st month, 2nd
humidity
material month, 3rd month, 6th month
60%±10%
Note: Compatibility test are conducted with 3 batches of vaccines corresponding to 1 batch of halogenated
rubber plugs and 1 batch of vaccine corresponding to 3 batches
halogenated butyl rubber plugs, all placed upside down. Extractions are studied using validated methods.
Puncture exfoliation function of rubber plugs after stability test is
tested simultaneously.

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In conclusion, the PCV13 manufactured by Walvax uses neutral borosilicate glass vial for injection
manufactured by Ningbo Zhengli Pharmaceutical Packaging Co., LTD and halogenated butyl
rubber plugs of inner package injection syringes manufactured by Jiangsu Hualan Pharmaceutical
New Materials Co., Ltd. Compatibility studies of drug substances and packaging materials showed
that no high risk extractum was migrated to the drug substance, and no destruction to the packaging
materials by the drug substance was observed, quality and stability of PCV13 were maintained.
Walvax will follow compatibility research protocols and continue conducting research for 24
months and 36 months under 5±3 ■, then provide an objective and scientific basis for selecting
halogenated butyl rubber plugs for inner package injection syringe manufactured by Ningbo
Zhengli Pharmaceutical Packaging Co., LTD and Jiangsu Hualan Pharmaceutical New Materials
Co., Ltd.

More details please refer to 3.2.P.2.4.2

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2.3.P.2.4.3 Primary Packaging (Pre-filled Syringe)

Table 106 Brief Profile of Packaging Materials
Packaging material Vendor Specification Standard
Shandong Weigao Group
Rubber plug for pre-filled Medical Polymer Company 1-3 mL
YBB00072004-2015
syringe Limited
BD 1-3 mL
Shandong Weigao Group
Medical Polymer Company 1 mL
Pole for pre-filled syringe YBB00062004-2015
Limited
BD 1 mL

2.3.P.2.4.4 Pre-filled Syringes Compatibility
The compatibility research of pre-filled syringe packaging was conducted on Rubber plug for pre-

filled syringe (produced by Shandong Weigao Group Medical Polymer Co., Ltd) and Pole for pre-filled syringe (produced by Shandong Weigao Group Medical Polymer Co., Ltd).

Compatibility summary and conclusion on pre-filled syringe
Walvax conducted compatibility studies of pre-filling syringes manufactured by Shandong Weigao Group Medical Polymer Products Co. LTD and PCV13 to investigate the influence of packaging materials on formulation stability and potential migration, as well as safety and applicability in use.

So far, migration test and compatibility study of packaging material for 24 months under 5±3■ and 6 months under 25±2■ had completed. The detailed research is shown as follows.

Objective

The test is to investigate compatibility of PCV13 and pre-filled syringes manufactured by Shandong Weigao Group Medical Polymer Products Co. LTD to evaluate potential migration and its safety, and the influence of vaccine on packaging systems, then to provide basis about selection of packaging material and safety of drug usage.

Basis

- 1 guideline for compatibility for pharmaceutical packaging material and drug (YBB00142002-2015)
- 2 Compatibility research technology guideline for chemical pharmaceutical injection and pharmaceutical packaging container (trial implementation)
- 3 Compatibility research technology guideline between chemical pharmaceutical and elastomeric seal (exposure draft)
- 4 Chinese pharmacopoeia, 4th edition, 2015, <9621>: guidelines for general requirements of pharmaceutical packaging materials
- 5 Chinese pharmacopoeia, 4th edition, 2015, <9622>: guidelines for pharmaceutical packaging materials and containers

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- 6 Chinese pharmacopoeia, 4th edition, 2015, <9001>: guidelines for stability tests of raw drugs and preparations
- 7 Technical guidelines for stability studies of biological products
- 8 ICH Q3D, Introduction of elements and impurities
- 9 United States Pharmacopeia, <1660> and <1664>
- 10 FDA Guidelines for Industry: Container closure systems for packaging pharmaceuticals and biologics for human use (May 1999)
- 11 European Union: Guideline for plastic packaging material (2015)
- 12 E/L Recommendations, safety threshold and best practice of oral inhalation and nasal substance quality (September 2006)
- 13 ICH Q1A, Guideline for stability tests of raw drugs and formulations
- 14 ICH Q5C, Guideline for stability studies of biological products or biotechnological products.

Experimental design

Migration tests were conducted with 3 batches of PCV13 and the corresponding 1 batch of packaging material, to evaluate migration of glass tubing to PCV13, and to evaluate corrosion of the surface of the pre-filled syringe glass tubing by PCV13, and to conduct safety assessment on the migration quantity. The tests used 3 batches of PCV13 and corresponding 1 batch of packaging material to evaluate the corrosion of pre-filled syringe glass tubing by PCV13. Detailed experimental design is shown in Table 107.

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Table 107 Experimental Design for Compatibility Study of Rubber Plug for Pre-filled Syringe

Batch No. of packaging	
Batch No. of Content of material Storage	
Product name Categories Sampling time Test items	
vaccine research Batch No. of Batch No. of condition	
glass tubing rubber plugs	
0 month, 3rd month, 6th Metal elements testing, degraded	
month, 9th month, 12th oxidants, oxidants, sulfur,	
3 batches of -	
month, 18th month, 24th decmethylcyclopentaneoxosilane,	
PCV13 and	
GK201709004 month, 36th month 2,6-dit-Bu-4-methylphenol,	
corresponding Migration	
GK201709005 X201709003 X201709006 dibutyl phthalate, bis-(2-	
1 batch of substances	
GK201709010 25±2 ■ 0 month, 1st month, 2nd ethylhexyl) phthalate, erucamide,	
13-valent packaging	
humidity month, 3rd month, 6th stearic acid, palmitic acid,	
pneumococcal material	
60%±10% month polydimethylsiloxane, isobutene-	
polysaccharide	
isoprene oligomers	
conjugate	
0 month, 3rd month, 6th	
vaccine 3 batches of	
month, 9th month, 12th	
PCV13 and Stability 2-8 ■	
GK201709004 X201709003 X201709006 month, 18th month, 24th Appearance, staining with	
corresponding of packa	
GK201709008 X201709004 X201709007 month, 36th month methylene blue, inspection with	
1 batch of ging	
GK201709009 X201709005 X201709008 25±2 ■ 0 month, 1st month, 2nd DIC, inspection with SEM,	
packaging material	
humidity month, 3rd month, 6th	
material	
60%±10% month	

Conclusion:

PCV13 manufactured by Walvax uses pre-filled syringes packaging manufactured by Shandong Weigao Group Medical Polymer Products Co. Ltd.

Compatibility studies showed no high-risk extractum in PCV13 and no destruction to the packaging material, PCV13 maintained good quality and safety.

Walvax will continue conducting research following compatibility research protocols of 36 months under 5±3■, providing an objective and scientific basis of selecting Shangdong Weigao's pre-filled syringes packaging.

More details please refer to 3.2.P.2.4.4

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2.3.P.2.5 Microbiological Attributes

This product is sealed packages of sterile suspension. As the stability results in the "2.3.P.8 Stability" section show that during the entire period of the study, the index of all safety remains constant and qualified. It can be approved that the tightness and integrity of the packaging system of this product meet the design and quality requirements. Therefore, the packaging can fully guarantee that products will not be contaminated by microorganisms during the validity period.

2.3.P.2.6 Compatibility

Please refer to 2.3.P.2.4

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2.3.P.3 Manufacture

2.3.P.3.1 Information on the Manufacturer

Name: Yuxi Walvax Biotechnology Co., Ltd.

Address: No. 83 South Dongfeng Road, High ■ New Technology Industries Development Zone, Yuxi, Yunnan Province, China

Function: vaccine manufacture

Tel: 0877-2076210

Fax: 0877-2076918

Manufacturing site: 1st floor, Fill & Finish Centre, K Building 203#, No. 83 South Dongfeng Road, High & New Technology Industries Development Zone, Yuxi, Yunnan Province, China

Tel: 0877-2076210

Fax: 0877-2076918

QC site: 3rd and 4th floor, QC laboratory, RD&QC building 308#, No. 83 South Dongfeng Road, High & New Technology Industries Development Zone, Yuxi, Yunnan Province, China

Tel: 0877-2076210

Fax: 0877-2076918

2.3.P.3.2 Batch Formula

Table 108 Batch Formula Information

Active Substances Nominal Composition Function

Pneumococcal polysaccharide serotype 1 2.6 µg Antigen

Pneumococcal polysaccharide serotype 3 2.5 µg Antigen

Pneumococcal polysaccharide serotype 4 3.0 µg Antigen

Pneumococcal polysaccharide serotype 5 2.5 µg Antigen

Pneumococcal polysaccharide serotype 6A 2.5 µg Antigen

Pneumococcal polysaccharide serotype 6B 6.0 µg Antigen

Pneumococcal polysaccharide serotype 7F 2.85 µg Antigen

Pneumococcal polysaccharide serotype 9V 2.5 µg Antigen

Pneumococcal polysaccharide serotype 14 2.75 µg Antigen

Pneumococcal polysaccharide serotype 18C 3.25 µg Antigen

Pneumococcal polysaccharide serotype 19A 2.6 µg Antigen

Pneumococcal polysaccharide serotype 19F 2.75 µg Antigen

Pneumococcal polysaccharide serotype 23F 3.0 µg Antigen

Aluminum phosphate 0.5 mg Adjuvant

Sodium chloride 4.25 mg Excipient

Sodium dihydrogen phosphate 44.35 µg Excipient

Disodium hydrogen phosphate 19.0 µg Excipient

Water for injection Qs to 0.5 mL Excipient

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2.3.P.3.3 Description of Manufacturing Process

2.3.P.3.3.1 Flow Chart of Manufacturing Process and Process Control

Calculate the amount of excipients and

13 serotypes of monovalent bulk conjugate

based on formulation protocol

Mix WFI, 10% NaCl solution and phosphate-buffered saline to obtain excipients mixture

Add 13 serotypes of monovalent bulk conjugate into excipients mixture

Stir for more than 10 min

Quality Control points and items:

Sterile filtration Sterility test

Wash and sterilization of vials and rubber plugs

Sterile filling

Capping with sterile
aluminum caps Quality Control points and items:
Appearance, extractable volume,
osmolality, pH, NaCl content,
Polysaccharide content, conjugate
Visual inspection antigen content, conjugate protein
content, total protein content,
sterility test, abnormal toxicity test,
Labelling aluminium content

Packaging

Storage

Figure 28 Manufacturing Process Flow Chart for Fill and Finish of Finished Product (Vial)

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Calculate the amount of excipients and
13 serotypes of monovalent bulk conjugate
based on formulation protocol

Mix WFI, 10% NaCl solution
and phosphate-buffered saline to
obtain excipients mixture

Add 13 serotypes of monovalent bulk
conjugate into excipients mixture

Stir for more than 10 min

Quality Control points and items:
Sterile filtration Sterility test

Wash and sterilization of
pre-filled syringes

Sterile filling

Quality Control points and items:
Visual inspection Appearance, extractable volume,
osmolality, pH, NaCl content,
Polysaccharide content, conjugate
antigen content, conjugate protein
Labelling content, total protein content,
sterility test, abnormal toxicity test,
aluminium content
Packaging

Storage

Figure 29 Manufacturing Process Flow Chart for Fill and Finish of Finished Product (Pre-filled Syringe)

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2.3.P.3.3.2 Discription of Manufacturing Process

Walvax issued Standard Operation Procedures for fill and finish of 13-valent Pneumococcal Polysaccharide Conjugate Vaccine according to Regulation of registration for 13-valent Pneumococcal Conjugate Vaccine approved by National Medical Products Administration (Standard No: YBS00392019). The product is a finished product after formulation, filling, capping, visual inspection, labelling, and packaging.

Manufacturing process steps include:

(1) Perpetration the excipients stock solution

Weigh the medicinal grade sodium chloride in the container according to the formula amount, dry heat sterilization and remove the pyrogen; weigh the medicinal grade phosphate in another container according to the formula amount, dry heat sterilization and remove the pyrogen; add the formula amount of WFI into the container, mix well, autoclave and obtain the excipients stock solution, place at 2-8 °C for use;

(2) Weigh the formula amount of each serotype polysaccharide bulk conjugate, mix well to obtain the mixture.

(3) Measure the formula amount of aluminium phosphate adjuvant, phosphate stock solution (pH 6.5), sodium chloride stock solution and WFI, mix well to obtain the excipients mixture.

(4) Pump the polysaccharide bulk conjugate mixture into the excipients mixture while stirring, sterile filtrate to obtain the final bulk.

(5) Fill the final bulk into vials or pre-filled syringes, after labelling and packaging to obtain finished products.

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2.3.P.3.4 Controls of Critical Steps and Intermediates

Table 109 Controls of Critical Steps of Formulation and Fill & Finish Process for 13-valent

Pneumococcal Polysaccharide Conjugate Vaccine

Step Critical manufacturing steps Parameter range

Washing and sterilization Sterilization time 30 min

for rubber plug Sterilization temperature 121 ■
 Ultrasonic power 50 (300W)
 Circulating water pressure 0.20-0.50 Mpa
 Purified water pressure 0.20-0.50 Mpa
 WFI pressure 0.20-0.50 Mpa
 Compressed air pressure 0.40-0.60 Mpa
 Sterilization temperature 320±10 ■
 Washing and sterilization Sterilization time ≥ 5 min
 for vials Belt frequency 20 HZ
 Preheating ≥ 60 Pa
 Heating 1 ≥ 160 Pa
 Heating ≥ 160 Pa
 Differential pressure
 Cooling ≥ 50 Pa
 Cooling 2 ≥ 50 Pa
 Cooling 3 ≥ 50 Pa
 Final bulk mixing speed 25 rpm
 Stirring speed in 10 L bottle 250 rpm
 Sterile filling
 Extractable volume 0.50-0.60 mL
 Filling room temperature 18±2 ■
 Capping Capping temperature 18±2 ■
 Labelling Room temperature < 25 ■
 Packaging temperature ≤ 25 ■
 The information of packaging material
 Packaging
 Packaging information batch number, production date and
 expiry date are correct and clear
 Warehousing Store temperature 2-8 ■

2.3.P.3.5 Process Validation and/or Evaluation

The 13-valent Pneumococcal Polysaccharide Conjugate Vaccine production process includes six process steps: final bulk formulation, filling, capping, physical appearance inspection, labelling, and packaging. Through evaluate the critical process parameters and control of each step, in 2013, a total of 6 batches of process validation were carried out for 13-valent Pneumococcal Conjugate vaccines, including 3 batches of vials packaging (GI201708001, GI201708002, GI201708001), and 3 batches of pre-filled syringe packaging (GK201709001, GK201709002, GK201709003). The validation of the production process of the batch of 13-valent Pneumococcal Polysaccharide Conjugate Vaccine shows that the production process of 13-valent pneumococcal conjugate vaccine can continuously produce products that meet the quality standards within the specified process parameters in a controlled state, the process is stable.

2.3.P.4 Control of Excipients

Excipients included in the manufacturing of 13-valent Pneumococcal Polysaccharide Conjugate Vaccine are: sodium chloride, disodium hydrogen phosphate, sodium dihydrogen phosphate, aluminium phosphate, in which sodium chloride, disodium hydrogen phosphate, sodium dihydrogen phosphate are tested according to CP. Aluminium phosphate is not included in CP,

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which was tested according to Standard Operation Procedures and Specification for Aluminium Phosphate (WSYX-QC-SOP-00-09-0608) issued by Walvax. The specifications of all excipients are showed as following tables, all excipients are released after qualified.

Table 110 Specification of Aluminium Phosphate

Test items Specification Analytical method
 Should be a white suspension, which can
 Appearance layer after long standing, back to suspension Visual detection
 after shaking.
 pH 6.0-8.0 pH meter
 Chloride NMT 0.5% Chemical reaction
 Sulphate Should be less than 0.1% Chemical reaction
 The color of test tube should not be deeper
 Heavy metal than standard solution tube. The content of Thioacetamide method
 heavy metal should not more than 0.0002%.
 Arsenic salt NMT 0.00004 Gutzeit
 Aluminium
 Should not less than 20 mg/mL. Chemical reaction
 phosphate
 Sodium chloride NMT 10 g/L Chemical reaction
 Bacterial endotoxin Should less than 5.0 EU/mg Gel-clot method
 Sterility Should comply with the requirements Membrane filtration method

Table 111 Specification of Sodium Chloride

Item specification Analytical method
 This product is colourless, transparent cubic
 Appearance crystal or white crystalline powder, odorless. Physical examination
 Soluble in water and almost insoluble in ethanol.
 1.Sodium Salt:
 (1) Take platinum wire, wet with hydrochloric acid, dip test product, burn in a colorless flame, and the flame is bright yellow.
 (2) Take 100 mg the test product, put it in a 10 ml test tube, dissolve with 2 ml injection water, add 15% potassium carbonate solution 2 ml, heated to boil, should not produce precipitation; add 4 ml potassium pyroantimonate test solution, Identify heated to boil; cool in ice water, if necessary, rub Chemical reaction
 tube wall with glass stick, should produce dense precipitation.
 2.Chloride

(1) Add silver nitrate solution, produce a white coagulate precipitate, which was dissolved by adding ammonia solution, add dilute nitric acid, the precipitate reproduces.

(2) Chlorine can be produced; wet potassium iodide starch test paper turns blue.

Add 2 drops of bromothymol blue indicator solution, if show yellow, add sodium hydroxide titration solution (0.02mol/L) 0.10ml, turn blue; Acidity and alkalinity Physical examination if show blue or green, add hydrochloric acid titration solution (0.02mol/L) 0.20ml, turn yellow.

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Clarity and color of

Should be colorless and clear Physical examination solution

Under the sunlight (or under the fluorescent iodide lamp), no blue marks can be seen in the crystals Starch reaction within 5 minutes.

Ultraviolet visible

Bromide NMT 0.01%

spectrophotometric method

Sulfate NMT 0.002% Chemical reaction

Ultraviolet visible

Nitrite Absorbance at 354nm should not more than 0.01

spectrophotometric method

Phosphate NMT 0.0025% Chemical reaction

Ferrocyanide Should not show blue within 10 min Chemical reaction

Barium salt Both of the solution should be clear Chemical reaction

Calcium salt No opalescence is produced within 5 min Chemical reaction

Magnesium salt NMT 0.001% Chemical reaction

Potassium salt NMT 0.02% Chemical reaction

Loss on drying NMT 0.5% Drying method

Ferric salt NMT 0.0003% Chemical reaction

heavy metal NMT 0.0002% Thioacetamide method

Arsenic salt 0.00004% Gutzeit

Bacterial endotoxin test Less than 5 EU/g Semi quantitative gel method

Assay NLT 99.5% Titration

Table 112 Specification of Sodium Dihydrogen Phosphate

Item Specification Analytical method

This product is colorless or white crystal or

lump: odorless; it is easy to weathering at room

Appearance Physical method

temperature air. Soluble in water, almost insoluble in ethanol.

1.the aqueous solution of test product

effervescent on addition of sodium carbonate.

2.1. Sodium Salt:

(1) Take platinum wire, wet with hydrochloric acid, dip test product, burn in a colorless flame, and the flame is bright yellow.

(2) Take 100 mg the test product, put it in a 10 ml test tube, dissolve with 2 ml injection water, add 15% potassium carbonate solution 2 ml, heated to boil, should not produce precipitation; add 4 ml potassium pyroantimonate test solution, heated to boil; cool in ice water, if Identification necessary, rub tube wall with glass stick, should Chemical reaction produce dense precipitation.

2.2. Phosphate:

(1) The neutral solution of the test product, add silver nitrate solution, produce light yellow precipitation. Separate, precipitation is easy to dissolve in ammonia solution or dilute nitric acid.

(2) add magnesium ammonium chloride solution in the test product, produce white crystalline precipitate.

(3) take test product, add ammonium molybdate solution and nitric acid, produce yellow

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precipitate, separate, which can be dissolved in ammonia solution.

Alkalinity 4.1-4.5 pH meter

Clarity of solution Should be clear and colorless Physical method

Chloride NMT 0.01% Chemical reaction

Sulphate NMT 0.05% Chemical reaction

Loss on drying 10.0-15.0% Drying method

Aluminium salt NMT 0.01% Chemical reaction

Calcium salt NMT 0.01% Chemical reaction

Heavy metal NMT 10ppm Thioacetamide method

Arsenic salt NMT 0.0005% Gutzeit

Assay ≥98% Titration

Table 113 Specification of Dodecahydrate Disodium Hydrogen Phosphate

Item Specification Analytical method

This product is colorless or white crystal or lump, odorless; it is easy to weathering at room temperature air. Soluble in water, almost insoluble in ethanol.

1.Sodium Salt:

(1) Take platinum wire, wet with hydrochloric acid, dip test product, burn in a colorless flame, and the flame is bright yellow.

(2) Take 100 mg the test product, put it in a 10 ml test tube, dissolve with 2 ml injection water, add 15% potassium carbonate solution 2 ml, heated to boil, should not produce precipitation; add 4 ml potassium pyroantimonate test solution, heated to boil; cool in ice water, if necessary, rub tube wall with glass stick, should produce dense precipitation.

Identification 2.Phosphate: Chemical reaction

(1) The neutral solution of the test product, add silver nitrate solution, produce light yellow precipitation. Separate, precipitation is easy to dissolve in ammonia solution or dilute nitric acid.

(2) add magnesium ammonium chloride solution in the test product, produce white crystalline precipitate.

(3) take test product, add ammonium molybdate solution and nitric acid, produce yellow precipitate, separate, which can be dissolved in ammonia solution.

Alkalinity 9.1-9.4 pH meter

Clarity of solution Should be clear and colorless Physical method

Chloride NMT 0.001% Chemical reaction

Sulphate NMT 0.002% Chemical reaction

Carbonate Should not produce bubbles Chemical reaction

Insoluble particles in

NMT 10mg (0.05%) Chemical reaction, weighing water

The violet red color of the solution will not

Reductive substance Chemical reaction disappear.

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Loss on drying 55.0-64.0% Drying method

Ferric salt NMT 0.002% Chemical reaction

heavy metal NMT 5ppm Thioacetamide method

Arsenic salt NMT 0.0002% Gutzeit

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2.3.P.5 Control of Drug Product

2.3.P.5.1 Justification of Specification

Table 114 Justification of Specification for Finished Product

Test items Test method Specifications Ph. Eur. WHO Rationale for specifications

An identity test should be

performed that demonstrates Double immunodiffusion test

Each polysaccharide

that all the intended confirms the presence of certain

present in the vaccine is

Double pneumococcal polysaccharide serotype of polysaccharide. The

Identity / identified by a suitable

immunodiffusion serotypes and carrier protein (s) identity of tetanus toxoid is tested

immunochemical

are present in the final product, in the tetanus toxoid bulk method.

unless this test has been production with flocculant test.

performed on the final bulk.

The limit is set according to

Quality Standard for

Visual Milky white suspension after Pneumococcal Polysaccharide

Appearance //

determination shaking. Conjugate Vaccine (WSYX-QC-

TS-0006-0721), which is

approved by NMPA.

The limit is set according to

Quality Standard for

Visual Pneumococcal Polysaccharide

Volume No less than 0.50 mL/dose //

determination Conjugate Vaccine (WSYX-QC-

TS-0006-0721), which is

approved by NMPA.

The limit is set according to

Quality Standard for

Osmolality Pneumococcal Polysaccharide

Osmolality 220 - 310 mOsmol/kg / /
meter Conjugate Vaccine (WSYX-QC-
TS-0006-0721), which is
approved by NMPA.

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The limit is set according to
Quality Standard for
Pneumococcal Polysaccharide
The amount of aluminum
Aluminum Maximum 1.25 mg per Conjugate Vaccine (WSYX-QC-
Titration 0.18 - 0.27 mg/mL should not exceed 1.25 mg per
content single human dose. TS-0006-0721), which is
single human dose.
approved by NMPA and is a
stricter requirement than both Ph.
Eur. and WHO requirements.
The pH of each final lot should The limit is set according to
be tested and shown to be Quality Standard for
within the range of values found Pneumococcal Polysaccharide
pH pH meter 5.5 - 7.5 /
for vaccine lots shown to be Conjugate Vaccine (WSYX-QC-
safe and effective in clinical TS-0006-0721), which is
trials and in stability studies. approved by NMPA.
The limit is set according to
Quality Standard for
Sodium Pneumococcal Polysaccharide
Titration 7.5 - 9.5 g/L / /
chloride Conjugate Vaccine (WSYX-QC-
TS-0006-0721), which is
approved by NMPA.
Range The polysaccharide
Serotype
(µg/mL) content for each
1 3.6 - 6.8 serotype is determined
3 3.5 - 6.5 by a suitable
4 2.0 - 7.8 immunochemical
5 3.5 - 6.5 The amount of each The limit is set according to
method. The vaccine
6A 3.5 - 6.5 pneumococcal polysaccharide Quality Standard for
contains not less than 70
Polysaccharide in the final containers should be Pneumococcal Polysaccharide
Turbidimetry 6B 8.4 - 15.6 per cent and not more
content determined and shown to be Conjugate Vaccine (WSYX-QC-
7F 4.0 - 7.4 than 130 per cent of the
within the specifications agreed TS-0006-0721), which is
9V 3.5 - 6.5 quantity stated on the
by the NRA. approved by NMPA.
14 3.8 - 7.2 label for each
18C 4.6 - 8.4 polysaccharide. The
19A 3.6 - 6.8 confidence limits
19F 3.8 - 7.2 (P=0.95) are not less
23F 4.2 - 7.8 than 80 per cent and not

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more than 120 per cent
of the estimated content.
No less than the limit as below.
Standard
Release in
Serotype standard validity
(%) period
(%)
1 30 25
3 80 75 The limit is set according to
4 75 70 Quality Standard for
Conjugate
Gel 5 75 70 Pneumococcal Polysaccharide
antigen / /
chromatography 6A 80 75 Conjugate Vaccine (WSYX-QC-
content
TS-0006-0721), which is
6B 80 75
approved by NMPA.
7F 75 70
9V 80 75
14 80 75
18C 80 75
19A 70 65
19F 65 60
23F 65 60
The limit is set according to
Quality Standard for
No less than 80% for release
Conjugate Gel Pneumococcal Polysaccharide
standard, no less than 75% in / /

protein content chromatography Conjugate Vaccine (WSYX-QC- validity period.
TS-0006-0721), which is
approved by NMPA.
The limit is set according to
Quality Standard for
Total protein Pneumococcal Polysaccharide
Lowry method 41 - 150 µg/dose / /
content Conjugate Vaccine (WSYX-QC-
TS-0006-0721), which is
approved by NMPA.
It complies with the test Should be tested for bacterial The limit is set according to
Sterility Film filtration Meets the requirement of the test
for sterility. and mycotic sterility in Quality Standard for
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accordance with the Pneumococcal Polysaccharide
requirements of the revised Conjugate Vaccine (WSYX-QC-
General requirements for the TS-0006-0721), which is
sterility of biological substances approved by NMPA.
or by a method approved by the
NRA.
The limit is set according to
The requirement to test lots of
No adverse reactions, healthy and Quality Standard for
pneumococcal conjugate
Abnormal weight gain for each guinea pig Pneumococcal Polysaccharide
Animal test / vaccine for unexpected toxicity
toxicity and mouse during observation Conjugate Vaccine (WSYX-QC-
(abnormal toxicity) should be
period. TS-0006-0721), which is
agreed with the NRA.
approved by NMPA.
Endotoxin content or pyrogenic
Less than 12.5 EU per activity should be consistent
Bacterial single human dose, with levels found to be The specification is required by
Gel-clot method No more than 12.5 EU/dose
endotoxin unless otherwise acceptable in vaccine lots used Ph. Eur. (2150).
justified and authorized. in clinical trials and approved
by the NRA.

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2.3.P.5.2 Analytical Procedures

Table 115 SOP List for Finished Product Test
Test items SOP Document Code Method
Visual
Appearance SOP for Appearance Check WSYX-QC-SOP-00-09-0108
determination
SOP for the Extractable Volume Weight
Volume WSYX-QC-SOP-00-09-0109
Test measurement
pH SOP for pH Value Test WSYX-QC-SOP-00-09-0110 pH meter
SOP for Aluminum Hydroxide
Aluminum
(or Aluminum phosphate) WSYX-QC-SOP-00-09-0117 Titration
content
Content Test
Sodium chloride SOP for Sodium Chloride
WSYX-QC-SOP-00-09-0120 Titration
content Content Determination
Conjugate SOP for Conjugate Antigen Biochemical
WSYX-QC-SOP-06-09-0003
antigen content Content Determination method

Osmolality SOP for Osmolality Test WSYX-QC-SOP-00-09-0135 Osmometer

Total protein SOP for Protein Content
WSYX-QC-SOP-0009-0102 Lowry method
content Determination

SOP of Conjugate Protein
Conjugate Content Test for 13-valent Gel
WSYX-QC-SOP-06-09-0006
protein content Pneumococcal Polysaccharide chromatography
Conjugate Vaccine

Abnormal SOP for Abnormal Toxicity
WSYX-AC-SOP-00-09-0021 Animal test
toxicity Test
Bacterial SOP for Bacterial Endotoxin
WSYX-QC-SOP-00-09-0203 Gel-clot method
Endotoxin Test
Membrane
Sterility SOP for Sterility Test WSYX-QC-SOP-00-09-0202
filtration method
SOP for Polysaccharide Content

Polysaccharide Rate nephelometry
Determination with Rate WSYX-QC-SOP-00-09-0141
content method
nephelometry Method
Conjugate SOP for Conjugate Antigen Gel
WSYX-QC-SOP-06-09-0003
antigen content Content Determination chromatography

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2.3.P.5.3 Validation of Analytical Procedures

Table 116 Validation or Qualification List of Analytical Procedures
Validation/Qualification
Test items Document Code Test Method
protocol
Qualification Protocol of
Aluminum Content
Aluminum content Determination for 13- WSYX-QC-TS-0006-0661 Titration
valent Pneumococcal
Conjugate Vaccine
Qualification Protocol of
Sodium chloride content
Sodium chloride content Determination for 13- WSYX-QC-TS-0006-0660 Titration
valent Pneumococcal
Conjugate Vaccine
Qualification Protocol of
Total Protein content
Total protein content Determination for 13- WSYX-QC-TS-0006-0747 Lowry method
valent Pneumococcal
Conjugate Vaccine
Validation Protocol of
Conjugate Antigen
Conjugate antigen Gel
Content Test for13-valent WSYX-QC-TS-0006-0728
content chromatography
Pneumococcal Conjugate
Vaccine
Qualification Protocol of
Bacterial Endotoxin Test
Bacterial endotoxin for 13-valent WSYX-QC-TS-0006-0764 Gel-clot method
Pneumococcal Conjugate
Vaccine
Applicability Test for
Sterility Test Method of Membrane
Sterility SYX-QC-TS-0006-0543
13-valent Pneumococcal filtration method
Conjugate Vaccine
Qualification Protocol of
Rate
Rate Nephelometry
Polysaccharide content WSYX-QC-TS-0006-0647 nephelometry
Method for Detection of
method
Polysaccharide Content

2.3.P.5.4 Impurities
According to the manufacturing process, there will be no impurities added in finished product
process, please refer to 2.3.S.3 Characterization for detailed information of impurities.

2.3.P.5.5 Batch analyses
Different batches of 13-valent Pneumococcal Polysaccharide Conjugate Vaccine were analyzed
based on the analytical procedures. The data is showing in Table 117.

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Table 117 Batch Analyses Data Statistics for Finished Products
3 batches of pilot sample 3 batches for IND
Item Acceptable criteria
20110304 20110305 20110406 20110707 20110708 20110709
Should be milky white
Appearance Complied Complied Complied Complied Complied
suspension after shaking.
Volume Not less than 0.5 ml/dose Complied Complied Complied Complied Complied
pH 5.5-7.5 6.6 6.6 6.6 6.5 6.5
Aluminium
0.18-1.2 mg/mL 1.1 1.0 0.9 1.0 1.0 0.9
content
Sodium
chloride 7.5-9.5 g/L 8.7 8.6 8.9 8.4 8.4 8.5
content
1: 2.4 µg 1: 2.3 µg 1: 2.2 µg 1: 2.2 µg 1: 2.3 µg
3: 2.5 µg 3: 2.4 µg 3: 2.3 µg 3: 2.2 µg 3: 2.2 µg 3: 2.2 µg
In each human dose, the 4: 2.4 µg 4: 2.4 µg 4: 2.3 µg 4: 2.2 µg 4: 2.2 µg
polysaccharide content of 5: 2.5 µg 5: 2.4 µg 5: 2.4 µg 5: 2.3 µg 5: 2.2 µg 5: 2.3 µg
serotypes 1, 3, 4, 5, 6A, 7F, 6A: 2.4 µg 6A: 2.4 µg 6A: 2.3 µg 6A: 2.2 µg 6A: 2.2 µg
9V, 14, 18C, 19A, 19F and 6B: 4.8 µg 6B: 4.8 µg 6B: 4.5 µg 6B: 4.4 µg 6B: 4.5 µg
Polysaccharide

23F should be not less than 7F: 2.4 µg 7F: 2.4 µg 7F: 2.2 µg 7F: 2.2 µg 7F: 2.2 µg 7F: 2.2 µg content
2.2 µg respectively, and the 9V: 2.3 µg 9V: 2.3 µg 9V: 2.3 µg 9V: 2.2 µg 9V: 2.2 µg 9V: 2.2 µg polysaccharide content of 14: 2.4 µg 14: 2.4 µg 14: 2.4 µg 14: 2.2 µg 14: 2.3 µg 14: 2.3 µg serotype 6B should be not 18C: 2.4 µg 18C: 2.3 µg 18C: 2.4 µg 18C: 2.2 µg 18C: 2.3 µg 18C: 2.3 µg less than 4.4 µg. 19A: 2.4 µg 19A: 2.3 µg 19A: 2.3 µg 19A: 2.2 µg 19A: 2.2 µg 19A: 2.3 µg 19F: 2.5 µg 19F: 2.5 µg 19F: 2.4 µg 19F: 2.3 µg 19F: 2.3 µg 19F: 2.2 µg 23F: 2.4 µg 23F: 2.3 µg 23F: 2.3 µg 23F: 2.2 µg 23F: 2.3 µg 23F: 2.3 µg
During observation period,
guinea pigs and mice shall
all be healthy, no adverse
Abnormal
reactions, and each guinea Complied Complied Complied Complied Complied Complied Complied toxicity
pig and mouse may gain
weight at the end of the
experiment.
Bacterial
NMT 12.5 EU/dose Complied Complied Complied Complied Complied Complied Complied
endotoxin

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Test according to
Sterility requirements, should Complied Complied Complied Complied Complied Complied Complied comply.

1 batch for clinical 6 batches for process validation
Item Acceptable criteria
201511003 GI201708001 GI201708002 GI201709003 GK201709001 GK201709002 GK201709003
Should be milky white
Appearance Complied Complied Complied Complied Complied Complied Complied
suspension after shaking.
Volume Not less than 0.5ml/dose Complied Complied Complied Complied Complied Complied Complied
pH 5.5-7.5 6.42 6.2 6.25 6.22 6.22 6.20 6.22
Aluminium
0.18-1.2 mg/mL 1.01 1.03 1.08 0.90 0.94 0.99 1.07
content
Sodium
chloride 7.5-9.5 g/L 8.31 8.48 8.46 8.815 8.81 8.01 8.96
content
1: 3.0 µg 1: 3.2 µg 1: 2.8 µg 1: 2.9 µg 1: 3.0 µg 1: 3.0 µg 1: 3.1 µg
3: 3.1 µg 3: 2.4 µg 3: 2.6 µg 3: 2.4 µg 3: 2.7 µg 3: 2.8 µg 3: 2.8 µg
In each human dose, the 4: 2.8 µg 4: 2.8 µg 4: 2.4 µg 4: 2.67 µg 4: 2.6 µg 4: 2.6 µg 4: 2.6 µg
polysaccharide content 5: 2.4 µg 5: 2.7 µg 5: 2.4 µg 5: 2.7 µg 5: 2.6 µg 5: 2.6 µg 5: 2.6 µg
of serotypes 1, 3, 4, 6A: 3.0 µg 6A: 2.9 µg 6A: 2.9 µg 6A: 2.6 µg 6A: 2.6 µg 6A: 2.7 µg 6A: 2.6 µg
5, 6A, 7F, 9V, 14, 18C,
6B: 5.8 µg 6B: 6.0 µg 6B: 5.4 µg 6B: 4.8 µg 6B: 4.8 µg 6B: 4.6 µg 6B: 4.7 µg
Polysaccharide 19A, 19F and 23F
7F: 3.0 µg 7F: 3.1 µg 7F: 2.8 µg 7F: 2.7 µg 7F: 2.8 µg 7F: 2.8 µg 7F: 2.8 µg
content should be 2.2-3.3 µg
9V: 3.2 µg 9V: 2.6 µg 9V: 2.9 µg 9V: 2.4 µg 9V: 3.1 µg 9V: 3.1 µg 9V: 3.0 µg
respectively, and the
polysaccharide content 14: 2.9 µg 14: 2.7 µg 14: 2.6 µg 14: 2.7 µg 14: 2.6 µg 14: 2.6 µg 14: 2.7 µg
of serotype 6B should be 18C: 2.9 µg 18C: 2.7 µg 18C: 2.8 µg 18C: 2.7 µg 18C: 2.6 µg 18C: 2.9 µg 18C: 2.6 µg
4.4-6.6 µg. 19A: 3.1 µg 19A: 2.5 µg 19A: 2.5 µg 19A: 2.8 µg 19A: 3.0 µg 19A: 3.0 µg 19A: 3.1 µg
19F: 2.9 µg 19F: 2.5 µg 19F: 2.6 µg 19F: 2.8 µg 19F: 3.0 µg 19F: 2.9 µg 19F: 2.9 µg
23F: 3.1 µg 23F: 2.6 µg 23F: 2.9 µg 23F: 3.0 µg 23F: 2.4 µg 23F: 2.4 µg 23F: 2.5 µg
During observation
period, guinea pigs and
Abnormal mice shall all be healthy,
Complied Complied Complied Complied Complied Complied Complied
toxicity no adverse reactions,
and each guinea pig and
mouse may gain weight
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at the end of the
experiment.
Bacterial
NMT 12.5 EU/dose Complied Complied Complied Complied Complied Complied Complied
endotoxin
Test according to
Sterility requirements, should Complied Complied Complied Complied Complied Complied Complied comply.

Conclusion:
Through the analysis of the data of the above batches of products, it is shown that: appearance, volume,
pH value, aluminium content, sodium chloride
content, polysaccharide content, abnormal toxicity test, bacterial endotoxin and sterility test all met
the Quality Standard of 13-Valent Pneumococcal
Polysaccharide Conjugate Vaccine.

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2.3.P.6 Reference Standards or Materials
Table 118 List of Reference Standards or Materials used for Finished Products

Storage	Test Item	Reference name	Manufacturer	Unit	Condition
National standard sample	19.73	Protein content NIFDC	Below -20	■	
for protein content test	mg/tube				
Room					
Dextran GE 10	g/vial				
temperature					
Conjugate protein	Sinopharm				
Room					
Vitamin B12 Chemical Reagent	1 g/vial				
temperature					
Co., Ltd.					
Pneumococcal specific					
antiserum (serotypes 1, 3, Danish National					
1 ml/vial	2-8	■			
4, 5, 6A, 6B, 7F, 9V, 14, Serum Institute					
Polysaccharide 18C, 19A, 19F, 23F)					
content/conjugate Pneumococcal					
antigen content polysaccharide reference					
10 mg/vial					
(serotypes 1, 3, 4, 5, 6A, ATCC	Below -20	■			
2 mg/vial					
6B, 7F, 9V, 14, 18C, 19A,					
19F, 23F)					
Standard bacterial Zhanjiang Andusi					
Bacterial endotoxin 10 EU/tube	2-8	■			
endotoxin sample Biology Co., Ltd.					
Sterility Test Staphylococcus aureus NIFDC /	Below -20	■			

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2.3.P.7 Container Closure System
There are two kinds of packaging forms for 13-valent Pneumococcal Polysaccharide Conjugate Vaccine, which are the packaging of vials and pre-filled syringes. The packaging materials for direct contact drugs used in vials are neutral borosilicate glass vial, halogenated butyl rubber plugs for injection (chlorination) and aluminum-plastic combined caps; the packaging materials for direct contact drugs used in pre-filled syringes are rubber plug for pre-filled syringe and pole for pre-filled syringe.

2.3.P.7.1 Primary Packaging (Vial)

Table 119 Brief Description of the Container Closure System for Vial			
Name	Manufacturer	Specification	Standard
Shuangfeng Gerresheimer		Neutral pharmaceutical glass (Danyang) 1 mL/vial	
borosilicate Co., Ltd	YBB00292005-2-2015		
glass vial Ningbo zhengli pharmaceutical		1 mL/vial	
packaging Co. Ltd			
Halogenated Shengzhou pharmaceutical		butyl rubber packaging material technology	11G002-A17-D2T-S Q/SRPYBB10-2019
plugs for (China) Co., Ltd			
injection Jiangsu Hualan New	YBB00042005-2015		
11-B			
(chlorination) Pharmaceutical Material Co., Ltd	YBB00052005-2015		
Aluminium- Hebei Jinhuan Packaging Co., Ltd	11.3×6.4 mm		
plastics Jingjiang Huaxin Capping Co.,	YBB00372003-2015		
11×6.3 mm			
combined caps Ltd			

Table 120 Specification of Neutral Borosilicate Glass Vial	
Test Items	Specification
It should be colorless and transparent; the surface should be clean and smooth, and Appearance there should be no obvious defects. No crack should be found in any part. Crack detection condition (n/Ac=500/7); other inspection cases (n/Ac=315/21)	
Internal surface water The volume of 0.01mol/L hydrochloric acid consumed by 100m1 of leaching resistivity solution should be less than 2.0 MI.	
Heat resistance test shall not be broken (n/Ac=80/1)	
Cold resistance test shall not be broken (n/Ac=80/1)	
121■particles water Volume of hydrochloric acid titrant (0.02mol/L) consumed by each gram of glass resistance particles should be ≤ 0.10 mL.	
outer diameter of bottle body: Φ13.7±0.15	
outer diameter of lip: Φ10.7~11.2	
outer diameter of bottle neck≤Φ9.3	
Size inner diameter of lipΦ7.05±0.20	
bottle height:27.0±0.50	
bottle edge thickness3.6±0.20	
bottom thickness ≥ 0.7	

Table 121 Specification of Halogenated Butyl Rubber Plugs for Injection (Chlorination)	
Test Items Specification	

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According to the following Unqualified classification inspection method, Class A, B and C should not appear.

Class A: There are stains, impurities, bubbles and cracks in the needle ring or there are bubbles and cracks on surface of sealing face. N/AC=315/3

Appearance

Class B: The surface is stained, impurities, rubber filaments, rubber chips, spongy, wool edge, neck plug rough, obvious glue deficiency. N/AC=315/10

Class C: incomplete and zigzag edges, uneven marks and colors caused by moulds. N/AC=200/21

(1) produce white or yellow sediment

Identification

(2) coincident with reference chromatogram

Puncture and crumb The number of chips should not exceed 5.

Self-sealability Methylene blue solution must not penetrate into the bottle.

Adhesion between

rubber stopper and Methylene blue solution must not penetrate into the bottle.

container

Puncture force The force needed to puncture the rubber stopper should not exceed 10N

Insoluble particles Particles larger than 10um per ml: NMT 30 particles, (photoresist method) Particles larger than 25um per ml: NMT 3 particles.

Residue on ignition NMT45%

Should not form sulfur spots or color of sulfur spots should not be more strength

Volatile sulfide

than standard sulfur spots. (less than 50ugNa₂S/20cm²)

The solution should be clear. If it is muddy, it should not be more strength than

Clarity and color No. 2 standard solution. if it shows color, it should not be more strength than No.5 standard yellow-green solution.

pH difference between test solution and reference solution should not more than

pH variation

1.0

UV absorbance The maximum absorbance at 220-360nm should not exceed 0.1.

Nonvolatile matter

The difference between the two should not exceed 4.0mg

(mg)

Readily oxidizable The difference of the consumption of titrant between the two should not exceed substance (ml) 3.0ml.

The color displayed in the third tube is not lighter than that in the first tube, and

Heavy metal the color displayed in the second tube should not be more strength than that in the first tube. NMT 1ppm.

The color of the test solution should not be more strength than that of the standard

Ammonium ion

solution, ammonium ion should be not more than 0.0002%.

Zinc ion The color of the test solution is not more strength than that of standard solution.

The conductivity of blank control solution should not exceed 3.0uS/cm (20■ ±

Conductivity (uS/cm) 1■).

The conductivity of test solution should not exceed 40.0uS/cm (20■±1■).

in the preliminary test. the body temperature of all the 3 rabbits increased less

than 0.6■, sum of elevated temperature was less than 1.4■, or in the retest, there

Pyrogen is not more than 1 rabbit of the 5 whose body temperature raise for 0.6■ or more

than 0.6■, the sum of elevated temperature of preliminary test and retest study is

about 3.5■ or less than 3.5■. should meet the requirements.

Hemolysis

Hemolysis rate should be less than 5%.

(hemolysis test)

Acute systemic During the 72 hours observation period, the response of the animals in the test toxicity test group should not be greater than that in the control group.

Table 122 Specification of Aluminium-plastics Combined Caps

Test Items Specification

Take appropriate amount of the cap, visual inspection in the natural light, should

Appearance

be clean, with no residual lubricant, burrs, damage and injection molding flying

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edge, plastic parts should be fully integrated with aluminum parts.

M/AC=315/21

Convex edge NMT3% H/AC=50/3

Opening quality There is no damage on the opening site. E/AC=13/1

After sterilization, the plastic part of the bottle cover can withstand the steam sterilization temperature of 130■ without deformation and discoloration, and

Sterilizing resistance

the surface of the aluminum part should not have any obvious changes. There

should be no fracture and abnormal deformation of the cover. E/AC=13/1

Coating firmness There is no damage on the coating. E/AC=13/1

Inner diameter of aluminum part:

The aluminum parts can pass through the gauge side of the limit plug gauge

cannot pass the stop gauge side completely. K/AC=125/5

Total height of aluminum and plastic

7.3±0.4. K/AC=125/10

Size Outer diameter of plastic part

13.2±0.25. K/AC=125/10

Inner height of aluminum part

6.4±0.20. K/AC=125/5

Thickness of aluminum part

0.19±0.02. E/AC=13/1

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2.3.P.7.2 Primary Packaging (Pre-filled Syringe)

Table 123 Brief Description of the Container Closure System for Pre-filled Syringe

Name Manufacturer Specification Standard
Rubber plug for Shandong Weigao Group Medical
1-3 mL
pre-filled Polymer products Co., Ltd. YBB00072004-2015
syringe BD 1-3 mL
Shandong Weigao Group Medical
Pole for pre- 1 mL
Polymer products Co., Ltd. YBB00062004-2015
filled syringe
BD 1 mL

Table 124 Specification of Rubber Plug for Pre-filled Syringe

Test Items Specification

The surface color should be uniform, without stains, impurities, bubbles and cracks, impurities, rubber filaments, rubber chips, spongy and wool edge; Appearance incomplete and zigzag edges caused by cutting, uneven marks and colors caused by mould. If there is a sprue, it should not be it should not protrude from the surface of the rubber plug. N/AC=125/2
Particles larger than 10um per ml: NMT 60 particles,
Insoluble particles
Particles larger than 25um per ml: NMT 6 particles.
(1) produce white or yellow sediment
Identification
(2) coincident with reference chromatogram
Residue on ignition NMT50%
1. White precipitates should be produced
Identification
2. It should be consistent with the standard chromatography.
The lead acetate test paper for the test solution should not form sulfur spots, if Volatile sulfide sulfur spot is formed, color of sulfur spots should not be more strength than that of standard lead acetate test paper. (less than 50ug/20cm²)
Clarity and color The solution should be clear and colorless.
pH difference between test solution and reference solution should not more than pH variation
1.0
UV absorbance The maximum absorbance at 220-360nm should not exceed 0.1.
Nonvolatile matter
The difference between the two should not exceed 4.0mg
(mg)
Readily oxidizable The difference of the consumption of titrant between the two should not exceed substance (ml) 3.0ml.
The color displayed in the third tube is not lighter than that in the first tube, and Heavy metal the color displayed in the second tube should not be more strength than that in the first tube. NMT 1ppm.
The color of the test solution should not be more strength than that of the standard Ammonium ion
solution, ammonium ion should be not more than 0.0002%.
Zinc ion The color of the test solution is not more strength than that of standard solution.
The conductivity of blank control solution should not exceed 3.0uS/cm (20±1 ■).
Conductivity (uS/cm)
The conductivity of test solution should not exceed 40.0uS/cm (20±1 ■).
Sterility Should comply with requirements.
Bacterial endotoxin Should be less than 0.5 EU/mL
in the preliminary test. the body temperature of all the 3 rabbits increased less than 0.6 ■, sum of elevated temperature was less than 1.4 ■, or in the retest, there Pyrogen is not more than 1 rabbit of the 5 whose body temperature raise for 0.6 ■ or more than 0.6 ■, the sum of elevated temperature of preliminary test and retest study is about 3.5 ■ or less than 3.5 ■. should meet the requirements.
Hemolysis
Hemolysis rate should be less than 5%.
(hemolysis test)
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Acute systemic During the 72 hours observation period, the response of the animals in the test toxicity test group should not be greater than that in the control group.

Table 125 Specification of Pole for Pre-filled Syringe

Test Items Specification

It should be colorless and transparent; the surface should be clean and smooth, and Appearance there should be no obvious defects. No crack should be found in any part.
(N/AC=125/2)
Internal surface water
The volume of consumed should be less than 2.0ml
resistivity
Pinhole patency The needle with a diameter of 0.18 should be able to pass freely. N/AC=20/2
121■ particles water Volume of hydrochloric acid titrant (0.02mol/L) consumed by each gram of glass resistance particles should be ≤ 0.10 mL.
Sterility Should comply with requirements.
Bacterial endotoxin Should be less than 0.5 EU/mL

2.3.P.7.3 Secondary Packaging

The secondary packaging form of 13-valent Pneumococcal Polysaccharide Conjugate Vaccine (packed with vial and pre-filled syringe) is 0.5ml/vial (syringe) per box. Each small box is marked with product name, specification, batch number, manufacturing date, expiration date, indication, eligibles, adverse reactions, contraindications, precautions, etc. at the same time, each small box is marked with the vaccine traceability code, which can be queried through the system to each vaccine information.

2.3.P.8 Stability

In accordance with the requirements of GMP, Walvax has compiled the document Product Stability Inspection Management rules (Document code: WSYX-QA-SMP-0009-0030). According to this document, we have placed accelerated stability test and long-term stability test on six batches of 13-valent Pneumococcal Polysaccharide Conjugate Vaccine.

Stability summary and conclusion on finished products

Table 126 Batch Information of 13-valent Pneumococcal Polysaccharide Conjugate Vaccine used in Stability Study

Batch No. Packaging Batch Quantity Storage condition Mfg. Date

GI201708001 87923 2019.08.17

GI201708002 Vial 85722 2018.08.25

GI201709003 85162 2019.09.01

2-8 ■

GK201709001 84697 2019.08.30

Pre-filled

GK201709002 84851 2019.09.03

syringe

GK201709003 82354 2019.09.13

Test condition:

Accelerated stability test condition: Temperature: 25±2 ■, humidity: 60±10%

Long-term stability test condition: Temperature: 5±3 ■, humidity: 60±10%

Testing period:

Accelerated stability test: 0th, 1st month, 2nd month, 3rd month, 6th month, 9th month.

Long-term stability test: 0th, 3rd month, 6th month, 9th month, 12th month, 18th month, 24th month, 30th month.

Table 127 Test Items and Specification of Accelerated Stability Study

Test items Specification

Appearance Should be milky uniform suspension after shaking

Extractable volume Should be not less than 0.5ml/vial or syringe

pH 5.5-7.5

Aluminium phosphate

0.8-1.2 g/L

content

Sodium chloride

7.5-9.5 g/L

content

1: 1.8-3.4 µg/dose, 3: 1.75-3.25 µg/dose, 4: 2.1-3.9 µg/dose, 5: 1.75-3.25 µg/dose,

6A: 1.75-3.25 µg/dose, 6B: 4.2-7.8 µg/dose, 7F: 2-3.7 µg/dose, 9V: 1.75-3.25

Polysaccharide content

µg/dose, 14: 1.9-3.6 µg/dose, 18C: 2.3-4.2 µg/dose, 19A: 1.8-3.4 µg/dose, 19F:

1.9-3.6 µg/dose, 23F: 2.1-3.9 µg/dose.

Abnormal toxicity Should comply with the requirements.

Bacterial endotoxin Should be not more than 12.5 EU/dose

Sterility test Should comply with the requirements

Osmolarity 220-310 mOsm/Kg

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Table 128 Test Items and Specification of Long-term Stability Study

Test items Specification

Appearance Should be milky uniform suspension after shaking

Extractable volume Should be not less than 0.5 ml/vial or syringe

pH 5.5-7.5

Aluminium content 0.8-1.2 g/L

Sodium chloride

7.5-9.5 g/L

content

0th, 3rd, 6th, 9th, 12th, 18th month:

1: 1.8-3.4 µg/dose, 3: 1.75-3.25 µg/dose, 4: 2.1-3.9 µg/dose, 5: 1.75-3.25 µg/dose,

6A: 1.75-3.25 µg/dose, 6B: 4.2-7.8 µg/dose, 7F: 2-3.7 µg/dose, 9V: 1.75-3.25

µg/dose, 14: 1.9-3.6 µg/dose, 18C: 2.3-4.2 µg/dose, 19A: 1.8-3.4 µg/dose, 19F:

1.9-3.6 µg/dose, 23F: 2.1-3.9 µg/dose.

Polysaccharide content

24th month:

1: 3.6-6.8 µg/dose, 3: 3.5-6.5 µg/dose, 4: 4.2-7.8 µg/dose, 5: 3.5-6.5 µg/dose, 6A:

3.5-6.5 µg/dose, 6B: 8.4-15.6 µg/dose, 7F: 4.0-7.4 µg/dose, 9V: 3.5-6.5 µg/dose,

14: 3.8-7.2 µg/dose, 18C: 4.6-8.4 µg/dose, 19A: 3.6-6.8 µg/dose, 19F: 3.8-7.2

µg/dose, 23F: 4.2-7.8 µg/dose.

Serotype 1: release standard is ≥ 30%, standard in expiry date is ≥ 25%;

Serotype 3, 6A, 6B, 9V, 14, 18C: release standard is ≥ 80%, standard in expiry

Conjugate antigen date is ≥ 75%;

content Serotype 4, 5, 7F: release standard is ≥ 75%, standard in expiry date is ≥ 70%;

Serotype 19A: release standard is ≥ 70%, standard in expiry date is ≥ 65%;

Type 19F, 23F release standard is ≥ 65%, standard in expiry date is ≥ 60%.

Conjugate protein Release standard shall be not less than 85%, standard in expiry date shall be not content less than 80%

Total protein content 41-150 µg/dose

Abnormal toxicity Should comply with requirements.

Bacterial endotoxin Should be not more than 12.5 EU/dose

Sterility test Should comply with requirements

Osmolarity 220-310 mOsm/Kg

Conclusion:

According to the analysis of stability study test results, it showed that the six batches (three vials and three pre-filled syringe) of 13-valent Pneumococcal Polysaccharide Conjugate Vaccine were stored at 25±2 °C for 9 months accelerated stability study met the 13-valent Pneumococcal Polysaccharide Conjugate Vaccine registration standards (code: YBS00392019) approved by NMPA.

The long-term stability experiment has now completed the 24th month study. The test results showed that six batches (three vials and three pre-filled syringe) of products were stored at 5±3 °C for long-term stability study met the 13-valent Pneumococcal Polysaccharide Conjugate Vaccine registration standards (code: YBS00392019) approved by NMPA.

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2.3.A APPENDICES

2.3.A.1 Facilities and Equipment

Walvax has established the Plant and Facilities Management Control Procedures (document code: WSYX-P-03-0003) specifying the maintenance cycle and scope of the plant and facilities. The engineering supply department conducts a monthly inspection on the periphery of the clean plant to see whether the floor, walls, ceiling, doors and windows of the plant are free of cracks and shedding and are intact; and whether the main bearing structure is overload, corroded and so on. Each manufacturing workshop shall inspect the inside of the clean plant every month to see whether the ground, wall, ceiling, doors and windows of the clean plant are free of cracks and shedding and are intact; and whether the main bearing structure is overload, corroded and so on. During the inspection process, it shall be timely reported and handled in case of abnormalities according to the relevant requirements. After inspection, the inspection records shall be filed monthly to the engineering supply department, so as to know about the situations inside the plant in a timely manner.

Main equipment lists are showed as following:

Table 129 Main Equipment List of Water System

Serial	Device name	model	Manufacturer
number			
1	Purified water Pretreatment and Osmotron 10000 A	Best water technology	
2	machine select (Shanghai) Co., Ltd.		
3	Purified water Pretreatment and Osmotron 10000 B	Best water technology	
4	multi-effect water	Best water technology	
5	MULTITRON MT7500-6 A	distillator (Shanghai) Co., Ltd.	
6	multi-effect water	Best water technology	
7	MULTITRON MT7500-6 B	distillator (Shanghai) Co., Ltd.	
8	Best water technology		
9	Pure steam generator Vapotron 4000 A	(Shanghai) Co., Ltd.	
10	Best water technology		
11	Pure steam generator Vapotron 4000 B	(Shanghai) Co., Ltd.	

Table 130 Main Equipment List of Compressed Air System

Serial	Device name	model	Manufacturer
number			
1	Oil free rotary Atlas Copco	1 ZT37VSD-8.6	
2	Air compressor Airpower/ Antwerp, Belgium		
3	Oil free rotary Atlas Copco	2 ZT37-8.6	
4	Air compressor Airpower/ Antwerp, Belgium		
5	Oil free rotary Atlas Copco	3 ZT37-8.6	
6	Air compressor Airpower/ Antwerp, Belgium		

Table 131 Main Equipment List of Air Conditioning Purification System

Serial	Device name	Equipment model	Manufacturer
number			
1	Air treatment unit JK2-		
2	DMA1918 Yantai Dunham-Bush Industry Co., Ltd.		
3	MGa		

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4	Air treatment unit JK2-		
5	DMA1314 Yantai Dunham-Bush Industry Co., Ltd.		
6	MGb		
7	Air treatment unit JK2-		
8	DMA1822 Yantai Dunham-Bush Industry Co., Ltd.		
9	MGc		
10	Air treatment unit JK2-Ea DMA0910 Yantai Dunham-Bush Industry Co., Ltd.		
11	Air treatment unit JK2-Ec DMA1109 Yantai Dunham-Bush Industry Co., Ltd.		
12	Air treatment unit JK2-		
13	DMA1818 Yantai Dunham-Bush Industry Co., Ltd.		

Table 132 Main Equipment for the Production of Pneumococcal Polysaccharide Conjugate Vaccine

Serial number	Device name	Model	Manufacturer
Carbon dioxide			
1	Thermo Fisher incubator		
2	Biosafety cabinet	BSC-1600IIA2	Suzhou Antai Air Technology Co., Ltd.
	Shanghai Divoxin Bioengineering Co.,		
3	Bioreactor BIO-DVT-3000F		Ltd.
	Shanghai Baidian instrument and		
4	Biochemical incubator LRH-70		Equipment Co., Ltd.
5	Centrifuge AvantiJ-26SXP		Beckman Kurt trading (China) Co., Ltd.
	High speed landing		
6	Lynx4000 Thermo Fisher		
	centrifuge		
	Large capacity		
7	RC 12BP+ Thermo Fisher		
	centrifuge		
8	laminar hood -----	MayAir (China)	
9	Ultrafiltration system UFC10CC	Pol filter (Beijing) Co., Ltd.	
	Serial number	Device name	Model Manufacturer
Carbon dioxide			
1	Thermo Fisher incubator		
2	Biosafety cabinet BSC-1600IIA2		Suzhou Antai Air Technology Co., Ltd.
	Shanghai Divoxin Bioengineering Co.,		
3	Bioreactor BIO-DVT-3000F		Ltd.
4	Biosafety cabinet BSC-1600IIA2		Suzhou Antai Air Technology Co., Ltd.
	Large capacity high		
5	CR 7 Hitachi Corporation		
	speed centrifuge		
	Large capacity high		
6	CR 7 Hitachi Corporation		
	speed centrifuge		
	High speed landing		
7	Lynx4000 Thermo Fisher		
	centrifuge		
	Large capacity		
8	RC 12BP+ Thermo Fisher		
	centrifuge		
	Shanghai Nelly fluid equipment Co.,		
9	Precipitator 3000L		Ltd.
10	Ultrafiltration system UFC10CC	Pol filter (Beijing) Co., Ltd.	
	High speed landing		
11	Lynx4000 Thermo Fisher		
	centrifuge		
	Large capacity		
12	RC 12BP+ Thermo Fisher		
	centrifuge		
	High speed landing		
13	Lynx4000 Thermo Fisher		
	centrifuge		
	Large capacity		
14	RC 12BP+ Thermo Fisher		
	centrifuge		
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15 Ultrafiltration system FS013K05C Pol filter (Beijing) Co., Ltd.

Table 133 Main Equipment for Fill and Finish

Serial number	Device name	Model	Manufacturer
1	Liquid dispensing tank 100L		Nanjing Bojian Technology Co., Ltd.
	Automatic filling and		
2	SV125 Optima Packaging Machinery Co., Ltd.		
	capping machine		
5	Lamp inspector DGG--1		Nanjing Bojian Technology Co., Ltd.
	Colamark(Guangzhou) Labeling		
6	Sticker machine A210N		Equipment Limited
7	Packing line KHP40T		Shanghai Kangsong Packaging Co., Ltd.

Table 134 Main Equipment for QC Analytical

Serial number	Device name	Model	Manufacturer
	Constant temperature and STIK (Shanghai) instrument equipment		
1	CTHI-250B		
	humidity incubator limited		
	Shanghai Keelrein Scientific		
2	Electric heat dryness box DHG-9070A		Instruments Co., Ltd.
	Nanjing Jinheng experimental		
3	Electric heat dryness box HG101-3AS		
	instrument factory		
	Beijing Sedorius instrument system		
4	Electronic balance BS124S		

Co., Ltd.
5 Electronic balance CP225D Sartorius
Beijing Sedorius instrument system
6 Electronic balance BS2202S
Co., Ltd.
Ultraviolet SHIMADZU enterprise management
7 UV-2600
spectrophotometer (China) Co., Ltd.
Electrothermal constant Shanghai Keelrein Scientific
8 DK-600A
temperature water tank Instruments Co., Ltd.
Electrothermal constant
Beijing Changfeng instrument
9 temperature water bath HW.SY21-K
Company
pot
Electrothermal constant Shanghai Yiheng Scientific Instruments
10 DK-600A
temperature water tank Co., Ltd.
AKTA purification
11 AKTA Pure 25 GE Life Science
system
12 Step collector Frac-920 Amersham Biosciences
Electrothermal constant Shanghai Yiheng Scientific Instruments
13 DK-600A
temperature water tank Co., Ltd.
14 Moisture meter AQV-2100SP AQUA COUNTER
15 Osmotic pressure meter Advanced 3250 ADVANCED INSTRUMENTS, INC
High speed refrigerated
16 CR21G Hitachi
centrifuge
Sujing group Suzhou Antai Air
17 Bioclean worktable BCM-1600A
Technology Co., Ltd.
18 Bacteria collector HTY-602A Zhejiang Tailin bioengineering Co.,Ltd.
Sujing group Suzhou Antai Air
19 Biosafety cabinet BSC-1600 II A2
Technology Co., Ltd.
STIK (Shanghai) instrument equipment
20 climate chamber CTHI-250B
limited
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Tianjin Tiandatianfa Technology Co.,
21 Intelligent pyrometer ZRY-2D
Ltd.
22 Constant current pump P-1 Amersham Biosciences
23 Chromatography column XK16/100 Amersham Biosciences
Automatic Kjeldahl
24 Kjelttec 2200 FOSS
nitrogen meter
Temperature controlled TecatorTM
25 FOSS
digester Digestor
SHIMADZU enterprise management
26 Gas chromatograph GC-2014C
(China) Co., Ltd.
Specific protein analysis
27 IMMAGE 800 American Beckman Kurt Co., Ltd.
system
28 pH meter Thermo American Beckman Kurt Co., Ltd.

2.3.A.2 Adventitious Agents Safety Evaluation
Walvax establishes a comprehensive and scientific material management and storage system.
Developed Material Management Control Procedure (Document code: WSYX-P-05-0005). For
animal source materials, in addition to strict quality test, the suppliers should provide proof of
origin or safety certification. This vaccine's production is involved with animal-derived materials
of sheep blood. Detailed information as below:

Table 135 Control Measures for Animal-derived Materials
Name Usage Risk control description
strain recovery of the fist Product quality test, Safety proof documents from
Sheep blood
passage supplier

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Code WSYX-QC-SP-0609-0099 Version 00
Protocol of Product Stability Test Quality Control Laboratory

Protocol of Product Stability Test

Sample Name■Pneumococcal Polysaccharide Derivatives of 13-valent

Pneumococcal Polysaccharide Conjugate Vaccine

Revision History

Version number Revision date Reason for Revision

00 2019.08.23 New Drafting Document

Document Review

Department Position Signature Date

Drafted by Quality Control Lab Staff Ting Huang 2019.08.25

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Code WSYX-QC-SP-0609-0099 Version 00
Protocol of Product Stability Test Quality Control Laboratory

(Deputy)Manager 2019.08.25
Quality Control Lab Pengyang Li

(Deputy)Manager 2019.08.26
Quality Assurance Xiuli Wu

2019.08.26
Reviewed by Quality Management (Deputy)Manager Feng Shen

Head of Quality 2019.08.26
—— Guoliang Fang
Management
2019.08.26
General Manager Jing Shi

s

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Protocol of Product Stability Test Quality Control Laboratory

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I Product Information	
1. Product information (2-8■)	
Serotype Batch No. First Test Date Estimated completion time Note	
9V SP9VC201908001	
2019.08.26 2019.09.30	
1 SP1C201908001	
9V SP9VC201908002	
2019.08.26 2019.09.30	
3 SP3C201908001	
9V SP9VC201908003	
2019.08.28 2019.10.02	
23F SP23FC201908002	
19A SP19AC201908001	

2019.08.30 2019.10.04
 18C SP18CC201908001
 19A SP19AC201908002
 2019.09.02 2019.10.07
 18C SP18CC201908002
 19A SP19AC201909003
 2019.09.04 2019.10.09
 18C SP18CC201909003
 5 SP5C201909001
 2019.09.06 2019.10.11 2-8■
 19F SP19FC201909001
 5 SP5C201909002
 2019.09.09 2019.10.14
 19F SP19FC201909002
 5 SP5C201909003
 2019.09.11 2019.10.16
 19F SP19FC201909003
 6A SP6AC201909001
 2019.09.13 2019.10.18
 1 SP1C201909002
 6A SP6AC201909002
 2019.09.16 2019.10.21
 1 SP1C201909003
 6A SP6AC201909003
 2019.09.18 2019.10.23
 3 SP3C201909002
 23F SP23FC201909003
 2019.09.20 2019.10.25
 3 SP3C201909003

Note: 1. The order in the above table is divided in the order of the expected delivery date of the samples. The specific delivery date is filled in according to the actual delivery; each type of Pneumococcal Polysaccharide

Derivatives will be tested for 3 samples.

2. The test samples were taken from the on-site inspection production batch, including 3 batches of Type 4, Type 6B, Type 7F, Type 14 and Type 23F and 1 batch of Type 23F. These batches were sent by August 25, 2019.

Therefore the five serotype of 13 batches will be used for stability studies with late-production Pneumococcal

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Polysaccharide Derivatives; specific test batches (Batch No.), information such as the date of the first test date, the date of the first submission, the estimated completion time, etc. shall be included in the table in Annex 1.

II Purpose

1. In order to monitor the quality of the 13 serotype of Pneumococcal Polysaccharide Derivatives used in the 13-valent Pneumococcal Polysaccharide Conjugate Vaccine within the validity period, and to determine that it can or is expected to meet the requirements of the quality standard under the labeled storage conditions, Yuxi Walvax conducts the stability test of 113 serotype of Pneumococcal Polysaccharide Derivatives.

III Basis for the stability test

1. Good Manufacturing Practice for Pharmaceutical Products (Amended in 2010).
2. The Pharmacopoeia of the People's Republic of China 2015 Edition Volume II: Stability Testing of Drug Substances and Products (XIX C).
3. Standard Operation Procedure for Continuous Stability Study of Products, code: WSYX-QA-SMP-0009-0030.
4. Quality Standard of 13-valent Pneumococcal Polysaccharide Conjugate Vaccine, code: WSYX-QC-TS-0609-0005.
5. Standard Operation Procedure for Determination of Molecular Size by SEC-MALS Method, code: WSYX-QC-SOP-00-09-0143.

IV Responsibility

1. Quality Assurance Department:

- Provide the Application of Conducting the Stability Study for Products and Plan of Conducting the Stability Study for Products to Quality Assurance Laboratory;
- Review the Protocol of Product Stability Test;
- Review the Report of Product Stability Study;

2. Quality Control Laboratory:

- Draft the Protocol of Product Stability Test;
- QC manager review the Protocol of Product Stability Test;
- Store the test sample under the 2-8■ according to the Protocol of Product Stability Test;
- Sampling at the time point according to the Protocol of Product Stability Test;
- Test the sample and issue the report for results of test;
- Analysis and summarize the product quality according to test results, draft and review the periodical

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and finalized Study Report of Product Stability Test;

3. Production Management Department:

- According to the documents and Protocol of Product Stability Test approved by Quality Assurance Department, issue the manufacture order to related workshop;

4. Bacterial Vaccine Workshop II:

- According to the manufacture order approved by Production Management Department, arrange the manufacture for the samples required in Protocol of Product Stability Test, send the samples to Quality Control Laboratory;

5. Quality Management Department:

-Review the Protocol of Product Stability Test

6. Quality Responsible Person:

-Review and approval the Protocol of Product Stability Test;

-Review and approval the Study Report of Product Stability Test;

V Study design

1. Sample storage condition: 2-8■

2. Test time: 0 day, 10 days, 20 days, 30 days, 35 days;

3. Test items:

The Pneumococcal Polysaccharide Derivatives is liquid form and stored at 2-8■. According to the requirements of product quality standards, the indicators that need to be tested include: total solids, derivation rate, residual cyanide, and molecular size.

The total solids detection result is to provide data support for the combination process. This indicator monitors the weight of the substance contained in each 1ml of the Pneumococcal Polysaccharide Derivatives, which is less affected by the storage time.

The derivation rate is the concentration of the Pneumococcal Polysaccharide Derivatives Adipic acid hydrazide (hereinafter referred to as ADH) divided by the polysaccharide content (glucose measured by the anthrone method). Adipic acid hydrazide (ADH) is chemically stable under normal temperature and pressure. In addition, the derivatization rate is monitored as a ratio of total ADH content to polysaccharide content in the derivative homogenate, and this method does not identify free ADH and Polysaccharide-derived ADH, so it is of little significance to evaluate the stability of derivatives by this index.

The residual amount of cyanide is the residual impurity in the sample and will not increase

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with the increase of the sample storage time.

The molecular size will affect the conjugation of the conjugate: the degradation of the derivatives will reduce the molecular weight of the conjugate, which will affect the quality of the product. Therefore, the molecular size of the derivative should be monitored.

The specific test items are as follows:

Time Test Items

0 day, 10 days, 20 days, 30 days, 35 days Molecular Size by SEC-MALS Method

4. Description of the container closure system: the test samples are packed in the stopper glass bottles.

5. Test method:

Standard Operation Procedure for WSYX-QC-SOP-00-09-0143.

Stability Test Study of

Determination of Molecular Size by

Molecular Size

SEC-MALS Method

VI Amount and batch No. of testing samples

Name Temperature Test Time Amount for Total Amount

single test(batch) (batch)

All Serotypes of 2-8■ 1ml 1+3*4=13ml

0 day

Pneumococcal

Polysaccharide 10 days, 20 days, 30 days, 35 3ml

Derivatives days

VII Evaluation of test results

According to Quality Standard of 13-valent Pneumococcal Polysaccharide Conjugate Vaccine (WSYX-QC-TS-03-009-0002), Yuxi Walvax conducts test for all Serotypes of Pneumococcal Polysaccharide Derivatives placed under 2-8 ■, different periods of time were tested, results were evaluated and Study Report of Product Stability Test was issued.

VIII Annexes

1. Please find the Annex 1 for sample information on the Pneumococcal Polysaccharide Derivatives for 3 batches of Types 4, 6B, 7F, 14 and 1 batch of Type 23F;

2. Please find the Annex-2 for The molecular size (SEC-MALS method) records.

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Annex-1

Additional information on batches of post-test records

The 3 batches of Type 4, Type 6B, Type 7F, Type 14 and Type 23F and 1 batch of Type 23F were sent by August 25, 2019. Therefore, the five serotypes of 13 batches will be used for stability studies with late-production Pneumococcal Polysaccharide Derivatives; specific test batches (Batch No.), information please find the following table:

Serotype Batch No. First Test Date Estimated completion time Note

4

6B

2-8■

7F

14

23F

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Protocol of Product Stability Test Quality Control Laboratory

Protocol of Product Stability Test

Sample Name■Pneumococcal Polysaccharide Derivatives of 13-valent

Pneumococcal Polysaccharide Conjugate Vaccine

(Supplementary)

Revision History

Version number Revision date Reason for Revision

00 2019.08.23 New Drafting Document
This version of the program involves changes to the batch
01 2020.01.09 testing items and the addition of a full testing items at the time
of the final inspection.

Document Review

Department Position Signature Date

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Code WSYX-QC-SP-0609-0099 Version 01
Protocol of Product Stability Test Quality Control Laboratory

Drafted by Quality Control Lab Staff Ting Huang 2020.01.09

(Deputy)Manager 2020.01.09
Quality Control Lab Pengyang Li

(Deputy)Manager 2020.01.09
Quality Assurance Xiuli Wu
Reviewed by
2020.01.09
Quality Management (Deputy)Manager Feng Shen

Head of Quality 2020.01.09
—— Feng Shen
Management

S

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Code WSYX-QC-SP-0609-0099 Version 01
Protocol of Product Stability Test Quality Control Laboratory

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Protocol of Product Stability Test Quality Control Laboratory

I Product Information

1. Product information (2-8■)

Serotype Batch No. First Test Date Estimated completion time Note

23F SP23FD 202001001 2020.01.09 2020.02.13

7F SP7FD 202001001 2020.01.14 2020.02.18

6B SP6BD 202001001 2020.01.14 2020.02.18

4 SP4D 202001001 2020.01.16 2020.02.20

14 SP14D 202001001 2020.01.16 2020.02.20

6B SP6BD 202001002 2020.01.24 2020.02.28

7F SP7FD 202002002 2020.02.06 2020.03.12 2-8■

14 SP14D 202002002 2020.02.10 2020.03.16

7F SP7FD 202002003 2020.02.15 2020.03.21

4 SP4D 202002002 2020.02.24 2020.03.30

6B SP6BD 202002003 2020.02.24 2020.03.30

14 SP14D 202002003 2020.02.27 2020.04.02
4 SP4D 202005003 2020.05.12 2020.06.16

II Purpose

In order to monitor the quality of the 13 serotype of Pneumococcal Polysaccharide Derivatives used in the 13-valent Pneumococcal Polysaccharide Conjugate Vaccine within the validity period, and to determine that it can or is expected to meet the requirements of the quality standard under the labeled storage conditions, Yuxi Walvax conducts the stability test of 113 serotype of Pneumococcal Polysaccharide Derivatives.

According to the version 00 of the protocol, except for 3 batches each of type 4, 6B, 7F, 14 and 1 batch of type 23F pneumococcal polysaccharide derivatives, the stability of the other types of derivatives have been done, therefore the 01 version of protocol is only for the stability of pneumococcal polysaccharide derivatives of 3 batches each of type 4, 6B, 7F, 14 and 1 batch of type 23F, a total of 13 batches of pneumococcal polysaccharide derivatives.

III Basis for the stability test

1. Good Manufacturing Practice for Pharmaceutical Products (Amended in 2010).
2. The Pharmacopoeia of the People's Republic of China 2015 Edition Volume II: Stability Testing of Drug Substances and Products (XIX C).

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Protocol of Product Stability Test Quality Control Laboratory

3. Standard Operation Procedure for Continuous Stability Study of Products, code: WSYX-QA-SMP-0009-0030.
4. Quality Standard of 13-valent Pneumococcal Polysaccharide Conjugate Vaccine, code: WSYX-QC-TS-0609-0005.
5. Standard Operation Procedure for Determination of Molecular Size by SEC-MALS Method, code: WSYX-QC-SOP-00-09-0143.

IV Responsibility

1. Quality Assurance Department:

- Provide the Application of Conducting the Stability Study for Products and Plan of Conducting the Stability Study for Products to Quality Assurance Laboratory;
- Review the Protocol of Product Stability Test;
- Review the Report of Product Stability Study;

2. Quality Control Laboratory:

- Draft the Protocol of Product Stability Test;
- QC manager review the Protocol of Product Stability Test;
- Store the test sample under the 2-8℃ according to the Protocol of Product Stability Test;
- Sampling at the time point according to the Protocol of Product Stability Test;
- Test the sample and issue the report for results of test;
- Analysis and summarize the product quality according to test results, draft and review the periodical and finalized Study Report of Product Stability Test;

3. Production Management Department:

- According to the documents and Protocol of Product Stability Test approved by Quality Assurance Department, issue the manufacture order to related workshop;

4. Bacterial Vaccine Workshop II:

- According to the manufacture order approved by Production Management Department, arrange the manufacture for the samples required in Protocol of Product Stability Test, send the samples to Quality Control Laboratory;

5. Quality Management Department:

- Review the Protocol of Product Stability Test

6. Quality Responsible Person:

- Review and approval the Protocol of Product Stability Test;
- Review and approval the Study Report of Product Stability Test;

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Protocol of Product Stability Test Quality Control Laboratory

V Study design

1. Sample storage condition: 2-8℃
2. Test time: 0 day, 10 days, 20 days, 30 days, 35 days;
3. Test items:

Time Test Items

0 day, 35 days Full test: total solid, derivation rate, residual cyanide content, molecular size determination, molecular size (SEC-MALS method)

10 days, 20 days, 30 days Molecular Size by SEC-MALS Method

4. Description of the container closure system: the test samples are packed in the stopper glass bottles.

5. Test method:

Test Items Standard Criterion

SOP for Determination of

Total solid Total solid: WSYX-QC- ——(mg/ml)

SOP-00-09-0111

Type 1 2.6-4.4 Type 3 5.0-14.0 Type 4 3.0-11.0

SOP for Determination of Type Type

ADH Content: WSYX- Type 5 3.5-18.0 3.0-6.0 1.5-3.4

6A 6B

QC-SOP-00-09-0105 Type Type Type

Derivation 2.9-5.3 3.3-7.4 2.5-5.9

SOP for glucose 7F 9V 14

rate (%)

measurement by anthrone Type Type Type

method: WSYX-QC- 2.0-4.8 3.5-8.0 3.2-6.2

18C 19A 19F

SOP-00-09-0126 Type

1.1-3.7

23F

SOP for Determination of

Residual

Residual Cyanide
cyanide Should no more than 10ng/mg
Content: WSYX-QC-
content
SOP-00-09-0106
Serotype KD value Recovery rate at KD 0.2-0.6
Type 1 $\leq 0.37 \geq 45\%$
Type 3 $\leq 0.45 \geq 60\%$
Type 4 $\leq 0.42 \geq 65\%$
Before KD reached 0.6,
Molecular Type 5 ≤ 0.90
recovery rate $\geq 37\%$
size
SOP for Molecular Size Type 6A $\leq 0.62 \geq 60\%$
determinati
Determination: WSYX- Type 6B $\leq 0.33 \geq 50\%$
on (CL-4B
QC-SOP-00-09-0104
Gel Type 7F $\leq 0.43 \geq 60\%$
filtration) Type 9V $\leq 0.46 \geq 70\%$
Type 14 $\leq 0.40 \geq 65\%$
Type 18C $\leq 0.41 \geq 65\%$
Type 19A $\leq 0.48 \geq 65\%$
Type 19F $\leq 0.56 \geq 70\%$
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Protocol of Product Stability Test Quality Control Laboratory

Type 23F $\leq 0.46 \geq 75\%$

Test Items Standard

SOP for Molecular Size WSYX-QC-SOP-
Study of molecular size
stability methods Determination by SEC-MALS Method 00-09-0974

VI Amount and batch No. of testing samples
Name Temperature Test Time Amount for Total Amount
single test(batch) (batch)
Serotype ____ of 2-8 ■ 13ml 13+1*4=17ml
35 days
Pneumococcal
Polysaccharide 0 day, 10 days, 20 days, 30 1ml
Derivatives days

VII Evaluation of test results
According to Quality Standard of 13-valent Pneumococcal Polysaccharide Conjugate Vaccine
(WSYX-QC-TS-03-009-0002), Yuxi Walvax conducts test for all Serotypes of Pneumococcal
Polysaccharide Derivatives placed under 2-8 ■, different periods of time were tested, results were
evaluated and Study Report of Product Stability Test was issued.

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Protocol of Product Stability Test Quality Control Laboratory

Protocol of Product Stability Test

Sample Name:■Pneumococcal polysaccharide serotype 6B

Batch No.: 6B-PVPV15090201
6B-PVPV15110202
6B-PVPV15120203

Revision History

Version number Revision date Reason for Revision

00 2015.09.07 New Drafting Document

1 / 6
Code WSYX-QC-SPR-0609-0011 Version 00
Protocol of Product Stability Test Quality Control Laboratory

Document Review

Department Position Signature Date

Staff
Drafted by Quality Control Lab Guangmin Na 2015.09.07

(Deputy)Manager 2015.09.07
Quality Control Lab Pengyang Li

(Deputy)Manager 2015.09.08
Quality Assurance Xiuli Wu
Reviewed by
(Deputy)Manager 2015.09.08
Quality Management Xiuli Wu

Head of Quality
—— Guoliang Fang 2015.09.09
Management

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Code WSYX-QC-SPR-0609-0011 Version 00
Protocol of Product Stability Test Quality Control Laboratory

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Protocol of Product Stability Test Quality Control Laboratory

I Product Information
1.Product information■below -20■■■
Name Batch No. First Test Date Estimated completion time
Pneumococcal 6B-PVPV15090201 2015.09.19 2021.03.199
polysaccharide serotype 6B 6B-PVPV15110202 2015.11.24 2021.05.24
6B-PVPV15120203 2015.12.24 2021.06.24

II Purpose
To conduct the stability test of Process Validation Products, Yuxi Walvax draft the Protocol of Product Stability Test according to the Good Manufacturing Practice for Pharmaceutical Products (Amended in 2010) and its implementation guidelines, Technical Guidelines for Biological Product Stability Research (pilot edition) and the Standard Operation Procedure for Continuous Stability Study of Products.

III Basis for the stability text
1. Good Manufacturing Practice for Pharmaceutical Products (Amended in 2010).
2. The Pharmacopoeia of the People's Republic of China 2015 Edition Volume ■: Stability Testing of Drug Substances and Products (9001).
3. Standard Operation Procedure for Continuous Stability Study of Products, code: WSYX-QA-SMP-0009-0030.

4. Quality Standard of 23- valent Pneumococcal Polysaccharide Vaccine, code: WSYX-QC-TS-0609-0004.
5. Quality Standard of 13-valent Pneumococcal Conjugate Vaccine, code: WSYX-QC-TS-0609-0005.

IV Responsibility

1. Quality Control Laboratory:

- Draft the Protocol of Product Stability Test after receiving the Protocol of Continuous Stability Study of Products draft by Quality Assurance Department;
- Taking sample at the time point and store the test sample under the -20℃ according to the Protocol of Product Stability Test;
- Test the sample and issue the report for results of test;

2. Quality Assurance:

- Taking sample according to the Protocol of Product Stability Test;
- Analysis and summarize the product quality according to test results, draft and review the periodical and finalized Study Report of Product Stability Test;

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Code WSYX-QC-SPR-0609-0011 Version 00

Protocol of Product Stability Test Quality Control Laboratory

- Review the Protocol of Product Stability Test;

3. Quality Management Department:

- Approval the Study Report of Product Stability Test;

4. Quality Responsible person:

- Review and approval the Protocol of Product Stability Test;

V Study design

1. Storage condition: below -20℃.

2. Test points: 0th month, 1 year, 2 years, 3 years, 4 years, 4 years and 6 months, 5 years, 5 years and 6 months.

3. Test items

Identification test, total solid content, nucleic acid content, protein content, total nitrogen content, phosphorus content, methylpentose, molecular size, organic solvent (ethanol, ether, acetone) residues, bacterial endotoxin, sodium deoxycholate residues.

4. Description of container closure system: test sample are packaged in glass bottles with rubber plugs for good sealing.

5. Test standard and criterion

Test Items Standard Criterion

Identification test Appear positive response to specific

SOP for Identification Test: WSYX-QC- antiserum type 6B

SOP-0009-0201

Total solid ——

SOP for Total Solid Determination: WSYX-

QC-SOP-0009-0111

SOP for Protein Content Determination: ≤2%

Protein content

WSYX-QC-SOP-0009-0102

SOP for Nucleic Acid Content ≤2%

Nucleic acid content

Determination: WSYX-QC-SOP-0009-0103

Total nitrogen SOP for Total Nitrogen Content 0-2.0%

content Determination: WSYX-QC-SOP-0009-0751

SOP for Phosphorus Content Determination:

Phosphorus content 2.5-5.0%

WSYX-QC-SOP-0009-0101

SOP for Methylpentose Determination:

Methylpentose ≥15%

WSYX-QC-SOP-0009-0127

Sephadex CL-2B gel≤0.50, more

SOP for Molecular size Determination: than 85% of serotype 6B

Molecular size (KD) polysaccharide should be recovered

WSYX-QC-SOP-0009-0104

before the KD value of 0.6 is

reached.

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Protocol of Product Stability Test Quality Control Laboratory

Bacterial SOP for Bacterial Endotoxin Test: WSYX-

≤0.5EU/μg

endotoxin** QC-SOP-0009-0203

Ethanol residual ■0.08%

SOP for Organic Solvent Residue

Acetone residual ■0.03%

Determination: WSYX-QC-SOP-0009-0138

Ether residual ■0.01%

SOP for Sodium Deoxycholate Test:

Sodium deoxycholate ■0.20

WSYX-QC-SOP-0009-0137

VI Amount and batch No. of testing samples

Product name Batch amount Amount for single test Total amount

Pneumococcal 6B-PVPV15090201 0.19g 1.52g
polysaccharide serotype 6B-PVPV15110202 0.19g 1.52g
6B 6B-PVPV15120203 0.19g 1.52g

Code WSYX-QC-SPR-0609-0005 Version 00
Protocol of Product Stability Test Quality Control Laboratory

Protocol of Product Stability Test

Sample Name:■Pneumococcal polysaccharide serotype 1

Batch No.■ 1-PVPV15101201
1-PVPV15121202
1-PVPV16011203

Revision History

Version number Revision date Reason for Revision

00 2015.09.07 New Drafting Document

1 / 6
Code WSYX-QC-SPR-0609-0005 Version 00
Protocol of Product Stability Test Quality Control Laboratory

Document Review

Department Position Signature c

Staff
Drafted by Quality Control Lab Guangmin Na 2015.09.07

(Deputy)Manager 2015.09.07
Quality Control Lab Pengyang Li

(Deputy)Manager 2015.09.07
Quality Assurance Xiuli Wu
Reviewed by
(Deputy)Manager 2015.09.07
Quality Management Xiuli Wu

Head of Quality
—— Guoliang Fang 2015.09.09
Management

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Code WSYX-QC-SPR-0609-0005 Version 00
Protocol of Product Stability Test Quality Control Laboratory

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Code WSYX-QC-SPR-0609-0005 Version 00
Protocol of Product Stability Test Quality Control Laboratory

I Product Information
1.Product information■below -20■■■
Name Batch No. First Test Date Estimated completion
time
Pneumococcal 1-PVPV15101201 2015.10.29 2021.04.29
polysaccharide serotype 1 1-PVPV15121202 2015.12.28 2021.06.28
1-PVPV16011203 2016.01.19 2021.07.19

II Purpose
To conduct the stability test of Process Validation Products, Yuxi Walvax draft the Protocol of Product Stability Test according to the Good Manufacturing Practice for Pharmaceutical Products (Amended in 2010) and its implementation guidelines, Technical Guidelines for Biological Product Stability Research (pilot edition) and the Standard Operation Procedure for Continuous Stability Study of Products.

III Basis for the stability test
1. Good Manufacturing Practice for Pharmaceutical Products (Amended in 2010).
2. The Pharmacopoeia of the People's Republic of China 2015 Edition Volume ■: Stability Testing of Drug Substances and Products (9001).
3. Standard Operation Procedure for Continuous Stability Study of Products, code: WSYX-QA-SMP-0009-0030.

4. Quality Standard of 23- valent Pneumococcal Polysaccharide Vaccine, code: WSYX-QC-TS-0609-0004.
5. Quality Standard of 13-valent Pneumococcal Conjugate Vaccine, code: WSYX-QC-TS-0609-0005.

IV Responsibility

1. Quality Control Laboratory:

- Draft the Protocol of Product Stability Test after receiving the Protocol of Continuous Stability Study of Products draft by Quality Assurance Department;
 - Taking sample at the time point and store the test sample under the -20℃ according to the Protocol of Product Stability Test;
 - Test the sample and issue the report for results of test;
- ##### 2. Quality Assurance:
- Taking sample according to the Protocol of Product Stability Test;

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Code WSYX-QC-SPR-0609-0005 Version 00

Protocol of Product Stability Test Quality Control Laboratory

- Analysis and summarize the product quality according to test results, draft and review the periodical and finalized Study Report of Product Stability Test;
 - Review the Protocol of Product Stability Test;
- ##### 3. Quality Management Department:
- Approval the Study Report of Product Stability Test;
- ##### 4. Quality Responsible person:
- Review and approval the Protocol of Product Stability Test;

V Study design

1. Storage condition: below -20℃.
2. Test points: 0th month, 1 year, 2 years, 3 years, 4 years, 4 years and 6 months, 5 years, 5 years and 6 months.
3. Test items
Identification test, total solid content, nucleic acid content, protein content, total nitrogen content, phosphorus content, uronic acid content, O-acetyl content, molecular size, organic solvent (ethanol, ether, acetone) residues, bacterial endotoxin, sodium deoxycholate residues.
4. Description of container closure system: test sample are packaged in glass bottles with rubber plugs for good sealing.
5. Test standard and criterion
Test Items Standard Criterion
Identification test Appear positive response to specific
SOP for Identification Test: WSYX-QC- antiserum type 1
SOP-0009-0201

Total solid ——

SOP for Total Solid Determination: WSYX-QC-SOP-0009-0111

SOP for Protein Content Determination: ≤2%

Protein content

WSYX-QC-SOP-0009-0102

Nucleic acid content

SOP for Nucleic Acid Content ≤2%

Determination: WSYX-QC-SOP-0009-0103

Total nitrogen SOP for Total Nitrogen Content 3.5-6.0%

content Determination: WSYX-QC-SOP-0009-0751

SOP for Phosphorus Content Determination: 0-1.5%

Phosphorus content

WSYX-QC-SOP-0009-0101

≥45%

SOP for Uronic Acid Content

Uronic acid content Determination:

WSYX-QC-SOP-0009-0129

≥1.8

O-acetyl group SOP for O-acetyl Content Determination:

content WSYX-QC-SOP-0009-0112

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Code WSYX-QC-SPR-0609-0005 Version 00

Protocol of Product Stability Test Quality Control Laboratory

Sepharose CL-4B gels ≤0.15, more

than 75% of serotype 1

SOP for Molecular size Determination:

Molecular size (KD) polysaccharide should be recovered

WSYX-QC-SOP-0009-0104

before the KD value of 0.3 is

reached.

Bacterial SOP for Bacterial Endotoxin Test: WSYX-

≤0.5EU/μg

endotoxin** QC-SOP-0009-0203

Ethanol residual ■0.08%

SOP for Organic Solvent Residue

Acetone residual ■0.03%

Determination: WSYX-QC-SOP-0009-0138

Ether residual ■0.01%

SOP for Sodium Deoxycholate Test:

Sodium deoxycholate ■0.20
WSYX-QC-SOP-0009-0137

VI Amount and batch No. of testing samples
Product name Batch amount Amount for single test Total amount

Pneumococcal 1-PVPV15101201 0.19g 1.52g
polysaccharide serotype 1 1-PVPV15121202 0.19g 1.52g
1-PVPV16011203 0.19g 1.52g

Stability Data of Purified Liquid of Tetanus Toxoid Yuxi Walvax Biotechnology Co.,Ltd.

Stability Data of Purified Liquid of Tetanus Toxoid
1■Batch no. :CTL201909018(Bottle No.1) , Storage Condition: 2-8■
Test items and Protein content Sterility Bacterial endotoxin Purity standard
Shall comply with No more than 10 No less than
No less than 16 mg/ml
Storage time requirements EU/mg 90.0%

0th month 26 Complied Complied 100.0%
1st month 24 ——— 100.0%
2nd month 25 ——— 100.0%
3rd month 25 ——— 100.0%
6th month 25 ——— 100.0%

2■Batch no. :CTL201909019(Bottle No.1) , Storage Condition: 2-8■
Test items and Protein content Sterility Bacterial endotoxin Purity standard
Shall comply with No more than 10 No less than
No less than 16 mg/ml
Storage time requirements EU/mg 90.0%

0th month 24 Complied Complied 99.9%
1st month 24 ——— 100.0%
2nd month 24 ——— 100.0%
3rd month 25 ——— 100.0%
6th month 26 ——— 100.0%

3■Batch no. :CTL201909020(Bottle No.1), Storage Condition: 2-8■
Test items and Protein content Sterility Bacterial endotoxin Purity standard
Shall comply with No more than 10 No less than
No less than 16 mg/ml
Storage time requirements EU/mg 90.0%

0th month 26 Complied Complied 100.0%
1st month 26 ——— 100.0%
2nd month 25 ——— 100.0%
3rd month 25 ——— 96.4%
6th month 27 ——— 97.2%

3 batches of Purified Liquid of Tetanus Toxoid CTL201909018(Bottle No.1), CTL201909019(Bottle No.1), CTL201909020(Bottle No.1) were kept at 2-8■ for 6 months, tested at 0th month, 1st month, 2nd month, 3rd month, 6th month, test results were all complied with specification of Diphtheria, Tetanus and Acellular Pertussis Combined Vaccine, Adsorbed (Code■

Version: July. 2020
Stability Data of Purified Liquid of Tetanus Toxoid Yuxi Walvax Biotechnology Co.,Ltd.

WSYX-QC-TS-00-09-0001)■It shows that the purified liquid of tetanus toxoid produced by the Yuxi Walvax can meet the requirement of 6 months stored at 2-8■.

Version: July. 2020

Code WSYX-QC-SPR-0609-0012 Version 00
Protocol of Product Stability Test Quality Control Laboratory

Protocol of Product Stability Test

Sample Name:■Pneumococcal polysaccharide serotype 7F

Batch No.: 7F-PVPV15100701
7F-PVPV15110702
7F-PVPV16010703

Revision History

Version number Revision date Reason for Revision

00 2015.09.07 New Drafting Document

1 / 6
Code WSYX-QC-SPR-0609-0012 Version 00
Protocol of Product Stability Test Quality Control Laboratory

Document Review

Department Position Signature Date

Staff
Drafted by Quality Control Lab Guangmin Na 2015.09.07

(Deputy)Manager 2015.09.07
Quality Control Lab Pengyang Li

(Deputy)Manager 2015.09.08
Quality Assurance Xiuli Wu
Reviewed by
(Deputy)Manager 2015.09.08
Quality Management Xiuli Wu

Head of Quality
—— Guoliang Fang 2015.09.09
Management

2 / 6
Code WSYX-QC-SPR-0609-0012 Version 00
Protocol of Product Stability Test Quality Control Laboratory

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Code WSYX-QC-SPR-0609-0012 Version 00
Protocol of Product Stability Test Quality Control Laboratory

I Product Information
1.Product information■below -20■■■
Name Batch No. First Test Date Estimated completion
time
Pneumococcal 7F-PVPV15100701 2015.10.09 2021.04.09
polysaccharide serotype 7F 7F-PVPV15110702 2015.11.30 2021.05.30
7F-PVPV16010703 2015.01.13 2021.07.13

II Purpose
To conduct the stability test of Process Validation Products, Yuxi Walvax sdraft the Protocol of Product Stability Test according to the Good Manufacturing Practice for Pharmaceutical Products (Amended in 2010) and its implementation guidelines, Technical Guidelines for Biological Product Stability Research (pilot edition) and the Standard Operation Procedure for Continuous Stability Study of Products.

III Basis for the stability text
1. Good Manufacturing Practice for Pharmaceutical Products (Amended in 2010).
2. The Pharmacopoeia of the People's Republic of China 2015 Edition Volume ■: Stability Testing of Drug Substances and Products (9001).
3. Standard Operation Procedure for Continuous Stability Study of Products, code: WSYX-QA-SMP-0009-0030.

4. Quality Standard of 23- valent Pneumococcal Polysaccharide Vaccine, code: WSYX-QC-TS-0609-0004.
5. Quality Standard of 13-valent Pneumococcal Conjugate Vaccine, code: WSYX-QC-TS-0609-0005.

IV Responsibility

1. Quality Control Laboratory:

- Draft the Protocol of Product Stability Test after receiving the Protocol of Continuous Stability Study of Products draft by Quality Assurance Department;
- Taking sample at the time point and store the test sample under the -20℃ according to the Protocol of Product Stability Test;
- Test the sample and issue the report for results of test;

2. Quality Assurance:

- Taking sample according to the Protocol of Product Stability Test;
- Analysis and summarize the product quality according to test results, draft and review the periodical and finalized Study Report of Product Stability Test;

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Code WSYX-QC-SPR-0609-0012 Version 00

Protocol of Product Stability Test Quality Control Laboratory

- Review the Protocol of Product Stability Test;

3. Quality Management Department:

- Approval the Study Report of Product Stability Test;

4. Quality Responsible person:

- Review and approval the Protocol of Product Stability Test;

V Study design

1. Storage condition: below -20℃.

2. Test points: 0th month, 1 year, 2 years, 3 years, 4 years, 4 years and 6 months, 5 years, 5 years and 6 months.

3. Test items

Identification test, total solid content, nucleic acid content, protein content, total nitrogen content, phosphorus content, methylpentose, molecular size, organic solvent (ethanol, ether, acetone) residues, bacterial endotoxin, sodium deoxycholate residues.

4. Description of container closure system: test sample are packaged in glass bottles with rubber plugs for good sealing.

5. Test standard and criterion

Test Items Standard Criterion

Identification test Appear positive response to specific

SOP for Identification Test: WSYX-QC- antiserum type 7F

SOP-0009-0201

Total solid ——

SOP for Total Solid Determination: WSYX-

QC-SOP-0009-0111

SOP for Protein Content Determination: ≤5%

Protein content

WSYX-QC-SOP-0009-0102

SOP for Nucleic Acid Content ≤2%

Nucleic acid content

Determination: WSYX-QC-SOP-0009-0103

Total nitrogen SOP for Total Nitrogen Content 1.5-4.0%

content Determination: WSYX-QC-SOP-0009-0751

SOP for Phosphorus Content Determination:

Phosphorus content 2.5-5.0%

WSYX-QC-SOP-0009-0101

SOP for Methylpentose Determination:

Methylpentose ≥15%

WSYX-QC-SOP-0009-0128

Sephacrose CL-2B gel≤0.50, more

SOP for Molecular size Determination: than 85% of serotype 6B

Molecular size (KD) polysaccharide should be recovered

WSYX-QC-SOP-0009-0104

before the KD value of 0.6 is

reached.

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Code WSYX-QC-SPR-0609-0012 Version 00

Protocol of Product Stability Test Quality Control Laboratory

Bacterial SOP for Bacterial Endotoxin Test: WSYX-

≤0.5EU/μg

endotoxin** QC-SOP-0009-0203

Ethanol residual ■0.08%

SOP for Organic Solvent Residue

Acetone residual ■0.03%

Determination: WSYX-QC-SOP-0009-0138

Ether residual ■0.01%

SOP for Sodium Deoxycholate Test:

Sodium deoxycholate ■0.20

WSYX-QC-SOP-0009-0137

VI Amount and batch No. of testing samples

Product name Batch amount Amount for single test Total amount

Pneumococcal 6B-PVPV15090201 0.19g 1.52g
polysaccharide serotype 6B-PVPV15110202 0.19g 1.52g
6B 6B-PVPV15120203 0.19g 1.52g

Stability Data of Tetanus Toxoid Bulk Yuxi Walvax Biotechnology Co.,Ltd.

Stability Data of Tetanus Toxoid Bulk
1■Batch no.:201511004 (Bottle No.1) , Storage Condition: 6±2■
Residu
Floccula
Appearanc Toxicity Detoxification al Protein
pH tion Purity Sterility Specific toxicity
e reversal test tests ammon content
units
Test items ium
Observation on
and Observation on
the 7th, 14th and Observation on
standard the 7th, 14th and
21st day showing the 7th, 14th and
21st day
It shall be no local 21st day
showing no
yellow or Shall suppuration and showing no
No less tetanus NMT
brown —— comply necrosis, no tetanus ——
6.6-7.4 than 1500 symptoms, 0.25m
yellow (Lf/ml) with tetanus symptoms, (ug/ml)
Storage Lf/mg weight gain and g/ml
transparen requirement symptoms. weight gain for
time alive for each
t liquid. Weight gain for each guinea pig
guinea pig at
each guinea pig at the end of
the end of the
at the end of the the period.
period.
period.
0th month Complied 7.30 1540 1672.1 Complied Complied Complied Complied 0.033 5689.1
3rd month Complied 7.30 1540 1758.0 —— —— —— —— 0.027 5075.0
Not
6th month Complied 7.28 1260 1649.2 Complied Complied Complied —— detecte 5324.8
d
9th month Complied 7.30 1320 1639.8 —— —— —— —— 0.04 5713.6
12th month Complied 7.32 1400 1605.5 Complied Complied Complied —— 0.013 5256.8
18th month Complied 7.32 1520 1755.2 Complied Complied Complied —— 0.01 5057.2
24th month Complied 7.3 1320 1553 Complied Complied Complied —— 0.09 5983.6
Not
36th month Complied 7.3 1040 1520 Complied Complied Complied —— detecte 5602.9
d
42th
Complied 7.3 1160 1538 Complied Complied Complied —— 0.08 4905.8
month
48th month Complied 6.8 1290 1600 Complied Complied Complied —— 0.01 5785.2

Version: July. 2020 1
Stability Data of Tetanus Toxoid Bulk Yuxi Walvax Biotechnology Co.,Ltd.

2■Batch no.:CTB201606005 (Bottle No.1) , Storage Condition: 6±2■
Residu
Floccula
Appearanc Toxicity Detoxification al Protein
pH tion Purity Sterility Specific toxicity
e reversal test tests ammon content
units
Test items ium
Observation on
and Observation Observation on
the 7th, 14th and
standard on the 7th, the 7th, 14th and
21st day showing
14th and 21st 21st day
It shall be no local
day showing showing no
yellow or suppuration and
No less Shall comply no tetanus tetanus NMT
brown —— necrosis, no ——
6.6-7.4 than 1500 with symptoms, symptoms, 0.25m
yellow (Lf/ml) tetanus (ug/ml)
Storage Lf/mg requirement weight gain weight gain and g/ml
transparen symptoms.
time for each alive for each
t liquid. Weight gain for
guinea pig at guinea pig at
each guinea pig
the end of the the end of the
at the end of the
period. period.
period.
Not
0th month Complied 7.27 800 1720.4 Complied Complied Complied Complied detecte 2983.1
d
3rd month Complied 7.34 860 1676.4 —— —— —— —— 0.027 3379.0
6th month Complied 7.31 800 1581.0 Complied Complied Complied —— 0.020 3430.8

9th month Complied 7.28 800 1687.8 ——— 0.013 3387.8
 12th month Complied 7.32 720 1555.1 Complied Complied Complied ——— 0.07 3467.4
 18th month Complied 7.1 800 1681 Complied Complied Complied ——— 0.07 3378.8
 Not
 24th month Complied 7.2 900 1899 Complied Complied Complied ——— detecte 3301.2
 d
 Not
 36th month Complied 7.0 795 1725 Complied Complied Complied ——— detecte 3277.8
 d
 42th
 Complied 7.0 840 1783 Complied Complied Complied ——— 0 3314.5
 month

Version: July. 2020 2
 Stability Data of Tetanus Toxoid Bulk Yuxi Walvax Biotechnology Co.,Ltd.

3■Batch no.:CTB201610006 (Bottle No.1) , Storage Condition: 6±2■

Residu
 Floccula
 Appearance Toxicity Detoxification al Protein
 pH tion Purity Sterility Specific toxicity
 e reversal test tests ammon content
 units

Test items ium
 Observation on
 and Observation Observation on
 the 7th, 14th and
 standard on the 7th, the 7th, 14th and
 21st day showing

14th and 21st 21st day
 It shall be no local
 day showing showing no
 yellow or suppuration and
 No less Shall comply no tetanus tetanus NMT
 brown ——— necrosis, no ———
 6.6-7.4 than 1500 with symptoms, symptoms, 0.25m
 yellow (Lf/ml) tetanus (ug/ml)
 Storage Lf/mg requirement weight gain weight gain and g/ml
 transparen symptoms.
 time for each alive for each
 t liquid. Weight gain for
 guinea pig at guinea pig at
 each guinea pig
 the end of the the end of the
 at the end of the
 period. period.
 period.

0th month Complied 6.88 800 1698.5 Complied Complied Complied Complied 0.007 3057.3
 3rd month Complied 7.02 800 1834.9 ——— ——— ——— 0.040 3002.2
 6th month Complied 7.11 800 1830.7 Complied Complied Complied ——— 0.030 2974.7
 9th month Complied 7.38 800 1793.7 ——— ——— ——— 0.05 2952.15
 12th month Complied 7.3 920 2000 Complied Complied Complied ——— 0.07 2920.8
 18th month Complied 7.2 900 2059 Complied Complied Complied ——— 0.04 2721.0
 24th month Complied 7.0 840 2005 Complied Complied Complied ——— 0.01 2780.5
 36th month Complied 7.3 660 1568 Complied Complied Complied ——— 0.04 2998.8

Conclusion:
 3 batches of Tetanus Toxoid Bulk 201511004(Bottle No.1), CTB201606005(Bottle No.1), CTB201610006(Bottle
 No.1) were kept at 2-8■ for
 48 months, 42months and 36months respectively■test results were all complied with specification of
 Diphtheria, Tetanus and Acellular Pertussis
 Combined Vaccine, Adsorbed (Code■WSYX-QC-TS-00-09-0001)■It shows that the tetanus toxoid bulk produced by
 the Yuxi Walvax can meet the
 requirement of 36 months stored at 2-8■.

Version: July. 2020 3

Code WSYX-QC-SPR-0609-0009 Version 00
Protocol of Product Stability Test Quality Control Laboratory

Protocol of Product Stability Test

Sample Name:■Pneumococcal polysaccharide serotype 5

Batch No.: 5-PVPV15101301
5-PVPV15121302
5-PVPV16011303

Revision History

Version number Revision date Reason for Revision

00 2015.09.07 New Drafting Document

1 / 6
Code WSYX-QC-SPR-0609-0009 Version 00
Protocol of Product Stability Test Quality Control Laboratory

Document Review

Department Position Signature Date

Staff
Drafted by Quality Control Lab Guangmin Na 2015.09.07

(Deputy)Manager 2015.09.07
Quality Control Lab Pengyang Li

(Deputy)Manager 2015.09.08
Quality Assurance Xiuli Wu
Reviewed by
(Deputy)Manager 2015.09.08
Quality Management Xiuli Wu

Head of Quality
—— Guoliang Fang 2015.09.09
Management

2 / 6
Code WSYX-QC-SPR-0609-0009 Version 00
Protocol of Product Stability Test Quality Control Laboratory

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Code WSYX-QC-SPR-0609-0009 Version 00
Protocol of Product Stability Test Quality Control Laboratory

I Product Information
1.Product information■below -20■■■
Name Batch No. First Test Date Estimated completion time
Pneumococcal 5-PVPV15101301 2015.11.02 2021.05.02
polysaccharide serotype 5 5-PVPV15121302 2016.01.08 2021.07.08
5-PVPV16011303 2016.01.17 2021.07.17

II Purpose
To conduct the stability test of Process Validation Products, Yuxi Walvax draft the Protocol of Product Stability Test according to the Good Manufacturing Practice for Pharmaceutical Products (Amended in 2010) and its implementation guidelines, Technical Guidelines for Biological Product Stability Research (pilot edition) and the Standard Operation Procedure for Continuous Stability Study of Products.

III Basis for the stability text
1. Good Manufacturing Practice for Pharmaceutical Products (Amended in 2010).
2. The Pharmacopoeia of the People's Republic of China 2015 Edition Volume ■: Stability Testing of Drug Substances and Products (9001).
3. Standard Operation Procedure for Continuous Stability Study of Products, code: WSYX-QA-SMP-0009-0030.

4. Quality Standard of 23- valent Pneumococcal Polysaccharide Vaccine, code: WSYX-QC-TS-0609-0004.
5. Quality Standard of 13-valent Pneumococcal Conjugate Vaccine, code: WSYX-QC-TS-0609-0005.

IV Responsibility

1. Quality Control Laboratory:

- Draft the Protocol of Product Stability Test after receiving the Protocol of Continuous Stability Study of Products draft by Quality Assurance Department;
- Taking sample at the time point and store the test sample under the -20℃ according to the Protocol of Product Stability Test;
- Test the sample and issue the report for results of test;

2. Quality Assurance:

- Taking sample according to the Protocol of Product Stability Test;
- Analysis and summarize the product quality according to test results, draft and review the periodical and finalized Study Report of Product Stability Test;

4 / 6

Code WSYX-QC-SPR-0609-0009 Version 00

Protocol of Product Stability Test Quality Control Laboratory

- Review the Protocol of Product Stability Test;

3. Quality Management Department:

- Approval the Study Report of Product Stability Test;

4. Quality Responsible person:

- Review and approval the Protocol of Product Stability Test;

V Study design

1. Storage condition: below -20℃.

2. Test points: 0th month, 1 year, 2 years, 3 years, 4 years, 4 years and 6 months, 5 years, 5 years and 6 months.

3. Test items

Identification test, total solid content, nucleic acid content, protein content, total nitrogen content, phosphorus content, hexosamines, molecular size, organic solvent (ethanol, ether, acetone) residues, bacterial endotoxin, sodium deoxycholate residues.

4. Description of container closure system: test sample are packaged in glass bottles with rubber plugs for good sealing.

5. Test standard and criterion

Test Items Standard Criterion

Identification test Appear positive response to specific

SOP for Identification Test: WSYX-QC- antiserum type 5

SOP-0009-0201

Total solid ——

SOP for Total Solid Determination: WSYX-

QC-SOP-0009-0111

SOP for Protein Content Determination: ≤7.5%

Protein content

WSYX-QC-SOP-0009-0102

SOP for Nucleic Acid Content ≤2%

Nucleic acid content

Determination: WSYX-QC-SOP-0009-0103

Total nitrogen SOP for Total Nitrogen Content 2.5-6.0%

content Determination: WSYX-QC-SOP-0009-0751

SOP for Phosphorus Content Determination: ≤2.0%

Phosphorus content

WSYX-QC-SOP-0009-0101

≥20%

SOP for Hexosamines Determination:

Hexosamines

WSYX-QC-SOP-0009-0127

Sephacrose CL-2B gel≤0.60, more

SOP for Molecular size Determination: than 70% of serotype 5

Molecular size (KD) polysaccharide should be recovered

WSYX-QC-SOP-0009-0104

before the KD value of 0.6 is

reached.

5 / 6

Code WSYX-QC-SPR-0609-0009 Version 00

Protocol of Product Stability Test Quality Control Laboratory

Bacterial SOP for Bacterial Endotoxin Test: WSYX-

≤0.5EU/μg

endotoxin** QC-SOP-0009-0203

Ethanol residual ■0.08%

SOP for Organic Solvent Residue

Acetone residual ■0.03%

Determination: WSYX-QC-SOP-0009-0138

Ether residual ■0.01%

SOP for Sodium Deoxycholate Test:

Sodium deoxycholate ■0.20

WSYX-QC-SOP-0009-0137

VI Amount and batch No. of testing samples

Product name Batch amount Amount for single test Total amount

Pneumococcal 5-PVPV15101301 0.19g 1.52g
polysaccharide serotype 5 5-PVPV15121302 0.19g 1.52g
5-PVPV16011303 0.19g 1.52g

Code WSYX-QC-SPR-0609-0019 Version 00
Protocol of Product Stability Test Quality Control Laboratory

Protocol of Product Stability Test

Sample Name:■Pneumococcal polysaccharide serotype 14

Batch No.: 14-PVPV15101001
14-PVPV15121002
14-PVPV16011003

Revision History

Version number Revision date Reason for Revision

00 2015.09.07 New Drafting Document

1 / 6
Code WSYX-QC-SPR-0609-0019 Version 00
Protocol of Product Stability Test Quality Control Laboratory

Document Review

Department Position Signature Date

Staff
Drafted by Quality Control Lab Guangmin Na 2015.09.07

(Deputy)Manager 2015.09.07
Quality Control Lab Pengyang Li

(Deputy)Manager 2015.09.08
Quality Assurance Xiuli Wu
Reviewed by
(Deputy)Manager 2015.09.08
Quality Management Xiuli Wu

Head of Quality
—— Guoliang Fang 2015.09.09
Management

2 / 6
Code WSYX-QC-SPR-0609-0019 Version 00
Protocol of Product Stability Test Quality Control Laboratory

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3 / 6
Code WSYX-QC-SPR-0609-0019 Version 00
Protocol of Product Stability Test Quality Control Laboratory

I Product Information
1.Product information■below -20■■■
Name Batch No. First Test Date Estimated completion
time
Pneumococcal 14-PVPV15101001 2015.10.21 2021.04.21
polysaccharide serotype 14 14-PVPV15121002 2015.12.30 2021.06.30
14-PVPV16011003 2015.01.11 2021.07.11

II Purpose
To conduct the stability test of Process Validation Products, Yuxi Walvax sdraft the Protocol of Product Stability Test according to the Good Manufacturing Practice for Pharmaceutical Products (Amended in 2010) and its implementation guidelines, Technical Guidelines for Biological Product Stability Research (pilot edition) and the Standard Operation Procedure for Continuous Stability Study of Products.

III Basis for the stability text
1. Good Manufacturing Practice for Pharmaceutical Products (Amended in 2010).
2. The Pharmacopoeia of the People's Republic of China 2015 Edition Volume ■: Stability Testing of Drug Substances and Products (9001).
3. Standard Operation Procedure for Continuous Stability Study of Products, code: WSYX-QA-SMP-0009-0030.

4. Quality Standard of 23- valent Pneumococcal Polysaccharide Vaccine, code: WSYX-QC-TS-0609-0004.
5. Quality Standard of 13-valent Pneumococcal Conjugate Vaccine, code: WSYX-QC-TS-0609-0005.

IV Responsibility

1. Quality Control Laboratory:

- Draft the Protocol of Product Stability Test after receiving the Protocol of Continuous Stability Study of Products draft by Quality Assurance Department;
- Taking sample at the time point and store the test sample under the -20℃ according to the Protocol of Product Stability Test;
- Test the sample and issue the report for results of test;

2. Quality Assurance:

- Taking sample according to the Protocol of Product Stability Test;
- Analysis and summarize the product quality according to test results, draft and review the periodical and finalized Study Report of Product Stability Test;

4 / 6

Code WSYX-QC-SPR-0609-0019 Version 00

Protocol of Product Stability Test Quality Control Laboratory

- Review the Protocol of Product Stability Test;

3. Quality Management Department:

- Approval the Study Report of Product Stability Test;

4. Quality Responsible person:

- Review and approval the Protocol of Product Stability Test;

V Study design

1. Storage condition: below -20℃.

2. Test points: 0th month, 1 year, 2 years, 3 years, 4 years, 4 years and 6 months, 5 years, 5 years and 6 months.

3. Test items

Identification test, total solid content, nucleic acid content, protein content, total nitrogen content, phosphorus content, hexosamines, molecular size, organic solvent (ethanol, ether, acetone) residues, bacterial endotoxin, sodium deoxycholate residues.

4. Description of container closure system: test sample are packaged in glass bottles with rubber plugs for good sealing.

5. Test standard and criterion

Test Items Standard Criterion

Identification test Appear positive response to specific

SOP for Identification Test: WSYX-QC- antiserum type 14

SOP-0009-0201

Total solid ——

SOP for Total Solid Determination: WSYX-

QC-SOP-0009-0111

SOP for Protein Content Determination: ≤5%

Protein content

WSYX-QC-SOP-0009-0102

SOP for Nucleic Acid Content ≤2%

Nucleic acid content

Determination: WSYX-QC-SOP-0009-0103

Total nitrogen SOP for Total Nitrogen Content 1.5-4.0%

content Determination: WSYX-QC-SOP-0009-0751

SOP for Phosphorus Content Determination:

Phosphorus content 0-1.0s%

WSYX-QC-SOP-0009-0101

SOP for Hexosamines Determination:

Hexosamines ≥20%

WSYX-QC-SOP-0009-0127

Sepharose CL-4B gel≤0.30, more

SOP for Molecular size Determination: than 55% of serotype 14

Molecular size (KD) polysaccharide should be recovered

WSYX-QC-SOP-0009-0104

before the KD value of 0.3 is

reached.

5 / 6

Code WSYX-QC-SPR-0609-0019 Version 00

Protocol of Product Stability Test Quality Control Laboratory

SOP for Bacterial Endotoxin Test: WSYX-

Bacterial endotoxin ≤0.5EU/μg

QC-SOP-0009-0203

Ethanol residual ■0.08%

SOP for Organic Solvent Residue

Acetone residual ■0.03%

Determination: WSYX-QC-SOP-0009-0138

Ether residual ■0.01%

SOP for Sodium Deoxycholate Test:

Sodium deoxycholate ■0.20

WSYX-QC-SOP-0009-0137

VI Amount and batch No. of testing samples

Product name Batch amount Amount for single test Total amount

Pneumococcal 14-PVPV15101001 0.19g 1.52g
polysaccharide serotype 14-PVPV15121002 0.19g 1.52g
14 14-PVPV16011003 0.19g 1.52g

Code WSYX-QC-SPR-0609-0007 Version 00
Protocol of Product Stability Test Quality Control Laboratory

Protocol of Product Stability Test

Sample Name:■Pneumococcal polysaccharide serotype 3

Batch No.: 3-PVPV15100901
3-PVPV15110902
3-PVPV15120903

Revision History

Version number Revision date Reason for Revision

00 2015.09.07 New Drafting Document

1 / 6
Code WSYX-QC-SPR-0609-0007 Version 00
Protocol of Product Stability Test Quality Control Laboratory

Document Review

Department Position Signature Date

Staff
Drafted by Quality Control Lab Guangmin Na 2015.09.07

(Deputy)Manager 2015.09.07
Quality Control Lab Pengyang Li

(Deputy)Manager 2015.09.08
Quality Assurance Xiuli Wu
Reviewed by
(Deputy)Manager 2015.09.08
Quality Management Xiuli Wu

Head of Quality
—— Guoliang Fang 2015.09.09
Management

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I Product Information
1.Product information■below -20■■■
Name Batch No. First Test Date Estimated completion time
Pneumococcal 3-PVPV15100901 2015.10.17 2021.04.17
polysaccharide serotype 3 3-PVPV15110902 2015.12.06 2021.06.06
3-PVPV15120903 2016.01.01 2021.07.01

II Purpose
To conduct the stability test of Process Validation Products, Yuxi Walvax draft the Protocol of Product Stability Test according to the Good Manufacturing Practice for Pharmaceutical Products (Amended in 2010) and its implementation guidelines, Technical Guidelines for Biological Product Stability Research (pilot edition) and the Standard Operation Procedure for Continuous Stability Study of Products.

III Basis for the stability text
1. Good Manufacturing Practice for Pharmaceutical Products (Amended in 2010).
2. The Pharmacopoeia of the People's Republic of China 2015 Edition Volume ■: Stability Testing of Drug Substances and Products (9001).
3. Standard Operation Procedure for Continuous Stability Study of Products, code: WSYX-QA-SMP-0009-0030.

4. Quality Standard of 23- valent Pneumococcal Polysaccharide Vaccine, code: WSYX-QC-TS-0609-0004.
5. Quality Standard of 13-valent Pneumococcal Conjugate Vaccine, code: WSYX-QC-TS-0609-0005.

IV Responsibility

1. Quality Control Laboratory:

- Draft the Protocol of Product Stability Test after receiving the Protocol of Continuous Stability Study of Products draft by Quality Assurance Department;
- Taking sample at the time point and store the test sample under the -20■ according to the Protocol of Product Stability Test;
- Test the sample and issue the report for results of test;

2. Quality Assurance:

- Taking sample according to the Protocol of Product Stability Test;
- Analysis and summarize the product quality according to test results, draft and review the periodical and finalized Study Report of Product Stability Test;

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Code WSYX-QC-SPR-0609-0007 Version 00

Protocol of Product Stability Test Quality Control Laboratory

- Review the Protocol of Product Stability Test;

3. Quality Management Department:

- Approval the Study Report of Product Stability Test;

4. Quality Responsible person:

- Review and approval the Protocol of Product Stability Test;

V Study design

1. Storage condition: below -20■.

2. Test points: 0th month, 1 year, 2 years, 3 years, 4 years, 4 years and 6 months, 5 years, 5 years and 6 months.

3. Test items

Identification test, total solid content, nucleic acid content, protein content, total nitrogen content, phosphorus content, uronic acid content, O-acetyl content, molecular size, organic solvent (ethanol, ether, acetone) residues, bacterial endotoxin, sodium deoxycholate residues.

4. Description of container closure system: test sample are packaged in glass bottles with rubber plugs for good sealing.

5. Test standard and criterion

Test Items Standard Criterion

Identification test Appear positive response to specific

SOP for Identification Test: WSYX-QC- antiserum type 3

SOP-0009-0201

Total solid ——

SOP for Total Solid Determination: WSYX-

QC-SOP-0009-0111

SOP for Protein Content Determination: ≤5%

Protein content

WSYX-QC-SOP-0009-0102

SOP for Nucleic Acid Content ≤2%

Nucleic acid content

Determination: WSYX-QC-SOP-0009-0103

Total nitrogen SOP for Total Nitrogen Content 0-1.0%

content Determination: WSYX-QC-SOP-0009-0751

SOP for Phosphorus Content Determination: 0-1.0%

Phosphorus content

WSYX-QC-SOP-0009-0101

≥40%

SOP for Uronic Acid Content

Uronic acid content Determination:

WSYX-QC-SOP-0009-0129

Sepharose CL-4B gel≤0.15, more

SOP for Molecular size Determination: than 78% of serotype 3

Molecular size (KD) polysaccharide should be recovered

WSYX-QC-SOP-0009-0104

before the KD value of 0.3 is

reached.

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Protocol of Product Stability Test Quality Control Laboratory

Bacterial SOP for Bacterial Endotoxin Test: WSYX-

≤0.5EU/μg

endotoxin** QC-SOP-0009-0203

Ethanol residual ■0.08%

SOP for Organic Solvent Residue

Acetone residual ■0.03%

Determination: WSYX-QC-SOP-0009-0138

Ether residual ■0.01%

SOP for Sodium Deoxycholate Test:

Sodium deoxycholate ■0.20

WSYX-QC-SOP-0009-0137

VI Amount and batch No. of testing samples

Product name Batch amount Amount for single test Total amount

Pneumococcal 3-PVPV15100901 0.19g 1.52g

polysaccharide serotype 3 3-PVPV15110902 0.19g 1.52g
3-PVPV15120903 0.19g 1.52g

Code WSYX-QC-SPR-0609-0023 Version 00
Protocol of Product Stability Test Quality Control Laboratory

Protocol of Product Stability Test

Sample Name:■Pneumococcal polysaccharide serotype 19A

Batch No.: 19A-PVPV15090401
19A-PVPV15110402
19A-PVPV15120403

Revision History

Version number Revision date Reason for Revision

00 2015.09.07 New Drafting Document

1 / 6
Code WSYX-QC-SPR-0609-0023 Version 00
Protocol of Product Stability Test Quality Control Laboratory

Document Review

Department Position Signature Date

Staff
Drafted by Quality Control Lab Guangmin Na 2015.09.07

(Deputy)Manager 2015.09.07
Quality Control Lab Pengyang Li

(Deputy)Manager 2015.09.07
Quality Assurance Xiuli Wu
Reviewed by
(Deputy)Manager 2015.09.07
Quality Management Xiuli Wu

Head of Quality
—— Guoliang Fang 2015.09.09
Management

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Code WSYX-QC-SPR-0609-0023 Version 00
Protocol of Product Stability Test Quality Control Laboratory

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Protocol of Product Stability Test Quality Control Laboratory

I Product Information
1.Product information■below -20■■■
Name Batch No. First Test Date Estimated completion
time
Pneumococcal 19A-PVPV15090401 2015.09.27 2021.03.27
polysaccharide serotype 19A-PVPV15110402 2015.11.27 2021.05.27
19A 19A-PVPV15120403 2015.12.26 2021.06.26

II Purpose
To conduct the stability test of Process Validation Products, Yuxi Walvax sdraft the Protocol of Product Stability Test according to the Good Manufacturing Practice for Pharmaceutical Products (Amended in 2010) and its implementation guidelines, Technical Guidelines for Biological Product Stability Research (pilot edition) and the Standard Operation Procedure for Continuous Stability Study of Products.

III Basis for the stability text
1. Good Manufacturing Practice for Pharmaceutical Products (Amended in 2010).
2. The Pharmacopoeia of the People's Republic of China 2015 Edition Volume ■: Stability Testing of Drug Substances and Products (9001).
3. Standard Operation Procedure for Continuous Stability Study of Products, code: WSYX-QA-SMP-0009-0030.

4. Quality Standard of 23- valent Pneumococcal Polysaccharide Vaccine, code: WSYX-QC-TS-0609-0004.
5. Quality Standard of 13-valent Pneumococcal Conjugate Vaccine, code: WSYX-QC-TS-0609-0005.

IV Responsibility

1. Quality Control Laboratory:

- Draft the Protocol of Product Stability Test after receiving the Protocol of Continuous Stability Study of Products draft by Quality Assurance Department;
- Taking sample at the time point and store the test sample under the -20■ according to the Protocol of Product Stability Test;
- Test the sample and issue the report for results of test;

2. Quality Assurance:

- Taking sample according to the Protocol of Product Stability Test;
- Analysis and summarize the product quality according to test results, draft and review the periodical and finalized Study Report of Product Stability Test;

4 / 6

Code WSYX-QC-SPR-0609-0023 Version 00

Protocol of Product Stability Test Quality Control Laboratory

- Review the Protocol of Product Stability Test;

3. Quality Management Department:

- Approval the Study Report of Product Stability Test;

4. Quality Responsible person:

- Review and approval the Protocol of Product Stability Test;

V Study design

1. Storage condition: below -20■.

2. Test points: 0th month, 1 year, 2 years, 3 years, 4 years, 4 years and 6 months, 5 years, 5 years and 6 months.

3. Test items

Identification test, total solid content, nucleic acid content, protein content, total nitrogen content, phosphorus content, hexosamines, methylpentose, molecular size, organic solvent (ethanol, ether, acetone) residues, bacterial endotoxin, sodium deoxycholate residues.

4. Description of container closure system: test sample are packaged in glass bottles with rubber plugs for good sealing.

5. Test standard and criterion

Test Items Standard Criterion

Identification test Appear positive response to specific

SOP for Identification Test: WSYX-QC- antiserum type 19A

SOP-0009-0201

Total solid ——

SOP for Total Solid Determination: WSYX-

QC-SOP-0009-0111

SOP for Protein Content Determination: ≤2%

Protein content

WSYX-QC-SOP-0009-0102

SOP for Nucleic Acid Content ≤2%

Nucleic acid content

Determination: WSYX-QC-SOP-0009-0103

Total nitrogen SOP for Total Nitrogen Content 0.6-3.5%

content Determination: WSYX-QC-SOP-0009-0751

SOP for Phosphorus Content Determination:

Phosphorus content 3.0-7.0%

WSYX-QC-SOP-0009-0101

SOP for Hexosamines Determination:

Hexosamines ≥12%

WSYX-QC-SOP-0009-0127

SOP for Methylpentose Determination:

Methylpentose ≥20%

WSYX-QC-SOP-0009-0128

Sepharose CL-4B gels≤0.45, more

SOP for Molecular size Determination:

Molecular size (KD) than 80% of serotype 19A

WSYX-QC-SOP-0009-0104

polysaccharide should be recovered

5 / 6

Code WSYX-QC-SPR-0609-0023 Version 00

Protocol of Product Stability Test Quality Control Laboratory

before the KD value of 0.5 is reached.

SOP for Bacterial Endotoxin Test: WSYX-

Bacterial endotoxin ≤0.5EU/μg

QC-SOP-0009-0203

Ethanol residual ■0.08%

SOP for Organic Solvent Residue

Acetone residual ■0.03%

Determination: WSYX-QC-SOP-0009-0138

Ether residual ■0.01%

SOP for Sodium Deoxycholate Test:

Sodium deoxycholate ■0.20

VI Amount and batch No. of testing samples
Product name Batch amount Amount for single test Total amount

Pneumococcal 19A-PVPV15090401 0.19g 1.52g
polysaccharide serotype 19A-PVPV15110402 0.19g 1.52g
19A 19A-PVPV15120403 0.19g 1.52g

Code WSYX-QC-SPR-0609-0024 Version 00
Protocol of Product Stability Test Quality Control Laboratory

Protocol of Product Stability Test

Sample Name:■Pneumococcal polysaccharide serotype 19F

Batch No.: 19F-PVPV15090501
19F-PVPV15110502
19F-PVPV15120503

Revision History

Version number Revision date Reason for Revision

00 2015.09.07 New Drafting Document

1 / 6
Code WSYX-QC-SPR-0609-0024 Version 00
Protocol of Product Stability Test Quality Control Laboratory

Document Review

Department Position Signature Date

Staff
Drafted by Quality Control Lab Guangmin Na 2015.09.07

(Deputy)Manager 2015.09.07
Quality Control Lab Pengyang Li

(Deputy)Manager 2015.09.07
Quality Assurance Xiuli Wu
Reviewed by
(Deputy)Manager 2015.09.07
Quality Management Xiuli Wu

Head of Quality
—— Guoliang Fang 2015.09.09
Management

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Code WSYX-QC-SPR-0609-0024 Version 00
Protocol of Product Stability Test Quality Control Laboratory

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Protocol of Product Stability Test Quality Control Laboratory

I Product Information
1.Product information■below -20■■■
Name Batch No. First Test Date Estimated completion
time
Pneumococcal 19F-PVPV15090501 2015.10.01 2021.04.01
polysaccharide serotype 19F-PVPV15110502 2015.11.27 2021.05.27
19F 19F-PVPV15120503 2015.12.26 2021.06.26

II Purpose
To conduct the stability test of Process Validation Products, Yuxi Walvax sdraft the Protocol of Product Stability Test according to the Good Manufacturing Practice for Pharmaceutical Products (Amended in 2010) and its implementation guidelines, Technical Guidelines for Biological Product Stability Research (pilot edition) and the Standard Operation Procedure for Continuous Stability Study of Products.

III Basis for the stability text
1. Good Manufacturing Practice for Pharmaceutical Products (Amended in 2010).
2. The Pharmacopoeia of the People's Republic of China 2015 Edition Volume ■: Stability Testing of Drug Substances and Products (9001).
3. Standard Operation Procedure for Continuous Stability Study of Products, code: WSYX-QA-SMP-0009-0030.

4. Quality Standard of 23- valent Pneumococcal Polysaccharide Vaccine, code: WSYX-QC-TS-0609-0004.
5. Quality Standard of 13-valent Pneumococcal Conjugate Vaccine, code: WSYX-QC-TS-0609-0005.

IV Responsibility

1. Quality Control Laboratory:

- Draft the Protocol of Product Stability Test after receiving the Protocol of Continuous Stability Study of Products draft by Quality Assurance Department;
- Taking sample at the time point and store the test sample under the -20■ according to the Protocol of Product Stability Test;
- Test the sample and issue the report for results of test;

2. Quality Assurance:

- Taking sample according to the Protocol of Product Stability Test;
- Analysis and summarize the product quality according to test results, draft and review the periodical and finalized Study Report of Product Stability Test;

4 / 6

Code WSYX-QC-SPR-0609-0024 Version 00

Protocol of Product Stability Test Quality Control Laboratory

- Review the Protocol of Product Stability Test;

3. Quality Management Department:

- Approval the Study Report of Product Stability Test;

4. Quality Responsible person:

- Review and approval the Protocol of Product Stability Test;

V Study design

1. Storage condition: below -20■.

2. Test points: 0th month, 1 year, 2 year, 3 year, 4 year, 4 year and 6 months, 5 year, 5 year and 6 months.

3. Test items

Identification test, total solid content, nucleic acid content, protein content, total nitrogen content, phosphorus content, hexosamines, methylpentose, molecular size, organic solvent (ethanol, ether, acetone) residues, bacterial endotoxin, sodium deoxycholate residues.

4. Description of container closure system: test sample are packaged in glass bottles with rubber plugs for good sealing.

5. Test standard and criterion

Test Items Standard Criterion

Identification test Appear positive response to specific

SOP for Identification Test: WSYX-QC- antiserum type 19F

SOP-0009-0201

Total solid ——

SOP for Total Solid Determination: WSYX-

QC-SOP-0009-0111

SOP for Protein Content Determination: ≤3%

Protein content

WSYX-QC-SOP-0009-0102

SOP for Nucleic Acid Content ≤2%

Nucleic acid content

Determination: WSYX-QC-SOP-0009-0103

Total nitrogen SOP for Total Nitrogen Content 1.4-3.5%

content Determination: WSYX-QC-SOP-0009-0751

SOP for Phosphorus Content Determination:

Phosphorus content 3.0-5.5%

WSYX-QC-SOP-0009-0101

SOP for Hexosamines Determination:

Hexosamines ≥12.5%

WSYX-QC-SOP-0009-0127

SOP for Methylpentose Determination:

Methylpentose ≥20%

WSYX-QC-SOP-0009-0128

Sepharose CL-4B gels≤0.20, more

SOP for Molecular size Determination:

Molecular size (KD) than 75% of serotype 19F

WSYX-QC-SOP-0009-0104

polysaccharide should be recovered

5 / 6

Code WSYX-QC-SPR-0609-0024 Version 00

Protocol of Product Stability Test Quality Control Laboratory

before the KD value of 0.3 is reached.

SOP for Bacterial Endotoxin Test: WSYX-

Bacterial endotoxin ≤0.5EU/μg

QC-SOP-0009-0203

Ethanol residual ■0.08%

SOP for Organic Solvent Residue

Acetone residual ■0.03%

Determination: WSYX-QC-SOP-0009-0138

Ether residual ■0.01%

SOP for Sodium Deoxycholate Test:

Sodium deoxycholate ■0.20

VI Amount and batch No. of testing samples
Product name Batch amount Amount for single test Total amount

Pneumococcal 19F-PVPV15090501 0.19g 1.52g
polysaccharide serotype 19F-PVPV15110502 0.19g 1.52g
19F 19F-PVPV15120503 0.19g 1.52g

Code WSYX-QC-SPR-0209-0003 Version 00
Protocol of Product Stability Test Quality Control Laboratory

Protocol of Product Stability Test

Sample Name■Purified Liquid of Tetanus Toxoid

Batch No.■CTL201909018■Bottle No.1■
CTL201909019■Bottle No.1■
CTL201909020■Bottle No.1■

Revision History

Version number Revision date Reason for Revision

00 2019.08.06 New Drafting Document

Document Review

Department Position Signature Date

Drafted by Quality Control Lab Staff Ting Huang 2019.08.06

1 / 6
Code WSYX-QC-SPR-0209-0003 Version 00
Protocol of Product Stability Test Quality Control Laboratory

(Deputy)Manager
Quality Control Lab Pengyang Li 2019.08.06

(Deputy)Manager
Quality Assurance Xiuli Wu 2019.08.06
Reviewed by
Quality Management (Deputy)Manager Shen Feng 2019.08.06

Head of Quality
—— Guoliang Fang 2019.08.06
Management

s

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Code WSYX-QC-SPR-0209-0003 Version 00
Protocol of Product Stability Test Quality Control Laboratory

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Code WSYX-QC-SPR-0209-0003 Version 00
Protocol of Product Stability Test Quality Control Laboratory

I Product Information	
1. Product information	
Name Batch No. First Test Date Estimated Note completion time	
Purified Liquid CTL201909018 (Bottle No.1)	2019.09.22 2022.03.22
of Tetanus CTL201909020 (Bottle No.1)	2019.09.24 2022.03.24 2-8■
Toxoid CTL201909019 (Bottle No.1)	2019.10.02 2022.04.02

II Purpose
1. In order to monitor the quality of the tetanus toxoid purified liquid within the validity period, and to determine that it can or is expected to meet the requirements of the quality standards under the labelled storage conditions, Yuxi Walvax is conducting a stability test of the three batches of tetanus toxoid purified liquid .

III Basis for the stability test
1. Good Manufacturing Practice for Pharmaceutical Products (Amended in 2010).
2. The Pharmacopoeia of the People's Republic of China 2015 Edition Volume ■: Stability Testing of Drug Substances and Products (9001).

3. Standard Operation Procedure for Continuous Stability Study of Products, code: WSYX-QA-SMP-00-09-0010.
4. Quality Standard of Purified Liquid of Tetanus Toxoid, code: WSYX-QC-TS-06-09-0002.

IV Responsibility

1. Quality Assurance Department:

- Provide the Application of Conducting the Stability Study for Products and Plan of Conducting the Stability Study for Products to Quality Assurance Laboratory;
- Review the Protocol of Product Stability Test;
- Review the Report of Product Stability Study;
- In January of each year, integrate the results of the continuous stability test of all finished products completed in the previous year, and issue the Review of Product Stability Test;

2. Quality Control Laboratory:

- Draft the Protocol of Product Stability Test;
- QC manager review the Protocol of Product Stability Test;
- Store the test sample under the 2-8℃ according to the Protocol of Product Stability Test;

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Code WSYX-QC-SPR-0209-0003 Version 00

Protocol of Product Stability Test Quality Control Laboratory

- Sampling at the time point according to the Protocol of Product Stability Test;
 - Test the sample and issue the report for results of test;
 - Analysis and summarize the product quality according to test results, draft and review the periodical and finalized Study Report of Product Stability Test;
- ##### 3. Production Management Department:
- According to the documents and Protocol of Product Stability Test approved by Quality Assurance Department, issue the manufacture order to related workshop;
- ##### 4. Bacterial Vaccine Workshop III
- According to manufacture order and Protocol of Product Stability Test, manufacture the required sample;
- ##### 5. Quality Management:
- Review the Protocol of Product Stability Test;
- ##### 6. Quality Responsible person:
- Review and approval the Protocol of Product Stability Test;
 - Review and approval the Study Report of Product Stability Test;

V Study design

1. Sample storage condition: 2-8℃

2. Test time:

0 month, 1 month, 2 months, 3 months, 6 months, 9 months, 12 months, 15 months, 18 months, 24 months, 30 months;

3. Test items

Time Test Items

Full test: purity, sterility, protein content, bacterial

0 month, 12 months, 30 months

endotoxin

1 month, 2 months, 3 months, 6

months, 9 months, 15 months, 18 months, Purity, protein content

months, 24 months

4. Description of container closure system

The test sample are packed in 1ml sterility pre-filling syringe, and the container is well sealed.

5. Test method and criterion

Purified Liquid of Tetanus Toxoid

Test Items Standard Criterion

5 / 6

Code WSYX-QC-SPR-0209-0003 Version 00

Protocol of Product Stability Test Quality Control Laboratory

SOP for Protein Content

Protein content Determination (Lowry) Method $\geq 16\text{mg/ml}$

WSYX-QC-SOP-00-09-0134

Sterility SOP for Sterility Test Should comply with requirements

WSYX-QC-SOP-00-09-0202

SOP for Bacterial Endotoxin Test

Bacterial endotoxin Should be not more than 10EU/mg protein

WSYX-QC-SOP-0009-0203

SOP for Purity Test of Purified Liquid Calculate the purity of the carrier protein

Purity of Tetanus Toxoid according to the Area Normalization Method,

WSYX-QC-SOP-00-09-0143 which should not be less than 90%

VI Amount and batch No. of testing samples

Total amount for

Product Name Batch No. Amount for single test

single batch

CTL201909018 (Bottle

No.1) 0 month, 12 months, 30

48 ml

CTL201909020 (Bottle months

Purified Liquid of

No.1) 1 month, 2 months, 3 months, $48 \times 3 + 4 \times 8 + 48 = 224$

Tetanus Toxoid CTL201909019 (Bottle 6 months, 9 months, 15

No.1) 4ml

months, 18 months, 24

months

Note: Total amount: 48ml package form: 42ml/bottle $\times 1$ (Sterility)

3ml/bottle $\times 1$ (Protein content)

1ml/bottle $\times 2$ (Bacterial endotoxin)

1ml/bottle×1 (Purity)
4ml package form: 3ml/bottle×1 (Protein content)
1ml/bottle×1 (Purity)

VII Evaluation of test results

According to Quality Standard of Purified Liquid of Tetanus Toxoid, code: WSYX-QC-TS-06-09-0002 the samples of tetanus toxoid purified liquid placed at different time points under 2-8■ were tested. The results will be evaluated and a stability study report will be issued.

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Code WSYX-QC-SPR-0609-0010 Version 00
Protocol of Product Stability Test Quality Control Laboratory

Protocol of Product Stability Test

Sample Name:■Pneumococcal polysaccharide serotype 6A

Batch No.: 6A-PVPV15090101
6A-PVPV15110102
6A-PVPV15120103

Revision History

Version number Revision date Reason for Revision

00 2015.09.07 New Drafting Document

1 / 6
Code WSYX-QC-SPR-0609-0010 Version 00
Protocol of Product Stability Test Quality Control Laboratory

Document Review

Department Position Signature Date

Staff
Drafted by Quality Control Lab Guangmin Na 2015.09.07

(Deputy)Manager 2015.09.07
Quality Control Lab Pengyang Li

(Deputy)Manager 2015.09.08
Quality Assurance Xiuli Wu
Reviewed by
(Deputy)Manager 2015.09.08
Quality Management Xiuli Wu

Head of Quality
—— Guoliang Fang 2015.09.09
Management

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Code WSYX-QC-SPR-0609-0010 Version 00
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Code WSYX-QC-SPR-0609-0010 Version 00
Protocol of Product Stability Test Quality Control Laboratory

I Product Information
1.Product information■below -20■■■
Name Batch No. First Test Date Estimated completion time
Pneumococcal 6A-PVPV15090101 2015.09.15 2021.03.15
polysaccharide serotype 6A 6A-PVPV15110102 2015.11.24 2021.05.24
6A-PVPV15120103 2015.12.24 2021.06.24

II Purpose
To conduct the stability test of Process Validation Products, Yuxi Walvax draft the Protocol of Product Stability Test according to the Good Manufacturing Practice for Pharmaceutical Products (Amended in 2010) and its implementation guidelines, Technical Guidelines for Biological Product Stability Research (pilot edition) and the Standard Operation Procedure for Continuous Stability Study of Products.

III Basis for the stability text
1. Good Manufacturing Practice for Pharmaceutical Products (Amended in 2010).
2. The Pharmacopoeia of the People's Republic of China 2015 Edition Volume ■: Stability Testing of Drug Substances and Products (9001).
3. Standard Operation Procedure for Continuous Stability Study of Products, code: WSYX-QA-SMP-0009-0030.

4. Quality Standard of 23- valent Pneumococcal Polysaccharide Vaccine, code: WSYX-QC-TS-0609-0004.
5. Quality Standard of 13-valent Pneumococcal Conjugate Vaccine, code: WSYX-QC-TS-0609-0005.

IV Responsibility

1. Quality Control Laboratory:

- Draft the Protocol of Product Stability Test after receiving the Protocol of Continuous Stability Study of Products draft by Quality Assurance Department;
- Taking sample at the time point and store the test sample under the -20℃ according to the Protocol of Product Stability Test;
- Test the sample and issue the report for results of test;

2. Quality Assurance:

- Taking sample according to the Protocol of Product Stability Test;
- Analysis and summarize the product quality according to test results, draft and review the periodical and finalized Study Report of Product Stability Test;

4 / 6

Code WSYX-QC-SPR-0609-0010 Version 00

Protocol of Product Stability Test Quality Control Laboratory

- Review the Protocol of Product Stability Test;

3. Quality Management Department:

- Approval the Study Report of Product Stability Test;

4. Quality Responsible person:

- Review and approval the Protocol of Product Stability Test;

V Study design

1. Storage condition: below -20℃.

2. Test points: 0th month, 1 year, 2 years, 3 years, 4 years, 4 years and 6 months, 5 years, 5 years and 6 months.

3. Test items

Identification test, total solid content, nucleic acid content, protein content, total nitrogen content, phosphorus content, methylpentose, molecular size, organic solvent (ethanol, ether, acetone) residues, bacterial endotoxin, sodium deoxycholate residues.

4. Description of container closure system: test sample are packaged in glass bottles with rubber plugs for good sealing.

5. Test standard and criterion

Test Items Standard Criterion

Identification test Appear positive response to specific

SOP for Identification Test: WSYX-QC- antiserum type 6A

SOP-0009-0201

Total solid ——

SOP for Total Solid Determination: WSYX-

QC-SOP-0009-0111

SOP for Protein Content Determination: ≤1.5%

Protein content

WSYX-QC-SOP-0009-0102

SOP for Nucleic Acid Content ≤1.5%

Nucleic acid content

Determination: WSYX-QC-SOP-0009-0103

Total nitrogen SOP for Total Nitrogen Content 0-2.0%

content Determination: WSYX-QC-SOP-0009-0751

SOP for Phosphorus Content Determination:

Phosphorus content 2.4-5.0%

WSYX-QC-SOP-0009-0101

SOP for Methylpentose Determination:

Methylpentose ≥15%

WSYX-QC-SOP-0009-0128

Sepharose CL-2B gel≤0.50, more

SOP for Molecular size Determination: than 50% of serotype 6A

Molecular size (KD) polysaccharide should be recovered

WSYX-QC-SOP-0009-0104

before the KD value of 0.6 is

reached.

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Code WSYX-QC-SPR-0609-0010 Version 00

Protocol of Product Stability Test Quality Control Laboratory

Bacterial SOP for Bacterial Endotoxin Test: WSYX-

≤0.5EU/μg

endotoxin** QC-SOP-0009-0203

Ethanol residual ≤0.08%

SOP for Organic Solvent Residue

Acetone residual ≤0.03%

Determination: WSYX-QC-SOP-0009-0138

Ether residual ≤0.01%

SOP for Sodium Deoxycholate Test:

Sodium deoxycholate ≤0.20

WSYX-QC-SOP-0009-0137

VI Amount and batch No. of testing samples

Product name Batch amount Amount for single test Total amount

Pneumococcal 6A-PVPV15090101 0.19g 1.52g
polysaccharide serotype 6A-PVPV15110102 0.19g 1.52g
6A 6A-PVPV15120103 0.19g 1.52g

Code WSYX-QC-SPR-0609-0015 Version 00
Protocol of Product Stability Test Quality Control Laboratory

Protocol of Product Stability Test

Sample Name:■Pneumococcal polysaccharide serotype 9V

Batch No.: 9V-PVPV15090301
9V-PVPV15110302
9V-PVPV15120303

Revision History

Version number Revision date Reason for Revision

00 2015.09.07 New Drafting Document

1 / 6
Code WSYX-QC-SPR-0609-0015 Version 00
Protocol of Product Stability Test Quality Control Laboratory

Document Review

Department Position Signature Date

Staff
Drafted by Quality Control Lab Guangmin Na 2015.09.07

(Deputy)Manager 2015.09.07
Quality Control Lab Pengyang Li

(Deputy)Manager 2015.09.08
Quality Assurance Xiuli Wu
Reviewed by
(Deputy)Manager 2015.09.08
Quality Management Xiuli Wu

Head of Quality
—— Guoliang Fang 2015.09.09
Management

2 / 6
Code WSYX-QC-SPR-0609-0015 Version 00
Protocol of Product Stability Test Quality Control Laboratory

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Code WSYX-QC-SPR-0609-0015 Version 00
Protocol of Product Stability Test Quality Control Laboratory

I Product Information
1.Product information■below -20■■■
Name Batch No. First Test Date Estimated completion
time
Pneumococcal 9V-PVPV15090301 2015.09.23 2021.03.23
polysaccharide serotype 9V 9V-PVPV15110302 2015.11.30 2021.05.30
9V-PVPV15120303 2015.12.28 2021.06.28

II Purpose
To conduct the stability test of Process Validation Products, Yuxi Walvax sdraft the Protocol of Product Stability Test according to the Good Manufacturing Practice for Pharmaceutical Products (Amended in 2010) and its implementation guidelines, Technical Guidelines for Biological Product Stability Research (pilot edition) and the Standard Operation Procedure for Continuous Stability Study of Products.

III Basis for the stability text
1. Good Manufacturing Practice for Pharmaceutical Products (Amended in 2010).
2. The Pharmacopoeia of the People's Republic of China 2015 Edition Volume ■: Stability Testing of Drug Substances and Products (9001).
3. Standard Operation Procedure for Continuous Stability Study of Products, code: WSYX-QA-SMP-0009-0030.

4. Quality Standard of 23- valent Pneumococcal Polysaccharide Vaccine, code: WSYX-QC-TS-0609-0004.
5. Quality Standard of 13-valent Pneumococcal Conjugate Vaccine, code: WSYX-QC-TS-0609-0005.

IV Responsibility

1. Quality Control Laboratory:

- Draft the Protocol of Product Stability Test after receiving the Protocol of Continuous Stability Study of Products draft by Quality Assurance Department;
- Taking sample at the time point and store the test sample under the -20℃ according to the Protocol of Product Stability Test;
- Test the sample and issue the report for results of test;

2. Quality Assurance:

- Taking sample according to the Protocol of Product Stability Test;
- Analysis and summarize the product quality according to test results, draft and review the periodical and finalized Study Report of Product Stability Test;

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Code WSYX-QC-SPR-0609-0015 Version 00

Protocol of Product Stability Test Quality Control Laboratory

- Review the Protocol of Product Stability Test;

3. Quality Management Department:

- Approval the Study Report of Product Stability Test;

4. Quality Responsible person:

- Review and approval the Protocol of Product Stability Test;

V Study design

1. Storage condition: below -20℃.

2. Test points: 0th month, 1 year, 2 years, 3 years, 4 years, 4 years and 6 months, 5 years, 5 years and 6 months.

3. Test items

Identification test, total solid content, nucleic acid content, protein content, total nitrogen content, phosphorus content, uronic acid content, hexosamines, molecular size, organic solvent (ethanol, ether, acetone) residues, bacterial endotoxin, sodium deoxycholate residues.

4. Description of container closure system: test sample are packaged in glass bottles with rubber plugs for good sealing.

5. Test standard and criterion

Test Items Standard Criterion

Identification test Appear positive response to specific

SOP for Identification Test: WSYX-QC- antiserum type 9V

SOP-0009-0201

Total solid ——

SOP for Total Solid Determination: WSYX-

QC-SOP-0009-0111

SOP for Protein Content Determination: ≤2%

Protein content

WSYX-QC-SOP-0009-0102

SOP for Nucleic Acid Content ≤2%

Nucleic acid content

Determination: WSYX-QC-SOP-0009-0103

Total nitrogen SOP for Total Nitrogen Content 0.5-3.0%

content Determination: WSYX-QC-SOP-0009-0751

SOP for Phosphorus Content Determination:

Phosphorus content 0-1.0s%

WSYX-QC-SOP-0009-0101

Uronic acid SOP for Uronic Acid Content

≥15%

content Determination: WSYX-QC-SOP-0009-0129

SOP for Hexosamines Determination:

Hexosamines ≥13%

WSYX-QC-SOP-0009-0127

Sepharose CL-2B gels≤0.45, more

SOP for Molecular size Determination:

Molecular size (KD) than 80% of serotype 9V

WSYX-QC-SOP-0009-0104

polysaccharide should be recovered

5 / 6

Code WSYX-QC-SPR-0609-0015 Version 00

Protocol of Product Stability Test Quality Control Laboratory

before the KD value of 0.6 is reached.

SOP for Bacterial Endotoxin Test: WSYX-

Bacterial endotoxin ≤0.5EU/μg

QC-SOP-0009-0203

Ethanol residual ■0.08%

SOP for Organic Solvent Residue

Acetone residual ■0.03%

Determination: WSYX-QC-SOP-0009-0138

Ether residual ■0.01%

SOP for Sodium Deoxycholate Test:

Sodium deoxycholate ■0.20

VI Amount and batch No. of testing samples
Product name Batch amount Amount for single test Total amount

Pneumococcal 9V-PVPV15090301 0.19g 1.52g
polysaccharide serotype 9V-PVPV15110302 0.19g 1.52g
9V 9V-PVPV15120303 0.19g 1.52g

Code WSYX-QC-SPR-0609-0008 Version 00
Protocol of Product Stability Test Quality Control Laboratory

Protocol of Product Stability Test

Sample Name:■Pneumococcal polysaccharide serotype 4

Batch No.: 4-PVPV15101101
4-PVPV15121102
4-PVPV16011103

Revision History

Version number Revision date Reason for Revision

00 2015.09.07 New Drafting Document

1 / 6
Code WSYX-QC-SPR-0609-0008 Version 00
Protocol of Product Stability Test Quality Control Laboratory

Document Review

Department Position Signature Date

Staff
Drafted by Quality Control Lab Guangmin Na 2015.09.07

(Deputy)Manager 2015.09.07
Quality Control Lab Pengyang Li

(Deputy)Manager 2015.09.08
Quality Assurance Xiuli Wu
Reviewed by
(Deputy)Manager 2015.09.08
Quality Management Xiuli Wu

Head of Quality
—— Guoliang Fang 2015.09.09
Management

2 / 6
Code WSYX-QC-SPR-0609-0008 Version 00
Protocol of Product Stability Test Quality Control Laboratory

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Code WSYX-QC-SPR-0609-0008 Version 00
Protocol of Product Stability Test Quality Control Laboratory

I Product Information
1.Product information■below -20■■■
Name Batch No. First Test Date Estimated completion
time
Pneumococcal 4-PVPV15101101 2015.10.25 2021.04.25
polysaccharide serotype 4 4-PVPV15121102 2016.01.03 2021.07.03
4-PVPV16011103 2016.01.15 2021.07.15

II Purpose
To conduct the stability test of Process Validation Products, Yuxi Walvax draft the Protocol of Product Stability Test according to the Good Manufacturing Practice for Pharmaceutical Products (Amended in 2010) and its implementation guidelines, Technical Guidelines for Biological Product Stability Research (pilot edition) and the Standard Operation Procedure for Continuous Stability Study of Products.

III Basis for the stability text
1. Good Manufacturing Practice for Pharmaceutical Products (Amended in 2010).
2. The Pharmacopoeia of the People's Republic of China 2015 Edition Volume ■: Stability Testing of Drug Substances and Products (9001).
3. Standard Operation Procedure for Continuous Stability Study of Products, code: WSYX-QA-SMP-0009-0030.

4. Quality Standard of 23- valent Pneumococcal Polysaccharide Vaccine, code: WSYX-QC-TS-0609-0004.
5. Quality Standard of 13-valent Pneumococcal Conjugate Vaccine, code: WSYX-QC-TS-0609-0005.

IV Responsibility

1. Quality Control Laboratory:

- Draft the Protocol of Product Stability Test after receiving the Protocol of Continuous Stability Study of Products draft by Quality Assurance Department;
- Taking sample at the time point and store the test sample under the -20℃ according to the Protocol of Product Stability Test;
- Test the sample and issue the report for results of test;

2. Quality Assurance:

- Taking sample according to the Protocol of Product Stability Test;
- Analysis and summarize the product quality according to test results, draft and review the periodical and finalized Study Report of Product Stability Test;

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Code WSYX-QC-SPR-0609-0008 Version 00

Protocol of Product Stability Test Quality Control Laboratory

- Review the Protocol of Product Stability Test;

3. Quality Management Department:

- Approval the Study Report of Product Stability Test;

4. Quality Responsible person:

- Review and approval the Protocol of Product Stability Test;

V Study design

1. Storage condition: below -20℃.

2. Test points: 0th month, 1 year, 2 years, 3 years, 4 years, 4 years and 6 months, 5 years, 5 years and 6 months.

3. Test items

Identification test, total solid content, nucleic acid content, protein content, total nitrogen content, phosphorus content, hexosamines, molecular size, organic solvent (ethanol, ether, acetone) residues, bacterial endotoxin, sodium deoxycholate residues.

4. Description of container closure system: test sample are packaged in glass bottles with rubber plugs for good sealing.

5. Test standard and criterion

Test Items Standard Criterion

Identification test Appear positive response to specific

SOP for Identification Test: WSYX-QC- antiserum type 4

SOP-0009-0201

Total solid ——

SOP for Total Solid Determination: WSYX-

QC-SOP-0009-0111

SOP for Protein Content Determination: ≤3%

Protein content

WSYX-QC-SOP-0009-0102

SOP for Nucleic Acid Content ≤2%

Nucleic acid content

Determination: WSYX-QC-SOP-0009-0103

Total nitrogen SOP for Total Nitrogen Content 4.0-6.0%

content Determination: WSYX-QC-SOP-0009-0751

SOP for Phosphorus Content Determination: 0-1.5%

Phosphorus content

WSYX-QC-SOP-0009-0101

≥40%

SOP for Hexosamines Determination:

Hexosamines

WSYX-QC-SOP-0009-0127

Sepharose CL-4B gel≤0.15, more

SOP for Molecular size Determination: than 75% of serotype 4

Molecular size (KD) polysaccharide should be recovered

WSYX-QC-SOP-0009-0104

before the KD value of 0.3 is

reached.

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Code WSYX-QC-SPR-0609-0008 Version 00

Protocol of Product Stability Test Quality Control Laboratory

Bacterial SOP for Bacterial Endotoxin Test: WSYX-

≤0.5EU/μg

endotoxin** QC-SOP-0009-0203

Ethanol residual ■0.08%

SOP for Organic Solvent Residue

Acetone residual ■0.03%

Determination: WSYX-QC-SOP-0009-0138

Ether residual ■0.01%

SOP for Sodium Deoxycholate Test:

Sodium deoxycholate ■0.20

WSYX-QC-SOP-0009-0137

VI Amount and batch No. of testing samples

Product name Batch amount Amount for single test Total amount

Pneumococcal 4-PVPV15101101 0.19g 1.52g
polysaccharide serotype 4 4-PVPV15121102 0.19g 1.52g
4-PVPV16011103 0.19g 1.52g

Standard operating procedures of Yuxi Walvax Biotechnology Co., Ltd.
WSYX-QC-SOP-00-09-
Coding
0774
SOP for protein nitrogen
content test Version 03 Pages 1/6
Confidential documents, no copying.

1 Purpose
To standardize the determination of protein nitrogen content (Kjeldahl method).
2 Definitions/abbreviations
No
3 Scope of application
Suitable for the determination of protein nitrogen content.
4 Responsibilities
Inspectors strictly follow this procedure.
5 Equipment and materials
5.1 Using the device

Serial number Device name Device model Manufacturers Note
Temperature-
TecatorTM or equivalent
1 controlled FOSS
Digester equipment
digesters
Kai's
nitrogen or equivalent
2 kjltecTM 2200 FOSS
fixed equipment
instrument
5.2 Use of reagents
Serial number Reagent Name Level
1 Digestants No

2 Sulfate AR
5.3 Using solution
Serial number Solution name Category
1 10% trichloroacetic acid solution Solution
2 Mixed indicator solution Indication fluid
Standard operating procedures of Yuxi Walvax Biotechnology Co., Ltd.
WSYX-QC-SOP-00-09-
Coding
0774
SOP for protein nitrogen
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3 2% boric acid absorption solution Solution
4 40% sodium hydroxide solution Solution
Hydrochloric acid titration solution
5 Titrant
(0.1 mol/L)
Hydrochloric acid titrant (0.02
6 Titrant
mol/L)

6 Work procedures
6.1 preparation method of digestive agent: weigh 0.5g copper sulfate and 4.5g potassium sulfate respectively, mix them in a mortar, pour the finely ground mixture into the reagent bottle, label it, and reserve it.
6.2 the first method: trichloroacetic acid precipitation and filtrate determination of non protein nitrogen method
6.2.1 preparation of test solution
6.2.1.1 preparation of total nitrogen determination solution
The appropriate amount of the test sample was accurately diluted with water to 1.0-2.0 mg of nitrogen per 1 ml.
6.2.1.2 preparation of non-protein nitrogen determination solution
Accurately measure 10.0 ml of the test sample, add 10.0 ml of 10% trichloroacetic acid, mix well, stand for 30 minutes, filter, and take the filtrate as the non protein nitrogen test sample.
6.2.2 Digestion
6.2.2.1 accurately measure a certain volume of the test solution and reference substance (equivalent to the nitrogen content of 1.0-2.0 mg), put them into the digestion tube, add about 5 g of digestive agent, add 10ml of sulfuric acid, and digest at 420■ for 40 minutes until the solution is clear and bright green.
6.2.2.2 take 14.0 ml of non-protein nitrogen test sample and 14.0 ml of water (blank control) into digestion tube of Kjeldahl nitrogen analyzer. Make 2 tubes of each test sample in parallel, Standard operating procedures of Yuxi Walvax Biotechnology Co., Ltd.
WSYX-QC-SOP-00-09-
Coding
0774
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Confidential documents, no copying.

add about 5 g of digestive agent, 10 ml of sulfuric acid, and digest at 420■ for 40 minutes until the solution is clear and bright green.
6.2.3 Distillation
Before distillation, steam and wash the distiller for more than 15 minutes. When the receiving liquor is purple red and the cooking solution is reduced to room temperature, take a triangular flask to the right of Kjeldahl nitrogen analyzer, insert the condenser tube and boric

acid absorption solution feeding tube, and then install the digested digestion tube on the evaporator on the left side of the Kjeldahl nitrogen analyzer. Run the program, the instrument starts automatic distillation. After stopping the distillation, remove the triangular flask and use the Rinse the condenser tube with water, collect the eluent in a triangular flask, and prepare for titration.

6.2.4 Titration

Titrate with hydrochloric acid titration solution (0.02 mol/L) until the solution changes from blue-green to gray purple, and the titration results are corrected by blank test.

6.2.5 Result calculation

$$(\text{VX1} - \text{VX0}) \times c \times 14.01 \times \frac{1}{n}$$

Non protein nitrogen content (mg/ml) =

■1

$$(\text{VX2} - \text{VX0}) \times c \times 14.01 \times \frac{1}{n}$$

Total protein nitrogen content (mg/ml) =

■2

Protein nitrogen content (mg / ml) = total nitrogen content - non protein nitrogen content

Non protein nitrogen content of test sample (% , g/ml)

protein nitrogen content $\times 6.25 \times 100$

=

1000

Where: VX1 is the volume of hydrochloric acid titrant consumed by the non protein nitrogen determination solution of the test sample, ml;

VX2 is the volume of hydrochloric acid titrant consumed by the total nitrogen determination solution of the test sample, ml;

V0 is the volume of hydrochloric acid titration solution consumed by blank, ml;

c is the concentration of hydrochloric acid titration solution, mol/L;

Standard operating procedures of Yuxi Walvax Biotechnology Co., Ltd.

WSYX-QC-SOP-00-09-

Coding

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SOP for protein nitrogen

content test Version 03 Pages 4/6

Confidential documents, no copying.

n is the dilution multiple of the test sample;

V1 is the volume of non protein nitrogen test solution, ml;

V2 is the volume of total nitrogen test solution, ml;

14.01 is the relative atomic mass of nitrogen;

6.25 is a constant (1 g nitrogen is equivalent to 6.25 g protein).

6.3 the second method: trichloroacetic acid precipitation to determine protein nitrogen detection method

6.3.1 Preparation of protein nitrogen

Accurately measure the appropriate volume of test sample (protein content of each 1ml is about 4-10 mg) into a suitable sharp bottom centrifuge tube, add equal volume of 10% trichloroacetic acid, and mix well. After standing for 30 minutes, centrifuge at 4000 rpm for 10 minutes, discard the supernatant, and wash the precipitate into the digestion tube of Kjeldahl analyzer with about 3ml water. At the same time, blank control was made.

6.3.2 Digestion

In the blank and the digestion tube of Kjeldahl nitrogen determinator with test sample, add about 5 g digestive agent, 10ml sulfuric acid, and digest at 420■ for 40 minutes until the solution is clear and bright green.

6.3.3 Distillation

Before distillation, steam and wash the distiller for more than 15 minutes. When the receiving liquor is purple red and the cooking solution is reduced to room temperature, take a triangular flask to the right of Kjeldahl nitrogen analyzer, insert the condenser tube and boric acid absorption solution feeding tube, and then install the digested digestion tube on the evaporator on the left side of the Kjeldahl nitrogen analyzer. Run the program, the instrument starts automatic distillation. After stopping the distillation, remove the triangular flask and use the Rinse the condenser tube with water, collect the eluent in a triangular flask, and prepare for titration.

6.3.4 Titration

Standard operating procedures of Yuxi Walvax Biotechnology Co., Ltd.

WSYX-QC-SOP-00-09-

Coding

0774

SOP for protein nitrogen

content test Version 03 Pages 5/6

Confidential documents, no copying.

Titrate with hydrochloric acid titration solution (0.02 mol/L) until the solution changes from blue-green to gray purple, and the titration results are corrected by blank test.

6.3.5 Result calculation

$$(\text{VX} - \text{VX0}) \times c \times 14.01$$

Protein nitrogen content of test sample (mg/ml) =

■

Protein nitrogen content of test sample (% , mg/ml)

protein nitrogen content $\times 6.25 \times 100$

=

1000

Where: VX is the volume of hydrochloric acid titrant consumed by the test sample, ml;

V0 is the volume of hydrochloric acid titration solution consumed by blank, ml;

c is the concentration of hydrochloric acid titration solution, mol/L;

V is the volume of the test solution, ml;

14.01 is the relative atomic mass of nitrogen;

6.25 is a constant (1 g nitrogen is equivalent to 6.25 g protein).

7 Flow chart

No

8 Precautions

8.1 the range of blank 2 should not be greater than 0.05 ml, the blank value is usually 0.02-0.3 ml, and the blank value of 0 ml is normal.

8.2 the distilled ammonia receiving solution should be titrated as soon as possible, so as to avoid

the influence of too long storage time on the determination results.

8.3 if there is no black CuO precipitated in the distillation process, it means that the amount of alkali added is insufficient, and the alkali amount should be supplemented or the experiment should be repeated.

8.4 after adding 2% boric acid absorption solution into the triangular flask, it should be purple red. If it turns green, it indicates that the triangular flask is polluted by alkaline substances.

8.5 the interference of ammonia and alkaline gas should be avoided in the operation

Standard operating procedures of Yuxi Walvax Biotechnology Co., Ltd.

WSYX-QC-SOP-00-09-

Coding

0774

SOP for protein nitrogen

content test Version 03 Pages 6/6

Confidential documents, no copying.

environment.

9 References or standard basis

9.1 Chinese Pharmacopoeia, Volume 3, General rule, 0731 determination of protein content, page 50, 2015

9.2 Current edition of Chinese standard operating procedures for drug inspection

10 Revision history

Basis, reasons and content of

Version Release Effective Revision

the amendment Note

No. Time time Department

00 2014.01.21 2014.01.22 QC 2015 edition of Pharmacopoeia

Laboratory changed

01 2015.11.24 2015.12.01 QC Instead of sodium tungstate

Laboratory precipitation method, add

trichloroacetic acid

precipitation, take the filtrate to

measure non protein nitrogen

method

02 2016.05.31 2016.06.01 QC Standard test record

Laboratory

Code WSYX-QC-SPR-0209-0001 Version 00
Protocol of Product Stability Test Quality Control Laboratory

Protocol of Product Stability Test

Sample Name■Tetanus Toxoid Bulk

Batch No.■201511004■Bottle No.1■
CTB201606005■Bottle No.1■
CTB201610006■Bottle No.1■

Revision History

Version number Revision date Reason for Revision

00 2015.12.31 New Drafting Document

Document Review

Department Position Signature Date

Drafted by Quality Control Lab Staff Guangmin Na 2015.12.31

1 / 6
Code WSYX-QC-SPR-0209-0001 Version 00
Protocol of Product Stability Test Quality Control Laboratory

(Deputy)Manager 2015.12.31
Quality Control Lab Pengyang Li

(Deputy)Manager 2015.12.31
Quality Assurance Xiuli Wu
Reviewed by
2015.12.31
Quality Management (Deputy)Manager Xiuli Wu

Head of Quality 2015.12.31
—— Guoliang Fang
Management

s

2 / 6
Code WSYX-QC-SPR-0209-0001 Version 00
Protocol of Product Stability Test Quality Control Laboratory

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Code WSYX-QC-SPR-0209-0001 Version 00
Protocol of Product Stability Test Quality Control Laboratory

I Product Information
1. Product information (6±2■)
Name Batch No. First Test Date Estimated Note
completion time
Tetanus Toxoid 201511004 (Bottle No.1) 2016.01.04 2020.01.04
Bulk CTB201606005 (Bottle No.1) 2016.08.02 2020.08.02 /
CTB201610006 (Bottle No.1) 2016.12.11 2020.12.11

II Purpose
1. According to the Good Manufacturing Practice for Pharmaceutical Products (Amended in 2010) and Technical Guidelines for Biological Product Stability Research and Standard Operation Procedure for Continuous Stability Study of Products Yuxi Walvax is conducting a stability test study of an intermediate for three consecutive batches of the Diphtheria, Tetanus and Acellular pertussis Combined Vaccine, Adsorbed.

III Basis for the stability test
1. Good Manufacturing Practice for Pharmaceutical Products (Amended in 2010).
2. The Pharmacopoeia of the People's Republic of China 2015 Edition Volume ■: Stability Testing of Drug Substances and Products (9001).
3. Standard Operation Procedure for Continuous Stability Study of Products, code: WSYX-QA-

SMP-0009-0030.

4. Quality Standard of Diphtheria, Tetanus and Acellular pertussis Combined Vaccine, Adsorbed, code: WSYX-QC-TS-0609-0003.

IV Responsibility

1. Quality Control Laboratory:

- Draft the Protocol of Product Stability Test;
- Store the test sample under the $6\pm 2^{\circ}\text{C}$ according to the Protocol of Product Stability Test;
- Sampling at the time point according to the Protocol of Product Stability Test;
- Test the sample and issue the report for results of test;

2. Quality Assurance Department:

- Taking sample according to the Protocol of Product Stability Test;
- Analysis and summarize the product quality according to test results,
- Review the Protocol of Product Stability Test;

3. Quality Management Department:

4 / 6

Code WSYX-QC-SPR-0209-0001 Version 00

Protocol of Product Stability Test Quality Control Laboratory

- Review the Protocol of Product Stability Test;

4. Quality Responsible person:

- Review and approval the Protocol of Product Stability Test;

V Study design

1. Sample storage condition: $6\pm 2^{\circ}\text{C}$

2. Test time:

0 month, 3 months, 6 months, 9 months, 12 months, 18 months, 24 months, 30 months, 36 months, 42 months, 48 months;

3. Test items

Time Test Items

Appearance, pH, flocculent unit, purity, sterility test.

0 month abnormal toxicity, toxicity reverse test, protein

content, residual ammonium, detoxification test

6 months, 12 months, 18 months, 24 Appearance, pH, flocculent unit, purity, sterility test,

months, 36 months, 42 months, 48 abnormal toxicity, toxicity reverse test, protein

months content, residual ammonium

Appearance, pH, flocculent unit, purity, protein

3 months, 9 months

content, residual ammonium

4. Test method and criterion

Tetanus Toxoid Bulk

Test Items Standard Criterion

Should be yellow or brownish yellow

Appearance ——

transparent liquid

SOP for pH Value

pH 6.6-7.4

WSYX-QC-SOP-0009-0110

SOP for Toxoid Flocculent Unit

Flocculent unit Should comply with the requirements.

WSYX-QC-SOP-0209-0002

SOP for Purity Test The antigenic purity is not less than 1500Lf

Purity

WSYX-QC-SOP-0009-0209 per milligram of protein nitrogen.

SOP for Sterility Test

Sterility test Should comply with the requirements.

WSYX-QC-SOP-0009-0202

Observed on the 7th, 14th, and 21st days

after injection. There was no local

SOP for Abnormal Toxicity Test of suppuration or necrosis. Animals should not

Abnormal toxicity Tetanus Toxoid Bulk

WSYX-QC-SOP-0009-0302 have symptoms of tetanus. Those who

expired and the weight of each animal

increased than before injection were eligible.

SOP for Toxicity Reverse Test of The guinea pigs must not have symptoms of

Toxicity reverse

Tetanus Toxoid Bulk tetanus. Each guinea pig whose weight was

test WSYX-QC-SOP-0009-0303 more than that before the injection by the end

5 / 6

Code WSYX-QC-SPR-0209-0001 Version 00

Protocol of Product Stability Test Quality Control Laboratory

of the period was qualified.

Animals should not have symptoms of

SOP for Detoxification Test of

tetanus. Each animal should not lose more

Detoxification test Tetanus Toxoid Bulk

weight than before injection, and the

WSYX-QC-SOP-0009-0304

survivors are eligible.

Residual SOP for Residual Ammonium

Determination $\leq 0.25\text{mg/ml}$

ammonium WSYX-QC-SOP-0009-0754

SOP for Protein Content

Protein content Determination ——

WSYX-QC-SOP-0009-0102

VI Amount and batch No. of testing samples

Total test

Product amount

Batch No. Amount for single test
Name
201511004 (Bottle 0 month 162ml 1292ml
No.1) 6 months,12 months,18 months, 24 142ml×7
months, 36 months, 42 months, 48 months
3 months, 9 months 68ml×2
CTB201606005 0 month 162ml 1292ml
(Bottle No.1)
Tetanus 6 months,12 months,18 months, 24 142ml×7
Toxoid Bulk months, 36 months, 42 months, 48 months
3 months, 9 months 68ml×2
0 month 162ml 1292ml
6 months,12 months,18 months, 24 142ml×7
CTB201610006
(Bottle No.1) months, 36 months, 42 months, 48 months
3 months, 9 months 68ml×2

Code WSYX-QC-SPR-0609-0027 Version 00
Protocol of Product Stability Test Quality Control Laboratory

Protocol of Product Stability Test

Sample Name:■Pneumococcal polysaccharide serotype 23F

Batch No.: 23F-PVPV15090601
23F-PVPV15120602
23F-PVPV16010603

Revision History

Version number Revision date Reason for Revision

00 2015.09.07 New Drafting Document

1 / 6
Code WSYX-QC-SPR-0609-0027 Version 00
Protocol of Product Stability Test Quality Control Laboratory

Document Review

Department Position Signature Date

Staff
Drafted by Quality Control Lab Guangmin Na 2015.09.07

(Deputy)Manager 2015.09.07
Quality Control Lab Pengyang Li

(Deputy)Manager 2015.09.08
Quality Assurance Xiuli Wu
Reviewed by
(Deputy)Manager 2015.09.08
Quality Management Xiuli Wu

Head of Quality
—— Guoliang Fang 2015.09.09
Management

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Code WSYX-QC-SPR-0609-0027 Version 00
Protocol of Product Stability Test Quality Control Laboratory

Content

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Code WSYX-QC-SPR-0609-0027 Version 00
Protocol of Product Stability Test Quality Control Laboratory

I Product Information
1.Product information■below -20■■■
Name Batch No. First Test Date Estimated completion
time
Pneumococcal 23F-PVPV15090601 2015.10.05 2021.04.05
polysaccharide serotype 23F-PVPV15120602 2015.12.15 2021.06.15
23F 23F-PVPV16010603 2016.01.19 2021.07.19

II Purpose
To conduct the stability test of Process Validation Products, Yuxi Walvax sdraft the Protocol of Product Stability Test according to the Good Manufacturing Practice for Pharmaceutical Products (Amended in 2010) and its implementation guidelines, Technical Guidelines for Biological Product Stability Research (pilot edition) and the Standard Operation Procedure for Continuous Stability Study of Products.

III Basis for the stability text
1. Good Manufacturing Practice for Pharmaceutical Products (Amended in 2010).
2. The Pharmacopoeia of the People's Republic of China 2015 Edition Volume ■: Stability Testing of Drug Substances and Products (9001).
3. Standard Operation Procedure for Continuous Stability Study of Products, code: WSYX-QA-SMP-0009-0030.

4. Quality Standard of 23- valent Pneumococcal Polysaccharide Vaccine, code: WSYX-QC-TS-0609-0004.
5. Quality Standard of 13-valent Pneumococcal Conjugate Vaccine, code: WSYX-QC-TS-0609-0005.

IV Responsibility

1. Quality Control Laboratory:

- Draft the Protocol of Product Stability Test after receiving the Protocol of Continuous Stability Study of Products draft by Quality Assurance Department;
- Taking sample at the time point and store the test sample under the -20■ according to the Protocol of Product Stability Test;
- Test the sample and issue the report for results of test;

2. Quality Assurance:

- Taking sample according to the Protocol of Product Stability Test;
- Analysis and summarize the product quality according to test results, draft and review the periodical and finalized Study Report of Product Stability Test;

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Code WSYX-QC-SPR-0609-0027 Version 00

Protocol of Product Stability Test Quality Control Laboratory

- Review the Protocol of Product Stability Test;

3. Quality Management Department:

- Approval the Study Report of Product Stability Test;

4. Quality Responsible person:

- Review and approval the Protocol of Product Stability Test;

V Study design

1. Storage condition: below -20■.

2. Test points: 0th month, 1 year, 2 years, 3 years, 4 years, 4 years and 6 months, 5 years, 5 years and 6 months.

3. Test items

Identification test, total solid content, nucleic acid content, protein content, total nitrogen content, phosphorus content, methylpentose, molecular size, organic solvent (ethanol, ether, acetone) residues, bacterial endotoxin, sodium deoxycholate residues.

4. Description of container closure system: test sample are packaged in glass bottles with rubber plugs for good sealing.

5. Test standard and criterion

Test Items Standard Criterion

Identification test Appear positive response to specific

SOP for Identification Test: WSYX-QC- antiserum type 23F

SOP-0009-0201

Total solid ——

SOP for Total Solid Determination: WSYX-

QC-SOP-0009-0111

SOP for Protein Content Determination: ≤2%

Protein content

WSYX-QC-SOP-0009-0102

SOP for Nucleic Acid Content ≤2%

Nucleic acid content

Determination: WSYX-QC-SOP-0009-0103

Total nitrogen SOP for Total Nitrogen Content 0-1.0%

content Determination: WSYX-QC-SOP-0009-0751

SOP for Phosphorus Content Determination:

Phosphorus content 3.0-4.5%

WSYX-QC-SOP-0009-0101

SOP for Methylpentose Determination:

Methylpentose ≥37%

WSYX-QC-SOP-0009-0128

Sepharose CL-4B gel≤0.15, more

than 85% of serotype 23F

SOP for Molecular size Determination:

Molecular size (KD) polysaccharide should be recovered

WSYX-QC-SOP-0009-0104

before the KD value of 0.3 is

reached.

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Code WSYX-QC-SPR-0609-0027 Version 00

Protocol of Product Stability Test Quality Control Laboratory

SOP for Bacterial Endotoxin Test: WSYX-

Bacterial endotoxin ≤0.5EU/μg

QC-SOP-0009-0203

Ethanol residual ■0.08%

SOP for Organic Solvent Residue

Acetone residual ■0.03%

Determination: WSYX-QC-SOP-0009-0138

Ether residual ■0.01%

SOP for Sodium Deoxycholate Test:

Sodium deoxycholate ■0.20

WSYX-QC-SOP-0009-0137

VI Amount and batch No. of testing samples

Product name Batch amount Amount for single test Total amount

Pneumococcal 23F-PVPV15090601 0.19g 1.52g
polysaccharide serotype 23F-PVPV15120602 0.19g 1.52g
23F 23F-PVPV16010603 0.19g 1.52g

SOP of Yuxi Walvax Biotechnology Co., Ltd.

Code WSYX-QC-SOP-0009-0134

SOP for the protein content 1/6

Version 02 Page

test (Lowry method 2)

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1 Objective

Standardize the operation of protein content test (Lowry method 2).

2 Definitions/Abbreviations

N/A

3 Scope of application

3.1 It is applicable to determine the protein content of acellular pertussis purified liquid, tetanus toxoid bulk and tetanus toxoid purified liquid other test samples manufactured by Yuxi Walvax.

4 Responsibilities

All the operators shall strictly conduct the SOP.

5 Materials and equipment

5.1 Use of equipment

Equipment Equipment code

No. Equipment name Manufacturer Notes

model

U/V WYQCF0444 Shimadzu instrument Or equivalent

1 UV-2600

spectrophotometer WYQCF0445 (Suzhou) Co., Ltd equipment

Or equivalent

2 Electronic balance BS124S WYQCF0036 Sartorius

equipment

Shanghai Qixin

WYQCF0061

Electric constant Scientific Instruments

Or equivalent

3 temperature water DK-600A WYQCF0287 Co., Ltd. / Shanghai

equipment

tank Yiheng Scientific

WYQCF0293

Instruments Co., Ltd.

High speed Thermo Fisher

Multifuge WYQCF0472 Or equivalent

4 freezing Technology (China)

X1R WYQCF0473 equipment

centrifuge Co., Ltd

SOP of Yuxi Walvax Biotechnology Co., Ltd.

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SOP for the protein content 2/6

Version 02 Page

test (Lowry method 2)

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5.2 Use reagent

No. Reagent name Grade

1 Sodium hydroxide AR

2 Anhydrous sodium carbonate AR

3 Disodium tartrate dihydrate AR

4 Copper sulfate pentahydrate AR

5 Sodium deoxycholate Import reagent

6 Trichloroacetic acid AR

7 2N foline-phenol Import reagent

5.3 Use of materials

No. Material name number Material name

1 Pipette 2 Pipette tip

3 Volumetric flask 4 test tube

5 powderless latex glove 6 web

7 Triangular flasks 8 Measuring cylinder

6 Working procedure

6.1 Principle

According to this method, the peptide bond contained in protein molecules is chelated with Cu^{2+} in alkaline solution to form a protein copper complex. The complex reduces the phosphomolybdic acid of phenol reagent to produce blue compound. At the same time, under alkaline conditions, phenol reagent is easily reduced by tyrosine, tryptophan and cysteine in protein to produce blue reaction. In a certain range, the color depth is directly proportional to the protein concentration. The protein content in the test sample was determined by colorimetry with the protein standard solution as the standard curve. Before the determination, deoxycholate trichloroacetic acid was added into the sample, and the interfering substances were removed by protein precipitation. This method can also be used to concentrate protein in dilute solution.

6.2 preparation of reagents

SOP of Yuxi Walvax Biotechnology Co., Ltd.

Code WSYX-QC-SOP-0009-0134

SOP for the protein content 3/6

Version 02 Page

test (Lowry method 2)
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6.2.1 1% sodium hydroxide solution

Weigh 1g sodium hydroxide into a beaker, dissolve it with purified water and transfer it to a 100ml volumetric flask. Add a small amount of purified water to moisten the beaker for three times and transfer it to the volumetric flask. Fix the volume to the scale line and mix well.

6.2.2 5% sodium carbonate solution

Weigh 5g anhydrous sodium carbonate into a beaker, dissolve it with purified water and transfer it to a 100ml volumetric flask. Add a small amount of purified water to moisten the beaker for three times and transfer it to the volumetric flask. Fix the volume to the scale line and mix well.

6.2.3 test solution A

Take 200ml of 1% sodium hydroxide solution and 200ml of 5% sodium carbonate solution into a 500ml volumetric flask, add purified water to the scale line, mix well, label and set aside.

6.2.4 2.98% sodium tartrate dihydrate solution

Weigh 2.98g of disodium tartrate dihydrate into a beaker, dissolve with purified water and transfer to a 100ml volumetric flask. Add a small amount of purified water to moisten the beaker for 3 times and transfer to the volumetric flask. Fix the volume to the scale line, mix well, label and reserve.

6.2.5 1.25% copper sulfate solution

Weigh 1.25g copper sulfate pentahydrate into a beaker, dissolve it with purified water and transfer it to a 100ml volumetric flask. Add a small amount of purified water to moisten the beaker for 3 times and transfer it to the volumetric flask. Fix the volume to the scale line, mix well, label and reserve.

6.2.6 test solution B (ready to use)

Take 2.98% sodium tartrate dihydrate solution, 1.25% copper sulfate solution and purified water according to the volume ratio of 2:2:1, and mix well. (according to the amount needed in the experiment, it is prepared in proportion).

6.2.7 test solution c (ready to use)

Take the test solution a and test solution B according to the volume ratio of 50:1, and mix SOP of Yuxi Walvax Biotechnology Co., Ltd.

Code WSYX-QC-SOP-0009-0134

SOP for the protein content 4/6

Version 02 Page

test (Lowry method 2)

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well. (according to the amount needed in the experiment, it is prepared in proportion).

6.2.8 Folin phenol test solution (ready to use)

Take the stock solution (2mol / L acid concentration) 1 → 2 (i.e. take 1000 μl of Folin phenol into a triangular flask, add purified water to make 2000 μl Folin phenol test solution, and mix well. According to the experimental requirement, the mixture was prepared according to 1 → 2. The prepared Folin phenol test solution should meet the following requirements: 1000 μl of test sample, 5000 μl of test solution and 500 μl of prepared Folin phenol test solution, and the pH value of the solution obtained should be 10.3 ± 0.3).

6.2.9 sodium deoxycholate test solution

Weigh 0.15g sodium deoxycholate into a beaker, dissolve it with purified water and transfer it to a 100ml volumetric flask. Add a small amount of purified water to moisten the beaker for 3 times and transfer it to the volumetric flask. Fix the volume to the scale line, mix well, label and reserve.

6.2.10 72% trichloroacetic acid solution

Weigh 7.2g trichloroacetic acid in a beaker, add purified water to dissolve it into a 10ml volumetric flask, add a small amount of purified water, repeatedly wash the beaker for three times, transfer it to the volumetric flask, and fix the volume to the scale line, mix well, then get the product, label it for use.

6.2.11 200 μg / ml protein national standard solution

Please refer to wsyx-qc-sop-0009-0801 for preparation details

6.3 test procedure

6.3.1 treatment of test articles

A pipette with a range of 100-1000 μl was used to accurately measure 1000 μl of the test sample (containing about 20-80 μg of protein) into the test tube in duplicate.

6.3.2 preparation of standard curve

0 1 2 3 4 5

200 μg / ml protein national standard solution

0 100 200 300 400 500

(μl)

SOP of Yuxi Walvax Biotechnology Co., Ltd.

Code WSYX-QC-SOP-0009-0134

SOP for the protein content 5/6

Version 02 Page

test (Lowry method 2)

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Purified water (μl) 1000 900 800 700 600 500

Protein content (μg/ml) 0 20 40 60 80 100

6.3.3 add 100 μL sodium deoxycholate test solution into the above tubes, mix well, and place at room temperature for 10min.

6.3.4 add 100 μl of 72% trichloroacetic acid solution into the above tubes, mix well, centrifugate at 3000g for 30min, gently pour out the supernatant, remove the remaining liquid with a pipette, and retain the protein precipitate.

6.3.5 protein precipitate: dissolve each tube with C 1000 μl of test solution, then add 5000 μl of test solution, and mix well.

6.3.6 place the above tubes in the electric constant temperature water bath at 25■ for 10 minutes, add 500 μl of Folin phenol test solution, mix one tube immediately after adding one tube, and place the electric heating constant temperature water bath at 25■ for 30 minutes.

6.3.7 read the absorbance at 750 nm.

6.4 result calculation

6.4.1 take standard 0 tube as blank, calculate the linear regression equation with the

concentration of standard solution and its corresponding absorbance, and then substitute the absorbance of the test solution into the linear regression equation to calculate the protein concentration of the test solution.

Protein concentration of test sample = average protein concentration of test sample × dilution multiple

6.4.2 Results judgment

Carry out according to corresponding quality standard.

7 Flow chart

No

8 Matters need attention

8.1 Temperature and time have influence on protein content detection and should be strictly controlled.

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SOP for the protein content 6/6

Version 02 Page

test (Lowry method 2)

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8.2 avoid centrifugal precipitation when taking the supernatant.

9 Reference or standard basis

9.1 Pharmacopoeia of People's Republic of China, volume 3, 2015. General rule 0731.

10 Revision history

Issuing Time in Revision Revision basis, reason and

Version Note

time force Dept. content

QC Complete some parts of the

00 2016.08.12 2016.08.13

laboratory content

QC Complete some parts of the

01 2017.09.30 2017.10.01

laboratory content

Code WSYX-QC-SPR-0609-0055 Version 02
Protocol of Product Stability Test Quality Control Laboratory

Protocol of Product Stability Test

Sample Name■13-valent Pneumococcal Conjugate Vaccine

Dosage Form: Injections

Batch No.■GK201709001
GK201709002
GK201709003

Package Form■1ml Pre-filling syringe■BD■

Revision History

Version number Revision date Reason for Revision

00 2017.08.28 New Drafting Document

01 2017.11.15 Terminate the 40■-stability test
According to the Quality Standards for 13-valent
Pneumococcal Polysaccharide Conjugate Vaccine, version 04,
02 2019.07.04
WSYX-QC-TS-06-09-0002, adjust the test items at some time
points in the protocol

Document Review

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Code WSYX-QC-SPR-0609-0055 Version 02
Protocol of Product Stability Test Quality Control Laboratory

Department Position Signature Date

Drafted by Quality Control Lab Staff Ting Huang 2019.07.04

(Deputy)Manager
Quality Control Lab Pengyang Li 2019.07.04

(Deputy)Manager
Animal laboratory Yongqing Yan 2019.07.04

(Deputy)Manager
Reviewed by Quality Assurance Xiuli Wu 2019.07.04

Quality Management (Deputy)Manager Shen Feng 2019.07.09

Head of Quality
—— Guoliang Fang 2019.07.09
Management

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Code WSYX-QC-SPR-0609-0055 Version 02
Protocol of Product Stability Test Quality Control Laboratory

Content

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Code WSYX-QC-SPR-0609-0055 Version 02
Protocol of Product Stability Test Quality Control Laboratory

I Product Information
1. Product information
Name Batch No. First Test Date Estimated Note
completion time

13-valent GK201709001 2017.09.02 2021.03.02 5±3■
Pneumococcal 2018.03.02 25±2■
Conjugate Vaccine GK201709002 2017.09.06 2021.03.06 5±3■
2018.03.06 25±2■
GK201709003 2017.09.17 2021.03.17 5±3■
2018.03.17 25±2■

Note: According to the version 00 and 01 of the protocol (WSYX-QC-SPR-0609-0055), the acceleration stability test under 25■ has been completed. Therefore, the version 02 will not conduct the acceleration stability test under 25■, only remain this part of acceleration stability test under 25■ to ensure the integrity of the protocol.

II Purpose

1. According to the Good Manufacturing Practice for Pharmaceutical Products (Amended in 2010), Technical Guidelines for Biological Product Stability Research, The Pharmacopoeia of the People's Republic of China 2015 Edition, ICH Q1A Stability Testing of New Drug Substances and Productions, ICH Q5C Stability Testing of Biotechnological/Biological Products and Standard Operation Procedure for Continuous Stability Study of Products, Yuxi Walvax conducts stability studies on 3 batches of 13-valent pneumococcal polysaccharide conjugate vaccines for vaccines to provide a basis for the storage validity period.

2. According to the Manufacturing and Test Procedure of 13-valent Pneumococcal Polysaccharide Conjugate Vaccine effective on July 03, 2019: and Quality Standard of 13-valent Pneumococcal Conjugate Vaccine, code: WSYX-QC-TS-06-09-0002, Yuxi Walvax adjusted the test items of the 13-valent pneumococcal polysaccharide conjugate vaccine. The detection of the adsorption rate will be cancelled, and the detection of the content of conjugate protein and total protein will be added, so the content of the protocol needs to be modified.

III Basis for the stability test

1. Good Manufacturing Practice for Pharmaceutical Products (Amended in 2010).
2. The Pharmacopoeia of the People's Republic of China 2015 Edition Volume ■: Stability Testing of Drug Substances and Products (9001).

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Code WSYX-QC-SPR-0609-0055 Version 02
Protocol of Product Stability Test Quality Control Laboratory

3. Technical Guiding Principles for Compatibility Research of Chemical Injection and Pharmaceutical Glass Packaging Containers (Trial)
4. ICH Q1A Stability Testing of New Drug Substances and Productions
5. ICH Q5C Stability Testing of Biotechnological/Biological Products
6. Standard Operation Procedure for Continuous Stability Study of Products, code: WSYX-QA-SMP-0009-0030.
7. Quality Standard of 13-valent Pneumococcal Conjugate Vaccine, code: WSYX-QC-TS-06-09-0002.

IV Responsibility

1. Quality Assurance Department:
- Provide the Application of Conducting the Stability Study for Products and Plan of Conducting the Stability Study for Products to Quality Assurance Laboratory;
- Review the Protocol of Product Stability Test;
- Review the Report of Product Stability Study;
- In January of each year, integrate the results of the continuous stability test of all finished products completed in the previous year, and issue the Review of Product Stability Test;

2. Quality Control Laboratory:
- Draft the Protocol of Product Stability Test;
- QC manager review the Protocol of Product Stability Test;
- Store the test sample under the 5±3■, 25■ according to the Protocol of Product Stability Test;
- Sampling at the time point according to the Protocol of Product Stability Test;
- Test the sample and issue the report for results of test;
- Analysis and summarize the product quality according to test results, draft and review the periodical and finalized Study Report of Product Stability Test;

3. Animal Laboratory
- Review the Protocol of Product Stability Test;
- Test the sample and report the results of test to Quality Control Laboratory;
- Review the Report of Product Stability Study;

4. Production Management Department:
- According to the documents and Protocol of Product Stability Test approved by Quality Assurance Department, issue the manufacture order to related workshop;

5. Fill & Finish Center

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Code WSYX-QC-SPR-0609-0055 Version 02
Protocol of Product Stability Test Quality Control Laboratory

- According to manufacture order and Protocol of Product Stability Test, manufacture the required sample;

6. Quality Supervision Department
- According to Protocol of Product Stability Test, send the required sample amount to Quality Control Laboratory;

7. Quality Management:
- Review the Protocol of Product Stability Test;

8. Quality Responsible person:
- Review and approval the Protocol of Product Stability Test;
- Review and approval the Study Report of Product Stability Test;

V Study design

1. Package information
Pre-filling syringe
Manufacturer: BD

2. Sample storage condition: 5±3■, 25±2■

3. Sample placement method: Horizontal

4. Test time:

5±3■ continuous stability: 0 months, 3 months, 6 months, 9 months, 12 months, 18 months, 24 months, 30 months;

25±2■ accelerated stability: 0 months, 1 month, 2 months, 3 months, 6 months.

Note: a. The expected purpose of this stability test is to investigate under the above conditions until the product expires, that is, to the time point when there is at least one inconsistency in the product quality parameter. However, due to the limitation of the planned sample volume, if the continuous stability under 5±3■, the parameters still meet the requirements after 30 months, it needs to be delayed to 36 months, 42 months, and 48 months; Accelerated stability need to postpone 9 months after 6 months;

b. As of July 03, 2019, the continuous stability of the 3 batches for 13-valent pneumococcal polysaccharide conjugate vaccines involved in the protocol at 2-8°C has been investigated to 18 months, so this version of the protocol is only for 24 months and after the inspection time point (The test protocol at each time point before 24 months is subject to the

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Code WSYX-QC-SPR-0609-0055 Version 02

Protocol of Product Stability Test Quality Control Laboratory

version 01 protocol).

5. Test items

Items Appearance, extractable volume, Polysaccharide content, conjugate pH, aluminum phosphate content, Time antigen content, conjugate protein sodium chloride content, Sterility content, total protein content osmolality, abnormal toxicity, bacterial endotoxin

24 months Test■Y N Y N Y N

30 months Test■Y N Y N Y N

36 months Test■Y N

42 months Test■Y N Refer to the note

48 months Test■Y N

Note: For the 36 months, 42 months, and 48 months test, besides the above test items, other test items conduct as per following requirements:

a. The above test items, under the condition that the three time points are qualified, then the other test

items should conduct full test at 48 months.

b. The above test items, no matter at which time point there is a unqualified, it request conducting full test for all other items at that time point; and stopping the observation at subsequent time points.

6. Description of container closure system

The test sample are packed in 1ml sterility pre-filling syringe, and the container is well sealed.

6. Test method and criterion

13-valent Pneumococcal Conjugate Vaccine

Test Items Standard Criterion

SOP for Appearance check Should be milky uniform suspension after

Appearance

WSYX-QC-SOP-0009-0108 shaking

SOP for the extractable volume test

Extractable volume Should be not less than 0.5ml/vial or syringe

WSYX-QC-SOP-0009-0109

SOP for pH Value test

pH 5.5-7.5

WSYX-QC-SOP-0009-0110

SOP for Aluminum hydroxide (or

Aluminum

Aluminum phosphate) content test 0.8-1.2g/L

phosphate content

WSYX-QC-SOP-0009-0117

SOP for sodium chloride content

Sodium chloride

determination 7.5-9.5g/L

content

WSYX-QC-SOP-0009-0120

SOP Osmolality test

Osmolality 220-310mOsm/Kg

WSYX-QC-SOP-0009-0135

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Code WSYX-QC-SPR-0609-0055 Version 02

Protocol of Product Stability Test Quality Control Laboratory

1: 1.8-3.4μg/dose, 3: 1.75-3.25μg/dose, 4:

2.1-3.9μg/dose, 5: 1.75-3.25μg/dose, 6A:

SOP of Rate nephelometry method for 1.75-3.25μg/dose, 6B: 4.2-7.8μg/dose, 7F: 2-

Polysaccharide

detection of polysaccharide content 3.7μg/dose, 9V: 1.75-3.25μg/dose, 14: 1.9-

content

WSYX-QC-SOP-0009-0414 3.6μg/dose, 18C: 2.3-4.2μg/dose, 19A: 1.8-

3.4μg/dose, 19F: 1.9-3.6μg/dose, 23F: 2.1-

3.9μg/dose.

Type 1: release standard is ≥30%, standard

in expiry date is ≥25%;

Type 3, 6A, 6B, 9V, 14, 18C: release standard

is ≥80%, standard in expiry date is ≥75%;

Conjugate antigen SOP for Conjugate antigen content Type 4, 5, 7F: release standard is ≥75%,

determination

content standard in expiry date is ≥70%;

WSYX-QC-SOP-0609-0003
Type 19A: release standard is $\geq 70\%$,
standard in expiry date is $\geq 65\%$;
Type 19F, 23F release standard is $\geq 65\%$,
standard in expiry date is $\geq 60\%$.
SOP for conjugate protein content test
Release standard shall be not less than 85%,
Conjugate protein for 13-valent pneumococcal conjugate
standard in expiry date shall be not less than
content vaccine
80%.
WSYX-QC-SOP-06-09-0006
SOP for the protein content
Total protein determination 41-150 μg /dose
content
WSYX-QC-SOP-06-09-0006
All guinea pigs and mice should be healthy
and no abnormal reaction during the
SOP for Abnormal Toxicity Test
Abnormal toxicity observation period, and the weight of each
WSYX-QC-SOP-0009-0306
guinea pig and mouse should be increased at
the end of the period.
SOP for Bacterial Endotoxin Test
Bacterial endotoxin Should be not more than 12.5EU/dose
WSYX-QC-SOP-0009-0203
SOP for Sterility Test
Sterility Should comply with requirements
WSYX-QC-SOP-0009-0202

VI Amount and batch No. of testing samples

Product Name Batch No. Amount for single test Total amount Note

GK201709001 165 syringes 1650 syringes
13-valent
GK201709002 165 syringes 1650 syringes 5 $\pm$ 3■
Pneumococcal
Conjugate Vaccine GK201709003 165 syringes 1650 syringes

GK201709001 165 syringes 825 syringes 25 $\pm$ 2■

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Code WSYX-QC-SPR-0609-0055 Version 02
Protocol of Product Stability Test Quality Control Laboratory

GK201709002 165 syringes 825 syringes

GK201709003 165 syringes 825 syringes

VII Evaluation of test results
According to Quality Standard of 13-valent Pneumococcal Conjugate Vaccine, code: WSYX-QC-TS-06-09-0002 samples of 13-valent pneumococcal polysaccharide conjugate vaccine were tested at different temperatures and time points, results were evaluated and Study Report of Product Stability Test was issued.

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Code WSYX-QC-SPR-0609-0056 Version 02
Protocol of Product Stability Test Quality Control Laboratory

Protocol of Product Stability Test

Sample Name■13-valent Pneumococcal Conjugate Vaccine

Dosage Form: Injections

Batch No.■GI201708001
GI201708002
GI201709003
Package Form■1ml Vial■Gerresheimer Shuangfeng■

Revision History

Version number Revision date Reason for Revision

00 2017.08.19 New Drafting Document

01 2017.11.15 Terminate the 40■-stability test
According to the Quality Standards for 13-valent
Pneumococcal Polysaccharide Conjugate Vaccine, version 04,
02 2019.07.04
WSYX-QC-TS-06-09-0002, adjust the test items at some time
points in the protocol

Document Review

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Code WSYX-QC-SPR-0609-0056 Version 02
Protocol of Product Stability Test Quality Control Laboratory

Department Position Signature Date

Drafted by Quality Control Lab Staff Ting Huang 2019.07.04

(Deputy)Manager
Quality Control Lab Pengyang Li 2019.07.04

(Deputy)Manager
Animal laboratory Yongqing Yan 2019.07.04

(Deputy)Manager
Reviewed by Quality Assurance Xiuli Wu 2019.07.04

Quality Management (Deputy)Manager Shen Feng 2019.07.09

Head of Quality
—— Guoliang Fang 2019.07.09
Management

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Protocol of Product Stability Test Quality Control Laboratory

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I Product Information				
1. Product information				
Name	Batch No.	First Test Date	Estimated Note	
completion time				
13-valent	GI201708001	2017.08.19	2021.02.19	5±3■

Pneumococcal 2018.02.19 25±2■
Conjugate Vaccine G1201708002 2017.08.27 2021.02.27 5±3■
2018.02.27 25±2■
G1201709003 2017.09.02 2021.03.02 5±3■
2018.03.02 25±2■

Note: According to the version 00 and 01 of the protocol (WSYX-QC-SPR-0609-0056), the acceleration stability test under 25■ has been completed. Therefore, the version 02 will not conduct the acceleration stability test under 25■, only remain this part of acceleration stability test under 25■ to ensure the integrity of the protocol.

2. VVM information

Manufacturer: Temptime Corporation

This stability test involves vaccine VVM label heat stability data.

VVM Type No. of days to end point at +25°C Time to end point at +5°C

VVM 30: high stability 193 days > 4 years

VVM 14: medium stability 90 days > 3 years

II Purpose

1. According to the Good Manufacturing Practice for Pharmaceutical Products (Amended in 2010), Technical Guidelines for Biological Product Stability Research, 2. The Pharmacopoeia of the People's Republic of China 2015 Edition, ICH Q1A Stability Testing of New Drug Substances and Productions, ICH Q5C Stability Testing of Biotechnological/Biological Products and Standard Operation Procedure for Continuous Stability Study of Products, Yuxi Walvax conducts stability studies on 3 batches of 13-valent pneumococcal polysaccharide conjugate vaccines for vaccines to provide a basis for the storage validity period.

2. Based on the stability data of Vaccine Vial Monitor (VVM), the stability test of vaccine VVM (VVM14, VVM30) was carried out at 5±3■ and 25±2■, and the optical density value of VVM on the sample vial was detected. Combined with the test results of various quality test of the stability sample, to confirm the stability of the thermal sensitivity of the vaccine heat label and the stability of the 13-valent pneumococcal polysaccharide conjugate vaccine, to provide basis for the selection of VVM for 13-valent pneumococcal polysaccharide conjugate vaccine.

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3. According to the Manufacturing and Test Procedure of 13-valent Pneumococcal Polysaccharide Conjugate Vaccine effective on July 03, 2019: and Quality Standard of 13-valent Pneumococcal Conjugate Vaccine, code: WSYX-QC-TS-06-09-0002, Yuxi Walvax adjusted the test items of the 13-valent pneumococcal polysaccharide conjugate vaccine. The detection of the adsorption rate will be cancelled, and the detection of the content of conjugate protein and total protein will be added, so the content of the protocol needs to be modified.

III Basis for the stability test

1. Good Manufacturing Practice for Pharmaceutical Products (Amended in 2010).
2. The Pharmacopoeia of the People's Republic of China 2015 Edition Volume ■: Stability Testing of Drug Substances and Products (9001).
3. Technical Guiding Principles for Compatibility Research of Chemical Injection and Pharmaceutical Glass Packaging Containers (Trial)
4. ICH Q1A Stability Testing of New Drug Substances and Productions
5. ICH Q5C Stability Testing of Biotechnological/Biological Products
6. Standard Operation Procedure for Continuous Stability Study of Products, code: WSYX-QA-SMP-0009-0030.
7. Quality Standard of 13-valent Pneumococcal Conjugate Vaccine, code: WSYX-QC-TS-06-09-0002.

IV Responsibility

1. Quality Assurance Department:

- Provide the Application of Conducting the Stability Study for Products and Plan of Conducting the Stability Study for Products to Quality Assurance Laboratory;
- Review the Protocol of Product Stability Test;
- Review the Report of Product Stability Study;
- In January of each year, integrate the results of the continuous stability test of all finished products completed in the previous year, and issue the Review of Product Stability Test;

2. Quality Control Laboratory:

- Draft the Protocol of Product Stability Test;
- QC manager review the Protocol of Product Stability Test;
- Store the test sample under the 5±3■, 25■ according to the Protocol of Product Stability Test;
- Sampling at the time point according to the Protocol of Product Stability Test;

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- Test the sample and issue the report for results of test;
- Analysis and summarize the product quality according to test results, draft and review the periodical and finalized Study Report of Product Stability Test;

3. Animal Laboratory

- Review the Protocol of Product Stability Test;
- Test the sample and report the results of test to Quality Control Laboratory;
- Review the Report of Product Stability Study;

4. Production Management Department:

- According to the documents and Protocol of Product Stability Test approved by Quality Assurance Department, issue the manufacture order to related workshop;

5. Fill & Finish Center

- According to manufacture order and Protocol of Product Stability Test, manufacture the required sample;

6. Quality Supervision Department

- According to Protocol of Product Stability Test, send the required sample amount to Quality Control Laboratory;

7. Quality Management:

-Review the Protocol of Product Stability Test;

8. Quality Responsible person:

-Review and approval the Protocol of Product Stability Test;

-Review and approval the Study Report of Product Stability Test;

V Study design

1. Package information

a. Neutral borosilicate glass injection vial

Manufacturer: Shuangfeng Gerresheimer pharmaceutical glass (Danyang) Co., Ltd;

b. Halogenated butyl rubber plug for injection (chlorination)

Manufacturer: Shengzhou pharmaceutical packaging material technology (China) Co., Ltd;

c. Aluminium-plastics combined caps

Manufacturer: Jingjiang Huaxin Capping Co., Ltd;

2. Sample storage condition: $5\pm 3^\circ\text{C}$, $25\pm 2^\circ\text{C}$

3. Sample placement method: Vial inversion

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4. Test time:

$5\pm 3^\circ\text{C}$ continuous stability: 0 months, 3 months, 6 months, 9 months, 12 months, 18 months, 24 months, 30 months;

$25\pm 2^\circ\text{C}$ accelerated stability: 0 months, 1 month, 2 months, 3 months, 6 months.

Note: a. The expected purpose of this stability test is to investigate under the above conditions until the product expires, that is, to the time point when there is at least one inconsistency in the product quality parameter. However, due to the limitation of the planned sample volume, if the continuous stability under $5\pm 3^\circ\text{C}$, the parameters still meet the requirements after 30 months, it needs to be delayed to 36 months, 42 months, and 48 months; Accelerated stability need to postpone 9 months after 6 months;

b. As of July 03, 2019, the continuous stability of the 3 batches for 13-valent pneumococcal polysaccharide conjugate vaccines involved in the protocol at $2-8^\circ\text{C}$ has been investigated to 18 months, so this version of the protocol is only for 24 months and after the inspection time point (The test protocol at each time point before 24 months is subject to the version 01 protocol).

5. Test items

Items Appearance, extractable volume,

Polysaccharide content, conjugate pH, aluminum phosphate content,

Time antigen content, conjugate protein sodium chloride content, Sterility

content, total protein content osmolality, abnormal toxicity,

bacterial endotoxin

24 months Test: Y N Y N Y N

30 months Test: Y N Y N Y N

36 months Test: Y N

42 months Test: Y N Refer to the note

48 months Test: Y N

Note: For the 36 months, 42 months, and 48 months test, besides the above test items, other test items conduct as per following requirements:

a. The above test items, under the condition that the three time points are qualified, then the other test

items should conduct full test at 48 months.

b. The above test items, no matter at which time point there is a unqualified, it request conducting full test for all other items at that time point; and stopping the observation at subsequent time points.

6. Description of container closure system

The test sample are packed in 1ml sterility vial, with rubber stopper and aluminum-plastic

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combination cap and the container is well sealed.

7. Test method and criterion

13-valent Pneumococcal Conjugate Vaccine

Test Items Standard Criterion

SOP for Appearance check Should be milky uniform suspension after

Appearance

WSYX-QC-SOP-0009-0108 shaking

SOP for the extractable volume test

Extractable volume Should be not less than 0.5ml/vial or syringe

WSYX-QC-SOP-0009-0109

SOP for pH Value test

pH 5.5-7.5

WSYX-QC-SOP-0009-0110

SOP for Aluminum hydroxide (or

Aluminum

Aluminum phosphate) content test 0.8-1.2g/L

phosphate content

WSYX-QC-SOP-0009-0117

SOP for sodium chloride content

Sodium chloride

determination 7.5-9.5g/L

content

WSYX-QC-SOP-0009-0120

SOP Osmolality test

Osmolality 220-310mOsm/Kg

WSYX-QC-SOP-0009-0135

1: 1.8-3.4 $\mu\text{g}/\text{dose}$, 3: 1.75-3.25 $\mu\text{g}/\text{dose}$, 4:

2.1-3.9 $\mu\text{g}/\text{dose}$, 5: 1.75-3.25 $\mu\text{g}/\text{dose}$, 6A:

SOP of Rate nephelometry method for 1.75-3.25µg/dose, 6B: 4.2-7.8µg/dose, 7F: 2-
Polysaccharide
detection of polysaccharide content 3.7µg/dose, 9V: 1.75-3.25µg/dose, 14: 1.9-
content
WSYX-QC-SOP-0009-0414 3.6µg/dose, 18C: 2.3-4.2µg/dose, 19A: 1.8-
3.4µg/dose, 19F: 1.9-3.6µg/dose, 23F: 2.1-
3.9µg/dose.

Type 1: release standard is ≥30%, standard
in expiry date is ≥25%;
Type 3, 6A, 6B, 9V, 14, 18C: release standard
is ≥80%, standard in expiry date is ≥75%;
Conjugate antigen SOP for Conjugate antigen content Type 4, 5, 7F: release standard is ≥75%,
determination
content standard in expiry date is ≥70%;
WSYX-QC-SOP-0609-0003
Type 19A: release standard is ≥70%,
standard in expiry date is ≥65%;
Type 19F, 23F release standard is ≥65%,
standard in expiry date is ≥60%.
SOP for conjugate protein content test
Release standard shall be not less than 85%,
Conjugate protein for 13-valent pneumococcal conjugate
standard in expiry date shall be not less than
content vaccine
80%.
WSYX-QC-SOP-06-09-0006
SOP for the protein content
Total protein determination 41-150µg/dose
content
WSYX-QC-SOP-06-09-0006
Abnormal toxicity SOP for Abnormal Toxicity Test All guinea pigs and mice should be healthy

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WSYX-QC-SOP-0009-0306 and no abnormal reaction during the
observation period, and the weight of each
guinea pig and mouse should be increased at
the end of the period.
SOP for Bacterial Endotoxin Test
Bacterial endotoxin Should be not more than 12.5EU/dose
WSYX-QC-SOP-0009-0203
SOP for Sterility Test
Sterility Should comply with requirements
WSYX-QC-SOP-0009-0202

VI Amount and batch No. of testing samples
1.Vaccine
Product Name Batch No. Amount for single test Total amount Note

GK201709001 165 syringes 1650 syringes

GK201709002 165 syringes 1650 syringes 5±3■
13-valent
GK201709003 165 syringes 1650 syringes
Pneumococcal
Conjugate Vaccine GK201709001 165 syringes 825 syringes

GK201709002 165 syringes 825 syringes 25±2■

GK201709003 165 syringes 825 syringes
2. VVM
Product Name Batch No. Amount for test Note
YG170/1
100 5±3■
VVM 14
YG170/1
100 25±2■
YG161/1
100 5±3■
VVM 30
YG161/1
100 25±2■

VII Evaluation of test results
According to Quality Standard of 13-valent Pneumococcal Conjugate Vaccine, code: WSYX-QC-TS-
06-09-0002 samples of 13-valent pneumococcal polysaccharide conjugate vaccine were tested at
different temperatures and time points, results were evaluated and Study Report of Product Stability
Test was issued.

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Protocol of Product Stability Test

Sample Name■Bulk Conjugate of 13-valent Pneumococcal Polysaccharide

Conjugate Vaccine

Revision History

Version number Revision date Reason for Revision

00 2017.05.23 New Drafting Document

Document Review

Department Position Signature Date

Drafted by Quality Control Lab Supervisor Ran Yang 2017.05.23

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Protocol of Product Stability Test Quality Control Laboratory

(Deputy)Manager 2017.05.24
Quality Control Lab Pengyang Li

(Deputy)Manager 2017.05.25
Quality Assurance Xiuli Wu
Reviewed by
2017.05.26
Quality Management (Deputy)Manager Feng Shen

Head of Quality 2017.05.27
—— Guoliang Fang
Management

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I Product Information
1. Product information (2-8■)
Name Batch No. First Test Date Estimated completion time

RSP1C201706001
2017.06.27 2018.06.27
Serotype 1
RSP1C201707002
Pneumococcal 2017.07.28 2018.07.28
Bulk Conjugate
RSP1C201709003
2017.09.13 2018.09.13
RSP3C201706001
2017.06.30 2018.06.30
Serotype 3
RSP3C201707002
Pneumococcal 2017.08.01 2018.08.01
Bulk Conjugate
RSP3C201708003
2017.08.26 2018.08.26
RSP4C201706001
2017.06.22 2018.06.22

Serotype 4
 RSP4C201707002
 Pneumococcal 2017.08.01 2018.08.01
 Bulk Conjugate
 RSP4C201709003
 2017.09.10 2018.09.10
 RSP5C201705001
 2017.06.01 2018.06.01
 Serotype 5
 RSP5C201707002
 Pneumococcal 2017.07.19 2018.07.19
 Bulk Conjugate
 RSP5C201708003
 2017.08.22 2018.08.22
 RSP6AC201705001
 2017.05.26 2018.05.26
 Serotype 6A
 RSP6AC201707002
 Pneumococcal 2017.07.07 2018.07.07
 Bulk Conjugate
 RSP6AC201708003
 2017.08.28 2018.08.28
 RSP6BC201705001
 2017.05.26 2018.05.26
 Serotype 6B
 RSP6BC201707002
 Pneumococcal 2017.07.07 2018.07.07
 Bulk Conjugate
 RSP6BC201708003
 2017.09.01 2018.09.01
 RSP7FC201705001
 2017.06.12 2018.06.12
 Serotype 7F
 RSP7FC201706002
 Pneumococcal 2017.07.03 2018.07.03
 Bulk Conjugate
 RSP7FC201708003
 2017.09.03 2018.09.03
 RSP9VC201705001
 2017.05.29 2018.05.29
 Serotype 9V
 RSP9VC201707002
 Pneumococcal 2017.07.11 2018.07.11
 Bulk Conjugate
 RSP9VC201709003
 2017.09.10 2018.09.10
 Serotype 14 RSP14C201706001
 2017.06.30 2018.06.30
 Pneumococcal
 RSP14C201707002
 Bulk Conjugate 2017.07.28 2018.07.28

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RSP14C201709003
 2017.09.13 2018.09.13
 RSP18CC201706001
 2017.06.28 2018.06.28
 Serotype 18C
 RSP18CC201707002
 Pneumococcal 2017.07.24 2018.07.24
 Bulk Conjugate
 RSP18CC201708003
 2017.08.20 2018.08.20
 RSP19AC201705001
 2017.05.29 2018.05.29
 Serotype 19A
 RSP19AC201707002
 Pneumococcal 2017.07.11 2018.07.11
 Bulk Conjugate
 RSP19AC201708003
 2017.09.03 2018.09.03
 RSP19FC201706001-1
 2017.07.03 2018.07.03
 Serotype 19F
 RSP19FC201707002
 Pneumococcal 2017.07.23 2018.07.23
 Bulk Conjugate
 RSP19FC201708003
 2017.09.07 2018.09.07
 RSP23FC201705001
 2017.06.04 2018.06.04
 Serotype 23F
 RSP23FC201707002
 Pneumococcal 2017.07.19 2018.07.19
 Bulk Conjugate
 RSP23FC201708003
 2017.09.07 2018.09.07

II Purpose

1. To conduct stability test on 3 batches of process validation for each serotype pneumococcal bulk conjugate of 13-valent Pneumococcal Polysaccharide Conjugate Vaccine.

III Basis for the stability test

1. Good Manufacturing Practice for Pharmaceutical Products (Amended in 2010).
2. The Pharmacopoeia of the People's Republic of China 2015 Edition Volume II: Stability Testing of Drug Substances and Products (XIX C).
3. Standard Operation Procedure for Continuous Stability Study of Products, code: WSYX-QA-SMP-0009-0030.
4. Quality Standard of 13-valent Pneumococcal Polysaccharide Conjugate Vaccine, code: WSYX-QC-TS-0609-0005.

IV Responsibility

1. Quality Control Laboratory:
-Draft the Protocol of Product Stability Test after receiving Plan of Conducting the Product Stability Test;

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- Sampling at the time point according to the Protocol of Product Stability Test;
- Test the sample and issue the report for results of test;

2. Quality Assurance Department:

- Taking sample according to the Protocol of Product Stability Test;
- Analysis and summarize the product quality according to test results,
- Review the Protocol of Product Stability Test;

3. Quality Management Department:

- Review the Protocol of Product Stability Test;

4. Quality Responsible person:

- Review and approval the Protocol of Product Stability Test;

V Study design

1. Sample storage condition: 2-8■

2. Test plan:

Time Test Items

Full test: Sterility, bacterial endotoxin, polysaccharide content, protein content, EDAC residue, polysaccharide-protein ratio, 0 month, 12 months

free polysaccharide content, free protein content, molecular size, identification, ADH content

Polysaccharide content, protein content, polysaccharide-

1 month, 2 months, 3 months, 6 months, 9

protein ratio, free polysaccharide content, free protein content, months

molecular size

3. Description of the container closure system: the test samples are packed in the stopper glass bottles.

Physical and chemical testing 45ml/bottle×1 bottle

Sterility test 42ml/bottle×1 bottle

Bacterial endotoxin test 1ml/bottle×2 bottles

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4. Test method and criterion

Polysaccha Polysaccha Free

Protein Free EDAC Bacterial

ride ride- polysaccha Molecular Identificati

Serotype content protein residue endotoxin Sterility

content protein ride size on

µg/ml content µmol /L EU/µg

µg/ml ratio content

Appear

positive Should meet

KD≤0.2

response to the

1 ≥45 ≥70 0.15-0.60 ≤25% ≤4.5% ≤5 ≤0.75 recovery

specific requirement

rate≥80%

antiserum s

type 1

Appear

positive Should meet

KD≤0.2

response to the

3 ≥50 ≥60 0.40-0.90 ≤20% ≤4.0% ≤5 ≤0.75 recovery

specific requirement

rate≥80%

antiserum s

type 3

Appear

positive Should meet

KD≤0.2

response to the

4 ≥45 ≥40 0.30-1.10 ≤25% ≤3.5% ≤5 ≤0.75 recovery

specific requirement

rate≥80%

antiserum s

type 4
Appear
positive Should meet
 $KD \leq 0.2$
response to the
 $5 \geq 30 \geq 100 \quad 0.10-0.30 \leq 20\% \leq 4.5\% \leq 5 \leq 0.75$ recovery
specific requirement
rate $\geq 80\%$
antiserum s
type 5

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Appear
positive Should meet
 $KD \leq 0.2$
response to the
 $6A \geq 40 \geq 80 \quad 0.10-0.50 \leq 20\% \leq 4.5\% \leq 5 \leq 0.75$ recovery
specific requirement
rate $\geq 80\%$
antiserum s

type 6A
Appear
positive Should meet
 $KD \leq 0.2$
response to the
 $6B \geq 40 \geq 60 \quad 0.20-0.65 \leq 25\% \leq 4.0\% \leq 5 \leq 0.75$ recovery
specific requirement
rate $\geq 80\%$
antiserum s

type 6B
Appear
positive Should meet
 $KD \leq 0.2$
response to the
 $7F \geq 80 \geq 180 \quad 0.15-0.45 \leq 20\% \leq 4.5\% \leq 5 \leq 0.75$ recovery
specific requirement
rate $\geq 80\%$
antiserum s

type 7F
Appear
positive Should meet
 $KD \leq 0.2$
response to the
 $9V \geq 30 \geq 50 \quad 0.10-0.50 \leq 25\% \leq 3.5\% \leq 5 \leq 0.75$ recovery
specific requirement
rate $\geq 80\%$
antiserum s

type 9V
Appear
positive Should meet
 $KD \leq 0.2$
response to the
 $14 \geq 40 \geq 60 \quad 0.15-0.70 \leq 25\% \leq 4.5\% \leq 5 \leq 0.75$ recovery
specific requirement
rate $\geq 80\%$
antiserum s
type 14

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Appear
positive Should meet
 $KD \leq 0.2$
response to the
 $18C \geq 30 \geq 50 \quad 0.30-0.60 \leq 30\% \leq 3.5\% \leq 5 \leq 0.75$ recovery
specific requirement
rate $\geq 80\%$
antiserum s

type 18C
Appear
positive Should meet
 $KD \leq 0.2$
response to the
 $19A \geq 80 \geq 120 \quad 0.10-0.65 \leq 30\% \leq 3.5\% \leq 5 \leq 0.75$ recovery
specific requirement
rate $\geq 80\%$
antiserum s

type 19A
Appear
positive Should meet
 $KD \leq 0.2$
response to the
 $19F \geq 40 \geq 100 \quad 0.10-0.40 \leq 35\% \leq 3.5\% \leq 5 \leq 0.75$ recovery
specific requirement
rate $\geq 80\%$
antiserum s
type 19F

Appear
positive Should meet
 $KD \leq 0.2$
response to the
23F $\geq 40 \geq 60$ 0.25-0.65 $\leq 25\% \leq 4.0\% \leq 5 \leq 0.75$ recovery
specific requirement
rate $\geq 80\%$
antiserum s
type 23F

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VI Amount and batch No. of testing samples
Name Temperature Test Time Amount for Total Amount
single test(batch) (batch)
2-8■ 89ml 178ml
Serotype 1 0 month, 12 months
Pneumococcal 2-8■ 1 month, 2 months, 3 months, 45ml 225ml
Bulk Conjugate
6 months, 9months
2-8■ 89ml 178ml
Serotype 3 0 month, 12 months
Pneumococcal 2-8■ 1 month, 2 months, 3 months, 45ml 225ml
Bulk Conjugate
6 months, 9months
2-8■ 89ml 178ml
Serotype 4 0 month, 12 months
Pneumococcal 2-8■ 1 month, 2 months, 3 months, 45ml 225ml
Bulk Conjugate
6 months, 9months
2-8■ 89ml 178ml
Serotype 5 0 month, 12 months
Pneumococcal 2-8■ 1 month, 2 months, 3 months, 45ml 225ml
Bulk Conjugate
6 months, 9months
2-8■ 89ml 178ml
Serotype 6A 0 month, 12 months
Pneumococcal 2-8■ 1 month, 2 months, 3 months, 45ml 225ml
Bulk Conjugate
6 months, 9months
2-8■ 89ml 178ml
Serotype 6B 0 month, 12 months
Pneumococcal 2-8■ 1 month, 2 months, 3 months, 45ml 225ml
Bulk Conjugate
6 months, 9months
2-8■ 89ml 178ml
Serotype 7F 0 month, 12 months
Pneumococcal 2-8■ 1 month, 2 months, 3 months, 45ml 225ml
Bulk Conjugate
6 months, 9months
2-8■ 89ml 178ml
Serotype 9V 0 month, 12 months
Pneumococcal 2-8■ 1 month, 2 months, 3 months, 45ml 225ml
Bulk Conjugate
6 months, 9months
2-8■ 89ml 178ml
Serotype 14 0 month, 12 months
Pneumococcal 2-8■ 1 month, 2 months, 3 months, 45ml 225ml
Bulk Conjugate
6 months, 9months
2-8■ 89ml 178ml
Serotype 18C 0 month, 12 months
Pneumococcal 2-8■ 1 month, 2 months, 3 months, 45ml 225ml
Bulk Conjugate
6 months, 9months
2-8■ 89ml 178ml
Serotype 19A 0 month, 12 months
Pneumococcal 2-8■ 1 month, 2 months, 3 months, 45ml 225ml
Bulk Conjugate
6 months, 9months
2-8■ 89ml 178ml
Serotype 19F 0 month, 12 months
Pneumococcal 2-8■ 1 month, 2 months, 3 months, 45ml 225ml
Bulk Conjugate
6 months, 9months

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2-8■ 89ml 178ml
Serotype 23F 0 month, 12 months
Pneumococcal 2-8■ 1 month, 2 months, 3 months, 45ml 225ml
Bulk Conjugate
6 months, 9months

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Protocol of Product Stability Test

Sample Name:■Pneumococcal polysaccharide serotype 18C

Batch No.: 18C-PVPV15100801
18C-PVPV15110802
18C-PVPV16010803

Revision History

Version number Revision date Reason for Revision

00 2015.09.07 New Drafting Document

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Protocol of Product Stability Test Quality Control Laboratory

Document Review

Department Position Signature Date

Staff
Drafted by Quality Control Lab Guangmin Na 2015.09.07

(Deputy)Manager 2015.09.07
Quality Control Lab Pengyang Li

(Deputy)Manager 2015.09.07
Quality Assurance Xiuli Wu
Reviewed by
(Deputy)Manager 2015.09.07
Quality Management Xiuli Wu

Head of Quality
—— Guoliang Fang 2015.09.09
Management

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III Basis for the stability text 4

IV Responsibility 4

V Study design 5

VI Amount and batch No. of testing samples 6

3 / 6
Code WSYX-QC-SPR-0609-0022 Version 00
Protocol of Product Stability Test Quality Control Laboratory

I Product Information
1.Product information■below -20■■■
Name Batch No. First Test Date Estimated completion
time
Pneumococcal 18C-PVPV15100801 2015.10.13 2021.04.13
polysaccharide serotype 18C-PVPV15110802 2015.12.03 2021.06.03
18C 18C-PVPV16010803 2015.01.09 2021.07.09

II Purpose
To conduct the stability test of Process Validation Products, Yuxi Walvax sdraft the Protocol of Product Stability Test according to the Good Manufacturing Practice for Pharmaceutical Products (Amended in 2010) and its implementation guidelines, Technical Guidelines for Biological Product Stability Research (pilot edition) and the Standard Operation Procedure for Continuous Stability Study of Products.

III Basis for the stability text
1. Good Manufacturing Practice for Pharmaceutical Products (Amended in 2010).
2. The Pharmacopoeia of the People's Republic of China 2015 Edition Volume ■: Stability Testing of Drug Substances and Products (9001).
3. Standard Operation Procedure for Continuous Stability Study of Products, code: WSYX-QA-SMP-0009-0030.

4. Quality Standard of 23- valent Pneumococcal Polysaccharide Vaccine, code: WSYX-QC-TS-0609-0004.
5. Quality Standard of 13-valent Pneumococcal Conjugate Vaccine, code: WSYX-QC-TS-0609-0005.

IV Responsibility

1. Quality Control Laboratory:

- Draft the Protocol of Product Stability Test after receiving the Protocol of Continuous Stability Study of Products draft by Quality Assurance Department;
- Taking sample at the time point and store the test sample under the -20℃ according to the Protocol of Product Stability Test;
- Test the sample and issue the report for results of test;

2. Quality Assurance:

- Taking sample according to the Protocol of Product Stability Test;
- Analysis and summarize the product quality according to test results, draft and review the periodical and finalized Study Report of Product Stability Test;

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Code WSYX-QC-SPR-0609-0022 Version 00

Protocol of Product Stability Test Quality Control Laboratory

- Review the Protocol of Product Stability Test;

3. Quality Management Department:

- Approval the Study Report of Product Stability Test;

4. Quality Responsible person:

- Review and approval the Protocol of Product Stability Test;

V Study design

1. Storage condition: below -20℃.

2. Test points: 0th month, 1 year, 2 years, 3 years, 4 years, 4 years and 6 months, 5 years, 5 years and 6 months.

3. Test items

Identification test, total solid content, nucleic acid content, protein content, total nitrogen content, phosphorus content, methylpentose, molecular size, organic solvent (ethanol, ether, acetone) residues, bacterial endotoxin, sodium deoxycholate residues.

4. Description of container closure system: test sample are packaged in glass bottles with rubber plugs for good sealing.

5. Test standard and criterion

Test Items Standard Criterion

Identification test Appear positive response to specific

SOP for Identification Test: WSYX-QC- antiserum type 18C

SOP-0009-0201

Total solid ——

SOP for Total Solid Determination: WSYX-

QC-SOP-0009-0111

SOP for Protein Content Determination: ≤3%

Protein content

WSYX-QC-SOP-0009-0102

SOP for Nucleic Acid Content ≤2%

Nucleic acid content

Determination: WSYX-QC-SOP-0009-0103

Total nitrogen SOP for Total Nitrogen Content 0-1.0%

content Determination: WSYX-QC-SOP-0009-0751

SOP for Phosphorus Content Determination:

Phosphorus content 2.4-4.9%

WSYX-QC-SOP-0009-0101

SOP for Methylpentose Determination:

Methylpentose ≥14%

WSYX-QC-SOP-0009-0128

Sephacrose CL-4B gel≤0.15, more

SOP for Molecular size Determination: than 70% of serotype 18C

Molecular size (KD) polysaccharide should be recovered

WSYX-QC-SOP-0009-0104

before the KD value of 0.3 is

reached.

5 / 6

Code WSYX-QC-SPR-0609-0022 Version 00

Protocol of Product Stability Test Quality Control Laboratory

SOP for Bacterial Endotoxin Test: WSYX-

Bacterial endotoxin ≤0.5EU/μg

QC-SOP-0009-0203

Ethanol residual ■0.08%

SOP for Organic Solvent Residue

Acetone residual ■0.03%

Determination: WSYX-QC-SOP-0009-0138

Ether residual ■0.01%

SOP for Sodium Deoxycholate Test:

Sodium deoxycholate ■0.20

WSYX-QC-SOP-0009-0137

VI Amount and batch No. of testing samples

Product name Batch amount Amount for single test Total amount

Pneumococcal 18C-PVPV15100801 0.19g 1.52g
polysaccharide serotype 18C-PVPV15110802 0.19g 1.52g
18C 18C-PVPV16010803 0.19g 1.52g

SOP of Yuxi Walvax Biotechnology Co., Ltd
Code WSYX-QC-SOP-00-09-0110
SOP for pH Value test Version 07 Page 1/4
Confidential files No copy

1 Objective

Standardize operating procedure for pH value test.

2 Definition/abbreviation

None

3 Scope of application

It applies to test operation for pH value of solution.

4 Responsibility

All operators shall perform the tests in strict accordance with this SOP.

5 Equipments and Materials

5.1 Equipments

Equipment

No. Equipment name Manufacturer Notes

model

1 PH meter φ390 BECKMAN Or equivalent equipment

5.2 Reagents

No. Reagent name Grade

1 PH4 RED BUFFER Imported reagent

2 PH7 GREEN BUFFER Imported reagent

3 PH10 BLUE BUFFER Imported reagent

5.3 Materials

No. Material name No. Material name

1 Beaker 2 Plastic tube

3 Filter paper

6 Working procedure

6.1 Power-on preheating: Plug in the power, open the pH meter switch, preheat at least 30 minutes.

6.2 Calibrate pH meter: Operate according <φ390 pH meter use, clean and maintenance SOP>.

SOP of Yuxi Walvax Biotechnology Co., Ltd

Code WSYX-QC-SOP-00-09-0110

SOP for pH Value test Version 07 Page 2/4

Confidential files No copy

6.3 Test sample preparation

6.3.1 Intermediate Product: Take the sample from the refrigerator, and put in room temperature, the appropriate sample is taken out into the plastic tube, pending test.

6.3.2 Vial/prefilled syringe: Take the sample from the refrigerator, and put in room temperature, the appropriate sample is taken out into the plastic tube, pending test.

6.3.3 Freeze dried preparation for injection: Take the sample from the refrigerator, and put in room temperature. Reconstitute the final product with diluents of 0.5ml/vial. Mix well, then the appropriate sample is taken out into the plastic tube, pending test.

6.4 Test the sample, the operation follows <φ390 pH meter use, clean and maintenance SOP>.

7 Flowchart

N/A

8 Matters need attention

8.1 The standard buffer is taken from the refrigerator until its temperature rises to room temperature for use.

8.2 Before test, according to each product item stipulation, chooses two kinds of pH value difference vary approximate 3 units of standard buffer solution, and make the pH value of test product solution is in the correction pH range.

8.3 Taking the first buffer solution which is close to the pH value of the test sample solution to calibrate the instrument, make that the instrument display value is consistent with the buffer value.

8.4 After the instrument positioning, and then check instrument value, the error should not be larger than the ±0.02pH unit, if greater than this deviation, it should be carefully adjusted the slope, so that the value and the second table column values match. Repeat the above positioning and slope adjustment operation, to the instrument value and standard buffer solution of the specified value difference is not greater than ±0.02pH unit. Otherwise, it is necessary to check the instrument or replace the electrode, then to be corrected to comply with the regulations.

SOP of Yuxi Walvax Biotechnology Co., Ltd

Code WSYX-QC-SOP-00-09-0110

SOP for pH Value test Version 07 Page 3/4

Confidential files No copy

8.5 The buffer correction slope should be between 90%-110% (internal control standard 96%-102%). Exit the correction display, enter the reading display, select the buffer solution close to the test solution pH value to retest. The instrument value and standard buffer value difference is not greater than ±0.02pH unit.

8.6 Each time of change buffer or test solution, use purified water to fully wash the electrodes, and make the water suction completely, also can use standard buffer or test solution for washing.

8.7 To determine the high pH value of the test sample and standard buffer, it should pay attention to the problem of alkali error, if necessary, select the appropriate glass electrode to test.

8.8 For the determination of the pH value of the solution without buffer effect, unless otherwise specified, the pH4.00 standard buffer is used to calibrate the instrument and to test the solution, and then retest the test solution until the pH reading changes within one minute not more than ±0.05; then use the borax standard buffer to calibrate the instrument and test again. The difference of two pH readings should be no more than 0.1, and the average of two readings should be the pH value of the test sample.

8.9 The membrane of the glass electrode is used for ion exchange and is fragile, so it can not be collided, touched, and the new electrode should be soaked for 24 hours by injection water

before use.

8.10 In the electrode cap, the 3mol/L potassium chloride solution should always keep the electrode immersed to prevent the electrode from drying up and damaging.

8.11 If the sample is concentrated protein, it can be used saline to immerse electrode after test, and then rinse with purified water.

9 Reference or standard basis

9.1 Chinese Pharmacopeia, Volume ■, current version

9.2 <Beckman φ390 pH meter use guide>

10 Revision history

SOP of Yuxi Walvax Biotechnology Co., Ltd

Code WSYX-QC-SOP-00-09-0110

SOP for pH Value test Version 07 Page 4/4

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Issuing Time in Revision Revision basis, reason and
Version Note

time force Dept. content

03 01 Nov. 01 Dec. QC Changed when file expired

2011 2011

laboratory

2 Sep. 16 Sep. QC Complete some parts of the

04

2014 2014 laboratory content

05 18 Mar. 20 Mar. QC Complete some parts of the

2016 2016 laboratory content

06 30 Sep. 01 Oct. QC File to upgrade

2017 2017 laboratory

Standard Operating Procedures of Yuxi Walvax Biotechnology Co., Ltd.

Code WSYX-QC-SOP-00-09-0209
SOP for Purity Test Version 05 Pages 1/2
Confidential documents, no copying.

1 Purpose
Specification the operation of purity determination.
2 Definitions/abbreviations
No
3 Scope of application
For operations for purity determination.
4 Responsibilities
Quality control laboratory personnel in strict accordance with this standard operating procedures.
5 Materials and equipment
No
6 Work procedures
6.1 Action
6.1.1 Calculation: Purity - Flocculent units/protein nitrogen content
6.1.2 Flocculent units as determined according to the "Toxin Flocculation Unit Detection Operating Procedure".
6.1.3 Protein Nitrogen Content - Total Nitrogen - Non-Protein Nitrogen
Total nitrogen according to the "total nitrogen content determination operating procedures", non-protein nitrogen according to the "non-protein nitrogen content determination operating procedures".
7 Flow chart
No
8 Precautions
8.1 The repair of the valid number.
9 Related documents
9.1 SOP for detection of flocculent units of toxoid. Code: WSYX-QC-SOP-0009-0011
9.2 SOP for determination of total nitrogen content. Code: WSYX-QC-SOP-0009-0751
Standard Operating Procedures of Yuxi Walvax Biotechnology Co., Ltd.

Code WSYX-QC-SOP-00-09-0209
SOP for Purity Test Version 05 Pages 2/2
Confidential documents, no copying.

9.3 SOP for determination of non-protein nitrogen. Code: WSYX-QC-SOP-0009-0752
10 Related records
10.1 Test record of tetanus toxoid. Code: QCR -rTT (t)
10.2 The batch test record of diphtheria toxoid bulk. Code: QCR-(DTaP)d
11 References or standard basis
No
12 Appendix
12.1 Purity Test Record. Code: WSYX-QC-SOP-0009-0209-R01
13 Revision history

Revision Basis, reasons and content of the
Version Release Effective
Departme amendment Note
No. Time time
nt
03 2011.11.01 2011.12.01 QC File expiration
Laboratory
04 2014.11.01 2014.12.01 QC File expiration
Laboratory

Standard Operating Procedures of Yuxi Walvax Biotechnology Co., Ltd.

Code WSYX-QC-SOP-02-09-0002

SOP for toxicity reversal

Version 07 Pages 1/4

test of tetanus toxoid bulk

Confidential documents, no copying.

1 Purpose

Standardize the operation of toxicity reversal test of tetanus toxoid bulk.

2 Definition/abbreviation

No

3 Scope of application

Suitable for toxicity reversal test of tetanus toxoid bulk.

4 Responsibilities

4.1 Quality Control Laboratory

- The operator is responsible for the implementation of this procedure

- Department manager is responsible for overseeing the implementation of this procedure

4.2 Quality Supervision Department

- Quality supervisors are responsible for overseeing the implementation of this code

5 Equipment and materials

5.1 Using the equipment

Serial Equipment Equipment Equipment

Manufacturers

number name model coding

Electronic

1 PMA7501 WYQCF0263 Sartorius

balance

Purification Suzhou Antai Air Technology Co.,

2 BLB-1300 WYQCF0233

table Ltd.

Constant

Shanghai Qixin Scientific

3 temperature CHP-9270 WYQCF0018

Instrument Co., Ltd.

incubator

5.2 Use of reagents

Serial Level/Categ Serial Level/Categ

Reagent Name Reagent Name

number ory number ory

1 NaHPO₄.12H₂O AR 2 KH₂PO₄ AR

Standard Operating Procedures of Yuxi Walvax Biotechnology Co., Ltd.

Code WSYX-QC-SOP-02-09-0002

SOP for toxicity reversal

Version 07 Pages 2/4

test of tetanus toxoid bulk

Confidential documents, no copying.

3 NaCl AR 4 Injection water ---

5.3 Materials

Serial Serial

Material name Material name

number number

250-350 g guinea pigs above

1 2 Graduated pipette

clean grade

3 Washing ear ball 4 50 ml vials

5 Alcohol lamp 6 Bottle stopper

7 Parafilm 8 key

9 normal saline 10 Measuring cylinder

11 Beaker 12 Triangular flask

13 Disposable syringe 14 Disposable latex gloves

15 Hemostatic forceps

6 Work procedures

6.1 PBS solution preparation (pH 7.0-7.4)

Weigh 0.106 g disodium hydrogen phosphate (Na₂HPO₄.12H₂O), 0.0144 g potassium dihydrogen phosphate (KH₂PO₄) and 0.9 g sodium chloride (NaCl) into suitable triangular flasks, add water for injection to dissolve and dilute to 100 ml, adjust pH 7.0-7.4, pack separately, sterilize at 121.1■ for 20min, store at 2-8■ for standby, and the validity period is 25 days.

6.2 Test procedure

Samples were taken from each bottle, diluted with PBS solution (pH 7.0-7.4) to 10 Lf/ml, and placed in 37■ incubator for 42 days. Four guinea pigs weighing 250-350 g were injected subcutaneously with 5 ml each. The observation was carried out on the 7th, 14th and 21st days after injection, and the results were recorded according to the regulations.

6.3 Criteria

Standard Operating Procedures of Yuxi Walvax Biotechnology Co., Ltd.

Code WSYX-QC-SOP-02-09-0002

SOP for toxicity reversal

Version 07 Pages 3/4

test of tetanus toxoid bulk

Confidential documents, no copying.

The animals were observed on the 7th, 14th and 21st days after injection, and the animals were not allowed to have tetanus symptoms; the weight of each animal at the end of the period was qualified if the body weight increased compared with that before injection.

7 Flowchart

No

8 Precautions

8.1 Experimental animals must have a certificate of conformity.

8.2 Samples rarely must be strictly in accordance with the requirements of sterile operation

9 References or standard basis

9.1 Chinese Pharmacopoeia, Volume III, Page 89, China Pharmaceutical Science and Technology Press, 2015

10 Revision history

Basis, reasons and content of the

Revision amendment

Version Effective

Departme (Note: If the space is not written Note

No. time

nt enough, a separate page

description may be attached)

According to the corresponding

product names in Chinese

Pharmacopoeia (2010 Edition), it

is standardized and unified. The

QC names of protein carrier

03 2011.12.01 No

Laboratory fermentation broth (TT), protein

carrier refined liquid (TT) and

protein carrier purified liquid (TT)

were changed to tetanus toxoid

fermentation broth, tetanus toxoid

Standard Operating Procedures of Yuxi Walvax Biotechnology Co., Ltd.

Code WSYX-QC-SOP-02-09-0002

SOP for toxicity reversal

Version 07 Pages 4/4

test of tetanus toxoid bulk

Confidential documents, no copying.

original liquid and tetanus toxoid

purified liquid.

Experimen

04 2012.11.01 tal Animal File expiration No

Center

Experimen

05 2015.10.29 tal Animal Document revision and upgrading No

Center

The organizational structure of

laboratory animal center and

quality control laboratory was

QC changed (change control No.: CC-

06 2018.08.01

Laboratory 201909012), so the responsible

department was revised from

"Laboratory Animal Center" to

"quality control laboratory".

Stability data of 13 serotypes of pneumococcal polysaccharides Yuxi Walvax Biotechnology C., Ltd.

Stability data of 13 serotypes of pneumococcal polysaccharides

1. Pneumococcal polysaccharide serotype 1
1.1 Stability Test Results of batch 1-PVPV15101201 pneumococcal polysaccharide (storage condition:-20■)

Nucle Organic solvent residual
Protei Total
Test ic Phosph Uronic O-acety Sodium
n nitroge Bacterial
items Identity acid oruscon acid I Molecular size deoxych
conte n ether acetone ethanol endotoxin
and conte tent content content olate
Total nt content Phenol
standar nt
Solid residual
ds KD■0.3
Positive mg/ml g/L
recovery
response to 3.5 K D≤ ■ ■ ■ ≤0.5EU/μg, ■
Storage ≤2% ≤2% 0-1.5% ≥45% ≥1.8% rate
antiserum type -6.0% 0.15 0.01% 0.03% 0.08% ≤50EU/ml 0.2%
time should
1
be≥75%
Not Not
0th
Complied 3.245 1.9% 0.8% 5.0% 0.3% 55.9% 5.8% 0 ——— detecte detecte 0.007% Complied 0.02% ———
month
d d
Not Not
1st year Complied 2.735 1.3% 0.9% 4.61% 0.33% 53.0% 5.84% 0.04 ——— detecte detecte 0.010% Complied 0.03%
——
d d
Not Not Not
2nd Not
Complied 2.220 1.6% 1.0% 4.9% 0.4% 57% 4.8% 0.07 ——— detecte detecte detecte Complied 0.0205
year detected
d d d
Not Not Not
3rd year Complied 2.790 0.1% 0.9% 4.6% 0.4% 64% 6.6% 0.07 ——— detecte detecte 0.004% Complied 0.1276%
detecte
d d d
4th year Complied 2.190 1% 0% 4.6% 0.3% 66% 6.1% 0.04 96% 0% 0% 0% Complied 0% 0

Version: July 2020 1
Stability data of 13 serotypes of pneumococcal polysaccharides Yuxi Walvax Biotechnology C., Ltd.

1.2 Stability Test Results of batch 1-PVPV15121202 pneumococcal polysaccharide (storage condition:-20■) ■

Nucle Organic solvent residual
Protei Total
Test ic Phosph Uronic O-acety Sodium
n nitroge Bacterial
items Identity acid oruscon acid I Molecular size deoxychol
conte n ether acetone ethanol endotoxin
and conte tent content content ate
Total nt content Phenol
standar nt
Solid residual
ds KD■
Positive mg/ml g/L
0.3recove
response to 3.5 K D≤ ■ ■ ■ ≤0.5EU/μg, ■
Storage ≤2% ≤2% 0-1.5% ≥45% ≥1.8% ry rate
antiserum type -6.0% 0.15 0.01% 0.03% 0.08% ≤50EU/ml 0.2%
time should
1
be≥75%
Not Not
0th
Complied 2.93 1.5% 1.1% 3.7% 0.3% 60.8% 6.1% 0 ——— detecte 0.001% detecte Complied 0.09% ———
month
d d
Not Not
1st year Complied 2.620 1.5% 1.1% 4.73% 0.34% 65.2% 6.14% 0.04 ——— detecte detecte 0.001 Complied 0.06% ———
d d
Not Not Not
2nd Not
Complied 2.200 1.2% 0.8% 5.4% 0.3% 52% 6.4% 0.07 ——— detecte detecte 0.004% Complied detecte
year detected
d d d
Not Not Not
3rd year Complied 2.730 1.5% 1.0% 4.6% 0.4% 51% 6.6% 0.04 ——— detecte detecte 0.005% Complied 0.03549%
detecte
d d d
4th year Complied 2.875 1.5% 1% 4.1% 0.3% 54% 4.7% 0.04 96% 0% 0% 0% Complied 0.04% 0.0434

Version: July 2020 2
Stability data of 13 serotypes of pneumococcal polysaccharides Yuxi Walvax Biotechnology C., Ltd.

1.3 Stability Test Results of batch 1-PVPV16011203 pneumococcal polysaccharide (storage condition:-20■) ■

Nucle Organic solvent residual
Protei Total
Test ic Phosph Uronic 0-acety Sodium
n nitroge Bacterial
items Identity acid oruscon acid l Molecular size deoxychol
conte n ether acetone ethanol endotoxin
and conte tent content content ate Phenol
Total nt content
standar nt residua
Solid
ds KD■ l
Positive mg/ml
0.3recove g/L
response to 3.5 K D≤ ■ ■ ■ ≤0.5EU/μg, ■
Storage ≤2% ≤2% 0-1.5% ≥45% ≥1.8% ry rate
antiserum type -6.0% 0.15 0.01% 0.03% 0.08% ≤50EU/ml 0.2%
time should
1
be≥75%
Not Not Not
0th
Complied 3.080 1.6% 0.9% 5.6% 0.6% 57.4% 5.7% 0.04 ——— detecte detecte detecte Complied 0.10% ———
month
d d d
Not Not Not
1st year Complied 2.720 1.9% 1.3% 4.63% 0.47% 57.2% 5.38% 0.12 ——— detecte detecte detecte Complied 0.06%
——
d d d
Not Not
2nd
Complied 2.850 1.8% 1.1% 4.5% 0.6% 51% 5.6% 0.10 ——— detecte detecte 0.004% Complied 0.07165% 0.027
year
d d
Not Not
3rd year Complied 2.890 1.6% 1.4% 5.0% 0.5% 54% 5.6% 0.11 ——— detecte detecte 0.001% Complied 0.03512%
0.0262
d d
4th year Complied 2.920 1.8% 1% 5.1% 0.6% 60% 5.4% 0.08 96% 0% 0% 0% Complied 0% 0.0322

Version: July 2020 3
Stability data of 13 serotypes of pneumococcal polysaccharides Yuxi Walvax Biotechnology C., Ltd.

2. Pneumococcal polysaccharide serotype 3
2.1 Stability Test Results of batch 3-PVPV15100901 pneumococcal polysaccharide (storage condition:-20■) ■
Protei Total Organic solvent residual
Test Nucleic Phosph Uronic Sodium
n nitroge Bacterial
items Identity acid oruscon acid Molecular size deoxych
conte n ether acetone ethanol endotoxin
and content tent content olate
Total nt content Phenol
standar
Solid KD■ residual
ds Positive
mg/ml 0.3recov g/L
response to 0- KD≤0.1 ■ ■ ■ ≤0.5EU/μg, ■
≤5% ≤2% 0-1.0% ≥40% ery rate
Storage antiserum 1.0% 5 0.01% 0.03% 0.08% ≤50EU/ml 0.2%
should
time type 3
be≥78%
Not Not
0th
Complied 3.105 0.1% 0.04% 0.68% 0.00% 50.5% 0 ——— detecte detecte 0.002% Complied 0.030% ———
month
d d
Not Not
1st year Complied 1.935 0.1% 0.1% 0.47% 0.02% 51.7% 0.08 ——— detecte detecte 0.054% Complied 0.008% ———
d d
Not Not
2nd
Complied 1.948 0.1% 0.3% 0.5% 0.0% 50% 0.03 ——— detecte detecte 0.002% Complied 0.0124% 0.004
year
d d
Not Not Not Not
3rd Not Not
Complied 2.515 0.1% 0.0% 0.1% detecte 48% 0.04 ——— detecte detecte detecte Complied
year detected detected
d d d d
4th
Complied 2.075 0% 0% 0.0% 0% 52% 0.07 97% 0% 0% 0% Complied 0% 0
year

Version: July 2020 4
Stability data of 13 serotypes of pneumococcal polysaccharides Yuxi Walvax Biotechnology C., Ltd.

2.2 Stability Test Results of batch 3-PVPV15110902 pneumococcal polysaccharide (storage condition:-20■) ■
Protei Total Organic solvent residual
Test Nucleic Phosph Uronic Sodium
n nitroge Bacterial
items Identity acid oruscon acid Molecular size deoxychol
conte n ether acetone ethanol endotoxin

and content tent content e
Total nt content Phenol
standar
Solid KD residual
ds Positive
mg/ml 0.3recov g/L
response to 0- KD≤0.1 ≤0.5EU/μg, ≤5% ≤2% 0-1.0% ≥40% ery rate
Storage antiserum 1.0% 5 0.01% 0.03% 0.08% ≤50EU/ml 0.2%
should
time type 3
be≥78%
Not Not Not
0th
Complied 3.065 0.4% 0.09% 0.65% 0.00% 52.90% 0 — detecte detecte detecte Complied 0.16% —
month
d d d
Not Not Not Not
Not
1st year Complied 2.74 0.1% 0.05% 0.18% detecte 50.8% 0 — detecte detecte detecte Complied —
detected
d d d d
Not Not Not Not
2nd Not
Complied 2.465 0.1% 0.0% 0.1% detecte 51% 0 — detecte detecte detecte Complied 0.04499%
year detected
d d d d
Not Not Not Not
3rd
Complied 3.315 0.1% 0.0% detecte 0.0% 51% 0.04 — detecte detecte 0.001% Complied 0.009140% 0.00815
year
d d d
4th
Complied 2.700 0% 0% 0.1% 0% 48% 0 98% 0% 0% 0% Complied 0% 0.0386
year

Version: July 2020 5
Stability data of 13 serotypes of pneumococcal polysaccharides Yuxi Walvax Biotechnology C., Ltd.

2.3 Stability Test Results of batch 3-PVPV15120903 pneumococcal polysaccharide (storage condition:-20) Protei Total Organic solvent residual
Test Nucleic Phosph Uronic Sodium
n nitroge Bacterial
items Identity acid oruscon acid Molecular size deoxychola
conte n ether acetone ethanol endotoxin
and content tent content te
Total nt content Phenol
standar
Solid KD residual
ds Positive
mg/ml 0.3recov g/L
response to 0- KD≤0.1 ≤0.5EU/μg, ≤5% ≤2% 0-1.0% ≥40% ery rate
Storage antiserum 1.0% 5 0.01% 0.03% 0.08% ≤50EU/ml 0.2%
should
time type 3
be≥78%
Not Not Not
0th
Complied 2.325 0.5% 1.99% 0.86% 0.10% 47.8% 0.04 — detecte detecte detecte Complied 0.170% —
month
d d d
Not Not Not Not
1st year Complied 3.190 0.1% 0.04% 0.06% detecte 46.6% 0.04 — detecte detecte detecte Complied 0.10% —
d d d d
Not Not Not Not
2nd
Complied 2.480 0.1% 0.1% 0.1% detecte 48% 0.03 — detecte detecte detecte Complied 0.005181% 0.0205
year
d d d d
Not Not Not Not
3rd
Complied 3.125 0.0% 0.0% 0.1% detecte 50% 0.04 — detecte detecte detecte Complied 0.03194% 0.01795
year
d d d d
4th
Complied 2.945 0% 0% 0.1% 0% 50% 0.07 90% 0% 0% 0% Complied 0.02% 0.0145
year

Version: July 2020 6
Stability data of 13 serotypes of pneumococcal polysaccharides Yuxi Walvax Biotechnology C., Ltd.

3. Pneumococcal polysaccharide serotype 4
3.1 Stability Test Results of batch 4-PVPV15101101 pneumococcal polysaccharide (storage condition:-20) Total Organic solvent residual
Test Nucleic Phosph Hexosa Sodium
Protein nitroge Bacterial
items Identity acid oruscon mineco Molecular size deoxychol
content n ether acetone ethanol endotoxin
and content tent ntent ate
Total content Phenol
standar

Solid KD■ residual
ds Positive
mg/ml 0.3recov ≤0.5EU/μg g/L
response to 4.0- KD≤0.1 ■ ■ ■ ■
≤3% ≤2% 0-1.5% ≥40% ery rate ,
Storage antiserum 6.0% 5 0.01% 0.03% 0.08% 0.2%
should ≤50EU/ml
time type 4
be≥75%
Not
0th
Complied 2.600 1.30% 0.20% 5.73% 0.20% 45.2% 0 ——— detecte 0.002% 0.018% Complied 0.110% ———
month
d
Not
1st year Complied 2.880 1.1% 0.5% 5.00% 0.20% 41.9% 0.12 ——— detecte 0.001 0.020 Complied 0.08% ———
d
Not
2nd Not
Complied 2.820 1.1% 0.3% 4.9% 0.2% 40% 0.08 ——— detecte 0.001 0.012 Complied 0.005
year detected
d
Not Not
3rd
Complied 2.800 1.2% 0.2% 4.6% 0.2% 41% 0.11 ——— detecte detecte 0.011 Complied 0.08446% 0.004
year
d d
4th
Complied 2.575 1% 0% 4.3% 0.2% 43% 0.04 95% 0% 0% 0.01% Complied 0% 0
year

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Stability data of 13 serotypes of pneumococcal polysaccharides Yuxi Walvax Biotechnology C., Ltd.

3.2 Stability Test Results of batch 4-PVPV15121102 pneumococcal polysaccharide (storage condition:-20■) ■
Total Organic solvent residual
Test Nucleic Phosph Hexosa Sodium
Protein nitroge Bacterial
items Identity acid oruscon mineco Molecular size deoxychol
content n ether acetone ethanol endotoxin
and content tent ntent ate
Total content Phenol
standar
Solid KD■ residual
ds Positive
mg/ml 0.3recov ≤0.5EU/μg g/L
response to 4.0- KD≤0.1 ■ ■ ■ ■
≤3% ≤2% 0-1.5% ≥40% ery rate ,
Storage antiserum 6.0% 5 0.01% 0.03% 0.08% 0.2%
should ≤50EU/ml
time type 4
be≥75%
Not Not
0th
Complied 3.140 1.3% 0.3% 5.73% 0.29% 46.4% 0.04 detecte detecte 0.006% Complied 0.16%
month
d d ———
Not Not
1st year Complied 2.720 1.3% 0.5% 4.45% 0.27% 44.7% 0.08 detecte detecte 0.004% Complied 0.19
d d ———
Not Not Not
2nd
Complied 2.265 1.6% 0.6% 4.3% 0.2% 42% 0.07 detecte detecte 0.001% Complied 0.08671% detecte
year
d d d
Not Not
3rd
Complied 2.770 1.3% 0.5% 4.7% 0.3% 44% 0.11 0.001 detecte 0.007% Complied 0.03630% detecte
year
d d
4th
Complied 2.890 2% 0% 4.4% 0.3% 43% 0.04 96% 0% 0% 0.01% Complied 0% 0.0128
year

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Stability data of 13 serotypes of pneumococcal polysaccharides Yuxi Walvax Biotechnology C., Ltd.

3.3 Stability Test Results of batch 4-PVPV16011103 pneumococcal polysaccharide (storage condition:-20■) ■
Total Organic solvent residual
Test Nucleic Phosph Hexosa Sodium
Protein nitroge Bacterial
items Identity acid oruscon mineco Molecular size deoxychol
content n ether acetone ethanol endotoxin
and content tent ntent ate
Total content Phenol
standar
Solid KD■ residual
ds Positive
mg/ml 0.3recov ≤0.5EU/μg g/L
response to 4.0- KD≤0.1 ■ ■ ■ ■
≤3% ≤2% 0-1.5% ≥40% ery rate ,
Storage antiserum 6.0% 5 0.01% 0.03% 0.08% 0.2%

should ≤50EU/ml
time type 4
be≥75%
Not Not
0th
Complied 2.710 2.7% 0.6% 5.83% 0.28% 42.0% 0 ——— detecte detecte 0.001% Complied 0.11% ———
month
d d
Not Not
Not
1st year Complied 1.975 1.5% 0.6% 4.81% 0.30% 45.5% 0.08 ——— detecte detecte 0.004% Complied ———
detected
d d
Not Not
2nd Not
Complied 1.980 1.3% 0.7% 4.8% 0.3% 41% 0.07 ——— detecte detecte 0.002% Complied 0.0015
year detected
d d
Not Not
3rd Not
Complied 2.640 1.0% 0.5% 5.2% 0.2% 42% 0.14 ——— detecte detecte 0.010% Complied 0.00165
year detected
d d
4th
Complied 1.785 2% 0% 4.3% 0.2% 43% 0.04 90% 0% 0% 0.01% Complied 0% 0.0016
year

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Stability data of 13 serotypes of pneumococcal polysaccharides Yuxi Walvax Biotechnology C., Ltd.

4. Pneumococcal polysaccharide serotype 5
4.1 Stability Test Results of batch 5-PVPV15101301 pneumococcal polysaccharide (storage condition:-20■) ■
Total Organic solvent residual
Test Nucleic Phosph Uronic Hexosa Sodium
Protein nitroge Bacterial
items Identity acid oruscon acid mineco Molecular size deoxychol
content n ether acetone ethanol endotoxin
and content tent content ntent ate
Total content Phenol
standar
Solid KD■ residual
ds Positive
mg/ml 0.6recov ≤0.5EU/μg g/L
response to ≤ 2.5-6.0 KD≤0 ■ ■ ■ ■
≤2% ≤2.0% ≥12% ≥20% ery rate ,
Storage antiserum 7.5% % .60 0.01% 0.03% 0.08% 0.2%
should ≤50EU/ml
time type 5
be≥70%
Not
0th
Complied 3.080 3.90% 1.70% 3.64% 0.20% 13.60% 21.70% 0.35 ——— detecte 0.0045% 0.020% Complied 0.13% ———
month
d
Not
1st year Complied 2.965 0.11% 0.04% 3.44% 0.01% 23.7% 20.7% 0.33 ——— detecte 0.001% 0.023% Complied 0.01%
———
d
Not
2nd Not
Complied 3.320 0.09% 0.03% 3.7% 0.0% 24% 23% 0.43 ——— detecte 0.002% 0.044% Complied 0.01813%
year detected
d
Not
3rd Not
Complied 2.980 0.1% 0.0% 4.2% 0.0% 22% 21% 0.44 ——— detecte 0.001% 0.041% Complied 0.01039%
year detected
d
4th
Complied 2.900 0.2% 0% 4.2% 0% 23% 21% 0.46 88% 0% 0% 0.02% Complied 0% 0.0096
year

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Stability data of 13 serotypes of pneumococcal polysaccharides Yuxi Walvax Biotechnology C., Ltd.

4.2 Stability Test Results of batch 5-PVPV15121302 pneumococcal polysaccharide (storage condition:-20■) ■
Total Organic solvent residual
Test Nucleic Phosph Uronic Hexosa Sodium
Protein nitroge Bacterial
items Identity acid oruscon acid mineco Molecular size deoxyc
content n ether acetone ethanol endotoxin
and content tent content ntent holate
Total content Phenol
standar
Solid KD■ residual
ds Positive
mg/ml 0.6recov ≤0.5EU/μg g/L
response to ≤ 2.5-6.0 KD≤0 ■ ■ ■ ■
≤2% ≤2.0% ≥12% ≥20% ery rate ,
Storage antiserum 7.5% % .60 0.01% 0.03% 0.08% 0.2%
should ≤50EU/ml
time type 5

be≥70%
Not Not
0th Not
Complied 3.185 6.40% 1.50% 5.90% 0.30% 13.3% 23.5% 0.39 ——— detecte detecte Complied 0.17% ———
month detected
d d
Not Not
Not
1st year Complied 2.290 0.13% 0.05% 5.72% 0.02% 22.7% 23.9% 0.33 ——— detecte 0.024% Complied detecte ———
detected
d d
Not Not Not
2nd Not
Complied 2.540 0.1% 0.1% 4.3% 0.0% 28% 22% 0.33 ——— detecte 0.018% Complied detecte detecte
year detected
d d d
Not Not Not
3rd Not
Complied 3.010 0.1% 0.0% 4.2% 0.0% 24% 22% 0.37 ——— detecte 0.019% Complied detecte detecte
year detected
d d d
4th
Complied 2.235 0.3% 0% 4.2% 0% 22% 22% 0.50 77% 0% 0% 0% Complied 0% 0.0306
year

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Stability data of 13 serotypes of pneumococcal polysaccharides Yuxi Walvax Biotechnology C., Ltd.

4.3 Stability Test Results of batch 5-PVPV16011303 pneumococcal polysaccharide (storage condition:-20■) ■
Total Organic solvent residual
Test Nucleic Phosph Uronic Hexosa Sodium
Protein nitroge Bacterial
items Identity acid oruscon acid mineco Molecular size deoxych
content n ether acetone ethanol endotoxin
and content tent content ntent olate
Total content Phenol
standar
Solid KD■ residual
ds Positive
mg/ml 0.6recov ≤0.5EU/μg g/L
response to ≤ 2.5-6.0 KD≤0 ■ ■ ■ ■
≤2% ≤2.0% ≥12% ≥20% ery rate ,
Storage antiserum 7.5% % .60 0.01% 0.03% 0.08% 0.2%
should ≤50EU/ml
time type 5
be≥70%
Not
0th Not 0.0045
Complied 3.090 4.5% 1.1% 3.82% 0.2% 14.8% 20.9% 0.35 ——— detecte Complied 0.12% ———
month detected %
d
Not
Not
1st year Complied 2.715 0.3% 0.1% 3.98% 0.02% 26.6% 23.1% 0.33 ——— detecte 0.012% Complied 0.10% ———
detected
d
Not Not
2nd
Complied 2.525 0.3% 0.0% 3.6% 0.0% 22% 21% 0.38 ——— detecte 0.0005% 0.026% Complied 0.07426% detecte
year
d d
Not Not
3rd Not
Complied 2.975 0.2% 0.1% 3.1% 0.0% 24% 21% 0.33 ——— detecte 0.021% Complied 0.03615% detecte
year detected
d d
4th
Complied 2.965 0.2% 0% 4.2% 0% 23% 21% 0.44 92% 0% 0% 0.02% Complied 0% 0.0032
year

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Stability data of 13 serotypes of pneumococcal polysaccharides Yuxi Walvax Biotechnology C., Ltd.

5. Pneumococcal polysaccharide serotype 6A
5.1 Stability Test Results of batch 6A-PVPV15090101 pneumococcal polysaccharide (storage condition:-20■) ■
Protei Total Organic solvent residual
Test Nucleic Phosph Methyl Sodium
n nitroge Bacterial
items Identity acid oruscon pentose Molecular size deoxyc
conten n ether acetone ethanol endotoxin
and content tent content holate
Total t content Phenol
standar
Solid KD■ residual
ds Positive
mg/ml 0.6recov ≤0.5EU/μg g/L
response to ≤ 0- 2.5- KD≤0 ■ ■ ■ ■
≤1% ≥15% ery rate ,
Storage antiserum type 1.5% 2.0% 5.0% .50 0.01% 0.03% 0.08% 0.2%
should ≤50EU/ml
time 6A

be≥50%
0th Not
Complied 2.020 0.36% 0.082% 1.3% 4.5% 23.7% 0.39 —— 0.0025% 0.0565% Complied 0% ——
month detected
Not
Not
1st year Complied 2.305 0.33% 0.05% 0.13% 4.21% 20.7% 0.48 —— 0.002% 0.035% Complied detecte ——
detected
d
2nd Not
Complied 2.370 0.38% 0.05% 0.30% 4.42% 24.0% 0.42 —— 0.002% 0.051% Complied 0.01% 0.0175
year detected
Not
3rd
Complied 2.370 0.28% 0.08% 0.3% 4.5% 23% 0.42 —— 0.002% 0.009% 0.050% Complied detecte 0.0319
year
d
4th
Complied 2.590 0% 0% 0.1% 4.0% 21% 0.46 78% 0% 0% 0.02% Complied 0% 0.0032
year

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Stability data of 13 serotypes of pneumococcal polysaccharides Yuxi Walvax Biotechnology C., Ltd.

5.2 Stability Test Results of batch 6A-PVPV15110102 pneumococcal polysaccharide (storage condition:-20■)

Protei Total Organic solvent residual
Test Nucleic Phosph Methyl Sodium
n nitroge Bacterial
items Identity acid oruscon pentose Molecular size deoxyc
conten n ether acetone ethanol endotoxin
and content tent content holate
Total t content Phenol
standar
Solid KD■ residual
ds Positive
mg/ml 0.6recov ≤0.5EU/μg g/L
response to ≤ 0- 2.5- KD≤0 ■ ■ ■ ■
≤1% ≥15% ery rate ,
Storage antiserum type 1.5% 2.0% 5.0% .50 0.01% 0.03% 0.08% 0.2%
should ≤50EU/ml
time 6A
be≥50%

0th Not Not
Complied 2.165 1.16% 0.08% 1.5% 4.17% 22.6% 0.39 —— 0.061% Complied 0.02% ——
month detected detected
Not Not
1st year Complied 2.125 1.0% 0.08% 0.47% 4.30% 20.9% 0.25 —— 0.048% Complied 0.11% ——
detected detected
Not
2nd Not Not
Complied 2.080 0.9% 0.2% 0.3% 4.1% 20% 0.38 —— 0.013% Complied detecte 0.0055%
year detected detected
d
Not
3rd Not Not
Complied 2.165 0.8% 0.1% 0.1% 4.1% 20% 0.32 —— 0.063% Complied detecte ——
year detected detected
d
4th
Complied 2.100 1% 0% 0.3% 4.2% 20% 0.31 91% 0% 0% 0.05% Complied 0% 0.0129
year

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Stability data of 13 serotypes of pneumococcal polysaccharides Yuxi Walvax Biotechnology C., Ltd.

5.3 Stability Test Results of batch 6A-PVPV15120103 pneumococcal polysaccharide (storage condition:-20■)

Protei Total Organic solvent residual
Test Nucleic Phosph Methyl Sodium
n nitroge Bacterial
items Identity acid oruscon pentose Molecular size deoxychola
conten n ether acetone ethanol endotoxin
and content tent content te
Total t content Phenol
standar
Solid KD■ residual
ds Positive
mg/ml 0.6recov ≤0.5EU/μg g/L
response to ≤ 0- 2.5- KD≤0 ■ ■ ■ ■
≤1% ≥15% ery rate ,
Storage antiserum type 1.5% 2.0% 5.0% .50 0.01% 0.03% 0.08% 0.2%
should ≤50EU/ml
time 6A
be≥50%
0th Not
Complied 2.825 1.3% 0.1% 1.4% 4.17% 23.6% 0.39 —— 0.002% 0.026% Complied 0% ——
month detected
Not Not
1st year Complied 2.315 0.87% 0.16% 0.35% 4.03% 21.4% 0.29 —— 0.043% Complied 0.16% ——
detected detected

2nd Not Not Not
Complied 2.065 0.9% 0.2% 0.5% 4.0% 19% 0.29 ——— 0.014% Complied 0.019
year detected detected detected
3rd Not Not
Complied 1.885 1.1% 0.1% 0.2% 4.3% 21% 0.32 ——— 0.046% Complied 0.01631% ———
year detected detected
4th
Complied 1.430 1% 0% 0.4% 4.5% 19% 0.31 96% 0% 0% 0.02% Complied 0% 0.0290
year

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Stability data of 13 serotypes of pneumococcal polysaccharides Yuxi Walvax Biotechnology C., Ltd.

6. Pneumococcal polysaccharide serotype 6B
6.1 Stability Test Results of batch 6B-PVPV15090201 pneumococcal polysaccharide (storage condition:-20■)

Test Nuclei Total Organic solvent residual
Phosph Methyl Sodium
items Protein c acid nitroge Bacterial
Identity oruscon pentose Molecular size deoxyc
and content conten n ether acetone ethanol endotoxin
Total tent content holate Phenol
standar t content
Solid residual
ds KD■
Positive response mg/ml ≤0.5EU/μg g/L
0- 2.5- KD≤0 0.6recovery ■ ■ ■ ■
to antiserum type ≤2% ≤2% ≥15% ,
Storage 2.0% 5.0% .50 rate should 0.01% 0.03% 0.08% 0.2%
6B ≤50EU/ml
time be≥85%
Not
0th
Complied 2.505 0.10% 0.10% 0.96% 4.40% 23.40% 0.35 ——— detecte 0.002% 0.066% Complied 0.050% ———
month
d
Not Not Not
1st year Complied 2.145 0.20% 0.05% 0.05% 4.09% 20.8% 0.35 ——— detecte detecte 0.040% Complied detecte ———
d d d
Not
2nd
Complied 1.900 0.2% 0.1% 0.11% 4.38% 26.2% 0.29 ——— detecte 0.001% 0.074% Complied 0.004% 0.013
year
d
Not Not Not
3rd Not
Complied 2.250 0.1% 0.0% 0.2% 4.5% 22% 0.42 ——— detecte detecte 0.005% Complied detecte
year detected
d d d
4th
Complied 2.610 0% 0% 0% 3.5% 20% 0.39 90% 0% 0% 0.02% Complied 0% 0.0146
year

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Stability data of 13 serotypes of pneumococcal polysaccharides Yuxi Walvax Biotechnology C., Ltd.

6.2 Stability Test Results of batch 6B-PVPV15110202 pneumococcal polysaccharide (storage condition:-20■)

Test Nuclei Total Organic solvent residual
Phosph Methyl Sodium
items Protein c acid nitroge Bacterial
Identity oruscon pentose Molecular size deoxyc
and content conten n ether acetone ethanol endotoxin
Total tent content holate Phenol
standar t content
Solid residual
ds KD■
Positive response mg/ml ≤0.5EU/μg g/L
0- 2.5- KD≤0 0.6recovery ■ ■ ■ ■
to antiserum type ≤2% ≤2% ≥15% ,
Storage 2.0% 5.0% .50 rate should 0.01% 0.03% 0.08% 0.2%
6B ≤50EU/ml
time be≥85%
Not
0th
Complied 2.105 0.40% 0.10% 1.09% 4.40% 23.80% 0.3 ——— detecte 0.004% 0.064% Complied 0.110% ———
month
d
Not
1st year Complied 2.315 0.4% 0.1% 0.26% 4.35% 21.6% 0.4 ——— detecte 0.001% 0.032% Complied 0.12% ———
d
Not Not Not
2nd
Complied 2.400 0.4% 0.2% 0.4% 4.1% 23% 0.29 ——— detecte detecte 0.008% Complied detecte 0.005
year
d d d
Not Not Not
3rd
Complied 2.555 0.5% 0.1% detecte 4.3% 22% 0.28 ——— detecte 0.002% 0.016% Complied detecte 0.0041
year
d d d
4th

Complied 2.120 0% 0% 0.2% 4.4% 21% 0.31 93% 0% 0% 0.01% Complied 0% 0.0016 year

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Stability data of 13 serotypes of pneumococcal polysaccharides Yuxi Walvax Biotechnology C., Ltd.

6.3 Stability Test Results of batch 6B-PVPV15120203 pneumococcal polysaccharide (storage condition:-20■)

Test Nuclei Total Organic solvent residual
Phosph Methyl Sodium
items Protein c acid nitroge Bacterial
Identity oruscon pentose Molecular size deoxychola
and content conten n ether acetone ethanol endotoxin
Total tent content te Phenol
standar t content
Solid residual
ds KD■
Positive response mg/ml ≤0.5EU/μg g/L
0- 2.5- KD≤0.6recovery ■ ■ ■ ■
to antiserum type ≤2% ≤2% ≥15% ,
Storage 2.0% 5.0% .50 rate should 0.01% 0.03% 0.08% 0.2%
6B ≤50EU/ml
time be≥85%
Not
0th 0.0275
Complied 2.87 1.40% 0.10% 0.73% 4.10% 23.50% 0.3 ——— detecte 0.002% Complied 0.180% ———
month %
d
Not Not
1st year Complied 2.865 0.9% 0.1% 0.31% 4.14% 21.7% 0.19 ——— detecte detecte 0.027% Complied 0.15% ———
d d
Not Not
2nd Not Not
Complied 2.655 0.9% 0.1% 0.4% 4.1% 20% 0.21 ——— detecte detecte 0.007% Complied
year detected detected
d d
Not
3rd
Complied 3.080 0.8% 0.1% 0.3% 4.3% 21% 0.30 ——— detecte 0.001% 0.034% Complied 0.02623% 0.0082
year
d
4th
Complied 2.885 1% 0% 0.3% 4.5% 20% 0.35 94 0% 0% 0.02% Complied 0% 0.0144
year

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Stability data of 13 serotypes of pneumococcal polysaccharides Yuxi Walvax Biotechnology C., Ltd.

7. Pneumococcal polysaccharide serotype 7F
7.1 Stability Test Results of batch 7F-PVPV15100701 pneumococcal polysaccharide (storage condition:-20■)

Nucle Organic solvent residual
Protei Total
Test ic Phosph Methyl Sodium
n nitroge Bacterial
items Identity acid oruscon pentose Molecular size deoxychol
conte n ether acetone ethanol endotoxin
and conte tent content ate
Total nt content Phenol
standard nt
Solid residual
s KD■
Positive mg/ml g/L
0.3recov ≤0.5EU/μg
response to 1.5- 0- KD≤0 ■ ■ ■ ■
Storage ≤5% ≤2% ≥13% ery rate ,
antiserum type 4.0% 1.0% .20 0.01% 0.03% 0.08% 0.2%
time should ≤50EU/ml
7F
be≥70%
Not
0th
Complied 2.830 0.2% 0.4% 3.07% 0.1% 19.4% 0.15 ——— detecte 0.001% 0.022% Complied 0.04% ———
month
d
Not
Not
1st year Complied 2.915 0.7% 0.3% 2.16% 0.05% 19.2% 0.2 ——— detecte 0.0005% 0.022% Complied ———
detected
d
Not
Not
2nd year Complied 3.020 0.6% 0.3% 2.09% 0.07% 21.4% 0.13 ——— detecte 0.009% Complied 0.01% 0.0130
detected
d
Not
Not
3rd year Complied 3.375 0.6% 0.3% 2.2% 0.1% 19% 0.19 ——— detecte 0.005% Complied 0.002770% 0.008
detected
d
4th year Complied 2.175 1% 0% 2.5% 0.1% 20% 0.18 84% 0% 0% 0.01% Complied 0% 0.0097

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Stability data of 13 serotypes of pneumococcal polysaccharides Yuxi Walvax Biotechnology C., Ltd.

7.2 Stability Test Results of batch 7F-PVPV15110702 pneumococcal polysaccharide (storage condition:-20■)

■
Nucle Organic solvent residual
Protei Total
Test ic Phosph Methyl Sodium
n nitroge Bacterial
items Identity acid oruscon pentose Molecular size deoxychol
conte n ether acetone ethanol endotoxin
and conte tent content ate
Total nt content Phenol
standard nt
Solid residual
s KD■
Positive mg/ml g/L
0.3recov ≤0.5EU/μg
response to 1.5- 0- KD≤0 ■ ■ ■ ■
Storage ≤5% ≤2% ≥13% ery rate ,
antiserum type 4.0% 1.0% .20 0.01% 0.03% 0.08% 0.2%
time should ≤50EU/ml
7F
be≥70%
Not
0th 0.014
Complied 3.430 1.2% 0.2% 3.38% 0.5% 19.0% 0.12 —— detecte 0.002% Complied 0.12% ——
month %
d
Not
0.010 Not
1st year Complied 2.830 1.5% 0.2% 2.69% 0.5% 18.0% 0.15 —— detecte 0.002% Complied ——
% detected
d
Not
Not 0.006 Not
2nd year Complied 2.855 1.4% 0.2% 2.9% 0.4% 17% 0.13 —— detecte Complied 0.0035
detected % detected
d
0.001 0.011 Not
3rd year Complied 2.565 1.6% 0.3% 2.5% 0.4% 18% 0.14 —— 0.001% Complied 0.00815
% % detected
4th year Complied 2.995 2% 0% 2.5% 0.4% 16% 0.11 93% 0% 0% 0.01% Complied 0% 0

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Stability data of 13 serotypes of pneumococcal polysaccharides Yuxi Walvax Biotechnology C., Ltd.

7.3 Stability Test Results of batch 7F-PVPV16010703 pneumococcal polysaccharide (storage condition:-20■)

■
Nucle Organic solvent residual
Protei Total
Test ic Phosph Methyl Sodium
n nitroge Bacterial
items Identity acid oruscon pentose Molecular size deoxychol
conte n ether acetone ethanol endotoxin
and conte tent content ate
Total nt content Phenol
standard nt
Solid residual
s KD■
Positive mg/ml g/L
0.3recov ≤0.5EU/μg
response to 1.5- 0- KD≤0 ■ ■ ■ ■
Storage ≤5% ≤2% ≥13% ery rate ,
antiserum type 4.0% 1.0% .20 0.01% 0.03% 0.08% 0.2%
time should ≤50EU/ml
7F
be≥70%
Not
0th Not
Complied 2.585 1.1% 0.2% 3.87% 0.40% 17.3% 0 —— detecte 0.005% Complied 0.05% ——
month detected
d
Not
Not Not
1st year Complied 2.425 2.0% 0.3% 2.39% 0.41% 16.6% 0.2 —— detecte 0.011% Complied ——
detected detected
d
Not
Not
2nd year Complied 2.870 1.6% 0.2% 2.6% 0.3% 15% 0.19 —— detecte 0.006% Complied 0.06672% 0.0105
detected
d
Not
Not Not
3rd year Complied 1.850 2.4% 0.2% 2.4% 0.5% 19% 0.18 —— detecte 0.024% Complied 0.0098
detected detected
d
4th year Complied 2.450 1% 0% 2.2% 0.4% 16% 0.19 87% 0% 0% 0.02% Complied 0.02% 0.0209

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Stability data of 13 serotypes of pneumococcal polysaccharides Yuxi Walvax Biotechnology C., Ltd.

8. Pneumococcal polysaccharide serotype 9V
8.1 Stability Test Results of batch 9V-PVPV15090301 pneumococcal polysaccharide (storage condition:-20■)
■
Nucle Organic solvent residual
Protei Total
Test ic Phosph Uronic Hexosa Sodium
n nitroge Bacterial
items Identity acid oruscon acid mineco Molecular size deoxyc
conte n ether acetone ethanol endotoxin
and conte tent content ntent holate
Total nt content Phenol
standar nt
Solid residual
ds KD■
mg/ml g/L
Positive response 0.6recov ≤0.5EU/μg
0.5-3.0 KD≤0 ■ ■ ■ ■ ■
Storage to antiserum type ≤2% ≤2% 0-1.0% ≥15% ≥13% ery rate ,
% .45 0.01% 0.03% 0.08% 0.2%
time 9V should ≤50EU/ml
be≥80%
Not Not
0th
Complied 3.075 0.4% 0.2% 1.95% 0.09% 22.2% 13.0% 0.35 ——— detecte detecte 0.007% Complied 0.04% ———
month
d d
Not Not
1st year Complied 2.885 0.2% 0.1% 1.32% 0.10% 21.9% 14.2% 0.25 ——— detecte detecte 0.007% Complied 0.03%
———
d d
Not Not Not
2nd
Complied 3.000 0.2% 0.2% 1.47% 0.11% 22.0% 14.7% 0.25 ——— detecte detecte 0.004% Complied detecte 0.006
year
d d d
Not Not Not
3rd
Complied 3.125 0.1% 0.2% 1.5% 0.1% 22% 14% 0.42 ——— detecte detecte 0.007% Complied detecte 0.0159
year
d d d
4th
Complied 1.925 0% 0% 2.3% 0.2% 22% 14% 0.35 97% 0% 0% 0.02% Complied 0% 0.0065
year

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Stability data of 13 serotypes of pneumococcal polysaccharides Yuxi Walvax Biotechnology C., Ltd.

8.2 Stability Test Results of batch 9V-PVPV15110302 pneumococcal polysaccharide (storage condition:-20■)
■
Nucle Organic solvent residual
Test Protei Total
ic Phosph Uronic Hexosa Sodium
items n nitroge Bacterial
Identity acid oruscon acid mineco Molecular size deoxychol
and conte n ether acetone ethanol endotoxin
conte tent content ntent e
standar Total nt content Phenol
nt
ds Solid residual
KD■
mg/ml g/L
Positive response 0.6recov ≤0.5EU/μg
0.5-3.0 KD≤0 ■ ■ ■ ■ ■
to antiserum type ≤2% ≤2% 0-1.0% ≥15% ≥13% ery rate ,
Storage % .45 0.01% 0.03% 0.08% 0.2%
9V should ≤50EU/ml
time
be≥80%
Not Not
0th 0.05
Complied 2.920 0.3% 2.36% 0.03% 26.1% 14.6% 0.30 ——— detecte detecte 0.020% Complied 0.14% ———
month %
d d
Not Not
Not
1st year Complied 3.065 0.2% 0.1% 1.53% 0.04% 26.3% 14.3% 0.25 ——— detecte detecte 0.012% Complied ———
detected
d d
Not Not
2nd Not
Complied 2.770 0.1% 0.1% 1.6% 0.0% 25% 14% 0.21 ——— detecte detecte 0.005% Complied 0.004
year detected
d d
Not Not
3rd 0.0006056
Complied 3.220 0.2% 0.1% 1.3% 0.0% 25% 14% 0.28 ——— detecte detecte 0.013% Complied 0.0033
year %
d d
4th
Complied 2.990 0% 0% 1.2% 0% 23% 14% 0.35 97% 0% 0% 0.01% Complied 0% 0.0080
year

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Stability data of 13 serotypes of pneumococcal polysaccharides Yuxi Walvax Biotechnology C., Ltd.

8.3 Stability Test Results of batch 9V-PVPV15120303 pneumococcal polysaccharide (storage condition:-20■)

■

Nucle Organic solvent residual
Test Protei Total
ic Phosph Uronic Hexosa Sodium
items n nitroge Bacterial
Identity acid oruscon acid mineco Molecular size deoxychol
and conte n ether acetone ethanol endotoxin
conte tent content ntent ate
standar Total nt content Phenol
nt
ds Solid residual
KD■
mg/ml g/L
Positive response 0.6recov ≤0.5EU/μg
0.5-3.0 KD≤0 ■ ■ ■ ■
to antiserum type ≤2% ≤2% 0-1.0% ≥15% ≥13% ery rate ,
Storage % .45 0.01% 0.03% 0.08% 0.2%
9V should ≤50EU/ml
time
be≥80%
Not Not
0th 0.06
Complied 2.625 0.3% 2.48% 0.07% 24.3% 15.7% 0.30 ——— detecte 0.001% detecte Complied 0.180% ———
month %
d d
Not Not
1st year Complied 3.145 0.2% 0.1% 1.49% 0.03% 23.8% 13.7% 0.24 ——— detecte detecte 0.009 Complied 0.001%
———
d d
Not Not
2nd Not
Complied 3.320 0.2% 0.1% 0.80% 0.10% 22% 14% 0.25 ——— detecte detecte 0.003 Complied 0.0095
year detected
d d
Not Not
3rd
Complied 2.800 0.2% 0.1% 1.40% 0.00% 25% 18% 0.32 ——— detecte detecte 0.011 Complied 0.01071% 0.0049
year
d d
4th
Complied 2.920 0% 0% 0.8% 0% 24% 14% 0.35 99% 0% 0% 0.01% Complied 0% 0.0096
year

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Stability data of 13 serotypes of pneumococcal polysaccharides Yuxi Walvax Biotechnology C., Ltd.

9. Pneumococcal polysaccharide serotype 14

9.1 Stability Test Results of batch 14-PVPV15101001 pneumococcal polysaccharide (storage condition:-20■)

■

Nucle Organic solvent residual
Protei Total
Test ic Phosph Hexosa Sodium
n nitroge Bacterial
items Identity acid oruscon mineco Molecular size deoxycho
conten n ether acetone ethanol ■■
and conte tent ntent late
Total t content Phenol
standard nt
Solid residual
s KD■
Positive mg/ml g/L
0.3recov
response to 1.5- KD≤0 ■ ■ ■ ■≤0.5EU/μg ■
Storage ≤5% ≤2% 0-1.0% ≥20% ery rate
antiserum type 4.0% .30 0.01% 0.03% 0.08% ,≤50EU/ml 0.2%
time should
14
be≥55%
Not
0th Not
Complied 3.085 0.8% 1.2% 2.82% 0.20% 22.2% 0.19 ——— detecte 0.009% Complied 0.06% ———
month detected
d
Not
Not
1st year Complied 2.455 4.7% 0.1% 2.57% 0.35% 26.3% 0.11 ——— detecte 0.023% Complied 0.08% ———
detected
d
Not
2nd Not
Complied 2.550 3.9% 0.2% 2.4% 0.3% 26% 0.07 ——— detecte 0.0002% 0.020% Complied 0.004
year detected
d
Not
Not Not
3rd year Complied 2.635 3.6% 0.1% 2.2% 0.4% 26% 0.14 ——— detecte 0.017% Complied 0.07391%
detected detected

d
4th year Complied 1.845 4% 0% 2.1% 0.3% 25% 0.08 77% 0% 0% 0.02% Complied 0% 0.01455

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Stability data of 13 serotypes of pneumococcal polysaccharides Yuxi Walvax Biotechnology C., Ltd.

9.2 Stability Test Results of batch 14-PVPV15121002 pneumococcal polysaccharide (storage condition:-20■)
■
Nucle Organic solvent residual
Protei Total
Test ic Phosph Hexosa Sodium
n nitroge Bacterial
items Identity acid oruscon mineco Molecular size deoxycho
conten n ether acetone ethanol ■■
and conte tent ntent late
Total t content Phenol
standard nt
Solid residual
s KD■
Positive mg/ml g/L
0.3recov
response to 1.5- KD≤0 ■■■ ≤0.5EU/μg ■
Storage ≤5% ≤2% 0-1.0% ≥20% ery rate
antiserum type 4.0% .30 0.01% 0.03% 0.08% ,≤50EU/ml 0.2%
time should
14
be≥55%
Not
0th Not 0.0095
Complied 2.385 1.9% 0.1% 3.35% 0.30% 23.6% 0.08 ——— detecte Complied 0.1800% ———
month detected %
d
Not Not
Not
1st year Complied 2.940 1.9% 0.7% 2.35% 0.34% 23.4% 0.11 ——— detecte detecte Complied 0.08% ———
detected
d d
Not
2nd Not Not
Complied 2.550 1.8% 0.6% 2.3% 0.3% 23% 0.29 ——— detecte 0.003% Complied 0.07906%
year detected detected
d
Not
Not Not
3rd year Complied 3.050 1.4% 0.7% 2.0% 0.3% 22% 0.27 ——— detecte 0.003% Complied 0.07644%
detected detected
d
4th year Complied 2.925 2% 1% 2.2% 0.4% 23% 0.14 77% 0% 0% 0% Complied 0.04% 0.0080

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Stability data of 13 serotypes of pneumococcal polysaccharides Yuxi Walvax Biotechnology C., Ltd.

9.3 Stability Test Results of batch 14-PVPV16011003 pneumococcal polysaccharide (storage condition:-20■)
■
Nucle Organic solvent residual
Protei Total
Test ic Phosph Hexosa Sodium
n nitroge Bacterial
items Identity acid oruscon mineco Molecular size deoxycho
conten n ether acetone ethanol ■■
and conte tent ntent late
Total t content Phenol
standard nt
Solid residual
s KD■
Positive mg/ml g/L
0.3recov
response to 1.5- KD≤0 ■■■ ≤0.5EU/μg ■
Storage ≤5% ≤2% 0-1.0% ≥20% ery rate
antiserum type 4.0% .30 0.01% 0.03% 0.08% ,≤50EU/ml 0.2%
time should
14
be≥55%
Not
0th Not 0.0045
Complied 2.750 1.5% 0.7% 3.02% 0.30% 22.9% 0 ——— detecte Complied 0.0900% ———
month detected %
d
Not
Not Not
1st year Complied 2.760 3.1% 0.7% 2.43% 0.36% 24.0% 0.14 ——— detecte 0.001% Complied ———
detected detected
d
Not
2nd Not
Complied 2.655 1.7% 0.7% 2.3% 0.4% 24% 0.14 ——— detecte 0.002% Complied 0.09849% 0.019
year detected
d
Not
Not
3rd year Complied 2.810 1.6% 0.7% 2.1% 0.3% 25% 0.09 ——— detecte 0.001% Complied 0.05480% 0.01965
detected

d
4th year Complied 2.815 1% 1% 2.2% 0.3% 25% 0.15 82% 0% 0% 0% Complied 0% 0.0257

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Stability data of 13 serotypes of pneumococcal polysaccharides Yuxi Walvax Biotechnology C., Ltd.

10. Pneumococcal polysaccharide serotype 18C
10.1 Stability Test Results of batch 18C-PVPV15100801 pneumococcal polysaccharide (storage condition:-20■) ■
Nucle Total Organic solvent residual
Protei
Test ic nitrog Phosph Methyl Sodium
n Bacterial
items Identity acid en oruscon pentose Molecular size deoxycho
conte ether acetone ethanol endotoxin
and conte conten tent content late
Total nt Phenol
standar nt t
Solid residual
ds KD■
Positive mg/ml g/L
0.3recov ≤0.5EU/μg
response to 0- 2.4- KD≤0 ■ ■ ■ ■
Storage ≤3% ≤2% ≥14% ery rate ,
antiserum type 1.0% 4.9% .15 0.01% 0.03% 0.08% 0.2%
time should ≤50EU/ml
18C
be≥70%
Not
0th Not
Complied 2.725 1.0% 0.2% 0.92% 3.00% 14.5% 0.07 ——— detecte 0.020% Complied 0.090% ———
month detected
d
Not
Not Not
1st year Complied 2.915 0.5% 0.1% 0.34% 2.91% 15.6% 0.12 ——— detecte 0.017% Complied ———
detected detected
d
Not
2nd Not Not
Complied 2.815 0.4% 0.1% 0.21% 3.01% 15.1% 0.03 ——— detecte 0.010% Complied 0.0075
year detected detected
d
Not
3rd Not
Complied 2.795 0.5% 0.1% 0.2% 3.1% 17% 0.14 ——— detecte 0.011% Complied 0.04174% 0.0064
year detected
d
4th
Complied 2.865 1% 0% 0.2% 2.9% 15% 0.04 89% 0% 0% 0.01% Complied 0% 0
year

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Stability data of 13 serotypes of pneumococcal polysaccharides Yuxi Walvax Biotechnology C., Ltd.

10.2 Stability Test Results of batch 18C-PVPV15110802 pneumococcal polysaccharide (storage condition:-20■) ■
Nucle Total Organic solvent residual
Protei
Test ic nitrog Phosph Methyl Sodium
n Bacterial
items Identity acid en oruscon pentose Molecular size deoxycho
conte ether acetone ethanol endotoxin
and conte conten tent content late
Total nt Phenol
standar nt t
Solid residual
ds KD■
Positive mg/ml g/L
0.3recov ≤0.5EU/μg
response to 0- 2.4- KD≤0 ■ ■ ■ ■
Storage ≤3% ≤2% ≥14% ery rate ,
antiserum type 1.0% 4.9% .15 0.01% 0.03% 0.08% 0.2%
time should ≤50EU/ml
18C
be≥70%
0th Not
Complied 2.130 0.8% 0.2% 0.89% 2.80% 17.5% 0.08 ——— 0.002% 0.034% Complied 0.060% ———
month detected
Not
Not Not
1st year Complied 3.630 0.6% 0.1% 0.28% 3.11% 15.7% 0.07 ——— detecte 0.001% Complied ———
detected detected
d
Not
2nd Not
Complied 3.415 0.6% 0.2% 0.3% 2.9% 14% 0.10 ——— detecte 0.010% Complied 0.03540% 0.0005
year detected
d
Not
3rd Not Not Not
Complied 3.345 0.6% 0.2% 0.2% 2.9% 17% 0.04 ——— detecte 0.005% Complied

year detected detected detected
d
4th
Complied 2.880 1% 0% 0.3% 3.1% 16% 0.07 92% 0% 0% 0% Complied 0% 0.0113
year

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Stability data of 13 serotypes of pneumococcal polysaccharides Yuxi Walvax Biotechnology C., Ltd.

10.3 Stability Test Results of batch 18C-PVPV16010803 pneumococcal polysaccharide (storage condition:-20■) ■
Nucle Total Organic solvent residual
Protei
Test ic nitrog Phosph Methyl Sodium
n Bacterial
items Identity acid en oruscon pentose Molecular size deoxycho
conte ether acetone ethanol endotoxin
and conte conten tent content late
Total nt Phenol
standar nt t
Solid residual
ds KD■
Positive mg/ml g/L
0.3recov ≤0.5EU/μg
response to 0- 2.4- KD≤0 ■ ■ ■ ■
Storage ≤3% ≤2% ≥14% ery rate ,
antiserum type 1.0% 4.9% .15 0.01% 0.03% 0.08% 0.2%
time should ≤50EU/ml
18C
be≥70%
0th Not
Complied 1.990 1.3% 0.2% 0.95% 2.70% 14.6% 0.08 ——— 0.002% 0.046% Complied 0.110% ———
month detected
Not
Not
1st year Complied 2.360 0.9% 0.2% 0.42% 2.95% 14.4% 0.04 ——— 0.001% detecte 0.023% Complied ———
detected
d
Not
2nd Not Not
Complied 2.680 0.9% 0.2% 0.4% 2.5% 15% 0.03 ——— detecte 0.010% Complied 0.006
year detected detected
d
Not
3rd Not Not
Complied 2.065 1.0% 0.3% 0.4% 3.2% 15% 0.04 ——— detecte 0.029% Complied 0.0057
year detected detected
d
4th
Complied 2.310 1% 0% 0.4% 2.7% 15% 0.04 96% 0% 0% 0.03% Complied 0% 0.0016
year

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Stability data of 13 serotypes of pneumococcal polysaccharides Yuxi Walvax Biotechnology C., Ltd.

11. Pneumococcal polysaccharide serotype 19A
11.1 Stability Test Results of batch 19A-PVPV15090401 pneumococcal polysaccharide (storage condition:-20■) ■
Identity Nucle Organic solvent residual
Test Protei Total Sodi
ic Phosph Hexosa Methyl
items n nitroge Bacterial m
acid oruscon mineco pentose Molecular size acetone
and conte n ether ethanol endotoxin deoxyc
conte tent ntent content e
standar Positive Total nt content holate Phenol
nt
ds response to Solid residual
KD■
antiserum type mg/ml g/L
0.5recov ≤0.5EU/μg
19A 0.6- 3.0- KD≤0 ■ ■ ■ ■
≤2% ≤2% ≥12% ≥20% ery rate ,
Storage 3.5% 7.0% .45 0.01% 0.03% 0.08% 0.2%
should ≤50EU/ml
time
be≥80%
Not
0th 0.002
Complied 2.210 0.4% 0.3% 3.26% 4.90% 12.6% 29.0% 0.18 ——— detecte 0.015% Complied 0.07% ———
month %
d
Not Not
1st year Complied 2.875 0.3% 0.3% 2.05% 4.78% 13.4% 26.4% 0.08 ——— detecte detecte 0.008% Complied 0.07%
———
d d
Not Not
2nd 0.001
Complied 3.425 0.3% 0.4% 2.28% 4.69% 13.1% 29.3% 0.07 ——— detecte 0.014% Complied detecte 0.0095
year %
d d
Not Not Not

3rd
Complied 3.075 0.3% 0.3% 2.2% 4.9% 13% 26% 0.30 ——— detecte detecte 0.008% Complied detecte 0.0159
year
d d d
4th
Complied 3.235 0% 1.4% 0% 4.8% 14% 24% 0.08 99% 0% 0% 0.01% Complied 0% 0.0048
year

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Stability data of 13 serotypes of pneumococcal polysaccharides Yuxi Walvax Biotechnology C., Ltd.

11.2 Stability Test Results of batch 19A-PVPV15110402 pneumococcal polysaccharide (storage condition:-20■) ■
Identity Nucle Organic solvent residual
Test Protei Total Sodi
ic Phosph Hexosa Methyl
items n nitroge Bacterial m
acid oruscon mineco pentose Molecular size acetone
and conte n ether ethanol endotoxin deoxyc
conte tent ntent content e
standar Positive Total nt content holate Phenol
nt
ds response to Solid residual
KD■
antiserum type mg/ml g/L
0.5recov ≤0.5EU/μg
19A 0.6- 3.0- KD≤0 ■ ■ ■ ■ ■
≤2% ≤2% ≥12% ≥20% ery rate ,
Storage 3.5% 7.0% .45 0.01% 0.03% 0.08% 0.2%
should ≤50EU/ml
time
be≥80%
Not
0th 0.02 0.002 0.004
Complied 2.445 0.7% 3.31% 4.7% 13.7% 31.3% 0.04 ——— detecte 0.0145 Complied ———
month % % %
d
Not Not
0.05
1st year Complied 2.885 0.7% 2.43% 4.77% 12.5% 24.6% 0.07 ——— detecte detecte 0.009% Complied 0.09% ———
%
d d
Not Not Not
2nd
Complied 2.410 0.7% 0.1% 2.4% 4.8% 13% 25% 0.03 ——— detecte detecte 0.003% Complied detecte 0.0065
year
d d d
Not Not
3rd 0.001
Complied 2.710 0.7% 0.1% 2.3% 4.9% 13% 26% 0.00 ——— detecte 0.007% Complied detecte 0.0082
year %
d d
4th
Complied 3.270 1% 2.0% 0% 4.2% 13% 23% 0 99% 0% 0% 0.01% Complied 0% 0.0161
year

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Stability data of 13 serotypes of pneumococcal polysaccharides Yuxi Walvax Biotechnology C., Ltd.

11.3 Stability Test Results of batch 19A-PVPV15120403 pneumococcal polysaccharide (storage condition:-20■) ■
Identity Nucle Organic solvent residual
Test Protei Total Sodi
ic Phosph Hexosa Methyl
items n nitroge Bacterial m
acid oruscon mineco pentose Molecular size acetone
and conte n ether ethanol endotoxin deoxyc
conte tent ntent content e
standar Positive Total nt content holate Phenol
nt
ds response to Solid residual
KD■
antiserum type mg/ml g/L
0.5recov ≤0.5EU/μg
19A 0.6- 3.0- KD≤0 ■ ■ ■ ■ ■
≤2% ≤2% ≥12% ≥20% ery rate ,
Storage 3.5% 7.0% .45 0.01% 0.03% 0.08% 0.2%
should ≤50EU/ml
time
be≥80%
0th 0.05 0.002
Complied 2.850 0.5% 3.44% 4.9% 15.0% 33.8% 0.04 ——— 0.001% 0.005% Complied 0.15% ———
month % %
Not Not
0.05
1st year Complied 3.430 0.4% 2.04% 4.20% 12.4% 23.5% 0.04 ——— detecte detecte 0.009% Complied 0.19% ———
%
d d
Not Not Not
2nd
Complied 3.090 0.5% 0.0% 2.2% 4.6% 14% 24% 0.11 ——— detecte detecte 0.003% Complied detecte 0.016
year

d d d
Not Not
3rd 0.001
Complied 2.945 0.5% 0.0% 2.1% 5.0% 14% 24% 0.23 ——— detecte 0.032% Complied detecte 0.0049
year %
d d
4th
Complied 2.065 1% 1.3% 0% 4.9% 14% 24% 0.15 97% 0% 0% 0% Complied 0% 0.0129
year

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Stability data of 13 serotypes of pneumococcal polysaccharides Yuxi Walvax Biotechnology C., Ltd.

12. Pneumococcal polysaccharide serotype 19F
12.1 Stability Test Results of batch 19F-PVPV15090501 pneumococcal polysaccharide (storage
condition:-20■) ■
Identity Protei Total Organic solvent residual
Test Nucleic Phosph Hexosa Methyl Sodium
n nitroge Bacterial
items acid oruscon mineco pentose Molecular size aceton deoxychola
conte n ether ethanol endotoxin
and content tent ntent content e te
Positive Total nt content Phenol
standar
response to Solid KD■ residual
ds
mg/ml 0.3recov ≤0.5EU/μg g/L
antiserum type 1.4-3.5 3.0- ≥ KD≤0. ■ ■ ■ ■
19F ≤3% ≤2% ≥20% ery rate ,
Storage % 5.5% 12.5% 20 0.01% 0.03% 0.08% 0.2%
should ≤50EU/ml
time
be≥75%
Not
0th 0.002
Complied 2.750 0.4% 0.1% 3.27% 4.60% 19.10% 28.1% 0.04 ——— detecte 0.013% Complied 0.10% ———
month %
d
Not Not
Not
1st year Complied 2.570 0.4% 0.04% 2.37% 5.35% 13.22% 27.1% 0.12 ——— detecte detecte 0.022% Complied ———
detected
d d
Not Not
2nd
Complied 2.645 0.3% 0.2% 2.19% 5.12% 15.14% 31.4% 0.14 ——— detecte detecte 0.001% Complied 0.02% 0.0115
year
d d
Not Not
3rd
Complied 2.735 0.3% 0.1% 2.0% 5.0% 14% 26% 0.07 ——— detecte detecte 0.023% Complied 0.007057% 0.00955
year
d d
4th
Complied 2.565 0% 0% 2.1% 5.1% 13.3% 29% 0.04 94% 0% 0% 0.03% Complied 0% 0.0146
year

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Stability data of 13 serotypes of pneumococcal polysaccharides Yuxi Walvax Biotechnology C., Ltd.

12.2 Stability Test Results of batch 19F-PVPV15110502 pneumococcal polysaccharide (storage
condition:-20■) ■
Identity Protei Total Organic solvent residual
Test Nucleic Phosph Hexosa Methyl Sodium
n nitroge Bacterial
items acid oruscon mineco pentose Molecular size deoxychol
conte n ether acetone ethanol endotoxin
and content tent ntent content ate Phenol
Positive Total nt content
standar residua
response to Solid KD■
ds l
mg/ml 0.3recov ≤0.5EU/μg
antiserum type 1.4-3.5 3.0- ≥ KD≤0. ■ ■ ■ ■ g/L
19F ≤3% ≤2% ≥20% ery rate ,
Storage % 5.5% 12.5% 20 0.01% 0.03% 0.08% 0.2%
should ≤50EU/ml
time
be≥75%
Not
0th
Complied 2.535 0.2% 0.1% 3.00% 4.80% 20.3% 27.9% 0 ——— detecte 0.0030% 0.0145% Complied 0.00% ———
month
d
Not
1st year Complied 2.555 0.2% 0.1% 2.39% 4.86% 13.1% 28.5% 0.04 ——— detecte 0.0007% 0.028% Complied 0.01%
——
d
Not
2nd Not Not
Complied 2.485 0.2% 0.1% 2.3% 4.8% 12.5% 28% 0.03 ——— detecte 0.011% Complied 0.004
year detected detected

d
Not Not
3rd
Complied 2.475 0.2% 0.1% 2.1% 5.1% 14.2% 28% 0.04 ——— detecte 0.001% 0.036% Complied 0.02269% detecte
year
d d
4th
Complied 3.010 0% 0% 1.8% 4.5% 13.4% 25% 0.04 89% 0% 0% 0.03% Complied 0% 0
year

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Stability data of 13 serotypes of pneumococcal polysaccharides Yuxi Walvax Biotechnology C., Ltd.

12.3 Stability Test Results of batch 19F-PVPV15120503 pneumococcal polysaccharide (storage
condition:-20■) ■
Identity Protei Total Organic solvent residual
Test Nucleic Phosph Hexosa Methyl Sodium
n nitroge Bacterial
items acid oruscon mineco pentose Molecular size deoxyc
conte n ether acetone ethanol endotoxin
and content tent ntent content holate Phenol
Positive Total nt content
standar residua
response to Solid KD■
ds l
mg/ml 0.3recov ≤0.5EU/μg
antiserum type 1.4-3.5 3.0- ≥ KD≤0. ■ ■ ■ ■ g/L
19F ≤3% ≤2% ≥20% ery rate ,
Storage % 5.5% 12.5% 20 0.01% 0.03% 0.08% 0.2%
should ≤50EU/ml
time
be≥75%
Not
0th
Complied 3.130 0.3% 0.1% 3.42% 4.80% 18.30% 24.2% 0 ——— detecte 0.0005% 0.0035% Complied 0.13% ———
month
d
Not
1st year Complied 1.960 0.2% 0.2% 2.40% 4.94% 12.83% 27.9% 0.04 ——— detecte 0.001% 0.029% Complied 0.10%
——
d
Not Not
2nd Not
Complied 2.215 0.2% 0.2% 2.7% 5.4% 13.6% 31% 0.07 ——— detecte 0.011% Complied detecte 0.0125
year detected
d d
Not Not
3rd
Complied 2.910 0.2% 0.0% 2.2% 3.7% 14.5% 25% 0.07 ——— detecte 0.001% 0.034% Complied detecte 0.0016
year
d d
4th
Complied 2.750 0% 0% 2.0% 4.6% 13.5% 24% 0.11 89% 0% 0% 0.03% Complied 0.06% 0.0402
year

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Stability data of 13 serotypes of pneumococcal polysaccharides Yuxi Walvax Biotechnology C., Ltd.

13. Pneumococcal polysaccharide serotype 23F
13.1 Stability Test Results of batch 23F-PVPV15090601 pneumococcal polysaccharide (storage
condition:-20■) ■
Nucle Total
Protei Organic solvent residual
Test ic nitrog Phosp Methyl Sodium
n Bacterial
items Identity acid en horusc pentose Molecular size deoxyc
conte endotoxin
and conte conten ontent content ether acetone ethanol holate
Total nt Phenol
standar nt t
Solid residual
ds KD■
mg/ml g/L
Positive response 0.3recov ≤0.5EU/μg
0- 3.0- KD≤0 ■ ■ ■ ■
Storage to antiserum type ≤2% ≤2% ≥37% ery rate ,
1.0% 4.5% .15 0.01% 0.03% 0.08% 0.2%
time 23F should ≤50EU/ml
be≥85%
Not
0th
Complied 2.585 0.3% 0.2% 0.89% 3.80% 41.5% 0.04 ——— detecte 0.002% 0.017% Complied 0.04% ———
month
d
Not Not
1st year Complied 2.520 0.3% 0.2% 0.28% 3.83% 37.1% 0.04 ——— detecte 0.001% 0.040% Complied detecte ———
d d
Not
2nd
Complied 2.425 0.2% 0.2% 0.29% 3.75% 41.9% 0.06 ——— detecte 0.001% 0.031% Complied 0.03% 0.021
year
d

Not Not
3rd
Complied 2.625 0.3% 0.2% 0.3% 3.9% 37% 0.04 ——— detecte 0.0002% 0.010% Complied detecte 0.0207
year
d d
4th
Complied 2.670 0% 0% 0.2% 3.8% 39% 0.04 91% 0% 0% 0.02% Complied 0% 0.0048
year

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Stability data of 13 serotypes of pneumococcal polysaccharides Yuxi Walvax Biotechnology C., Ltd.

13.2 Stability Test Results of batch 23F-PVPV15120602 pneumococcal polysaccharide (storage condition:-20■) ■
Nucle Total Organic solvent residual
Protei
Test ic nitrog Phosp Methyl Sodium
n Bacterial
items Identity acid en horusc pentose Molecular size deoxychol
conte ether acetone ethanol endotoxin
and conte conten ontent content ate
Total nt Phenol
standar nt t
Solid residual
ds KD■
mg/ml g/L
Positive response 0.3recov ≤0.5EU/μg
0- 3.0- KD≤0 ■ ■ ■ ■ ■
Storage to antiserum type ≤2% ≤2% ≥37% ery rate ,
1.0% 4.5% .15 0.01% 0.03% 0.08% 0.2%
time 23F should ≤50EU/ml
be≥85%

Not
0th
Complied 3.415 0.2% 0.1% 0.88% 3.90% 41.8% 0.08 ——— detecte 0.003% 0.010% Complied 0.17% ———
month
d
Not
Not
1st year Complied 2.775 0.2% 0.1% 0.29% 3.57% 43.3% 0.00 ——— detecte 0.002% 0.003% Complied ———
detected
d
Not
2nd Not
Complied 2.650 0.2% 0.2% 0.3% 4.0% 42% 0.03 ——— detecte 0.005% Complied 0.02796% 0.01895
year detected
d
Not
3rd
Complied 2.700 0.2% 0.1% 0.0% 4.3% 42% 0.04 ——— detecte 0.002% 0.010% Complied 0.01083% 0.00325
year
d
4th
Complied Complied
year 2.960 0% 0% 0.2% 3.9% 40% 0.07 98% 0% 0% 0% 0% 0.0450

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Stability data of 13 serotypes of pneumococcal polysaccharides Yuxi Walvax Biotechnology C., Ltd.

13.3 Stability Test Results of batch 23F-PVPV16010603 pneumococcal polysaccharide (storage condition:-20■) ■
Nucle Total Organic solvent residual
Protei
Test ic nitrog Phosp Methyl Sodium
n Bacterial
items Identity acid en horusc pentose Molecular size deoxychol
conte ether acetone ethanol endotoxin
and conte conten ontent content ate
Total nt Phenol
standar nt t
Solid residual
ds KD■
mg/ml g/L
Positive response 0.3recov ≤0.5EU/μg
0- 3.0- KD≤0 ■ ■ ■ ■ ■
Storage to antiserum type ≤2% ≤2% ≥37% ery rate ,
1.0% 4.5% .15 0.01% 0.03% 0.08% 0.2%
time 23F should ≤50EU/ml
be≥85%

Not
0th Not 0.0015
Complied 3.180 0.3% 0.1% 0.97% 3.30% 38.8% 0.00 ——— detecte Complied 0.08% ———
month detected %
d
Not
1st year Complied 2.820 0.4% 0.2% 0.21% 3.94% 40.6% 0.00 ——— detecte 0.0008% 0.011% Complied 0.07% ———
d
Not
2nd
Complied 2.960 0.4% 0.1% 0.2% 3.9% 37% 0.04 ——— detecte 0.0005% 0.006% Complied 0.06478% 0.0205
year
d

Not
3rd
Complied 2.645 0.4% 0.1% 0.2% 4.3% 41% 0.04 ——— detecte 0.001% 0.013% Complied 0.03703% 0.0196
year
d
4th
Complied 2.975 0% 0% 0.3% 3.3% 40% 0.04 90% 0% 0% 0.01% Complied 0% 0
year

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Stability data of 13 serotypes of pneumococcal polysaccharides Yuxi Walvax Biotechnology C., Ltd.

Conclusion:
pneumococcal polysaccharide serotype 1, 3,4,5,6A,6B,7F,9V,14,18C,19A,19F,23F manufactured by Yuxi Walvax
were kept at■20■, test at 0th
month■1st year■2nd year■3rd year■4th year■test results were all complied with specification of 13-valent
pneumococcal polysaccharide conjugate
vaccine (WSYX-QC-TS-06-09-0005).
This experiment was conducted by the quality control laboratory strictly sampled at each test point,
could provide evidence for safety,
effectiveness and validity of pneumococcal polysaccharide stored at -20■. In conclusion, shelf life of 13
serotypes of pneumococcal polysaccharide
manufactured by Yuxi Walvax stored at -20■ is 4th year.

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1 Objective

Standardize operating procedure for sterility test.

2 Definition/abbreviation

Sterility test: Test for sterility is a method to detect whether medicines, biologics, medical devices, raw materials, excipients or other articles, which are required to be sterile according to the Chinese Pharmacopoeia, are aseptic.

3 Scope of application

Membrane filtration test is applicable for intermediate products, final products and related excipients of Haemophilus Influenzae Type b Conjugate Vaccine, Group A and C Meningococcal Polysaccharide Vaccine, Meningococcal Group A, C Bivalent Polysaccharide Conjugate Vaccine, Group ACYW135 Meningococcal Polysaccharide Vaccine, Diphtheria, Tetanus and Acellular Pertussis Combined Vaccine, adsorbed, 23-valent Pneumococcal Polysaccharide Vaccine and 13-valent Pneumococcal Polysaccharide Conjugate Vaccine and other test samples.

4 Responsibility

Personnel of QC Laboratory shall perform the tests in strict accordance with this SOP.

5 Equipment and materials

5.1 Equipment

No. Equipment name Equipment model Manufacturer
Biological clean Suzhou Antai Airtech Co., Ltd of Su Jing

1 BCM-1600A

bench Group LLC

HTY-601/HTY/H Hangzhou Tailin Biotechnology

2 Sterility Test System

TY602A Equipment Co, Ltd

Suzhou Antai Airtech Co., Ltd of Su Jing

3 Biosafety cabinet BSC-1600IIA2

Group LLC

Constant temperature

STIK (Shanghai) Instrument Equipment

4 and humidity CTHI-250B

Limited

chamber

30-35℃ constant

5 temperature 2QC407 -

incubation room

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20-25℃ constant

6 temperature 2QC408 -

incubation room

5.2 Materials

No. Material name

1 Disposable sealed sterility test system

2 sterile syringes

3 pipettes sterilized by hot air

5.3 Reagents

No. Reagent name Type

1 Fluid thioglycollate medium Medium

2 Trypticase soy broth medium Medium

3 NaCl-peptone buffer Buffer

4 0.9% NaCl solution solution

5 Staphylococcus aureus Positive bacteria

6 Working procedure

6.1 Medium, diluent and rinsing solution

6.1.1 Fluid thioglycollate medium (dehydrated medium)

6.1.1.1 Preparation method: take 1000ml of preparation as an example

a) Composition: fluid thioglycollate medium 30g

Water 1000ml

b) Preparation: mix fluid thioglycollate medium with the water, heat the solution to boil.

Dispense the medium in suitable containers which provide a ratio of surface to depth of medium such that no more than the upper half of the medium has undergone a color change (pink color) indicative of oxygen uptake at the end of the incubation period. Sterilize using a validated process (at 121℃ for 15min), and adjust the pH so that after sterilization the solution will have a pH of 7.1±0.2. Reserve the prepared solution for use. Unless otherwise specified, cultivate in fluid thioglycollate medium at 30-35℃.

6.1.2 Trypticase soy broth medium (dehydrated medium)

6.1.2.1 Preparation method: take 1000ml of preparation as an example

a) Composition: Trypticase soy broth medium 30g

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Water 1000ml

b) Preparation: Mildly heat until dissolved, dispense the media into containers and sterilize using a validated process (at 121℃ for 15min). Adjust the pH so that after sterilization the solution will have a pH of 7.3±0.2. Reserve the prepared solution for use. Cultivate in trypticase soy broth medium at 20-25℃.

6.1.3 Traptcase soy agar medium (dehydrated medium)

6.1.3.1 Preparation method: take 1000ml of preparation as an example

a) Composition: Traptcase soy agar medium 40g

Water 1000ml

b) Preparation: Mildly heat until dissolved, dispense the media into containers and sterilize using a validated process (at 121■ for 15min). Adjust the pH so that after sterilization the solution will have a pH of 7.3±0.2. Reserve the prepared solution for use.

6.1.4 Sabouraud dextrose agar medium

6.1.4.1 Preparation method: take 1000ml of preparation as an example

a) Composition: Sabouraud dextrose agar medium 65 g

Water 1000ml

b) Mildly heat until dissolved, dispense the media into containers and sterilize using a validated process (at 121■ for 15min). pH of the solution shall be 5.6±0.2 after cooling to room temperature. Reserve the prepared solution for use.

6.1.5 Sabouraud glucose broth medium

6.1.5.1 Preparation method: take 1000ml of preparation as an example

a) Composition: Sabouraud glucose broth medium 30 g

Water 1000ml

b) Preparation: Mildly heat until dissolved, dispense the media into containers and sterilize using a validated process (at 121■ for 15min). pH of the solution shall be 5.6±0.2 after cooling to room temperature. Reserve the prepared solution for use.

6.1.6 pH 7.0 sodium chloride-peptone buffer

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a) Composition: take 1000ml of preparation as an example

Potassium dihydrogen phosphate 3.56g

Anhydrous sodium hydrogen phosphate 5.77g

Sodium chloride 4.3g

peptone 1.0g

b) Preparation method: mix the above components in 1000ml of water, mildly heat to dissolve, filter and dispense into containers. Sterilize using a validated process (at 121■ for 15min).

6.1.6.1 pH 7.0 sodium chloride-peptone buffer (According to the finished product formula ratio)

a) Composition: Sodium chloride-peptone buffer 14.63g

water 1000ml

b) Preparation method: mildly heat to dissolve, filter and dispense into containers. Sterilize using a validated process (at 121■ for 15min).

6.1.7 0.9% sodium chloride solution (rinsing solution). Dissolve 9g of sodium chloride in 1000ml of water, filter and dispense into containers. Sterilize using a validated process(at 121■ for 15min).

6.2 Sterility test of samples

Techniques of sterility test includes membrane filtration method and direct inoculation method. As long as the character of test sample permitted, the technique of membrane filtration shall be used. The test methods and conditions used for sterile examination of test samples shall be the same as those of validation test of the applicable method.

If any reagents, such as surfactants, inactivators and neutralizers, are used in the process of the sterility test, their effectiveness and non-toxic to microorganisms shall be confirmed.

6.2.1 Number of products to be tested

It is the number of minimum package (ampoule/vial/bottle) of products to be tested in one sterility test. Each single sub-lot of the final product shall be examined for sterility test. Unless otherwise specified, the minimum number of the products to be tested is specified in SOP of Yuxi Walvax Biotechnology Co., Ltd

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the following table, which is not included the quantity of test samples for positive control test.

Minimum number of product to be tested from each batch (final product, final bulk and bulk)

Product to be No. of items in

Minimum number of items to be tested

tested the batch (N)

N≤100 10% or 4 containers (choose the more one)

Parenteral 100<N≤500 10 containers

preparations 2% or 4 containers (choose the less one)

N>500

20 containers (biological products)

Each container (the minimum quantity to be tested

Bulk or final

shall be 0.1% of the whole quantity in each container

bulk of

or not less than 10ml. Bulk and final bulk shall be

biological

sampled as described above at each opening of the

products

container.)

6.2.2 Quantity of product to be tested

It is the minimum quantity (g or ml) of each container of the product in test for sterility once.

Unless otherwise specified, the minimum quantity of each container of the product for each medium is defined in the following table. When the amount of each ampoule (vial) for the test sample is sufficient to fill the two kinds of culture medium, it shall be incubated to fluid thioglycollate medium and traptcase soy broth medium respectively. When using the technique of membrane filtration■as long as the character of test sample permitted, the whole contents of each container shall be filtered for testing.

Minimum quantity of product to be tested

Product to be tested Quantity per container Minimum inoculum quantity per container

V≤1ml The whole contents of each container

Half the contents of each container, but not
Aqueous solutions 1ml■V≤40ml
less than 1 ml
(ml)
40ml■V≤100ml 20ml
V■100 10%, but not less than 20ml
Bulk or final bulk of
Half the contents of each container
biological products
6.3 Membrane filtration method
Sealed sterility testing system shall be used in the technique of membrane filtration. The
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membrane used in the test for sterility has a nominal pore size which is not greater than 0.45μm and a diameter about 50mm. Selection of filtration membrane type shall be performed according to the characters of product to be tested and solvent used. Integrity of the membrane in the apparatus shall be kept during the process filtration.

When filtering aqueous solution, the membrane in the filter shall be pre-wetted with a small quantity of rinsing fluids. When the samples tested are oily preparations, the filtration apparatus and membrane must be thoroughly dried before use. The sample solution and rinsing fluid shall be kept always to cover the whole surface of the membrane throughout the operation for maximal filtration efficiency. If necessary, wash the membrane by filtering through it with 100ml of rinsing fluid each time, and the total volume used shall not exceed 1000ml for one single membrane to avoid any damage to the microorganisms held by the membrane, after filtering the sample solution.

So far, the rinsing fluid amount used in the test related to production is 0 ml.

6.3.1 Measure specific quantity of test samples, filter directly, or transfer the samples into aseptic vessels containing not less than 100ml of the sterile diluent, mix up and filter immediately. If the products being examined contain preservatives or have inherent antimicrobial activity, the membrane shall be washed not less than 3 times and the quantity of the rinsing fluid and the method to wash the membrane shall be the same as that described Validation test.

6.3.2 After washing the biological product samples, add 100ml of fluid thioglycolate medium into two filters respectively and 100ml of Trapticase soy broth medium into one of the filter, stick the sterility test sample label.

6.3.3 The above-mentioned culture medium containers after inoculating the test samples shall be cultured for 14 days under the specified temperature; the containers of thioglycolate fluid culture medium inoculating the test samples of biological products shall be divided into two parts, one cultured at 30-35■ and the other cultured at 20-25■.

6.4 Direct inoculation method
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6.4.1 Equally inoculate the specified quantity of test sample into a container of thioglycollate liquid medium and a container of trapticase soy broth medium respectively. The volume of fluid thioglycollate medium shall not be less than 15ml/tube, and the volume of trapticase soy broth medium shall not be less than 10ml/tube. The quantity of containers with fluid thioglycollate medium and trapticase soy broth medium shall be 2:1.

6.4.2 The containers of thioglycollate liquid medium which inoculated should be divided into two parts, one cultured at 30-35■ for 14 days and the other cultured at 20-25■ for 14 days. Observe and record the results day by day.

6.5 Incubation and observation

Culture the test samples inoculated in the above medium at the specified temperature of each medium for 14 days. For biological products, the thioglycollate liquid medium containers shall be divided into two equal parts, half of the containers are incubated at 30-35■ and another half at 20-25■. Observe and record each medium container for evidence of microbial growth every day during incubation period. If the medium which is inoculated test samples appears turbid during the incubation period, and it cannot be readily determined by visual examination after 14-day of incubation, the defined amount medium can transfer to the same fresh medium. Observe whether the medium inoculated appears turbid after 3-day of incubation. If necessary, bacterial identification test, such as microscopic examination of stained smears, should be conducted.

6.6 Positive control

6.6.1 Strain recovery: Swab one ring of Staphylococcus aureus to trapticase soy broth medium. Culture at 30-35■ for 18-24 hours.

6.6.2 Preparation and counting of bacterial liquid: Dilute it with 0.9% sterile sodium chloride solution to 10-6 times, obtain the bacteria suspension which is containing bacteria less than 100cfu/1ml.

6.6.3 Measure 0.1ml of 10-5 bacteria suspension respectively and inoculate respectively in two culture dishes of trapticase soy broth medium for counting. Count the colony counts after
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cultured at 30-35■ for 18-24h hours. The colony counts in the two culture dishes shall be among 10-100 cfu.

6.6.4 Take one of the thioglycollate liquid medium, inoculate less than 100CFU positive bacterial in it as the positive control. Culture it at 30-35■ for 48-72h, the bacterial colony should growth well.

6.6.5 Except the above positive control preparation, it also can be carried out that the less than 100CFU positive bacterial is inoculated into the one of the thioglycollate liquid medium cultured after 14-day test sample sterility test as the positive control. Culture it at 30-35■ for 48-72h, the bacterial colony should growth well.

6.7 Negative control

Take the corresponding solvent, diluent and rinsing fluid as a negative control for sterility test. No bacterial should be found in negative control.

6.8 Results judgment

Microbial growth shall be well in the positive control container, and must be not found in the negative control container. Otherwise, the test may be considered invalid.

Except the positive control container, if no evidence of microbial growth is found in all of the test containers described above, the test sample complies with the requirement of sterility test; if evidence of microbial growth is found in any one of the test containers, the test sample does not comply with the requirement of sterility test, unless it can be clearly demonstrated that the test was invalid for causes unrelated to the product to be examined. The test may be considered invalid only when one or more of the following conditions are fulfilled:

(1) The result of the microbiological monitoring of the sterility testing facility shows that it does not meet the requirements of the test for sterility.

(2) A review of the testing procedure used during the test in question reveals a fault.

(3) After determination of the identification of the microorganisms isolated from the test, the growth of this species (or these species) may be ascribed unequivocally to faults in respect with the material or the technique used in conducting the sterility test procedure.

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If the test is validated to be invalid, it shall be repeated with the same number of the samples and procedure as in the original test. If no evidence of microbial growth is found in the test repeated, the product examined complied with the test for sterility; if microbial growth is found, the product examined does not comply with the test for sterility.

6.9 Dynamic monitoring

During each sterility test operation, the sedimentated bacteria in the sterile room, the sedimentated bacteria and dust particles in the purification platform are monitored and recorded. Test personnel should be sampled in every sterility test, sampling points are the left hand, right hand, left forearm, right forearm, left shoulder, right shoulder, forehead and masks. Record the monitoring results of sedimentated bacteria in sterile rooms and sedimentated bacteria and dust particles on the clean benches.

6.9.1 Monitoring petri dishes placed as the following stationing figure:

Sterile Room

6.9.2 Criteria

Monitoring items Sampling position Criteria Comments

Sterile room If the alarm of dust particle

Sedimentated

(3QC406, ■5CFU/petri dish numbers does not continue bacteria

3QC409) more than 10 minutes, the

Sedimentated dynamic monitoring of

Clean bench ■1CFU/petri dish

bacteria sterility test of dust particles

Dust particle meets the requirements.

Clean bench ≤3520

(0.5um) Otherwise the results do not

Dust particle (5um) Clean bench ≤20 meet the requirements, and

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Surface the test is not valid.

Personnel ≤5CFU/petri dish microorganism

7 Flow chart

7.1 The flow chart of thioglycollate liquid medium preparation

Flow chart Operating instructions Control point/CCP

Start

30g medium,

Weigh 30g of thioglycollate Increase the corresponding

1L purified

liquid medium, add 1000ml of amount proportionally

water

purified water

Boiling to the color of the

medium change from red to

Heat to dissolve until

yellow.

boiled

Subpackage with the saline

bottle to 300ml/bottle

Subpackage

Plug the top of the bottle with

an inverted cap, encase it with

Sealing

a layer of aluminum foil and a

layer of sulfate paper

Sterilize at 121■ for 15 min 121■, 15 min
Autoclaved sterilization

Mark the information of
Label batch No., validity period,
operator

Place in the 20-25 ■constant
Pre-incubation temperature incubating room to
pre-incubate 48h

7.2 The flow chart of trapticase soy broth medium preparation
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Flow chart Operating instructions Control point/CCP

Start

Increase the corresponding 30g medium,
Weigh 30g of trapticase soy
amount proportionally 1L purified
broth medium, add 1000ml of
water
purified water

The particulate matters
Heat to dissolve until
dissolve completely
boiled

Subpackage with the saline
Subpackage bottle to 300ml/bottle

Plug the top of the bottle with
an inverted cap, encase it with
Sealing a layer of aluminum foil and a
layer of sulfate paper

Sterilize at 121■ for 15 min 121■, 15 min
Autoclaved sterilization

Mark the information of
batch No., validity period,
Label
operator

Place in the 20-25 ■constant
Pre-incubation temperature incubating room to
pre-incubate 48h

7.3 The flow chart of pH7.0 NaCl – peptone buffer preparation
Flow chart Operating instructions Control point/CCP
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Start

Weigh 3.56g of potassium
dihydrogen phosphate, 5.77g of
anhydrous disodium hydrogen
phosphate, 4.3g of sodium Increase the corresponding
chloride and 1.0g of peptone, amount proportionally
add 1000ml of purified water

The particulate matters
Mildly heat to
dissolve completely
dissolve

Extraction and filtration with
peristaltic pump
Filter

Subpackage with the saline
bottle to 300ml/bottle
Subpackage

Plug the top of the bottle with
an inverted cap, encase it with
Sealing
a layer of aluminum foil and a
layer of sulfate paper

Autoclaved sterilization

Sterilize at 121■ for 15 min 121■, 15 min

Label

Mark the information of
batch No., validity period,
Pre-incubation
operator

7.4 The flow chart of sterility test
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Flow chart Operating instructions Control point/CCP

Start

Wipe the surface of the sample
with 75% ethanol and write the
Test samples preparation sample label

Put it into the airlock for Self-purify
self-purification, transfer it to for 20min,
Sample self-purification irradiate for
the transfer window for
irradiation 30min

Preparation of Sterilize the medium and
experimental materials equipment, and then put them
into the transfer window for
Enter the operation room to
irradiation
Environmental monitor Sedimentated bacteria
monitoring and dust particles

Sterility test Prepare and filter the samples

Personnel surface microbial
sampling, cleaning and
Experiment finished
disinfection

Thioglycollate liquid medium 1
Culture and observation is cultured at 33 ■ , the Culture for 14
thioglycollate liquid medium 2 days
and TSB4 are cultured at 23■

Results Y
judgement

N Comply The medium is clear.

Not The medium is turbid.
comply

8 Matters need attention
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8.1 All the procedures from the preparation of the equipment to the end of the experiment
shall be strictly sterile according to the standard operation.
8.2 Equipment related to sterility test shall be sterilized regularly and placed in a clean
environment.
8.3 The environmental monitoring shall be carried out throughout the experimental process.
Microbiological monitoring of personnel should be carried out before and after each sterility
test.
8.4 Sterility test should ensure that the upper interface of the culture device is connected to
3mm and above.
8.5 When observing the sterile samples, the operator should hold the bottle body of the culture
incubator, not the connecting hose at the upper end of the culture device.
9 References and standard basis
9.1 Pharmacopoeia of People's Republic of China, Volume III, 2015. general rule 1101
10 Revision history
Versio Issuing Time in Revision
Revision basis, reason and content Notes
n date force Dept.

According to the CP, improve the
01 Nov. 01 Dec. QC
03 sterility test record and observation
2011 2011 Laboratory
record.

According to the CP, improve the
06 Feb. 10 Feb. QC
04 sterility test record and observation

2012 2012 Laboratory
record.

According to the GMP on-site
11 May 14 May QC verification rectification, add the
05
2012 2012 Laboratory dynamic monitoring of suspended
particles during sterility test.
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06
2014 2014 Laboratory edition of the CP
19 Nov. 01 Dec. QC Revised the document because of the
07
2015 2015 Laboratory change in the name of the product
17 Mar. 20 Mar. QC Complete the content of the
08
2016 2016 Laboratory document
29 Aug. 01 Sep. QC Complete the contents of the
09
2016 2016 Laboratory appendix to the document

Complete the content of the
17 Mar. 20 Mar. QC
10 document, delete personnel dynamic
2017 2017 Laboratory
pre-monitoring

13 Oct. 15 Oct. QC Refine content of the document,
11
2017 2017 Laboratory revised the appendix

SOP of Yuxi Walvax Biotechnology Co., Ltd

Code WSYX-QC-SOP-00-09-0108

Appearance check SOP Version 10 Page 1/6

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1 Objective

Standardize operating procedure for appearance check.

2 Definition/abbreviation

Colorless: It means the color of test solution is similar to water or the used reagent.

3 Scope of application

It applies to the tests of all products produced in the company.

4 Responsibility

All the operators shall perform the tests in strict accordance with this SOP.

5 Equipment and Materials

5.1 Equipment

Equipment Equipment Equipment

No. Manufacturer Notes

name model code

Tianjin Optical Or equivalent

1 Clarity detector YB-II WYQCF0078

Instrument Factory equipment

6 Working procedure

6.1 Requirements of inspectors

By both near and distant vision, the visual acuity of inspectors should be equal to or above 4.9 (the corrected visual acuity should be equal to or above 5.0), and the inspectors shall not have color blindness, check for every three months.

6.2 Checking illuminance requirements

Test sample variety Illuminance Illuminance

requirements in stipulated in lab

Pharmacopeia

Colorless sample solution 1000-1500lx 1300lx

Transparent plastic container or color test solution 2000-3000lx 2300lx

Suspension type of test sample or milky solution About 4000 lx 4000lx

6.3 Visible foreign matter: metal chips, glass chips, fibers with a length of more than 2mm, lumps with a maximum particle size of more than 2 mm, smoke like particle deposits, uncountable particle groups or unshakable precipitates visible to the naked eye when rotating
SOP of Yuxi Walvax Biotechnology Co., Ltd

Code WSYX-QC-SOP-00-09-0108

Appearance check SOP Version 10 Page 2/6

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slightly after a certain period of time, and protein floccules which are difficult to count within a specified period of time, etc. Slight visible foreign matters: dots, short fibers and lumps below 2mm, etc., and semitransparent small protein floccules or protein particles less than 1mm of biochemical drugs or biological products.

6.4 Description of slight visible foreign matter

Name of foreign matter Specific character

White dot It means the white particle without clear plane or edges.

Slight protein floccule or protein It means a translucent protein floccules or protein particles particles of less than approximate 1mm.

Small amount of floccule or It means the protein floccules or protein particle that is protein particles difficult to count within the specified time limit for the examination.

micro sediment It means a trace of precipitate formed after stationary standing, which appears as a floating fog-like precipitate by gently inverting the container and disappears on gently shaking.

Precipitate not dispersed on It means a trace of precipitate formed in protein shaking solutions after storage for a long time, which can not be dispersed on gently shaking.

6.5 Preparation before examine

6.5.1 Before the examine, start-up, preheating, adjust illuminance according to the clarity detector's requirement.

6.5.2 According to the test sample type to take the corresponding amount, remove the container label, clean the stain on outer wall of the bottle, place at room temperature for 30 minutes. The specific sample number is in the table below:

Test sample type Sample number Retest number

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Injections 20 vials (syringe) Another 20 vials (syringe)

Lyophilization preparation

Lyophilization preparations

for 5 vials, diluents for 5 Another 10 vials

for injection

vials

Diluent 20 vials Another 20 vials

Pneumococcal

polysaccharide conjugate 5ml Another 5ml

bulk

6.6 Test method

6.6.1 Injection: Takes 2 vials (syringes) each time, hold the neck of the containers at the edge of light baffle in an appropriate distance to inspector's eyes (usually 25 cm), and view against black and white backgrounds respectively by gently rotating and inverting the containers to suspend the visible foreign particles possibly existing in solution (care shall be taken to avoid the production of air bubbles). The time limit of visual examination is 20 seconds. Record the observation result.

6.6.2 Lyophilization preparation for injection: Take corresponding diluents for test sample to reconstitute in indicated volume (0.5ml/vial). Rotate and shake gently to dissolve the test sample, keep container wall clean, and observe the reconstitution appearance. Hold the neck of the containers at the edge of light baffle in an appropriate distance to inspector's eyes (usually 25 cm), and view against black and white backgrounds respectively by gently rotating and inverting the containers to suspend the visible foreign particles possibly existing in solution (care shall be taken to avoid the production of air bubbles). The time limit of visual examination is 20 seconds. Record the observation result and determine according to the corresponding sample specification.

6.6.3 Pneumococcal polysaccharide conjugate bulk: Take the test sample 5ml into 10ml Nessler colorimetric tube and keep the tube wall clean. Holding the upper end of Nessler SOP of Yuxi Walvax Biotechnology Co., Ltd

Code WSYX-QC-SOP-00-09-0108

Appearance check SOP Version 10 Page 4/6

Confidential files No copy

colorimetric tube, gently rotate the tube to suspend the visible foreign matters in the test object under the background of black and white respectively at the distance from the test object to the human eye (usually 25cm) for visual inspection. The inspection time limit is 20 seconds, and the test object shall be clear and transparent liquid or slightly white floccules after shaking.

6.6.4 Solution color examination method

6.6.4.1 This method is applicable to the test sample which is specified in the quality standard to use solution color examination method.

6.6.4.2 Examine method: Take the appropriate amount of the test sample into 10ml Nessler colorimetric tube. Take the same amount of water into another 10ml Nessler colorimetric tube as control tube. The two tubes are placed under white background, take the top-to-down perspective, or head-up observation, compare the test tube and control tube color.

6.7 Result evaluation

6.7.1 Injections: according to the corresponding specification

Sample name Specification

23-valent Pneumococcal polysaccharide

It should be colorless transparent liquid.

vaccine

After shaking, it is milky white uniform

13-valent Pneumococcal conjugate vaccine

suspension

There are no more than 3 slight visible foreign matters in each test sample (vial). If only one sample (vial) exceeds the limit, it meets the requirements. If two samples (vials) exceed the limit, another 20 samples (vials) shall be taken for retest in the same way, 40 samples (vials) for primary and retest. If more than two samples (vials) exceed the limit, it does not meet the requirements.

6.7.2 Lyophilization preparation for injection

The test sample should be crisp cake, which reconstitute rapidly once adding diluents. It shall be colorless and transparent solution without foreign matters after the test sample is

SOP of Yuxi Walvax Biotechnology Co., Ltd

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Appearance check SOP Version 10 Page 5/6

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completely reconstituted. No visible foreign particles shall be found in all the 5 containers. If one container is found with visible particle, the test shall be repeated with additional 10 containers in the same procedures. In both primary and repeat tests the total number of containers with visible particles shall be not more than one.

6.7.3 Pneumococcal polysaccharide conjugate bulk

It shall be clear and transparent liquid or slightly white floccules after shaking.

6.7.4 Solution color examination method

The color in test sample tube shall not be darker than the control tube.

7 Flowchart

N/A

8 Matters need attention

8.1 Requirements of inspectors: by both near and distant vision, the visual acuity of inspectors should be equal to or above 4.9 (the corrected visual acuity should be equal to or above 5.0), and the inspectors shall not have color blindness.

8.2 Check the illuminance before examine.

8.3 When checking the Lyophilization preparation, the rubber crumb produced by the needle punching latex plug is not counted as visible foreign matters.

9 Reference or standard basis

9.1 Chinese Pharmacopeia, Volume IV, general rule 0904, page 42, China medical technology press, 2015 version

9.2 <Use instruction of YB-II clarity detector>

10 Revision history

Time in Revision

Version Revision basis, reason and content Notes

force Dept.

13 May QC

04 Improve file content /

2012 laboratory

05 25 April QC Improve file content /

SOP of Yuxi Walvax Biotechnology Co., Ltd

Code WSYX-QC-SOP-00-09-0108

Appearance check SOP Version 10 Page 6/6

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2015 laboratory /
20 March QC
06 Improve file content /
2016 laboratory
Add appearance check and solution
31 May QC color check method for 23-valent
07 /
2016 laboratory Pneumococcal Polysaccharide
Vaccine

28 April QC
08 Improve file content /
2017 laboratory

01 Sep. QC Add appearance check method for
09 /
2018 laboratory Pneumococcal conjugate bulk

Standard Operating Procedures of Yuxi Walvax Biotechnology Co., Ltd.

Code WSYX-QC-SOP-02-09-0004
SOP of Detoxification
Inspection of Tetanus Version 07 Pages 1/3
Toxoid Bulk
Confidential documents, no copying.

1 Purpose

Standardize the operation of detoxification inspection of tetanus toxoid bulk.

2 Definition / acronym

No

3 Scope of application

Suitable for detoxification inspection of tetanus toxoid bulk.

4 Responsibilities

4.1 Quality Control Laboratory

- The operator is responsible for the implementation of this procedure

- Department manager is responsible for overseeing the implementation of this procedure

4.2 Quality Supervision Department

- Quality supervisors are responsible for overseeing the implementation of this code

5 Equipment and materials

5.1 Using the equipment

Serial

Equipment Equipment Equipment

numb Manufacturers

name model coding

er

Electronic

1 PMA7501 WYQCF0263 Sartorius

balance

Purification Suzhou Antai Air Technology Co.,

2 BLB-1300 WYQCF0233

table Ltd.

5.2 Use of reagents

Serial Level/Categ Serial Level/Categ

Reagent Name Reagent Name

number ory number ory

0.9% sodium Medical Medical

1 2 75% ethanol

chloride injection grade grade

5.3 Materials

Standard Operating Procedures of Yuxi Walvax Biotechnology Co., Ltd.

Code WSYX-QC-SOP-02-09-0004

SOP of Detoxification

Inspection of Tetanus Version 07 Pages 2/3

Toxoid Bulk

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Serial Serial

Material name Material name

number number

250-350 g guinea pigs above

1 2 Graduated pipette

clean grade

3 Washing ear ball 4 50 ml vials

5 Alcohol lamp 6 Bottle stopper

7 Parafilm 8 Disposable syringe

9 Disposable latex gloves 10 Hemostatic forceps

6 Work procedures

6.1 Test procedure:

At least two guinea pigs weighing 300-400 g were injected subcutaneously with 500 Lf per bottle. It can also be diluted into 100Lf/ml with sodium chloride injection and injected subcutaneously with 5 ml. The observation was carried out on the 7th, 14th and 21st days after injection.

6.2 Criteria

The animals were observed on the 7th, 14th and 21st days after injection, and the animals should not have tetanus symptoms. At the end of the period, the body weight of each animal should not be reduced compared with that before injection, and the healthy ones were qualified. Weight loss should double the number of animals for retest. In case of tetanus symptoms, the original solution should continue to be detoxified.

7 Flow chart

No

8 Precautions

8.1 Experimental animals must have a certificate of conformity.

8.2 Samples rarely must be strictly in accordance with the requirements of sterile operation

9 References or standard basis

Standard Operating Procedures of Yuxi Walvax Biotechnology Co., Ltd.

Code WSYX-QC-SOP-02-09-0004

SOP of Detoxification

Inspection of Tetanus Version 07 Pages 3/3

Toxoid Bulk

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10 Revision history

Basis, reasons and content of
Version Effective Revision
the amendment Note
No. time Department

According to the corresponding
product names in Chinese
Pharmacopoeia (2010 Edition), it
is standardized and unified. The
names of protein carrier
fermentation broth (TT), protein
03 2011.12.01 QC Laboratory No
carrier refined liquid (TT) and
protein carrier purified liquid (TT)
were changed to tetanus toxoid
fermentation broth, tetanus toxoid
original liquid and tetanus toxoid
purified liquid.
Experimental
04 2012.11.01 File expiration No
Animal Center
Experimental
05 2015.10.21 Document revision and upgrading No
Animal Center
The organizational structure of
laboratory animal center and
06 2018.08.01 QC Laboratory quality control laboratory was No
changed (change control No.: CC-
201909012), so the responsible
Standard Operating Procedures of Yuxi Walvax Biotechnology Co., Ltd.

Code WSYX-QC-SOP-02-09-0004
SOP of Detoxification
Inspection of Tetanus Version 07 Pages 4/3
Toxoid Bulk
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department was revised from
"Laboratory Animal Center" to
"quality control laboratory".

Stability Data of PCV13-TT Yuxi Walvax Biotechnology Co.,Ltd.

Stability data of
13-valent Pneumococcal Polysaccharide Conjugate vaccine

Note■According to TS requirements of 13-valent Pneumococcal Polysaccharide Conjugate vaccine■
1. According to specification version 01-03, specification of polysaccharide content is■
Each dose of vaccine (0.5ml) contains:
For Serotype 1, 3, 4, 5, 6A, 7F, 9V, 14, 18C, 19A, 19F and 23F: 2.2-3.3µg, for serotype 6B: 4.4-6.6µg.
2. According to specification version 04, specification of polysaccharide content is■
Serotype 1: 3.6-6.8µg/ml, Serotype 3: 3.5-6.5µg/ml, Serotype 4:4.2-7.8µg/ml, Serotype 5:3.5-6.5µg/ml,
Serotype 6A:3.5-6.5µg/ml, Serotype 6B:8.4-15.6µg/ml, Serotype 7F:4.0-7.4µg/ml, Serotype 9V:3.5-6.5µg/ml,
Serotype 14:3.8-7.2µg/ml, Serotype 18C:4.6-8.4µg/ml, Serotype 19A:3.6-6.8µg/ml, Serotype
19F:3.8-7.2µg/ml,
Serotype 23F:4.2-7.8µg/ml
3. Conjugate antigen content, Conjugate protein content and Total protein content were added in
specification of 13-valent Pneumococcal
Polysaccharide Conjugate vaccine version 04
4. Based on the time frame for the entry into force of each version, the quality standard for the
polysaccharide content of the following six
batches of 13-valent pneumococcal polysaccharide conjugate vaccine, from 1 month to 18 months, the
standards used for the period are those
specified in item 1; for the period 24 months and later, the standards used are those specified in item
2.

Version:July, 2020 1
Stability Data of PCV13-TT Yuxi Walvax Biotechnology Co.,Ltd.

Part 1: Stability data of 13-valent Pneumococcal Polysaccharide Conjugate vaccine■Vial■
Long-term Stability data of 13-valent Pneumococcal Polysaccharide Conjugate vaccine■Vial■
1. Batch no.:GI201708001, Storage condition: 6±2■ for 30months
Alumi Sodi
Bacter
Appe num m
volu Osmolali Abnormal ial Steril
aranc pH phosph chlorid Polysaccharide content■µg/ml■
me ty toxicity endoto ity
e ate e
xin
content content
All guinea
Test 18 19 19 23 pigs and mice
1 3 4 5 6A 6B 7F 9V 14
items C A F F should be
and Shoul healthy and
standar d be no abnormal Shou
Shoul
d milky reaction Id
d be
unifo during the NMT comp
not 0.8 270±50
rm 5.5 7.5-9.5 observation 12.5E ly
less -1.2mg mOsmol/
Storage suspe -7.5 g/L period, and U/dos with
than /ml Kg Serotype 1, 3, 4, 5, 6A, 7F, 9V, 14, 18C, 19A, 19F and 23F: 2.2-3.3µg/dose
time nsion the weight of e requi
0.5ml 6B: 4.4-6.6µg/dose
after each guinea reme
/vial
shaki pig and nts
ng mouse should
be increased
at the end of
the period.
0th Com Com Compl Com
6.2 1.03 8.48 253 3.2 2.4 2.8 2.7 2.9 6.0 3.1 2.6 2.7 2.7 2.5 2.5 2.6 Complied
month plied plied ied plied
3rd Com Com Compl Com
6.2 1.12 8.60 255.5 2.5 3.0 2.5 2.4 3.1 6.0 2.4 2.9 3.0 2.6 2.6 2.4 2.7 Complied
month plied plied ied plied

Version:July, 2020 2
Stability Data of PCV13-TT Yuxi Walvax Biotechnology Co.,Ltd.
6th Com Com Compl Com
6.0 1.0 8.5 252 2.4 2.6 2.5 2.5 2.7 5.2 2.5 2.5 2.7 2.6 2.6 2.8 2.4 Complied
month plied plied ied plied
9th Com Com Compl Com
6.1 1.0 7.9 253 2.5 2.3 2.7 2.9 2.8 4.9 2.5 2.3 2.5 2.5 2.4 2.8 2.5 Complied
month plied plied ied plied
12th Com Com Compl Com
6.1 1.0 8.4 254 2.8 2.5 3.3 3.2 3.0 5.5 3.0 2.9 3.2 3.1 2.7 2.9 2.8 Complied
month plied plied ied plied
18th Com Com Compl Com
6.1 1.1* 8.3 254 2.7 3.1 2.7 2.8 2.6 4.9 2.4 2.5 2.7 2.7 2.5 2.7 2.4 Complied
month plied plied ied plied
*Note■At 18th month, data on aluminium phosphate content converted from aluminium content;

Formula: Aluminium Phosphate Content = Aluminium Content ×(121.95÷26.98).
Alumi
Sodium

Appearance
volume pH chloride Osmolality Polysaccharide content $\mu\text{g/ml}$
Test e Content
content
items t
and Each ml of vaccine contains:
standard Should be Serotype 1: 3.6-6.8 $\mu\text{g/ml}$, Serotype 3: 3.5-6.5 $\mu\text{g/ml}$, Serotype 4: 4.2-7.8 $\mu\text{g/ml}$,
Should
milky Serotype 5: 3.5-6.5 $\mu\text{g/ml}$, Serotype 6A: 3.5-6.5 $\mu\text{g/ml}$, Serotype 6B: 8.4-15.6 $\mu\text{g/ml}$,
be not 0.18-
uniform Serotype 7F: 4.0-7.4 $\mu\text{g/ml}$, Serotype 9V: 3.5-6.5 $\mu\text{g/ml}$, Serotype 14: 3.8-7.2 $\mu\text{g/ml}$,
less than 5.5-7.5 0.27 7.5-9.5g/L 200-310
Storage suspension Serotype 18C: 4.6-8.4 $\mu\text{g/ml}$, Serotype 19A: 3.6-6.8 $\mu\text{g/ml}$, Serotype 19F: 3.8-7.2 $\mu\text{g/ml}$,
0.5ml/via mg/ml mOsmol/Kg
time after Serotype 23F: 4.2-7.8 $\mu\text{g/ml}$
I
shaking
1 3 4 5 6A 6B 7F 9V 14 18C 19A 19F 23F

24th Complied
Complied 6.2 0.23 8.48 253 3.2 2.4 2.8 2.7 2.9 6.0 3.1 2.6 2.7 2.7 2.5 2.5 2.6
month d
30th Complied
Complied 6.2 0.25 8.60 255.5 2.5 3.0 2.5 2.4 3.1 6.0 2.4 2.9 3.0 2.6 2.6 2.4 2.7
month d
Test Conjugat Total
Abnormal Bacterial
items Conjugate antigen content e protein protein Sterility
toxicity endotoxin
and content content
Version: July. 2020 3
Stability Data of PCV13-TT Yuxi Walvax Biotechnology Co., Ltd.
standard Each dose contains: All guinea
Type 1: release standard is $\geq 30\%$, standard in expiry date is $\geq 25\%$; pigs and
Type 3, 6A, 6B, 9V, 14, 18C: release standard is $\geq 80\%$, standard in expiry date is $\geq 75\%$; mice should
Storage Type 4, 5, 7F: release standard is $\geq 75\%$, standard in expiry date is $\geq 70\%$; Release be healthy
time Type 19A: release standard is $\geq 70\%$, standard in expiry date is $\geq 65\%$; standard and no
Type 19F, 23F release standard is $\geq 65\%$, standard in expiry date is $\geq 60\%$. shall be abnormal
not less reaction
Should
than 41- during the
NMT comply
85%, 150 observation
12.5EU/d with
standard $\mu\text{g/dos}$ period, and
ose requireme
in expiry e the weight
nts
date shall of each
1 3 4 5 6A 6B 7F 9V 14 18C 19A 19F 23F
be not guinea pig
less than and mouse
80%. should be
increased at
the end of
the period.
24th 100 100 100 100 94% 72 Complied Complied Complied
45% 82% 80% 80% 100% 100% 78% 73% 80%
month % % % %
30th 100 100 100 100 100 100 100 87% 71 Complied Complied Complied
81% 100% 100% 100% 86% 100%
month % % % % % %

Version: July. 2020 4
Stability Data of PCV13-TT Yuxi Walvax Biotechnology Co., Ltd.

2. Batch no.: GI201708002 Storage condition: $6\pm 2^\circ\text{C}$ for 30 months
Alumi Sodi
Bacter
Appe num m
Volu Osmolali Abnormal ial Steril
aranc pH phosph chlorid Polysaccharide content $\mu\text{g/ml}$
me ty toxicity endoto ity
e ate e
xin
content content
All guinea
Test 18 19 19 23 pigs and mice
1 3 4 5 6A 6B 7F 9V 14
items C A F F should be
and Shoul healthy and
standar d be no abnormal Shou
Shoul
d milky reaction Id
d be
unifo during the NMT comp
not 0.8
rm 5.5 7.5-9.5 270 $\pm$ 50 observation 12.5E ly
less -1.2mg
Storage suspe -7.5 g/L mOsmol/ period, and U/dos with
than /ml Serotyoe 1, 3, 4, 5, 6A, 7F, 9V, 14, 18C, 19A, 19F and 23F: 2.2-3.3 $\mu\text{g/dose}$
time nsion Kg the weight of e requi

0.5ml 6B: 4.4-6.6µg/dose
after each guinea reme
/vial
shaki pig and nts
ng mouse should
be increased
at the end of
the period.
0th Com Com Compl Com
6.25 1.08 8.46 255.5 2.8 2.6 2.4 2.4 2.9 5.4 2.8 2.9 2.6 2.8 2.5 2.6 2.9 Complied
month plied plied ied plied
3rd Com Com Compl Com
6.2 1.10 8.30 254.5 2.5 2.9 2.7 2.6 2.7 5.0 2.2 2.9 2.7 2.7 3.0 2.2 2.5 Complied
month plied plied ied plied
6th Com Com Compl Com
6.1 1.0 8.3 254.5 2.4 2.4 2.5 2.5 2.6 4.6 2.9 2.6 2.6 2.3 2.4 2.7 2.4 Complied
month plied plied ied plied

9th Com Com 6.1 0.9 8.2 254 2.9 3.1 2.9 2.5 3.0 5.5 2.5 2.5 2.8 3.2 2.5 2.5 2.6 Complied Compl Com
Version:July. 2020 5
Stability Data of PCV13-TT Yuxi Walvax Biotechnology Co.,Ltd.
month plied plied ied plied

12th Com Com Compl Com
6.0 1.0 8.3 252 2.9 2.4 2.4 2.5 2.4 5.6 2.2 2.5 2.5 2.5 2.8 2.6 2.8 Complied
month plied plied ied plied
18th Com Com Compl Com
6.0 1.1* 8.4 259 2.4 2.9 2.6 2.8 2.5 4.8 2.7 2.3 2.6 2.9 2.8 2.8 2.5 Complied
month plied plied ied plied
*Note■At 18th month, data on aluminium phosphate content converted from aluminium content;

Formula: Aluminium Phosphate Content = Aluminium Content ×(121.95÷26.98).
Alumi Sodium
Appearanc
Volume pH nium chlorideco Osmolality ■■ content
Test e
content ntent
items
Each ml of vaccine contains:
and
Should be Serotype 1: 3.6-6.8µg/ml, Serotype 3: 3.5-6.5µg/ml, Serotype 4:4.2-7.8µg/ml,
standard Should
milky Serotype 5:3.5-6.5µg/ml, Serotype 6A:3.5-6.5µg/ml, Serotype 6B:8.4-15.6µg/ml,
be not 0.18-
uniform Serotype 7F:4.0-7.4µg/ml, Serotype 9V:3.5-6.5µg/ml, Serotype 14:3.8-7.2µg/ml,
less than 5.5-7.5 0.27 7.5-9.5g/L 200-310
suspension Serotype 18C:4.6-8.4µg/ml, Serotype 19A:3.6-6.8µg/ml, Serotype 19F:3.8-7.2µg/ml,
Storage 0.5ml/via mg/ml mOsmol/Kg
after Serotype 23F:4.2-7.8µg/ml
time l
shaking
1 3 4 5 6A 6B 7F 9V 14 18C 19A 19F 23F

24th Complie
Complied 6.1 0.22 8.7 258 4.8 4.6 5.0 3.9 4.2 8.8 7.1 5.7 5.0 4.7 5.3 4.3 4.5
month d
30th Complie
Complied 6.1 0.20 8.3 253 4.8 4.2 5.1 5.3 5.6 9.2 5.8 5.1 5.1 6.2 4.7 4.8 4.6
month d
Test Conjugat Total
Abnormal Bacterial
items Conjugate antigencontent e protein protein Sterility
toxicity endotoxin
and content content
standard Each dose contains: Release 41- All guinea NMT Should
Type 1: release standard is ≥30%, standard in expiry date is≥25%; standard 150 pigs and 12.5EU/d comply
Type 3, 6A, 6B, 9V, 14, 18C: release standard is ≥80%, standard in expiry date is≥75%; shall be µg/dos
mice should ose with
Version:July. 2020 6
Stability Data of PCV13-TT Yuxi Walvax Biotechnology Co.,Ltd.
Storage Type 4, 5, 7F: release standard is ≥75%, standard in expiry date is ≥70%; not less e be healthy
requireme
time Type 19A: release standard is ≥70%, standard in expiry date is ≥65%; than and no nts
Type 19F, 23F release standard is ≥65%, standard in expiry date is ≥60%. 85%, abnormal
standard reaction
in expiry during the
date shall observation
be not period, and
less than the weight
80%. of each
1 3 4 5 6A 6B 7F 9V 14 18C 19A 19F 23F
guinea pig
and mouse
should be
increased at
the end of
the period.
24th 100 100 99% 68 Complied Complied Complied
71% 78% 78% 84% 79% 76% 75% 79% 80% 77% 84%
month % %
30th 100 100 100 100 100 97% 68 Complied Complied Complied
79% 86% 91% 100% 100% 100% 82% 100%

month % % % % %

Version:July. 2020 7
Stability Data of PCV13-TT Yuxi Walvax Biotechnology Co.,Ltd.

3. Batch no.:GI201709003 Storage condition: 6±2■ for 24 months
Alumi Sodi
Bacter
Appe num m
Volu Osmolali Abnormal ial Steril
aranc pH phosph chlorid Polysaccharide content■μg/ml■
me ty toxicity endoto ity
e ate e
xin
content content
All guinea
Test 18 19 19 23 pigs and mice
1 3 4 5 6A 6B 7F 9V 14
items C A F F should be
and Shoul healthy and
standar d be no abnormal Shou
Shoul
d milky reaction Id
d be
unifo during the NMT comp
not 0.8
rm 5.5 7.5-9.5 270±50 observation 12.5E ly
less -1.2mg
Storage suspe -7.5 g/L mOsmol/ period, and U/dos with
than /ml Serotyoe 1, 3, 4, 5, 6A, 7F, 9V, 14, 18C, 19A, 19F and 23F: 2.2-3.3μg/dose
time nsion Kg the weight of e requi
0.5ml 6B: 4.4-6.6μg/dose
after each guinea reme
/vial
shaki pig and nts
ng mouse should
be increased
at the end of
the period.
0th Com Com Compl Com
6.22 0.90 8.815 268.5 2.9 2.4 2.7 2.7 2.6 4.8 2.7 2.4 2.7 2.7 2.8 2.8 3.0 Complied
month plied plied ied plied
3rd Com Com Compl Com
6.3 0.96 8.00 268.5 2.5 3.0 2.6 2.5 2.7 5.0 2.3 2.8 2.8 2.5 2.9 2.3 2.6 Complied
month plied plied ied plied
6th Com Com Compl Com
6.1 1.0 8.1 255 2.4 2.8 2.4 2.7 2.6 5.5 2.3 2.6 2.7 2.5 2.4 2.9 2.9 Complied
month plied plied ied plied

9th Com Com 6.1 0.9 8.5 254 2.6 2.9 2.5 2.5 2.6 5.4 2.2 2.2 2.5 2.6 2.8 2.5 2.5 Complied Compl Com
Version:July. 2020 8
Stability Data of PCV13-TT Yuxi Walvax Biotechnology Co.,Ltd.
month plied plied ied plied

12th Com Com Compl Com
6.1 0.9 8.5 254 2.7 3.0 2.4 2.4 3.0 4.7 2.8 2.6 2.8 2.7 2.8 2.7 3.0 Complied
month plied plied ied plied
18th Com Com Compl Com
6.0 1.0* 8.4 254 2.7 2.3 2.9 2.9 2.4 5.0 2.9 2.6 2.7 2.9 2.9 2.9 2.6 Complied
month plied plied ied plied
*Note■At 18th month, data on aluminium phosphate content converted from aluminium content;

Formula: Aluminium Phosphate Content = Aluminium Content ×(121.95÷26.98).
Alumi Sodium
Appearanc
Volume pH nium chlorideco Osmolality Polysaccharide content
Test e
content ntent
items
Each ml of vaccine contains:
and
Should be Serotype 1: 3.6-6.8μg/ml, Serotype 3: 3.5-6.5μg/ml, Serotype 4:4.2-7.8μg/ml,
standard Should
milky Serotype 5:3.5-6.5μg/ml, Serotype 6A:3.5-6.5μg/ml, Serotype 6B:8.4-15.6μg/ml,
be not 0.18-
uniform Serotype 7F:4.0-7.4μg/ml, Serotype 9V:3.5-6.5μg/ml, Serotype 14:3.8-7.2μg/ml,
less than 5.5-7.5 0.27 7.5-9.5g/L 200-310
suspension Serotype 18C:4.6-8.4μg/ml, Serotype 19A:3.6-6.8μg/ml, Serotype 19F:3.8-7.2μg/ml,
Storage 0.5ml/via mg/ml mOsmol/Kg
after Serotype 23F:4.2-7.8μg/ml
time l
shaking
1 3 4 5 6A 6B 7F 9V 14 18C 19A 19F 23F

24th Complie
Complied 6.0 0.20 8.2 257 6.3 5.0 6.2 5.8 5.7 12.9 5.4 5.9 6.4 7.7 6.0 6.1 5.4
month d
Test Conjugat Total
Abnormal Bacterial
items Conjugate antigen content e protein protein Sterility
toxicity endotoxin
and content content

standard Each dose contains: Release All guinea Should
41-
Type 1: release standard is ≥30%, standard in expiry date is≥25%; standard pigs and NMT comply
150
Type 3, 6A, 6B, 9V, 14, 18C: release standard is ≥80%, standard in expiry date is≥75%; shall be mice
should 12.5EU/d with
µg/dos
Storage Type 4, 5, 7F: release standard is ≥75%, standard in expiry date is ≥70%; not less be healthy ose
requireme
e
time Type 19A: release standard is ≥70%, standard in expiry date is ≥65%; than and no nts
Version:July. 2020 9
Stability Data of PCV13-TT Yuxi Walvax Biotechnology Co.,Ltd.
Type 19F, 23F release standard is ≥65%, standard in expiry date is ≥60%. 85%, abnormal
standard reaction
in expiry during the
date shall observation
be not period, and
less than the weight
80%. of each
1 3 4 5 6A 6B 7F 9V 14 18C 19A 19F 23F guinea pig
and mouse
should be
increased at
the end of
the period.
24th 100 100 100 100 100 100% 72 Complied Complied Complied
72% 85% 92% 100% 100% 86% 100% 81%
month % % % % %

Conclusion:
3 batches of 13-valent pneumococcal polysaccharide conjugate vaccine GI201708001, GI201708002,
GI201709003 were kept at 6±2■ for
30months, 30months and 24months respectively. Test results at all the test point were all complied with
requirements of Specification of
13-valent pneumococcal polysaccharide conjugate vaccine Code■WSYX-QC-TS-06-09-0002, in conclusion,
13-valent pneumococcal
polysaccharide conjugate vaccine could be kept at 6±2■ for 24months.

Version:July. 2020 10
Accelerated Stability data of 13-valent Pneumococcal Polysaccharide Conjugate vaccine■Vial■
1. Batch no.:GI201708001 Storage condition: 25±2■
Alumi Sodi
Bacter
Appe num m
Volu Osmolali Abnormal ial Steril
aranc pH phosph chlorid Polysaccharide content■µg/ml■
me ty toxicity endoto ity
e ate e
xin
content content
All guinea
Test 18 19 19 23 pigs and mice
1 3 4 5 6A 6B 7F 9V 14
items C A F F should be
and Shoul healthy and
standar d be no abnormal Shou
Shoul
d milky reaction ld
d be
unifo during the NMT comp
not 0.8
rm 5.5 7.5-9.5 270±50 observation 12.5E ly
less -1.2
Storage suspe -7.5 g/L mOsmol/ period, and U/dos with
than mg/ml Serotyoe 1, 3, 4, 5, 6A, 7F, 9V, 14, 18C, 19A, 19F and 23F: 2.2-3.3µg/dose
time nsion Kg the weight of e requi
0.5ml 6B: 4.4-6.6µg/dose
after each guinea reme
/vial
shaki pig and nts
ng mouse should
be increased
at the end of
the period.
0th Com Com Compl Com
6.2 253 8.48 1.03 3.2 2.4 2.8 2.7 2.9 6.0 3.1 2.6 2.7 2.7 2.5 2.5 2.6 Complied
month plied plied ied plied
1st Com Com Compl Com
6.4 255.5 7.83 1.00 2.5 2.8 2.6 2.5 2.7 5.5 2.6 2.7 2.6 3.1 3.2 2.8 2.6 Complied
month plied plied ied plied
2nd Com Com Compl Com
6.2 255 8.49 1.16 2.6 2.8 2.6 2.6 2.8 4.7 2.4 2.7 3.0 2.3 3.1 2.4 2.6 Complied
month plied plied ied plied

Version:July. 2020 11
3rd Com Com Compl Com
6.1 255.5 8.00 1.10 2.5 3.2 2.5 2.5 2.9 6.2 2.5 3.0 3.0 2.8 2.6 2.5 2.6 Complied
month plied plied ied plied
6th Com Com Compl Com
6.1 256 8.2 1.0 2.6 2.4 2.4 2.5 2.8 5.0 2.4 3.0 2.6 2.9 2.6 2.9 2.4 Complied
month plied plied ied plied

9th Com Com Compl Com
6.1 256 8.1 1.0 2.7 2.9 2.8 2.4 2.9 4.7 2.9 2.8 3.1 2.6 2.6 2.7 2.7 Complied
month plied plied ied plied

Version:July. 2020 12
2. Batch no.:GI201708002 Storage condition: 25±2■
Alumi
Sodiu
num Bacter
Appe m
Volu phosp Osmolali Abnormal ial Steril
aranc pH chlorid Polysaccharide content■μg/ml■
me hate ty toxicity endoto ity
e e
conte xin
content
nt
All guinea
Test 18 19 19 23
1 3 4 5 6A 6B 7F 9V 14 pigs and mice
items C A F F should be
and
Shoul healthy and
standar
d be no abnormal Shou
d Shoul
milky reaction Id
d be
unifo during the NMT comp
not 0.8
rm 5.5 7.5-9.5 270±50 observation 12.5E ly
Storage less -1.2
suspe -7.5 g/L mOsmol/ period, and U/dos with
time than mg/ml Serotyoe 1, 3, 4, 5, 6A, 7F, 9V, 14, 18C, 19A, 19F and 23F: 2.2-3.3μg/dose
nsion Kg the weight of e requi
0.5ml 6B: 4.4-6.6μg/dose
after each guinea reme
/vial
shaki pig and nts
ng mouse should
be increased
at the end of
the period.

0th Com Com Compl Com
6.25 255.5 8.46 1.08 2.8 2.6 2.4 2.4 2.9 5.4 2.8 2.9 2.6 2.8 2.5 2.6 2.9 Complied
month plied plied ied plied
1st Com Com Compl Com
6.6 255 7.93 0.97 2.5 2.6 2.5 2.4 2.8 5.5 2.6 2.9 2.7 3.1 3.1 2.8 2.6 Complied
month plied plied ied plied
2nd Com Com Compl Com
6.2 255.5 8.11 1.14 2.6 2.9 2.7 2.4 2.8 4.8 2.5 2.9 2.8 2.4 3.1 2.3 2.5 Complied
month plied plied ied plied

Version:July. 2020 13
3rd Com Com Compl Com
6.2 256 8.10 1.00 2.7 2.6 2.4 2.4 2.8 5.0 2.3 2.6 2.9 2.8 2.6 2.8 2.6 Complied
month plied plied ied plied
6th Com Com Compl Com
6.1 256.5 8.2 0.9 2.5 2.5 2.4 2.6 2.6 5.1 2.3 3.0 2.5 2.7 2.5 2.9 2.3 Complied
month plied plied ied plied
9th Com Com Compl Com
6.1 254 8.1 1.0 2.7 2.9 2.7 2.4 2.7 4.7 3.0 2.9 3.1 2.6 2.6 2.7 2.6 Complied
month plied plied ied plied

Version:July. 2020 14
3. Batch no.: GI201709003 Storage condition: 25±2■
Alumi
Sodiu
num Bacter
Appe m
Volu phosp Osmolali Abnormal ial Steril
aranc pH chlorid Polysaccharide content■μg/ml■
me hate ty toxicity endoto ity
e e
conte xin
content
nt
All guinea
Test 18 19 19 23
1 3 4 5 6A 6B 7F 9V 14 pigs and mice
items C A F F should be
and
Shoul healthy and
standar
d be no abnormal Shou
d Shoul
milky reaction Id
d be
unifo during the NMT comp
not 0.8
rm 5.5 7.5-9.5 270±50 observation 12.5E ly
Storage less -1.2

suspe -7.5 g/L mOsmol/ period, and U/dos with
time than mg/ml Serotype 1, 3, 4, 5, 6A, 7F, 9V, 14, 18C, 19A, 19F and 23F: 2.2-3.3µg/dose
nsion Kg the weight of e requi
0.5ml 6B: 4.4-6.6µg/dose
after each guinea reme
/vial
shaki pig and nts
ng mouse should
be increased
at the end of
the period.
0th Com Com Compl Com
6.22 268.5 8.815 0.9 2.9 2.4 2.7 2.7 2.6 4.8 2.7 2.4 2.7 2.7 2.8 2.8 3.0 Complied
month plied plied ied plied
1st Com Com Compl Com
6.4 269.5 8.29 0.99 2.6 2.9 2.5 2.5 2.7 5.8 2.6 2.8 2.7 3.1 3.1 2.8 2.7 Complied
month plied plied ied plied
2nd Com Com Compl Com
6.2 255 8.05 1.16 2.9 3.0 2.3 2.5 2.7 5.4 2.8 2.8 2.7 3.0 2.8 2.6 2.5 Complied
month plied plied ied plied

Version:July. 2020 15
3rd Com Com Compl Com
6.3 269.5 8.20 1.10 2.6 2.7 2.4 2.5 2.9 5.4 2.4 2.8 2.8 2.7 2.7 2.8 2.7 Complied
month plied plied ied plied
6th Com Com Compl Com
6.1 255.5 8.2 1.0 3.2 3.2 2.9 2.8 2.5 5.0 2.3 2.5 2.5 2.9 2.6 2.7 2.7 Complied
month plied plied ied plied
9th Com Com Compl Com
6.1 254 8.2 1.0 2.4 2.8 2.6 2.9 2.4 5.7 2.7 2.8 2.8 2.9 2.5 2.8 2.7 Complied
month plied plied ied plied
Conclusion:
3 batches of 13-valent pneumococcal polysaccharide conjugate vaccine GI201708001, GI201708002,
GI201709003 were kept at 25±2■ for 9months.
Test results at all the test point were all complied with requirements of Specification of 13-valent
pneumococcal polysaccharide conjugate vaccine
Code■WSYX-QC-TS-06-09-0002, in conclusion, 13-valent pneumococcal polysaccharide conjugate vaccine could
be kept at 25±2■ for 9months.

Version:July. 2020 16
Part 2: Stability data of 13-valent Pneumococcal Polysaccharide Conjugate vaccine■Pre-filled Syringe■
Long-term Stability data of 13-valent Pneumococcal Polysaccharide Conjugate vaccine■Pre-filled Syringe■
1. Batch no.: GK201709001 Storage condition: 6±2■
Alumi Sodi
Bacter
Appe num m
Volu Osmolali Abnormal ial Steril
aranc pH phosph chlorid Polysaccharide content■µg/ml■
me ty toxicity endoto ity
e ate e
xin
content content
All guinea
Test 18 19 19 23 pigs and mice
1 3 4 5 6A 6B 7F 9V 14
items C A F F should be
and Shoul healthy and
standar d be Shoul no abnormal Shou
d milky d be reaction ld
unifo not during the NMT comp
0.8
rm less 5.5 7.5-9.5 270±50 observation 12.5E ly
-1.2mg
Storage suspe than -7.5 g/L mOsmol/ period, and U/dos with
/ml Serotype 1, 3, 4, 5, 6A, 7F, 9V, 14, 18C, 19A, 19F and 23F: 2.2-3.3µg/dose
time nsion 0.5ml Kg the weight of e requi
after /syrin 6B: 4.4-6.6µg/dose each guinea reme
shaki ge pig and nts
ng mouse should
be increased
at the end of
the period.
0th Com Com Compl Com
6.22 0.94 8.81 272.5 3.0 2.7 2.6 2.6 2.6 4.8 2.8 3.1 2.6 2.6 3.0 3.0 2.4 Complied
month plied plied ied plied
3rd Com Com Compl Com
6.3 0.94 8.10 272.5 2.4 2.8 2.5 2.4 2.4 5.0 2.2 2.7 2.9 2.6 2.7 2.6 2.4 Complied
month plied plied ied plied

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6th Com Com Compl Com
6.1 1.0 8.1 258.5 2.4 2.6 2.4 2.9 2.8 6.0 2.4 2.4 2.6 2.4 2.5 2.8 2.4 Complied
month plied plied ied plied
9th Com Com Compl Com
6.1 1.0 8.5 258 2.7 2.8 2.6 2.5 2.7 5.7 2.6 2.7 2.6 2.8 2.9 2.5 2.7 Complied
month plied plied ied plied
12th Com Com Compl Com
6.2 1.0 8.1 257 2.9 3.0 2.6 2.7 2.9 4.9 2.8 3.0 2.9 2.7 2.8 2.8 3.1 Complied
month plied plied ied plied
18th Com Com Compl Com
6.0 1.0* 8.5 259 2.5 3.0 2.8 2.8 2.9 5.6 2.6 3.2 2.6 2.9 2.8 2.9 2.5 Complied
month plied plied ied plied

*Note■At 18th month, data on aluminium phosphate content converted from aluminium content;

Formula: Aluminium Phosphate Content = Aluminium Content ×(121.95÷26.98).
Alumi Sodium
Appearanc
Volume pH nium chlorideco Osmolality Polysaccharide content
Test e
content ntent
items
Each ml of vaccine contains:
and
Should be Serotype 1: 3.6-6.8µg/ml, Serotype 3: 3.5-6.5µg/ml, Serotype 4:4.2-7.8µg/ml,
standard Should
milky Serotype 5:3.5-6.5µg/ml, Serotype 6A:3.5-6.5µg/ml, Serotype 6B:8.4-15.6µg/ml,
be not 0.18-
uniform Serotype 7F:4.0-7.4µg/ml, Serotype 9V:3.5-6.5µg/ml, Serotype 14:3.8-7.2µg/ml,
less than 5.5-7.5 0.27 7.5-9.5g/L 200-310
suspension Serotype 18C:4.6-8.4µg/ml, Serotype 19A:3.6-6.8µg/ml, Serotype 19F:3.8-7.2µg/ml,
Storage 0.5ml/via mg/ml mOsmol/Kg
after Serotype 23F:4.2-7.8µg/ml
time l
shaking
1 3 4 5 6A 6B 7F 9V 14 18C 19A 19F 23F

24th Complie
Complied 6.1 0.22 9.0 262 4.2 4.4 5.7 5.7 5.6 9.2 6.2 5.8 5.1 5.1 4.2 5.1 4.7
month d
Test Conjugat Total
Abnormal Bacterial
items Conjugate antigencontent e protein protein Sterility
toxicity endotoxin
and content content
standard Each dose contains: Release 41- All guinea NMT Should
Type 1: release standard is ≥30%, standard in expiry date is≥25%; standard 150 pigs and 12.5EU/d comply
Type 3, 6A, 6B, 9V, 14, 18C: release standard is ≥80%, standard in expiry date is≥75%; shall be µg/dos
mice should ose with

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Storage Type 4, 5, 7F: release standard is ≥75%, standard in expiry date is ≥70%; not less e be healthy
requireme
time Type 19A: release standard is ≥70%, standard in expiry date is ≥65%; than and no nts
Type 19F, 23F release standard is ≥65%, standard in expiry date is ≥60%. 85%, abnormal
standard reaction
in expiry during the
date shall observation
be not period, and
less than the weight
80%. of each
1 3 4 5 6A 6B 7F 9V 14 18C 19A 19F 23F
guinea pig
and mouse
should be
increased at
the end of
the period.
24th 100 100 100 100 92% 71 Complied Complied Complied
59% 85% 84% 82% 100% 100% 100% 75% 83%
month % % % %

Version:July. 2020 19
2. Batch no.: GK201709002 Storage condition: 6±2■
Alumi Sodi
Bacter
Appe num m
Volu Osmolali Abnormal ial Steril
aranc pH phosph chlorid Polysaccharide content■µg/ml■
me ty toxicity endoto ity
e ate e
xin
content content
All guinea
Test 18 19 19 23 pigs and mice
1 3 4 5 6A 6B 7F 9V 14
items C A F F should be
and Shoul healthy and
standar d be Shoul no abnormal Shou
d milky d be reaction ld
unifo not during the NMT comp
0.8
rm less 5.5 7.5-9.5 270±50 observation 12.5E ly
-1.2mg
Storage suspe than -7.5 g/L mOsmol/ period, and U/dos with
/ml Serotyoe 1, 3, 4, 5, 6A, 7F, 9V, 14, 18C, 19A, 19F and 23F: 2.2-3.3µg/dose
time nsion 0.5ml Kg the weight of e requi
after /syrin 6B: 4.4-6.6µg/dose each guinea reme
shaki ge pig and nts
ng mouse should
be increased
at the end of
the period.
0th Com Com Compl Com
6.2 0.99 8.01 253 3.0 2.8 2.6 2.6 2.7 4.6 2.8 3.1 2.6 2.9 3.0 2.9 2.4 Complied

month plied plied ied plied
3rd Com Com Compl Com
6.2 0.93 8.60 253 3.1 3.0 2.6 2.4 2.4 4.9 2.9 2.9 2.8 2.6 2.7 2.8 2.6 Complied
month plied plied ied plied
6th Com Com Compl Com
6.1 1.0 8.2 259 2.4 2.7 2.4 2.8 2.8 4.9 2.3 2.4 2.4 2.4 2.3 2.7 2.3 Complied
month plied plied ied plied
9th Com Com Compl Com
6.1 1.0 8.5 258 2.5 2.8 2.6 2.3 2.9 5.0 2.4 2.3 2.6 2.6 2.8 2.6 2.6 Complied
month plied plied ied plied

Version:July. 2020 20
12th Com Com Compl Com
6.0 1.1 8.6 258 2.7 3.1 2.4 2.4 2.5 4.9 2.4 2.9 2.7 2.7 2.7 2.9 2.5 Complied
month plied plied ied plied
18th Com Com Compl Com
6.0 1.0* 8.5 259 2.8 2.6 2.6 2.5 2.6 6.0 2.4 2.4 2.4 2.5 2.6 2.7 2.3 Complied
month plied plied ied plied
*Note■At 18th month, data on aluminium phosphate content converted from aluminium content;

Formula: Aluminium Phosphate Content = Aluminium Content ×(121.95÷26.98).
Alumi Sodium
Appearanc
Volume pH nium chlorideco Osmolality Polysaccharide content
Test e
content ntent
items
Each ml of vaccine contains:
and
Should be Serotype 1: 3.6-6.8µg/ml, Serotype 3: 3.5-6.5µg/ml, Serotype 4:4.2-7.8µg/ml,
standard Should
milky Serotype 5:3.5-6.5µg/ml, Serotype 6A:3.5-6.5µg/ml, Serotype 6B:8.4-15.6µg/ml,
be not 0.18-
uniform Serotype 7F:4.0-7.4µg/ml, Serotype 9V:3.5-6.5µg/ml, Serotype 14:3.8-7.2µg/ml,
less than 5.5-7.5 0.27 7.5-9.5g/L 200-310
suspension Serotype 18C:4.6-8.4µg/ml, Serotype 19A:3.6-6.8µg/ml, Serotype 19F:3.8-7.2µg/ml,
Storage 0.5ml/syr mg/ml mOsmol/Kg
after Serotype 23F:4.2-7.8µg/ml
time inge
shaking
1 3 4 5 6A 6B 7F 9V 14 18C 19A 19F 23F

24th Complie
Complied 6.1 0.21 7.9 261 4.9 4.2 4.9 4.1 5.6 9.4 5.3 5.8 4.7 6.2 5.4 4.7 4.4
month d
Conjugat Total
Abnormal Bacterial
Test Conjugate antigencontent e protein protein Sterility
toxicity endotoxin
items content content
and Each dose contains: Release All guinea
standard Type 1: release standard is ≥30%, standard in expiry date is≥25%; standard pigs and Should
41-
Type 3, 6A, 6B, 9V, 14, 18C: release standard is ≥80%, standard in expiry date is≥75%; shall be mice
should NMT comply
150
Type 4, 5, 7F: release standard is ≥75%, standard in expiry date is ≥70%; not less be healthy 12.5EU/d
with
µg/dos
Storage Type 19A: release standard is ≥70%, standard in expiry date is ≥65%; than and no ose requireme
e
time Type 19F, 23F release standard is ≥65%, standard in expiry date is ≥60%. 85%, abnormal nts
1 3 4 5 6A 6B 7F 9V 14 18C 19A 19F 23F standard reaction

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in expiry during the
date shall observation
be not period, and
less than the weight
80%. of each
guinea pig
and mouse
should be
increased at
the end of
the period.
24th 100 100 100 100 100 97% 61 Complied Complied Complied
76% 83% 88% 100% 100% 85% 100% 77%
month % % % % %

Version:July. 2020 22
3. Batch no.:GK201709003 Storage condition: 6±2■
Alumi Sodi
Bacter
Appe num m
Volu Osmolali Abnormal ial Steril
aranc pH phosph chlorid Polysaccharide content■µg/ml■
me ty toxicity endoto ity
e ate e
xin
content content
All guinea

Test 18 19 19 23 pigs and mice
1 3 4 5 6A 6B 7F 9V 14
items C A F F should be
and Shoul healthy and
standar d be Shoul no abnormal Shou
d milky d be reaction ld
unifo not during the NMT comp
0.8
rm less 5.5 7.5-9.5 270±50 observation 12.5E ly
-1.2mg
Storage suspe than -7.5 g/L mOsmol/ period, and U/dos with
/ml Serotyoe 1, 3, 4, 5, 6A, 7F, 9V, 14, 18C, 19A, 19F and 23F: 2.2-3.3µg/dose
time nsion 0.5ml Kg the weight of e requi
after /syrin 6B: 4.4-6.6µg/dose each guinea reme
shaki ge pig and nts
ng mouse should
be increased
at the end of
the period.
0th Com Com Compl Com
6.22 1.07 8.96 288.5 3.1 2.8 2.6 2.6 2.6 4.7 2.8 3.0 2.7 2.6 3.1 2.9 2.5 Complied
month plied plied ied plied
3rd Com Com Compl Com
6.0 1.0 8.8 289 2.4 3.0 2.6 2.5 2.5 5.6 2.2 3.1 2.7 2.6 2.7 2.4 2.5 Complied
month plied plied ied plied
6th Com Com Compl Com
6.1 1.0 8.6 258 2.5 2.5 2.2 2.7 2.4 4.7 3.0 2.7 2.4 2.3 2.3 2.6 2.2 Complied
month plied plied ied plied
9th Com Com Compl Com
6.1 1.0 8.5 256 2.6 2.8 2.6 2.4 2.8 6.2 2.3 2.9 2.7 2.9 2.5 2.8 2.7 Complied
month plied plied ied plied

Version:July. 2020 23
12th Com Com Compl Com
6.1 1.0 8.3 257 2.9 2.7 2.5 2.5 2.5 4.9 2.5 2.8 2.5 2.7 2.7 2.8 2.5 Complied
month plied plied ied plied
18th Com Com Compl Com
6.1 1.0* 8.3 261 2.8 2.9 2.5 3.0 2.5 5.2 2.5 2.8 2.9 3.0 3.0 2.3 2.5 Complied
month plied plied ied plied
*Note■At 18th month, data on aluminium phosphate content converted from aluminium content;

Formula: Aluminium Phosphate Content = Aluminium Content ×(121.95÷26.98).
Alumi Sodium
Appearanc
Volume pH nium chlorideco Osmolality Polysaccharide content
Test e
content ntent
items
Each ml of vaccine contains:
and
Should be Serotype 1: 3.6-6.8µg/ml, Serotype 3: 3.5-6.5µg/ml, Serotype 4:4.2-7.8µg/ml,
standard Should
milky Serotype 5:3.5-6.5µg/ml, Serotype 6A:3.5-6.5µg/ml, Serotype 6B:8.4-15.6µg/ml,
be not 0.18-
uniform Serotype 7F:4.0-7.4µg/ml, Serotype 9V:3.5-6.5µg/ml, Serotype 14:3.8-7.2µg/ml,
less than 5.5-7.5 0.27 7.5-9.5g/L 200-310
suspension Serotype 18C:4.6-8.4µg/ml, Serotype 19A:3.6-6.8µg/ml, Serotype 19F:3.8-7.2µg/ml,
Storage 0.5ml/syr mg/ml mOsmol/Kg
after Serotype 23F:4.2-7.8µg/ml
time inge
shaking
1 3 4 5 6A 6B 7F 9V 14 18C 19A 19F 23F

24th Complie
Complied 6.1 0.21 8.2 260 6.0 5.0 6.4 5.3 5.8 12.0 6.0 5.7 6.6 7.6 6.3 6.0 5.3
month d
Conjugat Total
Abnormal Bacterial
Test Conjugate antigencontent e protein protein Sterility
toxicity endotoxin
items content content
and Each dose contains: Release All guinea
standard Type 1: release standard is ≥30%, standard in expiry date is≥25%; standard pigs and Should
41-
Type 3, 6A, 6B, 9V, 14, 18C: release standard is ≥80%, standard in expiry date is≥75%; shall be mice
should NMT comply
150
Type 4, 5, 7F: release standard is ≥75%, standard in expiry date is ≥70%; not less be healthy 12.5EU/d
with
µg/dos
Storage Type 19A: release standard is ≥70%, standard in expiry date is ≥65%; than and no ose requireme
e
time Type 19F, 23F release standard is ≥65%, standard in expiry date is ≥60%. 85%, abnormal nts
1 3 4 5 6A 6B 7F 9V 14 18C 19A 19F 23F standard reaction

Version:July. 2020 24
in expiry during the
date shall observation
be not period, and
less than the weight
80%. of each
guinea pig

and mouse
should be
increased at
the end of
the period.
24th 100 100 100 100 100 99% 111 Complied Complied Complied
83% 86% 91% 100% 100% 88% 100% 81%
month % % % % %

Conclusion:
3 batches of 13-valent pneumococcal polysaccharide conjugate vaccine GK201709001, GK201709002, GK201709003 were kept at 6±2 for 24months. Test results at all the test point were all complied with requirements of Specification of 13-valent pneumococcal polysaccharide conjugate vaccine Code WSYX-QC-TS-06-09-0002, in conclusion, 13-valent pneumococcal polysaccharide conjugate vaccine could be kept at 6±2 for 24months.

Version:July. 2020 25
Accelerated Stability data of 13-valent Pneumococcal Polysaccharide Conjugate vaccinePre-filled Syringe

1. Batch no.: GK201709001 Storage condition: 25±2
Alumi Sodi
um Bacter
um m
Volu Osmolali Abnormal ial Steril
aranc pH phosph chlorid Polysaccharide contentµg/ml
me ty toxicity endoto ity
e ate e
xin
content content
All guinea
Test 18 19 19 23 pigs and mice
1 3 4 5 6A 6B 7F 9V 14
items C A F F should be
and Shoul healthy and
standar d be Shoul no abnormal Shou
d milky d be reaction ld
unifo not during the NMT comp
0.8
rm less 5.5 7.5-9.5 270±50 observation 12.5E ly
-1.2
Storage suspe than -7.5 g/L mOsmol/ period, and U/dos with
mg/ml Serotyoe 1, 3, 4, 5, 6A, 7F, 9V, 14, 18C, 19A, 19F and 23F: 2.2-3.3µg/dose
time nsion 0.5ml Kg the weight of e requi
after /syrin 6B: 4.4-6.6µg/dose each guinea reme
shaki ge pig and nts
ng mouse should
be increased
at the end of
the period.
0th Com Com Compl Com
6.22 272.5 8.81 0.94 3.0 2.7 2.6 2.6 2.6 4.8 2.8 3.1 2.6 2.6 3.0 3.0 2.4 Complied
month plied plied ied plied
1st Com Com Compl Com
6.1 273 8.85 1.07 2.8 2.8 2.5 2.4 3.2 5.5 2.5 3.1 2.9 2.6 2.8 2.3 2.7 Complied
month plied plied ied plied
2nd Com Com Compl Com
6.2 272 8.6 1.16 2.7 2.5 2.6 2.3 2.5 5.2 2.4 2.7 2.8 2.6 3.1 2.4 2.6 Complied
month plied plied ied plied

Version:July. 2020 26
3rd Com Com Compl Com
6.2 273 8.2 1.0 2.5 2.9 2.5 2.4 3.1 5.2 2.5 2.6 2.7 2.8 2.5 2.4 2.6 Complied
month plied plied ied plied
6th Com Com Compl Com
6.1 258.5 8.3 1.0 2.5 2.5 2.4 2.4 2.6 5.0 2.7 2.4 2.5 2.6 2.5 2.8 3.1 Complied
month plied plied ied plied
9th Com Com Compl Com
6.1 260 8.5 1.0 2.7 2.6 2.6 2.5 2.9 5.8 2.5 3.0 2.7 2.8 2.7 2.6 2.7 Complied
month plied plied ied plied

Version:July. 2020 27
2. Batch No.: GK201709002 Storage condition: 25±2

Alumi
Sodi
um Bacter
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Volu phosp Osmolali Abnormal ial Steril
aranc pH chlorid Polysaccharide contentµg/ml
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e e
conte xin
content
nt
All guinea
Test 18 19 19 23
1 3 4 5 6A 6B 7F 9V 14 pigs and mice
items C A F F should be
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standar
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milky d be reaction Id
unifo not during the NMT comp
0.8
rm less 5.5 7.5-9.5 270±50 observation 12.5E ly
Storage -1.2
suspe than -7.5 g/L mOsmol/ period, and U/dos with
time mg/ml Serotyoe 1, 3, 4, 5, 6A, 7F, 9V, 14, 18C, 19A, 19F and 23F: 2.2-3.3µg/dose
nsion 0.5ml Kg the weight of e requi
after /syrin 6B: 4.4-6.6µg/dose each guinea reme
shaki ge pig and nts
ng mouse should
be increased
at the end of
the period.
0th Com Com Compl Com
6.20 253 8.01 0.99 3.0 2.8 2.6 2.6 2.7 4.6 2.8 3.1 2.6 2.9 3.0 2.9 2.4 Complied
month plied plied ied plied
1st Com Com Compl Com
6.1 253.5 8.05 0.99 2.8 2.7 2.5 2.4 3.1 5.3 2.5 2.3 2.9 2.7 2.8 2.4 2.6 Complied
month plied plied ied plied
2nd Com Com Compl Com
6.1 253 8.1 1.14 2.9 2.6 2.6 2.5 2.4 5.8 2.4 2.6 2.9 2.9 3.1 2.4 2.7 Complied
month plied plied ied plied

Version:July. 2020 28
3rd Com Com Compl Com
6.2 253 8.1 1 2.4 2.8 2.4 2.3 2.9 5.2 2.3 2.8 2.6 2.9 2.5 2.3 2.7 Complied
month plied plied ied plied
6th Com Com Compl Com
6.1 259 8.4 1.1 2.7 2.4 2.7 2.8 2.7 4.8 2.8 2.7 2.5 3.2 2.7 2.6 2.8 Complied
month plied plied ied plied
9th Com Com Compl Com
6.1 261 8.5 1.0 2.8 2.6 2.6 2.7 2.9 5.6 2.7 3.0 2.7 3.2 2.6 2.8 2.8 Complied
month plied plied ied plied

Version:July. 2020 29
3. Batch no.: GK201709003 Storage condition: 25±2■
Alumi
Sodiu
num Bacter
Appe m
Volu phosp Osmolali Abnormal ial Steril
aranc pH chlorid Polysaccharide content■µg/ml■
me hate ty toxicity endoto ity
e e
conte xin
content
nt
All guinea
Test 18 19 19 23
1 3 4 5 6A 6B 7F 9V 14 pigs and mice
items C A F F should be
and
Shoul healthy and
standar
d be Shoul no abnormal Shou
d

milky d be reaction Id
unifo not during the NMT comp
0.8
rm less 5.5 7.5-9.5 270±50 observation 12.5E ly
Storage -1.2
suspe than -7.5 g/L mOsmol/ period, and U/dos with
time mg/ml Serotyoe 1, 3, 4, 5, 6A, 7F, 9V, 14, 18C, 19A, 19F and 23F: 2.2-3.3µg/dose
nsion 0.5ml Kg the weight of e requi
after /syrin 6B: 4.4-6.6µg/dose each guinea reme
shaki ge pig and nts
ng mouse should
be increased
at the end of
the period.
0th Com Com Compl Com
6.22 288.5 8.96 1.07 3.1 2.8 2.6 2.6 2.6 4.7 2.8 3.0 2.7 2.6 3.1 2.9 2.5 Complied
month plied plied ied plied
1st Com Com Compl Com
6.1 289 9.1 1.0 2.9 2.8 2.6 2.5 2.6 5.4 2.4 2.3 2.8 2.6 2.8 2.3 2.5 Complied
month plied plied ied plied
2nd Com Com Compl Com
6.1 288 8.9 1.14 2.8 2.9 2.4 2.5 2.6 5.6 2.4 2.7 2.9 2.7 3.1 2.4 2.6 Complied
month plied plied ied plied

Version:July. 2020 30
3rd Com Com Compl Com
6.1 289 8.3 1.0 2.5 3.1 2.6 2.4 2.6 5.6 2.8 3.0 2.7 2.8 2.6 2.4 2.6 Complied
month plied plied ied plied
6th Com Com Compl Com
6.1 258 8.0 1.1 2.8 2.5 2.9 2.8 3.2 5.4 2.6 2.7 2.4 2.7 3.1 2.5 3.0 Complied
month plied plied ied plied
9th Com Com Compl Com
6.1 260 8.5 1.0 2.6 3.2 2.8 2.5 2.7 5.4 2.6 2.6 2.8 2.9 2.7 2.8 2.8 Complied
month plied plied ied plied

Conclusion:

3 batches of 13-valent pneumococcal polysaccharide conjugate vaccine GK201709001, GK201709002, GK201709003 were kept at $25\pm 2^{\circ}\text{C}$ for 9months. Test results at all the test point were all complied with requirements of Specification of 13-valent pneumococcal polysaccharide conjugate vaccine Code WSYX-QC-TS-06-09-0002, in conclusion, 13-valent pneumococcal polysaccharide conjugate vaccine could be kept at $25\pm 2^{\circ}\text{C}$ for 9months.

Version: July, 2020 31

SOP of Yuxi Walvax Biotechnology Co., Ltd

Code WSYX-QC-SOP-00-09-0120

SOP for sodium chloride

Version 06 page 1/4

content determination

Confidential files No copy

1 Objective

Standardize operating procedure for sodium chloride content determination.

2 Definition/abbreviation

No

3 Applicable scope

This SOP is applicable to the product manufactured by Yuxi Walvax.

4 Responsibility

All the QC lab operators shall strictly conduct the SOP

5 Materials and equipment

5.1 Use of equipment

Equipment Equipment

No. Equipment Code Manufacturer Notes

Name model

electronic BS124S/ WYQCF0036/ Or equivalent

1 Sartorius

balance BSA124S WYQCF0357 equipment

5.2 Use of reagents

No. Reagent name Grade Notes

1 nitric acid AR Content 65-68%

2 silver nitrate AR /

3 ammonium rhodanate AR /

4 ammonium ferric sulfate AR /

5.3 Use of materials

No. Material name No. Material name

1 triangular flask 2 beaker

3 acid burette 4 volumetric flask

5 One-mark pipette 6 pipette

7 pipette tips 8

SOP of Yuxi Walvax Biotechnology Co., Ltd

Code WSYX-QC-SOP-00-09-0120

SOP for sodium chloride

Version 06 page 2/4

content determination

Confidential files No copy

6 Procedures

6.1 Principle

In this method, the protein in the test sample is destroyed by nitric acid. Then, excess silver nitrate is added to reacts with the chloride ion contained in the test sample completely, and silver chloride precipitation is generated. The excess silver nitrate is titrated with ammonium thiocyanate titrant. According to the consumption of ammonium thiocyanate titrant, the content of sodium chloride in the test sample can be calculated.

6.2 Reagent preparation

6.2.1 Ammonium thiocyanate titrating solution (0.1mol/L)

Weigh 8.0g of ammonium thiocyanate, dissolve with water and dilute to 1000ml. Shake well. Precisely measure 25ml of silver nitrate titration solution (0.1mol/L), add water 50ml, nitric acid 2ml, 2ml of 8% Ammonium ferric sulfate indicator solution, titrate with this solution until the solution is slightly reddish brown; it still does not fade after a violent shake, that is the end. The concentration of the liquid is calculated according to the consumption of the liquid.

6.2.2 Preparation of ammonium thiocyanate titration solution (0.05mol/L)

Precisely measure 100ml of ammonium thiocyanate titrating solution (0.1mol/l) with one-mark pipette and put it into a 200ml volumetric flask, add purified water to dilute it to the marking line, shake well (it can be prepared according to the required amount).

6.2.3 0.1mol/L silver nitrate solution

Weigh 17.0g of silver nitrate into a beaker, dissolve with purified water, transfer it to a 1000ml volumetric flask and fix the volume to the marking line, mix well. Transfer to a 1000ml brown reagent bottle, labelled standby.

6.2.4 Saturated potassium permanganate solution

Weigh at least 6.38g of potassium permanganate, add 100ml of purified water and boil for 15 minutes to make it dissolve into saturated solution, put it into a brown reagent bottle, and close the plug. Take the supernatant to use.

SOP of Yuxi Walvax Biotechnology Co., Ltd

Code WSYX-QC-SOP-00-09-0120

SOP for sodium chloride

Version 06 page 3/4

content determination

Confidential files No copy

6.2.5 8.0mol/L nitric acid solution

In the 100ml volumetric flask, add the proper amount of purified water. Measure 50ml of concentrated nitric acid and lowly inject into volumetric flask, shake while adding. Place to cool before to scale, then add purified water to 100ml, mix well, labelled, stand by.

6.2.6 8% Ammonium ferric sulfate indicator solution

Weigh 8g of ammonium ferric sulfate dodecahydrate into beaker, dissolve with purified water to 100ml and transfer to a 100ml volumetric bottle, wash the beaker with purified water for 3 times and transfer the lotion to the volumetric flask, fix the volume to the scale, mix well,

filter it and transfer it to the reagent flask, label it for standby.

6.3 Test procedure

6.3.1 Digestion

Accurately measure 1000μl of test samples using a pipette with a measuring range of 100-1000μl (two parallel samples for each test sample), add 5000μl of 0.1mol/L silver nitrate solution (if the protein content is high, add 2.0ml of saturated potassium permanganate solution) using a pipette with a measuring range of 0.5-5ml, mix well. Add 10ml of 8.0mol/L nitrate solution, heat and digest until the solution is clear, place to cool.

6.3.2 Titration

Add 50ml purified water into the above solution, mix well. Add 1000μl of 8% ammonium ferric sulfate indicator solution, mix well, titrate with ammonium thiocyanate titration solution (0.05mol/L) until the solution is light brown red, and still does not fade after shaking, that is the end point. The results of titration were corrected by blank test (not digested, two parallel).

6.4 Result calculation

Sodium chloride content (g/L) = $(V_0 - V_x) \times C \times 58.45$

V_0 : volume of ammonium thiocyanate titrating solution consumed in blank test, ml;

V_x : Volume of ammonium thiocyanate titrate consumed by test sample, ml;

C: Concentration of ammonium thiocyanate titrating solution, mol/L;

SOP of Yuxi Walvax Biotechnology Co., Ltd

Code WSYX-QC-SOP-00-09-0120

SOP for sodium chloride

Version 06 page 4/4

content determination

Confidential files No copy

Relative molecular weight of sodium chloride:58.45

6.5 Result judgment

Carry out according to corresponding quality standard

7 Flow chart

No

8 Notes

8.1 When digesting, use small fire, in case it is boiled dry. During the digestion process, the triangular bottle should be picked up and shake to make it evenly heated and prevent the solution from ejecting.

8.2 When digesting, please take precautions, wear masks and cotton gloves

8.3 When titrating, do not drop the titrating solution too fast; when it is near the end, slow down the titration rate, pay attention to the operation of half drop (if the titrant remains on the bottleneck of the triangular flask, wash it with appropriate amount of purified water).

9 Reference or standard basis

9.1 CP2015 volume III, general rule 3107

10 Revision history

Issuing Time in Revision

Version Revision basis, reason and content Note

time force Dept.

01 Nov. 01 Dec. Changed when file expired

03 QC Lab

2011 2011 Complete some parts of the content

1 Nov. 01 Dec. Changed when file expired

04 QC Lab

2014 2014 Complete some parts of the content

30 Sep. 01 Oct.

05 QC Lab Complete some parts of the content

2017 2017

SOP of Yuxi Walvax Biotechnology Co., Ltd.

Code WSYX-QC-SOP-00-09-0135

Edition

Osmolality test SOP 02 Page 1/3

No.

Confidential files No copy

1 Objective

Standardize operating procedure for osmolality test.

2 Definition/abbreviation

None.

3 Scope of application

It applies to osmolality test operation.

4 Responsibility

All operators shall perform the tests in strict accordance with this SOP.

5 Equipments and Materials

5.1 Equipments

Equipment Equipment

No. Equipment code Manufacturer Notes

name model

Fiske Or equivalent

1 Osmometer Advanced 3250 WYQCF0392

Associates equipment

5.2 Materials

No. Material name No. Material name

1 Sample tube 2 Pipette

Clinitrol™ 290 mOsm/kg

3 4 Pipette tips

reference solution

6 Working procedure

6.1 principle

In general, the drop of freezing point of the solution is used to indirectly measure its osmolality.

6.2 Test procedure

6.2.1 Verification calibration: take 250μl Clinitrol™ 290 mOsm/kg reference solution into sample tube and put into the freezing chamber. Press START key and input 290, press the ENTER button and START button. The readings are displayed on the screen. The standard for SOP of Yuxi Walvax Biotechnology Co., Ltd.

Code WSYX-QC-SOP-00-09-0135

Edition

Osmolality test SOP 02 Page 2/3

No.

Confidential files No copy

verifying calibration is 290±2mOsm/kg and should be tested at least for 2 reference solution.

6.2.2 Sample test: Take sample 250μl to clean sample tube and put into the freezing chamber. Press START key, then enter sample batch number, press ENTER key, press START button. The reading is displayed on the screen while the printer prints the results. At least 2 samples should be tested for testing.

6.2.3 After each test, gently wipe the probe, stirring/freezing line, mandrel and freezing chamber top with a soft, dry, cotton-free, non-ionic paper towel in order to avoid contamination of the next sample to be tested.

6.2.4 After testing, insert an empty sample tube in the freezing chamber to avoid the scrap falling into the chamber.

6.2.5 Press TEST key, the screen displays "Select Test Item". Then press < key to scroll the screen menu, select "Head Up/Down Test", press START key to show "[START]Test[STOP]", press START key and the probe runs up and down, press the STOP key when the probe runs to bottom, shut up the Osmometer when the probe stops.

6.3 Results judgment:

Judge according to the corresponding quality standard.

7 Flowchart

N/A

8 Matters need attention

8.1 Be careful not to bend the probe or stir/freezing line after each test cleaning.

8.2 Cold enzymes contain harmful chemicals, and appropriate personal protective measures should be taken when testing.

9 Reference or standard basis

9.1 Chinese Pharmacopeia, Volume III, general rule 0632 osmolality test method. China medical technology press, 2015 version.

10 Revision history

SOP of Yuxi Walvax Biotechnology Co., Ltd.

Code WSYX-QC-SOP-00-09-0135

Edition

Osmolality test SOP 02 Page 3/3

No.

Confidential files No copy

Issuing Time in Revision Revision basis, reason and

Version Notes

time force Dept. content

31 Oct. 10 Nov. QC Complete some parts of the

00

2014 2014 laboratory content

06 Nov. 07 Nov. QC Complete some parts of the

01

2017 2017 laboratory content

Standard Operating Procedures of Yuxi Walvax Biotechnology Co., Ltd.

Coding WSYX-QC-SOP-00-09-0754
SOP for Residual
Version 06 Pages 1/3
Ammonium Content Test
Confidential documents, no copying.

1 Purpose
Standardize the determination of residual ammonium content.
2 Definitions/abbreviations
No
3 Scope of application
Suitable for the detection of residual ammonium content.
4 Responsibilities
Inspectors follow this procedure strictly.
5 Equipment and materials
5.1 Equipment

Serial Equipment
Equipment model Manufacturers Note
number name
Kai's nitrogen
1 kjltectTM 2200 FOSS or equivalent equipment
fixed instrument
Kai's nitrogen
2 Kjeltect 8200 FOSS or equivalent equipment
fixed instrument
5.2 Solution
Serial
Solution name Category
number
1 10% sodium tungstate solution Solution

2 Sulfuric acid solution (0.33 mol/L) Solution
3 6 mg/ml ammonium sulfate solution Solution
4 4% sodium hydroxide solution Solution
5 Mixed indicator solution Indication fluid
6 2% boric acid absorption solution Solution
7 40% sodium hydroxide solution Solution

8 Hydrochloric acid titration solution (0.1 mol/L) Titrant
Standard Operating Procedures of Yuxi Walvax Biotechnology Co., Ltd.

Coding WSYX-QC-SOP-00-09-0754
SOP for Residual
Version 06 Pages 2/3
Ammonium Content Test
Confidential documents, no copying.

9 Hydrochloric acid titrant (0.02 mol/L) Titrant
6 Work procedures
6.1 Preparation of test solution
Accurately measure 4.0 mL of the test sample, add 28.0 ml of water, add 4.0 mL of 10% sodium tungstate solution and 4.0 mL of sulfuric acid solution (0.33 mol/L), shake well, stand for 30 minutes, and then filter.
6.2 precisely measure 10 ml of protein filtrate and put it into the digestion tube. Make two parallel tubes. Then add 1.0 ml of 4% sodium hydroxide solution and 10 ml of water to each tube.
6.3 Distillation
Before distillation, steam and wash the distiller for more than 15 minutes. When the receiving liquor is purple red, take a triangular flask on the right side of Kjeldahl nitrogen analyzer, insert the condenser tube and boric acid absorption solution feeding tube, and then install the digester tube with added sample to the distiller on the left side of Kjeldahl nitrogen analyzer. Run the program, the instrument will start automatic distillation. After stopping the distillation, take down the triangular flask, rinse the condenser tube with water, and collect the eluent in a triangular flask and prepare for titration.
6.4 Titration
Titrate with hydrochloric acid titration solution (0.02 mol/L) until the solution changes from blue-green to gray purple, and the titration results are corrected by blank test.
6.5 Result calculation
$$(\text{V}_1 - \text{V}_0) \times c \times 14.01 \times 4.715$$

Ammonium sulfate content (%) = $\frac{\text{V}_1 - \text{V}_0}{\text{V}_1} \times 100$
1000
Where: V1 is the volume of hydrochloric acid titrant consumed by the test sample, ml;
V0 is the volume of hydrochloric acid titrant consumed by blank control, ml;
c is the concentration of hydrochloric acid titration solution, mol/L;
4.715 is a constant (1 g nitrogen is equivalent to 4.715 g ammonium sulfate);
14.01 is the relative atomic mass of nitrogen.
Standard Operating Procedures of Yuxi Walvax Biotechnology Co., Ltd.

Coding WSYX-QC-SOP-00-09-0754
SOP for Residual
Version 06 Pages 3/3
Ammonium Content Test
Confidential documents, no copying.

7 Flow chart
No
8 Precautions

8.1 Distilled ammonia receiving fluid should be titrated as soon as possible, so as to avoid placing it too long, affecting the measurement results.
8.2 In the distillation process, if there is no black CuO, indicating that the amount of alkali added is insufficient, the amount of alkali should be replenished or redo experiment.
8.3 The triangular bottle should be fuchsia after adding 2% boric acid absorption, such as green, indicating that the triangle bottle has alkaline substance pollution.
8.4 Interference with ammonia and alkaline gases should be avoided in the operating environment.
9 References or standard basis
9.1 Chinese Pharmacopoeia, volume 4, 3104 Ammonium Sulphate Measurements, China Pharmaceutical Science and Technology Press, 2015, Page 227
10 Revision history

Revision Basis, reasons and content of the

Version Release Effective

Department amendment Note

No. Time time

nt

03 2011.11.01 2011.12.01 QC File expiration

Laboratory

04 2014.11.01 2014.12.01 QC According to the 2015 edition of

Laboratory the Pharmacopoeia requirements,

improve the text

Pieces

05 2016.08.04 2016.08.05 QC Improve the verification record

Laboratory

SOP of Yuxi Walvax Biotechnology Co., Ltd.

Code WSYX-QC-SOP-00-09-0109
SOP for the extractable 1/2
Version 07 Page
volume test
Confidential files No copy

1 Objective

Standardize the operation of extractable volume test.

2 Definitions/Abbreviations

N/A

3 Scope of application

It is applicable to test the extractable volume of small volume injections manufactured by Yuxi Walvax.

4 Responsibilities

All the operators shall strictly conduct the SOP.

5 Materials and equipment

5.1 Use of equipment

No

5.2 Use reagent

No

5.3 Use of materials

number Material name number Material name

1 1ml Disposable Syringe 2 1ml measuring cylinder

6 Working procedure

6.1 Test procedure

6.1.1 Take 1 vial (Syringe) of the test product, take out the contents by a 1ml disposable syringe, then Inject into the standard measuring cylinder (not drain out the liquid in the needle), inspect at room temperature, and check the other samples one by one.

6.1.2 Take out 5 vials (or Syringe) of the test sample, use a disposable syringe to take the contents of one vial (Syringe) of the test product and inject into a 1ml measuring cylinder (not drain out the SOP of Yuxi Walvax Biotechnology Co., Ltd.

Code WSYX-QC-SOP-00-09-0109

SOP for the extractable 2/2

Version 07 Page

volume test

Confidential files No copy

liquid in the needle), inspect at room temperature and check the rest of test products.

6.2 Result Judgement

6.2.1 Read out the volume of each vial (Syringe), the volume of injection should not be less than label amount.

7 Flow chart

No

8 Matters need attention

8.1 The size of the measuring tool shall be such that the volume to be measured shall account for at least 40% of its rated volume.

9 Reference or standard basis

9.1 "Chinese Pharmacopoeia" version 2015, volume III, general rules 0102

10 Revision history

Issuing Time in Revision Revision basis, reason and

Version Note

time force Dept. content

03 01 Nov. 01 Dec. QC Changed when file expired

2011 2011 laboratory

23 Apr. 25 Apr. QC Complete some parts of the

04

2015 2015 laboratory content

05 18 Mar. 20 Mar. QC Complete some parts of the

2016 2016 laboratory content

06 31 July 02 Aug. QC Complete some parts of the

2016 2016 laboratory content

Standard Operation Procedures of Yuxi Walvax Biotechnology Co., Ltd.

Code WSYX-QC-SOP-00-09-0011
SOP for Flocculent Units
Version 08 Pages 1/5
of Toxoid Test
Confidential documents, no copying.

1 Purpose
Standardize the determination of flocculent units of toxoid.
2 Definitions/abbreviations
No
3 Scope of application
It is suitable for the flocculent determination of tetanus toxoid fermentation liquid, tetanus toxin seed, tetanus toxin raw liquid, diphtheria toxin fermentation liquid, diphtheria toxin seed liquid, diphtheria toxin progenitor fluid.

4 Responsibilities
Quality control laboratory personnel in strict accordance with this standard operating procedures.
5 Materials and equipment
5.1 Using the device
Serial Device model
Device name Manufacturers
number
Three use electric
Tianjin Tester Instruments Co.,
1 constant temperature SHH.W21.600
Ltd.
water tank
5.2 Use of materials/reagents

Serial
Material/Reagent Name Level/Category
number
National standard for tetanus flocculent
1 Standard
antitoxin
National standard for diphtheria flocculent
2 Standard
antitoxin
3 Borate saline buffer Buffer
5.3 Materials used

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Standard Operation Procedures of Yuxi Walvax Biotechnology Co., Ltd.

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SOP for Flocculent Units
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Serial Serial
Materials Materials
number number
1 Tube 3 Test tube rack
2 Pipette 4 Beaker
5 Volumetric flask
6 Work procedures
6.1 Working principle: according to the toxoid and the corresponding antitoxin in the appropriate content, proportion, temperature, reaction time and other conditions, antigen antibody binding occurs in the test tube to produce flocculent reaction visible to the naked eye.
6.2 Reagents
6.2.1 national standard for tetanus flocculent antitoxin: China Academy of food and drug control
6.2.2 national standard for diphtheria flocculent antitoxin: China Academy of food and drug control
6.2.3 Borate buffer solution
6.2.3.1 Preparation method: weigh 0.5 g sodium tetraborate (Na₂B₄O₇·10H₂O), 4.5 g boric acid and 8.5 g sodium chloride, dissolve them with water and dilute to 1000 ml, adjust the pH value to 7.0-7.2
6.3 Test procedure:
6.3.1 preparation of standard solution: take diphtheria or tetanus flocculent antitoxin national standard, and accurately dilute it with borate buffer solution to a solution containing 100 Lf per 1 ml.
6.3.2 Preparation of test solution: take appropriate amount of test sample, add borate buffer solution to dilute to appropriate flocculent unit.
6.3.3 Determination
According to the need, accurately measure 300, 400, 500, 600 and 700 µl of antitoxin flocculent reaction standard solution per 1 ml, and add them into the flocculent reaction tube respectively.

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Accurately measure 1000 µl of the test solution, quickly and accurately add it into the flocculent reaction tubes, shake well, place in a 45-50℃ water bath for continuous observation, and record the order and time of flocculent appearance (if flocculent occurs first) The volume of antibody was not in the middle tube and could be adjusted before detection; Then take 5 flocculent reaction tubes and repeat the above test. Put the first flocculent tube in the middle, add two tubes of antitoxin flocculent reaction standard solution in front of and after each tube, with an interval of 50 µl, and then add 1000 µl of test sample to each tube to observe the flocculent appearance; According to the results, repeat the test again. Put the first flocculent tube in the middle, add two tubes of different amount of antitoxin flocculent reaction standard solution before and after each tube, with an interval of 20 µl, and then add 1000 µl of test sample to each tube to observe the flocculent appearance, and take the same value 2-3 times as the final measurement value.

6.4 Result calculation

6.4.1 calculate according to the following formula:

Flocculent unit (LF/ml) = $V \times n \times 100$

Where V is the volume of antitoxin flocculent reaction standard solution containing

100Lf per 1 ml when flocculent occurs first, ml;

n is the dilution ratio of the test sample.

7 Flow chart

No

8 Precautions

8.1 during the determination, the reaction condition of the sample should be observed at any time to avoid missing the optimal reaction volume.

8.2 for the first flocculent antibody volume, whether the highest tube or the lowest tube appears, the dilution ratio of the test sample can be adjusted for detection, and the sample volume of antibody volume can be adjusted for detection (if the flocculent antibody volume is 300 µl, the antibody volume of 100, 200, 300, 400 and 500 µl can be detected again).

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SOP for Flocculent Units

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of Toxoid Test

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9 References or standard basis

9.1 China Pharmacopeia, 2015 edition, General rules 3506

10 Revision history

Basis, reasons and content of the

Version Effective Revision amendment

Note

No. time Department (Note: If the space is not written enough, a

separate page description may be attached)

03 2011.12.01 QC File expiration

Laboratory

04 2014.12.01 QC Revised references

Laboratory

05 2016.09.01 QC Revise the scope of application,

Laboratory materials and equipment, work

procedures and relevant records

06 2017.10.15 QC Revised appendix

Laboratory

07 2018.09.01 QC Increase the measurement operation: if

Laboratory the volume of the antibody with

flocculent appearance is not in the

intermediate tube, the detection can be

carried out after adjustment; add the

precautions 8.2 for the antibody volume

with the first flocculent appearance,

whether the highest tube or the lowest

tube, the dilution ratio of the test

sample can be adjusted for detection,

and the sample volume of the antibody

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of Toxoid Test

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volume can be adjusted for detection (if

the antibody volume with flocculent

appearance is 300 µl, the antibody

volume of 100, 200, 300, 400 and 500

µl can be detected again).

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Code WSYX-QC-SOP-00-09-0143
SOP for purity test of
tetanus toxoid purified Version 00 Pages 1/3
liquid
Confidential documents, no copying.

1 Purpose

Standardize the method of determining the purity of tetanus toxoid purified liquid.

2 Definitions/abbreviations

No

3 Scope of application

Suitable for the determination of the purity of tetanus toxoid purification fluid.

4 Responsibilities

Quality control laboratory inspectors strictly in accordance with this SOP for sample inspection.

5 Equipment and materials

5.1 Equipment

Serial Equipment name Equipment model Manufacturers Note
number

1 Liquid or equivalent

LC-20A Shimadzu
chromatography equipment

5.2 Reagents

Serial number Reagent Name Level

1 Disodium hydrogen phosphate AR

2 Sodium dihydrogen phosphate AR

3 Dimethyl sulfoxide (DMSO) AR

5.3 Materials

Serial number Material name Serial number Material name

1 Gunhead 3 In-sample bottle

2 Pipette 4 Columns

6 Work procedures

6.1 Reagents and formulation methods

——0.1 mol/L PB buffer (pH7.5)

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SOP for purity test of
tetanus toxoid purified Version 00 Pages 2/3
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964g sodium dihydrogen phosphate ($\text{NaH}_2\text{PO}_4 \cdot 2\text{H}_2\text{O}$) and 29.0 g sodium hydrogen phosphate ($\text{Na}_2\text{HPO}_4 \cdot 12\text{H}_2\text{O}$) were accurately weighed and dissolved in 1000 ml of water for injection or ultra-pure water. After mixing, the mixture was filtered with 0.45 μM filter membrane and ultrasonic was obtained.

——0.05% DMSO solution (i.e. 0.05% DMSO)

First take about 50 ml of water for injection or ultrapure water into a 100 ml volumetric flask, then precisely add 50 μL dimethyl sulfoxide, and finally fix the volume with water for injection or ultra pure water, and mix evenly. This solution should be prepared immediately.

6.2. Test procedure

6.2.1 column efficiency determination

6.2.1.2 determination parameters of column efficiency

Mobile phase: water for injection or ultrapure water, filtered with 0.45 μM filter membrane, and obtained by ultrasonic.

Injection volume: 20 μL , detection wavelength: 210 nm, flow rate: 1.0ml/min

Column temperature: 30 $\square$, sample: 0.05% DMSO, running time: 30min

6.2.1.2 column efficiency determination steps: after confirming that the instrument is stable, inject a blank needle and a 0.05% DMSO solution, record the chromatogram, and calculate the number of theoretical plates with 0.05% DMSO chromatographic peak, and the number of theoretical plates should not be less than 5000.

6.2.2 test article detection

6.2.2.1 preparation of test sample: dilute the test sample with water for injection to the protein concentration of 2000-3000 $\mu\text{g}/\text{ml}$, filter it with 0.45 μM filter membrane, bottle and load the sample; if the protein concentration is lower than 2000-3000 $\mu\text{g}/\text{ml}$, filter directly with 0.45 μM filter membrane, bottle and load the sample.

6.2.2.2 test parameters

Chromatographic column: TSK-gel G5000PW (7.5 mm * 300 mm)

Mobile phase: 0.1 mol/LPB buffer (pH 7.5)

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tetanus toxoid purified Version 00 Pages 3/3
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Injection volume: 10 μL , detection wavelength: 280nm

Flow rate: 1.0 ml/min column temperature: 30 $\square$, running time: 30min

6.2.2.3 detection steps: after confirming that the instrument is stable, inject a blank needle, and then inject the test sample. The purity of purified tetanus toxin was calculated by area normalization method.

6.3 result determination

According to the corresponding quality standards.

7 Flow chart

No

8 Precautions

8.1 after use, wash the chromatographic column and pipeline with filtered water for injection or ultra pure water for at least 1 h to prevent salt out from blocking the column and pipeline and corrosion of chloride ion. When the chromatographic column needs to be stored for a long time (more than 30 days), wash the column with 0.05% sodium azide aqueous solution and save the column.

9 References or standard basis

No

10 Revision history

No

SOP of Yuxi Walvax Biotechnology Co., Ltd.

Code WSYX-QC-SOP-0009-0102
SOP for the protein content 1/5
Version 09 Page
determination
Confidential files No copy

1 Objective

Standardize the operation of protein content test.

2 Definitions/Abbreviations

N/A

3 Scope of application

3.1 It is applicable to determine the protein content of:
polysaccharide, bulk (bulk conjugate) of Haemophilus Influenzae Type b Conjugate Vaccine,
Group A and C Meningococcal Polysaccharide Vaccine, Meningococcal Group A, C Bivalent
Polysaccharide Conjugate Vaccine, Group ACYW135 Meningococcal Polysaccharide Vaccine,
23-valent Pneumococcal Polysaccharide Vaccine and 13-valent Pneumococcal Conjugate
Vaccine and other test samples manufactured by Yuxi Walvax.

4 Responsibilities

All the operators shall strictly conduct the SOP.

5 Materials and equipment

5.1 Use of equipment

Equipment

No. Equipment name Manufacturer Notes
model

Analytical Or equivalent

1 BS124S Sartorius

balance equipment

Electro- Shanghai Qixin Scientific Instruments

Or equivalent

2 thermostatic water DK-600A Co., Ltd. / Shanghai Yiheng Scientific
equipment

tank Instruments Co., Ltd.

U/V Or equivalent

3 UV-2600 Shimadzu instrument (Suzhou) Co., Ltd

spectrophotometer equipment

5.2 Use reagent

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determination

Confidential files No copy

No. Reagent name Grade

1 Sodium hydroxide AR

2 Anhydrous sodium carbonate AR

3 Potassium tartrate AR

4 Copper sulfate pentahydrate AR

5 2N foline-phenol Import reagent

5.3 Use of materials

No. Material name number Material name

1 Thermometer 2 15*150 test tube

3 Triangular flask 4 5000μl pipette tips

5 5000μl pipette 6 1000μl pipette tips

7 1000μl pipette 8 200μl pipette tips

9 200μl pipette 10 Beaker

6 Working procedure

6.1 Principle

In this method, the peptide bond contained in the protein molecule is chelated with Cu²⁺ in the alkaline solution to form the protein-copper complex, which can reduce the phosphomolybdic acid of the phenol reagent and produce a blue compound. At the same time, under alkaline condition, the phenol reagent is easily reduced by the tyrosine, tryptophan and cysteine in the protein to form blue. In a certain range, the strength of color is directly proportional to the protein concentration. The protein content in the test sample is determined by colorimetry with the protein reference solution as the standard curve.

6.2 Solution

6.2.1 Protein detection solution A

Respectively weigh 10g of sodium hydroxide and 50g of anhydrous sodium carbonate into beaker, add 400ml of purified water, dissolve and mix together, pour it into plastic bottle, labeled as protein detection solution A, standby;

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SOP for the protein content 3/5

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determination

Confidential files No copy

6.2.2 Protein detection solution B

Weigh 0.5g of potassium tartrate into beaker, add 50ml of purified water to dissolve it.

Weigh 0.25g of copper sulfate pentahydrate into beaker, add 30ml of purified water to dissolve it, then mix the two solutions as protein detection solution B, labeled, standby;

6.2.3 Alkaline copper test solution (Prepare the solution immediately before use)

Before use, mix well according to the volume proportion of protein detection solution A:

protein detection solution B: purified water (20:4:1), labeled, standby.

6.2.4 Foline-phenol test solution (Prepare the solution immediately before use)

Take the foline-phenol stock solution (2mol/L acid concentration) 1→16 (take 1000μl foline-phenol stock solution and put it into a triangular flask, add the purified water to 16ml, mix well, or prepare according to the experiment required amount).

6.2.5 200μg/ml national standard protein working solution

refer to the <operation procedures for preparation of standard solution> WSYX-QC-SOP-0009-0801

6.3 Test procedure

6.3.1 Treatment of test samples

6.3.1.1 Dilute the test sample with purified water to a solution contains 20-40μg/ml of protein, to be tested.

6.3.1.2 Group A and C meningococcal polysaccharide conjugate vaccine: measure 2000μl

purified water with a pipette and put it into a 3K ultrafiltration cup, mark the lower liquid level scale as 1 with a mark pen, pour out the purified water, take 10 vials of lyophilized vaccines and reconstitute with 0.5ml/vial of diluent, take 4000μl and put it into the ultrafiltration cup, mark the lower liquid level scale as 2, and carry out ultrafiltration concentration at 7000rpm/min, 5■ for 30min; Centrifuge again under the same conditions after adding water to the scale line 2; after centrifugation, add water to the scale line 1, after fully blow wash, and dilute 8 times before use (take 0.5ml of the solution after ultrafiltration and mixing, add 3.5ml of purified water, and mix well), to be tested.

6.3.2 preparation of standard curve

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Version 09 Page

determination

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0 1 2 3 4 5

National standard protein working solution (μl) 0 50 100 150 200 250

Purified water (μl) 1000 950 900 850 800 750

Protein content (μg/ml) 0 10 20 30 40 50

6.3.3 Accurately measure 1000μl of the test sample into the test tube using a pipette with a range of 100-1000μl, in duplicate. After the above steps, add 1000μl of alkaline copper test solution to each tube, mix well, and place it in a 25■ Electro-thermostatic water tank for 10 minutes. After taking it out, quickly add 4000μl of folin-phenol test solution to each tube, mix the tube immediately. After mixing, place tubes in a 25■ electro-thermostatic water tank for 30 minutes, and measure the absorbance at the wavelength of 650nm using the UV-VIS spectrophotometer.

6.4 Results calculation

6.4.1 Take tube 0 as the blank, calculate the linear regression equation with the national standard working solution series concentration and its corresponding absorbance, then substitute the absorbance of the test solution into the linear regression equation, and calculate the protein concentration.

Protein concentration of test sample (μg/ml) = Average protein concentration of test sample × dilution multiple

Protein content of test sample (μg/ml) = (Average protein concentration of test sample/2) × dilution multiple

Protein content of test sample (mg/g) = protein concentration of test sample/total solid

Protein content of test sample (%) = (Protein concentration of test sample/1000/total solid) × 100

6.4.2 Results judgment

Carry out according to corresponding quality standard.

7 Flow chart

No

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Code WSYX-QC-SOP-0009-0102

SOP for the protein content 5/5

Version 09 Page

determination

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8 Matters need attention

8.1 Temperature and time have influence on protein content detection and should be strictly controlled.

9 Reference or standard basis

9.1 "Chinese Pharmacopoeia" edition 2015, volume III, general rule 0731

10 Revision history

Issuing Time in Revision Revision basis, reason and

Version Note

time force Dept. content

01 Nov. 01 Dec. QC Changed when file expired and

03

2011 2011 laboratory complete some parts of the content

01 Nov. 01 Dec. QC Complete some parts of the

04

2014 2014 laboratory content

18 Mar. 20 Mar. QC Complete some parts of the

05

2016 2016 laboratory content
31 July 02 Aug. QC Complete some parts of the
06
2016 2016 laboratory content
28 Oct. 15 Nov. QC Complete some parts of the
07
2016 2016 laboratory content
30 Sep. 01 Oct. QC Complete some parts of the
08
2017 2017 laboratory content

Standard Operating Procedures of Yuxi Walvax Biotechnology Co., Ltd.

Code WSYX-QC-SOP-02-09-0003
SOP for specific toxicity
Version 07 Pages 1/3
test of tetanus toxoid bulk
Confidential documents, no copying.

1 Purpose

Standardize the operation of specific toxicity test of tetanus toxoid bulk.

2 Definition/abbreviation

No

3 Scope of application

Suitable for tetanus toxoid progenitor-specific toxicity testing.

4 Responsibilities

4.1 Quality Control Laboratory

- The operator is responsible for the implementation of this procedure
- Department manager is responsible for overseeing the implementation of this procedure

4.2 Quality Supervision Department

- Quality supervisors are responsible for overseeing the implementation of this code

5 Equipment and materials

5.1 Using the equipment

Serial Equipment Equipment Equipment

Manufacturers

number name model coding

Electronic

1 PMA7501 WYQCF0263 Sartorius

balance

Purification Suzhou Antai Air Technology Co.,

2 BLB-1300 WYQCF0233

table Ltd.

5.2 Use of reagents

Serial Level/Cate Serial Level/Categ

Reagent Name Reagent Name

number gory number ory

Sodium chloride Medical Medical

1 2 75% ethanol

injection grade grade

5.3 Materials

Serial Material name Serial Material name

Standard Operating Procedures of Yuxi Walvax Biotechnology Co., Ltd.

Code WSYX-QC-SOP-02-09-0003

SOP for specific toxicity

Version 07 Pages 2/3

test of tetanus toxoid bulk

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number number

250-350 g guinea pigs above

1 2 Graduated pipette

clean grade

3 Washing ear ball 4 50 ml vials

5 Alcohol lamp 6 Bottle stopper

7 Parafilm 8 Disposable syringe

9 Disposable latex gloves 10 Hemostatic forceps

6 Work procedures

6.1 Test procedure:

Samples were taken from each bottle, diluted to 250 Lf/ml with sodium chloride injection.

Four guinea pigs weighing 250-350 g were injected subcutaneously into abdomen of each guinea pig. The observation was carried out on the 7th, 14th and 21st days after injection, and the results were recorded according to the regulations.

6.2 Criteria

On the 7th, 14th and 21st days after injection, there was no local suppuration and necrosis.

The animals should not have tetanus symptoms, and the weight of each animal at the end of the injection was qualified.

7 Flow chart

No

8 Precautions

8.1 Experimental animals must have a certificate of conformity.

8.2 Samples rarely must be strictly in accordance with the requirements of sterile operation

9 References or standard basis

9.1 Chinese Pharmacopeia, Volume III, Page 89, China Pharmaceutical Science and Technology Press, 2015

10 Revision history

Standard Operating Procedures of Yuxi Walvax Biotechnology Co., Ltd.

Code WSYX-QC-SOP-02-09-0003

SOP for specific toxicity

Version 07 Pages 3/3

test of tetanus toxoid bulk

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Basis, reasons and content of the

Version Effective Revision
amendment Note
No. time Department

According to the corresponding
product names in Chinese
Pharmacopoeia (2010 Edition), it is
standardized and unified. The names
of protein carrier fermentation broth
QC

03 2011.12.01 (TT), protein carrier refined liquid No
Laboratory

(TT) and protein carrier purified liquid
(TT) were changed to tetanus toxoid
fermentation broth, tetanus toxoid
original liquid and tetanus toxoid
purified liquid.

Laboratory
04 2012.11.01 Animal File expiration No
Center

Laboratory
05 2015.10.29 Animal Document revision and upgrading No
Center

he organizational structure of
laboratory animal center and quality
control laboratory was changed

QC
06 2018.08.01 (change control No.: CC-201909012), No
Laboratory

so the responsible department was
revised from "Laboratory Animal
Center" to "quality control laboratory".

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Code WSYX-QC-SOP-02-09-0003

SOP for specific toxicity

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test of tetanus toxoid bulk

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SOP of Walvax Biotechnology Co., Ltd.

Code WSYX-QC-SOP-00-09-0203
Bacterial Endotoxin
Version 12 Page 1/13
Test SOP
Confidential Photocopy Prohibited

1 Objective

Standardize operating procedure for bacterial endotoxin test.

2 Definitions and Abbreviations

2.1 bacterial endotoxin test method: using limulus reagent to detect or quantify bacterial endotoxin produced by gram-negative bacteria, in order to judge whether the limit of bacterial endotoxin in the sample is in accordance with the prescribed method.

2.2 standard product of bacterial endotoxin: calibration of its titer based on the national standard of bacterial endotoxin, used for the reexamination of the sensitivity, interference test and positive control in the test.

2.3 water for bacterial endotoxin test: Sterile Water for Injection with less than 0.015EU/ml (gel method) or 0.005 EU/ml (used for photometry) and no interfering effect on endotoxin test.

2.4 gel method: the method of limit detection or semi quantitative detection of endotoxin is based on the principle of agglutination reaction between limulus reagent and endotoxin.

3 Scope of application

This SOP applies to detect the bacterial endotoxin of:

intermediate products, final products and related excipients of Haemophilus Influenzae Type b Conjugate Vaccine, Group A and C Meningococcal Polysaccharide Vaccine, Meningococcal Group A, C Bivalent Polysaccharide Conjugate Vaccine, Group ACYW135 Meningococcal Polysaccharide Vaccine, Diphtheria, Tetanus and Acellular Pertussis Combined Vaccine, adsorbed, 23-valent Pneumococcal Polysaccharide Vaccine and 13-valent Pneumococcal Polysaccharide Conjugate Vaccine and other test samples.

4 Responsibility

The quality control laboratory personnel strictly operate according to the SOP.

5 Materials and Equipment

5.1 Equipment

No. Equipment name Equipment model Manufacturer
SOP of Walvax Biotechnology Co., Ltd.

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Bacterial Endotoxin
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Electrothermal constant Shanghai Qixin Scientific

1 DK-600A

temperature water bath Instruments Co., Ltd.

5.2 Materials/reagent

No. Materials/reagent name Grade/classification

1 Limulus reagent reagent

2 Bacterial endotoxin working standard standard

3 Bacterial endotoxin test water Diluent

5.3 Consumables

No. Consumables No. Consumables

1 Test tube 2 Test tube rack

3 Straw 4 Ear suction ball

5 Timer 6 Alcohol cotton ball

6 Working procedure

6.1 Sensitivity test of limulus reagent

6.1.1 under the conditions stipulated in the bacterial endotoxin test method, the lowest concentration of the agglutinating endotoxin produced by the limulus reagent, that is, the labeling sensitivity of the limulus reagent, is expressed as EU/ml. When the new batch number of limulus reagent or test conditions has any effect that may affect the test results, the sensitivity test of limulus lysate should be carried out.

6.1.2 Preparation of bacterial endotoxin standard solution

Take one of the bacterial endotoxin standard products, light the wall of the bottle, make the powder fall into the bottom of the bottle, then use the grinding wheel to gently scratch the top of the bottle neck. After the 75% alcohol cotton ball is wiped and open, the glass chips should be prevented from falling into the bottle.

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In accordance with the standard specification, add the prescribed amount of bacterial endotoxin to check water to dissolve its contents, seal the bottle with the sealing membrane and mix the swirl mixer for 15 minutes. Then the standard solution of four concentrations of bacterial endotoxin was prepared, that is, 2λ , λ , 0.5, and 0.25 λ (λ as the labeling sensitivity of the reagents of the reagents), and each dilution should be mixed on the vortex mixer for 30 seconds each.

6.1.3 Preparation of limulus reagent to be rechecked

Take the 18 branches of the limulus reagent with the specification of 0.1ml/tube, light the wall of the bottle, make the powder fall into the bottom of the bottle, gently scratch the bottle with the grinding wheel in the bottle neck, and open the spare after the 75% alcohol cotton ball

wiping, and prevent the glass from falling into the bottle. Each 0.1ml is added with test water to dissolve, gently rotating the bottle wall, so that the contents are fully dissolved and avoid bubbles.

6.1.4 adding sample

The 18 limulus reagents, which were fully dissolved, were divided into 5 rows, of which 4 were listed in 4 columns, 1 in 2. The 4 column was added each with 0.1ml of 2λ , λ , 0.5λ , and 0.25λ of the endotoxin standard solution, and the other 1 column with 2 tubes were added with 0.1ml bacterial endotoxin test water as negative control.

6.1.5 After added, the limulus reagent was sealed with a sealing membrane and mixed gently to avoid producing bubbles. It was placed vertically into the thermostat at $37\pm 1^\circ\text{C}$ or in a suitable thermostat. The heat preservation was 60 ± 2 minutes.

6.1.6 Observe and record the results. Remove the test tube gently from the thermostat or thermostat to avoid vibration, and change each tube slowly to 180 degrees. If the gel is formed in the tube, the gel is not deformed and the glide from the tube is positive, recorded as (+); the gel is not formed or the gel is not solid, deformed and negative from the tube wall, recorded as (-).

6.1.7 Results calculation

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Bacterial Endotoxin

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Test SOP

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When the 4 tubes with maximum concentration of 2λ were all positive, the 4 tubes with minimum concentration of 0.25λ were negative, negative tube was negative, the test side was effective. The geometric mean value of the reaction end point concentration was calculated as the rechecking result, which is the sensitivity of the limulus reagent (λ_c).

$\lambda_c = \text{antilg}(\sum X/4)$

The value of X in the formula is the value of the end point of the reaction (lg). The end point of the reaction is the concentration of the last positive result in a series of reduced endotoxin concentrations.

6.1.8 Result judgment

When λ_c was between 0.5λ to 2λ (including 0.5λ and 2λ), the sensitivity of this batch of limulus reagent was determined to be qualified, which could be used for bacterial endotoxin test and the sensitivity of the limulus reagent was determined by λ (marked sensitivity)

6.2 Interference test

6.2.1 Test method

6.2.1.1 The purpose is to test whether the test product at a certain concentration has any interference effect on the reaction of limulus reagent and endotoxin. The test solution should be a solution which does not detect endotoxin and does not exceed the maximum effective dilution factor (MVD).

6.2.1.2 Limulus reagent labeling sensitivity control series preparation

1 bacterial endotoxin standard sample was obtained and diluted with bacterial endotoxins test water with 4 concentrations, namely 2λ , λ , 0.5λ and 0.25λ .

6.2.1.3 Preparation of interference test series solution

Dilute the test sample, and then the diluted test sample solution is used to dilute the same bacterial endotoxin standard in sensitivity control series of limulus reagent to 4 concentrations, namely, bacterial endotoxin with concentrations of 2λ , λ , 0.5λ and 0.25λ .

6.2.1.4 Preparation limulus reagent

The 28 tubes of the limulus reagent with the specification of 0.1ml/tube were taken. The light SOP of Walvax Biotechnology Co., Ltd.

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Bacterial Endotoxin

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projectile wall makes the powder fall into the bottom of the bottle, gently scratches the bottle with the grinding wheel in the bottle neck. After the 75% alcohol cotton ball is wiped, it opens reserve to prevent the glass from falling into the bottle. Each 0.1ml is added with test water dissolved, gently rotating the bottle wall, so that the contents are fully dissolved and avoid bubbles.

6.2.1.5 Adding samples

The 10 limulus reagents, which were fully dissolved, were divided into 5 rows, of which 4 were listed in 4 columns, 1 in 2. The 4 column was added each with 0.1ml of 2λ , λ , 0.5λ , and 0.25λ of the endotoxin standard solution, and the other 1 column with 2 tubes were added with 0.1ml bacterial endotoxin test water as negative control.

The other 18 limulus reagents, which were fully dissolved, were divided into 5 rows, of which 4 were listed in 4 columns, 1 in 2. The 4 column was added each with 0.1ml of 2λ , λ , 0.5λ , and 0.25λ of the endotoxin standard solution, and the other 1 column with 2 tubes were added with 0.1ml bacterial endotoxin test water as negative control.

After the addition, the limulus reagent was sealed with a sealing membrane and mixed gently to avoid producing bubbles. It was placed vertically into the thermostat at $37\pm 1^\circ\text{C}$ or in a suitable thermostat.

6.2.2 Results calculation

All the parallel tubes in negative control and negative control were negative, and the maximum concentration of 2λ tube in the standard solution of endotoxin was positive, and the test was effective when the lowest concentration of 0.25λ tube was negative. The reaction end point calculated by the geometric concentration of standard solution made of endotoxin test water of the average value (E_s) and endotoxin test with geometric solution made of diluent for the reaction end point average concentration value (E_t) are calculated by the formula.

$E_s = \text{antilg}(\sum X_s/2)$

$E_t = \text{antilg}(\sum X_t/4)$

The X_s and X_t are numeric values of the test water and end point concentration of the SOP of Walvax Biotechnology Co., Ltd.

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endotoxin solution (lg), which is made from the diluent, respectively.

6.2.3 Result judgment

When Es is 0.5 λ to 2 λ (including 0.5 λ and 2 λ), and Et is 0.5 Es to 2.0Es (including 0.5 Es and 2.0Es), the test product is considered to have no interference effect at this concentration, and the bacterial endotoxin test can be carried out at this concentration. If the test solution is disturbed by the dilution of less than MVD, the test solution should be further diluted with no more than MVD and repeat the interference test.

6.3 Test method

The test methods are divided into two methods: gel limit test and gel semi quantitative test.

6.3.1 Gel limit test

6.3.1.1 determine the maximum valid dilution multiple (MVD) with the following formula.

$$MVD = \frac{C \times L}{\lambda}$$

Where: L is the bacterial endotoxin limit for the test products

C is the concentration of the solution. When L is expressed in EU/ml, C equals 1.0ml/ml.

When L is expressed as EU/mg or EU/U, the unit of C needs to be mg/ml or U/ml.

λ is the labelling sensitivity (EU/ml) of the Limulus amebocyte lysate in gel method, or the lowest endotoxin concentration on the standard curve used in photometric determination.

6.3.1.2 take a bacterial endotoxin work standard, break from the top of the ampoule, add a proper amount of bacterial endotoxin to check water according to the label, mix for 15 minutes on the vortex mixer.

6.3.1.3 diluted the test product to MVD according to the limit of the tested product and the sensitivity of the limulus reagent.

6.3.1.4 The preparation of test solution: use the solution with Dilution Times of MVD and has eliminated the interfering solution to prepare the solution in the following table:

Preparation of gel limit test solution

Endotoxin concentration /

Solution name Parallel tube number

solution added to endotoxin

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Test sample solution No/test sample solution 2

Test sample positive control 2λ/test sample solution 2

Positive control 2λ/test water 2

Negative control No/ test water 2

Test sample solution The content of polysaccharide or protein content of the sample is diluted to 1μg/ml or 1mg/ml, and then diluted to the maximum valid dilution multiple (MVD) by 1μg/ml or 1mg/ml, and the finished product can be diluted directly to the maximum valid dilution multiple (MVD).

e.g. 1. Group A meningococcal polysaccharide bacterial endotoxin standard: not more than 10Eu/ug

MVD value (maximum valid dilution multiple) = $\frac{CL}{\lambda} = 1 \times 10 / 0.25 = 40$ (C=1μg/ml)

Group A meningococcal polysaccharide content is 4610.4μg/ml

0.2ml 0.2ml

0.2ml

4610.4μg/ml 461.04μg/ml 46.104μg/ml

1.8ml 1.8ml

1.8ml

0.8ml

0.2ml 0.2ml 0.8ml

4.6104μg/ml 1μg/ml 10 20 40 folds

0.72ml 1.8ml 0.8ml 0.8ml

Take the 40-fold diluted solution for testing

e.g. 2.Diphtheria, Tetanus and Acellular Pertussis Combined Vaccine, Adsorbed

bacterial endotoxin standard: not more than 1500Eu/dose

MVD value (maximum valid dilution multiple) = $\frac{CL}{\lambda} = 1 \times 3000 / 0.25 = 12000$ (C=1ml/ml)

5 bottles of the test samples were combined and mixed thoroughly, then under 3000rpm

10min centrifugally precipitated and the supernatant was diluted.

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0.2ml 0.2ml 0.5ml

Test sample 10-fold dilution 100-fold dilution

1.8ml 1.8ml 1.8ml

0.8ml

400-fold dilution 800-fold dilution

1.5ml

mix well, take the 800-fold diluted for testing

Test sample positive control Take the maximum valid dilution multiple (MVD) 1/2 of the test sample, add the equivalent 4λ endotoxin standard solution, and mix it well.

Positive control the standard solution of 2λ endotoxin was used as a positive control

6.3.1.5 takes the specification of 0.1ml/tube limulus reagent and 0.1ml limulus reagent reconstituted by bacterial endotoxin test water.

6.3.1.6 according to the above table, add two samples to each test solution and add 0.1ml to each one.

6.3.1.7 preserved at 37 ±1■ for 60 ±2 minutes then observe.

6.3.1.8 Result judgment

a) if the parallel tubes of the negative control solution were all negative, the parallel tubes of test samples control solution were all positive, and the parallel tubes in the positive control solution were all positive, and the test was effective.

b) if the two parallel tubes of the sample solution were all negative, the test samples were found to be in accordance with the standard. If the two parallel tubes of the test samples were all positive, the test samples were not in accordance with the standard.

c) if a tube in the two parallel pipes of the test solution is positive and the other tube is negative, it is necessary to have a reexamination. During the reexamination, 4 parallel tubes should be made for the test solution. If all parallel tubes are negative, the sample should be SOP of Walvax Biotechnology Co., Ltd.

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found to be in conformity with the standard. Otherwise, the sample will not meet the requirements.

6.3.2 Gel semi quantitative test

6.3.2.1 determine the maximum valid dilution multiple (MVD) with the following formula.
 $MVD = \frac{C}{L}$
Where, L is the bacterial endotoxin limit for the test products
C is the concentration of the solution. When L is expressed in EU/ml, C equals 1.0ml/ml.
When L is expressed as EU/mg or EU/U, the unit of C needs to be mg/ml or U/ml.
 λ is the labelling sensitivity (EU/ml) of the Limulus reagent in gel method, or the lowest endotoxin concentration on the standard curve used in photometric determination.

6.3.2.2 take a bacterial endotoxin work standard, break from the top of the ampoule, add a proper amount of bacterial endotoxin with test water according to the label, mix for 15 minutes on the vortex mixer.

6.3.2.3 The preparation of the test solution Uses a dilution multiple of no more than MVD and an interfered test solution to prepare the solution in the follow table. It is diluted to 1 times, 2 times, 4 times and 8 times by the dilution multiple of the interference test, and the final dilution multiple must not exceed MVD:

Endotoxin	The Parallel	concentration / Dilution	Solution name	Diluent concentration	tube	solution added	multiple	of endotoxin	number	to endotoxin
1	2									
Test sample	No/test sample	test 2	2							

solution	solution	water	4	2						
8	2									
Test sample	2 λ /test sample	1	2 λ	2						
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positive control solution	Limulus reagent	1	2 λ	2	marked test	2	λ	2	2 λ /test water	sensitivity water	4	0.5 λ	2	control series	8	0.25 λ	2	Negative	No/test water	-----	-----	-----	2	control

6.3.2.4 takes the specification of 0.1ml/tube limulus reagent and 0.1ml limulus reagent reconstituted by bacterial endotoxin test water.

6.3.2.5 according to the above table, add two samples to each test solution and add 0.1ml to each one.

6.3.2.6 preserved at 37 ±1■ for 60 ±2 minutes then observe.

6.3.2.7 Result judgment

a) if the parallel tubes of the negative control solution were all negative, the parallel tubes of the test sample positive control solutions were all positive, and the geometric mean of the reaction point concentration of the control series of the limulus reagent sensitivity of the control series was between 0.5 and 2 λ , then the test was effective.

b) the end-point dilution multiple of each series of parallel tubes in the sample solution multiplied by λ . For each series of reaction endpoints, the geometric mean of the end point concentration of all parallel tubes was the endotoxin concentration of the sample solution [according to formula $CE = \text{anti}lg(\sum X/2)$]. If the diluent for the test is used, the result should be multiplied by the dilution multiple when calculating the concentration of endotoxin in the original solution.

c) if all the parallel tubes of the test solution in the test are negative, it should be noted that the concentration of endotoxin is less than λ (if the tested is the diluted test sample, it is calculated as less than λ multiplied by the initial dilution multiple of the semi quantitative SOP of Walvax Biotechnology Co., Ltd.

test). If all parallel tubes in the test sample solution are positive, it should be noted that the concentration of endotoxin is greater than or equal to the maximum dilution multiplied by λ .
d) if the concentration of endotoxin is less than the specified limit, the product is determined to conform to the requirement. If the concentration of endotoxin is greater than or equal to the specified limit, it is judged that the test sample does not meet the requirements.

7 Flow chart

None

8 Matters need attention

8.1 The pH value of the solution is suitable for the range is 6 to 8 after the mixture of common sample solution and limulus reagent. The pH value of the tested solution (or its diluent) should be adjusted for the test products with excess acid, excess alkali or its own buffer capacity. The pH value can be adjusted by acid, alkali solution or suitable buffer solution. Acid or alkaline solutions must be tested with bacterial endotoxins for use in containers containing endotoxin. The buffer must be validated without endotoxin and interfering factors.

8.2 Interference test should be carried out before establishing the endotoxin test before or after the new drug has been tested. When any changes in the test results are likely to be affected by the lysate reagent, the prescription of the test product, the production process change or the test environment, the interference test must be carried out again.

8.3 The utensils used in the test should be treated to remove potential exogenous endotoxins. The usual method is to bake at 180 degrees for at least 120 minutes, and other suitable methods to confirm the test of bacterial endotoxins.

8.4 If plastic appliances are used, such as microporous plates and suction heads matching with micro sample feeder, it should be noted that to use instruments that indicate no endotoxin and interfere with the test.

8.5 Test operation should prevent contamination of endotoxin.

8.6 If there is a test sample that has interference with the limulus reagent, such as the adsorption of acellular pertussis combined vaccine, the test should be removed after the SOP of Walvax Biotechnology Co., Ltd.

interference (3000rpm, 20min, 10 minutes' centrifugation, and the supernatant is tested).

8.7 When the new batch number of limulus reagent or test conditions has any influence on the test results, the sensitivity test of limulus lysate reagent should be carried out.

8.8 Preparation of test samples for prefilling syringe: take at least 3 samples, take water or sodium chloride injection for bacterial endotoxin test to the marked capacity, pull the push rod back to the opening, vibrate twice, and seal it in a 37±2℃ incubator for 2 hours. After taking out, collect the test solution in prefilling syringe in a glassware without pyrogen, and carry out bacterial endotoxin test within 2 hours.

9 References and basis

9.1 Pharmacopoeia of People's Republic of China, Volume III, 2015. General rule 1143.

10 Revision history

Revising

Basis, reason and revision

Version Valid from department Note

nt

According to the Pharmacopoeia, add

01 Dec. QC

03 the sensitivity test and improve the

2011 Laboratory

content of the document.

According to the Pharmacopoeia, add

16 May, QC

04 the gel interference test, improve the

2012 Laboratory

content of the document.

9 Sep. QC Revised according to the 2015 edition

05

2013 Laboratory of the Pharmacopoeia

01 Dec. QC Modifying the document as the product

06

2015 Laboratory name change

20 Mar. QC

07 Improve the content of the document

2016 Laboratory

QC

08 01 Sep. Improve the content of the appendix to

Laboratory

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2016 the document

20 Mar. QC Refine the preparation of the test

09

2017 Laboratory sample, add appendix IV.

15 Oct. QC
10 Revise appendix
2017 Laboratory

According to the IP (2015 edition),
revise the bacterial endotoxin standard
01 Sep. QC of Diphtheria, Tetanus and Acellular
11
2018 Laboratory Pertussis Combined Vaccine form not
higher than 1500EU/vial to not higher
than 100EU/vial.

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Code WSYX-QC-SOP-00-09-0111
SOP for Total Solids
Version 07 Pages 1/5
Content Test
Non-photocopying, confidential documents

1 Purpose
Standardize the detection method of the total solids determination.
2 Definitions/abbreviations
Constant weight: unless otherwise provided, refers to the weight difference of less than 0.3 mg after two consecutive dryings or scorching test products; the second and subsequent weighing of drying to constant weight shall be carried out after 1 consecutive drying under specified conditions; and the second weighing to constant weight shall be carried out after 30 minutes of continuous burning.
3 Scope of application
Suitable for the company's production products haemophilus influenzae type b polysaccharide, group A meningococcal polysaccharides (bulk), group C meningococcal polysaccharides (bulk), group Y meningococcal polysaccharides (bulk), group W135 meningococcal polysaccharides (bulk), pneumococcal polysaccharides (bulk), pneumococcal polysaccharides derivatives.
4 Responsibilities
Quality control laboratory personnel operate strictly in accordance with this SOP.
5 Equipment and materials
5.1 Using the device
Serial Device Device model Device
Manufacturers Note
number name coding
WYQCF00
Electric hot
53/ Shanghai Qixin
constant or
DHG-9070A/ WYQCF04 Scientific
1 temperature equivalent
DHG-9075A 46/ Instrument Co.,
blowwind equipment
WYQCF02 Ltd.
drying box
88/
2 Electronic CP225D/ WYQCF00 Sedolis or
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balance MSU225S-1CE- 37/ equivalent
DU WYQCF03 equipment
59
Tianjin Otseins or
Ultrasonic WYQCF01
3 AS10200A Instrument Co., equivalent
cleaners 62
Ltd. equipment
5.2 Materials used
Serial number Material name Serial number Material name

1 Weighing bottles 2 Silk gloves
3 Dryer (built-in silicone) 4 Pipette
5 Pipette, tip.
6 Work procedures
6.1 Principles
This method is based on a certain temperature, so that the liquid components of the test products evaporated, using the remaining solid components to calculate the total amount of solids for test products.
6.2 Dry roasting at 105■
6.2.1 Constant weight of weighing bottle
6.2.2 Put the weighing bottle simply in the ultrasonic cleaner for 15 minutes and repeat 3 times (replace the new purified water after each cleaning) by placing the weighing bottle in the ultrasonic cleaner for 15 minutes.
6.2.3 Put the washed weighing bottle in a 105■ tank, after baking drying (the first baking time is not less than 24 hours), cool in a dryer and weigh. Until two consecutive weighing result errors of 0.0001 g, and record the constant weight result M1.
6.2.4 Pipette with a dosage range of 100-1000 μl to extract the exact amount of the test product 1000 μl in a weighing bottle that has been dried to a constant weight, and after baking and drying in a drying tank at 105■ (the first baking time is not less than 24 hours), cooled in the
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dryer and weighed. Until two consecutive weighing result errors of 0.0001 g, and record the constant weight result M2.
6.3 Dry roasting at 50■

6.3.1 Constant weight of weighing bottle

6.3.2 Put the weighing bottle simply in the ultrasonic cleaner for 15 minutes and repeat 3 times (replace the new purified water after each cleaning) by placing the weighing bottle in the ultrasonic cleaner for 15 minutes.

6.3.3 Put the washed weighing bottle in a 50℃ tank, after baking drying (the first baking time is not less than 72 hours), set the dryer to cool and weigh. Until two consecutive weighing result errors of 0.0001 g, and record the constant weight result M1.

6.3.4 Pipette with a dosage range of 100-1000 µl is accurately taken for the test 1000 µl in a weighing bottle that has been dried to a constant weight, and after baking and drying at 50℃ (the first baking time is not less than 72 hours), cooled in the dryer and weighed. Until two consecutive weighing result the error is 0.0001 g, and the constant weight result M2 is recorded.

6.4 Press calculation:
Total weight of solid weigh $M(\text{mg/ml}) = (M2 - M1) \times 1000 \div 1$
where:
The weight of the test product and weighing bottle is M2, g;
The weight of the weighing bottle is M1, g;
1 is the volume of 1 ml of precision quantity for test products.

7 Flow chart
N/A

8 Precautions

8.1 Note that during operation, it is forbidden to touch the weighing bottle directly by hand, and you must wear clean silk gloves to prevent sweat from staining on the bottle to affect the test results.

8.2 During operation, the action should be gentle, weighing bottle out of the dry box bottle cap

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should be tight, cover tight bottle cap, take care not to put the fiber on the gloves in the weighing bottle.

8.3 Group A meningococcal polysaccharides (bulk) solid scant scans were tested by 50℃ dry-roasting, while the rest of the products were tested by the 105℃ dry-roasting method.

8.4 Weighing bottle constant weight end, when the sample is sampled when the pipette suction head can not remain sample, the sample must be completely transferred to the weighing bottle.

8.5 Schematic (e.g. there are three samples when the total weight of solids is tested, each with two parallels.)

Oven, electronic balance use record sample information order: sample one, sample two, sample three;
The dryer weighs the order in which the bottle is placed: from the triangle to put the first, and then placed clockwise.

9 References or standard basis

11.1 Chinese Pharmacopeia, volume III, General Principles 3101 Solid Total Measurement, China Pharmaceutical Science and Technology Press, 2015 Edition

10 Revision history
Version Release Effective Revision Basis, reasons and content of the Note
No. Time time Departme amendment
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nt
01 01
QC File expiration, change
03 November December
Laboratory Improve part of the file content
2011 2011
02 01
QC
04 September December Improve part of the file content
Laboratory
2014 2014
March 18, March 20, QC
05 Improve part of the file content
2016 2016 Laboratory

September 01 October QC
06 Improve part of the file content
30, 2017 2017 Laboratory

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Code WSYX-QC-SOP-00-09-0106

SOP for Residual Cyanide Version 09 Pages 1/5

Content Test

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1 Objective

Standardizing the determination method of cyanide residue.

2 Definitions/abbreviations

N/A

3 Scope of application

This SOP is applicable to the cyanide residue determination of haemophilus influenzae type b polysaccharide bulk derivative, group A meningococcal polysaccharide bulk conjugate, group C meningococcal polysaccharide bulk conjugate, pneumococcal polysaccharide derivative and other products manufactured by Yuxi Walvax.

4 Responsibility

All operators operate strictly in accordance with this SOP.

5 Equipment and materials

5.1 Using the device

Serial Equipment Note

Equipment name Equipment code Manufacturers

number model

Or

1 Electronic balance BS124S WYQCF0036 Starorius equivalent equipment

WYQCF0444/ Or

Ultraviolet Shimadzu Enterprise

WYQCF0445/ equivalent

2 Visible Light UV-2600 Management (China)

WYQCF0529/ equipment

Spectrometer Co., Ltd.

WYQCF0530

5.2 Use of reagents

Serial Note

Reagent name/material Level

number

1 Biphenylamine AR /

2 Pyridine AR /

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Code WSYX-QC-SOP-00-09-0106

SOP for Residual Cyanide Version 09 Pages 2/5

Content Test

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3 Hydrochloride AR content (HCL)%≥36.0-38.0

4 Brominated cyanide AR /

5.3 Materials

Serial

Materials Serial number Materials

number

1 Capacity bottle 2 Conical bottle

Powder-free latex

3 4 Gunhead

gloves

5 Pipette 6 Sealing film

6 Work procedures

6.1 Principle

Under acidic condition, cyanogen bromide reacts with pyridine benzidine, and the content of cyanogen bromide in the sample is determined by ultraviolet visible spectrophotometry.

6.2 Reagents

6.2.1 Pyridine benzidine solution (ready to use)

According to the preparation amount of 30ml: weigh 0.25g benzidine into a 100ml triangular flask, add 5ml of 2% hydrochloric acid solution and 25ml of 60% pyridine solution to dissolve and mix them evenly.

6.2.2 2% hydrochloric acid solution (ready to use)

According to the preparation amount of 5ml: take 0.25ml hydrochloric acid, add 4.75ml purified water, and mix well.

6.2.3 60% pyridine solution (ready to use)

According to 25 ml preparation amount: take 15 ml pyridine, add purified water to 25ml, mix well.

6.2.4 0.1 mg/ml CNBr reference stock solution

Weigh 0.01 g of cyanogen bromide in a beaker, dissolve it with purified water immediately and transfer it to a 100ml volumetric flask. Rinse the beaker three times and transfer the lotion Standard Operating Procedures of Yuxi Walvax Biotechnology Co., Ltd.

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Content Test

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to the volumetric flask. Fix the volume with purified water to the scale mark and mix well to obtain 0.1 mg/ml cyanogen bromide reference stock solution. Prepare it before use.

6.2.5 500 ng/ml CNBr control working solution

Pipette 1000 UL of 0.1 mg/ml cyanogen bromide reference stock solution into a 200 ml volumetric flask, add purified water to constant volume to the scale line, and mix well to obtain 500 ng/ml cyanogen bromide control working solution.

6.3 Test procedure

6.3.1 test article treatment

A pipette with a range of 100-1000 ul was used to precisely transfer 2000 ul of the test sample into the test tube in duplicate.

6.3.2 Preparation of standard curve:

Tube No.

Reagent 0 1 2 3 4 5 6

CNBr control

working solution 0 100 200 400 600 800 1000

(ul)

Purified water (ul) 2000 1900 1800 1600 1400 1200 1000

CNBr concentrate

0 25 50 100 150 200 250

(ng/ml)

6.3.3 Add pyridine-based phenylamine solution 2000 ul in the above tubes, mix well, below 20°C, and place disthere for 15 minutes, and measure the absorbance at the wavelength of 520 nm wavelength of ultraviolet visible light spectrometer.

6.4 Result calculation

6.4.1 Then the absorbance of the test solution was substituted into the linear regression equation to calculate the concentration of cyanogen bromide.

B (ng/ml) = the concentration of cyanogen bromide × dilution ratio calculated by linear regression equation

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6.4.1.1 cyanide content of Haemophilus influenzae type b polysaccharide bulk derivative

■
cyanide content of derivative (ng/mg)=10

B: cyanide content of test sample (ng/ml);

10: polysaccharide bulk derivative of test sample, mg/ml;

6.4.1.2 cyanide content of Pneumococcal polysaccharides bulk derivative

■
cyanide content of Pneumococcal polysaccharides bulk derivative (ng/mg)=■■■■■■ ■■■■■■

6.4.1.3 cyanide content of Meningococcal polysaccharide bulk conjugate

B
cyanide content of Meningococcal polysaccharide bulk conjugate (ng/mg)=1000

6.4.2 Result judgment

According to the appropriate quality standards.

7 Flow chart

No

8 Precautions

8.1 CNBr is highly toxic and volatile, pyridine benzidine has a certain smell, so safety protection measures should be taken.

8.2 the whole test process is carried out in the fume hood.

8.3 CNBr is easy to volatilize, so it should be dissolved immediately after weighing. In the process of the experiment, the standard curve should be made after taking the supply solution. When adding the working solution of the reference substance, it should be added from high concentration to low concentration. After adding, pyridine benzidine solution should be added immediately for sealing, so as to avoid the linear unqualified curve caused by the volatilization of cyanogen bromide.

8.4 type 1, 4, 6A, 6B, 7F, 18C, 19A, 19F and 23F pneumococcal polysaccharides were activated by 1-cyano-4-dimethylaminopyridine tetrafluoroborate (CDAP). The molecular weight of cyanogen bromide and CDAP was 105.92 and 234.99 respectively.

9 References or standard basis

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Content Test

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9.1 China Pharmacopoeia, Supplemental, General Principle 0806, China Pharmaceutical Science and Technology Press, 2015

10 Revision history

Revision Basis, reasons and content of the

Version Effective

Departme amendment Note

No. time

nt

03 2011.12.01 QC File expiration, change

Laboratory Improve part of the file content

QC

04 2014.12.01 Improve part of the file content

Laboratory

05 2016.11.16 QC Improve part of the file content

Laboratory

06 2017.10.01 QC Improve part of the file content

Laboratory

07 2018.09.01 QC Calculating the formula based on

Laboratory the revised results of the 2015

supplement

08 2019.02.11 QC The relative molecular weights of

Laboratory different types of 13 valent
activators and their activators are
described

SOP of Yuxi Walvax Biotechnology Co., Ltd.

Code WSYX-QC-SOP-00-09-0117
Aluminum hydroxide (or
Aluminum phosphate) Version 07 Page 1/4
content test SOP
Confidential files No copy

1 Objective

Standardize the operation of the content test of aluminum hydroxide, aluminum phosphate and aluminum.

2 Definitions/Abbreviations

N/A

3 Scope of application

It is applicable to the products containing aluminum produced by Yuxi Walvax.

4 Responsibilities

The QC lab operators shall strictly conduct the SOP during the sample testing.

5 Materials and equipment

5.1 Use of equipment

Equipment

No. Equipment name Manufacturer Notes

model

Or equivalent

1 Electronic balance BS124S Sartorius

equipment

Electrothermal constant Shanghai Qixin Scientific Or equivalent

2 HWS-24

temperature water bath Instruments Co., Ltd. equipment

5.2 Use reagent

No. Reagent name Grade/classification

1 Phosphoric acid AR

2 EDTA PT

3 Hydrochloric acid AR

4 Ammonium acetate AR

5 Glacial acetic acid AR

6 Xylenol orange AR

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Aluminum phosphate) Version 07 Page 2/4

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5.3 Use of materials

No. Material name number Material name

1 Volumetric flask 2 pipette

3 Conical flask 4 Acid burette

5 One-mark pipettes

6 Working procedure

6.1 In this method, the excessive EDTA-2Na reacts with aluminum ion firstly, and the residual EDTA-2Na is titrated with zinc titrant. According to the consumption of zinc titrant, the content of aluminum hydroxide (or aluminum phosphate) in the test sample can be calculated.

6.2 Reagent preparation

6.2.1 Phosphoric acid solution (6→100)

Take 6ml of phosphoric acid to 100ml volumetric flask, dilute it with purified water to the scale, mix it evenly.

6.2.2 EDTA-2Na titrant (0.05mol/L)

Calibration according to the <Operation procedures of titrant preparation>, WSYX-QC-SOP-0009-0802.

6.2.3 Zinc titrant (0.05mol/L)

Calibrating according to the <Operation procedures of titrant preparation >, WSYX-QC-SOP-0009-0802.

6.2.4 Zinc titrant (0.025mol/L)

Accurately measure 50ml of zinc titration solution (0.05mol/L), add the purified water to 100ml, shake well.

6.2.5 Acetate acid-ammonium acetate buffer (pH4.5)

Weigh 7.7g of ammonium acetate, dilute with 50ml of purified water, add 6ml of glacial acetic acid and appropriate amount of purified water to constant volume of 100ml.

6.2.6 Xylenol orange indicate solution

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content test SOP

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Weigh 0.2g of xylenol orange, add 100ml of purified water to dissolve it, and mix well.

(Prepare the solutions immediately before use)

6.3 Test procedures

6.3.1 Accurately measure the appropriate amount of the test sample to make the aluminum content about 1-10mg (generally, 1000μl is taken when the test sample is aluminum hydroxide solution; 5000μl is taken when the test sample is 13-valent pneumococcal conjugate vaccine),

place it in a conical flask (set two parallel), add 1500μl of phosphoric acid solution (6→100) to make it completely dissolved. If necessary, warm it up in a water bath (increase the amount of phosphoric acid if it is difficult to dissolve).

6.3.2 Add 10ml of EDTA-2Na titrating solution (0.05mol/L) and 10ml of acetic acid-ammonium acetate buffer solution (pH4.5) precisely, heat it on boiling water bath for 10min, and take it out and cool to room temperature.

6.3.3 Add 1000μl xylenol orange indicator solution, titrate with zinc titrant (0.025mol/L), when the solution changes from bright yellow to orange, it is the end point, and calibrate the titration result with blank test (test two parallel).

6.3.4 Results calculation

6.3.4.1 Calculation formula

Aluminum hydroxide content (mg/ml) = $(V_0 - V_1) \times c \times 78.01 / V_2$

Aluminum phosphate content (mg/ml) = $(V_0 - V_1) \times c \times 121.95 / V_2$

Aluminum content (mg/ml) = $(V_0 - V_1) \times c \times 26.98 / V_2$

V_0 : the volume of zinc titrant consumed in blank test, ml;

V_1 : the volume of zinc titrant consumed by the test samples, ml;

C : the concentration of zinc titrant, mol/L;

V_2 : the volume of the test samples, ml;

78.01, 121.95 and 26.98 are the molecular weight or relative molecular mass of aluminum hydroxide, aluminum phosphate and aluminum, respectively.

6.3.4.2 Results judgment

According to the corresponding quality standard.

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Aluminum hydroxide (or

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content test SOP

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7 Flow chart

No

8 Matters need attention

8.1 The temperature and humidity should be controlled during titration;

8.2 The burette and the volumetric flask shall be checked for leakage before use;

8.3 Take the titrant in strict accordance with the management of the titrant.

9 Reference or standard basis

9.1 "Chinese Pharmacopoeia" edition 2015, volume III, general rule 54

10 Revision history

Time in Revision

Version Revision basis, reason and content Notes

force Dept.

Application scope adding: test the aluminium

01 Dec. QC

03 phosphate content, add the test record of /

2011 laboratory

aluminium phosphate content

21 Jan. QC

04 Changed when file expired /

2014 laboratory

17 Jan. QC Detailed operation steps, completed some

05 /

2017 laboratory parts of the content

01 Sep. QC Change the test item for 13-valent

06 /

2018 laboratory Pneumococcal conjugate vaccine

Stability data of 13 serotypes of Pneumococcal Polysaccharide Derivatives Yuxi Walvax Biotechnology Co., Ltd.

Stability data of 13 serotypes of Pneumococcal Polysaccharide Derivatives
1.Stability data of Serotype 1 Pneumococcal Polysaccharide Derivatives (Storage condition:2-8■)
Derivation Residual
Total solid Molecular size Molecular size■SEC-MALS method■
rate cyanide
Storage Mn■Number Mw■Weight
Batch no.
time KD0.2-0.6, average average Mw/Mn
——mg/ml 2.6%-4.4% ≤10nl/mg KD≤0.37
recovery rate≥45% molecular molecular ■Dispersion■
weight■ weight■
SP1C201908001 7.900 3.3% 6 0.32 88% 258.8 334.6 1.292
0th day SP1C201909002 7.180 3.9% 2 0.26 68% 237.8 298.6 1.255
SP1C201909003 8.500 3.6% 7 0.33 81% 463.3 695.8 1.502
SP1C201908001 249.05 321.45 1.2905
10th
SP1C201909002 205.05 254.5 1.242
day
SP1C201909003 484.55 660.95 1.3645
SP1C201908001 244.25 311.15 1.2745
20th
SP1C201909002 245.3 286.7 1.1705
day
SP1C201909003 409.8 646.6 1.5785
——
SP1C201908001 226.8 285.6 1.2595
30th
SP1C201909002 211.65 253.35 1.197
day
SP1C201909003 387.55 601.8 1.5525
SP1C201908001 213.8 263.05 1.2315
35th
SP1C201909002 216.7 280.25 1.2935
day
SP1C201909003 461.5 656.55 1.4235

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Stability data of 13 serotypes of Pneumococcal Polysaccharide Derivatives Yuxi Walvax Biotechnology Co., Ltd.

2.Stability data of Serotype 3 Pneumococcal Polysaccharide Derivatives (Storage condition:2-8■)
Derivation Residual
Total solid Molecular size Molecular size■SEC-MALS method■
rate cyanide
Storage Mn■Number Mw■Weight
Batch no.
time KD0.2-0.6, average average Mw/Mn
——mg/ml 5.0%-14.0% ≤10nl/mg KD≤0.45
recovery rate≥60% molecular molecular ■Dispersion■
weight■ weight■
SP3C201908001 14.700 10.6% 1 0.14 64% 296.4 590.0 1.991
0th day SP3C201909002 14.440 13.0% 1 0.18 67% 412.55 1207 2.9255
SP3C201909003 15.530 12.9% 1 0.15 65% 362.9 988.05 2.7235
SP3C201908001 295.8 671.0 2.268
10th
SP3C201909002 385.3 926.2 2.406
day
SP3C201909003 333.85 731.5 2.191
SP3C201908001 299.55 625.05 2.0885
20th
SP3C201909002 368.6 928.3 2.5195
day
SP3C201909003 407.45 973.95 2.3905
——
SP3C201908001 272.15 674.1 2.4775
30th
SP3C201909002 460.15 1300 2.828
day
SP3C201909003 459.25 1244.5 2.7115
SP3C201908001 288.95 641.85 2.2255
35th
SP3C201909002 450.65 1426.5 3.166
day
SP3C201909003 317.15 907.55 2.864

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Stability data of 13 serotypes of Pneumococcal Polysaccharide Derivatives Yuxi Walvax Biotechnology Co., Ltd.

3.Stability data of Serotype 4 Pneumococcal Polysaccharide Derivatives (Storage condition:2-8■)
Derivation Residual
Total solid Molecular size Molecular size■SEC-MALS method■
rate cyanide
Storage Mn■Number Mw■Weight
Batch no. KD0.2-0.6,
time average average Mw/Mn
——mg/ml 3.0%-11.0% ≤10nl/mg KD≤0.42 recovery

molecular molecular ■■Dispersity■
rate≥65%
weight■ weight■
SP4D 202001001 13.055 5.8% 3 0.33 76% 163.15 210.4 1.2895
0th day SP4D 202002002 10.855 6.0% 0 0.37 77% 155.5 191.35 1.2305
SP4D 202005003 9.750 6.4% 1 0.29 65.2% 193.0 255.4 1.3235
SP4D 202001001 188.85 228.5 1.2095
10th
SP4D 202002002 153.2 196.9 1.286
day
SP4D 202005003 200.15 256.55 1.282
SP4D 202001001 189.5 238.3 1.2575
20th
SP4D 202002002 — 155.65 199.95 1.285
day
SP4D 202005003 204.25 261.55 1.2805
SP4D 202001001 184.6 239.0 1.2945
30th
SP4D 202002002 160.9 198.6 1.2345
day
SP4D 202005003 207.8 266.85 1.284
SP4D 202001001 12.860 6.0% 2 0.33 74% 193.8 242.75 1.2525
35th
SP4D 202002002 10.950 6.5% 0 0.36 72% 158.45 198.65 1.254
day
SP4D 202005003 207.55 265.85 1.2805

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Stability data of 13 serotypes of Pneumococcal Polysaccharide Derivatives Yuxi Walvax Biotechnology Co., Ltd.

4.Stability data of Serotype 5 Pneumococcal Polysaccharide Derivatives (Storage condition:2-8■)

Derivation Residual
Total solid Molecular size Molecular size■SEC-MALS method■
rate cyanide
Storage Mn■Number
Batch no. KD■0.6, Mw■Weight
time average Mw/Mn
—mg/ml 3.5%-18.0% NMT10ng/mg KD≤0.90 recovery average molecular
molecular ■■Dispersity■
rate≥37% weight■
weight■
SP5C201909001 8.365 12.6% 1 0.58 51% 27.01 34.45 1.275
0th day SP5C201909002 9.065 11.9% 0 0.63 44% 24.86 31.95 1.285
SP5C201909003 9.945 12.6% 1 0.68 44% 27.61 40.76 1.477
SP5C201909001 31.72 38.70 1.220
10th
SP5C201909002 28.295 32.76 1.1575
day
SP5C201909003 28.15 35.065 1.2465
SP5C201909001 27.40 32.51 1.186
20th
SP5C201909002 27.83 43.41 1.560
day
SP5C201909003 26.76 31.73 1.1855
—
SP5C201909001 25.48 33.29 1.3065
30th
SP5C201909002 26.025 35.21 1.354
day
SP5C201909003 27.21 34.31 1.261
SP5C201909001 28.45 37.725 1.326
35th
SP5C201909002 25.275 34.345 1.359
day
SP5C201909003 27.86 42.29 1.518

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Stability data of 13 serotypes of Pneumococcal Polysaccharide Derivatives Yuxi Walvax Biotechnology Co., Ltd.

5.Stability data of Serotype 6A Pneumococcal Polysaccharide Derivatives (Storage condition:2-8■)

Derivation Residual
Total solid Molecular size Molecular size■SEC-MALS method■
rate cyanide
Storage Mn■Number Mw■Weight
Batch no.
time KD0.2-0.6, average average Mw/Mn
—mg/ml 3.0%-6.0% ≤10nl/mg KD≤0.62
recovery rate≥60% molecular molecular ■■Dispersity■
weight■ weight■
SP6AC201909001 11.875 5.2% 0 0.39 83% 217.7 251.4 1.155
0th day SP6AC201909002 10.460 5.2% 0 0.35 79% 256.7 303.0 1.181
SP6AC201909003 10.020 5.5% 0 0.41 85% 214.4 247.6 1.155
SP6AC201909001 208.15 237.4 1.1405
10th
SP6AC201909002 265.0 295.0 1.1135
day
SP6AC201909003 201.1 226.05 1.124
SP6AC201909001 197.55 217.15 1.099
20th
SP6AC201909002 255.55 308.2 1.206

day
SP6AC201909003 194.6 222.55 1.1435
——
SP6AC201909001 202.4 229.05 1.131
30th
SP6AC201909002 259.7 289.0 1.113
day
SP6AC201909003 196.95 232.05 1.1785
SP6AC201909001 203.85 235.0 1.1535
35th
SP6AC201909002 277.35 333.25 1.2015
day
SP6AC201909003 189.35 221.9 1.172

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Stability data of 13 serotypes of Pneumococcal Polysaccharide Derivatives Yuxi Walvax Biotechnology Co., Ltd.

6.Stability data of Serotype 6B Pneumococcal Polysaccharide Derivatives (Storage condition:2-8■)

Derivation Residual

Total solid Molecular size Molecular size■SEC-MALS method■

rate cyanide

Storage Mn■Number Mw■Weight

Batch no.

time KD0.2-0.6, average average Mw/Mn

——mg/ml 1.5%-3.4% ≤10nl/mg KD≤0.33

recovery rate≥50% molecular molecular ■Dispersion■

weight■ weight■

SP6BD 202001001 11.150 2.7% 3 0.30 74% 278.05 350.95 1.263

0th day SP6BD 202001002 10.760 2.7% 0 0.30 78% 218.9 256.15 1.170

SP6BD 202002003 8.315 2.8% 0 0.30 74% 224.65 267.3 1.1895

SP6BD 202001001 297.85 351.0 1.1785

10th

SP6BD 202001002 216.4 250.9 1.159

day

SP6BD 202002003 225.35 267.25 1.186

SP6BD 202001001 290.4 340.45 1.1725

20th

SP6BD 202001002 —— 206.05 249.25 1.2095

day

SP6BD 202002003 219.55 268.75 1.224

SP6BD 202001001 288.45 347.3 1.2040

30th

SP6BD 202001002 215.7 257.2 1.1925

day

SP6BD 202002003 216.7 265.3 1.2245

SP6BD 202001001 11.225 2.6 6 0.30 74% 283.75 349.55 1.2315

35th

SP6BD 202001002 10.500 2.6 1 0.27 78% 210.35 253.95 1.207

day

SP6BD 202002003 8.400 3.0 1 0.32 79% 223.0 264.1 1.1845

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Stability data of 13 serotypes of Pneumococcal Polysaccharide Derivatives Yuxi Walvax Biotechnology Co., Ltd.

7.Stability data of Serotype 7F Pneumococcal Polysaccharide Derivatives (Storage condition:2-8■)

Derivation Residual

Total solid Molecular size Molecular size■SEC-MALS method■

rate cyanide

Storage Mn■Number Mw■Weight

Batch no.

time KD0.2-0.6, average average Mw/Mn

——mg/ml 2.9%-5.3% ≤10nl/mg KD≤0.43

recovery rate≥60% molecular molecular ■Dispersion■

weight■ weight■

SP7FD 202001001 9.925 4.3% 1 0.37 78% 404.6 607.2 1.5015

0th day SP7FD 202002002 9.760 4.0% 0 0.32 76% 367.55 571.45 1.5545

SP7FD 202002003 7.220 4.3% 1 0.37 79% 357.75 472.4 1.3205

SP7FD 202001001 410.15 657.7 1.6035

10th

SP7FD 202002002 388.5 583.9 1.504

day

SP7FD 202002003 359.25 472.4 1.315

SP7FD 202001001 407.05 605.6 1.4875

20th

SP7FD 202002002 396.4 540.05 1.3625

day

SP7FD 202002003 361.35 471.8 1.3055

SP7FD 202001001 391.9 567.5 1.4485

30th

SP7FD 202002002 390.75 538.7 1.3785

day

SP7FD 202002003 345.5 473.3 1.370

SP7FD 202001001 9.905 4.0% 2 0.37 77% 406.65 629.15 1.5475

35th

SP7FD 202002002 10.085 4.4% 2 0.35 74% 365.55 542.7 1.4845

day

SP7FD 202002003 7.360 4.3% 4 0.35 75% 334.0 473.15 1.4165

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Stability data of 13 serotypes of Pneumococcal Polysaccharide Derivatives Yuxi Walvax Biotechnology Co.,

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8.Stability data of Serotype 9V Pneumococcal Polysaccharide Derivatives (Storage condition:2-8℃)

Derivation Residual
Total solid Molecular size Molecular size■SEC-MALS method■
rate cyanide
Storage Mn■Number Mw■Weight
Batch no.
time KD0.2-0.6, average average Mw/Mn
——mg/ml 3.3%-7.4% ≤10nl/mg KD≤0.46
recovery rate≥70% molecular molecular ■Dispersity■
weight■ weight■
SP9VC201908001 8.750 4.7% 0 0.30 81% 636.4 1059 1.664
0th day SP9VC201908002 10.085 5.0% 1 0.28 92% 959.3 2067.5 2.155
SP9VC201908003 8.850 4.6% 1 0.35 82% 685.2 1266 1.848
SP9VC201908001 636.1 1121.5 1.763
10th
SP9VC201908002 918.4 1766 1.922
day
SP9VC201908003 711.2 1292 1.818
SP9VC201908001 654.7 1160 1.772
20th
SP9VC201908002 867.3 1822 2.1
day
SP9VC201908003 686.8 1362 1.982
——
SP9VC201908001 641.3 1089.5 1.699
30th
SP9VC201908002 824.55 1737.5 2.107
day
SP9VC201908003 645.75 1267 1.963
SP9VC201908001 618.4 1094 1.769
35th
SP9VC201908002 862.4 1838 2.131
day
SP9VC201908003 610.9 1229 2.0115

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Stability data of 13 serotypes of Pneumococcal Polysaccharide Derivatives Yuxi Walvax Biotechnology Co., Ltd.

9.Stability data of Serotype 14 Pneumococcal Polysaccharide Derivatives (Storage condition:2-8℃)

Derivation Residual
Total solid Molecular size Molecular size■SEC-MALS method■
rate cyanide
Storage Mn■Number Mw■Weight
Batch no.
time KD0.2-0.6, average average Mw/Mn
——mg/ml 2.5%-5.9% ≤10nl/mg KD≤0.40
recovery rate≥65% molecular molecular ■Dispersity■
weight■ weight■
SP14D 202001001 11.775 3.9% 0 0.29 79% 923.35 2168.5 2.3485
0th day SP14D 202002002 10.470 4.3% 0 0.33 72% 1048 2007.5 1.916
SP14D 202002003 11.695 4.3% 0 0.30 70% 1272.5 2185 1.717
SP14D 202001001 990.75 2135 2.1545
10th
SP14D 202002002 1029 2067.5 2.0095
day
SP14D 202002003 1203.5 2464 2.0485
SP14D 202001001 1003 2533.5 2.526
20th
SP14D 202002002 —— 1090.5 2097 1.19235
day
SP14D 202002003 1196.5 2497 2.0865
SP14D 202001001 985.3 2171.5 2.204
30th
SP14D 202002002 1042.5 2177.5 2.0885
day
SP14D 202002003 1236.5 2456 1.986
SP14D 202001001 11.510 4.1% 1 0.31 78% 995.4 2176.5 2.186
35th
SP14D 202002002 10.255 4.5% 4 0.35 67% 1014 2220.5 2.189
day
SP14D 202002003 11.065 4.8% 0 0.29 70% 1267.5 2416 1.906

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Stability data of 13 serotypes of Pneumococcal Polysaccharide Derivatives Yuxi Walvax Biotechnology Co., Ltd.

10.Stability data of Serotype 18C Pneumococcal Polysaccharide Derivatives (Storage condition:2-8℃)

Derivation Residual
Total solid Molecular size Molecular size■SEC-MALS method■
rate cyanide
Storage Mn■Number Mw■Weight
Batch no.
time KD0.2-0.6, average average Mw/Mn
——mg/ml 2.0%-4.8% ≤10nl/mg KD≤0.41
recovery rate≥65% molecular molecular ■Dispersity■
weight■ weight■
SP18CC201908001 10.540 3.7% 1 0.32 74% 202.5 245.2 1.211
0th day SP18CC201908002 11.670 4.0% 0 0.36 74% 209.0 250.4 1.198
SP18CC201909003 10.710 4.0% 0 0.35 73% 206.4 246.4 1.194

SP18CC201908001 202.2 260.1 1.286
 10th
 SP18CC201908002 208.6 268.8 1.288
 day
 SP18CC201909003 214.7 264.4 1.232
 SP18CC201908001 225.4 289.8 1.286
 20th
 SP18CC201908002 215.05 273.95 1.274
 day
 SP18CC201909003 208.95 266.25 1.274
 ———
 SP18CC201908001 205.05 263.05 1.283
 30th
 SP18CC201908002 213.3 260.4 1.2205
 day
 SP18CC201909003 204.6 237.55 1.161
 SP18CC201908001 220.8 244.7 1.108
 35th
 SP18CC201908002 194.5 266.2 1.3685
 day
 SP18CC201909003 190.1 255.7 1.345

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11. Stability data of Serotype 19A Pneumococcal Polysaccharide Derivatives (Storage condition: 2-8℃)

Derivation Residual
 Total solid Molecular size Molecular size ■ SEC-MALS method ■
 rate cyanide
 Storage Mn ■ Number Mw ■ Weight
 Batch no.
 time KD 0.2-0.6, average average Mw/Mn
 —mg/ml 3.5%-8.0% ≤10nl/mg KD ≤0.48
 recovery rate ≥65% molecular molecular ■ Dispersity ■
 weight ■ weight ■
 SP19AC201908001 14.170 6.4% 1 0.44 79% 161.3 191.4 1.186
 0th day SP19AC201908002 13.335 6.9% 0 0.44 82% 163.0 191.4 1.175
 SP19AC201909003 14.025 7.1% 0 0.40 82% 179.6 215.2 1.199
 SP19AC201908001 181.4 215.6 1.189
 10th
 SP19AC201908002 146.6 188.2 1.284
 day
 SP19AC201909003 206.25 254.8 1.236
 SP19AC201908001 201.0 238.5 1.186
 20th
 SP19AC201908002 150.35 190.95 1.270
 day
 SP19AC201909003 189.1 239.1 1.2645
 SP19AC201908001 163.65 204.55 1.250
 30th
 SP19AC201908002 143.1 183.25 1.281
 day
 SP19AC201909003 194.1 220.3 1.1345
 SP19AC201908001 174.6 200.6 1.1495
 35th
 SP19AC201908002 143.75 184.15 1.281
 day
 SP19AC201909003 184.15 238.7 1.2965

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Stability data of 13 serotypes of Pneumococcal Polysaccharide Derivatives Yuxi Walvax Biotechnology Co., Ltd.

12. Stability data of Serotype 19F Pneumococcal Polysaccharide Derivatives (Storage condition: 2-8℃)

Derivation Residual
 Total solid Molecular size Molecular size ■ SEC-MALS method ■
 rate cyanide
 Storage Mn ■ Number Mw ■ Weight
 Batch no.
 time KD 0.2-0.6, average average Mw/Mn
 —mg/ml 3.2%-6.2% NMT10ng/mg KD ≤0.56
 recovery rate ≥70% molecular molecular ■ Dispersity ■
 weight ■ weight ■
 SP19FC201909001 11.095 4.5% 0 0.37 76% 247.2 272.4 1.102
 0th day SP19FC201909002 11.220 4.8% 2 0.37 81% 195.7 227.6 1.163
 SP19FC201909003 11.580 5.7% 2 0.37 76% 223.8 264.8 1.182
 SP19FC201909001 250.5 301.7 1.204
 10th
 SP19FC201909002 215.7 257.8 1.195
 day
 SP19FC201909003 233.45 271.2 1.1615
 SP19FC201909001 246.35 286.3 1.1625
 20th
 SP19FC201909002 208.5 242.0 1.1605
 day
 SP19FC201909003 219.3 262.85 1.199
 SP19FC201909001 237.8 286.05 1.203
 30th
 SP19FC201909002 191.7 243.8 1.2715
 day
 SP19FC201909003 229.05 269.55 1.1765

SP19FC201909001 256.3 295.05 1.1515
35th
SP19FC201909002 222.9 251.9 1.130
day
SP19FC201909003 235.5 271.55 1.153

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Stability data of 13 serotypes of Pneumococcal Polysaccharide Derivatives Yuxi Walvax Biotechnology Co., Ltd.

13.Stability data of Serotype 23F Pneumococcal Polysaccharide Derivatives (Storage condition:2-8℃)

Derivation Residual
Total solid Molecular size Molecular size■SEC-MALS method■
rate cyanide
Storage Mn■Number Mw■Weight
Batch no.
time KD0.2-0.6, average average Mw/Mn
——mg/ml 1.1%-3.7% NMT10ng/mg KD≤0.46
recovery rate≥75% molecular molecular ■Dispersity■
weight■ weight■
SP23FC201908002 11.790 2.5% 1 0.37 79% 194.6 223.8 1.150
0th day SP23FC201909003 12.975 2.5% 0 0.36 78% 223.15 254.5 1.1405
SP23FD 202001001 10.425 2.6% 2 0.41 90% 224.75 275.8 1.227
SP23FC201908002 204.3 230.4 1.128
10th
SP23FC201909003 224.6 254.85 1.1345
day
SP23FD 202001001 215.6 270.9 1.2565
SP23FC201908002 206.8 239.0 1.156
20th
SP23FC201909003 211.55 249.95 1.1815
day
SP23FD 202001001 —— 237.35 273.0 1.1515
SP23FC201908002 198.45 227.2 1.1445
30th
SP23FC201909003 213.95 250.2 1.1695
day
SP23FD 202001001 230.95 277.5 1.202
SP23FC201908002 196.65 227.55 1.157
35th
SP23FC201909003 206.1 247.9 1.203
day
SP23FD 202001001 10.510 2.6 5 0.37 79% 220.9 268.55 1.2155

Conclusion■According to stability protocol of Pneumococcal Polysaccharide Derivatives
WSYX-QC-SP-0609-0099 (version 00, 01)■13 serotypes of
Pneumococcal Polysaccharide Derivatives were kept at 2-8℃ for 35 days■except batch SP4D202005003 at 35th
day, the tests were not totally
finished■All the test results of the other batches at each time point were in accordance with the
specification of the 13-valent pneumococcal
polysaccharide conjugate vaccine■Code■WSYX-QC-TS-06-09-0002■.

Version: July 2020 13

Standard Operation Procedures of Yuxi Watson Biotechnology Co., Ltd.

Operating procedures Code WSYX-QC-SOP-0009-0138
for determination of
Version 04 Page 1/4
ethanol, ether and
acetone residues Non-photocopying, confidential documents

1 Objective
standardize the test Method for the determination of ethanol, ether and acetone residues.
2 Definitions/Abbreviations
No.
3 Scope of application
Applicable to the determination of ethanol, ether and acetone residues in intermediate
product and finished product of vaccine.
4 Responsibilities
All the operators shall strictly conduct the SOP
5 Materials and equipment
5.1 Use of equipment
No. Name of equipment style Manufacturer Remark

1 Shimadzu Or equivalent
gas chromatograph GC-2014C
equipment
5.2 Use of reagents
No. Name of reagent Grade
1 ethanol AR
2 acetone AR
3 diethyl [ethyl] ether AR
5.3 Use of materials
No. Material name No. Material name
1 pipette tips 4 Sampling bottle
2 pipette 5 chromatographic
column
3 Volumetric flask
6 Procedures
6.1 Solution preparation
Standard Operation Procedures of Yuxi Watson Biotechnology Co., Ltd.

Operating procedures Code WSYX-QC-SOP-0009-0138
for determination of
Version 04 Page 2/4
ethanol, ether and
acetone residues Non-photocopying, confidential documents

6.1.1 Mixed reference solution (Containing 0.08% ethanol, 0.03% acetone, 0.01% ether)
Respectively take 80 µl of Ethanol, 30 µl of acetone, 10 µl of ether (Note: after adding
one kind of solvent, the volumetric flask should be sealed immediately and mixed quickly),
diluted with purified water to 100 ml. mix well. This solution should be used immediately.
6.2 According to the following instrument parameter settings, the residual amount of ethanol,
ether and acetone was detected on the gas chromatograph.
Residual organic solvent detection parameters
detector FID■flame ionization ditector■
terminal time 8min
SPL temperature 230■
Temperature of Column temperature box 40■
Detector temperature 250■
Carrier gas type nitrogen
Sampling mode split-flow
pressure 74.3kPa
total flow 14 ml/min
Chromatographic column flow 1 ml/min
Nitrogen flow rate 40 ml/min
Hydrogen flow rate 40 ml/min
Air velocity 400 ml/min
Sample volume 0.5 µl
chromatographic column Wondacap-WAX(30 m×0.25 mm×0.25 µm)
6.3 test procedure: operating procedures: take about 1 ml to 1.5 ml of mixed reference
solution into the injection bottle, and inject two parallel needles; take parallel samples of the
test sample, take about 1ml to 1.5 ml in two injection bottles respectively, inject 1 needle for
each, and test according to the above detection parameters.
6.4 results
6.4.1 The peak area was regressed by the concentration of each component in the reference
solution. Then the peak area of the sample solution is calculated with the equation■get the
concentration of each organic solvent.
Concentration of test product = $Cr \times \frac{Ax}{Ar}$ ■
Cx: Concentration of test product, %; Cr: Concentration of external standard, %
Standard Operation Procedures of Yuxi Watson Biotechnology Co., Ltd.

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for determination of
Version 04 Page 3/4
ethanol, ether and
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Ax: peak area of test product; Ar: peak area of external standard.
6.4.2 Result judgment
Carry out according to corresponding quality standard
7 Flow chart
No

8 Notes

8.1 Hydrogen is flammable gas, when the test is finished, should immediately close the main valve of the hydrogen tank.

8.2 After the detection, stop GC, stopping the gas supply after the detector temperature below 100■.

8.3 Alcohol tampons are prohibited during the experiment.

9 Reference or standard basis

The test record of ethanol, ether and acetone residues. Record code:

WSYX-QC-SOP-00-09-0138-R-01

10 Revision history

Revising

Basis, reason and revision

Version Valid from departme Remarks

nt

Standardize the name of the document,

specify the purpose of the document,

QC

00 2015.08.01 the scope of use, and the preparation

laboratory

details of the mixture

Improve the document.

QC Improve the content of documents

01 2018.06.01

laboratory

New chromatographic management

QC system was added in the laboratory,

02 2018.09.01

laboratory and the operation method was changed

according to the new chromatographic

Standard Operation Procedures of Yuxi Watson Biotechnology Co., Ltd.

Operating procedures Code WSYX-QC-SOP-0009-0138

for determination of

Version 04 Page 4/4

ethanol, ether and

acetone residues Non-photocopying, confidential documents

management system.

03 QC Alcohol tampons are prohibited during

2019.06.10

laboratory the experiment.

Standard Operation Procedures of Yuxi Watson Biotechnology Co., Ltd.

code WSYX-QC-SOP-0009-0201
Operation procedures
determination of Version 09 Page 1/6
Identification
Non-photocopying, confidential documents

1...Objective
This SOP describes the method for identification test
2 Definition/abbreviation
No
3 Applicable scope
This SOP is applicable to the identification test of haemophilus influenzae type b conjugate vaccine, meningococcal group A, C bivalent polysaccharide conjugate vaccine, group A and C meningococcal polysaccharide vaccine, intermediate product and finished product of ACYW135 meningococcal polysaccharide vaccine, 23-valent pneumococcal polysaccharide Vaccine, intermediate product of 13-valent pneumococcal polysaccharide conjugate vaccine.
4 Responsibility
All the operators shall strictly conduct the SOP
5 Materials and equipment
5.1 Use of equipment
Equipment
No. Device Name Production Factory
model
Constant
Stoker instruments and equipment
1 temperature and GTHI-280B
(Shanghai) Co., Ltd
humidity incubator
Beijing saiduolis Instrument
2 Electronic balance BS124S
System Co., Ltd
5.2 Use of reagents
number Material/Name number Material/Name

1 Slide slides 2 Hole Punching device
3 Wet box 4 Pipette
5 Dye VAT
5.3 Use of materials
Standard Operation Procedures of Yuxi Watson Biotechnology Co., Ltd.

code WSYX-QC-SOP-0009-0201
Operation procedures
determination of Version 09 Page 2/6
Identification
Non-photocopying, confidential documents

No. Name Grade
1 Physiological saline Solution
2 Agarose Reagent pure
3 Specific antiserum Standard

4 Positive control Standard solution
6 Procedures
6.1 Principle: when the antigen and antibody correspond to each other, and the concentration and proportion are appropriate, the precipitation line of immune complex will be formed between the antigen and antibody after a certain period of time, so as to check the specificity of the test sample.
6.2 Solution preparation
6.2.1 1.5% agarose gel
agarose 1.5 g
0.9% sodium chloride injection 100 ml, heating and swelling
6.2.2 Distaining solution: 1000 ml dispensing quantity
Acetic acid 100 ml
Ethanol 450 ml
Purified water 450 ml
6.2.3 0.5% Coomassie brilliant blue staining fluid: 100 ml
Coomassie brilliant blue (10B) 0.5 g
Methyl alcohol 50 ml
Acetic acid 10 ml
Water 40 ml
6.3 Testing procedure
6.3.1 Preparation of phalanx figure AGAR plate: pour 1% agarose solution on the level of slides, 0.19 ml of agarose per cm², after solidification, punch diameter 3 mm, the distance between neighboring wells is 3 mm, as shown in figure:
Standard Operation Procedures of Yuxi Watson Biotechnology Co., Ltd.

code WSYX-QC-SOP-0009-0201
Operation procedures
determination of Version 09 Page 3/6
Identification
Non-photocopying, confidential documents

1
2 5 4
3
6.3.2 Preparation of the test sample: dilute the protein concentration or polysaccharide content of the test sample to the appropriate concentration with physiological sodium chloride solution.
6.3.2.1 For the test sample of polysaccharide, the polysaccharide content was diluted to about

100 ug/ml (generally, the test sample was diluted by 30 or 40 times).

6.3.2.2 The bulk conjugate was loaded twice

6.3.2.3 The finished product can be concentrated or diluted for sample loading:

Six samples of haemophilus influenzae type b conjugate vaccine were mixed and centrifuged at 8000 rpm/min for 15 minutes, then ultrafiltration was detected.

The meningococcal polysaccharide vaccine was detected after three lyophilized preparations were dissolved with one diluent.

The meningococcal polysaccharide conjugate vaccine was detected after 7 lyophilized preparations were dissolved with 1 diluent

6.3.2.4 For tetanus toxoid immune serum, the flocculent unit was diluted to 30 Lf/ml for sample loading.

6.3.3 According to the different test samples, the central well shall add the corresponding specific antisera, in the surrounding wells, add test substance and positive reference substances, every well added 20 ul of sample.

6.3.4 Wet box spread of 37■ for 24 hours; fully immerse in 0.9% sodium chloride solution the agarose gel board to remove the unbound protein

6.3.5 Remove immersed agarose gel plate, dry, stain 5 minutes with coomassie brilliant blue , then use destaining solution to discolorize until the background is colorless and clear blue precipitation line appear.

6.3.6 Make gel: lay a layer of cellophane on the glass plate, put the decolorized film on the cellophane, then lay a layer of cellophane, clamp and dry it.

Standard Operation Procedures of Yuxi Watson Biotechnology Co., Ltd.

code WSYX-QC-SOP-0009-0201

Operation procedures

determination of Version 09 Page 4/6

Identification

Non-photocopying, confidential documents

6.4 The results decision

The test is valid if the reference substance forms a corresponding precipitation line. If a precipitation line forms between the test sample and antibody wells the test sample is judged as corresponding to the antibody.

6.5 Positive control information

The store

The content of The validity

Positive condition of

Sample name positive of positive

control name positive

control control

control

Haemophilus

100 ug

Haemophilus influenzae influenzae type

polysaccharide below - 20■ 5 years

type b polysaccharide b

per 1ml

polysaccharide

Haemophilus

100 ug

Haemophilus influenzae influenzae type

polysaccharide below - 20■ 5 years

type b bulk conjugate■ b

per 1ml

Haemophilus influenzae polysaccharide

type b conjugate vaccine Tetanus toxoid 100 ug protein

2-8■ 3 years

purified per 1ml

Meningococcal

100 ug

polysaccharide (bulk)■ Meningococcal

polysaccharide below - 20■ 5 years

Meningococcal polysaccharide

per 1ml

polysaccharide vaccine

Meningococcal 100 ug

Meningococcal

polysaccharide bulk polysaccharide below - 20■ 5 years

polysaccharide

conjugate■ per 1ml

Standard Operation Procedures of Yuxi Watson Biotechnology Co., Ltd.

code WSYX-QC-SOP-0009-0201

Operation procedures

determination of Version 09 Page 5/6

Identification

Non-photocopying, confidential documents

Meningococcal

Tetanus toxoid 100 ug potein

polysaccharide conjugate 2-8■ 3 years

purified per 1ml

vaccine

100 ug 5 years

Pneumococcal Pneumococcal

polysaccharide below - 60■

polysaccharide (bulk) polysaccharide

per 1ml

Pneumococcal 100 ug 5 years

Pneumococcal

polysaccharide bulk polysaccharide below - 60
polysaccharide
conjugate per 1ml

■Pneumococcal Tetanus toxoid 100 ug protein
2-8■ 3 years
polysaccharide vaccine purified per 1ml

6.6 Selection of positive control

When a batch of products are selected as the positive control, it should be confirmed: three people independently operate (except for personnel, other operations should be consistent), and carry out serotype identification test, a total of 10 times. If the test results can show obvious precipitation line, it can be used as the positive control of identification test.

7 Flow chart

N/A

8 General considerations

8.1 Agarose should not be overheated, in order to avoid excessive moisture evaporates.

8.2 If the positive control is within the validity period

9 References

9.1 Pharmacopoeia of People's Republic of China, 2015. General requirement 3403.

10 Revision history

Revision

Version Basis, reason and content of Revision

Issuing date Valid from Department

Revision marks

t

Standard Operation Procedures of Yuxi Watson Biotechnology Co., Ltd.

code WSYX-QC-SOP-0009-0201

Operation procedures

determination of Version 09 Page 6/6

Identification

Non-photocopying, confidential documents

QC Document expiration

03 2011.11.01 2011.12.01 Laboratory Improve the content of documents

QC Revised according to the

04 2014.11.01 2014.12.01 Laboratory 2015 edition of Pharmacopoeia

QC Document modification due

05 2015.11.01 2015.12.01

Laboratory to change of product name

QC Add the relevant provisions

06 2016.03.17 2016.03.20

Laboratory of positive control

QC Increase the preparation steps

07 2016.06.30 2016.07.01

Laboratory of test sample

QC Add the selection of positive

08 2016.08.29 2016.09.01

Laboratory control and appendix

Standard Operation Procedures of Yuxi Walvax Biotechnology Co., Ltd

Code WSYX-QC-SOP-0009-0103
Operation procedures
determination of Version 06 Page 1/2
nucleic acid content
Non-photocopying, confidential documents

1 Objective

This SOP describes the method for nucleic acid content determination.

2 Definitions/abbreviations

N/A

3 Scope of application

This SOP is applicable to the nucleic acid determination of haemophilus influenzae type b polysaccharide (bulk), Group A Meningococcal Polysaccharide (bulk), Group C Meningococcal Polysaccharide (bulk), Group Y Meningococcal Polysaccharide (bulk), Group W135 Meningococcal bulk Polysaccharide, Pneumococcal Polysaccharide (bulk), and other test samples manufactured by Yuxi Walvax.

4 Responsibility

All the operators shall strictly conduct the SOP

5 Equipment and materials

5.1 Equipment

No. Name Model Code manufacturer Notes

ultraviolet-visible Or equivalent

1 CP225D WYQCF0037 Shimadzu

spectrophotometer equipment

6 Working procedure

6.1 testing procedure

Take about 3.5 ml of the test solution (the concentration should not be lower than 3 mg/ml), UV-2600 ultraviolet-visible spectrophotometer was used to determine the absorbance a value at 26 0nm wavelength, and calculate the nucleic acid content according to

= 200 .

1%

E 1cm

6.2 result judgment

According to the corresponding sample quality standard.

6.2 result decision

Standard Operation Procedures of Yuxi Walvax Biotechnology Co., Ltd

Code WSYX-QC-SOP-0009-0103
Operation procedures
determination of Version 06 Page 2/2
nucleic acid content
Non-photocopying, confidential documents

According to the relative quality standard.

7 Flow chart

N/A

8 General considerations

8.1 the sample was recovered to room temperature before detection.

8.2 be careful when using the cuvette.

9 References

9.1 Pharmacopoeia of People's Republic of China, volume III, 2010. appendix, IIA.

10 Revision history

Revising

Issuing Basis, reason and revision Rem

Version Valid from departme

date arks

nt

QC Document expiration

03 2011.11.01 2011.12.01

laboratory Improve the content of documents

QC

04 2014.11.01 2014.12.01 Improve the content of documents

laboratory

QC

05 2017.09.30 2017.10.01 Improve the content of documents

laboratory

SOP of Walvax Biotechnology Co., Ltd.

Code WSYX-AC-SOP-00-09-0021

Abnormal toxicity test

Version 06 Page 1/3

SOP

Confidential files No copy

1 Objective

Standardize operating procedure for abnormal toxicity test.

2 Definitions and Abbreviations

None

3 Scope of application

This SOP applies to detect the abnormal toxicity.

4 Responsibility

4.1 Experimental animal center

-Experimental Animal Center staff strictly follow the operating rules.

4.2 Quality supervision department

-The quality supervision department supervises the implementation of this SOP.

5 Equipment and Materials

5.1 Equipment

5.2 Materials

No. Materials name No. Materials name

Clean grade and SPF grade guinea

1 Mice above the clean grade 18-22g 2

pigs 250-350g

3 Disposable syringe 4 Disposable latex gloves

5 ethanol cotton ball

6 Working procedure

6.1 Test procedure

In addition to the other regulation, the test of abnormal toxicity should include the mice test and the guinea pigs test. In the test, the same batch of animals should be set in blank control. The test is valid, if all animals survived without any abnormal reaction during the observation period, and the weight of each animal should be increased at the end of the test, slowly inject vaccine into the animal body according to the prescribed route of administration.

SOP of Walvax Biotechnology Co., Ltd.

Code WSYX-AC-SOP-00-09-0021

Abnormal toxicity test

Version 06 Page 2/3

SOP

Confidential files No copy

6.1.1 Test in mice

Five mice are used for the test on each batch of product. Weigh and number each mouse before test. Record the data. The body weight of each mouse shall be 18-22g. Each mouse was injected with 0.5ml test sample and observed for 7 days. During the observation period, all mice should be kept intact without any abnormal reaction, and the weight of each mouse should be increased at the time become due, so that the test product is in accordance with the requirements. If it does not meet the above requirements, another 10 mice weighed 19-21g should be retested, and the result shall be judged as described above.

6.1.2 Test in guinea pigs

Two guinea pigs are used for the test on each batch of product. Weigh and number each guinea pig before test. Record the data. The body weight of each guinea pig shall be 250-350g. Each guinea pig was injected intraperitoneally with 5.0ml and observed for 7 days. During the observation period, guinea pigs should be all healthy and no abnormal reaction. When the time become due, the weight of each guinea pig should be increased. If the above requirements cannot be met, 4 guinea pigs can be retested, and the result shall be judged as described above.

7 Flow chart

None

8 Matters need attention

None

9 References

9.1 Pharmacopoeia of People's Republic of China, Volume III, 2015. general rule 1141, page 94.

10 Revision history

Revising

Issuing Time in Revision basis, reason and

Version departme Notes

date force content

nt

SOP of Walvax Biotechnology Co., Ltd.

Code WSYX-AC-SOP-00-09-0021

Abnormal toxicity test

Version 06 Page 3/3

SOP

Confidential files No copy

01 Nov. 01 Dec. QC

03 Changed when file expired /

2011 2011 Laboratory

01 Nov. 01 Dec. Experimen Revision of the 2015 edition

04 tal animal /

2014 2014 center of Pharmacopoeia

15 Nov. 01 Dec. Experimen Add the test record of

05 tal animal /

2015 2015 center abnormal toxicity test

SOP of Yuxi Walvax Biotechnology Co., Ltd.

Code WSYX-QC-SOP-0009-0134
SOP for the protein content 1/6
Version 02 Page
test (Lowry method 2)
Confidential files No copy

1 Objective

Standardize the operation of protein content test (Lowry method 2).

2 Definitions/Abbreviations

N/A

3 Scope of application

3.1 It is applicable to determine the protein content of acellular pertussis purified liquid, tetanus toxoid bulk and tetanus toxoid purified liquid other test samples manufactured by Yuxi Walvax.

4 Responsibilities

All the operators shall strictly conduct the SOP.

5 Materials and equipment

5.1 Use of equipment

Equipment Equipment code
No. Equipment name Manufacturer Notes
model

U/V WYQCF0444 Shimadzu instrument Or equivalent
1 UV-2600
spectrophotometer WYQCF0445 (Suzhou) Co., Ltd equipment

Or equivalent

2 Electronic balance BS124S WYQCF0036 Sartorius
equipment
Shanghai Qixin
WYQCF0061

Electric constant Scientific Instruments

Or equivalent

3 temperature water DK-600A WYQCF0287 Co., Ltd. / Shanghai
equipment
tank Yiheng Scientific
WYQCF0293

Instruments Co., Ltd.

High speed Thermo Fisher

Multifuge WYQCF0472 Or equivalent

4 freezing Technology (China)

X1R WYQCF0473 equipment

centrifuge Co., Ltd

SOP of Yuxi Walvax Biotechnology Co., Ltd.

Code WSYX-QC-SOP-0009-0134
SOP for the protein content 2/6
Version 02 Page
test (Lowry method 2)
Confidential files No copy

5.2 Use reagent

No. Reagent name Grade

1 Sodium hydroxide AR

2 Anhydrous sodium carbonate AR

3 Disodium tartrate dihydrate AR

4 Copper sulfate pentahydrate AR

5 Sodium deoxycholate Import reagent

6 Trichloroacetic acid AR

7 2N foline-phenol Import reagent

5.3 Use of materials

No. Material name number Material name

1 Pipette 2 Pipette tip

3 Volumetric flask 4 test tube

5 powderless latex glove 6 web

7 Triangular flasks 8 Measuring cylinder

6 Working procedure

6.1 Principle

According to this method, the peptide bond contained in protein molecules is chelated with Cu^{2+} in alkaline solution to form a protein copper complex. The complex reduces the phosphomolybdic acid of phenol reagent to produce blue compound. At the same time, under alkaline conditions, phenol reagent is easily reduced by tyrosine, tryptophan and cysteine in protein to produce blue reaction. In a certain range, the color depth is directly proportional to the protein concentration. The protein content in the test sample was determined by colorimetry with the protein standard solution as the standard curve. Before the determination, deoxycholate trichloroacetic acid was added into the sample, and the interfering substances were removed by protein precipitation. This method can also be used to concentrate protein in dilute solution.

6.2 preparation of reagents

SOP of Yuxi Walvax Biotechnology Co., Ltd.

Code WSYX-QC-SOP-0009-0134
SOP for the protein content 3/6
Version 02 Page

test (Lowry method 2)
Confidential files No copy

6.2.1 1% sodium hydroxide solution

Weigh 1g sodium hydroxide into a beaker, dissolve it with purified water and transfer it to a 100ml volumetric flask. Add a small amount of purified water to moisten the beaker for three times and transfer it to the volumetric flask. Fix the volume to the scale line and mix well.

6.2.2 5% sodium carbonate solution

Weigh 5g anhydrous sodium carbonate into a beaker, dissolve it with purified water and transfer it to a 100ml volumetric flask. Add a small amount of purified water to moisten the beaker for three times and transfer it to the volumetric flask. Fix the volume to the scale line and mix well.

6.2.3 test solution A

Take 200ml of 1% sodium hydroxide solution and 200ml of 5% sodium carbonate solution into a 500ml volumetric flask, add purified water to the scale line, mix well, label and set aside.

6.2.4 2.98% sodium tartrate dihydrate solution

Weigh 2.98g of disodium tartrate dihydrate into a beaker, dissolve with purified water and transfer to a 100ml volumetric flask. Add a small amount of purified water to moisten the beaker for 3 times and transfer to the volumetric flask. Fix the volume to the scale line, mix well, label and reserve.

6.2.5 1.25% copper sulfate solution

Weigh 1.25g copper sulfate pentahydrate into a beaker, dissolve it with purified water and transfer it to a 100ml volumetric flask. Add a small amount of purified water to moisten the beaker for 3 times and transfer it to the volumetric flask. Fix the volume to the scale line, mix well, label and reserve.

6.2.6 test solution B (ready to use)

Take 2.98% sodium tartrate dihydrate solution, 1.25% copper sulfate solution and purified water according to the volume ratio of 2:2:1, and mix well. (according to the amount needed in the experiment, it is prepared in proportion).

6.2.7 test solution c (ready to use)

Take the test solution a and test solution B according to the volume ratio of 50:1, and mix SOP of Yuxi Walvax Biotechnology Co., Ltd.

Code WSYX-QC-SOP-0009-0134

SOP for the protein content 4/6

Version 02 Page

test (Lowry method 2)

Confidential files No copy

well. (according to the amount needed in the experiment, it is prepared in proportion).

6.2.8 Folin phenol test solution (ready to use)

Take the stock solution (2mol / L acid concentration) 1 → 2 (i.e. take 1000 μl of Folin phenol into a triangular flask, add purified water to make 2000 μl Folin phenol test solution, and mix well. According to the experimental requirement, the mixture was prepared according to 1 → 2. The prepared Folin phenol test solution should meet the following requirements: 1000 μl of test sample, 5000 μl of test solution and 500 μl of prepared Folin phenol test solution, and the pH value of the solution obtained should be 10.3 ± 0.3).

6.2.9 sodium deoxycholate test solution

Weigh 0.15g sodium deoxycholate into a beaker, dissolve it with purified water and transfer it to a 100ml volumetric flask. Add a small amount of purified water to moisten the beaker for 3 times and transfer it to the volumetric flask. Fix the volume to the scale line, mix well, label and reserve.

6.2.10 72% trichloroacetic acid solution

Weigh 7.2g trichloroacetic acid in a beaker, add purified water to dissolve it into a 10ml volumetric flask, add a small amount of purified water, repeatedly wash the beaker for three times, transfer it to the volumetric flask, and fix the volume to the scale line, mix well, then get the product, label it for use.

6.2.11 200 μg / ml protein national standard solution

Please refer to wsyx-qc-sop-0009-0801 for preparation details

6.3 test procedure

6.3.1 treatment of test articles

A pipette with a range of 100-1000 μl was used to accurately measure 1000 μl of the test sample (containing about 20-80 μg of protein) into the test tube in duplicate.

6.3.2 preparation of standard curve

0 1 2 3 4 5

200 μg / ml protein national standard solution

0 100 200 300 400 500

(μl)

SOP of Yuxi Walvax Biotechnology Co., Ltd.

Code WSYX-QC-SOP-0009-0134

SOP for the protein content 5/6

Version 02 Page

test (Lowry method 2)

Confidential files No copy

Purified water (μl) 1000 900 800 700 600 500

Protein content (μg/ml) 0 20 40 60 80 100

6.3.3 add 100 μL sodium deoxycholate test solution into the above tubes, mix well, and place at room temperature for 10min.

6.3.4 add 100 μl of 72% trichloroacetic acid solution into the above tubes, mix well, centrifugate at 3000g for 30min, gently pour out the supernatant, remove the remaining liquid with a pipette, and retain the protein precipitate.

6.3.5 protein precipitate: dissolve each tube with C 1000 μl of test solution, then add 5000 μl of test solution, and mix well.

6.3.6 place the above tubes in the electric constant temperature water bath at 25■ for 10 minutes, add 500 μl of Folin phenol test solution, mix one tube immediately after adding one tube, and place the electric heating constant temperature water bath at 25■ for 30 minutes.

6.3.7 read the absorbance at 750 nm.

6.4 result calculation

6.4.1 take standard 0 tube as blank, calculate the linear regression equation with the

concentration of standard solution and its corresponding absorbance, and then substitute the absorbance of the test solution into the linear regression equation to calculate the protein concentration of the test solution.

Protein concentration of test sample = average protein concentration of test sample × dilution multiple

6.4.2 Results judgment

Carry out according to corresponding quality standard.

7 Flow chart

No

8 Matters need attention

SOP of Yuxi Walvax Biotechnology Co., Ltd.

Code WSYX-QC-SOP-0009-0134

SOP for the protein content 6/6

Version 02 Page

test (Lowry method 2)

Confidential files No copy

8.1 Temperature and time have influence on protein content detection and should be strictly controlled.

8.2 avoid centrifugal precipitation when taking the supernatant.

9 Reference or standard basis

9.1 Pharmacopoeia of People's Republic of China, volume 3, 2015. General rule 0731.

10 Revision history

Issuing Time in Revision Revision basis, reason and

Version Note

time force Dept. content

QC Complete some parts of the

00 2016.08.12 2016.08.13

laboratory content

QC Complete some parts of the

01 2017.09.30 2017.10.01

laboratory content

Standard Operation Procedures of Yuxi Walvax Biotechnology Co., Ltd
WSYX-QC-SOP-0009-075
Code
Operation procedures 1
determination of total version 07 Page 1/5
nitrogen content
Non-photocopying, confidential documents

1 Objective

This SOP describes the method for total nitrogen content determination, establishing an accurate method for experimenters to master, and conducting the test.

2 Definitions/abbreviations

N/A

3 Scope of application

This SOP is applicable to the total nitrogen content determination of products manufactured by Yuxi Walvax.

4 Responsibility

All the operators shall strictly conduct the SOP

5 Equipment and materials

5.1 Equipment

No. Name Model manufacturer Remarks

Temperature Or equivalent

1 TecatorTM Digestor FOSS

control digester equipment

Temperature Or equivalent

2 SCINO DT220 FOSS

control digester equipment

Kjeldahl Or equivalent

3 kjeltecTM 2200 FOSS

apparatus equipment

Kjeldahl Or equivalent

4 Kjeltec 8200 FOSS

apparatus equipment

Electronic Or equivalent

5 BSA124S Starorius equipment

balance

Electronic MSU225S-1CU-D Or equivalent

6 Starorius equipment

balance U

5.2 reagents

Standard Operation Procedures of Yuxi Walvax Biotechnology Co., Ltd

WSYX-QC-SOP-0009-075

Code

Operation procedures 1

determination of total version 07 Page 2/5

nitrogen content

Non-photocopying, confidential documents

No. Material/reagent Grade

1 digestant N/A

2 sulphuric acid analytical reagent

5.3 materials

No. Name of solution Category

1 Mixed indicator liquid solution

2 2% Boric acid absorption solution solution

3 40% Sodium hydroxide solution solution

4 5 mg/ml tryptone solution solution

5 Hydrochloric acid titration solution (0.1 mol/L) Titrating solution

Hydrochloric acid titration solution (0.02

mol/L)

mol/L)

6 Working procedure

6.1 Test reference substance: 5 mg/ml tryptone solution

Weigh 0.05 g tryptone (total nitrogen content 12.7%, source: Oxoid, article number: LP0042), dissolve with water, dilute to 10 ml, shake well. Prepare freshly.

6.2 The preparation of test solution: accurately weigh proper amount of the test product, the nitrogen content of the solution is 0.1 mg.

6.3 Digestion

Take a certain volume of test solution and tryptone 1.0 ml, put in the digesting and boiling tube, add about 5 g of the digestion agent, add sulfuric acid 10 ml, digestion at 420 ■ for 40 minutes, until the solution is clear, show bright green color.

6.4 Distillation

Steam the distiller for more than 15 minutes before distillation. When receiving liquid was purplish red, and digesting and boiling solution cool to room temperature. a triangle

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nitrogen content

Non-photocopying, confidential documents

bottle was put at the right side of the Kjeldahl nitrogen meter and insert condensing pipe and the boric acid absorbent adding pipe. Then install the digested boiling pipe on the left side of the Kjeldahl nitrogen meter and turn on the machine. The instrument begins to distillation automatically, after stopping distillation, take off the triangle bottle, wash the condensation

tube with the water for injection, collect the eluent in the triangle bottle and prepare for titration.

6.5 Titration

The distilled solution was titrated with hydrochloric acid titration solution (0.02 mol/L) until the solution turn from blue-green to grey purple, and the titration result was corrected by blank test.

6.6 Data processing

$$(VX - V0) \times c \times 14.01 \times n$$

$$\text{mg / ml} = \frac{\text{Total nitrogen content}}{V}$$

Total nitrogen content V

$$\text{Total nitrogen content (\%)} = \frac{C}{(C0 \times 12.7\%)} \times 100\%$$

where VX is volume of hydrochloric acid titration solution consumed by test solution, ml;

V0 is volume of hydrochloric acid titration solution consumed by blank control, ml;

c is concentration of hydrochloric acid titration solution, mol/L;

n is dilution multiple of test solution;

V is volume of test solution, ml;

14.01 is relative atomic mass of nitrogen;

C is total nitrogen content of test product, mg/ml;

C0 is concentration of control sample, mg/ml;

12.7% is total nitrogen content of tryptone.

7 Flow chart

N/A

8 Notes

8.1 5 mg/ml tryptone solution was used as the reference substance, and the recovery rate

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should be in the range of 80% - 120%.

8.2 The range of 2 blank tests shall not be greater than 0.05ml, blank value should be 0.02ml, blank value of 0 ml is also normal.

8.3 The distilled ammonia receiving liquid should be titrated as soon as possible to avoid too long storage time and affect the determination results.

8.4 The determination of total nitrogen and non-protein nitrogen can be controlled by the same blank.

8.5 If there is no black CuO precipitation in the distillation process, it shows that the addition of alkali is insufficient, so the experiment should be made up or redone.

8.6 Add 2% boric acid absorption solution to the triangle bottle should show purplish red, if it show green, indicating that the triangle bottle pollute by alkaline substance.

8.7 Operating environment should be free from ammonia and alkaline gas interference.

9 Reference

9.1 0704 Nitrogen test, CP2015 volume III China Pharmaceutical Science and Technology Press, 2015

9.2 operation manual of FOSS Kjeldahl nitrogen analyzer

10 Revision history

Revising

Issuing Basis, reason and revision Rem

Version Valid from departme

date arks

nt

03 2011.11.01 2011.12.01 QC Document expiration

laboratory

Improve the content of

documents

04 QC Improved the documents

2014.11.01 2014.12.01 laboratory according to the requirements of Pharmacopoeia 2015

05 QC Added internal parameters in the

2016.05.31 2016.06.01

laboratory inspection process

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Code

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nitrogen content

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06 QC Standard verification record

2016.08.04 2016.08.05

laboratory

Stability data Pneumococcal Monovalent Bulk Conjugate Yuxi Walvax Biotechnology Co., Ltd.

Stability data of 13 serotypes of Pneumococcal Monovalent Bulk Conjugate
1. Stability Data of Serotype 1 Pneumococcal Bulk Conjugate (Storage condition: 2-8■)
1.1 Batch No.:RSP1C201706001
Test Free Free
Polysacc
items Protein PS/protein Molecul Residual polysac carrier Bacterial ADH
haride Identity Sterility
and content ratio ar size EDAC charide protein endotoxin Content
content
standard content content
s NMT
Positive response Shall meet
≥40ug/m ≥70ug/m ≤5.0μmol/ 0.75EU/μg
0.20-1.80 ≥80% ≤25.0% ≤5.0% to antiserum type requireme ■0.15ug/ml
Storage I I L polysaccha
1 nts.
time ride
0th <0.75EU/
245.5 239.1 1.03 91.4 2.8 10.8 4.1 Complied Complied 0.05
month μg
1th
243.6 246.9 0.99 100.0 — 13.6 1.9 — — — —
month
2nd
230.0 226.8 1.01 100.0 — 10.8 0.7 — — — —
month
3rd
241.2 254.2 0.95 96.4 — 12.4 1.97 — — — —
month
6th
261.0 267.1 0.98 99.6 — 11 1.0 — — — —
month
9th
280.0 222.6 1.26 99.2 — 16 1.2 — — — —
month
12th <0.75EU/
241.4 261.5 0.92 99.5 2.766 16 1.3 Complied Complied 0.10
month μg

Version: July 2020 1
Stability data Pneumococcal Monovalent Bulk Conjugate Yuxi Walvax Biotechnology Co., Ltd.

1.2 Batch No.:RSP1C201707002
Test Free Free
Polysacc
items Protein PS/protein Molecul Residual polysac carrier Bacterial ADH
haride Identity Sterility
and content ratio ar size EDAC charide protein endotoxin Content
content
standard content content
s NMT
Positive response Shall meet
≥40ug/m ≥70ug/m ≤5.0μmol/ 0.75EU/μg
0.20-1.80 ≥80% ≤25.0% ≤5.0% to antiserum type requireme ■0.15ug/ml
Storage I I L polysaccha
1 nts.
time ride
Not
0th <0.75EU/
258.0 170.5 1.51 99.4 2.9 15.8 detecte Complied Complied 0.06
month μg
d
1th
235.5 166.1 1.42 99.1 — 15.6 0.99 — — — —
month
2nd
228.5 184.5 1.24 99.1 — 15.6 3.6 — — — —
month
3rd
260.4 160.7 1.62 98.3 — 16 0.1 — — — —
month
6th
265.5 160.4 1.66 99.6 — 20 4.3 — — — —
month
9th
265.0 177.3 1.49 99.4 — 16 1.5 — — — —
month
12th <0.75EU/
270.0 174.7 1.55 94.0 2.040 16 1.7 Complied Complied 0.12
month μg

1.3 Batch No.:RSP1C201709003
Test Polysacc Protein PS/protein Molecul Residual Free Free Bacterial ADH
Identity Sterility
items haride content ratio ar size EDAC polysac carrier endotoxin Content

Version: July 2020 2
Stability data Pneumococcal Monovalent Bulk Conjugate Yuxi Walvax Biotechnology Co., Ltd.

and content charide protein
standard content content
s NMT
Positive response Shall meet
≥40ug/m ≥70ug/m ≤5.0μmol/ 0.75EU/μg
0.20-1.80 ≥80% ≤25.0% ≤5.0% to antiserum type requireme ■0.15ug/ml
Storage I I L polysaccha
1 nts.
time ride
0th Not <0.75EU/
423.8 462.2 0.92 96.2 14.0 1.3 Complied Complied 0.086
month detected μg
1th
445.0 486.8 0.91 98.1 — 13.8 3.3 — — — —
month
2nd
417.0 497.8 0.84 99.0 — 14 2.4 — — — —
month
3rd
453.8 507.6 0.89 97.7 — 14 3.4 — — — —
month
6th
422.5 433.6 0.97 97.8 — 16 1.5 — — — —
month
9th
427.8 441.2 0.97 97.7 — 14 3.0 — — — —
month
12th <0.75EU/
388.8 479.5 0.81 90.2 2.151 14 2.2 Complied Complied 0.11
month μg

2. Stability data of serotype 3 Pneumococcal Bulk Conjugate (Storage condition: 2-8■)
2.1 Batch No.:RSP3C201706001
Test Free Free
Polysacc
items Protein PS/protein Molecul Residual polysac carrier Bacterial ADH
haride Identity Sterility
and content ratio ar size EDAC charide protein endotoxin Content
content
standard content content

Version: July 2020 3
Stability data Pneumococcal Monovalent Bulk Conjugate Yuxi Walvax Biotechnology Co., Ltd.

s NMT
Positive response Shall meet
≥20ug/m ≥70ug/m ≤5.0μmol/ 0.75EU/μg
0.10-0.90 ≥80% ≤20.0% ≤5.0% to antiserum type requireme ■0.25ug/ml
Storage I I L polysaccha
3 nts.
time ride
0th <0.75EU/
61.6 141.6 0.44 98.5 1.1 15.7 3.8 Complied Complied 0.016
month μg
1th
55.0 153.7 0.36 86.8 — 17.4 2.1 — — — —
month
2nd
67.0 150.3 0.45 95.3 — 6.4 2.4 — — — —
month
3rd
64.5 166.7 0.39 95.3 — 15.7 0.7 — — — —
month
6th
68.1 152.0 0.45 99.6 — 15 0.4 — — — —
month
9th
56.8 154.2 0.37 98.3 — 15 3.9 — — — —
month
12th <0.75EU/
58.5 166.4 0.35 98.5 2.680 15 3.6 Complied Complied 0.07
month μg

2.2 Batch No.:RSP3C201707002
Test Free Free
Polysacc
items Protein PS/protein Molecul Residual polysac carrier Bacterial ADH
haride Identity Sterility
and content ratio ar size EDAC charide protein endotoxin Content
content
standard content content

Version: July 2020 4
Stability data Pneumococcal Monovalent Bulk Conjugate Yuxi Walvax Biotechnology Co., Ltd.

s NMT
Positive response Shall meet
≥20ug/m ≥70ug/m ≤5.0μmol/ 0.75EU/μg
0.10-0.90 ≥80% ≤20.0% ≤5.0% to antiserum type requireme ■0.25ug/ml
Storage I I L polysaccha
3 nts.

time ride
0th <0.75EU/
77.2 355.8 0.22 97.8 4.4 6.2 0.48 Complied Complied 0.160
month µg
1th
72.6 336.6 0.22 98.3 — 6.3 0.57 — — — —
month
2nd
85.0 333.6 0.25 99.6 — 6.0 1.69 — — — —
month
3rd
73.8 328.5 0.22 100.0 — 6 0.8 — — — —
month
6th
75.9 329.2 0.23 98.9 — 8 0.8 — — — —
month
9th
71.8 329.2 0.22 97.9 — 7 0.9 — — — —
month
12th <0.75EU/
77.4 321.1 0.24 97.7 1.964 6 0.9 Complied Complied 0.08
month µg

2.3 Batch No.:RSP3C201708003
Test Free Free
Polysacc
items Protein PS/protein Molecul Residual polysac carrier Bacterial ADH
haride Identity Sterility
and content ratio ar size EDAC charide protein endotoxin Content
content
standard content content
s NMT
Positive response Shall meet
≥20ug/m ≥70ug/m ≤5.0µmol/ 0.75EU/µg
Storage 0.10-0.90 ≥80% ≤20.0% ≤5.0% to antiserum type requireme ■0.25ug/ml
I I L polysaccha
time 3 nts.
ride

Version: July 2020 5
Stability data Pneumococcal Monovalent Bulk Conjugate Yuxi Walvax Biotechnology Co., Ltd.

0th <0.75EU/
63.6 285.3 0.22 97.6 3.8 12.2 1.20 Complied Complied 0.128
month µg
1th
60.3 265.2 0.23 95.8 — 6.1 1.7 — — — —
month
2nd
55.2 275.8 0.20 98.5 — 6 1.6 — — — —
month
3rd
70.0 343.3 0.20 100.0 — 6 0.0 — — — —
month
6th
63.0 268.4 0.23 98.0 — 6 0.1 — — — —
month
9th
66.8 304.6 0.22 97.6 — 5 1.5 — — — —
month
12th <0.75EU/
58.4 283.7 0.21 99.0 2.398 5 0.9 Complied Complied Not detected
month µg

3. Stability data of serotype 4 Pneumococcal Bulk Conjugate (Storage condition: 2-8■)

3.1 Batch No.:RSP4C201706001
Test Free Free
Polysacc
items Protein PS/protein Molecul Residual polysac carrier Bacterial ADH
haride Identity Sterility
and content ratio ar size EDAC charide protein endotoxin Content
content
standard content content
s NMT
Positive response Shall meet
≥40ug/m ≥70ug/m ≤5.0µmol/ 0.75EU/µg
0.30-0.90 ≥80% ≤25.0% ≤5.0% to antiserum type requireme ■0.15ug/ml
Storage I I L polysaccha
4 nts.
time ride
0th <0.75EU/
154.3 254.1 0.61 98.9 0.9 19.6 4.7 Complied Complied 0.082
month µg
1th 153.9 264.6 0.58 95.4 — 19.6 2.89 — — — —

Version: July 2020 6
Stability data Pneumococcal Monovalent Bulk Conjugate Yuxi Walvax Biotechnology Co., Ltd.

month
2nd
146.0 271.5 0.54 97.3 — 17.1 0.70 — — — —
month

3rd
158.0 250.4 0.63 95.2 — 20.5 2.1 — — — —
month
6th
172.0 301.8 0.57 97.5 — 19 4.4 — — — —
month
9th
150.0 267.0 0.56 97.7 — 18 1.4 — — — —
month
12th <0.75EU/
148.0 258.2 0.57 98.3 1.485 20 1.7 Complied Complied 0.07
month µg

3.2 Batch No.:RSP4C201707002
Test Free Free
Polysacc
items Protein PS/protein Molecul Residual polysac carrier Bacterial ADH
haride Identity Sterility
and content ratio ar size EDAC charide protein endotoxin Content
content
standard content content
s NMT
Positive response Shall meet
≥40ug/m ≥70ug/m ≤5.0µmol/ 0.75EU/µg
0.30-0.90 ≥80% ≤25.0% ≤5.0% to antiserum type requireme ■0.15ug/ml
Storage I I L polysaccha
4 nts.
time ride
0th <0.75EU/
196.0 276.2 0.71 98.0 3.6 23.6 2.2 Complied Complied 0.110
month µg
1th Not
170.0 283.8 0.599 96.9 19.7 2.1 — — — —
month detected
2nd
197.4 286.6 0.689 97.2 0.22 19.4 1.62 — — — —
month
3rd
174.5 276.7 0.63 100.0 0.3640 19 0.6 — — — —
month
Version: July 2020 7
Stability data Pneumococcal Monovalent Bulk Conjugate Yuxi Walvax Biotechnology Co., Ltd.

6th
192.6 277.8 0.69 97.0 — 24 1.2 — — — —
month
9th
204.8 271.7 0.75 97.9 — 19 0.7 — — — —
month
12th <0.75EU/
212.2 274.4 0.77 96.3 1.532 19 1.7 Complied Complied 0.09
month µg

3.3 Batch No.:RSP4C201709003
Test Free Free
Polysacc
items Protein PS/protein Molecul Residual polysac carrier Bacterial ADH
haride Identity Sterility
and content ratio ar size EDAC charide protein endotoxin Content
content
standard content content
s NMT
Positive response Shall meet
≥40ug/m ≥70ug/m ≤5.0µmol/ 0.75EU/µg
0.30-0.90 ≥80% ≤25.0% ≤5.0% to antiserum type requireme ■0.15ug/ml
Storage I I L polysaccha
4 nts.
time ride
0th <0.75EU/
190.8 288.0 0.66 99.4 3.5 14.3 0.9 Complied Complied 0.082
month µg
1th
161.5 296.2 0.55 97.1 2.6455 14.2 0.52 — — — —
month
2nd
169.0 268.5 0.63 98.3 3.084 14 0.6 — — — —
month
3rd
169.5 261.9 0.65 98.4 — 13 1.1 — — — —
month
6th
179.4 259.5 0.69 98.1 — 12 1.1 — — — —
month
9th
187.0 247.0 0.76 98.4 — 14 1.1 — — — —
month
12th 179.4 251.6 0.71 96.3 2.052 11 1.1 Complied <0.75EU/ Complied 0.08

Version: July 2020 8
Stability data Pneumococcal Monovalent Bulk Conjugate Yuxi Walvax Biotechnology Co., Ltd.

month µg

4. Stability data of serotype 5 Pneumococcal Bulk Conjugate (Storage condition: 2-8■)

4.1 Batch No.:RSP5C201705001

Test Free Free

Polysacc

items Protein PS/protein Molecul Residual polysac carrier Bacterial ADH

haride Identity Sterility

and content ratio ar size EDAC charide protein endotoxin Content

content

standard content content

s NMT

Positive response Shall meet

≥40ug/m ≥70ug/m ≤5.0μmol/ 0.75EU/μg

0.10-0.60 ≥80% ≤20.0% ≤5.0% to antiserum type requireme ■0.25ug/ml

Storage I I L polysaccha

5 nts.

time ride

0th <0.75EU/

91.1 431.0 0.21 94.3 0.7 18.0 3.2 Complied Complied 0.139

month μg

1th

101.8 400.7 0.25 89.0 — 12 2.7 — — — —

month

2nd

82.2 450.9 0.18 97.0 — 17.5 3.9 — — — —

month

3rd

90.3 404.9 0.22 99.1 — 12.6 2.01 — — — —

month

6th

92.6 424.5 0.22 97.8 — 12 2.4 — — — —

month

9th

84.2 436.9 0.19 97.8 — 15 2.1 — — — —

month

12th <0.75EU/

92.4 418.7 0.22 99.3 1.400 13 2.0 Complied Complied 0.10

month μg

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Stability data Pneumococcal Monovalent Bulk Conjugate Yuxi Walvax Biotechnology Co., Ltd.

4.2 Batch No.:RSP5C201707002

Test Free Free

Polysacc

items Protein PS/protein Molecul Residual polysac carrier Bacterial ADH

haride Identity Sterility

and content ratio ar size EDAC charide protein endotoxin Content

content

standard content content

s NMT

Positive response Shall meet

≥40ug/m ≥70ug/m ≤5.0μmol/ 0.75EU/μg

0.10-0.60 ≥80% ≤20.0% ≤5.0% to antiserum type requireme ■0.25ug/ml

Storage I I L polysaccha

5 nts.

time ride

Not

0th <0.75EU/

57.8 172.1 0.34 98.9 0.8 17.4 detecte Complied Complied 0.102

month μg

d

1th

59.6 204.8 0.291 95.2 — 12.3 1.98 — — — —

month

2nd

63.6 180.9 0.352 96.0 — 17.0 3.4 — — — —

month

Not

3rd

70.6 180.6 0.39 97.9 — 17 detecte — — — —

month

d

Not

6th

64.8 178.3 0.36 98.3 — 18 detecte — — — —

month

d

9th

59.8 183.6 0.33 97.7 — 17 0.0 — — — —

month

12th <0.75EU/

54.8 187.3 0.29 97.7 0.7090 18 2.1 Complied Complied 0.08

month μg

Version: July 2020 10

Stability data Pneumococcal Monovalent Bulk Conjugate Yuxi Walvax Biotechnology Co., Ltd.

4.3 Batch No.:RSP5C201708003

Test Free Free

Polysacc

items Protein PS/protein Molecul Residual polysac carrier Bacterial ADH

haride Identity Sterility
and content ratio ar size EDAC charide protein endotoxin Content
content
standard content content
s NMT
Positive response Shall meet
≥40ug/m ≥70ug/m ≤5.0μmol/ 0.75EU/μg
0.10-0.60 ≥80% ≤20.0% ≤5.0% to antiserum type requireme ■0.25ug/ml
Storage I I L polysaccha
5 nts.
time ride
0th <0.75EU/
99.8 224.6 0.44 96.8 3.7 15.4 0.5 Complied Complied 0.102
month μg
1th
94.5 173.4 0.54 98.1 — 15.0 2.2 — — — —
month
2nd
94.0 267.4 0.35 96.7 — 16 0.0 — — — —
month
3rd
96.0 178.2 0.54 97.3 — 15 0.8 — — — —
month
6th
98.7 234.1 0.42 98.4 — 16 1.3 — — — —
month
9th
102.6 229.0 0.45 98.2 — 15 2.1 — — — —
month
12th <0.75EU/
96.4 203.4 0.47 94.6 1.762 11 2.4 Complied Complied 0.12
month μg

5. Stability data of serotype 6A Pneumococcal Bulk Conjugate (Storage condition: 2-8■)
5.1 Batch No.:RSP6AC201705001

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Stability data Pneumococcal Monovalent Bulk Conjugate Yuxi Walvax Biotechnology Co., Ltd.

Test Free Free
Polysacc
items Protein PS/protein Molecul Residual polysac carrier Bacterial ADH
haride Identity Sterility
and content ratio ar size EDAC charide protein endotoxin Content
content
standard content content
s NMT
Positive response Shall meet
≥40ug/m ≥70ug/m ≤5.0μmol/ 0.75EU/μg
0.10-0.50 ≥80% ≤20.0% ≤5.0% to antiserum type requireme ■1.20ug/ml
Storage I I L polysaccha
6A nts.
time ride
0th <0.75EU/
139.1 547.4 0.25 95.6 3.6 13.7 3.6 Complied Complied 0.163
month μg
1th
145.6 540.0 0.27 97.8 — 16.1 3.3 — — — —
month
Not
2nd
172.6 516.1 0.33 99.5 — 16.5 detecte — — — —
month
d
3rd
137.5 486.6 0.28 98.9 — 8.2 2.0 — — — —
month
6th
155.6 564.9 0.28 99.0 — 10 0.7 — — — —
month
9th
152.5 523.0 0.29 98.1 — 10 2.4 — — — —
month
12th <0.75EU/
139.5 500.5 0.28 98.1 2.279 10 1.8 Complied Complied 0.02
month μg

5.2 Batch No.:RSP6AC201707002
Test Free Free
Polysacc
items Protein PS/protein Molecul Residual polysac carrier Bacterial ADH
haride Identity Sterility
and content ratio ar size EDAC charide protein endotoxin Content
content
standard content content

Version: July 2020 12
Stability data Pneumococcal Monovalent Bulk Conjugate Yuxi Walvax Biotechnology Co., Ltd.

s NMT
Positive response Shall meet
≥40ug/m ≥70ug/m ≤5.0μmol/ 0.75EU/μg

0.10-0.50 ≥80% ≤20.0% ≤5.0% to antiserum type requireme ■1.20ug/ml
Storage I I L polysaccha
6A nts.
time ride
0th <0.75EU/
154.6 493.0 0.31 88.1 2.6 19.4 0.9 Complied Complied 0.081
month µg
1th
148.5 479.9 0.31 96.6 3.3315 18.4 1.41 — — — —
month
2nd
149.0 555.3 0.268 97.9 2.519 19.0 1.19 — — — —
month
3rd
158.0 476.4 0.33 100.0 1.798 18 1.4 — — — —
month
6th
149.0 563.5 0.26 100.0 — 20 2.0 — — — —
month
9th
155.7 512.6 0.30 98.9 — 19 0.7 — — — —
month
12th <0.75EU/
170.0 517.0 0.33 98.9 1.968 19 0.7 Complied Complied 0.90
month µg
5.3 Batch No.:RSP6AC201708003
Test Free Free
Polysacc
items Protein PS/protein Molecul Residual polysac carrier Bacterial ADH
haride Identity Sterility
and content ratio ar size EDAC charide protein endotoxin Content
content
standard content content
s NMT
Positive response Shall meet
≥40ug/m ≥70ug/m ≤5.0µmol/ 0.75EU/µg
0.10-0.50 ≥80% ≤20.0% ≤5.0% to antiserum type requireme ■1.20ug/ml
Storage I I L polysaccha
6A nts.
time ride
0th <0.75EU/
86.8 191.5 0.45 97.7 2.3 12.8 0.5 Complied Complied 0.227
month µg

Version: July 2020 13
Stability data Pneumococcal Monovalent Bulk Conjugate Yuxi Walvax Biotechnology Co., Ltd.

1th
90.5 251.8 0.36 98.2 2.324 13.0 0.57 — — — —
month
2nd
88.2 200.0 0.44 98.1 2.154 13 0.4 — — — —
month
3rd
87.3 205.9 0.42 98.7 — 13 1.7 — — — —
month
6th
86.7 190.6 0.45 98.9 — 18 1.8 — — — —
month
9th
84.0 196.5 0.43 98.2 — 13 1.1 — — — —
month
12th <0.75EU/
80.8 189.4 0.43 98.1 1.956 13 1.7 Complied Complied 0.11
month µg

6. Stability data of serotype 6B Pneumococcal Bulk Conjugate (Storage condition: 2-8■)
6.1 Batch No.:RSP6BC201705001
Test Free Free
Polysacc
items Protein PS/protein Molecul Residual polysac carrier Bacterial ADH
haride Identity Sterility
and content ratio ar size EDAC charide protein endotoxin Content
content
standard content content
s NMT
Positive response Shall meet
≥40ug/m ≥70ug/m ≤5.0µmol/ 0.75EU/µg
0.30-0.60 ≥80% ≤25.0% ≤5.0% to antiserum type requireme ■0.25ug/ml
Storage I I L polysaccha
6B nts.
time ride
0th <0.75EU/
152.9 327.4 0.47 98.6 2.8 21.6 4.0 Complied Complied 0.012
month µg
1th
141.8 309.2 0.46 96.8 — 19.4 2.9 — — — —
month
2nd 139.2 275.8 0.50 98.0 — 22.4 0.77 — — — —

Version: July 2020 14
Stability data Pneumococcal Monovalent Bulk Conjugate Yuxi Walvax Biotechnology Co., Ltd.

month
3rd
150.8 264.1 0.57 98.7 — 16.8 2.4 — — — —
month
6th
164.0 320.1 0.51 97.8 — 18 1.2 — — — —
month
9th
144.8 314.4 0.46 98.4 — 17 1.5 — — — —
month
12th <0.75EU/
146.0 313.0 0.47 99.3 2.378 18 1.3 Complied Complied 0.09
month µg

6.2 Batch No.:RSP6BC201707002
Test Free Free
Polysacc
items Protein PS/protein Molecul Residual polysac carrier Bacterial ADH
haride Identity Sterility
and content ratio ar size EDAC charide protein endotoxin Content
content
standard content content
s NMT
Positive response Shall meet
≥40ug/m ≥70ug/m ≤5.0µmol/ 0.75EU/µg
0.30-0.60 ≥80% ≤25.0% ≤5.0% to antiserum type requireme ■0.25ug/ml
Storage I I L polysaccha
6B nts.
time ride
0th <0.75EU/
170.2 349.9 0.49 93.5 0.5 21.1 0.9 Complied Complied 0.094
month µg
1th
167.8 334.4 0.50 92.9 — 21.1 2.50 — — — —
month
2nd
166.5 394.0 0.423 95.6 — 21.2 1.34 — — — —
month
3rd
160.0 412.1 0.39 89.7 — 18 1.0 — — — —
month
6th
161.6 404.3 0.40 95.5 — 20 0.5 — — — —
month
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Stability data Pneumococcal Monovalent Bulk Conjugate Yuxi Walvax Biotechnology Co., Ltd.

9th
156.0 348.3 0.45 95.5 — 21 2.8 — — — —
month
12th <0.75EU/
179.2 372.4 0.48 98.4 1.943 21 0.9 Complied Complied 0.06
month µg

6.3 Batch No.:RSP6BC201708003
Test Free Free
Polysacc
items Protein PS/protein Molecul Residual polysac carrier Bacterial ADH
haride Identity Sterility
and content ratio ar size EDAC charide protein endotoxin Content
content
standard content content
s NMT
Positive response Shall meet
≥40ug/m ≥70ug/m ≤5.0µmol/ 0.75EU/µg
0.30-0.60 ≥80% ≤25.0% ≤5.0% to antiserum type requireme ■0.25ug/ml
Storage I I L polysaccha
6B nts.
time ride
0th <0.75EU/
124.8 296.5 0.42 96.1 0.1 9.2 2.1 Complied Complied 0.142
month µg
1th
119.7 264.5 0.45 97.5 — 12.8 2.33 — — — —
month
2nd
123.0 322.4 0.38 97.9 — 10 3.3 — — — —
month
3rd
112.5 312.4 0.36 99.7 — 9 2.8 — — — —
month
6th
117.6 312.0 0.38 98.3 — 13 1.7 — — — —
month
9th
119.6 288.5 0.41 91.8 — 9 1.2 — — — —
month
12th <0.75EU/
130.5 288.5 0.45 97.6 1.557 9 2.1 Complied Complied 0.12
month µg

7. Stability data of serotype 7F Pneumococcal Bulk Conjugate (Storage condition: 2-8■)

7.1 Batch No.:RSP7FC201705001
Test Free Free
Polysacc
items Protein PS/protein Molecul Residual polysac carrier Bacterial ADH
haride Identity Sterility
and content ratio ar size EDAC charide protein endotoxin Content
content
standard content content
s NMT
Positive response Shall meet
≥40ug/m ≥70ug/m ≤5.0μmol/ 0.75EU/μg
0.20-0.90 ≥80% ≤20.0% ≤5.0% to antiserum type requireme ■0.20ug/ml
Storage I I L polysaccha
7F nts.
time ride
0th <0.75EU/
284.1 643.5 0.44 97.2 3.2 18.4 4.0 Complied Complied 0.066
month μg
1th
262.5 620.3 0.42 99.2 — 14.0 0.72 — — — —
month
2nd
249.8 557.6 0.45 94.2 — 14.2 1.03 — — — —
month
3rd
280.0 555.1 0.50 92.0 — 18.5 1.9 — — — —
month
6th
282.0 524.7 0.54 98.0 — 19 2.0 — — — —
month
9th
256.5 588.2 0.44 97.3 — 19 1.0 — — — —
month
12th <0.75EU/
271.0 445.7 0.61 97.8 3.069 18 2.2 Complied Complied 0.09
month μg

7.2 Batch No.:RSP7FC201706002
Test Polysacc Free Free
Protein PS/protein Molecul Residual Bacterial ADH
items haride polysac carrier Identity Sterility
content ratio ar size EDAC endotoxin Content
and content charide protein

standard content content
s
NMT
Positive response Shall meet
Storage ≥40ug/m ≥70ug/m ≤5.0μmol/ 0.75EU/μg
0.20-0.90 ≥80% ≤20.0% ≤5.0% to antiserum type requireme ■0.20ug/ml
time I I L polysaccha
7F nts.
ride
0th <0.75EU/
303.0 362.8 0.84 96.6 1.2 18.5 0.3 Complied Complied 0.121
month μg
1th
280.0 416.0 0.67 98.6 — 16.5 1.55 — — — —
month
2nd
310.0 452.8 0.685 99.1 — 15.4 1.44 — — — —
month
3rd
305.0 412.1 0.74 97.1 — 16.5 1.6 — — — —
month
6th
288.8 354.7 0.81 99.0 — 16 1.0 — — — —
month
9th
319.0 370.4 0.86 95.0 — 17 1.0 — — — —
month
12th <0.75EU/
304.8 346.3 0.88 95.2 4.013 17 1.2 Complied Complied 0.06
month μg

7.3 Batch No.:RSP7FC201708003
Test Free Free
Polysacc
items Protein PS/protein Molecul Residual polysac carrier Bacterial ADH
haride Identity Sterility
and content ratio ar size EDAC charide protein endotoxin Content
content
standard content content
s NMT
Positive response Shall meet
≥40ug/m ≥70ug/m ≤5.0μmol/ 0.75EU/μg
0.20-0.90 ≥80% ≤20.0% ≤5.0% to antiserum type requireme ■0.20ug/ml

Storage I I L polysaccha
7F nts.
time ride

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Stability data Pneumococcal Monovalent Bulk Conjugate Yuxi Walvax Biotechnology Co., Ltd.

0th <0.75EU/
310.8 494.3 0.63 96.6 3.7 18.9 1.3 Complied Complied 0.088
month µg
1th
272.0 397.0 0.69 96.2 — 18.4 0.5 — — — —
month
2nd
318.6 389.8 0.82 97.2 — 19 3.0 — — — —
month
3rd
346.0 435.6 0.79 98.3 — 19 3.0 — — — —
month
6th
314.0 398.0 0.79 99.9 — 19 3.1 — — — —
month
9th
313.6 389.1 0.81 99.8 — 18 3.1 — — — —
month
12th <0.75EU/
312.8 420.0 0.74 95.0 2.422 18 1.3 Complied Complied 0.11
month µg

8. Stability data of serotype 9V Pneumococcal Bulk Conjugate (Storage condition: 2-8■)

8.1 Batch No.:RSP9VC201705001

Test Free Free

Polysacc

items Protein PS/protein Molecul Residual polysac carrier Bacterial ADH

haride Identity Sterility

and content ratio ar size EDAC charide protein endotoxin Content

content

standard content content

s NMT

Positive response Shall meet

≥40ug/m ≥70ug/m ≤5.0µmol/ 0.75EU/µg

0.10-0.35 ≥80% ≤25.0% ≤5.0% to antiserum type requireme ■0.70ug/ml

Storage I I L polysaccha

9V nts.

time ride

0th <0.75EU/

148.6 513.5 0.29 100.0 2.5 6.8 4.3 Complied Complied 0.176

month µg

1th 129.5 528.8 0.24 96.5 — 0.8 3.20 — — — —

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Stability data Pneumococcal Monovalent Bulk Conjugate Yuxi Walvax Biotechnology Co., Ltd.

month

2nd

148.4 578.1 0.26 81.1 — 6.1 0.3 — — — —

month

3rd

143.2 522.2 0.27 96.5 — 0.8 0.89 — — — —

month

6th

130.5 554.1 0.24 98.3 — 1 1.2 — — — —

month

9th

150.8 528.4 0.29 97.3 — 3 1.1 — — — —

month

12th <0.75EU/

146.4 565.1 0.26 97.4 2.814 1 1.4 Complied Complied 0.02

month µg

8.2 Batch No.:RSP9VC201707002

Test Free Free

Polysacc

items Protein PS/protein Molecul Residual polysac carrier Bacterial ADH

haride Identity Sterility

and content ratio ar size EDAC charide protein endotoxin Content

content

standard content content

s NMT

Positive response Shall meet

≥40ug/m ≥70ug/m ≤5.0µmol/ 0.75EU/µg

0.10-0.35 ≥80% ≤25.0% ≤5.0% to antiserum type requireme ■0.70ug/ml

Storage I I L polysaccha

9V nts.

time ride

0th <0.75EU/

159.3 614.1 0.26 99.7 2.2 12.0 1.6 Complied Complied 0.153

month µg

1th

159.5 518.7 0.31 98.8 — 11.8 1.41 — — — —

month

2nd

156.4 732.6 0.213 97.7 — 11.8 0.1 — — — —
month
3rd
156.8 709.3 0.221 95.7 — 12.9 2.22 — — — —
month
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Stability data Pneumococcal Monovalent Bulk Conjugate Yuxi Walvax Biotechnology Co., Ltd.

6th
148.0 621.8 0.24 99.2 — 12 2.5 — — — —
month
9th
153.6 577.0 0.27 95.7 — 12 0.6 — — — —
month
12th <0.75EU/
159.5 637.1 0.25 95.4 1.609 12 0.4 Complied Complied 0.32
month µg
8.3 Batch No.:RSP9VC201709003
Test Free Free
Polysacc
items Protein PS/protein Molecul Residual polysac carrier Bacterial ADH
haride Identity Sterility
and content ratio ar size EDAC charide protein endotoxin Content
content
standard content content
s NMT
Positive response Shall meet
≥40ug/m ≥70ug/m ≤5.0µmol/ 0.75EU/µg
0.10-0.35 ≥80% ≤25.0% ≤5.0% to antiserum type requireme ■0.70ug/ml
Storage I I L polysaccha
9V nts.
time ride
0th <0.75EU/
157.5 722.3 0.22 98.8 4.1 11.8 0.7 Complied Complied 0.203
month µg
1th
163.4 588.8 0.28 96.6 — 13.7 0.16 — — — —
month
2nd
155.6 562.0 0.28 98.1 — 12 0.0 — — — —
month
3rd
162.0 630.4 0.26 96.5 — 12 0.3 — — — —
month
6th
144.5 648.1 0.22 97.3 — 14 0.8 — — — —
month
9th
149.4 544.8 0.27 98.4 — 12 0.5 — — — —
month
12th <0.75EU/
145.5 496.1 0.29 96.3 2.190 12 0.6 Complied Complied 0.25
month µg

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Stability data Pneumococcal Monovalent Bulk Conjugate Yuxi Walvax Biotechnology Co., Ltd.

9. Stability data of serotype 14 Pneumococcal Bulk Conjugate (Storage condition: 2-8■)

9.1 Batch No.:RSP14C201706001
Test Free Free
Polysacc
items Protein PS/protein Molecul Residual polysac carrier Bacterial ADH
haride Identity Sterility
and content ratio ar size EDAC charide protein endotoxin Content
content
standard content content
s NMT
Positive response Shall meet
≥40ug/m ≥70ug/m ≤5.0µmol/ 0.75EU/µg
0.20-0.60 ≥80% ≤25.0% ≤5.0% to antiserum type requireme ■1.15ug/ml
Storage I I L polysaccha
14 nts.
time ride
0th <0.75EU/
143.2 480.7 0.30 95.9 2.4 14.6 3.1 Complied Complied 0.016
month µg
1th
139.6 479.4 0.29 99.3 — 17.3 3.12 — — — —
month
2nd
142.8 411.4 0.35 98.3 — 13.5 1.94 — — — —
month
3rd
133.0 467.2 0.28 98.6 — 14.5 1.44 — — — —
month
6th
138.0 451.5 0.31 93.3 — 14 0.7 — — — —
month
9th
138.0 397.2 0.35 96.3 — 14 0.5 — — — —
month
12th <0.75EU/

146.2 464.8 0.31 94.1 4.493 11 0.9 Complied Complied 0.08
month µg

9.2 Batch No.:RSP14C201707002

Version: July 2020 22
Stability data Pneumococcal Monovalent Bulk Conjugate Yuxi Walvax Biotechnology Co., Ltd.

Test Free Free
Polysacc
items Protein PS/protein Molecu Residual polysac carrier Bacterial ADH
haride Identity Sterility
and content ratio ar size EDAC charide protein endotoxin Content
content
standard content content
s NMT
Positive response Shall meet
≥40ug/m ≥70ug/m ≤5.0µmol/ 0.75EU/µg
0.20-0.60 ≥80% ≤25.0% ≤5.0% to antiserum type requireme ■1.15ug/ml
Storage I I L polysaccha
14 nts.
time ride
0th <0.75EU/
134.9 340.4 0.4 99.0 3.8 12.6 1.0 Complied Complied 0.120
month µg
1th
129.9 290.4 0.45 98.7 — 12.6 2.12 — — — —
month
2nd
126.6 331.8 0.382 99.7 — 14.4 2.43 — — — —
month
3rd
142.0 304.9 0.47 98.9 — 14 0.4 — — — —
month
6th
141.2 333.3 0.42 95.7 — 15 0.9 — — — —
month
9th
134.4 344.6 0.39 95.8 — 14 0.6 — — — —
month
12th <0.75EU/
121.5 343.3 0.35 97.8 2.168 14 0.5 Complied Complied 0.08
month µg

9.3 Batch No.:RSP14C201709003
Test Free Free
Polysacc
items Protein PS/protein Molecu Residual polysac carrier Bacterial ADH
haride Identity Sterility
and content ratio ar size EDAC charide protein endotoxin Content
content
standard content content

Version: July 2020 23
Stability data Pneumococcal Monovalent Bulk Conjugate Yuxi Walvax Biotechnology Co., Ltd.

s NMT
Positive response Shall meet
≥40ug/m ≥70ug/m ≤5.0µmol/ 0.75EU/µg
0.20-0.60 ≥80% ≤25.0% ≤5.0% to antiserum type requireme ■1.15ug/ml
Storage I I L polysaccha
14 nts.
time ride
0th <0.75EU/
125.0 305.8 0.41 97.2 1.0 23.9 1.0 Complied Complied 0.114
month µg
1th
127.0 332.2 0.38 98.5 — 23.5 0.84 — — — —
month
2nd
114.8 333.9 0.34 98.9 — 23 0.9 — — — —
month
3rd
123.6 368.7 0.34 98.4 — 24 1.7 — — — —
month
6th
118.4 349.2 0.34 99.4 — 24 1.0 — — — —
month
9th
126.8 343.3 0.37 93.9 — 24 1.1 — — — —
month
12th <0.75EU/
102.8 347.1 0.30 96.6 1.948 23 1.0 Complied Complied 0.10
month µg

10. Stability data of serotype 18C Pneumococcal Bulk Conjugate (Storage condition: 2-8■)
10.1 Batch No.:RSP18CC201706001
Test Free Free
Polysacc
items Protein PS/protein Molecu Residual polysac carrier Bacterial ADH
haride Identity Sterility
and content ratio lar size EDAC charide protein endotoxin Content
content

standard content content

Version: July 2020 24
Stability data Pneumococcal Monovalent Bulk Conjugate Yuxi Walvax Biotechnology Co., Ltd.

s NMT
Positive response Shall meet
≥40ug/m ≥70ug/m 0.75EU/μg
0.40-0.80 ≥80% ≤5μmol/L ≤30.0% ≤5.0% to antiserum type requireme ■0.20ug/ml
Storage I I polysaccha
18C nts.
time ride
0th <0.75EU/
106.6 195.8 0.54 97.7 0.9 7.7 2.7 Complied Complied 0.097
month μg
1th
102.6 203.2 0.50 100.0 — 12.0 0.9 — — — —
month
2nd
113.0 187.6 0.60 100.0 — 7.6 2.1 — — — —
month
3rd
97.8 198.0 0.49 98.9 — 7.6 1.47 — — — —
month
6th
96.0 209.2 0.46 98.9 — 8 1.5 — — — —
month
9th
99.6 182.0 0.55 96.6 — 7 1.1 — — — —
month
12th <0.75EU/
106.0 202.6 0.52 98.0 2.650 8 0.4 Complied Complied 0.03
month μg

10.2 Batch No.:RSP18CC201707002
Test Free Free
Polysacc
items Protein PS/protein Molecu Residual polysac carrier Bacterial ADH
haride Identity Sterility
and content ratio lar size EDAC charide protein endotoxin Content
content
standard content content
s NMT
Positive response Shall meet
≥40ug/m ≥70ug/m 0.75EU/μg
Storage 0.40-0.80 ≥80% ≤5μmol/L ≤30.0% ≤5.0% to antiserum type requireme ■0.20ug/ml
I I polysaccha
time 18C nts.
ride

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Stability data Pneumococcal Monovalent Bulk Conjugate Yuxi Walvax Biotechnology Co., Ltd.

0th <0.75EU/
153.3 263.1 0.58 86.9 1.5 26.2 0.9 Complied Complied 0.108
month μg
1th
148.8 279.8 0.53 97.3 — 26.1 1.97 — — — —
month
Not
2nd
147.7 264.7 0.558 97.8 — 26.1 detecte — — — —
month
d
3rd
156.8 219.5 0.71 93.7 — 26 2.8 — — — —
month
6th
155.2 262.8 0.59 98.7 — 26 1.7 — — — —
month
9th
163.8 270.9 0.60 97.5 — 26 1.1 — — — —
month
12th <0.75EU/
155.2 269.6 0.58 94.8 1.480 26 1.5 Complied Complied 0.11
month μg

10.3 Batch No.:RSP18CC201708003
Test Free Free
Polysacc
items Protein PS/protein Molecu Residual polysac carrier Bacterial ADH
haride Identity Sterility
and content ratio lar size EDAC charide protein endotoxin Content
content
standard content content
s NMT
Positive response Shall meet
≥40ug/m ≥70ug/m 0.75EU/μg
0.40-0.80 ≥80% ≤5μmol/L ≤30.0% ≤5.0% to antiserum type requireme ■0.20ug/ml
Storage I I polysaccha
18C nts.
time ride
0th <0.75EU/

144.4 199.8 0.72 96.9 0.7 22.6 0.7 Complied Complied 0.015
month µg
1th
146.3 227.7 0.64 98.4 — 22.8 1.31 — — — —
month
2nd 149.4 223.0 0.67 97.8 — 23 Not — — — —

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Stability data Pneumococcal Monovalent Bulk Conjugate Yuxi Walvax Biotechnology Co., Ltd.

month detecte
d
3rd
125.0 252.2 0.50 99.3 — 22 1.4 — — — —
month
6th
143.0 201.5 0.71 98.3 — 23 2.5 — — — —
month
9th
148.0 212.0 0.70 98.4 — 22 2.0 — — — —
month
12th <0.75EU/
130.0 229.3 0.57 98.7 1.897 23 1.2 Complied Complied 0.11
month µg

11. Stability data of serotype 19A Pneumococcal Bulk Conjugate (Storage condition: 2-8■)
11.1 Batch No.:RSP19AC201705001
Test Free Free
Polysacc
items Protein PS/protein Molecu Residual polysac carrier Bacterial ADH
haride Identity Sterility
and content ratio lar size EDAC charide protein endotoxin Content
content
standard content content
s NMT
Positive response Shall meet
≥40ug/m ≥70ug/m ≤5.0µmol/ 0.75EU/µg
0.20-0.70 ≥80% ≤30.0% ≤5.0% to antiserum type requireme ■0.20ug/ml
Storage I I L polysaccha
19A nts.
time ride
0th <0.75EU/
131.7 409.8 0.32 95.4 1.2 27.4 2.9 Complied Complied 0.108
month µg
1th
127.5 385.2 0.33 92.7 — 19.1 3.5 — — — —
month
2nd
131.0 409.6 0.32 91.4 — 20.3 3.2 — — — —
month
3rd 131.8 391.1 0.34 96.2 — 20.5 0.99 — — — —

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Stability data Pneumococcal Monovalent Bulk Conjugate Yuxi Walvax Biotechnology Co., Ltd.

month
6th
123.9 394.0 0.31 98.0 — 21 1.2 — — — —
month
9th
127.2 400.0 0.32 98.1 — 19 1.6 — — — —
month
12th <0.75EU/
129.8 418.5 0.31 96.1 1.752 21 1.3 Complied Complied 0.02
month µg

11.2 Batch No.:RSP19AC201707002
Test Free Free
Polysacc
items Protein PS/protein Molecu Residual polysac carrier Bacterial ADH
haride Identity Sterility
and content ratio lar size EDAC charide protein endotoxin Content
content
standard content content
s NMT
Positive response Shall meet
≥40ug/m ≥70ug/m ≤5.0µmol/ 0.75EU/µg
0.20-0.70 ≥80% ≤30.0% ≤5.0% to antiserum type requireme ■0.20ug/ml
Storage I I L polysaccha
19A nts.
time ride
0th <0.75EU/
163.6 435.2 0.38 93.3 1.3 26.0 2.8 Complied Complied 0.130
month µg
1th
154.0 404.5 0.38 91.0 — 26.2 1.80 — — — —
month
2nd
157.0 504.4 0.311 94.7 — 26.3 0.20 — — — —
month
3rd
152.8 590.9 0.259 84.4 — 25.4 1.06 — — — —

month
6th
170.4 443.4 0.38 98.7 — 26 0.9 — — — —
month
9th
144.5 432.6 0.33 99.6 — 27 1.4 — — — —
month
Version: July 2020 28
Stability data Pneumococcal Monovalent Bulk Conjugate Yuxi Walvax Biotechnology Co., Ltd.

12th <0.75EU/
161.0 428.4 0.38 98.3 1.644 26 1.1 Complied Complied 0.10
month µg
11.3 Batch No.:RSP19AC201708003
Test Free Free
Polysacc
items Protein PS/protein Molecu Residual polysac carrier Bacterial ADH
haride Identity Sterility
and content ratio lar size EDAC charide protein endotoxin Content
content
standard content content
s NMT
Positive response Shall meet
≥40ug/m ≥70ug/m ≤5.0µmol/ 0.75EU/µg
0.20-0.70 ≥80% ≤30.0% ≤5.0% to antiserum type requireme ■0.20ug/ml
Storage I I L polysaccha
19A nts.
time ride
0th <0.75EU/
202.0 444.9 0.45 99.4 2.3 28.0 0.7 Complied Complied 0.103
month µg
1th
201.0 448.4 0.45 98.3 — 27.6 0.55 — — — —
month
2nd
212.2 386.8 0.55 93.5 — 28 0.6 — — — —
month
3rd
178.8 434.4 0.41 93.9 — 29 0.4 — — — —
month
6th
190.2 439.0 0.43 97.5 — 20 0.6 — — — —
month
9th
194.4 442.2 0.44 97.3 — 28 0.6 — — — —
month
12th <0.75EU/
199.6 430.5 0.46 96.1 2.350 27 0.6 Complied Complied 0.12
month µg

12. Stability data of serotype 19F Pneumococcal Bulk Conjugate (Storage condition: 2-8■)
12.1 Batch No.:RSP19FC201706001

Version: July 2020 29
Stability data Pneumococcal Monovalent Bulk Conjugate Yuxi Walvax Biotechnology Co., Ltd.

Test Free Free
Polysacc
items Protein PS/protein Molecu Residual polysac carrier Bacterial ADH
haride Identity Sterility
and content ratio lar size EDAC charide protein endotoxin Content
content
standard content content
s NMT
Positive response Shall meet
≥40ug/m ≥70ug/m ≤5.0µmol/ 0.75EU/µg
0.10-0.70 ≥80% ≤35.0% ≤5.0% to antiserum type requireme ■0.20ug/ml
Storage I I L polysaccha
19F nts.
time ride
0th <0.75EU/
195.7 500.3 0.39 98.8 3.9 24.4 3.2 Complied Complied 0.073
month µg
1th
174.6 494.3 0.35 95.2 — 27.3 0.73 — — — —
month
2nd
199.5 497.6 0.40 96.3 — 24.2 0.9 — — — —
month
3rd
190.5 505.6 0.38 96.0 — 20.2 0.55 — — — —
month
6th
195.0 500.0 0.39 96.6 — 24 1.2 — — — —
month
9th
184.8 513.1 0.36 98.3 — 25 0.8 — — — —
month
12th <0.75EU/
186.0 499.3 0.37 98.7 4.105 24 1.0 Complied Complied 0.07
month µg

12.2 Batch No.:RSP19FC201707002
Test Free Free
Polysacc
items Protein PS/protein Molecu Residual polysac carrier Bacterial ADH
haride Identity Sterility
and content ratio lar size EDAC charide protein endotoxin Content
content
standard content content

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Stability data Pneumococcal Monovalent Bulk Conjugate Yuxi Walvax Biotechnology Co., Ltd.

s NMT
Positive response Shall meet
≥40ug/m ≥70ug/m ≤5.0μmol/ 0.75EU/μg
0.10-0.70 ≥80% ≤35.0% ≤5.0% to antiserum type requireme ■0.20ug/ml
Storage I I L polysaccha
19F nts.
time ride
0th <0.75EU/
145.6 237.8 0.61 98.1 1.4 30.1 2.4 Complied Complied 0.126
month μg
1th
145.4 257.6 0.56 90.7 — 30.2 2.24 — — — —
month
2nd
153.0 240.2 0.637 99.3 — 30.2 1.16 — — — —
month
3rd
153.5 279.5 0.55 96.8 — 30 0.3 — — — —
month
6th
151.5 234.8 0.65 97.7 — 29 0.8 — — — —
month
9th
150.5 239.8 0.63 96.9 — 30 0.7 — — — —
month
12th <0.75EU/
148.5 265.1 0.56 86.4 0.9545 30 0.8 Complied Complied 0.09
month μg

12.3 Batch No.:RSP19FC201708003
Test Free Free
Polysacc
items Protein PS/protein Molecu Residual polysac carrier Bacterial ADH
haride Identity Sterility
and content ratio lar size EDAC charide protein endotoxin Content
content
standard content content
s NMT
Positive response Shall meet
≥40ug/m ≥70ug/m ≤5.0μmol/ 0.75EU/μg
0.10-0.70 ≥80% ≤35.0% ≤5.0% to antiserum type requireme ■0.20ug/ml
Storage I I L polysaccha
19F nts.
time ride
0th <0.75EU/
134.0 300.3 0.45 96.4 3.1 25.2 1.1 Complied Complied 0.066
month μg

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Stability data Pneumococcal Monovalent Bulk Conjugate Yuxi Walvax Biotechnology Co., Ltd.

1th
140.0 239.6 0.58 97.8 — 25.5 1.31 — — — —
month
2nd
138.5 266.6 0.52 88.9 — 26 1.5 — — — —
month
3rd
146.0 409.2 0.36 94.8 — 25 2.0 — — — —
month
6th
136.8 319.5 0.43 99.0 — 25 2.4 — — — —
month
9th
126.5 301.4 0.42 98.0 — 25 1.9 — — — —
month
12th <0.75EU/
128.0 395.3 0.32 94.1 1.810 25 1.8 Complied Complied 0.09
month μg

13. Stability data of serotype 23F Pneumococcal Bulk Conjugate (Storage condition: 2-8■)
13.1 Batch No.:RSP23FC201705001
Test Free Free
Polysacc
items Protein PS/protein Molecu Residual polysac carrier Bacterial ADH
haride Identity Sterility
and content ratio lar size EDAC charide protein endotoxin Content
content
standard content content
s NMT
Positive response Shall meet

≥40ug/m ≥70ug/m ≤5.0μmol/ 0.75EU/μg
0.25-0.65 ≥80% ≤25.0% ≤5.0% to antiserum type requireme ■0.45ug/ml
Storage I I L polysaccha
23F nts.
time ride
0th <0.75EU/
85.4 232.8 0.37 98.7 0.7 13.0 1.6 Complied Complied 0.162
month μg
1th
86.2 252.2 0.34 98.0 — 14.0 0.03 — — — —
month
2nd 80.6 269.5 0.30 97.4 — 11.4 3.55 — — — —

Version: July 2020 32
Stability data Pneumococcal Monovalent Bulk Conjugate Yuxi Walvax Biotechnology Co., Ltd.

month
3rd
90.0 312.5 0.29 94.3 — 11.6 1.56 — — — —
month
6th
81.2 288.0 0.28 96.3 — 11 2.7 — — — —
month
9th
88.5 245.9 0.36 96.7 — 13 2.7 — — — —
month
12th <0.75EU/
87.2 289.6 0.30 97.7 1.802 12 2.1 Complied Complied 0.06
month μg

13.2 Batch No.:RSP23FC201707002
Test Free Free
Polysacc
items Protein PS/protein Molecu Residual polysac carrier Bacterial ADH
haride Identity Sterility
and content ratio lar size EDAC charide protein endotoxin Content
content
standard content content
s NMT
Positive response Shall meet
≥40ug/m ≥70ug/m ≤5.0μmol/ 0.75EU/μg
0.25-0.65 ≥80% ≤25.0% ≤5.0% to antiserum type requireme ■0.45ug/ml
Storage I I L polysaccha
23F nts.
time ride
0th <0.75EU/
91.3 215.0 0.42 100.0 2.5 24.9 2.1 Complied Complied 0.174
month μg
1th
100.0 250.2 0.40 97.6 — 21.2 2.2 — — — —
month
2nd
90.0 256.4 0.351 96.8 — 24.4 3.7 — — — —
month
Not
3rd
85.6 238.2 0.359 100.0 — 24.4 detecte — — — —
month
d

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Stability data Pneumococcal Monovalent Bulk Conjugate Yuxi Walvax Biotechnology Co., Ltd.

6th
87.6 238.0 0.37 99.2 — 24 3.4 — — — —
month
9th
107.6 214.6 0.50 97.8 — 25 1.3 — — — —
month
12th <0.75EU/
88.5 256.7 0.34 97.3 1.021 25 1.3 Complied Complied 0.10
month μg
13.3 Batch No.:RSP23FC201708003
Test Free Free
Polysacc
items Protein PS/protein Molecu Residual polysac carrier Bacterial ADH
haride Identity Sterility
and content ratio lar size EDAC charide protein endotoxin Content
content
standard content content
s NMT
Positive response Shall meet
≥40ug/m ≥70ug/m ≤5.0μmol/ 0.75EU/μg
0.25-0.65 ≥80% ≤25.0% ≤5.0% to antiserum type requireme ■0.45ug/ml
Storage I I L polysaccha
23F nts.
time ride
0th <0.75EU/
75.8 212.3 0.36 97.5 2.1 18.7 0.6 Complied Complied 0.160
month μg
1th
69.4 271.1 0.26 96.7 — 17.8 1.00 — — — —

month
Not
2nd
85.8 275.6 0.31 98.5 — 18 detecte — — — —
month
d
3rd
89.2 288.4 0.31 97.9 — 18 1.2 — — — —
month
6th
85.8 217.7 0.39 94.5 — 18 2.9 — — — —
month
9th
80.5 214.2 0.38 93.7 — 19 2.5 — — — —
month
12th 81.3 273.2 0.30 95.3 2.268 18 1.6 Complied <0.75EU/ Complied 0.22

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Stability data Pneumococcal Monovalent Bulk Conjugate Yuxi Walvax Biotechnology Co., Ltd.

month µg

Type 1, 3, 4, 5, 6A, 6B, 7F, 9V, 14, 18C, 19A, 19F, 23F Pneumococcal Bulk Conjugate were stored at 2-8■, tested at 0th month, 1th month, 2nd month, 3rd month, 6th month, 9th month, 12th month■test results were all complied with specification of 13-valent pneumococcal polysaccharide conjugate vaccine (WSYX-QC-TS-06-09-0005).

This experiment was conducted by the quality control laboratory strictly sampled at each test point, could provide evidence for safety, effectiveness and validity of pneumococcal polysaccharide stored at 2-8■. In conclusion, shelf life of 13 serotypes of pneumococcal polysaccharide manufactured by Yuxi Walvax stored at 2-8■ is 12 months.

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SOP of Yuxi Walvax Biotechnology Co., Ltd
WSYX-QC-SOP-0009-012
Code
Operating procedures 7
for determination version 09 Page 1/5
hexosamine content
Non-photocopying, confidential documents

1 Objective
This SOP describes the method for total hexosamine content determination, establishing an accurate method for experimenters to master, and conducting the test.
2 Definitions/abbreviations
N/A
3 Scope of application
This SOP is applicable to the total nitrogen content determination of products manufactured by Yuxi Walvax.
4 Responsibility
All the operators shall strictly conduct the SOP
5 Equipment and materials
5.1 Equipment
No. Name Model manufacturer Remarks

Electronic CP225D Or equivalent
1 Starorius
balance MSU225S-1DU equipment
Electronic BS124S Or equivalent
2 Starorius
balance BSA124S equipment
Shanghai Qixin Or equivalent
equipment
Chemical Instrument
Temperature HWS—24/
3 Co., Ltd /Shanghai
control digester HWSY21-K
Yiheng Chemical
Instrument Co., Ltd
Ultraviolet-visi Or equivalent
ble equipment
4 spectrophotome UV-2600 Shimadzu

ter

5.2 reagents
SOP of Yuxi Walvax Biotechnology Co., Ltd
WSYX-QC-SOP-0009-012
Code
Operating procedures 7
for determination version 09 Page 2/5
hexosamine content
Non-photocopying, confidential documents

No. Material/reagent Grade Remarks
1 Glucosamine hydrochloride analytical reagent /
concentrated hydrochloric Content: ≥36.0■38.0
2 Import reagent
acid
3 thymolphthalein analytical reagent /

4 absolute ethyl alcohol analytical reagent /
5 sodium hydroxide analytical reagent /
6 anhydrous sodium carbonate analytical reagent /
7 acetylacetone Import reagent /
4-dimethylaminobenzaldehy Import reagent /
8
de
5.3 materials
No. material No. material

1 pipette 2 Pipette tips
3 Volumetric flask 4 Tube with cover
5 powder free latex gloves 6 web paper
6 Working procedure
6.1 preparation of reagents
6.1.1 8 mol/L hydrochloric acid solution
Take a 100 ml volumetric flask, add appropriate amount of purified water, accurately measure 67.0 ml hydrochloric acid, slowly pour into the volumetric flask, and add purified water to the scale line of 100 ml and mix well.
6.1.2 0.5% (M/V) thymol phenolphthalein ethanol solution (ready to use)
Accurately weigh 0.05 g of thymol phthalein, dissolve in 10 ml of absolute ethanol, and mix well.
6.1.3 5 mol/L sodium hydroxide solution
SOP of Yuxi Walvax Biotechnology Co., Ltd
WSYX-QC-SOP-0009-012
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for determination version 09 Page 3/5
hexosamine content
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Accurately weigh 20.0 g sodium hydroxide into a beaker, dissolve it with purification,

pour it into a 100 ml volumetric flask, and then moisten the beaker with purified water for three times, then inject the moistening solution into the volumetric flask, and finally make the volume to the scale, and mix evenly.

6.1.4 5.3% anhydrous sodium carbonate solution

Accurately weigh 5.30 g anhydrous sodium carbonate in a beaker, dissolve it with purified water and fix the volume to 100ml, mix well.

6.1.5 acetylacetone reagent (ready for use)

Use a pipette gun to accurately measure 1 volume of acetylacetone and 50 volume of 5.3% anhydrous sodium carbonate solution, and mix well.

6.1.6 0.1 mol/L hydrochloric acid solution

Take a 100 ml volumetric flask, add appropriate amount of purified water, accurately measure 850 μ l hydrochloric acid, slowly pour into the volumetric flask, add purified water and fix the volume to 100 ml scale mark, mix well.

6.1.7 4-dimethylaminobenzaldehyde reagent (ready for use)

Accurately weigh 0.80g of 4-dimethylaminobenzaldehyde in a beaker, measure 15 ml of absolute ethanol and hydrochloric acid with a pipette gun to dissolve them, and then mix them evenly.

6.1.8 glucosamine standard working solution (0.5 g/L)

For details, please refer to the operation procedures for preparation of standard solution, code: WSYX-QC-SOP-00-09-0801

6.2 test procedure

6.2.1 use a 5000 μ l pipette to measure 10 ml of purified water into a plug calibrated tube, and mark the scale accurately.

6.2.2 dilute the sample with purified water to make the content of hexosamine about 100-400 μ g/ml.

6.2.3 use a pipette gun with a range of 100-1000 μ l to accurately suck the working solution of SOP of Yuxi Walvax Biotechnology Co., Ltd

WSYX-QC-SOP-0009-012

Code

Operating procedures 7

for determination version 09 Page 4/5

hexosamine content

Non-photocopying, confidential documents

glucosamine standard into each labeled plug calibration tube.

name Stand Stand Stand Stand Stand Stand Parall Parall

tube No. ard 0 ard 1 ard 2 ard 3 ard 4 ard 5 el 1 el 2

Reference working - -

0.0 200 400 600 800 1000

solution

Water for injection - -

1000 800 600 400 200 0.0

(μ l)

Test sample (μ l) - - - - - 1000 1000

Hexosamine content - -

0 100 200 300 400 500

(μ g/ml)

6.2.4 add 1000 μ l 8 mol/L hydrochloric acid solution to each plug calibration tube, mix well, plug and seal; place in boiling water bath for 60 minutes, take out and place at room temperature for 20 minutes.

6.2.5 add 50 μ l 0.5% (M/V) thymol phenolphthalein ethanol solution into each test tube, mix well; titrate with 5 mol / L sodium hydroxide solution until the solution turns blue; and then titrate with 0.1 mol/l hydrochloric acid solution until the solution is colorless.

6.2.6 fix the volume of each tube to 10ml with purified water, prepare neutral hydrolysate, and mix well.

6.2.7 transfer 1000 μ l of neutral hydrolysate in each tube to the new test tube with plug, add 1000 μ l acetylacetone reagent to each tube, mix well, plug, place in boiling water bath for 45 min, and then place at room temperature for 20 min.

6.2.8 add 2500 μ l anhydrous ethanol into each tube, mix well, then add 1000 μ l 4-dimethylaminobenzaldehyde reagent, plug and mix well, and place in dark place at room temperature for 90 min.

6.2.9 read the absorbance at 530 nm.

6.2.10 calculate the content of hexosamine.

6.2.11 content of hexosamine in test sample = content of hexosamine in test sample $\times$ dilution multiple

7 Flow chart

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WSYX-QC-SOP-0009-012

Code

Operating procedures 7

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hexosamine content

Non-photocopying, confidential documents

N/A

8 Notes

8.1 hydrochloric acid will be used in the experiment, and safety protection measures shall be taken during operation.

8.2 pay attention to avoid scalding when using water bath.

8.3 pay more attention to the process of water bath, and the test tube with plug should be tightly covered to avoid the reagent being heated and ejected during the water bath.

9 References or Standard Basis

N/A

10 Revision history

Revising Basis, reason and revision

Version Valid from Remarks

department

Document expiration

03 2014.12.01 QC laboratory Improve the content of / documents

Improve the content of

04 2014.12.01 QC laboratory /

documents
Improve the content of
05 2015.05.01 QC laboratory /
documents
Improve the content of
06 2017.10.01 QC laboratory /
documents
Improve the description of
07 2018.09.01 QC laboratory /
operating procedures
The reference substance of
08 2019.06.19 QC laboratory glucosamine was revised to /
glucosamine standard

SOP of Yuxi Walvax Biotechnology Co., Ltd.

SOP of conjugate protein Code WSYX-QC-SOP-06-09-0006
content test for 13-valent 1/8
Version 02 Page
pneumococcal conjugate
vaccine Confidential files No copy

1 Objective

Standardize the operation of conjugate protein content test method for PCV13.

2 Definitions/Abbreviations

N/A

3 Scope of application

3.1 It is applicable to determine the conjugate protein content of 13-valent pneumococcal conjugate vaccine manufactured by Yuxi Walvax.

4 Responsibilities

4.1 All the operators shall strictly conduct the SOP.

5 Materials and equipment

5.1 Use of equipment

Equipment Equipment Equipment

No. Manufacturer Notes

name model code

Analytical Or equivalent

1 BS124S WYQCF0036 Sartorius

balance equipment

Shanghai Qixin Scientific

Electro- Instruments Co., Ltd. /

Or equivalent

2 thermostatic DK-600A WYQCF0293 Shanghai Yiheng

equipment

water tank Scientific Instruments Co.,

Ltd.

U/V

Shimadzu instrument Or equivalent

3 spectrophotomet UV-2600 WYQCF0529

(Suzhou) Co., Ltd equipment

er

Frac-920

Or equivalent

4 automatic Frac-920 WYQCF0081 Amersham Biosciences

equipment

collector

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content test for 13-valent 2/8

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P-1 constant Or equivalent

5 P-1 WYQCF0080 Amersham Biosciences

flow pump equipment

Chromatographi Or equivalent

6 XK16-100 WYQCF0082 Amersham Biosciences

c column equipment

AKTA Pure25 AKTA Or equivalent

7 WYQCF0455 GE

system purifier Pure25 equipment

5.2 Use reagent

No. Reagent name Grade

1 Sodium hydroxide AR

2 Anhydrous sodium carbonate AR

3 Potassium tartrate AR

4 Copper sulfate pentahydrate AR

5 2N foline-phenol Imported reagent

6 Citric acid AR

7 Sodium dihydrogen phosphate AR

8 Disodium hydrogen phosphate AR

9 Sodium chloride AR

10 0.9% sodium chloride injection /

5.3 Use of materials

No. Material name number Material name

1 Thermometer 2 15*150 test tube

3 Triangular flask 4 5000μl pipette tips

5 5000μl pipette 6 1000μl pipette tips

7 1000μl pipette 8 200μl pipette tips

9 200μl pipette 10 Beaker

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11 0.45μm needle filter 12 3K ultrafiltration cup
13 Sepharose CL-4B gel 14 Web

6 Working procedure

6.1 Solution

6.1.1 0.6mol/L sodium hydroxide solution

Weigh 2.4g of sodium hydroxide, dissolve it with purified water and dilute it to 100ml, mix evenly.

6.1.2 0.2mol/L citric acid solution

Weigh 4.2g of citric acid, dissolve it with purified water and dilute to 100ml, mix well.

6.1.3 Protein detection solution A

Weigh 10g of sodium hydroxide and 50g of anhydrous sodium carbonate into beakers respectively, add 400ml of purified water, dissolve and mix together, pour it into plastic bottle, label as protein detection solution A, and use for standby.

6.1.4 Protein detection solution B

Weigh 0.5g of potassium tartrate into the beaker, add 50ml of purified water to dissolve it. Weigh 0.25g of copper sulfate pentahydrate into the beaker, add 30ml of purified water to dissolve it, and then mix the two solutions as protein detection solution B, label and reserve.

6.1.5 Alkaline copper test solution (Prepare the solutions immediately before use)

Before use, mix according to the volume proportion of protein detection solution A: protein detection solution B: purified water (20:4:1), and then reserve.

6.1.6 Foline-phenol test solution (Prepare the solutions immediately before use)

Take Foline-phenol (2mol/L acid concentration) 1→16 (take 1000μl foline-phenol test solution and put it into a triangular flask, add the purified water to 16ml, mix well, and prepare according to the experiment requirement)

6.1.7 0.9% sodium chloride injection (commercial purchase)

6.1.8 PBS

Take proper amount of sodium dihydrogen phosphate, disodium hydrogen phosphate and SOP of Yuxi Walvax Biotechnology Co., Ltd.

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sodium chloride, dissolve them in purified water to make a solution contains 88.7μg of sodium dihydrogen phosphate, 38.0μg of disodium hydrogen phosphate and 8.5mg of sodium chloride per ml, respectively, and mix evenly.

6.1.9 200μg/ml national standard protein working solution

refer to the <operation procedures for preparation of standard solution> WSYX-QC-SOP-0009-0801

6.2 Test procedure

6.2.1 Free protein reference substance

The tetanus toxoid carrier protein was diluted with PBS to the protein content of about 2000 μg/ml, and filtered with 0.45μm needle filter to obtain the free protein reference, which was recorded as S1.

6.2.2 Desorption solution reference

Take appropriate amount of aluminum phosphate adjuvant, sodium dihydrogen phosphate, disodium hydrogen phosphate and sodium chloride, dissolve in purified water to make a solution contains 1.0mg of aluminum phosphate (adjuvant), 88.7μg of sodium dihydrogen phosphate, 38.0μg of disodium hydrogen phosphate and 8.5mg of sodium chloride per ml, to obtain the finished product diluent. Take 2000μl of finished product diluent, add 100μl of 0.6mol/L sodium hydroxide solution, and add 100μl of 0.2mol/L citric acid solution within 10s, mix well. 0.45μm needle filter is used for filtration to obtain desorption solution reference substance.

6.2.3 Test samples preparation

6.2.3.1 Take a 3K ultrafiltration cup, add 1500μl purified water into each cup, mark the concave liquid surface with a mark pen, and then discard 1500μl purified water

6.2.3.2 Take 10 vials ■syringes■ of the finished product of 13-valent pneumococcal conjugate vaccine and mix well. Take 5000μl, add 250μl of 0.6mol/L sodium hydroxide solution, quickly add 250μl of 0.2mol/L citric acid solution within 10s, mix with vortex mixer, and obtain desorption solution of finished product, which is recorded as S0.

6.2.3.3 Take 4500μl of the desorption solution of finished product and put it into the marked ultrafiltration cup, and centrifuge it at 5■, 7000rpm for 30min. After centrifugation, take out the SOP of Yuxi Walvax Biotechnology Co., Ltd.

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ultrafiltration cup and add PBS to 1500μl. Fully blow wash the solution and mix well and filter it with 0.45μm needle filter to obtain the test sample.

6.2.4 Column chromatography

6.2.4.1 Take the 1ml of free protein reference to load into column (XK16/100 column filled with Sepharose CL-4B gel), eluted with 0.9% sodium chloride solution, the flow rate is 0.3ml/min, the automatic collector shall collect eluant in tubes one by one, and the 4.5ml is collected from each tube. 1.2 times of column volume is collected, the collected solution is detected at the 214nm wavelength by UV-VIS spectrophotometer, and the elution map is drawn with the tube number as the X-axis and OD214nm as the Y-axis. The number of tubes corresponding to the separation peak of free protein reference (the first tube when OD value is from less than 0.01 to greater than or equal to 0.01) is recorded as n1, at the end of chromatography, the corresponding number of tubes - (OD value is from greater than or equal to 0.01 to less than 0.01) is recorded as n2. Collect the eluant in each tube from n1 to n2, take 1000μl from each tube and mix them together then get solution S2.

6.2.4.2 Determine the protein concentration of S0 in accordance with the <SOP of protein content determination>, which multiply by the dilution multiple 1.1 in the desorption step to obtain the total protein concentration S0 in the finished product; determine the protein

concentration of S1 and S2 solutions in accordance with the same method, record s1 and s2 respectively.

6.2.4.3 Take 1ml of desorption solution reference substance and load into column, elute it with 0.9% sodium chloride solution, 0.3ml/min. Collect it tube by tube by automatic partial collector, collect 4.5ml in each tube. The volume of 1.2 column bed volumes is collected; the collected solution is detected with UV-VIS spectrophotometer at the wavelength of 214nm, take the tube number as the X-axis and OD214nm as the Y-axis. to draw the elution map. The number of tubes corresponding to the separation peak of desorption solution reference (the first tube whose OD value changed from less than 0.01 to greater than or equal to 0.01) is recorded as n3.

6.2.4.4 Take 1ml of the treated test sample to load into the column. Elute, collect, UV spectrophotometer detect with the same method, and draw the elution map.

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6.3 Results calculation

6.3.1 Experimental effectiveness (%) = $4.5 \times (n2 - n1 + 1) \times s2 / s1 / V \times 100$, should be 80%-120%.

4.5: the collected volume from each tube, ml;

n1: the corresponding tube number at the beginning of the separation peak of free protein reference;

n2: the corresponding tube number at the end of the separation peak of free protein reference;

s1: the protein concentration of free protein reference, before loading into the column, µg/ml;

s2: the protein concentration of free protein reference after mixing the eluents from n1 to n2, µg/ml;

V: the volume of free protein reference loading into the column, ml.

6.3.2 Conjugate protein content of 13-valent pneumococcal conjugate vaccine:

Conjugate protein content (%) = $(1 - \Sigma OD2 \times 1.1 / 3 / \Sigma OD1 \times s1 / s0) \times 100$

If $n3 < n2$,

ΣOD1: the sum of the OD values of free protein reference from n1 to n3 tubes;

ΣOD2: the sum of OD values of the test sample from n1 to n3 tubes;

If $N3 \geq N2$,

ΣOD1: the sum of OD values of the free protein reference from n1 to n2 tubes;

ΣOD2: the sum of OD values of the test samples from n1 to n2 tubes;

s0: the protein concentration of the finished product, µg/ml;

s1: the protein concentration of free protein reference before loading column, µg/ml;

3: the concentrated multiple of the finished product desorption sample;

1.1: the multiple of the finished product desorption sample diluted by desorption dilution.

6.3.3 Results judgment

Carry out according to corresponding quality standard.

7 Flow chart

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No

8 Matters need attention

8.1 Make sure that the joints of the gel column tightly connected without leakage before use.

8.2 Plug the liquid outlet end with a plug before plugging the liquid inlet end, and the sequence shall not be reversed.

8.3 Every month, a column volume of 0.5mol/L NaOH solution is used as the mobile phase to clean the gel. After washing, it was cleaned with the corresponding eluent to the pH of elute is less than 7 in which the gel column was calibrated.

8.4 The self-prepared eluent must be degassed by steam sterilization or appropriate methods (such as negative pressure).

8.5 The collecting tubes shall be marked by mark pen to prevent confusion.

8.6 Numbering principle of free protein reference and desorption solution reference:

The fixed number of chromatographic column is A, B, C, D and so on. The date when the column starts to be calibrated is the figure number. If the free protein reference loading into column A is on August 5, 2018, the number is column A 201808005, which is valid for 30 days and expires on September 3, 2018.

Number based on the AKTA pure25 purifier system and column starting to be calibrated. If the WYQCF0455 column is loaded the free protein reference on August 5, 2018, the number is WYQCF04155 201808005, the validity period is 30 days, and the validity period is until September 3, 2018.

8.7 Temperature and time have influence on protein content detection, which should be strictly controlled.

8.8 The free protein reference, desorption solution reference and test sample should be detected in the same column and completed within the validity period of the chromatographic column.

8.9 The calibration standard on chromatography column: calibration consistency test standard in different time periods is V0, V1 shall not exceed 1 tube, that is, the tube number of V0, V1 of SOP of Yuxi Walvax Biotechnology Co., Ltd.

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the chromatographic column shall not move forward or backward more than 1 tube in

continuous twice calibration.
9 Reference or standard basis

9.1 "Chinese Pharmacopoeia" edition 2015, volume III, general rule 0731

10 Revision history
Time in Revision
Version Revision basis, reason and content Notes
force Dept.
03 Aug. QC Add the appendix of free protein reference
00
2019 laboratory valid

25 Sep. QC Revise the conjugate protein content
01
2019 laboratory calculation formula

Standard Operating Procedures of Yuxi Walvax Biotechnology Co., Ltd.

Code WSYX-QC-SOP-00-09-0105
Operating Procedures
determination of adipic Version 08 Pages 1/5
dihydrazide (ADH) content
Confidential documents, no copying.

1 Objective
Standardize the method of determination the content of adipic dihydrazide (ADH).
2 Definitions/abbreviations
No
3 Scope of application
This SOP is applicable to the adipic dihydrazide determination of group A meningococcal polysaccharide bulk derivatives, Group C meningococcal polysaccharides polysaccharide bulk derivatives. Group Y meningococcal polysaccharides polysaccharide bulk derivatives, Group W135 meningococcal polysaccharides polysaccharide bulk derivatives, haemophile influenza type b bulk conjugate, 13-valent pneumococcal polysaccharide bulk derivative
4 Responsibility
All the operators shall strictly conduct the SOP
5 Equipment and materials
5.1 Using the equipment
Serial Equipment Equipment Equipment Manufacturer
Note
number name model coding s
WYQCF0444
Ultraviolet
WYQCF0445 or equivalent
1 Visible Light UV-2600 Shimadzu
WYQCF0529 equipment
Spectrometer
WYQCF0530

CP225D Sartorius
Electronic WYQCF0037 or equivalent
2 MSu225S-
balance WYQCF0359 equipment
1DU
Electronic BS124S WYQCF0036 Sartorius or equivalent
3
balance BSA124S WYQCF0357 equipment
Bidirectional Jintan Fuhua or equivalent
4 79—2 WYQCF0026
magnetic Instrument equipment
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heating Co., Ltd.
agitator

5.2 Use of materials/reagents
Serial
numb Name of material/reagent Level/Category Note
er
1 Adipic dihydrazide (ADH) Imported reagents /

Trinitrobenzene Sulfonate /
2 Imported reagents
(TNBS)
Sodium tetraborate /
3 Analysis pure
(Na2B4O7·10H2O)
5.3 Materials used
Serial Serial
Materials Materials
number number
1 Latex Gloves 2 Pipette head
3 Pipette gun 4 Capacity bottle
5 Beaker 6 Rotor

6 Work procedures
6.1 Principle
Based on the fact that the amino group in ADH can react with TNBS in the presence of sodium tetraborate, the content of adipic hydrazide in derivatives is determined by UV Vis spectrophotometry.
6.2 Reagents
6.2.1 Adipic hydrazide (ADH) reference working solution (20 µg/ml)
For details, please refer to the operation procedures for preparation of standard solution, code: WSYX-QC-SOP-00-09-0801.
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6.2.2 3% TNBS solution

Accurately measure 5 ml of TNBS solution into a volumetric flask, add purified water to the volume of 50 ml, mix well, and then 3% TNBS solution is obtained. Pack it into about 7 ml/vial, label it and store it at -20℃.

6.2.3 5% sodium tetraborate solution

Weigh 47.35 g sodium tetraborate ($\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O}$), dissolve with purified water and fix the volume to 500 ml, mix well, transfer to 500 ml reagent bottle, label, and store at room temperature.

6.3 Test procedure

6.3.1 Treatment of test articles

Use a pipette gun with a range of 100-1000 μl to dilute the test sample with purified water until the concentration of adipic hydrazide is not higher than 20 $\mu\text{g}/\text{ml}$ as the test solution, and accurately measure 2000 μl of the test solution into the test tube in duplicate.

Original solution of pneumococcal polysaccharide conjugate: Take 8 ml-10 ml of the test sample, and evenly pack it into two 15 ml 3K ultrafiltration centrifuge tubes, 7000 g, 5℃, 30 min, take the filtrate of the lower layer of two tubes, and mix them evenly.

6.3.2 Standard curve preparation:

Tube No.	Stand	Stand	Stand	Stand	Stand	Stand	Stand
Reagents	ard 0	ard 1	ard 2	ard 3	ard 4	ard 5	ard 6

ADH Reference working

0.0 200 400 800 1200 1600 2000

solution (μl)

purified water (μl) 2000 1800 1600 1200 800 400 0.0

ADH concentrate ($\mu\text{g}/\text{ml}$) 0 2 4 8 12 16 20

6.3.3 Add 2000 μL 5% sodium tetraborate solution into each tube of the reference substance and the test sample, mix well, then add 600 μl of 3% TNBS solution, mix and place for 15 minutes, and measure the absorbance at the wavelength of 500 nm of UV-Vis spectrophotometer.

6.4 Result calculation

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6.4.1 Make linear regression equation with the series concentration and corresponding absorbance of adipic hydrazide reference solution, and then substitute the absorbance of the test solution into the linear regression equation to calculate the concentration of adipic hydrazide.

The content of ADH ($\mu\text{g}/\text{ml}$) = average concentration of ADH $\times$ dilution ratio

6.5 Result determination

According to the corresponding sample quality standard.

7 Flow chart

No

8 Precautions

8.1 provide sufficient local exhaust during operation.

8.2 operators must receive special training and strictly abide by the operation regulations.

8.3 operators should wear latex gloves.

8.4 keep away from fire and heat source.

8.5 use explosion-proof ventilation system and equipment.

8.6 avoid contact with oxidant and alkali.

8.7 fire fighting equipment and leakage emergency treatment equipment of corresponding variety and quantity shall be provided.

8.8 the empty container may contain harmful substances.

9 References or standard basis

9.1 Pharmacopoeia of People's Republic of China, 2015. Volume ■, General requirement 3118.

10 Revision history

Basis, reasons and content of

Version Effective Revision

the amendment Note

No. time Department

QC File expiration, change

03 2011.12.01 /

Laboratory Improve part of the file content

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Code WSYX-QC-SOP-00-09-0105

Operating Procedures

determination of adipic Version 08 Pages 5/5

dihydrazide (ADH) content

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QC

04 2014.12.01 Improve part of the file content /

Laboratory

QC

05 2016.11.15 Improve part of the file content /

Laboratory

QC

06 2017.10.01 Improve part of the file content /

Laboratory

Revised pneumococcal

QC

07 2018.09.01 polysaccharide conjugate /

Laboratory

treatment

SOP of Yuxi Walvax Biotechnology Co., Ltd

Code WSYX-QC-SOP-0009-0115

Operating procedures

for determination of Version 06 page 1/5

phenol content

Non-photocopying, confidential documents

1 Objective

This SOP describes the method for determination of phenol content.

2 Definition/abbreviation

No

3 Applicable scope

This SOP is applicable to the phenol content determination of meningococcal group A, bulk polysaccharide, group C meningococcal bulk polysaccharide, group Y meningococcal bulk polysaccharide, group W135 meningococcal bulk polysaccharide, 23-valent pneumococcal polysaccharide (bulk) except sterotype 11A, 15B, 20, 23-valent pneumococcal polysaccharide vaccine.

4 Responsibility

All the operators shall strictly conduct the SOP

5 Materials and equipment

5.1 Use of equipment

Equipment Production Remark

No. Device Name Device Code

model Factory

electronic BS124S WYQCF0036 Or equivalent

1 Sartorius

balance equipment

5.2 Use of reagents

No. Reagent name Grade

1 soluble starch CP

2 concentrated hydrochloric acid AR

3 potassium bromate AR

4 sodium thiosulfate AR

5 potassium iodide AR

6 Potassium bromide AR

5.3 Use of materials

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Code WSYX-QC-SOP-0009-0115

Operating procedures

for determination of Version 06 page 2/5

phenol content

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No. Material name No. Material name

1 transfer liquid gun 2 conical flask with cover

3 pipette tips 4 acid burette

5 powder free latex gloves 6 beaker

7 Glass rod 8 weighing scoop

9 weighing paper 10 time-meter

6 Procedures

6.1 Principle

This method is based on the reaction of bromate solution with hydrochloric acid to produce bromine and phenol to tribromophenol.. Excess bromine reacts with potassium iodide to release iodine. The precipitated iodine is titrated with sodium thiosulfate titrant. According to the consumption of sodium thiosulfate titrant, the content of phenol in the test sample can be calculated.

6.2 Reagent preparation

6.2.1 Starch indicator

Weigh Soluble starch 0.5 g, add 5 ml of water, suspend it, slowly added into 100 ml of boiling water, stir while adding, Keep boiling for 2 min, cool, take upper liquid, this solution should use immediately.

6.2.2 0.02 mol/L Bromine solution

Weigh potassium bromate 0.56 g, add potassium iodide 3 g, dissolve and dilute with water for injection to 1000 ml, shake well, that is.

6.2.3 6 mol/L Hydrochloric acid solution

Take a 100 ml volumetric flask, add proper amount of water for injection, take 50 ml of concentrated hydrochloric acid, Slowly inject into a capacity bottle, shake while adding, add water for injection to near volume, Shake and cool down to room temperature, add water for injection to volume, mix well and label, that is.

6.2.4 25% KI solution

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Weigh 25 g of potassium iodide, dissolve with injection water and dilute volume to 100 ml, mix well, pour into brown bottle, label, that is.

6.2.5 Sodium thiosulfate titration solution (0.02 mol/L)

6.2.5.1 preparation of Sodium thiosulfate titration solution (0.1mol/L)

For details of preparation and calibration, please refer to operation procedures for

titration solution preparation (WSYX-QC-SOP-0009-0802).

6.2.5.2 preparation of Sodium thiosulfate titration solution (0.02 mol/L)
 accurately measure sodium thiosulfate titration solution (0.1 mol/L) 100 ml, dilute with water to 500 ml, shake well, that is.

6.3 test procedure
 Accurately measure test product 1 ml in a conical flask with cover, add 50 ml of water, accurately add 15 ml of 0.02 mol/L Bromine solution, add 10 ml of 6 mol/L hydrochloric acid solution along the wall of bottle, shake well, cover the conical flask, place at the dark place for 30 min, take 2 ml of 25% potassium iodide solution at the bottle-neck of the conical flask with cover, slightly open the cover a little, let the solution flow down, tight the cover, shake well. Wash the bottle neck with a little water, titrate with sodium thiosulfate(0.02 mol/L) to the end point, add about 0.5 ml of starch indicator solution until blue color disappear. The results of titration were corrected by blank test.

6.4 Result calculation

Content of phenol=

Content of phenol=average phenol content (g/L)/solid weight×1000
 V0: Volume of sodium thiosulfate titration solution consumed by blank control, ml
 V1: Volume of sodium thiosulfate titration solution consumed by test product, ml
 C: Concentration of sodium thiosulfate titration solution, mol/L
 15.69: 1/6 of Phenol molecular weight.

6.5 Result judgment
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Carry out according to corresponding quality standard

7 Flow chart
 No

8 Notes
 8.1 The temperature and humidity of the test should be strictly controlled.
 8.2 The burette and volumetric flask should be checked for leakage before use.
 8.3 Control the time of placing in the dark place (30min) accurately.
 8.4 Strictly follow the management of titration solution.

9 Relevant document
 N/A

10 Relevant records
 10.1 The record of solution preparation. Record code: WSYX-QC-SOP-0009-0804-R01
 10.2 The record of titration solution preparation and calibration. Record code: WSYX-QC-SOP-0009-0802-R01
 10.3 The record of indicator solution preparation. Record code: WSYX-QC-SOP-0009-0806-R01

11 Reference or standard basis
 11.1 China Pharmacopoeia, Part 4, Page 230 China Medical Science and Technology Press, 2015

12 Revision history

Revising
 Issuing Basis, reason and revision Rem
 Version Valid from department
 date
 03 2011.11.01 2011.12.01 QC Document expiration
 laboratory
 Improve the content of documents
 04 2014.11.01 2014.12.01 QC Improve the content of documents
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2 laboratory

05 QC Improve the content of documents
 2017.08.31 2016.00.01
 laboratory

Standard Operating Procedures of Yuxi Walvax Biotechnology Co., Ltd.

Code WSYX-QC-SOP-00-09-0104
Operating procedures
determination of molecular Version 13 Pages 1/21
size
Confidential documents, no copying.

- 1 Purpose
Standardize the operation of molecular size determination.
- 2 Definitions/abbreviations
No
- 3 Scope of application
This SOP is applicable to the molecular size determination of haemophilus influenzae type b polysaccharide, haemophilus influenzae type b bulk conjugate, haemophilus influenzae type b conjugate vaccine, group A meningococcal polysaccharides (bulk), group A meningococcal polysaccharides bulk conjugate, group C meningococcal polysaccharides (bulk), group C meningococcal polysaccharides bulk conjugate, group Y meningococcal polysaccharides (bulk), group W135 meningococcal polysaccharides (bulk), group W135 meningococcal polysaccharides bulk conjugate, group A and C meningococcal polysaccharides vaccine, group ACYW135 meningococcal polysaccharides vaccine, pneumococcal polysaccharide (bulk), pneumococcal polysaccharide derivative, pneumococcal polysaccharide bulk conjugate, degradation polysaccharide.
- 4 Responsibilities
All operators operate strictly in accordance with this SOP.
- 5 Equipment and materials
5.1 Equipment

Serial Note
Equipment Equipment
numb Name Manufacturers
model coding
er
Electronic or equivalent
1 CP225D WYQCF0037 Sartorius
balance equipment
Desktop high- Hitachi Japan or equivalent
2 CR21G WYQCF0042
speed centrifuges Corporation equipment
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High-speed Thermo Fisher
Multifuge WYQCF0072/ or equivalent
3 freezing Technology
X1R WYQCF0073 equipment
Centrifuges (China) Co., Ltd.
WYQCF0081/
Frac-920
WYQCF0084/ Amersham or equivalent
4 automatic Frac-920
WYQCF0087/ Biosciences equipment
collector
WYQCF0090
WYQCF0080/
P-1 Constant WYQCF0083/ Amersham or equivalent
5 P-1
Current Pump WYQCF0086/ Biosciences equipment
WYQCF0089

WYQCF0082/
WYQCF0085/
WYQCF0088/
WYQCF0091/
WYQCF0411/
Tomography Amersham or equivalent
6 XK16-100 WYQCF0467/
Column Biosciences equipment
WYQCF0468/
WYQCF0469/
WYQCF0470/
WYQCF0510/
WYQCF0511

WYQCF0444/
Ultraviolet Shizu Enterprise
WYQCF0445/ or equivalent
7 Visible Light UV-2600 Management
WYQCF0529/ equipment
Spectrometer (China) Co., Ltd.
WYQCF0530
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AKTA Purifier 10 AKTA Purifier or equivalent
8 WYQCF0192 GE
System 10 equipment

WYQCF0455/
WYQCF0456/
AKTA Pure25 WYQCF0457/ or equivalent
9 AKTA Pure25 GE
System Purifier WYCF0458/ equipment
WYCF0506/
WYCF0507/
Shanghai
Guangzheng
Rack drug or equivalent
10 JEZREEL-5 WYQCF0076 Medical
balance equipment
Instruments Co.,
Ltd.

Shanghai Qixin
WYQCF0586
Electric heated Scientific or equivalent
11 HWS-24 WYQCF0517
water bath Instrument Co., equipment
WYQCF0580
Ltd.
Shanghai Qixin
WYQCF0061
Electric heated Scientific or equivalent
12 DK-600A WYQCF0293
water tank Instrument Co., equipment
WYQCF0287
Ltd.
5.2 Reagents

Serial
numb Reagent Name Level/Category
er
1 Sodium chloride AR
2 absolute ethyl alcohol AR
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Code WSYX-QC-SOP-00-09-0104
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3 Sodium hydroxide AR
4 0.9% sodium chloride injection /
5.3 Materials used
Serial Serial
Name Name
number number
1 Sepharose CL-4B(2B) gel 2 Collection tube
3 Needle filter 4 Syringe

5 Capacity bottle 6 2L transit bottle
7 Roll paper 8 Volume
6 Work procedures
6.1 Principles
Gel filter tomography is a tomography technique that separates biological molecules
according to the size of molecules.
6.2 Preparation of reagents
6.2.1 0.2 mol/L sodium chloride solution
Called sodium chloride 11.7 g, stacked sodium nitrogen 0.1 g, dissolved to 1000 ml with
purified water, adjusted pH 7.0 with 0.1 mol/L sodium hydroxide solution, mixed, filtered,
fill&finish, autoclave (121■, 15 minutes), labeled, spare.
6.2.2 0.2 mol/L sodium chloride solution (used in pneumococcal polysaccharide binding
sodium)
Called take sodium chloride 11.7 g in the beaker, purified water dissolved to 1000 ml
capacity bottle, added purified water washing beaker repeat three times, will be purified water
to the capacity bottle, add purified water to the scale, mixed, mixed, high-pressure sterilization
(121 ■, 15 minutes), label, backup.
6.2.3 Molecular size control solution
Vitamin B12 Solution: Preparation details see Standard Solution Preparation Procedure
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Code WSYX-QC-SOP-00-09-0104
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Blue glucan 2000 solution: preparation details see "Standard Solution Preparation
Operating Procedure" WSYX-QC-SOP-00-09-0801
6.2.4 0.5 mol/L Sodium Hydroxide solution

According to the preparation amount of 500 ml, said sodium hydroxide 10 g in the beaker, dissolved with the appropriate amount of purified water to room temperature, and transferred to 500 ml capacity bottle, and then with the appropriate amount of purified water 3 times after the washing beaker will be remixed into the capacity bottle, re-filled purity water to the scale line mixed, that is.

6.2.5 Ethanol-aqueous solution (20:80)

Take 20ml of waterless ethanol, add 80ml of purified water, mix well.

6.3 Trial steps

6.3.1 Choice of Gel:

6.3.1.1 Products with Sepharose CL-4B gel column

Haemophilus influenzae type b polysaccharides, haemophilus influenzae type b bulk conjugate, haemophilus influenzae type b conjugate vaccine, group A meningococcal polysaccharides bulk conjugate, group C meningococcal polysaccharides (bulk), group C meningococcal polysaccharides bulk conjugate, group Y meningococcal polysaccharides (bulk), group W135 meningococcal polysaccharides (bulk), group W135 meningococcal polysaccharides bulk conjugate, group A and C meningococcal polysaccharides vaccine, group ACYW135 meningococcal polysaccharides vaccine, pneumococcal polysaccharide (bulk) (stereotype 1, 2, 3, 4, 7F, 8, 9N, 12F, 14, 18C, 19A, 19F, 23F), pneumococcal polysaccharide derivative, pneumococcal polysaccharide bulk conjugate, degradation polysaccharide.

6.3.1.2 Products with Sepharose CL-2B gel column

Pneumococcal polysaccharide (bulk) (stereotype 5, 6A, 6B, 9V, 10A, 11A, 15B, 17F, 20, 22F, 33F)

6.3.2 Sepharose CL-4B (CL-2B) Calibration of gel columns

Take 1.5 ml of 2 mg/ml blue dextran 2000 solution and 1 mg/ml vitamin B12 solution

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respectively, mix them well, filter them through 0.45 μM filter membrane and use them as molecular size reference solution. Take 1 ml of test sample on the column and elute at the flow rate of 0.3 ml/min. Collect a tube every 15 min, that is, 4.5 ml/tube, and wash off at least 1.2 column volume. After elution, the absorbance of the collected solution was determined at the wavelength of 260 nm and 370 nm. The fifth and sixth tubes of the collected solution were mixed and used as blank control to mark zero. After the zero mark, the absorbance of the eighth to the 48th tubes were read to obtain the elution peaks of blue dextran 2000 solution and vitamin B12 solution. At 260 nm, the volume of blue dextran 2000 elution peak is V0 of the column. At 370nm, the volume of vitamin B12 solution eluting the peak corresponds to the Vi of the column. The post calibration period is 30 days.

According to the following formula, calculate the number of tubes in which KD0.2, KD

0.3, KD0.5 and KD0.6 are located.

Partition coefficient $KD = (V_e - V_0) / (V_i - V_0)$

0.2 $\frac{V_i - V_0}{V_i - V_0} + V_0$

Number of tubes where KD0.2 is located=

4.5

0.3 $\frac{V_i - V_0}{V_i - V_0} + V_0$

Number of tubes where KD0.3 is located=

4.5

0.5 $\frac{V_i - V_0}{V_i - V_0} + V_0$

Number of tubes where KD0.5 is located =

4.5

0.6 $\frac{V_i - V_0}{V_i - V_0} + V_0$

Number of tubes where KD0.6 is located =

4.5

6.3.3 Determination of KD values and recovery rates for each product

6.3.3.1 Haemophilus influenzae type b polysaccharides molecular size test:

a) Sampling volume and collection method: use 0.45 m membrane filtration for test products, take 1 ml of test products, wash with 0.2 mol/L sodium chlorate solution, fluid lotion amount is at least 1.2 times the volume of the column bed, automatic collector tube-by-tube collection, 4.5 ml per tube collection.

b) Mix the liquids collected by the step-by-step collector with the 5th and 6th tubes as blank controls.

c) The number of tubes in which KD is located was found out. From the number of tubes where Standard Operating Procedures of Yuxi Walvax Biotechnology Co., Ltd.

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KD0.5 is located, every two tubes are merged from the back to the front, and the seventh tube is merged. At the same time, the collected liquid after the number of KD0.5 tubes is merged in turn. When the number of tubes KD0.5 is even, tubes 7 to 48 can be combined into one tube. When the number of tubes KD0.5 is odd, tubes 7 and 48 can not be combined with the rest. At this time, tube 7 and tube 48 can not be combined with other tubes. At this time, tube 7 and tube 48 can not be combined as one tube.

d) Take 1 ml of blank control and combination tubes (when KD0.5 is in odd number, including the 7th and 48th tubes not combined), Add 5000 μ l hydrochloric acid solution of 0.1% FeCl3, mix well, add 400 μ l of 10% lichen phenol ethanol solution, mix them evenly, put them into ice water bath for 5 minutes after boiling water bath for 5 minutes, measure the absorbance at the wavelength of 670 nm of ultraviolet visible spectrophotometer, and draw the flow wash diagram.

c) Calculation method:

The sum of ribose contents in each tube (or the sum of absorbance A1) before KD value ≤ 0.5 and the sum of ribose contents (or the sum of absorbance a) of all tubes before KD value ≤ 0.5 were calculated according to the flow elution map drawn, so as to obtain the recovery rate of bulk polysaccharide with KD ≤ 0.5. Namely:

$\frac{\sum A_1}{\sum a} \leq 0.5$

$\frac{\sum A_1}{\sum a} \leq 0.5$ (1)

[illegible]

a) Sampling volume and collection method: with 0.45 μm membrane filtration for test products, for test ingenuity 1 ml column, with 0.2 mol/L sodium chloride solution flow wash, flow lot wash amount at least 1.2 times the volume of the column bed, automatic collector tube-by-tube collection, 4.5 ml per tube collection.

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c) Calculation method:

The area (or the sum of absorbance values A1) of the components with KD value less than

[illegible]

XXXXXXXXXX XXXXXX (XXXXXXXXXXXXXXXXXXXX XXXXXXXX XX) XXX XXXX XXXXXXXXXXXXXXX

a) Take 16 vials (bottles) of finished products, mix and remove, with 3K ultrafiltration cup, 5000 rpm centrifugation 30 min, concentrated to 1.5 ml, fully blown and mixed. Filtered with 0.45 µm membrane, 1 ml of the test product, flow washed with 0.2 mol/L sodium chloride solution, the amount of fluid lotion is at least 1.2 times the volume of the column bed, automatic collector tube-by-tube collection, 4.5 ml per tube collection.

c) Calculation method:

The recovery rate of haemophilus influenzae type b conjugate vaccine KD < 0.2 was obtained.

$\frac{1}{\sqrt{n}} \sum_{i=1}^n (\bar{X}_i - \mu) = O_p(1)$

XXXXXXXXXX XXXXXX (XXXX XXXXXXXXXXXXXXXX XXXXXXX X) XX XXXX XXXXXXXXXXXXXXX

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a) Sampling volume and collection method: with 0.45 µm membrane filtration for test products, for testing 1 ml column, with 0.2 mol/L sodium chloride solution flow wash, fluid lotion amount is at least 1.2 times the volume of the column bed, automatic collector tube-by-tube collection, each tube collection 4.5 ml.

c) Calculation method:

The area of the component peak of the KD value of 0.5 (or the sum of the absorbance value A1) is calculated according to the resulting flow map, divided by the total area of all component peaks (or the sum of the absorbance values A). The recovery rate of Group A meningococcal polysaccharides (bulk) KD $\leq$ 0.5 was obtained. That is:

□□□□ □□□□ (□□ □□□□□□□□ □□□□ □1) □□ □□□□□□□□ □□□□□□ □□ ≤ 0.5

XXXXXXXXXX (**XX XXXXXXXXXXXXXXXX XXXXXX X**) **XX XX**

a) Sampling volume and collection method: with 0.45 μm membrane filtration for test products, for testing 1 ml column, with 0.2 mol/L sodium chloride solution flow wash, fluid lotion amount is at least 1.2 times the volume of the column bed, automatic collector tube-by-tube collection, each tube collection 4.5 ml.

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control to mark zero. After zero, the absorbance of tube 8 to 48 was read. The number of collecting tubes was taken as the abscissa, and the absorbance value was used as the ordinate to draw the flow elution map. The flow washing volume V_e (flow washing volume $V_e = 4.5 \text{ ml} \times \text{tube number}$) from the sample adding time to the peak was calculated.

c) Calculation method:

The area of the component peak of the KD value of 0.2 (or sum A1 of the absorbance value) is calculated according to the drawn flow map, divided by the total area of all component peaks (or sum of absorbance values A), i.e. the recovery rate of Group A meningococcal polysaccharide conjugate $KD \leq 0.2$. That is:

$$\frac{\sum_{i=1}^n A_{0.2}}{\sum_{i=1}^n A_i} \times 100 \leq 0.2$$

6.3.3.6 The determination of molecular size of group C meningococcal polysaccharides (bulk):

a) Sampling volume and collection method: with $0.45 \mu\text{m}$ membrane filtration for test products, for testing 1 ml column, with 0.2 mol/L sodium chloride solution flow wash, fluid lotion amount is at least 1.2 times the volume of the column bed, automatic collector tube-by-tube collection, each tube collection 4.5 ml.

b) Determine the absorbance of the liquid collected by the automatic collector at the wavelength of 206 nm. The fifth and sixth tubes of the collected solution were mixed and used as blank control to mark zero. After zero, the absorbance of tube 8 to 48 was read. The number of collecting tubes was taken as the abscissa, and the absorbance value was used as the ordinate to draw the flow elution map. The flow washing volume V_e (flow washing volume $V_e = 4.5 \text{ ml} \times \text{tube number}$) from the sample adding time to the peak was calculated.

c) Calculation method:

Partition coefficient $KD = (V_e - V_0) / (V_i - V_0)$

The area of the component peak of the KD value of 0.5 (or the sum of the absorbance value A1) is calculated according to the resulting flow map, divided by the total area of all component peaks (or the sum of the absorbance values A). The recovery rate of Group A meningococcal Standard Operating Procedures of Yuxi Walvax Biotechnology Co., Ltd.

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polysaccharides (bulk) $KD \leq 0.5$ was obtained. That is:

$$\frac{\sum_{i=1}^n A_{0.5}}{\sum_{i=1}^n A_i} \times 100 \leq 0.5$$

6.3.3.7 The determination of molecular size of group C meningococcal bulk conjugate:

Sampling volume and collection method: with $0.45 \mu\text{m}$ membrane filtration for test products, for testing 1 ml column, with 0.2 mol/L sodium chloride solution flow wash, fluid lotion amount is at least 1.2 times the volume of the column bed, automatic collector tube-by-tube collection, each tube collection 4.5 ml.

b) Determine the absorbance of the liquid collected by the automatic collector at the wavelength of 206 nm. The fifth and sixth tubes of the collected solution were mixed and used as blank control to mark zero. After zero, the absorbance of tube 8 to 48 was read. The number of collecting tubes was taken as the abscissa, and the absorbance value was used as the ordinate to draw the flow elution map. The flow washing volume V_e (flow washing volume $V_e = 4.5 \text{ ml} \times \text{tube number}$) from the sample adding time to the peak was calculated.

c) Calculation method:

Partition coefficient $KD = (V_e - V_0) / (V_i - V_0)$

The area of the component peak of the KD value of 0.5 (or the sum of the absorbance value A1) is calculated according to the resulting flow map, divided by the total area of all component peaks (or the sum of the absorbance values A). The recovery rate of Group A meningococcal polysaccharides bulk conjugate $KD \leq 0.5$ was obtained. That is:

$$\frac{\sum_{i=1}^n A_{0.5}}{\sum_{i=1}^n A_i} \times 100 \leq 0.5$$

6.3.3.8 The determination of molecular size of group Y meningococcal polysaccharides (bulk):

a) Sampling volume and collection method: with $0.45 \mu\text{m}$ membrane filtration for test products, for testing 1 ml column, with 0.2 mol/L sodium chloride solution flow wash, fluid lotion amount is at least 1.2 times the volume of the column bed, automatic collector tube-by-tube collection, Standard Operating Procedures of Yuxi Walvax Biotechnology Co., Ltd.

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each tube collection 4.5 ml.

b) Determine the absorbance of the liquid collected by the automatic collector at the wavelength of 206 nm. The fifth and sixth tubes of the collected solution were mixed and used as blank control to mark zero. After zero, the absorbance of tube 8 to 48 was read. The number of collecting tubes was taken as the abscissa, and the absorbance value was used as the ordinate to draw the flow elution map. The flow washing volume V_e (flow washing volume $V_e = 4.5 \text{ ml} \times \text{tube number}$) from the sample adding time to the peak was calculated.

c) Calculation method:

Partition coefficient $KD = (V_e - V_0) / (V_i - V_0)$

The area of the component peak of the KD value of 0.5 (or the sum of the absorbance value A1) is calculated according to the resulting flow map, divided by the total area of all component peaks (or the sum of the absorbance values A). The recovery rate of Group A meningococcal polysaccharides (bulk) $KD \leq 0.5$ was obtained. That is:

$$\frac{\sum_{i=1}^n A_{0.5}}{\sum_{i=1}^n A_i} \times 100 \leq 0.5$$

= × 100

■■■■■ ■■■■ (■■ ■■■■■■■■■■ ■■■■ ■) ■■ ■■■ ■■■■■■■■■■

6.3.3.9 The determination of molecular size of group W135 meningococcal polysaccharides (bulk):

- a) Sampling volume and collection method: with 0.45 μm membrane filtration for test products, for testing 1 ml column, with 0.2 mol/L sodium chloride solution flow wash, fluid lotion amount is at least 1.2 times the volume of the column bed, automatic collector tube-by-tube collection, each tube collection 4.5 ml.
- b) Determine the absorbance of the liquid collected by the automatic collector at the wavelength of 206 nm. The fifth and sixth tubes of the collected solution were mixed and used as blank control to mark zero. After zero, the absorbance of tube 8 to 48 was read. The number of collecting tubes was taken as the abscissa, and the absorbance value was used as the ordinate to draw the flow elution map. The flow washing volume V_e (flow washing volume $V_e = 4.5 \text{ ml} \times \text{tube number}$) from the sample adding time to the peak was calculated.
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size

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c) Calculation method:

Partition coefficient $KD = (V_e - V_0) / (V_i - V_0)$

The area of the component peak of the KD value of 0.5 (or the sum of the absorbance value A1) is calculated according to the resulting flow map, divided by the total area of all component peaks (or the sum of the absorbance values A). The recovery rate of Group A meningococcal polysaccharides (bulk) $KD \leq 0.5$ was obtained. That is:

■■ ≤ 0.5 ■■■■■■■■ ■■ ■■■■■■■■■■

■■■■ ■■■■ (■■ ■■■■■■■■■■ ■■■■ ■1) ■■ ■■■■■■■■■■ ■■■■■ ■■ ≤ 0.5

= × 100

■■■■■ ■■■■ (■■ ■■■■■■■■■■ ■■■■ ■) ■■ ■■■ ■■■■■■■■■■

6.3.3.10 The determination of molecular size of group A and C meningococcal polysaccharide vaccine:

- a) Sampling volume and collection method: take 20 bottles of group A and C meningococcal polysaccharide vaccine lyophilizer, with 20 bottles of corresponding diluents after removal and mixing, with 3K ultrafiltration cup, 5■■, 7000 rpm centrifugal 30 min (repeated centrifugation twice), concentrated to 2 ml, fully mixed, that is, to be used for testing. With 0.45 m membrane filtration for test products, for test ingenuity 1 ml column, with 0.2 mol/L sodium chloride solution to elude, the amount of fluid lotion at least 1.2 times the volume of the column bed, automatic collector tube-by-tube collection, each tube collection 4.5 ml.
- b) The determination of molecular size of group A: automatically collect liquid fifth, sixth pipes and mix together and take 1ml as blank for care, from eighth tubes to 48 tubes, each collection of 1ml collection solution, add 150 μl of mineralized reagent to each tube, mix it up, cover the test tube with aluminum foil, place the test tube in an electrothermal constant temperature blower oven, adjust the temperature to 150■■, and dry the oven for 1 hours after the temperature arrives, adjust the temperature. After the temperature reached 180■■, dry bake for 30 minutes, adjust the temperature to 250■■, dry bake for 1 hour after the temperature reached, so as to mineralize and inorganic the test sample, and then add 1850 μl purified water while hot. Add 2000 μl chromogenic agent into each tube, mix well, and soak in water for 2 hours in 37 ■■ electric constant temperature water bath. Detect OD value of each tube at 825 nm wavelength, Standard Operating Procedures of Yuxi Walvax Biotechnology Co., Ltd.

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draw flow washing map with tube number as abscissa and OD value as ordinate.

Detection of molecular size of group C: the 5th and 6th tubes of the automatic collection liquid were combined and evenly mixed, and 2 ml was taken as the blank control tube. From the 8th tube to the 48th tube, 2 ml of collecting solution was taken from each tube, and 2000 μl resorcinol chromogenic agent was added into each tube. After mixing, the solution was placed in boiling water bath for 15 minutes, and then put into ice water for 10 minutes The OD value of each tube was detected at 585 nm. The flow elution map was drawn with the number of tubes as abscissa and OD value as ordinate.

c) Calculation method:

Partition coefficient $KD = (V_e - V_0) / (V_i - V_0)$

■■ ≤ 0.5 ■■■■■■■■ ■■ ■■■■ ■ ■■■■■■■■■■

■■■■ ■■ ■■■■■■■■■■ ■■■■■■■■ (■■ ■■■■■■■■■■ ■■■■■) ■■ ■■■■■■■■■■ ■■■■■ ■■ ■■■■ ■■■■ ■■■■ ■■ 0.5

=

■■■■ ■■■■ ■■ ■■■■■■■■■■ ■■■■■■■■ (■■ ■■■■■■■■■■ ■■■■■) ■■ ■■■■■■■■■■ ■■■■■ ■■ ■■■■ ■■■■ ■■■■ ■■ ■■■■ ■■■■ ■■■■ ■■

× 100

■■ ≤ 0.5 ■■■■■■■■ ■■ ■■■■ ■ ■■■■■■■■■■

■■■■ ■■ ■■■■■■■■■■ ■■■■■■■■ (■■ ■■■■■■■■■■ ■■■■■) ■■ ■■■■■■■■■■ ■■■■■ ■■ ■■■■ ■■■■ ■■■■ ■■ 0.5

=

■■■■ ■■■■ ■■ ■■■■■■■■■■ ■■■■■■■■ (■■ ■■■■■■■■■■ ■■■■■) ■■ ■■■■■■■■■■ ■■■■■ ■■ ■■■■ ■■■■ ■■■■ ■■ ■■■■ ■■■■ ■■■■ ■■

× 100

6.3.3.11 The determination of molecular size of group ACYW135 meningococcal polysaccharide vaccine:

- a) Samples and collection methods: Take 10 group ACYW135 meningococcal polysaccharide vaccine lyophilizer, and dissolve 2 ml with 4 corresponding diluents. With 0.45 m membrane filtration, for test ingenuity 1 ml column, with 0.2 mol/L sodium chloride solution eluted, the amount of fluid lotion at least 1.2 times the volume of the column bed, automatic collector tube-by-tube collection, 4.5 ml per tube collection.
- b) Liquid collected by tube by automatic collector, measured the height or concentration of each rocket peak in accordance with the Rocket Immuno-Electrophoretic Operating Procedure, with the number of tubes as horizontal coordinates, and the height of the rocket peak or concentration

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as the ordinate of the flow wash map.

c) Calculation method:
Partition coefficient $KD = (V_e - V_o) / (V_i - V_o)$
 $KD \leq 0.5$ $\frac{\text{Peak area of } KD \leq 0.5}{\text{Total area of } KD \leq 0.5} \times 100$
=
 $\frac{\text{Peak area of } KD \leq 0.5}{\text{Total area of } KD \leq 0.5} \times 100$
 $\times 100$

6.3.3.12 The determination of molecular size pneumococcal polysaccharide (bulk):
a) Sampling volume and collection method: take pneumococcal polysaccharide (bulk) with 0.45 m membrane filtration, take the test 1 ml column, with 0.9% sodium chloride injection elution, fluid lotion amount at least 1.2 times the volume of the column bed, automatic collector tube-by-tube collection, 4.5 ml per tube collection.

b) The liquid collected by the automatic collector, 5th and 6th tubes, is combined and mixed and mixed to take 1 ml as blank care, from the 11th tube to the 40th tube, 1 ml per tube, 1ml. Add 4000 μ l ketone mixture in each tube, mix well, boil in the boiling water bath in the water bath for 20 minutes, remove and put in the 40 degrees C water bath for 10 minutes, measure the absorbance at 620 nm, with the number of tubes for horizontal coordinates, the corresponding absorbance for the coordinates of the polysaccharide molecular spectrum, to find the corresponding number of tubes, calculate the peak volume of the corresponding volume V_e at the peak (flow washing volume $V_e = 4.5\text{ml} \times \text{tube number}$).

c) How to calculate:
Partition coefficient $KD = (V_e - V_o) / (V_i - V_o)$
According to the different types, the recoveries were calculated according to the following requirements: stereotype 1, 2, 3, 4, 7F, 8, 9N, 12F, 14, 18C, 19F, 23F of pneumococcal polysaccharides (bulk) to calculate the recovery rate of $KD < 0.3$; stereotype 19A of pneumococcal polysaccharide (bulk) calculated $KD < 0.5$ rate of recovery; stereotype 5, 6A, 6B, 9V, 10A, 11A, 15B, 17F, 20, 22F, 33F of pneumococcal polysaccharide (bulk) calculated $KD < 0.6$ rate of recovery.
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0.6 rate of recovery. According to the molecular map of the polysaccharide (bulk) of this type of pneumococcus, the sum of the contents (or absorbance) of the polysaccharide with $KD < 0.3$, < 0.5 and < 0.6 was calculated and divided by the sum of the contents (or absorbance) of all the polysaccharides (or absorbance) of the type of pneumococcal polysaccharide (bulk), the recoveries of $KD < 0.3$, < 0.5 and < 0.6 were obtained. Namely:

$KD < 0.3$ $\frac{\text{Peak area of } KD < 0.3}{\text{Total area of } KD < 0.3} \times 100$
=
 $\frac{\text{Peak area of } KD < 0.3}{\text{Total area of } KD < 0.3} \times 100$
 $\times 100$

$KD < 0.5$ $\frac{\text{Peak area of } KD < 0.5}{\text{Total area of } KD < 0.5} \times 100$
=
 $\frac{\text{Peak area of } KD < 0.5}{\text{Total area of } KD < 0.5} \times 100$
 $\times 100$

$KD < 0.6$ $\frac{\text{Peak area of } KD < 0.6}{\text{Total area of } KD < 0.6} \times 100$
=
 $\frac{\text{Peak area of } KD < 0.6}{\text{Total area of } KD < 0.6} \times 100$
 $\times 100$

6.3.3.13 The determination of molecular size of pneumococcal polysaccharide derivative:
a) Sampling volume and collection method: take pneumococcal polysaccharide derivatives with 0.45 m membrane filtration, take the test 1 ml column, wash away with 0.9% sodium chloride injection, the amount of fluid lotion is at least 1.2 times the volume of the column bed, automatic collector tube collection, 4.5 ml per tube collection.

b) The liquid collected by the automatic collector, 5th and 6th tubes, is combined and mixed and mixed to take 1 ml as blank care, from the 11th tube to the 40th tube, 1 ml per tube, 1ml. Add 4000 μ l ketone mixture in each tube, mix well, boil in the boiling water bath in the water bath for 20 minutes, remove and put in the 40 degrees C water bath for 10 minutes, measure the absorbance at 620 nm, with the number of tubes for horizontal coordinates, the corresponding absorbance for the coordinates of the polysaccharide molecular spectrum, to find the corresponding number of tubes, calculate the peak volume of the corresponding V_e volume
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V_e at the peak (flow washing volume $V_e = 4.5\text{ml} \times \text{tube number}$).

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8.3 Wash the gel with a column volume of 0.5 mol/L sodium hydroxide solution per month as a flow phase, and use phase after cleaning. The delude should be cleaned, until the pH of the outflow fluid after cleaning is less than 7 and then the gel column is demarcated.

8.4 If the self-matching eraser must be degassed by steam sterilization or suitable methods (e.g. negative pressure).

8.5 The collection pipe of the branch collector shall be marked with a marker to prevent confusion.

8.6 Tomography column standard, free protein control numbering principle:
To A, B, C, D... As a fixed number for the tomography column, the date on which the bar is marked is a numeric number, if the A-pillar is standard on August 05, 2018, the number is A-pillar 201808005, which is valid for 30 days and is valid until September 03, 2018.
The device number of the AKTA pure25 purifier system is digitized by the date of the start of the bar: if The WYQCF0455 bar is on August 05, 2018, the number is WYQCF455 201808005, which is valid for 30 days and is valid until September 03, 2018.

8.7 Each test tube used to collect eluent on the step-by-step collector needs to be numbered, and when the eluate content of each tube is determined, the test tube containing the reaction solution also needs to be numbered to ensure the accuracy of molecular size detection results.

8.8 Under 260nm, the blue glucan 2000 solution elution peak (first peak) top corresponds to the Standard Operating Procedures of Yuxi Walvax Biotechnology Co., Ltd.

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volume of the post V0. At 370 nm, the volume of the vitamin B12 solution eludes the peak (peak) top corresponding to the Vi of the post.

8.9 from the addition of samples to polysaccharide peak flow wash volume Ve flow wash volume Ve 4.5 ml, the number of tubes (if there are two peaks, take the second peak corresponding tube number).

8.10 About the standard of column chromatography calibration: V0, Vi, the judgment standard of calibration consistency in different time periods. It should be no more than 1 tube, that is, the chromatographic column V0, Vi calibrated continuously before and after two times, and the number of detection tubes should not move forward and backward more than 1 tube. If it does not meet the requirements, analyze the cause and conduct investigation.

8.11 Gel column filling: If the refill gel column, to be filled, after the completion of calibration, the last sample before the gel column filling needs to be retested, the two test results will be compared, KD shall not exceed 0.06, the recovery rate should not exceed 10%, if the requirements are not met to analyze the reasons, investigate.

9 References or standard basis

9.1 Chinese Pharmacopoeia, volume III, General rule 3419, China Pharmaceutical Science and Technology Press, 2015

10 Revision history

Revision Basis, reasons and content of the

Version Effective

Departme amendment Note

No. time

nt

QC File expiration, change

03 2011.12.01

Laboratory Improve part of the file content

QC

04 2014.12.01 Improve part of the file content

Laboratory

QC

05 2016.11.16 Improve part of the file content

Laboratory

QC

06 2017.10.01 Improve part of the file content

Laboratory

QC Increase the recovery rate of

07 2018.09.01

Laboratory pneumococcal polysaccharide

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polysaccharides (bulk), pneumococcal
polysaccharide derivatives, and
increase the recovery rate of
degradation polysaccharides testing
projects for the manufacture of 13-
valent pneumococcal polysaccharides
Increase the numbering of each test
tube used to collect the elution on the
QC step-by-step collector, number the test

08 2019.02.11

Laboratory tube to undertake the test tube, and
describe the peak pick-up position in
detail.

Increase the recovery rate of degraded
QC polysaccharide molecules, revise the

09 2019.06.19

Laboratory range of molecular size recovery rate
of polysaccharide derivatives
Describe the specific description of
QC the description of the reference to

10 2019.07.31
Laboratory other operating procedures as actual
operation
QC Increase gel column filling
11 2019.09.25
Laboratory management
Increase 0.45 µm membrane
QC
12 2019.10.31 specification
Laboratory
Add spreadsheets

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Code WSYX-QC-SOP-06-09-0003

Conjugate antigen content

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determination SOP

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1 Objective

Standardize operating procedure for test of conjugate antigen content in 13-valent Pneumococcal conjugate vaccine.

2 Definition/abbreviation

N/A

3 Applicable scope

It applies to testing conjugate antigen content of 13-valent pneumococcal conjugate vaccine.

4 Responsibility

All operators should in strict accordance with this SOP operation.

5 Equipment and Materials

5.1 Equipment

Equipment

No. Equipment name Manufacturer Notes

model

Specific protein IMAGE BECKMAN Or equivalent

1

Analysis System 800 COULTER equipment

CR21G/Multi HITACHI/Therm Or equivalent

2 Centrifuge

fuge X1R o Scientific equipment

BS124S/BSA Or equivalent

3 Electronic balance Sartorius

124S equipment

5.2 Using reagents

No. Reagents Name Grade Notes

1 Diluent 1 — /

2 Buffer 1 — /

3 Wash solution — /

4 Pneumococcal polysaccharide Standard substance — /

5 Anti-serum of pneumococcal polysaccharide — /

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Conjugate antigen content

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6 Sodium hydroxide AR /

7 Citric acid AR /

8 Sodium deoxycholate — /

9 Hydrochloric acid AR /

10 Sodium Chloride AR /

11 Absolute ethyl alcohol AR /

5.3 Use of materials

No. Material name No. Material name

1 Sample rack 2 Sample Cups

3 Kit 4 0.6ml/2ml/15ml/50ml Centrifuge Tube

5 100-1000ul Pipette 6 20-200ul Pipette

7 500-5000ul Pipette 8 5000ul/1000ul/200ul Pipette tip

6 Working procedure

6.1 Principle

6.1.1 Method one: this method uses high concentration salt and high concentration ethanol to precipitate free polysaccharide and conjugate glycoprotein in low temperature environment, and then add water to redissolve free polysaccharide. Conjugate glycoprotein cannot be redissolved due to irreversible denaturation of protein, so the purpose of separation of conjugate glycoprotein and free polysaccharide can be realized.

6.1.2 Method two: sodium deoxycholate forms protonated micelles under acidic conditions. With the increase of hydrogen ion concentration in the solution, protonated micelles gradually increase through the hydrogen bond of acid salt structure and finally form stable high-grade micelles. The micelles have the characteristics of protein adsorption, so the purpose of separation of bound glycoproteins and free polysaccharides can be realized.

6.2 Reagent preparation

6.2.1 5mol/L sodium chloride solution

Accurately weigh 14.61g of sodium chloride, put it into a 50ml volumetric flask, add a proper amount of purified water to dissolve and volume it to the scale, and mix it evenly.

6.2.2 1% sodium deoxycholate solution

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b) Add 50ul of 1mol/L hydrochloric acid solution into each centrifuge tube. mix well, and

place at 2-8℃ for 30min.

c) Centrifuge the tubes at 8000g, 5min for 30min.

d) Carefully take 800μl of supernatant using a pipette respectively, which is recorded as DS and YS.

6.3.3.2 Determination of polysaccharide content

a) Take 300μl of solution D, add 900μl of diluent 1 (DIL1), mix well to obtain 4 times dilute solution, determine the polysaccharide content on the IMMAGE 800, input 4.4 (i.e. 4×1.1) of off-line dilution multiple. If the result is beyond the inspection range, adjust the dilution multiple to A, and input A×1.1 of off-line dilution multiple.

b) Take 300μl of solution Y, add 60μl of diluent 1 (DIL1), mix well to obtain 3 times dilute solution, determine the polysaccharide content on the IMMAGE 800, input 3.3 (i.e. 3×1.1) of off-line dilution multiple.

c) Take 400μl of DS solution, add 800μl of diluent 1 (DIL1), mix well to obtain 3 times dilute solution, and determine polysaccharide content on the IMMAGE 800. If the result is beyond the detection range, adjust the dilution multiple. The final result is calculated SOP of Yuxi Walvax Biotechnology Co., Ltd.

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manually by the tester (i.e. test result × dilute multiple).

d) Take solution YS and directly put it on the IMMAGE 800 to determine the polysaccharide content. The final result is calculated manually by the tester (i.e. test result × dilution multiple).

6.3.3.3 Result calculation
Calculate the conjugate antigen content and test validity as following formula:

$$H = (1 - P) \times 100\%$$

$$H = \frac{A}{B} \times 100\%$$

In the formula,
H: the conjugate antigen content of test samples, %;
P: the test validity, which shall be 70-130%.

7 Flow chart

N/A

8 Matters need attention

8.1 In the process of free polysaccharide separating from 13-valent pneumococcal polysaccharide vaccine samples(except type 3), the supernatant in the centrifuge tube should be directly poured out after centrifuging at 6000g, 5min for 90min, and the supernatant in the centrifuge tube should be completely sucked by inverted on the absorbent paper.

8.2 When removing supernatant from the centrifuge tube, pay special attention that the pipette tip is not allowed to touch the tube wall, and it is not allowed to take precipitate or floating substance.

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9 Reference or standard basis

9.1 The production and test procedure for 13-valent pneumococcal conjugate vaccine

10 Revise history

Time in Revision
Version Revision basis, reason and content Notes
force Dept.
13 Feb. QC
00 Complete the content
2019 Laboratory
20 Jun. QC Complete the content and adjust the
01
2019 Laboratory sequence of operation steps

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SOP of Polysaccharide Version 05 Pages 1/2

Protein Ratio

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1 Purpose

The method of determining polysaccharide protein ratio is standardized.

2 Definitions/abbreviations

No

3 Scope of application

This SOP is applicable to the free protein determination of Haemophilus Influenzae Type b bulk Conjugate, pneumococcal polysaccharide bulk conjugate, Group A meningococcal polysaccharide bulk conjugate, Group C meningococcal polysaccharide bulk conjugate.

4 Responsibilities

All operators operate strictly in accordance with this SOP.

5 Materials and equipment

No

6 Work procedures

6.1 Experimental steps

The determination method of polysaccharide content was carried out according to the corresponding method of polysaccharide content of each variety.

The protein content was determined according to the operation procedure for protein content determination. (Code: WSYX-QC-SOP-0009-0102).

6.2 Result judgment

Polysaccharide content

Polysaccharide protein ratio =

Protein content

7 Flow chart

N/A

8 Precautions

No

9 References or standard basis

No

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Protein Ratio

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10 Revision history

Basis, reasons and content of the
Version Release Effective Revision
amendment Note

No. time time Department

QC File expiration, change

03 2011.11.01 2011.12.01

Laboratory Improve part of the file content

QC

04 2014.11.01 2014.12.01 Improve part of the file content

Laboratory

MODULE 3 QUALITY
Version No. Walvax-202007-00 Yuxi Walvax Biotechnology Co., Ltd.

PRODUCT REGIDTRATION DOSSIER

13-valent Pneumococcal Polysaccharide Conjugate Vaccine
(Tetanus Toxoid Protein)

MODULE 3 QUALITY

Yuxi Walvax Biotechnology Co., Ltd.

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MODULE 3 QUALITY
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3.2 BODY OF DATA

INTRODUCTION

The Applicant has developed the 13-valent Pneumococcal Polysaccharide Conjugate Vaccine (Weuphoria), for use in infants and young children to prevent pneumococcal disease, caused by the 13 pneumococcal serotypes contained in the vaccine. Weuphoria is a sterile liquid formulation of pneumococcal capsular polysaccharides of serotypes 1, 3, 4, 5, 6A, 6B, 7F, 9V, 14, 18C, 19A, 19F, and 23F, individually conjugated to tetanus toxoid. Phosphate-buffered Saline is also included in the Weuphoria formulation to improve process control and to provide further pH control following the addition of aluminum phosphate.

Manufacturer: Yuxi Walvax Biotechnology Co., Ltd.

Common name of drug product: 13-valent Pneumococcal Polysaccharide Conjugate Vaccine.

This vaccine is produced by culture, extraction and purification of polysaccharide antigens of 13 epidemic and infectious Streptococcus pneumoniae of serotypes of 1, 3, 4, 5, 6A, 6B, 7F, 9V, 14, 18C, 19A, 19F and 23F. After being activated and derived, the polysaccharide conjugates with tetanus toxoid carrier protein to form polysaccharide protein conjugate, which is prepared by mixing with a certain proportion and adding aluminum phosphate adjuvant. The vaccine is milky even suspension after shaking. After storage, white precipitates and supernatant can be seen due to the subsidence of adjuvant. No preservatives contained.

Table 1 Product Composition (each dose/0.5mL contains)

Active Substances	Nominal Composition	Function
Pneumococcal polysaccharide serotype 1	2.6 µg	Antigen
Pneumococcal polysaccharide serotype 3	2.5 µg	Antigen
Pneumococcal polysaccharide serotype 4	3.0 µg	Antigen
Pneumococcal polysaccharide serotype 5	2.5 µg	Antigen
Pneumococcal polysaccharide serotype 6A	2.5 µg	Antigen
Pneumococcal polysaccharide serotype 6B	6.0 µg	Antigen
Pneumococcal polysaccharide serotype 7F	2.85 µg	Antigen
Pneumococcal polysaccharide serotype 9V	2.5 µg	Antigen

Pneumococcal polysaccharide serotype 14 2.75 µg Antigen
Pneumococcal polysaccharide serotype 18C 3.25 µg Antigen
Pneumococcal polysaccharide serotype 19A 2.6 µg Antigen
Pneumococcal polysaccharide serotype 19F 2.75 µg Antigen
Pneumococcal polysaccharide serotype 23F 3.0 µg Antigen
Aluminum phosphate 0.5 mg Adjuvant
Sodium chloride 4.25 mg Excipient
Sodium dihydrogen phosphate 44.35 µg Excipient
Disodium hydrogen phosphate 19.0 µg Excipient
Water for injection Qs to 0.5 mL Excipient

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Proprietary name:

Route of administration: The vaccine should be shaken well immediately prior to use, and be given intramuscularly. The preferred sites for injection are the anterolateral aspect of the thigh in infants and the deltoid muscle of the upper arm in toddlers and children. Each dose (0.5 mL) is to be injected. The vaccine should not be injected in and/or near the areas where there may be a nerve trunk and/or blood vessel.

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3.2.S DRUG SUBSTANCE

3.2.S.1 General Information

3.2.S.1.1 Nomenclature

Table 2 Name of each Serotype of Pneumococcal Polysaccharide

Serotypes Name Chemical Name

Pneumococcal

-4-(D-galacturonopyranosyl-1-α)-3-(D-galacturonopyranosyl-1-α)-3-(2-1 polysaccharide

acetamido-4-amino-2,4,6-trideoxy-D-galactopyranosyl-1-α) residues serotype 1

Pneumococcal

3 polysaccharide -3-(D-glucuronopyranosyl-1-β)-4-(D-glucopyranosyl-1-β) residues serotype 3

-3-(D-2-acetamido-2-deoxy-D-mannopyranosyl-1-β)-3-(L-2-acetamido-

Pneumococcal

2-deoxy-D-fucopyranosyl-1-α)-3-(D-2-acetamido-2-deoxy-D-4 polysaccharide

galactopyranosyl-1-α)-4-(D-2,3-S-pyruvyl-galactopyranosyl-1-α) serotype 4

residues

-4-(D-glucopyranosyl-1-β)-4-[L-2-acetamido-2-deoxy-3-(2-acetamido-

Pneumococcal

2,6-dideoxy-L-talopyranosyl-1-α-2-D-glucuronopyranosyl-1-β)-5 polysaccharide

fucopyranosyl-1-α]-3-(D-2-acetamido-2,6-dideoxy-D-xylo-serotype 5

hexopyranos-4-ulosyl-1-β) residues

Pneumococcal

2-(D-galactopyranosyl-1-α)-3-(D-glucopyranosyl-1-α)-3-(L-6A polysaccharide

rhamnopyranosyl-1-α)-3-(D-ribitol-1-phosphate) residues serotype 6A

Pneumococcal

-2-(D-galactopyranosyl-1-α)-3-(D-glucopyranosyl-1-α)-3-(L-6B polysaccharide

rhamnopyranosyl-1-α)-4-(D-ribitol-5-phosphate) residues serotype 6B

-6-[D-2-(galactopyranosyl-1-β)-galactopyranosyl-1-α]-3-(L-2-O-

Pneumococcal

acetyl-rhamnopyranosyl-1-β)-4-(D-glucopyranosyl-1-β)-3-[D-2-7F polysaccharide

acetamido-2-deoxy-4-(2-acetamido-2-deoxyglucopyranosyl-1-α)-2-(L-serotype 7F

rhamnopyranosyl-1-β)-galactopyranosyl-1-β] residues

Pneumococcal -4-(D-glucopyranosyl-1-β)-4-(D-glucuronopyranosyl-1-α)-3-(D-9V polysaccharide

galactopyranosyl-1-α)-3-(D-2-acetamido-2-deoxy-mannopyranosyl-1-serotype 9V β)-4-(D-glucopyranosyl-1-β) residues

Pneumococcal -4-(D-glucopyranosyl-1-β)-6-[D-2-acetamido-2-deoxy-4-(D-14 polysaccharide

galactopyranosyl-1-β)-glucopyranosyl-1-β]-3-(D-galactopyranosyl-1-β) serotype 14 residues

Pneumococcal -4-(D-glucopyranosyl-1-β)-4-[(2-glucopyranosyl-1-α-3-18C polysaccharide

glycerophospho)-D-galactopyranosyl-1-β]-4-(D-glucopyranosyl-1-α)-3-serotype 18C (L-rhamnopyranosyl-1-α) residues

Pneumococcal

-4-(D-2-acetamido-2-deoxy-D-mannopyranosyl-1-β)-4-(D-19A polysaccharide

glucopyranosyl-1-α)-3-(L-rhamnopyranosyl-1-α-phosphate) residues serotype 19A

Pneumococcal

-4-(D-2-acetamido-2-deoxy-D-mannopyranosyl-1-β)-4-(D-19F polysaccharide

glucopyranosyl-1-α)-2-(L-rhamnopyranosyl-1-α-phosphate) residues serotype 19F

Pneumococcal -4-(D-glucopyranosyl-1-β)-4-[D-2-(L-rhamnopyranosyl-1-α)-3-

23F polysaccharide (glycerophospho)-galactopyranosyl-1-β]-4-(L-rhamnopyranosyl-1-β)
serotype 23F residues

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3.2.S.1.2 Structure
1. Streptococcus Pneumoniae serotype 1 was obtained from CMCC (No. 31404)
For pneumococcal polysaccharide serotype 1:
The galacturonopyranosyl residues are partially O-acetylated. Each mole of the saccharide repeat unit carries a non-stoichiometric amount of O-acetyl groups per repeating unit.

* O
4 6 COOH
O HO HOOC
6 O
5 4 H2N H3C 6
HO 2 5
2 O
4 2
1 O 5 1
3 OH 3 OH 1 O 3 NHAC
*

n
Molecular size: determined by gel filtration through Sepharose CL-4B, release specification is
KD ≤ 0.15, recovery rate ≥ 75% before KD value reached 0.3.

2. Streptococcus Pneumoniae serotype 3 was obtained from CMCC (No. 31441)
For pneumococcal polysaccharide serotype 3:

OH
4 HOOC 6
6 O 4
O
HO 2 2
* 5
O 5 *
O 3 1 HO
OH 3 1
OH
n

Molecular size: determined by gel filtration through Sepharose CL-4B, release specification is
KD ≤ 0.15, recovery rate ≥ 78% before KD value reached 0.3.

3. Streptococcus Pneumoniae serotype 4 was obtained from CMCC (No. 31448)
For pneumococcal polysaccharide serotype 4:

HO
4 6 OH 5
H3C
4 OH
HO O 4 3 1
O 6 O
* 5 NHAc O
HO 5
O O NHAc 2 1
3 2 1 2 3 NHAC
O OH
6
4
O
2
5
O 1
3
O
*
H3C n
COOH

Molecular size: determined by gel filtration through Sepharose CL-4B, release specification is
KD ≤ 0.15, recovery rate ≥ 73% before KD value reached 0.3.

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4. Streptococcus Pneumoniae serotype 5 was obtained from CMCC (No. 31457)
For pneumococcal polysaccharide serotype 5:

6 4
H3C 5
OH 6 CH3
4 1 O
4 6
O O O
2 3
* O 5 O O
HO O 2 NHAc 2 1
3 OH 1 3
NHAc
O
HOOC 6 2
1

*
4 5 n
3
O
CH3
HO 1 3
2

NHAC 5 OH
O
6 4
HO
H3C

Molecular size: determined by gel filtration through Sepharose CL-2B, release specification is
KD ≤ 0.60, recovery rate ≥ 70% before KD value reached 0.6.

5. Streptococcus Pneumoniae serotype 6A was obtained from CMCC (No. 31476)
For pneumococcal polysaccharide serotype 6A:
HO
4 6
OH

HO O
5 2
3
4 OH
HO 6
1
5 O
2 OH
O O
* 3 H3C
4 6
OH 1
5 O P O *
O 3
O OH HO
OH 1
2 O
n
OH

OH

Molecular size: determined by gel filtration through Sepharose CL-2B, release specification is
KD ≤ 0.50, recovery rate ≥ 50% before KD value reached 0.6.

6. Streptococcus Pneumoniae serotype 6B was obtained from CMCC (No. 31492)
For pneumococcal polysaccharide serotype 6B:
HO

4
OH
6
HO O
5 2
3
4 OH
6
1 HO
5 O
2 OH
O
* 3 H3C
4 6
OH 1
5 O
O 3
O O P O *
OH 1
2
O HO
n
OH
OH

OH

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Molecular size: determined by gel filtration through Sepharose CL-2B, release specification is
KD ≤ 0.50, recovery rate ≥ 85% before KD value reached 0.6.

7. Streptococcus Pneumoniae serotype 7F was obtained from CMCC (No. 31510)
For pneumococcal polysaccharide serotype 7F:
OH
OH
HO H3C
O

HO HO
HO
O H3C NHAc O
O
* HO O
O OH
O O O
O OH *
HO
O CH4
AcO
O NHAc
O O
HO OH HO OH
HO

HO

n

Molecular size: determined by gel filtration through Sepharose CL-4B, release specification is
KD ≤ 0.20, recovery rate ≥ 70% before KD value reached 0.3.

8. Streptococcus Pneumoniae serotype 9V was obtained from CMCC (No. 31547)

For pneumococcal polysaccharide serotype 9V:

OH
O
* O
AcO HOOC
OH
O HO
O OH
O OAc
AcO O OH
OH *
HO NHAc O
O O
OH
O
HO OH
n

Molecular size: determined by gel filtration through Sepharose CL-2B, release specification is
KD ≤ 0.45, recovery rate ≥ 80% before KD value reached 0.6.

9. Streptococcus Pneumoniae serotype 14 was obtained from CMCC (No. 31605)

For pneumococcal polysaccharide serotype 14:

HO
OH
OH
O
O O *

* O
O
HO O OH
HO OH
OH
O O NHAc
HO
HO OH

n

Molecular size: determined by gel filtration through Sepharose CL-2B, release specification is
KD ≤ 0.30, recovery rate ≥ 55% before KD value reached 0.3.

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10. Streptococcus Pneumoniae serotype 18C was obtained from CMCC (No. 31689)

For pneumococcal polysaccharide serotype 18C:

O OH
O
O OH
* HO OH OH
O O O H3C
OH
O HO
HO
O OH OH O
P
OAc O
O O OH
O
HO OH
*
OH
HO
n

Molecular size: determined by gel filtration through Sepharose CL-4B, release specification is
KD ≤ 0.15, recovery rate ≥ 70% before KD value reached 0.3.

11. Streptococcus Pneumoniae serotype 19A was obtained from CMCC (No. 31708)

For pneumococcal polysaccharide serotype 19A:

OH
O
OH
AcHN O OH
* O H3C
O
HO
HO OH
O
O O
HO O P *
HO

n

Molecular size: determined by gel filtration through Sepharose CL-4B, release specification is
KD ≤ 0.45, recovery rate ≥ 80% before KD value reached 0.5.

12. Streptococcus Pneumoniae serotype 19F was obtained from CMCC (No. 31696)

For pneumococcal polysaccharide serotype 19F:

OH

OH H3C
O
OH
AcHN O
HO
* O
O
HO O
HO OH O

O
O
P *

OH

n

Molecular size: determined by gel filtration through Sepharose CL-4B, release specification is
KD ≤ 0.20, recovery rate ≥ 75% before KD value reached 0.3.

13. Streptococcus Pneumoniae serotype 23F was obtained from CMCC (No. 31761)

For pneumococcal polysaccharide serotype 23F:

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OH
O
OH
O
O
* H3C
HO OH O O O
*
HO P O HO O
O
O OH

HO OH O
OH

OH
H3C

HO n

Molecular size: determined by gel filtration through Sepharose CL-4B, release specification is
KD ≤ 0.15, recovery rate ≥ 85% before KD value reached 0.3.

3.2.S.1.3 General Properties

Appearance of each serotype of polysaccharide: Clear to slightly turbid, colorless to pale yellow liquid.

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3.2.S.2 Manufacture

3.2.S.2.1 Manufacturer(s)

Name: Yuxi Walvax Biotechnology Co., Ltd.

Address: No. 83 South Dongfeng Road, High ■ New Technology Industries Development Zone, Yuxi, Yunnan Province, China

Function: Vaccine manufacture (All manufacturing procedures)

Tel: 0877-2076210

Fax: 0877-2076918

Manufacturing site: 4th floor, pneumococcal vaccine workshop, J building 301# (vaccine building); 1st floor, Tetanus vaccine workshop, K building 201#, No. 83 South Dongfeng Road, High ■ New Technology Industries Development Zone, Yuxi, Yunnan Province, China

Tel: 0877-2076210

Fax: 0877-2076918

QC site: 3rd and 4th floor, QC laboratory, G building 308# (Research & QC building), No. 83 South Dongfeng Road, High ■ New Technology Industries Development Zone, Yuxi, Yunnan Province, China
Tel: 0877-2076210
Fax: 0877-2076918

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3.2.S.2.2 Description of Manufacturing Process and Process Controls 3.2.S.2.2.1 Flow Chart of Manufacturing Process and Process Control

Lyophilized WB thawed with media

Cultivate for 20±4 h

Inoculate to solid medium for cultivating the 1st generation (1 Kolle flask)

Cultivate for 20±4 h

Inoculate to solid medium for cultivating the 2nd generation (5 Kolle flasks)

Bacterial colony is washed and collected for fermentation

Go into fermentation

Figure 1 Manufacturing Process Flow Chart for Pneumococcal Strain Recovery

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In-process control items:
Inoculate collected liquid to OD694
10 L cell tank (3-6 L) Purity test
Gram stain
Cultivate for 3-10 h

In-process control items:
Inoculate to 100 L OD694
amplification tank (30-60 L) Purity test
Gram stain
Cultivate for ≤ 6 h

In-process control items:
Inoculate to 900L OD694
fermentation tank (300-600 L) Purity test
Gram stain
Cultivate for 2-8 h

Killing bacteria with 10% DOC for 30 min

Harvesting fermentation broth

Centrifuge for 15 min in 7000 rpm & collect supernatant

UF concentration of supernatant
10±2 fold, 100 kDa

1st Ethanol precipitation
(Concentration: 17-50%)
Centrifuge for 60 min in 10000 rpm & collect supernatant

2nd Ethanol precipitation
(Concentration: 30-78%)
Centrifuge for 15 min in 4000 rpm & collect precipitate

Precipitate washed with ethanol / acetone / ether

Go into
Pump dry & collect crude polysaccharide
polysaccharide
(Store at -20 ■) purification

Figure 2 Manufacturing Process Flow Chart for Polysaccharide Preparation Process I:
Crude Polysaccharide Preparation

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Crude polysaccharide dissolution

Cold phenol extraction
(3-7 times)

Centrifuge for 40 min in 10000
rpm & collect supernatant

UF/DF to remove phenol
MWCO: 50/100 kDa

1st Ethanol precipitation
(Concentration: 15-30%)

Centrifuge for 60 min in 10000
rpm & collect supernatant

UF/DF to remove nucleic acid
MWCO: 30/50/100 kDa

2nd Ethanol precipitation
(Concentration: 40-80%)

Centrifuge for 15 min in 4000 In-process control items:
rpm & collect precipitate Identification, Total solid, Protein
content, Nucleic acid content, Total
Precipitate washed with ethanol / nitrogen content, Phosphorus
acetone / ether content, Molecular weight, Residual
ethanol content, Residual ether
content, Residual acetone content,
Bacterial endotoxin test, Uronic acid
content, O-Acetyl content,
Pump dry & collect refined
Hexosamine content, Methyl
polysaccharide pentose content, Sodium
(Store at -20 ■) deoxycholate

Go into conjugation

Figure 3 Manufacturing Process Flow Chart for Polysaccharide Preparation Process II:
Refined Polysaccharide Preparation (Serotype 1, 2, 4, 5, 6A, 6B, 7F, 9N, 9V, 10A, 12F, 14,
17F, 18C, 19A, 19F, 22F, 23F, 33F)

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Crude polysaccharide dissolution

Cold phenol extraction
(3-7 times)
Centrifuge for 40 min in 10000
rpm & collect supernatant

UF/DF to remove phenol
MWCO: 50/100 kDa

CTAB precipitation
(Concentration: 2%)
Centrifuge for 40 min in 10000
rpm & collect precipitate

Re-dissolving with NaCl

Centrifuge for 30 min in 10000
rpm & collect supernatant

Ethanol precipitation
(Concentration: 18-33.3%)
In-process control items:
Centrifuge for 15 min in 4000 Identification, Total solid,
rpm & collect precipitate Protein content, Nucleic acid
content, Total nitrogen content,
Precipitate washed with ethanol / Phosphorus content, Molecular
acetone / ether weight, Residual ethanol
content, Residual ether content,
Residual acetone content,
Pump dry & collect refined Bacterial endotoxin test, Uronic
acid content, Sodium
polysaccharide
deoxycholate
(Store at -20 ■)

Go into conjugation

Figure 4 Manufacturing Process Flow Chart for Polysaccharide Preparation Process II:
Refined Polysaccharide Preparation (Serotype 3)

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Strains open

In-process control items:
Cultivation of the 1st generation Microscopy

Cultivate for 48±2 h
at 34.5±1 ■

In-process control items:
Cultivation of the 2nd generation Microscopy

Cultivate for 48±2 h
at 34.5±1 ■

In-process control items:
Cultivation of the 3rd generation Microscopy

Cultivate for 48±2 h
at 34.5±1 ■

In-process control items:
Fermentation & Cultivation Microscopy,
Flocculent unit ≥ 40 Lf/mL
Cultivate for 60-72 h
at 34.5±1 ■

Killing bacteria with
formaldehyde solution for 30 min

Go into purification

Figure 5 Manufacturing Process Flow Chart for Clostridium Tetani Fermentation

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Harvesting Add formaldehyde
fermentation broth solution
Centrifuge for 15
min in 7000 rpm & Sterile filtration
collect supernatant
(0.45+0.22 μm)
Filtration
(0.45 μm)
Detoxification
(37±1 ■)
UF concentration
(1/8-1/10 fold) Filtration (0.45 μm)
Centrifuge for 15 & UF to remove
min in 10000 rpm & formaldehyde
collect supernatant
Filtration
(0.22 μm) Add NaCl & adjust
pH with NaHCO₃
Adjust pH with Na₂CO₃

1st (NH₄)₂SO₄ Sterile filtration
fractionation (0.45+0.22 μm)
(46-48 h) In-process control items:
Sterility test,
Centrifuge for 20
Bulk tetanus toxoid detoxification test,
min in 10000rpm &
stored at 2-8 ■ abnormal toxicity, toxicity
collect supernatant
reversion, flocculent unit,
Filtration protein content, purity,
(0.22 μm) Filtration (0.45 μm) & appearance, pH
UF to final concentration
2nd (NH₄)₂SO₄ at 50-100 mg/mL
fractionation
(18-24 h)
Filtration (0.45 μm)
Centrifuge for 15
min in 10000 rpm &
collect precipitate Column chromatography
Precipitate dissolution
& filtration
Concentration & sterile
filtration (0.45+0.22 μm)
UF to remove
ammonium In-process control items:
(15 times) Purified liquid of tetanus Protein content, sterility,
bacterial endotoxin, purity
Centrifuge for 15 toxoid stored at 2-8 ■
min in 10000 rpm &
collect precipitate
Add NaCl & adjust Go into conjugation
pH with NaHCO₃
Figure 6 Manufacturing Process Flow Chart for Purification of Tetanus Toxoid Carrier
Protein

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Refined polysaccharide dissolution
Concentration: 3-8 mg/mL

Polysaccharide activation with CNBR or
CDAP & derivation with ADH

Polysaccharide derivatives filtration
In-process control
items:
Total solid
Derivation rate
Derivatives UF to remove cyanide and ADH
Residual cyanide
MWCO: 10 kDa Molecular weight

Conjugation with tetanus toxoid
(TT: 0.1-1.3 mg/mg polysaccharide
EDAC: 1.5-10.1 mg/mL)

Reaction condition: 2-6
h, 2-8 °C, pH at 5-6.

Polysaccharide conjugate
(Store at 2-8 °C)

Conjugate dialysis
NMT 8 h

In-process control items:
Identification, Protein content,
Conjugate purified with column chromatography Polysaccharide content,
Molecular weight,
Appearance, Bacterial
endotoxin test, Free
Final sterile filtration polysaccharide content, Free
protein content,
Polysaccharide/protein ratio,
Residual ADH content,
Residual EDAC content,
Monovalent bulk conjugate Sterility test

Figure 7 Manufacturing Process Flow Chart and Quality Control for Monovalent Bulk
Conjugate

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3.2.S.2.2.2 Brief Description of Manufacturing Process

Walvax carries out production in accordance with the Drug Registration Standard for 13-valent
Pneumococcal Conjugate Vaccine (Standard No.: YBS00392019) approved by the National
Medical Products Administration.
The manufacturing process includes the following process: establishment and passage of the seed
batches, amplification of strain and large tank cultivation, sterilization of bacteria and harvest of
fermentation liquor, preparation of the crude polysaccharide and refined polysaccharide,
polysaccharide derivatization, polysaccharide conjugation with tetanus toxoid carrier protein.
After purification, sterilization and filtration, the conjugate bulk was obtained and stored at 2-8 °C.
The preparation of tetanus toxoid protein carrier includes the establishment and passage of seed
batches, amplification of bacteria and large tank fermentation, sterilization and harvest of
fermentation liquid after culture, ultrafiltration, two-stage salting out and formaldehyde
detoxification to obtain tetanus toxoid original liquid, and further purification to obtain the purified
tetanus toxoid liquid.

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3.2.S.2.2.3 Batch and Scale Definition
According to the "Batch Number Management Procedure" (Code: WSYX-PM-SMP-0008-0001)
made by Yuxi Walvax, standardize formulation and management of batch number principles
requirements to ensure the traceability of product information. preparation principles of the
working seed lot and intermediate product batch number are shown in the Table 3.

Table 3 Definition of Batch Number
Item Category Establishment principle
Streptococcus
SP N/A
pneumoniae
Clostridium tetani CT N/A
strain Strain code + serotype + category of seed lot + Year +
Mater seed M
Month + Yearly serial number
Strain code + serotype + category of seed lot + Year +
Work seed W
Month + Yearly serial number
Crude Strain code + serotype + intermediate code + Year +
CP
polysaccharide Month + Yearly serial number + bottle number
Strain code + serotype + intermediate code + Year +
Polysaccharide P

Month + Yearly serial number + bottle number
 Interim Strain code + serotype + intermediate code + Year +
 Bulk derivative D
 ediate Month + Yearly serial number + bottle number
 produc Strain code + serotype + intermediate code + Year +
 Bulk conjugate C
 t Month + Yearly serial number + bottle number
 Strain code + serotype + intermediate code + Year +
 Bulk B
 Month + Yearly serial number + bottle number
 Purified tetanus Strain code + serotype + intermediate code + Year +
 L
 toxoid liquid Month + Yearly serial number + bottle number
 Finished product code + year + month + annual serial
 Finished product I/K
 number

For example:

1. SP1W201901001 means the first batch of type 1 pneumococcal working seed produced in Jan. 2019.
2. SP1P201909001 means the first batch of type 1 pneumococcal polysaccharide produced in Sep. 2019.
3. SP1D201908001 means the first batch of type 1 pneumococcal polysaccharide derivative bulk produced in Aug. 2019.
4. SP1C201909002 means the second batch of type 1 pneumococcal polysaccharide bulk conjugate produced in Sep, 2019.
5. CTB201510003 means the third batch of tetanus toxoid bulk produced in Oct. 2015.
6. CTL201510003 means the third batch of purified tetanus toxoid liquid manufactured in Oct, 2015.
7. I201909002 means the second batch of 13-valent Pneumococcal Conjugate Vaccine (vial) manufactured in Sep. 2019.
8. K201909001 means the first batch of 13-valent Pneumococcal Conjugate Vaccine (pre-filling syringe) manufactured in Sep. 2019.

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3.2.S.2.3 Control of Materials

2.3.S.2.3.1 Strain for Production

(1) Source of Strain

The strains used in the manufacturing are pneumococcus serotype 1, 3, 4, 5, 6A, 6B, 7F, 9V, 14, 18C, 19A, 19F and 23F strains, which are purchased from National Centre for Medical Culture Collections of NIFDC.

Table 4 The name and CMCC code of pneumococcal strains

Serotypes Strain Name CMCC Code

1	Streptococcus pneumoniae serotype 1	31404
3	Streptococcus pneumoniae serotype 3	31441
4	Streptococcus pneumoniae serotype 4	31448
5	Streptococcus pneumoniae serotype 5	31457
6A	Streptococcus pneumoniae serotype 6A	31476
6B	Streptococcus pneumoniae serotype 6B	31492
7F	Streptococcus pneumoniae serotype 7F	31510
9V	Streptococcus pneumoniae serotype 9V	31547
14	Streptococcus pneumoniae serotype 14	31605
18C	Streptococcus pneumoniae serotype 18C	31689
19A	Streptococcus pneumoniae serotype 19A	31708
19F	Streptococcus pneumoniae serotype 19F	31696
23F	Streptococcus pneumoniae serotype 23F	31761

Clostridium tetani (CMCC 64008), which is used for producing the tetanus toxoid carrier protein, was purchased from National Centre for Medical Culture Collections of NIFDC.

(2) Establishment of Seed Lot

According to the Chinese Pharmacopoeia, Walvax established the Bacteria Management procedure (WSYX-QA-SMP-00-05-0001) for the bacteria bank establishment and the three-level seed bank management. The three-level seed bank includes original seed lot, master seed lot and working seed lot. The original seed is lyophilized after culturing two generations, as well as the working seed lot. The working seed lot is used for vaccine production. The biological characteristics of the working seed lot is consistent with the original seed lot, and each seed lot is stored below -60 °C.

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Original Master seed Working
 seed (the 0th (the 2nd seed (the 4th
 generation) generation) generation)

Strain open Strain open
 Inoculated on solid Inoculated on solid
 medium medium
 Culture & Culture &
 amplification at amplification at
 37 °C 37 °C

Sub-cultivate on solid Sub-cultivate on solid
 medium medium
 Culture &
 amplification at Culture &
 37 °C amplification at 37 °C

Collection of Collection of
the bacteria the bacteria

Packaging Packaging

Lyophilization Lyophilization

Seal Seal

Test Test

Figure 8 Establishment Flow Chart of Pneumococcal Three-level Seed Bank

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Table 5 Three-level Seed Bank Information of 13 Serotypes of Pneumococcus														
Serotype Generation														
1 3 4 5 6A 6B 7F 9V 14 18C 19A 19F 23F														
Strain Info. n														
CMCC														
31404 31441 31448 31457 31476 31492 31510 31547 31605 31689 31708 31696 31761														
Code														
Date of														
Original Purchas 2008.10 2009.04 2003.09 2008.10 2013.06 2003.09 2009.06 2003.09 2003.09 2003.09 2003.09 2009.06														
2003.09 2003.09 0														
e														
Quantit														
1 1 1 1 1 1 1 1 1 1 1 1 1 1 1														
y (vial)														
2008102 2009051 2004040 2008101 2013060 2004042 2009062 2004050 2004051 2004052 2009072 2004060 2004061														
Lot No.														
8 0 1 5 8 3 1 3 4 4 6 2 5														
Master 2nd														
Quantit														
30 40 30 30 40 30 40 30 30 40 30 30														
y (vial)														
2008110 2009051 2004040 2008102 2013061 2004043 2009062 2004051 2004052 2004053 2009072 2004060 2004062														
Lot No.														
Workin 2 3 8 0 3 0 4 0 1 1 9 9 2														
4th														
g Quantit														
100 50 100 100 50 100 50 100 100 100 50 100 100														
y (vial)														

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Original seed (the Master seed (the Working seed (the
0th generation) 2nd generation) 4th generation)

Strain open Strain open
Inoculated on semi-solid Inoculated on semi-solid
medium medium
Culture & Culture &
amplification at amplification at
34.5±1 ■ 34.5±1 ■
Inoculated on liquid Inoculated on liquid
medium medium
Culture & Culture &
amplification at amplification at
34.5±1 ■ 34.5±1 ■

Collection of Collection of
the bacteria the bacteria

Packaging Packaging

Lyophilization Lyophilization

Seal Seal

Test Test

Figure 9 Establishment Flow Chart of Clostridium Tetani Three-level Seed Bank

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Table 6 Three-level Seed Bank Information of Clostridium Tetani Strain	
Clostridium tetani strain info.	Generation
CMCC Code	64008
Original Date of Purchase	2003.06.27 0th
Quantity (Vial)	3
Lot No.	200403001
Master 2nd	
Quantity (Vial)	16
Lot No.	20040320
Working 4th	

Quantity (Vial) 38

(3) Strain detection

Based on Bergey's Manual of Determinative Bacteriology (the 8th Edition), Testing Methods for Streptococcus Pneumoniae on Experimental Animals (GB/T 14926.15-2001), and Vaccines (2004 Edition), Walvax established testing items and standards for the Streptococcus Pneumoniae strains and Clostridium tetani strains.

According to the test items and specifications in the Table 7, the three-level seed lot of 13 types Streptococcus Pneumoniae were tested. The results showed from the Table 8 to Table 20, which indicate that the master seed lot and working seed lot have the same biological, biochemical and serological characteristics as the original seed lot.

Table 7 Testing Items and Standards of Pneumococcal Strains

Test items Specifications

No growth on the common nutrient agar slant and good growth on the 10% Culture and Staining sheep blood agar slant. Colonies isolated and cultured on blood agar appear Microscopy grey and translucent, with neat edges and smooth, moist surfaces. Alpha hemolytic, G+ aureus, a double ball or short chain arrangement.
Glucose Positive
Raffinose Positive
Biochemical Melibiose Positive
Reactions
Inulin Positive
Sorbitol Negative
Serum Agglutination Agglutination with corresponding serum
Bile Salt Dissolution Test Positive
Aupu Tosin Test Positive
Capsule Swelling Test Capsule observable

Table 8 Test Results for Three-level Seed Lot of Serotype 1 Pneumococcal Strain

Original seed Master seed Working seed

Test items Standards

lot lot lot
No growth on the common nutrient agar
Culture characteristic slant and good growth Comply Comply Comply
on the 10% sheep blood agar slant.
Colonies isolated and cultured on blood agar
Colonial morphology Comply Comply Comply
appear grey and translucent, with neat

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edges and smooth, moist
surfaces, α hemolytic
G+ aureus, a double ball
Gram staining or short chain Comply Comply Comply
arrangement
Capsule staining Capsule observable Comply Comply Comply
Bile Salt Dissolution Test + + + +
Aupu Tosin Test + + + +
Glucose + + + +
Raffinose + + + +
Biochemical
Melibiose + + + +
reaction
Inulin + + + +
Sorbitol - - -
Agglutination Agglutination Agglutination
Agglutination with
Serum Agglutination with serotype 1 with serotype 1 with serotype 1
corresponding serum
serum serum serum
Meet the Meet the Meet the
Conclusion
requirement requirement requirement

Table 9 Test Results for Three-level Seed Lot of Serotype 3 Pneumococcal Strain

Original seed Master seed Working seed

Test items Standards

lot lot lot
No growth on the common nutrient agar
Culture characteristic slant and good growth Comply Comply Comply
on the 10% sheep blood agar slant.
Colonies isolated and cultured on blood agar
appear grey and translucent, with neat
edges and smooth, moist
surfaces, α hemolytic
G+ aureus, a double ball
Gram staining or short chain Comply Comply Comply
arrangement
Capsule staining Capsule observable Comply Comply Comply

Bile Salt Dissolution Test + + + +
Aupu Tosin Test + + + +
Glucose + + + +
Raffinose + + + +
Biochemical
Melibiose + + + +
reaction
Inulin + + + +
Sorbitol - - - -
Agglutination Agglutination Agglutination
Agglutination with
Serum Agglutination with serotype 3 with serotype 3 with serotype 3
corresponding serum
serum serum serum
Meet the Meet the Meet the
Conclusion
requirement requirement requirement

Table 10 Test Results for Three-level Seed Lot of Serotype 4 Pneumococcal Strain
Original seed Master seed Working seed
Test items Standards
lot lot lot
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No growth on the
common nutrient agar
Culture characteristic slant and good growth Comply Comply Comply
on the 10% sheep blood
agar slant.
Colonies isolated and
cultured on blood agar
appear grey and
Colonial morphology Comply Comply Comply
translucent, with neat
edges and smooth, moist
surfaces, α hemolytic
G+ aureus, a double ball
Gram staining or short chain Comply Comply Comply
arrangement
Capsule staining Capsule observable Comply Comply Comply
Bile Salt Dissolution Test + + + +
Aupu Tosin Test + + + +
Glucose + + + +
Raffinose + + + +
Biochemical
Melibiose + + + +
reaction
Inulin + + + +
Sorbitol - - - -
Agglutination Agglutination Agglutination
Agglutination with
Serum Agglutination with serotype 4 with serotype 4 with serotype 4
corresponding serum
serum serum serum
Meet the Meet the Meet the
Conclusion
requirement requirement requirement

Table 11 Test Results for Three-level Seed Lot of Serotype 5 Pneumococcal Strain
Original seed Master seed Working seed
Test items Standards
lot lot lot
No growth on the
common nutrient agar
Culture characteristic slant and good growth Comply Comply Comply
on the 10% sheep blood
agar slant.
Colonies isolated and
cultured on blood agar
appear grey and
Colonial morphology Comply Comply Comply
translucent, with neat
edges and smooth, moist
surfaces, α hemolytic
G+ aureus, a double ball
Gram staining or short chain Comply Comply Comply
arrangement
Capsule staining Capsule observable Comply Comply Comply
Bile Salt Dissolution Test + + + +
Aupu Tosin Test + + + +
Glucose + + + +
Raffinose + + + +
Biochemical
Melibiose + + + +
reaction
Inulin + + + +
Sorbitol - - - -

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Agglutination Agglutination Agglutination
Agglutination with
Serum Agglutination with serotype 5 with serotype 5 with serotype 5
corresponding serum
serum serum serum
Meet the Meet the Meet the
Conclusion
requirement requirement requirement

Table 12 Test Results for Three-level Seed Lot of Serotype 6A Pneumococcal Strain
Original seed Master seed Working seed

Test items Standards
lot lot lot
No growth on the
common nutrient agar
Culture characteristic slant and good growth Comply Comply Comply
on the 10% sheep blood
agar slant.
Colonies isolated and
cultured on blood agar
appear grey and
Colonial morphology translucent, with neat Comply Comply Comply
edges and smooth,
moist surfaces, α
hemolytic
G+ aureus, a double
Gram staining ball or short chain Comply Comply Comply
arrangement
Capsule staining Capsule observable Comply Comply Comply
Bile Salt Dissolution Test + + + +
Aupu Tosin Test + + + +
Glucose + + + +
Raffinose + + + +
Biochemical
Melibiose + + + +
reaction
Inulin + + + +
Sorbitol - - -
Agglutination Agglutination Agglutination
Agglutination with
Serum Agglutination with serotype with serotype with serotype
corresponding serum
6A serum 6A serum 6A serum
Meet the Meet the Meet the
Conclusion
requirement requirement requirement

Table 13 Test Results for Three-level Seed Lot of Serotype 6B Pneumococcal Strain
Original seed Master seed Working seed

Test items Standards
lot lot lot
No growth on the
common nutrient agar
Culture characteristic slant and good growth Comply Comply Comply
on the 10% sheep blood
agar slant.
Colonies isolated and
cultured on blood agar
appear grey and
Colonial morphology translucent, with neat Comply Comply Comply
edges and smooth,
moist surfaces, α
hemolytic
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G+ aureus, a double
Gram staining ball or short chain Comply Comply Comply
arrangement
Capsule staining Capsule observable Comply Comply Comply
Bile Salt Dissolution Test + + + +
Aupu Tosin Test + + + +
Glucose + + + +
Raffinose + + + +
Biochemical
Melibiose + + + +
reaction
Inulin + + + +
Sorbitol - - -
Agglutination Agglutination Agglutination
Agglutination with
Serum Agglutination with serotype with serotype with serotype
corresponding serum
6B serum 6B serum 6B serum
Meet the Meet the Meet the
Conclusion
requirement requirement requirement

Table 14 Test Results for Three-level Seed Lot of Serotype 7F Pneumococcal Strain

Original seed Master seed Working seed
Test items Standards
lot lot lot
No growth on the
common nutrient agar
Culture characteristic slant and good growth Comply Comply Comply
on the 10% sheep blood
agar slant.
Colonies isolated and
cultured on blood agar
appear grey and
Colonial morphology translucent, with neat Comply Comply Comply
edges and smooth,
moist surfaces, α
hemolytic
G+ aureus, a double
Gram staining ball or short chain Comply Comply Comply
arrangement
Capsule staining Capsule observable Comply Comply Comply
Bile Salt Dissolution Test + + + +
Aupu Tosin Test + + + +
Glucose + + + +
Raffinose + + + +
Biochemical
Melibiose + + + +
reaction
Inulin + + + +
Sorbitol - - - -
Agglutination Agglutination Agglutination
Agglutination with
Serum Agglutination with serotype with serotype with serotype
corresponding serum
7F serum 7F serum 7F serum
Meet the Meet the Meet the
Conclusion
requirement requirement requirement

Table 15 Test Results for Three-level Seed Lot of Serotype 9V Pneumococcal Strain
Original seed Master seed Working seed
Test items Standards
lot lot lot

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No growth on the
common nutrient agar
Culture characteristic slant and good growth Comply Comply Comply
on the 10% sheep blood
agar slant.
Colonies isolated and
cultured on blood agar
appear grey and
Colonial morphology Comply Comply Comply
translucent, with neat
edges and smooth, moist
surfaces, α hemolytic
G+ aureus, a double ball
Gram staining or short chain Comply Comply Comply
arrangement
Capsule staining Capsule observable Comply Comply Comply
Bile Salt Dissolution Test + + + +
Aupu Tosin Test + + + +
Glucose + + + +
Raffinose + + + +
Biochemical
Melibiose + + + +
reaction
Inulin + + + +
Sorbitol - - - -
Agglutination Agglutination Agglutination
Agglutination with
Serum Agglutination with serotype with serotype with serotype
corresponding serum
9V serum 9V serum 9V serum
Meet the Meet the Meet the
Conclusion
requirement requirement requirement

Table 16 Test Results for Three-level Seed Lot of Serotype 14 Pneumococcal Strain
Original seed Master seed Working seed
Test items Standards
lot lot lot
No growth on the
common nutrient agar
Culture characteristic slant and good growth Comply Comply Comply
on the 10% sheep blood
agar slant.
Colonies isolated and
cultured on blood agar
appear grey and

Colonial morphology Comply Comply Comply
translucent, with neat
edges and smooth, moist
surfaces, α hemolytic
G+ aureus, a double ball
Gram staining or short chain Comply Comply Comply
arrangement
Capsule staining Capsule observable Comply Comply Comply
Bile Salt Dissolution Test + + + +
Aupu Tosin Test + + + +
Glucose + + + +
Raffinose + + + +
Biochemical
Melibiose + + + +
reaction
Inulin + + + +
Sorbitol - - - -

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Agglutination Agglutination Agglutination
Agglutination with
Serum Agglutination with serotype with serotype with serotype
corresponding serum
14 serum 14 serum 14 serum
Meet the Meet the Meet the
Conclusion
requirement requirement requirement

Table 17 Test Results for Three-level Seed Lot of Serotype 18C Pneumococcal Strain
Original seed Master seed Working seed
Test items Standards
lot lot lot
No growth on the
common nutrient agar
Culture characteristic slant and good growth Comply Comply Comply
on the 10% sheep blood
agar slant.
Colonies isolated and
cultured on blood agar
appear grey and
Colonial morphology Comply Comply Comply
translucent, with neat
edges and smooth, moist
surfaces, α hemolytic
G+ aureus, a double ball
Gram staining or short chain Comply Comply Comply
arrangement
Capsule staining Capsule observable Comply Comply Comply
Bile Salt Dissolution Test + + + +
Aupu Tosin Test + + + +
Glucose + + + +
Raffinose + + + +
Biochemical
Melibiose + + + +
reaction
Inulin + + + +
Sorbitol - - - -
Agglutination Agglutination Agglutination
Agglutination with
Serum Agglutination with serotype with serotype with serotype
corresponding serum
18C serum 18C serum 18C serum
Meet the Meet the Meet the
Conclusion
requirement requirement requirement

Table 18 Test Results for Three-level Seed Lot of Serotype 19A Pneumococcal Strain
Original seed Master seed Working seed
Test items Standards
lot lot lot
No growth on the
common nutrient agar
Culture characteristic slant and good growth Comply Comply Comply
on the 10% sheep blood
agar slant.
Colonies isolated and
cultured on blood agar
appear grey and
Colonial morphology Comply Comply Comply
translucent, with neat
edges and smooth, moist
surfaces, α hemolytic

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G+ aureus, a double ball
Gram staining or short chain Comply Comply Comply

arrangement
Capsule staining Capsule observable Comply Comply Comply
Bile Salt Dissolution Test + + + +
Aupu Tosin Test + + + +
Glucose + + + +
Raffinose + + + +
Biochemical
Melibiose + + + +
reaction
Inulin + + + +
Sorbitol - - - -
Agglutination Agglutination Agglutination
Agglutination with
Serum Agglutination with serotype with serotype with serotype
corresponding serum
19A serum 19A serum 19A serum
Meet the Meet the Meet the
Conclusion
requirement requirement requirement

Table 19 Test Results for Three-level Seed Lot of Serotype 19F Pneumococcal Strain

Original seed Master seed Working seed
Test items Standards
lot lot lot
No growth on the
common nutrient agar
Culture characteristic slant and good growth Comply Comply Comply
on the 10% sheep blood
agar slant.
Colonies isolated and
cultured on blood agar
appear grey and
Colonial morphology Comply Comply Comply
translucent, with neat
edges and smooth, moist
surfaces, α hemolytic
G+ aureus, a double ball
Gram staining or short chain Comply Comply Comply
arrangement
Capsule staining Capsule observable Comply Comply Comply
Bile Salt Dissolution Test + + + +
Aupu Tosin Test + + + +
Glucose + + + +
Raffinose + + + +
Biochemical
Melibiose + + + +
reaction
Inulin + + + +
Sorbitol - - - -
Agglutination Agglutination Agglutination
Agglutination with
Serum Agglutination with serotype with serotype with serotype
corresponding serum
19F serum 19F serum 19F serum
Meet the Meet the Meet the
Conclusion
requirement requirement requirement

Table 20 Test Results for Three-level Seed Lot of Serotype 23F Pneumococcal Strain

Original seed Master seed Working seed
Test items Standards
lot lot lot
No growth on the
Culture characteristic Comply Comply Comply
common nutrient agar
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slant and good growth
on the 10% sheep blood
agar slant.
Colonies isolated and
cultured on blood agar
appear grey and
Colonial morphology Comply Comply Comply
translucent, with neat
edges and smooth, moist
surfaces, α hemolytic
G+ aureus, a double ball
Gram staining or short chain Comply Comply Comply
arrangement
Capsule staining Capsule observable Comply Comply Comply
Bile Salt Dissolution Test + + + +
Aupu Tosin Test + + + +
Glucose + + + +
Raffinose + + + +
Biochemical
Melibiose + + + +
reaction
Inulin + + + +
Sorbitol - - - -

Agglutination Agglutination Agglutination
Agglutination with
Serum Agglutination with serotype with serotype with serotype
corresponding serum
23F serum 23F serum 23F serum
Meet the Meet the Meet the
Conclusion
requirement requirement requirement

According to the test items and specifications in the Table 21, the three-level seed lot of Clostridium Tetani were tested. The results showed in the Table 22, which indicate that the master seed lot and working seed lot have the same biological, biochemical and serological characteristics as the original seed lot.

Table 21 Testing Items and Standards of Clostridium Tetani Strain
Test items Specifications
Obligate anaerobic, preferential temperature at 37 ■, turbid with gas produced and rotten stench in cooked
Growth Characteristic meat culture, permeated bacteria colony in blood agar medium, flagellar motility in semi-solid puncture
Cultural culture.
Characteristic
Gram negative, rod-shaped bacteria with little amount of spores at the beginning of cultivation; transformed to
Staining Microscopy
Gram positive, drumstick-shaped bacteria with round spores on the top after 48 h of cultivation.
Glucose Negative
Lactose Negative
Xylose Negative
Saccharose Negative
Biochemical Galactose Negative
Reactions Fructose Negative
Nitrate Negative
Gelatin liquefaction Positive
Hydrogen sulfide Positive
Dynamic experiment Positive
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Symptoms such as stiffened tail, cramp of hind legs or
Toxin production test muscle spasm of the whole body, or even death, should
Zoology appear.
experiment Mice in the positive control group should appear
Specific neutralization test obvious tetanus symptoms and die, while mice in the test group survive.

Table 22 Test Results for Three-level Seed Lot of Clostridium Tetani Strain
Original seed Master seed Working seed
Test items Standards
lot lot lot
Obligate anaerobic, preferential temperature at 37 ■, turbid with gas produced and rotten stench in cooked meat
Growth Characteristic culture, permeated Comply Comply Comply
bacteria colony in blood agar medium, flagellar motility in semi-solid puncture culture.
Gram negative, rod-shaped bacteria with little amount of spores at the beginning of cultivation; transformed
Staining Microscopy Comply Comply Comply
to Gram positive, drumstick-shaped bacteria with round spores on the top after 48 h of cultivation.
Glucose - - - -
Lactose - - - -
Xylose - - - -
Saccharose - - - -
Galactose - - - -
Fructose - - - -
Biochemical reaction Nitrate - - - -
Gelatin + + + +
liquefaction
Hydrogen + + + +
sulfide
Dynamic + + + +
experiment

Symptoms such as
stiffened tail, cramp of
Toxin
hind legs or muscle
production Comply Comply Comply
spasm of the whole
test
Zoology body, or even death,
experiment should appear.
Mice in the positive
Specific
control group should
neutralization Comply Comply Comply
appear obvious tetanus
test
symptoms and die,

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while mice in the test
group survive.
Meet the Meet the Meet the
Conclusion
requirement requirement requirement

(4) Study on the heredity stability of pneumococcal strains
[Pneumococcal strain]
In order to ensure the stability of both process and product quality, Walvax carried out heredity
stability studies for strains of all 13 serotypes. Each strain is passed continuously to the 20th
generation. During the process, bacteria from the 10th, 15th, and 20th generations are
comprehensively tested. The process diagram is as follows:

Open a vial of lyophilized Working Seed Lot

Culture on solid medium for 20±4 h, then
↓ transfer the bacteria
2nd Generation

Culture on solid medium for 20±4 h, then
↓ transfer the bacteria

3rd Generation

Culture on solid medium for 20±4 h, then
↓ transfer the bacteria
4th Generation

Culture on liquid medium for 6 h, then
↓ transfer the bacteria

5th Generation
Culture on liquid medium for 6 h, then
transfer the bacteria
↓

6th Generation

Culture on liquid medium for 6 h/generation, then
↓ transfer the bacteria
10th Generation, sampling for tests
Culture on liquid medium for 6 h/generation,
↓ then transfer the bacteria

15th Generation, sampling for tests
Culture on liquid medium for 6 h/generation,
then transfer the bacteria
↓

20th Generation, sampling for tests

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↓
Terminate culturing, commence sterilization
Figure 10 The Process of Heredity Stability for Pneumococcal Strains

Testing results of strains from all generations have been constant with the Working Seed Lot.
There has been no change in biological, biochemical, or serological characteristics of the strains.
The stability of the strains used in production is proven.

[Clostridium tetani strain]
In order to ensure the stability of vaccine production process and product quality, Walvax carried
out heredity stability studies of Clostridium tetani strain. Each strain is passed and cultured in the
cooked meat semi-solid medium every year. The storage strain has been passed three times, and
the production strain has been passed five times, without any decline in virulence and
morphological variation. The test results of the strain show that its culture characteristics, staining
microscopy, biochemical reaction and specific neutralization tests are same as the seed lot, which
confirmed that the production strains of Clostridium tetani had good stability.

(5) Study on the virulence stability of pneumococcal strains
[Study Objective]

To investigate the virulence and virulence stability of pneumococcal serotype 1, 3, 4, 5, 6A, 6B, 7F, 9V, 14, 18C, 19A, 19F, and 23F production strain.

[Reference]
Manual of medical biological products, 235-241

[Experiment material]
The 3rd generation and the 20th generation of pneumococcal serotype 1, 3, 4, 5, 6A, 6B, 7F, 9V, 14, 18C, 19A, 19F, and 23F production strain.

[Experiment method]
The strain was diluted with normal saline and inoculated on solid medium containing 10% sheep blood. After incubation at 37±1 ■ and CO2 for 20±4 hours, the coating of pneumococcus was washed with normal saline and gradient diluted. The dilution was 1:10, 1:100, 1:103 until 1:108, the last three dilutions (1:106, 1:107, 1:108) were used to inject 18-22 g healthy mice (3 mice for each dilutions) into abdominal cavity at 0.5 ml/mouse.

[Criteria]
In 72 hours, if 2 of the 3 mice injected with 1:107 and 1:108 dilutions died, the strain is a strong virulent strain, otherwise it is a weak virulent strain.

[Experiment result]

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Table 23 The Virulence Test Result of the 3rd and the 20th Generation of all 13 Serotypes of Pneumococcal Strain

Generation	1	3	4	5	6A	6B	7F	9V	14	18C	19A	19F	23F
Serotypes the 3rd the 20th													
Items													
Dilutability 10-6 10-7 10-8 10-6 10-7 10-8													
1													
Mortality/totality 0/3 0/3 0/3 0/3 0/3 0/3													
Dilutability 10-6 10-7 10-8 10-6 10-7 10-8													
3													
Mortality/totality 1/3 0/3 0/3 1/3 0/3 0/3													
Dilutability 10-6 10-7 10-8 10-6 10-7 10-8													
4													
Mortality/totality 1/3 0/3 0/3 1/3 0/3 0/3													
Dilutability 10-6 10-7 10-8 10-6 10-7 10-8													
5													
Mortality/totality 0/3 0/3 0/3 0/3 0/3 0/3													
Dilutability 10-6 10-7 10-8 10-6 10-7 10-8													
6A													
Mortality/totality 1/3 0/3 0/3 0/3 0/3 0/3													
Dilutability 10-6 10-7 10-8 10-6 10-7 10-8													
6B													
Mortality/totality 0/3 0/3 0/3 1/3 0/3 0/3													
Dilutability 10-6 10-7 10-8 10-6 10-7 10-8													
7F													
Mortality/totality 1/3 1/3 0/3 1/3 1/3 0/3													
Dilutability 10-6 10-7 10-8 10-6 10-7 10-8													
9V													
Mortality/totality 1/3 0/3 0/3 0/3 0/3 0/3													
Dilutability 10-6 10-7 10-8 10-6 10-7 10-8													
14													
Mortality/totality 1/3 0/3 0/3 1/3 0/3 0/3													
Dilutability 10-6 10-7 10-8 10-6 10-7 10-8													
18C													
Mortality/totality 1/3 0/3 0/3 1/3 0/3 0/3													
Dilutability 10-6 10-7 10-8 10-6 10-7 10-8													
19A													
Mortality/totality 1/3 0/3 0/3 1/3 0/3 0/3													
Dilutability 10-6 10-7 10-8 10-6 10-7 10-8													
19F													
Mortality/totality 1/3 1/3 0/3 1/3 0/3 0/3													
Dilutability 10-6 10-7 10-8 10-6 10-7 10-8													
23F													
Mortality/totality 1/3 0/3 0/3 1/3 0/3 0/3													

From Table 23, it can be seen that the 13 strains (pneumococcus serotypes 1, 3, 4, 5, 6A, 6B, 7F, 9V, 14, 18C, 19A, 19F, and 23F) used for production are attenuated vaccine strain. After 20 generations of continuous transplanting, the strain virulence did not change and the strain was still attenuated vaccine strain. It showed that the virulence of the strain is stable and the strains are suitable for vaccine development and production.

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3.2.S.2.3.2 Other Raw Materials

The company has formulated the Material Management Control Procedure (WSYX-P-05-0005) to guide and control the management and storage of the company's materials. All purchased materials are from qualified suppliers approved by the quality management director. For each kind of material, the company has established the inspection standard operating procedures according to the legal standards of the material to check whether the quality indicators of the material meet the quality requirements of the product production. At the same time, collect documents to prove whether the raw materials used in production contain ingredients of animal derived materials. If the materials contain ingredients of animal derived materials, further collect animal safety certification materials of the materials to ensure the safety of materials used in production and further ensure the safety of products. See the Table 24 and Table 25 for the composition of culture medium.

Table 24 Culture Medium Components for Production of 13 Serotypes of Pneumococcal Polysaccharides

Medium name No. Material name

1 Tryptone
2 Soybean peptone
3 Dipotassium hydrogen phosphate
Composition of pneumococcal comprehensive solid 4 L-cysteine hydrochloride
medium 5 Tryptophan
6 Tyrosine
7 Agar
8 Sheep blood
9 Tryptone
10 Soybean peptone
11 L-cysteine hydrochloride
12 L-tryptophan
13 L-tyrosine
Dipotassium hydrogen phosphate
14
trihydrate
15 Sodium chloride
16 Catalase
17 Sodium bicarbonate
18 Mercaptoacetic acid
19 Nicotinic acid
20 Biotin
Composition of pneumococcal liquid medium 21 Vitamin B1
22 Vitamin B6
23 D-calcium Pantothenate
24 Uracil
25 Adenine sulfate
26 L-glutamine
27 L-asparagine
28 Magnesium sulfate anhydrous
Ferrous sulfate heptahydrate
29
(FeSO₄·7H₂O)
30 Riboflavin
31 Zinc sulfate (ZnSO₄·7H₂O)
32 Manganese Sulfate (MnSO₄·H₂O)
33 Choline chloride

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Table 25 Toxin Production Medium Components of Clostridium Tetani

Medium name No. Material name

1 Vitamin B1
2 Vitamin B2
3 Vitamin B6
Factor ■
4 Calcium pantothenate
5 Biotin
6 Absolute ethanol
7 Nicotinic acid
8 Uracil
9 Cystine
Factor ■
10 Zinc sulphate
11 Magnesium sulphate anhydrous
12 Hydrochloric acid
13 Beef stomach enzyme digestive fluid
14 Casein digestive fluid
15 Sodium acetate
16 Disodium hydrogen phosphate
17 Potassium dihydrogen phosphate
Production medium 18 Glycerine
19 Glucose
20 Factor ■
21 Factor ■
22 Yeast extract
23 Ferric chloride

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3.2.S.2.4 Controls of Critical Steps and Intermediates

Table 26 Critical Manufacture Steps of Serotypes 1, 4, 5, 6A, 6B, 7F, 9V, 14, 18C, 19A, 19F, 23F Pneumococcal Polysaccharide and Bulk Conjugate

Process Process variable
Medium volume used for dissolving strains (mL)
Number of opened Kirschner bottles
Temperature of the first-generation culture (■)
Carbon dioxide concentration of the first-generation culture
Time of the 1st generation culture (h)
Strain Amplification Process The number of the Kirschner flask which is bacteria passing into
Temperature of the 2nd generation culture (■)
Carbon dioxide concentration of the 2nd generation culture
Time of the 2nd generation culture (h)
Bacterial liquid volume collected (mL)
OD value
Seed tank culture volume (L)

Seed tank culture temperature (■)
 Seed tank culture pH
 Seed tank culture time (h)
 Amplification tank culture volume (L)
 Amplification tank culture temperature (■)
 Amplification tank culture pH
 Amplification tank culture time (h)
 Fermentation tank culture volume (L)
 Fermentation tank culture temperature (■)
 Fermentation tank culture pH
 Fermentation tank culture time (h)
 Polysaccharide preparation process
 Sterilant concentration
 I:
 Sterilization time (min)
 Crude polysaccharide preparation
 Centrifugal speed of fermentation liquid (rpm)
 Centrifugal time of fermentation liquid (min)
 Centrifugal temperature of fermentation liquid (min)
 Ultrafiltration concentration fold
 Final concentration of sodium acetate in the 1st ethanol precipitation
 Concentration of ethanol in the 1st ethanol precipitation
 Final concentration of sodium acetate in the 2nd ethanol precipitation
 Concentration of ethanol in the 2nd ethanol precipitation
 Times of crude polysaccharide washed by absolute ethanol
 Times of crude polysaccharide washed by acetone
 Times of crude polysaccharide washed by ether
 Crude polysaccharide dried time (h)
 Concentration of crude polysaccharide after dissolving (mg/mL)
 Times of cold phenol extraction
 Final concentration of ethanol for precipitating nucleic acid
 Polysaccharide preparation process
 Final concentration of ethanol for precipitating polysaccharide
 II:
 Times of refined polysaccharide washed by absolute ethanol
 Refined polysaccharide preparation
 Times of refined polysaccharide washed by acetone
 Times of refined polysaccharide washed by ether
 Time of refined polysaccharide dried (h)
 Polysaccharide quantity used for conjugation (g)
 Polysaccharide conjugate process Polysaccharide concentration after dissolving (mg/mL)
 CNBR or CDAP quantity for per mg of polysaccharide (mg)
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pH value of activation
 Activation temperature (■)
 Activation time (min)
 ADH quantity for per mg of polysaccharide (mg)
 pH value of derivatization
 Derivatization temperature (■)
 Derivatization reaction time (min)
 Polysaccharide derivative standing time (h)
 Standing temperature of polysaccharide derivative (■)
 Volume ratio of ultrafiltrate and derivative reaction fluid
 Ultrafiltration reaction times
 Storage temperature (■)
 Storage time (month)
 Carrier protein quantity for per mg of polysaccharide derivative (mg)
 Sodium chloride concentration (mol/L)
 pH value of conjugation
 EDAC quantity for per ml of mixed liquid (mg)
 Conjugation reaction time (h)
 Final pH of conjugation reaction
 Conjugate standing time (h)
 Conjugate standing temperature (■)
 Chromatographic column filler
 Collection KD value
 Pore size of sterilization & filtration membrane

Table 27 Critical Manufacture Steps of Serotypes 3 Pneumococcal Polysaccharide and Bulk Conjugate

Process Process variable
 Medium volume used for dissolving strains (mL)
 Number of opened Kirschner bottles
 Temperature of the 1st generation culture (■)
 Carbon dioxide concentration of the 1st generation culture
 Time of the 1st generation culture (h)
 Strain Amplification Process Number of the Kirschner flask which is bacteria passing into
 Temperature of the 2nd generation culture (■)
 Carbon dioxide concentration of the 2nd generation culture
 Time of the 2nd generation culture (h)
 Volume of bacterial liquid collected (mL)
 OD value
 Seed tank culture volume (L)
 Seed tank culture temperature (■)
 Seed tank culture pH
 Seed tank culture time (h)
 Amplification tank culture volume (L)
 Polysaccharide preparation process Amplification tank culture temperature (■)
 I: Amplification tank culture pH

Crude polysaccharide preparation Amplification tank culture time (■)
Fermentation tank culture volume (L)
Fermentation tank culture temperature (■)
Fermentation tank culture pH
Fermentation tank culture time (h)
Sterilant concentration

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Sterilization time (min)
Centrifugal speed of fermentation liquid (rpm)
Centrifugal time of fermentation liquid (min)
Centrifugal temperature of fermentation liquid (■)
Ultrafiltration concentration fold
Final concentration of sodium acetate in the 1st ethanol precipitation
Final concentration of ethanol in the 1st ethanol precipitation
Final concentration of sodium acetate in the 2nd ethanol precipitation
Final concentration of ethanol in the 2nd ethanol precipitation
Times of crude polysaccharide washed by absolute ethanol
Times of crude polysaccharide washed by acetone
Times of crude polysaccharide washed by ether
Crude polysaccharide dried time (h)
Crude polysaccharide concentration after dissolving (mg/mL)
Times of cold phenol extraction
Final concentration of CTAB for precipitating nucleic acid
Polysaccharide preparation process Final concentration of sodium acetate for precipitating polysaccharide
II: Final concentration of ethanol for precipitating polysaccharide
Refined polysaccharide preparation Times of refined polysaccharide washed by absolute ethanol
Times of refined polysaccharide washed by acetone
Times of refined polysaccharide washed by ether
Refined polysaccharide dried time (h)
Polysaccharide quantity used for conjugation (g)
Polysaccharide concentration after dissolving (mg/mL)
CNBR quantity for per mg of polysaccharide (mg)
pH value of activation
Activation temperature (■)
Activation reaction time (min)
ADH quantity for per mg of polysaccharide (mg)
pH value of derivatization
Derivatization temperature (■)
Derivatization reaction time (min)
Polysaccharide derivative standing time (h)
Standing temperature of polysaccharide derivative (■)
Volume ratio of ultrafiltrate and derivative reaction fluid
Polysaccharide conjugate process Ultrafiltration times
Storage temperature (■)
Storage time (month)
Carrier protein quantity for per mg of polysaccharide derivative (mg)
Sodium chloride concentration (mol/L)
pH value of conjugation
EDAC quantity for per ml of mixed liquid (mg)
Conjugation reaction time (h)
Final pH of conjugation reaction
Conjugate standing time (h)
Conjugate standing temperature (■)
Chromatographic column filler
Collection KD value
Pore size of sterilization & filtration membrane

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3.2.S.2.5 Process Validation and/or Evaluation

The process validation of 13-valent Pneumococcal Polysaccharide Conjugate Vaccine was divided into 2 stages, (1) process design to define the commercial manufacturing process, and (2) process qualification to determine if the process is capable of reproducible commercial manufacturing. The purposes of this process validation were to find (1) whether quality of the product is built into and assured by the manufacturing process, and (2) whether the designed process parameters can produce products with sufficient safety and efficacy.

Validation Protocols

The validation and production of validation samples comply with Standard Operation Procedure of Production and Testing of 13-valent Pneumococcal Polysaccharide Conjugate Vaccine. The validation of monovalent polysaccharide bulk production completed in 2015, and the validation of monovalent polysaccharide conjugate production as well as final product production (vial and pre-filled syringe) completed in 2017.

During monovalent polysaccharide bulk production validation and monovalent polysaccharide conjugate bulk production validation, continuous 3 batches of polysaccharide and polysaccharide conjugate were manufactured and tested to see if the manufacturing process for monovalent polysaccharide bulk and monovalent polysaccharide conjugate bulk can produce homogeneous monovalent polysaccharide and monovalent polysaccharide conjugate that complies with specifications.

In Walvax's Pneumococcal Vaccines Workshop, there are in total 2 production lines for pneumococcal polysaccharide conjugate, and in total 4 purification systems with BPG300/750 purification column. During the process validation, 1 of the 4 purification systems was used for 1 single batch.

Documented validation protocols and reports are listed as below.

Table 28 List of Process Validation Protocol and Report for Intermediate Products

Categories Process validation protocol and report Code

Process validation protocol and report of serotype 1 WSYX-BPVW-TS-pneumococcal polysaccharide 0006-0190
Process validation protocol and report of serotype 3 WSYX-BPVW-TS-pneumococcal polysaccharide 0006-0197
Process validation protocol and report of serotype 4 WSYX-BPVW-TS-pneumococcal polysaccharide 0006-0198
Process validation protocol and report of serotype 5 WSYX-BPVW-TS-pneumococcal polysaccharide 0006-0199
Bulk Process validation protocol and report of serotype 6A WSYX-BPVW-TS-Polysaccharide pneumococcal polysaccharide 0006-0240
Process validation protocol and report of serotype 6B WSYX-BPVW-TS-pneumococcal polysaccharide 0006-0200
Process validation protocol and report of serotype 7F WSYX-BPVW-TS-pneumococcal polysaccharide 0006-0201
Process validation protocol and report of serotype 9V WSYX-BPVW-TS-pneumococcal polysaccharide 0006-0204
Process validation protocol and report of serotype 14 WSYX-BPVW-TS-pneumococcal polysaccharide 0006-0208

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Process validation protocol and report of serotype 18C WSYX-BPVW-TS-pneumococcal polysaccharide 0006-0211
Process validation protocol and report of serotype 19A WSYX-BPVW-TS-pneumococcal polysaccharide 0006-0212
Process validation protocol and report of serotype 19F WSYX-BPVW-TS-pneumococcal polysaccharide 0006-0213
Process validation protocol and report of serotype 23F WSYX-BPVW-TS-pneumococcal polysaccharide 0006-0216
Process validation protocol and report of serotype 1 WSYX-BV2-TS-0006-pneumococcal polysaccharide bulk conjugate 0110
Process validation protocol and report of serotype 3 WSYX-BV2-TS-0006-pneumococcal polysaccharide bulk conjugate 0111
Process validation protocol and report of serotype 4 WSYX-BV2-TS-0006-pneumococcal polysaccharide bulk conjugate 0112
Process validation protocol and report of serotype 5 WSYX-BV2-TS-0006-pneumococcal polysaccharide bulk conjugate 0113
Process validation protocol and report of serotype 6A WSYX-BV2-TS-0006-pneumococcal polysaccharide bulk conjugate 0114
Process validation protocol and report of serotype 6B WSYX-BV2-TS-0006-Monovalent pneumococcal polysaccharide bulk conjugate 0115
Polysaccharide Process validation protocol and report of serotype 7F WSYX-BV2-TS-0006-Bulk pneumococcal polysaccharide bulk conjugate 0116
Conjugate Process validation protocol and report of serotype 9V WSYX-BV2-TS-0006-pneumococcal polysaccharide bulk conjugate 0117
Process validation protocol and report of serotype 14 WSYX-BV2-TS-0006-pneumococcal polysaccharide bulk conjugate 0118
Process validation protocol and report of serotype 18C WSYX-BV2-TS-0006-pneumococcal polysaccharide bulk conjugate 0119
Process validation protocol and report of serotype 19A WSYX-BV2-TS-0006-pneumococcal polysaccharide bulk conjugate 0120
Process validation protocol and report of serotype 19F WSYX-BV2-TS-0006-pneumococcal polysaccharide bulk conjugate 0121
Process validation protocol and report of serotype 23F WSYX-BV2-TS-0006-pneumococcal polysaccharide bulk conjugate 0122

Process Design

The manufacturing process is designed to consist 3 stages, which are (1) production of monovalent polysaccharide bulk, (2) production of monovalent polysaccharide conjugate bulk, and (3) production of final product of PCV13 (vial and pre-filled syringe).

Production of bulk polysaccharide

The production process of pneumococcal polysaccharides can be concluded to fit into 2 types of processes, one is the manufacturing process of serotypes 1, 4, 5, 6A, 6B, 7F, 9V, 14, 18C, 19A, 19F and 23F polysaccharides (Type 1 process), the other is the manufacturing process of serotype 3 polysaccharide (Type 2 process). The 2 types of processes are mostly similar with small variations.

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Brief description of polysaccharide production Type 1

1. Strain amplification

When working seed lot is opened, it was inoculated to a kolle bottle containing Streptococcus pneumoniae comprehensive solid culturing media with 10% sheep blood. Culture for 20±4 hours under 37±1 ■ with 5 - 10% carbon dioxide.

It was then inoculated to a kolle bottle containing Streptococcus pneumoniae comprehensive solid culturing media with 10% sheep blood. Culture for 20±4 hours under 37±1 ■ with 5% - 10% carbon dioxide.

2. Fermentation

After strain amplification, inoculate to a 10 L seed tank, fermenting for 3 - 10 hours at 36.5±1 ■. Then it is transferred to a 100 L tank to ferment for no more than 6 hours at 36.5±1 ■ before being transferred to a 900 L tank to ferment for 6±2 hours at 36.0±1 ■.

3. Crude polysaccharide production

Adding 10% sodium deoxycholate solution till final concentration (w/v) reaches 1% ■ hold for 30 minutes to terminate Streptococcus pneumoniae.

3.1. Removal of culturing media

After termination of bacteria, the fermentation liquid is centrifugated at 7000 rpm for 15 minutes.

Supernatant was collected and ultrafiltrated to 10±2 times concentration. To the concentrate is added anhydrous sodium acetate till a final concentration (w/v) of 3 - 8%. pH is adjusted with glacial acetic acid to 5.6±0.3, and pre-cooled ethanol is added to a final concentration (v/v) of 14 - 50%, hold for no less than 8 hours under 2 - 8 °C. Detailed concentration of sodium acetate and ethanol is shown in the table below.

Table 29 Concentration of Sodium Acetate and Ethanol in Crude Polysaccharide Production Process

Serotypes	Sodium acetate (%)	Ethanol (%)	Serotypes	Sodium acetate (%)	Ethanol (%)
1	5±0.5	20±3	9V	6±0.5	33±3
4	6±0.5	37.5±3	14	5±0.5	20±3
5	7.5±0.5	25±3	18C	5±0.5	28.5±3
6A	5±0.5	25±3	19A	5±0.5	27±3
6B	5±0.5	25±3	19F	5±0.5	27±3
7F	7±0.5	45.7±3	23F	5±0.5	28±3

3.2. Polysaccharide precipitation
After removal of culturing media, the mixture is centrifugated at 10000 rpm for 60 minutes. Supernatant is collected and anhydrous sodium acetate is added to a final concentration of 5 - 13% (no need to add sodium acetate for serotype 5). pH is adjusted with glacial acetic acid to 5.6±0.3. Pre-cooled ethanol is added to a final concentration of 30 - 78%. Hold at 2 - 8 °C for no less than 3 hours. Detailed concentration of sodium acetate and ethanol is shown in the table below.

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Table 30 Concentration of Sodium Acetate and Ethanol in Refined Polysaccharide Production Process

Serotypes	Sodium acetate (%)	Ethanol (%)	Serotypes	Sodium acetate (%)	Ethanol (%)
1	6±0.5	44±3	9V	9±0.5	55.5±3
4	12±0.5	75±3	14	10±0.5	60±3
5	70±3	18C	10±0.5	63.6±3	
6A	10±0.5	50±3	19A	10±0.5	63.5±3
6B	10±0.5	50±3	19F	10±0.5	63.5±3
7F	12±0.5	75±3	23F	10±0.5	60±3

3.3. Crude polysaccharide washing
After polysaccharide precipitation, the mixture is centrifugated at 4000 rpm for 15 minutes. Precipitate is collected and washed in turn with anhydrous ethanol, acetone and ether. The crude polysaccharide is then lyophilized and stored below -20 °C.

4. Refined polysaccharide production
4.1. Removal of protein
Crude polysaccharide is dissolved into neutral 0.3 mol/L sodium acetate solution to a final concentration of 5 - 20 mg/mL. Cold phenol solution is added at a volume ratio of 1:0.5 - 1:1, details of number of extraction times is listed in the table below. The mixture is then centrifugated at 10000 rpm for 40 minutes. Supernatant is collected and ultrafiltrated with no less than 40 times water for injection.

Table 31 Number of Extraction (Times) in Protein Removal Process

Serotypes	Number of extraction (times)	Serotypes	Number of extraction (times)
1	5±1	9V	4±1
4	6±1	14	5±1
5	4±1	18C	5±1
6A	5±1	19A	4±1
6B	5±1	19F	4±1
7F	4±1	23F	5±1

4.2. Removal of nucleic acid
To the mixture is added calcium chloride solution to a final concentration of 0.5 mol/L, then pre-cooled ethanol to a concentration of 15 - 30%. Stir for 30 - 120 minutes, hold for no less than 8 hours at 2 - 8 °C. Centrifugate at 10000 rpm for 60 minutes and collect supernatant, ultrafiltrate for no less than 10 times and collect the concentrate. To the mixture is then added sodium chloride till a final concentration of 0.3 mol/L, then ethanol till a final concentration of 40 - 80%. Stir for 30 minutes and hold for no less than 8 hours at 2 - 8 °C. Detailed ethanol concentration is shown in the table below.

Table 32 Ethanol Concentration used in the Refined Polysaccharide Production Process

Serotypes	1st ethanol (%)	2nd ethanol (%)	Serotypes	1st ethanol (%)	2nd ethanol (%)
1	18±3	44.4±3	9V	25±3	55.5±3
4	25±3	75±3	14	20±3	60±3
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5	25±3	69±3	18C	25±3	63.6±3
6A	25±3	55.5±3	19A	27±3	67±3
6B	25±3	55.5±3	19F	25±3	63.5±3
7F	25±3	75±3	23F	25±3	60±3

4.3. Refined polysaccharide washing
After removal of nucleic acid, the mixture is centrifugated at 4000 rpm for 15 minutes. Precipitate is collected and washed in turn with anhydrous ethanol, acetone twice and ether. The refined polysaccharide is then lyophilized and stored under -20 °C.

Brief description of polysaccharide production Type 2
1. Strain amplification
When working seed lot is opened, it was inoculated to a kolle bottle containing Streptococcus

pneumoniae comprehensive solid culturing media with 10% sheep blood. Culture for 20±4 hours under 37±1 ■ with 5% - 10% carbon dioxide.
It was then inoculated to a kolle bottle containing Streptococcus pneumoniae comprehensive solid culturing media with 10% sheep blood. Culture for 20±4 hours under 37±1 ■ with 5 - 10% carbon dioxide.

2. Fermentation

After strain amplification, inoculate to 10L seed tank, fermenting for 3 - 10 hours at 36.5±1 ■. Then it is transferred to a 10 L tank to ferment for no more than 6 hours at 36.5±1 ■ before being transferred to a 900 L tank to ferment for 6±2 hours at 36.0±1 ■.

3. Crude polysaccharide production

Adding 10% sodium deoxycholate solution till final concentration (w/v) reaches 1%■ hold for 30 minutes to terminate Streptococcus pneumoniae.

3.1. Removal of culturing media

After termination of bacteria, the fermentation liquid is centrifugated at 7000 rpm for 15 minutes. Supernatant was collected and ultrafiltrated to 8±4 times concentration. To the concentrate is added anhydrous sodium acetate till a final concentration (w/v) of 5%. pH is adjusted with glacial acetic acid to 5.4±0.5, and pre-cooled ethanol is added to a final concentration (v/v) of 33.3±1%, hold for no less than 8 hours under 2 - 8 ■.

3.2. Polysaccharide precipitation

After removal of culturing media, the mixture is centrifugated at 4000 rpm for 15 minutes.

Precipitation is collected.

3.3. Crude polysaccharide washing

After collecting the precipitate. It was washed in turn with anhydrous ethanol, acetone and ether. The crude polysaccharide is then lyophilized and stored under -20 ■.

4. Refined polysaccharide production

4.1 Removal of protein

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Crude polysaccharide is dissolved into neutral 0.3 mol/L sodium acetate till a final concentration of 5 - 20 mg/mL. Cold phenol solution is added at a volume ratio of 1:0.5 - 1:1, extract for 4±1 times. The mixture is then centrifugated and supernatant is collected. the supernatant is then ultrafiltrated with no less than 40 times water for injection.

4.2. Removal of nucleic acid

pH is adjusted with sodium bicarbonate solution to 7.0 - 7.5, CTAB solution is added till a final concentration of 1 - 3%, hold for no less than 8 hours at 2 - 8 ■. The mixture is then centrifugated and precipitate is collected. to the precipitate is added 0.2 - 0.4 mol/L sodium chloride solution and stir for 3 - 6 hours. The mixture is then centrifugated at 10000 rpm for 30 minutes and supernatant is collected. Sodium acetate is added till a final concentration of 4 - 10%, and pre-cooled ethanol is added till a final concentration of 33.3±3. Stir for 30 - 120 minutes, hold for no less than 8 hours at 2 - 8 ■.

4.3. Refined polysaccharide washing

The mixture is centrifugated at 4000 rpm for 15 minutes, and precipitate is collected. the precipitate is washed in turn with anhydrous ethanol twice, acetone twice and ether. The refined polysaccharide is then lyophilized and stored at below -20 ■.

Production of monovalent polysaccharide conjugate bulk

1. Derivation of polysaccharide

Refined polysaccharide is dissolved in 0.2 mol/L sodium chloride solution to a final concentration of 3 - 8 mg/ml. If polysaccharide needs molecule degradation, then use ultrasonic method to degrade.

2. Preparation of polysaccharide-ADH derivative

2.1. Derivatives of serotypes 3, 5, 9V and 14

To the solution is added cyanogen bromide at 0.8±0.6 mg per milligram polysaccharide, keep the reaction going at pH 10.5±0.5 under 23±3■ for 12±7 minutes. To the mixture is added adipic acid dihydrazide (ADH) at 4±1 mg per milligram polysaccharide. Keep the reaction going for 20±10 minutes. The reaction is quenched and hold for no less than 8 hours under 5±3■. After ultrafiltration and sterilization filtration, polysaccharide-ADH derivative is stored under 5±3■.

2.2. Derivatives of serotypes 1, 4, 6A, 6B, 7F, 18C, 19A, 19F and 23F

To the solution is added 1-cyano-4-dimethylaminopyridinium tetrafluoroborate (CDAP) as activation agent at 0.5±0.4 mg per milligram polysaccharide. Keep the reaction going at pH 9.0±1.0 under 20±3 ■ for 5±2 minutes. To the mixture is added adipic acid dihydrazide (ADH) at 4±1 mg per milligram polysaccharide. Keep the reaction going for 20±10 minutes. The reaction is quenched and hold for no less than 8 hours under 5±3 ■. After ultrafiltration and sterilization filtration, polysaccharide-ADH derivative is stored under 5±3 ■.

2.3. Preparation of polysaccharide-protein conjugate

To the derivative is added tetanus toxoid at 0.8±0.5 mg per milligram of derivative and sodium chloride till a final concentration of 0.2 mol/L. pH is adjusted with hydrochloric acid to 5.5±0.5. To the mixture is added 1-ethyl-3-(3-dimethylaminopropyl) carbodiimide (EDAC) at 6±5 mg per milliliter of mixture. Keep the reaction going under 5±3 ■ for 4±2 hours. The reaction is quenched with pH adjustment to 7.0±0.5. Hold for no less than 5 hours and dialyze with sodium chloride solution for no less than 8 hours.

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3. Purification of conjugate

The conjugate is purified by column chromatography with Sepharose 4FF column. Use normal saline (for serotype 3) or 0.2 mol/L sodium chloride solution (for all other serotypes) as elution solution. pH at elute side is 5.0 - 7.0, pressure is no more than 0.3 MPa, and loading size is no more than 8% of column volume.

Monovalent polysaccharide conjugate bulk is collected after sterilization filtration of elute around VO.

Production of Final Product (Vial and Pre-filled Syringe)

1. Dilution

Monovalent polysaccharide conjugate bulk, sodium chloride solution and aluminum phosphate are opened and changed to siphonages in a access-restricted isolated system. Water for injection, sodium chloride solution, aluminum phosphate and phosphate buffer are linked to the liquid preparation tank via sterilized connector. Injection water, sodium chloride solution, aluminum

phosphate and phosphate buffer are pumped into the liquid preparation tank per dilution plan. Monovalent polysaccharide conjugate bulks are linked to the bulk weighing bottle, and the weighing bottle is linked to the liquid preparation tank. The monovalent polysaccharide bulks are weighed in the weighing bottle and then pumped into the liquid preparation bottle. Dilution protocol is as below.

There should be 8.8 - 13.2 µg of serotype 6B polysaccharide conjugate, 4.4 - 6.6 µg of serotypes 1, 3, 4, 5, 6A, 7F, 9V, 14, 18C, 19A, 19F and 23F polysaccharide conjugates each, 1.0 mg aluminum phosphate as adjuvant, 88.7 µg monosodium phosphate, 38.0 µg disodium phosphate and 8.5 mg sodium chloride per milliliter final bulk.

2. Sterilized filling

Table 33 Process Parameter of Sterilized Filling

Process Parameter Vial Pre-filled syringe

Volume (mL) 0.5 - 0.6 0.5 - 0.6

Filling workshop temperature (■) 18±2 18±2

Capping workshop temperature (■) 18±2 18±2

Lamp inspection workshop temperature (■) < 25 < 25

Process Qualification

Process qualification data are as shown below.

Monovalent bulk polysaccharide

Table 34 Qualification Data of Serotype 1 Polysaccharide

Batch No.

1-PVPV15101201 1-PVPV15121202 1-PVPV16011203

Items and Specifications

Positive reaction with

Identity Complies Complies Complies

serotype 1 antiserum

Total solid

- 3.245 2.93 3.08

amount

Protein

≤ 2% 1.9 1.5 1.6

content

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Nucleic acid

≤ 2% 0.8 1.1 0.9

content

Total nitrogen

3.5-6.0% 5.0 3.7 5.6

content

Phosphorus

0-1.5% 0.3 0.3 0.6

content

Uronic acid

≥ 45% 55.9 60.8 57.4

content

O-acetyl

≥ 1.8 5.8 6.1 5.7

content

Molecular size

≤ 0.15 0 0 0.04

(KD)

Bacterial

≤ 0.5 EU/µg < 0.5 EU/µg < 0.5 EU/µg < 0.5 EU/µg

endotoxin

Residual

< 0.08% 0.007 Not detected Not detected

ethanol

Residual

< 0.08% Not detected 0.001 Not detected

acetone

Residual ether < 0.08% Not detected Not detected Not detected

Sodium

< 0.2 0.02 0.09 0.10

deoxycholate

Conclusion

The batch size of serotype 1 pneumococcal polysaccharide used in process validation is 16 g/batch, which is sufficient to fill to 1.2 million doses of 13-valent pneumococcal polysaccharide conjugate vaccine and conforms to the batch size of commercial manufacture. The manufacturing process complies with Standard Operation Procedure for Production and Testing of 13-valent Pneumococcal Polysaccharide Conjugate Vaccine issued by Center for Drug Evaluation, China Food and Drug Administration.

Table 35 Qualification Data of Serotype 3 Polysaccharide

Batch No.

3-PVPV15100901 3-PVPV15110902 3-PVPV15120903

Items and Specifications

Positive reaction with

Identity Complies Complies Complies

serotype 3 antiserum

Total solid

- 3.105 3.065 2.325

amount

Protein

≤ 5% 0.1 0.4 0.5

content

Nucleic acid

≤ 2% 0.04 0.09 1.99

content
Total nitrogen
0-1.0% 0.68 0.65 0.86
content
Phosphorus
0-1.0% 0 0 0.10
content
Uronic acid
≥ 40% 50.5 52.9 47.8
content
Molecular
≤ 0.15 0 0 0.04
size (KD)
Bacterial
≤ 0.5 EU/μg < 0.5 EU/μg < 0.5 EU/μg < 0.5 EU/μg
endotoxin
Residual
< 0.08% 0.002 Not detected Not detected
ethanol
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Residual
< 0.03% Not detected Not detected Not detected
acetone
Residual
< 0.01% Not detected Not detected Not detected
ether
Sodium
< 0.2 0.03 0.16 0.17
deoxycholate
Conclusion

The batch size of serotype 3 pneumococcal polysaccharide used in process validation is 25 g/batch, which is sufficient to fill to 0.49 million doses of 13-valent pneumococcal polysaccharide conjugate vaccine and conforms to the batch size of commercial manufacture. The manufacturing process complies with Standard Operation Procedure for Production and Testing of 13-valent Pneumococcal Polysaccharide Conjugate Vaccine issued by Center for Drug Evaluation, China Food and Drug Administration.

Table 36 Qualification Data for Serotype 4 Polysaccharide
Batch No.

4-PVPV15101101 4-PVPV15121102 4-PVPV16011103

Items and Specification
Positive reaction with
Identity Complies Complies Complies
serotype 4 antiserum
Total solid
- 2.600 3.140 2.710

amount
Protein
≤ 3% 1.3 1.3 2.7
content
Nucleic acid
≤ 2% 0.2 0.3 0.6
content
Total nitrogen
4.0-6.0% 5.73 5.73 5.83

content
Phosphorus
0-1.5% 0.20 0.29 0.28
content
Hexosamine
≥ 40% 45.2 46.4 42.0
content
Molecular
≤ 0.15 0 0.04 0
size (KD)
Bacterial
≤ 0.5 EU/μg < 0.5E U/μg < 0.5 EU/μg < 0.5 EU/μg
endotoxin
Conclusion

The batch size of serotype 4 pneumococcal polysaccharide used in process validation is 15 g/batch, which is sufficient to fill to 0.84 million doses of 13-valent pneumococcal polysaccharide conjugate vaccine and conforms to the batch size of commercial manufacture. The manufacturing process complies with Standard Operation Procedure for Production and Testing of 13-valent Pneumococcal Polysaccharide Conjugate Vaccine issued by Center for Drug Evaluation, China Food and Drug Administration.

Table 37 Qualification Data for Serotype 5 Polysaccharide
Batch No.

5-PVPV15101301 5-PVPV15121302 5-PVPV16011303

Items and Specification
Positive reaction with
Identity Complies Complies Complies
serotype 5 antiserum

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Total solid

- 3.08 3.185 3.09
amount
Protein
≤ 7.5% 3.9 6.4 4.5
content
Nucleic acid
≤ 2% 1.7 1.5 1.1
content
Total nitrogen
2.5-6.0% 3.64 5.90 3.82
content
Phosphorus
≤ 2.0% 0.2 0.3 0.2
content
Uronic acid
≥ 12% 13.6 13.3 14.8
content
Hexosamine
≥ 20% 21.7 23.5 20.9
content
Molecular
≤ 0.60 0.35 0.39 0.35
size (KD)
Bacterial
≤ 0.5EU/μg < 0.5 EU/μg < 0.5 EU/μg
endotoxin
Residual
< 0.08% 0.020 0.005 0.0045
ethanol
Residual
< 0.03% 0.0045 Not detected Not detected
acetone
Residual
< 0.01% Not detected Not detected Not detected
ether
Sodium
< 0.2 0.13 0.17 0.12
deoxycholate
Conclusion
The batch size of serotype 5 pneumococcal polysaccharide used in process validation is 17 g/batch, which is sufficient to fill to 0.34 million doses of 13-valent pneumococcal polysaccharide conjugate vaccine and conforms to the batch size of commercial manufacture. The manufacturing process complies with Standard Operation Procedure for Production and Testing of 13-valent Pneumococcal Polysaccharide Conjugate Vaccine issued by Center for Drug Evaluation, China Food and Drug Administration.

Table 38 Qualification Data for Serotype 6A Polysaccharide

Batch No.
6A-PVPV15090101 6A-PVPV15110102 6A-PVPV15120103
Items and Specification
Positive reaction
Identity with serotype 6A Complies Complies Complies
antiserum
Total solid
- 2.020 2.165 2.825
amount
Protein
≤ 1.5% 0.36 1.16 1.3
content
Nucleic acid
≤ 1.5% 0.0082 0.08 0.1
content
Total nitrogen
0-2.0% 1.3 1.5 1.4
content
Phosphorus
2.4-5.0% 4.5 4.17 4.17
content

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Methyl
pentose ≥ 15% 23.7 22.6 23.6
content
Molecular
≤ 0.50 0.39 0.39 0.39
size (KD)
Bacterial
≤ 0.5 EU/μg < 0.5 EU/μg < 0.5 EU/μg
endotoxin
Residual
< 0.08% 0.0565 0.061 0.026
ethanol
Residual
< 0.03% 0.0025 0.002 Not detected
acetone
Residual
< 0.01% Not detected Not detected Not detected
ether
Sodium

< 0.2 0 0.02 0

deoxycholate

Conclusion

The batch size of serotype 6A pneumococcal polysaccharide used in process validation is 25 g/batch, which is sufficient to fill to 0.46 million doses of 13-valent pneumococcal polysaccharide conjugate vaccine and conforms to the batch size of commercial manufacture. The manufacturing process complies with Standard Operation Procedure for Production and Testing of 13-valent Pneumococcal Polysaccharide Conjugate Vaccine issued by Center for Drug Evaluation, China Food and Drug Administration.

Table 39 Qualification Data for Serotype 6B Polysaccharide

Batch No.

6B-PVPV15090201 6B-PVPV15110202 6B-PVPV15120203

Items and Specification

Positive reaction

Identity with serotype 6B Complies Complies Complies

antiserum

Total solid

- 2.505 2.105 2.870

amount

Protein content ≤ 2% 0.1 0.4 1.4

Nucleic acid

≤ 2% 0.1 0.1 0.1

content

Total nitrogen

0-2.0% 0.96 1.09 0.73

content

Phosphorus

2.5-5.0% 4.4 4.4 4.1

content

Methyl pentose

≥ 15% 23.4 23.8 23.5

content

Molecular size

≤ 0.50 0.35 0.30 0.30

(KD)

Bacterial

≤ 0.5 EU/μg < 0.5 EU/μg < 0.5 EU/μg < 0.5 EU/μg

endotoxin

Residual

< 0.08% 0.066 0.064 0.0275

ethanol

Residual

< 0.03% 0.002 0.004 0.002

acetone

Residual ether < 0.01% Not detected Not detected Not detected

Sodium

< 0.20 0.05 0.11 0.18

deoxycholate

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Conclusion

The batch size of serotype 6B pneumococcal polysaccharide used in process validation is 25 g/batch, which is sufficient to fill to 0.53 million doses of 13-valent pneumococcal polysaccharide conjugate vaccine and conforms to the batch size of commercial manufacture. The manufacturing process complies with Standard Operation Procedure for Production and Testing of 13-valent Pneumococcal Polysaccharide Conjugate Vaccine issued by Center for Drug Evaluation, China Food and Drug Administration.

Table 40 Qualification Data for Serotype 7F Polysaccharide

Batch No.

7F-PVPV15100701 7F-PVPV15110702 7F-PVPV16010703

Items and Specification

Positive reaction

Identity with serotype 7F Complies Complies Complies

antiserum

Total solid

- 2.830 3.430 2.585

amount

Protein content ≤ 5% 0.2 1.2 1.1

Nucleic acid

≤ 2% 0.4 0.2 0.2

content

Total nitrogen

1.5-4.0% 3.07 3.38 3.87

content

Phosphorus 0-1.0% 0.1 0.5 0.4

Methyl pentose ≥ 13% 19.4 19.0 17.3

Molecular size

≤ 0.20 0.15 0.12 0

(KD)

Bacterial

≤ 0.5 EU/μg < 0.5 EU/μg < 0.5 EU/μg < 0.5 EU/μg

endotoxin

Residual

< 0.08% 0.022 0.014 0.005

ethanol

Residual

< 0.03% 0.001 0.002 Not detected

acetone
Residual ether < 0.01% Not detected Not detected Not detected
Sodium
< 0.2 0.04 0.12 0.05
deoxycholate
Conclusion
The batch size of serotype 7F pneumococcal polysaccharide used in process validation is 20 g/batch, which is sufficient to fill to 0.93 million doses of 13-valent pneumococcal polysaccharide conjugate vaccine and conforms to the batch size of commercial manufacture. The manufacturing process complies with Standard Operation Procedure for Production and Testing of 13-valent Pneumococcal Polysaccharide Conjugate Vaccine issued by Center for Drug Evaluation, China Food and Drug Administration.

Table 41 Qualification Data for Serotype 9V antiserum
Batch No.
9V-PVPV15090301 9V-PVPV15110302 9V-PVPV15120303
Items and Specification
Positive reaction
Identity with serotype 9V Complies Complies Complies
antiserum

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Total solid
- 3.075 2.920 2.625
amount
Protein content ≤ 2% 0.4 0.3 0.3
Nucleic acid
≤ 2% 0.2 0.05 0.06
content
Total nitrogen
0.5-3.0% 1.95 2.36 2.48
content
Phosphorus
0-1.0% 0.09 0.03 0.07
content
Uronic acid
≥ 15% 22.2 26.1 24.3
content
Hexosamine
≥ 13% 13.0 14.6 15.7
content
Molecular size
≤ 0.45 0.35 0.30 0.30
(KD)
Bacterial
≤ 0.5 EU/μg < 0.5 EU/μg < 0.5 EU/μg <0.5 EU/μg
endotoxin
Residual
< 0.08% 0.007 0.0195 Not detected
ethanol
Residual
< 0.03% Not detected Not detected 0.001
acetone
Residual ether < 0.01% Not detected Not detected Not detected
Sodium
< 0.2 0.04 0.14 0.18
deoxycholate
Conclusion
The batch size of serotype 9V pneumococcal polysaccharide used in process validation is 15 g/batch, which is sufficient to fill to 0.59 million doses of 13-valent pneumococcal polysaccharide conjugate vaccine and conforms to the batch size of commercial manufacture. The manufacturing process complies with Standard Operation Procedure for Production and Testing of 13-valent Pneumococcal Polysaccharide Conjugate Vaccine issued by Center for Drug Evaluation, China Food and Drug Administration.

Table 42 Qualification Data for Serotype 14 Polysaccharide
Batch No.
14-PVPV15101001 14-PVPV15121002 14-PVPV16011003
Items and Specification
Positive reaction
Identity with serotype 14 Complies Complies Complies
antiserum
Total solid
- 3.085 2.385 2.750
amount
Protein content ≤ 5% 0.8 1.9 1.5
Nucleic acid
≤ 2% 1.2 0.1 0.7
content
Total nitrogen
1.5-4.0% 2.82 3.35 3.02
content
Phosphorus
0-1.0% 0.2 0.3 0.3
content
Hexosamine
≥ 20% 22.2 23.6 22.9
content
Molecular size

≤ 0.30 0.19 0.08 0
(KD)

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Bacterial
≤ 0.5 EU/μg < 0.5 EU/μg < 0.5 EU/μg < 0.5 EU/μg
endotoxin
Residual
< 0.08% 0.009 0.0095 0.0045
ethanol
Residual
< 0.03% Not detected Not detected Not detected
acetone
Residual ether < 0.01% Not detected Not detected Not detected
Sodium
< 0.2 0.06 0.18 0.09
deoxycholate
Conclusion
The batch size of serotype 14 pneumococcal polysaccharide used in process validation is 20 g/batch, which is sufficient to fill to 0.53 million doses of 13-valent pneumococcal polysaccharide conjugate vaccine and conforms to the batch size of commercial manufacture. The manufacturing process complies with Standard Operation Procedure for Production and Testing of 13-valent Pneumococcal Polysaccharide Conjugate Vaccine issued by Center for Drug Evaluation, China Food and Drug Administration.

Table 43 Qualification Data for Serotype 18C Polysaccharide
Batch No.
18C-PVPV15100801 18C-PVPV15110802 18C-PVPV16010803
Items and Specification
Positive
reaction with
Identity Complies Complies Complies
serotype 18C
antiserum
Total solid
- 2.725 2.130 1.990
amount
Protein
≤ 3% 1.0 0.8 1.3
content
Nucleic acid
≤ 2% 0.2 0.2 0.2
content
Total nitrogen
0-1.0% 0.92 0.89 0.95
content
Phosphorus
2.4-4.0% 3.0 2.8 2.7
content
Methyl
pentose ≥ 14% 14.5 17.5 14.6
content
Molecular
≤ 0.15 0.07 0.08 0.08
size (KD)
Bacterial
≤ 0.5 EU/μg < 0.5 EU/μg < 0.5 EU/μg < 0.5 EU/μg
endotoxin
Residual
< 0.08% 0.020 0.034 0.046
ethanol
Residual
< 0.03% Not detected 0.002 0.002
acetone
Residual
< 0.01% Not detected Not detected Not detected
ether
Sodium
< 0.2 0.09 0.06 0.11
deoxycholate
Conclusion

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The batch size of serotype 18C pneumococcal polysaccharide used in process validation is 15 g/batch, which is sufficient to fill to 0.86 million doses of 13-valent pneumococcal polysaccharide conjugate vaccine and conforms to the batch size of commercial manufacture. The manufacturing process complies with Standard Operation Procedure for Production and Testing of 13-valent Pneumococcal Polysaccharide Conjugate Vaccine issued by Center for Drug Evaluation, China Food and Drug Administration.

Table 44 Qualification Data for Serotype 19A Polysaccharide
Batch No.
19A-PVPV15090401 19A-PVPV15110402 19A-PVPV15120403
Items and Specification
Positive
reaction with

Identity Complies Complies Complies
serotype 19A
antiserum
Total solid
- 2.210 2.445 2.850
amount
Protein
≤ 2% 0.4 0.7 0.5
content
Nucleic acid
≤ 2% 0.3 0.02 0.05
content
Total nitrogen
0.6-3.5% 3.26 3.31 3.44
content
Phosphorus
3.0-7.0% 4.9 4.7 4.9
content
Hexosamine
≥ 12% 12.6 13.7 15.0
content
Methyl
pentose ≥ 20% 29.0 31.3 31.8
content
Molecular
≤ 0.45 0.18 0.04 0.04
size (KD)
Bacterial
≤ 0.5 EU/μg < 0.5 EU/μg < 0.5 EU/μg < 0.5 EU/μg
endotoxin
Residual
< 0.08% 0.015 0.0145 0.005
ethanol
Residual
< 0.03% 0.002 0.002 0.002
acetone
Residual
< 0.01% Not detected Not detected 0.001
ether
Sodium
< 0.2 0.07 0.004 0.15
deoxycholate
Conclusion
The batch size of serotype 19A pneumococcal polysaccharide used in process validation is 15 g/batch, which is sufficient to fill to 0.64 million doses of 13-valent pneumococcal polysaccharide conjugate vaccine and conforms to the batch size of commercial manufacture. The manufacturing process complies with Standard Operation Procedure for Production and Testing of 13-valent Pneumococcal Polysaccharide Conjugate Vaccine issued by Center for Drug Evaluation, China Food and Drug Administration.

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Table 45 Qualification Data for Serotype 19F Polysaccharide
Batch No.
19F-PVPV15090501 19F-PVPV15110502 19F-PVPV15120503
Items and Specification

Positive
reaction with
Identity Complies Complies Complies
serotype 19F
antiserum
Total solid
- 2.750 2.535 3.130
amount
Protein
≤ 3% 0.4 0.2 0.3
content
Nucleic acid
≤ 2% 0.1 0.1 0.09
content
Total nitrogen
1.4-3.5% 3.27 3.00 3.42
content
Phosphorus
3.0-5.5% 4.6 4.8 4.8
content
Hexosamine
≥ 12.5% 19.1 20.3 18.3
content
Methyl
pentose ≥ 20% 28.1 27.9 24.2
content
Molecular
≤ 0.40 0.04 0 0
size (KD)
Bacterial
≤ 0.5 EU/μg < 0.5 EU/μg < 0.5 EU/μg < 0.5 EU/μg
endotoxin
Residual
< 0.08% 0.013 0.0145 0.0035

ethanol
Residual
< 0.03% 0.002 0.003 0.0005
acetone
Residual
< 0.01% Not detected Not detected Not detected
ether
Sodium
< 0.2 0.10 0 0.13
deoxycholate
Conclusion
The batch size of serotype 19F pneumococcal polysaccharide used in process validation is 20 g/batch, which is sufficient to fill to 0.63 million doses of 13-valent pneumococcal polysaccharide conjugate vaccine and conforms to the batch size of commercial manufacture. The manufacturing process complies with Standard Operation Procedure for Production and Testing of 13-valent Pneumococcal Polysaccharide Conjugate Vaccine issued by Center for Drug Evaluation, China Food and Drug Administration.

Table 46 Qualification Data for Serotype 23F Polysaccharide
Batch No.
23F-PVPV15090601 23F-PVPV15120602 23F-PVPV16010603
Items and Specification
Positive
reaction with
Identity Complies Complies Complies
serotype 23F
antiserum
Total solid
- 2.585 3.145 3.180
amount

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Protein
≤ 2% 0.3 0.2 0.3
content
Nucleic acid
≤ 2% 0.2 0.1 0.1
content
Total nitrogen
0-1.0% 0.89 0.88 0.97
content
Phosphorus
3.0-4.5% 3.8 3.9 3.3
content
Methyl
pentose ≥ 37% 41.5 41.8 38.8
content
Molecular
≤ 0.15 0.04 0.08 0
size (KD)
Bacterial
≤ 0.5 EU/μg < 0.5 EU/μg < 0.5 EU/μg < 0.5 EU/μg
endotoxin
Residual
< 0.08% 0.017 0.010 0.0015
ethanol
Residual
< 0.03% 0.002 0.003 Not detected
acetone
Residual
< 0.01% Not detected Not detected Not detected
ether
Sodium
< 0.2 0.04 0.17 0.08
deoxycholate
Conclusion
The batch size of serotype 23F pneumococcal polysaccharide used in process validation is 20 g/batch, which is sufficient to fill to 0.49 million doses of 13-valent pneumococcal polysaccharide conjugate vaccine and conforms to the batch size of commercial manufacture. The manufacturing process complies with Standard Operation Procedure for Production and Testing of 13-valent Pneumococcal Polysaccharide Conjugate Vaccine issued by Center for Drug Evaluation, China Food and Drug Administration.

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Monovalent bulk conjugate

Table 47 Qualification Data for Serotype 1 Bulk Conjugate
Batch No.
RSP1C201706001 RSP1C201707002 RSP1C201709003
Items and Specification
Polysaccharide
- 245.4 μg/mL 258.0 μg/mL 423.8 μg/mL
content
Protein content - 239.1 μg/mL 170.5 μg/mL 462.2 μg/mL
PS:protein 0.20 - 1.80 1.03 1.51 0.92
Recovery at KD

Molecular size ≤0.20 should be no 91.4% 99.4% 96.2%
less than 80%
Residual EDAC Less than 5 µmol/L 2.8 µmol/L 2.9 µmol/L Not detected
Free
No more than 25.0% 10.8% 15.8% 14.0%
Polysaccharide
Free protein No more than 5.0% 4.1% Not detected 1.3%
ADH content < 0.15 µg/mL 0.048 µg/mL 0.062 µg/mL 0.086 µg/mL
Positive reaction with
Identity Complies Complies Complies
serotype 1 antiserum
Bacterial No more than 0.75
< 0.75 EU/µg < 0.75 EU/µg < 0.75 EU/µg
endotoxin EU per ug PS
Should comply CP
Sterilization Complies Complies Complies
2015
Yield - 13.22% 23.10% 33.48%

Table 48 Qualification Data for Serotype 3 Bulk Conjugate
Batch No.

RSP3C201706001 RSP3C201707002 RSP3C201708003

Items and Specification

Polysaccharide

- 61.6 µg/mL 77.2 µg/mL 63.6 µg/mL
content

Protein content - 141.6 µg/mL 355.8 µg/mL 285.3 µg/mL

PS:protein 0.10 - 0.90 0.44 0.22 0.22

Recovery at KD

Molecular size ≤0.20 should be no 98.5% 97.8% 97.6%

less than 80%

Residual EDAC Less than 5µmol/L 1.1 µmol/L 4.4 µmol/L 3.8 µmol/L

Free

No more than 20.0% 15.7% 6.2% 12.2%

Polysaccharide

Free protein No more than 5.0% 3.8% 0.48% 1.2%

ADH content < 0.25 µg/mL 0.016 µg/mL 0.160 µg/mL 0.128 µg/mL

Positive reaction with

Identity Complies Complies Complies

serotype 3 antiserum

Bacterial No more than 0.75

< 0.75 EU/µg < 0.75 EU/µg < 0.75 EU/µg

endotoxin EU per ug PS

Should comply CP

Sterilization Complies Complies Complies

2015

Yield ----- 5.88% 6.37% 5.54%

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Table 49 Qualification Data for Serotype 4 Bulk Conjugate

Batch No.

RSP4C201706001 RSP4C201707002 RSP4C201709003

Items and Specification

Polysaccharide

- 154.3 µg/mL 196.0 µg/mL 190.8 µg/mL

content

Protein content - 254.1 µg/mL 276.2 µg/mL 288.0 µg/mL

PS:protein 0.30 - 0.90 0.61 0.71 0.66

Recovery at KD ≤

Molecular size 0.20 should be no 98.9% 98.0% 99.4%

less than 80%

Residual EDAC Less than 5 µmol/L 0.9 µmol/L 3.6 µmol/L 3.5 µmol/L

Free

No more than 25.0% 19.6% 23.6% 14.3%

Polysaccharide

Free protein No more than 5.0% 4.7% 2.2% 0.9%

ADH content < 0.15 µg/mL 0.082 µg/mL 0.110 µg/mL 0.082 µg/mL

Positive reaction with

Identity Complies Complies Complies

serotype 4 antiserum

Bacterial No more than 0.75

<0.75 EU/µg <0.75 EU/µg <0.75 EU/µg

endotoxin EU per ug PS

Should comply CP

Sterilization Complies Complies Complies

2015

Yield ----- 20.12% 16.35% 13.90%

Table 50 Qualification Data for Serotype 5 Bulk Conjugate

Batch No.

RSP5C201705001 RSP5C201707002 RSP5C201708003

Items and Specification

Polysaccharide

- 91.1 µg/mL 57.8 µg/mL 99.8 µg/mL

content

Protein content - 431.0 µg/mL 172.1 µg/mL 224.6 µg/mL

PS:protein 0.10 - 0.60 0.21 0.34 0.44

Recovery at KD ≤

Molecular size 0.20 should be no less 94.3% 98.9% 96.8%

than 80%
 Residual EDAC Less than 5 µmol/L 0.7 µmol/L 0.8 µmol/L 3.7 µmol/L
 Free
 No more than 20.0% 18.0% 17.4% 15.4%
 Polysaccharide
 Free protein No more than 5.0% 3.2% Not detected 0.5%
 ADH content < 0.25 µg/mL 0.139 µg/mL 0.102 µg/mL 0.102 µg/mL
 Positive reaction with
 Identity Complies Complies Complies
 serotype 5 antiserum
 Bacterial No more than 0.75
 < 0.75 EU/µg < 0.75 EU/µg < 0.75 EU/µg
 endotoxin EU per ug PS
 Should comply CP
 Sterilization Complies Complies Complies
 2015
 Yield - 6.67% 4.83% 6.36%

Table 51 Qualification Data for Serotype 6A Bulk Conjugate
 Batch No.
 RSP6AC201705001 RSP6AC201707002 RSP6AC201708003
 Items and Specification
 Polysaccharide
 - 139.1 µg/mL 154.6 µg/mL 86.8 µg/mL
 content
 Protein content - 547.4 µg/mL 493.0 µg/mL 191.5 µg/mL

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PS:protein 0.10 - 0.50 0.25 0.31 0.45
 Recovery at
 KD ≤ 0.20
 Molecular size 95.6% 88.1% 97.7%
 should be no
 less than 80%
 Less than 5
 Residual EDAC 3.6 µmol/L 2.6 µmol/L 2.3 µmol/L
 µmol/L
 Free No more than
 13.7% 19.4% 12.8%
 Polysaccharide 20.0%
 No more than
 Free protein 3.6% 0.9% 0.5%
 5.0%
 ADH content < 1.20 µg/mL 0.163 µg/mL 0.081 µg/mL 0.227 µg/mL
 Positive
 reaction with
 Identity Complies Complies Complies
 serotype 6A
 antiserum
 No more than
 Bacterial
 0.75 EU per < 0.75 EU/µg < 0.75 EU/µg < 0.75 EU/µg
 endotoxin
 ug PS
 Should
 Sterilization comply CP Complies Complies Complies
 2015
 Yield - 6.06% 6.44% 3.97%

Table 52 Qualification Data for Serotype 6B Bulk Conjugate
 Batch No.
 RSP6BC201705001 RSP6BC201707002 RSP6BC201708003
 Items and Specification
 Polysaccharide
 - 152.9 µg/mL 170.2 µg/mL 124.8 µg/mL
 content
 Protein
 - 327.4 µg/mL 349.9 µg/mL 296.5 µg/mL
 content
 PS:protein 0.30 - 0.60 0.47 0.49 0.42
 Recovery at KD ≤
 Molecular size 0.20 should be no 98.6% 93.5% 96.1%
 less than 80%
 Residual
 Less than 5 µmol/L 2.8 µmol/L 0.5 µmol/L 0.1 µmol/L
 EDAC
 Free
 No more than 25.0% 21.6% 21.1% 9.2%
 Polysaccharide
 Free protein No more than 5.0% 4.0% 0.9% 2.1%
 ADH content < 0.25 µg/mL 0.012 µg/mL 0.094 µg/mL 0.142 µg/mL
 Positive reaction
 Identity with serotype 6B Complies Complies Complies
 antiserum
 Bacterial No more than 0.75
 < 0.75 EU/µg <0.75 EU/µg <0.75 EU/µg
 endotoxin EU per ug PS
 Should comply CP
 Sterilization Complies Complies Complies

2015
Yield - 11.92% 14.97% 11.31%

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Table 53 Qualification Data for Serotype 7F Bulk Conjugate
Batch No.

RSP7FC201705001 SP7FC201706002 SP7FC201708003
Items and Specification

Polysaccharide

- 284.1 µg/mL 303.0 µg/mL 310.8 µg/mL
content

Protein content - 643.5 µg/mL 362.8 µg/mL 494.3 µg/mL

PS:protein 0.20 - 0.90 0.44 0.84 0.63

Recovery at KD ≤0.20

Molecular size should be no less than 97.2% 96.6% 96.6%
80%

Residual

Less than 5 µmol/L 3.2 µmol/L 1.2 µmol/L 3.7 µmol/L

EDAC

Free

No more than 25.0% 18.4% 18.5% 18.9%

Polysaccharide

Free protein No more than 5.0% 4.0% 0.3% 1.3%

ADH content < 0.20 µg/mL 0.066 µg/mL 0.121 µg/mL 0.088 µg/mL

Positive reaction with

Identity Complies Complies Complies

serotype 7F antiserum

Bacterial No more than 0.75 EU

<0.75 EU/µg <0.75 EU/µg <0.75 EU/µg

endotoxin per ug PS

Should comply CP

Sterilization Complies Complies Complies

2015

Yield ----- 7.46% 18.13% 16.39%

Table 54 Qualification Data for Serotype 9V Bulk Conjugate
Batch No.

RSP9VC201705001 RSP9VC201707002 RSP9VC201709003
Items and Specification

Polysaccharide

- 148.6 µg/mL 159.3 µg/mL 157.5 µg/mL
content

Protein content - 513.5 µg/mL 614.1 µg/mL 722.3 µg/mL

PS:protein 0.10 - 0.35 0.29 0.26 0.22

Recovery at KD

Molecular size ≤ 0.20 should be 100.0% 99.7% 98.8%
no less than 80%

Less than 5

Residual EDAC 2.5 µmol/L 2.2 µmol/L 4.1 µmol/L
µmol/L

Free No more than

6.8% 12.0% 11.8%

Polysaccharide 25.0%

No more than

Free protein 4.3% 1.6% 0.7%

5.0%

ADH content < 0.70 µg/mL 0.176 µg/mL 0.153 µg/mL 0.203 µg/mL

Positive reaction

Identity with serotype 9V Complies Complies Complies
antiserum

No more than

Bacterial

0.75 EU per ug <0.75 EU/µg <0.75 EU/µg <0.75 EU/µg

endotoxin

PS

Should comply

Sterilization Complies Complies Complies

CP 2015

Yield - 11.55% 11.49% 12.3%

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Table 55 Qualification Data for Serotype 14 Bulk Conjugate
Batch No.

RSP14C201706001 RSP14C201707002 RSP14C201709003
Items and Specification

Polysaccharide

- 143.2 µg/mL 134.9 µg/mL 125.0 µg/mL
content

Protein content - 480.7 µg/mL 340.4 µg/mL 305.8 µg/mL

PS:protein 0.20 - 0.60 0.30 0.40 0.41

Recovery at KD ≤

Molecular size 0.20 should be no 95.9% 99.0% 97.2%
less than 80%

Residual EDAC Less than 5 µmol/L 2.4 µmol/L 3.8 µmol/L 1.0 µmol/L

Free No more than

14.6% 12.6% 23.9%

Polysaccharide 25.0%
Free protein No more than 5.0% 3.1% 1.0% 1.0%
ADH content < 1.15 µg/mL 0.016 µg/mL 0.120 µg/mL 0.114 µg/mL
Positive reaction
Identity with serotype 14 Complies Complies Complies
antiserum
Bacterial No more than 0.75
<0.75 EU/µg <0.75 EU/µg <0.75 EU/µg
endotoxin EU per ug PS
Should comply CP
Sterilization Complies Complies Complies
2015
Yield - 8.91% 7.76% 7.10%

Table 56 Qualification Data for Serotype 18C Bulk Conjugate
Batch No.
RSP18CC201706001 RSP18CC201707002 RSP18CC201708003
Items and Specification
Polysaccharide
- 106.6 µg/mL 153.3 µg/mL 144.4 µg/mL
content
Protein content - 195.8 µg/mL 263.1 µg/mL 199.8 µg/mL
PS:protein 0.40 - 0.80 0.54 0.58 0.72
Recovery at
KD ≤ 0.20
Molecular size 97.7% 86.9% 96.9%
should be no
less than 80%
Less than 5
Residual EDAC 0.9 µmol/L 1.5 µmol/L 0.7 µmol/L
µmol/L
Free No more than
7.7% 26.2% 22.6%
Polysaccharide 30.0%
No more than
Free protein 2.7% 0.9% 0.7%
5.0%
ADH content < 0.20 µg/mL 0.097 µg/mL 0.108 µg/mL 0.015 µg/mL
Positive
reaction with
Identity Complies Complies Complies
serotype 18C
antiserum
No more than
Bacterial
0.75 EU per ug < 0.75 EU/µg < 0.75 EU/µg < 0.75 EU/µg
endotoxin
PS
Should comply
Sterilization Complies Complies Complies
CP 2015
Yield - 12.25% 21.82% 17.60%

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Table 57 Qualification Data for Serotype 19A Bulk Conjugate
Batch No.
RSP19AC201705001 RSP19AC201707002 RSP19AC201708003
Items and Specification
Polysaccharide
- 131.7 µg/mL 163.6 µg/mL 202.0 µg/mL
content
Protein content - 409.8 µg/mL 435.2 µg/mL 444.9 µg/mL
PS:protein 0.20 - 0.70 0.32 0.38 0.45
Recovery at
KD ≤0.20
Molecular size 95.4% 93.3% 99.4%
should be no
less than 80%
Less than 5
Residual EDAC 1.2 µmol/L 1.3 µmol/L 2.3 µmol/L
µmol/L
Free No more than
27.4% 26.0% 28.0%
Polysaccharide 30.0%
No more than
Free protein 2.9% 2.8% 0.7%
5.0%
ADH content < 0.20 µg/mL 0.108 µg/mL 0.130 µg/mL 0.103 µg/mL
Positive
reaction with
Identity Complies Complies Complies
serotype 19A
antiserum
No more than
Bacterial
0.75 EU per < 0.75 EU/µg < 0.75 EU/µg < 0.75 EU/µg
endotoxin
ug PS
Should

Sterilization comply CP Complies Complies Complies
2015
Yield ----- 9.20% 13.05% 16.48%

Table 58 Qualification Data for Serotype 19F Bulk Conjugate

Batch No.
RSP19FC201706001 RSP19FC201707002 RSP19FC201708003
Items and Specification
Polysaccharide
- 195.7 µg/mL 145.6 µg/mL 134.0 µg/mL
content
Protein content - 500.3 µg/mL 237.8 µg/mL 300.3 µg/mL
PS:protein 0.10 - 0.70 0.39 0.61 0.45
Recovery at
KD ≤0.20
Molecular size 88.8% 98.1% 96.4%
should be no
less than 80%
Less than 5
Residual EDAC 3.9 µmol/L 1.4 µmol/L 3.1 µmol/L
µmol/L
Free No more than
24.4% 30.1% 25.2%
Polysaccharide 35.0%
No more than
Free protein 3.2% 2.4% 1.1%
5.0%
ADH content < 0.20 µg/mL 0.073 µg/mL 0.126 µg/mL 0.066 µg/mL
Positive
reaction with
Identity Complies Complies Complies
serotype 19F
antiserum

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No more than
Bacterial
0.75 EU per ug < 0.75 EU/µg < 0.75 EU/µg < 0.75 EU/µg
endotoxin
PS
Should comply
Sterilization Complies Complies Complies
CP 2015
Yield ----- 11.61% 8.62% 8.08%

Table 59 Qualification Data for Serotype 23F Bulk Conjugate

Batch No.
RSP23FC201705001 RSP23FC201707002 RSP23FC201708003
Items and Specification
Polysaccharide
- 85.4 µg/mL 91.3 µg/mL 75.8 µg/mL
content
Protein content - 232.8 µg/mL 215.0 µg/mL 212.3 µg/mL
PS:protein 0.25 - 0.65 0.37 0.42 0.36
Recovery at KD
Molecular size ≤ 0.20 should be 98.7% 100.0% 97.5%
no less than 80%
Residual Less than 5
0.7 µmol/L 2.5 µmol/L 2.1 µmol/L
EDAC µmol/L
Free No more than
13.0% 24.9% 18.7%
Polysaccharide 35.0%
No more than
Free protein 1.6% 2.1% 0.6%
5.0%
ADH content < 0.45 µg/mL 0.162 µg/mL 0.174 µg/mL 0.160 µg/mL
Positive reaction
Identity with serotype Complies Complies Complies
23F antiserum
No more than
Bacterial
0.75 EU per ug < 0.75 EU/µg < 0.75 EU/µg < 0.75 EU/µg
endotoxin
PS
Should comply
Sterilization Complies Complies Complies
CP 2015
Yield - 8.80% 9.02% 4.28%
Conclusion

The production process of monovalent pneumococcal polysaccharide conjugate bulk under pre-determined process parameters can continuously manufacture products that comply with Standard Operation Procedure for Production and Testing of 13-valent Pneumococcal Polysaccharide Conjugate Vaccine, critical process parameters and critical quality attributes are well in control during manufacturing process and the process is stable.

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Final product (vial)
Table 60 Qualification Data for 13-valent Pneumococcal Polysaccharide Conjugate Vaccine (Vial)
Osmotic Sodium Aluminum Bacterial Abnormal
Items Appearance Volume pH Sterilization
pressure chloride phosphate endotoxin toxicity
Healthy and
weight gain for
Milky white
No less No more each guinea pig
Specs 270±50 turbid liquid 7.5 - 9.5 Should comply
than 0.5 5.5 - 7.5 0.8 - 1.2 mg/mL than 12.5 and mouse
Batch mOsmol/kg after g/L with regulation
mL/vial EU/dose during
shaking
observation
period
< 12.5
GI201708001 253 Complies Complies 6.2 8.48 1.03 Complies Complies
EU/dose
< 12.5
GI201708002 255.5 Complies Complies 6.25 8.46 1.08 Complies Complies
EU/dose
< 12.5
GI201708003 268.5 Complies Complies 6.22 8.815 0.90 Complies Complies
EU/dose
Items Polysaccharide content

Serotypes 1 3 4 5 6A 6B 7F 9V 14 18C 19A 19F 23F
Specs
4.4 - 6.6 µg/dose for serotype 6B; 2.2 - 3.3 µg/dose for all other serotypes
Batch
GI201708001 3.2 2.4 2.8 2.7 2.9 6.0 3.1 2.6 2.7 2.7 2.5 2.5 2.6

GI201708002 2.8 2.6 2.4 2.4 2.9 5.4 2.8 2.9 2.6 2.8 2.5 2.6 2.9

GI201708003 2.9 2.4 2.7 2.7 2.6 4.8 2.7 2.4 2.7 2.7 2.8 2.8 3.0

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MODULE 3 QUALITY
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Final product (pre-filled syringe)
Table 61 Qualification Data for 13-valent Pneumococcal Polysaccharide Conjugate Vaccine (Pre-filled Syringe)
Osmotic Sodium Aluminum Bacterial
Items Appearance Volume pH Sterilization Abnormal toxicity
pressure chloride phosphate endotoxin
Healthy and weight
Milky white
No less gain for each
Specs 270±50 turbid liquid 7.5 - 9.5 No more than Should comply
than 0.5 5.5 - 7.5 0.8 - 1.2 mg/mL guinea pig and
Batch mOsmol/kg after g/L 12.5 EU/dose with regulation
mL/vial mouse during
shaking
observation period
< 12.5
GI201708001 272.5 Complies Complies 6.22 8.81 0.94 Complies Complies
EU/dose
< 12.5
GI201708002 253 Complies Complies 6.20 8.01 0.99 Complies Complies
EU/dose
< 12.5
GI201708003 288.5 Complies Complies 6.22 8.96 1.07 Complies Complies
EU/dose
Items Polysaccharide content

Serotypes 1 3 4 5 6A 6B 7F 9V 14 18C 19A 19F 23F
Specs
4.4 - 6.6 µg/dose for serotype 6B; 2.2 - 3.3 µg/dose for all other serotypes
Batch
GI201708001 3.0 2.7 2.6 2.6 2.6 4.8 2.8 3.1 2.6 2.6 3.0 3.0 2.4

GI201708002 3.0 2.8 2.6 2.6 2.7 4.6 2.8 3.1 2.6 2.9 3.0 2.9 2.4

GI201708003 3.1 2.8 2.6 2.6 2.6 4.7 2.8 3.0 2.7 2.6 3.1 2.9 2.5

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Conclusion
The production process of 13-valent pneumococcal polysaccharide conjugate vaccine (vial) under pre-determined process parameters can continuously manufacture products that comply with Standard Operation Procedure for Production and Testing of 13-valent Pneumococcal Polysaccharide Conjugate Vaccine (Vial), critical process parameters and critical quality attributes are well in control during manufacturing process and the process is stable.
The production process of 13-valent pneumococcal polysaccharide conjugate vaccine (syringe) under pre-determined process parameters can continuously manufacture products that comply with Standard Operation Procedure for Production and Testing of 13-valent Pneumococcal Polysaccharide Conjugate Vaccine (Syringe), critical process parameters and critical quality

attributes are well in control during manufacturing process and the process is stable. During process validation of Packaging Center, manufacturing capability of vial (production line 1 and 2) is between 80000 - 150000 vials, and batch size during process validation was 80000 vials. Thus, the commercial manufacturing batch size is 100000 vials, which falls within the manufacturing capability. During process validation of Packaging Center, manufacturing capability of pre-filled syringe is between 80000 - 140000 pre-filled syringes, and the batch size during process validation was 80000 pre-filled syringes. Thus, the commercial manufacturing batch size is 100000 pre-filled syringes, which falls within the manufacturing capability.

Continued Process Verification

Walvax will continue to monitor the manufacturing process of PCV13. Based on statistical analysis of process data and critical quality attributes of the product, the manufacturing process is constantly monitored, and dynamically ensures critical quality attributes are in control in the whole life cycle of the product.

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3.2.S.2.6 Manufacturing process Development

1. Polysaccharide preparation process

The company has independently developed a pneumococcal synthetic medium, which is suitable for the growth of pneumococcus. Growth of bacteria is monitored by testing OD at 694nm. Systematic condition screening and optimizing are carried out for critical parameters including temperature, pH, pressure, speed of stirring, inoculation concentration, etc. Batches of fermentation and polysaccharide purification process validation for all serotypes have been completed. According to the results of validation, the process is feasible and repeatable; the yield of purified polysaccharide is stable. All the quality testing results are in compliance with registration standards, Chinese Pharmacopeia (2010 Edition), European Pharmacopeia 7.0 (2011 Edition), and WHO guidelines.

2. Production Process of Carrier Protein (Tetanus Toxoid)

The production process of Yuxi Walvax's tetanus toxoid is as follows: recovery and amplification of tetanus clostridia, fermentation, precipitation with ammonium sulfate, purification, detoxification with formaldehyde, and purification using chromatography. The purified tetanus toxoid bulk is then used as carrier proteins.

Yuxi Walvax's production process of tetanus toxoid is mature and stable; fermentation produces high yields of toxin (floc unit of ferment is over 100 lf/ml, which is much higher than Chinese Pharmacopeia specifications of 40lf/ml); the quality of the products is good and stable; all testing results comply with requirements in Chinese Pharmacopeia.

3. Conjugation Process

The company carries out comprehensive molecular structural analysis for purified polysaccharide.

Conjugation is carried out via indirect method: the polysaccharide is activated and derivated to yield polysaccharide-ADH; derivation is conjugated to the carrier protein (tetanus toxoid), with EDAC as the catalyst to yield polysaccharide-TT conjugate; conjugate is purified by gel column chromatography; collect the absorption peak near V0 to yield conjugate bulk. Stability of the process is validated.

Results of process validation demonstrate that the processes of activation, derivation, and conjugation are stable; that the yield of conjugation is stable; and that the quality of conjugate bulk complies with in-house specifications.

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3.2.S.3 Characterization

3.2.S.3.1 Elucidation of Structure and other Characteristics

Walvax carried out 1H-NMR tests of the 13 serotypes of pneumococcal polysaccharides, and analyzed the corresponding spectra by comparison with ATCC standard polysaccharides. The results showed that the structures of Walvax's polysaccharides are in full accord with ATCC standard polysaccharides.

Table 62 General Information of 1H-NMR Tests of the 13 Serotypes of Pneumococcal Polysaccharides

Test information Walvax ATCC

NMR machine Varian Inova 500 Varian Inova 500

Frequency 500 MHz 500 MHz

For serotypes 1, 3, 4, 5, 6A, 6B, 7F, 9V, 18C,

For all 13 serotypes: D2O + 19A, 19F, 23F: D2O + 0.01%DMSO + 0.01%

Solution system 0.05%wt TSP + 0.5 mg/mL DSS-d6

maleic acid For serotype 14: D2O + 0.05%wt TSP +

0.5mg/ml maleic acid

Polysaccharide

5 mg/mL 5 mg/mL

concentration

Given the information in Table 62, test instrument and test frequency of Walvax's polysaccharides tests match those of ATCC standard polysaccharides tests. Walvax's polysaccharides used a solution system of D2O + 0.05% wt TSP + 0.5 mg/mL maleic acid, while ATCC serotypes 1, 3, 4, 5, 6A, 7F, 9V, 18C, 19A, 19F, 23F polysaccharides used a solution system of D2O + 0.01% DMSO + 0.01% DSS-d6, and serotype 14 used a solution system of D2O + 0.05% wt TSP + 0.5 mg/mL maleic acid. The two systems used maleic acid and DMSO respectively as quantitative and TSP and DSS-d6 respectively as internal standard, which has no influence on test results.

Table 63 Abbreviations for Polysaccharide 1H-NMR Spectra

Abbreviations Name Abbreviations Name

D2O Heavy water Gal Galactose

HDO half-heavy water GlcA Glucuronic acid

Sodium 3 - (trimethylsilicon)

TSP GlcNAc N-acetyl glucose

deuteropropionate

DMSO Dimethyl sulfoxide ManNAc N-acetyl mannose

Sodium 3,3,4,4,5,5-hexadeuteriu-2,2-
DSS-d6 FucNAc N-acetyl fucose
dimethyl-2-cyclopentane-5-sulfonate
2-acetyl amino-2,6-dideoxy-D-
C-poly C-polysaccharide Sug
glutaraldehyde-hexose-4-ketone
GalA Galacturonic acid Pyr Pyruvic acid
GalNAc N-acetyl galactose Rha Rhamnolipid
Glc Glucose -PO3 Phosphate radical

1
H-NMR spectra and corresponding comparisons with ATCC standard polysaccharides are shown as follows.

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(1) Structure of Serotype 1 Pneumococcal Polysaccharide:

→ 4) – α – GalpA(1 → 3) – α – GalpA(1 → 3) – α – 6dGalpNAc4 N (1 →

* O
4 6 COOH
O HO HOOC
6 O
5 4 H2N H3C 6
HO 2 5
2 O
4 2
1 O 5 1
3 OH 3 OH 1 O 3 NHAC
*

n
Molecular size: determined by gel filtration through Sepharose CL-4B, release specification is
KD ≤ 0.15, recovery rate ≥ 75% before KD value reached 0.3.

(a) Walvax

(b) ATCC

Figure 11 1H-NMR Spectra of Native Serotype 1 Pneumococcal Polysaccharide from (a) Walvax and (b) ATCC

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Comparison of 1H-NMR spectra showed Walvax's polysaccharide is the same as ATCC polysaccharide, apart from quantitative DMSO peak. Both spectra showed broad and overlapped signals at anomeric carbon region, which indicated the structure of Walvax's serotype 1 polysaccharide is in full accord with ATCC standard polysaccharide.

(a) Walvax

(b) ATCC

Figure 12 1H-NMR Spectra of De-O-acetylated Serotype 1 Pneumococcal Polysaccharide from (a) Walvax and (b) ATCC

Comparison of 1H-NMR spectra showed the structural peaks (including GalA, Sug) and residual C polysaccharide peak (C-poly) of Walvax's de-O-acetyled serotype 1 polysaccharide completely matched the peaks of the de-O-acetyled serotype 1 polysaccharide of ATCC.

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(2) Structure of Serotype 3 Pneumococcal Polysaccharide:

→ 3) – ■ – D – GlcpA(1 → 4) – ■ – D – Glcp(1 →

OH
4 HOOC 6
6 O 4
O
HO 2 2
* 5
O 5 *
O 3 1 HO
OH 3 1
OH
n

Molecular size: determined by gel filtration through Sepharose CL-4B, release specification is
KD ≤ 0.15, recovery rate ≥ 78% before KD value reached 0.3.

(a) Walvax

(b) ATCC

Figure 13 1H-NMR Spectra of Native Serotype 3 Pneumococcal Polysaccharide from (a) Walvax and (b) ATCC

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Comparison of 1H-NMR spectra showed Walvax's polysaccharide is the same as ATCC polysaccharide, apart from quantitative DMSO peak and residual ethanol peak. Both spectra showed structural peaks of Glc, GlcA and Acetate, which indicated the structure of Walvax's serotype 3 polysaccharide is in full accord with ATCC standard polysaccharide.

(3) Structure of Serotype 4 Pneumococcal Polysaccharide:



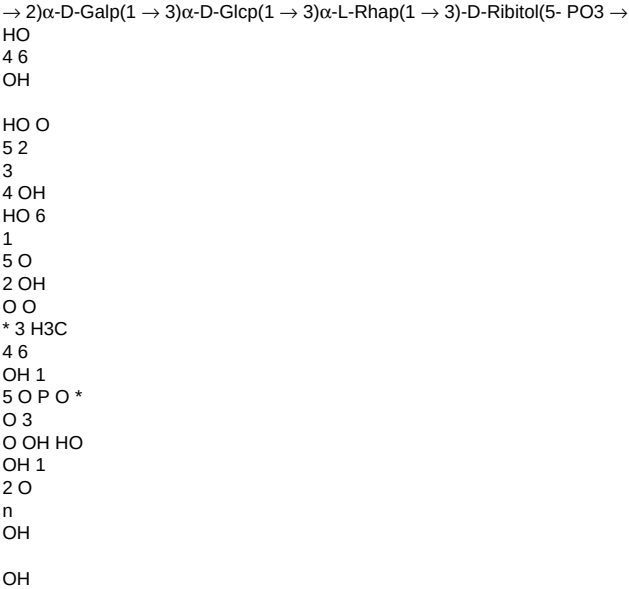
- (a) Walvax
(b) ATCC

Figure 15 1H-NMR Spectra of Native Serotype 5 Pneumococcal Polysaccharide from (a) Walvax and (b) ATCC

Comparison of 1H-NMR spectra showed Walvax's polysaccharide is the same as ATCC polysaccharide, apart from quantitative DMSO peak and residual ethanol peak. Both spectra showed structural peaks of N-Acetate and residual C-polysaccharide, which indicated the structure of Walvax's serotype 5 polysaccharide is in full accord with ATCC standard polysaccharide.

(5) Structure of Serotype 6A Pneumococcal Polysaccharide:

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Molecular size: determined by gel filtration through Sepharose CL-2B, release specification is KD ≤ 0.50, recovery rate ≥ 50% before KD value reached 0.6.

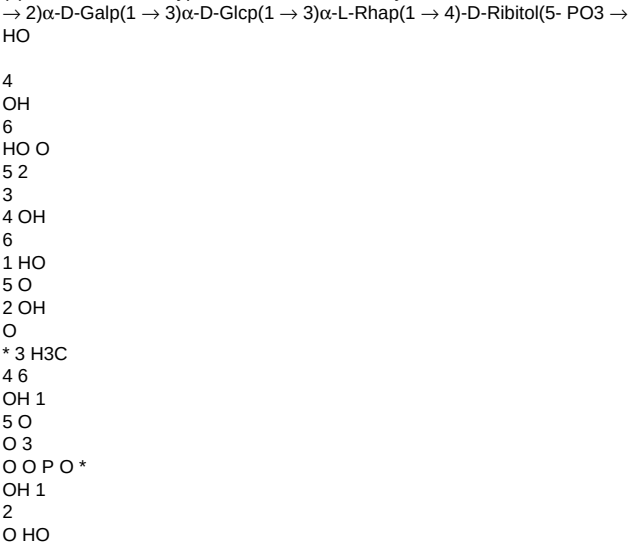
- (a) Walvax
(b) ATCC

Figure 16 1H-NMR Spectra of Native Serotype 6A Pneumococcal Polysaccharide from (a) Walvax and (b) ATCC

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Comparison of 1H-NMR spectra showed Walvax's polysaccharide is the same as ATCC polysaccharide, apart from quantitative DMSO peak and residual ethanol peak. Both spectra showed structural peaks of Gal, Glc, Rha and residual C-polysaccharide, which indicated the structure of Walvax's serotype 6A polysaccharide is in full accord with ATCC standard polysaccharide.

(6) Structure of Serotype 6B Pneumococcal Polysaccharide:



n
OH
OH

OH

Molecular size: determined by gel filtration through Sepharose CL-2B, release specification is $KD \leq 0.50$, recovery rate $\geq 85\%$ before KD value reached 0.6.

(a) Walvax

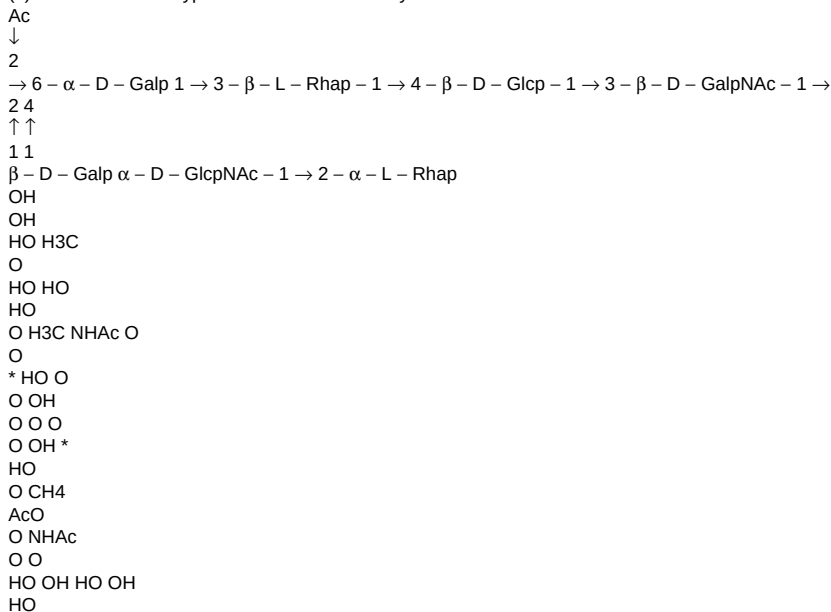
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(b) ATCC

Figure 17 1H-NMR Spectra of Native Serotype 6B Pneumococcal Polysaccharide from (a) Walvax and (b) ATCC

Comparison of 1H-NMR spectra showed Walvax's polysaccharide is the same as ATCC polysaccharide, apart from quantitative DMSO peak and residual ethanol peak. Both spectra showed structural peaks of Gal, Glc, Rha and residual C-polysaccharide, which indicated the structure of Walvax's serotype 6B polysaccharide is in full accord with ATCC standard polysaccharide.

(7) Structure of Serotype 7F Pneumococcal Polysaccharide:



HO

n
Molecular size: determined by gel filtration through Sepharose CL-4B, release specification is $KD \leq 0.20$, recovery rate $\geq 70\%$ before KD value reached 0.3.

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(a) Walvax

(b) ATCC

Figure 18 1H-NMR Spectra of Native Serotype 7F Pneumococcal Polysaccharide from (a) Walvax and (b) ATCC

Comparison of 1H-NMR spectra showed Walvax's polysaccharide is the same as ATCC polysaccharide, apart from quantitative DMSO peak. Both spectra showed broad and overlapped signals at the anomeric carbon region, which indicated the structure of Walvax's serotype 7F polysaccharide is in full accord with ATCC standard polysaccharide.

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(a) Walvax

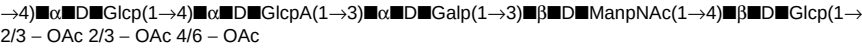
(b) ATCC

Figure 19 1H-NMR Spectra of De-O-acetylated Serotype 7F Pneumococcal Polysaccharide from (a) Walvax and (b) ATCC

Comparison of 1H-NMR spectra showed the structural peaks (including Gal, Rha, GlcNAc, Acetate and residual C-polysaccharide) of Walvax's de-O-acetylated serotype 7F polysaccharide completely matched the peaks of the de-O-acetylated serotype 7F polysaccharide of ATCC.

(8) Structure of Serotype 9V Pneumococcal Polysaccharide:

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OH
O
* O
AcO HOOC
OH
O HO
O OH
O OAc
AcO O OH
OH *
HO NHAC O
O O
OH
O
HO OH
n

Molecular size: determined by gel filtration through Sepharose CL-2B, release specification is
KD ≤ 0.45, recovery rate ≥ 80% before KD value reached 0.6.

(a) Walvax

(b) ATCC

Figure 20 1H-NMR Spectra of Native Serotype 9V Pneumococcal Polysaccharide from (a)
Walvax and (b) ATCC

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Comparison of 1H-NMR spectra showed Walvax's polysaccharide is the same as ATCC
polysaccharide, apart from quantitative DMSO peak. Both spectra showed broad and overlapped
signals at the anomeric carbon region, which indicated the structure of Walvax's serotype 9V
polysaccharide is in full accord with ATCC standard polysaccharide.

(a) Walvax

(b) ATCC

Figure 21 1H-NMR Spectra of De-O-acetylated Serotype 9V Pneumococcal Polysaccharide
from (a) Walvax and (b) ATCC

Comparison of 1H-NMR spectra showed the structural peaks (including Glc, Gal, GlcA, ManNAc,
Acetate) of Walvax's de-O-acetyled serotype 9V polysaccharide completely matched the peaks of
the de-O-acetyled serotype 9V polysaccharide of ATCC.

(9) Structure of Serotype 14 Pneumococcal Polysaccharide:

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4
↑
1
β – D – Galp
HO
OH
OH
O
O O *
* O
O
HO O OH
HO OH
OH
O O NHAc
HO
HO OH
n

Molecular size: determined by gel filtration through Sepharose CL-4B, release specification is
KD ≤ 0.30, recovery rate ≥ 55% before KD value reached 0.3.

(a) Walvax

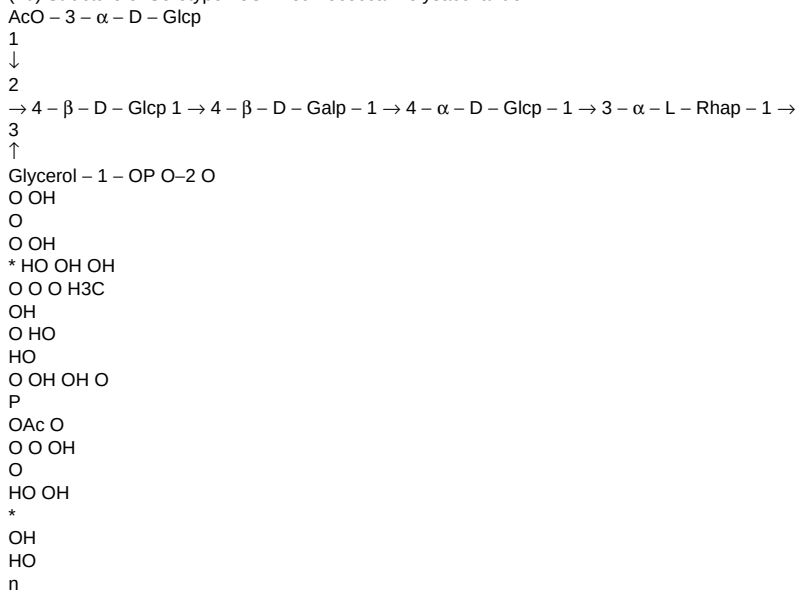
(b) ATCC

Figure 22 1H-NMR Spectra of Native Serotype 14 Pneumococcal Polysaccharide from (a)
Walvax and (b) ATCC

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Comparison of ¹H-NMR spectra showed Walvax's polysaccharide is the same as ATCC polysaccharide, apart from residual ethanol peak. Both spectra showed structural peaks of GlcNAc, Gal and residual C-polysaccharide, which indicated the structure of Walvax's serotype 14 polysaccharide is in full accord with ATCC standard polysaccharide.

(10) Structure of Serotype 18C Pneumococcal Polysaccharide:



Molecular size: determined by gel filtration through Sepharose CL-4B, release specification is KD ≤ 0.15, recovery rate ≥ 70% before KD value reached 0.3.

(a) Walvax

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(b) ATCC

Figure 23 ¹H-NMR Spectra of Native Serotype 18C Pneumococcal Polysaccharide from (a) Walvax and (b) ATCC

Comparison of ¹H-NMR spectra showed Walvax's polysaccharide is the same as ATCC polysaccharide, apart from quantitative DMSO peak. Both spectra showed broad and overlapped signals at the anomeric carbon region, which indicated the structure of Walvax's serotype 18C polysaccharide is in full accord with ATCC standard polysaccharide.

(a) Walvax

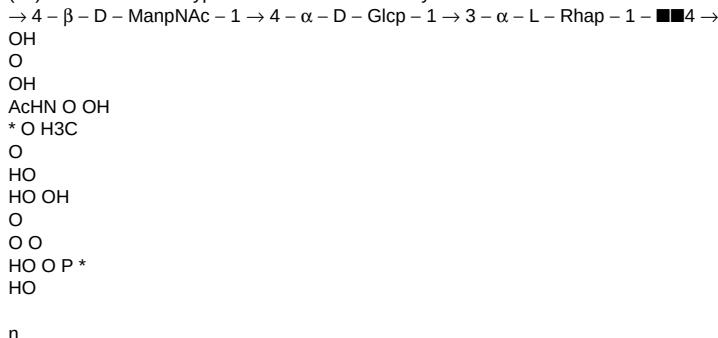
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MODULE 3 QUALITY
Version No. Walvax-202007-00 Yuxi Walvax Biotechnology Co., Ltd.

(b) ATCC

Figure 24 ¹H-NMR Spectra of De-O-acetylated Serotype 18C Pneumococcal Polysaccharide from (a) Walvax and (b) ATCC

Comparison of ¹H-NMR spectra showed the structural peaks (including Glc, Rha, Gal, Acetate) and residual C-polysaccharide peak of Walvax's de-O-acetyled serotype 18C polysaccharide completely matched the peaks of the de-O-acetyled serotype 18C polysaccharide of ATCC.

(11) Structure of Serotype 19A Pneumococcal Polysaccharide:



Molecular size: determined by gel filtration through Sepharose CL-4B, release specification is KD ≤ 0.45, recovery rate ≥ 80% before KD value reached 0.5.

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- (a) Walvax
- (b) ATCC

Figure 25 1H-NMR Spectra of Native Serotype 19A Pneumococcal Polysaccharide from (a) Walvax and (b) ATCC

Comparison of 1H-NMR spectra showed Walvax's polysaccharide is the same as ATCC polysaccharide, apart from quantitative DMSO peak and residual ethanol peak. Both spectra showed structural peaks of Rha, Glc, ManNAc, N-Acetateand and residual C-polysaccharide, which indicated the structure of Walvax's serotype 19A polysaccharide is in full accord with ATCC standard polysaccharide.

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(12) Structure of Serotype 19F Pneumococcal Polysaccharide:
→ 4) – ■ – D – ManpNAc(1 → 4) – ■ – D – Glcp(1 → 2) – α – L – Rhap(1 – PO3 → OH

OH H3C
O
OH
AcHN O
HO
* O
O
HO O
HO OH O

O
O
P *

OH

n
Molecular size: determined by gel filtration through Sepharose CL-4B, release specification is KD ≤ 0.20, recovery rate ≥ 75% before KD value reached 0.3.

- (a) Walvax
- (b) ATCC

Figure 26 1H-NMR Spectra of Native Serotype 19F Pneumococcal Polysaccharide from (a) Walvax and (b) ATCC

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Comparison of 1H-NMR spectra showed Walvax's polysaccharide is the same as ATCC polysaccharide, apart from quantitative DMSO peak and residual ethanol peak. Both spectra showed structural peaks of Rha, Glc, ManNAc, N-Acetateand residual C-polysaccharide, which indicated the structure of Walvax's serotype 19F polysaccharide is in full accord with ATCC standard polysaccharide.

(13) Structure of Serotype 23F Pneumococcal Polysaccharide:
α – L – Rhap

1
↓
2
→ 4 – β – D – Glcp – 1 → 4 – β – D – Galp – 1 → 4 – β – L – Rhap – 1 →
3
PO2 OH
↑
HOCH2 – CH – CH2 OH
OH
O
OH
O
O
* H3C
HO OH O O O
*
HO P O HO O
O
O OH

HO OH O
OH

OH
H3C

HO n
Molecular size: determined by gel filtration through Sepharose CL-4B, release specification is KD ≤ 0.15, recovery rate ≥ 85% before KD value reached 0.3.

(a) Walvax

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(b) ATCC

Figure 27 1H-NMR Spectra of Native Serotype 23F Pneumococcal Polysaccharide from (a) Walvax and (b) ATCC

Comparison of 1H-NMR spectra showed Walvax's polysaccharide is the same as ATCC polysaccharide, apart from quantitative DMSO peak and residual ethanol peak. Both spectra showed structural peaks of Rha, Gal, Glc and residual C-polysaccharide, which indicated the structure of Walvax's serotype 23F polysaccharide is in full accord with ATCC standard polysaccharide.

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3.2.S.3.2 Impurities

3.2.S.3.2.1 Impurities and their Limits in Polysaccharide Purification Process

(1) C-polysaccharide, nucleic acid and protein are the main impurities in refined polysaccharide production.

The quality of pneumococcal polysaccharides manufactured by Walvax are controlled according to the European Pharmacopeia. Control of C-polysaccharide are implemented via various purification steps. Table x shows residual C-polysaccharide contents in 12 serotypes of polysaccharides derived from NMR spectrum analysis, which are all below 20%, indicating the purification process can effectively remove C-polysaccharide impurities.

Table 64 Residual C-polysaccharide Content in 13 Serotypes of Pneumococcal Polysaccharide

Serotypes Batch No. Residual C-polysaccharide content % (W/W)

1 PN1201609001 20
3 PN3201507003-1 None detected
4 4-PVPV16011103 11.4
5* PN5201610005 -
6A PN6A201503001 1.4
6B PN6B201408001 0.5
7F 7F-PVPV15100701 None detected
9V 9V-PVPV15090301 0.7
14 PN14201403001 7.5
18C 18C-PVPV15100801 0.9
19A PN19A201408001 1.6
19F PN19F201508002 1.2
23F 23F-PVPV15090601 2.7

*: Since signature peak(s) and content of C-polysaccharide cannot be analyzed in NMR spectrum of serotype 5 polysaccharide due to its structural features, no result for serotype 5 is available.

During approval process of new drug of PPSV23, Walvax completed process optimization and validation studies, including impurities removal process validation. According to the established purification process of serotypes 1, 3, 4, 5, 6B, 7F, 9V, 14, 18C, 19A, 19F and 23F, 3 continuous batches of the above serotypes are purified, sampled in each step, and impurities removal rate was calculated with nucleic acid and protein content derived from polysaccharide content, and viability, reproducibility and stability was evaluated.

Crude polysaccharide solution, concentrated phenol extraction ultrafiltration solution and concentrated solution after precipitation of nucleic acid of serotypes 1, 4, 5, 6B, 7F, 9V, 14, 18C, 19A, 19F and 21F were sampled and analyzed for nucleic acid, protein and polysaccharide content, percentages of nucleic acid and protein were calculated, and impurities removal rate of each step was evaluated by comparison of data from before/after each steps. Table X shows the removal rate of nucleic acid and protein.

Table 65 Impurities Removal Rate of Purification Process of Serotypes 1, 4, 5, 6B, 7F, 9V, 14, 18C, 19A, 19F and 23F
After phenol extraction After nucleic acid extraction and precipitation and

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ultrafiltration ultrafiltration
concentration concentration
Nucleic acid (%) 33.2 90.5
RPN1201310001
Protein (%) 85.4 60.4
Nucleic acid (%) 29.4 92.5
1 RPN1201311002
Protein (%) 88.2 57.2
Nucleic acid (%) 35.6 95.1
RPN1201312003
Protein (%) 90.3 56.8
Nucleic acid (%) 36.1 93.4
RPN4201311001
Protein (%) 87.2 61.5
Nucleic acid (%) 34.5 91.7
4 RPN4201311002
Protein (%) 83.6 59.3
Nucleic acid (%) 39.3 89.1

RPN4201312003
 Protein (%) 88.2 63.6
 Nucleic acid (%) 36.2 92.2
 RPN5 201310001
 Protein (%) 87.3 52.3
 Nucleic acid (%) 40.6 90.4
 5 RPN5 201311002
 Protein (%) 89.1 49.7
 Nucleic acid (%) 35.4 87.4
 RPN5 201312003
 Protein (%) 84.4 58.5
 Nucleic acid (%) 38.5 91.6
 RPN6B201311001
 Protein (%) 88.4 53.4
 Nucleic acid (%) 35.2 87.8
 6B RPN6B201312002
 Protein (%) 93.5 58.8
 Nucleic acid (%) 33.6 94.6
 RPN6B201312003
 Protein (%) 95.2 62.7
 Nucleic acid (%) 41.3 90.7
 RPN7F201311001
 Protein (%) 87.1 52.7
 Nucleic acid (%) 39.5 93.5
 7F RPN7F201312002
 Protein (%) 90.6 57.7
 Nucleic acid (%) 35.7 89.6
 RPN7F201312003
 Protein (%) 88.5 61.4
 Nucleic acid (%) 34.1 90.9
 RPN9V201311001
 Protein (%) 89.5 56.6
 Nucleic acid (%) 34.1 90.9
 9V RPN9V201311001
 Protein (%) 89.5 56.6
 Nucleic acid (%) 34.1 90.9
 RPN9V201311001
 Protein (%) 89.5 56.6
 Nucleic acid (%) 34.5 89.8
 RPN14 201311001
 Protein (%) 87.3 54.9
 Nucleic acid (%) 32.6 91.4
 14 RPN14 201311002
 Protein (%) 85.3 63.5
 Nucleic acid (%) 30.7 90.3
 RPN14 201312003
 Protein (%) 89.2 58.5
 Nucleic acid (%) 38.4 89.4
 RPN18C201311001
 Protein (%) 84.7 58.8
 Nucleic acid (%) 41.8 86.3
 18C RPN18C201312002
 Protein (%) 88.3 60.6
 Nucleic acid (%) 36.6 93.2
 RPN18C201312003
 Protein (%) 86.8 61.7
 Nucleic acid (%) 40.5 89.6
 RPN19A201311001
 Protein (%) 87.3 58.3
 19A
 Nucleic acid (%) 85.7 63.4
 RPN19A201312002
 Protein (%) 36.3 91.3
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Nucleic acid (%) 87.3 60.7
 RPN19A201312003
 Protein (%) 34.5 93.8
 Nucleic acid (%) 40.6 91.4
 RPN19F201311001
 Protein (%) 88.4 61.7
 Nucleic acid (%) 37.7 89.4
 19F RPN19F201312002
 Protein (%) 90.4 59.8
 Nucleic acid (%) 35.6 92.3
 RPN19F201312003
 Protein (%) 92.7 63.2
 Nucleic acid (%) 36.6 93.5
 RPN23F201311001
 Protein (%) 85.4 54.4
 Nucleic acid (%) 35.2 91.2
 23F RPN23F201312002
 Protein (%) 87.7 58.8
 Nucleic acid (%) 33.1 89.6
 RPN23F201312003
 Protein (%) 84.1 57.2
 Note: Removal rate (%) = (impurities content before processing - impurities content after processing)/impurities content before processing ×100%

Table 65 shows that the nucleic acid and protein removal rate of 3 batches of 11 serotypes of polysaccharide was relatively stable, which indicates the purification process is stable.

Table 66 Impurities Removal Rate of 3 Batches of Serotype 3 Polysaccharide

After phenol	
After CTAB	
extraction and	
Serotype Batch No. Index precipitation and NaCl	
ultrafiltration	
dissociation	
concentration	
Nucleic acid (%)	30.2 95.4
RPN3 201310001	
Protein (%)	85.7 40.3
Nucleic acid (%)	29.8 95.8
3 RPN3 201311002	
Protein (%)	87.5 39.8
Nucleic acid (%)	32.4 96.4
RPN3 201312003	
Protein (%)	88.7 41.2
Note: Removal rate (%) = (impurities content before processing - impurities content after processing)/impurities content before processing ×100%	

Table 66 shows that the nucleic acid and protein removal rate of 3 batches of serotype 3 polysaccharide was relatively stable, which indicates the purification process is stable. Nucleic acid and protein impurity contents were monitored in the established purification process of serotypes 1, 3, 4, 5, 6B, 7F, 9V, 14, 18C, 19A, 19F and 23F, and removal rates of each step was calculated. The test results show the removal rates were high and stable, content homogeneity was well controlled, which indicates good reproducibility and stability for the purification process in pneumococcal polysaccharide vaccine manufacture.

(2) Studies on organic solvents (reagents) removal in polysaccharide production
I. Types and purposes of organic reagents used in polysaccharide manufacture
In the production process of pneumococcal polysaccharide, organic solvents (reagents) are used. According to the Chinese Pharmacopeia (2010 version), similar reagents are used in pneumococcal polysaccharide vaccine, Vi capsular polysaccharide vaccine and Hib conjugate vaccine
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productions, which indicates this as a feature related to bacterium capsular polysaccharide vaccines.
Table 67 shows organic reagents used in 13-Valent Pneumococcal Polysaccharide Conjugate Vaccine.

Table 67 Usage of Organic Reagents and their Purposes
Organic solvent
Steps Used Objective
(reagent)
DOC Sterilization after fermentation Neutralization of pneumococcus
Purification of pneumococcal Removal of protein impurities by extraction with
Phenol
polysaccharide cold phenol
Crude polysaccharide production Ethanol is used as precipitation agent to separate
95% Ethanol
and purification polysaccharides and impurities
Washing of crude polysaccharide Anhydrous ethanol is used as dehydration agent,
Anhydrous ethanol
and refined polysaccharide and to remove alcohol-soluble impurities
Washing of crude polysaccharide Acetone is used as dehydration agent, and to
Acetone
and refined polysaccharide remove acetone-soluble impurities
Washing of crude polysaccharide Ether is used as dehydration agent, and to
Ether
and refined polysaccharide remove ether-soluble impurities
Due to its ability of precipitating capsular
Purification of serotype 3
CTAB polysaccharide and nucleic acid, CTAB is used
polysaccharide
to remove nucleic acid.

II. Organic solvents (reagents) removal process
Table 68 Description of Organic Solvents (Reagents) Removal Process
Organic solvent
Removal process
(reagent)
After fermentation, DOC is added for sterilization, after which several steps are done, including ultrafiltration of crude polysaccharide, ethanol precipitation, ultrafiltration of DOC refined polysaccharide and ethanol precipitation. Due to low molecular size and good water solubility, DOC can be effectively removed from capsular polysaccharide.
Phenol is used in the removal of protein, after extraction with phenol, the mixture goes through ultrafiltration to remove phenol, after which several stages of ethanol precipitation and ultrafiltration were conducted. In the ultrafiltration and ethanol Phenol precipitation process, phenol can be effectively removed due to its low molecular size and good solubility in both water and organic solution. Moreover, in the process of washing with ethanol, acetone and ether, phenol was further separated and removed. 95% ethanol is used in precipitation of crude and refined polysaccharide and can be 95% ethanol effectively removed with ultrafiltration. Vacuum drying and washing with acetone and ether of refined polysaccharide can further remove ethanol.

Anhydrous Anhydrous ethanol is used to wash refined polysaccharides, it can be removed with ethanol acetone, ether and vacuum drying.
Acetone is used to wash refined polysaccharide; it can be removed by washing with Acetone ether and vacuum drying.
Ether Ether is used to wash refined polysaccharide; it can be removed by vacuum drying.
CTAB is used for removal of nucleic acid in serotype 3 polysaccharide; freezing, CTAB centrifugation and ethanol precipitation can remove most of the CTAB used, residual CTAB can be removed with washing using anhydrous ethanol.

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III. Residual organic solvents (reagents) content
Table 69 shows the test results of residual organic reagents content (DOC, phenol, CTAB, ethanol, ether and acetone) of continuous 3 batches of polysaccharide.

Table 69 Residual Organic Reagents Content of Continuous 3 Batches of 13 serotypes of Polysaccharide

DOC	Phenol	CTAB	Ethanol	Ether	Acetone
Serotypes Batch No.					
µg/mL	µg/mL	µg/mL	(%)	(%)	(%)
20090401	0	0	0.0050	0.0016	0.0015
Serotype 1					
20090402	0	0	0.0035	0.0017	0.0015
polysaccharide					
20090503	0	0	0.0033	0.0017	0.0015
20090602	0.09	0	0.0045	0.0003	0.0019
Serotype 3					
20090603	0	0	0.0040	0.0003	0.0017
polysaccharide					
20090604	0	0	0.0040	0.0003	0.0017
20090101	0	0	0.0089	0.0013	0.0059
Serotype 4					
20090102	0	0	0.0088	0.0013	0.0060
polysaccharide					
20090103	0	0	0.0089	0.0012	0.0059
20090401	1.1	0	0.0098	0.0003	0.0059
Serotype 5					
20090402	2.5	0	0.0102	0.0003	0.0060
polysaccharide					
20090403	9.3	0	0.0103	0.0003	0.0058
None					
6A-PVPV15090101	0	0	0.0565	0.0025	
detected					
Serotype 6A None None					
6A-PVPV15110102	0.02	0	0.061		
polysaccharide* detected detected					
None					
6A-PVPV15120103	0	0	0.026	0.002	
detected					
20090201	0	0	0.0182	0.0007	0.0125
Serotype 6B					
20090202	0.5	0	0.0178	0.0007	0.0125
polysaccharide					
20090203	2.3	0	0.0175	0.0006	0.0121
20090802	0.1	0	0.0045	0.0011	0.0116
Serotype 7F					
20090803	1.5	0	0.0045	0.0011	0.0118
polysaccharide					
20090804	2.3	0	0.0046	0.0011	0.0117
20090201	0	0	0.0069	0.0042	0.0064
Serotype 9V					
20090302	0	0	0.0070	0.0041	0.0064
polysaccharide					
20090303	0	0	0.0066	0.0068	0.0079
20090301	0	0	0.0021	0.0009	0.0043
Serotype 14					
20090302	0	0	0.0020	0.0009	0.0043
polysaccharide					
20090303	0	0	0.0020	0.0010	0.0043
20090301	0.09	0	0.0046	0.0018	0.0062
Serotype 18C					
20090302	0	0	0.0046	0.0018	0.0062
polysaccharide					
20090303	2.64	0	0.0046	0.0018	0.0063
20090902	0	0	0.0535	0.0004	0.0035
Serotype 19A					
20090903	0	0	0.0534	0.0004	0.0035
polysaccharide					
20090904	0	0	0.0544	0.0004	0.0035
20090301	0	0	0.0571	0.0002	0.0031
Serotype 19F					
20090302	0	0	0.0559	0.0002	0.0031
polysaccharide					
20090403	0	0	0.0555	0.0002	0.0032
20090401	0	0	0.0018	0.0025	0.0140
Serotype 23F					
20090402	0	0	0.0019	0.0024	0.0137
polysaccharide					
20090403	0	0	0.0019	0.0026	0.0140

Notes: “-” indicates this item was not tested for this sample, “*” indicates the 3 continuous batches of serotype 6A polysaccharide produced in process validation approval.

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Those test results show good reproducibility of ethanol, ether and acetone concentration between continuous 3 batches. There's difference between different serotypes, but residual contents remain low, the average residual ethanol is 119.22 µg/mL (polysaccharide concentration at 3.0 mg/mL), the average residual ether is 8.87 µg/mL (polysaccharide concentration at 3.0 mg/mL), the average residual acetone is 47.09 µg/mL (polysaccharide concentration at 3.0 mg/mL). According to the chemical handbook, the boiling points of ethanol, ether and acetone at 1 atm is 78.37 ■, 34.6 ■ and 56.48 ■ respectively, the lower the boiling point, the easier it is to be removed with vacuum drying, which accounted for the lowest average residual of ether and highest average residual of ethanol.

The results of residual DOC, phenol and CTAB show that no residual phenol was present in 13 serotypes of polysaccharide. No CTAB was detected in the only serotype that used CTAB in the process, serotype 3. The average residual DOC concentration was low at 0.575 µg/mL (polysaccharide concentration at 3 mg/mL). The production process can effectively remove DOC, phenol and CTAB.

According to ICH and Chinese Pharmacopeia (2010 version), ethanol, ether and acetone are low toxicity solvents (type 3 solvents) and have limited impact on human health. Acute toxicity and short-term toxicity tests show they're almost non-toxic and non-genotoxic. No discussion is needed if content intake is 50 mg/day or lower.

Walvax conducted further validation of removal of product-related and process-related impurities through planning to be listed commercial-scale process validation during clinical studies. The results showed the removal of the above impurities is effective. The results are shown in Table 70 to Table 82.

Table 70 Test Result of Serotype 1 Pneumococcal Polysaccharide Impurities

Nucleic acid types	Sodium	Protein	Ethanol	Acetone	Ether
Sample					
Item					
Protein					
Batch No.	≤ 2%	< 0.08%	< 0.03%	< 0.01%	
Validation	≤ 2%	< 0.2%			
Clinical					
PN1201504002	0.67	1.5	0.02	0.007	- -
Sample					
Process					
PN1201609001	1.1	1.3	0.09	0.001	-
Validation					
Process					
PN1201610002	0.5	1.1	0.1	- - -	
Validation					
Process					
PN1201611004	0.1	0.5	0.017	0.0006	-
Validation					
Process					
PN1201702001	0.8	1.3	0.07	- - -	
Validation					

Table 71 Test Result of Serotype 3 Pneumococcal Polysaccharide Impurities

Nucleic acid types	Sodium	Protein	Ethanol	Acetone	Ether
Sample					
Item					
Protein					
Batch No.	≤ 2%	< 0.08%	< 0.03%	< 0.01%	
Validation	≤ 2%	< 0.2%			
Clinical					
PN3201504001	0.10	0.1	0.14	0.002	- -
Sample					
Process					
3-PVPV15100901	0.04	0.1	0.03	0.002	- -
Validation					
Process					
3-PVPV15110902	0.09	0.4	0.16	- - -	
Validation					

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Process					
3-PVPV15120903	1.99	0.5	0.17	- - -	
Validation					
Process					
SP3P201606002	0.10	0.02	0.002	- -	
Validation					
Process					
SP3P201608003	0.10	0.03	0.07	0.002	- -
Validation					

Table 72 Test Result of Serotype 4 Pneumococcal Polysaccharide Impurities

Nucleic acid types	Sodium	Protein	Ethanol	Acetone	Ether
Sample					
Item					
Protein					
Batch No.	≤ 2%	< 0.08%	< 0.03%	< 0.01%	
Validation	≤ 2%	< 0.2%			
Clinical					
PN4 201503001	0.17	1.5	0.13	0.009	0.001 -
Sample					

Process
4-PVPV15121102 0.3 1.3 0.16 0.006 - -
validation
Process
4-PVPV16011103 0.6 2.7 0.11 0.001 - -
validation
Process
SP4P201605001 0.4 0.9 0.04 0.016 0.004 -
validation

Table 73 Test Result of Serotype 5 Pneumococcal Polysaccharide Impurities

Nucleic Sodium
Sample Item Protein Ethanol Acetone Ether
acid deoxycholate
types Batch No. $\leq 2\% < 0.08\% < 0.03\% < 0.01\%$
 $\leq 2\% < 0.2\%$
Clinical
PN5201503001 0.2 5.89 0.13 0.02 0.0045 -
sample
Process
PN5201701001 0.7 7.30 0.17 0.005 - -
validation
Process
PN5201701002 0.8 6.42 0.12 0.0045 - -
validation
Process
PN5201701003 0.5 5.72 0.12 0.005 - -
validation

Table 74 Test Result of Serotype 6A Pneumococcal Polysaccharide Impurities

Nucleic Protei Sodium
Sample Item Ethanol Acetone Ether
acid n deoxycholate
types Batch No. $< 0.08\% < 0.03\% < 0.01\%$
 $\leq 2\% \leq 2\% < 0.2\%$
Clinical
PN6A201503001 0.17 0.95 0.15 0.062 0.006 0.004
sample
Process
6A-PVPV15090101 0.082 0.36 0 0.0565 0.0025 -
validation
Process
6A-PVPV15110102 0.08 1.16 0.02 0.061 - -
validation
Process
6A-PVPV15120103 0.10 1.30 0 0.026 0.002 -
validation

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Table 75 Test Result of Serotype 6B Pneumococcal Polysaccharide Impurities

Nucleic Sodium
Sample Item Protein Ethanol Acetone Ether
acid deoxycholate
types Batch No. $\leq 2\% < 0.08\% < 0.03\% < 0.01\%$
 $\leq 2\% < 0.2\%$
Clinical
PN6B201503001 0.2 0.86 0.05 0.066 0.002 -
sample
Process
6B-PVPV15110202 0.1 0.4 0.11 0.064 0.004 -
validation
Process
6B-PVPV15120203 0.1 1.4 0.18 0.0275 0.002 -
validation
Process
SP6BP201605001 0.1 0.2 0.05 0.044 - -
validation

Table 76 Test Result of Serotype 7F Pneumococcal Polysaccharide Impurities

Nuclei Sodium
Sample Item Protein Ethanol Acetone Ether
c acid deoxycholate
types Batch No. $\leq 2\% < 0.08\% < 0.03\% < 0.01\%$
 $\leq 2\% < 0.2\%$
Clinical
PN7F201504001 0.44 2.5 0.05 0.001 0.002 -
sample
Process
7F-PVPV15100701 0.4 0.2 0.04 0.022 0.001 -
validation
Process
7F-PVPV16010703 0.2 1.1 0.05 0.005 - -
validation
Process
SP7FP201606001 0.2 0.6 0.08 0.031 0.003 -
validation
Process
SP7FP201608003 0.4 0.9 0.04 0.016 0.007 0.003
validation

Nucleic Sodium
Sample Item Protein Ethanol Acetone Ether
acid deoxycholate
types Batch No. $\leq 2\% < 0.08\% < 0.03\% < 0.01\%$
 $\leq 2\% < 0.2\%$
Clinical
PN9V201503001 0.04 0.2 0.04 0.007 - -
sample
Process
9V-PVPV15120303 0.06 0.3 0.18 - 0.001 -
validation
Process
SP9VP201605001 0.10 0.1 0.03 0.001 - -
validation
Process
SP9VP201607002 0.10 0.2 0.05 0.013 0.0006 -
validation

Nucleic Sodium
Sample Item Protein Ethanol Acetone Ether
acid deoxycholate
types Batch No. $\leq 2\% < 0.08\% < 0.03\% < 0.01\%$
 $\leq 2\% < 0.2\%$
Clinical
PN14201503001 0.22 1.6 0.06 0.009 - -
sample
Process
PN14201610001 0.2 3.1 0.18 0.0095 - -
validation

```
Process
PN14201610002 0.3 2.7 0.09 0.0045 - -
validation
Process
PN14201703001 0.4 1.1 0.04 - - -
validation
```

Nucleic Sodium
Sample Item Protein Ethanol Acetone Ether
acid deoxycholate
types Batch No. $\leq 2\% < 0.08\% < 0.03\% < 0.01\%$
 $\leq 2\% < 0.2\%$
Clinical
PN18C201504001 0.14 0.96 0.09 0.02 - -
sample
Process
18C-PVPV15110802 0.2 0.8 0.06 0.034 0.002 -
validation
Process
18C-PVPV16010803 0.2 1.3 0.11 0.046 0.002 -
validation
Process
SP18CP201608003 0.7 1.2 0.003 0.006 - -
validation

Nucleic Sodium
Sample Item Protein Ethanol Acetone Ether
acid deoxycholate
types Batch No. $\leq 2\% < 0.08\% < 0.03\% < 0.01\%$
 $\leq 2\% < 0.2\%$
Clinical
PN19A201504001 0.05 0.5 0.07 0.015 0.002 -
sample
Process
19A-PVPV15110402 0.02 0.7 0.004 0.0145 0.002 -
validation
Process
19A-PVPV15120403 0.05 0.5 0.15 0.005 0.002 0.0001
validation
Process
SP19AP201606001 0.10 0.3 0.03 0.011 0.001 -
validation

Sample Item	Protein	Ethanol	Acetone	Ether
acid deoxycholate				
types Batch No.	≤ 2%	< 0.08%	< 0.03%	< 0.01%
	≤ 2%	< 0.2%		
Clinical				
PN19F201504001	0.09	0.6	0.1	0.013
sample	0.002	-		
Process				

19F-PVPV15110502 0.10 0.2 0 0.0145 - -
validation
Process
19F-PVPV15120503 0.10 0.3 0.13 0.0035 0.0005 -
validation
Process
SP19FP201606001 0.2 0.1 0.1 0.01 0.002 -
validation

Table 82 Test Result of Serotype 23F Pneumococcal Polysaccharide Impurities
Nucleic Sodium
Sample Item Protein Ethanol Acetone Ether
acid deoxycholate
types Batch WNo. ≤ 2% < 0.08% < 0.03% < 0.01%
≤ 2% < 0.2%

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Clinical
PN23F201504001 0.18 0.6 0.04 0.017 0.002 -
sample
Process
23F-PVPV15120602 0.1 0.2 0.17 0.01 0.003 -
validation
Process
23F-PVPV16010603 0.1 0.3 0.08 0.0015 - -
validation
Process
SP23FP201606001 0.3 0.2 0.1 0.025 0.001 -
validation

Conclusion: the results of impurities of 13 types of polysaccharide indicate the removal of product-related and process-related impurities is indeed effective.

2.3.S.3.2.2 Impurities and their Limits in Conjugation Process
Walvax uses CNBR or CDAP for activation of polysaccharide, the activated polysaccharides are derived using excessive ADH to ensure complete removal of cyanide aliphatic groups. PS-ADH and carrier protein are conjugated with EDAC to form PS-ADH-protein conjugate. The non-conjugate activation points in the conjugation process mainly come from un-conjugated hydrazide base of the PS-ADH derivative and the hydrazide base of EDAC in the bulk. Those activation points have relatively low reactivity, and can only go through conjugation with the presence of medium to strong acid and EDAC. Thus, in near neutral atmosphere in human body and without a condensation catalyst, the reactivity of hydrazine base groups is similar to that of amino and acylamino groups, theoretically it is with minimal risk.
Production of clinical samples and commercial-scale validation show good sealing of non-conjugate excitation points in Walvax's product, low residual ADH, which fall well in the range of quality control.
The test results of impurities of polysaccharide derivative and monovalent bulk conjugate are shown in the Table 83 to Table 95.

Table 83 Test Results of 5 Batches of Serotype 1 Polysaccharide Derivative and Bulk Conjugate
Polysaccharide
Bulk conjugate
derivative
Items Residual ADH Residual Free-
Derivation Free-
Batch No. cyanide content EDAC protein
rate polysaccharide
(≤ 10 (< 0.15 (≤ 5 (≤
(2.0-7.0%) (≤ 25%)
ng/mg) μg/mL) μmol/L) 5.0%)
PN1201504002 1.10 3.82 0.1100 0.7490 16.9 0.16
PN1201612001 8.50 3.62 0.1225 2.2515 14.3 1.80
RSP1C201706001 5.90 3.06 0.0480 2.8000 10.8 4.10
None
RSP1C201707002 5.40 3.63 0.0620 2.9000 15.8
detected
None
RSP1C201709003 4.20 3.23 0.0860 14.0 1.30
detected

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Table 84 Test Results of 5 Batches of Serotype 3 Polysaccharide Derivative and Bulk Conjugate
Polysaccharide
Bulk conjugate
derivative
Items Residual ADH Residual
Derivation Free- Free-
Batch No. cyanide content EDAC
rate polysaccharide protein
(≤ 10 (< 0.25 (≤ 5
(6.5-14%) (≤ 20%) (≤ 4.0%)
ng/mg) μg/mL) μmol/L)
PN3201505004
(polysaccharide

derivative) 5.2 6.75 0.2045 2.0230 6.5 0.62
 PN3201505001
 (bulk conjugate)
 PN3 201612001 3.8 13.03 0.1040 1.5305 14.7 2.50
 RSP3C201706001 2.8 11.39 0.0160 1.1000 15.7 3.80
 RSP3C201707002 1.6 10.23 0.1600 4.4000 6.2 0.48
 RSP3C201708003 2.0 9.26 0.1280 3.8000 12.2 1.20

Table 85 Test Results of 5 Batches of Serotype 4 Polysaccharide Derivative and Bulk Conjugate

Polysaccharide
 Bulk conjugate
 derivative
 Items Residual ADH Residual
 Derivation Free- Free-
 Batch No. cyanide content EDAC
 rate polysaccharide protein
 (≤ 10 (< 0.15 (≤ 5
 (3.0-11%) ($\leq 25\%$) ($\leq 5.0\%$)
 ng/mg) $\mu\text{g/mL}$ $\mu\text{mol/L}$)
 PN4201505002 2.8 5.45 0.1140 2.4575 11.0 0.14
 PN4201612001 2.8 5.00 0.1225 2.0435 11.8 1.60
 None
 RSP4C201706001 9.70 0.0820 0.9000 19.6 3.10
 detected
 RSP4C201707002 0.3 6.66 0.1100 3.6000 23.6 2.20
 RSP4C201709003 2.8 3.31 0.0820 3.5000 14.3 0.90

Table 86 Test Results of 5 Batches of Serotype 5 Polysaccharide Derivative and Bulk Conjugate

Polysaccharide
 Bulk conjugate
 derivative
 Items Residual ADH Residual
 Derivation Free- Free-
 Batch No. cyanide content EDAC
 rate polysaccharide protein
 (≤ 10 (< 0.25 (≤ 5
 (5.4-18.0%) ($\leq 20\%$) ($\leq 4.5\%$)
 ng/mg) $\mu\text{g/mL}$ $\mu\text{mol/L}$)
 PN5201505002 3.6 5.31 0.240 1.9575 19.5 3.3
 PN5201703002 0.7 13.19 0.012 0.3270 8.2 2.6
 RSP5C201705001 2.1 17.89 0.139 0.7000 18.0 3.2
 None
 RSP5C201707002 2.5 16.08 0.102 0.8000 17.4
 detected
 RSP5C201708003 1.1 10.01 0.102 3.7000 15.4 0.5

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Table 87 Test Results of 5 Batches of Serotype 6A Polysaccharide Derivative and Bulk Conjugate

Polysaccharide
 Bulk conjugate
 derivative
 Items Residual ADH Residual
 Derivation Free- Free-
 Batch No. cyanide content EDAC
 rate polysaccharide protein
 (≤ 10 (< 1.2 (≤ 5
 (4.0-6.4%) ($\leq 20\%$) ($\leq 4.5\%$)
 ng/mg) $\mu\text{g/mL}$ $\mu\text{mol/L}$)
 PN6A201504001 0.61 3.90 0.932 0.4355 19.7 1.97
 PN6A201611001 0.80 4.90 0.215 3.213 17.4 3.3
 None
 RSP6AC201705001 4..27 0.163 3.6 13.7 3.6
 detected
 RSP6AC201707002 2.10 5.09 0.081 2.6 19.4 0.9
 RSP6AC201708003 2.60 4.28 0.227 2.3 12.8 0.5

Table 88 Test Results of 5 Batches of Serotype 6B Polysaccharide Derivative and Bulk Conjugate

Polysaccharide
 Bulk conjugate
 derivative
 Items Residual ADH Residual
 Derivation Free- Free-
 Batch No. cyanide content EDAC
 rate polysaccharide protein
 (≤ 10 (< 0.25 (≤ 5
 (1.5-3.4%) ($\leq 25\%$) ($\leq 4.0\%$)
 ng/mg) $\mu\text{g/mL}$ $\mu\text{mol/L}$)
 PN6B 201504001 0.63 2.92 0.15050 0.637 17.3 0.13
 PN6B 201611001 1.90 2.71 0.1740 2.409 17.9 1.4
 None
 RSP6BC201705001 2.33 0.0120 2.800 21.6 4.0
 detected
 RSP6BC201707002 3.10 2.57 0.0940 0.500 21.1 0.9
 RSP6BC201708003 6.70 2.42 0.1412 0.100 9.2 2.1

Table 89 Test Results of 5 Batches of Serotype 7F Polysaccharide Derivative and Bulk

Conjugate
 Polysaccharide
 Bulk conjugate
 derivative
 Items Residual ADH Residual
 Derivation Free- Free-
 Batch No. cyanide content EDAC
 rate polysaccharide protein
 (≤ 10 (< 0.2 (≤ 5
 (2.9-5.3%) ($\leq 20\%$) ($\leq 4.5\%$)
 ng/mg) $\mu\text{g/mL}$ $\mu\text{mol/L}$)
 PN7F201504002 1.4 4.28 0.150 0.149 16.7 1.69
 None
 PN7F201704003 3.35 0.137 2.362 13.6 3.80
 detected
 None
 RSP7FC201705001 4.76 0.066 3.200 18.4 4.00
 detected
 RSP7FC201706002 3.8 3.85 0.121 1.200 18.5 0.30
 RSP7FC201708003 6.10 4.44 0.088 3.700 18.9 1.30

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Table 90 Test Results of 5 Batches of Serotype 9V Polysaccharide Derivative and Bulk

Conjugate
 Polysaccharide
 Bulk conjugate
 derivative
 Items Residual ADH Residual
 Derivation Free- Free-
 Batch No. cyanide content EDAC
 rate polysaccharide protein
 (≤ 10 (< 0.7 (≤ 5
 (3.3-7.4%) ($\leq 25\%$) ($\leq 3.5\%$)
 ng/mg) $\mu\text{g/mL}$ $\mu\text{mol/L}$)
 PN9V201506002 0.6 3.94 0.582 3.2035 2.0 0.7
 None
 PN9V201612001 6.41 0.177 4.6130 1.1 2.2
 detected
 RSP9VC201705001 0.1 5.30 0.176 2.5000 6.8 2.9
 RSP9VC201707002 1.5 6.09 0.153 2.2000 12.0 1.6
 RSP9VC201709003 1.8 4.98 0.203 4.1000 11.8 0.7

Table 91 Test Results of 5 Batches of Serotype 14 Polysaccharide Derivative and Bulk

Conjugate
 Polysaccharide
 Bulk conjugate
 derivative
 Items Residual ADH Residual
 Derivation Free- Free-
 Batch No. cyanide content EDAC Free-protein
 rate polysaccharide
 (≤ 10 (< 1.15 (≤ 5 ($\leq 4.5\%$)
 (2.5-5.9%) ($\leq 25\%$)
 ng/mg) $\mu\text{g/mL}$ $\mu\text{mol/L}$)
 PN14201505003 4.20 4.28 0.832 1.9098 8.1 0.71
 PN14201704003 0.05 3.93 0.208 3.6940 12.7 3.80
 None
 RSP14C201706001 5.11 0.016 2.4000 14.6 3.10
 detected
 None
 RSP14C201707002 4.86 0.120 3.8000 12.6 1.00
 detected
 RSP14C201709003 1.60 3.52 0.114 1.0000 23.9 1.00

Table 92 Test Results of 5 Batches of Serotype 18C Polysaccharide Derivative and Bulk

Conjugate
 Polysaccharide
 Bulk conjugate
 derivative
 Items Residual ADH Residual
 Derivation Free- Free-
 Batch No. cyanide content EDAC
 rate polysaccharide protein
 (≤ 10 (< 0.2 (≤ 5
 (2.0-5.4%) ($\leq 30\%$) ($\leq 3.5\%$)
 ng/mg) $\mu\text{g/mL}$ $\mu\text{mol/L}$)
 PN18C201505002 3.3 3.61 0.170 0.9155 10.5 2.01
 PN18C201612001 0.9 2.58 0.122 2.1130 12.0 0.60
 RSP18C201706001 3.5 3.39 0.097 0.9000 7.7 2.70
 None
 RSP18CC201707002 3.84 0.108 1.5000 26.2 0.9
 detected
 RSP18CC201708003 1.0 3.38 0.015 0.7000 22.6 0.7

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Table 93 Test Results of 5 Batches of Serotype 19A Polysaccharide Derivative and Bulk Conjugate

Polysaccharide

Bulk conjugate derivative

Items Residual ADH Residual

Derivation Free- Free-

Batch No. cyanide content EDAC

rate polysaccharide protein

(≤ 10 (< 0.2 (≤ 5

(3.5-8.0%) (≤ 30%) (≤ 3.5%)

ng/mg) μg/mL) μmol/L)

PN19A201504002 2.6 6.21 0.1700 0.3630 21.6 0.30

PN19A201611001 2.8 5.63 0.1575 2.8845 10.2 2.00

None

RSP19AC201705001 3.97 0.1080 1.2000 27.4 2.90

detected

RSP19AC201707002 4.7 5.58 0.1300 1.3000 26.0 2.80

RSP19AC201708003 7.0 6.36 0.1030 2.300 28.0 0.13

Table 94 Test Results of 5 Batches of Serotype 19F Polysaccharide Derivative and Bulk Conjugate

Polysaccharide

Bulk conjugate derivative

Items Residual ADH Residual

Derivation Free- Free-

Batch No. cyanide content EDAC

rate polysaccharide protein

(≤ 10 (< 0.2 (≤ 5

(2.9-6.2%) (≤ 35%) (≤ 3.5%)

ng/mg) μg/mL) μmol/L)

PN19F201504001 6.5 5.09 0.174 1.3340 14.6 1.39

PN19F201611001 0.5 5.07 0.089 2.5945 6.1 2.00

RSP19FC201706001 3.8 3.92 0.073 3.9000 24.4 3.20

RSP19FC201707002 1.3 5.26 0.126 1.4000 30.1 2.40

RSP19FC201708003 5.9 5.00 0.066 3.1000 25.2 1.10

Table 95 Test Results of 5 Batches of Serotype 23F Polysaccharide Derivative and Bulk Conjugate

Polysaccharide

Bulk conjugate derivative

Items Residual ADH Residual

Derivation Free- Free-

Batch No. cyanide content EDAC

rate polysaccharide protein

(≤ 10 (< 0.45 (≤ 5

(0.9-3.7%) (≤ 25%) (≤ 4.0%)

ng/mg) μg/mL) μmol/L)

PN23F201505002 2.6 2.95 0.4065 2.2840 7.4 0.15

PN23F201611001 2.1 2.29 0.2600 2.4775 4.4 3.00

None

RSP23FC201705001 3.12 0.1620 0.7000 13.0 1.60

detected

RSP23FC201707002 3.7 2.35 0.1740 2.5000 24.9 2.10

RSP23FC201708003 2.3 2.00 0.1600 2.1000 18.7 0.60

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3.2.S.4 Control of Drug Substance

3.2.S.4.1 Specification

Table 96 Specification for Pneumococcal Polysaccharide

Sodium deoxycholate

Acetone residual (%)

Ethanol residual (%)

Nucleic acid content

Molecular size (KD)

Protein content (%)

Uronic acid content

Bacterial endotoxin

O-acetyl group (%)

Total nitrogen (%)

Ether residual (%)

Recovery rate at

Recovery rate at

Phosphorus (%)

Methyl pentose

KD≤0.30 (%)

KD≤0.60 (%)

Hexosamine
Serotypes

Pyrogen
Identity

(EU/μg)
(%)

(%)

(%)

1 ≤ 2 ≤ 2 3.5-6.0 0-1.5 // ≥ 45 ≥ 1.8 ≤ 0.15 ≥ 75 /
Positive response to corresponding antiserum

3 ≤ 5 ≤ 2 0-1.0 0-1.0 // ≥ 40 // ≤ 0.15 ≥ 78 /
4 ≤ 3 ≤ 2 4.0-6.0 0-1.5 ≥ 40 // // ≤ 0.15 ≥ 75 /
5 ≤ 7.5 ≤ 2 2.5-6.0 ≤ 2.0% ≥ 20 // ≥ 12 // ≤ 0.60 // ≥ 70
6A ≤ 2 ≤ 1 0-2.0 2.4-5.0 // ≥ 15 // ≤ 0.50 // ≥ 50
6B ≤ 2 ≤ 2 0-2.0 2.5-5.0 // ≥ 15 // ≤ 0.50 // ≥ 85
7F ≤ 5 ≤ 5 1.5-4.0 0-1.0 // ≥ 13 // ≤ 0.20 ≥ 70 // ≤ 0.5 < 0.08 < 0.03 < 0.01 < 0.02 /
9V ≤ 2 ≤ 2 0.5-3.0 0-1.0 ≥ 13 // ≥ 15 // ≤ 0.45 // ≥ 80
14 ≤ 5 ≤ 2 1.5-4.0 0-1.0 ≥ 20 // // ≤ 0.30 ≥ 55 /
18C ≤ 3 ≤ 2 0-1.0 2.4-4.9 // ≥ 14 // // ≤ 0.15 ≥ 70 /
19A ≤ 2 ≤ 2 0.6-3.5 3.0-7.0 ≥ 12 ≥ 20 // // ≤ 0.15 ≥ 75 /
19F ≤ 3 ≤ 2 1.4-3.5 3.0-5.5 ≥ 12.5 ≥ 20 // // ≤ 0.15 ≥ 75 /
23F ≤ 2 ≤ 2 0-1.0 3.0-4.5 // ≥ 37 // // ≤ 0.15 ≥ 75 /

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Table 97 Specification for Polysaccharide Derivatives

Test items Specifications

Serotype Rate (%)

1 2.6 – 4.4

3 5.0 – 14.0

4 3.0 – 11.0

5 3.5 – 18.0

6A 3.0 – 6.0

6B 1.5 – 3.4

Derivation rate

7F 2.9 – 5.3

9V 3.3 – 7.4

14 2.5 – 5.9

18C 2.0 – 4.8

19A 3.5 – 8.0

19F 3.2 – 6.2

23F 1.1 – 3.7

Residual cyanide No more than 10 ng/mg

KD = 0.2 – KD > 0.6

Serotype KD

0.6 recovery recovery

1 ≤ 0.37 ≥ 45% -

3 ≤ 0.45 ≥ 60% -

4 ≤ 0.42 ≥ 65% -

5 ≤ 0.90 - ≥ 37%

6A ≤ 0.62 ≥ 60% -

Molecular size 6B ≤ 0.33 ≥ 50% -

7F ≤ 0.43 ≥ 60% -

9V ≤ 0.46 ≥ 70% -

14 ≤ 0.40 ≥ 65% -

18C ≤ 0.41 ≥ 65% -

19A ≤ 0.48 ≥ 65% -

19F ≤ 0.56 ≥ 70% -

23F ≤ 0.46 ≥ 75% -

Table 98 Specification for Bulk Tetanus Toxoid

Test items Specifications

pH 6.6 – 7.4

Purity No less than 1500 Lf/mg

Sterility Should be in accordance with the regulations

Observation on the 7th, 14th and 21st day showing no local suppuration and necrosis, no

Specific toxicity

tetanus symptoms. Weight gain for each guinea pig at the end of the period.

Toxicity reversal Observation on the 7th, 14th and 21st day showing no tetanus symptoms, weight gain for test each guinea pig at the end of the period.

Table 99 Specification for Purified Liquid of Tetanus Toxoid

Test items Specifications

Identity /

Protein content No less than 16 mg/mL

Bacterial endotoxin No more than 10 EU/mg

Purity No less than 90%

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Table 100 Specification for Monovalent Bulk Conjugate

Test items Specifications

Appearance Clear, transparent liquid or slightly white floccule after shaking

A clear precipitation line should appear with the corresponding pneumococcal

Identity
specific antiserum and tetanus toxoid immune serum.
No less than the limit below.

Limit
Serotype
($\mu\text{g/mL}$)
1 40
3 20
4 40
5 40
6A 40
Polysaccharide content
6B 40
7F 40
9V 40
14 40
18C 40
19A 40
19F 40
23F 40

Protein content No less than 70 $\mu\text{g/mL}$.

Serotype Range
1 0.20 – 1.80
3 0.10 – 0.73
4 0.30 – 0.90
5 0.10 – 0.60
6A 0.10 – 0.50
6B 0.30 – 0.60
PS: protein ratio
7F 0.20 – 0.90
9V 0.10 – 0.35
14 0.20 – 0.60
18C 0.40 – 0.80
19A 0.20 – 0.70
19F 0.20 – 0.70
23F 0.25 – 0.60
Serotype KD KD < 0.2 recovery
1 $\geq 85\%$
3 $\geq 95\%$
4 $\geq 80\%$
5 $\geq 80\%$
6A $\geq 80\%$
6B $\geq 90\%$

Molecular size (KD)
7F $\leq 0.2 \geq 90\%$
9V $\geq 95\%$
14 $\geq 90\%$
18C $\geq 80\%$
19A $\geq 85\%$
19F $\geq 80\%$
23F $\geq 90\%$

No more than the limit below.

Free polysaccharide
Serotype Limit (%)
content
1 25.0

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3 20.0
4 25.0
5 20.0
6A 20.0
6B 25.0
7F 20.0
9V 25.0
14 25.0
18C 30.0
19A 30.0
19F 35.0
23F 25.0
Free carrier protein
No more than 5.0%
content
Residual EDAC No more than 5.0 $\mu\text{mol/L}$
No more than the limit below.
Serotype Limit
($\mu\text{g/mL}$)
1 0.15
3 0.25
4 0.15
5 0.25
6A 1.20
ADH content
6B 0.25
7F 0.20
9V 0.70
14 1.15
18C 0.20

19A 0.20
19F 0.20
23F 0.45
Sterility Meets the requirement of the test.
Bacterial endotoxin No more than 0.75 EU/μg
Observation on the 7th, 14th and 21st day showing no local suppuration and
Specific toxin of carrier
necrosis, no tetanus symptoms. Weight gain for each guinea pig at the end of the
protein
period.

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3.3.S.4.2 Analytical Procedures 3.2.S.4.2.1 SOP for Polysaccharide Test

Table 101 SOP List of Pneumococcal Polysaccharides Test
Tests SOP Document code Test method
Double
Identity SOP for Identity Test WSYX-QC-SOP-00-09-0201
immunodiffusion
Bacterial
SOP for Bacterial Endotoxin WSYX-QC-SOP-00-09-0203 Gel-clot method
endotoxin
Heat-dry and weighing
Total solid SOP for Total Solid Content WSYX-QC-SOP-00-09-0111
method
SOP for Protein Content
Protein content WSYX-QC-SOP-00-09-0102 Lowry method
Determination
Nucleic acid SOP for Nucleic Acid UV-vis
WSYX-QC-SOP-00-09-0103
content Content Determination spectrophotometry
Phosphorus SOP for Phosphorus Content
WSYX-QC-SOP-00-09-0101 Chromogenic method
content Determination
Hexosamine SOP for Hexosamine
WSYX-QC-SOP-00-09-0127 Chromogenic method
content Content Determination
Uronic acid SOP for Uronic Acid
WSYX-QC-SOP-00-09-0129 Chromogenic method
content Content Determination
SOP for O-acetyl group
O-acetyl group WSYX-QC-SOP-00-09-0112 Chromogenic method
Content Determination
Methyl pentose SOP for Methyl pentose
WSYX-QC-SOP-00-09-0128 Chromogenic method
content Content Determination
SOP for Molecular Size
Molecular size WSYX-QC-SOP-00-09-0104 Gel chromatography
Determination
SOP for Phenol Content
Phenol content WSYX-QC-SOP-00-09-0115 Titration
Determination
Residual
Gas chromatography
ethanol SOP for Residual Ethanol,
Residual Ether, Acetone Content WSYX-QC-SOP-00-09-0138
Gas chromatography
acetone Determination
Residual ether Gas chromatography
SOP for Total Nitrogen
Total nitrogen WSYX-QC-SOP-00-09-0751 Kjeldahl method
Content Determination
SOP for Residual Sodium
Sodium UV-vis
Deoxycholate Content WSYX-QC-SOP-00-09-0137
deoxycholate spectrophotometry
Determination

3.2.S.4.2.2 SOP for Polysaccharide Derivative Test

Table 102 SOP List of Pneumococcal Polysaccharide Derivatives test
Tests SOP Document code Test method
Heat-dry and weighing
Total solid SOP for Total Solid Content WSYX-QC-SOP-00-09-0111
method
SOP for Adipic Dihydrazide
WSYX-QC-SOP-00-09-0105 Chromogenic method
Derivation Content Determination
rate SOP for Glucose Analysis
WSYX-QC-SOP-00-09-0126 Anthrone method
with Anthrone

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Residual SOP for Residual Cyanide
WSYX-QC-SOP-00-09-0106 Chromogenic method

cyanide Content Determination
Molecular SOP for Molecular Size
WSYX-QC-SOP-00-09-0104 Gel chromatography
size Determination

3.2.S.4.2.3 SOP for Carrier Protein

Table 103 SOP List of Bulk Tetanus Toxoid Test
Tests SOP Document code Test method
Specification of Diphtheria,
WSYX-QC-TS-00-09-
Appearance tetanus, and acellular pertussis Visual determination
0001
combined vaccine, adsorbed
SOP for pH Value WSYX-QC-SOP-00-09-
pH pH meter
Determination 0110
Residual SOP for Residual Ammonium WSYX-QC-SOP-00-09-
Kjeldahl method
ammonium Determination 0754
Protein nitrogen SOP for Protein Nitrogen WSYX-QC-SOP-00-09-
Kjeldahl method
content Content Determination 0774
Flocculation SOP for Toxoid Flocculation WSYX-QC-SOP-00-09-
Serological method
units Units Determination 0011
Calculation
WSYX-QC-SOP-00-09- (flocculation
Purity SOP for Purity Test
0209 units/protein nitrogen
content)
WSYX-QC-SOP-00-09- Membrane filtration
Sterility SOP for Sterility Test
0202 method
SOP for Protein Content - WSYX-QC-SOP-00-09-
Protein content Lowry method
Lowry Method 0134
SOP for Specific Toxicity Test WSYX-QC-SOP-02-09-
Specific toxicity Animal Test
of Tetanus Toxoid Bulk 0003
Toxicity SOP for Toxicity Reverse Test WSYX-QC-SOP-02-09-
Animal Test
Reversal of Tetanus Toxoid Bulk 0002
Detoxification SOP for Detoxification Test of WSYX-QC-SOP-02-09-
Animal Test
test Tetanus Toxoid Bulk 0004

Table 104 SOP List of Purified Liquid of Tetanus Toxoid Test
Tests SOP Document code Test method
SOP for Protein Content WSYX-QC-SOP-00-09-
Protein content Lowry method
- Lowry Method 0134
WSYX-QC-SOP-00-09- Membrane filtration
Sterility SOP for Sterility Test
0202 method
SOP for Bacterial WSYX-QC-SOP-00-09-
Bacterial endotoxin Gel-clot method
Endotoxin Test 0203
SOP for Purity Test of
WSYX-QC-SOP-00-09-
Purity Tetanus Toxoid Purified HPLC
0143
Liquid

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3.2.S.4.2.4 SOP for Monovalent Bulk Conjugate

Table 105 SOP List of Monovalent Bulk Conjugate Test
Tests SOP Document code Test method
SOP for Appearance WSYX-QC-SOP-00-09-
Appearance Visual determination
Determination 0108
WSYX-QC-SOP-00-09- Double
Identity SOP for Identity Test
0201 immunodiffusion
SOP for Bacterial WSYX-QC-SOP-00-09-
Bacterial endotoxin Gel-clot method
Endotoxin 0203
WSYX-QC-SOP-00-09-
Sterility SOP for Sterility Test Membrane filtration
0202
SOP for Polysaccharide
Content Determination WSYX-QC-SOP-00-09- Rate Nephelometry
Polysaccharide content
with Rate Nephelometry 0141 Method
Method
SOP for Protein Content WSYX-QC-SOP-00-09-
Protein content Lowry method
Determination 0102

SOP for
Polysaccharide/protein WSYX-QC-SOP-00-09-
Polysaccharide/Protein Calculation
ratio 0125
Ratio Determination
SOP for Free
Polysaccharide
Free polysaccharide Determination of WSYX-QC-SOP-00-09-
Chromogenic method
content Pneumococcal 0123
Polysaccharide bulk
Conjugate
SOP for Free Protein WSYX-QC-SOP-00-09-
Free protein content Gel chromatography
Determination 0114
SOP for Molecular Size WSYX-QC-SOP-00-09-
Molecular size Gel chromatography
Determination 0104
SOP for Residual EDAC WSYX-QC-SOP-00-09-
Residual EDAC Chromogenic method
Determination 0107
SOP for ADH Content WSYX-QC-SOP-00-09-
ADH content Chromogenic method
Determination 0105

Please refer to Annex 1 - SOP for 13-valent Pneumococcal Polysaccharide Vaccine intermediates

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3.2.S.4.3 Validation or Qualification of Analytical Procedures

Table 106 List of Test Method Validation/Qualification Protocol for Intermediate Product
Test items Validation/Qualification protocol Document ode Test Method
Validation Protocol of Identification Immuno-double
Identity WSYX-QC-TS-0006-0697
test for Pneumococcal Polysaccharide diffusion
Qualification Protocol of Protein
Protein content WSYX-QC-TS-0006-0592 Lowry method
Content Determination
Qualification Protocol of Nucleic
Nucleic Acid UV visible
Acid Content Determination for WSYX-QC-TS-0006-0719
Content spectrophotometry
Pneumococcal Polysaccharide
Qualification Protocol of Total
Total Nitrogen Nitrogen Content Determination for WSYX-QC-TS-0006-0587 Kjeldahl method
Pneumococcal Polysaccharide
Phosphorus Qualification Protocol of Total Solid Chromogenic
WSYX-QC-TS-0006-0598
Content content Determination method
Qualification Protocol of Uronic Acid
Uronic Acid Chromogenic
Determination for Pneumococcal WSYX-QC-TS-0006-0589
Content method
Polysaccharide
Qualification Protocol of Hexosamine
Hexosamine Chromogenic
Determination for Pneumococcal WSYX-QC-TS-0006-0591
Content method
Polysaccharide
Methyl Qualification Protocol of
Chromogenic
pentoses Methylpentoses Determination for WSYX-QC-TS-0006-0590
method
Content Pneumococcal Polysaccharide
Qualification Protocol of O-acetyl
O-acetyl Chromogenic
Determination for Pneumococcal WSYX-QC-TS-0006-0599
Content method
Polysaccharide
Qualification Protocol of Molecular Gel
Molecular Size WSYX-QC-TS-0006-0597
Size Determination chromatography
Organic
Solvent Validation Protocol of Ethanol, Ether,
Gas
(ethanol, ether, Acetone Residues Determination for WSYX-QC-TS-0006-0694
chromatography
acetone) Pneumococcal Polysaccharide
residues
Qualification Protocol of Bacterial
Bacterial
Endotoxin Test Method for
Endotoxin WSYX-QC-TS-0006-07785 Gel-clot method
Pneumococcal Polysaccharide and
Content
Bulk Polysaccharide
Qualification Protocol of Sodium
Sodium UV-vis
Deoxycholate Content Determination WSYX-QC-TS-0006-0721

Deoxycholate spectrophotometry
for Pneumococcal Polysaccharide
Qualification Protocol of Phenol
Phenol content Content Determination for WSYX-QC-TS-0006-0717 Titration
Pneumococcal Polysaccharide
Validation Protocol of Free
Free
Polysaccharide Content Chromogenic
Polysaccharide WSYX-QC-TS-0006-0668
Determination for Pneumococcal method
Content
Bulk Polysaccharide
Validation Protocol of Free Protein
Free Protein Gel
Content Determination for WSYX-QC-TS-0006-0679
Content chromatography
Pneumococcal Bulk Polysaccharide
Bacterial
Interference Test / /
Endotoxin Test
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Method Applicability Test for
Microbial
Microbial Limit Test of 13-valent WSYX-QC-TS-0006-0553 /
Limit
Pneumococcal Bulk Conjugate
Qualification Protocol of Rate
Polysaccharide
Polysaccharide Content WSYX-QC-TS-0006-0647 Nephelometry
Content
Determination Method
Method Applicability Test for
Membrane
Sterility Test Microbial Limit Test of 13-valent WSYX-QC-TS-0006-0543
filtration
Pneumococcal Conjugate Vaccine
Validation Protocol of Purity Test for
Purity WSYX-QC-TS-0006-0740 HPLC
purified Tetanus Toxoid
Protein Content Determination
Protein WSYX-QC-TS-0006-0564 Lowry Method
Method
Method Applicability Test for Membrane
Sterility Test WSYX-QC-TS-006-0551
Sterility Test of Excipients filtration

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3.2.S.4.4 Batch Analyses
Different batches of 13 serotypes of pneumococcal polysaccharide, 13 serotypes of polysaccharide
derivatives, bulk tetanus toxoid, purified liquid of
tetanus toxoid, 13 serotypes of bulk conjugate were analyzed based on the analytical procedures. The data
is showing in table below.

Table 107 Batch Analyses Data Statistics for Pneumococcal Polysaccharide
Test O-
Total Bacteria Sodium
items Identification Total Nucleic Phosph Uronic acetyl Hexosa Methyl Phenol
Protein nitrogen Molecular size I Ether Acetone Ethanol m
ation solid acid orus acid group mine pentose residue
content n (KD) endotoxin residual residual residual deoxyc
Batch test mg/mL content content content conten content content g/L
content in holate
No. t
Appear
While
positive ≤ 0.5
KD <0.3,
response EU/μg
Specific 3.5- KD ≤ the < < < <
to - ≤ 2% ≤ 2% 0-1.5% ≥ 45% ≥ 1.8% / / /
ation 6.0% 0.15 recovery 0.01% 0.03% 0.08% 0.2%
specific ≤ 50
rate
antiserum EU/mL
≥75%
m type 1
1-
PVPV15 3.245 1.9% 0.8% 5.0% 0.3% 0 / ND ND 0.007% 0.02% 55.9% 5.8% / / /
101201
Comply Comply
1-
with the with the
PVPV15 2.93 1.5% 1.1% 3.7% 0.3% 0 / ND 0.001% ND 0.09% 60.8% 6.1% / / /
requirement
121202
ents ents

1-
PVPV16 3.080 1.6% 0.9% 5.6% 0.6% 0.04 / ND ND ND 0.10% 57.4% 5.7% / / /
011203
Appear
While
positive ≤ 0.5
KD <
response EU/μg
Specific 0- KD≤0.1 0.3, the < < < <
to - ≤5% ≤2% ≤5% / / / /
ation 1.0% 5 recovery 0.01% 0.03% 0.08% 0.2%
specific ≤ 50 ≥40%
rate
antiseru EU/mL
≥78%
m type 3
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Test O-
Total Bacteria Sodi
items Identific Total Nucleic Phosph Uronic acetyl Hexosa Methyl Phenol
Protein nitroge Molecular size I Ether Acetone Ethanol m
ation solid acid orus acid group mine pentose residue
content n (KD) endotox residual residual residual deoxyc
Batch test mg/mL content content content conten content content g/L
content in holate
No. t
3-
PVPV15 3.105 0.1% 0.04% 0.68% 0.00% 0 / ND ND 0.002% 0.030% 50.5% / / / /
100901
Comply Comply
3-
with the with the
PVPV15 3.065 0.4% 0.09% 0.65% 0.00% 0 / ND ND ND 0.16% 52.90% / / / /
requirem requirem
110902
ents ents
3-
PVPV15 2.325 0.5% 1.99% 0.86% 0.10% 0.04 / ND ND ND 0.170% 47.8% / / / /
120903
Appear
While
positive ≤ 0.5
KD <
response EU/μg
Specific 4.0- KD ≤ 0.3, the < < < <
to - ≤ 3% ≤ 2% 0-1.5% / / ≥ 40% / /
ation 6.0% 0.15 recovery 0.01% 0.03% 0.08% 0.2%
specific ≤ 50
rate
antiseru EU/mL
≥75%
m type 4
4-
PVPV15 2.600 1.30% 0.20% 5.73% 0.20% 0 / ND 0.002% 0.018% 0.110% / / 45.2% / /
101101
Comply Comply
4-
with the with the
PVPV15 3.140 1.3% 0.3% 5.73% 0.29% 0.04 / ND ND 0.006% 0.16% / / 46.4% / /
requirem requirem
121102
ents ents
4-
PVPV16 2.710 2.7% 0.6% 5.83% 0.28% 0 / ND ND 0.001% 0.11% / / 42.0% / /
011103
Appear ≤ 0.5
While
positive EU/μg
Specific 2.5- KD ≤ KD < < < < <
response - ≤ 7.5% ≤ 2% ≤ 2.0% / ≥ 20% / /
ation 6.0% 0.60 0.6, the 0.01% 0.03% 0.08% 0.2% ≥ 12%
to ≤50EU/
recovery
specific mL

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Version No. Walvax-202007-00 Yuxi Walvax Biotechnology Co., Ltd.

Test O-
Total Bacteria Sodi
items Identific Total Nucleic Phosph Uronic acetyl Hexosa Methyl Phenol
Protein nitroge Molecular size I Ether Acetone Ethanol m
ation solid acid orus acid group mine pentose residue
content n (KD) endotox residual residual residual deoxyc
Batch test mg/mL content content content conten content content g/L
content in holate
No. t
antiseru rate

m type 5 ≥70%
5-
PVPV15 3.080 3.90% 1.70% 3.64% 0.20% 0.35 / ND 0.0045% 0.020% 0.13% 13.60% / 21.70% / /
101301
Comply Comply
5-
with the with the
PVPV15 3.185 6.40% 1.50% 5.90% 0.30% 0.39 / ND ND ND 0.17% 13.3% / 23.5% / /
requirem requirem
121302
ents ents
5-
PVPV16 4.5% 1.1% 3.82% 0.2% 4.5% 0.35 / ND ND 0.0045% 0.12% 14.8% / 20.9% / /
011303
Appear
positive While
≤ 0.5
response KD <
EU/μg
Specific to 0- 2.5- KD ≤ 0.6, the < < < <
- ≤ 1.5% ≤ 1% / / ≥ 15% /
ation specific 2.0% 5.0% 0.50 recovery 0.01% 0.03% 0.08% 0.2% /
≤ 50
antiseru rate ≥
EU/mL
m type 50%
6A
6A-
PVPV15 2.020 0.36% 0.082% 1.3% 4.5% 0.39 / ND 0.0025% 0.0565% 0% / / / 23.7% /
090101
Comply Comply
6A-
with the with the
PVPV15 2.165 1.16% 0.08% 1.5% 4.17% 0.39 / ND ND 0.061% 0.02% / / / 22.6% /
requirem requirem
110102
ents ents
6A-
PVPV15 2.825 1.3% 0.1% 1.4% 4.17% 0.39 / ND 0.002% 0.026% 0% / / / 23.6% /
120103
Specific Appear 0- 2.5- KD≤0.5 While ≤0.5EU/ < < < <
≤2% ≤2% / / ≥15% /
ation positive 2.0% 5.0% 0 KD<0.6, μg 0.01% 0.03% 0.08% 0.2% /

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MODULE 3 QUALITY
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Test O-
Total Bacteria Sodi
items Identific Total Nucleic Phosph Uronic acetyl Hexosa Methyl Phenol
Protein nitroge Molecular size I Ether Acetone Ethanol m
ation solid acid orus acid group mine pentose residue
content n (KD) endotox residual residual residual deoxyc
Batch test mg/mL content content content conten content content g/L
content in holate
No. t
response the
to recovery ≤ 50
specific rate ≥ EU/mL
antiseru 85%
m type
6B
6B-
PVPV15 2.505 0.10% 0.10% 0.96% 4.40% 0.35 / ND 0.002% 0.066% 0.050% / / 23.40% /
/
090201
Comply Comply
6B-
with the with the
PVPV15 2.105 0.40% 0.10% 1.09% 4.40% 0.3 / ND 0.004% 0.064% 0.110% / / 23.80% /
requirem requirem /
110202
ents ents
6B-
PVPV15 2.87 1.40% 0.10% 0.73% 4.10% 0.3 / ND 0.002% 0.0275% 0.180% / / 23.50% /
/
120203
Appear
positive While
≤ 0.5
response KD <
EU/μg
Specific to 1.5- 0- KD ≤ 0.3, the < < < <
- ≤ 5% ≤ 2% / / / ≥ 13%
ation specific 4.0% 1.0% 0.20 recovery 0.01% 0.03% 0.08% 0.2%
≤ 50
antiseru rate ≥
EU/mL
m type 70%
7F
7F-

PVPV15 Comply 2.830 0.2% 0.4% 3.07% 0.1% 0.15 / Comply ND 0.001% 0.022% 0.04% / / / 19.4% /
100701 with the with the
7F- requirem requirem
PVPV15 ents 3.430 1.2% 0.2% 3.38% 0.5% 0.12 / ents ND 0.002% 0.014% 0.12% / / / 19.0% /
110702

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MODULE 3 QUALITY
Version No. Walvax-202007-00 Yuxi Walvax Biotechnology Co., Ltd.

Test O-
Total Bacteria Sodi
items Identific Total Nucleic Phosph Uronic acetyl Hexosa Methyl Phenol
Protein nitroge Molecular size I Ether Acetone Ethanol m
ation solid acid orus acid group mine pentose residue
content n (KD) endotox residual residual residual deoxyc
Batch test mg/mL content content content conten content content g/L
content in holate
No. t
7F-
<
PVPV16 2.585 1.1% 0.2% 3.87% 0.40% 0 / ND ND 0.005% / / 17.3% /
0.2% /
010703
Appear
positive While
≤ 0.5
response KD <
EU/μg
Specific to 0.5- KD ≤ 0.6, the < < < <
- ≤ 2% ≤ 2% 0-1.0% ≥ 15% / ≥ 13% / /
ation specific 3.0% 0.45 recovery 0.01% 0.03% 0.08% 0.2%
≤ 50
antiseru rate ≥
EU/mL
m type 80%
9V
9V-
PVPV15 3.075 0.4% 0.2% 1.95% 0.09% 0.35 / ND ND 0.007% 0.04% 22.2% / 13.0% / /
090301
Comply Comply
9V-
with the with the
PVPV15 2.920 0.3% 0.05% 2.36% 0.03% 0.30 / ND ND 0.020% 0.14% 26.1% / 14.6% / /
requirem requirem
110302
ents ents
9V-
PVPV15 2.625 0.3% 0.06% 2.48% 0.07% 0.30 / ND 0.001% ND 0.180% 24.3% / 15.7% / /
120303
Appear
positive While
≤ 0.5
response KD <
EU/μg
Specific to 1.5- KD ≤ 0.3, the < < < <
- ≤ 5% ≤ 2% 0-1.0% / / ≥ 20% / /
ation specific 4.0% 0.30 recovery 0.01% 0.03% 0.08% 0.2%
≤ 50
antiseru rate ≥
EU/mL
m type 55%
14
14-
Comply Comply
PVPV15 3.085 0.8% 1.2% 2.82% 0.20% 0.19 / ND ND 0.009% 0.06% / / 22.2% / /
with the with the
101001
Confidential 642
MODULE 3 QUALITY
Version No. Walvax-202007-00 Yuxi Walvax Biotechnology Co., Ltd.

Test O-
Total Bacteria Sodi
items Identific Total Nucleic Phosph Uronic acetyl Hexosa Methyl Phenol
Protein nitroge Molecular size I Ether Acetone Ethanol m
ation solid acid orus acid group mine pentose residue
content n (KD) endotox residual residual residual deoxyc
Batch test mg/mL content content content conten content content g/L
content in holate
No. t
14- requirem requirem
0.1800
PVPV15 ents 2.385 1.9% 0.1% 3.35% 0.30% 0.08 / ents ND ND 0.0095% / / 23.6% / /
%
121002
14-
0.0900
PVPV16 2.750 1.5% 0.7% 3.02% 0.30% 0 / ND ND 0.0045% / / 22.9% / /
%
011003
Appear

positive While
≤ 0.5
response KD <
EU/μg
Specific to 0- 2.4- KD ≤ 0.3, the < < < <
- ≤ 3% ≤ 2% / / / ≥ 14% /
ation specific 1.0% 4.9% 0.15 recovery 0.01% 0.03% 0.08% 0.2%
≤ 50
antiseru rate ≥
EU/mL
m type 70%
18C
18C-
PVPV15 2.725 1.0% 0.2% 0.92% 3.00% 0.07 / ND ND 0.020% 0.090% / / / 14.5% /
100801
Comply Comply
18C-
with the with the
PVPV15 2.130 0.8% 0.2% 0.89% 2.80% 0.08 / ND 0.002% 0.034% 0.060% / / / 17.5% /
requirem requirem
110802
ents ents
18C-
PVPV16 1.990 1.3% 0.2% 0.95% 2.70% 0.08 / ND 0.002% 0.046% 0.110% / / / 14.6% /
010803
Appear
positive While
≤ 0.5
response KD <
EU/μg
Specific to 0.6- 3.0- KD ≤ 0.5, the < < < <
- ≤ 2% ≤ 2% / / / ≥ 12% ≥ 20% /
ation specific 3.5% 7.0% 0.45 recovery 0.01% 0.03% 0.08% 0.2%
≤ 50
antiseru rate ≥
EU/mL
m type 80%
19A
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Test O-
Total Bacteria Sodium
items Identific Total Nucleic Phosph Uronic acetyl Hexosa Methyl Phenol
Protein nitroge Molecular size I Ether Acetone Ethanol m
ation solid acid orus acid group mine pentose residue
content n (KD) endotox residual residual residual deoxyc
Batch test mg/mL content content content conten content content g/L
content in holate
No. t
19A-
PVPV15 2.210 0.4% 0.3% 3.26% 4.90% 0.18 / ND 0.002% 0.015% 0.07% / / 12.6% 29.0% /
090401
Comply Comply
19A-
with the with the
PVPV15 2.445 0.7% 0.02% 3.31% 4.7% 0.04 / ND 0.002% 0.0145 0.004% / / 13.7% 31.3% /
requirem requirem
110402
ents ents
19A-
PVPV15 2.850 0.5% 0.05% 3.44% 4.9% 0.04 / 0.001% 0.002% 0.005% 0.15% / / 15.0% 33.8% /
120403
Appear
positive While
≤ 0.5
response KD <
EU/μg
Specific to 1.4- 3.0- KD ≤ 0.3, the < < < < ≥
- ≤ 3% ≤ 2% / / / ≥ 20%
ation specific 3.5% 5.5% 0.20 recovery 0.01% 0.03% 0.08% 0.2% 12.5%
≤ 50
antiseru rate ≥
EU/mL
m type 75%
19F
19F-
PVPV15 2.750 0.4% 0.1% 3.27% 4.60% 0.04 / ND 0.002% 0.013% 0.10% / / 19.10% 28.1% /
090501
Comply Comply
19F-
with the with the
PVPV15 2.535 0.2% 0.1% 3.00% 4.80% 0 / ND 0.0030% 0.0145% 0.00% / / 20.3% 27.9% /
requirem requirem
110502
ents ents
19F-
PVPV15 3.130 0.3% 0.1% 3.42% 4.80% 0 / ND 0.0005% 0.0035% 0.13% / / 18.30% 24.2% /
120503
Appear While
≤ 0.5

Specific positive 0- 3.0- KD ≤ KD < < < < <
- ≤ 2% ≤ 2% EU/μg / / / ≥ 37%
ation response 1.0% 4.5% 0.15 0.3, the 0.01% 0.03% 0.08% 0.2%
to recovery

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Test O-
Total Bacteria Sodi
items Identific Total Nucleic Phosph Uronic acetyl Hexosa Methyl Phenol
Protein nitroge Molecular size I Ether Acetone Ethanol m
ation solid acid orus acid group mine pentose residue
content n (KD) endotox residual residual residual deoxyc
Batch test mg/mL content content content conten content content g/L
content in holate
No. t
specific rate ≥ ≤ 50
antiseru 85% EU/mL
m type
23F-
23F-
PVPV15 2.585 0.3% 0.2% 0.89% 3.80% 0.04 / ND 0.002% 0.017% 0.04% / / / 41.5% /
090601
Comply Comply
23F-
with the with the
PVPV15 3.415 0.2% 0.1% 0.88% 3.90% 0.08 / ND 0.003% 0.010% 0.17% / / / 41.8% /
requiirem requiirem
120602
ents ents
23F-
0.0015
PVPV16 3.180 0.3% 0.1% 0.97% 3.30% 0.00 / ND ND 0.0015% ND ND / 38.8% /
%
010603

Table 108 Batch Analyses Data Statistics for Polysaccharide Derivatives

Residual
Total solid Derivation Molecular Size Determination by SEC-MALS
cyanide Molecular size (KD)
Test item and mg/mL rate (%) Method
content
Serotypes specification
While KD = 0.2- Mn (Number- Mw (Mass
Batch No. Mw/Mn
- 2.6-4.4% ≤ 10 ng/mg KD ≤ 0.37 0.6, the recovery average Molecular average molar
(polydispersity)
rate ≥ 45% Weight) mass)
SP1C201908001 7.900 3.3% 6 0.32 88% 258.8 334.6 1.292
1 SP1C201909002 7.180 3.9% 2 0.26 68% 237.8 298.6 1.255
SP1C201909003 8.500 3.6% 7 0.33 81% 463.3 695.8 1.502
SP3C201908001 14.700 10.6% 1 0.14 64% 296.4 590.0 1.991
3 SP3C201909002 14.440 13.0% 1 0.18 67% 412.55 1207 2.9255
SP3C201909003 15.530 12.9% 1 0.15 65% 362.9 988.05 2.7235
SP4D202001001 13.055 5.8% 3 0.33 76% 163.15 210.4 1.2895
4
SP4D 202002002 10.855 6.0% 0 0.37 77% 155.5 191.35 1.2305
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SP4D 202005003 9.750 6.4% 1 0.29 65.2% 193.0 255.4 1.3235
SP5C201909001 8.365 12.6% 1 0.58 51% 27.01 34.45 1.275
5 SP5C201909002 9.065 11.9% 0 0.63 44% 24.86 31.95 1.285
SP5C201909003 9.945 12.6% 1 0.68 44% 27.61 40.76 1.477
SP6AC201909001 11.875 5.2% 0 0.39 83% 217.7 251.4 1.155
6A SP6AC201909002 10.460 5.2% 0 0.35 79% 256.7 303.0 1.181
SP6AC201909003 10.020 5.5% 0 0.41 85% 214.4 247.6 1.155
SP6BD 202001001 11.150 2.7% 3 0.30 74% 278.05 350.95 1.263
6B SP6BD 202001002 10.760 2.7% 0 0.30 78% 218.9 256.15 1.170
SP6BD 202002003 8.315 2.8% 0 0.30 74% 224.65 267.3 1.1895
SP7FD 202001001 9.925 4.3% 1 0.37 78% 404.6 607.2 1.5015
7F SP7FD 202002002 9.760 4.0% 0 0.32 76% 367.55 571.45 1.5545
SP7FD 202002003 7.220 4.3% 1 0.37 79% 357.75 472.4 1.3205
SP9VC201908001 8.750 4.7% 0 0.30 81% 636.4 1059 1.664
9V SP9VC201908002 10.085 5.0% 1 0.28 92% 959.3 2067.5 2.155
SP9VC201908003 8.850 4.6% 1 0.35 82% 685.2 1266 1.848
SP14D 202001001 11.775 3.9% 0 0.29 79% 923.35 2168.5 2.3485
14C SP14D 202002002 10.470 4.3% 0 0.33 72% 1048 2007.5 1.916
SP14D 202002003 11.695 4.3% 0 0.30 70% 1272.5 2185 1.717
SP18CC201908001 10.540 3.7% 1 0.32 74% 202.5 245.2 1.211
18C SP18CC201908002 11.670 4.0% 0 0.36 74% 209.0 250.4 1.198
SP18CC201909003 10.710 4.0% 0 0.35 73% 206.4 246.4 1.194
SP19AC201908001 14.170 6.4% 1 0.44 79% 161.3 191.4 1.186
19A SP19AC201908002 13.335 6.9% 0 0.44 82% 163.0 191.4 1.175
SP19AC201909003 14.025 7.1% 0 0.40 82% 179.6 215.2 1.199
SP19FC201909001 11.095 4.5% 0 0.37 76% 247.2 272.4 1.102
19F SP19FC201909002 11.220 4.8% 2 0.37 81% 195.7 227.6 1.163
SP19FC201909003 11.580 5.7% 2 0.37 76% 223.8 264.8 1.182
SP23FC201908002 11.790 2.5% 1 0.37 79% 194.6 223.8 1.150

23F SP23FC201909003 12.975 2.5% 0 0.36 78% 223.15 254.5 1.1405
SP23FD 202001001 10.425 2.6% 2 0.41 90% 224.75 275.8 1.227

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Table 109 Batch Analyses Data Statistics for Bulk Tetanus Toxoid
Residual
Test items Flocculent Detoxification Protein
Appearance pH Purity Sterility test Abnormal toxicity Toxicity reverse test ammoniu
Batch No. unit test content
m
Observed on the 7th,
14th, and 21st days
Animals should
after injection. There The guinea pigs must
not have
The antigenic was no local not have symptoms
Should be symptoms of
purity is not suppuration or of tetanus. Each
yellow or tetanus. Each
less than Should comply necrosis. Animals guinea pig whose
brownish - animal should not ≤ 0.25 -
Specification 6.6-7.4 1500Lf per with the should not have weight was more
yellow (Lf/mL) lose more weight mg/mL ($\mu\text{g/mL}$)
milligram of requirements. symptoms of tetanus. than that before the
transparent than before
protein Those who expired injection by the end
liquid injection, and the
nitrogen. and the weight of of the period was
survivors are
each animal increased qualified.
eligible.
than before injection
were eligible.
Comply with
201511004 Comply with the Comply with the Comply with the Comply with the
the 7.30 1540 1672.1 0.033 5689.1
(Bottle No.1) requirements. requirements. requirements. requirements.
requirements.
Comply with
CTB201606005 Comply with the Comply with the Comply with the Comply with the
the 7.27 800 1720.4 ND 2983.1
(Bottle No.1) requirements. requirements. requirements. requirements.
requirements.
Comply with
CTB201610006 Comply with the Comply with the Comply with the Comply with the
the 6.88 800 1698.5 0.007 3057.3
(Bottle No.1) requirements. requirements. requirements. requirements.
requirements.

Table 110 Batch Analyses Data Statistics for Purified Liquid of Tetanus Toxoid
Test items
Protein content Sterility Bacterial endotoxin Purity
Batch No.
Specification ≥ 16 mg/mL Should comply with requirements Should be not more than 10 EU/mg protein Should
not be less than 90%
CTL201909018 (Bottle No.1) 26 Comply with the requirements. Comply with the requirements. 100.0%
CTL201909020 (Bottle No.1) 24 Comply with the requirements. Comply with the requirements. 99.9%
CTL201909019 (Bottle No.1) 26 Comply with the requirements. Comply with the requirements. 100.0%
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Table 111 Batch Analyses Data Statistics for Monovalent Bulk Conjugate
Free Free
Polysaccharide Protein Polysaccharide- Molecular EDAC Bacterial ADH
polysaccharide protein Identification Sterility
content content protein ratio size (KD) residue endotoxin content
Test item and content content
specification Appear
Serotypes Should
positive
 $\geq 70 \leq 5.0 \leq 0.75$ EU/ μg comply with < 0.15
Batch No. ≥ 40 $\mu\text{g/mL}$ 0.20-1.80 $\geq 80\% \leq 25.0\% \leq 5.0\%$ response to
 $\mu\text{g/mL}$ $\mu\text{mol/L}$ polysaccharide the $\mu\text{g/mL}$
specific
requirements.
antiserum
Comply with Comply with
RSP1C201706001 245.5 239.1 1.03 91.4 2.8 10.8 4.1 the < 0.75 EU/ μg the 0.05
requirements. requirements.
Comply with Comply with
1 RSP1C201707002 258.0 170.5 1.51 99.4 2.9 15.8 ND the < 0.75 EU/ μg the 0.06
requirements. requirements.
Comply with Comply with
RSP1C201709003 423.8 462.2 0.92 96.2 ND 14.0 1.3 the < 0.75 EU/ μg the 0.086
requirements. requirements.
Comply with Comply with
RSP3C201706001 61.6 141.6 0.44 98.5 1.1 15.7 3.8 the < 0.75 EU/ μg the 0.016
requirements. requirements.

Comply with Comply with
3 RSP3C201707002 77.2 355.8 0.22 97.8 4.4 6.2 0.48 the < 0.75 EU/μg the 0.160
requirements. requirements.
Comply with Comply with
RSP3C201708003 63.6 285.3 0.22 97.6 3.8 12.2 1.20 the < 0.75 EU/μg the 0.128
requirements. requirements.
Comply with Comply with
RSP4C201706001 154.3 254.1 0.61 98.9 0.9 19.6 4.7 the < 0.75 EU/μg the 0.082
requirements. requirements.
4
Comply with Comply with
RSP4C201707002 196.0 276.2 0.71 98.0 3.6 23.6 2.2 the < 0.75 EU/μg the 0.110
requirements. requirements.

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Comply with Comply with
RSP4C201709003 190.8 288.0 0.66 99.4 3.5 14.3 0.9 the < 0.75 EU/μg the 0.082
requirements. requirements.
Comply with Comply with
RSP5C201705001 91.1 431.0 0.21 94.3 0.7 18.0 3.2 the < 0.75 EU/μg the 0.139
requirements. requirements.
Comply with Comply with
5 RSP5C201707002 57.8 172.1 0.34 98.9 0.8 17.4 ND the < 0.75 EU/μg the 0.102
requirements. requirements.
Comply with Comply with
RSP5C201708003 99.8 224.6 0.44 96.8 3.7 15.4 0.5 the < 0.75 EU/μg the 0.102
requirements. requirements.
Comply with Comply with
RSP6AC201705001 139.1 547.4 0.25 95.6 3.6 13.7 3.6 the < 0.75 EU/μg the 0.163
requirements. requirements.
Comply with Comply with
6A RSP6AC201707002 154.6 493.0 0.31 88.1 2.6 19.4 0.9 the < 0.75 EU/μg the 0.081
requirements. requirements.
Comply with Comply with
RSP6AC201708003 86.8 191.5 0.45 97.7 2.3 12.8 0.5 the < 0.75 EU/μg the 0.227
requirements. requirements.
Comply with Comply with
RSP6BC201705001 152.9 327.4 0.47 98.6 2.8 21.6 4.0 the < 0.75 EU/μg the 0.012
requirements. requirements.
Comply with Comply with
6B RSP6BC201707002 170.2 349.9 0.49 93.5 0.5 21.1 0.9 the < 0.75 EU/μg the 0.094
requirements. requirements.
Comply with Comply with
RSP6BC201708003 124.8 296.5 0.42 96.1 0.1 9.2 2.1 the < 0.75 EU/μg the 0.142
requirements. requirements.
Comply with Comply with
7F RSP7FC201705001 284.1 643.5 0.44 97.2 3.2 18.4 4.0 the < 0.75 EU/μg the 0.066
requirements. requirements.

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Comply with Comply with
RSP7FC201706002 303.0 362.8 0.84 96.6 1.2 18.5 0.3 the < 0.75 EU/μg the 0.121
requirements. requirements.
Comply with Comply with
RSP7FC201708003 310.8 494.3 0.63 96.6 3.7 18.9 1.3 the < 0.75 EU/μg the 0.088
requirements. requirements.
Comply with Comply with
RSP9VC201705001 148.6 513.5 0.29 100.0 2.5 6.8 4.3 the < 0.75 EU/μg the 0.176
requirements. requirements.
Comply with Comply with
9V RSP9VC201707002 159.3 614.1 0.26 99.7 2.2 12.0 1.6 the < 0.75 EU/μg the 0.153
requirements. requirements.
Comply with Comply with
RSP9VC201709003 157.5 722.3 0.22 98.8 4.1 11.8 0.7 the < 0.75 EU/μg the 0.203
requirements. requirements.
Comply with Comply with
RSP14C201706001 143.2 480.7 0.30 95.9 2.4 14.6 3.1 the < 0.75 EU/μg the 0.016
requirements. requirements.
Comply with Comply with
14C RSP14C201707002 134.9 340.4 0.4 99.0 3.8 12.6 1.0 the < 0.75 EU/μg the 0.120
requirements. requirements.
Comply with Comply with
RSP14C201709003 125.0 305.8 0.41 97.2 1.0 23.9 1.0 the < 0.75 EU/μg the 0.114
requirements. requirements.
Comply with Comply with
RSP18CC201706001 106.6 195.8 0.54 97.7 0.9 7.7 2.7 the < 0.75 EU/μg the 0.097
requirements. requirements.
Comply with Comply with
18C RSP18CC201707002 153.3 263.1 0.58 86.9 1.5 26.2 0.9 the < 0.75 EU/μg the 0.108
requirements. requirements.
Comply with Comply with
RSP18CC201708003 144.4 199.8 0.72 96.9 0.7 22.6 0.7 the < 0.75 EU/μg the 0.015
requirements. requirements.

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Comply with Comply with
RSP19AC201705001 131.7 409.8 0.32 95.4 1.2 27.4 2.9 the < 0.75 EU/μg the 0.108
requirements. requirements.
Comply with Comply with
19A RSP19AC201707002 163.6 435.2 0.38 93.3 1.3 26.0 2.8 the < 0.75 EU/μg the 0.130
requirements. requirements.
Comply with Comply with
RSP19AC201708003 202.0 444.9 0.45 99.4 2.3 28.0 0.7 the < 0.75 EU/μg the 0.103
requirements. requirements.
Comply with Comply with
RSP19FC201706001-
195.7 500.3 0.39 88.8 3.9 24.4 3.2 the < 0.75 EU/μg the 0.073
1
requirements. requirements.
Comply with Comply with
19F RSP19FC201707002 145.6 237.8 0.61 98.1 1.4 30.1 2.4 the < 0.75 EU/μg the 0.126
requirements. requirements.
Comply with Comply with
RSP19FC201708003 134.0 300.3 0.45 96.4 3.1 25.2 1.1 the < 0.75 EU/μg the 0.066
requirements. requirements.
Comply with Comply with
RSP23FC201705001 85.4 232.8 0.37 98.7 0.7 13.0 1.6 the < 0.75 EU/μg the 0.162
requirements. requirements.
Comply with Comply with
23F RSP23FC201707002 91.3 215.0 0.42 100.0 2.5 24.9 2.1 the < 0.75 EU/μg the 0.174
requirements. requirements.
Comply with Comply with
RSP23FC201708003 75.8 212.3 0.36 97.5 2.1 18.7 0.6 the < 0.75 EU/μg the 0.160
requirements. requirements.

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MODULE 3 QUALITY

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3.2.S.4.5 Justification of Specification for Drug Substance

3.2.S.4.5.1 Justification of Specification for Pneumococcal Polysaccharide

Table 112 Justification of Specification for Serotype 1 Pneumococcal Polysaccharide

Test items Test method Specifications Ph. Eur. WHO Rationale for specifications

Double immunodiffusion test is an

Double Appear positive By an immunochemical A test should be performed on

immunochemical method which

Identity test immunodiffusion response to specific method or other suitable the purified polysaccharide to

confirms the presence of serotype 1

test antiserum type 1 methods verify its identity.

polysaccharide.

Not more than the limit

Each lot of purified

approved for the

polysaccharide should typically The specification is required by Ph.

Protein content Lowry method ≤ 2% product, calculated with

contain no more Eur. (2150).

reference to the dried

than 3% by weight of protein.

substance

Sufficient polysaccharide shall

UV-vis be assayed to detect 2% nucleic The specification is required by Ph.

Nucleic acid content ≤ 2% ≤ 2%

spectrometry acid Eur. (2150).

contamination accurately.

The specification is required by Ph.

Total nitrogen content Kjeldahl method 3.5-6.0% 3.5-6.0% 3.5-6.0%

Eur. (2150).

Chromogenic The specification is required by Ph.

Phosphorus content 0-1.5% 0-1.5% 0-1.5%

method Eur. (2150).

Chromogenic The specification is required by Ph.

Uronic acid content ≥ 45% ≥ 45% ≥ 45%

method Eur. (2150).

Chromogenic The specification is required by Ph.

O-acetyl group ≥ 1.8 ≥ 1.8 ≥ 1.8

method Eur. (2150).

An acceptable level of

Recovery rate is calculated as per

KD ≤ 0.15, recovery Appropriate detection consistency should be agreed

Polysaccharide Molecular Size

Gel rate at KD ≤ 0.30 system, an acceptable with the NRA and can be

Molecular size (KD) Distribution (WSYX-QC-TS-0006-

chromatography should be no less than value is established for established either by process

0597), which is approved by

75% each Ps. validation or by measurement

NMPA.

on each lot

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Bacteria endotoxin is tested to limit

pyrogenic effect of the product. It is
 Bacterial endotoxin Gel-clot method ≤ 0.5 EU/ μ g // an additional quality control
 according to in-house standard
 (WSYX-QC-TS-0609-0004).
 Gas Ethanol, acetone and ether are used
 Ethanol residual $< 0.08\%$ //
 chromatography to control quality of bulk
 Gas polysaccharide. These test items are
 Acetone residual $< 0.03\%$ //
 chromatography additional quality control according
 Gas to in-house standard (WSYX-QC-
 Ether residual $< 0.01\%$ //
 chromatography TS-0609-0004).
 The limit is set according to Quality
 Standard for Pneumococcal
 UV-vis
 Sodium deoxycholate < 0.20 // Polysaccharide Conjugate Vaccine
 spectrometry
 (WSYX-QC-TS-0006-0721), which
 is approved by NMPA.
 The pyrogen content of the
 purified polysaccharide should
 Pyrogen test required by WHO can
 Pyrogen // be determined and shown to be
 be represented by endotoxin test
 within acceptable limits agreed
 by the NRA.

Table 113 Justification of Specification for Serotype 3 Pneumococcal Polysaccharide
 Test items Test method Specifications Ph. Eur. WHO Rationale for specifications
 Double immunodiffusion test is a
 Double Appear positive By an immunochemical A test should be performed on
 immunochemical method which
 Identity test immunodiffusion response to specific method or other suitable the purified polysaccharide
 to
 confirms the presence of serotype 3
 test antiserum type 3 methods verify its identity.
 polysaccharide.
 Not more than the limit
 Each lot of purified
 approved for the
 polysaccharide should typically The specification is required by Ph.
 Protein content Lowry method $\leq 5\%$ product, calculated with
 contain not more Eur. (2150).
 reference to the dried
 than 3% by weight of protein.
 substance

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Sufficient polysaccharide shall
 UV-vis The specification is required by Ph.
 Nucleic acid content $\leq 2\%$ $\leq 2\%$ be assayed to detect 2% nucleic
 spectrometry Eur. (2150).
 acid contamination accurately.
 The specification is required by Ph.
 Total nitrogen content Kjeldahl method 0-1.0% 0-1.0% 0-1%
 Eur. (2150).
 Chromogenic The specification is required by Ph.
 Phosphorus content 0-1.0% 0-1.0% 0-1%
 method Eur. (2150).
 Chromogenic The specification is required by Ph.
 Uronic acid content $\geq 40\%$ $\geq 40\%$ $\geq 40\%$
 method Eur. (2150).
 An acceptable level of
 Recovery rate is calculated as per
 KD ≤ 0.15 , recovery Appropriate detection consistency should be agreed
 Polysaccharide Molecular Size
 Gel rate at KD ≤ 0.30 system, an acceptable with the NRA and can be
 Molecular size (KD) Distribution (WSYX-QC-TS-0006-
 chromatography should be no less than value is established for established either by process
 0597), which is approved by
 78% each Ps. validation or by measurement
 NMPA.
 on each lot.
 Bacteria endotoxin is tested to limit
 pyrogenic effect of the product. It is
 Bacterial endotoxin Gel-clot method ≤ 0.5 EU/ μ g // an additional quality control
 according to in-house standard
 (WSYX-QC-TS-0609-0004).
 Gas Ethanol, acetone and ether are used
 Ethanol residual $< 0.08\%$ //
 chromatography to control quality of bulk
 Gas polysaccharide. These test items are
 Acetone residual $< 0.03\%$ //
 chromatography additional quality control according
 Gas to in-house standard (WSYX-QC-
 Ether residual $< 0.01\%$ //
 chromatography TS-0609-0004).
 The limit is set according to Quality

Standard for Pneumococcal
UV-vis
Sodium deoxycholate < 0.2 // Polysaccharide Conjugate Vaccine
spectrometry
(WSYX-QC-TS-0006-0721), which
is approved by NMPA.
The pyrogen content of the Pyrogen test result required by
Pyrogen // purified polysaccharide should WHO can be represented by
be determined and shown to be endotoxin test.

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within acceptable limits agreed
by the NRA.

Table 114 Justification of Specification for Serotype 4 Pneumococcal Polysaccharide
Test items Test method Specifications Ph. Eur. WHO Rationale for specifications
Double immunodiffusion test is an
Double Appear positive By an immunochemical A test should be performed on
immunochemical method which
Identity test immunodiffusion response to specific method or other suitable the purified polysaccharide
to
confirms the presence of serotype 4
test antiserum type 4 methods verify its identity.
polysaccharide.
Each lot of purified
polysaccharide should typically The specification is required by Ph.
Protein content Lowry method $\leq 3\% \leq 3\%$
contain no more than 3% by Eur. (2150).
weight of protein.
Sufficient polysaccharide shall
UV-vis The specification is required by Ph.
Nucleic acid content $\leq 2\% \leq 2\%$ be assayed to detect 2% nucleic
spectrometry Eur. (2150).
acid contamination accurately.
The specification is required by Ph.
Total nitrogen content Kjeldahl method 4.0-6.0% 4.0-6.0% 4-6%
Eur. (2150).
Chromogenic The specification is required by Ph.
Phosphorus content 0-1.5% 0-1.5% 0-1.5%
method Eur. (2150).
Chromogenic The specification is required by Ph.
Hexosamine content $\geq 40\% \geq 40\% \geq 40\%$
method Eur. (2150).
Chromogenic The specification is required by Ph.
Methyl pentose // $\geq 10\%$
method Eur. (2150).
An acceptable level of
Recovery rate is calculated as per
KD ≤ 0.15 , recovery Appropriate detection consistency should be agreed
Polysaccharide Molecular Size
Gel rate at KD ≤ 0.30 system, an acceptable with the NRA and can be
Molecular size (KD) Distribution (WSYX-QC-TS-0006-
chromatography should be no less than value is established for established either by process
0597), which is approved by
75% each Ps. validation or by measurement
NMPA.
on each lot
Bacteria endotoxin is tested to limit
Bacterial endotoxin Gel-clot method $\leq 0.5 \text{ EU}/\mu\text{g}$ // pyrogenic effect of the product. It is
an additional quality control

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according to in-house standard
(WSYX-QC-TS-0609-0004).
Gas Ethanol, acetone and ether are used
Ethanol residual < 0.08% //
chromatography to control quality of bulk
Gas polysaccharide. These test items are
Acetone residual < 0.03% //
chromatography additional quality control according
Gas to in-house standard (WSYX-QC-
Ether residual < 0.01% //
chromatography TS-0609-0004).
The limit is set according to Quality
Standard for Pneumococcal
UV-vis
Sodium deoxycholate < 0.20 // Polysaccharide Conjugate Vaccine
spectrometry
(WSYX-QC-TS-0006-0721), which
is approved by NMPA.
The pyrogen content of the
purified polysaccharide should
Pyrogen test required by WHO can
Pyrogen // be determined and shown to be
be represented by endotoxin test
within acceptable limits agreed

by the NRA.

Table 115 Justification of Specification for Serotype 5 Pneumococcal Polysaccharide

Test items Test method Specifications Ph. Eur. WHO Rationale for specifications

Double immunodiffusion test is an

Double Appear positive By an immunochemical A test should be performed on immunochemical method which

Identity test immunodiffusion response to specific method or other suitable the purified polysaccharide to

confirms the presence of serotype 5

test antiserum serotype 5 methods verify its identity.

polysaccharide.

Each lot of purified

polysaccharide should typically The specification is required by Ph.

Protein content Lowry method $\leq 7.5\% \leq 7.5\%$

contain no more than 3% by Eur. (2150).

weight of protein.

Sufficient polysaccharide shall

UV-vis The specification is required by Ph.

Nucleic acid content $\leq 2\% \leq 2\%$ be assayed to detect 2% nucleic spectrometry Eur. (2150).

acid contamination accurately.

The specification is required by Ph.

Total nitrogen content Kjeldahl method 2.5-6.0% 2.5-6.0% 2.5-6% Eur. (2150).

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Chromogenic The specification is required by Ph.

Phosphorus content $\leq 2.0\% \leq 2.0\% < 2\%$ method Eur. (2150).

Chromogenic The specification is required by Ph.

Uronic acid content $\geq 12\% \geq 12\% \geq 12\%$ method Eur. (2150).

Chromogenic The specification is required by Ph.

Hexosamine content $\geq 20\% \geq 20\% \geq 20\%$ method Eur. (2150).

An acceptable level of

Recovery rate is calculated as per

$KD \leq 0.60$, recovery Appropriate detection consistency should be agreed

Polysaccharide Molecular Size

Gel rate at $KD \leq 0.60$ system, an acceptable with the NRA and can be

Molecular size (KD) Distribution (WSYX-QC-TS-0006-

chromatography should be no less than value is established for established either by process 0597), which is approved by

70% each Ps. validation or by measurement

NMPA.

on each lot

Bacteria endotoxin is tested to limit

pyrogenic effect of the product. It is

Bacterial endotoxin Gel-clot method $\leq 0.5\text{EU}/\mu\text{g}$ // an additional quality control

according to in-house standard

(WSYX-QC-TS-0609-0004).

Gas Ethanol, acetone and ether are used

Ethanol residual $< 0.08\%$ //

chromatography to control quality of bulk

Gas polysaccharide. These test items are

Acetone residual $< 0.03\%$ //

chromatography additional quality control according

Gas to in-house standard (WSYX-QC-

Ether residual $< 0.01\%$ //

chromatography TS-0609-0004).

The limit is set according to Quality

Standard for Pneumococcal

UV-vis

Sodium deoxycholate < 0.20 // Polysaccharide Conjugate Vaccine

spectrometry

(WSYX-QC-TS-0006-0721), which

is approved by NMPA.

The pyrogen content of the

purified polysaccharide should

Pyrogen test required by WHO can

Pyrogen /// be determined and shown to be

be represented by endotoxin test

within acceptable limits agreed

by the NRA.

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Table 116 Justification of Specification for Serotype 6A Pneumococcal Polysaccharide

Test items Test method Specifications Ph. Eur. WHO Rationale for specifications

Double immunodiffusion test is an

Double Appear positive By an immunochemical A test should be performed on

immunochemical method which

Identity test immunodiffusion response to specific method or other suitable the purified polysaccharide to

confirms the presence of serotype

test antiserum type 6A methods verify its identity.

6A polysaccharide.

Each lot of purified polysaccharide should typically The specification is required by Ph. Protein content Lowry method $\leq 2\% \leq 7.5\%$ contain no more than 3% by Eur. (2150). weight of protein. Sufficient polysaccharide shall UV-vis The specification is required by Ph. Nucleic acid content $\leq 1\% \leq 2\%$ be assayed to detect 2% nucleic acid contamination accurately. The specification is required by Ph. Total nitrogen content Kjeldahl method 0-2.0% 2.5-6.0% / Eur. (2150). Chromogenic The specification is required by Ph. Phosphorus content 2.4-5.0% $\leq 2.0\%$ / method Eur. (2150). Chromogenic The specification is required by Ph. Methyl pentose content $\geq 15\% \geq 12\%$ / method Eur. (2150). An acceptable level of Recovery rate is calculated as per $KD \leq 0.50$, recovery Appropriate detection consistency should be agreed Polysaccharide Molecular Size Gel rate at $KD \leq 0.60$ system, an acceptable with the NRA and can be Molecular size (KD) Distribution (WSYX-QC-TS-0006-chromatography should be no less than value is established for established either by process 0597), which is approved by 50% each Ps. validation or by measurement NMPA. on each lot. Bacteria endotoxin is tested to limit pyrogenic effect of the product. It is Bacterial endotoxin Gel-clot method $\leq 0.5EU/\mu g$ / / an additional quality control according to in-house standard (WSYX-QC-TS-0609-0004). Gas Ethanol, acetone and ether are used Ethanol residual $< 0.08\%$ / / chromatography to control quality of bulk Gas polysaccharide. These test items are Acetone residual $< 0.03\%$ / / chromatography additional quality control according

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Gas to in-house standard (WSYX-QC-Ether residual $< 0.01\%$ / / chromatography TS-0609-0004). The limit is set according to Quality Standard for Pneumococcal UV-vis Sodium deoxycholate < 0.20 / / Polysaccharide Conjugate Vaccine spectrometry (WSYX-QC-TS-0006-0721), which is approved by NMPA. The pyrogen content of the purified polysaccharide should Pyrogen test required by WHO can Pyrogen / / be determined and shown to be be represented by endotoxin test. within acceptable limits agreed by the NRA.

Table 117 Justification of Specification for Serotype 6B Pneumococcal Polysaccharide
Test items Test method Specifications Ph. Eur. WHO Rationale for specifications
Double immunodiffusion test is an Double Appear positive By an immunochemical A test should be performed on immunochemical method which Identity test immunodiffusion response to specific method or other suitable the purified polysaccharide to confirms the presence of serotype test antiserum type 6B methods verify its identity. 6B polysaccharide. Each lot of purified polysaccharide should typically The specification is required by Ph. Protein content Lowry method $\leq 2\% \leq 7.5\%$ contain no more than 3% by Eur. (2150). weight of protein. Sufficient polysaccharide shall UV-vis The specification is required by Ph. Nucleic acid content $\leq 2\% \leq 2\%$ be assayed to detect 2% nucleic acid contamination accurately. The specification is required by Ph. Total nitrogen content Kjeldahl method 0-2.0% 2.5-6.0% 0-2.0% Eur. (2150). Chromogenic The specification is required by Ph. Phosphorus content 2.5-5.0% $\leq 2.0\%$ 2.5-5.0% method Eur. (2150). Chromogenic The specification is required by Ph. Methyl pentose content $\geq 15\% \geq 12\% \geq 15\%$

method Eur. (2150).
Gel KD ≤ 0.50 , recovery Appropriate detection An acceptable level of Recovery rate is calculated as per
Molecular size (KD)
chromatography rate at KD ≤ 0.60 system, an acceptable consistency should be agreed Polysaccharide
Molecular Size
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should be no less than value is established for with the NRA and can be Distribution (WSYX-QC-TS-0006-
85% each Ps. established either by process 0597), which is approved by
validation or by measurement NMPA.
on each lot.
Bacteria endotoxin is tested to limit
pyrogenic effect of the product. It is
Bacterial endotoxin Gel-clot method $\leq 0.5\text{EU}/\mu\text{g}$ // an additional quality control
according to in-house standard
(WSYX-QC-TS-0609-0004).
Gas Ethanol, acetone and ether are used
Ethanol residual $< 0.08\%$ //
chromatography to control quality of bulk
Gas polysaccharide. These test items are
Acetone residual $< 0.03\%$ //
chromatography additional quality control according
Gas to in-house standard (WSYX-QC-
Ether residual $< 0.01\%$ //
chromatography TS-0609-0004).
The limit is set according to Quality
Standard for Pneumococcal
UV-vis
Sodium deoxycholate < 0.20 // Polysaccharide Conjugate Vaccine
spectrometry
(WSYX-QC-TS-0006-0721), which
is approved by NMPA.
The pyrogen content of the
purified polysaccharide should
Pyrogen test required by WHO can
Pyrogen /// be determined and shown to be
be represented by endotoxin test
within acceptable limits agreed
by the NRA.

Table 118 Justification of Specification for Serotype 7F Pneumococcal Polysaccharide
Test items Test method Specifications Ph. Eur. WHO Rationale for specifications
Double immunodiffusion test is an
Double Appear positive By an immunochemical A test should be performed on
immunochemical method which
Identity test immunodiffusion response to specific method or other suitable the purified polysaccharide
to
confirms the presence of serotype
test antiserum type 7F methods verify its identity.
7F polysaccharide.
Each lot of purified The specification is required by Ph.
Protein content Lowry method $\leq 5\% \leq 5\%$
polysaccharide should typically Eur. (2150).
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contain no more than 3% by
weight of protein.
Sufficient polysaccharide shall
UV-vis The specification is required by Ph.
Nucleic acid content $\leq 2\% \leq 2\%$ be assayed to detect 2% nucleic
spectrometry Eur. (2150).
acid contamination accurately.
The specification is required by Ph.
Total nitrogen content Kjeldahl method 1.5-4.0% 1.5-4.0% 1.5-4.0%
Eur. (2150).
Chromogenic The specification is required by Ph.
Phosphorus content 0-1.0% 0-1.0% 0-1.0%
method Eur. (2150).
Chromogenic The specification is required by Ph.
Methyl pentose content $\geq 13\% \geq 13\% \geq 13\%$
method Eur. (2150).
Chromogenic The specification is required by Ph.
Hexosamine content // 33.09%
method Eur. (2150).
Chromogenic The specification is required by Ph.
O-acetyl group content // 3.5%
method Eur. (2150).
An acceptable level of
Recovery rate is calculated as per
KD ≤ 0.20 , recovery Appropriate detection consistency should be agreed
Polysaccharide Molecular Size
Gel rate at KD ≤ 0.30 system, an acceptable with the NRA and can be
Molecular size (KD) Distribution (WSYX-QC-TS-0006-
chromatography should be no less than value is established for established either by process
0597), which is approved by
70% each Ps. validation or by measurement
NMPA.
on each lot.

Bacteria endotoxin is tested to limit pyrogenic effect of the product. It is Bacterial endotoxin Gel-clot method $\leq 0.5 \text{ EU}/\mu\text{g}$ // an additional quality control according to in-house standard (WSYX-QC-TS-0609-0004). Gas Ethanol, acetone and ether are used Ethanol residual $< 0.08\%$ // chromatography to control quality of bulk Gas polysaccharide. These test items are Acetone residual $< 0.03\%$ // chromatography additional quality control according Gas to in-house standard (WSYX-QC-Ether residual $< 0.01\%$ // chromatography TS-0609-0004). The limit is set according to Quality UV-vis Sodium deoxycholate < 0.20 // Standard for Pneumococcal spectrometry Polysaccharide Conjugate Vaccine

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(WSYX-QC-TS-0006-0721), which is approved by NMPA. The pyrogen content of the purified polysaccharide should Pyrogen test required by WHO can Pyrogen /// be determined and shown to be be represented by endotoxin test. within acceptable limits agreed by the NRA.

Table 119 Justification of Specification for Serotype 9V Pneumococcal Polysaccharide
Test items Test method Specifications Ph. Eur. WHO Rationale for specifications
Double immunodiffusion test is an Double Appear positive By an immunochemical A test should be performed on immunochemical method which Identity test immunodiffusion response to specific method or other suitable the purified polysaccharide to confirms the presence of serotype test antiserum type 9V methods verify its identity. 9V polysaccharide. Each lot of purified polysaccharide should typically The specification is required by Ph. Protein content Lowry method $\leq 2\% \leq 2\%$ contain not more than 3% by Eur. (2150). weight of protein. Sufficient polysaccharide shall UV-vis The specification is required by Ph. Nucleic acid content $\leq 2\% \leq 2\%$ be assayed to detect 2% nucleic spectrometry Eur. (2150). acid contamination accurately. The specification is required by Ph. Total nitrogen content Kjeldahl method 0.5-3.0% 0.5-3.0% 0.5-3.0% Eur. (2150). Chromogenic The specification is required by Ph. Phosphorus content 0-1.0% 0-1.0% 0-1.0% method Eur. (2150). Chromogenic The specification is required by Ph. Uronic acid content $\geq 15\% \geq 15\% \geq 15\%$ method Eur. (2150). Chromogenic The specification is required by Ph. Hexosamine content $\geq 13\% \geq 13\% \geq 13\%$ method Eur. (2150). Recovery rate is calculated as per $\text{KD} \leq 0.45$, recovery Appropriate detection An acceptable level of Polysaccharide Molecular Size Gel rate at $\text{KD} \leq 0.60$ system, an acceptable consistency should be agreed Molecular size (KD) Distribution (WSYX-QC-TS-0006- chromatography should be no less than value is established for with the NRA and can be 0597), which is approved by 80% each Ps. established either by process NMPA.

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validation or by measurement on each lot. Bacteria endotoxin is tested to limit pyrogenic effect of the product. It is Bacterial endotoxin Gel-clot method $\leq 0.5 \text{ EU}/\mu\text{g}$ // an additional quality control according to in-house standard (WSYX-QC-TS-0609-0004). Gas Ethanol, acetone and ether are used Ethanol residual $< 0.08\%$ // chromatography to control quality of bulk Gas polysaccharide. These test items are Acetone residual $< 0.03\%$ // chromatography additional quality control according

Gas to in-house standard (WSYX-QC-Ether residual < 0.01% // chromatography TS-0609-0004). The limit is set according to Quality Standard for Pneumococcal UV-vis Sodium deoxycholate < 0.20 // Polysaccharide Conjugate Vaccine spectrometry (WSYX-QC-TS-0006-0721), which is approved by NMPA. The pyrogen content of the purified polysaccharide should Pyrogen test required by WHO can Pyrogen /// be determined and shown to be be represented by endotoxin test within acceptable limits agreed by the NRA.

Table 120 Justification of Specification for Serotype 14 Pneumococcal Polysaccharide
 Test items Test method Specifications Ph. Eur. WHO Rationale for specifications
 Double immunodiffusion test is an Double Appear positive By an immunochemical A test should be performed on immunochemical method which Identity test immunodiffusion response to specific method or other suitable the purified polysaccharide to confirms the presence of serotype test antiserum type 14 methods verify its identity. 14 polysaccharide. Each lot of purified polysaccharide should typically The specification is required by Ph. Protein content Lowry method $\leq 5\% \leq 5\%$ contain no more than 3% by Eur. (2150). weight of protein. Confidential 663
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Sufficient polysaccharide shall UV-vis The specification is required by Ph. Nucleic acid content $\leq 2\% \leq 2\%$ be assayed to detect 2% nucleic spectrometry Eur. (2150). acid contamination accurately. The specification is required by Ph. Total nitrogen content Kjeldahl method 1.5-4.0% 1.5-4.0% 1.5-4.0% Eur. (2150). Chromogenic The specification is required by Ph. Phosphorus content 0-1.0% 0-1.0% 0-1.0% method Eur. (2150). Chromogenic The specification is required by Ph. Hexosamine content $\geq 20\% \geq 20\% \geq 20\%$ method Eur. (2150). An acceptable level of Recovery rate is calculated as per $KD \leq 0.30$, recovery Appropriate detection consistency should be agreed Polysaccharide Molecular Size Gel rate at $KD \leq 0.30$ system, an acceptable with the NRA and can be Molecular size (KD) Distribution (WSYX-QC-TS-0006- chromatography should be no less than value is established for established either by process 0597), which is approved by 55% each Ps. validation or by measurement NMPA. on each lot. Bacteria endotoxin is tested to limit pyrogenic effect of the product. It is Bacterial endotoxin Gel-clot method $\leq 0.5 \text{ EU}/\mu\text{g}$ // an additional quality control according to in-house standard (WSYX-QC-TS-0609-0004). Gas Ethanol, acetone and ether are used Ethanol residual < 0.08% // chromatography to control quality of bulk Gas polysaccharide. These test items are Acetone residual < 0.03% // chromatography additional quality control according Gas to in-house standard (WSYX-QC-Ether residual < 0.01% // chromatography TS-0609-0004). The limit is set according to Quality Standard for Pneumococcal UV-vis Sodium deoxycholate < 0.20 // Polysaccharide Conjugate Vaccine spectrometry (WSYX-QC-TS-0006-0721), which is approved by NMPA. The pyrogen content of the Pyrogen test required by WHO can Pyrogen /// purified polysaccharide should be represented by endotoxin test be determined and shown to be

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within acceptable limits agreed
by the NRA.

Table 121 Justification of Specification for Serotype 18C Pneumococcal Polysaccharide

Test items Test method Specifications Ph. Eur. WHO Rationale for specifications

Double immunodiffusion test is an

Double Appear positive By an immunochemical A test should be performed on
immunochemical method which

Identity test immunodiffusion response to specific method or other suitable the purified polysaccharide
to

confirms the presence of serotype

test antiserum type 18C methods verify its identity.

18C polysaccharide.

Each lot of purified

polysaccharide should typically The specification is required by Ph.

Protein content Lowry method $\leq 3\% \leq 3\%$

contain not more than 3% by Eur. (2150).

weight of protein.

Sufficient polysaccharide shall

UV-vis The specification is required by Ph.

Nucleic acid content $\leq 2\% \leq 2\%$ be assayed to detect 2% nucleic
spectrometry Eur. (2150).

acid contamination accurately.

The specification is required by Ph.

Total nitrogen content Kjeldahl method 0-1.0% 0-1.0% 0-1.0%

Eur. (2150).

Chromogenic The specification is required by Ph.

Phosphorus content 2.4-4.9% 2.4-4.9% 2.4-4.9%

method Eur. (2150).

Chromogenic The specification is required by Ph.

Methyl pentose content $\geq 14\% \geq 14\% \geq 14\%$

method Eur. (2150).

Chromogenic Complies the requirement of Ph.

O-acetyl group content / / 4.24%

method Eur. (2150).

An acceptable level of

Recovery rate is calculated as per

KD ≤ 0.15 , recovery Appropriate detection consistency should be agreed

Polysaccharide Molecular Size

Gel rate at KD ≤ 0.30 system, an acceptable with the NRA and can be

Molecular size (KD) Distribution (WSYX-QC-TS-0006-

chromatography should be no less than value is established for established either by process
0597), which is approved by

70% each Ps. validation or by measurement

NMPA.

on each lot.

Bacteria endotoxin is tested to limit

Bacterial endotoxin Gel-clot method $\leq 0.5 \text{ EU}/\mu\text{g}$ / / pyrogenic effect of the product. It is
an additional quality control

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according to in-house standard

(WSYX-QC-TS-0609-0004).

Gas Ethanol, acetone and ether are used

Ethanol residual $< 0.08\%$ / /

chromatography to control quality of bulk

Gas polysaccharide. These test items are

Acetone residual $< 0.03\%$ / /

chromatography additional quality control according

Gas to in-house standard (WSYX-QC-

Ether residual $< 0.01\%$ / /

chromatography TS-0609-0004).

The limit is set according to Quality

Standard for Pneumococcal

UV-vis

Sodium deoxycholate < 0.20 / / Polysaccharide Conjugate Vaccine

spectrometry

(WSYX-QC-TS-0006-0721), which

is approved by NMPA.

The pyrogen content of the

purified polysaccharide should

Pyrogen test required by WHO can

Pyrogen / / be determined and shown to be

be represented by endotoxin test

within acceptable limits agreed

by the NRA.

Table 122 Justification of Specification for Serotype 19A Pneumococcal Polysaccharide

Test items Test method Specifications Ph. Eur. WHO Rationale for specifications

Double immunodiffusion test is an

Double Appear positive By an immunochemical A test should be performed on

immunochemical method which

Identity test immunodiffusion response to specific method or other suitable the purified polysaccharide
to

confirms the presence of serotype

test antiserum type 19A methods verify its identity.

19A polysaccharide.

Each lot of purified

polysaccharide should typically The specification is required by Ph.
 Protein content Lowry method $\leq 2\% \leq 2\%$
 contain no more than 3% by Eur. (2150).
 weight of protein.
 Sufficient polysaccharide shall
 UV-vis The specification is required by Ph.
 Nucleic acid content $\leq 2\% \leq 2\%$ be assayed to detect 2% nucleic
 spectrometry Eur. (2150).
 acid contamination accurately.
 The specification is required by Ph.
 Total nitrogen content Kjeldahl method 0.6-3.5% 0.6-3.5% 0.6-3.5%
 Eur. (2150).
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Chromogenic The specification is required by Ph.
 Phosphorus content 3.0-7.0% 3.0-7.0% 3.0-7.0%
 method Eur. (2150).
 Chromogenic The specification is required by Ph.
 Hexosamine content $\geq 12\% \geq 12\% \geq 12\%$
 method Eur. (2150).
 Chromogenic The specification is required by Ph.
 Methyl pentose content $\geq 20\% \geq 20\% \geq 20\%$
 method Eur. (2150).
 An acceptable level of
 Recovery rate is calculated as per
 $KD \leq 0.15$, recovery Appropriate detection consistency should be agreed
 Polysaccharide Molecular Size
 Gel rate at $KD \leq 0.30$ system, an acceptable with the NRA and can be
 Molecular size (KD) Distribution (WSYX-QC-TS-0006-
 chromatography should be no less than value is established for established either by process
 0597), which is approved by
 75% each Ps. validation or by measurement
 NMPA.
 on each lot.
 Bacteria endotoxin is tested to limit
 pyrogenic effect of the product. It is
 Bacterial endotoxin Gel-clot method $\leq 0.5 \text{ EU}/\mu\text{g}$ // an additional quality control
 according to in-house standard
 (WSYX-QC-TS-0609-0004).
 Gas Ethanol, acetone and ether are used
 Ethanol residual $< 0.08\%$ //
 chromatography to control quality of bulk
 Gas polysaccharide. These test items are
 Acetone residual $< 0.03\%$ //
 chromatography additional quality control according
 Gas to in-house standard (WSYX-QC-
 Ether residual $< 0.01\%$ //
 chromatography TS-0609-0004).
 The limit is set according to Quality
 Standard for Pneumococcal
 UV-vis
 Sodium deoxycholate < 0.20 // Polysaccharide Conjugate Vaccine
 spectrometry
 (WSYX-QC-TS-0006-0721), which
 is approved by NMPA.
 The pyrogen content of the
 purified polysaccharide should
 Pyrogen test required by WHO can
 Pyrogen /// be determined and shown to be
 be represented by endotoxin test
 within acceptable limits agreed
 by the NRA.

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Table 123 Justification of Specification for Serotype 19F Pneumococcal Polysaccharide
 Test items Test method Specifications Ph. Eur. WHO Rationale for specifications
 Double immunodiffusion test is an
 Double Appear positive By an immunochemical A test should be performed on
 immunochemical method which
 Identity test immunodiffusion response to specific method or other suitable the purified polysaccharide
 to
 confirms the presence of serotype
 test antiserum type 19F methods verify its identity.
 19F polysaccharide.
 Each lot of purified
 polysaccharide should typically The specification is required by Ph.
 Protein content Lowry method $\leq 3\% \leq 3\%$
 contain not more than 3% by Eur. (2150).
 weight of protein.
 Sufficient polysaccharide shall
 UV-vis The specification is required by Ph.
 Nucleic acid content $\leq 2\% \leq 2\%$ be assayed to detect 2% nucleic
 spectrometry Eur. (2150).
 acid contamination accurately.
 The specification is required by Ph.
 Total nitrogen content Kjeldahl method 1.4-3.5% 1.4-3.5% 1.4-3.5%
 Eur. (2150).

Chromogenic The specification is required by Ph.
 Phosphorus content 3.0-5.5% 3.0-5.5% 3.0-5.5%
 method Eur. (2150).
 Chromogenic The specification is required by Ph.
 Hexosamine content $\geq 12.5\%$ $\geq 12.5\%$ $\geq 12.5\%$
 method Eur. (2150).
 Chromogenic The specification is required by Ph.
 Methyl pentose content $\geq 20\%$ $\geq 20\%$ $\geq 20\%$
 method Eur. (2150).
 An acceptable level of
 Recovery rate is calculated as per
 $KD \leq 0.15$, recovery Appropriate detection consistency should be agreed
 Polysaccharide Molecular Size
 Gel rate at $KD \leq 0.30$ system, an acceptable with the NRA and can be
 Molecular size (KD) Distribution (WSYX-QC-TS-0006-
 chromatography should be no less than value is established for established either by process
 0597), which is approved by
 75% each Ps. validation or by measurement
 NMPA.
 on each lot
 Bacteria endotoxin is tested to limit
 pyrogenic effect of the product. It is
 Bacterial endotoxin Gel-clot method $\leq 0.5 \text{ EU}/\mu\text{g}$ // an additional quality control
 according to in-house standard
 (WSYX-QC-TS-0609-0004).
 Gas Ethanol, acetone and ether are used
 Ethanol residual $< 0.08\%$ //
 chromatography to control quality of bulk

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Gas polysaccharide. These Test itemss
 Acetone residual $< 0.03\%$ //
 chromatography are additional quality control
 Gas according to in-house standard
 Ether residual $< 0.01\%$ //
 chromatography (WSYX-QC-TS-0609-0004).
 The limit is set according to Quality
 Standard for Pneumococcal
 UV-vis
 Sodium deoxycholate < 0.20 // Polysaccharide Conjugate Vaccine
 spectrometry
 (WSYX-QC-TS-0006-0721), which
 is approved by NMPA.
 The pyrogen content of the
 purified polysaccharide should
 Pyrogen test required by WHO can
 Pyrogen // be determined and shown to be
 be represented by endotoxin test
 within acceptable limits agreed
 by the NRA.

Table 124 Justification of Specification for Serotype 23F Pneumococcal Polysaccharide
 Test items Test method Specifications Ph. Eur. WHO Rationale for specifications
 Double immunodiffusion test is an
 Double Appear positive By an immunochemical A test should be performed on
 immunochemical method which
 Identity test immunodiffusion response to specific method or other suitable the purified polysaccharide
 to

confirms the presence of serotype 1
 test antiserum type 23F methods verify its identity.
 polysaccharide.
 Each lot of purified
 polysaccharide should typically The specification is required by Ph.
 Protein content Lowry method $\leq 2\%$ $\leq 2\%$
 contain not more than 3% by Eur. (2150).
 weight of protein.
 Sufficient polysaccharide shall
 UV-vis The specification is required by Ph.
 Nucleic acid content $\leq 2\%$ $\leq 2\%$ be assayed to detect 2% nucleic
 spectrometry Eur. (2150).
 acid contamination accurately.
 The specification is required by Ph.
 Total nitrogen content Kjeldahl method 0-1.0% 0-1.0% 0-1%
 Eur. (2150).
 Chromogenic The specification is required by Ph.
 Phosphorus content 3.0-4.5% 3.0-4.5% 3.0-4.5%
 method Eur. (2150).
 Chromogenic The specification is required by Ph.
 Methyl pentose content $\geq 37\%$ $\geq 37\%$ $\geq 37\%$
 method Eur. (2150).
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Chromogenic The specification is required by Ph.
 O-acetyl group content // 4.24%
 method Eur. (2150).
 An acceptable level of
 Recovery rate is calculated as per

KD ≤ 0.15, recovery Appropriate detection consistency should be agreed

Polysaccharide Molecular Size

Gel rate at KD ≤ 0.30 system, an acceptable with the NRA and can be

Molecular size (KD) Distribution (WSYX-QC-TS-0006-chromatography should be no less than value is established for established either by process 0597), which is approved by

75% each Ps. validation or by measurement

NMPA.

on each lot

Bacteria endotoxin is tested to limit

pyrogenic effect of the product. It is

Bacterial endotoxin Gel-clot method ≤ 0.5 EU/μg // an additional quality control

according to in-house standard

(WSYX-QC-TS-0609-0004).

Gas Ethanol, acetone and ether are used

Ethanol residual < 0.08% //

chromatography to control quality of bulk

Gas polysaccharide. These test items are

Acetone residual < 0.03% //

chromatography additional quality control according

Gas to in-house standard (WSYX-QC-

Ether residual < 0.01% //

chromatography TS-0609-0004).

The limit is set according to Quality

Standard for Pneumococcal

UV-vis

Sodium deoxycholate < 0.20 // Polysaccharide Conjugate Vaccine

spectrometry

(WSYX-QC-TS-0006-0721), which

is approved by NMPA.

The pyrogen content of the

purified polysaccharide should

Pyrogen test required by WHO can

Pyrogen /// be determined and shown to be

be represented by endotoxin test

within acceptable limits agreed

by the NRA.

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3.2.S.4.5.2 Justification of Specification for Pneumococcal Polysaccharide Derivatives

Table 125 Justification of Specification for Pneumococcal Polysaccharide Derivatives

Test items Test method Specifications Ph. Eur. WHO Rationale for specifications

The manufacturer should

demonstrate consistency

of the degree of

modification of the

Serotype Rate (%) polysaccharide, either by

1 2.6 - 4.4 an assay of each batch of

3 5.0 - 14.0 the polysaccharide or by

4 3.0 - 11.0 validation of the

5 3.5 - 18.0 manufacturing process. The limit is set according to

6A 3.0 - 6.0 Depending on the Quality Standard for

Chromogenic 6B 1.5 - 3.4 conjugation chemicals Pneumococcal Polysaccharide

Derivation rate /

method 7F 2.9 - 5.3 used, consistency in Conjugate Vaccine (WSYX-

9V 3.3 - 7.4 degree of polysaccharide QC-TS-0006-0721), which is

14 2.5 - 5.9 activation may be approved by NMPA.

18C 2.0 - 4.8 determined as part of

19A 3.5 - 8.0 process validation or

19F 3.2 - 6.2 reflected by

23F 1.1 - 3.7 characteristics of vaccine

lots shown to have

adequate safety and

immunogenicity in

clinical trials.

Several methods for the

The limit is set according to

chemical modification of

Quality Standard for

polysaccharides prior to

Residual Pneumococcal Polysaccharide

Lowry method No more than 10 ng/mg / conjugation may be

cyanide Conjugate Vaccine (WSYX-

satisfactory. The chosen

QC-TS-0006-0721), which is

method should be

approved by NMPA.

approved by the NRA.

KD = 0.2 - In the case of a The degree of size The limit is set according to

Gel KD > 0.6

Molecular size Serotype KD 0.6 size-reduced reduction of the Quality Standard for

chromatography recovery

recovery modified polysaccharide will Pneumococcal Polysaccharide

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1 ≤ 0.37 ≥ 45% - pneumococcal depend upon the Conjugate Vaccine (WSYX-

3 ≤ 0.45 ≥ 60% - polysaccharide, manufacturing process. QC-TS-0006-0721), which is

$4 \leq 0.42 \leq 65\%$ - the molecular- The average size approved by NMPA.
 $5 \leq 0.90 \geq 37\%$ size or distribution (degree of
 $6A \leq 0.62 \geq 60\%$ - molecular-mass polymerization) of the
 $6B \leq 0.33 \geq 50\%$ - distributions modified polysaccharide
 $7F \leq 0.43 \geq 60\%$ - determined by should be determined by a
 $9V \leq 0.46 \geq 70\%$ - size-exclusion suitable method and
 $14 \leq 0.40 \geq 65\%$ - chromatography shown to be consistent.
 $18C \leq 0.41 \geq 65\%$ - combined with The molecular size
 an appropriate distribution should be
 $19A \leq 0.48 \geq 65\%$ -
 detection specified for each
 $19F \leq 0.56 \geq 70\%$ -
 system. An serotype, with appropriate
 $23F \leq 0.46 \geq 75\%$ -
 acceptable limits for consistency, as
 value is the size may affect the
 established for reproducibility of the
 each modified conjugation process.
 pneumococcal
 polysaccharide.

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3.2.S.4.5.3 Justification of Specification for Bulk Tetanus Toxoid

Table 126 Justification of Specification for Bulk Tetanus Toxoid
 Test items Test method Specifications Ph. Eur. WHO Rationale for specifications
 Specification is set according to
 Quality Standard for Pneumococcal
 Polysaccharide Conjugate Vaccine
 (WSYX-QC-TS-0006-0721), which
 pH pH meter 6.6 - 7.4 //
 is approved by NMPA, and
 supported by process capability and
 is well within range of safety
 concern.
 Limit of The antigenic purity is
 flocculation No less than 1500 no less than 1500Lf per The specification is required by Ph.
 Purity /
 Lf/mg milligram of protein Eur. (2150).
 Kjeldahl method nitrogen.
 Specification is set according to
 Quality Standard for Pneumococcal
 Polysaccharide Conjugate Vaccine
 Should be in
 (WSYX-QC-TS-0006-0721), which
 Sterility Film filtration accordance with the //
 is approved by NMPA, and
 regulations
 supported by process capability and
 is well within range of safety
 concern.
 Observation on the
 Specification is set according to
 7th, 14th and 21st day
 Quality Standard for Pneumococcal
 showing no local
 Polysaccharide Conjugate Vaccine
 suppuration and
 (WSYX-QC-TS-0006-0721), which
 Specific toxicity Animal test necrosis, no tetanus //
 is approved by NMPA, and
 symptoms. Weight
 supported by process capability and
 gain for each guinea
 is well within range of safety
 pig at the end of the
 concern.
 period.

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Specification is set according to
 Observation on the
 Quality Standard for Pneumococcal
 7th, 14th and 21st day
 Polysaccharide Conjugate Vaccine
 showing no tetanus
 (WSYX-QC-TS-0006-0721), which
 Toxicity reversal test Animal test symptoms, weight //
 is approved by NMPA, and
 gain for each guinea
 supported by process capability and
 pig at the end of the
 is well within range of safety
 period.
 concern.

3.2.S.4.5.4 Justification of Specification for Purified Liquid of Tetanus Toxoid
Table 127 Justification of Specification for Purified Liquid of Tetanus Toxoid
Test items Test method Specifications Ph. Eur. WHO Rationale for specifications
The identity of tetanus toxoid is
The identity may be determined tested in tetanus toxoid bulk
Identity / / /
serologically, production with flocculant test,
which is a serological method.
The limit is set according to Quality
Standard for Pneumococcal
No less than 16 Should be approved by the
Protein content Lowry method / Polysaccharide Conjugate Vaccine
mg/mL NRA.
(WSYX-QC-TS-0006-0721), which
is approved by NMPA.
The limit is set according to Quality
Standard for Pneumococcal
No more than 10
Bacterial endotoxin Gel-clot method / / Polysaccharide Conjugate Vaccine
EU/mg
(WSYX-QC-TS-0006-0721), which
is approved by NMPA.
The limit is set according to Quality
Liquid
Standard for Pneumococcal
chromatography
Purity No less than 90% / / Polysaccharide Conjugate Vaccine
(UV absorption
(WSYX-QC-TS-0006-0721), which
at 280 nm)
is approved by NMPA.

3.2.S.4.5.5 Justification of Specification for Monovalent Bulk Conjugate
Table 128 Justification of Specification for Monovalent Bulk Conjugate
Test items Test method Specifications Ph. Eur. WHO Rationale for specifications
Significant color change may be
indicative of contamination or
Clear, transparent liquid or
Visual degradation. Levels of color are
Appearance slightly white floccule after / /
determination set to ensure consistency with
shaking
clinical, production and stability
experience.
A test should be performed on
the monovalent bulk to verify
A clear precipitation line
its identity. The method Double immunodiffusion test is an
should appear with the
Double should be validated to show immunochemical method which
corresponding
Identity immunodiffusion / that it distinguishes the confirms the presence of a specific
pneumococcal specific
test desired monovalent material serotype of polysaccharide-tetanus
antiserum and tetanus
from all other polysaccharides toxoid conjugate.
toxoid immune serum.
and conjugates produced on
that manufacturing site.
No less than the limit
below.
Serotype Limit
(µg/mL) The polysaccharide
1 40 content is determined
3 20 by a suitable physical The limit is set according to
4 40 or chemical method or Quality Standard for
Polysaccharide 5 40 by an Pneumococcal Polysaccharide
Turbidimetry /
content 6A 40 immunochemical Conjugate Vaccine (WSYX-QC-
6B 40 method. The value TS-0006-0721), which is approved
7F 40 complies with the by NMPA.
9V 40 requirement approved
14 40 for each serotype.
18C 40
19A 40
19F 40

23F 40
The protein content is The protein content of the
determined by a conjugate should be The limit is set according to
suitable physical or determined by means of an Quality Standard for

chemical method. The appropriate validated assay Pneumococcal Polysaccharide Protein content Lowry method No less than 70 µg/mL value complies with and comply with limits for the Conjugate Vaccine (WSYX-QC- the requirement product. Each batch should be TS-0006-0721), which is approved approved for each tested for conjugated and by NMPA.

serotype. unbound protein.

Serotype Range

1 0.20 - 1.80

3 0.10 - 0.73

4 0.30 - 0.90

5 0.10 - 0.60 Determine the For each conjugate, the ratio The specification is required by

6A 0.10 - 0.50 polysaccharide to should be within the range Ph. Eur. (2150). The limit is set protein ratio by approved for that particular according to Quality Standard for

6B 0.30 - 0.60

PS: protein ratio Calculation calculation. The value conjugate by the NRA and Pneumococcal Polysaccharide

7F 0.20 - 0.90

compiles with the should be consistent with Conjugate Vaccine (WSYX-QC-

9V 0.10 - 0.35

requirement approved vaccine shown to be effective TS-0006-0721), which is approved

14 0.20 - 0.60 for each serotype. in clinical trials. by NMPA.

18C 0.40 - 0.80

19A 0.20 - 0.70

19F 0.20 - 0.70

23F 0.25 - 0.60

KD < Molecular-size or The relative molecular size of

Serotype KD 0.2 molecular-mass the polysaccharide-protein

recovery distribution is conjugate should be

The specification is required by

1 ≥ 85% determined by size- determined for each bulk,

Ph. Eur. (2150). The limit is set

3 ≥ 95% exclusion using a gel matrix appropriate

according to Quality Standard for

Gel chromatography to the size of the conjugate.

Molecular size (KD) 4 ≥ 80% Pneumococcal Polysaccharide

chromatography combined with an The method should be

5 ≤ ≥ 80% Conjugate Vaccine (WSYX-QC-

appropriate detection validated with an emphasis on

6A 0.2 ≥ 80% TS-0006-0721), which is approved

system. An acceptable specificity to distinguish the

6B ≥ 90% value is established for polysaccharide-protein

by NMPA.

7F ≥ 90% the bulk conjugate of conjugate from other

9V ≥ 95% each polysaccharide. components that may be

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14 ≥ 90% Each batch must be present, e.g. unbound protein

18C ≥ 80% shown to comply with or polysaccharide. The size

19A ≥ 85% this limit. distribution specifications will

19F ≥ 80% be vaccine-specific and

23F ≥ 90% should be consistent with lots

shown to be immunogenic in

clinical trials.

Unbound

No more than the limit polysaccharide is

below. determined after

Serotype Limit (%) removal of the

1 25.0 conjugate, for example

Each batch of conjugate

3 20.0 by anion-exchange

should be tested for unbound

4 25.0 size-exclusion or The specification is required by

or free polysaccharide in

5 20.0 hydrophobic Ph. Eur. (2150). The limit is set

order to establish consistency

6A 20.0 chromatography, according to Quality Standard for

Free polysaccharide Anthrone-sulfate of production and to ensure

ultrafiltration, or other Pneumococcal Polysaccharide

content method 6B 25.0 that the amount present in the

validated methods. A Conjugate Vaccine (WSYX-QC-

7F 20.0 purified bulk is within the

value consistent with TS-0006-0721), which is approved

9V 25.0 limits agreed by the NRA

adequate by NMPA.

14 25.0 based on lots shown to be

immunogenicity as

18C 30.0 clinically safe and efficacious.

shown in clinical trials

19A 30.0 is established foreach

19F 35.0 serotype and each lot

23F 25.0 must be shown to

comply with this limit

Determine the content

by a suitable method,

either directly or by The limit is set according to

deriving the content Quality Standard for

Free carrier protein Gel by calculation from Pneumococcal Polysaccharide

No more than 5.0% /

content chromatography the results of other Conjugate Vaccine (WSYX-QC-

tests. The value TS-0006-0721), which is approved

complies with the by NMPA.

requirement approved
for each serotype.
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Where applicable,
suitable tests are
carried out to
determine residues of
The conjugate purification
reagents used during
procedures should remove
inactivation and
residual reagents used for
purification. An
conjugation and capping. The
acceptable value for
removal of reagents and The limit is set according to
each reagent is
reaction by products such as Quality Standard for
established for the
Chromogenic cyanide, 1-ethyl-3,3-(3- Pneumococcal Polysaccharide
Residual EDAC No more than 5.0 µmol/L product and each
method dimethylaminopropyl)- Conjugate Vaccine (WSYX-QC-
batch of conjugate
carbodiimide (EDAC) and TS-0006-0721), which is approved
must be shown to
others, depending on the by NMPA.
comply with this limit
conjugation chemistry, should
Where validation
be confirmed by suitable tests
studies have
or by validation of the
demonstrated removal
purification process.
of residual reagents,
the test on conjugate
polysaccharides
maybe omitted.
No more than the limit
below.
Limit
Serotype
(µg/mL)
1 0.15 The limit is set according to
3 0.25 Quality Standard for
Chromogenic 4 0.15 Pneumococcal Polysaccharide
ADH content / /
method 5 0.25 Conjugate Vaccine (WSYX-QC-
6A 1.20 TS-0006-0721), which is approved
6B 0.25 by NMPA.
7F 0.20
9V 0.70
14 1.15
18C 0.20

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19A 0.20
19F 0.20
23F 0.45
The bulk purified conjugate
should be tested for bacterial
and mycotic sterility in
It complies with the accordance with the
The specification is required by
test for sterility, requirements of the revised
Ph. Eur. (2150). The limit is set
carried out using 10 General requirements for the
according to Quality Standard for
Meets the requirement of ml for each medium or sterility of biological
Sterility Film filtration Pneumococcal Polysaccharide
the test. the equivalent of 100 substances or by a method
Conjugate Vaccine (WSYX-QC-
doses for each approved by the NRA. If a
TS-0006-0721), which is approved
medium, whichever is preservative has been added
by NMPA.
less to the product, appropriate
measures should be taken to
prevent it from interfering
with the test.
To ensure an acceptable level
of endotoxin in the final
Should less than 0.75 product, the endotoxin content
The specification is required by
Bacterial endotoxin Gel-clot method No more than 0.75 EU/µg IU per microgram of of the monovalent bulk
may

Ph. Eur. (2150).
 polysaccharide. be determined and shown to
 be within acceptable limits
 agreed by the NRA.
 The bulk conjugate should be
 Observation on the 7th, 14th
 tested for the absence of The limit is set according to
 and 21st day showing no
 specific toxicity of the carrier Quality Standard for
 local suppuration and
 Specific toxin of protein where appropriate. Pneumococcal Polysaccharide
 Animal test necrosis, no tetanus /
 carrier protein Absence of specific toxicity Conjugate Vaccine (WSYX-QC-
 symptoms. Weight gain for
 of the carrier protein may also TS-0006-0721), which is approved
 each guinea pig at the end
 be assessed through validation by NMPA.
 of the period.
 of the production process.

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3.2.S.5 Reference Standards or Materials

Table 129 List of Reference Standards or Materials used

Storage
 Test Items Reference name Manufacturer Unit
 Condition
 National standard
 19.73
 Protein content sample for protein NIFDC Below -20 ■
 mg/tube
 content test
 Standard Phosphorus
 Room
 Phosphorus content (potassium NIFDC 100 mg/vial
 temperature
 dihydrophosphate)
 Room
 Uronic acid content Sodium glucuronate SIGMA 10 g/vial
 temperature
 D-glucosamine Room
 Hexosamine content NIFDC 200 mg/vial
 hydrochloride temperature
 Room
 Methyl pentose content Rhamnose SIGMA 10 g/vial
 temperature
 O-acetyl content Acetylcholine chloride SIGMA 25 g/vial Below 20 ■
 Room
 Dextran GE 10 g/vial
 temperature
 Molecule size Sinopharm
 Room
 Vitamin B12 Chemical Reagent 1 g/vial
 temperature
 Co., Ltd.
 Residual sodium Room
 Sodium deoxycholate SIGMA 500 g/vial
 deoxycholate content temperature
 Yunnan Yanglin
 Shandian 500
 Ethanol Store in shade
 Pharmaceutical mL/vial
 Co., Ltd.
 Yunnan Yanglin
 Residual organic solvent Shandian 500
 Ether Store in shade
 content Pharmaceutical mL/vial
 Co., Ltd.
 Yunnan Yanglin
 Shandian 500
 Acetone Store in shade
 Pharmaceutical mL/vial
 Co., Ltd.
 Standard bacterial Zhanjiang Andusi
 Bacterial endotoxin 10 EU/tube 2-8 ■
 endotoxin sample Biology Co., Ltd.
 ADH Adipic dihydrazide SIGMA 100 g/vial Below -20 ■
 Sinopharm
 Room
 Glucose content Anhydrous glucose Chemical Reagent 500 g/vial
 temperature
 Co., Ltd.
 J&K Scientific
 Residual Cyanide Cyanogen bromide 100 g/vial 2-8 ■
 Ltd.
 Pneumococcal
 specific antiserum
 Danish National
 (serotypes 1, 3, 4, 5, 1 ml/vial 2-8 ■

Serum Institute
6A, 6B, 7F, 9V, 14,
Identity / Polysaccharide
18C, 19A, 19F, 23F)
content
Pneumococcal
polysaccharide 10 mg/vial
ATCC Below -20 ■
reference (serotypes 1, 2 mg/vial
3, 4, 5, 6A, 6B, 7F,
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9V, 14, 18C, 19A,
19F, 23F)
EDAC Carbodiimide SIGMA 10 g/vial Below -20 ■
Sterility Test Staphylococcus aureus NIFDC / Below -20 ■

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3.2.S.6 Container Closure System

The 13 types polysaccharides are stored in centrifuge bottles, and the information of the centrifuge bottles is shown in Table 130.

Table 130 Information Regarding the Container Closure System for Pneumococcal Polysaccharide

Item Information
Container name Centrifuge bottle
Manufacturer Backman & Thermo Fisher Scientific
Container material PP
Container specification 250 mL/bottle and 50mL/bottle
Container structure Cap + body

The 13 types pneumococcal polysaccharide bulk conjugates are stored in 10 L glass bottles, and the tetanus toxoid carrier protein bulk was stored in 500 mL glass bottles, the information of the glass bottles is shown in Table 131

Table 131 Information Regarding the Container Closure System for Bulk

Item
conjugate protein
Container name Serum bottle Saline bottle
Chongzhou Shubo Scientific Instruments Sichuan Jianwei Hesheng Glass Co.,
Manufacturer
Co., Ltd Ltd
Container material Glass Soda-lime glass
Container
10 L/bottle 500 mL/bottle
specification
Container structure Body + rubber plug Body + rubber plug

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3.2.S.7 Stability
According to the requirements of preventive vaccine 13-valent Pneumococcal Conjugate Vaccine. Walvax has performed the stability studies of intermediate products, including 13 types of pneumococcal polysaccharide, pneumococcal polysaccharide bulk conjugate and carrier protein.

3.2.S.7.1 Stability Summary and Conclusion
3.2.S.7.1.1 Stability Summary and Conclusion on Polysaccharide

Table 132 Batch information of Pneumococcal Polysaccharide used in Stability Study

Batch	Storage
Serotype Batch No.	Packaging Mfg.Date
quantity condition	
Pneumococcal 1-PVPV15101201	11.12 g 2015.10.21
polysaccharide 1-PVPV15121202	13.74 g 2015.10.20
serotype 1 1-PVPV16011203	14.51 g 2016.01.11
Pneumococcal 3-PVPV15100901	35.24 g 2015.10.09
polysaccharide 3-PVPV15110902	24.06 g 2015.11.28
serotype 3 3-PVPV15120903	29.26 g 2015.12.24
Pneumococcal 4-PVPV15101101	28.21 g 2015.10.17
polysaccharide 4-PVPV16011102	24.64 g 2015.12.26
serotype 4 4-PVPV15121103	29.82 g 2016.01.07
Pneumococcal 5-PVPV15101301	11.81 g 2015.10.25
polysaccharide 5-PVPV15121302	12.27 g 2015.12.28
serotype 5 5-PVPV160111303	10.38 g 2016.01.09
Pneumococcal 6A-PVPV15090101	67.33 g 2015.09.07
polysaccharide 6A-PVPV15110102	52.67 g 2015.11.06
serotype 6A 6A-PVPV15120103	59.43 g 2015.12.16
Pneumococcal 6B-PVPV15090201	59.95 g 2015.09.11
polysaccharide 6B-PVPV15110202	63.18 g 2015.11.16
serotype 6B 6B-PVPV15120203	48.67 g 2015.12.16
Pneumococcal 7F-PVPV15100701	33.94 g 2015.10.01
Centrifuge	
polysaccharide 7F-PVPV15110702	25.72 g below -20 ■ 2015.11.22

tube
serotype 7F 7F-PVPV16010703 36.22 g 2016.01.05
Pneumococcal 9V-PVPV15090301 36.34 g 2015.09.15
polysaccharide 9V-PVPV15110302 39.68 g 2015.11.22
serotype 9V 9V-PVPV15120303 46.54 g 2015.12.20
Pneumococcal 14-PVPV15101001 17.33 g 2015.10.13
polysaccharide 14-PVPV15121002 22.73 g 2015.12.22
serotype 14 14-PVPV16011003 19.55 g 2016.01.03
Pneumococcal 18C-PVPV15100801 21.97 g 2015.10.05
polysaccharide 18C-PVPV15110802 27.01 g 2015.11.25
serotype 18C 18C-PVPV16010803 25.06 g 2016.01.01
Pneumococcal 19A-PVPV15090401 84.01 g 2015.09.19
polysaccharide 19A-PVPV15110402 66.76 g 2015.11.19
serotype 19A 19A-PVPV15120403 74.89 g 2015.12.18
Pneumococcal 19F-PVPV15090501 83.41 g 2015.09.23
polysaccharide 19F-PVPV15110502 90.62 g 2015.11.19
serotype 19F 19F-PVPV15120503 74.85 g 2015.12.18
Pneumococcal 23F-PVPV15090601 68.03 g 2015.09.27
polysaccharide 23F-PVPV15120602 62.44 g 2015.11.19
serotype 23F 23F-PVPV16010603 63.49 g 2015.12.18

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Basis of Study
1. GMP
2. Chinese Pharmacopeia 2015, volume IV 9000, Guiding principles for stability test of bulk drugs and pharmaceutical product.
3. Management Procedures for Continuous Stability Investigation of Products (Document code: WSYX-QA-SMP-0009-0030)
4. Quality Standard of 13-Valent Pneumococcal Polysaccharide Conjugate Vaccine (Document code: WSYX-QC-TS-0609-0005).

Test protocol
1. storage condition: < -20 ■
2. test intervals: initial, 1st year, 2nd year, 3rd year, 4th year, the 4th and a half year, 5th year, the 5th and a half year.
3. Test items: full test in each time point.

Test items and standard
See the Table 133.

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Table 133 Test Items and Specifications of Pneumococcal Polysaccharides
Serotype
1 3 4 5 6A 6B 7F 9V 14 18C 19A 19F 23F
Test items
Identification Should form the obvious precipitation line with corresponding specific antiserum.
Total solid -
Nucleic acid
≤ 2% ≤ 2% ≤ 2% ≤ 2% ≤ 1.5% ≤ 2% ≤ 2% ≤ 2% ≤ 2% ≤ 2% ≤ 2% ≤ 2% ≤ 2%
content
Protein content ≤ 2% ≤ 5% ≤ 3% ≤ 7.5% ≤ 1.5% ≤ 2% ≤ 5% ≤ 2% ≤ 3% ≤ 2% ≤ 3% ≤ 2%
Total nitrogen 3.5-6.0% 0-1.0% 4.0-6.0% 2.5-6.0% 0-2.0% 0-2.0% 1.5-4.0% 0.5-3.0% 1.5-4.0% 0-1.0% 0.6-3.5%
1.4-3.5% 0-1.0%
Phosphorous 3.0-
0-1.5% 0-1.0% 0-1.5% 0-2.0% 2.4-5.0% 2.5-5.0% 0-1.0% 0-1.0% 0-1.0% 2.4-4.9% 3.0-7.0% 3.0-5.5%
content 4.5%
Uronic acid ≥ 45% ≥ 40% / ≥ 12% / / / ≥ 15% / / / /
O-acetyl
≥ 1.8% / / / / / / / / / / /
content
Hexosamine
/ / ≥ 40% ≥ 20% / / / ≥ 13% ≥ 20% / ≥ 12% ≥ 12.5% /
content
Methyl pentose
/ / / / ≥ 15% ≥ 15% ≥ 13% / / ≥ 14% / ≥ 2 0% ≥ 37%
content
KD ≤ KD ≤ KD ≤ KD ≤ KD ≤ KD ≤ KD ≤ KD ≤ KD ≤ KD ≤ KD ≤ KD
Molecular size KD ≤ 0.60 KD ≤ 0.50
0.15 0.15 0.15 0.50 0.20 0.45 0.30 0.15 0.45 0.20 0.15
Bacterial
≤ 0.5 EU/μg
endotoxin
Sodium
< 0.2%
deoxycholate
Ether content ≤ 0.01%
Acetone
≤ 0.03%
content
Ethanol content ≤ 0.08%

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Conclusion

The pneumococcal polysaccharides of serotypes 1, 3, 4, 5, 6A, 6B, 7F, 9V, 14, 18C, 19A, 19F and 23F produced by Walvax were stored at -20 °C for 0 months, 1 year, 2 years, 3 years and 4 years respectively. The detection values of each test item at each test time point were in accordance with the Quality Standard for 13-Valent Pneumococcal Polysaccharide Conjugate Vaccine (WSYX-QC-TS-06-09-0005).

This stability study was conducted by the quality control laboratory in strict accordance with the stability study protocol, which can provide the basis for the safety, immunogenicity and validity of pneumococcal polysaccharide stored at -20 °C. To sum up, the pneumococcal polysaccharide produced by Walvax can meet the requirements of 4 years storage at -20 °C.

3.2.S.7.1.2 Stability Summary and Conclusion on Polysaccharide Derivatives

Table 134 Batch information of Polysaccharide Derivatives used in Stability Study

Batch Storage

Serotypes Batch No. Packaging Mfg.Date

quantity condition

Pneumococcal SP1C201908001 1476 mL 2019.08.23

polysaccharide SP1C201909002 1480 mL 2019.09.13

derivative serotype 1 SP1C201909003 1459 mL 2019.09.16

Pneumococcal SP3C201908001 1429 mL 2019.08.26

polysaccharide SP3C201909002 1519 mL 2019.09.18

derivative serotype 3 SP3C201909003 1469 mL 2019.09.20

Pneumococcal SP4D202001001 1800 mL 2020.01.16

polysaccharide SP4D202002002 2480 mL 2020.02.24

derivative serotype 4 SP4D202005003 1970 mL 2020.05.12

Pneumococcal SP5C201909001 1650 mL 2019.09.06

polysaccharide SP5C201909002 1690 mL 2019.09.09

derivative serotype 5 SP5C201909003 1470 mL 2019.09.11

Pneumococcal SP6AC201909001 1430 mL 2019.09.13

polysaccharide SP6AC201909002 1440 mL 2019.09.16

derivative serotype 6A SP6AC201909003 1490 mL 2019.09.18

Pneumococcal SP6BD202001001 2500 mL 2020.01.14

polysaccharide SP6BD202001002 Glass bottle 2491 mL 2-8 °C 2020.01.24

derivative serotype 6B SP6BD202002003 3290 mL 2020.02.24

Pneumococcal SP7FD202001001 1070 mL 2020.01.14

polysaccharide SP7FD202002002 2460 mL 2020.02.06

derivative serotype 7F SP7FD202002003 2620 mL 2020.02.15

Pneumococcal SP9VC201908001 1790 mL 2019.08.23

polysaccharide SP9VC201908002 1810 mL 2019.08.26

derivative serotype 9V SP9VC201908003 1790 mL 2019.08.28

Pneumococcal SP14D202001001 2000 mL 2020.01.16

polysaccharide SP14D202002002 3630 mL 2020.02.09

derivative serotype 14 SP14D202002003 3391 mL 2020.02.27

Pneumococcal SP18CC201908001 1469 mL 2019.08.30

polysaccharide SP18CC201908002 1469 mL 2019.09.02

derivative serotype 18C SP18CC201908003 1489 mL 2019.09.04

Pneumococcal SP19AC201908001 1500 mL 2019.08.30

polysaccharide SP19AC201908002 1530 mL 2019.09.02

derivative serotype 19A SP19AC201909003 1549 mL 2019.09.04

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Pneumococcal SP19FC201909001 1389 mL 2019.09.06

polysaccharide SP19FC201909002 1369 mL 2019.09.09

derivative serotype 19F SP19FC201909003 1389 mL 2019.09.11

Pneumococcal SP23FC201908002 1389 mL 2019.08.20

polysaccharide SP23FC201908003 1410 mL 2019.08.28

derivative serotype 23F SP23FD202001001 1611 mL 2019.09.20

Basis of Study

1. GMP

2. Chinese Pharmacopeia 2015, volume IV 9000, Guiding principles for stability test of bulk drugs and pharmaceutical product.

3. Management Procedures for Continuous Stability Investigation of Products (Document code: WSYX-QA-SMP-0009-0030)

4. Quality Standard of 13-Valent Pneumococcal Polysaccharide Conjugate Vaccine (Document code: WSYX-QC-TS-0609-0005).

5. Operation Procedure for Determination of Molecular Size with SEC-MALS Method (Document code: WSYX-QC-SOP-00-09-0143).

Test protocol

1. storage condition: 2-8 °C;

2. test intervals: initial, 10th day, 20th day, 30th day, 35th day;

3. test items and corresponding time:

0th day, 35th day: full test;

10th day, 20th day, 30th day: molecular size (SEC-MALS method);

35th day: add test items of 3 batches of serotypes 4, 6B, 7F, 14, and 1 batch of serotype 23F in the 35th day, more detail please see the stability report of polysaccharide derivatives.

Test items and standard

See the Table 135.

Table 135 Test Items and Specifications of Polysaccharide Derivatives

Test items Specification

Total solid —

5.0-

Serotype 1 2.6-4.4 Serotype 3 Serotype 4 3.0-11.0

Derivation rate 14.0

(%) Serotype 5 3.5-18.0 Serotype 6A 3.0-6.0 Serotype 6B 1.5-3.4

(ADH Serotype 7F 2.9-5.3 Serotype 9V 3.3-7.4 Serotype 14 2.5-5.9 /polysaccharide) Serotype 18C 2.0-4.8 Serotype 19A 3.5-8.0 Serotype 19F 3.2-6.2 Serotype 23F 1.1-3.7
Residual cyanide Should not more than 10 ng/mg
Serotype KD value Recovery rate at KD of 0.2-0.6
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$1 \leq 0.37 \leq 45\%$
 $3 \leq 0.45 \leq 60\%$
 $4 \leq 0.42 \leq 65\%$
Recovery rate before KD reached 0.6
 $5 \leq 0.90$
 $\geq 37\%$
 $6A \leq 0.62 \leq 60\%$
Molecular size
 $6B \leq 0.33 \leq 50\%$
(Sepharose CL-
 $7F \leq 0.43 \leq 60\%$
4B filter)
 $9V \leq 0.46 \leq 70\%$
 $14 \leq 0.40 \leq 65\%$
 $18C \leq 0.41 \leq 65\%$
 $19A \leq 0.48 \leq 65\%$
 $19F \leq 0.56 \leq 70\%$
 $23F \leq 0.46 \leq 75\%$
Molecular size
(SEC-MALS -
method)

Conclusion
According to Stability Protocol of Pneumococcal Polysaccharide Derivatives (Document code: WSYX-QC-SP-0609-0099, version 00, 01), 13 serotypes of Pneumococcal Polysaccharide Derivatives were kept at 2-8 °C for 35 days, except batch SP4D202005003 at 35th day, the tests were not totally finished, all the test results of the other batches at each time point were in accordance with the Quality Standard of 13-Valent Pneumococcal Polysaccharide Conjugate Vaccine (Document code: WSYX-QC-TS-06-09-0002).

3.2.S.7.1.3 Stability Summary and Conclusion on Bulk Tetanus Toxoid

Table 136 Batch Information of Bulk Tetanus Toxoid used in Stability Study
Batch Storage Mfg Date
Sample Batch No. Packaging
quantity condition (Strain opening)
201511004 10000 mL 2015.11.17
Bulk tetanus toxoid CTB201606005 Glass bottle 59900 mL 2-8 °C 2016.06.15
CTB201610006 59800 mL 2016.10.25

Basis of Study
1. GMP
2. Chinese Pharmacopeia 2015, volume IV, Guiding principles for stability test of bulk drugs and pharmaceutical product.
3. Management Procedure for Continuous Stability Inspection of Products (Document Code: WSYX-QA-SMP-0009-0030)
4. Quality Standard of Diphtheria, Tetanus and Acellular Pertussis Combined Vaccine, Adsorbed (Document Code: WSYX-QC-TS-0609-0003).

Test protocol
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MODULE 3 QUALITY
Version No. Walvax-202007-00 Yuxi Walvax Biotechnology Co., Ltd.

1. storage condition: $6 \pm 2^\circ\text{C}$
2. test intervals: initial, 3rd month, 6th month, 9th month, 12th month, 18th month, 24th month, 36th month, 42nd month, 48th month.
3. test items and corresponding time:
Initial: detoxification test, appearance, pH, flocculation unit, purity, sterility test, specific toxicity
test, toxicity reversal test, protein content, residual ammonium;
6th month, 12th month, 18th month, 24th month, 36th month, 42nd month, 48th month: appearance, pH, limit of flocculation, purity, sterility test, specific toxicity test, toxicity reversal test, protein content, residual ammonium;
3rd month, 9th month: appearance, pH, limit of flocculation, purity, protein content, residual ammonium.

Test items and specification
See the Table 137
Table 137 Test Items and Specifications of Bulk Tetanus Toxoid
Test items Specifications
Appearance Should be yellow or brown yellow transparent liquid.
pH 6.6 - 7.4
Limit of flocculation
-
(Lf/mL)

Purity No less than 1500 Lf/mg

Sterility Should be in accordance with the regulations
Observation on the 7th, 14th and 21st day showing no local suppuration and
Specific toxicity necrosis, no tetanus symptoms. Weight gain for each guinea pig at the end of the

period.
Observation on the 7th, 14th and 21st day showing no tetanus symptoms, weight
Toxicity reversal test
gain for each guinea pig at the end of the period.
Observation on the 7th, 14th and 21st day showing no tetanus symptoms, weight
Detoxification test
gain for each guinea pig at the end of the period.
Residual ammonium Not more than 0.25 mg/mL
Protein content (µg/mL) -

Conclusion
3 batches of Bulk Tetanus Toxoid, which are 201511004 (Bottle No.1), CTB201606005 (Bottle No.1), CTB201610006 (Bottle No.1) were kept at 2-8 ■ for 48 months, 42 months and 36 months respectively, test results were all complied with specification of Diphtheria, Tetanus and Acellular Pertussis Combined Vaccine, Adsorbed (Document code: WSYX-QC-TS-00-09-0001), the test results show that the bulk tetanus toxoid produced by Walvax can meet the requirement of 36 months stored at 2-8 ■.

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Version No. Walvax-202007-00 Yuxi Walvax Biotechnology Co., Ltd.

3.2.S.7.1.4 Stability Summary and Conclusion on Purified Liquid of Tetanus Toxoid

Table 138 Batch Information of Purified Liquid of Tetanus Toxoid used in Stability Study
Batch Storage
Sample Batch No. Packaging Mfg Date
quantity condition
CTL201909018 3000 mL 2019.09.20
Purified liquid of
CTL201909019 Glass bottle 3000 mL 2-8 ■ 2019.09.22
tetanus toxoid
CTL201909020 2500 mL 2019.09.30

Basis of Study
1. GMP
2. Chinese Pharmacopeia 2015, volume IV, Guiding principles for stability test of bulk drugs and pharmaceutical product.
3. Management Procedure for Continuous Stability Inspection of Products (Document Code: WSYX-QA-SMP-00-09-0010)
4. Quality Standard of Purified Liquid of Tetanus Toxoid (Document Code: WSYX-QC-TS-02-09-0002).

Test protocol
1. storage condition: 2-8 ■
2. test intervals: initial, 1st month, 2nd month, 3rd month, 6th month, 9th month, 12th month, 15th month, 18th month, 24th month, 30th month.
3. test items and corresponding time:
Initial, 12th month, 30th month: full test (purity, sterility test, protein content, bacterial endotoxin);
1st month, 2nd month, 3rd month, 6th month, 9th month, 15th month, 18th month, 24th month: purity, protein content;

Test items and specification
See the Table 139.
Table 139 Test Items and Specifications of Purified Liquid of Tetanus Toxoid
Test items Specifications
Protein content No less than 16 mg/mL
Bacterial endotoxin No more than 10 EU/mg
Purity No less than 90%
Sterility test Should comply with the requirement

Conclusion
3 batches of Purified Liquid of Tetanus Toxoid, which are CTL201909018 (Bottle No.1), CTL201909019 (Bottle No.1), CTL201909020 (Bottle No.1) were kept at 2-8 ■ for 6 months, tested at 0th month, 1st month, 2nd month, 3rd month, 6th month. Test results were all complied with Quality standard of Diphtheria, Tetanus and Acellular Pertussis Combined Vaccine, Adsorbed
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Version No. Walvax-202007-00 Yuxi Walvax Biotechnology Co., Ltd.

(Code: WSYX-QC-TS-00-09-0001). Test results show that the purified liquid of tetanus toxoid produced by Walvax can meet the requirement of 6 months stored at 2-8 ■.

3.2.S.7.1.5 Stability Summary and Conclusion on Monovalent Bulk Conjugate

Table 140 Batch Information of Monovalent Bulk Conjugate used in Stability Study
Batch Storage
Serotypes Batch No. Packaging Mfg. Date
quantity condition
Serotype 1 RSP1C201706001 9286 mL 2017.06.21
pneumococcal bulk RSP1C201707002 12302 mL 2017.07.21
conjugate RSP1C201709003 12660 mL 2017.09.06
Serotype 3 RSP3C201706001 22700 mL 2017.06.24
pneumococcal bulk RSP3C201707002 22700 mL 2017.07.26
conjugate RSP3C201708003 20600 mL 2017.08.19
Serotype 4 RSP4C201706001 20800 mL 2017.06.03
pneumococcal bulk RSP4C201707002 12300 mL 2017.07.26
conjugate RSP4C201709003 11400 mL 2017.09.03
Serotype 5 RSP5C201705001 11702 mL 2017.05.25
pneumococcal bulk RSP5C201707002 12200 mL 2017.07.13
conjugate RSP5C201708003 10700 mL 2017.08.16

Serotype 6A RSP6AC201705001 11496 mL 2017.05.19
pneumococcal bulk RSP6AC201707002 11491 mL 2017.07.01
conjugate RSP6AC201708003 10000 mL 2017.08.22
Serotype 6B RSP6BC201705001 21800 mL 2017.05.19
pneumococcal bulk RSP6BC201707002 22600 mL 2017.07.01
conjugate RSP6BC201708003 22900 mL 2017.08.25
Serotype 7F RSP7FC201705001 5255 mL 2017.05.30
pneumococcal bulk RSP7FC201706002 Glass bottle 11500 mL 2-8 ■ 2017.06.27
conjugate RSP7FC201708003 10500 mL 2017.08.28
Serotype 9V RSP9VC201705001 11400 mL 2017.05.22
pneumococcal bulk RSP9VC201707002 11000 mL 2017.07.05
conjugate RSP9VC201709003 11900 mL 2017.09.03
Serotype 14 RSP14C201706001 12156 mL 2017.06.24
pneumococcal bulk RSP14C201707002 11400 mL 2017.07.21
conjugate RSP14C201709003 11546 mL 2017.09.06
Serotype 18C RSP18CC201706001 20800 mL 2017.06.21
pneumococcal bulk RSP18CC201707002 22000 mL 2017.07.17
conjugate RSP18CC201708003 21800 mL 2017.08.13
Serotype 19A RSP19AC201705001 11686 mL 2017.05.22
pneumococcal bulk RSP19AC201707002 12131 mL 2017.07.05
conjugate RSP19AC201708003 11700 mL 2017.08.28
Serotype 19F RSP19FC201706001 12031 mL 2017.06.27
pneumococcal bulk RSP19FC201707002 12311 mL 2017.07.17
conjugate RSP19FC201708003 12000 mL 2017.08.31
Serotype 23F RSP23FC201705001 22162 mL 2017.05.28
pneumococcal bulk RSP23FC201707002 22600 mL 2017.07.13
conjugate RSP23FC201708003 11000 mL 2017.08.31

Basis of Study

1. GMP
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MODULE 3 QUALITY
Version No. Walvax-202007-00 Yuxi Walvax Biotechnology Co., Ltd.

2. Chinese Pharmacopeia 2015, volume IV, Guiding principles for stability test of bulk drugs and pharmaceutical product.
3. Management Procedure for Continuous Stability Inspection of Products (Document Code: WSYX-QA-SMP-0009-0030)
4. Quality Standard of 13-valent Pneumococcal Polysaccharide Conjugate Vaccine (Document Code: WSYX-QC-TS-0609-0005).

Test protocol

1. storage condition: 2-8 ■
2. test intervals: initial, 1st month, 2nd month, 3rd month, 6th month, 9th month, 12th month.
3. test items and corresponding time:
Initial, 12th month: full test (sterility test, bacterial endotoxin, polysaccharide content, protein content, residual EDAC, polysaccharide/protein ratio, free-polysaccharide content, free-protein content, molecular size, identity test, ADH content);
1st month, 2nd month, 3rd month, 6th month, 9th month: polysaccharide content, protein content, polysaccharide/protein ratio, free-polysaccharide content, free-protein content, molecular size.

Test items and specification
See the Table 141

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Table 141 Test Items and Specifications of Monovalent Bulk Conjugate				
Polysaccharide Protein Free- Free- Residual Bacterial				
Polysaccharide Molecular				
Serotype	content	content	polysaccharide protein	EDAC endotoxin Identity test Sterility test
/ protein ratio	size			
μg/mL	μg/mL	content	content	μmol/L EU/μg
KD ≤ 0.2, Should comply				
Should be positive				
1	≥ 45	≥ 70	0.15-0.60	≤ 25% ≤ 4.5% ≤ 5 ≤ 0.75 recovery rate with the
with antiserum type 1				
≥ 80% requirement				
KD ≤ 0.2, Should comply				
Should be positive				
3	≥ 50	≥ 60	0.40-0.90	≤ 20% ≤ 4.0% ≤ 5 ≤ 0.75 recovery rate with the
with antiserum type 3				
≥ 80% requirement				
KD ≤ 0.2, Should comply				
Should be positive				
4	≥ 45	≥ 40	0.30-1.10	≤ 25% ≤ 3.5% ≤ 5 ≤ 0.75 recovery rate with the
with antiserum type 4				
≥ 80% requirement				
KD ≤ 0.2, Should comply				
Should be positive				
5	≥ 30	≥ 100	0.10-0.30	≤ 20% ≤ 4.5% ≤ 5 ≤ 0.75 recovery rate with the
with antiserum type 5				
≥ 80% requirement				
KD ≤ 0.2, Should be positive Should comply				
6A	≥ 40	≥ 80	0.10-0.50	≤ 20% ≤ 4.5% ≤ 5 ≤ 0.75 recovery rate with antiserum type with the
≥ 80% 6A requirement				
KD ≤ 0.2, Should be positive Should comply				
6B	≥ 40	≥ 60	0.20-0.65	≤ 25% ≤ 4.0% ≤ 5 ≤ 0.75 recovery rate with antiserum type with the
≥ 80% 6B requirement				
KD ≤ 0.2, Should be positive Should comply				
7F	≥ 80	≥ 180	0.15-0.45	≤ 20% ≤ 4.5% ≤ 5 ≤ 0.75 recovery rate with antiserum type with the

≥ 80% 7F requirement
KD ≤ 0.2, Should be positive Should comply
9V ≥ 30 ≥ 50 0.10-0.50 ≤ 25% ≤ 3.5% ≤ 5 ≤ 0.75 recovery rate with antiserum type with the
≥ 80% 9V requirement

KD ≤ 0.2, Should be positive Should comply
14 ≥ 40 ≥ 60 0.15-0.70 ≤ 25% ≤ 4.5% ≤ 5 ≤ 0.75 recovery rate with antiserum type with the
≥ 80% 14 requirement

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KD ≤ 0.2, Should be positive Should comply
18C ≥ 30 ≥ 50 0.30-0.60 ≤ 30% ≤ 3.5% ≤ 5 ≤ 0.75 recovery rate with antiserum type with the
≥ 80% 18C requirement
KD ≤ 0.2, Should be positive Should comply
19A ≥ 80 ≥ 120 0.10-0.65 ≤ 30% ≤ 3.5% ≤ 5 ≤ 0.75 recovery rate with antiserum type with the
≥ 80% 19A requirement
KD ≤ 0.2, Should be positive Should comply
19F ≥ 40 ≥ 100 0.10-0.40 ≤ 35% ≤ 3.5% ≤ 5 ≤ 0.75 recovery rate with antiserum type with the
≥ 80% 19F requirement
KD ≤ 0.2, Should be positive Should comply
23F ≥ 40 ≥ 60 0.25-0.65 ≤ 25% ≤ 4.0% ≤ 5 ≤ 0.75 recovery rate with antiserum type with the
≥ 80% 23F requirement

Conclusion
Serotypes 1, 3, 4, 5, 6A, 6B, 7F, 9V, 14, 18C, 19A, 19F, 23F Pneumococcal Bulk Conjugate were stored at 2-8 °C, tested at 0th month, 1st month, 2nd month, 3rd month, 6th month, 9th month, 12th month, test results were all complied with Quality Standard of 13-valent Pneumococcal Polysaccharide Conjugate Vaccine (Document code: WSYX-QC-TS-06-09-0005).
This stability study was conducted by the quality control laboratory strictly sampled at each test point, could provide evidence for safety, effectiveness and validity of pneumococcal bulk conjugate stored at 2-8 °C. In conclusion, shelf life of 13 serotypes of pneumococcal bulk conjugate manufactured Walvax stored at 2-8 °C is 12 months.

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3.2.S.7.2 Post-approval Stability Protocol and Stability Commitment
Please refer to Annex 2 - Stability study protocol for 13-valent Pneumococcal Polysaccharide Conjugate Vaccine intermediates
3.2.S.7.3 Stability Data
Please refer to Annex 3 - Stability study data for 13-valent Pneumococcal Polysaccharide Conjugate Vaccine intermediates

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3.2.P DRUG PRODUCT

3.2.P.1 Description and Composition of the Drug Product

13-Valent Pneumococcal Polysaccharide Conjugate Vaccine is a sterile liquid suspension of capsular polysaccharide antigens of *Streptococcus pneumoniae* serotypes 1, 3, 4, 5, 6A, 6B, 7F, 9V, 14, 18C, 19A, 19F, and 23F, with each saccharide individually conjugated to Tetanus Toxoid protein. The vaccine contains 2.6 µg/dose of serotype 1 and 19A, 2.5 µg/dose of serotype 3, 5, 6A, 9V, 3.0 µg/dose of serotype 4 and 23F, 6 µg/dose of serotype 6B, 2.85 µg/dose of serotype 7F, 2.75 µg/dose of serotype 14 and 19F, 3.25 µg/dose of serotype 18C. The vaccine is formulated in Phosphate-buffered Saline containing 4.25 mg/dose NaCl, 44.35 µg/dose NaH₂PO₄, 19.0 µg/dose Na₂HPO₄, and contains 0.5 mg/dose of aluminum phosphate as an adjuvant. Each 1 mL syringe or vial contains a single 0.5 mL dose of vaccine for intramuscular administration, with no preservative.

3.2.P.2 Pharmaceutical Development

3.2.P.2.1 Components of the Drug Product

2.3.P.2.1.1 Drug Substances

Tetanus toxoid was chosen as the carrier protein for the 13-valent Pneumococcal Polysaccharide Vaccine based on the previous Walvax experience with tetanus toxoid in the *Haemophilus influenzae* type b conjugate vaccine.

Activation and derivation are two key steps in polysaccharide protein conjugation. After a thorough investigation on the 13 polysaccharide structures, different activation agents are used according to different structures of capsular polysaccharides, and polysaccharide-protein conjugates are prepared with indirect conjugation method. The method uses cyanide as activation agent to activate capsular polysaccharides, the activated polysaccharides are bonded with the linker, ADH, through covalent bonding to form polysaccharide-ADH derivatives. The derivatives are bonded with carrier protein through covalent bonding with condensation agent EDAC to form polysaccharide-ADH-carrier protein conjugates. The immunogenicity of the combination containing 13 polysaccharide-protein conjugates and aluminum phosphate was compared to single antigen during compatibility test in pre-clinical research. Test results showed no difference between the mitogenicity of the combination and single antigen, which proves good immunogenicity of the combination.

3.2.P.2.1.2 Excipients

The excipients in the 13-valent Pneumococcal Polysaccharide Vaccine are NaCl at 8.5 mg/mL, sodium dihydrogen at 88.7 µg/mL, disodium hydrogen phosphate at 38 µg/mL and aluminum phosphate at 0.5 mg/mL. The phosphate/saline buffer was added to the formulation to provide pH control and suitable osmolality. Aluminum phosphate was added as an adjuvant to enhance immunogenicity.

2.3.P.2.2 Drug Product
2.3.P.2.2.1 Formulation Development
(1) Adjuvant selection

Aluminium adjuvant has been used in clinic for more than 80 years, which is the first classical adjuvant approved by FDA for human vaccine. In the development of 9-valent pneumococcal conjugate vaccine preparation process, Walvax referred to the successful experience of Wyeth 7-valent pneumococcal conjugate vaccine containing aluminium phosphate adjuvant, and the requirements of European Pharmacopoeia Adsorbed Pneumococcal Conjugate Vaccine (Annex 3-6) for aluminium adjuvant, and determined to add aluminium adjuvant in the preparation of 9-valent Pneumococcal Polysaccharide Conjugate Vaccine final bulk.

Walvax used aluminium hydroxide and aluminium phosphate as adjuvants to formulate 9-valent Pneumococcal Polysaccharide Conjugate Vaccine final bulk. After visual inspection, the appearance of final bulk with aluminium hydroxide adjuvant is milky uneven suspension with coagulum after shaking, and the appearance of final bulk with aluminium phosphate adjuvant is milky homogeneous suspension after shaking. Therefore, in the preparation of 13-valent Pneumococcal Polysaccharide Conjugate Vaccine final bulk, we choose aluminium phosphate as the adjuvant of the vaccine.

In addition, we checked the instructions of PCV10 (GSK) and PCV13 (Pfizer) which were launched in 2009 and 2010 respectively, and both of them also adopted the formula containing aluminium phosphate adjuvant. Up to now, the vaccine composition of pneumococcal conjugate vaccine approved for marketing in the world is shown in the table below:

Table 142 Vaccine Composition of PCV7, PCV13 and PCV10	
Product	PCV7 (Pfizer) PCV13 (Pfizer) PCV10 (GSK)
Composition	18C: TT, 19F: DT, other
Carrier protein	CRM197 CRM197
serotypes: PD	6B: 4.4 µg/vial
6B: 4 µg/vial 4, 18C, 19F: 3 µg/vial	
Polysaccharide 1, 3, 4, 5, 6A, 7F, 9V, 14, 4, 9V, 14, 18C, 19F 23F: 1, 5, 6B, 7F, 9V, 14, 23F: content 18C, 19A, 19F, 23F: 2.2 4µg/vial 1 µg/vial µg/vial	
0.5 mg of aluminium Aluminium phosphate, Aluminium phosphate, 0.5	
Adjuvant	phosphate 0.125 mg of aluminium mg of aluminium
100 µg of Tween-80	
Other excipients	Sodium chloride Sodium chloride
295 µg of succinate	

It can be seen from the Table 142 that all the three approved pneumococcal conjugate vaccines in the world are adsorbed with aluminium adjuvant without exception, and all of them use aluminium phosphate adjuvant, which can effectively induce the production of high-efficient protective antibodies.

(2) Specification and dose study

In the pre-clinical study of 13-valent Pneumococcal Polysaccharide Conjugate Vaccine, Walvax referred to the actual situation that the content of each antigen in all kinds of listed conjugate vaccines was 1-10 µg, and referred to the specification of PCV7 (Pfizer) and PCV13 (Pfizer), our company designed and carried out the specification and dose research comparing to PCV7 (Pfizer), and determined the product specification and dose through the immunogenicity test.

3.2.P.2.2.2 Overages
Not Applicable

3.2.P.2.2.3 Physicochemical and Biological Properties
Appearance: should be milky uniform suspension.

3.2.P.2.3 Manufacturing Process Development
3.2.P.2.3.1 Determination of Adjuvants

According to the European Pharmacopoeia (5.0) Pneumococcal polysaccharide conjugate vaccine (adsorption) and other related literature, the characteristics (structure, charge, molecular size, etc.) of pneumococcal polysaccharide conjugate antigen, aluminium adjuvant is proposed. Aluminium hydroxide and aluminium phosphate commonly used in vaccines as adjuvant. After compounding with aluminium hydroxide and aluminium phosphate, respectively, the appropriate adjuvant can be determined through observing the appearance of complex.

Table 143 Appearance after Compounding with Adjuvant

Sample Appearance

Aluminium hydroxide milky uneven suspension with coagulum after shaking

Aluminium phosphate milky homogeneous suspension after shaking

The results in the Table 143 shows that aluminium phosphate has good compatibility with polysaccharides, while aluminium hydroxide is incompatible. Aluminium phosphate is selected as the adjuvant for this vaccine.

3.2.P.2.3.2 Determination of Buffer System

Antigenicity of an antigen is affected by its primary (molecular formula) and higher order (secondary and higher) structures. Higher order structures of antigen are dramatically affected by the solution system it's in, factors include pH, osmotic pressure, ionic strength, and so on, among which pH more drastically affects its higher order structures. Apart from controlling the osmotic

pressure the same to that of body fluid and ionic strength within bearing range of human body, a suitable buffer system should be developed as well to keep constant pH.

In reference to Chinese Pharmacopeia 2010 Version, Volume III, Manufacturing and Testing Procedure of Diphtheria, Tetanus and Acellular Pertussis Combined Vaccine, Adsorbed, and Walvax's Production Process of Final Bulk Diphtheria, Tetanus and Acellular Pertussis Combined Vaccine, adsorbed, final bulk was added to a phosphate buffer solution of pH 6.5, then filled to vials and conducted accelerated experiment at 37 °C. Appearance and pH of the sample were constantly tested to ensure a constant pH, thus evaluate the phosphate buffer system.

The test results showed that while formulating the final bulk, the NaH₂PO₄-Na₂HPO₄ buffer can stabilize the pH value effectively and ensure that the final bulk maintains a good physical appearance. The buffer system for final bulk is as follow:

NaH₂PO₄ 88.7 µg/mL
Na₂HPO₄ 38.0 µg/mL
NaCl 8.5 mg/mL

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3.2.P.2.3.3 Establishment of Final Bulk Preparation Process

In reference to Chinese Pharmacopeia 2010 Version, Volume III, Manufacturing and Testing Procedure of Diphtheria, Tetanus and Acellular Pertussis Combined Vaccine, Adsorbed, and Walvax's Manufacturing Process of Final Bulk Diphtheria, Tetanus and Acellular Pertussis Combined Vaccine, adsorbed, and Walvax's Production and Testing Procedure of 9-valent Pneumococcal Conjugate Vaccine, Walvax established the operation procedure of final bulk. Through the detection of appearance, pH, antigen content, aluminium phosphate content, sodium chloride content, sterility, pyrogen, abnormal toxicity index, to investigate whether it meets the predetermined standard.

The preparation process of final bulk is determined as follows:

(1) Perpetration the excipients stock solution

Weigh the medicinal grade sodium chloride in the container according to the formula amount, dry heat sterilization and remove the pyrogen; weigh the medicinal grade phosphate in another container according to the formula amount, dry heat sterilization and remove the pyrogen; add the formula amount of WFI into the container, mix well, autoclave and obtain the excipients stock solution, place at 2-8 °C for use;

(2) Weigh the formula amount of each serotype polysaccharide bulk conjugate, mix well to obtain the mixture.

(3) Measure the formula amount of aluminium phosphate adjuvant, phosphate stock solution (pH 6.5), sodium chloride stock solution and WFI, mix well to obtain the excipients mixture.

(4) Pump the polysaccharide bulk conjugate mixture into the excipients mixture while stirring to obtain the final bulk.

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3.2.P.2.4 Container Closure System

3.2.P.2.4.1 Primary Packaging (Vial)

Table 144 Brief Profile of Packaging Materials

Packaging material Vendor Specification Standard

Shuangfeng Gerresheimer Pharmaceutical

1 mL/vial

Neutral borosilicate Glass (Danyang)Co., Ltd. YBB00292005-2-

glass vial Ningbo Zhengli Pharmaceutical Packing 2015

1 mL/vial

Co.ltd.

Halogenated butyl Shengzhou Pharmaceutical Packing 11G002-A17-

Q/SRPYBB10-2019

rubber plugs for Material (China) Co., Ltd. D2T-S

injection Jiangsu Hualan New Pharmaceutical YBB00042005-2015

11-B

(chlorinated) Material Co., Ltd. YBB00052005-2015

Aluminium-plastics Hebei Jinhuan Packaging Co., Ltd 11.3×6.4 mm

YBB00372003-2015

combined caps Jingjiang Huaxin Capping Co., Ltd. 11×6.3 mm

3.2.P.2.4.2 Vial Compatibility Research

The compatibility research of vial packaging was conducted on neutral borosilicate glass vial (produced by Ningbo Zhengli Pharmaceutical Packaging Co., LTD) and halogenated butyl rubber plugs for injection (chlorinated) (produced by Jiangsu Hualan Pharmaceutical New Materials Co., Ltd).

3.2.P.2.4.2.1 Compatibility Study on Neutral Borosilicate Glass Vial

Objective:

This test is to investigate the compatibility of PCV13 and middle-boro glass injection vials (hereinafter referred to as vials) manufactured by Ningbo Zhengli as inner packing material, to evaluate the migration of the packaging material and its safety, the influence of vaccine on packing systems, and then to provide basis of selection of packaging materials and safety of drug usage.

Basis:

1 The compatibility trail guidelines between pharmaceutical packaging materials and drug substances (YBB00142002-2015)

2 The compatibility research technology guidelines of chemical pharmaceutical injection and glass pharmaceutical packaging container (trial implementation)

3 The criterion of neutral borosilicate glass vial (YBB00292005-2-2015)

4 Chinese pharmacopoeia, 4th edition, 2015

5 ICH Q3D, Introduction of elements and impurities, Step 4, 16 December 2014

6 United States Pharmacopeia <1660>

Experimental design:

1.The influence of glass container on drug substances

1.1 Migration tests: 1 batch of neutral borosilicate glass vial were used to package 3 batches of

vaccine to conduct compatibility test and placed upright or upside down. At the test points of

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accelerated test and long-term test, elements of extraction were studied with validated method, estimating possibly migration of elements to drug substance.
1.2 Migration of organics of rubber plugs: testing small molecules antioxidants content such as 2, 6-tert-butyl-4-methyl phenol (BHT) in every point. (Test results are used to evaluate compatibility and whether to continue the test.)
2 The test items of influences of pharmaceuticals on inner surface of glass containers.
2.1 The trend of packaging materials damage and erosion: 1 batch of vaccine were filled to 3 batches of neutral borosilicate glass vial placed upright or upside down to test compatibility. At the test points of accelerated test and long-term, samples were taken to study inner surface erosion and silicon, boron elements in the drug by staining, atomic absorption spectrum, etc. The influence of drug substances on vials was evaluated by analysis of trends of silicon and silicon/boron rate.

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Table 145 Experimental Design for Compatibility Study of Neutral Borosilicate Glass Vial
Packaging material batch
Content of No. Storage
Product name Batch No. Categories Sampling time Test items
research Batch No. Batch No. of condition
of vial rubber plug
0 month, 1st month, 2nd
month, 3rd month, 6th
month, 9th month, Cadmium, Lead,
3 batches of PCV13 2-8 ■
GI201709004 12th month, Arsenic, Silicon,
and corresponding 1 Migration 18th month,
GI201709009 201705006 X201709009 Boron, Antimony,
batch of packaging substances
GI201709010 24th month, 36th month Calcium,
material
25±2 ■, 0 month, 1st month, 2nd Aluminum
13-valent
humidity month, 3rd month, 6th
pneumococcal
60%±10% month
polysaccharide
conjugate 0 month, 1st month, 2nd
Glass peeling
month, 3rd month, 6th
vaccine (staining method),
month, 9th month,
1 batch of PCV13 Stability 2-8 ■ visible foreign
GI201709004 201705006 X201709009 12th month,
and corresponding 3 of packa bodies, changes of
GI201709006 201706001 X201709010 18th month,
batches of ging silicon Si
GI201709008 201706044 X201709011 24th month, 36th month
packaging material material concentration and
25±2 ■, 0 month, 1st month, 2nd
silicon / boron
humidity month, 3rd month, 6th
ratio
60%±10% month

Safety assessment of migration test results
Migration test is based on confirming potential targets and validating relevant methodology. Target substances in samples are tested and compared to related threshold value, Safety of migrated substance is then assessed.

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Experimental results
1. Package stability assessment results of 3 batches samples under 5±3 ■ condition.

Table 146 Silicon Migration Amount (µg/Vial)
18th
Batch No. criterion 0 month 3rd month 6th month 9th month 12th month
month
GI201709004 0.730 1.153 1.581 1.294 1.454 1.464
GI201709006 - 0.705 1.159 1.839 1.220 1.286 1.318
GI201709008 0.781 1.115 1.997 1.220 1.281 1.254

Table 147 Boron Migration Amount (µg/Vial)
12th 18th
Batch No. criterion 0 month 3rd month 6th month 9th month
month month
GI201709004 0.133 0.243 0.331 0.196 0.228 0.242
GI201709006 - 0.083 0.226 0.419 0.226 0.251 0.268
GI201709008 0.144 0.225 0.399 0.220 0.231 0.336

Table 148 Staining with Methylene Blue
12th 18th

Batch No. criterion 0 month 3rd month 6th month 9th month
month month
No No No No No No
GI201709004 obvious obvious obvious obvious obvious
staining staining staining staining staining staining
No No No No No No
GI201709006 obvious obvious obvious obvious obvious obvious
staining staining staining staining staining staining
No No No No No No
GI201709008 obvious obvious obvious obvious obvious
staining staining staining staining staining staining

Table 149 Visible Foreign Body
12th 18th
Batch No. criterion 0 month 3rd month 6th month 9th month
month month
Smaller Less than Less than Less than Less than Less than Less than
GI201709004
than the the limit the limit the limit the limit the limit the limit
slightly Less than Less than Less than Less than Less than Less than
GI201709006
visible the limit the limit the limit the limit the limit the limit
foreign Less than Less than Less than Less than Less than Less than
GI201709008
body the limit the limit the limit the limit the limit the limit

Conclusion:
3 batches of samples were placed under 5±3 ■ for 18 months, test results shows no obvious
peeling or visible foreign body larger than limit.

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2. Migration test results of 3 batches samples under 5±3 ■
Table 150 Migration Elements Test Results (µg/Vial)
Criterion or/AET
Batch No. 0 month 3rd month 6th month 9th month 12th month 18th month
value
As: < 0.006 As: < 0.006 As: < 0.006 As: < 0.006 As: < 0.006 As: < 0.006
Sb: 0.066 Sb: 0.075 Sb: 0.086 Sb: 0.100 Sb: 0.109 Sb: 0.087
GI201709004 Pb: 0.114 Pb: 0.059 Pb: 0.093 Pb: 0.035 Pb: 0.059 Pb: 0.073
Cd: 0.023 Cd: 0.007 Cd: 0.033 Cd: 0.016 Cd: 0.016 Cd: 0.020
Ca: 0.767 Ca: 0.915 Ca: 0.899 Ca: 1.001 Ca: 1.003 Ca: 1.013
arsenic: 15 As: < 0.006 As: < 0.006 As: < 0.006 As: < 0.006 As: < 0.006 As: < 0.006
antimony: 90 Sb: 0.140 Sb: 0.137 Sb: 0.146 Sb: 0.134 Sb: 0.138 Sb: 0.124
GI201709009 lead: 5 Pb: 0.122 Pb: 0.065 Pb: 0.110 Pb: 0.053 Pb: 0.068 Pb: 0.089
cadmium: 2 Cd: 0.024 Cd: 0.011 Cd: 0.038 Cd: 0.020 Cd: 0.018 Cd: 0.018
calcium: - Ca: 0.815 Ca: 0.974 Ca: 0.879 Ca: 0.977 Ca: 1.010 Ca: 0.914
As: < 0.006 As: < 0.006 As: < 0.006 As: < 0.006 As: < 0.006 As: < 0.006
Sb: 0.081 Sb: 0.081 Sb: 0.085 Sb: 0.090 Sb: 0.097 Sb: 0.078
GI201709010 Pb: 0.133 Pb: 0.053 Pb: 0.087 Pb: 0.076 Pb: 0.059 Pb: 0.069
Cd: 0.025 Cd: 0.009 Cd: 0.037 Cd: 0.017 Cd: 0.012 Cd: 0.016
Ca: 0.797 Ca: 0.936 Ca: 0.841 Ca: 1.035 Ca: 1.000 Ca: 0.992

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Table 151 Organic in Rubber Plug Migration Test Results (µg/Vial)
Criterion or 12th 18th
Batch No. 0 month 3rd month 6th month 9th month
AET value month month
Not
GI201709004 2, 6-tert- 0.04 0.03 0.15 0.11 0.10
detected
butyl, 4-
Not
GI201709009 methyl 0.04 0.03 0.22 0.10 0.10
detected
phenol(BHT):
Not
GI201709010 - 0.04 0.04 0.17 0.09 0.10
detected
Conclusion:
Migration test results showed arsenic, antimony, lead, cadmium, calcium content for the 3 batches
of PCV13 samples are far lower than AET value.

3. Packaging material stability results of 3 batches of samples under 25±2 ■

Table 152 Silicon Migration Amount (µg/Vial)
Batch No. Criterion 0 month 1st month 2nd month 3rd month 6th month
GI201709004 0.730 1.153 1.581 2.015 3.477
GI201709006 - 0.705 1.159 1.839 2.005 3.969
GI201709008 0.781 1.115 1.997 2.183 3.693

Table 153 Boron Migration Amount (µg/Vial)
Batch No. Criterion 0 month 1st month 2nd month 3rd month 6th month
GI201709004 0.133 0.243 0.331 0.511 0.704
GI201709006 - 0.083 0.226 0.419 0.564 0.657
GI201709008 0.144 0.225 0.399 0.611 0.781

Table 154 Staining with Methylene Blue
Batch No. Criterion 0 month 1st month 2nd month 3rd month 6th month
No obvious No obvious No obvious No obvious No obvious
GI201709004
staining staining staining staining staining
No obvious No obvious No obvious No obvious No obvious No obvious
GI201709006
staining staining staining staining staining staining
No obvious No obvious No obvious No obvious No obvious
GI201709008
staining staining staining staining staining

Table 155 Visible Foreign BoDy
Batch No. Criterion 0 month 1st month 2nd month 3rd month 6th month
Smaller Less than Less than Less than Less than Less than
GI201709004
than the the limit the limit the limit the limit the limit
slightly Less than Less than Less than Less than Less than
GI201709006
visible the limit the limit the limit the limit the limit
foreign Less than Less than Less than Less than Less than
GI201709008
body the limit the limit the limit the limit the limit

Conclusion:
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Test results showed no obvious peeling or visible foreign body larger than limit in the samples.

4. Migration substances results of 3 batches samples under 25±2 ■ condition

Table 156 Rubber Plug Organic Migration Test Results (µg/vial)
3rd month 6th month
Criterion or 1st 2nd
Batch No. 0 month Upside Upside
AET value month month upright upright
down down
Not Not
GI201709004 0.03 0.04 0.04 0.03 0.03
detected detected
2, 6-tert-
butyl, 4-
Not
GI201709009 methyl 0.01 0.02 0.03 0.04 0.03 0.04
detected
phenol
(BHT): -
Not Not
GI201709010 0.03 0.04 0.03 0.03 0.02
detected detected

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Table 157 Migration Elements Test Results (µg/vial)
2nd month 3rd month
Criterion or
Batch No. 0 month 0 month 1st month
AET value Upright Upside down upright Upside down
As: < 0.006 As: < 0.006 As: < 0.006 As: < 0.006 As: < 0.006 As: < 0.006 As: < 0.006
Sb: 0.066 Sb: 0.111 Sb: 0.129 Sb: 0.127 Sb: 0.082 Sb: 0.101 Sb: 0.100
GI201709004 Pb: 0.114 Pb: 0.136 Pb: 0.071 Pb: 0.056 Pb: 0.055 Pb: 0.121 Pb: 0.093
Cd: 0.023 Cd: 0.017 Cd: 0.017 Cd: 0.010 Cd: 0.015 Cd: 0.037 Cd: 0.035
Ca: 0.767 Ca: 1.047 Ca: 0.947 Ca: 1.019 Ca: 0.989 Ca: 0.964 Ca: 1.009
arsenic:15 As: < 0.006 As: < 0.006 As: < 0.006 As: < 0.006 As: < 0.006 As: < 0.006 As: < 0.006
antimony:90 Sb: 0.140 Sb: 0.153 Sb: 0.176 Sb: 0.156 Sb: 0.155 Sb: 0.153 Sb: 0.169
GI201709009 lead:5 Pb: 0.122 Pb: 0.127 Pb: 0.067 Pb: 0.058 Pb: 0.073 Pb: 0.082 Pb: 0.188
cadmium:2 Cd: 0.024 Cd: 0.023 Cd: 0.018 Cd: 0.013 Cd: 0.015 Cd: 0.035 Cd: 0.035
calcium: - Ca: 0.815 Ca: 0.961 Ca: 0.952 Ca: 1.033 Ca: 1.079 Ca: 1.035 Ca: 1.011
As: < 0.006 As: < 0.006 As: < 0.006 As: < 0.006 As: < 0.006 As: < 0.006 As: < 0.006
Sb: 0.081 Sb: 0.080 Sb: 0.093 Sb: 0.082 Sb: 0.083 Sb: 0.084 Sb: 0.093
GI201709010 Pb: 0.133 Pb: 0.135 Pb: 0.60 Pb: 0.053 Pb: 0.073 Pb: 0.060 Pb: 0.104
Cd: 0.025 Cd: 0.027 Cd: 0.017 Cd: 0.015 Cd: 0.017 Cd: 0.036 Cd: 0.035
Ca: 0.797 Ca: 0.945 Ca: 0.953 Ca: 0.975 Ca: 1.057 Ca: 0.998 Ca: 1.114

Conclusion:
Migration test results showed arsenic, antimony, lead, cadmium, calcium content of 3 batches of PCV13 samples are far lower than AET value.

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3.2.P.2.4.2.1 Compatibility Study on Halogenated Butyl Rubber Plugs

Objectives:
The test is conducted to evaluate the compatibility of PCV13 with halogenated butyl rubber plugs which used to injection were made by Jiangsu Hualan Pharmaceutical New Materials Co., Ltd, and to evaluate possible migration substances and its safety, test the influence of vaccine to

package system, and then provide bases of selecting package materials and using drug safely.

Basis:

1 Compatibility test guideline for pharmaceutical packaging material and drug substance (YBB00142002-2015)

2 Compatibility research technology guidelines for chemical pharmaceutical and elastomeric seal (trial implementation)

3 Standard of halogenated rubber plugs (YBB00042005-2015)

4 Chinese pharmacopoeia, 4th edition, 2015

5 ICH Q3D, Introduction of elements and impurities, Step 4, dated 16 December 2014

6 United States Pharmacopeia <1663>

7 United States Pharmacopeia <1664>

Experimental design

1 Extraction test

1.1 Elements Extraction: Elements are the easiest extractable substances in rubber plugs.

Referencing Compatibility research technology guideline for chemical pharmaceutical and elastomeric seal (trial implementation), and ICH Q3D, we focus on detection of the following elements: magnesium, silicon, titanium, cadmium, lead, arsenic, mercury, cobalt, vanadium, nickel, antimony, lithium and copper. If high content of other elements, they shall be listed as test items for compatibility test.

1.2 Organics extraction: organics are mainly used as assistant agents during rubber plugs manufacturing process. Referencing Compatibility research technology guideline for chemical pharmaceutical and elastomeric seal (trial implementation), major assistant agents are antioxidants (BHT and its degradants), sulfides (disulfuric alkyl phenol and its degradants), volatile organic compounds (C13, C21 rubber oligomers), lubricants (dimethyl silicone oil).

2 Rubber plugs will age as time goes by, physical properties will change, and it may influence fragments for the rubber plugs, thus it was investigated.

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Table 158 Experimental Design for Compatibility Study of Halogenated Butyl Rubber Plugs

Batch No. of package

Test Storage

Product name Batch No. Categories Sampling time Test items

content Batch No. Batch No. of condition

of vials rubber plugs

0 month, 3rd month, 6th

3 batches of month, 9th month, 12th month,

PCV13 and 2-8 ■ metal elements: Zn, Ca, Fe,

GI201709004 18th month, 24th month, 36th

corresponding Migration month Si, Mg, Cu, Li, Sb, Ni, V, Co,

GI201709009 201705006 X201709009 Hg, As, Pb, Cd, Ti;

1 batch of substances

GI201709010 organic: C13 oligomers, C13

packaging

13-valent material chlorine oligomer, C21

pneumococcal oligomers, stearic acid,

polysaccharide 25±2 ■ palmitic acid, BHT,

conjugate 0 month, 1st month, 2nd antioxidant 1076, antioxidant

humidity

vaccine month, 3rd month, 6th month 1010, silicone oil, Vulcanizer

60%±10%

fragments (sulfur, p-tert-amyl

0 month, 3rd month, 6th phenol), 2-Chloro-4-(tert-

1 batch of month, 9th month, 12th month,

2-8 ■ pentyl)phenol);

PCV13 and Stability 18th month, 24th month, 36th Puncture exfoliation

GI201709004 201705006 X201709009 month

corresponding of packa

GI201709006 201706001 X201709010

3 batches of ging

GI201709008 201706044 X201709011 25±2 ■

packaging material 0 month, 1st month, 2nd

humidity

material month, 3rd month, 6th month

60%±10%

Note: Compatibility test are conducted with 3 batches of vaccines corresponding to 1 batch of halogenated rubber plugs and 1 batch of vaccine corresponding to 3 batches

halogenated butyl rubber plugs, all placed upside down. Extractions are studied using validated methods.

Puncture exfoliation function of rubber plugs after stability test is

tested simultaneously.

Safety assessment of migration test results

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1 Based on the monitoring test results, making an initial estimation of the possibility of actual drug's

ionic migration in rubber plugs; through inspecting

pharmaceutical regular items, the categories and content of migration substances in migration test,

evaluating the influences of rubber plugs to drug and

absorption experiment, then analyzing the package system have compatibility with PCV13 or not.

2 Analytical and evaluation threshold (AET): AET value is calculated based on Compatibility research

technology guideline for chemical pharmaceutical

injection and pharmaceutical packaging container (trial implementation), Annex 6, metal catalyst or metal

agent residue approved by EMEA/CHMP the

PDE value of ICH Q3D, the safety carcinogen threshold (SCT) of 1.5 µg/day recommended by European

Medicine Agency and EPA database. The AET

values may vary for different packaging and dosage. In order to meet AET value, the test method may be adjusted.

Calculation method: AET= PDE (or SCT)/maximum dose for one day/specification of each formulation.

Test results
1 Packaging material stability results of 3 batches of samples under 5■±3■
1.1 Test results of 1 batch of PCV13 and the corresponding 3 batches of packaging material

Table 159 Antioxidants Test Results (μg/mL)
Detection
Batch No. Test items AET 0 month 3rd month 6th month 9th month 12th month 18th month
limit
GI201709004 Antioxidant1010 Not detected Not detected Not detected Not detected Not detected Not detected
0.026 197520
GI201709006 Antioxidant Not detected Not detected Not detected Not detected Not detected Not detected
0.014 3600
BHT
GI201709008 0.058 0.058 Not detected Not detected Not detected Not detected Not detected Not detected
Antioxidant1076

Table 160 Stearic Acid and Palmitic Acid Test Results (μg/mL)
Detection
Batch No. Test items AET 0 month 3rd month 6th month 9th month 12th month 18th month
limit
GI201709004 Not detected Not detected Not detected Not detected Not detected Not detected
GI201709006 stearic acid 0.063 3600 Not detected Not detected Not detected Not detected Not detected Not
detected
palmitic acidtest 0.100 3600
GI201709008 Not detected Not detected Not detected Not detected Not detected Not detected
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Table 161 Vulcanizing Agent and Fragments Test Results (μg/mL)
Detection
Batch No. Test items AET 0 month 3rd month 6th month 9th month 12th month 18th month
limit
GI201709004 sulfur Not detected Not detected Not detected Not detected Not detected Not detected
0.006 180
GI201709006 p-tert-pentylphenol Not detected Not detected Not detected Not detected Not detected Not
detected
0.039 3600
2-chloro-p-tert-
GI201709008 0.032 180 Not detected Not detected Not detected Not detected Not detected Not detected
pentylphenol

Table 162 Oligomers Test Results (μg/mL)
Detection
Batch No. Test items AET 0 month 3rd month 6th month 9th month 12th month 18th month
limit
GI201709004 C13 oligomer 0.304 3600 Not detected Not detected Not detected Not detected Not detected Not
detected
GI201709006 C21 oligomer 0.304 3600 Not detected Not detected Not detected Not detected Not detected Not
detected
GI201709008 C13 chlorine oligomer 0.304 3 Not detected Not detected Not detected Not detected Not
detected Not detected

Table 163 Lubricant Test Results (μg/mL)
Detection
Batch No. Test items AET 0 month 3rd month 6th month 9th month 12th month 18th month
limit
GI201709004 Not detected < 3.49 4.94 < 3.49 Not detected 72.77
GI201709006 silicone oil 1.05 4800 Not detected < 3.49 < 3.49 4.20 Not detected 54.93
GI201709008 Not detected < 3.49 < 3.49 5.06 Not detected 46.79

Table 164 Puncture Exfoliation Test Results
Batch No. Test items criterion 0 month 3rd month 6th month 9th month 12th month 18th month
GI201709004 1 1 0 2 3 2
Puncture No more than
GI201709006 1 0 0 1 3 4
exfoliation 5 particles
GI201709008 1 2 0 3 4 5

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Table 165 Metal Elements Test Results (μg/mL)
Detection
Batch No. Test items AET 0 month 3rd month 6th month 9th month 12th month 18th month
limit
Li: Not detected
Mg: 0.4788
Li: 0.00167 Li: < 0.00149 Li: 0.00163 Li: 0.00169
Li: 0.08699 Ca: Not detected
Mg: 0.82339 Mg: 0.98546 Mg: 0.42548 Mg: 0.73601
Mg: 0.77376 Ti: 0.13570
Ca: 0.54453 Ca: 1.08148 Ca: 0.16099 Ca: 0.94202
Ca: 0.17784 V: Not detected
Ti: 0.00379 Ti: 0.04391 Ti: 0.00235 Ti: 0.00950
Ti: Not detected Fe: 0.03478
V: 0.00083 V: 0.00100 V: 0.00092 V: 0.00188
V: Not detected Co: Not
Fe: 0.21540 Fe: 0.10618 Fe: 0.04825 Fe: 0.11657
Fe: 0.11331 detected

Co: 0.00113 Co: 0.00118 Co: 0.00113 Co: 0.00077
Co: Not detected Ni: Not detected
lithium 0.00045 500 Ni: 0.00941 Ni: 0.00814 Ni: 0.00548 Ni: 0.01239
Ni: Not detected Cu: Not
GI201709004 magnesium 0.00229 N/A Cu: 0.03895 Cu: 0.02329 Cu: 0.00180 Cu: 0.01654
Cu: Not detected detected
calcium 0.00466 N/A Zn: 0.08842 Zn: 0.14528 Zn: < 0.00669 Zn: 0.22055
Zn: 7.86236 Zn: 29.88296
titanium 0.00038 N/A As: 0.00041 As: 0.00057 As: 0.00049 As: < 0.00033
As: Not detected As: Not detected
vanadium 0.00005 20 Cd: 0.00007 Cd: 0.0012 Cd: 0.00028 Cd: Not detected
Cd: Not detected Cd: Not
ferrum 0.00524 N/A Sb: 0.14036 Sb: 0.12261 Sb: 0.07899 Sb: 0.12662
Sb: 0.15182 detected
cobalt 0.00002 10 Hg: Not Hg: Not Hg: Not Hg: Not
Hg: 0.01677 Sb: 0.02345
nickel 0.00012 40 detected detected detected detected
Pb: 0.00143 Hg: Not
cuprum 0.00020 600 Pb: 0.00458 Pb: 0.00479 Pb: 0.00058 Pb: 0.00299
Si: 0.7891 detected
zinc 0.00201 N/A Si: 0.9182 Si: 0.6418 Si: 0.7108 Si: 0.7941
Pb: Not detected
arsenic 0.00010 30
Si: 1.172
cadmium 0.00002 4
Li: 0.00174 Li: 0.00178 Li: 0.00286 Li: 0.00242 Li: 0.06881 Li: Not detected
antimony 0.00003 180
Mg: 0.68776 Mg: 1.49981 Mg: 0.68476 Mg: 0.75734 Mg: 0.68625 Mg: 0.59680
mercury 0.00217 6
Ca: 0.46144 Ca: 17.37447 Ca: 0.32255 Ca: 0.91556 Ca: 0.18008 Ca: Not detected
plumbum 0.00008 10
Ti: 0.00343 Ti: 0.07633 Ti: 0.00515 Ti: 0.00860 Ti: Not detected Ti: 0.13919
silicon 0.0258 N/A
V: 0.00086 V: 0.00163 V: 0.00131 V: 0.00158 V: Not detected V: Not detected
Fe: 0.18029 Fe: 0.32853 Fe: 0.23299 Fe: 0.13734 Fe: 0.13273 Fe: 0.06252
GI201709006 Co: 0.00097 Co: 0.00139 Co: 0.00155 Co: 0.00078 Co: Not detected Co: Not
Ni: 0.01262 Ni: 0.02192 Ni: 0.01115 Ni: 0.01926 Ni: Not detected detected
Cu: 0.02227 Cu: 0.08201 Cu: 0.00444 Cu: 0.01800 Cu: Not detected Ni: Not detected
Zn: 0.19164 Zn: 0.23800 Zn: 0.09644 Zn: 0.20072 Zn: 7.39028 Cu: Not
As: 0.00044 As: 0.00063 As: 0.00040 As: < 0.00033 As: Not detected detected
Cd: < 0.00007 Cd: 0.00019 Cd: 0.00034 Cd: < 0.00008 Cd: Not detected Zn: 32.66239
Sb: 0.12669 Sb: 0.13346 Sb: 0.13921 Sb: 0.14038 Sb: 0.16046 As: Not detected
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Hg: Not Hg: Not Hg: < 0.00722 Hg: Not Hg: 0.01174 Cd: Not
detected detected Pb: 0.00095 detected Pb: 0.00245 detected
Pb: 0.00322 Pb: 0.01477 Si: 0.7222 Pb: 0.00322 Si: 0.9076 Sb: 0.03309
Si: 0.7380 Si: 0.6242 Si: 0.7458 Hg: Not
detected
Pb: Not detected
Si: 1.290
Li: Not detected
Mg: 0.61977
Li: 0.00199 Li: < 0.00149 Li: 0.00163 Li: 0.00351
Li: 0.17980 Ca: Not detected
Mg: 0.69517 Mg: 1.20801 Mg: 0.25839 Mg: 0.73223
Mg: 0.70120 Ti: 0.13932
Ca: 0.39993 Ca: 1.62270 Ca: 0.07178 Ca: 0.87837
Ca: 0.14203 V: Not detected
Ti: 0.00334 Ti: 0.06252 Ti: 0.00173 Ti: 0.00804
Ti: Not detected Fe: 0.08505
V: 0.00082 V: 0.00136 V: 0.00079 V: 0.00146
V: Not detected Co: Not
Fe: 0.21704 Fe: 0.28330 Fe: 0.03018 Fe: 0.11824
Fe: 0.25342 detected
Co: 0.00101 Co: 0.00119 Co: 0.00095 Co: 0.00069
Co: Not detected Ni: Not detected
Ni: 0.00802 Ni: 0.05017 Ni: 0.00304 Ni: 0.01224
Ni: 0.00721 Cu: Not
GI201709008 Cu: 0.01584 Cu: 0.02039 Cu: 0.00111 Cu: 0.01622
Cu: Not detected detected
Zn: 0.08191 Zn: 0.13340 Zn: 0.01116 Zn: 0.18354
Zn: 10.43730 Zn: 33.33363
As: 0.00038 As: 0.00058 As: < 0.00033 As: 0.00034
As: Not detected As: Not detected
Cd: < 0.00007 Cd: 0.00046 Cd: 0.00025 Cd: < 0.00008
Cd: Not detected Cd: Not
Sb: 0.13308 Sb: 0.10490 Sb: 0.04576 Sb: 0.13709
Sb: 0.15488 detected
Hg: Not Hg: Not Hg: Not Hg: Not
Hg: 0.01099 Sb: 0.06017
detected detected detected detected
Pb: 0.00076 Hg: Not
Pb: 0.00322 Pb: 0.00769 Pb: 0.00031 Pb: 0.00282
Si: 0.7053 detected
Si: 0.7380 Si: 0.6434 Si: 0.7044 Si: 0.7134
Pb: Not detected
Si: 1.428

Conclusion:

Test results showed that no precipitate beyond threshold value hadn't been found in PCV13, and puncture exfoliation results also meet requirement of the test.

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2 Migration test results of 3 batches of samples under 5±3 ■
2.1 Test results of 3 batches of PCV13 and the corresponding 1 batch of packaging material

Table 166 Antioxidants Test Results (µg/mL)

Detection	Batch No.	Test items	AET	0 month	3rd month	6th month	9th month	12th month	18th month
limit									
GI201709004	Antioxidant	1010	0.026	197520	Not detected	Not detected	Not detected	Not detected	Not detected
GI201709009	Antioxidant	BHT	0.014	3600	Not detected	Not detected	Not detected	Not detected	Not detected
GI201709010	Antioxidant	1076	0.058	0.058	Not detected	Not detected	Not detected	Not detected	Not detected

Table 167 Stearic Acid and Palmitic Acid Test Results (µg/mL)

Detection	Batch No.	Test items	AET	0 month	3rd month	6th month	9th month	12th month	18th month
limit									
GI201709004	stearic acid	0.063	3600	Not detected	Not detected	Not detected	Not detected	Not detected	Not detected
GI201709009	palmitic acid	0.100	3600	Not detected	Not detected	Not detected	Not detected	Not detected	Not detected
GI201709010				Not detected	Not detected	Not detected	Not detected	Not detected	Not detected

Table 168 Vulcanizing Agent and Fragments Test Results (µg/mL)

Detection	Batch No.	Test items	AET	0 month	3rd month	6th month	9th month	12th month	18th month
limit									
GI201709004	Sulfur	Not detected		Not detected	Not detected	Not detected	Not detected	Not detected	Not detected
GI201709009	sulfur p-tert-pentylphenol	0.039	3600	Not detected	Not detected	Not detected	Not detected	Not detected	Not detected
GI201709010	p-tert-pentylphenol	0.131		Not detected	Not detected	Not detected	Not detected	Not detected	Not detected
	2-chloro-p-tert-2-chloro-p-tert-	0.032	180	Not detected	Not detected	Not detected	Not detected	Not detected	Not detected
	pentylphenol pentylpheno	Not detected		Not detected	Not detected	Not detected	Not detected	Not detected	Not detected

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Table 169 Oligomers Test Results (µg/mL)

Detection	Batch No.	Test items	AET	0 month	3rd month	6th month	9th month	12th month	18th month
limit									
GI201709004	C13 oligomer	0.304	3600	Not detected	Not detected	Not detected	Not detected	Not detected	Not detected
GI201709009	C21 oligomer	0.304	3600	Not detected	Not detected	Not detected	Not detected	Not detected	Not detected
GI201709010	C13 chlorine oligomer	0.304	3	Not detected	Not detected	Not detected	Not detected	Not detected	Not detected

Table 170 Lubricant Test Results (µg/mL)

Detection	Batch No.	Test items	AET	0 month	3rd month	6th month	9th month	12th month	18th month
limit									
GI201709004	silicone oil	1.05	4800	Not detected	< 3.49	< 3.49	41.95	Not detected	40.62
GI201709010				Not detected	< 3.49	< 3.49	4.19	Not detected	36.99

Table 171 Puncture Exfoliation Test Results

Batch No.	Test items	criterion	0 month	3rd month	6th month	9th month	12th month	18th month
GI201709004	puncture	1 1 1 1 4 2						
GI201709009	exfoliation	1 2 0 5 3 3						
GI201709010		0 2 1 1 3 5						

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Table 172 Metal Elements Test Results (µg/mL)

Detection	Batch No.	Test items	AET	0 month	3rd month	6th month	9th month	12th month	18th month
limit									
Li	0.00190	Li	0.00206	Li	Not detected				
Mg	0.92928	Mg	1.05677	Mg	0.74966	Mg	0.55598		
	0.69697		0.60233						

Ca: 0.61799 Ca: 1.61352 Ca: 0.91213 Ca: Not detected
Ca: 0.26763 Ca: 0.13472
Ti: 0.00440 Ti: 0.04682 Ti: 0.00786 Ti: 0.13707
Ti: 0.00351 Ti: Not detected
V: 0.00096 V: 0.00096 V: 0.00168 V: Not detected
V: 0.00111 V: Not detected
Fe: 0.23557 Fe: 0.10607 Fe: 0.12099 Fe: 0.04525
Fe: 0.08204 Fe: 0.08903
Co: 0.00101 Co: 0.00124 Co: 0.00085 Co: Not detected
Co: 0.00153 Co: Not detected
Ni: 0.01101 Ni: 0.01021 Ni: 0.01297 Ni: Not detected
Ni: 0.00761 Ni: Not detected
GI201709004 Cu: 0.06096 Cu: 0.02098 Cu: 0.01708 Cu: Not detected
lithium 0.00045 500 Cu: 0.00387 Cu: Not detected
Zn: 0.30880 Zn: 0.09065 Zn: 0.19601 Zn: 31.55723
magnesium 0.00229 N/A Zn: 0.02561 Zn: 7.34990
As: 0.00043 As: 0.00049 As: 0.00037 As: Not detected
calcium 0.00466 N/A As: 0.00062 As: Not detected
Cd: 0.00034 Cd: 0.00008 Cd: < 0.00008 Cd: Not detected
titanium 0.00038 N/A Cd: 0.00033 Cd: Not detected
Sb: 0.11579 Sb: 0.13016 Sb: 0.15399 Sb: 0.04400
vanadium 0.00005 20 Sb: 0.14694 Sb: 0.10271
Hg: Not Hg: Not Hg: Not Hg: Not
ferrum 0.00524 N/A Hg: < 0.00722 Hg: 0.01344
detected detected detected detected
cobalt 0.00002 10 Pb: 0.00061 Pb: 0.00262
Pb: 0.00486 Pb: 0.00451 Pb: 0.00281 Pb: Not detected
nickel 0.00012 40 Si: 0.7094 Si: 0.5650
Si: 0.9196 Si: 0.6052 Si: 0.8001 Si: 1.245
cuprum 0.00020 600
Li: 0.00261 Li: 0.00157 Li: 0.00347 Li: 0.00291 Li: 1.46484 Li: Not detected
zinc 0.00201 N/A
Mg: 0.97062 Mg: 1.08589 Mg: 0.72365 Mg: 0.76223 Mg: 0.65955 Mg: 0.71261
arsenic 0.00010 30
Ca: 0.54020 Ca: 6.81065 Ca: 0.30925 Ca: 0.90153 Ca: 0.33666 Ca: Not detected
cadmium 0.00002 4
Ti: 0.00678 Ti: 0.03204 Ti: 0.00384 Ti: 0.00806 Ti: Not detected Ti: 0.14160
antimony 0.00003 180
V: 0.00094 V: 0.00135 V: 0.00116 V: 0.00147 V: Not detected V: Not detected
mercury 0.00217 6
Fe: 0.82670 Fe: 0.14309 Fe: 0.10332 Fe: 0.08826 Fe: 2.28585 Fe: 0.10238
plumbum 0.00008 10
Co: 0.00165 Co: 0.00145 Co: 0.00186 Co: 0.00114 Co: Not detected Co: Not detected
silicon 0.0258 N/A
Ni: 0.00750 Ni: 0.01583 Ni: 0.00828 Ni: 0.01308 Ni: 0.30623 Ni: Not detected
GI201709009
Cu: 0.04502 Cu: 0.02328 Cu: 0.00446 Cu: 0.01827 Cu: Not detected Cu: Not detected
Zn: 0.08886 Zn: 1.02877 Zn: 0.00710 Zn: 0.21319 Zn: 47.15704 Zn: 39.77655
As: 0.00050 As: 0.00073 As: 0.00059 As: < 0.00033 As: Not detected As: Not detected
Cd: 0.00008 Cd: 0.00031 Cd: 0.00031 Cd: < 0.00008 Cd: Not detected Cd: Not detected
Sb: 0.21526 Sb: 0.19966 Sb: 0.23129 Sb: 0.23628 Sb: 0.19407 Sb: 0.15011
Hg: Not Hg: Not Hg: Not Hg: Not Hg: 0.01046 Hg: Not
detected detected detected detected Pb: 0.00549 detected
Pb: 0.00856 Pb: 0.01583 Pb: 0.00062 Pb: 0.00302 Si: 0.8204 Pb: Not detected
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Si: 1.1522 Si: 0.6778 Si: 0.8194 Si: 0.8214 Si: 1.301
Li: 0.00247 Li: 0.00154 Li: 0.00286 Li: 0.00230 Li: Not detected
Li: 0.05634
Mg: 0.75731 Mg: 0.89661 Mg: 0.74069 Mg: 0.64910 Mg: 1.03342
Mg: 0.71267
Ca: 0.40346 Ca: 4.61420 Ca: 0.31590 Ca: 0.85770 Ca: 0.35216
Ca: 0.16681
Ti: 0.00708 Ti: 0.02098 Ti: 0.00293 Ti: 0.00762 Ti: 0.15365
Ti: Not detected
V: 0.00077 V: 0.00134 V: 0.00130 V: 0.00129 V: Not detected
V: Not detected
Fe: 0.18003 Fe: 0.12366 Fe: 0.12727 Fe: 0.10889 Fe: 0.43800
Fe: 0.08003
Co: 0.00115 Co: 0.00126 Co: 0.00181 Co: 0.00081 Co: Not detected
Co: Not detected
Ni: 0.01135 Ni: 0.01149 Ni: 0.01419 Ni: 0.01398 Ni: Not detected
Ni: Not detected
GI201709010 Cu: 0.02714 Cu: 0.01456 Cu: 0.00606 Cu: 0.02058 Cu: Not detected
Cu: Not detected
Zn: 0.08161 Zn: 0.27265 Zn: 0.05863 Zn: 0.19646 Zn: 35.69463
Zn: 6.0088
As: 0.00032 As: 0.00073 As: 0.00057 As: < 0.00033 As: Not detected
As: Not detected
Cd: < 0.00007 Cd: 0.00034 Cd: 0.00032 Cd: < 0.00008 Cd: Not detected
Cd: Not detected
Sb: 0.12549 Sb: 0.11255 Sb: 0.13852 Sb: 0.11573 Sb: 0.03417
Sb: 0.13850
Hg: Not Hg: Not Hg: Not Hg: Not Hg: Not
Hg: 0.00969
detected detected detected detected detected
Pb: 0.0087
Pb: 0.00187 Pb: 0.01583 Pb: 0.00062 Pb: 0.00310 Pb: Not detected
Si: 0.6027
Si: 0.9666 Si: 0.6778 Si: 0.8194 Si: 0.7590 Si: 1.352

Conclusion: Test results showed that no precipitate beyond threshold value hadn't been found in PCV13, and puncture exfoliation results also meet requirement of the test.

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3. Packaging material stability results of 3 batches of samples under 25±2 ■
3.1 Test results of 1 batch of PCV13 and the corresponding 3 batches of packaging material

Table 173 Antioxidant Test Results (μg/mL)

Detectio 0 1st 2nd 3rd 6th
Batch No. Test items AET
n limit month month month month
Not Not Not Not Not
GI20170900
detecte detecte detecte detecte detecte
4 antioxidant101
d d d d d
0 19752
0.026 Not Not Not Not Not
GI20170900 antioxidantBH 0
0.014 detecte detecte detecte detecte detecte
6 T 3600
0.058 d d d d d
antioxidant107 0.058
Not Not Not Not Not
GI20170900 6
detecte detecte detecte detecte detecte
8
d d d d d

Table 174 Stearic Acid and Palmitic Acid Test Results (μg/mL)

Detection 1st 2nd 3rd 6th
Batch No. Test items AET 0 month
limit month month month month
Not Not Not Not Not
GI201709004
stearic detected detected detected detected detected
acid 0.063 3600 Not Not Not Not Not
GI201709006
palmitic 0.100 3600 detected detected detected detected detected
acid Not Not Not Not Not
GI201709008
detected detected detected detected detected

Table 175 Vulcanizing Agent Test Results (μg/mL)

Test Detectio AE 0 1st
Batch No. 2nd month 3rd month 6th month
items n limit T month month
sulphur: Not sulphur: Not
detected detected
p-tert- p-tert-
Amylphenol Amylphenol
Not Not
GI2017090 Not : 0.171 : < 0.131
detecte detecte
04 detected 2-Chloro-4- 2-Chloro-4-
d d
sulphur (tert- (tert-
p-tert- pentyl)phen pentyl)phen
Amylphen 180 ol: Not ol: Not
0.006
ol 360 detected detected
0.039
2-Chloro- 0 sulphur: Not sulphur: Not sulphur: Not
0.032
4-(tert- 180 detected detected detected
pentyl) p-tert- p-tert- p-tert-
phenol Amylphenol Amylphenol Amylphenol
Not Not
GI2017090 : < 0.131 : 0.162 : < 0.131
detecte detecte
06 2-Chloro-4- 2-Chloro-4- 2-Chloro-4-
d d
(tert- (tert- (tert-
pentyl)phen pentyl)phen pentyl)phen
ol: Not ol: Not ol: Not
detected detected detected

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sulphur: Not sulphur: Not
detected detected
p-tert- p-tert-
Amylphenol Amylphenol
Not Not
GI2017090 Not : 0.152 : < 0.131
detecte detecte

08 detected 2-Chloro-4- 2-Chloro-4-
d d
(tert- (tert-
pentyl)phen pentyl)phen
ol: Not ol: Not
detected detected

Table 176 Volatile Organics (µg/mL)
Detection 1st 2nd 3rd 6th
Batch No. Test items AET 0 month
limit month month month month

C13 Not Not Not Not Not
GI201709004
oligomer detected detected detected detected detected
0.304 3600
C21 Not Not Not Not Not
GI201709006 0.304 3600
oligomer detected detected detected detected detected
0.304 3
C13 chlorine Not Not Not Not Not
GI201709008
oligomer detected detected detected detected detected

Table 177 Lubricant Test Results (µg/mL)
Test Detection 1st 2nd 3rd 6th
Batch No. AET 0 month
items limit month month month month
Not Not
GI201709004 < 3.49 < 3.49 < 3.49
detected detected
silicone Not Not
GI201709006 1.05 4800 < 3.49 < 3.49 < 3.49
oil detected detected
Not
GI201709008 < 3.49 < 3.49 < 3.49 < 3.49
detected

Table 178 Puncture Exfoliation Test Results
Test
Batch No. criterion 0 month 1st month 2nd month 3rd month 6th month
items
GI201709004 1 1 1 1 1 1
No more
Puncture
GI201709006 than 5 1 1 0 0 1
exfoliation
particles
GI201709008 1 1 1 0 0

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Table 179 Metal Elements Test Results (µg/mL)
Detection
Batch No. Test items AET 0 month 1st month 2nd month 3rd month 6th month
limit

Li: 0.00167 Li: 0.00240 Li: 0.00226 Li: < 0.00149 Li: 0.00209
Mg: 0.82339 Mg: 0.90117 Mg: 0.97493 Mg: 1.22284 Mg: 0.90635
Ca: 0.54453 Ca: 0.47905 Ca: 0.51136 Ca: 1.00356 Ca: 0.27672
Ti: 0.00379 Ti: 0.00329 Ti: 0.00368 Ti: 0.00557 Ti: 0.00292
V: 0.00083 V: 0.00075 V: 0.00097 V: 0.00106 V: 0.00114
Fe: 0.21540 Fe: 0.19961 Fe: 0.13817 Fe: 0.06612 Fe: 0.02735
Co: 0.00113 Co: 0.00087 Co: 0.0091 Co: 0.00095 Co: 0.00121
lithium 0.00045 500
Ni: 0.00941 Ni: 0.00737 Ni: 0.00879 Ni: 0.00754 Ni: 0.00885
GI201709004 magnesium 0.00229 N/A
Cu: 0.03895 Cu: 0.03725 Cu: 0.03424 Cu: 0.00805 Cu: 0.00296
calcium 0.00466 N/A
Zn: 0.08842 Zn: 0.07511 Zn: 0.11602 Zn: 0.07462 Zn: 0.13958
titanium 0.00038 N/A
As: 0.00041 As: 0.00038 As: 0.00059 As: 0.00044 As: < 0.00033
vanadium 0.00005 20
Cd: 0.00007 Cd: < 0.00007 Cd: 0.00012 Cd: < 0.00008 Cd: 0.00026
ferrum 0.00524 N/A
Sb: 0.14036 Sb: 0.15623 Sb: 0.17862 Sb: 0.15575 Sb: 0.15792
cobalt 0.00002 10
Hg: Not detected Hg: Not detected Hg: Not detected Hg: Not detected Hg: Not detected
nickel 0.00012 40
Pb: 0.00458 Pb: 0.00211 Pb: 0.00521 Pb: 0.00255 Pb: < 0.00026
cuprum 0.00020 600
Si: 0.9182 Si: 1.1376 Si: 1.4446 Si: 1.1904 Si: 2.5635
zinc 0.00201 N/A
arsenic 0.00010 30
cadmium 0.00002 4 Li: 0.00174 Li: 0.00169 Li: 0.00197 Li: < 0.00149 Li: 0.00209
antimony 0.00003 180 Mg: 0.68776 Mg: 0.91236 Mg: 0.94681 Mg: 0.95775 Mg: 0.59137
mercury 0.00217 6 Ca: 0.46144 Ca: 0.56712 Ca: 0.56130 Ca: 0.58988 Ca: 0.14942
plumbum 0.00008 10 Ti: 0.00343 Ti: 0.00468 Ti: 0.00373 Ti: 0.00356 Ti: Not detected
silicon 0.0258 N/A V: 0.00086 V: 0.00072 V: 0.00086 V: 0.00104 V: 0.00086
Fe: 0.18029 Fe: 0.20687 Fe: 0.15864 Fe: 0.04923 Fe: 0.02439

GI201709006 Co: 0.00097 Co: 0.00084 Co: 0.00080 Co: 0.00087 Co: 0.00101
Ni: 0.01262 Ni: 0.00818 Ni: 0.00843 Ni: 0.00765 Ni: 0.00625
Cu: 0.02227 Cu: 0.03536 Cu: 0.02238 Cu: 0.01085 Cu: 0.00220
Zn: 0.19164 Zn: 0.92156 Zn: 1.57732 Zn: 0.06969 Zn: 0.01481
As: 0.00044 As: 0.00033 As: 0.00043 As: 0.00039 As: Not detected
Cd: < 0.00007 Cd: < 0.00007 Cd: 0.00007 Cd: < 0.00008 Cd: 0.00020
Sb: 0.12669 Sb: 0.14143 Sb: 0.14404 Sb: 0.13408 Sb: 0.09454
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Hg: Not detected Hg: Not detected Hg: Not detected Hg: Not detected Hg: Not detected
Pb: 0.00322 Pb: 0.00295 Pb: 0.00349 Pb: 0.00157 Pb: 0.00030
Si: 0.7380 Si: 1.1562 Si: 1.6236 Si: 1.6188 Si: 2.0319
Li: 0.00199 Li: 0.00217 Li: 0.00211 Li: 0.00045 Li: 0.00163
Mg: 0.69517 Mg: 1.12927 Mg: 0.83200 Mg: 0.00229 Mg: 0.74750
Ca: 0.39993 Ca: 2.57973 Ca: 0.51310 Ca: 0.00466 Ca: 0.27134
Ti: 0.00334 Ti: 0.01142 Ti: 0.00500 Ti: 0.00038 Ti: 0.00300
V: 0.00082 V: 0.00128 V: 0.00098 V: 0.00005 V: 0.00118
Fe: 0.21704 Fe: 0.12481 Fe: 0.48830 Fe: 0.00524 Fe: 0.06105
Co: 0.00101 Co: 0.00093 Co: 0.00110 Co: 0.00002 Co: 0.00132
Ni: 0.00802 Ni: 0.00930 Ni: 0.00929 Ni: 0.00012 Ni: 0.00717
GI201709008
Cu: 0.01584 Cu: 0.02020 Cu: 0.08739 Cu: 0.00020 Cu: 0.00384
Zn: 0.08191 Zn: 0.32218 Zn: 0.07980 Zn: 0.00201 Zn: 0.64944
As: 0.00038 As: 0.00080 As: 0.00057 As: 0.00010 As: < 0.00033
Cd: < 0.00007 Cd: 0.00018 Cd: 0.00008 Cd: 0.00002 Cd: 0.00031
Sb: 0.13308 Sb: 0.17798 Sb: 0.21061 Sb: 0.00003 Sb: 0.17506
Hg: Not detected Hg: Not detected Hg: Not detected Hg: 0.00217 Hg: Not detected
Pb: 0.00322 Pb: 0.00522 Pb: 0.00344 Pb: 0.00008 Pb: 0.00047
Si: 0.7380 Si: 2.92 Si: 2.05 Si: 0.0258 Si: 3.0090
Conclusion:
3 batches of PCV were placed under 25±2 ■ for 6 months, the migration test results showed no precipitate beyond threshold value hadn't been found in PCV13, and puncture exfoliation results also meet requirement of the test.

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MODULE 3 QUALITY
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4. Migration test results of 3 batches of samples under 25±2 ■

Table 180 Antioxidan Test Results (µg/mL)
Detection 1st 2nd 3rd 6th
Batch No. Test items AET 0 month
limit month month month month
Antioxidant Not Not Not Not Not
GI201709004
1010 detected detected detected detected
0.026 197520
Antioxidant Not Not Not Not Not
GI201709009 0.014 3600
BHT detected detected detected detected detected
0.058 0.058
Antioxidant Not Not Not Not Not
GI201709010
1076 detected detected detected detected detected

Table 181 Stearic Acid and Palmitic Acid Test Results (µg/mL)
Detection 1st 2nd 3rd 6th
Batch No. Test items AET 0 month
limit month month month month
Not Not Not Not Not
GI201709004
stearic detected detected detected detected
acid 0.063 3600 Not Not Not Not Not
GI201709009
palmitic 0.100 3600 detected detected detected detected
acid Not Not Not Not Not
GI201709010
detected detected detected detected detected

Table 182 Vulcanizing agent test results(µg/ml)
Test Detectio AE 0 1st
Batch No. 2nd month 3rd month 6th month
items n limit T month month
sulphur: Not sulphur: Not sulphur: Not
detected detected detected
p-tert- p-tert- p-tert-
Amylphenol Amylphenol Amylphenol
Not Not
GI2017090 : < 0.131 : 0.159 : < 0.131
detecte detecte
04 2-Chloro-4- 2-Chloro-4- 2-Chloro-4-
d d
(tert- (tert- (tert-
penty)phen penty)phen penty)phen
ol: Not ol: Not ol: Not
sulphur
detected detected detected
p-tert-
sulphur: Not sulphur: Not sulphur: Not

Amylphen 180
0.006 detected detected detected
ol 360
0.039 p-tert- p-tert- p-tert-
2-Chloro- 0
0.032 Amylphenol Amylphenol Amylphenol
4-(tert- 180 Not Not
GI2017090 : < 0.131 : 0.162 : < 0.131
pentyl) detecte detecte
09 2-Chloro-4- 2-Chloro-4- 2-Chloro-4-
phenol d d
(tert- (tert- (tert-
pentyl)phen pentyl)phen pentyl)phen
ol: Not ol: Not ol: Not
detected detected detected
sulphur: Not sulphur: Not sulphur: Not
Not Not detected detected detected
GI2017090
detecte detecte p-tert- p-tert- p-tert-
10
d d Amylphenol Amylphenol Amylphenol
: < 0.131 : 0.178 : < 0.131

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2-Chloro-4- 2-Chloro-4- 2-Chloro-4-
(tert- (tert- (tert-
pentyl)phen pentyl)phen pentyl)phen
ol: Not ol: Not ol: Not
detected detected detected

Table 183 Volatile Organics (µg/mL)
Detection 1st 2nd 3rd 6th
Batch No. Test items AET 0 month
limit month month month month
Not Not Not Not Not
GI201709004
C13 oligomer detected detected detected detected detected
0.304 3600
C21 oligomer Not ■Not Not Not Not
GI201709009 0.304 3600
C13 chlorine detected detected detected detected detected
0.304 3
oligomer Not Not Not Not Not
GI201709010
detected detected detected detected detected

Table 184 Lubricant Test Results (µg/mL)
Test Detection 1st 2nd 3rd 6th
Batch No. AET 0 month
items limit month month month month
Not Not Not
GI201709004 < 3.49 < 3.49
detected detected detected
silicone Not
GI201709009 1.05 4800 < 3.49 < 3.49 < 3.49 < 3.49
oil detected
Not
GI201709010 < 3.49 < 3.49 < 3.49 < 3.49
detected

Table 185 Puncture Exfoliation Test Results
Test
Batch No. criterion 0 month 1st month 2nd month 3rd month 6th month
items
GI201709004 No more 1 1 1 0 0
Puncture
GI201709009 than 5 1 1 1 0 0
exfoliation
GI201709010 particles 0 0 1 0 0

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Table 186 Metal Elements Test Results (µg/mL)
Batch No. Test items Detection limit AET 0 month 1st month 2nd month 3rd month 6th month

Li: 0.00190 Li: 0.00209 Li: 0.00220 Li: < 0.00149 Li: 0.00225
Mg: 0.92928 Mg: 0.88909 Mg: 0.90515 Mg: 1.09261 Mg: 0.88357
Ca: 0.61799 Ca: 0.48068 Ca: 0.56266 Ca: 1.87660 Ca: 0.29935
Ti: 0.00440 Ti: Not detected Ti: < 0.00127 Ti: 0.02891 Ti: Not detected
V: 0.00096 V: 0.00080 V: 0.00075 V: 0.00113 V: 0.00102
Fe: 0.23557 Fe: 0.20860 Fe: 0.11534 Fe: 0.09954 Fe: 0.02091
Co: 0.00101 Co: 0.00084 Co: 0.00071 Co: 0.00088 Co: 0.00116
GI201709004 Ni: 0.01101 Ni: 0.00832 Ni: 0.00829 Ni: 0.00883 Ni: 0.01029
lithium 0.00045 500 Cu: 0.06096 Cu: 0.02769 Cu: 0.03686 Cu: 0.02288 Cu: 0.00324
magnesium 0.00229 N/A Zn: 0.30880 Zn: 0.10679 Zn: 0.09322 Zn: 0.10533 Zn: 0.01201
calcium 0.00466 N/A As: 0.00043 As: 0.00043 As: 0.00033 As: 0.00048 As: 0.00038
titanium 0.00038 N/A Cd: 0.00034 Cd: < 0.00007 Cd: Not detected Cd: 0.00009 Cd: 0.00027

vanadium 0.00005 20 Sb: 0.11579 Sb: 0.13598 Sb: 0.13440 Sb: 0.11813 Sb: 0.12718
ferrum 0.00524 N/A Hg: Not detected Hg: Not detected Hg: Not detected Hg: Not detected Hg: Not detected
cobalt 0.00002 10 Pb: 0.00486 Pb: 0.00297 Pb: 0.00217 Pb: 0.00517 Pb: 0.00032
nickel 0.00012 40 Si: 0.9196 Si: 1.2772 Si: 1.3672 Si: 1.4340 Si: 2.2770
cuprum 0.00020 600
zinc 0.00201 N/A Li: 0.00261 Li: 0.00247 Li: 0.00240 Li: 0.00157 Li: 0.00271
arsenic 0.00010 30 Mg: 0.97062 Mg: 0.87379 Mg: 0.97189 Mg: 1.06145 Mg: 0.83504
cadmium 0.00002 4 Ca: 0.54020 Ca: 0.48725 Ca: 0.51798 Ca: 4.35584 Ca: 0.30956
antimony 0.00003 180 Ti: 0.00678 Ti: < 0.00127 Ti: 0.00276 Ti: 0.02022 Ti: 0.00411
mercury 0.00217 6 V: 0.00094 V: 0.00081 V: 0.00126 V: 0.00118 V: 0.00110
plumbum 0.00008 10 Fe: 0.82670 Fe: 0.25048 Fe: 0.24102 Fe: 0.08752 Fe: 0.04505
silicon 0.0258 N/A Co: 0.00165 Co: 0.00111 Co: 0.00108 Co: 0.00111 Co: 0.00136
GI201709009 Ni: 0.00750 Ni: 0.00640 Ni: 0.00718 Ni: 0.00704 Ni: 0.00661
Cu: 0.04502 Cu: 0.03339 Cu: 0.09114 Cu: 0.02689 Cu: 0.00330
Zn: 0.08886 Zn: 0.06317 Zn: 0.00521 Zn: 0.14692 Zn: 0.06988
As: 0.00050 As: 0.00038 As: 0.00073 As: 0.00064 As: 0.00055
Cd: 0.00008 Cd: < 0.00007 Cd: 0.00041 Cd: 0.00017 Cd: 0.00024
Sb: 0.21526 Sb: 0.24964 Sb: 0.25398 Sb: 0.24493 Sb: 0.24349
Hg: Not detected Hg: Not detected Hg: Not detected Hg: Not detected Hg: Not detected
Pb: 0.00856 Pb: 0.00228 Pb: 0.00960 Pb: 0.01106 Pb: 0.00052
Si: 1.1522 Si: 1.268 Si: 1.9292 Si: 1.6506 Si: 2.6697

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Li: 0.00247 Li: 0.00215 Li: 0.00243 Li: 0.00219 Li: 0.00347
Mg: 0.75731 Mg: 0.78656 Mg: 1.00745 Mg: 0.95209 Mg: 0.96376
Ca: 0.40346 Ca: 0.47605 Ca: 1.36673 Ca: 1.88359 Ca: 0.38382
Ti: 0.00708 Ti: 0.00373 Ti: 0.00857 Ti: 0.00934 Ti: 0.00372
V: 0.00077 V: 0.00097 V: 0.00095 V: 0.00109 V: 0.00111
Fe: 0.18003 Fe: 0.19662 Fe: 0.64817 Fe: 0.14541 Fe: 0.05959
Co: 0.00115 Co: 0.00100 Co: 0.00095 Co: 0.00111 Co: 0.00147
GI201709010 Ni: 0.01135 Ni: 0.00968 Ni: 0.01165 Ni: 0.01044 Ni: 0.01565
Cu: 0.02714 Cu: 0.01764 Cu: 0.09213 Cu: 0.01191 Cu: 0.00414
Zn: 0.08161 Zn: 0.11089 Zn: 0.12032 Zn: 0.09048 Zn: 0.13292
As: 0.00032 As: 0.00056 As: 0.00054 As: 0.00054 As: 0.00046
Cd: < 0.00007 Cd: 0.00015 Cd: < 0.00007 Cd: 0.00011 Cd: 0.00033
Sb: 0.12549 Sb: 0.13067 Sb: 0.13510 Sb: 0.13447 Sb: 0.14418
Hg: Not detected Hg: Not detected Hg: Not detected Hg: Not detected Hg: Not detected
Pb: 0.00187 Pb: 0.00833 Pb: 0.00465 Pb: 0.00472 Pb: 0.00047
Si: 0.9666 Si: 1.2282 Si: 1.9352 Si: 1.6401 Si: 2.8173

Conclusion:
Migration test results showed no precipitate beyond threshold value had not been found in PCV13, and
puncture exfoliation results also meet requirement
of the test.
Summary: In conclusion, the PCV13 manufactured by Walvax uses middle-boro glass vials for injection
manufactured by Ningbo Zhengli
Pharmaceutical Packaging Co., LTD and halogenated butyl rubber plugs of inner package injection syringes
manufactured by Jiangsu Hualan
Pharmaceutical New Materials Co., Ltd. Compatibility studies of drug substances and packaging materials
showed that no high risk extractum was
migrated to the drug substance, and no destruction to the packaging materials by the drug substance was
observed, quality and stability of PCV13 were
maintained. Walvax will follow compatibility research protocols and continue conducting research for 24
months and 36 months under 5■±3■, then
provide an objective and scientific basis for selcting halogenated butyl rubber plugs for inner package
injection syringe manufactured by Ningbo Zhengli
Pharmaceutical Packaging Co., LTD and Jiangsu Hualan Pharmaceutical New Materials Co., Ltd.

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3.2.P.2.4.3 Primary Packaging (Pre-filled Syringe)
Table 187 Brief Profile of Packaging Materials
Packaging material Vendor Specification Standard
Shandong Weigao Group
Rubber plug for pre- Medical Polymer Company 1-3 mL
YBB00072004-2015
filling syringe Limited
BD 1-3 mL
Shandong Weigao Group
Pole for pre-filling Medical Polymer Company 1 mL
YBB00062004-2015
syringe Limited
BD 1 mL

3.2.P.2.4.4 Pre-filled Syringes Compatibility

Walvax conducted compatibility studies of pre-filling syringes manufactured by Shandong
Weigao Group Medical Polymer Products Co. LTD and PCV13 to investigate the influence of
packaging materials on formulation stability and potential migration, as well as safety and
applicability in use.
So far, migration test and compatibility study of packaging material for 24 months under 5±3 and
6 months under 25±2 ■ had completed. The detailed research is shown as follows:
Objective
The test is to investigate compatibility of PCV13 and pre-filled syringes manufactured by
Shandong Weigao Group Medical Polymer Products Co. LTD to evaluate potential migration and
its safety, and the influence of vaccine on packaging systems, then to provide basis about selection
of packaging material and safety of drug usage.
Basis

- 1 guideline for compatibility for pharmaceutical packaging material and drug (YBB00142002-2015)
- 2 Compatibility research technology guideline for chemical pharmaceutical injection and pharmaceutical packaging container (trial implementation)
- 3 Compatibility research technology guideline between chemical pharmaceutical and elastomeric seal (exposure draft)
- 4 Chinese pharmacopoeia, 4th edition, 2015, <9621>: guidelines for general requirements of pharmaceutical packaging materials
- 5 Chinese pharmacopoeia, 4th edition, 2015, <9622>: guidelines for pharmaceutical packaging materials and containers
- 6 Chinese pharmacopoeia, 4th edition, 2015, <9001>: guidelines for stability tests of raw drugs and preparations
- 7 Technical guidelines for stability studies of biological products
- 8 ICH Q3D, Introduction of elements and impurities
- 9 United States Pharmacopoeia, <1660> and <1664>

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- 10 FDA Guidelines for Industry: Container closure systems for packaging pharmaceuticals and biologics for human use (May 1999)
 - 11 European Union: Guideline for plastic packaging material (2015)
 - 12 E/L Recommendations, safety threshold and best practice of oral inhalation and nasal substance quality (September 2006)
 - 13 ICH Q1A, Guideline for stability tests of raw drugs and formulations
 - 14 ICH Q5C, Guideline for stability studies of biological products or biotechnological products.
- Experimental design
- Migration tests were conducted with 3 batches of PCV13 and the corresponding 1 batch of packaging material, to evaluate migration of glass tubing to PCV13, and to evaluate corrosion of the surface of the pre-filled syringe glass tubing by PCV13, and to conduct safety assessment on the migration quantity. The tests used 3 batches of PCV13 and corresponding 1 batch of packaging material to evaluate the corrosion of pre-filled syringe glass tubing by PCV13. Detailed experimental design is shown in Table 188.

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Table 188 Experimental Design for Compatibility Study of Rubber Plug for Pre-filled Syringe

Batch No. of packaging	Batch No. of Content of material	Storage	Product name	Categories	Sampling time	Test items
vaccine research	Batch No. of Batch No. of condition	glass tubing rubber plugs				
0 month, 3rd month, 6th						Metal elements testing, degraded
month, 9th month, 12th						oxidants, oxidants, sulfur,
3 batches of -						
month, 18th month, 24th						decmethylcyclopentaneoxosilane,
PCV13 and						
GK201709004	month, 36th month					2,6-dit-Bu-4-methylphenol,
corresponding	Migration					
GK201709005	X201709003	X201709006				dibutyl phthalate, bis-(2-
1 batch of substances						
GK201709010	25±2	0 month, 1st month, 2nd				ethylhexyl) phthalate, erucamide,
13-valent packaging						
humidity month, 3rd month, 6th						stearic acid, palmitic acid,
pneumococcal material						
60%±10% month						polydimethylsiloxane, isobutene-
polysaccharide						
isoprene oligomers						
conjugate						
0 month, 3rd month, 6th						
vaccine 3 batches of						
month, 9th month, 12th						
PCV13 and Stability 2-8						
GK201709004	X201709003	X201709006				month, 18th month, 24th Appearance, staining with
corresponding of packa						
GK201709008	X201709004	X201709007				month, 36th month methylene blue, inspection with
1 batch of ging						
GK201709009	X201709005	X201709008				25±2 0 month, 1st month, 2nd DIC, inspection with SEM,
packaging material						
humidity month, 3rd month, 6th						
material						
60%±10% month						

Safety evaluation of migration test results

- 1 Permissible daily exposure (PDE value): According to the guidelines, authorities and references, we can find permissible daily exposure (PDE value) for human of elements and organics. Permissible daily intake (ADI value) and ICH Q3D are referenced if PDE cannot be obtained, the NO(A)EL values provided by data from toxicological studies, PDE is calculated as:

$$PDE = NO(A)EL \times \text{weight(kg)} / [F1 \times F2 \times F3 \times F4 \times F5]$$
- 2 Analytical evaluation threshold (AET): AET implicates that leachates above this threshold should be qualitatively and quantitatively assessed and reported for potential toxicological assessment. AET value = PDE/maximum daily use.
- 3 Maximum dosage of PCV13 is 0.5 mL/day

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Test results
1 Stability (destruction and erosion) test results of packaging materials to 3 batches of samples under 5±3 ■
Table 189 Appearance Test Results of Packaging Material under 5±3 ■ Condition
glass tubing rubber plug
rd th
Batch No. criterion 0 3 6 9th 12th 18 th
24 th
0 3 rd
6 th
9th 12th 18th 24th
month month month month month month month month month month month month month month
GK201709 glass Qualifi Qualifi Qualifi Qualifi Qualifi Qualifi Qualifi Qualifi Qualifi Qualifi Qualifi Qualifi Qualifi
Qualifi Qualifi Qualifi
004 tubing: ed ed ed ed ed ed ed ed ed ed ed ed ed ed ed
clean and
GK201709 Qualifi Qualifi Qualifi Qualifi Qualifi Qualifi Qualifi Qualifi Qualifi Qualifi Qualifi Qualifi Qualifi
Qualifi Qualifi
smooth,
008 ed ed ed ed ed ed ed ed ed ed ed ed ed ed ed
with no
crack or
bubbles
were
detected
GK201709 rubber Qualifi Qualifi Qualifi Qualifi Qualifi Qualifi Qualifi Qualifi Qualifi Qualifi Qualifi Qualifi Qualifi
Qualifi Qualifi Qualifi
009 plug: ed ed ed ed ed ed ed ed ed ed ed ed ed ed ed
without
crack or
deformati
on

Table 190 Staining with Methylene Blue Test Results of Packaging Material under 5±3 ■ Condition
Batch No. criterion 0 month 3rd month 6th month 9th month 12th month 18th month 24th month
GK201709004 Qualified Qualified Qualified Qualified Qualified Qualified Qualified
GK201709008 without staining Qualified Qualified Qualified Qualified Qualified Qualified Qualified
GK201709009 Qualified Qualified Qualified Qualified Qualified Qualified Qualified

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Table 191 Test Results of Inspection with DIC of Packaging Materials under 5±3 ■ Condition
Batch No. criterion 0 month 3rd month 6th month 9th month 12th month 18th month 24th month
GK201709004 the inner surface of the Qualified Qualified Qualified Qualified Qualified Qualified
Qualified
glass syringe should be
GK201709008 Qualified Qualified Qualified Qualified Qualified Qualified Qualified
not visible hole, crack or
GK201709009 bubbles and erosion Qualified Qualified Qualified Qualified Qualified Qualified Qualified

Table 192 Test Results of Inspection with SEM of Packaging Materials under 5±3 ■ Condition
Batch No. criterion 0 month 3rd month 6th month 9th month 12th month 18th month 24th month
GK201709004 Qualified Qualified Qualified Qualified Qualified Qualified Qualified
the inner surface of the glass
GK201709008 syringe should be not visible Qualified Qualified Qualified Qualified Qualified Qualified
Qualified
bubbles and erosion
GK201709009 Qualified Qualified Qualified Qualified Qualified Qualified Qualified

Table 193 Test Results of Visible Foreign Body under 5±3 ■ Condition
Batch No. criterion 0 month 3rd month 6th month 9th month 12th month 18th month 24th month
GK201709004 Qualified Qualified Qualified Qualified Qualified Qualified Qualified
GK201709008 No visible foreign body Qualified Qualified Qualified Qualified Qualified Qualified Qualified
GK201709009 Qualified Qualified Qualified Qualified Qualified Qualified Qualified

Conclusion:
Test results showed that the inner surface of the glass syringe was clean and smooth, with no crack or
bubbles; the surface of the rubber plug was without
crack or deformation. Inspection with DIC and SEM found no distinct corrosion of pre-filled syringe glass
tubing.

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2. Migration test results of 3 batches of samples under 5±3 ■
Table 194 Test Results of Metal Elements in the Samples under 5±3■ Condition
criterion
Batch No. 0 month 3rd month 6th month 9th month 12th month 18th month 24th month
elements PDE AET
Li 250 500 Li: Not Li: Not Li:Not Li:Not Li:Not
B 10000 20000 detected; detected; detected; detected; detected;
Li: Not
Mg 25000 50000 B: Not B: Not B:Not B:Not B:< 0.2815;
Li: Not detected;
Si 60000 120000 detected; detected; detected; detected; Mg:0.6366;
detected; B: Not
K 750000 1500000 Mg: 0.9884; Mg: 0.6163; Mg:0.7969; Mg:0.7301; Si:Not
B: < 0.5917; detected;
Ca 150000 300000 Si: Not Si: <11.0169; Si:3.9924; Si:< 6.2311; detected;

Mg: 1.0184; Mg: 0.7617;
Ti 750 1500 detected; K: <0.6789; K:< 0.7710; K:0.9570; K:0.5704;
Si: 7.5558; Si: < 3.0354;
Cr 1100 2200 K: < 0.9979; Ca: 1.2867; Ca:1.3386; Ca:1.4657; Ca:< 1.1209;
K: 1.0782; K: 0.5891;
Fe 1300 2600 Ca: 3.4260; Ti: Not Ti:0.0120; Ti:< 0.0207; Ti:Not
Ca: 5.6118; Ca: 1.7875;
Co 5 10 Ti: Not detected; Cr:< 0.0154; Cr:Not detected;
Ti: < 0.0124; Ti: < 0.0124;
Ni 20 40 detected; Cr: Not Fe:1.1891; detected; Cr:Not
Cr: Not Cr: < 0.0206;
Cu 300 600 Cr: Not detected; Co:< 0.0043; Fe:< 0.7973; detected;
detected; Fe: 0.9010;
Zn 1300 2600 detected; Fe: 0.6663; Ni:0.0207; Co:Not Fe:< 0.6653;
Fe: 1.3091; Co: Not
As 15 30 Fe: < 0.9386; Co: Not Cu:0.0178; detected; Co:Not
Co: Not detected;
Mo 1500 3000 Co: Not detected; Zn:1.1463; Ni:0.0209; detected;
GK201709004 detected; Ni: 0.0221;
Cd 2 4 detected; Ni: 0.0133; As:Not Cu:< 0.0221; Ni:0.0194;
Ni: 0.0223; Cu: Not
Sn 600 1200 Ni: Not Cu: <0.0114; detected; Zn:0.7976; Cu:< 0.0440;
Cu: < 0.0281; detected;
Sb 90 180 detected; Zn: 0.4316; Mo:Not As:Not Zn:0.7021;
Zn: 0.4774; Zn: 0.5684;
Ba 700 1400 Cu: Not As: Not detected; detected; As:Not
As: Not As: Not
W 50 100 detected; detected; Cd:< 0.0009; Mo:Not detected;
detected; detected;
Pb 5 10 Zn: 0.7892; Mo: Not Sn:Not detected; Mo:Not
Mo: 0.0972; Mo: < 0.0931;
As: Not detected; detected; Cd:Not detected;
Cd: < 0.0015; Cd: < 0.0015;
detected; Cd: Not Sb:0.1804; detected; Cd:Not
Sn: < 0.0364; Sn: Not
Mo: Not detected; Ba:0.0672; Sn:Not detected;
Sb: 0.1944; detected;
detected; Sn: Not W:0.2945; detected; Sn:Not
Ba: 0.0985; Sb: 0.1839;
Cd: Not detected; Pb:0.0222; Sb:0.1661; detected;
W: 0.4277; Ba: 0.0450;
detected; Sb: 0.1664; Ba:0.0414; Sb:0.1701;
Pb: 0.0167; W: 0.6436;
Sn: Not Ba: 0.0322; W:< 0.3266; Ba:0.0725;
Pb: 0.0214;
detected; W: 0.6512; Pb:< 0.0288; W:0.9932;
Sb: 0.1667; Pb: <0.0126; Pb:< 0.0320;

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Ba: 0.0617;
W: 1.1584;
Pb: < 0.0261;

Table 195 Test Results of Metal Elements in the Samples under 5±3 ■ Condition
criterion
Batch No. 0 month 3rd month 6th month 9th month 12th month 18th month 24th month
elements PDE AET
Li: Not Li: Not Li: Not Li: Not Li:Not Li:Not
Li 250 500 Li:Not
detected; detected; detected; detected; detected; detected;
B 10000 20000 detected;
B: Not B: Not B: Not B: <0.2064; B:Not B:Not
Mg 25000 50000 B:< 0.2815;
detected; detected; detected; Mg: 0.9848; detected; detected;
Si 60000 120000 Mg:1.0181;
Mg: 1.2136; Mg: 1.2254; Mg: 1.3967; Si: <11.0169; Mg:0.6278; Mg:1.3058;
K 750000 1500000 Si:< 6.3446;
Si: 10.1457; Si: 5.0977; Si: < 14.3379; K: 0.8298; Si:3.8064; Si:< 6.2311;
Ca 150000 300000 K:0.7365;
K: 1.1611; K: 1.0943; K: 1.3558; Ca: 2.7004; K:< 0.7710; K:1.2632;
Ti 750 1500 Ca:2.2330;
Ca: 5.5318; Ca: 3.5349; Ca: 6.2286; Ti: <0.0106; Ca:< 0.9237; Ca:2.7710;
Cr 1100 2200 Ti:Not
Ti: 0.0160; Ti: < 0.0124; Ti: < 0.0309; Cr: Not Ti:0.0101; Ti:< 0.0207;
Fe 1300 2600 detected;
Cr: < 0.0206; Cr: < 0.0206; Cr: Not detected; Cr:Not Cr:< 0.0403;
Co 5 10 Cr:Not
Fe: 1.4953; Fe: 1.3814; detected; Fe: 1.1393; detected; Fe:1.2859;
GK201709005 Ni 20 40 detected;
Co: < 0.0062; Co: < 0.0062; Fe: < 0.9386; Co: Not Fe:< 0.9569; Co:Not
Cu 300 600 Fe:0.8592;
Ni: 0.0200; Ni: 0.0366; Co: < 0.0121; detected; Co:< 0.0043; detected;
Zn 1300 2600 Co:Not
Cu: < 0.0281; Cu: < 0.0281; Ni: Not Ni: 0.0134; Ni:< 0.0081; Ni:0.0229;
As 15 30 detected;
Zn: 0.6339; Zn: 1.8972; detected; Cu: 0.0204; Cu:< 0.0099; Cu:< 0.0221;
Mo 1500 3000 Ni:0.0191;
As: Not As: Not Cu: Not Zn: 0.6293; Zn:0.4119; Zn:0.9547;
Cd 2 4 Cu:< 0.0440;

detected; detected; detected; As: Not As:Not As:Not
Sn 600 1200 Zn:0.8854;
Mo: Not Mo: Not Zn: 0.6766; detected; detected; detected;
Sb 90 180 As:Not
detected; detected; As: Not Mo: Not Mo:Not Mo:Not
Ba 700 1400 detected;
Cd: 0.0019; Cd: < 0.0015; detected; detected; detected; detected;
W 50 100 Mo:Not
Sn: Not Sn: Not Mo: Not Cd: Not Cd:Not Cd:Not
Pb 5 10 detected;
detected; detected; detected; detected; detected; detected;
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Sb: 0.5396; Sb: 0.5651; Cd: Not Sn: Not Sn:Not Sn:Not Cd:Not
Ba: 0.1126; Ba: 0.0785; detected; detected; detected; detected; detected;
W: 0.8729; W: 0.7439; Sn: Not Sb: 0.5304; Sb:0.4489; Sb:0.4894; Sn:Not
Pb: 0.0570; Pb: 0.0598; detected; Ba: 0.0653; Ba:0.0248; Ba:0.0791; detected;
Sb: 0.4829; W: 0.3892; W:< 0.1059; W:Not Sb:0.5392;
Ba: 0.0647; Pb: 0.0603; Pb:0.0184; detected; Ba:0.1105;
W: < 0.2121; Pb:0.0719; W:0.9539;
Pb: 0.0556; Pb:0.0472;

Table 196 Test Results of Metal Elements in the Samples under 5±3 ■ Condition
criterion
Batch No. 0 month 3rd month 6th month 9th month 12th month 18th month 24th month
elements PDE AET
Li 250 500 Li: Not Li: Not Li:Not
Li: Not Li: Not Li:Not Li:Not
B 10000 20000 detected; detected; detected;
detected; detected; detected; detected;
Mg 25000 50000 Li: Not B: <0.2064; B:Not
Li: Not Li: Not B:Not B: < 0.2815;
Si 60000 120000 detected; Mg: 4.0707; detected;
detected; detected; detected; Mg:1.0181;
K 750000 1500000 Mg: 0.7406; Si: 13.0884; Mg:2.8406;
Mg: 0.6995; Mg: 0.9703; Mg:0.4378; Si:< 6.3446;
Ca 150000 300000 Si: 4.2235; K: 2.0386; Si:< 6.2311;
Si: 12.9563; Si: < 14.3379; Si:8.0268; K:0.7365;
Ti 750 1500 K: 0.9199; Ca: 13.8437; K:1.8963;
K: 1.0389; K: 1.1416; K:< 0.7710; Ca:2.2330;
Cr 1100 2200 Ca: 2.2976; Ti: 0.0323; Ca:7.3743;
Ca: 3.9272; Ca: 5.1615; Ca:< 0.9237; Ti:Not
Fe 1300 2600 Ti: < 0.0124; Cr: 0.0499; Ti:0.0245;
Ti: < 0.0124; Ti: Not Ti:< 0.0070; detected;
Co 5 10 Cr: < 0.0206; Fe: 3.9320; Cr:< 0.0403;
Cr: < 0.0206; detected; Cr:< 0.0154; Cr:Not
Ni 20 40 Fe: 1.3119; Co: <0.0091; Fe:2.7365;
GK201709010 Fe: 1.3253; Cr: < 0.0444; Fe:< 0.9569; detected;
Cu 300 600 Co: Not Ni: 0.0618; Co:Not
Co: Not Fe: < 0.9386; Co:< 0.0043; Fe:1.0741;
Zn 1300 2600 detected; Cu: 0.0706; detected;
detected; Co: Not Ni:0.0086; Co:Not
As 15 30 Ni: 0.0159; Zn: 1.3867; Ni:0.0420;
Ni: 0.0095; detected; Cu:0.0127; detected;
Mo 1500 3000 Cu: Not As: Not Cu:0.0410;
Cu: < 0.0281; Ni: Not Zn:0.6587; Ni:0.0172;
Cd 2 4 detected; detected; Zn:1.5570;
Zn: 0.4764; detected; As:Not Cu:< 0.0440;
Sn 600 1200 Zn: 1.7488; Mo: Not As:Not
As: Not Cu: Not detected; Zn:0.8110;
Sb 90 180 As: Not detected; detected;
detected; detected; Mo:Not As:Not
Ba 700 1400 detected; Cd: 0.0051; Mo:Not
Mo: Not Zn: 0.6896; detected; detected;
W 50 100 Mo: Not Sn: <0.0099; detected;
detected; As: Not Cd:Not Mo:Not
Pb 5 10 detected; Sb: 0.6428; Cd:Not
Cd: < 0.0015; detected; detected; detected;
Cd: < 0.0015; Ba: 0.3839; detected;
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Sn: Not Sn: Not Mo: Not W: 0.7297; Sn:Not Sn:Not Cd:Not
detected; detected; detected; Pb: 0.2427; detected; detected; detected;
Sb: 0.3186; Sb: 0.3242; Cd: Not Sb:0.2639; Sb:0.4834; Sn:Not
Ba: 0.0858; Ba: 0.0392; detected; Ba:0.0204; Ba:0.2269; detected;
W: 0.3186; W: 0.8202; Sn: Not W:< 0.1059; W:Not Sb:0.3426;
Pb: 0.0174; Pb: 0.0196; detected; Pb:0.0102; detected; Ba:0.0809;
Sb: 0.1632; Pb:0.1407; W:0.6131;
Ba: 0.0654; Pb:0.0466;
W: 0.3400;
Pb: 0.0838;

Table 197 Test Results of Antioxidant Degradation in the Samples under 5±3 ■ Condition (HPLC) (µg/mL)
Instrument Method
3rd 6th 9th 12th 18th 24th
Batch No. Test items detection detection PDE AET 0 month
month month month month month
limit limit

3,5- 3,5- 3,5- 3,5- 3,5- 3,5- 3,5-
DTBHA: DTBHA: DTBHA: DTBHA: DTBHA: DTBHA: DTBHA:
Not Not Not Not Not Not Not
detected detected detected detected detected detected detected
3, 5-di-tert-butyl-4
2,4-DTB: 2,4-DTB: 2,4-DTB: 2,4-DTB: 2,4-DTB: 2,4-DTB: 2,4-DTB:
hydroxyphenylpropionic
Not Not Not Not Not Not Not
GK201709004 acid(3,5-DTBHA) 0.08918 22.29500 300 600
detected detected detected detected detected detected detected
2,6- 2,6- 2,6- 2,6- 2,6- 2,6- 2,6-
DTBQ: DTBQ: DTBQ: DTBQ: DTBQ: DTBQ: DTBQ:
2,4-di-tert-
Not Not Not Not Not Not Not
butylphenol(2,4-DTB)
detected detected detected detected detected detected detected

3,5- 3,5- 3,5- 3,5- 3,5- 3,5- 3,5-
2,6-di-tert-butyl-1,4-
DTBHA: DTBHA: DTBHA: DTBHA: DTBHA: DTBHA: DTBHA:
benzoquinone(2,6-
Not Not Not Not Not Not Not
DTBQ)
GK201709005 0.07558 18.89500 2500 5000 detected detected detected detected detected detected detected
2,4-DTB: 2,4-DTB: 2,4-DTB: 2,4-DTB: 2,4-DTB: 2,4-DTB: 2,4-DTB:
Not Not Not Not Not Not Not
detected detected detected detected detected detected detected

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2,6- 2,6- 2,6- 2,6- 2,6- 2,6- 2,6-
DTBQ: DTBQ: DTBQ: DTBQ: DTBQ: DTBQ: DTBQ:
Not Not Not Not Not Not Not
detected detected detected detected detected detected detected
3,5- 3,5- 3,5- 3,5- 3,5- 3,5- 3,5-
DTBHA: DTBHA: DTBHA: DTBHA: DTBHA: DTBHA: DTBHA:
Not Not Not Not Not Not Not
detected detected detected detected detected detected detected
2,4-DTB: 2,4-DTB: 2,4-DTB: 2,4-DTB: 2,4-DTB: 2,4-DTB: 2,4-DTB:
GK201709010 0.06028 15.07000 2500 5000 Not Not Not Not Not Not Not
detected detected detected detected detected detected detected
2,6- 2,6- 2,6- 2,6- 2,6- 2,6- 2,6-
DTBQ: DTBQ: DTBQ: DTBQ: DTBQ: DTBQ: DTBQ:
Not Not Not Not Not Not Not
detected detected detected detected detected detected detected

Table 198 Test Results of Antioxidant and Sulfur in the Samples under 5±3 ■ Condition (HPLC)
Instru Meth
ment od
Batch PD AE
Migration substances detecti detect 0 month 3rd month 6th month 9th month 12th month 18th month 24th
month
No. E T
on ion
limit limit
0.0245 3.067 sulfur: Not sulfur: Not sulfur: Not sulfur: Not sulfur: Not sulfur: Not
sulfur 9 18 detected detected detected detected detected detected detected
4 50
antioxidan antioxidant antioxidant antioxidant antioxidant antioxidant antioxidant
t1010: Not t1010: Not t1010: Not t1010: Not t1010: Not t1010: Not t1010: Not
detected detected detected detected detected detected detected
GK2017 Pentaerythritol tetrakis(3- antioxidant antioxidant antioxidant antioxidant antioxidant antioxidant
antioxidan
09004 (3,5-di-tert-butyl-4- 0.0701 8.767 100 200 t1076: Not t1076: Not t1076: Not t1076: Not
t1076: Not t1076: Not
hydroxyphenyl)propionate)(a 4 50 00 00 detected detected detected detected detected detected detected
ntioxidant1010) antioxidant antioxidant antioxidant antioxidant antioxidant antioxidant antioxidant
t168 t168 t168 t168 t168 t168 t168
phosphate: phosphate: phosphate: phosphate: phosphate: phosphate: phosphate:

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Not Not Not Not Not Not Not
detected detected detected detected detected detected detected
antioxidan antioxidant antioxidant antioxidant antioxidant antioxidant antioxidant
t168: Not t168: Not t168: Not t168: Not t168: Not t168: Not
detected detected detected detected detected detected detected
Octadecyl 3-(3,5-di-tert- sulfur: Not sulfur: Not sulfur: Not sulfur: Not sulfur: Not sulfur:
Not
butyl-4- detected detected detected detected detected detected detected
0.2401 30.01
hydroxyphenyl)propionate(an 300 600 antioxidant antioxidant antioxidant antioxidant antioxidant antioxidant
antioxidan
5 875
tioxidant1076) t1010: Not t1010: Not t1010: Not t1010: Not t1010: Not t1010: Not
detected detected detected detected detected detected detected
antioxidan antioxidant antioxidant antioxidant antioxidant antioxidant antioxidant
t1076: Not t1076: Not t1076: Not t1076: Not t1076: Not t1076: Not

GK2017 detected detected detected detected detected detected detected detected
09005 antioxidan antioxidan antioxidan antioxidan antioxidan antioxidan antioxidan
Tris(2,4-di-tert- t168 t168 t168 t168 t168 t168 t168
0.3605 45.07 250 500
butylphenyl)phosphate(antiox phosphate: phosphate: phosphate: phosphate: phosphate: phosphate: phosphate:
6 000 0 0
idant168 phosphate) Not Not Not Not Not Not Not
detected detected detected detected detected detected detected detected
antioxidan antioxidan antioxidan antioxidan antioxidan antioxidan antioxidan
t168: Not t168: Not t168: Not t168: Not t168: Not t168: Not
detected detected detected detected detected detected detected detected
sulfur: Not sulfur: Not sulfur: Not sulfur: Not sulfur: Not sulfur: Not sulfur: Not
detected detected detected detected detected detected detected detected
antioxidan antioxidan antioxidan antioxidan antioxidan antioxidan antioxidan
t1010: Not t1010: Not t1010: Not t1010: Not t1010: Not t1010: Not
detected detected detected detected detected detected detected detected
antioxidan antioxidan antioxidan antioxidan antioxidan antioxidan antioxidan
GK2017 Tris(2,4-ditert-butylphenyl) 0.2379 29.73 250 500
t1076: Not t1076: Not t1076: Not t1076: Not t1076: Not t1076: Not
09010 phosphite (antioxidant168) 0 750 0 0
detected detected detected detected detected detected detected detected
antioxidan antioxidan antioxidan antioxidan antioxidan antioxidan antioxidan
t168 t168 t168 t168 t168 t168 t168 t168
phosphate: phosphate: phosphate: phosphate: phosphate: phosphate: phosphate:
Not Not Not Not Not Not Not Not
detected detected detected detected detected detected detected detected

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antioxidan antioxidan antioxidan antioxidan antioxidan antioxidan antioxidan
t168: Not t168: Not t168: Not t168: Not t168: Not t168: Not
detected detected detected detected detected detected detected detected

Table 199 Test Results of Stearic Acid and Palmitic Acid in the Samples under 5±3 ■ Condition (GC)

Instrument Method
migration 12th 18th 24th
Batch No. detection detection PDE AET 0 month 3rd month 6th month 9th month
substance month month month
limit limit
palmitic palmitic palmitic palmitic palmitic palmitic palmitic
acid: acid: acid: acid: acid: acid: acid:
Not Not Not Not Not Not Not
detected detected detected detected detected detected detected
GK201709004
stearic stearic stearic stearic stearic stearic stearic
acid: acid: acid: acid: acid: acid: acid:
Not Not Not Not Not Not Not
palmitic detected detected detected detected detected detected detected
1.60328 22.44592 5000 10000
acid palmitic palmitic palmitic palmitic palmitic palmitic
acid: acid: acid: acid: acid: acid: acid:
Not Not Not Not Not Not Not
detected detected detected detected detected detected detected
GK201709005
stearic stearic stearic stearic stearic stearic stearic
acid: acid: acid: acid: acid: acid: acid:
Not Not Not Not Not Not Not
detected detected detected detected detected detected detected
palmitic palmitic palmitic palmitic palmitic palmitic palmitic
acid: acid: acid: acid: acid: acid: acid:
Not Not Not Not Not Not Not
stearic detected detected detected detected detected detected detected
GK201709010 1.60988 22.53832 6250 12500
acid stearic stearic stearic stearic stearic stearic
acid: acid: acid: acid: acid: acid: acid:
Not Not Not Not Not Not Not
detected detected detected detected detected detected detected

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Table 200 Test Results of Dimethylpolysiloxane under 5±3 ■ Condition (HPLC)

Instrument Method
Migration 3rd 6th 9th 12th 18th 24th
Batch No. detection detection PDE AET 0 month
substances month month month month month month
limit limit
Not Not Not Not Not Not Not
GK201709004
detected detected detected detected detected detected detected
Not Not Not Not Not Not Not
GK201709005 Dimethylpolysiloxane 0.16418 2.46270 9 18
detected detected detected detected detected detected detected
Not Not Not Not Not Not Not
GK201709010
detected detected detected detected detected detected detected

Table 201 Test Results about Isobutene-isoprene Oligomers (C13H24, C13H23CL, C21H40) in the Samples under 5±3 ■

Batch No. Test method 0 month 3rd month 6th month 9th month 12th month 18th month 24th month
Not
GK201709004 Not detected Not detected Not detected Not detected Not detected Not detected
detected
GC full-scan(Scan)
Not
GK201709005 Characteristic ion Not detected Not detected Not detected Not detected Not detected Not
detected
detected
scanning(SIM)
Not
GK017090010 Not detected Not detected Not detected Not detected Not detected Not detected
detected

Conclusion: Migration tests showed no degraded antioxidant, no antioxidant, no sulfur, no stearic acid,
no palmitic acid, no erucamide, no plasticizer,
no epoxy silane and no isobutene-isoprene oligomers in the samples, residual metal elements were well
below PDE.

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3 Stability (destruction and erosion) test results of packaging materials to 3 batches of samples
under 25±2 ■

Table 202 Appearance Test Results of Packaging Material under 25±2 ■ Condition
glass tubing rubber plug
Batch criterio
0 1st 2nd 3rd 6th 0 1st 2nd 3rd 6th

No. n
mont mont mont mont mont mont mont mont mont
h h h h h h h h h
glass
GK2017 tubing: Quali Quali Quali Quali Quali Quali Quali Quali Quali Quali
09004 clean fied fied fied fied fied fied fied fied fied fied
and
smooth,
GK2017 with no Quali Quali Quali Quali Quali Quali Quali Quali Quali Quali
09008 crack fied fied fied fied fied fied fied fied fied fied
or
bubbles
were
detecte
d
GK2017 rubber Quali Quali Quali Quali Quali Quali Quali Quali Quali Quali
09009 plug: fied fied fied fied fied fied fied fied fied fied
without
crack
or
deform
ation

Table 203 Staining with Methylene Blue Test Results of Packaging Material under 25±2■
Condition
Batch No. criterion 0 month 1st month 2nd month 3rd month 6th month

GK201709004 Qualified Qualified Qualified Qualified Qualified
Without
GK201709008 Qualified Qualified Qualified Qualified Qualified
staining
GK201709009 Qualified Qualified Qualified Qualified Qualified

Table 204 Test Results of Inspection with DIC of Packaging Materials under 25±2 ■
Condition

Batch No. criterion 0 month 1st month 2nd month 3rd month 6th month
the inner
GK201709004 Qualified Qualified Qualified Qualified Qualified
surface of the
glass syringe
GK201709008 should be not Qualified Qualified Qualified Qualified Qualified
visible hole,
crack or
GK201709009 bubbles and Qualified Qualified Qualified Qualified Qualified
erosion

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Table 205 Test Results of Inspection with SEM of Packaging Materials under 25±2 ■
Condition

Batch No. criterion 0 month 1st month 2nd month 3rd month 6th month
the inner
GK201709004 Qualified Qualified Qualified Qualified Qualified
surface of the
glass syringe
GK201709008 should be not Qualified Qualified Qualified Qualified Qualified
visible
GK201709009 bubbles and Qualified Qualified Qualified Qualified Qualified
erosion

Table 206 Test Results of Visible Foreign Body under 25±2 ■ Condition
Batch No. criterion 0 month 1st month 2nd month 3rd month 6th month

GK201709004 Qualified Qualified Qualified Qualified Qualified
No visible
GK201709008 Qualified Qualified Qualified Qualified Qualified
foreign body
GK201709009 Qualified Qualified Qualified Qualified Qualified

Conclusion:
Test results showed that the inner surface of the glass syringe was clean and smooth, with no crack or bubbles; the surface of the rubber plug was without crack or deformation. Inspection with DIC and SEM found no distinct corrosion of pre-filled syringe glass tubing.

4. Migration test results of 3 batches of samples under 25±2 ■

Table 207 Test Results of Metal Elements in the Samples under 25±2 ■ Condition
criterion 2nd 3rd
Batch No. 0 month 1st month 6th month
elements PDE AET month month
Li 250 500 Li: Not Li: Not Li: Not Li: Not Li: Not
B 10000 20000 detected detected detected detected detected
Mg 25000 50000 B: < B: B: Not B: < B: Not
Si 60000 120000 0.5917 1.4846 detected 0.5917 detected
K 750000 1500000 Mg: Mg: Mg: Mg: Mg:
Ca 150000 300000 1.0184 0.6158 0.8255 0.8236 1.5084
Ti 750 1500 Si: Si: Si: Si: Si: Not
Cr 1100 2200 7.5558 14.7814 12.6825 3.8549 detected
GK201709004 Fe 1300 2600 K: K: K: K:
Co 5 10 1.0782 1.4417 1.1660 0.8777 1.0965
Ni 20 40 Ca: Ca: Ca: Ca: Ca:
Cu 300 600 5.6118 2.4192 4.5840 4.1787 6.8747
Zn 1300 2600 Ti: < Ti: < Ti: < Ti: Not Ti: Not
As 15 30 0.0124 0.0124 0.0124 detected detected
Mo 1500 3000 Cr: Not Cr: Not Cr: < Cr: < Cr: <
Cd 2 4 detected detected 0.0206 0.0206 0.0444
Sn 600 1200
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Sb 90 180 Fe: Fe: Fe: Fe: Fe: <
Ba 700 1400 1.3091 1.4309 1.5772 0.8766 0.9386
W 50 100 Co: Not Co: Not Co: Not Co: Not Co: Not
Pb 5 10 detected detected detected detected detected
Ni: Ni: Ni: Ni: Not Ni: Not
0.0223 0.0111 0.0170 detected detected
Cu: < Cu: < Cu: < Cu: < Cu: Not
0.0281 0.0281 0.0281 0.0281 detected
Zn: Zn: Zn: Zn: Zn:
0.4774 0.3482 0.7009 0.6239 0.9498
As: Not As: Not As: Not As: Not As: Not
detected detected detected detected detected
Mo: Mo: < Mo: Not Mo: Not Mo: Not Mo: Not
0.0972 0.0931 detected detected detected
Cd: < Cd: < Cd: < Cd: Not Cd: Not
0.0015 0.0015 0.0015 detected detected
Sn: < Sn: Not Sn: Not Sn: Not Sn: Not
0.0364 detected detected detected detected
Sb: Sb: Sb: Sb: Sb:
0.1944 0.1838 0.1901 0.1860 0.1671
Ba: Ba: Ba: Ba: Ba:
0.0985 0.7016 0.0948 0.0579 0.0674
W: W: < W: W: W:
0.4277 0.2323 0.3134 0.9648 0.8926
Pb: Pb: < Pb: Pb: Pb: <
0.0167 0.0094 0.0159 0.0129 0.0261

Table 208 Test Results of Metal Elements in the Samples under 25±2 ■ Condition
criterion 1st 2nd 3rd 6th
Batch No. 0 month
elements PDE AET month month month month
Li 250 500 Li: Not Li: Not Li: Not Li: Not Li: Not
B 10000 20000 detected detected detected detected detected
Mg 25000 50000 B: Not B: Not B: Not B: < B: Not
Si 60000 120000 detected detected detected 0.5917 detected
K 750000 1500000 Mg: Mg: Mg: Mg: Mg:
Ca 150000 300000 1.2136 1.3112 1.3861 1.4336 1.4357
Ti 750 1500 Si: Si: Si: Si: Si: Not
Cr 1100 2200 10.1457 12.4074 15.1994 7.7407 detected
Fe 1300 2600 K: K: K: K: K:
Co 5 10 1.1611 1.3188 1.4149 1.2862 1.3233
Ni 20 40 Ca: Ca: Ca: Ca: Ca:
Cu 300 600 5.5318 5.9096 6.0567 6.5071 6.6891
GK201709005
Zn 1300 2600 Ti: Ti: Ti: Ti: < Ti: <
As 15 30 0.0160 0.0144 0.0208 0.0124 0.0309
Mo 1500 3000 Cr: < Cr: < Cr: < Cr: < Cr: <
Cd 2 4 0.0206 0.0206 0.0206 0.0206 0.0444
Sn 600 1200 Fe: Fe: Fe: Fe: Fe:
Sb 90 180 1.4953 1.7799 1.8608 1.4024 1.0605
Ba 700 1400 Co: < Co: < Co: < Co: Not Co: <
W 50 100 0.0062 0.0062 0.0062 detected 0.0121
Pb 5 10 Ni: Ni: Ni: Ni: Not Ni: Not

0.0200 0.0178 0.0160 detected detected
Cu: < Cu: < Cu: < Cu: Not Cu: Not
0.0281 0.0281 0.0281 detected detected

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Zn: Zn: Zn: Zn: Zn:
0.6339 0.7968 0.8708 0.8388 0.9826
As: Not As: Not As: Not As: Not As: Not
detected detected detected detected detected
Mo: Not Mo: Not Mo: Not Mo: Not Mo: Not
detected detected detected detected detected
Cd: Cd: < Cd: < Cd: Not Cd: Not
0.0019 0.0015 0.0015 detected detected
Sn: Not Sn: Not Sn: Not Sn: Not Sn: Not
detected detected detected detected detected
Sb: Sb: Sb: Sb: Sb:
0.5396 0.5812 0.6002 0.5676 0.5407
Ba: Ba: Ba: Ba: Ba:
0.1126 0.1067 0.1086 0.0946 0.0875
W: W: W: W: W:
0.8729 0.8980 0.8388 0.8056 0.4122
Pb: Pb: Pb: Pb: Pb:
0.0570 0.0551 0.0534 0.0607 0.0722

Table 209 Test Results of Metal Elements in the Samples under 25±2 ■ Condition
criterion 1st 2nd 3rd 6th

Batch No. 0 month
elements PDE AET month month month month
Li 250 500 Li: Not Li: Not Li: Not Li: Not
B 10000 20000 detected detected detected detected detected
Mg 25000 50000 B: Not B: Not B: Not B: < B: Not
Si 60000 120000 detected detected detected detected 0.5917 detected
K 750000 1500000 Mg: Mg: Mg: Mg: Mg:
Ca 150000 300000 0.6995 1.1502 0.9803 0.9597 1.0230
Ti 750 1500 Si: Si: Si: Si: Si: Not
Cr 1100 2200 12.9563 13.4163 14.3601 6.4160 detected
Fe 1300 2600 K: K: K: K: K: <
Co 5 10 1.0389 1.3028 1.3437 1.1466 0.9979
Ni 20 40 Ca: Ca: Ca: Ca: Ca:
Cu 300 600 3.9272 5.7021 5.0209 6.9680 3.7619
Zn 1300 2600 Ti: < Ti: < Ti: < Ti: Not
As 15 30 0.0124 0.0124 0.0124 0.0124 detected
Mo 1500 3000 Cr: < Cr: < Cr: < Cr: Not
Cd 2 4 0.0206 0.0206 0.0206 0.0206 detected
GK2017090010
Sn 600 1200 Fe: Fe: Fe: Fe: Fe: <
Sb 90 180 1.3253 1.6540 1.7039 1.3111 0.9386
Ba 700 1400 Co: Not Co: Not Co: Not Co: Not Co: Not
W 50 100 detected detected detected detected detected
Pb 5 10 Ni: Ni: Ni: Ni: Not Ni: Not
0.0095 0.0248 0.0188 detected detected
Cu: < Cu: < Cu: < Cu: Not Cu: Not
0.0281 0.0281 0.0281 detected detected
Zn: Zn: Zn: Zn: Zn:
0.4764 0.7442 0.7335 0.6782 1.0264
As: Not As: Not As: Not As: Not As: Not
detected detected detected detected detected
Mo: Not Mo: Not Mo: Not Mo: Not Mo: Not
detected detected detected detected detected
Cd: < Cd: < Cd: < Cd: Not Cd: Not
0.0015 0.0015 0.0015 detected detected

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Sn: Not Sn: Not Sn: Not Sn: Not Sn: Not
detected detected detected detected detected
Sb: Sb: Sb: Sb: Sb:
0.3186 0.2012 0.1971 0.3127 0.2825
Ba: Ba: Ba: Ba: Ba:
0.0858 0.1332 0.1049 0.0604 0.0459
W: W: W: W: W:
0.3186 1.6763 0.4941 0.5038 2.5966
Pb: Pb: Pb: Pb: Pb: <
0.0174 0.0246 0.0124 0.0200 0.0261

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Table 210 Test Results of Antioxidant Degradation in the Samples under 25±2 ■ Condition (HPLC)
Instrument Method
Batch No. reference substance detection detection PDE AET 0 month 1st month 2nd month 3rd month 6th month
limit limit
3,5-DTBHA: 3,5-DTBHA: 3,5-DTBHA: 3,5-DTBHA:
Not detected Not detected Not detected Not detected Not detected
3,5-di-tert-butyl-4-
2,4-DTB: Not 2,4-DTB: Not 2,4-DTB: Not 2,4-DTB: Not 2,4-DTB: Not

GK201709004 hydroxylphenylpropanoic 0.08918 22.29500 300 600
detected detected detected detected detected
acid(3,5-DTBHA)
2,6-DTBQ: 2,6-DTBQ: 2,6-DTBQ: 2,6-DTBQ:
Not detected Not detected Not detected Not detected Not detected
3,5-DTBHA: 3,5-DTBHA: 3,5-DTBHA: 3,5-DTBHA: 3,5-DTBHA:
Not detected Not detected Not detected Not detected Not detected
2,4-di-tert- 2,4-DTB: Not 2,4-DTB: Not 2,4-DTB: Not 2,4-DTB: Not 2,4-DTB: Not
GK201709005 0.07558 18.89500 2500 5000
butylphenol(2,4-DTB) detected detected detected detected detected
2,6-DTBQ: 2,6-DTBQ: 2,6-DTBQ: 2,6-DTBQ: 2,6-DTBQ:
Not detected Not detected Not detected Not detected Not detected
3,5-DTBHA: 3,5-DTBHA: 3,5-DTBHA: 3,5-DTBHA: 3,5-DTBHA:
Not detected Not detected Not detected Not detected Not detected
2,6-di-tert-butyl-1,4-
2,4-DTB: Not 2,4-DTB: Not 2,4-DTB: Not 2,4-DTB: Not 2,4-DTB: Not
GK201709010 benzoquinone(2,6- 0.06028 15.07000 2500 5000
detected detected detected detected detected
DTBQ)
2,6-DTBQ: 2,6-DTBQ: 2,6-DTBQ: 2,6-DTBQ: 2,6-DTBQ:
Not detected Not detected Not detected Not detected Not detected

Table 211 Test Results of Antioxidant and Sulfur in the Samples under 25±2 ■ Condition (HPLC)
Metho
Instrum
d
ent AE
Batch No. Migration substances detecti PDE 0 month 1st month 2nd month 3rd month 6th month
detection T
on
limit
limit
GK2017090 3.0675 sulfur: Not sulfur: Not sulfur: Not sulfur: Not sulfur: Not
sulfur 0.02454 9 18
04 0 detected detected detected detected detected

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antioxidant1 antioxidant1 antioxidant1 antioxidant1 antioxidant1
010: Not 010: Not 010: Not 010: Not
detected detected detected detected detected
antioxidant1 antioxidant1 antioxidant1 antioxidant1 antioxidant1
Pentaerythritol tetrakis(3-(3,5-di- 076: Not 076: Not 076: Not 076: Not
tert-butyl-4- detected detected detected detected detected
8.7675 1000 2000
hydroxyphenyl)propionate) 0.07014 antioxidant1 antioxidant1 antioxidant1 antioxidant1 antioxidant1
0 0 0
(antioxidant 68 68 68 68 68
1010) phosphate: phosphate: phosphate: phosphate: phosphate:
Not detected Not detected Not detected Not detected Not detected
antioxidant1 antioxidant1 antioxidant1 antioxidant1 antioxidant1
68: Not 68: Not 68: Not 68: Not
detected detected detected detected detected
Octadecyl 3-(3,5-di-tert-butyl-4- sulfur: Not sulfur: Not sulfur: Not sulfur: Not
30.018
hydroxyphenyl)propionate(antioxid 0.24015 300 600 detected detected detected detected detected
75
ant1076) antioxidant1 antioxidant1 antioxidant1 antioxidant1 antioxidant1
010: Not 010: Not 010: Not 010: Not 010: Not
detected detected detected detected detected
antioxidant1 antioxidant1 antioxidant1 antioxidant1 antioxidant1
076: Not 076: Not 076: Not 076: Not 076: Not
GK2017090
detected detected detected detected detected
05 Tris(2,4-di-tert-butylphenyl)
45.070 antioxidant antioxidant1 antioxidant1 antioxidant1 antioxidant1
phosphate (antioxidant ` 2500 5000
00 168 68 68 68 68
168 phosphate)
phosphate: phosphate: phosphate: phosphate: phosphate:
Not detected Not detected Not detected Not detected Not detected
antioxidant1 antioxidant1 antioxidant1 antioxidant1 antioxidant1
68: Not 68: Not 68: Not 68: Not
detected detected detected detected detected
sulfur: Not sulfur: Not sulfur: Not sulfur: Not sulfur: Not
detected detected detected detected detected
antioxidant1 antioxidant1 antioxidant1 antioxidant1 antioxidant1
GK2017090 Tris(2,4-ditert-butylphenyl) 29.737 010: Not 010: Not 010: Not 010: Not
0.23790 2500 5000
010 phosphite(antioxidant 168) 50 detected detected detected detected detected
antioxidant1 antioxidant1 antioxidant1 antioxidant1 antioxidant1
076: Not 076: Not 076: Not 076: Not 076: Not
detected detected detected detected detected
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antioxidant1 antioxidant1 antioxidant1 antioxidant1 antioxidant1
68 68 68 68 68
phosphate: phosphate: phosphate: phosphate: phosphate:

Not detected Not detected Not detected Not detected Not detected
antioxidant1 antioxidant1 antioxidant1 antioxidant1 antioxidant1
68: Not 68: Not 68: Not 68: Not 68: Not
detected detected detected detected detected

Table 212 Test Results of D5, D6 and Erucamide in the Samples under 25±2 ■ Condition (GC)
Instrument Method
0

Batch No. Migration substances detection detection PDE AET 1st month 2nd month 3rd month 6th month
month
limit limit
decamethylcyclopentane D5: Not
0.46700 4.67000 160 320 detected
siloxane(D5)
D6: Not D5: Not D5: Not D5: Not
detected detected detected detected detected
BHT: D6: Not D6: Not D6: Not D6: Not
Not detected detected detected detected
detected BHT: Not BHT: Not BHT: Not BHT: Not
DBP: detected detected detected detected
GK201709004
Not DBP: Not DBP: Not DBP: Not DBP: Not
dodecamethylcyclhexasiloxane(D6) 0.47628 4.76280 9 18 detected detected detected detected detected
DEHP: DEHP: Not DEHP: Not DEHP: Not DEHP: Not
Not detected detected detected detected
detected EM: Not EM: Not EM: Not EM: Not
EM: detected detected detected detected
Not
detected
D5: Not D5: Not D5: Not D5: Not
2,6-dit-Bu-4-methylphenol(BHT) 0.24502 2.45020 120 240 D5: Not detected detected detected detected
detected D6: Not D6: Not D6: Not D6: Not
GK201709005
D6: Not detected detected detected detected
dibutyl phthalate(DBP) 0.30793 3.07930 240 480 detected BHT: Not BHT: Not BHT: Not BHT: Not
detected detected detected detected

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BHT: DBP: Not DBP: Not DBP: Not DBP: Not
Not detected detected detected detected
detected DEHP: Not DEHP: Not DEHP: Not DEHP: Not
DBP: detected detected detected detected
Not EM: Not EM: Not EM: Not EM: Not
detected detected detected detected detected
DEHP:
Not
detected
EM:
Not
detected
D5: Not
bis-(2-ethylhexyl)phthalate(DEHP) 0.36214 3.62140 240 480 detected
D6: Not D5: Not D5: Not D5: Not
detected detected detected detected detected
BHT: D6: Not D6: Not D6: Not D6: Not
Not detected detected detected detected
detected BHT: Not BHT: Not BHT: Not BHT: Not
DBP: detected detected detected detected
GK201709010
Not DBP: Not DBP: Not DBP: Not DBP: Not
erucamide(EM) 0.53254 5.32540 120 240 detected detected detected detected detected
DEHP: DEHP: Not DEHP: Not DEHP: Not DEHP: Not
Not detected detected detected detected
detected EM: Not EM: Not EM: Not EM: Not
EM: detected detected detected detected
Not
detected

Table 213 Test Results of Stearic Acid and Palmitic Acid in the Samples under 25±2 ■ Condition (GC)
Instrument Method
Migration

Batch No. detection detection PDE AET 0 month 1st month 2nd month 3rd month 6th month
substances
limit limit

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palmitic acid: palmitic acid: palmitic acid: palmitic acid: palmitic acid:
Not detected Not detected Not detected Not detected Not detected
GK201709004
stearic acid: stearic acid: stearic acid: stearic acid: stearic acid:
palmitic Not detected Not detected Not detected Not detected Not detected
1.60328 22.44592 5000 10000
acid palmitic acid: palmitic acid: palmitic acid: palmitic acid:
Not detected Not detected Not detected Not detected Not detected
GK201709005
stearic acid: stearic acid: stearic acid: stearic acid: stearic acid:

Not detected Not detected Not detected Not detected Not detected
palmitic acid: palmitic acid: palmitic acid: palmitic acid: palmitic acid:
Not detected Not detected Not detected Not detected Not detected
GK201709010 stearic acid 1.60988 22.53832 6250 12500
stearic acid: stearic acid: stearic acid: stearic acid: stearic acid:
Not detected Not detected Not detected Not detected Not detected

Table 214 Test Results about Polydimethylsiloxane in the Samples under 25±2 ■ Condition (HPLC)

Instrument Method									
Migration									
Batch No.	detection	detection	PDE	AET	0 month	1st month	2nd month	3rd month	6th month
substances									
limit limit									
GK201709004	Not detected	Not detected	Not detected	Not detected	Not detected	Not detected	Not detected	Not detected	
GK201709005	polydimethylsiloxane	0.16418	2.46270	9	18	Not detected	Not detected	Not detected	Not detected
Not detected									
GK201709010	Not detected	Not detected	Not detected	Not detected	Not detected	Not detected	Not detected	Not detected	

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Table 215 Test Results about Isobutene-isoprene Oligomers (C13H24, C13H23CL, C21H40) in the Samples under 25±2 ■ Condition
Batch No. Test method 0 month 1st month 2nd month 3rd month 6th month
Not Not Not Not Not
GK201709004 GC full- detected detected detected detected detected
scan(Scan)
Not Not Not Not Not
GK201709005 Characteristic
detected detected detected detected detected
ion
scanning(SIM) Not Not Not Not Not
GK201709010
detected detected detected detected detected

Conclusion:
Migration tests showed no degraded antioxidant, no antioxidant, no sulfur, no stearic acid, no palmitic acid, no erucamide, no plasticizer, no epoxy silane and no isobutene-isoprene oligomers in the samples, residual metal elements were well below PDE.
Summary: In conclusion, the PCV13 manufactured by Walvax uses pre-filling syringes packaging manufactured by Shandong Weigao Group Medical Polymer Products Co. LTD. Compatibility studies showed no high-risk extractum in PCV13 and no destruction to the packaging material, PCV13 maintained good quality and safety. Walvax will continue conducting research following compatibility research protocols of 36 months under 5±3 ■, providing an objective and scientific basis of selecting Shangdong Weigao's pre-filling syringes packaging.

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3.2.P.2.5 Microbiological Attributes
This product is sealed packages of sterile suspension. As the stability results in the "2.3.P.8 Stability" section show that during the entire period of the study, the index of all safety remains constant and qualified. It can be approved that the tightness and integrity of the packaging system of this product meet the design and quality requirements. Therefore, the packaging can fully guarantee that products will not be contaminated by microorganisms during the validity period.

3.2.P.2.6 Compatibility
Please refer to 2.3.P.2.4

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3.2.P.3 Manufacture
3.2.P.3.1 Information on the Manufacturer

Name: Yuxi Walvax Biotechnology Co., Ltd.
Address: No. 83 South Dongfeng Road, High ■ New Technology Industries Development Zone, Yuxi, Yunnan Province, China
Function: vaccine manufacture
Tel: 0877-2076210
Fax: 0877-2076918
Manufacturing site: 1st floor, Fill & Finish Centre, K Building 203#, No. 83 South Dongfeng Road, High & New Technology Industries Development Zone, Yuxi, Yunnan Province, China
Tel: 0877-2076210
Fax: 0877-2076918
QC site: 3rd and 4th floor, QC laboratory, RD&QC building 308#, No. 83 South Dongfeng Road, High & New Technology Industries Development Zone, Yuxi, Yunnan Province, China
Tel: 0877-2076210
Fax: 0877-2076918

3.2.P.3.2 Batch Formula
Table 216 Batch Formula Information
Active Substances Nominal Composition Function
Pneumococcal polysaccharide serotype 1 2.6 µg Antigen
Pneumococcal polysaccharide serotype 3 2.5 µg Antigen
Pneumococcal polysaccharide serotype 4 3.0 µg Antigen
Pneumococcal polysaccharide serotype 5 2.5 µg Antigen

Pneumococcal polysaccharide serotype 6A 2.5 µg Antigen
Pneumococcal polysaccharide serotype 6B 6.0 µg Antigen
Pneumococcal polysaccharide serotype 7F 2.85 µg Antigen
Pneumococcal polysaccharide serotype 9V 2.5 µg Antigen
Pneumococcal polysaccharide serotype 14 2.75 µg Antigen
Pneumococcal polysaccharide serotype 18C 3.25 µg Antigen
Pneumococcal polysaccharide serotype 19A 2.6 µg Antigen
Pneumococcal polysaccharide serotype 19F 2.75 µg Antigen
Pneumococcal polysaccharide serotype 23F 3.0 µg Antigen
Aluminum phosphate 0.5 mg Adjuvant
Sodium chloride 4.25 mg Excipient
Sodium dihydrogen phosphate 44.35 µg Excipient
Disodium hydrogen phosphate 19.0 µg Excipient
Water for injection Qs to 0.5 mL Excipient

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3.2.P.3.3 Description of Manufacturing Process

3.2.P.3.3.1 Flow Chart of Manufacturing Process and Process Control

Calculate the amount of excipients and
13 serotypes of monovalent bulk conjugate
based on formulation protocol

Mix WFI, 10% NaCl solution
and phosphate-buffered saline to
obtain excipients mixture

Add 13 serotypes of monovalent bulk
conjugate into excipients mixture

Stir for more than 10 min

Quality Control points and items:
Sterile filtration Sterility test

Wash and sterilization of
vials and rubber plugs

Sterile filling

Capping with sterile
aluminum caps

Visual inspection
Quality Control points and items:
Appearance, extractable volume,
osmolality, pH, NaCl content,
Labelling Polysaccharide content, conjugate
antigen content, conjugate protein
content, total protein content,
Packaging sterility test, abnormal toxicity test,
aluminium content

Storage

Figure 28 Manufacturing Process Flow Chart for Fill and Finish of Finished Product (Vial)

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Calculate the amount of excipients and
13 serotypes of monovalent bulk conjugate
based on formulation protocol

Mix WFI, 10% NaCl solution
and phosphate-buffered saline to
obtain excipients mixture

Add 13 serotypes of monovalent bulk
conjugate into excipients mixture

Stir for more than 10 min

Quality Control points and items:
Sterile filtration Sterility test

Wash and sterilization of
pre-filled syringes

Sterile filling

Visual inspection Quality Control points and items:
Appearance, extractable volume,
osmolality, pH, NaCl content,
Polysaccharide content, conjugate
Labelling antigen content, conjugate protein
content, total protein content,
sterility test, abnormal toxicity test,
Packaging aluminium content

Storage

Figure 29 Manufacturing Process Flow Chart for Fill and Finish of Finished Product (Pre-filled Syringe)

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3.2.P.3.3.2 Discription of Manufacturing Process

Walvax issued Standard Operation Procedures for fill and finish of 13-valent Pneumococcal Polysaccharide Conjugate Vaccine according to Regulation of registration for 13-valent Pneumococcal Conjugate Vaccine approved by National Medical Products Administration (Standard No: YBS00392019). The product is a finished product after formulation, filling, capping, visual inspection, labelling, and packaging.

Manufacturing process steps include:

(1) Perpetration the excipients stock solution

Weigh the medicinal grade sodium chloride in the container according to the formula amount, dry heat sterilization and remove the pyrogen; weigh the medicinal grade phosphate in another container according to the formula amount, dry heat sterilization and remove the pyrogen; add the formula amount of WFI into the container, mix well, autoclave and obtain the excipients stock solution, place at 2-8 °C for use;

(2) Weigh the formula amount of each serotype polysaccharide bulk conjugate, mix well to obtain the mixture.

(3) Measure the formula amount of aluminium phosphate adjuvant, phosphate stock solution (pH 6.5), sodium chloride stock solution and WFI, mix well to obtain the excipients mixture.

(4) Pump the polysaccharide bulk conjugate mixture into the excipients mixture while stirring, sterile filtrate to obtain the final bulk.

(5) Fill the final bulk into vials or pre-filled syringes, after labelling and packaging to obtain finished products.

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3.2.P.3.4 Controls of Critical Steps and Intermediates

Table 217 Controls of Critical Steps of Formulation and Fill & Finish Process for 13-valent

Pneumococcal Polysaccharide Conjugate Vaccine

Step Critical manufacturing steps Parameter range

Washing and sterilization Sterilization time 30 min

for rubber plug Sterilization temperature 121 °C

Ultrasonic power 50 (300W)

Circulating water pressure 0.20-0.50 Mpa

Purified water pressure 0.20-0.50 Mpa

WFI pressure 0.20-0.50 Mpa

Compressed air pressure 0.40-0.60 Mpa

Sterilization temperature 320±10 °C

Washing and sterilization Sterilization time ≥ 5 min

for vials Belt frequency 20 HZ

Preheating ≥ 60 Pa

Heating 1 ≥ 160 Pa

Heating 2 ≥ 160 Pa

Differential pressure

Cooling ≥ 50 Pa

Cooling 2 ≥ 50 Pa

Cooling 3 ≥ 50 Pa

Final bulk mixing speed 25 rpm

Stirring speed in 10 L bottle 250 rpm

Sterile filling

Extractable volume 0.50-0.60 mL

Filling room temperature 18±2 °C

Capping Capping temperature 18±2 °C

Labelling Room temperature < 25 °C

Packaging temperature ≤ 25 °C

The information of packaging material

Packaging

Packaging information batch number, production date and

expiry date are correct and clear

Warehousing Store temperature 2-8 °C

3.2.P.3.5 Process Validation and/or Evaluation

The 13-valent Pneumococcal Polysaccharide Conjugate Vaccine production process includes six process steps: final bulk formulation, filling, capping, physical appearance inspection, labelling, and packaging. Through evaluate the critical process parameters and control of each step, in 2013, a total of 6 batches of process validation were carried out for 13-valent Pneumococcal Conjugate vaccines, including 3 batches of vials packaging (G1201708001, G1201708002, G1201708001), and 3 batches of pre-filled syringe packaging (GK201709001, GK201709002, GK201709003). The validation of the production process of the batch of 13-valent Pneumococcal Polysaccharide Conjugate Vaccine shows that the production process of 13-valent pneumococcal conjugate vaccine can continuously produce products that meet the quality standards within the specified process parameters in a controlled state, the process is stable. More detail please refer to 3.2.S.2.5.

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3.2.P.4 Control of Excipients

Excipients included in the manufacturing of 13-valent Pneumococcal Polysaccharide Conjugate Vaccine are: sodium chloride, disodium hydrogen phosphate, sodium dihydrogen phosphate,

aluminium phosphate, in which sodium chloride, disodium hydrogen phosphate, sodium dihydrogen phosphate are tested according to CP. Aluminium phosphate is not included in CP, which was tested according to Standard Operation Procedures and Specification for Aluminium Phosphate (WSYX-QC-SOP-00-09-0608) issued by Walvax. The specifications of all excipients are showed as following tables, all excipients are released after qualified.

3.2.P.4.1 Specifications

Table 218 Specification of Aluminium Phosphate

Test items Specification Analytical method

Should be a white suspension, which can

Appearance layer after long standing, back to suspension Visual detection after shaking.

pH 6.0-8.0 pH meter

Chloride NMT 0.5% Chemical reaction

Sulphate Should be less than 0.1% Chemical reaction

The color of test tube should not be deeper

Heavy metal than standard solution tube. The content of Thioacetamide method

heavy metal should not more than 0.0002%.

Arsenic salt NMT 0.00004 Gutzeit

Aluminium

Should not less than 20 mg/mL. Chemical reaction

phosphate

Sodium chloride NMT 10 g/L Chemical reaction

Bacterial endotoxin Should less than 5.0 EU/mg Gel-clot method

Sterility Should comply with the requirements Membrane filtration method

Table 219 Specification of Sodium Chloride

Item specification Analytical method

This product is colourless, transparent cubic

Appearance crystal or white crystalline powder, odorless. Physical examination

Soluble in water and almost insoluble in ethanol.

1.Sodium Salt:

(1) Take platinum wire, wet with hydrochloric acid, dip test product, burn in a colorless flame, and the flame is bright yellow.

(2) Take 100 mg the test product, put it in a 10 ml test tube, dissolve with 2 ml injection water, add 15% potassium carbonate solution 2 ml, Identify heated to boil, should not produce precipitation; Chemical reaction add 4 ml potassium pyroantimonate test solution, heated to boil; cool in ice water, if necessary, rub tube wall with glass stick, should produce dense precipitation.

2.Chloride

(1) Add silver nitrate solution, produce a white coagulate precipitate, which was dissolved by

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adding ammonia solution, add dilute nitric acid, the precipitate reproduces.

(2) Chlorine can be produced; wet potassium iodide starch test paper turns blue.

Add 2 drops of bromothymol blue indicator solution, if show yellow, add sodium hydroxide titration solution (0.02mol/L) 0.10ml, turn blue;

Acidity and alkalinity Physical examination

if show blue or green, add hydrochloric acid

titration solution (0.02mol/L) 0.20ml, turn yellow.

Clarity and color of

Should be colorless and clear Physical examination solution

Under the sunlight (or under the fluorescent

Iodide lamp), no blue marks can be seen in the crystals Starch reaction within 5 minutes.

Ultraviolet visible

Bromide NMT 0.01%

spectrophotometric method

Sulfate NMT 0.002% Chemical reaction

Ultraviolet visible

Nitrite Absorbance at 354nm should not more than 0.01 spectrophotometric method

Phosphate NMT 0.0025% Chemical reaction

Ferrocyanide Should not show blue within 10 min Chemical reaction

Barium salt Both of the solution should be clear Chemical reaction

Calcium salt No opalescence is produced within 5 min Chemical reaction

Magnesium salt NMT 0.001% Chemical reaction

Potassium salt NMT 0.02% Chemical reaction

Loss on drying NMT 0.5% Drying method

Ferric salt NMT 0.0003% Chemical reaction

heavy metal NMT 0.0002% Thioacetamide method

Arsenic salt 0.00004% Gutzeit

Bacterial endotoxin test Less than 5 EU/g Semi quantitative gel method

Assay NLT 99.5% Titration

Table 220 Specification of Sodium Dihydrogen Phosphate

Item Specification Analytical method

This product is colorless or white crystal or

lump; odorless; it is easy to weathering at room temperature
Appearance Physical method
temperature air. Soluble in water, almost insoluble in ethanol.

1.the aqueous solution of test product effervescent on addition of sodium carbonate.

2.1. Sodium Salt:

(1) Take platinum wire, wet with hydrochloric acid, dip test product, burn in a colorless flame, and the flame is bright yellow.

(2) Take 100 mg the test product, put it in a 10 ml test tube, dissolve with 2 ml injection water, add 15% potassium carbonate solution 2 ml, heated to boil, should not produce precipitation; add 4 ml potassium pyroantimonate test solution, heated to boil; cool in ice water, if necessary, rub tube wall with glass stick, should produce dense precipitation.

2.2. Phosphate:

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(1) The neutral solution of the test product, add silver nitrate solution, produce light yellow precipitation. Separate, precipitation is easy to dissolve in ammonia solution or dilute nitric acid.

(2) add magnesium ammonium chloride solution in the test product, produce white crystalline precipitate.

(3) take test product, add ammonium molybdate solution and nitric acid, produce yellow precipitate, separate, which can be dissolved in ammonia solution.

Alkalinity 4.1-4.5 pH meter

Clarity of solution Should be clear and colorless Physical method

Chloride NMT 0.01% Chemical reaction

Sulphate NMT 0.05% Chemical reaction

Loss on drying 10.0-15.0% Drying method

Aluminium salt NMT 0.01% Chemical reaction

Calcium salt NMT 0.01% Chemical reaction

Heavy metal NMT 10ppm Thioacetamide method

Arsenic salt NMT 0.0005% Gutzeit

Assay ≥98% Titration

Table 221 Specification of Dodecahydrate Disodium Hydrogen Phosphate

Item Specification Analytical method

This product is colorless or white crystal or

lump; odorless; it is easy to weathering at room

Appearance Physical method

temperature air. Soluble in water, almost insoluble in ethanol.

1.Sodium Salt:

(1) Take platinum wire, wet with hydrochloric acid, dip test product, burn in a colorless flame, and the flame is bright yellow.

(2) Take 100 mg the test product, put it in a 10 ml test tube, dissolve with 2 ml injection water, add 15% potassium carbonate solution 2 ml, heated to boil, should not produce precipitation; add 4 ml potassium pyroantimonate test solution, heated to boil; cool in ice water, if necessary, rub tube wall with glass stick, should produce dense precipitation.

Identification 2.Phosphate: Chemical reaction

(1) The neutral solution of the test product, add silver nitrate solution, produce light yellow precipitation. Separate, precipitation is easy to dissolve in ammonia solution or dilute nitric acid.

(2) add magnesium ammonium chloride solution in the test product, produce white crystalline precipitate.

(3) take test product, add ammonium molybdate solution and nitric acid, produce yellow precipitate, separate, which can be dissolved in ammonia solution.

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Alkalinity 9.1-9.4 pH meter

Clarity of solution Should be clear and colorless Physical method

Chloride NMT 0.001% Chemical reaction

Sulphate NMT 0.002% Chemical reaction

Carbonate Should not produce bubbles Chemical reaction

Insoluble particles in

NMT 10mg (0.05%) Chemical reaction, weighing water

The violet red color of the solution will not

Reductive substance Chemical reaction

disappear.

Loss on drying 55.0-64.0% Drying method

Ferric salt NMT 0.002% Chemical reaction

heavy metal NMT 5ppm Thioacetamide method

Arsenic salt NMT 0.0002% Gutzeit

3.2.P.4.2 Analytical Procedures

The analysis of excipients is carried out according to CP, please refer to the analytical procedures in CP2015.

3.2.P.4.3 Validation of Analytical Procedures

Not Applicable.

3.2.P.4.4 Justification of Specification

The excipients used in the production of 23-valent Pneumococcal Polysaccharide Vaccine include lactose, sodium chloride, water for injection, they are tested according to the Chinese Pharmacopoeia.

3.2.P.4.5 Excipients of Human or Animal-origin

Not Applicable.

3.2.P.4.6 Novel Excipients

Not Applicable.

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3.2.P.5 Control of Drug Product

3.2.P.5.1 Specification(s)

Table 222 Specification for Finished Product

Test items Specifications

Identity /

Appearance Milky white suspension after shaking.

Volume No less than 0.50 ml/dose

Osmolality 220 – 310 mOsm/kg

Aluminum content 0.18 – 0.27 mg/ml

pH 5.5 – 7.5

Sodium chloride 7.5 – 9.5 g/L

Range (μ

Serotype

g/ml

1 3.6 – 6.8

3 3.5 – 6.5

4 2.0 – 7.8

5 3.5 – 6.5

6A 3.5 – 6.5

Polysaccharide content 6B 8.4 – 15.6

7F 4.0 – 7.4

9V 3.5 – 6.5

14 3.8 – 7.2

18C 4.6 – 8.4

19A 3.6 – 6.8

19F 3.8 – 7.2

23F 4.2 – 7.8

No less than the limit as below.

Standard

Release in

Serotype standard validity

(% period

(%)

1 30 25

3 80 75

4 75 70

Conjugate antigen content 5 75 70

6A 80 75

6B 80 75

7F 75 70

9V 80 75

14 80 75

18C 80 75

19A 70 65

19F 65 60

23F 65 60

No less than 80% for release standard, no less than 75% in validity

Conjugate protein content

period.

Total protein content 41 – 150 μg/dose

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Sterility Meets the requirement of the test

No adverse reactions, healthy and weight gain for each guinea pig and

Abnormal toxicity

mouse during observation period.

Bacterial endotoxin No more than 12.5 EU/dose

Please refer to Annex 4 SOP for Finished Product.

3.2.P.5.2 Analytical Procedures

Table 223 SOP List for Finished Product Test

Test items SOP Document Code Method

Visual

Appearance SOP for Appearance Check WSYX-QC-SOP-00-09-0108 determination

SOP for the Extractable Volume Weight

Volume WSYX-QC-SOP-00-09-0109

Test measurement

pH SOP for pH Value Test WSYX-QC-SOP-00-09-0110 pH meter

SOP for Aluminum Hydroxide

Aluminum

(or Aluminum phosphate) WSYX-QC-SOP-00-09-0117 Titration content

Content Test

Sodium chloride SOP for Sodium Chloride

WSYX-QC-SOP-00-09-0120 Titration

content Content Determination

Conjugate SOP for Conjugate Antigen Biochemical

WSYX-QC-SOP-06-09-0003

antigen content Content Determination method

Osmolality SOP for Osmolality Test WSYX-QC-SOP-00-09-0135 Osmometer

Total protein SOP for Protein Content

WSYX-QC-SOP-0009-0102 Lowry method

content Determination

SOP of Conjugate Protein

Conjugate Content Test for 13-valent Gel

WSYX-QC-SOP-06-09-0006

protein content Pneumococcal Polysaccharide chromatography

Conjugate Vaccine

Abnormal SOP for Abnormal Toxicity

WSYX-AC-SOP-00-09-0021 Animal test

toxicity Test

Bacterial SOP for Bacterial Endotoxin

WSYX-QC-SOP-00-09-0203 Gel-clot method

Endotoxin Test

Membrane

Sterility SOP for Sterility Test WSYX-QC-SOP-00-09-0202

filtration method

SOP for Polysaccharide Content

Polysaccharide Rate nephelometry

Determination with Rate WSYX-QC-SOP-00-09-0141

content method

nephelometry Method

Conjugate SOP for Conjugate Antigen Gel

WSYX-QC-SOP-06-09-0003

antigen content Content Determination chromatography

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3.2.P.5.3 Validation of Analytical Procedures

Table 224 Validation or Qualification List of Analytical Procedures

Validation/Qualification

Test items Document Code Test Method

protocol

Qualification Protocol of

Aluminum Content

Aluminum content Determination for 13- WSYX-QC-TS-0006-0661 Titration

valent Pneumococcal

Conjugate Vaccine

Qualification Protocol of

Sodium chloride content

Sodium chloride content Determination for 13- WSYX-QC-TS-0006-0660 Titration

valent Pneumococcal

Conjugate Vaccine

Qualification Protocol of

Total Protein content

Total protein content Determination for 13- WSYX-QC-TS-0006-0747 Lowry method

valent Pneumococcal

Conjugate Vaccine

Validation Protocol of

Conjugate Antigen

Conjugate antigen Gel

Content Test for 13-valent WSYX-QC-TS-0006-0728

content chromatography

Pneumococcal Conjugate

Vaccine

Qualification Protocol of

Bacterial Endotoxin Test

Bacterial endotoxin for 13-valent WSYX-QC-TS-0006-0764 Gel-clot method

Pneumococcal Conjugate

Vaccine

Applicability Test for

Sterility Test Method of Membrane

Sterility SYX-QC-TS-0006-0543
13-valent Pneumococcal filtration method
Conjugate Vaccine
Qualification Protocol of
Rate
Rate Nephelometry
Polysaccharide content WSYX-QC-TS-0006-0647 nephelometry
Method for Detection of
method
Polysaccharide Content

3.2.P.5.4 Batch analyses
Different batches of 13-valent Pneumococcal Polysaccharide Conjugate Vaccine were analyzed based on the analytical procedures. The data is showing in Table 225.

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Table 225 Batch Analyses Data Statistics for Finished Products
3 batches of pilot sample 3 batches for IND
Item Acceptable criteria
20110304 20110305 20110406 20110707 20110708 20110709
Should be milky white
Appearance Complied Complied Complied Complied Complied
suspension after shaking.
Volume Not less than 0.5 ml/dose Complied Complied Complied Complied Complied
pH 5.5-7.5 6.6 6.6 6.6 6.5 6.5 6.5
Aluminium
0.18-1.2 mg/mL 1.1 1.0 0.9 1.0 1.0 0.9
content
Sodium
chloride 7.5-9.5 g/L 8.7 8.6 8.9 8.4 8.4 8.5
content
1: 2.4 µg 1: 2.3 µg 1: 2.2 µg 1: 2.2 µg 1: 2.3 µg
3: 2.5 µg 3: 2.4 µg 3: 2.3 µg 3: 2.2 µg 3: 2.2 µg
In each human dose, the 4: 2.4 µg 4: 2.4 µg 4: 2.3 µg 4: 2.2 µg 4: 2.2 µg
polysaccharide content of 5: 2.5 µg 5: 2.4 µg 5: 2.4 µg 5: 2.3 µg 5: 2.2 µg 5: 2.3 µg
serotypes 1, 3, 4, 5, 6A, 7F, 6A: 2.4 µg 6A: 2.4 µg 6A: 2.3 µg 6A: 2.2 µg 6A: 2.2 µg
9V, 14, 18C, 19A, 19F and 6B: 4.8 µg 6B: 4.8 µg 6B: 4.5 µg 6B: 4.4 µg 6B: 4.5 µg
Polysaccharide
23F should be not less than 7F: 2.4 µg 7F: 2.4 µg 7F: 2.2 µg 7F: 2.2 µg 7F: 2.2 µg
content
2.2 µg respectively, and the 9V: 2.3 µg 9V: 2.3 µg 9V: 2.2 µg 9V: 2.2 µg 9V: 2.2 µg
polysaccharide content of 14: 2.4 µg 14: 2.4 µg 14: 2.2 µg 14: 2.3 µg 14: 2.3 µg
serotype 6B should be not 18C: 2.4 µg 18C: 2.3 µg 18C: 2.4 µg 18C: 2.2 µg 18C: 2.3 µg
less than 4.4 µg. 19A: 2.4 µg 19A: 2.3 µg 19A: 2.3 µg 19A: 2.2 µg 19A: 2.2 µg
19F: 2.5 µg 19F: 2.5 µg 19F: 2.4 µg 19F: 2.3 µg 19F: 2.3 µg
23F: 2.4 µg 23F: 2.3 µg 23F: 2.3 µg 23F: 2.2 µg 23F: 2.3 µg
During observation period,
guinea pigs and mice shall
all be healthy, no adverse
Abnormal
reactions, and each guinea Complied Complied Complied Complied Complied
toxicity
pig and mouse may gain
weight at the end of the
experiment.
Bacterial
NMT 12.5 EU/dose Complied Complied Complied Complied Complied
endotoxin

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Test according to
Sterility requirements, should Complied Complied Complied Complied Complied
comply.

1 batch for clinical 6 batches for process validation
Item Acceptable criteria
201511003 GI201708001 GI201708002 GI201709003 GK201709001 GK201709002 GK201709003
Should be milky white
Appearance Complied Complied Complied Complied Complied Complied
suspension after shaking.
Volume Not less than 0.5ml/dose Complied Complied Complied Complied Complied
pH 5.5-7.5 6.42 6.2 6.25 6.22 6.22 6.20 6.22
Aluminium
0.18-1.2 mg/mL 1.01 1.03 1.08 0.90 0.94 0.99 1.07
content
Sodium
chloride 7.5-9.5 g/L 8.31 8.48 8.46 8.815 8.81 8.01 8.96
content
1: 3.0 µg 1: 3.2 µg 1: 2.8 µg 1: 2.9 µg 1: 3.0 µg 1: 3.0 µg 1: 3.1 µg
3: 3.1 µg 3: 2.4 µg 3: 2.6 µg 3: 2.4 µg 3: 2.7 µg 3: 2.8 µg 3: 2.8 µg
In each human dose, the 4: 2.8 µg 4: 2.8 µg 4: 2.4 µg 4: 2.67 µg 4: 2.6 µg 4: 2.6 µg
polysaccharide content 5: 2.4 µg 5: 2.7 µg 5: 2.4 µg 5: 2.7 µg 5: 2.6 µg 5: 2.6 µg
of serotypes 1, 3, 4, 6A: 3.0 µg 6A: 2.9 µg 6A: 2.9 µg 6A: 2.6 µg 6A: 2.7 µg 6A: 2.6 µg
5, 6A, 7F, 9V, 14, 18C,
6B: 5.8 µg 6B: 6.0 µg 6B: 5.4 µg 6B: 4.8 µg 6B: 4.8 µg 6B: 4.6 µg 6B: 4.7 µg
Polysaccharide 19A, 19F and 23F

7F: 3.0 µg 7F: 3.1 µg 7F: 2.8 µg 7F: 2.7 µg 7F: 2.8 µg 7F: 2.8 µg 7F: 2.8 µg
content should be 2.2-3.3 µg
9V: 3.2 µg 9V: 2.6 µg 9V: 2.9 µg 9V: 2.4 µg 9V: 3.1 µg 9V: 3.1 µg 9V: 3.0 µg
respectively, and the
polysaccharide content 14: 2.9 µg 14: 2.7 µg 14: 2.6 µg 14: 2.7 µg 14: 2.6 µg 14: 2.6 µg 14: 2.7 µg
of serotype 6B should be 18C: 2.9 µg 18C: 2.7 µg 18C: 2.8 µg 18C: 2.7 µg 18C: 2.6 µg 18C: 2.9 µg 18C: 2.6
µg
4.4-6.6 µg. 19A: 3.1 µg 19A: 2.5 µg 19A: 2.5 µg 19A: 2.8 µg 19A: 3.0 µg 19A: 3.0 µg 19A: 3.1 µg
19F: 2.9 µg 19F: 2.5 µg 19F: 2.6 µg 19F: 2.8 µg 19F: 3.0 µg 19F: 2.9 µg 19F: 2.9 µg
23F: 3.1 µg 23F: 2.6 µg 23F: 2.9 µg 23F: 3.0 µg 23F: 2.4 µg 23F: 2.4 µg 23F: 2.5 µg
During observation
period, guinea pigs and
Abnormal mice shall all be healthy,
Complied Complied Complied Complied Complied Complied Complied
toxicity no adverse reactions,
and each guinea pig and
mouse may gain weight
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at the end of the
experiment.
Bacterial
NMT 12.5 EU/dose Complied Complied Complied Complied Complied Complied Complied Complied
endotoxin
Test according to
Sterility requirements, should Complied Complied Complied Complied Complied Complied Complied
comply.

Conclusion:
Through the analysis of the data of the above batches of products, it is shown that: appearance, volume,
pH value, aluminium content, sodium chloride
content, polysaccharide content, abnormal toxicity test, bacterial endotoxin and sterility test all met
the Quality Standard of 13-Valent Pneumococcal
Polysaccharide Conjugate Vaccine.

3.2.P.5.5 Impurities
According to the manufacturing process, there will be no impurities added in finished product process,
please refer to 2.3.S.3 Characterization for detailed
information of impurities.

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3.2.P.5.6 Justification of Specification

Table 226 Justification of Specification for Finished Product
Test items Test method Specifications Ph. Eur. WHO Rationale for specifications
An identity test should be
performed that demonstrates Double immunodiffusion test
Each polysaccharide
that all the intended confirms the presence of certain
present in the vaccine is
Double pneumococcal polysaccharide serotype of polysaccharide. The
Identity / identified by a suitable
immunodiffusion serotypes and carrier protein (s) identity of tetanus toxoid is tested
immunochemical
are present in the final product, in the tetanus toxoid bulk
method.
unless this test has been production with flocculant test.
performed on the final bulk.
The limit is set according to
Quality Standard for
Visual Milky white suspension after Pneumococcal Polysaccharide
Appearance //
determination shaking. Conjugate Vaccine (WSYX-QC-
TS-0006-0721), which is
approved by NMPA.
The limit is set according to
Quality Standard for
Visual Pneumococcal Polysaccharide
Volume No less than 0.50 mL/dose //
determination Conjugate Vaccine (WSYX-QC-
TS-0006-0721), which is
approved by NMPA.
The limit is set according to
Quality Standard for
Osmolality Pneumococcal Polysaccharide
Osmolality 220 - 310 mOsm/kg //
meter Conjugate Vaccine (WSYX-QC-
TS-0006-0721), which is
approved by NMPA.
The limit is set according to
The amount of aluminum
Aluminum Maximum 1.25 mg per Quality Standard for
Titration 0.18 - 0.27 mg/mL should not exceed 1.25 mg per
content single human dose. Pneumococcal Polysaccharide
single human dose.
Conjugate Vaccine (WSYX-QC-
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Standard Conjugate Vaccine (WSYX-QC-Release in TS-0006-0721), which is Serotype standard validity approved by NMPA.

(%) period

(%)

1 30 25

3 80 75

4 75 70

5 75 70

6A 80 75

6B 80 75

7F 75 70

9V 80 75

14 80 75

18C 80 75

19A 70 65

19F 65 60

23F 65 60

The limit is set according to Quality Standard for No less than 80% for release

Conjugate Gel Pneumococcal Polysaccharide standard, no less than 75% in // protein content chromatography Conjugate Vaccine (WSYX-QC-validity period.

TS-0006-0721), which is approved by NMPA.

The limit is set according to Quality Standard for Total protein Pneumococcal Polysaccharide Lowry method 41 - 150 µg/dose // content Conjugate Vaccine (WSYX-QC-TS-0006-0721), which is approved by NMPA.

Should be tested for bacterial The limit is set according to and mycotic sterility in

It complies with the test Quality Standard for

Sterility Film filtration Meets the requirement of the test accordance with the for sterility. Pneumococcal Polysaccharide requirements of the revised Conjugate Vaccine (WSYX-QC- General requirements for the Confidential 770
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sterility of biological substances TS-0006-0721), which is or by a method approved by the approved by NMPA.
NRA.
The limit is set according to
The requirement to test lots of
No adverse reactions, healthy and Quality Standard for pneumococcal conjugate
Abnormal weight gain for each guinea pig Pneumococcal Polysaccharide
Animal test / vaccine for unexpected toxicity
toxicity and mouse during observation Conjugate Vaccine (WSYX-QC- (abnormal toxicity) should be
period. TS-0006-0721), which is
agreed with the NRA.
approved by NMPA.
Endotoxin content or pyrogenic
Less than 12.5 EU per activity should be consistent
Bacterial single human dose, with levels found to be The specification is required by
Gel-clot method No more than 12.5 EU/dose
endotoxin unless otherwise acceptable in vaccine lots used Ph. Eur. (2150).
justified and authorized. in clinical trials and approved
by the NRA.

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3.2.P.6 Reference Standards or Materials
Table 227 List of Reference Standards or Materials used for Finished Products
Storage
Test Item Reference name Manufacturer Unit
Condition
National standard sample 19.73
Protein content NIFDC Below -20 ■
for protein content test mg/tube
Room
Dextran GE 10 g/vial
temperature
Conjugate protein Sinopharm
Room
Vitamin B12 Chemical Reagent 1 g/vial
temperature
Co., Ltd.
Pneumococcal specific
antiserum (serotypes 1, 3, Danish National
1 ml/vial 2-8 ■
4, 5, 6A, 6B, 7F, 9V, 14, Serum Institute
Polysaccharide 18C, 19A, 19F, 23F)
content/conjugate Pneumococcal
antigen content polysaccharide reference
10 mg/vial
(serotypes 1, 3, 4, 5, 6A, ATCC Below -20 ■
2 mg/vial
6B, 7F, 9V, 14, 18C, 19A,
19F, 23F)
Standard bacterial Zhanjiang Andusi
Bacterial endotoxin 10 EU/tube 2-8 ■
endotoxin sample Biology Co., Ltd.
Sterility Test Staphylococcus aureus NIFDC / Below -20 ■

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3.2.P.7 Container Closure System
There are two kinds of packaging forms for 13-valent Pneumococcal Polysaccharide Conjugate Vaccine, which are the packaging of vials and pre-filled syringes. The packaging materials for direct contact drugs used in vials are neutral borosilicate glass vial, halogenated butyl rubber plugs for injection (chlorination) and aluminum-plastic combined caps; the packaging materials for direct contact drugs used in pre-filled syringes are rubber plug for pre-filled syringe and pole for pre-filled syringe.

3.2.P.7.1 Primary Packaging (Vial)

Table 228 Brief Description of the Container Closure System for Vial
Name Manufacturer Specification Standard
Shuangfeng Gerresheimer
Neutral pharmaceutical glass (Danyang) 1 mL/vial
borosilicate Co., Ltd YBB00292005-2-2015
glass vial Ningbo zhengli pharmaceutical
1 mL/vial
packaging Co. Ltd
Halogenated Shengzhou pharmaceutical
butyl rubber packaging material technology 11G002-A17-D2T-S Q/SRPYBB10-2019

plugs for (China) Co., Ltd
injection Jiangsu Hualan New YBB00042005-2015
11-B
(chlorination) Pharmaceutical Material Co., Ltd YBB00052005-2015
Aluminium- Hebei Jinhuan Packaging Co., Ltd 11.3×6.4 mm
plastics Jingjiang Huaxin Capping Co., YBB00372003-2015
11×6.3 mm
combined caps Ltd

Table 229 Specification of Neutral Borosilicate Glass Vial

Test Items Specification
It should be colorless and transparent; the surface should be clean and smooth, and Appearance there should be no obvious defects. No crack should be found in any part. Crack detection condition (n/Ac=500/7); other inspection cases (n/Ac=315/21)
Internal surface water The volume of 0.01mol/L hydrochloric acid consumed by 100m1 of leaching resistivity solution should be less than 2.0 Ml.
Heat resistance test shall not be broken (n/Ac=80/1)
Cold resistance test shall not be broken (n/Ac=80/1)
121■particles water Volume of hydrochloric acid titrant (0.02mol/L) consumed by each gram of glass resistance particles should be ≤ 0.10 mL.
outer diameter of bottle body: Φ13.7±0.15
outer diameter of lip: Φ10.7-11.2
outer diameter of bottle neck≤Φ9.3
Size inner diameter of lipΦ7.05±0.20
bottle height:27.0±0.50
bottle edge thickness3.6±0.20
bottom thickness ≥ 0.7

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Table 230 Specification of Halogenated Butyl Rubber Plugs for Injection (Chlorination)

Test Items Specification
According to the following Unqualified classification inspection method, Class A, B and C should not appear.
Class A: There are stains, impurities, bubbles and cracks in the needle ring or there are bubbles and cracks on surface of sealing face. N/AC=315/3
Appearance
Class B: The surface is stained, impurities, rubber filaments, rubber chips, spongy, wool edge, neck plug rough, obvious glue deficiency. N/AC=315/10
Class C: incomplete and zigzag edges, uneven marks and colors caused by moulds. N/AC=200/21
(1) produce white or yellow sediment
Identification
(2) coincident with reference chromatogram
Puncture and crumb The number of chips should not exceed 5.
Self-sealability Methylene blue solution must not penetrate into the bottle.
Adhesion between
rubber stopper and Methylene blue solution must not penetrate into the bottle.
container
Puncture force The force needed to puncture the rubber stopper should not exceed 10N
Insoluble particles Particles larger than 10um per ml: NMT 30 particles, (photoresist method) Particles larger than 25um per ml: NMT 3 particles.
Residue on ignition NMT45%
Should not form sulfur spots or color of sulfur spots should not be more strength
Volatile sulfide
than standard sulfur spots. (less than 50ugNa2S/20cm2)
The solution should be clear. If it is muddy, it should not be more strength than
Clarity and color No. 2 standard solution. if it shows color, it should not be more strength than No.5 standard yellow-green solution.
pH difference between test solution and reference solution should not more than
pH variation
1.0
UV absorbance The maximum absorbance at 220-360nm should not exceed 0.1.
Nonvolatile matter
The difference between the two should not exceed 4.0mg
(mg)
Readily oxidizable The difference of the consumption of titrant between the two should not exceed substance (ml) 3.0ml.
The color displayed in the third tube is not lighter than that in the first tube, and
Heavy metal the color displayed in the second tube should not be more strength than that in the first tube. NMT 1ppm.
The color of the test solution should not be more strength than that of the standard
Ammonium ion
solution, ammonium ion should be not more than 0.0002%.
Zinc ion The color of the test solution is not more strength than that of standard solution.
The conductivity of blank control solution should not exceed 3.0uS/cm (20■ ±
Conductivity (uS/cm) 1■).
The conductivity of test solution should not exceed 40.0uS/cm (20■±1■).
in the preliminary test. the body temperature of all the 3 rabbits increased less
than0.6■, sum of elevated temperature was less than 1.4■,or in the retest, there
Pyrogen is not more than 1 rabbit of the 5 whose body temperature raise for 0.6■ or more
than 0.6■, the sum of elevated temperature of preliminary test and retest study is
about 3.5■ or less than 3.5■. should meet the requirements.
Hemolysis
Hemolysis rate should be less than 5%.
(hemolysis test)
Acute systemic During the 72 hours observation period, the response of the animals in the test
toxicity test group should not be greater than that in the control group.

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Table 231 Specification of Aluminium-plastics Combined Caps

Test Items Specification

Take appropriate amount of the cap, visual inspection in the natural light, should be clean, with no residual lubricant, burrs, damage and injection molding flying Appearance

edge, plastic parts should be fully integrated with aluminum parts.

M/AC=315/21

Convex edge NMT3% H/AC=50/3

Opening quality There is no damage on the opening site. E/AC=13/1

After sterilization, the plastic part of the bottle cover can withstand the steam sterilization temperature of 130 without deformation and discoloration, and Sterilizing resistance

the surface of the aluminum part should not have any obvious changes. There should be no fracture and abnormal deformation of the cover. E/AC=13/1

Coating firmness There is no damage on the coating. E/AC=13/1

Inner diameter of aluminum part:

The aluminum parts can pass through the gauge side of the limit plug gauge cannot pass the stop gauge side completely. K/AC=125/5

Total height of aluminum and plastic

7.3±0.4. K/AC=125/10

Size Outer diameter of plastic part

13.2±0.25. K/AC=125/10

Inner height of aluminum part

6.4±0.20. K/AC=125/5

Thickness of aluminum part

0.19±0.02. E/AC=13/1

3.2.P.7.2 Primary Packaging (Pre-filled Syringe)

Table 232 Brief Description of the Container Closure System for Pre-filled Syringe

Name Manufacturer Specification Standard

Rubber plug for Shandong Weigao Group Medical

1-3 mL

pre-filled Polymer products Co., Ltd. YBB00072004-2015

syringe BD 1-3 mL

Shandong Weigao Group Medical

Pole for pre- 1 mL

Polymer products Co., Ltd. YBB00062004-2015

filled syringe

BD 1 mL

Table 233 Specification of Rubber Plug for Pre-filled Syringe

Test Items Specification

The surface color should be uniform, without stains, impurities, bubbles and cracks; impurities, rubber filaments, rubber chips, spongy and wool edge;

Appearance incomplete and zigzag edges caused by cutting, uneven marks and colors caused by mould. If there is a sprue, it should not be it should not protrude from the surface of the rubber plug. N/AC=125/2

Particles larger than 10um per ml: NMT 60 particles,

Insoluble particles

Particles larger than 25um per ml: NMT 6 particles.

(1) produce white or yellow sediment

Identification

(2) coincident with reference chromatogram

Residue on ignition NMT50%

1. White precipitates should be produced

Identification

2. It should be consistent with the standard chromatography.

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The lead acetate test paper for the test solution should not form sulfur spots, if

Volatile sulfide sulfur spot is formed, color of sulfur spots should not be more strength than that of standard lead acetate test paper. (less than 50ug/20cm²)

Clarity and color The solution should be clear and colorless.

pH difference between test solution and reference solution should not more than pH variation

1.0

UV absorbance The maximum absorbance at 220-360nm should not exceed 0.1.

Nonvolatile matter

The difference between the two should not exceed 4.0mg

(mg)

Readily oxidizable The difference of the consumption of titrant between the two should not exceed substance (ml) 3.0ml.

The color displayed in the third tube is not lighter than that in the first tube, and

Heavy metal the color displayed in the second tube should not be more strength than that in the first tube. NMT 1ppm.

The color of the test solution should not be more strength than that of the standard

Ammonium ion

solution, ammonium ion should be not more than 0.0002%.

Zinc ion The color of the test solution is not more strength than that of standard solution.

The conductivity of blank control solution should not exceed 3.0uS/cm (20±1 ■).

Conductivity (uS/cm)

The conductivity of test solution should not exceed 40.0uS/cm (20±1 ■).

Sterility Should comply with requirements.

Bacterial endotoxin Should be less than 0.5 EU/mL

in the preliminary test. the body temperature of all the 3 rabbits increased less

than 0.6 °C, sum of elevated temperature was less than 1.4 °C, or in the retest, there
Pyrogen is not more than 1 rabbit of the 5 whose body temperature raise for 0.6 °C or more
than 0.6 °C, the sum of elevated temperature of preliminary test and retest study is
about 3.5 °C or less than 3.5 °C. should meet the requirements.

Hemolysis

Hemolysis rate should be less than 5%.

(hemolysis test)

Acute systemic During the 72 hours observation period, the response of the animals in the test
toxicity test group should not be greater than that in the control group.

Table 234 Specification of Pole for Pre-filled Syringe

Test Items Specification

It should be colorless and transparent; the surface should be clean and smooth, and
Appearance there should be no obvious defects. No crack should be found in any part.

(N/AC=125/2)

Internal surface water

The volume of consumed should be less than 2.0ml

resistivity

Pinhole patency The needle with a diameter of 0.18 should be able to pass freely. N/AC=20/2

121 particles water Volume of hydrochloric acid titrant (0.02mol/L) consumed by each gram of glass
resistance particles should be ≤ 0.10 mL.

Sterility Should comply with requirements.

Bacterial endotoxin Should be less than 0.5 EU/mL

3.2.P.7.3 Secondary Packaging

The secondary packaging form of 13-valent Pneumococcal Polysaccharide Conjugate Vaccine
(packed with vial and pre-filled syringe) is 0.5ml/vial (syringe) per box. Each small box is marked
with product name, specification, batch number, manufacturing date, expiration date, indication,
eligibles, adverse reactions, contraindications, precautions, etc. at the same time, each small box
is marked with the vaccine traceability code, which can be queried through the system to each
vaccine information.

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3.2.P.8 Stability

In accordance with the requirements of GMP, Walvax has compiled the document Product

Stability Inspection Management rules (Document code: WSYX-QA-SMP-0009-0030).

According to this document, we have placed accelerated stability test and long-term stability test
on six batches of 13-valent Pneumococcal Polysaccharide Conjugate Vaccine.

3.2.P.8.1 Stability summary and conclusion

Table 235 Batch Information of 13-valent Pneumococcal Polysaccharide Conjugate

Vaccine used in Stability Study

Batch No. Packaging Batch Quantity Storage condition Mfg. Date

GI201708001 87923 2019.08.17

GI201708002 Vial 85722 2018.08.25

GI201709003 85162 2019.09.01

2-8 °C

GK201709001 84697 2019.08.30

Pre-filled

GK201709002 84851 2019.09.03

syringe

GK201709003 82354 2019.09.13

Test condition:

Accelerated stability test condition: Temperature: 25 ± 2 °C, humidity: $60 \pm 10\%$

Long-term stability test condition: Temperature: 5 ± 3 °C, humidity: $60 \pm 10\%$

Testing period:

Accelerated stability test: 0th, 1st month, 2nd month, 3rd month, 6th month, 9th month.

Long-term stability test: 0th, 3rd month, 6th month, 9th month, 12th month, 18th month, 24th month,
30th month.

Table 236 Test Items and Specification of Accelerated Stability Study

Test items Specification

Appearance Should be milky uniform suspension after shaking

Extractable volume Should be not less than 0.5ml/vial or syringe

pH 5.5-7.5

Aluminium phosphate

0.8-1.2 g/L

content

Sodium chloride

7.5-9.5 g/L

content

1: 1.8-3.4 µg/dose, 3: 1.75-3.25 µg/dose, 4: 2.1-3.9 µg/dose, 5: 1.75-3.25 µg/dose,

6A: 1.75-3.25 µg/dose, 6B: 4.2-7.8 µg/dose, 7F: 2-3.7 µg/dose, 9V: 1.75-3.25

Polysaccharide content

µg/dose, 14: 1.9-3.6 µg/dose, 18C: 2.3-4.2 µg/dose, 19A: 1.8-3.4 µg/dose, 19F:

1.9-3.6 µg/dose, 23F: 2.1-3.9 µg/dose.

Abnormal toxicity Should comply with the requirements.

Bacterial endotoxin Should be not more than 12.5 EU/dose

Sterility test Should comply with the requirements

Osmolarity 220-310 mOsm/Kg

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MODULE 3 QUALITY

Version No. Walvax-202007-00 Yuxi Walvax Biotechnology Co., Ltd.

Table 237 Test Items and Specification of Long-term Stability Study

Test items Specification

Appearance Should be milky uniform suspension after shaking

Extractable volume Should be not less than 0.5 ml/vial or syringe
pH 5.5-7.5
Aluminium content 0.8-1.2 g/L
Sodium chloride
7.5-9.5 g/L
content
0th, 3rd, 6th, 9th, 12th, 18th month:
1: 1.8-3.4 µg/dose, 3: 1.75-3.25 µg/dose, 4: 2.1-3.9 µg/dose, 5: 1.75-3.25 µg/dose,
6A: 1.75-3.25 µg/dose, 6B: 4.2-7.8 µg/dose, 7F: 2-3.7 µg/dose, 9V: 1.75-3.25
µg/dose, 14: 1.9-3.6 µg/dose, 18C: 2.3-4.2 µg/dose, 19A: 1.8-3.4 µg/dose, 19F:
1.9-3.6 µg/dose, 23F: 2.1-3.9 µg/dose.
Polysaccharide content
24th month:
1: 3.6-6.8 µg/dose, 3: 3.5-6.5 µg/dose, 4: 4.2-7.8 µg/dose, 5: 3.5-6.5 µg/dose, 6A:
3.5-6.5 µg/dose, 6B: 8.4-15.6 µg/dose, 7F: 4.0-7.4 µg/dose, 9V: 3.5-6.5 µg/dose,
14: 3.8-7.2 µg/dose, 18C: 4.6-8.4 µg/dose, 19A: 3.6-6.8 µg/dose, 19F: 3.8-7.2
µg/dose, 23F: 4.2-7.8 µg/dose.
Serotype 1: release standard is ≥ 30%, standard in expiry date is ≥ 25%;
Serotype 3, 6A, 6B, 9V, 14, 18C: release standard is ≥ 80%, standard in expiry
Conjugate antigen date is ≥ 75%;
content Serotype 4, 5, 7F: release standard is ≥ 75%, standard in expiry date is ≥ 70%;
Serotype 19A: release standard is ≥ 70%, standard in expiry date is ≥ 65%;
Type 19F, 23F release standard is ≥ 65%, standard in expiry date is ≥ 60%.
Conjugate protein Release standard shall be not less than 85%, standard in expiry date shall be not
content less than 80%
Total protein content 41-150 µg/dose
Abnormal toxicity Should comply with requirements.
Bacterial endotoxin Should be not more than 12.5 EU/dose
Sterility test Should comply with requirements
Osmolarity 220-310 mOsm/L/kg

Conclusion:

According to the analysis of stability study test results, it showed that the six batches (three vials
and three pre-filled syringe) of 13-valent Pneumococcal Polysaccharide Conjugate Vaccine were
stored at 25±2 °C for 9 months accelerated stability study met the 13-valent Pneumococcal
Polysaccharide Conjugate Vaccine registration standards (code: YBS00392019) approved by
NMPA.

The long-term stability experiment has now completed the 24th month study. The test results
showed that six batches (three vials and three pre-filled syringe) of products were stored at 5±3 °C
for long-term stability study met the 13-valent Pneumococcal Polysaccharide Conjugate Vaccine
registration standards (code: YBS00392019) approved by NMPA.

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MODULE 3 QUALITY

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3.2.P.8.2 Post-approval Stability Protocol and Stability Commitment

Please refer to Annex 5 Stability study protocol for Finished product.

3.2.P.8.3 Stability Data

Please refer to Annex 6 Stability study data for Finished product.

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MODULE 3 QUALITY

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3.2.A APPENDICES

3.2.A.1 Facilities and Equipment

Walvax has established the Plant and Facilities Management Control Procedures (document
code: WSYX-P-03-0003) specifying the maintenance cycle and scope of the plant and facilities.
The engineering supply department conducts a monthly inspection on the periphery of the clean
plant to see whether the floor, walls, ceiling, doors and windows of the plant are free of cracks
and shedding and are intact; and whether the main bearing structure is overload, corroded and so
on. Each manufacturing workshop shall inspect the inside of the clean plant every month to see
whether the ground, wall, ceiling, doors and windows of the clean plant are free of cracks and
shedding and are intact; and whether the main bearing structure is overload, corroded and so
on. During the inspection process, it shall be timely reported and handled in case of
abnormalities according to the relevant requirements. After inspection, the inspection records
shall be filed monthly to the engineering supply department, so as to know about the situations
inside the plant in a timely manner.

Main equipment lists are showed as following:

Table 238 Main Equipment List of Water System

Serial

Device name model Manufacturer

number

Purified water Pretreatment and Osmotron 10000 A Best water technology

1

machine select (Shanghai) Co., Ltd.

Purified water Pretreatment and Osmotron 10000 B Best water technology

2

machine select (Shanghai) Co., Ltd.

multi-effect water Best water technology

3 MULTITRON MT7500-6 A

distillator (Shanghai) Co., Ltd.

multi-effect water Best water technology

4 MULTITRON MT7500-6 B

distillator (Shanghai) Co., Ltd.

Best water technology

5 Pure steam generator Vapotron 4000 A

(Shanghai) Co., Ltd.

Best water technology
6 Pure steam generator Vapotron 4000 B
(Shanghai) Co., Ltd.

Table 239 Main Equipment List of Compressed Air System

Serial
Device name model Manufacturer
number
Oil free rotary Atlas Copco
1 ZT37VSD-8.6
Air compressor Airpower/ Antwerp, Belgium
Oil free rotary Atlas Copco
2 ZT37-8.6
Air compressor Airpower/ Antwerp, Belgium
Oil free rotary Atlas Copco
3 ZT37-8.6
Air compressor Airpower/ Antwerp, Belgium

Table 240 Main Equipment List of Air Conditioning Purification System

Serial
Device name Equipment model Manufacturer
number
Air treatment unit JK2-
1 DMA1918 Yantai Dunham-Bush Industry Co., Ltd.
MGa
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Air treatment unit JK2-
2 DMA1314 Yantai Dunham-Bush Industry Co., Ltd.
MGb
Air treatment unit JK2-
3 DMA1822 Yantai Dunham-Bush Industry Co., Ltd.
MGc
4 Air treatment unit JK2-Ea DMA0910 Yantai Dunham-Bush Industry Co., Ltd.
5 Air treatment unit JK2-Ec DMA1109 Yantai Dunham-Bush Industry Co., Ltd.
Air treatment unit JK2-
6 DMA1818 Yantai Dunham-Bush Industry Co., Ltd.
MGd

Table 241 Main Equipment for the Production of Pneumococcal Polysaccharide Conjugate Vaccine

Serial number Device name Model Manufacturer
Carbon dioxide
1 311 Thermo Fisher
incubator
2 Biosafety cabinet BSC-1600IIA2 Suzhou Antai Air Technology Co., Ltd.
Shanghai Divoxin Bioengineering Co.,
3 Bioreactor BIO-DVT-3000F
Ltd.
Shanghai Baidian instrument and
4 Biochemical incubator LRH-70
Equipment Co., Ltd.
5 Centrifuge AvantiJ-26SXP Beckman Kurt trading (China) Co., Ltd.
High speed landing
6 Lynx4000 Thermo Fisher
centrifuge
Large capacity
7 RC 12BP+ Thermo Fisher
centrifuge
8 laminar hood ----- MayAir (China)
9 Ultrafiltration system UFC10CC Pol filter (Beijing) Co., Ltd.
Serial number Device name Model Manufacturer
Carbon dioxide
1 311 Thermo Fisher
incubator
2 Biosafety cabinet BSC-1600IIA2 Suzhou Antai Air Technology Co., Ltd.
Shanghai Divoxin Bioengineering Co.,
3 Bioreactor BIO-DVT-3000F
Ltd.
4 Biosafety cabinet BSC-1600IIA2 Suzhou Antai Air Technology Co., Ltd.
Large capacity high
5 CR 7 Hitachi Corporation
speed centrifuge
Large capacity high
6 CR 7 Hitachi Corporation
speed centrifuge
High speed landing
7 Lynx4000 Thermo Fisher
centrifuge
Large capacity
8 RC 12BP+ Thermo Fisher
centrifuge
Shanghai Nelly fluid equipment Co.,
9 Precipitator 3000L
Ltd.
10 Ultrafiltration system UFC10CC Pol filter (Beijing) Co., Ltd.
High speed landing
11 Lynx4000 Thermo Fisher
centrifuge
Large capacity

12 RC 12BP+ Thermo Fisher
centrifuge
High speed landing
13 Lynx4000 Thermo Fisher
centrifuge
Large capacity
14 RC 12BP+ Thermo Fisher
centrifuge
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MODULE 3 QUALITY
Version No. Walvax-202007-00 Yuxi Walvax Biotechnology Co., Ltd.

15 Ultrafiltration system FS013K05C Pol filter (Beijing) Co., Ltd.

Table 242 Main Equipment for Fill and Finish
Serial number Device name Model Manufacturer
1 Liquid dispensing tank 100L Nanjing Bojian Technology Co., Ltd.
Automatic filling and
2 SV125 Optima Packaging Machinery Co., Ltd.
capping machine
5 Lamp inspector DGG--1 Nanjing Bojian Technology Co., Ltd.
Colamark(Guangzhou) Labeling
6 Sticker machine A210N
Equipment Limited
7 Packing line KHP40T Shanghai Kangsong Packaging Co., Ltd.

Table 243 Main Equipment for QC Analytical
Serial number Device name Model Manufacturer
Constant temperature and STIK (Shanghai) instrument equipment
1 CTHI-250B
humidity incubator limited
Shanghai Keelrein Scientific
2 Electric heat dryness box DHG-9070A
Instruments Co., Ltd.
Nanjing Jinheng experimental
3 Electric heat dryness box HG101-3AS
instrument factory
Beijing Sedorius instrument system
4 Electronic balance BS124S
Co., Ltd.
5 Electronic balance CP225D Sartorius
Beijing Sedorius instrument system
6 Electronic balance BS2202S
Co., Ltd.
Ultraviolet SHIMADZU enterprise management
7 UV-2600
spectrophotometer (China) Co., Ltd.
Electrothermal constant Shanghai Keelrein Scientific
8 DK-600A
temperature water tank Instruments Co., Ltd.
Electrothermal constant
Beijing Changfeng instrument
9 temperature water bath HW.SY21-K
Company
pot
Electrothermal constant Shanghai Yiheng Scientific Instruments
10 DK-600A
temperature water tank Co., Ltd.
AKTA purification
11 AKTA Pure 25 GE Life Science
system
12 Step collector Frac-920 Amersham Biosciences
Electrothermal constant Shanghai Yiheng Scientific Instruments
13 DK-600A
temperature water tank Co., Ltd.
14 Moisture meter Aqv-2100SP AQUA COUNTER
15 Osmotic pressure meter Advanced 3250 ADVANCED INSTRUMENTS, INC
High speed refrigerated
16 CR21G Hitachi
centrifuge
Sujing group Suzhou Antai Air
17 Bioclean worktable BCM-1600A
Technology Co., Ltd.
18 Bacteria collector HTY-602A Zhejiang Tailin bioengineering Co.,Ltd.
Sujing group Suzhou Antai Air
19 Biosafety cabinet BSC-1600 II A2
Technology Co., Ltd.
STIK (Shanghai) instrument equipment
20 climate chamber CTHI-250B
limited
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Tianjin Tiandatanfa Technology Co.,
21 Intelligent pyrometer ZRY-2D
Ltd.
22 Constant current pump P-1 Amersham Biosciences
23 Chromatography column XK16/100 Amersham Biosciences
Automatic Kjeldahl
24 Kjelttec 2200 FOSS
nitrogen meter

Temperature controlled TecatorTM
25 FOSS
digester Digester
SHIMADZU enterprise management
26 Gas chromatograph GC-2014C
(China) Co., Ltd.
Specific protein analysis
27 IMMAGE 800 American Beckman Kurt Co., Ltd.
system
28 pH meter Thermo American Beckman Kurt Co., Ltd.

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3.2.A.2 Adventitious Agents Safety Evaluation
Walvax establishes a comprehensive and scientific material management and storage system.
Developed Material Management Control Procedure (Document code: WSYX-P-05-0005). For
animal source materials, in addition to strict quality test, the suppliers should provide proof of
origin or safety certification. This vaccine's production is involved with animal-derived materials
of sheep blood. Detailed information as below:

Table 244 Control Measures for Animal-derived Materials
Name Usage Risk control description
strain recovery of the fist Product quality test, Safety proof documents from
Sheep blood
passage supplier

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SOP of Yuxi Walvax Biotechnology Co., Ltd

WSYX-QC-SOP-0009-013
Code
SOP for Residual 7
Content of Sodium
Version 02 Page 1/5
Deoxycholate
Non-photocopying, confidential documents

1 Objective
This SOP describes the method for sodium deoxycholate residual content.
2 Definitions/abbreviations
N/A
3 Scope of application
This SOP is applicable to the sodium deoxycholate residual content determination of products manufactured by Yuxi Walvax.
4 Responsibility
All the operators shall strictly conduct the SOP.
5 Equipment and materials
5.1 Equipment
No. Name Model Code manufacturer Remarks
Electronic Or equivalent
1 BS124S WYQCF0036 Sartorius
balance equipment
ultraviolet-visi WYQCF0444/ Or equivalent
ble WYQCF0445/ equipment
2 UV-2600 Shimadzu
spectrophotom WYQCF0529/
eter WYQCF0530
Shanghai Or equivalent
Qixin equipment
Electric Chemical
constant WYQCF0061/ Instrument
3 DK-600A
temperature WYQCF0287/ Co., Ltd
water tank /Shanghai
Yiheng
Chemical
SOP of Yuxi Walvax Biotechnology Co., Ltd

WSYX-QC-SOP-0009-013
Code
SOP for Residual 7
Content of Sodium
Version 02 Page 2/5
Deoxycholate
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Instrument
Co., Ltd

5.2 Reagents
No. Material/reagent Grade Remarks
1 concentrated sulfuric acid GR Content: ≥95.0■98.0
2 Glacial acetic acid AR None
3 Sodium deoxycholate Import reagent None
5.3 Materials
No. material No. material
1 Tips 2 Latex gloves
3 Tube 4 Pipette
5 Rubber dropper 6 Beaker

7 Volumetric flask 8 Glass rod
6 Working procedure
6.1 principle
This method is based on the formation of colored compound of sodium deoxycholate under acidic conditions, and the residual amount of sodium deoxycholate in the test sample is determined by colorimetry.
6.2 solution
6.2.1 sodium deoxycholate reference stock solution (1000ug/ml)
Accurately weigh 0.05g sodium deoxycholate dried to constant weight in a beaker, add 30ml glacial acetic acid to dissolve it, transfer it into a 50ml volumetric flask, moisten the beaker with purified water for three times, transfer the washing solution to the volumetric flask, and fix the volume to the scale with purified water.
6.2.2 sodium deoxycholate reference solution (100ug/ml)
Before use, 500 UL sodium deoxycholate reference solution (1000 UG/ml) was precisely
SOP of Yuxi Walvax Biotechnology Co., Ltd

WSYX-QC-SOP-0009-013
Code
SOP for Residual 7
Content of Sodium
Version 02 Page 3/5
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measured, 4500ul purified water was added, and then 100 UG / ml sodium deoxycholate reference solution was obtained.
6.2.2 60% acetic acid solution

Measure 150ml glacial acetic acid, add 100ml purified water, and mix well.

6.2.3 43.5% sulfuric acid solution

Measure 435ml of concentrated sulfuric acid in the fume cupboard, slowly add it into the beaker containing 500ml purified water, stir while adding. After the solution is cooled, transfer it into a 1000ml volumetric flask, moisten the beaker with a small amount of purified water for three times, transfer the washing solution to the volumetric flask, add purified water to the scale, and mix well.

6.3 Test procedure

6.3.1 Test article treatment

A pipette with a range of 100-1000 ul was used to precisely transfer 1000 ul of the test sample into the test tube in duplicate.

6.3.2 preparation of standard curve

name Standard Standard Standard Standard Standard

tube No. 0 1 2 3 4

100ug/ml sodium

deoxycholate reference 0 100 200 400 800

solution (UL)

60% acetic acid

1000 900 800 600 200

solution (UL)

Sodium deoxycholate

concentration (UG / 0 10 20 40 80

ml)

6.3.3 add 14.ml of 43.5% sulfuric acid solution into the above tubes respectively, mix them while adding them, heat them at 70■ for 20 minutes, cool them to room temperature, and measure the absorbance at 387nm wavelength of ultraviolet visible spectrophotometer.

6.4 result calculation

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Code

SOP for Residual 7

Content of Sodium

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Deoxycholate

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6.4.1 the concentration of sodium deoxycholate reference solution was used for linear regression to the corresponding absorbance, and the absorbance of the test solution was substituted into the regression equation to calculate the content of sodium deoxycholate in the test sample.

6.4.2 take the average value of the test results of the two tubes as the concentration of sodium deoxycholate of the test sample, and calculate according to the following formula:

Sodium deoxycholate content (%) = sodium deoxycholate concentration of the test sample × 1000 × total solids of the test sample × 100

6.4.3 Result judgment

According to the quality standard of relevant samples.

7 Flow chart

N/A

8 Notes

8.1 Concentrated hydrochloric acid will be used in the experiment, and safety precautions should be made during the operation.

8.2 Carefully use water bath to avoid scalding.

8.3 Pay more attention to water bath process, cover the tubes tightly, avoid reagents spraying out.

9 References or Standard Basis

9.1 Chinese Pharmacopoeia, volume III, page 124, China Pharmaceutical Science and Technology Press, 2015

10 Revision history

Release time Time in Revision Revision basis, reason and

Version Note

force Department content

Complete some parts of the

00 2015.07.30 2015.08.01 QC laboratory /

content

SOP of Yuxi Walvax Biotechnology Co., Ltd

WSYX-QC-SOP-0009-013

Code

SOP for Residual 7

Content of Sodium

Version 02 Page 5/5

Deoxycholate

Non-photocopying, confidential documents

Complete some parts of the

01 2017.09.30 2017.10.01 QC laboratory /

content

Standard Operation Procedures of Yuxi Walvax Biotechnology Co.,Ltd

Code WSYX-QC-SOP-0009-0101
Operation procedures
determination of Version 08 Page 1/6
phosphorous content
Non-photocopying, confidential documents

1 Objective
This SOP describes the method for phosphorous content determination.
2 Definitions/abbreviations
N/A
3 Scope of application
This SOP is applicable to the phosphorous content determination. It is applicable to the identification of haemophilus influenzae type b conjugate polysaccharide, group A meningococcal polysaccharide (bulk), group A meningococcal polysaccharide bulk derivative, group A meningococcal polysaccharide bulk conjugate, group A and C meningococcal polysaccharide vaccine, pneumococcal polysaccharide (bulk).
4 Responsibility
All the operators shall strictly conduct the SOP.
5 Equipment and materials
5.1 Equipment
No. Name Model Code manufacturer Notes
1 Electronic balance BS124S/ WYQCF0036/ Sartorius Or BSA124S
WYQCF0357 equivalent
equipment
2 Centrifuge CR21G/ WYQCF0068/ Hitachi/Ther Or Multifuge equivalent
X1R WYQCF0472/ mo Fisher equipment
WYQCF0473 Technology
(China) Co.,
Ltd
3 Constant HG-101-3AS WYQCF0451/ Nanjing Or temperature /DHG-9145A WYQCF0452/ equivalent
Jinheng equipment
blower oven WYQCF0437
experimental
instrument
Standard Operation Procedures of Yuxi Walvax Biotechnology Co.,Ltd

Code WSYX-QC-SOP-0009-0101
Operation procedures
determination of Version 08 Page 2/6
phosphorous content
Non-photocopying, confidential documents

factory/Shan
ghai Yiheng
Chemical
Instrument
Co., Ltd
4 Ultraviolet-visible UV-2600 WYQCF0444/ Shimadzu Or spectrophotometer WYQCF0445 equivalent
enterprise equipment
management
(China) Co.,
Ltd
5.2 Reagent
No. Name Grade Notes
1 Concentrated sulfuric acid GR mass fraction 95.0■
98.0
2 ammonium molybdate AR None
3 Anhydrous sodium sulfite AR None
4 potassium dihydrogen National drug None
standard
phosphate
substances
5 perchloric acid AR (content■
70.0■
72.0%)
6 sodium bisulfite AR None
7 4-amino-3-naphthol-1-sulfonic Imported reagent None
acid

5.3 materials
No Name No Name
1 Pipette tips 2 latex gloves
Standard Operation Procedures of Yuxi Walvax Biotechnology Co.,Ltd

Code WSYX-QC-SOP-0009-0101
Operation procedures
determination of Version 08 Page 3/6
phosphorous content
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3 Test tube 4 pipette
5 Stainless steel test
tube frame

6 Working procedure

6.1 Principle

Phosphorus content was determined by converting the organic phosphorus into inorganic phosphorus by this method. Phosphate is formed into ammonium phosphomolybdate with ammonium molybdate in acidic solution. When there is reducing agent, blue substance (a mixture of molybdenum trioxide and molybdenum pentoxide) is produced. It is called "molybdenum blue". The content of phosphorus was determined by colorimetry Measure.

6.2 Reagent

6.2.1 Mineralizing agent (prepare freshly)

Measure equal volume of concentrated sulfuric acid and perchloric acid, mix well and obtain.

6.2.2 0.04 mol/L ammonium molybdate

Based on 100 ml preparation amount, weigh ammonium molybdate 5 g, and add water for

injection to dissolve to 100 ml, mix well and transfer into a 100 ml reagent glass. Attach label and place at room temperature for further use.

6.2.3 reducing agent (prepare freshly)

Based on 5 ml preparation amount, weigh sodium bisulfite 0.6 g, anhydrous sodium sulfite 0.12 g, 4-amino-3-naphthol-1-sulfonic acid 0.01 g, add water for injection to dissolve to 5ml, mix well.

6.2.4 20µg/ml phosphorous reference working solution

See "standard solution preparation procedure" Code: WSYX-QC-SOP-0009-0801

6.3 testing procedure

6.3.1 The standard curve was prepared by accurately measuring 20 µg/ml phosphorus standard working solution according to the table below:

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Operation procedures

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phosphorous content

Non-photocopying, confidential documents

name Standar Standard Standard Standard Standard Standard

tube No. d0 1 2 3 4 5

20µg/ml

Phosphorous reference 0.0 200 400 600 800 1000

working solution

Water for

1000 800 600 400 200 0.0

injection (µl)

Phosphorous

0 4 8 12 16 20

content (µg/ml)

6.3.2 Pre-treatment of test sample

6.3.2.1 30 times dilution of haemophilus influenzae type b conjugate polysaccharide. 30 times dilution of group A meningococcal polysaccharide (bulk). 80 times dilution of group A meningococcal polysaccharide bulk derivative. 3.5 times dilution of group A meningococcal polysaccharide bulk conjugate. See "standard solution preparation procedure" Code: WSYX-QC-SOP-0009-0007

6.3.2.2 Group A and C meningococcal polysaccharide vaccine: take 6 tubes of freeze-dried vaccines reconstituted with 0.5 ml/tube diluent and mixed in 1 ultrafiltration cups, and the liquid level was marked for ultrafiltration lactose treatment and centrifuged for 30 minutes at 7000 rpm/min and 5■. After the centrifuge is finished, add water to the liquid level and blow washes, then wait to be tested.

6.3.3 Inorganic treatment

Measure diluted test sample 1.0ml, and apply parallel test. Add mineralizing agent into standard working solutions and each tube of test sample of 0.05 ml, mix well and place in the oven at 150■ for at least 1h, 180■ for at least 2 h, or 230■ for at least 30min, enabling test samples to be mineralized, inorganically treated, and leaving only a small quantity of liquid, add water for injection 2.0 ml while the test samples are still hot, mix well and place at room temperature; add 0.04 mol/L ammonium molybdate 0.4 ml, mix well, add reducing agent 0.2 ml, mix well, keep adding water for injection 3.4 ml to the final volume of 6 ml, Standard Operation Procedures of Yuxi Walvax Biotechnology Co.,Ltd

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Operation procedures

determination of Version 08 Page 5/6

phosphorous content

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place at room temperature for 15 min.

6.3.4 Measure the absorbance at the wavelength of 820 nm.

6.4 Result

6.4.1 Establish regression line using concentrations of phosphorous standard solution against absorbance, then substitute the absorbance value of test sample into the regression line, the concentration of phosphorous is determined.

Phosphorous content=mean concentration of phosphorous × dilution factor

6.4.2 The content of polysaccharide is calculated according to the content of phosphorus■

Haemophilus influenzae type b conjugate polysaccharide: phosphorus content(mg/g) = phosphorus content/total solid

Group A polysaccharide (bulk): content of group A polysaccharide(µg/ml) =phosphorus content (µg/ml)/0.08 phosphorus content of group A(mg/g) = phosphorus content (mg)/total solid

Group A meningococcal polysaccharide bulk derivative and group A meningococcal polysaccharide bulk conjugate: content of group A polysaccharide (µg/ml) =phosphorus content (µg/ml)/0.08

Group A, C meningococcal polysaccharide vaccine: Content of group A polysaccharide (µg/dose)= phosphorus content(mg)/0.075×0.5

Pneumococcal polysaccharide (bulk):

phosphorus content(%)=content of pneumococcal polysaccharide(µg/ml)/(total solid×1000)

×100
 6.4.3 result determination
 According to the corresponding sample quality standard.
 7 Flow chart
 N/A
 8 General considerations
 8.1 When using ammonium molybdate colorimetry method to test phosphorous content, the
 Standard Operation Procedures of Yuxi Walvax Biotechnology Co.,Ltd

 Code WSYX-QC-SOP-0009-0101
 Operation procedures
 determination of Version 08 Page 6/6
 phosphorous content
 Non-photocopying, confidential documents

 carbonization time and temperature of test sample impact itself to some extent. Therefore, the
 SOP should be strictly followed.
 8.2 Before adding 0.04 mol/L ammonium molybdate and reducing agent, add some water for
 injection and cool the solution to room temperature.
 9 References
 9.1 Pharmacopoeia of People's Republic of China, volume 3, 2015. General rule 3103.
 10 Revision history
 Revising
 Issuing Basis, reason and revision Rem
 Version Valid from departme
 date arks
 nt
 03 2011.11.01 2011.12.01 QC Document expiration
 laboratory
 Improve the content of
 documents
 04 QC Improve the content of documents
 2014.09.02 2014.09.16
 laboratory
 05 QC Improve the content of documents
 2016.03.18 2016.03.20
 laboratory
 06 QC Improve the content of documents
 2017.09.30 2017.10.01
 laboratory

SOP of Yuxi Watson Biotechnology Co., Ltd.

Coding WSYX-QC-SOP-00-09-0141
SOP of Rate nephelometry
method for detection of Version 08 Page 1/15
polysaccharide content
Confidential files No copy

1 Objective

Standardize operating detection of pneumococcal polysaccharide content.

2 Definition/abbreviation

N/A

3 Applicable scope

It applies to the polysaccharide content test of 23-valent pneumococcal polysaccharide vaccine and monovalent polysaccharide bulk; 13-valent pneumococcal conjugate vaccine and monovalent polysaccharide conjugate bulk.

4 Responsibility

All the OQ lab operators should in strict accordance with this SOP.

5 Equipment and Materials

5.1 Equipment
Equipment
No. Equipment name Manufacturer Notes
model
Specific protein IMAGE BECKMAN Or equivalent

1
Analysis System 800 COULTER equipment

5.2 Using reagents
No. Reagents Name Product Code Specifications Grade
1 Diluent 1 (DIL1) 447640 4×120ml —
2 Buffer 1 (BUF1) 447650 4×120ml —

3 Wash solution 447060 10L —

Pneumococcal polysaccharide —
4 — 10mg, 2mg
Standard substance
Anti-serum of pneumococcal —
5 — 1ml/bottle
polysaccharide
6 Sodium hydroxide — 500g/bottle AR
7 Citric acid — 500g/bottle AR
5.3 Use of materials
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No. Material name No. Material name
1 Sample rack 2 Sample Cups
3 Kit 4 0.6ml/2ml/15ml centrifuge tubes
5 100-1000μl Pipette 6 20-200μl Pipette
7 500-5000μl Pipette 8 5000μl/1000μl/200μl Pipette tips
9 volumetric flask

6 Working procedure

6.1 Principle
Rate nephelometry is a kinetic method. Under certain conditions, the antigen and the corresponding antibody can quickly form the antigen-antibody complex particles. Rate nephelometry is to determine the maximum reaction speed (peak value of the reaction) of antigen and antibody in a certain period. The so-called rate is the rate of the combination of antigen and antibody in a unit time. Rate nephelometry method is to determine the maximum reaction speed, that is, determine the amount of its complex formation, when the antigen antibody reaction reaches the peak, which is usually for tens of seconds. The peak value is directly proportional to the amount of antigen in the case of antibody excess. The peak time was directly related to antibody concentration and affinity. The peak velocity of different antigen content was different and the antigen content can calculate by microcomputer.

6.2 Reagent preparation
6.2.1 Pneumococcal polysaccharide standard substance stock solution (500μg/ml) (Validity period: 5 years, storage conditions: -60■ to -80■)
Accurately measure 2ml of sterile water for injection (WFI) to dissolve lyophilized pneumococcal polysaccharide standard substance, mix well and place at 2-8■ overnight (at least 8h), then dilute the polysaccharide standard substance concentration using sterile WFI to 500μg/ml in the next day. Refill it into 0.6ml centrifuge tubes in the specification of 400μl/tube. Labeling. Freeze-thaw cycles not more than 3 times.

6.2.2 Pneumococcal polysaccharide Standard working solution (50μg/ml)(Validity period:
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Prepare the solutions immediately before use)
6.2.2.1 Pneumococcal polysaccharide Standard working solution used for calibrating standard
curve (50µg/ml) (Validity period: Prepare the solutions immediately before use)

Accurately measure 100µl standard substance stock solution into 2ml centrifuge tube,
add defined amount Diluent 1 (according to the actual concentration of standard substance
stock solution) and mix well to obtain 50µg/ml Pneumococcal Polysaccharide Standard
substance working solution.

6.2.2.2 Pneumococcal polysaccharide Standard working solution used for preparing the

internal control reference (50µg/ml) (Validity period: Prepare the solutions

immediately before use)

a) Type 1, 3, 4, 5, 6B, 7F, 9V, 14, 18, 19A, 19F and 23F Pneumococcal polysaccharide

Standard working solution (50µg/ml) (Validity period: Prepare the solutions immediately
before use)

Accurately measure the required volume in the 45ml centrifuge tube according to the
actual concentration of the standard stock solution, add defined amount Diluent 1 (DIL1) to
obtain 15ml of 50µg/ml Pneumococcal Polysaccharide Standard working solution.

b) Type 2, 6A, 8, 9N, 10A, 11A, 12F, 15B, 17F, 20, 22F and 33F Pneumococcal

polysaccharide Standard working solution (50µg/ml) (Validity period: Prepare the

solutions immediately before use)

Accurately measure the required volume in the 45ml centrifuge tube according to the
actual concentration of the standard stock solution, add defined amount Diluent 1 (DIL1) to
obtain 9ml of 50µg/ml Pneumococcal Polysaccharide Standard working solution.

6.2.3 Internal control reference of 13-valent Pneumococcal polysaccharide
(1.0µg/ml/serotype), (Validity period: not beyond the shelf life of stock solution, storage
conditions: -60 to -80)

Accurately measure 2.0ml polysaccharide standard working solution (50µg/ml) of each
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13 Pneumococcal serotypes, respectively, add 74.0ml of Diluent 1 (DIL1), mix well and then
refill them into 2ml centrifuge tubes in the specification of 1ml/tube, labeling. Freeze-thaw
cycles not more than 3 times.

6.2.4 Internal control reference of 13-valent Pneumococcal polysaccharide
(2.0µg/ml/serotype), (Validity period: not beyond the shelf life of stock solution,, storage
conditions: -60 to -80)

Accurately measure 4.0ml polysaccharide standard working solution (50µg/ml) of each
13 Pneumococcal serotypes, respectively, add 48.0ml of Diluent 1(DIL1), mix well and then
refill them into 2ml centrifuge tubes in the specification of 1ml/tube, labeling. Freeze-thaw
cycles not more than 3 times.

6.2.5 Internal control reference of 23-valent Pneumococcal polysaccharide
(1.0µg/ml/serotype), (Validity period: not beyond the shelf life of stock solution,, storage
conditions: -60 to -80)

Accurately measure 2.0ml polysaccharide standard working solution (50µg/ml) of each
23 Pneumococcal serotypes, respectively, add 54.0ml of Diluent 1(DIL1), mix well and then
refill them into 2ml centrifuge tubes in the specification of 1ml/tube, labeling. Freeze-thaw
cycles not more than 3 times.

6.2.6 Internal control reference of 23-valent Pneumococcal polysaccharide
(2.0µg/ml/serotype), (Validity period: not beyond the shelf life of stock solution,, storage
conditions: -60 to -80)

Accurately measure 4.0ml polysaccharide standard working solution (50µg/ml) of each
23 Pneumococcal serotypes, respectively, add 8.0ml of Diluent 1(DIL1), mix well and then
refill them into 2ml centrifuge tubes in the specification of 1ml/tube, labeling. Freeze-thaw
cycles not more than 3 times.

6.2.7 0.6mol/L Sodium Hydroxide Solution
Measure sodium hydroxide 2.4g, dilute with purified water and fixed to 100ml, mix
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well.

6.2.8 0.2mol/L Citric Acid Solution

Measure Citric acid 4.2g, dilute with purified water and fixed to 100ml, mix well.

6.2.9 Anti-serum working solution of pneumococcal polysaccharide

6.2.9.1 The antiserum of pneumococcal polysaccharide was diluted appropriately. Reference
dilution multiples of antiserum are shown in the following table:

Type 1 3 4 5 6A 6B 7F 9V 14 18C 19A 19F

Dilution

2 3 4 3 3 6 1 4 2 3 6 4

multiple

Type 2 8 9N 10A 11A 12F 15B 17F 20 22F 33F 23F

Dilution

2 2 1 1.5 1 1 3 3 1 2 5 2

multiple

The antiserum dilution multiples of the above types are for reference only and can be adjusted appropriately according to different batches of antiserum. The purpose of antiserum dilution is to ensure that the value of internal control reference substance is within the specified range.

6.2.9.2 Pneumococcal antiserum working solution preparation: during the preparation of antiserum working solution, special pipette shall be used. The preparation amount shall be estimated according to the daily inspection amount before preparation.

6.2.9.3 Antiserum storage: The original antiserum shall be put back to 2-8℃ for storage in time after used, and the prepared antiserum working solution shall be stored in 2-8℃ if it is not detected in time.

6.2.9.4 Antiserum usage: Before the preparation of antiserum, check whether its appearance is clear. If there is turbidity or foreign matter, stop using. After the preparation of antiserum, test the internal control reference substance first, and test the samples after the internal control reference substance is qualified. For the same dilute multiples of antiserum working solution, the internal control test shall be carried out according to the preparation times (e.g. if the same antiserum is prepared twice in the same day, the internal control reference test shall be carried out separately, and the sample test can only be carried out after the internal control reference test is qualified).

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6.2.9.5 In order to avoid confusion, two different batches of the same type of antiserum working solution should not appear at the same time.

6.3 Test steps

6.3.1 Power on: Turn on the printer, display, computer host, UPS power, instrument power successively.

6.3.2 Calibration of standard curves

6.3.2.1 Prepare polysaccharide standard solution according to the following steps, take solution B, solution C, solution D and solution E for calibrating standard curves:

1) Take 100μl of standard working solution (50μg/ml), add 900μl of diluent 1 (DIL1), mix well and obtain solution A (5μg/ml);

2) Take 300μl of solution A, add 300μl of diluent 1 (DIL1), mix well and obtain solution B

■2.5μg/ml■;

3) Take 200μl of solution A, add 300μl of diluent 1 (DIL1), mix well and obtain solution C

■2.0μg/ml■;

4) Take 200μl solution A, add 600μl of diluent 1 (DIL1), mix well and obtain solution D

■1.25μg/ml■;

5) Take 100μl solution A, add 700μl of diluent1 (DIL1), mix well and obtain solution E

■0.625μg/ml■.

6.3.2.2 Setup→<18>User defined chemistries, choose the calibration item (type) → F1

Define/Edit → F9 Save,

Set the item (type) and calibration parameters according to the following table methods:

Name Set up Name Set up

Units ug/ml Test per cartridge 300

Non-competitive

Protocol

Nephelometric.

Buffer BUF1 Diluent DIL1

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Sample or Reaction Buffer

20* 250*

Dilution Volume Volume

Compartment A Compartment B

20* 0

Volume Volume

Gain 1* Cal Dilution 1

Sample Dilution 1 Reaction time 1.5*

Levels 4 Replicates 2

Update levels 4 Replicates 2

Level 1 0.625 Level 2 1.25

Level 3 2.00 Level 4 2.50

Note: The item parameter with “*” is setting for reference only, appropriately adjust according to the calibration of the actual curve.

6.3.2.3 Setup→<18>User defined chemistries, Selected calibration items→F4 Request Cal.

Check that the selected shelf is available, if not available, select F8, and enter the shelf number →ok to be cleared.

6.3.2.4 Enter the shelf number →F9 Save.

6.3.2.5 Select F6 Cal Ldist to review the project information for the requested calibration and make a final check.

6.3.2.6 Put the antiserum working solution of pneumococcal polysaccharide placed in UDR

kit), polysaccharide standard solution placed in sample cup) and buffer 1 in the

corresponding position of instrument, respectively.

6.3.2.7 Main → Run, Start the calibration.

6.3.2.8 Select the calibration items from User defined chemistries → F5 Approve Cal, select F1 Approve Update, to approve the calibrated curve.

6.3.2.9 Each concentration of standard curve should be repeated 2 times. CV value should not be higher than 15%, and the standard curve correlation coefficient R2 should not be less than 0.985. Determinate the internal control reference under the calibrated curve at the same time. The recovery rate calculated should be in 85%-115%, that is, concentration is in 1.0μg/ml SOP of Yuxi Watson Biotechnology Co., Ltd.

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(0.85-1.15μg/ml) and 2.0μg/ml (1.70-2.30μg/ml).

6.3.2.10 Principle that the standard curves need to be recalibrated

Items Standard Test results Corresponding measurement

The CV of each Not higher than 15% No

point in twice Not higher than Calibrate again under the curves 15% Higher than 15% same condition as the calibration previous.

The CV of each point

in twice curves

calibration is not

higher than 15% and

No

the correlation

coefficient of the

curves R2 is not less

than 0.985.

① Re-prepare the new

standard solution, and do the

recalibration to exclude the

reason that the operator

Correlation

prepare the inaccurate

coefficient of the Not less than 0.985

The CV of each point diluent. ② If the CV is still

curves, R2

in twice curves not higher than 15% and the

calibration is not R2 is still less than 0.985, it

higher than 15% and means that the antigen is

the correlation excess rather than antibody

coefficient of the during the antigen-antibody

curves R2 is less than reaction (the ideal condition

0.985. is antibody excess). The

curves should be

recalibrated under the

condition of high antibody

concentration (reducing the

antiserum dilute multiples).

Test the 85-115% The CV of each point No

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recovery rate of [The concentration in twice curves

the calibrated should be 1.0μg/ml calibration is not

internal control (0.85-1.15μg/ml), higher than 15% and

reference 2.0μg/ml the R2 is not less than

(1.70-2.30μg/ml)] 0.985. The recovery

rate of the calibrated

internal control

reference is in the

range of 85-115%.

The CV of each point

in twice curves

Adjust the parameter of

calibration is not

curve calibration (reaction

higher than 15% and

time, gain coefficient, the

the R2 is not less than

buffer capacity of reaction,

0.985. The recovery

sample capacity and

rate of the calibrated

antiserum working solution

internal control

capacity) and recalibrate.

reference is not in the range of 85-115%.

6.3.2.11 The original record of standard curves should be filed. And the standard curves used for sample test should be attached in the batch test record.

6.3.3 Preparation of test samples

6.3.3.1 The polysaccharide bulk solution is normally diluted 2000 times to use. Take 100μl polysaccharide bulk solution to the 100ml volumetric flask, add purified water to scale, in which dilute 1000 times. Take 1000μl of the 1000 times test sample diluent, add 1000μl of purified water to obtain 2000 times diluent (The volume of sample can be adjusted according to the proportion of testing dosage). Determine the polysaccharide content in the IMMAGE 800, enter 2000 for off-line dilution multiple. If the result is beyond the detection range, adjust the dilution ratio to A, and enter A for off-line dilution multiple.

6.3.3.2 When the test sample is conjugate bulk, directly dilute with Diluent 1 (DIL1) for appropriate dilution (record the exact dilution multiple) so that its polysaccharide content in SOP of Yuxi Watson Biotechnology Co., Ltd.

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the 0.625~2.5μg/ml, mix well.

6.3.3.3 If test sample is 13-valent pneumococcal conjugate vaccine, take 5 vaccines and transfer them into a 15ml centrifuge tube, mix well and take 2000μl into another clean centrifuge tube by pipette, add 100μl of sodium hydroxide solution (0.6mol/L), mix well, then add 100μl of citric acid solution (0.2mol/L) within 10s, mix well. Take 300μl of the desorption solution, add 600μl of Diluent 1 (DIL1) and mix well to obtain 3 times diluent (The volume of sample can be adjusted according to the proportion of testing dosage), determine polysaccharide content (except for type 6B) on the IMMAGE 800, enter 3.3 (i.e. 3×1.1) for off-line dilution ratio. Take another 200μl of desorption solution, add 1000μl of Diluent 1 (DIL1) and mix well to obtain 6 times diluent (The volume of sample can be adjusted according to the proportion of testing dosage), determine polysaccharide content (type 6B) on the IMMAGE 800, enter 6.6 (i.e. 3×1.1) for off-line dilution ratio. If the result is beyond the detection range, adjust the dilution ratio to A, and enter A×1.1 for off-line dilution multiple.

6.3.3.4 If test sample is 23-valent pneumococcal polysaccharide vaccine, take 5 vaccines into 15ml centrifuge tube and mix well. Take 200μl of it, add 1800μl Diluent 1 (DIL1) and mix well to obtain 10 times test sample diluent. Take 1500μl of the above diluent and dilute with 3000μl Diluent 1 (DIL1) and mix well to obtain 30 times diluent, determine polysaccharide content on the IMMAGE 800, enter 30 for off-line dilution ratio.

6.3.4 Sample test

6.3.4.1 Enter sample display, select a test item (type) in the menu. Enter the shelf number at "Rack", enter the location number at "Pos", input samples number at "Sample ID", each sample ID number is unique in the same equipment and cannot be reused.

6.3.4.2 Enter the batch number, sample for stability test and temperature and time of storage at "Sample Comment", select a test item (type) in the menu → click F3 "Sample Options" dialog box, enter the repeat times at "Sample Replicate", enter the off-line dilution multiples at "Off-line Dilution Factor" → OK.

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6.3.4.3 F10 Save/Next save and program the next sample.

6.3.4.4 Place the diluted sample (placed in the sample cup) in samples rack, then move it into the sample turn plate.

6.3.4.5 Main → Run to start testing.

6.3.5 Results view and print

6.3.5.1 Enter Results display: search the results through the any information of sample number (Sample ID), shelf number (Rack), location number (Pos), running time etc.

6.3.5.2 F1 Display Results, view the results on the computer and print the results.

6.3.6 Power-off: The instrument must be in the Standby state for system shut down.

6.3.6.1 Utils → F12 Shutdown → Shutdown Complete power off the Immage and Reboot the system. Turn off the printer, display, computer host, UPS system and instrument and so on.

7 Flow chart

N/A

8 Matters need attention

8.1 Before use, all mechanical parts of the instrument must be free of obstruction. Check remain of flushing liquid, waste liquid barrel; Check if there is any liquid leakage on syringe valve and instrument bottom etc., and clean outside surface of reagent needle/stirring needle, sample needle/agitator needle.

8.2 The pipeline should be rinsed every day before making samples. Utils → F1 Prime, enter flushing times → OK.

8.3 Any external operations in the test should be in the Standby or Pausing state of instrument.

8.4 When use the new batch antiserum, the standard curve should be calibrated again.

8.5 Concentration confirming of the pneumococcal polysaccharide standard substance stock solution

8.5.1 When preparing the new batch of pneumococcal polysaccharide standard stock solution, it is necessary to take the newly prepared pneumococcal polysaccharide standard stock solution with the concentration of 1.5μg/ml, and test its concentration with the curve

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calibrated by the previous batch of pneumococcal polysaccharide standard stock solution.

8.5.2 The same tester shall repeat the operation "8.5.1" for 10 times in different time periods (no more than 3 times in the same day), calculate the concentration of the newly prepared pneumococcal polysaccharide standard stock solution according to each test result, take the average value of the 10 test results as the final concentration of the newly prepared pneumococcal polysaccharide standard stock solution, and then subpackage the stock solution into centrifugation tubes according to 400μl/vial, paste the label.

8.5.3 For each batch, two pieces of stock solution for pneumococcal polysaccharide standard need to be reserved to confirm the concentration of stock solution for next batch.

8.6 Before each sample test, the corresponding type of internal control reference should be tested. Only when the test results of internal control reference meet the standard requirements of 1.0μg/ml (0.85-1.15μg/ml/type), 2.0μg/ml (1.70-2.30μg/ml/type), can the sample be tested. When the test results of internal control reference do not meet the standard: ① in the test of internal control reference with two concentrations, one of which does not meet the standard the same antiserum and the same internal control can be used to directly and repeatedly detect the concentration of the internal control reference that do not meet the requirements; ② when the both internal control reference of two concentrations do not meet the requirements, the antiserum working solution (or use another internal control reference of the same concentration) can be reconstituted for testing; ③ the curve need to be recalibrated, and the sample test can be conducted after the test results of the internal control reference meet the requirements.

8.6.1 When testing the content of polysaccharide in the pneumococcal polysaccharide bulk or pneumococcal polysaccharide conjugate bulk: when the polysaccharide bulk or conjugate bulk of the same batch number is less than 5 bottles, the frequency of internal control test is: before and after sample test; if the test results of two concentrations of internal control reference in the process of sample test or after sample test do not meet the requirements, then the corresponding previous stage The result of polysaccharide content in the pneumococcal SOP of Yuxi Watson Biotechnology Co., Ltd.

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polysaccharide bulk should be judged as invalid. After the internal control test results meet the requirements, carry out the original sample retest, and issue the test data based on the retest results. If there is a concentration of internal control reference whose one test result does not meet the requirements in the sample testing process or after the sample testing, the internal control reference of the corresponding concentration can be retested once. If the repeated test results of the control reference meet the requirements, then the corresponding results of the polysaccharide content in the original pneumococcal polysaccharide solution in the previous stage do not need to be retested; if the repeated test results of the internal control reference products do not meet the requirements, then the results of the polysaccharide content in the original pneumococcal polysaccharide solution in the corresponding previous stage are invalid. After the results of the internal control test meet the requirements, carry out the original retest to issue the test results Data, the original report of internal control test results after the end of sample test is attached to the verification record of the last sample.

8.7 The polysaccharide content of each type of sample shall be determined 5 times

respectively, and the RSD of the 5 results shall not be more than 15%

$RSD = SD / \text{arithmetic mean of calculation result (x)} \times 100\%$

SD = arithmetic square root of variance

8.7.1 When the relative standard deviation (RSD) of the results of 5 determinations is not more than 15%, according to the general rule 198 of the volume III of the Chinese Pharmacopoeia, the peculiar value eliminated in the peculiar reaction elimination. When the special value can be eliminated according to the rules, the peculiar value shall be eliminated, and the average value after the missing items are supplemented shall be issued with the result, and the qualified report shall be issued when the result is qualified. {The general rule 198 of the volume III of the Chinese Pharmacopoeia, the peculiar value is eliminated in the peculiar reaction elimination: 5 values, from large to small (or from small to large) Y1, Y2, Y3, Y4, Y5, $(Y2-Y1)/(Y5-Y1) > 0.73$, then Y1 can be eliminated. For example, $RSD=10\%$ of the value 66.7, 54.8, 54.2, 53.6, $(54.8-66.7)/(53.6-66.7) = 0.91$, in this case, the value 66.7 can be eliminated
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under the condition that the regulations are met. At the same time, the average value 54.4 of the results 54.8, 54.8, 54.2, 53.6 can be used as the deficiency item to make up, and the report will be issued according to the average value 54.4 of 54.8, 54.8, 54.2, 53.6, 54.4.

8.7.2 When the relative standard deviation (RSD) of the 5 test results is higher than 15%, whether the test results meet the requirements or not, it shall be determined that the test results are invalid and shall be re-measured; if the relative standard deviation (RSD) of the re-measured results is not higher than 15%, and the results meet the requirements, the test data shall be issued according to the re-measured results.

8.8 For the determination of polysaccharide content in the polysaccharide bulk or the conjugate bulk, the difference between any bottles of polysaccharide content in the polysaccharide bulk or the conjugate bulk of the same batch shall not exceed 15%. The formula for calculating the difference between bottles:

■-■
Difference between bottles (%) = $|\frac{A-B}{A}| \times 100$

A: the average value of polysaccharide content in all bottles of the same batch of polysaccharide bulk or conjugate bulk;

B: the content of polysaccharide in any bottle of the same batch of polysaccharide bulk or conjugate bulk.

8.9 In the process of sample testing, if the kit batch number is used up, and the test can only be carried out after re-calibration, the shelf number of the corresponding type of sample shall be deleted, and the unfinished test shall be terminated. After the internal control test of the curve calibration is completed and meets the requirements, the unfinished sample book shall be edited again for testing.

9 Reference or standard basis

9.1 <IMAGE@800 User standardization Training manual for specific protein analysis systems 2.0>

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10 Revise history

Time in Revision

Edition Revision basis, reason and content Notes

force Dept.

01 Aug. QC

00 Refine operations, complete content /

2015 Laboratory

25 July QC

01 Refine operations, complete content /

2016 Laboratory

30 Sep. QC

02 Refine operations, complete content /

2018 Laboratory

01 Sep. QC

03 Refine operations, complete content /

2018 Laboratory

20 Jun. QC Add the management of antiserum

04 /

2019 Laboratory working solution and test data

24 July QC Refine operations, add the peculiar value

05 /

2019 Laboratory elimination of results judgment

Revise the standard of difference between

23 Sep. QC different bottle of bulk (conjugate bulk).

06 /

2019 Laboratory Standardize the operation of standard reference working solution preparation.

Standard Operating Procedure of Yuxi Walvax Biotechnology Co., Ltd.

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polysaccharide content test
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1 Purpose

The method of free polysaccharide content test is standardized.

2 Definitions/abbreviations

No

3 Scope of application

This SOP is applicable to the free polysaccharide content test of group A meningococcal polysaccharide bulk conjugate, group C meningococcal polysaccharide bulk conjugate, group Y meningococcal polysaccharide bulk conjugate, group W135 meningococcal polysaccharide bulk conjugate, Meningococcal group A, C bivalent polysaccharides conjugate vaccine.

4 Responsibilities

All operators operate strictly in accordance with this SOP.

5 Equipment and materials

5.1 Using the equipment

Serial Equipment Equipment Equipment Note

Manufacturers

number name model coding

Electronic or equivalent

1 CP225D WYQCF0037 Sartorius

balance equipment

Shimadzu

Ultraviolet WYQCF0444

Enterprise

Visible Light WYQCF0445 or equivalent

2 UV-2600 Management

Separation WYQCF0529 equipment

(China) Co.,

Photometer WYQCF0530

Ltd.

Electronic or equivalent

3 BSA124S WYQCF0357 Sartorius

balance equipment

Shanghai Qixin

Scientific

Instruments

Electric WYQCF0287 Co.,

or equivalent

4 heated water DK-600A WYQCF0061 Ltd./Shanghai

equipment

bath WYQCF0293 Evergrande

Scientific

Instruments Co.,

Ltd.

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Electric Shanghai Qixin

constant DHG-9145A WYQCF0452 Scientific or equivalent

6

temperature HG101-3AS WYQCF0437 Instruments Co., equipment

drum dry box Ltd.

Herlith

Heraeus or equivalent

7 desktop WYQCF0042 Biofuge/Fresco

(Heris) equipment

centrifuges

Thermo Fisher

High-speed

multifugeXI WYQCF0473 Technology or equivalent

8 refrigeration

R WYQCF0472 (China) Co., equipment

centrifuges

Ltd.

Hitachi High

or equivalent

9 Speed CR21G WYQCF0068 Hitachi, Japan

equipment

Centrifuge

Beijing

Changfeng

Instrument

Electric WYQCF0517

HWSY21-K Company / or equivalent

10 heated water WYQCF0580

HH-8 Changzhou equipment

bath pot WYQCF0586

Aohua

Instrument Co.,

Ltd.

5.2 Use of reagents

Serial

Reagent Name Level, don't. Note
number

Potassium

1 dihydrogen Standard No

phosphate

Concentrated sulfuric

2 GR Content (w/%) 95.0~98.0
acid

3 Perchloric acid AR No

Ammonium

4 AR No

molybdate

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5 Ascorbic acid (VC) Imported reagents No

6 Phenol AR No

N-hexyl neuraminic

7 Imported reagents No

acid

8 resorcinol AR No

Copper sulfate

9 AR No

pentahydrate

Concentrated

10 AR No

hydrochloric acid

11 Butyl acetate AR No

12 N-butanol AR No

13 Sodium acetate AR No

5.3 Materials

Serial number Material name Serial number Material name

1 Tube 2 Pipette tip

3 Capacity bottle 4 Ultrafiltration Cup

5 Centrifugal tube 6 Syringe

6 Work procedures

6.1 Principle

6.1.1 Group A free polysaccharides: the protein bound polysaccharide in the original solution of the conjugate was precipitated with cold phenol, and the phosphorus content in the original solution before precipitation and the supernatant after precipitation was determined respectively. The detection of phosphorus content is to determine the content of phosphorus after transforming organic phosphorus into inorganic phosphorus. In acid solution, phosphate reacts with ammonium molybdate to form ammonium molybdate, which forms blue substance (mixture of molybdenum trioxide and molybdenum pentoxide) in the presence of reducing Standard Operating Procedure of Yuxi Walvax Biotechnology Co., Ltd.

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agent, which is called "molybdenum blue". The content of phosphorus in the test sample is determined by colorimetry.

6.1.2 Group C, Y and W135 free polysaccharides: precipitate the polysaccharide bound with protein in the original solution of conjugate with cold phenol, and determine the content of sialic acid in the original solution before precipitation and the supernatant after precipitation. Sialic acid content detection is to change the bound sialic acid into free state by acid hydrolysis method. The free sialic acid reacts with resorcinol to form colored compounds, and then extracts with organic acids to determine sialic acid content.

6.2 Reagents

6.2.1 10% saturated sodium acetate solution (pH 7.0 ± 0.02): weigh excess sodium acetate, heat and dissolve in purified water (430g → 1L), filter while hot; cool down, the solution crystallizes; mix 1 part of supernatant with 9 parts of purified water (such as 100ml of supernatant and 900 ml of purified water), adjust the pH value to 7.0 ± 0.02.

6.2.2 cold phenol solution: put phenol in water bath at 50℃ to liquefy; weigh 100 g of liquefied phenol, dissolve it in 40ml 10% saturated sodium acetate solution (pH 7.0 ± 0.02), and place it at 2-8℃ for more than 1 h.

6.3 test procedure

6.3.1 Test article treatment

6.3.1.1 Group A meningococcal polysaccharide bulk conjugate: dilute the conjugate solution and corresponding derivatives to about 20-30 µg/ml with purified water to obtain test sample P1. Take P1600 µl derivative test sample and P1600 µl conjugate test sample respectively and add them into a 2000 µl centrifuge tube. Make 6 tubes for each test sample in parallel, and then add 1200 µl cold phenol solution to each tube (the cold phenol solution must be fully mixed before each absorption), shake violently for 20 min, place in ice water bath for 10 min, centrifuge at 1500 and 10000 rpm for 15 min. After centrifugation, the liquid was divided into two layers, and obvious precipitates were observed at the boundary. Carefully absorb the centrifuged supernatant (aqueous phase) and transfer it to a clean labeled test tube to obtain P2. Standard Operating Procedure of Yuxi Walvax Biotechnology Co., Ltd.

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Mix well before use.

6.3.1.2 Group C meningococcal polysaccharide bulk conjugate: dilute the conjugate solution and corresponding derivatives with purified water. P1 was obtained when the polysaccharide content was about 40-60 µg/ml. Take P1600 µl derivative test sample and P1600 µl conjugate test sample respectively and add them into 2000 µl centrifugal tube. Each test sample is made into 12 tubes in parallel, and then each tube is added with 1200 µl cold phenol solution (the cold phenol solution must be fully mixed before each absorption), shake violently for 20 min, place in ice water bath for 10 min, centrifuge at 15000 and 10000 rpm for 15 min. After centrifugation, the liquid was divided into two layers, and obvious precipitates were observed at the boundary. Carefully absorb the centrifuged supernatant (aqueous phase) and transfer it to a clean labeled test tube to obtain P2. Mix well before use.

6.3.1.3 Group Y meningococcal polysaccharide bulk conjugate: dilute the conjugate solution and corresponding derivatives to 40-60 µg/ml with purified water to obtain P1. Take P1600 µl derivative test sample and P1600 µl conjugate test sample respectively and add them into 2000 µl centrifugal tube. Each test sample is made into 12 tubes in parallel, and then each tube is added with 1200 µl cold phenol solution (the cold phenol solution must be fully mixed before each absorption), shake violently for 20 min, place in ice water bath for 10 min, centrifuge at 15000 and 10000 rpm for 15 min. After centrifugation, the liquid was divided into two layers, and obvious precipitates were observed at the boundary. Carefully absorb the centrifuged supernatant (aqueous phase) and transfer it to a clean labeled test tube to obtain P2. Mix well before use.

6.3.1.4 Group W135 meningococcal polysaccharide bulk conjugate: dilute the original solution of conjugate to be tested and its corresponding derivatives with purified water to about 40-60 µg/ml of polysaccharide to obtain P1. Take P1600 µl derivative test sample and P1600 µl conjugate test sample respectively and add them into 2000 µl centrifugal tube. Each test sample is made into 12 tubes in parallel, and then each tube is added with 1200 µl cold phenol solution (the cold phenol solution must be fully mixed before each absorption), shake violently for 20 min, place in ice water bath for 10 min, centrifuge at 15000 and 10000 rpm for 15 min. After centrifugation, the liquid was divided into two layers, and obvious precipitates were observed at the boundary. Carefully absorb the centrifuged supernatant (aqueous phase) and transfer it to a clean labeled test tube to obtain P2. Mix well before use.

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min, place in ice water bath for 10 min, centrifuge at 15000 and 10000 rpm for 15 min. After centrifugation, the liquid was divided into two layers, and obvious precipitates were observed at the boundary. Carefully absorb the centrifuged supernatant (aqueous phase) and transfer it to a clean labeled test tube to obtain P2. Mix well before use.

6.3.1.5 Group A and C meningococcal polysaccharide conjugate vaccine:

a) Group A free polysaccharides: Take the finished 14 bottles with the corresponding thinner to 0.5 ml/bottle resolvable, mixed and removed, with 2 3K ultrafiltration cups. Each take about 3.5 ml, the sample into the ultrafiltration cup, with the mark note under the liquid surface scale, balance after 500, 7000 rpm centrifugal 30min, centrifugation after taking out water to the scale line, and then in the same conditions after centrifugation to rehydrate to the tick mark, fully blow-washed evenly mixed, as a finished P1. Dilute the corresponding derivative saccharine to contain polysaccharides with purified water about 20 to 30 µg/ml, as a derivative P1. Take derivatives for test products P1600 µl and finished products P1600 µl each added to 2000 µl centrifuge tubes, each for test parallel to do 6 tubes, and then each tube to add 1200 µl cold phenol solution (cold phenol solution must be fully mixed before each absorption), severe shock 20 min, ice water bath 10 min, 15000, 10000 rpm centrifugal 20 min. After the centrifuge of the finished product for the test product, the liquid is divided into two layers, and the more obvious sediment can be seen at the demarcation boundary. Carefully absorb the centrifugal upper clear (water phase) and transfer to a clean marked test tube, as P2, mixed with the front.

b) Group C free polysaccharides: take the finished 28 bottles with the corresponding thinner with 0.5 ml / bottle resolvable, mixed and removed the combined, that is, the finished solution. P1: Take the finished solution at least 4.5 ml into a 3K ultrafiltration cup, the sample into the ultrafiltration cup, with the mark note under the liquid surface scale, balance after 500, 7000 rpm centrifugation 30 min, after centrifugation to remove the rehydration to the tick mark, and then in the same conditions after centrifugation to rehydrate to the tick mark, fully blow-out evenly for the finished product P1. Dilute the corresponding derivative to contain polysaccharides of about 40 to 60 µg/ml, as a derivative P1.

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P2: The finished solution, take the derivative for the test P1600 µl and the finished solution 600 µl each added to the 2000 µl centrifuge tube, each for the test parallel to do 12 tubes, and then add 1200 µl cold phenol (cold phenol solution must be fully mixed before each absorption), violent lysing 20 min, ice bath water 10 min, 15000, 10000 rpm centrifugal 20 min. After the centrifuge of the finished product for the test product, the liquid is divided into two layers, and the more obvious sediment can be seen at the demarcation boundary. Carefully absorb the centrifugal upper clear (water phase) and transfer to a clean marked test tube, i.e. derivative P2 and finished P2 and mixed together before. Take 7 3K ultrafiltration cups, pipette to remove 750 µl purified water in ultrafiltration cups, take a marker note under the scale, add purified water 4000 µl, add the upper clear 250 µl, balance after 500, 7000 rpm centrifugation 30min, after centrifugation, add the upper clear 250 µl in 7 ultrafiltration cups, add purified water 4000 µl, continue to centrifuge in the same conditions, and then add the upper clear 250 µl to the 7 ultrafiltration cups, and add the purified water 4000 µl, continue to centrifuge in the same conditions, add purified water 4000 µl at the end, still centrifuge in the same conditions, rehydrate to 750 µl after the centrifugation, and remove the liquid from the 7 ultrafiltration cups well after the full blow-up is mixed with P2.

6.3.2 Take a clean glass test tube with a name, lot number and disposal number, where P1 represents an unprocessed test product and P2 represents a centrifugal liquid after treatment. Each sample marking and dilution can be found in the table below:

Group A free polysaccharides (Group A meningococcal polysaccharide bulk conjugate,

Meningococcal group A, C bivalent polysaccharide conjugate vaccine) cold phenol solution after treatment of sample fluid table (μl)
Conjugate (finished product) Derivatives
Samples
P1 P1 P2 P2 P1 P1 P2 P2
Volume of sample 1000 1000 1000 1000 1000 1000 1000 1000
(μl)
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Purified water (μl) 0 0 0 0 0 0 0 0
Dilution multiple 1 1 1 1 1 1 1 1

Group C, Y, W135 free polysaccharides (C, Y, W135 group meningococcal polysaccharides bulk conjugate), C group free polysaccharides (Meningococcal group A, C bivalent polysaccharide conjugate vaccine) cold phenol solution treatment after the sample fluid add (μl).

Conjugate (finished product) Derivatives
Samples
P1 P1 P2 P2 P1 P1 P2 P2
Volume of sample 2000 2000 2000 2000 2000 2000 2000 2000
(μl)
Purified water (μl) 0 0 0 0 0 0 0 0
Dilution multiple 1 1 1 1 1 1 1 1

6.3.3 Group A free polysaccharides in meningococcal group A, C bivalent polysaccharide conjugate vaccine according to the "phosphorus content (micro-method) operation procedures" code encoding: WSYX-QC-SOP-00-09-0121 for phosphorus content detection, in the test tube of each tube is added mineralized reagent 150 μl, mix, aluminium-foil cover test tube, put the test tube placed in the electric heating constant temperature blower drying oven, the temperature is 150℃, the temperature reaches after drying for 1 hour, adjust the temperature to 180℃, 30 minutes after reaching the temperature, adjust the temperature to 250℃, and dry bake for 1 hour after the temperature reaches, so as to mineralize and inorganic the test sample. After taking out, add 1850 μl purified water while it is hot (this step is not required for the reference working solution). Add 2000 μl of chromogenic agent into each tube, mix well, and take water bath in 37℃ constant temperature water bath for 2 hours. The absorbance was measured at 825 nm.

6.3.4 Group A free polysaccharide in group A meningococcal polysaccharide bulk conjugate in accordance with the "phosphorus content (trace method) operating procedures" code encoding: WSYX-QC-SOP-00-09-0121 for phosphorus content detection, in the test tubes, all the Standard Operating Procedure of Yuxi Walvax Biotechnology Co., Ltd.

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mineralized reagent 150 μl, mix together, aluminium-foil covered test tube, put the test tube in the electric heating thermostat blower oven, the temperature is 150℃, after the temperature reaches, dry bake for 1 hour, adjust the temperature to 180℃, 30 minutes after reaching the temperature, adjust the temperature to 250℃, dry bake for 1 hour after reaching the temperature, make the test sample mineralized and inorganic, and add 1850 μl purified water while taking it out (this step is not required for the reference working solution). Add 2000 μl of chromogenic agent into each tube, mix well, and take water bath in 37℃ constant temperature water bath for 2 hours. The absorbance was measured at 825 nm.

6.3.5 Group C free polysaccharides in meningococcal group A, C bivalent polysaccharide conjugate vaccine, group C free polysaccharides in group C meningococcal polysaccharide bulk conjugate, group Y free polysaccharides in group Y meningococcal polysaccharide bulk conjugate, group W135 free polysaccharides in group W135 meningococcal polysaccharide bulk conjugate shall be in accordance with the document code of operation procedures for determination of sialic acid content: WSYX-QC-SOP-00-09-0122, and the content of sialic acid was detected. Add 2000 μl resorcinol chromogenic agent into each tube, mix well, place it in boiling water bath for 15 minutes, take it out and put it in ice water for 10 minutes, add 4000 μl organic phase to each tube, mix well, and let it stand for 10 minutes at room temperature. The solution is naturally stratified. Take the upper organic phase and measure the absorbance at 585 nm wavelength.

6.4 Result calculation
Experimental validity% = derivative P2 ÷ derivative P1 × 100
Free polysaccharide% = conjugate (finished product) P2 ÷ conjugate (finished product) P1 × 100

6.5 Result determination
According to the corresponding sample quality standard.
7 Flow chart
No
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8 Precautions
8.1 when centrifuging, pay attention to the actual temperature of the centrifuge. If the temperature before use fails to meet the experimental requirements, the centrifuge can be idled in advance.
8.2 take the supernatant carefully, and avoid stirring the liquid and inhaling the sediment by the pipette.
8.3 pay attention to the refrigeration temperature of cold phenol solution and keep it at a low

temperature when using. If it spills, it should be treated with absolute alcohol, and gloves should be worn during operation to avoid skin injury.

8.4 the time of each step should be strictly controlled.

9 References or standard basis

9.1 Chinese Pharmacopoeia, Volume 3, General rule 3102, Determination of sialic acid, China Pharmaceutical Science and Technology Press, 2015 Edition

9.2 Chinese Pharmacopoeia, Volume 3, General rule 3119, determination of polymer conjugates content, China Pharmaceutical Science and Technology Press, 2015 Edition

13 Revision history

Basis, reasons and content of

Version Effective Revision

the amendment Note

No. time Department

QC File expiration, change

03 2011.12.01

Laboratory Improve part of the file content

QC

04 2014.09.17 Improve part of the file content

Laboratory

QC

05 2016.03.20 Improve part of the file content

Laboratory

06 2017.10.01 QC Improve part of the file content

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Laboratory

Increase detection step of free

QC polysaccharide in group Y, W135

07 2018.09.01

Laboratory meningococcal polysaccharide

bulk conjugate

Revise the file and code of group

QC A free polysaccharide in the

08 2020.01.31

Laboratory meningococcal polysaccharide

bulk conjugate

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Operation procedures of Code WSYX-QC-SOP-00-09-0114
determination free protein
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1 Purpose

The method of determining the content of free protein is standardized.

2 Definitions/abbreviations

No

3 Scope of application

This SOP is applicable to the free protein determination of Group A, C bivalent meningococcal polysaccharide conjugate vaccine, pneumococcal polysaccharide bulk conjugate.

4 Responsibilities

All operators operate strictly in accordance with this SOP.

5 Materials and equipment

5.1 Using the equipment

Serial Equipment Equipment

Equipment code Manufacturers Note

number name model

WYQCF0082/WYQCF00

85

WYQCF0088/WYQCF00

91/WYQCF0467/WYQC

Tomography Amersham or equivalent

1 XK16/100 F0468

Column Biosciences equipment

WYQCF0469/WYQCF04

70

WYQCF0510/WYQCF05

11

WYQCF0081/WYQCF00

Step-by-step 84 Amersham or equivalent

2 Frac920

collector WYQCF0087/WYQCF00 Biosciences equipment

90

WYQCF0080/WYQCF00

Constant 83 Amersham or equivalent

3 P-1

current pump WYQCF0086/WYQCF00 Biosciences equipment

89

AKTA system WYQCF0455/WYQCF04

4 AKTA GE or equivalent

Purifier 56

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pure25 WYQCF0457/WYQCF04 equipment

58

WYQCF0506/WYQCF05

07

AKTA system AKTA or equivalent

5 WYQCF0192 GE

Purifier Purifier 10 equipment

Shimadzu

Ultraviolet or equivalent

WYQCF0444/WYQCF04 Enterprise

6 Visible Light μ V-2600

45 Management equipment

Spectrometer

(China) Co., Ltd.

High-speed Thermo Firth or equivalent

Multifuge WYQCF0472/

7 freezing Technology

X1R WYQCF0473 equipment

Centrifuges (China) Co., Ltd.

Hitachi High

or equivalent

8 Speed CR21G WYQCF0068 Hitachi, Japan

equipment

Centrifuge

Shanghai

Guangzheng or equivalent

Rack drug

9 JEZREEL-5 WYQCF0076 Medical

balance equipment

Instruments Co.,

Ltd.

Analysis of or equivalent

10 CP225D WYQCF0037 Sartorius

balances equipment

Shanghai Qixin

Scientific
Electric heated WYQCF0061/ Instrument Co., or equivalent
11 DK-600A WYQCF0287/ Ltd./Shanghai
water tank WYQCF0293 Yiheng Scientific equipment
Instrument Co.,
Ltd.
5.2 Use of reagents
Serial
Reagent Name Level/Category
number

1 Sodium hydroxide AR

2 Sodium carbonate anhydrous AR
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3 Potassium tartrate AR
4 Copper sulfate pentahydrate AR
5 2N foline-phenol Imported reagents
6 Sodium chloride AR

5.3 Materials used
Serial Serial
Material name Material name
number number
Tomography Media: Sepharose CL-
1 2 Needle filter
4B
3 Disposable sterilized syringe 4 Plastic collection tube
5 Pipette tip 6 Sealing film
7 Ultrafiltration Cup 8 15x150 test tube

9 Triangle bottle
6 Work procedures
6.1 Experimental steps
6.1.1 Preparation of solution
6.1.1.1 0.2 mol/L sodium chloride solution
1 mg sodium hydroxide solution was mixed with 0.1 mg sodium chloride solution at pH
of 0.1 ml, and then mixed with 0.1 ml sodium chloride solution.
6.1.1.2 molecular size reference solution
For details of preparation, please refer to WSYX-QC-SOP-00-09-0801
6.1.2 calibration column
The column was calibrated with molecular size reference solution to determine the empty
flow volume V_0 and total column bed volume V_i . the column calibration period was 30 days.
6.1.3 test article treatment
6.1.3.1 free protein control (tetanus toxoid purification solution): dilute the carrier protein
tetanus toxoid purification solution with normal saline until the protein content is about 2000
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$\mu\text{g/ml}$ to obtain solution S1.
6.1.3.2 concentration of group A, C bivalent meningococcal polysaccharide conjugate vaccine:
a) Take a 3K ultrafiltration cup, add 2000 μl purified water, mark the concave surface with a
marking pen, and then discard 2000 μl purified water.
b) Take 10 bottles of finished products, dissolve them with 0.5 ml/bottle with corresponding
diluent, mix them well and take them out. Put 4000 μl of the above solution into the marked
ultrafiltration cup and mark the scale on the concave surface with a marking pen.
c) After balancing with a shelf plate drug balance, centrifuge for 30 min at 5000 and 7000 rpm.
After centrifugation, take out the water to make up to the sample scale line; after the second
centrifugation under the same conditions, make up water to the 2000 μl scale mark, and then
fully purge and filter with a 0.45 μm needle filter.
6.1.3.3 concentration of pneumococcal polysaccharide conjugate solution:
a) Take two 3K ultrafiltration cups, add 1000 μl purified water into each cup, mark the concave
surface with a marking pen, and then discard 1000 μl purified water.
b) Put 5000 μl test sample into 3K ultrafiltration cup, make two parallel tubes, balance with
shelf plate drug balance, centrifuge at 5000 and 7000 rpm for 30 min, and the liquid level after
centrifugation should be below 1000 μl scale mark.
c) At the end of centrifugation, take out the make-up water to 1000 μl scale mark, fully purge
and mix, and filter with 0.45 μm needle filter to obtain the test sample. 6.1.4
chromatographic column loading
6.1.4.1 Take 1 ml free protein control column, elute with 0.2 mol/l sodium chloride solution;
collect 4.5 ml each tube by automatic collector, collect 1.2 column bed volume; detect the
collected liquid at 214 nm wavelength with UV visible spectrophotometer, draw the flow
elution map with the flow washing volume as the abscissa, and take the absorbance as the
ordinate, and record the separation peak of downstream free protein control (OD value from
small to small) The number of tubes corresponding to the number of tubes at the end (OD value
changed from greater than or equal to 0.01 to less than 0.01) was N1, and the number of tubes
Standard Operating Procedures of Yuxi Walvax Biotechnology Co., Ltd.

Operation procedures of Code WSYX-QC-SOP-00-09-0114
determination free protein

corresponding to the end (OD value changed from greater than or equal to 0.01 to less than 0.01) was N2; the sum of OD of each tube from the appearance of separation peak to the end of separation peak $\sum OD1$ was calculated; the free protein control was collected from the beginning to the end of each tube, and 1000 μ l of collecting solution was mixed to obtain solution S2. The validity period of free protein control was 30 days

6.1.4.2 determine the protein concentration of S0, S1 and S2 solutions according to the operation procedures of protein content determination. According to 6.1.5.1, the validity standard should be 80% - 120%.

6.1.4.3 take 1 ml of the treated sample on the column, elute, collect and colorimetry with the same method; calculate the sum of OD values $\sum OD2$ of each tube from the appearance to the end of the corresponding free protein control separation peak.

6.1.5 Result calculation:

6.1.5.1 experimental validity $P(\%) = \text{collection volume of each tube} \times \text{number of combined tubes} \times S2 \div S1 \times 100$

6.1.5.2 meningococcal polysaccharide conjugate vaccine group A and group C:

Free protein content $P(\%) = (\sum OD2 \div \sum od1) \times S1 \div S0 \times 2 \times 100$

6.1.5.3 original solution of pneumococcal polysaccharide conjugate:

Free protein content $P(\%) = (\sum OD2 \div \sum od1) \times S1 \div S0 \times 5 \times 100$

S0: protein concentration of the test sample (μ g/ml);

S1: Pre column protein concentration of free protein control (tetanus toxoid purification solution) (μ g/ml);

S2: free protein control separation peak from the beginning to the end of each tube mixture protein concentration (μ g/ml);

6.1.6 Result judgment

According to the quality standard of corresponding samples.

7 Flow chart

N/A

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determination free protein

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8 Precautions

8.1 The corresponding tube absorption to the end of the free protein control separation peak should be 0.010.

9 References or standard basis

No

10 Revision history

Basis, reasons and content of the

Version Effective Revision

amendment Note

No. time Department

QC File expiration, change

03 2011.12.01

Laboratory Improve part of the file content

QC

04 2014.09.16 Improve part of the file content

Laboratory

QC

05 2016.03.20 Improve part of the file content

Laboratory

QC

06 2017.10.01 Improve part of the file content

Laboratory

2018.09.01 QC Describe in detail where separation

07

Laboratory peaks appear and end

2019.09.25 QC

08 Add spreadsheets

Laboratory

MODULE 4 NONCLINICAL STUDY REPORTS
Version No. Walvax-202007-00 Yuxi Walvax Biotechnology Co., Ltd.

PRODUCT REGISTRATION DOSSIER

13-valent Pneumococcal Polysaccharide Conjugate Vaccine
(Tetanus Toxoid Protein)

MODULE 4 NONCLINICAL STUDY REPORTS

Yuxi Walvax Biotechnology Co., Ltd.

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MODULE 4 NONCLINICAL STUDY REPORTS
Version No. Walvax-202007-00 Yuxi Walvax Biotechnology Co., Ltd.

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4.2. STUDY REPORTS
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4.2.1.2. Secondary Pharmacodynamics
N/A
4.2.1.3. Safety Pharmacology
N/A
4.2.1.4. Pharmacodynamic Drug Interactions
N/A
4.2.2. Pharmacokinetics
N/A
4.2.3. Toxicology
4.2.3.1. Single-Dose Toxicity
N/A
4.2.3.2. Repeat-Dose Toxicity
N/A
4.2.3.3. Genotoxicity
N/A
4.2.3.4. Carcinogenicity
N/A
4.2.3.5. Reproductive and Developmental Toxicity
N/A
4.2.3.6. Local Tolerance

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4.2.3.6.1. Muscle Irritation Test

Experiment Report

Experiment Name Muscle Irritation Test through Single
Intramuscular Injection of 13-JHYM in
Japanese Big-ear Rabbits
A201118-T073-01
Experiment Number
WestChina-Frontier PharmaTech Co. (National
Research Institution Name
Chengdu Center for Safety Evaluation of Drugs)
No. 28 Gaopeng Avenue, High & New Tech Dev.
Research Institution Address
Zone, Chengdu, Sichuan
Yunnan Walvax Biotechnology Co., Ltd.
Client Name
4th Floor, building A3, Phase 2 Science and
Client Address
Technology Park of Yunnan National University,
Kunming, Yunnan
Wei Guo
Experiment Director
Chaoxuan Han
Members of Experiment Group
Jianjun Zhao

Chenxi Li

Hao Wang

Ke Mu

Lihua Duan

Xiankui Gou
Tao Chen
Qin Zhang
June, 2011-October, 2011
Duration of Experiment
Archives Room of WestChina-Frontier
Original Data Depository
PharmaTech Co.

WestChina-Frontier PharmaTech Co. (WCFP)
National Chengdu Center for Safety Evaluation of Drugs (NCCSED)
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Confirmation of Experiment Report

Experiment Name: Muscle Irritation Test through Single
Intramuscular Injection of 13-JHYM in
Japanese Big-ear Rabbits
Experiment Number: A201118-T073-01

Experiment Director: Wei Guo

Signature: Date: 08/10/2011

QAU Director: Feng Cheng

Signature: Date: 08/10/2011

Institution Director: Li Wang

Signature: Date: 09/10/2011

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Compliance Statement

Muscle Irritation Test through Single

Experiment Name: Intramuscular Injection of 13-JHYM in Japanese

Big-ear Rabbits

Experiment Number: A201118-T073-01

This experiment was completed in accordance with Non-clinical Good Laboratory Practice formulated by China Food and Drug Administration (China Food and Drug Administration rule No. 2, effective from September 1st, 2003). The experiment procedure was carried out in accordance with the experiment protocol and the relevant standard operating procedure of the research institution. No deviation that affected the reliability of experiment results and conclusions was observed. This report faithfully reflects the experiment materials, experiment methods and experiment results.

Experiment Director: Wei Guo

Signature of Experiment Director: Date: 08/10/2011

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QAU Statement

Experiment Name: Muscle Irritation Test through Single

Intramuscular Injection of 13-JHYM in

Japanese Big-ear Rabbits

Experiment Number: A201118-T073-01

Experiment Director: Wei Guo

In order to ensure that the contents of the experiment protocol can be strictly implemented and the experiment can be operated in accordance with the SOP of this institution, QAU has checked the experiment protocol, the experiment implementation process and the experiment record, and has reviewed the original data and the experiment report. The operating procedures, experiment methods and results described in the report faithfully reflect the contents of the original data. The QAU inspection items, inspection dates and report dates are as follows:

Date of reporting to Date of reporting

Inspection items Inspection date the experiment to the institution

director director

Experiment 2011.06.10 2011.06.10 2011.06.10

protocol 2011.06.14 2011.06.14 2011.06.14

Test article

2011.06.21 2011.06.21 2011.06.21

preparation

Grouping 2011.06.21 2011.06.21 2011.06.21

Administration 2011.06.21 2011.06.21 2011.06.21

Observation 2011.06.21 2011.06.21 2011.06.21

Anatomy 2011.06.23 2011.06.23 2011.06.23

Original data 2011.09.26-2011.09.28 2011.09.28 2011.09.28

Experiment 2011.09.26-2011.09.28 2011.09.28 2011.09.28

report 2011.09.28 2011.09.30 2011.09.30

Experiment QA: Xinxin Li

Signature: Date: 08/10/2011

QAU director: Feng Cheng

Signature: Date: 08/10/2011

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Abstract

The 13-valent pneumococcal polysaccharide conjugate vaccine (13-JHYM) is made from pneumococcus capsule polysaccharide antigens of serotype 1, 3, 4, 5, 6A, 6B, 7F, 9V, 14, 18C, 19A, 19F and 23F covalently combined with tetanus toxoid carrier protein and adsorbed by aluminum phosphate adjuvant. The clinical use of this vaccine is for infants and children aged 2 months to 2 years old to prevent the invasive infection caused by the above-mentioned serotype pneumococcus. The proposed dosage is 0.5 mL per person, and the proposed route of use is intramuscular injection.

In accordance with the relevant requirements of new drug declaration redefined by Provisions for Drug Registration, the irritation test of 13-JHYM provided by Yunnan Walvax Biotechnology Co., Ltd. was carried out through single intramuscular injection in rabbits by WestChina-Frontier PharmaTech Co., with a purpose to observe the irritant reactions induced by 13-JHYM to the quadriceps femoris muscle of rabbits, and to provide a reference for evaluating the safety of clinical use.

A total of 11 groups were set in this experiment, including a negative control group, a low-dose and a high-dose group for aluminum phosphate, a low-dose and a high-dose group for 7-valent pneumococcal conjugate vaccine, as well as a low-dose and a high-dose group for 13-JHYM of 3 different batches respectively (batch No. 20110304, 20110305 and 20110406), with 4 rabbits included in each group (half male and half female). The rabbits in each low-dose group were injected with sample or comparison product at corresponding concentration in the volume of 0.5 mL per side through quadriceps femoris muscle for one time, the rabbits in each high-dose group were injected with sample or comparison product at corresponding concentration in the volume of 1.5 mL per side, and the rabbits in the negative control group were given 0.9% sodium chloride through single injection in the volume of 1.5 mL/ per side. The day of administration was determined as Day 1 in this experiment. Two rabbits were taken from each group and were euthanized (CO₂ inhalation) and anatomized respectively around 48 hours after administration and on the 16th day of the experiment. The situation of the quadriceps femoris muscle at the injection site was observed and graded, and histopathology was performed. Main results are as below:

General Conditions and Gross Anatomy Results Observed by Naked Eye:

No local reactions such as swelling and ecchymosis were observed at the injection site of rabbits in each group during both the administration and recovery period.

According to the anatomy results obtained around 48 hours after administration, mild hyperaemia was observed in the right quadriceps femoris muscle at the injection site of 1 rabbit and moderate hyperaemia was observed in the left quadriceps femoris muscle at the injection site of 1 rabbit in the high-dose group of 7-valent pneumococcal conjugate vaccine; moderate hyperaemia was observed in the right quadriceps femoris muscle at the injection site of 1 rabbit in the high-dose group of 13-JHYM (batch No. 20110304); moderate hyperaemia was observed in the right quadriceps femoris muscle at the injection site of 1 rabbit in the high-dose group of 13-JHYM (batch No. 20110406). No obvious abnormalities were observed at the injection site for the rest rabbits.

On the 16th day of the experiment, no obvious abnormalities were observed at the injection site for all rabbits.

Histopathology Results:

Around 48 hours after administration, changes including mild to severe degeneration and necrosis of muscle fibers/muscular stroma, hemorrhage as well as inflammatory cell infiltration were

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observed in the high-dose group of 7-valent pneumococcal conjugate vaccine, the high-dose group of 13-JHYM of 3 batches and the low-dose group of 13-JHYM (batch No.20110406), suggesting muscle irritation at the injection site. On the 16th day of the experiment, except that mild interstitial inflammatory cell infiltration could still be observed at the injection site in the high-dose and low-dose group of aluminum phosphate as well as the high-dose group of 13-JHYM (batch No. 20110305), no obvious abnormalities were observed at the injection site for the rest groups, suggesting that the irritation could be partially recovered after the 16-day observation.

In conclusion, under the conditions of this experiment, after the low and high doses (0.5 mL/side, 1.5 mL/side, clinical dosage) of aluminum phosphate, 7-valent pneumococcal conjugate vaccine and 13-JHYM (batch No. 2011030, 20110305 and 20110406) were given to the rabbits through single injection on the quadriceps femoris muscle, the low-dose and high-dose aluminum phosphate, high-dose 7-valent pneumococcal conjugate vaccine, high-dose 13-JHYM of 3 batches, and the low-dose 13-JHYM of batch No. 20110406 had obvious irritation to the quadriceps femoris muscle of rabbits. The above mentioned muscle irritation could be partially recovered after the 16-day observation.

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1. BASIC INFORMATION

1.1 EXPERIMENT NAME AND NUMBER

Experiment name: Muscle Irritation Test through Single Intramuscular Injection of 13-JHYM in Japanese Big-ear Rabbits

Experiment number: A201118-T073-01

1.2 OBJECTIVES

The irritation test of 13-valent pneumococcal polysaccharide conjugate vaccine (13-JHYM) was carried out through single intramuscular injection in rabbits, with a purpose to observe the irritant reactions induced by 13-JHYM to the quadriceps femoris muscle of rabbits, and to provide a reference for evaluating the safety of clinical use.

1.3 RESEARCH INSTITUTE

Name: WestChina-Frontier PharmaTech Co. (National Chengdu Center for Safety Evaluation of Drugs)

Address: No. 28 Gaopeng Avenue, High & New Tech Dev. Zone, Chengdu, Sichuan

Zip code: 610041

Contact person: Xiaobo Cen

Telephone: 028-85178767

Fax: 028-85173043

Email: xbcen@glpcd.com

1.4 CLIENT INSTITUTE

Name: Yunnan Walvax Biotechnology Co., Ltd.

Address: 4th Floor, building A3, 2nd Phase Science and Technology Park of Yunnan National University, Kunming, Yunnan

Zip code: 650106

Contact person: Kai Wu

Telephone: 0871-8351253-8851

Fax: 0871-8312123

1.5 MEMBERS OF EXPERIMENT GROUP

1.5.1 EXPERIMENT DIRECTOR

Name: Wei Guo

Telephone: 028-85154334-551

Fax: 028-85173043

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1.5.2 GROUP MEMBERS

Experiment operation: Ke Mu, Hao Wang, Jianjun Zhao, Chenxi Li, Chaoxuan Han;

Veterinarian: Lihua Duan;

Animal rearing and management: Qing Liao, Yongshao Cai;

Test articles preparation and management: Qin Zhang;

Histopathology examination: Tao Chen.

2. MATERIALS

2.1 EXPERIMENT ANIMALS

2.1.1 BASIC INFORMATION

Species: Japanese big-eared rabbit;

Grade: normal;

Amount and gender of animals purchased: 46 animals in total, half male and half female;

Amount and gender of animals used: 44 animals in total, half male and half female;

Body weight: 2.135-3.300 kg;

Source: Sichuan Laboratory Animal Special Committee farm (Production License No. SCXK (Sichuan) 2008-014; Certificate of Quality No. 0024847, issuing unit: Sichuan Department of Science and Technology);

Entry date: 10/06/2011;

Environmental adaptation period after entry: Before the experiment, the rabbits adapted to the environment for 11 days (10/06/2011-20/06/2011). Healthy (female must not be pregnant) rabbits were selected as test animals. Observation measurements during adaptation period: general conditions, body weight etc.

2.1.2 ANIMAL REARING CONDITIONS

2.1.2.1 REARING SITES

The ordinary animal house of WestChina-Frontier PharmaTech Co. (laboratory animal room license No. SYXK (Sichuan) 2009-123).

2.1.2.2 REARING CONDITIONS

Rearing cage: stainless steel rabbit cage (L*W*H: 550 mm*550 mm*380 mm);

Rearing density: 1 animal per cage.

2.1.2.3 REARING ENVIRONMENT

Standard: the National Standard of the People's Republic of China GB14925-2001;

Control system: the EBI400 automatic control fresh air air conditioning system produced by Honeywell;

Temperature: room temperature 22.98-25.61°C (with a diurnal temperature range of 0.00-1.11°C);

Humidity: relative humidity 44.8-62.9%;

Illumination: artificial lighting, 12 hours of alternating light and dark (lights on at 07:30, lights off at 19:30);

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Ventilation: 8-10 times per hour, 100% fresh air.

2.1.2.4 FEED

Type: complete pellet feed for rabbits;

Manufacturer: Beijing Ke'ao Xieli Feed Co., Ltd. (Feed Production Permit: SCXK (Beijing) 2009-0012, batch No.10122603 and 11044111);

Feeding method: free intake;

Nutrient analysis: provided by the feed provider, of which the routine nutritional measurements of crude protein, crude fat, crude fiber, crude ash, water, calcium and phosphorus were reported for each batch(referring to the National Standard of the People's Republic of China GB14924.5-2001); Confirmation of contaminant content in feed: provided by the feed provider, indicators for chemical contaminants including arsenic, lead, cadmium, mercury, hexachlorocyclohexane, DDT, aflatoxin B₁, total colony count, coliform group, mold and yeast count, pathogenic bacteria (Salmonella) were provided at least once a year (referring to the National Standard of the People's Republic of China GB14924.2-2001).

2.1.2.5 DRINKING

Type: drinking water for laboratory animals (reverse osmosis water);

Method of water supply: held in drinking water bottles, free intake;

Colony count detection: referring to the National Standard of the People's Republic of China GB5749-2006, sampling self-examination (once per week);

Detection of conventional indicators of water quality: referring to the National Standard of the People's Republic of China GB5749-2006, samples were sent to units with relevant qualification for testing at least 2 times per year.

2.1.3 ANIMAL GROUPING AND MARKING METHODS

2.1.3.1 ANIMAL GROUPING

Group design: a total of 11 groups, including a negative control group, a low-dose and a high-dose group for aluminum phosphate, a low-dose and a high-dose group for 7-valent pneumococcal conjugate vaccine, as well as a low-dose and a high-dose group for 13-JHYM of 3 different batches respectively (batch No. 20110304, 20110305 and 20110406);

Amount: 4 animals in each group;

Gender: half male and half female;

Grouping method: the rabbits were randomly divided into groups according to their body weight;

Disposal of surplus animals: handed over to the department of laboratory animal administration.

The specific group information is shown in Table 1.

Table 1 Grouping Table

Animal amount Animal number

Group

Female Male Female Male

1F01 1M01

Negative control group 2 2

1F02 1M02

2F01 2M01

Low-dose aluminum phosphate group 2 2

2F02 2M02

High-dose aluminum phosphate group 2 2 3F01 3M01

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3F02 3M02

4F01 4M01

Low-dose 7-valent pneumococcal conjugate vaccine group 2 2

4F02 4M02

5F01 5M01

High-dose 7-valent pneumococcal conjugate vaccine group 2 2

5F02 5M02

6F01 6M01

Low-dose 13-JHYM (batch 1) group 2 2

6F02 6M02

7F01 7M01

High-dose 13-JHYM (batch 1) group 2 2

7F02 7M02

8F01 8M01

Low-dose 13-JHYM (batch 2) group 2 2

8F02 8M02

9F01 9M01

High-dose 13-JHYM (batch 2) group 2 2

9F02 9M02

10F01 10M01

Low-dose 13-JHYM (batch 3) group 2 2

10F02 10M02

11F01 11M01

High-dose 13-JHYM (batch 3) group 2 2

11F02 11M02

Notes: The first digit of the animal number represents the group (e.g. 2 and 3 represent the low-dose and high-dose aluminum phosphate group respectively); the second letter stands for gender (M for male, F for female); the third and fourth digits represent cage number.

2.1.3.2 MARKING METHOD

Adaptation period: cage card was used as animal identification markers;

Post-grouping period: The cage card and ear number were used as animal identification markers;

Marking method of cage card:

Adaptation period: unified use of white cage card, with the test number, animal species, animal number, entry time and notes indicated;

Post-grouping: cage cards in different colors were used to distinguish the groups, and the cards were marked with test number, animal species, animal number, treatment factors, expected start and end time of the experiment and notes;

Colors of cage cards after grouping: white represented the negative control group as well as the high-dose and low-dose aluminum phosphate group; yellow represented the low-dose 7-valent pneumococcal conjugate vaccine group, and blue represented the high-dose 7-valent pneumococcal conjugate vaccine group; blue represented the low-dose 13-JHYM group, and red represented the high-dose 13-JHYM group.

2.2 TEST ARTICLES

2.2.1. TEST ARTICLE INFORMATION

Name: 13-valent pneumococcal polysaccharide conjugate vaccine (13-JHYM);

Source: Yunnan Walvax Biotechnology Co., Ltd.;

Character: suspension with opalescence;

Specification: 0.5 mL: 32.2µg per vial;

Batch No.: 20110304 (batch 1), 20110305 (batch 2), 20110406 (batch 3);

Preservation conditions and notes: stored under 2 - 8■;

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Expiration date: February, 2012 for batch No. 20110304 and 20110305; March, 2013 for 20110406;

Preservation site: the Samples Management Department of WestChina-Frontier PharmaTech Co.

2.2.2 TEST ARTICLE PREPARATION

Preparation methods: direct separation of the bulk;

Marking methods after preparation: use the original bottle label;

Preservation condition and period of validity after preparation: stored in ice box for use within 24 hours.

2.2.3 RETURN AND DISPOSAL OF REMAINING SAMPLES

The remaining samples after administration were sterilized by high temperature and collected in a waste liquid tank. The reserved samples were deposited in the sample room referring to the SOP for the preservation of test articles. The remaining samples at the end of the experiment were returned to the client.

2.3 COMPARISON PRODUCT

2.3.1 COMPARISON PRODUCT INFORMATION

Name: 0.9% sodium chloride injection (normal saline);

Source: Sichuan Kelun Pharmaceutical Co., Ltd.;

Character: colorless transparent liquid;

Specification: 100 mL per bottle;

Batch No.: C100928;

Preservation conditions and notes: sealed storage at room temperature;

Expiration date: August, 2012;

Preparation methods: direct separation of the bulk;

Marking methods after preparation: test articles were marked with white labels with experiment number, substance name and concentration used in the experiment, volume, storage condition, expiration date, responsible person and preparation date;

Preservation condition and period of validity after preparation: stored in ice box for use within 24 hours.

2.3.2 ADJUVANT CONTROL

Name: aluminum phosphate;

Source: Yunnan Walvax Biotechnology Co., Ltd.;

Character: milky suspension;

Concentration: 1 mg/mL;

Specification: 50 mL per bottle;

Batch No.: 8169402;
Preservation conditions and notes: stored under 2-8■;
Term of validity: long term;
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Preparation methods: direct separation of the bulk;
Marking methods after preparation: test articles were marked with yellow labels with experiment number, substance name and concentration used in the experiment, volume, storage condition, expiration date, responsible person and preparation date;
Preservation condition and period of validity after preparation: stored in ice box for use within 24 hours.

2.3.3 SIMILAR LICENSED COMPARISON PRODUCT

Name: 7-valent pneumococcal conjugate vaccine;
Source: Wyeth Pharmaceutical Division of Wyeth Holdings Corporation;
Character: white suspension;
Specification: 0.5 mL: 16μg per vial;
Batch No.: E61516E52444, E11660D57298;
Preservation conditions and notes: stored under 2-8■;
Term of validity: December, 2011 for batch No. E61516E52444; March, 2013 for E11660D57298;
Preparation methods: direct separation of the bulk;
Marking methods after preparation: use the original bottle label;
Preservation condition and period of validity after preparation: stored in ice box for use within 24 hours.

2.3.4 RETURN AND DISPOSAL OF REMAINING COMPARISON PRODUCTS

After administration, the remaining negative control was directly discharged into the sewer, the adjuvant control was collected in the waste liquid barrel, and the licensed similar control was collected in the waste liquid barrel after sterilization.

2.4 MAIN INSTRUMENTS

Electronic platform scale, injector, etc.

3. METHODS AND OBSERVATION MEASUREMENTS

3.1 METHODS

3.1.1 DOSAGE DESIGN

Referring to the Technical Guidelines for the Study of Irritation, Hypersensitivity and Hemolysis of Chemical Drugs (March 2005), the dosage design of the irritation test should mainly consider the concentration of the sample and the overall dosage. Normally the same concentration as the clinical preparations was used and one dose group was established. Based on the information provided by the client, the clinical proposed dose of 13-JHYM was 0.5 mL per person. A single serving contained 32.2 μg of pneumococcal polysaccharide and 0.5 mg of aluminum phosphate, which meant the total amount of polysaccharide was 64.4μg/mL and that of aluminum phosphate was 1 mg/mL.

In this experiment, the clinical proposed dosage of 0.5 mL per side muscle was determined as the low-dose group of 13-JHYM (3 batches) and the high-dose group was set up using 3 times of this dosage, i.e. 1.5ml per side muscle. At the same time, the dosage of low-dose and high-dose group of aluminum phosphate and 7-valent pneumococcal conjugate vaccine were designed in the same

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way. Animals in the negative control group were given normal saline in the volume of 1.5 mL per side muscle (Table 2).

Table 2 Table of Dosage Design

Volume	Animal amount	Administration Dosage Concentration (mL/kg)
Group	route (mL/side) (μg/mL) Left Right	Male Female
(mL) (mL)		
Negative control Intramuscular		
1.5 / 1.5	1.5 2 2	
group injection		
Low-dose		
Intramuscular		
aluminum 0.5 I mg/mL	0.5 0.5 2 2	
injection		
phosphate group		
High-dose		
Intramuscular		
aluminum 1.5 I mg/mL	1.5 1.5 2 2	
injection		
phosphate group		
Low-dose 7-valent		
pneumococcal Intramuscular		
0.5 32 0.5 0.5 2 2		
conjugate vaccine injection		
group		
High-dose 7-valent		
pneumococcal Intramuscular		
1.5 32 1.5 1.5 2 2		
conjugate vaccine injection		
group		
Low-dose 13-		
Intramuscular		
JHYM (batch 1) 0.5 64.4 0.5 0.5 2 2		
injection		
group		
High-dose 13-		
Intramuscular		
JHYM (batch 1) 1.5 64.4 1.5 1.5 2 2		

injection
group
Low-dose 13-
Intramuscular
JHYM (batch 2) 0.5 64.4 OS 05 2 2
injection
group
High-dose 13-
Intramuscular
JHYM (batch 2) 1.5 64.4 1.5 1.5 2 2
injection
group
Low-dose 13-
Intramuscular
JHYM (batch 3) 0.5 64.4 0.5 0.5 2 2
injection
group
High-dose 13-
Intramuscular
JHYM (batch 3) 1.5 64.4 1.5 1.5 2 2
injection
group

3.1.2 ADMINISTRATION

Administration route: intramuscular injection, injected into both hind limbs at the same time;
consistent with clinical administration route;
Frequency: once;
Administration method: based on the dosage described in Table 2, rabbits in each group were given sample or comparison product at corresponding concentration through single injection on the quadriceps femoris muscle and rabbits in the negative control group were given normal saline. The day of administration was determined as Day 1 of the experiment.

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3.2 OBSERVATION AND EXAMINATIONS

3.2.1 OBSERVATION OF GENERAL CONDITIONS

3.2.1.1 OBSERVATION TIME AND FREQUENCY

Rearing personnel: twice a day;
Experiment personnel: once a day;
Veterinarian: once a week;
Observation Period: 16 days after administration.

3.2.1.2 OBSERVATION MEASUREMENTS

Rearing personnel: death and other abnormalities of the rabbits;
Experiment personnel: including but not limited to the appearance, the general behavior activity, the mental state, the aspiration status, the excrement character, the genitals, the death and other toxic performance of the rabbits;
Veterinarian: including but not limited to the appearance, the general behavior activity, the mental state, the aspiration status, the excrement character, the genitals, the death and other toxic performance of the rabbits.

3.2.1.3 OBSERVATION OF IRRITATION AT INJECTION SITE

Observation time and frequency: the injection site was observed once 48 hours after administration;
Observation animal: all surviving animals;
Observation method: the animals were removed from the cage and the injection site was observed for irritant reactions such as redness, swelling, ecchymosis, induration, suppuration and ulceration.

3.2.2 HISTOPATHOLOGY EXAMINATION

Some animals (2 in each group, half male and half female) were euthanized (CO₂ inhalation) 48 hours after administration to observe the local muscle irritant reactions at the injection site, such as hyperemia, redness, degeneration, necrosis, etc. The quadriceps femoris was removed during anatomy and incised longitudinally. The irritant reactions of the injected muscle were observed and graded according to the standard in Table 3. The histopathological examination of local muscle tissue was performed.

The remaining animals were observed, graded and histopathologically examined on the 16th day of the experiment.

Table 3 Grading Standard for Muscle Irritant Reactions

Symptoms of reactions Grade

No obvious change 0

Mild hyperemia, with a range below 0.5*1.0 cm 1

Moderate hyperemia, with a range over 0.5*1.0 cm 2

Severe hyperemia, with degeneration of muscles 3

Necrosis with brown degeneration 4

Extensive necrosis 5

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4. RESULTS

4.1 GENERAL CONDITIONS, GROSS ANATOMY OBSERVATION AND GRADING

No local reactions such as swelling and ecchymosis were observed at the injection site of rabbits in each group during both the administration and recovery period.

According to the anatomy results obtained around 48 hours after administration, mild hyperaemia was observed in the right quadriceps femoris muscle at the injection site of 1 rabbit and moderate hyperaemia was observed in the left quadriceps femoris muscle at the injection site of 1 rabbit in the high-dose group of 7-valent pneumococcal conjugate vaccine; moderate hyperaemia was observed in the right quadriceps femoris muscle at the injection site of 1 rabbit in the high-dose group of 13-JHYM (batch No. 20110304); moderate hyperaemia was observed in the right quadriceps femoris muscle at the injection site of 1 rabbit in the high-dose group of 13-JHYM (batch No. 20110406). No obvious abnormalities were observed at the injection site for the rest rabbits.

On the 16th day of the experiment, no obvious abnormalities were observed at the injection site

for all rabbits.
Please see Table 4 for specific grading.
Table 4 Grading of Irritant Reactions on Injected Muscles
48 hours after administration On the 16th day of the experiment
Group Animal Muscle Animal Muscle
Grade Total Grade Total
No. position No. position
Left 0 Left 0
1M01 1M02
Negative Right 0 Right 0
0 0
control group Left 0 Left 0
1F01 1F02
Right 0 Right 0
Left 0 Left 0
Low-dose 2M01 2M02
aluminum Right 0 Right 0
0 0
phosphate Left 0 Left 0
group 2F01 2F02
Right 0 Right 0
Left 0 Left 0
High-dose 3M01 3M02
aluminum Right 0 Right 0
0 0
phosphate Left 0 Left 0
group 3F01 3F02
Right 0 Right 0

Low-dose 7- Left 0 Left 0
4M01 4M02
valent Right 0 Right 0
pneumococcal 0 0
conjugate Left 0 Left 0
4F01 4F02
vaccine group Right 0 Right 0
Left 2 Left 0
High-dose 7- 5M01 5M02
valent Right 0 3 Right 0 0
pneumococcal
5F01 Left 0 5F02 Left 0
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conjugate
Right 1 Right 0
vaccine group
Left 0 Left 0
6M01 6M02
Low-dose 13- Right 0 Right 0
JHYM (batch 1) 0 0
group Left 0 Left 0
6F01 6F02
Right 0 Right 0
Left 0 Left 0
7M01 7M02
High-dose 13- Right 2 Right 0
JHYM (batch 1) 2 0
group Left 0 Left 0
7F01 7F02
Right 0 Right 0
Left 0 Left 0
8M01 8M02
Low-dose 13- Right 0 Right 0
JHYM (batch 2) 0 0
group Left 0 Left 0
8F01 8F02
Right 0 Right 0
Left 0 Left 0
9M01 9M02
High-dose 13- Right 0 Right 0
JHYM (batch 2) 0 0
group Left 0 Left 0
9F01 9F02
Right 0 Right 0
Left 0 Left 0
10M01 10M02
Low-dose 13- Right 0 Right 0
JHYM (batch 3) 0 0
group Left 0 Left 0
10F01 10F02
Right 0 Right 0
Left 0 Left 0
11M01 11M02
High-dose 13- Right 0 Right 0
JHYM (batch 3) 2 0
group Left 0 Left 0
11F01 11F02
Right 2 Right 0

4.2 HISTOPATHOLOGY EXAMINATION RESULTS

4.2.1 HISTOPATHOLOGIC RESULTS OBTAINED 48 HOURS AFTER ADMINISTRATION

High-dose 7-valent pneumococcal conjugate vaccine: moderate degeneration and necrosis of muscle fibers/muscle stroma, hemorrhage as well as inflammatory cell infiltration with purulent fluid formation were observed in the quadriceps femoris at the right injection site for 1 animal (5F01); slight degeneration and necrosis of muscle fibers/muscle stroma as well as inflammatory cell infiltration were observed in the quadriceps femoris at the left injection site for 1 animal (5M01);

High-dose 13-JHYM (batch 1) group: slight degeneration and necrosis of muscle fibers/muscle stroma as well as inflammatory cell infiltration were observed in the quadriceps femoris at the right injection site for 1 animal (7M01);

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High-dose 13-JHYM (batch 2) group: mild degeneration and necrosis of muscle fibers/muscle stroma, hemorrhage as well as inflammatory cell infiltration with purulent fluid formation were observed in the quadriceps femoris at the right injection site for 1 animal (9F01);

Low-dose 13-JHYM (batch 3) group: slight degeneration and necrosis of muscle fibers/muscle stroma as well as inflammatory cell infiltration were observed in the quadriceps femoris at the right injection site for 1 animal (10M01);

High-dose 13-JHYM (batch 3) group: severe degeneration and necrosis of muscle fibers/muscle stroma, hemorrhage as well as inflammatory cell infiltration with purulent fluid formation were observed in the quadriceps femoris at the right injection site for 1 animal (11F01).

No histopathologic abnormalities were observed in the quadriceps femoris of the rest rabbits.

Please see Appendix I for specific results.

4.2.2 HISTOPATHOLOGIC RESULTS OBTAINED ON THE 16TH DAY OF THE EXPERIMENT

Low-dose aluminum phosphate group: slight interstitial inflammatory cell infiltration was observed in the quadriceps femoris at the bilateral injection sites for 1 animal (2M02);

High-dose aluminum phosphate group: mild interstitial inflammatory cell infiltration was observed in the quadriceps femoris at the right injection site for 1 animal (3F02);

High-dose 13-JHYM (batch 2) group: slight interstitial inflammatory cell infiltration was observed in the quadriceps femoris at the right injection site for 1 animal (9M02).

No histopathologic abnormalities were observed in the quadriceps femoris of the rest rabbits.

Please see Appendix I for specific results.

5. DISCUSSION

5.1 IRRITATION INDUCED BY ALUMINUM PHOSPHATE

Although no obvious abnormalities were observed in the quadriceps femoris at the injection site 48 hours after aluminum phosphate administration, on the 16th day of the experiment, slight to mild interstitial inflammatory cell infiltration was observed in the quadriceps femoris at the injection site of rabbits in aluminum phosphate group. The pathological changes were dose-related, and the incidence and degree of pathological changes were significantly higher than those in the control group. Common adverse reactions of aluminum adjuvant such as sterile abscesses, erythema, subcutaneous nodules and granulomatous inflammation were often reported in clinical use. Therefore, the above mentioned pathological changes were related the injection of aluminum phosphate.

5.2 IRRITATION INDUCED BY LICENSED COMPARISON PRODUCT AND 13-JHYM

Around 48 hours after administration, changes including mild to severe degeneration and necrosis of muscle fibers/muscular stroma, hemorrhage as well as inflammatory cell infiltration were observed in the high-dose group of 7-valent pneumococcal conjugate vaccine, the high-dose group of 13-JHYM of 3 batches and the low-dose group of 13-JHYM (batch No.20110406), suggesting muscle irritation at the injection site, suggesting obvious irritation to the muscle at the injection site. Except that the muscle irritation at the injection site was relatively strong in the high-dose 13-JHYM (batch No.20110406) group, no significant difference was observed in the severity of the pathological changes among the batches.

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On the 16th day of the experiment, except that mild interstitial inflammatory cell infiltration could still be observed at the injection site in the high-dose group of 13-JHYM (batch No. 20110305), no obvious abnormalities were observed at the injection site for the rest groups, suggesting that the irritation could be partially recovered after the 16-day observation.

6. CONCLUSIONS

In conclusion, under the conditions of this experiment, after the low and high doses (0.5 mL/side, 1.5 mL/side, clinical dosage) of aluminum phosphate, 7-valent pneumococcal conjugate vaccine and 13-JHYM (batch No. 2011030, 20110305 and 20110406) were given to the rabbits through single injection on the quadriceps femoris muscle, the low-dose and high-dose aluminum phosphate, high-dose 7-valent pneumococcal conjugate vaccine, high-dose 13-JHYM of 3 batches, and the low-dose 13-JHYM of batch No. 20110406 had obvious irritation to the quadriceps femoris muscle of rabbits. The above mentioned muscle irritation could be partially recovered after the 16-day observation.

7. ORIGINAL DATA DEPOSITORY

7.1 PAPER FILE AND ELECTRONIC DATA

Preservation site: the archives room of WestChina-Frontier PharmaTech Co.;

Preservation conditions: Standard;

Preservation period: The paper file were kept for at least 5 years after the drug got licensed, and the electronic data should be kept permanently;

Contact number: 028-85154334-511;

Contact person: Ling Chang.

7.2 PATHOLOGICAL SPECIMEN

Preservation site: the archives room of WestChina-Frontier PharmaTech Co.;

Preservation conditions: Standard;

Preservation period: maximum period of time without affecting the quality;

Contact number: 028-85154334-511;

Contact person: Ling Chang.

8. DEVIATIONS

No deviation that affected the reliability of the experiment results and conclusions was observed

during the experiment process.

9. REFERENCES

1. Drug Good Laboratory Practices, issued by National Medical Products Administration, No.2, September 2003.
2. Provisions for Drug Registration, issued by National Medical Products Administration, No.28, October 2007.
3. Technical Guidelines for the Study of Irritation, Hypersensitivity and Hemolysis of Chemical Drugs, issued by National Medical Products Administration, March 2005.

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Appendix I Muscle Irritation Test through Single Intramuscular Injection of 13-JHYM in Japanese Big-ear Rabbits (Pathological Report)

Muscle Irritation Test through Single Intramuscular Injection of 13-JHYM in Japanese Big-ear Rabbits

(48 hours after administration and on the 16th day of the experiment)

A201118-T073-01

Pathological Examination Report

Report issued (signature and date):

Histological observation: Tao Chen

Histological observation review: Yinzhu Zhang

Histological specimen preparation: Han Zhong, Xianfu Shu, Yanyan Lu, Yiming Hu

Report preparation: Tao Chen

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Muscle Irritation Test through Single Intramuscular Injection of 13-JHYM in Japanese Big-ear Rabbits

(48 hours after administration and on the 16th day of the experiment)

Pathological Examination Report

Experiment name and Muscle Irritation Test through Single Intramuscular Injection of 13-JHYM in experiment No. Japanese Big-ear Rabbits/ A201118-T073-01

Test article 13-JHYM

Test animals and amount 44 Japanese big-ear rabbits

Inspection department Toxicology Department

Experiment director Wei Guo

Animal anatomy time 23/06/2011, 06/07/2011

Report time 07/09/2011

1. EXPERIMENT DESIGN

The Japanese big-ear rabbits were given 13JHYM (64.4 µg/mL) through single intramuscular injection. Histopathological changes at the injection site of bilateral quadriceps femoris of 44 Japanese big-ear rabbits in the negative control group and each administration group were observed 48 hours after administration and on the 16th day of the experiment, respectively. Please see Table 1 for animal grouping and dosage design.

Table 1 Table of Animal Grouping and Dosage Design in Pathological Examination

Volume

Animal amount

Administration Concentration (mL/kg)

Group

route (µg/mL) Left Right

Male Female

(mL) (mL)

Negative control Intramuscular

/ 1.5 1.5 2 2

group injection

Low-dose

Intramuscular

aluminum I mg/mL 05 0.5 2 2

injection

phosphate group

High-dose

Intramuscular

aluminum I mg/mL 1.5 1.5 2 2

injection

phosphate group

Low-dose 7-valent

pneumococcal Intramuscular

32 0.5 0.5 2 2

conjugate vaccine injection

group

High-dose 7-valent

pneumococcal Intramuscular

32 1.5 1.5 2 2

conjugate vaccine injection

group

Low-dose 13-

Intramuscular

JHYM (batch 1) 64.4 0.5 0.5 2 2

High-dose 13-
Intramuscular
JHYM (batch 1) 64.4 1.5 1.5 2 2
injection
group
Low-dose 13-
Intramuscular
JHYM (batch 2) 64.4 OS 05 2 2
injection
group
High-dose 13-
Intramuscular
JHYM (batch 2) 64.4 1.5 1.5 2 2
injection
group
Low-dose 13-
Intramuscular
JHYM (batch 3) 64.4 0.5 0.5 2 2
injection
group
High-dose 13-
Intramuscular
JHYM (batch 3) 64.4 1.5 1.5 2 2
injection
group

2. METHOD

The animals were euthanized (CO2 inhalation) 48 hours after administration and on the 16th day of the experiment respectively. The quadriceps femoris was removed and incised longitudinally. The irritant reactions of the injected muscle were observed and graded according to the standard predefined by the experiment protocol. After that, the injected quadriceps femoris muscle was fixed in 10% neutral formalin and routine paraffin section (thickness of 3-5µm) was performed. The paraffin sections were stained with HE and observed under light microscope.

3. RESULTS

3.1 GROSS ANATOMY RESULTS OBSERVED BY NAKED EYE

According to the Muscle Stimulation Response Grading Standard, 48 hours after administration, in the high-dose group of 7-valent pneumococcal conjugate vaccine, the reaction grade of quadriceps femoris muscle of 1 animal (5F01) at the right injection site was grade 1, and that of 1 animal (5M01) at the left injection site was grade 2; the reaction grade of injected quadriceps femoris muscle on the right side of 1 animal (7M01) in the high-dose group of 13-JHYM (batch 1) was grade 2; the reaction grade of injected quadriceps femoris muscle on the right side of 1 animal (11F01) in the high-dose group of 13-JHYM (batch 3) was grade 2. The reaction grades of the rest animals were all 0. Please see Table 2 for specific results. On the 16th day of the experiment, the response grade of injected quadriceps femoris muscle was 0 in all animals.

Table 2 Table of Gross Anatomy Results Observed by Naked Eye

Number of observed	
Observation pathological Reaction	
Group	Animal No.
time changes/ 2 inspected grade	
sites	
High-dose 7- 5F01	1 1
valent	
pneumococcal	
5M01	1 2
48 hours after conjugate	
administration vaccine group	
High-dose 13-	
JHYM group 7M01	1 2
(batch 1)	

High-dose 13-
JHYM group 11F01 1 2
(batch 3)

3.2 HISTOPATHOLOGICAL OBSERVATION

48 hours after administration: slight interstitial inflammatory cell infiltration was observed in the quadriceps femoris at the right injection site for 1 animal (1F01) in the negative control group; moderate degeneration and necrosis of muscle fibers/muscle stroma, hemorrhage as well as inflammatory cell infiltration with purulent fluid formation were observed in the quadriceps femoris at the right injection site for 1 animal (5F01) in the high-dose group of 7-valent pneumococcal conjugate vaccine, and mild degeneration and necrosis of muscle fibers/muscle stroma as well as inflammatory cell infiltration were observed in the quadriceps femoris at the left injection site for 1 animal (5M01); slight degeneration and necrosis of muscle fibers/muscle stroma and inflammatory cell infiltration were observed in the quadriceps femoris at the right injection site for 1 animal (7M01) in the high-dose 13-JHYM group (batch 1); mild degeneration and necrosis of muscle fibers/muscle stroma, hemorrhage as well as inflammatory cell infiltration with purulent fluid formation were observed in the quadriceps femoris at the right injection site for 1 animal (9F01) in the high-dose 13-JHYM group (batch 2); slight degeneration and necrosis of muscle fibers/muscle stroma and inflammatory cell infiltration were observed in the quadriceps femoris at the right injection site for 1 animal (10M01) in the low-dose 13-JHYM group (batch 3); sever degeneration and necrosis of muscle fibers/muscle stroma, hemorrhage as well as inflammatory cell infiltration with purulent fluid formation were observed in the quadriceps

femoris at the right injection site for 1 animal (11F01) in the high-dose13-JHYM group (batch 3). On the 16th day of the experiment: slight interstitial inflammatory cell infiltration was observed in quadriceps femoris at both injection sites for 1 animal (2M01) in the low-dose aluminum phosphate group; mild interstitial inflammatory cell infiltration was observed in quadriceps femoris at the right injection site for 1 animal (3F02) in the high-dose aluminum phosphate group; slight interstitial inflammatory cell infiltration was observed in quadriceps femoris at the right injection site for 1 animal (9M02) in the high-dose 13-JHYM group (batch 2). No abnormalities were observed for the rest animals in the histopathological observation of quadriceps femoris under microscope. The incidence and classification of reactions observed 48 hours after administration are shown in the attached Table 1. The incidence and classification of reactions observed on the 16th day of the experiment are shown in the attached Table 2. The pictures of possible drug-related pathological changes in each stage are shown in the attached pictures.

4. DISCUSSION AND CONCLUSIONS

The irritation of 13-JHYM on quadriceps femoris was observed 48 hours after a single intramuscular injection of 13-JHYM was given to the Japanese big-ear rabbits and on the 16th day of the experiment. The main pathological changes observed 48 hours after administration were degeneration and necrosis of muscle fibers/muscle stroma, hemorrhage as well as inflammatory cell infiltration with purulent fluid formation; interstitial inflammatory cell infiltration. The incidence and degree of pathological changes in each group are shown in Table 3.

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Table 3 Table of Results of Histopathological Observation under Microscope 48 Hours after Administration through Intramuscular Injection

Degeneration and necrosis of muscle fibers/muscle stroma, hemorrhage as Interstitial inflammatory cell well as inflammatory cell infiltration infiltration with purulent fluid formation	
Group Number of Number of observed observed pathological Reaction grade pathological Reaction grade changes/ 4 changes/ 4 inspected sites inspected sites	
Negative control group	0 0 1 1
Low-dose aluminum	0 0 0 0
phosphate group	
High-dose aluminum	0 0 0 0
phosphate group	
Low-dose 7-valent pneumococcal conjugate	0 0 0 0
vaccine group	
High-dose 7-valent pneumococcal conjugate	2 1-3 0 0
vaccine group	
Low-dose 13-JHYM	0 0 0 0
(batch 1) group	
High-dose 13-JHYM	1 1 0 0
(batch 1) group	
Low-dose 13-JHYM	0 0 0 0
(batch 2) group	
High-dose 13-JHYM	1 2 0 0
(batch 2) group	
Low-dose 13-JHYM	1 1 0 0
(batch 3) group	
High-dose 13-JHYM	1 4 0 0
(batch 3) group	
Grading: 0 = normal, 1 = slight, 2 = mild, 3 = moderate, 4 = severe, 5 = serious	

The above results showed that, 48 hours after the administration:
1) Possible vaccine-related reactions: slight to moderate degeneration and necrosis of muscle fibers/muscle stroma, hemorrhage as well as inflammatory cell infiltration with purulent fluid formation were observed in 2 inspected sites (2/4) in the high-dose 7-valent pneumococcal conjugate vaccine group; slight to severe degeneration and necrosis of muscle fibers/muscle stroma, hemorrhage as well as inflammatory cell infiltration with purulent fluid formation were observed in 1 inspected sites (1/4) in the13-JHYM (batch 1) group, the high-dose 13-JHYM group (batch 2), as well as both the high-dose and low-dose 13-JHYM (batch 3) groups respectively. The above-mentioned pathological changes only appeared in the administration groups, and the incidence was high. The degree of pathological changes was severe and showed obvious dose correlation. Therefore, the above mentioned pathological changes were probably caused by vaccine irritation. Besides, slight interstitial inflammatory cell infiltration was observed in 1 inspected site (1/4) in the negative control group, which should be probably caused by mechanical irritation.

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2) Comparison of vaccine irritation: among the three groups of 13-JHYM with different batch numbers, the high-dose 13-JHYM (batch 3) group had the strongest irritation on the muscle at the

injection site, and it was slightly stronger than that of the high-dose 7-valent pneumococcal conjugate vaccine group; the irritation of the high-dose 13-JHYM (batch 2) group was slightly weaker, but still stronger than that of the high-dose 13-JHYM (batch 1) group; the low-dose 13-JHYM (batch 3) group and high-dose 13-JHYM (batch 1) group had the similar irritation on the injected muscle.

On the 16th day of the experiment, histopathological examination under microscope showed that the main pathological change was interstitial inflammatory cell infiltration. The incidence and degree of pathological changes in each group are shown in Table 4.

Table 4 Table of Results of Histopathological Observation under Microscope on the 16th Day of the Experiment

Interstitial inflammatory cell infiltration

Group

Number of observed pathological changes/ 4 Reaction

inspected sites grade

Negative control group 0 0

Low-dose aluminum phosphate group 2 1

High-dose aluminum phosphate group 1 2

Low-dose 7-valent pneumococcal conjugate

0 0

vaccine group

High-dose 7-valent pneumococcal conjugate

0 0

vaccine group

Low-dose 13-JHYM (batch 1) group 0 0

High-dose 13-JHYM (batch 1) group 0 0

Low-dose 13-JHYM (batch 2) group 0 0

High-dose 13-JHYM (batch 2) group 1 1

Low-dose 13-JHYM (batch 3) group 0 0

High-dose 13-JHYM (batch 3) group 0 0

Grading: 0 = normal, 1 = slight, 2 = mild, 3 = moderate, 4 = severe, 5 = serious

The above results showed that, 48 hours after the administration:

1) Possible vaccine-related reactions: slight to mild interstitial inflammatory cell infiltration was observed respectively in 2 inspected sites (2/4) in the low-dose aluminum phosphate group and 1 inspected site in the high-dose aluminum phosphate group; slight interstitial inflammatory cell infiltration was observed respectively in 1 inspected site (1/4) in the high-dose 13-JHYM (batch 2) group. The above-mentioned pathological changes only appeared in the administration groups, and the incidence was high. The vaccine-related possibility cannot be ruled out.

2) Comparison with results obtained 48 hours after administration: Only one case of slight interstitial inflammation was observed in the high-dose 13-JHYM (batch 2) group and the pathological changes were significantly reduced compared with the previous pathological changes; degeneration and necrosis of muscle fibers/muscle stroma, hemorrhage as well as inflammatory cell infiltration with purulent fluid formation observed 48 hours after administration in the high-

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dose 7-valent pneumococcal conjugate vaccine group, the high-dose 13-JHYM (batch 1) group as well as both the high-dose and low-dose 13-JHYM (batch 3) groups were no longer observed during the recovery period (on the 16th day of the experiment), which could be determined as fully recovered.

In conclusion, various degrees of degeneration and necrosis of muscle fibers/muscle stroma, hemorrhage as well as inflammatory cell infiltration with purulent fluid formation were observed in the 13-JHYM (batch 1) group, the high-dose 13-JHYM (batch 2) group, as well as both the high-dose and low-dose 13-JHYM (batch 3) groups 48 hours after Japanese big-ear rabbits were given 13-JHYM (64.4 µg/mL) through single intramuscular injection, suggesting obvious irritation to the muscle at the injection site. Among them, except for the high-dose 13-JHYM (batch 3) group, the degree of pathological changes in other groups was mainly slight to mild. On the 16th day of the experiment, pathological changes observed in the high-dose 13-JHYM (batch 2) group were significantly relieved, and the pathological changes in other administration groups were completely recovered. In addition, the possibility of the infiltration of interstitial inflammatory cells observed on the 16th day in the aluminum phosphate group was vaccine-related could not be ruled out.

5. REFERENCES

1. Animal Histopathology Atlas. Edited by Ning Su, Southeast University Press, April 2005.

2. Handbook of Toxicologic Pathology. WANDA M.HASCHEK, CCLIN G. R OUSSEAUX, MATTHEW A. WALLIG,525 B Street, Suit 1900, San Diego, California 92101-4495, USA. Second edition 2002.

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Attached Table 1 Table of Incidence and Grading of Pathological Changes (48 Hours after Administration)

Incidence (number of affected animals)

Group Low-dose 7-valent

High-dose 7-valent Low-dose 13- High-dose 13- High-dose 13- High-dose 13-

Low-dose aluminum High-dose aluminum pneumococcal Low-dose 13-JHYM Low-dose 13-JHYM

Negative control group pneumococcal conjugate JHYM (batch 1) JHYM (batch 1) JHYM (batch 2) JHYM (batch 3)

phosphate group phosphate group conjugate vaccine (batch 2) group (batch 3) group

vaccine group group group group group

group

Observation under

Gender Male Female Male Female Male Female Male Female Male Female Male Female Male Female Male Female

Male Female Male Female Male Female

microscope
Number of
inspected 2
sites

Degeneration and Grade: 0 2 2 2 2 2 2 2 2 1 1 2 2 2 1 2 2 1 2 2 1 1 2
necrosis of muscle
fibers/muscle stroma, 1 0 0 0 0 0 0 0 0 0 1 0 0 0 1 0 0 0 0 0 1 0 0
hemorrhage as well as
inflammatory cell
2 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0
infiltration with purulent
fluid formation
3 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0

4 0 1 0

Grade: 0 1 2
Interstitial inflammatory
cell infiltration
1 1 0

Grading: 0 = normal, 1 = slight, 2 = mild, 3 = moderate, 4 = severe, 5 = serious
Attached Table 2 Table of Incidence and Grading of Pathological Changes (on the 16th Day of the
Experiment)
Incidence (number of affected animals)

Group Low-dose 7-valent
High-dose 7-valent Low-dose 13- High-dose 13- High-dose 13- High-dose 13-
Low-dose aluminum High-dose aluminum pneumococcal Low-dose 13-JHYM Low-dose 13-JHYM
Negative control group pneumococcal conjugate JHYM (batch 1) JHYM (batch 1) JHYM (batch 2) JHYM (batch 3)
phosphate group phosphate group conjugate vaccine (batch 2) group (batch 3) group
vaccine group group group group group group
group
Observation under
Gender Male Female Male Female Male Female Male Female Male Female Male Female Male Female Male Female
Male Female Male Female Male Female
microscope
Number of
inspected 2
sites

Interstitial inflammatory Grade: 0 2 2 2 0 1 2 2 2 2 2 2 2 2 2 2 2 1 2 2 2 2
cell infiltration
1 0 0 0 2 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

2 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Grading: 0 = normal, 1 = slight, 2 = mild, 3 = moderate, 4 = severe, 5 = serious

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Attached Pictures-Possible Vaccine-Related Pathological Changes (48 Hours after Administration)

Negative control group High-dose 7-valent pneumococcal
conjugate vaccine group

Normal muscle at the injection site (1F01-L-26), HE*100 Degeneration and necrosis of muscle fibers/muscle
stroma,
hemorrhage as well as inflammatory cell infiltration with
purulent fluid formation at the injection site (5F01-R-26),
HE*100

High-dose 13-JHYM (batch 1) High-dose 13-JHYM (batch 2)
group group

Degeneration and necrosis of muscle fibers/muscle stroma Degeneration and necrosis of muscle
fibers/muscle stroma,
with inflammatory cell infiltration at the injection site (7M01- hemorrhage as well as inflammatory cell
infiltration with
R-26), HE*100 purulent fluid formation at the injection site (9F01-R-26),
HE*100

High-dose 13-JHYM (batch 3)
group

Degeneration and necrosis of muscle fibers/muscle stroma,
hemorrhage as well as inflammatory cell infiltration with
purulent fluid formation at the injection site (11F01-R-26),
HE*100

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Attached Pictures-Possible Vaccine-Related Pathological Changes (on the 16th Day of the
Experiment)

Negative control group High-dose aluminum phosphate group

Normal muscle at the injection site (1F02-R-26), HE*100 Interstitial inflammatory cell infiltration at the injection site (3F02-R-26), HE*100

High-dose 13-JHYM (batch 2) group

Interstitial inflammatory cell infiltration at the injection site (9M02-R-26), HE*100

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4.2.3.6.2. Passive Cutaneous Anaphylaxis Test
Experiment Report

Experiment Name Passive Cutaneous Anaphylaxis Test of 13-JHYM in Guinea Pigs
A201118-T075-01
Experiment Number
WestChina-Frontier PharmaTech Co. (National Research Institution Name
Chengdu Center for Safety Evaluation of Drugs)
No. 28 Gaopeng Avenue, High & New Tech Dev.
Research Institution Address
Zone, Chengdu, Sichuan
Yunnan Walvax Biotechnology Co., Ltd.
Client Name
4th Floor, building A3, Phase 2 Science and
Client Address
Technology Park of Yunnan National University,
Kunming, Yunnan
Wei Guo
Experiment Director
Mu Ke
Members of Experiment Group
Xuemei Huang

Hong Yuan

Xi Wang

Quanli Xu

Lihua Duan

Wulin Zheng

June, 2011-October, 2011
Duration of Experiment
Archives Room of WestChina-Frontier
Original Data Depository
PharmaTech Co.

WestChina-Frontier PharmaTech Co. (WCFP)
National Chengdu Center for Safety Evaluation of Drugs (NCCSED)

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Confirmation of Experiment Report

Experiment Name: Passive Cutaneous Anaphylaxis Test of
13-JHYM in Guinea Pigs
Experiment Number: A201118-T075-01

Experiment Director: Wei Guo

Signature: Date: 08/10/2011

QAU Director: Feng Cheng

Signature: Date: 08/10/2011

Institution Director: Li Wang

Signature: Date: 09/10/2011

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Compliance Statement

Passive Cutaneous Anaphylaxis Test of 13-
Experiment Name:
JHYM in Guinea Pigs
Experiment Number: A201118-T075-01

This experiment was completed in accordance with Non-clinical Good Laboratory Practice formulated by China Food and Drug Administration (China Food and Drug Administration rule No. 2, effective from September 1st, 2003). The experiment procedure was carried out in accordance with the experiment protocol and the relevant standard operating procedure of the research institution. No deviation that affected the reliability of experiment results and conclusions was observed. This report faithfully reflects the experiment materials, experiment methods and experiment results.

Experiment Director: Wei Guo

Signature of Experiment Director: Date: 08/10/201

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QAU Statement

Experiment Name: Passive Cutaneous Anaphylaxis Test of
13-JHYM in Guinea Pigs
Experiment Number: A201118-T075-01
Experiment Director: Wei Guo

In order to ensure that the contents of the experiment protocol can be strictly implemented and the experiment can be operated in accordance with the SOP of this institution, QAU has checked the experiment protocol, the experiment implementation process and the experiment record, and has reviewed the original data and the experiment report. The operating procedures, experiment methods and results described in the report faithfully reflect the contents of the original data. The QAU inspection items, inspection dates and report dates are as follows:

Date of reporting to Date of reporting

Inspection items Inspection date the experiment to the institution
director director

Experiment 2011.06.09-2011.06.10 2011.06.10 2011.06.10

protocol 2011.06.10 2011.06.10 2011.06.10

2011.06.27 2011.06.27 2011.06.27

Grouping

2011.07.20 2011.07.20 2011.07.20

Test article

2011.06.28 2011.06.28 2011.06.28

preparation

Body weight

2011.06.28 2011.06.28 2011.06.28

measurement

Antiserum 2011.06.28 2011.06.28 2011.06.28

preparation 2011.07.16 2011.07.16 2011.07.16

Observation 2011.06.28 2011.06.28 2011.06.28

Sensitization 2011.07.21 2011.07.21 2011.07.21

Provocation 2011.07.21 2011.07.21 2011.07.21

Locus coeruleus

2011.07.21 2011.07.21 2011.07.21

determination

Original data 2011.09.27-2011.09.28 2011.09.28 2011.09.28

Experiment 2011.09.27-2011.09.28 2011.09.28 2011.09.28

report 2011.09.29 2011.09.29 2011.09.29

Experiment QA: Yun Yue

Signature: Date: 08/10/2011

QAU director: Feng Cheng

Signature: Date: 08/10/2011

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Abstract

The 13-valent pneumococcal polysaccharide conjugate vaccine (13-JHYM) is made from pneumococcus capsule polysaccharide antigens of serotype 1, 3, 4, 5, 6A, 6B, 7F, 9V, 14,18C, 19A, 19F and 23F covalently combined with tetanus toxoid carrier protein and adsorbed by aluminum phosphate adjuvant. The clinical use of this vaccine is for infants and children aged 2 months to 2 years old to prevent the invasive infection caused by the above-mentioned serotype pneumococcus. The proposed dosage is 0.5 mL per person, and the proposed route of use is intramuscular injection.

In accordance with the relevant requirements of new drug declaration redefined by Provisions for Drug Registration, the passive systemic anaphylaxis test of 13-JHYM provided by Yunnan Walvax Biotechnology Co., Ltd. was carried out in guinea pigs by WestChina-Frontier PharmaTech Co., with a purpose to observe whether 13-JHYM could induce passive systemic anaphylaxis in guinea pigs, and to provide a reference for evaluating the safety of clinical use.

This experiment was divided into 2 stages: (1) antiserum preparation stage; A total of 20 guinea pigs (half male and half female) were randomly divided into 10 groups, including a negative control group, a positive control group, a low-dose and a high-dose group for 7-valent pneumococcal conjugate vaccine, as well as a low-dose and a high-dose group for 13-JHYM of 3 different batches respectively (batch No. 20110304, 20110305 and 20110406). The sensitization dosage for the low-dose groups was 0.05 mL bulk/kg, which was equal to the clinical proposed dosage, and the sensitization dosage for the high-dose groups was 0.5 mL bulk/kg, which was 10 times of the clinical proposed dosage. The guinea pigs in the 7-valent pneumococcal conjugate vaccine groups and 13-JHYM groups were intramuscularly injected with sample or control at the corresponding concentration based on the above mentioned dosage, guinea pigs in the positive control group were intraperitoneally injected with 10mg/mL ovalbumin solution in the volume of 0.5 mL per animal, and guinea pigs in the negative control group were intramuscularly injected with 0.9% sodium chloride injection (normal saline) in the volume of 0.5 mL/kg. The administration was carried out every other day, and on the 10th day after the 5th consecutive administration, the antiserum was prepared using the abdominal aorta blood, which was mixed with the serum of 2 animals in the same group. The serums of two animals in the same group were mixed together and kept in reserve. (2) Locus coeruleus test: A total of 60 guinea pigs (half male and half female) were randomly divided into 10 groups at the same settings as the antiserum preparation stage. Animals in each group were intracutaneously injected with the antiserum of each group at the back (3 pre-epilated points of 3*4 cm²) after the antiserum was diluted at the ratio of 1:2, 1:8 and 1:32. Each animal was given 0.1 mL/point of antiserum in each dilution degree. The animals were intravenously injected with provoking antigen at the same dosage as phase 1 along with equivalent 1% Evans Blue dye (1 mL in total) 4 hours after being intracutaneously injected with antiserum. After 30 minutes, the guinea pigs were anesthetized (around 60 mg/kg pentobarbital sodium, intraperitoneally injection) and euthanized by abdominal aorta bloodletting. The length and diameter of the locus coeruleus were measured and the locus coeruleus were photographed.

The results showed that, in the negative control group, the diameter of intradermal locus coeruleus of each dilution degree at the injection site was all 0 mm; in the positive control group, the mean diameter of locus coeruleus was 15.24 mm at the dilution degree of 1:32, 19.17 mm at the dilution degree of 1:8 and 22.62 mm at the dilution degree of 1:2; the diameter of intradermal locus coeruleus of each dilution degree at the injection site was all 0 mm in the high-dose and low-dose groups of both 7-valent pneumococcal conjugate vaccine and 13-JHYM of 3 batches.

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In conclusion, under the conditions of this experiment, the guinea pigs were intramuscularly injected with 13JHYM of 3 batches (batch No. 20110304, 20110305 and 20110406) at the dosage of 0.05 mL/kg and 0.5 mL/kg respectively, and the dilution degrees of the antiserum were 1:2, 1:8 and 1:32, which all obtained negative results in the passive systemic anaphylaxis in guinea pigs.

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1. BASIC INFORMATION

1.1 EXPERIMENT NAME AND NUMBER

Experiment name: Passive Cutaneous Anaphylaxis Test of 13-JHYM in Guinea Pigs

Experiment number: A201118-T075-01

1.2 OBJECTIVES

The passive systemic anaphylaxis test of 13-valent pneumococcal polysaccharide conjugate vaccine (13-JHYM) was carried out in guinea pigs, with a purpose to observe whether 13-JHYM could induce passive systemic anaphylaxis in guinea pigs, and to provide a reference for evaluating the safety of clinical use.

1.3 RESEARCH INSTITUTE

Name: WestChina-Frontier PharmaTech Co. (National Chengdu Center for Safety Evaluation of Drugs)

Address: No. 28 Gaopeng Avenue, High & New Tech Dev. Zone, Chengdu, Sichuan

Zip code: 610041

Contact person: Xiaobo Cen

Telephone: 028-85178767

Fax: 028-85173043

Email: xbcen@glpcd.com

1.4 CLIENT INSTITUTE

Name: Yunnan Walvax Biotechnology Co., Ltd.

Address: 4th Floor, building A3, 2nd Phase Science and Technology Park of Yunnan National University, Kunming, Yunnan

Zip code: 650106

Contact person: Kai Wu

Telephone: 0871-8351253-8851

Fax: 0871-8312123

1.5 MEMBERS OF EXPERIMENT GROUP

1.5.1 EXPERIMENT DIRECTOR

Name: Wei Guo

Telephone: 028-85154334-551

Fax: 028-85173043

Email: wguo@glpcd.com

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1.5.2 GROUP MEMBERS

Experiment operation: Ke Mu, Xuemei Huang, Hong Yuan, Xi Wang, Quanli Xu;

Veterinarian: Lihua Duan;

Animal rearing and management: Qing Liao, Yongshao Cai;
Test articles preparation and management: Wulin Zheng.

2. MATERIALS

2.1 EXPERIMENT ANIMALS

2.1.1 BASIC INFORMATION

Species: British guinea pigs;

Grade: normal;

Amount and gender of animals purchased: 90 animals in total, half male and half female;

Amount and gender of animals used: 20 animals in total, half male and half female (stage 1); 60

animals in total, half male and half female (stage 2);

Body weight: 264.3-335.6g (stage 1); 303.2-360.7g (stage 2);

Source: Sichuan Laboratory Animal Special Committee farm (Production License No. SCXK

(Sichuan) 2008-014; Certificate of Quality No. 0024849 and 0021850, issuing unit: Sichuan

Department of Science and Technology);

Entry date: 14/06/2011 (stage 1), 05/07/2011 (stage 2);

Environmental adaptation period after entry: Before the experiment, the guinea pigs included in

stage 1 adapted to the environment for 14 days (14/06/2011-27/06/2011) and the guinea pigs

included in stage 2 adapted to the environment for 16 days (05/07/2011-20/07/2011). Healthy

guinea pigs (female must not be pregnant) were selected as formal test animals. Observation

measurements during adaptation period: general conditions, body weight etc.

2.1.2 ANIMAL REARING CONDITIONS

2.1.2.1 REARING SITES

The ordinary animal house of WestChina-Frontier PharmaTech Co. (laboratory animal room

license No. SYXK (Sichuan) 2009-123).

2.1.2.2 REARING CONDITIONS

Rearing cage: guinea pig cage (L*W*H: 515 mm*400 mm*260 mm);

Rearing density: no more than 3 animal per cage.

2.1.2.3 REARING ENVIRONMENT

Standard: the National Standard of the People's Republic of China GB14925-2001;

Control system: the EBI400 automatic control fresh air air conditioning system produced by

Honeywell;

Temperature: room temperature 22.19-26.32°C (with a diurnal temperature range of 0.07-1.66°C);

Humidity: relative humidity 37.63-74.43%;

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Illumination: artificial lighting, 12 hours of alternating light and dark (lights on at 07:30, lights off at 19:30);

Ventilation: 8-10 times per hour, 100% fresh air.

2.1.2.4 FEED

Type: pellet feed for guinea pigs;

Manufacturer: Beijing Ke'ao Xieli Feed Co., Ltd. (Feed Production Permit: SCXK (Beijing) 2009-

0012);

Batch No.: 11030551;

Feeding method: free intake;

Nutrient analysis: provided by the feed provider, of which the routine nutritional measurements of

crude protein, crude fat, crude fiber, crude ash, water, calcium and phosphorus were reported for

each batch(referring to the National Standard of the People's Republic of China GB14924.5-2001);

Confirmation of contaminant content in feed: provided by the feed provider, indicators for

chemical contaminants including arsenic, lead, cadmium, mercury, hexachlorocyclohexane, DDT,

aflatoxin B₁, total colony count, coliform group, mold and yeast count, pathogenic bacteria

(Salmonella) were provided at least once a year (referring to the National Standard of the People's

Republic of China GB14924.2-2001).

2.1.2.5 DRINKING

Type: drinking water for laboratory animals (reverse osmosis water);

Method of water supply: held in drinking water bottles, free intake;

Colony count detection: referring to the National Standard of the People's Republic of China

GB5749-2006, sampling self-examination (once per week);

Detection of conventional indicators of water quality: referring to the National Standard of the

People's Republic of China GB5749-2006, samples were sent to units with relevant qualification

for testing at least 2 times per year.

2.1.3 ANIMAL GROUPING AND MARKING METHODS

2.1.3.1 ANIMAL GROUPING

Group design: a total of 10 groups, including a negative control group, a positive control group, a

low-dose and a high-dose group for 7-valent pneumococcal conjugate vaccine, as well as a low-

dose and a high-dose group for 13-JHYM of 3 different batches;

Amount: 2 animals in each group for the antiserum preparation stage and 6 animals in each group

for the passive sensitization stage;

Gender: half male and half female;

Grouping method: the guinea pigs were randomly divided into groups according to their body

weight;

Disposal of surplus animals: handed over to the department of laboratory animal administration;

The specific group information is shown in Table 1.

Table 1 Grouping Table

Group Stage Animal amount Animal number

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Female Male Female Male

Antiserum

1 1 1F0101 1M0101

preparation

Negative control group

Passive 1F0102, 1M0102,

3 3

sensitization 1F0103, 1F0104 1M0103, 1M0104

Antiserum

1 1 2F0101 2M0101

preparation
Positive control group
Passive 2F0102, 2M0102,
3 3
sensitization 2F0103, 2F0104 2M0103, 2M0104
Antiserum
Low-dose 7-valent 1 1 3F0101 3M0101
preparation
pneumococcal conjugate
vaccine group Passive 3F0102, 3M0102,
3 3
sensitization 3F0103, 3F0104 3M0103, 3M0104
Antiserum
High-dose 7-valent 1 1 4F0101 4M0101
preparation
pneumococcal conjugate
vaccine group Passive 4F0102, 4M0102,
3 3
sensitization 4F0103, 4F0104 4M0103, 4M0104
Antiserum
1 1 5F0101 5M0101
Low-dose 13-JHYM (batch 1) preparation
group
Passive 5F0102, 5M0102,
3 3
sensitization 5F0103, 5F0104 5M0103, 5M0104
Antiserum
1 1 6F0101 6M0101
High-dose 13-JHYM (batch 1) preparation
group
Passive 6F0102, 6M0102,
3 3
sensitization 6F0103, 6F0104 6M0103, 6M0104
Antiserum
1 1 7F0101 7M0101
Low-dose 13-JHYM (batch 2) preparation
group
Passive 7F0102, 7M0102,
3 3
sensitization 7F0103, 7F0104 7M0103, 7M0104
Antiserum
1 1 8F0101 8M0101
High-dose 13-JHYM (batch 2) preparation
group
Passive 8F0102, 8M0102,
3 3
sensitization 8F0103, 8F0104 8M0103, 8M0104
Antiserum
1 1 9F0101 9M0101
Low-dose 13-JHYM (batch 3) preparation
group
Passive 9F0102, 9M0102,
3 3
sensitization 9F0103, 9F0104 9M0103, 9M0104
Antiserum
1 1 10F0101 10M0101
preparation
High-dose 13-JHYM (batch 3)
10F0102, 10M0102,
group Passive
3 3 10F0103, 10M0103,
sensitization
10F0104 10M0104
Notes: The first digit of the animal number represents the group (e.g. 1 and 2 represent the negative control group and positive control group respectively); the second letter stands for gender (M for male, F for female); the third and fourth digits represent cage number, and the fifth and sixth digits represent animal serial number.

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2.1.3.2 MARKING METHOD

Adaptation period: cage card and red mark were used as animal identification markers;
Post-grouping period: The cage card and individual ear number were used as animal identification markers;

Marking method of cage card:

Adaptation period: unified use of white cage card, with the test number, animal species, animal number, entry time and notes indicated;

Post-grouping: cage cards in different colors were used to distinguish the groups, and the cards were marked with test number, animal species, animal number, treatment factors, expected start and end time of the experiment and notes;

Colors of cage cards after grouping: white represented the negative control group; yellow represented the positive control group; blue represented the low-dose 7-valent pneumococcal conjugate vaccine group, and orange represented the high-dose 7-valent pneumococcal conjugate vaccine group; green represented the low-dose 13-JHYM group, and red represented the high-dose 13-JHYM group.

2.2 TEST ARTICLES

2.2.1 TEST ARTICLE INFORMATION

Name: 13-valent pneumococcal polysaccharide conjugate vaccine (13-JHYM);

Source: Yunnan Walvax Biotechnology Co., Ltd.;
Character: suspension with opalescence;
Specification: 0.5 mL: 32.2µg per vial;
Batch No.: 20110304 (batch 1), 20110305 (batch 2), 20110406 (batch 3);
Preservation conditions and notes: kept under 2-8■;
Expiration date: February, 2013 for batch No. 20110304 and 20110305; March, 2013 for 20110406;
Preservation site: the Samples Management Department of WestChina-Frontier PharmaTech Co.

2.2.2 TEST ARTICLE PREPARATION

Menstruum: 0.9% sodium chloride injection;
Preparation methods: samples for the low-dose groups were prepared with 0.9% sodium chloride injection under aseptic condition, and samples for the high-dose groups used direct separation of the bulk;
Marking methods after preparation: the high-dose groups were marked with red labels and the low-dose group were marked with green labels, with experiment number, substance name and concentration used in the experiment, volume, storage condition, expiration date, responsible person and preparation date;
Preservation condition and period of validity after preparation: stored in ice box for use within 24 hours.

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2.2.3 RETURN AND DISPOSAL OF REMAINING SAMPLES

The remaining samples after administration were collected in the waste liquid bucket after sterilized with high temperature and high pressure. The reserved samples were deposited in the sample room referring to the SOP for the preservation of test articles. The remaining samples at the end of the experiment were returned to the client.

2.3 COMPARISON PRODUCT

2.3.1 NEGATIVE COMPARISON PRODUCT INFORMATION

Name: 0.9% sodium chloride injection;
Source: Sichuan Kelun Pharmaceutical Co., Ltd.;
Character: colorless transparent liquid;
Specification: 100 mL per bottle;
Batch No.: C100928;
Preservation conditions and notes: sealed storage at room temperature;
Expiration date: August, 2012;
Preparation methods: direct separation of the bulk on the day of use;
Marking methods after preparation: test articles were marked with white labels with experiment number, substance name used in the experiment, volume, storage condition, expiration date, responsible person and preparation date;
Preservation condition and period of validity after preparation: stored in the ice box for use within 24 hours.

2.3.2 POSITIVE COMPARISON PRODUCT INFORMATION

Name: ovalbumin;
Source: Shanghai Yuanju Bio-tech Co., Ltd.;
Character: white crystal;
Specification: 1g per bottle;
Batch No.: 091221;
Preservation conditions and notes: kept under 2-8■;
Period of validity: 2 years;
Preparation methods: prepared into 10mg/mL solution with 0.9% sodium chloride injection when use;
Preservation condition and period of validity after preparation: stored in the ice box for use within 24 hours.

2.3.3 SIMILAR LICENSED COMPARISON PRODUCT

Name: 7-valent pneumococcal conjugate vaccine;
Source: Wyeth Pharmaceutical Division of Wyeth Holdings Corporation;
Character: white suspension;

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Specification: 0.5 mL: 16µg per vial;
Batch No.: E11660D57298, E19166D94429 and E61516E52444;
Preservation conditions and notes: stored under 2-8■;
Term of validity: December, 2011 for batch No.E11660D57298; April, 2012 for batch No. No. E19166D94429; March, 2013 for E61516E52444;
Preparation methods: samples for the low-dose groups were prepared with 0.9% sodium chloride injection under aseptic condition, and samples for the high-dose groups used direct separation of the bulk;
Marking methods after preparation: the high-dose groups were marked with red labels and the low-dose group were marked with green labels, with experiment number, substance name and concentration used in the experiment, volume, storage condition, expiration date, responsible person and preparation date;
Preservation condition and period of validity after preparation: stored in ice box for use within 24 hours.

2.3.4 RETURN AND DISPOSAL OF REMAINING COMPARISON PRODUCTS

After each administration, the remaining negative control was directly discharged into the sewer, the positive control was collected in the waste liquid barrel, and the licensed control was collected in the waste liquid barrel after sterilization.

2.4 MAIN REAGENT

Name: Evans Blue;
Source: Sigma;
Character: brown metallic luster powder;
Specification: 10g per bottle;
Batch No.: MKBB1167;
Preservation conditions and notes: sealed storage at room temperature;
Expiration date: March, 2013;
Preparation methods: dissolved with 0.9% sodium chloride injection to the required concentration

when use;
Marking methods: marked with blue label, with experiment number, substance name and concentration used in the experiment, volume, storage condition, expiration date, responsible person and preparation date;
Preservation condition and period of validity after preparation: sealed storage at room temperature, validate within the same day of preparation.
After use, the remaining Evans Blue solution was collected into the waste liquid bucket.

Name: pentobarbital sodium;
Source: Sigma;
Character: white powder;

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Specification: 25g per bottle;
Batch No.: 127K1005;
Preservation conditions and notes: sealed storage at room temperature;
Expiration date: March, 2013 (re-inspection date);
Preparation methods: dissolved with 0.9% sodium chloride injection to the required concentration when use;
Marking methods after preparation: marked with white label;
Preservation condition and period of validity after preparation: sealed storage under 2-25 °C, validate within 14 days.
After use, the remaining pentobarbital sodium was returned to the Samples Management Department and handled according to the corresponding SOP.

2.5 MAIN INSTRUMENTS

Injector, test tube, pipette gun, hair shears, vernier scale, digital camera, electronic balance, refrigerator etc.

3. METHODS AND OBSERVATION MEASUREMENTS

3.1 METHODS

3.1.1 ANTISERUM PREPARATION STAGE

Referring to the Technical Guidelines for the Study of Irritation, Hypersensitivity and Hemolysis of Chemical Drugs (March 2005), in a passive systemic anaphylaxis test, animals in the negative control group should be given menstuum in the same volume, animals in the positive control group should be given bovine serum albumin or ovalbumin or known sensitizing positive substance in the volume of 1-5 mg per animal, animals in the low-dose groups should be given samples at the maximum clinical dosage, and animals in the high-dose groups should be given samples at a dosage which was several times of the dosage used in the low-dose group. Based on the information provided by the client, the clinical proposed dosage of 13-JHYM was 0.5 mL per person (a single serving contained 32.2 µg pneumococcal polysaccharides of various types. The dosage for infants whose weight was determined as 10 kg was converted into a clinical proposed dosage of 0.05 mL/kg, i.e. 3.22 µg/kg).

On the basis of above mentioned premises, 0.05 mL bulk/kg was determined as low dosage for 13-JHYM of 3 different batches in the antiserum preparation stage, and 0.5 mL bulk/kg was determined as high dosage, which was equal to and 10 times of the clinical proposed dosage for human beings respectively. The low-dose and high-dose 7-valent pneumococcal conjugate vaccine group were set at the same dosage as 13-JHYM groups. The dosage of 5 mg per animal was set for the positive control group (intraperitoneal injection, 10 mg/mL ovalbumin), and the normal saline was given to the animals in the negative control group in the volume of 0.5 mL/kg through intramuscular injection (Table 2).

Table 2 Table of Sensitizing Dosage Design

Dosage	Animal
Administration	Volume
Group (mL)	Concentration amount
route (mL/kg)	
bulk/kg)	Male Female
Negative control	Intramuscular
// 0.5	1 1
group injection	

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Positive control	Intraperitoneal	5 mg per 0.5 mL per
10 mg/mL	1 1	
group injection	animal animal	
Low-dose 7-valent		
pneumococcal	Intramuscular	
0.05 bulk/10	0.5 1 1	
conjugate vaccine	injection	
group		
High-dose 7-valent		
pneumococcal	Intramuscular	
0.5 bulk	0.5 1 1	
conjugate vaccine	injection	
group		
Low-dose 13-		
Intramuscular		
JHYM (batch 1)	0.05 bulk/10	0.5 1 1
injection		
group		
High-dose 13-		
Intramuscular		
JHYM (batch 1)	05 bulk	0.5 1 1
injection		
group		
Low-dose 13-		
Intramuscular		
JHYM (batch 2)	0.05 bulk/10	0.5 1 1

injection
group
High-dose 13-
Intramuscular
JHYM (batch 2) 0.5 bulk 0.5 1 1
injection
group
Low-dose 13-
Intramuscular
JHYM (batch 3) 0.05 bulk/10 0.5 1 1
injection
group
High-dose 13-
Intramuscular
JHYM (batch 3) 0.5 bulk 0.5 1 1
injection
group

The proposed clinical administration route for the samples used in this experiment was intramuscular injection, therefore intramuscular injection on the hind limbs was adopted in this experiment. The sensitization was carried out on the 1st, 3rd, 5th, 7th and 9th day of the experiment respectively, 5 times in total continuously. On the 10th day after the last sensitization, the animals were anesthetized using pentobarbital sodium solution (around 60 mg/kg, intraperitoneal injection) and blood samples were collected from abdominal aortic. The blood sample was centrifuged at room temperature for 10 minutes at 2000 rpm and the serum was separated. The serum of 2 animals in the same dosage group was stored together at -20 °C and kept in reserve for 2 weeks.

3.1.2 PASSIVE SENSITIZATION STAGE

Antiserum preparation: the antiserum prepared from each group was diluted to 3 proportions of 1:2, 1:8 and 1:32 respectively with 0.9% sodium chloride injection.

Depilation: after the guinea pigs in each group were anesthetized (pentobarbital sodium, around 30mg/kg, intraperitoneal injection), the back area from neck to buttocks of the guinea pigs was depilated.

Passive sensitization: animals in each group were intradermally injected at 3 injection points in the thorax-back, low back, abdomen back respectively according to the dosage design in Table 3.

Antiserum diluted in the proportion of 1:2, 1:8 and 1:32 was injected in the order from chest to abdomen (0.1 mL/point). The distance between two points was at least 1-2 cm.

Method of intradermal injection: after sterilized with 75% alcohol, the tip of the needle pressed against the skin and stabbed flatly into the skin with the pinhole up forward. The solution was injected with a great resistance during the injection process. Translucent lump bulge was observed at the injection site after injection.

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Table 3 Table of Antiserum Dosage Design in Passive Sensitization

Animal	Administration Volume (mL/point)	Group amount	route
1:2 1:8 1:32 Male Female			
Negative control	Intradermal	0.1 0.1 0.1 3 3	group injection
Positive control	Intradermal	0.1 0.1 0.1 3 3	group injection
Low-dose 7-valent			
pneumococcal	Intradermal	0.1 0.1 0.1 3 3	conjugate vaccine injection
High-dose 7-valent			
pneumococcal	Intradermal	0.1 0.1 0.1 3 3	conjugate vaccine injection
Low-dose 13-			
Intradermal			
JHYM (batch 1)	0.1 0.1 0.1 3 3		injection
High-dose 13-			
Intradermal			
JHYM (batch 1)	0.1 0.1 0.1 3 3		injection
Low-dose 13-			
Intradermal			
JHYM (batch 2)	0.1 0.1 0.1 3 3		injection
High-dose 13-			
Intradermal			
JHYM (batch 2)	0.1 0.1 0.1 3 3		injection
Low-dose 13-			
Intradermal			
JHYM (batch 3)	0.1 0.1 0.1 3 3		injection
High-dose 13-			
Intradermal			

JHYM (batch 3) 0.1 0.1 0.1 3 3

injection

group

Guinea pigs in each group were injected with provoking antigen at the same dosage as phase 1 along with equivalent 1% Evans Blue dye (1 mL in total) 4 hours after sensitization through dorsum pedis or any other appropriate part (if the provoking antigen was insufficient, normal saline was added).

Table 4 Table of Provocation Administration Schedule

Animal

Processing Administration Volume

Group amount

factor route (mL)

Male Female

Normal saline

Negative control Intravenous

+ 1% Evans 1 3 3

group injection

Blue

Ovalbumin +

Positive control Intravenous

1% Evans 1 3 3

group injection

Blue

Low-dose 7- 7-valent

valent pneumococcal

Intravenous

pneumococcal conjugate 1 3 3

injection

conjugate vaccine vaccine+ 1%

group Evans Blue

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High-dose 7- 7-valent

valent pneumococcal

Intravenous

pneumococcal conjugate 1 3 3

injection

conjugate vaccine vaccine+ 1%

group Evans Blue

Low-dose 13- 13-JHYM +

Intravenous

JHYM (batch 1) 1% Evans 1 3 3

injection

group Blue

High-dose 13- 13-JHYM +

Intravenous

JHYM (batch 1) 1% Evans 1 3 3

injection

group Blue

Low-dose 13- 13-JHYM +

Intravenous

JHYM (batch 2) 1% Evans 1 3 3

injection

group Blue

High-dose 13- 13-JHYM +

Intravenous

JHYM (batch 2) 1% Evans 1 3 3

injection

group Blue

Low-dose 13- 13-JHYM +

Intravenous

JHYM (batch 3) 1% Evans 1 3 3

injection

group Blue

High-dose 13- 13-JHYM +

Intravenous

JHYM (batch 3) 1% Evans 1 3 3

injection

group Blue

3.2 OBSERVATION MEASUREMENTS

The guinea pigs in each group were anesthetized (around 60 mg/kg pentobarbital sodium, intraperitoneally injection) and euthanized by abdominal aorta bloodletting 30 minutes after provoking injection. The skin on the back was cut off, the length and diameter of the locus coeruleus were measured as well as the locus coeruleus were photographed.

3.3 RESULTS DETERMINATION

The results were determined using the method of fixed dilution ratio. The result at a certain dilution degree could be determined as positive if: the average diameter of the locus coeruleus in the positive control group was more than 5mm at that dilution degree, no locus coeruleus was observed in the negative control group at that dilution degree, and the diameter of locus coeruleus was observed to be more than 5mm in the samples' groups at that dilution degree.

4. RESULTS

4.1 PHOTOGRAPHS OF LOCUS COERULEUS

No locus coeruleus was observed at the injection site in the negative control group, the high-dose and low-dose groups of both 7-valent pneumococcal conjugate vaccine and 13-JHYM of 3 batches (batch No.20110304, 20110305 and 20110406) at all dilution degrees. Regular locus coeruleus was observed at the injection site in the positive control group at each dilution degree. The photographs of locus coeruleus were as below (the locus coeruleus in the photographs are in the order of dilution degrees, i.e. 1:2, 1:8, 1:32 from top to bottom):

4.2 DIAMETERS OF LOCUS COERULEUS

In the negative control group, as well as the high-dose and low-dose groups of both 7-valent pneumococcal conjugate vaccine and 13-JHYM of 3 batches (batch No.20110304, 20110305 and 20110406), the diameters of locus coeruleus in each group at all dilution degrees were all 0 mm. The diameters of locus coeruleus in the positive control group were all larger than 5mm at the dilution degree of 1:2, 1:8 and 1:32 (Table 5).

Table 5 Table of Mean Diameter of Locus Coeruleus in Each Group (mm, $\bar{X} \pm SD$)

Mean diameter of locus coeruleus at each

Amount of

Group dilution degree (mm)

animals

1:2 1:8 1:32

Negative control group 6 0 0 0

Positive control group 6 22.62 ± 2.23 19.17 ± 3.39 15.24 ± 2.10

Low-dose 7-valent

pneumococcal conjugate 6 0 0 0

vaccine group

High-dose 7-valent

pneumococcal conjugate 6 0 0 0

vaccine group

Low-dose 13-JHYM (batch 1)

6 0 0 0

group

High-dose 13-JHYM (batch 1)

6 0 0 0

group

Low-dose 13-JHYM (batch 2)

6 0 0 0

group

High-dose 13-JHYM (batch 2)

6 0 0 0

group

Low-dose 13-JHYM (batch 3)

6 0 0 0

group

High-dose 13-JHYM (batch 3)

6 0 0 0

group

5. CONCLUSIONS

In conclusion, under the conditions of this experiment, the guinea pigs were intramuscularly injected with 13JHYM of 3 batches (batch No. 20110304, 20110305 and 20110406) at the dosage of 0.05 mL/kg and 0.5 mL/kg respectively, and the dilution degrees of the antiserum were 1:2, 1:8 and 1:32, which all obtained negative results in the passive systemic anaphylaxis in guinea pigs.

6. ORIGINAL DATA DEPOSITORY

Preservation site: the archives room of WestChina-Frontier PharmaTech Co.;

Preservation conditions: Standard;

Preservation period: The paper file were kept for at least 5 years after the drug got licensed, and the electronic data should be kept permanently;

Contact number: 028-85154334-511;

Contact person: Ling Chang.

7. DEVIATION

Type: experiment protocol deviation;

Description: Module 4.2.3 in the original protocol required the relative humidity to be 40-70%, while the temperature range in this experiment was 22.19-26.32°C (with a diurnal temperature range of 0.07-1.66°C) and the range of the relative humidity was 37.63-74.43%;

Processing: None;

Evaluation: the range of the relative humidity which was not within the required range from the protocol was only observed during a part of time in 3 days, which should not have impacts on the results evaluation in this experiment.

Apart from the above-mentioned deviation, no deviation which had impact on the reliability of the experiment results and conclusions as well as other abnormalities were observed in this experiment.

8. REFERENCES

1. Drug Good Laboratory Practices, issued by National Medical Products Administration, No.2, September 2003.

2. Provisions for Drug Registration, issued by National Medical Products Administration, No.28, October 2007.

3. General Principles for Technical Review of Preclinical Safety Assessment of Prophylactic Biological Products, issued by National Medical Products Administration, December 2005.

4. Technical Guidelines for the Study of Irritability, Anaphylaxis and Hemolysis of Chemical Drugs, issued by National Medical Products Administration, March 2005.

4.2.3.6.3. In Vitro Hemolysis Test

Experiment Report

Study Name Study on In Vitro Hemolysis of 13-JHYM

Study No. A201118-T071 -01

Name of study facility WestChina-Frontier PharmaTech Co., Ltd. (National Chengdu Center for Safety Evaluation of Drugs)

Address of study facility No. 28 Gaopeng Avenue, High-tech Development Zone, Chengdu, Sichuan

Client Name Yunnan Walvax Biotechnology Co., Ltd.

Address of client 4F, Building A3, Phase II, Yunnan National University Science Park, High-tech Zone, Kunming

Study Director Guo Wei

Members of study team: Huang Xuemei, Xu Quanli, Wang Xi, Yuan Hong, Yi Bing, Mu Ke

Date of study initiation and August 2011 - October 2011 completion:

Location for storing initial Archives Room, WestChina-Frontier PharmaTech Co., Ltd. data

WestChina-Frontier PharmaTech Co. (WCFP)
National Chengdu Center for Safety Evaluation of Drugs (NCCSED)

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EXPERIMENT REPORT CONFIRMATION

Study Name: Study on In Vitro Hemolysis of 13-JHYM
Study No.: A201118-T071-01

Study Director: Guo Wei

Signature: Date: October 8, 2011

QAU director: Cheng Feng

Signature: Date: October 8, 2011

Facility Director: Wang Li

Signature: Date: October 9, 2011

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COMPLIANCE STATEMENT

Study Name: Study on In Vitro Hemolysis of 13-JHYM
Study No.: A201118-T071-01

This study was completed in accordance with the Good Laboratory Practice issued by the State Food and Drug Administration of the People's Republic of China (Order No.2 of the State Food and Drug Administration, which came into effect on September 1, 2003). The study process complied with the study protocol and relevant standard operating procedures of the study facility, and there were no deviations or other abnormalities affecting the reliability of study results and conclusions. This report truthfully reflected the study materials, study methods and study results.

Study Director: Guo Wei

Signature: Date: October 8, 2011

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QAU STATEMENT

Study Name: Study on In Vitro Hemolysis of 13-JHYM
Study No.: A201118-T071-01
Study Director: Guo Wei

In order to ensure that the study could be conducted strictly in accordance with the study protocol and SOP of the facility, QAU inspected the study protocol, study implementation process and study records and reviewed the original data and study report. The operating procedures, study methods and study results described in the report truthfully reflected the contents of original data of the study.

QAU test items, test dates and report dates are as follows:

Date of Reporting to the Study Date of Reporting to the

Test Item Test Date

Director Facility Director

2011.08.03 2011.08.03 2011.08.03

Study Protocol

2011.08.04 2011.08.04 2011.08.04

Blood collection 2011.08.18 2011.08.18 2011.08.18

Preparation of RBC

2011.08.18 2011.08.18 2011.08.18

suspension

Sample injection 2011.08.18 2011.08.18 2011.08.18

Observation 2011.08.18 2011.08.18 2011.08.18

Original data 2011.09.29 2011.09.29 2011.09.29

Study Report 2011.09.29 2011.09.29 2011.09.29

Study QA: Yue Yun

Signature: Date: October 8, 2011

QAU director: Cheng Feng

Signature: Date: October 8, 2011

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ABSTRACT

The 13-valent pneumococcal polysaccharide conjugate vaccine (code: 13-JHYM) is prepared by adsorption by aluminum phosphate adjuvant after covalent binding between types 1, 3, 4, 5, 6A, 6B, 7F, 9V, 14, 18C, 19A, 19F and 23F pneumococcal capsular polysaccharides antigen and tetanus toxoid carrier protein. It is clinically intended for the prevention of invasive infections caused by the above serotypes of pneumococci in infants aged 2 months to 2 years and children. The intended clinical dose is 0.5 mL/time/person, and the intended route of administration is intramuscular injection.

WestChina-Frontier PharmaTech Co., Ltd. conducted study on in vitro hemolysis of 13-JHYM provided by Yunnan Walvax Biotechnology Co., Ltd. according to the relevant requirements for new drug application in the Provisions for Drug Registration, and observed the effect of 13-JHYM on rabbit red blood cell, providing a reference for evaluating the safety of clinical use.

The study included negative control group (0.9% sodium chloride injection), positive control group (sterile water for injection), adjuvant control group (aluminum phosphate concentration of 1 mg/mL) and 13-JHYM (clinical preparation concentration) groups from three batches (20110304, 20110305 and 20110406). Five dose groups (0.5, 0.4, 0.3, 0.2 and 0.1 mL/tube) were set up for aluminum phosphate and 13-JHYM of each batch. The 2% rabbit red blood cell suspension was mixed with aluminum phosphate adjuvant, 13-JHYM, 0.9% sodium chloride injection and sterile water for injection at a certain proportion, then placed in a $37^{\circ}\text{C} \pm 0.5^{\circ}\text{C}$ incubator and observed at 0, 15, 30, 45, 60 minutes and 2, 3 hours respectively. Three replicates were prepared for each dose.

The results showed that after incubation at $37^{\circ}\text{C} \pm 0.5^{\circ}\text{C}$ for 3 hours, no hemolysis and aggregation reaction occurred in the negative control group; hemolysis reaction occurred in the positive control group; red blood cell aggregation occurred in each tube (0.5, 0.4, 0.3, 0.2 and 0.1 mL/tube) in aluminum phosphate adjuvant group, and the degree of aggregation was significantly dose-related, but no hemolysis reaction was observed; red blood cell aggregation occurred in each tube (0.5, 0.4, 0.3, 0.2 and 0.1 mL/tube) of 13-JHYM from 3 batches, and the degree of aggregation was lower than that in the aluminum phosphate group and was not significantly dose-related, and no hemolysis reaction was observed.

In conclusion, under the conditions of this study, aluminum phosphate adjuvant (at the concentration of 1 mg/mL, the same concentration as aluminum phosphate in 13-JHYM formulation) and 13-JHYM (20110304, 20110305 and 20110406) had aggregation effect on rabbit red blood cell at different doses (0.5, 0.4, 0.3, 0.2 and 0.1 mL/tube), but no hemolysis reaction was observed.

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MODULE 4 NONCLINICAL STUDY REPORTS

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1 GENERAL INFORMATION

1.1 NAME AND NO. OF THE EXPERIMENT

Study name: Study on In Vitro Hemolysis of 13-JHYM

Study No.: A201118-T071

1.2 OBJECTIVE

To conduct study on in vitro hemolysis of 13-JHYM and observe the effect of 13-JHYM on rabbit red blood cell, so as to provide reference for evaluating the safety of clinical use.

1.3 EXPERIMENT FACILITY

Name: WestChina-Frontier PharmaTech Co., Ltd. (National Chengdu Center for Safety Evaluation of Drugs)

Address: No. 28 Gaopeng Avenue, High-tech Development Zone, Chengdu, Sichuan

Postal code: 610041

Contact: Cen Xiaobo

Tel.: 028-85178767

Fax: 028-85173043

Email: xbcen@glpcd.commailto:xbcen@glpcd.com

1.4 CLIENT

Name: Yunnan Walvax Biotechnology Co., Ltd.

Address: 4F, Building A3, Phase II, Yunnan National University Science Park, High-tech Zone, Kunming

Postal code: 650106

Contact: Wu Kai

Tel.: 0871-8351253-8851

Fax: 0871-8312123

1.5 MEMBERS OF EXPERIMENT TEAM

1.5.1 STUDY DIRECTOR

Study Director: Guo Wei

Tel: 028-85154334-551

Fax: 028-85173043

Email: wguo@glpcd.commailto:wguo@glpcd.com

1.5.2 MEMBERS OF STUDY TEAM:

Study operators: Mu Ke, Huang Xuemei, Yuan Hong, Wang Xi, Xu Quanli

Veterinarian: Yi Bing

Management and preparation of test article: Zheng Wulin

Animal housing and management: Liao Qing

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2 MATERIALS

2.1 LABORATORY ANIMALS

2.1.1 GENERAL INFORMATION

Species: Japanese white rabbit;

Class: conventional;

Quantity and sex of animals purchased: 1, male;

Weight: 2.825 kg;

Source: Farm of Sichuan Laboratory Animal Specialized Committee, laboratory animal production

license No.: SCXK(C) 2008-014;

House entry time: August 2, 2011;

Acclimation period after entering animal house: Before the study, the rabbit was acclimated to the environment for 14 days (August 2, 2011 ~ August 15, 2011), and the study animal was a healthy

rabbit. Main inspection items during the acclimation period: General state of animals, body weight, etc.

2.1.2 ANIMAL HOUSING CONDITIONS

2.1.2.1 HOUSING PLACE

General Animal House (Building 1) of WestChina-Frontier PharmaTech Co., Ltd. (No. of license for use of laboratory animal house: SYXK (C) 2009-123).

2.1.2.2 HOUSING CONDITIONS

Type of housing cage: stainless steel rabbit cage (L × W × H: 550 mm × 550 mm × 380 mm);

Housing density: ≤1 animal/cage.

2.1.2.3 HOUSING ENVIRONMENTAL CONDITIONS

Standard for housing environmental conditions: National Standard of the People's Republic of China GB14925-2001;

Housing environmental control system: Honeywell EBI400 automatic control all-fresh air central air conditioning system;

Temperature: room temperature 23.11 ~ 24.49°C (daily range of temperature 0.53 ~ 1.15°C);

Humidity: relative humidity 52.3 ~ 61.6%;

Illumination: Artificial lighting with a 12 hour light/dark cycle (lights on at 07:30 and off at 19:30);

Ventilation rate: 8~10 times/h, 100% all fresh air.

2.1.2.4 FEED

Type: complete formula pellet feeds for rabbits

Manufacturer: Beijing Keao Xieli Feed Co., Ltd. Production license No. of feed: SCXK (J) 2009-0012;

Batch No.: 11044111;

Feeding method: Ad libitum;

Test of nutritional ingredients: Provided by the feed supplier. Test reports for conventional nutritional ingredient indicators (crude protein, crude fat, crude fiber, crude ash, moisture, calcium and

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phosphorus) were provided for each batch (refer to the national standard of the People's Republic of China GB14924.4-2001);

Confirmation of contaminant content of feed: Provided by the feed supplier. Chemical contaminant indicators such as arsenic, lead, cadmium, mercury, BHC, DDT, aflatoxin B1, total bacterial count, coliform group, mold and yeast count and pathogenic bacteria (salmonella) were provided at least once a year (refer to the national standard of the People's Republic of China GB14924.2-2001).

2.1.2.5 DRINKING WATER

Type: Drinking water for laboratory animals (reverse osmosis water)

Water supply method: Supplied in drinking bottle with water available ad libitum;

Test of total bacterial count: Refer to the national standard of the People's Republic of China GB5749-2006. Samples were taken for self-test (1 time/week).

Test of conventional indicators of water quality: Refer to the national standard of the People's Republic of China GB5749-2006. Samples were taken and sent for study.

2.1.2.6 ANIMAL IDENTIFICATION METHOD

Acclimation period and study period: Cage card was used as animal identification mark;

Marking method of cage card:

Acclimation period: White cage card was used, the study number, animal species, animal serial number, house entry time, remarks and other information were indicated.

2.2 TEST ARTICLE

2.2.1 INFORMATION OF TEST ARTICLE

Name or designation: 13-valent pneumococcal polysaccharide conjugate vaccine (code: 13-JHYM);

Source: Yunnan Walvax Biotechnology Co., Ltd.;

Description: Suspension with opalescence;

Strength: 0.5 mL: 32.2 µg/vial;

Batch No.: 20110304 (batch 1), 20110305 (batch 2), 20110406 (batch 3);

Storage conditions and precautions: Store at 2 ~ 8°C;

Shelf life: batch 20110304 and 20110305 were valid until February 2013, and batch 20110406 was valid until March 2013;

Place of storage: Test Article Management Department of WestChina-Frontier PharmaTech Co., Ltd.

2.1.2 PREPARATION OF TEST SOLUTION

Preparation method: bulk was directly used;

Method of identification after preparation: Follow the label on the original vial;

Temporary storage conditions and shelf life after preparation: Temporarily stored in the ice box and used within 24 h.

2.1.3 TREATMENT OF RETURNED AND REMAINING TEST ARTICLES

After being sterilized under high temperature and high pressure, the remaining test articles were uniformly collected into the waste liquid bucket. The retained samples were stored at the storage room of test articles in accordance with the SOP Storage of Test Articles. After completion of the test items, the remaining test articles were returned to the Client.

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2.3 CONTROL ARTICLE

2.3.1 GENERAL INFORMATION

Name or designation: aluminum phosphate;

Source: Yunnan Walvax Biotechnology Co., Ltd.

Description: Suspension with opalescence;

Concentration: 1 mg/mL;

Strength: 50 mL/vial;

Batch No.: 8169402;

Storage conditions and precautions: Store at 2 ~ 8°C;

Shelf life: long term;

Preparation method: Bulk was sub-packaged and directly used;

Method of identification after preparation: Yellow label was used for identification, and the study number, name of study substance, preparation volume, storage conditions, shelf life, responsible person and preparation date were indicated.

Temporary storage conditions and shelf life after preparation: Temporarily stored in the ice box and used within 24 h.

Name or designation: Sterile water for injection
Source: Sichuan Kelun Pharmaceutical Co., Ltd.;
Description: colorless clear liquid;
Strength: 500ml/vial;
Batch No.: M10120817;
Storage condition: sealed storage at room temperature;
Valid until: November 2012;
Preparation method: Bulk was sub-packaged and directly used;
Method of identification after preparation: Yellow label was used for identification, and the study number, name and concentration of study substance, volume, storage conditions, shelf life, responsible person and preparation date were indicated.
Temporary storage conditions and shelf life after preparation: Temporarily stored in the ice box and used within 24 h.

Name or designation: 0.9% sodium chloride injection;
Source: Sichuan Kelun Pharmaceutical Co., Ltd.;
Description: colorless clear liquid;
Strength: 100 ml/vial;
Concentration: 0.9%;
Batch No.: C100928;

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Storage condition: sealed storage at room temperature;
Valid until: August, 2012;
Preparation method: bulk was directly used;
Method of identification after preparation: White label was used for identification, and the study number, name and concentration of study substance, volume, storage conditions, shelf life, responsible person and preparation date were indicated.
Temporary storage conditions and shelf life after preparation: Temporarily stored in the ice box and used within 24 h.

2.3.2 DISPOSAL OF RETURNED AND REMAINING CONTROL ARTICLES

The returned control articles were discharged directly into the sewer.

2.4 OTHER MAIN REAGENTS

Name: pentobarbital sodium
Source: Sigma;
Description: white powder;
Strength: 25 g/vial;
Batch No.: 127K1005;
Storage condition: sealed storage at room temperature;
Retest time: March 2013;
Preparation method: 0.9% sodium chloride injection was dissolved to the required concentration immediately before use;
Temporary storage condition and shelf life after preparation: Temporarily store at 2 ~ 25°C for future use, with shelf life of 14 days;
The remaining pentobarbital sodium after use was returned to the Test Article Department.

2.5 MAIN INSTRUMENTS AND DEVICES REQUIRED FOR THE STUDY

Equipment Name Manufacturer Model Authority No.

Centrifuge Eppendorf 5810R MEC-01-13

Incubator Binder BD115 MEH-08-04

Electronic platform

scale, etc.

3 EXPERIMENT METHOD

3.1 DOSE DESIGN AND BASIS

According to the information provided by the Client, the concentration of 13-JHYM for clinical use was the concentration of formulation (0.5 mL/dose, one dose containing 32.2 µg of various pneumococcal polysaccharide, 0.5 mg of aluminum phosphate, i.e. 64.4 µg/mL of various pneumococcal polysaccharide, 1 mg/mL of aluminum phosphate). In this study, the clinical preparation concentration of the test article was taken as the study concentration.

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3.2 PREPARATION OF RBC SUSPENSION

After the rabbit was anesthetized with pentobarbital sodium (about 30mg/kg, injected intravenously into ear), collect blood from the abdominal aorta until the animal was euthanized, take appropriate amount of blood and remove fibrinogen to form defibrinated blood. Add about 10 times of 0.9% sodium chloride injection, shake well, dispense into centrifuge tubes, centrifuge at 1,000 - 2,000 rpm for 15 minutes, and discard the supernatant. Wash the precipitated red blood cells with 0.9% sodium chloride injection for at least 2 - 3 times using the above method until the supernatant did not show a red color. Prepare the obtained red blood cells into 2% suspension with 0.9% sodium chloride injection for study.

3.3 SAMPLE INJECTION AND OBSERVATION

Take clean tubes and number them: tubes No. 1 ~ 15 for test article 13-JHYM (batches 1, 2 and 3), tubes No. 16 ~ 20 for aluminum phosphate control, tube No. 21 for negative control, and tube No. 22 for positive control. Add 2% red blood cell suspension, 0.9% sodium chloride injection, 13-JHYM formulation bulk or aluminum phosphate, sterile water for injection in turn at the proportion set out in Table 1, mix well, place the mixture in an incubator at 37 ± 0.5°C for standing, and observe once at 15, 30, 45, 60, 120 and 180 minutes, respectively. Conduct 3 times of parallel determinations. See Table 1 for specific sample injection methods.

Table 1 Proportion of Preparation for Study on In Vitro Hemolysis of 13-JHYM

0.9% Sodium 13-JHYM

2% RBC Sterile Water for

Group Tube No. Chloride Injection Bulk/Aluminum

Suspension (mL) Injection (mL)

(mL) Phosphate (mL)

1-x 2.5 2.0 0.5 —

2-x 2.5 2.1 0.4 —

13-JHYM
3-x 2.5 2.2 0.3 —
(Batch 1)
4-x 2.5 2.3 0.2 —
5-x 2.5 2.4 0.1 —
6-x 2.5 2.0 0.5 —
7-x 2.5 2.1 0.4 —
13-JHYM
8-x 2.5 2.2 0.3 —
(Batch 2)
9-x 2.5 2.3 0.2 —
10-x 2.5 2.4 0.1 —
11-x 2.5 2.0 0.5 —
12-x 2.5 2.1 0.4 —
13-JHYM
13-x 2.5 2.2 0.3 —
(Batch 3)
14-x 2.5 2.3 0.2 —
15-x 2.5 2.4 0.1 —
16-x 2.5 2.0 0.5 —
17-x 2.5 2.1 0.4 —
Aluminum
18-x 2.5 2.2 0.3 —
phosphate
19-x 2.5 2.3 0.2 —
20-x 2.5 2.4 0.1 —
Negative
21-x 2.5 2.5 — —
Control
Positive
22-x 2.5 — — 2.5
Control

Note: x represents 1 ~ 3 replicates.
3.4 DETERMINATION OF RESULTS
3.4.1 RESULTS OBSERVATION
Complete hemolysis - The solution is clear and red, with no residual cells at the bottom of the tube;

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Partial hemolysis - The solution is red or brown, with a small amount of residual red blood cells at the bottom of the tube;
No hemolysis: All red blood cells sink to the bottom, and the supernatant is colorless and transparent;
Condensation-No hemolysis, but there is brownish red or reddish flocculent precipitates in the solution. The precipitates do not disperse after shaking. This indicates there is condensation of red blood cells.

3.4.2 RESULTS INTERPRETATION
When there are no hemolysis and condensation shown in the negative control tube and there is hemolysis in the positive control tube, if no hemolysis and condensation occur within 3 h for the solution in the test article tube, the result of test is negative; if hemolysis and (or) condensation occur within 3h for the solution in the test article tube, the result of test is positive.

4 RESULTS
After incubation at 37 °C ± 0.5 °C for 3 hours, all the red blood cells in the negative control group (saline) sank, and the supernatant was colorless ~ transparent. After appropriate shaking, the sinking red blood cells were re-dispersed, without hemolysis or aggregation; the solution in the positive control group (sterile water for injection) was clear and red, without cell residue at the bottom of tube, with hemolysis, indicating that the study system is normal and reliable.
After incubation at 37°C ± 0.5°C for 3 hours, aggregation was observed in each tube (0.1, 0.2, 0.3, 0.4 and 0.5 mL/tube) in aluminum phosphate adjuvant group. A large number of red blood cells aggregated and adsorbed on the tube wall and tube bottom, and the degree of aggregation was significantly dose-related; in each tube (0.1, 0.2, 0.3, 0.4 and 0.5 mL/tube) of 13-JHYM from batches 1, 2 and 3, only a small amount of red blood cells aggregated and adsorbed on the tube bottom. The supernatant in the above tubes was clear and colorless, and no hemolysis was observed.
See Table 2 and Figures 1, 2, 3 and 4 for specific results.

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Table 2 Results of Study on Hemolysis of 13-JHYM (Clinical Preparation Concentration)

0.9% Sodium 13-JHYM Sterile									
Group Tube No. 2% RBC(mL)									
Suspension Chloride Injection Bulk/Aluminum Water for									
Injection 0 min 15 min 30 min 45 min 1h 2h 3h									
(mL) Phosphate (mL) (mL)									
1-1	2.5	2.0	0.5	—	—	—	—	—	*
2-1	2.5	2.1	0.4	—	—	—	—	—	*
3-1	2.5	2.2	0.3	—	—	—	—	—	*
4-1	2.5	2.3	0.2	—	—	—	—	—	*
5-1	2.5	2.4	0.1	—	—	—	—	—	*
1-2	2.5	2.0	0.5	—	—	—	—	—	*
13-JHYM 2-2	2.5	2.1	0.4	—	—	—	—	—	*
(Batch 1) 3-2	2.5	2.2	0.3	—	—	—	—	—	*
4-2	2.5	2.3	0.2	—	—	—	—	—	*
5-2	2.5	2.4	0.1	—	—	—	—	—	*
1-3	2.5	2.0	0.5	—	—	—	—	—	*
2-3	2.5	2.1	0.4	—	—	—	—	—	*
3-3	2.5	2.2	0.3	—	—	—	—	—	*
4-3	2.5	2.3	0.2	—	—	—	—	—	*

5-3 2.5 2.4 0.1 —————*
6-1 2.5 2.0 0.5 —————*
7-1 2.5 2.1 0.4 —————*
8-1 2.5 2.2 0.3 —————*
9-1 2.5 2.3 0.2 —————*
10-1 2.5 2.4 0.1 —————*
6-2 2.5 2.0 0.5 —————*
13-JHYM 7-2 2.5 2.1 0.4 —————*
(Batch 2) 8-2 2.5 2.2 0.3 —————*
9-2 2.5 2.3 0.2 —————*
10-2 2.5 2.4 0.1 —————*
6-3 2.5 2.0 0.5 —————*
7-3 2.5 2.1 0.4 —————*
8-3 2.5 2.2 0.3 —————*
9-3 2.5 2.3 0.2 —————*
10-3 2.5 2.4 0.1 —————*
11-1 2.5 2.0 0.5 —————*
12-1 2.5 2.1 0.4 —————*
13-1 2.5 2.2 0.3 —————*
14-1 2.5 2.3 0.2 —————*
13-JHYM 15-1 2.5 2.4 0.1 —————*
(Batch 3) 11-2 2.5 2.0 0.5 —————*
12-2 2.5 2.1 0.4 —————*
13-2 2.5 2.2 0.3 —————*
14-2 2.5 2.3 0.2 —————*

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0.9% Sodium 13-JHYM Sterile
Group Tube No. 2% RBC(mL)
Suspension Chloride Injection Bulk/Aluminum Water for
Injection 0 min 15 min 30 min 45 min 1 h 2 h 3h
(mL) Phosphate (mL) (mL)
15-2 2.5 2.4 0.1 —————*
11-3 2.5 2.0 0.5 —————*
12-3 2.5 2.1 0.4 —————*
13-3 2.5 2.2 0.3 —————*
14-3 2.5 2.3 0.2 —————*
15-3 2.5 2.4 0.1 —————*
16-1 2.5 2.0 0.5 —————*
17-1 2.5 2.1 0.4 —————*
18-1 2.5 2.2 0.3 —————*
19-1 2.5 2.3 0.2 —————*
20-1 2.5 2.4 0.1 —————*
16-2 2.5 2.0 0.5 —————*
Aluminum 17-2 2.5 2.1 0.4 —————*
phosphate 18-2 2.5 2.2 0.3 —————*
19-2 2.5 2.3 0.2 —————*
20-2 2.5 2.4 0.1 —————*
16-3 2.5 2.0 0.5 —————*
17-3 2.5 2.1 0.4 —————*
18-3 2.5 2.2 0.3 —————*
19-3 2.5 2.3 0.2 —————*
20-3 2.5 2.4 0.1 —————*
Negative 21-1 2.5 2.5 —————
Control 21-2 2.5 2.5 —————
21-3 2.5 2.5 —————
Positive 22-1 2.5 — 2.5 ++ ++ ++ ++ ++ ++ ++
Control 22-2 2.5 — 2.5 ++ ++ ++ ++ ++ ++ ++
22-3 2.5 — 2.5 ++ ++ ++ ++ ++ ++ ++
Notes: 1: In the results, "++" represents "complete hemolysis"; "+" represents "partial hemolysis"; "-" represents "no hemolysis" and "*" represents "aggregation".
2: min represents minute and h represents hour.

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Fig. 1 Result of study on in vitro hemolysis of 13-JHYM (13-JHYM batch 1, first replicate, standing at constant temperature for 3 hours)

Fig. 2 Result of study on in vitro hemolysis of 13-JHYM (13-JHYM batch 2, first replicate, standing at constant temperature for 3 hours)

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Fig. 3 Result of study on in vitro hemolysis of 13-JHYM (13-JHYM batch 3, first replicate, standing at constant temperature for 3 hours)

Fig. 4 Result of study on in vitro hemolysis of 13-JHYM (aluminum phosphate, first replicate, standing at constant temperature for 3 hours)
(The glass tubes in the above pictures were numbered in X-Y, where X represents the dose number: 1 ~ 5 for the 13-JHYM (batch 1) group, 6 ~ 10 for 13-JHYM (batch 2) group, 11 ~ 15 for the 13-JHYM (batch 3) group, 16 ~ 20 for

the aluminum phosphate group, 21 for the negative control group, 22 for the negative control group; Y represents the replicate number of each dose.)

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5 DISCUSSION

13-JHYM was prepared through the covalent binding of pneumococcal capsular polysaccharide antigen with tetanus toxoid carrier protein and the adsorption by aluminum phosphate adjuvant. In the study, red blood cells aggregated and attached to different degrees to the tube wall in aluminum phosphate adjuvant and 13-JHYM (three batches) groups, showing a certain dose-related relation. Aluminum phosphate has strong adsorption effect and can adsorb viruses, bacteria and toxins produced by bacteria. Therefore, it is speculated that the above phenomenon may be caused by the adsorption effect of aluminum phosphate.

6 CONCLUSION

In conclusion, under the conditions of this study, aluminum phosphate adjuvant (at the concentration of 1 mg/mL, the same concentration as aluminum phosphate in 13-JHYM formulation) and 13-JHYM (20110304, 20110305 and 20110406, clinical preparation concentration) had aggregation effect on rabbit red blood cell at different doses (0.5, 0.4, 0.3, 0.2 and 0.1 mL/tube), but no hemolysis reaction was observed.

7 LOCATION FOR STORING INITIAL DATA

Place of storage: Archives Room, WestChina-Frontier PharmaTech Co., Ltd.;

Storage conditions: Conventional conditions;

Retention period: Paper archives should be kept at least 5 years after the drug is marketed, and electronic files should be permanently kept;

Tel.: 028-85154334-511;

Contact: Chang Ling.

8 DEVIATION

This study has no deviations or other abnormalities affecting the reliability of study results and conclusions.

9 KEY REFERENCES

Good Laboratory Practice, No. 2 Order of State Food and Drug Administration, September 1, 2003;

Provisions for Drug Registration, No. 28 Order of State Food and Drug Administration, October 1, 2007;

General Principles for Pre-clinical Safety Technical Review of Preventive Biological Products (State Food and Drug Administration, December 2005);

Technical Guidelines for Irritation, Allergic and Hemolytic Research of Chemical Drugs (State Food and Drug Administration, March 2005).

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4.2.3.7. Other Toxicity Studies

4.2.3.7.1. Antigenicity

N/A

4.2.3.7.2. Immunotoxicity

N/A

4.2.3.7.3. Mechanistic studies (if not included elsewhere)

N/A

4.2.3.7.4. Dependence

N/A

4.2.3.7.5. Metabolites

N/A

4.2.3.7.6. Impurities

N/A

4.2.3.7.7. Other

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4.2.3.7.7.1. Acute Toxicity Test

Experiment Report

Experiment Name Acute Toxicity Test of 13-JHYM through Intramuscular Injection in Kunming Mice

A201118-T001-01

Experiment Number

WestChina-Frontier PharmaTech Co. (National

Research Institution Name

Chengdu Center for Safety Evaluation of Drugs)

No. 28 Gaopeng Avenue, High & New Tech Dev.

Research Institution Address

Zone, Chengdu, Sichuan

Yunnan Walvax Biotechnology Co., Ltd.

Client Name

4th Floor, building A3, Phase 2 Science and

Client Address

Technology Park of Yunnan National University,

Kunming, Yunnan

Wei Guo

Experiment Director

Ke Mu

Members of Experiment Group

Xuemei Huang

Hong Yuan

Xi Wang

Quanli Xu

Bin Yi

Qin Zhang
Qing Ju
Xiaoyan Gao
June, 2011-October, 2011
Duration of Experiment
Archives Room of WestChina-Frontier
Original Data Depository
PharmaTech Co.

WestChina-Frontier PharmaTech Co. (WCFP)
National Chengdu Center for Safety Evaluation of Drugs (NCCSED)

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Confirmation of Experiment Report

Experiment Name: Acute Toxicity Test of 13-JHYM
through Intramuscular Injection in
Kunming Mice
Experiment Number: A201118-T001-01

Experiment Director: Wei Guo

Signature: Date: 30/09/2011

QAU Director: Feng Cheng

Signature: Date: 30/09/2011

Institution Director: Li Wang

Signature: Date: 09/10/2011

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MODULE 4 NONCLINICAL STUDY REPORTS

Version No. Walvax-202007-00 Yuxi Walvax Biotechnology Co., Ltd.

Compliance Statement

Acute Toxicity Test of 13-JHYM Through

Experiment Name:

Intramuscular Injection in Kunming Mice

Experiment Number: A201118-T001-01

This experiment was completed in accordance with Non-clinical Good Laboratory Practice formulated by China Food and Drug Administration (China Food and Drug Administration rule No. 2, effective from September 1st, 2003). The experiment procedure was carried out in accordance with the experiment protocol and the relevant standard operating procedure of the research institution (please see Module 7 for relevant deviation in this experiment). No deviation that affected the reliability of experiment results and conclusions was observed. This report faithfully reflects the experiment materials, experiment methods and experiment results.

Experiment Director: Wei Guo

Signature of Experiment Director: Date: 30/09/2011

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MODULE 4 NONCLINICAL STUDY REPORTS

Version No. Walvax-202007-00 Yuxi Walvax Biotechnology Co., Ltd.

QAU Statement

Experiment Name: Acute Toxicity Test of 13-JHYM through

Intramuscular Injection in Kunming Mice

Experiment Number: A201118-T001-01

Experiment Director: Wei Guo

In order to ensure that the contents of the experiment protocol can be strictly implemented and the experiment can be operated in accordance with the SOP of this institution, QAU has checked the experiment protocol, the experiment implementation process and the experiment record, and has reviewed the original data and the experiment report. The operating procedures, experiment methods and results described in the report faithfully reflect the contents of the original data. The QAU inspection items, inspection dates and report dates are as follows:

Date of reporting to Date of reporting

Inspection items Inspection date the experiment to the institution
director director

Experiment 2011.06.09-2011.06.10 2011.06.10 2011.06.10

protocol 2011.06.14 2011.06.14 2011.06.14

Test article

2011.06.20 2011.06.21 2011.06.21

preparation

Grouping 2011.06.20 2011.06.21 2011.06.21

Body weight 2011.06.20 2011.06.21 2011.06.21

Administration 2011.06.20 2011.06.21 2011.06.21

Observation 2011.06.20 2011.06.21 2011.06.21

Food intake 2011.06.21 2011.06.21 2011.06.21

Anatomy 2011.07.04 2011.07.04 2011.07.04

Collection and

examination of

blood samples

2011.07.04 2011.07.04 2011.07.04

(hematology and

blood

biochemistry)

Original data 2011.09.20-2011.09.23 2011.09.23 2011.09.23

Experiment 2011.09.20-2011.09.23 2011.09.23 2011.09.23

report 2011.09.28 2011.09.28 2011.09.28

Experiment QA: Xinxin Li

Signature: Date: 30/09/2011

QAU director: Feng Cheng

Signature: Date: 30/09/2011

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Version No. Walvax-202007-00 Yuxi Walvax Biotechnology Co., Ltd.

Abstract

The 13-valent pneumococcal polysaccharide conjugate vaccine (13-JHYM) is made from pneumococcus capsule polysaccharide antigens of serotype 1, 3, 4, 5, 6A, 6B, 7F, 9V, 14, 18C, 19A, 19F and 23F covalently combined with tetanus toxoid carrier protein and adsorbed by aluminum phosphate adjuvant. The clinical use of this vaccine is for infants and children aged 2 months to 2 years old to prevent the invasive infection caused by the above-mentioned serotype pneumococcus. The proposed dosage is 0.5 mL per person, and the proposed route of use is intramuscular injection.

In accordance with the relevant requirements of new drug declaration, the acute toxicity test of JHYM provided by Yunnan Walvax Biotechnology Co., Ltd. was carried out through

intramuscular injection in Kunming mice by WestChina-Frontier PharmaTech Co., with a purpose to evaluate the nature, degree, time-effect relationship and reversibility of the possible toxic reaction, to determine the toxic target organ or target tissue, and to provide a reference for the safe range of clinical use.

A total of 3 groups were set for the test article 13-JHYM in this experiment, including test articles of Batch No.20110304, 20110305 and 20110406 respectively. The 13-JHYM of the corresponding batch number for each group was injected in the concentration of 64.4 µg/mL (in terms of the total amount of pneumococcal polysaccharide contained) in the volume of 10 mL/kg. A negative control group was set and mice in the control group were given 0.9% sodium chloride injection in the volume of 10 mL/kg. Each group included 10 mice (half male and half female). The mice in each group were given a single intramuscular injection with the corresponding sample or control in the maximum volume of 10 mL/kg (simultaneous injection into both hind limb muscles). The day of administration was determined as the first day in this experiment.

After administration, the general conditions of mice in each group was observed daily, meanwhile the body weight and food intake were measured twice a week. On the 15th day of the experiment, all the surviving mice were anesthetized (about 60 mg/kg pentobarbital sodium was injected intraperitoneally) and blood samples were collected from abdominal aorta for examination of hematology and blood biochemical measurements. A systematic autopsy was also performed to observe the appearance and internal organs of the animals. The main results are as below:

General Conditions

During the 14-day observation after administration, the mice in each group showed normal physical, mental and motor state. The natural pores were clean, and no abnormal symptoms were observed. No irritant reactions such as redness, swelling, ecchymosis, induration, suppuration and ulceration were observed at the injection site.

Body Weight and Food Intake

No abnormalities were observed in food intake and body weight for mice in each group at each detection time points during the 14-day observation after administration.

Hematology and Blood Biochemical Examination

On the 15th day of the experiment, no abnormalities were observed in hematology and blood biochemical measurements for mice in each group.

Gross Anatomy Results

On the 15th day of the experiment, no abnormal changes were observed with naked eyes in the size, shape, color and texture of the major organs such as heart, liver, spleen, lung and kidney for the

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mice in each group. No irritant reactions such as redness, swelling, and induration were observed at the injection site.

In conclusion, under the conditions of this experiment, the mice were in good condition and no toxic reactions were observed during the 14-day observation after the mice were given 13-JHYM (Batch No. 20110304, 20110305 and 20110406) through single muscular injection in the maximum volume of 10mL/kg.

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MODULE 4 NONCLINICAL STUDY REPORTS

Version No. Walvax-202007-00 Yuxi Walvax Biotechnology Co., Ltd.

1 BASIC INFORMATION

1.1 EXPERIMENT NAME AND NUMBER

Experiment name: Acute Toxicity Test of 13-JHYM through Intramuscular Injection in Kunming Mice

Experiment number: A201118-T001-01

1.2 OBJECTIVES

The Kunming mice in this experiment were intramuscularly injected with 13-JHYM for acute toxicity test, with a purpose to evaluate the nature, degree, time-effect relationship and reversibility of the possible toxic reaction, to determine the toxic target organ or target tissue, and to provide a reference for the safe range of clinical use.

1.3 RESEARCH INSTITUTE

Name: WestChina-Frontier PharmaTech Co. (National Chengdu Center for Safety Evaluation of Drugs)

Address: No. 28 Gaopeng Avenue, High & New Tech Dev. Zone, Chengdu, Sichuan

Zip code: 610041

Contact person: Xiaobo Cen

Telephone: 028-85178767

Fax: 028-85173043

Email: xbcen@glpcd.com

1.4 CLIENT INSTITUTE

Name: Yunnan Walvax Biotechnology Co., Ltd.

Address: 4th Floor, building A3, 2nd Phase Science and Technology Park of Yunnan National University, Kunming, Yunnan

Zip code: 650106

Contact person: Kai Wu

Telephone: 0871-8351253-8851

Fax: 0871-8312123

1.5 MEMBERS OF EXPERIMENT GROUP

1.5.1 EXPERIMENT DIRECTOR

Name: Wei Guo

Telephone: 028-85154334-551

Fax: 028-85173043

Email: wguo@glpcd.com

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1.5.2 GROUP MEMBERS

Experiment operation: Ke Mu, Xuemei Huang, Hong Yuan, Xi Wang, Quanli Xu
Veterinarian: Bin Yi
Animal rearing and management: Junyan Li, Qinghua Yang, Bo Zhang
Test articles preparation and management: Qin Zhang
Clinical examination: Qing Ju
Statistical Analysis: Xiaoyan Gao

2 MATERIALS

2.1 EXPERIMENT ANIMALS

2.1.1 BASIC INFORMATION

Species: Kunming mice;
Grade: SPF;
Amount and gender of animals purchased: 46 animals in total, half male and half female;
Amount and gender of animals used: 40 animals in total, half male and half female;
Body weight (at the time of grouping): 25.0-27.9g for females, 25.3-29.1g for males;
Source: Shanghai SLAC Laboratory Animal Co., Ltd. (Production License No. SCXK (Shanghai) 2007-0005; Certificate of Quality No. 2007000521239, Serial number 0049180, issuing unit: Shanghai Laboratory Animal Administration Office);
Entry date: 14/06/2011;
Environmental adaptation period after entry: Before the experiment, the Kunming mice adapted to the environment for 7 days (14/06/2011-20/06/2011) and went through relevant examinations. The healthy mice were selected to enter the formal experiment. Observation measurements during adaptation period: Is it consistent with the quality measurements required at the time of ordering? Is the animal in general condition? Is the body weight within the range required by the experiment?

2.1.2 ANIMAL REARING CONDITIONS

2.1.2.1 REARING SITES

SPF small animal house of WestChina-Frontier PharmaTech Co. (laboratory animal room license No. SYXK (Sichuan) 2009-123).

2.1.2.2 REARING CONDITIONS

Rearing cage: plastic mouse cage (Upper L*W: 295 mm*175 mm, Lower L*W: 270 mm*155 mm, H: 140mm);
Padding material: corncob, provided by Beijing Ke'ao Xieli Feed Co., Ltd.;
Rearing density: 55 animals per cage.

2.1.2.3 REARING ENVIRONMENT

Standard: the National Standard of the People's Republic of China GB14925-2001;
Control system: the EBI400 automatic control fresh air air conditioning system produced by Honeywell;

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Temperature: room temperature 21.24-24.14°C (with a diurnal temperature range of 0.55-1.78°C);
Humidity: relative humidity 48.9-69.5%;
Illumination: artificial lighting, 12 hours of alternating light and dark (lights on at 07:30, lights off at 19:30);
Ventilation: 10-20 times per hour, 100% fresh air.

2.1.2.4 FEED

Type: Cobalt 60 irradiation sterilized complete feed for mice;
Manufacturer: Shanghai SLAC Laboratory Animal Co., Ltd. (Feed Production Permit: Shanghai Feed Audit (2008)04002);
Batch No.: M02-F-20110316003;
Feeding method: free intake;
Nutrient analysis: provided by the feed provider, of which the routine nutritional measurements of crude protein, crude fat, crude fiber, crude ash, water, calcium and phosphorus were reported for each batch; amino acid measurements of threonine, cysteine + methionine, satin, L-Isoleucine, leucine, tyrosine, L-Phenylalanine, histidine, lysine and arginine were reported at least once a year (referring to the National Standard of the People's Republic of China GB14924.5-2001);
Confirmation of contaminant content in feed: provided by the feed provider, indicators for chemical contaminants including arsenic, lead, cadmium, mercury, hexachlorocyclohexane, DDT, aflatoxin B1, total colony count, coliform group, mold and yeast count, pathogenic bacteria (Salmonella) were provided at least once a year (referring to the National Standard of the People's Republic of China GB14924.2-2001).

2.1.2.5 DRINKING

Type: drinking water for laboratory animals (reverse osmosis water);
Method of water supply: held in drinking water bottles, free intake;
Colony count detection: referring to the National Standard of the People's Republic of China GB5749-2006, sampling self-examination (once per week);
Detection of conventional indicators of water quality: referring to the National Standard of the People's Republic of China GB5749-2006, samples were sent to units with relevant qualification for testing at least 2 times per year.

2.1.3 ANIMAL GROUPING AND MARKING METHODS

2.1.3.1 ANIMAL GROUPING

Group design: a total of 4 groups, including a negative control group and three 13-JHYM groups with batch No. 20110304, 20110305 and 20110406 respectively;
Amount: 10 animals in each group;
Gender: half male and half female;
Grouping method: Kunming mice were randomly divided into groups according to their body weight;
Disposal of surplus animals: handed over to the department of laboratory animal administration. The specific group information is shown in Table 1.

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Table 1 Grouping Table

Animal amount	Animal number
Group	
Female Male Female Male	
Negative 1F0101, 1F0102, 1F0103, 1M0101, 1M0102, 1M0103,	
5 5	
control group 1F0104, 1F0105 1M0104, 1M0105	

13-JHYM 2F0101, 2F0102, 2F0103, 2M0101, 2M0102, 2M0103,

5 5

group 1 2F0104, 2F0105 2M0104, 2M0105

13-JHYM 3F0101, 3F0102, 3F0103, 3M0101, 3M0102, 3M0103,

5 5

group 2 3F0104, 3F0105 3M0104, 3M0105

13-JHYM 4F0101, 4F0102, 4F0103, 4M0101, 4M0102, 4M0103,

5 5

group 3 4F0104, 4F0105 4M0104, 4M0105

Notes: The first digit of the animal number represents the group (1 represented the control group; 2, 3 and

4 represent 13-JHYM group 1, 2 and 3); the second letter stands for gender (M for male, F for female); the

third and fourth digits represent cage number; the fifth and sixth digits represent animal serial number.

2.1.3.2 MARKING METHOD

Adaptation period: cage card and tail mark were used as animal identification markers;

Post-grouping period: The cage card, ear number and tail mark were used as animal identification markers;

Experiment period: the cage card and ear markers were used as animal identification marks;

Marking method of cage card:

Adaptation period: unified use of white cage card, with the test number, animal species, animal number, entry time and notes indicated;

Post-grouping: cage cards in different colors were used to distinguish the groups, and the cards were marked with test number, animal species, animal number, treatment factors, expected start and end time of the experiment and notes;

Colors of cage cards after grouping: white represented the negative control group, green represented 13-JHYM group 1, blue represented 13-JHYM group 2 and red represented 13-JHYM group 3.

2.2 TEST ARTICLES

2.2.1 TEST ARTICLE INFORMATION

Name: 13-valent pneumococcal polysaccharide conjugate vaccine (13-JHYM);

Source: Yunnan Walvax Biotechnology Co., Ltd.;

Character: suspension with opalescence;

Specification: 0.5 mL: 32.2μg per vial;

Batch No.: 20110304, 20110305, 20110406;

Preservation conditions and notes: kept under 2-8■;

Expiration date: February, 2012 for batch No. 20110304 and 20110305; March, 2013 for 20110406;

Preservation site: the Samples Management Department of WestChina-Frontier PharmaTech Co.

2.2.2 TEST ARTICLE PREPARATION

Preparation methods: direct separation of the bulk;

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Marking methods after preparation: use the original bottle label;

Preservation condition and period of validity after preparation: stored in ice box for use within 24 hours.

2.2.3 RETURN AND DISPOSAL OF REMAINING SAMPLES

The remaining samples after administration were sterilized by high temperature and high pressure and diluted with a large amount of water before being discharged into the sewer. The reserved samples were deposited in the sample room referring to the SOP for the preservation of test articles. The remaining samples at the end of the experiment were returned to the client.

2.3 COMPARISON PRODUCT

2.3.1 COMPARISON PRODUCT INFORMATION

Name: 0.9% sodium chloride injection (normal saline);

Source: Sichuan Kelun Pharmaceutical Co., Ltd.;

Character: colorless transparent liquid;

Specification: 100 mL per bottle;

Batch No.: C100928;

Preservation conditions and notes: sealed storage at room temperature;

Expiration date: August, 2012.

2.3.2 COMPARISON PRODUCT PREPARATION

Preparation methods: direct separation of the bulk;

Marking methods after preparation: test articles were marked with red labels with experiment number, substance name and concentration used in the experiment, volume, storage condition, expiration date, responsible person and preparation date;

Preservation condition and period of validity after preparation: sealed storage under 2■8°C, use within the same day.

2.2.3 RETURN AND DISPOSAL OF REMAINING COMPARISON PRODUCTS

After administration, the remaining negative control was directly discharged into the sewer, the adjuvant control was collected in the waste liquid barrel, and the listed control was collected in the waste liquid barrel after sterilization.

2.4 MAIN INSTRUMENTS

Instrument name Manufacturer Model type Instrument number

Automatic five-category

Siemens ADVIA2120 MEB-01-14

blood analyzer

Automatic blood

Roche Cobas C501 MEB-02-05

biochemical analyzer

Centrifugal machine,

electronic balance, Eppendorf 5810R MEC-01-13

injector etc.

2.5 MAIN REAGENT

Name: pentobarbital sodium;

Source: Sigma;

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Character: white powder;
Specification: 25g per bottle;
Batch No.: 127K1005;
Preservation conditions and notes: sealed storage at room temperature;
Expiration date: March, 2013 (re-inspection date);
Preparation methods: dissolved with 0.9% sodium chloride injection to the required concentration when use;
Marking methods: marked with white label;
Preservation condition and period of validity after preparation: sealed storage under 2-25 °C, validate within 14 days.
After use, the remaining pentobarbital sodium was returned to the Samples Management Department.

3. METHODS AND OBSERVATION MEASUREMENTS

3.1 METHODS

3.1.1 DOSAGE DESIGN

Referring to the General Principles for the Technical Review of the Preclinical Safety Evaluation of Prophylactic Biological Products (December 2005) and Technical Guidelines for Acute Toxicity Testing of Chemicals (March 2005), the dosage in acute toxicity test can be determined according to the maximum administration capacity of different animals and routes of administration. Based on the anatomical characteristics of the mice, the site for intramuscular injection should be on bilateral hind limbs and the total volume of the injection should not exceed 10 mL/kg.

In this experiment, test articles prepared using 13-JHYM of the 3 different batch numbers were given to mice in the maximum volume of 10 mL/kg through intramuscular injection (weight of baby mice was determined as 10kg, which meant 200 times of the clinical proposed dosage for human). Mice in the negative control group were injected intramuscularly with normal saline of equal volume.

The specific dosage information is shown in Table 2.

Table 2 Table of Dosage Design

Administration Dosage Concentration Volume Animal amount

Group
route (µg/kg) (µg/mL) (mL/kg) Male Female

Negative control Intramuscular

// 10 5 5

group injection

Intramuscular

13-JHYM group 1 644 64.4 10 5 5

injection

Intramuscular

13-JHYM group 2 644 64.4 10 5 5

injection

Intramuscular

13-JHYM group 3 644 64.4 10 5 5

injection

3.1.2 ADMINISTRATION

Administration route: intramuscular injection;

Volume: 10 mL/kg;

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Frequency: all groups of mice were injected with sample or control into the muscle of both hind limbs at one time. The single injection amount was about 1/2 of the total amount. The day of administration was defined as the first day in this experiment.

3.2 OBSERVATION MEASUREMENTS

3.2.1 GENERAL CONDITIONS

3.2.1.1 OBSERVATION TIME AND FREQUENCY

Rearing personnel: twice a day;

Experiment personnel: once a day;

Veterinarian: once a week;

Observation Period: 14 days after administration.

3.2.1.2 OBSERVATION MEASUREMENTS

Rearing personnel: death and near-death situation of the mice;

Experiment personnel: including but not limited to the appearance, the general behavior activity, the mental state, the aspiration status, the excrement character, the genitals, the death and other toxic performance of the mice;

Veterinarian: including but not limited to the appearance, the general behavior activity, the mental state, the aspiration status, the excrement character, the genitals, the death and other toxic performance of the mice.

3.2.2 OBSERVATION OF IRRITATION AT INJECTION SITE

Observation time and frequency: once a day within 1 week after administration (i.e. Day 1 to Day 7 of the experiment);

Observation animal: all surviving animals;

Observation method: the animals were removed from the cage and the injection site was observed for irritant reactions such as redness, swelling, ecchymosis, induration, suppuration and ulceration.

3.2.3 BODY WEIGHT

Measurement time: once before administration and twice a week thereafter;

Measurement animal: all surviving mice in each group.

3.2.4 FOOD INTAKE

Measurement time: once a week;

Measurement animal: all mice in each group;

Measurement method: the feed quantity of each cage was determined on the first day, and the remaining feed was measured at approximately the same time on Day 2. The difference between the two measurements was the total food intake of the animals in each cage in 24 hours, and the total food intake for each cage was divided by the number of animals in each cage to calculate the average food intake for each animal.

3.2.5 HEMATOLOGY EXAMINATION

Measurement time: on the 15th day of the experiment;

Measurement animal: all surviving mice to be anatomized in each group;

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Method of blood collection: the mice were fasting for at least 12 hours before blood collection but were not deprived of water. The mice to be anatomized were anesthetized with pentobarbital sodium (about 60 mg/kg, intraperitoneal injection), and blood samples were collected from the abdominal aorta (about 0.5 mL);
Blood sample preparation: blood samples (about 0.5 mL) for hematologic examination used EDTA as anticoagulant, and the whole blood samples were used for direct examination.

Measurement item: please see Table 3.

Table 3 Measurement Items of Hematology Examination

Measurement item Measurement method Equipment

RBC Two-dimensional laser scanning ADVIA2120

HGB Hemoglobin nitride method ADVIA2120

HCT Calculation: $MCV \times RBC$ ADVIA2120

MCV Two-dimensional laser scanning ADVIA2120

MCH HGB/RBC ADVIA2120

MCHC $HGB/(MCV \times RBC) \times 1000$ ADVIA2120

RET Oxazine 750 staining ADVIA2120

RET% $(RET/RBC) \times 100$ ADVIA2120

WBC POX/Two-dimensional laser scanning ADVIA2120

NEUT (NEUT%) POX ADVIA2120

LYMPH (LYMPH%) POX ADVIA2120

MONO (MONO%) POX ADVIA2120

EOS (EOS%) POX ADVIA2120

BASO (BASO%) POX ADVIA2120

PLT Two-dimensional laser scanning ADVIA2120

3.2.6 BLOOD BIOCHEMICAL EXAMINATION

Measurement time: on the 15th day of the experiment;

Measurement animal: all surviving mice to be anatomized in each group;

Method of blood collection: the mice were fasting for at least 12 hours before blood collection but were not deprived of water. The mice to be anatomized were anesthetized with pentobarbital sodium (about 60 mg/kg, intraperitoneal injection), and blood samples were collected from the abdominal aorta (about 0.5 mL);

Blood sample preparation: the blood sample (not anticoagulated) was centrifuged for 10 minutes at 3000RP (centrifugal force about 1800g). The serum was separated and examined. The remaining serum was stored at -20 °C.

Measurement item: please see Table 4.

Table 4 Measurement Items of Blood Biochemical Examination

Measurement item Measurement method Equipment

AST IFCC P-5'-P Cobas C501

ALT IFCC P-5'-P Cobas C501

CK HK-G6PD Cobas C501

ALP 4-NPP Cobas C501

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Urea Urease-GLDH-Kinetic Cobas C501

TP Bmret Cobas C501

ALB BCG Cobas C501

A/G Calculation Cobas C501

GLU GLK-G6PDH Cobas C501

TBIL Diazo Cobas C501

Crea Enzyme method Cobas C501

CHOL CHOD-P Cobas C501

TG GPO-PAP Cobas C501

+

NA ISE Cobas C501

K+ ISE Cobas C501

Cl- ISE Cobas C501

3.2.7 SYSTEMATIC AUTOPSY (GROSS ANATOMY)

Autopsy time: on the 15th day of the experiment;

Autopsy animal: all surviving mice in each group;

Methods of anesthesia and autopsy: before autopsy, the mice were fasting for at least 12 hours.

The mice were anesthetized with pentobarbital sodium depending on their weight (about 60mg/kg, intraperitoneal injection), and mice were taken in parallel in each group. After observing the appearance of mice, the mice were euthanized by bleeding from the abdominal aorta, and the systematic autopsy (gross anatomy) was performed.

3.3 STATISTICAL ANALYSIS AND RESULTS DETERMINATION

Mean $\pm$ standard deviation ($\bar{X} \pm SD$) was used to describe quantitative indices including weight, hematology and blood biochemical measurements. Raw data was presented directly for food intake. When the sample size was less than 3, raw data was presented directly and the data in that group was not included in statistical analysis.

ONE-WAY ANOVA was used to test the group differences in the mean of quantitative indices of Kunming mice. When the mean difference was statistically significant ($p \leq 0.05$), LSD method was used to compare the difference between the 13-JHYM group of different batch and the negative control group. The KRUSKAL-H rank sum test (K-W test) was used to test the group differences of the above indices when the homogeneity of variance ($p \leq 0.05$) was indicated by the Levene test. MANN-WHITNEY U rank sum test (m-w method) was used to compare the mean of 13-JHYM group of different batch and that of negative control group when the mean difference was statistically significant ($p \leq 0.05$).

All tests were bilateral tests with $\alpha = 0.05$. All statistical analyses were performed in SPSS (Window s 13.0).

4. RESULTS

4.1 GENERAL CONDITIONS

During the 14-day observation after administration, each mouse was in normal physical and mental state, and the natural holes were clean. No obvious abnormalities were observed. No local reactions

at the injection site such as redness, swelling, ecchymosis, induration, suppuration and suppuration were observed.

4.2 BODY WEIGHT

No obvious abnormalities were observed in the body weight of the mice in each 13-JHYM group at each detection time point during the 14-day observation after administration.

Please see Table 1 in Appendix I for specific results.

4.3 FOOD INTAKE

No obvious abnormalities were observed in the food intake of the mice in each 13-JHYM group at each detection time point during the 14-day observation after administration.

Please see Table 2 in Appendix I for specific results.

4.4 HEMATOLOGY EXAMINATION

On the 15th day of the experiment, EOS of 13-JHYM batch No. 20110305 and 20110406 groups were significantly higher than those of the control group ($p \leq 0.05$). Apart from that, no abnormal changes were observed in hematological measurement for 13-JHYM batch no. 20110304, 20110305 and 20110406 groups.

In terms of the individual data, the above changes were mainly caused by the abnormal rise of EOS values (EOS values of 0.25 and 0.6) in certain individual animals (animal No.3M0103 and 3M0105). The abnormal rise of EOS values (EOS value of 0.24) was also observed in certain individual animal (animal No.1F0101) in the control group. Moreover, no obvious consistency was observed between the changes of EOS values among the three batches. Therefore, the changes of EOS values were not related to the injected samples.

Please see Table 3 in Appendix I for specific results.

4.5 BLOOD BIOCHEMICAL EXAMINATION

On the 15th day of the experiment, abnormal rise of ALT value (ALT value of 419.7) was observed for certain individual animal (animal No.3M0104) in 13-JHYM batch No.20110305 group. Apart from that, no abnormal changes in blood biochemical measurements were observed for 13-JHYM batch 20110304, 20110305 and 20110406 groups. These changes were only seen in individual animals in 13-JHYM batch No.20110305 group, while no similar changes were observed in the other groups. Therefore, the above change was individual difference and was not related to the injected samples.

Please see Table 4 in Appendix I for specific results.

4.6 GROSS ANATOMY OBSERVATION

On the 15th day of the experiment, the mice in each 13-JHYM group were anatomized. No abnormalities were observed with naked eyes in the size, shape, color and texture of main organs including heart, liver, spleen, lung and kidney. No irritant reactions such as redness, swelling and induration were observed in the muscles at the injection site.

5. CONCLUSIONS

In conclusion, under the experimental conditions, the Kunming mice were in good condition and no obvious toxic reactions were observed in the 14-day observation after the mice were injected with 13-JHYM (batch no. 20110304, 20110305 and 20110406) through a single intramuscular injection in the maximum volume of 10 mL/kg.

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6. ORIGINAL DATA DEPOSITORY

Preservation site: the archives room of WestChina-Frontier PharmaTech Co.

Preservation conditions: Standard;

Preservation period: The paper file were kept for at least 5 years after the drug got licensed, and the electronic data should be kept permanently;

Contact number: 028-85154334-511;

Contact person: Ling Chang.

7. DEVIATIONS

Type: experiment protocol deviation;

Description: The Hematology Examination (Module 8.4) and Blood Biochemical Examination (Module 8.5) in the original protocol required hematology and blood biochemical examination for all surviving mice to be anatomized in each group. At the time of blood collection, no biochemical sample for animal No.1F0101 and hematologic sample for animal No.2F0104 were collected. Blood coagulation was detected in samples of animal No.1F0103, 2F0103 and 4M0102, which resulted in the failure of hematological and biochemical examination for these animals;

Processing: None;

Evaluation: Due to the restriction of mouse blood volume and blood sample collection method, resampling and reexamination were not possible. Besides, only a certain part of the measurements for certain individual animals could not be detected, which had no obvious influence on statistical analysis and therefore had no impact on the experiment results.

Apart from the above-mentioned deviation, no deviation which had impact on the reliability of the experiment results and conclusions as well as other abnormalities were observed in this experiment.

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Appendix I Summary Tables of Experiment Results in Acute Toxicity Test of 13-JHYM through Intramuscular Injection in Kunming Mice

Table 1 Effects of 13-JHYM on Body Weight of Female Kunming Mice (g)

Negative control

13-JHYM group 1 13-JHYM group 2 13-JHYM group 3

Measurement group

time n ■X ±SD n ■X ±SD n ■X ±SD n ■X ±SD

Before 5 25.5 ±1.0 5 25.6 ±1.0 5 25.7 ±1.1 5 25.5 ±0.9

administration

Day 4 of the 5 28.3 ±1.5 5 28.1 ±1.5 5 28.2 ±1.8 5 28.8 ±2.1

experiment

Day 8 of the 5 31.3 ±1.8 5 31.1 ±1.7 5 30.8 ±1.9 5 31.5 ±2.2

experiment

Day 11 of the 5 32.8 ±1.8 5 32.2 ±1.6 5 32.1 ±2.3 5 33.5 ±1.9

experiment

Day 15 of the 5 32.5 ±1.3 5 31.2 ±1.8 5 32.1 ±2.2 5 32.6 ±2.0
experiment
Table 1 (Continued Table) Effects of 13-JHYM on Body Weight of Male Kunming Mice (g)
Negative control
13-JHYM group 1 13-JHYM group 2 13-JHYM group 3
Measurement group
time n ■X ±SD n ■X ±SD n ■X ±SD n ■X ±SD

Before 5 26.9 ±1.0 5 27.2 ±0.6 5 26.6 ±1.3 5 26.8 ±0.4
administration
Day 4 of the 5 31.8 ±1.7 5 31.0 ±1.1 5 31.2 ±1.6 5 30.1 ±1.9
experiment
Day 8 of the 5 37.2 ±2.8 5 34.9 ±2.0 5 35.9 ±1.7 5 33.9 ±3.1
experiment
Day 11 of the 5 38.7 ± 3.2 5 36.7 ±2.7 5 37.8 ±2.0 5 35.5 ±3.4
experiment
Day 15 of the 5 38.7 ±2.4 5 37.2 ±3.8 5 38.3 ±2.1 5 35.9 ±2.8
experiment
Table 2 Effects of 13-JHYM on Food Intake of Kunming Mice (g)
Negative control group 13-JHYM group 1 13-JHYM group 2 13-JHYM group 3
Measurement
time n ■X n ■X n ■X n ■X
Female

Day 2 of the 1 9.8 1 8.4 1 7.1 1 8.3
experiment
Day 9 of the 1 5.7 1 6.0 1 6.2 1 7.2
experiment
Male

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Day 2 of the 1 8.8 1 10.0 1 7.3 1 9.0
experiment
Day 9 of the 1 8.4 1 8.2 1 7.4 1 9.2
experiment
Table 3 Effects of 13-JHYM on Hematology Measurements of Kunming Mice
Negative control
13-JHYM group 1 13-JHYM group 2 13-JHYM group 3
Measurement group
n1) ■X ±SD n2) ■X ±SD n ■X ±SD n3) ■X ±SD
WBC (109/L) 9 1.81 ±0.52 8 1.87 ±0.80 10 2.20 ±1.28 9 1.58 ±0.33
9
NEU (10 /L) 9 0.40 ±0.16 8 0.29 ±0.10 10 0.40 ±0.18 9 0.29 ±0.13
LYM (109/L) 9 1.30 ±0.33 8 1.46 ±0.73 10 1.58 ±1.06 9 1.17 ±0.25
MONO (109/L) 9 0.02 ±0.01 8 0.02 ±0.01 10 0.03 ±0.02 9 0.02 ±0.01
EOS (109/L) 9 0.07 ±0.07 8 0.08 ±0.04 10 0.17 ±0.16* 9 0.09 ±0.04*
BASO (109/L) 9 0.01 ±0.00 8 0.01 ±0.01 10 0.01 ±0.00 9 0.01 ±0.01
NEU% (%) 9 21.5 ±3.4 8 16.7 ±5.3 10 20.0 ±10.7 9 18.0 ±6.3
LYM% (%) 9 72.9 ±4.1 8 76.1 ±6.9 10 70.7 ±12.7 9 73.8 ±8.1
MONO% (%) 9 0.9 ±0.3 8 1.1 ±0.3 10 1.2 ±0.5 9 1.4 ±0.6
EOS% (%) 9 3.7 ±2.9 8 5.1 ±3.1 10 7.1 ±3.2 9 5.9 ±2.2
BASO% (%) 9 0.5 ±0.2 8 0.5 ±0.2 10 0.4 ±0.2 9 0.4 ±0.2
RBC (1012/L) 9 8.64 ±0.57 8 8.11 ±0.85 10 8.63 ±0.72 9 7.96 ±0.62
HGB (g/L) 9 138 ±9 8 129 ±14 10 137 ±10 9 128 ±11
HCT (%) 9 43.6 ±2.8 8 41.4 ±4.1 10 43.3 ±3.5 9 40.4 ±2.8
MCV (fL) 9 50.5 ±1.9 8 51.1 ±1.5 10 50.2 ±1.5 9 50.8 ±1.3
MCH (pg) 9 15.9 ±0.6 8 16.0 ±0.3 10 15.9 ±0.5 9 16.1 ±0.4
MCHC (g/L) 9 316 ±7 8 313 ±6 10 316 ±6 9 317 ±10
0.252 ± 0.215 ± 0.275 ±
RET (1012/L) 9 0.197 ±0.030 8 10 9
0.174 0.059 0.184
RET% (%) 9 2.29 ±0.36 8 3.36 ±3.04 10 2.48 ±0.65 9 3.61 ±2.82
PLT (109/L) 9 1068 ±219 8 1132 ±203 10 953 ±213 9 1186 ±322

Notes:
1) Due to the fail in collecting blood sample from animal 1F0103, the sample size of the negative control group was 9;
2) Due to the fail in collecting blood sample from animal 2F0103 and 2F0104, the sample size of 13-JHYM group 1 was 8;
3) Due to the fail in collecting blood sample from animal 4M0102, the sample size of 13-JHYM group 3 was 9;
* The difference in the mean was significant (p≤0.05) when compared with the negative control group.

Table 4 Effects of 13-JHYM on Blood Biochemical Measurements of Kunming Mice
Negative control
13-JHYM group 1 13-JHYM group 2 13-JHYM group 3
Measurement group
n1) ■X ±SD n2) ■X ±SD n ■X ±SD n3) ■X ±SD

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150.9 ± 172.6 ± 152.4 ±
ALP (U/L) 9 191.0 ±69.7 10 10 10
40.9 46.2 37.8
28.2 ±
ALT (U/L) 9 21.0 ±4.2 10 9 32.2 ±13.5 10 19.2 ±3.7
14.0
71.9 ±

AST (U/L) 9 50.8 ±3.5 10 10 79.3 ±63.5 10 52.6 ±11.1
33.9
CK (U/L) 9 292 ±88 10 365 ±272 10 436 ±319 10 304 ±240
5.80 ±
Urea (mmol/L) 9 6.67 ±0.89 10 10 7.49 ±3.79 10 5.69 ±0.97
1.01
Crea (umol/L) 9 9.3 ±3.7 10 6.2 ±1.6 10 10.1 ±8.2 10 6.0 ±1.5
TP (g/L) 9 43.8 ±2.9 10 43.0 ±2.1 10 44.9 ±4.1 10 42.4 ±2.1
ALB (g/L) 9 32.2 ±3.5 10 31.4 ±3.1 10 33.0 ±2.3 10 30.6 ±3.1
2.80 ±
A/G 9 2.93 ±0.91 10 10 2.86 ±0.56 10 2.66 ±0.54
0.71
12.74 ± 11.76 ± 11.19 ±
GLU (mmol/L) 9 11.80 ±1.34 10 10 10
2.31 2.50 1.97
TBIL (umol/L) 9 1.3 ±0.3 10 1.3 ±0.3 10 1.1 ±0.4 10 1.4 ±0.4
CHOL 1.56 ±
9 1.66 ±0.25 10 10 1.66 ±0.30 10 1.62 ±0.25
(mmol/L) 0.32
1.17 ±
TG (mmol/L) 9 1.03 ±0.31 10 10 1.24 ±0.31 10 1.10 ±0.31
0.29
4.88 ±
K+ (mmol/L) 9 4.72 ±0.60 10 10 5.00 ±0.74 10 4.96 ±0.54
0.65
150.0 ±
Na+ (mmol/L) 9 149.4 ±1.0 10 10 150.2 ±2.0 10 149.7 ±1.6
1.3
115.5 ±
Cl- (mmol/L) 9 114.9 ±1.7 10 10 115.2 ±3.3 10 115.0 ±1.1
2.9

Notes:
1) Due to the fail in collecting blood sample from animal 1F0103, the sample size of the negative control group was 9;
2) The ALT value of animal 3M0104 was not include in the statistical analysis, therefore the sample size of 13-JHYM group 2 for ALT measurement was 9.

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Appendix II Summary Tables of Individual Animal Data in Acute Toxicity Test of 13-JHYM through Intramuscular Injection in Kunming Mice

Table 1.1 Individual Data of Mouse Weight in the Negative Control Group (g)
Measurement time

Animal
Group Gender Before
No. Day 4 Day 8 Day 11 Day 15
administration
1F0101 F 24.0 26.8 29.2 30.9 31.1
Negative 1F0102 F 25.0 26.4 29.5 30.8 31.1
control 1F0103 F 25.8 29.2 32.1 34.7 33.8
group 1F0104 F 26.4 29.6 33.0 34.0 32.7
1F0105 F 26.1 29.3 32.7 33.5 33.7
Mean 25.5 28.3 31.3 32.8 32.5
SD 1.0 1.5 1.8 1.8 1.3
Sample
5 5 5 5 5
size

Table 1.1 (Continued Table) Individual Data of Mouse Weight in the Negative Control Group (g)
Measurement time

Animal
Group Gender Before
No. Day 4 Day 8 Day 11 Day 15
administration
1M0101 M 25.3 29.8 35.2 36.3 36.4
Negative 1M0102 M 26.6 30.6 34.7 36.2 38.0
control 1M0103 M 27.2 31.6 36.0 37.4 37.5
group 1M0104 M 27.1 33.8 41.5 43.7 42.6
1M0105 M 28.1 33.1 38.5 39.8 39.2
Mean 26.9 31.8 37.2 38.7 38.7
SD 1.0 1.7 2.8 3.2 2.4
Sample
5 5 5 5 5
size

Table 1.2 Individual Data of Mouse Weight in 13-JHYM Group 1 (g)
Measurement time

Animal
Group Gender Before
No. Day 4 Day 8 Day 11 Day 15
administration
2F0101 F 24.0 25.6 28.2 29.4 28.1
13- 2F0102 F 25.5 29.2 32.3 32.9 31.3
JHYM 2F0103 F 26.0 28.8 31.3 33.1 32.4
Group 1 2F0104 F 26.2 28.3 31.7 32.2 32.8
2F0105 F 26.5 28.8 32.1 33.3 31.2
Mean 25.6 28.1 31.1 32.2 31.2
SD 1.0 1.5 1.7 1.6 1.8
Sample

5 5 5 5
size
Table 1.2 (Continued Table) Individual Data of Mouse Weight in 13-YHJM Group 1 (g)
Measurement time
Animal
Group Gender Before
No. Day 4 Day 8 Day 11 Day 15
administration
2M0101 M 26.5 30.5 34.7 35.8 35.0
13- 2M0102 M 26.8 29.9 33.1 34.6 34.3
JHYM 2M0103 M 27.3 32.7 38.3 41.2 43.7
Group 1 2M0104 M 27.4 31.3 34.7 36.9 37.0
2M0105 M 28.1 30.5 33.8 34.8 35.8

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Mean 27.2 31.0 34.9 36.7 37.2
SD 0.6 1.1 2.0 2.7 3.8
Sample
5 5 5 5
size
Table 1.3 Individual Data of Mouse Weight in 13-JHYM Group 2 (g)
Measurement time
Animal
Group Gender Before
No. Day 4 Day 8 Day 11 Day 15
administration
3F0101 F 24.6 26.3 29.2 29.7 30.1
13- 3F0102 F 24.7 26.7 29.0 29.8 30.8
JHYM 3F0103 F 26.1 28.7 31.2 32.9 32.5
Group 2 3F0104 F 25.9 28.9 31.0 33.0 31.3
3F0105 F 27.3 30.6 33.6 35.1 35.6
Mean 25.7 28.2 30.8 32.1 32.1
SD 1.1 1.8 1.9 2.3 2.2
Sample
5 5 5 5
size

Table 1.3 (Continued Table) Individual Data of Mouse Weight in 13-YHJM Group 2 (g)
Measurement time
Animal
Group Gender Before
No. Day 4 Day 8 Day 11 Day 15
administration
3M0101 M 24.4 28.6 33.1 35.2 35.8
13- 3M0102 M 26.6 32.3 37.7 40.4 40.5
JHYM 3M0103 M 27.1 30.9 35.5 36.8 36.6
Group 2 3M0104 M 27.5 31.9 36.4 37.8 38.7
3M0105 M 27.4 32.5 36.7 38.8 40.0
Mean 26.6 31.2 35.9 37.8 38.3
SD 1.3 1.6 1.7 2.0 2.1
Sample
5 5 5 5
size

Table 1.4 Individual Data of Mouse Weight in 13-JHYM Group 3 (g)
Measurement time
Animal
Group Gender Before
No. Day 4 Day 8 Day 11 Day 15
administration
4F0101 F 24.2 25.7 27.9 30.1 29.3
13- 4F0102 F 24.9 28.1 31.5 34.7 33.9
JHYM 4F0103 F 26.0 30.1 33.5 34.3 32.4
Group 3 4F0104 F 25.9 28.9 31.7 34.4 32.8
4F0105 F 26.5 31.3 32.8 33.8 34.5
Mean 25.5 28.8 31.5 33.5 32.6
SD 0.9 2.1 2.2 1.9 2.0
Sample
5 5 5 5
size

Table 1.4 (Continued Table) Individual Data of Mouse Weight in 13-YHJM Group 3 (g)
Measurement time
Animal
Group Gender Before
No. Day 4 Day 8 Day 11 Day 15
administration
4M0101 M 26.2 28.0 30.4 31.7 32.8
13- 4M0102 M 26.5 29.2 32.4 33.9 35.1
JHYM 4M0103 M 27.0 32.2 36.9 38.7 38.9
Group 3 4M0104 M 26.9 29.3 32.3 33.9 33.8
4M0105 M 27.2 32.0 37.3 39.5 38.8
Mean 26.8 30.1 33.9 35.5 35.9
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SD 0.4 1.9 3.1 3.4 2.8
Sample
5 5 5 5
size
Table 2.1 Individual Data of Female Mouse Food Intake (g)

Measurement time
Group Animal No. Gender
Day 2 Day 9
Negative
1F01 F 9.8 5.7
control group
13-JHYM
2F01 F 8.4 6.0
group 1
13-JHYM
3F01 F 7.1 6.2
group 2
13-JHYM
4F01 F 8.3 7.2
group 3
Table 2.1 (Continued Table) Individual Data of Male Mouse Food Intake (g)
Measurement time
Group Animal No. Gender
Day 2 Day 9
Negative
1M01 M 8.8 8.4
control group
13-JHYM
2M01 M 10.0 8.2
group 1
13-JHYM
3M01 M 7.3 7.4
group 2
13-JHYM
4M01 M 9.0 9.2
group 3

Attached Table 3.1 Individual Data of Hematology in Mice (Negative Control Group, Day 15)

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Attached Table 3.2 Individual Data of Hematology in Mice (13-JHYM Batch Number Group 1, Day 15)

Attached Table 3.3 Individual Data of Hematology in Mice (13-JHYM Batch Number Group 2, Day 15)

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MODULE 4 NONCLINICAL STUDY REPORTS
Version No. Walvax-202007-00 Yuxi Walvax Biotechnology Co., Ltd.

Attached Table 3.4 Individual Data of Hematology in Mice (13-JHYM Batch Number Group 3, Day 15)

Attached Table 4.1 Individual Data of Blood Chemistry in Mice (Negative Control Group, Day 15)

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MODULE 4 NONCLINICAL STUDY REPORTS
Version No. Walvax-202007-00 Yuxi Walvax Biotechnology Co., Ltd.

Attached Table 4.2 Individual Data of Blood Chemistry in Mice (13-JHYM Batch Number Group 1, Day 15)

Attached Table 4.3 Individual Data of Blood Chemistry in Mice (13-JHYM Batch Number Group 2, Day 15)

Attached Table 4.4 Individual Data of Blood Chemistry in Mice (13-JHYM Batch Number Group 3, Day 15)

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MODULE 4 NONCLINICAL STUDY REPORTS
Version No. Walvax-202007-00 Yuxi Walvax Biotechnology Co., Ltd.

4.2.3.7.2 Abnormal Toxicity Test
Experiment Report

Study Name Study on Abnormal Toxicity of 13-JHYM by Intraperitoneal Injection in Kunming Mice

Study No. A201118-O001-01

Name of study facility WestChina-Frontier PharmaTech Co., Ltd.
(National Chengdu Center for Safety Evaluation of Drugs)

Address of study facility No. 28 Gaopeng Avenue, High-tech Development Zone, Chengdu, Sichuan

Client Name Yunnan Walvax Biotechnology Co., Ltd.

Address of client 4F, Building A3, Phase II, Yunnan National University Science Park, High-tech Zone, Kunming

Study Director Guo Wei

Main members of study team Zhu Xia, Kang Yufan, Yi Bing, Zheng Wulin

Date of study initiation and August 2011 - October 2011
completion:

Location for storing initial Archives Room, WestChina-Frontier PharmaTech Co., Ltd.
data

WestChina-Frontier PharmaTech Co. (WCFP)
National Chengdu Center for Safety Evaluation of Drugs (NCCSED)

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Version No. Walvax-202007-00 Yuxi Walvax Biotechnology Co., Ltd.

Experiment Report Confirmation

Study Name: Study on Abnormal Toxicity of 13-JHYM by Intraperitoneal Injection in Kunming Mice
Study No.: A201118-O001-01

Study Director: Guo Wei

Signature: Date: October 8, 2011

Facility Director: Wang Li

Signature: Date: October 9, 2011

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MODULE 4 NONCLINICAL STUDY REPORTS

Version No. Walvax-202007-00 Yuxi Walvax Biotechnology Co., Ltd.

Abstract

The 13-valent pneumococcal polysaccharide conjugate vaccine (code: 13-JHYM) is prepared by adsorption by aluminum phosphate adjuvant after covalent binding between types 1, 3, 4, 5, 6A, 6B, 7F, 9V, 14, 18C, 19A, 19F and 23F pneumococcal capsular polysaccharides antigen and tetanus toxoid carrier protein. It is clinically intended for the prevention of invasive infections caused by the above serotypes of pneumococci in infants aged 2 months to 2 years and children. The intended clinical dose is 0.5 mL/time/person, and the intended route of administration is intramuscular injection.

WestChina-Frontier PharmaTech Co., Ltd. has carried out a study on abnormal toxicity of 13-JHYM provided by Yunnan Walvax Biotechnology Co., Ltd. via intraperitoneal injection in mice in accordance with the relevant requirements for new drug application, so as to check whether the test article is contaminated by exogenous substances and whether there is any unexpected risk factor about safety, and to provide a reference for safety assessment on clinical application. In this study, 1 dose group (0.5 mL/animal) was set up based on the intended clinical concentration for the test article 13-JHYM, with 5 male mice in total. Each mouse was intraperitoneally injected with 0.5 mL of the test article at one time, and the appearance and signs, general behavior and activity, mental status, respiration, genitalia and crissum, feces and urine, moribundity or death, and other conditions of the animals were observed and recorded once a day after administration for a total of 7 days. Then the body weight was measured at the end of the observation on day 7.

Study results:

General Conditions

During the period from post-administration to the end of observation, each mouse was normal in terms of the appearance and signs, mental and motor status with natural holes all clean and no obvious abnormal reaction symptoms.

Body Weight

On Day 7 after administration, the body weight of each mouse increased compared with that before administration.

In conclusion, under the conditions of this study, the study on abnormal toxicity of 13-JHYM by single intraperitoneal injection of 0.5 mL/animal (1 dose) in Kunming mice is qualified.

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MODULE 4 NONCLINICAL STUDY REPORTS

Version No. Walvax-202007-00 Yuxi Walvax Biotechnology Co., Ltd.

1 GENERAL INFORMATION

1.1 NAME AND NO. OF THE STUDY

Study name: Study on Abnormal Toxicity of 13-JHYM by Intraperitoneal Injection in Kunming Mice

Study No.: A201118-O001-01

1.2 OBJECTIVE

To carry out a study on abnormal toxicity of 13-JHYM by intraperitoneal injection and observe for 7 days to check whether the test article is contaminated by exogenous substances and whether there are unexpected unsafe factors, and to provide a reference for evaluating the safety of clinical application.

1.3 STUDY FACILITY

Name: WestChina-Frontier PharmaTech Co., Ltd. (National Chengdu Center for Safety Evaluation of Drugs)

Address: No. 28, Gaopeng Avenue, High-tech Development Zone, Chengdu, Sichuan

Postal code: 610041

Contact: Cen Xiaobo

Tel.: 028-85178767

Fax: 028-85173043

Email: xbcen@glpcd.commailto:xbcen@glpcd.com

1.4 CLIENT

Name: Yunnan Walvax Biotechnology Co., Ltd.

Address: 4F, Building A3, Phase II, Yunnan National University Science Park, High-tech Zone, Kunming

Postal code: 650106

Contact: Wu Kai

Tel.: 0871-8351253-8851

Fax: 0871-8312123

1.5 MEMBERS OF STUDY TEAM

1.5.1 STUDY DIRECTOR

Study Director: Guo Wei

Tel: 028-85154334-551

Fax: 028-85173043

Email: wguo@glpcd.com

1.5.2 MEMBERS OF STUDY TEAM:

Study operators: Zhu Xia, Kang Yufan

Veterinarian: Yi Bing

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MODULE 4 NONCLINICAL STUDY REPORTS

Version No. Walvax-202007-00 Yuxi Walvax Biotechnology Co., Ltd.

Animal feeding and management: Gou Xiankui

Management and preparation of test article: Zheng Wulin

2 MATERIALS

2.1 LABORATORY ANIMALS

2.1.1 GENERAL INFORMATION

Species: Kunming mice;

Grade: SPF;

Quantity and sex of animals purchased: 16, male;

Quantity and sex of animals used: 5, male;
Body weight: 21.1~23.8 g;
Source: Beijing Vital River Lab Animal Technology Co., Ltd., production license No.: SCXK (J) 2006-0005. Quality certificate No.: 0226898, issued by: Beijing Laboratory Animal Management Office;
Animal house entry time: August 10, 2011;
Acclimation period after entering animal house: Kunming mice that entered the animal house were acclimatized to the environment for 7 days (August 10, 2011-August 16, 2011) before the study and were examined accordingly. All the mice that entered the animal house were healthy animals and were included in the formal study. Main examination items during the acclimation period: if the quality indicators are consistent with that required upon ordering; general status of animals: if the body weight reaches the weight range required by the study.

2.1.2 ANIMAL HOUSING CONDITIONS

2.1.2.1 HOUSING PLACE

SPF Animal House (Building 1) of WestChina-Frontier PharmaTech Co., Ltd. (No. of license for use of laboratory animal house: SYXK (C) 2009-123).

2.1.2.2 HOUSING CONDITIONS

Type of housing cage: Plastic mouse cage (top L x W: 295 mm x 175 mm, bottom: L x W: 270 mm x 155 mm, H: 140 mm);

Padding: Corncob, supplied by Beijing Keao Xieli Feed Co., Ltd.;

Housing density: ≤5 animals/cage

2.1.2.3 HOUSING ENVIRONMENTAL CONDITIONS

Standard for housing environmental conditions: National Standard of the People's Republic of China GB14925-2001;

Housing environmental control system: Honeywell EBI400 automatic control all-fresh air central air conditioning system;

Temperature: room temperature 21.84 ~ 23.63 °C (daily temperature difference 0.55 ~ 1.18 °C);

Humidity: relative humidity 47.28 ~ 61.92%;

Illumination: Artificial lighting with a 12 hour light/dark cycle (lights on at 07:30 and off at 19:30);

Ventilation rate: 10~20 times/h, 100% all fresh air.

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2.1.2.4 FEED

Type: Rat complete pellet feed sterilized by cobalt-60 irradiation;

Manufacturer: Shanghai SLAC Laboratory Animal Co., Ltd. No. of certificate for inspection of feed production enterprise: HSS (2008) 04002;

Batch No.: M02-F-20110506005, M02-F-20110713006;

Feeding method: Ad libitum;

Test of nutritional ingredients: Nutritional ingredient test result: Provided by the feed supplier.

Test reports for conventional nutritional ingredient indicators (crude protein, crude fat, crude fiber, crude ash, moisture, calcium and phosphorus) were provided for each batch; test reports for amino acid indicators (threonine, cystine + methionine, valine, isoleucine, leucine, tyrosine, phenylalanine, histidine, lysine and arginine) were provided at least once a year (refer to the national standard of the People's Republic of China GB14924.3-2001);

Confirmation of contaminant content of feed: Provided by the feed supplier. Chemical contaminant indicators such as arsenic, lead, cadmium, mercury, BHC, DDT, aflatoxin B1, total bacterial count, coliform group, mold and yeast count and pathogenic bacteria (salmonella) were provided at least once a year (refer to the national standard of the People's Republic of China GB14924.2-2001).

2.1.2.5 DRINKING WATER

Type: Drinking water for laboratory animals (reverse osmosis water)

Water supply method: Supplied in drinking bottle with water available ad libitum;

Test of total bacterial count: Refer to the national standard of the People's Republic of China GB5749-2006. Samples were taken for self-test (1 time/week).

Test of conventional indexes of water quality: Refer to the national standard of the People's Republic of China GB5749-2006. Samples should be taken and sent to the unit with relevant qualifications for test at least twice a year.

2.1.3 ANIMAL IDENTIFICATION METHOD

Acclimation period and study period: Cage card and tail mark were used as animal identification marks;

Marking method of cage card:

Acclimation period: White cage cards were uniformly used, and the study number, animal species, cage number, number of animals, animal house entry time, remark information and other information were indicated;

After grouping: Red cage cards were used for 13-JHYM group, with the study number, animal species, cage number, treatment factors, number of animals, expected starting and ending time of the study, remark information and other information indicated on the cage cards.

2.2 TEST ARTICLE

2.2.1 INFORMATION OF TEST ARTICLE

Name or designation: 13-valent pneumococcal polysaccharide conjugate vaccine (code: 13-JHYM);

Source: Yunnan Walvax Biotechnology Co., Ltd.;

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Version No. Walvax-202007-00 Yuxi Walvax Biotechnology Co., Ltd.

Description: Suspension with opalescence;

Strength: 0.5 mL: 32.2 µg/vial;

Batch No.: 20110304;

Storage conditions and precautions: Store at 2 ~ 8°C;

Shelf life: Expired in February 2013;

Place of storage: Test Article Management Department of WestChina-Frontier PharmaTech Co., Ltd.

2.2.2 PREPARATION OF TEST SOLUTION

Preparation method: The bulk can be directly used.

Identification method after preparation: Red label was used for identification, and the study number, name and concentration of test substance, volume, storage condition, shelf life, responsible person and preparation date were indicated.

Temporary storage conditions and shelf life after preparation: Temporarily stored in the ice box

and used within 24 h.

2.2.3 TREATMENT OF RETURNED AND REMAINING TEST ARTICLES

The test articles returned after administration were autoclaved and then poured into the waste liquid bucket. The retained samples were stored at the storage room of test articles in accordance with the SOP Storage of Test Articles. After completion of the test items, the remaining test articles were returned to the Client.

2.3 MAIN INSTRUMENTS AND DEVICES REQUIRED FOR THE STUDY

Electronic balance, syringe, etc.

3 STUDY METHOD AND OBSERVATION INDICATORS

3.1 ADMINISTRATION

Five mice weighing 21 ~ 24 g were used. Each mouse was intraperitoneally injected with 0.5 mL (1 dose) of the test article at one time.

3.2 OBSERVATION

The appearance and signs, general behavior and activities, mental status, respiration, genitalia and crissum, feces and urine, moribundity or death and other conditions of the animals were observed and recorded once a day after administration for a total of 7 days.

3.3 BODY WEIGHT

Body weight was measured once before administration, and once on Day 7 after administration.

4 DETERMINATION OF RESULTS

The study can be judged as qualified if no abnormal reaction is observed during the observation period, and the body weight at the end of the observation period is higher than that before administration for each animal.

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Version No. Walvax-202007-00 Yuxi Walvax Biotechnology Co., Ltd.

5 RESULTS

5.1 GENERAL CONDITIONS

During the period from post-administration to the end of observation, each mouse was normal in terms of the appearance and signs, mental and motor status with natural holes all clean and no obvious abnormal reaction symptoms.

5.2 EFFECT OF 13-JHYM ON THE BODY WEIGHT OF MOUSE

On Day 7 after administration, the body weight of each mouse increased compared with that before administration (Table 1)

Table 1 Effect of 13-JHYM on the Body Weight of Mouse (g)

Test Time

Animal No.

Before administration Day 7 after administration

1M0101 21.7 24.8

1M0102 21.1 24.2

1M0103 23.1 25.3

1M0104 22.8 26.4

1M0105 23.8 28.9

Note: The first number of the animal number represents the group (1 represents 13-JHYM group). The second letter represents sex (M indicates male), the third and fourth numbers represent animal cage number, and the

fifth and sixth numbers represent animal serial number.

6 CONCLUSION

In conclusion, under the conditions of this study, the study on abnormal toxicity of 13-JHYM by single intraperitoneal injection of 0.5 mL/animal (1 dose) in Kunming mice is qualified.

7 LOCATION FOR STORING INITIAL DATA

Place of storage: Archives Room, WestChina-Frontier PharmaTech Co., Ltd.;

Storage conditions: Conventional conditions;

Retention period: Paper archives should be kept at least 5 years after the drug is marketed, and electronic files should be permanently kept;

Tel.: 028-85154334-511;

Contact: Chang Ling.

8 KEY REFERENCES

Good Laboratory Practice, No. 2 Order of State Food and Drug Administration, September 1, 2003

Provisions for Drug Registration, No. 28 Order of State Food and Drug Administration, October 1, 2007

General Principles for Pre-clinical Safety Technical Review of Preventive Biological Products,

State Food and Drug Administration, December 2005

Chinese Pharmacopoeia Volume III, Chinese Pharmacopoeia Commission, 2010.

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MODULE 4 NONCLINICAL STUDY REPORTS

Version No. Walvax-202007-00 Yuxi Walvax Biotechnology Co., Ltd.

Experiment Report

Study title Study on Abnormal Toxicity of 13-JHYM by Intraperitoneal Injection in Guinea Pigs

Study No. A201118-O002-01

Name of study facility WestChina-Frontier PharmaTech Co., Ltd.
(National Chengdu Center for Safety Evaluation of Drugs)

Address of study facility No. 28 Gaopeng Avenue, High-tech Development Zone, Chengdu, Sichuan

Client Name Yunnan Walvax Biotechnology Co., Ltd.

Address of client 4F, Building A3, Phase II, Yunnan National University Science Park, High-tech Zone, Kunming

Study Director Guo Wei

Main members of study Mu Ke, Huang Xuemei, Yuan Hong, Wang Xi, Xu Quanli, Zheng Wulin, Yi team Bing

Date of study initiation August 2011 - October 2011
and completion:

Location for storing initial Archives Room, WestChina-Frontier PharmaTech Co., Ltd.
data

WestChina-Frontier Pharma Tech Co.(WCFP)
National Chengdu Center for Safety Evaluation of Drugs(NCCSED)

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Version No. Walvax-202007-00 Yuxi Walvax Biotechnology Co., Ltd.

Experiment Report Confirmation

Study title: Study on Abnormal Toxicity of 13-JHYM by Intraperitoneal Injection in Guinea Pigs
Study No.: A201118-O002-01

Study Director: Guo Wei

Signature: Date: October 8, 2011

Facility Director: Wang Li

Signature: Date: October 9, 2011

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MODULE 4 NONCLINICAL STUDY REPORTS

Version No. Walvax-202007-00 Yuxi Walvax Biotechnology Co., Ltd.

Abstract

The 13-valent pneumococcal polysaccharide conjugate vaccine (code: 13-JHYM) is prepared by adsorption by aluminum phosphate adjuvant after covalent binding between types 1, 3, 4, 5, 6A, 6B, 7F, 9V, 14, 18C, 19A, 19F and 23F pneumococcal capsular polysaccharides antigen and tetanus toxoid carrier protein. It is clinically intended for the prevention of invasive infections caused by the above serotypes of pneumococci in infants aged 2 months to 2 years and children. The intended clinical dose is 0.5 mL/time/person, and the intended route of administration is intramuscular injection.

WestChina-Frontier PharmaTech Co., Ltd. has carried out a study on abnormal toxicity of 13-JHYM provided by Yunnan Walvax Biotechnology Co., Ltd. by intraperitoneal injection in Guinea Pigs in accordance with the relevant requirements for new drug application, so as to check whether the test article is contaminated by exogenous substances and whether there is any unexpected risk factor about safety, and to provide a reference for safety assessment on clinical application.

In this study, 1 dose group of 5 mL/animal (10 doses) was set up based on the intended clinical concentration for the test article 13-JHYM, with 6 guinea pigs in total of half males and half females. Each guinea pig was intraperitoneally injected with 5 mL of the test article at one time, and the appearance and signs, general behavior and activities, mental status, respiration, genitalia and crissum, feces and urine, moribundity or death, and other conditions of the animals were observed and recorded once a day after administration for a total of 7 days.

Then the body weight was measured at the end of the observation on day 7. Study results:

General Conditions

During the period from post-administration to the end of observation, each guinea pig was normal in terms of the appearance and signs, mental and motor status with natural holes all clean and no obvious abnormal reaction symptoms.

Body Weight

On Day 7 after administration, the body weight of each guinea pig increased compared with that before administration.

In conclusion, under the conditions of this study, the study on abnormal toxicity of 13-JHYM by single intraperitoneal injection of 5 mL/animal (10 doses) in guinea pig is qualified.

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MODULE 4 NONCLINICAL STUDY REPORTS

Version No. Walvax-202007-00 Yuxi Walvax Biotechnology Co., Ltd.

1 GENERAL INFORMATION

1.1 NAME AND NO. OF THE STUDY

Study name: Study on Abnormal Toxicity of 13-JHYM by Intraperitoneal Injection in Guinea Pigs

Study No.: A201118-O002-01

1.2 OBJECTIVE

To carry out a study on abnormal toxicity of 13-JHYM by intraperitoneal injection and conduct observation for 7 days to check whether the test article is contaminated by exogenous substances and whether there is any unexpected risk factor about safety, and to provide a reference for safety assessment on clinical application.

1.3 STUDY FACILITY

Name: WestChina-Frontier PharmaTech Co., Ltd. (National Chengdu Center for Safety Evaluation of Drugs)

Address: No. 28 Gaopeng Avenue, High-tech Development Zone, Chengdu, Sichuan

Postal code: 610041

Contact: Cen Xiaobo

Tel.: 028-85178767

Fax: 028-85173043

Email: xbcen@glpcd.com

1.4 CLIENT

Name: Yunnan Walvax Biotechnology Co., Ltd.

Address: 4F, Building A3, Phase II, Yunnan National University Science Park, High-tech Zone, Kunming

Postal code: 650106

Contact: Wu Kai

Tel.: 0871-8351253-8851

Fax: 0871-8312123

1.5 MEMBERS OF THE STUDY TEAM

1.5.1 STUDY DIRECTOR

Name: Guo Wei

Tel: 028-85154334-551

Fax: 028-85173043

Email: wguo@glpcd.com

1.5.2 MEMBERS OF STUDY TEAM:

Study operators: Mu Ke, Huang Xuemei, Yuan Hong, Wang Xi, Xu Quanli

Veterinarian: Yi Bing

Animal housing and management: Liao Qing

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MODULE 4 NONCLINICAL STUDY REPORTS

Version No. Walvax-202007-00 Yuxi Walvax Biotechnology Co., Ltd.

Management and preparation of test article: Zheng Wulin

2 MATERIALS

2.1 LABORATORY ANIMALS

2.1.1 GENERAL INFORMATION

Species: British guinea pig;

Class: conventional;

Quantity and sex of animals used: 6, females and males;

Body weight: 274.5~355.4g;

Source: Farm of Sichuan Laboratory Animal Specialized Committee, laboratory animal

production license No.: SCXK(C)2008-014.
Quality certificate No.: No.0019271, issued by: Science & Technology Department of Sichuan Province;
Animal house entry time: August 16, 2011;
Acclimation period after entering animal house: The guinea pigs were acclimated to the environment for 14 days before the study (August 16, 2011~August 29, 2011), and healthy guinea pigs were selected to enter the formal study. Main inspection items during the acclimation period: General state of animals, body weight, etc.
2.1.2 ANIMAL HOUSING CONDITIONS
2.1.2.1 HOUSING PLACE
General Animal House (Building 1) of WestChina-Frontier PharmaTech Co., Ltd. (No. of license for use of laboratory animal house: SYXK (C) 2009-123).
2.1.2.2 HOUSING CONDITIONS
Type of housing cage: guinea pig cage (L×W×H: 515 mm × 400 mm × 260 mm)
Housing density: ≤3 animals/cage
2.1.2.3 HOUSING ENVIRONMENTAL CONDITIONS
Standard for housing environmental conditions: National Standard of the People's Republic of China GB14925-2001;
Housing environmental control system: Honeywell EBI400 automatic control all-fresh air central air conditioning system;
Temperature: room temperature 23.25 ~ 25.95°C (daily temperature difference 0.14 ~ 1.7°C);
Humidity: relative humidity 43.33 ~ 59.47%;
Illumination: Artificial lighting with a 12 hour light/dark cycle (lights on at 07:30 and off at 19:30);
Ventilation rate: 8~10 times/h, 100% all fresh air.
2.1.2.4 FEED
Type: pellet feeds for guinea pigs
Manufacturer: Beijing Keao Xieli Feed Co., Ltd. Production license No. of feed: SCXK (J) 2009-0012;

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Batch No.: 11030551;
Feeding method: Ad libitum;
Test of nutritional ingredients: Provided by the feed supplier. Test reports for conventional nutritional ingredient indicators (crude protein, crude fat, crude fiber, crude ash, moisture, calcium and phosphorus) were provided for each batch (refer to the national standard of the People's Republic of China GB14924.5-2001);
Confirmation of contaminant content of feed: Provided by the feed supplier, chemical contaminant indexes such as arsenic, lead, cadmium, mercury, BHC, DDT, aflatoxin B1, total bacterial count, coliform group, mold and yeast count and pathogenic bacteria (salmonella) should be provided at least once a year (refer to the national standard of the People's Republic of China GB14924.2-2001)
2.1.2.6 DRINKING WATER
Type: Drinking water for laboratory animals (reverse osmosis water)
Water supply method: Supplied in drinking bottle with water available ad libitum;
Test of total bacterial count: Refer to the national standard of the People's Republic of China GB5749-2006. Samples were taken for self-test (1 time/week).
Test of conventional indexes of water quality: Refer to the national standard of the People's Republic of China GB5749-2006. Samples should be taken and sent to the unit with relevant qualifications for test at least twice a year.
2.1.3 ANIMAL IDENTIFICATION METHOD
Acclimation period: Cage cards and marks of red marker were used as animal identification marks;
Study period: Cage cards and ear marks were used as the animal identification marks;
Marking method of cage card:
Acclimation period: White cage cards were uniformly used, and the study number, animal species, animal serial number, animal house entry time, remark information and other information were indicated;
After grouping: Red cage cards were used for 13-JHYM group, with the study number, animal species, cage number, treatment factors, number of animals, expected starting and ending time of the study, remark information and other information indicated on the cage cards;

2.2 TEST ARTICLE

2.2.1 INFORMATION OF TEST ARTICLE

Name or designation: 13-valent pneumococcal polysaccharide conjugate vaccine (code: 13-JHYM);
Source: Yunnan Walvax Biotechnology Co., Ltd.;
Description: Suspension with opalescence;
Strength: 0.5 mL; 32.2 µg/vial;
Batch No.: 20110304;
Storage conditions and precautions: Store at 2 ~ 8°C;
Shelf life: Expired in February 2013;
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Place of storage: Test Article Management Department of WestChina-Frontier PharmaTech Co., Ltd.

2.2.2 PREPARATION OF TEST SOLUTION

Preparation method: The bulk can be directly used.
Identification method after preparation: Red label was used for identification, and the study number, name and concentration of test substance, volume, storage condition, shelf life, responsible person and preparation date were indicated.
Temporary storage conditions and shelf life after preparation: Temporarily stored in the ice box and used within 24 h.

2.2.3 TREATMENT OF RETURNED AND REMAINING TEST ARTICLES

The test articles returned after administration were autoclaved and then poured into the waste liquid bucket. The retained samples were stored at the storage room of test articles in accordance with the SOP Storage of Test Articles. After completion of the test items, the remaining test articles were returned to the Client.

2.3 MAIN INSTRUMENTS AND DEVICES REQUIRED FOR THE STUDY

Syringe, electronic balance, etc.

3 STUDY METHOD AND OBSERVATION INDICATORS

3.1 ADMINISTRATION

6 guinea pigs weighing 274.5–355.4 g were selected (4 guinea pigs were used to repeat the test once again since the results for 2 guinea pigs used in the 1st test were unqualified at the end of observation period), and each guinea pig was intraperitoneally injected with 5 mL (10 doses) of the test article at one time.

3.2 OBSERVATION

The appearance and signs, general behavior and activities, mental status, respiration, genitalia and crissum, feces and urine, moribundity or death and other conditions of the animals were observed and recorded once a day after administration for a total of 7 days.

3.3 WEIGHT

Body weight was measured once before administration, and once on Day 7 after administration.

4 DETERMINATION OF RESULTS

The study can be judged as qualified if no abnormal reaction is observed during the observation period, and the body weight at the end of the observation period is higher than that before administration for each animal.

5 RESULTS

5.1 GENERAL CONDITIONS

During the period from post-administration to the end of observation, each guinea pig was normal in terms of the appearance and signs, mental and motor status with natural holes all clean and without obvious abnormal reaction symptoms.

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5.2 EFFECT OF 13-JHYM ON THE BODY WEIGHT OF GUINEA PIG

5.2.1 THE 1ST TEST

On Day 7 after administration, the body weight of one guinea pig decreased compared with that before administration (Table 1).

Table 1 Effect of 13-JHYM on the Body Weight of Guinea Pig (g)

Test Time

Animal No.

Before administration Day 7 after administration

1M0101 293.9 258.6

1M0102 274.5 305.2

Remarks: The first number of the animal number represents the group (1 represents 13-JHYM group). The second letter represents sex (M indicates male), the third and fourth numbers represent animal cage number, and the fifth and sixth numbers represent animal serial number.

5.2.2 THE 2ND TEST

On Day 7 after administration, the body weight of each guinea pig increased compared with that before administration (Table 2).

Table 2 Effect of 13-JHYM on the Body Weight of Guinea Pig (g)

Test Time

Animal No.

Before administration Day 7 after administration

1M0103 355.4 408.0

1M0104 310.9 361.8

1F0101 275.7 300.4

1F0102 291.3 298.5

Remarks: The first number of the animal number represents the group (1 represents 13-JHYM group). The second letter represents sex (M indicates male, F indicates female), the third and fourth numbers represent animal cage number, and the fifth and sixth numbers represent animal serial number.

6 CONCLUSION

In conclusion, under the conditions of this study, the study on abnormal toxicity of 13-JHYM by single intraperitoneal injection of 5 mL/animal (10 doses) in guinea pig is qualified.

7 LOCATION FOR STORING INITIAL DATA

Place of storage: Archives Room, WestChina-Frontier PharmaTech Co., Ltd.;

Storage conditions: Conventional conditions;

Retention period: Paper archives should be kept at least 5 years after the drug is marketed, and electronic files should be permanently kept;

Tel.: 028-85154334-511;

Contact: Chang Ling.

8 KEY REFERENCES

Good Laboratory Practice, No. 2 Order issued by State Food and Drug Administration, September 01, 2003

Provisions for Drug Registration, No. 28 Order of State Food and Drug Administration, October 1, 2007

General Principles for Pre-clinical Safety Technical Review of Preventive Biological Products, State Food and Drug Administration, December 2005

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Chinese Pharmacopoeia (Volume III), Chinese Pharmacopoeia Commission, 2010.

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4.2.3.7.7.3. Specific Toxicity Test

Experiment Report

Experiment Name

Specific Toxicity Test of JHYM through

Subcutaneous Injection in Guinea Pigs

Experiment Number A200932-O006

Research Institution Name WestChina-Frontier PharmaTech Co. (National

Chengdu Center for Safety Evaluation of Drugs)

Research Institution Address No. 28 Gaopeng Avenue, High & New Tech Dev.
Zone, Chengdu, Sichuan
Client Name Yunnan Walvax Biotechnology Co., Ltd.
Client Address 4th Floor, building A3, Phase 2 Science and
Technology Park of Yunnan National University,
Kunming, Yunnan
Contact Person Kai Wu
Telephone 0871-8351253-8851
Experiment Director Wei Guo
Members of Experiment Group Chunyan Zhang (Experiment Operation)
Xuejun Liu (Experiment Operation)
Ting Li (Experiment Operation)
Qin Zhang (Test articles Management)
Fengcai Zhang (Test articles Management)
Qing Lan (Veterinarian)
Duration of Experiment December, 2009-March, 2010
Original Data Depository Archives Room of WestChina-Frontier
PharmaTech Co.

WestChina-Frontier PharmaTech Co. (WCFP)
National Chengdu Center for Safety Evaluation of Drugs (NCCSED)

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Compliance Statement

Experiment Number: A200932-O006

Specific Toxicity Test of JHYM Through Subcutaneous Injection in
Experiment Name:
Guinea Pigs
Test Article: JHYM

Experiment Director: Wei Guo

This experiment was completed in accordance with Non-clinical Good Laboratory Practice formulated by China Food and Drug Administration (August 2003). The experiment procedure was carried out in accordance with the experiment protocol and the relevant standard operating procedure of the research institution. No deviation that affected the reliability of experiment results and conclusions was observed. This report faithfully reflects the experiment materials, experiment methods and experiment results.

Signature of Experiment Director: Date: 17/03/2010

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Version No. Walvax-202007-00 Yuxi Walvax Biotechnology Co., Ltd.

Abstract

JHYM is a prophylactic vaccine manufactured by Yunnan Walvax Biotechnology Co., Ltd. It is used to prevent pneumonia, meningitis, otitis media and other diseases in infants and children aged 2 months to 2 years old and 2 to 9 years old. The clinical proposed dosage is 0.5 mL per person and is injected intramuscularly.

In accordance with the relevant requirements of new drug declaration, the specific toxicity test of JHYM provided by Yunnan Walvax Biotechnology Co., Ltd. was carried out through subcutaneous injection in guinea pigs by WestChina-Frontier PharmaTech Co., with a purpose to observe whether the animals have tetanus symptoms or local reactions at the injection site such as suppuration and necrosis, and to provide a reference for evaluating the safety of clinical use.

A total of 9 groups were set in this experiment and each group included 4 guinea pigs (half male and half female). The guinea pigs in each group were injected with the corresponding serotypes of JHYM (pneumococcus bulk of serotype 1, 4, 5, 6B, 9V, 14, 18C, 19F, and 23F). Each guinea pig was subcutaneously injected with 2 ml of the corresponding sample at multiple points. The animals were observed for a total of 21 days, and the appearance of tetanus symptoms or local reactions at the injection site such as suppuration and necrosis were observed on the 7th, 14th and 21st day after administration. The body weight was measured 7, 14, and 21 days before and after administration. The results are described below:

General Conditions

Each Guinea pig was in normal physical and mental state, and the natural holes were clean. No tetanus symptoms such as muscle spasm, convulsion, and opisthotonos were observed. No local reactions at the injection site such as suppuration and necrosis were observed.

Body Weight

On the 21st day after administration, the body weight of each guinea pig increased normally compared with that obtained before administration.

In conclusion, the specific toxicity test of JHYM (pneumococcus bulk of serotype 1, 4, 5, 6B, 9V, 14, 18C, 19F and 23F) through single subcutaneous injection in guinea pigs was up to standard.

In conclusion, the specific toxicity test of JHYM (pneumococcus bulk of serotype 1, 4, 5, 6B, 9V, 14, 18C, 19F and 23F) through single subcutaneous injection in guinea pigs was up to standard.

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1. OBJECTIVES

In accordance with the relevant requirements for new drug declaration predefined by the Provisions for Drug Registration, referring to the General Principles for Technical Review of Non-Clinical Safety of Biological Products for Preventive use (December 2005) and Pharmacopoeia of the People's Republic of China (2005), the specific toxicity test of carrier protein (tetanus toxoid) through subcutaneous injection in guinea pigs was carried out, with a purpose to observe whether the animals have tetanus symptoms or local reactions at the injection site such as suppuration and necrosis, and to provide a reference for evaluating the safety of clinical use.

2. MATERIALS

2.1 TEST ARTICLES

Pneumococcus serotype 1 polysaccharide conjugate bulk, suspension with opalescence, batch No. 20090501, 10 mL per vial, valid for 5 years;

Pneumococcus serotype 4 polysaccharide conjugate bulk, suspension with opalescence, batch No. 20090501, 10 mL per vial, valid for 5 years;

Pneumococcus serotype 5 polysaccharide conjugate bulk, suspension with opalescence, batch No. 20090501, 10 mL per vial, valid for 5 years;

Pneumococcus serotype 6B polysaccharide conjugate bulk, suspension with opalescence, batch No. 20090501, 10 mL per vial, valid for 5 years;

Pneumococcus serotype 9V polysaccharide conjugate bulk, suspension with opalescence, batch No. 20090501, 10 mL per vial, valid for 5 years;

Pneumococcus serotype 14 polysaccharide conjugate bulk, suspension with opalescence, batch No. 20090501, 10 mL per vial, valid for 5 years;

Pneumococcus serotype 18C polysaccharide conjugate bulk, suspension with opalescence, batch No. 20090501, 10 mL per vial, valid for 5 years;

Pneumococcus serotype 19F polysaccharide conjugate bulk, suspension with opalescence, batch No. 20090501, 10 mL per vial, valid for 5 years;

Pneumococcus serotype 23F polysaccharide conjugate bulk, suspension with opalescence, batch No. 20090501, 10 mL per vial, valid for 5 years.

The above test articles were provided by Yunnan Walvax Biotechnology Co., Ltd.

Sample preparation: Immediate access when use. Test articles were kept in ice box and out of light until the end of administration.

The reserved samples were deposited in the Samples Management Department of WestChina-Frontier PharmaTech Co. The remaining samples after administration were sterilized by high temperature and high pressure and diluted with a large amount of water before being discharged into the sewer. The remaining samples at the end of the experiment were handled according to the client's opinions.

2.2 EXPERIMENT ANIMALS

A total of 36 normal healthy British guinea pigs were included in this experiment (half male and

half female, 268.0-348.9g), provided by Sichuan Laboratory Animal Special Committee farm (Production License No. SCXK (Sichuan) 2008-014). Entry Date: December 1st, 2009. The guinea pigs were reared in the ordinary animal house of WestChina-Frontier PharmaTech Co. (Laboratory

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Animal Room License No. SYXK (Sichuan) 2009-123). The animals were mated and reared in stainless steel cages for guinea pigs (rearing density: 3 animals per cage). The EBI400 automatic control fresh air conditioning system produced by Honeywell was adopted to ensure that the feeding conditions in line with the National Standard of the People's Republic of China GB14925-2001: room temperature 21±5℃ (with a diurnal temperature range within 4℃), relative humidity 55±15%, 12 hours of alternating light and dark, and ventilate 8-10 times per hour. The guinea pigs were fed with pellet feed provided by Beijing Ke'ao Xieli Feed Co., Ltd. (Batch No. 091016, 091202). The animals were allowed to ingest food and drink water freely with 0.1g Vitamin C per 500 mL per day adding in the drinking water. The animals were fed with fresh vegetables in the meantime.

The animals were distinguished from each other by cage number and individual number (ear number method).

3. METHODS AND OBSERVATION MEASUREMENTS

3.1 ADMINISTRATION

A total of 36 guinea pigs were included (250-350g). Each guinea pig was subcutaneously injected with 2 mL sample at multiple points (no more than 1 mL per points).

3.2 OBSERVATION

The animals were observed for a total of 21 days. The appearance of tetanus symptoms or local reactions at the injection site such as suppuration and necrosis were observed once respectively on the 7th, 14th and 21st day after administration.

3.3 BODY WEIGHT

The body weight of guinea pigs was measured once before administration and once respectively 7, 13, 21 days after administration.

4. RESULTS DETERMINATION

At the end of observation, animals with no tetanus symptoms, no suppuration or necrosis at the injection site and whose body weight increased compared with that obtained before administration were determined as qualified.

5. DEVIATIONS

No deviation that could affect the results of this experiment was observed.

6. RESULTS

6.1 GENERAL CONDITIONS

Each Guinea pig was in normal physical and mental state, and the natural holes were clean. No tetanus symptoms such as muscle spasm, convulsion, and opisthotonos were observed. No local reactions at the injection site such as suppuration and necrosis were observed.

6.2 EFFECTS ON BODY WEIGHT

On the 21st day after administration, the body weight of each guinea pig increased normally compared with that obtained before administration.

Table 1 Effects of JHYM on Body Weight of Guinea Pigs (g)
Time of Measurement

Group Before 7 days after 14 days after 21 days after
administration administration administration administration
Pneumococcus serotype 1 polysaccharide conjugate bulk group
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1M0101	287.8	285.3	340.3	386.7
1M0102	302.3	347.1	354.8	408.2
1F0101	302.4	355.3	393.5	457.5
1F0102	285.8	310.8	342.1	373.6
Pneumococcus serotype 4 polysaccharide conjugate bulk group				
2M0101	297.9	332.5	354.5	425.0
2M0102	316.9	344.9	388.0	453.0
2F0101	395.7	315.8	344.8	331.3
2F0102	312.5	347.0	395.4	406.1
Pneumococcus serotype 5 polysaccharide conjugate bulk group				
3M0101	275.7	315.3	364.2	404.4
3M0102	322.5	355.7	410.7	469.2
3F0101	268.3	285.9	312.7	364.5
3F0102	285.0	324.9	375.1	417.4
Pneumococcus serotype 6B polysaccharide conjugate bulk group				
4M0101	283.6	328.9	371.7	427.7
4M0102	304.0	344.6	395.1	428.7
4F0101	322.3	334.3	405.6	455.2
4F0102	306.4	352.8	355.8	373.8
Pneumococcus serotype 9V polysaccharide conjugate bulk group				
5M0101	294.8	341.1	372.9	442.9
5M0102	306.3	361.9	395.6	444.7
5F0101	285.0	325.0	363.0	432.7
5F0102	291.7	322.2	365.4	400.3
Pneumococcus serotype 14 polysaccharide conjugate bulk group				
6M0101	351.0	393.8	353.1	407.9
6M0102	286.6	303.3	427.9	498.3
6F0101	269.2	295.8	321.0	348.3
6F0102	311.1	333.0	368.7	409.4
Pneumococcus serotype 18C polysaccharide conjugate bulk group				
7M0101	282.4	306.7	360.5	388.1
7M0102	351.5	385.2	438.6	486.9
7F0101	294.5	324.6	355.9	401.0
7F0102	296.3	333.3	368.7	409.4
Pneumococcus serotype 19F polysaccharide conjugate bulk group				
8M0101	294.7	338.1	389.5	439.4
8M0102	358.1	390.5	427.4	489.7
8F0101	280.2	328.1	362.4	394.2

8F0102 303.0 306.4 343.6 383.9
Pneumococcus serotype 23F polysaccharide conjugate bulk group
9M0101 294.0 325.1 377.2 481.1
9M0102 339.2 385.4 437.4 490.5
9F0101 272.2 317.0 359.2 405.9
9F0102 316.6 343.8 375.7 401.6

7. CONCLUSIONS
In conclusion, the specific toxicity test of JHYM (pneumococcus bulk of serotype 1, 4, 5, 6B, 9V, 14, 18C, 19F and 23F) through single subcutaneous injection in guinea pigs was up to standard.
8. ORIGINAL DATA DEPOSITORY
All data related to this experiment (including written data, electronic data, etc.) were kept in the archives room of WestChina-Frontier PharmaTech Co. The data would be kept for at least five years after the vaccine got licensed and electronic data would be eternally retained. Standard archive preservation conditions were adopted for data storage.

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9. REFERENCES
1. Non-clinical Good Laboratory Practice (GLP), China Food and Drug Administration rule NO.2, Sep.1st, 2003.
2. Drug Registration Regulation, China Food and Drug Administration rule NO.28, Oct. 1st, 2007.
3. Technical Guideline for Stimulation, Anaphylaxis and Haemolytic Study of Drugs, China Food and Drug Administration, May 2014.

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Experiment Report

Experiment Name
Specific Toxicity Test of 13-JHYM through
Subcutaneous Injection in Guinea Pigs
Experiment Number A201118-O0060-01
Research Institution Name WestChina-Frontier PharmaTech Co. (National
Chengdu Center for Safety Evaluation of Drugs)
Research Institution Address No. 28 Gaopeng Avenue, High & New Tech Dev.
Zone, Chengdu, Sichuan
Client Name Yunnan Walvax Biotechnology Co., Ltd.
Client Address 4th Floor, building A3, Phase 2 Science and
Technology Park of Yunnan National University,
Kunming, Yunnan
Experiment Director Wei Guo
Members of Experiment Group Ke Mu
Xuemei Huang
Hong Yuan
Xi Wang
Quanli Xu
Bin Yi
Wulin Zheng
Duration of Experiment August, 2011-October, 2011
Original Data Depository Archives Room of WestChina-Frontier
PharmaTech Co.

WestChina-Frontier PharmaTech Co. (WCFP)
National Chengdu Center for Safety Evaluation of Drugs (NCCSED)

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Confirmation of Experiment Report

Experiment Name: Specific Toxicity Test of 13-JHYM
through Subcutaneous Injection in
Guinea Pigs
Experiment Number: A201118-O006-01

Experiment Director: Wei Guo

Signature: Date: 08/10/2011

Institution Director: Li Wang

Signature: Date: 09/10/2011

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Abstract

The 13-valent pneumococcal polysaccharide conjugate vaccine (13-JHYM) is made from pneumococcus capsule polysaccharide antigens of serotype 1, 3, 4, 5, 6A, 6B, 7F, 9V, 14,18C, 19A, 19F and 23F covalently combined with tetanus toxoid carrier protein and adsorbed by aluminum phosphate adjuvant. The clinical use of this vaccine is for infants and children aged 2 months to 2 years old to prevent the invasive infection caused by the above-mentioned serotype pneumococcus. The proposed dosage is 0.5 mL per person, and the proposed route of use is intramuscular injection.

In accordance with the relevant requirements of new drug declaration, the specific toxicity test of JHYM provided by Yunnan Walvax Biotechnology Co., Ltd. was carried out through subcutaneous injection in guinea pigs by WestChina-Frontier PharmaTech Co., with a purpose to observe whether the animals have tetanus symptoms or local reactions at the injection site such as suppuration and necrosis, and to provide a reference for evaluating the safety of clinical use.

A total of 4 groups were set for the test article 13-JHYM in this experiment, including serotype 3 pneumococcus polysaccharide conjugate group, serotype 6A pneumococcus polysaccharide conjugate group, serotype 7F pneumococcus polysaccharide conjugate group and serotype 19A pneumococcus polysaccharide conjugate group. Each group included 4 guinea pigs (half male and half female). Each guinea pig was subcutaneously injected with 2 ml of the corresponding sample at multiple points. The animals were observed for a total of 21 days, and the appearance of tetanus symptoms or local reactions at the injection site such as suppuration and necrosis were observed

on the 7th, 14th and 21st day after administration. The body weight was measured 7, 14, and 21 days before and after administration. The results are described below:

General Conditions

Each Guinea pig was in normal physical and mental state, and the natural holes were clean. No tetanus symptoms such as muscle spasm, convulsion, and opisthotonos were observed. No local reactions at the injection site such as suppuration and necrosis were observed.

Body Weight

On the 21st day after administration, the body weight of each guinea pig increased normally compared with that obtained before administration.

In conclusion, the specific toxicity test of 13-JHYM (pneumococcus bulk of serotype3, 6A, 7F and 19A) through single subcutaneous injection in guinea pigs was up to standard.

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1. BASIC INFORMATION

1.1 EXPERIMENT NAME AND NUMBER

Experiment name: Specific Toxicity Test of 13-JHYM through Subcutaneous

Injection in Guinea Pigs

Experiment number: A201118-O006-01

1.2 OBJECTIVES

The guinea pigs in this experiment were subcutaneously injected with 13-JHYM for specific toxicity test and were observed for 21 days, with a purpose to observe whether the animals have tetanus symptoms or local reactions at the injection site such as suppuration and necrosis, and to provide a reference for evaluating the safety of clinical use.

1.3 RESEARCH INSTITUTE

Name: WestChina-Frontier PharmaTech Co. (National Chengdu Center for Safety Evaluation of Drugs)

Address: No. 28 Gaopeng Avenue, High & New Tech Dev. Zone, Chengdu, Sichuan

Zip code: 610041

Contact person: Xiaobo Cen

Telephone: 028-85178767

Fax: 028-85173043

Email: xbcen@glpcd.com

1.4 CLIENT INSTITUTE

Name: Yunnan Walvax Biotechnology Co., Ltd.

Address: 4th Floor, building A3, 2nd Phase Science and Technology Park of Yunnan National University, Kunming, Yunnan

Zip code: 650106

Contact person: Kai Wu

Telephone: 0871-8351253-8851

Fax: 0871-8312123

1.5 MEMBERS OF EXPERIMENT GROUP

1.5.1 EXPERIMENT DIRECTOR

Name: Wei Guo

Telephone: 028-85154334-551

Fax: 028-85173043

Email: wguo@glpcd.com

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1.5.2 GROUP MEMBERS

Experiment operation: Ke Mu, Xuemei Huang, Hong Yuan, Xi Wang, Quanli Xu;

Veterinarian: Bin Yi;

Animal rearing and management: Qing Liao;

Test articles preparation and management: Wulin Zheng.

2. MATERIALS

2.1 EXPERIMENT ANIMALS

2.1.1 BASIC INFORMATION

Species: British guinea pigs;

Grade: normal;

Amount and gender: 16 animals in total, half male and half female;

Body weight: 243.2-307.2g;

Source: Sichuan Laboratory Animal Special Committee farm (Production License No. SCXK (Sichuan) 2008-014; Certificate of Quality No. 0024594, issuing unit: Sichuan Department of Science and Technology);

Entry date: 16/08/2011;

Environmental adaptation period after entry: Before the experiment, the guinea pigs adapted to the environment for 14 days (16/08/2011-29/08/2011), and the healthy guinea pigs were selected to enter the formal experiment. Observation measurements during adaptation period: general conditions of animals, body weight, etc.

2.1.2 ANIMAL REARING CONDITIONS

2.1.2.1 REARING SITES

The ordinary animal house of WestChina-Frontier PharmaTech Co. (laboratory animal room license No. SYXK (Sichuan) 2009-123).

2.1.2.2 REARING CONDITIONS

Rearing cage: guinea pigs cages (L*W*H: 515 mm*400 mm*260 mm);

Rearing density: no more than 3 animals per cage.

2.1.2.3 REARING ENVIRONMENT

Standard: the National Standard of the People's Republic of China GB14925-2001;

Control system: the EBI400 automatic control fresh air air conditioning system produced by Honeywell;

Temperature: room temperature 21.04-25.95°C (with a diurnal temperature range of 0.14-2.3°C);

Humidity: relative humidity 43.07-61.7%;

Illumination: artificial lighting, 12 hours of alternating light and dark (lights on at 07:30, lights off at 19:30);
Ventilation: 8-10 times per hour, 100% fresh air.

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2.1.2.4 FEED
Type: pellet feed for guinea pigs provided by Beijing Ke'ao Xieli Feed Co., Ltd. (Feed Production Permit: SCKK (Beijing) 2009-0012);
Batch No.: 11030551;
Feeding method: free intake;
Nutrient analysis: provided by the feed provider, of which the routine nutritional indexes of crude protein, crude fat, crude fiber, crude ash, water, calcium and phosphorus were reported for each batch (referring to the National Standard of the People's Republic of China GB14924.5-2001);
Confirmation of contaminant content in feed: provided by the feed provider, indicators for chemical contaminants including arsenic, lead, cadmium, mercury, hexachlorocyclohexane, DDT, aflatoxin B₁, total colony count, coliform group, mold and yeast count, pathogenic bacteria (Salmonella) were provided at least once a year (referring to the National Standard of the People's Republic of China GB14924.2-2001).

2.1.2.5 DRINKING
Type: drinking Water for laboratory animals (reverse osmosis water);
Method of water supply: held in drinking water bottles, free intake;
Colony count detection: referring to the National Standard of the People's Republic of China GB5749-2006, sampling self-examination (once per week);
Detection of conventional indicators of water quality: referring to the National Standard of the People's Republic of China GB5749-2006, samples were sent to units with relevant qualification for testing at least 2 times per year.

2.1.3 ANIMAL GROUPING AND MARKING METHODS
2.1.3.1 ANIMAL GROUPING
Group design: the guinea pigs were randomly divided into pneumococcus serotype 3, 6A, 7F and 19A polysaccharide conjugate group respectively, a total of 4 groups, with 4 animals in each group (half male and half female);
Amount: 16 animals in total;
Gender: half male and half female;
Grouping method: Guinea pigs were randomly divided into groups according to their body weight. The specific group information is shown in Table 1.

Table 1 Grouping Table

Animal amount	Animal number
Group	
Female Male Female Male	
Pneumococcus serotype 3	
polysaccharide 2 2	1F0101, 1F0102 1M0101, 1M0102
conjugate bulk group	
Pneumococcus serotype 6A	
polysaccharide 2 2	2F0101, 2F0102 2M0101, 2M0102
conjugate bulk group	

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Pneumococcus serotype 7F
polysaccharide 2 2 3F0101, 3F0102 3M0101, 3M0102
conjugate bulk group
Pneumococcus serotype 19A
polysaccharide 2 2 4F0101, 4F0102 4M0101, 4M0102
conjugate bulk group
Notes: The first digit of the animal number represents the group (1, 2, 3 and 4 represent the pneumococcus serotype 3, 6A, 7f and 19A polysaccharide conjugate bulk group respectively); the second letter stands for gender (M for male, F for female); the third and fourth digits represent cage number; the fifth and sixth digits represent animal serial number.

2.1.3.2 MARKING METHOD
Adaptation period: cage cards and red mark were used as animal identification markers;
Experiment period: the cage card and ear mark were used as animal identification markers;
Marking method of cage card:
Adaptation period: unified use of white cage card, with the test number, animal species, animal number, entry time and notes indicated;
Post-grouping: cage cards in different colors were used to distinguish the groups, and the cards were marked with test number, animal species, animal number, treatment factors, expected start and end time of the experiment and notes;
Colors of cage cards after grouping: white represented pneumococcus serotype 3 polysaccharide conjugate bulk group, green represented pneumococcus serotype 6A polysaccharide conjugate bulk group, blue represented pneumococcus serotype 7F polysaccharide conjugate bulk group and red represented pneumococcus serotype 19A polysaccharide conjugate bulk group.

2.2 TEST ARTICLES
2.2.1 TEST ARTICLE INFORMATION
NAME: PNEUMOCOCCUS SEROTYPE 3 POLYSACCHARIDE CONJUGATE BULK;
Source: Yunnan Walvax Biotechnology Co., Ltd.;
Character: suspension with opalescence;

Specification: 10mL per vial;
Batch No.: 20101103;
Preservation conditions and notes: kept under 2-8■;
Expiration date: August, 2012.

Name: Pneumococcus serotype 6A polysaccharide conjugate bulk;
Source: Yunnan Walvax Biotechnology Co., Ltd.;
Character: suspension with opalescence;
Specification: 10mL per vial;
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Batch No.: 20101103;
Preservation conditions and notes: kept under 2-8■;
Expiration date: October, 2012.

Name: Pneumococcus serotype 7F polysaccharide conjugate bulk;
Source: Yunnan Walvax Biotechnology Co., Ltd.;
Character: suspension with opalescence;
Specification: 10mL per vial;
Batch No.: 20101003;
Preservation conditions and notes: kept under 2-8■;
Expiration date: September, 2012.

Name: Pneumococcus serotype 19A polysaccharide conjugate bulk;
Source: Yunnan Walvax Biotechnology Co., Ltd.;
Character: suspension with opalescence;
Specification: 10mL per vial;
Batch No.: 20101103;
Preservation conditions and notes: kept under 2-8■;
Expiration date: October, 2012;
Preservation site: the Samples Management Department of WestChina-Frontier PharmaTech Co.

2.2.2 TEST ARTICLE PREPARATION

Preparation methods: direct separation of the bulk;
Marking methods after preparation: test articles were marked with red labels with experiment number, substance name and concentration used in the experiment, volume, storage condition, expiration date, responsible person and preparation date;
Preservation condition and period of validity after preparation: stored in ice box for use within 24 hours.

2.2.3 RETURN AND DISPOSAL OF REMAINING SAMPLES

The remaining samples after administration were sterilized by high temperature and high pressure and diluted with a large amount of water before being discharged into the sewer. The reserved samples were deposited in the sample room referring to the SOP for the preservation of test articles. The remaining samples at the end of the experiment were returned to the client.

2.3 MAIN INSTRUMENTS REQUIRED FOR THE EXPERIMENT

Injector, electronic balance, etc.

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3. METHODS AND OBSERVATION MEASUREMENTS

3.1 ADMINISTRATION

A total of 36 guinea pigs were included (243.2-307.2g). Each guinea pig was subcutaneously injected with 2 mL sample.

3.2 OBSERVATION

The animals were observed for a total of 21 days. The appearance of tetanus symptoms or local reactions at the injection site such as suppuration and necrosis were observed once respectively on the 7th, 14th and 21st day after administration.

3.3 BODY WEIGHT

The body weight of guinea pigs was measured once before administration and once respectively 7, 13, 21 days after administration.

4. RESULTS DETERMINATION

At the end of observation, animals with no tetanus symptoms, no suppuration or necrosis at the injection site and whose body weight increased compared with that obtained before administration were determined as qualified.

5. RESULTS

5.1 GENERAL CONDITIONS

Each Guinea pig was in normal physical and mental state, and the natural holes were clean. No tetanus symptoms were observed. No local reactions at the injection site such as suppuration and necrosis were observed.

5.2 EFFECTS ON BODY WEIGHT

On the 21st day after administration, the body weight of each guinea pig increased normally compared with that obtained before administration.

Table 2 Effects of 13JHYM on Body Weight of Guinea Pigs (g)

Time of Measurement

Group Before 7 days after 14 days after 21 days after
administration administration administration administration

Pneumococcus serotype 3 polysaccharide conjugate bulk group

1M0101 237.3 274.4 305.9 330.3

1M0102 279.7 303.2 351.5 392.6

1F0101 256.0 286.7 333.3 351.7

1F0102 278.2 307.9 342.7 385.7

Pneumococcus serotype 6A polysaccharide conjugate bulk group

2M0101 245.8 271.0 316.3 336.7

2M0102 301.0 351.1 414.8 467.6

2F0101 264.4 315.5 331.4 386.7

2F0102 291.8 343.0 411.8 462.9

Pneumococcus serotype 7F polysaccharide conjugate bulk group

3M0101 300.3 306.2 377.8 383.0

3M0102 274.2 326.0 386.7 457.7

3F0101 278.6 311.1 355.0 386.9
3F0102 270.7 302.5 354.5 398.2
Pneumococcus serotype 19A polysaccharide conjugate bulk group
4M0101 269.0 320.7 375.1 435.3
4M0102 318.8 366.7 421.0 479.7
4F0101 264 .5 297.4 357.5 398.9
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4F0102 285.5 321.6 394.1 440.7
Notes: The first digit of the animal number represents the group (1, 2, 3 and 4 represent the pneumococcus serotype 3, 6A, 7f and 19A polysaccharide conjugate bulk group respectively); the second letter stands for gender (M for male, F for female); the third and fourth digits represent cage number; the fifth and sixth digits represent animal serial number.

6. CONCLUSIONS
In conclusion, the specific toxicity test of 13-JHYM (pneumococcus bulk of serotype 3, 6A, 7F and 19A) through single subcutaneous injection (2mL per animal) in guinea pigs was up to standard.
7. ORIGINAL DATA DEPOSITORY
Preservation site: the archives room of WestChina-Frontier PharmaTech Co.;
Preservation conditions: Standard;
Preservation period: The paper file were kept for at least 5 years after the drug got licensed, and the electronic data should be kept permanently;
Contact number: 028-85154334-511;
Contact person: Ling Chang.

8. REFERENCES
1. Non-clinical Good Laboratory Practice (GLP) , China Food and Drug Administration rule NO.2, Sep.1st, 2003.
2. Drug Registration Regulation, China Food and Drug Administration rule NO.28, Oct. 1st, 2007.
3. General Principles for Technical Review of preclinical safety assessment of biological products for preventive use, China Food and Drug Administration, December, 2005.
4. Pharmacopoeia of the People's Republic of China, National Pharmacopoeia Committees, 2010.

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4.2.3.7.7.4. Toxicity Reverse Test

Experiment Report

Study Name :
Study on Toxicity Reversal of JHYM Related Carrier Protein (Tetanus Toxoid)
Study No. : A200932-O004
Name of study facility : WestChina-Frontier PharmaTech Co., Ltd.
(National Chengdu Center for Safety Evaluation of Drugs)
Address of study : No. 28 Gaopeng Avenue, High-tech Development Zone, Chengdu, Sichuan facility
Client Name : Yunnan Walvax Biotechnology Co., Ltd.
Address of client : 4F, Building A3, Phase II, Yunnan National University Science Park
Contact Person : Wu Kai
Tel. : 0871-8351253-8851
Study Director : Guo Wei
Members of study : Zhang Chunyan Study Operator
team:
Liu Xuejun Study Operator
Li Ting Study Operator
Zhang Qin Management of Test Articles
Zhang Fengcai Management of Test Articles
Lan Qing Veterinarian
Date of study initiation December 2009 - March 2010
and completion:
Location for storing Archives Room, WestChina-Frontier PharmaTech Co., Ltd.
initial data

WestChina-Frontier PharmaTech Co. (WCFP)
National Chengdu Center for Safety Evaluation of Drugs (NCCSED)

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Version No. Walvax-202007-00 Yuxi Walvax Biotechnology Co., Ltd.

Study Name : Study on Toxicity Reversal of JHYM Related Carrier Protein (Tetanus Toxoid)
Study No. : A200932-O004
Report prepared
Study Director (SD) : March 17, 2010
on:
Report approved
Director : March 17, 2010
on:
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Version No. Walvax-202007-00 Yuxi Walvax Biotechnology Co., Ltd.

Compliance Statement
Study No.: A200932-O004
Study Name: Study on Toxicity Reversal of JHYM Related Carrier Protein (Tetanus Toxoid)
Name of Test Article: JHYM
Study Director: Guo Wei

The study was carried out in accordance with the Good Laboratory Practice (August 2003) issued by the State Food and Drug Administration. The study was carried out in accordance with the study protocol and relevant standard operating procedures of the study facility, with no deviation affecting the reliability of the test results and conclusions. This report truthfully reflected the test materials, test methods and test results.

Signature of the Study
Date:
Director:

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Abstract

JHYM is a preventive vaccine provided by Yunnan Walvax Biotechnology Co., Ltd. It is used for preventing pneumonia, meningitis, otitis media and other diseases in infants aged 2 months to 2 years and children aged 2 to 9 years. The intended clinical dose is 0.5 mL/time/person, and the intended route of administration is intramuscular injection.

WestChina-Frontier PharmaTech Co., Ltd. carried out the study on toxicity reversal of JHYM related carrier protein (tetanus toxoid) provided by Yunnan Walvax Biotechnology Co., Ltd. (subcutaneous injection of test article in guinea pigs) in accordance with the relevant requirements for new drug application to observe if the animals had tetanus symptoms and if there are suppuration, necrosis and other local reactions at the injection site, and to provide a reference for evaluating the safety of clinical application.

There were 9 groups in the study, with 4 guinea pigs in each group, 50% males and 50% females. Guinea pigs in each group were respectively injected with the corresponding types of JHYM (pneumococcus types 1, 4, 5, 6B, 9V, 14, 18C, 19F and 23F, respectively). After the bulk of the test article was diluted with PBS to the required concentration, the solution was stored at 37 °C for 42 days, each guinea pig was subcutaneously injected with 5 mL of the corresponding test article at one time at multiple sites. On Days 7, 14 and 21 after administration, animals were observed for muscle spasm, convulsion, opisthotonos and other tetanus symptoms, as well as local reactions such as suppuration and necrosis at the injection site for a total of 21 days. Body weights were measured before administration and on Days 7, 14, and 21 after administration. Study results were as follows:

During the period from post-administration to the end of observation, each guinea pig was normal in terms of the appearance and signs, mental and motor status with natural holes all clean and no tetanus symptoms; no local reactions such as suppuration and necrosis were observed at the injection site.

On Days 7, 14 and 21 after administration, the body weight of guinea pigs in all groups increased normally compared with that before administration.

In conclusion, under the conditions of the study, the study on toxicity reversal of JHYM (pneumococcus types 1, 4, 5, 6B, 9V, 14, 18C, 19F and 23F) related carrier protein (tetanus toxoid) by single subcutaneous injection of 5 mL/guinea pig is qualified.

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Version No. Walvax-202007-00 Yuxi Walvax Biotechnology Co., Ltd.

1 OBJECTIVE

To carry out the study on toxicity reversal of JHYM by subcutaneous injection of related carrier protein (tetanus toxoid) in guinea pigs in accordance with the relevant requirements for new drug application in the Provisions for Drug Registration with reference to the General Principles for Non-clinical Safety Technical Review of Preventive Biological Products (December 2005) and Chinese Pharmacopoeia Volume III (2005), to observe if the animals have tetanus symptoms and if there are local reactions such as suppuration and necrosis at the injection site, and to provide a reference for evaluating the safety of clinical application.

2 MATERIALS

2.1 TEST ARTICLE

Bulk of pneumococcal type 1 polysaccharide conjugate, suspension with opalescence, batch number 20090501, 25 mL/vial, shelf life: 5 years.

Bulk of pneumococcal type 4 polysaccharide conjugate, suspension with opalescence, batch number 20090201, 25 mL/vial, shelf life: 5 years.

Bulk of pneumococcal type 5 polysaccharide conjugate, suspension with opalescence, batch number 20090401, 25 mL/vial, shelf life: 5 years.

Bulk of pneumococcal type 6B polysaccharide conjugate, suspension with opalescence, batch number 20090301, 25 mL/vial, shelf life: 5 years.

Bulk of pneumococcal type 9V polysaccharide conjugate, suspension with opalescence, batch number 20090301, 25 mL/vial, shelf life: 5 years.

Bulk of pneumococcal type 14 polysaccharide conjugate, suspension with opalescence, batch number 20090301, 25 mL/vial, shelf life: 5 years.

Bulk of pneumococcal type 18C polysaccharide conjugate, suspension with opalescence, batch number 20090301, 25 mL/vial, shelf life: 5 years.

Bulk of pneumococcal type 19F polysaccharide conjugate, suspension with opalescence, batch number 20090401, 25 mL/vial, shelf life: 5 years.

Bulk of pneumococcal type 23F polysaccharide conjugate, suspension with opalescence, batch number 20090401, 25 mL/vial, shelf life: 5 years.

The above test articles were diluted to the required concentration of the study with PBS and stored at 37°C for 42 days, and were provided by Yunnan Walvax Biotechnology Co., Ltd. after treatment. Preparation of test articles: test articles were treated by the Client, and were taken and used directly. Test articles were stored in ice box, protected from light, till the end of administration.

Retained samples were stored at the Test Article Management Department of WestChina-Frontier PharmaTech Co., Ltd., and the remaining test articles of administration were diluted with a large amount of water and then discharged into the sewer after high temperature and high pressure sterilization. The remaining test articles after completion of the study were treated in accordance with the opinion of the Client.

2.2 LABORATORY ANIMALS

36 ordinary healthy British guinea pigs, 50% males and 50% females, weighing 250.2 ~ 337.4 g, supplied by the Farm of Sichuan Laboratory Animal Specialized Committee (Production License No.: SCXK (C) 2008-014), and entered animal house on December 1, 2009.

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Guinea pigs were housed in the General Animal House (Building 1) of WestChina-Frontier PharmaTech Co., Ltd. (No. of license for use of laboratory animal house: SYXK (C) 2009-123). Guinea pigs were housed in stainless steel cages (housing density: 3 animals/cage). Honeywell EBI400 automatic control all-fresh air central air conditioning system was used to ensure that the housing conditions met the national standard GB14925-2001: room temperature of 21 ±5°C (daily temperature difference ≤ 4°C), relative humidity of 55 ± 15%, 12-hour light-dark cycle, and ventilation rate of 8 ~ 10 times/h. Pellet feed for guinea pigs (supplied by Beijing Keao Xieli Feed Co., Ltd., batch number: 091016, 091202) was fed. The animals could eat and drink free (Vc, 0.1

g/500 mL H₂O/day was added to the drinking water), while fresh vegetables were fed. Cage number and individual number (ear number method) were used to distinguish laboratory animals

3 STUDY METHOD AND OBSERVATION INDICATORS

3.1 PREPROCESSING OF TEST ARTICLE

Bulk of test article was diluted to the required concentration with PBS, and stored at 37°C for 42 days (treatment had been completed by the Client).

3.2 ADMINISTRATION

36 guinea pigs (250 ~ 350 g) were selected. Each guinea pig received a single subcutaneous injection of 5 mL of test article by multiple injection sites, with not more than 1 mL at each site.

3.3 OBSERVATION

On Days 7, 14 and 21 after administration, animals were observed once for muscle spasm, convulsion, opisthotonos and other tetanus symptoms, as well as local reactions such as suppuration and necrosis at the injection site for a total of 21 days.

3.4 BODY WEIGHT

Body weight was determined once before administration, and once on Days 7, 14 and 21 respectively after administration.

4 DETERMINATION OF RESULTS

At the end of the observation period, the animals did not show tetanus symptoms, with no suppuration or necrosis at the injection site, with normal weight gain compared with that before administration.

5 DEVIATION FROM STUDY

During the study, no deviation affecting the study results was observed.

6 RESULTS

6.1 GENERAL CONDITIONS

During the period from post-administration to the end of observation, each guinea pig was normal in terms of the appearance and signs, mental and motor status with natural holes all clean and no tetanus symptoms such as muscle spasm, convulsion and opisthotonos; no local reactions such as suppuration and necrosis were observed at the injection site.

6.2 EFFECT OF JHYM ON THE BODY WEIGHT OF GUINEA PIG

On Day 21 after administration, the body weight of each guinea pig increased normally compared with that before administration (Table 1).

Table 1 Effect of JHYM on the Body Weight of Guinea Pig (g)

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Test Time

Group Before 7 days after 14 days after 21 days after
administration administration administration administration

Pneumococcal type 1 polysaccharide conjugate group

1M0101 281.5 321.7 349.5 382.6

1M0102 318.6 357.6 404.9 459.1

1F0101 262.8 316.3 376.8 417.5

1F0102 301.8 347.0 408.5 443.9

Pneumococcal type 4 polysaccharide conjugate group

2M0101 291.1 339.1 392.5 442.7

2M0102 327.1 378.7 429.1 495.2

2F0101 261.2 32.18 352.7 391.8

2F0102 285.8 345.4 391.8 454.5

Pneumococcal type 5 polysaccharide conjugate group

3M0101 269.6 297.9 341.1 347.2

3M0102 327.7 366.5 419.0 431.9

3F0101 252.6 269.8 300.5 330.3

3F0102 284.1 309.4 352.8 396.3

Pneumococcal type 6B polysaccharide conjugate group

4M0101 286.7 338.3 381.9 419.0

4M0102 306.3 347.7 382.6 422.9

4F0101 261.6 317.8 352.8 374.4

4F0102 279.4 324.2 363.1 402.0

Pneumococcal type 9V polysaccharide conjugate group

5M0101 291.3 317.2 385.7 402.2

5M0102 318.5 345.3 413.7 455.1

5F0101 258.8 290.5 322.6 371.4

5F0102 278.3 320.2 371.6 414.3

Pneumococcal type 14 polysaccharide conjugate group

6M0101 299.8 321.1 385.3 432.4

6M0102 322.2 363.3 393.4 429.8

6F0101 322.0 345.5 394.1 416.3

6F0102 285.2 327.8 350.4 394.2

Pneumococcal type 18C polysaccharide conjugate group

7M0101 278.0 312.2 359.4 415.7

7M0102 328.1 374.3 387.5 465.5

7F0101 246.8 290.7 328.5 377.7

7F0102 277.0 302.9 338.1 374.3

Pneumococcal type 19F polysaccharide conjugate group

8M0101 296.2 321.5 373.0 405.2

8M0102 364.3 383.9 435.2 488.2

8F0101 266.4 293.3 332.6 375.3

8F0102 260.7 311.9 342.8 394.5

Pneumococcal type 23F polysaccharide conjugate group

9M0101 292.5 342.5 379.3 441.7

9M0102 308.2 331.8 375.8 406.1

9F0101 286.3 317.9 355.7 389.9

9F0102 309.0 348.0 390.3 440.9

Note: For the animal number, M indicates male animal, and F indicates female animal.

7 CONCLUSION

In conclusion, under the conditions of the study, the study on toxicity reversal of JHYM (pneumococcus types 1, 4, 5, 6B, 9V, 14, 18C, 19F and 23F) related carrier protein (tetanus toxoid) by single subcutaneous injection of 5 mL/guinea pig is qualified.

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8 LOCATION FOR STORING INITIAL DATA
All the data related to the study (including written data, electronic data, etc.) will be kept in the Archives Room of WestChina-Frontier PharmaTech Co., Ltd. till at least five years after the drug is marketed, and the electronic data will be permanently kept. The keeping condition is conventional archives keeping condition.
9 KEY REFERENCES
9.1 Good Laboratory Practice issued by State Food and Drug Administration, August, 2003
9.2 Chinese Pharmacopoeia Commission. Chinese Pharmacopoeia (Volume III), 2005.
9.3 Center for Drug Evaluation, SFDA, General Principles for Non-clinical Safety Technical Review of Preventive Biological Products, December 2005.

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MODULE 4 NONCLINICAL STUDY REPORTS
Version No. Walvax-202007-00 Yuxi Walvax Biotechnology Co., Ltd.

Experiment Report

Study Name Study on Toxicity Reversal of 13-JHYM Related Carrier Protein (Tetanus Toxoid)

Study No. A201118-O004-01

Name of study facility WestChina-Frontier PharmaTech Co., Ltd.
(National Chengdu Center for Safety Evaluation of Drugs)

Address of study facility No. 28 Gaopeng Avenue, High-tech Development Zone, Chengdu, Sichuan

Client Name Yunnan Walvax Biotechnology Co., Ltd.

Address of client 4F, Building A3, Phase II, Yunnan National University Science Park, High-tech Zone, Kunming

Study Director Guo Wei

Main members of study Mu Ke, Huang Xuemei, Yuan Hong, Wang Xi, Xu Quanli, Yi Bing, Zheng Wulin team

Date of study initiation August 2011 - October 2011
and completion:

Location for storing Archives Room, WestChina-Frontier PharmaTech Co., Ltd.
initial data

WestChina-Frontier PharmaTech Co. (WCFP)
National Chengdu Center for Safety Evaluation of Drugs (NCCSED)

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Version No. Walvax-202007-00 Yuxi Walvax Biotechnology Co., Ltd.

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Experiment Report Confirmation

Study Name: Study on Toxicity Reversal of 13-JHYM Related Carrier Protein (Tetanus Toxoid)
Study No.: A201118-O004-01

Study Director: Guo Wei

Signature: Date: October 8, 2011

Facility Director: Wang Li

Signature: Date: October 9, 2011

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Abstract

The 13-valent pneumococcal polysaccharide conjugate vaccine (code: 13-JHYM) is prepared by adsorption by aluminum phosphate adjuvant after covalent binding between types 1, 3, 4, 5, 6A, 6B, 7F, 9V, 14, 18C, 19A, 19F and 23F pneumococcal capsular polysaccharides antigen and tetanus toxoid carrier protein. It is clinically intended for the prevention of invasive infections caused by the above serotypes of pneumococci in infants aged 2 months to 2 years and children. The intended clinical dose is 0.5 mL/time/person, and the intended route of administration is intramuscular injection.

WestChina-Frontier PharmaTech Co., Ltd. carried out the study on toxicity reversal of 13-JHYM related carrier protein (tetanus toxoid) provided by Yunnan Walvax Biotechnology Co., Ltd. (subcutaneous injection of test article in guinea pigs) in accordance with the relevant requirements for new drug application to observe if the animals had tetanus symptoms and if there are suppuration, necrosis and other local reactions at the injection site, and to provide a reference for evaluating the safety of clinical application.

In the study, 4 groups were set for the test article 13-JHYM: bulk groups of streptococcus pneumoniae types 3, 6A, 7F and 19A polysaccharide conjugates, with 4 guinea pigs in each group, 50% males and 50% females. After the bulk of the test article was diluted to the required concentration with PBS, the solution was stored at 37 °C for 42 days. Each guinea pig was subcutaneously injected with 5 mL at one time at multiple sites, respectively. On Days 7, 14 and 21 after administration, the animals were observed once for tetanus symptoms, suppuration, necrosis and other local reactions at the injection site for a total of 21 days. The body weight was measured before administration and on Days 7, 14 and 21 after administration. Study results:

General Conditions

During the period from post-administration to the end of observation, each guinea pig was normal in terms of the appearance and signs, mental and motor status with natural holes all clean and no tetanus symptoms; no local reactions such as suppuration and necrosis were observed at the injection site.

Body Weight

On Day 21 after administration, the body weight of each guinea pig increased compared with that before administration.

In conclusion, under the conditions of the study, the study on toxicity reversal of 13-JHYM (bulk of streptococcus pneumoniae types 3, 6A, 7F and 19A polysaccharide conjugates) related carrier protein (tetanus toxoid) by single subcutaneous injection of 5 mL/guinea pig is qualified.

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MODULE 4 NONCLINICAL STUDY REPORTS
Version No. Walvax-202007-00 Yuxi Walvax Biotechnology Co., Ltd.

1 GENERAL INFORMATION

1.1 NAME AND NO. OF THE STUDY

Study name: Study on Toxicity Reversal of 13-JHYM Related Carrier Protein (Tetanus Toxoid)
Study No.: A201118-O004-01

1.2 OBJECTIVE

To carry out the study on toxicity reversal of 13-JHYM related protein (tetanus toxoid) by subcutaneous injection in guinea pigs to observe if the animals had tetanus symptoms and if there are suppuration, necrosis and other local reactions at the injection site for 21 days, and to provide a reference for evaluating the safety of clinical application.

1.3 STUDY FACILITY

Name: WestChina-Frontier PharmaTech Co., Ltd. (National Chengdu Center for Safety Evaluation of Drugs)

Address: No. 28 Gaopeng Avenue, High-tech Development Zone, Chengdu, Sichuan

Postal code: 610041

Contact person: Cen Xiaobo

Tel.: 028-85178767

Fax: 028-85173043

Email: xbcen@glpcd.com

1.4 CLIENT

Name: Yunnan Walvax Biotechnology Co., Ltd.

Address: 4F, Building A3, Phase II, Yunnan National University Science Park, High-tech Zone, Kunming

Postal code: 650106

Contact person: Wu Kai

Tel.: 0871-8351253-8851

Fax: 0871-8312123

1.5 MEMBERS OF STUDY TEAM

1.5.1 STUDY DIRECTOR

Name: Guo Wei

Tel: 028-85154334-551

Fax: 028-85173043

Email: wguo@glpcd.com

1.5.2 MEMBERS OF STUDY TEAM:

Study operators: Mu Ke, Huang Xuemei, Yuan Hong, Wang Xi, Xu Quanli

Veterinarian: Yi Bing

Animal feeding and management: Gou Xiankui

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Management and preparation of test article: Zheng Wulin

2 MATERIALS

2.1 LABORATORY ANIMALS

2.1.1 GENERAL INFORMATION

Species: British guinea pig;

Class: conventional;

Quantity and sex of animals used: 16, 50% males and 50% females;

Body weight: 241.7~310g;

Source: Farm of Sichuan Laboratory Animal Specialized Committee, laboratory animal production license No.: SCXK(C)2008-014. Quality certificate No.: No.0024564, issued by:

Science & Technology Department of Sichuan Province;

Entry time: August 16, 2011;

Acclimation period after entering animal house: Guinea pigs were acclimated to the environment for 14 days before the study (August 16, 2011~August 29, 2011), and healthy guinea pigs were selected for the formal study. Main inspection items during the acclimation period: General state of animals, body weight, etc.

2.1.2 ANIMAL HOUSING CONDITIONS

2.1.2.1 HOUSING PLACE

General Animal House (Building 1) of WestChina-Frontier PharmaTech Co., Ltd. (No. of license for use of laboratory animal house: SYXK (C) 2009-123).

2.1.2.2 HOUSING CONDITIONS

Type of housing cage: guinea pig cage (L×W×H: 515mm×400mm×260mm)

Housing density: ≤3 animals/cage

2.1.2.3 HOUSING ENVIRONMENTAL CONDITIONS

Standard for housing environmental conditions: National Standard of the People's Republic of China GB14925-2001;

Housing environmental control system: Honeywell EBI400 automatic control all-fresh air central air conditioning system;

Temperature: Room temperature: 21.04~25.95°C (daily temperature difference: 0.14~2.3°C);

Humidity: Relative humidity: 43.07~61.7%;

Lighting: Artificial lighting, 12-hour light-dark cycle (lights on at 07:30 and off at 19:30);

Ventilation rate: 8~10 times/h, 100% all fresh air.

2.1.2.4 FEED

Type: pellet feeds for guinea pigs

Manufacturer: Beijing Keao Xieli Feed Co., Ltd. Production license No. of feed: SCXK (J) 2009-0012;

Batch No.: 11030551;

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Feeding method: Ad libitum;

Test of nutritional ingredients: Provided by the feed supplier. Test reports for conventional nutritional ingredient indicators (crude protein, crude fat, crude fiber, crude ash, moisture, calcium and phosphorus) were provided for each batch (refer to the national standard of the People's Republic of China GB14924.5-2001);

Confirmation of contaminant content of feed: Provided by the feed supplier, chemical contaminant indexes such as arsenic, lead, cadmium, mercury, BHC, DDT, aflatoxin B1, total bacterial count, coliform group, mold and yeast count and pathogenic bacteria (salmonella) should be provided at least once a year (refer to the national standard of the People's Republic of China GB14924.2-2001)

2.1.2.6 DRINKING WATER

Type: Drinking water for laboratory animals (reverse osmosis water)

Water supply method: Supplied in drinking bottle with water available ad libitum;

Test of total bacterial count: Refer to the national standard of the People's Republic of China

GB5749-2006. Samples were taken for self-test (1 time/week).

Test of conventional indexes of water quality: Refer to the national standard of the People's Republic of China GB5749-2006. Samples should be taken and sent to the unit with relevant qualifications for test at least twice a year.

2.1.3 ANIMAL GROUPING AND IDENTIFICATION METHOD

2.1.3.1 ANIMAL GROUPING

Group design: Guinea pigs were randomized according to the SOP Grouping of Laboratory Animals, including 4 groups, namely streptococcus pneumoniae types 3, 6A, 7F and 19A polysaccharide conjugate groups, with 4 animals in each group, 50% males and 50% females.

Number of animals: 16;

Sex ratio: 50% males and 50% females;

Grouping method: Randomized by body weight of guinea pigs.

See Table 1 for the specific grouping information.

Table 1 Grouping Information

Number of

Animal No.

Group Animals

Female Male Female Male

Streptococcus pneumoniae type 3

2 2 1F0101, 1F0102 1M0101, 1M0102

polysaccharide conjugate group

Streptococcus pneumoniae type

6A polysaccharide conjugate 2 2 2F0101, 2F0102 2M0101, 2M0102

group

Streptococcus pneumoniae type 7F

2 2 3F0101, 3F0102 3M0101, 3M0102

polysaccharide conjugate group

Streptococcus pneumoniae type

19A polysaccharide conjugate 2 2 4F0101, 4F0102 4M0101, 4M0102

group

Remarks: The first number of the animal number represents the group (1, 2, 3, 4 represent the

streptococcus

pneumoniae types 3, 6A, 7F, 19A polysaccharide conjugate groups, respectively). The second letter

represents

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sex (M indicates male, F indicates female), the third and fourth numbers represent animal cage number, and the

fifth and sixth numbers represent animal serial number.

2.1.3.2 IDENTIFICATION METHOD

Acclimation period: Cage cards and marks of red marker were used as animal identification marks;

Study period: Cage cards and ear marks were used as the animal identification marks;

Marking method of cage card:

Acclimation period: White cage cards were uniformly used, and the study number, animal species, animal serial number, animal house entry time, remark information and other information were indicated;

After grouping: Cage cards of different colors were used for group differentiation. The study number, animal species, animal number, treatment factor, expected starting and ending time of the study, remark information and other information were indicated on the cage cards.

Color of cage card after grouping: White for streptococcus pneumoniae type 3 polysaccharide conjugate group, green for streptococcus pneumoniae type 6A polysaccharide conjugate group, blue for streptococcus pneumoniae type 7F polysaccharide conjugate group, and red for streptococcus pneumoniae type 19A polysaccharide conjugate group.

2.2 TEST ARTICLE

2.2.1 INFORMATION OF TEST ARTICLE

Name or designation: Bulk of streptococcus pneumoniae type 3 polysaccharide conjugate;

Source: Yunnan Walvax Biotechnology Co., Ltd.;

Description: Suspension with opalescence;

Strength: 25 ml/vial;

Batch No.: 20100903;

Storage conditions and precautions: Store at 2 ~ 8°C;

Valid until: August, 2012;

Name or designation: Bulk of streptococcus pneumoniae type 6A polysaccharide conjugate;

Source: Yunnan Walvax Biotechnology Co., Ltd.;

Description: Suspension with opalescence;

Strength: 25 ml/vial;

Batch No.: 20101103;

Storage conditions and precautions: Store at 2 ~ 8°C;

Valid until: October 2012;

Name or designation: Bulk of streptococcus pneumoniae type 7F polysaccharide conjugate;

Source: Yunnan Walvax Biotechnology Co., Ltd.;

Description: Suspension with opalescence;

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Strength: 25 ml/vial;

Batch No.: 20101003;

Storage conditions and precautions: Store at 2 ~ 8°C;

Valid until: September 2012;

Name or designation: Bulk of streptococcus pneumoniae type 19A polysaccharide conjugate;

Source: Yunnan Walvax Biotechnology Co., Ltd.;

Description: Suspension with opalescence;

Strength: 25 ml/vial;

Batch No.: 20101103;

Storage conditions and precautions: Store at 2 ~ 8°C;

Valid until: October 2012;

Place of storage: Test Article Management Department of WestChina-Frontier PharmaTech Co., Ltd.

2.2.2 PREPARATION OF TEST SOLUTION

Preparation method: Bulk was filled and used directly. Bulk of test article was diluted to the required concentration with PBS, and stored at 37°C for 42 days (treatment had been completed by the Client).

Identification method: Red label was used for identification, and the study number, name and concentration of test substance, volume, storage conditions, shelf life, responsible person and preparation date were indicated.

Temporary storage conditions and shelf life: Temporarily stored in the ice box and used within 24 h.

2.2.3 TREATMENT OF RETURNED AND REMAINING TEST ARTICLES

The test articles returned after administration were autoclaved and then poured into the waste liquid bucket. The retained samples were stored at the storage room of test articles in accordance with the SOP Storage of Test Articles. After completion of the test items, the remaining test articles were returned to the Client.

2.3 MAIN INSTRUMENTS AND DEVICES REQUIRED FOR THE STUDY

Syringe, electronic balance, etc.

3 STUDY METHOD AND OBSERVATION INDICATORS

3.1 PREPROCESSING OF TEST ARTICLE

Bulk of test article was diluted to the required concentration with PBS, and stored at 37°C for 42 days (treatment had been completed by the Client).

3.2 ADMINISTRATION

16 guinea pigs (241.7 ~ 310g) were selected, and each guinea pig was subcutaneously injected with 5 mL of the test article at one time.

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3.3 OBSERVATION

On Days 7, 14 and 21 after administration, animals were observed once for tetanus symptoms, as well as local reactions such as suppuration and necrosis at the injection site for a total of 21 days.

3.4 BODY WEIGHT

Body weight was determined once before administration, and once on Days 7, 14 and 21 respectively after administration.

4 DETERMINATION OF RESULTS

At the end of the observation period, the animals did not show tetanus symptoms, with no suppuration or necrosis at the injection site, with normal weight gain compared with that before administration.

5 RESULTS

5.1 GENERAL CONDITIONS

During the period from post-administration to the end of observation, each guinea pig was normal in terms of the appearance and signs, mental and motor status with natural holes all clean and no tetanus symptoms; no local reactions such as suppuration and necrosis were observed at the injection site.

5.2 EFFECT OF 13-JHYM ON THE BODY WEIGHT OF GUINEA PIG

On Day 21 after administration, the body weight of each guinea pig increased compared with that before administration (Table 2).

Table 2 Effect of 13-JHYM Streptococcus Pneumoniae Types 3, 6A, 7F and 19A Polysaccharide Conjugates) on the Body Weight of Guinea Pig (g)

Test Time

Group Before Day 7 after Day 14 after Day 21 after
administration administration administration administration

Streptococcus pneumoniae type 3

polysaccharide conjugate group

1M0101 278.1 307.3 361.0 399.9

1M0102 288.3 312.6 354.4 393.0

1F0101 263.3 310.5 367.2 415.8

1F0102 295.5 348.4 410.3 466.8

Streptococcus pneumoniae type 6A

polysaccharide conjugate group

2M0101 265.0 310.4 367.0 417.1

2M0102 307.0 350.8 384.3 430.7

2F0101 269.3 288.9 355.4 400.7

2F0102 245.9 283.8 325.6 351.3

Streptococcus pneumoniae type 7F

polysaccharide conjugate group

3M0101 255.3 297.8 337.1 378.8

3M0102 300.7 343.0 411.4 459.6

3F0101 289.1 306.3 351.0 383.5

3F0102 275.9 300.5 360.8 421.2

Streptococcus pneumoniae type 19A

polysaccharide conjugate group

4M0101 290.3 337.0 392.0 451.9

4M0102 312.2 378.1 437.5 508.7

4F0101 275.5 323.0 357.9 419.2

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Test Time

Group Before Day 7 after Day 14 after Day 21 after
administration administration administration administration

4F0102 240.9 272.9 311.1 348.2

Remarks: The first number of the animal number represents the group (1, 2, 3, 4 represent the streptococcus

pneumoniae types 3, 6A, 7F, 19A polysaccharide conjugate groups, respectively). The second letter

represents

sex (M indicates male, F indicates female), the third and fourth numbers represent animal cage number,

and the

fifth and sixth numbers represent animal serial number.

6 CONCLUSION

In conclusion, under the conditions of the study, the study on toxicity reversal of 13-JHYM (bulk of streptococcus pneumoniae types 3, 6A, 7F and 19A polysaccharide conjugates) related carrier protein (tetanus toxoid) by single subcutaneous injection of 5 mL/guinea pig is qualified.

7 LOCATION FOR STORING INITIAL DATA

Place of storage: Archives Room, WestChina-Frontier PharmaTech Co., Ltd.;

Storage conditions: Conventional conditions;

Retention period: Paper archives should be kept at least till 5 years after the drug is marketed, and electronic data should be permanently kept;

Tel.: 028-85154334-511;

Contact: Chang Ling.

8 KEY REFERENCES

Good Laboratory Practice, No. 2 Order of State Food and Drug Administration, September 1, 2003

Provisions for Drug Registration, No. 28 Order of State Food and Drug Administration, October 1, 2007

General Principles for Pre-clinical Safety Technical Review of Preventive Biological Products, State Food and Drug Administration, December 2005

Chinese Pharmacopoeia (Volume III), Chinese Pharmacopoeia Commission, 2010.

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4.2.3.7.7.5. Active Systemic Anaphylaxis Test

Experiment Report

Experiment Name Active Systemic Anaphylaxis Test of 13-JHYM in Guinea Pigs

A201118-T074-01

Experiment Number

WestChina-Frontier PharmaTech Co. (National Research Institution Name

Chengdu Center for Safety Evaluation of Drugs)

No. 28 Gaopeng Avenue, High & New Tech Dev.

Research Institution Address

Zone, Chengdu, Sichuan

Yunnan Walvax Biotechnology Co., Ltd.

Client Name

4th Floor, building A3, Phase 2 Science and

Client Address

Technology Park of Yunnan National University, Kunming, Yunnan

Wei Guo

Experiment Director

Ke Mu

Members of Experiment Group

Xuemei Huang

Hong Yuan

Xi Wang

Quanli Xu

Lihua Duan

Wulin Zheng

June, 2011-October, 2011

Duration of Experiment

Archives Room of WestChina-Frontier

Original Data Depository

PharmaTech Co.

WestChina-Frontier PharmaTech Co. (WCFP)

National Chengdu Center for Safety Evaluation of Drugs (NCCSED)

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Confirmation of Experiment Report

Experiment Name: Active Systemic Anaphylaxis Test of
13-JHYM in Guinea Pigs

Experiment Number: A201118-T074-01

Experiment Director: Wei Guo

Signature: Date: 08/10/2011

QAU Director: Feng Cheng

Signature: Date: 08/10/2011

Institution Director: Li Wang

Signature: Date: 09/10/2011

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MODULE 4 NONCLINICAL STUDY REPORTS

Version No. Walvax-202007-00 Yuxi Walvax Biotechnology Co., Ltd.

Compliance Statement

Active Systemic Anaphylaxis Test of 13-JHYM

Experiment Name:

in Guinea Pigs

Experiment Number: A201118-T074-01

This experiment was completed in accordance with Non-clinical Good Laboratory Practice formulated by China Food and Drug Administration (China Food and Drug Administration rule No. 2, effective from September 1st, 2003). The experiment procedure was carried out in accordance with the experiment protocol and the relevant standard operating procedure of the research institution. No deviation that affected the reliability of experiment results and conclusions was observed. This report faithfully reflects the experiment materials, experiment methods and experiment results.

Experiment Director: Wei Guo

Signature of Experiment Director: Date: 08/10/201

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MODULE 4 NONCLINICAL STUDY REPORTS

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Abstract

The 13-valent pneumococcal polysaccharide conjugate vaccine (13-JHYM) is made from pneumococcus capsule polysaccharide antigens of serotype 1, 3, 4, 5, 6A, 6B, 7F, 9V, 14, 18C, 19A, 19F and 23F covalently combined with tetanus toxoid carrier protein and adsorbed by aluminum phosphate adjuvant. The clinical use of this vaccine is for infants and children aged 2 months to 2 years old to prevent the invasive infection caused by the above-mentioned serotype pneumococcus. The proposed dosage is 0.5 mL per person, and the proposed route of use is intramuscular injection.

In accordance with the relevant requirements of new drug declaration redefined by Provisions for Drug Registration, the active systemic anaphylaxis test of 13-JHYM provided by Yunnan Walvax Biotechnology Co., Ltd. was carried out in guinea pigs by WestChina-Frontier PharmaTech Co., with a purpose to observe whether 13-JHYM could induce active systemic anaphylaxis in guinea pigs, and to provide a reference for evaluating the safety of clinical use.

A total of 10 groups were set in this experiment, including a negative control group, a positive control group, a low-dose and a high-dose group for 7-valent pneumococcal conjugate vaccine, as well as a low-dose and a high-dose group for 13-JHYM of 3 different batches respectively (batch No. 20110304, 20110305 and 20110406), with 6 guinea pigs included in each group (half male and half female). The sensitized dosage of low-dose group was 0.05 mL bulk/kg, which was equal the clinical proposed dosage, and the sensitized dosage of high-dose group was 0.5 mL bulk/kg, which was equal to 10 times of the clinical proposed dosage.

When sensitizing, the guinea pigs in the 7-valent pneumococcal conjugate vaccine groups and 13-JHYM groups were given sample or comparison product at corresponding concentration based on the above mentioned dosage design. Guinea pigs in the positive control group were given 10 mg/mL ovalbumin solution through intraperitoneal injection in the volume of 0.5 mL/kg, while guinea pigs in the negative control group were given 0.9% sodium chloride (normal saline) through intramuscular injection in the volume of 0.5 mL/kg. Sensitization was done once every other day and 5 times in total continuously. On the 10th day after the last sensitization, guinea pigs in each group were provoked by single injection of 2 times sensitized dosage through dorsalis pedis vein. The systemic reactions and death of guinea pigs were observed at least 30 minutes after provocation injection. Main results are as below:

No anaphylaxis was observed in the negative control group, 7-valent pneumococcal conjugate vaccine groups and the low-dose 13-JHYM groups of 3 batches within 30 minutes after provocation injection. Anaphylaxes observed in the positive control group were all strong positive and guinea pigs with anaphylaxis all died within 4 minutes. Three Guinea pigs in the high-dose 13-JHYM (batch No.20110304) group showed strong positive anaphylaxis and the duration of the reaction was about 6-21 minutes; one guinea pig showed extremely positive anaphylaxis and died about 7 minutes after being provoked. One guinea pig in the high-dose 13-JHYM group (batch No.20110305) showed positive anaphylaxis and another showed strong positive anaphylaxis with a duration of 4-6 minutes; one guinea pig showed extremely positive anaphylaxis and died about 7 minutes after being provoked. Three guinea pigs in the high-dose 13-JHYM group (batch No.20110406) showed positive anaphylaxis and one showed strong positive anaphylaxis with a duration of 23-30 minutes.

In conclusion, under the conditions of this experiment, the guinea pigs were intramuscularly injected with 3 batches of 13-JHYM (batch No.20110304, 20110305 and 20110406) in the volume of 0.05 mL/kg and 0.5 mL/kg respectively. At the dosage of 0.5 mL/kg, the active systemic

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anaphylaxis test of guinea pigs was positive, while at the dosage of 0.05 mL/kg, the active systemic anaphylaxis test of guinea pigs was negative.

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1. BASIC INFORMATION

1.1 EXPERIMENT NAME AND NUMBER

Experiment name: Active Systemic Anaphylaxis Test of 13-JHYM in Guinea Pigs
Experiment number: A201118-T074-01

1.2 OBJECTIVES

The active systemic anaphylaxis test of 13-valent pneumococcal polysaccharide conjugate vaccine (13-JHYM) was carried out in guinea pigs, with a purpose to observe whether 13-JHYM could induce active systemic anaphylaxis in guinea pigs, and to provide a reference for evaluating the safety of clinical use.

1.3 RESEARCH INSTITUTE

Name: WestChina-Frontier PharmaTech Co. (National Chengdu Center for Safety Evaluation of Drugs)
Address: No. 28 Gaopeng Avenue, High & New Tech Dev. Zone, Chengdu, Sichuan
Zip code: 610041
Contact person: Xiaobo Cen
Telephone: 028-85178767
Fax: 028-85173043
Email: xbcen@glpcd.com

1.4 CLIENT INSTITUTE

Name: Yunnan Walvax Biotechnology Co., Ltd.
Address: 4th Floor, building A3, 2nd Phase Science and Technology Park of Yunnan National University, Kunming, Yunnan
Zip code: 650106
Contact person: Kai Wu
Telephone: 0871-8351253-8851
Fax: 0871-8312123

1.5 MEMBERS OF EXPERIMENT GROUP

1.5.1 EXPERIMENT DIRECTOR

Name: Wei Guo
Telephone: 028-85154334-551
Fax: 028-85173043
Email: wguo@glpcd.com

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1.5.2 GROUP MEMBERS

Experiment operation: Ke Mu, Xuemei Huang, Hong Yuan, Xi Wang, Quanli Xu;
Veterinarian: Lihua Duan;
Animal rearing and management: Qing Liao, Yongshao Cai;
Test articles preparation and management: Wulin Zheng.

2. MATERIALS

2.1 EXPERIMENT ANIMALS

2.1.1 BASIC INFORMATION

Species: British guinea pigs;
Grade: normal;
Amount and gender of animals purchased: 66 animals in total, half male and half female;
Amount and gender of animals used: 60 animals in total, half male and half female;
Body weight: female 269.6-389.6g, male 289.5-398.2g;
Source: Sichuan Laboratory Animal Special Committee farm (Production License No. SCXK (Sichuan) 2008-014; Certificate of Quality No. 0024848, issuing unit: Sichuan Department of Science and Technology);
Entry date: 14/06/2011;
Environmental adaptation period after entry: Before the experiment, the guinea pigs adapted to the environment for 21 days (14/06/2011-04/07/2011). Healthy guinea pigs were selected as formal test animals. Observation measurements during adaptation period: general conditions, body weight etc.

2.1.2 ANIMAL REARING CONDITIONS

2.1.2.1 REARING SITES

The ordinary animal house of WestChina-Frontier PharmaTech Co. (laboratory animal room license No. SYXK (Sichuan) 2009-123).

2.1.2.2 REARING CONDITIONS

Rearing cage: guinea pig cage (L*W*H: 515 mm*400 mm*260 mm);
Rearing density: no more than 3 animal per cage.

2.1.2.3 REARING ENVIRONMENT

Standard: the National Standard of the People's Republic of China GB14925-2001;
Control system: the EBI400 automatic control fresh air air conditioning system produced by Honeywell;
Temperature: room temperature 19.75-25.08°C (with a diurnal temperature range of 0.07 - 2.98°C);
Humidity: relative humidity 42.00-59.13%;
Illumination: artificial lighting, 12 hours of alternating light and dark (lights on at 07:30, lights off at 19:30);
Ventilation: 8-10 times per hour, 100% fresh air.

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2.1.2.4 FEED

Type: pellet feed for guinea pigs;
Manufacturer: Beijing Ke'ao Xieli Feed Co., Ltd. (Feed Production Permit: SCXK (Beijing) 2009-0012);
Batch No.: 11030551;
Feeding method: free intake;
Nutrient analysis: provided by the feed provider, of which the routine nutritional measurements of crude protein, crude fat, crude fiber, crude ash, water, calcium and phosphorus were reported for each batch(referring to the National Standard of the People's Republic of China GB14924.5-2001);
Confirmation of contaminant content in feed: provided by the feed provider, indicators for

chemical contaminants including arsenic, lead, cadmium, mercury, hexachlorocyclohexane, DDT, aflatoxin B₁, total colony count, coliform group, mold and yeast count, pathogenic bacteria (Salmonella) were provided at least once a year (referring to the National Standard of the People's Republic of China GB14924.2-2001).

2.1.2.5 DRINKING

Type: drinking water for laboratory animals (reverse osmosis water);

Method of water supply: held in drinking water bottles, free intake;

Colony count detection: referring to the National Standard of the People's Republic of China

GB5749-2006, sampling self-examination (once per week);

Detection of conventional indicators of water quality: referring to the National Standard of the People's Republic of China GB5749-2006, samples were sent to units with relevant qualification for testing at least 2 times per year.

2.1.3 ANIMAL GROUPING AND MARKING METHODS

2.1.3.1 ANIMAL GROUPING

Group design: a total of 10 groups, including a negative control group, a positive control group, a low-dose and a high-dose group for 7-valent pneumococcal conjugate vaccine, as well as a low-dose and a high-dose group for 13-JHYM of 3 different batches;

Amount: 6 animals in each group;

Gender: half male and half female;

Grouping method: the guinea pigs were randomly divided into groups according to their body weight;

Disposal of surplus animals: handed over to the department of laboratory animal administration.

The specific group information is shown in Table 1.

Table 1 Grouping Table

Animal

Animal number

Group amount

Female Male Female Male

1F0101, 1F0102, 1M0101, 1M0102,

Negative control group 3 3

1F0103 1M0103

2F0101, 2F0102, 2M0101, 2M0102,

Positive control group 3 3

2F0103 2M0103

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Low-dose 7-valent pneumococcal 3F0101, 3F0102, 3M0101, 3M0102,
3 3

conjugate vaccine group 3F0103 3M0103

High-dose 7-valent pneumococcal 4F0101, 4F0102, 4M0101, 4M0102,
3 3

conjugate vaccine group 4F0103 4M0103

5F0101, 5F0102, 5M0101, 5M0102,

Low-dose 13-JHYM (batch 1) group 3 3

5F0103 5M0103

6F0101, 6F0102, 6M0101, 6M0102,

High-dose 13-JHYM (batch 1) group 3 3

6F0103 6M0103

7F0101, 7F0102, 7M0101, 7M0102,

Low-dose 13-JHYM (batch 2) group 3 3

7F0103 7M0103

8F0101, 8F0102, 8M0101, 8M0102,

High-dose 13-JHYM (batch 2) group 3 3

8F0103 8M0103

9F0101, 9F0102, 9M0101, 9M0102,

Low-dose 13-JHYM (batch 3) group 3 3

9F0103 9M0103

10F0101, 10F0102, 10M0101, 10M0102,

High-dose 13-JHYM (batch 3) group 3 3

10F0103 10M0103

Notes: The first digit of the animal number represents the group (e.g. 1 and 2 represent the negative control

group and positive control group respectively); the second letter stands for gender (M for male, F for female);

the third and fourth digits represent cage number, and the fifth and sixth digits represent animal serial number.

2.1.3.2 MARKING METHOD

Adaptation period: cage card and marks were used as animal identification markers;

Post-grouping period: The cage card and individual ear number were used as animal identification markers.

Marking method of cage card:

Adaptation period: unified use of white cage card, with the test number, animal species, animal number, entry time and notes indicated;

Post-grouping: cage cards in different colors were used to distinguish the groups, and the cards were marked with test number, animal species, animal number, treatment factors, expected start and end time of the experiment and notes;

Colors of cage cards after grouping: white represented the negative control group; yellow represented the positive control group; blue represented the low-dose 7-valent pneumococcal conjugate vaccine group, and orange represented the high-dose 7-valent pneumococcal conjugate vaccine group; green represented the low-dose 13-JHYM group, and red represented the high-dose 13-JHYM group.

2.2 TEST ARTICLES

2.2.1 TEST ARTICLE INFORMATION

Name: 13-valent pneumococcal polysaccharide conjugate vaccine (13-JHYM);

Source: Yunnan Walvax Biotechnology Co., Ltd.;

Character: suspension with opalescence;

Specification: 0.5 mL: 32.2μg per vial;

Batch No.: 20110304 (batch 1), 20110305 (batch 2), 20110406 (batch 3);

Preservation conditions and notes: kept under 2-8℃;

Expiration date: February, 2012 for batch No. 20110304 and 20110305; March, 2013 for 20110406;
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Preservation site: the Samples Management Department of WestChina-Frontier PharmaTech Co.
2.2.2 TEST ARTICLE PREPARATION

Menstruum: 0.9% sodium chloride injection;
Preparation methods: samples for the low-dose groups were prepared with 0.9% sodium chloride injection under aseptic condition, and samples for the high-dose groups used direct separation of the bulk;
Marking methods after preparation: the high-dose groups were marked with red labels and the low-dose group were marked with green labels, with experiment number, substance name and concentration used in the experiment, volume, storage condition, expiration date, responsible person and preparation date;
Preservation condition and period of validity after preparation: stored in ice box for use within 24 hours.

2.2.3 RETURN AND DISPOSAL OF REMAINING SAMPLES

The remaining samples after administration were collected in the waste liquid bucket after sterilized with high temperature and high pressure. The reserved samples were deposited in the sample room referring to the SOP for the preservation of test articles. The remaining samples at the end of the experiment were returned to the client.

2.3 COMPARISON PRODUCT

2.3.1 NEGATIVE COMPARISON PRODUCT INFORMATION

Name: 0.9% sodium chloride injection;
Source: Sichuan Kelun Pharmaceutical Co., Ltd.;
Character: colorless transparent liquid;
Specification: 100 mL per bottle;
Batch No.: C100928;
Preservation conditions and notes: sealed storage at room temperature;
Expiration date: August, 2012;
Preparation methods: direct separation of the bulk on the day of use;
Marking methods after preparation: test articles were marked with white labels with experiment number, substance name used in the experiment, volume, storage condition, expiration date, responsible person and preparation date;
Preservation condition and period of validity after preparation: stored in the ice box for use within 24 hours.

2.3.2 POSITIVE COMPARISON PRODUCT INFORMATION

Name: ovalbumin;
Source: Shanghai Yuanju Bio-tech Co., Ltd.;
Character: white crystal;
Specification: 1g per bottle;
Batch No.: 091221;

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Preservation conditions and notes: kept under 2-8℃;
Period of validity: 2 years;
Preparation methods: prepared into 10mg/mL solution with 0.9% sodium chloride injection when use;
Preservation condition and period of validity after preparation: stored in the ice box for use within 24 hours.

2.3.3 SIMILAR LICENSED COMPARISON PRODUCT

Name: 7-valent pneumococcal conjugate vaccine;
Source: Wyeth Pharmaceutical Division of Wyeth Holdings Corporation;
Character: white suspension;
Specification: 0.5 mL: 16µg per vial;
Batch No.: E19166D94429, E35571D94426 and E61516E52444;
Preservation conditions and notes: stored under 2-8℃;
Term of validity: February, 2012 for batch No. E19166D94429 and E35571D94426; March, 2013 for E61516E52444;
Preparation methods: samples for the low-dose groups were prepared with 0.9% sodium chloride injection under aseptic condition, and samples for the high-dose groups used direct separation of the bulk;
Marking methods after preparation: the high-dose groups were marked with red labels and the low-dose group were marked with green labels, with experiment number, substance name and concentration used in the experiment, volume, storage condition, expiration date, responsible person and preparation date;
Preservation condition and period of validity after preparation: stored in ice box for use within 24 hours.

2.3.4 RETURN AND DISPOSAL OF REMAINING COMPARISON PRODUCTS

After each administration, the remaining negative control was directly discharged into the sewer, the positive control was collected in the waste liquid barrel, and the licensed control was collected in the waste liquid barrel after sterilization.

3. METHODS AND OBSERVATION MEASUREMENTS

3.1 METHODS

Referring to the Technical Guidelines for the Study of Irritation, Hypersensitivity and Hemolysis of Chemical Drugs (March 2005), in an active systemic anaphylaxis test, animals in the negative control group should be given menstruum in the same volume, animals in the positive control group should be given bovine serum albumin or ovalbumin or known sensitizing positive substance in the volume of 1-5 mg per animal, animals in the low-dose groups should be given samples at the maximum clinical dosage, and animals in the high-dose groups should be given samples at a dosage which was several times of the dosage used in the low-dose group. Based on the information provided by the client, the clinical proposed dosage of 13-JHYM was 0.5 mL per person (a single serving contained 32.2 µg pneumococcal polysaccharides of various types. The dosage for infants whose weight was determined as 10 kg was converted into a clinical proposed dosage of 0.05 mL/kg, i.e. 3.22 µg/kg).

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On the basis of above mentioned premises, 0.05 mL bulk/kg was determined as low dosage for 13-JHYM of 3 different batches when sensitizing, and 0.5 mL bulk/kg was determined as high dosage, which was equal to and 10 times of the clinical proposed dosage for human beings respectively. The low-dose and high-dose 7-valent pneumococcal conjugate vaccine group were set at the same dosage as 13-JHYM groups. The dosage of 5 mg per animal was set for the positive control group (intraperitoneal injection, 10 mg/mL ovalbumin), and the normal saline was given to the animals in the negative control group in the volume of 0.5 mL/kg through intramuscular injection.

Guinea pigs in each group were intramuscularly injected with corresponding sample or control for sensitization through hind limbs at the above dosage on the 1st, 3rd, 5th, 7th and 9th day of the experiment respectively, and guinea pigs in the positive control group were intraperitoneally injected with 10 mg/mL ovalbumin in the volume of 0.5 mL per animal. On the 10th day after the last sensitization (i.e. the 19th day of the experiment), guinea pigs in each group were all provoked through intravenous injection at a twice sensitizing dosage.

Please see Table 2 and Table 3 for the specific dosage design.

Table 2 Table of Sensitizing Dosage Design

Dosage	Animal	Administration	Volume	Group (mL)	Concentration	amount	route (mL/kg)	bulk/kg)	Male	Female
Intramuscular	Negative control group	/ /	0.5	3	3		injection			
Intraperitoneal	5 mg per	0.5 mL								
Positive control group	10 mg/mL	0.5	3	3		injection	animal per animal			
Low-dose 7-valent										
Intramuscular	pneumococcal conjugate	0.05	bulk/10	0.5	3	3	injection			
vaccine group										
High-dose 7-valent										
Intramuscular	pneumococcal conjugate	0.5	bulk	0.5	3	3	injection			
vaccine group										
Low-dose 13-JHYM	Intramuscular									
0.05	bulk/10	0.5	3	3		(batch 1) group	injection			
High-dose 13-JHYM	Intramuscular									
0.5	bulk	0.5	3	3		(batch 1) group	injection			
Low-dose 13-JHYM	Intramuscular									
0.05	bulk/10	0.5	3	3		(batch 2) group	injection			
High-dose 13-JHYM	Intramuscular									
0.5	bulk	0.5	3	3		(batch 2) group	injection			
Low-dose 13-JHYM	Intramuscular									
0.05	bulk/10	0.5	3	3		(batch 3) group	injection			
High-dose 13-JHYM	Intramuscular									
0.5	bulk	0.5	3	3		(batch 3) group	injection			

Table 3 Table of Provocation Dosage Design

Animal	Provocation	(on the 19th day of the	Group	amount	experiment)	Male	Female	Administration	route	Volume
Negative control group	3	3	Intravenous	injection	1	mL/kg				
1	mL per									
Positive control group	3	3	Intravenous	injection						
animal										
Low-dose 7-valent pneumococcal conjugate										
3	3	Intravenous	injection	1	mL/kg					
vaccine group										
High-dose 7-valent pneumococcal conjugate										
3	3	Intravenous	injection	1	mL/kg					
vaccine group										

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Low-dose 13-JHYM (batch 1) group 3 3 Intravenous injection 1 mL/kg
 High-dose 13-JHYM (batch 1) group 3 3 Intravenous injection 1 mL/kg
 Low-dose 13-JHYM (batch 2) group 2 2 Intravenous injection 1 mL/kg
 High-dose 13-JHYM (batch 2) group 3 3 Intravenous injection 1 mL/kg
 Low-dose 13-JHYM (batch 3) group 3 3 Intravenous injection 1 mL/kg
 High-dose 13-JHYM (batch 3) group 3 3 Intravenous injection 1 mL/kg

3.2 OBSERVATION MEASUREMENTS

3.2.1 OBSERVATION MEASUREMENTS DURING SENSITIZATION

3.2.1.1 OBSERVATION TIME AND FREQUENCY

Observation period: from the 1st sensitization to provocation;

Observation frequency:

Rearing personnel: twice a day;

Experiment personnel: at least once around 2 hours after each administration, and at least once a day for the rest time;

Veterinarian: at least once a week.

3.2.1.2 OBSERVATION MEASUREMENTS AND CONTENTS

Rearing personnel: death and other abnormalities of guinea pigs;

Experiment personnel: including but not limited to the appearance, the general behavior activity, the mental state, the aspiration status, the excrement character, the genitals, the death and other toxic performance of guinea pigs;
Veterinarian: including but not limited to the appearance, the general behavior activity, the mental state, the aspiration status, the excrement character, the genitals, the death and other toxic performance of guinea pigs.

3.2.2 OBSERVATION AFTER PROVOCATION

The response of each guinea pig as well as the time of occurrence and disappearance of symptoms was observed and recorded immediately to 30 minutes after intravenous provocation according to the symptoms specified in Table 4. The maximum observation time was 3 hours.

Table 4 Table of Anaphylaxis Symptoms

No. Symptom	No. Symptom	No. Symptom	No. Symptom
No Shortness of breath	7 14		
symptom	breath	Gait	
1 Alysms	8 Urination	15 Jumping	
Pilo-erection	2 9 Defecation	16 Gasping	
3 Shudder	10 Dacryorrhea	17 Spasm	
Scratching	4 11 Anhelation	18 Wingover	
nose			
Tidal	5 Sneezing	12 Rales	19
breathing	6 Cough	13 Hemacelinosis	20 Death

3.2.3 BODY WEIGHT

Measurement time: once respectively on the day of sensitization and provocation;

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Measurement animal: all surviving guinea pigs.

3.3 RESULTS DETERMINATION

The degree of the anaphylaxes were determined based on the criteria specified in Table 5, and a comprehensive judgment was made according to the incidence and degree of the anaphylaxes.

Table 5 Table of Standards for Evaluation of Systemic Sensitization in Guinea Pigs

Symptom No. Symbol Results Determination

- 0 - Solvent of anaphylaxis
- 1-4 + The anaphylaxis was weakly positive.
- 5-10 ++ The anaphylaxis was positive.
- 11-19 +++ The anaphylaxis was strong positive.
- 20 ++++ The anaphylaxis was extremely strong positive.

4. RESULTS

4.1 GENERAL CONDITIONS

The guinea pigs in each group were in good conditions before provocation, with normal activities, normal weight gain and no other abnormal reactions.

4.2 SYSTEMIC ANAPHYLAXIS

Within 30 minutes after provocation injection, no anaphylaxis was observed for the guinea pigs in the negative control group. The anaphylaxes of guinea pigs in the positive control group were all extremely strong positive and the guinea pigs all died within 4 minutes. No anaphylaxis was observed for guinea pigs in the high-dose and low-dose 7-valent pneumococcal conjugate vaccine group, as well as the low-dose 13-JHYM groups of 3 different batches. The anaphylaxes of 3 guinea pigs in the high-dose 13-JHYM group (batch 1) were strong positive with a duration of 6-21 minutes; the anaphylaxis of 1 guinea pig was extremely strong positive and the guinea pig died around 7 minutes after provocation; the positive rate was 50%. The anaphylaxes of 3 guinea pigs in the high-dose 13-JHYM group (batch 3) were positive; the anaphylaxis of 1 guinea pig was strong positive with a duration of 23-30 minutes; the positive rate was 83.3%.

Another 6 guinea pigs in the negative control group were intravenously injected with 13-JHYM in the volume of 1.0 mL bulk/kg (3 batches, 2 animals each batch, half male and half female) and observed for 30 minutes. No anaphylaxis symptoms were observed.

Please see Table 6 for the observation results of the anaphylaxes.

Table 6 Table of Observation Results of the Systemic Anaphylaxis Test

Symptom No. and the amount

Amount of animals with symptoms	Incidence of Degree of Group the
11- anaphylaxes	anaphylaxes
animals 0 1-4 5-10 20	
19	
Negative control group	6 6/6 0 0 0 0 Negative
Extremely	
Positive control group	6 0 0 0 0 6/6 100
strong positive	
Low-dose 7-valent pneumococcal	
6 6/6 0 0 0 0 Negative	
conjugate vaccine group	

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High-dose 7-valent
pneumococcal
6 6/6 0 0 0 0 0 Negative
conjugate vaccine
group
Low-dose 13-JHYM
6 6/6 0 0 0 0 0 Negative
(batch 1) group
High-dose 13-JHYM Extremely
6 2/6 0 0 1/6 1/6 66.7
(batch 1) group strong positive
Low-dose 13-JHYM
6 6/6 0 0 0 0 0 Negative
(batch 2) group
High-dose 13-JHYM Extremely
6 3/6 0 1/6 1/6 1/6 5
(batch 2) group strong positive
Low-dose 13-JHYM
6 6/6 0 0 0 0 0 Negative
(batch 3) group
High-dose 13-JHYM Extremely
6 1/6 0 3/6 0 0 83.3
(batch 3) group strong positive

5. CONCLUSIONS

In conclusion, under the conditions of this experiment, the guinea pigs were intramuscularly injected with 13-JHYM of 3 batches (batch No.20110304, 20110305 and 20110406) in the volume of 0.05 mL bulk/kg and 0.5 mL bulk/kg. When injected in the volume of 0.5 mL bulk/kg, the results of active systemic anaphylaxis test for guinea pigs were all positive, and when injected in the volume of 0.05 mL bulk/kg, the results of active systemic anaphylaxis test for guinea pigs were all negative.

6. ORIGINAL DATA DEPOSITORY

Preservation site: the archives room of WestChina-Frontier PharmaTech Co.;

Preservation conditions: Standard;

Preservation period: The paper file were kept for at least 5 years after the drug got licensed, and the electronic data should be kept permanently;

Contact number: 028-85154334-511;

Contact person: Ling Chang.

7. DEVIATIONS

No deviation that affected the reliability of the experiment results and conclusions was observed during the experiment process.

8. REFERENCES

1. Drug Good Laboratory Practices, issued by National Medical Products Administration, No.2, September 2003.
2. Provisions for Drug Registration, issued by National Medical Products Administration, No.28, October 2007.
3. General Principles for Technical Review of Preclinical Safety Assessment of Prophylactic Biological Products, issued by National Medical Products Administration, December 2005.
4. Technical Guidelines for Acute Toxicity Tests of Chemical Drugs, issued by National Medical Products Administration, March 2005.

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4.3. LITERATURE REFERENCES

- [1] Drug Good Laboratory Practices, issued by National Medical Products Administration, No.2, September 2003.
- [2] Provisions for Drug Registration, issued by National Medical Products Administration, No.28, October 2007.
- [3] Technical Guidelines for the Study of Irritability, Anaphylaxis and Hemolysis of Chemical Drugs, issued by National Medical Products Administration, March 2005.
- [4] Animal Histopathology Atlas. Edited by Ning Su, Southeast University Press, April 2005.
- [5] Handbook of Toxicologic Pathology. WANDA M.HASCHEK, CCLIN G. R OUSSEAU, MATTHEW A. WALLIG, 525 B Street, Suit 1900, San Diego, California 92101-4495, USA. Second edition 2002.
- [6] General Principles for Technical Review of Preclinical Safety Assessment of Prophylactic Biological Products, issued by National Medical Products Administration, December 2005.
- [7] Drug Registration Regulation, China Food and Drug Administration rule NO.28, Oct. 1st, 2007.
- [8] Pharmacopoeia of the People's Republic of China (Volume III), National Pharmacopoeia Committees, 2010.
- [9] Technical Guidelines for Acute Toxicity Tests of Chemical Drugs, issued by National Medical Products Administration, March 2005.

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PRODUCT REGISTRATION DOSSIER

13-valent Pneumococcal Polysaccharide Conjugate Vaccine
(Tetanus Toxoid Protein)

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Yuxi Walvax Biotechnology Co., Ltd.

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5.2 TABULAR LIST OF ALL CLINICAL STUDIES

Preliminary trial
Phase/Study No. Phase I PCV13-TT 001
Country China
Status Completed
Objective Primary To assess the safety of 13-Valent Pneumococcal Conjugate Vaccine in infants aged 2 months (≥6 weeks) and 3 months, children aged 2-5 years and adults aged 18 years old.
Secondary To explore the IgG immune response of 13-Valent Pneumococcal Polysaccharide Conjugate Vaccine in infants aged 2 months and 3 months, children aged 2-5 years and adults aged 18 years old.
Group PCV13-TT Group PCV7 Group
Vaccine Name PCV13-TT /
Manufacture Yuxi Walvax Biotechnology Co., Ltd /
Specification Pneumococcal polysaccharide serotype 1: 2.6 µg /
Pneumococcal polysaccharide serotype 3: 2.5 µg
Pneumococcal polysaccharide serotype 4:3.0 µg
Pneumococcal polysaccharide serotype 5:2.5 µg
Pneumococcal polysaccharide serotype 6A:2.5 µg
Pneumococcal polysaccharide serotype 6B:6.0 µg
Pneumococcal polysaccharide serotype 7F :2.85 µg
Pneumococcal polysaccharide serotype 9V:2.5 µg
Pneumococcal polysaccharide serotype 14:2.75 µg
Pneumococcal polysaccharide serotype 18C:3.25 µg
Pneumococcal polysaccharide serotype 19A:2.6 µg
Pneumococcal polysaccharide serotype 19F:2.75 µg
Pneumococcal polysaccharide serotype 23F:3.0 µg

Dosage
Items Age Primary Series Boost Dose Number Primary Series Boost Dose Number
Schedule dose Schedule dose Schedule dose Schedule dose
Group1 Adults aged NA 1 NA NA 30 NA NA NA NA
18 years and above
Group2 Children aged NA 1 NA NA 30 NA NA NA NA
2-5 years
Group3 Infants aged 3 0, 1, 2, 3 NA NA 30 NA NA NA NA
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months month
Group4 Infants aged 2 0, 2, 4, 3 NA NA 30 NA NA NA NA
months month
Pivotal trial
Phase/Study No. Phase III PCV13-TT 002
Country China
Status Completed
Objective Primary To assess the immunogenicity of Walvax PCV13-TT in healthy children aged 3 months and above. The proposed regimen consists of four

doses, with a single intramuscular injection at 0, 1, 2 month for primary immunisation schedule and a single intramuscular injection at 12 to 15 months for booster immunisation schedule. To assess the non-inferiority of 7 common pneumococcal serotypes compared to Pnevfromnar 7® and the efficacy of 6 additional pneumococcal serotypes with same immunization schedule.

Secondary (1) To assess the immunogenicity of Walvax PCV13-TT in healthy children aged 2 months (≥6 weeks). The proposed regimen consists of four doses, with a single intramuscular injection at 0, 1 and 2 month for primary immunisation schedule and a single intramuscular injection at 12 to 15 months for booster immunisation schedule.

(2) To assess the non-inferiority of 7 common pneumococcal serotypes compared to Pnevfromnar 7® and the efficacy of 6 additional pneumococcal serotypes with same immunization schedule.

(3) To assess antibody opsono-phagocytosis activity (OPA) Walvax PCV13-TT in healthy children aged 2 months (≥6 weeks) to 71months by different immunization procedures.

(4) To assess the safety of Walvax PCV13-TT in healthy children aged 2 months (≥6 weeks) to 71months. Evaluate safety of PCV13 in 6 weeks - 5 years old children

Group Test Group Control Group

Vaccine Name Walvax PCV13-TT Pnevfromnar 7® (Pfizer)

Manufacture Yuxi Walvax Biotechnology Co., Ltd Wyeth Pharmaceuticals Inc.

Specification Pneumococcal polysaccharide serotype 1: 2.6 µg Each 0.5 ml dose contains 2 µg of polysaccharide from pneumococcal

Pneumococcal polysaccharide serotype 3: 2.5 µg serotypes 4, 9V, 14, 18C,19F,23F and 4 µg polysaccharide for 6B.

Pneumococcal polysaccharide serotype 4:3.0 µg

Pneumococcal polysaccharide serotype 5:2.5 µg

Pneumococcal polysaccharide serotype 6A:2.5 µg

Pneumococcal polysaccharide serotype 6B:6.0 µg

Pneumococcal polysaccharide serotype 7F :2.85 µg

Pneumococcal polysaccharide serotype 9V:2.5 µg

Pneumococcal polysaccharide serotype 14:2.75 µg

Pneumococcal polysaccharide serotype 18C:3.25 µg

Pneumococcal polysaccharide serotype 19A:2.6 µg

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MODULE 5 CLINICAL STUDY REPORTS

Version No. Walvax-202007-00 Yuxi Walvax Biotechnology Co., Ltd.

Pneumococcal polysaccharide serotype 19F:2.75 µg

Pneumococcal polysaccharide serotype 23F:3.0 µg

Batch number 201511003 H29135H13880

Expiry Date 2017.11 2017.05

Items Age Primary Series Boost Dose Number Primary Series Boost Dose Number

Schedule dose Schedule dose Schedule dose

Group1 Infants aged 3 0, 1, 2 month 3 12-15month 1 520 0, 1, 2, month 3 12-15month 1 520 months

Infants aged 2 0, 2, 4 month 3 12-15month 1 520 N/A N/A N/A N/A months

Group2 Infants aged 7- 0, 2 month 2 12-15month(at least 2 1 200 0, 2 month 2 12-15month(at least 2 1 200 11 least 2 months months apart from the months apart from the second dose)

second dose)

Group3 Infants aged 0, 2 month 2 N/A N/A 200 0, 2 month 2 N/A N/A 200 12-23 months

Group4 Children aged 0 month 1 N/A N/A 200 0 month 1 N/A N/A 200 24-71 months

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MODULE 5 CLINICAL STUDY REPORTS

Version No. Walvax-202007-00 Yuxi Walvax Biotechnology Co., Ltd.

5.3 CLINICAL STUDY REPORTS

5.3.1 Reports of Biopharmaceutical Studies

N/A

5.3.2 Reports of Studies Pertinent to Pharmacokinetics Using Human Biomaterials

N/A

5.3.3 Reports of Human Pharmacokinetic (PK) Studies

N/A

5.3.4 Reports of Human Pharmacodynamic (PD) studies

N/A

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MODULE 5 CLINICAL STUDY REPORTS

Version No. Walvax-202007-00 Yuxi Walvax Biotechnology Co., Ltd.

5.3.5 Reports of Efficacy and Safety Studies

5.3.5.1 Study Reports of Controlled Clinical Studies Pertinent to the Claimed Indication

In China, the pivotal phase III clinical trial of 13-valent Pneumococcal Polysaccharide Conjugate Vaccine (Tetanus Toxoid Protein) (PCV13-TT) was a randomized and blinded non-inferiority study comparing to Pevnar. This study evaluated the immunogenicity and safety of PCV13-TT and Pevnar in 2760 subjects 2 through 71 months of age (at least 6 weeks of age). A total of 1040 infants aged 3 months were randomized in a 1:1 ratio as study group and control group, with PCV13-TT or Pevnar given at 3, 4, 5 and 12-15 months of age, respectively; for 520 infants aged 2 months (at least 6 weeks of age), PCV13-TT were vaccinated at 2, 4, 6 and 12-15 months of age. 1200 Children of 7 through 71 months of age, including children 7 months through 11 months of age, 12 months through 23 months of age and 24 months through 5 years of age (prior to the 6th birthday) who were naïve to pneumococcal conjugate vaccine, were split into three age groups with each containing 400 subjects randomized in a 1:1 ratio given 3, 2 or 1 dose of PCV13-TT or Pevnar, respectively, according to the age-appropriate schedules. Enzyme-linked immunosorbent assay (ELISA) was utilized to detect the IgG antibody concentration in all serum samples collected

from each group; additionally, a subsets of sera randomly obtained from 100 subjects in each age group for PCV13-TT and Prevnar, totaling 900, were tested by opsonophagocytic assay (OPA) to measure the ability of serotype-specific functional antibodies to eliminate corresponding pneumococci by promoting complement-mediated phagocytosis.

The details of phase III clinical trial of PCV13-TT please refer to Annex 5-1Phase III clinical study report of 13-valent Pneumococcal Polysaccharide Conjugate Vaccine (Tetanus Toxoid Protein)

5.3.5.2 Study Reports of Uncontrolled Clinical Studies

Phase I clinical trial of the PCV13-TT was conducted in China, including 120 subjects, to evaluate the safety of 13-valent pneumococcal polysaccharide conjugate vaccine (PCV13-TT) in healthy population aged 2-3 months (minimum age: 6 weeks), 2-5 years and 18 years and above and to explore the IgG immune responses of PCV13-TT vaccinated in subjects aged 2-3 months (minimum age: 6 weeks), 2-5 years and 18 years and older. For subjects have been vaccinated with at least one dose of PCV13-TT were observed for 30 minutes after each dose of PCV13-TT, and were under active observation for adverse events (AEs) within 0-7 days post vaccination, while for 8-30 days post vaccination, AEs were spontaneously reported and collected by subjects. Serious adverse events (SAEs) were monitored for 6 months after full immunization of PCV13-TT which was administered in accordance with the immunization procedure prescribed in the protocol.

The details of phase I clinical trial please refer to Annex 5-2 Phase I clinical study report of 13-valent Pneumococcal Polysaccharide Conjugate Vaccine (Tetanus Toxoid Protein)

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Version No. Walvax-202007-00 Yuxi Walvax Biotechnology Co., Ltd.

5.3.5.3 Reports of Analyses of Data from More than One Study

N/A

5.3.5.4 Other Study Reports

N/A

5.3.6 Reports of Post-Marketing Experience

The 13-valent Pneumococcal Polysaccharide Conjugate Vaccine (Tetanus Toxoid Protein) licensed on 30th 12 2019, the post-marketing data will be reported at the fourth quarter by Walvax in Dec. 2020. This is because there is only one major source for AEFI in China. Firstly, the Chinese CDC, which is the only allowed point of vaccination, collects the original AEFI and then reported to National Medical Products Administration (NMPA). Subsequently, NMPA will sort out the reported AEFI by province and distribute to Provincial MPA (PMPA). Finally, the PMPA will send the AEFIs of last year to each company according to the vaccine batch number in every July or August. The AEFI data for 2019 will only be issued by Chinese regulatory agency in July or August in 2020.

5.3.7 Case Report Forms and Individual Patient Listings

N/A

5.4 Literature References

Please refer to the Annex 5-1 and 5-2.

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Phase I Clinical Study Report of 13-valent Pneumococcal Polysaccharide Conjugate Vaccine
Protocol No.: 002152015002

Clinical Study Report
of the
13-valent Pneumococcal Polysaccharide
Conjugate Vaccine
(Tetanus Toxoid Protein)

Phase I

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Phase I Clinical Study Report of 13-valent Pneumococcal Polysaccharide Conjugate Vaccine
Protocol No.: 002152015002

Generic Name of Test Drug: 13-valent Pneumococcal Polysaccharide Conjugate Vaccine

Study Type: Single center, open label, non-control

Study No.: 022152015002

Start and End Date: June 2015 to December 2017

Principal investigator (signature):

Responsible Unit: (stamp) Beijing Chaoyang District Center for Disease Control and Prevention

Responsible Statistician (signature):

Statistical unit (stamp): Beijing Key Tech Statistical Consulting Co., Ltd.

Drug Registration Applicant (stamp): Yunnan Walvax Biotechnology Co., Ltd

Yuxi Walvax Biotechnology Co., Ltd

Contact person of the applicant: Yuan Lin

Tel: 138888316530

e-mail: ynwsyl@walvax.com

Address: No.83, Dongfeng South Road, High Tech Zone, Yuxi, Yunnan, China

Postal Code: 653100

Report Version: v2.0 Report Date: October 2018

Original data kept in:0 Beijing Chaoyang District Center for Disease Control and prevention

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SYNOPSIS OF PHASE I CLINICAL TRIAL

(Tabulated Clinical Trial Report)

Vaccine name 13-valent pneumococcal polysaccharide conjugate vaccine
Yunnan Walvax Biotechnology Co., Ltd.

Sponsor

Yuxi Walvax Biotechnology Co., Ltd

Study title Phase I clinical study of 13-valent pneumococcal polysaccharide conjugate vaccine
Clinical trial

Approval No.: 2015L00591 Approved on April, 2015

approval

Principal

investigator

and Nianming Shi, Beijing Chaoyang District Center for Disease Control and Prevention

responsible

units

Duration of

From: June, 2015 To: December, 2016

clinical trial

Number of

120

subjects

Appearance inspection:

appears as milky white homogeneous suspension after shaking, comply with the regulation.

Extractable volume:

not less than 0.5 mL/vial, comply with the regulation.

pH value:

6.3 (specification: 5.5-7.5), comply with the regulation.

Aluminum phosphate content (mg/mL):

1.0 (specification:0.8-1.2), comply with the regulation.

Sodium chloride content (g/L):

8.5 (specification: 7.5-9.5), comply with the regulation.

Polysaccharide content (µg /dose):

Results of

Serotype 1: 2.6; serotype 3: 2.5; serotype 4: 2.7; serotype 5: 2.6; serotype 6A: 2.8;

investigational

serotype 6B: 5.5; serotype 7F: 3.0; serotype 9V: 2.7; serotype 14: 2.3; serotype 18C: 2.8; vaccine
 serotype 19A: 2.9; serotype 19F: 2.9; serotype 23F: 2.5.
 (standard range: 2.2-3.3 for serotype 1,3, 4, 5, 6A, 7F, 9V, 14, 18C, 19A, 19F, 23F; 4.4-6.6 for 6B), comply with the regulation.
 Abnormal toxicity test:
 All mice survived during the observation period, and no abnormal reaction was observed; weight gain was observed in each mouse at the end of observation, comply with the regulation.
 All guinea pigs survived during the observation period, and no abnormal reaction was observed; weight gain was observed in each guinea pig at the end of observation, comply with the regulation.
 Bacterial endotoxin content (EU/dose): not higher than 12.5, comply with the regulation.
 Sterility test: No bacterial growth, comply with the regulation.
 Usage and dosage: 0.5 mL for
 13-valent pneumococcal
 intramuscular injection
 polysaccharide conjugate
 Adults aged
 vaccine, manufactured by
 18 years 1 dose
 Yuxi Walvax
 Batch No.: and above
 Biotechnology Co., Ltd,
 Investigational 201511003 Children
 for injection, 0.5 mL/vial,
 vaccine Expiration date: aged 2-5 1 dose
 each dose contains 2.2 µg
 23/11/2017 years
 of polysaccharides from
 pneumococcal serotype 1, 2
 One dose each at month
 3, 4, 5, 6A, 7F, 9V, 14, Infants aged
 0, 2 and 4, 3 doses in total
 18C, 19A, 19F, and 23F, 2 months

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and 4.4 µg polysaccharide
 for 6B. One dose each at month
 Infants aged
 0, 1 and 2 months, 3
 3 months
 doses in total

Safety evaluation measures:
 Local reactions: administration site pain, itching, redness, swelling, induration, rash (injection site) size and severity, and body movement disorders.
 Systemic reactions: observation of fever, allergy, headache, tiredness/fatigue, nausea/vomiting, diarrhea, myalgia, Cough and other discomfort or clinical adverse
 Efficacy reactions.
 measures Laboratory tests: hematology and urinalysis, hepatic and renal function correlates in blood biochemistry.
 Immunogenicity evaluation measures:
 Proportion of subjects with specific pneumococcal IgG antibody concentration ≥ 0.35 µg/mL (positive rate)
 Specific pneumococcal IgG geometric mean concentration (GMC).
 Safety:

The incidence of vaccine-related adverse events was 80.00%, 53.33%, 66.67% and 70.00% in adults, children, 2-month-old and PCV13-TT (3 months) group, respectively. A total of 6 cases in the adult group experienced grade 3 local adverse reactions, corresponding to an incidence rate of 20% (6 / 30), including 1 case of local redness, 1 case of induration, 2 cases of swelling, 2 cases of redness combined with swelling. The incidence rate of adverse reactions with grade 3 severity in other age groups was less than 15%.

The most common symptoms of systemic adverse reactions were fever in adults and children aged 2-5 years, and were fever and diarrhea in 2-month-old and 3-month-old infants. Common local adverse reactions in adults included redness, pain and swelling at the vaccination site, and were pain and redness in children aged 2-5 years, as compared to redness in infants 2 and 3 months of age.

Adverse events mainly occurred on Day 0-7. No adverse events related to the study vaccine were collected on Day 8-30, and no serious adverse events were collected during the observation period.

Clinical trial
 Immunogenicity:
 evaluation

High IgG antibody positive rates and antibody levels were observed in adults and children aged 2-5 years before immunization for all 13 serotypes. The IgG antibody positive rates of all serotypes reached 100% or over 80% after immunization, the antibody level increased more than 2 folds after immunization. In 2-month-old and PCV13-TT (3 months) group, all the 13 serotypes were with lower IgG antibody positive rates and antibody levels than those in the adult and children groups before immunization, while after immunization, the antibody positive rates all reached 100% except for serotype 3, 5, 6A and 6B, and the fold increase of antibody was significantly higher in the two infants groups compared to adult group and children group.

The positive rates and antibody levels of OPA were lower than that of IgG in each age PCV7 before immunization, while the level of OPA antibody after immunization significantly increased with a higher fold increase compared with IgG. The OPA positive rate of each serotype was $\geq 90\%$ in subjects aged 3 months after immunization.

The correlation analysis between IgG and OPA indicated high consistency between post-

immunization antibody positive rates of the two measures, while some bias might exist due to the small sample size.

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Phase I Clinical Study Report of 13-valent Pneumococcal Polysaccharide Conjugate Vaccine
Protocol No.: 002152015002

1. INTRODUCTION

The 13-valent pneumococcal polysaccharide conjugate vaccine, jointly developed by the Yunnan Walvax Biotechnology Co., Ltd. and the Yuxi Walvax Biotechnology Co., Ltd., is consistent with the marketed 13-valent pneumococcal conjugate vaccine with respect to serotypes, and is used for the prevention of related diseases caused by infections of the 13 pneumococcal serotypes contained in this vaccine (types 1, 3, 4, 5, 6A, 6B, 7F, 9V, 14, 18C, 19A, 19F, and 23F). Upon review and approval by the National Medical Products Administration (NMPA) in accordance with the Drug Administration Law of the P.R.C and Provisions for Drug Registration, this vaccine met the relevant provisions for drug registration, and therefore, the "Drug Clinical Trial Approval" was issued in Apr. 2015 (Approval No.: 2015L00591) for conduct of clinical trials.

Entrusted by the Yunnan Walvax Biotechnology Co., Ltd. and the Yuxi Walvax Biotechnology Co., Ltd., the Beijing Chaoyang Center for Disease Control and Prevention (CDC) was responsible for the clinical trial of this vaccine, with a view to evaluating its safety in healthy population aged 2 months and above. The Beijing Chaoyang CDC, based on the relevant requirements in the "Drug Clinical Trial Approval" by NMPA (Approval No.: 2015L00591) and the Provisions for Drug Registration and as per the protocol, conducted a phase I clinical trial on the 13-valent pneumococcal polysaccharide conjugate vaccine jointly developed by the Yunnan Walvax Biotechnology Co., Ltd. and Yuxi Walvax Biotechnology Co., Ltd., and issued this clinical trial report.

2. OBJECTIVES

2.1. PRIMARY OBJECTIVES

To evaluate the safety of 13-valent pneumococcal polysaccharide conjugate vaccine (PCV13-TT) in healthy population aged 2-3 months (minimum age: 6 weeks), 2-5 years and 18 years and above.

2.2. SECONDARY OBJECTIVES

To explore the IgG immune responses of PCV13-TT vaccinated in subjects aged 2-3 months (minimum age: 6 weeks), 2-5 years and 18 years and older.

3. CLINICAL TRIAL MANAGEMENT

3.1. MAIN UNITS AND COLLABORATORS OF CLINICAL STUDY

3.1.1. Main responsible units

This clinical study was undertaken by the Beijing Chaoyang CDC. This research institution obtained the CFDA (now NMPA) approval as "One-Time Certification for Vaccine Clinical Trial Institution" after review, and the clinical trial protocol, the informed consent form, and related forms were reviewed and approved by the ethics committee.

The Beijing Chaoyang CDC has carried out multiple clinical studies on preventive biological products successfully. The principle investigator has experience in clinical studies on preventive biological products, all staff in the Institute of Vaccine Clinical Study that participated in this study have attended GCP training and examinations organized by the national government or this institution, and after training and authorization, all the research participants had been aware of the responsibilities and obligations of clinical investigators and familiar with the nature, effects, efficacy, and safety of the investigational drugs to be used in the clinical trial. Effective measures and conditions were available to ensure the safety of subjects and the timely, accurate, and truthful recording of clinical research data. They were mainly responsible for:

- ✓ Formulating the vaccine clinical trial protocol, and organizing the implementation of such a protocol;

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- ✓ Preparing and printing such on-site application forms as informed consent forms, diary cards, and case report forms;
- ✓ Submitting ethical review materials to the ethics committee for approval;
- ✓ Choosing the clinical trial site meeting the requirements set forth in the protocol, and recruiting trial subjects;
- ✓ Organizing the implementation of on-site vaccination and on-site work related to clinical observations;
- ✓ Organizing the follow-up of subjects in the clinical trial site and the collection of adverse events, and reporting, investigating and handling abnormal vaccine reactions;
- ✓ Completing the collection of blood samples, laboratory processing and delivery to the designated laboratory as per the protocol;
- ✓ Completing all the forms and case report forms on the clinical trial site;
- ✓ Issuing a clinical trial report.

3.1.2. Technical collaboration unit

National Institutes for Food and Drug Control, which was responsible for laboratory testing of serum samples and issuing a testing report.

3.1.3. Research institutions on the trial site

The research institutions on this trial site included Heizhuanghu Community Health Service Center Affiliated to Shuangqiao Hospital in Chaoyang District of Beijing, Xiaohongmen Community Health Service Center in Chaoyang District of Beijing, and Nanmofang Community Health Service Center. The number of trained investigators, technical strengths, and environmental facilities on these clinical trial sites met the needs of the clinical trial, so were the resources of subjects. The direct reporting network system operated normally, and the monitoring system for side reactions arising from vaccination was complete. All the laboratories passed the metrological certification. The institutions on the trial site were responsible for subject recruitment, organization on the site, and handling and reporting of adverse events.

3.1.3.1. Beijing Chaoyang Center for Disease Control and Prevention

The Beijing Chaoyang CDC, which is located in the east of Beijing, covers a total area of 470.8 square kilometers, and has an actual resident population of near 5.1 million. Currently, the Beijing Chaoyang CDC has a total of 199 professional technical personnel, which account for 86% of the actual number of employees, and the personnel have an average age of 35.9 years. Specifically, there are 35 personnel with senior titles, 11 with Ph.D degrees, 86 with master's degrees, and 74 with bachelor's degrees. Since 2007, the center has undertaken scientific research projects from the Chinese Preventive Medicine Association, the NMPA 12th Five-Year Major Project, etc., and the phase III clinical trial of the EV71 vaccine. Since the promulgation of "Good Clinical Practice on Vaccine Clinical Trials (Pilot Edition)" by NMPA in 2013, it has undertaken the phase III clinical study involving the freeze-dried rabies vaccine (MRC-5 cells) for human use from Beijing

Minhai Biological Technology Co., Ltd., the randomized, blinded, parallel, controlled phase II clinical study involving the recombinant humanized rabies virus monoclonal antibody injection from the NCPIC, and the phase III clinical trial involving the recombinant human papillomavirus bivalent (types 16/18) vaccine (yeast) from the Shanghai Zerun Biotechnology Co., Ltd. The employees from the CDC have published more than 10 papers at home and abroad, including 6 papers in foreign languages, and published 5 monographs on vaccines. All staff in the Institute of

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Vaccine Clinical Study that participated in this study have attended GCP training and examinations organized by the national government or this institution.

(1) Heizhuanghu Community Health Service Center Affiliated to Shuangqiao Hospital in Chaoyang District of Beijing

Established in 1958, the Shuangqiao Hospital in Chaoyang District of Beijing, is affiliated to the Health Bureau of Chaoyang District, and has been approved by the Beijing Municipal Human Resources and Social Security Bureau as a designated medical institution for medical insurance in Beijing. Located in the southeast of the Chaoyang District, the hospital covers an area of 26 square kilometers and a building area of 3512 square meters. The hospital has 72 personnel of various professional and technical levels, including 3 with senior titles, 9 with intermediate titles, 2 with master's degrees, 30 with bachelor's degrees, and 20 with associated college degrees. There are altogether 9 community health service stations under the hospital, including Daludian, Xiaoludian, Wanzhiyong, Dingxinzhuan, Langgezhuang, Langxinzhuan, Shuangshu, Dongxu, and Yijingcheng, undertaking the six-in-one medical and health work for the 80,000 population in the Heizhuanghu area with respect to prevention, health care, medical treatment, rehabilitation, family planning, and health education. The hospital has such clinical departments as general practice, Traditional Chinese medicine (TCM), gynecology, stomatology, rehabilitation, and health care. The hospital undertakes the related work in the emergency work station in the southern area according to the "regionalization" and "gridding" goal as set by the Chaoyang District Health Bureau, and it has carried out the Sanofi Pasteur's HAF87 clinical trial study.

(2) Beijing Chaoyang Xiaohongmen Community Health Service Center

Established in 1953, the Beijing Chaoyang Xiaohongmen Community Health Service Center is affiliated to the Health Bureau of Chaoyang District, and has been approved by the Beijing Municipal Human Resources and Social Security Bureau as a designated medical institution for medical insurance in Beijing. Located at No. 1 Xiaohongmen Middle Street, Chaoyang District, the Xiaohongmen Community Health Service Center covers an area of 12.47 square kilometers and a building area of 3037 square meters. The hospital has 70 personnel of various professional and technical levels, including 2 with senior titles, 8 with intermediate titles, 8 with bachelor's degrees and 24 with associated college degrees. There are altogether 4 community health service stations under the hospital, including Paifang, Xiaohongmen, Xiaocun, and Longzhaoshu, undertaking the six-in-one medical and health work for the 80,000 population in the Xiaohongmen area with respect to prevention, health care, medical treatment, rehabilitation, family planning, and health education. The center has such clinical departments as internal medicine, surgery, TCM, gynecology, stomatology, ophthalmology, and health care. The phase III clinical study on the H1N1 influenza vaccine, the phase III clinical study on the EV71 vaccine, the phase III clinical study on the meningococcal polysaccharide ACYW135 vaccine, and the phase III clinical study on the lyophilized human rabies vaccine (MRC-5 cells) have been carried out here.

(3) Beijing Chaoyang District Nanmofang Community Health Service Center

The Beijing Chaoyang Nanmofang Community Health Service Center is affiliated to the Chaoyang Health and Family Planning Commission, representing a designated medical institution for medical insurance in Beijing. Located at No. 29 West Dawang Road, Chaoyang District and with a service area of 9.64 square kilometers, it serves a population of 160,000. The center has such departments as general medicine, stomatology, surgery, gynecology, TCM, orthopedic massage, physiotherapy, and preventive health care, and five community health service stations under it: Nanxinyuan, Shuanglongnanli, Huanlegu, Zinanjiayuan, and Baiziwan, with experts on TCM, endocrinology, gynecology, and surgery from the Chuiyangliu medical consortium (tertiary hospital) for specialist clinic service of the center on a regular basis. The center is equipped with

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such examination and treatment equipment as Color Ultrasound Doppler Apparatus, CR, Automatic Biochemical Analyzer, Fluorescent Immunoassay Analyzer, Lumbar and Cervical Spine Treatment Instrument, Infrared Polarized Light Treatment Instrument, and TCM Meridian Flow Injection Treatment Instrument. With these pieces of equipment, vascular ultrasonography, cardiac ultrasound examinations, testing of tumor markers, Helicobacter pylori testing, high and low frequency and infrared pain treatment and other physical therapy items are available. The center has 134 employees, including 105 professional technicians with associate college degrees or above and 6 with professional titles as associated chief physicians or above, and is open to an average of over 1000 outpatients per day. In areas under its jurisdiction, the health care department serves 4000 children 0–3 of age, and manages over 4000 children from 15 nursery schools, and 8 primary and secondary schools.

3.2. SPONSOR

Yunnan Walvax Biotechnology Co., Ltd. and Yuxi Walvax Biotechnology Co., Ltd., which were mainly responsible for:

(1) Providing the preliminary clinical trial protocol and finally confirming the clinical trial protocol;

(2) Providing such application documents on the site as the Clinical Trial Approval, the Investigator's Brochure for the clinical trial (pre-clinical product safety information), and the product implementation standards;

(3) Providing vaccines used in the clinical study and issuing test control reports;

(4) Dispatching monitors to assess and recognize the clinical trial site, perform the monitoring duty as per the GCP requirements, and verify the research data during the trial;

(5) Participating in the investigation and handling of cases of abnormal reactions arising from the vaccine, and providing medical and related compensation fees for cases of abnormal reactions which are clinically proven related to the vaccine as per relevant regulations;

(6) Providing the fund for the clinical study.

3.3. STUDY TEAM

Principal investigator, Shi Nianmin (Beijing Chaoyang District Center for Disease Control and prevention)

Medical Monitor, Wang Meng (Beijing Simoon Record Pharma Information Consulting Co., Ltd.)

Director of Statistics, Jiang Zhiwei (Beijing Key Tech Statistical Consulting Co., Ltd.)
Clinical Report Writer, Li Yanping (Center for Disease Control and prevention of Guangxi Zhuang Autonomous Region)
Clinical Report Reviewer, Shi Nianmin

3.4. ETHICS

This study complied with the "Technical Guidelines for Vaccine Clinical Trials" by NMPA and the ethical principles of medical study in humans set forth in the World Medical Association's "Helsinki Declaration", "Good Clinical Practice" (GCP), and all applicable regulatory requirements. Prior to this study, this clinical research protocol and the informed consent form had been approved by the Ethics Review Committees of the Beijing Chaoyang CDC. The subjects for the clinical trial were recruited in accordance with the principle of informed consent and voluntariness. All subjects/legal guardians were informed of matters related to the vaccine clinical Confidential 10 / 56

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trial in detail in order to obtain their voluntary signed informed consent forms. The subjects were screened in strict accordance with the inclusion criteria.

3.5 DATA MANAGEMENT

The study records and reports were managed in accordance with Articles 48–52 of the GCP issued by NMPA. All operations were performed as per the SOPs, and the clinical trial records and report data will be stored in the on-site CDC as required until five years after the clinical trial.

3.6 OBSERVATION AND REPORTING OF ADVERSE EVENTS

In strict accordance with the GCP principles, each subject was observed closely for local and systemic adverse reactions and adverse events. All adverse reactions and adverse events were recorded in the CRF fully, including the symptoms, time of occurrence, severity, duration, measures adopted and the status of recovery; the investigator was responsible for judging the relevance of the adverse events to the test vaccine.

If serious adverse events occurred during the study, appropriate treatment measures were taken immediately, whether they were associated with the test vaccine, and the events were reported to the sponsor, the ethics committee, and the local food and drug administration for the responsible institutions of each sub-center by telephone or fax within 24 hours after learning about such information.

4. STUDY DESIGN AND PROCEDURES

4.1. DESCRIPTION OF STUDY DESIGN AND PROTOCOL

This clinical trial was designed as a single center, open, non-controlled trial.

Healthy subjects aged 2 months and above were included. The planned sample size was 120, and subjects were recruited in three stages in the order of: adults (18 years and above), children (2-5 years) and infant and toddlers (2-3 months, minimum 6 weeks).

A total of 30 subjects were enrolled in the first stage. Each subject was administered with 1 dose of 13-valent pneumococcal polysaccharide conjugate vaccine (PCV13-TT). Blood samples were collected 3 times from all subjects before vaccination, on Day 3 post vaccination (interval 2 days, same below) and Day 30 post vaccination, respectively. The blood samples collected before vaccination and on Day 3 post vaccination were assessed by hematology, liver and kidney function tests, meanwhile, the serum IgG antibody concentration was measured using the blood sample taken before vaccination and on Day 30 post vaccination. Subjects who had no abnormal results of hematology, liver and kidney function tests before vaccination and met other inclusion criteria could proceed to get vaccinated. If no SAE related to study vaccine was identified by the investigators after 7 days' observation and no grade 4 vaccine-related abnormalities determined by the investigators as measured by hematology, liver and kidney function tests were detected on Day 3 after vaccination, then the study proceeded to the next stage.

A total of 30 subjects were enrolled in the second stage. Each subject was administered with 1 dose of PCV13-TT. Blood samples were collected 3 times from all subjects respectively before the vaccination, on Day 3 post vaccination (interval 2 days, same below) and Day 30 days post vaccination. The blood samples collected before vaccination and on Day 3 post vaccination were assessed by hematology, liver and kidney function tests, meanwhile, the serum IgG antibody concentration was measured using the blood sample taken before vaccination and on Day 30 post vaccination. Subjects with testing results within the normal range predefined in the protocol or not within the normal range but were determined by site investigators to be normal and in compliance with other inclusion criteria could be administered with the vaccine. If no SAE related to study

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vaccine was identified by the investigators after 7 days' observation and no grade 4 vaccine-related abnormalities determined by the investigators as measured by hematology, liver and kidney function tests were detected on Day 3 after vaccination, then the study proceeded to the next stage.

A total of 60 subjects were enrolled in the third stage and subjects were divided into 2 months (minimum age: 6 weeks) group and 3 months group with 30 subjects in each arm. Subjects in the 2 months group were administered with 3 doses of PCV13-TT according to the procedure of one dose each vaccinated at 0, 2 and 4 months and subjects in the 3 months group were administered with 3 doses of PCV13-TT following the procedure of one dose each at 0, 1 and 2 months. Blood samples were collected from all subjects twice respectively before the first dose and 30 days after the third dose. The serum was separated and the IgG antibody concentration was measured. All subjects were observed for 30 minutes after each dose of PCV13-TT, and were under active observation for adverse events (AEs) within 0-7 days post vaccination, while for 8-30 days post vaccination, AEs were spontaneously reported and collected by subjects. Serious adverse events (SAEs) were monitored for 6 months after full immunization of PCV13-TT which was administered in accordance with the immunization procedure prescribed in the protocol.
Flow chart of Phase I clinical trial of 13-valent pneumococcal conjugate vaccine

▲: No abnormal findings for measures associated with hematology, hepatic and renal functions

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■: No grade 4 and vaccine-related abnormal findings for measures associated with hematology, hepatic and renal functions as adjudicated by the investigator

●: No investigational vaccine-related SAE at Day 7 as adjudicated by the investigator

4.2. ENDPOINTS

4.2.1. Primary Endpoints

(1) The incidence of adverse reactions / events, and the incidence of adverse reactions of grade 3 and above at 30 days after vaccination of 13-valent pneumococcal conjugate vaccine in first, second and third stage following respective vaccination schedule.

(2) If death occurs during the clinical trial, then the clinical trial shall be interrupted / terminated.

4.2.2. Secondary Endpoint

The changes of hematology, hepatic and renal function were observed in the first and second stages after vaccination.

4.2.3. Exploratory Endpoints

(1) The proportion of subjects with vaccine serotypes-specific pneumococcal IgG antibody concentration $\geq 0.35 \mu\text{g/mL}$ (positive rate) at 30 days after vaccination of 13-valent pneumococcal conjugate vaccine in the first, second and third stage following respective vaccination schedule.

(2) The geometric mean concentration (GMC) of vaccine serotypes-specific pneumococcal IgG antibody at 30 days after vaccination of 13-valent pneumococcal conjugate vaccine in the first, second and third stage following respective vaccination schedule.

4.3. STUDY DURATION

Enrollment and study period for clinical trial subjects: a total of 18 months.

The first subject was enrolled in June 2015.

The study was completed in December 2017.

4.4. PROPOSED INDICATIONS AND RATIONALES

Indications: for the prevention of related diseases caused by infections of the 13 pneumococcal serotypes contained in this vaccine (types 1, 3, 4, 5, 7A, 6B, 7F, 9V, 14, 18C, 19A, 19F, and 23F).

Justification: "Pneumococcal polysaccharide vaccine" in the European Pharmacopoeia (version 5.0), and the instructions of use for similar vaccines that have been marketed abroad.

4.5. SELECTION OF SUBJECTS

4.5.1. Study Population

The eligible subjects were population 2 months of age (minimum age: 6 weeks) and above, without contraindications and no history of pneumococcal vaccination, and there were no gender and ethnicity restrictions. Subjects might participate in the study voluntarily, and the informed consent form must be signed by the subject or the guardian of the subject (for subjects aged 12-17 years, the signature from the recipient and the guardians should all be collected).

4.5.2. Inclusion Criteria:

- Healthy infants (aged 2-3 months [at least 6 weeks of age], 2-5 years, and 18 years and above), children or adults as evidenced by medical history, physical examination, and investigators' judgement;

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- Subjects and or legal guardians of subjects voluntarily participate in the study and must sign the informed consent;
- Subjects / legal guardians of subjects are able to follow requirements of the trial;
- No abnormal findings for pre-immunization hematology, hepatic and renal function-related measures for subjects aged 2-5 years and 18 years or above;
- Subjects who have not been vaccinated with pneumococcal vaccines, and have not been administered any other preventive products within 7 days;
- Axillary temperature $\leq 37^{\circ}\text{C}$.

4.5.3. Exclusion Criteria:

4.5.3.1. Exclusion Criteria for First Vaccination

- Previously vaccination listed or vaccinated with trial pneumococcal vaccine
- History of invasive disease caused by *Streptococcus pneumoniae*, which was confirmed by bacterial culture
- Prior history of allergic reactions to any drug, vaccine, or vaccine-related serotype
- Vaccination-related fever of 39°C or more has occurred with biological products used for vaccination
- Birth weight $< 2.5 \text{ kg}$
- There is a history or family history of allergies, convulsions, seizures, encephalopathy, and psychosis
- Abnormal births of infants (difficult labors, aided by instruments) or a history of asphyxia, neurological organ damage
- History of diagnosed thrombocytopenia or other coagulation disorders
- Infants with diagnosed pathological jaundice
- Known or suspected to suffer from diseases, including: respiratory system disease, acute infection or chronic disease, serious cardiovascular disease, kidney disease, malignant tumor, skin diseases, mothers with HIV infection (as evidenced by test report)
- Known serious congenital malformations or severe chronic disease; congenital deformities, developmental disorder, or a clinical diagnosis of severe chronic diseases (such as Down's syndrome, diabetes, sickle cell anemia, or neurological disorders, Guillain-Barre Syndrome)
- Known or suspected co-morbidities including: respiratory disorders, acute infections or active attack of chronic disease, serious cardiovascular disorders, liver and kidney disorders, malignant neoplasms, skin diseases.
- Received blood products or globulin. Hepatitis B hyper-immune globulin can be used
- Participating in other clinical trials
- Any condition that the investigator considered to have the potential to affect the evaluation of the study

4.5.3.2. Exclusion Criteria for 2nd, 3rd Vaccination

- Subjects who had severe allergic reactions during the first or second vaccination
- Subjects who experienced serious adverse reactions from the previous vaccination with causal relationship demonstrated
- After the first vaccination, the newly discovered or newly occurred conditions not conforming to the first dose of the inclusion criteria or to the exclusion criteria for the first dose
- Any other conditions that may affect the study evaluations as assessed by investigators

4.5.4. Criteria for Early Termination of the Trial

Early termination means that the subject terminates the study visit prior to the time set forth in clinical study protocol, and no more study procedures would be performed, the subject would not

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be followed and no more information should be collected from the subject. In case of any of the

following conditions, the study will be terminated in advance:

- The subject or the guardians may withdraw from the trial at any time at their own disposal;
- Subjects who leave the region and fail to complete the study procedures are deemed to have voluntarily withdrawn from the test;
- Occurrence of intolerable adverse events, whether or not related to the investigational vaccine;
- The health status restricts their ability to participate in the study, and the observation should be terminated according to the investigator's judgment;
- Any other reason considered by the investigators.

Any adverse event observed in subjects are adjudicated by the investigator to determine the relevance with investigational vaccine, and the investigator shall decide whether the clinical trial shall be terminated in advance; if the early-terminated subject has been vaccinated with the investigational vaccine, the clinical trial data of the subject will still be used for safety analysis; the subjects cannot be replaced during the conduct of the trial; the subjects who have been vaccinated with the vaccine used in the clinical trial shall be followed by investigator to monitor the adverse events occurred in the subject and related to the study after withdrawing or terminating the study.

4.5.5 Criteria for interruption or termination of the clinical trial

The clinical trial shall be interrupted or terminated under any of the following circumstances:

- ✓ Occurrence of subject deaths related to vaccination;
- ✓ Occurrence of collective serious adverse reactions;
- ✓ Interruption or termination of the clinical trial as required by the sponsor due to such reasons as the investigational vaccine; and
- ✓ Interruption or termination as required by the NMPA.

4.6. GROUPS

The total number of patients in phase I clinical trial was 120, and the study is divided into three stages in sequence as follows:

Clinical Study Groups and Number of Vaccine Recipients

Age at Number of subjects Name of

Stages Immunization schedule

enrollment vaccinated vaccination

18 years and

First stage 30 PCV13-TT 1 dose

above

Second

2-5 years 30 PCV13-TT 1 dose

stage

2 months

1 dose each at month 0, 2,

(minimum age: 6 30 PCV13-TT

4, 3 doses in total

Third weeks)

stage 1 dose eat at month 0, 1,

3 months 30 PCV13-TT 2,

3 doses in total

total 120 //

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4.7. INVESTIGATIONAL DRUG

4.7.1. Investigational Vaccine

Product Name: 13-valent pneumococcal polysaccharide conjugate vaccine

Manufacturer: Yuxi Walvax Biotechnology Co., Ltd

Strength and components: 0.5 mL/syringe, containing 30.8 µg of pneumococcal polysaccharides (total amount of polysaccharides), including 2.2 µg each of serotype 1, 3, 4, 5, 6A, 7F, 9V, 14, 18C, 19A, 19F and 23F, and 4.4 µg of serotype 6B polysaccharide.

Dosage form: injection (suspension for injection)

Batch No.: 201511003

Expiration date: November 23, 2017

4.7.2. Packaging

Yuxi Walvax is responsible for the preparation of the investigational vaccine, and Beijing Simoon Record Pharma Information Consulting Co., Ltd. (CRO) is responsible for printing the vaccine repackaging label. The Registration and Clinical Management Department of Yunnan Walvax is responsible for reviewing and grouping vaccine labels. Based on the "PCV13 Phase I Clinical Study Investigational Vaccine Repackaging Plan" and the clinical study protocol, the Registration and Clinical Management Department of Yunan Walvax completed the operation training, introduced in detail the repackaging process and related considerations for the personnel participating in the repackaging program before the implementation of the repackaging procedures, and organized to complete the repackaging of investigational vaccine.

4.7.3 Storage and transportation of the vaccine

The vaccine shall be stored and transported at 2–8 °C, away from light. The storage temperature needed to be monitored daily and recorded. If the storage and transportation conditions exceeded the specified ranges, the investigator on the site shall contact the staff of the unit responsible for the clinical study and the sponsor immediately, and the vaccine shall not be used until obtaining the permission of the sponsor.

4.7.4. Route of administration and dose

Route of administration: intramuscular injection. The lateral skin of the deltoid muscle of the upper arm or the anterolateral skin of the thigh was sterilized with 75% medical alcohol, the vaccine was shaken well, and 0.5 mL of vaccine was intramuscularly injected after the skin had become slightly dry; for infants under 12 months of age, the preferred vaccination site is the anterolateral thigh.

4.7.5 Concomitant medication

Allowed medication: If adverse events occurred in a subject during the trial, the subject shall be allowed to have access to necessary medications and medical treatment.

Allowed vaccines: The interval between other vaccines and the investigational vaccine shall not be less than 7 days, and that between the live attenuated vaccine and the investigational vaccine shall not be less than 14 days; emergency vaccination can be conducted at any time according to the need of vaccination (e.g., rabies or tetanus vaccine). If the concomitant vaccine vaccination led to the violation of the interval between the vaccination and the investigational vaccine vaccination against the requirements of this protocol, such a circumstance was handled as protocol violation and there was no need to terminate the trial. The investigator shall record any information on

concomitant medication, including the drug name, dosage form, dosage and administration, and time of administration.

Medication records: In order to learn about the impact of drug use on the safety and immunogenicity of the vaccine during the trial or to collect adverse events that may be related to vaccination without omission, the investigator needed to collect the medications of subjects during the observation period by taking measures, and guide the subjects or their guardians in recording their visits and medication in the diary cards as much as possible. The following drugs shall be transcribed into the Electronic Data Capture (EDC) system:

- (1) Oral or injectable hormones/steroids (e.g., dexamethasone);
- (2) Anti-allergic drugs (e.g., chlorpheniramine);
- (3) Antipyretic/analgesic/non-steroidal anti-inflammatory drugs (e.g., aspirin, ibuprofen, fenbid);
- (4) Any vaccines or biological products;
- (5) Immunosuppressants;
- (6) Antibiotics
- (7) Anti-viral drugs;
- (8) Other drugs

4.8 STUDY PROCEDURES

4.8.1 Recruitment of subjects

After the research site had been determined and approval had been secured from the ethics committee, the recruitment notices were issued to those meeting the requirements on age, vaccination history, and health status as per the protocol.

4.8.2 Informed consent

Prior to the enrollment, the investigator informed fully the subject or his guardian of information related to the clinical protocol, and the subject, his guardian, and the research doctor signed the informed consent form jointly in duplicate, with the subject keeping the copy.

4.8.3 Screening via physical examinations

The investigator carried out physical examinations on all subjects, including the height, body weight, body temperature, skin examinations, and hematology and indicators related to liver and kidney functions. The investigator carried out inquiries as per the "inclusion and exclusion criteria", observed the general state of subjects, examined the throat, heart, and lungs, and measured the blood pressure of subjects at 18 years old and above, thereby excluding subjects with a serious medical history, e.g., serious diseases, infectious diseases, and organ removal. The subjects were assigned to groups in accordance with the inclusion and exclusion criteria. The basic information of all the subjects was filled in the "Physical Examination Form". The "Source Document Book" was filled in.

4.8.4. Assignment of study numbers

Subjects were assigned with unique random numbers according to the order of their enrollment, and the study number were recorded in the "Source Document Book", the name abbreviations of the subjects were filled in the "Vaccine Allocation Form", which was then signed and dated by the investigator.

4.8.5. Blood collection before immunization

Subjects in the first and second stage were subjected to the collection of whole blood not less than 5.0 mL in volume by means of venipuncture before the first vaccination, and the blood samples were divided in two parts, the first part was on the same day sent to hospital of Grade A Level 2 and above for hematology, hepatic and renal function-related testing; another part was separated through centrifugation and then stored at -20 °C or below, and was intended for pre-immunization IgG antibody concentration testing.

Subjects in the third stage was subjected to the collection of whole blood not less than 3.0 mL, which was then centrifuged to separate the serum, and then stored at -20°C or below, to test the pre-immunization IgG antibody concentration.

4.8.6. Vaccination

Subjects in the first and second stage of clinical study may receive vaccination only when their hematology, hepatic and renal function measures are within the normal range specified in the protocol, or not within the normal range but are judged to be normal after comprehensive evaluation by the on-site investigators, and met other inclusion criteria, subjects in the third stage may receive vaccination only when they meet the inclusion criteria and have completed the pre-immunization blood collection. The corresponding vaccine number is obtained according to the study number, and number on the source document book is also checked to confirm the consistency, and then the name abbreviation of the subject is filled on the label of the outer package of the vaccine, the internal label of the vaccine is then filled and pasted at the designated position of the source document book. Double check the information of the subjects before vaccination. The vaccine should be shaken well before injection, 0.5 ml vaccine is injected into the lateral deltoid muscle of upper arm or anterolateral thigh. The preferred vaccination site for infants under 1 year old is anterolateral thigh.

4.8.7. Medical observation

After each dose of vaccination, a 30-min observation was required on the trial site, the "Diary Card", the thermometer, and the scale were issued, and the legal guardians of the subjects or the were trained to use the thermometer, observe the adverse events, and fill out the "Diary Card", and an appointment was made for the time to return the "Diary Card" (the "Diary Card" was returned at day 8 after the vaccination, and the "Contact Card" was received in the meantime).

4.8.8. Blood Collection at Day 3 Post Dose 1

For subjects in stage 1 and 2, no less than 3.5 mL of whole blood was collected through venipuncture at Day 3 post vaccination (with an interval of 2 days), the blood samples were sent to hospitals of Grade A Level Two or above on the same day for hematology and hepatic and renal function tests. The time window for blood collection was +1 day.

4.8.9. Vaccinations of the Dose 2 and Dose 3

For subjects in stage 3, the second and third doses were given as per corresponding vaccination schedule after the first dose, with a time window of 15 days. The contact card was collected and reviewed before of each dose. The contact card for the last dose of vaccination was collected and reviewed one month after the last dose.

4.8.10. Blood Collection after Complete Series

When the subjects in the three stages had completed the full series following the vaccination schedule specified in the clinical study protocol, no less than 3.0 mL whole blood was collected

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from each subject, and then the serum was separated, and serum IgG antibody concentration was detected. The time window for blood collection was + 10 days.

4.8.11. End of the Trial

The serum was sent to a designated laboratory for testing, the review for the entry of EDC data was completed, and after review of all materials and data clarification, the report on site closure was issued.

Scheduled Activities in Phase I Clinical Trial

First stage Second stage Third stage

Scheduled activities (adults 18 years and (2-5 years (2-3 months old, minimum older) old) age of 6 weeks)

Informed Consent ● ● ●

Screening for physical examination ● ● ●

Assigning randomization number ● ● ●

Pre-immunization blood collection ● ● ●

First vaccination ● ● ●

Blood collection at Day 3 post first dose ● ● ● \

Second and third vaccinations \ \ ●

Blood collection at Day 30 post complete

● ● ●

series

30-minute medical observation (after each

● ● ●

dose)

Day 0-7 systematic safety observation

● ● ●

(after each dose)

Day 8-30 safety observation (after each

● ● ●

dose)

SAE collection within 6 months post

immunization (after each dose of ● ● ●

vaccination)

4.9. OBSERVATIONS

4.9.1. Safety Observations

Safety observations included all the expected and unexpected medical events related or unrelated to vaccination occurred during the observation period of clinical trial in subjects.

Local reactions: pain, itching, redness, swelling, induration, rash (injection site) and their size and severity at injection site, and disturbance of body activity.

Systemic reactions: observe fever, allergic reactions, headache, fatigue / weakness, nausea / vomiting, diarrhea, muscle pain, cough, other discomfort or clinical adverse reactions.

Other events: any medical event other than the above stated reactions, such as acute diseases, accidental injury, etc.

Hematology, hepatic and renal function observations: hemoglobin content (HGB), white blood cell count (WBC), platelet count (PLT), alanine aminotransferase (ALT), aspartate aminotransferase (AST), total bilirubin (TBIL), creatinine (CR), urea nitrogen (BUN).

Grading standard: the systemic and local reactions following vaccination were classified according to the Guideline for Classification of Adverse Reactions in Clinical Trials of Preventive Vaccines promulgated and implemented by the National Medicinal Product Administration.

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4.9.2. Immunogenicity Observations

Blood samples were collected from all subject before immunization and 30 days post complete series to detect the vaccine serotype-specific pneumococcal IgG and opsonophagocytic antibody (OPA):

(1) The proportion of subjects with vaccine serotype-specific pneumococcal IgG antibody concentration $\geq 0.35 \mu\text{g/ml}$ (positive conversion rate).

(2) The geometric mean concentrations (GMCs) of pneumococcal IgG antibody were determined.

(3) The proportion of subjects with antibody titer of vaccine serotype-specific pneumococcal OPA $\geq 1:8$ (positive rate) in adults, children and 3-month-old infants.

(4) The geometric mean titers (GMTs) of the vaccine serotype-specific pneumococcal OPA antibody were detected in adults, children and 3-month-old infant.

4.10. LABORATORY TESTS

4.10.1. Clinical laboratory indicators

The hematology, liver and kidney function detection in this clinical trial were entrusted to the Laboratory of Shuangqiao Hospital in Chaoyang District, Beijing. See the tables below for methods used and the normal value reference range for each measurement.

(1) Detection method and reference range for hematology

Measurement Method Reference range

Male: 110-175 g/L

Hemoglobin (HGB) Automatic Blood Cell Analyzer

Female: 110-150 g/L

White blood cell count (WBC) Automatic Blood Cell Analyzer (3.5-9.5) $\times 10^9/\text{L}$

Platelet count (PLT) Automatic Blood Cell Analyzer (125-350) $\times 10^9/\text{L}$

(2) Detection method and reference range for liver function

Measurement Method Reference range

Male: 9.0-50.0 U/L

Alanine aminotransferase (ALT) Velocity method

Female: 7.0-40.0 U/L

Male: 15.0-40.0 U/L

Aspartate aminotransferase (AST) Velocity method

Female: 13.0-35.0 U/L
Total bilirubin (TBIL) Modified diazonium method 3.44-25.40 $\mu\text{mol/L}$

(3) Detection method and reference range for kidney function

Measurement Method Reference range

Male: 44.0-97.0 $\mu\text{mol/L}$

Creatinine (cr) Jaffe reaction

Female: 35.0-80.0 $\mu\text{mol/L}$

Blood urea nitrogen (BUN) Enzymatic method 2.9-8.2 mmol/L

4.10.2. Serum antibody detection

Serum samples detection was entrusted to National Institutes for Food and Drug Control.

The antibody concentration of IgG in serum samples was detected using enzyme-linked immunosorbent assay (ELISA), the subjects achieving IgG antibody concentration $\geq 0.35 \mu\text{g/mL}$ and the geometric mean concentration (GMCs) were calculated.

Opsonophagocytic Assay (OPA) was used to detect the antibody opsonophagocytic activity of serum samples, meanwhile the subjects achieving an OPA antibody $\geq 1:8$ and the geometric mean titers (GMTs) were calculated.

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Testing standard samples and quality control standards were provided by the laboratory. The method and quality control of ELISA and OPA can be found in the standard procedure of laboratory.

4.11. STATISTICAL ANALYSIS PLAN AND STATISTICAL ANALYSIS

4.11.1. Data management and Statistical Software

Establishment of Database: professional team of Medidata and data administrator of Sponsor constructed the clinical research database using Medidata Rave EDC system according to the study protocol. In the meantime, the logic verification program was set to verify the quality of data input in real time.

Data Entry: all safety data were entered into EDC system directly by investigators in real time; immunogenicity data were entered by two operators from data management team in duplicate.

Data Audit: the project data were managed and reviewed by data administrator according to the Data Management Plan of the project, and the data cleaning was performed through the process of "online query and online answer".

Database Locking: the databases was lock after the data were reviewed and approved by principal investigators, biostatisticians, data administrator and sponsor.

Statistical analysis was performed using SAS 9.13 software.

4.11.2. Statistical Analysis of Clinical Safety

The evaluation of the data mainly included the summary of clinical response endpoints, local and systemic clinical reactions during the observation period for all subjects, and the incidence rates of related adverse reactions / events.

Subjects with at least one adverse reaction (Day 0-7, Day 8-30, Day 0-30):

The percentages of adverse events / reactions of all subject populations were statistically analyzed;

The percentages of adverse events / reactions after each dose were statistically analyzed;

The percentages of local or systemic adverse reactions of subject populations were statistically analyzed;

The percentages of adverse reactions with different severity were statistically analyzed.

Analysis of symptoms of adverse reactions:

The percentages of any systemic and local abnormal symptoms after vaccination were statistically analyzed;

The percentages and severity of body temperature reactions after vaccination were statistically analyzed.

Events and serious adverse events resulted in withdrawal of subjects:

The cases that discontinued study due to severe or serious adverse events were identified, and the inter-group differences for frequency of occurrence were statistically compared between groups.

4.11.3. Statistical Analysis of Immunogenicity (based on FAS and PPS sets, respectively)

IgG concentration $\geq 0.35 \mu\text{g/mL}$ was considered as the positive level for vaccine serotype-specific pneumococcal IgG antibody. For different types of IgG antibody, the post-immunization positive rate and the positive conversion rate of IgG antibody in subjects with negative pre-

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immunization antibody level were calculated respectively, with the 95% confidence interval (CI) given by Clopper Pearson method.

For different types of IgG antibodies, the geometric mean and 95% CI were calculated to describe the post-immunization GMC and GMC fold increase of all groups, groups with negative pre-immunization antibody, and groups with positive pre-immunization antibody.

OPA titer $\geq 1:8$ was considered as the positive antibody level for vaccine serotype-specific pneumococcal OPA antibodies. For different types of OPA antibody, the post-immunization positive rate and the positive conversion rate of IgG antibody in subjects with negative pre-immunization antibody level, post-immunization 4-fold increase rates of antibody in subjects with pre-immunization positive antibody, as well as the post-immunization positive conversion rates (4-fold increase) in all subjects were calculated respectively.

For different types of OPA antibodies, the geometric mean and 95% confidence interval were calculated to describe the post-GMT and GMT growth multiple after immunization in all the population composed of 18 years old and above, 2-5 years old and 3 months old, negative before immunization and positive before immunization.

For different types of OPA antibodies, the geometric mean and 95% CI were calculated to statistically describe the post-immunization GMT and GMT fold increase in subjects aged 18 years and above, 2-5 years and 3 months, GMC and GMC fold increase of all groups.

4.11.4. Statistical Analysis Plan

4.11.4.1 Project plan

The personnel undertaking the statistical analysis task participated in the whole process, from trial design, implementation, formulation of the statistical analysis plan to the analyses and summary, established a database and carried out statistical analysis, wrote the statistical analysis plan and completed the clinical statistical report.

4.11.4.2 Selection of analysis sets

(1) Full Analysis Set (FAS) refers to all subjects who complied with the intention-to-treat (ITT) analysis principle, who have undergone randomization, and have received at least one dose of

vaccine and have finished pre-immunization blood collection with antibody detection result. Subjects with missing data of serum testing results were imputed using Last Observation Carried Forward (LOCF) method. Subjects with incorrect vaccinations were evaluated for immunogenicity by randomized groups following ITT rule.

(2) Per Protocol Set (PPS) refers to all subjects who met the inclusion/exclusion criteria, entered randomization, completed vaccinations, undergone pre- and post-immunization immunogenicity evaluations, and with valid titer values. Additionally, subjects should be excluded from PPS if the following criteria were met:

- Study protocol violation;
- Vaccination with wrong number;
- Use of vaccines or medications inhibited by the study protocol;
- Other conditions violating the study protocol.

FAS and PPS were mainly intended for evaluating the immunogenicity of vaccines. This clinical trial defined specific FAS and PPS for the IgG antibody and OPA antibody analysis of 13 individual serotypes.

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(3) Safety Analysis Set (SS): All subjects who received vaccination after randomization. Subjects vaccinated with wrong number were included in specific safety evaluation as per actual vaccine group following All Subjects As Treated (ASST) rules.

Safety population is mainly used for safety evaluations. The safety population included in this trial includes overall safety population, safety population of dose 1, safety population of dose 2, and safety population of dose 3.

4.11.4.3. Statistical methods

General principle:

Statistical description: Quantitative indicators were listed out with the mean $\pm$ standard deviation, median, maximum and minimum values; qualitative indicators or grade indicators were listed out with the frequency distribution table. The antibody titer and antibody concentration of the immunogenicity indicators in this trial were expressed as geometric mean, standard deviation, median, maximum and minimum values, and 95% confidence interval, and the antibody titer and antibody concentration should be logarithmically transformed.

Drop-out analysis: The case number of enrollment, drop-out, exclusion and completers were listed, as well as the number of cases in each analysis set, the total drop-out rate and the drop-out rate caused by adverse events.

Safety analysis: The safety analysis was based on the SS. The frequency and case number of adverse events and adverse reactions in each group were listed out for calculation of the incidence and corresponding severity; the detailed list of various adverse events was given.

4.12. MODIFICATION AND INTERIM ANALYSIS DURING THE STUDY

Not applicable.

5. STUDY RESULTS

5.1. ALLOCATION, DROPOUT AND EXCLUSION OF SUBJECTS

A total of 200 subjects were screened in this study, eventually 120 subjects were enrolled, including 30 subjects each in age groups of ≥ 18 years old, 2-5 years old, 2 months old and 3 months old.

Among the enrolled subjects, 117 cases (97.50%) completed the trial, and 3 cases (1 case in 2-5 years old group and 2 cases in 3-month-old group) dropped out. All age groups were analyzed for IgG immunogenicity, including 113 cases (94.17%) entering FAS and 103 cases (85.83%) entering PPS; immunogenicity OPA analysis was conducted for subjects aged ≥ 18 years, 2-5 years and 3 months, including 83 cases (92.22%) entering FAS and 78 cases (86.67%) entering PPS.

See Table 1 for the allocation, dropouts and exclusion of subjects.

Table 1. Allocation, Completers and Dropouts and Exclusion of Subjects

	≥ 18 yrs (N=30)	2-5 yrs (N=30)	2 mos (N=30)	3 mos (N=30)	Total
n	30	30	30	30	120
Enrolled	30	100.0	30	100.0	120
Completers	30	100.0	29	96.67	30
Dropout*	0	0.00	1	3.33	0
SAEs	0	0.00	0	0.00	0
non-SAEs	0	0.00	0	0.00	0
Voluntary withdrawal	0	0.00	0	0.00	0
Moved away from t	1	3.33	3	2.50	0
he region	0	0.00	0	0.00	2

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Other 0 0.00 0 0.00 0 0.00 0 0.00 0 0.00

Dose 1 30 100.0 30 100.0 30 100.0 30 100.0 120 100.0

SS Dose 2 0 0.00 0 0.00 30 100.0 29 96.67 59 98.33

Dose 3 0 0.00 0 0.00 30 100.0 28 93.33 58 96.66

IgG 30 100.0 27 90.00 29* 96.67 27 90.00 113 94.17

FAS

OPA 30 100.0 28 93.33 0 0.00 25 83.33 83 92.22

PPS IgG 30 100.0 27 90.00 26* 86.67 20 66.67 103 85.83

* One subject in the age group of 2 months was not included in FAS and PPS for missing values of serotypes

6B, 9V and 14 due to inadequate serum volumes.

5.2. DEMOGRAPHIC ANALYSIS OF SUBJECTS

The age and sex distribution of subjects are shown in Table 2.

Table 2. Age and gender distribution of subjects in phase I clinical trial

Age (months)	Male	Female	
Groups	N	N	
Mean $\pm$ SD	Median	Min, Max	
≥ 18 years	30	35.79 $\pm$ 12.53	30.67 18.04,64.02
2-5 years	30	4.04 $\pm$ 1.26	4.06 2.06,5.94
2 months	30	0.19 $\pm$ 0.03	0.19 0.13,0.24
3 months	30	0.28 $\pm$ 0.02	0.28 0.25,0.33

5.3. SAFETY ANALYSIS

5.3.1. Observation Method

Safety observations were performed in the sequence of adults, children and target population, the adults and children groups were vaccinated with one dose of investigational vaccine, and the subjects aged 2 or 3 months were vaccinated with 3 doses of investigational vaccines, while the vaccinations for 2 month olds were given at 2, 4, 6 months of age, and at 3, 4, 5 months of age for those aged 3 months.

In the adult group, one dose of vaccine was given before blood samples collection, subsequently at Day 3, hematology, urinalysis, and blood chemistry tests were performed, at Day 8, the diary cards were collected for preliminary 7-day safety evaluation after immunization, and the subjects continued to complete the safety observation of Day 8-30.

After the preliminary safety evaluation in adults, children subjects were then enrolled, after blood collection, one dose of investigational vaccine was inoculated. At Day 3, hematology, urinalysis, and blood chemistry tests were performed, at Day 8, the diary cards were collected for preliminary 7-day safety evaluation after immunization, and the subjects continued to complete the safety observation of Day 8-30.

After the preliminary safety evaluation in children, subjects aged 2 months and 3 months were enrolled, and were vaccinated with 3 doses as per prespecified immunization schedules, all adverse events within 30 days post each dose were observed.

5.3.2. Overall Incidence Rate of Adverse Events

Solicited adverse events in phase I clinical trials are by default associated with the study vaccine, and the relatedness of unsolicited adverse reactions was adjudicated by investigators. The overall adverse events in each stage is summarized in Table 3.

A total of 24 (80.00%) subjects in adults group experienced adverse events related to the study vaccine, of which 10 cases (33.33%) experienced systemic adverse reactions, but no grade 3 systemic adverse reaction was reported; 23 cases (76.67%) experienced local reactions, of which the severity of the reaction for 6 cases (20.00%) reached grade 3.

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A total of 16 (53.33%) subjects in children group experienced adverse events related to the study vaccine, of which 7 cases (23.33%) experienced systemic adverse reactions, and the severity of 2 cases (6.67%) reached grade 3; 8 cases (26.67%) experienced local reactions, of which the severity of the reaction for 1 case (3.33%) reached grade 3.

A total of 20 (66.67%) subjects in the 2-month age group experienced adverse events related to the study vaccine, of which 18 cases (60.00%) experienced systemic adverse reactions, and the severity of 2 cases (6.67%) reached grade 3; 12 cases (40.00%) experienced local reactions, of which the severity of the reaction for 2 cases (6.67%) reached grade 3. Another case of adverse event which was unrelated to the study vaccine was reported as herpangina and the severity reached grade 3.

A total of 21 (70.00%) subjects in the 3-month age group experienced adverse events related to the study vaccine, of which 17 cases (56.67%) experienced systemic adverse reactions, but no grade 3 systemic adverse reaction was reported; 11 cases (36.67%) experienced local reactions, of which the severity of the reaction for 1 case (3.33%) reached grade 3. Another case of adverse event which was unrelated to the study vaccine was reported as upper respiratory tract infection and the severity reached grade 3.

No serious adverse event was collected during the observation period, and no drop-outs due to adverse events were noted.

Table 3. Incidence of adverse reactions in phase I clinical trial
≥18 years (N=30) 2-5 years (N=30) 2 months (N=30) 3 months (N=30)

Events	Cases	%	Events	Cases	%	Events	Cases	%	Events	Cases	%	
Any AE	81	24	80.00	41	19	63.33	105	21	70.00	79	22	73.33
Grade 1	49	23	76.67	33	18	60.00	67	21	70.00	58	20	66.67
Grade 2	24	13	43.33	4	4	13.33	32	13	43.33	19	15	50.00
Grade 3	8	6	20.00	4	2	6.67	6	4	13.33	2	2	6.67
Related	35	16	53.33	66	21	70.00						
	69	24	80.00	82	20	66.67						
AE												
Grade 1	43	23	76.67	28	15	50.00	59	20	66.67	52	19	63.33
Grade 2	18	10	33.33	3	3	10.00	18	10	33.33	13	10	33.33
Grade 3	8	6	20.00	4	2	6.67	5	4	13.33	1	1	3.33
Solicited	81	20	66.67									
	63	23	76.67	28	13	43.33	64	21	70.00			
AE												
Grade 1	37	21	70.00	21	12	40.00	58	20	66.67	50	19	63.33
Grade 2	18	10	33.33	3	3	10.00	18	10	33.33	13	10	33.33
Grade 3	8	6	20.00	4	2	6.67	5	4	13.33	1	1	3.33
Systemic	13	7	23.33	40	17	56.67						
	15	10	33.33	54	18	60.00						
reactions												
Grade 1	10	7	23.33	8	6	20.00	38	17	56.67	34	16	53.33
Grade 2	5	4	13.33	3	3	10.00	14	8	26.67	6	6	20.00
Grade 3	0	0	0.00	2	2	6.67	2	2	6.67	0	0	0.00
Local	15	8	26.67	24	11	36.67						
	48	23	76.67	27	12	40.00						
reactions												
Grade 1	27	21	70.00	13	7	23.33	20	11	36.67	16	10	33.33
Grade 2	13	10	33.33	0	0	0.00	4	3	10.00	7	5	16.67
Grade 3	8	6	20.00	2	1	3.33	3	2	6.67	1	1	3.33
Unsolicited												
	6	5	16.67	7	5	16.67	1	1	3.33	2	2	6.67
AE												
Grade 1	6	5	16.67	7	5	16.67	1	1	3.33	2	2	6.67
Grade 2	0	0	0.00	0	0	0.00	0	0	0.00	0	0	0.00
Grade 3	0	0	0.00	0	0	0.00	0	0	0.00	0	0	0.00
Unrelated												
	12	8	26.67	6	6	20.00	23	11	36.67	13	10	33.33

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Grade 1 6 4 13.33 5 5 16.67 8 7 23.33 6 5 16.67
Grade 2 6 6 20.00 1 1 3.33 14 8 26.67 6 6 20.00
Grade 3 0 0 0.00 0 0 0.00 1 1 3.33 1 1 3.33

5.3.3. Adverse Events Occurred within 30 minutes, Day 0-7 and Day 8-30 after vaccination
Most adverse reactions were observed within Day 0-7 after vaccination in all groups, with local pain and redness being the most common symptoms, while the incidence rate of pain in adults was higher compared to that in children and infants. No adverse events related to the study vaccine were observed during Day 8-30.
The incidence of adverse events in 30 minutes, Day 0-7 and Day 8-30 after vaccination in each age group is shown in Table 4.

Table 4. Incidence of Adverse Reactions within Different Observation Time after Vaccination

	30 mins	Day 0-7	Day 8-30
Observation time	%	Cases	%
Subjects			
Any AE	9 9	30.00	80 24 80.00 1 1 3.33
Related AE	9 9	30.00	69 24 80.00 0 0 0.00
≥18 yrs Solicited AE	9 9	30.00	63 23 76.67 0 0 0.00
Unsolicited AE	0 0	0.00	6 5 16.67 0 0 0.00
Unrelated AE	0 0	0.00	11 8 26.67 1 1 3.33
Any AE	1 1	3.33	38 17 56.67 3 3 10.00
Related AE	1 1	3.33	35 16 53.33 0 0 0.00
2-5 yrs Solicited AE	1 1	3.33	28 13 43.33 0 0 0.00
Unsolicited AE	0 0	0.00	7 5 16.67 0 0 0.00
Unrelated AE	0 0	0.00	3 3 10.00 3 3 10.00
Any AE	21 13	43.33	84 20 66.67 10 6 20.00
Related AE	21 13	43.33	82 20 66.67 0 0 0.00
2 mos Solicited AE	21 13	43.33	81 20 66.67 0 0 0.00
Unsolicited AE	0 0	0.00	1 1 3.33 0 0 0.00
Unrelated AE	0 0	0.00	2 2 6.67 10 6 20.00
Any AE	11 8	26.67	69 21 70.00 8 7 23.33
Related AE	11 8	26.67	66 21 70.00 0 0 0.00
3 mos Solicited AE	11 8	26.67	64 21 70.00 0 0 0.00
Unsolicited AE	0 0	0.00	2 2 6.67 0 0 0.00
Unrelated AE	0 0	0.00	3 2 6.67 8 7 23.33

5.3.4. Adverse reactions after each vaccination in 2 months old group and 3 months old group
The incidence of adverse events in the 2 months old group (vaccinations were given at 0, 2, 4 months) and the 3 months old group (vaccinations were given at month 0, 1, 2) are shown in Table 5 to Table 6.

No growing tendency was observed for adverse reaction as the doses of vaccination increased in subjects administered with either immunization procedure. No significant difference was observed in the incidences of adverse events among each dose in the schedule of vaccination at 0, 2, 4 months, and a downward trend was noted as the doses increased in the schedule of vaccination at 0, 1, 2 months.

Table 5. Adverse reactions after each dose in subjects aged 2 months (schedule of vaccination at 0, 2, 4 months)

	1st Dose (N=30)	2nd Dose (N=30)	3rd Dose (N=30)
Events	Cases	%	Events Cases %
Any AE	34 16	53.33	34 16 53.33 37 14 46.67
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Related AE 26 14 46.67 24 14 46.67 32 14 46.67
Solicited AE 26 14 46.67 24 14 46.67 31 14 46.67
Systemic reactions 24 12 40.00 15 12 40.00 15 10 33.33
Local reactions 2 2 6.67 9 7 23.33 16 10 33.33
Unsolicited AE 0 0 0.00 0 0 0.00 1 1 3.33
AEs unrelated to vaccine
8 5 16.67 10 6 20.00 5 3 10.00

Table 6. Adverse reactions after each dose in subjects aged 3 months (schedule of vaccination at 0, 1, 2 months)

	1st Dose (N=30)	2nd Dose (N=29)	3rd Dose (N=28)
Events	Cases	%	Events Cases %
Any AE	39 17	56.67	27 15 51.72 13 11 39.29
Related AE	34 16	53.33	20 13 44.83 12 10 35.71
Solicited AE	33 16	53.33	19 13 44.83 12 10 35.71
Systemic reactions	22 11	36.67	11 10 34.48 7 7 25.00
Local reactions	11 7	23.33	8 7 24.14 5 4 14.29
Unsolicited AE	1 1	3.33	1 1 3.45 0 0 0.00
AEs unrelated to vaccine	5 5	16.67	7 5 17.24 1 1 3.57

5.3.5. Solicited adverse events – systemic and local reactions
See Table 7 to Table 10 for the incidences of systemic and local reactions.
The most common systemic reaction was fever in all age groups, the incidence of fever in infants and toddlers was higher than that in adults and children aged 2-5 years. The severity of fever was mainly grade 1, while 2 cases in 2-5 years old group and 1 case in 2 months old group reach grade 3. The incidences of vomiting, diarrhea, abnormal crying and other symptoms were also higher in subjects aged 2 and 3 months. Except for fever, one case in the 2 months old group had grade 3 fatigue / somnolence. The severity of other systemic reactions was mainly grade 1, and no grade 3 reaction was reported.
Common local reactions in adults and children aged 2-5 years were pain, redness and swelling. The severity of local pain for 1 case in the children group reached grade 3; the severity of local redness for 3 adults and 1 child reached grade 3; the severity of local swelling for 4 cases in the adults group reached grade 3; the severity of local induration for 1 case in the adults group reached grade 3. Other local adverse reactions were mainly grade 1. The most common symptoms were local tenderness, redness and induration in 2 months old group; tenderness, redness and swelling

were more common in 3 months old group. The severity of local redness and local induration for 1 case in 2 months old group reached grade 3, and the severity of swelling for 1 case in 3-month group reached grade 3.

Within 30 minutes after vaccination, the primarily observed adverse reactions were fever, fatigue (mainly observed among adults) and local redness, including 1 case of fever, one case of fatigue and 7 cases of local redness in the adults group who received 1 dose of vaccine; one case of fever was observed in the children group who received 1 dose of vaccine; eight cases of fever and 7 cases of local redness were observed in the 2 months old group who received three-dose series given at 0, 2, 4 months; four cases of fever and 4 cases of local redness were observed in the 3 months old group who received three-dose series at 0, 1, 2 months.

Table 7. Analysis of systemic reactions

≥18 years (N=30) 2-5 years (N=30) 2 months (N=30) 3 months (N=30)

Events Cases % Events Cases % Events Cases % Events Cases %

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fever 8 7 23.33 9 7 23.33 27 16 53.33 20 14 46.67

Grade 1 5 4 13.33 5 5 16.67 21 16 53.33 19 13 43.33

Grade 2 3 3 10.00 2 2 6.67 5 5 16.67 1 1 3.33

Grade 3 0 0 0.00 2 2 6.67 1 1 3.33 0 0 0.00

Fatigue / 4 4 13.33 1 1 3.33 2 2 6.67 1 1 3.33

somnolence

Grade 1 3 3 10.00 0 0 0.00 1 1 3.33 1 1 3.33

Grade 2 1 1 3.33 1 1 3.33 0 0 0.00 0 0 0.00

Grade 3 0 0 0.00 0 0 0.00 1 1 3.33 0 0 0.00

Nausea 0 0 0.00 0 0 0.00 1 1 3.33 3 3 10.00

vomiting

Grade 1 0 0 0.00 0 0 0.00 1 1 3.33 2 2 6.67

Grade 2 0 0 0.00 0 0 0.00 0 0 0.00 1 1 3.33

Grade 3 0 0 0.00 0 0 0.00 0 0 0.00 0 0 0.00

Diarrhea 0 0 0.00 1 1 3.33 6 5 16.67 5 5 16.67

Grade 1 0 0 0.00 1 1 3.33 4 4 13.33 3 3 10.00

Grade 2 0 0 0.00 0 0 0.00 2 2 6.67 2 2 6.67

Grade 3 0 0 0.00 0 0 0.00 0 0 0.00 0 0 0.00

Cough 3 3 10.00 1 1 3.33 6 3 10.00 4 3 10.00

Grade 1 2 2 6.67 1 1 3.33 5 2 6.67 4 3 10.00

Grade 2 1 1 3.33 0 0 0.00 1 1 3.33 0 0 0.00

Grade 3 0 0 0.00 0 0 0.00 0 0 0.00 0 0 0.00

Crying 0 0 0.00 1 1 3.33 11 7 23.33 7 7 23.33

Grade 1 0 0 0.00 1 1 3.33 5 4 13.33 5 5 16.67

Grade 2 0 0 0.00 0 0 0.00 6 5 16.67 2 2 6.67

Grade 3 0 0 0.00 0 0 0.00 0 0 0.00 0 0 0.00

Allergic 0 0 0.00 0 0 0.00 1 1 3.33 0 0 0.00

reactions

Grade 1 0 0 0.00 0 0 0.00 1 1 3.33 0 0 0.00

Grade 2 0 0 0.00 0 0 0.00 0 0 0.00 0 0 0.00

Grade 3 0 0 0.00 0 0 0.00 0 0 0.00 0 0 0.00

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Table 8. Analysis of local reactions

≥18 years (N=30) 2-5 years (N=30) 2 months (N=30) 3 months (N=30)

Events Cases % Events Cases % Events Cases % Events Cases %

Local

19 18 60.00 6 6 20.00 5 4 13.33 4 4 13.33

pain

Grade 1 14 13 43.33 5 5 16.67 4 4 13.33 3 3 10.00

Grade 2 5 5 16.67 0 0 0.00 1 1 3.33 1 1 3.33

Grade 3 0 0 0.00 1 1 3.33 0 0 0.00 0 0 0.00

Local

13 13 43.33 5 5 16.67 16 11 36.67 11 7 23.33

redness

Grade 1 7 7 23.33 4 4 13.33 11 8 26.67 8 6 20.00

Grade 2 3 3 10.00 0 0 0.00 3 3 10.00 3 2 6.67

Grade 3 3 3 10.00 1 1 3.33 2 1 3.33 0 0 0.00

Local

9 9 30.00 1 1 3.33 2 2 6.67 7 5 16.67

swelling

Grade 1 1 1 3.33 1 1 3.33 2 2 6.67 4 4 13.33

Grade 2 4 4 13.33 0 0 0.00 0 0 0.00 2 2 6.67

Grade 3 4 4 13.33 0 0 0.00 0 0 0.00 1 1 3.33

Local

4 4 13.33 0 0 0.00 4 3 10.00 2 2 6.67

induration

Grade 1 2 2 6.67 0 0 0.00 3 2 6.67 1 1 3.33

Grade 2 1 1 3.33 0 0 0.00 0 0 0.00 1 1 3.33

Grade 3 1 1 3.33 0 0 0.00 1 1 3.33 0 0 0.00

Local

3 3 10.00 3 3 10.00 0 0 0.00 0 0 0.00

pruritis

Grade 1 3 3 10.00 3 3 10.00 0 0 0.00 0 0 0.00

Grade 2 0 0 0.00 0 0 0.00 0 0 0.00 0 0 0.00

Grade 3 0 0 0.00 0 0 0.00 0 0 0.00 0 0 0.00

Table 9. Analysis of systemic and local reactions 30 mins post vaccination

≥18 years (N=30) 2-5 years (N=30) 2 months (N=30) 3 months (N=30)

Events Cases % Events Cases % Events Cases % Events Cases %

fever 1 1 3.33 1 1 3.33 11 8 26.67 5 4 13.33

Fatigue /

1 1 3.33 0 0 0.00 0 0 0.00 0 0 0.00
somnolence
Local
7 7 23.33 0 0 0.00 10 7 23.33 6 4 13.33
redness

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Table 10. Analysis of systemic and local reactions 0-7 days post vaccination

≥18 years (N=30) 2-5 years (N=30) 2 months (N=30) 3 months (N=30)
Events Cases % Events Cases % Events Cases % Events Cases %
fever 8 7 23.33 9 7 23.33 27 16 53.33 20 14 46.67

Fatigue /
4 4 13.33 1 1 3.33 2 2 6.67 1 1 3.33
somnolence
Nausea
0 0 0.00 0 0 0.00 1 1 3.33 3 3 10.00
vomiting
Diarrhea 0 0 0.00 1 1 3.33 6 5 16.67 5 5 16.67
Cough 3 3 10.00 1 1 3.33 6 3 10.00 4 3 10.00
Crying 0 0 0.00 1 1 3.33 11 7 23.33 7 7 23.33
Allergic
0 0 0.00 0 0 0.00 1 1 3.33 0 0 0.00
reactions
Local pain 19 18 60.00 6 6 20.00 5 4 13.33 4 4 13.33
Local
13 13 43.33 5 5 16.67 16 11 36.67 11 7 23.33
redness
Local
9 9 30.00 1 1 3.33 2 2 6.67 7 5 16.67
swelling
Local
4 4 13.33 0 0 0.00 4 3 10.00 2 2 6.67
induration
Local
3 3 10.00 3 3 10.00 0 0 0.00 0 0 0.00
pruritis

5.3.6. Unsolicited adverse events
A total of 70 unsolicited adverse events including laboratory tests were observed in 44 cases, including 13 vaccine-related cases (16 events) and 35 vaccine-independent cases (54 events). In the adult and 2-5-year-old children group, 9 cases (11 events) were observed with abnormal laboratory findings related to vaccination. The severity of abnormal laboratory results related to vaccine was grade 1, and were determined as clinically irrelevant events by investigators. Five other unsolicited adverse events, other than laboratory tests, were determined as "probably related to study vaccine" by the investigators, including one case of lymphadenopathy in adult group, one case of rhinorrhea (runny nose) in children group, 1 case of decreased appetite in 2 months old group, 1 case of dyspepsia and 1 case of vomiting in 3-month group. See Table 11-Table 14 for analysis of unsolicited events.

Table 11. Unsolicited adverse events
≥18 years (N=30) 2-5 years (N=30) 2 months (N=30) 3 months (N=30)
Event Case Event Case Event Case Event Case
% % % %

s s s s s s s s
Any
43.3 30.0 36.6 36.6
unsolicited 18 13 13 9 24 11 15 11
3 0 7 7
AE
Study
16.6 16.6
vaccine- 6 5 7 5 1 1 3.33 2 2 6.67
7 7
related
Investigatio 13.3 16.6
5 4 6 5 / / / / / /
ns 3 7
Other 1 1 3.33 1 1 3.33 1 1 3.33 2 2 6.67
Unrelated to
26.6 20.0 36.6 33.3
study 12 8 6 6 23 11 13 10
7 0 7 3
vaccine

Table 12. Abnormal laboratory findings
≥18 years (N=30) 2-5 years (N=30)
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Events Cases % Events Cases %
Leukocytopenia 1 1 3.33 0 0 0.00
Elevated white blood cell count 0 0 0.00 6 6 20.00
Increased alanine
1 1 3.33 0 0 0.00
aminotransferase
Increased aspartate
1 1 3.33 0 0 0.00
aminotransferase
Increased bilirubin 2 2 6.67 0 0 0.00
Increased hemoglobin 1 1 3.33 0 0 0.00
Elevated serum creatinine 1 1 3.33 0 0 0.00

Decreased blood urea 1 1 3.33 1 1 3.33
Elevated platelet count 1 1 3.33 1 1 3.33
Table 13. Unsolicited events related to study vaccine
≥18 years (N=30) 2-5 years (N=30) 2 months (N=30) 3 months (N=30)
Events Cases % Events Cases % Events Cases % Events Cases %
Decreased 1 1 3.33
0 0 0.00 0 0 0.00 0 0 0.00
appetite
Rhinorrhea 0 0 0.00 1 1 3.33 0 0 0.00 0 0 0.00
Dyspepsia 0 0 0.00 0 0 0.00 0 0 0.00 1 1 3.33
Infant vomiting 0 0 0.00 0 0 0.00 0 0 0.00 1 1 3.33
Lymphadenopathy 1 1 3.33 0 0 0.00 0 0 0.00 0 0 0.00
Table 14. Unsolicited events unrelated to study vaccine
≥18 years (N=30) 2-5 years (N=30) 2 months (N=30) 3 months (N=30)
Events Cases % Events Cases % Events Cases % Events Cases %
Nasopharyngitis 2 2 6.67 1 1 3.33 3 3 10.00 1 1 3.33
Viral infection 0 0 0.00 0 0 0.00 0 0 0.00 1 1 3.33
Infectious 0 0 0.00 0 0 0.00 0 0 0.00 1 1 3.33
diarrhea
Respiratory 0 0 0.00 1 1 3.33 0 0 0.00 0 0 0.00
tract infection
Candida 0 0 0.00 0 0 0.00 1 1 3.33 0 0 0.00
infection
Tracheitis 0 0 0.00 0 0 0.00 1 1 3.33 0 0 0.00
Upper 3 3 10.00 0 0 0.00 3 3 10.00 4 4 13.33
respiratory tract
infection
Rhinorrhea 1 1 3.33 1 1 3.33 0 0 0.00 1 1 3.33
Bronchitis 0 0 0.00 0 0 0.00 0 0 0.00 1 1 3.33
Herpangina 0 0 0.00 0 0 0.00 1 1 3.33 0 0 0.00
Cough 0 0 0.00 0 0 0.00 0 0 0.00 1 1 3.33
Rash 0 0 0.00 0 0 0.00 1 1 3.33 0 0 0.00
Papular rash 1 1 3.33 0 0 0.00 0 0 0.00 0 0 0.00
Eczema 0 0 0.00 0 0 0.00 1 1 3.33 0 0 0.00
Urticaria 0 0 0.00 0 0 0.00 1 1 3.33 0 0 0.00
Itching 0 0 0.00 1 1 3.33 0 0 0.00 0 0 0.00
fever 0 0 0.00 0 0 0.00 4 3 10.00 1 1 3.33
Limb pain 1 1 3.33 0 0 0.00 0 0 0.00 0 0 0.00
Diarrhea 0 0 0.00 0 0 0.00 4 4 13.33 0 0 0.00
Decreased 0 0 0.00 0 0 0.00 1 1 3.33 0 0 0.00
appetite
Dyspepsia 0 0 0.00 0 0 0.00 2 2 6.67 1 1 3.33
Infant vomiting 0 0 0.00 0 0 0.00 0 0 0.00 1 1 3.33

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5.3.7. Listing of Adverse Events
Table 15. Listing of Adverse Events Related to Vaccine
Days since
Subject Doses Duration
Gender Age Term of AE last Severity Relevance
No. vaccinated (days)
vaccination
43.7 Possibly
1 Male 1 Local pain 0 1 Grade 1
yrs related
43.7 Possibly
1 Male 1 Increased creatinine 3 20 Grade 1
yrs related
46.6 Possibly
3 Female 1 Local pain 1 1 Grade 1
yrs related
27.4 Possibly
4 Female 1 Fatigue/somnolence 1 2 Grade 1
yrs related
27.4 Possibly
4 Female 1 Cough 1 6 Grade 1
yrs related
27.4 Possibly
4 Female 1 Fever 2 1 Grade 1
yrs related
27.4 Possibly
4 Female 1 Local pain 0 4 Grade 1
yrs related
27.4 Possibly
4 Female 1 Local redness 1 3 Grade 3
yrs related
27.4 Possibly
4 Female 1 Local swelling 1 4 Grade 2
yrs related
27.4 Possibly
4 Female 1 Local induration 1 13 Grade 2
yrs related
27.4 Possibly
4 Female 1 Local itching 3 3 Grade 1
yrs related
19.2 Possibly
5 Male 1 Fever 0 2 Grade 2
yrs related
19.2 Possibly

5 Male 1 Local pain 1 1 Grade 2
 yrs related
 19.2 Possibly
 5 Male 1 Local induration 1 2 Grade 3
 yrs related
 44.1 Possibly
 6 Male 1 Increased bilirubin 3 28 Grade 1
 yrs related
 50.2 Possibly
 7 Male 1 Local pain 0 1 Grade 1
 yrs related
 50.2 Possibly
 7 Male 1 Local itching 0 1 Grade 1
 yrs related
 29.6 Possibly
 8 Male 1 Local redness 0 1 Grade 1
 yrs related
 24.1 Possibly
 9 Male 1 Local redness 0 1 Grade 1
 yrs related
 50.1 Possibly
 11 Female 1 Local redness 0 1 Grade 1
 yrs related
 Possibly
 12 Male 37 yrs 1 Local redness 0 1 Grade 1
 related
 Possibly
 13 Female 26 yrs 1 Local redness 0 1 Grade 1
 related
 Possibly
 13 Female 26 yrs 1 Local pain 1 2 Grade 1
 related
 Possibly
 13 Female 26 yrs 1 Local pain 6 4 Grade 1
 related
 Possibly
 13 Female 26 yrs 1 Local swelling 6 4 Grade 2
 related
 Possibly
 13 Female 26 yrs 1 Decreased WBC 3 20 Grade 1
 related
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29.3 Possibly
 14 Female 1 Cough 1 3 Grade 2
 yrs related
 29.3 Possibly
 14 Female 1 Fever 1 1 Grade 1
 yrs related
 29.3 Possibly
 14 Female 1 Local redness 0 4 Grade 3
 yrs related
 29.3 Possibly
 14 Female 1 Local pain 0 3 Grade 2
 yrs related
 29.3 Possibly
 14 Female 1 Local swelling 1 3 Grade 3
 yrs related
 29.3 Possibly
 14 Female 1 Local induration 3 20 Grade 1
 yrs related
 29.3 Possibly
 14 Female 1 Local itching 5 3 Grade 1
 yrs related
 Possibly
 16 Female 18 yrs 1 Fever 0 2 Grade 2
 related
 Possibly
 16 Female 18 yrs 1 Local pain 0 3 Grade 1
 related
 Possibly
 16 Female 18 yrs 1 Local redness 1 2 Grade 2
 related
 21.6 Possibly
 17 Male 1 Fever 1 1 Grade 1
 yrs related
 21.6 Possibly
 17 Male 1 Fever 6 6 Grade 1
 yrs related
 21.6 Possibly
 17 Male 1 Local pain 0 2 Grade 1
 yrs related
 31.2 Possibly
 18 Male 1 Fever 0 1 Grade 1
 yrs related
 31.2 Possibly
 18 Male 1 Local redness 0 1 Grade 1
 yrs related
 51.6 Possibly
 19 Male 1 Fatigue/somnolence 0 1 Grade 1

yrs related
 51.6 Possibly
 19 Male 1 Local pain 3 8 Grade 1
 yrs related
 Possibly
 20 Female 24 yrs 1 Cough 4 7 Grade 1
 related
 Possibly
 20 Female 24 yrs 1 Local pain 0 5 Grade 2
 related
 Possibly
 20 Female 24 yrs 1 Local swelling 0 5 Grade 3
 related
 Possibly
 20 Female 24 yrs 1 Local redness 1 2 Grade 1
 related
 Possibly
 21 Female 64 yrs 1 Local pain 0 1 Grade 1
 related
 40.6 Possibly
 22 Male 1 Fatigue/somnolence 0 1 Grade 1
 yrs related
 40.6 Possibly
 22 Male 1 Local pain 0 3 Grade 1
 yrs related
 40.6 Possibly
 22 Male 1 Local redness 0 4 Grade 2
 yrs related
 40.6 Possibly
 22 Male 1 Local swelling 0 4 Grade 2
 yrs related
 40.6 Possibly
 22 Male 1 Increased HGB 3 20 Grade 1
 yrs related
 40.6 Possibly
 22 Male 1 Increased bilirubin 3 29 Grade 1
 yrs related
 42.7 Possibly
 24 Female 1 Local pain 5 9 Grade 2
 yrs related
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42.7 Possibly
 24 Female 1 Local redness 5 9 Grade 3
 yrs related
 42.7 Possibly
 24 Female 1 Local swelling 5 13 Grade 3
 yrs related
 42.7 Possibly
 24 Female 1 Lymphadenopathy 4 14 Grade 1
 yrs related
 30.2 Possibly
 25 Male 1 Local pain 0 1 Grade 1
 yrs related
 23.7 Possibly
 27 Male 1 Local pain 0 3 Grade 1
 yrs related
 23.7 Possibly
 27 Male 1 Local swelling 0 3 Grade 1
 yrs related
 Possibly
 28 Female 24 yrs 1 Fever 0 6 Grade 2
 related
 Possibly
 28 Female 24 yrs 1 Fatigue/somnolence 1 1 Grade 2
 related
 Possibly
 28 Female 24 yrs 1 Local pain 0 5 Grade 1
 related
 Possibly
 28 Female 24 yrs 1 Local swelling 1 4 Grade 2
 related
 Possibly
 28 Female 24 yrs 1 Local redness 2 3 Grade 2
 related
 25.2 Possibly
 29 Male 1 Local pain 0 8 Grade 2
 yrs related
 25.2 Possibly
 29 Male 1 Local swelling 0 8 Grade 3
 yrs related
 25.2 Possibly
 29 Male 1 Local induration 0 2 Grade 1
 yrs related
 4.1 Possibly
 33 Female 1 Fever 0 6 Grade 3
 yrs related
 4.1 Possibly
 33 Female 1 Fatigue/somnolence 1 1 Grade 2
 yrs related

4.1 Possibly
 33 Female 1 Crying 1 1 Grade 1
 yrs related
 4.1 Possibly
 33 Female 1 Local pain 1 1 Grade 3
 yrs related
 4.1 Possibly
 33 Female 1 Local redness 1 1 Grade 3
 yrs related
 2.8 Possibly
 35 Female 1 Increased WBC 3 . Grade 1
 yrs related
 5.7 Possibly
 36 Female 1 Diarrhea 1 1 Grade 1
 yrs related
 5.7 Possibly
 36 Female 1 Cough 1 6 Grade 1
 yrs related
 5.7 Possibly
 36 Female 1 Fever 1 1 Grade 1
 yrs related
 5.7 Possibly
 36 Female 1 Local pain 0 2 Grade 1
 yrs related
 5.7 Possibly
 36 Female 1 Local redness 1 3 Grade 1
 yrs related
 5.7 Possibly
 36 Female 1 Local swelling 1 1 Grade 1
 yrs related
 5.7 Possibly
 36 Female 1 Local itching 1 3 Grade 1
 yrs related
 5.7 Possibly
 36 Female 1 Increased WBC 3 29 Grade 1
 yrs related
 5.7 Possibly
 36 Female 1 Rhinorrhea 1 6 Grade 1
 yrs related
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2.7 Possibly
 38 Female 1 Increased WBC 3 . Grade 1
 yrs related
 3.4 Possibly
 39 Female 1 Fever 2 1 Grade 1
 yrs related
 3.4 Possibly
 39 Female 1 Increased WBC 3 . Grade 1
 yrs related
 4.6 Possibly
 40 Male 1 Local pain 0 2 Grade 1
 yrs related
 4.6 Possibly
 40 Male 1 Local redness 0 5 Grade 1
 yrs related
 5.3 Possibly
 41 Male 1 Fever 0 3 Grade 2
 yrs related
 5.3 Possibly
 41 Male 1 Fever 6 1 Grade 1
 yrs related
 5.8 Possibly
 42 Male 1 Fever 3 5 Grade 3
 yrs related
 5.9 Possibly
 44 Male 1 Local redness 1 3 Grade 1
 yrs related
 5.3 Possibly
 45 Male 1 Fever 1 1 Grade 1
 yrs related
 5.9 Possibly
 48 Female 1 Local pain 0 2 Grade 1
 yrs related
 4.6 Possibly
 49 Male 1 Local pain 0 1 Grade 1
 yrs related
 4.6 Possibly
 49 Male 1 Local itching 0 1 Grade 1
 yrs related
 4.7 Possibly
 50 Male 1 Local pain 0 5 Grade 1
 yrs related
 4.7 Possibly
 50 Male 1 Local redness 0 5 Grade 1
 yrs related
 3.7 Possibly
 52 Male 1 Fever 0 2 Grade 2
 yrs related
 3.7 Possibly

52 Male 1 Fever 5 1 Grade 1
 yrs related
 5.3 Possibly
 54 Female 1 Local itching 0 1 Grade 1
 yrs related
 3.3 Possibly
 55 Male 1 Increased PLT 3 . Grade 1
 yrs related
 3.3 Possibly
 55 Male 1 Increased WBC 3 . Grade 1
 yrs related
 Possibly
 61 Male 4 mos 1 Diarrhea 0 2 Grade 1
 related
 Nausea and Possibly
 61 Male 4 mos 1 0 2 Grade 1
 vomiting related
 Possibly
 61 Male 4 mos 1 Crying 2 1 Grade 1
 related
 Possibly
 61 Male 4 mos 1 Cough 3 3 Grade 1
 related
 Possibly
 61 Male 4 mos 1 Fever 4 1 Grade 1
 related
 Possibly
 61 Male 4 mos 1 Local pain 2 1 Grade 1
 related
 Possibly
 61 Male 4 mos 1 Local swelling 2 2 Grade 1
 related
 Possibly
 61 Male 4 mos 2 Local redness 0 2 Grade 2
 related
 Possibly
 61 Male 4 mos 3 Local redness 0 1 Grade 2
 related
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3.5 Possibly
 62 Female 1 Local redness 1 1 Grade 1
 mos related
 3.5 Possibly
 62 Female 2 Fever 0 6 Grade 1
 mos related
 3.5 Possibly
 62 Female 2 Local redness 0 2 Grade 1
 mos related
 3.5 Possibly
 62 Female 3 Local redness 0 1 Grade 1
 mos related
 3.2 Nausea and Possibly
 64 Male 1 0 11 Grade 2
 mos vomiting related
 3.2 Possibly
 65 Female 2 Fever 0 1 Grade 1
 mos related
 3.2 Possibly
 65 Female 3 Fever 6 1 Grade 1
 mos related
 3.7 Possibly
 66 Female 1 Local swelling 0 6 Grade 3
 mos related
 3.7 Possibly
 66 Female 2 Fever 0 1 Grade 1
 mos related
 3.7 Possibly
 66 Female 2 Local swelling 0 5 Grade 1
 mos related
 3.7 Possibly
 66 Female 3 Crying 0 1 Grade 2
 mos related
 3.7 Possibly
 66 Female 3 Local swelling 0 8 Grade 2
 mos related
 3.7 Possibly
 67 Female 2 Diarrhea 5 4 Grade 1
 mos related
 3.7 Possibly
 67 Female 3 Fever 2 2 Grade 2
 mos related
 3.2 Possibly
 68 Female 2 Local redness 0 1 Grade 1
 mos related
 3.2 Possibly
 68 Female 3 Local redness 0 2 Grade 2
 mos related
 3.2 Possibly
 68 Female 3 Local induration 1 2 Grade 2

mos related
 3.7 Possibly
 69 Male 1 Fever 2 1 Grade 1
 mos related
 3.7 Possibly
 69 Male 2 Local induration 0 7 Grade 1
 mos related
 3.5 Possibly
 71 Male 2 Fever 1 1 Grade 1
 mos related
 3.5 Possibly
 71 Male 2 Local swelling 0 2 Grade 1
 mos related
 3.5 Possibly
 71 Male 3 Fever 7 2 Grade 1
 mos related
 3.1 Possibly
 72 Male 1 Local redness 0 1 Grade 1
 mos related
 3.2 Possibly
 73 Female 1 Crying 0 2 Grade 1
 mos related
 3.2 Possibly
 73 Female 1 Fever 4 1 Grade 1
 mos related
 3.7 Possibly
 76 Female 1 Fever 0 1 Grade 1
 mos related
 Possibly
 77 Female 3 mos 1 Crying 0 2 Grade 1
 related
 Possibly
 77 Female 3 mos 1 Fever 1 1 Grade 1
 related
 Possibly
 77 Female 3 mos 1 Fever 6 1 Grade 1
 related
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Possibly
 77 Female 3 mos 2 Fever 4 1 Grade 1
 related
 3.4 Possibly
 82 Female 1 Local swelling 1 1 Grade 2
 mos related
 Possibly
 83 Male 3 mos 1 Fever 3 1 Grade 1
 related
 Possibly
 83 Male 3 mos 1 Cough 3 4 Grade 1
 related
 Possibly
 83 Male 3 mos 3 Cough 3 3 Grade 1
 related
 3.1 Possibly
 84 Male 1 Diarrhea 0 2 Grade 2
 mos related
 3.1 Possibly
 84 Male 1 Cough 0 2 Grade 1
 mos related
 3.1 Possibly
 84 Male 1 Crying 0 2 Grade 1
 mos related
 3.1 Nausea and Possibly
 84 Male 1 1 2 Grade 1
 mos vomiting related
 3.1 Possibly
 84 Male 1 Local pain 0 3 Grade 1
 mos related
 3.1 Possibly
 84 Male 1 Local redness 1 2 Grade 1
 mos related
 3.1 Possibly
 84 Male 1 Local swelling 1 2 Grade 1
 mos related
 Possibly
 86 Male 3 mos 1 Crying 0 1 Grade 1
 related
 Possibly
 86 Male 3 mos 2 Fever 1 3 Grade 1
 related
 Possibly
 86 Male 3 mos 2 Diarrhea 1 4 Grade 2
 related
 3.5 Possibly
 87 Male 1 Local pain 0 3 Grade 1
 mos related
 3.5 Possibly
 87 Male 1 Local redness 1 2 Grade 1
 mos related

3.2 Possibly
 88 Female 1 Crying 4 2 Grade 2
 mos related
 3.2 Possibly
 88 Female 1 Infant vomiting 5 1 Grade 1
 mos related
 3.2 Possibly
 88 Female 2 Diarrhea 6 12 Grade 1
 mos related
 3.1 Possibly
 89 Female 2 Fever 0 1 Grade 1
 mos related
 3.1 Possibly
 89 Female 3 Fever 0 1 Grade 1
 mos related
 3.1 Possibly
 90 Male 1 Fever 1 1 Grade 1
 mos related
 3.1 Possibly
 90 Male 2 Fatigue/somnolence 1 2 Grade 1
 mos related
 3.1 Possibly
 90 Male 2 Local pain 0 2 Grade 2
 mos related
 3.1 Possibly
 90 Male 2 Local redness 0 2 Grade 1
 mos related
 3.1 Possibly
 90 Male 2 Dyspepsia 1 3 Grade 1
 mos related
 3.1 Possibly
 90 Male 3 Fever 0 1 Grade 1
 mos related
 2.5 Possibly
 91 Female 1 Fever 0 1 Grade 1
 mos related
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2.5 Possibly
 91 Female 2 Local pain 0 1 Grade 1
 mos related
 2.5 Possibly
 91 Female 2 Local redness 0 1 Grade 1
 mos related
 2.5 Possibly
 91 Female 3 Fever 0 1 Grade 1
 mos related
 2.5 Possibly
 91 Female 3 Fever 5 6 Grade 3
 mos related
 2.5 Possibly
 91 Female 3 Local redness 0 2 Grade 2
 mos related
 2.2 Possibly
 92 Female 2 Local induration 4 3 Grade 1
 mos related
 2.2 Possibly
 92 Female 3 Local redness 1 2 Grade 1
 mos related
 2.2 Possibly
 92 Female 3 Local induration 1 7 Grade 1
 mos related
 2.6 Possibly
 93 Female 1 Crying 0 1 Grade 1
 mos related
 2.6 Possibly
 93 Female 1 Fever 1 2 Grade 1
 mos related
 2.6 Possibly
 93 Female 1 Diarrhea 4 1 Grade 1
 mos related
 2.6 Possibly
 93 Female 2 Fever 0 4 Grade 2
 mos related
 2.6 Possibly
 93 Female 2 Local redness 0 1 Grade 1
 mos related
 2.6 Possibly
 93 Female 3 Local redness 0 1 Grade 1
 mos related
 2.2 Possibly
 94 Male 1 Diarrhea 0 1 Grade 1
 mos related
 2.2 Possibly
 94 Male 2 Fever 0 1 Grade 1
 mos related
 2.2 Possibly
 94 Male 2 Local pain 0 2 Grade 1
 mos related
 2.2 Possibly

94 Male 3 Crying 0 6 Grade 2
 mos related
 2.2 Possibly
 94 Male 3 Fever 1 5 Grade 2
 mos related
 2.2 Possibly
 94 Male 3 Diarrhea 2 4 Grade 2
 mos related
 2.2 Possibly
 94 Male 3 Local pain 1 2 Grade 2
 mos related
 2.2 Possibly
 94 Male 3 Local redness 1 1 Grade 2
 mos related
 2.2 Possibly
 94 Male 3 Local swelling 1 1 Grade 1
 mos related
 2.2 Possibly
 94 Male 3 Local induration 1 5 Grade 3
 mos related
 1.8 Possibly
 95 Male 1 Local redness 0 1 Grade 1
 mos related
 1.8 Possibly
 95 Male 2 Allergic reaction 2 2 Grade 1
 mos related
 1.8 Possibly
 95 Male 2 Local redness 0 1 Grade 1
 mos related
 1.8 Possibly
 95 Male 3 Local redness 0 1 Grade 1
 mos related
 1.8 Possibly
 96 Female 2 Fever 4 1 Grade 1
 mos related
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1.8 Possibly
 96 Female 3 Fever 1 1 Grade 1
 mos related
 1.8 Possibly
 96 Female 3 Local redness 0 2 Grade 1
 mos related
 2.4 Possibly
 98 Male 1 Local redness 0 1 Grade 1
 mos related
 2.4 Possibly
 98 Male 2 Fever 0 1 Grade 1
 mos related
 2.3 Possibly
 99 Female 3 Cough 4 1 Grade 2
 mos related
 2.3 Possibly
 99 Female 3 Local redness 0 1 Grade 1
 mos related
 2.3 Possibly
 99 Female 3 Local induration 0 3 Grade 1
 mos related
 1.7 Possibly
 100 Male 1 Fatigue/somnolence 0 7 Grade 3
 mos related
 1.7 Possibly
 100 Male 1 Cough 0 1 Grade 1
 mos related
 1.7 Possibly
 100 Male 1 Fever 4 1 Grade 1
 mos related
 1.7 Possibly
 100 Male 1 Crying 6 1 Grade 2
 mos related
 1.7 Possibly
 100 Male 1 Cough 6 2 Grade 1
 mos related
 1.7 Possibly
 100 Male 2 Cough 0 8 Grade 1
 mos related
 1.7 Possibly
 100 Male 2 Crying 0 8 Grade 2
 mos related
 1.7 Possibly
 100 Male 3 Cough 0 4 Grade 1
 mos related
 1.7 Possibly
 100 Male 3 Crying 0 2 Grade 1
 mos related
 1.7 Possibly
 100 Male 3 Local pain 0 3 Grade 1
 mos related
 1.7 Possibly
 100 Male 3 Decreased appetite 0 9 Grade 1

mos related
 1.6 Possibly
 102 Female 1 Fever 1 1 Grade 1
 mos related
 Possibly
 103 Male 2 mos 1 Fever 3 1 Grade 1
 related
 2.4 Possibly
 104 Male 3 Local redness 0 1 Grade 2
 mos related
 2.4 Possibly
 104 Male 3 Local pain 0 1 Grade 1
 mos related
 1.6 Possibly
 106 Male 1 Fever 0 7 Grade 1
 mos related
 2.5 Possibly
 110 Female 3 Fever 0 1 Grade 1
 mos related
 2.4 Possibly
 111 Female 1 Crying 0 7 Grade 1
 mos related
 2.4 Possibly
 111 Female 1 Diarrhea 1 2 Grade 1
 mos related
 2.4 Nausea and Possibly
 111 Female 1 2 1 Grade 1
 mos vomiting related
 2.4 Possibly
 111 Female 1 Cough 5 1 Grade 1
 mos related
 2.4 Possibly
 111 Female 2 Fever 0 1 Grade 1
 mos related
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2.4 Possibly
 111 Female 2 Crying 0 1 Grade 1
 mos related
 Possibly
 112 Male 2 mos 2 Fever 1 2 Grade 1
 related
 Possibly
 112 Male 2 mos 3 Fever 1 3 Grade 1
 related
 2.2 Possibly
 113 Female 1 Diarrhea 1 4 Grade 2
 mos related
 2.2 Possibly
 113 Female 1 Crying 1 4 Grade 2
 mos related
 2.2 Possibly
 113 Female 1 Fatigue/somnolence 2 3 Grade 1
 mos related
 2.2 Possibly
 113 Female 2 Fever 0 1 Grade 2
 mos related
 2.2 Possibly
 113 Female 3 Fever 2 6 Grade 1
 mos related
 2.6 Possibly
 115 Female 1 Fever 0 1 Grade 1
 mos related
 2.6 Possibly
 115 Female 2 Fever 0 1 Grade 1
 mos related
 2.6 Possibly
 115 Female 2 Local redness 1 1 Grade 3
 mos related
 2.6 Possibly
 115 Female 3 Fever 0 3 Grade 2
 mos related
 2.6 Possibly
 115 Female 3 Local redness 1 2 Grade 3
 mos related
 1.7 Possibly
 117 Male 1 Diarrhea 2 1 Grade 1
 mos related
 1.7 Possibly
 117 Male 1 Crying 3 1 Grade 1
 mos related
 1.7 Possibly
 117 Male 2 Fever 1 1 Grade 1
 mos related
 1.7 Possibly
 117 Male 2 Crying 1 1 Grade 2
 mos related
 1.7 Possibly
 117 Male 2 Local redness 1 3 Grade 1
 mos related

1.7 Possibly
 117 Male 2 Local swelling 1 3 Grade 1
 mos related
 2.6 Possibly
 118 Male 1 Fever 1 7 Grade 1
 mos related
 2.6 Possibly
 118 Male 2 Fever 0 1 Grade 1
 mos related
 2.6 Possibly
 118 Male 3 Fever 0 3 Grade 2
 mos related
 2.6 Possibly
 118 Male 3 Crying 0 1 Grade 2
 mos related
 Table 16. Listing of Adverse Reactions Unrelated to Vaccine
 Days since
 Subject Doses Duration
 Gender Age Term of AE last Severity Relevance
 No. vaccinated (Days)
 vaccination
 27.4 Upper respiratory tract Possibly
 4 Female 1 1 3 Grade 1
 yrs infection unrelated
 27.4 Possibly
 4 Female 1 Increased ALT 3 20 Grade 2
 yrs unrelated
 27.4 Possibly
 4 Female 1 Increased AST 3 20 Grade 1
 yrs unrelated

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27.4 Possibly
 4 Female 1 Increased PLT 3 20 Grade 1
 yrs unrelated
 29.3 Possibly
 14 Female 1 Nasopharyngitis 1 3 Grade 2
 yrs unrelated
 18 Upper respiratory tract Possibly
 16 Female 1 0 20 Grade 2
 yrs infection unrelated
 21.6 Definitely
 17 Male 1 Nasopharyngitis 5 7 Grade 2
 yrs unrelated
 31.2 Possibly
 18 Male 1 Decreased BUN 3 20 Grade 1
 yrs unrelated
 24 Upper respiratory tract Possibly
 20 Female 1 3 8 Grade 1
 yrs infection unrelated
 64 Possibly
 21 Female 1 Maculopapular rash 2 6 Grade 1
 yrs unrelated
 64 Possibly
 21 Female 1 Pain in extremities 23 90 Grade 2
 yrs unrelated
 23.7 Possibly
 27 Male 1 Rhinorrhea 6 2 Grade 2
 yrs unrelated
 2.7 Possibly
 38 Female 1 Decreased BUN 3 29 Grade 1
 yrs unrelated
 3.4 Definitely
 39 Female 1 Nasopharyngitis 28 8 Grade 1
 yrs unrelated
 5.8 Respiratory tract Possibly
 42 Male 1 4 3 Grade 2
 yrs infection unrelated
 3 Possibly
 46 Male 1 Increased WBC 3 28 Grade 1
 yrs unrelated
 4 Possibly
 51 Male 1 Rhinorrhea 26 1 Grade 1
 yrs unrelated
 2.4 Possibly
 60 Female 1 Itching 21 1 Grade 1
 yrs unrelated
 4 Definitely
 61 Male 1 Infant vomiting 5 1 Grade 1
 mos unrelated
 3.5 Upper respiratory tract Definitely
 62 Female 2 25 4 Grade 2
 mos infection unrelated
 3.2 Definitely
 65 Female 2 Nasopharyngitis 19 3 Grade 2
 mos unrelated
 3.5 Possibly
 71 Male 1 Bronchitis 22 13 Grade 2
 mos unrelated

3.2 Upper respiratory tract Possibly
 73 Female 1 25 8 Grade 2
 mos infection unrelated
 3.5 Possibly
 74 Female 2 Infective diarrhea 33 3 Grade 2
 mos unrelated
 3.5 Upper respiratory tract Possibly
 74 Female 2 35 8 Grade 1
 mos infection unrelated
 3 Possibly
 77 Female 1 Fever 30 1 Grade 1
 mos unrelated
 3.1 Possibly
 84 Male 2 cough 6 1 Grade 1
 mos unrelated
 3.1 Possibly
 84 Male 2 Rhinorrhea 7 2 Grade 1
 mos unrelated
 3 Possibly
 86 Male 2 Viral infection 10 8 Grade 2
 mos unrelated
 3 Upper respiratory tract Possibly
 86 Male 3 12 4 Grade 3
 mos infection unrelated
 3.1 Possibly
 90 Male 1 Dyspepsia 13 1 Grade 1
 mos unrelated
 2.5 Definitely
 91 Female 2 Fever 59 2 Grade 2
 mos unrelated
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2.5 Definitely
 91 Female 3 Herpangina 5 6 Grade 3
 mos unrelated
 2.5 Definitely
 91 Female 3 Dyspepsia 8 3 Grade 1
 mos unrelated
 2.6 Definitely
 93 Female 1 Nasopharyngitis 49 4 Grade 1
 mos unrelated
 2.2 Possibly
 94 Male 3 Nasopharyngitis 22 9 Grade 2
 mos unrelated
 2.2 Possibly
 94 Male 3 Diarrhea 11 10 Grade 2
 mos unrelated
 1.8 Upper respiratory tract Possibly
 95 Male 2 57 6 Grade 2
 mos infection unrelated
 1.8 Possibly
 95 Male 2 Diarrhea 57 6 Grade 1
 mos unrelated
 2.3 Possibly
 99 Female 3 Nasopharyngitis 4 1 Grade 2
 mos unrelated
 1.7 Definitely
 100 Male 1 Decreased appetite 48 15 Grade 1
 mos unrelated
 1.7 Definitely
 100 Male 2 Dyspepsia 42 59 Grade 1
 mos unrelated
 2.4 Upper respiratory tract Possibly
 104 Male 2 29 11 Grade 2
 mos infection unrelated
 2.4 Possibly
 104 Male 2 Urticaria 44 4 Grade 2
 mos unrelated
 2.5 Possibly
 110 Female 1 Candidiasis infection 55 4 Grade 1
 mos unrelated
 2.5 Upper respiratory tract Possibly
 110 Female 1 22 5 Grade 2
 mos infection unrelated
 2.5 Possibly
 110 Female 1 Fever 37 2 Grade 2
 mos unrelated
 2.5 Possibly
 110 Female 2 Fever 61 2 Grade 2
 mos unrelated
 2.4 Possibly
 111 Female 1 Tracheitis 15 19 Grade 1
 mos unrelated
 2.4 Possibly
 111 Female 1 Diarrhea 15 14 Grade 2
 mos unrelated
 2.6 Possibly
 115 Female 2 Rash 25 4 Grade 2
 mos unrelated
 2.6 Possibly

115 Female 2 Fever 22 4 Grade 2
mos unrelated
2.6 Possibly
115 Female 2 Diarrhea 22 3 Grade 2
mos unrelated
2.2 Possibly
116 Female 1 Eczema 37 3 Grade 1
mos unrelated

5.3.8. Serious Adverse Events and Important Events
Fourteen subjects experienced grade 3 adverse events, including 1 case of local redness, 1 case of induration, 2 cases of swelling, 2 cases of redness with swelling after one dose of vaccine in adults $\geq$ 18 years old; 1 case of fever and 1 case of fever with local reaction in children group aged 2-5 years; 1 case of local redness (2 episodes), 1 case of local induration, 1 case of somnolence, 1 case of herpangina with fever (herpangina was not related to vaccination, while fever was determined to be probably related to vaccination) in subjects of 2 months old; 1 case of local reaction and 1 case of upper respiratory tract infection in the 3 months olds. Except for 3 cases receiving outpatient treatment, most of the patients recovered spontaneously. No serious adverse events were observed in this trial.

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Adverse events with grade 3 severity were summarized in Table 16.

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Table 16. Adverse events with severity of grade 3

Days since No. of Duration Gender Age Dose Type Term of adverse events last Severity Treatment Relevance subjects (days) vaccination
4 Female 27.4 yrs 1 Local reactions Local redness 1 3 Grade 3 Untreated Possibly related
5 Male 19.2 yrs 1 Local reactions Local induration 1 2 Grade 3 Untreated Possibly related
14 Female 29.3 yrs 1 Local reactions Local redness 0 4 Grade 3 Untreated Possibly related
14 Female 29.3 yrs 1 Local reactions Local swelling 1 3 Grade 3 Untreated Possibly related
20 Female 24 yrs 1 Local reactions Local swelling 0 5 Grade 3 Untreated Possibly related
24 Female 42.7 yrs 1 Local reactions Local redness 5 9 Grade 3 Untreated Possibly related
24 Female 42.7 yrs 1 Local reactions Local swelling 5 13 Grade 3 Untreated Possibly related
29 Male 25.2 yrs 1 Local reactions Local swelling 0 8 Grade 3 Untreated Possibly related
33 Female 4.1 yrs 1 Systemic reactions Fever 0 6 Grade 3 Untreated Possibly related
33 Female 4.1 yrs 1 Local reactions Local pain 1 1 Grade 3 Untreated Possibly related
33 Female 4.1 yrs 1 Local reactions Local redness 1 1 Grade 3 Untreated Possibly related
42 Male 5.8 yrs 1 Systemic reactions Fever 3 5 Grade 3 Clinic visit Possibly related
66 Female 3.7 mos 1 Local reactions Local swelling 0 6 Grade 3 Untreated Possibly related
Infections and
86 Male 3 mos 3 Upper respiratory tract infection 12 4 Grade 3 Clinic visit Possibly unrelated
infestations
91 Female 2.5 mos 3 Systemic reactions Fever 5 6 Grade 3 Clinic visit Possibly related
Infections and
91 Female 2.5 mos 3 Herpangina 5 6 Grade 3 Clinic visit Definitely unrelated
infestations
94 Male 2.2 mos 3 Local reactions Local induration 1 5 Grade 3 Untreated Possibly related
100 Male 1.7 mos 1 Systemic reactions Fatigue / somnolence 0 7 Grade 3 Untreated Possibly related
115 Female 2.6 mos 2 Local reactions Local redness 1 1 Grade 3 Untreated Possibly related
115 Female 2.6 mos 3 Local reactions Local redness 1 2 Grade 3 Untreated Possibly related

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5.4. IMMUNOGENICITY ANALYSIS
5.4.1. Pre- and Post-Immunization Antibody Test Results in Adults Group
5.4.1.1. Immunogenicity Results of IgG Antibody
The positive rate of PCV13-TT IgG antibody was higher in adult group before immunization, the antibody positive rate was 100% for 13 serotypes after immunization except for serotype 3 and 6A (96% for both). The IgG antibody level post immunization were with at least 2-fold increase from baseline, with the lowest fold as 2.23 for serotype 3 and the highest fold as 10.92 for serotype 6B (Table 17 to Table 18).

Table 17. Pre- and Post-Immunization IgG Antibody Positive Rates in Adult Group (PPS)

Pre-immunization (N=30)	Post-immunization (N=30)
N of N of	
Serotypes subjects	subjects
with % 95%CI	with % 95%CI
positive positive	
antibody antibody	
1 30 100.0 88.43-100.0	30 100.0 88.43-100.0
3 27 90.00 73.47-97.89	29 96.67 82.78-99.92
4 20 66.67 47.19-82.71	30 100.0 88.43-100.0
5 29 96.67 82.78-99.92	30 100.0 88.43-100.0
6A 30 100.0 88.43-100.0	29 96.67 82.78-99.92
6B 30 100.0 88.43-100.0	30 100.0 88.43-100.0
7F 29 96.67 82.78-99.92	30 100.0 88.43-100.0
9V 28 93.33 77.93-99.18	30 100.0 88.43-100.0
14 30 100.0 88.43-100.0	30 100.0 88.43-100.0
18C 28 93.33 77.93-99.18	30 100.0 88.43-100.0
19A 30 100.0 88.43-100.0	30 100.0 88.43-100.0
19F 30 100.0 88.43-100.0	30 100.0 88.43-100.0

23F 30 100.0 88.43-100.0 30 100.0 88.43-100.0

Table 18. Pre- and Post-Immunization IgG Antibody Levels in Adult Group (PPS)
Pre-immunization (N=30) Post-immunization (N=30) Antibody fold increase
Serotypes

GMC	95%CI	GMC	95%CI	Fold	95%CI
1	0.84	0.69-1.03	5.10	3.73-6.98	6.07 4.65-7.92
3	0.57	0.48-0.69	1.28	0.99-1.65	2.23 1.77-2.82
4	0.56	0.43-0.73	4.12	2.95-5.75	7.37 5.16-10.54
5	0.93	0.78-1.11	6.06	4.23-8.67	6.52 4.70-9.03
6A	1.40	0.98-1.99	8.45	5.04-14.18	6.05 3.78-9.67
6B	2.31	1.83-2.91	25.21	16.34-38.91	10.92 7.41-16.11
7F	1.29	1.01-1.64	7.30	5.11-10.42	5.67 4.02-8.02
9V	1.59	1.16-2.17	6.96	5.05-9.58	4.38 3.29-5.83
14	5.15	3.84-6.92	16.67	12.15-22.87	3.23 2.46-4.25
18C	1.35	0.97-1.87	11.64	7.42-18.27	8.63 5.31-14.02
19A	4.13	3.14-5.43	17.93	13.16-24.43	4.34 3.37-5.60
19F	3.20	2.59-3.96	19.32	13.12-28.46	6.04 4.12-8.84
23F	1.11	0.93-1.33	7.66	5.58-10.51	6.88 5.09-9.29

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5.4.1.2. Immunogenicity Results of OPA Antibody
The positive rate of OPA antibody in adult group was 100% for serotype 9V, 18C, 19A, 19F, and 23F after immunization, and was $\geq 80\%$ for serotype 1 and 5, while it was $\geq 90\%$ for other serotypes. The OPA antibody titer in all serotypes increased by more than 3 folds compared with that before immunization. Except for the fold increase of GMT of serotypes 1, 3 and 9V OPA antibody being <10 , the fold increases of other serotypes were > 10 (Table 19 to Table 21).
Table 19. Pre- and Post-Immunization OPA Antibody Positive Rates in Adult Group (PPS)

Pre-immunization (N=30)	Post-immunization (N=30)
N of	N of
Serotypes	subjects
with % 95%CI	with % 95%CI
positive	positive
antibody	antibody
1	5 16.67 5.64-34.72
3	13 43.33 25.46-62.57
4	8 26.67 12.28-45.89
5	8 26.67 12.28-45.89
6A	19 63.33 43.86-80.07
6B	15 50.00 31.30-68.70
7F	12 40.00 22.66-59.40
9V	28 93.33 77.93-99.18
14	18 60.00 40.60-77.34
18C	18 60.00 40.60-77.34
19A	28 93.33 77.93-99.18
19F	23 76.67 57.72-90.07
23F	13 43.33 25.46-62.57

Table 20. Pre- and Post-Immunization OPA Antibody GMTs in Adult Group (PPS)
Pre-immunization (N=30) Post-immunization (N=30) Antibody fold increase
Serotypes

GMT	95%CI	GMT	95%CI	Fold	95%CI
1	5.66	4.11-7.79	43.65	23.22-82.06	7.72 4.43-13.43
3	10.79	6.65-17.50	42.17	26.40-67.34	3.91 2.45-6.23
4	7.73	4.83-12.37	695.03	392.7-1230.2	89.87 43.12-187.33
5	5.77	4.33-7.70	204.88	88.89-472.22	35.49 16.49-76.36
6A	47.82	19.51-117.23	3143.79	1682.3-5875.0	65.74 26.60-162.47
6B	21.69	10.31-45.62	1905.83	1016.3-3574.0	87.87 36.48-211.68
7F	18.65	8.68-40.09	1015.21	557.1-1849.9	54.43 23.15-127.93
9V	42.70	25.38-71.86	409.27	248.43-674.24	9.58 5.35-17.16
14	28.54	13.32-61.13	366.22	181.86-737.50	12.83 5.58-29.54
18C	23.30	10.96-49.51	1388.70	881.1-2188.8	59.60 26.45-134.32
19A	112.23	56.02-224.84	2665.96	1805.6-3936.4	23.75 12.93-43.64
19F	26.37	14.74-47.19	2083.98	1356.2-3202.4	79.03 42.42-147.21
23F	8.64	5.95-12.56	557.07	324.41-956.60	64.44 38.13-108.90

Table 21. Pre- and Post-Immunization Antibody Positive Rates and GMTs in Adult Group with Negative Pre-Immunization OPA Antibody (PPS)
Serotypes OPA Antibody Positive Rate OPA Antibody Level

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N of	Fold	subjects	n % 95%CI	GMT	95%CI	95%CI
increase	analyzed					
1	25	19	76.00	54.87-90.64	28.26	15.38-51.94
3	17	14	82.35	56.57-96.20	25.46	13.43-48.27
4	22	21	95.45	77.16-99.88	638.63	306.8-1329.4
5	22	18	81.82	59.72-94.81	116.86	44.24-308.68
6A	11	10	90.91	58.72-99.77	1253.42	288.5-5445.2
6B	15	14	93.33	68.05-99.83	1568.96	567.6-4337.2
7F	18	17	94.44	72.71-99.86	723.05	313.3-1668.8
9V	2	2	100.0	15.81-100.0	58.24	0.0289-117444
14	12	10	83.33	51.59-97.91	204.62	49.41-847.32

121.74-
 18C 12 12 100.0 73.54-100.0 1242.84 487.0-3172.0 310.71
 792.99
 19A 2 2 100.0 15.81-100.0 992.21 119.6-8233.2 248.05 29.9-2058.3
 157.69-
 19F 7 7 100.0 59.04-100.0 1281.64 630.8-2604.1 320.41
 651.03
 23F 17 17 100.0 80.49-100.0 270.69 139.28-526.12 67.67 34.82-131.53

5.4.2. Results of Antibody Detection Pre- and Post-Immunization in Subjects Aged 2-5 years

5.4.2.1. Immunogenicity Results of IgG Antibody

The antibody positive rates of 13 serotypes in children aged 2-5 years reached 100% except for serotype 3 and 6A, which were 88.89% and 96.30%, respectively. The IgG antibody levels post immunization increased by more than 3 folds compared with those before immunization. Except for the fold increase of GMT of serotypes 1, 3, 5, 6A, 6B, 14, and 19F being <10, the fold increases of other serotypes were > 10 (Table 22 to Table 23).

Table 22. Pre- and Post-Immunization IgG Antibody Positive Rates in Children Group Aged 2-5 Years (PPS)

Pre-immunization (N=27) Post-immunization (N=27)

N of N of

Serotypes subjects subjects

with % 95%CI with % 95%CI

positive positive

antibody antibody

1 17 62.96 42.37-80.60 27 100.0 87.23-100.0
 3 10 37.04 19.40-57.63 24 88.89 70.84-97.65
 4 10 37.04 19.40-57.63 27 100.0 87.23-100.0
 5 20 74.07 53.72-88.89 27 100.0 87.23-100.0
 6A 18 66.67 46.04-83.48 26 96.30 81.03-99.91
 6B 23 85.19 66.27-95.81 27 100.0 87.23-100.0
 7F 14 51.85 31.95-71.33 27 100.0 87.23-100.0
 9V 8 29.63 13.75-50.18 27 100.0 87.23-100.0
 14 25 92.59 75.71-99.09 27 100.0 87.23-100.0
 18C 7 25.93 11.11-46.28 27 100.0 87.23-100.0
 19A 23 85.19 66.27-95.81 27 100.0 87.23-100.0
 19F 26 96.30 81.03-99.91 27 100.0 87.23-100.0
 23F 17 62.96 42.37-80.60 27 100.0 87.23-100.0

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Table 23. Pre- and Post-Immunization IgG Antibody Levels in Children Group Aged 2-5 Years (PPS)

Pre-immunization (N=27) Post-immunization (N=27) Antibody fold increase

Serotypes

GMC 95%CI GMC 95%CI Fold 95%CI

1 0.41 0.31-0.54 2.28 1.80-2.88 5.55 4.14-7.44
 3 0.27 0.19-0.37 0.84 0.67-1.07 3.14 2.27-4.35
 4 0.18 0.09-0.37 5.26 3.94-7.02 28.88 12.53-66.56
 5 0.43 0.27-0.68 2.29 1.75-2.98 5.35 3.44-8.34
 6A 0.64 0.47-0.88 3.02 1.94-4.69 4.70 3.30-6.68
 6B 1.09 0.74-1.62 7.98 5.82-10.94 7.31 4.41-12.11
 7F 0.35 0.21-0.59 4.26 3.19-5.70 12.22 7.00-21.34
 9V 0.22 0.11-0.42 2.90 1.96-4.27 13.22 6.79-25.71
 14 1.70 1.21-2.40 10.36 6.55-16.39 6.09 3.83-9.66
 18C 0.15 0.06-0.38 6.70 4.81-9.34 44.19 17.66-110.57
 19A 1.04 0.59-1.83 11.18 7.86-15.89 10.78 6.27-18.55
 19F 1.80 1.21-2.67 10.98 8.09-14.91 6.11 3.83-9.75
 23F 0.40 0.23-0.70 5.82 3.85-8.79 14.49 7.89-26.62

5.4.2.2. Immunogenicity Results of OPA Antibody

In subjects aged 2-5 years, the positive rates of OPA antibody in 13 serotypes after immunization was 53.57% for serotype 1, and was > 80% for serotype 3, 5, 6B and 14, and was 100% for other serotypes. The OPA antibody titers of all serotypes increased by ≥ 3 folds compared with that before immunization. Except for serotypes 1 and 3 with fold increase < 10 folds, the fold increases of other serotypes were > 10 folds, and the highest fold was 345 for serotype 4 (Table 24 to Table 26).

Table 24. Pre- and Post-Immunization OPA Antibody Positive Rates in Children Group Aged 2-5 Years (PPS)

Pre-immunization (N=28) Post-immunization (N=28)

N of N of

Serotypes subjects subjects

with % 95%CI with % 95%CI

positive positive

antibody antibody

1 0 0.00 0.0-12.34 15 53.57 33.87-72.49
 3 5 17.86 6.06-36.89 23 82.14 63.11-93.94
 4 5 17.86 6.06-36.89 28 100.0 87.66-100.0
 5 0 0.00 0.0-12.34 26 92.86 76.50-99.12
 6A 9 32.14 15.88-52.35 28 100.0 87.66-100.0
 6B 10 35.71 18.64-55.93 26 92.86 76.50-99.12
 7F 6 21.43 8.30-40.95 28 100.0 87.66-100.0
 9V 11 39.29 21.50-59.42 28 100.0 87.66-100.0
 14 5 17.86 6.06-36.89 25 89.29 71.77-97.73
 18C 9 32.14 15.88-52.35 28 100.0 87.66-100.0
 19A 14 50.00 30.65-69.35 28 100.0 87.66-100.0
 19F 11 39.29 21.50-59.42 28 100.0 87.66-100.0
 23F 4 14.29 4.03-32.67 28 100.0 87.66-100.0

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Table 25. Pre- and Post-Immunization OPA Antibody GMTs in Children Group Aged 2-5 Years (PPS)

Pre-immunization (N=28)		Post-immunization (N=28)		Antibody fold increase	
Serotypes					
GMT 95%CI GMT 95%CI Fold 95%CI					
1	4.00	4.00-4.00	12.01	7.26-19.87	3.00 1.81-4.97
3	5.93	4.12-8.54	40.98	24.10-69.69	6.91 3.95-12.10
4	7.05	3.98-12.49	2434.47	1672.4-3543.9	345.31 160.18-744.37
5	4.00	4.00-4.00	75.62	41.67-137.23	18.90 10.42-34.31
6A	9.51	5.27-17.17	833.61	499.6-1390.9	87.66 38.34-200.40
6B	13.92	6.07-31.92	1017.06	448.3-2307.2	73.08 27.54-193.89
7F	17.43	5.56-54.68	2470.51	1594.3-3828.2	141.73 54.31-369.89
9V	10.21	5.85-17.80	696.49	401.6-1208.0	68.25 31.00-150.22
14	7.52	4.40-12.86	155.37	63.58-379.70	20.67 9.20-46.44
18C	9.94	5.05-19.55	1433.80	843.7-2436.8	144.26 65.29-318.73
19A	22.71	9.80-52.63	2384.08	1580.4-3596.5	104.96 40.33-273.16
19F	13.86	6.90-27.84	1223.14	759.4-1970.2	88.22 31.59-246.42
23F	5.81	3.88-8.69	1700.09	848.6-3405.9	292.78 135.09-634.54

Table 26. Pre- and Post-Immunization Antibody Positive Rates and GMTs in Children Group Aged 2-5 Years with Negative Pre-Immunization OPA Antibody (PPS)

OPA Antibody Positive Rate		OPA Antibody Level	
Serotypes N of 95%CI			
Fold			
subjects n % 95%CI GMT 95%CI			
increase			
analyzed			
1	28	15	53.57 33.87-72.49 12.01 7.26-19.87 3.00 1.81-4.97
3	23	18	78.26 56.30-92.54 35.39 18.90-66.26 8.85 4.72-16.57
4	23	23	100.0 85.18-100.0 2563.23 1731.1-3795.4 640.81 432.77-948.86
5	28	26	92.86 76.50-99.12 75.62 41.67-137.23 18.90 10.42-34.31
6A	19	19	100.0 82.35-100.0 1029.43 541.8-1955.8 257.36 135.46-488.96
6B	18	16	88.89 65.29-98.62 697.17 234.6-2071.9 174.29 58.65-517.97
7F	22	22	100.0 84.56-100.0 1798.09 1151.0-2809.1 449.52 287.74-702.27
9V	17	17	100.0 80.49-100.0 813.91 484.3-1367.9 203.48 121.07-341.99
14	23	20	86.96 66.41-97.22 96.66 36.69-254.65 24.16 9.17-63.66
18C	19	19	100.0 82.35-100.0 1255.25 677.7-2325.1 313.81 169.42-581.28
19A	14	14	100.0 76.84-100.0 2464.86 1463.6-4151.0 616.22 365.9-1037.7
19F	17	17	100.0 80.49-100.0 1982.59 1197.5-3282.5 495.65 299.36-820.62
23F	24	24	100.0 85.75-100.0 1689.08 758.7-3760.3 422.27 189.68-940.08

5.4.3. Results of IgG Antibody Detection Pre- and Post-Immunization in Subjects Aged 2 months

In subjects aged 2 months, the serum positive rate reached 100% for most serotypes, except for an IgG antibody positive rate of 84.62% for serotype 5, and 92.31% and 96.00% for serotype 3 and 6B, respectively; among those with pre-immunization negative antibody levels, all but serotype 3 and 5 reached 100% as seroconversion. Post-immunization fold increase of IgG antibody was ≥ 5, and it was <10 for serotypes 3, 5, 6B, 14, and 19A, while was >10 folds for other serotypes (Table 27 to Table 29).

Table 27. Pre- and Post-Immunization IgG Antibody Positive Rates in Infant Group Aged 2 Months (PPS)

Serotypes Pre-immunization (N=26)		Post-immunization (N=26)	
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N of N of	
subjects subjects	
with % 95%CI with % 95%CI	
positive positive	
antibody antibody	

1	4	15.38	4.36-34.87	26	100.0	86.77-100.0
3	1	3.85	0.10-19.64	24	92.31	74.87-99.05
4	0	0.00	0.0-13.23	26	100.0	86.77-100.0
5	2	7.69	0.95-25.13	22	84.62	65.13-95.64
6A	7	26.92	11.57-47.79	26	100.0	86.77-100.0
6B	12	48.00	27.80-68.69	24*	96.00	79.65-99.90
7F	5	19.23	6.55-39.35	26	100.0	86.77-100.0
9V	5	20.00	6.83-40.70	25*	100.0	86.28-100.0
14	21	84.00	63.92-95.46	25*	100.0	86.28-100.0
18C	7	26.92	11.57-47.79	26	100.0	86.77-100.0
19A	18	69.23	48.21-85.67	26	100.0	86.77-100.0
19F	17	65.38	44.33-82.79	26	100.0	86.77-100.0
23F	3	11.54	2.45-30.15	26	100.0	86.77-100.0

*One subject in 2 months old group was not included in FAS and PPS for lacking results of serotypes 6B, 9V

and 14 due to inadequate serum, thus the number of analyzed subjects was 25 instead of 26.

Table 28. Post-Immunization Antibody Positive Rates in Infant Group Aged 2 Months with Negative Pre-Immunization Antibody (PPS)

Post-immunization	
Serotypes N of subjects analyzed N of subjects	
with positive % 95%CI	
antibody	
1	22 22 100.0 84.56-100.0
3	25 23 92.00 73.97-99.02
4	26 26 100.0 86.77-100.0
5	24 21 87.50 67.64-97.34
6A	19 19 100.0 82.35-100.0
6B	13 13 100.0 75.29-100.0

7F 21 21 100.0 83.89-100.0
 9V 20 20 100.0 83.16-100.0
 14 4 4 100.0 39.76-100.0
 18C 19 19 100.0 82.35-100.0
 19A 8 8 100.0 63.06-100.0
 19F 9 9 100.0 66.37-100.0
 23F 23 23 100.0 85.18-100.0

Table 29. Pre- and Post-Immunization IgG Antibody Levels in Infant Group Aged 2 Months (PPS)

Pre-immunization (N=26) Post-immunization (N=26) Antibody fold increase

Serotypes

GMC 95%CI GMC 95%CI Fold 95%CI

1 0.22 0.17-0.28 2.19 1.53-3.13 10.06 6.33-15.98
 3 0.10 0.08-0.12 0.97 0.70-1.34 9.97 6.38-15.57
 4 0.02 0.01-0.05 2.49 1.85-3.35 131.02 47.12-364.30
 5 0.15 0.10-0.23 0.76 0.57-1.03 4.94 2.74-8.91
 6A 0.12 0.06-0.28 3.29 2.43-4.44 26.33 10.58-65.52
 6B 0.38 0.28-0.51 3.06 2.10-4.47 8.15 4.60-14.43

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7F 0.09 0.04-0.21 4.13 3.12-5.46 45.39 18.16-113.44
 9V 0.16 0.09-0.28 2.69 1.91-3.80 17.28 8.15-36.64
 14 1.04 0.66-1.64 10.19 7.19-14.43 9.83 5.05-19.13
 18C 0.18 0.09-0.34 1.87 1.36-2.59 10.66 5.09-22.33
 19A 0.46 0.27-0.76 3.15 2.37-4.18 6.90 3.55-13.44
 19F 0.30 0.15-0.61 4.74 3.56-6.33 15.87 7.10-35.45
 23F 0.23 0.18-0.30 3.41 2.36-4.93 14.74 9.17-23.68

5.4.4. Results of Antibody Detection Pre- and Post-Immunization in Subjects Aged 3 Months

5.4.4.1. Immunogenicity Results of IgG Antibody

In subjects aged 3 months, the IgG antibody positive rate and antibody seroconversion rate (for those with pre-immunization negative antibody) reached 100% except serotype 6A and 6B, the post-immunization fold increase of IgG antibody level was >10 except for 6B with 6 fold increase, and the highest fold was 469 for serotype 4 (Table 30 to Table 32).

Table 30. Pre- and Post-Immunization IgG Antibody Positive Rates in Infant Group Aged 3 Months (PPS)

Pre-immunization (N=20) Post-immunization (N=20)

N of N of

Serotypes subjects subjects

with % 95%CI with % 95%CI

positive positive

antibody antibody

1 2 10.00 1.23-31.70 20 100.0 83.16-100.0
 3 0 0.00 0.0-16.84 20 100.0 83.16-100.0
 4 1 5.00 0.13-24.87 20 100.0 83.16-100.0
 5 0 0.00 0.0-16.84 20 100.0 83.16-100.0
 6A 3 15.00 3.21-37.89 18 90.00 68.30-98.77
 6B 4 20.00 5.73-43.66 18 90.00 68.30-98.77
 7F 6 30.00 11.89-54.28 20 100.0 83.16-100.0
 9V 2 10.00 1.23-31.70 20 100.0 83.16-100.0
 14 17 85.00 62.11-96.79 20 100.0 83.16-100.0
 18C 2 10.00 1.23-31.70 20 100.0 83.16-100.0
 19A 8 40.00 19.12-63.95 20 100.0 83.16-100.0
 19F 11 55.00 31.53-76.94 20 100.0 83.16-100.0
 23F 1 5.00 0.13-24.87 20 100.0 83.16-100.0

Table 31. Post-Immunization IgG Antibody Positive Rates in Infant Group Aged 3 Months with Negative Pre-Immunization Antibody (PPS)

Post-immunization

N of subjects N of subjects

Serotypes

analyzed with positive % 95%CI

antibody

1 18 18 100.0 81.47-100.0
 3 20 20 100.0 83.16-100.0
 4 19 19 100.0 82.35-100.0
 5 20 20 100.0 83.16-100.0
 6A 17 15 88.24 63.56-98.54
 6B 16 14 87.50 61.65-98.45

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7F 14 14 100.0 76.84-100.0
 9V 18 18 100.0 81.47-100.0
 14 3 3 100.0 29.24-100.0
 18C 18 18 100.0 81.47-100.0
 19A 12 12 100.0 73.54-100.0
 19F 9 9 100.0 66.37-100.0
 23F 19 19 100.0 82.35-100.0

Table 32. Pre- and Post-Immunization IgG Antibody Levels in Infant Group Aged 3 Months (PPS)

Pre-immunization (N=20) Post-immunization (N=20) Antibody fold increase

Serotypes

GMC 95%CI GMC 95%CI Fold 95%CI

1 0.08 0.03-0.19 1.88 1.30-2.72 24.74 9.32-65.64

between the two methods.

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Since the antibody levels detected with the two methods were expressed as concentration and titer respectively, no comparison was made on the consistency rate for IgG antibody and OPA antibody level.

The comparison for the consistency of post-immunization IgG and OPA antibody positive rates in subjects 3 months of age after immunization is shown in Table 36.

Table 36. Consistency Between Pre- and Post-Immunization IgG Antibody Positive Rates and OPA Antibody Positive Rates in Infant Group Aged 3 Months (PPS)

Pre-immunization Post-immunization

Serotypes OPA Consistency OPA Consistency

IgG IgG

+ - (%) + - (%)

+ 0 3 + 23 2

1 88.00 92.00

- 0 22 - 0 0

+ 0 0 + 25 0

3 100.0 100.0

- 0 25 - 0 0

+ 1 0 + 25 0

4 100.0 100.0

- 0 24 - 0 0

+ 0 0 + 25 0

5 96.00 100.0

- 1 24 - 0 0

+ 3 0 + 23 0

6A 88.00 96.00

- 3 19 - 1 1

+ 2 3 + 21 2

6B 84.00 88.00

- 1 19 - 1 1

+ 5 1 + 24 1

7F 84.00 96.00

- 3 16 - 0 0

+ 1 1 + 25 0

9V 92.00 100.0

- 1 22 - 0 0

+ 7 15 + 24 1

14 40.00 96.00

- 0 3 - 0 0

+ 1 1 + 25 0

18C 84.00 100.0

- 3 20 - 0 0

+ 5 5 + 25 0

19A 76.00 100.0

- 1 14 - 0 0

+ 5 9 + 25 0

19F 60.00 100.0

- 1 10 - 0 0

+ 1 1 + 24 1

23F 96.00 96.00

- 0 23 - 0 0

6. DISCUSSIONS AND CONCLUSIONS

6.1. SAFETY

The incidence of adverse events related to the study vaccine in adults, children, 2 months and 3 months group was 80.00%, 53.33%, 66.67% and 70.00%, respectively. A total of 6 subjects in the adults group experienced local adverse reactions with severity reaching grade 3, including 1 case of local redness, 1 case of induration, 2 cases of swelling and 2 cases of redness with swelling, corresponding to an incidence of 20% (6/30) in the adults group. The incidence of adverse reactions with grade 3 severity in other age groups was less than 15%.

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The most common symptom of systemic adverse reactions was fever in adults and children aged 2-5 years, while fever and diarrhea were more common in 2 months and 3 months group. Local adverse reactions in adults were mainly redness, pain and swelling at the vaccination site; pain and redness were common local reactions in children aged 2-5 years, and redness was the most common local symptom in 2 months and 3 months group.

Adverse events mainly occurred within 0-7 days post vaccination and no adverse events related to the study vaccine were collected during 8-30 days post vaccination. No serious adverse events were reported during the extended observation period.

6.2. IMMUNOGENICITY

The IgG positive rates and antibody levels of 13 serotypes in adults and children 2-5 years of age were relatively high before immunization, and most of the IgG antibody positive rates of all serotypes in these groups reached 100% or > 80% post immunization, accompanied by more than 2 fold increases in post-immunization antibody level compared to those before vaccination. The IgG positive rates and antibody levels of 13 serotypes in subjects aged 2 or 3 months were lower than those in adults and children groups before immunization, and the positive rate of most IgG antibody reached 100% except for serotypes 3, 5, 6A and 6B, besides, the fold increase of antibody was significantly higher compared to adults and children groups.

The positive rates of OPA antibody and antibody levels were lower than that of IgG antibody before immunization in all age groups. After immunization, OPA antibody level increased significantly, with higher fold the increase compared to IgG antibody, and in 3 months age group, the OPA positive antibody rates in subjects aged 3 months were ≥ 90%.

The correlation analysis of IgG and OPA results demonstrated rather high consistency of the

positive rates of IgG and OPA after immunization, while bias might exist due to the small sample size.

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7. STATEMENTS ON OF NOTEWORTHY FINDINGS IN THE STUDY

In the observation period for adults, 6 subjects developed grade 3 adverse events, corresponding to an incidence rate of 20%, which was higher than the study interruption criteria (15%) set forth in the protocol. The adverse reactions of grade 3 were tolerable local symptoms such as redness and swelling, and no severe reactions such as abscess, death or serious reactions of limb movement disorder (reaching grade 4). After consulting the experts, it was analyzed that the syringe for vaccination was equipped with small needles specific for children, which did not meet the requirements specified in the protocol for deep injection, and the investigators might have not fully mixed the vaccine before vaccination, so it was suggested to improve the operating procedure. The preliminary summary of adverse events in adult group was reported to the ethics committee and then the next stage of observation initiated. After another round of training, the incidents never occurred in the following stage involving subjects 2 and 3 months of age.

8. SUMMARY OF ALL UNITS PARTICIPATING IN THE CLINICAL TRIAL

This clinical trial involves a single center, no available data for other sites.

9. REFERENCE

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- [3] Guidelines for Quality Control of Clinical Trials of Vaccines (Pilot Edition), National Medical Products Administration, Oct. 31, 2013.
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10. APPENDIX

N/A

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Protocol No.: 022152015003

Clinical Study Report
of the
13-valent Pneumococcal Polysaccharide
Conjugate Vaccine
(Tetanus Toxoid Protein)

Phase III

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General Information

Generic name of studied drug 13-valent Pneumococcal Conjugate Vaccine
Study Design Multi-center, Randomised, Double-Blind, Similar Vaccine-Controlled
Protocol No. 022152015003
Study time (start and end) 2015.06-2018.01
Primary Investigator (signature)
Responsible unit (stamp) Beijing Chaoyang District Center for Disease Control and Prevention
Sub-center Primary(signature)
Investigator
Sub-center unit (stamp) HeBei Center for Disease Control and Prevention
HeNan Center for Disease Control and Prevention
ShanXi Center for Disease Control and Prevention
Data Statistics Specialist (signature)
Sponsor (stamp) Yunnan Walvax Biotechnology Co., Ltd.
Yuxi Walvax Biotechnology Co., Ltd
Contact person of sponsor Lin Yuan
Tel 0871-68336070
Email ynwsl@walvax.com
Address No.83 South Dongfeng Road, High ■ New Technology Industries
Development Zone, Yuxi, Yunnan Province, CHINA
Zip code 653100
Report Version V2.0
Date 2018.10
Beijing Chaoyang District, Hebei, Henan and Shanxi Center for Disease Control and Prevention

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Tabulated Clinical Trail Report

Table 1.Tabulated Clinical Trial Report
Vaccine name 13-valent pneumococcal polysaccharide conjugate vaccine
Sponsor Yunnan Walvax Biotechnology Co., Ltd.
Yuxi Walvax Biotechnology Co., Ltd
Study title Phase III Clinical Trial of 13-valent pneumococcal polysaccharide conjugate vaccine
Clinical trial Approval No.: 2015L00591 Approved on: April, 2015
approval
Principal Nianming Shi Beijing Chaoyang District Center for Disease Control and Prevention
investigator Yuliang Zhao Hebei Province Center for Disease Control and Prevention
and Shengli Xia Henan Province Center for Disease Control and Prevention
responsible
units Guohua Li Shanxi Province Center for Disease Control and Prevention
Duration of From: June, 2015 To: January, 2018
clinical trial
Number of 2760
subjects
Result of the National Institutes for Food and Drug Control certified that this vaccine's quality
complied
investigational with the Enterprise Registration Standard of 13-valent pneumococcal polysaccharide
vaccine conjugate vaccine. Report No.: SZ201511716
Investigational13-valent pneumococcal polysaccharide conjugate Batch No.: 201511003
vaccine vaccine, manufactured by Yuxi Walvax Expiration date: November, 2017
Biotechnology Co., Ltd, for injection, 0.5 mL/vial,
each dose contains 2.2 µg of polysaccharide from
pneumococcal serotype 1, 3, 4, 5, 6A, 7F, 9V, 14,
18C, 19A, 19F, and 23F, and 4.4 µg polysaccharide
for 6B.
Comparator 7-valent pneumococcal conjugate vaccine, Batch No.: H29315H13880
vaccine manufactured by Pfizer (Wyeth) Pharmaceutical Inc., Expiration date: May, 2017
for injection, 0.5 mL/vial, each dose contains 2 µg of
polysaccharide from pneumococcal serotypes 4, 9V,
14, 18C, 19F, 23F and 4 µg polysaccharide for 6B.

Usage and dosage of Age Number of Primary series Booster dose
 investigational strata Groups
 subjects Schedule Dose(s) Schedule Dose
 vaccine and PCV13- Month 0,
 comparator 520 3 doses
 2-6 TT (3M) 1, 2
 vaccine months PCV13- Month 0, 12-15 1
 520 3 doses
 (≥ 6 TT (2M) 2, 4 months of age dose
 weeks) PCV7 Month 0,
 520 3 doses
 (3M) 1, 2
 PCV13- Month 0,
 200 2 doses 12-15 months of age
 7-11 TT 2 1
 (≥ 2 months apart
 months Month 0, dose
 PCV7 200 2 doses from the second dose)
 2
 PCV13- Month 0,
 200 2 doses / /
 12-23 TT 2
 months Month 0,
 PCV7 200 2 doses / /
 2
 PCV13-
 24-71 200 Month 0 1 dose / /
 TT
 months
 PCV7 200 Month 0 1 dose / /
 Efficacy Immunogenicity:
 measures
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■ The percentage of subjects with vaccine-type specific IgG antibody concentration ≥ 0.35 $\mu\text{g/mL}$ and ≥ 1.0 $\mu\text{g/mL}$ as well as the IgG geometric mean concentration (GMC) measured 30 days after primary series.

■ The percentage of subjects with IgG vaccine-type specific IgG antibody concentration ≥ 0.35 $\mu\text{g/mL}$ and ≥ 1.0 $\mu\text{g/mL}$, as well as the IgG geometric mean concentration (GMC) measured before and 30 days after booster dose.

■ The percentage of subjects with vaccine-type specific anti-pneumococcal OPA titer $\geq 1:8$ and the geometric mean titer (GMT) measured 30 days after primary series and 30 days after booster dose.

Safety:

■ Adverse events within 30 days after full primary immunizations and within 30 days after booster dose.

Clinical trial In the main target population (3 months of age)
 evaluation The common serotypes achieved noninferiority criteria for comparison between PCV13-TT and PCV7 group except for serotype 6B during primary series, meanwhile all common serotypes in PCV13-TT group were noninferior to that in PCV7 group after the booster dose; for additional serotypes, superiority criteria were met for PCV13-TT group vs PCV7 group after primary series; the common serotypes in PCV13-TT group aged 3-71 months were all noninferior to those in PCV7 group during primary series, besides, the additional serotypes in PCV13-TT group were all superior as compared to PCV7 group.

For PCV13-TT (2 months) group vs PCV7 (3 months) group
 Both common and additional serotypes had achieved noninferiority or superiority criteria post primary series. For PCV13-TT (2 months) group vs PCV13-TT (3 months) group, 12 serotypes reached noninferiority criteria during primary series except for additional serotype 5; and all 13 serotypes were noninferior to those in PCV13-TT (3 months) group post booster dose.

In PCV13-TT recipients aged 7-11 months, 12-23 months, and 24-71 months
 The proportions of subjects with IgG antibody concentration ≥ 0.35 $\mu\text{g/mL}$ for common serotypes were all noninferior to those in PCV7 group. The additional serotypes in PCV13-TT group of 7-11 months had reached superiority criteria after primary series; for PCV13-TT groups of 12-23 months and 24-71 months, superiority criteria were generally met after immunization.

The results before and after booster dose indicated decrease in the antibody levels of several serotypes in the 13 serotypes either in PCV13-TT group or PCV7 group to some extent, while the percentage of subjects with IgG antibody concentration ≥ 3.5 $\mu\text{g/mL}$ and antibody GMC after 1 dose of booster immunization both reached or exceeded the levels observed after primary series, demonstrating significant effect of the booster dose, reflecting the immunological memory effect of the conjugate vaccine adequately. The antibody GMC post booster dose had more than 3-fold increase in the 2-6 months olds, and greater than 1-fold increase in subjects aged 7-11 months as compared with the GMC before booster dose.

The overall incidence of adverse events in PCV13-TT group was lower than that in PCV7 group. No significant intergroup difference was observed in the incidence of vaccine-related adverse events. The main symptom of systemic adverse reaction was fever. The incidence of systemic reactions in PCV13-TT group was lower than that in PCV7 group. Local adverse reactions were primarily redness, swelling and pain at the injection site. No significant difference was observed in the incidence of local reactions between PCV13-TT group and PCV7 group.

The incidence of vaccine-related grade 3 adverse events in PCV13-TT group and PCV7 group was 4.21% and 2.41%, respectively. The incidence of local adverse reactions with severity reaching grade 3 in PCV13-TT group was higher than that in PCV7 group, while no significant intergroup difference was observed in subjects aged 3 months.
 A total of 80 cases (105 events) of SAEs were observed during primary series and 8 cases (10

events) of SAEs were noted during booster dose. No significant intergroup difference was
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observed in the incidence of SAEs. SAEs were not related to the study vaccine except for one case in PCV13-TT (3 months) group with fever determined to be probably related to the study vaccine.

To sum up, the immunogenicity of 7 common serotypes in main target population aged 3 months met the non-inferiority criteria after booster dose, and that of 6 new serotypes had already achieved the standard of superiority after primary immunizations in this clinical trial. The common and additional serotypes had fulfilled the non-inferiority or superiority standard in subjects 3-71 months of age after primary immunizations. The common and additional serotypes in PCV13-TT group (2 months) were noninferior and/or superior to that in PCV7 (3 months) group. In PCV13-TT (2 months) group, all 13 serotypes were noninferior to those in the PCV13-TT (3 months) group post booster dose. The incidence of adverse events/reactions in PCV13-TT group was not higher than that in PCV7 group. The results of immunogenicity and safety evaluation in this clinical trial had achieved the study objectives predefined in the protocol.

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Overview of the Clinical Study

1. INTRODUCTION

The 13-valent pneumococcal polysaccharide conjugate vaccine, jointly developed by the Yunnan Walvax Biotechnology Co., Ltd. and the Yuxi Walvax Biotechnology Co., Ltd., is consistent with the marketed 13-valent pneumococcal conjugate vaccine with respect to serotypes, and is used for the prevention of related diseases caused by infections of the 13 pneumococcal serotypes contained in this vaccine (types 1, 3, 4, 5, 6A, 6B, 7F, 9V, 14, 18C, 19A, 19F, and 23F). Upon review and approval by the National Medical Products Administration (NMPA) in accordance with the Drug Administration Law of the P.R.C and Provisions for Drug Registration, this vaccine met the relevant provisions for drug registration, and therefore, the "Drug Clinical Trial Approval" was issued in Apr. 2015 (Approval No.: 2015L00591) for conduct of clinical trials.

Entrusted by the Yunnan Walvax Biotechnology Co., Ltd. and the Yuxi Walvax Biotechnology Co., Ltd., the Beijing Chaoyang Center for Disease Control and Prevention (CDC), the Hebei Provincial CDC, the Henan Provincial CDC, and the Shanxi Provincial CDC were responsible for the clinical trial of this vaccine, with a view to evaluating its immunogenicity and safety in healthy children at 2–71 months of age (minimum, 6 weeks). The Beijing Chaoyang CDC, the Hebei Provincial CDC, the Henan Provincial CDC, and the Shanxi Provincial CDC, based on the relevant requirements in the "Drug Clinical Trial Approval" by NMPA (Approval No.: 2015L00591) and the Provisions for Drug Registration and as per the protocol, conducted a phase III clinical trial on the 13-valent pneumococcal polysaccharide conjugate vaccine jointly developed by the Yunnan Walvax Biotechnology Co., Ltd. and Yuxi Walvax Biotechnology Co., Ltd., and issued this clinical trial report.

2. OBJECTIVE

2.1. PRIMARY OBJECTIVE

To evaluate the safety of 13-valent pneumococcal polysaccharide conjugate vaccine (PCV13-TT) in healthy population aged 2-3 months (minimum age: 6 weeks), 2-5 years and 18 years and above. To demonstrate the IgG immune response to 13-valent pneumococcal polysaccharide conjugate vaccine in healthy children who were enrolled in the study group at the age of 3 months and were injected with the study vaccine in the schedule that one dose was administered in Month 0, 1, and 2 respectively and the booster dose was administered during Month 12-15. The results were compared with 7-valent pneumococcal conjugate vaccine in the same immunization schedule. The non-inferiority test of 7 common serotypes and the effectiveness evaluation of 6 additional serotypes were performed.

2.2. SECONDARY OBJECTIVE

(1) To observe the IgG immune response to 13-valent pneumococcal polysaccharide conjugate vaccine in healthy children who were enrolled in the study group at the age of 2 months (minimum age: 6 weeks) and were injected with the study vaccine in the schedule that one dose was administered in Month 0, 2, and 4 respectively and the booster dose was administered during Month 12-15. The results were compared with those obtained the control group where healthy children who were enrolled in the group at the age of 3 months and were injected with 7-valent

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pneumococcal conjugate vaccine in the schedule that one dose was administered in Month 0, 1, and 2 respectively and the booster dose was administered during Month 12-15. The non-inferiority test of 7 common serotypes and the effectiveness evaluation of 6 additional serotypes were performed.

(2) To observe and evaluate the serotype-specific opsonophagocytic activity (OPA) of pneumococcal antibodies in healthy children aged 2-71 months (minimum age: 6 weeks) vaccinated with 13-valent pneumococcal polysaccharide conjugate vaccine in different immunization schedules.

(3) To observe and evaluate the safety of vaccinating healthy children aged 2-71 months (minimum age: 6 weeks) with 13-valent pneumococcal polysaccharide conjugate vaccine.

3. STUDY MANAGEMENT

3.1. MAIN RESPONSIBLE UNITS AND TECHNICAL COLLABORATION UNIT

3.1.1. Main responsible units

This clinical study was undertaken by the Beijing Chaoyang CDC, the Hebei Provincial CDC, the Henan Provincial CDC, and the Shanxi Provincial CDC. These research institutions obtained the CFDA (now NMPA) approval as "One-Time Certification for Vaccine Clinical Trial Institution" after review, and the clinical trial protocol, the informed consent form, and related forms were reviewed and approved by the ethics committee.

The Beijing Chaoyang CDC, Hebei Provincial CDC, Henan Provincial CDC, and Shanxi Provincial CDC have carried out multiple clinical studies on preventive biological products

successfully. The principle investigator has experience in clinical studies on preventive biological products, all staff in the Institute of Vaccine Clinical Study that participated in this study have attended GCP training and examinations organized by the national government or this institution, and after training and authorization, all the research participants had been aware of the responsibilities and obligations of clinical investigators and familiar with the nature, effects, efficacy, and safety of the investigational drugs to be used in the clinical trial. Effective measures and conditions were available to ensure the safety of subjects and the timely, accurate, and truthful recording of clinical research data. They were mainly responsible for:

- Formulating the vaccine clinical trial protocol, and organizing the implementation of such a protocol;
- Preparing and printing such on-site application forms as informed consent forms, diary cards, and case report forms;
- Submitting ethical review materials to the ethics committee for approval;
- Choosing the clinical trial site meeting the requirements set forth in the protocol, and recruiting trial subjects;
- Organizing the implementation of on-site vaccination and on-site work related to clinical observations;
- Organizing the follow-up of subjects in the clinical trial site and the collection of adverse events, and reporting, investigating and handling abnormal vaccine reactions;
- Completing the collection of blood samples, laboratory processing and delivery to the designated laboratory as per the protocol;
- Completing all the forms and case report forms on the clinical trial site;
- Issuing a clinical trial report.

3.1.2. Technical collaboration unit

National Institutes for Food and Drug Control, which was responsible for laboratory testing of

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serum samples and issuing a testing report.

3.1.3. Research institutions on the trial site

The research institutions on this trial site included the Zhengding County Center for Disease Control and Prevention and the Dingxing County Center for Disease Control and Prevention in Hebei Province, the Puyang County Center for Disease Control and Prevention and the Wen County Center for Disease Control and Prevention in Henan Province, and the Xiangfen County Center for Disease Control and Prevention and the Wanrong County Center for Disease Control and Prevention in Shanxi Province. In addition, the Xiaohongmen Community Health Service Center from Chaoyang District, Beijing, the Heizhuanghu Community Health Service Center under the Shuangqiao Hospital, and the Nanmofang Community Health Service Center were used as standby clinical trial sites (not in operation). The number of trained investigators, technical strengths, and environmental facilities on this clinical trial site met the needs of the clinical trial, so were the resources of subjects. The direct reporting network system operated normally, and the monitoring system for side reactions arising from vaccination was complete. All the laboratories passed the metrological certification. The institutions on the trial site were responsible for subject recruitment, organization on the site, and handling and reporting of adverse events.

3.1.3.1. Beijing Chaoyang Center for Disease Control and Prevention

The Beijing Chaoyang CDC, which is located in the east of Beijing, covers a total area of 470.8 square kilometers, and has an actual resident population of near 5.1 million. Currently, the Beijing Chaoyang CDC has a total of 199 professional technical personnel, which account for 86% of the actual number of employees, and the personnel have an average age of 35.9 years. Specifically, there are 35 personnel with senior titles, 11 with Ph.D degrees, 86 with master's degrees, and 74 with bachelor's degrees. Since 2007, the center has undertaken scientific research projects from the Chinese Preventive Medicine Association, the NMPA 12th Five-Year Major Project, etc., and the phase III clinical trial of the EV71 vaccine. Since the promulgation of "Good Clinical Practice on Vaccine Clinical Trials (Pilot Edition)" by NMPA in 2013, it has undertaken the phase III clinical study involving the freeze-dried rabies vaccine (MRC-5 cells) for human use from Beijing Minhai Biological Technology Co., Ltd., the randomized, blinded, parallel, controlled phase II clinical study involving the recombinant humanized rabies virus monoclonal antibody injection from the NCPC, and the phase III clinical trial involving the recombinant human papillomavirus bivalent (types 16/18) vaccine (yeast) from the Shanghai Zerun Biotechnology Co., Ltd. The employees from the CDC have published more than 10 papers at home and abroad, including 6 papers in foreign languages, and published 5 monographs on vaccines. All staff in the Institute of Vaccine Clinical Study that participated in this study have attended GCP training and examinations organized by the national government or this institution.

(1) Heizhuanghu Community Health Service Center under the Shuangqiao Hospital in Chaoyang District, Beijing

Established in 1958, the Shuangqiao Hospital in Chaoyang District, Beijing, is affiliated to the Health Bureau of Chaoyang District, and has been approved by the Beijing Municipal Human Resources and Social Security Bureau as a designated medical institution for medical insurance in Beijing. Located in the southeast of the Chaoyang District, the hospital covers an area of 26 square kilometers and a building area of 3512 square meters. The hospital has 72 personnel of various professional and technical levels, including 3 with senior titles, 9 with intermediate titles, 2 with master's degrees, 30 with bachelor's degrees, and 20 with associated college degrees. There are altogether 9 community health service stations under the hospital, including Daludian, Xiaoludian, Wanziyang, Dingxinzhuang, Langgezhuang, Langxinzhuang, Shuangshu, Dongxu, and Yijingcheng, undertaking the six-in-one medical and health work for the 80,000 population in the Heizhuanghu area with respect to prevention, health care, medical treatment, rehabilitation, family

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planning, and health education. The hospital has such clinical departments as general practice, Traditional Chinese medicine (TCM), gynecology, stomatology, rehabilitation, and health care. The hospital undertakes the related work in the emergency work station in the southern area according to the "regionalization" and "gridding" goal as set by the Chaoyang District Health Bureau, and it has carried out the Sanofi Pasteur's HAF87 clinical trial study.

(2) Beijing Chaoyang Xiaohongmen Community Health Service Center

Established in 1953, the Beijing Chaoyang Xiaohongmen Community Health Service Center is affiliated to the Health Bureau of Chaoyang District, and has been approved by the Beijing Municipal Human Resources and Social Security Bureau as a designated medical institution for medical insurance in Beijing. Located at No. 1 Xiaohongmen Middle Street, Chaoyang District,

the Xiaohongmen Community Health Service Center covers an area of 12.47 square kilometers and a building area of 3037 square meters. The hospital has 70 personnel of various professional and technical levels, including 2 with senior titles, 8 with intermediate titles, 8 with bachelor's degrees and 24 with associated college degrees. There are altogether 4 community health service stations under the hospital, including Paifang, Xiaohongmen, Xiaocun, and Longzhaoshu, undertaking the six-in-one medical and health work for the 80,000 population in the Xiaohongmen area with respect to prevention, health care, medical treatment, rehabilitation, family planning, and health education. The center has such clinical departments as internal medicine, surgery, TCM, gynecology, stomatology, ophthalmology, and health care. The phase III clinical study on the H1N1 influenza vaccine, the phase III clinical study on the EV71 vaccine, the phase III clinical study on the meningococcal polysaccharide ACYW135 vaccine, and the phase III clinical study on the lyophilized human rabies vaccine (MRC-5 cells) have been carried out here.

(3) Beijing Chaoyang District Nanmofang Community Health Service Center

The Beijing Chaoyang District Nanmofang Community Health Service Center is affiliated to the Chaoyang Health and Family Planning Commission, representing a designated medical institution for medical insurance in Beijing. Located at No. 29 West Dawang Road, Chaoyang District and with a service area of 9.64 square kilometers, it serves a population of 160,000. The center has such departments as general medicine, stomatology, surgery, gynecology, TCM, orthopedic massage, physiotherapy, and preventive health care, and five community health service stations under it: Nanxinyuan, Shuanglongnanli, Huanlegu, Zinanjiayuan, and Baiziwan, with experts on TCM, endocrinology, gynecology, and surgery from the Chuiyangliu medical consortium (tertiary hospital) for specialist clinic service of the center on a regular basis. The center is equipped with such examination and treatment equipment as Color Ultrasound Doppler Apparatus, CR, Automatic Biochemical Analyzer, Fluorescent Immunoassay Analyzer, Lumbar and Cervical Spine Treatment Instrument, Infrared Polarized Light Treatment Instrument, and TCM Meridian Flow Injection Treatment Instrument. With these pieces of equipment, vascular ultrasonography, cardiac ultrasound examinations, testing of tumor markers, Helicobacter pylori testing, high and low frequency and infrared pain treatment and other physical therapy items are available. The center has 134 employees, including 105 professional technicians with associate college degrees or above and 6 with professional titles as associated chief physicians or above, and is open to an average of over 1000 outpatients per day. In areas under its jurisdiction, the health care department serves 4000 children 0–3 of age, and manages over 4000 children from 15 nursery schools, and 8 primary and secondary schools.

3.1.3.2. Hebei Provincial Center for Disease Control and Prevention

Based on the overall deployment of the country and Hebei on health system reforms, the Hebei Provincial CDC is a public institution directly under the provincial Health Department after disbanding and merging of the 6 former units, namely the Hebei Provincial Health and Epidemic Prevention Station, the Provincial Endemic Disease Control Center, the Provincial Center for

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Occupational Disease Control, the Provincial Institute of Radiological Hygiene, the Provincial Center for Tuberculosis Prevention, and the Provincial Medical Science Academy, and was officially established in Aug. 2001. The CDC is mainly responsible for the following 7 tasks: (1) disease prevention and control; (2) emergency handling of public health emergencies; (3) epidemic reporting and management of health related factors information; (4) monitoring and intervention of health hazards; (5) laboratory testing, analysis and evaluation; (6) health education and health promotion; and (7) technical management and guidance on application research. It is a center offering business technical guidance for disease prevention and control within the province.

At present, the center has 40 offices, and is attached to 6 provincial-level associations, such as the Hebei Provincial anti-Tuberculosis Association, the Endemic Disease Association, and those related to TCM, integrated TCM and western medicine, acupuncture, and nursing; the center has a total of 507 employees, including 171 senior professional technical employees, 108 employees with master's degrees and Ph.D degrees, 21 provincially-administered experts, middle-aged and young experts, experts on special government allowances, and 2 national and provincial model workers.

The Hebei Provincial CDC has a tertiary laboratory for biosafety, and over 10 laboratories for poliomyelitis, AIDS confirmation, virological testing, disinfection, and toxicological testing, etc. The CDC has three province-level key medical disciplines, including medical virology, food safety risk monitoring laboratories, and cardiovascular and cerebrovascular disease prevention and treatment. The center boasts a total of 587 large and high-precision instruments and pieces of equipment, including the Ultra-high Performance Liquid Chromatograph-Quadrupole Mass Spectrometer/Mass Spectrometer, Ultra-high Performance Liquid Chromatograph, ICP-MS, DNA Synthesizer, Quantitative PCR Instrument, Liquid-mass Spectrometer, Gas-mass Spectrometer, Atomic Absorption Spectrophotometer, Gas Chromatograph, Ion Chromatograph, Amino Acid Analyzer, and Automatic Microbe Identification/Drug Sensitivity Analysis System. The center has 17 department certifications at the national and provincial levels, which are represented by national laboratory accreditation and metrological certification, and these make it the testing institution with the widest testing items within the province's public health field. There are a total of 737 sanitary inspection and testing items that involve food, health food, cosmetics, water and water-related products, pesticides, and various pathogenic microorganisms, etc., and the inspection and testing results have been recognized by a total of over 50 countries and regions across the globe.

(1) Dingxing County Center for Disease Control and Prevention

Located in the middle of Baoding, Hebei Province, the Dingxing County covers a total area of 714 square kilometers. It has 5 towns, 11 townships, and 274 administrative villages under its jurisdiction, with a total population of 580,000; the number of urban populations reaches 95,927, and the average annual birth rate stands at about 14.8‰. With an area of 8867 square meters and a total construction area of 2822 square meters, the Dingxing County CDC has a three-story laboratory building which was put into use in 2004 with a construction area of 2054 square meters, a 2–8 °C room-temperature freezer with an area of 15.3 square meters, and 20 types of instruments and pieces of equipment. It has 85 employees, including 62 professional technical employees, 3 employees with deputy senior titles, 36 employees with junior titles, and 14 other medical technicians. The CDC has 14 departments, and its laboratory passed the metrological certification review by the Hebei Provincial Bureau of Quality and Technical Supervision in Apr. 2011. There are 26 sites for planned immunization and spot inoculation. The direct reporting network systems at the county, township, and town levels are operating normally, EPI is conducted in the form of centralized vaccination in the vaccination clinics of township health centers, and the center's cold

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chains are operating all year round. Both planned immunization and vaccination rate exceed 97%. A total of 6 employees from the CDC have currently attended the training by NMPA and obtained the GCP qualification certificates, and 29 employees have obtained the GCP qualification certificates issued by the Hebei Food and Drug Administration; they have undertaken and completed the Novartis's M37P2, M37P2E1, V37_06, V37_07, and V37_07E1, the influenza vaccine project by the Yunnan Kunming Institute, and the phase I clinical trial involving mumps, etc., gaining a wealth of experience and a good mass foundation.

(2) By Zhengding County Center for Disease Control and Prevention

Located in the north of Shijiazhuang, Hebei Province, the Zhengding County lies to the urban area by a river, and covers a total area of 468 square kilometers. It has jurisdiction over 1 street, 4 towns, 5 townships, and 174 administrative villages, with a total population of 480,000, of which the urban population is 113,778 and the average annual birth rate stands at about 14.13‰. The Zhengding County CDC is a business unit under the Hebei Provincial CDC. There are a total of 63 employees in the center, including 51 professional technical personnel, 3 with senior titles, 17 with intermediate titles, 44 majoring in health, and 38 with associate college degree. The CDC has 14 departments, is equipped with a central laboratory and testing laboratories which are 1142 square meters in area, and has 19 sites for both planned and spot immunizations. The direct reporting network systems at the county, township, and town levels are operating normally, EPI is conducted in the form of centralized vaccination in the vaccination clinics of township health centers, and the center's cold chains are operating all year round. Both planned immunization and vaccination rate exceed 98%. At present, a total of 13 employees from the CDC have attended the training by the NMPA and obtained the GCP qualification certificates, and 41 employees have obtained the GCP qualification certificates issued by the Hebei Food and Drug Administration. The Zhengding County has been established as a national research base on viral hepatitis since the 1980s, and has been undertaking national scientific and technological research projects or major projects since the beginning of the 7th "Five-Year Plan". So far, the center has completed a total of 7 national scientific and technological research projects or major projects involving hepatitis and another 6 scientific research projects, and has won 4 provincial awards for scientific and technological progresses. The center has undertaken and completed such clinical trials as Novartis's M37P2 and M37P2E1, gaining a wealth of experience and a good mass foundation.

3.1.3.3. Henan Provincial Center for Disease Control and Prevention

The Henan provincial CDC is a public welfare health institution with the qualifications for administering preventive biological products and immunization planning and vaccination, and its administrative and business fall under the jurisdiction of the Henan Provincial Health and Family Planning Commission. The center has 9 business institutes, including the Institute for the Prevention and Control of Infectious Diseases, the Institute for the Prevention and Control of Parasitic Diseases, the Institute for Immunization Prevention and Planning, the Center for Health Detection and Testing, the Institute for STD & AIDS Prevention and Research, the Institute for the Prevention and Control of Endemic Diseases, the Institute for the Prevention and Control of Tuberculosis, the Institute for the Prevention and Control of Health Education and Chronic Non-infectious Diseases, and the Institute for Research on Public Health. There are a total of 439 in-service employees and 375 professional technicians on health, including 27 chief physicians (technicians), 94 deputy chief physicians (technicians or superintendent nurses), 14 with Ph.D degrees, and 94 with master's degrees. There are 5 excellent experts at the provincial level, 1 expert on special government allowances, and over 20 employees assuming positions in the national professional academic groups. More than 30 business technical backbones have been abroad for research and studies since 2006. Specifically, the Vaccine Clinical Research Center is a dedicated department for the evaluation of vaccine safety and efficacy, and with the assistance of other

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departments in the center, it can carry out immunological, etiological, molecular biological, biochemical, and routine tests.

Composition of technical personnel: 3 with senior titles, 1 with associate senior titles, 6 with intermediate titles, and 1 for administrative affairs. Composition of technical personnel by educational background: 4 with master degrees, 4 with bachelor degrees, and 3 with associate college degrees. Composition of technical employees by profession: 4 epidemiologists, 2 clinicians, 4 testers, and 1 for other affairs. A total of 10 employees have obtained the national GCP training certificates. The management personnel and technicians from the Vaccine Clinical Research Center are familiar with the procedures of drug clinical trials, and there is a reasonable composition of management personnel and technicians and clear division of labor, meeting the corresponding job responsibilities. The Vaccine Clinical Research Center formulates and implements the systems for relevant regulations and technical training. The Vaccine Clinical Research Center has established the clinical trial quality control and quality assurance systems and various SOPs which are complete in content and operable and which run throughout the clinical trial process, ensuring the implementation of the quality control and quality assurance systems of clinical trials. Quality control can be carried out during various stages of data processing, to ensure the completeness, accuracy, truthfulness, and reliability of data. For many times, the center has obtained the qualifications for one-time vaccine clinical trial institutions by NMPA.

Experience in clinical trials: In recent years, the center has carried out clinical trials on over 10 vaccines, including the influenza vaccine, the H1N1 influenza vaccine, the rotavirus vaccine, the group A and C meningococcal polysaccharide vaccine, the 10MG recombinant hepatitis B vaccine, the ACYW135 meningococcal conjugate vaccine, the freeze-dried rabies vaccine for human use (VERO cells), the freeze-dried Haemophilus influenza type b conjugate vaccine, the 23-valent pneumonia vaccine, and the HPV cervical cancer vaccine, gaining a wealth of experience in clinical trials of vaccines.

(1) Puyang County Center for Disease Control and Prevention

Located in the northeast of Henan Province, Puyang County falls under the jurisdiction of Puyang City. Covering a total area of 1441.6 square kilometers, the County has jurisdiction over 14 townships, 6 towns, and 1005 administrative villages, with a population of 1.089 million. Specifically, the resident population in the urban area is 150,000. The Puyang County CDC has the qualifications for administering preventive biological products and the immunization planning, and vaccination. Among the 11 internal departments in the center, there are a total of 137 cadres, 104 of which are in service, and 75 professional technicians, 1 employee with senior title, 18 with intermediate titles, and 56 with junior titles. The HPV office is composed of three employees. With Ma Zihé, deputy director, as the director of the Project Office, a total of 14 professional technical personnel were transferred to attend the pre-clinical research project (2 personnel on clinical medicine and 4 laboratory employees). The laboratory passed the Henan metrological certification/metrological certification re-review in Oct. 2002. There are a total of 22 immunization

and vaccination clinics within the county, all of which belong to the provincial-level demonstration vaccination clinics. The vaccination rate exceeds 98%. In 2013, the Puyang County CDC participated in the work on the sites of such clinical trials as the Hualan Bio's phase III clinical study involving 10 µg recombinant hepatitis B vaccine (Hansenula polymorpha) for the maternal-neonatal hepatitis B blockage, and HPV cervical cancer vaccine, gaining work experience in clinical trials.

(2) Wen County Center for Disease Control and Prevention

Located in the northwest of Henan and closed to the Yellow River in the south and to the Qin River in the north, the Wen County is a typical agricultural county and an authentic production place of the "great four Huai medicines", e.g., yam, Rehmannia glutinosa, chrysanthemum, and achyranthes. The Wen county has a long development history and rich cultural origins. It is home to Sima Yi, a

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famous military strategist during the period of the Three Kingdoms, and the birthplace of Taijiquan, which is well-known both domestically and internationally. The whole county covers a total area of 462 square kilometers, and has jurisdiction over 10 townships and 262 administrative villages, with a total population of 460,000 and an annual birth rate of 12.5‰. Specifically, the urban area has a resident population of over 80,000. The county has a total of 16 clinics for immunization and vaccination, of which there are 7 provincial-level demonstration vaccination clinics, with a vaccination rate of more than 98%.

The Wen County CDC has the qualifications for administering preventive biological products and immunization planning, and vaccination. Among the 15 internal departments in the center, there are a total of 84 cadres, 68 of which are in service, and 54 professional technicians, 3 employee with senior title, 17 with intermediate titles, and 34 with junior titles. The HPV office is composed of three employees. A total of 15 professional technical personnel were transferred to attend the pre-clinical research project (3 personnel on clinical medicine and 5 laboratory employees). The center has a laboratory which covers an area of over 1500 square meters. The laboratory has a total of 53 (in 32 categories) instruments or pieces of equipment, including the Gas Chromatograph, High-performance Liquid Chromatograph, and Atomic Absorption Spectrophotometer, with which 393 monitoring items can be carried out, including microbiology, physical and chemical testing, clinical testing, and disease control. The laboratory passed the metrological certification review by the Henan Provincial Bureau of Quality and Technical Supervision in Jul. 2013. In 2012, the Wen County CDC undertook the phase III clinical research project of the III-valent rotavirus gene reassortment vaccine, which is about to complete now. In 2013, it undertook the work on the site of such clinical trials as the Hualan Bio's "Phase III clinical study involving 10 µg recombinant hepatitis B vaccine (Hansenula polymorpha) for the maternal-neonatal hepatitis B blockage", and HPV cervical cancer vaccine, gaining the work experience in carrying out vaccine clinical trials.

3.1.3.4. Shanxi Provincial Center for Disease Control and Prevention

The Shanxi Provincial CDC was established in Jun. 2001 based on the adjustment, optimization, and combination of the former Shanxi Provincial Health and Epidemic Prevention Station, the Shanxi Provincial Institute of Occupational Disease Control, and the Shanxi Provincial Institute for Tuberculosis Control. The center covers a total area of 17.17 Mu and a work housing area of 18,393 square meters, of which the laboratory housing area stands at 7794 square meters. In the laboratory housing, the areas of microbiology laboratory, physical and chemical laboratory, and toxicology laboratory are 5109, 1881, and 804 square meters, respectively. There are a total of 18 business departments, 12 functional departments, and 1 training center in the center. There are 262 formal employees in the center, and currently, there are 233 in-service employees, including 173 professional health technicians (accounting for 74.25% of the in-service employees) and 77 assuming senior professional technical positions (accounting for 44.51% of the in-service employees). Among the health professionals, there is 1 with Ph.D degree (0.57%), 41 with master degrees (23.70%), and 87 with bachelor degrees (50.29%).

The center's Vaccine Clinical Evaluation Room was established on Jun. 10, 2014 through the merging of two related business offices under the Department of Infectious Disease Prevention and Control and the Center for Biological Product Distribution, and is responsible for the organization, management, and implementation of vaccine clinical trials. There are 9 researchers in the Vaccine Clinical Evaluation Room, and by technical title, they are composed of 3 chief physicians, 1 deputy chief physician, 3 head physicians, 1 physician, and 1 chief technician. Composition of technical personnel by educational background: 7 with master degrees, 1 with bachelor degree, and 1 with other educational background. The center is headed by Li Guohua,

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chief of the Department of Infectious Disease Prevention and Control and chief physician. He has certain experience in vaccine clinical trials because he went to Canada for studies in the late 1990s as a visiting scholar, and attended training related to the national GCP and ethics review and obtained the training certificate. The Vaccine Clinical Evaluation Room now has 6 active management systems, 73 standard operating procedures (SOPs), 28 SOPs related to clinical institutions, 45 SOPs related to the trial site, 2 emergency plans, and 2 job responsibilities (1 on the institution, and 1 on the trial site). A total of 8 employees have received the national GCP trainings, and the institution has carried out such regulatory and technical trainings for internal GCP, SOPs, and clinical trial protocols according to the conduct of clinical trials.

The Vaccine Clinical Evaluation Room has a fixed office venue and necessary facilities and equipment for clinical trials, e.g., computers, printers, telephones, fax machines, scanners, and paper shredders. In addition, there is a relatively independent archive room which is equipped with fire-proof, anti-theft, moisture-proof, insect-proof, and rodent-proof facilities, and dedicated personnel are in place for archiving and management of the archives.

(1) Xiangfen County Center for Disease Control and Prevention

Located in the south of Shanxi Province, the Xiangfen County covers a total area of 1034 square kilometers. The county has jurisdiction over 7 towns and 6 townships (mainly Han nationality), with a population of over 500,000 and an average annual birth rate of about 11.23‰. The Xiangfen County CDC has 13 departments which are composed of 38 professional technicians, including 1 with senior professional title, 19 with intermediate professional titles, and 18 medical technicians. There are 8 employees in the Department of Epidemiology Prevention and Treatment and the Department of Planned Immunology Department, and 10 employees in the laboratory. The center's laboratory passed the metrological certification review by the Provincial Bureau of Quality and Technical Supervision in Jul. 2006, and passed the metrological certification re-review by the Bureau in Oct. 2009 and Apr. 2013. There are 346 sites for planned immunization and spot inoculation. The direct reporting network systems at the county, township, and town levels are operating normally, and EPI is carried out in the form of spot vaccination, with the center operating

in the cold chain mode once a month. Both planned immunization and vaccination rate exceed 95%.

(2) Wanrong County Center for Disease Control and Prevention

Located in the southwest of Shanxi Province and at the junction of the Yellow River and the Fenhe River, the Wanrong County covers a total area of 1082 square kilometers and a cultivated area of 1.02 million Mu. It administers 14 townships and towns and 281 administrative villages, with a total population of 440,000. The Wanrong County CDC has 35 formal employees, including 10 with bachelor degrees, 14 with college degrees, and 11 with other academic degrees. The center has over 10 departments, including Central Office, Finance, Business/Food, Information/Biological Products, Immunization Program/Infectious Diseases, Tuberculosis, STD/AIDS, Chronic Diseases/Disinfection and Sterilization, Vaccination Clinic, Clinical Laboratory, and Physical Examination. The county administers 14 townships and towns and 281 administrative villages. All the 14 townships and towns have established the department of prevention and protection. The chief of the department is administered by both the township/town health center (for administrative assessment) and the county CDC (for business assessment). The 281 administrative villages have established a total of 282 village health centers; all of the departments of the prevention and protection of health hospitals in 14 townships and towns and

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282 village health centers undertake vaccination, and all of them have the qualifications as vaccination clinics granted by the county health bureau. From Jul. 2013 to Feb. 2014, the center, in collaboration with the Beijing Chaoyang CDC and the Shanxi Provincial CDC, carried out the clinical trial site work on the "Research Project on Protective Effects of the 2-injection Varicella Vaccine Immunization Procedures". From Dec. 2014 to Mar. 2015, the center carried out the "Phase III Clinical Study on Lyophilized Rabies Vaccine for Human Use (MRC-5 cells)" as a trial site.

3.2 SPONSOR

Yunnan Walvax Biotechnology Co., Ltd. and Yuxi Walvax Biotechnology Co., Ltd., which were mainly responsible for:

- 1■ Providing the preliminary clinical trial protocol and finally confirming the clinical trial protocol;
- 2■ Providing such application documents on the site as the Clinical Trial Approval, the Investigator's Brochure for the clinical trial (pre-clinical product safety information), and the product implementation standards;
- 3■ Providing vaccines used in the clinical study and issuing test control reports;
- 4■ Dispatching monitors to assess and recognize the clinical trial site, perform the monitoring duty as per the GCP requirements, and verify the research data during the trial;
- 5■ Participating in the investigation and handling of cases of abnormal reactions arising from the vaccine, and providing medical and related compensation fees for cases of abnormal reactions which are clinically proven related to the vaccine as per relevant regulations;
- 6■ Providing the fund for the clinical study.

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3.3. INVESTIGATORS

Name

Nianmin Shi Beijing Chaoyang District Center for Disease Control and Prevention

Yuliang

Primary Hebei Provincial Center for Disease Control and Prevention

Zhao

Investigator

Shengli Xia Henan Provincial Center for Disease Control and Prevention

Guohua Li Shanxi Provincial Center for Disease Control and Prevention

Clinical Research

Meng Wang Simoon Record Pharm Information Consulting., Ltd

Associate

Medical

Zhiwei Jiang Beijing Key Tech Statistical Consulting Co., Ltd

statistician

The Guangxi Zhuang Autonomous Region Center for Disease Control

CSR Writing Yanping Li

and Preventio

CSR Review Nianming Shi, Yuliang Zhao, Shengli Xia, Guohua Li

3.4. ETHICS

This study complied with the "Technical Guidelines for Vaccine Clinical Trials" by NMPA and the ethical principles of medical study in humans set forth in the World Medical Association's "Helsinki Declaration", "Good Clinical Practice" (GCP), and all applicable regulatory requirements. Prior to this study, this clinical research protocol and the informed consent form had been approved by the Ethics Review Committees of the Beijing Chaoyang CDC, the Hebei Provincial CDC, the Henan Provincial CDC, and the Shanxi Provincial CDC. The subjects for the clinical trial were recruited in accordance with the principle of informed consent and voluntariness. All subjects/legal guardians were informed of matters related to the vaccine clinical trial in detail in order to obtain their voluntary signed informed consent forms. The subjects were screened in strict accordance with the inclusion criteria.

3.5. DATA MANAGEMENT

The study records and reports were managed in accordance with Articles 48–52 of the GCP issued by NMPA. All operations were performed as per the SOPs, and the clinical trial records and report data will be stored in the on-site CDC as required until five years after the clinical trial.

3.6. OBSERVATION AND REPORTING OF ADVERSE EVENTS

In strict accordance with the GCP principles, each subject was observed closely for local and systemic adverse reactions and adverse events. All adverse reactions and adverse events were recorded in the CRF fully, including the symptoms, time of occurrence, severity, duration, measures adopted and the status of recovery; the investigator was responsible for judging the relevance of the adverse events to the test vaccine.

If serious adverse events occurred during the study, appropriate treatment measures were taken immediately, whether they were associated with the test vaccine, and the events were reported to

the sponsor, the ethics committee, and the local food and drug administration for the responsible institutions of each sub-center by telephone or fax within 24 hours after learning about such information.

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4. STUDY DESIGN AND PROCEDURE

This clinical trial was designed as a randomized, blinded, comparable vaccine-controlled, multicenter, non-inferiority clinical trial.

The subjects were healthy children aged 2-71 months (minimum of 6 weeks), stratified into four age groups: 2-6 months (minimum of 6 weeks), 7-11 months, 12-23 months and 24-71 months. Subjects were divided into PCV13-TTs and PCV7 group, with the PCV13-TT receiving the 13-valent pneumococcal conjugate vaccine (PCV-13) from Yuxi Walvax Biotechnology Co., Ltd. and the control group receiving the Prevenar7 (PCV-7) from Pfizer (Wyeth), Inc.

Accordance to the 'Technical Guidelines for Clinical Trials of Vaccine' and 'Regulation of Drug Registration', the sample size of phase III clinical trial should not less than 500, a total of 2,760 subjects were recruited in this trial. The sponsor entrusted the Beijing Chaoyang District CDC, Hebei Province CDC, Henan Province CDC and Shanxi Province CDC to complete this clinical study. The enrolled subjects were completed by the business cooperation units or subordinate units of Zhengding County CDC, Dingxing County CDC, Puyang County CDC and Wenxian County CDC, Xiangfen County CDC and Wanrong County CDC. The Xiaohongmen Community Health Service Centre, the Heizhuang Family Community Health Service Centre of Shuangqiao Hospital, and the Nanmoufang Community Health Service Centre were used as alternate sites for the clinical trial. There was no alternative plan for the subjects who were dropped out during the study.

The study consisted of 4 subgroups enrolling total of 2760 healthy subjects of 2-71months old.

Subgroup one inclusion of 1560 subjects aged 2-6 months (minimum 6 weeks of age), divided into PCV13-TT 1 (520 subjects, aged 3 months), PCV13-TT 2 (520 subjects, aged 2 months) and control group (520 subjects, aged 3 months).The proposed regimen for PCV13-TT 1 consists of four doses of PCV-13, with a single intramuscular injection at months 0, 1 and 2 (3, 4 and 5 months of age) for primary series and at ages12-15months for booster dose. The proposed regimen for PCV13-TT 2 consists of four doses of PCV-13, with a single intramuscular injection at months 0, 2 and 4 (2, 4 and 6 months of age) for primary series and at ages12-15months for booster dose. The proposed regimen for control group consists of four doses of PCV-7, with a single intramuscular injection at months 0, 1 and 2 (3, 4 and 5 months of age) for primary series and at ages12-15months for booster dose.

Subgroup two inclusion of 400 subjects aged 7-11 months, divided into PCV13-TT (200 subjects) and control group (200 subjects). The proposed regimen for PCV13-TT and control group consists of three doses of PCV-13 and PCV-7 separately, with a single intramuscular injection at months 0 and 2 for primary series and at ages12-15months for booster dose. The vaccination interval between the 2nd dose for primary series and booster dose was at least 2 months.

Subgroup three inclusion of 400 subjects aged 12-23 months, divided into PCV13-TT (200 subjects) and control group (200 subjects). The proposed regimen for PCV13-TT and control group consists of two doses of PCV-13 and PCV-7 separately, with a single intramuscular injection at months 0 and 2.

Subgroup four inclusion of 400 subjects aged 24-71 months, divided into PCV13-TT (200 subjects) and control group (200 subjects).

The proposed regimen for PCV13-TT and control group inoculated with PCV-13 and PCV-7 at 0 months. separately.

Safety observations were performed after each dose of vaccination. The immunogenicity was evaluated before the first dose of vaccination and 30 days after primary series. Subjects with booster dose were also required to observe immunogenicity before and 30 days after booster dose. After the on-site blood sampling, the data collation, verification and clarification should be complete. The results of the blood tests were returned for statistical analysis, and a statistical analysis report was issued to write the clinical trial report.

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The details of the Phase III clinical trial plan of 13-valent pneumococcal conjugate vaccine were as follows:

Clinical trial schedule
Time 30 days
Aged
Age/Subgroup 0 1 2 3 4 5 after
12-15
month month month month month booster
months
dose
PCV13-TT ■
■ 2nd 3rd
group ■ Booster ■
1st dose dose dose
Aged 2-6 (3 months) dose
months PCV13-TT ■
■ 2nd 3rd
(Minimum group st ■ Booster ■
1 dose dose dose
age of 6 (2 months) dose
weeks) ■
PCV7 group ■ 2nd 3rd
■ Booster ■
(3 months) 1st dose dose dose
dose
■
PCV13-TT ■ 2nd
■ Booster ■
group 1st dose dose
Aged 7-11 dose
months ■

■ 2nd
PCV7 group ■ Booster ■
1st dose dose
dose
PCV13-TT
■ 2nd
group ■
Aged 12- 1st dose dose
23 months
PCV7 group ■ 2nd
■
1st dose dose
PCV13-TT
■
group ■
Aged 24-71 1st dose
months
PCV7 group ■
■
1st dose
■: Blood sampling

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4.2. SAMPLE SIZE ESTIMATION

4.2.1. Sample Size Consideration

4.2.1.1. Consideration based on 7 common serotypes

(1) It was assumed that for each vaccine serotype, the proportion of subjects achieving serum IgG antibody concentrations $\geq 0.35 \mu\text{g/mL}$ was not less than 85% in both PCV13-TT and control group. Set: $\alpha/2 = 0.0125$ (One side),

$\delta = 0.1$ (Threshold of Non-inferiority),

$1-\beta = 0.80$ (Power of test)■

To ensure that the total power of test is not less than 80%, the adjusted value is $1-0.2/7=0.97$.

Sample distribution ratio was 1:1 for the PCV13-TT and control group. The PASS11 was used to calculate the minimum number of subjects to be enrolled in each of the test and PCV7 group was 434 separately.

(2) It was assumed that after the GMC ratio logarithmically transformed (with 10 as the base), the covariance was not greater than 1.0 for each vaccine serotype,

Set: $\alpha/2 = 0.0125$ (One side)■

$\delta = 0.3$ (Threshold of Non-inferiority)

$1-\beta = 0.80$ (Power of test)

To ensure that the total power of test is not less than 80%, the adjusted value is $1-0.2/7=0.97$.

Sample distribution ratio was 1:1 for the PCV13-TT and control group. The PASS11 was used to calculate the minimum number of subjects to be enrolled in each of the test and PCV7 group was 379 separately.

4.2.1.2. Consideration based on the 6 additional serotypes after the immunogenicity non-inferiority test of 7 common serotypes confirmed

It was assumed that each vaccine serotype, the proportion of subjects achieving serum IgG antibody concentrations $\geq 0.35 \mu\text{g/mL}$ was not less than 80%. For the 6 additional vaccine serotype, the proportion of subjects achieving serum IgG antibody concentrations $\geq 0.35 \mu\text{g/mL}$ was lower than 50%

Set: $\alpha = 0.025$ (One side)

It was assumed that each target vaccine serotype, the proportion of subjects achieving serum IgG antibody concentrations $\geq 0.35 \mu\text{g/mL}$ was not less than 70%. The Lower limit of the one-sided 97.5% CI for the IgG seroresponse rates $\geq 0.35 \mu\text{g/mL}$ was $\geq 70\%$.

$1-\beta = 0.80$ (Power of test)

To ensure that the total power of test is not less than 80%, the adjusted value is $1-0.2/6=0.97$.

Sample distribution ratio was 1:1 for the PCV13-TT and control group. The PASS11 was used to calculate the minimum number of subjects to be enrolled in each of the test and PCV7 group was 274 separately.

Considering the cross-protective effects of 19F and 6B in 7 common serotypes on 19A and 6A in 6 additional serotypes, respectively, the effectiveness evaluation of 19A and 6A needs to be combined with the results of OPA.

4.2.1.3. Final consideration of sample size

According to the above situation, the PCV13-TT and the control group should be enrolled 434 subjects at least,

Considering the drop-out during the study, 520 subjects were enrolled in the main subgroup which aged 2-6 months in both PCV13-TT and control group separately.

At the same time, another three subgroups including 7-11 months old, 12-23 months old and 24-5 years old were investigated. Sample distribution ratio was 1:1 for the PCV13-TT and control group

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in these three subgroups. 200 subjects were enrolled in each age group for both PCV13-TT and control group.

Sample size allocation and immunogenicity test subgroup

Table 2. Sample size allocation
Age (Month) PCV13-TT group PCV13-TT group
PCV7 group Total
(3-71M) (2M)
2-6
520 520 520 1560
(Minimum age of 6 weeks)
7-11 200 / 200 400
12-23 200 / 200 400
24-71 200 / 200 400

4.2.2. Criterion for immunogenicity hypothesis testing

Common 7 serotypes: non-inferiority test

The evaluation of non-inferiority criterion for comparison of common 7 serotypes in PCV13-TT group and PCV7 group includes:

- (1) The lower limit of the 2-sided 97.5% CI for the rate difference of percentage of subjects with IgG antibody concentration ≥ 0.35 $\mu\text{g/mL}$ (PCV13-TT – PCV7) was greater than -10%
- (2) The lower limit of the 2-sided 97.5% CI for IgG antibody GMC ratio (GMR, PCV13-TT relative to PCV7) is greater than 0.5.

Any common serotype that meets the evaluation criteria 1) or 2) will be judged to be immunogenically non-inferior.

Additional 6 serotypes: superiority test

The evaluation of superiority criterion for comparison of additional 6 serotypes in PCV13-TT group and PCV7 group includes:

- (1) The lower limit of the 2-sided 95% CI for the rate difference of percentage of subjects with IgG antibody concentration ≥ 0.35 $\mu\text{g/mL}$ (PCV13-TT – PCV7 seroresponse or PCV7 reference) was greater than 0.
- (2) The lower limit of the 2-sided 95% CI for IgG antibody GMC ratio (GMR, PCV13-TT relative to PCV7 seroresponse or PCV7 reference) is greater than 1.
- (3) The lower limit of the 2-sided 95% CI for percentage of subjects with IgG antibody concentration ≥ 0.35 $\mu\text{g/mL}$ was greater than 70% in PCV13-TT group.

Any additional serotype, which simultaneously meets the evaluation criterion (1), (2) and (3), is judged to be superior.

PCV7 reference is referred to as the lowest response among the 7 common serotypes contained in PCV7 achieved by subjects who received PCV7.

The evaluation of non-inferiority criterion for secondary endpoints, including percentage of subjects with IgG antibody concentration ≥ 1.0 $\mu\text{g/mL}$, OPA titer $\geq 1:8$, GMT and GMT ratio were same as the above described evaluation criterion.

Sequential tests

In the hypothesis test, if the non-inferiority of PCV13-TT (3 months) group vs PCV7 group was established and additional 6 serotypes met the pre-specified superiority criterion, the non-inferiority test of PCV13-TT group (2 months) vs PCV7 group was performed. Then, if all the serotypes met the pre-specified evaluation criterion in PCV13-TT (2 months) group, the non-inferiority test of PCV13-TT group (2 months) vs PCV13-TT group (3 months) was performed.

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Strict statistical analysis was performed for the comparison of PCV13-TT (3 months) vs PCV7 (3 months), PCV13-TT (2 months) vs PCV7 (3 months), and PCV13-TT (2 months) vs PCV13-TT (3 months) in a sequential fashion, while other comparisons for non-primary populations (including age groups of 7-11, 12-23, 24-71 months) were statistically analyzed while without strict statistical inference.

The immunogenicity analysis would be performed in all eligible subjects with evaluable serum for testing as per the study protocol. Immunogenicity analysis was performed with the PPS data set as the primary analysis set.

4.3. STUDY ENDPOINTS

4.3.1. Immunogenicity Endpoints

4.3.1.1. Primary immunogenicity endpoints

- (1) Percentage of subjects with pneumococcal serotype-specific IgG antibody concentration ≥ 0.35 $\mu\text{g/mL}$ 30 days post-dose 3 in PCV13-TT group and PCV7 group
- (2) Pneumococcal serotype-specific IgG geometric mean concentration (GMC) 30 days post-dose 3 in PCV13-TT group and PCV7 group

4.3.1.2. Secondary immunogenicity endpoints

- (1) Percentage of subjects with pneumococcal serotype-specific OPA antibody titer $\geq 1:8$ one month post-dose 3 in PCV13-TT group and PCV7 group
- (2) Percentage of subjects with pneumococcal serotype-specific OPA GMT one month post-dose 3 in PCV13-TT group and PCV7 group
- (3) Percentage of subjects with pneumococcal serotype-specific IgG antibody concentration ≥ 1.0 $\mu\text{g/mL}$ 30 days post-dose 3 in PCV13-TT group and PCV7 group
- (4) Percentage of subjects with pneumococcal serotype-specific IgG antibody concentration ≥ 0.35 $\mu\text{g/mL}$ 30 days post-dose 4 in PCV13-TT group and PCV7 group
- (5) Pneumococcal serotype-specific IgG GMC 30 days post-dose 4 in PCV13-TT group and PCV7 group
- (6) Percentage of subjects with pneumococcal serotype-specific OPA antibody titer $\geq 1:8$ one month post-dose 4 in PCV13-TT group and PCV7 group
- (7) Percentage of subjects with pneumococcal serotype-specific OPA GMT one month post-dose 4 in PCV13-TT group and PCV7 group
- (8) Percentage of subjects with pneumococcal serotype-specific IgG antibody concentration ≥ 1.0 $\mu\text{g/mL}$ 30 days post-dose 4 in PCV13-TT group and PCV7 group.

4.3.2. Endpoint of Safety

The incidence of local and systemic adverse reactions 30 days after primary series and booster days was chosen as the primary endpoints for safety evaluation.

If there is an unexpected serious abnormal reaction during the study, or after any dose vaccination, if the proportion of subjects who had level 3 and above local or systemic reactions achieve 15% of the total subjects on the same day, the trial should be suspended. The sponsor should report to the Ethics Committee and National Medical Products Administration and other related NRA where necessary. Experts meeting would be held to demonstrate vaccine safety. Perform unblinding procedure to determine whether to continue the study where necessary.

4.4. RANDOM AND CODED BLINDING

In this clinical trial, randomized blindness was through the method of vaccine blinding.

4.5. PACKAGING AND LABEL

Packaging: Uniform outer packaging will be used for packaging the vaccines used in the clinical trial so that their appearance looks the same.

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Label on the vaccine vial: The label shall contain the number and name abbreviation. The pattern is as follows:

Dose 1

Phase III Clinical Study of PCV13-TT Dose 1

Dose 1

Vaccine No.:

Initials:

Walvax For Clinical Use Only Store at 2-8°C, protected from light

■■■■■■■■■■

The "number" was the study number from the random allocation table. Besides the labels of the same number attached to the outer packaging and inner packaging on the test vaccine, a pre-printed sticker label with the same number was placed on the original records book of subjects.

Vaccine's medium box label: The label shall contain such basic information as the number segment of the vaccine, name of the sponsor, and storage conditions, and specify "for use in clinical trials only". The pattern is as follows:

Phase III Clinical Study of PCV13-TT

Dose 1

Number of vaccines in this box (xxxx-1—xxxx-1)

Walvax For Clinical Use Only Store at 2-8°C, protected from light

Blood sample label: The blood sample label was the study number plus the suffix of the number of blood collections, and blinded submission for testing was performed:

Study No. -1 denotes the blood collection before immunization.

Study No. -2 denotes the second blood collection.

Study No. -3 denotes the third blood collection.

Study No. -4 denotes the forth blood collection.

For a study number requiring IgG testing only, three copies of the label were prepared, with the larger label attached to the blood collection tube, and two smaller labels attached to the separated serum 1 pending testing/standby serum, respectively, and priority was given to serum 1 pending testing. For a study number requiring IgG and OPA testing, four copies of the label were prepared, with the larger label attached to the blood collection tube, and three smaller labels attached to the separated serum 1 pending testing/serum 2 pending testing/standby serum; the serums were collected in the order of serum 1 pending testing, serum 2 pending testing, and the standby serum.

The style of the serum label was as follows:

Study No.: 0001-1

Initials:

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4.6. CONTROL VACCINE

In this clinical study, 7-valent pneumococcal conjugate vaccine (PCV7) manufactured by Pfizer (Wyeth) was used as the control vaccine. The route of administration, dosage and precautions were carried out according to the instructions of the vaccine.

4.7. STUDY PERIOD

The study period was 26 months.

The first subject was enrolled in April 2016.

Site closing time: March 2018

Clinical study report issued in January 2018

4.8 PROPOSED INDICATIONS AND RATIONALES

Indications: for the prevention of related diseases caused by infections of the 13 pneumococcal serotypes contained in this vaccine (types 1, 3, 4, 5, 7A, 6B, 7F, 9V, 14, 18C, 19A, 19F, and 23F). Justification: "Pneumococcal polysaccharide vaccine" in the European Pharmacopoeia (version 5.0), and the instructions of use for similar vaccines that have been marketed abroad.

4.9. SUBJECT CRITERIA

4.9.1. Populations

Subjects were healthy children aged 2-71 months (minimum 6 weeks of age). No contraindications to vaccination or history of pneumonia vaccination, no gender or ethnicity stipulations. On the principle of voluntary participation, the informed consent form (signed by the recipient and guardian for the recipients aged 12-17 years) should be signed by themselves or their guardians.

4.9.2. Inclusion Criteria

- Healthy infants (aged 2-71 months) as evidenced by medical history, physical examination, and investigators' judgement
- Subjects are in compliance with age requirements for this clinical trial (children from 6 weeks to 5 years old or adults over 18 years old)
- Parents / legal guardians of subjects must sign and date the informed consent in person
- Parents / legal guardians of subjects have been informed of all information related to the trial
- Subjects / legal guardians of subjects are able to follow requirements of the trial
- Subjects who have not been vaccinated with pneumococcal vaccines, and have not been administered any other preventive products within 7 days
- Axillary temperature $\leq 37^{\circ}\text{C}$

4.9.3. Exclusion Criteria

4.9.3.1. Exclusion Criteria for First Vaccination

- Previously vaccination listed or vaccinated with trial pneumococcal vaccine
- History of invasive disease caused by *Streptococcus pneumoniae*, which was confirmed by bacterial culture
- Prior history of allergic reactions to any drug, vaccine, or vaccine-related serotype
- Vaccination-related fever of 39°C or more has occurred with biological products used for vaccination
- Birth weight < 2.5 kg
- There is a history or family history of allergies, convulsions, seizures, encephalopathy, and psychosis
- Abnormal births of infants (difficult labors, aided by instruments) or a history of asphyxia, neurological organ damage
- History of diagnosed thrombocytopenia or other coagulation disorders
- Infants with diagnosed pathological jaundice

- Known or suspected to suffer from diseases, including: respiratory system disease, acute infection or chronic disease, serious cardiovascular disease, kidney disease, malignant tumor, skin diseases, mothers with HIV infection (as evidenced by test report)
- Known serious congenital malformations or severe chronic disease; congenital deformities, developmental disorder, or a clinical diagnosis of severe chronic diseases (such as Down's syndrome, diabetes, sickle cell anemia, or neurological disorders, Guillain-Barre Syndrome)
- Known or suspected co-morbidities including: respiratory disorders, acute infections or active attack of chronic disease, serious cardiovascular disorders, liver and kidney disorders, malignant neoplasms, skin diseases.
- Received blood products or globulin. Hepatitis B hyper-immune globulin can be used
- Participating in other clinical trials
- Any condition that the investigator considered to have the potential to affect the evaluation of the study

4.9.3.2. Exclusion Criteria for 2nd, 3rd Vaccination or Booster Dose

- Subjects who had severe allergic reactions during the first or second vaccination
- Subjects who experienced serious adverse effects from the previous vaccination with causal relationship demonstrated
- After the first vaccination, the newly discovered or newly occurred conditions not conforming to the first dose of the inclusion criteria or to the exclusion criteria for the first dose
- Other exclusion criteria affirmed by investigators

4.9.4. Criteria for Termination of the Study

4.9.5. Criteria for suspension or termination of the clinical trial

The clinical trial shall be suspended or terminated by the sponsor and the investigator jointly under any of the following circumstances:

- Occurrence of subject deaths related to vaccination;
- Occurrence of collective serious adverse reactions;
- Suspension or termination of the clinical trial as required by the sponsor due to such reasons as the test vaccine; and
- Suspension or termination as required by the NMPA.

4.10. PCV13-TT

A total of 2760 subjects were enrolled in the phase III clinical trial of PCV13-TT, including 1560 subjects in the age group of 2-6 months (the minimum age of 6 weeks), 1040 subjects in the PCV13-TT group and 520 subjects in the PCV7 group, 400 subjects in the 7-11 month age group, 12-23 months age group and 24-71 months age group. There were 200 cases in the PCV13-TT group and the PCV7 group respectively.

Summary of PCV13-TTs, Number of Subjects and Immunization Schedule

Primary Immunizations

Groups Booster Dose Schedule

Number of Schedule

Age strata (age at

Subjects Immunization Immunization

enrollment) Doses Dose

Schedule Schedule

PCV13-TT

group (3 520 Month 0, 1, 2 3 Doses

2-6 months months)

(minimum PCV13-TT 1

12-15 months

age: 6 group (2 520 Month 0, 2, 4 3 Doses Dose

weeks) months)

PCV7 group

520 Month 0, 1, 2 3 Doses

(3 months)

PCV13-TT

12-15 months (at

group (7-11 200 Month 0, 2 2 Doses

least 2 months 1

7-11 months months)

apart from 2nd Dose

PCV7 group

200 Month 0, 2 2 Doses dose)

(7-11 months)

PCV13-TT

group (12-23 200 Month 0, 2 2 Doses / /

12-23 months)

months PCV7 group

(12-23 200 Month 0, 2 2 Doses / /

months)

PCV13-TT

group (24-71 200 Month 0 1 Dose / /

24-71 months)

months PCV7 group

(24-71 200 Month 0 1 Dose / /

months)

4.11. GENERAL INFORMATION FOR VACCINE

4.11.1. Study Vaccine

4.11.2. Control Vaccine

4.11.3. Storage and Transportation of the Vaccine

The vaccine shall be stored and transported at 2–8 °C, away from light. The storage temperature needed to be monitored daily and recorded. If the storage and transportation conditions exceeded the specified ranges, the investigator on the site shall contact the staff of the unit responsible for the clinical study and the sponsor immediately, and the vaccine shall not be used until obtaining the permission of the sponsor.

4.11.4. Route of administration and dose

The lateral skin of the deltoid muscle of the upper arm or the anterolateral skin of the thigh was sterilized with 75% medical alcohol, the vaccine was shaken thoroughly, and 0.5 mL of vaccine was intramuscularly injected after the skin had become slightly dry; for infants under 12 months of age, the preferred vaccination site is the anterolateral thigh. The control vaccine was administered in the same way.

4.11.5 Concomitant medication

Allowed medication: If adverse events occurred in a subject during the trial, the subject shall be allowed to have access to necessary medications and medical treatment.

Allowed vaccines: The interval between other vaccines and the test vaccine shall not be less than 10 days, and that between the live attenuated vaccine and the test vaccine shall not be less than 14 days; emergency vaccination can be conducted at any time according to the need of vaccination (e.g., rabies or tetanus vaccine), and as for this, each sub-center shall implement as per the immunization program policies in local province. If the concomitant vaccine inoculation led to the violation of the interval between the vaccination and the test vaccine inoculation against the requirements of this protocol, such a circumstance was handled as violation of the protocol and there was no need to terminate the trial. The investigator shall record any information on concomitant medication, including the drug name, dosage form, dosage and administration, and time of administration.

Medication records: In order to learn about the impact of drug use on the safety and immunogenicity of the vaccine during the trial or to collect adverse events that may be related to vaccination without omission, the investigator needed to collect the medications of subjects during the observation period by taking measures, and guide the subjects or their guardians in recording their visits and medication in the diary cards as much as possible. The following drugs shall be transcribed into the case report form (CRF):

- (1) Oral or injectable hormones/steroids (e.g., dexamethasone);
- (2) Anti-allergic drugs (e.g., chlorpheniramine);
- (3) Antipyretic/analgesic/non-steroidal anti-inflammatory drugs (e.g., aspirin, ibuprofen, fenbid);
- (4) Any vaccines or biological products;
- (5) Immunosuppressants;
- (6) Antibiotics
- (7) Anti-viral drugs;
- (8) Other drugs

4.12 TRIAL PROCEDURES

4.12.1 Recruitment of subjects

After the research site had been determined and approval had been secured from the ethics

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committee, the recruitment notices were issued to those meeting the requirements on age, vaccination history, and health status as per the protocol.

4.12.2 Informed consent

Prior to the enrollment, the investigator informed fully the subject or his guardian of information related to the clinical protocol, and the subject, his guardian, and the research doctor signed the informed consent form jointly in duplicate, with the subject keeping the copy.

4.12.3 Screening via physical examinations

The investigator carried out physical examinations on all subjects, including the height, body weight, body temperature, skin examinations, and routine blood test and indicators related to liver and kidney functions. The investigator carried out inquiries as per the "inclusion and exclusion criteria", observed the general state of subjects, examined the throat, heart, and lungs, and measured the blood pressure of subjects at 18 years old and above, thereby excluding subjects with a serious medical history, e.g., serious diseases, infectious diseases, and organ removal. The subjects were assigned to groups in accordance with the inclusion and exclusion criteria. The basic information of all the subjects was filled in the "Physical Examination Form". The "Original Record Book" was filled in.

4.12.4 Assignment of study numbers

After the subjects had provided their informed consent, and underwent the inclusion and exclusion criteria and physical examinations, those who met the enrollment criteria were assigned unique random numbers (vaccine number/study number) each in the order of their enrollment, and the "Original Record Book" was filled in. The name abbreviations of the subjects were filled in the "Random Assignment Form", and the signature of the investigator and date were filled in, too.

4.12.5 Blood collection before immunization

Subjects requiring IgG and OPA testing simultaneously prior to immunization were subjected to the collection of whole blood not less than 5.0 mL in volume by means of venipuncture before the first vaccination, and the other subjects were subjected to collection of whole blood not less than 3.0 mL in volume by venipuncture. Failure of the volume of blood collection to reach that required by the protocol while which did not affect the testing shall not be regarded as a violation of the protocol. The serum was separated and stored at –20 °C or below.

4.12.6 Inoculation of the vaccine

The vaccines with the corresponding numbers were obtained according to the assigned random numbers. After the numbers on the original record book had been verified, the name abbreviation of the subject was filled in on the outer packaging label of the vaccine, the inner label of the vaccine was filled in and pasted in the designated place of the original record book, and the subject information was verified again before the vaccination. A total of 0.5 mL of vaccine was injected into the deltoid muscle in the lateral aspect of the upper arm or the muscle in the anterolateral aspect of the thigh after full shaking of the vaccine; for infants under 12 months of age, the preferred vaccination site was the anterolateral thigh.

4.12.7 Medical observation

After each dose of vaccination, a 30-min observation was required on the trial site, the "Diary Card", the thermometer, and the scale were issued, and the legal guardians of the subjects or the clients were trained to use the thermometer, observe the adverse events, and fill out the "Diary Card", and an appointment was made for the time to return the "Diary Card" (the "Diary Card" was returned at day 8 after the vaccination, and the "Contact Card" was received in the meantime).

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4.12.8 Inoculation of second and third doses of vaccines and the enhanced vaccine

The subjects were given the second and third doses of vaccines and/or the enhanced vaccines as per the corresponding immunization procedures in the protocol, with a time range of +15 days. The contact card was recovered and reviewed before each dose. The contact card for the last dose of vaccination was recovered and reviewed during the blood collection 30 days after the full immunization.

4.12.9 Blood collection after immunization

Subjects requiring IgG and OPA testing simultaneously after the immunization were subjected to whole blood not less than 5.0 mL in volume via venipuncture 30 days after the basic immunization procedure, before the enhanced immunization (for subjects requiring enhanced immunization) and 30 days after the enhanced immunization (for subjects requiring enhanced immunization); the remaining subjects were subjected to whole blood not less than 3.0 mL in volume via venipuncture 30 days after the basic immunization procedure, before the enhanced immunization (for subjects requiring enhanced immunization) and 30 days after the enhanced immunization (for subjects requiring enhanced immunization). Failure of the volume of blood collection to reach that required by the protocol while which did not affect the testing shall not be regarded as a violation of the protocol. The serum was separated and stored at -20 °C or below. The time range for blood collection was +10 days.

4.12.10 Termination of the trial

The serum was sent to a designated laboratory for testing, the review for the entry of EDC data was completed, and after review of all materials and data clarification, the report on site closure was issued.

Schedule of Activities of Phase III Clinical Trial

2-6 months (minimum age: 6 weeks)

7-11 12-23 24-71

Activities PCV13-TT PCV13-TT PCV7

M M M

(3 M) (2 M) (3 M)

Informed consent ●●●●●●●●

Screening and physical examination ●●●●●●●●

Randomization ●●●●●●●●

Blood collection before first dose ●●●●●●●●

First dose ●●●●●●●●

Second dose ●●●●●●●● /

Third dose ●●●●●● / / /

Blood collection on Day 30 post complete

●●●●●●●●

series / primary immunizations

SAEs surveillance for 6 months post complete

●●●●●●●●

series / primary immunizations

Blood collection pre-booster dose ●●●●●●

Booster dose ●●●●●● / /

Blood collection on Day 30 post-booster dose ●●●●●● / /

30 min medical observation (post each dose) ●●●●●●●●

Systemic safety observation for Day 0-7 (post

●●●●●●●●

each dose)

Safety observation for Day 8-30 (post each

●●●●●●●●

dose)

SAEs surveillance for 6 months post booster

●●●●●●●●

dose (post last dose)

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4.13. SAFETY FOLLOW-UP

4.13.1 Definition of adverse reaction / time

Adverse events (AE): An unintended medical event observed in the subjects in a clinical trial that is not necessarily causally related to the vaccine or the administration process of the study vaccine. Adverse reaction: An unintended or damaging reaction during the process of vaccination according to the prescribed dosages and schedules, usually associated with vaccination.

Serious adverse reaction (SAE): Events such as hospitalization, prolonged hospitalization, disability, impairment of working ability, life-threatening or death, and congenital malformations occurred in clinical trials.

4.13.2 Timing and method for safety observation

All the subjects shall be observed for immediate reactions at the vaccination site for 30 min following each vaccination. The trained investigator conducted systematic observations and physical examinations on each subject, and recorded in detail both the local and systemic reactions in the subjects that occurred within 30 min. At 0-7 days after each dose of vaccination, the adverse reactions were systematically observed through the "Diary Card" filled in by the subject's legal guardian or his client; at 8-30 days, adverse events were collected through the "Contact Card" filled in by the subject's legal guardian or his client; at 6 months after the full basic immunization and at 6 months after the full enhanced immunization, serious adverse events were collected by active reporting in combination with regular follow-ups.

After each vaccination and before the next vaccination, the investigator shall transcribe the information on adverse reactions and other original information on the "Diary Card" and the "Contact Card" to the "Original Record Book" in a timely manner, and enter it into the Medidata Rave electronic data collection and management system after review and confirmation.

The duration, seriousness, and severity of any adverse event and its causal relationship with the test vaccine shall be evaluated. In the meantime, the measures taken for the test vaccine, the possible concomitant treatment and the results, and whether the event leads to the termination of the study shall be recorded.

4.13.3 Content and indicators for the follow-ups

Safety observations included all expectations and unexpectations, any medical events related or unrelated to vaccination during clinical trial observation.

Local reactions: Vaccination site's pains, itching, redness, swelling, induration, and rashes (vaccination site) and their degrees, and the body movement disorders.

Systemic reactions: The subjects were observed for fever, allergy, headache, fatigue/weakness, nausea/vomiting, diarrhea, muscle pains, cough, other discomfort or clinical adverse reactions.

Other events: Any medical event other than the above-mentioned ones, e.g., acute illness and accidental injury.

Criteria for grading: The systemic and local reactions after vaccination were judged in accordance with the "Guidelines for Criteria of Grading of Adverse Reactions in Clinical Trials of Preventive Vaccines" issued by the NMPA.

4.13.4 Correlation between adverse events and the vaccine

Definitely irrelevant: Other factors lead to the adverse events, and therefore, there is sufficient evidence to show that the adverse reactions/events occur as a result of other reasons while are irrelevant to vaccination.

Possibly irrelevant: Adverse events may occur as a result of other factors, such as the clinical conditions, other treatments, or concomitant medications in the subject, and are inconsistent with the known adverse reactions to vaccination.

Possibly relevant: Adverse events are consistent with the known information on the test vaccine,

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occur in a reasonable time sequence compared with vaccination, and/or are adverse events that arose before vaccination. The adverse event shows a causal relationship with the test vaccine, but it may also be related to other factors.

Very likely relevant: Adverse events are consistent with the known information on the test vaccine, there is a causal relationship between it and the test vaccine, and the events cannot be explained by other factors, e.g., the clinical conditions, other treatments, or concomitant medication in the subject.

Definitely relevant: The adverse events are consistent with the known information on the test vaccine, there is a causal relationship between it and the test vaccine, and such a relationship cannot be explained by other factors, e.g., the clinical conditions, other treatments, or concomitant medication in the subject. In addition, the adverse event appears again when the subject resumes the use of the test vaccine.

4.14 Immunogenicity observation

The venous blood was collected from the subject before the first dose of immunization and 30 days after the basic immunization; for subjects requiring enhanced immunization, it was necessary to collect the venous blood before the enhanced immunization and 30 days after the enhanced immunization. By testing the 13 pneumococcal serotype-specific IgG antibodies in all subjects before the first dose of immunization, 30 days after the basic immunization, before the enhanced immunization, and 30 days after the enhanced immunization, the proportions of subjects with the concentration of the specific IgG antibody $\geq 0.35 \mu\text{g/mL}$ in the various serotypes and the geometric mean concentrations of the specific antibody after the immunization with the test vaccine vs. the control vaccine were compared.

There were 100 subjects in each of the PCV13-TT group and the PCV7 group by age stratification. A total of 900 subjects were subjected to testing of serum specific antibody opsonization phagocytic activity (OPA), of whom a total of 300 subjects at the 2–6 months of age (minimum, 6 weeks) were tested for the serum specific antibody OPA 30 days after the basic immunization, before the enhanced immunization and 30 days after the enhanced immunization, a total of 200 subjects at the 7–11 months of age were tested for serum specific antibody OPA before the first dose, 30 days after the basic immunization, before the enhanced immunization and 30 days after the enhanced immunization, and a total of 400 subjects at the 12–71 months of age were tested for the serum specific antibody OPA before the first dose, and 30 days after the full immunization. As per the proportion of the sample size of the enrolled subjects 2 months of age in each clinical trial site to the total sample size of enrolled subjects 2 months of age, the subjects 2 months of age enrolled who required OPA testing were allocated to each trial site. For the other subjects whose age on enrollment required OPA testing, references shall be made to the proportion of sample sizes of various clinical trials, and the requirements on randomization shall be met. The distribution of study numbers for subjects requiring OPA testing at different ages across different clinical trial sites is shown in Table 7.

For subjects to undergo the testing of pneumococcal serotype-specific IgG antibody only, the venous blood not less than 3.0 mL in volume was collected each time; for subjects to undergo testing of pneumococcal serotype-specific IgG antibody and serum-specific antibody OPA, venous blood not less than 5.0 mL in volume was collected each time. Failure of the volume of blood collection to reach that required by the protocol while which did not affect the testing shall not be regarded as a violation of the protocol.

Table of OPA Assay Number Allocation

OPA Assay number allocation

Age groups

Wanrong Xiangfen Zhengding Dingxing Wenxian Puyang

2 months* 0441-0455 0361-0375 0281-0295 0201-0215 0101-0120 0001-0020

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3 months 1401-1432 1241-1272 1081-1112 0921-0952 0721-0756 0521-0556
 7-11 months 1901-1932 1841-1872 1781-1812 1721-1752 1641-1676 1561-1596
 12-23 months 2301-2332 2241-2272 2181-2212 2121-2152 2041-2076 1961-1996
 24-71 months 2701-2732 2641-2672 2581-2612 2521-2552 2441-2476 2361-2396

*Minimum age, 6 weeks

4.15. SAMPLE PROCESSING AND TESTING IN CLINICAL TRIALS

4.15.1 Processing, aliquoting, storage, and transportation of the blood samples

After collection, the venous blood was allowed to stand at room temperature for at least 2 h or coagulate fully using other measures, and then the serum was separated within 24 h. If the blood sample collected could not be centrifuged on that day, it shall be stored in a refrigerator at 2–8 °C. For a study number requiring IgG testing only, the separated serum was aliquoted into 2 tubes: tube 1 pending testing and the standby tube, and for serum aliquoting, priority was given to tube 1 pending testing. For a study number requiring IgG and OPA testing, the separated serum was aliquoted into 3 tubes: tube 1 pending testing, tube 2 pending testing, and the standby tube, and the serum was collected in the order of tube 1, tube 2, and the standby tube.

All the aliquoted serums were frozen at –20 °C or below, and the serum samples shall be kept under the frozen state when being transported.

4.15.2 Rules for number assignment to the serum samples

In principle, the study numbers should be used for the serum samples of subjects, and preceded by different sampling time points. For specific number assignment, see the "SOPs for Number Assignment to Biological Samples in Clinical Trials".

Test standards and quality control standards were provided by the testing laboratory.

4.15.3. Serum antibody detection

All serum samples detection were entrusted to National Institutes for Food and Drug Control.

The antibody concentration of IgG in serum samples was detected using enzyme-linked immunosorbent assay (ELISA), the subjects achieving IgG antibody concentration ≥ 0.35 $\mu\text{g/mL}$ and the geometric mean concentration (GMCs) were calculated.

Opsonophagocytic Assay (OPA) was used to detect the antibody opsonophagocytic activity of serum samples, meanwhile the subjects achieving an OPA antibody $\geq 1:8$ and the geometric mean titers (GMTs) were calculated.

Testing standard samples and quality control standards were provided by the laboratory. The method and quality control of ELISA and OPA can be found in the standard procedure of laboratory.

4.16. STATISTICAL ANALYSIS

4.16.1. Statistical Project plan

4.16.1.1 Project plan

The personnel undertaking the statistical analysis task participated in the whole process, from trial design, implementation, formulation of the statistical analysis plan to the analyses and summary, established a database and carried out statistical analysis, wrote the statistical analysis plan and completed the clinical statistical report.

4.16.1.2 Selection of analysis sets

Full Analysis Set (FAS) refers to all subjects who complied with the intention-to-treat (ITT) analysis principle, who met the inclusion/exclusion criteria, and who have undergone randomization, and have received at least one dose of vaccine and at least one follow-up record.

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Completed Full Basic Immunization Per Protocol Set (PPS) refers to all subjects who met the inclusion/exclusion criteria, and who have undergone the full basic immunization and planned follow-up as per the protocol.

Completed Full Enhanced Immunization Per Protocol Set (PPS) refers to all subjects who met the inclusion/exclusion criteria and who have undergone full enhanced immunization and planned follow-up as per the protocol.

PPS exclusion cases: those who did not meet the inclusion criteria; those who failed to be followed up the data and information after vaccination; those who seriously missed information and data after randomization; those who met the criteria for suspension or withdrawal but did not withdraw; those who received the wrong vaccination or incorrect dosage.

If there was any inconsistency between the FAS and PPS analysis results, it should be discussed in the report.

Safety Set (SS): All subjects who received vaccination after randomization experienced the safety evaluation, and data that violated the protocol should not be excluded.

> Safety Set (SS)

All subjects who received at least one vaccination after randomization and had at least one information on safety evaluation should experience the safety evaluation. The SS was summarized according to overall situation, severity, and number of doses; and data that violated the protocol should not be excluded.

The safety analysis of each dose was based on the actual number of people vaccinated each time, that is, the population of the safety analysis of the first dose was subjects who were randomly assigned to receive the first dose of test vaccine; the population of the safety analysis of the second dose was subjects who were randomly assigned to receive the second dose of test vaccine; the population of the safety analysis of the third dose was subjects who were randomly assigned to receive the third dose of test vaccine; the population of the safety analysis of the enhanced dose was subjects who were randomly assigned to receive the enhanced dose of test vaccine.

Safety analysis PPS: subjects who were randomized and completed full immunization.

The safety analysis was mainly based on the results of the FAS and took the analysis results of the PPS as a reference.

> Immunogenicity Set

Completed Full Basic Immunization Immunogenicity PPS: All subjects who met the inclusion/exclusion criteria entered randomization, completed the full basic immunization and corresponding blood collection as per the protocol, and had the test results of serum immunogenicity. The investigator and laboratory testers should discuss and decide to accept or reject the subject whose serum antibody concentration was inverted before and after immunization.

Completed Full Enhanced Immunization Immunogenicity PPS: All subjects who met the inclusion/exclusion criteria entered randomization, completed the full enhanced immunization and corresponding blood collection as per the protocol, and had the test results of serum immunogenicity. The investigator and laboratory testers should discuss and decide to accept or reject the subject whose serum antibody concentration was inverted before and after immunization.

4.16.2 Statistical methods

General principle:

Statistical description: Quantitative indicators were listed out with the mean $\pm$ standard deviation, median, maximum and minimum values; qualitative indicators or grade indicators were listed out with the frequency distribution table. The antibody titer and antibody concentration of the

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immunogenicity indicators in this trial were expressed as geometric mean, standard deviation, median, maximum and minimum values, and 95% confidence interval, and the antibody titer and antibody concentration should be logarithmic transformed.

Statistical inference: Appropriate statistical analysis methods were used for inter-group comparison according to different applicable conditions. All statistical tests gave the test statistics and corresponding P values, and the P value was directly given when using the Fisher's exact probability test; all statistical tests were two-sided, and the P value less than or equal to 0.05 would be considered that the difference of the test was statistically significant (except for special instructions).

Drop-out analysis: The case number of enrollment, drop-out, and exclusion were listed out, as well as the number of completed case and the number of cases in each analysis set. The Fisher's exact probability test was used for inter-group comparison of the total drop-out rate and the drop-out rate caused by adverse events.

Homogeneity analysis: Appropriate statistical analysis methods were used for comparison between groups for the demographics of the selected vaccine recipients in each group and the characteristics of the relevant baselines, so as to investigate the equilibrium comparability between groups. Immunogenicity analysis: X2 test or Fisher's exact probability test was used to compare the differences of antibody positive seroconversion rates between the PCV13-TT group and the PCV7 group; the t test was used to compare the ratios of geometric mean concentration (GMC) of antibodies between the PCV13-TT group and the PCV7 group, and the test statistics and their corresponding P values were given. The P values were calculated directly by the Fisher's exact probability test, and $P \leq 0.05$ was taken as the standard with statistical significance. Safety analysis: The safety analysis was based on the SS. The frequency and case number of adverse events and adverse reactions in each group were listed out for calculation of the incidence and corresponding severity; the inter-group comparison of the adverse event (reaction) rates was performed using the Fisher's exact probability test, and the detailed list of various adverse events was given.

4.16.3 Statistical content

4.16.3.1 Demographic feature description

■ Numbers and reasons of enrollment, completion, and drop-out;

■ Demographic balance.

4.16.3.2 Statistical analysis of immunogenicity

■ The analysis was based on that all subjects met the requirements of the trial protocol and had a series of effective serums before and after immunization.

■ For a total of 7 serotypes, the proportion of samples with specific pneumococcal IgG antibody concentration $\geq 0.35 \mu\text{g/mL}$ of each serotype was counted in the PCV13-TT group and the PCV7 group, respectively, and chi-square test was used to statistically compare the difference between groups in each serotype, while the binomial distribution method was used to calculate the two-sided 97.5% (one-sided 98.75%) confidence interval of the difference of positive seroconversion rates (test group – control group), and a non-inferiority judgment of interval estimation was performed;

■ For a total of 7 serotypes, the GMC proportion (test group/control group) of specific pneumococcal IgG antibody of each serotype was calculated in the PCV13-TT group and the PCV7 group, respectively. After performing logarithmic conversion (based on 10), t test was used to statistically compare the difference between groups in each serotype, and the two-
 $X_1 - X_2$ ■ t df, 0.025S $X_1 - X_2$

sided 97.5% (one-sided 98.75%) confidence interval () of the

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difference of means was calculated after conversion (ratio after anti-logarithm), and a non-inferior judgment of interval estimation was performed;

■ For the newly added 6 serotypes, the proportion of samples with specific pneumococcal IgG antibody concentration $\geq 0.35 \mu\text{g/mL}$ of each serotype, and their two-sided 95% confidence intervals were calculated; chi-square test was used to statistically compare the differences between groups in each serotype, and to calculate the two-sided 95% confidence interval of the difference of positive seroconversion rates (test group – control group) (determination of the immunogenicity of the newly added 6 serotypes);

■ Subjects who were 2–6 months of age were the core target population of the study. Statistical comparisons were made for other age groups, and statistical descriptions but no statistical conclusions were given, and other age groups were incorporated into the 2- to 6-month-old PCV13-TT group for non-inferiority judgment between groups;

■ After a total of 7 serotypes of the test vaccine and the control vaccine passed the non-inferiority test, and if the newly added 6 serotypes of the test vaccine were better than the blank serotype of the control vaccine, then the positive seroconversion rate test of the specific pneumococcal IgG antibody of the newly added 6 serotypes was performed.

■ When the non-inferiority test of the test group 1 and the control group was established and the immunogenicity of the newly added 6 serotypes was determined, the non-inferiority test of the test group 2 and the control group was carried out; when the non-inferiority test of the test group 2 and the control group was established, the non-inferiority tests (sequential tests) of the test group 2 and the test group 1 were further performed.

4.16.3.3 Statistical analysis of clinical safety

The evaluation of the data would include a summary of the clinical response endpoints, the incidence of local and systemic clinical responses in all subjects during the observation period. Subjects with at least one adverse reaction in the test group and the control group within day 0–7, days 8–30, and days 0–30:

■ Statistical analysis of the percentage of adverse reactions in subjects;

■ Statistical analysis of the percentage of adverse reactions in each dose;

■ Statistical analysis of the percentage of local or systemic adverse reactions in subjects;

■ Statistical analysis of the percentage of adverse reactions of different severity levels in subjects;

■ The Fisher's exact probability test was used to statistically compare the incidence of various adverse events between the PCV13-TT group and the PCV7 group.

Analysis of adverse reaction symptoms in the PCV13-TT group and the PCV7 group:

■ Percent of occurrence of any systemic and local abnormal symptoms after vaccination of the subjects;

■ Percentage and intensity of body temperature reaction after each vaccination of the subjects;

■ The Fisher's exact probability test was used to statistically compare the incidence of various adverse events between the PCV13-TT group and the PCV7 group.

Adverse events leading to withdrawal and serious adverse events:

The cases with suspension in this study due to adverse events and occurrence of severe or serious adverse events were especially marked, and the incidence was statistically compared between groups.

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Usage of concomitant medication:

The frequency of concomitant medications was described in the PCV13-TT group and the PCV7 group, respectively, and the Fisher's exact probability test was used for statistical comparison.

4.17 MODIFICATION AND INTERIM ANALYSIS DURING THE TRIAL

This trial did not carry out modifications and interim analysis.

5 TRIAL COMPLETION AND BASELINE CONDITIONS

5.1 ALLOCATION OF SUBJECTS

This clinical trial was a multi-center study in three provinces. The phase I clinical trial was conducted in the Chaoyang District of Beijing Center for Disease Prevention and Control, and phase III clinical trial was completed in Hebei, Henan, and Shanxi provinces, respectively. A total of 2760 subjects were enrolled, 2753 of which completed at least one dose of vaccination. The number of subjects enrolled in each center is shown in Table 1.

Table 1 Subject Enrolment in Each Site

Site 1	Site 2	Site 3	Total
Groups Enrolle	Vaccin	Vaccin	Enroll
Enroll	Vaccinat	Enroll	Vaccinat
Enrolled			
d	ated	ated	d
ed	ed	ed	ed
2 months	160	160	200
198	160	159	520
517			
3 months	320	320	400
398	320	320	1040
1038			
7-11 months	120	119	160
159	120	120	400
398			
12-23 months	120	120	160
160	120	120	400
400			
24-71 months	120	120	160
160	120	120	400
400			
Total	840	839	1080
1075	840	839	2760
2753			
Site 1:	Hebei Provincial Center for Disease Control and Prevention.		
Site 2:	Henan Provincial Center for Disease Control and Prevention.		
Site 3:	Shanxi Provincial Center for Disease Control and Prevention.		

5.2 TRIAL COMPLETION

In this study, a total of 3240 subjects were screened, and 2760 subjects were enrolled in this study, 1640 of which were in the PCV13-TT group (including subjects at 2 months of age in 0, 2, 4 program groups and no control) and 1120 of which were in the PCV7 group. Among the 2760 subjects enrolled in the study, 7 subjects voluntarily withdrew without the first dose of vaccination, and it was not due to adverse events, so a total of 2753 (99.75%) subjects completed at least one dose of vaccination, 1634 (99.57%) of which were in the PCV13-TT group and 1119 (99.91%) of which in the PCV7 group. Among the 2760 subjects enrolled in the study, 94 subjects dropped out (including the above 7 enrolled subjects who voluntarily withdrew without the first dose of vaccination), 62 (3.78%) of which were in the PCV13-TT group and 32(2.86%) of which were in the PCV7 group, and there was no difference between groups in the drop-out rate, $p = 0.2010$; 2666 (96.59%) subjects completed the trial, 1578 (96.22%) of which were in the PCV13-TT group and 1088 (97.14%) of which were in the PCV7 group. Among the 2760 subjects enrolled in the study, 2753 subjects were included in the SS, 1634 (99.57%) of which were in the PCV13-TT group and 1119 (99.91%) of which were in the PCV7 group; except for the type 9V, 2753 subjects were included in the FAS for the other serotypes, 1634 (99.63%) of which were in the PCV13-TT group and 1119 (99.91%) of which were in the PCV7 group; because one patient in the PCV13-TT group had insufficient pre-immunization Confidential 39 / 310 Phase III Clinical Study Report of the 13-Valent Pneumococcal Polysaccharide Conjugate Vaccine Protocol No.: 022152015003

serum samples, there was no pre-immunization serum test result of type 9V, and because the second dose of this subject dropped out, there was no post-immunization serum test result, therefore, 2752 subjects were included in the FAS for the type 9V, 1633 (99.63%) of which were in the PCV13-TT group and 1119 (99.91%) of which were in the PCV7 group; 2589 subjects were included in the PPS, 1525 (92.99%) of which were in the PCV13-TT group and 1064 (95.00%) of which were in the PCV7 group. The enrollment and completion of all subjects and subjects of each age group are shown in Tables 2-1 to 2-6 and the trial flow chart. Table 2-1 Completers of Enrolled Subjects

PCV13-TT group	PCV7 group	Total
(N=1640)	(N=1120)	
n	n	n
Enrolled	1640	100.0
1120	100.0	2760
100.0		
Received at least 1 dose of	1634	99.57%
1119	99.91%	2753
99.75%		
vaccination		
Completers#	1578	96.22
1088	97.14	2666
96.59		
Dropouts*	62	3.78
32	2.86	94
3.41*		
SAEs	2	0.12
2	0.18	4
0.14		
Non-SAEs	11	0.67
13	1.16	24
0.87		
Protocol violation	0	0.00
0	0.00	0
0.00		
Voluntary withdrawal	41	2.50
15	1.34	56
2.03		
Outside, migrated or moved away	7	0.43
2	0.18	9
0.33		
from the area		
Lost to follow-up	1	0.06
0	0.00	1
0.04		
Other	0	0.00
0	0.00	0
0.00		
Safety analysis set	1634	99.57
1119	99.91	2753
99.75		
FAS (9V)	1633	99.63
1119	99.91	2752
99.71		
Immunogenicity FAS (other	1634	99.63
1119	99.91	2753
99.75		
analysis set serotypes)		
PPS	1525	92.99
1064	95.00	2589
93.80		
*Inter-group drop out rates, Fisher $P=0.2010$. #Completed safety and immunogenicity observations.		

Table 2-2 Completers of Enrolled Subjects 2 Aged Months

n	n	n
Enrolled	520	100.0
Received at least 1 dose of	517	99.42
vaccination		
Completers	491	94.42
Dropouts	29	5.58
SAEs	0	0.00
Non-SAEs	5	0.96
Protocol violation	0	0.00
Voluntary withdrawal (non-AE)	21	4.04
Outside, migrated or moved away	3	0.58
from the area		
Lost to follow-up	0	0.00
Other	0	0.00
Safety analysis set	517	99.42

Immunogenicity FAS 517 99.42
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analysis set PPS 482 92.69
#Completed safety and immunogenicity observations.

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Table 2-3 Completers, Dropouts and Removal of Enrolled Subjects Aged 3 Months
PCV13-TT
PCV7 group Total
group
n ■ n ■ n ■
Enrolled 520 100.0 520 100.0 1040 100.0
Received at least 1 dose of
519 99.81 519 99.81 1038 99.81
vaccination
Completers 501 96.35 503 96.73 1004 96.54
Dropouts* 19 3.65 17 3.27 36 3.46
SAEs 1 0.19 1 0.19 2 0.19
Non-SAEs 3 0.58 6 1.15 9 0.87
Protocol violation 0 0.00 0 0.00 0 0.00
Voluntary withdrawal (non-AE) 12 2.31 10 1.92 22 2.12
Outside, migrated or moved
2 0.38 0 0.00 2 0.19
away from the area
Lost to follow-up 1 0.19 0 0.00 1 0.10
Other 0 0.00 0 0.00 0 0.00
Safety analysis set 519 99.81 519 99.81 1038 99.81
Immunogenicity FAS 519 99.81 519 99.81 1038 99.81
analysis set PPS 471 90.58 487 93.65 958 92.12
*Inter-group drop out rates, Fisher P=0.8656. #Completed safety and immunogenicity observations.

Table 2-4 Completers, Dropouts and Removal of Enrolled Subjects Aged 7-11 Months
PCV13-TT
PCV7 group Total
group
n ■ n ■ n ■
Enrolled 200 100.0 200 100.0 400 100.0
Received at least 1 dose of
198 99.00 200 100.0 398 99.50
vaccination
Completers 193 96.50 193 96.50 386 96.50
Dropouts* 7 3.50 7 3.50 14 3.50
SAEs 1 0.50 0 0.00 1 0.25
Non-SAEs 1 0.50 2 1.00 3 0.75
Protocol violation 0 0.00 0 0.00 0 0.00
Voluntary withdrawal (non-AE) 5 2.50 3 1.50 8 2.00
Outside, migrated or moved
0 0.00 2 1.00 2 0.50
away from the area
Lost to follow-up 0 0.00 0 0.00 0 0.00
Other 0 0.00 0 0.00 0 0.00
Safety analysis
198 99.00 200 100.0 398 99.50
set
FAS 9V 197 98.50 200 100.0 397 99.25
Immunogenicity
FAS (other) 198 99.00 200 100.0 398 99.50
analysis set
PPS 185 92.57 189 94.50 374 93.50
*Inter-group drop out rates, Fisher P=1.0000. #Completed safety and immunogenicity observations.

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Table 2-5 Completers, Dropouts and Removal of Enrolled Subjects Aged 12-23 Months
PCV13-TT
PCV7 group Total
group
n ■ n ■ n ■
Enrolled 200 100.0 200 100.0 400 100.0
Received at least 1 dose of
200 100.0 200 100.0 400 100.0
vaccination
Completers 193 96.50 192 96.00 385 96.25
Dropouts* 7 3.50 8 4.00 15 3.75
SAEs 0 0.00 1 0.50 1 0.25
Non-SAEs 2 1.00 5 2.50 7 1.75
Protocol violation 0 0.00 0 0.00 0 0.00
Voluntary withdrawal (non-AE) 3 1.50 2 1.00 5 1.25
Outside, migrated or moved
2 1.00 0 0.00 2 0.50
away from the area
Lost to follow-up 0 0.00 0 0.00 0 0.00
Other 0 0.00 0 0.00 0 0.00
Safety analysis

200 100.0 200 100.0 400 100.0
set
Immunogenicity FAS 200 100.0 200 100.0 400 100.0
analysis set PPS 189 94.50 189 94.50 378 94.50
*Inter-group drop out rates, Fisher P=1.0000. #Completed safety and immunogenicity observations.

Table 2-6 Completers, Dropouts and Removal of Enrolled Subjects Aged 24-71 Months

PCV13-TT					
PCV7 group Total					
group					
n ■ n ■ n ■					
Enrolled	200	100.0	200	100.0	400 100.0
Received at least 1 dose of					
200 100.0 200 100.0 400 100.0					
vaccination					
Completers	200	100.0	200	100.0	400 100.0
Dropouts*	0	0.00	0	0.00	0 0.00
SAEs	0	0.00	0	0.00	0 0.00
Non-SAEs	0	0.00	0	0.00	0 0.00
Protocol violation	0	0.00	0	0.00	0 0.00
Voluntary withdrawal (non-					
0 0.00 0 0.00 0 0.00					
AE)					
Outside, migrated or					
0 0.00 0 0.00 0 0.00					
moved away from the area					
Lost to follow-up	0	0.00	0	0.00	0 0.00
Other	0	0.00	0	0.00	0 0.00
Safety analysis set	200	100.0	200	100.0	400 100.0
Immunogenicity FAS	200	100.0	200	100.0	400 100.0
analysis set PPS	198	99.00	199	99.50	397 99.25
#Completed safety and immunogenicity observations.					

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Flow Chart of Primary Immunizations for Phase III Clinical Trial of 13-valent Pneumococcal Conjugate Vaccine
Note: 2nd and 3rd doses are not given in population aged 24-71 months; 3rd dose is note given in population aged 12-23 months.

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5.3 COMPLIANCE ANALYSIS
Among the 2760 subjects, 2753 of which completed the first dose of vaccination and safety observation, and 2670 cases completed the full basic immunization/full immunization; 5402 blood samples were collected effectively after vaccination and before and after immunization. The compliance analysis of subjects is shown in Table 3.

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Table 3 Analysis of Compliance for Subjects					
Vaccinations and Safety Observation Immunogenicity Observation					
Pre-Immunization blood sampling Post-Immunization blood sampling					
Groups Dose(s) Planned Completed Completion rate					
Planned Completed Plannedn Completed					
n n (%) % %					
n* n*					
1 520 517 99.42					
2 months 2 517 496 95.94 520 517 99.42% 491 488 99.39					
3 496 491 98.99					
1 1040 1038 99.81					
3 months 2 1038 1014 97.69 1040 1037 99.71% 1006 1000 99.40					
3 1014 1006 99.21					
1 400 398 99.50					
7-11 months 400 395 98.75 387 382 98.70					
2 398 387 97.24					
1 400 400 100.0%					
12-23 months 400 400 100.00 386 384 99.48					
2 400 386 96.50					
24-71 months 1 400 400 100.0% 400 400 100.00 400 399 99.75					
*Number of blood sampling completed refers to the number of subjects that completed blood collection and had at least one serum sample for testing.					

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5.4 DEVIATION OF THE TRIAL PROTOCOL
There was no serious deviation of the protocol in this trial. A total of 312 cases of individual protocol violations/deviations occurred in the trial, including 91 cases in the 2-month-old test group, 83 cases in the 3-month-old test group and 55 cases in the control group, 34 cases in the 7- to 11-month-old PCV13-TT group and 28 cases in the PCV7 group, 10 cases in the 12- to 23-month-old PCV13-TT group and 7 cases in the PCV7 group, 1 case in the 24- to 71-month-old PCV13-TT group and 3 cases in the PCV7 group. Four subjects with protocol violations/deviations withdrew the trial because they met the "Termination and Withdrawal Criteria". The protocol

violations/deviations were mainly due to the time of vaccination or blood collection exceeding the window period, followed by the lack of enough interval for other vaccinations. Infants under 12 months of age had a higher incidence.

Table 4 Protocol Violation/Deviation

PCV13-TT group	PCV7 group
Observations Number	Observations Number
Groups Number of	Number of
of	of
Events	Events
Cases	Cases
2 months	520 77 91 / / /
3 months	520 70 83 520 53 55
7-11 months	200 30 34 200 23 28
12-23 months	200 9 10 200 7 7
24-71 months	200 1 1 200 3 3
Total	1640 187 219 1120 86 93

5.5. DEMOGRAPHIC BASELINE OF SUBJECTS

The proportions of age and sex of subjects aged 2 months (minimum age: 6 weeks) and above are shown in Table 3 to

Table 7. Considering the age at which the comparator vaccine was approved for use in China, no PCV7 group was set up for 2 months old group (minimum age: 6 weeks). No statistical intergroup difference was observed in the mean age and sex distribution of SS and FAS data sets between the PCV13-TT and PCV7 groups in subjects aged 3-71 months. The baseline demographics were balanced and comparable.

Table 3 Baseline demographics of subjects aged 2 months (at least 6 weeks of age)

Data	Age (months)	Gender
Observations		
set Mean±SD	Median (Min, Max)	Male (n%) Female (n%)
SS	517 2.27±0.40 2.23 (1.41,2.92)	236(45.65) 281(54.35)
FAS	517 2.27±0.40 2.23 (1.41,2.92)	236(45.65) 281(54.35)
*A total of 108 subjects in the age group of 2 months were 42-60 days of age, accounting for 20.89% of this age group.		

Table 4 Baseline demographics of subjects aged 3 months

Obse	Age (months)	Gender
Data		
Groups	rvati	Mediann P P
set Mean±SD	Malen%	Femalen%
ons (Min,Max)		
PCV13-		
519	3.33±0.30 3.25(2.73,3.91)	270(52.02) 249(47.98)
SS TT	0.4351 0.4939	
PCV7	519 3.34±0.30 3.29(2.63,3.91)	281(54.14) 238(45.86)
PCV13- 519		
3.33±0.30	3.25(2.73,3.91)	270(52.02) 249(47.98)
FAS TT	0.4351 0.4939	
PCV7	519 3.34±0.30 3.29(2.63,3.91)	281(54.14) 238(45.86)

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Table 5 Baseline demographics of subjects aged 7-11 months

Obse	Age (months)	Gender
Data		
Groups	rvati	P P
set Mean±SD	Median (Min,Max)	Malen% Femalen%
ons		
PCV13		
198	9.34±1.37 9.30(6.90,11.79)	97(48.99) 101(51.01)
SS -TT	0.9616 0.4236	
PCV7	200 9.33±1.33 9.23(6.93,11.76)	106(53.00) 94(47.00)
PCV13		
198	9.34±1.37 9.30(6.90,11.79)	97(48.99) 101(51.01)
FAS -TT	0.9616 0.4236	
PCV7	200 9.33±1.33 9.23(6.93,11.76)	106(53.00) 94(47.00)

Table 6 Baseline dmographics of subjects aged 12-23 months

Obse	Age (months)	Gender
Data		
Groups	rvati	P P
set Mean±SD	Median (Min,Max)	Malen% Femalen%
ons		
PCV13		
200	17.36±3.25 17.50(11.63,23.59)	112(56.00) 88(44.00)
SS -TT	0.3604 0.6862	
PCV7	200 17.66±3.27 17.38(11.89,23.62)	116(58.00) 84(42.00)
PCV13		
200	17.36±3.25 17.50(11.63,23.59)	112(56.00) 88(44.00)
FAS -TT	0.3604 0.6862	
PCV7	200 17.66±3.27 17.38(11.89,23.62)	116(58.00) 84(42.00)

Table 7 Baseline demogrphics of subjects aged 24-71 months

Obse	Age (months)	Gender
Data		
Groups	rvati	P P

set Mean±SD Median (Min,Max) Malen% Femalen%
 ons
 PCV13
 200 42.82±13.49 41.74(23.75,70.80) 99(49.50) 101(50.50)
 SS -TT 0.1110 0.2708
 PCV7 200 40.76±12.30 40.41(23.66,69.29) 110(55.00) 90(45.00)
 PCV13
 200 42.82±13.49 41.74(23.75,70.80) 99(49.50) 101(50.50)
 FAS -TT 0.1110 0.2708
 PCV7 200 40.76±12.30 40.41(23.66,69.29) 110(55.00) 90(45.00)

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5.6. PRE-IMMUNE ANTIBODY POSITIVE RATE AND ANTIBODY LEVEL BASELINE OF 13-VALENT PNEUMONIA SEROTYPES IN SUBJECTS

5.6.1. Pre-immune IgG Antibody Positive Rate of 13-valent Pneumonia Serotypes in Subjects (≥0.35 µg/mL)

The pre-immune antibody positive rates of subjects aged 2 months, 3 months, 7-11 months, 12-23 months and 24-71 months are shown in Table 6-1 to 6-6.

The pre-immune antibody positive rates of all 13-valent pneumonia serotypes in the 2-month study group were higher than those in the 3-month study group and control group.

No significant difference was observed in the antibody positive rates of all 13-valent pneumonia serotypes between the PCV13-TT group and PCV7 group of subjects aged 3 months, 7-11 months and 12-23 months.

The antibody positive rates in the study group of subjects aged 24-71 months were all higher than those in the control group. Except for serotypes 3, 6b, 7F and 14, the difference of antibody positive rate between the study group and the control group of other serotypes was statistically significant. The pre-immune antibody positive rates of 4 serotypes (6B, 14, 19A and 19F) in the study and control group of each age group were relatively higher than those of other serotypes.

Table 6-1 Pre-Immunization IgG Antibody Positive Rates in Subjects of PCV13-TT Group
 (2M) and PCV3-TT Group (3M) (FAS)

PCV13-TT (2M) group (N=517) PCV13-TT group (3M) (N=518)

number of number of

Serotypes P

subjects with % 95% CI subjects with % 95% CI

seroresponse seroresponse

1 168 32.50 28.47-36.72 56 10.81 8.27-13.81 <0.0001

3 30 5.80 3.95-8.18 9 1.74 0.80-3.27 0.0006

4 52 10.06 7.60-12.98 17 3.28 1.92-5.20 <0.0001

5 120 23.21 19.64-27.09 19 3.67 2.22-5.67 <0.0001

6A 274 53.00 48.59-57.37 156 30.12 26.19-34.27 <0.0001

6B 325 62.86 58.54-67.04 199 38.42 34.21-42.76 <0.0001

7F 166 32.11 28.10-36.32 94 18.15 14.92-21.74 <0.0001

9V 188 36.36 32.21-40.68 94 18.15 14.92-21.74 <0.0001

14 475 91.88 89.18-94.08 445 85.91 82.61-88.79 0.0022

18C 184 35.59 31.46-39.89 84 16.22 13.15-19.68 <0.0001

19A 426 82.40 78.84-85.58 391 75.48 71.54-79.13 0.0064

19F 428 82.79 79.25-85.94 350 67.57 63.35-71.59 <0.0001

23F 224 43.33 39.01-47.72 126 24.32 20.69-28.26 <0.0001

*FAS was based on post-immunization serum findings, there was 1 case lacking pre-immunization data in PCV13-TT group (3M).

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Table 6-2 Pre-Immunization IgG Antibody Positive Rates in Subjects of PCV13-TT Group
 (2M) and PCV7 Group (3M) (FAS)

PCV13-TT (2M) group (N=517) PCV7 (3M) group (N=516)

Serotypes number of number of P

subjects with % 95% CI subjects with % 95% CI

seroresponse seroresponse

1 168 32.50 28.47-36.72 54 10.47 7.96-13.43 <0.0001

3 30 5.80 3.95-8.18 17 3.29 1.93-5.22 0.0531

4 52 10.06 7.60-12.98 12 2.33 1.21-4.03 <0.0001

5 120 23.21 19.64-27.09 21 4.07 2.54-6.15 <0.0001

6A 274 53.00 48.59-57.37 150 29.07 25.18-33.20 <0.0001

6B 325 62.86 58.54-67.04 217 42.05 37.75-46.45 <0.0001

7F 166 32.11 28.10-36.32 106 20.54 17.14-24.29 <0.0001

9V 188 36.36 32.21-40.68 88 17.05 13.91-20.58 <0.0001

14 475 91.88 89.18-94.08 445 86.24 82.96-89.09 0.0037

18C 184 35.59 31.46-39.89 84 16.28 13.20-19.75 <0.0001

19A 426 82.40 78.84-85.58 393 76.16 72.25-79.78 0.0134

19F 428 82.79 79.25-85.94 351 68.02 63.81-72.03 <0.0001

23F 224 43.33 39.01-47.72 126 24.42 20.77-28.36 <0.0001

*FAS was based on post-immunization serum findings, there were 3 cases lacking pre-immunization data in PCV7 (3M) group.

Table 6-3 Pre-Immunization IgG Antibody Positive Rates in Subjects of PCV13-TT Group
 (3M) and PV7 Group (3M) (FAS)

PCV13-TT group (3M) (N=518) PCV7 (3M) group (N=516)

Serotypes P

n % 95%CI n % 95%CI

1 56 10.81 8.27-13.81 54 10.47 7.96-13.43 0.8569

3 9 1.74 0.80- 3.27 17 3.29 1.93-5.22 0.1098

4 17 3.28 1.92- 5.20 12 2.33 1.21-4.03 0.3517

5 19 3.67 2.22- 5.67 21 4.07 2.54-6.15 0.7376

6A 156 30.12 26.19-34.27 150 29.07 25.18-33.20 0.7125

6B 199 38.42 34.21-42.76 217 42.05 37.75-46.45 0.2330

7F 94 18.15 14.92-21.74 106 20.54 17.14-24.29 0.3294
 9V 94 18.15 14.92-21.74 88 17.05 13.91-20.58 0.6446
 14 445 85.91 82.61-88.79 445 86.24 82.96-89.09 0.8771
 18C 84 16.22 13.15-19.68 84 16.28 13.20-19.75 0.9781
 19A 391 75.48 71.54-79.13 393 76.16 72.25-79.78 0.7984
 19F 350 67.57 63.35-71.59 351 68.02 63.81-72.03 0.8754
 23F 126 24.32 20.69-28.26 126 24.42 20.77-28.36 0.9718
 *FAS was based on post-immunization serum findings, there was 1 and 3 cases lacking pre-immunization data in PCV13-TT group (3M) and PCV7 (3M) group, respectively.

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Table 6-4 Pre-Immunization IgG Antibody Positive Rates in Subjects of PCV13-TT Group and PCV7 Group Aged 7-11 Months (FAS)
 PCV13-TT group (N=195) PCV7 group (N=199)

Serotypes P
 n % 95%CI n % 95%CI
 1 52 26.67 20.60-33.46 46 23.12 17.45-29.60 0.4149
 3 11 5.64 2.85-9.87 14 7.04 3.90-11.52 0.5703
 4 7 3.59 1.46-7.26 8 4.02 1.75-7.77 0.8234
 5 21 10.77 6.79-15.99 19 9.55 5.85-14.51 0.6881
 6A 47 24.10 18.28-30.73 48 24.12 18.35-30.68 0.9967
 6B 86 44.10 37.01-51.37 95 47.74 40.63-54.92 0.4690
 7F 45 23.08 17.36-29.63 39 19.60 14.32-25.80 0.3992
 9V* 11 5.67 2.86-9.92 10 5.03 2.44-9.05 0.7762
 14 136 69.74 62.77-76.10 137 68.84 61.91-75.21 0.8466
 18C 5 2.56 0.84-5.88 7 3.52 1.43-7.11 0.5819
 19A 142 72.82 66.00-78.93 140 70.35 63.48-76.60 0.5870
 19F 133 68.21 61.17-74.67 130 65.33 58.27-71.92 0.5443
 23F 35 17.95 12.83-24.07 29 14.57 9.98-20.25 0.3637
 *There was 1 case without results for serotype 9V due to inadequate serum.

Table 6-5 Pre-Immunization IgG Antibody Positive Rates in Subjects of PCV13-TT Group and PCV7 Group Age 12-23 Months (FAS)
 PCV13-TT group (N=200) PCV7 group (N=200)

number of number of
 Serotypes subjects subjects P
 % 95% CI % 95% CI
 with with
 seroresponse seroresponse
 1 84 42.00 35.07-49.17 82 41.00 34.11-48.16 0.8392
 3 21 10.50 6.62-15.60 24 12.00 7.84-17.33 0.6350
 4 22 11.00 7.02-16.18 19 9.50 5.82-14.44 0.6209
 5 55 27.50 21.44-34.24 62 31.00 24.67-37.91 0.4417
 6A 101 50.50 43.36-57.63 100 50.00 42.87-57.13 0.9203
 6B 152 76.00 69.47-81.74 162 81.00 74.87-86.19 0.2236
 7F 88 44.00 37.01-51.17 95 47.50 40.41-54.66 0.4823
 9V 20 10.00 6.22-15.02 29 14.50 9.93-20.16 0.1699
 14 183 91.50 86.74-94.97 183 91.50 86.74-94.97 1.0000
 18C 18 9.00 5.42-13.85 18 9.00 5.42-13.85 1.0000
 19A 184 92.00 87.33-95.36 190 95.00 91.00-97.58 0.2236
 19F 184 92.00 87.33-95.36 181 90.50 85.56-94.18 0.5955
 23F 98 49.00 41.88-56.15 103 51.50 44.35-58.61 0.6171

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Table 6-6 Pre-Immunization IgG Antibody Positive Rates in Subjects of PCV13-TT Group and PCV7 Group Aged 24-71 Months (FAS)
 PCV13-TT group (N=200) PCV7 group (N=200)

number number
 Serotype of of
 subjects subject P
 s % 95% CI s % 95% CI
 with s with
 serorespo serores
 nse ponse
 1 101 50.50 43.36-57.63 78 39.00 32.20-46.13 0.0207
 3 62 31.00 24.67-37.91 51 25.50 19.61-32.13 0.2218
 4 71 35.50 28.88-42.56 51 25.50 19.61-32.13 0.0299
 5 118 59.00 51.84-65.89 93 46.50 39.44-53.67 0.0123
 6A 152 76.00 69.47-81.74 127 63.50 56.42-70.18 0.0065
 6B 187 93.50 89.14-96.49 174 87.00 81.53-91.33 0.0284
 7F 118 59.00 51.84-65.89 107 53.50 46.33-60.56 0.2676
 9V 97 48.50 41.39-55.65 77 38.50 31.72-45.62 0.0437
 14 196 98.00 94.96-99.45 189 94.50 90.37-97.22 0.0654
 18C 78 39.00 32.20-46.13 54 27.00 20.98-33.72 0.0107
 19A 199 99.50 97.25-99.99 189 94.50 90.37-97.22 0.0034
 19F 200 100.0 98.17-100.0 191 95.50 91.63-97.92 0.0036
 23F 145 72.50 65.76-78.56 125 62.50 55.39-69.23 0.0328

5.6.2. Pre-immune IgG Antibody Level (GMC) of 13-valent Pneumonia Serotypes in Subjects
 The analysis of pre-immune antibody levels of subjects aged 2 months, 3 months, 7-11 months, 12-23 months and 24-71 months are shown in Table 7-1 to 7-6.
 The pre-immune GMCs of all 13-valent pneumonia serotypes in the 2-month study group were higher than those in the 3-month study group and control group.
 No significant difference was observed in GMCs of all 13-valent pneumonia serotypes between the PCV13-TT group and PCV7 group of subjects aged 3 months, 7-11 months and 12-23 months.
 GMC in the study group of subjects aged 24-71 months were all higher than those in the control

group. Except for serotypes 7F, 9V and 14, the difference of GMC between the study group and the control group of other serotypes was statistically significant.
The pre-immune GMC of 4 serotypes (6B, 14, 19A and 19F) in the study and control group of each age group were relatively higher than those of other serotypes.

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Table 7-1 Pre-Immunization IgG GMC in Subjects of PCV13-TT Group (2M) and PCV13-TT Group (3M) (FAS)
PCV13-TT (2M) group (N=517) PCV13-TT group (3M) (N=518)

Serotypes P	GMC	95% CI	% GMC	95% CI	
1	0.22	0.20-0.24	0.10	0.09-0.11	<0.0001
3	0.12	0.11-0.13	0.06	0.06-0.07	<0.0001
4	0.07	0.06-0.08	0.02	0.02-0.02	<0.0001
5	0.19	0.18-0.21	0.05	0.05-0.06	<0.0001
6A	0.30	0.26-0.34	0.15	0.13-0.17	<0.0001
6B	0.44	0.40-0.48	0.28	0.25-0.31	<0.0001
7F	0.19	0.16-0.21	0.12	0.10-0.13	<0.0001
9V	0.20	0.18-0.23	0.08	0.07-0.10	<0.0001
14	1.36	1.24-1.49	0.86	0.79-0.95	<0.0001
18C	0.20	0.17-0.23	0.08	0.07-0.10	<0.0001
19A	0.81	0.74-0.88	0.50	0.45-0.56	<0.0001
19F	0.74	0.69-0.80	0.46	0.42-0.50	<0.0001
23F	0.33	0.30-0.36	0.18	0.17-0.21	<0.0001

*FAS was based on post-immunization serum findings, there was 1 case lacking pre-immunization data in PCV13-TT group (3M).

Table 7-2 Pre-Immunization IgG GMC in Subjects of PCV13-TT Group (2M) and PCV7 Group (3M) (FAS)
PCV13-TT (2M) group (N=517) PCV7 (3M) group (N=516)

Serotypes P	GMC	95% CI	% GMC	95% CI	
1	0.22	0.20-0.24	0.11	0.09-0.12	<0.0001
3	0.12	0.11-0.13	0.05	0.05-0.06	<0.0001
4	0.07	0.06-0.08	0.02	0.02-0.02	<0.0001
5	0.19	0.18-0.21	0.06	0.05-0.07	<0.0001
6A	0.30	0.26-0.34	0.14	0.12-0.16	<0.0001
6B	0.44	0.40-0.48	0.28	0.25-0.31	<0.0001
7F	0.19	0.16-0.21	0.11	0.10-0.13	<0.0001
9V	0.20	0.18-0.23	0.07	0.06-0.08	<0.0001
14	1.36	1.24-1.49	0.86	0.78-0.95	<0.0001
18C	0.20	0.17-0.23	0.08	0.07-0.10	<0.0001
19A	0.81	0.74-0.88	0.50	0.45-0.55	<0.0001
19F	0.74	0.69-0.80	0.47	0.43-0.52	<0.0001
23F	0.33	0.30-0.36	0.17	0.15-0.19	<0.0001

*FAS was based on post-immunization findings, there were 3 cases lacking pre-immunization data in PCV7 (3M) group.

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Table 7-3 Pre-Immunization IgG GMC in Subjects of PCV13-TT Group (3M) and PCV7 Group (3M) (FAS)
PCV13-TT group (3M)
PCV7 (3M) group (N=516)

Serotypes (N=518) P	GMC	95%CI	GMC	95%CI	
1	0.10	0.09-0.11	0.11	0.09-0.12	0.4735
3	0.06	0.06-0.07	0.05	0.05-0.06	0.0654
4	0.02	0.02-0.02	0.02	0.02-0.02	0.8132
5	0.05	0.05-0.06	0.06	0.05-0.07	0.5256
6A	0.15	0.13-0.17	0.14	0.12-0.16	0.4598
6B	0.28	0.25-0.31	0.28	0.25-0.31	0.8011
7F	0.12	0.10-0.13	0.11	0.10-0.13	0.7522
9V	0.08	0.07-0.10	0.07	0.06-0.08	0.1417
14	0.86	0.79-0.95	0.86	0.78-0.95	0.9882
18C	0.08	0.07-0.10	0.08	0.07-0.10	0.8102
19A	0.50	0.45-0.56	0.50	0.45-0.55	0.8934
19F	0.46	0.42-0.50	0.47	0.43-0.52	0.6131
23F	0.18	0.17-0.21	0.17	0.15-0.19	0.3647

*FAS was based on post-immunization findings, there was 1 and 3 cases lacking pre-immunization data in PCV13-TT group (3M) and PCV7 (3M) group, respectively.

Table 7-4 Pre-Immunization IgG GMC in Subjects of PCV13-TT Group and PCV7 Group Aged 7-11 Months (FAS)
PCV13-TT group (N=195) PCV7 group (N=199)

Serotypes P	GMC	95%CI	GMC	95%CI	
1	0.17	0.14-0.21	0.18	0.15-0.22	0.7154
3	0.06	0.05-0.08	0.07	0.06-0.09	0.3486
4	0.02	0.01-0.02	0.02	0.02-0.03	0.3056
5	0.10	0.08-0.13	0.13	0.10-0.15	0.1088
6A	0.07	0.05-0.09	0.08	0.06-0.11	0.3547
6B	0.30	0.25-0.36	0.33	0.29-0.38	0.3752
7F	0.07	0.05-0.10	0.08	0.06-0.11	0.7636
9V*	0.01	0.01-0.02	0.01	0.01-0.02	0.9929
14	0.47	0.39-0.56	0.51	0.44-0.60	0.4415
18C	0.01	0.01-0.01	0.01	0.01-0.01	0.9404
19A	0.42	0.34-0.54	0.51	0.41-0.63	0.2790

19F 0.52 0.46-0.60 0.49 0.43-0.57 0.5382
23F 0.14 0.11-0.17 0.16 0.14-0.19 0.2444
*There was 1 case without results for serotype 9V due to inadequate serum.

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Table 7-5 Pre-Immunization IgG GMC in Subjects of PCV13-TT Group and PCV7
Group Aged 12-23 Months (FAS)
PCV13-TT group (N=200) PCV7 group (N=200)

Serotypes P	GMC	95%CI	GMC	95%CI
1	0.29	0.25-0.34	0.27	0.23-0.32
3	0.11	0.09-0.13	0.12	0.10-0.14
4	0.06	0.04-0.08	0.07	0.05-0.09
5	0.21	0.18-0.25	0.23	0.20-0.27
6A	0.27	0.22-0.34	0.30	0.24-0.38
6B	0.60	0.53-0.68	0.65	0.58-0.73
7F	0.25	0.20-0.31	0.33	0.29-0.38
9V	0.04	0.03-0.06	0.05	0.04-0.07
14	1.01	0.87-1.17	1.07	0.92-1.24
18C	0.02	0.02-0.03	0.02	0.02-0.03
19A	1.05	0.90-1.22	1.15	1.02-1.30
19F	1.00	0.89-1.13	1.05	0.93-1.18
23F	0.35	0.30-0.41	0.38	0.32-0.44

Table 7-6 Pre-Immunization IgG GMC in Subjects of PCV13-TT Group and PCV7
Group Aged 24-71 Months (FAS)
PCV13-TT group (N=200) PCV7 group (N=200)

Serotypes P	GMC	95% CI	%	GMC	95% CI
1	0.35	0.30-0.40	0.28	0.24-0.32	0.0289
3	0.26	0.22-0.31	0.19	0.15-0.24	0.0353
4	0.27	0.22-0.34	0.17	0.14-0.22	0.0074
5	0.43	0.39-0.48	0.29	0.25-0.34	<0.0001
6A	0.58	0.49-0.69	0.38	0.31-0.47	0.0021
6B	1.09	0.97-1.22	0.87	0.76-1.01	0.0192
7F	0.45	0.39-0.52	0.41	0.34-0.49	0.3823
9V	0.26	0.19-0.34	0.19	0.14-0.26	0.1584
14	2.17	1.86-2.53	1.82	1.50-2.21	0.1653
18C	0.15	0.11-0.21	0.09	0.06-0.13	0.0386
19A	1.84	1.64-2.07	1.45	1.23-1.71	0.0211
19F	2.07	1.84-2.33	1.63	1.44-1.85	0.0064
23F	0.62	0.55-0.70	0.46	0.39-0.54	0.0039

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5.7 CONCOMITANT MEDICATIONS

During the observation period of basic immunization phase, among subjects at the 3–71 months of age, 812 subjects (3753 cases) in the PCV13-TT group and 840 subjects (3728 cases) in the PCV7 group recorded concomitant medications, and among subjects at the 2 months of age, 495 subjects (3418 cases) recorded concomitant medications. The most commonly used drugs were vaccines and biological products, especially EPI vaccines for subjects who were at the 2–23 months of age, followed by antipyretic/analgesic/non-steroidal anti-inflammatory drugs, antibiotics and antiallergic drugs. The incidence of concomitant medications was the highest in subjects at the 2 and 3 months of age, and the lowest in subjects at the 24–71 months of age. There were no statistical differences between the PCV13-TT group and the PCV7 group in the incidence of various concomitant medications. During the observation period, the concomitant medications of the subjects are shown in Tables 8-1 to 8-6.

Table 8-1 Concomitant Medications in Subjects 3-71 Months of Age During the Clinical Trial Period

PCV13-TT group	PCV7 group (N=1119)	(N=1117)	Numb	Numb	Medication Types	Numb	Numb	P
er of	er of	er of	%	%	er of	%	er of	%
Event	Event	Cases	Cases	s	s			
Any medication	3753	812	72.69	3728	840	75.07	0.2108	
Hormone/steroids	37	29	2.60	42	29	2.59	1.0000	
Allergy medications	160	126	11.28	158	130	11.62	0.8421	
Anti-pyretics/analgesics/NSAIDs	399	264	23.63	394	280	25.02	0.4599	
Any vaccines or biological products	2000	651	58.28	1981	663	59.25	0.6657	
Antibiotics	301	211	18.89	319	219	19.57	0.7073	
Antivirals	73	60	5.37	94	77	6.88	0.1581	
Other medications	723	304	27.22	687	309	27.61	0.8497	
Unspecified	60	48	4.30	53	40	3.57	0.3867	

Table 8-2 Concomitant Medications in Subjects 2 Aged Months
PCV13-TT group (N=517)

Medication Types	Number of	Number of	%
Events	Cases		
Any medication	3418	495	95.74

Hormone/steroids 10 7 1.35
Antivirals 34 29 5.61
Allergy medications 81 68 13.15
Antibiotics 123 94 18.18
Any vaccines or biological products 2680 483 93.42
Anti-pyretics/analgesics/NSAIDs 148 116 22.44
Other medications 314 148 28.63
Unspecified 28 19 3.68

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Table 8-3 Concomitant Medications in Subjects Aged 3 Months
PCV13-TT group (N=519) PCV7 group (N=519)
Numbe Numbe Numbe Numbe
Medication Types P
r of r of % r of r of %
Events Cases Events Cases
Any medication 2551 474 91.33 2442 468 90.17 0.5924
Hormone/steroids 14 10 1.93 26 18 3.47 0.1790
Antivirals 33 28 5.39 46 38 7.32 0.2521
Allergy medications 79 64 12.33 77 65 12.52 1.0000
Antibiotics 152 108 20.81 149 102 19.65 0.6993
Any vaccines or biological products 1599 439 84.59 1546 418 80.54 0.1016
Anti-pyretics/analgesics/NSAIDs 207 135 26.01 186 126 24.28 0.5671
Other medications 436 162 31.21 380 174 33.53 0.4656
Unspecified 31 24 4.62 32 20 3.85 0.6444

Table 8-4 Concomitant Medications in Subjects Aged 7-11 Months
PCV13-TT group (N=198) PCV7 group (N=200)
Numbe Numbe Numbe Numbe
Medication Types P
r of r of % r of r of %
Events Cases Events Cases
Any medication 601 163 82.32 705 178 89.00 0.0635
Hormone/steroids 9 8 4.04 6 4 2.00 0.2577
Antivirals 13 12 6.06 20 17 8.50 0.4412
Allergy medications 32 25 12.63 38 30 15.00 0.5619
Antibiotics 69 47 23.74 84 55 27.50 0.4225
Any vaccines or biological products 254 136 68.69 273 152 76.00 0.1168
Anti-pyretics/analgesics/NSAIDs 100 66 33.33 110 74 37.00 0.4637
Other medications 114 58 29.29 162 61 30.50 0.8271
Unspecified 10 9 4.55 12 11 5.50 0.8193

Table 8-5 Concomitant Medications in Subjects Aged 12-23 Months
PCV13-TT group (N=200) PCV7 group (N=200)
Numbe Numbe Numbe Numbe
Medication Types P
r of r of % r of r of %
Events Cases Events Cases
Any medication 481 125 62.50 419 129 64.50 0.7554
Hormone/steroids 9 7 3.50 6 3 1.50 0.3375
Antivirals 24 17 8.50 18 14 7.00 0.7090
Allergy medications 34 25 12.50 26 20 10.00 0.5272
Antibiotics 57 38 19.00 54 36 18.00 0.8976
Any vaccines or biological products 145 74 37.00 159 90 45.00 0.1272
Anti-pyretics/analgesics/NSAIDs 60 40 20.00 62 49 24.50 0.3362
Other medications 137 58 29.00 86 39 19.50 0.0354
Unspecified 15 11 5.50 8 8 4.00 0.6394

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Table 8-6 Concomitant Medications in Subjects Aged 24-71 Months
PCV13-TT group (N=200) PCV7 group (N=200)
Numbe Numbe Numbe Numbe
Medication Types P
r of r of % r of r of %
Events Cases Events Cases
Any medication 120 50 25.00 162 65 32.50 0.1218
Hormone/steroids 5 4 2.00 4 4 2.00 1.0000
Antivirals 3 3 1.50 10 8 4.00 0.2201
Allergy medications 15 12 6.00 17 15 7.50 0.6909
Antibiotics 23 18 9.00 32 26 13.00 0.2632
Any vaccines or biological products 2 2 1.00 3 3 1.50 1.0000
Anti-pyretics/analgesics/NSAIDs 32 23 11.50 36 31 15.50 0.3057
Other medications 36 26 13.00 59 35 17.50 0.2658
Unspecified 4 4 2.00 1 1 0.50 0.3719

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Study Results

6. IMMUNOGENICITY ANALYSIS

According to the protocol, the primary end point measurements of immunogenicity were the proportion of subjects whose serotype-specific pneumonia IgG antibody concentration ≥ 0.35 $\mu\text{g/mL}$ (positive rate) and the geometric mean concentration (GMC) of IgG obtained 30 days after subjects in the study and control group completed the primary immunization series.

The positive conversion rate was defined as the proportion of subjects whose serotype-specific IgG antibody concentration ≥ 0.35 $\mu\text{g/mL}$ after immunization in the original protocol. Since the pre-immune blood samples were examined before immunization in this clinical trial, and considering that the report would evaluate the antibody positive rate in the pre-immune antibody negative population, in order to distinguish the two descriptions, in this report, the proportion of subjects with serotype-specific IgG antibody concentration $\geq 0.35\mu\text{g/mL}$ after vaccination was described as "positive rate" in this report; the proportion of subjects whose pre-immune antibody concentration $< 0.35\mu\text{g/mL}$ while the post-immune antibody concentration was higher than 0.35 g/ml was described as "positive conversion rate" .

The 7 common serotypes in the study and control group all reached the standard of non-inferiority, which meant that the lower limit of bilateral 97.5% confidence interval (CI) of the positive rate difference (the study group minus the control group) was greater than -10%, or the lower limit of bilateral 97.5% confidence interval (CI) of the ratio of antibody GMC (the study group divided by the control group) was greater than 0.5.

The 6 additional serotypes were superior to the 7-valent vaccine, which meant that the lower limit of bilateral 95% confidence interval of antibody positive rate was greater than 0, the lower limit of bilateral 95% confidence interval of the ratio of antibody GMC was greater than 1, and the lower limit of bilateral 95% confidence interval of IgG antibody positive rate was no less than 70%.

The analysis would be based on the premise that all subjects met the requirements of the protocol, and effective serum samples needed to be collected from the subjects before and after immunization. Therefore, the PPS data set was used as the main analysis set in immunogenicity analysis.

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6.1. PRIMARY ENDPOINTS ANALYSIS

6.1.1. Positive rate of IgG antibody and the IgG antibody concentrations (GMCs) analysis for subjects aged 3-71 months after immunization in PCV13-TT and control group

6.1.1.1. Comparison of percentage of subjects achieving IgG antibody concentration ≥ 0.35 $\mu\text{g/mL}$ and the IgG antibody geometric mean concentrations (GMCs)

According to the protocol, the subjects aged 7-11 months, 12-23 months and 24-71 months were incorporate into the subjects aged 3 months for non-inferiority analysis (hereinafter referred to as "3-71 months").

Among the 7 common serotypes the proportion of responders at ≥ 0.35 $\mu\text{g/mL}$ in the PCV13-TT was more than 95%. The lower limit of 2-sided, 97.5% CI for the differences of positive rate between PCV13-TT and control group was greater than -10% (refer to table 9-1) in primary series. The GMCs for 6 of the 7 common serotypes (except serotype 6B) were greater than 3.0 $\mu\text{g/mL}$ in PCV13-TT and met the GMC ratio non-inferiority criterion (lower limit of 2-sided, 97.5% CI for the GMR>0.5) (refer to table 9-3). For serotype 6B, the GMC was 2.68 $\mu\text{g/mL}$. Therefore, the positive rate and GMC of 7 common serotype in the PCV13-TT were noninferior to the control group.

Among the 6 additional serotypes the proportion of responders at ≥ 0.35 $\mu\text{g/mL}$ in the PCV13-TT was at least 96%. The lower limit of 2-sided, 95% CI for the positive rate in PCV13-TT was greater than 70%. The lower limit of 2-sided, 95% CI differences of positive rate was greater than 0 and the GMC ratio (test group/control group) was greater than 1 between PCV13-TT and control group. Therefore, the positive rate and GMC of each additional serotype in the PCV13-TT were superiority to the control group.

The statistical analysis of positive rate of IgG antibody and the IgG antibody concentrations (GMCs) for subjects aged 3-71 months after immunization in PCV13-TT and control group was shown in tables 9-1 to 9-4.

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Table 9-1 Comparison of Subjects Aged 3-71 months Achieving a Pneumococcal IgG Antibody Concentration $>0.35\mu\text{g/mL}$ After immunization (PPS)

Noninferior / Superiority			
PCV13-TT (N=1043) Control group (N=1064)			
test			
Serotype			
Differenc 95%/97.5%CI			
Na % 95% CI Nb % 95% CI			
e(%) *			
104	100.	99.65-	105 99.5
4	98.91-	99.85	0.47 -0.00,0.94
3	0	100.0	9 3
93.62-	105	98.9	
6B	992	95.11	98.16-99.48 -3.86 -5.11,-2.21
96.34	3	7	
103	98.8	98.00-	105 99.1
9V	98.40-	99.61	-0.30 -1.28,0.67
1	5	99.40	5 5
104	99.9	99.47-	106 99.8
Common	14	99.32-	99.98 0.09 -0.27,0.46
2	0	100.0	2 1
103	98.9	98.12-	105 98.8
18C	98.04-	99.42	0.07 -0.94,1.09
2	5	99.47	2 7
104	100.	99.65-	106 100.
19F	99.65-	100.0	0.00 -0.41,0.42
3	0	100.0	4 0
102	97.9	96.94-	104 98.3
23F	97.34-	98.99	-0.32 -1.64,1.00
2	9	98.75	6 1

103 99.3 98.62- 64.6
1 688 61.70-67.54 34.67 31.75,37.58
6 3 99.73 6
100 96.3 95.03- 48.5
3 516 45.45-51.55 47.86 44.65,51.07
5 6 97.41 0
96.9 95.70- 36.9
Additional 5 1011 393 34.03-39.92 60.00 56.91,63.08
3 97.89 4

I
101 96.8 95.59- 83.5
6A 889 81.19-85.73 13.28 10.82,15.75
0 4 97.81 5
103 99.5 98.88- 45.3
7F 482 42.28-48.35 54.22 51.20,57.24
8 2 99.84 0
19 104 100. 99.65- 104 98.1
97.11-98.85 1.88 1.06,2.70
A 3 0 100.0 4 2

Na: Number of subjects for PCV13-TT.
Nb: Number of subjects for control group.
*CI■97.5% was CI for Common serotype■95% was CI for additional serotype.

Table 9-2 Comparison of Subjects Aged 3-71 months Achieving a Pneumococcal IgG Antibody Concentration >0.35µg/mL After immunization (FAS)
PCV13-TT (N=1117) Control group (N=1119) Noninferior / Superiority test

Serotype	Na	%	95% CI	Nb	%	95% CI	Difference(%)	95%/97.5%CI*
95.46-	1080	96.51	95.27-97.51	0.17				
4	1080	96.69	97.66	-1.54,1.89				
92.15-	1091	97.50	96.40-98.33	-3.76				
6B	1047	93.73	95.08	-5.70,-1.83				
95.46-	1080	96.51	95.27-97.51	0.17				
9V**	1079	96.68	97.66	-1.55,1.89				
98.83-	Common	1111	99.29	98.60-99.69	0.18			
14	1111	99.46	99.80	-0.57,0.93				
94.95-	1078	96.34	95.06-97.36	-0.10				
18C	1075	96.24	97.28	-1.89,1.70				
99.22-	1111	99.29	98.60-99.69	0.45				
19F	1114	99.73	99.94	-0.22,1.11				
94.04-	1076	96.16	94.86-97.21	-0.72				
23F	1066	95.43	96.58	-2.63,1.18				

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95.16-
709 63.36 60.46-66.19 33.06
1 1077 96.42 97.43 30.03,36.08
91.66-
536 47.90 44.94-50.87 45.39
3 1042 93.29 94.68 42.11,48.66
92.05-
401 35.84 33.02-38.72 57.81
5 1046 93.64 95.00 54.66,60.96
Additional
92.74-
917 81.95 79.57-84.16 12.32
6A 1053 94.27 95.56 9.69,14.96
96.29-
496 44.33 41.39-47.29 53.08
7F 1088 97.40 98.25 50.02,56.13
99.35-
1093 97.68 96.61-98.48 2.14
19A 1115 99.82 99.98 1.23,3.06

a
N : Number of subjects for PCV13-TT.
Nb: Number of subjects for control group.
*CI■97.5% was CI for Common serotype■95% was CI for additional serotype.

Table 9-3 Comparison of the IgG antibody geometric mean concentrations (GMCs) - Post dose 3- between PCV13-TT and PCV7 groups (PPS)
PCV13-TT (N=1043) Control group (N=1064) Noninferior / Superiority test

Serotype	GMC	95%CI	GMC	95%CI	Ratio#	95%/97.5%CI*
4	5.18	4.94-5.43	6.60	6.28-6.93	0.78	0.73,0.85
6B	2.68	2.46-2.92	3.11	2.92-3.30	0.86	0.77,0.97
9V	3.32	3.15-3.50	3.72	3.53-3.93	0.89	0.82,0.97
Common	14	9.78	9.22-10.37	16.04	15.07-17.07	0.61 0.55,0.67
18C	4.15	3.91-4.41	4.74	4.47-5.03	0.88	0.80,0.96
19F	8.03	7.65-8.42	6.27	5.96-6.60	1.28	1.18,1.39
23F	3.52	3.30-3.74	3.76	3.53-4.00	0.94	0.85,1.03
1	2.16	2.07-2.26	0.35	0.32-0.38	6.20	5.63,6.83
3	1.29	1.24-1.36	0.37	0.35-0.39	3.48	3.24,3.73
5	1.41	1.34-1.48	0.12	0.10-0.13	12.07	10.48,13.90

Additional
6A 2.08 1.94-2.23 0.75 0.68-0.83 2.76 2.45,3.12
7F 5.34 5.07-5.62 0.13 0.11-0.15 40.06 34.21,46.91
19A 5.46 5.19-5.73 1.64 1.53-1.76 3.32 3.05,3.62
#Ratio: PCV13-TTGMC/PCV7GMC*CI: 97.5% for common serotypes, 95% for additional serotypes.
*CI97.5% was CI for Common serotype95% was CI for additional serotype.

Table 9-4 Comparison of the IgG antibody geometric mean concentrations (GMCs) - Post dose 3- between PCV13-TT and PCV7 groups (FAS)
Noninferior / Superiority
PCV13-TT (N=1117) PCV7 (N=1119)

Serotype test
GMC 95%CI GMC 95%CI Ratio# 95%/97.5%CI*
4 4.14 3.81-4.50 5.42 4.99-5.90 0.76 0.67,0.87
6B 2.52 2.31-2.74 2.91 2.73-3.11 0.87 0.77,0.98
9V** 2.90 2.71-3.11 3.20 2.96-3.45 0.91 0.81,1.02
common 14 8.97 8.41-9.56 14.66 13.68-15.70 0.61 0.55,0.68
18C 3.47 3.20-3.77 4.04 3.72-4.38 0.86 0.75,0.98
19F 7.34 6.97-7.74 5.82 5.50-6.16 1.26 1.16,1.38
23F 3.17 2.96-3.40 3.40 3.17-3.66 0.93 0.83,1.05
1 1.91 1.79-2.03 0.33 0.31-0.36 5.71 5.13,6.34
Additional
3 1.17 1.11-1.24 0.35 0.33-0.38 3.31 3.05,3.59

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5 1.22 1.13-1.30 0.11 0.10-0.13 10.96 9.48,12.67
6A 1.92 1.78-2.06 0.72 0.65-0.79 2.67 2.36,3.02
7F 4.72 4.42-5.04 0.13 0.11-0.15 35.80 30.58,41.89
19A 5.06 4.80-5.33 1.59 1.48-1.70 3.19 2.92,3.47
#Ratio: PCV13-TTGMC/PCV7GMC*CIcommon 97.5%Additional 95%
**The subjects number for serotype 9V in PCV13-TT group was1116.
6.1.1.2. Pneumococcal Serotype Antibody Concentrations in the Subjects aged 3-71 months in PCV13-TT and PCV7 group
The results of antibody concentrations in the PCV13-TT and control group aged 3-71 months were shown in Table 10-1 to 10-13, and the reverse cumulative distribution curves were shown in Figures 9 to 22.

Table 10-1 Distribution of the pneumococcal serotype 4 IgG antibody concentrations for subjects aged 3-71 in PCV13-TT group and PCV7 group

FAS N (%) PPS N (%)
PCV13-TT PCV13-TT
concentration PCV7 (N=1119) PCV7 (N=1064)
(N=1117) (N=1043)
N % N % N % N %
≥0.01 1092 97.76 1094 97.77 1043 100.0 1063 99.91
≥0.1 1092 97.76 1094 97.77 1043 100.0 1063 99.91
≥1 1060 94.90 1071 95.71 1024 98.18 1050 98.68
≥10 204 18.26 315 28.15 201 19.27 308 28.95
≥100 0 0.00 0 0.00 0 0.00 0 0.00

Figure 9 Reverse cumulative distribution curves, pneumococcal serotype 4, IgG antibody concentrations for subjects in PCV13-TT) group and PCV7 group

Table 10-2 Distribution of the pneumococcal serotype 6B IgG antibody concentrations for subjects aged 3-71 in PCV13-TT group and PCV7 group

FAS N (%) PPS N (%)
PCV13-TT
concentration PCV7 (N=1119) PCV13-TT (N=1043) PCV7 (N=1064)
(N=1117)
N % N % N % N %
≥0.01 1105 98.93 1117 99.82 1032 98.95 1064 100.0

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≥0.1 1105 98.93 1117 99.82 1032 98.95 1064 100.0
≥1 852 76.28 948 84.72 822 78.81 927 87.12
≥10 156 13.97 132 11.80 150 14.38 126 11.84
≥100 4 0.36 2 0.18 4 0.38 2 0.19

Serotype 6B, After Full Immunization (FAS) Serotype 6B, After Full Immunization (PPS)

100 100
90 90
80 80
70 70

% of Subjects
% of Subjects

60 60
50 50
40 40
30 30
20 20
10 10
0 0
0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

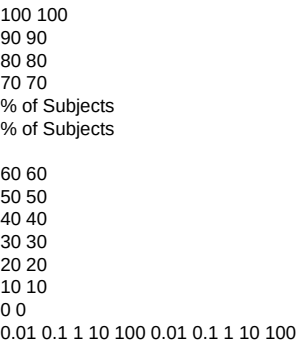
Figure 10. Reverse cumulative distribution curves, pneumococcal serotype 6B, IgG antibody concentrations for subjects in PCV13-TT) group and PCV7 group

Table 10-3 Distribution of the pneumococcal serotype 9V IgG antibody concentrations for subjects aged 3-71 in PCV13-TT group and PCV7 group

FAS N (%)	PPS N (%)	PCV13-TT	PCV13-TT
Concentration	PCV7 (N=1119)	PCV7 (N=1064)	(N=1117) (N=1043)
N %	N %	N %	N %
≥0.01	1102	98.75	1097 98.03 1042 99.90 1062 99.81
≥0.1	1102	98.75	1097 98.03 1042 99.90 1062 99.81
≥1	991	88.80	1023 91.42 959 91.95 1002 94.17
≥10	101	9.05	121 10.81 98 9.40 117 11.00
≥100	0	0.00	0 0.00 0 0.00 0 0.00

*The subjects for serotype 9Vfor PCV13-TT group in FAS was1116.

Serotype 9V, After Full Immunization (FAS) Serotype 9V, After Full Immunization (PPS)



Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

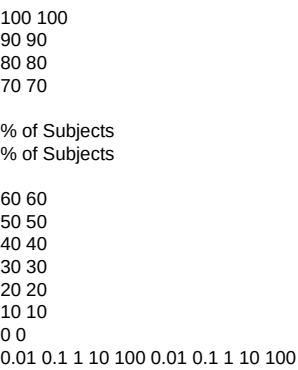
PCV13-TT PCV7 PCV13-TT PCV7

Figure 21. Reverse cumulative distribution curves, pneumococcal serotype 94, IgG antibody concentrations for subjects in PCV13-TT group and PCV7 group
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Table 10-4 Distribution of the pneumococcal serotype 14 IgG antibody concentrations for subjects aged 3-71 in PCV13-TT group and PCV7 group

FAS N (%)	PPS N (%)	PCV13-TT	PCV13-TT
Concentration	PCV7 (N=1119)	PCV7 (N=1064)	(N=1117) (N=1043)
N %	N %	N %	N %
≥0.01	1116	99.91	1116 99.73 1043 100.0 1062 99.81
≥0.1	1115	99.82	1116 99.73 1042 99.90 1062 99.81
≥1	1082	96.87	1093 97.68 1036 99.33 1059 99.53
≥10	543	48.61	780 69.71 527 50.53 764 71.80
≥100	17	1.52	41 3.66 14 1.34 39 3.67

Serotype 14, After Full Immunization (FAS) Serotype 14, After Full Immunization (PPS)



Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

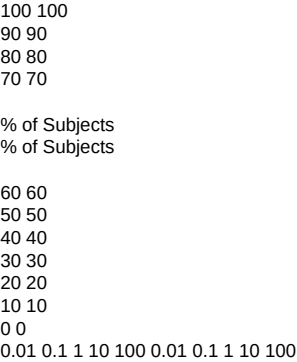
Figure 32. Reverse cumulative distribution curves, pneumococcal serotype 14, IgG antibody concentrations for subjects in PCV13-TT) group and PCV7 group

Table 10-5 Distribution of the pneumococcal serotype 18C IgG antibody concentrations for subjects aged 3-71 in PCV13-TT group and PCV7 group

FAS N (%)	PPS N (%)
PCV13-TT	PCV13-TT
Concentration	PCV7 (N=1119)
(N=1117)	(N=1043)
N %	N %
≥0.01	1095 98.03 1096 97.94 1040 99.71 1061 99.72
≥0.1	1095 98.03 1096 97.94 1040 99.71 1061 99.72
≥1	1023 91.58 1041 93.03 991 95.01 1020 95.86
≥10	172 15.40 205 18.32 168 16.11 200 18.80
≥100	1 0.09 5 0.45 1 0.10 5 0.47

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Serotype 18C, After Full Immunization (FAS) Serotype 18C, After Full Immunization (PPS)



Antibody Concentration (µg/mL)
Antibody Concentration (µg/mL)

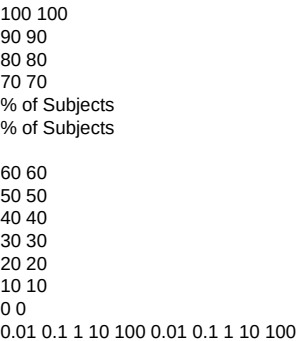
PCV13-TT PCV7 PCV13-TT PCV7

Figure 40. Reverse cumulative distribution curves, pneumococcal serotype 18C, IgG antibody concentrations for subjects in PCV13-TT) group and PCV7 group

Table 10-6 Distribution of the pneumococcal serotype 19F IgG antibody concentrations for subjects aged 3-71 in PCV13-TT group and PCV7 group

FAS N (%)	PPS N (%)
PCV13-TT	PCV13-TT
Concentration	PCV7 (N=1119)
(N=1117)	(N=1043)
N %	N %
≥0.01	1117 100.0 1118 99.91 1043 100.0 1064 100.0
≥0.1	1117 100.0 1118 99.91 1043 100.0 1064 100.0
≥1	1075 96.24 1072 95.80 1030 98.75 1041 97.84
≥10	410 36.71 321 28.69 400 38.35 314 29.51
≥100	4 0.36 1 0.09 4 0.38 1 0.09

Serotype 19F, After Full Immunization (FAS) Serotype 19F, After Full Immunization (PPS)



Antibody Concentration (µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

Figure 50. Reverse cumulative distribution curves, pneumococcal serotype 19F, IgG antibody concentrations for subjects in PCV13-TT) group and PCV7 group

Table 10-7 Distribution of the pneumococcal serotype 23F IgG antibody concentrations for subjects aged 3-71 in PCV13-TT group and PCV7 group

FAS N (%)	PPS N (%)
PCV13-TT	PCV13-TT
Concentration	PCV7 (N=1119)
(N=1117)	(N=1043)
N %	N %
≥0.01	1117 100.0 1118 99.91 1043 100.0 1064 100.0
≥0.1	1117 100.0 1118 99.91 1043 100.0 1064 100.0
≥1	1075 96.24 1072 95.80 1030 98.75 1041 97.84
≥10	410 36.71 321 28.69 400 38.35 314 29.51
≥100	4 0.36 1 0.09 4 0.38 1 0.09

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≥0.01 1112 99.55 1111 99.29 1041 99.81 1062 99.81
≥0.1 1112 99.55 1111 99.29 1041 99.81 1062 99.81
≥1 980 87.74 1002 89.54 949 90.99 983 92.39
≥10 142 12.71 165 14.75 134 12.85 160 15.04
≥100 0 0.00 5 0.45 0 0.00 4 0.38

Serotype 23F, After Full Immunization (FAS) Serotype 23F, After Full Immunization (PPS)

100 100
90 90
80 80
70 70

% of Subjects
% of Subjects

60 60
50 50
40 40
30 30
20 20
10 10
0 0
0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

Figure 60. Reverse cumulative distribution curves, pneumococcal serotype 23F, IgG antibody concentrations for subjects in PCV13-TT) group and PCV7 group

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Table 10-8 Distribution of the pneumococcal serotype 1 IgG antibody concentrations for subjects aged 3-71 in PCV13-TT group and PCV7 group

FAS N (%)	PPS N (%)						
PCV13-TT	PCV13-TT						
Concentration	PCV7 (N=1119)	PCV7 (N=1064)					
(N=1117)	(N=1043)						
N %	N %	N %	N %	N %	N %	N %	N %
≥0.01	1105	98.93	1047	93.57	1043	100.0	1000 93.98
≥0.1	1105	98.93	1044	93.30	1043	100.0	999 93.89
≥1	947	84.78	107	9.56	915	87.73	104 9.77
≥10	21	1.88	1	0.09	19	1.82	1 0.09
≥100	0	0.00	0	0.00	0	0.00	0 0.00

Serotype 1, After Full Immunization (FAS) Serotype 1, After Full Immunization (PPS)

100 100
90 90
80 80
70 70

% of Subjects
% of Subjects

60 60
50 50
40 40
30 30
20 20
10 10
0 0
0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

Figure 70. Reverse cumulative distribution curves, pneumococcal serotype 1, IgG antibody concentrations for subjects in PCV13-TT) group and PCV7 group

Table 10-9 Distribution of the pneumococcal serotype 3 IgG antibody concentrations for subjects aged 3-71 in PCV13-TT group and PCV7 group

FAS N (%)	PPS N (%)						
PCV13-TT	PCV13-TT						
Concentration	PCV7 (N=1119)	PCV7 (N=1064)					
(N=1117)	(N=1043)						
N %	N %	N %	N %	N %	N %	N %	N %
≥0.01	1114	99.73	1110	99.20	1043	100.0	1061 99.72
≥0.1	1090	97.58	1063	95.00	1042	99.90	1031 96.90
≥1	671	60.07	118	10.55	650	62.32	109 10.24
≥10	8	0.72	3	0.27	7	0.67	3 0.28
≥100	0	0.00	0	0.00	0	0.00	0 0.00

Serotype 3, After Full Immunization (FAS) Serotype 3, After Full Immunization (PPS)

100 100
90 90
80 80
70 70

% of Subjects
% of Subjects

60 60
50 50
40 40
30 30
20 20
10 10
0 0
0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

Figure 80. Reverse cumulative distribution curves, pneumococcal serotype 3, IgG antibody concentrations for subjects in PCV13-TT) group and PCV7 group

Table 10-10 Distribution of the pneumococcal serotype 5 IgG antibody concentrations for subjects aged 3-71 in PCV13-TT group and PCV7 group

FAS N (%)	PPS N (%)	PCV13-TT	PCV13-TT	PCV7	PCV7	
Concentration	(N=1119)	(N=1064)	(N=1117)	(N=1043)		
N	%	N	%	N	%	
≥0.01	1095	98.03	862	77.03	1040	99.71
≥0.1	1095	98.03	860	76.85	1040	99.71
≥1	699	62.58	38	3.40	682	65.39
≥10	10	0.90	2	0.18	10	0.96
≥100	0	0.00	0	0.00	0	0.00

Serotype 5, After Full Immunization (FAS) Serotype 5, After Full Immunization (PPS)

100 100
90 90
80 80
70 70

% of Subjects
% of Subjects

60 60
50 50
40 40
30 30
20 20
10 10
0 0
0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

Figure 90. Reverse cumulative distribution curves, pneumococcal serotype 5, IgG antibody concentrations for subjects in PCV13-TT) group and PCV7 group

Table 10-11 Distribution of the pneumococcal serotype 6A IgG antibody concentrations for subjects aged 3-71 in PCV13-TT group and PCV7 group

FAS N (%)	PPS N (%)	PCV13-TT	PCV13-TT	PCV7	PCV7	
Concentration	(N=1117)	(N=1043)	(N=1119)	(N=1064)	(N=1064)	
N	%	N	%	N	%	
≥0.01	1101	98.57	1056	94.37	1032	98.95
≥0.1	1100	98.48	1054	94.19	1032	98.95
≥1	859	76.90	476	42.54	825	79.10
≥10	62	5.55	21	1.88	58	5.56
≥100	0	0.00	2	0.18	0	0.00

Serotype 6A, After Full Immunization (FAS) Serotype 6A, After Full Immunization (PPS)

100 100
90 90
80 80

70 70

% of Subjects
% of Subjects

60 60
50 50
40 40
30 30
20 20
10 10
0 0
0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

Figure 100. Reverse cumulative distribution curves, pneumococcal serotype 6A, IgG antibody concentrations for subjects in PCV13-TT) group and PCV7 group

Table 10-12 Distribution of the pneumococcal serotype 7F IgG antibody concentrations for subjects aged 3-71 in PCV13-TT group and PCV7 group

FAS N (%)	PPS N (%)	PCV13-TT	PCV13-TT
Concentration	PCV7 (N=1119)	PCV7 (N=1064)	(N=1117) (N=1043)
N	%	N	%
≥0.01	1109 99.28 855 76.41 1041 99.81 811 76.22		
≥0.1	1107 99.10 848 75.78 1040 99.71 806 75.75		
≥1	1060 94.90 136 12.15 1021 97.89 135 12.69		
≥10	209 18.71 7 0.63 205 19.65 7 0.66		
≥100	1 0.09 0 0.00 0 0.00 0 0.00		

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Serotype 7F, After Full Immunization (FAS) Serotype 7F, After Full Immunization (PPS)

100 100
90 90
80 80
70 70

% of Subjects
% of Subjects

60 60
50 50
40 40
30 30
20 20
10 10
0 0
0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

Figure 110. Reverse cumulative distribution curves, pneumococcal serotype 7F, IgG antibody concentrations for subjects in PCV13-TT group and PCV7 group.

Table 1-13 Distribution of the pneumococcal serotype 19A IgG antibody concentrations for subjects aged 3-71 in PCV13-TT group and PCV7 group

FAS N (%)	PPS N (%)	PCV13-TT	PCV13-TT
Concentration	PCV7 (N=1119)	PCV7 (N=1064)	(N=1117) (N=1043)
N	%	N	%
≥0.01	1117 100.0 1105 98.75 1043 100.0 1052 98.87		
≥0.1	1117 100.0 1105 98.75 1043 100.0 1052 98.87		
≥1	1076 96.33 769 68.72 1029 98.66 741 69.64		
≥10	237 21.22 58 5.18 234 22.44 58 5.45		
≥100	1 0.09 2 0.18 1 0.10 2 0.19		

Serotype 19A, After Full Immunization (FAS) Serotype 19A, After Full Immunization (PPS)

100 100
90 90
80 80
70 70

% of Subjects
% of Subjects

60 60
50 50
40 40
30 30

20 20
10 10
0 0
0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

Figure 120. Reverse cumulative distribution curves, pneumococcal serotype 19A, IgG antibody concentrations for subjects in PCV13-TT) group and PCV7 group.

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6.1.2. Positive rate of IgG antibody and the IgG antibody concentrations (GMCs) analysis for subjects aged 2-6 months after immunization in PCV13-TT and control group

6.1.2.1. Positive rate of IgG antibody and the IgG antibody concentrations (GMCs) analysis for subjects aged 3 months after immunization in PCV13-TT and control group
Among the 7 common serotypes, the proportion of responders at ≥0.35 µg/mL in the PCV13-TT (except serotype 6B, the the proportion of responders at ≥0.35 µg/mL of 6B was 90.45%) was more than 97%. The positive rate and GMCs for 6 of the 7 common serotypes (except serotype 6B) were noninferior to the control group after primary series. The results were shown in tables 11-1 and 11-3.
Among the 6 additional serotypes the proportion of responders at ≥0.35 µg/mL in the PCV13-TT was at least 96%. The lower limit of 2-sided, 95% CI for the positive rate in PCV13-TT was greater than 70%. The lower limit of 2-sided, 95% CI differences of positive rate was greater than 0 and the GMC ratio (testgroup/control group) was greater than 1 between PCV13-TT and control group. Therefore, the positive rate and GMC of each additional serotype in the PCV13-TT were superiority to the control group.

The statistical analysis of positive rate of IgG antibody and the IgG antibody concentrations (GMCs) for subjects aged 3-71 months after immunization in PCV13-TT and control group was shown in tables 11-1 to 11-4.

Table 11-1 Comparison of Subjects Aged 3 months Achieving a Pneumococcal IgG Antibody Concentration >0.35µg/mL After immunization (PPS)

Tset group (N=471) Control group (N=487) Non-inferior / Superiority test

Serotype a	N	%	95% CI	Nb	%	95% CI	Difference(%)	95%/97.5%CI*
99.22- 97.91-	4	483	99.18	0	82			
471	100.0	100.0	99.78	-0.10,1.74				
87.42- 97.62-	6	482	98.97	-8.53				
6B	426	90.45	92.95	99.67	-11.73	■5.32		
97.84- 98.86-	9	486	99.79	-0.64				
9V	467	99.15	99.77	99.99	-1.70,0.41			
99.22- 98.86-	Common	14	486	99.79	0.21			
471	100.0	100.0	99.99	-0.25,0.67				
96.68- 95.99-	18	476	97.74	0.56				
18C	463	98.30	99.26	98.87	-1.45,2.57			
99.22- 99.25-	19	487	100.0	0.00				
19F	471	100.0	100.0	100.0	-0.90,0.93			
95.33- 97.06-	23	480	98.56	-1.32				
23F	458	97.24	98.52	99.42	-3.40,0.76			
98.47- 52.97-	1	280	57.49	42.08				
469	99.58	99.95	61.93	37.65,46.51				
96.40- 40.69-	3	220	45.17	52.91				
462	98.09	99.12	49.72	48.32,57.50				
Additional	96.96-	13.43-						
5	81	16.63	81.88					
464	98.51	99.40	20.24	78.40,85.36				
94.80- 74.52-	6	382	78.44	18.38				
6A	456	96.82	98.21	82.01	14.39,22.36			
98.47- 20.49-	7	118	24.23	75.35				
7F	469	99.58	99.95	28.29	71.49,79.20			
99.22- 95.22-	19	473	97.13	2.87				
19A	471	100.0	100.0	98.42	1.38,4.36			
Na: Number of subjects for PCV13-TT.								
Nb: Number of subjects for control group.								
*CI■97.5% was CI for Common serotype■95% was CI for additional serotype.								

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Table 11-2 Comparison of Subjects Aged 3 months Achieving a Pneumococcal IgG Antibody Concentration >0.35µg/mL After immunization (FAS)

PCV13-TT (N=519) Control group (N=519) Noninferior / Superiority test

Serotype	Na	%	95% CI	Nb	%	95% CI	Difference(%)	95%/97.5%CI*
----------	----	---	--------	----	---	--------	---------------	--------------

4 497 95.76 93.65-97.32 499 96.15 94.11-97.63 -0.39 -3.13,2.36
 6B 463 89.21 86.22-91.75 504 97.11 95.28-98.37 -7.90 -11.37,-4.43
 9V 502 96.72 94.81-98.08 505 97.30 95.52-98.52 -0.58 -2.95,1.79
 Common 14 516 99.42 98.32-99.88 517 99.61 98.61-99.95 -0.19 -1.16,0.77
 18C 493 94.99 92.75-96.70 496 95.57 93.42-97.17 -0.58 -3.53,2.37
 19F 516 99.42 98.32-99.88 513 98.84 97.50-99.57 0.58 -0.71,1.87
 23F 489 94.22 91.85-96.07 501 96.53 94.57-97.93 -2.31 -5.23,0.61
 1 498 95.95 93.88-97.48 294 56.65 52.26-60.96 39.31 34.72,43.89
 3 489 94.22 91.85-96.07 234 45.09 40.75-49.48 49.13 44.40,53.86
 5 489 94.22 91.85-96.07 86 16.57 13.47-20.05 77.65 73.87,81.43
 Additional
 6A 488 94.03 91.63-95.91 403 77.65 73.82-81.16 16.38 12.25,20.50
 7F 504 97.11 95.28-98.37 126 24.28 20.65-28.20 72.83 68.87,76.79
 19A 517 99.61 98.61-99.95 499 96.15 94.11-97.63 3.47 1.73,5.21
 *CI■common 97.5%■95% for additional serotypes.

Table 11-3 Comparison of the IgG antibody geometric mean concentrations (GMCs) - Post dose 3- between PCV13-TT and PCV7 groups (PPS)

Noninferior / Superiority
 PCV13-TT (N=471) PCV7 (N=487)
 Serotype test
 GMC 95%CI
 GMC 95%CI Ratio# 95%/97.5%CI*
 4 4.51 4.20-4.84 7.79 7.22-8.41 0.58 0.51,0.65
 6B 1.25 1.11-1.42 3.52 3.25-3.81 0.36 0.30,0.42
 9V 3.72 3.45-4.02 4.47 4.19-4.78 0.83 0.74,0.93
 common 14 10.38 9.57-11.27 16.53 15.09-18.11 0.63 0.55,0.72
 18C 2.91 2.68-3.15 4.80 4.37-5.28 0.61 0.52,0.70
 19F 6.95 6.49-7.45 9.00 8.43-9.60 0.77 0.69,0.86
 23F 2.93 2.65-3.23 4.44 4.08-4.82 0.66 0.57,0.76
 1 2.15 2.01-2.30 0.30 0.26-0.34 7.17 6.14,8.37
 3 1.67 1.56-1.79 0.33 0.31-0.36 5.02 4.55,5.53
 5 1.88 1.75-2.02 0.05 0.04-0.06 40.29 32.02,50.69
 Additional
 6A 1.96 1.77-2.16 0.58 0.50-0.67 3.38 2.83,4.03
 7F 5.47 5.06-5.92 0.05 0.04-0.07 105.38 82.43,134.72
 19A 3.85 3.62-4.09 1.06 0.97-1.16 3.63 3.25,4.06
 #Ratio: PCV13-TTGMC/PCV7GMC; *CI■common97.5%, Additional 95%

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Table 11-4 Comparison of the IgG antibody geometric mean concentrations (GMCs) - Post dose 3- between PCV13-TT and PCV7 groups (FAS)

Noninferior / Superiority
 PCV13-TT (N=519) PCV7 (N=519)
 ■■■■ ■■■■ test
 GMC 95%CI GMC 95%CI Ratio# 95%/97.5%CI*
 4 3.46 3.06-3.92 6.40 5.64-7.26 0.54 0.44,0.66
 6B 1.23 1.09-1.39 3.26 2.97-3.58 0.38 0.32,0.45
 9V 3.24 2.95-3.56 3.91 3.54-4.32 0.83 0.71,0.97
 common 14 9.32 8.49-10.22 15.45 14.01-17.03 0.60 0.52,0.70
 18C 2.42 2.18-2.70 4.29 3.84-4.79 0.57 0.47,0.67
 19F 6.26 5.80-6.76 8.12 7.48-8.81 0.77 0.68,0.88
 23F 2.59 2.32-2.90 3.95 3.54-4.41 0.65 0.55,0.78
 1 1.79 1.60-1.99 0.29 0.25-0.33 6.27 5.26,7.47
 3 1.45 1.33-1.58 0.32 0.29-0.34 4.58 4.06,5.16
 5 1.50 1.34-1.68 0.05 0.04-0.06 32.80 25.92,41.52
 Additional
 6A 1.80 1.62-2.00 0.57 0.49-0.65 3.18 2.66,3.81
 7F 4.66 4.19-5.19 0.05 0.04-0.07 87.39 68.38,111.69
 19A 3.57 3.35-3.81 1.03 0.94-1.13 3.46 3.09,3.88
 #Ratio: PCV13-TTGMC/PCV7GMC■*CI■common 97.5%■95% for additional serotypes.

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6.1.2.2. Positive rate of IgG antibody and the IgG antibody concentrations (GMCs) analysis for subjects aged 2 months (minimum age of 6 weeks) in PCV13-TT and control group aged 3 months

Among the 7 common serotypes the proportion of responders at $\geq 0.35 \mu\text{g/mL}$ in the PCV13-TT aged 2 months was more than 97% using 0, 2, and 4 month dosing schedule. The positive rate of 7 common serotype in the PCV13-TT aged 2 month was noninferior to the control group aged 3 months using 0, 1, and 2 month dosing schedule, the result was shown in Table 12-1. The GMCs for 5 of the 7 common serotypes (except serotype 4 and 18C) in the PCV13-TT aged 2 month were noninferior to the control group aged 3 months. For serotype 4 and 18C, the GMC was $2.43 \mu\text{g/mL}$ and $2.04 \mu\text{g/mL}$ separately, the result was shown in Table 12-3.

Among the 6 additional serotypes (except serotype 5) the proportion of responders at $\geq 0.35 \mu\text{g/mL}$ in the PCV13-TT aged 2 months was at least 96% and the lower limit of 2-sided, 95% CI for the positive rate in PCV13-TT was greater than 70%. For serotype 5, the proportion of responders at $\geq 0.35 \mu\text{g/mL}$ was 90.87%. The lower limit of 2-sided, 95% CI differences of positive rate was greater than 0 and the GMC ratio (PCV13-TT/control group) was greater than 1 between PCV13-TT aged 2 months and control group aged 3 months. Therefore, the positive rate and GMCs of each additional serotype in the PCV13-TT aged 2 months were superiority to the control group.

The statistical analysis of positive rate of IgG antibody and the IgG antibody concentrations (GMCs) for subjects in PCV13-TT aged 2 months n and control group aged 3 months after immunization was shown in tables 12-1 to 12-4.

Table 12-1 Comparison of Subjects Aged 3 months Achieving a Pneumococcal IgG Antibody Concentration $>0.35 \mu\text{g/mL}$ After immunization -between PCV13-TT (2M) and

PCV7(3M) groups (PPS)
PCV13-TT (2M)
PCV7(3M) (N=487) Noninferior / Superiority test
(N=482)
Serotype
Rate diff.
N % 95%CI N % 95%CI 95%/97.5%CI*
(%)
99.24-
482 100.0 483 99.18 97.91-99.78 0.82 -0.10,1.74
4 100.0
95.95-
471 97.72 482 98.97 97.62-99.67 -1.26 -3.09,0.58
6B 98.86
98.85-
481 99.79 486 99.79 98.86-99.99 -0.00 -0.66,0.65
9V 99.99
98.85-
common 481 99.79 486 99.79 98.86-99.99 -0.00 -0.66,0.65
14 99.99
97.60-
477 98.96 476 97.74 95.99-98.87 1.22 -0.61,3.05
18C 99.66
99.24-
482 100.0 487 100.0 99.25-100.0 0.00 -0.90,0.91
19F 100.0
98.85-
481 99.79 480 98.56 97.06-99.42 1.23 -0.07,2.52
23F 99.99
99.24-
482 100.0 280 57.49 52.97-61.93 42.51 38.11,46.90
1 100.0
93.91-
463 96.06 220 45.17 40.69-49.72 50.88 46.13,55.63
3 97.61
87.94-
Additional 438 90.87 81 16.63 13.43-20.24 74.24 70.05,78.43
5 93.29
97.89-
478 99.17 382 78.44 74.52-82.01 20.73 16.99,24.47
6A 99.77
98.51-
480 99.59 118 24.23 20.49-28.29 75.36 71.51,79.20
7F 99.95
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99.24-
482 100.0 473 97.13 95.22-98.42 2.87 1.39,4.36
19A 100.0
*CI■common97.5%, 95% for additional serotypes.

Table 12-2 Comparison of Subjects Achieving a Pneumococcal IgG Antibody
Concentration >0.35µg/mL After immunization PCV13-TT (2M) and PCV7(3M) groups
(FAS)

PCV13-TT (2M) Noninferior / Superiority
PCV7(3M) (N=519)
(N=517) test
Serotype Rate
N % 95%CI N % 95%CI diff. 95%/97.5%CI*
(%)
92.94-
499 96.15 94.11-97.63 -0.98
4 492 95.16
96.85 -3.82,1.86
94.55-
504 97.11 95.28-98.37 -0.59
6B 499 96.52 97.92 -3.04,1.85
95.50-
505 97.30 95.52-98.52 -0.01
9V 503 97.29 98.51 -2.27,2.25
97.49-
common 517 99.61 98.61-99.95 -0.78
14 511 98.84 99.57 -1.99,0.44
94.55-
496 95.57 93.42-97.17 0.95
18C 499 96.52 97.92 -1.76,3.66
97.76-
513 98.84 97.50-99.57 0.19
19F 512 99.03 99.69 -1.24,1.62
95.98-
501 96.53 94.57-97.93 1.15
23F 505 97.68 98.80 -1.19,3.48
95.02-
294 56.65 52.26-60.96 40.26
1 501 96.91 98.22 35.74,44.78
87.88-
234 45.09 40.75-49.48 45.63
3 469 90.72 93.08 40.67,50.59
84.26-
86 16.57 13.47-20.05 70.86

5 452 87.43 90.16 66.57,75.15
 Additional
 95.02-
 403 77.65 73.82-81.16 19.26
 6A 501 96.91 98.22 15.37,23.14
 95.50-
 126 24.28 20.65-28.20 73.01
 7F 503 97.29 98.51 69.07,76.96
 98.31-
 499 96.15 94.11-97.63 3.27
 19A 514 99.42 99.88 1.49,5.05

*CI: 97.5% for common serotypes, 95% for additional serotypes.

Table 12-3 Comparison of the IgG antibody geometric mean concentrations (GMCs) -

Post dose 3-PCV13-TT (2M) and PCV7(3M) groups (PPS)

PCV13-TT (2M) Noninferior / Superiority

PCV7(3M) (N=487)

Serotype Serotype (N=482) test

GMC 95%CI GMC 95%CI Ratio# 95%/97.5%CI*

4 2.43 2.28-2.60 7.79 7.22-8.41 0.31 0.28,0.35
 6B 3.72 3.37-4.10 3.52 3.25-3.81 1.06 0.91,1.22
 9V 3.66 3.40-3.94 4.47 4.19-4.78 0.82 0.73,0.92
 common 14 10.25 9.46-11.11 16.53 15.09-18.11 0.62 0.54,0.71
 18C 2.04 1.87-2.23 4.80 4.37-5.28 0.43 0.37,0.49
 19F 5.46 5.14-5.80 9.00 8.43-9.60 0.61 0.55,0.67
 23F 3.86 3.53-4.21 4.44 4.08-4.82 0.87 0.76,1.00

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1 2.09 1.95-2.23 0.30 0.26-0.34 6.96 5.97,8.12
 3 1.41 1.32-1.50 0.33 0.31-0.36 4.22 3.83,4.65
 5 0.79 0.73-0.85 0.05 0.04-0.06 16.88 13.42,21.24

Additional

6A 3.28 3.01-3.57 0.58 0.50-0.67 5.65 4.78-6.69
 7F 4.76 4.42-5.13 0.05 0.04-0.07 91.73 72.03,116.82
 19A 3.12 2.95-3.30 1.06 0.97-1.16 2.94 2.64,3.28

#Ratio: PCV13-TT (2M)/PCV7(3M)■*CI: 97.5% for common serotypes, 95% for additional serotypes.

Table 12-4 Comparison of the IgG antibody geometric mean concentrations (GMCs) -

Post dose 3-PCV13-TT (2M) and PCV7(3M) groups (FAS)

PCV13-TT (2M) Noninferior / Superiority

PCV7(3M) (N=519)

Serotype Serotype (N=517) test

GMC 95%CI GMC 95%CI Ratio# 95%/97.5%CI*

4 1.94 1.74-2.17 6.40 5.64-7.26 0.30 0.25,0.37
 6B 3.38 3.05-3.75 3.26 2.97-3.58 1.04 0.89,1.22
 9V 3.11 2.82-3.43 3.91 3.54-4.32 0.80 0.68,0.93
 common 14 8.73 7.87-9.68 15.45 14.01-17.03 0.57 0.48,0.66
 18C 1.82 1.65-2.00 4.29 3.84-4.79 0.42 0.36,0.50
 19F 4.92 4.60-5.27 8.12 7.48-8.81 0.61 0.54,0.69
 23F 3.40 3.10-3.74 3.95 3.54-4.41 0.86 0.73,1.01
 1 1.87 1.73-2.02 0.29 0.25-0.33 6.55 5.59,7.67
 3 1.21 1.11-1.32 0.32 0.29-0.34 3.82 3.39,4.30
 5 0.69 0.63-0.76 0.05 0.04-0.06 15.13 12.03,19.03

Additional

6A 2.85 2.57-3.16 0.57 0.49-0.65 5.04 4.22,6.03
 7F 4.04 3.67-4.43 0.05 0.04-0.07 75.71 59.54,96.28
 19A 2.93 2.76-3.11 1.03 0.94-1.13 2.84 2.54,3.17

#Ratio: PCV13-TT (2M)/PCV7(3M)■*CI: 97.5% for common serotypes, 95% for additional serotypes.

6.1.2.3. Positive rate of IgG antibody and the IgG antibody concentrations (GMCs) analysis for subjects aged 2 months (minimum age of 6 weeks) in PCV13-TT and control group aged 3months

The proportion of responders at ≥ 0.35 $\mu\text{g/mL}$ for 11 of 13 serotypes (except serotype 3 and 5) was more than 97% in PCV13-TT aged 2 months using 0, 2, and 4 month dosing schedule. For serotype 3 and 5, the proportion of responders at ≥ 0.35 $\mu\text{g/mL}$ was 96.06% and 90.87% separately. The lower limit of 2-sided, 97.5% CI differences of positive rate for 12 of 13 serotypes (except serotype 5) in PCV13-TT aged 2 months using 0, 2, and 4 month dosing schedule was less than -10%. For serotype 5, the lower limit of 2-sided, 97.5% CI differences of positive rate was -10.84%. And the positive rate of 12 serotype (except serotype 5) in the PCV13-TT aged 2 month using 0, 2, and 4 month dosing schedule was non-inferior to the PCV13-TT aged 3 months using 0, 1, and 2 month dosing schedule, the result was shown in Table 13-1. The lower limit of 2-sided, 97.5% CI of GMC ratio for 11 of 13 serotypes (except serotype 4 and 5) was greater than 0.5 in PCV13-TT aged 2 months.and PCV13-TT aged 3 months. For serotype 4 and 5, the lower limit of 2-sided, 97.5% CI differences of GMC ratio was 0.48 and 0.37 separately. The GMCs of 11 serotypes in PCV13-TT aged 2 months were non-inferior to the PCV13-TT aged 3 months, the result was shown in Table 13-3.

The statistical analysis of positive rate of IgG antibody and the IgG antibody concentrations (GMCs) for subjects in PCV13-TT aged 2 months and PCV13-TT aged 3 months after immunization was shown in tables 13-1 to 13-4.

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Table 13-1 Comparison of Subjects Achieving a Pneumococcal IgG Antibody

Concentration $>0.35\mu\text{g/mL}$ After immunization-between PCV13-TT (2M) and PCV13-TT

(3M) groups (PPS)

PCV13-TT (2M)

PCV13-TT (3M) (N=471) Noninferiority test

(N=482)

Serotype

Rate diff.

N % 95%CI N % 95%CI 97.5%CI
(%)
99.24-
471 100.0 99.22-100.0 0.00
4 482 100.0 100.0 -0.93,0.91
95.95-
87.42-92.95 7.27
6B 471 97.72 98.86 426 90.45 3.87,10.67
98.85-
97.84-99.77 0.64
9V 481 99.79 99.99 467 99.15 -0.41,1.70
98.85-
common 99.22-100.0 -0.21
14 481 99.79 99.99 471 100.0 -0.67,0.26
97.60-
96.68-99.26 0.66
18C 477 98.96 99.66 463 98.30 -1.03,2.35
99.24-
471 100.0 99.22-100.0 0.00
19F 482 100.0 100.0 -0.93,0.91
98.85-
95.33-98.52 2.55
23F 481 99.79 99.99 458 97.24 0.80,4.31
99.24-
98.47-99.95 0.42
1 482 100.0 100.0 469 99.58 -0.25,1.10
93.91-
96.40-99.12 -2.03
3 463 96.06 97.61 462 98.09 -4.47,0.41
87.94-
96.96-99.40 -7.64
5 438 90.87 93.29 464 98.51 -10.84,-4.45
Additional
97.89-
94.80-98.21 2.35
6A 478 99.17 99.77 456 96.82 0.32,4.39
98.51-
98.47-99.95 0.01
7F 480 99.59 99.95 469 99.58 -0.93,0.95
99.24-
471 100.0 99.22-100.0 0.00
19A 482 100.0 100.0 -0.93,0.91

Table 13-2 Comparison of Subjects Achieving a Pneumococcal IgG Antibody Concentration >0.35µg/mL After immunization-between PCV13-TT (2M) and PCV13-TT (3M) groups (FAS)
PCV13-TT (2M) (N=517) PCV13-TT (3M) (N=519) Noninferiority test
Serotype Rate diff.

N % 95%CI N % 95%CI 97.5%CI
(%)
4 492 95.16 92.94-96.85 497 95.76 93.65-97.32 -0.60 -3.50,2.30
6B 499 96.52 94.55-97.92 463 89.21 86.22-91.75 7.31 3.76,10.86
9V 503 97.29 95.50-98.51 502 96.72 94.81-98.08 0.57 -1.80,2.94
common 14 511 98.84 97.49-99.57 516 99.42 98.32-99.88 -0.58 -1.88,0.71
18C 499 96.52 94.55-97.92 493 94.99 92.75-96.70 1.53 -1.28,4.33
19F 512 99.03 97.76-99.69 516 99.42 98.32-99.88 -0.39 -1.61,0.83
23F 505 97.68 95.98-98.80 489 94.22 91.85-96.07 3.46 0.73,6.19
1 501 96.91 95.02-98.22 498 95.95 93.88-97.48 0.95 -1.63,3.53
3 469 90.72 87.88-93.08 489 94.22 91.85-96.07 -3.50 -7.17,0.16
Additional 5 452 87.43 84.26-90.16 489 94.22 91.85-96.07 -6.79 -10.79,-2.80
6A 501 96.91 95.02-98.22 488 94.03 91.63-95.91 2.88 -0.01,5.77
7F 503 97.29 95.50-98.51 504 97.11 95.28-98.37 0.18 -2.11,2.48
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19A 514 99.42 98.31-99.88 517 99.61 98.61-99.95 -0.19 -1.16,0.77

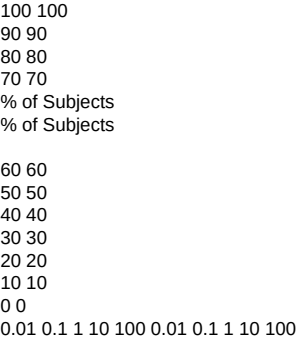
Table 13-3 Comparison of Subjects IgG GMCs between PCV13-TT (2M) and PCV13-TT (3M) groups (PPS)
PCV13-TT (2M) Noninferiority test
PCV13-TT (3M) (N=471)
Serotype (N=482)

GMC 95%CI GMC 95%CI Ratio# 97.5%CI
4 2.43 2.28-2.60 4.51 4.20-4.84 0.54 0.48,0.60
6B 3.72 3.37-4.10 1.25 1.11-1.42 2.97 2.47,3.56
9V 3.66 3.40-3.94 3.72 3.45-4.02 0.98 0.87,1.11
common 14 10.25 9.46-11.11 10.38 9.57-11.27 0.99 0.87,1.13
18C 2.04 1.87-2.23 2.91 2.68-3.15 0.70 0.61,0.81
19F 5.46 5.14-5.80 6.95 6.49-7.45 0.79 0.71,0.87
23F 3.86 3.53-4.21 2.93 2.65-3.23 1.32 1.13,1.53
1 2.09 1.95-2.23 2.15 2.01-2.30 0.97 0.87,1.09
3 1.41 1.32-1.50 1.67 1.56-1.79 0.84 0.76,0.94
5 0.79 0.73-0.85 1.88 1.75-2.02 0.42 0.37,0.47
Additional
6A 3.28 3.01-3.57 1.96 1.77-2.16 1.67 1.44,1.95
7F 4.76 4.42-5.13 5.47 5.06-5.92 0.87 0.77,0.98
19A 3.12 2.95-3.30 3.85 3.62-4.09 0.81 0.74,0.89
#Ratio: PCV13-TT (2M) GMC/PCV13-TT (3M) GMC.

Table 13-4 Comparison of Subjects IgG GMCs between PCV13-TT (2M) and PCV13-TT (3M) groups (FAS)

PCV13-TT (2M) Noninferiority test
PCV13-TT (3M) (N=519)
Serotype Serotype (N=517)
GMC 95%CI GMC 95%CI Ratio# 97.5%CI
4 1.94 1.74-2.17 3.46 3.06-3.92 0.56 0.46,0.68
6B 3.38 3.05-3.75 1.23 1.09-1.39 2.75 2.29,3.30
9V 3.11 2.82-3.43 3.24 2.95-3.56 0.96 0.82,1.12
common 14 8.73 7.87-9.68 9.32 8.49-10.22 0.94 0.80,1.10
18C 1.82 1.65-2.00 2.42 2.18-2.70 0.75 0.63,0.89
19F 4.92 4.60-5.27 6.26 5.80-6.76 0.79 0.70,0.88
23F 3.40 3.10-3.74 2.59 2.32-2.90 1.31 1.11,1.55
1 1.87 1.73-2.02 1.79 1.60-1.99 1.04 0.90,1.22
3 1.21 1.11-1.32 1.45 1.33-1.58 0.83 0.73,0.96
5 0.69 0.63-0.76 1.50 1.34-1.68 0.46 0.39,0.55
Additional
6A 2.85 2.57-3.16 1.80 1.62-2.00 1.59 1.34,1.88
7F 4.04 3.67-4.43 4.66 4.19-5.19 0.87 0.74,1.02
19A 2.93 2.76-3.11 3.57 3.35-3.81 0.82 0.74,0.91
#Ratio: PCV13-TT (2M) GMC/PCV13-TT (3M) GMC.
6.1.2.4. Pneumococcal Serotype Antibody Concentrations in the Subjects in PCV13-TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group
The results of pneumococcal serotype antibody concentrations for subjects in PCV13-TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group post primary series were shown in Table 8 to Table 20. The reverse cumulative distribution curves were shown in Figure 1 to 13.
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Table 8. Distribution of the pneumococcal serotype 4 IgG antibody concentrations for subjects in PCV13-TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group post primary series
FAS N (%) PPS N (%)
PCV13-TT PCV13-TT PCV13-TT PCV13-TT
PCV7 (3M) PCV7 (3M)
Concentration (2M) (3M) (2M) (3M)
(N=519) (N=487)
(N=517) (N=519) (N=482) (N=471)
n % n % n % n % n %
≥0.01 506 97.87 506 97.50 507 97.69 482 100.0 471 100.0 486 99.79
≥0.1 505 97.68 506 97.50 507 97.69 482 100.0 471 100.0 486 99.79
≥1 429 82.98 482 92.87 496 95.57 423 87.76 456 96.82 480 98.56
≥10 15 2.90 77 14.84 194 37.38 15 3.11 75 15.92 187 38.40
≥100 0 0.00 0 0.00 0 0.00 0 0.00 0 0.00 0 0.00
N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT group (2M), PCV13-TT group (3M) and PCV7 group (3M).
Serotype 4, After Primary Series (FAS) Serotype 4, After Primary Series (PPS)



Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)
PCV13-TT (2M) PCV13-TT (3M) PCV7 (3M) PCV13-TT (2M) PCV13-TT (3M) PCV7 (3M)

Figure 13 Reverse cumulative distribution curves, pneumococcal serotype 4, IgG antibody concentrations for subjects in PCV13-TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group - post primary series.
Table 9 Distribution of the pneumococcal serotype 6B IgG antibody concentrations for subjects in PCV13-TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group post primary series
FAS N (%) PPS N (%)
PCV13-TT PCV13-TT PCV13-TT PCV13-TT
PCV7 (3M) PCV7 (3M)
Concentration (2M) (3M) (2M) (3M)
(N=519) (N=487)
(N=517) (N=519) (N=482) (N=471)
n % n % n % n % n %
≥0.01 514 99.42 508 97.88 517 99.61 480 99.59 461 97.88 487 100.0
≥0.1 514 99.42 508 97.88 517 99.61 480 99.59 461 97.88 487 100.0
≥1 446 86.27 309 59.54 464 89.40 431 89.42 288 61.15 448 91.99
≥10 77 14.89 18 3.47 65 12.52 75 15.56 15 3.18 59 12.11
≥100 0 0.00 0 0.00 0 0.00 0 0.00 0 0.00 0 0.00
N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT group (2M), PCV13-TT group (3M) and PCV7 group (3M).

TT group (3M) and PCV7 group (3M).

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Serotype 6B, After Primary Series (FAS) Serotype 6B, After Primary Series (PPS)

100 100
90 90
80 80
70 70

% of Subjects
% of Subjects

60 60
50 50
40 40
30 30
20 20
10 10
0 0
0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration (µg/mL) Antibody Concentration ■µg/mL)

PCV13-TT (2M) PCV13-TT (3M) PCV7 (3M) PCV13-TT (2M) PCV13-TT (3M) PCV7 (3M)

Figure 14 Reverse cumulative distribution curves, pneumococcal serotype 6B, IgG antibody concentrations for subjects in PCV13-TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group - post primary series.

Table 10 Distribution of the pneumococcal serotype 9V IgG antibody concentrations for subjects in PCV13-TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group post primary series
FAS N (%) PPS N (%)
PCV13-TT PCV13-TT PCV13-TT
PCV7 (3M) PCV13-TT (3M) PCV7 (3M)
Concentration (2M) (3M) (2M)
(N=519) (N=471) (N=487)
(N=517) (N=519) (N=482)
n % n % n % n % n % n %
≥0.01 512 99.03 515 99.23 512 98.65 482 100.0 470 99.79 487 100.0
≥0.1 512 99.03 515 99.23 512 98.65 482 100.0 470 99.79 487 100.0
≥1 462 89.36 477 91.91 488 94.03 451 93.57 452 95.97 472 96.92
≥10 48 9.28 50 9.63 62 11.95 48 9.96 48 10.19 58 11.91
≥100 0 0.00 0 0.00 0 0.00 0 0.00 0 0.00 0 0.00
N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT group (2M), PCV13-TT group (3M) and PCV7 group (3M).
Serotype 9V, After Primary Series (FAS) Serotype 9V, After Primary Series (PPS)

100 100
90 90
80 80
70 70

% of Subjects
% of Subjects

60 60
50 50
40 40
30 30
20 20
10 10
0 0
0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT (2M) PCV13-TT (3M) PCV7 (3M) PCV13-TT (2M) PCV13-TT (3M) PCV7 (3M)

Figure 15 Reverse cumulative distribution curves, pneumococcal serotype 9V, IgG antibody concentrations for subjects in PCV13-TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group - post primary series.

Table 11 Distribution of the pneumococcal serotype 14 IgG antibody concentrations for subjects in PCV13-TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group post primary series
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months) group post primary series
FAS N (%) PPS N (%)
PCV13-TT PCV13-TT PCV13-TT
PCV7 (3M) PCV13-TT (3M) PCV7 (3M)
Concentration (2M) (3M) (2M)
(N=519) (N=471) (N=487)
(N=517) (N=519) (N=482)

n % n % n % n % n % n %
≥0.01 514 99.42 518 99.81 518 99.81 482 100.0 471 100.0 486 99.79
≥0.1 514 99.42 518 99.81 518 99.81 482 100.0 471 100.0 486 99.79
≥1 497 96.13 500 96.34 508 97.88 475 98.55 466 98.94 484 99.38
≥10 261 50.48 268 51.64 384 73.99 260 53.94 257 54.56 370 75.98
≥100 1 0.19 6 1.16 11 2.12 1 0.21 4 0.85 9 1.85
N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT group (2M), PCV13-TT group (3M) and PCV7 group (3M).

Serotype 14, After Primary Series (FAS) Serotype 14, After Primary Series (PPS)

100 100
90 90
80 80
70 70

% of Subjects
% of Subjects

60 60
50 50
40 40
30 30
20 20
10 10
0 0
0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT (2M) PCV13-TT (3M) PCV7 (3M) PCV13-TT (2M) PCV13-TT (3M) PCV7 (3M)

Figure 16 Reverse cumulative distribution curves, pneumococcal serotype 14, IgG antibody concentrations for subjects in PCV13-TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group - post primary series.

Table 12. Distribution of the pneumococcal serotype 18C IgG antibody concentrations for subjects in PCV13-TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group post primary series
FAS N (%) PPS N (%)
PCV13-TT PCV13-TT PCV13-TT PCV13-TT
PCV7 (3M) PCV7 (3M)
Concentration (2M) (3M) (2M) (3M)
(N=519) (N=487)
(N=517) (N=519) (N=482) (N=471)
n % n % n % n % n % n %
≥0.01 509 98.45 510 98.27 512 98.65 477 98.96 470 99.79 484 99.38
≥0.1 509 98.45 510 98.27 512 98.65 477 98.96 470 99.79 484 99.38
≥1 409 79.11 454 87.48 481 92.68 403 83.61 431 91.51 465 95.48
≥10 14 2.71 34 6.55 102 19.65 13 2.70 32 6.79 97 19.92
≥100 0 0.00 0 0.00 0 0.00 0 0.00 0 0.00 0 0.00
N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT group (2M), PCV13-TT group (3M) and PCV7 group (3M).

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Serotype 18C, After Primary Series (FAS) Serotype 18C, After Primary Series (PPS)

100 100
90 90
80 80
70 70

% of Subjects
% of Subjects

60 60
50 50
40 40
30 30
20 20
10 10
0 0
0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT (2M) PCV13-TT (3M) PCV7 (3M) PCV13-TT (2M) PCV13-TT (3M) PCV7 (3M)

Figure 17 Reverse cumulative distribution curves, pneumococcal serotype 18C, IgG antibody concentrations for subjects in PCV13-TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group - post primary series.

Table 13. Distribution of the pneumococcal serotype 19F IgG antibody concentrations for

subjects in PCV13-TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group post primary series
FAS N (%) PPS N (%)
PCV13-TT PCV13-TT PCV13-TT PCV13-TT
PCV7 (3M) PCV7 (3M)
Concentration (2M) (3M) (2M) (3M)
(N=519) (N=487)
(N=517) (N=519) (N=482) (N=471)
n % n % n % n % n % n %
≥0.01 517 100.0 519 100.0 518 99.81 482 100.0 471 100.0 487 100.0
≥0.1 517 100.0 519 100.0 518 99.81 482 100.0 471 100.0 487 100.0
≥1 497 96.13 497 95.76 499 96.15 478 99.17 466 98.94 479 98.36
≥10 85 16.44 149 28.71 224 43.16 85 17.63 143 30.36 217 44.56
≥100 0 0.00 0 0.00 0 0.00 0 0.00 0 0.00 0 0.00
N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT group (2M), PCV13-TT group (3M) and PCV7 group (3M).

Serotype 19F, After Primary Series (FAS) Serotype 19F, After Primary Series (PPS)

100 100
90 90
80 80
70 70
% of Subjects
% of Subjects

60 60
50 50
40 40
30 30
20 20
10 10
0 0
0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT (2M) PCV13-TT (3M) PCV7 (3M) PCV13-TT (2M) PCV13-TT (3M) PCV7 (3M)

Figure 18 Reverse cumulative distribution curves, pneumococcal serotype 19F, IgG antibody concentrations for subjects in PCV13-TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group - post primary series.

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Table 14. Distribution of the pneumococcal serotype 23F IgG antibody concentrations for subjects in PCV13-TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group post primary series
FAS N (%) PPS N (%)
PCV13-TT PCV13-TT PCV13-TT PCV13-TT
PCV7 (3M) PCV7 (3M)
Concentration (2M) (3M) (2M) (3M)
(N=519) (N=487)
(N=517) (N=519) (N=482) (N=471)
n % n % n % n % n % n %
≥0.01 516 99.81 514 99.04 513 98.84 481 99.79 469 99.58 486 99.79
≥0.1 516 99.81 514 99.04 513 98.84 481 99.79 469 99.58 486 99.79
≥1 458 88.59 436 84.01 487 93.83 450 93.36 412 87.47 472 96.92
≥10 79 15.28 53 10.21 92 17.73 78 16.18 48 10.19 87 17.86
≥100 0 0.00 0 0.00 1 0.19 0 0.00 0 0.00 0 0.00
N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT group (2M), PCV13-TT group (3M) and PCV7 group (3M).

Serotype 23F, After Primary Series (FAS) Serotype 23F, After Primary Series (PPS)

100 100
90 90
80 80
70 % of Subjects 70
% of Subjects

60 60
50 50
40 40
30 30
20 20
10 10
0 0
0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT (2M) PCV13-TT (3M) PCV7 (3M) PCV13-TT (2M) PCV13-TT (3M) PCV7 (3M)

60 60
50 50
40 40
30 30
20 20
10 10
0 0
0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT (2M) PCV13-TT (3M) PCV7 (3M) PCV13-TT (2M) PCV13-TT (3M) PCV7 (3M)

Figure 23 Reverse cumulative distribution curves, pneumococcal serotype 6A, IgG antibody concentrations for subjects in PCV13-TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group - post primary series.

Table 19. Distribution of the pneumococcal serotype 7F IgG antibody concentrations for subjects in PCV13-TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group post primary series

FAS N (%)	PPS N (%)	PCV13-TT (2M)	PCV13-TT (3M)	PCV7 (3M)
515	519	471	482	482
99.61	99.61	99.04	99.04	99.04
512	512	323	323	323
98.65	98.65	62.24	62.24	62.24
480	480	28	28	28
92.84	92.84	5.39	5.39	5.39
86	86	117	117	117
16.63	16.63	2	2	2
0.39	0.39	86	86	86
17.84	17.84	114	114	114
24.20	24.20	2	2	2
0.41	0.41	0	0	0
0.00	0.00	1	1	1
0.19	0.19	0	0	0
0.00	0.00	0	0	0
0.00	0.00	0	0	0
0.00	0.00	0	0	0

N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT group (2M), PCV13-TT group (3M) and PCV7 group (3M).

Figure 24 Reverse cumulative distribution curves, pneumococcal serotype 7F, IgG antibody concentrations for subjects in PCV13-TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group - post primary series.

Table 20. Distribution of the pneumococcal serotype 19A IgG antibody concentrations for subjects in PCV13-TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group post primary series

FAS N (%)	PPS N (%)	PCV13-TT (2M)	PCV13-TT (3M)	PCV7 (3M)
515	519	471	482	482
99.61	99.61	99.04	99.04	99.04
512	512	323	323	323
98.65	98.65	62.24	62.24	62.24
480	480	28	28	28
92.84	92.84	5.39	5.39	5.39
86	86	117	117	117
16.63	16.63	2	2	2
0.39	0.39	86	86	86
17.84	17.84	114	114	114
24.20	24.20	2	2	2
0.41	0.41	0	0	0
0.00	0.00	1	1	1
0.19	0.19	0	0	0
0.00	0.00	0	0	0
0.00	0.00	0	0	0
0.00	0.00	0	0	0

N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT group (2M), PCV13-TT group (3M) and PCV7 group (3M).

months) group post primary series

FAS N (%)	PPS N (%)	PCV13-TT (2M)	PCV13-TT (3M)	PCV7 (3M)
515	519	471	482	482
99.61	99.61	99.04	99.04	99.04
512	512	323	323	323
98.65	98.65	62.24	62.24	62.24
480	480	28	28	28
92.84	92.84	5.39	5.39	5.39
86	86	117	117	117
16.63	16.63	2	2	2
0.39	0.39	86	86	86
17.84	17.84	114	114	114
24.20	24.20	2	2	2
0.41	0.41	0	0	0
0.00	0.00	1	1	1
0.19	0.19	0	0	0
0.00	0.00	0	0	0
0.00	0.00	0	0	0
0.00	0.00	0	0	0

N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT group (2M), PCV13-TT group (3M) and PCV7 group (3M).

Serotype 19A, After Primary Series (FAS) Serotype 19A, After Primary Series (PPS)

100 100
90 90
80 80
70 70
% of Subjects
% of Subjects
60 60
50 50
40 40
30 30
20 20
10 10
0 0
0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT (2M) PCV13-TT (3M) PCV7 (3M) PCV13-TT (2M) PCV13-TT (3M) PCV7 (3M)

Figure 25 Reverse cumulative distribution curves, pneumococcal serotype 19A, IgG antibody

concentrations for subjects in PCV13-TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group - post primary series.

6.1.3 Summary of primary endpoints

For major subjects (aged 2-6 month, minimum 6 weeks old), the positive rate and GMCs for 6 of the 7 common serotypes (except serotype 6B) in the PCV13-TT aged 3 month were noninferior to the control group aged 3 months using 0, 1, and 2 month dosing schedule. Among the 6 additional serotypes the lower limit of 2-sided, 95% CI for the positive rate in PCV13-TT was greater than 70%, and the positive rate and GMCs of each additional serotype in the PCV13-TT were superiority to the control group.

For subjects aged 3-71 months, among the 7 common serotypes, the positive rate and GMC of 7 common serotype in the PCV13-TT were noninferior to the control group. Among the 6 additional serotypes, the lower limit of 2-sided, 95% CI for the positive rate in PCV13-TT was greater than 70%. And the positive rate and GMC of each additional serotype in the PCV13-TT were superiority to the control group. The immunogenicity of the PCV13-TT aged 3-71 months was non-inferior to the control group aged 3-71months.

The positive rate of 7 common serotype in the PCV13-TT aged 2 month(minimun age of 6 weeks) was noninferior to the control group aged 3 months using 0, 2, and 4 month dosing schedule. The GMCs for 5 of the 7 common serotypes (except serotype 4 and 18C) in the PCV13-TT aged 2 month were noninferior to the control group aged 3 months.

Among the 6 additional serotypes, the lower limit of 2-sided, 95% CI for the positive rate in PCV13-TT aged 2 months was greater than 70%. The positive rate and GMCs of each additional serotype in the PCV13-TT aged 2 months were superiority to the control group aged 3 months.

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And the positive rate of 12 serotype (except serotype 5) in the PCV13-TT aged 2 month (minimun age of 6 weeks) using 0, 2, and 4 month dosing schedule was non-inferior to the PCV13-TT aged 3 months using 0, 1, and 2 month dosing schedule.

The positive rate of the subjects aged 3-71 months and major subjects (aged 2-6 month, minimum 6 weeks old) was greater than 90%. For the 6 additional serotypes, the lower limit of 2-sided, 95% CI for the positive rate was greater than 70%. The GMCs of most serotypes was greater than 2.0 µg/mL, except serotype 3 and 5, 6B and 6A in PCV13-TT aged 3 months.

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6.2. SECONDARY ENDPOINT ANALYSIS

6.2.1. Positive rate of IgG antibody and the IgG antibody concentrations (GMCs) analysis for subjects aged 7-11months, 12-23 months and 24-71 months after immunization in PCV13-TT and control group

6.2.1.1 Positive rate of IgG antibody and the IgG antibody concentrations (GMCs) analysis for subjects aged 7-11 months after immunization in PCV13-TT and control group

(1)Positive rate of IgG antibody and the IgG antibody concentrations (GMCs) analysis for subjects aged 7-11 months

Among the 7 common serotypes the proportion of responders at ≥0.35 µg/mL in the PCV13-TT was more than 96%. The lower limit of 2-sided, 97.5% CI for the differences of positive rate between PCV13-TT and control group was greater than -10% (refer to table 15-1) in primary series.

The GMCs for the 7 common serotypes were greater than 2.0 µg/mL. The lower limit of 2-sided, 97.5% CI for the GMC ratio for 6 of the 7 common serotypes (except serotype 14) was greater than 0.5. For serotype 14, the lower limit of 2-sided, 97.5% CI for the GMC ratio was 0.45 and the GMC ration was 0.53.The result was shown in table 15-3

Among the 6 additional serotypes the lower limit of 2-sided, 95% CI for the positive rate in PCV13-TT was greater than 70%. The lower limit of 2-sided 95% CI differences of positive rate was greater than 0 and the GMC ratio (testgroup/control group) was greater than 1.

The statistical analysis of positive rate of IgG antibody and the IgG antibody concentrations (GMCs) for subjects aged 7-71 months after immunization in PCV13-TT and control group was shown in tables 15-1 to 15-4.

Table 15-1 Comparison of Subjects aged 7-11 months Achieving a Pneumococcal IgG Antibody Concentration >0.35µg/mL After immunization between PCV13-TT and PCV7 Group(PPS)

PCV13-TT (N=185) PCV7 (N=189) Rate diff.

Serotype 95%CI

N % 95% CI N % 95% CI (%)

4 185 100.0 98.03-100.0 189 100.0 98.07-100.0 0.00 -1.93,1.97
6B 179 96.76 93.07-98.80 188 99.47 97.09-99.99 -2.71 -5.47,0.04
co 9V 180 97.30 93.81-99.12 185 97.88 94.67-99.42 -0.59 -3.70,2.52
m
14 185 100.0 98.03-100.0 189 100.0 98.07-100.0 0.00 -1.93,1.97
m
on 18C 184 99.46 97.03-99.99 188 99.47 97.09-99.99 -0.01 -1.49,1.47
19F 185 100.0 98.03-100.0 189 100.0 98.07-100.0 0.00 -1.93,1.97
23F 180 97.30 93.81-99.12 183 96.83 93.22-98.83 0.47 -2.95,3.89
1 183 98.92 96.15-99.87 143 75.66 68.90-81.60 23.26 16.96,29.55
A 3 181 97.84 94.56-99.41 117 61.90 54.57-68.86 35.93 28.70,43.17
dd 169 91.35 86.34-94.98 84 44.44 37.23-51.83 46.91 38.75,55.07
5
iti
on 6A 177 95.68 91.66-98.11 147 77.78 71.17-83.49 17.90 11.29,24.51
al 7F 184 99.46 97.03-99.99 87 46.03 38.77-53.42 53.43 46.24,60.61
19A 185 100.0 98.03-100.0 185 97.88 94.67-99.42 2.12 0.06,4.17

Table 15-2 Comparison of Subjects aged 7-11 months Achieving a Pneumococcal IgG Antibody Concentration >0.35µg/mL After immunization between PCV13-TT and PCV7 Group(FAS)

Serotype PCV13-TT (N=198) PCV7 (N=200) Rate 95%CI

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diff.
N % 95% CI N %
(%)
92.85- 91.63-
191 96.46 191 95.50 0.96 -2.89,4.82
4 98.57 97.92
90.28- 94.26-
187 94.44 195 97.50 -3.06 -6.91,0.80
6B 97.19 99.18
90.86- 89.14-
187 94.92 187 93.50 1.42 -3.17,6.01
9V* 97.54 96.49
common 97.22- 95.68-
197 99.49 197 98.50 0.99 -0.96,2.95
14 99.99 99.69
92.19- 91.00-
190 95.96 190 95.00 0.96 -3.12,5.04
18C 98.24 97.58
98.15- 96.43-
198 100.0 198 99.00 1.00 -0.38,2.38
19F 100.0 99.88
90.28- 89.14-
187 94.44 187 93.50 0.94 -3.73,5.62
23F 97.19 96.49
92.19- 66.81-
190 95.96 147 73.50 22.46 15.76,29.16
1 98.24 79.48
89.65- 52.85-
186 93.94 120 60.00 33.94 26.38,41.50
3 96.83 66.85
83.08- 35.56-
175 88.38 85 42.50 45.88 37.71,54.06
Additional 5 92.49 49.67
88.42- 67.87-
184 92.93 149 74.50 18.43 11.41,25.45
6A 96.08 80.39
93.52- 37.98-
192 96.97 90 45.00 51.97 44.67,59.27
7F 98.88 52.18
98.15- 94.96-
198 100.0 196 98.00 2.00 0.06,3.94
19A 100.0 99.45
*The subjects for serotype 9V for PCV13-TT in FAS was N=197.

Table 15-3 Comparison of Subjects aged 7-11 months IgG antibody concentrations (GMCs) between PCV13-TT and PCV7 Group (PPS)
Noninferior / Superiority
PCV13-TT (N=185) PCV7 (N=189)
Serotype Serotype test
GMC 95%CI GMC 95%CI Ratio# 95%CI
4 4.26 3.86-4.71 4.23 3.88-4.62 1.01 0.88,1.15
6B 2.90 2.46-3.42 2.20 1.92-2.52 1.32 1.07,1.63
9V 2.38 2.09-2.71 2.76 2.43-3.12 0.86 0.72,1.03
common 14 8.54 7.60-9.60 16.17 14.53-17.99 0.53 0.45,0.62
18C 3.78 3.30-4.34 3.44 3.08-3.84 1.10 0.92,1.31
19F 7.21 6.43-8.08 4.09 3.70-4.52 1.76 1.52,2.05
23F 3.56 3.10-4.10 2.70 2.36-3.09 1.32 1.09,1.60
1 2.23 2.03-2.45 0.47 0.40-0.56 4.69 3.89,5.65
3 1.32 1.20-1.45 0.46 0.41-0.51 2.89 2.51,3.32
5 0.82 0.73-0.91 0.19 0.15-0.25 4.30 3.23,5.72
Additional
6A 1.79 1.51-2.12 0.54 0.41-0.72 3.31 2.38,4.62
7F 5.73 5.04-6.51 0.14 0.10-0.20 39.88 27.69,57.43
19A 5.06 4.54-5.64 1.50 1.29-1.74 3.37 2.81,4.05
#Ratio: PCV13-TTGMC/PCV7GMC.
Table 15-4 Comparison of Subjects aged 7-11 months IgG antibody concentrations (GMCs) between PCV13-TT and PCV7 Group (FAS)
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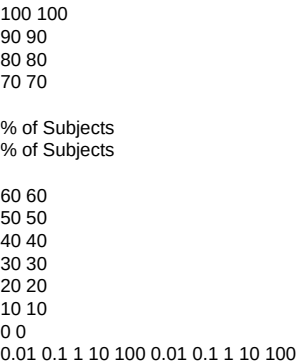
Noninferior / Superiority
PCV13-TT (N=198) PCV7 (N=200)
Serotype Serotype test
GMC 95%CI GMC 95%CI Ratio# 95%CI
common 4 3.34 2.73-4.10 3.08 2.47-3.85 1.09 0.80,1.47
6B 2.75 2.33-3.25 2.02 1.76-2.33 1.36 1.09,1.69
9V 2.04 1.70-2.45 2.11 1.70-2.61 0.97 0.73,1.28
14 7.82 6.85-8.93 13.54 11.52-15.91 0.58 0.47,0.71
18C 3.01 2.43-3.73 2.53 2.02-3.17 1.19 0.87,1.62
19F 6.59 5.80-7.48 3.75 3.36-4.18 1.76 1.49,2.08
23F 3.19 2.73-3.72 2.36 2.01-2.77 1.35 1.08,1.69
1 2.07 1.87-2.30 0.46 0.39-0.54 4.51 3.74,5.43
3 1.19 1.06-1.34 0.42 0.37-0.48 2.80 2.36,3.33
5 0.77 0.68-0.87 0.18 0.14-0.23 4.29 3.21,5.72
Additional
6A 1.68 1.41-2.00 0.51 0.39-0.67 3.31 2.39,4.57
7F 5.11 4.41-5.92 0.14 0.10-0.20 35.48 24.84,50.67
19A 4.75 4.24-5.33 1.46 1.27-1.68 3.26 2.72,3.91
#Ratio: PCV13-TTGMC/PCV7GMC■*PCV13-TT9V■FAS■N=197.

(2)Pneumococcal Serotype antibody concentrations in the subjects aged 7-11 months in PCV13-TT group and PCV7 group
The results of pneumococcal serotype antibody concentrations for subjects aged 7-11 months after primary series in PCV13-TT group and PCV7 group were shown in Table 21. to Table 33. The reverse cumulative distribution curves were shown in Figure 27 to 39.

Table 21. Distribution of the pneumococcal serotype 4 IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group post primary series
FAS N (%) PPS N (%)
PCV13-TT PCV13-TT
Concentration PCV7 (N=200) PCV7 (N=189)
(N=198) (N=185)
n % n % n % n %
≥0.01 192 96.97 192 96.00 185 100.0 189 100.0
≥0.1 192 96.97 192 96.00 185 100.0 189 100.0
≥1 188 94.95 187 93.50 182 98.38 185 97.88
≥10 20 10.10 12 6.00 19 10.27 12 6.35
≥100 0 0.00 0 0.00 0 0.00 0 0.00
N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

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Serotype 4, After Primary Series (FAS) Serotype 4, After Primary Series (PPS)



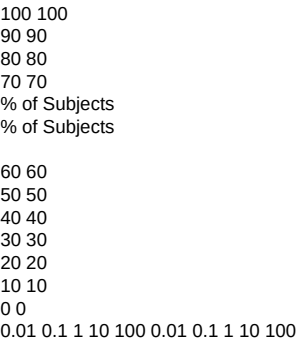
Antibody Concentration (µg/mL) Antibody Concentration ■µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

Figure 26 Reverse cumulative distribution curves, pneumococcal serotype 4, IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group - post primary series.

Table 22. Distribution of the pneumococcal serotype 6B IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group post primary series
FAS N (%) PPS N (%)
PCV13-TT PCV13-TT
Concentration PCV7 (N=200) PCV7 (N=189)
(N=198) (N=185)
n % n % n % n %
≥0.01 197 99.49 200 100.0 184 99.46 189 100.0
≥0.1 197 99.49 200 100.0 184 99.46 189 100.0
≥1 160 80.81 143 71.50 155 83.78 142 75.13
≥10 19 9.60 11 5.50 17 9.19 11 5.82
≥100 0 0.00 0 0.00 0 0.00 0 0.00
N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

Serotype 6B, After Primary Series (FAS) Serotype 6B, After Primary Series (PPS)



Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

Figure 27 Reverse cumulative distribution curves, pneumococcal serotype 6B, IgG antibody

concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group - post primary series.

Table 23. Distribution of the pneumococcal serotype 9V IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group post primary series

FAS N (%) PPS N (%)

PCV13-TT PCV13-TT

Concentration PCV7 (N=200) PCV7 (N=189)

(N=197) (N=185)

n % n % n % n %

≥0.01 193 97.97 193 96.50 185 100.0 189 100.0

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≥0.1 193 97.97 193 96.50 185 100.0 189 100.0

≥1 158 80.20 171 85.50 154 83.24 169 89.42

≥10 10 5.08 12 6.00 9 4.86 12 6.35

≥100 0 0.00 0 0.00 0 0.00 0 0.00

N=number of subjects included in FAS and PPS for vaccine group.

n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

Serotype 9V, After Primary Series (FAS) Serotype 9V, After Primary Series (PPS)

100 100

90 90

80 80

70 70

% of Subjects

% of Subjects

60 60

50 50

40 40

30 30

20 20

10 10

0 0

0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration ■µg/mL)

Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

Figure 28 Reverse cumulative distribution curves, pneumococcal serotype 9V, IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group - post primary series.

Table 24 Distribution of the pneumococcal serotype 14 IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group post primary series

FAS N (%) PPS N (%)

PCV13-TT PCV13-TT

Concentration PCV7 (N=200) PCV7 (N=189)

(N=198) (N=185)

n % n % n % n %

≥0.01 198 100.0 199 99.50 185 100.0 189 100.0

≥0.1 198 100.0 199 99.50 185 100.0 189 100.0

≥1 191 96.46 193 96.50 184 99.46 189 100.0

≥10 85 42.93 139 69.50 82 44.32 137 72.49

≥100 2 1.01 2 1.00 1 0.54 2 1.06

N=number of subjects included in FAS and PPS for vaccine group.

n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

Serotype 14, After Primary Series (FAS) Serotype 14, After Primary Series (PPS)

100 100

90 90

80 80

70 70

% of Subjects

% of Subjects

60 60

50 50

40 40

30 30

20 20

10 10

0 0

0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration ■µg/mL)

Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

Figure 29 Reverse cumulative distribution curves, pneumococcal serotype 14, IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group - post primary series.

Table 25. Distribution of the pneumococcal serotype 18C IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group post primary series

FAS N (%) PPS N (%)

PCV13-TT PCV13-TT

Concentration PCV7 (N=200) PCV7 (N=189)

(N=198) (N=185)

n % n % n % n %

≥0.01 191 96.46 191 95.50 184 99.46 189 100.0

≥0.1 191 96.46 191 95.50 184 99.46 189 100.0

≥1 181 91.41 179 89.50 176 95.14 177 93.65

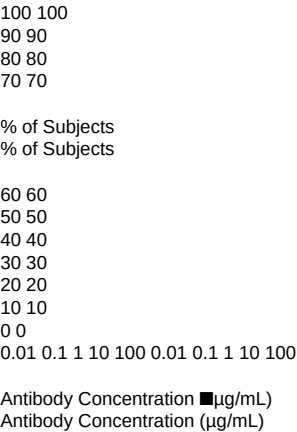
≥10 19 9.60 13 6.50 18 9.73 13 6.88

≥100 0 0.00 0 0.00 0 0.00 0 0.00

N=number of subjects included in FAS and PPS for vaccine group.

n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

Serotype 18C, After Primary Series (FAS) Serotype 18C, After Primary Series (PPS)



PCV13-TT PCV7 PCV13-TT PCV7

Figure 30 Reverse cumulative distribution curves, pneumococcal serotype 18C, IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group - post primary series.

Table 26. Distribution of the pneumococcal serotype 19F IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group post primary series

FAS N (%) PPS N (%)

PCV13-TT PCV13-TT

Concentration PCV7 (N=200) PCV7 (N=189)

(N=198) (N=185)

n % n % n % n %

≥0.01 198 100.0 200 100.0 185 100.0 189 100.0

≥0.1 198 100.0 200 100.0 185 100.0 189 100.0

≥1 187 94.44 186 93.00 179 96.76 182 96.30

≥10 66 33.33 20 10.00 64 34.59 20 10.58

≥100 0 0.00 0 0.00 0 0.00 0 0.00

N=number of subjects included in FAS and PPS for vaccine group.

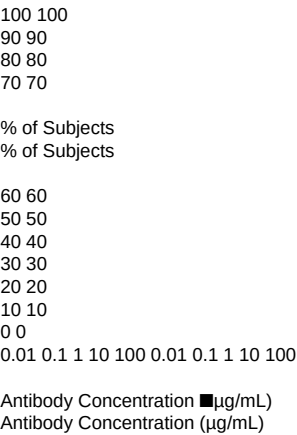
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

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Serotype 19F, After Primary Series (FAS) Serotype 19F, After Primary Series (PPS)



PCV13-TT PCV7 PCV13-TT PCV7

Figure 31 Reverse cumulative distribution curves, pneumococcal serotype 19F, IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group - post primary series.

Table 27. Distribution of the pneumococcal serotype 23F IgG antibody concentrations for subjects aged

7-11 months in PCV13-TT group and PCV7 group post primary series
FAS N (%) PPS N (%)
PCV13-TT PCV13-TT
Concentration PCV7 (N=200) PCV7 (N=189)
(N=198) (N=185)
n % n % n % n %
≥0.01 198 100.0 199 99.50 185 100.0 189 100.0
≥0.1 198 100.0 199 99.50 185 100.0 189 100.0
≥1 173 87.37 166 83.00 169 91.35 164 86.77
≥10 26 13.13 14 7.00 25 13.51 14 7.41
≥100 0 0.00 0 0.00 0 0.00 0 0.00
N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

Serotype 23F, After Primary Series (FAS) Serotype 23F, After Primary Series (PPS)

100 100
90 90
80 80
70 70
% of Subjects
% of Subjects

60 60
50 50
40 40
30 30
20 20
10 10
0 0
0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

Figure 32 Reverse cumulative distribution curves, pneumococcal serotype 23F, IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group - post primary series.

Table 28. Distribution of the pneumococcal serotype 1 IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group post primary series
FAS N (%) PPS N (%)
Concentration PCV13-TT PCV13-TT
PCV7 (N=200) PCV7 (N=189)
(N=198) (N=185)
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n % n % n % n %
≥0.01 198 100.0 195 97.50 185 100.0 184 97.35
≥0.1 198 100.0 193 96.50 185 100.0 183 96.83
≥1 176 88.89 25 12.50 170 91.89 25 13.23
≥10 3 1.52 0 0.00 3 1.62 0 0.00
≥100 0 0.00 0 0.00 0 0.00 0 0.00
N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

Serotype 1, After Primary Series (FAS) Serotype 1, After Primary Series (PPS)

100 100
90 90
80 80
70 70

% of Subjects
% of Subjects

60 60
50 50
40 40
30 30
20 20
10 10
0 0
0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

Figure 33 Reverse cumulative distribution curves, pneumococcal serotype 1, IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group - post primary series.

Table 29 Distribution of the pneumococcal serotype 3 IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group post primary series
FAS N (%) PPS N (%)
PCV13-TT PCV13-TT

Concentration PCV7 (N=200) PCV7 (N=189)
(N=198) (N=185)
n % n % n % n %
≥0.01 198 100.0 199 99.50 185 100.0 189 100.0
≥0.1 192 96.97 191 95.50 184 99.46 187 98.94
≥1 127 64.14 26 13.00 124 67.03 24 12.70
≥10 0 0.00 0 0.00 0 0.00 0 0.00
N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

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Serotype 3, After Primary Series (FAS) Serotype 3, After Primary Series (PPS)

100 100
90 90
80 80
70 70

% of Subjects
% of Subjects

60 60
50 50
40 40
30 30
20 20
10 10
0 0
0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

Figure 34. Reverse cumulative distribution curves, pneumococcal serotype 3, IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group - post primary series.

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Table 30. Distribution of the pneumococcal serotype 5 IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group post primary series
FAS N (%) PPS N (%)
PCV13-TT PCV13-TT
Concentration PCV7 (N=200) PCV7 (N=189)
(N=198) (N=185)
n % n % n % n %
≥0.01 196 98.99 172 86.00 184 99.46 164 86.77
≥0.1 196 98.99 172 86.00 184 99.46 164 86.77
≥1 72 36.36 6 3.00 69 37.30 6 3.17
≥10 0 0.00 0 0.00 0 0.00 0 0.00
N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

Serotype 5, After Primary Series (FAS) Serotype 5, After Primary Series (PPS)

100 100
90 90
80 80
70 70

% of Subjects
% of Subjects

60 60
50 50
40 40
30 30
20 20
10 10
0 0
0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

Figure 35 Reverse cumulative distribution curves, pneumococcal serotype 5, IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group - post primary series.

Table 31. Distribution of the pneumococcal serotype 6A IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group post primary series
FAS N (%) PPS N (%)
PCV13-TT PCV13-TT

Concentration PCV7 (N=200) PCV7 (N=189)
(N=198) (N=185)
n % n % n % n %
≥0.01 195 98.48 180 90.00 182 98.38 170 89.95
≥0.1 195 98.48 179 89.50 182 98.38 169 89.42
≥1 143 72.22 74 37.00 139 75.14 73 38.62
≥10 8 4.04 4 2.00 6 3.24 4 2.12
≥100 0 0.00 0 0.00 0 0.00 0 0.00
N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

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Serotype 6A, After Primary Series (FAS) Serotype 6A, After Primary Series (PPS)

100 100
90 90
80 80
70 70

% of Subjects
% of Subjects

60 60
50 50
40 40
30 30
20 20
10 10
0 0
0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

Figure 36 Reverse cumulative distribution curves, pneumococcal serotype 6A, IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group - post primary series.

Table 32 Distribution of the pneumococcal serotype 7F IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group post primary series
FAS N (%) PPS N (%)
PCV13-TT PCV13-TT
Concentration PCV7 (N=200) PCV7 (N=189)
(N=198) (N=185)
n % n % n % n %
≥0.01 197 99.49 159 79.50 184 99.46 149 78.84
≥0.1 197 99.49 158 79.00 184 99.46 149 78.84
≥1 189 95.45 19 9.50 183 98.92 19 10.05
≥10 34 17.17 1 0.50 33 17.84 1 0.53
≥100 0 0.00 0 0.00 0 0.00 0 0.00
N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

Serotype 7F, After Primary Series (FAS) Serotype 7F, After Primary Series (PPS)

100 100
90 90
80 80
70 70
% of Subjects
% of Subjects

60 60
50 50
40 40
30 30
20 20
10 10
0 0
0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

Figure 37 Reverse cumulative distribution curves, pneumococcal serotype 7F, IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group - post primary series.

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Table 33. Distribution of the pneumococcal serotype 23F IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group post primary series
FAS N (%) PPS N (%)

PCV13-TT PCV13-TT
Concentration PCV7 (N=200) PCV7 (N=189)
(N=198) (N=185)
n % n % n % n %
≥0.01 198 100.0 198 99.00 185 100.0 187 98.94
≥0.1 198 100.0 198 99.00 185 100.0 187 98.94
≥1 190 95.96 135 67.50 181 97.84 132 69.84
≥10 32 16.16 4 2.00 31 16.76 4 2.12
≥100 0 0.00 0 0.00 0 0.00 0 0.00
N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

Serotype 19A, After Primary Series (FAS) Serotype 19A, After Primary Series (PPS)

100 100
90 90
80 80
70 70

% of Subjects
% of Subjects

60 60
50 50
40 40
30 30
20 20
10 10
0 0
0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

Figure 38 Reverse cumulative distribution curves, pneumococcal serotype 19A, IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group - post primary series.

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6.2.1.2. Positive rate of IgG antibody and the IgG antibody concentrations (GMCs) analysis for subjects aged 12-23 months after immunization in PCV13-TT and control group
(1) Positive rate of IgG antibody and the IgG antibody concentrations (GMCs) analysis for subjects aged 12-23 months
Among the 7 common serotypes the proportion of responders at ≥0.35 µg/mL in the PCV13-TT was more than 99%. The lower limit of 2-sided, 95% CI for the differences of positive rate between PCV13-TT and control group was greater than -10% (refer to table 17-1) after Complete immunization. The GMCs for 12 of 13 serotypes (except serotype 9V) were greater than 4.0 µg/mL in PCV13-TT. For serotype 9V, the GMC was 2.97 µg/mL. The lower limit of 2-sided, 95% CI for the GMC ratio was great than > 0.5 (refer to table 17-3).
Among the 6 additional serotypes the proportion of responders at ≥0.35 µg/mL in the PCV13-TT was at least 96%. The lower limit of 2-sided, 95% CI for the positive rate in PCV13-TT was greater than 70%.
The lower limit of 2-sided, 95% CI differences of positive rate for 5 of 6 serotypes (except serotype 19) was greater than 0 and the GMC ratio (testgroup/control group) was greater than 1 between PCV13-TT and control group. For serotype 19, the lower limit of 2-sided, 95% CI differences of positive rate was -0.51 and the differences of positive rate was 0.53.
The statistical analysis of positive rate of IgG antibody and the IgG antibody concentrations (GMCs) for subjects aged 12-23 months after immunization in PCV13-TT and control group was shown in tables 17-1 to 17-4.

Table 17-1 Comparison of Subjects aged 12-23 months Achieving a Pneumococcal IgG Antibody Concentration >0.35µg/mL After immunization between PCV13-TT and PCV7 Group(PPS)

Noninferior / Superiority	
PCV13-TT (N=189) PCV7 (N=189)	
test	
Serotype	
Rate	
N % 95% CI N % 95% CI 95%CI	
diff.(%)	
4 189 100.0 98.07-100.0 188 99.47 97.09-99.99 0.53 -0.51,1.56	
6B 189 100.0 98.07-100.0 186 98.41 95.43-99.67 1.59 -0.19,3.37	
9V 189 100.0 98.07-100.0 187 98.94 96.23-99.87 1.06 -0.40,2.52	
common	
14 189 100.0 98.07-100.0 189 100.0 98.07-100.0 0.00 -1.93,1.93	
18C 189 100.0 98.07-100.0 189 100.0 98.07-100.0 0.00 -1.93,1.93	
19F 189 100.0 98.07-100.0 189 100.0 98.07-100.0 0.00 -1.93,1.93	
23F 188 99.47 97.09-99.99 188 99.47 97.09-99.99 0.00 -1.46,1.46	
1 189 100.0 98.07-100.0 133 70.37 63.31-76.78 29.63 23.12,36.14	
3 186 98.41 95.43-99.67 90 47.62 40.32-54.99 50.79 43.45,58.13	
5 183 96.83 93.22-98.83 101 53.44 46.06-60.71 43.39 35.85,50.92	
Additional	
6A 186 98.41 95.43-99.67 177 93.65 89.17-96.68 4.76 0.86,8.67	
7F 189 100.0 98.07-100.0 135 71.43 64.42-77.75 28.57 22.13,35.01	
19A 189 100.0 98.07-100.0 188 99.47 97.09-99.99 0.53 -0.51,1.56	

Table 17-2 Comparison of Subjects aged 12-23 months Achieving a Pneumococcal IgG Antibody Concentration >0.35µg/mL After immunization between PCV13-TT and PCV7

Group (FAS)
Noninferior /
PCV13-TT (N=200) PCV7 (N=200)
Superiority test
Serotype
Rate
N % 95% CI N % 95% CI 95%CI
diff.
common 4 193 96.50 92.92-98.58 191 95.50 91.63-97.92 1.00 -2.84,4.84
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6B 197 98.50 95.68-99.69 194 97.00 93.58-98.89 1.50 -1.40,4.40
9V 193 96.50 92.92-98.58 190 95.00 91.00-97.58 1.50 -2.45,5.45
14 199 99.50 97.25-99.99 198 99.00 96.43-99.88 0.50 -1.19,2.19
18C 194 97.00 93.58-98.89 192 96.00 92.27-98.26 1.00 -2.60,4.60
19F 200 100.0 98.17-100.0 200 100.0 98.17-100.0 0.00 -1.83,1.83
23F 193 96.50 92.92-98.58 192 96.00 92.27-98.26 0.50 -3.22,4.22
1 193 96.50 92.92-98.58 135 67.50 60.53-73.94 29.00 22.03,35.97
3 190 95.00 91.00-97.58 92 46.00 38.95-53.17 49.00 41.46,56.54
Additional 5 186 93.00 88.53-96.12 102 51.00 43.85-58.12 42.00 34.22,49.78
6A 189 94.50 90.37-97.22 181 90.50 85.56-94.18 4.00 -1.15,9.15
7F 195 97.50 94.26-99.18 137 68.50 61.57-74.87 29.00 22.21,35.79
19A 200 100.0 98.17-100.0 199 99.50 97.25-99.99 0.50 -0.48,1.48

Table 17-3 Comparison of Subjects aged 12-23months IgG antibody concentrations (GMCs) between PCV13-TT and PCV7 Group (PPS)

Noninferior / Superiority
PCV13-TT (N=189) PCV7 (N=189)
Serotype Serotype test
GMC 95%CI GMC 95%CI Ratio# 95%CI
4 5.01 4.60-5.45 5.08 4.63-5.58 0.99 0.87,1.12
6B 5.42 4.86-6.04 2.71 2.35-3.11 2.00 1.68,2.39
9V 2.97 2.68-3.30 2.95 2.64-3.30 1.01 0.87,1.17
common 14 9.20 8.28-10.22 14.98 13.51-16.61 0.61 0.53,0.71
18C 5.03 4.63-5.46 3.92 3.56-4.31 1.28 1.13,1.45
19F 8.80 8.10-9.55 4.20 3.79-4.65 2.09 1.84,2.39
23F 4.32 3.89-4.79 2.96 2.61-3.36 1.46 1.24,1.72
1 2.25 2.07-2.44 0.42 0.35-0.50 5.33 4.39,6.49
3 1.06 0.98-1.14 0.39 0.34-0.45 2.71 2.31,3.17
5 1.01 0.91-1.12 0.27 0.22-0.32 3.79 3.05,4.71
Additional
6A 2.46 2.08-2.92 1.17 1.01-1.35 2.11 1.69,2.64
7F 5.47 5.04-5.94 0.41 0.33-0.52 13.28 10.44,16.88
19A 7.81 7.08-8.61 2.67 2.28-3.12 2.93 2.43,3.52
#Ratio: PCV13-TTGMC/PCV7GMC

Table 17-4 Comparison of Subjects aged 12-23months IgG antibody concentrations (GMCs) between PCV13-TT and PCV7 Group (FAS)

Noninferior / Superiority
PCV13-TT (N=200) PCV7 (N=200)
Serotype Serotype test
GMC 95%CI GMC 95%CI Ratio# 95%CI
4 3.87 3.17-4.72 3.99 3.30-4.83 0.97 0.74,1.28
6B 4.92 4.33-5.59 2.52 2.19-2.91 1.95 1.61,2.36
9V 2.35 1.94-2.85 2.36 1.94-2.87 1.00 0.76,1.31
common 14 8.32 7.34-9.43 13.01 11.38-14.87 0.64 0.53,0.77
18C 3.99 3.32-4.79 3.02 2.47-3.69 1.32 1.01,1.73
19F 7.98 7.19-8.85 3.90 3.50-4.35 2.04 1.76,2.37
23F 3.90 3.43-4.44 2.66 2.32-3.06 1.46 1.21,1.77
Additional 1 1.98 1.73-2.26 0.39 0.33-0.47 5.03 4.01,6.31

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3 0.93 0.82-1.06 0.36 0.31-0.42 2.58 2.12,3.15
5 0.85 0.73-1.00 0.24 0.20-0.29 3.54 2.74,4.58
6A 2.13 1.73-2.62 1.05 0.89-1.24 2.02 1.55,2.63
7F 4.73 4.15-5.39 0.39 0.31-0.49 12.10 9.37,15.62
19A 7.07 6.32-7.91 2.53 2.17-2.94 2.80 2.32,3.38
#Ratio: PCV13-TTGMC/PCV7GMC.

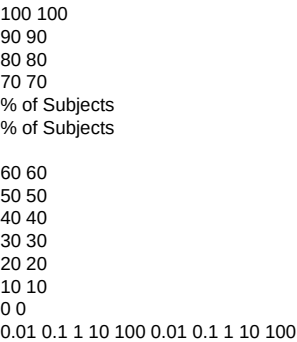
(2). Pneumococcal Serotype antibody concentrations in the subjects aged 12-23 months in PCV13-TT group and PCV7 group
The results of pneumococcal serotype antibody concentrations for subjects aged 12-23 months after full immunizations in PCV13-TT group and PCV7 group were shown in Table 34. to Table 46 The reverse cumulative distribution curves were shown in Figure 53 to 65.

Table 34. Distribution of the pneumococcal serotype 4 IgG antibody concentrations for subjects in PCV13-TT group and PCV7 group post full immunizations

FAS N (%) PPS N (%)
Concentration PCV13-TT (N=200) PCV7 (N=200) PCV13-TT (N=189) PCV7 (N=189)
n % n % n % n %
≥0.01 194 97.00 195 97.50 189 100.0 189 100.0
≥0.1 194 97.00 195 97.50 189 100.0 189 100.0
≥1 192 96.00 190 95.00 189 100.0 187 98.94
≥10 22 11.00 27 13.50 22 11.64 27 14.29
≥100 0 0.00 0 0.00 0 0.00 0 0.00

N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

Serotype 4, After Full Immunizations (FAS) Serotype 4, full immunizations (PPS)



Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

Figure 39 Reverse cumulative distribution curves, pneumococcal serotype 4, IgG antibody concentrations for subjects aged 12-23 months in PCV13-TT group and PCV7 group – post full immunizations

Table 35. Distribution of the pneumococcal serotype 6B IgG antibody concentrations for subjects in PCV13-TT group and PCV7 group post full immunizations

FAS N (%) PPS N (%)
Concentration PCV13-TT (N=200) PCV7 (N=200) PCV13-TT (N=189) PCV7 (N=189)
n % n % n % n %

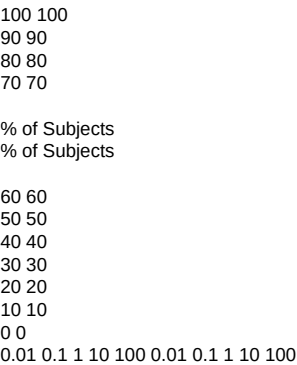
≥0.01	200	100.0	200	100.0	189	100.0	189	100.0
≥0.1	200	100.0	200	100.0	189	100.0	189	100.0
≥1	191	95.50	167	83.50	188	99.47	164	86.77

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≥10	38	19.00	19	9.50	37	19.58	19	10.05
≥100	0	0.00	0	0.00	0	0.00	0	0.00

N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

Serotype 6B, After Full Immunizations (FAS) Serotype 6B, After Full Immunizations (PPS)



Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

Figure 40 Reverse cumulative distribution curves, pneumococcal serotype 6B, IgG antibody concentrations for subjects aged 12-23 months in PCV13-TT group and PCV7 group post full immunizations

Table 36. Distribution of the pneumococcal serotype 9V IgG antibody concentrations for subjects in PCV13-TT group and PCV7 group post full immunizations

FAS N (%) PPS N (%)
Concentration PCV13-TT (N=200) PCV7 (N=200) PCV13-TT (N=189) PCV7 (N=189)
n % n % n % n %

≥0.01	194	97.00	194	97.00	189	100.0	189	100.0
≥0.1	194	97.00	194	97.00	189	100.0	189	100.0
≥1	175	87.50	176	88.00	173	91.53	173	91.53
≥10	9	4.50	14	7.00	9	4.76	14	7.41
≥100	0	0.00	0	0.00	0	0.00	0	0.00

N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

Serotype 9V, After Full Immunizations (FAS) Serotype 9V, After Full Immunizations (PPS)



70 70
% of Subjects
% of Subjects

60 60
50 50
40 40
30 30
20 20
10 10
0 0
0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

Figure 41 Reverse cumulative distribution curves, pneumococcal serotype 9V, IgG antibody concentrations for subjects aged 12-23 months in PCV13-TT group and PCV7 group post full immunizations

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Table 37. Distribution of the pneumococcal serotype 14 IgG antibody concentrations for subjects in PCV13-TT group and PCV7 group post full immunizations

FAS N (%)	PPS N (%)	Concentration	PCV13-TT (N=200)	PCV7 (N=200)	PCV13-TT (N=189)	PCV7 (N=189)
n %	n %	n %	n %	n %	n %	n %
≥0.01	200	100.0	200	100.0	189	100.0
≥0.1	200	100.0	200	100.0	189	100.0
≥1	193	96.50	194	97.00	189	100.0
≥10	91	45.50	139	69.50	89	47.09
≥100	0	0.00	2	1.00	0	0.00

N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

Serotype 14, After Full Immunizations (FAS) Serotype 14, After Full Immunizations (PPS)

100 100
90 90
80 80
70 70

% of Subjects
% of Subjects

60 60
50 50
40 40
30 30
20 20
10 10
0 0
0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

Figure 42 Reverse cumulative distribution curves, pneumococcal serotype 14, IgG antibody concentrations for subjects aged 12-23 months in PCV13-TT group and PCV7 group post full immunizations

Table 38. Distribution of the pneumococcal serotype 18C IgG antibody concentrations for subjects in PCV13-TT group and PCV7 group post full immunizations

FAS N (%)	PPS N (%)	Concentration	PCV13-TT (N=200)	PCV7 (N=200)	PCV13-TT (N=189)	PCV7 (N=189)
n %	n %	n %	n %	n %	n %	n %
≥0.01	195	97.50	193	96.50	189	100.0
≥0.1	195	97.50	193	96.50	189	100.0
≥1	191	95.50	186	93.00	189	100.0
≥10	21	10.50	17	8.50	21	11.11
≥100	0	0.00	0	0.00	0	0.00

N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

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Serotype 18C, After Full Immunizations (FAS) Serotype 18C, After Full Immunizations (PPS)

100 100
90 90
80 80
70 70

% of Subjects
% of Subjects

60 60
50 50
40 40
30 30
20 20
10 10
0 0
0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

Figure 43 Reverse cumulative distribution curves, pneumococcal serotype 18C IgG antibody concentrations for subjects aged 12-23 months in PCV13-TT group and PCV7 group - post full immunizations

Table 39. Distribution of the pneumococcal serotype 19F IgG antibody concentrations for subjects in PCV13-TT group and PCV7 group post full immunizations

FAS N (%)	PPS N (%)	Concentration	PCV13-TT (N=200)	PCV7 (N=200)	PCV13-TT (N=189)	PCV7 (N=189)
n %	n %	n %	n %	n %	n %	n %
≥0.01	200	100.0	200	100.0	189	100.0
≥0.1	200	100.0	200	100.0	189	100.0
≥1	193	96.50	191	95.50	189	100.0
≥10	85	42.50	23	11.50	84	44.44
≥100	0	0.00	0	0.00	0	0.00

N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

Serotype 19F, After Full Immunizations (FAS) Serotype 19F, After Full Immunizations (PPS)

100 100
90 90
80 80
70 70
% of Subjects
% of Subjects

60 60
50 50
40 40
30 30
20 20
10 10
0 0
0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

Figure 44 Reverse cumulative distribution curves, pneumococcal serotype 19F, IgG antibody concentrations for subjects aged 12-23 months in PCV13-TT group and PCV7 group - post full immunizations

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Table 40. Distribution of the pneumococcal serotype 23F IgG antibody concentrations for subjects in PCV13-TT group and PCV7 group post full immunizations

FAS N (%)	PPS N (%)	Concentration	PCV13-TT (N=200)	PCV7 (N=200)	PCV13-TT (N=189)	PCV7 (N=189)
n %	n %	n %	n %	n %	n %	n %
≥0.01	200	100.0	200	100.0	189	100.0
≥0.1	200	100.0	200	100.0	189	100.0
≥1	186	93.00	171	85.50	184	97.35
≥10	22	11.00	16	8.00	20	10.58
≥100	0	0.00	0	0.00	0	0.00

N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

Serotype 23F, After Full Immunizations (FAS) Serotype 23F, After Full Immunizations (PPS)

100 100
90 90
80 80
70 70
% of Subjects
% of Subjects

60 60

50 50
40 40
30 30
20 20
10 10
0 0
0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

Figure 45 Reverse cumulative distribution curves, pneumococcal serotype 23F, IgG antibody concentrations for subjects aged 12-23 months in PCV13-TT group and PCV7 group post full immunizations

Table 41. Distribution of the pneumococcal serotype 1 IgG antibody concentrations for subjects in PCV13-TT group and PCV7 group post full immunizations

FAS N (%)	PPS N (%)
Concentration PCV13-TT (N=200)	PCV7 (N=200) PCV13-TT (N=189) PCV7 (N=189)
n %	n % n % n %
≥0.01	198 99.00 190 95.00 189 100.0 181 95.77
≥0.1	198 99.00 190 95.00 189 100.0 181 95.77
≥1	179 89.50 25 12.50 176 93.12 25 13.23
≥10	1 0.50 0 0.00 1 0.53 0 0.00
≥100	0 0.00 0 0.00 0 0.00 0 0.00

N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

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Serotype 1, After Full Immunizations (FAS) Serotype 1, After Full Immunizations (PPS)

100 100
90 90
80 80
70 70

% of Subjects
% of Subjects

60 60
50 50
40 40
30 30
20 20
10 10
0 0
0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

Figure 46 Reverse cumulative distribution curves, pneumococcal serotype 1, IgG antibody concentrations for subjects aged 12-23 months in PCV13-TT group and PCV7 group - post full immunizations

Table 42. Distribution of the pneumococcal serotype 3 IgG antibody concentrations for subjects in PCV13-TT group and PCV7 group post full immunizations

FAS N (%)	PPS N (%)
Concentration PCV13-TT (N=200)	PCV7 (N=200) PCV13-TT (N=189) PCV7 (N=189)
n %	n % n % n %
≥0.01	198 99.00 198 99.00 189 100.0 188 99.47
≥0.1	194 97.00 190 95.00 189 100.0 186 98.41
≥1	105 52.50 19 9.50 103 54.50 18 9.52
≥10	0 0.00 1 0.50 0 0.00 1 0.53
≥100	0 0.00 0 0.00 0 0.00 0 0.00

N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

Serotype 3, After Full Immunizations (FAS) Serotype 3, After Full Immunizations (PPS)

100 100
90 90
80 80
70 70
% of Subjects
% of Subjects

60 60
50 50
40 40
30 30
20 20
10 10
0 0

0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

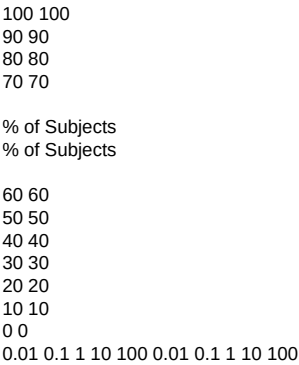
Figure 47 Reverse cumulative distribution curves, pneumococcal serotype 3, IgG antibody concentrations for subjects aged 12-23 months in PCV13-TT group and PCV7 group - post full immunizations

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Table 43 Distribution of the pneumococcal serotype 5 IgG antibody concentrations for subjects in PCV13-TT group and PCV7 group post full immunizations

FAS N (%)	PPS N (%)								
Concentration PCV13-TT (N=200) PCV7 (N=200) PCV13-TT (N=189) PCV7 (N=189)									
n %	n %	n %	n %	n %	n %				
≥0.01	194 97.00	181 90.50	188 99.47	175 92.59					
≥0.1	194 97.00	181 90.50	188 99.47	175 92.59					
≥1	93 46.50	5 2.50	93 49.21	5 2.65					
≥10	0 0.00	0 0.00	0 0.00	0 0.00					
N=number of subjects included in FAS and PPS for vaccine group.									
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.									

Serotype 5, After Full Immunizations (FAS) Serotype 5, After Full Immunizations (PPS)



Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

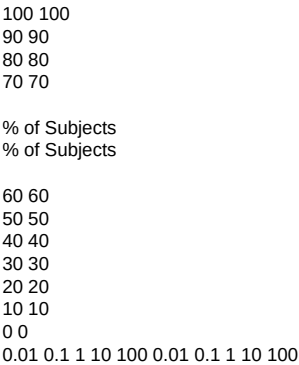
Figure 48 Reverse cumulative distribution curves, pneumococcal serotype 5, IgG antibody concentrations for subjects aged 12-23 months in PCV13-TT group and PCV7 group - post full immunizations

Table 44. Distribution of the pneumococcal serotype 6A IgG antibody concentrations for subjects in PCV13-TT group and PCV7 group post full immunizations

FAS N (%)	PPS N (%)								
Concentration PCV13-TT (N=200) PCV7 (N=200) PCV13-TT (N=189) PCV7 (N=189)									
n %	n %	n %	n %	n %	n %				
≥0.01	194 97.00	197 98.50	186 98.41	188 99.47					
≥0.1	194 97.00	197 98.50	186 98.41	188 99.47					
≥1	167 83.50	99 49.50	164 86.77	97 51.32					
≥10	13 6.50	3 1.50	12 6.35	3 1.59					
≥100	0 0.00	0 0.00	0 0.00	0 0.00					
N=number of subjects included in FAS and PPS for vaccine group.									
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.									

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Serotype 6A, After Full Immunizations (FAS) Serotype 6A, After Full Immunizations (PPS)



Antibody Concentration ■µg/mL)

Antibody Concentration (µg/mL)

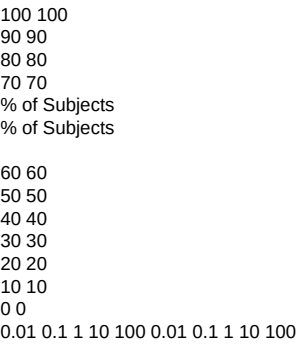
PCV13-TT PCV7 PCV13-TT PCV7

Figure 49 Reverse cumulative distribution curves, pneumococcal serotype 6A IgG antibody concentrations for subjects aged 12-23 months in PCV13-TT group and PCV7 group post full immunizations

Table 45. Distribution of the pneumococcal serotype 7F IgG antibody concentrations for subjects in PCV13-TT group and PCV7 group post full immunizations

FAS N (%)	PPS N (%)								
Concentration PCV13-TT (N=200) PCV7 (N=200) PCV13-TT (N=189) PCV7 (N=189)									
n %	n %	n %	n %	n %	n %	n %	n %	n %	n %
≥0.01	199	99.50	186	93.00	189	100.0	176	93.12	
≥0.1	199	99.50	186	93.00	189	100.0	176	93.12	
≥1	191	95.50	36	18.00	188	99.47	36	19.05	
≥10	25	12.50	1	0.50	25	13.23	1	0.53	
≥100	0	0.00	0	0.00	0	0.00	0	0.00	
N=number of subjects included in FAS and PPS for vaccine group.									
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.									

Serotype 7F, After Full Immunizations (FAS) Serotype 7F, After Full Immunizations (PPS)



Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

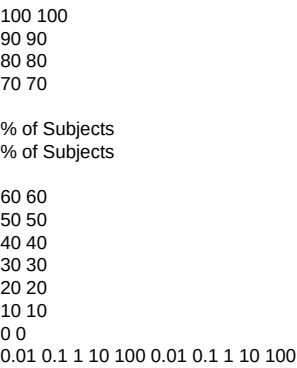
Figure 50 Reverse cumulative distribution curves, pneumococcal serotype 7F, IgG antibody concentrations for subjects aged 12-23 months in PCV13-TT group and PCV7 group - post full immunizations

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Table 46. Distribution of the pneumococcal serotype 19A IgG antibody concentrations for subjects in PCV13-TT group and PCV7 group post full immunizations

FAS N (%)	PPS N (%)								
Concentration PCV13-TT (N=200) PCV7 (N=200) PCV13-TT (N=189) PCV7 (N=189)									
n %	n %	n %	n %	n %	n %	n %	n %	n %	n %
≥0.01	200	100.0	199	99.50	189	100.0	188	99.47	
≥0.1	200	100.0	199	99.50	189	100.0	188	99.47	
≥1	194	97.00	174	87.00	189	100.0	168	88.89	
≥10	64	32.00	20	10.00	64	33.86	20	10.58	
≥100	1	0.50	2	1.00	1	0.53	2	1.06	
N=number of subjects included in FAS and PPS for vaccine group.									
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.									

Serotype 19A, After Full Immunizations (FAS) Serotype 19A, After Full Immunizations (PPS)



Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

Figure 51 Reverse cumulative distribution curves, pneumococcal serotype 19A, IgG antibody

concentrations for subjects aged 12-23 months in PCV13-TT group and PCV7 group - post full immunizations

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6.2.1.3 Positive rate of IgG antibody and the IgG antibody concentrations (GMCs) analysis for subjects aged 24-71 months after immunization in PCV13-TT and control group
(1)Positive rate of IgG antibody and the IgG antibody concentrations (GMCs) analysis for subjects aged 24-71 months
Among the 7 common serotypes the proportion of responders at $\geq 0.35 \mu\text{g/mL}$ in the PCV13-TT was more than 98%. The lower limit of 2-sided, 95% CI for the differences of positive rate between PCV13-TT and control group was greater than -10% (refer to table 19-1) after complete immunization. The GMCs for 13 serotypes was great than or equal to $3.84 \mu\text{g/mL}$. The lower limit of 2-sided, 95% CI for the GMC ratio for 12 of 13 serotypes (except serotype 14) was great than 0.5 (refer to table 19-3). For serotype 14, the lower limit of 2-sided, 95% CI for the GMC ratio was 0.49.
Among the 5 of 6 additional serotypes (except serotype 3) the proportion of responders at $\geq 0.35 \mu\text{g/mL}$ in the PCV13-TT was at least 96%. The lower limit of 2-sided, 95% CI for the positive rate in PCV13-TT was greater than 70%.
The lower limit of 2-sided, 95% CI differences of positive rate for 4 of 6 serotypes (except serotype 6A and 19A) was greater than 0 and the GMC ratio (test group/control group) was greater than 1 between PCV13-TT and control group. For serotype 6A and 19A, the lower limit of 2-sided, 95% CI differences of positive rate was -0.07 and 0.48 separately.
The statistical analysis of positive rate of IgG antibody and the IgG antibody concentrations (GMCs) for subjects aged 24-71 months after immunization in PCV13-TT and control group was shown in tables 19-1 to 19-4.
Table 19-1 Comparison of Subjects aged 24-71 months Achieving a Pneumococcal IgG Antibody Concentration $>0.35 \mu\text{g/mL}$ After immunization between PCV13-TT and PCV7 Group (PPS)
Noninferior / Superiority
PCV13-TT (N=198) PCV7 (N=199)
test
Serotype
Rate diff.
N % 95% CI N % 95% CI 95%CI
(%)
4 198 100.0 98.15-100.0 199 100.0 98.16-100.0 0.00 -1.84,1.85
c 6B 198 100.0 98.15-100.0 197 98.99 96.42-99.88 1.01 -0.38,2.39
o 9V 195 98.48 95.64-99.69 197 98.99 96.42-99.88 -0.51 -2.70,1.68
m
14 197 99.49 97.22-99.99 198 99.50 97.23-99.99 -0.00 -1.40,1.39
m
o 18C 196 98.99 96.40-99.88 199 100.0 98.16-100.0 -1.01 -2.40,0.38
n 19F 198 100.0 98.15-100.0 199 100.0 98.16-100.0 0.00 -1.84,1.85
23F 196 98.99 96.40-99.88 195 97.99 94.93-99.45 1.00 -1.40,3.40

A 1 195 98.48 95.64-99.69 132 66.33 59.31-72.86 32.15 25.37,38.94
d 3 176 88.89 83.66-92.90 89 44.72 37.69-51.92 44.17 35.99,52.34
di 195 98.48 95.64-99.69 127 63.82 56.72-70.50 34.67 27.78,41.56
5
ti
o 6A 191 96.46 92.85-98.57 183 91.96 87.27-95.33 4.50 -0.07,9.08
n 7F 196 98.99 96.40-99.88 142 71.36 64.54-77.53 27.63 21.20,34.07
al 198 100.0 98.15-100.0 198 99.50 97.23-99.99 0.50 -0.48,1.48
19A

Table 19-2 Comparison of Subjects aged 24-71 months Achieving a Pneumococcal IgG Antibody Concentration $>0.35 \mu\text{g/mL}$ After immunization between PCV13-TT and PCV7 Group(FAS)
Noninferior / Superiority
Serotype PCV13-TT (N=200) PCV7 (N=200)
test

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Rate
N % 95% CI N % 95% CI diff. 95%CI
(%)
97.25-
199 99.50 199 99.50 97.25-99.99 0.00 -1.38,1.38
4 99.99
98.17-
200 100.0 198 99.00 96.43-99.88 1.00 -0.38,2.38
6B 100.0
95.68-
197 98.50 198 99.00 96.43-99.88 -0.50 -2.68,1.68
9V 99.69
97.25-
Common 199 99.50 199 99.50 97.25-99.99 0.00 -1.38,1.38
14 99.99
96.43-
198 99.00 200 100.0 98.17-100.0 -1.00 -2.38,0.38
18C 99.88
98.17-
200 100.0 200 100.0 98.17-100.0 0.00 -1.83,1.83
19F 100.0
95.68-
197 98.50 196 98.00 94.96-99.45 0.50 -2.07,3.07
23F 99.69

94.96-
196 98.00 133 66.50 59.50-73.00 31.50 24.68,38.32
1 99.45
83.25-
177 88.50 90 45.00 37.98-52.18 43.50 35.31,51.69
3 92.57
94.96-
196 98.00 128 64.00 56.93-70.65 34.00 27.07,40.93
5 99.45
Additional
92.27-
192 96.00 184 92.00 87.33-95.36 4.00 -0.64,8.64
6A 98.26
95.68-
197 98.50 143 71.50 64.71-77.64 27.00 20.52,33.48
7F 99.69
98.17-
200 100.0 199 99.50 97.25-99.99 0.50 -0.48,1.48
19A 100.0

Table 19-3 Comparison of Subjects aged 24-71 months IgG antibody concentrations (GMCs) between PCV13-TT and PCV7 Group(PPS)
PCV13-TT (N=198) PCV7 (N=199) Noninferior / Superiority test

Serotype	Serotype	GMC	95%CI	GMC	95%CI	Ratio#	95%CI
4	8.90	7.95-9.96	8.58	7.63-9.66	1.04	0.88,1.22	
6B	7.75	6.58-9.12	3.63	3.04-4.32	2.14	1.68,2.71	
9V	3.84	3.35-4.41	3.94	3.36-4.62	0.97	0.79,1.20	
common	14	10.20	8.40-12.38	15.75	12.82-19.34	0.65	0.49,0.86
18C	8.82	7.56-10.29	7.47	6.45-8.66	1.18	0.95,1.46	
19F	11.45	10.05-13.04	5.71	4.98-6.54	2.01	1.66,2.42	
23F	4.42	3.80-5.14	4.29	3.56-5.17	1.03	0.81,1.31	
1	2.06	1.84-2.30	0.31	0.26-0.39	6.54	5.17,8.27	
3	0.84	0.75-0.94	0.38	0.33-0.45	2.19	1.80,2.66	
5	1.60	1.43-1.79	0.31	0.25-0.38	5.13	4.06,6.48	
Additional							
6A	2.35	2.00-2.76	1.29	1.05-1.60	1.82	1.39,2.37	
7F	4.60	4.00-5.28	0.43	0.33-0.56	10.80	8.00,14.58	
19A	9.56	8.47-10.79	3.32	2.85-3.88	2.87	2.36,3.49	

#Ratio: PCV13-TTGMC/PCV7GMC

Table 19-4 Comparison of Subjects aged 24-71 months IgG antibody concentrations (GMCs) between PCV13-TT and PCV7 Group (FAS)
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Noninferior /
PCV13-TT (N=200) PCV7 (N=200)
Serotype Serotype Superiority test
GMC 95%CI GMC 95%CI Ratio# 95%CI
4 8.71 7.73-9.81 8.43 7.46-9.53 1.03 0.87,1.22
6B 7.63 6.48-9.00 3.62 3.04-4.30 2.11 1.66,2.68
9V 3.82 3.33-4.38 3.91 3.34-4.59 0.98 0.79,1.20
common 14 10.01 8.24-12.16 15.60 12.70-19.15 0.64 0.48,0.85
18C 8.86 7.61-10.33 7.37 6.34-8.55 1.20 0.97,1.49
19F 11.39 10.01-12.97 5.69 4.97-6.51 2.00 1.66,2.42
23F 4.34 3.73-5.06 4.26 3.53-5.13 1.02 0.80,1.30
1 2.03 1.81-2.27 0.32 0.26-0.39 6.42 5.08,8.11
3 0.83 0.74-0.93 0.39 0.33-0.45 2.15 1.77,2.61
5 1.59 1.42-1.77 0.32 0.26-0.39 5.02 3.97,6.34
Additional
6A 2.31 1.97-2.72 1.29 1.04-1.60 1.79 1.37,2.34
7F 4.51 3.91-5.19 0.43 0.33-0.56 10.55 7.82,14.23
19A 9.47 8.40-10.68 3.31 2.84-3.86 2.86 2.36,3.48

#Ratio: PCV13-TTGMC/PCV7GMC

(2). Pneumococcal Serotype antibody concentrations in the subjects aged 24-71 months in PCV13-TT group and PCV7 group

The results of pneumococcal serotype antibody concentrations for subjects aged 24-71 months after full immunizations in PCV13-TT group and PCV7 group were shown in Table 47. to Table 59. The reverse cumulative distribution curves were shown in Figure 66 to 78.

Table 47. Distribution of the pneumococcal serotype 4 IgG antibody concentrations for subjects aged 24-71 months in PCV13-TT group and PCV7 group post full immunization

FAS N (%)	PPS N (%)
PCV13-TT	PCV13-TT
Concentration	PCV7 (N=200) PCV7 (N=199)
(N=200)	(N=198)
n %	n % n % n %
≥0.01	200 100.0 200 100.0 198 100.0 199 100.0
≥0.1	200 100.0 200 100.0 198 100.0 199 100.0
≥1	198 99.00 198 99.00 197 99.49 198 99.50
≥10	85 42.50 82 41.00 85 42.93 82 41.21
≥100	0 0.00 0 0.00 0 0.00 0 0.00

N=number of subjects included in FAS and PPS for vaccine group.

n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

Serotype 4, After Full Immunization (FAS) Serotype 4, full immunization (PPS)

100 100
90 90
80 80

70 70
% of Subjects
% of Subjects

60 60
50 50
40 40
30 30
20 20
10 10
0 0
0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration (µg/mL) Antibody Concentration ■µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

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Figure 52 Reverse cumulative distribution curves, pneumococcal serotype 4, IgG antibody concentrations for subjects aged 24-71 months in PCV13-TT group and PCV7 group - post full immunization

Table 48. Distribution of the pneumococcal serotype 6B IgG antibody concentrations for subjects aged 24-71 months in PCV13-TT group and PCV7 group post full immunization

FAS N (%)	PPS N (%)	PCV13-TT PCV13-TT	Concentration PCV7 (N=200)	PCV7 (N=199)
(N=200)	(N=198)			
n %	n %	n %	n %	n %
≥0.01	200 100.0	200 100.0	198 100.0	199 100.0
≥0.1	200 100.0	200 100.0	198 100.0	199 100.0
≥1	192 96.00	174 87.00	191 96.46	173 86.93
≥10	81 40.50	37 18.50	81 40.91	37 18.59
≥100	4 2.00	2 1.00	4 2.02	2 1.01

N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

Serotype 6B, After Full Immunization (FAS) Serotype 6B, After Full Immunization (PPS)

100 100
90 90
80 80
70 70
% of Subjects
% of Subjects

60 60
50 50
40 40
30 30
20 20
10 10
0 0
0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration (µg/mL) Antibody Concentration ■µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

Figure 53 Reverse cumulative distribution curves, pneumococcal serotype 6B, IgG antibody concentrations for subjects aged 24-71 months in PCV13-TT group and PCV7 group - post full immunization

Table 49. Distribution of the pneumococcal serotype 9V IgG antibody concentrations for subjects aged 24-71 months in PCV13-TT group and PCV7 group post full immunization

FAS N (%)	PPS N (%)	PCV13-TT PCV13-TT	Concentration PCV7 (N=200)	PCV7 (N=199)
(N=200)	(N=198)			
n %	n %	n %	n %	n %
≥0.01	200 100.0	198 99.00	198 100.0	197 98.99
≥0.1	200 100.0	198 99.00	198 100.0	197 98.99
≥1	181 90.50	188 94.00	180 90.91	188 94.47
≥10	32 16.00	33 16.50	32 16.16	33 16.58
≥100	0 0.00	0 0.00	0 0.00	0 0.00

N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

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Serotype 9V, After Full Immunization (FAS) Serotype 9V, After Full Immunization (PPS)

100 100
90 90
80 80
70 70
% of Subjects

% of Subjects

60 60
50 50
40 40
30 30
20 20
10 10
0 0
0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

Figure 54 Reverse cumulative distribution curves, pneumococcal serotype 9V, IgG antibody concentrations for subjects aged 24-71 months in PCV13-TT group and PCV7 group - post full immunization

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Table 50. Distribution of the pneumococcal serotype 14 IgG antibody concentrations for subjects aged 24-71 months in PCV13-TT group and PCV7 group post full immunization

FAS N (%)	PPS N (%)	PCV13-TT	PCV13-TT	Concentration	PCV7 (N=200)	PCV7 (N=199)	(N=200)	(N=198)
n %	n %	n %	n %	n %	n %	n %	n %	n %
≥0.01	200	100.0	199	99.50	198	100.0	198	99.50
≥0.1	199	99.50	199	99.50	197	99.49	198	99.50
≥1	198	99.00	198	99.00	197	99.49	197	98.99
≥10	99	49.50	118	59.00	99	50.00	118	59.30
≥100	9	4.50	26	13.00	9	4.55	26	13.07

N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

Serotype 14, After Full Immunization (FAS) Serotype 14, After Full Immunization (PPS)

100 100
90 90
80 80
70 70

% of Subjects
% of Subjects

60 60
50 50
40 40
30 30
20 20
10 10
0 0
0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

Figure 55 Reverse cumulative distribution curves, pneumococcal serotype 14, IgG antibody concentrations for subjects aged 24-71 months in PCV13-TT group and PCV7 group - post full immunization

Table 51. Distribution of the pneumococcal serotype 18C IgG antibody concentrations for subjects aged 24-71 months in PCV13-TT group and PCV7 group post full immunization

FAS N (%)	PPS N (%)	PCV13-TT	PCV13-TT	Concentration	PCV7 (N=200)	PCV7 (N=199)	(N=200)	(N=198)
n %	n %	n %	n %	n %	n %	n %	n %	n %
≥0.01	199	99.50	200	100.0	197	99.49	199	100.0
≥0.1	199	99.50	200	100.0	197	99.49	199	100.0
≥1	197	98.50	195	97.50	195	98.48	195	97.99
≥10	98	49.00	73	36.50	97	48.99	73	36.68
≥100	1	0.50	5	2.50	1	0.51	5	2.51

N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

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Serotype 18C, After Full Immunization (FAS) Serotype 18C, After Full Immunization (PPS)

100 100
90 90

80 80
70 70

% of Subjects
% of Subjects

60 60
50 50
40 40
30 30
20 20
10 10
0 0
0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

Figure 56 Reverse cumulative distribution curves, pneumococcal serotype 18C, IgG antibody concentrations for subjects aged 24-71 months in PCV13-TT group and PCV7 group - post full immunization

Table 52. Distribution of the pneumococcal serotype 19F IgG antibody concentrations for subjects aged 24-71 months in PCV13-TT group and PCV7 group post full immunization

FAS N (%)	PPS N (%)	PCV13-TT	PCV13-TT
Concentration	PCV7 (N=200)	PCV7 (N=199)	(N=200) (N=198)
n %	n %	n %	n %
≥0.01	200 100.0	200 100.0	198 100.0 199 100.0
≥0.1	200 100.0	200 100.0	198 100.0 199 100.0
≥1	198 99.00	196 98.00	196 98.99 195 97.99
≥10	110 55.00	54 27.00	109 55.05 54 27.14
≥100	4 2.00	1 0.50	4 2.02 1 0.50

N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

Serotype 19F, After Full Immunization (FAS) Serotype 19F, After Full Immunization (PPS)

100 100
90 90
80 80
70 70

% of Subjects
% of Subjects

60 60
50 50
40 40
30 30
20 20
10 10
0 0
0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

Figure 57 Reverse cumulative distribution curves, pneumococcal serotype 19F, IgG antibody concentrations for subjects aged 24-71 months in PCV13-TT group and PCV7 group - post full immunization

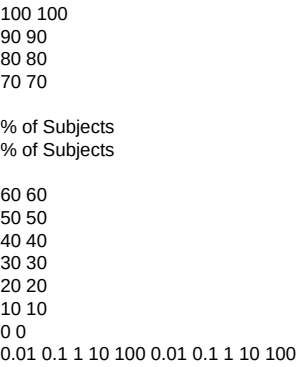
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Table 53. Distribution of the pneumococcal serotype 23F IgG antibody concentrations for subjects aged 24-71 months in PCV13-TT group and PCV7 group post full immunization

FAS N (%)	PPS N (%)	PCV13-TT	PCV13-TT
Concentration	PCV7 (N=200)	PCV7 (N=199)	(N=200) (N=198)
n %	n %	n %	n %
≥0.01	200 100.0	199 99.50	198 100.0 198 99.50
≥0.1	200 100.0	199 99.50	198 100.0 198 99.50
≥1	185 92.50	178 89.00	184 92.93 178 89.45
≥10	41 20.50	43 21.50	41 20.71 43 21.61
≥100	0 0.00	4 2.00	0 0.00 4 2.01

N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

Serotype 23F, After Full Immunization (FAS) Serotype 23F, After Full Immunization (PPS)



Antibody Concentration ■µg/mL
Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

Figure 58 Reverse cumulative distribution curves, pneumococcal serotype 23F, IgG antibody concentrations for subjects aged 24-71 months in PCV13-TT group and PCV7 group - post full immunization

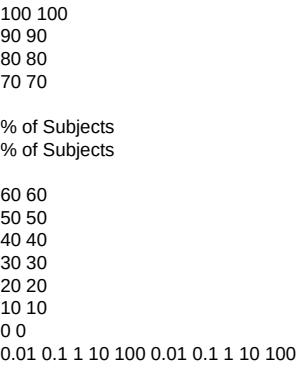
Table 54. Distribution of the pneumococcal serotype 1 IgG antibody concentrations for subjects aged 24-71 months in PCV13-TT group and PCV7 group post full immunization

FAS N (%)	PPS N (%)
PCV13-TT	PCV13-TT
Concentration PCV7 (N=200)	PCV7 (N=199)
(N=200)	(N=198)
n %	n %
≥0.01	200 100.0
186 93.00	198 100.0
185 92.96	
≥0.1	200 100.0
186 93.00	198 100.0
185 92.96	
≥1	165 82.50
13 6.50	165 83.33
13 6.53	
≥10	5 2.50
0 0.00	5 2.53
0 0.00	
≥100	0 0.00
0 0.00	0 0.00
0 0.00	

N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

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Serotype 1, After Full Immunization (FAS) Serotype 1, After Full Immunization (PPS)



Antibody Concentration ■µg/mL
Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

Figure 59 Reverse cumulative distribution curves, pneumococcal serotype 1, IgG antibody concentrations for subjects aged 24-71 months in PCV13-TT group and PCV7 group - post full immunization

Table 55. Distribution of the pneumococcal serotype 3 IgG antibody concentrations for subjects aged 24-71 months in PCV13-TT group and PCV7 group post full immunization

FAS N (%)	PPS N (%)
PCV13-TT	PCV13-TT
Concentration PCV7 (N=200)	PCV7 (N=199)
(N=200)	(N=198)
n %	n %
≥0.01	200 100.0
199 99.50	198 100.0
198 99.50	
≥0.1	199 99.50
189 94.50	198 100.0
188 94.47	
≥1	69 34.50
35 17.50	68 34.34
34 17.09	
≥10	2 1.00
1 0.50	2 1.01
1 0.50	
≥100	0 0.00
0 0.00	0 0.00
0 0.00	

N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

Serotype 3, After Full Immunization (FAS) Serotype 3, After Full Immunization (PPS)

100 100
90 90
80 80
70 70
% of Subjects
% of Subjects

60 60
50 50
40 40
30 30
20 20
10 10
0 0
0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

Figure 60 Reverse cumulative distribution curves, pneumococcal serotype 3, IgG antibody concentrations for subjects aged 24-71 months in PCV13-TT group and PCV7 group - post full immunization

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Table 56. Distribution of the pneumococcal serotype 5 IgG antibody concentrations for subjects aged 24-71 months in PCV13-TT group and PCV7 group post full immunization

FAS N (%)	PPS N (%)								
PCV13-TT	PCV13-TT								
Concentration	PCV7 (N=200)	PCV7 (N=199)							
(N=200)	(N=198)								
n	%	n	%	n	%	n	%	n	%
≥0.01	200	100.0	184	92.00	198	100.0	183	91.96	
≥0.1	200	100.0	184	92.00	198	100.0	183	91.96	
≥1	142	71.00	16	8.00	141	71.21	15	7.54	
≥10	4	2.00	0	0.00	4	2.02	0	0.00	
≥100	0	0.00	0	0.00	0	0.00	0	0.00	
N=number of subjects included in FAS and PPS for vaccine group.									
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.									

Serotype 5, After Full Immunization (FAS) Serotype 5, After Full Immunization (PPS)

100 100
90 90
80 80
70 70

% of Subjects
% of Subjects

60 60
50 50
40 40
30 30
20 20
10 10
0 0
0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

Figure 61 Reverse cumulative distribution curves, pneumococcal serotype 5, IgG antibody concentrations for subjects aged 24-71 months in PCV13-TT group and PCV7 group - post full immunization

FAS N (%)	PPS N (%)								
PCV13-TT	PCV13-TT								
Concentration	PCV7 (N=200)	PCV7 (N=199)							
(N=200)	(N=198)								
n	%	n	%	n	%	n	%	n	%
≥0.01	199	99.50	196	98.00	197	99.49	195	97.99	
≥0.1	199	99.50	196	98.00	197	99.49	195	97.99	
≥1	159	79.50	110	55.00	158	79.80	110	55.28	
≥10	19	9.50	12	6.00	19	9.60	12	6.03	
≥100	0	0.00	2	1.00	0	0.00	2	1.01	
N=number of subjects included in FAS and PPS for vaccine group.									
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.									

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Serotype 6A, After Full Immunization (FAS) Serotype 6A, After Full Immunization (PPS)

100 100
90 90
80 80
70 70

% of Subjects
% of Subjects

60 60
50 50
40 40
30 30
20 20
10 10
0 0
0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

Figure 64 Reverse cumulative distribution curves, pneumococcal serotype 19A, IgG antibody concentrations for subjects aged 24-71 months in PCV13-TT group and PCV7 group - post full immunization.

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6.2.2 Analysis of antibody positive seroconversion rate and antibody level in the pre-immunization antibody negative population
6.2.2.1 Antibody positive seroconversion rate and antibody level in the pre-immunization antibody negative population at the 3–71 months of age
After basic immunization, among the pre-immunization antibody negative population at 3–71 months of age, 7 common serotypes in the PCV13-TT group had antibody positive seroconversion rates greater than 91%. Compared with the PCV7 group, the lower limits of the two-sided 97.5% confidence interval of the differences in antibody positive seroconversion rates were all greater than –10%, except for type 6B (–10.40%) (as shown in Table 21-1); the GMC ratios of post-immune antibodies of 7 common serotypes were all greater than 0.5, and the lower limits of 97.5% confidence interval of antibody GMC ratios were all greater than 0.5, except for the type 14 (0.45) (as shown in Table 21-3).
For the newly added 6 serotypes, the antibody positive seroconversion rates in the PCV13-TT group were all greater than 95%, and the lower limits of the two-sided 95% confidence interval were all greater than 70%. Compared with the PCV7 group, the lower limits of the two-sided 95% confidence interval of the differences in antibody positive seroconversion rates in the PCV13-TT group were all greater than 0, and the lower limits of the two-sided 95% confidence interval of the antibody GMC ratios were all greater than 1.
Statistical analyses of post-immunization IgG antibody positive seroconversion rate and antibody level in the PCV13-TT group and PCV7 group of the pre-immunization IgG antibody negative population at the 3–71 months of age are shown in Tables 21-1 to 21-4.
Table 21-1 Post-Immunization IgG Antibody Positive Rates in Subjects with Negative Pre-Immunization Antibody Level in PCV13-TT Group and PCV7 Group Aged 3-71 Months (PPS)
PCV13-TT group PCV7 group Rate
Serotypes difference 95%/97.5%CI*
N n % 95% CI N n % 95% CI
(%)
99.60- 98.80-
4 923 923 100.0 970 965 99.48 0.52 0.00,1.03
100.0 99.83
88.27- 96.71-
6B 447 408 91.28 435 428 98.39 -7.12 -10.40,-3.83
93.72 99.35
97.80- 98.19-
9V 833 823 98.80 866 858 99.08 -0.28 -1.39,0.84
99.42 99.60
97.44- 97.60-
Common 14 142 142 100.0 152 152 100.0 0.00 -2.84,3.04
100.0 100.0
98.50- 98.56-
18C 865 859 99.31 904 898 99.34 -0.03 -0.91,0.85
99.75 99.76
98.42- 98.52-
19F 232 232 100.0 247 247 100.0 0.00 -1.76,1.87
100.0 100.0
95.69- 96.80-
23F 655 637 97.25 689 676 98.11 -0.86 -2.71,0.98
98.36 98.99
98.11- 52.68-
1 758 751 99.08 810 455 56.17 42.90 39.42,46.39
99.63 59.62
94.73- 40.79-
3 939 903 96.17 960 422 43.96 52.21 48.84,55.58
97.30 47.16
94.59- 23.17-
5 829 797 96.14 867 226 26.07 70.07 66.87,73.28
Additional 97.35 29.12
93.70- 73.41-
6A 599 573 95.66 648 498 76.85 19.81 15.17,22.44
97.15 80.05
98.57- 27.47-

7F 713 709 99.44 727 224 30.81 68.63 65.23,72.03
99.85 34.31
19A 186 186 100.0 98.04- 193 178 92.23 87.51- 7.77 3.99,11.55
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100.0 95.58
Note: N=number of subjects analyzed; n= number of subjects with seroresponse; *CI: 97.5% for common serotypes 97.5%, 95% for additional serotypes.

Table 21-2 Post-Immunization IgG Antibody Positive Rates in Subjects with Negative Pre-Immunization Antibody Level in PCV13-TT Group and PCV7 Group Aged 3-71 Months (FAS)

PCV13-TT group	PCV7 group	Rate	Serotypes	95%	95% difference	95%/97.5%CI*
N	n	%	N	n	%	(%)
CI	CI					
94.92-	94.84-					
4	996	959	96.29	1025	986	96.20 0.09 -1.81,1.99
97.37	97.28					
84.94-	92.45-					
6B	489	431	88.14	467	443	94.86 -6.72 -10.72,-2.72
90.87	96.68					
94.57-	94.32-					
9V	890	855	96.07	911	873	95.83 0.24 -1.84,2.32
97.25	97.03					
92.54-	92.07-					
Common	14	153	148	96.73	161	155 96.27 0.46 -4.19,5.10
98.93	98.62					
94.55-	94.92-					
18C	928	891	96.01	952	917	96.32 -0.31 -2.30,1.67
97.18	97.43					
96.48-	94.07-					
19F	246	243	98.78	262	254	96.95 1.83 -1.02,4.69
99.75	98.67					
91.12-	92.94-					
23F	709	661	93.23	732	694	94.81 -1.58 -4.38,1.22
94.97	96.30					
93.42-	51.10-					
1	820	780	95.12	855	466	54.50 40.62 36.97,44.27
96.49	57.88					
91.00-	40.13-					
3	1010	937	92.77	1009	436	43.21 49.56 46.11,53.01
94.29	46.33					
90.15-	22.44-					
Additional	5	900	829	92.11	920	232 25.22 66.89 63.58,70.21
93.79	28.15					
88.91-	70.92-					
6A	657	600	91.32	690	513	74.35 16.98 13.07,20.88
93.36	77.57					
94.77-	26.22-					
7F	768	740	96.35	768	226	29.43 66.93 63.44,70.41
97.56	32.79					
96.38-	84.62-					
19A	197	195	98.98	203	182	89.66 9.33 4.91,13.75
99.88	93.48					

Note: N=number of subjects analyzed; n= number of subjects with seroresponse; *CI: 97.5% for common serotypes 97.5%, 95% for additional serotypes.

Table 21-3 Post-Immunization IgG GMC in Subjects with Negative Pre-Immunization Antibody Level in PCV13-TT Group and PCV7 Group Aged 3-71 Months (PPS)

Noninferiority/superiority	PCV13-TT group	PCV7 group	Serotypes test	N	GMC	95%CI	N	GMC	95%CI	Ratio#	95%/97.5%CI*
6.13-											
4	923	5.04	4.80-5.29	970	6.46	0.78	0.72,0.85				
6.81											
2.39-											
Common	6B	447	1.58	1.39-1.80	435	2.62	0.60	0.50,0.73			
2.87											
3.35-											
9V	833	3.19	3.01-3.39	866	3.55	0.90	0.82,0.99				
3.75											

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13.63-											
14	142	8.90	7.68-10.30	152	15.71	0.57	0.45,0.71				
18.11											
4.37-											
18C	865	4.07	3.84-4.33	904	4.65	0.88	0.79,0.97				
4.94											
5.62-											
19F	232	6.90	6.22-7.64	247	6.26	1.10	0.93,1.31				
6.97											
3.20-											
23F	655	3.27	3.02-3.55	689	3.44	0.95	0.84,1.08				
3.71											

0.26-
1 758 2.10 1.99-2.21 810 0.29 7.31 6.50,8.22
0.32
0.32-
3 939 1.29 1.23-1.35 960 0.33 3.85 3.58,4.13
0.35
0.07-
5 829 1.35 1.27-1.44 867 0.08 16.03 13.62,18.87
0.10
Additional
0.48-
6A 599 1.80 1.63-1.99 648 0.55 3.25 2.74,3.85
0.63
0.06-
7F 713 5.26 4.93-5.61 727 0.07 72.09 59.09,87.95
0.09
0.66-
19A 186 4.20 3.80-4.64 193 0.80 5.28 4.23,6.58
0.97
Note: N=number of subjects analyzed; #Ratio=PCV13-TT / PCV7; *CI: 97.5% for common serotypes, 95% for additional serotypes.

Table 21-4 Post-Immunization IgG GMC in Subjects with Negative Pre-Immunization Antibody Level in PCV13-TT Group and PCV7 Group Aged 3-71 Months (FAS)
PCV13-TT group PCV7 group Noninferiority/superiority test

Serotypes
N GMC 95%CI N GMC 95%CI Ratio# 95%/97.5%CI*

3.60-
4 996 3.94 1025 5.22 4.77-5.71 0.75 0.65,0.87
4.31
1.33-
6B 489 1.52 467 2.38 2.14-2.65 0.64 0.53,0.77
1.73
2.56-
9V 890 2.78 911 2.95 2.71-3.22 0.94 0.82,1.08
3.01
6.36- 10.72-
Common 14 153 7.80 161 13.19 0.59 0.42,0.82
9.55 16.21
3.06-
18C 928 3.35 952 3.90 3.56-4.26 0.86 0.74,0.99
3.66
5.96-
19F 246 6.64 262 5.61 4.91-6.40 1.19 0.97,1.44
7.41
2.62-
23F 709 2.88 732 3.04 2.77-3.33 0.95 0.82,1.10
3.16
1.65-
1 820 1.79 855 0.27 0.25-0.30 6.53 5.73,7.45
1.94
1.09-
3 1010 1.15 1009 0.32 0.30-0.34 3.65 3.35,3.97
1.23
1.05-
5 900 1.13 920 0.08 0.07-0.09 14.10 11.93,16.66
1.23
Additional
1.43-
6A 657 1.59 690 0.52 0.46-0.60 3.05 2.56,3.63
1.77
4.19-
7F 768 4.56 768 0.07 0.06-0.09 62.73 51.43,76.52
4.97
3.65-
19A 197 4.05 203 0.74 0.61-0.91 5.45 4.34,6.84
4.50

Note: N=number of subjects analyzed; #Ratio=PCV13-TT / PCV7; *CI: 97.5% for common serotypes, 95% for additional serotypes.

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6.2.2.2 Antibody positive seroconversion rate and antibody level in the pre-immunization antibody negative population at the 2–6 months of age

(1) Antibody positive seroconversion rate and antibody level in the test group and control group of the pre-immunization antibody negative population at the 3 months of age
After basic immunization, among the pre-immunization antibody negative population at the 3 months of age, 7 common serotypes in the test group had relatively low antibody positive seroconversion rates, except for type 6B (88.58%), and the antibody positive seroconversion rates of other common serotypes were all greater than 96%. Compared with the control group, the lower limits of the two-sided 97.5% confidence interval of the differences in positive seroconversion rates were all greater than –10% (as shown in Table 22-1); the GMC ratios of antibodies were all greater than 0.5, except for type 6B (0.33), and the lower limits of 97.5% confidence interval of GMC ratios were all greater than 0.5, except for type 6B (0.26) and type 14 (0.46). (as shown in Table 22-3)

For the newly added 6 serotypes, the antibody positive seroconversion rates in the PCV13-TT group were all greater than 96%, and the lower limits of the two-sided 95% confidence interval of the positive seroconversion rates were all greater than 70%; compared with the PCV7 group, the lower limits of the two-sided 95% confidence interval of the differences in antibody positive seroconversion rates in the PCV13-TT group were all greater than 0, and the lower limits of the two-sided 95% confidence interval of the antibody GMC ratios were all greater than 1.

Statistical analyses of post-immunization IgG antibody positive seroconversion rate and antibody

level in the test group and control group of pre-immunization IgG antibody negative population at the 3 months of age are shown in Tables 22-1 to 22-4.

Table 22-1 Post-Immunization IgG Antibody Positive Rates in Subjects with Negative Pre-Immunization Antibody Level in PCV13-TT Group (3M) and PCV7 Group (3M) Aged 3 Months (PPS)
PCV13-TT (3M) group PCV7 (3M) group Rate
Serotypes 95% 95% difference 95%/97.5%CI*
N n % N n % (%)

CI	CI								
99.19-	97.84-								
4 453	453 100.0	472	468	99.15	0.85	-0.10,	1.79		
100.0	99.77								
84.34-	96.88-								
6B 289	256 88.58	278	275	98.92	-10.34	-14.76,-	5.92		
92.01	99.78								
97.76-	98.63-								
9V 389	386 99.23	405	404	99.75	-0.52	-1.66,	0.61		
99.84	99.99								
94.48-	94.72-								
Common 14	65 65 100.0	68	68	100.0	0.00	-6.24,	6.52		
100.0	100.0								
97.79-	96.81-								
18C 394	391 99.24	406	400	98.52	0.72	-0.95,	2.38		
99.84	99.46								
97.66-	97.62-								
19F 156	156 100.0	153	153	100.0	0.00	-2.82,	2.77		
100.0	100.0								
94.52-	97.23-								
23F 355	344 96.90	367	363	98.91	-2.01	-4.40,	0.38		
98.44	99.70								
98.29-	49.79-								
1 421	419 99.52	434	237	54.61	44.92	40.19,	49.65		
99.94	59.36								
96.33-	40.00-								
Additional 3	462 453 98.05	469	209	44.56	53.49	48.82,	58.16		
99.11	49.19								
96.84-	12.74-								
5 453	446 98.45	464	74	15.95	82.51	78.99,	86.03		
99.38	19.60								

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94.43-	68.33-								
6A 326	316 96.93	345	253	73.33	23.60	18.57,	28.63		
98.52	77.93								
98.16-	14.15-								
7F 390	388 99.49	387	69	17.83	81.66	77.78,	85.54		
99.94	22.01								
96.92-	84.46-								
19A 118	118 100.0	114	104	91.23	8.77	3.58,	13.96		
100.0	95.71								

Note: N=number of subjects analyzed; n=number of subjects with seroresponse; *CI: 97.5% for common serotypes, 95% for additional serotypes.

Table 22-2 Post-Immunization IgG Antibody Positive Rates in Subjects with Negative Pre-Immunization Antibody Level in PCV13-TT Group (3M) and PCV7 Group (3M) Aged 3 Months (FAS)
PCV13-TT (3M) group PCV7 (3M) group Rate
differ 95%/97.5%CI
Serotypes

N n % 95% CI	N n % 95% CI	ence *							
(%)									
4 501	479 95.61	93.43-	97.23	504	484	96.03	93.94-	97.56	-0.42 -3.25,2.41
6B 319	275 86.21	81.93-	89.80	299	286	95.65	92.68-	97.66	-9.45 -14.52,-4.37
Co 9V	424 408 96.23	93.94-	97.83	428	414	96.73	94.57-	98.20	-0.50 -3.33,2.33
mm	14 73 70 95.89	88.46-	99.14	71	70	98.59	92.40-	99.96	-2.70 -8.78,3.38
on 18C	434 413 95.16	92.70-	96.98	432	414	95.83	93.49-	97.51	-0.67 -3.83,2.49
19F 168	165 98.21	94.87-	99.63	165	159	96.36	92.25-	98.65	1.85 -2.14,5.84
23F 392	364 92.86	89.84-	95.20	390	375	96.15	93.74-	97.83	-3.30 -6.94,0.35
1 462	441 95.45	93.14-	97.16	462	247	53.46	48.80-	58.08	41.99 37.06,46.92
Add 3	509 479 94.11	91.69-	95.99	499	221	44.29	39.88-	48.77	49.82 45.00,54.63
ition 5	499 469 93.99	91.53-	95.91	495	78	15.76	12.66-	19.27	78.23 74.40,82.06
al 6A	362 336 92.82	89.65-	95.25	366	263	71.86	66.95-	76.41	20.96 15.64,26.28
7F 424	409 96.46	94.23-	98.01	410	70	17.07	13.56-	21.07	79.39 75.34,83.43
19A 127	125 98.43	94.43-	99.81	123	107	86.99	79.74-	92.38	11.43 5.11,17.76
Note: N=number of subjects analyzed; n=number of subjects with seroresponse; *CI: 97.5% for common serotypes, 95% for additional serotypes.									

Table 22-3 Post-Immunization IgG GMC in Subjects with Negative Pre-Immunization Antibody Level in Subjects Aged 3 Months (PPS)
PCV13-TT (3M) PCV7 (3M)
Serotypes group group Ratio# 95%/97,5%CI*
N GMC 95%CI N GMC 95%CI

4.25-	7.21-								
4 453	4.56 472	7.80	0.59	0.52,	0.66				
4.90	8.43								
0.98-	3.13-								
6B 289	1.15 278	3.47	0.33	0.26,	0.42				
1.36	3.85								
3.49-	4.26-								

9V 389 3.80 405 4.57 0.83 0.73,0.94
 4.13 4.91
 10.37- 17.15-
 Common 14 65 12.68 68 20.40 0.62 0.46,0.84
 15.52 24.27
 2.94- 4.74-
 18C 394 3.18 406 5.25 0.61 0.52,0.70
 3.44 5.81
 6.55- 7.85-
 19F 156 7.45 153 8.86 0.84 0.69,1.03
 8.48 9.99
 2.74- 4.22-
 23F 355 3.08 367 4.64 0.66 0.56,0.79
 3.47 5.10

Additional 1 421 2.20 2.05- 434 0.28 0.24- 7.85 6.64,9.29

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2.36 0.32
 1.58- 0.31-
 3 462 1.69 469 0.33 5.11 4.63,5.65
 1.81 0.36
 1.77- 0.04-
 5 453 1.90 464 0.04 43.01 33.99,54.42
 2.05 0.06
 1.66- 0.40-
 6A 326 1.89 345 0.48 3.92 3.12,4.92
 2.14 0.58
 5.06- 0.03-
 7F 390 5.52 387 0.04 151.85 115.63,199.42
 6.02 0.05
 3.60- 0.55-
 19A 118 4.05 114 0.71 5.71 4.34,7.52
 4.55 0.91

Note: N=number of subjects analyzed; #Ratio: PCV13-TT / PCV7; *CI: 97.5% for common serotypes, 95% for additional serotypes.

Table 22-4 Post-Immunization IgG GMC in Subjects with Negative Pre-Immunization

Antibody Level in Subjects Aged 3 Months (FAS)

PCV13-TT (3M) group PCV7 (3M) group

Serotypes Ratio# 95%/97.5%CI*

N GMC 95%CI N GMC 95%CI

4 501 3.47 3.05-3.94 504 6.36 5.59-7.25 0.55 0.44,0.67
 6B 319 1.15 0.97-1.35 299 3.11 2.73-3.55 0.37 0.29,0.47
 9V 424 3.35 3.02-3.73 428 3.91 3.48-4.38 0.86 0.72,1.03
 73 7.30-
 Common 14 10.33 71 19.65 15.90-24.29 0.53 0.33,0.84
 14.63
 18C 434 2.59 2.30-2.92 432 4.64 4.10-5.26 0.56 0.46,0.68
 19F 168 7.02 6.11-8.07 165 7.56 6.33-9.03 0.93 0.72,1.20
 23F 392 2.68 2.33-3.08 390 4.07 3.57-4.65 0.66 0.53,0.82
 1 462 1.80 1.60-2.03 462 0.26 0.23-0.31 6.82 5.62,8.26
 3 509 1.47 1.34-1.60 499 0.31 0.29-0.34 4.67 4.13,5.28
 5 499 1.50 1.34-1.69 495 0.04 0.03-0.05 34.88 27.39,44.43
 Additional
 6A 362 1.69 1.48-1.93 366 0.46 0.38-0.56 3.64 2.88,4.60
 7F 424 4.70 4.15-5.31 410 0.04 0.03-0.05 127.54 96.80,168.04
 19A 127 3.87 3.41-4.39 123 0.64 0.49-0.83 6.07 4.54,8.12

Note: N=number of subjects analyzed; #Ratio: PCV13-TT / PCV7; *CI: 97.5% for common serotypes, 95% for additional serotypes.

(2) Antibody positive seroconversion rate and antibody level of the pre-immunization antibody negative population in the test group at the 2 months of age and the control group at the 3 months of age

After basic immunization, among the pre-immunization negative population, 7 common serotypes in the test group at the 2 months of age (minimum, 6 weeks) had post-immunization antibody positive seroconversion rates greater than 98%. Compared with the control group at the 3 months of age, the lower limits of the two-sided 97.5% confidence interval of the differences in antibody positive seroconversion rates were all greater than -10% (as shown in Table 23-1); the GMC ratios of post-immunization antibodies were all greater than 2 µg/mL, the antibody GMC ratios were all greater than 0.5, except for type 4 and type 18C, and the lower limits of 97.5% confidence interval of antibody GMC ratios were greater than 0.5, except for type 4, type 14, type 18C, and type 19F (as shown in Table 23-3).

For the newly added 6 serotypes, the post-immunization antibody positive seroconversion rates in

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the test group at the 2 months of age were all greater than 96%, except for type 5 (89.73%), and the lower limits of the two-sided 95% confidence interval were all greater than 70%; compared with the control group, the lower limits of the two-sided 95% confidence interval of the differences in post-immunization antibody positive seroconversion rates were all greater than 0, and the lower limits of the 95% confidence interval of the antibody GMC ratios were all greater than 1.

Statistical analyses of post-immunization antibody positive seroconversion rate and antibody level of pre-immunization IgG antibody negative population in the test group at the 2 months of age and the control group at the 3 months of age are shown in Tables 23-1 to 23-4.

Table 23-1 Post-Immunization IgG Antibody Positive Rates in Subjects with Negative Pre-Immunization Antibody Level in PCV13-TT (2M) Group and PCV7 (3M) Group (PPS)

PCV13-TT (2M) group PCV7 (3M) group Rate

Serotypes 95% 95% difference 95%/97.5%CI*
N n % N n % (%)
CI CI
99.15- 97.84-
4 434 434 100.0 472 468 99.15 0.85 -0.10,1.79
100.0 99.77
95.21- 96.88-
6B 180 177 98.33 278 275 98.92 -0.59 -3.14,1.96
99.65 99.78
98.23- 98.63-
9V 313 312 99.68 405 404 99.75 -0.07 -0.98,0.83
99.99 99.99
90.26- 94.72-
Common 14 36 36 100.0 68 68 100.0 0.00 -6.24,11.46
100.0 100.0
98.84- 96.81-
18C 317 317 100.0 406 400 98.52 1.48 0.14,2.82
100.0 99.46
95.70- 97.62-
19F 84 84 100.0 153 153 100.0 0.00 -2.82,5.08
100.0 100.0
98.68- 97.23-
23F 278 278 100.0 367 363 98.91 1.09 -0.12,2.30
100.0 99.70
98.88- 49.79-
1 327 327 100.0 434 237 54.61 45.39 40.71,50.08
100.0 59.36
93.78- 40.00-
3 452 434 96.02 469 209 44.56 51.45 46.61,56.30
97.62 49.19
86.18- 12.74-
Additional 5 370 332 89.73 464 74 15.95 73.78 69.24,78.33
92.63 19.60
96.87- 68.33-
6A 228 226 99.12 345 253 73.33 25.79 20.97,30.61
99.89 77.93
97.87- 14.15-
7F 336 334 99.40 387 69 17.83 81.58 77.67,85.48
99.93 22.01
95.85- 84.46-
19A 87 87 100.0 114 104 91.23 8.77 3.58,13.96
100.0 95.71
Note: N=number of subjects analyzed, n=number of subjects with seroresponse; *CI: 97.5% for common serotypes, 95% for additional serotypes.

Table 23-2 Post-Immunization IgG Antibody Positive Rates in Subjects with Negative Pre-Immunization Antibody Level in PCV13-TT (2M) Group and PCV7 (3M) Group (FAS)
PCV13-TT (2M) group PCV7 (3M) group
Rate
95%/97.5%
Serotypes Antibody seroresponse Antibody seroresponse differen
N N CI*
n % 95% CI n % 95% CI ce (%)

C 4 465 440 94.62 92.17-96.49 504 484 96.03 93.94-97.56 -1.41 -4.46,1.64
o 6B 192 182 94.79 90.63-97.47 299 286 95.65 92.68-97.66 -0.86 -5.32,3.60

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m 9V 329 315 95.74 92.96-97.65 428 414 96.73 94.57-98.20 -0.98 -4.14,2.17
m 42 37 88.10 74.37-96.02 71 70 98.59 92.40-99.96 -10.50 -22.13,1.13
14
on
18C 333 320 96.10 93.42-97.91 432 414 95.83 93.49-97.51 0.26 -2.95,3.47
19F 89 84 94.38 87.37-98.15 165 159 96.36 92.25-98.65 -1.98 -8.35,4.39
23F 293 282 96.25 93.38-98.11 390 375 96.15 93.74-97.83 0.09 -3.22,3.40
1 349 333 95.42 92.66-97.36 462 247 53.46 48.80,58.08 41.95 36.90,47.00
A
dd 3 487 440 90.35 87.37-92.82 499 221 44.29 39.88-48.77 46.06 40.97,51.15
iti 5 397 338 85.14 81.25-88.49 495 78 15.76 12.66-19.27 69.38 64.63,74.13
on 6A 243 229 94.24 90.52-96.81 366 263 71.86 66.95-76.41 22.38 16.92,27.84
al 351 410
7F 337 96.01 93.40-97.80 70 17.07 13.56-21.07 78.94 74.76,83.12
19A 91 88 96.70 90.67-99.31 123 107 86.99 79.74-92.38 9.71 2.73,16.70
Note: N=number of subjects analyzed, n=number of subjects with seroresponse; *CI: 97.5% for common serotypes, 95% for additional serotypes.
Table 23-3 Post-Immunization IgG GMC in Subjects with Negative Pre-Immunization Antibody Level in PCV13-TT (2M) Group and PCV7 (3M) Group (PPS)
PCV7 (3M)
PCV13-TT (2M) group
group
95%/97.5%CI
Serotypes Analyze Analyze Ratio#
GM 95%C GM 95%C *
d d
C I C I
n n
2.28- 7.21-
4 434 2.45 472 7.80 0.31 0.28,0.35

2.63 8.43
 3.90- 3.13-
 6B 180 4.48 278 3.47 1.29 1.06,1.57
 5.15 3.85
 3.42- 4.26-
 9V 313 3.75 405 4.57 0.82 0.72,0.93
 4.10 4.91
 11.7 8.98- 20.4 17.15-
 Common 14 36 68 0.58 0.41,0.82
 7 15.43 0 24.27
 2.11- 4.74-
 18C 317 2.28 406 5.25 0.44 0.37,0.51
 2.47 5.81
 4.51- 7.85-
 19F 84 5.13 153 8.86 0.58 0.47,0.72
 5.84 9.99
 3.64- 4.22-
 23F 278 4.07 367 4.64 0.88 0.74,1.03
 4.54 5.10
 1.96- 0.24-
 1 327 2.14 434 0.28 7.65 6.34,9.23
 2.33 0.32
 1.33- 0.31-
 3 452 1.42 469 0.33 4.30 3.90,4.75
 1.52 0.36
 0.71- 0.04-
 5 370 0.78 464 0.04 17.54 13.51,22.77
 Additiona 0.85 0.06
 I 2.91- 0.40-
 6A 228 3.30 345 0.48 6.84 5.31,8.81
 3.74 0.58
 4.39- 0.03- 131.8
 7F 336 4.79 387 0.04 98.64,176.10
 5.22 0.05 0
 19 2.80- 0.55-
 87 3.21 114 0.71 4.53 3.31,6.20
 A 3.69 0.91
 Note: N=number of subjects analyzed; #Ratio: PCV13-TT (2M) / PCV7 (3M); *CI: 97.5% for common serotypes, 95% for additional serotypes.

Table 23-4 Post-Immunization IgG GMC in Subjects with Negative Pre-Immunization
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Antibody Level in PCV13-TT (2M) Group and PCV7 (3M) Group (FAS)
 PCV13-TT (2M)
 PCV7 (3M) group
 Serotypes group Ratio# 95%/97.5%CI*
 N GMC 95%CI N GMC 95%CI

1.71-
 4 465 1.93 504 6.36 5.59-7.25 0.30 0.25,0.37
 2.18
 3.35-
 6B 192 3.97 299 3.11 2.73-3.55 1.28 1.00,1.63
 4.71
 2.75-
 9V 329 3.15 428 3.91 3.48-4.38 0.81 0.66,0.99
 3.60
 2.31- 15.90-
 Common 14 42 5.04 71 19.65 0.26 0.12,0.54
 10.99 24.29
 1.80-
 18C 333 2.01 432 4.64 4.10-5.26 0.43 0.36,0.53
 2.24
 3.55-
 19F 89 4.31 165 7.56 6.33-9.03 0.57 0.41,0.79
 5.24
 3.21-
 23F 293 3.63 390 4.07 3.57-4.65 0.89 0.72,1.10
 4.10
 1.71-
 1 349 1.90 462 0.26 0.23-0.31 7.17 5.90,8.71
 2.10
 1.11-
 3 487 1.21 499 0.31 0.29-0.34 3.87 3.42,4.37
 1.33
 0.59-
 5 397 0.66 495 0.04 0.03-0.05 15.30 11.79,19.84
 0.74
 Additional
 2.24-
 6A 243 2.68 366 0.46 0.38-0.56 5.78 4.39,7.61
 3.20
 3.70-
 7F 351 4.15 410 0.04 0.03-0.05 112.82 84.33,150.94
 4.67
 2.50-
 19A 91 2.94 123 0.64 0.49-0.83 4.62 3.29,6.48
 3.46
 Note: N=number of subjects analyzed; #Ratio: PCV13-TT (2M) / PCV7 (3M); *CI: 97.5% for common serotypes, 95% for additional serotypes.

(3) Post-immunization antibody positive seroconversion rate and antibody level of pre-immunization antibody negative population in the test group at the 2 months of age and the test group at the 3 months of age

After basic immunization, among the pre-immunization negative population, except for type 5 (89.73%), the post-immunization antibody positive seroconversion rates of the remaining 12 serotypes in the test group at the 2 months of age (minimum, 6 weeks) were all greater than 96%. Compared with the test group at the 3 months of age, except for type 5, the lower limits of the two-sided 97.5% confidence interval of the differences in antibody positive seroconversion rates of the other serotypes were greater than -10% (as shown in Table 24-1); except for type 5 (0.41), the antibody GMC ratios were all greater than 0.5, and the lower limits of 97.5% confidence interval of antibody GMC ratios were all greater than 0.5, except for type 4 (0.48) and type 5 (0.36) (as shown in Table 24-3).

Statistical analyses of post-immunization antibody positive seroconversion rate and antibody level of pre-immunization IgG antibody negative population in the test group at the 2 months of age and the test group at the 3 months of age are shown in Tables 24-1 to 24-4.

Table 24-1 Post-Immunization IgG Antibody Positive Rates in Subjects with Negative Pre-Immunization Antibody Level in PCV13-TT (2M) Group and PCV13-TT (3M) Group (PPS)

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PCV13-TT (2M) group	PCV13-TT group (3M)	Rate	N	N diffe	Serotypes	97.5%CI	n	%	95% CI	n	%	95% CI	rence	(%)
4	434	434	100.0	99.15-100.0	453	453	100.0	99.19-100.0	0.00	-0.96	1.00			
6B	180	177	98.33	95.21-99.65	289	256	88.58	84.34-92.01	9.75	5.04	14.46			
C	9V	313	312	99.68	98.23-99.99	389	386	99.23	97.76-99.84	0.45	-0.77	1.68		
o	m	14	36	36	100.0	90.26-100.0	65	65	100.0	94.48-100.0	0.00	-6.52	11.46	
m	317	317	100.0	98.84-100.0	394	391	99.24	97.79-99.84	0.76	-0.22	1.74			
18C	on	19F	84	84	100.0	95.70-100.0	156	156	100.0	97.66-100.0	0.00	-2.77	5.08	
23F	278	278	100.0	98.68-100.0	355	344	96.90	94.52-98.44	3.10	1.04	5.16			
1	327	327	100.0	98.88-100.0	421	419	99.52	98.29-99.94	0.48	-0.28	1.23			
A	3	452	434	96.02	93.78-97.62	462	453	98.05	96.33-99.11	-2.03	-4.55	0.48		
dd	iti	5	370	332	89.73	86.18-92.63	453	446	98.45	96.84-99.38	-8.73	-12.49	-4.96	
on	228	226	99.12	96.87-99.89	326	316	96.93	94.43-98.52	2.19	-0.36	4.74			
6A	al	7F	336	334	99.40	97.87-99.93	390	388	99.49	98.16-99.94	-0.08	-1.32	1.16	
19A	8787	100.0	95.85-100.0	118	118	100.0	96.92-100.0	0.00	-3.65	4.91				

Note: N=number of subjects analyzed, n=number of subjects with seroresponse.

Table 24-2 Post-Immunization IgG Antibody Positive Rates in Subjects with Negative Pre-Immunization Antibody Level in PCV13-TT (2M) Group and PCV13-TT (3M) Group (FAS)

PCV13-TT (2M) group	PCV13-TT group (3M)	Rate
Serotypes difference 97.5%CI		
N n % 95% CI	N n % 95% CI	(%)
92.17- 93.43-		
4 465 440 94.62	501 479 95.61	-0.99 -4.10,2.13
96.49 97.23		
90.63- 81.93-		
6B 192 182 94.79	319 275 86.21	8.58 2.96,14.21
97.47 89.80		
92.96- 93.94-		
9V 329 315 95.74	424 408 96.23	-0.48 -3.73,2.76
97.65 97.83		
74.37- 88.46- -		
Common 14 42 37 88.10	73 70 95.89	-7.80
96.02 99.14	20.15,4.56	
93.42- 92.70-		
18C 333 320 96.10	434 413 95.16	0.93 -2.38,4.25
97.91 96.98		
87.37- 94.87-		
19F 89 84 94.38	168 165 98.21	-3.83 -9.76,2.10
98.15 99.63		
93.38- 89.84-		
23F 293 282 96.25	392 364 92.86	3.39 -0.44,7.22
98.11 95.20		
92.66- 93.14-		
1 349 333 95.42	462 441 95.45	-0.04 -3.36,3.28
97.36 97.16		
87.37- 91.69-		
3 487 440 90.35	509 479 94.11	-3.76 -7.56,0.05
92.82 95.99		
81.25- 91.53- -13.51,-		
Additional 5 397 338 85.14	499 469 93.99	-8.85
88.49 95.91	4.19	
90.52- 89.65-		
6A 243 229 94.24	362 336 92.82	1.42 -3.10,5.95
96.81 95.25		

93.40- 94.23-
 7F 351 337 96.01 424 409 96.46 -0.45 -3.54,2.64
 97.80 98.01
 19A 91 88 96.70 90.67- 127 125 98.43 94.43- -1.72 -6.59,3.15
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99.31 99.81
 Note: N=number of subjects analyzed, n=number of subjects with seroresponse.

Table 24-3 Post-Immunization IgG GMC in Subjects with Negative Pre-Immunization
 Antibody Level in PCV13-TT (2M) Group and PCV13-TT (3M) Group (PPS)
 PCV13-TT (2M) group PCV13-TT group (3M)
 Serotypes Ratio# 97.5%CI
 N GMC 95%CI N GMC 95%CI
 4 434 2.45 2.28-2.63 453 4.56 4.25-4.90 0.54 0.48,0.60
 6B 180 4.48 3.90-5.15 289 1.15 0.98-1.36 3.88 2.95,5.11
 9V 313 3.75 3.42-4.10 389 3.80 3.49-4.13 0.99 0.86,1.14
 Common 14 36 11.77 8.98-15.43 65 12.68 10.37-15.52 0.93 0.63,1.36
 18C 317 2.28 2.11-2.47 394 3.18 2.94-3.44 0.72 0.63,0.82
 19F 84 5.13 4.51-5.84 156 7.45 6.55-8.48 0.69 0.55,0.86
 23F 278 4.07 3.64-4.54 355 3.08 2.74-3.47 1.32 1.09,1.59
 1 327 2.14 1.96-2.33 421 2.20 2.05-2.36 0.97 0.86,1.10
 3 452 1.42 1.33-1.52 462 1.69 1.58-1.81 0.84 0.75,0.94
 5 370 0.78 0.71-0.85 453 1.90 1.77-2.05 0.41 0.36,0.47
 Additional
 6A 228 3.30 2.91-3.74 326 1.89 1.66-2.14 1.75 1.42,2.16
 7F 336 4.79 4.39-5.22 390 5.52 5.06-6.02 0.87 0.75,1.00
 19A 87 3.21 2.80-3.69 118 4.05 3.60-4.55 0.79 0.65,0.97
 Note: N=number of subjects analyzed; #Ratio: PCV13-TT (2M) / PCV13-TT (3M).

Table 24-4 Post-Immunization IgG GMC in Subjects with Negative Pre-Immunization
 Antibody Level in PCV13-TT (2M) Group and PCV13-TT (3M) Group (FAS)
 PCV13-TT (2M) group PCV13-TT group (3M)
 Serotypes Ratio# 97.5%CI
 N GMC 95%CI N GMC 95%CI
 4 465 1.93 1.71-2.18 501 3.47 3.05-3.94 0.56 0.45,0.68
 6B 192 3.97 3.35-4.71 319 1.15 0.97-1.35 3.46 2.60,4.61
 9V 329 3.15 2.75-3.60 424 3.35 3.02-3.73 0.94 0.78,1.14
 Common 14 42 5.04 2.31-10.99 73 10.33 7.30-14.63 0.49 0.21,1.14
 18C 333 2.01 1.80-2.24 434 2.59 2.30-2.92 0.78 0.64,0.94
 19F 89 4.31 3.55-5.24 168 7.02 6.11-8.07 0.61 0.47,0.81
 23F 293 3.63 3.21-4.10 392 2.68 2.33-3.08 1.35 1.09,1.69
 1 349 1.90 1.71-2.10 462 1.80 1.60-2.03 1.05 0.87,1.27
 3 487 1.21 1.11-1.33 509 1.47 1.34-1.60 0.83 0.72,0.95
 5 397 0.66 0.59-0.74 499 1.50 1.34-1.69 0.44 0.36,0.53
 Additional
 6A 243 2.68 2.24-3.20 362 1.69 1.48-1.93 1.59 1.24,2.04
 7F 351 4.15 3.70-4.67 424 4.70 4.15-5.31 0.88 0.73,1.08
 19A 91 2.94 2.50-3.46 127 3.87 3.41-4.39 0.76 0.60,0.96
 Note: N=number of subjects analyzed; #Ratio: PCV13-TT (2M) / PCV13-TT (3M).

6.2.2.3 Antibody positive seroconversion rate and antibody level in the pre-immunization
 antibody negative population at the 7–11 months of age
 After basic immunization, among the pre-immunization antibody negative population at 7–11
 months of age, 7 common serotypes in the PCV13-TT group had post-immunization antibody
 positive seroconversion rates greater than 94%. Compared with the PCV7 group, except for type
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6B (–10.55%), the lower limits of the two-sided 95% confidence interval of the differences in
 antibody positive seroconversion rates of the other common serotypes were all greater than –10%
 (as shown in Table 25-1); except for type 14, the lower limits of the two-sided 95% confidence
 interval of post-immunization antibody GMC ratios of the other common serotypes were greater
 than 0.5 (as shown in Table 25-3).
 For the newly added 6 serotypes, the post-immunization antibody positive seroconversion rates in
 the PCV13-TT group were all greater than 90%, and the lower limits of the two-sided 95%
 confidence interval were all greater than 70%; compared with the PCV7 group, the lower limits of
 the two-sided 95% confidence interval of the differences in post-immunization antibody positive
 seroconversion rates in the PCV13-TT group were all greater than 0, and the lower limits of the
 two-sided 95% confidence interval of the post-immunization antibody GMC ratios were all greater
 than 1.
 Statistical analyses of post-immunization IgG antibody positive seroconversion rate and antibody
 level of the pre-immunization antibody negative population at the 7–11 months of age are shown
 in Tables 25-1 to 25-4.

Table 25-1 Post-Immunization IgG Antibody Positive Rates in Subjects with Negative
 Pre-Immunization Antibody Level in PCV13-TT Group and PCV7 Group Aged 7-11
 Months (PPS)
 PCV13-TT group PCV7 group Rate
 Serotypes 95% 95% difference 95%CI
 N n % N n %
 CI CI
 97.91- 97.97-
 4 175 175 100.0 180 180 100.0 0.00 -2.03,2.09
 100.0 100.0
 87.52- 96.31- -10.55,-
 6B 101 95 94.06 98 98 100.0 -5.94
 97.79 100.0 1.33
 94.15- 94.35-

9V 172 168 97.67 178 174 97.75 -0.08 -3.21,3.05
 99.36 99.38
 93.73- 93.94-
 Common 14 57 57 100.0 59 59 100.0 0.00 -6.06,6.27
 100.0 100.0
 96.89- 97.98-
 18C 177 176 99.44 181 181 100.0 -0.56 -1.67,0.54
 99.99 100.0
 94.04- 94.56-
 19F 60 60 100.0 66 66 100.0 0.00 -5.44,5.96
 100.0 100.0
 92.44- 92.07-
 23F 151 146 96.69 161 155 96.27 0.42 -3.67,4.50
 98.92 98.62
 94.67- 62.45-
 1 133 131 98.50 146 103 70.55 27.95 20.27,35.63
 99.82 77.80
 94.96- 52.01-
 3 171 168 98.25 176 105 59.66 38.59 31.08,46.10
 99.64 66.97
 84.46- 32.22-
 Additional 5 162 146 90.12 169 67 39.64 50.48 41.79,59.17
 94.25 47.44
 88.82- 66.22-
 6A 137 129 94.16 140 104 74.29 19.87 11.64,28.11
 97.45 81.29
 96.06- 31.65- 51.91-
 7F 139 138 99.28 152 60 39.47 59.81
 99.98 47.72 67.70
 93.02- 83.54-
 19A 51 51 100.0 59 55 93.22 6.78 0.36,13.19
 100.0 98.12

Note: N=number of subjects analyzed, n=number of subjects with seroresponse.

Table 25-2 Post-Immunization IgG Antibody Positive Rates in Subjects with Negative Pre-Immunization Antibody Level in PCV13-TT Group and PCV7 Group Aged 7-11 Months (FAS)

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PCV13-TT group PCV7 group Rate

Serotypes N N difference 95%CI

n % 95% CI n % 95% CI (%)

188 92.48- 191 91.24-

181 96.28 182 95.29 0.99 -3.06,5.03

4 98.49 97.82

109 82.66- 104 90.44-

98 89.91 100 96.15 -6.25 -13.00,0.51

6B 94.85 98.94

183 90.87- 189 88.53-

174 95.08 176 93.12 1.96 -2.82,6.74

9V 97.73 96.29

59 90.91- 62 86.50-

Common 58 98.31 59 95.16 3.14 -3.13,9.42

14 99.96 98.99

190 91.87- 192 91.29-

182 95.79 183 95.31 0.48 -3.66,4.61

18C 98.16 97.83

62 94.22- 69 89.92-

62 100.0 67 97.10 2.90 -1.06,6.86

19F 100.0 99.65

160 88.03- 170 87.28-

149 93.13 157 92.35 0.77 -4.83,6.37

23F 96.52 95.87

143 89.27- 153 59.28-

135 94.41 103 67.32 27.09 18.75,35.42

1 97.55 74.68

184 89.56- 185 49.83-

173 94.02 106 57.30 36.72 28.82,44.63

3 96.98 64.53

174 80.83- 180 30.67-

Additional 151 86.78 68 37.78 49.00 40.32,57.69

5 91.43 45.29

148 84.64- 151 62.22-

134 90.54 106 70.20 20.34 11.66,29.03

6A 94.73 77.36

150 91.50- 160 29.98- 50.37,

144 96.00 60 37.50 58.50

7F 98.52 45.49 66.63

53 93.28- 59 83.54-

53 100.0 55 93.22 6.78 0.36,13.19

19A 100.0 98.12

Note: N=number of subjects analyzed, n=number of subjects with seroresponse.

Table 25-3 Post-Immunization IgG GMC in Subjects with Negative Pre-Immunization Antibody Level in PCV13-TT Group and PCV7 Group Aged 7-11 Months (PPS)

PCV13-TT group PCV7 group

Serotypes Ratio# 95%CI

N GMC 95%CI N GMC 95%CI

4 175 4.29 3.88-4.74 180 4.19 3.83-4.58 1.02 0.90,1.17

6B 101 2.31 1.81-2.96 98 1.79 1.49-2.17 1.29 0.95,1.76

9V 172 2.39 2.09-2.73 178 2.71 2.39-3.08 0.88 0.73,1.06

Common 14 57 6.77 5.57-8.23 59 15.99 13.17-19.42 0.42 0.32,0.56
 18C 177 3.80 3.30-4.38 181 3.48 3.12-3.88 1.09 0.92,1.31
 19F 60 5.64 4.62-6.90 66 3.97 3.37-4.67 1.42 1.10,1.84
 23F 151 3.38 2.89-3.96 161 2.58 2.23-2.99 1.31 1.06,1.63
 1 133 2.05 1.84-2.28 146 0.40 0.33-0.49 5.07 4.04,6.35
 3 171 1.33 1.20-1.46 176 0.43 0.39-0.47 3.10 2.71,3.56
 5 162 0.78 0.70-0.89 169 0.17 0.13-0.22 4.68 3.44,6.37
 Additional
 6A 137 1.63 1.32-2.02 140 0.48 0.35-0.65 3.41 2.33,4.99
 7F 139 5.41 4.63-6.32 152 0.11 0.07-0.16 49.30 31.80,76.45
 19A 51 4.02 3.25-4.97 59 0.92 0.65-1.31 4.36 2.85,6.65
 Note: N=number of subjects analyzed; #Ratio: PCV13-TT / PCV7.

Table 25-4 Post-Immunization IgG GMC in Subjects with Negative Pre-Immunization Antibody Level in PCV13-TT Group and PCV7 Group Aged 7-11 Months (FAS)
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PCV13-TT group PCV7 group
 Serotypes Ratio# 95%CI
 N GMC 95%CI N GMC 95%CI
 4 188 3.32 2.68-4.11 191 3.00 2.39-3.79 1.10 0.81,1.51
 6B 109 2.16 1.68-2.78 104 1.65 1.36-2.01 1.31 0.95,1.80
 9V 183 2.04 1.68-2.47 189 2.04 1.64-2.55 1.00 0.74,1.34
 Common 14 59 6.38 5.14-7.91 62 12.04 8.19-17.71 0.53 0.34,0.82
 18C 190 3.00 2.40-3.75 192 2.53 2.00-3.18 1.19 0.86,1.64
 19F 62 5.66 4.63-6.93 69 3.66 3.03-4.41 1.55 1.18,2.03
 23F 160 3.05 2.56-3.63 170 2.25 1.88-2.70 1.35 1.05,1.74
 1 143 1.87 1.65-2.12 153 0.39 0.32-0.47 4.79 3.82,6.01
 3 184 1.19 1.05-1.34 185 0.39 0.34-0.44 3.05 2.57,3.62
 5 174 0.74 0.64-0.85 180 0.16 0.12-0.21 4.66 3.42,6.35
 Additional
 6A 148 1.52 1.22-1.88 151 0.45 0.33-0.60 3.40 2.35,4.93
 7F 150 4.84 4.06-5.78 160 0.11 0.07-0.16 44.52 29.07,68.19
 19A 53 3.96 3.21-4.89 59 0.92 0.65-1.31 4.29 2.83,6.52
 Note: N=number of subjects analyzed; #Ratio: PCV13-TT / PCV7.

6.2.2.4 Antibody positive seroconversion rate and antibody level in the pre-immunization antibody negative population at the 12–23 months of age
 After full immunization, among the pre-immunization antibody negative population at the 12–23 months of age, 7 common serotypes in the PCV13-TT group had post-immune antibody positive seroconversion rates greater than 98%. Compared with the PCV7 group, except for type 14 and type 19F (100 % of the positive seroconversion rate), the lower limits of the two-sided 95% confidence interval of the differences in post-immunization antibody positive seroconversion rates of the other common serotypes in the PCV13-TT group were all greater than –10% (as shown in Table 26-1); except for type 14, the GMC ratios of post-immunization antibodies were all greater than 0.5, and the lower limits of the two-sided 95% confidence interval of post-immunization antibody GMC ratios of the other common serotypes were all greater than 0.5 (as shown in Table 26-3).
 For the newly added 6 serotypes, the post-immunization antibody positive seroconversion rates in the PCV13-TT group were all greater than 95%, and the lower limits of the two-sided 95% confidence interval of the antibody positive seroconversion rates were all greater than 70%; compared with the PCV7 group, the antibody positive seroconversion rates of the newly added 6 serotypes were all greater than 0, and the lower limits of the 95% confidence interval of the differences in antibody positive seroconversion rates were all greater than 0, except for type 6A (96.63%) and type 19A (100%); and the lower limits of the two-sided 95% confidence interval of the post-immunization antibody GMC ratios in the PCV13-TT group were all greater than 1. Statistical analyses of IgG antibody positive seroconversion rate and antibody level of pre-immunization antibody negative population at 12–23 months of age are shown in Tables 26-1 to 26-4.

Table 26-1 Post-Immunization IgG Antibody Positive Rates in Subjects with Negative Pre-Immunization Antibody Level in PCV13-TT Group and PCV7 Group Aged 12-23 Months (PPS)
 PCV13-TT group PCV7 group Rate
 Serotypes difference 95%CI
 N n % 95% CI N n % 95% CI (%)
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97.83- 96.77-
 4 168 168 100.0 170 169 99.41 0.59 -0.56,1.74
 100.0 99.99
 91.96- 79.77-
 6B 44 44 100.0 33 31 93.94 6.06 -2.08,14.20
 100.0 99.26
 97.85- 95.58-
 9V 170 170 100.0 161 159 98.76 1.24 -0.47,2.95
 100.0 99.85
 79.41- 76.84- -
 Common 14 16 16 100.0 14 14 100.0 0.00
 100.0 100.0 23.16,20.59
 97.89- 97.87-
 18C 173 173 100.0 171 171 100.0 0.00 -2.13,2.11
 100.0 100.0
 79.41- 82.35- -
 19F 16 16 100.0 19 19 100.0 0.00
 100.0 100.0 17.65,20.59
 94.27- 95.85-

23F 95 94 98.95 87 87 100.0 -1.05 -3.10,1.00
99.97 100.0
96.61- 42.94-
1 107 107 100.0 108 57 52.78 47.22 37.81,56.64
100.0 62.46
94.90- 33.40-
3 169 166 98.22 166 68 40.96 57.26 49.52,65.00
99.63 48.85
90.51- 28.61-
Additional 5 134 128 95.52 127 47 37.01 58.51 49.42,67.61
98.34 46.02
90.46- 81.86-
6A 89 86 96.63 90 81 90.00 6.63 -0.61,13.87
99.30 95.32
96.52- 40.07-
7F 104 104 100.0 95 48 50.53 49.47 39.42,59.53
100.0 60.95
79.41- 69.15- -
19A 16 16 100.0 10 10 100.0 0.00
100.0 100.0 30.85,20.59
Note: N=number of subjects analyzed, n=number of subjects with seroresponse.

Table 26-2 Post-Immunization IgG Antibody Positive Rates in Subjects with Negative Pre-Immunization Antibody Level in PCV13-TT Group and PCV7 Group Aged 12-23 Months (FAS)
Rate
PCV13-TT group PCV7 group 95%CI
Serotypes difference(%)
N n % 95% CI N n % 95% CI
92.07- 90.77-
4 178 171 96.07 181 172 95.03 1.04 -3.22,5.30
98.40 97.70
82.80- 71.91-
6B 48 45 93.75 38 33 86.84 6.91 -5.84,19.65
98.69 95.59
92.15- 89.51-
9V 180 173 96.11 171 161 94.15 1.96 -2.55,6.47
98.42 97.16
71.31- 63.56- -
Common 14 17 16 94.12 17 15 88.24 5.88
99.85 98.54 13.08,24.85
92.96- 91.52-
18C 182 176 96.70 182 174 95.60 1.10 -2.85,5.05
98.78 98.08
79.41- 82.35- -
19F 16 16 100.0 19 19 100.0 0.00
100.0 100.0 17.65,20.59
86.37- 85.70-
23F 102 95 93.14 97 90 92.78 0.35 -6.76,7.47
97.20 97.05
87.96- 39.83-
1 116 109 93.97 118 58 49.15 44.81 34.81,54.82
97.54 58.52
89.97- 31.95-
3 179 169 94.41 176 69 39.20 55.21 47.25,63.17
Additional 97.29 46.83
84.33- 26.88-
5 145 131 90.34 138 48 34.78 55.56 46.27,64.85
94.62 43.35
6A 99 88 88.89 80.99- 100 84 84.00 75.32- 4.89 -4.60,14.37
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94.32 90.57
89.89- 36.87-
7F 112 107 95.54 105 49 46.67 48.87 38.59,59.15
98.53 56.66
79.41- 69.15- -
19A 16 16 100.0 10 10 100.0 0.00
100.0 100.0 30.85,20.59
Note: N=number of subjects analyzed, n=number of subjects with seroresponse.

Table 26-3 Post-Immunization IgG GMC in Subjects with Negative Pre-Immunization Antibody Level in PCV13-TT Group and PCV7 Group Aged 12-23 Months (PPS)
PCV13-TT group PCV7 group
Serotypes Ratio# 95%CI
N GMC 95%CI N GMC 95%CI
4 168 5.02 4.59-5.50 170 5.00 4.53-5.52 1.00 0.88,1.15
6B 44 4.17 3.38-5.15 33 1.39 1.00-1.95 2.99 2.06,4.34
9V 170 2.89 2.59-3.23 161 2.85 2.53-3.21 1.02 0.86,1.19
Common 14 16 7.24 4.61-11.35 14 10.93 6.13-19.48 0.66 0.33,1.32
18C 173 5.02 4.61-5.46 171 3.81 3.44-4.21 1.32 1.16,1.50
19F 16 6.89 5.31-8.95 19 3.49 2.60-4.68 1.98 1.34,2.90
23F 95 3.73 3.24-4.29 87 2.46 2.05-2.96 1.51 1.20,1.90
1 107 2.12 1.89-2.38 108 0.30 0.24-0.38 7.12 5.48,9.24
3 169 1.04 0.96-1.13 166 0.35 0.30-0.40 2.96 2.51,3.50
5 134 0.88 0.78-1.00 127 0.20 0.15-0.26 4.43 3.35,5.85
Additional
6A 89 2.32 1.68-3.19 90 0.93 0.75-1.16 2.48 1.69,3.64
7F 104 5.26 4.70-5.89 95 0.26 0.18-0.38 20.49 13.98,30.03
19A 16 5.56 3.83-8.07 10 1.23 0.78-1.94 4.51 2.57,7.91

Note: N=number of subjects analyzed; #Ratio: PCV13-TT / PCV7.

Table 26-4 Post-Immunization IgG GMC in Subjects with Negative Pre-Immunization Antibody Level in PCV13-TT Group and PCV7 Group Aged 12-23 Months (FAS)

PCV13-TT group		PCV7 group		Serotypes Ratio#		95%CI	
N	GMC	95%CI	N	GMC	95%CI		
4	178	3.81	3.06-4.75	181	3.83	3.11-4.72	0.99 0.74,1.34
6B	48	3.53	2.68-4.65	38	1.27	0.90-1.80	2.78 1.81,4.27
9V	180	2.25	1.82-2.78	171	2.19	1.75-2.73	1.03 0.76,1.40
Common	14	17	5.81	3.10-10.88	17	6.94	3.14-15.32 0.84 0.32,2.21
18C	182	3.99	3.28-4.86	182	2.86	2.30-3.56	1.39 1.04,1.87
19F	16	6.89	5.31-8.95	19	3.49	2.60-4.68	1.98 1.34,2.90
23F	102	3.22	2.66-3.89	97	2.05	1.66-2.53	1.57 1.18,2.08
1	116	1.72	1.39-2.13	118	0.27	0.21-0.35	6.35 4.60,8.76
3	179	0.89	0.77-1.03	176	0.32	0.27-0.37	2.79 2.26,3.45
5	145	0.70	0.57-0.86	138	0.18	0.13-0.23	4.00 2.88,5.57
Additional							
6A	99	1.71	1.16-2.51	100	0.78	0.61-1.01	2.18 1.37,3.45
7F	112	4.34	3.54-5.32	105	0.24	0.17-0.35	17.87 11.95,26.71
19A	16	5.56	3.83-8.07	10	1.23	0.78-1.94	4.51 2.57,7.91

Note: N=number of subjects analyzed; #Ratio: PCV13-TT / PCV7.

6.2.2.5 Antibody positive seroconversion rate and antibody level in the pre-immunization antibody negative population at the 24–71 months of age
After full immunization, among the pre-immunization antibody negative population at 24–71 months of age, the antibody positive seroconversion rates of the other common serotypes were all Confidential 141 / 310
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greater than 97% in the 7 common serotypes in the PCV13-TT group, except that the type 19F had no detection value, and the antibody geometric mean concentrations (GMCs) were all greater than 3.0 µg/mL except for the type 19F. Compared with the PCV7 group, except for type 19F and type 14 (the antibody positive seroconversion rates were 100%, and the rate differences with the PCV7 group were 0), the lower limits of the 95% confidence interval of the differences in antibody positive seroconversion rates of the other common serotypes were all greater than –10%; except for type 14 (the GMC ratios were 0.69), the lower limits of the 95% confidence interval of the antibody GMC ratios of the other common serotypes were all greater than 0.5. (Tables 27-1 and 27-3)

Among the 6 newly added types, except for type 3 (84.67%) and type 6A (89.36%), the other types all had an antibody positive seroconversion rate greater than 96%; for the lower limit of the two-sided 95% confidence interval of the antibody positive seroconversion rate, except for type 19A (only one negative case before immunization, and the antibody concentration of the subject reached positive seroconversion after immunization), the other newly added types all had a rate greater than 70%. Compared with the PCV7 group, except for type 6A (difference of positive seroconversion rate being 7.17%) and type 19A (difference of positive seroconversion rate being 10.0%), the other newly added types all had a lower limit of 95% confidence interval of difference greater than 0%; except for type 6A (GMC ratio being 1.42) and type 19A (GMC ratio being 38.13), the other newly added types all had a lower limit of 95% confidence interval of GMC ratio greater than 1. (Tables 27-1 and 27-3)

The number of negative subjects before immunization in types 6B, 14, and 19A was relatively few, and no result was found in type 19F in the PCV13-TT group, resulting in the difficulty of accurate evaluation.

The statistical results of IgG antibody positive seroconversion rate of negative population before immunization at the 24–71 months of age are shown in Tables 27–1 to 27–4.

Table 27-1 Post-Immunization IgG Antibody Positive Rates in Subjects with Negative Pre-Immunization Antibody Level in PCV13-TT Group and PCV7 Group Aged 24-71 Months (PPS)

PCV13-TT group	PCV7 group	Rate	Serotypes	95%	95% difference	95%CI		
N	n	%	N	n	%	(%)		
CI	CI							
97.14-	97.54-							
4	127	127	100.0	148	148	100.0	0.00	-2.46,2.86
100.0	100.0							
75.29-	74.87-							
6B	13	13	100.0	26	24	92.31	7.69	-2.55,17.93
100.0	99.05							
91.64-	95.52-							
9V	102	99	97.06	122	121	99.18	-2.12	-5.77,1.53
99.39	99.98							
39.76-	71.51-							
Common	14	4	4	100.0	11	11	100.0	0.00
100.0	100.0							
94.16-	97.51-							
18C	121	119	98.35	146	146	100.0	-1.65	-3.92,0.62
99.80	100.0							
66.37-								
19F	///	0	9	100.0	0.00	/		
100.0								
90.11-	88.61-							
23F	54	53	98.15	74	71	95.95	2.20	-3.55,7.96
99.95	99.16							
91.23-	38.43-							
1	97	94	96.91	122	58	47.54	49.37	39.86,58.87
99.36	56.78							
Additional 77.53- 19.92-								
3	137	116	84.67	149	40	26.85	57.83	48.50,67.15
90.25	34.71							
5	80	77	96.25	89.43-	107	38	35.51	26.50- 60.74 50.76,70.71

99.22 45.35
76.90- 71.47-
6A 47 42 89.36 73 60 82.19 7.17 -5.27,19.61
96.45 90.16
93.23- 39.97- 37.76-
7F 80 79 98.75 93 47 50.54 48.21
99.97 61.07 58.66
2.50- 55.50-
19A 1 1 100.0 10 9 90.00 10.00 -8.59,28.59
100.0 99.75

Note: N=number of subjects analyzed, n=number of subjects with seroresponse.

Table 27-2 Post-Immunization IgG Antibody Positive Rates in Subjects with Negative Pre-Immunization Antibody Level in PCV13-TT Group and PCV7 Group Aged 24-71 Months (FAS)

PCV13-TT group	PCV7 group	Rate	Serotypes	95% difference	95%CI
N	n	%	N	n	%
CI	CI				
95.76-	96.32-				
4 129	128	99.22	149	148	99.33
		-0.10			-2.11,1.90
99.98	99.98				
75.29-	74.87-				
6B 13	13	100.0	26	24	92.31
		7.69			-2.55,17.93
100.0	99.05				
91.72-	95.55-				
9V 103	100	97.09	123	122	99.19
		-2.10			-5.71,1.51
99.40	99.98				
39.76-	71.51-				
Common 14	4	4	100.0	11	11
		100.0			28.49,60.24
94.20-	97.51-				
18C 122	120	98.36	146	146	100.0
		-1.64			-3.89,0.61
99.80	100.0				
66.37-					
19F	///	0	9	100.0	0.00
		100.0			
87.47-	88.75-				
23F 55	53	96.36	75	72	96.00
		0.36			-6.28,7.01
99.56	99.17				
89.98-	38.43-				
1 99	95	95.96	122	58	47.54
		48.42			38.75,58.09
98.89	56.78				
76.86-	19.92-				
3 138	116	84.06	149	40	26.85
		57.21			47.83,66.59
89.73	34.71				
87.98-	26.50-				
Additional 5	82	78	95.12	107	38
		35.51			59.61
98.66	45.35				49.41,69.80
74.75-	71.47-				
6A 48	42	87.50	73	60	82.19
		5.31			-7.52,18.14
95.27	90.16				
91.47-	39.97-				
7F 82	80	97.56	93	47	50.54
		47.02			36.33,57.72
99.70	61.07				
2.50-	58.72-				
19A 1	1	100.0	11	10	90.91
		9.09			-7.90,26.08
100.0	99.77				

Note: N=number of subjects analyzed, n=number of subjects with seroresponse.

Table 27-3 Post-Immunization IgG GMC in Subjects with Negative Pre-Immunization Antibody Level in PCV13-TT Group and PCV7 Group Aged 24-71 Months (PPS)

PCV13-TT group		PCV7 group	
Serotypes	Ratio# 95%CI	N	GMC 95%CI
N	GMC 95%CI	N	GMC 95%CI
7.79-			
4 127	8.97 148 8.08 7.04-9.27	1.11 0.91,1.35	
10.33			
6B 13	3.39 1.80-6.38	26 1.20 0.83-1.72	2.83 1.47,5.44
Common 9V	102 3.18 2.60-3.89	122 3.02 2.52-3.61	1.05 0.81,1.38
0.18-			
14 4	3.13 11 4.51 1.93-10.50	0.69 0.12,4.04	
53.25			
18C 121	7.50 6.03-9.34	146 6.00 5.11-7.05	1.25 0.96,1.63

19F /// 9 1.66 1.03-2.66 //
23F 54 3.50 2.58-4.75 74 2.18 1.72-2.76 1.61 1.11,2.34
1 97 1.77 1.49-2.10 122 0.20 0.15-0.27 8.67 6.06,12.39
3 137 0.65 0.58-0.73 149 0.25 0.22-0.29 2.61 2.18,3.12
5 80 1.22 1.02-1.47 107 0.17 0.12-0.24 7.16 4.70,10.91
Additional
6A 47 1.05 0.74-1.50 73 0.74 0.48-1.16 1.42 0.76,2.62
7F 80 3.96 3.07-5.11 93 0.19 0.12-0.30 21.04 12.26,36.12
19A 1 30.56 / 10 0.80 0.20-3.25 38.13 0.37,3944.91
Note: N=number of subjects analyzed; #Ratio: PCV13-TT / PCV7.

Table 27-4 Post-Immunization IgG GMC in Subjects with Negative Pre-Immunization Antibody Level in PCV13-TT Group and PCV7 Group Aged 24-71 Months (FAS)

PCV13-TT group		PCV7 group	
Serotypes		Ratio#	95%CI
N		GMC	95%CI
7.44-			
4	129	8.67	149 7.89 6.82-9.11 1.10 0.89,1.36
10.11			
6B	13	3.39	1.80-6.38 26 1.20 0.83-1.72 2.83 1.47,5.44
9V	103	3.21	2.63-3.93 123 2.99 2.50-3.57 1.07 0.82,1.40
Common		0.18-	
14	4	3.13	11 4.51 1.93-10.50 0.69 0.12,4.04
53.25			
18C	122	7.62	6.12-9.48 146 6.00 5.11-7.05 1.27 0.97,1.65
19F		///	9 1.66 1.03-2.66 //
23F	55	3.33	2.43-4.56 75 2.15 1.70-2.71 1.55 1.06,2.26
1	99	1.71	1.44-2.04 122 0.20 0.15-0.27 8.40 5.88,12.00
3	138	0.64	0.57-0.72 149 0.25 0.22-0.29 2.57 2.14-3.08
5	82	1.20	1.00-1.44 107 0.17 0.12-0.24 7.05 4.64,10.70
Additional			
6A	48	1.02	0.71-1.45 73 0.74 0.48-1.16 1.37 0.74,2.53
7F	82	3.79	2.92-4.92 93 0.19 0.12-0.30 20.13 11.75,34.49
19A	1	30.56	/ 11 0.84 0.24-2.93 36.45 0.48,2770.98

Note: N=number of subjects analyzed; #Ratio: PCV13-TT / PCV7.

6.2.3 The proportion of responders at $\geq 1.0 \mu\text{g/mL}$ for serotype-specific IgG response in PCV13-TT and control group

6.2.3.1. The proportion of responders at $\geq 1.0 \mu\text{g/mL}$ for serotype-specific IgG response in PCV13-TT and control group

Among the 7 common serotypes, the lower limit of 2-sided, 95% CI for the difference of the proportion of responders at $\geq 1.0 \mu\text{g/mL}$ in the PCV13-TT and control group was more than -10%, except serotype 6B in subgroup aged 3-71 months, serotype 6B and 23F in subgroup aged 3 months and serotype 9V in subgroup aged 7-11 months. The results were shown in table 28-1 to 28-4. Among the 6 additional serotypes, the lower limit of 2-sided, 95% CI for the ratio of the proportion of responders at $\geq 1.0 \mu\text{g/mL}$ in the PCV13-TT and control group was more than 0. The statistical analysis of proportion of responders at $\geq 1.0 \mu\text{g/mL}$ for serotype-specific IgG response in subjects aged 3-71 months after immunization in PCV13-TT and control group was shown in tables 28-1 to 28-10.

Table 28-1 Comparison of Subjects aged 3-71 months Achieving a Pneumococcal IgG
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Antibody Concentration $1.0 \mu\text{g/mL}$ $\mu\text{g/mL}$ After immunization between PCV13-TT and PCV7 Group(PPS)

PCV13-TT (N=1043)		PCV7 (N=1064)		Rate
Serotype		95%CI		
N		%	95% CI	N % 95% CI diff.(%)
97.17-		97.80-		
1024	98.18	1050	98.68	-0.51 -1.57,0.56
4	98.90	99.28		
76.20-	84.96-	-11.51,-		
822	78.81	927	87.12	-8.31
6B	81.25	89.08	5.12	
90.13-	92.59-			
959	91.95	1002	94.17	-2.23 -4.40,-0.06
9V	93.53	95.50		
98.62-	98.91-			
Common	1036	99.33	1059	99.53 -0.20 -0.84,0.44
14	99.73	99.85		
93.51-	94.49-			
991	95.01	1020	95.86	-0.85 -2.63,0.93
18C	96.25	96.98		
97.88-	96.77-			
1030	98.75	1041	97.84	0.92 -0.19,2.02
19F	99.33	98.62		
89.08-	90.63-			
949	90.99	983	92.39	-1.40 -3.76,0.96
23F	92.66	93.91		
85.58-	8.06-			
915	87.73	104	9.77	77.95 75.28,80.63
1	89.66	11.72		
59.30-	8.49-			
650	62.32	109	10.24	52.08 48.62,55.54
3	65.27	12.22		
62.41-	2.46-			
682	65.39	37	3.48	61.91 58.82,65.00
5	68.28	4.76		
Additional				
76.50-	40.23-			
825	79.10	460	43.23	35.87 32.00,39.73
6A	81.53	46.27		
96.82-	10.75-			
1021	97.89	135	12.69	85.20 83.02,87.38
7F	98.67	14.84		
97.76-	66.78-			
1029	98.66	741	69.64	29.01 26.17,31.86
19A	99.26	72.40		

Note: N=number of subjects analyzed, n=number of subjects with positive antibody.

Table 28-2 Comparison of Subjects aged 3-71 months Achieving a Pneumococcal IgG

Antibody Concentration 1.0µg/mL µg/mL After immunization between PCV13-TT and PCV7 Group (FAS)
PCV13-TT (N=1117) PCV7 (N=1119) Rate
Serotype 95%CI
N % 95% CI N % 95% CI diff.(%)
93.44- 94.35-
1060 94.90 1071 95.71 -0.81 -2.57,0.94
4 96.11 96.82
73.67- 82.47- -11.71,-
852 76.28 948 84.72 -8.44
6B 78.74 86.78 5.18
86.80- 89.62-
991 88.80 1023 91.42 -2.62 -5.09,-0.15
9V* 90.59 93.00
95.67- 96.61-
Common 1082 96.87 1093 97.68 -0.81 -2.16,0.54
14 97.81 98.48
89.80- 91.38-
1023 91.58 1041 93.03 -1.44 -3.65,0.76
18C 93.15 94.45
94.95- 94.45-
1075 96.24 1072 95.80 0.44 -1.18,2.06
19F 97.28 96.90
85.67- 87.60-
980 87.74 1002 89.54 -1.81 -4.44,0.82
23F 89.60 91.28
82.54- 7.90-
947 84.78 107 9.56 75.22 72.50,77.94
Additional 1 86.84 11.44
3 671 60.07 57.13- 118 10.55 8.81- 49.53 46.14,52.92

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62.96 12.49
59.67- 2.41-
699 62.58 38 3.40 59.18 56.15,62.21
5 65.42 4.63
74.32- 39.62-
859 76.90 476 42.54 34.36 30.56,38.17
6A 79.35 45.50
93.44- 10.30-
1060 94.90 136 12.15 82.74 80.43,85.05
7F 96.11 14.21
95.05- 65.91-
1076 96.33 769 68.72 27.61 24.68,30.54
19A 97.35 71.43
Note: N=number of subjects analyzed, N=number of subjects with positive antibody; *PCV13-TT9V■
FAS■■■■N■1116.

Table 28-3 Comparison of Subjects Achieving a Pneumococcal IgG Antibody Concentration 1.0 µg/mL After immunization between PCV13-TT (3M) and PCV7 (3M)Group PCV13(PPS)
PCV13-TT (N=471) PCV7 (N=487) Rate
Serotype 95%CI
N % 95% CI N % 95% CI diff.(%)
94.80- 97.06-
480 98.56 -1.75
4 456 96.82 98.21 99.42 -3.65,0.16
56.58- 89.21- -35.86,-
448 91.99 -30.85
6B 288 61.15 65.57 94.24 25.83
93.77- 94.97-
472 96.92 -0.95
9V 452 95.97 97.55 98.27 -3.30,1.39
97.54- 98.21-
Common 484 99.38 -0.45
14 466 98.94 99.65 99.87 -1.60,0.71
88.62- 93.24-
465 95.48 -3.98
18C 431 91.51 93.86 97.15 -7.10,-0.85
97.54- 96.79-
479 98.36 0.58
19F 466 98.94 99.65 99.29 -0.88,2.04
84.14- 94.97-
472 96.92 -9.45
23F 412 87.47 90.33 98.27 -12.81,-6.09
82.29- 6.11-
41 8.42 77.36
1 404 85.77 88.80 11.25 73.35,81.36
71.22-
33 6.78 4.71- 9.38 68.60
3 355 75.37 79.20 64.11,73.08
76.59-
11 2.26 1.13- 4.01 78.21
5 379 80.47 83.95 74.39,82.02
Additional
73.23- 32.66-
180 36.96 40.32
6A 364 77.28 80.99 41.42 34.60,46.04
95.59-

27 5.54 3.69- 7.96 91.91
 7F 459 97.45 98.68 89.43,94.39
 96.68- 48.85-
 260 53.39 44.91
 19A 463 98.30 99.26 57.89 40.33,49.50
 Note: N=number of subjects analyzed, n=number of subjects with positive antibody.

Table 28-4 Comparison of Subjects Achieving a Pneumococcal IgG Antibody Concentration 1.0 µg/mL After immunization between PCV13-TT (3M) and PCV7 (3M)Group PCV13((FAS)
 PCV13-TT (N=519) PCV7 (N=519) Rate
 Serotype 95%CI
 N % 95% CI N % 95% CI diff.(%)

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90.31- 93.42-
 496 95.57 -2.70
 4 482 92.87 94.93 97.17 -5.53,0.14
 55.17- 86.43- -34.85,-
 464 89.40 -29.87
 6B 309 59.54 63.79 91.92 24.88
 89.22- 91.63-
 488 94.03 -2.12
 9V 477 91.91 94.11 95.91 -5.23,0.99
 94.34- 96.24-
 Common 508 97.88 -1.54
 14 500 96.34 97.78 98.94 -3.58,0.49
 84.32- 90.09-
 481 92.68 -5.20
 18C 454 87.48 90.20 94.77 -8.83,-1.58
 93.65- 94.11-
 499 96.15 -0.39
 19F 497 95.76 97.32 97.63 -2.78,2.01
 80.57- 91.41-
 487 93.83 -9.83
 23F 436 84.01 87.06 95.74 -13.60,-6.05
 78.71- 6.23-
 44 8.48 73.80
 1 427 82.27 85.46 11.21 69.73,77.86
 67.19-
 38 7.32 5.23- 9.91 63.97
 3 370 71.29 75.15 59.48,68.46
 71.60-
 11 2.12 1.06- 3.76 73.41
 5 392 75.53 79.17 69.51,77.31
 Additional
 71.19- 33.02-
 193 37.19 37.96
 6A 390 75.14 78.81 41.50 32.38,43.54
 91.63-
 28 5.39 3.61- 7.70 88.63
 7F 488 94.03 95.91 85.82,91.45
 92.97- 49.17-
 278 53.56 41.62
 19A 494 95.18 96.86 57.92 36.95,46.29

Table 28-5 Comparison of Subjects aged 7-11 months Achieving a Pneumococcal IgG Antibody Concentration 1.0 µg/mL After immunization between PCV13-TT and PCV7 Group(PPS)
 PCV13-TT (N=185) PCV7 (N=189) Rate
 Serotype 95%CI
 N % 95% CI N % 95% CI diff.(%)
 95.33- 94.67-
 182 98.38 185 97.88 0.49 -2.25,3.24
 4 99.66 99.42
 77.67- 68.34-
 155 83.78 142 75.13 8.65 0.52,16.79
 6B 88.78 81.12
 77.07- 84.13-
 154 83.24 169 89.42 -6.17 -13.12,0.77
 9V 88.32 93.42
 97.03- 98.07-
 Common 184 99.46 189 100.0 -0.54 -1.60,0.52
 14 99.99 100.0
 90.97- 89.17-
 176 95.14 177 93.65 1.48 -3.17,6.14
 18C 97.75 96.68
 93.07- 92.52-
 179 96.76 182 96.30 0.46 -3.25,4.17
 19F 98.80 98.50
 86.34- 81.10-
 169 91.35 164 86.77 4.58 -1.72,10.88
 23F 94.98 91.25
 86.98- 8.75-
 170 91.89 25 13.23 78.66 72.44,84.89
 1 95.39 18.90
 59.75- 8.31-
 Additional 124 67.03 24 12.70 54.33 46.06,62.60
 3 73.75 18.30

30.31-
69 37.30 6 3.17 1.17- 6.78 34.12 26.72,41.53
5 44.70

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68.26- 31.65-
139 75.14 73 38.62 36.51 27.18,45.84
6A 81.18 45.96

96.15- 6.16-
183 98.92 19 10.05 88.87 84.33,93.40
7F 99.87 15.25

94.56- 62.76-
181 97.84 132 69.84 28.00 21.13,34.87
19A 99.41 76.29

Note: N=number of subjects analyzed, n=number of subjects with positive antibody.
Table 28-6 Comparison of Subjects aged 7-11 months Achieving a Pneumococcal IgG
Antibody Concentration 1.0 µg/mL After immunization between PCV13-TT and PCV7
Group (FAS)

PCV13-TT (N=198) PCV7 (N=200) Rate
Serotype 95%CI

N % 95% CI N % 95% CI diff.(%)

90.91- 89.14-
188 94.95 187 93.50 1.45 -3.13,6.03

4 97.55 96.49
74.62- 64.71-
160 80.81 143 71.50 9.31 0.99,17.63

6B 86.05 77.64
73.95- 79.84-
158 80.20 171 85.50 -5.30 -12.70,2.10

9V 85.53 90.07
92.85- 92.92-
Common 191 96.46 193 96.50 -0.04 -3.66,3.58

14 98.57 98.58
86.61- 84.40-
181 91.41 179 89.50 1.91 -3.85,7.68

18C 94.92 93.38
90.28- 88.53-
187 94.44 186 93.00 1.44 -3.32,6.21

19F 97.19 96.12
81.93- 77.06-
173 87.37 166 83.00 4.37 -2.59,11.34

23F 91.66 87.93
83.66- 8.26-
176 88.89 25 12.50 76.39 70.05,82.73

1 92.90 17.90
57.04- 8.67-
127 64.14 26 13.00 51.14 43.00,59.29

3 70.82 18.47
29.66-
72 36.36 6 3.00 1.11-6.42 33.36 26.26,40.47

5 43.48
Additional
65.43- 30.30-
143 72.22 74 37.00 35.22 26.07,44.37

6A 78.34 44.09
91.55- 5.82-
189 95.45 19 9.50 85.95 80.96,90.95

7F 97.90 14.44
92.19- 60.53-
190 95.96 135 67.50 28.46 21.41,35.51

19A 98.24 73.94
Note: N=number of subjects analyzed, N=number of subjects with positive antibody; *Number of subjects
included in FAS for serotype 9V in PCV13-TT was 194.

Table 28-7 Comparison of Subjects aged 12-23 months Achieving a Pneumococcal IgG
Antibody Concentration 1.0 µg/mL After immunization between PCV13-TT and PCV7
Group(PPS)

PCV13-TT (N=189) PCV7 (N=189) Rate
Serotype 95%CI

N % 95% CI N % 95% CI diff.(%)

98.07- 96.23-
189 100.0 187 98.94 1.06 -0.40,2.52
4 100.0 99.87

97.09- 81.10-
Common 188 99.47 164 86.77 12.70 7.76,17.64
6B 99.99 91.25

86.62- 86.62-
173 91.53 173 91.53 0.00 -5.61,5.61
9V 95.08 95.08

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98.07- 98.07-
189 100.0 189 100.0 0.00 -1.93,1.93
14 100.0 100.0

98.07- 93.22-
189 100.0 183 96.83 3.17 0.68,5.67
18C 100.0 98.83

98.07- 94.67-
 189 100.0 185 97.88 2.12 0.06,4.17
 19F 100.0 99.42
 93.93- 84.13-
 184 97.35 169 89.42 7.94 2.99,12.88
 23F 99.14 93.42
 88.53- 8.75-
 176 93.12 25 13.23 79.89 73.87,85.92
 1 96.29 18.90
 47.11- 5.74-
 103 54.50 18 9.52 44.97 36.73,53.21
 3 61.74 14.63
 41.88-
 93 49.21 5 2.65 0.86-6.07 46.56 39.08,54.05
 5 56.56
 Additional
 81.10- 43.96-
 164 86.77 97 51.32 35.45 26.84,44.06
 6A 91.25 58.64
 97.09- 13.71-
 188 99.47 36 19.05 80.42 74.73,86.12
 7F 99.99 25.38
 98.07- 83.52-
 189 100.0 168 88.89 11.11 6.63,15.59
 19A 100.0 92.99
 Note: N=number of subjects analyzed, n=number of subjects with positive antibody.

Table 28-8 Comparison of Subjects aged 12-23 months Achieving a Pneumococcal IgG Antibody Concentration 1.0 µg/mL After immunization between PCV13-TT and PCV7 Group(FAS)

PCV13-TT (N=200) PCV7 (N=200) Rate Serotype 95%CI

N % 95% CI N % 95% CI diff.(%)
 92.27- 91.00-
 192 96.00 190 95.00 1.00 -3.06,5.06
 4 98.26 97.58
 91.63- 77.62-
 191 95.50 167 83.50 12.00 6.11,17.89
 6B 97.92 88.36
 82.10- 82.67-
 175 87.50 176 88.00 -0.50 -6.93,5.93
 9V 91.74 92.16
 92.92- 93.58-
 Common 193 96.50 194 97.00 -0.50 -3.98,2.98
 14 98.58 98.89
 91.63- 88.53-
 191 95.50 186 93.00 2.50 -2.06,7.06
 18C 97.92 96.12
 92.92- 91.63-
 193 96.50 191 95.50 1.00 -2.84,4.84
 19F 98.58 97.92
 88.53- 79.84-
 186 93.00 171 85.50 7.50 1.47,13.53
 23F 96.12 90.07
 84.40- 8.26-
 179 89.50 25 12.50 77.00 70.75,83.25
 1 93.38 17.90
 45.34- 5.82-
 105 52.50 19 9.50 43.00 34.97,51.03
 3 59.59 14.44
 39.44-
 93 46.50 5 2.50 0.82- 5.74 44.00 36.76,51.24
 5 53.67
 Additional
 77.62- 42.37-
 167 83.50 99 49.50 34.00 25.37,42.63
 6A 88.36 56.64
 91.63- 12.94-
 191 95.50 36 18.00 77.50 71.45,83.55
 7F 97.92 24.04
 93.58- 81.53-
 194 97.00 174 87.00 10.00 4.77,15.23
 19A 98.89 91.33
 Note: N=number of subjects analyzed, n=number of subjects with positive antibody.

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Table 28-9 Comparison of Subjects aged 24-71 months Achieving a Pneumococcal IgG Antibody Concentration 1.0 µg/mL After immunization between PCV13-TT and PCV7 Group(PPS)

PCV13-TT (N=198) PCV7 (N=199) Rate Serotype 95%CI

N % 95% CI N % 95% CI diff.(%)
 97.22- 97.23-
 197 99.49 198 99.50 -0.00 -1.40,1.39
 4 99.99 99.99
 92.85- 81.44-
 191 96.46 173 86.93 9.53 4.19,14.87
 6B 98.57 91.28
 86.01- 90.32-

180 90.91 188 94.47 -3.56 -8.67,1.55
 9V 94.52 97.21
 97.22- 96.42-
 Common 197 99.49 197 98.99 0.50 -1.20,2.20
 14 99.99 99.88
 95.64- 94.93-
 195 98.48 195 97.99 0.49 -2.09,3.08
 18C 99.69 99.45
 96.40- 94.93-
 196 98.99 195 97.99 1.00 -1.40,3.40
 19F 99.88 99.45
 88.42- 84.32-
 184 92.93 178 89.45 3.48 -2.08,9.05
 23F 96.08 93.35
 77.40- 3.52-
 165 83.33 13 6.53 76.80 70.58,83.02
 1 88.24 10.91
 27.76- 12.13-
 68 34.34 34 17.09 17.26 8.83,25.69
 3 41.41 23.05
 64.37- 4.28-
 141 71.21 15 7.54 63.67 56.38,70.97
 5 77.41 12.13
 Additional
 73.52- 48.08-
 158 79.80 110 55.28 24.52 15.63,33.41
 6A 85.16 62.31
 92.85- 20.63-
 191 96.46 53 26.63 69.83 63.17,76.49
 7F 98.57 33.35
 96.40- 86.08-
 196 98.99 181 90.95 8.04 3.81,12.26
 19A 99.88 94.55

Note: N=number of subjects analyzed, n=number of subjects with positive antibody.
 Table 28-10 Comparison of Subjects aged 24-71 months Achieving a Pneumococcal IgG
 Antibody Concentration 1.0 µg/mL After immunization between PCV13-TT and PCV7
 Group(FAS)

PCV13-TT (N=200) PCV7 (N=200) Rate

Serotype 95%CI

N % 95% CI N % 95% CI diff.(%)

96.43- 96.43-
 198 99.00 198 99.00 0.00 -1.95,1.95
 4 99.88 99.88
 92.27- 81.53-
 192 96.00 174 87.00 9.00 3.61,14.39
 6B 98.26 91.33
 85.56- 89.75-
 181 90.50 188 94.00 -3.50 -8.73,1.73
 9V 94.18 96.86
 96.43- 96.43-
 Common 198 99.00 198 99.00 0.00 -1.95,1.95
 14 99.88 99.88
 95.68- 94.26-
 197 98.50 195 97.50 1.00 -1.74,3.74
 18C 99.69 99.18
 96.43- 94.96-
 198 99.00 196 98.00 1.00 -1.38,3.38
 19F 99.88 99.45
 87.93- 83.82-
 185 92.50 178 89.00 3.50 -2.17,9.17
 23F 95.74 92.98

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76.51- 3.51-
 165 82.50 13 6.50 76.00 69.72,82.28
 1 87.50 10.86
 27.94- 12.50-
 69 34.50 35 17.50 17.00 8.57,25.43
 3 41.53 23.49
 64.18- 4.64-
 142 71.00 16 8.00 63.00 55.67,70.33
 5 77.18 12.67
 Additional
 73.23- 47.82-
 159 79.50 110 55.00 24.50 15.62,33.38
 6A 84.87 62.02
 92.27- 20.52-
 192 96.00 53 26.50 69.50 62.81,76.19
 7F 98.26 33.19
 96.43- 86.15-
 198 99.00 182 91.00 8.00 3.80,12.20
 19A 99.88 94.58

Note: N=number of subjects analyzed, n=number of subjects with positive antibody.

6.2.3.2.The proportion of responders at ≥ 1.0 µg/mL for serotype-specific IgG response in
 PCV13-TT and control group for subjects whose antibody responder ≥ 0.35 µg/mL pre-
 immunization

Among the 7 common serotypes, the lower limit of 2-sided, 95% CI for the difference of the
 proportion of responders at ≥ 1.0 µg/mL in the PCV13-TT and control group for subjects whose

antibody responder ■ 0.35 µg/mL pre-immunization was more than -%10, except serotype 6B in subgroup aged 3-71 months, serotype 6B and 23F in subgroup aged 3 months, serotype 9V in subgroup aged 7-11 months, and serotype 19F in subgroup aged 12-23 months. There were a little subjects whose antibody responder ■ 0.35 µg/mL pre-immunization for serotype 14 and 19F in subgroup aged 12-23 months, serotype 6B, 14 and 19A in subgroup aged 24-71 months, and there was no available value for 19F. Therefore, it was difficult to evaluate accurately.

Among the 6 additional serotypes, the lower limit of 2-sided, 95% CI for the difference of for the ratio of the proportion of responders at ≥1.0 µg/mL was more than 0 between the PCV13-TT and control group, except serotype 6A in subgroup aged 24-71months. For serotype 6A in subgroup aged 24-71months, the lower limit of 2-sided, 95% CI for the difference of for the ratio of the proportion of responders at ≥1.0 µg/mL was -0.43%.

The statistical analysis of proportion of responders at ≥ 1.0 µg/mL for serotype-specific IgG response in subjects aged 3-71 months after immunization in PCV13-TT and control group was shown in tables 29-1 to 29-10.

Table 29-1 Comparison of Subjects aged 3-71 months Achieving a Pneumococcal IgG Antibody Concentration 1.0 µg/mL After immunization between PCV13-TT and PCV7 Group for subjects with pre-immunization negative antibody level (PPS)									
PCV13-TT PCV7 Rate									
Serotype 95%CI									
N n % 95% CI N n % 95% CI diff.(%)									
97.07- 97.72-									
4	923	906	98.16	970	957	98.66	-0.50	-1.63,0.63	
98.92 99.28									
65.54- 78.87- -18.27, -									
6B	447	313	70.02	435	360	82.76	-12.74		
74.24 86.19 7.20									
90.03- 91.81-									
9V	833	767	92.08	866	811	93.65	-1.57	-4.02,0.88	
Common 93.82 95.18									
96.14- 96.39-									
14	142	141	99.30	152	151	99.34	-0.05	-1.93,1.84	
99.98 99.98									
94.68- 95.04-									
18C	865	832	96.18	904	872	96.46	-0.28	-2.03,1.48	
97.36 97.57									
19F	232	227	97.84	95.04-	247	239	96.76	93.72- 1.08	-1.81,3.98
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99.30	98.59
87.53-	88.77-
23F	655
590	90.08
689	628
91.15	-1.07
-4.19,	2.05
92.26	93.16
83.76-	4.94-
1	758
655	86.41
810	53
6.54	79.87
76.89,	82.84
88.77	8.47
58.57-	4.89-
3	939
580	61.77
960	61
6.35	55.41
51.94,	58.88
64.89	8.09
59.95-	0.64-
Additional	5
829	525
63.33	867
11	1.27
62.06	58.70,
65.42	
66.62	2.26
72.16-	31.21-
6A	599
454	75.79
648	226
34.88	40.92
35.89,	45.94
79.17	38.69
96.38-	5.39-
7F	713
697	97.76
727	52
7.15	90.60
88.44,	92.77
98.71	9.27
95.36-	37.42-
19A	186
183	98.39
193	86
44.56	53.83
46.59,	61.07
99.67	51.87

Table 29-2 Comparison of Subjects aged 3-71 months Achieving a Pneumococcal IgG Antibody Concentration 1.0 µg/mL After immunization between PCV13-TT and PCV7 Group for subjects with pre-immunization negative antibody level (FAS)

PCV13-TT		PCV7		Rate	
Serotype		95%CI			
N	n	%	95% CI	N	n
		%		95% CI diff.(%)	
996	92.98-	1025	93.95-		
942	94.58	978	95.41	-0.84	-2.74,1.07
4	95.90	96.61			
489	63.34-	467	75.94-		
331	67.69	373	79.87	-12.18	-17.70,-6.67
6B	71.82	83.42			
890	86.99-	911	88.47-		
794	89.21	825	90.56	-1.35	-4.13,1.44
9V	91.18	92.38			
153	91.66-	161	91.25-		
Common	147	96.08	154	95.65	0.43 -3.98,4.83
14	98.55	98.23			
928	90.80-	952	91.84-		
860	92.67	891	93.59	-0.92	-3.21,1.37
18C	94.27	95.06			
246	93.69-	262	90.27-		
238	96.75	246	93.89	2.85	-0.79,6.50
19F	98.59	96.47			
709	83.57-	732	85.25-		
612	86.32	643	87.84	-1.52	-4.99,1.94
23F	88.76	90.12			

820 80.30- 855 4.99-
681 83.05 56 6.55 76.50 73.44,79.56
1 85.56 8.42
1010 56.20- 1009 5.09-
599 59.31 66 6.54 52.77 49.37,56.16
3 62.36 8.25
900 56.83- 920 0.60-
Additional 541 60.11 11 1.20 58.92 55.64,62.19
5 63.33 2.13
657 69.18- 690 30.95-
478 72.75 238 34.49 38.26 33.35,43.18
6A 76.13 38.17
768 92.97- 768 5.10-
728 94.79 52 6.77 88.02 85.65,90.39
7F 96.25 8.78
197 94.18- 203 36.91-
192 97.46 89 43.84 53.62 46.45■60.79
19A 99.17 50.96

Table 29-3 Comparison of Subjects aged 3 months Achieving a Pneumococcal IgG
Antibody Concentration 1.0 µg/mL After immunization between PCV13-TT and PCV7
Group for subjects with pre-immunization negative antibody level (PPS)
PCV13-TT PCV7 Rate
Serotype 95%CI
N n % 95% CI N n % 95% CI diff.(%)

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453 95.14- 472 96.97-
465 98.52 -1.39
4 440 97.13 98.46 99.40 -3.27,0.50
289 56.07- 278 89.11- -37.24,-
258 92.81 -30.87
6B 179 61.94 67.56 95.55 24.50
389 94.67- 405 95.51-
395 97.53 -0.62
9V 377 96.92 98.40 98.81 -2.90,1.67
65 94.48- 68 94.72-
Common 68 100.0 0.00
14 65 100.0 100.0 100.0 -5.28,5.52
394 91.67- 406 94.59-
393 96.80 -2.38
18C 372 94.42 96.47 98.28 -5.22,0.46
156 95.45- 153 93.44-
149 97.39 1.33
19F 154 98.72 99.84 99.28 -1.75,4.42
355 84.35- 367 96.11-
360 98.09 -9.92
23F 313 88.17 91.34 99.23 -13.56,-6.28
421 82.56- 434 5.68-
35 8.06 78.16
1 363 86.22 89.37 11.04 73.99,82.33
462 71.81- 469 4.53-
31 6.61 69.36
3 351 75.97 79.80 9.25 64.87,73.86
453 77.33- 464 1.04-
Additional 10 2.16 79.08
5 368 81.24 84.73 3.93 75.25,82.91
326 73.01- 345 28.65-
116 33.62 44.29
6A 254 77.91 82.30 38.88 37.57,51.01
390 95.34- 387 2.58-
17 4.39 93.04
7F 380 97.44 98.76 6.94 90.47,95.62
118 94.01- 114 29.63-
44 38.60 59.71
19A 116 98.31 99.79 47.17 50.47,68.94

Table 29-4 Comparison of Subjects aged 3 months Achieving a Pneumococcal IgG
Antibody Concentration 1.0 µg/mL After immunization between PCV13-TT and PCV7
Group for subjects with pre-immunization negative antibody level (FAS)
PCV13-TT PCV7 Rate
Serotype 95%CI
N n % 95% CI N n % 95% CI diff.(%)

501 90.42- 504 93.23-
481 95.44 -2.42
4 466 93.01 95.09 97.09 -5.30,0.46
319 55.22- 299 85.61- -35.19,-
268 89.63 -28.82
6B 194 60.82 66.21 92.85 22.44
424 91.14- 428 92.05-
405 94.63 -0.76
9V 398 93.87 95.96 96.56 -3.89,2.37
73 88.46- 71 92.40-
Common 70 98.59 -2.70
14 70 95.89 99.14 99.96 -8.02,2.61
434 87.14- 432 91.58-
407 94.21 -3.89
18C 392 90.32 92.94 96.22 -7.44,-0.34
168 93.19- 165 89.14-

155 93.94 3.08
19F 163 97.02 99.03 97.06 -1.37,7.54
392 80.74- 390 92.50- -14.59,-
371 95.13 -10.43
23F 332 84.69 88.11 97.04 6.28
462 79.15- 462 5.89-
38 8.23 74.68
1 383 82.90 86.22 11.11 70.43,78.93
509 67.78- 499 5.10-
36 7.21 64.69
3 366 71.91 75.77 9.85 60.17,69.21
Additional
499 72.16- 495 0.97-
10 2.02 74.13
5 380 76.15 79.83 3.68 70.19,78.07
362 70.64- 366 29.30-
125 34.15 41.26
6A 273 75.41 79.76 39.26 34.68,47.84

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424 91.97- 410 2.43-
17 4.15 90.43
7F 401 94.58 96.53 6.56 87.54,93.32
127 92.13- 123 28.84-
46 37.40 59.45
19A 123 96.85 99.14 46.58 50.38,68.53

Table 29-5 Comparison of Subjects aged 7-11 months Achieving a Pneumococcal IgG
Antibody Concentration 1.0 µg/mL After immunization between PCV13-TT and PCV7
Group for subjects with pre-immunization negative antibody level (PPS)

PCV13-TT PCV7 Rate
Serotype 95%CI
N n % 95% CI N n % 95% CI diff.(%)
175 95.07- 180 94.41-
176 97.78 0.51
4 172 98.29 99.65 99.39 -2.38,3.40
101 66.74- 98 56.07-
65 66.33 9.91
6B 77 76.24 84.14 75.56 -2.60,22.42
172 77.34- 178 83.18-
158 88.76 -5.04
9V 144 83.72 88.90 93.00 -12.25,2.17
57 90.61- 59 93.94-
Common 59 100.0 -1.75
14 56 98.25 99.96 100.0 -5.16,1.65
177 91.29- 181 90.07-
171 94.48 1.01
18C 169 95.48 98.03 97.32 -3.52,5.53
60 86.08- 66 89.48-
64 96.97 -1.97
19F 57 95.00 98.96 99.63 -8.86,4.92
151 84.15- 161 78.64-
137 85.09 4.97
23F 136 90.07 94.33 90.21 -2.31,12.26
133 84.77- 146 4.32-
12 8.22 82.76
1 121 90.98 95.25 13.92 76.16,89.36
171 59.07- 176 4.85-
15 8.52 58.14
3 114 66.67 73.68 13.67 49.96,66.33
162 27.28- 169 0.00-
0 0.00 34.57
5 56 34.57 42.43 2.16 27.24,41.89
137 63.97- 140 24.51-
45 32.14 40.12
6A 99 72.26 79.57 40.55 29.35,50.89
139 94.90- 152 4.63-
13 8.55 90.01
7F 137 98.56 99.83 14.18 85.14,94.88
51 89.55- 59 35.89-
29 49.15 48.89
19A 50 98.04 99.95 62.50 35.57,62.20

Table 29-6 Comparison of Subjects aged 7-11 months Achieving a Pneumococcal IgG
Antibody Concentration 1.0 µg/mL After immunization between PCV13-TT and PCV7
Group for subjects with pre-immunization negative antibody level (FAS)

PCV13-TT PCV7
Rate
Serotype N 95% N 95% 95%CI
n % n % diff.(%)
CI CI
188 90.44- 191 88.64-
178 93.19 1.49
4 178 94.68 97.42 96.33 -3.31,6.29
109 63.10- 104 53.45-
66 63.46 9.02
6B 79 72.48 80.60 72.69 -3.47,21.50
183 73.82- 189 78.71-
Common 160 84.66 -4.33

9V 147 80.33 85.83 89.48 -12.05,3.39
59 88.29- 62 86.50-
59 95.16 1.45
14 57 96.61 99.59 98.99 -5.61,8.51
190 86.68- 192 84.98-
173 90.10 1.47
18C 174 91.58 95.11 93.94 -4.31,7.26

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62 86.50- 69 85.82-
65 94.20 0.96
19F 59 95.16 98.99 98.40 -6.72,8.64
160 79.93- 170 75.13-
139 81.76 4.49
23F 138 86.25 91.18 87.26 -3.40,12.37
143 80.84- 153 4.12-
12 7.84 79.57
1 125 87.41 92.37 13.30 72.66,86.48
184 56.19- 185 4.61-
15 8.11 55.48
3 117 63.59 70.54 13.02 47.49,63.47
174 26.92- 180 0.00-
Additional 0 0.00 33.91
5 59 33.91 41.46 2.03 26.87,40.94
148 60.80- 151 23.24-
46 30.46 38.46
6A 102 68.92 76.26 38.47 27.99,49.92
150 90.62- 160 4.40-
13 8.13 87.21
7F 143 95.33 98.10 13.49 81.79,92.62
53 89.93- 59 35.89-
29 49.15 48.96
19A 52 98.11 99.95 62.50 35.69,62.23

Table 29-7 Comparison of Subjects aged 12-23 months Achieving a Pneumococcal IgG
Antibody Concentration 1.0 µg/mL After immunization between PCV13-TT and PCV7
Group for subjects with pre-immunization negative antibody level (PPS)

PCV13-TT PCV7 Rate
Serotype 95%CI
N n % 95% CI N n % 95% CI diff.(%)
168 97.83- 170 95.81-
168 98.82 1.18
4 168 100.0 100.0 99.86 -0.44,2.80
44 91.96- 33 45.12-
21 63.64 36.36
6B 44 100.0 100.0 79.60 19.95,52.78
170 86.57- 161 84.36-
145 90.06 1.70
9V 156 91.76 95.42 94.21 -4.50,7.90
16 79.41- 14 76.84- -
Common 14 100.0 0.00
14 16 100.0 100.0 100.0 23.16,20.59
173 97.89- 171 92.52-
165 96.49 3.51
18C 173 100.0 100.0 98.70 0.75,6.27
16 79.41- 19 82.35- -
19 100.0 0.00
19F 16 100.0 100.0 100.0 17.65,20.59
95 91.05- 87 75.80-
74 85.06 11.78
23F 92 96.84 99.34 91.80 3.51,20.06
107 84.63- 108 2.07-
6 5.56 86.03
1 98 91.59 96.08 11.70 79.23,92.84
169 44.27- 166 3.35-
11 6.63 45.44
3 88 52.07 59.80 11.55 37.02,53.87
134 33.33- 127 0.00-
Additional 0 0.00 41.79
5 56 41.79 50.62 2.86 33.44,50.14
89 76.32- 90 32.92-
39 43.33 42.06
6A 76 85.39 91.99 54.20 29.46,54.66
104 94.76- 95 4.42-
9 9.47 89.56
7F 103 99.04 99.98 17.22 83.38,95.75
16 79.41- 10 34.75-
7 70.00 30.00
19A 16 100.0 100.0 93.33 1.60,58.40

Table 29-8 Comparison of Subjects aged 12-23 months Achieving a Pneumococcal IgG
Antibody Concentration 1.0 µg/mL After immunization between PCV13-TT and PCV7
Group for subjects with pre-immunization negative antibody level (FAS)
Serotype PCV13-TT PCV7 Rate 95%CI

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N n % 95% CI N n % 95% CI diff.(%)

178 92.07- 181 90.07-
171 94.48 1.59
4 171 96.07 98.40 97.32 -2.79,5.98
48 82.80- 38 43.39-
23 60.53 33.22
6B 45 93.75 98.69 75.96 16.24,50.21
180 82.08- 171 79.84-
147 85.96 1.81
9V 158 87.78 92.18 90.80 -5.26,8.88
17 71.31- 17 63.56- -
Common 15 88.24 5.88
14 16 94.12 99.85 98.54 13.08,24.85
182 92.24- 182 87.43-
168 92.31 3.85
18C 175 96.15 98.44 95.73 -0.93,8.62
16 79.41- 19 82.35- -
19 100.0 0.00
19F 16 100.0 100.0 100.0 17.65,20.59
102 83.91- 97 68.83-
76 78.35 12.83
23F 93 91.18 95.89 86.07 2.95,22.70
116 78.57- 118 1.89-
6 5.08 81.12
1 100 86.21 91.91 10.74 73.70,88.54
179 42.18- 176 3.16-
11 6.25 43.47
3 89 49.72 57.28 10.91 35.32,51.62
145 30.66- 138 0.00-
Additional 0 0.00 38.62
5 56 38.62 47.06 2.64 30.70,46.55
99 69.42- 100 31.26-
41 41.00 37.79
6A 78 78.79 86.36 51.29 25.23,50.35
112 88.70- 105 3.99-
9 8.57 86.07
7F 106 94.64 98.01 15.65 79.28,92.86
16 79.41- 10 34.75-
7 70.00 30.00
19A 16 100.0 100.0 93.33 1.60,58.40

Table 29-9 Comparison of Subjects aged 24-71 months Achieving a Pneumococcal IgG Antibody Concentration 1.0 µg/mL After immunization between PCV13-TT and PCV7 Group for subjects with pre-immunization negative antibody level (PPS)

PCV13-TT PCV7 Rate

Serotype 95%CI

N n % 95% CI N n % 95% CI diff.(%)

127 95.69- 148 97.54-
148 100.0 -0.79
4 126 99.21 99.98 100.0 -2.32,0.75
13 75.29- 26 40.57-
16 61.54 38.46
6B 13 100.0 100.0 79.77 19.76,57.16
102 80.35- 122 86.46-
113 92.62 -4.39
9V 90 88.24 93.77 96.57 -12.17,3.40
4 39.76- 11 58.72-
Common 10 90.91 9.09
14 4 100.0 100.0 99.77 -7.90,26.08
121 92.93- 146 94.11-
143 97.95 -0.42
18C 118 97.52 99.49 99.57 -4.03,3.18
9 39.99-
/ 7 77.78 /
19F / / / 97.19 /
54 79.70- 74 65.79-
57 77.03 13.71
23F 49 90.74 96.92 86.01 1.40,26.03
97 65.46- 122 0.00-
0 0.00 75.26
1 73 75.26 83.46 2.98 66.67,83.85
137 13.41- 149 0.74-
4 2.68 17.02
Additional 3 27 19.71 27.36 6.73 9.87,24.17
80 44.70- 107 0.02-
1 0.93 55.32
5 45 56.25 67.32 5.10 44.29,66.34
6A 47 25 53.19 38.08- 73 26 35.62 24.75- 17.58 -0.43,35.58
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67.89 47.69
80 89.43- 93 7.66-
13 13.98 82.27
7F 77 96.25 99.22 22.72 74.09,90.46
1 2.50- 10 26.24-
6 60.00 40.00
19A 1 100.0 100.0 87.84 9.64,70.36

Table 29-10 Comparison of Subjects aged 24-71 months Achieving a Pneumococcal IgG Antibody Concentration 1.0 µg/mL After immunization between PCV13-TT and PCV7

Group for subjects with pre-immunization negative antibody level (FAS)

PCV13-TT PCV7 Rate

Serotype 95%CI

N n % 95% CI N n % 95% CI diff.(%)

129 94.51- 149 96.32-

148 99.33 -0.88

4 127 98.45 99.81 99.98 -3.38,1.62

13 75.29- 26 40.57-

16 61.54 38.46

6B 13 100.0 100.0 79.77 19.76,57.16

103 80.53- 123 85.56-

113 91.87 -3.52

9V 91 88.35 93.83 96.03 -11.38,4.34

4 39.76- 11 58.72-

Common 10 90.91 9.09

14 4 100.0 100.0 99.77 -7.90,26.08

122 92.98- 146 94.11-

143 97.95 -0.40

18C 119 97.54 99.49 99.57 -3.99,3.18

9 39.99-

/ 7 77.78 /

19F /// 97.19 /

55 77.75- 75 64.75-

57 76.00 13.09

23F 49 89.09 95.89 85.11 0.39,25.79

99 63.93- 122 0.00-

0 0.00 73.74

1 73 73.74 82.07 2.98 65.07,82.41

138 13.31- 149 0.74-

4 2.68 16.88

3 27 19.57 27.17 6.73 9.77,23.99

82 44.70- 107 0.02-

Additional 1 0.93 55.16

5 46 56.10 67.04 5.10 44.27,66.06

48 37.19- 73 24.75-

26 35.62 16.47

6A 25 52.08 66.71 47.69 -1.43,34.37

82 87.98- 93 7.66-

13 13.98 81.14

7F 78 95.12 98.66 22.72 72.69,89.59

1 2.50- 11 30.79-

7 63.64 36.36

19A 1 100.0 100.0 89.07 7.94,64.79

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6.2.4 Non-inferiority test between the serotype-specific IgG response (positive rate or GMC values) for 6 additional serotypes in PCV13-TT and the lowest serotype-specific IgG response for the 7 common serotype in control group

6.2.4.1 Analysis for serotype-specific IgG response (positive rate or GMC values) for 6 additional serotypes in PCV13-TT aged 3-71 months and the lowest serotype-specific IgG response for the 7 common serotype in control group aged 3-71 months

The the lowest serotype-specific IgG response for the 7 common serotypes of positive rate was 23F in control group after primary series, the positive rate was 98.31% for serotype 23F. The lower limit of 2-sided, 97.5% CI for the differences of positive rate between the IgG response for 6 additional serotypes and the lowest IgG response of serotypes 23F was greater than -10% (refer to table 30-1) in primary series.

The lowest GMC was serotype 6B (3.11) for the 7 common serotypes in control group. The lower limit of 2-sided, 97.5% CI for the GMC ratio (the 6 additional serotype/serotype 6B) for 4 of the 6 additional serotypes (including serotypes 1, 6A, 7F and 19A) and serotype 6B for the 7 common serotypes in control group was great than 0.5, the result was shown in table 30-3.

The statistical analysis of serotype-specific IgG response (positive rate or GMC values) for 6 additional serotypes in PCV13-TT aged 3-71 months and the lowest serotype-specific IgG response for the 7 common serotype in control group aged 3-71 months were shown in table 30-1 to 30.4.

Table 30-1 PCV13-TT comparison to the PCV7 seroresponse for the 6 additional serotypes of Subjects aged 3-71 months Achieving a Pneumococcal IgG Antibody Concentration 0.35 µg/mL After immunization (PPS)

PCV13-TT (N=1043) PCV7 (N=1064) Rate diff.

Serotype 97.5%CI

N % 95%CI N % 95%CI (%)

1 1036 99.33 98.62-99.73 1046 98.31 97.34-98.99 1.02 -0.03,2.07

3 1005 96.36 95.03-97.41 1046 98.31 97.34-98.99 -1.95 -3.53,-0.38

5 1011 96.93 95.70-97.89 1046 98.31 97.34-98.99 -1.38 -2.87,0.11

6A 1010 96.84 95.59-97.81 1046 98.31 97.34-98.99 -1.47 -2.98,0.03

7F 1038 99.52 98.88-99.84 1046 98.31 97.34-98.99 1.21 0.20,2.22

19A 1043 100.0 99.65-100.0 1046 98.31 97.34-98.99 1.69 0.81,2.58

Table 30-2 Table 30-1 PCV13-TT comparison to the PCV7 seroresponse for the 6 additional serotypes of Subjects aged 3-71 months Achieving a Pneumococcal IgG Antibody Concentration 0.35 µg/mL After immunization (FAS)

PCV13-TT (N=1117) PCV7 (N=1119) Rate diff.

Serotype 97.5%CI

N % 95%CI N % 95%CI (%)

1 1077 96.42 95.16-97.43 1076 96.16 94.86-97.21 0.26 -1.53,2.05

3 1042 93.29 91.66-94.68 1076 96.16 94.86-97.21 -2.87 -4.99,-0.76

5 1046 93.64 92.05-95.00 1076 96.16 94.86-97.21 -2.51 -4.60,-0.43

6A 1053 94.27 92.74-95.56 1076 96.16 94.86-97.21 -1.89 -3.91,0.14

7F 1088 97.40 96.29-98.25 1076 96.16 94.86-97.21 1.25 -0.43,2.92

19A 1115 99.82 99.35-99.98 1076 96.16 94.86-97.21 3.66 2.34,4.98

Table 30-3 PCV13-TT comparison to the PCV7 seroresponse for the 6 additional serotypes of Subjects aged 24-71 months IgG antibody concentrations (GMCs) (PPS)
PCV13-TT (N=1043) PCV7 (N=1064) PCV13-
Serotype 97.5%CI
GMC 95%CI GMC 95%CI TT/PCV7
1 2.16 2.07-2.26 3.11 2.92-3.30 0.70 0.64,0.76
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3 1.29 1.24-1.36 3.11 2.92-3.30 0.42 0.38,0.45
5 1.41 1.34-1.48 3.11 2.92-3.30 0.45 0.41,0.50
6A 2.08 1.94-2.23 3.11 2.92-3.30 0.67 0.60,0.74
7F 5.34 5.07-5.62 3.11 2.92-3.30 1.72 1.57,1.88
19A 5.46 5.19-5.73 3.11 2.92-3.30 1.76 1.61,1.92

Table 30-4 PCV13-TT comparison to the PCV7 seroresponse for the 6 additional serotypes of Subjects aged 24-71 months IgG antibody concentrations (GMCs) (FAS)
PCV13-TT (N=1117) PCV7 (N=1119) PCV13-
Serotype 97.5%CI
GMC 95%CI GMC 95%CI TT/PCV7
1 1.91 1.79-2.03 2.91 2.73-3.11 0.66 0.59,0.73
3 1.17 1.11-1.24 2.91 2.73-3.11 0.40 0.36,0.44
5 1.22 1.13-1.30 2.91 2.73-3.11 0.42 0.37,0.46
6A 1.92 1.78-2.06 2.91 2.73-3.11 0.66 0.59,0.74
7F 4.72 4.42-5.04 2.91 2.73-3.11 1.62 1.46,1.80
19A 5.06 4.80-5.33 2.91 2.73-3.11 1.74 1.58,1.91

6.2.4.2. Analysis for serotype-specific IgG response (positive rate or GMC values) for 6 additional serotypes in PCV13-TT aged 3 months and the lowest serotype-specific IgG response for the 7 common serotype in control group aged 3 months
The the lowest serotype-specific IgG response for the 7 common serotypes of positive rate was 18C in control group after primary series, the positive rate was 97.74% for serotype 18C. The lower limit of 2-sided, 97.5% CI for the differences of positive rate between the IgG response for 6 additional serotypes and the lowest IgG response for serotype 18C was greater than -10% (refer to table 31-1) in primary series.
The lowest GMC was serotype 6B (3.52) for the 7 common serotypes in control group. Aged 3 months. The GMC ratio (the 6 additional serotype/serotype 6B) was greater than 0.5 among the 6 additional serotype except serotype 3. For serotype 3, te GMC ratio (the 6 additional serotype/serotype 6B) was 0.48. The lower limit of 2-sided, 97.5% CI for the GMC ratio was great than 0.5 among 3 of 6 additional serotypes (except serotypes 3, 5 and 6A). For serotype 3, 5 and 6 A, the lower limit of 2-sided, 97.5% CI for the GMC ratio was 0.42, 0.47 and 0.48, respectively. The result was shown in table 31-3.
The statistical analysis of serotype-specific IgG response (positive rate or GMC values) for 6 additional serotypes in PCV13-TT aged 3 months and the lowest serotype-specific IgG response for the 7 common serotype in control group aged 3 months were shown in table 31-1 to 31-4.

Table 31-1 PCV13-TT comparison to the PCV7 seroresponse for the 6 additional serotypes of Subjects aged 3 months Achieving a Pneumococcal IgG Antibody Concentration 0.35 µg/mL After immunization (PPS)
PCV13-TT (N=471) PCV7 (N=487) Rate diff.
Serotype 97.5%CI
N % 95%CI N % 95%CI (%)
1 469 99.58 98.47-99.95 476 97.74 95.99-98.87 1.83 0.18,3.49
3 462 98.09 96.40-99.12 476 97.74 95.99-98.87 0.35 -1.72,2.42
5 464 98.51 96.96-99.40 476 97.74 95.99-98.87 0.77 -1.19,2.73
6A 456 96.82 94.80-98.21 476 97.74 95.99-98.87 -0.93 -3.29,1.43
7F 469 99.58 98.47-99.95 476 97.74 95.99-98.87 1.83 0.18,3.49
19A 471 100.0 99.22-100.0 476 97.74 95.99-98.87 2.26 0.75,3.77
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Table 31-2 PCV13-TT comparison to the PCV7 seroresponse for the 6 additional serotypes of Subjects aged 3 months Achieving a Pneumococcal IgG Antibody Concentration 0.35 µg/mL After immunization (FAS)
PCV13-TT (N=519) PCV7 (N=519) Rate diff.
Serotype 97.5%CI
N % 95%CI N % 95%CI (%)
1 498 95.95 93.88-97.48 496 95.57 93.42-97.17 0.39 -2.42,3.19
3 489 94.22 91.85-96.07 496 95.57 93.42-97.17 -1.35 -4.41,1.71
5 489 94.22 91.85-96.07 496 95.57 93.42-97.17 -1.35 -4.41,1.71
6A 488 94.03 91.63-95.91 496 95.57 93.42-97.17 -1.54 -4.63,1.55
7F 504 97.11 95.28-98.37 496 95.57 93.42-97.17 1.54 -1.07,4.15
19A 517 99.61 98.61-99.95 496 95.57 93.42-97.17 4.05 1.93,6.16

Table 31-3 PCV13-TT comparison to the PCV7 seroresponse for the 6 additional serotypes of Subjects aged 3 months IgG antibody concentrations (GMCs) (PPS)
PCV13-TT (N=471) PCV7 (N=487) PCV13-
Serotype 97.5%CI
GMC 95%CI GMC 95%CI TT/PCV7
1 2.15 2.01-2.30 3.52 3.25-3.81 0.61 0.54,0.69
3 1.67 1.56-1.79 3.52 3.25-3.81 0.48 0.42,0.54
5 1.88 1.75-2.02 3.52 3.25-3.81 0.54 0.47,0.61
6A 1.96 1.77-2.16 3.52 3.25-3.81 0.56 0.48,0.64
7F 5.47 5.06-5.92 3.52 3.25-3.81 1.56 1.37,1.77
19A 3.85 3.62-4.09 3.52 3.25-3.81 1.09 0.97,1.23
Table 31-4 PCV13-TT comparison to the PCV7 seroresponse for the 6 additional serotypes of Subjects aged 3 months IgG antibody concentrations (GMCs) (FAS)
PCV13-TT (N=519) PCV7 (N=519) PCV13-
Serotype 97.5%CI
GMC 95%CI GMC 95%CI TT/PCV7

1 1.79 1.60-1.99 3.26 2.97-3.58 0.55 0.47,0.65
 3 1.45 1.33-1.58 3.26 2.97-3.58 0.45 0.39,0.52
 5 1.50 1.34-1.68 3.26 2.97-3.58 0.46 0.39,0.54
 6A 1.80 1.62-2.00 3.26 2.97-3.58 0.55 0.47,0.65
 7F 4.66 4.19-5.19 3.26 2.97-3.58 1.43 1.22,1.68
 19A 3.57 3.35-3.81 3.26 2.97-3.58 1.10 0.96,1.25

6.2.4.3. Analysis for serotype-specific IgG response (positive rate or GMC values) for 6 additional serotypes in PCV13-TT aged 7-11 months and the lowest serotype-specific IgG response for the 7 common serotype in control group aged 7-11 months

The the lowest serotype-specific IgG response for the 7 common serotypes of positive rate was 23F in control group after primary series, the positive rate was 96.83% for serotype 23F. The lower limit of 2-sided, 95% CI for the differences of positive rate between the IgG response for 6 additional serotypes and the lowest IgG response for serotype 27F was greater than -10% (refer to table 32-1) in primary series.

The lowest GMC was serotype 6B (2.20) for the 7 common serotypes in control group. aged 3 months. The lower limit of 2-sided, 95% CI for the GMC ratio (the 6 additional serotype/serotype 6B) was great than 0.5 among 5 of 6 additional serotypes (except serotype 5). The result was shown Confidential 160 / 310

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in table 32-3.

The statistical analysis of serotype-specific IgG response (positive rate or GMC values) for 6 additional serotypes in PCV13-TT aged 7-11 months and the lowest serotype-specific IgG - response for the 7 common serotype in control group aged 7-11 months were shown in table 32-1 to 32-4.

Table 32-1 PCV13-TT comparison to the PCV7 seroresponse for the 6 additional serotypes of Subjects aged 7-11 months Achieving a Pneumococcal IgG Antibody Concentration 0.35 µg/mL After immunization(PPS)

PCV13-TT (N=185) PCV7 (N=189) Rate diff.

Serotype 95%CI

N % 95%CI N % 95%CI (%)

1 183 98.92 96.15-99.87 183 96.83 93.22-98.83 2.09 -0.82,5.00
 3 181 97.84 94.56-99.41 183 96.83 93.22-98.83 1.01 -2.25,4.27
 5 169 91.35 86.34-94.98 183 96.83 93.22-98.83 -5.47 -10.23,-0.71
 6A 177 95.68 91.66-98.11 183 96.83 93.22-98.83 -1.15 -5.00,2.70
 7F 184 99.46 97.03-99.99 183 96.83 93.22-98.83 2.63 -0.08,5.35
 19A 185 100.0 98.03-100.0 183 96.83 93.22-98.83 3.17 0.68,5.67

Table 32-2 PCV13-TT comparison to the PCV7 seroresponse for the 6 additional serotypes of Subjects aged 7-11 months Achieving a Pneumococcal IgG Antibody Concentration 0.35 µg/mL After immunization(FAS)

Seroty PCV13-TT (N=198) PCV7 (N=200) Rate diff.

95%CI

pe N % 95%CI N % 95%CI (%)

1 190 95.96 92.19-98.24 187 93.50 89.14-96.49 2.46 -1.92,6.84
 3 186 93.94 89.65-96.83 187 93.50 89.14-96.49 0.44 -4.33,5.21
 5 175 88.38 83.08-92.49 187 93.50 89.14-96.49 -5.12 -10.74,0.50
 6A 184 92.93 88.42-96.08 187 93.50 89.14-96.49 -0.57 -5.51,4.37
 7F 192 96.97 93.52-98.88 187 93.50 89.14-96.49 3.47 -0.70,7.64
 19A 198 100.0 98.15-100.0 187 93.50 89.14-96.49 6.50 3.08,9.92

Table 32-3 PCV13-TT comparison to the PCV7 seroresponse for the 6 additional serotypes of Subjects aged 7-11 months IgG antibody concentrations (GMCs) (PPS)

PCV13-TT (N=185) PCV7 (N=189) PCV13-

Serotype 95%CI

GMC 95%CI GMC 95%CI TT/PCV7

1 2.23 2.03-2.45 2.20 1.92-2.52 1.01 0.86,1.20
 3 1.32 1.20-1.45 2.20 1.92-2.52 0.60 0.51,0.71
 5 0.82 0.73-0.91 2.20 1.92-2.52 0.37 0.31,0.44
 6A 1.79 1.51-2.12 2.20 1.92-2.52 0.81 0.65,1.01
 7F 5.73 5.04-6.51 2.20 1.92-2.52 2.61 2.16,3.14
 19A 5.06 4.54-5.64 2.20 1.92-2.52 2.30 1.93,2.74

Table 32-4 PCV13-TT comparison to the PCV7 seroresponse for the 6 additional serotypes of Subjects aged 7-11 months IgG antibody concentrations (GMCs) (FAS)

PCV13-TT (N=198) PCV7 (N=200) PCV13-

Serotype 95%CI

GMC 95%CI GMC 95%CI TT/PCV7

1 2.07 1.87-2.30 2.02 1.76-2.33 1.02 0.86,1.22
 3 1.19 1.06-1.34 2.02 1.76-2.33 0.59 0.49,0.71
 5 0.77 0.68-0.87 2.02 1.76-2.33 0.38 0.31,0.46
 6A 1.68 1.41-2.00 2.02 1.76-2.33 0.83 0.66,1.04
 7F 5.11 4.41-5.92 2.02 1.76-2.33 2.52 2.06,3.09
 19A 4.75 4.24-5.33 2.02 1.76-2.33 2.35 1.96,2.81

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6.2.4.4. Analysis for serotype-specific IgG response (positive rate or GMC values) for 6 additional serotypes in PCV13-TT aged 12-23 months and the lowest serotype-specific IgG response for the 7 common serotype in control group aged 12-23 months

The the lowest serotype-specific IgG response for the 7 common serotypes of positive rate and GMC was 6B in control group after complete immunization, the positive rate and GMC was 98.43% and 2.71 separately for serotype 6B. The lower limit of 2-sided, 95% CI for the differences of positive rate between the IgG response for 6 additional serotypes and the lowest IgG response for serotype 6B was greater than -10% (refer to table 33-1) after complete immunization. The lower limit of 2-sided, 95% CI for the GMC ratio (the 6 additional serotype/serotype 6B) was great than 0.5 among 4 of 6 additional serotypes (except serotypes 3 and 5). The result was shown in table 33-3.

The statistical analysis of serotype-specific IgG response (positive rate or GMC values) for 6 additional serotypes in PCV13-TT aged 12-23 months and the lowest serotype-specific IgG - response for the 7 common serotype in control group aged 12-23 months were shown in table 33-1 to 33-4.

Table 33-1 PCV13-TT comparison to the PCV7 seroresponse for the 6 additional serotypes of Subjects aged 12-23 months Achieving a Pneumococcal IgG Antibody Concentration 0.35 µg/mL After immunization (PPS)

PCV13-TT (N=189) PCV7 (N=189) Rate

Serotype 95%CI

N % 95%CI N % 95%CI diff.

1 189 100.0 98.07-100.0 186 98.41 95.43-99.67 1.59 -0.19,3.37
 3 186 98.41 95.43-99.67 186 98.41 95.43-99.67 0.00 -2.52,2.52
 5 183 96.83 93.22-98.83 186 98.41 95.43-99.67 -1.59 -4.66,1.48
 6A 186 98.41 95.43-99.67 186 98.41 95.43-99.67 0.00 -2.52,2.52
 7F 189 100.0 98.07-100.0 186 98.41 95.43-99.67 1.59 -0.19,3.37
 19A 189 100.0 98.07-100.0 186 98.41 95.43-99.67 1.59 -0.19,3.37

Table 33-2 PCV13-TT comparison to the PCV7 seroresponse for the 6 additional serotypes of Subjects aged 12-23 months Achieving a Pneumococcal IgG Antibody Concentration 0.35 µg/mL After immunization (FAS)

Seroty PCV13-TT (N=200) PCV7 (N=200) Rate

95%CI

pe N % 95%CI N % 95%CI diff.

1 193 96.50 92.92-98.58 194 97.00 93.58-98.89 -0.50 -3.98,2.98
 3 190 95.00 91.00-97.58 194 97.00 93.58-98.89 -2.00 -5.84,1.84
 5 186 93.00 88.53-96.12 194 97.00 93.58-98.89 -4.00 -8.25,0.25
 6A 189 94.50 90.37-97.22 194 97.00 93.58-98.89 -2.50 -6.45,1.45
 7F 195 97.50 94.26-99.18 194 97.00 93.58-98.89 0.50 -2.70,3.70
 19A 200 100.0 98.17-100.0 194 97.00 93.58-98.89 3.00 0.64,5.36

Table 33-3 PCV13-TT comparison to the PCV7 seroresponse for the 6 additional serotypes of Subjects aged 12-23 months IgG antibody concentrations (GMCs) (PPS)

PCV13-TT (N=189) PCV7 (N=189) PCV13-

Serotype 95%CI

GMC 95%CI GMC 95%CI TT/PCV7

1 2.25 2.07-2.44 2.71 2.35-3.11 0.83 0.71,0.98
 3 1.06 0.98-1.14 2.71 2.35-3.11 0.39 0.33,0.46
 5 1.01 0.91-1.12 2.71 2.35-3.11 0.37 0.31,0.44

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6A 2.46 2.08-2.92 2.71 2.35-3.11 0.91 0.73,1.13

7F 5.47 5.04-5.94 2.71 2.35-3.11 2.02 1.72,2.38

19A 7.81 7.08-8.61 2.71 2.35-3.11 2.89 2.43,3.42

Table 33-4 PCV13-TT comparison to the PCV7 seroresponse for the 6 additional serotypes of Subjects aged 12-23 months IgG antibody concentrations (GMCs) (FAS)

PCV13-TT (N=200) PCV7 (N=200) PCV13-

Serotype 95%CI

GMC 95%CI GMC 95%CI TT/PCV7

1 1.98 1.73-2.26 2.52 2.19-2.91 0.78 0.64,0.95
 3 0.93 0.82-1.06 2.52 2.19-2.91 0.37 0.30,0.45
 5 0.85 0.73-1.00 2.52 2.19-2.91 0.34 0.27,0.42
 6A 2.13 1.73-2.62 2.52 2.19-2.91 0.84 0.66,1.08
 7F 4.73 4.15-5.39 2.52 2.19-2.91 1.87 1.54,2.27
 19A 7.07 6.32-7.91 2.52 2.19-2.91 2.80 2.34,3.36

6.2.4.5. Analysis for serotype-specific IgG response (positive rate or GMC) for 6 additional serotypes in PCV13-TT aged 24-71 months and the lowest serotype-specific IgG response for the 7 common serotype in control group aged 24-71 months.

The the lowest serotype-specific IgG response for the 7 common serotypes of positive rate and GMC was 23F and 6B separately in control group after complete immunization, the positive rate was 97.99% for serotype 23F and GMC was 3.63 serotype 6B. The lower limit of 2-sided, 95% CI for the differences of positive rate between the IgG response for 5 of 6 additional serotypes (except serotype 3) and the lowest IgG response for serotype 23F was greater than -10% (refer to table 34-1) after complete immunization. The GMC ratio (the 6 additional serotype/serotype 6B) was greater than 0.5 for 4 of 6 additional serotypes, except serotypes 3 and 5. The lower limit of 2-sided, 95% CI for the GMC ratio was great than 0.5 among 3 of 6 additional serotypes (except serotypes 1, 3 and 5). The result was shown in table 34-3.

The statistical analysis of serotype-specific IgG response (positive rate or GMC) for 6 additional serotypes in PCV13-TT aged 24-71 months and the lowest serotype-specific IgG -response for the 7 common serotype in control group aged 24-71 months were shown in table 34-1 to 34-4.

Table 34-1 PCV13-TT comparison to the PCV7 seroresponse for the 6 additional serotypes of Subjects aged 24-71 months Achieving a Pneumococcal IgG Antibody Concentration 0.35 µg/mL After immunization (PPS)

PCV13-TT (N=198) PCV7 (N=199) Rate

Serotype 95%CI

N % 95%CI N % 95%CI diff.

1 195 98.48 95.64-99.69 195 97.99 94.93-99.45 0.49 -2.09,3.08
 3 176 88.89 83.66-92.90 195 97.99 94.93-99.45 -9.10 -13.89,-4.31
 5 195 98.48 95.64-99.69 195 97.99 94.93-99.45 0.49 -2.09,3.08
 6A 191 96.46 92.85-98.57 195 97.99 94.93-99.45 -1.53 -4.75,1.70
 7F 196 98.99 96.40-99.88 195 97.99 94.93-99.45 1.00 -1.40,3.40
 19A 198 100.0 98.15-100.0 195 97.99 94.93-99.45 2.01 0.06,3.96

Table 34-2 PCV13-TT comparison to the PCV7 seroresponse for the 6 additional serotypes of Subjects aged 24-71 months Achieving a Pneumococcal IgG Antibody Concentration 0.35 µg/mL After immunization(FAS)

PCV13-TT (N=200) PCV7 (N=200) Rate

Serotype 95%CI

N % 95%CI N % 95%CI diff.

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1 196 98.00 94.96-99.45 196 98.00 94.96-99.45 0.00 -2.74,2.74
 3 177 88.50 83.25-92.57 196 98.00 94.96-99.45 -9.50 -14.33,-4.67
 5 196 98.00 94.96-99.45 196 98.00 94.96-99.45 0.00 -2.74,2.74
 6A 192 96.00 92.27-98.26 196 98.00 94.96-99.45 -2.00 -5.34,1.34
 7F 197 98.50 95.68-99.69 196 98.00 94.96-99.45 0.50 -2.07,3.07
 19A 200 100.0 98.17-100.0 196 98.00 94.96-99.45 2.00 0.06,3.94
 Table 34-3 PCV13-TT comparison to the PCV7 seroresponse for the 6 additional serotypes
 of Subjects aged 24-71 months IgG antibody concentrations (GMCs)(PPS)
 PCV13-TT (N=198) PCV7 (N=199) PCV13-
 Serotype 95%CI
 GMC 95%CI GMC 95%CI TT/PCV7
 1 2.06 1.84-2.30 3.63 3.04-4.32 0.57 0.46,0.70
 3 0.84 0.75-0.94 3.63 3.04-4.32 0.23 0.19,0.29
 5 1.60 1.43-1.79 3.63 3.04-4.32 0.44 0.36,0.54
 6A 2.35 2.00-2.76 3.63 3.04-4.32 0.65 0.51,0.82
 7F 4.60 4.00-5.28 3.63 3.04-4.32 1.27 1.01,1.58
 19A 9.56 8.47-10.79 3.63 3.04-4.32 2.64 2.13,3.26
 Table 34-4 PCV13-TT comparison to the PCV7 seroresponse for the 6 additional serotypes
 of Subjects aged 24-71 months IgG antibody concentrations (GMCs) (FAS)
 PCV13-TT (N=200) PCV7 (N=200) PCV13-
 Serotype 95%CI
 GMC 95%CI GMC 95%CI TT/PCV7
 1 2.03 1.81-2.27 3.62 3.04-4.30 0.56 0.46,0.69
 3 0.83 0.74-0.93 3.62 3.04-4.30 0.23 0.19,0.28
 5 1.59 1.42-1.77 3.62 3.04-4.30 0.44 0.36,0.54
 6A 2.31 1.97-2.72 3.62 3.04-4.30 0.64 0.50,0.81
 7F 4.51 3.91-5.19 3.62 3.04-4.30 1.25 1.00,1.56
 19A 9.47 8.40-10.68 3.62 3.04-4.30 2.62 2.12,3.23

6.3. BOOSTER DOSE ANALYSIS

The primary population of Per Protocol Set for booster dose analysis included subjects aged 2 months (minimum 6 weeks), subjects aged 3 months and subjects aged 7-11 months who completed immunization. The proportion of subjects achieving serum IgG antibody concentrations ≥ 0.35 $\mu\text{g/mL}$, was calculated, the IgG geometric mean antibody concentration (GMC) was measured and the antibody increase fold was calculated thirty days after the booster dose.

6.3.1. Status for booster dose

A total of 1,884 subjects should be enrolled to receive a booster dose, while 1833 subjects received a booster dose actually, including 482 subjects aged 2 months, 486 subjects aged 3 months in PCV13-TT and 490 subjects aged 3 months in control group. There were 51 subjects dropped off during the study. The primary reason for dropped off was voluntary withdrawal rather than adverse reaction. There were 9 subjects withdrawals due to serious adverse events or non-serious adverse events. The safety set (SS) included 1845 subjects. For immunogenicity analysis, the full analysis set (FAS) included 1840 subjects and the pre protocol set (PPS) included 1813 subjects.

The status for subjects

The details for completion and shedding of subjects with booster dose were shown in table 35-1 to 35-3.

Table 35-1 Booster dose vaccination in subjects aged 2 months (minimum age: 6 weeks)
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■■■■ N ■

Enrolled for booster dose 491 100.00
 Completers 482 98.17
 Dropouts 9 1.83
 SAEs 0 0.00
 non-SAEs 0 0.00
 Protocol violation 0 0.00
 Voluntary withdrawal (non-AE) 9 1.83
 Outside, migrated or moved away
 0 0.00
 from the region
 Lost to follow-up 0 0.00
 Other 0 0.00
 Safety analysis set SS 487 99.19
 FAS 486 98.98
 Immunogenicity analysis set
 PPS 476 96.95

Table 35-2 Booster dose vaccination of PCV13-TT and PCV7 in subjects aged 3 months
 PCV13-TT PCV7 ■■■

■■■■

N ■ N ■ N ■

Enrolled for booster dose 503 100.00 503 100.00 1006 100.00
 Completers 486 96.62 490 97.42 976 97.02
 Dropouts* 17 3.38 13 2.58 30 2.98
 SAEs 1 0.20 0 0.00 1 0.10
 non-SAEs 2 0.40 1 0.20 3 0.30
 Protocol violation 0 0.00 0 0.00 0 0.00
 Voluntary withdrawal (non-
 12 2.39 10 1.99 22 2.19
 AE)
 Outside, migrated or moved
 0 0.00 1 0.20 1 0.10
 away from the region
 Lost to follow-up 1 0.20 0 0.00 1 0.10
 Other 1 0.20 1 0.20 2 0.20
 Safety analysis SS
 488 97.02 493 98.01 981 97.51
 set

Immunogenicity FAS 486 96.62 492 97.81 978 97.22
analysis set PPS 475 94.43 487 96.82 962 95.63
*Comparison for intergroup dropout rates Fisher P=0.6990.

Table 35-3 7-11■■■PCV13-TT■PCV7■■■■■■■■■■

Table 35-3 Booster dose vaccination of PCV13-TT and PCV7 in subjects aged 7-11 months
PCV13-TT PCV7 ■■

■■■■
N ■■ N ■■ N ■■
Enrolled for booster dose 193 100.00 194 100.00 387 100.00
Completers 186 96.37 189 97.42 375 96.90
Dropouts* 7 3.63 5 2.58 12 3.10
SAEs 0 0.00 0 0.00 0 0.00
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non-SAEs 3 1.55 2 1.03 5 1.29
Protocol violation 0 0.00 0 0.00 0 0.00
Voluntary withdrawal (non-AE) 3 1.55 3 1.55 6 1.55
Outside, migrated or moved
1 0.52 0 0.00 1 0.26
away from the region
Lost to follow-up 0 0.00 0 0.00 0 0.00
Other 0 0.00 0 0.00 0 0.00
Safety analysis set SS 186 96.37 191 98.45 377 97.42
Immunogenicity FAS 186 96.37 190 97.94 376 97.16
analysis set PPS 186 96.37 189 97.42 375 96.90
*Comparison for intergroup dropout rates Fisher P=0.6803.
6.3.2. Positive rate of IgG antibody and the IgG antibody concentrations (GMCs) analysis
6.3.2.1. Positive rate of IgG antibody and the IgG antibody concentrations (GMCs) analysis
for subjects in PCV13-TT and control group aged 3 months before and after booster dose
(1).Positive rate of IgG antibody and the IgG antibody concentrations (GMCs) analysis for
subjects in PCV13-TT and control group aged 3 months before booster dose
Before booster dose, for 4 of the 7 common serotypes (except serotypes 4, 9V and 18C), the
proportion of responders at ≥ 0.35 $\mu\text{g/mL}$ was more than 96% in the PCV13-TT. The GMCs for
the 7 common serotypes were greater than 0.35 $\mu\text{g/mL}$ in PCV13-TT and control group (refer to
table 36-3).
For 4 of the 6 additional serotypes (except serotypes 1, 3 and 5), the proportion of responders at
 ≥ 0.35 $\mu\text{g/mL}$ was more than 94% in the PCV13-TT. The GMCs for the the 6 additional serotypes
were greater than 0.35 $\mu\text{g/mL}$ in PCV13-TT and control group.
Before booster dose, among the 13 serotypes, 6B, 14, 19F, 23F, 6A, 7F and 19A did not show a
significant decrease of positive rate after primary series, but the antibody levels of each serotypes
decreased in different degrees compares to levels when primary series finished. The trends for
positive rate and antibody level were the same for each serotype in PCV13-TT and control group
aged 3 months.
(2) Positive rate of IgG antibody and the IgG antibody concentrations (GMCs) analysis for
subjects in PCV13-TT and control group aged 3 months after booster dose
For subjects aged 3 months, among the 7 common serotypes, the positive rate and GMC of 7
common serotype in the PCV13-TT were non-inferior to the control group. After booster dose,
both the PCV13-TT and the control group had a 3.7-fold increase in antibody GMCs .(refer to
Table 36-4). Among the 7 common serotypes (except serotypes 4 and 18C), the proportion of
responders at ≥ 1.0 $\mu\text{g/mL}$ was greater than 95% in the PCV13-TT except for 93.26% for type 4
and 88.84% for type 18C. The lower limit of 2-sided, 95% CI for the difference of proportion ratio
of responders at ≥ 1.0 $\mu\text{g/mL}$ was greater than -10% after booster dose(refer to table 36-2).
Among the 6 additional serotypes, the positive rate for PCV13-TT was superiority to the control
group, except serotype 19A, the lower limit of 2-sided, 95% CI for the positive rate was -0.2%.
The GMCs for the 6 additional serotypes were superiority to the control group, the PCV13-TT had
a at least 3.5-fold increase in antibody GMCs after booster dose.
Among the 6 additional serotypes (except serotypes 1, 3 and 5), the proportion of responders at
 ≥ 1.0 $\mu\text{g/mL}$ was greater than 95% in the PCV13-TT. The lower limit of 2-sided, 95% CI for
difference of the proportion ratio of responders at ≥ 1.0 $\mu\text{g/mL}$ was greater than 0 after booster dose.
(3) Analysis for serotype-specific IgG response (positive rate or GMC values) for 6 additional
serotypes in PCV13-TT aged 3 months and the lowest serotype-specific IgG response for the

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7 common serotype in control group aged 3 months
The the lowest serotype-specific IgG response for the 7 common serotypes of positive rate was
18C in control group afterbooster dose, the positive rate was 99.59%.
The lower limit of 2-sided, 97.5% CI for the differences of positive rate between the IgG response
for 6 additional serotypes and the lowest IgG response for serotype 18C was greater than -10%
(refer to table 36-5) after booster dose.
The lowest GMC was serotype 9V (3.87) for the 7 common serotypes in control group aged 3
months. The GMC ratio (the 6 additional serotype/serotype 9V) was greater than 0.5 among the 6
additional serotypes except serotypes 3 and 5. For serotypes 3 and 5, the GMC ratio (the 6
additional serotype/serotype 6B) was 0.49 and 0.41. The lower limit of 2-sided, 97.5% CI for the
GMC ratio was great than 0.5 among 4 of 6 additional serotypes (except serotypes 3 and 5). For
serotype 3 and 5, the lower limit of 2-sided, 97.5% CI for the GMC ratio was 0.44 and 0.37,
respectively. The result was shown in table 36-6.
The analysis of positive rate of IgG antibody and the IgG antibody concentrations (GMCs) analysis
for subjects aged 3 months before and after booster dose in PCV13-TT and control group were
shown in table 36-1 to 36-4, and the forest chart was shown in Figures 159 to 162.
The statistical analysis of serotype-specific IgG response (positive rate or GMC values) for 6
additional serotypes in PCV13-TT aged 3 months and the lowest serotype-specific IgG response
for the 7 common serotype in control group aged 3 months were shown in table 36-5 to 36-6.

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Table 36-1 Pre- and post-booster dose antibody positive rates in PCV13-TT (3M) and PCV7 (3M) groups (PPS)

Pre-immunization Post-immunization												
PCV13-TT PCV7 P PCV13-TT PCV7												
Serotype Rate												
(N=474) (N=487) (N=475) (N=487) P 95%/97.5%CI*												
N N N N diff.												
% 95%CI % 95%CI % 95%CI % 95%CI												
70.30- 86.69- 98.83- 98.86- 0.9859												
4 353 74.47 437 89.73 <0.0001 474 99.79 486 99.79 -0.01 -0.66,0.65												
78.34 92.28 99.99 99.99												
98.48- 95.22- 99.23- 98.86- 0.3231												
6B 472 99.58 473 97.13 0.0030 475 100.0 486 99.79 0.21 -0.25,0.67												
99.95 98.42 100.0 99.99												
75.17- 78.00- 99.23- 99.25- 1.0000												
9V 375 79.11 398 81.72 0.3077 475 100.0 487 100.0 0.00 -0.90,0.92												
82.69 85.06 100.0 100.0												
99.22- 99.25- 98.83- 99.25- 0.3110												
Common 14 474 100.0 487 100.0 1.0000 474 99.79 487 100.0 -0.21 -0.68,0.26												
100.0 100.0 99.99 100.0												
55.13- 72.14- 98.83- 98.52- 0.5778												
18C 283 59.70 371 76.18 <0.0001 474 99.79 485 99.59 0.20 -0.60,1.00												
64.15 79.90 99.99 99.95												
97.85- 98.86- 99.23- 99.25- 1.0000												
19F 470 99.16 486 99.79 0.2116 475 100.0 487 100.0 0.00 -0.90,0.92												
99.77 99.99 100.0 100.0												
94.32- 91.56- 99.23- 98.86- 0.3231												
23F 457 96.41 458 94.05 0.0855 475 100.0 486 99.79 0.21 -0.25,0.67												
97.90 95.98 100.0 99.99												
83.78- 49.87- 98.83- 88.52- <0.0001												
1 413 87.13 265 54.41 <0.0001 474 99.79 445 91.38 8.41 5.89,10.94												
90.01 58.90 99.99 93.71												
50.46- 16.65- 97.86- 89.45- <0.0001												
3 261 55.06 98 20.12 <0.0001 471 99.16 449 92.20 6.96 4.44,9.48												
59.60 23.96 99.77 94.42												
56.63- 15.32- 96.16- 35.66- <0.0001												
5 290 61.18 91 18.69 <0.0001 465 97.89 195 40.04 57.85 53.31,62.39												
65.59 22.44 98.99 44.55												
Additional												
92.31- 76.04- 98.49- 95.99- 0.0136												
6A 449 94.73 389 79.88 <0.0001 473 99.58 476 97.74 1.84 0.40,3.28												
96.56 83.35 99.95 98.87												
94.58- 42.31- 99.23- 62.36- <0.0001												
7F 458 96.62 228 46.82 <0.0001 475 100.0 325 66.74 33.26 29.08,37.45												
98.06 51.36 100.0 70.91												
98.83- 94.72- 99.23- 98.86- 0.3231												
19A 473 99.79 471 96.71 0.0003 475 100.0 486 99.79 0.21 -0.20,0.61												
99.99 98.11 100.0 99.99												

Note: N=number of subjects analyzed, n=number of subjects with positive antibody; *CI: 97.5% for common serotypes, 95% for additional serotypes.

Table 36-2 Percentage of subjects with pre- and post-booster dose antibody concentration ≥ 1.0 µg/mL in PCV13-TT (3M) and PCV7 (3M) groups (PPS)

Pre-immunization		Post-immunization													
Serotyp	PCV13-TT	PCV7	PCV13-TT	PCV7											
Rate															
e (N=474) (N=487) P (N=475) (N=487) P 95%CI															
diff.															
N	% 95%CI	N	% 95%CI	N	% 95%CI	N	% 95%CI								
4	96	20.25	16.72-24.16	194	39.84	35.46-44.34	<0.0001	443	93.26	90.62-95.35	470	96.51	94.47-97.95	0.0221	
-3.25	-6.03,-0.46														
C	6B	402	84.81	81.26-87.92	318	65.3	60.88-69.52	<0.0001	475	100.0	99.23-100.0	478	98.15	96.52-99.15	0.0029
1.85	0.65,3.04														
o	9V	106	22.36	18.69-26.39	121	24.85	21.07-28.93	0.3649	455	95.79	93.57-97.41	471	96.71	94.72-98.11	0.4497
-0.93	-3.33,1.48														
m	14	450	94.94	92.56-96.73	479	98.36	96.79-99.29	0.0031	474	99.79	98.83-99.99	487	100.0	99.25-100.0	0.3110
-0.21	-0.62,0.20														
m	18C	48	10.13	7.56-13.20	102	20.94	17.41-24.83	<0.0001	422	88.84	85.66-91.53	460	94.46	92.04-96.31	0.0016
-5.61	-9.10,-2.13														
on	19F	404	85.23	81.71-88.30	376	77.21	73.22-80.86	0.0015	475	100.0	99.23-100.0	484	99.38	98.21-99.87	0.0867
0.62	-0.08,1.31														
23F	318	67.09	62.66-71.31	274	56.26	51.73-60.72	0.0006	469	98.74	97.27-99.54	484	99.38	98.21-99.87	0.2972	
-0.65	-1.87,0.57														
1	137	28.90	24.86-33.21	34	6.98	4.88-9.62	<0.0001	431	90.74	87.76-93.19	125	25.67	21.84-29.79	<0.0001	
65.07	60.40,69.74														
A															
3	34	7.17	5.02-9.88	23	4.72	3.02-7.00	0.1079	391	82.32	78.58-85.64	243	49.90	45.37-54.43	<0.0001	
26.81	38.03														
dd															
5	40	8.44	6.10-11.31	4	0.82	0.22-2.09	<0.0001	342	72.00	67.73-76.00	13	2.67	1.43- 4.52	<0.0001	
65.05	73.61														
iti															

6A 281 59.28 54.71-63.74 143 29.36 25.35-33.63 <0.0001 457 96.21 94.08-97.74 431 88.50 85.33-91.20
<0.0001 7.71 4.40,11.02
on
7F 314 66.24 61.79-70.49 45 9.24 6.82-12.17 <0.0001 466 98.11 96.43-99.13 69 14.17 11.19-17.59 <0.0001
83.94 80.61,87.27
al
19A 375 79.11 75.17-82.69 261 53.59 49.05-58.09 <0.0001 473 99.58 98.49-99.92 459 94.25 91.80-96.15
<0.0001 5.33 3.18,7.48
Note: N=number of subjects analyzed, n=number of subjects reaching critical value.

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Table 36-3 Pre- and post-booster dose antibody GMC in PCV13-TT (3M) and PCV7 (3M) groups (PPS)
Pre-immunization Post-immunization
PCV13-TT PCV7 PCV13-TT PCV7
Serotype
(N=474) (N=487) P (N=475) (N=487) P Ratio 95%/97.5%CI*
GMC 95%CI GMC 95%CI GMC 95%CI GMC 95%CI
4 0.57 0.53-0.60 0.83 0.78-0.89 <0.0001 3.19 2.96-3.44 4.27 3.98-4.59 <0.0001 0.75 0.66,0.84
6B 2.08 1.94-2.22 1.35 1.26-1.44 <0.0001 10.01 9.30-10.79 11.70 10.78-12.70 0.0059 0.86 0.75,0.97
9V 0.59 0.56-0.63 0.64 0.60-0.68 0.1195 4.22 3.90-4.56 3.87 3.62-4.14 0.0966 1.09 0.97,1.23
common 14 3.38 3.16-3.63 4.83 4.54-5.14 <0.0001 13.97 12.94-15.09 17.89 16.77-19.09 <0.0001 0.78
0.70,0.88
18C 0.38 0.35-0.42 0.51 0.46-0.56 <0.0001 2.44 2.27-2.63 3.90 3.59-4.23 <0.0001 0.63 0.55,0.71
19F 2.24 2.04-2.47 1.78 1.65-1.91 0.0002 12.18 11.25-13.20 7.84 7.33-8.38 <0.0001 1.56 1.38,1.75
23F 1.29 1.21-1.38 1.11 1.04-1.19 0.0021 6.9 6.39-7.45 10.81 9.91-11.80 <0.0001 0.64 0.56,0.73
1 0.71 0.67-0.75 0.36 0.34-0.39 <0.0001 2.79 2.61-2.99 0.67 0.62-0.72 <0.0001 4.18 3.76,4.65
3 0.39 0.37-0.42 0.20 0.18-0.21 <0.0001 1.89 1.77-2.01 1.02 0.95-1.09 <0.0001 1.86 1.69,2.04
5 0.43 0.41-0.45 0.17 0.15-0.19 <0.0001 1.58 1.47-1.70 0.16 0.14-0.19 <0.0001 9.63 8.06,11.51
Additional
6A 1.22 1.13-1.30 0.67 0.63-0.72 <0.0001 5.82 5.32-6.37 3.80 3.44-4.21 <0.0001 1.53 1.34,1.75
7F 1.29 1.20-1.38 0.30 0.27-0.33 <0.0001 4.59 4.29-4.91 0.31 0.26-0.36 <0.0001 15.01 12.58,17.91
19A 1.89 1.76-2.03 1.15 1.07-1.24 <0.0001 13.34 12.37-14.39 3.00 2.78-3.24 <0.0001 4.45 4.00,4.95
Note: N=number of subjects analyzed, n=number of subjects with positive antibody; *CI: 97.5% for common
serotypes, 95% for additional serotypes.

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Table 36-4 Post-booster dose antibody fold increase in PCV13-TT (3M) and PCV7
(3M) groups (PPS)
PCV13-TT (N=475) PCV7 (N=487)
Serotypes Serotype GMC GMC P
fold 95%CI fold 95%CI
increase increase
4 5.62 5.23-6.03 5.15 4.78-5.55 0.0980
6B 4.83 4.50-5.17 8.66 8.05-9.31 <0.0001
9V 7.11 6.57-7.69 6.06 5.63-6.52 0.0036
common 14 4.13 3.82-4.46 3.70 3.46-3.96 0.0356
18C 6.38 5.77-7.04 7.63 6.96-8.36 0.0090
19F 5.43 4.93-5.99 4.40 4.08-4.76 0.0009
23F 5.35 5.01-5.71 9.71 8.99-10.49 <0.0001
1 3.96 3.73-4.20 1.83 1.70-1.99 <0.0001
3 4.80 4.51-5.11 5.19 4.79-5.62 0.1335
5 3.67 3.45-3.92 0.95 0.82-1.09 <0.0001
Additional
6A 4.79 4.42-5.19 5.66 5.16-6.20 0.0073
7F 3.57 3.32-3.84 1.02 0.88-1.19 <0.0001
19A 7.08 6.51-7.70 2.61 2.42-2.82 <0.0001

Table 36-5 Comparison of antibody positive rates for additional and common
serotypes in subjects aged 3 months (PPS)
PCV13-TT (N=475) PCV7 (N=487) Rate
Serotype 95%CI diff. 97.5%CI
N % N % 95%CI (%)
1 474 99.79 98.83-99.99 485 99.59 98.52-99.95 0.20 -0.60,1.00
3 471 99.16 97.86-99.77 485 99.59 98.52-99.95 -0.43 -1.57,0.71
5 465 97.89 96.16-98.99 485 99.59 98.52-99.95 -1.69 -3.31,-0.08
6A 473 99.58 98.49-99.95 485 99.59 98.52-99.95 -0.01 -0.94,0.92
7F 475 100.0 99.23-100.0 485 99.59 98.52-99.95 0.41 -0.24,1.06
19A 475 100.0 99.23-100.0 485 99.59 98.52-99.95 0.41 -0.24,1.06

Table 36-6 Comparison of antibody GMC for additional and common serotypes in
subjects aged 3 months (PPS)
PCV13-TT (N=475) PCV7 (N=487) PCV13-
Serotype 97.5%CI
GMC 95%CI GMC 95%CI TT/PCV7
1 2.79 2.61-2.99 3.87 3.62-4.14 0.72 0.65,0.80
3 1.89 1.77-2.01 3.87 3.62-4.14 0.49 0.44,0.54
5 1.58 1.47-1.70 3.87 3.62-4.14 0.41 0.37,0.46
6A 5.82 5.32-6.37 3.87 3.62-4.14 1.50 1.32,1.71
7F 4.59 4.29-4.91 3.87 3.62-4.14 1.19 1.06,1.32
19A 13.34 12.37-14.39 3.87 3.62-4.14 3.45 3.07,3.87
6.3.2.2. Positive rate of IgG antibody and the IgG antibody concentrations (GMCs) analysis
for subjects in PCV13-TT aged 2 months and control group aged 3 months before and after

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-15.00 -18.85,-11.15
6B 416 87.58 84.27-90.41 318 65.3 60.88-69.52 <0.0001 472 99.16 97.86-99.77 478 98.15 96.52-99.15 0.1755
1.01 -0.44,2.46
9V 135 28.42 24.40-32.71 121 24.85 21.07-28.93 0.2097 451 94.75 92.34-96.57 471 96.71 94.72-98.11 0.1307
-1.97 -4.52,0.59
Common 14 457 96.21 94.08-97.74 479 98.36 96.79-99.29 0.0401 476 100.0 99.23-100.0 487 100.0 99.25-100.0
1.0000 0.00 -0.75,0.77
18C 48 10.11 7.55-13.17 102 20.94 17.41-24.83 <0.0001 384 80.67 76.83-84.12 460 94.46 92.04-96.31 <0.0001
-13.78 -17.87,-9.70
19F 428 90.11 87.06-92.64 376 77.21 73.22-80.86 <0.0001 476 100.0 99.23-100.0 484 99.38 98.21-99.87
0.0863 0.62 -0.08,1.31
23F 304 64.00 59.50-68.32 274 56.26 51.73-60.72 0.0143 465 97.69 95.90-98.84 484 99.38 98.21-99.87 0.0280
-1.69 -3.21,-0.18
1 122 25.68 21.81-29.86 34 6.98 4.88-9.62 <0.0001 387 81.30 77.50-84.71 125 25.67 21.84-29.79 <0.0001
55.64 50.41,60.86
3 30 6.32 4.30-8.89 23 4.72 3.02-7.00 0.2790 338 71.01 66.71-75.05 243 49.90 45.37-54.43 <0.0001 21.11
15.08,27.14
5 13 2.74 1.47-4.63 4 0.82 0.22-2.09 0.0242 256 53.78 49.19-58.33 13 2.67 1.43- 4.52 <0.0001 51.11
46.41,55.81
Additional
6A 285 60.00 55.44-64.44 143 29.36 25.35-33.63 <0.0001 454 95.38 93.09-97.08 431 88.50 85.33-91.20 0.0001
6.88 3.47,10.28
7F 309 65.05 60.58-69.34 45 9.24 6.82-12.17 <0.0001 472 99.16 97.86-99.77 69 14.17 11.19-17.59 <0.0001
84.99 81.79,88.20
19A 364 76.63 72.56-80.37 261 53.59 49.05-58.09 <0.0001 476 100.0 99.23-100.0 459 94.25 91.80-96.15
<0.0001 5.75 3.68,7.82
Note: N=number of subjects analyzed, n=number of subjects reaching critical value.

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Table 37-3 Pre- and post-booster dose antibody GMC in PCV13-TT (2M) and PCV7 (3M) groups (PPS)

Pre-immunization Post-immunization
PCV13-TT (2M) PCV7(3M) PCV13-TT (2M) PCV7(3M)
Serotype
(N=475) (N=487) P (N=476) (N=487) P Ratio# 95%/97.5%CI*
GMC 95%CI GMC 95%CI GMC 95%CI GMC 95%CI
4 0.50 0.47-0.53 0.83 0.78-0.89 <0.0001 2.07 1.92-2.22 4.27 3.98-4.59 <0.0001 0.48 0.43-0.54
6B 2.02 1.88-2.16 1.35 1.26-1.44 0.4442 8.60 7.93-9.32 11.70 10.78-12.70 <0.0001 0.73 0.64-0.84
9V 0.67 0.63-0.72 0.64 0.60-0.68 0.5593 3.83 3.56-4.12 3.87 3.62-4.14 0.8400 0.99 0.88-1.11
common 14 3.40 3.18-3.63 4.83 4.54-5.14 0.4938 11.61 10.85-12.44 17.89 16.77-19.09 <0.0001 0.65 0.58-0.72
18C 0.41 0.38-0.45 0.51 0.46-0.56 <0.0001 2.04 1.90-2.18 3.90 3.59-4.23 <0.0001 0.52 0.46-0.59
19F 2.08 1.96-2.21 1.78 1.65-1.91 1.0000 8.76 8.16-9.39 7.84 7.33-8.38 0.0252 1.12 1.00-1.25
23F 1.31 1.22-1.40 1.11 1.04-1.19 0.0379 5.70 5.25-6.18 10.81 9.91-11.80 <0.0001 0.53 0.46-0.60
1 0.66 0.63-0.70 0.36 0.34-0.39 <0.0001 1.98 1.86-2.12 0.67 0.62-0.72 <0.0001 2.97 2.68-3.30
3 0.39 0.37-0.42 0.20 0.18-0.21 <0.0001 1.53 1.44-1.64 1.02 0.95-1.09 <0.0001 1.51 1.37-1.66
5 0.32 0.30-0.35 0.17 0.15-0.19 <0.0001 1.13 1.06-1.21 0.16 0.14-0.19 <0.0001 6.92 5.80-8.25
Additional
6A 1.20 1.12-1.29 0.67 0.63-0.72 <0.0001 4.79 4.39-5.22 3.80 3.44-4.21 0.0007 1.26 1.10-1.44
7F 1.28 1.20-1.35 0.30 0.27-0.33 <0.0001 4.05 3.79-4.32 0.31 0.26-0.36 <0.0001 13.22 11.09-15.76
19A 1.64 1.54-1.75 1.15 1.07-1.24 0.0031 9.32 8.66-10.03 3.00 2.78-3.24 <0.0001 3.11 2.80-3.45
Note: N=number of subjects analyzed; #Ratio: PCV13-TT (2M)GMC/PCV13-TT (3M)GMC■*CI: 97.5% for common serotypes, 95% for additional serotypes.

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Table 37-4 Post-booster dose antibody fold increase in PCV13-TT (2M) and PCV7 (3M) groups (PPS)

PCV13-TT (2M) (N=476) PCV7(3M) (N=487)
Serotypes Serotype GMC fold GMC fold P
95%CI 95%CI
increase increase
4 4.17 3.91-4.44 5.15 4.78-5.55 <0.0001
6B 4.26 3.98-4.55 8.66 8.05-9.31 <0.0001
9V 5.69 5.30-6.10 6.06 5.63-6.52 0.2176
common 14 3.42 3.21-3.65 3.70 3.46-3.96 0.0959
18C 4.93 4.53-5.37 7.63 6.96-8.36 <0.0001
19F 4.20 3.92-4.51 4.40 4.08-4.76 0.3816
23F 4.35 4.08-4.64 9.71 8.99-10.49 <0.0001
1 3.00 2.83-3.17 1.83 1.70-1.99 <0.0001
3 3.91 3.70-4.14 5.19 4.79-5.62 <0.0001
5 3.50 3.26-3.75 0.95 0.82-1.09 <0.0001
Additional
6A 4.00 3.71-4.30 5.66 5.16-6.20 <0.0001
7F 3.17 3.00-3.35 1.02 0.88-1.19 <0.0001
19A 5.67 5.25-6.11 2.61 2.42-2.82 <0.0001

Note: N=number of subjects analyzed.

6.3.2.3. Positive rate of IgG antibody and the IgG antibody concentrations (GMCs) analysis for subjects in PCV13-TT aged 2 months and PCV13-TT aged 3 months before and after booster dose

Before booster dose, the positive rates of serotypes 4, 5 and 6B in PCV13-TT aged 2 months were significantly lower than the positive rate in PCV13-TT aged 3 months. and there was no statistically significant difference between the two PCV13-TTs for the other serotypes. After booster dose, the positive rate of 13 serotypes in PCV13-TT aged 2 months were non-inferiority to positive rate of 13 serotypes in PCV13-TT aged 3months (refer to table 38-1). Before booster dose, the GMCs for serotype 4,5,19A in subjects in PCV13-TT aged 2 months were lower than the GMCs for serotype 4,5,19A in subjects in PCV13-TT aged 3 months. While the GMC for serotype 9V in PCV13-TT aged 2 months was higher than PCV13-TT aged 3 months. And there was no statistically significant difference between the two PCV13-TTs for the other

serotypes. After booster dose, the GMCs of 13 serotypes in PCV13-TT aged 2 months were non--inferiority to the GMCs of 13 serotypes in PCV13-TT aged 3months (refer to table 38-3). After booster dose, antibody GMCs had at least 3-fold increase in PCV13-TT aged 2 months The analysis of positive rate of IgG antibody and the IgG antibody concentrations (GMCs) analysis for subjects in PCV13-TT aged 2 months and teatl group aged 3 months before and after booster dose were shown in table 38-1 to 38-4.

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Table 38-1 Pre- and post-booster dose antibody positive rates in PCV13-TT (2M) and PCV13-TT (3M) groups (PPS)

Pre-immunization Post-immunization														
PCV13-TT (2M) PCV13-TT (3M) PCV13-TT (2M) PCV13-TT (3M)														
Serotype Rate														
(N=475) (N=474) P (N=476) (N=475) P 97.5%CI														
diff.														
N % 95%CI N % 95%CI N % 95%CI N % 95%CI														
4	317	66.74	62.30-70.96	353	74.47	70.30-78.34	0.0089	473	99.37	98.17-99.87	474	99.79	98.83-99.99	0.3173
-0.42 -1.36,0.52														
C 6B	465	97.89	96.16-98.99	472	99.58	98.48-99.95	0.0203	476	100.0	99.23-100.0	475	100.0	99.23-100.0	
1.0000 0.00 -0.92,0.92														
o 9V	395	83.16	79.48-86.41	375	79.11	75.17-82.69	0.1113	476	100.0	99.23-100.0	475	100.0	99.23-100.0	
1.0000 0.00 -0.92,0.92														
m 14	474	99.79	98.83-99.99	474	100.0	99.22-100.0	1.0000	476	100.0	99.23-100.0	474	99.79	98.83-99.99	
0.3165 0.21 -0.26,0.68														
m 18C	291	61.26	56.72-65.67	283	59.70	55.13-64.15	0.6234	474	99.58	98.49-99.95	474	99.79	98.83-99.99	
0.5643 -0.21 -1.02,0.61														
on 19F	474	99.79	98.83-99.99	470	99.16	97.85-99.77	0.2173	476	100.0	99.23-100.0	475	100.0	99.23-100.0	
1.0000 0.00 -0.92,0.92														
23F	460	96.84	94.84-98.22	457	96.41	94.32-97.90	0.7146	475	99.79	98.84-99.99	475	100.0	99.23-100.0	0.3176
-0.21 -0.68,0.26														
1 400	84.21	80.61-87.37	413	87.13	83.78-90.01	0.1992	475	99.79	98.84-99.99	474	99.79	98.83-99.99	0.9988	
0.00 -0.67,0.67														
A														
3	269	56.63	52.04-61.14	261	55.06	50.46-59.60	0.6266	471	98.95	97.57-99.66	471	99.16	97.86-99.77	0.7401
-0.21 -1.62,1.20														
dd														
5	220	46.32	41.76-50.92	290	61.18	56.63-65.59	<0.0001	463	97.27	95.38-98.54	465	97.89	96.16-98.99	0.5299
-0.63 -2.86,1.61														
iti														
6A	456	96.00	93.82-97.57	449	94.73	92.31-96.56	0.3506	474	99.58	98.49-99.95	473	99.58	98.49-99.95	0.9983
0.00 -0.94,0.94														
on														
7F	465	97.89	96.16-98.99	458	96.62	94.58-98.06	0.2307	476	100.0	99.23-100.0	475	100.0	99.23-100.0	1.0000
0.00 -0.92,0.92														
al														
19A	472	99.37	98.17-99.87	473	99.79	98.83-99.99	0.6242	476	100.0	99.23-100.0	475	100.0	99.23-100.0	1.0000
0.00 -0.92,0.92														
Note: N=number of subjects analyzed, n=number of subjects with positive antibody.														

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Table 38-2 Percentage of subjects with pre- and post-booster dose antibody concentration ≥ 1.0 µg/mL in PCV13-TT (2M) and PCV13-TT (3M) groups (PPS)

Pre-immunization Post-immunization														
Seroty PCV13-TT (2M) PCV13-TT (3M) P PCV13-TT (2M) PCV13-TT (3M)														
Rate														
pe (N=475) (N=487) (N=476) (N=487) P 95%CI														
diff.														
N % 95%CI N % 95%CI N % 95%CI N % 95%CI														
4	68	14.32	11.29-17.79	96	20.25	16.72-24.16	0.0156	388	81.51	77.73-84.90	443	93.26	90.62-95.35	<0.0001
-11.75 -15.90,-7.60														
C														
6B	416	87.58	84.27-90.41	402	84.81	81.26-87.92	0.2163	472	99.16	97.86-99.77	475	100.0	99.23-100.0	0.0453
-0.84 -1.66,-0.02														
o														
9V	135	28.42	24.40-32.71	106	22.36	18.69-26.39	0.0320	451	94.75	92.34-96.57	455	95.79	93.57-97.41	0.4494
-1.04 -3.74,1.66														
m														
14	457	96.21	94.08-97.74	450	94.94	92.56-96.73	0.3401	476	100.0	99.23-100.0	474	99.79	98.83-99.99	0.3165
0.21 -0.20,0.62														
m														
18C	48	10.11	7.55-13.17	48	10.13	7.56-13.20	0.9913	384	80.67	76.83-84.12	422	88.84	85.66-91.53	0.0005
-8.17 -12.71,-3.63														
o														
19F	428	90.11	87.06-92.64	404	85.23	81.71-88.30	0.0224	476	100.0	99.23-100.0	475	100.0	99.23-100.0	1.0000
0.00 -0.77,0.77														
n														
23F	304	64.00	59.50-68.32	318	67.09	62.66-71.31	0.3168	465	97.69	95.90-98.84	469	98.74	97.27-99.54	0.2227
-1.05 -2.73,0.63														
A 1	122	25.68	21.81-29.86	137	28.9	24.86-33.21	0.2657	387	81.30	77.50-84.71	431	90.74	87.76-93.19	<0.0001
-9.43 -13.80,-5.07														
d 3	30	6.32	4.30-8.89	34	7.17	5.02-9.88	0.5985	338	71.01	66.71-75.05	391	82.32	78.58-85.64	<0.0001
-16.64,-5.98														
di 5	13	2.74	1.47-4.63	40	8.44	6.10-11.31	0.0001	256	53.78	49.19-58.33	342	72.00	67.73-76.00	<0.0001
-18.22 -24.25,-12.19														

ti 6A 285 60.00 55.44-64.44 281 59.28 54.71-63.74 0.8218 454 95.38 93.09-97.08 457 96.21 94.08-97.74
0.5226 -0.83 -3.38,1.72
o 7F 309 65.05 60.58-69.34 314 66.24 61.79-70.49 0.6990 472 99.16 97.86-99.77 466 98.11 96.43-99.13
0.1615 1.05 -0.42,2.53
n 375 79.11 75.17-82.69 0.1564
19A 364 76.63 72.56-80.37 0.3570 476 100.0 99.23-100.0 473 99.58 98.49-99.95 0.42 -0.16,1.00
al
Note: N=number of subjects analyzed, n=number of subjects reaching critical value.

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Table 38-3 Pre- and post-booster dose antibody GMC in PCV13-TT (2M) and PCV13-TT (3M) groups (PPS)
Pre-immunization Post-immunization
PCV13-TT (2M) PCV13-TT (3M) PCV13-TT (2M) PCV13-TT (3M)
Serotype
(N=475) (N=474) P (N=476) (N=475) P Ratio# 97.5%CI
GMC 95%CI GMC 95%CI GMC 95%CI GMC 95%CI
4 0.5 0.47-0.53 0.57 0.53-0.60 0.0027 2.07 1.92-2.22 3.19 2.96-3.44 <0.0001 0.65 0.58,0.73
6B 2.02 1.88-2.16 2.08 1.94-2.22 0.5436 8.6 7.93-9.32 10.01 9.30-10.79 0.0065 0.86 0.76,0.97
9V 0.67 0.63-0.72 0.59 0.56-0.63 0.0091 3.83 3.56-4.12 4.22 3.90-4.56 0.0739 0.91 0.80,1.03
Common 14 3.4 3.18-3.63 3.38 3.16-3.63 0.9320 11.61 10.85-12.44 13.97 12.94-15.09 0.0004 0.83 0.74,0.93
18C 0.41 0.38-0.45 0.38 0.35-0.42 0.2397 2.04 1.90-2.18 2.44 2.27-2.63 0.0004 0.83 0.74,0.94
19F 2.08 1.96-2.21 2.24 2.04-2.47 0.1936 8.76 8.16-9.39 12.18 11.25-13.20 <0.0001 0.72 0.64,0.81
23F 1.31 1.22-1.40 1.29 1.21-1.38 0.7818 5.7 5.25-6.18 6.9 6.39-7.45 0.0008 0.83 0.73,0.94
1 0.66 0.63-0.70 0.71 0.67-0.75 0.1082 1.98 1.86-2.12 2.79 2.61-2.99 <0.0001 0.71 0.64,0.79
3 0.39 0.37-0.42 0.39 0.37-0.42 0.9728 1.53 1.44-1.64 1.89 1.77-2.01 <0.0001 0.81 0.73,0.90
5 0.32 0.30-0.35 0.43 0.41-0.45 <0.0001 1.13 1.06-1.21 1.58 1.47-1.70 <0.0001 0.72 0.64,0.80
Additional
6A 1.2 1.12-1.29 1.22 1.13-1.30 0.8400 4.79 4.39-5.22 5.82 5.32-6.37 0.0023 0.82 0.71,0.95
7F 1.28 1.20-1.35 1.29 1.20-1.38 0.8574 4.05 3.79-4.32 4.59 4.29-4.91 0.0078 0.88 0.79,0.98
19A 1.64 1.54-1.75 1.89 1.76-2.03 0.0048 9.32 8.66-10.03 13.34 12.37-14.39 <0.0001 0.70 0.62,0.79
Note: N=number of subjects analyzed; #PCV13-TT (2M) GMC/PCV13-TT (3M) GMC.

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Table 38-4 Post-booster dose antibody fold increase in PCV13-TT (2M) and PCV13-TT (3M) groups (PPS)
PCV13-TT (2M) (N=476) PCV13-TT (3M) (N=475)
Serotypes Serotype GMC fold GMC fold P
95%CI 95%CI
increase increase
4 4.17 3.91-4.44 5.62 5.23-6.03 <0.0001
6B 4.26 3.98-4.55 4.83 4.50-5.17 0.0105
9V 5.69 5.30-6.10 7.11 6.57-7.69 <0.0001
common 14 3.42 3.21-3.65 4.13 3.82-4.46 0.0002
18C 4.93 4.53-5.37 6.38 5.77-7.04 0.0001
19F 4.20 3.92-4.51 5.43 4.93-5.99 <0.0001
23F 4.35 4.08-4.64 5.35 5.01-5.71 <0.0001
1 3.00 2.83-3.17 3.96 3.73-4.20 <0.0001
3 3.91 3.70-4.14 4.80 4.51-5.11 <0.0001
5 3.50 3.26-3.75 3.67 3.45-3.92 0.3075
Additional
6A 4.00 3.71-4.30 4.79 4.42-5.19 0.0010
7F 3.17 3.00-3.35 3.57 3.32-3.84 0.0112
19A 5.67 5.25-6.11 7.08 6.51-7.70 <0.0001
6.3.2.4. Analysis for antibody concentrations in the PCV13-TT aged 2 months and 3 months
and control group aged 3 months after booster dose
The results of pneumococcal serotype antibody concentrations for subjects in PCV13-TT (2
months) group, PCV13-TT (3 months) group and PCV7 (3 months) group post booster dose were
shown in Table 60. to Table 72. The reverse cumulative distribution curves were shown in Figure 14
to 26.
Table 60. Distribution of the pneumococcal serotype 4 IgG antibody concentrations for subjects in PCV13-
TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group post booster dose
FAS N (%) PPS N (%)
PCV13-TT PCV13-TT
Concen PCV13-TT PCV7 (3M) PCV13-TT PCV7 (3M)
(3M) (N=486) (2M)
tration (2M) (N=486) (N=492) (3M) (N=475) (N=487)
(N=476)
n % n % n % n % n % n %
≥0.01 485 99.79 486 100.0 492 100.0 476 100.0 475 100.0 487 100.0
≥0.1 485 99.79 486 100.0 492 100.0 476 100.0 475 100.0 487 100.0
≥1 390 80.25 451 92.80 472 95.93 388 81.51 443 93.26 470 96.51
≥10 14 2.88 47 9.67 71 14.43 14 2.94 47 9.89 71 14.58
≥100 0 0.00 0 0.00 0 0.00 0 0.00 0 0.00 0 0.00
N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT
group (2M), PCV13-
TT group (3M) and PCV7 group (3M).

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Serotype 4, After Booster Dose (FAS) Serotype 4, After Booster Dose (PPS)

100 100
90 90

80 80
70 70

% of Subjects
% of Subjects

60 60
50 50
40 40
30 30
20 20
10 10
0 0
0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration (µg/mL) Antibody Concentration ■µg/mL)

PCV13-TT (2M) PCV13-TT (3M) PCV7 (3M) PCV13-TT (2M) PCV13-TT (3M) PCV7 (3M)

Figure 65 Reverse cumulative distribution curves, pneumococcal serotype 4, IgG antibody concentrations for subjects in PCV13-TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group- post booster dose.

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Table 61. Distribution of the pneumococcal serotype 6B IgG antibody concentrations for subjects in PCV13-TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group post booster dose

FAS N (%)	PPS N (%)	PCV13-TT (2M) (N=486)	PCV13-TT (3M) (N=492)	PCV7 (3M) (N=475)
≥0.01	486	100.0	486	100.0
≥0.1	486	100.0	486	100.0
≥1	481	98.97	486	100.0
≥10	208	42.80	251	51.65
≥100	4	0.82	2	0.41

Concen PCV13-TT PCV7 (3M) PCV13-TT PCV7 (3M)
tration (2M) (N=486) (N=492) (3M) (N=475) (N=487)
(N=476)

n % n % n % n % n % n %

≥0.01 486 100.0 486 100.0 492 100.0 476 100.0 475 100.0 487 100.0

≥0.1 486 100.0 486 100.0 492 100.0 476 100.0 475 100.0 487 100.0

≥1 481 98.97 486 100.0 480 97.56 472 99.16 475 100.0 478 98.15

≥10 208 42.80 251 51.65 279 56.71 208 43.70 248 52.21 279 57.29

≥100 4 0.82 2 0.41 6 1.22 4 0.84 2 0.42 6 1.23

N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT group (2M), PCV13-TT group (3M) and PCV7 group (3M).

Serotype 6B, After Booster Dose (FAS) Serotype 6B, After Booster Dose (PPS)

100 100
90 90
80 80
70 70

% of Subjects
% of Subjects

60 60
50 50
40 40
30 30
20 20
10 10
0 0
0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration (µg/mL) Antibody Concentration ■µg/mL)

PCV13-TT (2M) PCV13-TT (3M) PCV7 (3M) PCV13-TT (2M) PCV13-TT (3M) PCV7 (3M)

Figure 66 Reverse cumulative distribution curves, pneumococcal serotype 6B, IgG antibody concentrations for subjects in PCV13-TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group - post booster dose.

Table 62. Distribution of the pneumococcal serotype 9V IgG antibody concentrations for subjects in PCV13-TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group post booster dose

FAS N (%)	PPS N (%)	PCV13-TT (2M) (N=486)	PCV13-TT (3M) (N=486)	PCV7 (3M) (N=492)
≥0.01	485	99.79	486	100.0
≥0.1	485	99.79	486	100.0
≥1	453	93.21	461	94.86
≥10	56	11.52	77	15.84
≥100	0	0.00	0	0.00

Concen PCV13-TT PCV13-TT PCV7 (3M) PCV13-TT (2M) PCV13-TT PCV7 (3M)
tration (2M) (N=486) (3M) (N=486) (N=492) (N=476) (3M) (N=475) (N=487)
(N=476)

n % n % n % n % n % n %

≥0.01 485 99.79 486 100.0 492 100.0 476 100.0 475 100.0 487 100.0

≥0.1 485 99.79 486 100.0 492 100.0 476 100.0 475 100.0 487 100.0

≥1 453 93.21 461 94.86 473 96.14 451 94.75 455 95.79 471 96.71

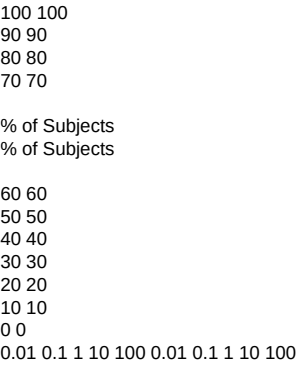
≥10 56 11.52 77 15.84 54 10.98 56 11.76 77 16.21 54 11.09

≥100 0 0.00 0 0.00 0 0.00 0 0.00 0 0.00 0 0.00

N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT group (2M), PCV13-TT group (3M) and PCV7 group (3M).

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Serotype 9V, After Booster Dose (FAS) Serotype 9V, After Booster Dose (PPS)



Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT (2M) PCV13-TT (3M) PCV7 (3M) PCV13-TT (2M) PCV13-TT (3M) PCV7 (3M)

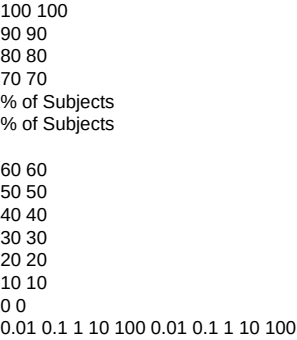
Figure 67 Reverse cumulative distribution curves, pneumococcal serotype 9V, IgG antibody concentrations for subjects in PCV13-TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group - post booster dose.

Table 63. Distribution of the pneumococcal serotype 14 IgG antibody concentrations for subjects in PCV13-TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group post booster dose

FAS N (%)	PPS N (%)	PCV13-TT (2M) (N=486)	PCV13-TT (3M) (N=492)	PCV7 (3M) (N=475)	PCV13-TT (2M) (N=486)	PCV13-TT (3M) (N=492)	PCV7 (3M) (N=475)					
≥0.01	486	100.0	485	99.79	492	100.0	476	100.0	474	99.79	487	100.0
≥0.1	486	100.0	485	99.79	492	100.0	476	100.0	474	99.79	487	100.0
≥1	485	99.79	485	99.79	492	100.0	476	100.0	474	99.79	487	100.0
≥10	280	57.61	326	67.08	385	78.25	279	58.61	322	67.79	383	78.64
≥100	1	0.21	2	0.41	5	1.02	1	0.21	2	0.42	5	1.03

N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT group (2M), PCV13-TT group (3M) and PCV7 group (3M).

Serotype 14, After Booster Dose (FAS) Serotype 14, After Booster Dose (PPS)



Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT (2M) PCV13-TT (3M) PCV7 (3M) PCV13-TT (2M) PCV13-TT (3M) PCV7 (3M)

Figure 68 Reverse cumulative distribution curves, pneumococcal serotype 14, IgG antibody concentrations for subjects in PCV13-TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group - post booster dose.

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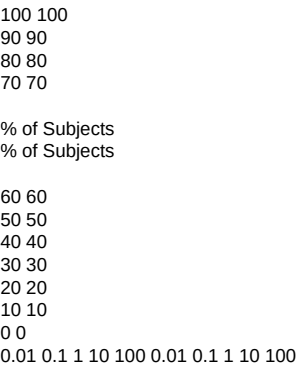
Table 64. Distribution of the pneumococcal serotype 18C IgG antibody concentrations for subjects in PCV13-TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group post booster dose

FAS N (%)	PPS N (%)	PCV13-TT (2M) (N=486)	PCV13-TT (3M) (N=492)	PCV7 (3M) (N=475)	PCV13-TT (2M) (N=486)	PCV13-TT (3M) (N=492)	PCV7 (3M) (N=475)					
≥0.01	484	99.59	485	99.79	490	99.59	476	100.0	474	99.79	486	99.79
≥0.1	484	99.59	485	99.79	490	99.59	476	100.0	474	99.79	486	99.79
≥1	385	79.22	430	88.48	462	93.90	384	80.67	422	88.84	460	94.46
≥10	9	1.85	20	4.12	71	14.43	9	1.89	18	3.79	71	14.58
≥100	0	0.00	0	0.00	0	0.00	0	0.00	0	0.00	0	0.00

N=number of subjects included in FAS and PPS for vaccine group.

n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT group (2M), PCV13-TT group (3M) and PCV7 group (3M).

Serotype 18C, After Booster Dose (FAS) Serotype 18C, After Booster Dose (PPS)



Antibody Concentration (µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT (2M) PCV13-TT (3M) PCV7 (3M) PCV13-TT (2M) PCV13-TT (3M) PCV7 (3M)

Figure 69 Reverse cumulative distribution curves, pneumococcal serotype 18C, IgG antibody concentrations for subjects in PCV13-TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group - post booster dose.

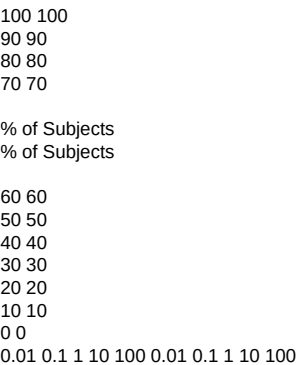
Table 65. Distribution of the pneumococcal serotype 19F IgG antibody concentrations for subjects in PCV13-TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group post booster dose

FAS N (%)	PPS N (%)	PCV13-TT (2M) (N=486)	PCV13-TT (3M) (N=492)	PCV7 (3M) (N=475)								
≥0.01	486	100.0	486	100.0	492	100.0	476	100.0	475	100.0	487	100.0
≥0.1	486	100.0	486	100.0	492	100.0	476	100.0	475	100.0	487	100.0
≥1	484	99.59	486	100.0	487	98.98	476	100.0	475	100.0	484	99.38
≥10	185	38.07	282	58.02	182	36.99	185	38.87	281	59.16	182	37.37
≥100	1	0.21	5	1.03	0	0.00	1	0.21	5	1.05	0	0.00

N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT group (2M), PCV13-TT group (3M) and PCV7 group (3M).

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Serotype 19F, After Booster Dose (FAS) Serotype 19F, After Booster Dose (PPS)



Antibody Concentration (µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT (2M) PCV13-TT (3M) PCV7 (3M) PCV13-TT (2M) PCV13-TT (3M) PCV7 (3M)

Figure 70 Reverse cumulative distribution curves, pneumococcal serotype 19F, IgG antibody concentrations for subjects in PCV13-TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group - post booster dose.

Table 66. Distribution of the pneumococcal serotype 23F IgG antibody concentrations for subjects in PCV13-TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group post booster dose													
FAS N (%)		PPS N (%)		PCV13-TT (2M) (N=486)		PCV13-TT (3M) (N=492)		PCV7 (3M) (N=475)		PCV13-TT (2M) (N=486)		PCV13-TT (3M) (N=492)	
n		%		n		%		n		%		n	
≥0.01		100.0		100.0		100.0		100.0		100.0		100.0	
≥0.1		100.0		100.0		100.0		100.0		100.0		100.0	
≥1		99.59		100.0		98.98		100.0		99.38		100.0	
≥10		38.07		58.02		36.99		38.87		37.37		59.16	
≥100		0.21		1.03		0.00		0.21		0.00		1.05	

≥0.01 485 99.79 486 100.0 492 100.0 475 99.79 475 100.0 487 100.0
≥0.1 485 99.79 486 100.0 492 100.0 475 99.79 475 100.0 487 100.0
≥1 472 97.12 478 98.35 485 98.58 465 97.69 469 98.74 484 99.38
≥10 122 25.10 159 32.72 262 53.25 122 25.63 158 33.26 262 53.80
≥100 1 0.21 3 0.62 7 1.42 1 0.21 3 0.63 7 1.44
N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT group (2M), PCV13-TT group (3M) and PCV7 group (3M).

Serotype 23F, After Booster Dose (FAS) Serotype 23F, After Booster Dose (PPS)

100 100
90 90
80 80
70 70
% of Subjects
% of Subjects

60 60
50 50
40 40
30 30
20 20
10 10
0 0
0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration ■µg/mL
Antibody Concentration (µg/mL)

PCV13-TT (2M) PCV13-TT (3M) PCV7 (3M) PCV13-TT (2M) PCV13-TT (3M) PCV7 (3M)

Figure 71 Reverse cumulative distribution curves, pneumococcal serotype 23F, IgG antibody concentrations for subjects in PCV13-TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group - post booster dose.

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Table 67. Distribution of the pneumococcal serotype 1 IgG antibody concentrations for subjects in PCV13-TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group post booster dose
FAS N (%) PPS N (%)
Concen PCV13-TT PCV13-TT PCV7 (3M) PCV13-TT (2M) PCV13-TT PCV7 (3M)
tration (2M) (N=486) (3M) (N=486) (N=492) (N=476) (3M) (N=475) (N=487)
n % n % n % n % n % n % n %
≥0.01 486 100.0 486 100.0 484 98.37 476 100.0 475 100.0 480 98.56
≥0.1 486 100.0 486 100.0 484 98.37 476 100.0 475 100.0 480 98.56
≥1 389 80.04 438 90.12 126 25.61 387 81.30 431 90.74 125 25.67
≥10 6 1.23 24 4.94 0 0.00 6 1.26 24 5.05 0 0.00
≥100 0 0.00 0 0.00 0 0.00 0 0.00 0 0.00 0 0.00
N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT group (2M), PCV13-TT group (3M) and PCV7 group (3M).

Serotype 1, After Booster Dose (FAS) Serotype 1, After Booster Dose (PPS)

100 100
90 90
80 80
70 70
% of Subjects
% of Subjects

60 60
50 50
40 40
30 30
20 20
10 10
0 0
0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration ■µg/mL
Antibody Concentration (µg/mL)

PCV13-TT (2M) PCV13-TT (3M) PCV7 (3M) PCV13-TT (2M) PCV13-TT (3M) PCV7 (3M)

Figure 72 Reverse cumulative distribution curves, pneumococcal serotype 1, IgG antibody concentrations for subjects in PCV13-TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group - post booster dose.

Table 68. Distribution of the pneumococcal serotype 3 IgG antibody concentrations for subjects in PCV13-TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group post booster dose
FAS N (%) PPS N (%)
PCV13-TT PCV13-TT
Concen PCV13-TT PCV7 (3M) PCV13-TT PCV7 (3M)
(3M) (N=486) (2M)
tration (2M) (N=486) (N=492) (3M) (N=475) (N=487)

(N=476)
n % n % n % n % n %
≥0.01 486 100.0 486 100.0 492 100.0 476 100.0 475 100.0 487 100.0
≥0.1 485 99.79 486 100.0 487 98.98 476 100.0 475 100.0 485 99.59
≥1 340 69.96 393 80.86 243 49.39 338 71.01 391 82.32 243 49.90
≥10 5 1.03 3 0.62 5 1.02 5 1.05 3 0.63 5 1.03
≥100 0 0.00 0 0.00 0 0.00 0 0.00 0 0.00 0 0.00
N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT group (2M), PCV13-TT group (3M) and PCV7 group (3M).

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Serotype 3, After Booster Dose (FAS) Serotype 3, After Booster Dose (PPS)

100 100
90 90
80 80
70 70

% of Subjects
% of Subjects

60 60
50 50
40 40
30 30
20 20
10 10
0 0
0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT (2M) PCV13-TT (3M) PCV7 (3M) PCV13-TT (2M) PCV13-TT (3M) PCV7 (3M)

Figure 73 Reverse cumulative distribution curves, pneumococcal serotype 3, IgG antibody concentrations for subjects in PCV13-TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group - post booster dose.

Table 69 Distribution of the pneumococcal serotype 5 IgG antibody concentrations for subjects in PCV13-TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group post booster dose
FAS N (%) PPS N (%)
PCV13-TT PCV13-TT
Concen PCV13-TT PCV7 (3M) PCV13-TT PCV7 (3M)
(3M) (N=486) (2M)
tration (2M) (N=486) (N=492) (3M) (N=475) (N=487)
(N=476)
n % n % n % n % n % n %
≥0.01 483 99.38 486 100.0 414 84.15 476 100.0 475 100.0 413 84.80
≥0.1 483 99.38 486 100.0 410 83.33 476 100.0 475 100.0 409 83.98
≥1 256 52.67 345 70.99 13 2.64 256 53.78 342 72.00 13 2.67
≥10 1 0.21 7 1.44 0 0.00 1 0.21 7 1.47 0 0.00
≥100 0 0.00 0 0.00 0 0.00 0 0.00 0 0.00 0 0.00
N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT group (2M), PCV13-TT group (3M) and PCV7 group (3M).

Serotype 5, After Booster Dose (FAS) Serotype 5, After Booster Dose (PPS)

100 100
90 90
80 80
70 70

% of Subjects
% of Subjects

60 60
50 50
40 40
30 30
20 20
10 10
0 0
0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT (2M) PCV13-TT (3M) PCV7 (3M) PCV13-TT (2M) PCV13-TT (3M) PCV7 (3M)

Figure 74 Reverse cumulative distribution curves, pneumococcal serotype 5, IgG antibody concentrations for subjects in PCV13-TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group - post booster dose.

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Table 70. Distribution of the pneumococcal serotype 6A IgG antibody concentrations for subjects in PCV13-TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group post booster dose

FAS N (%) PPS N (%)

PCV13-TT PCV13-TT

Concen PCV13-TT PCV7 (3M) PCV13-TT PCV7 (3M)

(3M) (N=486) (2M)

tration (2M) (N=486) (N=492) (3M) (N=475) (N=487)

(N=476)

n % n % n % n % n % n %

≥0.01 486 100.0 486 100.0 492 100.0 476 100.0 475 100.0 487 100.0

≥0.1 486 100.0 484 99.59 491 99.80 476 100.0 473 99.58 486 99.79

≥1 460 94.65 465 95.68 434 88.21 454 95.38 457 96.21 431 88.50

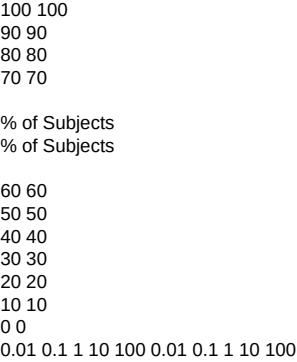
≥10 98 20.16 135 27.78 93 18.90 98 20.59 133 28.00 93 19.10

≥100 1 0.21 1 0.21 1 0.20 1 0.21 1 0.21 1 0.21

N=number of subjects included in FAS and PPS for vaccine group.

n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT group (2M), PCV13-TT group (3M) and PCV7 group (3M).

Serotype 6A, After Booster Dose (FAS) Serotype 6A, After Booster Dose (PPS)



Antibody Concentration ■µg/mL)

Antibody Concentration (µg/mL)

PCV13-TT (2M) PCV13-TT (3M) PCV7 (3M) PCV13-TT (2M) PCV13-TT (3M) PCV7 (3M)

Figure 75 Reverse cumulative distribution curves, pneumococcal serotype 6A, IgG antibody concentrations for subjects in PCV13-TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group - post booster dose.

Table 71. Distribution of the pneumococcal serotype 7F IgG antibody concentrations for subjects in PCV13-TT (2 months) group, PCV13-TT (3 months) group and PCV7 (3 months) group post booster dose

FAS N (%) PPS N (%)

PCV13-TT PCV13-TT

Concen PCV13-TT PCV7 (3M) PCV13-TT PCV7 (3M)

(3M) (N=486) (2M)

tration (2M) (N=486) (N=492) (3M) (N=475) (N=487)

(N=476)

n % n % n % n % n % n %

≥0.01 486 100.0 486 100.0 436 88.62 476 100.0 475 100.0 434 89.12

≥0.1 486 100.0 486 100.0 434 88.21 476 100.0 475 100.0 432 88.71

≥1 479 98.56 476 97.94 69 14.02 472 99.16 466 98.11 69 14.17

≥10 51 10.49 67 13.79 0 0.00 51 10.71 67 14.11 0 0.00

N=number of subjects included in FAS and PPS for vaccine group.

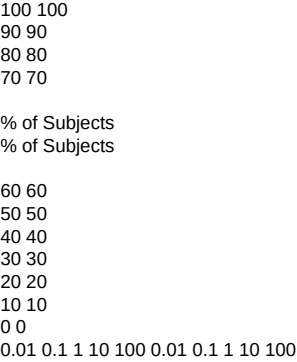
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT group (2M), PCV13-TT group (3M) and PCV7 group (3M).

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Serotype 7F, After Booster Dose (FAS) Serotype 7F, After Booster Dose (PPS)



Antibody Concentration ■µg/mL)

Antibody Concentration (µg/mL)

1.0000 0.00 -1.93,1.96
o 9V 178 96.22 92.36-98.47 187 99.47 97.07-99.99 0.0359 185 99.46 97.04-99.99 189 100.0 98.07-100.0
0.3128 -0.54 -1.59,0.51
m 14 185 100.0 98.03-100.0 187 99.47 97.07-99.99 1.0000 186 100.0 98.04-100.0 189 100.0 98.07-100.0
1.0000 0.00 -1.93,1.96
m 18C 182 98.38 95.33-99.66 187 99.47 97.07-99.99 0.3690 185 99.46 97.04-99.99 188 99.47 97.09-99.99
0.9909 -0.01 -1.48,1.47
on 19F 185 100.0 98.03-100.0 188 100.0 98.06-100.0 1.0000 186 100.0 98.04-100.0 189 100.0 98.07-100.0
1.0000 0.00 -1.93,1.96
23F 185 100.0 98.03-100.0 183 97.34 93.90-99.13 0.0609 186 100.0 98.04-100.0 189 100.0 98.07-100.0 1.0000
0.00 -1.93,1.96
1 185 100.0 98.03-100.0 128 68.09 60.91-74.68 <0.0001 186 100.0 98.04-100.0 156 82.54 76.36-87.67 <0.0001
17.46 12.05,22.87
A
3 179 96.76 93.07-98.80 77 40.96 33.86-48.35 <0.0001 184 98.92 96.17-99.87 130 68.78 61.65-75.31 <0.0001
30.14 23.37,36.91
dd
5 144 77.84 71.16-83.60 50 26.60 20.43-33.52 <0.0001 173 93.01 88.34-96.23 77 40.74 33.67-48.11 <0.0001
52.27 44.36,60.18
iti
6A 180 97.30 93.81-99.12 156 82.98 76.83-88.06 <0.0001 183 98.39 95.36-99.67 176 93.12 88.53-96.29 0.0117
5.27 1.23,9.30
on
7F 185 100.0 98.03-100.0 110 58.51 51.11-65.63 <0.0001 186 100.0 98.04-100.0 138 73.02 66.09-79.20
<0.0001 26.98 20.66,33.31
al
19A 184 99.46 97.03-99.99 188 100.0 98.06-100.0 0.4960 186 100.0 98.04-100.0 189 100.0 98.07-100.0 1.0000
0.00 -1.93,1.96
Note: N=number of subjects analyzed, n=number of subjects with positive antibody.

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Table 40-2 Percentage of subjects with pre- and post-booster dose antibody concentration ≥ 1.0 $\mu\text{g/mL}$ in PCV13-TT and PCV7 groups aged 7-11 months (PPS)

Pre-immunization Post-immunization
PCV13-TT PCV7 PCV13-TT PCV7
Serotype Rate
(N=474) (N=487) P (N=186) (N=189) P 95%CI
diff.
N % 95%CI N % 95%CI N % 95%CI N % 95%CI
4 166 89.73 84.43-93.70 167 88.83 83.43-92.95 0.7788 178 95.70 91.70-98.13 186 98.41 95.43-99.67 0.1194
-2.71 -6.13,0.70
6B 165 89.19 83.80-93.27 143 76.06 69.31-81.97 0.0008 180 96.77 87.05-95.42 182 96.30 92.52-98.50 0.8003
0.48 -3.22,4.18
9V 143 77.30 70.58-83.12 155 82.45 76.24-87.60 0.2147 171 91.94 87.83-96.16 181 95.77 91.83-98.16 0.1221
-3.83 -8.68,1.02
Common 14 185 100.0 98.03-100.0 187 99.47 97.07-99.99 1.0000 186 100.0 98.04-100.0 189 100.0 98.07-100.0
1.0000 0.00 -1.93,1.96
18C 160 86.49 80.70-91.06 157 83.51 77.42-88.51 0.4211 174 93.55 89.00-96.62 183 96.83 93.22-98.83 0.1377
-3.28 -7.60,1.05
19F 183 98.92 96.15-99.87 175 93.09 88.47-96.27 0.0041 186 100.0 98.04-100.0 186 98.41 95.43-99.67 0.0845
1.59 -0.19,3.37
23F 170 91.89 86.98-95.39 157 83.51 77.42-88.51 0.0138 180 96.77 93.11-98.81 181 95.77 91.83-98.16 0.6070
1.01 -2.83,4.84
1 137 74.05 67.11-80.21 26 13.83 9.24-19.60 <0.0001 168 90.32 85.14-94.16 35 18.52 13.25-24.80 <0.0001
71.80 64.82,78.78
3 78 42.16 34.95-49.63 14 7.45 4.13-12.18 <0.0001 136 73.12 66.14-79.34 38 20.11 14.64-26.54 <0.0001
53.01 44.45,61.57
5 35 18.92 13.55-25.32 2 1.06 0.13-3.79 <0.0001 75 40.32 33.21-47.75 3 1.59 0.33-4.57 <0.0001 38.74
31.46,46.01
Additional
6A 137 74.05 67.11-80.21 71 37.77 30.81-45.11 <0.0001 174 93.55 89.00-96.62 133 70.37 63.31-76.78 <0.0001
23.18 15.77,30.58
7F 179 96.76 93.07-98.80 25 13.30 8.79-19.00 <0.0001 185 99.46 97.04-99.99 33 17.46 12.33-23.64 <0.0001
82.00 76.49,87.52
19A 180 97.3 93.81-99.12 147 78.19 71.60-83.87 <0.0001 184 98.92 96.17-99.87 163 86.24 80.50-90.81
<0.0001 12.68 7.55,17.81
Note: N=number of subjects analyzed, n=number of subjects reaching critical value.

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Table 40-3 Pre- and post-booster dose antibody GMC in PCV13-TT and PCV7 groups aged 7-11 months (PPS)

Pre-immunization Post-immunization
PCV13-TT PCV7 PCV13-TT PCV7
Serotype
(N=185) (N=188) P (N=186) (N=189) P Ratio# 95%CI
GMC 95%CI GMC 95%CI GMC 95%CI GMC 95%CI
4 2.41 2.19-2.65 2.38 2.16-2.61 0.8571 3.06 2.76-3.38 3.12 2.85-3.41 0.7779 0.98 0.86,1.12
6B 2.87 2.56-3.22 1.69 1.51-1.89 <0.0001 5.62 5.01-6.32 4.57 4.08-5.11 0.0113 1.23 1.05,1.45
9V 1.77 1.57-2.00 2.11 1.88-2.36 0.0409 2.91 2.58-3.29 3.67 3.31-4.07 0.0044 0.79 0.68,0.93
common 14 8.50 7.67-9.42 13.50 12.18-14.95 <0.0001 13.94 12.54-15.51 19.49 17.94-21.17 <0.0001 0.72
0.63,0.82
18C 2.25 2.02-2.52 2.15 1.93-2.39 0.5420 2.97 2.62-3.36 3.32 3.00-3.67 0.1702 0.89 0.76,1.05
19F 5.33 4.82-5.89 2.83 2.57-3.12 <0.0001 7.54 6.84-8.31 3.92 3.56-4.31 <0.0001 1.92 1.68,2.20
23F 2.71 2.43-3.03 2.08 1.86-2.32 0.0008 4.53 4.01-5.12 4.21 3.77-4.71 0.3838 1.08 0.91,1.27
1 1.50 1.38-1.63 0.50 0.46-0.56 <0.0001 1.98 1.82-2.15 0.58 0.53-0.65 <0.0001 3.38 2.96,3.86

3 0.94 0.86-1.02 0.32 0.29-0.35 <0.0001 1.51 1.38-1.66 0.53 0.48-0.59 <0.0001 2.85 2.48,3.28
5 0.57 0.52-0.63 0.27 0.25-0.30 <0.0001 0.86 0.78-0.95 0.24 0.20-0.29 <0.0001 3.61 2.90,4.50
Additional
6A 1.60 1.43-1.79 0.78 0.68-0.90 <0.0001 2.99 2.64-3.37 1.63 1.43-1.87 <0.0001 1.83 1.53,2.19
7F 3.79 3.43-4.18 0.44 0.39-0.51 <0.0001 4.91 4.46-5.39 0.46 0.39-0.55 <0.0001 10.61 8.71,12.93
19A 3.95 3.51-4.46 1.92 1.70-2.17 <0.0001 6.28 5.71-6.91 2.17 1.96-2.40 <0.0001 2.90 2.52,3.33
Note: N=number of subjects analyzed; #Ratio: PCV13-TTGMC/PCV7GMC.

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Table 40-4 Post-booster dose antibody fold increase in PCV13-TT and PCV7 groups aged 7-11 months (PPS)

PCV13-TT (N=185)		PCV7 (N=188)		
Serotypes	Serotype	GMC fold	GMC fold	P
95%CI	95%CI			
increase	increase			
4	1.27	1.20-1.34	1.31	1.24-1.38 0.4037
6B	1.96	1.82-2.11	2.70	2.44-2.98 <0.0001
9V	1.63	1.53-1.74	1.73	1.62-1.86 0.2039
common	14	1.64	1.51-1.78	1.45 1.32-1.59 0.0445
18C	1.31	1.22-1.41	1.54	1.44-1.65 0.0010
19F	1.41	1.30-1.54	1.38	1.28-1.49 0.7036
23F	1.66	1.54-1.78	2.02	1.84-2.21 0.0010
1	1.31	1.24-1.39	1.15	1.07-1.25 0.0091
3	1.61	1.52-1.71	1.64	1.51-1.78 0.7019
5	1.51	1.41-1.61	0.87	0.73-1.04 <0.0001
Additional				
6A	1.86	1.68-2.06	2.09	1.82-2.40 0.1793
7F	1.28	1.18-1.39	1.04	0.89-1.22 0.0239
19A	1.59	1.42-1.78	1.13	1.01-1.26 <0.0001

6.3.2.6. Pneumococcal Serotype antibody concentrations in the subjects aged 7-11 months in PCV13-TT group and PCV7 group after booster dose
The results of pneumococcal serotype antibody concentrations for subjects aged 7-11 months after booster dose in PCV13-TT group and PCV7 group were shown in Table 73. to Table 85. The reverse cumulative distribution curves were shown in Figure 40 to 52.
Table 73. Distribution of the pneumococcal serotype 4 IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group post booster dose

FAS N (%)	PPS N (%)							
PCV13-TT	PCV13-TT							
Concentration	PCV7 (N=190)	PCV7 (N=189)						
(N=186)	(N=186)							
n	%	n	%	n	%			
≥0.01	186	100.0	190	100.0	186	100.0	189	100.0
≥0.1	186	100.0	190	100.0	186	100.0	189	100.0
≥1	178	95.70	187	98.42	178	95.70	186	98.41
≥10	9	4.84	9	4.74	9	4.84	9	4.76
≥100	0	0.00	0	0.00	0	0.00	0	0.00
N=number of subjects included in FAS and PPS for vaccine group.								
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.								

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Serotype 4, After Booster Dose (FAS) Serotype 4, After Booster Dose (PPS)

100	100
90	90
80	80
70	70
% of Subjects	
% of Subjects	
60	60
50	50
40	40
30	30
20	20
10	10
0	0
0.01	0.1
1	10
100	0.01
0.1	1
10	100

Antibody Concentration (µg/mL) Antibody Concentration ■µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

Figure 78 Reverse cumulative distribution curves, pneumococcal serotype 4, IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group - post booster dose

Table 74. Distribution of the pneumococcal serotype 6B IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group post booster dose									
FAS N (%)		PPS N (%)							
PCV13-TT		PCV13-TT							
Concentration		PCV7 (N=190)		PCV7 (N=189)					
(N=186)		(N=186)							
n	%	n	%	n	%	n	%	n	%
≥0.01	186	100.0	190	100.0	186	100.0	189	100.0	

≥0.1 186 100.0 190 100.0 186 100.0 189 100.0
≥1 180 96.77 183 96.32 180 96.77 182 96.30
≥10 37 19.89 29 15.26 37 19.89 29 15.34
≥100 0 0.00 0 0.00 0 0.00 0 0.00
N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

Serotype 6B, After After Booster Dose (FAS) Serotype 6B, After Booster Dose (PPS)

100 100
90 90
80 80
70 70
% of Subjects
% of Subjects

60 60
50 50
40 40
30 30
20 20
10 10
0 0
0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration (µg/mL) Antibody Concentration ■µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

Figure 79 Reverse cumulative distribution curves, pneumococcal serotype 6B, IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group - post booster dose

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Table 75. Distribution of the pneumococcal serotype 9V IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group post booster dose
FAS N (%) PPS N (%)
PCV13-TT PCV13-TT
Concentration PCV7 (N=190) PCV7 (N=189)
(N=186) (N=186)
n % n % n % n %
≥0.01 186 100.0 190 100.0 186 100.0 189 100.0
≥0.1 186 100.0 190 100.0 186 100.0 189 100.0
≥1 171 91.94 182 95.79 171 91.94 181 95.77
≥10 10 5.38 13 6.84 10 5.38 13 6.88
≥100 0 0.00 0 0.00 0 0.00 0 0.00
N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

Serotype 9V, After Booster Dose (FAS) Serotype 9V, After Booster Dose (PPS)

100 100
90 90
80 80
70 70
% of Subjects
% of Subjects
60 60
50 50
40 40
30 30
20 20
10 10
0 0
0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

Figure 80 Reverse cumulative distribution curves, pneumococcal serotype 9V, IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group - post booster dose

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Table 76. Distribution of the pneumococcal serotype 14 IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group post booster dose
FAS N (%) PPS N (%)
PCV13-TT PCV13-TT
Concentration PCV7 (N=190) PCV7 (N=189)
(N=186) (N=186)
n % n % n % n %
≥0.01 186 100.0 190 100.0 186 100.0 189 100.0

≥0.1 186 100.0 190 100.0 186 100.0 189 100.0
≥1 186 100.0 190 100.0 186 100.0 189 100.0
≥10 121 65.05 163 85.79 121 65.05 162 85.71
≥100 1 0.54 0 0.00 1 0.54 0 0.00
N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

Serotype 14, After Booster Dose (FAS) Serotype 14, After Booster Dose (PPS)

100 100
90 90
80 80
70 70

% of Subjects
% of Subjects

60 60
50 50
40 40
30 30
20 20
10 10
0 0
0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

Figure 81 Reverse cumulative distribution curves, pneumococcal serotype 14, IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group - post booster dose

Table 77. Distribution of the pneumococcal serotype 18C IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group post booster dose

FAS N (%) PPS N (%)
PCV13-TT PCV13-TT
Concentration PCV7 (N=190) PCV7 (N=189)
(N=186) (N=186)
n % n % n % n %
≥0.01 185 99.46 190 100.0 185 99.46 189 100.0
≥0.1 185 99.46 190 100.0 185 99.46 189 100.0
≥1 174 93.55 184 96.84 174 93.55 183 96.83
≥10 8 4.30 8 4.21 8 4.30 8 4.23
≥100 0 0.00 0 0.00 0 0.00 0 0.00
N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

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Serotype 18C, After Booster Dose (FAS) Serotype 18C, After Booster Dose (PPS)

100 100
90 90
80 80
70 70

% of Subjects
% of Subjects

60 60
50 50
40 40
30 30
20 20
10 10
0 0
0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

Figure 82 Reverse cumulative distribution curves, pneumococcal serotype 18C, IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group - post booster dose

Table 78. Distribution of the pneumococcal serotype 19F IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group post booster dose

FAS N (%) PPS N (%)
PCV13-TT PCV13-TT
Concentration PCV7 (N=190) PCV7 (N=189)
(N=186) (N=186)
n % n % n % n %
≥0.01 186 100.0 190 100.0 186 100.0 189 100.0
≥0.1 186 100.0 190 100.0 186 100.0 189 100.0
≥1 186 100.0 187 98.42 186 100.0 186 98.41

≥10 55 29.57 15 7.89 55 29.57 15 7.94
≥100 0 0.00 0 0.00 0 0.00 0 0.00
N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

Serotype 19F, After Booster Dose (FAS) Serotype 19F, After Booster Dose (PPS)

100 100
90 90
80 80
70 70
% of Subjects
% of Subjects

60 60
50 50
40 40
30 30
20 20
10 10
0 0
0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

Figure 83 Reverse cumulative distribution curves, pneumococcal serotype 19F, IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group - post booster dose

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Table 79. Distribution of the pneumococcal serotype 23F IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group post booster dose
FAS N (%) PPS N (%)
PCV13-TT PCV13-TT
Concentration PCV7 (N=190) PCV7 (N=189)
(N=186) (N=186)
n % n % n % n %
≥0.01 186 100.0 190 100.0 186 100.0 189 100.0
≥0.1 186 100.0 190 100.0 186 100.0 189 100.0
≥1 180 96.77 181 95.26 180 96.77 181 95.77
≥10 28 15.05 19 10.00 28 15.05 19 10.05
≥100 0 0.00 0 0.00 0 0.00 0 0.00
N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

Serotype 23F, After Booster Dose (FAS) Serotype 23F, After Booster Dose (PPS)

100 100
90 90
80 80
70 70

% of Subjects
% of Subjects

60 60
50 50
40 40
30 30
20 20
10 10
0 0
0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

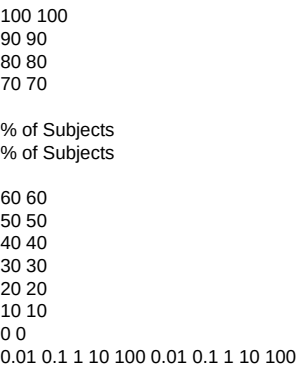
Figure 84 Reverse cumulative distribution curves, pneumococcal serotype 23F, IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group - post booster dose

Table 80. Distribution of the pneumococcal serotype 1 IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group post booster dose
FAS N (%) PPS N (%)
PCV13-TT PCV13-TT
Concentration PCV7 (N=190) PCV7 (N=189)
(N=186) (N=186)
n % n % n % n %
≥0.01 186 100.0 189 99.47 186 100.0 188 99.47
≥0.1 186 100.0 189 99.47 186 100.0 188 99.47
≥1 168 90.32 35 18.42 168 90.32 35 18.52
≥10 1 0.54 0 0.00 1 0.54 0 0.00
≥100 0 0.00 0 0.00 0 0.00 0 0.00
N=number of subjects included in FAS and PPS for vaccine group.

n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

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Serotype 1, After Booster Dose (FAS) Serotype 1, After Booster Dose (PPS)



Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

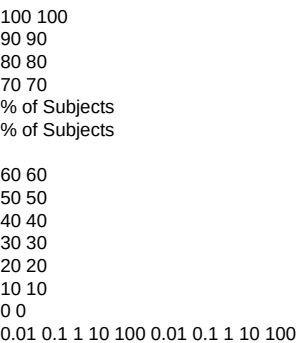
Figure 85 Reverse cumulative distribution curves, pneumococcal serotype 1, IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group - post booster dose

Table 81. Distribution of the pneumococcal serotype 3 IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group post booster dose

FAS N (%)	PPS N (%)	PCV13-TT	PCV13-TT
Concentration	PCV7 (N=190)	PCV7 (N=189)	(N=186) (N=186)
n %	n %	n %	n %
≥0.01	186 100.0	190 100.0	186 100.0 189 100.0
≥0.1	186 100.0	190 100.0	186 100.0 189 100.0
≥1	136 73.12	38 20.00	136 73.12 38 20.11
≥10	0 0.00	1 0.53	0 0.00 1 0.53
≥100	0 0.00	0 0.00	0 0.00 0 0.00

N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

Serotype 3, After Booster Dose (FAS) Serotype 3, After Booster Dose (PPS)



Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

Figure 86 Reverse cumulative distribution curves, pneumococcal serotype 3, IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group - post booster dose

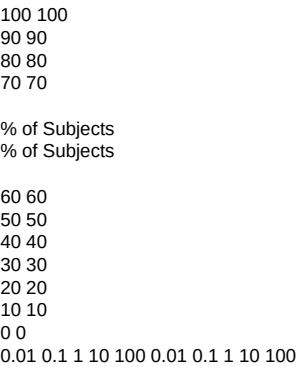
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Table 82. Distribution of the pneumococcal serotype 5 IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group post booster dose

FAS N (%)	PPS N (%)	PCV13-TT	PCV13-TT
Concentration	PCV7 (N=190)	PCV7 (N=189)	(N=186) (N=186)
n %	n %	n %	n %
≥0.01	186 100.0	176 92.63	186 100.0 175 92.59
≥0.1	186 100.0	175 92.11	186 100.0 174 92.06
≥1	75 40.32	3 1.58	75 40.32 3 1.59
≥10	1 0.54	0 0.00	1 0.54 0 0.00
≥100	0 0.00	0 0.00	0 0.00 0 0.00

N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

Serotype 5, After Booster Dose (FAS) Serotype 5, After Booster Dose (PPS)



Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

Figure 87 Reverse cumulative distribution curves, pneumococcal serotype 5, IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group - post booster dose

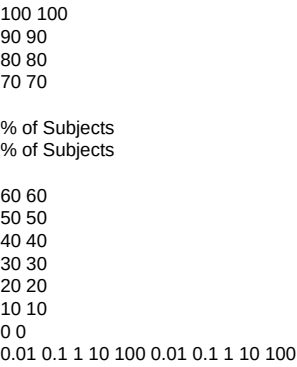
Table 83. Distribution of the pneumococcal serotype 6A IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group post booster dose

FAS N (%)	PPS N (%)	PCV13-TT PCV13-TT
Concentration	PCV7 (N=190)	PCV7 (N=189)
(N=186)	(N=186)	
n %	n %	n %
≥0.01	186 100.0	190 100.0
≥0.1	185 99.46	185 99.46
≥1	174 93.55	174 93.55
≥10	10 5.38	10 5.38
≥100	0 0.00	0 0.00

N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

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Serotype 6A, After Booster Dose (FAS) Serotype 6A, After Booster Dose (PPS)



Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

Figure 88 Reverse cumulative distribution curves, pneumococcal serotype 6A, IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group - post booster dose

Table 84. Distribution of the pneumococcal serotype 7F IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group post booster dose

FAS N (%)	PPS N (%)	PCV13-TT PCV13-TT
Concentration	PCV7 (N=190)	PCV7 (N=189)
(N=186)	(N=186)	
n %	n %	n %
≥0.01	186 100.0	185 97.37
≥0.1	186 100.0	183 96.32
≥1	185 99.46	174 92.09
≥10	21 11.29	0 0.00
≥100	0 0.00	0 0.00

N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

PCV7 group.

Serotype 7F, After Booster Dose (FAS) Serotype 7F, After Booster Dose (PPS)

100 100
90 90
80 80
70 70
% of Subjects
% of Subjects

60 60
50 50
40 40
30 30
20 20
10 10
0 0
0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

Figure 89 Reverse cumulative distribution curves, pneumococcal serotype 7F, IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group - post booster dose

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Table 85. Distribution of the pneumococcal serotype 19A IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group post booster dose

FAS N (%)	PPS N (%)								
PCV13-TT	PCV13-TT								
Concentration PCV7 (N=190)	PCV7 (N=189)								
(N=186)	(N=186)								
n	%	n	%	n	%	n	%	n	%
≥0.01	186	100.0	190	100.0	186	100.0	189	100.0	
≥0.1	186	100.0	190	100.0	186	100.0	189	100.0	
≥1	184	98.92	164	86.32	184	98.92	163	86.24	
≥10	38	20.43	5	2.63	38	20.43	5	2.65	
≥100	0	0.00	0	0.00	0	0.00	0	0.00	

N=number of subjects included in FAS and PPS for vaccine group.
n=number of subjects with a determinate IgG antibody concentration to the given serotype for PCV13-TT and PCV7 group.

Serotype 19A, After Booster Dose (FAS) Serotype 19A, After Booster Dose (PPS)

100 100
90 90
80 80
70 70

% of Subjects
% of Subjects

60 60
50 50
40 40
30 30
20 20
10 10
0 0
0.01 0.1 1 10 100 0.01 0.1 1 10 100

Antibody Concentration ■µg/mL)
Antibody Concentration (µg/mL)

PCV13-TT PCV7 PCV13-TT PCV7

Figure 90 Reverse cumulative distribution curves, pneumococcal serotype 19A, IgG antibody concentrations for subjects aged 7-11 months in PCV13-TT group and PCV7 group - post booster dose

6.3. SUMMARY OF BOOSTER DOSE
6.3.1. Summary of booster dose for subjects aged 2-6 months
In subjects aged 3 months, after booster dose of PCV13-TT, the OPA positive rates of all common serotypes exceeded 98%, reaching the lower limit defined in non-inferiority test compared to PCV7 group. The OPA positive rates of additional serotypes all exceeded 95% and reached the lower limit defined in superiority test except for serotype 6A as compared with PCV7 group. The positive rate of serotype 6A was 95.83% in PCV13-TT group and 92.78% in PCV7 group, which were both at a relatively high level. The lower limit of 95% CI for difference of positive rates was -3.47%, which was consistent with the reported evidence of possible cross-protection between pneumococcal serotype 6B and 6A in prior literature. The GMT of serotype 7F in PCV13-TT (3 months) group post booster dose rose to a similar level as that after primary series, while GMTs of other serotypes after booster dose were higher than that after primary series to varied degrees. In subjects aged 2 months, after booster dose of PCV13-TT, the OPA positive rates of 10 serotypes reached 100%, meanwhile the positive rates of other 3 serotypes exceeded 95%. The OPA positive rates of 7 common serotypes all reached the lower limit defined in non-inferiority test when compared with PCV7 (3 months) group. Five of the 6 additional serotypes reached the lower limit defined in superiority test except for 6A. The positive rate for serotype 6A was 97.83% after

booster dose in PCV13-TT (2 months) group. The lower limit of 95% CI for the difference between the PCV13-TT (2 months) group and PCV7 (3 months) group was -0.91%, which was similar to the result obtained in PCV13-TT (3 months) group for serotype 6A, consistent with the reported Confidential 203 / 310

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evidence of possible cross-protection between pneumococcal serotype 6B and 6A. The overall trend of the OPA positive rates for all serotypes was similar in PCV13-TT (2 months) group vs PCV7 (3 months) and PCV13-TT (3 months). Through booster dose, the GMT of serotype 3 rose to a similar level as that after primary series, while GMTs of other serotypes after booster dose were higher than that after primary series to varied degrees.

6.3.2. Summary of booster dose for subjects aged 7-11 months

In subjects aged 7-11 months, after booster dose of PCV13-TT, the OPA positive rates of 13 serotypes all exceeded 90%, among which the positive rates of 8 serotypes reached 100%. For OPA positive rates, the lower limit of 95% CI for the difference of OPA positive rates (PCV13-TT - PCV7) was greater than -10% (lower reference limit) for the 7 common serotypes in PCV13-TT group vs PCV7 group. The lower limit of 95% CI for the difference of OPA positive rates all exceeded 0% in all 6 additional serotypes. For the GMT of OPA, the lower limit of 95% CI for GMT ratio (PCV13-TT / PCV7) was greater than 0.5 (lower reference limit) for 5 of the 7 common serotypes in PCV13-TT group vs PCV7 group. The only two serotypes with lower limit of 95% CI for GMT ratio lower than 0.5 (lower reference limit) in PCV13-TT group were serotype 6B and 23F, while GMTs (PPS) of serotype 6B and 23F in PCV13-TT group had achieved 110- and 197-fold increase as compared with the positive threshold (1:8), respectively. The lower limits of 95% CI for intergroup GMT ratio in 6 additional serotypes were all higher than 1.

7. SAFETY ANALYSIS

7.1. INCIDENCE OF ADVERSE EVENTS

7.1.1. Incidence of Adverse Events for subjects aged 3-71 months

In Phase III clinical trial, 790 (2282 events) cases of adverse events were observed in the 3-71 month group, and 851 (2440 events) cases of adverse events were observed in the control group. The overall incidence of adverse events was 70.73% and 76.05% in the PCV13-TT and the control group, respectively. There was significant difference in the incidence of adverse events between groups, and the overall incidence of adverse events in the PCV13-TT was lower than that in the control group.

The incidence of adverse events related to the study vaccine was 60.79% in the PCV13-TT and 64.79% in the control group. The PCV13-TT was slightly lower than the control group but no significant difference was observed.

The incidence of vaccine-independent adverse events was 35.00% in the PCV13-TT and 36.82% in the control group. No significant difference was observed between groups.

The incidence of adverse events with grade 3 severity was 7.07% in the PCV13-TT and 5.36% in the control group. The grade 3 adverse events related to the study vaccine were 4.21% and 2.41% respectively, and the grade 3 adverse events unrelated to the study vaccine were 3.13% and 3.13% respectively. No significant difference was observed.

Since the subjects in this study were all infants and toddlers, in order to determine the severity of SAE in time for risk control, this study assessed the severity of the solicited symptoms or the most severe symptoms in SAE by referring to the Guidelines for The Classification of Adverse Reactions in Clinical Trials of Prophylactic Vaccines. A total of 31 subjects with 41 cases of grade 4 adverse events were observed (all included as SAE and detailed analysis was given in 7.1.7), of which 15 cases were in PCV13-TT and 16 in control group. The incidence was 1.34% and 1.43% respectively. Only 1 case of adverse event in the PCV13-TT was probably related to the study vaccine. The incidence of grade 4 adverse events related to the study vaccine were 0.09% and 0.00% in the PCV13-TT and the control group, and the incidence of grade 4 adverse events unrelated to the study vaccine was 1.25% and 1.43%, respectively. No significant intergroup difference was observed for grade 4.adverse event.

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The incidences of adverse events were higher in 3 months and 7-11 months age groups, with lower rates noted in PCV13-TT than in control group.

The incidences of adverse events in the 3-71 month group and the control group are shown in Table 42-1 to 42-5.

Table. 42-1 Overall incidence of adverse reactions in PCV13-TT and PCV7 groups of subjects aged 3 through 71 months

PCV13-TT (N=1117) PCV7 (N=1119)

P									
Events	Cases	%	Events	Cases	%				
Any AE	2282	790	70.73	2440	851	76.05	0.0047		
Grade 1	1251	603	53.98	1370	650	58.09	0.0552		
Grade 2	908	486	43.51	976	530	47.36	0.0680		
Grade 3	101	79	7.07	72	60	5.36	0.0968		
Grade 4	22	15	1.34	22	16	1.43	1.0000		
AEs related to vaccine									
Grade 1	1038	544	48.70	1144	590	52.73	0.0627		
Grade 2	467	312	27.93	511	344	30.74	0.1500		
Grade 3	59	47	4.21	33	27	2.41	0.0182		
Grade 4	2	1	0.09	0	0	0.00	0.4996		
Solicited AE									
Grade 1	1006	540	48.34	1114	583	52.10	0.0829		
Grade 2	425	294	26.32	460	327	29.22	0.1309		
Grade 3	59	47	4.21	33	27	2.41	0.0182		
Grade 4	1	1	0.09	0	0	0.00	0.4996		
Unsolicited AE									
Grade 1	32	27	2.42	30	26	2.32	0.8905		
Grade 2	42	37	3.31	51	43	3.84	0.5694		
Grade 3	//////								
Grade 4	1	1	0.09	0	0	0.00	0.4996		
AEs unrelated to vaccine									
Grade 1	213	159	14.23	226	170	15.19	0.5506		
Grade 2	441	270	24.17	465	292	26.09	0.3061		

Grade 3 42 35 3.13 39 35 3.13 1.0000
 Grade 4 20 14 1.25 22 16 1.43 0.8546
 Solicited AE 150 109 9.76 163 122 10.90 0.4045
 Grade 1 57 49 4.39 56 51 4.56 0.9186
 Grade 2 87 72 6.45 94 73 6.52 1.0000
 Grade 3 6 6 0.54 13 13 1.16 0.1653
 Grade 4 / / / / / / / /
 Unsolicited AE 566 352 31.51 589 365 32.62 0.5868
 Grade 1 156 131 11.73 170 139 12.42 0.6497
 Grade 2 354 239 21.40 371 262 23.41 0.2646
 Grade 3 36 30 2.69 26 23 2.06 0.3348
 Grade 4 20 14 1.25 22 16 1.43 0.8546

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Table. 42-2 Overall incidence of adverse reactions in PCV13-TT and PCV7 groups of subjects aged 3 months

PCV13-TT (N=519) PCV7 (N=519)
 P
 Events Cases % Events Cases %
 Any AE 1306 415 79.96 1447 440 84.78 0.0504
 Grade 1 760 338 65.13 867 365 70.33 0.0842
 Grade 2 489 243 46.82 538 271 52.22 0.0937
 Grade 3 46 35 6.74 32 27 5.20 0.3594
 Grade 4 11 8 1.54 10 8 1.54 1.0000
 AEs related to vaccine 939 371 71.48 1068 392 75.53 0.1594
 Grade 1 661 315 60.69 748 344 66.28 0.0710
 Grade 2 258 160 30.83 311 188 36.22 0.0758
 Grade 3 18 15 2.89 9 8 1.54 0.2049
 Grade 4 2 1 0.19 0 0 0.00 1.0000
 Solicited AE 885 365 70.33 1012 385 74.18 0.1877
 Grade 1 637 313 60.31 725 338 65.13 0.1234
 Grade 2 229 146 28.13 278 178 34.30 0.0378
 Grade 3 18 15 2.89 9 8 1.54 0.2049
 Grade 4 1 1 0.19 0 0 0.00 1.0000
 Unsolicited AE 54 44 8.48 56 44 8.48 1.0000
 Grade 1 24 19 3.66 23 21 4.05 0.8721
 Grade 2 29 26 5.01 33 25 4.82 1.0000
 Grade 3 / / / / / / / /
 Grade 4 1 1 0.19 0 0 0.00 1.0000
 AEs unrelated to
 367 197 37.96 379 211 40.66 0.4088
 vaccine
 Grade 1 99 78 15.03 119 90 17.34 0.3540
 Grade 2 231 132 25.43 227 144 27.75 0.4397
 Grade 3 28 21 4.05 23 20 3.85 1.0000
 Grade 4 9 7 1.35 10 8 1.54 1.0000
 Solicited AE 72 55 10.60 81 56 10.79 1.0000
 Grade 1 27 24 4.62 32 30 5.78 0.4850
 Grade 2 42 34 6.55 45 31 5.97 0.7980
 Grade 3 3 3 0.58 4 4 0.77 1.0000
 Grade 4 / / / / / / / /
 Unsolicited AE 295 181 34.87 298 191 36.80 0.5602
 Grade 1 72 65 12.52 87 72 13.87 0.5823
 Grade 2 189 124 23.89 182 135 26.01 0.4733
 Grade 3 25 19 3.66 19 17 3.28 0.8656
 Grade 4 9 7 1.35 10 8 1.54 1.0000

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Table. 42-3 Overall incidence of adverse reactions in PCV13-TT and PCV7 groups of subjects aged 7 through 11 months

PCV13-TT (N=198) PCV7 (N=200)
 P
 Events Cases % Events Cases %
 Any AE 443 150 75.76 505 168 84.00 0.0455
 Grade 1 224 114 57.58 252 128 64.00 0.2180
 Grade 2 194 102 51.52 232 115 57.50 0.2682
 Grade 3 23 19 9.60 12 12 6.00 0.1953
 Grade 4 2 2 1.01 9 5 2.50 0.4490
 AEs related to vaccine 287 130 65.66 303 142 71.00 0.2815
 Grade 1 173 99 50.00 198 111 55.50 0.3154
 Grade 2 99 70 35.35 99 74 37.00 0.7551
 Grade 3 15 13 6.57 6 6 3.00 0.1053
 Grade 4 0 0 0.00 0 0 0.00 1.0000
 Solicited AE 277 128 64.65 293 141 70.50 0.2390
 Grade 1 169 98 49.49 197 111 55.50 0.2695
 Grade 2 93 67 33.84 90 71 35.50 0.7527
 Grade 3 15 13 6.57 6 6 3.00 0.1053
 Grade 4 / / / / / / / /
 Unsolicited AE 10 9 4.55 10 9 4.50 1.0000
 Grade 1 4 4 2.02 1 1 0.50 0.2140
 Grade 2 6 5 2.53 9 9 4.50 0.4156
 Grade 3 / / / / / / / /
 Grade 4 / / / / / / / /
 AEs unrelated to
 156 87 43.94 202 95 47.50 0.4831
 vaccine

Grade 1 51 37 18.69 54 39 19.50 0.8987
 Grade 2 95 59 29.80 133 70 35.00 0.2853
 Grade 3 8 8 4.04 6 6 3.00 0.5993
 Grade 4 2 2 1.01 9 5 2.50 0.4490
 Solicited AE 35 25 12.63 44 34 17.00 0.2594
 Grade 1 16 13 6.57 10 9 4.50 0.3896
 Grade 2 17 15 7.58 29 23 11.50 0.2324
 Grade 3 2 2 1.01 5 5 2.50 0.4490
 Grade 4 / / / / / / /
 Unsolicited AE 121 75 37.88 158 82 41.00 0.5397
 Grade 1 35 29 14.65 44 33 16.50 0.6789
 Grade 2 78 49 24.75 104 60 30.00 0.2619
 Grade 3 6 6 3.03 1 1 0.50 0.0669
 Grade 4 2 2 1.01 9 5 2.50 0.4490

Table. 42-4 Overall incidence of adverse reactions in PCV13-TT and PCV7 groups of subjects aged 12 through 23 months
 PCV13-TT (N=200) PCV7 (N=200)

P
 Events Cases % Events Cases %
 Any AE 344 136 68.00 295 138 69.00 0.9143
 Grade 1 169 95 47.50 146 90 45.00 0.6884
 Grade 2 150 84 42.00 128 82 41.00 0.9192
 Grade 3 17 13 6.50 18 14 7.00 1.0000
 Grade 4 8 4 2.00 3 3 1.50 1.0000
 AEs related to vaccine 197 106 53.00 178 114 57.00 0.4818
 Grade 1 121 79 39.50 108 77 38.50 0.9184
 Grade 2 65 47 23.50 62 49 24.50 0.9069
 Grade 3 11 7 3.50 8 6 3.00 1.0000
 Grade 4 0 0 0.00 0 0 0.00 1.0000
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Solicited AE 188 105 52.50 168 111 55.50 0.6160
 Grade 1 118 78 39.00 102 76 38.00 0.9182
 Grade 2 59 47 23.50 58 47 23.50 1.0000
 Grade 3 11 7 3.50 8 6 3.00 1.0000
 Grade 4 / / / / / / /
 Unsolicited AE 9 8 4.00 10 8 4.00 1.0000
 Grade 1 3 3 1.50 6 4 2.00 1.0000
 Grade 2 6 5 2.50 4 4 2.00 1.0000
 Grade 3 / / / / / / /
 Grade 4 / / / / / / /
 AEs unrelated to
 147 72 36.00 117 62 31.00 0.3404
 vaccine
 Grade 1 48 32 16.00 38 28 14.00 0.6747
 Grade 2 85 54 27.00 66 45 22.50 0.3540
 Grade 3 6 6 3.00 10 9 4.50 0.6002
 Grade 4 8 4 2.00 3 3 1.50 1.0000
 Solicited AE 31 21 10.50 28 23 11.50 0.8732
 Grade 1 10 9 4.50 11 9 4.50 1.0000
 Grade 2 20 17 8.50 13 12 6.00 0.4411
 Grade 3 1 1 0.50 4 4 2.00 0.3719
 Grade 4 / / / / / / /
 Unsolicited AE 116 67 33.50 89 52 26.00 0.1255
 Grade 1 38 28 14.00 27 33 11.50 0.5491
 Grade 2 65 46 23.00 53 37 18.50 0.3240
 Grade 3 5 5 2.50 6 5 2.50 1.0000
 Grade 4 8 4 2.00 3 3 1.50 1.0000

Table. 42-5 Overall incidence of adverse reactions in PCV13-TT and PCV7 groups of subjects aged 24 through 71 months
 PCV13-TT (N=200) PCV7 (N=200)

P
 Events Cases % Events Cases %
 Any AE 189 89 44.50 193 105 52.50 0.1333
 Grade 1 98 56 28.00 105 67 33.50 0.2786
 Grade 2 75 57 28.50 78 62 31.00 0.6619
 Grade 3 15 12 6.00 10 7 3.50 0.3475
 Grade 4 1 1 0.50 0 0 0.00 1.0000
 AEs related to vaccine 143 72 36.00 139 77 38.50 0.6792
 Grade 1 83 51 25.50 90 58 29.00 0.5006
 Grade 2 45 35 17.50 39 33 16.50 0.8942
 Grade 3 15 12 6.00 10 7 3.50 0.3475
 Grade 4 0 0 0.00 0 0 0.00 1.0000
 Solicited AE 141 72 36.00 134 75 37.50 0.8357
 Grade 1 82 51 25.50 90 58 29.00 0.5006
 Grade 2 44 34 17.00 34 31 15.50 0.7865
 Grade 3 15 12 6.00 10 7 3.50 0.3475
 Grade 4 / / / / / / /
 Unsolicited AE 2 2 1.00 5 5 2.50 0.4490
 Grade 1 1 1 0.50 0 0 0.00 1.0000
 Grade 2 1 1 0.50 5 5 2.50 0.2152
 Grade 3 / / / / / / /
 Grade 4 / / / / / / /
 AEs unrelated to
 46 35 17.50 54 44 22.00 0.3150
 vaccine
 Grade 1 15 12 6.00 15 13 6.50 1.0000
 Grade 2 30 25 12.50 39 33 16.50 0.3202

Grade 3 0 0 0.00 0 0 0.00 1.0000
Grade 4 1 1 0.50 0 0 0.00 1.0000
Solicited AE 12 8 4.00 10 9 4.50 1.0000
Grade 1 4 3 1.50 3 3 1.50 1.0000
Grade 2 8 6 3.00 7 7 3.50 1.0000
Grade 3 / / / / / / / /
Grade 4 / / / / / / / /
Unsolicited AE 34 29 14.50 44 40 20.00 0.1854
Grade 1 11 9 4.50 12 11 5.50 0.8192
Grade 2 22 20 10.00 32 30 15.00 0.1732
Grade 3 / / / / / / / /
Grade 4 1 1 0.50 0 0 0.00 1.0000

7.1.2. Adverse events within 30 minutes time period, during the first 7 days and withing 30 days after vaccination

7.1.2.1. Adverse events occurred within 30 minutes after vaccination

Adverse events within a 30 minutes time period
The incidence of adverse events within a 30 minutes time period after vaccination for each subgroup was shown in Tables 43-1 to 43-5.
The incidence of adverse events within a 30 minutes time period after vaccination was 14.95% and 16.71% in the test and PCV7 group, respectively. The severity of adverse events was predominantly level 1, with no significant differences in the incidence of adverse events and in the incidence of adverse events among the subgroups.

Table 43-1 Adverse events within 30 minutes post vaccination in subjects aged 3-71 months

PCV13-TT (N=1117) PCV7 (N=1119)			
P			
Events	Cases	%	Events Cases %
Any AE	197	167	14.95 213 187 16.71 0.2711
Vaccine-related AE	196	166	14.86 212 186 16.62 0.2700
Solicited AE	195	166	14.86 212 186 16.62 0.2700
unsolicited AE	1	1	0.09 0 0 0.00 0.4996
Vaccine-unrelated AE	1	1	0.09 1 1 0.09 1.0000
Solicited AE	1	1	0.09 1 1 0.09 1.0000

Table 43-2 Adverse reactions within 30 minutes post vaccination in subjects aged 3 months

PCV13-TT (N=519) PCV7 (N=519)			
P			
Events	Cases	%	Events Cases %
Any AE	135	110	21.19 142 121 23.31 0.4556
Vaccine-related AE	135	110	21.19 142 121 23.31 0.4556
Solicited AE	134	110	21.19 142 121 23.31 0.4556
unsolicited AE	1	1	0.19 0 0 0.00 1.0000
Vaccine-unrelated AE	0	0	0.00 0 0 0.00 1.0000

Table 43-3 Adverse reactions within 30 minutes post vaccination in subjects aged 7-11 months

PCV13-TT (N=198) PCV7 (N=200)			
P			
Events	Cases	%	Events Cases %
Any AE	34	32	16.16 34 31 15.50 0.8913
Vaccine-related AE	33	31	15.66 34 31 15.50 1.0000
Solicited AE	33	31	15.66 34 31 15.50 1.0000
Vaccine-unrelated AE	1	1	0.51 0 0 0.00 0.4975
Solicited AE	1	1	0.51 0 0 0.00 0.4975

Table 43-4 Adverse reactions within 30 minutes post vaccination in subjects aged 12-23 months

months
PCV13-TT (N=200) PCV7 (N=200)

P			
Events	Cases	%	Events Cases %
Any AE	20	17	8.50 24 22 11.00 0.5006
Vaccine-related AE	20	17	8.50 23 21 10.50 0.6095
Solicited AE	20	17	8.50 23 21 10.50 0.6095
Vaccine-unrelated AE	0	0	0.00 1 1 0.50 1.0000
Solicited AE	0	0	0.00 1 1 0.50 1.0000

Table 43-5 Adverse reactions within 30 minutes post vaccination in subjects aged 24-71 months

PCV13-TT (N=200) PCV7 (N=200)			
P			
Events	Cases	%	Events Cases %
Any AE	8	8	4.00 13 13 6.50 0.3703
Vaccine-related AE	8	8	4.00 13 13 6.50 0.3703
Solicited AE	8	8	4.00 13 13 6.50 0.3703
Vaccine-unrelated AE	0	0	0.00 0 0 0.00 1.0000

7.1.2.2. Adverse events during the first 7 days after vaccination

Adverse events mainly occurred during the first 7 days after vaccination. The incidence of adverse events 7 days after vaccination was 63.56% and 67.20% in the test and PCV7 group, respectively. The incidence of adverse events related to the study vaccine was 60.25% in the PCV13-TT and 63.90% in the control group, respectively. There was no significant differences in the incidence of adverse events and in the incidence of adverse events among the subgroups.

The incidence of adverse events 7 days after vaccination in test and PCV7 group was shown in Tables 44-1 to 44-5.

Table 44-1 Adverse reactions during Day 0-7 post vaccination in subjects aged 3-71 months

PCV13-TT (N=1117) PCV7 (N=1119)

P

Events Cases % Events Cases %

Any AE 1780 710 63.56 1927 752 67.20 0.0754

Vaccine-related AE 1542 673 60.25 1658 715 63.90 0.0812

Solicited AE 1491 670 59.98 1607 712 63.63 0.0817

unsolicited AE 51 46 4.12 51 43 3.84 0.7469

Vaccine-unrelated AE 238 130 11.64 269 153 13.67 0.1617

Solicited AE 150 109 9.76 163 122 10.90 0.4045

unsolicited AE 88 77 6.89 106 94 8.40 0.2029

Table 44-2 Adverse reactions during Day 0-7 post vaccination in subjects aged 3 months

PCV13-TT (N=519) PCV7 (N=519)

P

Events Cases % Events Cases %

Any AE 1030 378 72.83 1174 394 75.92 0.2862

Vaccine-related AE 919 367 70.71 1046 386 74.37 0.2106

Solicited AE 885 365 70.33 1012 385 74.18 0.1877

unsolicited AE 34 30 5.78 34 29 5.59 1.0000

Vaccine-unrelated AE 111 61 11.75 128 70 13.49 0.4547

Solicited AE 72 55 10.60 81 56 10.79 1.0000

unsolicited AE 39 36 6.94 47 41 7.90 0.6359

Table 44-3 Adverse reactions during Day 0-7 post vaccination in subjects aged 7-11 months

PCV13-TT (N=198) PCV7 (N=200)

P

Events Cases % Events Cases %

Any AE 338 138 69.70 373 152 76.00 0.1766

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Vaccine-related AE 283 128 64.65 299 141 70.50 0.2390

Solicited AE 277 128 64.65 293 141 70.50 0.2390

unsolicited AE 6 6 3.03 6 5 2.50 0.7704

Vaccine-unrelated AE 55 31 15.66 74 42 21.00 0.1955

Solicited AE 35 25 12.63 44 34 17.00 0.2594

unsolicited AE 20 15 7.58 30 27 13.50 0.0719

Table 44-4 Adverse reactions during Day 0-7 post vaccination in subjects aged 12-23 months

PCV13-TT (N=200) PCV7 (N=200)

P

Events Cases % Events Cases %

Any AE 250 117 58.50 226 123 61.50 0.6099

Vaccine-related AE 197 106 53.00 175 112 56.00 0.6157

Solicited AE 188 105 52.50 168 111 55.50 0.6160

unsolicited AE 9 8 4.00 7 5 2.50 0.5746

Vaccine-unrelated AE 53 26 13.00 51 31 15.50 0.5675

Solicited AE 31 21 10.50 28 23 11.50 0.8732

unsolicited AE 22 20 10.00 23 20 10.00 1.0000

Table 44-5 Adverse reactions during Day 0-7 post vaccination in subjects aged 24-71 months

PCV13-TT (N=200) PCV7 (N=200)

P

Events Cases % Events Cases %

Any AE 162 77 38.50 154 83 41.50 0.6099

Vaccine-related AE 143 72 36.00 138 76 38.00 0.7561

Solicited AE 141 72 36.00 134 75 37.50 0.8357

unsolicited AE 2 2 1.00 4 4 2.00 0.6851

Vaccine-unrelated AE 19 12 6.00 16 10 5.00 0.8270

Solicited AE 12 8 4.00 10 9 4.50 1.0000

unsolicited AE 7 6 3.00 6 6 3.00 1.0000

7.1.2.3. Adverse events within 8-30 days after vaccination

The incidence of adverse events within 8-30 days after vaccination was 23.01% and 26.36% in the test and PCV7 group, respectively. The adverse events that occurred 8-30 days after were mainly unrelated to the study vaccine. The incidence of adverse events related to the study vaccine was 1.61% in the PCV13-TT and 2.32% in the control group, respectively. There was no significant differences in the incidence of adverse events and in the incidence of adverse events among the subgroups.

The incidence of adverse events within 8-30 days after vaccination in test and PCV7 group was shown in Tables 45-1 to 45-5.

Table 45-1 Adverse reactions during Day 8-30 post vaccination in subjects aged 3-71 months

PCV13-TT (N=1117) PCV7 (N=1119)

P

Events Cases % Events Cases %

Any AE 373 257 23.01 418 295 26.36 0.0696

Vaccine-related AE 24 18 1.61 30 26 2.32 0.2864

unsolicited AE 24 18 1.61 30 26 2.32 0.2864

Vaccine-unrelated AE 349 248 22.20 388 279 24.93 0.1350

unsolicited AE 349 248 22.20 388 279 24.93 0.1350

Table 45-2 Adverse reactions during Day 8-30 post vaccination in subjects aged 3 months

PCV13-TT (N=519) PCV7 (N=519) P

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Events Cases % Events Cases %
Any AE 237 155 29.87 252 176 33.91 0.1828
Vaccine-related AE 20 15 2.89 22 18 3.47 0.7240
unsolicited AE 20 15 2.89 22 18 3.47 0.7240
Vaccine-unrelated AE 217 147 28.32 230 165 31.79 0.2498
unsolicited AE 217 147 28.32 230 165 31.79 0.2498

Table 45-3 Adverse reactions during Day 8-30 post vaccination in subjects aged 7-11 months

PCV13-TT (N=198) PCV7 (N=200)

P

Events Cases % Events Cases %
Any AE 59 42 21.21 86 52 26.00 0.2889
Vaccine-related AE 4 3 1.52 4 4 2.00 1.0000
unsolicited AE 4 3 1.52 4 4 2.00 1.0000
Vaccine-unrelated AE 55 41 20.71 82 50 25.00 0.3404
unsolicited AE 55 41 20.71 82 50 25.00 0.3404

Table 45-4 Adverse reactions during Day 8-30 post vaccination in subjects aged 12-23 months

PCV13-TT (N=200) PCV7 (N=200)

P

Events Cases % Events Cases %
Any AE 50 36 18.00 42 33 16.50 0.7914
Vaccine-related AE 0 0 0.00 3 3 1.50 0.2481
unsolicited AE 0 0 0.00 3 3 1.50 0.2481
Vaccine-unrelated AE 50 36 18.00 39 31 15.50 0.5925
unsolicited AE 50 36 18.00 39 31 15.50 0.5925

Table 45-5 Adverse reactions during Day 8-30 post vaccination in subjects aged 24-71 months

PCV13-TT (N=200) PCV7 (N=200)

P

Events Cases % Events Cases %
Any AE 27 24 12.00 38 34 17.00 0.2009
Vaccine-related AE 0 0 0.00 1 1 0.50 1.0000
unsolicited AE 0 0 0.00 1 1 0.50 1.0000
Vaccine-unrelated AE 27 24 12.00 37 33 16.50 0.2524
unsolicited AE 27 24 12.00 37 33 16.50 0.2524

7.1.2.4. Adverse events 30 days after vaccination

The incidence of adverse events 30 days after vaccination was 8.33% and 7.06% in the test and PCV7 group, respectively. The adverse events that occurred 8-30 days after were unrelated to the study vaccine. There were no significant differences between groups.

The incidence of adverse events 30 days after vaccination in test and PCV7 group was shown in Tables 46-1 to 46-5.

Table 46-1 Adverse reactions at Day 30 post vaccination in subjects aged 3-71 months

PCV13-TT (N=1117) PCV7 (N=1119)

P

Events Cases % Events Cases %
Any AE 129 93 8.33 95 79 7.06 0.2676
Vaccine-related AE 0 0 0.00 0 0 0.00 1.0000
Vaccine-unrelated AE 129 93 8.33 95 79 7.06 0.2676

Table 46-2 Adverse reactions at Day 30 post vaccination in subjects aged 3 months

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PCV13-TT (N=519) PCV7 (N=519)

P

Events Cases % Events Cases %
Any AE 39 26 5.01 21 18 3.47 0.2807
Vaccine-related AE 0 0 0.00 0 0 0.00 1.0000
Vaccine-unrelated AE 39 26 5.01 21 18 3.47 0.2807

Table 46-3 Adverse reactions at Day 30 post vaccination in subjects aged 7-11 months

PCV13-TT (N=198) PCV7 (N=200)

P

Events Cases % Events Cases %
Any AE 46 36 18.18 46 41 20.50 0.6124
Vaccine-related AE 0 0 0.00 0 0 0.00 1.0000
Vaccine-unrelated AE 46 36 18.18 46 41 20.50 0.6124

Table 46-4 Adverse reactions at Day 30 post vaccination in subjects aged 12-23 months

PCV13-TT (N=200) PCV7 (N=200)

P

Events Cases % Events Cases %
Any AE 44 31 15.50 27 19 9.50 0.0956
Vaccine-related AE 0 0 0.00 0 0 0.00 1.0000
Vaccine-unrelated AE 44 31 15.50 27 19 9.50 0.0956

Table 46-5 Adverse reactions at Day 30 post vaccination in subjects aged 24-71 months

PCV13-TT (N=200) PCV7 (N=200)

P

Events Cases % Events Cases %
Any AE 0 0 0.00 1 1 0.50 1.0000
Vaccine-related AE 0 0 0.00 0 0 0.00 1.0000
Vaccine-unrelated AE 0 0 0.00 1 1 0.50 1.0000

7.1.3. Adverse events related to study vaccine

7.1.3.1. Solicited adverse events

(1) Systemic reactions in 3-71 months old PCV13-TT and control group

The incidence of systemic adverse reactions in the study and the control group was 54.52% and 58.98%, respectively, and the incidence of systemic adverse reactions in the PCV13-TT was lower than that in the control group (p=0.0364). The main symptom of systemic adverse reaction was fever, which was less in PCV13-TT than in control group (p=0.0035). No significant difference was observed for the incidence of other systemic reactions between groups. The systemic reactions were primarily grade 1 or 2, similar incidence rates of grade 3 systemic reactions were noted between PCV13-TT and control group. Systemic reactions and their incidence at different time observed in study and control group of the 3-71 months olds are shown in Table 47-1.

Table 47-1 Analysis of systemic reactions in PCV13-TT and PCV7 groups of subjects aged 3 through 71 months

PCV13-TT (N=1117) PCV7 (N=1119)											
P											
Events Cases % Events Cases %											
Systemic reactions											
Grade 1											
Grade 2											
Grade 3											
Grade 4											
Fever											
Grade 1											
Grade 2											
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Grade 3											
Grade 4											
Fatigue / somnolence											
Grade 1											
Grade 2											
Grade 3											
Nausea / vomiting											
Grade 1											
Grade 2											
Grade 3											
Diarrhea											
Grade 1											
Grade 2											
Grade 3											
Myalgia											
Grade 1											
Grade 2											
Grade 3											
Cough											
Grade 1											
Grade 2											
Grade 3											
Crying											
Grade 1											
Grade 2											
Grade 3											
Allergic reactions											
Grade 1											
Grade 2											
Grade 3											

(2)Local reactions in 3-71 months old PCV13-TT and control group
The incidence of local adverse reaction was 15.67% in the PCV13-TT and 14.66% in the control group, indicating no significant difference. The most common symptom of local adverse reaction was redness at injection site, followed by swelling and pain. The severity of local reactions was mainly grade 1. The incidence of local reactions with severity of grade 3 in the PCV13-TT was higher than that in the control group (p=0.0007). The symptoms of local reactions achieving grade 3 in the PCV13-TT were mainly redness, swelling and induration, however, no intergroup difference was observed with respect to redness, swelling and induration in 3-month-old population. The symptoms mainly occurred in the population over 7-month-old. Local reactions and their incidence at different time observed in study and control group of the 3-71 months olds are shown in Table 47-2.

Table 47-2 Analysis of local reactions in PCV13-TT and PCV7 groups of subjects aged 3 through 71 months

PCV13-TT (N=1117) PCV7 (N=1119)											
P											
Events Cases % Events Cases %											
Local reactions											
Grade 1											
Grade 2											
Grade 3											
Local pain											
Grade 1											
Grade 2											
Grade 3											
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Local redness											
Grade 1											
Grade 2											
Grade 3											
Local swelling											
Grade 1											
Grade 2											

Grade 3 17 16 1.43 5 5 0.45 0.0163
 Local rash 1 1 0.09 0 0 0.00 0.4996
 Grade 1 0 0 0.00 0 0 0.00 /
 Grade 2 1 1 0.09 0 0 0.00 0.4996
 Grade 3 0 0 0.00 0 0 0.00 /
 Local induration 30 27 2.42 22 21 1.88 0.3861
 Grade 1 9 9 0.81 12 11 0.98 0.8230
 Grade 2 16 14 1.25 8 8 0.71 0.2075
 Grade 3 5 5 0.45 2 2 0.18 0.2880
 Local pruritis 15 15 1.34 5 5 0.45 0.0258
 Grade 1 15 15 1.34 5 5 0.45 0.0258
 Grade 2 0 0 0.00 0 0 0.00 /
 Grade 3 0 0 0.00 0 0 0.00 /

(3)The time of occurrence of systemic and local reactions in 3-71 months old PCV13-TT and control group

The occurrence of systemic and local adverse reactions in the PCV13-TT and the control group 30 minutes and 7 days after vaccination are shown in Table 48-1 and 48-2. Within 30 minutes, only fever was observed as systemic reaction. The incidence of fever in the control group was higher than that in the PCV13-TT. Redness was the main local reaction, and was similar between groups in terms of incidence rate.

Table 48-1 Systemic and local reactions 30 min after vaccination in subjects aged 3 through 71 months

PCV13-TT (N=1117) PCV7 (N=1119)

P

Events Cases % Events Cases %
 Fever 147 127 11.37 175 162 14.48 0.0320
 Local pain 1 1 0.09 0 0 0.00 0.4996
 Local redness 46 44 3.94 37 32 2.86 0.1634
 Local swelling 1 1 0.09 0 0 0.00 0.4996

Table 48-2 Systemic and local reactions 7 days min after vaccination in subjects aged 3 through 71 months

PCV13-TT (N=1117) PCV7 (N=1119)

P

Events Cases % Events Cases %
 Fever 816 531 47.54 972 602 53.80 0.0035
 Fatigue / somnolence 47 42 3.76 39 37 3.31 0.5692
 Nausea / vomiting 55 46 4.12 40 37 3.31 0.3166
 Diarrhea 103 91 8.15 100 87 7.77 0.7552
 Myalgia 2 2 0.18 4 4 0.36 0.6871
 Cough 69 66 5.91 72 65 5.81 0.9285
 Abnormal crying 89 72 6.45 81 67 5.99 0.6625
 Allergic reactions 9 9 0.81 19 18 1.61 0.1198
 Local pain 59 55 4.92 43 40 3.57 0.1170
 Local redness 128 123 11.01 155 126 11.26 0.8931
 Local swelling 68 63 5.64 55 47 4.20 0.1188
 Local rash 1 1 0.09 0 0 0.00 0.4996

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Local induration 30 27 2.42 22 21 1.88 0.3861
 Local pruritis 15 15 1.34 5 5 0.45 0.0258

(4)Systemic reactions and local reactions observed in study and control group of different age groups

No significant difference was observed in the incidence of systemic and local adverse reactions between study and control group of the 3 months olds, the 7-11 months olds, the 12-23 months olds and the 24-71 months olds.

Fever was the most common systemic reaction in all age groups. The incidence of fever in 3-month age PCV13-TT was lower than that in the control group ($p=0.0072$), and the incidence of allergies in 7-11 month age PCV13-TT was lower than that in the control group ($p=0.0301$), the incidence of local itching in 24-71 months old PCV13-TT was higher than that in the control group ($p=0.0358$). No significant difference was observed between study and control group of other ages in fever and other systemic or local symptoms

The incidence of systemic and local reactions in the study and control group of each age stratification is shown in Table 49-1 to 49-4.

Table 49-1 Systemic and local reactions after vaccination in subjects aged 3 months

PCV13-TT (N=519) PCV7 (N=519)

P

Events Cases % Events Cases %
 Systemic reactions 778 352 67.82 862 371 71.48 0.2243
 Fever 515 301 58.00 628 344 66.28 0.0072
 Fatigue / 37 32 6.17 29 27 5.20 0.5922
 somnolence
 Nausea / vomiting 38 30 5.78 22 19 3.66 0.1426
 Diarrhea 76 67 12.91 74 62 11.95 0.7068
 Myalgia 1 1 0.19 2 2 0.39 1.0000
 Cough 40 38 7.32 41 34 6.55 0.7142
 Abnormal crying 68 54 10.40 62 49 9.44 0.6781
 Allergic reactions 3 3 0.58 4 3 0.58 1.0000
 Local reactions 107 70 13.49 150 81 15.61 0.3787
 Local pain 16 12 2.31 12 10 1.93 0.8300
 Local redness 59 55 10.60 92 71 13.68 0.1538
 Local swelling 12 10 1.93 25 18 3.47 0.1790
 Local induration 18 15 2.89 19 18 3.47 0.7240
 Local pruritis 2 2 0.39 2 2 0.39 1.0000

Table 49-2 Systemic and local reactions after vaccination in subjects aged 7 through 11 months

PCV13-TT (N=198) PCV7 (N=200)

P

Events Cases % Events Cases %

Systemic reactions 206 116 58.59 245 132 66.00 0.1476

Fever 149 106 53.54 184 123 61.50 0.1281

Fatigue / somnolence 3 3 1.52 5 5 2.50 0.7238

Nausea vomiting 12 11 5.56 8 8 4.00 0.4912

Diarrhea 15 14 7.07 19 18 9.00 0.5810

Myalgia 0 0 0.00 1 1 0.50 1.0000

Cough 11 10 5.05 9 9 4.50 0.8188

Crying 16 13 6.57 13 12 6.00 0.8393

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Allergic reactions 0 0 0.00 6 6 3.00 0.0301

Local reactions 71 42 21.21 48 29 14.50 0.0894

Local pain 14 14 7.07 8 7 3.50 0.1221

Local redness 34 33 16.67 30 25 12.50 0.2580

Local swelling 16 15 7.58 8 7 3.50 0.0830

Local induration 6 6 3.03 1 1 0.50 0.0669

Local pruritis 1 1 0.51 1 1 0.50 1.0000

Table 49-3 Systemic and local reactions after vaccination in subjects aged 12 through 23 months

PCV13-TT (N=200) PCV7 (N=200)

P

Events Cases % Events Cases %

Systemic reactions 143 93 46.50 141 100 50.00 0.5483

Fever 110 83 41.50 110 88 44.00 0.6861

Fatigue / somnolence 2 2 1.00 2 2 1.00 1.0000

Nausea vomiting 3 3 1.50 5 5 2.50 0.7238

Diarrhea 10 8 4.00 3 3 1.50 0.2201

Cough 10 10 5.00 11 11 5.50 1.0000

Crying 5 5 2.50 3 3 1.50 0.7238

Allergic reactions 3 3 1.50 7 7 3.50 0.3375

Local reactions 45 21 10.50 27 18 9.00 0.7364

Local pain 10 10 5.00 3 3 1.50 0.0870

Local redness 16 16 8.00 17 14 7.00 0.8498

Local swelling 15 13 6.50 6 6 3.00 0.1569

Local rash 1 1 0.50 0 0 0.00 1.0000

Local induration 1 1 0.50 1 1 0.50 1.0000

Local pruritis 2 2 1.00 0 0 0.00 0.4987

Table 49-4 Systemic reactions after vaccination in subjects aged 24 through 71 months

PCV13-TT (N=200) PCV7 (N=200)

P

Events Cases % Events Cases %

Systemic reactions 63 48 24.00 79 57 28.50 0.3633

Fever 42 41 20.50 50 47 23.50 0.5463

Fatigue / 5 5 2.50 3 3 1.50 0.7238

somnolence

Nausea vomiting 2 2 1.00 5 5 2.50 0.4490

Diarrhea 2 2 1.00 4 4 2.00 0.6851

Myalgia 1 1 0.50 1 1 0.50 1.0000

Cough 8 8 4.00 11 11 5.50 0.6394

Crying 0 0 0.00 3 3 1.50 0.2481

Allergic reactions 3 3 1.50 2 2 1.00 1.0000

Local reactions 78 42 21.00 55 36 18.00 0.5282

Local pain 19 19 9.50 20 20 10.00 1.0000

Local redness 19 19 9.50 16 16 8.00 0.7239

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Local swelling 25 25 12.50 16 16 8.00 0.1867

Local induration 5 5 2.50 1 1 0.50 0.2152

Local pruritis 10 10 5.00 2 2 1.00 0.0358

7.1.3.2 Unsolicited events related to study vaccine

Unsolicited adverse events (coincidental) that occurred 0-30 days after vaccination collected from the study and PCV7 group during the observation period are shown in Table 50-1 to 50-7.

Vaccine-related unsolicited events in the study and PCV7 group were mainly respiratory, thoracic and mediastinal disorders, gastrointestinal disorders, infections and infestations, general disorders and administration site conditions. General disorders and administration site conditions occurred mainly in 8-30 days after vaccination, and the most common symptom was fever. Considering from the time of occurrence, unsolicited adverse events related to the study vaccine occurred mainly within 7 days after vaccination. In view of age groups, more types and higher incidence rates of unsolicited adverse events related to the vaccine were reported in the 3-month age group, and a decreasing trend was observed in 7-11 month age group, 12-23 months age group and 24-71 months age group sequentially.

No significant difference was observed between the study and control group in the incidence of any vaccine-related unsolicited adverse events.

Table. 50-1 Vaccine-related unsolicited events in PCV13-TT and PCV7 groups of subjects aged 3 through 71 months

PCV13-TT (N=1117) PCV7 (N=1119)

P

Events Cases Incidence rate Events Cases Incidence rate

Unsolicited adverse reactions 75 63 5.64 81 66 5.90 0.8561

Metabolism and nutrition disorders 1 1 0.09 0 0 0.00 0.4996

Decreased appetite 1 1 0.09 0 0 0.00 0.4996

Infections and infestations 16 16 1.43 12 12 1.07 0.4551
Nasopharyngitis 3 3 0.27 1 1 0.09 0.3742
Tonsillitis 1 1 0.09 0 0 0.00 0.4996
Infectious pneumonia 0 0 0.00 1 1 0.09 1.0000
Upper respiratory tract infection 10 10 0.90 10 10 0.89 1.0000
Bronchitis 1 1 0.09 0 0 0.00 0.4996
Herpes virus infection 1 1 0.09 0 0 0.00 0.4996
Respiratory, thoracic and mediastinal disorders 28 25 2.24 34 29 2.59 0.6798
Rhinorrhea 22 19 1.70 30 27 2.41 0.2970
Nasal obstruction 3 3 0.27 1 1 0.09 0.3742
Cough 2 2 0.18 0 0 0.00 0.2494
Sneezing 1 1 0.09 2 2 0.18 1.0000
Increased production of sputum 0 0 0.00 1 1 0.09 1.0000
Psychiatric disorders 1 1 0.09 0 0 0.00 0.4996
Irritability 1 1 0.09 0 0 0.00 0.4996
Skin and subcutaneous tissue disorders 2 2 0.18 0 0 0.00 0.2494
Hyperhidrosis 1 1 0.09 0 0 0.00 0.4996
Eczema 1 1 0.09 0 0 0.00 0.4996
General disorders and administration site conditions 13 12 1.07 16 15 1.34 0.6994
Fever 13 12 1.07 16 15 1.34 0.6994
Gastrointestinal disorders 13 13 1.16 18 17 1.52 0.5821
Aphthous ulcer 1 1 0.09 0 0 0.00 0.4996

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White stool 0 0 0.00 1 1 0.09 1.0000
Abdominal pain 0 0 0.00 1 1 0.09 1.0000
Diarrhea 5 5 0.45 10 10 0.89 0.3001
Abdominal distension 1 1 0.09 1 1 0.09 1.0000
Oral ulcer 2 2 0.18 1 1 0.09 0.6245
Increased frequency of defecation 1 1 0.09 0 0 0.00 0.4996
Gastrointestinal pain 1 1 0.09 0 0 0.00 0.4996
Dyspepsia 1 1 0.09 3 3 0.27 0.6247
Infant vomiting 1 1 0.09 0 0 0.00 0.4996
Hard stool 0 0 0.00 1 1 0.09 1.0000
Blood and lymphatic system disorders 1 1 0.09 0 0 0.00 0.4996
Cold extremities 1 1 0.09 0 0 0.00 0.4996
Eye disorders 0 0 0.00 1 1 0.09 1.0000
Swollen eyelids 0 0 0.00 1 1 0.09 1.0000

Table. 50-2 Unsolicited events possibly related to study vaccine 0-7 days after vaccination
in subjects aged 3 through 71 months
PCV13-TT (N=1117) PCV7 (N=1119)
P

Events	Cases	Incidence rate	Events	Cases	Incidence rate
Unsolicited adverse reactions	51	4.12	51	43	3.84
Metabolism and nutrition disorders	1	1	0.09	0	0.00
Decreased appetite	1	1	0.09	0	0.00
Infections and infestations	15	1.34	11	11	0.98
Nasopharyngitis	2	2	0.18	1	0.09
Tonsillitis	1	1	0.09	0	0.00
Infectious pneumonia	0	0	0.00	1	0.09
Upper respiratory tract infection	10	10	0.90	9	0.80
Bronchitis	1	1	0.09	0	0.00
Herpes virus infection	1	1	0.09	0	0.00
Respiratory, thoracic and mediastinal disorders	24	2.22	31	28	2.50
Rhinorrhea	20	1.61	27	26	2.32
Nasal obstruction	3	3	0.27	1	0.09
Sneezing	1	1	0.09	2	0.18
Increased production of sputum	0	0	0.00	1	0.09
Psychiatric disorders	1	1	0.09	0	0.00
Irritability	1	1	0.09	0	0.00
Skin and subcutaneous tissue disorders	1	1	0.09	0	0.00
Hyperhidrosis	1	1	0.09	0	0.00
Gastrointestinal disorders	8	8	0.72	8	0.63
Aphthous ulcer	1	1	0.09	0	0.00
White stool	0	0	0.00	1	0.09
Abdominal pain	0	0	0.00	1	0.09
Abdominal distension	1	1	0.09	1	0.09
Oral ulcer	2	2	0.18	1	0.09
Increased frequency of defecation	1	1	0.09	0	0.00
Gastrointestinal pain	1	1	0.09	0	0.00
Dyspepsia	1	1	0.09	3	0.27
Infant vomiting	1	1	0.09	0	0.00
Hard stool	0	0	0.00	1	0.09
Blood and lymphatic system disorders	1	1	0.09	0	0.00

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Cold extremities 1 1 0.09 0 0 0.00 0.4996
Eye disorders 0 0 0.00 1 1 0.09 1.0000
Swollen eyelids 0 0 0.00 1 1 0.09 1.0000

Table. 50-3 Unsolicited events possibly related to study vaccine 8-30 days after vaccination
in subjects aged 3 through 71 months
PCV13-TT (N=1117) PCV7 (N=1119)
P

Events	Cases	Incidence rate	Events	Cases	Incidence rate
Unsolicited adverse reactions	24	1.61	30	26	2.32

Infections and infestations 1 1 0.09 1 1 0.09 1.0000
 Nasopharyngitis 1 1 0.09 0 0 0.00 0.4996
 Upper respiratory tract infection 0 0 0.00 1 1 0.09 1.0000
 Respiratory, thoracic and mediastinal disorders 4 4 0.36 3 2 0.18 0.4522
 Rhinorrhea 2 2 0.18 3 2 0.18 1.0000
 Cough 2 2 0.18 0 0 0.00 0.2494
 Skin and subcutaneous tissue disorders 1 1 0.09 0 0 0.00 0.4996
 Eczema 1 1 0.09 0 0 0.00 0.4996
 General disorders and administration site conditions 13 12 1.07 16 15 1.34 0.6994
 Fever 13 12 1.07 16 15 1.34 0.6994
 Gastrointestinal disorders 5 5 0.45 10 10 0.89 0.3001
 Diarrhea 5 5 0.45 10 10 0.89 0.3001

Table. 50-4 Unsolicited events related to study vaccine in subjects aged 3 months
 PCV13-TT (N=519) PCV7 (N=519)

P

Events	Cases	Incidence rate	Events	Cases	Incidence rate
Unsolicited adverse reactions	54	44	8.48	56	44
Metabolism and nutrition disorders	1	1	0.19	0	0.00
Decreased appetite	1	1	0.19	0	0.00
Infections and infestations	9	9	1.73	7	1.35
Nasopharyngitis	1	1	0.19	0	0.00
Tonsillitis	1	1	0.19	0	0.00
Infectious pneumonia	0	0	0.00	1	1
Upper respiratory tract infection	7	7	1.35	6	1.16
Respiratory, thoracic and mediastinal disorders	20	17	3.28	27	22
Rhinorrhea	14	11	2.12	23	20
Nasal obstruction	3	3	0.58	1	1
Cough	2	2	0.39	0	0.00
Sneezing	1	1	0.19	2	2
Increased production of sputum	0	0	0.00	1	1
Psychiatric disorders	1	1	0.19	0	0.00
Irritability	1	1	0.19	0	0.00
Skin and subcutaneous tissue disorders	1	1	0.19	0	0.00
Hyperhidrosis	1	1	0.19	0	0.00
General disorders and administration site conditions	11	10	1.93	10	9
Fever	11	10	1.93	10	9
Gastrointestinal disorders	10	10	1.93	12	12
Aphthous ulcer	1	1	0.19	0	0.00
Diarrhea	5	5	0.96	9	9
Abdominal distension	1	1	0.19	1	1
Increased frequency of defecation	1	1	0.19	0	0.00

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Dyspepsia 1 1 0.19 2 2 0.39 1.0000
 Infant vomiting 1 1 0.19 0 0 0.00 1.0000
 Blood and lymphatic system disorders 1 1 0.19 0 0 0.00 1.0000
 Cold extremities 1 1 0.19 0 0 0.00 1.0000

Table. 50-5 Unsolicited events related to study vaccine in subjects aged 7 through 11 months
 PCV13-TT (N=198) PCV7 (N=200)

P

Events	Cases	Incidence rate	Events	Cases	Incidence rate
Unsolicited adverse reactions	10	9	4.55	10	9
Infections and infestations	3	3	1.52	2	2
Nasopharyngitis	0	0	0.00	1	1
Upper respiratory tract infection	2	2	1.01	1	1
Bronchitis	1	1	0.51	0	0.00
Respiratory, thoracic and mediastinal disorders	3	3	1.52	3	3
Rhinorrhea	3	3	1.52	3	3
Skin and subcutaneous tissue disorders	1	1	0.51	0	0.00
Eczema	1	1	0.51	0	0.00
General disorders and administration site conditions	2	2	1.01	3	3
Fever	2	2	1.01	3	3
Gastrointestinal disorders	1	1	0.51	2	2
Abdominal pain	0	0	0.00	1	1
Gastrointestinal pain	1	1	0.51	0	0.00
Dyspepsia	0	0	0.00	1	1

Table. 50-6 Unsolicited events related to study vaccine in subjects aged 12 through 23 months
 PCV13-TT (N=200) PCV7 (N=200)

P

Events	Cases	Incidence rate	Events	Cases	Incidence rate
Unsolicited adverse reactions	9	8	4.00	10	8
Infections and infestations	3	3	1.50	1	1
Nasopharyngitis	2	2	1.00	0	0.00
Upper respiratory tract infection	1	1	0.50	1	1
Respiratory, thoracic and mediastinal disorders	4	4	2.00	3	3
Rhinorrhea	4	4	2.00	3	3
General disorders and administration site conditions	0	0	0.00	2	2
Fever	0	0	0.00	2	2
Gastrointestinal disorders	2	2	1.00	3	2
White stool	0	0	0.00	1	1
Diarrhea	0	0	0.00	1	1
Oral ulcer	2	2	1.00	0	0.00
Hard stool	0	0	0.00	1	1
Eye disorders	0	0	0.00	1	1
Swollen eyelids	0	0	0.00	1	1

Table. 50-7 Unsolicited events related to study vaccine in subjects aged 24 through 71 months

PCV13-TT (N=200) PCV7 (N=200)

P

Events Cases Incidence rate Events Cases Incidence rate

Unsolicited adverse reactions 2 2 1.00 5 5 2.50 0.4490

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Infections and infestations 1 1 0.50 2 2 1.00 1.0000

Upper respiratory tract infection 0 0 0.00 2 2 1.00 0.4987

Herpes virus infection 1 1 0.50 0 0 0.00 1.0000

Respiratory, thoracic and mediastinal disorders 1 1 0.50 1 1 0.50 1.0000

Rhinorrhea 1 1 0.50 1 1 0.50 1.0000

General disorders and administration site conditions 0 0 0.00 1 1 0.50 1.0000

Fever 0 0 0.00 1 1 0.50 1.0000

Gastrointestinal disorders 0 0 0.00 1 1 0.50 1.0000

Oral ulcer 0 0 0.00 1 1 0.50 1.0000

7.1.4. Adverse events unrelated to study vaccine

7.1.4.1. Solicited events unrelated to study vaccine

A total of 231 cases (313 events) were determined by the investigators as solicited adverse events unrelated to study vaccine, including 109 cases in the PCV13-TT and 122 cases in the control group, and the incidence rate was 9.76% and 10.90% respectively. No significant intergroup difference was observed for the incidence of vaccine-independent solicited adverse events.

Symptoms of solicited adverse events unrelated to study vaccine were mainly fever, cough, diarrhea, and nausea / vomiting.

The severity of vaccine-independent solicited adverse events was concentrated on grade 1 and 2.

The severity of 6 cases (0.54%) in the PCV13-TT and 13 cases (1.16%) in the control group were determined to be grade 3. No significant intergroup difference was observed in the incidence.

Fever, cough, and diarrhea were the most common vaccine-independent solicited adverse events in all age groups.

Incidences of solicited adverse events unrelated to study vaccine are shown in Table 51-1 to 51-5.

Table. 51-1 Solicited adverse events unrelated to study vaccine in PCV13-TT and PCV7

groups of subjects aged 3 through 71 months

PCV13-TT (N=1117) PCV7 (N=1119)

P

Events Cases % Events Cases %

Solicited AE unrelated

150 109 9.76 163 122 10.90 0.4045

to study

Grade 1 57 49 4.39 56 51 4.56 0.9186

Grade 2 87 72 6.45 94 73 6.52 1.0000

Grade 3 6 6 0.54 13 13 1.16 0.1653

Systemic reactions 150 109 9.76 161 120 10.72 0.4856

Grade 1 57 49 4.39 54 49 4.38 1.0000

Grade 2 87 72 6.45 94 73 6.52 1.0000

Grade 3 6 6 0.54 13 13 1.16 0.1653

Fever 56 54 4.83 67 62 5.54 0.5046

Grade 1 23 23 2.06 21 20 1.79 0.6482

Grade 2 29 28 2.51 37 36 3.22 0.3748

Grade 3 4 4 0.36 9 9 0.80 0.2654

Fatigue / somnolence 1 1 0.09 1 1 0.09 1.0000

Grade 1 0 0 0.00 1 1 0.09 1.0000

Grade 2 1 1 0.09 0 0 0.00 0.4996

Grade 3 0 0 0.00 0 0 0.00 /

Nausea / vomiting 11 10 0.90 11 11 0.98 1.0000

Grade 1 5 5 0.45 10 10 0.89 0.3001

Grade 2 6 5 0.45 1 1 0.09 0.1242

Grade 3 0 0 0.00 0 0 0.00 /

Diarrhea 23 22 1.97 24 23 2.06 1.0000

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Grade 1 7 6 0.54 7 7 0.63 1.0000

Grade 2 15 15 1.34 13 12 1.07 0.5690

Grade 3 1 1 0.09 4 4 0.36 0.3744

Cough 53 52 4.66 54 53 4.74 1.0000

Grade 1 20 20 1.79 15 15 1.34 0.4010

Grade 2 32 31 2.78 39 38 3.40 0.4634

Grade 3 1 1 0.09 0 0 0.00 0.4996

Abnormal crying 1 1 0.09 1 1 0.09 1.0000

Grade 1 0 0 0.00 0 0 0.00 /

Grade 2 1 1 0.09 1 1 0.09 1.0000

Grade 3 0 0 0.00 0 0 0.00 /

Allergic reactions 5 5 0.45 3 3 0.27 0.5067

Grade 1 2 2 0.18 0 0 0.00 0.2494

Grade 2 3 3 0.27 3 3 0.27 1.0000

Grade 3 0 0 0.00 0 0 0.00 /

Vaccination reactions /

0 0 0.00 2 2 0.18 0.4998

site redness

Grade 1 0 0 0.00 2 2 0.18 0.4998

Grade 2 0 0 0.00 0 0 0.00 /

Grade 3 0 0 0.00 0 0 0.00 /

Table. 51-2 Solicited adverse events unrelated to study vaccine in subjects aged 3 months

PCV13-TT (N=519) PCV7 (N=519)

P

Events	Cases	%	Events	Cases	%		
Fever	27	26	5.01	34	30	5.78	0.6806
Fatigue / somnolence	1	1	0.19	0	0	0.00	1.0000
Nausea / vomiting	3	2	0.39	5	5	0.96	0.4515
Diarrhea	11	11	2.12	17	16	3.08	0.4360
Cough	27	26	5.01	23	22	4.24	0.6579
Abnormal crying	1	1	0.19	0	0	0.00	1.0000
Allergic reactions	2	2	0.39	2	2	0.39	1.0000

Table. 51-3 Solicited adverse events unrelated to study vaccine in subjects aged 7 through 11 months

PCV13-TT (N=198) PCV7 (N=200)
P

Events	Cases	%	Events	Cases	%		
Fever	14	14	7.07	15	14	7.00	1.0000
Nausea / vomiting	3	3	1.52	4	4	2.00	1.0000
Diarrhea	7	6	3.03	5	5	2.50	0.7704
Cough	10	10	5.05	17	17	8.50	0.2314
Abnormal crying	0	0	0.00	1	1	0.50	1.0000
Allergic reactions	1	1	0.51	1	1	0.50	1.0000
Vaccination site redness	0	0	0.00	1	1	0.50	1.0000

Table. 51-4 Solicited adverse events unrelated to study vaccine in subjects aged 12 through

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23 months

PCV13-TT (N=200) PCV7 (N=200)
P

Events	Cases	%	Events	Cases	%		
Fever	12	11	5.50	14	14	7.00	0.6803
Fatigue / somnolence	0	0	0.00	1	1	0.50	1.0000
Nausea / vomiting	3	3	1.50	2	2	1.00	1.0000
Diarrhea	3	3	1.50	2	2	1.00	1.0000
Cough	13	13	6.50	8	8	4.00	0.3703
Vaccination site redness	0	0	0.00	1	1	0.50	1.0000

Table. 51-5 Solicited adverse events unrelated to study vaccine in subjects aged 24 through 71 months

PCV13-TT (N=200) PCV7 (N=200)
P

Events	Cases	%	Events	Cases	%		
Fever	3	3	1.50	4	4	2.00	1.0000
Nausea / vomiting	2	2	1.00	0	0	0.00	0.4987
Diarrhea	2	2	1.00	0	0	0.00	0.4987
Cough	3	3	1.50	6	6	3.00	0.5028
Allergic reactions	2	2	1.00	0	0	0.00	0.4987

7.1.4.2. Unsolicited events unrelated to study vaccine

A total of 717 cases (1155 events) of unsolicited adverse events unrelated to study vaccine were collected during the observation period, including 325 cases in the PCV13-TT and 365 cases in the control group, with incidence rates being 31.51% and 32.62%, respectively. No significant intergroup difference was observed for the incidence of vaccine-independent unsolicited adverse events.

Unsolicited adverse events unrelated to study vaccine were primarily infections and infestations, followed by gastrointestinal disorders, respiratory, thoracic and mediastinal disorders, general disorders and administration site conditions. Fever was observed with the highest proportion within the class of general disorders and administration site conditions, and was of relatively higher incidence rates among subjects aged 3 months, 7-11 months, and 12-23 months that were within 2 years old. No significant difference was observed for unsolicited adverse events between the PCV13-TT and the control group.

Incidences of vaccine-independent unsolicited adverse events in the study and control group of each age group are shown in Table 52-1-1 to 52-5.

Table. 52-1-1 Unsolicited events unrelated to study vaccine in subjects aged 3 through 71 months

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PCV13-TT (N=1117) PCV7 (N=1119)
Incidence Incidence
P

Events	Cases	Events	Cases	rate	rate		
Unsolicited adverse reactions	566	352	31.51	589	365	32.62	0.5868
Hepatobiliary disorders	0	0	0.00	1	1	0.09	1.0000
Abnormal liver function	0	0	0.00	1	1	0.09	1.0000
Infections and infestations	274	213	19.07	293	223	19.93	0.6311
Upper respiratory tract infection	163	135	12.09	182	150	13.40	0.3747
Nasopharyngitis	28	26	2.33	24	20	1.79	0.3764
Rhinitis	0	0	0.00	1	1	0.09	1.0000
Nasosinusitis	1	1	0.09	0	0	0.00	0.4996
Tonsillitis	3	3	0.27	4	4	0.36	1.0000
Viral encephalitis	1	1	0.09	1	1	0.09	1.0000
Viral rash	0	0	0.00	2	2	0.18	0.4998
Viral gastroenteritis	0	0	0.00	3	3	0.27	0.2497
Angular cheilitis	1	1	0.09	2	2	0.18	1.0000
Rubella	0	0	0.00	1	1	0.09	1.0000
Infectious enteritis	1	1	0.09	0	0	0.00	0.4996
Infectious pneumonia	16	16	1.43	10	10	0.89	0.2447

Infectious diarrhea 0 0 0.00 1 1 0.09 1.0000
 Anal abscess 0 0 0.00 1 1 0.09 1.0000
 Laryngitis 2 2 0.18 1 1 0.09 0.6245
 Respiratory tract infection 1 1 0.09 0 0 0.00 0.4996
 Conjunctivitis 0 0 0.00 1 1 0.09 1.0000
 Oral herpes 2 2 0.18 3 3 0.27 1.0000
 Measles 1 1 0.09 0 0 0.00 0.4996
 Urethritis 0 0 0.00 1 1 0.09 1.0000
 Sepsis 1 1 0.09 2 2 0.18 1.0000
 Abscess 1 1 0.09 0 0 0.00 0.4996
 Bacterial skin infections 1 1 0.09 0 0 0.00 0.4996
 Tracheitis 2 2 0.18 2 2 0.18 1.0000
 Upper respiratory tract viral
 infection 0 0 0.00 2 2 0.18 0.4998

Table. 52-1-2 Unsolicited events unrelated to study vaccine in subjects aged 3 through 71 months

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PCV13-TT (N=1117) PCV7 (N=1119)
 Incidence Incidence P
 Events Cases Events Cases
 rate rate
 Upper respiratory and digestive
 tract infections 1 1 0.09 0 0 0.00 0.4996
 Hand-foot-mouth disease 9 9 0.81 12 12 1.07 0.6621
 Gastroenteritis 1 1 0.09 2 2 0.18 1.0000
 Bacterial tonsillitis 1 1 0.09 0 0 0.00 0.4996
 Laryngeal tonsillitis 2 2 0.18 1 1 0.09 0.6245
 Laryngitis 1 1 0.09 2 2 0.18 1.0000
 Eye infections 1 1 0.09 0 0 0.00 0.4996
 Roseola infantum 3 3 0.27 8 8 0.71 0.2254
 Bronchitis 16 15 1.34 11 11 0.98 0.4391
 Mycoplasma pneumonia 1 1 0.09 0 0 0.00 0.4996
 Otitis media 2 2 0.18 2 1 0.09 0.6245
 Herpes virus infection 1 1 0.09 2 2 0.18 1.0000
 Herpangina 10 10 0.90 8 8 0.71 0.6460
 Herpes laryngitis 0 0 0.00 1 1 0.09 1.0000
 Investigations 1 1 0.09 7 6 0.54 0.1244
 Abnormal electromyographic
 findings 0 0 0.00 1 1 0.09 1.0000
 Increased biological markers of
 myocardial necrosis 1 1 0.09 6 6 0.54 0.1244
 Nervous system disorders 2 2 0.18 2 2 0.18 1.0000
 Febrile convulsion 1 1 0.09 1 1 0.09 1.0000
 Psychomotor retardation 0 0 0.00 1 1 0.09 1.0000
 Epilepsy 1 1 0.09 0 0 0.00 0.4996
 Injury, poisoning and procedural
 complications 1 1 0.09 0 0 0.00 0.4996
 Animal bite 1 1 0.09 0 0 0.00 0.4996
 Respiratory, thoracic and
 mediastinal disorders 94 80 7.16 86 72 6.43 0.5027
 Rhinorrhea 54 51 4.57 50 45 4.02 0.5334
 Nasal obstruction 2 2 0.18 0 0 0.00 0.2494
 Epistaxis 1 1 0.09 1 1 0.09 1.0000
 Tonsillar hypertrophy 1 1 0.09 0 0 0.00 0.4996
 Dysphonia 1 1 0.09 0 0 0.00 0.4996
 Pneumonia 2 2 0.18 7 6 0.54 0.2882
 Cough 31 30 2.69 23 21 1.88 0.2060
 Sneezing 1 1 0.09 2 2 0.18 1.0000
 Increased production of sputum 1 1 0.09 1 1 0.09 1.0000
 Pharyngeal erythema 0 0 0.00 1 1 0.09 1.0000
 Pharyngeal edema 0 0 0.00 1 1 0.09 1.0000
 Immune system disorders 1 1 0.09 0 0 0.00 0.4996
 Drug-induced hypersensitivity
 reactions 1 1 0.09 0 0 0.00 0.4996
 Skin and subcutaneous tissue
 disorders 7 6 0.54 10 10 0.89 0.4529
 Maculopapular rash 0 0 0.00 1 1 0.09 1.0000
 Allergic dermatitis 0 0 0.00 1 1 0.09 1.0000
 Erythema multiforme 1 1 0.09 0 0 0.00 0.4996

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Table. 52-1-3 Unsolicited events unrelated to study vaccine in subjects aged 3 through 71 months

PCV13-TT (N=1117) PCV7 (N=1119)
 Incidence Incidence P
 Events Cases Events Cases

	rate	rate
Rash	0 0 0.00	1 1 0.09 1.0000
Eczema	0 0 0.00	5 5 0.45 0.0622
Blister rash	2 1 0.09	0 0 0.00 0.4996
Urticaria	4 4 0.36	2 2 0.18 0.4522
General disorders and		
72 65 5.82 85 77 6.88		0.3402
administration site conditions		
Fever	71 64 5.73 84 76 6.79	0.3370
Crying	1 1 0.09	0 0 0.00 0.4996
Inflammation	0 0 0.00	1 1 0.09 1.0000
Reproductive system and breast		
0 0 0.00	1 1 0.09	1.0000
disorders		
Vulvar erythema	0 0 0.00	1 1 0.09 1.0000
Gastrointestinal disorders	109 97 8.68 99 86 7.69	0.3971
Aphthous ulcer	3 3 0.27	1 1 0.09 0.3742
Constipation	1 1 0.09	0 0 0.00 0.4996
Enteritis	4 4 0.36	2 2 0.18 0.4522
Intestinal obstruction	1 1 0.09	0 0 0.00 0.4996
Inguinal hernia	1 1 0.09	0 0 0.00 0.4996
Abdominal pain	2 2 0.18	1 1 0.09 0.6245
Diarrhea	69 66 5.91 70 65 5.81	0.9285
Abdominal distension	3 3 0.27	7 6 0.54 0.5069
Functional gastrointestinal disorders	1 1 0.09	1 1 0.09 1.0000
Oral ulcer	3 3 0.27	5 5 0.45 0.7261
Gastritis	1 1 0.09	0 0 0.00 0.4996
Dyspepsia	19 18 1.61 11 9 0.80	0.0851
Infant vomiting	1 1 0.09	1 1 0.09 1.0000
Blood and lymphatic system		
4 4 0.36	4 3 0.27	0.7260
disorders		
Granulopenia	2 2 0.18	0 0 0.00 0.2494
Lymphadenitis	1 1 0.09	0 0 0.00 0.4996
Anemia	1 1 0.09	3 3 0.27 0.6247
Neutropenia	0 0 0.00	1 1 0.09 1.0000
Eye disorders	1 1 0.09	1 1 0.09 1.0000
Eye congestion	1 1 0.09	0 0 0.00 0.4996
Blepharitis	0 0 0.00	1 1 0.09 1.0000

Table. 52-2-1 Unsolicited events unrelated to study vaccine in subjects aged 3 months

PCV13-TT (N=519) PCV7 (N=519)

Incidence Incidence P

Events Cases Events Cases

rate rate

Unsolicited adverse reactions 295 181 34.87 298 191 36.80 0.5602

Infections and infestations 140 109 21.00 143 113 21.77 0.8204

Nasopharyngitis 14 14 2.70 9 7 1.35 0.1847

Rhinitis 0 0 0.00 1 1 0.19 1.0000

Tonsillitis 0 0 0.00 1 1 0.19 1.0000

Viral rash 0 0 0.00 2 2 0.39 0.4995

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Viral gastroenteritis 0 0 0.00 1 1 0.19 1.0000

Angular cheilitis 0 0 0.00 1 1 0.19 1.0000

Rubella 0 0 0.00 1 1 0.19 1.0000

Infectious enteritis 1 1 0.19 0 0 0.00 1.0000

Infectious pneumonia 8 8 1.54 9 9 1.73 1.0000

Anal abscess 0 0 0.00 1 1 0.19 1.0000

Laryngitis 0 0 0.00 1 1 0.19 1.0000

Respiratory tract infection 1 1 0.19 0 0 0.00 1.0000

Measles 1 1 0.19 0 0 0.00 1.0000

Urethritis 0 0 0.00 1 1 0.19 1.0000

Sepsis 1 1 0.19 0 0 0.00 1.0000

Abscess 1 1 0.19 0 0 0.00 1.0000

Tracheitis 2 2 0.39 2 2 0.39 1.0000

Upper respiratory tract viral

0 0 0.00 1 1 0.19 1.0000

infection

Upper respiratory tract infection 91 74 14.26 92 78 15.03 0.7923

Hand-foot-mouth disease 1 1 0.19 1 1 0.19 1.0000

Laryngitis 1 1 0.19 1 1 0.19 1.0000

Eye infections 1 1 0.19 0 0 0.00 1.0000

Roseola infantum 3 3 0.58 7 7 1.35 0.3414

Bronchitis 10 9 1.73 6 6 1.16 0.6046

Otitis media 1 1 0.19 2 1 0.19 1.0000

Herpes virus infection 0 0 0.00 1 1 0.19 1.0000

Herpangina 3 3 0.58 1 1 0.19 0.6243

Herpes laryngitis 0 0 0.00 1 1 0.19 1.0000

Investigations 1 1 0.19 2 2 0.39 1.0000

Increased biological markers of

1 1 0.19 2 2 0.39 1.0000

myocardial necrosis

Nervous system disorders 2 2 0.39 1 1 0.19 1.0000

Febrile convulsion 1 1 0.19 0 0 0.00 1.0000

Psychomotor retardation 0 0 0.00 1 1 0.19 1.0000

Epilepsy 1 1 0.19 0 0 0.00 1.0000

Respiratory, thoracic and

45 38 7.32 40 33 6.36 0.6231

mediastinal disorders

Rhinorrhea 24 23 4.43 22 20 3.85 0.7558
Dysphonia 1 1 0.19 0 0 0.00 1.0000
Pneumonia 1 1 0.19 6 5 0.96 0.2174
Cough 17 16 3.08 9 8 1.54 0.1468

Table. 52-2-2. Unsolicited events unrelated to study vaccine in subjects aged 3 months

PCV13-TT (N=519) PCV7 (N=519)

Incidence Incidence P

Events Cases Events Cases

rate rate

Sneezing 1 1 0.19 1 1 0.19 1.0000

Increased production of sputum 1 1 0.19 0 0 0.00 1.0000

Pharyngeal erythema 0 0 0.00 1 1 0.19 1.0000

Pharyngeal edema 0 0 0.00 1 1 0.19 1.0000

Skin and subcutaneous tissue

2 2 0.39 5 5 0.96 0.4515

disorders

Maculopapular rash 0 0 0.00 1 1 0.19 1.0000

Eczema 0 0 0.00 4 4 0.77 0.1243

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Urticaria 2 2 0.39 0 0 0.00 0.4995

General disorders and administration

33 31 5.97 37 35 6.74 0.7030

site conditions

Fever 32 30 5.78 36 34 6.55 0.6990

Crying 1 1 0.19 0 0 0.00 1.0000

Inflammation 0 0 0.00 1 1 0.19 1.0000

Reproductive system and breast

0 0 0.00 1 1 0.19 1.0000

disorders

Vulvar erythema 0 0 0.00 1 1 0.19 1.0000

Gastrointestinal disorders 68 60 11.56 66 57 10.98 0.8445

Aphthous ulcer 2 2 0.39 0 0 0.00 0.4995

Enteritis 4 4 0.77 2 2 0.39 0.6866

Abdominal pain 1 1 0.19 1 1 0.19 1.0000

Diarrhea 43 41 7.90 48 45 8.67 0.7357

Abdominal distension 3 3 0.58 5 4 0.77 1.0000

Functional gastrointestinal disorders 1 1 0.19 0 0 0.00 1.0000

Oral ulcer 1 1 0.19 1 1 0.19 1.0000

Gastritis 1 1 0.19 0 0 0.00 1.0000

Dyspepsia 11 10 1.93 8 6 1.16 0.4509

Infant vomiting 1 1 0.19 1 1 0.19 1.0000

Blood and lymphatic system

3 3 0.58 3 2 0.39 1.0000

disorders

Granulopenia 2 2 0.39 0 0 0.00 0.4995

Anemia 1 1 0.19 2 2 0.39 1.0000

Neutropenia 0 0 0.00 1 1 0.19 1.0000

Eye disorders 1 1 0.19 0 0 0.00 1.0000

Eye congestion 1 1 0.19 0 0 0.00 1.0000

Table. 52-3 Unsolicited events unrelated to study vaccine in subjects aged 7 through 11 months

PCV13-TT (N=198) PCV7 (N=200)

Incidence Incidence P

Events Cases Events Cases

rate rate

Unsolicited adverse reactions 121 75 37.88 158 82 41.00 0.5397

Infections and infestations 56 44 22.22 75 49 24.50 0.6363

Nasopharyngitis 5 5 2.53 7 6 3.00 1.0000

Tonsillitis 2 2 1.01 1 1 0.50 0.6222

Viral gastroenteritis 0 0 0.00 1 1 0.50 1.0000

Angular cheilitis 1 1 0.51 1 1 0.50 1.0000

Infectious pneumonia 3 3 1.52 0 0 0.00 0.1222

Infectious diarrhea 0 0 0.00 1 1 0.50 1.0000

Laryngitis 2 2 1.01 0 0 0.00 0.2469

Oral herpes 0 0 0.00 2 2 1.00 0.4988

Sepsis 0 0 0.00 2 2 1.00 0.4988

Upper respiratory tract infection 32 26 13.13 47 36 18.00 0.2137

Hand-foot-mouth disease 2 2 1.01 2 2 1.00 1.0000

Gastroenteritis 0 0 0.00 1 1 0.50 1.0000

Bacterial tonsillitis 1 1 0.51 0 0 0.00 0.4975

Laryngeal tonsillitis 1 1 0.51 1 1 0.50 1.0000

Roseola infantum 0 0 0.00 1 1 0.50 1.0000

Bronchitis 2 2 1.01 2 2 1.00 1.0000

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Herpes virus infection 0 0 0.00 1 1 0.50 1.0000

Herpangina 5 5 2.53 5 5 2.50 1.0000

Investigations 0 0 0.00 1 1 0.50 1.0000

Increased biological markers of

0 0 0.00 1 1 0.50 1.0000

myocardial necrosis

Respiratory, thoracic and

21 18 9.09 25 20 10.00 0.8649

mediastinal disorders

Rhinorrhea 15 14 7.07 18 15 7.50 1.0000

Nasal obstruction 2 2 1.01 0 0 0.00 0.2469
 Cough 4 4 2.02 5 5 2.50 1.0000
 Sneezing 0 0 0.00 1 1 0.50 1.0000
 Increased production of sputum 0 0 0.00 1 1 0.50 1.0000
 Skin and subcutaneous tissue
 4 3 1.52 3 3 1.50 1.0000
 disorders
 Allergic dermatitis 0 0 0.00 1 1 0.50 1.0000
 Erythema multiforme 1 1 0.51 0 0 0.00 0.4975
 Blister rash 2 1 0.51 0 0 0.00 0.4975
 Urticaria 1 1 0.51 2 2 1.00 1.0000
 General disorders and
 17 15 7.58 30 24 12.00 0.1770
 administration site conditions
 Fever 17 15 7.58 30 24 12.00 0.1770
 Gastrointestinal disorders 22 20 10.10 23 19 9.50 0.8675
 Aphthous ulcer 1 1 0.51 1 1 0.50 1.0000
 Diarrhea 17 16 8.08 16 14 7.00 0.7083
 Abdominal distension 0 0 0.00 1 1 0.50 1.0000
 Oral ulcer 0 0 0.00 2 2 1.00 0.4988
 Dyspepsia 4 4 2.02 3 3 1.50 0.7231
 Blood and lymphatic system
 1 1 0.51 1 1 0.50 1.0000
 disorders
 Lymphadenitis 1 1 0.51 0 0 0.00 0.4975
 Anemia 0 0 0.00 1 1 0.50 1.0000

Table. 52-4-1 Unsolicited events unrelated to study vaccine in subjects aged 12 through 23 months

PCV13-TT (N=200) PCV7 (N=200)
 Incidence Incidence P
 Events Cases Events Cases
 rate rate
 Unsolicited adverse reactions 116 67 33.50 89 52 26.00 0.1255
 Hepatobiliary disorders 0 0 0.00 1 1 0.50 1.0000
 Abnormal liver function 0 0 0.00 1 1 0.50 1.0000
 Infections and infestations 56 40 20.00 52 38 19.00 0.8996
 Nasopharyngitis 4 3 1.50 5 4 2.00 1.0000
 Tonsillitis 1 1 0.50 2 2 1.00 1.0000
 Viral encephalitis 1 1 0.50 1 1 0.50 1.0000
 Infectious pneumonia 5 5 2.50 1 1 0.50 0.2152
 Conjunctivitis 0 0 0.00 1 1 0.50 1.0000
 Oral herpes 2 2 1.00 1 1 0.50 1.0000
 Bacterial skin infections 1 1 0.50 0 0 0.00 1.0000
 Upper respiratory tract viral
 0 0 0.00 1 1 0.50 1.0000
 infection
 Upper respiratory tract infection 27 22 11.00 26 19 9.50 0.7420

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Upper respiratory and digestive
 1 1 0.50 0 0 0.00 1.0000
 tract infections
 Hand-foot-mouth disease 6 6 3.00 8 8 4.00 0.7868
 Gastroenteritis 0 0 0.00 1 1 0.50 1.0000
 Laryngeal tonsillitis 1 1 0.50 0 0 0.00 1.0000
 Laryngitis 0 0 0.00 1 1 0.50 1.0000
 Bronchitis 3 3 1.50 2 2 1.00 1.0000
 Mycoplasma pneumonia 1 1 0.50 0 0 0.00 1.0000
 Herpes virus infection 1 1 0.50 0 0 0.00 1.0000
 Herpangina 2 2 1.00 2 2 1.00 1.0000
 Investigations 0 0 0.00 4 3 1.50 0.2481
 Abnormal electromyographic
 0 0 0.00 1 1 0.50 1.0000
 findings
 Increased biological markers of
 0 0 0.00 3 3 1.50 0.2481
 myocardial necrosis
 Nervous system disorders 0 0 0.00 1 1 0.50 1.0000
 Febrile convulsion 0 0 0.00 1 1 0.50 1.0000
 Injury, poisoning and procedural
 1 1 0.50 0 0 0.00 1.0000
 complications
 Animal bite 1 1 0.50 0 0 0.00 1.0000
 Respiratory, thoracic and
 21 18 9.00 10 10 5.00 0.1692
 mediastinal disorders
 Rhinorrhea 14 13 6.50 7 7 3.50 0.2509
 Epistaxis 0 0 0.00 1 1 0.50 1.0000
 Tonsillar hypertrophy 1 1 0.50 0 0 0.00 1.0000
 Pneumonia 1 1 0.50 1 1 0.50 1.0000
 Cough 5 5 2.50 1 1 0.50 0.2152
 Immune system disorders 1 1 0.50 0 0 0.00 1.0000
 Drug-induced hypersensitivity
 1 1 0.50 0 0 0.00 1.0000
 reactions
 Skin and subcutaneous tissue
 1 1 0.50 0 0 0.00 1.0000
 disorders

Urticaria 1 1 0.50 0 0 0.00 1.0000
 General disorders and
 19 16 8.00 13 13 6.50 0.7004
 administration site conditions
 Fever 19 16 8.00 13 13 6.50 0.7004

Table. 52-4-2 Unsolicited events unrelated to study vaccine in subjects aged 12 through 23 months

PCV13-TT (N=200) PCV7 (N=200)
 Incidence Incidence P
 Events Cases Events Cases
 rate rate
 Gastrointestinal disorders 17 15 7.50 7 7 3.50 0.1230
 Constipation 1 1 0.50 0 0 0.00 1.0000
 Intestinal obstruction 1 1 0.50 0 0 0.00 1.0000
 Inguinal hernia 1 1 0.50 0 0 0.00 1.0000
 Diarrhea 9 9 4.50 4 4 2.00 0.2589
 Abdominal distension 0 0 0.00 1 1 0.50 1.0000
 Functional gastrointestinal
 0 0 0.00 1 1 0.50 1.0000
 disorders
 Oral ulcer 1 1 0.50 1 1 0.50 1.0000
 Dyspepsia 4 4 2.00 0 0 0.00 0.1231
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Eye disorders 0 0 0.00 1 1 0.50 1.0000
 Blepharitis 0 0 0.00 1 1 0.50 1.0000

Table. 52-5 Unsolicited events unrelated to study vaccine in subjects aged 24 through 71 months

PCV13-TT (N=200) PCV7 (N=200)
 Incidence Incidence P
 Events Cases Events Cases
 rate rate
 Unsolicited adverse reactions 34 29 14.50 44 40 20.00 0.1854
 Infections and infestations 22 20 10.00 23 23 11.50 0.7472
 Nasopharyngitis 5 4 2.00 3 3 1.50 1.0000
 Nasosinusitis 1 1 0.50 0 0 0.00 1.0000
 Viral gastroenteritis 0 0 0.00 1 1 0.50 1.0000
 Upper respiratory tract infection 13 13 6.50 17 17 8.50 0.5697
 Hand-foot-mouth disease 0 0 0.00 1 1 0.50 1.0000
 Gastroenteritis 1 1 0.50 0 0 0.00 1.0000
 Bronchitis 1 1 0.50 1 1 0.50 1.0000
 Otitis media 1 1 0.50 0 0 0.00 1.0000
 Respiratory, thoracic and
 7 6 3.00 11 9 4.50 0.6002
 mediastinal disorders
 Rhinorrhea 1 1 0.50 3 3 1.50 0.6231
 Epistaxis 1 1 0.50 0 0 0.00 1.0000
 Cough 5 5 2.50 8 7 3.50 0.7710
 Skin and subcutaneous tissue
 0 0 0.00 2 2 1.00 0.4987
 disorders
 Rash 0 0 0.00 1 1 0.50 1.0000
 Eczema 0 0 0.00 1 1 0.50 1.0000
 General disorders and
 3 3 1.50 5 5 2.50 0.7238
 administration site conditions
 Fever 3 3 1.50 5 5 2.50 0.7238
 Gastrointestinal disorders 2 2 1.00 3 3 1.50 1.0000
 Abdominal pain 1 1 0.50 0 0 0.00 1.0000
 Diarrhea 0 0 0.00 2 2 1.00 0.4987
 Oral ulcer 1 1 0.50 1 1 0.50 1.0000

7.1.5. The Incidence of Adverse Events Observed in 2-month and 3-month Subjects in Different Immunization Schedules

The comparison of overall incidence of adverse events in the 2-month study group (0, 2, 4 schedule) with that in the 3-month study and control group was shown in Table 53-1.

No significant difference was observed in the overall adverse events, vaccine-related adverse events and vaccine-independent adverse events between the 2-month study group and the 3-month groups. The incidence of adverse events with a severity reaching grade 3 in the 2-month study group was lower than that in the 3-month study and control group.

The incidence of hypodynamia/somnolence and local redness in the 2-month study group was lower than that in the 3-month study and control group. The incidence of fever and redness in the 2-month study group and 3-month study group was lower than that in the 3-month control group. No significant difference was observed in the incidence of other adverse reactions among the three groups.

The incidence of adverse events in the 2-month study group and the 3-month study/control group in different immunization schedules are shown in Table 53-1 to 53-2.

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Table 53-1 Incidence rates of adverse reactions after vaccination in PCV13-TT (2M) and

PCV13-TT (3M) and PCV7 groups
 PCV13-TT (2M) PCV13-TT (3M) PCV7(3M)
 (N=517) (N=519) P
 Events Cases % Events Cases % Events Cases %
 Any AE 1278 412 79.69 1306 415 79.96 1447 440 84.78 0.0557
 Grade 1 775 339 65.57 760 338 65.13 867 365 70.33 0.1416

Grade 2 484 258 49.90 489 243 46.82 538 271 52.22 0.2192
 Grade 3 13 11 2.13 46 35 6.74 32 27 5.20 0.0009
 Grade 4 6 5 0.97 11 8 1.54 10 8 1.54 0.6802
 Vaccine-related AE 874 361 69.83 939 371 71.48 1068 392 75.53 0.1071
 Grade 1 633 310 59.96 661 315 60.69 748 344 66.28 0.0707
 Grade 2 235 154 29.79 258 160 30.83 311 188 36.22 0.0613
 Grade 3 5 5 0.97 18 15 2.89 9 8 1.54 0.0651
 Grade 4 1 1 0.19 2 1 0.19 0 0 0.00 0.7771
 Solicited AE 854 360 69.63 885 365 70.33 1012 385 74.18 0.2155
 Grade 1 621 310 59.96 637 313 60.31 725 338 65.13 0.1611
 Grade 2 227 150 29.01 229 146 28.13 278 178 34.30 0.0676
 Grade 3 5 5 0.97 18 15 2.89 9 8 1.54 0.0651
 Grade 4 1 1 0.19 1 1 0.19 0 0 0.00 0.7771
 unsolicited AE 20 17 3.29 54 44 8.48 56 44 8.48 0.0003
 Grade 1 12 10 1.93 24 19 3.66 23 21 4.05 0.1136
 Grade 2 8 7 1.35 29 26 5.01 33 25 4.82 0.0009
 Grade 3 ///////////////
 Grade 4 /// 1 1 0.19 0 0 0.00
 1.0000
 Vaccine-unrelated
 AE 404 227 43.91 367 197 37.96 379 211 40.66 0.1524
 Grade 1 142 105 20.31 99 78 15.03 119 90 17.34 0.0825
 Grade 2 249 163 31.53 231 132 25.43 227 144 27.75 0.0909
 Grade 3 8 6 1.16 28 21 4.05 23 20 3.85 0.0057
 Grade 4 5 5 0.97 9 7 1.35 10 8 1.54 0.7830
 Solicited AE 56 50 9.67 72 55 10.60 81 56 10.79 0.8237
 Grade 1 28 25 4.84 27 24 4.62 32 30 5.78 0.6600
 Grade 2 24 22 4.26 42 34 6.55 45 31 5.97 0.2329
 Grade 3 4 4 0.77 3 3 0.58 4 4 0.77 0.9342
 Grade 4 ///////////////
 unsolicited AE 348 209 40.43 295 181 34.87 298 191 36.80 0.1760
 Grade 1 114 91 17.60 72 65 12.52 87 72 13.87 0.0606
 Grade 2 225 154 29.79 189 124 23.89 182 135 26.01 0.0954
 Grade 3 4 3 0.58 25 19 3.66 19 17 3.28 0.0009
 Grade 4 5 5 0.97 9 7 1.35 10 8 1.54 0.7830

Table 53-2 Analysis of symptoms of systemic and local reactions in PCV13-TT (3M),
 PCV13-TT (2M) and PCV7 groups
 PCV13-TT (2M) PCV13-TT (3M) PCV7(3M)
 (N=517) (N=519) (N=519) P
 Events Cases % Events Cases % Events Cases %
 Fever 533 316 61.12 515 301 58.00 628 344 66.28 0.0211
 Fatigue/Somnolence 17 16 3.09 37 32 6.17 29 27 5.20 0.0562
 Nausea/vomiting 35 31 6.00 38 30 5.78 22 19 3.66 0.1637

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Diarrhea 71 57 11.03 76 67 12.91 74 62 11.95 0.6515
 Myalgia 1 1 0.19 1 1 0.19 2 2 0.39 1.0000
 Cough 32 29 5.61 40 38 7.32 41 34 6.55 0.5477
 Abnormal crying 69 54 10.44 68 54 10.40 62 49 9.44 0.8394
 Allergic reactions 3 3 0.58 3 3 0.58 4 3 0.58 1.0000
 Local pain 10 10 1.93 16 12 2.31 12 10 1.93 0.9248
 Local redness 48 41 7.93 59 55 10.60 92 71 13.68 0.0113
 Local swelling 19 17 3.29 12 10 1.93 25 18 3.47 0.2730
 Local induration 16 14 2.71 18 15 2.89 19 18 3.47 0.8058
 Local itching 0 0 0.00 2 2 0.39 2 2 0.39 0.5547

7.1.6. The Comparison of Adverse Events after Vaccination of Each Dose
 7.1.6.1. The Incidence of Adverse Events after Vaccination of Each Dose in the 2-month
 Group and 3-month Study/Control Group
 The number of adverse events did not increase with the number of vaccination in 2-month group
 and 3-month study/control group. The incidence of vaccine-related adverse events after
 vaccination of each dose in the 2-month and 3-month PCV13-TT groups was lower than that in
 the 3-month control group.
 The incidence of vaccine-related adverse events after the first dose of vaccination in the 2-month
 and 3-month study groups was lower than that in the 3-month control group and the difference was
 statistically significant ($P = 0.0344$). The incidence of fever and local redness in the 2-month and
 3-month PCV13-TT group was lower than that in the 3-month control group, and the incidence of
 hypodynamia/somnolence in the 2-month study group was lower than that in the 3-month study
 and control group. The difference was statistically significant ($P = 0.0064$). No intergroup
 significant difference was observed for other symptoms.
 The incidence of vaccine-related adverse events after the second dose of vaccination in the 2-
 month and 3-month PCV13-TT group was lower than that in the 3-month control group and the
 difference was statistically significant ($P = 0.0242$). The incidence of local redness in the 2-month
 and 3-month PCV13-TT group was lower than that in the 3-month control group and the difference
 was statistically significant ($P = 0.0160$). No intergroup significant difference was observed for
 other symptoms.
 The incidence of vaccine-related adverse events after the third dose of vaccination in the 2-month
 and 3-month PCV13-TT group was 41.35% lower than that in the 3-month control group and the
 difference was statistically significant ($P = 0.0492$). The incidence of local redness in the 2-month
 study group was lower than that in the 3-month study and control group and the difference was
 statistically significant ($P = 0.0015$). The incidence of local induration in the 2-month study group
 was lower than that of the 3-month study group. No intergroup significant difference was observed
 for other symptoms.
 The analysis of incidence of adverse events after the vaccination of each dose in the 2-month study
 group and 3-month study and control group are shown in Table 54-1 to 54-6.
 Table 54-1 Adverse events after first vaccination in PCV13-TT (2M), PCV13-TT (3M)
 and PCV7 groups
 PCV13-TT (2M) PCV13-TT (3M) PCV7(3M)

(N=517) (N=519) (N=519) P
Events Cases % Events Cases % Events Cases %
Any AE 568 295 57.06 567 281 54.14 624 316 60.89 0.0892
Vaccine-related AE 383 232 44.87 425 237 45.66 474 271 52.22 0.0344
Solicited AE 372 230 44.49 398 231 44.51 453 266 51.25 0.0424
unsolicited AE 11 9 1.74 27 22 4.24 21 18 3.47 0.0520
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Vaccine-unrelated
AE 185 134 25.92 142 94 18.11 150 111 21.39 0.0098
Solicited AE 24 22 4.26 29 24 4.62 35 28 5.39 0.7050
unsolicited AE 161 121 23.40 113 81 15.61 115 94 18.11 0.0053
Table 54-2 Analysis of symptoms of systemic and local reactions after first vaccination in
PCV13-TT (2M), PCV13-TT (3M) and PCV7 groups
PCV13-TT (2M) PCV13-TT (3M) PCV7(3M)
(N=517) (N=519) (N=519)
P
Eve Cas Even Case Even Case
% % %
nts es ts s ts s
Fever 190 171 33.08 177 162 31.21 238 213 41.04 0.0022
Fatigue/Som
nolence 10 10 1.93 29 29 5.59 23 23 4.43 0.0064
Nausea/vom
iting 22 21 4.06 31 27 5.20 13 13 2.50 0.0753
Diarrhea 40 37 7.16 46 46 8.86 41 41 7.90 0.6073
Myalgia 0 0 0.00 1 1 0.19 2 2 0.39 0.7773
Cough 20 20 3.87 26 26 5.01 23 22 4.24 0.6847
Abnormal
crying 41 39 7.54 46 46 8.86 38 38 7.32 0.6325
Allergic
reactions 1 1 0.19 2 2 0.39 2 2 0.39 1.0000
Local pain 6 6 1.16 9 9 1.73 9 9 1.73 0.7126
Local
redness 18 18 3.48 16 16 3.08 34 34 6.55 0.0162
Local
swelling 12 12 2.32 5 5 0.96 12 12 2.31 0.1690
Local
induration 12 12 2.32 8 8 1.54 16 16 3.08 0.2624
Local
itching 0 0 0.00 2 2 0.39 2 2 0.39 0.5547

Table 54-3 Adverse events after second vaccination in PCV13-TT (2M), PCV13-TT (3M)
and PCV7 groups
PCV13-TT (2M) PCV13-TT (3M) PCV7(3M)
(N=496) (N=506) (N=508) P
Events Cases % Events Cases % Events Cases %
Any AE 404 254 51.21 371 230 45.45 449 260 51.18 0.1088
Vaccine-related AE 278 193 38.91 253 180 35.57 322 223 43.90 0.0242
Solicited AE 274 193 38.91 236 173 34.19 303 219 43.11 0.0142
unsolicited AE 4 3 0.60 17 16 3.16 19 18 3.54 0.0018
Vaccine-unrelated
AE 126 94 18.95 118 89 17.59 127 88 17.32 0.7828
Solicited AE 15 13 2.62 22 21 4.15 26 18 3.54 0.4205
unsolicited AE 111 86 17.34 96 82 16.21 101 78 15.35 0.6973

Table 54-4 Analysis of symptoms of systemic and local reactions after second vaccination
in PCV13-TT (2M), PCV13-TT (3M) and PCV7 groups
PCV13-TT (2M) PCV13-TT (3M) PCV7(3M)
(N=496) (N=506) (N=508) P
Events Cases % Events Cases % Events Cases %

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Fever 186 168 33.87 164 152 30.04 194 181 35.63 0.1535
Fatigue/Somnolence 5 5 1.01 4 4 0.79 4 4 0.79 0.8853
Nausea/vomiting 8 8 1.61 5 5 0.99 7 7 1.38 0.6660
Diarrhea 22 22 4.44 16 16 3.16 24 23 4.53 0.4624
Myalgia 1 1 0.20 0 0 0.00 0 0 0.00 0.3285
Cough 5 5 1.01 6 6 1.19 10 10 1.97 0.4705
Abnormal crying 13 13 2.62 15 15 2.96 15 14 2.76 0.9611
Allergic reactions 1 1 0.20 0 0 0.00 1 1 0.20 0.7744
Local pain 1 1 0.20 4 4 0.79 2 2 0.39 0.4689
Local redness 23 23 4.64 16 16 3.16 36 36 7.09 0.0160
Local swelling 5 5 1.01 2 2 0.40 7 7 1.38 0.2607
Local induration 4 4 0.81 4 4 0.79 3 3 0.59 0.8682
Local itching 0 0 0.00 0 0 0.00 0 0 0.00 /

Table 54-5 Adverse events after third vaccination in PCV13-TT (2M), PCV13-TT (3M)
and PCV7 groups
PCV13-TT (2M) PCV13-TT (3M) PCV7(3M)
(N=491) (N=503) (N=503) P
Events Cases % Events Cases % Events Cases %
Any AE 306 209 42.57 368 232 46.12 374 254 50.50 0.0427
Vaccine-related AE 213 166 33.81 261 190 37.77 272 208 41.35 0.0492
Solicited AE 208 163 33.20 251 189 37.57 256 204 40.56 0.0536
unsolicited AE 5 5 1.02 10 9 1.79 16 12 2.39 0.2596
Vaccine-unrelated

93 65 13.24 107 74 14.71 102 72 14.31 0.7857
 AE
 Solicited AE 17 15 3.05 21 13 2.58 20 14 2.78 0.9051
 unsolicited AE 76 61 12.42 86 69 13.72 82 68 13.52 0.8174

Table 54-6 Analysis of symptoms of systemic and local reactions after third vaccination in
 PCV13-TT (2M), PCV13-TT (3M) and PCV7 groups
 PCV13-TT (2M) PCV13-TT (3M) PCV7(3M)
 (N=491) (N=503) (N=503) P

Events	Cases	%	Events	Cases	%	Events	Cases	%		
Fever	157	147	29.94	174	159	31.61	196	183	36.38	0.0805
Fatigue/Somnolence	2	2	0.41	4	4	0.80	2	2	0.40	0.7429
Nausea/vomiting	5	5	1.02	2	2	0.40	2	2	0.40	0.4500
Diarrhea	9	9	1.83	14	14	2.78	9	9	1.79	0.4943
Cough	7	7	1.43	8	7	1.39	8	8	1.59	1.0000
Abnormal crying	15	14	2.85	7	6	1.19	9	9	1.79	0.1789
Allergic reactions	1	1	0.20	1	1	0.20	1	1	0.20	1.0000
Local pain	3	3	0.61	3	3	0.60	1	1	0.20	0.6447
Local redness	7	7	1.43	27	27	5.37	22	22	4.37	0.0015
Local swelling	2	2	0.41	5	5	0.99	6	6	1.19	0.4660
Local induration	0	0	0.00	6	6	1.19	0	0	0.00	0.0040
Local itching	0	0	0.00	0	0	0.00	0	0	0.00	/

7.1.6. Adverse Events after the Vaccination of Each Dose in Other Age Groups

7.1.6.2. Adverse Events after the Vaccination of Each Dose in Other Age Groups in Groups
 of Subjects Aged 7-11 Months and 12-23 Months

No significant difference was observed in the incidence of adverse events after each dose of
 vaccination between the study group and the control group of subjects aged 7-11 months and 12-
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23 months. Except that the incidence of local swelling after the first dose of vaccination in the
 study group of subjects aged 7-11 months was higher than that in the control group, no significant
 intergroup difference was observed for other symptoms. The incidence of adverse events after the
 first dose of vaccination was higher than that of the second dose, as shown in Table 55-1 to 55-8.

Table 55-1 Adverse events after first vaccination in subjects aged 7-11 months
 PCV13-TT (N=198) PCV7 (N=200)

Events	Cases	%	Events	Cases	%		
Any AE	307	132	66.67	332	141	70.50	0.4501
Vaccine-related AE	184	99	50.00	189	117	58.50	0.1073
Solicited AE	181	99	50.00	184	117	58.50	0.1073
unsolicited AE	3	3	1.52	5	5	2.50	0.7238
Vaccine-unrelated AE	123	78	39.39	143	75	37.50	0.7573
Solicited AE	29	21	10.61	30	23	11.50	0.8733
unsolicited AE	94	65	32.83	113	67	33.50	0.9154

Table 55-2 Analysis of symptoms of systemic and local reactions after first vaccination in
 subjects aged 7-11 months

PCV13-TT (N=198) PCV7 (N=200)
 P

Events	Cases	%	Events	Cases	%		
Fever	85	75	37.88	104	90	45.00	0.1557
Fatigue/Somnolence	2	2	1.01	5	5	2.50	0.4490
Nausea/vomiting	11	11	5.56	7	7	3.50	0.3462
Diarrhea	9	9	4.55	11	11	5.50	0.8193
Myalgia	0	0	0.00	1	1	0.50	1.0000
Cough	7	7	3.54	7	7	3.50	1.0000
Abnormal crying	13	13	6.57	11	11	5.50	0.6795
Allergic reactions	0	0	0.00	5	5	2.50	0.0609
Local pain	10	10	5.05	5	5	2.50	0.1997
Local redness	24	24	12.12	22	22	11.00	0.7559
Local swelling	13	13	6.57	4	4	2.00	0.0269
Local induration	6	6	3.03	1	1	0.50	0.0669
Local itching	1	1	0.51	1	1	0.50	1.0000

Table 55-3 Adverse events after second vaccination in subjects aged 7-11 months
 PCV13-TT (N=193) PCV7 (N=194)

Events	Cases	%	Events	Cases	%		
Any AE	136	92	47.67	173	110	56.70	0.0839
Vaccine-related AE	103	76	39.38	114	82	42.27	0.6055
Solicited AE	96	72	37.31	109	80	41.24	0.4667
unsolicited AE	7	6	3.11	5	4	2.06	0.5429
Vaccine-unrelated AE	33	26	13.47	59	45	23.20	0.0177
Solicited AE	6	4	2.07	14	11	5.67	0.1113
unsolicited AE	27	25	12.95	45	39	20.10	0.0748

Table 55-4 Analysis of symptoms of systemic and local reactions after second vaccination
 in subjects aged 7-11 months

PCV13-TT (N=193) PCV7 (N=194)
 P

Events	Cases	%	Events	Cases	%		
Fever	64	57	29.53	80	68	35.05	0.2773
Fatigue/Somnolence	1	1	0.52	0	0	0.00	0.4987
Nausea/vomiting	1	1	0.52	1	1	0.52	1.0000
Diarrhea	6	6	3.11	8	8	4.12	0.7866

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Cough 4 4 2.07 2 2 1.03 0.4489
 Abnormal crying 3 2 1.04 2 2 1.03 1.0000
 Allergic reactions 0 0 0.00 1 1 0.52 1.0000
 Local pain 4 4 2.07 3 3 1.55 0.7237
 Local redness 10 10 5.18 8 8 4.12 0.6391
 Local swelling 3 3 1.55 4 4 2.06 1.0000
 Local induration 0 0 0.00 0 0 0.00 /
 Local itching 0 0 0.00 0 0 0.00 /

Table 55-5 Adverse events after first vaccination in subjects aged 12-23 months
 PCV13-TT (N=200) PCV7 (N=200)

P
 Events Cases % Events Cases %
 Any AE 245 112 56.00 214 114 57.00 0.9197
 Vaccine-related AE 131 80 40.00 120 87 43.50 0.5430
 Solicited AE 124 79 39.50 110 83 41.50 0.7600
 unsolicited AE 7 6 3.00 10 8 4.00 0.7868
 Vaccine-unrelated AE 114 62 31.00 94 56 28.00 0.5837
 Solicited AE 25 19 9.50 22 19 9.50 1.0000
 unsolicited AE 89 57 28.50 72 48 24.00 0.3633

Table 55-6 Analysis of symptoms of systemic and local reactions after first vaccination in subjects aged 12-23 months

PCV13-TT (N=200) PCV7 (N=200)
 P
 Events Cases % Events Cases %
 Fever 61 56 28.00 65 62 31.00 0.5837
 Fatigue/Somnolence 2 2 1.00 2 2 1.00 1.0000
 Nausea/vomiting 3 3 1.50 4 4 2.00 1.0000
 Diarrhea 8 8 4.00 2 2 1.00 0.1050
 Cough 8 8 4.00 8 8 4.00 1.0000
 Abnormal crying 5 5 2.50 3 3 1.50 0.7238
 Allergic reactions 3 3 1.50 7 7 3.50 0.3375
 Local pain 7 7 3.50 2 2 1.00 0.1747
 Local redness 13 13 6.50 12 12 6.00 1.0000
 Local swelling 10 10 5.00 4 4 2.00 0.1719
 Local rash 1 1 0.50 0 0 0.00 1.0000
 Local induration 1 1 0.50 1 1 0.50 1.0000
 Local itching 2 2 1.00 0 0 0.00 0.4987

Table 55-7 Adverse events after second vaccination in subjects aged 12-23 months
 PCV13-TT (N=194) PCV7 (N=192)

P
 Events Cases % Events Cases %
 Any AE 99 66 34.02 81 62 32.29 0.7464
 Vaccine-related AE 66 52 26.80 58 49 25.52 0.8173
 Solicited AE 64 52 26.80 58 49 25.52 0.8173
 unsolicited AE 2 2 1.03 0 0 0.00 0.4987
 Vaccine-unrelated AE 33 22 11.34 23 16 8.33 0.3935
 Solicited AE 6 3 1.55 6 4 2.08 0.7230
 unsolicited AE 27 21 10.82 17 14 7.29 0.2877

Table 55-8 Analysis of symptoms of systemic and local reactions after second vaccination in subjects aged 12-23 months

PCV13-TT (N=194) PCV7 (N=192)
 P
 Events Cases % Events Cases %
 Fever 49 45 23.20 45 43 22.40 0.9036
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Fatigue/Somnolence 0 0 0.00 0 0 0.00 /
 Nausea/vomiting 0 0 0.00 1 1 0.52 0.4974
 Diarrhea 2 2 1.03 1 1 0.52 1.0000
 Cough 2 2 1.03 3 3 1.56 0.6842
 Abnormal crying 0 0 0.00 0 0 0.00 /
 Allergic reactions 0 0 0.00 0 0 0.00 /
 Local pain 3 3 1.55 1 1 0.52 0.6231
 Local redness 3 3 1.55 5 5 2.60 0.5015
 Local swelling 5 5 2.58 2 2 1.04 0.4489
 Local induration 0 0 0.00 0 0 0.00 /
 Local itching 0 0 0.00 0 0 0.00 /

7.1.7. Serious Adverse Events

A total of 80 cases (105 events) of SAE were collected in the primary immunization stage, including 34 cases (49 events) in the PCV13-TT and 34 cases (43 events) in the control group of subjects aged 3-71 months. No significant difference was observed with respect to SAE incidence between the study and control group; a total of 12 cases (13 events) of SAE were reported in the subjects 2 months of age.

With the exception of 1 case of SAE in 3-month-old PCV13-TT whose fever symptom was determined to be possibly related to the study vaccine, other SAEs were all independent of study vaccine. The SAE incidence in subjects aged 2 months and 3 months subjects was higher than that of other age groups. The major SAEs were "infections and infestations".

The severity of solicited symptoms or the most severe symptoms in SAEs was assessed in accordance with Guidelines for The Classification of Adverse Reactions in Clinical Trials of Prophylactic Vaccines, with a purpose to timely understand and analyze the possible risks of SAE. Results showed that the severity was grade 4 in 32 cases, grade 3 in 34 cases, and grade 2 in 2 cases, notwithstanding no intergroup difference was observed for SAEs with different severity. See Table 56-1 to 56-9 for analysis of SAE incidence.

Table. 56-1 Incidence of serious adverse events in PCV13-TT and PCV7 groups of subjects aged 3 through 71 months

PCV13-TT (N=1117) PCV7 (N=1119)

P

Events Cases % Events Cases %

Any SAE 49 34 3.04 43 34 3.04 1.0000

SAE related to study vaccine 1 1 0.09 0 0 0.00 0.4996

SAE unrelated to study

48 33 2.95 43 34 3.04 1.0000

vaccine

Table. 56-2 Incidence rate of serious adverse events in subjects aged 2 months

PCV13-TT (N=517)

Events Cases %

Any SAE 13 12 2.32

SAE related to study vaccine 0 0 0.00

SAE unrelated to study vaccine 13 12 2.32

Infections and infestations 10 10 1.93

Infectious pneumonia 5 5 0.97

Upper respiratory tract infection 1 1 0.19

Roseola infantum 1 1 0.19

Bronchitis 3 3 0.58

Injury, poisoning and procedural

1 1 0.19

complications

Contusion 1 1 0.19

Gastrointestinal disorders 2 2 0.39

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Enteritis 1 1 0.19

Diarrhea 1 1 0.19

Table. 56-3 Incidence of serious adverse events in subjects aged 3 months

PCV13-TT (N=519) PCV7 (N=519)

P

Events Cases % Events Cases %

Any SAE 28 18 3.47 25 21 4.05 0.7445

SAE related to study

1 1 0.19 0 0 0.00 1.0000

vaccine

SAE unrelated to study

27 17 3.28 25 21 4.05 0.6206

vaccine

Infections and infestations 22 16 3.08 20 17 3.28 1.0000

Viral rash 0 0 0.00 1 1 0.19 1.0000

Infectious enteritis 1 1 0.19 0 0 0.00 1.0000

Infectious pneumonia 9 8 1.54 9 8 1.54 1.0000

Laryngitis 0 0 0.00 1 1 0.19 1.0000

Respiratory tract infection* 2 2 0.39 0 0 0.00 0.4995

Measles 1 1 0.19 0 0 0.00 1.0000

Sepsis 1 1 0.19 0 0 0.00 1.0000

Abscess 1 1 0.19 0 0 0.00 1.0000

Upper respiratory tract

5 3 0.58 5 5 0.96 0.7255

infection

Bronchitis 2 2 0.39 4 4 0.77 0.6866

Investigations 1 1 0.19 0 0 0.00 1.0000

Increased biological

markers of myocardial 1 1 0.19 0 0 0.00 1.0000

necrosis

Nervous system disorders 2 2 0.39 1 1 0.19 1.0000

Febrile convulsion 1 1 0.19 0 0 0.00 1.0000

Psychomotor retardation 0 0 0.00 1 1 0.19 1.0000

Epilepsy 1 1 0.19 0 0 0.00 1.0000

Respiratory, thoracic and

1 1 0.19 4 3 0.58 0.6243

mediastinal disorders

Pneumonia 1 1 0.19 4 3 0.58 0.6243

Gastrointestinal disorders 1 1 0.19 0 0 0.00 1.0000

Enteritis 1 1 0.19 0 0 0.00 1.0000

Blood and lymphatic

1 1 0.19 0 0 0.00 1.0000

system disorders

Granulopenia 1 1 0.19 0 0 0.00 1.0000

*1 case of respiratory tract infection is "possibly related" to study vaccine.

*1 case of respiratory tract infection is "possibly related" to study vaccine.

Table. 56-4 Incidence of serious adverse events in subjects aged 7 through 11 months

PCV13-TT (N=198) PCV7 (N=200)

P

Events Cases % Events Cases %

Any SAE 5 5 2.53 10 6 3.00 1.0000

SAE related to study 1.0000

0 0 0.00 0 0 0.00

vaccine

SAE unrelated to study 1.0000

5 5 2.53 10 6 3.00

vaccine

Infections and infestations 5 5 2.53 7 5 2.50 1.0000

Infectious pneumonia 3 3 1.52 0 0 0.00 0.1222

Sepsis 0 0 0.00 2 2 1.00 0.4988

Upper respiratory tract 0 0 0.00 3 3 1.50 0.2481

infection

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Hand-foot-mouth disease 1 1 0.51 0 0 0.00 0.4975
Bacterial tonsillitis 1 1 0.51 0 0 0.00 0.4975
Bronchitis 0 0 0.00 2 2 1.00 0.4988
Investigations 0 0 0.00 1 1 0.50 1.0000
Increased biological markers of myocardial necrosis 0 0 0.00 1 1 0.50 1.0000
Skin and subcutaneous tissue disorders 0 0 0.00 1 1 0.50 1.0000
Urticaria 0 0 0.00 1 1 0.50 1.0000
Gastrointestinal disorders 0 0 0.00 1 1 0.50 1.0000
Oral ulcer 0 0 0.00 1 1 0.50 1.0000

Table. 56-5 Incidence of serious adverse events in subjects aged 12 through 23 months
PCV13-TT (N=200) PCV7 (N=200)

P						
Events	Cases	%	Events	Cases	%	
Any SAE	14	9	4.50	5	4	2.00 0.2589
SAE related to study	1	0.00	0	0	0.00	
vaccine	0	0	0.00	0	0	0.00
SAE unrelated to study	14	9	4.50	5	4	2.00
vaccine	10	7	3.50	5	4	2.00 0.5431
Infections and infestations	10	7	3.50	5	4	2.00 0.5431
Tonsillitis	1	1	0.50	0	0	0.00 1.0000
Infectious pneumonia	4	4	2.00	1	1	0.50 0.3719
Bacterial skin infections	1	1	0.50	0	0	0.00 1.0000
Upper respiratory and digestive tract infections	1	1	0.50	0	0	0.00 1.0000
Hand-foot-mouth disease	1	1	0.50	3	3	1.50 0.6231
Bronchitis	0	0	0.00	1	1	0.50 1.0000
Mycoplasma pneumonia	1	1	0.50	0	0	0.00 1.0000
Herpangina	1	1	0.50	0	0	0.00 1.0000
Immune system disorders	1	1	0.50	0	0	0.00 1.0000
Drug-induced hypersensitivity reactions	1	1	0.50	0	0	0.00 1.0000
Gastrointestinal disorders	3	3	1.50	0	0	0.00 0.2481
Intestinal obstruction	1	1	0.50	0	0	0.00 1.0000
Inguinal hernia	1	1	0.50	0	0	0.00 1.0000
Oral ulcer	1	1	0.50	0	0	0.00 1.0000

Table. 56-6. Incidence of serious adverse events in subjects aged 24 through 71 months
PCV13-TT (N=200) PCV7 (N=200)

P						
Events	Cases	%	Events	Cases	%	
Any SAE	2	2	1.00	3	3	1.50 1.0000
SAE related to study	0	0	0.00	0	0	0.00
vaccine	0	0	0.00	0	0	0.00
SAE unrelated to study	2	2	1.00	3	3	1.50
vaccine	2	2	1.00	3	3	1.50 1.0000
Infections and infestations	2	2	1.00	3	3	1.50 1.0000
Viral encephalitis	0	0	0.00	1	1	0.50 1.0000
Infectious pneumonia	1	1	0.50	0	0	0.00 1.0000
Bacterial tonsillitis	0	0	0.00	1	1	0.50 1.0000
Bronchitis	1	1	0.50	1	1	0.50 1.0000

Table. 56-7 Proportion of serious adverse events in PCV13-TT and PCV7 groups of subjects aged 3 through 71 months by severity

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PCV13-TT (N=1117)	PCV7 (N=1119)
Severity	P
Cases	%
Grade 2	1 0.09 1 0.09
Grade 3	17 1.52 17 1.52 0.9940
Grade 4	16 1.43 16 1.43
Note: Severity assessment is merely applicable for solicited symptoms or most severe symptoms of SAE.	

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Table 56-8. Listing of serious adverse events in subjects during primary series

Days since						
Study Vaccine	Age	System	organ	Term of	Duration	
Gender	Site	Dose	Description	last	Severity	SAE leads to Relevance
No. groups	(mos)	classification	adverse events	(days)		
vaccination						
The subject had diarrhea on 20160903, 5-6 times / day,						

all were yellow watery stool. On 20160906, the was taken to Puyang County People's Hospital Hospitalization for hospitalization and Group Gastrointestinal or prolonged Possibly 69 Male 1.5 Henan Third dose Enteritis examination, revealing 4 7 Grade 3 A disorders length of unrelated active bowel sounds. hospital stay Vomiting persisted from 20160908 to 20160909, 2 times / day, and was gastric contents, not projectile vomiting. Diarrhea ended on September 12. The subject had fever on 20160905 at 39.6 ■. The subject was hospitalized in Puyang County People's Hospital on September 6 for treatment. The subject was examined on admission to hospital with heavy breathing sounds of both Hospitalization lungs. On September 6, the Group Infections and Roseola or prolonged Possibly 69 Male 1.5 Henan Third dose subject was transferred to 4 7 Grade 3 A infestations infantum length of unrelated Puyang City People's hospital stay Hospital, physical examination showed hyperemia of pharynx and heavy breathing sounds. On September 9, body temperature recovered to normal. On 20160910, the subject had new systemic rash. Confidential 243 / 310 Phase III Clinical Study Report of the 13-Valent Pneumococcal Polysaccharide Conjugate Vaccine Protocol No.: 022152015003

The subject had fever on October 14, with a maximum body temperature of 39.2 ■, accompanied by rhinorrhea. The subject received 1 dose of intramuscular treatment in Qingzu hospital, and treated with Ankahuangnamin granules, cefixime granules and other drugs. The symptom of high body temperature still repeated. On October 15, the subject was treated in the outpatient department of Qingzu town Hospitalization Upper Group Infections and hospital. The hematology or prolonged Possibly 98 Female 2.8 Henan Third dose respiratory tract 55 5 Grade 4 A infestations test showed increased length of unrelated infection WBC. On October 16, the hospital stay subject was hospitalized for "acute upper respiratory infection" was admitted to the hospital. Physical examination on admission to hospital showed: T: 37.2 ■, P: 100 times / min, R: 100 times / min. WBC: $11.5 \times 10^9/L$, symptomatic and supportive treatment such as anti-inflammatory, heat clearing and detoxification after admission, and was discharged on October 20. On April 14, the subject had cough, gurgling in throat, Hospitalization more obvious after Group Infections and Infectious or prolonged Possibly 115 Male 2.1 Henan First dose suckling, after treatment of 6 7 Grade 4 A infestations pneumonia length of unrelated oral medication in the hospital stay

department of outpatient in
Maternal and Child Health

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Hospital, poor effects were observed, accompanied with foaming at mouth, so the subject was taken to Wenxian People's Hospital for treatment on April 19, and was hospitalized for "bronchopneumonia", physical examination on admission findings: poor mental state, paleness, dry lips, hyperemia of pharynx, heavy breathing sounds in both lungs, crackles and stridor in lungs, chest X-ray indicated increased pulmonary texture. Cough with sputum, occasional wheezing, slight Hospitalization shortness of breath, Group Infections and Infectious or prolonged Definitely 142 Male 2.2 Henan Third dose hyperemia of pharynx, 100 6 Grade 4 A infestations pneumonia length of unrelated heavy breathing sounds of hospital stay both lungs, and rhonchi and crackles in the lung Cough and rhinorrhea, slight shortness of breath, heavy breathing sounds in both lungs, wheezing and Hospitalization fine crackles were noted. Group Infections and or prolonged Possibly 150 Female 2.6 Henan First dose Bronchitis Chest X-ray showed: 66 6 Grade 4 A infestations length of unrelated increased lung increased, hospital stay and Acinetobacter baumannii infection is found in throat swab culture. Cough and fever at 39.3 ■. Physical examination on Hospitalization Group Infections and Infectious admission findings: heavy or prolonged Possibly 155 Male 2 Henan First dose 37 8 Grade 4 A infestations pneumonia breathing sounds of both length of unrelated lungs, with crackles, and hospital stay poor mental state. Chest X-

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ray showed: both lungs with increased markings, thickening and disorder, and widened mediastinal shadow. Hematologic findings: WBC: $7.76 \times 10^9/L$, RBC: $3.25 \times 10^{12}/L$, N: 43.4%, L: 44.1%. Rhinorrhea, cough, asthmatic suffocation, Hospitalization fever, heavy breathing Group Second Infections and Infectious or prolonged Definitely 190 Male 1.9 Henan sounds in both lungs, with 46 6 Grade 4 A dose infestations pneumonia length of unrelated fine crackles and stridor. hospital stay Chest X-ray: bronchopneumonia The subject had fever on 20160625, with the highest temperature being 37.5 ■, accompanied by paroxysmal continuous cough and gurgling in throat; 2 episodes of vomiting were recorded. In the outpatient department of Dingxing county hospital,

the subject was treated with Hospitalization oral medication, while the Group Infections and or prolonged Possibly 228 Female 2.7 Hebei First dose Bronchitis cough worsened. The 38 9 Grade 4 A infestations length of unrelated subject was examined in hospital stay Dingxing county hospital on 20160630 for a second time. Physical examination revealed: heavy breathing sound of both lungs, with gurgling in throat; hematologic findings included elevated markers of infection, which primarily were lymph; the chest X-ray showed

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increased and enhanced markings of both lungs, with visible spots, which traveled along the bronchi.

The subject developed cough without obvious causes on 20170224, the cough was mild initially and gradually aggravating, presenting as non-spasmodic, nonproductive cough with expectoration, wheezing appeared on 20170227, exacerbating after physical activities, with fever (highest temperature of 38.9 ■), Hospitalization after oral medication Group Infections and or prolonged Possibly 301 Male 2.8 Hebei Third dose Bronchitis treatment (not specified), 160 7 Grade 3 A infestations length of unrelated the conditions were not hospital stay resolved. On 20170228, the subject was taken to Zhengding County People's Hospital, and underwent hematology test in outpatient department, with the following findings: WBC $8.87 \times 10^9/L$, N% 34.7%, L% 52.3%, HGB 107g/L, PLT $429 \times 10^9/L$, CRP 7.5 mg/L; chest X-ray: thickening markings in both lungs

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Nearly one month ago (estimated date: 20170123), the subject developed cough without obvious causes, presenting as non-spasmodic cough, with sputum, not easy to cough up, not treated in the beginning, on 20170218, the subject was treated with unspecified medication for about 3 days, without poor response, vomiting occurred on 20170221, with Hospitalization non-projectile gastric Group Infections and Infectious or prolonged Possibly 323 Female 2.2 Hebei Third dose contents, with high 155 12 Grade 3 A infestations pneumonia length of unrelated frequency, loose stools, hospital stay aggravated cough with wheezing, and poor mental diet appeared in the past two days, on 20170223, the

subject was hospitalized in the People's Hospital of Zhengding County for "bronchopneumonia and gastrointestinal dysfunction". Physical examination findings: hyperemia of pharynx, heavy breathing sounds in both lungs, with wheezing
20161004 Fall from bed, had fever at 40.0 ■ two days later, without Injury, vomiting, rhinorrhea or Hospitalization
Group poisoning and cough. On 1007, swelling or prolonged Definitely
350 Female 2.8 Hebei Third dose Contusion 12 8 Grade 3
A procedural and protrusion on the right length of unrelated complications top head without rupture. hospital stay
On 1008, in the provincial children's hospital, head CT showed hematoma of right

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parietal scalp and fracture of right parietal bone. Physical examination on admission to hospital findings: body temperature 36.7 ■, heart rate 120 / min, 34 breaths / min, conscious, in good mental state, swelling and protrusion of the top right scalp of 10 cm×8 cm, with palpable fluctuation, no rupture or tenderness, red maculopapules scattered in torso and both lower limbs. Hyperemia of pharynx, heavy breathing sounds in both lungs, positive Babinski sign on both sides. Puncture of scalp hematoma was performed on 20161015
20160922 Fever without causes, with peak frequency of 5-6 times / day, the Hospitalization
Group Second Gastrointestinal highest temperature was or prolonged Possibly
424 Female 1.8 Shanxi Diarrhea 62 7 Grade 3
A dose disorders 39.4 ■, accompanied with length of unrelated 8-9 times / day of diarrhea hospital stay
as yellow green and watery stool
Fever, body temperature 38.2 ■, hyperemia of Hospitalization
Group Infections and Respiratory pharynx, heavy breathing or prolonged Definitely
545 Male 3.9 Henan First dose 41 4 Grade 3
A infestations tract infection sounds in both lungs, length of unrelated cough, sneezing, hospital stay
paroxysmal crying
Increased Creatine, kinase, isoenzyme
Hospitalization
biological 48 M/L↑, lactate
Group or prolonged Definitely
545 Male 3.9 Henan First dose Investigations markers of dehydrogenase 379↑, A 41 4 Grade 3
A length of unrelated myocardial hydroxybutyrate hospital stay
necrosis dehydrogenase 330↑

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Blood and Neutrophil count was Hospitalization
Group lymphatic $0.7 \times 10^9/L$, hematology or prolonged Definitely
545 Male 3.9 Henan First dose Granulopenia 41 4 Grade 3
A system test showed a lymphocyte length of unrelated disorders ratio of 66.9%. hospital stay
The subject had fever on June 8, with the highest temperature of 38.0 ■, coughing with sputum. On June 11, the subject was Hospitalization
Group Second Infections and Infectious treated in Puyang people's or prolonged Possibly
572 Male 3.5 Henan 19 7 Grade 3
A dose infestations pneumonia hospital and was admitted. length of unrelated
Physical examination hospital stay

showed: mild hyperemia of pharynx, heavy breathing sounds in both lungs, with wheezing and crackles. Three hours before admission to hospital on June 11, the subject had Hospitalization Nervous Group Second Febrile another episode of fever, or prolonged Possibly 572 Male 3.5 Henan system 19 7 Grade 3 A dose convulsion with body temperature of length of unrelated disorders 37.8 ■, the eyes turned up, hospital stay lasted for nearly 10 minutes, and relieved then. The subject had fever on July 3, with a maximum body temperature of 39.2 ■, cough and rhinorrhea, and was Hospitalization Upper hospitalized in Puyang Group Infections and or prolonged Definitely 589 Male 3 Henan Third dose respiratory tract County maternal and child 11 7 Grade 3 B infestations length of unrelated infection health care hospital on July hospital stay 4. Physical examination findings: slight hyperemia of pharynx and heavy breathing sounds of both lungs. Hospitalization Upper Fever, heavy breathing Group Second Infections and or prolonged Definitely 603 Female 3.5 Henan respiratory tract sounds in both lungs, red 14 4 Grade 3 A dose infestations length of unrelated infection and swollen pharynx, the hospital stay

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highest fever temperature was 39.0 ■. The subject had fever on 20160508 after getting a cold, in the afternoon of the same day, the was taken to the outpatient department of Xuzhen Health Center for treatment, receiving ibuprofen suspension orally 2 ML for one dose, and another dose each on 0509 and 0510, respectively, with poor response. On 20160510 the subject was admitted to Xuzhen health center, with body Hospitalization Upper Group Infections and temperature on admission or prolonged Possibly 603 Female 3.5 Henan First dose respiratory tract 16 4 Grade 2 A infestations being 37.4 ■, and red length of unrelated infection pharynx. WBC hospital stay $16.11 \times 10^9/L$, diagnosis on admission: acute upper respiratory tract infection. 20160511 Diarrhea 3-4 times/day, with yellow loose stool. After anti-infective antipyretic treatment and symptomatic treatment, the subject was discharged from the hospital on 20160513. Discharge diagnosis: acute upper respiratory tract infection Fever at 39.0 ■, combined with diarrhea, pharyngeal Hospitalization Upper redness and swelling, heavy Group Infections and or prolonged Possibly 612 Female 3.6 Henan First dose respiratory tract breathing sounds of both 32 5 Grade 4 A infestations length of unrelated infection lungs, without rhonchi or hospital stay crackles, WBC count $16.33 \times 10^9/L$

Pharyngeal redness and swelling, clear breath sounds in both lungs, no Hospitalization
Upper
Group Infections and crackles, fever at 40.0 ■, or prolonged Possibly
612 Female 3.6 Henan First dose respiratory tract 16 5 Grade 4
A infestations WBC count 14.63×10^9 ^ length of unrelated infection
9/L, hsCRP > 10 mg/L, hospital stay
CRP 48.9 mg/L, 19.62×10^9 ^
9/L, L 47.3%
The subject on 20160610 had cough with sputum, vomiting of gastric contents. On June 13, the subject developed dyspnea.
On June 16, the was taken Hospitalization
to Puyang County People's
Group Second Infections and Infectious or prolonged Possibly
624 Male 3.8 Henan Hospital for treatment. 21 7 Grade 3
B dose infestations pneumonia length of unrelated
Physical examination on hospital stay
admission findings: heavy breathing sounds, with rhonchi and crackles. The highest temperature was 37.8 ■, and was discharged on June 22.
On 20160623, the subject had cough, fever, and asthma with a maximum body temperature of 39.5 ■. n 20160627, the subject was hospitalized in Xuzhen health hospital.
Respiratory, Hospitalization
Physical examination on Group Second thoracic and or prolonged Possibly
637 Male 3.1 Henan Pneumonia admission findings: red and 32 8 Grade 3
A dose mediastinal length of unrelated swelling pharynx, heavy disorders hospital stay
breathing sounds of both lungs, with rhonchi and stridor. The subject was discharged on 20160704, recovered on July 9, and the cough symptoms disappeared.

Fever > 38.5 ■, WBC
Hospitalization
count 26.20×10^9 /L, CRP
Group Infections and or prolonged Definitely
646 Female 3.9 Henan First dose Sepsis 82 mg/L, increased 42 7 Grade 3
A infestations length of unrelated respiratory rate and heart hospital stay
rate
Fever ■ 38.5 ■,
Hospitalization
Upper hyperemia of pharynx,
Group Infections and or prolonged Definitely
646 Female 3.9 Henan First dose respiratory tract heavy breathing sounds in 42 7 Grade 4
A infestations length of unrelated infection both lungs, increased WBC hospital stay
count and CRP
The subject had fever of 38.7 ■ on 20160624, with rhinorrhea and cough with sputum, on June 25 was treated in Puyang Hospital
Hospitalization
of traditional Chinese
Group Infections and or prolonged Definitely
646 Female 3.9 Henan First dose Bronchitis Medicine. Physical 58 6 Grade 3
A infestations length of unrelated examination on admission hospital stay
to hospitaL: red pharynx: T
38.1 ■ with heavy

breathing sounds in both lungs, with gurgling in throat.
The subject had fever and cough on 20160702, with the maximum body temperature of 38.4 ■. The subject was hospitalized in Respiratory, Xuzhen health center on Hospitalization Group Second thoracic and July 21, physical or prolonged Possibly 657 Male 3.3 Henan Pneumonia 43 7 Grade 3
B dose mediastinal examination on admission length of unrelated disorders indicated heavy breathing hospital stay sounds with few gurgling in throat. The subject was discharged on July 27, and the cough symptoms disappeared on July 30.
The main symptoms and Hospitalization Group Second Infections and Infectious signs were cough, gurgling or prolonged Possibly 667 Female 3.7 Henan 8 9 Grade 3
B dose infestations pneumonia in throat, heavy breathing length of unrelated sounds, with fine crackles, hospital stay

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diagnosed as bronchopneumonia, and was hospitalized during 20160611-0619.
The subject had fever of 38.0 ■ on 20160713, diarrhea of 5 times / day on July 15, with a small Hospitalization amount of yellow mucus
Group Second Infections and Infectious or prolonged Possibly 686 Female 3.5 Henan stool. The subject was 40 7 Grade 3
A dose infestations enteritis length of unrelated hospitalized in Xuzhen hospital stay
hospital on July 20 and discharged from the hospital and cured on July 26.
The subject had fever of 38.0 ■ on 20160713. On the same day, the subject was given intramuscular injection once in Xuzhenji clinic. On July 16, an induration was observed on Hospitalization the injection site of right
Group Second Infections and or prolonged Possibly 686 Female 3.5 Henan Abscess hip. On July 20, the subject 40 7 Grade 3
A dose infestations length of unrelated was hospitalized in Xuzhen hospital stay
health hospital. On July 22, the subject underwent abscess incision and drainage and was discharged on July 26. On August 5, the surgical wound healed.
The subject developed paroxysmal cough with gurgling in throat on Hospitalization 20160624. The subject was
Group Second Infections and Infectious or prolonged Possibly 698 Male 3.3 Henan hospitalized in Puyang 27 15 Grade 3
B dose infestations pneumonia length of unrelated maternal and child health hospital stay
care hospital on 20160704.
Physical examination on admission indicated heavy

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breathing sounds in both lungs and fixed crackles.
The subject was discharged on 20160718.
Rhinorrhea, cough, physical examination on admission

indicated: poor mental state,
heavy breathing sounds in
both lungs during Hospitalization
Group Infections and Infectious auscultation, with crackles. or prolonged Possibly
752 Male 3.2 Henan First dose 23 7 Grade 4
A infestations pneumonia Chest X-ray showed: length of unrelated
increased and thickened hospital stay
markings in both lungs, and
flocculent shadow was seen
in the middle of the right
lung. T: 38.2■

The subject had fever on
April 15, with body
temperature of 38.3 ■.

After symptomatic
treatment (unspecified), the
symptoms combined with
Hospitalization
fever and cough still

Group Infections and or prolonged Possibly

766 Male 3 Henan First dose Bronchitis persisted intermittently. The 39 7 Grade 4

B infestations length of unrelated
subject was hospitalized in
hospital stay

the health care hospital on
May 21. Physical

examination on admission
indicated: hyperemia of
pharynx, heavy breathing
sounds in both lungs.

Cough occurred on June 6,
aggravated on June 21, with
Hospitalization

sputum, occasional asthma,
Group Second Infections and Infectious or prolonged Possibly

780 Female 3 Henan hyperemia of pharynx, 41 7 Grade 4

B dose infestations pneumonia length of unrelated

heavy breathing sounds in
hospital stay
both lungs, with rhonchi
and crackles.

Fever on the evening of

Group Infections and Infectious Hospitalization Possibly

801 Male 3.9 Henan First dose April 19, with the 2 7 Grade 4

B infestations pneumonia or prolonged unrelated
maximum temperature of

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38.7 ■, accompanied by length of
cough and vomiting (3 hospital stay
times / day, milk curds); on
April 20, dyspnea, poor
mental state, dry lips,
hyperemia of pharynx,
heavy breathing sounds in
both lungs, fine crackles
and wheezing.

Fever occurred on June 24.

The village health center

gave compound zinc

ibuprofen and other drugs

(unspecified details) with

poor response. Fever

persisted intermittently. On

June 27, the body

Hospitalization

temperature was 38.0 ■.

Group Second Infections and or prolonged Possibly

801 Male 3.9 Henan Bronchitis The subject was 30 6 Grade 4

B dose infestations length of unrelated

hospitalized in Wenxian

hospital stay

County People's Hospital

on the same day. Physical

examination on admission

indicated: poor mental state,

hyperemia of pharynx,

heavy breathing sounds in

both lungs, and occasional

gurgling in throat.

Cough occurred on October Hospitalization

Group Infections and Infectious 10, fever occurred on or prolonged Definitely

801 Male 3.9 Henan Third dose 98 5 Grade 4

B infestations pneumonia October 12, the highest length of unrelated

temperature was 39.0 ■. hospital stay

On April 24, the subject

with wheezing developed

cough, treated on April 25

Hospitalization

in outpatient clinic with
Group Infections and Infectious or prolonged Possibly
809 Female 3.9 Henan First dose poor response, resulting in 10 9 Grade 4
B infestations pneumonia length of unrelated
aggravating symptoms,
hospital stay
accompanied with
wheezing and occasionally
choking cough. On April

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29, the subject was
diagnosed as asthmatic
bronchopneumonia in the
outpatient clinic, and was
admitted to the hospital for
intermittent cough. Physical
examination on admission
indicated: T: 36.5 ■, poor
mental state, slight
shortness of breath,
hyperemia of pharynx,
heavy breathing sounds in
both lungs during
auscultation, with wheezing
and crackles, chest X-ray
showed: increased lung
markings, blood RT:
WBC: $17.17 \times 10^9/L$, L:
42.5%, N: 50.90%, HGB:
118 g/L indicative of
bacterial infection, CRP 34
mg/L

The subject had cough on
May 1st, and fever with
39.0 ■ on May 25th, the
coughing persisted, and in
poor mental state, slight
shortness of breath, dry lips,
Hospitalization

hyperemia of pharynx,
Group Infections and Infectious or prolonged Possibly
828 Female 3.8 Henan First dose heavy breathing sounds in 36 6 Grade 4
A infestations pneumonia length of unrelated
both lungs, with rhonchi
hospital stay
and fine crackles.

Hematology test indicated:
WBC: $5.03 \times 10^9/L$, N:
7.72%, L: 86.91%. Chest
X-ray showed increased
lung markings.

Hospitalization

Loose and watery stool, 5-6

Group Gastrointestinal or prolonged Possibly

828 Female 3.8 Henan First dose Enteritis times / day, stool routine 40 2 Grade 4

A disorders length of unrelated

test: 6-7 leukocytes / HP

hospital stay

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On December 8, 2016, the
subject developed
paroxysmal cough, with
sputum, and combined with
asthma on December 9,
2016. The subject was
given symptomatic
treatment in the village
health center (unspecified Hospitalization
Group Infections and Infectious details) without remission. or prolonged Definitely
844 Male 3.1 Henan Third dose 167 8 Grade 4
A infestations pneumonia On December 11, 2016, the length of unrelated
subject was hospitalized in hospital stay
Wenxian maternal and child
health care hospital for
"bronchopneumonia". The
chest X-ray showed
increased and thickened
lung markings in both
lungs, with slightly blurred
and flocculent sign
The body temperature was
37.3 ■ in 30 minutes after
immunization and 39.5 ■
on May 3. After treatment

with oral medicine and outpatient clinical therapy, the body temperature decreased to about 37.5 ■, and reached 38.6 ■ on July Hospitalization

30. On July 31, the subject Group Second Infections and Respiratory or prolonged Possibly 877 Female 3 Henan was hospitalized in 62 10 Grade 4 A dose infestations tract infection length of related Zhengzhou children's hospital stay Hospital. Physical examination on admission indicated: T: 38.0 ■, slight hyperemia of pharynx, heavy breathing sounds in both lungs. T cells (CD3) were decreased to 48%. CD4 / CD8 was increased to 2.74, indicative of

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cellular immune dysfunction. Vitamin B12 117 pg/mL, decreased. cytomegalovirus IgG positive. Complement C3 0.75 g/L, decreased. On May 19, subjects had developed cough with sputum due to cold, and occasionally had shortness of breath. The subject was given oral medication in the local clinic while with poor response. On May 27, the cough worsened, and on Hospitalization Group Infections and Infectious May 29, the subject was or prolonged Possibly 904 Female 3 Henan First dose 20 8 Grade 4 A infestations pneumonia hospitalized in Wenxian length of unrelated people's hospital for hospital stay "bronchopneumonia". Physical examination on admission indicated poor mental state, dry lips, hyperemia of pharynx, heavy breathing sounds of both lungs, with rhonchi and crackles. On December 9, the subject had paroxysmal cough induced by cold, with sputum and shortness of breath. Oral medication in the street clinic showed Hospitalization poor response. On Group Infections and Infectious or prolonged Definitely 904 Female 3 Henan Third dose December 10, the subject 152 11 Grade 4 A infestations pneumonia length of unrelated was hospitalized in hospital stay Wenxian people's hospital was for "bronchopneumonia". Physical examination on admission indicated hyperemia of pharynx, Confidential 259 / 310
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heavy breathing sounds of both lungs, with crackles and wheezing.

Fever 38.2 ■, cough, Hospitalization Group Infections and Infectious heavy breathing sounds in or prolonged Possibly 907 Female 3.3 Henan First dose 38 7 Grade 4 A infestations pneumonia both lungs, with rhonchi length of unrelated and crackles. hospital stay During 20160715-0724, the subject suffered from upper respiratory tract infection, fever (39.0 ■), cough with expectoration, and rhinorrhea. All symptoms were relieved after

treatment. Fever occurred again on 0728, the highest temperature was 39.5 ■, and cough symptoms were aggravated. On 0728 the subject was diagnosed as "pneumonia" in the outpatient department of the Hospitalization county hospital while was Group Infections and Infectious or prolonged Possibly 984 Male 3.2 Hebei Third dose not treated. On 0729 the 37 4 Grade 4 B infestations pneumonia length of unrelated subject was taken to hospital stay Baoding children's Hospital for further treatment. At 0729 13:50, the subject was hospitalized for "bronchopneumonia". Physical examination showed: heavy breathing sounds of both lungs, with fine crackles and gurgling in the throat, hyperemia of pharynx, herpes was present in the pharyngeal isthmus. After treatment, the subject recovered and was discharged at 0801 16:00.

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The subject had cough and fever on 20170113, with the highest temperature of 38.4 ■. No response was observed after home Hospitalization treatment (unspecified Group Infections and or prolonged Possibly 1048 Female 3.1 Hebei Third dose Bronchitis medication). On 20170114, 183 5 Grade 4 B infestations length of unrelated the subject was hospitalized hospital stay and treated in the outpatient department of Dingxing Yunfeng maternity hospital and was discharged on 20170121. The subject was admitted to hospital on 20161122 for cough with wheezing lasting for 10 days, and aggravation with suffocation for 3 days. On 20161112, the subject had cough with wheezing with no obvious cause, and was treated with medications (unspecified) in the local outpatient clinic for 10 Respiratory, days, and the symptoms Hospitalization Group thoracic and were relieved. On or prolonged Possibly 1067 Male 3.2 Hebei Third dose Pneumonia 123 11 Grade 4 B mediastinal 20161119, the subject had length of unrelated disorders cough with wheezing and hospital stay mild suffocation due to cold, and was admitted to hospital on 20161122. After physical examination and auxiliary examination, the subject was diagnosed as asthmatic suffocating pneumonia, and was then treated with medications. After admission, examinations indicated elevated myocardial

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enzymes and myocardial cell damage, supplementary diagnosis showed myocardial injury, all symptoms were resolved and the subject was

discharged after hospitalization on 20161202, with medicines prescribed after discharge. Since 20160901, the subject had gurgling in throat without obvious cause, no evident cough or fever, and the parents did not pay attention to it. On 20161015 the subject was taken by parents to the village clinic for examination while without definite diagnosis. After 3 days of treatment with oral medicines (unspecified type and Respiratory, dosage), there was little Hospitalization Group thoracic and improvement observed. For or prolonged Possibly 1067 Male 3.2 Hebei Third dose Pneumonia 105 12 Grade 4 B mediastinal further diagnosis and length of unrelated disorders treatment, the subject was hospital stay taken to Dingxing county hospital on 20161104. At 20161104 12:00, the subject was hospitalized for "pneumonia". Physical examination on hospital admission indicated hyperemia of pharynx, heavy breathing sounds of both lungs, with crackles. Small patchy opacities were observed in the right upper lung field.

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On 20160721 the subject had cough and fever with no obvious cause. Antipyretic treatment (unspecified) was administered by village health room, followed by oral medication and enema therapy which resulted in poor response. On 0726, another episode of fever occurred (as high as 40.0 ■), with frequent coughing and evident wheezing, and in poor mental state. On 20160727 the subject was examined in Hospitalization Zhengding County People's Group Infections and Infectious or prolonged Possibly 1107 Female 3.6 Hebei Third dose Hospital, with body 30 6 Grade 3 A infestations pneumonia length of unrelated temperature being 40.0 ■, hospital stay poor mental state, slight shortness of breath, small bubbling sound in both lungs, a small amount of wheezing sound, CRP 19.5 mg/L, chest X-ray showed thickened markings and disorder in both lungs, with scattered high-density patchy opacity in both lung field, diagnosed as bronchopneumonia, and was hospitalized. On 20160801, the symptoms resolved and the subject was discharged. On 0804, cough
On July 25, the subject's Hospitalization Group Infections and body temperature was or prolonged Possibly 1146 Male 3.3 Hebei Third dose Measles 27 14 Grade 3 A infestations 39.6 ■, breath was 40 length of unrelated times / min, heart rate was hospital stay

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120 beats / min, blood

pressure was 85 / 50 mmHg. During the course of the disease, there were Kolik spots in buccal mucosa, full body rash, with normal skin during the occurrence of rashes, positive IgM antibody of measles virus. The subject was discharged on August 7.

The subject had cough from June 26 to July 7, and was not recovered after treatment with outpatient medication. On July 9, the subject was hospitalized in the Department of Pediatrics of Zhengding County People's hospital. The subject was found by physical examination to have cough with sputum Hospitalization Group Second Infections and Infectious and hyperemia of pharynx, or prolonged Possibly 1153 Female 3.3 Hebei 31 8 Grade 4 B dose infestations pneumonia heavy breathing sounds in length of unrelated both lungs, with gurgling in hospital stay throat, the chest X-ray DR indicated thickened and increased marking in both lungs, with blurred presentation, and the diagnosis was bronchopneumonia. The subject recovered on July 16, and the event was considered to be possibly unrelated to vaccination.

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From 0520 to 0526, the highest temperature of fever in the subject was 39.1 ■, with cough, and was not resolved after outpatient treatment, the subject was hospitalized in Zhengding County People's Hospital on May 27. The body temperature was 38.4 ■, Hospitalization with hyperemia of pharynx, Group Infections and or prolonged Possibly 1162 Male 3.2 Hebei First dose Bronchitis heavy breathing sounds in 18 5 Grade 3 A infestations length of unrelated both lungs, with some hospital stay wheezing and gurgling in throat, WBC $7.36 \times 10^9/L$, L% 25.4%, CRP 7.8 mg/L, chest X-ray indicated thickened lung markings in both lungs, and it was diagnosed as bronchitis, anti-infectives were given, and the subject was discharged on May 31. Cough for 4-5 days, asthma for 1 day, hospitalized on June 21, 2016, physical examination indicated: body temperature 37.3 ■, slightly fast breathing, hyperemia of pharynx, Hospitalization Group Second Infections and heavy breathing sounds in or prolonged Possibly 1175 Female 3.3 Hebei Bronchitis 13 10 Grade 3 B dose infestations both lungs, with rhonchi length of unrelated and crackles, WBC hospital stay $5.0 \times 10^9/L$, L% 70.8%, N% 21.3%, chest X-ray indicated bronchitis, recovered and discharged on June 30, 2016 after anti-infection treatment.

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On 20160719 There was cough without obvious cause, presenting as paroxysmal continuous cough with sputum, which was not easy to cough up, accompanied by rhinorrhea. On 20160721, the subject was hospitalized, physical examination indicated: hyperemia of pharynx, Hospitalization Group Second Infections and Infectious heavy breathing sounds, or prolonged Possibly 1188 Male 3.2 Hebei 30 6 Grade 3

A dose infestations pneumonia gurgling in throat both length of unrelated lungs can hear sputum hospital stay sound, hematology test: WBC $8.6 \times 10^9/L$, L% 75.1%, GR% 14.0%, HGB 122 g/L, PLT $440 \times 10^9/L$, CRP 1.3 mg/L, chest X-ray showed bronchopneumonia.

After symptomatic treatment, the subject recovered and was discharged on 20160726. On 20160815, the subject developed fever without obvious cause, the highest temperature was 40.2 ■, with mild cough, and did not have enough response to the treatment

Hospitalization

Upper administered in local clinic.

Group Infections and or prolonged Possibly

1200 Male 3.8 Hebei Third dose respiratory tract The subject was 25 4 Grade 3

B infestations length of unrelated infection hospitalized on 20160816.

hospital stay

Physical examination

showed slightly swollen

tonsils on both sides and

congestion. After 3 days,

the fever subsided and rash

appeared. WBC

$5.88 \times 10^9/L$, W 31.5%,

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with the diagnosis of upper respiratory tract infection, symptomatic anti-infection treatment was administered, and then the subject was cured and discharged on 20160819.

The subject's eyeballs did not follow the movement of object, and may not remain stable when raising head, unresponsive, and cannot Hospitalization

Nervous crawl, the subject had to be

Group Second Psychomotor or prolonged Possibly

1260 Male 3 Shanxi system supported and might not 10 7 Grade 3

B dose retardation length of unrelated

disorders make steps while walking,

hospital stay

and would not actively

grasp objects by either

hand, bilateral Babinski

sign, and could not roll

over.

Fever at 39.0 ■, cough,

Hospitalization

Upper poor mental state, heavy

Group Infections and or prolonged Possibly

1405 Male 3.8 Shanxi Third dose respiratory tract breathing sounds in both 26 7 Grade 3

B infestations length of unrelated

infection lungs, with hyperemia of

hospital stay

pharynx

Fever at 38.3 ■, cough,

groan, dyspnea, hyperemia

of pharynx, positive

inspiratory three concave

Respiratory, sign, heavy breathing Hospitalization

Group thoracic and sounds in both lungs, with or prolonged Possibly

1485 Female 3.1 Shanxi First dose Pneumonia 4 7 Grade 3

B mediastinal crackles, chest X-ray in length of unrelated

disorders posteroanterior position hospital stay
revealed increased and
disordered lung markings,
especially in the right lower
lung.
Convulsion, double eyes Hospitalization
Nervous
Group turned up, askew mouth, or prolonged Possibly
1499 Female 3.4 Shanxi Third dose system Epilepsy 24 6 Grade 3
A limb rigidity, hyperemia of length of unrelated
disorders
pharynx, heavy breathing hospital stay

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sounds of both lungs,
bilateral positive Babbitt
sign
Fever at 38.7 ■,
accompanied by cough,
hoarseness, barking cough,
inspiratory laryngeal Hospitalization
Group Infections and stridor, circumoral or prolonged Possibly
1537 Male 3.5 Shanxi Third dose Laryngitis 215 5 Grade 3
B infestations cyanosis, hyperemia of length of unrelated
pharynx, positive hospital stay
inspiratory three concave
sign, heavy breathing
sounds in both lungs.
Fever at 38.0 ■, poor sleep
at night, irritable, reduced
Hospitalization
Upper milk intake. Poor mental
Group Infections and or prolonged Possibly
1559 Male 3.1 Shanxi Third dose respiratory tract state, red face, red lips, 80 7 Grade 3
B infestations length of unrelated
infection throat congestion, heavy
hospital stay
breathing sounds in both
lungs.
Hospitalization
Many dark red herpes and
Group Infections and or prolonged Possibly
1560 Male 3.2 Shanxi Third dose Viral rash maculopapular rashes 216 5 Grade 3
B infestations length of unrelated
present on the whole body
hospital stay
Body temperature was
Hospitalization
Upper 37.2 ■, hyperemia of
Group Infections and or prolonged Possibly
1560 Male 3.2 Shanxi Third dose respiratory tract pharynx, swollen tonsils on 216 5 Grade 3
B infestations length of unrelated
infection both sides, heavy breathing
hospital stay
sounds of both lungs.
Fever at T 39.1 ■,
vomiting several times per
Hospitalization
day, hyperemia of pharynx,
Group Infections and or prolonged Possibly
1580 Male 8 Henan First dose Bronchitis heavy breathing sounds of 16 7 Grade 4
B infestations length of unrelated
both lungs, bronchitis as
hospital stay
indicated by chest X-ray in
posteroanterior view.
Fever with shiver, WBC Hospitalization
Group Infections and $12.22 \times 10^9/L$, C-reactive or prolonged Possibly
1580 Male 8 Henan First dose Sepsis 16 7 Grade 4
B infestations protein 31.47 mg/L, T length of unrelated
38.5 ■ hospital stay

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Hospitalization
Skin and Rash mainly on chest and
Group or prolonged Possibly
1580 Male 8 Henan First dose subcutaneous Urticaria back, and reddish 16 7 Grade 4
B length of unrelated
tissue disorders maculopapular rash.
hospital stay
Increased
Abnormal myocardial Hospitalization
biological
Group zymogram CKMB-M 26 or prolonged Possibly
1580 Male 8 Henan First dose Investigations markers of 16 7 Grade 4
B μ/L , LDH 301 μ/L , HBDH length of unrelated

myocardial
 270 μ L hospital stay
 necrosis
 The subject had fever on
 20160602, the highest body
 temperature was 39.2 ■,
 accompanied by cough,
 wheezing on June 7, the
 clinical symptoms were
 relieved after medication in
 local clinic, coughing on
 June 15, with wheezing
 aggravated, accompanied
 by fever of 38.0 ■, after
 clinical medication the
 body temperature was
 Hospitalization
 reduced to normal, the
 Group Second Infections and Infectious or prolonged Possibly
 1582 Female 8.3 Henan subject was hospitalized in 2 8 Grade 3
 A dose infestations pneumonia length of unrelated
 Puyang people's Hospital
 hospital stay
 on June 17 for treatment,
 physical examination on
 admission indicated:
 hyperemia of pharynx,
 heavy breathing sounds in
 both lungs, with apparent
 wheezing sounds, gurgling
 in throat and scattered
 coarse crackles, indicating
 positive serum mycoplasma
 pneumoniae antibody, and
 was discharged on
 20160624
 Fever, maximum
 Group Infections and Hand-foot- Hospitalization Possibly
 1590 Male 10.8 Henan First dose temperature of 38.9 ■, rash 3 6 Grade 4
 A infestations mouth disease or prolonged unrelated
 on hip, hands and feet,

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multiple herpes and ulcer in length of
 oral mucosa hospital stay
 Fever at 39.8 ■, C-reactive
 Hospitalization
 protein 22 mg/L, WBC
 Group Infections and or prolonged Definitely
 1606 Female 9.9 Henan First dose Sepsis $15.50 \times 10^9/L$, P: 136 19 6 Grade 4
 B infestations length of unrelated
 times / min, R 32 times /
 hospital stay
 min
 Increased respiratory rate,
 congestion of pharynx,
 heavy breathing sounds of
 both lungs during the
 auscultation, on May 5, the
 subject had paroxysmal
 cough accompanied with
 minor gurgling in the
 throat, the on May 6, the Hospitalization
 Upper
 Group Infections and maximum temperature was or prolonged Definitely
 1606 Female 9.9 Henan First dose respiratory tract 19 6 Grade 4
 B infestations 39.8 ■, the fever type was length of unrelated
 infection
 irregular, the laboratory test hospital stay
 results showed positive
 mycoplasma pneumonia,
 WBC count $15.50 \times 10^9/L$,
 neutrophil count
 $9.99 \times 10^9/L$, lymphocyte
 count $4.28 \times 10^9/L$, and the
 C-reactive protein was 22
 mg/L.
 Fever occurred on
 20160630, with a high body
 temperature of 38.9 ■, the
 subject and was
 hospitalized in Qingzu town Hospitalization
 Group Second Gastrointestinal hospital on 20160703. or prolonged Possibly
 1615 Male 10.4 Henan Oral ulcer 3 7 Grade 3
 B dose disorders Physical examination length of unrelated
 indicated hyperemia of hospital stay
 pharynx, and two ulcer sites
 were found in oral cavity,
 the subject was discharged

on 20160709

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Fever, body temperature
39.2 ■, hyperemia of
pharynx, heavy breathing Hospitalization
Upper
Group Infections and sounds in both lungs during or prolonged Possibly
1689 Male 8 Henan First dose respiratory tract 5 5 Grade 4
B infestations auscultation. WBC: length of unrelated
infection
3.42×10⁹/L, N: 46.81%, hospital stay
L: 38.04%, HB: 94 g/L,
PLT: 150×10⁹/L
Fever (39.5 ■), poor
mental state, hyperemia of
pharynx, heavy breathing
Hospitalization
Upper sounds of both lungs during
Group Second Infections and or prolonged Definitely
1702 Female 8 Henan respiratory tract auscultation, flushing face, 28 5 Grade 4
B dose infestations length of unrelated
infection dry lips, hematology test:
hospital stay
WBC: 12.29×10⁹/L, N:
70.14%, L: 22.44%, HGB:
90.0 g/L.
On 20160419 the subject
had sneezing, and on
20160421 paroxysmal
cough, with sputum that
was difficult to cough up.
Wheezing symptoms
occurred on 20160422, with
the highest severity
observed in the morning
and during the night, the
highest fever was 38.1 ■. Hospitalization
Group Infections and On 20160423, the subject or prolonged Possibly
1775 Male 8.8 Hebei First dose Bronchitis 8 8 Grade 4
B infestations was taken to the outpatient length of unrelated
department of Yunfeng hospital stay
maternity hospital in
Dingxing for treatment,
intravenous dripping was
administered in the
outpatient department for
one day, while no response
was noted. On 20160424,
the subject was hospitalized
in Dingxing Yunfeng
maternity hospital and was

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diagnosed as "bronchitis".
Physical examination
indicated: hyperemia of
pharynx, heavy breathing
sounds in both lungs, with
moderate gurgling in throat
and wheezing. After
treatment of anti-infection,
expectorant and relieving
asthma, the subject was
discharged on 20160501.
On 20160622, the subject
developed cough without
evident cause, accompanied
by rhinorrhea, untreated; on
20160627, the subject had
fever (the maximum
temperature was 38.8 ■),
one episode of spiking fever
per day, cough was more
severe than before,
presenting as paroxysmal
continuous cough, with
sputum that was difficult to
cough up, and symptoms Hospitalization
Group Infections and Infectious worsened at night. The or prolonged Possibly
1806 Female 10.2 Hebei First dose 78 7 Grade 3
A infestations pneumonia subject was treated in length of unrelated
personal clinic, with normal hospital stay
body temperature, and the
cough did not improve
significantly. For further

diagnosis and treatment, the subject was treated in Hebei children's Hospital on 20160702 and was examined with findings of heart rate 126 beats / min, 46 breaths / min, hyperemia of pharynx, slight shortness of breath, heavy breathing sounds in both lungs, with

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crackles and wheezing, lymphocyte percentage 48.4%, chest X-ray: bronchitis.

On July 20, 2016, the subject developed fever without obvious cause, with body temperature of 38.5 °C and rhinorrhea. The subject was treated on 20160721 in outpatient department with poor response, and was hospitalized on 20160722. Physical examination indicated: body temperature Hospitalization 36.7 °C, heart rate 100 beats Group Second Infections and Bacterial or prolonged Possibly 1819 Female 8 Hebei / min, 32 breaths / min, 32 5 Grade 3 A dose infestations tonsillitis length of unrelated hyperemia of pharynx, hospital stay swollen tonsil, with purulent coating, clear breath sounds in both lungs, rhonchi and crackles were absent, hemocyte count $5.02 \times 10^9/L$, C-reactive protein 1.90 mg/L, diagnosed as acute suppurative tonsillitis, and was discharged on 20160726 after symptomatic treatment. Parents reported that the subject had fever, Hospitalization rhinorrhea and other Group Infections and Infectious or prolonged Possibly 1837 Female 7.9 Hebei First dose symptoms on 20160518, 27 7 . A infestations pneumonia length of unrelated which were not resolved hospital stay after medication in outpatient department. On

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20160605, the subject was treated in the children's Hospital of Hebei Province, and was admitted to the hospital for "bronchopneumonia". Several red herpes with the size of rice grains distributed in bilateral palms and feet soles as well as in the hip, with the highest temperature of 38.0 °C. The throat was Hospitalization congested and swollen, with Group Infections and Hand-foot- or prolonged Possibly 1966 Female 21 Henan First dose grade 1 swollen tonsils of 50 5 Grade 4 B infestations mouth disease length of unrelated both sides, the red spots hospital stay were present in oral palate of the oral mucosa, and soft neck without nuchal rigidity. The lab tests showed positive enterovirus EV71 infection and positive

EV infection.
 The subject had fever on June 24, with the maximum temperature of 38.5 ■. On June 27, the subject vomited. The subject vomited immediately after eating, accompanied by Upper poor mental state. On June Hospitalization Group Second Infections and respiratory and 27, the subject was treated or prolonged Possibly 2004 Male 19.4 Henan 11 4 Grade 3
 A dose infestations digestive tract in and admitted to Puyang length of unrelated infections Oilfield General Hospital, hospital stay physical examination indicated hyperemia of pharynx, tonsil swelling of grade 1, hyperemia of pharynx, heavy breathing sounds in both lungs, and active bowel sounds.

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When abdominal pressure Hospitalization Group Gastrointestinal was increased, a 3×4 cm or prolonged Definitely 2019 Female 23.2 Henan First dose Inguinal hernia 12 4 Grade 2
 A disorders mass was noted in the right length of unrelated superficial inguinal ring. hospital stay
 Fever at T 39.0 ■, buccal mucosa and upper palate Hospitalization Group Gastrointestinal were with multiple ulcer or prolonged Possibly 2026 Male 17.9 Henan First dose Oral ulcer 5 6 Grade 4
 A disorders sites, coated with purulence length of unrelated and surrounded by hospital stay erythema
 The subject had intermittent fever on 20160712, cough with sputum, the highest temperature was 39.6 ■.
 The subject was treated in Hospitalization Group Second Infections and Qingzu town hospital on or prolonged Possibly 2028 Male 17.3 Henan Bronchitis 23 6 Grade 3
 B dose infestations July 19, and discharged length of unrelated from hospital on July 24, hospital stay still coughing, with sputum, coughing persisted on July 25, and the presence of sputum resolved.
 The body temperature was Hospitalization Group Infections and Hand-foot- as high as 39.3 ■, red or prolonged Possibly 2028 Male 17.3 Henan First dose 6 9 Grade 4
 B infestations mouth disease maculopapular rash was length of unrelated present in both feet and hip hospital stay Red brown maculopapular rash scattered in both hands Hospitalization Group Infections and Hand-foot- and feet as well as hip, or prolonged Possibly 2034 Male 16.5 Henan First dose 21 14 Grade 4
 B infestations mouth disease fever at 38.3 ■, herpes and length of unrelated ulcer scattered in buccal hospital stay mucosa, positive EV71
 On June 15, the subject had rhinorrhea, with fever at 38.5 ■, and had cough on Hospitalization 18th, wheezing on 21st, in Group Infections and Infectious or prolonged Possibly 2052 Female 20.8 Henan First dose poor mental state, with 72 7 Grade 4
 A infestations pneumonia length of unrelated slight shortness of breath, hospital stay heavy breathing sounds in both lungs, with rhonchi and fine crackles.

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Fever at 39.0 ■, heavy Hospitalization breathing sounds in both Group Second Infections and Infectious or prolonged Definitely 2065 Male 22.5 Henan lungs, with no obvious 14 6 Grade 4
 A dose infestations pneumonia length of unrelated crackles, lung DR showed hospital stay bronchopneumonia.
 Fever at 39.0 ■, hyperemia Hospitalization of pharynx, red herpes

Group Second Infections and or prolonged Definitely
 2065 Male 22.5 Henan Herpangina scattered in soft palate and 14 6 Grade 4
 A dose infestations length of unrelated
 pharynx, surrounded by
 hospital stay
 erythema.
 Hospitalization
 Fever 39.0 ■, hyperemia of
 Group Second Infections and or prolonged Definitely
 2065 Male 22.5 Henan Tonsillitis pharynx, swollen tonsils of 14 6 Grade 4
 A dose infestations length of unrelated
 grade 3 on both sides.
 hospital stay
 After pharmacotherapy in
 Hospitalization
 Immune Drug-induced outpatient department, the
 Group Second or prolonged Definitely
 2065 Male 22.5 Henan system hypersensitivity fever symptoms were 14 6 Grade 4
 A dose length of unrelated
 disorders reactions recurrent, and red rash was
 hospital stay
 scattered all over the body.
 There are two masses on
 Hospitalization
 the inner side of the right
 Group Second Infections and Bacterial skin or prolonged Definitely
 2065 Male 22.5 Henan arm, one of which is white 14 6 Grade 4
 A dose infestations infections length of unrelated
 at the top. It is painful to
 hospital stay
 touch.
 Cough, fever, body
 temperature of 40 ■,
 hyperemia of pharynx,
 heavy breathing sounds of Hospitalization
 Group Infections and Infectious both lungs during or prolonged Possibly
 2097 Female 19.6 Henan First dose 17 6 Grade 4
 A infestations pneumonia auscultation, with crackles, length of unrelated
 chest X-ray showed hospital stay
 increased and thickened
 markings in both lungs,
 with blurred structure.
 Xerocheilia, hyperemia of
 pharyngeal cavity, heavy Hospitalization
 Group Infections and Mycoplasma breathing sounds of both or prolonged Definitely
 2114 Male 12 Henan First dose 66 3 Grade 2
 A infestations pneumonia lungs during auscultation, length of unrelated
 mycoplasma pneumoniae: hospital stay
 weak positive

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Vomiting, 5-6 times / day,
 with gastric content, the
 subject was hospitalized
 and treated, physical
 Hospitalization
 examination on admission:
 Group Gastrointestinal Intestinal or prolonged Definitely
 2114 Male 12 Henan First dose poor mental state, dry lips. 66 3 Grade 3
 A disorders obstruction length of unrelated
 The abdominal plain film
 hospital stay
 showed partial intestinal
 gas in the abdomen, and
 several liquid gas plane
 shadows of different sizes.
 On 20160516 the subject
 had rash with a maximum
 body temperature of
 39.0 ■ and no vomiting.
 On 20160519, the subject
 was treated in Hebei
 children's Hospital.
 Physical examination
 indicated small herpes in Hospitalization
 Group Infections and Hand-foot- both hands and feet, or prolonged Possibly
 2197 Female 13.8 Hebei First dose 36 9 Grade 3
 A infestations mouth disease hyperemia of pharynx, and length of unrelated
 a few herpes were present hospital stay
 in oral cavity, hands and
 feet, and around anus. Heart
 rate 112 beats / min.
 Enterovirus showed
 positive EV71 and EVM
 PCR, indicative of HFMD.
 After treatment, the subject
 was discharged on May 27.
 On May 18, the subject had
 fever for 4 hours, the

highest temperature was
40.5 ■, the sudden Hospitalization
Group Infections and Infectious twitching lasted for about 1 or prolonged Possibly
2209 Male 16.6 Hebei First dose 35 20 Grade 3
B infestations pneumonia minute, and then resolved length of unrelated
spontaneously, with hospital stay
intermittent cough.
Hematology test indicated
WBC $14.3 \times 10^9/L$, CPR

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4.8 mg/L, ALT 86 μ/L ,
AST 63 μ/L , LDH 472 μ/L ;
myocardial enzyme
indicated HBDH 379 μ/L ,
CK 1286 μ/L , CKMB 53
 μ/L ; hyperemia of pharynx,
which was treated in
Zhengding County People's
Hospital from May 18 to
May 23 during
hospitalization. After
symptomatic treatment,
hyperemia of pharynx, with
herpes, liver function and
high myocardial enzyme
remained high, and the
subject was then transferred
to the provincial children's
hospital for treatment.
On June 2, 2016, the
subject had cough and
fever, and was hospitalized
on June 12, 2016. The main
symptoms were cough for
10 days and aggravating for
1 day, with slight
congestion of pharynx, with
wheezing and some small
Hospitalization
or medium bubbling sounds
Group Infections and Infectious or prolonged Possibly
2233 Male 11.6 Hebei First dose in the lung. After 52 6 Grade 3
A infestations pneumonia length of unrelated
admission, the chest X-ray
hospital stay
showed thickened markings
in both lungs, the subjects
were given antitussive
treatment as well as
therapies for thinning
sputum and relieving
asthma. The condition was
improved and discharged.
Discharge diagnosis:

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bronchopneumonia, June
17, 2016.

Heavy breathing sounds
Hospitalization
WBC: $8.6 \times 10^9/L$, fever
Group Infections and or prolonged Possibly
2411 Male 33.7 Henan First dose Bronchitis 39.4 ■, L: 0.138, N: 0.709, 1 5 Grade 4
A infestations length of unrelated
MP-IgM positive cough
hospital stay
with sputum
On September 9, the subject
had cough, and was treated
in the village health center
with enema and oral
medicine (unspecified),
while with poor response.
On June 21, cough
aggravated with sputum and
occasional shortness of
breath, intramuscular
"cefoperazone sulbactam"
and atomization
(unspecified) were
Hospitalization
1-6 months administered in Wenquan
Group Infections and Infectious or prolonged Possibly
2506 Female 23.8 Henan after last Town Hospital, while with 71 7 Grade 4

A infestations pneumonia length of unrelated vaccination poor response, on June 24, hospital stay "bronchopneumonia" was hospitalized in county hospital. Physical examination indicated poor mental state, hyperemia of pharynx, heavy breathing sounds in both lungs, with rhonchi and crackles. WBC: $11.28 \times 10^9/L$, N: 23%, L: 66.2%, HGB: 93 g/L, indicating viral infection and mild anemia. Weakly positive MP-IgM.

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On 20160624 the subject had cough, fever at $37.3^\circ C$, with sputum difficult to cough up, combined with low-grade fever at $37.3^\circ C$, the subject was treated in the individual outpatient treatment of the village, while poor response was achieved, and then the subject was admitted to Hospitalization 1-6 months hospital on 20160709, Group Infections and or prolonged Possibly 2581 Male 27.6 Hebei after last Bronchitis physical examination 89 9 Grade 3 B infestations length of unrelated vaccination indicated slight pharyngeal hospital stay congestion, heavy breathing sounds in both lungs, with mild gurgling in throat, chest X-ray showed thickened markings in both lungs, which was diagnosed as acute bronchitis, the subject recovered on 20160717 after symptomatic treatment and was discharged. On 20160722 the subject developed fever of $38.6^\circ C$ without obvious cause, combined with headache which was most significant in the forehead, while might be relieved in standing Hospitalization 1-6 months Group Infections and Viral position, the subject had or prolonged Possibly 2589 Female 63.5 Hebei after last 103 17 Grade 3 B infestations encephalitis projectile vomiting, and length of unrelated vaccination was gastric content. The hospital stay subject was treated in the neighboring village health center, but still had recurrent fever and headache, and vomiting was not obviously relieved, the

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subject was further treated in hospital on 20160723. Physical examination indicated hyperemia of pharynx, WBC $16.2 \times 10^9/L$, cerebrospinal fluid findings showed WBC $150 \times 10^6/L$, biochemical pressure 130 mmH₂O, diagnosed as viral encephalitis. After antiviral, anti-infection, reducing cranial pressure, anti-inflammatory, nutritional support for brain and organs, the subject recovered on 20160808 and was discharged.

Fever at 38.9 ■, throat swelling and pain, mental malaise, poor appetite, yellow tongue with greasy Hospitalization
1-6 months yellow tongue coating, Group Infections and Bacterial or prolonged Possibly
2752 Male 32.1 Shanxi after last floating pulse, hyperemia of 176 10 Grade 2
B infestations tonsillitis length of unrelated vaccination oral and pharyngeal cavity, hospital stay
hyperemia of bilateral tonsils in bright red, swollen, with pus on the surface.

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7.1.8. Safety Analysis for Booster Dose
7.1.8.1. Overall incidence of adverse events
The subjects with booster immunization included subjects aged 2 months and 3 months receiving 3-dose series, and those aged 7-11 months who received 2 doses of vaccination during primary immunization stage, and booster dose was given as 1 dose.
The incidence of adverse events was 35.46% in the PCV13-TT and 34.50% in the control group for 3-11 month of age who received booster immunization. The severity was mainly grade 1 and 2. Thirteen and seven grade 3 adverse events were reported in the PCV13-TT and the control group, and the incidence rate was respectively 1.93% and 1.02%. No significant difference was observed between the groups. One case of grade 4 adverse event was observed in each group (reported as SAE), which were both not related to the study vaccine.
There was no significant difference in the incidence of adverse events among study and PCV7 group in subjects 3 months and 7-11 month of age.
Adverse events after booster immunization in the study and control group were shown in Table 57-1 to 57-3.

Table. 57-1 Overall incidence in PCV13-TT and PCV7 groups post booster dose in subjects aged 3 through 11 months
PCV13-TT (N=674) PCV7 (N=684)

P	
Events Cases %	Events Cases %
Any AE 328 239 35.46	346 236 34.50 0.7329
Grade 1 158 135 20.03	172 141 20.61 0.8397
Grade 2 153 127 18.84	164 120 17.54 0.5737
Grade 3 15 13 1.93	9 7 1.02 0.1831
Grade 4 2 1 0.15	1 1 0.15 1.0000
AEs related to vaccine 242 197 29.23 241 193 28.22 0.7190	
Grade 1 132 119 17.66	151 126 18.42 0.7247
Grade 2 101 90 13.35	85 74 10.82 0.1576
Grade 3 9 8 1.19	5 4 0.58 0.2614
Grade 4	///////
Solicited AE 237 196 29.08 232 190 27.78 0.6303	
Grade 1 131 119 17.66	145 124 18.13 0.8321
Grade 2 97 88 13.06	82 73 10.67 0.1801
Grade 3 9 8 1.19	5 4 0.58 0.2614
Grade 4	///////
Unsolicited AE 5 5 0.74 9 9 1.32 0.4217	
Grade 1 1 1 0.15	6 6 0.88 0.1241
Grade 2 4 4 0.59	3 3 0.44 0.7240
Grade 3	///////
Grade 4	///////
AEs unrelated to vaccine 86 65 9.64 105 71 10.38 0.7178	
Grade 1 26 22 3.26	21 19 2.78 0.6368
Grade 2 52 44 6.53	79 56 8.19 0.2544
Grade 3 6 5 0.74	4 3 0.44 0.5035
Grade 4 2 1 0.15	1 1 0.15 1.0000

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Solicited AE 18 12 1.78 28 18 2.63 0.3566	
Grade 1 7 5 0.74	4 3 0.44 0.5035
Grade 2 9 8 1.19	23 17 2.49 0.1048
Grade 3 2 2 0.30	1 1 0.15 0.6220
Grade 4	///////
Unsolicited AE 68 64 9.50 77 63 9.21 0.9258	
Grade 1 19 19 2.82	17 16 2.34 0.6108
Grade 2 43 42 6.23	56 48 7.02 0.5867
Grade 3 4 4 0.59	3 2 0.29 0.4489
Grade 4 2 1 0.15	1 1 0.15 1.0000

Table. 57-2 Overall incidence in PCV13-TT and PCV7 groups post booster dose in subjects aged 3 months
PCV13-TT (N=488) PCV7 (N=493)

P	
Events Cases %	Events Cases %
Any AE 260 186 38.11	270 179 36.31 0.5972
AEs related to vaccine 190 153 31.35 187 149 30.22 0.7296	
Solicited AE 186 152 31.15	179 146 29.61 0.6272
Unsolicited AE 4 4 0.82	8 8 1.62 0.3848
AEs unrelated to	

70 51 10.45 83 51 10.34 1.0000
vaccine
Solicited AE 17 11 2.25 28 18 3.65 0.2579
Unsolicited AE 53 50 10.25 55 43 8.72 0.4464

Table. 57-3 Overall incidence in PCV13-TT and PCV7 groups post booster dose in subjects aged 7 through 11 months
PCV13-TT (N=186) PCV7 (N=191)
P

Events	Cases	%	Events	Cases	%		
Any AE	68	53	28.49	76	57	29.84	0.8210
AEs related to vaccine	52	44	23.66	54	44	23.04	0.9036
Solicited AE	51	44	23.66	53	44	23.04	0.9036
Unsolicited AE	1	1	0.54	1	1	0.52	1.0000
AEs unrelated to vaccine	16	14	7.53	22	20	10.47	0.3705
Solicited AE	1	1	0.54	0	0	0.00	0.4934
Unsolicited AE	15	14	7.53	22	20	10.47	0.3705

7.1.8..2. Adverse events occurred 30 minutes and 7 days after vaccination
The incidence of adverse events occurred 30 minutes and 7 days after vaccination is shown in tables 58-1 to 58-12.
No significant difference was observed in the incidence of adverse events within 30 minutes after vaccination between the PCV13-TT and the control group of 3 month of age and 7-11 month of age.
Table. 58-1 Incidence of adverse reactions 30 min post booster dose in subjects aged 3

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through 11 months
PCV13-TT (N=674) PCV7 (N=684)
P

Events	Cases	%	Events	Cases	%		
Any AE	30	30	4.45	34	34	4.97	0.7017
AEs related to vaccine	30	30	4.45	34	34	4.97	0.7017
Solicited AE	30	30	4.45	34	34	4.97	0.7017
AEs unrelated to vaccine	0	0	0.00	0	0	0.00	

Table. 58-2 Incidence of adverse reactions 30 min post booster dose in subjects aged 3 months
PCV13-TT (N=488) PCV7 (N=493)
P

Events	Cases	%	Events	Cases	%		
Any AE	25	25	5.12	24	24	4.87	0.8843
AEs related to vaccine	25	25	5.12	24	24	4.87	0.8843
Solicited AE	25	25	5.12	24	24	4.87	0.8843
AEs unrelated to vaccine	0	0	0.00	0	0	0.00	

Table. 58-3 7-11 months■■■■■■■■30■■■■■■■■
Table 2.7.4.2.2.2.3. Incidence of adverse reactions 30 min post booster dose in subjects aged 7 through 11 months
PCV13-TT (N=186) PCV7 (N=191)
P

Events	Cases	%	Events	Cases	%		
Any AE	5	5	2.69	10	10	5.24	0.2925
AEs related to vaccine	5	5	2.69	10	10	5.24	0.2925
Solicited AE	5	5	2.69	10	10	5.24	0.2925
AEs unrelated to vaccine	0	0	0.00	0	0	0.00	

Table. 58-4 Incidence of adverse reactions 0-7 days post booster dose in subjects aged 3 through 11 months
PCV13-TT (N=674) PCV7 (N=684)
P

Events	Cases	%	Events	Cases	%		
Any AE	273	208	30.86	281	203	29.68	0.6371
AEs related to vaccine	240	197	29.23	238	191	27.92	0.6309
Solicited AE	237	196	29.08	232	190	27.78	0.6303
Unsolicited AE	3	3	0.45	6	6	0.88	0.5065
AEs unrelated to vaccine	33	16	2.37	43	23	3.36	
Solicited AE	18	12	1.78	28	18	2.63	0.3566
Unsolicited AE	15	15	2.23	15	14	2.05	0.8530

Table. 58-5 Incidence of adverse reactions 0-7 days post booster dose in subjects aged 3 months
PCV13-TT (N=488) PCV7 (N=493)
P

Events	Cases	%	Events	Cases	%		
Any AE	219	163	33.40	225	158	32.05	0.6832
AEs related to vaccine	188	153	31.35	184	147	29.82	0.6278
Solicited AE	186	152	31.15	179	146	29.61	0.6272

Unsolicited AE 2 2 0.41 5 5 1.01 0.4515
 AEs unrelated to 31 15 3.07 0.3964
 41 21 4.26
 vaccine
 Solicited AE 17 11 2.25 28 18 3.65 0.2579
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Unsolicited AE 14 14 2.87 13 12 2.43 0.6959

Table. 58-6 Incidence of adverse reactions 0-7 days post booster dose in subjects aged 7 through 11 months

PCV13-TT (N=186) PCV7 (N=191)

P

Events Cases % Events Cases %

Any AE 54 45 24.19 56 45 23.56 0.9043

AEs related to vaccine 52 44 23.66 54 44 23.04 0.9036

Solicited AE 51 44 23.66 53 44 23.04 0.9036

Unsolicited AE 1 1 0.54 1 1 0.52 1.0000

AEs unrelated to

2 1 0.54 2 2 1.05 1.0000

vaccine

Solicited AE 1 1 0.54 0 0 0.00 0.4934

Unsolicited AE 1 1 0.54 2 2 1.05 1.0000

Table. 58-7. Incidence of adverse reactions 8-30 days post booster dose in subjects aged 3 through 11 months

PCV13-TT (N=674) PCV7 (N=684)

P

Events Cases % Events Cases %

Any AE 52 52 7.57 61 51 7.46 1.0000

AEs related to vaccine 2 2 0.30 3 3 0.44 1.0000

Unsolicited AE 2 2 0.30 3 3 0.44 1.0000

AEs unrelated to 0.9162

50 49 7.27 58 48 7.02

vaccine

Unsolicited AE 50 49 7.27 58 48 7.02 0.9162

Table. 58-8 Incidence of adverse reactions 8-30 days post booster dose in subjects aged 3 months

PCV13-TT (N=488) PCV7 (N=493)

P

Events Cases % Events Cases %

Any AE 39 39 7.99 42 34 6.90 0.5444

AEs related to vaccine 2 2 0.41 3 3 0.61 1.0000

Unsolicited AE 2 2 0.41 3 3 0.61 1.0000

AEs unrelated to 0.4524

37 37 7.58 39 31 6.29

vaccine

Unsolicited AE 37 37 7.58 39 31 6.29 0.4524

Table. 58-9 Incidence of adverse reactions 8-30 days post booster dose in subjects aged 7 through 11 months

PCV13-TT (N=186) PCV7 (N=191)

P

Events Cases % Events Cases %

Any AE 13 12 6.45 19 17 8.90 0.4412

AEs related to vaccine 0 0 0.00 0 0 0.00 1.0000

AEs unrelated to 0.4412

13 12 6.45 19 17 8.90

vaccine

Unsolicited AE 13 12 6.45 19 17 8.90 0.4412

Table. 58-10 Incidence of adverse reactions 30 days post booster dose in subjects aged 3 through 11 months

PCV13-TT (N=674) PCV7 (N=684)

P

Events Cases % Events Cases %

Any AE 3 3 0.45 4 4 0.58 1.0000

AEs related to vaccine 0 0 0.00 0 0 0.00 1.0000

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AEs unrelated to

vaccine 3 3 0.45 4 4 0.58 1.0000

Unsolicited AE 3 3 0.45 4 4 0.58 1.0000

Table. 58-11 Incidence of adverse reactions 30 days post booster dose in subjects aged 3 months

PCV13-TT (N=488) PCV7 (N=493)

P

Events Cases % Events Cases %

Any AE 2 2 0.41 3 3 0.61 1.0000

AEs related to vaccine 0 0 0.00 0 0 0.00 1.0000

AEs unrelated to

vaccine 2 2 0.41 3 3 0.61 1.0000

Unsolicited AE 2 2 0.41 3 3 0.61 1.0000

Table. 58-12 Incidence of adverse reactions 30 days post booster dose in subjects aged 7

through 11 months
PCV13-TT (N=186) PCV7 (N=191)
P
Events Cases % Events Cases %
Any AE 1 1 0.54 1 1 0.52 1.0000
AEs related to vaccine 0 0 0.00 0 0 0.00 1.0000
AEs unrelated to
1 1 0.54 1 1 0.52 1.0000
vaccine
Unsolicited AE 1 1 0.54 1 1 0.52 1.0000

7.1.8.3. Analysis for Adverse Events Related to Vaccine

(1) Systemic reactions and local reactions

No significant difference was observed in the incidence of systemic and local reactions among all 3-11 months old subjects and between the study and control group of different age groups, as shown in Table 59-1 to 59-3.

Table. 59-1 Systemic and local reactions post booster dose in subjects aged 3 through 11 months

PCV13-TT (N=674) PCV7 (N=684)
P
Events Cases % Events Cases %
Fever 186 175 25.96 190 175 25.58 0.9013
Fatigue / somnolence 1 1 0.15 1 1 0.15 1.0000
Nausea / vomiting 8 8 1.19 6 6 0.88 0.6029
Diarrhea 15 15 2.23 13 13 1.90 0.7064
Cough 5 5 0.74 8 8 1.17 0.5793
Crying 6 6 0.89 3 3 0.44 0.3388
Allergic reactions 0 0 0.00 1 1 0.15 1.0000
Local pain 2 2 0.30 1 1 0.15 0.6220
Local redness 6 6 0.89 5 5 0.73 0.7720
Local swelling 7 7 1.04 4 4 0.58 0.3825
Local induration 1 1 0.15 0 0 0.00 0.4963

Table. 59-2 Systemic and local reactions post booster dose in subjects aged 3 months

PCV13-TT (N=488) PCV7 (N=493)
P
Events Cases % Events Cases %
Fever 146 137 28.07 149 135 27.38 0.8308
Fatigue / somnolence 1 1 0.20 1 1 0.20 1.0000
Nausea / vomiting 8 8 1.64 4 4 0.81 0.2617
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Diarrhea 12 12 2.46 11 11 2.23 0.8362
Cough 4 4 0.82 6 6 1.22 0.7527
Crying 4 4 0.82 3 3 0.61 0.7243
Local pain 1 1 0.20 0 0 0.00 0.4975
Local redness 5 5 1.02 3 3 0.61 0.5040
Local swelling 4 4 0.82 2 2 0.41 0.4496
Local induration 1 1 0.20 0 0 0.00 0.4975

Table. 59-3 Systemic and local reactions post booster dose in subjects aged 7 through 11 months

PCV13-TT (N=186) PCV7 (N=191)
P
Events Cases % Events Cases %
Fever 40 38 20.43 41 40 20.94 1.0000
Nausea / vomiting 0 0 0.00 2 2 1.05 0.4988
Diarrhea 3 3 1.61 2 2 1.05 0.6817
Cough 1 1 0.54 2 2 1.05 1.0000
Crying 2 2 1.08 0 0 0.00 0.2427
Allergic reactions 0 0 0.00 1 1 0.52 1.0000
Local pain 1 1 0.54 1 1 0.52 1.0000
Local redness 1 1 0.54 2 2 1.05 1.0000
Local swelling 3 3 1.61 2 2 1.05 0.6817

(2). Unsolicited adverse events related to the study vaccine

Unsolicited adverse events related to the study vaccine as determined by the investigators are listed in Table 60-1 to 60-3.

Five and nine cases of unsolicited adverse events in the PCV13-TT and control group were judged to be related to the study vaccine. No significant difference was observed in the intergroup incidences.

Table. 60-1 Unsolicited events related to study vaccine post booster dose in subjects aged 3 through 11 months

PCV13-TT (N=674) PCV7 (N=684)
Even Cas Incidence Even Cas Incidence P
ts es rate ts es rate
0.42
Unsolicited adverse reactions 5 5 0.74 9 9 1.32
17
0.28
Infections and infestations 2 2 0.30 6 6 0.88
77
1.00
Nasopharyngitis 0 0 0.00 1 1 0.15
00
0.45
Upper respiratory tract infection 2 2 0.30 5 5 0.73
20
Respiratory, thoracic and mediastinal 1.00

2 2 0.30 2 2 0.29
disorders 00
0.49
Rhinorrhea 0 0 0.00 2 2 0.29
97
0.49
Nasal obstruction 1 1 0.15 0 0 0.00
63
0.49
Expectoration 1 1 0.15 0 0 0.00
63
General disorders and administration site 0.49
1 1 0.15 0 0 0.00
conditions 63
0.49
Fever 1 1 0.15 0 0 0.00
63

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1.00
Gastrointestinal disorders 0 0 0.00 1 1 0.15
00
1.00
Dyspepsia 0 0 0.00 1 1 0.15
00

Table. 60-2 Unsolicited events related to study vaccine post booster dose in subjects aged 3 months

PCV13-TT (N=488) PCV7 (N=493)
Even Cas Incidence Even Cas Incidence P
ts es rate ts es rate
0.38
Unsolicited adverse reactions 4 4 0.82 8 8 1.62
48
0.21
Infections and infestations 1 1 0.20 5 5 1.01
74
1.00
Nasopharyngitis 0 0 0.00 1 1 0.20
00
0.37
Upper respiratory tract infection 1 1 0.20 4 4 0.81
38
Respiratory, thoracic and mediastinal 1.00
2 2 0.41 2 2 0.41
disorders 00
0.49
Rhinorrhea 0 0 0.00 2 2 0.41
95
0.49
Nasal obstruction 1 1 0.20 0 0 0.00
75
0.49
Sputum 1 1 0.20 0 0 0.00
75
General disorders and administration site 0.49
1 1 0.20 0 0 0.00
conditions 75
0.49
Fever 1 1 0.20 0 0 0.00
75
1.00
Gastrointestinal disorders 0 0 0.00 1 1 0.20
00
1.00
Dyspepsia 0 0 0.00 1 1 0.20
00

Table. 60-3 Unsolicited events related to study vaccine post booster dose in subjects aged 7 through 11 months

PCV13-TT (N=186) PCV7 (N=191)
Incidence Incidence P
Events Cases Events Cases
rate rate
Unsolicited adverse
1 1 0.54 1 1 0.52 1.0000
reactions
Infections and
1 1 0.54 1 1 0.52 1.0000
infestations
Upper respiratory
1 1 0.54 1 1 0.52 1.0000
tract infection

7.1.8.4. Analysis for Adverse Events Unrelated to Vaccine
(1). Solicited events unrelated to study vaccine
Solicited adverse events unrelated to study vaccine were all systemic reactions. Twelve solicited cases (occurred within 7 days after vaccination) were observed in the PCV13-TT of the 3-11 months olds and 18 cases in the control group, which mainly occurred in subjects 3 months of age,

difference was observed for vaccine-independent solicited adverse events between groups (Table 61-1 to 61-2).

Table. 61-1 Solicited events unrelated to study vaccine post booster dose in subjects aged 3 through 11 months

PCV13-TT (N=674) PCV7 (N=684)									
P									
Events Cases % Events Cases %									
Solicited adverse events									
unrelated to study 18 12 1.78 28 18 2.63 0.3566									
vaccine									
Fever 6 6 0.89 11 11 1.61 0.3295									
Nausea / vomiting 1 1 0.15 3 3 0.44 0.6245									
Diarrhea 0 0 0.00 3 3 0.44 0.2495									
Cough 11 11 1.63 10 10 1.46 0.8294									
Allergic reactions 0 0 0.00 1 1 0.15 1.0000									

Table. 61-2 Solicited events unrelated to study vaccine post booster dose in subjects aged 3 months

PCV13-TT (N=488) PCV7 (N=493)									
P									
Events Cases % Events Cases %									
Solicited adverse events									
unrelated to study 17 11 2.25 28 18 3.65 0.2579									
vaccine									
Fever 5 5 1.02 11 11 2.23 0.2065									
Nausea / vomiting 1 1 0.20 3 3 0.61 0.6243									
Diarrhea 0 0 0.00 3 3 0.61 0.2493									
Cough 11 11 2.25 10 10 2.03 0.8292									
Allergic reactions 0 0 0.00 1 1 0.20 1.0000									

(2). Unsolicited events unrelated to study vaccine
The incidence rate of unsolicited adverse events was respectively 9.5% and 9.21% in the study and control group during the observation period. No intergroup difference was observed for the incidence of unsolicited adverse events in each age group, which were mainly infections and infestations (Table 62-1 to 62-3).

Table. 62-1 Unsolicited events unrelated to study vaccine post booster dose in subjects aged 3 through 11 months

PCV13-TT (N=674) PCV7 (N=684)									
Even Cas Incidence Even Cas Incidence P									
ts es rate ts es rate									
0.92									
Unsolicited adverse reactions 68 64 9.50 77 63 9.21									
58									
0.61									
Infections and infestations 38 36 5.34 34 32 4.68									
96									
0.49									
Nasopharyngitis 1 1 0.15 0 0 0.00									
63									
0.49									
Tonsillitis 1 1 0.15 0 0 0.00									
63									
1.00									
Infectious pneumonia 1 1 0.15 1 1 0.15									
00									

0.49									
Rotavirus gastroenteritis 1 1 0.15 0 0 0.00									
63									
0.49									
Upper respiratory tract viral infection 1 1 0.15 0 0 0.00									
63									
0.78									
Upper respiratory tract infection 28 27 4.01 32 30 4.39									
73									
0.49									
Hand-foot-mouth disease 1 1 0.15 0 0 0.00									
63									
0.49									
Gastroenteritis 1 1 0.15 0 0 0.00									
63									
0.24									
Mycoplasma pneumonia 2 2 0.30 0 0 0.00									
61									
1.00									
Herpangina 1 1 0.15 1 1 0.15									
00									
Injury, poisoning and procedural 0.49									
0 0 0.00 3 2 0.29									
complications 97									
1.00									
Skull base fractures 0 0 0.00 1 1 0.15									
00									

1.00
 Subcutaneous hematoma 0 0 0.00 1 1 0.15
 00
 1.00
 Injury 0 0 0.00 1 1 0.15
 00
 Respiratory, thoracic and mediastinal 0.61
 19 19 2.82 16 16 2.34
 disorders 08
 0.68
 Rhinorrhea 13 13 1.93 11 11 1.61
 54
 0.77
 Cough 6 6 0.89 5 5 0.73
 20
 1.00
 Skin and subcutaneous tissue disorders 0 0 0.00 1 1 0.15
 00
 1.00
 Erythema 0 0 0.00 1 1 0.15
 00
 General disorders and administration site 0.77
 5 5 0.74 8 7 1.02
 conditions 35
 0.77
 Fever 5 5 0.74 8 7 1.02
 35
 0.11
 Gastrointestinal disorders 6 6 0.89 15 14 2.05
 28
 0.49
 Enteritis 1 1 0.15 0 0 0.00
 63
 0.14
 Diarrhea 3 3 0.45 9 9 1.32
 43
 1.00
 Abdominal distension 0 0 0.00 1 1 0.15
 00
 1.00
 Oral ulcer 0 0 0.00 1 1 0.15
 00
 1.00
 Vomiting 0 0 0.00 1 1 0.15
 00
 1.00
 Dyspepsia 2 2 0.30 3 3 0.44
 00

Table. 62-2 Unsolicited events unrelated to study vaccine post booster dose in subjects aged
 3 months
 PCV13-TT (N=488) PCV7 (N=493) P

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Even Cas Incidence Even Cas Incidence
 ts es rate ts es rate
 0.44
 Unsolicited adverse reactions 53 50 10.25 55 43 8.72
 64
 0.45
 Infections and infestations 27 26 5.33 22 21 4.26
 80
 0.49
 Nasopharyngitis 1 1 0.20 0 0 0.00
 75
 1.00
 Infectious pneumonia 1 1 0.20 1 1 0.20
 00
 0.49
 Rotavirus gastroenteritis 1 1 0.20 0 0 0.00
 75
 1.00
 Upper respiratory tract infection 21 20 4.10 21 20 4.06
 00
 0.49
 Gastroenteritis 1 1 0.20 0 0 0.00
 75
 0.49
 Mycoplasma pneumonia 1 1 0.20 0 0 0.00
 75
 0.49
 Herpangina 1 1 0.20 0 0 0.00
 75
 Injury, poisoning and procedural 0.49
 0 0 0.00 3 2 0.41
 complications 95
 1.00
 Skull base fractures 0 0 0.00 1 1 0.20
 00

1.00
 Subcutaneous hematoma 0 0 0.00 1 1 0.20
 00
 1.00
 Injury 0 0 0.00 1 1 0.20
 00
 Respiratory, thoracic and mediastinal 0.44
 16 16 3.28 12 12 2.43
 disorders 97
 0.37
 Rhinorrhea 12 12 2.46 8 8 1.62
 62
 1.00
 Cough 4 4 0.82 4 4 0.81
 00
 1.00
 Skin and subcutaneous tissue disorders 0 0 0.00 1 1 0.20
 00
 1.00
 Erythema 0 0 0.00 1 1 0.20
 00
 General disorders and administration site 1.00
 4 4 0.82 6 5 1.01
 conditions 00
 1.00
 Fever 4 4 0.82 6 5 1.01
 00
 0.45
 Gastrointestinal disorders 6 6 1.23 11 10 2.03
 08
 0.49
 Enteritis 1 1 0.20 0 0 0.00
 75
 0.50
 Diarrhea 3 3 0.61 6 6 1.22
 58
 1.00
 Oral ulcer 0 0 0.00 1 1 0.20
 00
 1.00
 Vomiting 0 0 0.00 1 1 0.20
 00
 1.00
 Dyspepsia 2 2 0.41 3 3 0.61
 00

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Table. 62-3 Unsolicited events unrelated to study vaccine post booster dose in subjects aged
 7 through 11 months

PCV13-TT (N=186) PCV7 (N=191)
 Incidence Incidence P
 Events Cases Events Cases
 rate rate
 Unsolicited adverse reactions 15 14 7.53 22 20 10.47 0.3705
 Infections and infestations 11 10 5.38 12 11 5.76 1.0000
 Tonsillitis 1 1 0.54 0 0 0.00 0.4934
 Upper respiratory tract viral
 infection 1 1 0.54 0 0 0.00 0.4934
 Upper respiratory tract infection 7 7 3.76 11 10 5.24 0.6215
 Hand-foot-mouth disease 1 1 0.54 0 0 0.00 0.4934
 Mycoplasma pneumonia 1 1 0.54 0 0 0.00 0.4934
 Herpangina 0 0 0.00 1 1 0.52 1.0000
 Respiratory, thoracic and
 mediastinal disorders 3 3 1.61 4 4 2.09 1.0000
 Rhinorrhea 1 1 0.54 3 3 1.57 0.6231
 Cough 2 2 1.08 1 1 0.52 0.6191
 General disorders and
 administration site conditions 1 1 0.54 2 2 1.05 1.0000
 Fever 1 1 0.54 2 2 1.05 1.0000
 Gastrointestinal disorders 0 0 0.00 4 4 2.09 0.1231
 Diarrhea 0 0 0.00 3 3 1.57 0.2481
 Abdominal distension 0 0 0.00 1 1 0.52 1.0000

7.1.8.5. Occurrence of adverse events with different procedures in the 2-month and 3-month trial / PCV7 group

The incidence of vaccine related adverse events after the booster immunization in the 2-month group was significantly lower than that in the 3-month-old group ($p=0.0180$). The incidence of fever and diarrhea in the 2-month PCV13-TT was lower than that in the 3-month-old trial and control group (fever: $p=0.0360$, diarrhea: $p=0.0156$). No significant difference was observed among other symptom groups, see table 63-1 to 63-2.

Table. 63-1 Incidence of adverse reactions 30 days post booster dose in PCV13-TT (2M), PCV13-TT (3M) and PCV7 (3M)

PCV13-TT (2M) PCV13-TT (3M) PCV7 (3M)
 (N=487) (N=488) (N=493) P
 Events Cases % Events Cases % Events Cases %
 Any AE 202 151 31.01 260 186 38.11 270 179 36.31 0.0523

Grade 1 115 95 19.51 124 105 21.52 128 107 21.70 0.6516
Grade 2 86 74 15.20 122 104 21.31 133 92 18.66 0.0461
Grade 3 1 1 0.21 14 12 2.46 9 7 1.42 0.0051
AEs related to
vaccine 130 116 23.82 190 153 31.35 187 149 30.22 0.0180
Grade 1 90 83 17.04 105 93 19.06 116 98 19.88 0.4929
Grade 2 40 39 8.01 76 70 14.34 66 56 11.36 0.0070
Grade 3 0 0 0.00 9 8 1.64 5 4 0.81 0.0075
Solicited AE 125 113 23.20 186 152 31.15 179 146 29.61 0.0126
Grade 1 88 81 16.63 104 93 19.06 111 96 19.47 0.4657
Grade 2 37 36 7.39 73 68 13.93 63 55 11.16 0.0039
Grade 3 0 0 0.00 9 8 1.64 5 4 0.81 0.0075
Unsolicited AE 5 5 1.03 4 4 0.82 8 8 1.62 0.5354
Grade 1 2 2 0.41 1 1 0.20 5 5 1.01 0.2677
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Grade 2 3 3 0.62 3 3 0.61 3 3 0.61 1.0000
Grade 3 ///////////////
AEs unrelated to
vaccine 72 48 9.86 70 51 10.45 83 51 10.34 0.9523
Grade 1 25 20 4.11 19 15 3.07 12 11 2.23 0.2535
Grade 2 46 37 7.60 46 39 7.99 67 45 9.13 0.6700
Grade 3 1 1 0.21 5 4 0.82 4 3 0.61 0.4635
Solicited AE 24 14 2.87 17 11 2.25 28 18 3.65 0.4287
Grade 1 11 8 1.64 7 5 1.02 4 3 0.61 0.2567
Grade 2 12 8 1.64 8 7 1.43 23 17 3.45 0.0797
Grade 3 1 1 0.21 2 2 0.41 1 1 0.20 0.8504
Unsolicited AE 48 42 8.62 53 50 10.25 55 43 8.72 0.6248
Grade 1 14 14 2.87 12 12 2.46 8 8 1.62 0.3935
Grade 2 34 31 6.37 38 37 7.58 44 37 7.51 0.7195
Grade 3 0 0 0.00 3 3 0.61 3 2 0.41 0.2981

Table. 63-2 Incidence of systemic and local reactions post booster dose in subjects aged 7 through 11 months

PCV13-TT (2M) PCV13-TT (3M) 3 monthsPCV7											
(N=487) (N=488) (N=493) P											
Events Cases % Events Cases % Events Cases %											
Fever	114	105	21.56	146	137	28.07	149	135	27.38	0.0360	
Fatigue /	0	0	0.00								
somnolence	1	1	0.20	1	1	0.20	1	0.0000			
Nausea /											
vomiting	1	1	0.21	8	8	1.64	4	4	0.81	0.0523	
Diarrhea	2	2	0.41	12	12	2.46	11	11	2.23	0.0156	
Cough	1	1	0.21	4	4	0.82	6	6	1.22	0.2075	
Abnormal											
crying	1	1	0.21	4	4	0.82	3	3	0.61	0.4635	
Local pain	1	1	0.21	1	1	0.20	0	0	0.00	0.5536	
Local											
redness	2	2	0.41	5	5	1.02	3	3	0.61	0.5375	
Local											
swelling	3	3	0.62	4	4	0.82	2	2	0.41	0.6556	
Local	0	0	0.00								
induration	1	1	0.20	0	0	0.00	0	0.6642			

7.1.8.6. Serious Adverse Events (SAE)
A total of 8 cases (10 events) of SAE were observed after the booster immunization, including 3 cases in the PCV13-TT and 4 cases in the control group of subjects aged 3-11 months (no significant intergroup difference was observed); and 1 case in PCV13-TT of the 2 months olds. SAEs after booster immunization were not related to the study vaccine (Table 64-1 to 64-6).

Table. 64-1 Incidence of adverse events post booster dose in each PCV13-TT group											
Observations Any SAE Related SAE Unrelated SAE											
Groups											
Events Cases % Events Cases % Events Cases %											
3-11 PCV13- 674											
4	3	0.45	0	0	0.00	4	3	0.45			
months TT											
PCV7	684	5	4	0.58	0	0	0.00	5	4	0.58	
P	1.0000	1.0000	1.0000								
3 PCV13-											
488	0	0	0.00	0	0	0.00	0	0	0.00		
months TT											
PCV7	493	3	2	0.41	0	0	0.00	3	2	0.41	
P	0.4995	1.0000	0.4995								
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7-11 PCV13-
186 4 3 1.61 0 0 0.00 4 3 1.61
months TT
PCV7 191 2 2 1.05 0 0 0.00 2 2 1.05
P 0.6817 1.0000 0.6817
2 PCV13-
487 1 1 0.21 0 0 0.00 1 1 0.21
months TT

Table. 64-2 Incidence of adverse events post booster dose in subjects aged 2 months
PCV13-TT (N=487)
Events Cases Incidence rate
Infections and infestations 1 1 0.21

Infectious pneumonia 1 1 0.21

Table. 64-3 Incidence of adverse events post booster dose in subjects aged 3 through 11 months

PCV13-TT (N=674)		PCV7 (N=684)		Incidence P	
Events	Cases	Events	Cases	rate	rate
Infections and infestations	4 3	0.45 3 3	0.44 1.0000		
Infectious pneumonia	2 2	0.30 2 2	0.29 1.0000		
Hand-foot-mouth disease	1 1	0.15 0 0	0.00 0.4963		
Mycoplasma pneumonia	1 1	0.15 0 0	0.00 0.4963		
Herpangina	0 0	0.00 1 1	0.15 1.0000		
Injury, poisoning and procedural complications	0 0	0.00 2 1	0.15 1.0000		
Skull base fractures	0 0	0.00 1 1	0.15 1.0000		
Subcutaneous hematoma	0 0	0.00 1 1	0.15 1.0000		

Table. 64-4 Incidence of adverse events post booster dose in subjects aged 3 months

PCV13-TT (N=488)		PCV7 (N=493)		Incidence P	
Events	Cases	Events	Cases	rate	rate
Infections and infestations	0 0	0.00 1 1	0.20 1.0000		
Infectious pneumonia	0 0	0.00 1 1	0.20 1.0000		
Injury, poisoning and procedural complications	0 0	0.00 2 1	0.20 1.0000		
Skull base fractures	0 0	0.00 1 1	0.20 1.0000		
Subcutaneous hematoma	0 0	0.00 1 1	0.20 1.0000		

Table. 64-5 Incidence of adverse events post booster dose in subjects aged 7 through 11 months

PCV13-TT (N=186)		PCV7 (N=191)		P	
Events	Cases	Incidence rate	Events	Cases	Incidence rate
Infections and infestations	4 3	1.61 2 2	1.05 0.6817		
Infectious pneumonia	2 2	1.08 1 1	0.52 0.6191		
Hand-foot-mouth disease	1 1	0.54 0 0	0.00 0.4934		
Mycoplasma pneumonia	1 1	0.54 0 0	0.00 0.4934		
Herpangina	0 0	0.00 1 1	0.52 1.0000		

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Table 64-6. Listing of serious adverse events in subjects post booster dose

Term of Days since
Study Vaccine Age System organ Duration
Gender Site adverse Description last Severity SAE leads to Relevance
No. groups (mos) classification (days)
events vaccination

The subject coughed without apparent cause on 20170522. The cough was non-spasmodic and paroxysmal, with sputum difficult to cough up. The cough alternated between severe and mild and became apparent during night time. The subject experienced fever on 20170603. The highest body temperature was 39.2■ and the patterns of fever was irregular. The subject Hospitalization

Infections
Group Infectious was given oral drugs and atomization or prolonged Possibly 325 Male 2.7 Hebei and 78 6 Grade 3
A pneumonia treatment (unspecified) while with poor length of unrelated infestations response, therefore was treated in outpatient hospital stay department of People's Hospital of Zhengding County on 20170605.
Hematology results: WBC 10.40×10^9/L, CRP2.90 mg/L. Chest X-ray results: thickened lung markings in both lungs, with scattered patchy opacities in both lung fields. The subject was admitted for "bronchopneumonia" for further treatment. The subject experienced headache and Injury, dizziness, with interruption of bone Hospitalization poisoning
Group Skull base continuity in the right occipital. No or prolonged Definitely 1497 Male 3.2 Shanxi and 3 6 Grade 3
B fractures abnormality was observed in the residual length of unrelated procedural brain parenchyma. The brain middle was hospital stay complications structurally centered. Injury, Hospitalization poisoning A 4 cm×5 cm ecchymoma was observed in

Group Subcutaneous or prolonged Definitely
1497 Male 3.2 Shanxi and the right nuchal occipital region, with 3 6 Grade 3
B hematoma length of unrelated
procedural tenderness.
hospital stay
complications

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The subject experienced cough and
expectoration on April 12th. The cough
worsened on April 15th. Physical Hospitalization
Infections
Group Infectious examination on admission results: heavy or prolonged Possibly
1536 Female 3.3 Shanxi and 28 5 Grade 3
B pneumonia breathing sounds in both lungs with gurgling length of unrelated
infestations
in chest and bubbling sound. X-ray results hospital stay
showed thickened, increased and blurred
lung markings.
The subject developed a fever of 38.5■ on
November 27th due to a cold, with
paroxysmal and productive cough, and
occasional dyspnea. The subject was given
Hospitalization
Infections oral medicine treatment (unspecified) at the
Group Infectious or prolonged Definitely
1659 Female 10.6 Henan and village clinic and the People's hospital in the 113 9 Grade 4
B pneumonia length of unrelated
infestations county but with poor response. The cough
hospital stay
aggravated on November 30th and the
subject was admitted into the People's
Hospital of Wen County on December 2nd
for bronchopneumonia.
Herpes was identified on hands, feet, and hip
with a body temperature of 39.2■. Physical
examination on admission: The subject was
in poor spirits. Scattered red papules were
identified on hands, feet, and hip while
individual herpes was identified. The
Hospitalization
Infections Hand-foot- diameter of the papules was approximately
Group or prolonged Definitely
1663 Male 11.3 Henan and mouth 2-4 mm, with thick wall. The papules were 25 5 Grade 4
A length of unrelated
infestations disease opaque, with turbid papular liquid.
hospital stay
Xerocheilia and pharyngeal cavity
congestion were observed. A tiny herpes
was identified in the palatopharyngeal
arches with periphery redness. WBC:
8.76×10⁹/L, L: 59.40%, N: 25.20%, HB:
113G/L, indicative of viral infection.
Hospitalization
Infections Heavy breathing sounds in both lungs. Fever
Group Mycoplasma or prolonged Definitely
1663 Male 11.3 Henan and of 39.2■. Mycoplasma pneumonia weakly 25 5 Grade 4
A pneumonia length of unrelated
infestations positive.
hospital stay

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The subject experienced fever of 38.5■
with salivation in the mouth and crying
while eating. Physical examination results
Hospitalization
Infections on admission: poor spirit and xerocheilia
Group or prolonged Definitely
1714 Male 10.9 Henan and Herpangina and pharyngeal cavity congestion were 15 6 Grade 4
B length of unrelated
infestations observed. Several tiny herpes was identified
hospital stay
in both palatopharyngeal arches with
periphery redness. Heavy breathing sounds
were identified in both lungs.
The subject coughed without apparent cause
on around 20161120 (more than 10 days
before admission on 20161201). The cough
was non-spasmodic and paroxysmal, with
sputum difficult to cough up and wheezing
aggravated after exercise. The subject was
treated in the village clinic (unspecified) for
more than 10 days but with poor response. Hospitalization
Infections
Group Infectious The subject was treated in People's Hospital or prolonged Possibly
1817 Male 11.4 Hebei and 85 9 Grade 3

A pneumonia of Zhengding County on 20161128. length of unrelated infestations

Hematology results: WBC 16.77×10⁹/L, hospital stay HGB 87G/L, CRP 8.5 MG/L, chest X-ray results: thickened lung markings in both lungs, with scattered patchy opacities in both lung fields. The subject received intravenous drip treatment at the outpatient clinic, and the symptoms had slightly improved.

The subject coughed without apparent cause on 20161114. The cough was non-spasmodic and paroxysmal, with sputum difficult to cough up and rhinorrhea. The highest body temperature was 39.2■ and the patterns of fever was irregular, with Hospitalization

Infections
Group Infectious wheezing aggravated after exercise. The or prolonged Possibly 1823 Male 8.8 Hebei and 75 5 Grade 3

A pneumonia subject received An'erning, cephalosporin, length of unrelated infestations pholcodine and enema at the local village hospital stay clinic for around 3 days but with poor response, therefore was taken to the outpatient department of People's Hospital of Zhengding County treatment on 20161120. Hematology results: WBC

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10.13×10⁹/L, CRP 1.76 MG/L, chest X-ray results: thickened and disordered lung markings in both lungs. The outpatient diagnosis was bronchopneumonia, which was treated using cefoxitin, Reduning and doxofylline through intravenous drip.

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7.1.8.7 Analysis of concomitant medications
Concomitant medications were recorded in 239 subjects (414 cases) in the PCV13-TT group at the 3–11 months of age and in 237 cases (447 cases) in the PCV7 group during the period of enhanced immunization. The most commonly used drugs were vaccine, followed by antipyretic/analgesics/nonsteroidal anti-inflammatory drugs, antiallergic drugs, and antibiotics. There were no statistical differences between the PCV13-TT group and the PCV7 group in the incidence of various concomitant medications. During the period of enhanced immunization, the concomitant medications of the subjects are shown in Tables 65-1 to 65-4.

Table 65-1 Concomitant Medications Post-Booster Dose in Subjects Aged 2 Months
PCV13-TT group (N=487)

Medication Types	Number	Number of Incidence
Events Cases rate		
Any medication	334	209 42.92
Antivirals	2	2 0.41
Allergy medications (chlorpheniramine, etc.)	30	26 5.34
Antibiotics	22	22 4.52
Other medications	36	29 5.95
Any vaccines or biological products	205	159 32.65
Anti-pyretics/analgesics/NSAIDs (aspirin, ibuprofen, Fenbid, etc.)	39	33 6.78

Table. 65-2Concomitant Medications Post-Booster Dose in Subjects Aged 3-11 Months
PCV13-TT group (N=674) PCV7 group (N=684)

Number	Number	Number	Number
Medication Types P			
of of % of of %			
Events Cases Events Cases			
Any medication	414	239 35.46	447 237 34.65 0.7762
Hormone/steroids	4	4 0.59	3 3 0.44 0.7240
Allergy medications	34	31 4.60	32 26 3.80 0.5003
Anti-pyretics/analgesics/NSAIDs	70	61 9.05	60 52 7.60 0.3767
Any vaccines or biological products	175	147 21.81	188 149 21.78 1.0000
Antibiotics	41	35 5.19	50 41 5.99 0.5561
Antivirals	10	9 1.34	7 7 1.02 0.6247
Other medications	69	49 7.27	94 54 7.89 0.6829
Unspecified	11	9 1.34	13 11 1.61 0.8226

Table 65-2 Concomitant Medications Post-Booster Dose in Subjects Aged 3 Months
PCV13-TT group (N=488) PCV7 group (N=493)

Number	Number	Number	Number
Medication Types P			
of of % of of %			
Events Cases Events Cases			
Any medication	323	184 37.70	360 189 38.34 0.8439
Hormone/steroids	2	2 0.41	2 2 0.41 1.0000
Allergy medications	29	26 5.33	23 19 3.85 0.2886
Anti-pyretics/analgesics/NSAIDs	56	48 9.84	52 44 8.92 0.6620
Any vaccines or biological	136	112 22.95	157 121 24.54 0.5996

products
 Antibiotics 31 27 5.53 36 29 5.88 0.8908
 Antivirals 5 5 1.02 7 7 1.42 0.7731

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Other medications 55 38 7.79 75 41 8.32 0.8148
 Unspecified 9 7 1.43 8 7 1.42 1.0000

Table 65-4 Concomitant Medications Post-Booster Dose in Subjects Aged 7-11 Months

PCV13-TT group (N=186)	PCV7 group (N=191)
Number	Number
Medication Types	P
of	%
Events	Cases
Any medication	91 55 29.57 87 48 25.13 0.3564
Hormone/steroids	2 2 1.08 1 1 0.52 0.6191
Allergy medications	5 5 2.69 9 7 3.66 0.7710
Anti-pyretics/analgesics/NSAIDs	14 13 6.99 8 8 4.19 0.2672
Any vaccines or biological products	39 35 18.82 31 28 14.66 0.3340
Antibiotics	10 8 4.30 14 12 6.28 0.4924
Antivirals	5 4 2.15 0 0 0.00 0.0583
Other medications	14 11 5.91 19 13 6.81 0.8338
Unspecified	2 2 1.08 5 4 2.09 0.6852

7.2. SUMMARY OF SAFETY

Significant difference was observed in the incidence of adverse events between the study group and the control group. The overall incidence of adverse events in the study group was lower than that in the control group. The incidence of vaccine-related adverse events in the study group was slightly lower than that in the control group, but the difference was not significant.

The main symptom of systemic adverse reaction was fever. The incidence of systemic reactions in study group was lower than that in the control group. The main symptoms of local adverse reactions were redness, swelling and pain at the injection sites.

The incidence of adverse events with a severity reaching grade 3 was 7.07% and 5.36% in the study group and the control group respectively. The incidence of vaccine-related grade 3 adverse events was 4.21% and 2.41% respectively. Significant difference was observed between groups ($p = 0.0182$).

A total of 80 cases of SAEs were observed in the primary immunization stage, and 8 cases were observed in the booster immunization stage. No significant difference was observed in the inter-group incidence. Except for one case in the 3-month study group during the primary immunization stage which might be related to the study vaccine, the rest of SAEs were all not related to the study vaccine.

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8. DISCUSSION AND CONCLUSION

8.1 DISCUSSION OF IMMUNOGENICITY

8.1.1 Analysis of common serotypes

8.1.1.1 Antibody positive rate and antibody level

After primary immunization series in test group, the antibody positive rate (antibody concentration $\geq 0.35 \mu\text{g/mL}$) was high, though in common serotypes, the positive rate of 6B was 90.45% in 3-month age group, besides this, the positive rate of all other serotypes in remaining age groups was above 95%, with antibody GMC consistently over $2.0 \mu\text{g/mL}$.

Apart from the serotype 6B in subjects aged 3-71 months, 6B and 23F in subjects aged 3 months, and 9V in subjects aged 7-11 months, the lower limit of two-sided 95% CI for percentage difference (test group – control group) of other IgG antibody with concentration $\geq 1.0 \mu\text{g/mL}$ was $> -10\%$. The percentage of IgG antibody concentration $\geq 1.0 \mu\text{g/mL}$ in subjects of test group aged 12-23 months and 24-71 months were higher than that in control group.

8.1.1.2 Noninferiority Analysis of Primary Endpoint and the Antibody Positive rate and

Antibody Level in target population

The lower limit of 97.5% CI was $> -10\%$ for the rate difference of antibody positive rate of 7 common serotypes (test group – control group) in subjects aged 3-71 months after primary immunization series, the lower limit of 97.5% CI for antibody GMC ratio (test group/control group) was > 0.5 , demonstrating the Noninferiority between test group and control group for common serotypes.

In the 3-month main target population, 6 out of 7 common serotypes

8.1.1.3 Sequential comparison of different immunization procedures

For test group subjects aged 2 months versus control group subjects aged 3 months, the lower limit of 97.5% CI was $> -10\%$ for the rate difference of antibody positive rate of 7 common serotypes; the lower limit of 2-sided 97.5% CI was > 0.5 for the antibody GMC ratio of serotypes 6B, 9V, 14, 19F, and 23F. The 7 common serotypes in test group subjects aged 2 months were noninferior to control group subjects aged 3 months.

For test group aged 2 months versus test group aged 3 months, the additional serotype 5 was observed with a lower limit of 2-sided 97.5% CI $> -10\%$ (-10.84%) for the rate difference of antibody positive rate, and a lower limit of 2-sided 97.5% CI < 0.5 for antibody GMC ratio, besides, the common serotype 4 was with a lower limit of 2-sided 97.5% CI < 0.5 for antibody GMC, apart from the exceptions above, all other serotypes achieved noninferiority criteria in this comparison.

8.1.1.4. Antibody positive rate and antibody level analysis for other age groups

For 7-11 months age group, the lower limit of 95% CI was $> -10\%$ for the rate difference of antibody positive rate (test group – control group), the lower limit of 95% CI was > 0.5 for GMC ratio (test group/control group) of 6 common serotypes except serotype 14. In 12-23 months age group and 24-71 months age group, the lower limit of 95% CI was $> -10\%$ for rate difference of antibody positive rate (test group – control group), and the lower limit of 95% CI was > 0.5 for GMC ratio (test group / control group) for 7 common serotypes.

8.1.1.5 Antibody seroconversion rate and antibody level analysis for population with negative antibody before immunization

In 3-71 months age group, the lower limit of 97.5% CI was $> -10\%$ for antibody seroconversion

rate (test group – control group) of 6 common serotypes except serotype 6B; the antibody GMC ratio of 7 common serotypes was all > 0.5.
The antibody seroconversion rates of common serotypes in test group subjects aged 2 months were noninferior to control group subjects aged 3 months, while serotype 6B did not meet the
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noninferiority criteria, which was consistent with the findings of total population.
Accurate analysis might not be performed for 12-23 months age group and 24-71 months age group due to lack of adequate subjects with negative antibody before immunization.

8.1.2 Analysis of additional serotypes

8.1.2.1 Efficacy analysis

In 3-71 months age groups, the antibody positive rates of all additional serotypes were > 90%, and the lower limit of 95% CI was > 0 for the rate difference (test group – test group), besides, the lower limit of 95% CI was > 1 for GMC ratio (test group / control group) of each additional serotype, demonstrating the achievement of superiority of each additional serotype in PCV13-TT group.

8.1.2.2 Noninferiority analysis of antibody positive rate for common serotypes and serotypes with lowest GMC

The serotypes with lower antibody positive rate were 23F in 3-71 months age group, 18C in 3 months age group, 23F in 7-11 months age group, 6B in 12-23 months age group, and 23F in 24-71 months age group; serotype 6B was confirmed to be with the lowest antibody GMC in all groups.

Of all comparisons, apart from the serotype 5 in 7-11 months and serotype 3 in 24-71 months age group, the lower limit of 97.5% / 95% CI was > -10% for the rate difference of antibody positive rates of additional serotypes with lowest antibody positive rates, and thus were considered to be noninferior to common serotypes with lowest antibody positive rates according to noninferiority criteria.

8.1.3. Discussion of booster immunizations

8.1.3.1. Antibody positive rate and antibody level before booster immunization

Before booster dose, the antibody positive rates of 4 common serotypes (6B, 14, 19F, and 23F) and 2 additional serotypes (7F and 19A) in test group subjects aged 2 months and 3 months remained above 95% of the effects after primary doses; the antibody GMC of common serotypes 4, 9V, 18C and additional serotypes 1, 3 and 5 were <1.0 µg/mL, and all were > 0.35 µg/mL but serotype 5 in test group subjects aged 2 months with GMC of 0.32 µg/mL. Similar trend of change was noted for common serotypes between PCV13-TT group and PCV7 group.

In 7-11 months age group, a total of 7 serotypes had sustained relatively high antibody positive rates and antibody GMC before booster dose, with no significant inter-group differences observed for antibody positive rates between test group and control group, the GMC in test group were higher than that in control group except for serotypes 9V and 14. Among the 6 additional serotypes, antibody rates were all above 95% apart from serotype 5 with a rate of 77.84%.

8.1.3.2. After booster immunization

After booster dose, in test group subjects aged 3 months, the antibody positive rates and antibody GMC of 7 common serotypes were noninferior to those in control group, both the test group and control group had more than 3.5-fold increase in antibody GMC compared to that before booster dose. As to the 6 additional serotypes, the lower limit of 95% CI for antibody positive rate of 19A was -0.2%, though other serotypes were superior to control group; all the 6 additional serotypes attained superiority criteria with respect to antibody GMC, with > 3.5-fold increase compared to that before booster dose.

The antibody positive rates of 7 common serotypes in test group subjects aged 2 months were noninferior to control group subjects aged 3 months. As to the 6 additional serotypes, despite the lower limit of 95% CI for antibody positive rate of 19A was -0.2%, other serotypes were superior to control group; all the 6 additional serotypes attained superiority criteria with respect to antibody GMC.

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GMC. The antibody positive rates and antibody GMC of 13 serotypes in test group subjects aged 2 months were noninferior to those in test group subjects aged 3 months, with > 3.5-fold increase in GMC after booster dose compared to that before.

The antibody positive rates and antibody GMC of 7 common serotypes in test group subjects aged 7-11 months were noninferior to those in control group, for the 6 additional serotypes, the lower limit of 95% CI for the rate difference of serotype 19A was -1.93%, though all other serotypes were superior to control group; all the 6 additional serotypes attained superiority criteria with respect to antibody GMC. The antibody GMC of 13 serotypes in test group and 7 serotypes in control group were with more than 1-fold increase after booster dose.

8.2 SAFETY ANALYSIS

8.2.1. Primary immunization period

A total of 790 cases (2282 events) of adverse events were observed in test group (3-71 months), compared to 851 cases (2440 events) in control group, with the overall incidence of adverse events being 70.73% and 76.05% in test group and control group, and the difference between groups was of statistical significance, the overall incidence of adverse reactions was notably lower compared to control group. Incidence rates of investigational vaccine-related adverse reactions in test group and control group were 60.79% and 64.79%, respectively, with only numerical difference observed. Incidence of adverse reactions were relatively higher in 2-6 months and 7-11 months age groups, and major inter-group differences of adverse reactions were noted among subjects aged 7-11 months.

The incidence of systemic adverse reactions were 54.52% and 58.98% in test group and control group, and it was obviously lower in test group versus control group ($p=0.0364$). As to incidence of local adverse reactions, 15.67% and 14.66% were observed in test group and control group, respectively, demonstrating no significant differences.

The most common systemic adverse reaction was fever, which occurred in 47.54% and 53.80% of subjects in test group and control group, respectively, with lower incidence rate in test group compared to control group ($p=0.0035$), no evident differences regarding other systemic reactions or symptoms between test group and control group were generated. The most frequent local adverse reaction was redness of injection site, followed by swelling and pain.

The severity of adverse reactions in 3-71 months age group was primarily grade 1 or 2, and the investigational vaccine-related adverse reactions of grade 3 affected 4.21% and 2.41% in test group and control group subjects. No significant differences were seen between test group and control group with regard to systemic reactions of grade 3. Incidence of grade 3 local reactions were higher in test group versus control group ($p=0.0007$), and were predominantly redness, swelling and pain

in subjects aged 7 months and above, though there were no evidence of significant differences for grade 3 adverse events between test group and control group in subjects aged months. Totally, 80 cases with 105 events of SAE were collected during primary immunizations, in subjects aged 2 months, 12 cases with 13 events were observed, in test group subjects aged 3-71 months, there were 34 cases with 49 events, compared to 34 cases with 43 events in control group, while the incidence of SAE was of no statistical significance between test group and control group. All SAE, other than 1 case of SAE (fever) in subjects aged 3 months was adjudicated to be likely related to investigational vaccine, were irrelevant to the vaccine under investigation. The incidence of adverse events-related dropout and solicited/unsolicited events unrelated to investigational vaccine showed no difference between test group and control group.

8.2.2 Booster immunization period

In 3-11 months age group, the incidence of adverse events in test group and control group were 35.46% and 34.50%, with grade 3 adverse events occurred in 1.93% and 1.02% of each group, though the difference was not significant. Absence of significance regarding the incidence of

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adverse events between PCV13-TT group and PCV7 group in 3 months and 7-11 months age groups. The incidence of investigational vaccine-related adverse events in subjects aged 2 months was lower compared to test group and control group subjects aged 3 months.

Similar to what was observed during primary immunization period, main adverse reactions in booster immunization period were fever and local swelling, the incidence of any symptoms were similar across PCV13-TT group and PCV7 group in 3 months and 7-11 months age groups, while the incidence of symptoms such as fever and diarrhea were notably lower in 2 months age group compared to test group and control group subjects aged 3 months.

Adverse events were mainly grade 1 and 2, and there were 13 cases and 7 cases of grade 3 adverse events in test group and control group, respectively, corresponding to an incidence of 1.93% and 1.02%, with no inter-group differences observed.

After booster immunization, 8 cases with 10 events of SAE were recorded in all, with 3 cases in 3-11 months age group, 4 cases in 3-11 months age group, and 1 case in 2 months age group, the difference of incidence among these groups were insignificant, besides, SAE were not related to investigational vaccine.

8.3. CONCLUSION

This clinical trial is designed to evaluate the immunogenicity and safety of a 13-valent pneumococcal conjugate vaccine, jointly developed by Yunnan Walvax Biotechnology Co., Ltd and Yuxi Walvax Biotechnology Co., Ltd, as administered in subjects aged 3 months (dosed at 0, 1, 2 months), 7-11 months (dosed at 0, 2 months), 12-23 months (2 doses) and 24-71 months (1 dose), with marketed 7-valent pneumococcal conjugate vaccine as control, and to assess the immunization procedures of 13-valent pneumococcal conjugate vaccine in subjects aged 2 months (at least 6 weeks) and 3 months.

In main target population (3 months), during primary immunization period, all common serotypes, except 6B, were noninferior to control group, the serotypes in test group were noninferior to control group after booster immunization, meanwhile, additional serotypes in test group were superior to control group after primary immunization; in subjects aged 7-31 months, the common serotypes were noninferior to control group during primary immunization period, with all additional serotypes being superior to control group.

For subjects aged 2 months versus control group subjects aged 3 months, all common and additional serotypes after primary immunization achieved noninferiority or superiority criteria.

For subjects aged 2 months versus test group subjects aged 3 months, 12 serotypes, apart from serotype 5 in subjects aged 2 months during primary immunization period, were noninferior to test group subjects aged 3 months; all 13 serotypes after booster immunization were noninferior to test group aged 3 months.

The antibody positive rates of common serotypes in subjects aged 7-11 months, 12-23 months, and 24-71 months were noninferior to that in corresponding PCV7 group. For additional serotypes, the antibody positive rates in 7-11 months reached superiority criteria after primary immunization; and were generally superior to control group for test group subjects aged 12-23 months and 24-71 months.

The findings of before and after booster immunization indicated that, several serotypes were with moderately decreased antibody levels before booster immunization in subjects of either PCV13-TT group or PCV7 group, while the antibody positive rates and antibody GMC following 1 dose of booster immunization had reached or even exceeded the antibody response after primary immunization, with significant booster effects, demonstrating evident immune memory effects of conjugate vaccines. Subjects aged 2-6 months had more than 3-fold increase in antibody GMC after booster dose, and higher than 1-fold increase was noted among subjects aged 7-11 months. The overall incidence of adverse reactions were lower in test group compared to control group,

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though there was no significant inter-group difference with regard to investigational vaccine-related adverse reactions. Fever was the main systemic adverse reaction, and the incidence of systemic reactions in test group was lower than control group. Local adverse reactions included mainly redness, swelling and pain in injection site. No obvious differences were observed between test group and control group regarding the incidence of local adverse reactions.

In PCV13-TT group and PCV7 group, investigational vaccine-related adverse events of grade 3 occurred in 4.21% and 2.41% of subjects, respectively. Higher incidence of local adverse reactions of grade 3 was noted in test group versus control group, while there was no obvious difference observed in the subjects aged 3 months.

During primary immunization period, 80 cases with 105 events of SAE were collected, while in booster immunization period, 8 cases with 10 events of SAE were recorded, no differences were noted between PCV13-TT group and PCV7 group for the incidence of SAE. All SAE, other than 1 case of SAE (fever) in subjects aged 3 months was adjudicated to be likely related to investigational vaccine, were irrelevant to the vaccine under investigation.

In conclusion, the main population aged 3 months had achieved the noninferiority criteria for immunogenicity for 7 common serotypes after booster immunization, and reached the superiority criteria for immunogenicity for 6 additional serotypes after primary immunization. All common serotypes and additional serotypes in 3-71 months age groups had met the criteria of noninferiority or superiority. The common serotypes and additional serotypes in test group subjects aged 2 months were noninferior and/or superior to control group subjects aged 3 months. In test group subjects aged 2 months, 13 serotypes were noninferior to test group subjects aged 3 months after

booster immunization. The incidence of adverse events / reactions in test group was no higher than in control group. Evaluations of immunogenicity and safety in this clinical trial achieved study objective as defined in protocol.

9. DESCRIPTION OF SPECIAL CONDITIONS IN THE STUDY

N/A.

10. SUMMARY FROM RESEARCH CENTERS OF PARTICIPATING UNITS

Entrusted by Yunnan Walvax Biotechnology Co., LTD. and Yuxi Walvax Biotechnology Co., LTD., the Hebei Provincial CDC, the Henan Provincial CDC, and the Shanxi Provincial CDC, as the sub-centers, were responsible for the phase III clinical trial of 13-valent pneumococcal polysaccharide conjugate vaccine (PCV13).

10.1 HEBEI SUB-CENTER

10.1.1 Overview of Hebei sub-center

The Hebei Provincial CDC and its subordinate Zhengding County CDC and Dingxing County CDC were reviewed and approved by Ethics Committee of the Center for Disease Control and Prevention of Hebei Province to obtain "Ethical Approval Certificate" on Sep. 9, 2015; these research institutions obtained the "One-Time Certification for Vaccine Clinical Trial Institution" after review and approval by Center for Food and Drug Inspection of NMPA on Dec. 8, 2015; and these institutions convened this launch meeting and training session on Apr. 2, 2016. The major timelines of the trial are shown in Table 66-1.

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Table 66-1. Major timelines of the clinical trial in Hebei sub-center

Item Site 1 Site 2

Number of subjects 420 420

Start time of enrollment 20160408 20160409

Completion time of basic

20161122 20161109

immunization

Completion time of blood

20161222 20161209

collection after basic immunization

Completion time of enhanced

20170414 20170414

immunization

Completion time of blood

collection after enhanced 20170515 20170514

immunization

Completion time of safety

20171012 20171019

observation

Site 1: Zhengding CDC; Site 2: Dingxing CDC.

10.1.2 Enrollment and vaccination in Hebei sub-center

Hebei sub-center was planned to receive 840 subjects as the samples, including 160 subjects at the 2 months of age, 320 subjects at the 3 months of age, and each 120 subjects at the 7–11 months, 12–23 months, and 24–71 months of age. The enrollment and vaccination are shown in Table 66-2.

10.1.3 Completion of the trial in Hebei sub-center

In Hebei sub-center, 1030 subjects were screened and 840 subject were enrolled, 839 of which completed the first dose of vaccination and safety observation, and 784 subjects completed the full vaccination, with the compliance rate being 93.33%; 753 subjects completed vaccination and effective blood collection before and after immunization, with the compliance rate being 89.64%.

A total of 5405 serum samples were collected, including 2708 samples pending testing and 2697 standby samples. The samples pending testing were sent to the Institute for Biological Product Control, NIFDC (National Institutes for Food and Drug Control) for test on Jan. 13, 2017 and Jun. 8, 2017. The number of serum samples of each age group is shown in Table 66-3.

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10.1.4. Protocol violations/deviations in Hebei sub-center

There was no serious deviation of the trial protocol in Hebei sub-center. A total of 159 cases of individual protocol violations/deviations occurred, including 42 cases in the 2-month-old test group, 78 cases in the 3-month-old test group, 28 cases in the 7- to 11-month-old PCV13-TT group, 10 cases in the 12- to 23-month-old PCV13-TT group, and 1 case in the 24- to 71-month-old PCV13-TT group. The protocol violations/deviations were mainly due to failure of blood collection, lack of enough interval for other vaccinations, and visits exceeding the time window.

10.1.5 Serious adverse events (SAEs) in Hebei sub-center

In Hebei sub-center, 27 SAEs were observed. All SAEs were reported using "Report Form of Serious Adverse Event" within the 24 hours of being noticed by the investigator and followed periodically. The follow-up reports were submitted until the submission of final report when the event ended. All the SAEs were not related to the test vaccine. No suspicious and unexpected serious adverse reactions (SUSAR) were observed.

10.2 HENAN SUB-CENTER

10.2.1 Overview of Henan sub-center

The Henan Provincial CDC and its subordinate Puyang County CDC and Wen County CDC were reviewed and approved by Ethics Committee of the Center for Disease Control and Prevention of Henan Province to obtain "Ethical Approval Certificate" on Oct. 12, 2015; these research institutions obtained the "One-Time Certification for Vaccine Clinical Trial Institution" after review and approval by Center for Food and Drug Inspection of NMPA on Apr. 5, 2016; and these institutions convened this launch meeting and training session on Apr. 6, 2016. The major timelines of the trial are shown in Table 67-1.

10.2.2 Enrollment and vaccination in Henan sub-center

Henan sub-center was planned to receive 1080 subjects as the samples, including 200 subjects at the 2 months of age, 400 subjects at the 3 months of age, and each 160 subjects at the 7–11 months, 12–23 months, and 24–71 months of age.. The enrollment and vaccination are shown in Table 67-2.

10.2.3 Completion of the trial in Henan sub-center

In Henan sub-center, 1278 subjects were screened and 1080 subjects were enrolled, 1075 of which completed the first dose of vaccination and safety observation, and 1037 subjects completed the full vaccination, with the compliance rate being 96.02%; 1022 subjects completed vaccination and effective blood collection before and after immunization, with the compliance rate being 94.63%.

A total of 7056 serum samples were collected, including 3552 samples pending testing and 3504 standby samples. The samples pending testing were sent to the Institute for Biological Product Control, NIFDC (National Institutes for Food and Drug Control) for test on Jan. 16, 2017 and Jun. 8, 2017. The number of serum samples of each age group is shown in Table 67-3.

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10.2.4 Protocol violations/deviations in Henan sub-center

There was no serious deviation of the trial protocol in Henan sub-center. A total of 116 cases of individual protocol violations/deviations occurred, including 31 cases in the 2-month-old test group, 50 cases in the 3-month-old test group, 26 cases in the 7- to 11-month-old PCV13-TT group, 7 cases in the 12- to 23-month-old PCV13-TT group, and 2 case in the 24- to 71-month-old PCV13-TT group. Four subjects with protocol violations/deviations withdrew the trial because they met the "Termination and Withdrawal Criteria". The protocol violations/deviations were mainly due to vaccinations to subjects who met the exclusion criteria, lack of enough interval for other vaccinations, enrollment of subjects who violated the inclusion criteria, and visits exceeding the time window.

10.2.5 Serious adverse events (SAEs) in Henan sub-center

In Henan sub-center, 58 SAEs were observed. All SAEs were reported using "Report Form of Serious Adverse Event" within the 24 hours of being noticed by the investigator and followed periodically. The follow-up reports were submitted until the submission of final report when the event ended.

Except for one SAE (occurrence of fever in subject) in the 3-month-old test group that might be related to the test vaccine, all the other SAEs were not related. No suspicious and unexpected serious adverse reactions (SUSAR) were observed.

10.3 SHANXI SUB-CENTER

10.3.1 Overview of Shanxi sub-center

The Shanxi Provincial CDC and its subordinate Xiangfen County CDC and Wanrong County CDC were reviewed and approved by Ethics Committee of the Center for Disease Control and Prevention of Shanxi Province to obtain "Ethical Approval Certificate" on Aug. 12, 2015; these research institutions obtained the "One-Time Certification for Vaccine Clinical Trial Institution" after review and approval by Center for Food and Drug Inspection of NMPA on Nov. 20, 2015; and these institutions convened this launch meeting and training session on Apr. 9, 2016. The major timelines of the trial are shown in Table 68-1.

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10.3.2. Enrollment and vaccination in Shanxi sub-center

Shanxi sub-center was planned to receive 840 subjects as the samples, including 160 subjects at the 2 months of age, 320 subjects at the 3 months of age, and each 120 subjects at the 7–11 months, 12–23 months, and 24–71 months of age. The enrollment and vaccination are shown in Table 68-2.

10.3.3 Completion of the trial in Shanxi sub-center

In Shanxi sub-center, 932 subjects were screened and 840 subjects were enrolled, 807 of which completed the first dose of vaccination and safety observation, and 810 subjects completed the full vaccination, with the compliance rate being 96.43%; 805 subjects completed vaccination and effective blood collection before and after immunization, with the compliance rate being 95.83%.

A total of 6277 serum samples were collected, including 3557 samples pending testing and 2720 standby samples. The samples pending testing were sent to the Institute for Biological Product Control, NIFDC (National Institutes for Food and Drug Control) for test on Jan. 16, 2017 and Jun. 8, 2017. The number of serum samples of each age group is shown in Table 68-3.

10.3.4 Protocol violations/deviations in Shanxi sub-center

There was no serious deviation of the trial protocol in Shanxi sub-center. A total of 37 cases of individual protocol violations/deviations occurred, including 18 cases in the 2-month-old test group, 10 cases in the 3-month-old test group, 8 cases in the 7- to 11-month-old PCV13-TT group, 0 case in the 12- to 23-month-old PCV13-TT group, and 1 case in the 24- to 71-month-old PCV13-TT group. The protocol violations/deviations were mainly due to sampling exceeding the time window, vaccination exceeding the time window, lack of enough interval for other vaccinations, participating in other drug clinical trials during this trial, and failure to collect blood according to trial protocol.

10.3.5 Serious adverse events (SAEs) in Shanxi sub-center

In Shanxi sub-center, 11 SAEs were observed. All SAEs were reported using "Report Form of Serious Adverse Event" within the 24 hours of being noticed by the investigator and followed periodically. The follow-up reports were submitted until the submission of final report when the event ended. All the SAEs were not related to the test vaccine. No suspicious and unexpected serious adverse reactions (SUSAR) were observed.

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11. REFERENCE

- [1] Good Clinical Practice (GCP), National Medical Products Administration, Jun. 4, 2003
- [2] Technical Guidelines for Clinical Trials of Vaccines (NMPA Note [2004] No. 575), National Medical Products Administration, Dec. 3, 2004
- [3] Guidelines for Quality Control of Clinical Trials of Vaccines (Pilot Edition), National Medical Products Administration, Oct. 31, 2013.
- [4] Regulation for Drug Registration (Revised Draft), National Medical Products Administration,

Jul. 22, 2016.

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1.0 |Objective |
 |To lay down a procedure for the preparation, approval and control of |
 |Standard Operating Procedures. |
 2.0 |Scope |
 |This Standard Operating Procedure is applicable for the preparation and |
 |implementation of Standard Operating Procedures to be followed at Company |
 |Name XYZ. |
 3.0 |Responsibility |
3.1	Staff, Initiator Department shall be responsible for:	
	3.1.1	Preparation of SOP as per the procedure given in this SOP
	3.1.2	Updation of the existing SOP whenever required
3.2	Reviewing person shall be responsible for:	
	3.2.1	The correctness and technical contents of SOP
	3.2.2	Formatting of the SOP as per the procedure contained in SOP
3.3	HOD/designee, Initiator department shall be responsible for:	
	3.3.1	Review and verification of SOP
	3.3.2	Training of the concerned staff
3.4	HOD/designee, QA shall be responsible for:	
	3.4.1	Review of SOP for its correctness and adequacy with respect to
		regulations/ guidelines and as per procedure mentioned in SOP
	3.4.2	Final approval and subsequently compliance of the SOP
4.0	Abbreviations and Definitions	
4.1	Initiator	Department who shall initiate the preparation of SOP
	Department :	
4.2	SOP	Standard Operating Procedure; a document where step
		by step instructions are cited to serve as support
		for methods or manners of fulfilling a function or
		functions reliably and consistently
4.3	Corporate SOP	An SOP which shall be applicable for corporate
4.4	CQA	Corporate Quality Assurance
4.5	QA	Quality Assurance
5.0	Procedure	
5.1	Preparation of SOP:	
	5.1.1	All SOPs shall be written in clear and easy to understand
		language.
	5.1.2	All the SOPs shall be generated on computer in the format as
		defined in Annexure-1 and as followed in SOP. The text of the
		SOP shall be typed in font type 'Times New Roman' with font
		size '12'.
	5.1.3	Text of header and footer of SOP shall be typed in font size
		and type as mentioned below:
		S.No.
		1.
		2.
		3.
	5.1.4	An SOP shall be printed in black color on standard A-4 size
		sheet of xxx color of good quality.
	5.1.5	All SOPs shall be prepared using Microsoft Word application
		software
		5.1.5.1
	5.1.6	Description of the format for SOP preparation (Refer
		Annexure-1)
	5.1.6.1	Company logo shall be pasted on the top left hand side
		of the page
	5.1.6.2	Name of the initiator department e.g. Quality
		Assurance, Production, etc. shall be typed in bold on
		the top right hand side of the page.
	5.1.6.3	'STANDARD OPERATING PROCEDURE' shall be typed in bold
		capital letters.
	5.1.6.4	'Title' of the SOP shall be brief and self descriptive
		in nature. It shall be typed keeping the first letter
		of the key words in capitals (i.e. title case shall be
		typed in unbold).
	5.1.6.5	SOP No. - Denotes the number allotted for the
		particular SOP by QA Department. SOP No. shall be
		typed in Capital. The numbering of an SOP shall be
		carried out as follows:
		The departments / major activities are represented by
		alphabets as enlisted in annexure 2 e.g. PR for
		Production, QC for Quality Control and QA for Quality

			Assurance.	
			In case the SOPs are general in nature i.e.	
			irrespective of the product the departmental code	
			shall be followed by a slash and three-digit numeral	
			assigned to the SOP e.g. No. PR/001 represents the	
			first SOP of Production department which is general in	
			nature.	
			In case the SOP is product specific, the departmental	
			code shall be followed by a slash (/) followed by	
			product code of two digit number followed by a slash	
			(/) and followed by the three digits numeral assigned	
			to the SOP e.g. the No. represents the first SOP	
			pertaining to Production Department for a product with	
			product code '00'.	
			5.1.6.6	'Revision No.' shall be denoted in two numeric
			characters. New SOP shall carry Revision No. '00' and	
			the subsequent revisions shall be denoted by 01, 02,	
			03 and so on.	
			5.1.6.7	'Supersedes': Superseded number of the SOP along with
			the revision no. shall be given in this column. For	
			example, PR/001 Rev.00 (the first general SOP of	
			Production department) when prepared for the first	
			time (first revision), 'Supersedes' column shall be	
			filled with the word 'None' and when the same is	
			revised for the first time, 'Supersedes' shall be	
			given as 'PR/001, Rev.00' and the 'Revision No.' for	
			this SOP shall be filled as '01'.	
			5.1.6.8	'Page No.' shall be numbered as specific page number
			of the total pages of the SOP e.g. first page of SOP	
			of total 5 pages shall be 1 of 5.	
			5.1.6.9	Standard Operating Procedure prepared by the user
			department(person well versed with the procedure) and	
			reviewed by the User supervisor and accordingly shall	
			write their names and sign with date in columns	
			'Prepared by' and 'Reviewed by'. 'Prepared by' and	
			'Reviewed by' shall be typed in bold.	
			5.1.6.1	Head of Department/designee of concerned department
			shall sign with date in the column 'Approved by'.	
			'Approved by' shall be typed in bold text.	
			5.1.6.1	Head/designee, QA shall also sign with date in the
			1 column 'Approved by' hence approving the proposed SOP	
			for implementation.	
			5.1.6.1	Front page of an SOP shall exhibit the columns of
			2 Prepared By, Reviewed By and Approved By (departmental	
			and QA) also 'Effective Date' and 'Next Review date',	
			while the subsequent pages shall not bear these	
			columns.	
			5.1.6.1	'Effective Date' is the date when the content of SOP
			3 becomes operational. It shall be given by QA	
			department in consultation with concerned HOD/designee	
			after completion of training on the approved SOP, in	
			DD-MM-YY or DD.MM.YY format, by hand.	
			5.1.6.1	Once the concerned SOP is approved by
			4 Head/designee-QA, the approved copy shall be sent to	
			initiator department for the training of concerned	
			personnel and initiator department shall conduct	
			training of the SOP and evaluation of training	
			thereupon, as per SOP .	
			Note: If a department prepares an SOP that involves other	
			departments(s) in its execution, then the persons from this	
			(these) department(s) shall also be trained as per the	
			approved SOP and the training records generated thereupon	
			shall be compiled by the initiator department and shall be	
			submitted to QA.	
			5.1.6.1	After completion of training on SOP, the initiator
			5 department shall forward the QA approved SOP and SOP	
			training record to QA, for assigning 'Effective Date'	
			on it.	
			5.1.6.1	The 'Next Review Date' shall be after two years from
			6 the Effective Date of the SOP given by QA department	
			in DD-MM-YY or DD.MM.YY format, by hand (e.g. Next	
			review date assigned for SOP effective on	
			DD.MM.YYshall be DD.MM.YY). Every SOP shall be	
			reviewed within 60 days of the Next Review Date, or	
			earlier, if any amendment is required.	
			Note: All the entries made in hand by the concerned personnel shall be	
			in black ink, using ballpoint pen. Ink of good quality shall be used	
			for the purpose.	
			5.1.7	The contents of the SOP shall be covered under 09 subheadings,
			which are compulsory for each SOP. In case a subheading is not	
			applicable, mention 'Not applicable' under that subheading.	
			All the subheadings shall be written in bold with first letter	
			in capital. The narrative text of each subheading shall be	
			aligned below the subheading, not after the length of the	
			heading. In case the insertion of a 'Note' is absolutely	
			necessary, Note shall be typed in Times New Roman and unbold	
			text.	
			5.1.7.1	Objective as first point; an overview of the intention
			of preparation of Standard Operating Procedure shall	
			be briefly mentioned under this heading. The	
			'Objective' shall always start with the word 'To'.	
			5.1.7.2	Scope as second point; describes the site(s),

			location(s), plant(s) and/ or department(s) to which	
			the Standard Operating Procedure is applicable.	
			5.1.7.3	Responsibility as third point; specifies designation
			of the personnel and name of department that shall be	
			responsible for implementation and compliance of the	
			Standard Operating Procedure.	
			5.1.7.4	Abbreviations and Definitions as fourth point;
			describes the meaning or explanation of term(s) which	
			is (are) not common or having any specific meaning and	
			expansion of abbreviation(s) used in the SOP.	
			5.1.7.5	Procedure as fifth point; describes the detailed, step
			by step procedure in chronological sequence to be	
			followed in simple and clear sentences. It may include	
			the details of all operations, operational conditions	
			and precautions to be taken, if any, etc.	
			5.1.7.6	Forms and Records as sixth point; it shall enlist all
			specimen formats, specimen labels, flow charts or any	
			other attachment with the SOP as Annexure.	
			5.1.7.7	References as seventh point; References must be
			included for key documents appropriate to the SOP.	
			Enlist the references relevant to the procedure. While	
			listing the references, the SOPs shall be listed first	
			followed by other guidelines, pharmacopoeia etc. First	
			write the SOP number followed by its title. Reference	
			article(s) consulted, if any, while preparing the SOP	
			shall also be mentioned.	
			5.1.7.8	Distribution as eighth point shall describe the
			distribution of the SOP to concerned	
			departments/locations. QA shall issue the controlled	
			copies as per distribution list.	
			5.1.7.9	History as ninth point; history shall contain a
			chronological record of significant changes in an SOP	
			in brief, mentioned in three columns, namely: 'Date',	
			'Revision Number' and 'Reason for Revision'. Only last	
			five histories shall be mentioned in the SOP.	
			Note: Under 'Date' put Effective Date of the SOP.	
			Under 'Revision Number' current revision number of the	
			SOP shall be mentioned. Under 'Reason for Revision'	
			give reasons why the SOP at previous revision number	
			is revised. In case of new SOP write 'New SOP' under	
			'Reason for Revision' and in case of routine revision,	
			write 'Routine Revision' under this column. (For more	
			clarity refer 'History' at point no. 9.0 of this SOP.)	
		5.1.8	Main headings of the SOP as discussed shall be numbered 1.0.	
		2.0, 3.0,and so on. Each heading shall be typed in bold		
		keeping first letter capital and rest in small for each word.		
		The subsequent numbering for headings or steps shall be typed		
		with outline numbering system through five levels. For		
		instance, first level number for the heading at fifth position		
		shall be '5.0' and its second level shall be 5.1, 5.2.....so on		
		and the fifth level shall be '5.1.1.1.1'against the first step		
		of every subsequent level. After fifth level the sub-numbering		
		can be marked as 'a', 'b' and so on or 'i', 'ii' and so on.		
		5.1.9	Recordings, if required, as a result of implementation of an	
		SOP shall be discussed in the relevant SOP by defining a		
		format for the recording purpose which is termed as 'Form' of		
		the SOP and shall be annexed to the SOP.		
		5.1.10	Formatting and Contents of the Annexure:	
		Specimen forms for recording of data during performance of a		
		procedure, flow charts, diagrammatic illustrations etc. shall		
		be attached to SOP as an annexure.		
		5.1.10.1	Formatting:	
		The font type shall be 'Times New Roman' of preferable		
		size 12.However a font size of 10-12 is acceptable		
		depending upon the nature of annexure where relevant		
		information is to be accommodated in the same page for		
		better presentation. In such cases all the pages of		
		annexure shall be of uniform font size.		
		5.1.10.2	Header:	
		The header shall contain the following information		
		repeating on each page:		
		5.1.10.2.1	Logo of the company, on Top Left hand side	
		of the page and not surrounded by the		
		border.		
		5.1.10.2.2	'Title' of the Annexure.	
		5.1.10.2.3	Annexure No., on Top Right hand side of the	
		page and not surrounded by the border.		
		Annexure No. shall be mentioned on the		
		first page of the annexure only.		
		5.1.10.3	Contents of Annexure:	
		Following aspects shall be covered depending upon the		
		Nature of activity		
		5.1.10.3.1	Include the brief instructions which	
		correspond to SOP.		
		5.1.10.4.2	Establish the flow of information in such a	
		way so as to have a chronology in relation		
		to the operations/information established		
		in the procedure.		
		5.1.10.2.3	Reduce the information to be filled to a	
		minimum. For example: fill in the blanks,		
		check list from a list, circle the correct		
		one etc.		

				5.1.10.2.4	The materials solutions, equipment etc. to be used during the process are to be written along with their part numbers/ID No. and shall be enlisted preferably on the first page of the form, as applicable	
				5.1.10.2.5	Record form shall include space to fill in the required data, date(s), time and space for signature/initials, wherever appropriate, of the person(s) performing the activity and those who supervise and/or approve it.	
				5.1.10.4	Footer:	
				The Footer shall contain the following information		
				repeating on each page:		
				5.1.10.4.1	Form No. on bottom left hand side of the	
				all pages and not surrounded by the		
				border.(Form No. shall be typed with font		
				size 10)		
				5.1.10.4.2	'Page No.' on bottom Right hand side of the	
				page. Page shall be numbered as specific		
				page number of the total pages of the SOP		
				e.g. first page of annexure of total 5		
				pages shall be 1 of 5. .(Page No. shall be		
				typed with font size 10)		
				5.1.10.5	Revision no. of annexure/Forms shall be the same as	
				that of respective SOP.		
				5.1.11	SOP(s) prepared for the	
				operation/calibration/cleaning/maintenance of equipment or		
				instrument of one make and model number can be applied for the		
				operation/calibration/cleaning/maintenance of other equipment		
				for instrument(s) of same make or model number.		
				5.1.12	The initiator department shall forward the draft(s) of SOP	
				under preparation (putting "DRAFT" word on every page of the		
				SOP as water mark of appropriate size) to QA department for		
				comments and necessary input, if any.		
				5.1.13	Once the Draft SOP is finalized, the SOP shall be forwarded to	
				Documentation section of QA for final printing on xxx color		
				paper.		
				5.1.14	Documentation section shall send the SOP to the initiator	
				department for signatures on the relevant columns of the SOP		
				and then initiator department shall forward the same to QA for		
				final approval.		
				Note: The drafts reviewed by QA, if available with the		
				concerned department, shall be sent to Documentation section		
				for the destruction.		
				5.2	Review and/ or Revision and Distribution of SOPs:	
				5.2.1	Review of SOP	
				5.2.1.1	The SOP shall be reviewed by a person very well	
				acquainted with the activity and of supervisory level		
				as determined by the Head of the Department.		
				5.2.1.2	The reviewer shall ensure:	
				5.2.1.2.1	Technical contents are correct and	
				complete.		
				5.2.1.2.2	That the language is clear and easy to	
				understand.		
				5.2.1.2.3	The formatting is as mentioned in SOP xxxx	
				5.2.1.2.4	Data record forms are numbered correctly	
				and have provision to record all relevant		
				data.		
				5.2.2	Biennial review of an SOP shall be undertaken as per SOP xxx.	
				However change in any SOP shall be undertaken through SOP xxx.		
				5.2.3	Master Copies and Controlled Copies (for distribution) of the	
				SOPs shall be prepared by Documentation section of QA as per		
				SOP xxx. Other copies (controlled and uncontrolled) of the		
				master copy for distribution purpose, shall be xerox on white		
				paper of A-4 size of good quality. The controlled and		
				uncontrolled copy shall be stamped and signed by concerned		
				personnel in the documentation section of QA before		
				distribution.		
				5.3	Vernacular SOPs:	
				If an SOP is required to be prepared in vernacular for operational		
				purpose, the same shall be prepared as that of English version by the		
				concerned department. The 'Master Copy' of the vernacular SOP shall be		
				archived in Quality Assurance Department and controlled copy shall be		
				issued to concerned department by QA.		
				6.0	Forms and Records	
				7.0	References	
				8.0	Distribution	
				8.1	Master copy	
				8.2	Controlled	
				copies		
				9.0	History	
				Date	Revision Number	Reason for Revision

|Department: Date: |

S.	Document	Title	No. of	Reason for	Remarks
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Rev. No.TitleEffective date Periodic Review dateReviewed
On
(Sign/Date)Change
required
(Yes/No)
HOD Approval
(Sign / Date)Remarks
QA
Approval
(Sign/Date)

Department		Area/Lab/Section	
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Document type		Document No	
Revision		Effective date	
Title			
Reason(s) for			
cancellation			
Other documents			
affected			
	Requisitioned by	Reviewed by (HOD)	
Name:			
Signature			
Date			
REVIEW BY QUALITY ASSURANCE			
Remarks, if any:			
Proposed cancellation is: Approved / Rejected			
Name	Signature	Date	

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|DEPARTMENT | : | |

|UPDATE AS ON |: ||

S.	Document	Title	Revision No.	Area/Lab/Section
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| Prepared by | Reviewed by | Approved by |

Name				
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Edition NoTitleEffective / Approval/ Manufacture- DateDate*Due Date of DestructionRemarks

1.0 |Objective |
 |To lay down the overall procedure for organization of documentation and |
 |numbering system. |
 2.0 |Scope |
 |This Standard Operating Procedure is applicable to the organization of |
 |documentation and Numbring system at Company Name XYZ for various |
 |documentation aspects such as: |
2.1	Issue and control of documents
2.2	Review of documents
2.3	Changes in documents
2.4	Archiving of documents
2.5	Cancellation of documents
2.6	Destruction of documents
3.0	Responsibility
3.1	Quality Assurance Department is responsible for organization of
	documentation system.
3.2	The department receiving any document shall be responsible for its
	proper preservation and custody.
3.3	HOD/designee, Quality Assurance shall ensure the implementation of
	this procedure.
4.0	Abbreviations and Definitions
HOD	Head of the Department
:	
No.	Number
:	
QA	Quality Assurance
:	
Rev.	Revision
:	
SOP	Standard Operating Procedure
:	
OG	Operating Guideline
:	
Documents	An official written/printed text that can be used to
:	furnish evidence, information and taking decisions such
	as SOPs, Specifications and log books etc.
Master Copy	The original set of controlled documents or written
:	procedures which are retained at quality assurance
	department.
Controlled Copy	The photocopy of master copy of controlled documents or
:	written procedures which shall be distributed to user
	department(s) as a part of document distribution
	procedure and shall bear a stamp as 'Controlled copy'.
	These documents are retrievable by QA.
Uncontrolled Copy	The photocopy of master copy of controlled documents or
:	written procedures which shall be given to any department
	for person for reference purpose only. These documents are
	not controlled documents; hence are not retrievable by
	QA.
5.0	Procedure
5.1	Preparation and maintenance of master documents:
	5.1.1
	5.1.2
	5.1.3
	5.1.4
	5.1.5
	5.1.6
	5.1.7
	5.1.8
5.2	Issuance of controlled and uncontrolled copies of master documents:
	The procedure for issue and control of documents is as described
	below:
	5.2.1
5.2.2	Issuance of Controlled Copies:
	5.2.2.1

				Quality Assurance Department and can be retrieved	
				as and when required.	
				5.2.2.2	All controlled copies shall be reproduced from
				the approved Master Copy (Original) by	
				photocopying by Quality Assurance Department.	
				5.2.2.3	Quality of photocopy of master document should be
				legible and photocopies shall be taken by	
				concerned QA person.	
				5.2.2.4	Controlled copy (s) shall be issued against a
				requisition (justifying reason for its	
				requirement) which is duly approved by the	
				concerned HOD and Quality Assurance Department	
				except those which are mentioned in section	
				5.2.2.5.	
				5.2.2.5	The first controlled copy of Specifications/SOPs
				and subsequent controlled copies of SOPs as per	
				distribution list.	
				5.2.2.6	At the time of issue of any controlled copy, the
				document shall be stamped as 'Controlled copy' in	
				blue ink on right hand side bottom of each page	
				not covering the text part.	
				The stamp shall bear Copy No., Date of issue and	
				Signature of the issuing official of QA	
				Department.	
				5.2.2.7	The copy no. shall be a two digit consecutive
				number. e.g.	
				the first controlled copy of a document (SOP,	
				Specification) shall be numbered as Copy No. 01.	
				5.2.2.8	In case of approved changes in a document, the
				issued controlled copies shall be retrieved from	
				circulation and replaced with changed version.	
				5.2.3	Issuance of Uncontrolled Copies:
				5.2.3.1	Uncontrolled copies are those on which QA
				Department has no retrieval control.	
				5.2.3.2	An uncontrolled copy shall be issued against a
				requisition	
				duly approved by HOD, QA or his/her designee,	
				justifying the reason for its requirement.	
				5.2.3.2	At the time of issue of an uncontrolled copy, the
				document shall be stamped as 'Uncontrolled copy'	
				in blue ink on right hand side bottom of each	
				page with Issued by signature and date filled by	
				QA Department.	
				5.2.4	Issuance of forms:
				5.2.4.1	Data record forms related to batch production
				records, forms related to QC analytical file and	
				forms related to Quality management system shall	
				be controlled.	
				5.2.5	Issuance of Log books:
				5.2.5.1	The routine operation activity shall be logged in
				a log books and log books shall be issued to all	
				concerned departments as per SOP QA/000.	
				5.2.6	Unauthorized photocopying of any document (Master Copy/
				Controlled copy) is prohibited.	
				5.2.7	Wastage of photocopy (ies), if identified due to bad quality
				of photocopy shall be undertaken by destroying photocopy (ies)	
				by shredding or by incineration.	
				Note:	While stamping the documents, the documents should be stamped in such
				a way that the text part of document should not be covered with	
				stamp.	
				5.3	Maintenance of soft copies of approved documents:
				5.3.1	The back up data of documents in electronic soft copies shall
				be updated.	
				5.4	Review and Revision of documents:
				5.4.1	Documents such as Standard Operating Procedures,
				Specifications, Production Orders and Batch Records etc. shall	
				be periodically reviewed every two years from effective date.	
				However the document review is need based and can be carried	
				out earlier also by the concerned department if the situation	
				so demands.	
				5.4.2	Documents which do not have any specified effective date shall
				also be reviewed every two years.	
				5.4.3	The QA department shall plan and organize the periodic review
				exercise so as to complete the reviewing activity on time.	
				5.4.4	The QA department shall send review form to concerned
				department.	
				5.4.5	The documentation section of QA department shall fill the
				relevant portion of 'Document Review Form' and forward it to	
				the HOD of the concerned department at least 45 days in	
				advance of 'review date'.	
				5.4.6	The HOD of the concerned department or his/her designee shall
				get the documents reviewed and return the review form duly	
				filled in to Quality Assurance Department with in the	
				specified time (maximum 20 days) with their opinion whether	
				the documents require a change or no change is needed.	
				5.4.7	If change(s) are suggested, the same shall be carried out as
				per the procedure mentioned in this SOP.	
				5.4.8	During periodic review of documents, in case no change in text
				of document the same shall be checked for compliance for the	
				format of document, if format is a part of any SOP.	
				5.4.9	If the format matches with that mentioned in relevant SOP,
				document shall be stamped as mentioned below.	

|| 5.4.10|Stamping procedure: |

||| 5.4.10.|The concerned master copy shall be stamped in blue ink|

||| 1 |on the front side (Top middle) of the document bearing|

||| |following |

||| |Reviewed and Approved by : |

||| |Sign/Date : |

||| |Next Review Date : |

||| 5.4.10.| QA person shall review, approve the document and |

||| 2 |shall sign with date in column 'Sign/Date'. |

||| 5.4.10.|'Next Review Date' shall be the date on which review |

||| 3 |of same document shall be undertaken. |

||| 5.4.10.|Controlled copies of stamped documents shall be |

||| 4 |prepared and distributed/retrieved as per 5.2.2.1 to |

||| 5.2.2.8 |

|| 5.4.11|With the option of stamping procedure, review of relevant |

|| |document shall be done once only. Next review of same shall be|

|| |done through reprinting with the change in its revision number|

|| |as per SOP QA/xxx. |

|| 5.4.12|If the format does not match with that mentioned in current |

|| |relevant SOP, the document shall be reprinted in the current |

|| |applicable format changing its 'Revision number' wherever |

|| |applicable. |

|| 5.4.13|Documents shall be reviewed 60 days before or after the |

|| |'Review Date'. |

|| 5.4.14|Changes in documents: |

|| |Changes in a document can be as a result of a periodic review |

|| |or as a result of any variation to be carried out in a |

|| |process, analytical technique, procedure or acceptance |

|| |criteria etc. |

|| 5.4.14.|Any change to be made in a document shall be carried |

||| 1 |as per SOP QA/000. |

||| 5.4.14.|In case two or more documents (e.g. SOPs) of |

||| 2 |interrelated subjects are to be merged to make single |

||| |document with or without any change in its contents, |

||| |the same shall be processed through change control. |

||| |The newly formed document shall bear a new document |

||| |number as decided by QA and shall have revision number|

||| |00 and supersedes column shall bear None. The change |

||| |shall be mentioned in brief in history of the |

||| |document. The abandoned numbers shall stand vacant. |

||| |For the abandoned numbers, a request for cancellation |

||| |of document shall be filled by the concerned |

||| |department. |

|| 5.4.14.|In case there is change in the code (category) of a |

||| 3 |document the same shall be processed through change |

||| |control. The document shall bear new code and shall |

||| |have revision number 00; the abandoned number shall |

||| |stand vacant. The change shall be mentioned in brief |

||| |in history of the document, as applicable and the |

||| |supersedes column shall bear previous document number |

||| |with its revision. For the abandoned number, a request|

||| |for cancellation of document shall be filled by the |

||| |concerned department. |

|| 5.4.14.|In case of new and revised versions of SOPs, the HOD |

||| 4 |of concerned department or his/her designee requesting|

||| |the change (in case of revised SOPs) shall ensure that|

||| |the concerned personnel are trained as per the amended|

||| |SOP and the training records are submitted to Quality |

||| |Assurance Department before the effective date is |

||| |assigned to the SOPs by QA Department. |

|| 5.4.14.|The defunct Master Copy (Original) shall be stamped as|

||| 5 |'Not In Force' in red ink on all pages on the right |

||| |hand side bottom and shall be archived; the stamp |

||| |shall bear signature and date. |

|| 5.4.14.|The controlled copies of the previous version already |

||| 6 |issued shall be retrieved, destroyed and record of |

||| |destruction shall be filled. |

|| 5.5 |Control of Documents: |

|| 5.5.1 |A list of approved documents shall be prepared and updated at |

|| |least once a year (from the date of last update) by the |

|| |Quality Assurance Department. |

|| 5.5.2 |SOPs, Specifications, Batch Records and other documents shall |

|| |be enlisted separately. |

|| 5.5.3 |The list shall exhibit the following: |

||| 5.5.3.1|Document type |-Mention the type of document, i.e. |

||| |SOPs, Specification etc. |

||| 5.5.3.2|Department |-Mention the department to which the |

||| |document pertains to. |

||| 5.5.3.3|Update as on |-This is the date on which the List |

||| |has been updated and shall be |

||| |depicted on first page of document |

||| |list. |

||| 5.5.3.4|Document number |-Mention the document number |

||| 5.5.3.5|Revision number |-Mention the revision number |

||| 5.5.3.6|Area/Lab/Section|-Mention the Area/Lab/Section to |

||| |which document pertains to |

||| 5.5.3.7|Approvals |-List of document shall bear name |

||| |signature and date of personnel |

||| |performing activity of preparation, |

||| |review and approval of list and shall|

||| |be depicted on last page of the list.|

|| 5.6 |Archiving of documents: |

	5.6.1	Documents shall be archived in such a way that they can be		
		readily retrieved as and when required and protected from		
		deterioration.		
	5.6.2	Master copy of all documents which are 'In Force' and which		
		are 'Not in Force' shall be archived separately in Quality		
		Assurance Department with restricted access.		
	5.6.3	Documents and records related to manufactured batches of drug		
		substance such as Batch Records, Analytical files shall be		
		archived at least for a period of one year beyond the expiry		
		date of the last batch of drug product manufactured from that		
		batch of drug substance. However in case of documents and		
		records such as batch associated SOPs, Specifications,		
		logbooks the archiving period shall be considered from the		
		concerned not in force date or the date of last record, as		
		applicable (Category I).		
	5.6.4	Documents and records, not related directly with the batch or		
		process, shall be archived for at least seven years from the		
		concerned Not in force date or after the date of the record,		
		as applicable (Category II).		
	5.6.5	Documents related to preliminary process performance and its		
		integration into manufacturing such as Product Development,		
		Clinical studies, Technology transfers, Scale up, Consistency		
		batches, Stability data used in assigning shelf life,		
		Specifications pertaining to critical stage, ARM and final		
		bulk, Process Validation, initial qualification/validation of		
		facility, equipment, systems, Site master file, Policy		
		document, Quality Manual and any such document shall be		
		archived forever (Category III).		
	5.6.5	All documents shall be archived in a year-wise manner, as		
		applicable.		
	5.7	Cancellation of documents:		
	5.7.1	An approved document pertaining to an activity, product or		
		procedure if no longer necessary can be cancelled through a		
		change control.		
	5.7.2	The department desiring cancellation of a document shall fill		
		the 'Requisition for cancellation of a document' after the		
		approval of change control and submit it to Quality Assurance		
		Department.		
	5.7.3	Quality Assurance Department shall review this requisition and		
		submit it to HOD, Quality Assurance or his/her designee for		
		approval.		
	5.7.4	Once the cancellation request is approved, the defunct master		
		copy shall be stamped as 'Cancelled' and 'Not in Force' in red		
		ink and shall be archived.		
	5.7.5	All the controlled copies shall be retrieved from circulation,		
		destroyed and record of destruction shall be filled.		
	5.7.6	The number of the cancelled document shall stand vacant and		
		shall not be re-assigned.		
	5.7.7	Quality Assurance Department shall fill the 'Record of		
		cancelled documents'.		
	5.8	Destruction of documents:		
		Documents to be destroyed shall either be destroyed by shredding or		
		by incineration and the details shall be filled in its corresponding		
		data record form.		
	6.0	Forms and Records		
	6.1	Document control record	- Annexure 1	
	6.2	Requisition for copy of document	- Annexure 2	
	6.3	Record of uncontrolled copies issued	- Annexure 3	
	6.4	Document review form	- Annexure 4	
	6.5	Requisition for cancellation of a document	- Annexure 5	
	6.6	Record of cancelled documents	- Annexure 6	
	6.7	List of Documents	- Annexure 7	
	6.8	List of Obsolete documents for destruction	- Annexure 8	
	6.9	Requisition for Document number	- Annexure 9	
	7.0	References		
	7.1			
	7.2			
	8.0	Distribution		
	8.1	Master Copy		
	8.2	Controlled copies		
	9.0	History		
		Date	Revision Number	Reason for Revision

[1.0]Objective: |
| |To lay down the procedure for an effective training of all the personnel |
| |engaged in the manufacture and analysis of the products manufactured. |
[2.0]Scope: |
| |This Standard Operating Procedure is applicable for training of personnel |
| |working in xxx company |
[3.0]Responsibility: |
	3.1	Human Resource Department shall be responsible for the induction of new	
		joinees.	
	3.2	Training section in QA shall be responsible for conducting GMP training	
		of new joinee, issuance and verification of individual training	
		records, preparation of cGMP annual training planner, coordination of	
		trainings and its documentation.	
	3.3	Trainer shall be responsible for conducting training as per the	
		training plan, its evaluation and submitting the evaluation sheets to	
		trainees or training coordinator of concerned department.	
	3.4	Concerned personnel shall be responsible for updating and archiving	
		individual training file.	
	3.5	Concerned Area in charge shall be responsible for the preparation of on	
		the job training programme for new joinee and compilation of on the job	
		training report.	
	3.6	Training coordinator of respective department shall be responsible to	
		ensure the updation of training documents and all the training	
		activities pertaining to his/her department.	
	3.7	Head of each department shall be responsible	
		3.7.1	To nominate training coordinator(s).
		3.7.2	To nominate SOP trainers
		3.7.3	For approval of On the Job Training Programme and its report.
		3.7.4	To ensure each individual in department has been imparted
			required training.
		3.7.5	For identification of department training needs.
	3.8	Head, QA or designee shall be responsible for approval of training	
		planners and overall compliance to the SOP.	
[4.0]Abbreviations and Definitions:			
4.1 cGMP : Current Good Manufacturing Practices			
4.2 HOD : Head of the Department			
4.3 HRD : Human Resource Development			
4.4 JR : Job Responsibility			
4.5 OJT : On the job training			
4.6 QA : Quality Assurance			
4.7 SOP : Standard Operating Procedure			
4.8 QMS : Quality Management System			
4.9 CQA : Corporate Quality Assurance			

On the Job Training:

Training on Standard Operating Procedures applicable in the different department for the adequate and accurate operations. This training shall be imparted to those, who directly or indirectly fall in the scope of execution of a SOP.

cGMP Training:

Training pertaining to current Good Manufacturing Practices which are supposed to be followed at different stages of manufacturing of drug product. This type of training shall be given to all those who are directly or indirectly associated with the quality of the drug product.

Job Responsibility:

A job responsibility sets out the purpose of a job, where the job fits into the Organization structure, the main accountabilities and responsibilities of the job and the key task to be performed. The JR of the person shall be assigned by the HOD of the concerned department and duly signed by HOD and respective person, and authorized by HRD department and finally JR shall be archived in HR. Person who shall be designee in absence shall give the acceptance that he / she shall do the activity in the absence of the job card holder.

[5.0 | Procedure |
	5.1	Training		
		Following types of training shall be imparted to the personnel :-		
		a.	Induction to new joinee	
		b.	GMP training to new joinee	
		c.	On the job training	
		d.	cGMP training for existing employees as per annual training	
			planner	
		5.1.1	Induction to new joinee	
			5.1.1.	Human Resource Development shall conduct induction of
			1	new joinee which covers the overview of the
				organisation, unit and department as per approved
				procedure.
			5.1.1.	After completion of induction, HRD shall give a copy of
			2	induction program to new joinee to file it in his/her
				individual training file.
			5.1.1.	New joinee shall be issued Individual training record by
			3	training section,QA at the time of induction where in
				he/she shall record each training attended.
		5.1.2	GMP training to new joinee	
			5.1.2.	GMP training includes introduction to basic concepts of
			1	GMP to new joinee so as to provide overview of GMP.
			5.1.2.	GMP training shall be given by training section, QA to
			2	new joinee after the completion of induction program so

			as to acquaint the new joinee with the GMP, which shall
			be followed by evaluation.
			5.1.2. The GMP training shall be conducted as class room
			3 training or computer based training by showing the
			trainee different videos on the topics mentioned below ,
			selection of videos to be shown shall be based on the
			job requirement of the individual.
			5.1.2.3. Introduction to GMP
			1
			5.1.2.3. First day at work with GMP
			2
			5.1.2.3. Understanding Sterile Production
			3
			5.1.2.3. Basic Contamination Control
			4
			5.1.2.3. GMP of Cleaning & Disinfecting Clean Rooms
			5
			5.1.2.3. Cleaning Procedure & Techniques
			6
			5.1.2.3. Validating Cleaning
			7
			5.1.2.3. Laboratory Documentation
			8
			5.1.2.3. Understanding Training
			9
			5.1.2.3. Training and Presentation Techniques
			10
			5.1.2. Trainer can conduct training in vernacular language if
			4 required.
			5.1.2. GMP training planner shall be prepared by training
			5 section – QA and shall be given to new joinee to file it
			in individual training file.
			5.1.3. On the Job Training
			5.1.3. For new joinee
			1
			5.1.3.1. On the job training for new joinee shall include
			1 the training on SOPs in the concerned and cross
			functional departments as per his/her job
			requirement before he/she is allowed for the
			execution of any activity.
			5.1.3.1. Area in charge in consultation with the HOD
			2 shall prepare the on the job training program
			based on the job to be performed by new joinee
			as per on the job training program.
			5.1.3.1. The mode of imparting training shall be through
			3 Standard Operating Procedure, or practical
			demonstration for subject such as aseptic
			practices, gowning procedure, sterility testing,
			aseptic connection, wherever applicable.
			5.1.3.1. In addition to training on SOP, on the job
			4 training shall include cGXP training related to
			each department. (For e.g. GLP training for QC
			department, GEP for engineering department.
			5.1.3.1. Evaluation of training shall be done (as
			5 applicable) and recorded in Training attendance
			and evaluation sheet).
			5.1.3.1. After successful completion of training and
			6 evaluation of on the job training, on the job
			training report shall be prepared, reviewed and
			finally approved by Head of the concerned
			department. On the job training report shall be
			submitted to training section, QA.
			5.1.3.1. On the basis of the on the job training report
			7 and successful evaluation new joinee is
			certified by Head QA to perform activities in
			their department
			5.1.3.1. However if new joinee is having experience of
			8 more than two years then it could be decided by
			HOD to first impart on the job training on
			specific process activity. Once new joinee has
			submitted OJT report and relevant training
			records after due approvals from department, for
			specific activity, then he/she shall be
			certified as "trained" for performing that
			activity. However he/she shall complete full
			training as per on the job training program.
			5.1.3. For existing employees
			2
			5.1.3.2. Employee has to be trained on the current
			1 version of SOP' s as per their job
			responsibility and if there is any updation in
			SOP or new SOP is approved training on the same
			shall be given to the concerned before making
			it effective. However, after two years at the
			time of review of SOP, all concerned shall be
			trained on SOP even if there is no change in
			SOP.
			5.1.3.2. Trainee shall update all the trainings attended
			2 (internal as well as external), Individual
			training record.
			5.1.3.2. If any employee is transferred from one area of
			3 operation to other area of operation, he/she

			For supervisor level (fresher	
			to 05 years of experience)	
			Level-II (Specific cGMP) : For supervisor	
			(having experience	
			more than 05 years)	
			Level-III (Advanced cGMP) : For managers and above	
			5.1.4.5	Training section, QA shall prepare the training agenda
			to inform the trainees and HODs about the date, time	
			and location of training session which shall be	
			approved by Head QA or his/her designee.	
			5.1.4.6	Feedback of trainer and training program for cGMP
			training shall be taken by providing the Course	
			evaluation and feedback form to trainees to check the	
			effectiveness of training and performance of the	
			trainer.	
			5.1.4.7	HOD, QA/CQA shall decide the schedule of external
			training which will be conducted, as and when,	
			required.	
			5.1.4.8	The training planned as per cGMP annual training
			planner can be conducted $\pm$ 15 days of the month .	
			5.1.4.9	Although training topics for the month are specified
			in planner but monthly training topics can be	
			re-shuffled as per the requirement but it is required	
			that all the training topics are covered by the end of	
			calendar year. However in case the training needs are	
			identified for the topics other than those mentioned	
			in training planner, revision of planner can be taken	
			in such case.	
			5.1.4.10	All documents pertaining to cGMP and external
			trainings shall be archived in training section, QA.	
			5.2	Selection of Trainer
			5.2.1	Trainer selection shall be based on knowledge and skills to
			effectively teach and evaluate others. Trainers shall be	
			categorised as SOP (includes protocols etc.) and GMP trainers	
			5.2.1.	SOP's trainers
			1	
			5.2.1.	For selection of SOP trainer, concerned HODs shall
			1.1	identify personnel from their departments based on
			his / her qualification, knowledge and experience	
			in relevant areas.	
			5.2.1.	The personnel shall be regular permanent employees
			1.2	of the company, at least graduate, with an
			experience of at least of 3 years in the relevant	
			field.	
			5.2.1.	For certification of SOP trainer shall be
			1.3	submitted to training section, QA, and after the
			approval of QA Head, the personnel shall be	
			certified as trainer for SOPs.	
			5.2.1.	GMP trainer
			2	
			5.2.1.	Selection of GMP trainers shall be done by the
			2.1	committee consisting of HOD's/ designees and
			training section shall coordinate for selection	
			and qualification of GMP trainers. The candidates	
			to be qualified as a trainer shall be asked to	
			give mock training, if required.	
			5.2.1.	The personnel shall be regular, permanent employee
			2.2	of the company, at least graduate with an
			experience of at least 5 years in the relevant	
			field with good knowledge. He/she shall possess	
			good communication, presentation skills and shall	
			have exposure to trainings and audits.	
			5.2.1.	For certification of GMP trainer shall be filled
			2.3	by training section, QA and after the approval of
			QA Head the personnel shall be certified as GMP	
			trainer.	
			5.2.2	Train the trainer
			Trainers shall be trained for upgrading of different training	
			skills through train the trainer which shall be done once in a	
			year.	
			5.2.3	Training section, QA shall prepare list of internal trainers
			which shall be approved by Head – Quality Assurance, internal	
			trainer List. The list shall be updated as and when required and	
			archived in training section QA.	
			5.2.4	External trainer selection shall be based on his experience,
			qualification, skills and expertise on the training topics.	
			External trainers shall be selected by CQA/QA HOD.	
			5.3	Evaluation of Training
			5.3.1	Employee undergoing training shall be evaluated to check the
			effectiveness of training. A questionnaire shall be prepared by	
			the trainer	
			5.3.2	Training attendance and evaluation shall be recorded
			5.3.3	Written evaluation of training shall be done by means of a
			questionnaire as per the training topic. Questionnaire can be	
			descriptive / multiple choice true or false / choose the correct	

			the individual training file for each new joinee and handover	
			the file to the individual so that he/she can archive all	
			training documents in the file.	
		5.5.6	Each employee (technical staff/worker/operators) shall archive	
			the following documents in Individual training file:	
			5.5.6.1	Copy of final educational qualification
			5.5.6.2	Any other qualification/training certificate
			5.5.6.3	Copy of induction program
			5.5.6.4	GMP training planner for new joinee and questionnaire
			5.5.6.5	On the job training program
			5.5.6.6	On the job training report
			5.5.6.7	Copy of Job Responsibility
			5.5.6.8	Individual training Record
			5.5.6.9	Evaluation sheet (Except for cGMP trainings)
			5.5.6.10	Any other certificate.
		5.5.7	Whenever Job responsibility of any personnel is changed, he/she	
		shall archive the copy of changed JR in the individual training		
			file.	
		5.5.8	Any employee resigning the organization shall submit his	
		individual training file in QA which shall be archived in		
			training section.	
		5.6	Training Coordinator	
		5.6.1	HOD shall nominate training coordinator (s) from concerned	
			department.	
		5.6.2	Training coordinator shall be responsible for ensuring updation	
			of Individual training files of all personnel's in concerned	
			department and shall ensure the archival of documents given in	
			section 5.5.6 in training files of every individual of	
			respective department.	
		5.6.3	Training coordinators shall coordinate for cGXP training of new	
			joinee and on time submission of on job training documents.	
		5.6.4	Training coordinators shall ensure that schedules of all	
			training conveyed from QA are conveyed to concerned individuals.	
		5.6.5	Training coordinators shall ensure the submission of training	
			need identification	
			in Training section, QA.	
		5.6.6	Training coordinator shall provide the questionnaires to	
			personnel's in there department	
		6.0	Form and Records:	
		7.0	Reference	
		8.0	Distribution	
		Master Copy :		
		Controlled Copy :		
		9.0	History	
		Date	Revision No.	Reason for Revision

1.0	Objective	
	To lay down the procedure for Design qualification of Equipment/System.	
2.0	Scope	
	This Standard Operating Procedure is applicable for Design Qualification of	
	Equipment / instrument / system in all departments of Company Name XYZ.	
3.0	Responsibility	
	3.1	It is the responsibility of QA in execution of qualification,
		validation / revalidation with the support of user department.
	3.2	Head QA/ his designee shall be responsible for implementation and
		compliance of this SOP.
	3.3	Head-QA or his/her designee shall be responsible for review and
		authorization of qualification of documents
4.0	Abbreviations and Definitions	
	4.1	Operating System: An organization of components that have a defined
		operational function (e.g. piping, instrumentation, equipment,
		facilities, computer hardware/ software etc.)
	4.2	Calibration : The performance of tests and retests to ensure that
		measuring equipment (e.g. for temperature, weight, pH) used in a
		manufacturing process or analytical procedures (in production or
		quality control) give measurements that are correct within established
		limits.
		Design Qualification (DQ): Design Qualification is documented
	4.3	evidence that the proposed design of the facilities, systems and
		equipment are suitable for intended purpose.
	4.4	Factory Acceptance Test (FAT): Testing conducted at the supplier's
		factory usually involving the user to determine whether or not a
		system satisfies its acceptance criteria and to enable the customer to
		determine whether or not to accept the system.
	4.5	Installation Qualification (IQ): Installation Qualification is a
		documented verification that the facilities, systems and equipment as
		installed or modified; comply with the approved design and
		manufacturer's recommendations.
	4.6	Operational Qualification (OQ): The documented action of
		demonstrating that process equipment and ancillary systems work
		correctly and operate consistently in accordance with established
		specifications for the anticipated range of operation.
	4.7	Performance Qualification (PQ): Performance Qualification is the
		documented evidence, which verified that the equipment or system
		operates consistently, and gives reproducibility with in defined
		specifications and parameters for prolonged period.
	4.8	Qualification: Qualification is the act of planning, carrying out and
		recording of tests on equipment and systems, which form part of the
		validated process, to demonstrate that it will perform as intended.
	4.9	Re-validation: validation exercise to validate the system /equipment
		/process / method which is in state of validation & changes done which
		may impact it's earlier state of validation.
	4.10	User Requirement Specification (URS): A requirement specification
		that describes, what the equipment or system is supposed to do, thus
		containing at least asset of criteria or conditions that have to be
		met.
	4.11	Functional Design Specification(FDS):It is the technical Specification
		provided by the vendor against the User Requirement Specification
	4.12	Validation: Validation is the establishment of documented evidence
		that a system does what it is supposed to do. It is the collection and
		evaluation of data, which ensure that the manufacturing process
		including equipment, building, materials are capable of achieving the
		intended results on the consistent and continuous basis.
	4.13	Validation Protocol: The validation protocol is a written plan
		stating how validation will be conducted, including test parameters,
		product characteristics, and production equipment and decision points
		on what constitutes acceptable test results.
	4.14	Validation Report: The validation report is a written report on the
		validation activities, validation data and the conclusions drawn.
5.0	Procedure	
	Equipment qualification shall be performed as per the steps defined in	
	Validation master plan of respective block. Following shall be the	
	generalized plan for equipment qualification	
	Equipment URS	
	Design Qualification	
	Risk Analysis	
	IQ	
	FAT	
	PQ	

	SAT/IQ			
	Re-Qualification			
	5.1	Certain qualification activities can be combined depending upon size		
		and operation of the equipment. In some equipment like weighing		
		balance, plug and play Equipment, Table top centrifuge, conductivity		
		meter etc. commissioning report along with the IQ and OQ Checks shall		
		hold good as qualification document.		
	5.2	User Requirement Specification(URS)		
		Users Requirement Specifications are base documents that describe the		
		basic quality, intended parameters and shall be prepared by the		
		concerned personnel of the department tendering the equipment, with		
		co-ordination of Quality Assurance and Engineering departments. URS		
		shall include list of requirements from the system and define what		
		the equipment to perform. Users Requirement specification shall		
		include but not be limited to:		
		5.2.1	Description of equipment.	
		5.2.2	Description of concerned process steps.	
		5.2.3	Required Inputs/ Outputs. (utilities)	
		5.2.4	Change parts	
		5.2.5	Capacity, speed of the equipment	
		5.2.6	Operational parameters required	
		5.2.7	Space required and space available (length,width,height)	
		5.2.8	GMP consideration	
		5.2.9	Specifications (MOC)	
		5.2.10	Print out required for outputs.	
		5.2.11	Specific requirement like temperature, moisture ,	
		sensitivity, protection from light etc.		
		5.2.12	Safety parameters	
		5.2.13	Online sampling	
		Note: Functional Design Specification (FDS) Shall be provided by the		
		Vendor Against the Approved Users Requirement Specification (if		
		applicable)		
	5.3	Equipment Design Qualification (DQ)		
		Design qualification shall follow the below mentioned activities:		
		5.3.1	Design qualification shall verify that premises, facilities,	
		equipment and processes have been designed in accordance with		
		URS.		
		5.3.2	This document shall be used as a specified checklist for the	
		verification of design specification of the facility,		
		equipment.		
		5.3.3	The DQ shall be prepared by vendor, review by user; gap	
		analysis shall be performed with respect to URS and its impact		
		shall be concluded in DQ and finally approved by Head Quality		
		Assurance/Designee. For example:		
		5.3.3.1	Suitability of the equipment with site	
		5.3.3.2	Modification in basic design (if needed) matched with	
		URS.		
		5.3.3.3	Acceptability of risk level	
		5.3.4	The DQ document shall include the following details but not	
		limited to:		
		Pre-Approval		
		Objective		
		Scope		
		Overview		
		Qualification team and responsibility		
		Pre-requisites		
		Equipment / System description		
		Equipment / System / Technical Specification		
		Review / Evaluation of vendor design against user requirement		
		Deficiency Sheet		
		Review Inclusive of follow up action		
		Abbreviations		
		Recommendations / Conclusions		
		Post Approval		
		5.3.5	The DQ document must cover	
		All the necessary diagrams		
		Layouts		
		Location suitability desired special feature of components		
		Equipment and their specification		
		Desired material of construction		
		Location of control panel		
		Electrical requirement and utility requirement		
		Note: Any changes after approval of DQ of Equipment which may have		
		direct or indirect impact on quality, safety and efficacy of product		
		shall be evaluated and incorporated through change control procedure.		
	6.0	Forms and Records		
	6.1	Installation Qualification		
		Annexure 1		
	7.0	References		
	8.0	Distribution		
		Master Copy		
	:			
		Controlled Copies		
	:			
	9.0	History		
		Date	Revision Number	Reason for Revision



Protocol Prepared by: |Departmental Review: |
Name: |Name: |
Designation: |Designation: |
Signature: |Signature: |
Date: |Date: |
Engineering review |Quality Assurance review |
Name: |Name: |
Designation: |Designation: |
Signature: |Signature: |
Date: |Date: |
Departmental Approval: |Vendor Approval (If Required): |
Name: |Name: |
Designation: |Designation: |
Signature: |Signature: |
Date: |Date: |
Engineering Approval: |Quality Assurance Approval: |
Name: |Name: |
Designation: |Designation: |
Signature: |Signature: |
Date: |Date: |

Detail of Equipment

Installation Location :

Date of Installation :

Name of Equipment / System:

Equipment ID Number:

Production Block/Department:

Area (if any) :

Serial Number (if any):

Equipment ID issuance date:

Study Start Date:

Study End Date:

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| |Equipment | |
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15. |References | |

PERSONS PERFORMING INSTALLATION QUALIFICATION:

Work Group |
Department : |

Equipment/System|

Head of the Work Group |
| |
Name : _____ Signature: |

Designation : _____ Date : |

| |
Members |
Name : _____ Signature: |

Designation : _____ Date : |

| |
Name : _____ Signature: |

Designation : _____ Date : |

| |
Name : _____ Signature: |

|Designation : _____ Date : |
 | _____ |
 ||
 |Vendor |
 |Name : _____ Signature: |
 | _____ |
 |Designation : _____ Date : |
 | _____ |
 ||
 |Name : _____ Signature: |
 | _____ |
 |Designation : _____ Date : |
 | _____ |
 ||
 |Engineering |
 |Name : _____ Signature: |
 | _____ |
 |Designation : _____ Date : |
 | _____ |
 ||
 |Quality Assurance |
 |Name : _____ Signature: |
 | _____ |
 |Designation : _____ Date : |
 | _____ |

EQUIPMENT DETAILS

DESCRIPTION & FUNCTION OF EQUIPMENT:

LIST OF COMPONENTS

EQUIPMENT COMPONENT SPECIFICATION DATA SHEET

DESCRIPTION OF ANY OTHER COMPONENT

RECOMMENDED SPARE PART LIST

MATERIAL IN PRODUCT CONTACT REVIEW

INSTRUMENT CALIBRATION LIST

PREVENTIVE MAINTENANCE REQUIREMENTS FOR THE EQUIPMENT

INSTALLATION QUALIFICATION ATTACHMENTS

List of Drawings

(ii) List of Certificates

(iii) LIST OF ANY OTHER ATTACHMENTS:

DOCUMENTATION PROVIDED BY VENDOR:

DISCREPANCY/JUSTIFICATION AND CORRECTIVE ACTION:

INSTALLATION QUALIFICATION SUMMARY

CONCLUSION:

REFERENCES:

Protocol Prepared by:	Departmental Review:
Name:	Name:
Designation:	Designation:
Signature:	Signature:
Date:	Date:
Engineering review	Quality Assurance review
Name:	Name:
Designation:	Designation:
Signature:	Signature:
Date:	Date:
Departmental Approval:	Vendor Approval (If Required):
Name:	Name:
Designation:	Designation:
Signature:	Signature:
Date:	Date:
Engineering Approval:	Quality Assurance Approval:
Name:	Name:
Designation:	Designation:
Signature:	Signature:
Date:	Date:

1.0	Objective	
	To lay down the procedure for Installation qualification of	
	Equipment/System.	
2.0	Scope	
	This Standard Operating Procedure is applicable for Installation	
	Qualification of Equipment / instrument / system in all departments of	
	Company Name XYZ.	
3.0	Responsibility	
	3.1	It is the responsibility of QA in execution of qualification,
		Validation / revalidation with the support of user department.
	3.2	Head QA/ his designee shall be responsible for implementation and
		compliance of this SOP.
	3.3	Head-QA or his/her designee shall be responsible for review and
		authorization of qualification of documents
4.0	Abbreviations and Definitions	
	4.1	Operating System: An organization of components that have a defined
		operational function (e.g. piping, instrumentation, equipment,
		facilities, computer hardware/ software etc.)
	4.2	Calibration : The performance of tests and retests to ensure that
		measuring equipment (e.g. for temperature, weight, pH) used in a
		manufacturing process or analytical procedures (in production or
		quality control) give measurements that are correct within established
		limits.
		Design Qualification (DQ): Design Qualification is documented
	4.3	evidence that the proposed design of the facilities, systems and
		equipment are suitable for intended purpose.
	4.4	Factory Acceptance Test (FAT): Testing conducted at the supplier's
		factory usually involving the user to determine whether or not a
		system satisfies its acceptance criteria and to enable the customer to
		determine whether or not to accept the system.
	4.5	Installation Qualification (IQ): Installation Qualification is a
		documented verification that the facilities, systems and equipment as
		installed or modified; comply with the approved design and
		manufacturer's recommendations.
	4.6	Operational Qualification (OQ): The documented action of
		demonstrating that process equipment and ancillary systems work
		correctly and operate consistently in accordance with established
		specifications for the anticipated range of operation.
	4.7	Performance Qualification (PQ): Performance Qualification is the
		documented evidence, which verified that the equipment or system
		operates consistently, and gives reproducibility with in defined
		specifications and parameters for prolonged period.
	4.8	Qualification: Qualification is the act of planning, carrying out and
		recording of tests on equipment and systems, which form part of the
		validated process, to demonstrate that it will perform as intended.
	4.9	Re-validation: validation exercise to validate the system /equipment
		/process / method which is in state of validation & changes done which
		may impact it's earlier state of validation.
	4.10	User Requirement Specification (URS): A requirement specification
		that describes, what the equipment or system is supposed to do, thus
		containing at least asset of criteria or conditions that have to be
		met.
	4.11	Functional Design Specification(FDS):It is the technical Specification
		provided by the vendor against the User Requirement Specification
	4.12	Validation: Validation is the establishment of documented evidence
		that a system does what it is supposed to do. It is the collection and
		evaluation of data, which ensure that the manufacturing process
		including equipment, building, materials are capable of achieving the
		intended results on the consistent and continuous basis.
	4.13	Validation Protocol: The validation protocol is a written plan
		stating how validation will be conducted, including test parameters,
		product characteristics, and production equipment and decision points
		on what constitutes acceptable test results.
	4.14	Validation Report: The validation report is a written report on the
		validation activities, validation data and the conclusions drawn.
5.0	Procedure	
	Equipment qualification shall be performed as per the steps defined in	
	Validation master plan of respective block. Following shall be the	
	generalized plan for equipment qualification	
	Equipment URS	
	Design Qualification	
	Risk Analysis	
	IQ	
	FAT	

||PQ |
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 ||
 ||SAT/IQ |
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 ||
 ||Re-Qualification |
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	5.1	Certain qualification activities can be combined depending upon size
	and operation of the equipment. In some equipment like weighing	
	balance, plug and play Equipment, Table top centrifuge, conductivity	
	meter etc. commissioning report along with the IQ and OQ Checks shall	
	hold good as qualification document.	
	5.2	User Requirement Specification(URS)
	Users Requirement Specifications are base documents that describe the	
	basic quality, intended parameters and shall be prepared by the	
	concerned personnel of the department tendering the equipment, with	
	co-ordination of Quality Assurance and Engineering departments. URS	
	shall include list of requirements from the system and define what	
	the equipment to perform. Users Requirement specification shall	
	include but not be limited to:	
	5.2.1	Description of equipment.
	5.2.2	Description of concerned process steps.
	5.2.3	Required Inputs/ Outputs. (utilities)
	5.2.4	Change parts
	5.2.5	Capacity, speed of the equipment
	5.2.6	Operational parameters required
	5.2.7	Space required and space available (length,width,height)
	5.2.8	GMP consideration
	5.2.9	Specifications (MOC)
	5.2.10	Print out required for outputs.
	5.2.11	Specific requirement like temperature, moisture ,
	sensitiveness, protection from light etc.	
	5.2.12	Safety parameters
	5.2.13	Online sampling
	Note: Functional Design Specification (FDS) Shall be provided by the	
	Vendor Against the Approved Users Requirement Specification (if	
	applicable)	
	5.3	Equipment Design Qualification (DQ)
	Design qualification shall follow the below mentioned activities:	
	5.3.1	Design qualification shall verify that premises, facilities,
	equipment and processes have been designed in accordance with	
	URS.	
	5.3.2	This document shall be used as a specified checklist for the
	verification of design specification of the facility,	
	equipment.	
	5.3.3	The DQ shall be prepared by vendor, review by user; gap
	analysis shall be performed with respect to URS and its impact	
	shall be concluded in DQ and finally approved by Head Quality	
	assurance/Designee. For example:	
	5.3.3.1	Suitability of the equipment with site
	5.3.3.2	Modification in basic design (if needed) matched with
	URS.	
	5.3.3.3	Acceptability of risk level
	5.4	Factory Acceptance Test(FAT)
	5.4.1	FAT shall be conducted and documented at the supplier/
	manufacturer site, to take care problems and resolution of the	
	equipment prior to shipment, to assure that the system will	
	meet its specifications and operational functions. Before	
	doing FAT, the FAT protocol shall be prepared by vendor,	
	review by user and finally approved by Head Quality	
	assurance/Designee.	
	5.4.1.1	Verification as per FAT protocol
	5.4.1.2	Verification of equipment documents with respect to
	design review.	
	5.4.1.3	Operational verification or any feasible test
	5.4.1.4	Verification of alarm and safety measures.
	5.4.1.5	Verification of critical instrument calibration
	certificates or any other certificates.	
	5.5	Site Acceptance Test(SAT)
	5.5.1	The site acceptance test (SAT) shall be performed on site to
	determine that the system and any associated equipment has not	
	been damaged and function correctly in its operating	
	environment. Before doing SAT, the SAT protocol shall be	
	prepared by vendor, review by user and finally approved by	
	Quality assurance.	
	Site acceptance test shall consider the following elements but	
	not limited to:	
	5.5.1.1	Adequate availability of space (personnel and material
	movement)	
	5.5.1.2	Verification of recommended utilities as per FAT
	report.	
	5.6	Head Quality Assurance or his/ her designee in consultation with Head
	of concerned department shall decide whether to use IQ and OQ	
	document of vendor or In house IQ OQ document will be sufficient. If	
	IQ, OQ document of vendor has to be used, the Head of concerned	
	department/designee shall review and provide the departmental	
	approval and final approval shall be provided by Head Quality	
	assurance/Designee.	
	5.7	Equipment installation qualification (IQ)
	Installation Qualification activity shall be crosschecked with	
	respect to adherence to approved design and that manufacturer's	

|| recommendations and documented as follows: |

|| 5.7.1 |General: |

|| 5.7.1.1|Before doing IQ of an equipment, Equipment ID shall be |

|| |generated as per QA/000 (Equipment Identification |

|| |number) |

|| 5.7.1.2|The Installation Qualification document (annexure 1) |

|| |shall be filled for each equipment as identified in |

|| |Validation Master Plan. |

|| 5.7.2 |The IQ document shall preferably have the following contents: |

|| |The first page shall show the installation location of the |

|| |Equipment with date of its installation and the index. |

|| 5.7.2.1|Persons performing installation qualification: |

|| |The work group as formed by the Head of concerned |

|| |area shall be documented. |

|| 5.7.2.2|Equipment details: |

|| |Equipment description including the manufacturer, |

|| |supplier, Model, serial no, dimensions; location, |

|| |working capacity and supporting utilities etc. shall |

|| |be mentioned. |

|| 5.7.2.3|Description & Function of Equipment: |

|| |A brief description about the equipment and its |

|| |function shall be stated. |

|| 5.7.2.4|List of Components: |

|| |Serial No., vendor tag no. if any, component name and |

|| |location of the main components shall be mentioned. |

|| 5.7.2.5|Equipment Component Specification Data Sheet: |

|| |Description of the component (as specified, actual as |

|| |well as its verification) for which the IQ protocol |

|| |is to be filled shall be mentioned in the equipment |

|| |component specification data sheet. Depending upon |

|| |the equipment, relevant data sheet shall be filled. |

|| 5.7.2.6|Description of other components, if any: |

|| |Description of any other component, if any (as |

|| |specified, actual as well as its verification) shall |

|| |also be mentioned. |

|| 5.7.2.7|Recommended Spare Part List: |

|| |Enlist spare parts received under this heading. |

|| 5.7.2.8|Material in Product Contact Review: |

|| |Identity of the material of construction of |

|| |equipment/components shall be mentioned and their |

|| |verification that they confirm to specifications and |

|| |will not adversely affect the product shall also be |

|| |stated. |

|| 5.7.2.9|Instrument Calibration List: |

|| |The list of instruments requiring calibration shall |

|| |be mentioned under this heading. |

|| 5.7.2.1|Preventive maintenance Requirements for the |

|| |Equipment: |

|| |Components that need preventive maintenance and the |

|| |frequency for checks shall be mentioned under this |

|| |heading. |

|| 5.7.2.1|Installation Qualification Attachments |

|| |1 |

|| |List of Drawings, List of Certificates and any other |

|| |attachments shall be mentioned under this heading. |

|| 5.7.2.1|Documentation Provided By Vendor: |

|| |2 |

|| |List of documents provided by the vendor shall be |

|| |mentioned. |

|| 5.7.2.1|Discrepancy/Justification and Corrective Action: |

|| |3 |

|| |Any discrepancy during installation shall be |

|| |documented in the IQ document itself with proper |

|| |justification and Corrective Action. |

|| 5.7.2.1|Installation Qualification Summary & Conclusion: |

|| |4 |

|| |Installation qualification Summary & Conclusion shall |

|| |be filled by the person associated with the |

|| |Installation of the respective equipment and got |

|| |verified by the departmental Head and approved by |

|| |QA/designee. |

|| 5.7.2.1|References: |

|| |5 |

|| |At the last page the References shall be mentioned. |

|| Note: Risk Analysis Shall be done before the Installation |

|| Qualification as per SOP QA/000 |

|| 5.8 |Flow of updation of equipment/instrument in the calibration planner |

|| 5.8.1 |Whenever new equipment will be installed in the facility the |

|| |calibration planner (Refer Sop EN/000) shall be updated to |

|| |incorporate new equipment/instrument. Also when old equipment |

|| |will be shifted from one area to other, the same shall be |

|| |updated. Following is the flow for updation of |

|| |equipment/instrument in calibration planner. |

|| 5.8.2 |Before execution of IQ issuance of Equipment ID shall be given |

|| |as per SOP QA/000, Equipment identification number. |

|| 5.8.3 |System level impact analysis, as per SOP EN/000 Part number |

|| |system for component of Equipment/ Instrument/Devices |

|| 5.8.4 |For the instrument & component which require calibration or |

|| |replacement, Part number shall be given as per SOP EN/000 Part |

|| |number system for component of Equipment/ Instrument/Devices |

|| 5.8.5 |Updation of Equipment/Instrument in calibration planner as per |

|| |SOP xxx Calibration of instrument/ equipment |

| |5.9 |Equipment transfer |

| |5.9.1 |Transfer on temporary basis: In case of temporary transfer the|

| | |equipment ID of the equipment shall remain same and hence |

| | |there will be no updation in the calibration planner |

| |5.9.2 |Transfer on permanent basis: In case of permanent transfer of |

| | |equipment |

| | |from one area to other there shall be a change in the |

| | |equipment ID of the |

| | |equipment hence calibration planner shall be updated as per |

| | |above said flow |

| |5.9.3 |Based on the type of transfer, QA in consultation with user |

| | |and engineering shall evaluate the transfer process and its |

| | |further usage. |

| |5.9.4 |Based on evaluation, risk assessment shall be conducted (if |

| | |required) for finalizing requirement of qualification. |

|5.10 |Revalidation |

| |5.10.1 |Revalidation of any equipment/system shall be performed as per|

| | |revalidation planner on a defined frequency. |

| |5.10.2 |Revalidation of any equipment/system shall be performed in |

| | |case of any major change in the existing validated |

| | |system/equipment or on permanent shifting of equipment. |

|5.11 |The hard copies of annexure 3 shall be archived in Quality assurance |

| |at an interval of every six month; however the soft copy shall be |

| |updated in routine basis. |

|6.0 |Forms and Records |

|6.1 |Installation Qualification |

| |Annexure 1 |

|7.0 |References |

| |

|8.0 |Distribution |

| |Master Copy | |

| |:

| |Controlled Copies | |

| |:

|9.0 |History |

|Date |Revision Number |Reason for Revision |

| | |

| | |

| | |

Detail of Equipment

Installation Location :

Date of Installation :

Name of Equipment / System:

Equipment ID Number:

Production Block/Department:

Area (if any) :

Serial Number (if any):

Equipment ID issuance date:

Study Start Date:

Study End Date:

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1.	DESCRIPTION & FUNCTION OF EQUIPMENT	
2.	PERSON PERFORMING OPERATIONAL QUALIFICATION	
3.	LIST OF SOP's USED FOR OPERATIONAL QUALIFICATION & TRAINING	
RECORD		
4.	ACCEPTANCE CRITERIA	
5.	QUALIFICATION TEST COMPONENTS	
6.	MATERIAL AND EQUIPMENT REQUIRED FOR CARRYING OUT OPERATIONAL	
QUALIFICATION		
7.	CALIBRATION/ VERIFICATION DOCUMENTATION	
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10.	QUALIFICATION TEST	
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13.	ACCEPTANCE OF EQUIPMENT	
14.	SUMMARY OF OPERATIONAL QUALIFICATION	
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DESCRIPTION & FUNCTION OF EQUIPMENT:

[illegible]

- 5.0 QUALIFICATION TEST COMPONENTS:
- 6.0 MATERIAL AND EQUIPMENTS REQUIRED FOR CARRYING OUT OPERATIONAL QUALIFICATION:
- 7.0 CALIBRATION/ VERIFICATION DOCUMENTATION:
- 8.0 INITIAL SET POINTS (if applicable):
- 9.0 OPERATIONAL SETTINGS:
- 10. QUALIFICATION TESTS
- 11.0 OPERATIONAL QUALIFICATION ATTACHMENTS:
- 12.0 DISCREPANCY/JUSTIFICATION AND CORRECTIVE ACTION:
- 13.0 ACCEPTANCE OF EQUIPMENT:
- 14.0 OPERATIONAL QUALIFICATION SUMMARY
- CONCLUSION:
- 15.0 REFERENCES:

Protocol Prepared by:	Departmental Review:
Name:	Name:
Designation:	Designation:
Signature:	Signature:
Date:	Date:
Engineering review	Quality Assurance review
Name:	Name:
Designation:	Designation:
Signature:	Signature:
Date:	Date:
Departmental Approval:	Vendor Approval (If Required):
Name:	Name:
Designation:	Designation:
Signature:	Signature:
Date:	Date:
Engineering Approval:	Quality Assurance Approval:
Name:	Name:
Designation:	Designation:
Signature:	Signature:
Date:	Date:

	PQ	
	SAT/IQ	
	Re-Qualification	
	5.1	Certain qualification activities can be combined depending upon size
		and operation of the equipment. In some equipment like weighing
		balance, plug and play Equipment, Table top centrifuge, conductivity
		meter etc. commissioning report along with the IQ and OQ Checks shall
		hold good as qualification document.
	5.2	User Requirement Specification(URS)
		Users Requirement Specifications are base documents that describe the
		basic quality, intended parameters and shall be prepared by the
		concerned personnel of the department tendering the equipment, with
		co-ordination of Quality Assurance and Engineering departments. URS
		shall include list of requirements from the system and define what
		the equipment to perform. Users Requirement specification shall
		include but not be limited to:
	5.2.1	Description of equipment.
	5.2.2	Description of concerned process steps.
	5.2.3	Required Inputs/ Outputs. (utilities)
	5.2.4	Change parts
	5.2.5	Capacity, speed of the equipment
	5.2.6	Operational parameters required
	5.2.7	Space required and space available (length,width,height)
	5.2.8	GMP consideration
	5.2.9	Specifications (MOC)
	5.2.10	Print out required for outputs.
	5.2.11	Specific requirement like temperature, moisture ,
		sensitivity, protection from light etc.
	5.2.12	Safety parameters
	5.2.13	Online sampling
		Note: Functional Design Specification (FDS) Shall be provided by the
		Vendor Against the Approved Users Requirement Specification (if
		applicable)
	5.3	Equipment Design Qualification (DQ)
		Design qualification shall follow the below mentioned activities:
	5.3.1	Design qualification shall verify that premises, facilities,
		equipment and processes have been designed in accordance with
		URS.
	5.3.2	This document shall be used as a specified checklist for the
		verification of design specification of the facility,
		equipment.
	5.3.3	The DQ shall be prepared by vendor, review by user; gap
		analysis shall be performed with respect to URS and its impact
		shall be concluded in DQ and finally approved by Head Quality
		assurance/Designee. For example:
	5.3.3.1	Suitability of the equipment with site
	5.3.3.2	Modification in basic design (if needed) matched with
		URS.
	5.3.3.3	Acceptability of risk level
	5.4	Factory Acceptance Test (FAT)
	5.4.1	FAT shall be conducted and documented at the supplier/
		manufacturer site, to take care problems and resolution of the
		equipment prior to shipment, to assure that the system will
		meet its specifications and operational functions. Before
		doing FAT, the FAT protocol shall be prepared by vendor,
		review by user and finally approved by Head Quality
		assurance/Designee.
	5.4.1.1	Verification as per FAT protocol
	5.4.1.2	Verification of equipment documents with respect to
		design review.
	5.4.1.3	Operational verification or any feasible test
	5.4.1.4	Verification of alarm and safety measures.
	5.4.1.5	Verification of critical instrument calibration
		certificates or any other certificates.
	5.5	Site Acceptance Test (SAT)
	5.5.1	The site acceptance test (SAT) shall be performed on site to
		determine that the system and any associated equipment has not
		been damaged and function correctly in its operating
		environment. Before doing SAT, the SAT protocol shall be
		prepared by vendor, review by user and finally approved by
		Quality assurance.
		Site acceptance test shall consider the following elements but
		not limited to:
	5.5.1.1	Adequate availability of space (personnel and
		material movement)
	5.5.1.2	Verification of recommended utilities as per FAT
		report.
	5.6	Head Quality Assurance or his/ her designee in consultation with Head
		of concerned department shall decide whether to use IQ and OQ
		document of vendor or In house IQ OQ document will be sufficient. If
		IQ, OQ document of vendor has to be used, the Head of concerned
		department/designee shall review and provide the departmental
		approval and final approval shall be provided by Head Quality
		assurance/Designee.
	5.7	Equipment Operational Qualification (OQ)
		Operational Qualification activity shall be undertaken in after

		successful completion of Installation qualification. The Equipment	
		shall be crosschecked with anticipated range of operation, control	
		parameters and tolerances routinely used in manufacturing, testing	
		and critical utilities.	
		5.7.1	General:
		The Operational Qualification document (annexure 2) shall be	
		filled for each equipment as identified in Validation Master	
		Plan.	
		5.7.2	The OQ protocol shall preferably have the following contents
		The first page of the OQ protocol shall mention the initiation	
		and completion date of the study and the index.	
		5.7.2.2	Description & function of equipment:
		A brief description about the equipment and its	
		function shall be stated.	
		5.7.2.2	Persons performing operational qualification:
		A work group to perform the activity as constituted	
		by the Head of the department shall be documented.	
		5.7.2.3	A draft SOP for operation of the equipment shall be
		prepared before carrying out the operational	
		qualification.	
		5.7.2.4	Staff personnel shall be trained on the operational
		attributes of the equipment by the vendor service	
		engineer, for carrying out the qualification study.	
		5.7.2.5	Qualification test components:
		Qualification test components including vendor tag	
		no., component name and location shall be mentioned.	
		5.7.2.6	Material and equipment required for carrying out
		operational qualification: Material and equipment	
		required with purpose and quantity/specification of	
		material required shall be stated.	
		5.7.2.7	Calibration/ verification documentation:
		Calibration details for measuring	
		instrument/equipment before and after adjustment	
		shall be stated.	
		5.7.2.8	Initial set points:
		Checking of actual operating procedure and recording	
		of all initial set points of the utility shall be	
		stated.	
		5.7.2.9	Operational settings: Operational settings of
		components with tag No., parameters to be checked and	
		their verification shall be stated.	
		5.7.2.1	Qualification tests:
		0	
		Qualification tests for the equipment for which the	
		OQ document is to be filled shall be mentioned in the	
		qualification test sheets. It should include results	
		of the operational checks applied on the control	
		points of the equipment and alarms, specific	
		challenge of the equipment under normal condition as	
		well as under worst case shall be detailed. Depending	
		upon the equipment, relevant qualification test	
		sheets shall be filled.	
		5.7.2.1	Operational qualification attachments:
		1	
		The type and details of the attachments to OQ	
		document shall be mentioned.	
		5.7.2.1	Discrepancy /justification and corrective action:
		2	
		Any discrepancy observed during the operational	
		qualification shall be documented in the OQ document	
		itself with proper justification and Corrective	
		Action.	
		5.7.2.1	Summary of operational qualification:
		3	
		Operational qualification summary and conclusion	
		shall be filled by the person performing the	
		operational qualification, verified by the	
		departmental Head and approved by QA.	
		5.7.2.1	The SOP will be made effective for routine purpose
		4	after confirming in performance qualification.
		5.7.2.1	After completion of OQ, a training record shall be
		5	prepared as per respective SOP like operation,
		cleaning and preventive maintenance.	
		5.8	Flow of updation of equipment/instrument in the calibration planner:
		5.8.1	Whenever new equipment will be installed in the facility the
		calibration planner (xxx) shall be updated to incorporate new	
		equipment/instrument. Also when old equipment will be shifted	
		from one area to other, the same shall be updated. Following	
		is the flow for updation of equipment/instrument in	
		calibration planner.	
		5.8.2	Before execution of IQ issuance of Equipment ID shall be given
		as per SOP QA/xxx, Equipment identification number.	
		5.8.3	System level impact analysis, as per SOP EN/xxx Part number
		system for component of Equipment/ Instrument/Devices	
		5.8.4	For the instrument & component which require calibration or
		replacement, Part number shall be given as per SOP EN/xxx Part	
		number system for component of Equipment/ Instrument/Devices	
		5.8.5	Updation of Equipment/Instrument in calibration planner as per
		SOP EN/xxx Calibration of instrument/ equipment	
		5.9	Equipment transfer
		Equipment transfer can be of two type's viz. Temporary and Permanent.	
		5.9.1	Transfer on temporary basis: In case of temporary transfer the

			equipment ID of the equipment shall remain same and hence	
			there will be no	
			update in the calibration planner.	
		5.9.2	Transfer on permanent basis: In case of permanent transfer of	
			equipment	
			from one area to other there shall be a change in the	
			equipment ID of the	
			equipment hence calibration planner shall be updated as per	
			above said flow	
		5.9.3	Based on the type of transfer, QA in consultation with user	
			and engineering shall evaluate the transfer process and its	
			further usage.	
		5.9.4	Based on evaluation, risk assessment shall be conducted (if	
			required) for finalizing requirement of qualification.	
		5.10	Revalidation	
		5.10.1	Revalidation of any equipment/system shall be performed as per	
			revalidation planner on a defined frequency.	
		5.10.2	Revalidation of any equipment/system shall be performed in	
			case of any major change in the existing validated	
			system/equipment or on permanent shifting of equipment.	
		6.0	Forms and Records	
		6.1	Operational Qualification	
		Annexure 1		
		7.0	References	
		8.0	Distribution	
		Master Copy		
		:		
		Controlled Copies		
		:		
		9.0	History	
		Date	Revision Number	Reason for Revision

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	SAT/IQ	
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	5.2.8	GMP consideration
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		assurance/Designee. For example:
	5.3.3.1	Suitability of the equipment with site
	5.3.3.2	Modification in basic design (if needed) matched with
		URS.
	5.3.3.3	Acceptability of risk level
	5.4	Factory Acceptance Test(FAT)
	5.4.1	FAT shall be conducted and documented at the supplier/
		manufacturer site, to take care problems and resolution of the
		equipment prior to shipment, to assure that the system will
		meet its specifications and operational functions. Before
		doing FAT, the FAT protocol shall be prepared by vendor,
		review by user and finally approved by Head Quality
		assurance/Designee.
	5.4.1.1	Verification as per FAT protocol
	5.4.1.2	Verification of equipment documents with respect to
		design review.
	5.4.1.3	Operational verification or any feasible test
	5.4.1.4	Verification of alarm and safety measures.
	5.4.1.5	Verification of critical instrument calibration
		certificates or any other certificates.
	5.5	Site Acceptance Test(SAT)
	5.5.1	The site acceptance test (SAT) shall be performed on site to
		determine that the system and any associated equipment has not
		been damaged and function correctly in its operating
		environment. Before doing SAT, the SAT protocol shall be
		prepared by vendor, review by user and finally approved by
		Quality assurance.
		Site acceptance test shall consider the following elements but
		not limited to:
	5.5.1.1	Adequate availability of space (personnel and
		material movement)
	5.5.1.2	Verification of recommended utilities as per FAT
		report.
	5.6	Head Quality Assurance or his/ her designee in consultation with Head
		of concerned department shall decide whether to use IQ and OQ
		document of vendor or In house IQ OQ document will be sufficient. If
		IQ, OQ document of vendor has to be used, the Head of concerned
		department/designee shall review and provide the departmental
		approval and final approval shall be provided by Head Quality
		assurance/Designee.
	5.7	Equipment Installation qualification (IQ)
	5.8	Equipment Operational Qualification (OQ)

			system/equipment or on permanent shifting of equipment.
	5.13	The hard copies of annexure 3 shall be archived in Quality assurance	
	at an interval of every six month; however the soft copy shall be		
	updated in routine basis.		
6.0	Forms and Records		
	6.1	Performance Qualification (Annexure to be framed based on annexure	
	for Operational Qualification)		
7.0	References		
8.0	Distribution		
	Master Copy		
	:		
	Controlled Copies		
	:		
9.0	History		
	Date	Revision Number	Reason for Revision

STANDARD OPERATING PROCEDURE SOP No. : QMS 008
Revision No.: 00

Title: Requalification of Equipment Supersedes : None
Page No. : 1 of 3

1.0 Objective

To lay down the procedure for Re-qualification of Equipment

2.0 Scope

This SOP applies to all equipment that requires re-qualification within the organization.

3.0 Responsibility

3.1 Equipment Owner/Operator: Responsible for initiating and overseeing the re-qualification process for the equipment

3.2 Quality Assurance: Responsible for reviewing and approving the re-qualification plan and report

3.3 Maintenance Personnel: Responsible for performing any maintenance or adjustments required before re-qualification.

3.4 Operations Personnel: Responsible for executing the re-qualification activities as per the SOP.

4.0 Abbreviations and Definitions

QA Quality Assurance

SOP Standard operating procedure

5.0 Procedure

5.1 Pre-Re-qualification Preparations:

1. Review the equipment's usage history, any recent maintenance, and any changes made to its components or configuration.

2. Identify the need for re-qualification based on factors such as changes in process requirements, maintenance activities, or a predetermined time interval

5.2 Re-qualification Plan:

1. Develop a re-qualification plan detailing the objectives, scope, methodology, acceptance criteria, and schedule.

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2. Include details about the equipment's current status, maintenance activities, and any adjustments needed before re-qualification

5.3 Maintenance and Adjustment:

1. Perform any required maintenance activities or adjustments to ensure that the equipment is in optimal working condition.

2. Calibrate the equipment as necessary, using traceable standards and tools.

5.4 Data Collection:

1. Start the re-qualification process according to the plan

2. Collect relevant data during the re-qualification process, including performance parameters, measurements, and observations.

5.5 Data Analysis:

1. Use appropriate software and tools to analyze the collected data.

2. Compare the obtained data with the equipment's original specifications and acceptance criteria to determine its compliance.

5.6 Reporting:

1. Prepare a comprehensive re-qualification report that includes:

■ Re-qualification objectives and scope

■ Re-qualification plan and methodology

■ Equipment maintenance and calibration details

■ Data collected during re-qualification

■ Data analysis results

■ Deviations from specifications (if any)

■ Conclusions and recommendations

5.7 Review and Approval

STANDARD OPERATING PROCEDURE SOP No. : QMS 008

Revision No.: 00

Title: Requalification of Equipment Supersedes : None
Page No. : 3 of 3

1. Submit the re-qualification report to the Quality Assurance department for review

2. Upon approval, the report should be archived for future reference.

6.0 Documentation

Re-qualification plan

Maintenance and calibration records

Re-qualification report

Quality Assurance approvals

7.0 Post-Re-qualification Actions:

- 1 Implement corrective actions if deviations from specifications are identified.
- 2 Update the equipment's status and maintenance records as needed.

8.0 References

List any relevant standards, guidelines, or references used in developing this SOP.

9.0 Distribution

Master Copy :

Controlled Copies :

10. History: : Maintain a record of any revisions made to this SOP, along with the reasons for the revisions.

This SOP provides a guideline for re-qualifying equipment to ensure continued compliance with specifications and performance standards. Adapt the procedure to fit the specific requirements of your organization and the equipment being re-qualified while adhering to industry regulations and standards.

Date Revision Number Reason for Revision

GENERAL CHECKS:

INSTRUMENT/EQUIPMENT CALIBRATION STATUS: (Attach extra sheet if required)

Note: Enclose all the relevant Certificates of Calibration and Traceability documents.

Differential pressure: Date:

Observation table: (for light intensity table)

Conclusion:

Conclusion:

Equipment/System | | Date |

Department | | Location |

S/No. | Checkpoint | Remarks

(Yes/No)

1 | Training imparted to all the concerned personnel on OQ protocol |

2 | Calibration status of the components of system/equipment updated as on date |

3 | Ensure the availability of the testing instruments as per the protocol |

4 | Calibration status of the Testing Instruments updated as on date |

5 | Ensure the availability of the Material and Reagents as per the protocol |

S/No. | Instrument /Equipment Name | Identification No. | Calibration Done Date | Calibration Due Date

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Remarks (if any): | Remarks (if any): | Remarks (if any): | Remarks (if any): | Remarks (if any):

Compiled By : | Checked By :

Designation : | Designation :

Department : | Department :

Signature : | Signature :

Date : | Date :

Purpose: To determine the average supply airflow velocity in HEPA filter. | Purpose: To determine the

average supply airflow velocity in HEPA filter. | Purpose: To determine the average supply airflow

velocity in HEPA filter. | Purpose: To determine the average supply airflow velocity in HEPA filter. |

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average supply airflow velocity in HEPA filter. | Purpose: To determine the average supply airflow

velocity in HEPA filter. | Purpose: To determine the average supply airflow velocity in HEPA filter.

DEPARTMENT DATA | DEPARTMENT DATA | DEPARTMENT DATA | DEPARTMENT DATA | DEPARTMENT DATA

| DEPARTMENT DATA | DEPARTMENT DATA

Department | | Area/Location | | Date of Test | Date of Test |

INSTRUMENT DETAILS | INSTRUMENT DETAILS | INSTRUMENT DETAILS | INSTRUMENT DETAILS | INSTRUMENT DETAILS |

INSTRUMENT DETAILS | INSTRUMENT DETAILS | INSTRUMENT DETAILS

Instrument Name | Instrument Name | | Make | Make | |

Model | Model | | Serial Number | Serial Number | |

Calibrated On | Calibrated On | | Calibration Due On | Calibration Due On | |

OBTAINED TEST RESULTS | OBTAINED TEST RESULTS | OBTAINED TEST RESULTS | OBTAINED TEST RESULTS | OBTAINED

TEST RESULTS | OBTAINED TEST RESULTS | OBTAINED TEST RESULTS | OBTAINED TEST RESULTS | OBTAINED TEST

RESULTS | OBTAINED TEST RESULTS | OBTAINED TEST RESULTS | OBTAINED TEST RESULTS | OBTAINED TEST RESULTS |

OBTAINED TEST RESULTS | OBTAINED TEST RESULTS | OBTAINED TEST RESULTS | OBTAINED TEST RESULTS | OBTAINED

TEST RESULTS | OBTAINED TEST RESULTS

Room Name | Room ID | Room ID | HEPA Filter ID Number | Filter Area (Feet²) | Air Velocity in Feet/Minute

(FPM) | Air Velocity in Feet/Minute (FPM) | Air Velocity in Feet/Minute (FPM) | Air Velocity in

Feet/Minute (FPM) | Air Velocity in Feet/Minute (FPM) | Air Velocity in Feet/Minute (FPM) | Average

Velocity (FPM) | Average Velocity (FPM) | Acceptable Range

(±20% of Average) | Result(Pass /Fail) | CFM | Room Volume (ft³) | ACPH

Room Name | Room ID | Room ID | HEPA Filter ID Number | Filter Area (Feet²) | V1 | V2 | V3 | V3 | V4 | V5

| Average Velocity (FPM) | Average Velocity (FPM) | Acceptable Range

(±20% of Average) | Result(Pass /Fail) | CFM | Room Volume (ft³) | ACPH

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Acceptance Criteria | 90 ± 20 % of the Average Velocity, Observed velocities should be within the

specified limits of ±20% of Average velocity as calculated. | 90 ± 20 % of the Average Velocity, Observed

velocities should be within the specified limits of ±20% of Average velocity as calculated. | 90 ± 20 %

of the Average Velocity, Observed velocities should be within the specified limits of ±20% of Average

velocity as calculated. | 90 ± 20 % of the Average Velocity, Observed velocities should be within the

specified limits of ±20% of Average velocity as calculated. | 90 ± 20 % of the Average Velocity, Observed

velocities should be within the specified limits of ±20% of Average velocity as calculated. | 90 ± 20 %

Purpose: To confirm that the HEPA filter system is properly installed by verifying the absence of bypass leakage in the installation and that the filters are free from defects and small leaks. | Purpose: To confirm that the HEPA filter system is properly installed by verifying the absence of bypass leakage in the installation and that the filters are free from defects and small leaks. | Purpose: To confirm that the HEPA filter system is properly installed by verifying the absence of bypass leakage in the installation and that the filters are free from defects and small leaks. | Purpose: To confirm that the HEPA filter system is properly installed by verifying the absence of bypass leakage in the installation and that the filters are free from defects and small leaks. | Purpose: To confirm that the HEPA filter system is properly installed by verifying the absence of bypass leakage in the installation and that the filters are free from defects and small leaks. | Purpose: To confirm that the HEPA filter system is properly installed by verifying the absence of bypass leakage in the installation and that the filters are free from defects and small leaks. | Purpose: To confirm that the HEPA filter system is properly installed by verifying the absence of bypass leakage in the installation and that the filters are free from defects and small leaks. | Purpose: To confirm that the HEPA filter system is properly installed by verifying the absence of bypass leakage in the installation and that the filters are free from defects and small leaks.

DEPARTMENT DATA | DEPARTMENT DATA | DEPARTMENT DATA | DEPARTMENT DATA | DEPARTMENT DATA | DEPARTMENT DATA
| DEPARTMENT DATA | DEPARTMENT DATA

Department || Area/Location || Date of Test | Date of Test |
INSTRUMENT DETAILS | INSTRUMENT DETAILS | INSTRUMENT DETAILS | INSTRUMENT DETAILS | INSTRUMENT DETAILS |
INSTRUMENT DETAILS | INSTRUMENT DETAILS | INSTRUMENT DETAILS

Instrument Name | Instrument Name || | Make | Make ||
Model | Model || | Serial Number | Serial Number ||
Calibrated On | Calibrated On || | Calibration Due On | Calibration Due On ||

OBTAINED TEST RESULTS | OBTAINED TEST RESULTS | OBTAINED TEST RESULTS | OBTAINED TEST RESULTS | OBTAINED
TEST RESULTS | OBTAINED TEST RESULTS | OBTAINED TEST RESULTS

Room Name | Room ID | HEPA Filter ID Number | Upstream Concentration (%) and/or mg/m3 | Acceptance
Criteria (%) | Obtained Result in downstream (% Leakage) | Result (Pass/Fail)

Name : | Name :
Designation : | Designation :
Signature : | Signature :
Date : | Date :
Purpose: To provide the documented evidence that installed HEPA are capable to maintain the desired level of particulate concentration as per regulatory standards. | Purpose: To provide the documented evidence that installed HEPA are capable to maintain the desired level of particulate concentration as per regulatory standards. | Purpose: To provide the documented evidence that installed HEPA are capable to maintain the desired level of particulate concentration as per regulatory standards. | Purpose: To provide the documented evidence that installed HEPA are capable to maintain the desired level of particulate concentration as per regulatory standards. | Purpose: To provide the documented evidence that installed HEPA are capable to maintain the desired level of particulate concentration as per regulatory standards.

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S/ No. | Location (Sampling) | Count/m3
>5.0 µm | Sample Volume (m3) | Acceptance Limit | 95% UCL Count | Average Count | Result (Pass/Fail)

TEST CONDUCTED BY/ RECORDED BY | APPROVED BY

Name : | Name :
Designation : | Designation :
Signature : | Signature :
Date : | Date :

Purpose: To ensure that the differential pressures, temp & Humidity of area are within the acceptable range to maintain the differential gradient wherever required.

Room No. | Time | Differential Pressure | Acceptance criteria | Result (Pass/Fail) | Checked by |
Verified By | Remarks

Room No. | Time | Temperature | Acceptance criteria | Result (Pass/Fail) | Humidity | Acceptance criteria | Result (Pass/Fail) | Checked / Verified by

Purpose: To determine performance and efficiency of HVAC system to eliminate airborne particulates. |
Purpose: To determine performance and efficiency of HVAC system to eliminate airborne particulates. |
Purpose: To determine performance and efficiency of HVAC system to eliminate airborne particulates. |
Purpose: To determine performance and efficiency of HVAC system to eliminate airborne particulates. |
DEPARTMENT DATA | DEPARTMENT DATA | DEPARTMENT DATA | DEPARTMENT DATA

Department || |
Area/Room name || | Date of Test |
Room ID || | Condition | Rest / Operation
Classification/Grade || |
INSTRUMENT DETAILS | INSTRUMENT DETAILS | INSTRUMENT DETAILS | INSTRUMENT DETAILS
Instrument Name || | Make |
Model || | Serial Number |
Calibrated On || | Calibration Due On |
Room Name | Case Study | Time | Count/m3
0.5-5.0 µm | Count/m3
>5.0 µm | Recovery Time (Min) | Result (Pass /Fail)
| Base Line |||| |
| Worst Case |||| |
| Class Regain |||| |
| Particles/m3 (In Operation) | Particles/m3 (In Operation) | Particles/m3 (In Operation) | Particles/m3 (In Operation)
| Target Cleanliness Level | Target Cleanliness Level | Elevated Count (100 Times) | Elevated Count (100 Times)
Grade | ≥0.5µm | ≥5.0µm | ≥0.5µm | ≥5.0µm
B | 3520 | 20 | 352000 | 2000
Acceptance Criteria | Recovery time should not be more than 15 Minutes

TEST CONDUCTED BY/ RECORDED BY | APPROVED BY
Name : | Name :
Designation : | Designation :
Signature : | Signature :
Date : | Date :
Purpose: To determine performance and efficiency of HVAC system to control the temperature and humidity.
| Purpose: To determine performance and efficiency of HVAC system to control the temperature and humidity. | Purpose: To determine performance and efficiency of HVAC system to control the temperature and humidity. | Purpose: To determine performance and efficiency of HVAC system to control the temperature and humidity.
Room Name || | Room ID |
Date | Time | Initial Temperature | Start time | End time | Observed
Temperature | Recovery time | Verified

By

Remarks : | Remarks : | Remarks : | Remarks : | Remarks : | Remarks : | Remarks : | Remarks :
Date | Time | Initial humidity | Start time | End time | Observed
Humidity | Recovery time | Verified

By
|||||||
|||||||

Background/Introduction:

HVAC system function is to condition (heating and cooling), replace (makeup, fresh air, oxygen replacement), pressurize (containment), and clean (filter) the air in the environment to meet the required operational conditions.

HVAC system operational qualification program is designed to provide assurances that all the components of the HVAC system meet the environmental specification in a consistent and reproducible way. In one word it should demonstrate the robustness of the system.

Components of an HVAC system:

Air-Handling Units

Ductwork-Air Distribution Network

Air Filters

Purpose:

The purpose of this protocol is to establish documentary evidence to ensure that the Heating, Ventilation and Air Conditioning Systems installed at different areas and supporting facilities are performing as per their intended use and the predetermined specifications.

Scope:

This protocol is applicable to the Heating ventilation and air conditioning systems installed in xxx facility.

Responsibility:

It is the responsibility of Biobridge Healthcare solutions Pvt. Ltd. to provide the Operational qualification protocol and carryout the activity.

Safety considerations:

Follow the safety consideration as per the equipment specific SOPs.

Follow the gowning procedures of the respective areas before entering the classified areas.

Use personal protective equipment during the study, as required.

Allow the test results reading to stabilize adequately prior to reporting.

While handling aerosols, use synthetic gloves if necessary to prevent excessive skin contact. Do not wear contact lenses and always use ANSI approved safety glasses or splash shield.

Inhalation of vapors or spray mist may cause headaches, and/or nose and throat irritation.

Pre Requisites:

Training:

Training shall be imparted to the concerned personnel on this protocol before the execution of the activity. All necessary aspects related to the operational activity shall be explained during training program.

A copy of the training record shall be attached to the operational qualification report.

Equipment:

Non-viable particle counter

Digital Anemometer (Vane/Wire type)

Digital Thermo-hygrometer

PAO (Poly Alpha Olefin) Aerosol Generator or suitable alternate

Aerosol Photometer

Smoke Generator

Digital Video Camera

Lux Meter

Sound Level Meter (dBA meter)

Material and Reagents:

Poly Alpha Olefin Aerosol or suitable alternate

Flow Marker / WFI or suitable reagent for Smoke generation

70% v/v IPA (Filtered)

Prior to conducting/ executing the operational qualification activity following conditions must be fulfilled:

Ensure that all the test and measuring instruments are in calibrated state before initiation of activity.

System should be safe for execution.

Check availability of materials and reagents.

Record the pre-requisite details in Annexure-1.

Procedure:

Test Parameter and Frequency:

Air Velocity Test and Calculation of Air changes per hour:

Objective:

To demonstrate that air system is balanced and capable of delivering sufficient air velocity / volumes to maintain a desired cleanliness class/ hygiene level at operational condition.

Use Digital Anemometer to measure air velocity of the HEPA filter at five different locations with four locations at corners and one location at center of HEPA filter.

Measure the filter velocity downstream of the filter as a distance of about 150mm as shown in Figure 1.

Measuring time at each position should be at least 10 seconds duration, and the average, maximum and minimum values should be recorded. Allow the reading to stabilize completely.

Figure-1

Calculate the average air velocity as per formula given below:

Average velocity of each HEPA filter:

Calculate the Average Velocity (Vav, in Feet per Minute) of HEPA filter as:

V1, V2 etc. = Velocity observed at different points

Calculate the $\pm 20\%$ of Average as mentioned below:

Review all the observed velocities are within the specified limits of $\pm 20\%$ of Average velocity as calculated by above formulas.

Calculation of Air changes per hour:

7.2.7.1 Calculate the area of the filter by multiplying the length and width of the filter in feet

$A = l \times w$

l = length of HEPA filter

w = Width of HEPA filter

7.2.7.2 Calculate the total air volume per minute supplied in the clean room by following formula:

$T = A \times V$

A = Area of HEPA filter in square feet

V = Average air velocity in feet per minute

7.2.7.3 Calculate the total air in the room by multiplying length, width and height of the room in feet:

Volume = L x W x H

7.2.7.4 Calculate the air change per hour by using formula:

Air Change per Hour = $T \times 60$

Volume

Record the results as per the Annexure-2.

Installed HEPA Filter Integrity Testing:

Objective:

To ensure that the HEPA filters installed are not damaged during operation and/ or fixed properly in the appropriate place, such that there is no leakage from the filter media and the other joint gaskets, frames etc.

The filter testing shall be performed after operational velocities have been verified and adjusted where ever necessary.

The measuring and testing instruments being used shall be calibrated prior to usage with reference to the NPL or any other traceable reference standard.

Adjust the aerosol generator to achieve the PAO challenge concentration of approximately 20-80 µg/L of air (Ref: ISO 14644 Part III). Adjust this challenge to 100% scale of the upstream concentration at upstream of the HEPA filter and check for detection in photometer.

Select downstream scanning mode from the control panel. Scan the subjected HEPA filter by holding the photometer probe approximately 3-5 cm from the filter face and passing the probe in slightly overlapping strokes, at a scan rate of not more than 5cm/sec so that the entire filter face is sampled.

The entire periphery of the filter, the bond between the filter medium and the frame, all other joints shall be scanned.

Record the results as per the Annexure-3.

Air Borne (Non-viable) Particle count

Objective:

To provide the documented evidence that installed equipment's are capable to maintain the desired level of air borne particulate concentration as per regulatory standards.

Particulate cleanliness of the critical and controlled areas shall be evaluated as per the guideline of ISO 14644 Clean rooms and associated controlled environments, Part 1 (Classification of airborne particulates) and Part 2 (Specifications for testing and monitoring to prove continued compliance with ISO 14644-1, 2 & 3).

Air borne (Non –Viable) particle counts for HVAC system in Classified Area shall be tested at Rest Condition.

Determine the minimum number of sampling point locations from the following equation:

Where,

'NL' is the minimum number of sampling locations (rounded off to a whole number)

'A' is the area of the cleanroom or clean zone (cross section of the moving air perpendicular to the direction of the airflow) in square meters (1m=3.2808 Feet and 1Feet =0.3048m).

Ensure that the sampling locations are evenly distributed throughout the area of the cleanroom or clean zone and positioned at the height of the work activity.

Minimum single sample volume VS per location is determined by using equation:

Where,

'VS' is the minimum single sample volume per location, expressed in liters (except see

'Cn,m' is the class limit (number of particles per cubic meter) for the largest considered particle size specified for the relevant class.

'20' is the defined number of particles that could be counted if the particle concentration were at the class limit.

Based upon the above equation following sampling volume shall be considered for the different grades as follows:

Set up the particle counter at a working height according to appropriate test equipment operating procedures at the sample point as identified.

The sampling probe shall be positioned pointing into the airflow.

Identification of the sampling locations as per the attached sheet.

Collect the print and compile the data and attach with respective Annexures.

Interpretation of data shall be done as per the Acceptance Criteria.

Original along with a photocopy of the location wise non-viable particles counts prints out shall be attached with the report and annexure.

Record the results as per as Annexure-4.

Differential Pressure, Temperature & Humidity Monitoring:

Objective:

To ensure that the differential pressures, temp and humidity of the area are within the acceptable range to maintain the cleanroom environment.

Record the temperature and humidity readings from digital thermo hygrometers and monitor the parameters of areas for three consecutive readings at 4hrs interval in operation condition.

Review the data with respect to the acceptance criteria as specified in Clean room data sheet and record the results as per the Annexure-5.

Recovery Test:

This test is to be performed for determining operation and efficiency of HVAC system to eliminate airborne particulates.

Prior to recovery testing, airflow velocity and the uniformity of velocity must be in compliance to the acceptance criteria.

Setup the non-viable particle counter in the clean zone where the recovery time has to be determined.

Particle counter should not be placed directly under the air outlet. Adjust the single sample volume, number of cycles and delay time of the counter should be adjusted to not more than 10 seconds.

Adjust the single sample volume to the same value used for determining the cleanliness class.

In the case two different particle sizes are used for the cleanliness classification both shall be chosen for the study (e.g. >0.5 µm and > 5.0 µm).

Place the Particle Counter Probe at the test point and the probe should not be placed directly under the air outlet.

First, check and record the actual particle count level in the clean area, generate the smoke using the Aerosol Generator at the air inlet to the clean area in such a way that the generated smoke distributed uniformly, until the particle count level in the clean area becomes approximately >100 times (worst case) the target cleanliness level for 0.5µm particle (i.e. Acceptance Level)

Start measuring & recording the particle count each minute. Particle count level gradually comes down and particle count readings to be taken until the 100:1 ratio in the clean area is restored to the original condition recorded & record the recovery time.

Plot the data of decreasing particle concentration on a rectangular graph, with particle concentration in Y Axis and time in X Axis.

Based on the obtained particle concentration with respect to the time, plot the graph till the particle concentration in the clean area reaches the specified cleanliness level (i.e. Acceptance Level).

Cleanliness recovery rate is obtained from the slope value of the line.

In case it is difficult to raise the particle concentration 100 times the target cleanliness level, recovery time shall be calculated on the basis of the slope calculated from the graph plotted as follows:

The overall recovery time shall be calculated for the clean zone as follows:

Where,

TR is Recovery Time

N is number of tested location

TRI is the recovery time at each location

Record the results as per as Annexure-6.

Recovery test for humidity and temperature:

7.6.16.1 The humidity and temperature are checked at the off position of the HVAC system.

7.6.16.2 The humidity is increased to 75% and temperature to 40 °C.

7.6.16.3 The HVAC system is switched ON and humidity and temperature

Readings are checked at 1 min interval until the original condition of room restored as per clean room data sheet.

7.6.16.4 Record the time required to stabilize the humidity and temperature in Annexure-7.

Air Flow Visualization Test:

Objective:

Air Flow Visualization is performed to demonstrate that the air flow pattern below the HEPA filter to prevent air turbulence or stagnant air in clean zone.

Air Flow Visualization is performed to demonstrate that the airflow direction and its uniformity conform to the design and operational specifications

Place the smoke generator discharge port below the HEPA filter and observe the flow of smoke.

Videography and/or still photography shall be done during this test to record the activity.

Review the flow of the air and the direction shall be as per the area design and pressure cascade.

A copy of videography/photography shall be attached to the report.

Record the results as per the Annexure-8.

Light Intensity Test:

Objective:

To determine the light intensity level and its uniformity throughout an area at pre-determined points.

Record the light intensity levels on a pre-determined grid pattern/layout of lux level or on a 3m X 3m grid pattern in the absence of a specification.

Divide the room into four equal locations.

Record the light intensity level at working height or 1 meter above finished floor height.

The area under test must have all fitted lights working and the area structure should be complete. Also, if required, emergency lighting levels can be recorded.

Record the light intensity levels in lux for each position on the grid in Annexure-9.

Note down all variations from specification, if any.

Acceptance Criteria: It should meet the acceptance criteria as mentioned in the attached room data sheet.

Sound Level Test:

Objective:

To determine the Sound level and its uniformity throughout an area at pre- determined points.

Verify that the air conditioning system is operating in the stable condition and in operational condition.

Map out the area under test as required by the specification in a 2400 mm x 2400 mm x grid or on a previously agreed location. Divide the area into four equal parts.

Set the instrumentation to work and ensure that unauthorized entry and exits from the room are prevented.

Measure the sound levels at 750 mm from the finished floor level and away 1 meter from equipment.

Record the results of sound level test in Annexure-10.

Note down all variations from the specifications, if any.

Acceptance Criteria:

Operational Qualification Report:

A report shall be prepared based on the results obtained in study. Report shall be in a narrative form which describes the works as well as conclusion regarding acceptability.

All the results shall be recorded in the annexure of protocol and attached with the report; the following data (but not limited to) will be archived and attached with the report (as applicable) for HVAC system.

Calibration certificates of components and testing equipment

Air Velocity results

Installed HEPA Filter Integrity Testing results

Differential Pressure, Temperature and Humidity Monitoring data

Non-Viable particle Count Test data

Air Flow Visualization Record

Recovery Test data

Videography/photography

Light Intensity Test

Sound Level Test

References:

ISO14644-1, titled "Cleanrooms and associated controlled environments —Part 1: Classification of air cleanliness"

ISO14644-1, titled "Cleanrooms and associated controlled environments —Part 2: Specifications for testing and monitoring to prove continued compliance with ISO 14644-1

ISO14644-3, titled "Cleanrooms and associated controlled environments —Part 3: Metrology and test methods"

WHO TRS 961, titled "WHO expert committee on specifications for Pharmaceutical preparations"

Supplementary guidelines on Good manufacturing practices for Heating, ventilation and air conditioning (HVAC) Systems for non-sterile dosage forms

Annexure:

Abbreviations:

HEPA : High Efficiency Particulate Air

µm : Micron Meter

nm : nanometer

mm : Millimeter

IPA : Iso-Propyl Alcohol

WFI : Water for Injection

µg/L : Micro gram per Liter

ISO : International Organization for Standardization

m3 : Cubic Meter

WHO : World Health Organization

NMT : Not More Than

ND : Not Defined

UNC : Un-Classified

BSC : Bio-safety cabinet

Test Parameter | Frequency /Duration

Air Velocity | Once

Installed HEPA Filter Integrity Testing | Once

Differential Pressure, Temperature and RH monitoring | Three (3) consecutive readings after every four hours- In Operation

Non-Viable Particle Count Test | Once

Air Flow Visualization Test | Once

Recovery Test | Once

Light Intensity Test | Once

Sound Level Test | Once
 S. No. | Grade | Sampling Volume
 1 | Grade A | ≥ 1000 Liters
 2 | Grade B | ≥ 700 Liters
 3 | Grade C | ≥ 2 Liters
 4 | Grade D | ≥ 2 Liters
 S. No. | Test Parameter | Acceptance Criteria
 | Air Velocity | 90 ± 20 % of the Average Velocity
 Observed velocities should be within the specified limits of $\pm 20\%$ of Average velocity of installed filters.
 Unidirectional flow velocity should not disrupt the sensitivity of balances in weighing areas.

Refer Room data sheet for designed air volume & ACPH
 | Installed HEPA Filter Integrity Testing | Leak should not be more than 0.01% of total upstream concentration
 | Differential Pressure, Temp & Humidity | As per clean room data sheet
 | Non-Viable Particulate Count Test |
 | Air Flow Visualization Test | Air Flow shall be unidirectional and perpendicular from the HEPA Filter.
 Air should pass through positive to negative area
 | Recovery Test | Recovery time should not be more than 15 Minutes
 | Light Intensity Test | As per specification in room data sheet
 | Sound Level Test | Not more than 85 dB
 Annexure No. | Annexure Title
 Annexure 1 | Pre- Requisite for operational qualification of HVAC system
 Annexure 2 | Air Velocity Test Report
 Annexure 3 | Installed HEPA Filter Leak Integrity Test Report
 Annexure 4 | Non-Viable Particle Count Test Report
 Annexure 5 | Differential Pressure ,Temperature and Humidity record
 Annexure 6 | Recovery Test Record
 Annexure 7 | Recovery Test Record for Humidity and Temperature
 Annexure 8 | Air Flow Visualization Record
 Annexure 9 | Light Intensity Test Record
 Annexure 10 | Sound Level Test Record

Annexure 1

|Requisition for Production Order |Folio No.: |

Product		
---------	--	--

|Activity| |

ii

It has been planned to begin the

$$| \text{---} |$$

Batch No -----for -----

The tentative date of beginning of this Batch is:-----

S.	Name of the Activity/stage	Production Order No.	Remarks
----	----------------------------	----------------------	---------

No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
-----	---	---	---	---	---	---	---	---	---	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	-----

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1	2	3	4	5
6	7	8	9	10

1	2	3	4	5
6	7	8	9	10

1	2	3	4	5
6	7	8	9	10

1	2	3	4	5
1	2	3	4	5

1	2	3	4	5
1	2	3	4	5

1	2	3	4	5
6	7	8	9	10

|||

|||

.....

| Requisitioned by | Departmental Approval | QA Authorization |

Prerequisites	
Name	Grade
Math 101	101
Math 102	102
Math 103	103
Math 104	104
Math 105	105
Math 106	106
Math 107	107
Math 108	108
Math 109	109
Math 110	110
Math 111	111
Math 112	112
Math 113	113
Math 114	114
Math 115	115
Math 116	116
Math 117	117
Math 118	118
Math 119	119
Math 120	120
Math 121	121
Math 122	122
Math 123	123
Math 124	124
Math 125	125
Math 126	126
Math 127	127
Math 128	128
Math 129	129
Math 130	130
Math 131	131
Math 132	132
Math 133	133
Math 134	134
Math 135	135
Math 136	136
Math 137	137
Math 138	138
Math 139	139
Math 140	140
Math 141	141
Math 142	142
Math 143	143
Math 144	144
Math 145	145
Math 146	146
Math 147	147
Math 148	148
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Math 239	239
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Math 241	241
Math 242	242
Math 243	243
Math 244	244
Math 245	245
Math 246	246
Math 247	247
Math 248	248
Math 249	249
Math 250	250

|Signature| | | |

Date | | | |

Annexure 3

Record of Issue of Production Order	Folio No.:
Product	
Requisition	Production
	Order No.

Date |Folio ||||||||||||||||||||||||||||

Annexure 5

Destruction Record of Production Order Folio No.:				
Product				
/Activity				
S.No. Production Order No. No. of copies returned Issued for Batch No.				
Returned by				
(Sign/Date) Remarks Destroyed by (Sign/Date) Checked by (Sign/Date)				

Annexure 6

Destruction Record of Batch Record/Solution Preparation Record/Cleaning Record/Forms Folio No.:				
Product				
/Activity				
S.No.	Document No.	No. of copies returned	Issued for Batch No.	Returned
by				
(Sign/Date)	Remarks	Destroyed by (Sign/Date)	Checked by (Sign/Date)	

1.0 Objective |
 |To lay down the procedure for issuance of controlled copies of Production |
 |Order, Batch Record, Solution Preparation Record and Cleaning Record. |
 2.0 Scope |
 |This Standard Operating Procedure is applicable for the issuance of |
 |controlled copies of Production Order, Batch Record, Solution Preparation |
 |Record, Cleaning Record and forms which shall be part of batch production |
 |records. |
 3.0 Responsibility |
 |3.1 The Production department shall give requisition duly approved by the |
 |head of production or his/her designee for the required Production |
 |Order, Batch Record, Solution Preparation Record, Cleaning Record and |
 |forms which shall be part of batch production records (hence-forth |
 |referred as forms) when a batch/activity is planned in production. |
 |3.2 The Head QA or his/her designee shall be responsible for authorizing |
 |the issue of requisitioned documents. |
 |3.3 The Documentation section of QA Department shall be responsible for |
 |issue of controlled copies of the Production Order, Batch Record, |
 |Solution Preparation Record, Cleaning Record and forms as per QA |
 |authorized requisition. |
 |3.4 The Head of the production department or his/her designee shall be |
 |responsible to return unused Production Order, Batch Record, Solution |
 |Preparation Record, Cleaning Record and forms if any. |
 |3.5 Head, QA Department shall ensure the implementation of this SOP. |
 4.0 Abbreviations and Definitions |
 |4.1 HOD : Head of the Department |
 |4.2 SOP : Standard Operating Procedure |
 |4.3 QA : Quality Assurance |
 |4.5 No. Number |
 | : |
 |4.6 PO Production Order |
 | : |
 |4.7 BR Batch Record |
 |4.8 SPR Solution Preparation record |
 | : |
 |4.9 CR Cleaning Record |
 5.0 Procedure |
 |5.1 Request for controlled copies of Production Order, Batch Record, |
 |Solution Preparation Record, Cleaning Record and forms: |
 |5.1.1 A requisition duly approved by the Head of the Production |
 |Department or his/her designee for the issue of controlled |
 |copies of PO/BR/SPR/CR/forms shall be submitted to Quality |
 |Assurance Department prior to the start of the activity. |
 |5.1.2 After authorization of the said requisition by the HOD QA or |
 |his/her designee, folio number shall be assigned to the |
 |requisition. |
 |5.1.3 Assigning of Folio Number: |
 |5.1.3.1 The folio number shall be assigned such as xx/zzz. |
	For example: 23/001, where
	23 corresponds to last digits of the year 2023
	followed by a slash followed by, 001 (three digit
	consecutive number) indicates the first folio number
	of the year.
	For different products; product having different
	independent entity and different serotypes/strains of
	a product, the folio no. and records shall be
	maintained separately.
5.1.4 Controlled copies of Production Order, Batch Record, Solution	
Preparation Record, Cleaning Record and Forms shall be	
reproduced by photocopying from approved copies archived in	
Quality Assurance Department. All pages of the Production	
Order, Batch Record, Solution Preparation Record, Cleaning	
Record and forms shall be stamped and signed by a person from	
QA-Docmentation section.	
Specimen of Controlled Copy stamp:	
5.1.5 Assigning of date:	
	In the date XX YYY ZZZZ format xx correponds to the day, YYY
	corresponds to abbreviation for the month and zzzz for the
	year.
5.1.6 Assigning Controlled Copy Number:	
5.1.6.1 The controlled copy number for Production Order,	
Batch Records and Solution Preparation Records,	
Cleaning Record, Forms pertaining to batch production	
activities shall bear the block code (single digit	
numeral), followed by product code (of two digits)	
followed by last two digits of the current year,	
followed by a three-digit consecutive number assigned	
to the controlled copy.	
	For example:
	Controlled Copy No. 10012001
	Where numeral '1' depicts department name, '00' is
	product code assigned to the product say X, digits
	'23' corresponds to last two digits of the year 2023
	and 001 represents the first control copy number
	assigned to a Production Order, Batch Record,

				Solution Preparation Record, Cleaning Record and	
				Forms of a batch in the year.	
				5.1.6.2	The controlled copy number for Solution Preparation
				Record/Cleaning Record for general activities (i.e	
				not pertaining to production batch) shall bear the	
				block code (single digit numeral), followed by last	
				two digits of the current year, and followed by a	
				three digit consecutive number assigned to the	
				controlled copy.	
				For example:	
				Controlled Copy No. 123001	
				Where numeral '1' depicts department name, digits	
				'23' corresponds to last two digits of the year 2023	
				and 001 represents the first control copy number	
				assigned to a Sample Preparation Record/Cleaning	
				Record for general activities of the year.	
				5.1.6.3	In case of product(s) having different independent
				entities, the product code in the controlled copy	
				number shall bear a capital alphabet in subscript	
				form (Handwritten) for distinction of its entity.	
				For example in:	
				Controlled Copy No. 30012001	
				A	
				The alphabet A is a distinct independent entity of	
				the product X (bearing product code 00) distinguished	
				by its independent batch no.	
				5.1.6.4	In case of product with different serotypes/strains,
				the product code in controlled copy number shall bear	
				a capital alphabet in subscript form (as per 5.1.6.2)	
				for distinction of its entity followed by (in	
				bracket) the code of serotype/strains (Handwritten),	
				as applicable.	
				For example in:	
				Controlled Copy No. 30012001	
				A(1)	
				The alphabet A is a distinct independent entity of	
				product X, followed by (1) which is serotype/strain	
				of the product X.	
				5.1.6.5	The same copy number shall be applicable to all the
				PO/BR/CR/SPR/Forms of a batch of the product.	
				5.1.6.6	However in case the batch is to be distributed later,
				the parent controlled copy number shall be followed	
				by a hyphen (-), followed by capital alphabet 'D'	
				(Handwritten) depicting distribution followed by two	
				digits numeral 01,02 and so on indicating number of	
				times the distribution activity being carried out for	
				the said batch.	
				5.1.6.7	In case re-issuance of PO/BR/SPR/CR/Forms is
				necessitated for the same activity, the initially	
				assigned controlled copy number shall be followed by	
				hyphen (handwritten), followed by capital letter	
				'R'(depicting reissuance), followed by numeral 01,02	
				and so on (depicting number of the times the	
				reissuance activity being performed for the said	
				document). The requisition for these documents shall	
				also bear reason for reissuance in remarks column.	

	5.2	Record of issued controlled copies of Production Order, Batch		
		Record, Solution Preparation Record and Cleaning Record:		
		5.2.1	The requisitions shall be kept separately for different	
		products; for independent entities and for different		
		serotypes/strains of the product and shall be assigned folio		
		number.		
		5.2.2	Folio Number:	
			5.2.2.1	The folio number shall be assigned such as xx/zzz.
				For example:
				23/001, where 23 corresponds to last digits of the
				year 2023 followed by a slash followed by, 001
				(three-digit consecutive number) indicates the first
				folio number of the year.

	5.3	Return of unused Production Order, Batch Record, Solution Preparation	
		Record, Cleaning Record and Forms:	
		5.3.1	The Head of the production department or his/her designee
			shall be responsible for returning the unused Production
			Order, Batch Record, Solution Preparation Record and Cleaning
			Record, if any. Reason for return shall be mentioned in
			remarks column.

	5.4	Destruction of unused Production Order, Batch Record, Solution	
		Preparation Record Cleaning Record and Forms:	
		5.4.1	The documentation section in QA Department is responsible for
			destroying the Production Order, Batch Record, Solution
			Preparation Record, Cleaning Record and Forms which were not
			used.

||5.4.2 |The destruction will be carried out either by incineration or |
 ||||by shredding. |

||6.0 |Forms and Records |

||6.1 |Requisition for Production order |
 |||-Annexure 1 |

	6.2	Requisition for Batch Record/Solution Preparation Record/Cleaning
		Record
		Others
		-Annexure 2

6.3	Issue record of Production order		
		-Annexure 3	
6.4	Issue record of Batch record/ Solution Preparation Record/ Cleaning		
		Record	
		/Others	
		-Annexure 4	
6.5	Destruction Record of Production Order		
		-Annexure 5	
6.6	Destruction Record of Batch Record/Solution Preparation		
		Record/Cleaning Record/Others	
		-Annexure 6	
6.7	Subscript record Sheet		
		-Annexure 7	
7.0	References		
8.0	Distribution		
9.0	History		
	Date	Revision Number	Reason for Revision

Company name XYZ.

CONTROLLED COPY

Date XX YYY ZZZZ Sign.

Copy

No.

1.0 |Objective |
 |To lay down the procedure for handling of returned products. |
 2.0 |Scope |
 |This Standard Operating Procedure is applicable for handling of returned |
 |products manufactured at (Name of Organization). |
 3.0 |Responsibility |
3.1	The Quality Assurance department is responsible for proceeding
	according to the procedure laid down in this SOP.
3.2	Head, QA department is responsible for checking the compliance of
	this procedure and shall ensure that the SOP is reviewed every two
	years or if required earlier.
4.0	Abbreviations and Definitions
	QA : Quality Assurance
	QAR : Quality Assurance Record
	QC : Quality Control
	QE : Quality Engineering
	SOP : Standard Operating Procedure
5.0	Procedure
5.1	The QA department shall handle returned products.
5.2	The Supply and Distribution department (Warehouse) shall communicate
	in writing to the QA department about the arrival of returned
	products.
5.3	The following information shall be given in the communication:
	5.3.1
	5.3.2
	5.3.3
	5.3.4
5.4	QA Personnel shall assign a code and shall fill the form XXXX
	"Returned Products"
5.5	QA Inspector shall carry out a visual inspection of the returned
	product.
	5.5.1
	5.5.2
5.6	This report shall be sent to QA.
5.7	The inspector shall mark the returned product with the established
	(Returned) label.
5.8	Top management must be informed of any returned product and
	conclusions about its final destiny.
5.9	QA shall investigate the causes of the return, taking into
	consideration the original complaint or any other circumstance about
	the returned product.
	5.9.1
5.10	QA shall investigate, as well, the conditions in which the product
	was stored and handled, starting from its delivery to the client
	until the moment that the return was processed and shall also be
	included in the above report.
5.11	If it is necessary, sampling and testing shall be performed on the
	returned batch, as well as a checking of the batch production record.
5.12	The QA shall, also investigate the causes of deficiencies, if any, to
	plan corrective actions as per SOP XXXXXXXX.
5.13	Returned products shall be destroyed in case they do not meet the
	specified quality or if there are any doubts about the safety,
	identity, strength, quality or purity of the product due to storage
	conditions, shipment and handling.
5.14	They shall, also, be destroyed if the state of the product, its
	container and labeling indicates some doubt to this respect.
5.15	If the final decision is destruction, QA shall send a copy of Form
	XXXXXX Returned Products (Annexure 2) to the EOHS so that they
	proceed as per SOP XXXXXX.
5.16	In case the investigations show that the product complies with
	specifications for safety, identity, strength, quality or purity of
	the product, it could be sold or dedicated to other uses (reference

		material, analyst qualification etc.)	
	5.17	Documentation of Returned Products:	
		5.17.1	Returned products shall be documented in Form XXXXX
			Returned Products (Annexure 2).
		5.17.2	All inspections, records, etc. shall be annexed to this
			form.
6.0	Forms and Records		
	6.1	Return product intimation form	
			: Annexure 1
	6.2	Returned product	
			: Annexure 2
7.0	References		
8.0	Distribution		
	8.1		
9.0	History		
	Date	Revision Number	Reason for Revision

S. No. |
Incident
No. |
Initiated by /Date |Initiator Department |Product Name & Batch Number (If
Applicable) |Brief description of Incident |Graduated to Deviation
(Yes/No)
If Yes
(Deviation No.) |CAPA No.(if applicable) |Incident
closed on and by
(Sign./ Date) |
Remarks | |

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| | | | | Write 'N.A', if not applicable
To be filled by QA

Incident Report Form
Annexure - 2
|Section A: To be filled by QA |
|Incident Report No. |Issued by (Sign/Date) |
|Department: |
|Section B: To be filled by User Department |
|Reported by: |
|Date of Occurrence: |
|Product Name: (If Applicable) |Batch No. |
|Description of Incidence: (Attach extra sheet if required) |
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|
|Initiator |
|Sign/Date: |
|Immediate action/Correction (Attach extra sheet if required): |
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|
|Sign/Date |
|Sign/Date |
|Immediate Supervisor |
|Department Head |
|Section C: To be filled by QA |
|Any previous incident of the same type reported – Yes/No |
|(If Yes Give Details): |
|
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|
|Sign/Date |
|Section In Charge (QA) |
|Investigation Required/Not Required |
|
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|
|Sign/Date |
|Head -QA |
|Investigation findings: |
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|
|Sign/Date |
|
|Section In Charge Quality Assurance |
|Section D: |
|Reference CAPA number (if applicable): |
|Corrective and preventive actions: |
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|Sign/Date Sign/Date |
|Sign/Date |
|Concerned Department Other concerned dept. |
|Quality Assurance |
|Section E: |
|Conclusion by Head QA: |
|
|
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|

S No. |Source & reference |Category of Deviation (Planned/ Unplanned)
|Deviation
No. |Initiated by & Date |Initiator Department |Product Name & Batch Number
(If applicable) |Brief description of Deviation |Class of Deviation
(Critical/ Major/Minor) |CAPA No.(if applicable) |Deviation closed on and
by
(Sign./ Date) |Remarks | |

| | | | | | | | | | | Write 'N.A', if not applicable
To be filled by QA

Section A: To be filled by QA | Section A: To be filled by QA
Unplanned Deviation No: | Issued By :
Sign/date:
Deviation triggered from incident: (tick mark if any) Yes No

Reference Incident number: | Deviation triggered from incident: (tick mark if any) Yes No

Reference Incident number:
Section B: To be filled by Initiator Department | Section B: To be filled by Initiator Department
Name of Department: | Name of Department:
Reported By: | Sign & Date:
Deviation noticed for (mark tick")

Facility

Process

Product

Material

Document

Equipment/Instrument

Others

(specify):..... | Specify Area:

Master/Packaging Formula No.:

Doc. No./Title:
Product Details (If applicable)

Product Name:..... Product Code:.....B.No:..... Market:..... | Product Details (If applicable)

Product Name:..... Product Code:.....B.No:..... Market:.....
Description of Deviation (attach extra sheet if required)

Status of the Operation(s) (mark tick"): Stopped Continue | Description of Deviation (attach extra sheet if required)

Status of the Operation(s) (mark tick"): Stopped Continue | Description of Deviation (attach extra sheet if required)

Status of the Operation(s) (mark tick"): Stopped Continue
| Sign | Date
Initiator | |
User department Head | |
QA (In process) | |
Section C: | Section C: | Section C:
Investigation/Findings (attach extra sheet if required)

Concerned Deptt. Other Deptt. (If applicable) Q.A
(Sign/Date) (Sign/Date) (Sign/Date) | Investigation/Findings (attach extra sheet if required)

Concerned Deptt. Other Deptt. (If applicable) Q.A
(Sign/Date) (Sign/Date) (Sign/Date) | Investigation/Findings (attach extra sheet if required)

Concerned Deptt. Other Deptt. (If applicable) Q.A
(Sign/Date) (Sign/Date) (Sign/Date)
Impact of Deviation on Product Quality (Use impact assessment form as per annexure 6)

Concerned Deptt. Other Deptt. (If applicable) Q.A
(Sign/Date) (Sign/Date) (Sign/Date) | Impact of Deviation on Product Quality (Use impact assessment form as per annexure 6)

Concerned Deptt. Other Deptt. (If applicable) Q.A
(Sign/Date) (Sign/Date) (Sign/Date) | Impact of Deviation on Product Quality (Use impact assessment form as per annexure 6)

Concerned Deptt. Other Deptt. (If applicable) Q.A
(Sign/Date) (Sign/Date) (Sign/Date)
Section D: | Section D: | Section D:
Class of Deviation (mark tick"): Critical Major Minor

Head QA
Sign/Date: | Class of Deviation (mark tick"): Critical Major Minor

Head QA
Sign/Date: | Class of Deviation (mark tick"): Critical Major Minor

Head QA
Sign/Date:
Section E: | Section E: | Section E:
Corrective and Preventive Action (attach extra sheet if required or attach CAPA form if applicable)

Concerned Deptt. Other Deptt. (If applicable) Q.A
(Sign/Date) (Sign/Date) (Sign/Date) | Corrective and Preventive Action (attach extra sheet if required or attach CAPA form if applicable)

Concerned Deptt. Other Deptt. (If applicable) Q.A
(Sign/Date) (Sign/Date) (Sign/Date) | Corrective and Preventive Action (attach extra sheet if required or
attach CAPA form if applicable)

Concerned Deptt. Other Deptt. (If applicable) Q.A
(Sign/Date) (Sign/Date) (Sign/Date)
Section F: | Section F: | Section F:
Conclusion and approval :

Head – QA
(Sign/Date) | Conclusion and approval :

Head – QA
(Sign/Date) | Conclusion and approval :

Head – QA
(Sign/Date)
Section G: | Section G: | Section G:
Deviation closure:

List of Attachments (If Any):

Sign/Date
Closed by QA | Deviation closure:

List of Attachments (If Any):

Sign/Date
Closed by QA | Deviation closure:

List of Attachments (If Any):

Sign/Date
Closed by QA

Section A: To be filled by QA | Section A: To be filled by QA
Planned Deviation No: | Issued By :
Sign/date:
Section B: To be filled by Initiator Department | Section B: To be filled by Initiator Department
Name of Department: | Name of Department:
Reported By: | Sign & Date:
Deviation noticed for (mark tick")

Facility

Process

Product

Material

Document

Equipment/Instrument

Others

(specify):..... | Specify Area:

Master/Packaging Formula No.:

Doc. No./Title:
Product Details (If applicable)

Product Name:.....Product Code:.....B.No.:.....Market..... | Product Details (If applicable)

Product Name:.....Product Code:.....B.No.:.....Market..... | Product Details (If applicable)

Product Name:.....Product Code:.....B.No.:.....Market.....
Description of Planned Deviation (attach extra sheet if required)

Reason/justification for Planned Deviation

Proposed By Reviewed By:
(Initiator) (Head – Initiator deptt.)
Sign/Date Sign/Date | Description of Planned Deviation (attach extra sheet if required)

Reason/justification for Planned Deviation

Proposed By Reviewed By:
(Initiator) (Head – Initiator deptt.)
Sign/Date Sign/Date | Description of Planned Deviation (attach extra sheet if required)

Reason/justification for Planned Deviation

Proposed By Reviewed By:
(Initiator) (Head – Initiator deptt.)
Sign/Date Sign/Date
Impact of Deviation on Product Quality (Use impact assessment form as per annexure 6)

Concerned Deptt. Other Related Deptt.(if any) QA
(Sign./Date) (Sign/Date) (Sign/Date) | Impact of Deviation on Product Quality (Use impact assessment form as per annexure 6)

Concerned Deptt. Other Related Deptt.(if any) QA
(Sign./Date) (Sign/Date) (Sign/Date) | Impact of Deviation on Product Quality (Use impact assessment form as per annexure 6)

Concerned Deptt. Other Related Deptt.(if any) QA
(Sign./Date) (Sign/Date) (Sign/Date)
Proposed Action plan for Planned Deviation (Attach extra sheet if required) | Proposed Action plan for
Planned Deviation (Attach extra sheet if required) | Proposed Action plan for Planned Deviation (Attach
extra sheet if required)
Action Plans | Responsibility Center | Timeline

||
||
||
||
||
||
||
||
Preapred By
(Initiator)
Sign/Date: | Reveiewed By:
(Initiator department-HOD)
Sign/Date: | Reveiewed By:
(Initiator department-HOD)
Sign/Date:
Section C: To be filled by QA | Section C: To be filled by QA | Section C: To be filled by QA
Planned Deviation: Approved Rejected

Class of Deviation (mark tick"): Critical Major Minor

Head QA Sign/Date: | Planned Deviation: Approved Rejected

Class of Deviation (mark tick"): Critical Major Minor

Head QA Sign/Date: | Planned Deviation: Approved Rejected

Class of Deviation (mark tick"): Critical Major Minor

Head QA Sign/Date:
Comments:

Head QA
Sign/Date: | Comments:

Head QA
Sign/Date: | Comments:

Head QA
Sign/Date:
Section D: To be filled by QA | Section D: To be filled by QA | Section D: To be filled by QA
Planned Deviation Review and Closure: | Planned Deviation Review and Closure: | Planned Deviation Review
and Closure:
Reverted to original procedure on (date): | Reverted to original procedure on (date): | Reverted to
original procedure on (date):
Permanent change request submitted on (date):

Change control number (if any) | Permanent change request submitted on (date):

Change control number (if any) | Permanent change request submitted on (date):

Change control number (if any)
Remarks: | Remarks: | Remarks:
Reviewed on: Reviewed by: Sign/Date | Reviewed on: Reviewed by: Sign/Date | Reviewed on: Reviewed by:
Sign/Date
List of Attachments (If Any):

Closed on and by:
(Sign/Date) | List of Attachments (If Any):

Closed on and by:
(Sign/Date) | List of Attachments (If Any):

Closed on and by:
(Sign/Date)

Impact of Deviation on Product Quality (Attach extra sheet if required): | Impact of Deviation on Product Quality (Attach extra sheet if required): | Impact of Deviation on Product Quality (Attach extra sheet if required): | Impact of Deviation on Product Quality (Attach extra sheet if required):
Impact on | Assessment | Assessment | Remark
RM Specifications | Yes ☐ | No ☐ |
PM Specifications | Yes ☐ | No ☐ |
FP Specifications | Yes ☐ | No ☐ |
Manufacturing Documents | Yes ☐ | No ☐ |
New product introduction | Yes ☐ | No ☐ |
Packing Process | Yes ☐ | No ☐ |
Equipment | Yes ☐ | No ☐ |
Instrument | Yes ☐ | No ☐ |
Systems | Yes ☐ | No ☐ |
Documentation | Yes ☐ | No ☐ |
Qualification/ Validation | Yes ☐ | No ☐ |
Stability Study | Yes ☐ | No ☐ |
Statutory | Yes ☐ | No ☐ |
Regulatory | Yes ☐ | No ☐ |
Facility | Yes ☐ | No ☐ |
Product Quality | Yes ☐ | No ☐ |
Safety | Yes ☐ | No ☐ |
Any other | Yes ☐ | No ☐ |
Brief Description of Assessment:

(Sign/Date) (Sign/Date)
Department Head QA- Head | Brief Description of Assessment:

(Sign/Date) (Sign/Date)
Department Head QA- Head | Brief Description of Assessment:

(Sign/Date) (Sign/Date)
Department Head QA- Head | Brief Description of Assessment:

(Sign/Date) (Sign/Date)
Department Head QA- Head

[illegible]

3.0 | Responsibility | Responsibility | Responsibility | Responsibility | Responsibility | Responsibility
| Responsibility | Responsibility | Responsibility | Responsibility | Responsibility

[illegible][illegible][illegible][illegible]

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report after the receiving the final conclusion from QA head. | Designated QA personnel shall close the incident report after the receiving the final conclusion from QA head. | Designated QA personnel shall close the incident report after the receiving the final conclusion from QA head. | Designated QA personnel shall close the incident report after the receiving the final conclusion from QA head.

| 5.2 | 5.2 | Handling of Deviation:

The Deviations are categorized as planned and unplanned. | Handling of Deviation:

The Deviations are categorized as planned and unplanned. | Handling of Deviation:

The Deviations are categorized as planned and unplanned. | Handling of Deviation:

The Deviations are categorized as planned and unplanned. | Handling of Deviation:

The Deviations are categorized as planned and unplanned. | Handling of Deviation:

The Deviations are categorized as planned and unplanned. | Handling of Deviation:

The Deviations are categorized as planned and unplanned. | Handling of Deviation:

The Deviations are categorized as planned and unplanned. | Handling of Deviation:

The Deviations are categorized as planned and unplanned.

|| | 5.2.1 | 5.2.1 | Unplanned deviations: | Unplanned deviations: | Unplanned deviations: | Unplanned

deviations: | Unplanned deviations: | Unplanned deviations: | Unplanned deviations:

|| || | 5.2.1.1 | 5.2.1.1 | Unplanned deviation forms shall QA on request of user department as per SOP

XXXXX | Unplanned deviation forms shall QA on request of user department as per SOP XXXXX | Unplanned

deviation forms shall QA on request of user department as per SOP XXXXX | Unplanned deviation forms shall

QA on request of user department as per SOP XXXXX | Unplanned deviation forms shall QA on request of user

department as per SOP XXXXX

|| || | 5.2.1.2 | 5.2.1.2 | When any deviation happens in facility, manufacturing operation /process or

written procedures occurs or is noticed, the employee shall immediately inform, verbally or

telephonically to the concerned Supervisor /Shift-in-charge, who shall decide whether to continue the

process or stop the process. | When any deviation happens in facility, manufacturing operation /process

or written procedures occurs or is noticed, the employee shall immediately inform, verbally or

telephonically to the concerned Supervisor /Shift-in-charge, who shall decide whether to continue the

process or stop the process. | When any deviation happens in facility, manufacturing operation /process

or written procedures occurs or is noticed, the employee shall immediately inform, verbally or

telephonically to the concerned Supervisor /Shift-in-charge, who shall decide whether to continue the

process or stop the process. | When any deviation happens in facility, manufacturing operation /process

or written procedures occurs or is noticed, the employee shall immediately inform, verbally or

telephonically to the concerned Supervisor /Shift-in-charge, who shall decide whether to continue the

process or stop the process. | When any deviation happens in facility, manufacturing operation /process

or written procedures occurs or is noticed, the employee shall immediately inform, verbally or

telephonically to the concerned Supervisor /Shift-in-charge, who shall decide whether to continue the

process or stop the process.

|| || | 5.2.1.3 | 5.2.1.3 | The initiator shall fill up section B of Unplanned deviation report from

(annexure 5) and submit to QA within 24 hours from the time of occurrence. | The initiator shall fill up

section B of Unplanned deviation report from (annexure 5) and submit to QA within 24 hours from the time

of occurrence. | The initiator shall fill up section B of Unplanned deviation report from (annexure 5)

and submit to QA within 24 hours from the time of occurrence. | The initiator shall fill up section B of

Unplanned deviation report from (annexure 5) and submit to QA within 24 hours from the time of

occurrence. | The initiator shall fill up section B of Unplanned deviation report from (annexure 5) and

submit to QA within 24 hours from the time of occurrence.

|| || | 5.2.1.4 | 5.2.1.4 | Designated QA personnel shall issue the unplanned deviation number and fill

the section A of the unplanned deviation form. Same shall be mentioned in deviation log as per annexure

4. If deviation is triggered from incidence then incidence number shall be recorded in section A of

annexure 5. | Designated QA personnel shall issue the unplanned deviation number and fill the section A

of the unplanned deviation form. Same shall be mentioned in deviation log as per annexure 4. If deviation

is triggered from incidence then incidence number shall be recorded in section A of annexure 5. |

Designated QA personnel shall issue the unplanned deviation number and fill the section A of the

unplanned deviation form. Same shall be mentioned in deviation log as per annexure 4. If deviation is

triggered from incidence then incidence number shall be recorded in section A of annexure 5. | Designated

QA personnel shall issue the unplanned deviation number and fill the section A of the unplanned deviation

form. Same shall be mentioned in deviation log as per annexure 4. If deviation is triggered from

incidence then incidence number shall be recorded in section A of annexure 5. | Designated QA personnel

shall issue the unplanned deviation number and fill the section A of the unplanned deviation form. Same

shall be mentioned in deviation log as per annexure 4. If deviation is triggered from incidence then

incidence number shall be recorded in section A of annexure 5.

|| || | 5.2.1.5 | 5.2.1.5 | All unplanned deviations shall be assigned a unique reference code for

tracking purposes.

The designated person in QA Department shall assign a code to the Deviation as

LUD/DC/YY/ZZZ where:

L stands for site

UD stands for Unplanned deviation

DC- stands for Department code-

ZZ - Last two digits of the year

YYY - Three digit consecutive numbers assigned to a particular type of unplanned deviation and planned

deviation. | All unplanned deviations shall be assigned a unique reference code for tracking purposes.

The designated person in QA Department shall assign a code to the Deviation as

LUD/DC/YY/ZZZ where:

L stands for site

UD stands for Unplanned deviation

DC- stands for Department code-

ZZ - Last two digits of the year

YYY - Three digit consecutive numbers assigned to a particular type of unplanned deviation and planned

deviation. | All unplanned deviations shall be assigned a unique reference code for tracking purposes.

The designated person in QA Department shall assign a code to the Deviation as

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deviation. | All unplanned deviations shall be assigned a unique reference code for tracking purposes.

The designated person in QA Department shall assign a code to the Deviation as

LUD/DC/YY/ZZZ where:

L stands for site

UD stands for Unplanned deviation

DC- stands for Department code-

ZZ - Last two digits of the year

YYY - Three digit consecutive numbers assigned to a particular type of unplanned deviation and planned deviation. | All unplanned deviations shall be assigned a unique reference code for tracking purposes.

The designated person in QA Department shall assign a code to the Deviation as

LUD/DC/YY/ZZZ where:

L stands for site

UD stands for Unplanned deviation

DC- stands for Department code–

ZZ - Last two digits of the year

YYY - Three digit consecutive numbers assigned to a particular type of unplanned deviation and planned deviation.

||||| 5.2.1.6 | 5.2.1.6 | Head of Department or his/her designee and Quality Assurance who will take a prompt review of the deviation, its impact on the product /process. | Head of Department or his/her designee and Quality Assurance who will take a prompt review of the deviation, its impact on the product /process. | Head of Department or his/her designee and Quality Assurance who will take a prompt review of the deviation, its impact on the product /process. | Head of Department or his/her designee and Quality Assurance who will take a prompt review of the deviation, its impact on the product /process. | Head of Department or his/her designee and Quality Assurance who will take a prompt review of the deviation, its impact on the product /process.

||||| 5.2.1.7 | 5.2.1.7 | The concerned department supervisor/department Head along with QA and other relevant departments shall investigate as per XXXX. The deviation to record the investigation findings and perform the Impact Assessment on product quality as per Annexure 7. | The concerned department supervisor/department Head along with QA and other relevant departments shall investigate as per XXXX. The deviation to record the investigation findings and perform the Impact Assessment on product quality as per Annexure 7. | The concerned department supervisor/department Head along with QA and other relevant departments shall investigate as per XXXX. The deviation to record the investigation findings and perform the Impact Assessment on product quality as per Annexure 7. | The concerned department supervisor/department Head along with QA and other relevant departments shall investigate as per XXXX. The deviation to record the investigation findings and perform the Impact Assessment on product quality as per Annexure 7. | The concerned department supervisor/department Head along with QA and other relevant departments shall investigate as per XXXX. The deviation to record the investigation findings and perform the Impact Assessment on product quality as per Annexure 7.

||||| 5.2.1.8 | 5.2.1.8 | Head QA or his/her designee shall classify the deviation as Critical, major or Minor and mention in section D of annexure 5. | Head QA or his/her designee shall classify the deviation as Critical, major or Minor and mention in section D of annexure 5. | Head QA or his/her designee shall classify the deviation as Critical, major or Minor and mention in section D of annexure 5. | Head QA or his/her designee shall classify the deviation as Critical, major or Minor and mention in section D of annexure 5. | Head QA or his/her designee shall classify the deviation as Critical, major or Minor and mention in section D of annexure 5.

||||| 5.2.1.9 | 5.2.1.9 | Based on investigation and findings, the concerned department head shall propose the corrective and preventive action (CAPA), if applicable, as per SOP of "Corrective and Preventive Action" XXXX. The proposed corrective and preventive actions shall be counter signed by concerned departments. | Based on investigation and findings, the concerned department head shall propose the corrective and preventive action (CAPA), if applicable, as per SOP of "Corrective and Preventive Action" XXXX. The proposed corrective and preventive actions shall be counter signed by concerned departments. | Based on investigation and findings, the concerned department head shall propose the corrective and preventive action (CAPA), if applicable, as per SOP of "Corrective and Preventive Action" XXXX. The proposed corrective and preventive actions shall be counter signed by concerned departments. | Based on investigation and findings, the concerned department head shall propose the corrective and preventive action (CAPA), if applicable, as per SOP of "Corrective and Preventive Action" XXXX. The proposed corrective and preventive actions shall be counter signed by concerned departments. | Based on investigation and findings, the concerned department head shall propose the corrective and preventive action (CAPA), if applicable, as per SOP of "Corrective and Preventive Action" XXXX. The proposed corrective and preventive actions shall be counter signed by concerned departments.

||||| 5.2.1.10 | 5.2.1.10 | During investigation if it is found that product quality / efficacy / safety / purity is affected due to the deviation then material / product shall be rejected as per the respective procedures of rejections. | During investigation if it is found that product quality / efficacy / safety / purity is affected due to the deviation then material / product shall be rejected as per the respective procedures of rejections. | During investigation if it is found that product quality / efficacy / safety / purity is affected due to the deviation then material / product shall be rejected as per the respective procedures of rejections. | During investigation if it is found that product quality / efficacy / safety / purity is affected due to the deviation then material / product shall be rejected as per the respective procedures of rejections. | During investigation if it is found that product quality / efficacy / safety / purity is affected due to the deviation then material / product shall be rejected as per the respective procedures of rejections.

||||| 5.2.1.11 | 5.2.1.11 | Final disposition of the affected part/whole batch shall be decided based on the outcome of the investigation. | Final disposition of the affected part/whole batch shall be decided based on the outcome of the investigation. | Final disposition of the affected part/whole batch shall be decided based on the outcome of the investigation. | Final disposition of the affected part/whole batch shall be decided based on the outcome of the investigation. | Final disposition of the affected part/whole batch shall be decided based on the outcome of the investigation.

||||| 5.2.1.12 | 5.2.1.12 | Head QA shall conclude and approve the deviation based on investigation and impact analysis performed. Comments of Head QA shall be recorded in section F of annexure 5. | Head QA shall conclude and approve the deviation based on investigation and impact analysis performed. Comments of Head QA shall be recorded in section F of annexure 5. | Head QA shall conclude and approve the deviation based on investigation and impact analysis performed. Comments of Head QA shall be recorded in section F of annexure 5. | Head QA shall conclude and approve the deviation based on investigation and impact analysis performed. Comments of Head QA shall be recorded in section F of annexure 5. | Head QA shall conclude and approve the deviation based on investigation and impact analysis performed. Comments of Head QA shall be recorded in section F of annexure 5.

||||| 5.2.1.13 | 5.2.1.13 | Concerned personnel in QA shall close the deviation form after thorough review and compliance of Head QA comments and conclusions. All the documents generated during the process shall be listed out and attached with the report. | Concerned personnel in QA shall close the deviation form after thorough review and compliance of Head QA comments and conclusions. All the documents generated during the process shall be listed out and attached with the report. | Concerned personnel in QA shall close the deviation form after thorough review and compliance of Head QA comments and conclusions. All the documents generated during the process shall be listed out and attached with the report. | Concerned personnel in QA shall close the deviation form after thorough review and compliance of Head QA comments and conclusions. All the documents generated during the process shall be listed out and attached with the report. | Concerned personnel in QA shall close the deviation form after thorough review and compliance of Head QA comments and conclusions. All the documents generated during the process shall be listed out and attached with the report.

||| 5.2.2 | 5.2.2 | Planned Deviation: | Planned Deviation: | Planned Deviation: | Planned Deviation: |

Planned Deviation: | Planned Deviation: | Planned Deviation:

||| | 5.2.2.1 | 5.2.2.1 | 5.2.2.1 | The concerned supervisor /shift-in-charge shall propose the deviation through the form "request for planned deviation" well in advance and before the operation is started.(Annexure-6)

Note: Planned deviations shall not be used to circumvent existing quality management system. | The concerned supervisor /shift-in-charge shall propose the deviation through the form "request for planned deviation" well in advance and before the operation is started.(Annexure-6)

Note: Planned deviations shall not be used to circumvent existing quality management system. | The concerned supervisor /shift-in-charge shall propose the deviation through the form "request for planned deviation" well in advance and before the operation is started.(Annexure-6)

Note: Planned deviations shall not be used to circumvent existing quality management system. | The concerned supervisor /shift-in-charge shall propose the deviation through the form "request for planned deviation" well in advance and before the operation is started.(Annexure-6)

Note: Planned deviations shall not be used to circumvent existing quality management system.

||| | 5.2.2.2 | 5.2.2.2 | 5.2.2.2 | QA shall issue the planned deviation from with deviation number and log the same in annexure 4. | QA shall issue the planned deviation from with deviation number and log the same in annexure 4. | QA shall issue the planned deviation from with deviation number and log the same in annexure 4. | QA shall issue the planned deviation from with deviation number and log the same in annexure 4.

||| | 5.2.2.3 | 5.2.2.3 | 5.2.2.3 | All planned deviations shall be assigned a unique reference code for tracking purposes.

The designated person in QA Department shall assign a code to the Deviation as

L/PD/DC/YY/ZZZ where:

L stands for site

PD for Planned deviation

DC- stands for Department code--

ZZ - Last two digits of the year

YYY - Three digit consecutive numbers assigned to a particular type of unplanned deviation and planned deviation. | All planned deviations shall be assigned a unique reference code for tracking purposes.

The designated person in QA Department shall assign a code to the Deviation as

L/PD/DC/YY/ZZZ where:

L stands for site

PD for Planned deviation

DC- stands for Department code--

ZZ - Last two digits of the year

YYY - Three digit consecutive numbers assigned to a particular type of unplanned deviation and planned deviation. | All planned deviations shall be assigned a unique reference code for tracking purposes.

The designated person in QA Department shall assign a code to the Deviation as

L/PD/DC/YY/ZZZ where:

L stands for site

PD for Planned deviation

DC- stands for Department code--

ZZ - Last two digits of the year

YYY - Three digit consecutive numbers assigned to a particular type of unplanned deviation and planned deviation. | All planned deviations shall be assigned a unique reference code for tracking purposes.

The designated person in QA Department shall assign a code to the Deviation as

L/PD/DC/YY/ZZZ where:

L stands for site

PD for Planned deviation

DC- stands for Department code--

ZZ - Last two digits of the year

YYY - Three digit consecutive numbers assigned to a particular type of unplanned deviation and planned deviation.

||| | 5.2.2.4 | 5.2.2.4 | 5.2.2.4 | Supervisor of initiator department shall fill Section B of

Annexure 6 to provide the description of planned deviation, reason/justification, impact analysis and proposed action plan. | Supervisor of initiator department shall fill Section B of Annexure 6 to provide the description of planned deviation, reason/justification, impact analysis and proposed action plan. | Supervisor of initiator department shall fill Section B of Annexure 6 to provide the description of planned deviation, reason/justification, impact analysis and proposed action plan. | Supervisor of initiator department shall fill Section B of Annexure 6 to provide the description of planned deviation, reason/justification, impact analysis and proposed action plan.

||| | 5.2.2.5 | 5.2.2.5 | 5.2.2.5 | Supervisor /Head or In-charge of the concerned department along with other relevant department shall assess the impact of the proposed planned deviation on the product quality or the process (as per annexure 7) | Supervisor /Head or In-charge of the concerned department along with other relevant department shall assess the impact of the proposed planned deviation on the product quality or the process (as per annexure 7) | Supervisor /Head or In-charge of the concerned department along with other relevant department shall assess the impact of the proposed planned deviation on the product quality or the process (as per annexure 7) | Supervisor /Head or In-charge of the concerned department along with other relevant department shall assess the impact of the proposed planned deviation on the product quality or the process (as per annexure 7)

||| | 5.2.2.6 | 5.2.2.6 | 5.2.2.6 | Head of user department shall verify section B of Annexure 6 and forward the form along with supporting documents to QA department for approval. | Head of user department shall verify section B of Annexure 6 and forward the form along with supporting documents to QA department for approval. | Head of user department shall verify section B of Annexure 6 and forward the form along with supporting documents to QA department for approval.

||| | 5.2.2.7 | 5.2.2.7 | 5.2.2.7 | Head QA or his designee shall approve or reject the planned deviation after thorough review of planned deviation, reason/justification, and impact risk assessment and proposed action plan. | Head QA or his designee shall approve or reject the planned deviation after thorough review of planned deviation, reason/justification, and impact risk assessment and proposed action plan. | Head QA or his designee shall approve or reject the planned deviation after thorough review of planned deviation, reason/justification, and impact risk assessment and proposed action plan.

||| | 5.2.2.8 | 5.2.2.8 | 5.2.2.8 | If the planned deviation is rejected, Head-QA shall provide his/her comment and the deviation shall be closed as per section E of the form. | If the planned deviation is rejected, Head-QA shall provide his/her comment and the deviation shall be closed as per section E of the form. | If the planned deviation is rejected, Head-QA shall provide his/her comment and the deviation shall be closed as per section E of the form. | If the planned deviation is rejected, Head-QA shall provide his/her comment and the deviation shall be closed as per section E of the form.

||| | 5.2.2.9 | 5.2.2.9 | 5.2.2.9 | In case of approved deviation Head-QA shall classify the

Two bar charts showing the distribution of 'Yes' and 'No' responses for two different questions. The top chart shows 10 'Yes' and 10 'No' responses. The bottom chart shows 10 'Yes' and 10 'No' responses.

|If No; Tick (✓) |
 |Incomplete CAPA/ action plan ||
 |■ ||
 |User Department shall fill the extension request on Annexure 3 ■ |
 ||
 ||
 |Sign & Date |
 ||
 |Note: Proceed with section E if this section is selected |

[illegible][illegible]

1.0 |Objective |
 |To lay down the procedure for implementation of effective CAPA system. |
 2.0 |Scope |
 |This SOP is applicable to all the departments of Name of organization where |
 |Corrective and Preventive actions are planned to address identified issues |
 |or to implement quality improvements. These requirements apply to all GMP |
 |operations and areas which have been recommended for quality improvements. |
 3.0 |Responsibility |
 |3.1 |Personnel of all concerned department shall be responsible to follow |
 |the procedure mentioned in this SOP. |
 |3.2 |Head-QA shall be responsible for approval of proposed CAPA, extension |
 |of existing CAPA. |
 |3.3 |Heads of the concerned departments or their designee, shall be |
 |responsible for implementation of this SOP. |
 |3.4 |Head-Quality Assurance or his/her designee shall be responsible for |
 |overall compliance of this SOP. |
 4.0 |Abbreviations and Definitions |
CAPA	: Corrective and Preventive Action
QA	: Quality Assurance
CQA	: Corporate Quality Assurance
MQR	: Monthly Quality Report
SOP	: Standard Operating Procedure
Operating	: A set of instructions which give the information to do the
guideline A	particular activity.
set of	
instructions	
which give	
the	
information	
to do the	
particular	
activity.	
Correction	: An action taken to eliminate a detected non-conformity. A
correction can be made in conjunction with a corrective	
action.	
Corrective	: The action taken to eliminate or minimize the causes of a
action	detected non- conformity, defects or other undesirable
situation in order to prevent recurrence of the	
non-conformity.	
Preventive	: An action taken to eliminate or minimize the cause(s) of a
action	potential non-conformity or other undesirable situation. A
preventive action is taken to prevent the initial occurrence	
of a non-conformity.	
Non-Conformit	: The non-fulfillment of a specified requirement
y	
5.0	Procedure
5.1	Personnel from initiator department shall request QA for the issuance
of CAPA form which can be triggered from various sources.	
CAPA sources can be as follows (but not limited to):	
5.1.1	Complaints
5.1.2	Deviations
5.1.3	Product rejections
5.1.4	Recalls
5.1.5	Self Inspection
5.1.6	Regulatory inspections & findings
5.1.7	External audits (Consultant or customer)
5.1.8	Out of specifications and Out of limits
5.1.9	Trends of process performance
5.1.10	Outcome of Risk assessment
5.1.11	Investigation of non-conformance
5.1.12	Product quality review recommendations
5.1.13	Management Review Actions(e.g. MQR)
5.1.14	Ineffective CAPA
5.1.15	Incidents
5.2	QA personnel shall allocate twelve digit number to CAPA. The number
shall be in LCAPA/Department/ YY/ XXX format, where :	
L-stands for the site	
CAPA - Denotes Corrective action and Preventive action	
followed by a slash	
Department represents the departmental code	
	YY - Indicates last two digits of the year e.g. 23 for 2023, A
	new series of CAPA shall start at the beginning of New Year,
	followed by slash (/)
	XXX - Three digit consecutive numeral (001, 002, 003, etc)
	assigned to particular CAPA
	Each deficiency/discrepancy shall be assigned a separate CAPA
	number e.g. In case of Self Inspection/Audits, each deficiency
	shall be assigned a separate CAPA number.
5.3	Designated QA personnel shall log CAPA details in annexure 2 as per
the CAPA process (Sr. No, Source of CAPA, CAPA number etc.)	
5.4	Initiator shall fill the section B of CAPA Form (Annexure 1) which
includes source name and document no., brief details of	
discrepancies/deficiencies, Details of root cause, description of	
CAPA.	
5.5	Planning of corrective and preventive actions shall be based on the
root cause identified through investigations (carried out as per	
respective SOP No: XXXXX) for systems like deviation, incident, OOS,	
product recall etc. or based on audit reports, risk assessment	
report, etc.	

	5.5.1	The CAPA shall be planned by concerned personnel from user		
		department. Wherever required the CAPA shall be planned by a		
		cross-functional team consisting of designated personnel from		
		Production, Engineering, QC, QA, RA etc. coordinated by the		
		concerned personnel from user department.		
	5.5.2	During planning of CAPA actions, team shall assess the		
		objective of proposed CAPA and decide on best possible CAPA		
		actions.		
	5.5.3	The concerned personnel from user department shall list all		
		activities, tasks, documents and training requirements that		
		must be accomplished to prevent recurrence of an existing		
		non-conformity or occurrence of potential non-conformance.		
	5.5.4	Team shall fix CAPA target completion date based on action		
		points and the same shall be approved by QA personnel. However		
		if timeline is not proposed by the user then same shall be		
		considered as 30 days from the date of submission of CAPA.		
	5.5.5	Team shall identify the need to raise change control based on		
		proposed CAPA e.g. CAPAs that require changes to process,		
		procedures or equipment etc. as per Change control SOP		
		no.XXXXX.		
	5.5.6	The concerned department head shall forward the CAPA plan with		
		responsibilities and target completion date within 20 days		
		after the issuance of CAPA Form from QA.		
	5.5.7	A Unique Number shall be assigned to each CAPA action by the		
		initiator. The unique number shall be a combination of "CAPA		
		No. followed by dot followed by Serial Number" i.e.		
		LCAPA/Department/YY/XXX (.) Serial number (e.g. LCAPA/		
		QC/23/001.1,LCAPA/QC/23/001...2 so on.).		
	5.5.8	QA shall review the CAPA plan on the basis of root cause		
		analysis and send to Head-QA for approval. If CAPA actions are		
		justified then Head QA approve the plan otherwise reject the		
		plan and suggest initiating department to re-define the CAPA		
		plans.		
	5.6	Implementation of CAPA action plan:		
	5.6.1	Initiator shall ensure the implementation of corrective and		
		preventive action plans within defined timelines.		
	5.6.2	This section should contain a complete record of the actions		
		that were taken to correct the problem. This includes changes,		
		preventive measures, process controls, training, etc.		
	5.6.3	If initiator identifies that the completion of CAPA plan is		
		not achievable during the defined time period, then he/she		
		shall request for extension of CAPA target timelines with		
		justification (as per annexure 3).		
	5.6.4	Designated personnel in QA department shall monitor the agreed		
		timelines as per the CAPA action plan. In case agreed		
		timelines are not met, QA shall escalate the details to Head –		
		QA.		
	5.7	Verification of CAPA plan(s):		
	5.7.1	Designated QA personnel shall verify the CAPA completion		
		status as per respective target dates of CAPA completion and		
		record the verification activity as per section D of annexure		
		1.		
	5.7.2	While QA personnel reviewing the documentary evidences of CAPA		
		completion, if they find the CAPA is incomplete then they		
		shall recommend user department personnel to raise CAPA		
		extension form (annexure 3) in order to avail revised target		
		dates of CAPA completion.		
	5.7.3	In case of extended CAPA plan, designated QA personnel shall		
		verify the CAPA completion status as per revised target dates		
		of CAPA completion and record the verification activity as per		
		section E of annexure 1.		
	5.8	Incase, the CAPA is found to be ineffective, the same shall be		
		investigated for the cause, assessment of the impact shall be done		
		and the CAPA process shall be re-followed. When the Follow up is		
		complete, the document shall be made in place which shall be reviewed		
		by the concerned and approved by QA.		
	5.9	CAPA monitoring for effectiveness evaluation and CAPA closure		
	5.9.1	Designated QA personnel shall monitor the ongoing		
		effectiveness of the CAPA implemented through monthly quality		
		reviews, product quality reviews, monitoring of change		
		control, deviations of similar nature, anticipated potential		
		failures and other sources as described above.		
	5.9.2	Designated QA personnel to close the CAPA as per section F of		
		annexure 1.		
	6.0	Forms and Records		
	6.1	Annexure 1 – Corrective and preventive action form		
	6.2	Annexure 2 – Corrective and preventive action log		
	6.3	Annexure 3 – CAPA extension form		
	7.0	References:		
	7.1			
	8.0	Distribution		
	8.1			
	9.0	History		
		Date	Revision Number	Reason for Revision

Product Recall intimation to Top Management

1. Product Name :

2. Batch Number :

3. Batch Size :

4. Mfg. Date :

5. Retest/Expiry Date :

6. Shelf Life of product :

7. Packaging Details :

8. Antigen Content :

9. Client details :

:

10. Remarks, if any :

Name		
Designation		
Sign/date		
Department	Quality Assurance	Top Management

(Communication shall be sent to top management from the site QA Head once decision of recall has been taken and the strategy of recall has been decided by the recall committee)

Extra sheets shall be attached wherever required.

Specimen for recall letter

Product,

Batch no,

Expiry date:

Date:

To:

From:

Dear Sir/Madam,

We wish to advise you that product with detail mentioned above has been identified as requiring a product recall.

The reason for this recall is that Company has been notified that there has been the defect In the subject product.

(State actions that should be taken if a problem is discovered on checking the product, such who to inform, include name and address).

We apologies for this inconvenience this request may cause. Should you have any queries please contact (Insert contact details).

Yours sincerely,

Head Warehouse.

Name /Sign/Date

Product Information

Product Name	:		Batch No.	:	
Mfg. Date	:		Retest/Exp. Date	:	
Pack Details	:		Market (/s)	:	
Recall No. Assigned	:		Product Complaint No.,	:	
Date of release	:		if any	:	
			Total quantity Released		
Current Batch Quantity in stock	:		Product Recall started	:	
			on		

Reconciliation Record

A	B	C	D	E	F	G
S. No.	Name of	Address (With contact	Batch No.	Qty.	Qty. Recovered	Difference (E
	Customer (numbers)		dispatched		- F)	
Total Qty. Distributed		Total Qty.	Total			
ΣE =		Recovered	Difference			
		ΣF =	Σ G =			
Remarks, if any						
Product Recall completed on :						
	Prepared By	Checked By	Approved By			
Sign						
Name						
Designation			Head - Warehouse	Head - Quality Assurance		
Date						

Checklist of Product Recall Effectiveness

Information/ details:

1. Product Name :

2. Batch Number :

3. Batch Size :

4. Mfg. Date :

5. Retest/Expiry Date :

6. Shelf Life of product :

7. Type of Packaging :

8. Pack Size :

9. Antigen Content :

10. License Number/ registration No. :

11. Manufacturer's Name & Address :

12. Details and date how problem occurred :

13. Defect of product :

14. Does the defect affect the performance :
_____ and safety of the product

15. Do the product recall is due to presence of :
_____ contaminant i.e. cleaning fluid, machine oil
_____ etc. explain level of contaminant and attach
_____ MSDS for the contaminant

16. Do the product recall is due to failure of product :
_____ With reference to the specification. Attach COA
_____ and specifications.
17. Do the complaints received earlier also :
_____ for the same product
18. If yes, give details of complaint e.g. Date of :
_____ compliant, description of
complaint, lot no., _____
etc.

19. Details of recalled product:

- a. Total qty manufactured :

- b. Date of Manufacturing :

- c. Quantity dispatched :

- d. Date of dispatch :

- e. Locations where the product is dispatched :

20. Estimate amount remaining :

with customers

21. Details of distribution of product in market :

:

22. Copies/script of intimation given by through :

email, phone, newspaper, letters etc. are
available.

23. Effectiveness of product recall with comments :

Remarks if any:

Name		
Designation		
Sign/Date		
Department		

Extra sheets shall be attached wherever required.

| TITLE | PRODUCT RECALL RECORD |

S. No. |Product Recall No. |Recall initiated By |Name of the Product
Recalled |Reason for recall |Recall Completed On |Recall Closed By/Dare
Remarks				Name	Sign	Date				

Flow Chart: Product Recall and Handling of Recalled Products

Recall to be completed within 15 days

Voluntary Product recall / Recall on instruction of Regulatory authorities

In case of products sold/dispatched to any external customer, Drug
Authorities intimated by RA about recall

Head, QA and Committee members

Customers intimated

Review of distribution/dispatch records

Final authorization of recall by Head corporate Quality assurance

Recall strategy by committee

Initiation of Product Recall form

Investigation and decision

Investigation of health risk involved

Decision on product recall by head QA intimation sent to top management.

Further Distribution and dispatch of product to be stopped immediately

Customers Intimated

Receipt and verification of recalled products at the Warehouse

Storage of recalled products in the designated and secured area in
Warehouse

Reconciliation between delivered & recovered quantities

Evaluation of effectiveness of recall by means of internal audit

Warehouse to handle recovered recalled products as per decision on form
"Product recall - Strategy & Disposition"

1.0 |Objective |
 |To lay down the procedure for product recall and handling of recalled |
 |product. |
 2.0 |Scope |
 |This Standard Operating Procedure is applicable to handling of product |
 |recalls and recalled products at company name. |
 3.0 |Responsibility |
3.1	Head, Quality Assurance/ designee shall be responsible for the
	coordination of product recall, final disposition of recalled product
	and must keep top management informed of progress in product recall.
3.2	Head, Quality Assurance/ designee shall be responsible to decides if
	the batch shall be brought back to premises or destroyed in client's
	premises.
3.3	Head, Warehouse/ designee shall be responsible for providing the
	relevant information to the Head, Quality Assurance and execution of
	the product recall activity. He shall also be responsible for handling
	of recalled products as per this SOP.
3.4	Head administration shall be responsible to provide the necessary
	support for handling of recalled product and other administrative
	support if required.
3.5	Head of corporate quality assurance shall be responsible to authorize
	the decision of the batch recall taken by the QA head.
4.0	Abbreviations and Definitions
HOD	:
QA	:
RA	:
R&D	:
SOP	:
Product Recall: A system designed to remove the product from market/clients	
known or suspected to be defective promptly and effectively. Recall may be	
voluntary or directed by the regulatory agencies.	
Product Recall and Review Team: A group comprising heads of Quality	
Assurance, Production, Quality Control, Warehouse, Regulatory Affairs and	
this committee shall decide the recall strategy.	
Recall Strategy: Planned course of action to be taken in conduction of a	
specific recall which addresses depth & level of recall and extent of	
effectiveness of checks for recall it shall be customized strategy w.r.t	
case to case.	
5.0	Procedure
5.1	Dispatch or sale of the batch shall be stopped immediately once recall
	has been decided. There shall be designated QA person handling the
	product recall.
5.2	On receipt of any complaint on the product safety, efficacy, purity,
	identity, etc. the same shall be handled as per SOP XXXXX "Handling of
	product complaint". During the course of investigation and assessment
	of degree of seriousness of health hazard, if it emerges that the
	product is defective (or) and if company undertakes a voluntary recall
	for recall directed by the regulatory body(s), then the procedure
	mention in the SOP shall be followed.
	Note: If a batch has been dispatched under "ON TEST" status, return of
	that product shall not be considered as recall.
5.3	Apart from the complaint the decision to recall the product may be
	taken due to other concern such as by the results from the ongoing
	stability testing report, annual product review report, and detection
	of GMP failure after release, directions from the national authorities
	/ regulatory authorities, result of an inspection.
5.4	Head, Quality Assurance/ designee shall initiate Annexure-1 'Product
	Recall-Strategy & Disposition' by filling product details, nature of
	the product defect/complaint and name of the committee members (HODs
	of concerned departments to be provided by Head-QA/ designee).
5.5	Each product recall shall be coded as defined below by means of an
	example: LREC/00/12001
	L : Corresponds to Site name
	REC : Corresponds to product recall
	00 : Corresponds to the product code
	12 : last two digits of the year (XXXX) in which the recall took
	place.
	001: a three digit consecutive number of the recall in that year.
	In this example, the code corresponds to the first recall of product
	XX in year 2012 at company site.
5.6	Any batch of a product for which partial distribution has already
	begun and there is a product defect, the remaining quantity, if any
	shall be put under HOLD by Quality Assurance till the committee takes
	a final decision.
5.8	Head, QA/designee shall coordinate with other members of the committee
	of product recall.
5.9	Depending on the investigation, the QA head decides that the product
	shall be recalled or not and its execution.
5.10	Final approval and authorization for a product recall shall be
	obtained from Head CQA along with Head QA/ designee.
5.11	Recall committee shall devise a recall strategy for conducting product
	recall
	5.11.1
	5.11.2

		destruction shall be inspected by the QA person deputed by the
		head QA.
	5.12	In case of products which shall be dispatched/sold to any external
	customer, Head, RA on behalf of company shall inform the drug	
	authorities regarding decided product recall with following details.	
	5.12.1	Product Name, Batch No., Batch Size, Manufacturing Date, Retest
		Expiry Date.
	5.12.2	Reason for recall.
	5.13	The distribution and dispatch records shall be checked to list down
	the name, addresses and contact numbers of customers with quantities	
	distributed to each customer.	
	5.14	Inform Head QA of the concerned clients for retention of the batch if
	it has not been used or initiation of recall of those final products	
	made from the affected batch that has already been distributed by the	
	client.	
	5.15	All customers to whom products have been distributed / dispatched
	shall be informed promptly by Head, Warehouse. The list of contacts of	
	all customers, and delivered batch numbers and quantities of product	
	delivered shall be maintained by Warehouse.	
	5.16	The head warehouse shall send the recall letter to customers as per
	Annexure -3.	
	5.17	Recalled products shall be received by Warehouse personnel. Warehouse
	personnel shall physically verify the details of the recalled products	
	like Product name, Batch No., Mfg. Date, Expiry Date and packing	
	details. The product shall be stored in a designated area under the	
	recommended storage conditions in the Warehouse and shall be marked	
	with established labels while awaiting a decision on further course of	
	action. Remaining stock, if any, of the suspected batch of the product	
	in the warehouse shall also be moved into this secure storage area and	
	entry to the area shall be restricted.	
	5.17.1	Reconciliation between the delivered and recovered quantities
	of the product shall be recorded on form 'Product Recall –	
	Reconciliation' (Annexure-4) by Warehouse personnel and	
	approved by Head, Warehouse and Head, Quality Assurance.	
	5.17.2	The concerned person from Warehouse shall handle the recalled
	products as per the decision taken by the committee, if the	
	committee decide that the product is to be destroyed it shall	
	be destroyed as per SOP No. XXXX "Waste Disposal Management".	
	5.18	In case of product recall, the batch preceding and succeeding the
	recalled batch shall be Investigated as suggested by the investigation	
	team e.g. if batch number XX of product is recalled, then the samples	
	and documents of the previous manufactured batch and post manufactured	
	batch shall be reviewed and proper investigation undertaken.	
	5.19	An internal audit shall be carried out after reconciliation of
	recalled product to check the effectiveness of the product recall.	
	5.20	Product recall effectiveness shall be recorded as per Annexure- 5.
	5.21	Recall committee shall finally review the complete recall operation
	including all documentation and QA head shall close the recall putting	
	a stamp RECALL COMPLETED on form Product Recall Reconciliation.	
	Product recall should be completed with in 15 days after product	
	recall has been decided.	
	5.22	Record shall be maintained for the recalled product as per Annexure-6.
	5.23	Recall communication should be brief and to the point. It should
	clearly identify the product, lot number(s), contain a concise	
	statement of the reason for the recall, the known or potential	
	hazard(s), the initial shipping date, and instructions for consignees	
	to follow in handling the recall.	
	Communication pertaining to the recall with the customers or drug	
	authority shall be having the complete information about the product	
	under the recall, such as	
	5.23.1	Product name
	5.23.2	Description of the product
	5.23.3	Type of packaging (i.e. glass and bottle capacity).
	5.23.4	Product content details
	5.23.5	Registration number
	5.23.6	Expiration date(s) or Retest date(s).
	5.23.7	Manufacturer details, address, city, state, zip code
	5.23.8	Type of defect
	5.23.9	Explain in detail how product is defective
	5.23.1	If the recall is due to the presence of a foreign object,
	0	description of the foreign objects' size, composition.
	5.23.1	If the recall is due to the presence of a contaminant
	1	(cleaning fluid, machine oil), explain level of contaminant in
	the product.	
	5.23.1	If the recall is due to failure of the product to meet product
	2	specifications, provide the specifications and report all test
	results If the recall is due to a label/ingredient issue,	
	provide and identify the correct and incorrect label(s),	
	description(s).	
	5.23.1	The date on which problem was identified.
	3	
	5.24	The intimation of the recall to the top management shall be given
	through Annexure-2 (Product recall intimation to top management)	
6.0	Forms and Records	
6.1	Product Recall strategy and Disposition form	- Annexure-1
6.2	Product Recall intimation to top management	- Annexure-2
6.3	Specimen for recall letter	- Annexure-3
6.4	Product Recall – Reconciliation	- Annexure-4
6.5	Checklist for product recall effectiveness	- Annexure-5
6.6	Product Recall Record	- Annexure-6
6.7	Flow Chart – Product Recall and Handling of	- Annexure-7
	Recalled Products	

7.0	References		
8.0	Distribution		
9.0	History		
	Date	Revision Number	Reason for Revision

Company |Name of Organization |SOP: XXXXXX |
 Logo | |
 |Quality Assurance Department |Form No: XXXXX |Page No: 1 of
 | |1 |
 |Authorized signature records of technical persons |

[illegible]

Departmental Approval |
 |Name |Designation |Signature |Date |
 |||||
 QA Authorization |
 |||||

[Company |Name of Organization |SOP: XXXXXX |
[Logo | |

Department			
Name	Designation	Long signature	Short
		signature	

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180	181	182	183	184	185	186	187	188	189	190	191	192	193	194	195	196	197	198	199	200	201	202	203	204	205	206	207	208	209	210	211	212	213	214	215	216	217	218	219	220	221	222	223	224	225	226	227	228	229	230	231	232	233	234	235	236	237	238	239	240	241	242	243	244	245	246	247	248	249	250	251	252	253	254	255	256	257	258	259	260	261	262	263	264	265	266	267	268	269	270	271	272	273	274	275	276	277	278	279	280	281	282	283	284	285	286	287	288	289	290	291	292	293	294	295	296	297	298	299	300	301	302	303	304	305	306	307	308	309	310	311	312	313	314	315	316	317	318	319	320	321	322	323	324	325	326	327	328	329	330	331	332	333	334	335	336	337	338	339	340	341	342	343	344	345	346	347	348	349	350	351	352	353	354	355	356	357	358	359	360	361	362	363	364	365	366	367	368	369	370	371	372	373	374	375	376	377	378	379	380	381	382	383	384	385	386	387	388	389	390	391	392	393	394	395	396	397	398	399	400	401	402	403	404	405	406	407	408	409	410	411	412	413	414	415	416	417	418	419	420	421	422	423	424	425	426	427	428	429	430	431	432	433	434	435	436	437	438	439	440	441	442	443	444	445	446	447	448	449	450	451	452	453	454	455	456	457	458	459	460	461	462	463	464	465	466	467	468	469	470	471	472	473	474	475	476	477	478	479	480	481	482	483	484	485	486	487	488	489	490	491	492	493	494	495	496	497	498	499	500	501	502	503	504	505	506	507	508	509	510	511	512	513	514	515	516	517	518	519	520	521	522	523	524	5
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Remarks	

Departmental Approval |
 Name | Designation | Signature | Date |
 | | | |
 QA Authorization |
 | | | |

[Company | Name of Organization | SOP: XXXXXX |
 | Logo | |]

Quality Assurance Department | Form No: XXXXX | Page No: 1 of 1
 1
 Authorized Signature Records of Technical Persons |

S.No	Name of the Person	Signature	Date	Training imparted by
	trained			
		Signature	Date	

[illegible]

|Manufacturer's address: |
| |
|Folio No. | |
|Name of Product : |
|Batch No: |Date of Manufacture: |Date of Retest/Expiry : |
|In house Specification No: |Analytical File No: |

CHARACTERISTICS	ACCEPTANCE CRITERIA	RESULTS	REMARKS
Conclusions: Released / rejected			
Remarks:			
STORAGE CONDITIONS :			
Released by : Signature :			
Designation : Date :			
Issued by : Signature :			
Designation : Date :			

Name of product :					
Batch No :		Total volume :		Total antigen :	
Container No. /		Container Weight		Gross weight	Net Volume
		Antigen Content		Total Antigen	
capacity (Empty + cap)					
Remarks, if any:					
Compiled By		Reviewed By			
Approved By					
Name					
Designation					
Signature					
Date					

[This addendum is applicable to Certificate of Analysis pertaining to : |
Product	:	
Batch No.	:	
Date of Manufacture	:	
Date of Retest / Expiry	:	
Released Quantity	:	
No. of container(s) /	:	
capacity		
Release date	:	
Reason for addendum:	:	
Description	:	
Remarks, if any	:	
Approved by:		Signatur
Designation	:	
Issued by	:	
Designation	:	

1.0	Objective		
	To lay down the procedure for the release of a batch(s) of a biological		
	product(s).		
2.0	Scope		
	This Standard Operating Procedure is applicable to the release of batches of		
	different biological products manufactured at name of the organization.		
3.0	Responsibility		
	3.1	Head, Production shall be responsible for forwarding the Batch	
		Production Record (BPR) to Quality Assurance.	
	3.2	Head, Quality Control shall be responsible for forwarding the	
		Analytical file to Quality Assurance.	
	3.3	Lot Release Section of QA Department shall be responsible for review	
		of lot dossier.	
	3.4	Head, Quality Assurance shall be responsible for taking final decision	
		regarding the release / rejection of the batch.	
4.0	Abbreviations and Definitions		
	CoA	:	Certificate of Analysis
	GMP	:	Good Manufacturing Practice
	OOT	:	Out of Trend
	QA	:	Quality Assurance
	QC	:	Quality Control
	SOP	:	Standard Operating Procedure
	WHO	:	World Health Organization
	BPR	:	Batch production record
	CAPA	:	Corrective and Preventive Actions
5.0	Procedure		
	General considerations:		
	The Lot Release section in QA department shall review the Lot Dossier i.e.		
	Batch Production Record (BPR) and Analytical file (Quality Control tests		
	records) and based on review, assessment shall made to recommend release /		
	rejection of the batch to the Head, Quality Assurance / designee and final		
	decision shall be taken within 21 days from date of receiving of Lot		
	Dossier. Decision for release / rejection of the batch can only be extended		
	if is required.		
	5.1	Once the production activity of the batch is over, head Production	
		issue a Certificate of Integral Evaluation and forward the BPR to QA	
		for review.	
	5.2	The BPR shall be submitted to QA completed in all respects, within	
		five working days of the last record attached in the BPR.	
	5.3	Head, Quality Control or his designee shall forward to QA, the	
		Analytical File containing the results with supportive relevant data	
		record sheets on the tests put up on the product. The QC- Summary of	
		analysis of results also carries his opinion regarding compliance to	
		the specifications of the product, as a whole entity including the	
		tests outsourced, if any. The Analytical file shall be submitted to QA	
		within three days from the last record attached.	
	5.4	During the checking of the Lot Dossier, if any excursion(s) come to	
		the knowledge of Head, Production / QC, it shall be ensured that they	
		are documented, investigated and deviation reports are filed as per	
		the established procedure (in SOP XXXX "Handling of Deviation" and	
		attached with BPR and Analytical file before the documents are	
		submitted to QA for review).	
	5.5	Lot Dossier once received in QA, is reviewed taking into consideration	
		the following	
		aspects :	
		5.5.1	Manufacturing and testing done as per approved procedures.
		5.5.2	Raw materials used have been released by Quality Control
		5.5.3	Solutions have been used before their expiry date.
		5.5.4	Part numbers and Batch numbers are mentioned.
		5.5.5	Performance of in process controls.
		5.5.6	Calculations are correct.
		5.5.7	Conformity of intermediate products with specifications.
		5.5.8	The print outs as per activity requirements are attached, if
		any.	
		5.5.9	Results of environmental and critical system monitoring are
		considered.	
		5.5.10	Records are complete and signed by authorized persons.
		5.5.11	Corrections, if any are adequately made and signed with date.
		5.5.12	Compliance of Critical Quality Attributes with specifications.
		5.5.13	Compliance the control limits derived from control charts,
		wherever applicable.	
		5.5.14	Compliance w.r.t established trend.
		5.5.15	Investigation of OOT, out of control limits, deviations and out
		of specification results, if any.	
		5.5.16	cGMP compliance.
		5.5.17	IPQA spot check.
	5.6	During the review of BPR and Analytical file of respective batch, the	
		checklist shall be filled taking into consideration the production	
		process, testing parameters, acceptance criteria, wherever applicable	
		and cGMP compliance as per product specific SOP for review of lot	
		dossier.	
	5.7	Conclusions of the review :	
		5.7.1	If the Lot Dossier review reveals compliance with respect to
			points mentioned under 5.6 and there is no excursion which may
			affect the quality of the product, the conclusion is that the
			batch can be released.
		5.7.2	If the Batch has not been manufactured according to the
			established parameters as per established procedures or does not
			comply with the established specifications and investigation
			shows that Batch is not of the intended quality, it shall be

		recommended to be rejected. Investigation shall be carried to
		trace the root cause of excursion in the batch which shall be
		extended to nearby batches and CAPA shall be taken as per SOP
		XXXXX accordingly.
	5.7.3	The conclusion of lot review shall be documented in "Report on
	Lot Dossier" (Annexure 1) which includes the following :	
	5.7.3.	Product accounting: includes the details of manufactured,
	1	sampled or rejected quantity.
	5.7.3.	Observations on: Raw Materials, GMP inspection,
	2	Environmental monitoring, Critical systems monitoring,
		Batch Production Records , Analytical file, critical
		quality attribute test results , investigation of
		deviations, out of control limit and Out of
		Specification (OOS) results, if any.
	5.7.3.	Conclusion: State if the batch is recommended for release
	3	rejection.
	5.7.4	If the batch is not manufactured up to the final stage i.e.
		abandoned at any stage, Report on Lot Dossier shall be filled in
		"Report on Lot Dossier".
	5.7.5	All this documentation concerning the Batch (Batch Production
		Record, Analytical
		file, checklist, Distribution detail, Record of review of Lot
		Dossier and Report on Lot Dossier) shall be placed for review
		before the Head, Quality Assurance Department / designee in
		absence who takes the decision to release or reject the
		Batch.
		After the approval of Report on lot dossier & checklist, CoA &
		Summary protocol shall be prepared. In case batch is rejected
		only CoA shall be prepared.
	5.7.6	'Certificate of Analysis' (CoA) includes the general information
		about the batch i.e. manufacturer's address, product name, Batch
		No., Date of manufacture, Date of Retest/Expiry, In-house
		specification no., Analytical file no. etc. The characteristics
		(tested parameters) and its acceptance criteria are considered
		as per the specification pertaining to the product and results
		are recorded from the analytical file. Conclusion of CoA shall
		include whether the batch is released / rejected. Remarks column
		should include quantity / volume, no. of containers / container
		capacity, in case batch release for examination, test & analysis
		and any other remark related to batch if required.
	5.7.7	In-house No. for Certificate of Analysis shall be assigned
	5.7.8	Addendum to CoA shall be issued as per template in Annexure 5 in
		case any additional information w.r.t. the contents of CoA have
		to be updated. Additional information may include the updation
		of retest date / assigning shelf life / change in released no.
		of containers due to distribution of material / updation of
		characteristic(s) etc.
	5.7.9	Inhouse Number. for 'Addendum to Certificate of Analysis' shall
		remain the same as for that of the original CoA followed by one
		numeral value (1, 2, 3) where numeral value denotes the no.
		of times the addendum is issued for particular batch and
		activity. Suffix (alphabet) shall be assigned as follows :
		Example: 12/00(XXX)/001/1 where 12/00(XXX)/001 is In-house no.
		of the original CoA and 1 denotes the first addendum to the CoA.
	5.7.10	Summary protocol of the batch shall be prepared as per SOP XXXX.
		Head, Production and Quality Control certifies the manufacturing
		and testing procedure of a particular batch respectively and
		issues compliance certificates accordingly. Head, QA issues
		certificate of internal release. The quantities / result values
		in Summary Protocol and CoA shall be quoted up to two decimal
		points. Round off policy shall be followed according to which if
		the third digit after decimal is 5 to 9, the preceding figure is
		increased by 1, if it is less than 4 then value shall be quoted
		as such up to two decimal points.
	5.7.11	In case the batch is conditionally release or release for
		examination, test & analysis it shall be mentioned in the
		remark.
	5.7.12	Lot Release section shall give a copy of the "Certificate of
		Analysis" to IPQA, Production and Warehouse in charge.
	5.7.13	Lot Release section shall give uncontrolled copy of CoA and
		Summary Protocol to Warehouse in charge for the batches to be
		dispatched. However for critical intermediate product like PRP
		which is under joint custody of Production and QA, copy of CoA
		shall be provided to IPQA & Production department only.
	5.7.14	The IPQA personnel shall label the product as 'Approved' /
		'Rejected'. Rejected batch shall be handled as per SOP XXXX, so
		as to proceed with the control of non-conforming products.
6.0	Forms and Records	
6.1	Report on Lot Dossier	
	: Annexure 1	
6.2	Certificate of Analysis	
	: Annexure 2	
6.3	Record of Review of Lot Dossier	
	: Annexure 3	
6.4	Distribution Detail (Batch Wise)	
	: Annexure 4	
6.5	Addendum to Certificate of Analysis	
	: Annexure 5	
7.0 | References |
7.1 | |
8.0 | Distribution |
8.1 | |

9.0	History	
Date	Revision Number	Reason for Revision

[SHEET: Sheet1]

RISK ASSESSMENT FORM (FAILURE MODE EFFECT ANALYSIS) |||||

[illegible]

Team Leader ||||| Document No: |||||

Team Leader | | | | | Document No. | | | | |
Workflow Team | | | | | FMEA Date (Org.) | | Rev. | | | |

Sr No.	Potential Failure Mode	Potential Effect(s) of Failure	Severity	Potential Cause(s) of
--------	------------------------	--------------------------------	----------	-----------------------

Sl. No.	Potential Failure Mode	Failure	Occurrence	Current
---------	------------------------	---------	------------	---------

	Failure Occurrence Current
	Control Measures Detectability RPN = SxOxD Recommended Action(s) Responsibility Center Revised

Control Measures	Detectability	R/N = SxSxD Score	Risk Accepted ? (Y/N)	Address Via
1. All personnel must wear appropriate PPE (hard hats, safety glasses, gloves, etc.) at all times.	High	10	N	Safety Training
2. Regular safety meetings and toolbox talks are conducted to reinforce safety protocols.	Medium	8	N	Safety Meetings
3. All equipment is inspected and maintained according to manufacturer's specifications.	Medium	8	N	Equipment Maintenance Log
4. Clear communication channels are established between all team members.	Low	6	N	Team Briefing
5. Emergency procedures are clearly defined and practiced regularly.	Low	6	N	Emergency Drills
6. Work areas are kept clean and free of clutter to prevent tripping hazards.	Low	6	N	Safety Inspections
7. All workers receive specific training for their assigned tasks.	Medium	8	N	Task-Specific Training
8. A designated safety officer is responsible for monitoring compliance.	Medium	8	N	Safety Officer Report
9. Immediate reporting of incidents or near-misses is required.	Low	6	N	Incident Reporting Form
10. Safety barriers and signage are used to restrict access to hazardous areas.	Medium	8	N	Safety Barriers & Signage

Score ||| Risk Acc
(Documents Name)

(Documents Name)	Severity	Occurrence	Detectability	RPN
------------------	----------	------------	---------------	-----

Seventy
000000

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						0					0	

						0				0	
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						0				0	
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						0				0	
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				0				0	

1	1	1	1	1	0	1	1	1	1	0	1
1	1	1	1	1	0	1	1	1	1	0	1

				0				0	
				0				0	

1	1	1	1	1	0	1	1	1	1	0	1
1	1	1	1	1	0	1	1	1	1	0	1

	0		0	
	0		0	

1	1	1	1	1	0	1	1	1	1	0	1
1	1	1	1	1	0	1	1	1	1	0	1

[Sheet: Sheet2]

[SHEET: Sheet2]
[SHEET: Sheet3]

Prepared By (Team Member) Checked By (HOD: User Dept.) Approved By (Head QA)
 Hazard Analysis and Critical Control Points (HACCP) | Hazard Analysis and Critical Control Points (HACCP)
 | Hazard Analysis and Critical Control Points (HACCP) | Hazard Analysis and Critical Control Points
 (HACCP) | Hazard Analysis and Critical Control Points (HACCP) | Hazard Analysis and Critical Control
 Points (HACCP) | Hazard Analysis and Critical Control Points (HACCP) | Hazard Analysis and Critical
 Control Points (HACCP) | Hazard Analysis and Critical Control Points (HACCP) | Hazard Analysis and
 Critical Control Points (HACCP) | Hazard Analysis and Critical Control Points (HACCP) | Hazard Analysis
 and Critical Control Points (HACCP) | Hazard Analysis and Critical Control Points (HACCP) | Hazard
 Analysis and Critical Control Points (HACCP) | Hazard Analysis and Critical Control Points (HACCP)

|||||

Process step	Hazard	P/B/C	Severity	Occurrence	Control measures	S	O	RPN	CCP	Critical limit	Monitoring procedure	Corrective actions	Verification procedure	Record keeping

|||||

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30

[illegible]

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100

[illegible]

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100

|||||

Risk Assessment (RA) of
Area: -----
Effective date:-----

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APPROVALs

The Risk Assessment Document For -----
-- has been initiated, Checked, Approved and Authorized by the
following functional heads.

INITIATED BY (USER)

DESIGNATION	NAME	SIGNATURE	DATE

CHECKED BY

DESIGNATION	NAME	SIGNATURE	DATE
Engineering Head			
Production Head			
QC Head			

APPROVED BY

DESIGNATION	NAME	SIGNATURE	DATE
Quality Assurance			
Head			

Introduction

According to the definition, given in Annex 15 to the EU-GMP-Guide, a risk assessment is a method to assess and characterize the critical parameters in the functionality of an equipment / system. Therefore, risk assessment is a key element in the qualification and validation approach.

In the project context, risk analyses are performed as basic GMP/EHS-Risk assessment, which shall help to identify important GMP/EHS-requirements.

Aim of the Risk ASSESSMENT

At the very basic stage of design the risk assessment is to verify that all features are taken into consideration to avoid the risk of failure of critical GMP and EHS parameter in the commissioned Facility or System.

During study all GMP and EHS parameters will be identified and assessed for the risk if not considered in the design or requirements.

The Risk assessment report is produced to provide the documented evidence that design concepts or requirement are complete in considering all GMP and EHS risks.

Reference Documents

|S. No |Document Title |Document Number |
| | | | |
Risk ASSESSMENT PERFORMA

Risk Assessment shall be carried out as per annexure-II (FMEA) & III

(HACCP)

Risk ANALYSIS RESULTS

(Risk profile: A graphical representation of risks with high RPN number.

Comparison of Minor, moderate, major and critical risks before and after implementation of corrective actions. Summary of risk acceptance cases and justification

.

MAJOR FINDINGS AND OBSERVATIONS

Any critical or major risk which may affect operations at other locations.

Any risk that has been accepted in spite of having high RPN and management has to take call on that

SUMMARY & CONCLUSION

[1.0]Objective |
	This standard operating procedure aims to provide systematic approach to
	achieve quality standards by assessing, controlling, communicating and
	reviewing risks related to process, product, equipment, utilities,
	facility and environment.
[2.0]Scope	
	This Standard Operating Procedure is applicable to identification,
	assessment, control, communication and review of system level impact and
	risk to the quality of vaccines throughout their lifecycle Company Name
	XYZ
[3.0]Responsibility	
	3.1
	to follow and perform impact and risk assessment.
	3.2
	for tracking of risk assessment activities.
	3.3
	for the implementation of this procedure.
	3.4
	of this procedure.
[4.0]Abbreviations and Definitions	
[4.1]Abbreviations:	
	4.1.1
	4.1.2
	4.1.3
	4.1.4
	4.1.5
	4.1.6
	4.1.7
[4.2]Definitions:	
	4.2.1
	4.2.2
	4.2.3
	4.2.4
	4.2.5
	4.2.6
[5.0]Procedure:	
[5.1]System Level Impact Assessment	
	System level impact assessment shall be performed only for system
	and equipment.
	5.1.1
	5.1.2
	5.1.3
	5.1.4
	5.1.5
	5.1.6
	5.1.7

		generator).
		purging is given to the product during filling to avoid
		the oxidation of product by oxygen)
		reject product (e.g. Electronic batch recording system, or
		critical process parameter chart recorder).
		DCS) that may affect product quality and for which there
		is no system for independent verification of control
		system performance in place.
	5.1.8	
		system/equipment than the same system/equipment comes
		under "Direct Impact".
	5.1.9	
		than it comes under "Indirect Impact system"
	5.1.10	
		system" then it comes under "No impact system" criteria.
	5.1.11	
	5.1.12	
		decided based on their impact assessment & criticality of
		system/equipment with respect to product/product nature/
		route of administration and justification for the defined
		frequency shall be provided in the respective column of
		annexure-I
	5.2	
	5.2.1	
		criticality of the system function to product quality,
		strength, purity, identity and efficacy. These
		requirements are determined in a system specific risk
		assessment.
	5.2.2	
		critical parameters or conditions that could affect the
		product quality.
	5.2.3	
		be involved with logical, systematic and science-based
		approach to improve the effectiveness and efficiency of
		decision making.
	5.2.4	
		equipment/system/process/products.
	5.2.5	
		Risk Initiation
		Team Selection
		Identification of Equipment /system/ process/ products
		analysis
		Risk Assessment
		Risk Identification
		Risk Analysis
		Risk Evaluation
		Risk Control
		Risk Reduction
		Risk Acceptance
		Risk Communication
		Risk Review
	5.2.6	
		Prior to commencement of risk assessment, it is necessary
		to ask why the study is being initiated.
	5.2.6.1	
		formed to carry out the risk assessment. It
		provides a level of assurance that the
		qualitative or quantitative evaluation of
		potential hazards will be as objective as
		possible.
		Team member to carry out the risk assessment
		shall be selected from user department,
		Production, Engineering, Quality Control and
		Quality Assurance. Team member should have
		appropriate skill, knowledge & expertise
		necessary to correctly assess and quantify risk
		in team environment.
	5.2.6.2	
		It is important that critical quality
		function/attribute of the equipment, system,
		process & products are identified so that risk
		can be managed or reduces.
	5.6	
	Following tools shall be used to carry out risk assessment.	
	5.6.1	
		used to assess, control and review product quality risks
		related to biological product, process, equipment,
		utilities, facility and environment
	5.6.2	
		exists in controls over the raw materials, environment,
		personnel, storage and distribution.
	5.7	

		carry out as per annexure-III
	5.8	Document number for the FMEA & HACCP shall be assign by QA as
		Company short name /LV/A/YYYY/NNN
		Where...
		A = F (code for FMEA) or H (code for HACCP)
		YYYY= Year
		NNN= Digits
	5.9	Performing FMEA (Failure Mode Effects Analysis):
	5.9.1	FMEA provides for an evaluation of potential failure modes
		for products, equipment, facilities, utilities,
		environment and their likely effect on outcomes product
		quality
	5.9.2	Describe the process/ product/ equipment/ utility/
		facility and its intended use.
	5.9.3	Articulate a clear and specific Title of the risk
		assessment activity undergoing FMEA.
	5.9.4	Breakdown complex processes into manageable steps
		methodically.
	5.9.5	Block diagram, Flow charts, process mapping, mind mapping
		etc tools can be used to derive logical relationships
		between the components of the product or the steps/stages
		of the process.
	5.9.6	Enumerate the items (components, functions, steps, etc.)
		that make up the product or process and then break down
		the equipment / process being subjected to FMEA into its
		major components or steps.
	5.9.7	Final process diagram or output of the activity shall be
		signed by SRM or his/her designee and approved by QA
		representative present in the meeting.
	5.10	List down each of the components or steps identified in step
		number 5.9.6 in the FMEA table in logical manner.
	5.11	Discuss and identify all potential failure modes to define as how
		a system or process is failing to the specified conditions.
	5.12	List down each potential Failure Mode using its technical term
		into the specified format to avoid confusion.
	5.13	Describe the effects of each of the failure modes listed and
		assess the severity of each of these effects of a scale of 1- 10
		as per the table mentioned below.
		Effect
		Scale
		Description of 'Potential effect of system failure'
		No
		1
		No effect on output
		Very Slight
		2
		Stakeholder /customer not annoyed
		Slight
		3
		Slight
		Minor
		4
		Minor effect on performance
		Moderate
		5
		Moderate Effect on performance
		Significant
		6
		Partial Failure but operable
		Major
		7
		Product performance severely affected, but some operability and
		safe
		Extreme
		8
		Very dissatisfied, product Inoperable but safe.
		Serious
		9
		Potentially hazardous effect, time - dependent failure
		Hazardous
		10
		Hazardous Effect, Safety related sudden failure
	5.14	Identify the possible cause(s) of each failure mode identified. A
		potential cause should be something that can actually trigger the
		failure to occur.
	5.15	Quantify the probability of occurrence of each of the failure
		mode causes as per the scale of 1-10 mentioned below:
		Occurrence
		Scale
		Description of 'occurrence'

		Remote
		9
		Detection tools most likely will not detect failure
		Impossible
		10
		Failure not detected
	5.18 Calculate the Risk Priority Numbers (RPN):	
	5.18.1 The Risk Priority Number (RPN) is simply the product the	
		Failure Mode Severity, Occurrence and Control Detection
		Effectiveness ratings.
		$RPN = \text{Severity} \times \text{Probability of Occurrence} \times \text{Level of}$
		detection
	5.18.2 Acceptance Criteria: In case the calculated RPN rating is	
		greater than 27 for a particular identified risk then
		failures are not acceptable.
		Sr. No.
		RPN Rating
		RPN Category
		Action Status
		01
		≥ 76
		Critical
		CAPA is required
		02
		51 to 75
		Major
		CAPA is required
		03
		28 to 50
		Moderate
		5.18.2.1 For RPN rating ≤ 27 , no action plan is required.
		5.18.2.2 Action plan is required if severity, occurrence
		and detection is ≥ 7 (even if RPN is within
		acceptance criteria) for a particular identified
		risk.
	5.18.3 After completion of RPN scoring activity, sort the	
		potential risks in descending order in terms of RPN and
		assign actions against each failure mode until the RPN
		score reaches 27.
	5.18.4 For each recommended action, assign responsibility center	
		based on the expertise and discussion.
	5.18.5 Revised scores shall be filled up by expert team after due	
		consideration of proposed actions.
	5.18.6 If the revised RPN score falls within the range of 27, the	
		risk shall be accepted by default.
	5.18.7 If RPN score doesn't fall within the range of 27 even	
		after risk mitigation actions, then team shall discuss and
		perform either of the following
		5.18.7.1 Team shall propose new recommended action and
		then re-perform all the steps starting from step
		number 5.18.3
		5.18.7.2 Team shall discuss and reaches to a conclusion to
		accept the risk.
	5.18.8 QA shall take custody of the following documents after	
		completion of FMEA activity and prepare a file for every
		risk assessment activity.
		5.18.5.1 FMEA Sheet
		5.18.5.2 Final process map or activity output (QA approved)
		5.18.5.3 Scoring rational
		5.18.5.4 Modified FMEA sheet along with Change controls.
	5.18.9 For each recommended action, CAPA shall be raised as per	
		SOP xxx
	5.18.10 QA personnel shall track the progress & closure of	
		corrective or preventive actions using the CAPA handling
		system.
	5.18.11 User department shall prepare the Risk Assessment Report	
		as per annexure No. IV based on the FMEA and submit to the
		Quality Assurance Department for their review and
		approval.
	5.18.12 Head: Site QA shall forward the risk assessment report to	
		Corporate QA for approval & organizational communication
		through Monthly Quality Review meetings.
	5.18.13 Change control system shall be followed in order to change	
		any information on existing FMEA sheet.
	5.19 Risk Review & monitoring:	
	5.19.1 All critical & major risks shall be reviewed once in 6	
		month & moderate & minor risks shall be reviewed once in
		12 months by review team.
	5.19.2 The expert team shall re-assess the risk based on the	
		present conditions and revise the scores. Scoring
		rationale shall be recorded as per annexure -V
	5.20 Performing Hazard Analysis and Critical Control Points:	
	5.20.1 HACCP is a systematic, proactive, and preventive tool for	
		assuring product quality, reliability, and safety. A
		structured approach applies technical and scientific
		principles to analyze, evaluate, prevent, and control the
		risk or adverse consequence of hazard due to the design,

			development, production and nature of products.	
		5.20.2	Principles of HACCP:	
		5.20.2.1	Conduct a hazard analysis.	
		5.20.2.2	Determine the Critical Control Points (CCPs).	
		5.20.2.3	Establish critical limit(s).	
		5.20.2.4	Establish a system to monitor control of the CCP.	
		5.20.2.5	Establish the corrective action to be taken when	
			monitoring indicates that a particular CCP is not	
			under control.	
		5.20.2.6	Establish procedures for verification to confirm	
			that the HACCP system is working effectively.	
		5.20.2.7	Establish documentation concerning all procedures	
			and records appropriate to these principles and	
			their application.	
		5.20.3	Describe the product and process:	
			The team shall describe the product and process with all	
			relevant quality information such as composition,	
			physical/chemical properties, pH, temperature, cleaning	
			methods, bactericidal/ bacteriostatic treatments,	
			inoculation, fermentation, acid precipitation,	
			purification, filtration etc.	
		5.20.4	Identify the intended use:	
			The team shall describe the intended use of the product	
			with consideration to vulnerable populations like infants.	
		5.20.5	Construct a flow diagram of the process including all	
			relevant steps by the team members using the Batch Record.	
			The flow diagram shall consider all processes and decision	
			points.	
		5.19.5.1	In case of any change in the process flow	
			diagram, shall be routed through formal change	
			control procedure	
		5.20.6	HACCP shall be conducted in two stages...	
		5.20.6.1	1st Stage: List out all potential quality hazards	
			associated with each step by reviewing the	
			materials, activities, equipment, storage,	
			distribution and intended use the process.	
		5.20.6.2	A list of the potential hazards (biological,	
			chemical and physical), which may be introduced,	
			increased or controlled in each step, should be	
			drawn up.	
		5.20.6.3	In the hazard analysis, the following should be	
			included wherever possible:	
			5.20.6.3	The probable occurrence of hazards and
			.1	the severity of their adverse health
			effects.	
			5.20.6.3	The qualitative and/or quantitative
			.2	evaluation of the presence of hazards
			5.20.6.3	The survival or multiplication of
			.3	microorganisms of concern.
			5.20.6.3	The production or persistence in drugs
			.4	of toxins, chemicals or physical
			agents.	
			5.20.6.3	The conditions leading to the above
			.5	
		5.20.6.4	Identify the control measures apply against each	
			hazard. If required more than one control may be	
			required.	
		5.20.7	2nd stage: Team shall assign score against each hazard in	
			terms of the severity of the potential hazards and the	
			probability of their occurrence.	
		5.20.7.1	Score rationale shall be recorded in annexure -V	
		5.20.7.2	Each factor is being measure by giving	
			5-different ratings as below	
			Measures	
			Severity Rating	
			Occurrences Rating	
			Very High	
			5	
			5	
			High	
			4	
			4	
			Moderate	
			3	
			3	
			Low	
			2	
			2	
			None	
			1	
			1	
		5.20.7.3	Calculate the Risk Priority Number (RPN) number	
			by multiplication of Severity and Occurrence.	
			RPN = S x O	
		5.20.7.4	The team shall decide which potential hazard to	
			address, based on RPN score and recommend control	

				measures against that.	
		5.20.8	Determine the critical control points using the		
			questioning technique below.		
		[pic]			
		5.20.8.1	If a hazard has been identified at a step where		
				control is necessary for safety, and no control	
				measure exists at that step, or any other, the	
				product or process should be modified at that	
				step, or at an earlier or later stage, to include	
				such a control measure	
		5.20.9	Establish critical limits for each CCP using the expertise		
			and knowledge base. Regulatory guidelines, historical data		
				shall be used to establish critical limits.	
		5.20.10	The team shall establish the monitoring system of critical		
				control points. If monitoring system exists, then system	
				shall be modified to match recommended criteria. If	
				monitoring system does not exists then corrective action	
				is required which will be managed through CAPA system.	
				Please refer to CAPA SOP for further details.	
		5.20.10.	Team members shall decide the frequency of CCP		
				1	monitoring based on the criticality of the
				control. Regulatory guideline and experience with	
				the process shall be used to reach to the	
				conclusion.	
		5.20.11	Identify and list down the corrective actions to be		
				followed if CCP parameters fail to remain within specified	
				range. If corrective actions are not established during	
				the meeting, in that case expert is requested to provide	
				the corrective actions. Follow the CAPA system to track	
				the activities.	
		5.20.11.	In case of deviation of control parameters, same		
				1	shall be handled through deviation handling
				system.	
		5.20.12	Head: Validation or his/her designee along with team		
				members shall decide the review parameters, methodology	
				and frequency to check adequacy of the improvement system.	
				The expert team shall check the following parameter during	
				system review:	
				5.20.12.	Review of scientific rationale on having the
				1	monitoring tool.
				5.20.12.	Review of the HACCP system and its records
				2	
				5.20.12.	Review of deviations and product dispositions
				3	
				5.20.12.	Confirmation that CCPs are kept under control
				4	
				5.20.12.	In plant observation, measurement and evaluation.
				5	
		5.20.13	Quality Assurance Department takes custody of the		
				following documents after completion of HACCP activity.	
				5.20.13.	Process flow diagram (signed and approved by QA)
				1	
				5.20.13.	Scoring rationale
				2	
				5.20.13.	Critical limit determination.
				3	
				5.20.13.	Verification procedures and schedule
				4	
				5.20.13.	Deviations
				5	
				5.20.13.	Associated corrective actions
				6	
				5.20.13.	Modifications to the HACCP system with change
				7	controls.
		5.20.14	User department shall prepare the risk assessment report		
				after completion of risk assessment activity and submit to	
				site QA head as per format annexure-IV	
		5.20.15	QA head shall review the report and forward to Corporate		
				QA for organizational communication	
6.0	Forms and Records				
	6.1	Annexure-I: System Level Impact Assessment			
	6.2	Annexure-II: Risk Assessment Form FMEA			
	6.3	Annexure-III: Risk Assessment Form HACCP			
	6.4	Annexure-IV: Risk Assessment Report			
	6.5	Annexure-V: Risk Score Rationale			
	6.6	Annexure-VI: Risk Management Process Flow			
7.0	References				
	7.1				
8.0	Distribution				
	8.1				
9.0	History				
	Date	Revision Number	Reason for Revision		

|Department: Quality Assurance |
|Job Summary: Activities pertaining to Quality Assurance of Vaccines following |
|cGMP/cGLP norms in respect of multiple products. |
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|Organizational Relationship: |
|Management Representative |
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|Head of Department -QA |
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|Section In –Charge |
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||
|Supervisor |
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|Doer |

Introduction:

The document is divided into three parts, Part 1- "Position wise Roles/Functions, Skills, Education Qualification and Experience Requirement" describes the expected roles from various positions in the area concerned, minimum competency level, qualification and experience required for the successful completion of the roles of the position, while part 2 and 3 of the document focuses on the general and lab specific responsibilities of the department.

The intent of this job description document is to provide a representative summary of roles/functions and responsibilities that will be required of positions and shall not be construed as a declaration of specific roles/functions and responsibilities of any particular position. However employees may be required to perform job –related tasks other than those specifically presented in this document.

As Quality Assurance personnel, one has to keep abreast with the latest developments of cGMPs, current industry practices and applicable quality legislations.

1. Position wise Roles/functions, Skills, Education Qualification and Experience Requirements:

Roles/Functions of Doer:

Position	Roles/Functions
Doer	To learn Principles of cGMP/cGLP as per training provided to them
	To record information in the record forms as established
	To document the concerned activity concurrently
	To communicate any deviation to the set procedures promptly to the
	supervisor/section in-charge
	To assist in maintenance of records

Position Specifications:

Skills, Qualification and Experience Requirements of Doer:

Position	Soft Skills	Level required		
	Foundation	Proficiency	Mastery	
Doer	Learning	-	-	Required
	Skills			
	Creativity and	Required	-	-
	Innovation			
	Communication	Required	-	-
	Skills			
	Team Work	-	Required	-
	Technical Skills Required			
	Basic knowledge of cGMP/cGLP principles			
	Knowledge of MS words, MS Excel			
	Record Keeping			
	For Fresher's			
	Knowledge of Microbiological concepts			
	Basic knowledge of lab work			
	Knowledge of computers including MS words, MS Excel etc.			
	Education Qualification and Experience Requirements			
	Qualification: MSc. Microbiology/Biotechnology			
	Experience: Minimum 1 Yr Industrial Experience in the related			
	field.			
	For Fresher's			
	Qualification: MSc. Microbiology/Biotechnology			
	Experience: At least 4-6 months Lab experience which can be			
	summer training, Dissertation work etc.			

Roles/Functions of Supervisor:

Position	Roles/Functions
Supervisor	To supervise the activities of the section
	To develop SOPs, Protocols, Master documents as per the
	requirements of the section and to ensure that the QA documents
	reflect current requirements and best practices.
	To ensure that the established procedures are being followed in
	the concerned section
	To ensure that the activities are being carried out by the trained

	staff
	To ensure implementation of cGMP/cGLP principles, as applicable
	To ensure that the records are being filled concurrently
	To ensure that the study reports duly supported with raw data and
	supportive documentation are properly reviewed and archived after
	necessary corrections

Position Specifications:

Skills, Qualification and Experience Requirements of Supervisor:

Position	Soft Skills	Level required		
		Foundation	Proficiency	Mastery
Supervisor	Customer Focus	-	Required	-
	Creativity and	-	-	Required
	Innovation			
	Communication	-	Required	-
	Skills			
	Team Work	-	-	Required
	Technical Skills Required			
	Execution of SOPs/Protocols as a supervisor			
	Capability of verifying Records/Reports etc			
	Knowledge of cGMP/cGLP principles			
	Faced at-least 2 National / International regulatory audits			
	Education Qualification and Experience Requirements			
	Qualification: MSc. Microbiology/Biotechnology			
	Experience: Minimum 3 Yrs. Industrial Experience in the related			
	field			

Roles/Functions of Section In- charge:

Position	Roles/Functions
Section -In	To Plan, develop and coordinate the activities of the section.
charge	To direct the staff to develop procedures/ protocols/plans etc,
	as required.
	To evaluate results submitted by various departments.
	To participate in departmental major program or policy decisions
	concerned with the section.
	To impart training to the staff.
	To review the SOPs, Protocols, Plans etc and any amendments to
	the established procedures;
	To ensure that the procedures specified are followed, assess and
	document the impact of any deviations from the established
	procedures and to take/devise appropriate corrective action if
	necessary; acknowledge deviations from Standard Operating
	Procedures/Protocols etc.
	To review the final report for validity of the data and to
	indicate the extent to which the activity complies with the
	Principles of cGMP/cGLP;
	To solve technical problems or complex situations.
	To maintain complete records of activities.
	To evaluate staff performance.
	To ensure the security of confidential material at all times.
	To foster a team environment and facilitate the building of
	positive, productive relationships.

Position Specifications:

Skills, Qualification and Experience Requirements of Section In- charge:

Position	Soft Skills	Level required		
		Foundation	Proficiency	Mastery
Section-In	Managerial	-	-	Required
charge	skills			
	Decision Making	-	Required	-
	Customer Focus	-	Required	-
	Team Work	-	-	Required
	Technical Skills Required			
	Knowledge of Regulatory Aspects			
	In-depth knowledge of cGMP/cGLP principles			
	Ability to Independent handling of section			
	Faced at-least 3 National / International regulatory audits			
	Education Qualification and Experience Requirements			
	Qualification: MSc. Microbiology/Biotechnology			
	Experience: Minimum 5 Yrs. Industrial Experience in the			
	related field.			

|2 General Responsibilities/Activities QA Department: |

|To ensure that products are designed and developed in a way that takes account |
|of the requirements of GMP and other associated codes such as those of good |
|laboratory practices (GLP) ,good clinical practice (GCP), GXP ; |
|To ensure Production and control operations are clearly specified in a written |
|form and GMP requirements are adopted; |
|To ensure Job Responsibilities are clearly specified; |
|To ensure arrangements are made for the manufacture, supply, and use of the |
|correct starting and packaging materials; |
|To ensure all necessary controls on starting materials, intermediates, bulk |
|products and other in-process controls, calibrations, and validations are |
|carried out; |
|To ensure the finished product is correctly processed and checked, according to |
|the defined procedures; |
|To ensure products are not sold or supplied before the authorized persons have |
|certified that each production batch has been produced and controlled in |
|accordance with the requirements of the marketing authorization and other |
|regulations relevant to the production, control and release of products; |

[To ensure satisfactory arrangements exist to ensure, as far as possible, that |
the products are stored, distributed, and subsequently handled so that quality |
is maintained throughout their shelf-life; |
To ensure there is a procedure for self-inspection and/or quality audit that |
regularly appraises the effectiveness and applicability of the quality assurance |
system. |
To ensure all manufacturing processes are clearly defined, systematically |
reviewed in the light of experience, and shown to be capable of consistently |
manufacturing products of the required quality that comply with their |
specifications; |
To ensure critical steps of manufacturing processes and any significant changes |
made to the processes are validated; |
To ensure all necessary facilities are provided under the responsibility of the |
production management: |
appropriately qualified and trained personnel; |
adequate premises and space; |
suitable equipment and services; |
correct materials, containers, and labels; |
approved procedures and instructions; |
suitable storage and transport; and |
adequate personnel, laboratories, and equipment for in-process controls; |
To ensure that the instructions and procedures are written in clear and |
unambiguous language, specifically applicable to the facilities provided; |
To ensure operators are trained to carry out assigned activities; |
To ensure records are made (manually and/or by recording instruments) during |
manufacture to show that all the steps required by the defined procedures and |
instructions have in fact been taken and that the quantity and quality of the |
product are as expected; any significant deviations are fully recorded and |
investigated; |
To ensure records covering manufacture and distribution, which enable the |
complete history of a batch to be traced, are retained in a comprehensible and |
accessible form; |
To ensure the proper storage and distribution of the products minimizes any risk |
to their quality; |
To ensure a system is available to recall any batch of product from sale or |
supply; |
To ensure complaints about marketed products are examined; the causes of quality |
defects investigated, and appropriate measures taken in respect of the defective |
products and to prevent recurrence. |

[3. QA Responsibilities/Activities: As applicable to the specific section, but |
not limited to : |

AUDITS AND INSPECTIONS |

[To plan, organize and coordinate Quality Audits & Self Inspection to streamline |
the system. |
To assist in the classification of audit findings. |
To coordinate External audits during the course of audit. |
To plan and perform inspections of product destruction. |
To ensure follow up of Quality Audits and check that corrective actions are |
taken well in time. |
To plan and perform corrective and preventive actions for external audit report |
emitted by Drug authorities or External agency. |
To perform Vendor Audits. |
To coordinate with concerned departments in case of deviation and perform follow |
up visits. |
To handle out of specification results, its investigation and to perform follow |
up of corrective action taken. |
To evaluate and process change control. |
To ensure strict compliance of cGMP and requirements of World Health |
Organization and Indian Pharmacopoeia. |
To support regulatory affairs department for registration of products, review of |
documentation to comply with license and regulatory requirements. |
Lot release |
To review batch production records and analytical files for their correctness, |
legibility, non-deviation, to prescribed control limits/product specifications |
and regulatory compliance to ensure the quality of product/s. |
To prepare checklists of the production process and analytical test result |
parameters. |
To coordinate in Lot release activities. |
To archive batch production record and analytical file as per established |
procedure. |
To prepare annual product review of products manufactured. |
To review trend analysis reports of products manufactured. |
To review stability study reports. |
Documentation |
To ensure organization of documentation system. |
To assist in preparation and issuance of all technical documents such as |
Production Order, Master Formulae, Specifications, Standard operating procedures |
and their timely updation. |
To prepare templates for Standard Operating Procedures, Specifications, |
Production Orders, Master Formulae, Record forms etc. |
Control of documents viz. Layouts, Logbooks and Labels. |
To archive Master copies of SOPs, Specifications, Production Orders, Master |
Formulae and other controlled documents. |
To organize the periodic review activity for timely updation of documents. |
To assign codes to raw materials, intermediates, final product and critical/ |
non-critical equipments components being used in the facility. |
To ensure that cGMP guidelines are being followed in documentation with respect |
to updated regulatory requirements. |
Validation |
To establish Validation Master Plans. |
To establish general guidelines for governing Process validation. |

|To provide guidelines and coordinate in the preparation of Validation Protocols |
 |for various equipments / processes. |
 |To coordinate in the Installation, Operational and Performance Qualification of |
 |equipments. |
 |To participate in the validation activities for the different equipments |
 |installed in Production as |
 |well as Quality Control departments. |
 |To coordinate the Revalidation of equipments / processes as and when required. |
 |To co-ordinate in validation of analytical methods and development of protocols |
 |for the same and review of validation reports. |
 |To co-ordinate in the preparation of campaign change protocol and review of |
 |campaign change reports. |
 |To assist in conducting impact analysis and risk analysis for different |
 |equipment / systems. |
 |To co-ordinate in in-house/contractual calibration activities and review of |
 |calibration reports. |
 |To ensure that all the equipments are timely calibrated and all the processes |
 |and analytical techniques are validated. |
 |To conduct experimental analysis and coordinate in developing the production |
 |process validation for the assigned product. |
 |To keep proper record of calibrations done and validations accomplished. |
 |Quality Engineering |
 |To analyze results of environment and critical system monitoring. |
 |To perform trend analysis on the environment, critical systems and product |
 |parameters. |
 |Coordination in the implementation of environment monitoring program. |
 |To analyze out of control results and coordinate investigation of out of control|
 |results. |
 |Preparation of control charts. |
 |To perform trend analysis on the manufactured product. |
 |To analyze complaints and document the records. |
 |To co-ordinate and maintain records of returned/recalled products. |
 |To ensure controlled areas in production facility meet established environmental|
 |parameters and that the production activities are being carried out under |
 |prescribed conditions. |
 |To plan monitoring schedules during production of vaccine. |
 |To keep regular checks on implementation of monitoring schedules for environment|
 |and critical systems. |
 |Training |
 |To manage the Internal training of Technical and non-technical staff according |
 |to the training needs identified through a need based approach. |
 |To plan, design and prepare training program against the stated QA objectives |
 |and provide support in the implementation of the training program according to |
 |the Training calendar. |
 |To compile and archive individual training files of employees. |
 |To maintain the evaluation system in close co-ordination with HR department and |
 |prepare regular reports for senior management. |
 |To organize, monitor and maintain records for cGMP trainings. |
 |To archive CV's of the external qualified trainers. |

	Name	Designation	Signature	Date
Compiled By				
HOD				
HRD				
HOD QA				

Complaint Receipt Date	
Code of Complaint	
Name of the Product:	
Batch Number of the Product:	
Mfg. Date	
Retest/Expiry Date:	
Way in which it was communicated	
Name of the person that received the	
communication	
DATA OF THE COMPLAINT	
Complainant Name	
Designation	
Complaint received on	
Name and Address	
Description of the Complaint:	
Reported Previously	Yes () No ()
Code of previous complaint, if any:	
Complaint Sample for Quality Assessment	Yes No
purpose (If Required)	
Quantity of the samples:	
Complaint Informed to CQA By Site QA on	
:	
Complaint acknowledgement form sent to	
complainant on:	
Investigation Started On:	
Investigation Completed On:	
Classification of complaint:	
Complaint Closed By (Head QA)	
(sign/date):	

Annexure-3

S. No. |Complaint No. |Date of Receipt of complaint |Source of Complaint
|Type Classification of Complaint* |Product,
B. No.,
Mfg. Date,
Retest/Exp. Date |Acknowledgement sent on |CAPA
No. (If applicable) |Investigation completed on |Remarks |||||

Date of receipt of Complaint:	
Code of Complaint:	
Name of the product :	
Batch Number :	
Investigation start date	
(Tentative):	
Additional Information required from the complainant, if any:	
Head QA:	
Sign/date	
Thanks for raising the concern, our investigation team aims to investigate the	
root cause of the complaint within days of the date we received your	
complaint. We will get back to you with the details of the outcome of the	
investigation	
Acknowledgement given By:	
Head QA	
Sign/Date:	

Annexure-3

Complaint	Sample Analysis	Retained Sample Analysis
1. Complaint: The product is defective.	Analysis: The product was found to be defective due to a manufacturing error.	Analysis: The product was found to be defective due to a manufacturing error.
2. Complaint: The product is damaged.	Analysis: The product was found to be damaged due to handling during shipping.	Analysis: The product was found to be damaged due to handling during shipping.
3. Complaint: The product is expired.	Analysis: The product was found to be expired due to a labeling error.	Analysis: The product was found to be expired due to a labeling error.
4. Complaint: The product is contaminated.	Analysis: The product was found to be contaminated due to a cleaning error.	Analysis: The product was found to be contaminated due to a cleaning error.
5. Complaint: The product is missing.	Analysis: The product was found to be missing due to a packaging error.	Analysis: The product was found to be missing due to a packaging error.

1	2	3	4
5	6	7	8
9	10	11	12
13	14	15	16
17	18	19	20

| Immediate action taken (if any) : |
| Complaint circulated to: COA : Yes/No |

|||
| Sign/Date |
| Authorized Signatory |
|||

PRODUCT COMPLAINT INVESTIGATION FORM

(D) Investigation / Findings /Comments by Investigation Team: |
| (Attach separate Annexures, if required) |

[illegible]

Department	Name	Designation	Sign / Date

PRODUCT COMPLAINT INVESTIGATION FORM

|(E)|Quality Assurance Report and Disposition: |
|Root Cause Analysis (Attach extra sheet if required): |

Proposed corrective and Preventive action: Reference CAPA No. _____
--

1	1		
1	1	1	1
1	1	1	1
1	1	1	1

(F) Approval & Disposition By Head-CQA (For Major and Critical Complaints):
Comments:

Note: All the complaint forms, correspondence and investigation annexure shall be attached with this Investigation form.

1.0 |Objective |
 |To lay down the procedure for handling of product complaints. |
 2.0 |Scope |
 |This SOP is applicable to receive, acknowledgement, record, investigate and|
 |respond to all types of products related complaints at xxxxx. |
 3.0 |Responsibility |
3.1	Production Head or designee:	
	3.1.1	Investigation of complaint.
	3.1.2	To provide all required support to Quality Assurance and other
		departments for handling of product complaints.
	3.1.3	Training to all concerned personnel whenever required.
	3.1.4	Suggesting and Implementation of corrective and preventive
		actions.
3.2	Head QC or designee:	
	To participate in related complaint investigation with complaint	
	investigation team.	
	Analysis of complaint sample or control samples whenever required.	
3.3	Any Department Head if receive product complaint, shall be	
	responsible for forwarding the same to QA department and to provide	
	required support for investigation of product complaint.	
3.4	Head of Clinical Research department (Global Pharmacovigilance	
	Division):	
	Responsible for coordinating, processing and responding to all	
	product complaints related to Adverse Reactions / Adverse Events.	
	Global Pharmacovigilance Division (GPVD) shall reply to the	
	complainant for complaints related to adverse events (AE), or adverse	
	reaction (AR). GPVD if receive the Product Quality Complaint shall be	
	responsible to communicate immediately to the QA Head.	
3.5	Head of R&D department:	
	To participate in related complaint investigation with complaint	
	investigation team.	
	Recommendation of any change in process based on the investigation	
	outcome of the complaint.	
3.6	Head of RA department:	
	Responsible for coordinating & processing to all product complaints	
	related to or from regulatory agencies.	
3.7	QA Head or designee:	
	3.7.1	QA Head or designee shall be responsible for sending all
		intimation regarding Product complaints to Corporate Quality
		Assurance Department (CQA) as soon as complaint is received.
	3.7.2	QA Head or designee shall be responsible for coordinating and
		processing all product complaints, to carry out the recording,
		investigation review and approval, to fix root cause, identify
		appropriate CAPA and to ensure effective implementation of
		CAPA.
	3.7.3	QA Head or designee shall be responsible for prepare the
		complaint response and sent to complainant (the source from
		where the complaint received except Adverse Event/Adverse
		Reaction/Regulatory complaints) & for Periodic review and
		trending of complaints.
	3.7.4	Head-Quality Assurance or his/her designee shall be
		responsible for compliance to this SOP.
3.8	Head Corporate QA:	
	Responsible to ensure the complaints are addressed in timely manner	
	and send the final response regarding complaints received from	
	regulatory agencies.	
4.0	Abbreviations and Definitions	
CAPA	:	Corrective and Preventive Actions
CQA	:	Corporate Quality Assurance
CRD	:	Clinical Research Department
GPVD	:	Global Pharmacovigilance Division
PQC	:	Product Quality Complaint
OG	:	Operating Guideline
SOP	:	Standard Operating Procedure
QA	:	Quality Assurance
RA	:	Regulatory Affairs
WHO	:	World Health Organization
Product Complaint: - An expression of dissatisfaction by a customer that a		
product has not met their expectations in some way.		
Adverse Reaction (AR):- A response to a medicinal product which is noxious		
and unintended and which occurs at doses normally used in human for the		
prophylaxis, diagnosis or therapy of disease or for modification of a		
physiological function.		
Adverse Event (AE):- Any untoward medical occurrence in a patient or		
clinical investigation subject administered a pharmaceutical product and		
which does not necessarily have to have a casual relationship with this		
treatment (An adverse event, can therefore, be any unfavorable and		
unintended sign (e.g. an abnormal laboratory findings), symptom, or disease		
temporally associated with the use of a medicinal product, whether or not		
considered related to the medicinal product).		
Product Quality Complaint :- Product quality complaints are unwanted		
alterations to the physical or chemical attributes to the product, which		
includes but are not limited to :		
Breakage		
Leakage		
Microbiological contaminations / presence of foreign matter		
Change of colors or appearance		
Packaging or labeling issues		
Precipitation / flocculation in product		
Critical Complaints: - It includes those which put patient at life		

| threatening risk and may cause death of the patient, for example but not |
| limited to: |
| Mix – ups |
| Microbiological contaminations / presence of foreign matter |
| Wrong product (label and Content are different) |
| Major Complaints: - It includes those which may put patient at some risk |
| but are not life threatening and are quality related. For example: |
| Assay is low, precipitation / flocculation in product, change in |
| description / color of product. |
| Minor Complaint: - It includes those which may not pose a significant |
| hazard to health. It include those which are not life threatening, for |
| example: |
| Shortage or excess quantity received |
5.0 Procedure |
5.1 Receiving of complaint : |
5.1.1 Product complaint receipt from customer by any way of |
| communication to any employee shall be immediately notified in |
| duly filled Complaint form as per annexure 1 and shall be |
| forwarded to Quality Assurance Head. In case the complaint is |
| about Adverse Event/Adverse Reaction the same shall be sent to |
| GPVD. (A scanned copy of form would be sent immediately) |
5.2 Logging of complaint : |
5.2.1 Each complaint shall be assigned a unique number by QA. |
| Each complaint shall be coded as defined below by means of an |
| example: |
| Code of the complaint: |
| L : Corresponds to xxxx Site |
| COM: Corresponds to complaint |
| 00: Corresponds to product code of xxxx Vaccine |
| 23: last two digits of the year (2023) in which the complaint |
| was logged. |
| 001: a three digit consecutive number of the complaint in that |
| year. |
| In this example, the code corresponds to the first complaint |
| of xxxx vaccine in year 2023 at xxxx site. |
| A new series of the complaint number shall be started at the |
| beginning of a new calendar year. |
5.2.2 QA shall log the complaint as per Annexure 2. |
5.2.3 QA shall intimate CQA about the details of the complaint |
| received as per annexure 1. |
5.3 Acknowledgement of the Product complaint: |
5.3.1 QA shall provide the acknowledgement to the complainant after |
| the complaint is logged in as per annexure 3 "Complaint |
| Acknowledgement form". |
5.4 Complaint Classification: |
5.4.1 Based on the complaint, Head QA shall classify the complaint |
| as Minor, Major and Critical in Annexure 1. |
5.5 Formation of the Product Complaint investigation team: |
5.5.1 QA Head shall nominate the team members from different |
| departments like Production, Quality Control, Ware house, |
| Engineering, R&D etc. as required for investigation purpose |
| based on the nature of the complaint received. |
5.6 Investigation/Findings: |
5.6.1 QA shall initiate the complaint investigation form as per |
| Annexure 4. |
5.6.2 For critical complaint, the initial reply to the complainant |
| shall be given by QA within 72 hours with initial |
| investigation. In case the timeline is exceeded the same shall |
| be appropriately justified. |
5.6.3 Investigation of the product complaint shall include, but |
| shall not be restricted to investigation of batch |
| manufacturing records, control samples, analysis reports and |
| deviation reports, impact on the previous batches manufactured |
| if any. |
5.6.4 Based on the nature of investigation retained sample |
| /complaint samples shall also be examined/ tested (if |
| required) and all the related batch production records and |
| documentation shall be reviewed. |
5.6.5 Detailed findings of the investigation relevant to the |
| retained sample/complaint samples, production, storage and |
| distribution records shall be clearly mentioned. |
5.6.6 The investigation of product complaint shall be carried out as |
| per SOP xxxx and the extent of investigation shall depend upon |
| the particular case of product complaint received. |
5.7 CAPA, feedback to the Complainant/ Regulatory bodies and closing of |
| the Product Complaint: |
5.7.1 Based on the investigation and root cause of the Product |
| complaint corrective and preventive action shall be initiated. |
5.7.2 In case regulatory complaints the response shall be send by |
| CQA marking copy to Site QA. |
5.7.3 In case of Adverse Reaction/Adverse Event, response shall be |
| sent by GPVD marking copy to QA. All other complaints except |
| Adverse Reaction/Adverse Event /Regulatory complaints, |
| response shall be sent by QA to complainant. |
5.7.4 During the course of investigation and final disposition, if |
| it emerges that there is a need of product recall then SOP of |
| Product Recall and handling of recalled products shall be |
| followed. |
5.7.5 All forms, detail complaint samples, documents and |
| correspondences associated with the complaint shall be filed |
| in the Product Complaint File. |
5.7.6 One photocopy of product complaint shall be attached in the |

			Batch Manufacturing Record of respective batch.
		5.7.7	The complaint shall be shared with business partner if any,
		when received at site directly from the complainant.	
		5.7.8	If required, legal advice can be taken through legal
		department of the company.	
		5.7.9	Any product complaint received shall be closed by QA Head in a
		period of not more than 45 days from the date of logging. In	
		case the timeline is exceeded the same shall be justified and	
		approved by QA Head.	
		5.8	Periodic Review and trends:
		5.8.1	The product complaint trends shall be prepared at least
		Quarterly and shall be considered in Annual product Review.	
		5.8.2	There shall be annual compilation and review of all the
		complaints product wise.	
6.0	Forms and Records		
6.1	Product Complaint form	- Annexure-1	
6.2	Product Complaint log	- Annexure-2	
6.3	Product complaint Acknowledgement form	- Annexure-3	
6.4	Product Complaint Investigation form	- Annexure-4	
7.0	References:		
	7.1		
8.0	Distribution		
	8.1		
9.0	History		
Date	Revision Number	Reason for Revision	

VMP
(VALIDATION MASTER PLAN)
FOR
VACCINE FILL & FORMULATION FACILITY

Document No. :
Effective date :

APPROVALS

PREPARED BY:
|Name : |Signature : |
| Designation : |Date : |

REVIEWED BY:

PRODUCTION DEPARTMENT

|Name : |Signature : |
|Designation : |Date : |

QUALITY ASSURANCE APPROVAL:

|Name : |Signature : |
|Designation : |Date : |

MAINTENANCE DEPARTMENT

|Name : |Signature : |
|Designation : |Date : |

QUALITY ASSURANCE DEPARTMENT

|Name : |Signature : |
|Designation : |Date : |

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1.	INTRODUCTION	
	1.1	Manufacturer Details
		The firm M/s XXXX company have filling and formulation facility for
		vaccine. The fill and formulation facility of M/s XXXX company is
		situated inside the larger complex occupies over acres of land.
		This is located in the non-polluting clean surroundings in the
		industrial area of xxxxx.
	1.2	Project Details
		M/s XXXX company is constructing an integrated Vaccine
		manufacturing unit consisting of xxxx manufacturing and fill finish
		facility for vials with Lyophilization capability.
		Vaccine facility is a state of the art cGMP and BSL 2
		enhanced compliant Vaccine filling and formulation Facility. The
		facility consists of environmentally controlled rooms for
		formulation, filling, storage areas. The facility is spread over
		xxxx square meter area. The offices and document storage area is

	located on the	
	Vaccine filling and formulation facility	
	The production areas of fill finish facility have been designed as	
	per cGMP requirement as CNC, Grade D (Class 100000), Grade C (Class	
	10000), Grade B (1000) respectively & provided with individual AHUs	
	for individual process areas. All sterile operations are conducted	
	under LAF Grade A (Class 100).	
	The construction material used for walls and ceiling has been	
	selected as GI powder coated clean room wall panels and ceiling	
	panels. The material is easily cleanable and provides tremendous	
	resistance to repetitive cleaning through disinfectants. The floor	
	finish is Epoxy which is a monolithic joint-less floor. Man &	
	Material flow is unidirectional. All entries & exits is through	
	interlocked doors and access to the facility is restricted through	
	access control system. The air exhaust from all areas shall be	
	passed through HEPA filters. Facility has been provided with EN	
	certified A2 Bio safety cabinets for protection of product as well	
	as the person. Dedicated decontamination autoclave has been	
	provided for the decontamination of the biowaste. For process	
	material sterilization dedicated production autoclave & Dry Heat	
	Sterilizer with bio-shield has been provided in the facility. For	
	defumigation of area, dedicated / auto control defumigation system	
	are provided for respective areas and exhaust of the same is ducted	
	before discharging to ambient area. There will be a centralized	
	laundry facility for clean room garments.	
	Installed Capacities	
	The facility is designed for filling of liquid vaccine in vials,	
	half stoppering, lyophilization and aluminium sealing. In case of	
	fully liquid vaccines the filled vials can be stoppered and sealed	
	on a continuous basis.	
	The filling machine will have a filling speed of vials per min	
	and the lyophilizer will have a shelf size of sqm. Operation	
	areas consist of material preparation area, Bottle washing & drying	
	area, formulation area, washing and sterilization area, vial fill	
	and half stoppering, lyophilization, aluminum capping, Labeling &	
	Packing, Vial collection area	
	Dedicated critical utilities have been provided for the facility.	
	PW & WFI loops are provided with on line TOC sensors. Pure steam	
	generator has an additional provision for removal of	
	non-condensable gases. All inter/intra room transfers are affected	
	through a closed loop system. All process equipments are either	
	controlled through SCADA or through DCS. Cold Storage facilities	
	are available.	
	1.3	Objective
	The objective of Validation Plan is to outline the requirements	
	that will demonstrate and document that all the	
	equipment/systems/facility/control systems and their associated	
	functionality in the facility are appropriate for the cGMP	
	regulated process. It serves as the basis for the entire validation	
	effort and defines the overall validation philosophy of M/s XXXX	
	Recombinant Protein company for Pentavalent filling and formulation	
	facility. It delineates equipment, systems and procedures to be	
	validated, outlines the general acceptance criteria for validation	
	and establishes the responsibilities for performing the validation.	
	1.4	Scope
	The Validation Plan shall cover in its scope the utilities,	
	critical equipment, manufacturing processes, cleaning and	
	sanitizing procedures, analytical techniques, laboratory equipment,	
	automated systems (SCADA, DCS &BMS) to be kept in validated state	
	so as to produce drug substances of consistent quality following	
	cGMP, with high degree of confidence through established documented	
	evidence as per requirements of the marketing authorization. It is	
	applicable for the designing, commissioning, installation,	
	operational and performance qualification activities of the	
	facility. These activities play a significant role in delivering	
	and maintaining operationally safe and efficient validated	
	processes, utilities and qualified equipment.	
	This Validation Plan document shall describe the process	
	validation, facility qualification and validation, utility	
	qualification and validation ,equipment qualification and	
	validation as per current Good Manufacturing Practices (cGMP) norms	
	as advocated by World Health Organization (WHO),US Food and Drug	
	Administration, European guidelines, Pharmaceutical Inspection	
	Convention –PIC/S , Parenteral Drug Association -PDA and to	
	register its products for UNICEF supplies through WHO	
	prequalification, as well as US Food and Drug Administration	
	approval in case of its selected products to move in International	
	Commerce. Last but not the least the provisions applicable as per	
	Law of the Land shall also be upheld.	
	1.5	Organizational Structure
		Quality Assurance Department
	1.5.1	
		Quality Assurance Department shall establish the validation
	policy, general guidelines and approve all general documents	
	that govern process validation.	
		Review and approve all the documentation relating to the
	validation work. This will include the Validation Master	
	Plan, Product specific Validation Plan, Supplementary	
	Validation Plan(s), Validation Protocols, SOPs, executed	
	complied test results and validation reports, etc.	
		Coordinate in the execution of different programs supporting
	the validation work(s), such as Preventive Maintenance, Staff	

||| Training, Calibration, Environmental Monitoring and Internal
		Audits.	
		Authorization of validation documents.	
		1.5.2	Validation Section in Quality Assurance Department
		Validation Section coordinates with the Validation Work	
		Groups/ Project Validation Team formed to carry out the	
		various validation activities. As such the responsibilities	
		of Validation Section are combined in scope with validation	
		work groups in many cases:	
		Development and review of Validation Master Plan, Product	
		specific.	
		Supplementary Validation Plan(s).	
		Review of validation documentation, which includes SOPs,	
		Protocols (DQ, IQ, OQ and PQ as applicable), Study and	
		Summary reports, etc.	
		Coordination and participation in the execution of validation	
		activities.	
		Review and approval of Impact assessment and Risk analysis	
		documentation.	
		1.5.3	Project Validation Team
		The Project Validation Team or the Specific Validation	
		Commission (SVC) shall be formed only for the validations to	
		be carried out in case of establishment of a new project,	
		while in case of an already established manufacturing unit	
		the validations shall only be conducted by the concerned	
		department independently.	
		Project Validation Team shall be formed by the heads of all	
		the departments viz. Production, Quality Control, Quality	
		Assurance and Maintenance. Project validation Team shall be	
		formed to carry out the validation of a particular product,	
		process or system and is constituted from qualified and	
		experienced personnel from Production and Quality Unit to	
		carry out validation work.	
		Responsibilities of Validation Team -	
		Carry out validation work as per QA approved protocols.	
		Draws up the Validation Protocols.	
		Formulate necessary report forms to be filled up for the	
		tests to be performed during validation activity.	
		Fill the data forms related to the activities and tests	
		performed.	
		Elaborate Discrepancy, if any, and its impact.	
		1.5.4	Validation Work Group
		In case of an already established manufacturing unit the	
		validations shall only be conducted by the work group	
		independently. The work group shall be constituted by persons	
		from the concerned department viz.	
		Production Department:	
		Participation in the preparation of Validation Protocols for	
		production related processes and equipment.	
		Participation in validation activities, review and approval	
		of validation documents as required. Each document may have	
		different approval requirements, which is determined on	
		case-to-case basis.	
		Operation of equipment as per requirements in the validation	
		activity.	
		Participation in preparation of validation reports with	
		complete supporting data.	
		Execution of process/equipment/system associated risk	
		analysis and impact analysis.	
		Quality Control Department:	
		Preparation of Validation Protocols and getting them approved	
		from Quality Assurance department before starting the	
		validation activity.	
		Participation in validation activities, new method	
		development, review and approval of validation documents as	
		required. Each document may have different approval	
		requirements, which is determined on case-to-case basis.	
		Providing analytical testing facilities in support of	
		validation activities.	
		Ensuring that all relevant analytical laboratory techniques	
		are validated.	
		Preparing all necessary SOPs required for the validation	
		exercise and to ensure that the concerned technicians are	
		trained in cGMP and GLP as per requirements.	
		Engineering / Maintenance Department:	
		Participation in the development, review and approval of	
		validation documents, as and if required. Each document may	
		have different requirements of approval at different levels,	
		which is determined on case-to-case basis.	
		Support as required for drawings, documentation and operation	
		of equipment.	
		1.6	Validation Philosophy
		Validation is "Establishing documented evidence which provides a	
		high degree of assurance that a specific process will consistently	
		produce a product meeting its predetermined specifications and	
		quality attributes."	
		The validation philosophy focuses on a few questions like Does the	
		system/equipment requires validation. What will be the extent of	
		validation? What aspects of the system or process are critical to	
		product and patient safety?	
		The answers to these questions are found by evaluating the	
		criticality of the cGxP elements, their associated risks and impact	

	non-functional requirements, and what deliverables are to be	
	supplied by the vendor. The user requirement shall be prepared for	
	each piece of equipment in terms of function, throughput,	
	operability and applicable standards.	
	2.3	System Specifications:
	The information is the Basis of Design documents is used to develop	
	Proposed / purchase specifications such as speed, time, temperature	
	and volume for specific processes. The product quality	
	characteristics are the requirements that the equipment and process	
	performance can be measured against.	
	2.4	Factory Acceptance Test (FAT):
	An acceptance test in the supplier's factory, usually involving the	
	customer. Testing conducted to determine whether or not a system	
	satisfies its acceptance criteria and to enable the customer to	
	determine whether or not to accept the system.	
	2.5	Site Acceptance Test (SAT):
	An acceptance test in the customer's factory, usually involving the	
	customer. Testing conducted to determine whether or not a system	
	satisfies its acceptance criteria and to enable the customer to	
	determine whether or not to accept the system.	
	2.6	Commissioning:
	Commissioning is the phase of a project which includes: mechanical	
	completion, installation qualification, operational qualification,	
	start-up, and turn over. It incorporates systematic method of	
	testing and documenting of systems and equipments at the conclusion	
	of a project, but prior to validation. It involves the proper	
	documentation of facility construction and installation.	
	2.7	Design Qualification (DQ):
	Documented evidence that the premises, supporting utilities,	
	equipment and processes have been designed in accordance with the	
	requirements of GMP.	
	2.8	Installation Qualification (IQ):
	The performance of tests to ensure that the installations (such as	
	machines, measuring devices, utilities, manufacturing areas) used	
	in a manufacturing process are appropriately selected and correctly	
	installed and operate in accordance with established specifications.	
	2.9	Operational Qualification (OQ):
	Documented verification that the system or subsystem performs as	
	intended over all anticipated operating ranges.	
	2.10	Performance Qualification (PQ):
	PQ is the documentary evidence which verifies that the equipment or	
	system operates consistently and gives reproducibility within	
	defined specifications and parameters for prolonged periods.	
	2.11	Equipment Qualification:
	Studies that establish that the process equipment and ancillary	
	systems are capable of consistently operating within established	
	limits and tolerances. The studies must include equipment	
	specifications, installation qualification, and operational	
	qualification of all major equipment to be used in the manufacture	
	of commercial scale batches. Equipment Qualification should include	
	simulation of actual production conditions, including / stressed	
	conditions and is a part of validation (initial stage).	
	2.12	Validation Protocol:
	A written plan of actions stating how validation activity will be	
	conducted; will specify the various tasks and define testing	
	parameters; sampling plans, testing methods and specifications and	
	equipment(s) to be used. It will also mention the minimum number of	
	runs/batches to be used for the study; it must specify the	
	acceptance criteria and the authority who will approve/disapprove	
	the study.	
	2.13	Validation Report:
	A document in which the records, results and evaluation of a	
	completed validation program are assembled and summarized. It may	
	also contain proposals for the improvement of processes and/or	
	equipment.	
	2.14	Validation Team:
	A multi-disciplinary team of personnel primarily responsible for	
	conducting validation activities/studies. Such workgroup shall be	
	constituted by trained and experienced personnel in relevant	
	disciplines.	
	2.15	Critical Process Parameter:
	A parameter which if not controlled will contribute to the	
	variability of the end product.	
	2.16	Critical Systems:
	Critical Systems are those that generate utilities which become	
	part of the final product, or come in contact with the final	
	product during production or with the surfaces of processing	
	equipment and the quality of which may adversely affect the	
	efficacy, safety and quality of the final product.	
	2.17	Worst Case Condition:
	A set of conditions encompassing upper and lower limits and	
	circumstances that poses the greatest chance of failure when	
	compared with ideal conditions. These highest and lowest values of	
	a given parameter shall actually be evaluated in the validation	
	exercise.	
	2.18	Analytical Method Validation:
	Analytical Method validation is the process to confirm that the	
	analytical procedure employed for a specific test is suitable for	
	its intended use.	
	2.19	Linearity and Range:
	The linearity of an analytical procedure is its ability to produce	
	results that are directly proportional to the concentration of	

	analyte in the samples. The range of the procedure is an	
	expression of the lowest and highest levels of analyte that have	
	been demonstrated to be determinable with acceptable precision,	
	accuracy, and linearity.	
	2.20	Specificity:
		The specificity or selectivity of a procedure is its ability to
	measure the analyte in a manner that is free from interference from	
	other components in the sample being examined (for example,	
	impurities arising from manufacture or degradation or ingredients	
	other than the analyte).	
	2.21	Sensitivity:
		The specificity or sensitivity is the capacity of the test
	procedure to record small deviations in concentration. It is the	
	slope of the calibration curve. A more general use of the term to	
	encompass limit of detection and or limit of quantization should be	
	avoided.	
	2.22	Limit of Detection:
		The limit of detection is the lowest level of analyte that can be
	detected, but not necessarily determined in a quantitative fashion.	
	2.23	Limit of Quantitation:
		The limit of quantitation is the lowest concentration of analyte in
	a sample that may be determined with acceptable accuracy and	
	precision when the required procedure is applied.	
	2.24	Accuracy:
		The accuracy of the procedure is the closeness of the results
	obtained by the procedure to the true value. Accuracy may be	
	determined by applying the procedure to samples of the material to	
	be examined that have been prepared with quantitative accuracy.	
	Wherever possible, these samples should contain all the components	
	of the material, including the analyte.	
	2.25	Precision:
		The precision of the procedure is the degree of agreement among
	individual test results. It is measured by the scatter of	
	individual results from the mean and it is usually expressed as the	
	standard deviation or as the coefficient of variation.	
	2.26	Repeatability (within-laboratory variation):
		This is the precision of the procedure when repeated by the same
	analyst under the same set of conditions (same reagents, equipment,	
	settings, and laboratory) and within a short interval of time.	
	2.27	Reproducibility:
		This is the precision of the procedure when it is carried out under
	different conditions—usually in different laboratories—on separate,	
	putatively identical samples taken from the same homogeneous batch	
	of material.	
	3.	VALIDATION LIFE CYCLE
	[pic]	
		This life cycle shall provide guidance in preparation of a
	validation plan/validation report. This includes planning of	
	validation activities and predefinition of the life cycle (including	
	possible exemptions and possible deviations).	
	3.1	User Requirement Specification:
		The user requirement specification (URS) is typically created at the
	beginning of a qualification project and describes the initial user	
	requirements (e.g. process-, construction- or functionality-related)	
	regarding a system to be commissioned. The user requirement shall be	
	prepared for each piece of equipment in terms of function,	
	throughput, operability and applicable standards.	
		The URS describes what a system shall be capable of, and not how
	this is achieved! The "how" will be defined in collaboration with	
	the supplier of a system and will be documented in the Functional	
	Specific The functional specification is usually compiled by the	
	supplier of a system and describes how it is intended to fulfill the	
	requirements defined in the URS.	
	3.2	Design:
		The technical documentation is compiled by the supplier during the
	technical design phase and is a dossier containing e.g. a P&ID,	
	wiring diagrams, part lists, a measuring point list, etc. Purpose of	
	the technical documentation is to provide all documents relevant for	
	the construction of a system and to provide technical specifications	
	relevant for the qualification.	
		The user will carry review the design to check for availability of
	all construction-relevant documents as e.g. P&I diagrams or part	
	lists, but is mainly understood during the design qualification	
	stage. This review should assure that all relevant requirements have	
	been implemented in the design before the construction phase is	
	initiated (to avoid unnecessary changes afterwards). The design	
	review should provide enough assurance that the elaborated design	
	will lead to a system which is capable of fulfilling the user	
	requirements as defined in the URS.	
	3.3	System Level Impact Assessment (SLIA):
		SLIA shall be carried out for the System, Equipment or utility which
	shall document the performance of a Product Impact Assessment in	
	accordance with the recommendations of the ISPE. The intent of	
	performing SLIA shall be evaluating the impact of a system,	
	equipment or utility on product quality.	
	3.3.1	Direct impact systems:
		Direct Impact systems are expected to have an impact on
	product quality. Examples of such systems are when they are	
	either in direct physical contact with the drug product; a	
	system that produces data that is used to accept or reject	
	products; or the system is a process control system that may	
	affect product quality.	

	3.3.2	Indirect impact systems:
		Indirect Impact systems are not are not expected to have an
		impact on product quality. Examples here include support
		systems such as heat transfer systems, electric power, and
		non-process water sources. These are non-critical and need
		not be qualified. However, the monitoring and control of
		critical parameters, which these support systems affect,
		shall be validated. Both types of systems will require
		commissioning. However, as per the current regulations the
		"direct impact" shall be subject to supplementary
		qualification. Some indirect systems may also require
		supplementary qualification practices to meet the additional
		regulatory requirements.
	3.3.3	No impact systems:
		No Impact systems will not have any impact, either directly
		or indirectly, on product quality. These systems shall be
		designed and commissioned following Good Engineering Practice
		(GEP) only.
	3.3.4	The next step of the impact assessment involves the
		criticality assessment of the system/equipment components.
		The components shall be classified as either critical or non
		critical based on the reply to a set of questions specified
		in ISPE Pharmaceutical Engineering Guide for New and
		Renovated Facilities.
	3.3.5	As a result of the SLIA and Component Criticality Assessment
		the direct impact systems shall bear all critical components
		and may be a few non- critical components while the indirect
		impact systems shall not bear any critical component.
	3.3.6	Depending upon the impact status of the system as decided by
		the impact assessment, the extent of qualification required
		is determined. Those classified as Direct Impact shall
		subject to qualification practices i.e. IQ, OQ and PQ while
		those classified as Indirect Impact or No Impact shall be
		designed, installed and commissioned only.
	3.3.7	The Equipment like peristaltic pumps, Filtration assemblies
		Housings which are part of the process e.g. sterile
		filtration, shall be considered as components of sterile
		filtration process therefore shall undergo components
		criticality analysis instead of impact analysis.

|Fill Finish facility |
 |Table B - Validation matrix depicting the equipment impact status and required|
 |qualification activities |
Equipment	Impact Status	Required Qualification Phases	Revalidation
/System			
Description			
Direct			

	4.1	Prospective Process Validation:
		It is the established documented evidence that a specific process
		used in the manufacture of a product does what it purports to do
		based on the preplanned protocol, before the product is put to
		commercial use.
		This type of validation shall be carried out:
		When the optimization of the manufacturing process have been
		completed
		The operators and supervising personnel, who shall be 'running' the
		validation batches, are fully conversant with the process and its
		requirements
		The batch size of the product under validation shall be as close as
		possible with the contemplated production batch size
		Three successful consecutive batches/runs shall be considered
		acceptable for the validation of the process
		All the measuring equipment/devices have been calibrated
		All the equipment, used have been qualified and
		The analytical testing methods used shall be validated.
	4.2	Retrospective Process Validation:
		Conducted for a product already being marketed, and is based on
		historical data accumulated over several lots and over time.
		Retrospective Validation may be used for older products which were
		not validated by the manufacturer at the time they were first
		marketed, and which is now to be validated to conform to the
		requirements established by regulatory agencies.
		The historic data retrieved from the records (Batch Manufacturing
		Records) of the completed production batches shall be used to
		provide documented evidence that the process has been in a state of
		control.
		The retrospective validation shall be conducted in the following
		manner:
		Gather the numerical values from the completed batch record and
		include assay values, end product test and in process data.
		Organize these data in a chronological sequence, according to batch
		manufacturing data using a spread sheet format.
		Include data from at least last 15 manufactured batches for
		analysis. If the number of manufactured batches is less than 15,
		then include all the manufactured batches in the analysis.
		If appropriate, subject the resultant data to statistical analysis
		and evaluation.
		Conclusions shall be drawn on the state of control of the
		manufacturing process, based upon the analysis of retrospective
		validation data taking into consideration the in- process and final
		product specifications.
	4.3	Concurrent Process Validation:

	Concurrent process validation shall establish documented evidence					
	that the process is in a state of control during the actual					
	implementation of the process. In exceptional circumstances it may					
	be acceptable not to complete a prospective validation program					
	before routine production starts. In such a case the decision to					
	carry out concurrent validation shall be justified, documented and					
	approved by heads of Production, Quality Control and Quality					
	Assurance departments. The in-process as well as finished product					
	parameter requirements to be evaluated for the Process Validation					
	is those, which are mentioned, in the approved Specifications of					
	the product. The acceptance limits shall be part of the approved					
	protocol.					
	4.4	Acceptance Criteria for Process Validation				
	In prospective and concurrent validation, the main goal is to					
	demonstrate through the analysis of the critical parameters of					
	minimum three consecutive batches, that the process is consistently					
	performing and the final product accomplish the established quality					
	standards. In all cases the selected parameters must comply with					
	the approved Quality Specifications.					
	As Process Validation can be product dependent, this aspect shall					
	be detailed in product specific Validation Plan(s).					
	5.	ANALYTICAL TEST VALIDATION				
	The main objective of analytical techniques validation is to demonstrate					
	that the technique is not only suitable for its intended purpose but also					
	gives reliable consistent results with in the acceptable statistical					
	limits. An analytical validation protocol shall be prepared by Quality					
	Control department which shall be established by laboratory studies, that					
	the performance characteristics of the analytical method shall meet the					
	intended analytical application. The validation study shall be conducted					
	as applied for detection of impurities, potency, identity and composition					
	of the product as per QA approved protocols, prepared by Quality Control					
	department.					
	5.1	Categories of Assays Requiring Analytical Test Validation:				
	Most common categories of assays for which analytical validation					
	data shall be required are as follows:					
	Category I: Analytical methods for quantization of major components					
	of bulk drug substances or active ingredients (including					
	preservatives) in finished pharmaceutical products.					
	Category II: Analytical methods for determination of impurities in					
	bulk drug substances or degradation compounds in finished					
	pharmaceutical products. These methods include quantitative assays					
	and limit tests.					
	Category III: Analytical methods for determination of performance					
	characteristics (e.g., dissolution, drug release).					
	Category IV: Identification tests.					
	The following table indicates the characteristics to consider during					
	analytical validation:					
	Analytical	Assay	Assay	Assay	Assay	
	Performance	Category	Category II	Category	Category	
	Characteristics	I	III	IV		
		Quantitati	Limits			
		ve Tests	Tests			
	Accuracy	Yes	Yes	*	*	No
	Precision	Yes	Yes	No	Yes	No
	Specificity	Yes	Yes	Yes	*	Yes
	Detection Limit	No	No	Yes	*	No
	Quantization	No	Yes	No	*	No
	Limit					
	Linearity	Yes	Yes	No	*	No
	Range	Yes	Yes	*	*	No
	* May be required, depending on the nature of the specific test.					
	In case of compendia methods, the suitability of testing methods					
	used should be verified, documented and shall be approved by					
	Quality Assurance department.					
	A method is considered acceptable if:					
	• Linearity of the substance assay can be demonstrated.					
	• Reproducibility can be demonstrated.					
	• Accuracy can be demonstrated.					
	• Repeatability can be demonstrated.					
	• Robustness can be demonstrated.					
	• Limits of detection & limits of quantification and known					
	impurities are identifiable.					
	The revalidation of the critical assays involved in the release					
	specifications of established products will be carried out					
	whenever there is any change in the critical assays with respect to					
	manufacturing process , Testing process , change in the testing					
	kits , change in the equipment involved in the testing procedures/					
	assays and change in the specification					
	The strategy and plan for revalidation of critical assays will be					
	prepared and it will be part the change control and CAPA approval					
	process if any.					
	6.	COMPUTERIZED SYSTEM VALIDATION				
	All computerized systems within the facility shall be reviewed from a					
	validation perspective. Depending upon the extent of the GMP					
	functionality performed by the computerized system and the criticality of					
	those functions, the extent of the validation performed shall vary. There					
	will be two categories of validation performed on computerized systems					
	within the scope of this project.					
		Hardware:				
	As part of the validation process appropriate tests and challenges					
	to the hardware shall be performed.					

	The validation/qualification of the hardware should prove:
	That the capacity of the hardware matches its assigned function;
	That it operates within the operational limits;
	That it performs acceptably under worst-case conditions (e.g. long
	hours,
	temperature extremes); and
	Reproducibility/consistency (e.g. by performing at least three runs
	under
	different conditions).
	The validation should be done in accordance with written
	qualification protocols and the results should be recorded in the
	qualification reports. Validation may be performed by the computer
	vendor. However, the ultimate responsibility for the suitability of
	equipment used shall lie with the user department. When validation
	shall be done by outside firm, such as computer vendor, the records
	should be sufficiently complete (including general results and
	protocols) for adequacy of the validation.
	Software:
	Software is the term used to describe the complete set of programs
	used by a computer. Records are considered as software; focus is
	placed on accuracy, security, access, retention of records, review,
	double checks, documentation and accuracy of reproduction. The
	company should identify the key computer programs:
	language, name, function (purpose of the programs), input
	(determine inputs), output (determine outputs), fixed set point
	(process variable that cannot be changed by the operator), variable
	set point (entered by the operator), edits (reject input/output
	that does not conform to limits and minimize errors, e.g. four- or
	five-character number entry), input manipulation (and equations)
	and program overrides (e.g. to stop a mixer before time).
	The personnel who have been authorized to write, alter or have
	access to programs should be identified.
	The automated systems in the production are as follows-
	Tank systems
	Chromatography systems
	Filtration systems
	Water system
	HVAC system
	Centrifugation system
	Tunnel Sterilizer
	Filling, Stoppering and Sealing Machine
	Lyophilizer
	Visual Inspection machine
	SCADA – It stands for supervisory control and data acquisition. It
	is a computer system that monitors and controls the process. A
	Human – Machine Interface (HMI) is an apparatus in it which present
	process data to a human operator and through this the human
	operator monitors and controls the process. Remote terminal units
	connecting to sensor in the process converting sensor signals to
	digital data and sending digital data to the supervisory system. In
	the production, Chromatography system, Water generation system,
	Filtration system, and Centrifugation is regulated through the
	SCADA.
	PLC - Programmable logic controller is a digital computer used for
	automation of electromechanical process. PLC is designed for
	multiple inputs & outputs arrangements, extended temperature,
	immunity to electrical noise, resistance to impact of vibrations.
	In the production, Dry Heat Sterilizer is regulated through the
	PLC.

7. FACILITY DESCRIPTION :

Fill Finish Production Facility Description

7.1 Preparation area:

7.1.1 Rotary vial washing machine:

|| Glass vials for filling of vaccine are washed in an automatic |
 || rotary vial washing machine. |

7.1.2 Sterilizing and Depyrogenating tunnel:

	The washed vials are sterilized in Sterilizing and
	Depyrogenating tunnel, which is online with the rotary vial
	washing machine ensuring minimum handling, proper
	sterilization and depyrogenation.

7.1.3 HPHV steam sterilizer :

	A Double Door HPHV steam sterilizer is used for preparation
	of rubber stoppers, aluminium seals and steam sterilization
	of all materials and components like glassware, solutions and
	other components etc.

7.2 Blending area:

7.2.1 Jacketed stir vessel:

|| Electro polished Jacketed Stir Vessels (SS 316L) are in used |
 || for blending activities: |

7.3 Filling area:

7.3.1 Automatic Vial Filling and half Stoppering machine :

	Vaccine is filled in USP Type I tubular glass vials in an
	automatic filling machine which comprises an automatic
	filling machine.

7.3.2 Lyophilization

	The filled and half stoppered vials are taken to the
	lyophilizer protected under mobile Laminar flow chamber
	After the completion of the lyo cycle the vials are fully
	stoppered automatically within the chamber.
	These vials are then unloaded from the lyo chamber and taken

			into the next room for aluminium sealing.
			The lyophilizer is equipped with in-built CIP-SIP system.
		7.3.3	Vial sealing machine:
			Vials are sealed in an automatic vial sealing machine.
		7.4	Visual Inspection:
		7.4.1	Automatic visual inspection:
			The lyophilized vials are then inspected for integrity,
			absence of any foreign particles, deformity etc., using an
			automatic visual inspection machine.
			Rejected vials from the Visual inspection machine are further
			inspected manually and accepted or rejected based on the
			established criteria.
		7.5	Labeling
		Automatic sticker label application machine:	
		7.5.1	Printed sticker labels are applied on filled and sealed glass
		vials using high-speed Automatic sticker application	
			(Labeling) machine.
		7.6	Laminar flow units :
		7.6.1	All the critical processes are carried out in Grade A with
			laminar flow units having HEPA filters with a Grade B
			background.
		7.7	Facility monitoring System (FMS):
		7.7.1	The Facility Monitoring System (FMS) is a self-contained
			system to monitor the airborne particles 0.5µm and larger
			within those critical areas. Alarm levels are pre-programmed
			to indicate out of specification conditions. Graphs and data
			spreadsheets are also produced for trend analysis.
		7.8	Building monitoring System (BMS):
			Complete HVAC System is monitored with BMS system by placing
			sensors for Temperature, RH, and Differential Pressure as
			required to log the values in BMS works station for trend
			analysis.
		8.	CRITICAL UTILITIES
		Critical Systems are those that generate utilities, which are in direct	
		contact with the product or with the surfaces of processing equipment.	
		The Critical Utility systems comprise of Purified Water, Water for	
		Injection, Pure Steam, Heating Ventilation & Air Conditioning,	
		Compressed Air.	
		8.1	Pre Treatment Plant:
		The Pre Treatment Plant will provide soft water to Purified Water	
		generation System it have drinking water quality and have chemical	
		dosing for control the microbial growth.	
		8.2	Purified Water Generation System (PW):
		The Purified Water Plant will provide the total amount of Purified	
		Water required. The quality of Purified Water will correspond to	
		USP 31. The feed water to the Purified Water Plant would have	
		drinking water quality. The Purified Water plant includes a Reverse	
		Osmosis (RO) stage and an Electric De-ionization (EDI) stage.	
		The Purified Water Generation System includes the following steps,	
		which produces water which meets USP 31 requirements:	
		Pre-Filtration: The feed water is filtered through two filtration	
		stages (10 µm and 5 µm) using successive filter cartridges before	
		it is led into the RO membrane elements, thus reducing the number	
		of particles and colloids.	
		Further details of the Pure Water distribution loop are listed	
		below:	
		Elements	Description
		Piping material	0.065" wall thickness, 316L stainless steel ASTM
		A270	
		Ra (0.6 (m inside (240 grit); Ra (0.9 (m outside	
		)(150 grit) finishing.	
		Valves	Diaphragm sanitary type, with tri-clamp
		connections. Body AISI 316L diaphragm EPDM.	
		Fittings	0.065" wall, 316L stainless steel ASTM A270
		Ra (0.6 (m inside (240 grit), Ra (0.9 (m outside	
		)(150 grit) finishing.	
		Purified water	AISI 316L stainless steel sanitary. Simple
		circulation	mechanical seals tungsten carbide
		pump	AISI 316L stainless steel sanitary section valve
		AISI 316L stainless steel bottom discharge valve	
		Sanitary manometer.	
		Storage Tank to supply the Purified Water to different user points	
		through a distribution loop. The growth of microbes in the system	
		is controlled using continuous ultraviolet irradiation of the PW	
		stream. Deeper sanitation of the system will be carried out	
		periodically, to prevent the growth of microbes / to reduce the	
		microbial count.	
		Purified Water Distribution Loop:	
		The Purified Water distribution loop has points of use with	
		pneumatic sanitary diaphragm valves. The temperature of water at	
		the point of use shall be room temperature and each point of use is	
		provided with pneumatic sanitary diaphragm valves.	
		8.3	Water for Injection System (WFI):
		The Water for Injection System will provide a total amount of WFI,	
		required in the production facility. WFI quality will correspond to	
		USP31.	
		Water Still is fed with Purified Water. The generated WFI is stored	
		in storage tank and supplied to different user points through a	
		distribution loop. WFI is maintained at a minimum temperature of	
		80°C to prevent the growth of microbes and to reduce the microbial	
		count. Sanitation of the system will be carried out with Pure Steam	
		(PS) obtained from a Pure Steam Generator fed with PW.	

	WFI Generator Still consisting of:	
	Four Evaporation columns with condenser.	
	Pre-heaters and coolers manufactured in AISI 316L stainless steel,	
	Ra (0.6 (m (240 grit) interior, with pickled and passivated active	
	surfaces.	
	Condenser, pre-heaters and cooling chamber are insulated and	
	protected by AISI 304 stainless steel sheet in satin-finish.	
	Break tank and pressurizing multi-stage pump made of AISI 316L	
	stainless steel with process valve and accessories.	
	Control instruments incorporated in one main control board provided	
	with synoptic panel and operator interface terminal to indicate the	
	correct working of the whole process and the eventual abnormal	
	operating conditions.	
	8.4	Storage and Distribution Loop:
	The storage tank for WFI has a capacity of 2000L & 1000L and is	
	constructed of AISI 316L,	
	Ra (0.6 (m (240grit) interior and mirror polished. The WFI	
	distribution loop piping has the following characteristics:	
	Elements	Description
	Piping material	0.065" wall thickness, 316L stainless steel
		ASTM A270
		Ra(0.6 (m (240 grit) inside, Ra (0.9 (m (150
		grit) outside finishing
	Valves	Diaphragm sanitary type, with tri-clamp
		connections.
	WFI circulation	AISI 316L stainless steel sanitary
	pump	Pure Steam sterilizing at 3 bar
		Simple mechanical seals tungsten carbide
		AISI 316L stainless steel sanitary section valve
		AISI 316L stainless steel bottom discharge
		valve.
		Sanitary manometer
		Temperature: 121°C
	8.5	Heat Exchanger Cooler:
	The distribution loop has user points where WFI will be used	
	directly at 85°C and 15 points where it will be used at 35°C. Each	
	of these cold points will require heat exchanger cooler for removal	
	of heat.	
	The cold points of use are provided with automatic valves and the	
	control panel for operating logic of the heat exchanger cooler. The	
	heat exchangers are manufactured in AISI 316L stainless steel with	
	double tube sheet and are provided with automatic temperature	
	regulation system and control panels.	
	The cylindrical parts of the heat exchangers are insulated by 30-mm	
	of mineral wool covered by AISI 304 stainless steel sheets in satin	
	finish.	
	8.6	Pure Steam Generating System (PS):
	The Pure Steam is supplied to autoclaves for product materials and	
	containers sterilization, and is also used for the SIP of different	
	equipments and tanks.	
	The pure steam generator is fed with Purified Water (USP) and its	
	operation is fully automatic ensuring that the pure steam pressure	
	remains constant while meeting all demands, as well as the feeding	
	water level in the steam generator would remain constant. The pure	
	steam generator comprises of two parallel bodies: Double tube sheet	
	heat exchanger and Evaporator/ Decontamination chamber. The plant	
	is manufactured in AISI 316L stainless steel, with pickled and	
	passivated active surfaces. Its Exchanger and Evaporator/	
	Decontamination chamber are insulated and protected by AISI 304	
	stainless steel sheet, satin-finished. All the control instruments	
	of the whole plant are incorporated in one main control board	
	provided with synoptic panel and operator interface terminal to	
	indicate the correct performance of the whole process and the	
	eventual abnormal operating conditions.	
	8.7	Pure Steam Distribution System:
	Design details of the PS distribution system are listed below:	
	Elements	Description
	Piping	0.065" wall thickness, 316L stainless steel ASTM
	material	A270
		Ra (0.6 (m inside (240 grit or better),
		Ra (0.9 (m outside (150 grit) finished.
	Valves	Diaphragm valves with forged body and end
		connections with 3-piece construction and all
		wetted parts of AISI 316L stainless steel. All
		other functional parts of AISI 316 stainless steel.
		Internal wetted surfaces roughness of Ra(0.6 (m
		(240 grit or better). Body design includes bottom
		entry stem with double seals, and single piece PTFE
		seats. The ball valve is in line repairable.
		Connections are tri-clamp or flange type.
	Fittings	0.065" wall, 316L stainless steel ASTM A270
		Ra (0.6 (m internal (240 grit), Ra (0.9 (m
		outside (150 grit) finished.
	Thermostatic	Self-adjusting, balanced pressure type steam trap
	Steam Trap	capable of operating close to saturated steam
		temperature. All wetted parts are AISI 316L
		stainless steel with self-draining. Connections are
		screwed or tri-clamped.
	Pressure	Stainless steel direct acting pressure reducing
	Reducing	valves provide accurate downstream pressure control

	Valves	with tight shut off at the set pressure.	
			Connections are screwed or flange.
		Pressure	All Pressure Indicators are sanitary
		Indicator	
		Materials and other general design considerations of PW, WFI and PS	
		systems:	
		All materials in contact with PW, WFI and PS are of stainless steel	
		in AISI 316L with internal surface mechanically polished (Ra	
		(0.6µm) to prevent deposit of particles, undesirable chemicals and	
		substances that could support microbial growth.	
		All parts of the systems are dead leg free design as far as	
		possible and will keep a slope of 1% with valve for drainage at the	
		lowest point of the piping system.	
		Orbital welding (certified tungsten inert gas) and sanitary clamp	
		fittings (Tri clamp) to be used for joining the entire system	
		components.	
		Conductivity and temperature are continuously monitored and	
		recorded to control the quality of the generated and circulated	
		water (both PW and WFI), 24 hours per day. TOC is periodically	
		measured off line also.	
		Care has been taken that the dead leg length should be less than 6D	
		(where D is the inner diameter of the pipe).	
		8.8	Compressed Air System:
		The compressed air should accomplish the ISO 8573-1 Class 1.4.1	
		standard:	
		Particle maximum size (µm) 0.1	
		Polluting solids maximum concentration (mg/m3) 0.1	
		Dew point temperature (°C) +3	
		Oil maximum concentration (mg/m3) 0.01	
		Filtration through activated carbon to remove oil vapor and	
		hydrocarbon odors	
		limiting the maximum oil remaining content to <0.003 mg/m3 @ 21°C.	
		The pressure of processed air shall be reduced to 2 bars and	
		filtered through hydrophobic filters of 0.2 µm before each point of	
		use. This filter shall be sterilized periodically with pure steam	
		or by autoclaving.	
		The compressed air for instruments shall be used at 6 ~ 7 bar	
		pressure.	
		The compressed air will be produced in the Production Plant and	
		would be supplied to different points of use in the Production.	
		9.	HVAC SYSTEM
		The air is drawn by the blower through 10 micron filter and the air is	
		passed through cooling coil and heating coil and then passes through 5	
		µm & 3 micron filter. Thus chilled air passes through the supply duct and	
		enters into the respective rooms. The air is further filtered to 0.3	
		µm HEPA filter supply terminal and distributed in the room.	
		The return air is sucked by the blower through the return air raiser	
		and the air is filtered upto 10 micron filtration and mixes with the	
		fresh air and process continues.	
		Whenever defumigation required the Air Handling Unit will convert from	
		recirculation to Once Through mode, this is done by motorized damper	
		Actuators. In normal mode fresh air VCD for defumigation purpose will	
		be in closed mode and exhaust damper of defumigation unit will be in	
		closed mode, only the return air damper of the AHU will be in open	
		mode.	
		When the AHU is to be converted to defumigation mode both the fresh air	
		VCD and Exhaust air VCD will be in opened. The return air will be	
		closed; this operation is done through motorized damper actuators.	
		9.1	Air Handling Unit Plenum:
		The Structure of the Air Handling Unit has been constructed with	
		extruded aluminum profile of 2mm thick with thermal break. The	
		panels of the Air Handling Unit are of Double Skin. The inner	
		skin of the panel is plain G.I sheet of thickness 0.6mm and the	
		outer skin is of 0.6mm powder coated. The panels are sandwiched	
		and insulated with 40mm thick poly urethane foam.	
		9.2	Blower & Motor:
		The Air Handling Unit consists of SISW backward curved blower	
		driven by motor.	
		The De fumigation Unit consists of SISW backward curved blower	
		driven by motor.	
		9.3	Mixing Chamber:
		The mixing chamber for return and fresh air is built with 2mm	
		thick extruded profile with thermal break. The air from the rooms	
		through the return duct and the fresh air from TFO will be mixed	
		in the mixing chamber	
		9.4	Filter Plenum:
		The filter plenum consists of filter frame made of GI sheet duly	
		powder coated. The plenum for filter is made up of 2mm thick	
		extruded aluminium profile with double skin doors. The filter	
		frame holds the filters. The Stage – 1 (EU4) is of 10 micron	
		filter and the Stage – 2 (EU6) is of 5 micron filter, the	
		velocity across the filters are 500 fpm In the Stage – 3 (EU9) is	
		of 3 micron filter, Stage –4 (EU13) there are supply terminal	
		HEPA filters of 0.3 micron which are kept in the terminal boxes,	
		the filtered air passes through the grills and evenly distributed	
		in to rooms. The velocity across the filters is 500 fpm. In Stage	
		– 5 (EU4) 10 micron filters are fixed in the return air terminal	
		boxes and the return air is sucked by the supply blower through	
		return ducts, the velocity across the filter is 500 fpm.The	
		Differential pressures across the filters will be monitored by	
		the Magnehelic gauges which are placed on the panels of the Air	

|| Handling Unit. |

9.5 Cooling Coil Section: |

|| The cooling coil section is built with 2mm thick aluminium |

|| structure with thermal break profile of 0.6 mm thick inner skin |

|| and 0.6mm thick outer skin which is powder coated. The cooling |

|| coil is placed inside the drain pan which is made up of SS-304 |

|| sheet with drain outlet to drain off the condensate water. The |

|| coil is made up of Copper tubes and aluminium fins. The coil |

|| header is of 60mm NB. The velocity across the coil is less than |

|| 500 fpm. |

9.6 Chilled Water Manifold: |

|| The chilled water manifold consists of isolation valves, 3 – way |

|| mixing valve with actuator, balancing valve, 'Y' strainer, |

|| temperature gauges & Pressure gauges. The chilled water is fed in |

|| to the chilled water coil through this manifold at temperature of |

|| 7°C with a pressure of 1.5 kg/cm² - 2 kg/cm². The temperature |

|| sensor senses the temperature of the return air and actuates the |

|| actuator of the 3 – way mixing valve. Whenever the temperature |

|| reaches the desired condition the 3 – way mixing valve closes and |

|| the chilled water is by passed. |

|| The Chilled Water manifold is insulated with 50mm thick Thermocol |

|| sections with 0.6mm thick aluminium cladding as vapour barrier. |

9.7 Heating Coil Section: |

|| The Heating Coil is made up of copper tube and aluminium for the |

|| velocity across the coil is less than 500 fpm. The heating coil |

|| is kept in drain pan of the Air Handling Unit. The header of the |

|| heating coil is of 35NB. The hot water to be given at a |

|| temperature of 50°C at the suction end of the coil. |

9.8 Hot Water Manifold: |

|| The Hot water manifold consists of isolation valves, 3 – way |

|| mixing valve with actuator, balancing valve, 'Y' strainer, |

|| temperature gauges & Pressure gauges. The Hot water is fed in to |

|| the chilled water coil through this manifold at temperature of |

|| 50°C with a pressure of 1.5 kg/cm² - 2 kg/cm². The Humidity |

|| sensor senses the humidity of the return air and actuates the |

|| actuator of the 3 – way mixing valve. Whenever the humidity |

|| reaches the desired condition the 3 – way mixing valve closes |

|| The Hot Water manifold is insulated with 50mm thick glass wool |

|| with 0.6mm thick aluminium cladding as vapour barrier. |

9.9 Ducting: |

|| The Supply Duct is connected to the Supply Air Volume Control |

|| Damper, which is insulated with Aluminium Nitrile Rubber duly |

|| pasted with Aluminium Foil that not to transfer any heat loses. |

|| The supply ducts are connected to the terminal boxes which are |

|| holding 0.3 micron terminal HEPA filter through which the |

|| filtered air enters in to the rooms. From the return air raisers |

|| the return air duct which is insulated with nitrile rubber which |

|| is prelaminated with aluminium foil not to transfer any heat |

|| loses. |

|| The Supply Ducting has been made from galvanized steel sheet of |

|| 275 GSM coating thickness. The branches and the termination to |

|| the terminals has been done by galvanized sheets only. The supply |

|| ducting has been insulated by 19mm Nitrile rubber which is pre |

|| insulated with aluminium foil as vapor barrier. |

|| The return ducting has been made from galvanized steel sheets of |

|| 275 GSM coating thickness. The branches & termination to the |

|| terminals has been done by galvanized sheets only. The return |

|| ducts are also insulated with Nitrile Rubber of 9mm thickness |

|| which is also prelaminated with aluminium foil. |

9.10 Volume Control Dampers : |

|| The unit is provided with Volume control dampers at the supply |

|| end of the duct and after the end of return duct. These control |

|| dampers are used for controlling the air flow. |

|| The outer casing for the Volume Control Dampers are manufactured |

|| by Galvanized Iron Sheets duly powder coated. The Damper Profiles |

|| used are extruded aluminium aerofoil type. The Velocity for |

|| supply air volume control damper is 1400 fpm, for return air |

|| volume control damper is 1000 fpm and for fresh air volume |

|| control damper is 500 fpm. |

9.11 Fire Dampers: |

|| The Fire Dampers which are kept in the return air duct are made |

|| up of Galvanized Iron Sheets. The type of fire dampers are |

|| fusible link type. When the fire exists in the room due to the |

|| heat generated by the fire the fusible link will be fused when |

|| the temperature reaches 70°C. When the fusible link fuses the |

|| fire damper automatically closes. |

9.12 Air Filtration System and Room's Pressure Level: |

|| The air filtration system was designed to guarantee the optimum |

|| working conditions for filters. It is composed of pre filters of |

|| 85% and 90% Efficiency ASHRAE (EU-3), fine filters of 95% |

|| Efficiency Dust Spot (EU-8) and HEPA terminal filters of 99.97% |

|| Efficiency DOP (EU-12). Pre and fine filters will be placed in |

|| all the AHU. |

|| HEPA filters have been placed in all the terminals at the clean |

|| rooms ceiling. |

|| To ensure prospective working of fresh air intake and exhaust air |

|| outlet system elements which are in contact with the external |

|| environment, common protective grills have been provided. |

|| The pressure gradients have been established as follows: |

|| 5 Pa between external non-classified rooms and external |

|| environment. |

|| 15 Pa between external clean rooms and non-classified rooms. |

	13 Pa between adjacent clean rooms of different grade.	
	Measures have been taken for continuous monitoring of Relative	
	Pressure Values in the critical working rooms.	
	9.13	Dry Bulb Temperature and Relative Humidity Control:
		Dry Bulb Temperature (DBT) and Relative Humidity (RH) ranges are
		as follows:
		Clean Rooms Working Condition: DBT: 22±3°C, NMT 55±5%
		Conventional Rooms Condition: DBT: 24±3°C, RH: Not Controlled
		Relative Humidity Control was conceived by means of a three-way
		electric valve to regulate the chilled water flow to be passed
		through Air Handling Units cooling coils. A humidifier device is
		placed in system 00 to provide the correct set parameters to the
		fresh air in dry conditions. Relative Humidity sensors are placed
		in the air return main branch of all the systems.
		Two-way and three-way electrical valves would be used to regulate
		the chilled water and industrial steam flows respectively, and in
		this manner control the Dry Bulb temperature in the AHU units
		through cooling and heating coils.
	10.	CLEANING /CAMPAIGN CHANGE
		The objectives of Good Manufacturing Practices (GMP) include the
		prevention of possible contamination and cross-contamination of
		pharmaceutical starting materials and products. Adequate cleaning
		procedures play an important role in preventing contamination and
		cross-contamination. Contamination can be through a variety of agents
		such as microbes, with previous products, residues of cleaning agents,
		airborne matter such as dust and particulates lubricants and ancillary
		material such as disinfectants. Validation of cleaning methods provides
		documented evidence that an approved cleaning procedure will provide
		clean equipment/clean premises suitable for use.
		Cleaning validation should be considered important and should be
		performed for equipments, sanitization procedures, garment laundering
		and area cleaning.
		There should be written SOPs detailing the cleaning process for
		equipment and apparatus. Cleaning procedures shall be validated.
		Cleaning procedures shall be based upon established and validated
		methods.
		At least three consecutive applications of the cleaning procedure should
		be performed and shown to be successful in order to prove that the
		method is validated.
		If automated procedures are utilized (Clean-In-Place: CIP),
		consideration should be given to monitoring the critical control points
		and the parameters with appropriate sensors and alarm points to ensure
		the process is highly controlled.
	11.	STORAGE FACILITIES
		Typically the different storage facilities include:
		Cold Rooms (2 to 80C) – For storage of Bulk, In process samples, filled
		vaccine vials, Intermediate product and Support Area solutions.
		Deep freezer (-200C) – For Storage of specific raw materials and
		solutions, Bulk (which require low temperature storage).
	12	SHIPPING VALIDATION
		The shipping procedure of final product bulk shall be validated in
		order to ensure product quality is unharmed during transportation.
		Shipping Validation is the measurement and/or confirmation that
		temperatures inside the shipping containers remain within the
		defined temperature range (for the specific vaccine/s being
		transported) for a period of 48 hours minimum. The shipping
		validation typically includes :
		Container Closure/Packaging Studies
		Temperatures within the container should be continuously monitored
		within an accuracy of ± 0.5°C; the sensors used for this purpose
		must not be allowed to influence the test in any way. A minimum of
		8 simultaneous temperature measurements are required for an
		insulated box. The ambient temperatures at which the insulated box
		is tested must remain within a tolerance of ± 1°C. Every packaging
		size shall be validated.
		Insulated boxes should be fully loaded with vaccine vials
		conditioned at +5°C ±3°C. Loads can be pre-fitted with
		thermocouples. Loading should be carried out as quickly as
		possible and the time taken for loading should be recorded for
		each container.
	13.	PERSONNEL
		Procedure for conducting training of personnel and their evaluation is
		laid down. Every personnel undergoes training session depending on the
		job to be performed, for up gradation of skills and knowledge. Every
		departmental Head has to ensure that each individual in his department
		has been imparted training, relevant to the activities performed by the
		individual. Quality assurance department shall be responsible for
		preparation of annual training plan and maintaining the documents
		related to training.
	14.	CALIBRATION
		All the measuring equipments, instruments and devices shall be
		calibrated on an established schedule as per the established standards,
		which shall be traceable to National Accreditation Testing & Calibration
		Laboratories, Russia or any other standards maintained at National or
		International Standards Laboratory.
		The calibration program shall be implemented by Quality Assurance
		Department in coordination with Maintenance and concerned departments.
	15.	VALIDATION STRATEGY
		The validation of the equipments / system to be used for the products
		manufactured or to be manufactured should be carried out as per
		validation strategy provided in the validation master plan. The
		validation strategy define the equipments to be validated , the

| parameters to be included for the validation , acceptance criteria for |
 | validation studies performed , frequency of the validation planned for |
 | the current year. The progress made based on this plan will be reviewed |
 | by the upper and mid management periodically. |
 | The Validation protocols should include the following criteria, but |
 | should not be limited to them. Other additional criteria could be |
 | included in the specific validation protocols, which will help to |
 | support the validation studies on case to case basis. |
15.1	Process Equipments in Bulk vaccine and fill finish Facility:	
15.1.1	Autoclave:	
		Bowie Dick test should be carried out to demonstrate the
		removal of air pockets and heat penetration. Three
		consecutive runs in the empty chamber should be carried out
		to demonstrate the uniform heat distribution using
		temperature probes/thermocouples and data loggers.
		Three consecutive runs should be carried out by placing
		temperature probes/ thermocouples within the load to find
		out the coolest points during the sterilization cycles.
		Three consecutive runs should be carried out by placing
		Biological Indicators (spore count 10-6) at the coolest
		points detected above along with the temperature probes/
		thermocouples to achieve 12 D lethality.
15.1.2	Dry Heat Sterilizer:	
		Three consecutive runs in the empty chamber should be
		carried out to demonstrate the heat distribution using
		temperature probes/ thermocouples with data loggers. Three
		consecutive runs should be carried out with temperature
		probes/ thermocouples with in the load to find out the
		coolest points with in the load. Three consecutive runs
		should be carried out with Endotoxin Indicators vials (with
		known amount of endotoxin units) along with temperature
		probes/ thermocouples placed at the above deducted coolest
		points with in the load should be carried out. There should
		be more than 3 log reduction in the exposed vials.
15.1.3	Incubator:	
		It should be demonstrated that the temperature is
		homogeneous inside the equipment, within acceptable limits.
		During performance qualification of incubators, temperature
		distribution study (Empty chamber), temperature distribution
		study (loaded chamber) and variation in temperature on stock
		replenishment should be performed.
15.1.4	Cold Room:	
		During performance qualification of cold room, temperature
		distribution study (empty cold room), temperature
		distribution study (loaded cold room) and variation in
		temperature on stock replenishment (three times) should be
		done.
15.1.5	Laminar Air Flow Cabinets and Laminar Air Flow Ceiling:	
		It should be demonstrated that Laminar Air Flow Cabinets and
		LAF Ceiling remains in continued compliance with current
		regulatory norms. The validation shall include DOP testing
		(installed filter leak test), air velocity testing, ACPH
		testing, Smoke profile and non viable particle counting,
		which shall be carried out biannually. In addition other
		parameters should be monitored as per frequency mentioned in
		the Environment Monitoring Program.
15.1.6	Chromatography System:	
		The chromatography system validation should be performed to
		ensure consistent performance of the system. Chromatographic
		profiles for a specific chromatography obtained repeatedly
		in each case show a reproducible pattern.
		Performance qualification of chromatography system should
		include the integrity testing of the column and system so as
		to check for any leakage due to loose connections , checking
		of the flow rate (with and without column) to check pressure
		drop and HETP of column.
15.1.7	Purified Water System:	
		Purified water system should comply with Specifications
		incorporating requirements of pharmaceutical waters as per
		the compendia. The Performance Qualification of purified
		water system shall be carried out in three phases.
		During phase-1, verification of the operating range of the
		system shall be done. This phase shall involve daily
		testing of pH, conductivity, TOC and microbiology from all
		user points for 21 consecutive days. After successful
		completion of phase-1 study, phase-2 study of purified
		water system shall be initiated.
		During phase-2, the demonstration of the system to produce
		desired quality of purified water consistently shall be
		done. This phase shall involve daily testing of pH,
		conductivity, TOC and microbiology from all user points for
		next 21 consecutive days. After successful completion of
		phase-2 study, phase-3 study of purified water system shall
		be initiated.
		During phase-3, potential seasonal variations affecting the
		water quality shall be checked. This phase shall involve
		testing of pH, conductivity, TOC and microbiology from all
		user points as per sampling plan for a period of one year.
		For new system and for existing system whenever any major
		change is undertaken which warrants the revalidation of the
		water system, the trending at each phase shall be performed
		to decide alert and action levels. The alert and action

|||level obtained from Phase I will be implemented for phase I
 |||II, alert and action levels obtained for phase II will be I
 |||used for Phase III. The alert and action level obtained I
 |||from phase III will be used for monitoring the quality of I
 |||the water for long term usage till any deviation is I
 |||noticed. I
 |||15.1.8|Water for Injection System: I
 |||Water for Injection system should comply with I
 |||Specifications incorporating requirements of pharmaceutical
 |||waters as per the compendia. The Performance Qualification I
 |||of water for injection system shall be carried out in three
 |||phases. I
 |||During phase-1, verification of the operating range of the I
 |||system shall be done. This phase shall involve daily I
 |||testing of pH, conductivity, TOC, microbiology and I
 |||endotoxin from all user points for 21 consecutive days. I
 |||After successful completion of Phase-1 study, Phase-2 study
 |||of water of injection system shall be initiated. I
 |||During phase-2, the demonstration of the system to produce I
 |||desired quality of water of injection consistently shall be
 |||done. This phase shall involve daily testing of pH, I
 |||conductivity, TOC, microbiology and endotoxin from all user
 |||points for next 21 consecutive days. After successful I
 |||completion of phase-2 study, phase-3 study of water of I
 |||injection system shall be initiated. I
 |||During phase-3, potential seasonal variations affecting the
 |||water quality shall be checked. This phase shall involve I
 |||testing of pH, conductivity, TOC, microbiology and I
 |||endotoxin from all user points as per sampling plan for a I
 |||period of one year. I
 |||For new system and for existing system whenever any major I
 |||change is undertaken which warrants the revalidation of the
 |||water system, the trending at each phase shall be performed
 |||to decide alert and action levels. The alert and action I
 |||level obtained from Phase I will be implemented for phase I
 |||II, alert and action levels obtained for phase II will be I
 |||used for Phase III. The alert and action level obtained I
 |||from phase III will be used for monitoring the quality of I
 |||the water for long term usage till any deviation is I
 |||noticed. I
 |||15.1.9|Pure Steam System: I
 |||Pure Steam System should comply with Specifications I
 |||incorporating requirements of the compendia. The I
 |||Performance Qualification of pure steam system shall be I
 |||carried out in three phases. I
 |||During phase-1, verification of the operating range of the I
 |||system shall be done. This phase shall involve daily I
 |||testing of pH, conductivity, TOC, microbiology and I
 |||endotoxin from all user points for 21 consecutive days. I
 |||After successful completion of phase-1 study, phase-2 study
 |||of water of injection system shall be initiated. I
 |||During phase-2, the demonstration of the system to produce I
 |||desired quality of pure steam consistently shall be done. I
 |||This phase shall involve daily testing of pH, conductivity,
 |||TOC, microbiology and endotoxin from all user points for I
 |||next 21 consecutive days. After successful completion of I
 |||phase-2 study, phase-3 study of pure steam system shall be I
 |||initiated. I
 |||During phase-3, potential seasonal variations affecting the
 |||quality of pure steam shall be checked. This phase shall I
 |||involve testing of pH, conductivity, TOC, microbiology and I
 |||endotoxin from all user points as per sampling plan for a I
 |||period of one year. I
 |||For new system and for existing system whenever any major I
 |||change is undertaken which warrants the revalidation of the
 |||water system, the trending at each phase shall be performed
 |||to decide alert and action levels. The alert and action I
 |||level obtained from Phase I will be implemented for phase I
 |||II, alert and action levels obtained for phase II will be I
 |||used for Phase III. The alert and action level obtained I
 |||from phase III will be used for monitoring the quality of I
 |||the water for long term usage till any deviation is I
 |||noticed. I
 |||15.1.1|Compressed Air System: I
 |||0 I
 |||Compressed Air System should comply with specifications. I
 |||The performance qualification of compressed air system I
 |||shall be carried out for different parameters monitored I
 |||daily for pressure, microbiology and pressure dew point for
 |||a period of 21 consecutive days. I
 |||15.1.1|Deep Freezer: I
 |||1 I
 |||During performance qualification of deep freezer, I
 |||temperature distribution study for 7 days, variation in I
 |||temperature on opening the door for 30 seconds, three times
 |||for the door shall be done. I
 |||15.1.1|Vial Washer I
 |||2 I
 |||Operational Qualification I
 |||1. The system shall retain less than 100 ml of WFI after 15
 |||minutes of drainage (all normal drain valves open. I
 |||2. The washing machine can operate properly throughout its I

			operating range.
			3. Vial washing room is free of condensate and visible
			water vapor from the machine throughout its operating
			range.
			4. Glass breakage through the machine shall not exceed 1%
			of the number of bottles processed.
			5. Water at the last rinse station shall contain not more
			than 10% of USP SVP particulate matter limits [NMT 1000
			particles ~10 microns and NMT 100 particles >25 microns per
			100 ml].
			6. Air used to blow vials free of residual moisture shall
			meet FS 209 D Class 100particulate requirements.
			Performance Qualification
			1. The washing machine shall discharge vials containing
			less than 10% of the USP Large Volume Parenteral
			particulate matter limit [NMT 500 particles > 10microns and
			NMT 50 particles >25 microns per 100 ml] for bottles of 100
			ml and 200 ml. The number of particulates present in vials
			shall be reduced by passage through the machine.
			2. Any residual water in the washer after draining shall
			contain less than 0.25 EU/ml of residual endotoxin.
			4. Vials washed in the machine shall contain less than 1
			EU/vial after processing. There shall be no increase in
			vial endotoxin content as a result of passage through the
			machine.
		15.1.1	Depyrogenation Tunnel
		3	
			Installation Qualification
			1. The HEPA filters installed in the machine shall be
			certified to retain a minimum of 99.97% of particles larger
			than 0.5 micrometer.
			Operational Qualification
			1. Air within the tunnel shall meet FS 209 D Class 100
			requirements at operating temperature [circa 330°C].
			2. The tunnel shall be able to operate properly while
			maintaining a maximum 0.10" H2O pressure differential from
			discharge to indeed.
			3. The dry heat tunnel shall operate properly throughout
			its operating range of 0 to 100, 100 and 200 ml vials per
			minute.
			4. The tunnel shall maintain a distribution temperature
			range [belt edge to belt edge] of not more than 30°C
			throughout its length when empty.
			5. The tunnel shall maintain a distribution temperature
			range [belt edge to belt edge] of not more than 25°C
			throughout its length when full of 200 ml vials.
			Performance Qualification
			1. The tunnel shall be capable of a minimum 3.0 log
			reduction in endotoxin content for all vial sizes. The
			minimum challenge level shall be 1000 EU/vial
			[recoverable]]
			2. There shall be no increase in vial particulate matter
			content as a result of passage through the tunnel.
			3. The tunnel shall maintain a penetration temperature
			range [belt edge to belt edge] of not more than 20°C
			throughout its length when full of 200 ml vials.
		15.1.1	Vial Filler/Stoppering
		4	
			Operational qualification
			1. The vial filler / stoppering machine shall operate
			properly throughout its operating range.
			2. There shall be no increase in vial particulate matter
			content as a result of passage through the filling machine
			[without liquid addition].
			3. There shall be no more than a 10% increase in stopper
			particulate matter level as a result of passage through the
			stoppering bowl and discharge chute of the machine.
			Performance Qualification
			1. The machine shall maintain weight control within + 3% of
			set point for all fill weights. Weight control shall be
			demonstrated at 10, 50 and 100% of operating speed for all
			container sizes.
			2. The machine shall pass a series of media fill challenges
			as specified in the aseptic processing section of this
			document.
		15.1.1	Vial Capper
		5	
			Performance Qualification
			1. Vials of each size shall be tested for adequacy of
			container-closure integrity by immersion in a dye bath and
			pressurization at 15 psig for thirty ~30] minutes. Testing
			shall be performed at the extremes of seal pressure and
			operating speed.
		15.1.1	Vial Inspection
		6	
			Operational Qualification
			1. The system shall correctly reject the proper vial as
			indicated by the operators console selection.
			Reject vials shall not be discharged from the inspection
			machine.
			2. The inspection machine shall operate properly throughout
			its operating range.

			Performance Qualification
			Specific KNAPP test shall be performed on the inspection
			machine during PQ stage.
		15.1.1	Labeller/Label Verification
		7	
			Operational Qualification
			1. The inspection equipment shall demonstrate the ability
			to correctly identify all bar-codes imprinted on the label
			in the range of sizes to be utilized.
			Performance Qualification
			1. The labeller shall be capable of labeling with
			appropriate printing with positional accuracy of + 1/16".
			Printing shall be completed on each label with a positional
			accuracy of + 1 /32".
			2. The labeling and inspection machine shall operate
			properly throughout its operating range.
			3. The machine shall stop on the detection of a vial
			without a label or a label without a proper bar-code prior
			to the vial leaving the machine.
		15.1.1	Stopper Processor
		8	
			Operational Qualification
			1. At steady state [121°C] the unit shall exhibit a heat
			distribution of + 0.5°C about the set point temperature.
			2. Sterile air filters must pass manufacturers filter
			integrity test after completion of in-situ sterilization.
			3. Chamber vacuum leak rate shall not exceed 1 mm HG/minute
			when tested at 50 mm HG.
			Performance Qualification
			1. Biological challenge with Bacillus stearothermophilus
			spore strips [106 spores/strip] shall be inactivated in all
			locations when exposed to a minimum Fo of 15 minutes.
			2. The stopper washing machine shall discharge stoppers
			with less than 20% of the USP Large Volume Parenteral
			particulate matter limit [NMT 1000 particles > 10 microns
			and NMT 100 particles >25 microns per stopper] for stoppers
			to be utilized on vials of 100 and 200 ml. The number of
			particulates present on the stoppers shall be reduced by
			passage through the machine.
			3. Stoppers washed in the machine shall contain less than
			0.25 EU/stopper after processing. There shall be no
			increase in stopper endotoxin content as a result of
			passage- through the machine.
			4. The stopper washing machine shall be capable of a
			minimum 3.0 log reduction in endotoxin content for all
			stopper formulations. The minimum challenge. level shall be
			1000 EU/vial [recoverable].
		15.1.1	HVAC
		9	
			Installation Qualification
			1. Verification of cleaning of all ducts [supply and
			return] in the system.
			2. Confirmation of HVAC system [including all filters]
			conformance to design specifications and construction
			drawings.
			3. Review of documentation on air system testing to
			include: balancing reports, supply and return air
			velocities, air distribution, and air changes per hour.
			4. Initial cleaning of each room shall have been completed
			in accordance with the expected routine SOP.
			Operational Qualification
			1. Air changes per hour, pressure differentials between
			rooms, temperature and relative humidity must conform to
			the design criteria.
			2. Air velocities, pressures and temperatures throughout
			each HVAC system shall conform to the design
			specifications.
		HVAC System [Class 1,000, 10,000 and 100,000]	
			The criteria established in this section are for use during
			the PQ of the systems, Final limits will be developed upon
			a review of the PQ environmental monitoring results.
		Installation Qualification	
			1. Verification of cleaning of all ducts [supply and
			return] in each system.
			2. Confirmation of HVAC system [including all filters]
			conformance to design specifications are construction
			drawings.
			3. Review of documentation on air system testing to
			include: room pressurization, damper adjustments, balancing
			reports, supply and return air velocities, air distribution
			[smoke tests or equivalent], air changes per hour, and
			differential pressures.
			4. Documentation of duct leak testing for all controlled
			environment HVAC systems [supply and return ducts].
			5. Initial sanitization of each room shall have been
			completed in accordance with the expected routine SOP.
			6. All HEPA filter installations shall have been tested
			with DOP [or an equivalent challenge material] to confirm
			their integrity.
		Operational Qualification	
			1. Design criteria for air changes per hour, air flow
			patterns pressure differentials between rooms, temperature

		and relative humidity must conform.
		2. Air velocities, pressures and temperatures throughout
		each HVAC system shall conform to the design
		specifications.
		3. Airborne particulate testing performed under static [no
		equipment in operation, no components present,
		environmental monitoring personnel only] conditions shall
		be not more than 10% of the applicable FS 209 D particulate
		requirement for each controlled environment [i.e. Class 100
		must demonstrate less than 10 particles larger than 0.5
		microns per cubic foot of air sampled]. This monitoring
		shall be performed daily for a period of ten [10] days.
		4. Microbial testing under static conditions shall
		demonstrate an average microbial content of not more than
		25% of the proposed USP microbial levels for each
		controlled environment. No individual sample shall exceed
		75% of the proposed USP microbial limits. This monitoring
		shall be performed daily for a period of ten [10] days.
	Performance Qualification	
		1. Air flow patterns, pressure differentials between rooms,
		temperature and relative humidity must conform to the
		design criteria for load conditions.
		2. Airborne particulate testing performed under routine
		operation [equipment in operation, components present,
		normal complement of personnel] conditions shall be not
		more than 25% of the applicable FS 209 D particulate
		requirement for each controlled environment [i.e. Class 100
		must demonstrate less than 25 particles larger than 0.5
		micron per cubic foot of air sampled]. Particulate counts
		shall be taken in the absence of ZZZ powder. Data shall be
		accumulated for a period of ninety (90) days before the
		start of filling operations.
		3. Microbial testing under routine operating conditions
		shall demonstrate an average microbial content of not more
		than 75% of the proposed USP microbial levels for each
		controlled environment. No individual sample shall exceed
		200% of the proposed USP Microbial Limits. Microbial
		sampling will utilize neutralizing agents to assure that
		the results are valid. Data shall be accumulated for a
		period of ninety (90) days before the start of operations.
	15.1.2	Fixed Environmental Monitoring System
	0	
		Operational Qualification
		1. Particle measurement equipment shall be calibrated in
		accordance with manufacturer's recommendations prior to use
		in validation studies.
	15.1.2	Portable Environmental Monitoring Units
	1	
		Operational Qualification
		1. Particle measurement equipment shall be calibrated in
		accordance with manufacturer's recommendations prior to use
		in validation studies.
	15.1.2	Building Monitoring System
	2	
		Performance Qualification
		1. All data recorded by the Building Monitoring system
		shall be within +2% of independent measurements taken
		during the validation of the individual systems. Critical
		measurements shall be within +0.5% of the value reported by
		the local device.
		2. All alarms reportable by the Building Monitoring System
		shall be activated within 5 seconds of that alarm being
		reported by the local device.
	15.1.2	Plant Steam
	3	
		Installation Qualification
		1. Verification of hydrostatic testing, and cleaning of the
		distribution system.
		2. Confirmation of system conformance to design
		specifications and construction drawings.
	15.1.2	Gowning Procedures
	4	
		Operational Qualification
		1. The sterilization of all uniform components shall be
		validated [either on-site or at the vendors location as
		appropriate.
		2. Individual shall demonstrate their ability to do sterile
		gowns without undue difficulty. Each individual shall be
		monitored for their ability to properly put on sterile
		gowns by being subjected to microbial monitoring.
		Individuals shall demonstrate on three separate occasions
		that they can meet the USP proposed microbial limits for
		gown surfaces.
	15.1.2	Media Fills
	5	
		Performance Qualification
		A minimum of three process simulation tests will be
		conducted for each different vial size produced in the
		sterile powder facility. Initially, the process simulation
		tests should bracket the fill volumes in each size
		container with two maximum fills and one minimum fill. A
		minimum of three thousand (3,000) vials per test must be

			run with a failure rate of less than 0.1%. Any
			micro-organisms found must be identified to the extent
			possible. The media must be capable of supporting microbial
			growth for low numbers (less than one hundred [100]
			organisms per vial) of the USP growth promotion organisms.
			The medium to be utilized is Soybean-Casein Digest Medium
			(SCDM) [also known as Trypticase Soy Broth (TSB)]. The
			powder hopper shall be filled with a radiation sterilized
			excipient. The test units are to be incubated at 30-35°C
			for fourteen (14) days in an inverted position. During the
			course of the media fill process, environmental monitoring
			of viable and non-viable particulates in the environmental
			air should be performed. Personnel and surface monitoring
			of critical locations is also required. The monitoring
			should include the set-up of the line prior to the start of
			the fill.
16	DEVIATION		
		Any discrepancy during a validation exercise must be analyzed to	
		identify the origin of the discrepancy. Results and parameters out	
		of specified limits constitute a deviation.	
		Any incident regarding deviation from the written procedures shall	
		be documented. Investigations shall be carried out in case of	
		critical discrepancies which may affect the quality of the product.	
		Concerned departmental personnel along with personnel from Quality	
		Assurance department shall perform investigation needed to	
		establish if the deviation has an impact on the quality of the	
		product and validation status, as per established procedures	
17	CHANGE CONTROL		
		Written procedures should be in place to describe the actions to be	
		taken if a change is proposed to a design process equipment,	
		starting material, product component, process environment (or	
		site), method of production or testing or any other change that may	
		affect product quality or reproducibility of the process. Change	
		control procedures should ensure that sufficient supporting data	
		are generated to demonstrate that the revised process will result	
		in a product of the desired quality, consistent with the approved	
		specifications. The potential impact of the proposed change on the	
		quality of the product shall be evaluated and the level of testing,	
		validation, and documentation needed to justify changes to a	
		validated process shall be determined. Significant changes to	
		process need appropriate supporting data obtained through	
		revalidation. Through the provisions of this procedure of change	
		control a proposed change shall be formally requested, documented,	
		evaluated and approved internally on its scientific merit and	
		appropriate external authorization shall be sought for those that	
		have impact on regulatory submissions.	
18	REVALIDATION		
		Facilities, systems, equipment and processes, including cleaning,	
		should be periodically evaluated to confirm that they remain valid.	
		Where no significant changes have been made to the validated	
		status, a review with evidence that facilities, systems, equipment	
		and processes meet the prescribed requirements should be performed.	
		It shall be done in accordance with a defined revalidation program.	
		It has to be performed in the following cases:	
		Periodic revalidation: It is performed to guarantee that there has	
		been no drift in the equipment/system, process performance due to	
		passage of time. Periodic revalidation is performed even if no	
		changes have been implemented.	
		For product contact equipments/system the data of regular	
		production batches shall be compiled and shall hold good for	
		periodic revalidation.	
		All the equipments/systems shall be revalidated before starting	
		regular batch production after production shutdowns/breaks.	
		Revalidation due to changes: It is performed when a change in the	
		equipment/system, procedure, quality of the product and /or product	
		characteristics, process or facility is implemented.	
		Changes requiring revalidation may include:	
		Changes in starting materials,	
		Change in the location/site,	
		Changes of primary packaging material,	
		Changes in the manufacturing process,	
		Changes in the equipment*,	
		Relocation of equipment to other vaccine block / facility,	
		Any change in the support system,	
		Any change in the facility	
		*Changes of equipment which involve the replacement of equipment on	
		a "like-for-like" basis would not normally require a revalidation.	
		For example, installation of a new centrifugal pump to replace an	
		older model would not necessarily require revalidation. The extent	
		of revalidation will depend on the nature and significance of the	
		change(s). The Revalidation program shall be approved by the Heads	
		of Production, Quality Control, Maintenance and Quality	
		Assurance and the revalidation activity shall be performed in	
		accordance with approved protocols.	
		List of equipment/system that require periodic revalidation	
		Equipment/System Description	Revalidation
		Frequency	
		HVAC System and LAF cabinet/ Biosafety cabinet,	Biannually
		LAF ceiling, Powder containment station, Dynamic	
		pass box	
		Purified Water system	Annually

		Water for Injection system	Annually
		Pure Steam system	Annually
		Compressed Air Generation System	Annually
		Double Door Autoclave	Annually
		Dry Heat Sterilizer	Annually
		Cold Room	Annually
		Area Cleaning of Classified Areas	Three years
		Decontamination Autoclave	Annually
		Deep Freezer	Annually
		Filling/ stoppering / sealing machine	Annually
		Lyophilizer	Annually

	19	ABBREVIATIONS	
		AHU	: Air Handling Unit
		BMS	: Building Management System
		CIP	: Cleaning In Place
		cGMP	: current Good Manufacturing Practice
		DCS	: Digital Control System
		DHS	: Dry Heat Sterilizer
		DSP	: Downstream Processing
		DQ	: Design Qualification
		FAT	: Factory Acceptance Test
		FMEA	: Failure Mode Effect Analysis
		ETP	: Effluent Treatment Plant
		EU	: European Union
		GEP	: Good Engineering Practices
		GLP	: Good Laboratory Practices
		ICH	:International Conference on Harmonization
		IPQC	:In Process Quality Control
		IQ	: Installation Qualification
		OQ	: Operational Qualification
		PIC	: Pharmaceutical Inspection Convention
		PIC/S	: Pharmaceutical Inspection Co-operation scheme
		PQ	: Performance Qualification
		PDA	: Parentral Drug Association
		SCADA	: Supervisory Control And Data Acquisition
		SAT	: Site Acceptance Test
		SLIA	: System Level Impact Assessment
		SOP	: Standard Operating Procedure
		SVC	: Specific Validation Commission
		URS	:User Requirement Specification
		USFDA	:United States Food and Drug Administration
		USP	:Upstream Processing
		VMP	: Validation Master Plan
		WHO	: World Health Organization

20	REFERENCES		
		20.1	ISPE Pharmaceutical Engineering Guide for New and Renovated
		Facilities. Volume 5, Commissioning and Qualification (First	
		Edition, March 2001)	
		20.2	World Health Organization (WHO TRS 897, 2000)
		20.3	US Food and Drug Administration
		20.4	Pharmaceutical Inspection Convention –PIC/S , 3 August'2001
		20.5	Parentral Drug Association -PDA
		20.6	WHO recommendation on Environmental Monitoring of Clean Rooms in
		Vaccine Manufacturing Facilities 2012	
		20.7	WHO good manufacturing practices for active pharmaceutical
		Ingredients WHO-TRS 957 Annex 2 (2010)	
		20.8	WHO good manufacturing practices for pharmaceutical products: main
		principles WHO-TRS 961 Annex 3 (2011)	

|21. |HISTORY |

|| |

|22. |ANNEXURES |

|| |

Catalogue of Questions on GMP-FMEA

End of Risk Analysis

Description of Process/product

Determination of Potential Failure Implications

GMP- FMEA Report

Risk Priority Number

RPN =O X S X D

Find

Probability of Occurrence (O)

Severity of the Failures (S)

Probability of Detection (D)

Critical System

Collection of Process Information

Determination of Potential Failure Modes and Causes

No

Recommend Corrective Action & Initiate Measures

RPN<RPN Min/Max

User Requirement

Functional Design
(i.e. How as Schematic)

Detail Design
(i.e. How to make)

Operational
Qualification

Performance
Qualification

PQ Test Plan

OQ Test Plan

IQ Test Plan

Implementation

Impact Assessment

Design Development

Installation Qualification

RM:

- Wt of each empty box (PUFF Lined)
- Boxes acclimatized at vaccine storage temperature by placing at 5+ 30 C
- Load the bottles containing placebo in the PUFF lined box, as quickly as possible (15 Minutes)
- Pack the bottles with Soft Ice pack, Frozen soft Ice packs and then Hard Ice packs
- Number of Ice packs shall be as per the table
- Place 3 thermocouples in each box and 1 thermocouple in the environment wherever the study shall be carried out i.e. incubator for + 43 °C and deep freezer for -5°C. Each thermocouple should be positioned to a distance of 2.5/3.0 cm from the nearest ice pack
- Record the temperature at an interval of 10 minutes from thermocouples and every 2 Hours from digital display of incubator / deep freezer for 48 hours
- Place the packed boxes in the incubator at + 43 oC and in a deep freezer at -5 oC
- Carry out 3 Runs for each packing

[1. |Purpose: |

	The purpose is to validate the temperature during shipping of ARM /
	bulk vaccine, transported at 5 + 3°C so as to maintain the desired
	cold chain conditions during transportation.

[2. |Scope: |

	2.1	Applicable to the ARM /bulk vaccine (that comes under class C as
	per WHO Guidelines on the packaging and shipping of vaccines	
	WHO/IVB/R/05.23 December 2005) to be transported at the	
	temperature of 5 + 3°C.	

[4. |Safety considerations: |

	4.1	The ARM / bulk vaccine is to be maintained at recommended
	temperature at 5 + 3 °C. The ARM / bulk vaccine should not	
	freeze.	

| |4.2 |The containers of the ARM / bulk vaccine should be properly |
| |packed |

| |to avoid any breakage. |

| |4.3 |Ensure proper connections of thermocouples and their |
| |positioning. |

[5. |Equipments: |

| |5.1 |Data logger with thermocouples (accuracy ± 0.5°C) |

| |5.2 |BOD Incubator and Deep Freezer |

[6. |Materials and reagents: |

| |6.1 |Materials: |

| |6.1.1 |Glass bottles of appropriate capacity (20L, 10L, 5L, 2L |
| |and 1L)containing Placebo |

| |6.1.2 |PUF lined corrugated boxes (to house 10L/20L glass |
| |bottle) |

| |6.1.3 |PUF lined corrugated boxes (to house 1L/2L /5L glass |
| |bottle) |

| |6.1.4 |Ice packs hard / soft (stored at 5± 3°C and -20°C) |

| |6.1.5 |Polypropylene caps for sealing bottles |

| |6.1.6 |Thermocol sheets for immobilization |

| |6.1.7 |Adhesive tape |

| |6.2 |Reagents: |

| |6.2.1 |3% Alhydrogel |

| |6.2.2 |Di-sodium Hydrogen phosphate anhydrous |

| |6.2.3 |Sodium di-hydrogen phosphate dehydrate |

| |6.2.4 |Sodium Chloride |

| |6.2.5 |Thiomersal |

| |6.2.6 |Water for Injection |

[7. |Preparation of solutions: |

| |7.1 |Preparation of Placebo (50L): |

| |7.1.1|Take 3.0L of 3% alhydrogel in a 50 L glass bottle and add|
| |27 L of water for injection. |

| |7.1.2|Weigh Di-sodium Hydrogen phosphate anhydrous = 56gm, |
| |Sodium di-hydrogen phosphate dehydrate = 62 gm and Sodium|

| |Chloride = 400 gm. |

| |Dissolve the above weighed chemicals in 22L of water for |
| |injection. |

| |7.1.3|Weigh 20 gm Thiomersal and dissolve in 1.0 L of water for|
| |injection. |

| |7.1.4|Add 22L of phosphate buffer and 1L of Thiomersal |

| |solutions to 27L of alhydrogel in 50L glass bottle and |

| |homogenize by manual shaking. |

[8. |Procedure: |

| |8.1|Validation Study Plan: |

| |The study shall be carried out by keeping the bottles filled |

| |with Placebo (containing all the constituents of the vaccine |

| |except the ARM, having concentration equivalent to the |

| |concentration present in the vaccines) in incubator and deep |

| |freezer at + 43 °C and -5°C respectively, for a period of 48 |
| |hours. |

| |For monitoring of temperature 03 thermocouples (as per |

| |annexure-1) connected to the data logger shall be placed in |

| |side the PUF lined corrugated box and 01 thermocouple in the |

| |environment wherever the study shall be carried out i.e. |

| |incubator for + 43 °C and deep freezer for -5°C.The above study|

| |shall be repeated 03 times. These temperatures (+43 °C and |

| |-5°C) are representative of worst case environment. The |

| |temperature monitoring from the display of incubator and deep |

|| |freezer shall be done after 02 hours. |

||8.2 |Take appropriate capacity bottles containing Placebo properly |

|| |capped and sealed. |

||8.3 |Store the Placebo bottles at 5°C + 3°C for acclimatization and |

|| |Ice packs (Hard/Soft) at -20°C overnight. |

||8.4 |Take the packaging materials of required specifications, which |

|| |are to be validated and make a record of these. |

||8.5 |Pack the bottles in the boxes as per the standard packaging |

|| |procedure(FR/012) as follows: |

||8.5.1 |Glass bottles of pack size 20.0 and 10.0 L |

|| | |Check the weight of each empty box and load the 20 L and|

|| | |10 L bottles containing Placebo individually in the PUF |

|| | |Lined Corrugated boxes which shall be placed at 5 + 3°C |

|| | |to get acclimatize to vaccine storage temperature. |

|| | |Pack the bottles with the soft and hard ice packs. The |

|| | |soft ice packs of 5 + 3°C shall be placed adjacent to |

|| | |the bottle followed by frozen soft ice packs and hard |

|| | |ice packs. Immobilize the bottle by filling the |

|| | |remaining space, if any, with the thermocol sheets. |

|| | |Ice packs used for packing (Box Size) |

|| | |Number of Ice Packs |

|| | |

|| | |20L bottle |

|| | |Soft: 20-22no. |

|| | |Hard: 22-24 no. |

|| | |

|| | |10 L Bottle |

|| | |Soft: 16-18no. |

|| | |Hard: 18-20 no. |

|| | |

|| | |5 L Bottle |

|| | |Soft: 14-16no. |

|| | |Hard: 16-18 no. |

|| | |

|| | |2L Bottle |

|| | |Soft: 12-14no. |

|| | |Hard: 14-16 no. |

|| | |

|| | |1 L Bottle |

|| | |Soft: 10-12no. |

|| | |Hard: 08-10 no. |

|| | |

|| | |Check the weight of the each loaded box. Loading should |

|| | |be carried out as quickly as possible (less than 15 min)|

|| | |and the time for loading should be recorded for each |

|| | |container. |

|| | |Place 3 thermocouples (as per annexure-1) in each box |

|| | |through the seal taking care not to affect the quality |

|| | |of the seal and one thermocouple in the environment |

|| | |wherever the study shall be carried out i.e. incubator |

|| | |for + 43 °C and deep freezer for -5°C. Each thermocouple|

|| | |should be positioned to a distance of 2.5/3.0 cm from |

|| | |the nearest ice pack surface. |

|| | |Close the individual corrugated box with its lid and |

|| | |seal the box with the adhesive tape. |

|| | |Start the data logger to record the temperature at an |

|| | |interval of 10 minutes for 48 hours. Also record the |

|| | |temperature from digital display of incubator and deep |

|| | |freezer at every 2 Hours interval. |

|| | |Place the packed boxes containing in the incubator at + |

|| | |43 oC and in a deep freezer at -5 oC. |

|| | |Carry out 3 Runs for each packing by repeating 8.5.1.1 |

|| | |to 8.5.1.7. |

Annexure 1

Location of temperature sensors inside PUF lined corrugated box

a) PUF lined corrugated box: top view

[[pic] |

b) PUF lined corrugated box: top view

[[pic] |

Annexure 2

Positioning of Ice packs inside PUF lined corrugated box

[pic]

REPORT

1. EXECUTIVE SUMMARY OF THE STUDY

The validation program has been carried out to validate the packaging method for the shipping of ARM/Bulk vaccine (which comes under class C as per WHO Guidelines on the packaging and shipping of vaccines WHO/IVB/R/05.23 December 2005) transported at temperature 5 + 3°C so as to maintain the desired cold chain conditions during transportation. The study shall be carried out by keeping the Placebo (containing all the constituents of the vaccine except the active raw material having concentration equivalent to the concentration present in the vaccines) in incubator and deep freezer at + 43 °C and -5°C respectively, for a period of 48 hours. For monitoring of temperature three thermocouples connected to the data logger shall be placed in side the PUF lined corrugated box and one thermocouple in the environment wherever the study shall be carried out i.e. incubator for + 43 °C and deep freezer for -5°C. The above study shall be repeated three times. These temperatures (+43 °C and -5°C) are

representative of worst case environment. The temperature monitoring from the display of incubator and deep freezer shall be done after two hours. Two types of PUF lined corrugated boxes were used in the validation study were:

- PUF lined corrugated box to house 10L/20L glass bottle (PMS.18000)
- PUF lined corrugated box to house 1L/2L /5L glass bottle (PMS.18004)

7. TESTS PERFORMED & ACCEPTANCE CRITERIA CONSIDERED:
[Sr. No |Tests Performed |Acceptance criteria |
[1. |The packed box containing Placebo |The warmest temperature in the |
| |placed in environmental |packed box containing Placebo does|
| |temperature of 43 ±1 oC for a |not rise above |
| |period 48 hours. |5 ± 3oC. |
[2. |The packed box containing Placebo |The coolest temperature in the |
| |placed in environmental |packed box containing Placebo |
| |temperature of -5 ± 1 oC for a |does not fall below |
| |period of 48 hours. |+ 2oC |

8. STATISTICAL ANALYSIS REPORT (if any) :
8.1 Result of environmental temperature study at +43 °C:
Temperature recording interval: 10 minutes

Cycle No. 1 (04.03.2007 to 06.03.2007)
[Package details |Empty Box weight|Packed |No. of soft |No. of hard |Time taken |
	(Kg)	box	ice packs	ice packs	for packing
		weight	used	used	
		(Kg)			
[Box with 20L glass| | | | |
[bottle | | | | |
[Box with 10L glass| | | | |
[bottle | | | | |
[Box with 5L | | | | |
[glass bottle | | | | |
[Box with 2L | | | | |
[glass bottle | | | | |
[Box with 1L | | | | |
[glass bottle | | | | |

[Position of Data|Thermocouple |Minimum |Maximum |Temp. |Average |
logger	(Probe) No.	Temp.(°C)	Temp. (°C)	reading	Temp. (°C)
			at 48 hrs		
			(°C)		
[In Incubator	Probe # 16			43.4	
[In Box with 20L	Probe # 1			7.1	
[glass bottle					
	Probe # 2			7.1	
	Probe # 3			7.2	
[In Box with 10L	Probe # 4			7.1	
[glass bottle					
	Probe # 5			7.2	
	Probe # 6			7.1	
[In Box with 5L	Probe # 7			7.1	
[glass bottle					
	Probe # 8			7.1	
	Probe # 9			7.2	
[In Box with 2L	Probe # 10			7.1	
[glass bottle					
	Probe # 11			7.2	
	Probe # 12			7.1	
[In Box with 1L	Probe # 13			7.2	
[glass bottle					
	Probe # 14			7.1	
	Probe # 15			7.1	
[Digital display |--- | | | |

Cycle No. 2 (06.03.2007 to 08.03.2007)
Details as above

Cycle No. 3 (08.03.2007 to 10.03.2007)
Details as above

8.2 Results of environmental temperature study at -5°C:
Temperature recording interval: 10 minutes

Cycle No. 1(11.04.2007 to 13.04.2007)
[Package |Empty Box |Packed box |No. of soft |No. of hard |Time taken |
|details |weight (Kg) |weight (Kg) |ice packs |ice packs |for packing |
| | |used |used | |
[Box with 20L | | | | |
[glass bottle | | | | |
[Box with 10L | | | | |
[glass bottle | | | | |
[Box with 5L | | | | |
[glass bottle | | | | |
[Box with 2L | | | | |
[glass bottle | | | | |
[Box with 1L | | | | |
[glass bottle | | | | |

[Position of |Thermocouple |Minimum |Maximum |Temp. |Average |
[Data logger |(Probe) No. |Temp.(°C) |Temp. (°C)|reading |Temp. (°C) |
| | | |at 48 hrs | |
| | | |(°C) | |
[In deep |Probe # 16 | | | -5.7 | |

freezer				
In Box with	Probe # 1			4.5
20L glass				
bottle				
Probe # 2			4.6	
Probe # 3			4.7	
In Box with	Probe # 4			4.5
10L glass				
bottle				
Probe # 5			4.5	
Probe # 6			4.9	
In Box with 5L	Probe # 7			4.8
glass bottle				
Probe # 8			4.8	
Probe # 9			4.7	
In Box with 2L	Probe # 10			4.7
glass bottle				
Probe # 11			4.6	
Probe # 12			4.6	
In Box with 1L	Probe # 13			4.7
glass bottle				
Probe # 14			4.7	
Probe # 15			4.8	
Digital		---		
display				

Cycle No. 2 (13.03.2007 to 15.03.2007)
Details as above

Cycle No. 3 (15.03.2007 to 17.03.2007)
Details as above

8.3 Summary of Average temperature record for three cycles at + 43 °C
Position of Average Temp. Average Temp Average Temp. Average Temp. |
Data logger (°C) of Cycle 1(°C) of Cycle 2(°C) of Cycle 3(°C)of all 3 |
| | | |cycles |
In Incubator | | | |
In Box with | | | |
20L glass | | | |
bottle | | | |
In Box with | | | |
10L glass | | | |
bottle | | | |
In Box with 5L | | | |
glass bottle | | | |
In Box with 2L | | | |
glass bottle | | | |
In Box with 1L | | | |
glass bottle | | | |

4. Summary of Average temperature record for three cycles at -5°C:
Position of Average Temp. Average Temp Average Temp. Average Temp. |
Data logger (°C) of Cycle(°C) of Cycle(°C) of Cycle(°C) of all 3 |
| 1 | 2 | 3 |cycles |
In deep | | | |
freezer | | | |
In Box with | | | |
20L glass | | | |
bottle | | | |
In Box with | | | |
10L glass | | | |
bottle | | | |
In Box with 5L | | | |
glass bottle | | | |
In Box with 2L | | | |
glass bottle | | | |
In Box with 1L | | | |
glass bottle | | | |

10. DETAILS OF ATTACHMENTS TO REVALIDATION REPORT:
Sr. No. Type of Details |
Attachment (as |
applicable) | |
1 Record Forms Nil |
2 Calibration PR/EQP/RFG 03116, QC/EQP/FZR.11003, |
Certificates QC/EQP/FZR.11004, 10626000/116,106260001,106260002,|
| 106260003,106260004 |
3 Equipment Print Printouts of temperature |
outs | |
4 Strip/Chart Nil |
Records | |
5 Graphs/Trend Nil |
6 Diagrams/Drawings Location of temperature sensors inside PUF lined |
corrugated box |
Position of Ice packs inside PUF lined corrugated |
box |
7 Quality Control Nil |
Record forms | |
Details of any other attachments: |
Temperature records of digital display of incubator and deep freezer |

SHIPPING VALIDATION PROTOCOL

RM:

- Wt of each empty box (PUFF Lined)
- Boxes acclimatized at vaccine storage temperature by placing at 5+ 30 C
- Load the bottles containing placebo in the PUFF lined box, as quickly as possible (15 Minutes)
- Pack the bottles with Soft Ice pack, Frozen soft Ice packs and then Hard Ice packs
- Number of Ice packs shall be as per the table
- Place 3 thermocouples in each box and 1 thermocouple in the environment wherever the study shall be carried out i.e. incubator for + 43 °C and deep freezer for -5°C. Each thermocouple should be positioned to a distance of 2.5/3.0 cm from the nearest ice pack
- Record the temperature at an interval of 10 minutes from thermocouples and every 2 Hours from digital display of incubator / deep freezer for 48 hours
- Place the packed boxes in the incubator at + 43 oC and in a deep freezer at -5 oC
- Carry out 3 Runs for each packing

1. Purpose:

The purpose is to validate the temperature during shipping of ARM / bulk vaccine, transported at 5 + 3°C so as to maintain the desired cold chain conditions during transportation.

2. Scope:

2.1 Applicable to the ARM /bulk vaccine (that comes under class C as per WHO Guidelines on the packaging and shipping of vaccines WHO/IVB/R/05.23 December 2005) to be transported at the temperature of 5 + 3°C.

4. Safety considerations:

4.1 The ARM / bulk vaccine is to be maintained at recommended temperature at 5 + 3 °C. The ARM / bulk vaccine should not freeze.

4.2 The containers of the ARM / bulk vaccine should be properly packed to avoid any breakage.

4.3 Ensure proper connections of thermocouples and their positioning.

5. Equipments:

5.1 Data logger with thermocouples (accuracy $\pm 0.5^{\circ}\text{C}$)

5.2 BOD Incubator and Deep Freezer

6. Materials and reagents:

6.1 Materials:

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SHIPPING VALIDATION PROTOCOL

6.1.1 Glass bottles of appropriate capacity (20L, 10L, 5L, 2L and 1L) containing Placebo

6.1.2 PUF lined corrugated boxes (to house 10L/20L glass bottle)

6.1.3 PUF lined corrugated boxes (to house 1L/2L /5L glass bottle)

6.1.4 Ice packs hard / soft (stored at $5\pm 3^{\circ}\text{C}$ and -20°C)

6.1.5 Polypropylene caps for sealing bottles

6.1.6 Thermocol sheets for immobilization

6.1.7 Adhesive tape

6.2 Reagents:

6.2.1 3% Alhydrogel

6.2.2 Di-sodium Hydrogen phosphate anhydrous

6.2.3 Sodium di-hydrogen phosphate dehydrate

6.2.4 Sodium Chloride

6.2.5 Thiomersal

6.2.6 Water for Injection

7. Preparation of solutions:

7.1 Preparation of Placebo (50L):

7.1.1 Take 3.0L of 3% alhydrogel in a 50 L glass bottle and add 27 L of water for injection.

7.1.2 Weigh Di-sodium Hydrogen phosphate anhydrous = 56gm, Sodium di-hydrogen phosphate dehydrate = 62 gm and Sodium Chloride = 400 gm.

Dissolve the above weighed chemicals in 22L of water for injection.

7.1.3 Weigh 20 gm Thiomersal and dissolve in 1.0 L of water for injection.

7.1.4 Add 22L of phosphate buffer and 1L of Thiomersal solutions to 27L of alhydrogel in 50L glass bottle and homogenize by manual shaking.

8. Procedure:

8.1 Validation Study Plan:

The study shall be carried out by keeping the bottles filled with Placebo (containing all the constituents of the vaccine except the ARM, having concentration equivalent to the concentration present in the vaccines) in incubator and deep freezer at + 43 °C and -5°C respectively, for a period of 48 hours.

2 of 11

SHIPPING VALIDATION PROTOCOL

For monitoring of temperature 03 thermocouples (as per annexure-1) connected to the data logger shall be placed in side the PUF lined corrugated box and 01 thermocouple in the environment wherever the study shall be carried out i.e. incubator for + 43 °C and deep freezer for -5°C. The above study shall be repeated 03 times. These temperatures (+43 °C and -5°C) are representative of worst case environment. The temperature monitoring from the display of incubator and deep freezer shall be done after 02 hours.

8.2 Take appropriate capacity bottles containing Placebo properly capped and sealed.

8.3 Store the Placebo bottles at 5°C + 3°C for acclimatization and Ice packs (Hard/Soft) at -20°C overnight.

8.4 Take the packaging materials of required specifications, which are to be validated and make a record of these.

8.5 Pack the bottles in the boxes as per the standard packaging procedure(FR/012) as follows:

8.5.1 Glass bottles of pack size 20.0 and 10.0 L

■ Check the weight of each empty box and load the 20 L and 10 L bottles containing Placebo individually in the PUF Lined Corrugated boxes which shall be placed at $5 \pm 3^{\circ}\text{C}$ to get acclimatize to vaccine storage temperature.

■ Pack the bottles with the soft and hard ice packs. The soft ice packs of $5 \pm 3^{\circ}\text{C}$ shall be placed adjacent to the bottle followed by frozen soft ice packs and hard ice packs. Immobilize the bottle by filling the remaining space, if any, with the thermocol sheets.

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SHIPPING VALIDATION PROTOCOL

Ice packs used for packing Number of Ice Packs
(Box Size)

20L bottle Soft: 20-22no.

Hard: 22-24 no.

10 L Bottle Soft: 16-18no.

Hard: 18-20 no.

5 L Bottle Soft: 14-16no.

Hard: 16-18 no.

2L Bottle Soft: 12-14no.

Hard: 14-16 no.

1 L Bottle Soft: 10-12no.

Hard: 08-10 no.

■ Check the weight of the each loaded box. Loading should be carried out as quickly as possible (less than 15 min) and the time for loading should be recorded for each container.

■ Place 3 thermocouples (as per annexure-1) in each box through the seal taking care not to affect the quality of the seal and one thermocouple in the environment wherever the study shall be carried out i.e. incubator for $+43^{\circ}\text{C}$ and deep freezer for -5°C . Each thermocouple should be positioned to a distance of 2.5/3.0 cm from the nearest ice pack surface.

■ Close the individual corrugated box with its lid and seal the box with the adhesive tape.

■ Start the data logger to record the temperature at an interval of 10 minutes for 48 hours. Also record the temperature from digital display of incubator and deep freezer at every 2 Hours interval.

■ Place the packed boxes containing in the incubator at $+43^{\circ}\text{C}$ and in a deep freezer at -5°C .

■ Carry out 3 Runs for each packing by repeating 8.5.1.1 to 8.5.1.7.

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SHIPPING VALIDATION REPORT

Annexure 1

Location of temperature sensors inside PUF lined corrugated box

a) PUF lined corrugated box: top view

b) PUF lined corrugated box: top view

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SHIPPING VALIDATION REPORT

Annexure 2

Positioning of Ice packs inside PUF lined corrugated box

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SHIPPING VALIDATION REPORT

REPORT

1. EXECUTIVE SUMMARY OF THE STUDY

The validation program has been carried out to validate the packaging method for the shipping of ARM/Bulk vaccine (which comes under class C as per WHO Guidelines on the packaging and shipping of vaccines WHO/IVB/R/05.23 December 2005) transported at temperature $5 \pm 3^{\circ}\text{C}$ so as to maintain the desired cold chain conditions during transportation.

The study shall be carried out by keeping the Placebo (containing all the constituents of the vaccine except the active raw material having concentration equivalent to the concentration present in the vaccines) in incubator and deep freezer at $+43^{\circ}\text{C}$ and -5°C respectively, for a period of 48 hours. For monitoring of temperature three thermocouples connected to the data logger shall be placed inside the PUF lined corrugated box and one thermocouple in the environment wherever the study shall be carried out i.e. incubator for $+43^{\circ}\text{C}$ and deep freezer for -5°C . The above study shall be repeated three times. These temperatures ($+43^{\circ}\text{C}$ and -5°C) are representative of worst case environment. The temperature monitoring from the display of incubator and deep freezer shall be done after two hours.

Two types of PUF lined corrugated boxes were used in the validation study were:

■ PUF lined corrugated box to house 10L/20L glass bottle (PMS.18000)

■ PUF lined corrugated box to house 1L/2L /5L glass bottle (PMS.18004)

7. TESTS PERFORMED & ACCEPTANCE CRITERIA CONSIDERED:

Sr. No Tests Performed Acceptance criteria

1. The packed box containing Placebo placed The warmest temperature in the packed box

o in environmental temperature of $43 \pm 1^{\circ}\text{C}$ containing Placebo does not rise above for a period 48 hours. $5 \pm 3^{\circ}\text{C}$.

2. The packed box containing Placebo placed The coolest temperature in the packed box in environmental temperature of $-5 \pm 1^{\circ}\text{C}$ containing Placebo does not fall below for a period of 48 hours. $+2^{\circ}\text{C}$

8. STATISTICAL ANALYSIS REPORT (if any) :
8.1 Result of environmental temperature study at +43 °C:
Temperature recording interval: 10 minutes

Cycle No. 1 (04.03.2007 to 06.03.2007)

Package details Empty Box Packed No. of soft No. of hard Time taken
weight (Kg) box ice packs ice packs for packing
weight used used
(Kg)

Box with 20L glass
bottle

Box with 10L glass
bottle

Box with 5L
glass bottle

Box with 2L
glass bottle

Box with 1L
glass bottle

Position of Data Thermocouple Minimum Maximum Temp. Average
logger (Probe) No. Temp.(°C) Temp. reading Temp. (°C)
(°C) at 48 hrs
(°C)

In Incubator Probe # 16 43.4

Probe # 1 7.1

In Box with 20L

Probe # 2 7.1

glass bottle

Probe # 3 7.2

Probe # 4 7.1

In Box with 10L

Probe # 5 7.2

glass bottle

Probe # 6 7.1

Probe # 7 7.1

In Box with 5L

Probe # 8 7.1

glass bottle

Probe # 9 7.2

In Box with 2L Probe # 10 7.1

glass bottle Probe # 11 7.2

Probe # 12 7.1

Probe # 13 7.2

In Box with 1L

Probe # 14 7.1

glass bottle

Probe # 15 7.1

Digital display ----

Cycle No. 2 (06.03.2007 to 08.03.2007)

Details as above

Cycle No. 3 (08.03.2007 to 10.03.2007)

Details as above

8.2 Results of environmental temperature study at -5°C:
Temperature recording interval: 10 minutes

Cycle No. 1(11.04.2007 to 13.04.2007)

Package details Empty Box Packed box No. of soft No. of hard Time taken
weight (Kg) weight (Kg) ice packs ice packs for packing
used used

Box with 20L

glass bottle

Box with 10L

glass bottle

Box with 5L

glass bottle

Box with 2L

glass bottle

Box with 1L

glass bottle

Position of Thermocouple Minimum Maximum Temp. Average
Data logger (Probe) No. Temp.(°C) Temp. reading Temp. (°C)
(°C) at 48 hrs
(°C)

In deep freezer Probe # 16 -5.7

In Box with Probe # 1 4.5

20L glass bottle Probe # 2 4.6
Probe # 3 4.7
In Box with Probe # 4 4.5
10L glass bottle Probe # 5 4.5
Probe # 6 4.9
In Box with 5L Probe # 7 4.8
glass bottle Probe # 8 4.8
Probe # 9 4.7
In Box with 2L Probe # 10 4.7

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SHIPPING VALIDATION REPORT

glass bottle Probe # 11 4.6
Probe # 12 4.6
In Box with 1L Probe # 13 4.7
glass bottle Probe # 14 4.7
Probe # 15 4.8
Digital display ----

Cycle No. 2 (13.03.2007 to 15.03.2007)
Details as above

Cycle No. 3 (15.03.2007 to 17.03.2007)
Details as above

8.3 Summary of Average temperature record for three cycles at + 43 °C
Position of Average Temp. Average Temp Average Temp. Average Temp.
Data logger (°C) of Cycle 1 (°C) of Cycle 2 (°C) of Cycle 3 (°C) of all 3
cycles

In Incubator
In Box with
20L glass bottle
In Box with
10L glass bottle
In Box with 5L
glass bottle
In Box with 2L
glass bottle
In Box with 1L
glass bottle

8.4 Summary of Average temperature record for three cycles at -5°C:
Position of Average Average Temp Average Average Temp.
Data logger Temp. (°C) of (°C) of Cycle 2 Temp. (°C) of all 3 cycles
Cycle 1 (°C) of Cycle 3

In deep freezer
In Box with
20L glass bottle
In Box with
10L glass bottle
In Box with 5L
glass bottle
In Box with 2L
glass bottle
In Box with 1L
glass bottle

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SHIPPING VALIDATION REPORT

10. DETAILS OF ATTACHMENTS TO REVALIDATION REPORT:

Sr. No. Type of Attachment Details
(as applicable)

- 1 Record Forms Nil
- 2 Calibration PR/EQP/RFG 03116, QC/EQP/FZR.11003, QC/EQP/FZR.11004,
Certificates 10626000/116,106260001,106260002, 106260003,106260004
- 3 Equipment Print outs Printouts of temperature
- 4 Strip/Chart Records Nil
- 5 Graphs/Trend Nil
- 6 Diagrams/Drawings -Location of temperature sensors inside PUF lined corrugated box
-Position of Ice packs inside PUF lined corrugated box
- 7 Quality Control

Nil
Record forms
Details of any other attachments:
Temperature records of digital display of incubator and deep freezer

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Date: | Date: | Date: | Date:
Department/Area: Product Name: | Department/Area: Product Name: | Department/Area: Product Name: |
Department/Area: Product Name:
Batch No.: Stage: | Batch No.: Stage: | Batch No.: Stage: | Batch No.: Stage:
OBSERVATIONS | OBSERVATIONS | OBSERVATIONS | OBSERVATIONS
| | |
Written By: | | Reviewed by: |
Action taken by User Department: | Action taken by User Department: | Action taken by User Department: |
Action taken by User Department:
Head Of User Department
(Date/Sign) | Head Of User Department
(Date/Sign) | Head Of User Department
(Date/Sign) | Head Of User Department
(Date/Sign)
QA Comment: | QA Comment: | QA Comment: | QA Comment:
Reviewed By HOD QA
(Date/Sign) | Reviewed By HOD QA
(Date/Sign) | Reviewed By HOD QA
(Date/Sign) | Reviewed By HOD QA
(Date/Sign)

DATE: DEPARTMENT/ AREA:

PRODUCT NAME: BATCH NO. | DATE: DEPARTMENT/ AREA:

PRODUCT NAME: BATCH NO.

ASPECTS (Write the appropriate application If meets-Y for yes, N for No and NA for not applicable) |

ASPECTS (Write the appropriate application If meets-Y for yes, N for No and NA for not applicable)

Personnel: | Y/N/NA

Are the Personnel wearing appropriate protective clothing according to relevant SOP? |

Is there adequate use of safety devices by personnel, where applicable? |

Is there a compliance with health and hygiene requirements e.g.: no eating and drinking etc. in production classified areas? |

Are there sufficient numbers of persons available for performing each activity? |

Are the personnel working as per relevant SOP and following aseptic handling? if applicable. |

Check for the aseptic handling and clean room behavior at classified area. |

Personnel movement in classified area to avoid turbulence into the specified area. |

Areas, Equipment, Materials: | Areas, Equipment, Materials:

Sampling for environmental monitoring being followed, if applicable |

Are the manufacturing area cleaned according to established procedure and the cleaning data recorded? |

Are the materials adequately stored in the area and adequately labeled? |

Are the physical environmental conditions i.e. temperature, Differential pressure, Relative humidity within the control limits? |

Are the raw materials approved by QC and adequately labeled? |

Are the cleaning / sterilization records of Bottles and other accessories available? |

Are the identification, calibration and operation status label of equipment/instrument, available and correct? |

Check for the proper screw cap fitting & status label on solution bottles, ARM/bulk bottles. |

Check the physical condition of manufacturing facility i.e. peel off epoxy, cracks in wall, floor, coving, door damage etc. |

Documentation: | Documentation:

Does the correspondence of data, batch number and quantity of raw materials and other incoming materials with the information recorded in the Batch Record of the stage being inspected? |

Is there presence of the Master Formula and SOP in the area and are concurrent? |

Are log books being maintained by the user? |

Are the records of temperature, differential pressure, relative humidity and area cleaning, available and updated? |

Production and Process Controls: | Production and Process Controls: | Y/N/NA

Does the Batch Records filled as operations are performed? | Does the Batch Records filled as operations are performed? |

Are the filter used for sterile filtration tested for integrity before and after use? | Are the filter

used for sterile filtration tested for integrity before and after use? |

Check the identification and approved status of incoming material | Check the identification and approved status of incoming material |

Check the retest or expiry date of incoming material | Check the retest or expiry date of incoming material |

During aseptic operation, after every contact with sterile/non sterile surface ensure for disinfection of hands | During aseptic operation, after every contact with sterile/non sterile surface ensure for disinfection of hands |

Check the aseptic behavior of personnel while work under LAF station, and restricted movement of material

IN/OUT of LAF so as to avoid the turbulence. | Check the aseptic behavior of personnel while work under

LAF station, and restricted movement of material IN/OUT of LAF so as to avoid the turbulence. |

Check effective removal & decontamination of the used materials | Check effective removal & decontamination of the used materials |

Check transfer of materials IN/OUT to the clean room area through pass boxes only | Check transfer of materials IN/OUT to the clean room area through pass boxes only |

Ensure the sampling as per sampling schedule. | Ensure the sampling as per sampling schedule. |

Check the storage condition of final ARM/bulk? | Check the storage condition of final ARM/bulk? |

Remarks (If any): | Remarks (If any): | Remarks (If any):

Written By: | Reviewed By | Reviewed By

Name: | Name: | Name:

Signature: | Signature: | Signature:

Date: | Date: | Date:

Date : | Date : | Date : | Date :

ASPECTS (Write the appropriate application If meets-Y for yes, N for No and NA for not applicable) |

ASPECTS (Write the appropriate application If meets-Y for yes, N for No and NA for not applicable) |

ASPECTS (Write the appropriate application If meets-Y for yes, N for No and NA for not applicable) |

ASPECTS (Write the appropriate application If meets-Y for yes, N for No and NA for not applicable)

Serial. No. | Observation | Observation | Y/N/NA

| Are personnel wearing appropriate protective clothing according to relevant SOP? | Are personnel

wearing appropriate protective clothing according to relevant SOP? |

| Are the of status label available on ARM/bulk bottles stored in cold rooms? | Are the of status label

available on ARM/bulk bottles stored in cold rooms? |

| Check for the general cleanliness of the Material storage area/cold room area. | Check for the general

cleanliness of the Material storage area/cold room area. |

| Are the approved & under test materials separately placed with proper labels? | Are the approved &

under test materials separately placed with proper labels? |

| Are the rejected material segregated to approved and under test materials with appropriate labeling? |

Are the rejected material segregated to approved and under test materials with appropriate labeling? |

| Are the printed material kept in control at designated place? | Are the printed material kept in

control at designated place? |

| Are the specified temperature conditions maintained in different areas of warehouse? | Are the

specified temperature conditions maintained in different areas of warehouse? |

| Are the temperatures of cold rooms maintained within specified limits? | Are the temperatures of cold

rooms maintained within specified limits? |

| Are the temperature indicators of cold rooms calibrated? | Are the temperature indicators of cold rooms

calibrated? |

| Are log books updated? | Are log books updated? |

| Are packaging scheme followed as per standard procedure? | Are packaging scheme followed as per

standard procedure? |

Remarks (If any): | Remarks (If any): | Remarks (If any): | Remarks (If any):

Written By: | Written By: | Reviewed By | Reviewed By

Name: | Name: | Name: | Name:

Signature: | Signature: | Signature: | Signature:

Date: | Date: | Date: | Date:

Date : | Date : | Date : | Date :
ASPECTS (Write the appropriate application If meets-Y for yes, N for No and NA for not applicable) |
ASPECTS (Write the appropriate application If meets-Y for yes, N for No and NA for not applicable) |
ASPECTS (Write the appropriate application If meets-Y for yes, N for No and NA for not applicable) |
ASPECTS (Write the appropriate application If meets-Y for yes, N for No and NA for not applicable) |
Serial
No. | Observation | Observation | Y/N/NA
| Are the concerned SOPs present in place? | Are the concerned SOPs present in place? |
| Are the operational instructions and diagram present with the equipment? | Are the operational
instructions and diagram present with the equipment? |
| Are operating ranges of equipment as per the SOPs? | Are operating ranges of equipment as per the SOPs?
|
| Are WFI, PW and Pure Steam being generated as per the standard specifications? | Are WFI, PW and Pure
Steam being generated as per the standard specifications? |
| Are the pipelines being supplied to different areas proper coded? | Are the pipelines being supplied to
different areas proper coded? |
| Are the equipment/instruments coded with respective I.D No. /Serial No? | Are the equipment/instruments
coded with respective I.D No. /Serial No? |
| Are log books of equipment usage being maintained by the user? | Are log books of equipment usage being
maintained by the user? |
| Verify the record of sanitization and sterilization of water storage & distribution system. | Verify
the record of sanitization and sterilization of water storage & distribution system. |
| Check the burning hours of UV Lamp in case of water system. | Check the burning hours of UV Lamp in
case of water system. |
| Are measuring devices/instruments calibrated? | Are measuring devices/instruments calibrated? |
Remarks (If any): | Remarks (If any): | Remarks (If any): | Remarks (If any):
Written By: | Written By: | Reviewed By | Reviewed By
Name: | Name: | Name: | Name:
Signature: | Signature: | Signature: | Signature:
Date: | Date: | Date: | Date:

Date : | Date : | Date : | Date :
ASPECTS (Write the appropriate application If meets-Y for yes, N for No and NA for not applicable) |
ASPECTS (Write the appropriate application If meets-Y for yes, N for No and NA for not applicable) |
ASPECTS (Write the appropriate application If meets-Y for yes, N for No and NA for not applicable) |
ASPECTS (Write the appropriate application If meets-Y for yes, N for No and NA for not applicable) |
Serial
No. | Observation | Observation | Y/N/NA
| Are entry exit log books for quality control facility/laboratory area being maintained by user? | Are
entry exit log books for quality control facility/laboratory area being maintained by user? |
| Are the personnel wearing appropriate protective clothing for entry in to laboratories with as per
relevant SOP? | Are the personnel wearing appropriate protective clothing for entry in to laboratories
with as per relevant SOP? |
| Are the In-process samples and retention samples stored according to established procedure? | Are the
In-process samples and retention samples stored according to established procedure? |
| Check the aseptic handling of personnel while work under LAF station, | Check the aseptic handling of
personnel while work under LAF station. |
| Are the log books concurrently maintained by user? | Are the log books concurrently maintained by user?
|
| Are the stability chambers maintained as per established storage condition? | Are the stability
chambers maintained as per established storage condition? |
| Are the concerned SOP's available for the respective test and instruments? | Are the concerned SOP's
available for the respective test and instruments? |
| Are the physical environmental conditions i.e. temperature, Differential pressure, Relative humidity
within the control limits? | Are the physical environmental conditions i.e. temperature, Differential
pressure, Relative humidity within the control limits? |
| Use of sterilized material and its status, if applicable. | Use of sterilized material and its status,
if applicable. |
| Ensure the status label on instruments of Calibration / Qualification. | Ensure the status label on
instruments of Calibration / Qualification. |
| Adequate identification of reagents, solutions and materials used in laboratories and their uses log
book. | Adequate identification of reagents, solutions and materials used in laboratories and their uses
log book. |
| Disposal of used materials/media plates & its effective decontamination. | Disposal of used
materials/media plates & its effective decontamination. |
Remarks (If any): | Remarks (If any): | Remarks (If any): | Remarks (If any):
Written By: | Written By: | Reviewed By | Reviewed By
Name: | Name: | Name: | Name:
Signature: | Signature: | Signature: | Signature:
Date: | Date: | Date: | Date:

Total No. of Rounds. _____

*Vaccine Block-I and Vaccine Block-IV should thrice a week during non-manufacturing condition and Daily during manufacturing condition.

Block / Department | Frequency | Ist week | IInd Week | IIIrd Week | IVth Week

| Name | Designation | Signature | Date

Prepared By ||||

Checked By ||||

Approved By ||||

STANDARD OPERATING PROCEDURE SOP No. : QMS 032
Revision No.: 00

Title: IPQA Supersedes : None
Page No. : 1 of 7

1.0 Objective

To lay down the procedure for Quality Assurance overview on area/processes/activities of Production, Quality Control, Warehouse, Utility system and Animal House.

2.0 Scope

This Standard Operating Procedure is applicable for Quality Assurance overview of all the manufacturing facilities, Quality Control Laboratory, Warehouse, Utility system and Animal House at Company name.

3.0 Responsibility

3.1 IPQA personnel shall be responsible to visit and check the facility regarding cGMP compliance of Manufacturing area, Quality Control Laboratory, Warehouses, utility systems and Animal House.

3.2 IPQA personnel shall monitor and verify critical processes of Manufacturing.

3.3 Head, Quality Assurance Department shall ensure the implementation of this SOP.

4.0 Abbreviations and Definitions

4.1 AHU Air Handling unit

4.2 BR Batch Record

4.3 cGMP Current Good Manufacturing Practices

4.4 CR Cleaning Record

4.5 EMP Environmental Monitoring Program

4.6 GMP Good Manufacturing Practice

4.7 IPQA In Process Quality Assurance

4.8 IPA Iso Propyl Alcohol

4.9 LAF Laminar Air Flow

4.10 PW Purified Water

4.11 QA Quality Assurance

4.12 QC Quality Control

4.13 SOP Standard Operating Procedure

4.14 SS Stainless Steel

4.15 WFI Water for Injection

5.0 Procedure

5.1 IPQA Personnel shall overview all the areas/critical activities/processes to ensure

5.1.1 All the areas in a state of GMP compliance.

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Title: IPQA Supersedes : None
Page No. : 2 of 7

5.1.2 The activities/processes are carried out as per approved procedure/protocols.

5.1.3 IPQA personnel shall be trained on clean room behavior, aseptic handling and QMS SOPs but not limited to.

5.1.4 Apart from manufacturing processes IPQA person shall oversee other areas/ processes like Utility system, Warehouse, Animal House and Quality Control activities.

5.2 IPQA checks/overview during various operations/process, as follows:

5.2.1 IPQA personnel shall check in manufacturing area as follows, but not limited to:

5.2.1.1 Gowning of personnel present for entry in work area as per the respective SOP.

5.2.1.2 Personnel movement in classified area to avoid turbulence into the specified area.

5.2.1.3 General cleanliness of area, type of disinfectant used and its correspondence logbook.

5.2.1.4 Clean room behavior of personnel.

5.2.1.5 Cleaning and sterilization status of material and equipment to be used in processes.

5.2.1.6 The overview of physical environmental conditions i.e. Temperature, Differential pressure and Relative humidity

5.2.1.7 Status of reagent used for carrying manufacturing procedure.

5.2.1.8 Equipment, raw materials, in process materials and Solutions with proper status label & ensure they are within expiry period.

5.2.1.9 The use of calibrated measuring instruments and their identification

5.2.1.10 Verify the batch number of Raw Materials against the approved label.

5.2.1.11 Ensure that manufacturing and cleaning process are being carried as per approved BR and CR.

5.2.1.12 The presence of the Batch Record, SOP and relevant documents in the area.

5.2.1.13 Batch Records and logbooks shall be filled concurrently and accurately.

5.2.1.14 Check the physical condition of manufacturing facility i.e. peel off

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epoxy, cracks in wall, floor, coving, door damage etc.

5.2.1.15 IPQA personnel shall check the manufacturing process/area as per the annexure no.2

5.2.2 IPQA personnel shall check during campaign change process, but not limited to:

5.2.2.1 IPQA personnel shall verify the line clearance as per checklist of respective SOP of concerned manufacturing block.

5.2.2.2 Whether the unused raw materials of previous batch removed from area or returned to warehouse (When applicable).

5.2.2.3 Whether any Solutions, media or waste left over from previous production batch removed from area.

5.2.2.4 Whether batch processing records pertaining to previous batch has been remove from the area.

5.2.2.5 Whether the documents pertaining to the processing of previous campaign are removed from the area.

5.2.2.6 Whether the cleaning and sanitization of equipment /systems/Area has been carried out after previous campaign as per the campaign change protocol.

5.2.2.7 Are the cleaning and sanitization records available?

5.2.2.8 Whether the results of cleaning and sanitation are comply to recommended limits

5.2.2.9 Whether the records available for the campaign product.

5.2.2.10 IPQA personnel shall also check the area as per Annexure no.2 during the campaign change process.

5.2.3 IPQA personnel shall check over warehouses as follows but not limited to:

5.2.3.1 General cleanliness of the cold room area.

5.2.3.2 Segregation of approved and under test materials.

5.2.3.3 Operating temperature of the cold room and deep freezer as per the requirement.

5.2.3.4 Temperature and Relative Humidity monitoring in material storage area.

5.2.3.5 Segregation of materials with respect to their status.

5.2.3.6 Log books for opening & closing of cold room & verification of data

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recorded.

5.2.3.7 Ensure that all the materials are properly labeled.

5.2.3.8 IPQA personnel shall check the warehouse as per the annexure no. 3.

5.2.4 IPQA personnel shall check over utility system as follows but not limited to:

5.2.4.1 Operational SOP's in place.

5.2.4.2 Availability of operational instructions & diagrams at the site.

5.2.4.3 Operating ranges & operation modes of the equipment, as per the respective SOPs.

5.2.4.4 Verify record for sanitization status of water storage & distribution system.

5.2.4.5 Check calibration status of all measuring devices/instruments.

5.2.4.6 Proper coding of the supply pipeline to different areas.

5.2.4.7 Proper coding of the instruments with respective I.D Nos.

5.2.4.8 Log books are maintained by the user and are update.

5.2.4.9 Data recorded by the machine operators.

5.2.4.10 IPQA personnel shall check the utility system as per the annexure no. 4.

5.2.5 IPQA personnel shall check over clean room behavior & aseptic process as follows, but not limited to:

5.2.5.1 Entry of personnel in to clean room with proper gowning as per the SOP.

5.2.5.2 Disinfection of the hands with 70% IPA after final gowning & gloving.

5.2.5.3 Prior to commencement of work, disinfection of hands with 70% IPA.

5.2.5.4 Under LAF station, ensure for minimum movement of personnel hands.

5.2.5.5 Man and Material movement in facility.

5.2.5.6 Restricted number of personnel in a specified area.

5.2.5.7 Use of sterile forceps while picking up silicon tubes, caps during aseptic process, does not touch with gloved hands.

5.2.5.8 Disinfection of hands & forceps used after each intervention.

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5.2.5.9 Sterilization status of the materials being used in the manufacturing process.

5.2.5.10 Effective removal & decontamination of the used materials.

5.2.5.11 Temperature, relative humidity and differential pressure are within the acceptance limit.

5.2.5.12 Transfer of materials IN/OUT to the clean room area through pass boxes only.

5.2.6 IPQA personnel shall check over Quality control area as follows, but not limited to:

5.2.6.1 Entry of personnel in to laboratories with proper gowning as per the SOP.

5.2.6.2 Ensure the in process samples and retention samples stored according to established procedure.

5.2.6.3 Under LAF station, ensure for minimum movement of personnel hands.

5.2.6.4 Ensure the log books maintained by user are updated.

5.2.6.5 Check the stability chambers are maintained as per established storage condition.

5.2.6.6 Availability of SOP for the respective test and instruments.

5.2.6.7 Recording of physical parameters (Pressure differential, Humidity, Temperature)

5.2.6.8 Use of sterilized material and its status, as applicable.

5.2.6.9 Ensure the status label on instruments of Calibration / Qualification.

5.2.6.10 Adequate identification of reagents, solutions and materials used in

laboratories and their uses log.

5.2.6.11 Disposal of used materials/media plates & its effective decontamination.

5.2.6.12 Log books/records are filled correct and concurrent.

5.2.6.13 IPQA personnel shall check the Quality control as per Annexure-5.

5.2.7 Monthly schedule for IPQA checks shall be prepared as per Annexure No.6 and IPQA personnel shall perform these checks as per following schedule, but not limited to

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■ For Xxxx-I and Xxxx-IV:

During manufacturing: Each working day

During non-manufacturing: Thrice a week

■ For Xxxx-II ,Xxxx-III, QC, Warehouse and Animal House, Critical

Utility : Twice a week

5.3 In case of manufacturing activity IPQA personnel shall verify the activity following tentative batch schedule prepared as per SOP QA/000.

5.4 IPQA personnel shall record the exact observation in Form No. QA/000/00as per Annexure 1 for each check in manufacturing area, Critical Utilities area, Warehouse and Quality Control and Animal House if needed, will send to user department for necessary action. User department shall write the corrective measures in the form and submit to QA within two working days.

5.5 In case any observation related to physical condition of manufacturing facility as per point no. 5.2.1.14, IPQA shall share the observation with the user department as per annexure-1. Based on the observation user department and engineering department shall perform impact assessment as per SOP EN/000 prior to commencement of civil work.

5.6 Observations reported during the QA overview shall be dealt with applicable non-conformance/deviation handling procedures.

5.7 In case of facility shut down IPQA checks shall be performed on requirement basis.

6.0 Forms and Records

6.1 QA/000/00IPQA Observation Report Annexure - 1

6.2 QA/000/00IPQA check list for manufacturing area Annexure - 2

6.3 QA/000/00IPQA check list for Warehouse Annexure - 3

6.4 QA/000/00IPQA check list for Critical Utilities Annexure - 4

6.5 QA/000/00- 03 QA check list for Quality Control Annexure - 5

6.6 QA/000/00Monthly schedule for IPQA checks Annexure - 6

7.0 References:

7.1

8.0 Distribution

-

9.0 History

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Date Revision Number Reason for Revision

IPQA OBSERVATION REPORT Annexure-I

Date:

Department/Area: Product Name:

Batch No.: Stage:

OBSERVATIONS

Written By: Reviewed by:

Action taken by User Department:

Head Of User Department

(Date/Sign)

QA Comment:

Reviewed By HOD QA

(Date/Sign)

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IPQA CHECK LIST FOR MANUFACTURING AREA Annexure-2

DATE: DEPARTMENT/ AREA:

PRODUCT NAME: BATCH NO.

ASPECTS (Write the appropriate application If meets-Y for yes, N for No and NA for not applicable)

Personnel: Y/N/NA

Are the Personnel wearing appropriate protective clothing according to relevant SOP?

Is there adequate use of safety devices by personnel, where applicable?

Is there a compliance with health and hygiene requirements e.g.: no eating and drinking etc. in production classified areas?

Are there sufficient numbers of persons available for performing each activity?

Are the personnel working as per relevant SOP and following aseptic handling? if applicable.

Check for the aseptic handling and clean room behavior at classified area.

Personnel movement in classified area to avoid turbulence into the specified area.

Areas, Equipment, Materials:

Sampling for environmental monitoring being followed, if applicable

Are the manufacturing area cleaned according to established procedure and the cleaning data recorded?

Are the materials adequately stored in the area and adequately labeled?

Are the physical environmental conditions i.e. temperature, Differential pressure, Relative humidity within the control limits?
Are the raw materials approved by QC and adequately labeled?
Are the cleaning / sterilization records of Bottles and other accessories available?
Are the identification, calibration and operation status label of equipment/instrument, available and correct?
Check for the proper screw cap fitting & status label on solution bottles, ARM/bulk bottles.
Check the physical condition of manufacturing facility i.e. peel off epoxy, cracks in wall, floor, coving, door damage etc.
Documentation:
Does the correspondence of data, batch number and quantity of raw materials and other incoming materials with the information recorded in the Batch Record of the stage being inspected?
Is there presence of the Master Formula and SOP in the area and are concurrent?
Are log books being maintained by the user?
Are the records of temperature, differential pressure, relative humidity and area cleaning, available and updated?

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IPQA CHECK LIST FOR MANUFACTURING AREA

Production and Process Controls: Y/N/NA

Does the Batch Records filled as operations are performed?
Are the filter used for sterile filtration tested for integrity before and after use?
Check the identification and approved status of incoming material
Check the retest or expiry date of incoming material
During aseptic operation, after every contact with sterile/non sterile surface ensure for disinfection of hands
Check the aseptic behavior of personnel while work under LAF station, and restricted movement of material IN/OUT of LAF so as to avoid the turbulence.
Check effective removal & decontamination of the used materials
Check transfer of materials IN/OUT to the clean room area through pass boxes only
Ensure the sampling as per sampling schedule.
Check the storage condition of final ARM/bulk?

Remarks (If any):

Written By: Reviewed By

Name: Name:

Signature: Signature:

Date: Date:

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IPQA CHECK LIST FOR WAREHOUSE Annexure-3

Date :

ASPECTS (Write the appropriate application If meets-Y for yes, N for No and NA for not applicable)

Serial. Observation Y/N/NA

No.

1. Are personnel wearing appropriate protective clothing according to relevant SOP?
2. Are the of status label available on ARM/bulk bottles stored in cold rooms?
3. Check for the general cleanliness of the Material storage area/cold room area.
4. Are the approved & under test materials separately placed with proper labels?
5. Are the rejected material segregated to approved and under test materials with appropriate labeling?
6. Are the printed material kept in control at designated place?
7. Are the specified temperature conditions maintained in different areas of warehouse?
8. Are the temperatures of cold rooms maintained within specified limits?
9. Are the temperature indicators of cold rooms calibrated?
10. Are log books updated?
11. Are packaging scheme followed as per standard procedure?

Remarks (If any):

Written By: Reviewed By

Name: Name:

Signature: Signature:

Date: Date:

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IPQA CHECK LIST FOR CRITICAL UTILITIES Annexure-4

Date :

ASPECTS (Write the appropriate application If meets-Y for yes, N for No and NA for not applicable)

Serial Observation Y/N/NA

No.

1. Are the concerned SOPs present in place?
2. Are the operational instructions and diagram present with the equipment?
3. Are operating ranges of equipment as per the SOPs?
4. Are WFI, PW and Pure Steam being generated as per the standard specifications?
5. Are the pipelines being supplied to different areas proper coded?
6. Are the equipment/instruments coded with respective I.D No. /Serial No?
7. Are log books of equipment usage being maintained by the user?
8. Verify the record of sanitization and sterilization of water storage & distribution system.
9. Check the burning hours of UV Lamp in case of water system.
10. Are measuring devices/instruments calibrated?

Remarks (If any):

Written By: Reviewed By
Name: Name:
Signature: Signature:
Date: Date:

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IPQA CHECK LIST FOR QUALITY CONTROL AREA Annexure-5
Date :
ASPECTS (Write the appropriate application If meets-Y for yes, N for No and NA for not applicable)

- Serial Observation Y/N/NA
No.
1. Are entry exit log books for quality control facility/laboratory area being maintained by user?
 2. Are the personnel wearing appropriate protective clothing for entry in to laboratories with as per relevant SOP?
 3. Are the In-process samples and retention samples stored according to established procedure?
 4. Check the aseptic handling of personnel while work under LAF station,
 5. Are the log books concurrently maintained by user?
 6. Are the stability chambers maintained as per established storage condition?
 7. Are the concerned SOP's available for the respective test and instruments?
 8. Are the physical environmental conditions i.e. temperature, Differential pressure, Relative humidity within the control limits?
 9. Use of sterilized material and its status, if applicable.
 10. Ensure the status label on instruments of Calibration / Qualification.
 11. Adequate identification of reagents, solutions and materials used in laboratories and their uses log book.
 12. Disposal of used materials/media plates & its effective decontamination.

Remarks (If any):

Written By: Reviewed By
Name: Name:
Signature: Signature:
Date: Date:

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MONTHLY SCHEDULE FOR IPQA Annexure-6
Block / Frequency Ist week IIInd Week IIIrd Week IVth Week
Departme
nt

Total No. of Rounds. _____

*Vaccine Block-I and Vaccine Block-IV should thrice a week during non-manufacturing condition and Daily during manufacturing condition.

Name Designation Signature Date

Prepared By

Checked By

Approved By

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1.0 |Objective |
 |To lay down the procedure for Quality Assurance overview on |
 |area/processes/activities of Production, Quality Control, Warehouse, |
 |Utility system and Animal House. |
 2.0 |Scope |
 |This Standard Operating Procedure is applicable for Quality Assurance |
 |overview of all the manufacturing facilities, Quality Control Laboratory, |
 |Warehouse, Utility system and Animal House at Company name. |
 3.0 |Responsibility |
 |3.1 |IPQA personnel shall be responsible to visit and check the facility |
 |regarding cGMP compliance of Manufacturing area, Quality Control |
 |Laboratory, Warehouses, utility systems and Animal House. |
 |3.2 |IPQA personnel shall monitor and verify critical processes of |
 |Manufacturing. |
 |3.3 |Head, Quality Assurance Department shall ensure the implementation of |
 |this SOP. |
 4.0 |Abbreviations and Definitions |
4.1	AHU	Air Handling unit
4.2	BR	Batch Record
4.3	cGMP	Current Good Manufacturing Practices
4.4	CR	Cleaning Record
4.5	EMP	Environmental Monitoring Program
4.6	GMP	Good Manufacturing Practice
4.7	IPQA	In Process Quality Assurance
4.8	IPA	Iso Propyl Alcohol
4.9	LAF	Laminar Air Flow
4.10	PW	Purified Water
4.11	QA	Quality Assurance
4.12	QC	Quality Control
4.13	SOP	Standard Operating Procedure
4.14	SS	Stainless Steel
4.15	WFI	Water for Injection
5.0	Procedure	
5.1	IPQA Personnel shall overview all the areas/critical	
activities/processes to ensure		
5.1.1	All the areas in a state of GMP compliance.	
5.1.2	The activities/processes are carried out as per approved	
procedure/protocols.		
5.1.3	IPQA personnel shall be trained on clean room behavior,	
aseptic handling and QMS SOPs but not limited to.		
5.1.4	Apart from manufacturing processes IPQA person shall oversee	
other areas/ processes like Utility system, Warehouse, Animal		
House and Quality Control activities.		
5.2	IPQA checks/overview during various operations/process, as follows:	
5.2.1	IPQA personnel shall check in manufacturing area as follows,	
but not limited to:		
5.2.1.1	Gowning of personnel present for entry in work area	
as per the respective SOP.		
5.2.1.2	Personnel movement in classified area to avoid	
turbulence into the specified area.		
5.2.1.3	General cleanliness of area, type of disinfectant	
used and its correspondence logbook.		
5.2.1.4	Clean room behavior of personnel.	
5.2.1.5	Cleaning and sterilization status of material and	
equipment to be used in processes.		
5.2.1.6	The overview of physical environmental conditions	
i.e. Temperature, Differential pressure and Relative		
humidity		
5.2.1.7	Status of reagent used for carrying manufacturing	
procedure.		
5.2.1.8	Equipment, raw materials, in process materials and	
Solutions with proper status label & ensure they are		
with in expiry period.		
5.2.1.9	The use of calibrated measuring instruments and their	
identification		
5.2.1.10	Verify the batch number of Raw Materials against the	
approved label.		
5.2.1.11	Ensure that manufacturing and cleaning process are	
being carried as per approved BR and CR.		
5.2.1.12	The presence of the Batch Record, SOP and relevant	
documents in the area.		
5.2.1.13	Batch Records and logbooks shall be filled	
concurrently and accurately.		
5.2.1.14	Check the physical condition of manufacturing	
facility i.e. peel off epoxy, cracks in wall, floor,		
coving, door damage etc.		
5.2.1.15	IPQA personnel shall check the manufacturing	
process/area as per the annexure no.2		
5.2.2	IPQA personnel shall check during campaign change process, but	
not limited to:		
5.2.2.1	IPQA personnel shall verify the line clearance as per	
checklist of respective SOP of concerned		
manufacturing block.		
5.2.2.2	Whether the unused raw materials of previous batch	
removed from area or returned to warehouse (When		
applicable).		
5.2.2.3	Whether any Solutions, media or waste left over from	
previous production batch removed from area.		
5.2.2.4	Whether batch processing records pertaining to	
previous batch has been remove from the area.		

		5.2.2.5	Whether the documents pertaining to the processing of
			previous campaign are removed from the area.
		5.2.2.6	Whether the cleaning and sanitization of equipment
			systems/Area has been carried out after previous
			campaign as per the campaign change protocol.
		5.2.2.7	Are the cleaning and sanitization records available?
		5.2.2.8	Whether the results of cleaning and sanitation are
			comply to recommended limits
		5.2.2.9	Whether the records available for the campaign
			product.
		5.2.2.10	IPQA personnel shall also check the area as per
			Annexure no.2 during the campaign change process.
	5.2.3	IPQA personnel shall check over warehouses as follows but not	
			limited to:
		5.2.3.1	General cleanliness of the cold room area.
		5.2.3.2	Segregation of approved and under test materials.
		5.2.3.3	Operating temperature of the cold room and deep
			freezer as per the requirement.
		5.2.3.4	Temperature and Relative Humidity monitoring in
			material storage area.
		5.2.3.5	Segregation of materials with respect to their
			status.
		5.2.3.6	Log books for opening & closing of cold room &
			verification of data recorded.
		5.2.3.7	Ensure that all the materials are properly labeled.
		5.2.3.8	IPQA personnel shall check the warehouse as per the
			annexure no. 3.
	5.2.4	IPQA personnel shall check over utility system as follows but	
			not limited to:
		5.2.4.1	Operational SOP's in place.
		5.2.4.2	Availability of operational instructions & diagrams
			at the site.
		5.2.4.3	Operating ranges & operation modes of the equipment,
			as per the respective SOPs.
		5.2.4.4	Verify record for sanitization status of water
			storage & distribution system.
		5.2.4.5	Check calibration status of all measuring
			devices/instruments.
		5.2.4.6	Proper coding of the supply pipeline to different
			areas.
		5.2.4.7	Proper coding of the instruments with respective I.D
			Nos.
		5.2.4.8	Log books are maintained by the user and are update.
		5.2.4.9	Data recorded by the machine operators.
		5.2.4.10	IPQA personnel shall check the utility system as per
			the annexure no. 4.
	5.2.5	IPQA personnel shall check over clean room behavior & aseptic	
			process as follows, but not limited to:
		5.2.5.1	Entry of personnel in to clean room with proper
			gowning as per the SOP.
		5.2.5.2	Disinfection of the hands with 70% IPA after final
			gowning & gloving.
		5.2.5.3	Prior to commencement of work, disinfection of hands
			with 70% IPA.
		5.2.5.4	Under LAF station, ensure for minimum movement of
			personnel hands.
		5.2.5.5	Man and Material movement in facility.
		5.2.5.6	Restricted number of personnel in a specified area.
		5.2.5.7	Use of sterile forceps while picking up silicon
			tubes, caps during aseptic process, does not touch
			with gloved hands.
		5.2.5.8	Disinfection of hands & forceps used after each
			intervention.
		5.2.5.9	Sterilization status of the materials being used in
			the manufacturing process.
		5.2.5.10	Effective removal & decontamination of the used
			materials.
		5.2.5.11	Temperature, relative humidity and differential
			pressure are within the acceptance limit.
		5.2.5.12	Transfer of materials IN/OUT to the clean room area
			through pass boxes only.
	5.2.6	IPQA personnel shall check over Quality control area as	
			follows, but not limited to:
		5.2.6.1	Entry of personnel in to laboratories with proper
			gowning as per the SOP.
		5.2.6.2	Ensure the in process samples and retention samples
			stored according to established procedure.
		5.2.6.3	Under LAF station, ensure for minimum movement of
			personnel hands.
		5.2.6.4	Ensure the log books maintained by user are updated.
		5.2.6.5	Check the stability chambers are maintained as per
			established storage condition.
		5.2.6.6	Availability of SOP for the respective test and
			instruments.
		5.2.6.7	Recording of physical parameters (Pressure
			differential, Humidity, Temperature)
		5.2.6.8	Use of sterilized material and its status, as
			applicable.
		5.2.6.9	Ensure the status label on instruments of Calibration
			/ Qualification.
		5.2.6.10	Adequate identification of reagents, solutions and
			materials used in laboratories and their uses log.

			5.2.6.11	Disposal of used materials/media plates & its
			effective decontamination.	
			5.2.6.12	Log books/records are filled correct and concurrent.
			5.2.6.13	IPQA personnel shall check the Quality control as per
			Annexure-5.	
		5.2.7	Monthly schedule for IPQA checks shall be prepared as per	
			Annexure No.6 and IPQA personnel shall perform these checks as	
			per following schedule, but not limited to	
			For Xxxx-I and Xxxx-IV:	
			During manufacturing: Each working day	
			During non-manufacturing: Thrice a week	
			For Xxxx-II ,Xxxx-III, QC, Warehouse and Animal House,	
			Critical Utility : Twice a week	
		5.3	In case of manufacturing activity IPQA personnel shall verify the	
		activity following tentative batch schedule prepared as per SOP		
		QA/000.		
		5.4	IPQA personnel shall record the exact observation in Form No.	
		QA/000/00as per Annexure 1 for each check in manufacturing area,		
		Critical Utilities area, Warehouse and Quality Control and Animal		
		House if needed, will send to user department for necessary action.		
		User department shall write the corrective measures in the form and		
		submit to QA within two working days.		
		5.5	In case any observation related to physical condition of	
		manufacturing facility as per point no. 5.2.1.14, IPQA shall share		
		the observation with the user department as per annexure-1. Based on		
		the observation user department and engineering department shall		
		perform impact assessment as per SOP EN/000 prior to commencement of		
		civil work.		
		5.6	Observations reported during the QA overview shall be dealt with	
		applicable non-conformance/deviation handling procedures.		
		5.7	In case of facility shut down IPQA checks shall be performed on	
		requirement basis.		
	6.0	Forms and Records		
		6.1	QA/000/00IPQA Observation Report	Annexure - 1
		6.2	QA/000/00IPQA check list for manufacturing area	Annexure - 2
		6.3	QA/000/00IPQA check list for Warehouse	Annexure - 3
		6.4	QA/000/00IPQA check list for Critical Utilities	Annexure - 4
		6.5	QA/000/00- 03 QA check list for Quality Control	Annexure - 5
		6.6	QA/000/00Monthly schedule for IPQA checks	Annexure - 6
	7.0	References:		
	7.1			
	8.0	Distribution		
	9.0	History		
		Date	Revision Number	Reason for Revision

[illegible][illegible]

3.0 | Responsibility | Responsibility | Responsibility
| Responsibility | Responsibility | Responsibility

| 3.1 | 3.1 | |||||

4.0 | Abbreviations and Definitions | Abbreviations and Definitions | Abbreviations and Definitions |
Abbreviations and Definitions | Abbreviations and Definitions | Abbreviations and Definitions |
Abbreviations and Definitions | Abbreviations and Definitions | Abbreviations and Definitions

4.1 | | | | | | | |
5.0 | Procedure | Procedure | Procedure | Procedure | Procedure | Procedure | Procedure | Procedure |
Procedure

| 5.1 | 5.1 | Safety considerations: | Safety considerations: | Safety considerations: | Safety
considerations: | Safety considerations: | Safety considerations: | Safety considerations:

	Safety Considerations:	Safety Considerations:	Safety Considerations:	Safety Considerations:	Safety Considerations:	Safety Considerations:
5.1.1						
5.2 5.2 Equipment:	Equipment:	Equipment:	Equipment:	Equipment:	Equipment:	Equipment:
5.2.1						
5.3 5.3 Procedure:						

All personnel should avoid making entry errors
Information recorded should be legible, do not leave scope for misreading.
Print or use handwriting that is legible and easy to read
Always use permanent black ball pen for writing documents.
Short sentences preferred to long sentences
All defined spaces provided for entries that are not used must be filled as 'NA'. The spaces must be strike off starting from Top left to the bottom right corner with "NA" between the line. This should be followed by signature and date of the person at bottom right corner of the empty space. As this is not a quality impacting action, this should not be interpreted in a strict sense.
Correct

Wrong

In case of multiple entries in a single page, the intermittent empty spaces are to be filled by a Hyphen “-” where there is no scope for strike the space and put NA and Sign & Date. The situation can be clearly explained by the below mentioned Examples

Wrong

If an entry is missed on a controlled document, (e.g.) incase where there is a missing data that is retrievable from another document or database, make a copy or printout of the information, write an explanation, sign and date and attach it to the record.

When entries are made in record, these should be made indelibly in spaces provided for such entries, directly after performing the activities, and should identify the person making the entry.

If any error is found, immediately after recording the data cross out the incorrect information with single line (Horizontal / Vertical) and Enter correct data. Correction to entries should be dated and signed near the correction. Leave the original entry still readable

Correct

Wrong

Typographical error changes or additions do not require approval.

No page of a record or pre-numbered worksheet may be destroyed or removed. To invalidate a page, cross the page out with a single line, sign and date the cross out

If the date of correction is after the date of approval, a rationale must be provided

Backdating is prohibited. Neither signing nor initializing may be pre-dated nor post-dated.

Entries and observations must be recorded immediately after obtaining/verifying them. Do not proceed to the next step until the current step is documented.

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Do not use block letters for signatures

Correct

Wrong

- Photographs can be documented
- Flow charts provide substantial information at a momentary look
- Do not misrepresentation of data

Do not fabrication of data
Do not over-writing on the recorded data
The use of correction fluid, liquid paper, correction tape and the use of erasers are not allowed.
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Recording the data on document:
Use the following standards for writing letters and numbers, and for recording the date and time.
Use standard familiar English when writing information. Write clearly to avoid misunderstandings
Record numeric values using universal Arabic numbers (0,1,2,3,4,5...9)
Record time in a standard format with 24 hrs.
For e.g.: with traditional format, use the 24 hrs clock.
22:10 (for 10:10 PM)
Abbreviated date should be recorded as day/month/year (dd / mm / yyyy).
For e.g.: 21/07/2021
For all GMP activities, calendar year should be used as a reference.
Verification of data, information, calculations and tasks.

Attachments, printouts and charts:
Identify all printouts, charts attachments, etc. that are used to support a analytical record or process with the applicable Product name, batch number, time, date, and operator Full signature/Short signature.
Confirm there is enough ink supply for all print out and chart devices to make the printout easily readable. Print outs need to be continuous and if broken there needs to be an explanation included.
Do's and Dont's during Documentation
Do's
Use only Black, Indelible, Ball – point ink.
Make All entries legible (readable).
Sign and Date all entries.
Document each step in the respective sections of the document immediately before moving to the next step.
Put "NA", Sign & date empty spaces if it is not appropriate to fill them.
Record numbers less than one with a zero before the decimal point.
Dont's
Overwrite.
Use Liquid correction fluid.
Back date.
Record data before the action or event has occurred
Use ditto marks
Leave required data spaces blank.
Approve, Verify, review your own performance | Procedure:
All personnel should avoid making entry errors
Information recorded should be legible, do not leave scope for misreading.
Print or use handwriting that is legible and easy to read
Always use permanent black ball pen for writing documents.
Short sentences preferred to long sentences
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Do's and Dont's during Documentation

Do's

Use only Black, Indelible, Ball – point ink.

Make All entries legible (readable).

Sign and Date all entries.

Document each step in the respective sections of the document immediately before moving to the next step.

Put "NA". Sign & date empty spaces if it is not appropriate to fill them.

Record numbers less than one with a zero before the decimal point.

Dont's

Overwrite.

Use Liquid correction fluid.

Back date.

Record data before the action or event has occurred

Use ditto marks

Leave required data spaces blank.

Approve, Verify, review your own performance

6.0 | Forms and Records | Forms and Records | Forms and Records | Forms and Records | Forms and Records |

Forms and Records | Forms and Records | Forms and Records | Forms and Records

| NA | NA | NA | NA | NA | NA | NA | NA | NA

7.0 | References: | References: | References: | References: | References: | References: |

References: | References:
| 7.1 | 7.1 | | | | | |
| 7.2 | 7.2 | | | | | |
8.0 | Distribution | Distribution | Distribution | Distribution | Distribution | Distribution |
Distribution | Distribution | Distribution
| 8.1 | 8.1 | | | | | |
9.0 | History | History | History | History | History | History | History | History | History
9.0 | Date | Date | Date | Date | Revision Number | Revision Number | Revision Number | Reason for
Revision
| | | | | | | |

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Revision No.: 00

Title: Good Documentation Practices Supersedes : None
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1.0 Objective

To lay down the procedure for Good Documentation Practice (GDP)

2.0 Scope

The procedure is applicable to Good Documentation Practice (GDP)

3.0 Responsibility

3.1

4.0 Abbreviations and Definitions

4.1

5.0 Procedure

5.1 Safety considerations:

5.1.1

5.2 Equipment:

5.2.1

5.3 Procedure:

1.1.1 All personnel should avoid making entry errors

1.1.2 Information recorded should be legible, do not leave scope for misreading.

1.1.3 Print or use handwriting that is legible and easy to read

1.1.4 Always use permanent black ball pen for writing documents.

1.1.5 Short sentences preferred to long sentences

1.1.6 All defined spaces provided for entries that are not used must be filled as 'NA'. The spaces must be strike off starting from Top left to the bottom right corner with "NA" between the line. This should be followed by signature and date of the person at bottom right corner of the empty space. As this is not a quality impacting action, this should not be interpreted in a strict sense.

(a) Correct

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(b) Wrong

1.1.7 In case of multiple entries in a single page, the intermittent empty spaces are to be filled by a Hyphen "-" where there is no scope for strike the space and put NA and Sign & Date. The situation can be clearly explained by the below mentioned Examples

(a) Correct

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(b) Wrong

1.1.8 If an entry is missed on a controlled document, (e.g.) incase where there is a missing data that is retrievable from another document or database, make a copy or printout of the information, write an explanation, sign and date and attach it to the record.

1.1.9 When entries are made in record, these should be made indelibly in spaces provided for such entries, directly after performing the activities, and should identify the person making the entry.

1.1.10 If any error is found, immediately after recording the data cross out the incorrect information with single line (Horizontal / Vertical) and Enter correct data. Correction to entries should be dated and signed near the correction. Leave the original entry still readable

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Correct

Wrong

1.1.11 Typographical error changes or additions do not require approval.

1.1.12 No page of a record or pre-numbered worksheet may be destroyed or removed. To invalidate a page, cross the page out with a single line, sign and date the cross out

1.1.13 If the date of correction is after the date of approval, a rationale must be provided

1.1.14 Backdating is prohibited. Neither signing nor initializing may be pre-dated nor post-dated.

1.1.15 Entries and observations must be recorded immediately after obtaining/verifying them. Do not proceed to the next step until the current step is documented.

1.1.16 Documents or sections of documents shall not be changed or nullified except by following change control procedure

1.1.17 All documentation changes must be carefully documented, reviewed, and approved.

1.1.18 Original Information should be recorded on proper formats, immediately after action is completed. Entries of signatures or initials under Specified column must be made at the same time.

1.1.19 Signatures or initials must be unique.

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1.1.20 Your signature or initials mean that you take responsibility for the accuracy and correctness of the information you have written or verified.

1.1.21 There must be a signature record that identifies the signature and initials of an individual.

1.1.22 Pre-Inked stamps are not allowed for signature or initials on Documents.

1.1.23 Documentation lists are advisable as they assist personnel in retrieving exactly what was specified.

1.1.24 Every page of a document shall be marked with the page number as well as the total number of pages in the document.

1.1.25 Do not use block letters for signatures

Correct

Wrong

1.1.26 Photographs can be documented

1.1.27 Flow charts provide substantial information at a momentary look

1.1.28 Do not misrepresentation of data

1.1.29 Do not fabrication of data

1.1.30 Do not over-writing on the recorded data

1.1.31 The use of correction fluid, liquid paper, correction tape and the use of erasers are not allowed.

1.1.32 Documents or parts of documents may be copied for use as

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reference copies, but must be segregated from the official document

1.1.33 Documents must be stored or filed for the full retention period required by the company policy in a manner that allows convenient and easy retrieval of information for audit or inspection

1.2 Recording the data on document:

1.2.1 Use the following standards for writing letters and numbers, and for recording the date and time.

1.2.2 Use standard familiar English when writing information. Write clearly to avoid misunderstandings

1.2.3 Record numeric values using universal Arabic numbers (0,1,2,3,4,5...9)

1.2.4 Record time in a standard format with 24 hrs.

For e.g.: with traditional format, use the 24 hrs clock.

22:10 (for 10:10 PM)

1.2.5 Abbreviated date should be recorded as day/month/year (dd / mm / yyyy).

For e.g.: 21/07/2021

1.2.6 For all GMP activities, calendar year should be used as a reference.

1.2.7 Verification of data, information, calculations and tasks.

1.2.8 Attachments, printouts and charts:

1.2.8.1 Identify all printouts, charts attachments, etc. that are used to support a analytical record or process with the applicable Product name, batch number, time, date, and operator Full signature/Short signature.

1.2.8.2 Confirm there is enough ink supply for all print out and chart devices to make the printout easily readable.

Print outs need to be continuous and if broken there needs to be an explanation included.

1.3 Do's and Dont's during Documentation

1.3.1 Do's

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1.3.1.1 Use only Black, Indelible, Ball – point ink.

1.3.1.2 Make All entries legible (readable).

1.3.1.3 Sign and Date all entries.

1.3.1.4 Document each step in the respective sections of the document immediately before moving to the next step.

1.3.1.5 Put "NA", Sign & date empty spaces if it is not appropriate to fill them.

1.3.1.6 Record numbers less than one with a zero before the decimal point.

1.3.2 Dont's

1.3.2.1 Overwrite.

1.3.2.2 Use Liquid correction fluid.

1.3.2.3 Back date.

1.3.2.4 Record data before the action or event has occurred
1.3.2.5 Use ditto marks
1.3.2.6 Leave required data spaces blank.
1.3.2.7 Approve, Verify, review your own performance
6.0 Forms and Records
NA
7.0 References:
7.1
7.2
8.0 Distribution
8.1
9.0 History
Date Revision Number Reason for Revision

Vendor site audit Report Annexure - 16

1. Name of Organization :

2. Address :

3. Name & Address of Contact Person :

Date of Inspection :

Date of Preparation of Report :

Evaluation (use site audit checklist to support evaluation of vendor)

Title Grade*

A. General Information

B. Material Receipt, Storage and Handling

C. Manufacturing Operations, Equipment and Controls

D. Laboratories and Calibration

E. Document Control

F. Miscellaneous

(* Grading System: Excellent: E, Good: G, Satisfactory S, Not Satisfactory: N)

Overall Rating:

Recommendations of the Audit Team:

(Write the information in the assigned space or attach additional sheet s, if required)

Audit Team Representing xxxx Company.

Name Designation Department Signature

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Annexure – 17

Vendor approval report

1. Vendor details:

Name of the vendor

Category of the material

Addresses:

Address of Vendor

Address of Supplier

(if it is different from that of manufacturer)

2. Material(s)/Species of Animal to be procured/Testing to be performed:

3. Calculation of Valuation Index:

Valuation index : Applicable () Not applicable ()

Total marks: Marks obtained:

Valuation index = _____ x 100 =%

Page 1 of 2

Vendor approval report

4. Documents attached:

4.1 Site audit report ()

4.2 Filled checklist ()

4.3 Other users experience certificate ()

4.4 Copy of Quality System Certificate ()

4.5 Test reports / COA ()

4.6 CAPA report submitted by vendor ()

4.7 Other

5. Approvals:

5.1 The Vendor is previously approved

5.2 If yes, date of previous approval

5.3 Reason for preparing new report

5.4 Considered for the first time Yes / No / NA

6. Classification of Vendor:

Qualified () Conditional () Disqualified ()

7. Conclusion :

Following the above evaluation of the vendor, the vendor is approved / not approved.

Prepared by:

Name Designation Signature Date

Checked by:

Name Designation Signature Date

Remarks and Approval

Name Designation Signature Date

S. No. Name of Vendor (Manufacturer) Name of the Material Item Code

- 1.
- 2.
- 3.
- 4.
- 5.
- 6.
- 7.
- 8.
- 9.
- 10.

Prepared by Reviewed by Approved by

Name

Signature

Date

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Annexure - 19

List of approved contract testing laboratories / Service Providers

S. No. Name/Address of Service provider Service provided

Prepared by Reviewed by Approved by

Name

Signature

Date

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Annexure -20

QUALITY AGREEMENT

(Manufacture/Critical consumables)

Author Name Signature Date

Effective Date: Effective till

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QUALITY AGREEMENT

(Manufacture/Critical consumables)

Introduction

This document intends to provide expert guidance to the raw material supplier (Contract acceptor) and its customers (contract giver) for the implementation and maintenance of appropriate Quality agreements.

Scope

This Quality Agreement outlines the roles, responsibilities, deliverables and time requirements with respect to the Quality Assurance of the manufacture of the specified products (reference is made to the annexes) of Contract Giver (Customer) . It is an appendix to the Service Agreement executed by Contract Giver and Contract Acceptor (further referred as the "Agreement"). Contract Acceptor (Supplier) will operate a compliant GMP (including GDP) operation in all aspects of (where applicable) manufacture, testing, holding, packaging, labelling, shipping and/or distribution of the products of Contract Giver.

Parties to the agreement:

This Quality agreement is by and between the Contract giver i.e. xxxx and Contract Acceptor with office at (address), where the Contract Acceptor supplies Raw material suitable for pharmaceutical/biological use to xxxxx

The Quality Agreement may be revised on an "as-needed" basis or whenever the Agreement is renegotiated. All changes to this Quality Agreement will be reviewed and approved by authorised representatives of Quality Assurance from Contract Giver and Contract Acceptor ("Authorized Quality Representatives"). Any revision of this Quality Agreement must be documented, reviewed and approved by Authorised Quality Representatives from Contract Giver and Contract Acceptor and the Agreement shall be amended accordingly. Unless otherwise defined specifically in this Quality Agreement, all general terms used herein will be interpreted in accordance with the definitions provided in the Agreement. Any terms not so defined will be interpreted with the definitions proscribed in the Contract Acceptor Quality Policy, or when not so specified, will be interpreted with the definitions so stated in guidelines of the USA, Europe, WHO, etc (not limited) where applicable.

The Authorised Quality Representatives will resolve any disputes or conflicts relating to this Quality Agreement in a timely and equitable manner and in compliance with all applicable quality and regulatory requirements. Such resolutions shall be in writing and shall be signed by the authorised Quality Assurance representatives of each company. If any issue remains unresolved for more than twenty (20) business days, the senior Corporate Quality officials from each company should be contacted to resolve this issue.

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All communication affecting the contents of this Quality Agreement will be between the

Authorized Quality Representatives, as set forth below:

For Contract Giver: ____ (name) Quality Assurance

For Contract Acceptor : ____ (name) (Function)

In the event of a change, temporary or otherwise, of either Party's Authorized Quality Representative, the relevant Party shall notify the other in writing of such change.

Excipients Covered by the agreement

This Quality agreement pertains to the Raw material mentioned in Appendix I

Quality criteria:

The Contract Acceptor will manufacture, test and release the Raw material (mentioned in Appendix I) in accordance with the quality criteria / specifications as approved by the Quality Assurance personnel of Contract Giver enclosed as per Appendix II.

Term of Agreement:

This Agreement shall become effective and binding upon the date of the final signature and shall remain in effect until 3 years after the last delivery of Raw material by the Contract Acceptor to the Contract Giver unless the Contract Giver specifically requests an extension of the Agreement.

Revisions to Agreement:

No amendment to the terms of this Agreement shall be binding on the parties hereto unless made in writing and signed by an authorized representative of each of the parties.

Confidentiality:

The Contract Acceptor and Contract Giver should mutually undertake to observe secrecy on each other's know-how. This also applies after the agreement has ended. After the termination of the Agreement, neither of the parties to the contract is entitled to further commercially use the other party's know how disclosed under this agreement or let third parties do so without the other party's consent.

The obligation of confidentiality and of not using information does apply if that information:

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QUALITY AGREEMENT

(Manufacture/Critical consumables)

1. was already known to the other party before its disclosure under the Agreement and the other party declares this fact immediately or
2. has become common property through publication or by other means or
3. becomes known to one of the parties without originating directly or indirectly from the other party or
4. is made available to authorities due to legal requirements.

Responsibilities:

This quality agreement will outline the responsibilities of the Contract Giver and Contract Acceptor with regard to the quality activities described in the quality criteria tabulated below:

1. Regulatory Authorizations & GMP Compliance (where applicable)

Contract Contract
No Responsibilities
Giver Acceptor

1.1 Maintain all licenses, registrations and other authorizations as are required to operate a controlled facility and related functions under the Applicable Laws and guidelines ■
(including GMP where applicable)

1.2 Maintain facilities, utilities, equipment and conveyance and perform holding and distribution activities according to GMP guidance documents published by the

■
applicable authorities, approved internal policies and procedures and other regulatory requirements, as applicable.

1.3 Provide all necessary information pertinent to the Production of the Raw Material to ■
keep the regulatory filings current and up to date.

1.4 Prepare, maintain and update the Regulatory filings in accordance with all Applicable
■

Laws. Prepare and submit all required reports related to Product.

1.5 Provide all necessary information to Contract Giver concerning complaints, recalls, returned or salvaged drug products, and investigations associated with those activities,
■

where Contract Acceptor is involved, for the purpose of the issuing of Contract Giver's Annual Product Review.

2. Regulatory Actions & Inspections

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QUALITY AGREEMENT

(Manufacture/Critical consumables)

Contract Contract
Responsibilities
Giver Acceptor

2.1 Notify Contract Giver of any Regulatory Authority notice of inspection or inspection of the Facilities directly relating to the Raw Material(s) within two (2) business days.
■

Contract Giver may be on Contract Acceptor site during regulatory inspection as subject matter expert. Contract Acceptor manages all on site regulatory inspections.

2.2 Notify Contract Giver of any Regulatory Authority investigation relating to the Raw

Material(s).

2.3 Provide copies of any observation, FDA Form 483's, Warning Letters or the like from applicable Regulatory Authorities within two (2) business days of receipt, as such notices relate to the Raw Material(s) or Processes performed by the other party.

2.4 Reporting requirements to the respective regulatory agency(ies) with regard to

Contract Giver's registration documentation

2.4 Review of responses to inspection observations and questions prior to submission to

the concerned regulatory agency if pertinent to products distributed

2.5 Reporting requirements with regard to manufacturing site registration that may be required to support Contract Giver related activities, and will notify Contract Giver upon doing so.

2.6 Conduct investigations, prepare and approve corrective actions in response to regulatory observations. Provide copy of responses that could potentially impact

Contract Giver's Raw material(s) / Process to Contract Giver for review and approval prior to submission.

3. Premises and Equipment

Contract Contract

Responsibilities

Giver Acceptor

3.1 Maintaining facilities, utilities and equipment suitable for the production of Contract

Giver's raw material(s) and designed for their intended uses.

3.2 Maintaining facilities, equipment, computerised systems and utilities as are required to operate a manufacturing facility for Raw Material with Pharmaceutical applications under the Applicable Laws.

3.3 Calibrate measuring instruments used to monitor or control conditions of storage,

according to defined schedules.

3.4 Ongoing qualifications and/or verification activities according to Contract Acceptor

policies and procedures

3.5 Verify or validate equipment cleaning procedures to prevent cross-contamination from one raw material(s) to the next raw material(s) or from one batch to the next batch of the same raw material(s).

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4. Personnel

Contract Contract

Responsibilities

Giver Acceptor

4.1 Ensure that a management representative is appointed, who should be

responsible and qualified for maintaining the Quality System.

4.2 Ensure that all staff affected by this Agreement shall be appropriately trained.

4.3 Maintain a system to show training and qualification of personnel in procedures

relating to the Raw Material(s).

5. Manufacturing and batch documentation

Contract Contract

Responsibilities

Giver Acceptor

5.1 Establish procedures for change-over, cleaning, processing, packaging, labeling

and handling.

5.2 Establish provisions, parameters ranges or limits, test requirements and processing.

5.3 Maintain master batch records and ensure that master batch records, including test records, comply with the Regulatory filings, Clinical Trial Applications and/or the Agreed Supplies.

5.4 Contract Acceptor shall use Master batch records to document production process

of each batch in batch record.

6. Materials and consumables

Contract Contract

Responsibilities

Giver Acceptor

6.1 Procure, store, sample, test and release Materials and/or consumables used for the manufacturing of the Raw Material(s) in accordance with Contract Acceptor's written procedures.

6.2 Responsible for verification that materials are free of BSE/TSE (where

applicable) and providing regularly updated certification to Contract Giver.

6.3 Upon request, provide Contract Giver with documented evidence describing the qualifications of Contract Acceptors used to provide components, containers, labeling and packaging materials.

6.4 Develop test methods and specifications in case specifications are deficient or

■
unavailable

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QUALITY AGREEMENT
(Manufacture/Critical consumables)

Contract Contract
Responsibilities
Giver Acceptor
6.5 Store raw materials or consumables in accordance with the requirements set

■
forth in the specifications provided by Contract Giver.

7. Records

Contract Contract
No Responsibilities
Giver Acceptor
7.1 Ensure that records are made at each operation on Contract Giver's Product, in

■
such a way that all significant activities or events are traceable.
7.1 Store master record, batch records, distribution and transportation records and all other documentation related to the Product for a minimum of five (5) years.

■
Notify Contract Giver prior to destruction of any such records and provide the option to have the records sent to Contract Giver at Contract Giver's expense.
7.2 Contract Acceptor shall deliver batch record / CoA with every batch
7.3 Provide copies of all documentation necessary for the Contract Giver to respond
■
to enquiries by Regulatory Authorities.

8. Documentation for Safety

Contract Contract
Responsibilities
Giver Acceptor
8.1 Provide safety/hazard and handling data on the Raw Material(s) to Contract Giver ■

9. Samples

Contract Contract
Responsibilities
Giver Acceptor
9.1 Ship side samples and extra samples, where applicable, according to the
■
requirements of the Contract Giver
9.2 Retain reference samples of all materials, per Contract Acceptor procedures. ■
9.3 Store of retained samples under storage conditions in a secure area for 3 years
■
from the Date of Manufacture.

10. Testing & Analysis

Contract Contract
Responsibilities
Giver Acceptor
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(Manufacture/Critical consumables)

Contract Contract
Responsibilities
Giver Acceptor
10.1 Write protocols and perform verification or validation, as appropriate, of test
■
methods related to the release of the Raw Material(s).
10.2 Provide documented finished raw material(s) testing limits and specifications
■ Review
and other requirements.
10.3 Perform in-process testing and finished raw material(s) release testing according
■
to the requirements in 6.2.
10.4 Investigate all out-of-Specification test results ("OOS") and have available a report related to such investigations including a rationale for implemented ■
corrective and preventative actions.

11. Release

Contract Contract
Responsibilities
Giver Acceptor
11.1 Provide final release of Raw Materials for supply (COA) ■
11.2 Review the batch record within thirty (30) calendar days of receipt.
■
Communicate any issues identified to Contract Acceptor.

12. Storage and Shipment

Contract Contract
Responsibilities
Giver Acceptor

12.1 Specify finished raw material(s), packaging, storage and shipping requirements Approve ■
12.2 Store the finished raw material(s) in accordance with the specifications pending ■
■ release of the Raw Material(s).
12.3 Arrange shipments of Raw material(s) according to Contract Giver specified procedure to Contract Giver or Contract Giver's agent only upon written ■ approval from Contract Giver.

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QUALITY AGREEMENT

(Manufacture/Critical consumables)

13. Returned Goods

Contract Contract

Responsibilities

Giver Acceptor

13.1 Maintain a controlled and documented system and procedure for handling ■

returned goods and rejected materials.

13.2 Ensure that non-defective products returned from the market are stored separately from saleable stock, and are assessed according to defined procedures ■ before destruction or authorisation for re-use.

14. Deviations

Contract Contract

Responsibilities

Giver Acceptor

14.1 Investigate, resolve and document all abnormal events and deviations from the ■

master batch record and Specifications

14.2 Determine and document impact on raw material, plus corrective and ■

preventative actions related to the deviation(s)

14.3 Deviations will be managed through a written Procedure of the Contract ■

Acceptor

15. Complaints, Recalls and Withdrawals

Contract Contract

No Responsibilities

Giver Acceptor

15.1 Handling and reporting complaints related to the Raw Material. ■

15.2 Investigate complaints,. Issue reports and follow-up on corrective actions within ■

agreed timelines.

15.3 Notification within seven (7) days of any information coming into its possession ■

concerning the quality of previously released batch(es).

15.4 Contract Giver shall promptly notify Contract Acceptor of any complaint or ■

adverse event attributable to the quality of Product

15.2 Investigate complaints as requested by Contract Giver. Issue reports and follow- ■

up on corrective actions within agreed timelines.

15.6 In case of a Recall, Rejection or Withdrawal, Contract Acceptor will utilize its deviation procedure to investigate and manage the event and to the extent the ■ ■

issue directly or indirectly relates to other Contract Acceptor activities

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QUALITY AGREEMENT

(Manufacture/Critical consumables)

16. Change Management

Contract Contract

Responsibilities

Giver Acceptor

16.1 Ensure that changes related to

■ The manufacturing Process.

■ Reprocessing or rework

■ Changes in raw materials and consumables specified on the bill of materials. ■

■ Changes in layout, structure or site of manufacturing facility.

■ Changes in primary processing equipment that could affect Product.

■ Changes in testing

are documented including an Quality impact assessment on the quality of the Raw Materials

16.2 Notify Contract Giver of any proposed changes to Contract Acceptor processing Approve ■

that may impact the Product.

16.3 Contract Giver initiates such change by providing written notification through Contract Giver change control system to Contract Acceptor at least 30 days Inform ■ prior to the requested change.

16.4 Changes will be managed through a written Procedure of the Contract Acceptor ■

17. Contract Giver Audits

Contract Contract

No Responsibilities

Giver Acceptor

17.1 Provision of access to facilities and systems as they relate to the storage and

■
distribution of Products.

17.2 Provide timely notification for scheduling an audit. ■

17.3 "for cause" basis audits for product quality reasons as they apply to product distribution may be requested. Timing for a "for-cause" audit must be mutually ■ agreed upon but shall not be unreasonably delayed by both parties

17.4 Respond in writing to written audit observations within thirty (30) days of

■
receipt of report or as advised.

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QUALITY AGREEMENT

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18. Sub contracting

Contract Contract

No Responsibilities

Giver Acceptor

18.1 Select and approve the use of third party laboratories (subcontractors) for

Approve ■

intermediate and final Product testing, if applicable

18.2 Use of subcontractors for any part of the holding, distribution, packaging or

Approve ■

labelling of the product or intermediates only with the prior written approval.

18.3 Subcontractors will meet requirements for Approved Vendors. Contract Giver will accept Contract Acceptor's documented vendor compliance program ■ subject to outcome of Contract Giver audit.

Contracting of Third parties:

If Contract Acceptor does not do the work covered by the Agreement himself but contracts third parties to do it, the prior written consent of Contract Giver is required. In addition, Contract Acceptor must conclude a written Agreement on contracts manufacturing with the third party.

Also in case Contract Acceptor contracts third parties, Contract Acceptor is solely responsible towards Contract giver for the fulfillments of all contract obligations.

Term and Termination;

This Agreement shall come into force and effect once signed by the parties hereby. It shall be concluded for an indefinite period and may be terminated with due observance of a notice period of six months.

The right for an extraordinary termination for good cause is not affected

Each termination must be effected by registered mail

Definitions

Approval The term "Approval" is defined as concurrence between Contract Giver and Contract Acceptor, such as agreement on a proposed Changes, as evidenced in writing and signed by both companies' Authorized Quality Representatives.

Approved Vendor A vendor who has met minimum approval standards and who has been approved by both parties to provide required items or services that may impact product quality.

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QUALITY AGREEMENT

(Manufacture/Critical consumables)

Authorized Quality

Representative An individual named within the Quality Agreement with the authority to resolve any disputes or conflicts relating to this Quality Agreement in a timely and equitable manner and in compliance with all applicable quality and regulatory requirements.

Batch A specific quantity of material produced in a process or fraction of a process. Batches are defined as the material represented at the end of the intermediate processing steps and the material represented at the end of the processing step(s)

CFR Code of Federal Regulations

COA Certificate of Analysis: An authentic document that certifies that a specific batch of material has been evaluated in accordance with the Specification for that material.

Contract Acceptor Shall mean the contractor, which performs Manufacturing and Packaging Operations of the raw material .

Contract Giver Shall mean the contractor, which uses the raw material for the manufacturing of API (ARM) or finished product.

GMP Good Manufacturing Practice, including Current Good Manufacturing Practice: The essential rules governing the production, distribution and use of medicinal products, to safeguard public health. In this

Agreement GMP means both GMP and/or cGMP.

Date of Manufacture The date of fill into final container or packaging for intermediate or Finished Product

Disposition Decision by Contract Acceptor Quality on the suitability of the intermediate or Finished Product for further processing.

EMA European Medicines Evaluation Agency

Finished Product The product in it's agreed upon finished form. When the product is packed and labeled in it's market presentation it is referred to as "Drug Product" as well.

For Cause Visit The term "for cause visit" is used to describe site visits, other than audits or business discussions, for the purpose of reviewing

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QUALITY AGREEMENT
(Manufacture/Critical consumables)

documentation, facilities or processes related to a specific deviation affecting Product disposition.

Master Batch Record A detailed description of the production process for manufacturing of the Raw Material outlining the different actions an operator has to perform to complete the production process

OOS Out of Specification

Process Consumables Materials used as an aid in the manufacture of an intermediate or the Raw Material that do not themselves participate in a chemical or biological reaction. For the purposes of this document media, resins, raw materials, filters, membranes, product-contact materials or surfaces, disposable analytical test kits, disposable containers, product contact materials or surfaces, disposable lab supplies unique to the program or other items consumed and subcontracted analytical testing are considered to be process consumables.

Raw Material Any ingredient intended for use in the manufacture of the Raw Material, including those that may not appear in the final commercial presentation. These include chemicals used directly and/or indirectly in the manufacturing process.

Release Package Relevant documentation to be transferred to the Contract Giver to facilitate the release of Contract Giver's Finished Product. This package consists of copies of, Certificate of Analysis (COA) or analytical results, QA disposition of product statement, and where applicable, batch genealogy, and facility deviations.

Specification A set of criteria, limited to Critical In Process Tests or Finished Product, which must conform, to be considered acceptable for its intended use.

Page 13 of 16
QUALITY AGREEMENT
(Manufacture/Critical consumables)

Details of Contact Persons

Area Contract Giver Contract Acceptor

Quality Representative Name Name
Telephone:..... x... Telephone:..... x...
Fax:..... x... Fax:..... x...
e-mail: e-mail:

Product Recall Name Name
Telephone:..... x... Telephone:..... x...
Fax:..... x... Fax:..... x...
e-mail: e-mail:

Out of Office Person Name Name
Telephone:..... x... Telephone:..... x...
Fax:..... x... Fax:..... x...
e-mail: e-mail:

Name Name
Telephone:..... x... Telephone:..... x...
Fax:..... x... Fax:..... x...
e-mail: e-mail:

Page 14 of 16
QUALITY AGREEMENT
(Manufacture/Critical consumables)

APPENDICES:

I. List of Raw materials.

Page 15 of 16
QUALITY AGREEMENT
(Manufacture/Critical consumables)

SERVICE AGREEMENT

Service Agreement

This agreement made this ____ day of _____ at _____ is entered into by and between

Company Name XYZ, a Company incorporated and existing in accordance with the Companies Act, 1956 and having its Registered Office Location – Pin code, acting through Shri. Name , General Manager- Payment Cell & Compliances a representative duly authorized vide board resolution dated 2nd September, 2002 of the Company (hereinafter referred to as "Company Name") which expression shall unless be repugnant to the context include its successors and assigns of ONE PART.

AND

M/s. _____. Company incorporated and existing in accordance with the Companies Act. 1956 and having its Office at _____ acting through Mr. _____, _____ of M/s. _____, at _____ duly authorized vide General power of Attorney dated _____ (hereinafter referred to as "_____") which expression shall unless be repugnant to the context include its successors and assign of OTHER PART.

WHEREAS

a) Company Name is inter alia carrying on the business of manufacturing, marketing, selling and distributing bulk drugs/ medicines, pharmaceutical formulations, vaccines and other health related products and promotional items. (hereinafter referred to as the "Goods")

b) Company Name as and When required, has to deliver Goods, to and from its sales offices/ Loan Licensee manufacturers/ manufacturing locations/ Third party manufacturers/ clearing & forwarding Associates/ stockiest/ distributors or any associates in India.

c) _____ is carrying on the business of _____ and has represented to Company Name that they have all the necessary expertise, exposure and experience and having requisite infrastructure for _____ the Goods all over India (hereinafter referred to as the "Services") and is ready and willing to render the services to Company Name.

d) On the basis of the above representation of _____, Company name is Desirous of & has agreed to empanel/ engage _____ for the services as aforesaid.

Page 1 of 2

SERVICE AGREEMENT

e) The parties are desirous of recording their understanding into writing upon the terms and conditions hereinafter appearing.

NOW IT IS HEREBY AGREED BY AND BETWEEN THE PARTIES HERETO AS FOLLOWS:

1. Definition
2. Engagement
3. Scope of services
4. Duration/Termination
5. Delivery process
6. Service levels
7. Payment terms
8. Liability or Losses & Damages
9. Statutory compliance, undertaking, warranties and indemnity
10. Force majeure
11. Disputes, Arbitration and Jurisdiction
12. Miscellaneous

Sign for and behalf of

For M/s. Company Name For _____

Witnesses:

Name _____ Name _____

Address _____

List of Annexure:

Page 2 of 2

Annexure - 22

Approval report of additional material

1. Vendor details:

Name of the vendor

Category of the product

Address of Vendor

2. Material to be procured:

3. Document Review

History Satisfactory / Not Satisfactory

Delivery Schedule Satisfactory / Not Satisfactory

Vendor Capability Satisfactory / Not Satisfactory

4. Details of pre-purchase samples (If applicable)

Sample No. Sample Quantity Results Remarks (If any)

5. Conclusion:

On the basis of analysis of above information of the vendor, the Raw / Packaging material can be / can not be, included in approved Raw / Packaging materials of already approved Vendor.

Name Designation Signature Date

Prepared by:

Checked by:

Remarks and Approval

Name Designation Signature Date

Page 1 of 1
Caution Letter Annexure - 23

From,
Quality Assurance Department
Company name with location,

To,

Dear _____
This is related to the (Product details); it is an out of specification confirmed in analysis at our end for the said batch. Please investigate the matter at your end and share the results within 30 days. Following are the results of analysis at our end:-
Sr. No. Test performed Specification Results as per Results of analysis Remarks
COA received (Company name)

Further it is brought in your notice that Company name has temporarily stopped the purchase of the said material until detailed investigation report along with necessary CAPA report is submitted from you.

Regards,

Head – QA.
Cc;
CQA department.
Material Management department.
Warehouse department (Company name.)

Page 1 of 1
Vendor audit Schedule Annexure - 24

Schedule for the site audit of vendors for the year:-

Sr. Vendor Name and Planned / Audit schedule
Materials supplied
No. address actual Jan. Feb. March April May June July Aug. Sept. Oct. Nov. Dec.
Planned
1.
Actual
Planned
2.
Actual
Planned
3.
Actual
Planned
4.
Actual
Planned
5.
Actual
Planned
6.
Actual
Planned
7.
Actual
Plan updated by:- Sign/Date

Prepared by Reviewed by Approved by
Name
Designation
Signature/Date

Cc: Material Management department.

Page 1 of 1
Annexure - 25
Vendors Performance Report

Department: Report No: Year:

Name of the vendor :

Vendor category :

Observation

Prepared by Reviewed by Approved by
Name
Signature
Date

Page 1 of 1
Annexure - 26
Re-evaluation report of vendor

1. Vendor details:

Name of the vendor

Category of the product

Addresses:

Address of Vendor

Address of Vendor

(if it is different from that of manufacturer)

2. Material(s) / Services already being taken by the vendor:

3. Review of Vendor Performance Reports of last three years

Performance Report No. Year Status

Satisfactory / Not Satisfactory

Satisfactory / Not Satisfactory

Satisfactory / Not Satisfactory

Page 1 of 2

Re-evaluation report of vendor

4. Calculation of Valuation Index:

Valuation index : Applicable () Not applicable ()

Total marks: Marks obtained:

Valuation index = _____ x 100 =%

5. Documents attached:

5.1 Site audit report ()

5.2 Filled checklist ()

5.3 Other users experience certificate ()

5.4 Copy of Quality System Certificate ()

5.5 Test reports ()

5.6 Others ()

6. Conclusion :

Following the above evaluation of the vendor, the vendor is approved / conditionally approved / not approved to supply the materials.

Prepared by:

Name Designation Signature Date

Checked by:

Name Designation Signature Date

Remarks and Approval

Name Designation Signature Date

Page 2 of 2

Vendor evaluation and approval process flow chart Annexure - 27

Vendor evaluation and approval

RM/ PM/Culture media/testing Service providers / Contractual testing laboratories
animals/critical consumables

Already existing Service provider New Service provider, send

Already existing vendor Retrospective Retrospective evaluation every year pre-audit questionnaire

New Vendor, Send

evaluation every 3 years.

Send pre-audit Questionnaire. If $\geq 90\%$ pre -audit questionnaire marks obtained its approved.

Review and assessment of pre-audit questionnaire. If $\geq 90\%$ Site audit by vendor development team marks obtained - send 3 PP samples, to QC for analysis (except testing animals).

Sample Passes

Sample fails, Vendor Rejected

Non-Critical Materials Critical Materials

Site Yes

audit Site audit of manufacturing site, if required.
required

Site audit fail, Site audit pass, Vendor

No Vendor Rejected approved

Quality/Service agreement, if applicable.

Update approved vendor list

Page 1 of 1

Intimation of approval/Rejection of Vendor Annexure - 28

From,

Quality Assurance Department

Company name with Location

This is to inform you that QA has approved / re-approved /Rejected as conditional/Qualified vendor as a new / existing vendor on _____ as per below mentioned details:-

1. Name of the vendor :

2. Address of the vendor :

3. Type of material/Services :

4. Name of the material/Services :

5. Category of the material/Service :

The vendor shall remain approved / rejected for _____ years from the date of approval / rejected
i.e. till _____ or as per any further communication.

Regards,
Head – QA.

To;
CQA department.
Material Management department.
Warehouse department (Company Name.)

Page 1 of 1

Material Name:

Material Code: In

House Specification No.:

Material Type:

The Material is being used for manufacturing of following products:

ASSESSMENT

Assessment Criteria Assessment Criticality

|(Yes/No/NA) |(Critical/Non-critic|

|| |a| |

Does this material being used in final || |

bulk product? || |

If the product has direct contact with || |

final product? || |

Does it have impact on stability of || |

product? || |

If the material is a printing packaging || |

material? || |

Is the material is of biological origin? || |

Is the material sterile and / or || |

de-pyrogenated? || |

Conclusion: Based on this criticality assessment the material is classified as Critical/Non-critical.

|| Done by |Checked by |Approved by |

Name ||| |

Signature ||| |

Date ||| |

A. General Information |
	Name of Organization	:	
	Address	:	
	Name & Address of Contact	:	
	Person		
B. Building and Facility			
	Is the Facility easily accessible by suitable means of	:	Yes / No /NA
	transport		
		If yes, mention approximate distance and means to	
	reach the facility		
	Is the location free from possible source of air / water	:	Yes / No /NA
	/ soil, etc.	:	
	Pollution		
	Is ac current version of site plan / floor layout	:	Yes / No /NA
	available		
	Is the access to the facility restricted and controlled.	:	Yes / No /NA
	Are there adequate personnel facilities, like		
	a) Wash rooms,	:	Yes / No /NA
	b) Change rooms	:	Yes / No /NA
	c) Medical room, available	:	Yes / No /NA
	Is there a documented program for pest, rodent and bird	:	Yes / No /NA
	control		
		If yes, what is the frequency	:
	Are air curtains installed and functional	:	Yes / No /NA
	Are documented procedures available for cleaning,		Yes / No /NA
	disinfection or fumigation of the premises.	:	
	Is rotation of disinfectants followed	:	Yes / No /NA
	Is adequate lighting and ventilation provided	:	Yes / No /NA
	Do you have a separate Air Handling Unit at least for	:	Yes / No /NA
	finished product		
	Is quality of air supplied, especially to manufacturing		Yes / No /NA
	and packaging areas controlled, monitored and recorded	:	
	Is there a documented procedure for waste and scrap	:	Yes / No /NA
	disposal		
	Are there adequate utilities available	:	Yes / No /NA
		If yes attach their specifications	
	Is there a documented schedule to conduct preventive	:	Yes / No /NA
	maintenance of equipments		
C. Personnel			
	Attach qualification and experience record of key	:	Attach List
	personnel		
	Is there a documented training program for staff and		Yes / No /NA
	workers on cGMP, cGLP, SOPs, on-job activities,	:	
	personnel hygiene etc.		
		If yes, what is the frequency. Choose 3-4 employees	
	and check record		
	Is there a documented health care program (medical	:	Yes / No /NA
	examination)		
	Choose 3-4 employees and check record		
	Are the personnel provided with clean uniforms	:	Yes / No /NA
D. Stores			
	Are there physically segregated areas for:		
	Raw materials	:	Yes / No /NA
	Packaging materials	:	Yes / No /NA
	Are there physically segregated areas for:		
	Quarantine material (Under Test)	:	Yes / No /NA
	Approved material	:	Yes / No /NA
	Rejected material	:	Yes / No /NA
	Is the dispensing area separate	:	Yes / No /NA
	Is there any excise stores area	:	Yes / No /NA
	Do you have separate areas for non – beta lactum / beta –	:	Yes / No /NA
	lactum / other potent drugs (like cytotoxics)		
	Are there separate areas for materials requiring special	:	Yes / No /NA
	storage conditions controlled temperature & humidity with		
	monitoring devices like thermometers / hygrometers		
	Are the areas adequate for storage of the materials in	:	Yes / No /NA
	relation to their quantity		
	Is stock rotation program followed (like FIFO)	:	Yes / No /NA
E. Manufacturing			
	Is a process flow diagram for the manufacturing	:	Yes / No /NA
	operations available		
		If yes, attach a copy of process flow diagram	
	Is there a list of manufacturing equipment with	:	Yes / No /NA
	information like name, make capacity, Material of		
	Construction, date of procurement		
	Are the process equipment uniquely coded for	:	Yes / No /NA
	identification		
		If No, how are equipment identified during	
	documentation		
	Are the major equipment validated w. r. t. IQ, OQ & PQ	:	Yes / No /NA
	Are batches uniquely identified	:	Yes / No /NA
	Is there a system of line clearance before product or	:	Yes / No /NA
	batch change over		
	Is there any requirement of transportation of semi	:	Yes / No /NA
	finished product or material to other site. If Yes, how		
	is integrity of material guaranteed		
	What is the frequency of in process checks; please		

		exemplify			
			Process		Frequency
		i.	:		
		ii.	:		
		iii.	:		
		Are the sequence and timing of manufacturing operations	:	Yes / No /NA	
		recorded into a batch production record. Check and verify			
		form a current batch record			
		Is there a documented procedure for reprocessing	:	Yes / No /NA	
		What are the packing controls for (mention document	:	Yes / No /NA	
		name):			
			Printed packaging material reconciliation		
			Destruction of printed packaging material		
		Do you have procedure for printing of labels or printed	:	Yes / No /NA	
		material of similar appearance			
		Is there a documented procedure to monitor, control,	:	Yes / No /NA	
		record and prevent process deviations			
		Are the environmental conditions of finished goods	:	Yes / No /NA	
		storage area controlled w. r. t. temperature and			
		humidity?			
		Is there a documented procedure for rejection of finished	:	Yes / No /NA	
		goods			
		How many rejections have been recorded in the last	:	Yes / No /NA	
		quarter (for products offered for contract			
		manufacturing). Check a recent rejection record.			
		Has there been any product recall till date			
			If, yes provide details	:	Yes / No /NA
F	Quality Control				
		Is there a quality control unit	:	Yes / No /NA	
		Does Q.C. perform the following functions:			
		a.	Sampling of raw and packaging material	:	Yes / No /NA
		b.	Release of rejection of each batch of raw and	:	Yes / No /NA
			packaging material		
		c.	Development of specification for raw and packaging	:	Yes / No /NA
			material and in-process tests.		
		d.	Release or rejection of each batch or lot of	:	Yes / No /NA
			in-process bulk material		
		e.	Release or rejection of each batch of finished	:	Yes / No /NA
			product		
		f.	Issue, control and review of Batch Production Records	:	Yes / No /NA
		g.	Take control samples of raw materials and finished	:	Yes / No /NA
			goods		
		h.	Calibration of Q.C. Equipment	:	Yes / No /NA
		i.	Validation of test methods	:	Yes / No /NA
		j.	Evaluation of stability and quality of finished goods	:	Yes / No /NA
		Who performs the functions of Quality Assurance	:		
		Are there separate rooms / areas marked for			
			Storage RM Samples	:	Yes / No /NA
			Storage Control Samples	:	Yes / No /NA
			Stability Testing Equipment	:	Yes / No /NA
		Is there a system of Internal Audit	:	Yes / No /NA	
		Are there any reference books available	:	Yes / No /NA	
G	Documentation				
		Are all documents controlled i.e. bear a unique title,	:	Yes / No /NA	
		number, version number, effective date, etc.			
		Are all documents approved and authorized before	:	Yes / No /NA	
		distribution			
		Is the documents distribution record maintained	:	Yes / No /NA	
		Are the following documents available			
			Specifications (RM, PM, In process product, finished	:	Yes / No /NA
			product)		
			SOPs (RM, PM, In process product. Finished product)	:	Yes / No /NA
			Master Batch Production Record / Master formula	:	Yes / No /NA
			record		
			Raw Material Receipt and Issuance Records	:	Yes / No /NA
			Packaging Records	:	Yes / No /NA
			In-process Testing Records	:	Yes / No /NA
			Finished Goods Records	:	Yes / No /NA
			Microbiological Records (if applicable)	:	Yes / No /NA
			Calibration Records (Manufacturing and Q. C)	:	Yes / No /NA
			Operation , Cleaning & Preventive Maintenance Records	:	Yes / No /NA
			Complaint Records	:	Yes / No /NA
			Rejection Record	:	Yes / No /NA
			Product Recalls	:	Yes / No /NA
		Are SOPs available for the following operations			
			Quarantine and storage of material	:	Yes / No /NA
			Issuance of material	:	Yes / No /NA
			Quality control for In-process	:	Yes / No /NA
			Quality control for Finished Goods	:	Yes / No /NA
			Calibration of instruments	:	Yes / No /NA
			Product Recalls	:	Yes / No /NA
			Microbiological testing	:	Yes / No /NA
		Is a Certificate of Analysis issued for a batch released	:	Yes / No /NA	
		What are the rejection levels	:		
		What is the packaging material used for Primary Packaging	:		
		, Material of Construction			

Date of Inspection:

Audit Team Representing xxxx Company

Name	Designation	Department	Signature

Vendor's Representatives

Name	Designation	Department	Signature

A. General Information | | |
	Name of Organization	:	
	Address	:	
	Person(s) contacted at site during	:	
	audit		
B. Material Receipt, Storage and Handling			
	Is there a procedure for qualification of raw material	:	Yes / No /
	sourcing		NA
	Is a current list of approved vendors available and	:	Yes / No /
	circulated to relevant departments		NA
	Is adequate precautions taken to properly segregate	:	Yes / No /
	approved material from material under test of rejected		NA
	Is entry to storage area restricted	:	Yes / No /
			NA
	Is rejected material adequately controlled to prevent	:	Yes / No /
	misuse		NA
	Is there a separate or segregated area for sampling	:	Yes / No /
			NA
	Are there documented sampling procedures	:	Yes / No /
			NA
	Are materials properly identified w. r. t. status,	:	Yes / No /
	quantity etc.		NA
	Is there a retesting schedule	:	Yes / No /
			NA
	Where required, are materials stored in controlled	:	Yes / No /
	environment (e. g. humidity, temperature, etc.)		NA
C. Manufacturing Operations, Equipment and Controls			
	Are the items being manufactured identifiable throughout	:	Yes / No /
	the approval		NA
	Are the manufacturing operations orderly in flow	:	Yes / No /
			NA
	Are the manufacturing processing and facilities adequate	:	Yes / No /
	to produce high quality items		NA
	If gang printing is undertaken, are there adequate	:	Yes / No /
	controls to prevent mix-ups		NA
	Is there an adequate system of control of labels with	:	Yes / No /
	pre-printed lot number and / or expiration dates		NA
	Is the batch no. or lot no. reflective of homogenous	:	Yes / No /
	production run. If yes, exemplify		NA
	Is there a line clearance procedure	:	Yes / No /
			NA
	Are regular in-process tests performed by Quality	:	Yes / No /
	Control		NA
	Are manufacturing procedures formalized and controlled	:	Yes / No /
			NA
	Is there a documented procedure for handling process	:	Yes / No /
	deviations		NA
	Are there documented procedures for handling reworking	:	Yes / No /
	for rectification of errors		NA
	Are standard procedures of working and similar	:	Yes / No /
	functional documents readily available to employees for		NA
	reference		
	Is the batch or lot reconciled after end of	:	Yes / No /
	manufacturing operation		NA
	Are waste material clearly marked	:	Yes / No /
			NA
	Is there a pest control procedure	:	Yes / No /
			NA
	Are colour standards available, and are they checked on	:	Yes / No /
	a periodic basis for any change in appearance		NA
	Is there a system for 100% inspection of labels	:	Yes / No /
			NA
	In the event of in-process rejections, is there proper	:	Yes / No /
	investigation procedure to prevent recurrence		NA
	Are the equipment adequately identified	:	Yes / No /
			NA
	During operations, is product information displayed on	:	Yes / No /
	the equipment		NA
	Are there adequate procedures for final product	:	Yes / No /
	inspection, sampling and testing		NA
	Are the final inspection / test results recorded	:	Yes / No /
			NA
	Are the batch records reviewed before final dispatch of	:	Yes / No /
	products		NA
	Are the batch records and other related documents stored	:	Yes / No /
	for adequate length of time		NA
	Is the document retrieval system adequate	:	Yes / No /
			NA
	How is non conforming material controlled	:	Yes / No /
			NA
	Are samples of finished product retained	:	Yes / No /
			NA
	How adequate are security systems for drug labels	:	Yes / No /
			NA
	Is there a lot recall procedure	:	Yes / No /
			NA

D	Laboratories and Calibration				
		Is QC adequately staffed	:	Yes / No /	
				NA	
		Are sufficient equipment / instruments available for	:	Yes / No /	
		proper functioning of QC	NA		
		Are samples from incoming, in-process and final product	:	Yes / No /	
		materials properly controlled	NA		
		Is each lot checked against customer approved copy for	:	Yes / No /	
		correct printing	NA		
		Is each lot checked against a drawing for correct die	:	Yes / No /	
		cut	NA		
		Are certifications of inspection / test results	:	Yes / No /	
		available to customers	NA		
		Are tests / inspections properly documented	:	Yes / No /	
				NA	
		Are there acceptable procedures to assure that	:		
		instruments are calibrated and calibration recorded			Yes / No /
		Is calibration traceable to appropriate standards	:	NA	
				Yes / No /	
				NA	
		Are the instruments / equipments maintained in proper	:	Yes / No /	
		order	NA		
E	Document Control				
		Are functional documents like specification, Standard	:	Yes / No /	
		Operating Procedures, etc. comprehensive in content		NA	
		Are the documents in use 'current'	:	Yes / No /	
				NA	
		Are the documents approved and authorized by competent	:	Yes / No /	
		authority prior to distribution	NA		
		Is there a procedure to ensure retrieval and subsequent	:	Yes / No /	
		destruction of obsolete documents	NA		
F	Miscellaneous				
		Is there regular training program for employees,	:	Yes / No /	
		in-house or otherwise	NA		
		If yes, list type of training and frequency	:	Yes / No /	
				NA	
		Is there a program for training	:	Yes / No /	
				NA	
		Is there a documented procedure for performing internal	:	Yes / No /	
		audits	NA		
		Are internal audit reports prepared and circulated to			
		concerned department for compliance of observed	:	Yes / No /	
		deficiency	NA		
		Is there a 'Quality system' that addresses the			
		organizational structure, responsibilities, procedures,	:	Yes / No /	
		processes and resources for implementing quality		NA	
		management			
		Are there periodic management reviews of the 'Quality	:	Yes / No /	
		system'	NA		
		Are there sufficient engineering, production and quality			
		control support to requirements of Company Name XYZ.	:	Yes / No /	
				NA	

Date of Inspection:

Audit Team Representing Company Name XYZ.

|Name|Designation|Department|Signature|

Vendor's Representatives

|Name|Designation|Department|Signature|

A. General Information |
	Name of Organization	:	
	Address	:	
	Name & Address of Contact	:	
	Person		
B. Building and Facility			
	Is the Facility easily accessible by suitable means of	:	Yes / No /NA
	transport		
	If yes, mention approximate distance and means to		
	reach the facility		
	Is the location free from possible source of air / water	:	Yes / No /NA
	/ soil, etc.	:	
	Pollution		
	Is current version of site plan / floor layout available	:	Yes / No
	Is the access to the facility restricted and controlled.	:	Yes / No /NA
	Are there adequate personnel facilities, like		
	a) Wash rooms,	:	Yes / No /NA
	b) Change rooms	:	Yes / No /NA
	c) Medical room, available	:	Yes / No /NA
	Is there a documented program for pest, rodent and bird	:	Yes / No /NA
	control		
	Are documented procedures available for cleaning ,	:	Yes / No /NA
	disinfection or fumigation of the premises.	:	
	Is rotation of disinfectants followed	:	Yes / No /NA
	Is adequate lighting and ventilation provided	:	Yes / No /NA
	Is quality of air supplied, especially to manufacturing	:	Yes / No /NA
	and packaging areas controlled, monitored and recorded	:	
	Is there a documented procedure for waste and scrap	:	Yes / No /NA
	disposal		
	Are there adequate utilities available	:	Yes / No /NA
	Is there a schedule to conduct preventive maintenance of	:	Yes / No /NA
	equipments		
C. Personnel			
	Attach qualification and experience record of key	:	Attach List
	personnel		
	Is there a documented training program for staff and	:	Yes / No /NA
	workers?		
	Is there a documented health care program (medical	:	Yes / No /NA
	examination)		
	Are the personnel provided with clean uniforms	:	Yes / No /NA
D. Stores			
	Are there physically segregated areas for:		
	Raw materials	:	Yes / No /NA
	Packaging materials	:	Yes / No /NA
	Are there separate areas for materials requiring special	:	Yes / No /NA
	storage conditions: controlled temperature & humidity		
	with monitoring devices like thermometers / hygrometers		
	Are the areas adequate for storage of the materials in	:	Yes / No /NA
	relation to their quantity		
	Is stock rotation program followed (like FIFO)	:	Yes / No /NA
E. Manufacturing			
	Is a process flow diagram for the manufacturing	:	Yes / No /NA
	operations available		
	If yes, attach a copy of process flow diagram	:	Attach
	Is there a list of manufacturing equipment with	:	Yes / No /NA
	information like name, make capacity, Material of		
	Construction, etc.		
	Are the environmental conditions of finished goods	:	Yes / No /NA
	storage area controlled w. r. t. temperature and	:	
	humidity?		
	Is there a documented procedure for rejection of finished	:	
	goods?		
F. Quality Unit			
	Is there a quality control unit	:	Yes / No /NA
	Who performs the functions of Quality Assurance	:	
	Is there a system of Internal Audit	:	Yes / No /NA
	Are there any reference books available	:	Yes / No /NA
G. Documentation			
	Are all documents controlled	:	Yes / No /NA
	Is a Certificate of Analysis issued for the material	:	Yes / No /NA
	released		
	What are the rejection levels	:	

Date of Inspection:

Audit Team Representing xxxx Company

Name Designation Department Signature |
| | | |

|||||
|||||

Vendor's Representatives

Name	Designation	Department	Signature

A. [General Information |
	Name of Organization	:	
	Address	:	
	Name & Address of Contact	:	
	Person		
	Has CPCSEA/GLP approved (If	:	
	yes, Please attach a copy of		
	the certificate)		
B. [Building and Facility			
	Is the Facility easily accessible by suitable	:	Yes / No /NA
	means of transport		
	If yes, mention approximate distance and		
	means to reach the facility		
	Is the location free from possible source of		
	air / water / soil, etc.	:	
	Pollution		
	Is current version of site plan / floor	:	Check and
	layout available		attach copy
	Is the access to the facility restricted and	:	Yes / No /NA
	controlled.		
	Are there adequate personnel facilities, like		
	a) Wash rooms,	:	Yes / No /NA
	b) Change rooms	:	Yes / No /NA
	c) Medical room, available	:	Yes / No /NA
	Is there a documented program for pest,	:	Yes / No /NA
	rodent and bird control		
	Are documented procedures available for		
	cleaning , disinfection or fumigation of the	:	
	premises.		
	Is rotation of disinfectants followed	:	Yes / No /NA
	Is adequate lighting and ventilation provided	:	Yes / No /NA
	Is there an appropriate functioning		
	environmental control system regarding:		
	Temperature?		
	Humidity?		NA
	Adequate air changes /hour?		
	Time controlled lighting system?		NA
	Appropriate noise control system?		
	Is there a documented procedure for waste and	:	Yes / No /NA
	scrap disposal		
	Are there adequate utilities available	:	Yes / No /NA
	Is there a schedule to conduct preventive	:	Yes / No /NA
	maintenance of equipments		
	Are there enough animal rooms of appropriate		
	design to allow separate housing of:		
	The breeding colony		
	Different animal species		NA
	Sick animals		
	Is the Quality System enforced?		
	Are there facilities for collection and		
	disposal of animal waste and of dead animals		NA
	to minimize disease hazards and environmental		
	contamination?		
	Are there facilities for cleaning,		
	sanitizing, sterilizing and maintaining		NA
	supplies and equipments including animal		
	cages and racks?		
C [Personnel			
	Attach qualification and experience record of	:	Attach List
	key personnel		
	Is there a documented training program for	:	Yes / No /NA
	staff and workers?		
	Is there a documented health care program	:	Yes / No /NA
	(medical examination)		
	Are the personnel provided with clean	:	Yes / No /NA
	uniforms		
D [Stores			
	Are there separate storage areas for		
	equipments, animal feed, and bedding which		NA
	are protected from infection/contamination,		
	and with refrigeration where needed?		
	Are the areas adequate for storage of the	:	Yes / No /NA
	materials in relation to their quantity		
	Is stock rotation program followed (like	:	Yes / No /NA
	FIFO)		
E [Documentation			
	Are all documents controlled	:	Yes / No /NA
	Is a Certificate of Analysis issued for the	:	Yes / No /NA

|| |material released || |
|| |What are the rejection levels |: ||

Date of Inspection:

Audit Team Representing Company Name XYZ.

Name	Designation	Department	Signature

Vendor's Representatives

Name	Designation	Department	Signature

A. General Information |
	Name of Organization	
	Address	
	Name & Address of Contact	
	Person	
B. Questionnaire		
1 Quality System:		
1.1 Has the management established its quality policy in Yes / No / NA		
	writing?	
	If yes, please attach a copy.	
1.2 Is there an Organizational Chart? Yes / No / NA		
	If yes, please attach a copy.	
1.3 Are the personnel's responsibilities established in Yes / No / NA		
	their	
	job descriptions?	
1.4 Has the company / organization adequate resources in Yes / No / NA		
	terms of finance and personnel?	
1.5 Has the company / organization designated a responsible Yes / No / NA		
	person	
	with enough authority to assure the established quality	
	system?	
1.6 Is the company / organization approved by government Yes / No / NA		
	(State/Central/UT)	
	Please mention the nature of your organization:	
	(Government/ Semi government/Private)	
1.7 Does your organization carry any recognition or Yes / No / NA		
	accreditation from National / International agency like	
	ISO/WHO or any other agency?	
	If yes, please attach a copy of the latest certificate.	
1.8 Are personnel qualified to perform their tasks? Yes / No / NA		
	Please attach the list of educational qualifications of	
	personnel as a minimum of those working at key	
	positions.	
2. Documentation:		
2.1 Is the system of approval of documents established? Yes / No / NA		
2.2 Is there an established procedure for approving changes Yes / No / NA		
	in documents?	
2.3 Is there a system to keep records of approved documents Yes / No / NA		
	with their edition/revision number?	
2.4 Are procedures in force to assure the identification and Yes / No / NA		
	traceability of works/ tests carried out?	
2.5 Is there a procedure for identifying, coding and Yes / No / NA		
	archiving of records?	
3. Calibration, Maintenance and Validation:		
3.1 Is there a maintenance program in force for all Yes / No / NA		
	measuring and test equipments?	
3.2 Do you have a calibration program? Yes / No / NA		
3.3 Do you have procedures in force for performing Yes / No / NA		
	Calibrations and Maintenance work?	
3.4 Does the measuring and test equipments clearly Yes / No / NA		
	identified with calibration status?	
3.5 Are the standards used for Testing / Validation / Yes / No / NA		
	Calibration traceable to National / International	
	Reference Standards?	
3.6 Are the Instruments / Standards used for calibration Yes / No / NA		
	regularly updated?	
3.7 Do you have Standard Operating Procedures (SOPs) for Yes / No / NA		
	carrying out Testing / Validation / Calibration?	
	If so, can these be made available to the contracting Yes / No / NA	
	party on demand?	
4. Audit and Inspection:		
4.1 Is the facility subjected to quality audits? Yes / No / NA		
4.2 Are quality audits performed to verify the efficacy of Yes / No / NA		
	the Quality System?	
4.3 Is there a defined plan for audits? Yes / No / NA		
4.4 Is there a follow up of corrective actions taken? Yes / No / NA		
5. Training:		
5.1 Is there a procedure for training of staff in their Yes / No / NA		
	respective areas of working?	
5.2 Are training programs established? Yes / No / NA		
5.3 Are training records maintained? Yes / No / NA		
6. Testing:		
6.1 Are documented procedures in force for handling samples Yes / No / NA		
	and testing?	
6.2 Is there a traceability of the records for a test sample Yes / No / NA		
	/ work carried out? i.e. green sheets/logbooks/record	
	forms etc.	
6.3 Do you have required facilities for storage of samples Yes / No / NA		
	before (and after) testing?	
6.4 Are there separate areas for samples requiring special Yes / No / NA		
	storage conditions?	
6.5 Do you issue certificate of analysis for the Yes / No / NA		
	sample/product tested?	
6.6 Are there adequate facilities for testing of sterile Yes / No / NA		
	products including sterility and pyrogen testing?	

6.7	Is the sterility testing room classified?	Yes / No / NA
6.8	Do you maintain an animal house?	Yes / No / NA
6.9	Does the animal house have adequate area and required	Yes / No / NA
	facilities?	
6.10	Is there a qualified veterinarian to look after the	Yes / No / NA
	animal house?	
6.11	Is the animal house registered with regulatory	Yes / No / NA
	authorities?	
6.12	Is permission sought from the regulatory authorities for	Yes / No / NA
	using animals for various tests?	
6.13	Is periodical examination of animals for health status	Yes / No / NA
	carried out?	
6.14	Do you have arrangements to quarantine the animals after	Yes / No / NA
	reception?	
6.15	Do you have provision for isolation of sick animals as	Yes / No / NA
	well as animals under test?	
6.16	Are qualified and experienced personnel available for	Yes / No / NA
	carrying out testing in animals?	
6.17	Do you have an approval system for suppliers from whom	Yes / No / NA
	the testing animals are purchased? Please attach the	
	list of institution (s) from where the animals are	
	procured.	
7.	Remarks, if any:	

Date of Inspection:

Audit Team Representing xxxx Company

Name	Designation	Department	Signature

Vendor's Representatives

Name	Designation	Department	Signature

A. |General Information | | |
		Name of the service provider	:		
		Address & Contact No.	:		
		Nature of the service	:		
		Date audit conducted	:		
		Brief company profile	:		
		- Year of establishment	:		
		- Registration of	:		
		the agency			
		- No. of employees	:		
		working			
		- Certification	:		
		(if any)			
		- Any	:		
		accreditation from			
		reputed company			
		Whether credentials of the	:		
		company are impressive			
		Whether the firm was prosecuted	:		
		by law			
B	Building & facility :				
		Is the location suitable for the facility	:	Yes / No /	
		Is the access to the facility restricted and controlled	:	Yes / No /	
		Whether the facility is properly cleaned & maintained	:	Yes / No /	
		Is it devoid of cracks etc.	:	Yes / No /	
		Is it designed to keep it free from birds, pests, vermins	:	Yes / No /	
		& rodent.		NA	
		Are there adequate personnel facility like – wash rooms,	:	Yes / No /	
		canteen, medical rooms etc.		NA	
		Is adequate lighting & ventilation provided	:	Yes / No /	
		Are there adequate utilities available	:	Yes / No /	
		Whether the facility is properly cleaned & maintained	:	Yes / No /	
		Are periodic records of cleaning & maintenance kept	:	Yes / No /	
		properly		NA	
		Is there logical placement of equipments	:	Yes / No /	
		Is the warehouse area suitable & sufficient to the facility	:	Yes / No /	
		requirement		NA	
		Are the equipments properly maintained & calibrated	:	Yes / No /	
		Is there adequate facility for housing the laboratory	:	Yes / No /	
		animals		NA	
		(if Yes, please attach separate sheet)			
		Is there any system in place to address the complaint	:	Yes / No /	
		received against the service provider.		NA	
C	Personnel:				
		Is the organogram available & adequate	:	Yes / No /	
		Is the technical staff competent in qualification &	:	Yes / No /	
		experience		NA	
		Is the number of personnel employed adequate in proportion	:	Yes / No /	
		to the workload		NA	
		Are Personnel also provided with regular in-service	:	Yes / No /	
		training		NA	
		Whether the supervisory staff are whole time employees	:	Yes / No /	
D	Documentation :	:	Yes / No /		
		Are all documents controlled	:	Yes / No /	
		Are the documents approved signed & dated by authorized	:	Yes / No /	
		person before distribution		NA	
		Do the documents specify	:	Yes / No /	
		title		NA	
		nature			
		purpose			
		Is the documentation handled by electronic data processing	:	Yes / No /	
		methods		NA	
		Are the records maintained electronically protected by	:	Yes / No /	
		suitable means.		NA	
		Is record available for calibration & validation of	:	Yes / No /	
		equipments.		NA	

Date of Inspection:

Audit Team Representing Company Name .

|Name |Designation |Department |Signature |
 | | | | |

|||||
|||||

Vendor's Representatives

Name	Designation	Department	Signature

		Name of Organization	:	
		Address	:	
		Name & Address of Contact	:	
		Person		

Date of Inspection :

Date of Preparation of Report :

Evaluation (use site audit checklist to support evaluation of vendor)

		Title	Grade*
A.	General Information		
B.	Material Receipt, Storage and Handling		
C.	Manufacturing Operations, Equipment and		
		Controls	
D.	Laboratories and Calibration		
E.	Document Control		
F.	Miscellaneous		

(* Grading System: Excellent: E, Good: G, Satisfactory S, Not Satisfactory: N)

Overall Rating:

Recommendations of the Audit Team:

(Write the information in the assigned space or attach additional sheet s, if required)

Audit Team Representing xxxx Company.

Name	Designation	Department	Signature

1. Vendor details: |
| Name of the vendor | |
| Category of the material | |
| Addresses: | |
| Address of Vendor | |
| | |
| | |
| | |
| | |
| Address of Supplier | |
| (if it is different from | |
| that of manufacturer) | |
| | |
2. Material(s)/Species of Animal to be procured/Testing to be performed: |
| | |
| | |
| | |
| | |
| | |
| | |
| | |
| | |
| | |
| | |
| | |
| | |
3. Calculation of Valuation Index: |
| Valuation index : Applicable () Not applicable () |
| Total marks: | Marks obtained: | |
| | |
| Valuation index = _____ x 100 =% |
| | |
4. Documents attached: |
| 4.1 Site audit report () |
| 4.2 Filled checklist () |
| 4.3 Other users experience certificate () |
| 4.4 Copy of Quality System Certificate () |
| 4.5 Test reports / COA () |
| 4.6 CAPA report submitted by vendor () |
| 4.7 Other |
5. Approvals: |
| 5.1 The Vendor is previously | | |
| | approved | |
| 5.2 If yes, date of previous | |
| | approval | |
| 5.3 Reason for preparing new | |
| | report | |
| 5.4 Considered for the first | Yes / No / NA |
| | time | |
6. Classification of Vendor: |
| Qualified () Conditional () |
| Disqualified () |
7. Conclusion : |
| Following the above evaluation of the vendor, the vendor is approved / not |
| approved. |
Prepared by: |
Name | Designation | Signature | Date |
| | | |
Checked by: |
Name | Designation | Signature | Date |
| | | |
Remarks and Approval |
| |
Name | Designation | Signature | Date |
| | | |

S. No.	Name/Address of Service provider	Service provided

	Prepared by		Reviewed by		Approved by	
Name						
Signature						
Date						

Annexure – 2

1. Name of Organization :
2. Address of manufacturing site/Laboratory / service provider :
3. Name & contact details of person Responsible for commercial matters :
4. Name & contact details of person Responsible for technical matters :
5. Name of Parent Organization :

6. Address of Parent Organization :

7. Name of Key Products/Services (Attach :

Separate sheet, if required)

8. Name of Key Customers (Attach :

Separate sheet, if required)

9. Any material/Service being/previously :
supplied/provided to xxxx company (if yes, please specify)

10. Total Staff Strength (Technical & Others) :

Attach qualification and experience record
of key and approved personnel

11. Manufacturing License No. /Registration No. :

12. Is your organization approved /certified
from WHO/ISO/NABL/CPCSEA/any other
equivalent organization. If yes please
provide a copy of the certificate. :

13. Has management established a Quality Policy & Objectives? :

14. Do you have Site Master File? If yes, please provide a Copy :

15. Is there an Organizational chart? (if yes, please attach a copy):

16. What was the annual turnover of your Company
during last financial year? :

17. Are your supplier/dealers authorized? :

18. Do you have a procedure for auditing/monitoring of suppliers/dealers :

19. In case you change your supplier, how do you inform customers?

20. Provide name and address of the nearest suppliers/dealers
from xxxx company :

Checklist of the attachments:

1. List of the products : Attached/Not Attached/Not applicable.
2. List of the customers : Attached/Not Attached/Not applicable.
3. List of key personnel with qualification : Attached/Not Attached/Not applicable.
4. WHO/NABL/CPCSEA Certificate : Attached/Not Attached/Not applicable.
5. Copy of manufacturing license/registration: Attached/Not Attached/Not applicable.
6. Organizational chart : Attached/Not Attached/Not applicable.
7. List of authorized dealers/suppliers : Attached/Not Attached/Not applicable.
8. Site Master File : Attached/Not Attached/Not applicable.

Questionnaire filled by

Name :
Designation :
Signature :
Date :

Space to be used by company QA

Date of Questionnaire sent: Date of receipt of filled

Questionnaire:

All necessary attachments received: Yes/No

Comments:

Questionnaire Checked by: Reviewed by:

|Author |Name |Signature |Date |

|Effective Date: | |Effective till | |

|||||

Introduction

This document intends to provide expert guidance to the raw material supplier (Contract acceptor) and its customers (contract giver) for the implementation and maintenance of appropriate Quality agreements.

Scope

This Quality Agreement outlines the roles, responsibilities, deliverables and time requirements with respect to the Quality Assurance of the manufacture of the specified products (reference is made to the annexes) of Contract Giver (Customer) . It is an appendix to the Service Agreement executed by Contract Giver and Contract Acceptor (further referred as the "Agreement"). Contract Acceptor (Supplier) will operate a compliant GMP (including GDP) operation in all aspects of (where applicable) manufacture, testing, holding, packaging, labelling, shipping and/or distribution of the products of Contract Giver.

Parties to the agreement:

This Quality agreement is by and between the Contract giver i.e. xxxx and Contract Acceptor with office at (address), where the Contract Acceptor supplies Raw material suitable for pharmaceutical/biological use to xxxxx

The Quality Agreement may be revised on an "as-needed" basis or whenever the Agreement is renegotiated. All changes to this Quality Agreement will be reviewed and approved by authorised representatives of Quality Assurance from Contract Giver and Contract Acceptor ("Authorized Quality Representatives"). Any revision of this Quality Agreement must be documented, reviewed and approved by Authorised Quality Representatives from Contract Giver and Contract Acceptor and the Agreement shall be amended accordingly. Unless otherwise defined specifically in this Quality Agreement, all general terms used herein will be interpreted in accordance with the definitions provided in the Agreement. Any terms not so defined will be interpreted with the definitions proscribed in the Contract Acceptor Quality Policy, or when not so specified, will be interpreted with the definitions so stated in guidelines of the USA, Europe, WHO, etc (not limited) where applicable.

The Authorised Quality Representatives will resolve any disputes or conflicts relating to this Quality Agreement in a timely and equitable manner and in compliance with all applicable quality and regulatory requirements. Such resolutions shall be in writing and shall be signed by the authorised Quality Assurance representatives of each company. If any issue remains unresolved for more than twenty (20) business days, the senior Corporate Quality officials from each company should be contacted to resolve this issue.

All communication affecting the contents of this Quality Agreement will be between the Authorized Quality Representatives, as set forth below:

For Contract Giver: ____ (name) Quality Assurance

For Contract Acceptor : ____ (name) (Function)

In the event of a change, temporary or otherwise, of either Party's Authorized Quality Representative, the relevant Party shall notify the other in writing of such change.

Excipients Covered by the agreement

This Quality agreement pertains to the Raw material mentioned in Appendix I

Quality criteria:

The Contract Acceptor will manufacture, test and release the Raw material (mentioned in Appendix I) in accordance with the quality criteria / specifications as approved by the Quality Assurance personnel of Contract Giver enclosed as per Appendix II.

Term of Agreement:

This Agreement shall become effective and binding upon the date of the final signature and shall remain in effect until 3 years after the last delivery of Raw material by the Contract Acceptor to the Contract Giver unless the Contract Giver specifically requests an extension of the Agreement.

Revisions to Agreement:

No amendment to the terms of this Agreement shall be binding on the parties hereto unless made in writing and signed by an authorized representative of each of the parties.

Confidentiality:

The Contract Acceptor and Contract Giver should mutually undertake to observe secrecy on each other's know-how. This also applies after the agreement has ended. After the termination of the Agreement, neither of the parties to the contract is entitled to further commercially use the other party's know how disclosed under this agreement or let third parties do so without the other party's consent.

The obligation of confidentiality and of not using information does apply if that information:

1. was already known to the other party before its disclosure under the Agreement and the other party declares this fact immediately or
2. has become common property through publication or by other means or
3. becomes known to one of the parties without originating directly or indirectly from the other party or
4. is made available to authorities due to legal requirements.

Responsibilities:

This quality agreement will outline the responsibilities of the Contract Giver and Contract Acceptor with regard to the quality activities described in the quality criteria tabulated below:

1. Regulatory Authorizations & GMP Compliance (where applicable)

[No |Responsibilities |Contract|Contract|
		Giver	Acceptor
1.1	Maintain all licenses, registrations and other		(
	authorizations as are required to operate a controlled		
	facility and related functions under the Applicable Laws and		
	guidelines (including GMP where applicable)		
1.2	Maintain facilities, utilities, equipment and conveyance and		(
	perform holding and distribution activities according to GMP		
	guidance documents published by the applicable authorities,		
	approved internal policies and procedures and other		
	regulatory requirements, as applicable.		
1.3	Provide all necessary information pertinent to the		(
	Production of the Raw Material to keep the regulatory		
	filings current and up to date.		
1.4	Prepare, maintain and update the Regulatory filings in		(
	accordance with all Applicable Laws. Prepare and submit all		
	required reports related to Product.		
1.5	Provide all necessary information to Contract Giver		(
	concerning complaints, recalls, returned or salvaged drug		
	products, and investigations associated with those		
	activities, where Contract Acceptor is involved, for the		
	purpose of the issuing of Contract Giver's Annual Product		
	Review.		

2. Regulatory Actions & Inspections

	Responsibilities	Contract	Contract
		Giver	Acceptor
2.1	Notify Contract Giver of any Regulatory Authority notice of		(
	inspection or inspection of the Facilities directly relating		
	to the Raw Material(s) within two (2) business days.		
	Contract Giver may be on Contract Acceptor site during		
	regulatory inspection as subject matter expert. Contract		
	Acceptor manages all on site regulatory inspections.		
2.2	Notify Contract Giver of any Regulatory Authority		(
	investigation relating to the Raw Material(s).		
2.3	Provide copies of any observation, FDA Form 483's, Warning		(
	Letters or the like from applicable Regulatory Authorities		
	within two (2) business days of receipt, as such notices		
	relate to the Raw Material(s) or Processes performed by the		
	other party.		
2.4	Reporting requirements to the respective regulatory		(
	agency(ies) with regard to Contract Giver's registration		
	documentation		
2.4	Review of responses to inspection observations and questions		(
	prior to submission to the concerned regulatory agency if		
	pertinent to products distributed		
2.5	Reporting requirements with regard to manufacturing site		(
	registration that may be required to support Contract Giver		
	related activities, and will notify Contract Giver upon		
	doing so.		
2.6	Conduct investigations, prepare and approve corrective		(
	actions in response to regulatory observations. Provide		
	copy of responses that could potentially impact Contract		
	Giver's Raw material(s) / Process to Contract Giver for		
	review and approval prior to submission.		

3. Premises and Equipment

| |Responsibilities |Contract |Contract |
| | |Giver |Acceptor |

3.1 |Maintaining facilities, utilities and equipment suitable for| |(|
|the production of Contract Giver's raw material(s) and | | |
|designed for their intended uses. | | |
3.2 |Maintaining facilities, equipment, computerised systems and | |(|
utilities as are required to operate a manufacturing		
facility for Raw Material with Pharmaceutical applications		
under the Applicable Laws.		
3.3	Calibrate measuring instruments used to monitor or control	
conditions of storage, according to defined schedules.		
3.4	Ongoing qualifications and/or verification activities	
according to Contract Acceptor policies and procedures		
3.5	Verify or validate equipment cleaning procedures to prevent	
cross-contamination from one raw material(s) to the next raw		
material(s) or from one batch to the next batch of the same		
raw material(s).		

4. Personnel

| |Responsibilities |Contract |Contract |
| |Giver |Acceptor |
4.1 |Ensure that a management representative is appointed, who| |(|
|should be responsible and qualified for maintaining the | | |
|Quality System. | | |
4.2 |Ensure that all staff affected by this Agreement shall be| |(|
|appropriately trained. | | |
4.3 |Maintain a system to show training and qualification of | |(|
|personnel in procedures relating to the Raw Material(s). | | |

5. Manufacturing and batch documentation

| |Responsibilities |Contract|Contract|
| |Giver |Acceptor|
5.1 |Establish procedures for change-over, cleaning, | |(|
|processing, packaging, labeling and handling. | | |
5.2 |Establish provisions, parameters ranges or limits, test | |(|
|requirements and processing. | | |
5.3 |Maintain master batch records and ensure that master batch| |(|
records, including test records, comply with the		
Regulatory filings, Clinical Trial Applications and/or the		
Agreed Supplies.		
5.4	Contract Acceptor shall use Master batch records to	
document production process of each batch in batch record.		

6. Materials and consumables

| |Responsibilities |Contract |Contract |
| |Giver |Acceptor |
6.1 |Procure, store, sample, test and release Materials and/or| |(|
consumables used for the manufacturing of the Raw		
Material(s) in accordance with Contract Acceptor's		
written procedures.		
6.2	Responsible for verification that materials are free of	
BSE/TSE (where applicable) and providing regularly		
updated certification to Contract Giver.		
6.3	Upon request, provide Contract Giver with documented	
evidence describing the qualifications of Contract		
Acceptors used to provide components, containers,		
labeling and packaging materials.		
6.4	Develop test methods and specifications in case	
specifications are deficient or unavailable		
6.5	Store raw materials or consumables in accordance with the	
requirements set forth in the specifications provided by		
Contract Giver.		

7. Records

|No |Responsibilities |Contract |Contract |
| |Giver |Acceptor |
7.1 |Ensure that records are made at each operation on | |(|
|Contract Giver's Product, in such a way that all | | |
|significant activities or events are traceable. | | |
7.1 |Store master record, batch records, distribution and | |(|
transportation records and all other documentation		
related to the Product for a minimum of five (5) years.		
Notify Contract Giver prior to destruction of any such		
records and provide the option to have the records sent		
to Contract Giver at Contract Giver's expense.		
7.2	Contract Acceptor shall deliver batch record / CoA with	
every batch		
7.3	Provide copies of all documentation necessary for the	
Contract Giver to respond to enquiries by Regulatory		
Authorities.		

8. Documentation for Safety

| |Responsibilities |Contract |Contract |
| |Giver |Acceptor |
8.1 |Provide safety/hazard and handling data on the Raw | |(|
|Material(s) to Contract Giver | | |

9. Samples

| |Responsibilities |Contract |Contract |
| | |Giver |Acceptor |
9.1 |Ship side samples and extra samples, where applicable, | |(|
| |according to the requirements of the Contract Giver | | |
9.2 |Retain reference samples of all materials, per Contract | |(|
| |Acceptor procedures. | | |
9.3 |Store of retained samples under storage conditions in a | |(|
| |secure area for 3 years from the Date of Manufacture. | | |

10. Testing & Analysis

| |Responsibilities |Contract |Contract |
| | |Giver |Acceptor |
10.1 |Write protocols and perform verification or validation, | |(|
| |as appropriate, of test methods related to the release of | | |
| |the Raw Material(s). | | |
10.2 |Provide documented finished raw material(s) testing | |(Review |
| |limits and specifications and other requirements. | | |
10.3 |Perform in-process testing and finished raw material(s) | |(|
| |release testing according to the requirements in 6.2. | | |
10.4 |Investigate all out-of-Specification test results ("OOS") | |(|
	and have available a report related to such		
	investigations including a rationale for implemented		
	corrective and preventative actions.		

11. Release

| |Responsibilities |Contract |Contract |
| | |Giver |Acceptor |
11.1 |Provide final release of Raw Materials for supply (COA) | |(|
11.2 |Review the batch record within thirty (30) calendar days | |(|
| |of receipt. Communicate any issues identified to | | |
| |Contract Acceptor. | | |

12. Storage and Shipment

| |Responsibilities |Contract |Contract |
| | |Giver |Acceptor |
12.1 |Specify finished raw material(s), packaging, storage and |Approve |(|
| |shipping requirements | | |
12.2 |Store the finished raw material(s) in accordance with the | |(|
| |specifications pending release of the Raw Material(s). | | |
12.3 |Arrange shipments of Raw material(s) according to | |(|
	Contract Giver specified procedure to Contract Giver or		
	Contract Giver's agent only upon written approval from		
	Contract Giver.		

13. Returned Goods

| |Responsibilities |Contract |Contract |
| | |Giver |Acceptor |
13.1 |Maintain a controlled and documented system and procedure | |(|
| |for handling returned goods and rejected materials. | | |
13.2 |Ensure that non-defective products returned from the | |(|
	market are stored separately from saleable stock, and are		
	assessed according to defined procedures before		
	destruction or authorisation for re-use.		

14. Deviations

| |Responsibilities |Contract |Contract |
| | |Giver |Acceptor |
14.1 |Investigate, resolve and document all abnormal events and | |(|
| |deviations from the master batch record and | | |
| |Specifications | | |
14.2 |Determine and document impact on raw material, plus | |(|
| |corrective and preventative actions related to the | | |
| |deviation(s) | | |
14.3 |Deviations will be managed through a written Procedure of | |(|
| |the Contract Acceptor | | |

15. Complaints, Recalls and Withdrawals

|No |Responsibilities |Contract |Contract |
| | |Giver |Acceptor |
15.1 |Handling and reporting complaints related to the Raw | |(|
| |Material. | | |
15.2 |Investigate complaints,. Issue reports and follow-up on | |(|
| |corrective actions within agreed timelines. | | |
15.3 |Notification within seven (7) days of any information | |(|
| |coming into its possession concerning the quality of | | |
| |previously released batch(es). | | |
15.4 |Contract Giver shall promptly notify Contract Acceptor of | |(|
| |any complaint or adverse event attributable to the | | |
| |quality of Product | | |
15.2 |Investigate complaints as requested by Contract Giver. | |(|
| |Issue reports and follow-up on corrective actions within | | |
| |agreed timelines. | | |
15.6 |In case of a Recall, Rejection or Withdrawal, Contract | |(|
	Acceptor will utilize its deviation procedure to		
	investigate and manage the event and to the extent the		
	issue directly or indirectly relates to other Contract		

| | Accepter activities | | |

16. Change Management

| | Responsibilities | Contract | Contract |
| | | Giver | Acceptor |
16.1 | Ensure that changes related to | | (|
The manufacturing Process.		
Reprocessing or rework		
Changes in raw materials and consumables specified on the		
bill of materials.		
Changes in layout, structure or site of manufacturing		
facility.		
Changes in primary processing equipment that could affect		
Product.		
Changes in testing		
are documented including an Quality impact assessment on		
the quality of the Raw Materials		
16.2	Notify Contract Giver of any proposed changes to Contract	Approve
Acceptor processing that may impact the Product.		
16.3	Contract Giver initiates such change by providing written	Inform
notification through Contract Giver change control system		
to Contract Acceptor at least 30 days prior to the		
requested change.		
16.4	Changes will be managed through a written Procedure of	
the Contract Acceptor		

17. Contract Giver Audits

| No | Responsibilities | Contract | Contract |
| | | Giver | Acceptor |
17.1 | Provision of access to facilities and systems as they | | (|
| relate to the storage and distribution of Products. | | |
17.2 | Provide timely notification for scheduling an audit. | (| |
17.3 | "for cause" basis audits for product quality reasons as | (| |
they apply to product distribution may be requested.		
Timing for a "for-cause" audit must be mutually agreed		
upon but shall not be unreasonably delayed by both		
parties		
17.4	Respond in writing to written audit observations within	
thirty (30) days of receipt of report or as advised.		

18. Sub contracting

| No | Responsibilities | Contract | Contract |
| | | Giver | Acceptor |
18.1 | Select and approve the use of third party laboratories | Approve | (|
| (subcontractors) for intermediate and final Product | | |
| testing, if applicable | | |
18.2 | Use of subcontractors for any part of the holding, | Approve | (|
| distribution, packaging or labelling of the product or | | |
| intermediates only with the prior written approval. | | |
18.3 | Subcontractors will meet requirements for Approved | | (|
Vendors. Contract Giver will accept Contract Acceptor's		
documented vendor compliance program subject to outcome		
of Contract Giver audit.		

Contracting of Third parties:

If Contract Acceptor does not do the work covered by the Agreement himself but contracts third parties to do it, the prior written consent of Contract Giver is required. In addition, Contract Acceptor must conclude a written Agreement on contracts manufacturing with the third party.

Also in case Contract Acceptor contracts third parties, Contract Acceptor is solely responsible towards Contract giver for the fulfillments of all contract obligations.

Term and Termination;

This Agreement shall come into force and effect once signed by the parties hereby. It shall be concluded for an indefinite period and may be terminated with due observance of a notice period of six months.

The right for an extraordinary termination for good cause is not affected

Each termination must be effected by registered mail

Definitions

Approval The term "Approval" is defined as concurrence between Contract Giver and Contract Acceptor, such as agreement on a proposed Changes, as evidenced in writing and signed by both companies' Authorized Quality Representatives.

Approved Vendor A vendor who has met minimum approval standards and who has been approved by both parties to provide required items or services that may impact product quality.

Authorized Quality Representative An individual named within the Quality Agreement with the authority to resolve any disputes or conflicts

relating to this Quality Agreement in a timely and equitable manner and in compliance with all applicable quality and regulatory requirements.

Batch A specific quantity of material produced in a process or fraction of a process. Batches are defined as the material represented at the end of the intermediate processing steps and the material represented at the end of the processing step(s)

CFR Code of Federal Regulations

COA Certificate of Analysis: An authentic document that certifies that a specific batch of material has been evaluated in accordance with the Specification for that material.

Contract Acceptor Shall mean the contractor, which performs Manufacturing and Packaging Operations of the raw material .

Contract Giver Shall mean the contractor, which uses the raw material for the manufacturing of API (ARM) or finished product.

GMP Good Manufacturing Practice, including Current Good Manufacturing Practice: The essential rules governing the production, distribution and use of medicinal products, to safeguard public health. In this Agreement GMP means both GMP and/or cGMP.

Date of Manufacture The date of fill into final container or packaging for intermediate or Finished Product

Disposition Decision by Contract Acceptor Quality on the suitability of the intermediate or Finished Product for further processing.

EMA European Medicines Evaluation Agency

Finished Product The product in it's agreed upon finished form. When the product is packed and labeled in it's market presentation it is referred to as "Drug Product" as well.

For Cause Visit The term "for cause visit" is used to describe site visits, other than audits or business discussions, for the purpose of reviewing documentation, facilities or processes related to a specific deviation affecting Product disposition.

Master Batch Record A detailed description of the production process for manufacturing of the Raw Material outlining the different actions an operator has to perform to complete the production process

OOS Out of Specification

Process Consumables Materials used as an aid in the manufacture of an intermediate or the Raw Material that do not themselves participate in a chemical or biological reaction. For the purposes of this document media, resins, raw materials, filters, membranes, product-contact materials or surfaces, disposable analytical test kits, disposable containers, product contact materials or surfaces, disposable lab supplies unique to the program or other items consumed and subcontracted analytical testing are considered to be process consumables.

Raw Material Any ingredient intended for use in the manufacture of the Raw Material, including those that may not appear in the final commercial presentation. These include chemicals used directly and/or indirectly in the manufacturing process.

Release Package Relevant documentation to be transferred to the Contract Giver to facilitate the release of Contract Giver's Finished Product. This package consists of copies of, Certificate of Analysis (COA) or analytical results, QA disposition of product statement, and where applicable, batch genealogy, and facility deviations.

Specification A set of criteria, limited to Critical In Process Tests or Finished Product, which must conform, to be considered acceptable for its intended use.

Details of Contact Persons

Area	Contract Giver	Contract Acceptor
Quality Representative	Name	Name
	Telephone:..... x...	Telephone:..... x...
	Fax:..... x...	Fax:..... x...

	e-mail:	e-mail:
Product Recall	Name	Name
	Telephone:..... x...	Telephone:..... x...
	Fax:..... x...	Fax:..... x...
	e-mail:	e-mail:
Out of Office Person	Name	Name
	Telephone:..... x...	Telephone:..... x...
	Fax:..... x...	Fax:..... x...
	e-mail:	e-mail:
	Name	Name
	Telephone:..... x...	Telephone:..... x...
	Fax:..... x...	Fax:..... x...
	e-mail:	e-mail:

APPENDICES:

I. List of Raw materials.

Contract Giver Contract Acceptor

Service Agreement

This agreement made this ____ day of _____ at _____ is entered into by and between

Company Name XYZ, a Company incorporated and existing in accordance with the Companies Act, 1956 and having its Registered Office Location – Pin code, acting through Shri. Name , General Manager- Payment Cell & Compliances a representative duly authorized vide board resolution dated 2nd September, 2002 of the Company (hereinafter referred to as "Company Name") which expression shall unless be repugnant to the context include its successors and assigns of ONE PART.

AND

M/s. _____. Company incorporated and existing in accordance with the Companies Act. 1956 and having its Office at _____ acting through Mr. _____, _____ of M/s. _____, at _____ duly authorized vide General power of Attorney dated _____ (hereinafter referred to as "_____") which expression shall unless be repugnant to the context include its successors and assign of OTHER PART.

WHEREAS

a) Company Name is inter alia carrying on the business of manufacturing, marketing, selling and distributing bulk drugs/ medicines, pharmaceutical formulations, vaccines and other health related products and promotional items. (hereinafter referred to as the "Goods")

b) Company Name as and When required, has to deliver Goods, to and from its sales offices/ Loan Licensee manufacturers/ manufacturing locations/ Third party manufacturers/ clearing & forwarding Associates/ stockiest/ distributors or any associates in India.

c) _____ is carrying on the business of _____ and has represented to Company Name that they have all the necessary expertise, exposure and experience and having requisite infrastructure for _____ the Goods all over India (hereinafter referred to as the "Services") and is ready and willing to render the services to Company Name.

d) On the basis of the above representation of _____, Company name is Desirous of & has agreed to empanel/ engage _____ for the services as aforesaid.

e) The parties are desirous of recording their understanding into writing upon the terms and conditions hereinafter appearing.

NOW IT IS HEREBY AGREED BY AND BETWEEN THE PARTIES HERETO AS FOLLOWS:

1. Definition
2. Engagement
3. Scope of services
4. Duration/Termination
5. Delivery process
6. Service levels
7. Payment terms
8. Liability or Losses & Damages
9. Statutory compliance, undertaking, warranties and indemnity
10. Force majeure
11. Disputes, Arbitration and Jurisdiction
12. Miscellaneous

Sign for and behalf of

For M/s. Company Name For _____

Witnesses:

Name _____ Name _____

Address _____

List of Annexure:

1. Vendor details: |
| Name of the vendor | |
| Category of the product | |
| Address of Vendor |
2. Material to be procured: |
| |
3. Document Review |
| History | Satisfactory / Not Satisfactory |
| Delivery Schedule | Satisfactory / Not Satisfactory |
| Vendor Capability | Satisfactory / Not Satisfactory |
4. Details of pre-purchase samples (If applicable) |
| Sample No. | Sample Quantity | Results | Remarks (If any) |
| | | | |
| | | | |
| | | | |
5. Conclusion: |
| On the basis of analysis of above information of the vendor, the Raw / |
| Packaging material can be / can not be, included in approved Raw / Packaging |
| Materials of already approved Vendor. |
| Name | Designation | Signature | Date |
Prepared | | | |
by: | | | |
Checked by: | | | |
Remarks and Approval |
| |
Name | Designation | Signature | Date |
| | | |

From,
Quality Assurance Department
Company name with location,

To,

Dear _____
This is related to the (Product details); it is an out of specification confirmed in analysis at our end for the said batch. Please investigate the matter at your end and share the results within 30 days. Following are the results of analysis at our end:-
Sr. No.	Test performed	Specificatio	Results as per	Results of	Remarks
		n	COA received	analysis	
			(Company name)		

Further it is brought in your notice that Company name has temporarily stopped the purchase of the said material until detailed investigation report along with necessary CAPA report is submitted from you.

Regards,

Head – QA.
Cc;
CQA department.
Material Management department.
Warehouse department (Company name.)

Schedule for the site audit of vendors for the year:-

Sr.	Vendor Name	Materials supplied	Planned	Audit schedule
No.	and address	/	actual	
Name				
Designation				
Signature/Date				

Cc: Material Management department.

|Department: |Report No: |Year: |

Name of the vendor :
Vendor category :

Observation

	Prepared by	Reviewed by	Approved by
Name			
Signature			
Date			

1. Vendor details: |
| Name of the vendor | | |
| | | |
| Category of the product | |
| Addresses: | |
| Address of Vendor | |
| | | |
| | | |
| | | |
| | | |
| Address of Vendor | |
| (if it is different from | |
| that of manufacturer) | |
| | | |
| | | |
2. Material(s) / Services already being taken by the vendor: |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |
3. Review of Vendor Performance Reports of last three years |
| Performance Report No. | Year | Status |
| | | Satisfactory / Not Satisfactory |
| | | Satisfactory / Not Satisfactory |
| | | Satisfactory / Not Satisfactory |
4. Calculation of Valuation Index: |
| Valuation index : Applicable () Not applicable () |
| Total marks: | Marks obtained: | |
| | |
| Valuation index = _____ x 100 =% |
5. Documents attached: |
| 5.1 Site audit report () |
| 5.2 Filled checklist () |
| 5.3 Other users experience certificate () |
| 5.4 Copy of Quality System Certificate () |
| 5.5 Test reports () |
| 5.6 Others () |
6. Conclusion : |
| Following the above evaluation of the vendor, the vendor is approved / |
| conditionally approved / not approved to supply the materials. |
Prepared by: |
Name | Designation | Signature | Date |
| | | |
Checked by: |
Name | Designation | Signature | Date |
| | | |
Remarks and Approval |
| |
Name | Designation | Signature | Date |
| | | |

Yes

No

Vendor evaluation and approval

RM/ PM/Culture media/testing animals/critical consumables

Service providers / Contractual testing laboratories

New Service provider, send pre-audit questionnaire

Already existing Service provider
Retrospective evaluation every year

Already existing vendor Retrospective evaluation every 3 years.
Send pre-audit Questionnaire. If $\geq 90\%$ marks obtained its apxisting
vendor Retrospective evaluation every 3 years.
Send pre-audit Questionnaire. If $\geq 90\%$ marks obtained its approved.

New Vendor, Send
pre -audit questionnaire

Site audit by vendor development team

Review and assessment of pre-audit questionnaire. If $\geq 90\%$ marks obtained
- send 3 PP samples, to QC for analysis (except testing animals).

Sample Passes

Sample fails, Vendor Rejected

Site audit required

Non-Critical Materials

Critical Materials

Site audit of manufacturing site, if required.

Site audit fail,
Vendor Rejected

Site audit pass, Vendor approved

Quality/Service agreement, if applicable.

Update approved vendor list

From,
Quality Assurance Department
Company name with Location

This is to inform you that QA has approved / re-approved /Rejected as conditional/Qualified vendor as a new / existing vendor on _____ as per below mentioned details:-

- [1. |Name of the vendor |: | |
- [2. |Address of the vendor |: | |
- [3. |Type of material/Services |: | |
- [4. |Name of the |: | |
| |material/Services | | |
- [5. |Category of the |: | |
| |material/Service | | |

The vendor shall remain approved / rejected for _____years from the date of approval / rejected i.e. till _____ or as per any further communication.

Regards,
Head – QA.

To;
CQA department.
Material Management department.
Warehouse department (Company Name.)

Name of the organization | |
 Address of the organization | |
 Name of the material(s) to be | |
 approved, attach list if | |
 required | |

Note: Write the information in the assigned space. Put tick () in the answer wherever it is applicable and strike out wherever it is not applicable.

Name of Raw Materials (starting materials) used : | |
(Please use separate sheet for different products, if	
required)	
Are your suppliers approved?	Yes / No / NA
Do you have an approved DMF for the products to be	Yes / No/ NA
approved? If yes then:-	
If it is registered in any regulatory, semi-regulatory	
country?	
What is the DMF number?	
Please provide open part of DMF.	
Note: - Please attach extra sheet if required.	
Are procedures for controlling documentation in force?	Yes / No/ NA
What kind of documents do you use:	

Are Batch no(s) assigned to the products produced?	Yes / No / NA
Are there any in-process checks in the manufacturing	Yes / No/ NA
process?	
Are documented procedures available for cleaning,	Yes / No/ NA
disinfection/fumigation of premises?	
Are regular in-process tests performed for quality	Yes / No/ NA
control?	
Are all manufacturing procedures approved, documented	Yes / No/ NA
and controlled?	
Is there documented procedure for handling process	Yes / No/ NA
deviations?	
Is there a pest control procedure?	Yes / No/ NA
Is there documented procedure for waste and scrap	Yes / No/ NA
disposal?	
Are the equipments used in the manufacturing properly	Yes / No/ NA
identified and validated, (if required)?	
Do you have specifications and Standard Operating	Yes / No/ NA
Procedures for raw materials / in-process materials /	
finished products?	
Are the batch records reviewed and the batch released	Yes / No/ NA
before its final dispatch to the indenters?	
Do you keep records for Production process?	Yes / No/ NA
Is the system of line clearance followed before a batch	Yes / No/ NA
change over?	
Is there a lot recall procedure exists?	Yes / No/ NA
Are samples from incoming raw materials, in-process and	Yes / No/ NA
final product properly controlled?	
Is there any material of Animal Origin used during the	Yes / No/ NA
manufacturing procedure? If yes, then Please specify:	
Tissue of the animal used (i.e. brain, lung, blood etc.)	
Country where the animal originated and tissue was	
collected	
Feeding and Slaughtering method	
Means of controlling cross contamination (i.e. dedicated	
equipment, recognized cleaning/sanitization process	
etc.)	
Please provide declaration of BSE/TSE free products?	
Do you use organic solvents in manufacturing of	Yes / No/ NA
products? If yes, please provide declaration of OVI free	
certificate?	
Is there a procedure for return of unused material to	Yes / No/ NA
stores?	
Is there a procedure for retrieval and subsequent	Yes / No/ NA
destruction of obsolete documents?	
Is there a 'Quality system' that addresses the	Yes / No/ NA
organizational structure, responsibilities, procedures,	
processes and resources for implementing quality	
management?	
Do you perform inspection on incoming raw materials?	Yes / No/ NA
Are acceptance and rejection criteria established for	Yes / No/ NA
all raw materials?	
Do you keep records for such Inspections?	Yes / No/ NA
Do you issue certificate of analysis for each batch of	Yes / No/ NA
the product?	
Do you have calibration program?	Yes / No/ NA
Do you perform in-house calibration or it is contracted.	Yes / No/ NA
If contracted name the company?	
Is calibration traceable to appropriate standards?	Yes / No/ NA
Do you keep records for Calibration?	Yes / No/ NA
Do you have preventive maintenance program for	Yes / No/ NA
equipment?	
Do you keep records for preventive maintenance?	Yes / No/ NA
Do you perform control of non-conforming products?	Yes / No/ NA
Do you have adequate storage capacity & are procedures	Yes / No/ NA

	for storage established?	
	Are the environmental conditions of the finished goods	Yes / No/ NA
	storage area controlled w.r.t. temperature and humidity?	
	Do you have training program?	Yes / No/ NA
	Are there physically segregated areas for raw and	Yes / No/ NA
	packaging materials?	
	Do you have a process flow diagram for the manufacturing	Yes / No/ NA
	operations? (If 'Yes', attach copy)	
	Do you perform stability analysis? If yes please provide	Yes / No/ NA
	the result of stability of last one year for the	
	materials to be approved.	
	Do you perform impurity testing?	Yes / No/ NA
	Do you have a procedure for handling of rejection	Yes / No/ NA
	materials	
	Do you have a separate controlled area for rejection	Yes / No/ NA
	materials	

Please provide three pre-purchase samples (3 samples from different batches/lots) along with COA and Method of analysis (in case of non-pharmaceutical products).

Checklist of the attachments:

1. DMF (Open Part) : Attached/Not Attached/Not applicable.
 2. Declaration of BSE/TSE free material : Attached/Not Attached/Not applicable.
 3. Declaration of OVI free material : Attached/Not Attached/Not applicable.
 4. Process flow diagram : Attached/Not Attached/Not applicable.
 5. 3 pre-purchase samples with COA : Attached/Not Attached/Not applicable.
 6. Method of analysis (if applicable) : Attached/Not Attached/Not applicable.
 7. Results of stability testing : Attached/Not Attached/Not applicable.
- Authorized Signatory with stamp
 Name :
 Designation :
 Signature :
 Date :

Legends: Yes – Completely followed, No – Not followed, NA – Not Applicable.

Space to be used by QA

Calculation of valuation Index:

Total Marks: _____ Marks Obtained: _____

Valuation index = _____ x 100 =%

Comments:

Questionnaire Checked by: Reviewed by:

Name of the organization | |
 Address of the organization | |
 Name of the material(s) to be | |
 approved, attach list if | |
 required | |

Note: Write the information in the assigned space. Put tick (✓) in the answer wherever it is applicable and strike out wherever it is not applicable.

General information:

1. Name of Raw Materials (starting materials) used : | |
 | | |
 |(Please use separate sheet for different products, if | |
 required) | |
 | | |

2. Are your suppliers approved? :
 Yes / No / NA
 (Attach separate sheet, if required)

3. Do you have written procedure/instructions for testing of raw materials? :
 Yes / No / NA

4. Are Batch no(s) assigned to the products produced? :
 Yes / No / NA

5. Are there any in-process checks in the manufacturing process? :
 Yes / No / NA

6. Are documented procedures available for cleaning/ disinfection of premises? :
 Yes / No / NA

7. Are regular in-process tests performed for quality control? :
 Yes / No / NA

8. Are all manufacturing procedures approved, documented and controlled? : Yes /
 No / NA

9. Is there documented procedure for handling process deviations? :
 Yes / No / NA

10. Is there a pest control procedure? :
 Yes / No / NA

11. Is there documented procedure for waste and scrap disposal? :
 Yes / No / NA

12. Are the batch records reviewed and the batch released before its final dispatch to the indenters? :
 Yes / No / NA

13. Is the system of line clearance followed before a batch change over? : Yes / No / NA

14. Is there a lot recall procedure exists? :
 Yes / No / NA

15. Is there a procedure for retrieval and subsequent destruction of obsolete documents, artworks etc.? :
 Yes / No / NA

16. Do you issue certificate of analysis for each batch of the product? :
 Yes / No / NA

17. Do you have calibration program? :
 Yes / No / NA

18. Do you keep records for Calibration? :
 Yes / No / NA

19. Do you have preventive maintenance program for equipment? :
 Yes / No / NA

20. Do you keep records for preventive maintenance? :
 Yes / No / NA

21. Do you perform control of non-conforming products? :
 Yes / No / NA

22. Do you have adequate storage capacity & are procedures for storage established? :
 Yes / No / NA

23. Do you have training program? : Yes /
 No / NA

24. Are there physically segregated areas for raw and finished materials? :
 Yes / No / NA

25. Do you have a process flow diagram for the manufacturing operations? (If 'Yes', attach copy). :
 Yes / No / NA

26. Do you perform stability analysis? If yes, please provide the result of stability of last one year for the materials to be approved. : Yes / No / NA

27. Are the area for storage of materials have controlled temperature & humidity? : Yes / No / NA

28. Do you keep Rejection Records? : Yes / No / NA

29. Do you have a separate controlled area for rejection materials? :
 Yes / No / NA

30. Do you have written established procedure for destruction of rejected materials? : Yes / No / NA

Please provide three pre-purchase samples (3 samples from different batches/lots) along with COA.

Checklist of the attachments:

- 1. List of the products : Attached/Not Attached/Not applicable.
- 2. List of the suppliers of Raw Materials : Attached/Not Attached/Not applicable.
- 3. Process flow diagram : Attached/Not Attached/Not applicable.
- 4. Results of stability testing : Attached/Not Attached/Not applicable.

Authorized Signatory with stamp
Name :
Designation :
Signature :
Date :

Legends: Yes – Completely followed, No – Not followed, NA – Not Applicable.

Space to be used by xxxx company QA

Calculation of valuation Index:

Total Marks: _____ Marks Obtained: _____

Valuation index = _____ x 100 =%

Comments:

Questionnaire Checked by: Reviewed by:

Name of the organization | |
Address of the organization | |
Name of the material(s) to be | |
approved, attach list if | |
required | |

Material to be supplied (please tick): SPF eggs / Clean eggs / pre-incubated SPF eggs / pre-incubated clean eggs.

Note: Write the information in the assigned space. Put tick () in the answer wherever it is applicable and strike out wherever it is not applicable.

Are there enough animal rooms of appropriate design to allow	Yes / No /
separate housing of:	NA
The breeding colony	
Different animal species	
Sick animals	
Do you have health monitoring system?	Yes / No /
	NA
Do you have Filtered-air-positive-pressure (FAPP) poultry	Yes / No /
housing?	NA
Do you have Critical biosecurity measures?	Yes / No /
	NA
Do you have Careful selection of layer flocks?	Yes / No /
	NA
Does your establishment regularly inspected?	Yes / No /
	NA
Does your establishment satisfies the following criteria:	
The country is free from notified avian influenzae (both	
highly pathogenic and low pathogenic avian influenza)	Yes / No /
The eggs are drawn from an establishment where:	NA
New Castle disease (Ranikhet disease)	
Infectious Bursal disease (Gumboro disease)	
Marek's disease, Mycoplasmosis (M. gallisepticum)	Yes / No /
Fowl typhoid (Salmonella gallinarum)	NA
Pullorum disease (Salmonella pullorum)	
Fowl pox	
Avian infectious bronchitis	
Avian infectious laryngotracheitis	
Avian tuberculosis	
Psittacosis-ornithosis (Avian chlamydiosis)	
Fowl cholera (Pasteurellosis)	
Salmonella enteritidis	
West Nile Virus	
Salmonella typhimurium	
Egg drop syndrome (EDS)	
Avian encephalomyelitis	
Avian leucosis J. Virus	
Chicken anaemia	
Inclusion body hepatitis infection (IBH) has not been	
reported since past 12 months.	
Only clean eggs have been selected and fumigated and the	
eggs are packed from the establishment into new boxes for	
export and transported directly to the placement of shipment	
through an area where the diseases mentioned above have not	
been reported.	
The flocks from which the hatching eggs have been obtained	
are tested regularly and found negative for the diseases as	
under	
Haemagglutination inhibition (H1) test for New Castle	
disease.	
Haemagglutination inhibition (H1) / Agar gel immunodiffusion	
(AGID) test for Avian Influenza.	
Agar gel immunodiffusion (AGID) test for Infectious bursal	
disease (Gumboro disease).	
Agglutination (Agg)/ Haemagglutination inhibition (H1) test	Yes / No /
for Mycoplasmosis.	NA
Agglutination (Agg) test for Fowl Typhoid and Pullorum	
Disease.	
Haemagglutination inhibition (H1) / Virus Neutralization	
(VN) Enzyme linked immunosorbent assay (ELISA) for Avian	
infectious bronchitis.	
Agar gel immunodiffusion (AGID) test for Marek's disease /	Yes / No /
avian leucosis J. Virus.	NA
Agar gel immunodiffusion (AGID) / Virus Neutralization (VN)	
/ Enzyme linked immunosorbent assay (ELISA) for Avian	
infectious laryngotracheitis.	
Virus Neutralisation test for Chicken Anaemia.	
The birds of the supply flock have never been fed with feed	
of ruminant origin.	
The eggs are shipped in new, fresh, clean and disinfected	
packages.	

Name of the organization | |
Address of the organization | |
Name of the material(s) to be | |
approved, attach list if | |
required | |

Note: Write the information in the assigned space. Put tick () in the answer wherever it is applicable and strike out wherever it is not applicable.

Are there enough animal rooms of appropriate design to allow	
Separate housing of:	
The breeding colony	Yes / No /
Different animal species	NA
Sick animals	Yes / No /
	NA
	Yes / No /
	NA
Do you have health monitoring system? If yes, What type of	Yes / No /
Tests do you perform?	NA
.....	
.....	
.....	
.....	
Do you issue health certificate of animals you supply? If	Yes / No /
Yes please provide a specimen copy.	NA
Is proper veterinary care of animals being taken?	Yes / No /
	NA
Are there facilities for collection and disposal of animal	Yes / No /
Waste and of dead animals to minimize disease hazards and	NA
environmental contamination?	
Are there facilities for cleaning, sanitizing, sterilizing	Yes / No /
and maintaining supplies and equipments including animal	NA
cages and racks?	
Are there separate storage areas for equipments, animal	Yes / No /
feed, and bedding which are protected from	NA
infection/contamination, and with refrigeration where	
needed?	
Is there separate space for locker, shower, toilet and	Yes / No /
washing facilities for staff working in the animal	NA
facilities?	
Is there an appropriate functioning environmental control	
system regarding:	Yes / No /
Temperature?	NA
Humidity?	Yes / No /
Adequate air changes /hour?	NA
Time controlled lighting system?	Yes / No /
Appropriate noise control system?	NA
	Yes / No /
	NA
	Yes / No /
	NA
Is emergency power available in the event of a power	Yes / No /
failure?	NA
Is there an implemented pest control system that is	Yes / No /
documented, validated and showing absence of interference	NA
with maintaining the animal welfare?	
Do you have maintenance program for equipment?	Yes / No /
	NA
Do you keep records for Maintenance?	Yes / No /
	NA
Do you have system for marking inspection status?	Yes / No /
	NA
Do you keep records for Inspections?	Yes / No /
	NA
Do you keep records for Calibration?	Yes / No /
	NA
Do you perform quality audits?	Yes / No /
	NA
Do you have training program?	Yes / No /
	NA

Checklist of the attachments:

1. Health certificate of animals (specimen copy) :
Attached/Not Attached/Not applicable.

Questionnaire filled by

Name :
Designation :
Signature :
Date :

Space to be used by xxxx Company QA
Calculation of valuation Index:

Total Marks: _____ Marks Obtained:

Valuation index = _____ x 100 =%

Comments:

Questionnaire Checked by: Reviewed by:

Name of the organization | |
 Address of the organization | |
 | |
 Name of the material(s) to be | |
 approved, attach list if | |
 required | |

Note: Write the information in the assigned space. Put tick (✓) in the answer wherever it is applicable and strike out wherever it is not applicable.

1. Quality System: |
 - 1.1 Are the personnel's responsibilities established in | Yes / No / NA |
 - their | |
 - job descriptions? | |
 - 1.2 Has the company / organization designated a responsible | Yes / No / NA |
 - person | |
 - with enough authority to assure the established quality | |
 - system? | |
 - 1.3 Please mention the nature of your organization: | |
 - (Government/ Semi government/Private) | |
 - 1.4 Are personnel qualified to perform their tasks? | Yes / No / NA |
 - Please attach the list of educational qualifications of | |
 - personnel as a minimum of those working at key | |
 - positions. | |
2. Documentation: |
 - 2.1 Is the system of approval of documents established? | Yes / No / NA |
 - 2.2 Is there an established procedure for approving changes | Yes / No / NA |
 - in documents? | |
 - 2.3 Is there a system to keep records of approved documents | Yes / No / NA |
 - with their edition/revision number? | |
 - 2.4 Are procedures in force to assure the identification and | Yes / No / NA |
 - traceability of works/ tests carried out? | |
 - 2.5 Is there a procedure for identifying, coding and | Yes / No / NA |
 - archiving of records? | |
3. Calibration, Maintenance and Validation: |
 - 3.1 Is there a maintenance program in force for all | Yes / No / NA |
 - measuring and test equipments? | |
 - 3.2 Do you have a calibration program? | Yes / No / NA |
 - 3.3 Do you have procedures in force for performing | Yes / No / NA |
 - Calibrations and Maintenance work? | |
 - 3.4 Does the measuring and test equipments clearly | Yes / No / NA |
 - identified with calibration status? | |
 - 3.5 Are the standards used for Testing / Validation / | Yes / No / NA |
 - Calibration traceable to National / International | |
 - Reference Standards? | |
 - 3.6 Are the Instruments / Standards used for calibration | Yes / No / NA |
 - regularly updated? | |
 - 3.7 Do you have Standard Operating Procedures (SOPs) for | Yes / No / NA |
 - carrying out Testing / Validation / Calibration? | |
 - If so, can these be made available to the Company name | Yes / No / NA |
 - XYZ, on demand? | |
4. Audit and Inspection: |
 - 4.1 Is the facility subjected to quality audits? | Yes / No / NA |
 - 4.2 Are quality audits performed to verify the efficacy of | Yes / No / NA |
 - the Quality System? | |
 - 4.3 Is there a defined plan for audits? | Yes / No / NA |
 - 4.4 Is there a follow up of corrective actions taken? | Yes / No / NA |
5. Training: |
 - 5.1 Is there a procedure for training of staff in their | Yes / No / NA |
 - respective areas of working? | |
 - 5.2 Are training programs established? | Yes / No / NA |
 - 5.3 Are training records maintained? | Yes / No / NA |
6. Testing: |
 - 6.1 Are documented procedures in force for handling | Yes / No / NA |
 - samples and testing? | |
 - 6.2 Is there a traceability of the records for a test | Yes / No / NA |
 - sample / work carried out? i.e. green | |
 - sheets/logbooks/record forms etc. | |
 - 6.3 Do you have required facilities for storage of samples | Yes / No / NA |
 - before (and after) testing? Do you have a procedure | |
 - for destruction of samples after analysis? | |
 - 6.4 Are there separate areas for samples requiring special | Yes / No / NA |
 - storage conditions? | |
 - 6.5 Do you issue certificate of analysis for the | Yes / No / NA |
 - sample/product tested? | |
 - 6.6 Are there adequate facilities for testing of sterile | Yes / No / NA |
 - products including sterility and pyrogen testing? | |
 - 6.7 Is the sterility testing room classified? | Yes / No / NA |

Authorized Signatory with stamp

Name :

Designation :

Signature :

Date :

Legends: Yes – Completely followed, No – Not followed, NA – Not Applicable.

Space to be used by QA

Calculation of valuation Index:

Total Marks: _____ Marks Obtained:

Valuation index = _____ x 100 =%

Comments:

Questionnaire Checked by:
Reviewed by:

Name of the organization | |
Address of the organization | |
Name of the material(s) to be | |
approved, attach list if | |
required | |

Note: Write the information in the assigned space. Put tick () in the answer wherever it is applicable and strike out wherever it is not applicable.

1. Nature of the establishment : Private / Private Limited / Public Limited / Govt. Aided body etc

2. Registration Details of the Service Provider :

3. Details of nature of service provided :

(Attach seperate sheets if required)
4. Give brief details of training provided to the Staff for providing the particular service :

5. Give brief details of infrastructure available for providing the particular service :

6. Furnish brief details of quality norms/parameter followed by your company to provide the service:

7. Please attach credential list & list of service contract:

8. Has the company / organization adequate resources in terms of finance and personnel?
Yes / No / NA

9. Are personnel qualified to perform their tasks?
Yes / No / NA.

10. Are procedures in force to assure the identification and traceability of works/ tests carried out.
Yes / No / NA.

11. Is there a procedure for identifying, coding and archiving of records? Yes / No / NA.

12. Is there a procedure for training of staff in their respective areas of working? Yes / No / NA.

13. Any other details which you would like to state :

for consideration in placing your company on our approved vendor list

Authorized Signatory with stamp
Name :
Designation :
Signature :
Date :

Space to be used by xxxx Company QA

Date of Questionnaire sent: Date of receipt of filled Questionnaire:
All necessary attachments received: Yes/No

Comments:

Questionnaire Checked by:

Reviewed by:

Name of Vendor
(Manufacturer/Supplier): _____

Name of Material : _____

Material Type
(√ which is applicable) : RM/PM/Culture media/Critical consumable/Egg

Details of pre-purchase Samples: |

Sample No.	Vendor Batch	Number of	Quantity	Total Sample	Remarks
No.	containers	per	Quantity	(If any)	
	container				

Please perform the testing of pre-purchase samples provided by vendor as detailed above and report the results to Quality Assurance for further processing of Vendor Qualification.

| Name | Designation | Signature | Date |
Requested by | | | |
(QA): | | | |
Received by | | | |
(QC): | | | |

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1.0 Objective

To lay down the procedure for assessment and approval of vendors.

2.0 Scope

This Standard Operating Procedure is applicable to assess, qualify and approve the vendors of Raw Materials, culture media for testing, critical consumables, Packaging Materials, experimental animals, contractual testing laboratories and various service providers such as calibration agencies, pest control contractors at Company name.

3.0 Responsibility

3.1 Head, Global sourcing department or his/her designee shall be responsible for providing information from vendor, procurement, alternative vendor system, implementation and compliance of the procedure.

3.2 Head, Quality Control or his/her designee shall be responsible for arranging the sampling and testing/analysis of pre-purchase and commercial supply samples, as applicable

3.3 Head, Warehouse, Head, Quality Control and Head Animal House or his/her designee shall be responsible for providing the required testing data and delivery performance respectively for the raw material, culture media for testing, experimental animals and packaging material during performance control of vendors. In case of Service providers (Calibration and validation/revalidation), Engineering department shall provide the annual performance. Head – QC shall provide the annual performance of contractual testing laboratories. Head Administration shall be responsible for providing annual performance of Service provider for Pest Control.

3.4 A team constituted by Head of Quality Assurance department shall be responsible for evaluation/assessment of the vendors, as applicable.

4.0 Abbreviations and Definitions

4.1 SME - Subject Matter Expert

4.2 cGMP - Current Good Manufacturing Practices

4.3 VMT - Vendor Management Team

4.4 QMS - Quality Management System

4.5 WHO - World Health Organization

4.6 RM - Raw Material

4.7 PM - Packaging Material

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4.8 DMF - Drug Master File

4.9 VI - Valuation Index

4.10 CAPA - Corrective Action and Preventive Action

4.11 GPT - Growth Promotion Test

4.12 IOM - Inter Office Memorandum

4.13 OOS - Out of Specification

4.14 VAQ - Vendor Assessment Questionnaire

4.15 NRA - National Regulatory Authority

4.16 Vendor - The term shall include the manufacturer and/or supplier of raw and packaging materials, culture media for testing, experimental animals, contractual testing laboratories and service providers.

4.17 Approved Vendor - A vendor who has been approved by Company Name as per this SOP

5.0 Procedure

5.1 The manufacturers (Raw Materials, Packing Materials, Culture media and Lab. Chemicals etc.), suppliers (Eggs and animals), Contractual testing laboratories and service providers (Pest control and Calibration or validation service providers) shall be assessed and qualified before start purchasing material or availing services.

5.2 Selection of vendors

Vendors should be selected based upon following criteria:

5.2.1 Market reputation (global/local existence in the same field, business turnover and QMS etc.).

5.2.2 Information assess through internet (DMF holder/no. of accreditations/no. of clients etc.).

5.2.3 Certification / Qualification from WHO / other reputed regulatory authorities.

5.2.4 Easy to contact.

5.2.5 No. of years in business.

5.2.6 Already approval for other site of xxxxx company and recommendations from CQA.

5.3 Preparation of vendor development Team:

The vendor development team shall include representatives from Global sourcing

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department and Quality Assurance with support as appropriate from Production, Quality Control, and Engineering. Selection of the members of the vendor development team should be based upon the following criteria:

5.3.1 Experience of minimum 3 – 4 years.

5.3.2 Expertise in concerned activity.

5.3.3 Good knowledge of cGMP requirements.

5.3.4 Trainings on Vendor Qualification and training on cGMP.

5.4 Classification of Materials:

5.4.1 Critical Materials:

Criticality is decided on the basis of critical factors such as direct contact of the material with final product (ARM), impact on stability of product, printed packing materials, any material of biological origin, sterile materials and de-pyrogenated materials. Contract testing lab. shall be considered equivalent to critical materials. Define the material for critical or non-critical after evaluation as per Material assessment record (Annexure – 1).

5.4.2 Non-Critical Material:

Any material which is not classified as critical material.

5.4.3 Critical consumables:

Materials/consumables that are used during processing of final product, but are not a part of the finished product. For example:- Silicon tubing, 0.2 µm filters etc.

5.4.4 Minimum requisite for critical/non-critical Vendor and Service Provider:

Following is the minimum requisite for vendor approval of critical/non-critical Vendor:

Requirements Non Critical Critical Service

Vendor Vendor providers

TSE/BSE Assessment ✓

(1)

✓

(1)

NA

Questionnaire (General and vendor ✓ ✓ ✓

specific)

Site Audit - ✓

(2)

✓

(3)

Historical Performance (If ✓ ✓ ✓

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Applicable)

Quality Agreement - ✓ ✓

(4)

Service Agreement NA NA ✓

(1)

In case of material of Animal Origin

(2)

If the vendor is an approved DMF holder or Approved by WHO for the required material, the site audit may not be necessary for approval of critical vendor.

(3)

Site Audit is mandatory in case of Contractual Testing Laboratories only

(4)

Quality Agreement shall be prepared in case of Analytical Testing Lab only

5.5 Criteria for Evaluation/Assessment of vendors:

5.5.1 Upon request from Global sourcing department, Quality Assurance Department shall provide the "Vendor pre-audit assessment Questionnaire". The pre-audit questionnaire shall be given in two parts i.e. General Questionnaire, Annexure – 2 and Specific questionnaire based upon vendor to be qualified. Following specific questionnaire shall be used as required:-

"Pre-audit questionnaire for RM/Critical consumables vendors" – Annexure 3,

"Pre-audit questionnaire for PM vendors" – Annexure 4,

"Pre-audit questionnaire for eggs suppliers" – Annexure 5,

"Pre-audit questionnaire for animals suppliers" – Annexure 6,

"Pre-audit questionnaire for contractual testing laboratories" – Annexure 7, &

"Pre-audit questionnaire for service providers – Annexure 8.

5.5.2 Send the questionnaire through Global sourcing department and ask to provide the filled questionnaire and other supporting documents. Ask the vendor to provide pre-purchase samples (preferably 3 samples from different batches) along with certificate of analysis, as applicable. Quantity of each sample shall be sufficient to conduct at least one complete testing as per specifications.

Service providers and analytical testing laboratories shall not be subjected to any pre-purchase samples and only filled questionnaire is sufficient for preliminary evaluation.

Note: FAQ (Frequently asked Questions) provided by vendor can be accepted in place of filled checklist for preliminary evaluation if it is able to provide the

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information which can fulfils the minimum criteria of evaluation i.e. ≥ 90%.

5.5.3 If required, share the specification of required material(s) with vendor through Global sourcing department.

5.5.4 Upon receipt of filled form "Vendor pre-audit assessment Questionnaire" from the vendor, along with the samples, Quality Assurance Department shall evaluate the information furnished by the vendor as per evaluation of valuation index.

5.5.5 The filled checklist is scored as per the following criteria.

Where the activity mentioned in the Check-list is:

5.5.5.1 Followed (Yes) 1 points
5.5.5.2 Not followed (No) 0 point
5.5.5.3 Not applicable (NA) Not considered for evaluation.

Note: General information and fill in the blanks in the checklist shall not be assigned any point and points in checklist marked as not applicable shall not be considered for valuation index.

5.5.6 Valuation Index is calculated as under and is expressed in percentage:

Marks obtained

Valuation index (VI) = ----- X 100 (%)

Total Marks

5.5.7 In case VI is < 90% vendor should be rejected and samples should not be sent to QC and returned back to the vendor. If the valuation index (VI) is ≥ 90%, then the following steps shall be taken for approval of vendor.

5.5.8 Send the samples to QC along with the COA of respective samples along with the request for analysis (Annexure – 9) for vendor approval. The samples of culture media shall be subjected to Growth promotion Testing (GPT), and if required, additional testing should be carried out.

5.5.9 QC after testing as per in-house/pharmacopeia specifications, shall give results to QA and only those vendors shall be considered for approval whose all samples passes the testing.

5.5.10 After evaluation of the questionnaire and results of the samples analysis, Head – Quality Assurance or his designee or a designated team/trained personnel shall audit the manufacturing location to conduct the site audit of the vendor
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facility (in case of critical materials, wherever applicable).

5.5.11 The audit team visits the site of the vendor for evaluating his Quality Management System & material requirements as per vendor site audit checklist filled. Vendor site audit checklist can be used for reference only; site audit can go beyond the information contained in the checklist. Following specific audit checklist shall be used as required:-

"Audit Checklist for RM/Critical consumables vendors" – Annexure 10,

"Audit Checklist for Packaging Materials vendors" – Annexure 11,

"Audit Checklist for eggs suppliers" – Annexure 12,

"Audit Checklist for animals suppliers" – Annexure 13,

"Audit Checklist for contractual testing laboratories" – Annexure 14, &

"Audit Checklist for Service providers – Annexure 15.

5.5.12 After site audit, audit team shall prepare the Vendor Site Audit Report (Annexure – 16) and write recommendations/Observations.

5.5.13 Based upon observations made by auditors, vendor shall be asked to submit CAPA report within 4 weeks. In such case, vendor shall be approved conditionally for first three commercial consignments.

5.5.14 CAPA report submitted for compliance of the audit observations shall be evaluated. If the vendor comply all the recommendations, vendor shall be approved. If the CAPA report is not submitted within defined timeline or it is not effective, a warning letter to vendor shall be issued to vendor for submission of CAPA within one week after receipt of warning letter and if vendor still does not submits CAPA, The vendor shall be delisted and same shall be communicated to CQA, Global Sourcing, WH and other concerned departments.

5.5.15 QA shall prepare "Vendor approval report" – Annexure 17 after review of satisfactory CAPA. Prepare an addendum to the existing list of approved vendors and send information to warehouse, QC, CQA and global sourcing department. Include the name of the vendor in the list of approved vendors for RM/PM, Annexure – 18, & list of contract testing laboratories/service providers, Annexure – 19 on monthly basis.

5.5.16 After approval of vendor, QA department should send a Quality Agreement for STANDARD OPERATING PROCEDURE SOP No. : QMS 034
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manufacturer/Critical materials and contractual testing laboratories Annexure – 20 and Service agreement for service providers, Annexure – 21 to the vendor (preferably through Global sourcing department). The quality agreement shall be valid for three years, however the service agreement shall be prepared annually. The Quality /Service Agreement to be signed in between Company Name. and vendor, shall be prepared. The content of Quality/Service agreement may be prepared on case to case basis depending upon materials/services provided.

5.5.17 In case a vendor has its own developed quality agreement the same may be accepted if it is having following clauses:

- Manufacturing compliance as per cGMP, including batch records.

- Quality control including analysis, stability testing if applicable and OOS handling.

- Quality assurance including documentation, SOPs, product release, Deviation and Investigations, Change management, complaints and Batch disposition.

5.5.18 At least three service providers shall be approved for each type of service taken from external agencies (Except services of Pest Control). Similarly for critical Raw materials, Packaging materials and critical consumables at least one alternative vendor shall be approved as applicable.

5.5.19 Record of vendor approval:

The record of vendor approval contains the following documents (but not limited to):

5.5.19.1 Checklist for the vendor assessment (Pre-audit questionnaires).
5.5.19.2 Vendor's Quality Management System assessment by accredited agency for example:- GMP certificate, GLP approval, CPCSEA approval, NABL accreditation, WHO-GMP certificate or any other regulatory or semi regulatory certification.
5.5.19.3 Test reports.
5.5.19.4 Vendor Site Audit checklist.
5.5.19.5 Vendor Site Audit Report
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5.5.19.6 Vendor Approval Report.
5.5.19.7 Quality Agreement.
5.5.19.8 Any other relevant documents.
5.5.20 Updating of vendor approval record :
5.5.20.1 If an approved vendor's Quality Management System certificate is renewed, the renewed certificate is appended to the records.
5.5.20.2 A list of qualified and approved vendors prepared by QA Department shall be issued to Warehouse, QC, Production Department, Animal House, Global sourcing & list of approved service providers to respective functional departments like Engineering/administration on monthly basis.
5.6 When any new material is to be procured from already approved vendor, following steps shall be taken:
5.6.1 In case, where the already approved vendor has to be approved for a new material, Global sourcing department shall send "Pre-audit assessment questionnaire" for the specific questionnaire and procure pre purchase samples (preferably 3 samples from 3 different batches) from vendor.
5.6.2 Global sourcing department shall forward the questionnaire along with samples to QA. Quality Assurance department shall evaluate the questionnaire and if the valuation index (VI) is $\geq 90\%$, then forward the pre-purchase samples to QC along with the request for analysis (Annexure 9). QC shall give the results of analysis to QA. In case VI is $< 90\%$ samples should not be sent to QC and returned back to the vendor.
5.6.3 If pre-purchase samples passes the tests, the material shall be considered as approved material to purchase from already approved vendor and Quality Assurance Department shall prepare the Approval Report of additional material (Annexure – 22) which shall be appended to the record of vendor and the material shall be included in Approved vendor list, when the next review is due. An addendum to approved Vendor list for updating of new material shall be circulated to Warehouse, Quality Control and other concerned departments as per Annexure – 28.
5.6.4 In case, the vendor is already approved for any type of material from any other
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manufacturing location of the vendor, the vendor should be audited as per new vendor qualification procedure.
5.6.5 In case of change of Vendor for the same material / same type with same specification, a change control shall be initiated by user. The product stability and material comparability shall be carried out for critical materials. For non-critical materials, only material comparability should be done.
5.7 Disqualification of vendor:
An approved vendor may be disqualified based upon one or more of the following criteria, and following below mentioned procedure:
5.7.1 In case any single consignment of a specific material is rejected, then following an OOS procedure, if applicable, within a week a written communication shall be issued to the vendor through Global sourcing department, as per "Caution Letter" – Annexure 23 for thorough investigation and a detailed investigation report shall be sought from the vendor within the period of maximum 30 days. Global sourcing department shall not purchase same material from the vendor until clearance is given from QA. In case detailed investigation with effective CAPA is received from vendor end, QA shall give clearance for further purchase of the material.
5.7.2 In case any new consignment is received during investigation, it shall be tested as per procedure and shall only be used if material passes full testing parameters as per the specifications and if the consignment(s) get rejected in any of the parameter(s), the vendor shall be put on hold and final approval/rejection shall be made after following the complete requalification procedure.
5.7.3 If qualification of the vendor from WHO/other regulatory authority is cancelled, vendor shall be disqualified and same shall be intimated to corporate QA.
5.7.4 In case of raw materials including culture media, eggs, critical consumables and animals, more than 10% rejection in terms of number of consignments received annually from approved vendor, shall render a vendor unapproved for that specific material and in such case, site audit/visit (as appropriate) for re-approval shall be carried out. Similarly in case of packaging Materials, more than 15% rejections in terms of number of consignments received annually from approved vendor, shall render a vendor unapproved for that specific material
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and in such case, site audit/visit (as appropriate) for re-approval shall be carried out.

5.7.5 Three successive rejections of the materials shall render a vendor unapproved for that specific material and in such case, site audit/visit (as appropriate) for re-approval shall be carried out. Vendor shall be approved only after quality results of 3 lots of pre-purchased samples (as applicable). Upon successful evaluation of vendor, vendor should be included in approved vendor list.

5.7.6 If any vendor is disqualified, then CQA shall be informed to evaluate the impact for other sites.

5.7.7 In case, no material / service has been procured / provided respectively from an approved vendor for the last three years, the vendor shall be subjected to reassessment before restart the purchasing of material from the same vendor.

5.8 Quality dispute resolution

5.8.1 If there is any quality dispute related to material, being supplied by vendor, the following resolutions shall be done:

5.8.1.1 QC personnel of Company nameshall perform the quality test at vendor's site.

5.8.1.2 QC personnel of vendor shall perform the quality test at Company Name site.

5.8.1.3 Third party consultation or resolution may be taken into consideration.

5.9 Re-evaluation of vendor:

Each approved vendor, from which the company is procuring materials, shall be re-evaluated every 3rd year + three months. Contractual testing laboratories & service providers, from which the company is taking services, shall be evaluated every year to get updated information of the vendor. The following steps shall be taken for re-evaluation of vendor:

5.9.1 Prepare a schedule of vendor audit every year (preferably in first month) for the vendor's audit due in current year as per "Vendor audit Schedule", Annexure – 24. The audit schedule shall be shared with Global sourcing department for sharing audit information and asking for possible audit dates.

Update the audit schedule on monthly basis to check the actual performance of
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the audit.

5.9.2 Send "Vendor pre-audit assessment Questionnaire" to all the vendors before due dates of vendor approval. Send pre-audit questionnaire in two parts i.e. General Questionnaire Annexure – 2 and Specific questionnaire based upon vendor to be qualified. Following specific questionnaire shall be used as required:-

"Pre-audit questionnaire for RM/Critical consumables vendors" – Annexure 3,

"Pre-audit questionnaire for PM vendors" – Annexure 4,

"Pre-audit questionnaire for eggs suppliers" – Annexure 5,

"Pre-audit questionnaire for animals suppliers" – Annexure 6,

"Pre-audit questionnaire for contractual testing laboratories" – Annexure 7, &

"Pre-audit questionnaire for service providers" – Annexure 8.

5.9.3 The vendor whose Valuation Index (VI) is $\geq 90\%$ shall be considered to remain under approved vendor list.

5.9.4 After reviewing all the updated information and analysis of retrospective data, Quality assurance Department shall prepare the Re-evaluation Report of Vendor (Annexure – 26). Based upon the assessment of filled questionnaire and review of three years performance report, vendor shall be re-qualified for next three years.

5.9.5 Vendor, from which the critical material is procured, shall preferably be audited once in three years and asked to submit CAPA report within a period of 4 weeks. If the vendor is an approved DMF holder or approved form WHO for the required material and if the material supplies from a vendor have been consistently meeting the specifications and there are no quality issues, then the vendor re-audit may be exempted and in this case only the vendor assessment questionnaire is considered adequate.

5.9.6 CAPA report submitted for compliance of the audit observations shall be evaluated. If the vendor comply all the recommendations, vendor shall be approved. If the CAPA report is not submitted within defined timeline or it is not effective, vendor shall not be approved.

5.9.7 Upon successful re-approval of vendors, the quality / service agreement shall also be reviewed as per Annexure – 20 and Annexure – 21.

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5.10 Performance control of vendors:

5.10.1 The Quality Control Department gives annually, to QA Department, a retrospective report of the test based performance of each vendor, in respect of raw materials, used for production and culture media used for testing and in respect of testing of samples for water or any other outsource testing facility. Similarly for service providers for calibration and validation, Engineering department shall provide the performance of respective calibration/validation agencies. In case of test based performance of animal supplier, Animal house shall provide the performance of approved animal suppliers.

Similarly Warehouse shall also submit a report of the vendor's performance regarding the delivery related issues as per vendor performance report, (Annexure – 25) as follows:-

5.10.1.1 Warehouse

Warehouse shall provide the following information as performance of vendor:-

5.10.1.1.1 Name of the vendor.

5.10.1.1.2 Vendor category.

5.10.1.1.3 Products delivered.

5.10.1.1.4 Performance of vendor.
5.10.1.1.5 Observation.
5.10.1.2 Quality Control
5.10.1.2.1 Name of the vendor.
5.10.1.2.2 Vendor category.
5.10.1.2.3 Products delivered.
5.10.1.2.4 Performance of the vendors.
5.10.1.2.5 Results of tests performed with following information:
a. S. No.
b. Name of the Product.
c. Batch No.
d. Folio No.
e. Date of the test.
f. Total No. of Samples.
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g. Results (Pass/Fail).
h. Remarks.
5.10.1.3 Animal House
5.10.1.3.1 Name of Supplier.
5.10.1.3.2 Supplier category.
5.10.1.3.3 Animals delivered.
5.10.1.3.4 Performance of Supplier.
5.10.1.3.5 Observation.
5.10.1.4 Engineering department
5.10.1.4.1 Name of contractual testing facility.
5.10.1.4.2 Testing facility category.
5.10.1.4.3 Performance of the vendor.
5.10.1.4.4 Observation.
5.10.1.5 Quality Assurance
5.10.1.4.1 Name of vendor.
5.10.1.4.2 Vendor category.
5.10.1.4.3 Products delivered.
5.10.1.4.4 Reference reports (all reference report whosever applicable).
5.10.1.4.5 Reference report number and date along with name of concerned department.
5.10.1.4.6 Observations.
5.10.2 The QA Department will prepare Performance Control Report of the vendors every year, irrespective of the material purchased or not in last calendar year. The performance report shall be prepared for all the approved vendors/service providers.
5.10.3 The performance reports shall be assigned identification numbers as follows:
The first two letters correspond to the department which prepares the report e.g. QA for quality assurance department. This is followed by a slash (/) followed by three letters 'VPR' corresponding to the 'Vendor Performance Report' followed by (-) which is followed by a four digit number. The first two digits represent the last two digits of the year to which the report pertains and
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Revision No.: 00

Title: Assessment and approval of vendors Supersedes : None
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the next two digits represent the consecutive number of the vendor performance report in continuation.
For example QA/VPR-1201 represents the first performance report of a vendor pertaining to the year 2023 prepared by QA department.
This numbering system shall remain same for the report from Warehouse, Animal House, Engineering and Quality Control.
5.11 Refer to the vendor approval and assessment chart as per Annexure – 27.
5.12 After approval of vendor send intimation to CQA, Global sourcing department and warehouse departments as per Annexure – 28.
6. Forms and Records:
6.1 Material assessment record. Annexure 1
6.2 General questionnaire. Annexure 2
6.3 Pre-audit questionnaire for RM/Critical consumables vendors. Annexure 3
6.4 Pre-audit questionnaire for PM vendors. Annexure 4
6.5 Pre-audit questionnaire for eggs suppliers. Annexure 5
6.6 Pre-audit questionnaire for Animal Suppliers. Annexure 6
6.7 Pre-audit questionnaire for contractual testing laboratories. Annexure 7
6.8 Pre-audit questionnaire for Service providers. Annexure 8
6.9 Request for Analysis of pre-purchase samples. Annexure 9
6.10 Audit checklist for Raw materials/Critical consumables vendors. Annexure 10
6.11 Audit checklist for Packaging materials vendors. Annexure 11
6.12 Audit checklist for eggs suppliers. Annexure 12
6.13 Audit checklist for animal suppliers. Annexure 13
6.14 Audit checklist contractual testing laboratories. Annexure 14
6.15 Audit checklist for Service Providers. Annexure 15
6.16 Vendor Site Audit Report. Annexure 16
6.17 Vendor approval report. Annexure 17
6.18 List of approved vendors. Annexure 18
6.19 List of approved contract testing laboratories / Service providers. Annexure 19
6.20 Quality agreement for Manufacturers / Critical Materials vendors. Annexure 20
6.21 Service agreement for Suppliers. Annexure 21
6.22 Approval report of additional material. Annexure 22
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6.23 Caution Letter. Annexure 23
6.24 Vendor audit schedule. Annexure 24
6.25 Vendor performance report. Annexure 25
6.26 Re-evaluation report of vendor. Annexure 26
6.27 Vendor evaluation and approval process flow chart. Annexure 27
6.28 Intimation of vendor approval Annexure 28

7.0 References:

7.1

8.0 Distribution:

8.1

9.0 History:

Annexure-1

Assessment of Raw materials and Packaging materials

Material Name:

Material Code: In House Specification No.:

Material Type:

The Material is being used for manufacturing of following products:

ASSESSMENT

Assessment Criticality

Assessment Criteria

(Yes/No/NA) (Critical/Non-critical)

Does this material being used in final bulk product?

If the product has direct contact with final product?

Does it have impact on stability of product?

If the material is a printing packaging material?

Is the material is of biological origin?

Is the material sterile and / or de-pyrogenated?

Conclusion: Based on this criticality assessment the material is classified as Critical/Non-critical.

Done by Checked by Approved by

Name

Signature

Date

General Questionnaire

Annexure – 2

1. Name of Organization :

2. Address of manufacturing site/Laboratory

/ service provider :

3. Name & contact details of person

Responsible for commercial matters :

4. Name & contact details of person

Responsible for technical matters :

5. Name of Parent Organization :

6. Address of Parent Organization :

7. Name of Key Products/Services (Attach :

Separate sheet, if required)

8. Name of Key Customers (Attach :

Separate sheet, if required)

9. Any material/Service being/previously :

supplied/provided to xxxx company (if yes, please specify)

10. Total Staff Strength (Technical & Others) :

Attach qualification and experience record

of key and approved personnel

11. Manufacturing License No. /Registration No. :

12. Is your organization approved /certified
from WHO/ISO/NABL/CPCSEA/any other
equivalent organization. If yes please
provide a copy of the certificate. :

13. Has management established a Quality Policy & Objectives? :
General Questionnaire

14. Do you have Site Master File? If yes, please provide a Copy :
15. Is there an Organizational chart? (if yes, please attach a copy):
16. What was the annual turnover of your Company during last financial year? :
17. Are your supplier/dealers authorized? :
18. Do you have a procedure for auditing/monitoring of suppliers/dealers :
19. In case you change your supplier, how do you inform customers?
20. Provide name and address of the nearest suppliers/dealers from xxxx company :

Checklist of the attachments:

1. List of the products : Attached/Not Attached/Not applicable.
2. List of the customers : Attached/Not Attached/Not applicable.
3. List of key personnel with qualification : Attached/Not Attached/Not applicable.
4. WHO/NABL/CPCSEA Certificate : Attached/Not Attached/Not applicable.
5. Copy of manufacturing license/registration: Attached/Not Attached/Not applicable.
6. Organizational chart : Attached/Not Attached/Not applicable.
7. List of authorized dealers/suppliers : Attached/Not Attached/Not applicable.
8. Site Master File : Attached/Not Attached/Not applicable.

Questionnaire filled by

Name :
Designation :
Signature :
Date :

Space to be used by company QA

General Questionnaire

Date of Questionnaire sent: Date of receipt of filled Questionnaire:

All necessary attachments received: Yes/No

Comments: _____

Questionnaire Checked by: Reviewed by:

Annexure - 3

Pre-audit questionnaire for RM/Critical consumables vendors

Name of the organization

Address of the organization

Name of the material(s) to be approved, attach list if required

Note: Write the information in the assigned space. Put tick (■) in the answer wherever it is applicable and strike out wherever it is not applicable.

1. Name of Raw Materials (starting materials) used :
(Please use separate sheet for different products, if required)
2. Are your suppliers approved? Yes / No / NA
3. Do you have an approved DMF for the products to be approved? If yes Yes / No/ NA then:-
- If it is registered in any regulatory, semi-regulatory country?
- What is the DMF number?
- Please provide open part of DMF.
Note: - Please attach extra sheet if required.
4. Are procedures for controlling documentation in force? Yes / No/ NA
What kind of documents do you use: _____
5. Are Batch no(s) assigned to the products produced? Yes / No / NA
6. Are there any in-process checks in the manufacturing process? Yes / No/ NA
7. Are documented procedures available for cleaning, Yes / No/ NA
disinfection/fumigation of premises?
8. Are regular in-process tests performed for quality control? Yes / No/ NA
9. Are all manufacturing procedures approved, documented and controlled? Yes / No/ NA
10. Is there documented procedure for handling process deviations? Yes / No/ NA
11. Is there a pest control procedure? Yes / No/ NA
12. Is there documented procedure for waste and scrap disposal? Yes / No/ NA
13. Are the equipments used in the manufacturing properly identified and Yes / No/ NA validated, (if required)?
14. Do you have specifications and Standard Operating Procedures for raw Yes / No/ NA

Pre-audit questionnaire for RM/Critical consumables vendors

materials / in-process materials / finished products?

15. Are the batch records reviewed and the batch released before its final Yes / No/ NA dispatch to the indenters?
16. Do you keep records for Production process? Yes / No/ NA

17. Is the system of line clearance followed before a batch change over? Yes / No/ NA
18. Is there a lot recall procedure exists? Yes / No/ NA
19. Are samples from incoming raw materials, in-process and final product Yes / No/ NA properly controlled?
20. Is there any material of Animal Origin used during the manufacturing Yes / No/ NA procedure? If yes, then Please specify:
 - Tissue of the animal used (i.e. brain, lung, blood etc.)
 - Country where the animal originated and tissue was collected
 - Feeding and Slaughtering method
 - Means of controlling cross contamination (i.e. dedicated equipment, recognized cleaning/sanitization process etc.)
 - Please provide declaration of BSE/TSE free products?
21. Do you use organic solvents in manufacturing of products? If yes, please Yes / No/ NA provide declaration of OVI free certificate?
22. Is there a procedure for return of unused material to stores? Yes / No/ NA
23. Is there a procedure for retrieval and subsequent destruction of obsolete Yes / No/ NA documents?
24. Is there a 'Quality system' that addresses the organizational structure, Yes / No/ NA responsibilities, procedures, processes and resources for implementing quality management?
25. Do you perform inspection on incoming raw materials? Yes / No/ NA
26. Are acceptance and rejection criteria established for all raw materials? Yes / No/ NA
27. Do you keep records for such Inspections? Yes / No/ NA
28. Do you issue certificate of analysis for each batch of the product? Yes / No/ NA
29. Do you have calibration program? Yes / No/ NA

Page 2 of 4

Pre-audit questionnaire for RM/Critical consumables vendors

30. Do you perform in-house calibration or it is contracted. Yes / No/ NA
If contracted name the company?
31. Is calibration traceable to appropriate standards? Yes / No/ NA
32. Do you keep records for Calibration? Yes / No/ NA
33. Do you have preventive maintenance program for equipment? Yes / No/ NA
34. Do you keep records for preventive maintenance? Yes / No/ NA
35. Do you perform control of non-conforming products? Yes / No/ NA
36. Do you have adequate storage capacity & are procedures for Yes / No/ NA storage established?
37. Are the environmental conditions of the finished goods storage area Yes / No/ NA controlled w.r.t. temperature and humidity?
38. Do you have training program? Yes / No/ NA
39. Are there physically segregated areas for raw and packaging materials? Yes / No/ NA
40. Do you have a process flow diagram for the manufacturing operations? Yes / No/ NA (If 'Yes', attach copy)
41. Do you perform stability analysis? If yes please provide the result of Yes / No/ NA stability of last one year for the materials to be approved.
42. Do you perform impurity testing? Yes / No/ NA
43. Do you have a procedure for handling of rejection materials Yes / No/ NA
44. Do you have a separate controlled area for rejection materials Yes / No/ NA

Please provide three pre-purchase samples (3 samples from different batches/lots) along with COA and Method of analysis (in case of non-pharmaceutical products).

Checklist of the attachments:

1. DMF (Open Part) : Attached/Not Attached/Not applicable.
2. Declaration of BSE/TSE free material : Attached/Not Attached/Not applicable.
3. Declaration of OVI free material : Attached/Not Attached/Not applicable.
4. Process flow diagram : Attached/Not Attached/Not applicable.
5. 3 pre-purchase samples with COA : Attached/Not Attached/Not applicable.

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Pre-audit questionnaire for RM/Critical consumables vendors

6. Method of analysis (if applicable) : Attached/Not Attached/Not applicable.
7. Results of stability testing : Attached/Not Attached/Not applicable.

Authorized Signatory with stamp

Name :

Designation :

Signature :

Date :

Legends: Yes – Completely followed, No – Not followed, NA – Not Applicable.

Space to be used by QA

Calculation of valuation Index:

Total Marks: _____ Marks Obtained: _____

Valuation index = _____ x 100 =%

Comments:

Questionnaire Checked by: Reviewed by:

Name of the organization
Address of the organization
Name of the material(s) to be
approved, attach list if required

Note: Write the information in the assigned space. Put tick (■) in the answer wherever it is applicable and strike out wherever it is not applicable.

General information:

1. Name of Raw Materials (starting materials) used :
(Please use separate sheet for different products, if required)
2. Are your suppliers approved? : Yes / No / NA

(Attach separate sheet, if required)
3. Do you have written procedure/instructions for
testing of raw materials? : Yes / No / NA
4. Are Batch no(s) assigned to the products produced? : Yes / No / NA
5. Are there any in-process checks in the manufacturing process? : Yes / No / NA
6. Are documented procedures available for cleaning/ disinfection
of premises? : Yes / No / NA
7. Are regular in-process tests performed for quality control? : Yes / No / NA
8. Are all manufacturing procedures approved, documented and
controlled? : Yes / No / NA
9. Is there documented procedure for handling process deviations? : Yes / No / NA
10. Is there a pest control procedure? : Yes / No / NA
11. Is there documented procedure for waste and scrap disposal? : Yes / No / NA
12. Are the batch records reviewed and the batch released
before its final dispatch to the indenters? : Yes / No / NA
13. Is the system of line clearance followed before a batch change over? : Yes / No / NA
14. Is there a lot recall procedure exists? : Yes / No / NA
15. Is there a procedure for retrieval and subsequent destruction
of obsolete documents, artworks etc.? : Yes / No / NA
16. Do you issue certificate of analysis for each batch of the product? : Yes / No / NA
17. Do you have calibration program? : Yes / No / NA
18. Do you keep records for Calibration? : Yes / No / NA
19. Do you have preventive maintenance program for equipment? : Yes / No / NA
20. Do you keep records for preventive maintenance? : Yes / No / NA
21. Do you perform control of non-conforming products? : Yes / No / NA
22. Do you have adequate storage capacity & are procedures
for storage established? : Yes / No / NA
23. Do you have training program? : Yes / No / NA
24. Are there physically segregated areas for raw and finished materials? : Yes / No / NA
25. Do you have a process flow diagram for the manufacturing operations?
(If 'Yes', attach copy). : Yes / No / NA
26. Do you perform stability analysis? If yes, please provide the result of
stability of last one year for the materials to be approved. : Yes / No / NA
27. Are the area for storage of materials have controlled
temperature & humidity? : Yes / No / NA

Pre-audit Questionnaire for Packing Material Vendors

28. Do you keep Rejection Records? : Yes / No / NA

29. Do you have a separate controlled area for rejection materials? : Yes / No / NA

30. Do you have written established procedure for destruction of rejected materials? : Yes / No / NA

Please provide three pre-purchase samples (3 samples from different batches/lots) along with COA.

Checklist of the attachments:

1. List of the products : Attached/Not Attached/Not applicable.
2. List of the suppliers of Raw Materials : Attached/Not Attached/Not applicable.
3. Process flow diagram : Attached/Not Attached/Not applicable.
4. Results of stability testing : Attached/Not Attached/Not applicable.

Authorized Signatory with stamp

Name :

Designation :

Signature :

Date :

Legends: Yes – Completely followed, No – Not followed, NA – Not Applicable.

Space to be used by xxxx company QA

Calculation of valuation Index:

Total Marks: _____ Marks Obtained: _____

Valuation index = _____ x 100 =%

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Pre-audit Questionnaire for Packing Material Vendors

Comments:

Questionnaire Checked by: Reviewed by:

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Annexure - 5

Pre-audit Questionnaire for eggs suppliers

Name of the organization

Address of the organization

Name of the material(s) to be approved, attach list if required

Material to be supplied (please tick): SPF eggs / Clean eggs / pre-incubated SPF eggs / pre-incubated clean eggs.

Note: Write the information in the assigned space. Put tick (■) in the answer wherever it is applicable and strike out wherever it is not applicable.

1. Are there enough animal rooms of appropriate design to allow separate Yes / No / NA housing of:

- i. The breeding colony
- ii. Different animal species
- iii. Sick animals

2. Do you have health monitoring system? Yes / No / NA

3. Do you have Filtered-air-positive-pressure (FAPP) poultry housing? Yes / No / NA

4. Do you have Critical biosecurity measures? Yes / No / NA

5. Do you have Careful selection of layer flocks? Yes / No / NA

6. Does your establishment regularly inspected? Yes / No / NA

7. Does your establishment satisfies the following criteria:

■ The country is free from notified avian influenzae (both highly Yes / No / NA

pathogenic and low pathogenic avian influenza)

■ The eggs are drawn from an establishment where: Yes / No / NA

■ New Castle disease (Ranikhet disease)

■ Infectious Bursal disease (Gumboro disease)

■ Marek's disease, Mycoplasmosis (M. gallisepticum)

■ Fowl typhoid (Salmonella gallinarum)

■ Pullorum disease (Salmonella pullorum)

■ Fowl pox

Page 1 of 5

Pre-audit Questionnaire for eggs suppliers

- Avian infectious bronchitis
- Avian infectious laryngotracheitis
- Avian tuberculosis
- Psittacosis-ornithosis (Avian chlamydiosis)
- Fowl cholera (Pasteurellosis)
- Salmonella enteritidis
- West Nile Virus
- Salmonella typhimurium
- Egg drop syndrome (EDS)
- Avian encephalomyelitis
- Avian leucosis J. Virus
- Chicken anaemia
- Inclusion body hepatitis infection (IBH) has not been reported since past 12 months.
- Only clean eggs have been selected and fumigated and the eggs are Yes / No / NA packed from the establishment into new boxes for export and transported directly to the placement of shipment through an area where the diseases mentioned above have not been reported.
- The flocks from which the hatching eggs have been obtained are tested Yes / No / NA regularly and found negative for the diseases as under
- Haemagglutination inhibition (H1) test for New Castle disease.
- Haemagglutination inhibition (H1) / Agar gel immunodiffusion (AGID) test for Avian Influenza.
- Agar gel immunodiffusion (AGID) test for Infectious bursal disease (Gumboro disease).
- Agglutination (Agg)/ Haemagglutination inhibition (H1) test for Mycoplasmosis.
- Agglutination (Agg) test for Fowl Typhoid and Pullorum Disease.
- Haemagglutination inhibition (H1) / Virus Neutralization (VN)

Page 2 of 5
Pre-audit Questionnaire for eggs suppliers

Enzyme linked immunosorbent assay (ELISA) for Avian infectious bronchitis.

- Agar gel immunodiffusion (AGID) test for Marek's disease / avian leucosis J. Virus.
- Agar gel immunodiffusion (AGID) / Virus Neutralization (VN) / Enzyme linked immunosorbent assay (ELISA) for Avian infectious laryngotracheitis.
- Virus Neutralisation test for Chicken Anaemia.
- The birds of the supply flock have never been fed with feed of Yes / No / NA ruminant origin.
- The eggs are shipped in new, fresh, clean and disinfected packages. Yes / No / NA

8. Do you have a certificate indicating non-reporting of 20 specified diseases Yes / No / NA since the last 12 months?
9. Do you have a certificate for monitoring the listed diseases in the flocks on Yes / No / NA fortnight basis on the randomly picked up samples of flocks i.e. 1.25% or 50 birds per week?
10. Do you issue Certificate of Analysis along with the material you supply? Yes / No / NA
11. Is proper veterinary care of animals being taken? Yes / No / NA
12. Are there facilities for collection and disposal of animal waste and of dead Yes / No / NA animals to minimize disease hazards and environmental contamination?
13. Are there facilities for cleaning, sanitizing, sterilizing and maintaining supplies Yes / No / NA and equipments including animal cages and racks?
14. Are there separate storage areas for equipments, animal feed, and bedding Yes / No / NA which are protected from infection/contamination, and with refrigeration where needed?
15. Is there separate space for locker, shower, toilet and washing facilities for staff Yes / No / NA working in the animal facilities?
16. Is there an appropriate functioning environmental control system regarding: Yes / No / NA

Page 3 of 5
Pre-audit Questionnaire for eggs suppliers

- i. Temperature?
- ii. Humidity?
- iii. Adequate air changes /hour?
- iv. Time controlled lighting system?
- v. Appropriate noise control system?
17. Is emergency power available in the event of a power failure? Yes / No / NA
18. Do you have maintenance program for equipment? Yes / No / NA
19. Do you keep records for Calibration? Yes / No / NA
20. Do you perform quality audits? Yes / No / NA
21. Describe in brief the nature and identification of means of transport.
22. What type of containers used for transport?

Please provide three pre-purchase samples, (3 samples from different batches/lots) along with COA, if applicable.

Questionnaire filled by

Name :
Designation :
Signature :
Date :

Space to be used by QA

Calculation of valuation Index:

Total Marks: _____ Marks Obtained: _____

Valuation index = _____ x 100 =%

Comments:

Page 4 of 5
Pre-audit Questionnaire for eggs suppliers

Questionnaire Checked by: Reviewed by:

Page 5 of 5
Annexure - 6
Pre-audit Questionnaire for Animals suppliers

Name of the organization
Address of the organization
Name of the material(s) to be
approved, attach list if required
Note: Write the information in the assigned space. Put tick (■) in the answer wherever it is applicable
and
strike out wherever it is not applicable.

1. Are there enough animal rooms of appropriate design to allow separate housing of:
- i. The breeding colony Yes / No / NA
 - ii. Different animal species Yes / No / NA
 - iii. Sick animals Yes / No / NA
2. Do you have health monitoring system? If yes, What type of tests do you Yes / No / NA perform?
-
-
-
-
3. Do you issue health certificate of animals you supply? If yes please provide a Yes / No / NA specimen copy.
4. Is proper veterinary care of animals being taken? Yes / No / NA
5. Are there facilities for collection and disposal of animal waste and of dead Yes / No / NA animals to minimize disease hazards and environmental contamination?
6. Are there facilities for cleaning, sanitizing, sterilizing and maintaining Yes / No / NA supplies and equipments including animal cages and racks?
7. Are there separate storage areas for equipments, animal feed, and bedding Yes / No / NA which are protected from infection/contamination, and with refrigeration where needed?
8. Is there separate space for locker, shower, toilet and washing facilities for staff Yes / No / NA working in the animal facilities?

Page 1 of 3
Pre-audit Questionnaire for animals suppliers

9. Is there an appropriate functioning environmental control system regarding:
- i. Temperature? Yes / No / NA
 - ii. Humidity? Yes / No / NA
 - iii. Adequate air changes /hour? Yes / No / NA
 - iv. Time controlled lighting system? Yes / No / NA
 - v. Appropriate noise control system? Yes / No / NA
10. Is emergency power available in the event of a power failure? Yes / No / NA
11. Is there an implemented pest control system that is documented, validated and Yes / No / NA showing absence of interference with maintaining the animal welfare?
12. Do you have maintenance program for equipment? Yes / No / NA
13. Do you keep records for Maintenance? Yes / No / NA
14. Do you have system for marking inspection status? Yes / No / NA
15. Do you keep records for Inspections? Yes / No / NA
16. Do you keep records for Calibration? Yes / No / NA
17. Do you perform quality audits? Yes / No / NA
18. Do you have training program? Yes / No / NA

Checklist of the attachments:

1. Health certificate of animals (specimen copy) : Attached/Not Attached/Not applicable.

Questionnaire filled by

Name :
Designation :
Signature :
Date :

Space to be used by xxxx Company QA
Calculation of valuation Index:

Total Marks: _____ Marks Obtained: _____

Page 2 of 3
Pre-audit Questionnaire for animals suppliers

Valuation index = _____ x 100 =%

Comments:

Questionnaire Checked by: Reviewed by:

Page 3 of 3

Annexure - 7

Questionnaire for contractual testing laboratory

Name of the organization

Address of the organization

Name of the material(s) to be
approved, attach list if required

Note: Write the information in the assigned space. Put tick (■) in the answer wherever it is applicable
and strike
out wherever it is not applicable.

1. Quality System:

1.1 Are the personnel's responsibilities established in their Yes / No / NA
job descriptions?

1.2 Has the company / organization designated a responsible person Yes / No / NA
with enough authority to assure the established quality system?

1.3 Please mention the nature of your organization:

(Government/ Semi government/Private)

1.4 Are personnel qualified to perform their tasks? Yes / No / NA

Please attach the list of educational qualifications of personnel as a
minimum of those working at key positions.

2. Documentation:

2.1 Is the system of approval of documents established? Yes / No / NA

2.2 Is there an established procedure for approving changes in documents? Yes / No / NA

2.3 Is there a system to keep records of approved documents with their Yes / No / NA
edition/revision number?

2.4 Are procedures in force to assure the identification and traceability of Yes / No / NA
works/ tests carried out?

2.5 Is there a procedure for identifying, coding and archiving of records? Yes / No / NA

3. Calibration, Maintenance and Validation:

3.1 Is there a maintenance program in force for all measuring and test Yes / No / NA
equipments?

3.2 Do you have a calibration program? Yes / No / NA

3.3 Do you have procedures in force for performing Calibrations and Yes / No / NA
Maintenance work?

Page 1 of 3

Questionnaire for contractual testing laboratory

3.4 Does the measuring and test equipments clearly identified with Yes / No / NA
calibration status?

3.5 Are the standards used for Testing / Validation / Calibration traceable to Yes / No / NA
National / International Reference Standards?

3.6 Are the Instruments / Standards used for calibration regularly updated? Yes / No / NA

3.7 Do you have Standard Operating Procedures (SOPs) for carrying out Yes / No / NA
Testing / Validation / Calibration?

If so, can these be made available to the Company name XYZ, on Yes / No / NA
demand?

4. Audit and Inspection:

4.1 Is the facility subjected to quality audits? Yes / No / NA

4.2 Are quality audits performed to verify the efficacy of the Quality Yes / No / NA
System?

4.3 Is there a defined plan for audits? Yes / No / NA

4.4 Is there a follow up of corrective actions taken? Yes / No / NA

5. Training:

5.1 Is there a procedure for training of staff in their respective areas of Yes / No / NA
working?

5.2 Are training programs established? Yes / No / NA

5.3 Are training records maintained? Yes / No / NA

6. Testing:

6.1 Are documented procedures in force for handling samples and testing? Yes / No / NA

6.2 Is there a traceability of the records for a test sample / work carried Yes / No / NA
out? i.e. green sheets/logbooks/record forms etc.

6.3 Do you have required facilities for storage of samples before (and Yes / No / NA
after) testing? Do you have a procedure for destruction of samples
after analysis?

6.4 Are there separate areas for samples requiring special storage Yes / No / NA
conditions?

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Questionnaire for contractual testing laboratory

6.5 Do you issue certificate of analysis for the sample/product tested? Yes / No / NA

6.6 Are there adequate facilities for testing of sterile products Including Yes / No / NA
sterility and pyrogen testing?

6.7 Is the sterility testing room classified? Yes / No / NA

Authorized Signatory with stamp

Name :

Designation :

Signature :

Date :

Legends: Yes – Completely followed, No – Not followed, NA – Not Applicable.

Space to be used by QA

Calculation of valuation Index:

Total Marks: _____ Marks Obtained: _____

Valuation index = _____ x 100 =%

Comments:

Questionnaire Checked by: Reviewed by:

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Annexure - 8

Questionnaire for Service Providers

Name of the organization

Address of the organization

Name of the material(s) to be

approved, attach list if required

Note: Write the information in the assigned space. Put tick (■) in the answer wherever it is applicable and strike out wherever it is not applicable.

1. Nature of the establishment : Private / Private Limited / Public Limited / Govt. Aided body etc

2. Registration Details of the Service Provider : _____

3. Details of nature of service provided : _____
(Attach separate sheets if required)

4. Give brief details of training provided to the
Staff for providing the particular service : _____

5. Give brief details of infrastructure available
for providing the particular service : _____

6. Furnish brief details of quality norms/parameter
followed by your company to provide the service: _____

7. Please attach credential list & list of service contract: _____

8. Has the company / organization adequate resources
in terms of finance and personnel? Yes / No / NA

Page 1 of 2

Questionnaire for Service Providers

9. Are personnel qualified to perform their tasks? Yes / No / NA.

10. Are procedures in force to assure the identification
and traceability of works/ tests carried out. Yes / No / NA.

11. Is there a procedure for identifying, coding and archiving of records? Yes / No / NA.

12. Is there a procedure for training of staff in their respective areas of working? Yes / No / NA.

13. Any other details which you would like to state : _____
for consideration in placing your company on
our approved vendor list _____

Authorized Signatory with stamp

Name :

Designation :

Signature :

Date :

Space to be used by ynapmoC xppp QA

Date of Questionnaire sent: Date of receipt of filled Questionnaire:

All necessary attachments received: Yes/No

Comments:

Questionnaire Checked by: Reviewed by:

Request for Analysis of pre-purchase samples

Name of Vendor

(Manufacturer/Supplier): _____

Name of Material : _____

Material Type

(✓ which is applicable) : RM/PM/Culture media/Critical consumable/Egg

Details of pre-purchase Samples:

Sample No. Vendor Batch No. Number of Quantity per Total Sample Quantity Remarks
containers container (If any)

Please perform the testing of pre-purchase samples provided by vendor as detailed above and report the results to Quality Assurance for further processing of Vendor Qualification.

Name Designation Signature Date

Requested by (QA):

Received by (QC):

Page 1 of 1

Annexure - 10

Audit checklist for Raw materials/Critical consumables vendors

A. General Information

1. Name of Organization :

2. Address :

3. Name & Address of Contact Person :

B. Building and Facility

1. Is the Facility easily accessible by suitable means of transport : Yes / No /NA

■ If yes, mention approximate distance and means to reach the facility

2. Is the location free from possible source of air / water / soil, etc. Yes / No /NA

Pollution :

3. Is ac current version of site plan / floor layout available : Yes / No /NA

4. Is the access to the facility restricted and controlled. : Yes / No /NA

5. Are there adequate personnel facilities, like

a) Wash rooms, : Yes / No /NA

b) Change rooms : Yes / No /NA

c) Medical room, available : Yes / No /NA

6. Is there a documented program for pest, rodent and bird control : Yes / No /NA

■ If yes, what is the frequency :

7. Are air curtains installed and functional : Yes / No /NA

8. Are documented procedures available for cleaning, disinfection or fumigation of Yes / No /NA
the premises. :

9. Is rotation of disinfectants followed : Yes / No /NA

10. Is adequate lighting and ventilation provided : Yes / No /NA

11. Do you have a separate Air Handling Unit at least for finished product : Yes / No /NA

12. Is quality of air supplied, especially to manufacturing and packaging areas Yes / No /NA
controlled, monitored and recorded :

13. Is there a documented procedure for waste and scrap disposal : Yes / No /NA

14. ■ Are there adequate utilities available : Yes / No /NA

■ If yes attach their specifications

15. Is there a documented schedule to conduct preventive maintenance of equipments : Yes / No /NA

C Personnel

Page 1 of 5

Audit checklist for Raw materials/Critical consumables vendors

1. Attach qualification and experience record of key personnel : Attach List

2. Is there a documented training program for staff and workers on cGMP, cGLP, Yes / No /NA
SOPs, on-job activities, personnel hygiene etc. :

■ If yes, what is the frequency. Choose 3-4 employees and check record

3. Is there a documented health care program (medical examination) : Yes / No /NA

Choose 3-4 employees and check record

4. Are the personnel provided with clean uniforms : Yes / No /NA

D Stores

1. Are there physically segregated areas for:

■ Raw materials : Yes / No /NA

■ Packaging materials : Yes / No /NA

2. Are there physically segregated areas for:

■ Quarantine material (Under Test) : Yes / No /NA

■ Approved material : Yes / No /NA

■ Rejected material : Yes / No /NA

3. Is the dispensing area separate : Yes / No /NA

4. Is there any excise stores area : Yes / No /NA

5. Do you have separate areas for non – beta lactum / beta – lactum / other potent : Yes / No /NA
drugs (like cytotoxics)

6. Are there separate areas for materials requiring special storage conditions : Yes / No /NA
controlled temperature & humidity with monitoring devices like thermometers /
hygrometers

7. Are the areas adequate for storage of the materials in relation to their quantity : Yes / No /NA

8. Is stock rotation program followed (like FIFO) : Yes / No /NA

E Manufacturing

1. Is a process flow diagram for the manufacturing operations available : Yes / No /NA
■ If yes, attach a copy of process flow diagram Attach Diagram
2. Is there a list of manufacturing equipment with information like name, make : Yes / No /NA capacity, Material of Construction, date of procurement
3. Are the process equipment uniquely coded for identification : Yes / No /NA
■ If No, how are equipment identified during documentation
4. Are the major equipment validated w. r. t. IQ, OQ & PQ : Yes / No /NA
5. Are batches uniquely identified : Yes / No /NA

Page 2 of 5

Audit checklist for Raw materials/Critical consumables vendors

6. Is there a system of line clearance before product or batch change over : Yes / No /NA
 7. Is there any requirement of transportation of semi finished product or material to : Yes / No /NA other site. If Yes, how is integrity of material guaranteed
 8. What is the frequency of in process checks; please exemplify
Process Frequency
i. :
ii. :
iii. :
 9. Are the sequence and timing of manufacturing operations recorded into a batch : Yes / No /NA production record. Check and verify form a current batch record
 10. Is there a documented procedure for reprocessing : Yes / No /NA
 11. What are the packing controls for (mention document name): : Yes / No /NA
■ Printed packaging material reconciliation
■ Destruction of printed packaging material
 12. Do you have procedure for printing of labels or printed material of similar : Yes / No /NA appearance
 13. Is there a documented procedure to monitor, control, record and prevent process : Yes / No /NA deviations
 14. Are the environmental conditions of finished goods storage area controlled w. r. : Yes / No /NA t. temperature and humidity?
 15. Is there a documented procedure for rejection of finished goods : Yes / No /NA
 16. How many rejections have been recorded in the last quarter (for products offered : Yes / No /NA for contract manufacturing). Check a recent rejection record.
 17. Has there been any product recall till date
■ If, yes provide details : Yes / No /NA
- #### F Quality Control
1. Is there a quality control unit : Yes / No /NA
 2. Does Q.C. perform the following functions:
 - a. Sampling of raw and packaging material : Yes / No /NA
 - b. Release of rejection of each batch of raw and packaging material : Yes / No /NA
 - c. Development of specification for raw and packaging material and in- : Yes / No /NA process tests.
 - d. Release or rejection of each batch or lot of in-process bulk material : Yes / No /NA
 - e. Release or rejection of each batch of finished product : Yes / No /NA

Page 3 of 5

Audit checklist for Raw materials/Critical consumables vendors

- f. Issue, control and review of Batch Production Records : Yes / No /NA
- g. Take control samples of raw materials and finished goods : Yes / No /NA
- h. Calibration of Q.C. Equipment : Yes / No /NA
- i. Validation of test methods : Yes / No /NA
- j. Evaluation of stability and quality of finished goods : Yes / No /NA
3. Who performs the functions of Quality Assurance :
4. Are there separate rooms / areas marked for
■ Storage RM Samples : Yes / No /NA
■ Storage Control Samples : Yes / No /NA
■ Stability Testing Equipment : Yes / No /NA
5. Is there a system of Internal Audit : Yes / No /NA
6. Are there any reference books available : Yes / No /NA

G Documentation

1. Are all documents controlled i.e. bear a unique title, number, version number, : Yes / No /NA effective date, etc.
2. Are all documents approved and authorized before distribution : Yes / No /NA
3. Is the documents distribution record maintained : Yes / No /NA
4. Are the following documents available
■ Specifications (RM, PM, In process product, finished product) : Yes / No /NA
■ SOPs (RM, PM, In process product. Finished product) : Yes / No /NA
■ Master Batch Production Record / Master formula record : Yes / No /NA
■ Raw Material Receipt and Issuance Records : Yes / No /NA
■ Packaging Records : Yes / No /NA
■ In-process Testing Records : Yes / No /NA
■ Finished Goods Records : Yes / No /NA
■ Microbiological Records (if applicable) : Yes / No /NA
■ Calibration Records (Manufacturing and Q. C) : Yes / No /NA
■ Operation , Cleaning & Preventive Maintenance Records : Yes / No /NA
■ Complaint Records : Yes / No /NA
■ Rejection Record : Yes / No /NA
■ Product Recalls : Yes / No /NA
5. Are SOPs available for the following operations

Page 4 of 5

Audit checklist for Raw materials/Critical consumables vendors

- Quarantine and storage of material : Yes / No /NA
- Issuance of material : Yes / No /NA
- Quality control for In-process : Yes / No /NA

■ Quality control for Finished Goods : Yes / No /NA
■ Calibration of instruments : Yes / No /NA
■ Product Recalls : Yes / No /NA
■ Microbiological testing : Yes / No /NA
6. Is a Certificate of Analysis issued for a batch released : Yes / No /NA
7. What are the rejection levels :
8. What is the packaging material used for Primary Packaging , Material of :
Construction
Date of Inspection:

Audit Team Representing xxxx Company
Name Designation Department Signature

Vendor's Representatives

Name Designation Department Signature

Page 5 of 5
Audit checklist - Packaging materials Vendors Annexure - 11

A. General Information

1. Name of Organization :
2. Address :
3. Person(s) contacted at site during audit :

B. Material Receipt, Storage and Handling

1. Is there a procedure for qualification of raw material sourcing : Yes / No / NA
2. Is a current list of approved vendors available and circulated to relevant : Yes / No / NA departments
3. Is adequate precautions taken to properly segregate approved material from : Yes / No / NA material under test of rejected
4. Is entry to storage area restricted : Yes / No / NA
5. Is rejected material adequately controlled to prevent misuse : Yes / No / NA
6. Is there a separate or segregated area for sampling : Yes / No / NA
7. Are there documented sampling procedures : Yes / No / NA
8. Are materials properly identified w. r. t. status, quantity etc. : Yes / No / NA
9. Is there a retesting schedule : Yes / No / NA
10. Where required, are materials stored in controlled environment (e. g. humidity, : Yes / No / NA temperature, etc.)

C Manufacturing Operations, Equipment and Controls

1. Are the items being manufactured identifiable throughout the approval : Yes / No / NA
2. Are the manufacturing operations orderly in flow : Yes / No / NA
3. Are the manufacturing processing and facilities adequate to produce high quality : Yes / No / NA items
4. If gang printing is undertaken, are there adequate controls to prevent mix-ups : Yes / No / NA
5. Is there an adequate system of control of labels with pre-printed lot number and / : Yes / No / NA or expiration dates

Page 1 of 4
Audit checklist - Packaging materials Vendors

6. Is the batch no. or lot no. reflective of homogenous production run. If yes, : Yes / No / NA exemplify
7. Is there a line clearance procedure : Yes / No / NA
8. Are regular in-process tests performed by Quality Control : Yes / No / NA
9. Are manufacturing procedures formalized and controlled : Yes / No / NA
10. Is there a documented procedure for handling process deviations : Yes / No / NA
11. Are there documented procedures for handling reworking or rectification of : Yes / No / NA errors
12. Are standard procedures of working and similar functional documents readily : Yes / No / NA available to employees for reference
13. Is the batch or lot reconciled after end of manufacturing operation : Yes / No / NA
14. Are waste material clearly marked : Yes / No / NA
15. Is there a pest control procedure : Yes / No / NA
16. Are colour standards available, and are they checked on a periodic basis for any : Yes / No / NA change in appearance
17. Is there a system for 100% inspection of labels : Yes / No / NA
18. In the event of in-process rejections, is there proper investigation procedure to : Yes / No / NA prevent recurrence
19. Are the equipment adequately identified : Yes / No / NA
20. During operations, is product information displayed on the equipment : Yes / No / NA
21. Are there adequate procedures for final product inspection, sampling and testing : Yes / No / NA
22. Are the final inspection / test results recorded : Yes / No / NA
23. Are the batch records reviewed before final dispatch of products : Yes / No / NA
24. Are the batch records and other related documents stored for adequate length of : Yes / No / NA time
25. Is the document retrieval system adequate : Yes / No / NA
26. How is non conforming material controlled : Yes / No / NA
27. Are samples of finished product retained : Yes / No / NA
28. How adequate are security systems for drug labels : Yes / No / NA
29. Is there a lot recall procedure : Yes / No / NA

D Laboratories and Calibration

1. Is QC adequately staffed : Yes / No / NA

Page 2 of 4
Audit checklist - Packaging materials Vendors

2. Are sufficient equipment / instruments available for proper functioning of QC : Yes / No / NA

3. Are samples from incoming, in-process and final product materials properly : Yes / No / NA controlled
4. Is each lot checked against customer approved copy for correct printing : Yes / No / NA
5. Is each lot checked against a drawing for correct die cut : Yes / No / NA
6. Are certifications of inspection / test results available to customers : Yes / No / NA
7. Are tests / inspections properly documented : Yes / No / NA
8. Are there acceptable procedures to assure that instruments are calibrated and : calibration recorded Yes / No / NA
- Is calibration traceable to appropriate standards : Yes / No / NA
9. Are the instruments / equipments maintained in proper order : Yes / No / NA

E Document Control

1. Are functional documents like specification, Standard Operating Procedures, etc. : Yes / No / NA comprehensive in content
2. Are the documents in use 'current' : Yes / No / NA
3. Are the documents approved and authorized by competent authority prior to : Yes / No / NA distribution
4. Is there a procedure to ensure retrieval and subsequent destruction of obsolete : Yes / No / NA documents

F Miscellaneous

1. Is there regular training program for employees, in-house or otherwise : Yes / No / NA
If yes, list type of training and frequency : Yes / No / NA
2. Is there a program for training : Yes / No / NA
3. Is there a documented procedure for performing internal audits : Yes / No / NA
4. Are internal audit reports prepared and circulated to concerned department for compliance of observed deficiency : Yes / No / NA
5. Is there a 'Quality system' that addresses the organizational structure, responsibilities, procedures, processes and resources for implementing quality : Yes / No / NA management
6. Are there periodic management reviews of the 'Quality system' : Yes / No / NA

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Audit checklist - Packaging materials Vendors

7. Are there sufficient engineering, production and quality control support to requirements of Company Name XYZ. : Yes / No / NA

Date of Inspection:

Audit Team Representing Company Name XYZ.
Name Designation Department Signature

Vendor's Representatives

Name Designation Department Signature

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Audit Checklist for Eggs Suppliers Annexure - 12

A. General Information

1. Name of Organization :
2. Address :
3. Name & Address of Contact Person :

B. Building and Facility

1. Is the Facility easily accessible by suitable means of transport : Yes / No / NA
■ If yes, mention approximate distance and means to reach the facility
2. Is the location free from possible source of air / water / soil, etc. Yes / No / NA
Pollution :
3. Is current version of site plan / floor layout available : Yes / No
attach copy
4. Is the access to the facility restricted and controlled. : Yes / No / NA
5. Are there adequate personnel facilities, like
 - a) Wash rooms, : Yes / No / NA
 - b) Change rooms : Yes / No / NA
 - c) Medical room, available : Yes / No / NA
6. Is there a documented program for pest, rodent and bird control : Yes / No / NA
7. Are documented procedures available for cleaning , disinfection or fumigation of Yes / No / NA the premises. :
8. Is rotation of disinfectants followed : Yes / No / NA
9. Is adequate lighting and ventilation provided : Yes / No / NA
10. Is quality of air supplied, especially to manufacturing and packaging areas Yes / No / NA controlled, monitored and recorded :
11. Is there a documented procedure for waste and scrap disposal : Yes / No / NA
12. Are there adequate utilities available : Yes / No / NA
13. Is there a schedule to conduct preventive maintenance of equipments : Yes / No / NA

C Personnel

1. Attach qualification and experience record of key personnel : Attach List

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Audit Checklist for Eggs Suppliers

2. Is there a documented training program for staff and workers? : Yes / No / NA
3. Is there a documented health care program (medical examination) : Yes / No / NA
4. Are the personnel provided with clean uniforms : Yes / No / NA

D Stores

1. Are there physically segregated areas for:

■ Raw materials : Yes / No /NA

■ Packaging materials : Yes / No /NA

2. Are there separate areas for materials requiring special storage conditions: : Yes / No /NA
controlled temperature & humidity with monitoring devices like thermometers /
hygrometers

3. Are the areas adequate for storage of the materials in relation to their quantity : Yes / No /NA

4. Is stock rotation program followed (like FIFO) : Yes / No /NA

E Manufacturing

1. Is a process flow diagram for the manufacturing operations available : Yes / No /NA

■ If yes, attach a copy of process flow diagram Attach Diagram

2. Is there a list of manufacturing equipment with information like name, make : Yes / No /NA
capacity, Material of Construction, etc.

3. Are the environmental conditions of finished goods storage area controlled w. r. Yes / No /NA
t. temperature and humidity? :

4. Is there a documented procedure for rejection of finished goods?

F Quality Unit

1. Is there a quality control unit : Yes / No /NA

2. Who performs the functions of Quality Assurance :

3. Is there a system of Internal Audit : Yes / No /NA

4. Are there any reference books available : Yes / No /NA

G Documentation

Page 2 of 3

Audit Checklist for Eggs Suppliers

1. Are all documents controlled : Yes / No /NA

2. Is a Certificate of Analysis issued for the material released : Yes / No /NA

3. What are the rejection levels :

Date of Inspection:

Audit Team Representing xxxx Company

Name Designation Department Signature

Vendor's Representatives

Name Designation Department Signature

Page 3 of 3

Annexure - 13

Vendor Site Audit checklist for Animal Suppliers

A. General Information

1. Name of Organization :

2. Address :

3. Name & Address of Contact Person :

4. Has CPCSEA/GLP approved (If yes, :

Please attach a copy of the certificate)

B. Building and Facility

1. Is the Facility easily accessible by suitable means of transport : Yes / No /NA

■ If yes, mention approximate distance and means to reach the
facility

2. Is the location free from possible source of air / water / soil, etc. Yes / No /NA
Pollution :

3. Is current version of site plan / floor layout available : Check and
attach copy

4. Is the access to the facility restricted and controlled. : Yes / No /NA

5. Are there adequate personnel facilities, like

a) Wash rooms, : Yes / No /NA

b) Change rooms : Yes / No /NA

c) Medical room, available : Yes / No /NA

6. Is there a documented program for pest, rodent and bird control : Yes / No /NA

7. Are documented procedures available for cleaning , disinfection Yes / No /NA
or fumigation of the premises. :

8. Is rotation of disinfectants followed : Yes / No /NA

9. Is adequate lighting and ventilation provided : Yes / No /NA

10. Is there an appropriate functioning environmental control system
regarding:

i. Temperature? Yes / No / NA

ii. Humidity? Yes / No / NA

iii. Adequate air changes /hour? Yes / No / NA

Page 1 of 3

Vendor Site Audit checklist for Animal Suppliers

iv. Time controlled lighting system? Yes / No / NA

v. Appropriate noise control system? Yes / No / NA

11. Is there a documented procedure for waste and scrap disposal : Yes / No /NA

12. Are there adequate utilities available : Yes / No /NA

13. Is there a schedule to conduct preventive maintenance of : Yes / No /NA
equipments

14. Are there enough animal rooms of appropriate design to allow
separate housing of:

vi. The breeding colony Yes / No / NA

vii. Different animal species Yes / No / NA

viii. Sick animals Yes / No / NA

15. Is the Quality System enforced? Yes / No / NA
16. Are there facilities for collection and disposal of animal waste Yes / No / NA and of dead animals to minimize disease hazards and environmental contamination?
17. Are there facilities for cleaning, sanitizing, sterilizing and Yes / No / NA maintaining supplies and equipments including animal cages and racks?

C Personnel

1. Attach qualification and experience record of key personnel : Attach List
2. Is there a documented training program for staff and workers? : Yes / No /NA
3. Is there a documented health care program (medical : Yes / No /NA examination)
4. Are the personnel provided with clean uniforms : Yes / No /NA

D Stores

1. Are there separate storage areas for equipments, animal feed, Yes / No / NA and bedding which are protected from infection/contamination, and with refrigeration where needed?
2. Are the areas adequate for storage of the materials in relation to : Yes / No /NA

Page 2 of 3

Vendor Site Audit checklist for Animal Suppliers
their quantity

3. Is stock rotation program followed (like FIFO) : Yes / No /NA

E Documentation

1. Are all documents controlled : Yes / No /NA
2. Is a Certificate of Analysis issued for the material released : Yes / No /NA
3. What are the rejection levels :

Date of Inspection:

Audit Team Representing Company Name XYZ.
Name Designation Department Signature

Vendor's Representatives

Name Designation Department Signature

Page 3 of 3

Annexure - 14

Audit checklist of contractual testing laboratory

A. General Information

1. Name of Organization
2. Address
3. Name & Address of Contact Person

B. Questionnaire

1 Quality System:

- 1.1 Has the management established its quality policy in writing? Yes / No / NA
If yes, please attach a copy.
- 1.2 Is there an Organizational Chart? Yes / No / NA
If yes, please attach a copy.
- 1.3 Are the personnel's responsibilities established in their Yes / No / NA job descriptions?
- 1.4 Has the company / organization adequate resources in terms of finance Yes / No / NA and personnel?
- 1.5 Has the company / organization designated a responsible person Yes / No / NA with enough authority to assure the established quality system?
- 1.6 Is the company / organization approved by government Yes / No / NA (State/Central/UT)
Please mention the nature of your organization:
(Government/ Semi government/Private)
- 1.7 Does your organization carry any recognition or accreditation from Yes / No / NA National / International agency like ISO/WHO or any other agency?
If yes, please attach a copy of the latest certificate.

Page 1 of 4

Audit checklist of contractual testing laboratory

- 1.8 Are personnel qualified to perform their tasks? Yes / No / NA
Please attach the list of educational qualifications of personnel as a minimum of those working at key positions.

2. Documentation:

- 2.1 Is the system of approval of documents established? Yes / No / NA
- 2.2 Is there an established procedure for approving changes in documents? Yes / No / NA
- 2.3 Is there a system to keep records of approved documents with their Yes / No / NA edition/revision number?
- 2.4 Are procedures in force to assure the identification and traceability of Yes / No / NA works/ tests carried out?
- 2.5 Is there a procedure for identifying, coding and archiving of records? Yes / No / NA

3. Calibration, Maintenance and Validation:

- 3.1 Is there a maintenance program in force for all measuring and test Yes / No / NA equipments?
- 3.2 Do you have a calibration program? Yes / No / NA

3.3 Do you have procedures in force for performing Calibrations and Yes / No / NA
Maintenance work?
3.4 Does the measuring and test equipments clearly identified with Yes / No / NA
calibration status?
3.5 Are the standards used for Testing / Validation / Calibration traceable to Yes / No / NA
National / International Reference Standards?
3.6 Are the Instruments / Standards used for calibration regular
Yes / No / NA
3.7 Do you have Standard Operating Procedures (SOPs) for carrying out Yes / No / NA
Testing / Validation / Calibration?
If so, can these be made available to the contracting party on demand? Yes / No / NA

Page 2 of 4
Audit checklist of contractual testing laboratory

4. Audit and Inspection:
4.1 Is the facility subjected to quality audits? Yes / No / NA
4.2 Are quality audits performed to verify the efficacy of the Quality Yes / No / NA
System?
4.3 Is there a defined plan for audits? Yes / No / NA
4.4 Is there a follow up of corrective actions taken? Yes / No / NA
5. Training:
5.1 Is there a procedure for training of staff in their respective areas of Yes / No / NA
working?
5.2 Are training programs established? Yes / No / NA
5.3 Are training records maintained? Yes / No / NA
6. Testing:
6.1 Are documented procedures in force for handling samples and testing? Yes / No / NA
6.2 Is there a traceability of the records for a test sample / work carried out? Yes / No / NA
i.e. green sheets/logbooks/record forms etc.
6.3 Do you have required facilities for storage of samples before (and after) Yes / No / NA
testing?
6.4 Are there separate areas for samples requiring special storage conditions? Yes / No / NA
6.5 Do you issue certificate of analysis for the sample/product tested? Yes / No / NA
6.6 Are there adequate facilities for testing of sterile products Including Yes / No / NA
sterility and pyrogen testing?
6.7 Is the sterility testing room classified? Yes / No / NA
6.8 Do you maintain an animal house? Yes / No / NA
6.9 Does the animal house have adequate area and required facilities? Yes / No / NA
6.10 Is there a qualified veterinarian to look after the animal house? Yes / No / NA
6.11 Is the animal house registered with regulatory authorities? Yes / No / NA

Page 3 of 4
Audit checklist of contractual testing laboratory

6.12 Is permission sought from the regulatory authorities for using animals for Yes / No / NA
various tests?
6.13 Is periodical examination of animals for health status carried out? Yes / No / NA
6.14 Do you have arrangements to quarantine the animals after reception? Yes / No / NA
6.15 Do you have provision for isolation of sick animals as well as animals Yes / No / NA
under test?
6.16 Are qualified and experienced personnel available for carrying out testing Yes / No / NA
in animals?
6.17 Do you have an approval system for suppliers from whom the testing Yes / No / NA
animals are purchased? Please attach the list of institution (s) from where
the animals are procured.

7. Remarks, if any:

Date of Inspection:

Audit Team Representing xxxx Company
Name Designation Department Signature

Vendor's Representatives

Name Designation Department Signature

Page 4 of 4
Annexure - 15
Site audit checklist for Service Providers

A. General Information

:
1. Name of the service provider

:
2. Address & Contact No.

3. Nature of the service :
4. Date audit conducted :
5. Brief company profile :
6. - Year of establishment :
7. - Registration of the agency :
8. - No. of employees working :
9. - Certification (if any) :
10. - Any accreditation from :
reputed company
11. Whether credentials of the company are :
impressive

12. Whether the firm was prosecuted by law :

B Building & facility :

1. Is the location suitable for the facility : Yes / No / NA
2. Is the access to the facility restricted and controlled : Yes / No / NA
3. Whether the facility is properly cleaned & maintained : Yes / No / NA
4. Is it devoid of cracks etc. : Yes / No / NA
5. Is it designed to keep it free from birds, pests, vermins & rodent. : Yes / No / NA

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Site audit checklist for Service Providers

6. Are there adequate personnel facility like – wash rooms, canteen, medical : Yes / No / NA rooms etc.
7. Is adequate lighting & ventilation provided : Yes / No / NA
8. Are there adequate utilities available : Yes / No / NA
9. Whether the facility is properly cleaned & maintained : Yes / No / NA
10. Are periodic records of cleaning & maintenance kept properly : Yes / No / NA
11. Is there logical placement of equipments : Yes / No / NA
12. Is the warehouse area suitable & sufficient to the facility requirement : Yes / No / NA
13. Are the equipments properly maintained & calibrated : Yes / No / NA
14. Is there adequate facility for housing the laboratory animals : Yes / No / NA (if Yes, please attach separate sheet)
15. Is there any system in place to address the complaint received against the : Yes / No / NA service provider.

C Personnel:

1. Is the organogram available & adequate : Yes / No / NA
2. Is the technical staff competent in qualification & experience : Yes / No / NA
3. Is the number of personnel employed adequate in proportion to the workload : Yes / No / NA
4. Are Personnel also provided with regular in-service training : Yes / No / NA
5. Whether the supervisory staff are whole time employees : Yes / No / NA

D Documentation : : Yes / No / NA

6. Are all documents controlled : Yes / No / NA
7. Are the documents approved signed & dated by authorized person before : Yes / No / NA distribution

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Site audit checklist for Service Providers

8. Do the documents specify : Yes / No / NA
 - a) title
 - b) nature
 - c) purpose
9. Is the documentation handled by electronic data processing methods : Yes / No / NA
10. Are the records maintained electronically protected by suitable means. : Yes / No / NA
11. Is record available for calibration & validation of equipments. : Yes / No / NA

Date of Inspection:

Audit Team Representing aemoNmynapmoC.
Name Designation Department Signature

Vendor's Representatives

Name Designation Department Signature

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1.0 |Objective |
 |To lay down the procedure for assessment and approval of vendors. |

2.0 |Scope |
 |This Standard Operating Procedure is applicable to assess, qualify and |
 |approve the vendors of Raw Materials, culture media for testing, critical |
 |consumables, Packaging Materials, experimental animals, contractual |
 |testing laboratories and various service providers such as calibration |
 |agencies, pest control contractors at Company name. |

3.0 |Responsibility |
 |3.1 |Head, Global sourcing department or his/her designee shall be |
 |responsible for providing information from vendor, procurement, |
 |alternative vendor system, implementation and compliance of the |
 |procedure. |
 |3.2 |Head, Quality Control or his/her designee shall be responsible for |
 |arranging the sampling and testing/analysis of pre-purchase and |
 |commercial supply samples, as applicable |
 |3.3 |Head, Warehouse, Head, Quality Control and Head Animal House or |
 |his/her designee shall be responsible for providing the required |
 |testing data and delivery performance respectively for the raw |
 |material, culture media for testing, experimental animals and |
 |packaging material during performance control of vendors. In case of |
 |Service providers (Calibration and validation/revalidation), |
 |Engineering department shall provide the annual performance. Head – |
 |QC shall provide the annual performance of contractual testing |
 |laboratories. Head Administration shall be responsible for providing |
 |annual performance of Service provider for Pest Control. |
 |3.4 |A team constituted by Head of Quality Assurance department shall be |
 |responsible for evaluation/assessment of the vendors, as applicable. |

4.0 |Abbreviations and Definitions |
4.1	SME	Subject Matter Expert
4.2	cGMP	Current Good Manufacturing Practices
4.3	VMT	Vendor Management Team
4.4	QMS	Quality Management System
4.5	WHO	World Health Organization
4.6	RM	Raw Material
4.7	PM	Packaging Material
4.8	DMF	Drug Master File
4.9	VI	Valuation Index
4.10	CAPA	Corrective Action and Preventive Action
4.11	GPT	Growth Promotion Test
4.12	IOM	Inter Office Memorandum
4.13	OOS	Out of Specification
4.14	VAQ	Vendor Assessment Questionnaire
4.15	NRA	National Regulatory Authority
4.16	Vendor	The term shall include the manufacturer and/or
	supplier of raw and packaging materials, culture	
	media for testing, experimental animals,	
	contractual testing laboratories and service	
	providers.	

4.17 |Approved Vendor |A vendor who has been approved by Company Name as |
 | |per this SOP |

5.0 |Procedure |
 |5.1 |The manufacturers (Raw Materials, Packing Materials, Culture media |
 |and Lab. Chemicals etc.), suppliers (Eggs and animals), Contractual |
 |testing laboratories and service providers (Pest control and |
 |Calibration or validation service providers) shall be assessed and |
 |qualified before start purchasing material or availing services. |

5.2 |Selection of vendors |
 |Vendors should be selected based upon following criteria: |
5.2.1	Market reputation (global/local existence in the same field,
	business turnover and QMS etc.).
5.2.2	Information assess through internet (DMF holder/no. of
	accreditations/no. of clients etc.).
5.2.3	Certification / Qualification from WHO / other reputed
	regulatory authorities.
5.2.4	Easy to contact.
5.2.5	No. of years in business.
5.2.6	Already approval for other site of xxxxx company and
	recommendations from CQA.

5.3 |Preparation of vendor development Team: |
 |The vendor development team shall include representatives from |
 |Global sourcing department and Quality Assurance with support as |
 |appropriate from Production, Quality Control, and Engineering. |
 |Selection of the members of the vendor development team should be |
 |based upon the following criteria: |

		5.3.1	Experience of minimum 3 – 4 years.
		5.3.2	Expertise in concerned activity.
		5.3.3	Good knowledge of cGMP requirements.
		5.3.4	Trainings on Vendor Qualification and training on cGMP.
		5.4	Classification of Materials:
		5.4.1	Critical Materials:
		Criticality is decided on the basis of critical factors such	
		as direct contact of the material with final product (ARM),	
		impact on stability of product, printed packing materials,	
		any material of biological origin, sterile materials and	
		de-pyrogenated materials. Contract testing lab. shall be	
		considered equivalent to critical materials. Define the	
		material for critical or non-critical after evaluation as	
		per Material assessment record (Annexure – 1).	
		5.4.2	Non-Critical Material:
		Any material which is not classified as critical material.	
		5.4.3	Critical consumables:
		Materials/consumables that are used during processing of	
		final product, but are not a part of the finished product.	
		For example:- Silicon tubing, 0.2 µm filters etc.	
		5.4.4	Minimum requisite for critical/non-critical Vendor and
		Service Provider:	
		Following is the minimum requisite for vendor approval of	
		critical/non-critical Vendor:	
		Requirements	
		Non Critical Vendor	
		Critical Vendor	
		Service providers	
		TSE/BSE Assessment	
		(1)√	
		(1)√	
		NA	
		Questionnaire (General and vendor specific)	
		√	
		√	
		√	
		Site Audit	
		-	
		(2)√	
		(3)√	
		Historical Performance (If Applicable)	
		√	
		√	
		√	
		Quality Agreement	
		-	
		√	
		(4)√	
		Service Agreement	
		NA	
		NA	
		√	
		(1) In case of material of Animal Origin	
		(2) If the vendor is an approved DMF holder or Approved by	
		WHO for the required material, the site audit may not be	
		necessary for approval of critical vendor.	
		(3) Site Audit is mandatory in case of Contractual Testing	
		Laboratories only	
		(4) Quality Agreement shall be prepared in case of	
		Analytical Testing Lab only	
		5.5	Criteria for Evaluation/Assessment of vendors:
		5.5.1	Upon request from Global sourcing department, Quality
		Assurance Department shall provide the "Vendor pre-audit	
		assessment Questionnaire". The pre-audit questionnaire shall	
		be given in two parts i.e. General Questionnaire, Annexure –	
		2 and Specific questionnaire based upon vendor to be	
		qualified. Following specific questionnaire shall be used as	
		required:-	
		"Pre-audit questionnaire for RM/Critical consumables	
		vendors" – Annexure 3,	
		"Pre-audit questionnaire for PM vendors" – Annexure 4,	
		"Pre-audit questionnaire for eggs suppliers" – Annexure 5,	
		"Pre-audit questionnaire for animals suppliers" – Annexure	
		6,	
		"Pre-audit questionnaire for contractual testing	
		laboratories" – Annexure 7, &	
		"Pre-audit questionnaire for service providers – Annexure 8.	
		5.5.2	Send the questionnaire through Global sourcing department
		and ask to provide the filled questionnaire and other	
		supporting documents. Ask the vendor to provide pre-purchase	
		samples (preferably 3 samples from different batches) along	
		with certificate of analysis, as applicable. Quantity of	
		each sample shall be sufficient to conduct at least one	

		complete testing as per specifications. Service providers	
		and analytical testing laboratories shall not be subjected	
		to any pre-purchase samples and only filled questionnaire is	
		sufficient for preliminary evaluation.	
		Note: FAQ (Frequently asked Questions) provided by vendor	
		can be accepted in place of filled checklist for preliminary	
		evaluation if it is able to provide the information which	
		can fulfils the minimum criteria of evaluation i.e. $\geq 90\%$.	
	5.5.3	If required, share the specification of required material(s)	
		with vendor through Global sourcing department.	
	5.5.4	Upon receipt of filled form "Vendor pre-audit assessment	
		Questionnaire" from the vendor, along with the samples,	
		Quality Assurance Department shall evaluate the information	
		furnished by the vendor as per evaluation of valuation	
		index.	
	5.5.5	The filled checklist is scored as per the following	
		criteria.	
		Where the activity mentioned in the Check-list is:	
	5.5.5.1	Followed (Yes)	1 points
	5.5.5.2	Not followed (No)	0 point
	5.5.5.3	Not applicable (NA)	Not considered for
		evaluation.	
		Note: General information and fill in the blanks in	
		the checklist shall not be assigned any point and	
		points in checklist marked as not applicable shall	
		not be considered for valuation index.	
	5.5.6	Valuation Index is calculated as under and is expressed in	
		percentage:	
		Marks obtained	
		Valuation index (VI) = ----- X 100 (%)	
		Total Marks	
	5.5.7	In case VI is < 90% vendor should be rejected and samples	
		should not be sent to QC and returned back to the vendor. If	
		the valuation index (VI) is $\geq 90\%$, then the following steps	
		shall be taken for approval of vendor.	
	5.5.8	Send the samples to QC along with the COA of respective	
		samples along with the request for analysis (Annexure – 9)	
		for vendor approval. The samples of culture media shall be	
		subjected to Growth promotion Testing (GPT), and if	
		required, additional testing should be carried out.	
	5.5.9	QC after testing as per in-house/pharmacopeia	
		specifications, shall give results to QA and only those	
		vendors shall be considered for approval whose all samples	
		passes the testing.	
	5.5.10	After evaluation of the questionnaire and results of the	
		samples analysis, Head – Quality Assurance or his designee	
		or a designated team/trained personnel shall audit the	
		manufacturing location to conduct the site audit of the	
		vendor facility (in case of critical materials, wherever	
		applicable).	
	5.5.11	The audit team visits the site of the vendor for evaluating	
		his Quality Management System & material requirements as per	
		vendor site audit checklist filled. Vendor site audit	
		checklist can be used for reference only; site audit can go	
		beyond the information contained in the checklist. Following	
		specific audit checklist shall be used as required:-	
		"Audit Checklist for RM/Critical consumables vendors" –	
		Annexure 10,	
		"Audit Checklist for Packaging Materials vendors" – Annexure	
		11,	
		"Audit Checklist for eggs suppliers" – Annexure 12,	
		"Audit Checklist for animals suppliers" – Annexure 13,	
		"Audit Checklist for contractual testing laboratories" –	
		Annexure 14, &	
		"Audit Checklist for Service providers – Annexure 15.	
	5.5.12	After site audit, audit team shall prepare the Vendor Site	
		Audit Report (Annexure – 16) and write	
		recommendations/Observations.	
	5.5.13	Based upon observations made by auditors, vendor shall be	
		asked to submit CAPA report within 4 weeks. In such case,	
		vendor shall be approved conditionally for first three	
		commercial consignments.	
	5.5.14	CAPA report submitted for compliance of the audit	
		observations shall be evaluated. If the vendor comply all	
		the recommendations, vendor shall be approved. If the CAPA	
		report is not submitted within defined timeline or it is not	
		effective, a warning letter to vendor shall be issued to	
		vendor for submission of CAPA within one week after receipt	
		of warning letter and if vendor still does not submits CAPA,	
		The vendor shall be delisted and same shall be communicated	
		to CQA, Global Sourcing, WH and other concerned departments.	
	5.5.15	QA shall prepare "Vendor approval report" – Annexure 17	
		after review of satisfactory CAPA. Prepare an addendum to	
		the existing list of approved vendors and send information	
		to warehouse, QC, CQA and global sourcing department.	
		Include the name of the vendor in the list of approved	
		vendors for RM/PM, Annexure – 18, & list of contract testing	
		laboratories/service providers, Annexure – 19 on monthly	
		basis.	
	5.5.16	After approval of vendor, QA department should send a	
		Quality Agreement for manufacturer/Critical materials and	
		contractual testing laboratories Annexure – 20 and Service	

			agreement for service providers, Annexure – 21 to the vendor	
			(preferably through Global sourcing department). The quality	
			agreement shall be valid for three years, however the	
			service agreement shall be prepared annually. The Quality	
			/Service Agreement to be signed in between Company Name. and	
			vendor, shall be prepared. The content of Quality/Service	
			agreement may be prepared on case to case basis depending	
			upon materials/services provided.	
			5.5.17	In case a vendor has its own developed quality agreement the
			same may be accepted if it is having following clauses:	
			Manufacturing compliance as per cGMP, including batch	
			records.	
			Quality control including analysis, stability testing if	
			applicable and OOS handling.	
			Quality assurance including documentation, SOPs, product	
			release, Deviation and Investigations, Change management,	
			complaints and Batch disposition.	
			5.5.18	At least three service providers shall be approved for each
			type of service taken from external agencies (Except	
			services of Pest Control). Similarly for critical Raw	
			materials, Packaging materials and critical consumables at	
			least one alternative vendor shall be approved as	
			applicable.	
			5.5.19	Record of vendor approval:
			The record of vendor approval contains the following	
			documents (but not limited to):	
			5.5.19.	Checklist for the vendor assessment (Pre-audit
			1	questionnaires).
			5.5.19.	Vendor's Quality Management System assessment by
			2	accredited agency for example:- GMP certificate, GLP
			approval, CPCSEA approval, NABL accreditation,	
			WHO-GMP certificate or any other regulatory or semi	
			regulatory certification.	
			5.5.19.	Test reports.
			3	
			5.5.19.	Vendor Site Audit checklist.
			4	
			5.5.19.	Vendor Site Audit Report
			5	
			5.5.19.	Vendor Approval Report.
			6	
			5.5.19.	Quality Agreement.
			7	
			5.5.19.	Any other relevant documents.
			8	
			5.5.20	Updating of vendor approval record :
			5.5.20.	If an approved vendor's Quality Management System
			1	certificate is renewed, the renewed certificate is
			appended to the records.	
			5.5.20.	A list of qualified and approved vendors prepared by
			2	QA Department shall be issued to Warehouse, QC,
			Production Department, Animal House, Global sourcing	
			& list of approved service providers to respective	
			functional departments like	
			Engineering/administration on monthly basis.	
			5.6	When any new material is to be procured from already approved
			vendor, following steps shall be taken:	
			5.6.1	In case, where the already approved vendor has to be
			approved for a new material, Global sourcing department	
			shall send "Pre-audit assessment questionnaire" for the	
			specific questionnaire and procure pre purchase samples	
			(preferably 3 samples from 3 different batches) from vendor.	
			5.6.2	Global sourcing department shall forward the questionnaire
			along with samples to QA. Quality Assurance department shall	
			evaluate the questionnaire and if the valuation index (VI)	
			is $\geq 90\%$, then forward the pre-purchase samples to QC along	
			with the request for analysis (Annexure 9). QC shall give	
			the results of analysis to QA. In case VI is $< 90\%$ samples	
			should not be sent to QC and returned back to the vendor.	
			5.6.3	If pre-purchase samples passes the tests, the material shall
			be considered as approved material to purchase from already	
			approved vendor and Quality Assurance Department shall	
			prepare the Approval Report of additional material (Annexure	
			– 22) which shall be appended to the record of vendor and	
			the material shall be included in Approved vendor list, when	
			the next review is due. An addendum to approved Vendor list	
			for updating of new material shall be circulated to	
			Warehouse, Quality Control and other concerned departments	
			as per Annexure – 28.	
			5.6.4	In case, the vendor is already approved for any type of
			material from any other manufacturing location of the	
			vendor, the vendor should be audited as per new vendor	
			qualification procedure.	
			5.6.5	In case of change of Vendor for the same material / same
			type with same specification, a change control shall be	
			initiated by user. The product stability and material	
			comparability shall be carried out for critical materials.	
			For non-critical materials, only material comparability	
			should be done.	
			5.7	Disqualification of vendor:
			An approved vendor may be disqualified based upon one or more of the	
			following criteria, and following below mentioned procedure:	

	5.7.1	In case any single consignment of a specific material is
		rejected, then following an OOS procedure, if applicable,
		within a week a written communication shall be issued to the
		vendor through Global sourcing department, as per "Caution
		Letter" – Annexure 23 for thorough investigation and a
		detailed investigation report shall be sought from the vendor
		within the period of maximum 30 days. Global sourcing
		department shall not purchase same material from the vendor
		until clearance is given form QA. In case detailed
		investigation with effective CAPA is received from vendor
		end, QA shall give clearance for further purchase of the
		material.
	5.7.2	In case any new consignment is received during investigation,
		it shall be tested as per procedure and shall only be used if
		material passes full testing parameters as per the
		specifications and if the consignment(s) get rejected in any
		of the parameter(s), the vendor shall be put on hold and
		final approval/rejection shall be made after following the
		complete requalification procedure.
	5.7.3	If qualification of the vendor from WHO/other regulatory
		authority is cancelled, vendor shall be disqualified and same
		shall be intimated to corporate QA.
	5.7.4	In case of raw materials including culture media, eggs,
		critical consumables and animals, more than 10% rejection in
		terms of number of consignments received annually from
		approved vendor, shall render a vendor unapproved for that
		specific material and in such case, site audit/visit (as
		appropriate) for re-approval shall be carried out. Similarly
		in case of packaging Materials, more than 15% rejections in
		terms of number of consignments received annually from
		approved vendor, shall render a vendor unapproved for that
		specific material and in such case, site audit/visit (as
		appropriate) for re-approval shall be carried out.
	5.7.5	Three successive rejections of the materials shall render a
		vendor unapproved for that specific material and in such
		case, site audit/visit (as appropriate) for re-approval shall
		be carried out. Vendor shall be approved only after quality
		results of 3 lots of pre-purchased samples (as applicable).
		Upon successful evaluation of vendor, vendor should be
		included in approved vendor list.
	5.7.6	If any vendor is disqualified, then CQA shall be informed to
		evaluate the impact for other sites.
	5.7.7	In case, no material / service has been procured / provided
		respectively from an approved vendor for the last three
		years, the vendor shall be subjected to reassessment before
		restart the purchasing of material from the same vendor.
	5.8	Quality dispute resolution
	5.8.1	If there is any quality dispute related to material, being
		supplied by vendor, the following resolutions shall be done:
	5.8.1.1	QC personnel of Company nameshall perform the
		quality test at vendor's site.
	5.8.1.2	QC personnel of vendor shall perform the quality
		test at Company Name site.
	5.8.1.3	Third party consultation or resolution may be taken
		into consideration.
	5.9	Re-evaluation of vendor:
		Each approved vendor, from which the company is procuring materials,
		shall be re-evaluated every 3rd year + three months. Contractual
		testing laboratories & service providers, from which the company is
		taking services, shall be evaluated every year to get updated
		information of the vendor. The following steps shall be taken for
		re-evaluation of vendor:
	5.9.1	Prepare a schedule of vendor audit every year (preferably in
		first month) for the vendor's audit due in current year as
		per "Vendor audit Schedule", Annexure – 24. The audit
		schedule shall be shared with Global sourcing department for
		sharing audit information and asking for possible audit
		dates. Update the audit schedule on monthly basis to check
		the actual performance of the audit.
	5.9.2	Send "Vendor pre-audit assessment Questionnaire" to all the
		vendors before due dates of vendor approval. Send pre-audit
		questionnaire in two parts i.e. General Questionnaire
		Annexure – 2 and Specific questionnaire based upon vendor to
		be qualified. Following specific questionnaire shall be used
		as required:-
		"Pre-audit questionnaire for RM/Critical consumables
		vendors" – Annexure 3,
		"Pre-audit questionnaire for PM vendors" – Annexure 4,
		"Pre-audit questionnaire for eggs suppliers" – Annexure 5,
		"Pre-audit questionnaire for animals suppliers" – Annexure
		6,
		"Pre-audit questionnaire for contractual testing
		laboratories" – Annexure 7, &
		"Pre-audit questionnaire for service providers" – Annexure
		8.
	5.9.3	The vendor whose Valuation Index (VI) is $\geq 90\%$ shall be
		considered to remain under approved vendor list.
	5.9.4	After reviewing all the updated information and analysis of
		retrospective data, Quality assurance Department shall
		prepare the Re-evaluation Report of Vendor (Annexure – 26).
		Based upon the assessment of filled questionnaire and review
		of three years performance report, vendor shall be

		re-qualified for next three years.		
	5.9.5	Vendor, from which the critical material is procured, shall		
		preferably be audited once in three years and asked to		
		submit CAPA report within a period of 4 weeks. If the vendor		
		is an approved DMF holder or approved form WHO for the		
		required material and if the material supplies from a vendor		
		have been consistently meeting the specifications and there		
		are no quality issues, then the vendor re-audit may be		
		exempted and in this case only the vendor assessment		
		questionnaire is considered adequate.		
	5.9.6	CAPA report submitted for compliance of the audit		
		observations shall be evaluated. If the vendor comply all		
		the recommendations, vendor shall be approved. If the CAPA		
		report is not submitted within defined timeline or it is not		
		effective, vendor shall not be approved.		
	5.9.7	Upon successful re-approval of vendors, the quality /		
		service agreement shall also be reviewed as per Annexure –		
		20 and Annexure – 21.		
	5.10	Performance control of vendors:		
	5.10.1	The Quality Control Department gives annually, to QA		
		Department, a retrospective report of the test based		
		performance of each vendor, in respect of raw materials,		
		used for production and culture media used for testing and		
		in respect of testing of samples for water or any other		
		outsource testing facility. Similarly for service providers		
		for calibration and validation, Engineering department shall		
		provide the performance of respective calibration/validation		
		agencies. In case of test based performance of animal		
		supplier, Animal house shall provide the performance of		
		approved animal suppliers.		
		Similarly Warehouse shall also submit a report of the		
		vendor's performance regarding the delivery related issues		
		as per vendor performance report, (Annexure – 25) as		
		follows:-		
		5.10.1.	Warehouse	
		1		
		Warehouse shall provide the following information as		
		performance of vendor:-		
		5.10.1.1.	Name of the vendor.	
		1		
		5.10.1.1.	Vendor category.	
		2		
		5.10.1.1.	Products delivered.	
		3		
		5.10.1.1.	Performance of vendor.	
		4		
		5.10.1.1.	Observation.	
		5		
		5.10.1.	Quality Control	
		2		
		5.10.1.2.	Name of the vendor.	
		1		
		5.10.1.2.	Vendor category.	
		2		
		5.10.1.2.	Products delivered.	
		3		
		5.10.1.2.	Performance of the vendors.	
		4		
		5.10.1.2.	Results of tests performed with following	
		5	information:	
			a.	S. No.
			b.	Name of the Product.
			c.	Batch No.
			d.	Folio No.
			e.	Date of the test.
			f.	Total No. of Samples.
			g.	Results (Pass/Fail).
			h.	Remarks.
		5.10.1.	Animal House	
		3		
		5.10.1.3.	Name of Supplier.	
		1		
		5.10.1.3.	Supplier category.	
		2		
		5.10.1.3.	Animals delivered.	
		3		
		5.10.1.3.	Performance of Supplier.	
		4		
		5.10.1.3.	Observation.	
		5		
		5.10.1.	Engineering department	
		4		
		5.10.1.4.	Name of contractual testing facility.	
		1		
		5.10.1.4.	Testing facility category.	
		2		
		5.10.1.4.	Performance of the vendor.	
		3		
		5.10.1.4.	Observation.	
		4		
		5.10.1.	Quality Assurance	
		5		

			5.10.1.4.	Name of vendor.
			1	
			5.10.1.4.	Vendor category.
			2	
			5.10.1.4.	Products delivered.
			3	
			5.10.1.4.	Reference reports (all reference report
			4	whosever applicable).
			5.10.1.4.	Reference report number and date along
			5	with name of concerned department.
			5.10.1.4.	Observations.
			6	
	5.10.2	The QA Department will prepare Performance Control Report of		
		the vendors every year, irrespective of the material		
		purchased or not in last calendar year. The performance		
		report shall be prepared for all the approved		
		vendors/service providers.		
	5.10.3	The performance reports shall be assigned identification		
		numbers as follows:		
		The first two letters correspond to the department which		
		prepares the report e.g. QA for quality assurance		
		department. This is followed by a slash (/) followed by		
		three letters 'VPR' corresponding to the 'Vendor Performance		
		Report' followed by (-) which is followed by a four digit		
		number. The first two digits represent the last two digits		
		of the year to which the report pertains and the next two		
		digits represent the consecutive number of the vendor		
		performance report in continuation.		
		For example QA/VPR-1201 represents the first performance		
		report of a vendor pertaining to the year 2023 prepared by		
		QA department.		
		This numbering system shall remain same for the report from		
		Warehouse, Animal House, Engineering and Quality Control.		
	5.11	Refer to the vendor approval and assessment chart as per Annexure –		
	27.			
	5.12	After approval of vendor send intimation to CQA, Global sourcing		
		department and warehouse departments as per Annexure – 28.		
	6.	Forms and Records:		
	6.1	Material assessment record.	Annexure 1	
	6.2	General questionnaire.	Annexure 2	
	6.3	Pre-audit questionnaire for RM/Critical consumables	Annexure 3	
		vendors.		
	6.4	Pre-audit questionnaire for PM vendors.	Annexure 4	
	6.5	Pre-audit questionnaire for eggs suppliers.	Annexure 5	
	6.6	Pre-audit questionnaire for Animal Suppliers.	Annexure 6	
	6.7	Pre-audit questionnaire for contractual testing	Annexure 7	
		laboratories.		
	6.8	Pre-audit questionnaire for Service providers.	Annexure 8	
	6.9	Request for Analysis of pre-purchase samples.	Annexure 9	
	6.10	Audit checklist for Raw materials/Critical consumables	Annexure 10	
		vendors.		
	6.11	Audit checklist for Packaging materials vendors.	Annexure 11	
	6.12	Audit checklist for eggs suppliers.	Annexure 12	
	6.13	Audit checklist for animal suppliers.	Annexure 13	
	6.14	Audit checklist contractual testing laboratories.	Annexure 14	
	6.15	Audit checklist for Service Providers.	Annexure 15	
	6.16	Vendor Site Audit Report.	Annexure 16	
	6.17	Vendor approval report.	Annexure 17	
	6.18	List of approved vendors.	Annexure 18	
	6.19	List of approved contract testing laboratories / Service	Annexure 19	
		providers.		
	6.20	Quality agreement for Manufacturers / Critical Materials	Annexure 20	
		vendors.		
	6.21	Service agreement for Suppliers.	Annexure 21	
	6.22	Approval report of additional material.	Annexure 22	
	6.23	Caution Letter.	Annexure 23	
	6.24	Vendor audit schedule.	Annexure 24	
	6.25	Vendor performance report.	Annexure 25	
	6.26	Re-evaluation report of vendor.	Annexure 26	
	6.27	Vendor evaluation and approval process flow chart.	Annexure 27	
	6.28	Intimation of vendor approval	Annexure 28	
	7.0	References:		
	7.1			
	8.0	Distribution:		
	8.1			
	9.0	History:		

|Risk analysis for Preparation and Planning of Audit Program |
Year	Folio Number:			
Questionnaire,	VB I	VBII	VB III	VB IV
Block				
Written by:				
Checked by:				
Approved by:				

Date of audit:

Auditor:

Department: Auditee:

Audit report No.:- Audit

report no. assigned by (Sign/Date):

[S. No |DEPARTMENTS / SECTION CHECKPOINTS / CRITERIA |Yes |

| |

|1.0 |MANUFACTURING AREA | | | |

| |EQUIPMENT, MATERIAL / COMPONENT | | | |

|1.1.1 |Does all major equipments bear a unique I.D.? | | | |

|1.1.2 |Does the equipments and the facility have appropriate controls to | | | |

| |maintain required parameters e.g. temperature, relative humidity | | | |

| |pressure differentials etc.? | | | |

|1.1.3 |Is the equipment cleaned promptly after use? | | | |

|1.1.4 |Is the cleaning status of equipments is defined? | | | |

|1.1.5 |Are the rinse water samples sent for QC analysis? | | | |

|1.1.6 |Equipment /Instrument calibration & preventive Maintenance | | | |

|1.1.6.1 |Are written records maintained on equipment cleaning, sanitizing | | | |

| |and maintenance on or near each piece of equipment? | | | |

|1.1.6.2 |Does the facility have approved written procedure for checking | | | |

| |and calibration of each piece of measurement equipment? | | | |

|1.1.7 |MATERIAL /COMPONENT | | | |

|1.1.7.1 |All handling of material and product such as receipt, quarantine, | | | |

| |sample storage, tracking, labeling, dispensing, processing, | | | |

| |packaging and distribution is done in accordance with approved | | | |

| |written procedure, or instruction and recorded? | | | |

|1.1.7.2 |All incoming materials and finished products quarantined | | | |

| |immediately after receipt or processing until they released for | | | |

| |use or distribution? | | | |

|1.1.7.3 |Are all material handled in such a way to prevent contamination? | | | |

|1.2 |RAW MATERIALS | | | |

|1.2.1 |Does the store premise allow storage of raw materials at various | | | |

| |temperatures? | | | |

|1.2.2 |If answer yes to the above what are controls to log any | | | |

| |irregularities? | | | |

|1.2.3 |Are balance used are calibrated? | | | |

|1.2.4 |Are the standard weights used for calibration of balance are | | | |

| |calibrated from NABL? | | | |

|1.2.5 |Is any documentation done at the time of receiving the raw | | | |

| |material? | | | |

|1.2.6 |What is the process of communication with the laboratory for | | | |

| |sampling of the raw material? | | | |

|1.5 |STERILIZER/OVEN LOADING ROOM | | | |

|1.5.1 |Does the sterilizer meet the following criteria: | | | |

| |Is it capable of maintaining an empty chamber temperature | | | |

| |distribution of +/-1° C at a steady state? | | | |

| |Does the VLT passes regularly? | | | |

| |What is the size of vent filter? | | | |

| |Is the minimum and maximum load displayed? | | | |

| |Is the load pattern displayed? | | | |

| |Is the SOP for operation of sterilizer displayed? | | | |

| |Is it equipped with a clean steam? | | | |

| |Does the record of every sterilization cycle maintained? | | | |

|1.5.2 |Does the oven meet the following criteria? | | | |

| |Are the two interlocking doors working properly? | | | |

| |Are the re-circulating fans located upstream of the HEPA filters? | | | |

| |Does it have the resistance temperature device in the return air | | | |

| |duct for control temperature? | | | |

| |Is it equipped with temperature recorder? | | | |

|1.5.3 |Are the sterilizers and ovens calibrated and the results | | | |

| |documented? | | | |

|1.5.4 |Are the results of the testing documents? | | | |

|1.5.5 |Are the garments made of non- shredding material? | | | |

|1.5.6 |Are the garment washed and sterilized after use? | | | |

|1.5.7 |How frequently they sterilized? | | | |

|1.5.8 |Are biological or chemical indicator used? | | | |

|1.5.9 |Are the sterilizer and oven charts reviewed by IPQA prior to the | | | |

| |release of sterile products? | | | |

|1.5.10 |Are the procedures validated for the sterilized/depyrogenated | | | |

| |stoppers and components? | | | |

|1.6 |EQUIPMENT AIRLOCK | | | |

|1.6.1 |Is the air lock door interlock system working properly? | | | |

|1.6.2 |Is the air lock cleaning performed as per predefined procedure | | | |

| |and frequency? | | | |

|1.6.3 |Is the SOP for transfer of materials available? | | | |

|1.8 |MANUFACTURING OF STERILE PRODUCTS | | | |

|1.8.1 |Does the company have adequate sterilizing, depyrogenation, | | | |

| |sterile, filtration, and processing, filling and packaging | | | |

| |equipment available? | | | |

|1.8.2 |Is there an adequate control of viable, non-viable | | | |

| |micro-organisms and effective routine monitoring of sterile area | | | |

| |and aseptic practices through use of reliable equipment? | | | |

|1.8.3 |Do master/Working Cell Banks and Seed Stocks have detailed | | | |

| |records of: | | | |

|1.8.3.1 |History of cells including the number of generation doublings or | | | |

| |passages of virus? Is there a maximum limit? | | | |

|1.8.3.2 |Characterization according to the WHO TRS relevant to the | | | |

| |product? | | | |

1.8.3.3 |demonstration of purity? | | | |
 1.8.3.4 |manufacturing procedures? | | | |
 1.8.3.5 |appropriate storage and security with continuous monitoring of | | | |
 |temperature, alarms and backup power supply? | | | |
 1.8.3.6 |inventory log? | | | |
 1.8.3.7 |adequately segregated storage to avoid mix-up or | | | |
 |cross-contamination with other material? | | | |
 1.8.3.8 |routine monitoring of stability (viability/purity)? | | | |
 1.8.3.9 |Demonstration of identity? | | | |
 1.8.4 |Does the Manufacturing Formulae adequately describe the complete | | | |
 |production process? | | | |
 1.8.5 |Is the Manufacturing Formulae up-to-date and approved by QC/QA? | | | |
 1.8.6 |Is the Batch Production Record form an adequate representation of | | | |
 |the Manufacturing Formulae? | | | |
 1.9 |ENVIRONMENTAL MONITORING | | | |
 1.9.1 |Are laminar flow hood and HVAC system which serve the sterile | | | |
 |operation areas validated? | | | |
 1.9.2 |Is there an environmental monitoring program for the aseptic | | | |
 |batching area, sterile filling room, sterile filling line and | | | |
 |ancillary areas? | | | |
 1.9.3 |Is non-viable particulate sampling performed? | | | |
 1.9.4 |Are the results of environmental monitoring shared with | | | |
 |production department? | | | |
 1.10 |ASEPTIC BATCHING AREA | | | |
 1.10.1 |Are the light fixtures and motors totally sealed and enclosed? | | | |
 1.10.2 |Are the room air pressures, temperature, humidity and filter | | | |
 |pressure drop automatically monitored, alarmed and recorded? | | | |
 1.10.3 |Do personnel sanitize their gloved hands? | | | |
 1.10.4 |Are equipment status labeled? | | | |
 1.10.5 |Are the results documented for the equipments calibrated? | | | |
 1.10.6 |Are sterilization/ clarification filters integrity tested and | | | |
 |result documented? | | | |
 1.10.7 |What is the quality of water used for equipment cleaning? | | | |
 1.10.8 |Are tanks, hoses etc. sterilized prior to use and are these | | | |
 |procedure user validated? | | | |
 1.10.9 |Are tanks labeled during processing indicating status of product? | | | |
 1.10.10 |Are aseptic operations performed using laminar flow hoods? | | | |
 1.10.11 |Are sampling and testing complete prior to or concurrent with the | | | |
 |filling operation? | | | |
 1.11 |FILLING ROOMS | | | |
 1.11.1 |Is the product contact equipment sterilized by hot air or steam? | | | |
 1.11.2 |Are room air pressures, temperature, humidity and filter pressure | | | |
 |drop automatically monitored, alarmed and recorded? | | | |
 1.11.3 |What sanitizing agent is used to sanitize gloved hands? Is it | | | |
 |sterilized by filtration? | | | |
 1.11.4 |Are aseptic operations performed using laminar flow hoods? | | | |
 1.11.5 |Is the sterile filter integrity tested before and/or after use? | | | |
 1.11.6 |Is the porosity and type of filter included with the batch | | | |
 |records? | | | |
 1.11.7 |Are results of filter integrity test included with the batch | | | |
 |records? | | | |
 1.11.8 |What is the quality of water used for equipment cleaning? | | | |
 1.11.9 |Is filling line use record kept? | | | |
 2.0 |FINISHED 'ACTIVE RAW MATERIAL' CONTROL | | | |
 2.1 |Do written procedures indicate how and who verifies that correct | | | |
 |containers and packages are used for finished ARM product during | | | |
 |their finishing operation? | | | |
 2.2 |Are written sampling and testing procedures and acceptance | | | |
 |criteria available for each product to ensure conformance to | | | |
 |finished product specification? | | | |
 3.0 |Other Observations (Apart from the checklist) | | | |
 3.1 | | | | |
 3.2 | | | | |
 3.3 | | | | |
 3.4 | | | | |
 3.5 | | | | |
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 3.11 | | | | |

Remarks:

|Auditee's Sign/Date |Auditor's Sign/Date |
 | |

Each point of the checklist shall be filled as applicable and comments or remarks in addition shall be given under the remarks column in the checklist.

Note: Attach additional sheet if required for other observations.

Annexure-10

Date of audit:
Auditor:

Department: Auditee:

Audit report No.:- Audit
report no. assigned by (Sign/Date):

S. No	CHECKPOINTS/CRITERIA	Yes	No
1.0	WAREHOUSING / DISTRUBUTION		
1.2	Is there adequate space for the orderly storage of material?		
1.3	Is there a segregation of quarantined, approved & rejected materials?		
1.4	Are there separate stores for raw material, packaging material, labels and finished products?		
1.5	Are materials stored on pallets, shelves, or racks, off the floor and off the walls in all stores?		
1.6	Are minimum and maximum temperatures in the store recorded?		
1.7	Are air-conditioned and cold room facilities available?		
1.8	Are flammable and hazardous chemicals stored in safe- conditions?		
1.9	Are there no expired raw materials in the raw material store(s)?		
1.10	Are materials dispensed according to prescribed SOP meeting GMP requirements?		
1.11	After release by quality control, are raw materials and packaging materials released on a FIFO/FEFO (First in First Out / First Expiry First Out) basis?		
1.12	Do distribution records in the finished goods store enable specific batches to be traced?		
1.13	Is there updated list of approved vendors available with warehouse personnel?		
1.14	Is the shipping validation done for all the products?		
1.15	Is the list of approved transporters available with warehouse department and finished products are dispatched through approved transporters only?		
1.16	Is the data loggers sent with all shipments to monitor the temperature during transit?		
1.17	Is the data obtained from data logger downloaded and reported to QA?		
2.0	Other Observations (Apart from the checklist)		
2.1			
2.2			
2.3			
2.4			
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2.7			
2.8			

Remarks:

Auditee's Sign/Date Auditor's Sign/Date

Each point of the checklist shall be filled as applicable and comments or remarks in addition shall be given under the remarks column in the checklist.

Note: Attach additional sheet if required for other observations.

Date of audit:
Auditor:

Department: Auditee:

Audit report No.:- Audit
report no. assigned by (Sign/Date):

S. No.	CHECKPOINTS/CRITERIA	Yes	No
1.0	ANIMAL HOUSE		
1.2	Are animals kept in quarantine before shifting to dedicated rooms?		
1.3	Is the temperature/RH of quarantine area maintained?		
1.4	Are necessary tests for animals performed before shifting to dedicated rooms? Are the records maintained?		
1.5	Is the animal bedding changed routinely? Are the records maintained?		
1.6	Are animals tested prior to use? Are the records maintained?		
1.7	Are the animals used for testing separated and identified from other animals?		
1.8	Are animal cages autoclaved routinely? Are the records maintained?		
1.9	Is animal feed tested in house? Are the records maintained?		
1.10	Are the decadent animals kept separated? Are the records maintained?		
1.11	Is the autopsy performed of dead animals? Are the records of animal autopsy kept?		
1.12	Are animal tissues/organs archived (Wet and Dry)?		
1.13	Are the animals checked routinely for health checkups? Are the records maintained?		
1.14	Are the dead animals sent for incineration? Are the records maintained?		
1.15	Is there SOP displayed at the entrance to the animal house? Is the record maintained for every person visiting animal house?		
1.16	Is there a proper pest control system in place? Are the records maintained?		
2.0	Other Observations (Apart from the checklist)		
2.1			
2.2			
2.3			
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Remarks:

Auditee's Sign/Date Auditor's Sign/Date

Each point of the checklist shall be filled as applicable and comments or remarks in addition shall be given under the remarks column in the checklist.

Note: Attach additional sheet if required for other observations.

Annexure-12

Date of audit:
Auditor:

Department: Auditee:

Audit report No.:- Audit
report no. assigned by (Sign/Date):

S. No.	CHECKPOINTS/CRITERIA	Yes	No
1.0	Environment and Health		
1.2	Are all bio-hazard materials destroyed by incineration? Are records maintained?		
1.3	Is the Effluent Treatment Plant checked regularly and required chemical dose given as per requirement?		
1.4	Is the BOD & COD of the treated water determined routinely? Are the records maintained and values displayed on display board?		
1.5	Is the non hazardous solid waste handed over to approved waste management personnel?		
1.6	Is the preventive maintenance schedule for ETP available? Does the Preventive Maintenance performed as per the Schedule?		
2.0	Safety		
2.2	Is there is list of approved personnel available with contact details for handling of emergency situations?		
2.3	Is the personnel authorized for handling of emergency situations trained to handle the situation?		
2.4	Is the emergency exit checked routinely in all the departments for any obstacle? Are the records maintained?		
2.5	Is the contact number in case of emergency displayed in all critical areas including areas with maximum personnel?		
3.0	First Aid		
3.1	Is there a list of approved personnel for giving first aid?		
3.2	Are the personnel authorized for giving first aid trained for giving first aid?		
3.3	Is there the list of items kept within first aid box?		
3.4	Are the medicines kept within first aid box checked routinely for expiry date / other physical observations?		
3.5	Does all of the medicines kept in the first aid box are OTC category and not the drugs as per Schedule H?		
4.0	Other Observations (Apart from the checklist)		
4.1			
4.2			
4.3			
4.4			
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4.7			

Remarks:

Auditee's Sign/Date Auditor's Sign/Date
|||

Each point of the checklist shall be filled as applicable and comments or remarks in addition shall be given under the remarks column in the checklist.

Note: Attach additional sheet if required for other observations.

Annexure-13

|Audited Area: |Reporting Date: |
|Audit Report No. |Audit beginning date : |Audit ending date: |
|Responsible person in audited area |
Name	Designation
Objective and Scope:	
Reference Documents:	
Distribution of report	
Name	Designation
Total Deficiencies :	
Critical Deficiencies :	
Major Deficiencies :	
Minor Deficiencies :	
Conclusion:	
It is requested to_____that, taking to consideration the deficiencies found,	
determine the necessary actions to be taken and submit the audit compliance report	
(SOP No.:- QA/024) to Quality Assurance Department on or before _____.	
NAME OF AUDITORS	SIGNATURE / DATE

Audited Area:	Reporting Date:	
Audit Report No.	Audit beginning date :	Audit ending date:
S. No.	Deficiency	Classification
NAME OF AUDITORS	SIGNATURE / DATE	

|Department / Area |Year : |Folio No.: |

| Jan. | Feb. | Mar. | April |

|Written by: | | | | |

Checked by: | | | | |

Approved by: | | | |

|DEVIATION FROM THE SCHEDULED AUDIT PROGRAM |

|Year : |Folio : |

[illegible]

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Department: Area:
Reference to audit report No.:
Date of audit report:

To be filled by Auditee Department To be filled by QA	
Sr. No. of Immediate action (if any) and QA Comments Verification and Verified	
Audit action taken on closure by	
observation (Sign/Date	

Prepared by/date:
Checked by/date:
(Head/Designee auditee department)
(QA)
Remarks:-

Authorized by/date:-
(Head – QA)

|Department / Months|Year : |Folio No.: |

| |Jan. |Feb. |Mar. |April |

|Written by: | | | | |

Checked by: | | | |

Approved by: | | | |

|DEVIATION FROM THE SCHEDULED AUDIT PROGRAM |

|Year : |Folio : |

Month	Deviation number	Justification for Deviation	Signature
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[illegible]

Date of audit: Auditor:

Department: Auditee:

Audit report No.:- Audit
report no. assigned by (Sign/Date):

[S. No.]DEPARTMENT /SECTIONS CHECKPOINTS/CRITERIA [Yes |
|
1.0 |PERSONNEL | | | |
1.0 A |GENERAL | | | |
1 |Is there an organizational chart? | | | |
2 |Are there job descriptions for key personal? | | | |
|Are they Appropriate to the activities of the | | | |
|department? | | | |
3 |Number of staff _ | | | |
Number sufficient?			
Qualifications adequate?			
Experience sufficient?			
4	Is there a clear separation of job		
responsibilities of personnel?			
1.0 B	KEY PERSONNEL		
1	Are there sufficient key personnel to supervise		
assigned functions?			
2	Are they skilled / trained in their respective		
fields?			
1.0 C	TRAINING		
1	Are there on the job training procedures followed		
for new employees?			
2	Are training and education records available?		
Are they current?			
Are they filed with the supervisor?			
Is training schedule for current year is available			
with the department HOD?			
3	Does a GMP training program exist for annual		
update for all staff?			
4	Is there training in containment procedures?		
By written procedures?			
Are records maintained?			
1.0 D	PERSONAL HYGIENE		
1	Are appropriate protective apparel required?		
Is there a gowning SOP for production staff?			
2	Is staff instructed to report health or medical		
problems that may have an adverse effect on the			
product?			
3	Is there a medical monitoring program to ensure		
protection to ensure protection of staff and			
product?			
Vaccination where applicable?			
For all employees?			
4	Do controlled entry requirements exist?		
Do procedures exist for preventing unauthorized			
entry?			
Are the procedures in writing?			
2.0	PREMISES		
2.0 A	GENERAL		
1	Is there is appropriate cleaning, maintenance and		
proper operation?			
2	Are areas clearly defined and appropriately		
controlled:			
a	for quarantine and storage of starting materials?		
b	for storage of in-process material?		
c	for manufacturing and processing operations?		
d	for control and laboratory operations?		
e	for quarantine and storage of finished products?		
f	for holding of rejected material?		
g	for ancillary usage, e.g. rest rooms, maintenance		
workshops?			
h	for animal housing?		
3	Does the building design prevent the entry of		
insects, vermin and other animals?			
4	Do adequate drainage exist (Wherever required)?		
5	Does the design of the facility achieve a		
unidirectional flow of materials, personnel,			
product and waste so as to avoid cross-over of			
clean and dirty (infectious) material?			
6	Is the lighting provided adequate for the		
conditions necessary for the work being conducted			
in the area?			
7	Are facility layout drawings including mechanical,		
electrical and architectural kept up-to-date			
followings changes? Is revalidation of facilities			
performed following refurbishment?			
8	Are the premises satisfactory with respect to:		
a	neatness and cleanliness?		
b	state of repair, e.g. paint work, cracks in		
floors, ceilings or walls, door seals, etc?			
c	exposed piping or electrical wiring?		
3.0 B |CLEANING AND MAINTENANCE | | | |
1 |Is the equipment(s) suitably located to facilitate | | | |

|its use, cleaning and maintenance? ||||
 2 |Are equipments and utensils cleaned, maintained ||||
 |and sanitized as appropriate to prevent ||||
 |malfunction or cross contamination? ||||
 3 |Are piping systems, valves and vent filters ||||
properly designed to facilitate cleaning and				
sterilization? NOTE: Maintenance closed systems				
through the use of "clean in place" and "sterilize				
in place" is preferable.				
4	Are the valves on primary containment vessels			
(e.g. fermenters) steam sterilized?				
5	Are non-fiber releasing filters used for			
filtration?				
6	Are filters used for sterile filtration integrity			
tested before after use?				
7	Are calibrations and validation being performed			
adequately?				
8	Are autoclave and sterilizing ovens fitted with			
effective, proper air filters and is this				
integrity tested? Are sterilizers validated?				
9	Are supplies and equipment which are exposed to			
pathogens during processing kept separate from				
unused items to prevent cross-contamination?				
3.0 C	STANDARD OPERATING PROCEDURES (SOPs) AND RECORDS			
1	Are there written procedures (SOPs) for cleaning			
and maintenance of equipment and utensils and are				
they followed?				
2	Do these SOPs include:			
a	assignment of responsibility for cleaning?			
b	defined schedules for cleaning and maintenance?			
c	descriptions of methods, equipment and materials			
used?				
d	inspection of equipment for cleanliness			
immediately before use?				
e	assignment of identification number?			
f	documentation in record books?			
3	Are validated and approved cleaning and sanitizing			
agents used?				
4	Is clean equipment identified as such?			
5	Are calibrations and qualifications properly			
recorded?				
6	Are all certification and within date?			
7	Are there preventive maintenance programs and			
consistent record of work performed?				
4.0	DOCUMENTATION OF PROCESSING AND DISTRIBUTION			
4.0 A	GENERAL			
1	Are the records for:			
a	All material used?			
b	All standard operating procedures?			
c	Each lot and/or batch processing and distribution?			
d	All complaints and their investigation?			
e	All equipment, including cleaning, maintenance and			
validation?				
f	Cleaning, maintenance and environmental control of			
the premises?				
2	Are all records:			
a	Up dated?			
b	Signed by the person performing the task and for			
all critical steps by the person checking it?				
c	Kept at the work station during the entire			
operation?				
d	Retained and available for inspection at least 2			
years after the expiry date of the lot batch?				
4.0 B	DOCUMENTATION OF EQUIPMENT USED			
1	Are records on the use, cleaning, sterilization			
and maintenance of equipment kept in individual				
logs for each piece of equipment?				
2	Are these records dated and signed in			
chronological order?				
3	Do the records include information of the			
lot/batch including identification numbers and				
dates?				
5	Are standard procedures available for warehousing?			
6	Are standard procedures available that describe			
the shipping, final transit conditions and				
instruction for storage through the distribution				
chain, especially the cold chain?				
7	Are the shipping methods, especially the cold			
chain, validated and routinely monitored?				
8	Are records detailed and retrievable so that a			
rapid recall of any particular lot is achievable?				
Is the recall process delegated to the responsible				
person?				
9	Are records maintained for 2 years after the			
expiry date?				
5.0 C	OPERATIONAL PRACTICES AND PROCEDURES			
1	Are there standard operating procedures for			
decontamination of process equipment and				
facilities? Have these procedures been validated				
and is the performance monitored?				
2	Are standard operating procedures available and			
displayed outlining emergency procedures in the				

	event of a spill or accidental release of			
	contaminant?			
6.0	Any other observation			
6.1				
6.2				
6.3				

Remarks:

|Auditee's Sign/Date |Auditor's Sign/Date |
| | |

Each point of the checklist shall be filled as applicable and
comments or remarks in addition shall be given under the remarks
column in the checklist.

Note: Attach additional sheet if required for other observations.

Date of audit:

Auditor:

Department: Auditee:

Audit report No.:- Audit

report no. assigned by (Sign/Date):

|S. No |CHECKPOINTS/CRITERIA |Yes |
||
1.0	QUALITY CONTROL DEPARTMENT				
1.2	Are adequate facilities and equipment available for the physical,				
	chemical & microbiological testing?				
1.3	Are approved written procedures available for sampling, inspecting				
	and testing raw materials, packaging materials, in process and				
	finished products?				
1.4	Are samples of starting material , packaging material,				
	Intermediate products , bulk products and finished products taken				
	by methods and personnel's approved by QC department?				
1.5	Are in-process materials tested at appropriate phases for				
	identity, strengths quality, purity and are they approved or				
	rejected by Quality Control?				
1.6	Are records of in-process controls maintained?				
1.7	Does each batch of drug product prior to release confirms				
	compliance of finished product specification?				
1.8	Are validated & stability indicating assay methods used?				
1.9	Are reference standards used for the assay?				
1.10	Are accurate, clear & neat records maintained along with the raw				
	data for the entire analytical work?				
1.11	Are any deviations from these specifications, standard, etc.				
	recorded and justified?				
1.12	Do Laboratory controls include the establishment of scientifically				
	sound and appropriate,				
	- Specifications,				
	- Standards,				
	- Sampling plans,				
	- Test procedures				
	- Reference substances,				
	Are they designed to assure that tested materials conform to				
	appropriate standards of identity, strength, quality and purity?				
1.13	Do these laboratory controls include:				
1.13.1	Determination of compliance with written specifications for				
	acceptance of each lot within each shipment of materials or				
	holding of products?				
1.13.2	Description of sampling and testing procedure for in-process				
	materials?				
1.13.3	Retest policy, identifying the rationale and criteria for retests,				
	number of samples and the documentation required?				
1.13.4	A comprehensive calibration program that includes calibration /				
	certification intervals, acceptance criteria and provisions for				
	remedial action?				
1.13.5	Are reagents, culture media, etc. properly labeled, preparation				
	recorded in lab books with expiry dates given?				
1.13.6	Is appropriate testing done on each batch of product required to				
	be free of objectionable microorganisms?				
1.14	Are all reference reagents kept secure, properly stored,				
	identified and their integrity maintained?				
1.15	Are the tests results of all references and standard analyzed at				
	appropriate intervals for statistical variation from the expected				
	value?				
1.16	Is every sterilized lot of media is tested for Growth Promotion				
	Test (GPT)?				
1.17	Is the autoclave validated for maximum and minimum loading				
	pattern?				
1.18	Are the validated load pattern displayed near to the autoclaves?				
1.19	Are all the analysts performing microbiological analysis qualified				
	to perform analysis?				
1.20	Is the environmental monitoring performed every time along with				
	sterility test?				
1.21	Is a separate autoclave used for media decontamination? Are the				
	records maintained?				
1.22	Are the culture media plates used for environmental monitoring				
	pre-incubated prior to use?				
1.23	Is an approved sampling plan available for water sampling?				
1.24	Are environmental isolates are identified and used for Growth				
	Promotion Test? Are records maintained?				
1.25	Are the inventory records maintained for all dehydrated culture				
	media?				
1.26	Are the chemicals and media stored under controlled condition of				
	temperature and humidity?				
1.27	Is the pH meter calibrated on daily basis and records are				
	maintained?				
1.28	Is the analytical balance calibrated on daily basis and records				
	are maintained?				
1.29	Are the standard weights used for calibration of balance certified				
	from NABL?				
1.30	Is the inventory records of LAL reagent maintained?				
1.31	Is all the analysis results signed the analyst performing the				
	analysis and cross checked by another senior person?				
1.32	Is the temperature of Cell Bank maintained and recorded?				

1.33	Is the record of issuance of WCB (Working Cell Bank) maintained?					
	Does it matches with physical quantities?					
1.34	Is the WCB issued after QA approval and in presence of QA					
	personnel?					
1.35	Is the entry to Cell Bank is restricted to authorized personnel?					
2.0	Other Observations, if any (Apart from the checklist).					
2.1						
2.2						
2.3						
2.4						
2.5						
2.6						
2.7						
2.8						

Remarks:

|Auditee's Sign/Date |Auditor's Sign/Date |
| | |

Each point of the checklist shall be filled as applicable and comments or remarks in addition shall be given under the remarks column in the checklist.

Note: Attach additional sheet if required for other observations.

Annexure-6

Date of audit:

Auditor:

Department: Auditee:

Audit report No.:- Audit

report no. assigned by (Sign/Date):

[S. No. |CHECKPOINTS/CRITERIA |Yes |
 ||
 1.0 |QUALITY ASSURANCE DEPARTMENT | || ||
 1.1 |Is the company's quality policy statement available with all the || || ||
 | departments? || || ||
 1.2 |Is an updated organization chart available? || || ||
 1.3 |Are the previously audited records maintained? || || ||
 1.4 |Is the job responsibility is defined and made available to all || || ||
 | personnel? Are responsibilities clearly specified in job || || ||
 | description? || || ||
 1.5 |Are the designee defined in the job responsibility in absence of || || ||
 | concerned person? || || ||
 1.6 |Are the SOPs available for all operation carried out? || || ||
 1.7 |Are all necessary controls on starting materials, intermediate || || ||
 | products and other in-process control, calibrations and validation || || ||
 | carried out in supervision of QA? || || ||
 1.8 |Are finished products correctly processed and checked (In-Process || || ||
 | & Final document check) according to the written procedures? || || ||
 2.0 |VALIDATION || || ||
 2.1 |Is validation properly documented and include Validation Master || || ||
 | Plan? Is an approved validation Master Plan available? Is there || || ||
 | Validation/Calibration planner is available for current year? || || ||
 2.2 |Are validation studies conducted in accordance with pre-defined || || ||
 | protocols (VMP) and validation planner? || || ||
 2.3 |Are the traceability certificates of master instruments used for || || ||
 | validation area verified and archived? Are they cross checked by || || ||
 | QA person for their accuracy and suitability? || || ||
 2.4 |Are the personnel involved in validation trained prior to || || ||
 | execution of validation? || || ||
 2.5 |Validation of New Master Formula: || || ||
Is new master formula or method of preparation adopted and steps						
taken to demonstrate its suitability for routine processing,						
process defined materials and equipment specified?						
2.6	Is the validation report contains a summary and conclusion of					
validation activity?						
2.7	Are pre-qualified external agencies hired for validation					
activities?						
2.8	Are the validation reports cross checked by QA person to verify					
the results?						
3.0	DOCUMENTATION / RECORDS					
3.1	Are documentation meticulously maintained as per rules and					
regularly reviewed and kept up to date?						
3.2	Is documentation maintained in proper segregated area and easily					
available?						
3.3	Is master list of documentation maintained?					
3.4	Is documentation accurate, clear & neat? Does it define					
specifications and procedures for all materials & methods of						
manufacture & control?						
3.5	Are all the specifications, testing, procedures, master formulae,					
packing instructions and standard operating procedures (SOPs)						
available , current & being followed?						
3.6	Do the records provide existence of documented evidence,					
traceability, and an audit trail that will permit investigation?						
3.7	Does the record provide batch processing & packaging details					
including receiving sample, processing equipment, analytical						
testing and laboratory instrument records?						
3.8	Are the completed log books and formats retrieved in QA?					
3.9	Are the obsolete (Not in force) copies of controlled copies of SOP					
and other documents retrieved back in QA? Are the records						
available?						
3.10	Are the retrieved documents archived in document storage room? Is					
the entry to document storage room restricted to authorized						
personnel only?						
3.11	Are the uncontrolled documents issued after QA approval only?					
3.12	Are the documents stored as per defined time period and are					
destroyed after that as per established procedures?						
4.0	STANDARD OPERATING PROCEDURES (SOPs)					
4.1	Is the master list of all SOPs available with QA?					
4.2	Is the list of controlled copies of SOPs issued to concerned					
department maintained?						
4.3	Are all of the controlled copies of superseded SOPs retrieved					
before issuance of next version of SOPs?						
4.4	Is training provided before implementation of new version of any					
SOP?						
4.5	Are the Master copies of superseded version stamped "Not in force"					
and control copies of superseded version destroyed by shredding?						
Is the record maintained?						
4.6	Are all master copies of SOPs from all department is kept with QA					
department?						
 6.0 |VENDOR QUALIFICATION || || ||
 6.1 |Is the list of approved vendors updated? || || ||

6.2	Are the vendors supplying raw material (both active & inactive
	ingredients), primary packaging & printed components audited &
	found to be satisfactory?
7.0	CHANGE CONTROL PROGRAM
7.1	Is the change control procedure properly followed as per
	established Standard Operating Procedure?
7.2	Are all change controls approved by Head Quality Assurance
	department?
7.3	If proper justification given for proposed changes?
7.4	Are approved changes, assessed after implementation, to monitor
	their effectiveness/impact?
7.5	Are change controls closed within defined timelines?
8.0	DEVIATIONS
8.1	Is the deviation procedure properly followed as per established
	Standard Operating Procedure?
8.2	Are all deviations (planned and un-planned), approved by Head
	Quality Assurance department?
8.3	If proper justification given for deviations?
8.4	If investigation performed for all deviations?
8.5	Are approved deviations, assessed after implementation, to monitor
	their effectiveness/impact?
8.6	Is impact analysis performed for all deviations?
9.0	INCIDENTS
9.1	Are all incidents signed and logged by QA?
9.2	Is root cause analysis performed and proper corrective action(s)
	taken?
9.3	Is corrective action taken for all logged incidents?
10.0	ANNUAL PRODUCT REVIEW
10.1	Is the annual product review performed for all the products
	manufacture during the year?
11.2	Are all necessary parameters included as per SOP for Annual
	Product Review?
11.3	Does the APR identify the number of deviations and Change controls
	taken during the period of product review?
11.4	Does the APR identify number of product complaints received during
	the review period for the concerned products?
11.0	AUDIT / COMPLAINTS
11.1	INTERNAL GMP AUDITS
11.2	Is the internal audit/self inspection performed as per pre-defined
	schedule?
11.3	Is the report of self inspection circulated to concerned
	department heads to take necessary corrective actions found in
	internal audit?
11.4	Is there an audit planner available for all departments for the
	current year?
11.5	Does the compliance of all discrepancies found during audits, done
	and necessary CAPA taken as per requirement?
11.6	Are the lead auditor and support auditors qualified as per the
	defined SOP?
12.0	Other Observations, if any (Apart from the checklist).
12.1	
12.2	
12.3	
12.4	

Remarks:

|Auditee's Sign/Date |Auditor's Sign/Date |
 | |

Each point of the checklist shall be filled as applicable and comments or remarks in addition shall be given under the remarks column in the checklist.

Note: Attach additional sheet if required for other observations.

Annexure - 7

Date of audit:
Auditor:

Department:
Auditee:

Audit report No.:-
Audit report no. assigned by:

(Sign/Date)

S. No |DEPARTMENT /SECTIONS CHECKPOINTS/CRITERIA |Yes |
||
1.0 |SUPPORT SYSTEMS | || ||
1.1.1 |Are pre-filters present in heating, ventilation and | || ||
|air-conditioning (HVAC) systems cleaned/replaced on a routine | || ||
|basis? Is the record maintained? | || ||
1.1.2 |Is the log books for operation of HVAC systems maintained? | || ||
1.1.3 |Are the records of temp. / RH maintained? | || ||
1.1.4 |Is the schedule of preventive maintenance of HVAC system | || ||
|available? | || ||
1.1.5 |Is the preventive maintenance of HVAC done as per schedule? Are | || ||
|records maintained? | || ||
1.1.6 |Is the differential pressure of clean room areas recorded? | || ||
1.1.7 |In case of any major modification or change or maintenance work, | || ||
|if all related parameters reassessed and approved before starting | || ||
|production? | || ||
1.1.8 |Are the temperature / RH of returned air monitored and recorded? | || ||
1.2 |COMPRESSED AIR | || ||
1.2.1 |Is compressed air checked for presence of any oil? Are records | || ||
|maintained? | || ||
1.2.2 |Is the air supply filtered through a sterilizing grade air filter? | || ||
1.2.3 |Is humidity controlled? | || ||
1.3 |WATER AND PURE STEAM | || ||
1.3.1 |Is there a schedule available for sanitation of water and pure | || ||
|steam systems? | || ||
1.3.2 |Is the sanitation of water and pure steam system done as per | || ||
|schedule? Are records maintained? | || ||
1.3.3 |Is the area for water system clean and dry? | || ||
1.3.4 |Is there a schematic drawing available | || ||
1.3.5 |Is conductivity, pH and TOC monitored & recorded on routinely | || ||
|basis? | || ||
1.3.6 |Is the water for injection is stored at above 80 0C? | || ||
1.3.7 |Are water storage tanks cleaned / sanitized routinely? | || ||
1.3.8 |Are the vent filters of water storage tanks checked for integrity | || ||
|on routinely basis? | || ||
1.3.9 |Is the U.V. usage records maintained? | || ||
1.3.10 |Are U.V. tubes changed after consumption of usage hours? Are the | || ||
|records maintained? | || ||
1.3.11 |Are the steam traps sanitized and checked routinely for | || ||
|development of any bio-films? | || ||
2.0 |General | || ||
2.1 |Is the list of all equipments available? | || ||
2.2 |Is the schedule of preventive maintenance available? Is the | || ||
|preventive maintenance schedule available for monthly, six monthly | || ||
|& yearly preventive maintenance? | || ||
2.3 |Are the approved drawings available for water system, HVAC system | || ||
|and cleanrooms? | || ||
3.0 |Other Observations, if any (Apart from the checklist). | || ||
3.1 | || || ||
3.2 | || || ||
3.3 | || || ||
3.4 | || || ||
3.5 | || || ||
3.6 | || || ||
3.7 | || || ||

Remarks:

Auditee's Sign/Date |Auditor's Sign/Date |
|||

Each point of the checklist shall be filled as applicable and
comments or remarks in addition shall be given under the remarks column in
the checklist.

Note: Attach additional sheet if required for other
observations.

Date of audit:
Auditor:

Department:
Auditee:

Audit report No.:-
Audit report no. assigned by (Sign/Date):

S.	No	DEPARTMENT /SECTIONS CHECKPOINTS/CRITERIA	Yes	No
1.0		PERSONNEL CHECKLIST		
1.1		Is induction training is given to all employees before their joining into the department?		
1.2		Is medical fitness certificate is taken from all employees before joining?		
2.0		Personal Hygiene		
2.1		Is appropriate required protective apparel available?		
2.2		Are staffs instructed to report health or medical problems that may have an adverse effect on the product?		
2.3		Is there a medical monitoring programme to ensure protection of staff and product?		
		Vaccination where applicable?		
		For all employees?		
3.0		PREMISES CHECKLIST		
3.1		Are the manufacturing premises satisfactory with respect to:		
a)		Neatness and cleanliness?		
b)		State of repair, e.g. Paint work, cracks in floors, ceilings or walls, door seals etc.?		
c)		Exposed piping or electrical wiring?		
d)		Blocking of air ducts?		
e)		Equipment blocking corridors or exists?		
3.2		Are the wash rooms clean and disinfected routinely? Are the records maintained and cross checked daily?		
3.3		Are rodent bait stations, glue pads etc. used at all possible entry of rodents on daily basis?		
3.4		Are insect killers cleaned routinely?		
3.5		Is the pest control done on routinely basis? Are the records maintained?		
3.6		Is the person performing pest control trained to perform the pest control works?		
4.0		Other Observations, if any (Apart from the checklist).		
4.1				
4.2				
4.3				
4.4				
4.5				
4.6				
4.7				
4.8				

Remarks:

Auditee's Sign/Date Auditor's Sign/Date

Each point of the checklist shall be filled as applicable and comments or remarks in addition shall be given under the remarks column in the checklist.

Note: Attach additional sheet if required for other observations.

STANDARD OPERATING PROCEDURE SOP No. : QMS 037
Revision No.: 00

Title: Self Inspection Supersedes : None
Page No. : 1 of 13

1.0 Objective

To lay down the procedure for performing Self Inspections.

2.0 Scope

This Standard Operating Procedure is applicable to all facilities pertaining to vaccine bulk manufacturing and testing including support facilities of Company name.

3.0 Responsibility

3.1 Head, Quality Assurance (QA) / his or her designee shall be responsible for selection of audit team personnel from other departments.

3.2 Designated QA personnel shall prepare the Audit program based on the risk assessment.

3.3 Head, QA shall approve the Audit program.

3.4 Lead Auditor to execute the Audit as per Audit plan with support from support Auditors.

3.5 Lead Auditor to prepare & issue the Audit Report to all concerned departments.

3.6 Respective department HODs shall be responsible of Corrective Action And Preventive Action (CAPA) for audit findings (as applicable).

4.0 Abbreviations and Definitions

4.1 RM Raw Materials

4.2 PM Packing Materials

4.3 FG Finished Goods

4.4 HR Human Resources

4.5 EMP Environmental Monitoring Program

Self – inspection:- Self – inspection is an examination and assessment of all processes and procedures with respect to quality system with specific purpose of evaluating the compliance to established quality systems and cGMP requirements as well as continually improving the quality system.

5.0 Procedure

5.1 General principles:

5.1.1 Self Inspections are performed:

5.1.1.1 To determine conformity with Quality System with respect to Good Manufacturing Practices.

5.1.1.2 To ensure effective and on time corrective actions.

5.1.1.3 To prevent delivery of non-conforming products.

5.1.1.4 To address/check the deficiencies in the audited areas.

5.2 Scope of Audit

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The scope of the audit extends depending upon the criticality of the activity and events of nonconformance, deviations, change controls, product complaints, product rejection and pre & post external audits etc.

5.3 Self Inspections can be categorized into following:

5.3.1 Announced Audit:

Announced audit is planned and scheduled audit which is performed or conducted on regular frequency by giving prior intimation to respective department.

5.3.2 Unannounced Audit:

Unannounced audit is conducted as a surprise audit at any time to check the effective implementation and compliance of the quality system without prior intimation to the department under audit.

5.3.3 Unscheduled Announced Audit:

Unscheduled announced audits are need based audits, planned as the outcome of non-conformance such as complaints, incidents, deviation, external audit observations etc.

5.4 Following is the list of areas (but not limited to) to be inspected as a part of self-inspection:

5.4.1 Manufacturing, (Production, Packing, Process controls).

5.4.2 Warehouse (RM, PM), Sampling & Dispensing Areas and release product(FG)

5.4.3 Quality Control (Chemical and Biological area).

5.4.4 Animal House (Breeding and Experimentation).

5.4.5 Quality Assurance.

5.4.6 Engineering.

5.4.7 HR & Administration.

5.4.8 EOHS (Environment Occupation Health and Safety).

5.5 The general aspects for compliance (but not limited to) to be taken in consideration are:

5.5.1 Personnel and their training status.

5.5.2 Facilities.

5.5.3 Maintenance of buildings and equipment.

5.5.4 Storage of raw materials and final products.

5.5.5 Equipment.

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5.5.6 Manufacturing and process controls.

5.5.7 Quality Control.

5.5.8 Documentation.

5.5.9 Health and Hygiene (Medical).

5.5.10 Validation and Revalidation Programs.

5.5.11 Calibration of Measuring Instruments.

5.5.12 Complaints & Product Recall.

5.5.13 General upkeep of the surrounding open areas.

5.5.14 Label Control.

5.5.15 Pest Control.

5.5.16 Shipping validations.

5.5.17 Changes made like pertaining to Facility/Approved document/ Equipment/ Process/ Validations/EMP.

5.6 Audit Program:

5.6.1 The designated QA personnel shall prepare the audit program in the beginning of the year based upon risk analysis as per Annexure 1 (Form No.). Assign a folio number to the audit program as: XX/001, Where XX is the last two digits of the year, / is a separator and 001 is the first version of the audit program.

Risk analysis be carried out taking into account the below mentioned factors but not limited:

■ Existing requirements for Regulatory registration.

■ Overall compliance status with respect to observations made in External/Internal audit.

■ Consistency in the manufacturing process.

■ Number and significance of the quality defects (e.g. complaints, recalls)

■ Changes with respect to the facility, equipment, manufacturing process and key personnel.

■ Complexity of the process, product.

Each factor shall be assigned a number from 1 to 5 based on the risk as per following parameters :-

Assessment of the risk Risk Number

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Very Slight 1

Slight 2

Minor 3

Moderate 4

Major 5

The numbers assigned to each factor shall be averaged to get the final risk number for each block/department. The block getting the highest risk number shall be audited first and more than twice in the year. All the blocks/Departments shall be audited month by month in descending order of risk number; however all the departments shall be audited at least twice in a year.

Prepare audit program based upon risk analysis as per Annexure – 2. Mention the type of audit and the month during which audit is to be carried. The second page is for writing deviations from the program, if any. Distribute the approved audit plan to all departments head.

To elaborate the Audit Program, consideration to the following aspects is given:-

5.6.1.1 Quality System.

5.6.1.2 Facilities & Equipment System.

5.6.1.3 Materials System.

5.6.1.4 Production System.

5.6.1.5 Packaging & Labeling System.

5.6.1.6 Laboratory Control System.

5.6.2 Audit program shall be executed as per Form No. "Audit Program" (Annexure 2) and record the audit performance as per Form No. "Record of Audit Performance" (Annexure 3).

5.7 The designated QA personnel informs in writing to concerned departments to be audited about the date and scope of the audit at least 15 days before its beginning for Announced Audits. The duration of intimation can be lesser than 15 days in certain cases like sudden announcement of external audit, critical noncompliance's and product complaints etc.

5.8 Audits out of the program may be performed in special cases, as for example; repetitive rejections, any incident, non-compliance with GMP, before & after National Regulatory Authority audits/External audits, etc. to assess the completeness. Such audits shall be

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categorized as unscheduled announced audits.

5.9 Audit Team:

5.9.1 An audit team shall be formed by auditors selected from cross functional departments e.g. Quality Assurance, Quality Control, Production, Warehouse and Maintenance department etc. to perform the audit.

5.9.2 Head, Quality Assurance department or his / her designee shall select qualified and trained lead auditor and support auditors from various departments/sections based on the department to be audited. A minimum of 3 including the lead auditor will be part of the team for each area to be audited.

Lead Auditor shall not be from the department being audited. Selection of audit team shall be based upon their competency, knowledge, expertise and experience in the area to be audited.

5.9.3 A Lead auditor is one who leads the audit as per the scope of the audit and guides the support auditors.

5.9.4 Support auditors are those who are the members of the audit team.

5.10 Independence of Auditors:

5.10.1 Auditors should free from preferences or influences, which might affect the objectivity of audits. In specific, personnel routinely connected with the work in an area cannot participate in audits performed to that area.

5.11 Criteria for selection & Responsibilities of Lead Auditor:

5.11.1 All the Lead auditors shall be qualified based on selection criteria of Lead Auditor as per SOP and any qualified lead auditor may play role as support

auditor under guidance of another qualified lead auditor based on requirement.

5.11.2 Responsibilities of Lead Auditor:

5.11.2.1 Has authority to make final decisions in the management of the audit and audit findings.

5.11.2.2 To plan the audit, prepare work documents and give instructions to the audit team.

5.11.2.3 To represent the audit team before the Head of the audited area, as well as other levels of Company Name.

5.11.2.4 To perform the audit according to the approved plan.

5.11.2.5 To report immediately to the Head of Quality Assurance

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Department in case of any difficulty that might affect the performance of the audit.

5.11.2.6 To guide in preparation and finalization of the final audit report.

5.11.2.7 To keep confidentiality of information.

5.12 Criteria for selection & Responsibilities of support auditors of Audit Team:

5.12.1 All the support auditors shall be qualified based on selection criteria of support auditor as per SOP. Support auditor can be lead auditor only after if he/she qualify the selection criteria of lead auditor as per SOP.

5.12.2 Responsibilities of Support Auditors:

5.12.2.1 To participate in audit planning and in the preparation of work documents.

5.12.2.2 To perform assigned audit tasks in an objective way.

5.12.2.3 To collect and analyze objective evidences that is the base of audit results.

5.12.2.4 To report Leader Auditor of any difficulty found while performing the audit.

5.12.2.5 To verify the effectiveness of the corrective actions taken to solve deficiencies.

5.12.2.6 To keep confidentiality of information.

5.12.2.7 To keep custody of audit documentation.

5.12.2.8 To coordinate in the preparation of the final audit report.

5.13 Responsibilities of Auditee:

5.13.1 To inform the area personnel, about the objectives and scope of the audit.

5.13.2 To designate personnel who will accompany the auditors during the performance of an audit.

5.13.3 To allow access of auditors to facilities, documentation, etc.

5.13.4 To supply information and necessary resources to the auditors for efficient auditing.

5.13.5 To co-operate with auditors so as to obtain audit objectives.

5.13.6 To determine the causes of deficiencies found and to take corrective actions to eliminate them.

5.13.7 To ensure that all area personnel give due respect and support to the auditing

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team.

5.14 Performance of Announced Audit:

Announced audit shall be conducted as per approved Audit program by following three steps as below:

5.14.1 Preparation of the Audit Plan.

5.14.2 Execution of audit as per audit plan.

5.14.3 Work documents preparation during the audit.

5.15 Preparation of the Audit Plan.

5.15.1 The audit plan shall be established in Form No. "Audit Plan" (Annexure 4).

5.15.2 Audit planning typically consists of:

5.15.2.1 Opening discussion:

Opening Discussion shall include:

5.15.2.1.1 Roles of the Auditor.

5.15.2.1.2 Introduction and apportionment of tasks to auditors.

5.15.2.1.3 To define the reason, purpose and scope of the inspection.

5.15.2.1.4 To evaluate the list of documents to be seen and area

to be visited based on risk evaluation e.g. changes

made to facility/documents/process, any critical

observation during the last audit, changes in the key

personnel of the area, any abandoned batches, any

complaint from customer etc.

5.15.2.2 Site Inspection (Key areas to be focused).

5.15.2.3 Documentation to be checked.

5.15.2.4 Concluding Discussion.

5.15.3 Audit plan is prepared by Lead Auditor/designee of Lead auditor, approved by

Head of Quality Assurance and received by Head/designee of the area being audited.

5.15.4 Audit plan shall be circulated to auditee department at least 4 working days before the beginning of the self-inspection.

5.16 Execution of audit as per audit plan

5.16.1 During the site inspection, the auditors shall follow the flow of components

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from receipt, through the corresponding manufacturing steps, to the packaging, to storage and delivery. The quality control areas, critical system areas, e.g. water, steam and air etc. and other auxiliary areas (locks, change rooms etc.) must also be taken into consideration.

5.16.2 The scope of the site inspection is dependent on the objective and type of the inspection and can be adapted during the inspection at any time.

5.16.3 Documentation Check : The documentation check can include but not limited to the following:

5.16.3.1 Organization chart, job descriptions & its execution.

5.16.3.2 History of changes & Change control procedures.

5.16.3.3 Evaluation of personnel health & Hygiene status.

5.16.3.4 Evidence of training planning and execution.

5.16.3.5 Manufacturing instructions and records.

5.16.3.6 Test procedures and records.

5.16.3.7 Evidence of calibration, qualification and validation.

5.16.3.8 Records of maintenance and repair.

5.16.3.9 Records of environmental monitoring.

5.16.3.10 Batch release documentation.

5.16.3.11 Evidence for procurement of starting materials and packaging materials including their specifications.

5.16.3.12 Procedure for deviations from specifications.

5.16.3.13 Handling of OOS.

5.16.3.14 Handling complaints.

5.16.3.15 Records about recalls.

5.16.4 Concluding Discussion.

All auditors involved in the audit shall meet before the concluding discussion to coordinate results.

5.17 Work documents preparation during the audit.

5.17.1 The documents needed for auditing are:

5.17.1.1 Checklists

The checklist shall used during audit in such a way, as not to limit audit activities to the checklist only.

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A general checklist shall be considered for all the departments as per form no. QA/024/F5-10 (Annexure 5).

Department Specific Checklist shall be used along with general checklist for auditing concerned department as established in Form No. QA/024/F6-10 "Audit Checklist for Quality Control" (Annexure 6),

- Form No. 0 "Audit Checklist for Quality Assurance" (Annexure 7),

- Form No. QA/000/00 "Audit Checklist for Engineering" (Annexure 8),

- Form No. QA/000/00 "Audit Checklist for Human Resource and Administration" (Annexure 9),

- Form No. QA/000/00 "Audit Checklist for Production" (Annexure 10),

- Form No. QA/000/00 "Audit Checklist for Warehouse" (Annexure 11),

- Form No. QA/000/00 "Audit Checklist for Animal House" (Annexure 12) and

- Form No. QA/000/00 "Audit Checklist for EOHS" (Annexure 13).

5.18 Final Report of the Audit:

5.18.1 Preparing the final report:

5.18.1.1 The lead auditor along with other auditors shall summarize the observations/Recommendations in wrap-up meeting with all the auditees along with the Heads of the Audited area.

5.18.1.2 While demonstrating the deficiencies, these should be classified as Critical/ Major/ Minor according to their severity as defined:-

5.18.1.2.1 Critical: Non-conformance which have a direct impact on the quality of the product.

5.18.1.2.2 Major: Non-conformance which have an in-direct impact on the quality of the product.

5.18.1.2.3 Minor: Non-conformance which do not have an

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impact on the quality of the product.

5.18.1.2.4 Recommendations: Are the observations suggested for better understanding / clarity of any procedure or process, however does not reflects any non – conformance of the said procedure or process.

5.18.1.3 It is vital in all cases that the auditee shall understand all conclusions and deficiencies submitted.

5.18.1.4 The audit report must state clearly the deficiencies found/ recommendations as per Form No. QA/000/00 "Audit Report" (Annexure 14).

5.18.1.5 The Lead Auditor shall prepare & issue the final audit report to all concerned departments within 10 working days after completion of audit.

5.18.1.6 Numbers of deficiencies observed during an audit shall be clearly stated in the conclusions of the audit report.

5.18.2 Assigning Audit Report Number
5.19.1.5 5.18.2.1 For Announced Audit : WW/XX/YY/ZZ
Where, WW : Shall define the type of Audit i.e.
AN for Announced Audit.
UN for Unannounced Audit.
UA for Unscheduled Announced Audit
'/' : Separator.
XX : Last two digits of the year.
YY : Serial Number of the Audit i.e. 01 for first audit,
02 for second audit and so on.
ZZ : Department Code i.e.
VBI for Vaccine Block I
VB II for Vaccine Block II
VB III for Vaccine Block III
VB IV for Vaccine Block IV.
QC for Quality Control.
QA for Quality Assurance.
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EN for Engineering.
WH for Warehouse.
AH for Animal House.
ES for EOHS.
HR for Human Resource.
AD for Administration department.
5.18.3 The final audit report shall be kept with Audits and Inspection section of Quality Assurance department.
5.18.4 The final audit report (QA/000/00), filled checklist of the audit (QA/000/00 to QA/ QA/000/00) and audit compliance plan (QA/000/00) are confidential document and is not shown to any external personnel or auditing/inspecting agency.
5.19 Performance of Unannounced Audit:
Unannounced audit shall be conducted as per approved schedule for unannounced audit without prior intimation to auditee department. QA shall prepare audit schedule as per "Schedule for unannounced audit" – Annexure 15.
5.19.1 Head QA shall depute any internal Auditor(s) for execution of unannounced audit as per schedule and no audit team formation is required for the same.
5.19.2 The Auditor(s) selection for unannounced audit should not be from auditee department.
5.19.3 The auditor(s) shall plan and perform the audit by focusing any particular area/section of auditee department. During performance of un-announced audits, usage of audit checklists (as mentioned under section 5.17.1.1) is not applicable due to limited scope of the audit.
5.19.5 After Completion of unannounced audit the auditor(s) shall discuss the observations/non-compliances discussed with auditee and send the Audit Report as per Annexure – 14 (Audit Report) to Head of Auditee department and Head of QA within ten working days. In case of delay in issue audit report within ten days, a justification shall be given and audit report shall be issued not later than next ten working days.
5.20 Performance of Unscheduled Announced Audit:
5.20.1 Unscheduled announced audits are need based audits, planned as the outcome
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of non-conformance such as complaints, incidents, deviation, external audit observations etc.
5.20.2 Unscheduled announced audits shall be performed by following the same procedure as described for Announced Audits.
5.21 Corrective Actions:
5.21.1 The concerned department Head/designee of the audited area shall prepare an audit compliance report as per Annexure – 16 and submit it to QA within 10 (ten) working days after receipt of the audit report.
5.21.2 For all audit observations classified as Critical, investigation shall be performed as per SOP QA/000. In case of Major observations investigation shall be performed on case to case basis wherever required. In case of minor observations investigations are not required.
5.21.3 Designated person of QA shall verify the compliance as per Audit compliance report.
5.21.4 The concerned department Head/designee of the audited area shall take necessary corrective and preventive actions as per SOP for CAPA, SOP QA/025.
6.0 Forms and Records
6.1 Risk analysis for audit program - Annexure 1
6.2 Audit program - Annexure 2
6.3 Record of audit performance - Annexure 3
6.4 Audit plan - Annexure 4
6.5 General Audit checklist - Annexure 5
6.6 Audit Checklist for Quality Control - Annexure 6

- 6.7 Audit Checklist for Quality Assurance - Annexure 7
- 6.8 Audit Checklist for Engineering - Annexure 8
- 6.9 Audit Checklist for Human Resource and Administration - Annexure 9
- 6.10 Audit Checklist for Production department - Annexure 10

6.11 Audit Checklist for Warehouse - Annexure 11
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- 6.12 Audit Checklist for Animal House - Annexure 12
- 6.13 Audit Checklist for EOHS - Annexure 13
- 6.14 Audit Report - Annexure 14
- 6.15 Schedule for unannounced audit - Annexure 15

6.16 Audit compliance report - Annexure 16

7.0 References:

- 7.1
- 8.0 Distribution
- 8.1
- 9.0 History
- Date Revision Number Reason for Revision
- Annexure - 1

RISK ANALYSIS FOR AUDIT PROGRAM

Risk analysis for Preparation and Planning of Audit Program

Year Folio Number:

Questionnaire, Block
VB I VBII VB III VB IV QA QC AH WH EN EOHS ADM HR

Existing requirements for Regulatory registration

Overall compliance status with respect to observations made in
External/Internal audit in last one year.
Consistency in the manufacturing process.

Number and significance of the quality defects (e.g. complaints, recalls)

Changes with respect to the facility, equipment, manufacturing process and
key personnel.
Complexity of the process, product

Any other (if any)

Overall Risk and Priority of conducting the audit.

Name Designation Signature Date
Written by:
Checked by:
Approved by:
Page 1 of 1
Annexure-2

AUDIT PROGRAM

Year : Folio No.:
Department / Months
Jan. Feb. Mar. April May June July Aug. Sept. Oct. Nov. Dec.

Name Designation Signature Date

Written by:

Checked by:

Approved by:

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AUDIT PROGRAM

DEVIATION FROM THE SCHEDULED AUDIT PROGRAM
Year : Folio :

Month Deviation number Justification for Deviation Signature

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Annexure-3

RECORD OF AUDIT PERFORMANCE

Type of audit: _____

S. No. Date of Date of Remarks
Audit Report No. Audited Area Compiled By Checked By
Beginning Completion (if any)

Page 1 of 1
AUDIT PLAN Annexure-4

Area to be Audited: Audit No.:
Audit Type: Announced / Unannounced-Audited Dates:
LEAD AUDITOR
Name Designation Department

AUDIT TEAM MEMBERS
Name Designation Department Function (Lead Auditor
/ Support Auditor)

PERSONS RESPONSIBLE OF AREAS (AUDITEE)
Name Designation Department

OPENING DISCUSSION

SITE INSPECTION

Page 1 of 2
AUDIT PLAN

DOCUMENTATION CHECK

CONCLUDING DISCUSSION

Name Designation Signature Date
Prepared by :
(Lead auditor/Designee)
Approved by:
(Head QA/ designee)
Accepted by:
(Concerned HOD/Designee)

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Annexure - 5

GENERAL AUDIT CHECKLIST

Date of audit: Auditor:

Department: Auditee:

Audit report No.:- Audit report no. assigned by (Sign/Date):

Yes No NA REMARKS
S. No. DEPARTMENT /SECTIONS CHECKPOINTS/CRITERIA
(To be filled by Auditor)

1.0 PERSONNEL
1.0 A GENERAL
1 Is there an organizational chart?
2 Are there job descriptions for key personnel?
Are they Appropriate to the activities of the department?
3 Number of staff _
Number sufficient?
Qualifications adequate?
Experience sufficient?
4 Is there a clear separation of job responsibilities of personnel?
1.0 B KEY PERSONNEL
1 Are there sufficient key personnel to supervise assigned
functions?
2 Are they skilled / trained in their respective fields?
1.0 C TRAINING
1 Are there on the job training procedures followed for new
employees?
2 Are training and education records available?
Are they current?
Are they filed with the supervisor?
Is training schedule for current year is available with the
department HOD?
3 Does a GMP training program exist for annual update for all
staff?

Page 1 of 6
GENERAL AUDIT CHECKLIST

Yes No NA REMARKS
S. No. DEPARTMENT /SECTIONS CHECKPOINTS/CRITERIA
(To be filled by Auditor)

4 Is there training in containment procedures?
By written procedures?
Are records maintained?
1.0 D PERSONAL HYGIENE
1 Are appropriate protective apparel required?
Is there a gowning SOP for production staff?

2 Is staff instructed to report health or medical problems that may have an adverse effect on the product?
 3 Is there a medical monitoring program to ensure protection to ensure protection of staff and product?
 Vaccination where applicable?
 For all employees?
 4 Do controlled entry requirements exist?
 Do procedures exist for preventing unauthorized entry?
 Are the procedures in writing?
 2.0 PREMISES
 2.0 A GENERAL
 1 Is there is appropriate cleaning, maintenance and proper operation?
 2 Are areas clearly defined and appropriately controlled:
 a for quarantine and storage of starting materials?
 b for storage of in-process material?
 c for manufacturing and processing operations?
 d for control and laboratory operations?
 e for quarantine and storage of finished products?
 f for holding of rejected material?
 g for ancillary usage, e.g. rest rooms, maintenance workshops?

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 GENERAL AUDIT CHECKLIST

Yes No NA REMARKS
 S. No. DEPARTMENT /SECTIONS CHECKPOINTS/CRITERIA
 (To be filled by Auditor)

h for animal housing?
 3 Does the building design prevent the entry of insects, vermin and other animals?
 4 Do adequate drainage exist (Wherever required)?
 5 Does the design of the facility achieve a unidirectional flow of materials, personnel, product and waste so as to avoid cross-over of clean and dirty (infectious) material?
 6 Is the lighting provided adequate for the conditions necessary for the work being conducted in the area?
 7 Are facility layout drawings including mechanical, electrical and architectural kept up-to-date followings changes? Is revalidation of facilities performed following refurbishment?
 8 Are the premises satisfactory with respect to:
 a neatness and cleanliness?
 b state of repair, e.g. paint work, cracks in floors, ceilings or walls, door seals, etc?
 c exposed piping or electrical wiring?
 3.0 B CLEANING AND MAINTENANCE
 1 Is the equipment(s) suitably located to facilitate its use, cleaning and maintenance?
 2 Are equipments and utensils cleaned, maintained and sanitized as appropriate to prevent malfunction or cross contamination?
 3 Are piping systems, valves and vent filters properly designed to facilitate cleaning and sterilization? NOTE: Maintenance closed systems through the use of "clean in place" and "sterilize in place" is preferable.
 4 Are the valves on primary containment vessels (e.g. fermenters)

Page 3 of 6
 GENERAL AUDIT CHECKLIST

Yes No NA REMARKS
 S. No. DEPARTMENT /SECTIONS CHECKPOINTS/CRITERIA
 (To be filled by Auditor)

steam sterilized?
 5 Are non-fiber releasing filters used for filtration?
 6 Are filters used for sterile filtration integrity tested before after use?
 7 Are calibrations and validation being performed adequately?
 8 Are autoclave and sterilizing ovens fitted with effective, proper air filters and is this integrity tested? Are sterilizers validated?
 9 Are supplies and equipment which are exposed to pathogens during processing kept separate from unused items to prevent cross-contamination?
 3.0 C STANDARD OPERATING PROCEDURES (SOPs) AND RECORDS
 1 Are there written procedures (SOPs) for cleaning and maintenance of equipment and utensils and are they followed?
 2 Do these SOPs include:
 a assignment of responsibility for cleaning?
 b defined schedules for cleaning and maintenance?
 c descriptions of methods, equipment and materials used?
 d inspection of equipment for cleanliness immediately before use?
 e assignment of identification number?
 f documentation in record books?
 3 Are validated and approved cleaning and sanitizing agents used?
 4 Is clean equipment identified as such?
 5 Are calibrations and qualifications properly recorded?
 6 Are all certification and within date?
 7 Are there preventive maintenance programs and consistent record of work performed?

Yes No NA REMARKS

S. No. DEPARTMENT /SECTIONS CHECKPOINTS/CRITERIA
(To be filled by Auditor)

4.0 DOCUMENTATION OF PROCESSING AND
DISTRIBUTION

4.0 A GENERAL

1 Are the records for:

- a All material used?
- b All standard operating procedures?
- c Each lot and/or batch processing and distribution?
- d All complaints and their investigation?
- e All equipment, including cleaning, maintenance and validation?
- f Cleaning, maintenance and environmental control of the premises?

2 Are all records:

- a Up dated?
- b Signed by the person performing the task and for all critical steps by the person checking it?
- c Kept at the work station during the entire operation?
- d Retained and available for inspection at least 2 years after the expiry date of the lot batch?

4.0 B DOCUMENTATION OF EQUIPMENT USED

- 1 Are records on the use, cleaning, sterilization and maintenance of equipment kept in individual logs for each piece of equipment?
- 2 Are these records dated and signed in chronological order?
- 3 Do the records include information of the lot/batch including identification numbers and dates?
- 5 Are standard procedures available for warehousing?
- 6 Are standard procedures available that describe the shipping, final transit conditions and instruction for storage through the

Yes No NA REMARKS

S. No. DEPARTMENT /SECTIONS CHECKPOINTS/CRITERIA
(To be filled by Auditor)

distribution chain, especially the cold chain?

7 Are the shipping methods, especially the cold chain, validated and routinely monitored?

8 Are records detailed and retrievable so that a rapid recall of any particular lot is achievable? Is the recall process delegated to the responsible person?

9 Are records maintained for 2 years after the expiry date?

5.0 C OPERATIONAL PRACTICES AND PROCEDURES

1 Are there standard operating procedures for decontamination of process equipment and facilities? Have these procedures been validated and is the performance monitored?

2 Are standard operating procedures available and displayed outlining emergency procedures in the event of a spill or accidental release of contaminant?

6.0 Any other observation

6.1

6.2

6.3

Remarks: _____

Auditee's Sign/Date Auditor's Sign/Date

Each point of the checklist shall be filled as applicable and comments or remarks in addition shall be given under the remarks column in the checklist.

Note: Attach additional sheet if required for other observations.

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Audit Check List for Quality Control

Annexure-6

Date of audit: Auditor:

Department: Auditee:

Audit report No.:- Audit report no. assigned by (Sign/Date):

Yes No NA REMARKS

S. No CHECKPOINTS/CRITERIA
(To be filled by Auditor)

1.0 QUALITY CONTROL DEPARTMENT

1.1 Does the QC lab have SOPs to cover all the functional areas?

Are adequate facilities and equipment available for the physical, chemical &

1.2

microbiological testing?

Are approved written procedures available for sampling, inspecting and testing raw

1.3

materials, packaging materials, in process and finished products?

Are samples of starting material , packaging material, Intermediate products , bulk

1.4 products and finished products taken by methods and personnel's approved by QC department?

Are in-process materials tested at appropriate phases for identity, strengths quality, 1.5

purity and are they approved or rejected by Quality Control?

1.6 Are records of in-process controls maintained?

Page 1 of 6

Audit Check List for Quality Control

Yes No NA REMARKS

S. No CHECKPOINTS/CRITERIA

(To be filled by Auditor)

Does each batch of drug product prior to release confirms compliance of finished

1.7

product specification?

1.8 Are validated & stability indicating assay methods used?

1.9 Are reference standards used for the assay?

Are accurate, clear & neat records maintained along with the raw data for the entire

1.10

analytical work?

1.11 Are any deviations from these specifications, standard, etc. recorded and justified?

Do Laboratory controls include the establishment of scientifically sound and

appropriate,

- Specifications,

- Standards,

1.12 - Sampling plans,

- Test procedures

- Reference substances,

Are they designed to assure that tested materials conform to appropriate standards of identity, strength, quality and purity?

Do these laboratory controls include:

1.13

Page 2 of 6

Audit Check List for Quality Control

Yes No NA REMARKS

S. No CHECKPOINTS/CRITERIA

(To be filled by Auditor)

Determination of compliance with written specifications for acceptance of each lot

1.13.1

within each shipment of materials or holding of products?

1.13.2 Description of sampling and testing procedure for in-process materials?

1.13.3 Retest policy, identifying the rationale and criteria for retests, number of samples and the documentation required?

1.13.4 A comprehensive calibration program that includes calibration / certification intervals, acceptance criteria and provisions for remedial action?

1.13.5 Are reagents, culture media, etc. properly labeled, preparation recorded in lab books with expiry dates given?

1.13.6 Is appropriate testing done on each batch of product required to be free of objectionable microorganisms?

Are all reference reagents kept secure, properly stored, identified and their integrity

1.14

maintained?

Are the tests results of all references and standard analyzed at appropriate intervals for

1.15

statistical variation from the expected value?

1.16 Is every sterilized lot of media is tested for Growth Promotion Test (GPT)?

1.17 Is the autoclave validated for maximum and minimum loading pattern?

Page 3 of 6

Audit Check List for Quality Control

Yes No NA REMARKS

S. No CHECKPOINTS/CRITERIA

(To be filled by Auditor)

1.18 Are the validated load pattern displayed near to the autoclaves?

Are all the analysts performing microbiological analysis qualified to perform analysis?

1.19

1.20 Is the environmental monitoring performed every time along with sterility test?

1.21 Is a separate autoclave used for media decontamination? Are the records maintained?

Are the culture media plates used for environmental monitoring pre-incubated prior to

1.22

use?

1.23 Is an approved sampling plan available for water sampling?

Are environmental isolates are identified and used for Growth Promotion Test? Are

1.24

records maintained?

1.25 Are the inventory records maintained for all dehydrated culture media?

Are the chemicals and media stored under controlled condition of temperature and

1.26

humidity?

1.27 Is the pH meter calibrated on daily basis and records are maintained?

1.28 Is the analytical balance calibrated on daily basis and records are maintained?

Page 4 of 6

Audit Check List for Quality Control

Yes No NA REMARKS

S. No CHECKPOINTS/CRITERIA

(To be filled by Auditor)

1.29 Are the standard weights used for calibration of balance certified from NABL?

1.30 Is the inventory records of LAL reagent maintained?

Is all the analysis results signed the analyst performing the analysis and cross checked

1.31

by another senior person?

1.32 Is the temperature of Cell Bank maintained and recorded?

Is the record of issuance of WCB (Working Cell Bank) maintained? Does it matches

1.33

with physical quantities?

1.34 Is the WCB issued after QA approval and in presence of QA personnel?

1.35 Is the entry to Cell Bank is restricted to authorized personnel?

2.0 Other Observations, if any (Apart from the checklist).

2.1

2.2

2.3

Page 5 of 6

Audit Check List for Quality Control

Yes No NA REMARKS

S. No CHECKPOINTS/CRITERIA

(To be filled by Auditor)

2.4

2.5

2.6

2.7

2.8

Remarks:

Auditee's Sign/Date Auditor's Sign/Date

Each point of the checklist shall be filled as applicable and comments or remarks in addition shall be given under the remarks column in the checklist.

Note: Attach additional sheet if required for other observations.

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Audit Check List for Quality Assurance

Annexure - 7

Date of audit: Auditor:

Department: Auditee:

Audit report No.:- Audit report no. assigned by (Sign/Date):

Yes No NA REMARKS

S. No. CHECKPOINTS/CRITERIA

(To be filled by Auditor)

1.0 QUALITY ASSURANCE DEPARTMENT

1.1 Is the company's quality policy statement available with all the departments?

1.2 Is an updated organization chart available?

1.3 Are the previously audited records maintained?

Is the job responsibility is defined and made available to all personnel? Are

1.4

responsibilities clearly specified in job description?

1.5 Are the designee defined in the job responsibility in absence of concerned person?

1.6 Are the SOPs available for all operation carried out?

Page 1 of 9

Audit Check List for Quality Assurance

Yes No NA REMARKS

S. No. CHECKPOINTS/CRITERIA

(To be filled by Auditor)

Are all necessary controls on starting materials, intermediate products and other in-

1.7
process control, calibrations and validation carried out in supervision of QA?
Are finished products correctly processed and checked (In-Process & Final document
1.8
check) according to the written procedures?
2.0 VALIDATION
Is validation properly documented and include Validation Master Plan? Is an approved
2.1 validation Master Plan available? Is there Validation/Calibration planner is available
for current year?
Are validation studies conducted in accordance with pre-defined protocols (VMP) and
2.2
validation planner?
Are the traceability certificates of master instruments used for validation area verified
2.3
and archived? Are they cross checked by QA person for their accuracy and suitability?
2.4 Are the personnel involved in validation trained prior to execution of validation?
Validation of New Master Formula:
Is new master formula or method of preparation adopted and steps taken to
2.5
demonstrate its suitability for routine processing, process defined materials and
equipment specified?
2.6 Is the validation report contains a summary and conclusion of validation activity?

Page 2 of 9

Audit Check List for Quality Assurance

Yes No NA REMARKS

S. No. CHECKPOINTS/CRITERIA

(To be filled by Auditor)

2.7 Are pre-qualified external agencies hired for validation activities?
2.8 Are the validation reports cross checked by QA person to verify the results?
3.0 DOCUMENTATION / RECORDS
Are documentation meticulously maintained as per rules and regularly reviewed and
3.1
kept up to date?
3.2 Is documentation maintained in proper segregated area and easily available?
3.3 Is master list of documentation maintained?
Is documentation accurate, clear & neat? Does it define specifications and procedures
3.4
for all materials & methods of manufacture & control?
Are all the specifications, testing, procedures, master formulae, packing instructions
3.5
and standard operating procedures (SOPs) available , current & being followed?
Do the records provide existence of documented evidence, traceability, and an audit
3.6
trial that will permit investigation?
Does the record provide batch processing & packaging details including receiving
3.7
sample, processing equipment, analytical testing and laboratory instrument records?

Page 3 of 9

Audit Check List for Quality Assurance

Yes No NA REMARKS

S. No. CHECKPOINTS/CRITERIA

(To be filled by Auditor)

3.8 Are the completed log books and formats retrieved in QA?

Are the obsolete (Not in force) copies of controlled copies of SOP and other
3.9
documents retrieved back in QA? Are the records available?
Are the retrieved documents archived in document storage room? Is the entry to
3.10
document storage room restricted to authorized personnel only?
3.11 Are the uncontrolled documents issued after QA approval only?
Are the documents stored as per defined time period and are destroyed after that as per
3.12
established procedures?

4.0 STANDARD OPERATING PROCEDURES (SOPs)

4.1 Is the master list of all SOPs available with QA?
4.2 Is the list of controlled copies of SOPs issued to concerned department maintained?
Are all of the controlled copies of superseded SOPs retrieved before issuance of next
4.3
version of SOPs?
4.4 Is training provided before implementation of new version of any SOP?

Page 4 of 9

Audit Check List for Quality Assurance

Yes No NA REMARKS

S. No. CHECKPOINTS/CRITERIA

(To be filled by Auditor)

Are the Master copies of superseded version stamped "Not in force" and control
4.5
copies of superseded version destroyed by shredding? Is the record maintained?
4.6 Are all master copies of SOPs from all department is kept with QA department?
6.0 VENDOR QUALIFICATION
6.1 Is the list of approved vendors updated?
Are the vendors supplying raw material (both active & inactive ingredients), primary
6.2

packaging & printed components audited & found to be satisfactory?
7.0 CHANGE CONTROL PROGRAM
Is the change control procedure properly followed as per established Standard
7.1
Operating Procedure?
7.2 Are all change controls approved by Head Quality Assurance department?
7.3 If proper justification given for proposed changes?
Are approved changes, assessed after implementation, to monitor their
7.4
effectiveness/impact?
7.5 Are change controls closed within defined timelines?
8.0 DEVIATIONS

Page 5 of 9
Audit Check List for Quality Assurance
Yes No NA REMARKS
S. No. CHECKPOINTS/CRITERIA
(To be filled by Auditor)

Is the deviation procedure properly followed as per established Standard Operating
8.1
Procedure?
Are all deviations (planned and un-planned), approved by Head Quality Assurance
8.2
department?
8.3 If proper justification given for deviations?

8.4 If investigation performed for all deviations?
Are approved deviations, assessed after implementation, to monitor their
8.5
effectiveness/impact?
8.6 Is impact analysis performed for all deviations?
9.0 INCIDENTS
9.1 Are all incidents signed and logged by QA?
9.2 Is root cause analysis performed and proper corrective action(s) taken?
9.3 Is corrective action taken for all logged incidents?
10.0 ANNUAL PRODUCT REVIEW
Is the annual product review performed for all the products manufacture during the
10.1
year?

Page 6 of 9
Audit Check List for Quality Assurance
Yes No NA REMARKS
S. No. CHECKPOINTS/CRITERIA
(To be filled by Auditor)

11.2 Are all necessary parameters included as per SOP for Annual Product Review?
Does the APR identify the number of deviations and Change controls taken during the
11.3
period of product review?
Does the APR identify number of product complaints received during the review
11.4
period for the concerned products?

11.0 AUDIT / COMPLAINTS

11.1 INTERNAL GMP AUDITS

11.2 Is the internal audit/self inspection performed as per pre-defined schedule?
Is the report of self inspection circulated to concerned department heads to take
11.3
necessary corrective actions found in internal audit?
11.4 Is there an audit planner available for all departments for the current year?
Does the compliance of all discrepancies found during audits, done and necessary
11.5
CAPA taken as per requirement?
11.6 Are the lead auditor and support auditors qualified as per the defined SOP?

Page 7 of 9
Audit Check List for Quality Assurance
Yes No NA REMARKS
S. No. CHECKPOINTS/CRITERIA
(To be filled by Auditor)

12.0 Other Observations, if any (Apart from the checklist).

12.1

12.2

12.3

12.4

Remarks:

Auditee's Sign/Date Auditor's Sign/Date

Each point of the checklist shall be filled as applicable and comments or remarks in addition shall be given under the remarks column in the checklist.

Note: Attach additional sheet if required for other observations.

Date of audit: Auditor:

Department: Auditee:

Audit report No.:- Audit report no. assigned by:
(Sign/Date)

Yes No NA REMARKS

S. No DEPARTMENT /SECTIONS CHECKPOINTS/CRITERIA
(To be filled by Auditor)

1.0 SUPPORT SYSTEMS

1.2 HVAC SYSTEM

Are pre-filters present in heating, ventilation and air-conditioning (HVAC) systems

1.1.1

cleaned/replaced on a routine basis? Is the record maintained?

1.1.2 Is the log books for operation of HVAC systems maintained?

1.1.3 Are the records of temp. / RH maintained?

1.1.4 Is the schedule of preventive maintenance of HVAC system available?

Is the preventive maintenance of HVAC done as per schedule? Are records

1.1.5

maintained?

Yes No NA REMARKS

S. No DEPARTMENT /SECTIONS CHECKPOINTS/CRITERIA
(To be filled by Auditor)

1.1.6 Is the differential pressure of clean room areas recorded?

In case of any major modification or change or maintenance work, if all related

1.1.7

parameters reassessed and approved before starting production?

1.1.8 Are the temperature / RH of returned air monitored and recorded?

1.2 COMPRESSED AIR

1.2.1 Is compressed air checked for presence of any oil? Are records maintained?

1.2.2 Is the air supply filtered through a sterilizing grade air filter?

1.2.3 Is humidity controlled?

1.3 WATER AND PURE STEAM

1.3.1 Is there a schedule available for sanitation of water and pure steam systems?

Is the sanitation of water and pure steam system done as per schedule? Are records

1.3.2

maintained?

Yes No NA REMARKS

S. No DEPARTMENT /SECTIONS CHECKPOINTS/CRITERIA
(To be filled by Auditor)

1.3.3 Is the area for water system clean and dry?

1.3.4 Is there a schematic drawing available

1.3.5 Is conductivity, pH and TOC monitored & recorded on routinely basis?

1.3.6 Is the water for injection is stored at above 80 0C?

1.3.7 Are water storage tanks cleaned / sanitized routinely?

1.3.8 Are the vent filters of water storage tanks checked for integrity on routinely basis?

1.3.9 Is the U.V. usage records maintained?

Are U.V. tubes changed after consumption of usage hours? Are the records

1.3.10

maintained?

Yes No NA REMARKS

S. No DEPARTMENT /SECTIONS CHECKPOINTS/CRITERIA
(To be filled by Auditor)

1.3.11 Are the steam traps sanitized and checked routinely for development of any bio-films?

2.0 General

2.1 Is the list of all equipments available?

Is the schedule of preventive maintenance available? Is the preventive maintenance

2.2

schedule available for monthly, six monthly & yearly preventive maintenance?

Are the approved drawings available for water system, HVAC system and

2.3

cleanrooms?

3.0 Other Observations, if any (Apart from the checklist).

3.1

3.2

Page 4 of 6

Audit Check List for Engineering

Yes No NA REMARKS

S. No DEPARTMENT /SECTIONS CHECKPOINTS/CRITERIA

(To be filled by Auditor)

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Remarks:

Auditee's Sign/Date Auditor's Sign/Date

Page 5 of 6

Audit Check List for Engineering

Each point of the checklist shall be filled as applicable and comments or remarks in addition shall be given under the remarks column in the checklist.

Note: Attach additional sheet if required for other observations.

Page 6 of 6

Audit Check List for Human Resource and Administration

Annexure-9

Date of audit: Auditor:

Department: Auditee:

Audit report No.: Audit report no. assigned by (Sign/Date):

Yes No NA REMARKS

S. No DEPARTMENT /SECTIONS CHECKPOINTS/CRITERIA

(To be filled by Auditor)

1.0 PERSONNEL CHECKLIST

1.1 Is induction training is given to all employees before their joining into the department?

1.2 Is medical fitness certificate is taken from all employees before joining?

2.0 Personal Hygiene

2.1 Is appropriate required protective apparel available?

Are staffs instructed to report health or medical problems that may have an adverse

2.2

effect on the product?

Is there a medical monitoring programme to ensure protection of staff and product?

2.3 Vaccination where applicable?

For all employees?

3.0 PREMISES CHECKLIST

Page 1 of 4

Audit Check List for Human Resource and Administration

Yes No NA REMARKS

S. No DEPARTMENT /SECTIONS CHECKPOINTS/CRITERIA

(To be filled by Auditor)

3.1 Are the manufacturing premises satisfactory with respect to:

a) Neatness and cleanliness?

b) State of repair, e.g. Paint work, cracks in floors, ceilings or walls, door seals etc.?

c) Exposed piping or electrical wiring?

d) Blocking of air ducts?

e) Equipment blocking corridors or exists?

Are the wash rooms clean and disinfected routinely? Are the records maintained and

3.2

cross checked daily?

Are rodent bait stations, glue pads etc. used at all possible entry of rodents on daily

3.3

basis?

3.4 Are insect killers cleaned routinely?

3.5 Is the pest control done on routinely basis? Are the records maintained?

3.6 Is the person performing pest control trained to perform the pest control works?

Page 2 of 4

Audit Check List for Human Resource and Administration

Yes No NA REMARKS

S. No DEPARTMENT /SECTIONS CHECKPOINTS/CRITERIA

(To be filled by Auditor)

4.0 Other Observations, if any (Apart from the checklist).

4.1

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4.8

Remarks:

Page 3 of 4

Audit Check List for Human Resource and Administration

Auditee's Sign/Date Auditor's Sign/Date

Each point of the checklist shall be filled as applicable and comments or remarks in addition shall be given under the remarks column in the checklist.

Note: Attach additional sheet if required for other observations.

Page 4 of 4

Annexure-10

Audit Check List for Production Department

Date of audit: Auditor:

Department: Auditee:

Audit report No.:- Audit report no. assigned by (Sign/Date):

Yes No NA REMARKS

S. No DEPARTMENTS / SECTION CHECKPOINTS / CRITERIA

(To be filled by Auditor)

1.0 MANUFACTURING AREA

EQUIPMENT, MATERIAL / COMPONENT

1.1 EQUIPMENTS

1.1.1 Does all major equipments bear a unique I.D.?

1.1.2 Does the equipments and the facility have appropriate controls to maintain required parameters e.g. temperature, relative humidity pressure differentials etc.?

1.1.3 Is the equipment cleaned promptly after use?

1.1.4 Is the cleaning status of equipments is defined?

1.1.5 Are the rinse water samples sent for QC analysis?

1.1.6 Equipment /Instrument calibration & preventive Maintenance

Page 1 of 11

Audit Check List for Production Department

Yes No NA REMARKS

S. No DEPARTMENTS / SECTION CHECKPOINTS / CRITERIA

(To be filled by Auditor)

1.1.6.1 Are written records maintained on equipment cleaning, sanitizing and maintenance on or near each piece of equipment?

1.1.6.2 Does the facility have approved written procedure for checking and calibration of each piece of measurement equipment?

1.1.7 MATERIAL /COMPONENT

1.1.7.1 All handling of material and product such as receipt, quarantine, sample storage, tracking, labeling, dispensing, processing, packaging and distribution is done in accordance with approved written procedure, or instruction and recorded?

1.1.7.2 All incoming materials and finished products quarantined immediately after receipt or processing until they released for use or distribution?

1.1.7.3 Are all material handled in such a way to prevent contamination?

1.2 RAW MATERIALS

1.2.1 Does the store premise allow storage of raw materials at various temperatures?

Page 2 of 11

Audit Check List for Production Department

Yes No NA REMARKS

S. No DEPARTMENTS / SECTION CHECKPOINTS / CRITERIA

(To be filled by Auditor)

1.2.2 If answer yes to the above what are controls to log any irregularities?

1.2.3 Are balance used are calibrated?

1.2.4 Are the standard weights used for calibration of balance are calibrated from NABL?

1.2.5 Is any documentation done at the time of receiving the raw material?

1.2.6 What is the process of communication with the laboratory for sampling of the raw material?

1.5 STERILIZER/OVEN LOADING ROOM

1.5.1 Does the sterilizer meet the following criteria:

a. Is it capable of maintaining an empty chamber temperature distribution of +/- 1° C at a steady state?

b. Does the VLT passes regularly?

c. What is the size of vent filter?

d. Is the minimum and maximum load displayed?

Page 3 of 11

Audit Check List for Production Department

Yes No NA REMARKS

S. No DEPARTMENTS / SECTION CHECKPOINTS / CRITERIA

(To be filled by Auditor)

e. Is the load pattern displayed?

f. Is the SOP for operation of sterilizer displayed?

g. Is it equipped with a clean steam?

h. Does the record of every sterilization cycle maintained?

1.5.2 Does the oven meet the following criteria?

a. Are the two interlocking doors working properly?

b. Are the re-circulating fans located upstream of the HEPA filters?

c. Does it have the resistance temperature device in the return air duct for control temperature?

d. Is it equipped with temperature recorder?

1.5.3 Are the sterilizers and ovens calibrated and the results documented?

1.5.4 Are the results of the testing documents?

15.5 Are the garments made of non- shredding material?

Page 4 of 11

Audit Check List for Production Department

Yes No NA REMARKS

S. No DEPARTMENTS / SECTION CHECKPOINTS / CRITERIA

(To be filled by Auditor)

1.5.6 Are the garment washed and sterilized after use?

1.5.7 How frequently they sterilized?

1.5.8 Are biological or chemical indicator used?

1.5.9 Are the sterilizer and oven charts reviewed by IPQA prior to the release of sterile products?

1.5.10 Are the procedures validated for the sterilized/depyrogenated stoppers and components?

1.6 EQUIPMENT AIRLOCK

1.6.1 Is the air lock door interlock system working properly?

1.6.2 Is the air lock cleaning performed as per predefined procedure and frequency?

1.6.3 Is the SOP for transfer of materials available?

1.8 MANUFACTURING OF STERILE PRODUCTS

1.8.1 Does the company have adequate sterilizing, depyrogenation, sterile, filtration, and processing, filling and packaging equipment available?

Page 5 of 11

Audit Check List for Production Department

Yes No NA REMARKS

S. No DEPARTMENTS / SECTION CHECKPOINTS / CRITERIA

(To be filled by Auditor)

1.8.2 Is there an adequate control of viable, non-viable micro-organisms and effective routine monitoring of sterile area and aseptic practices through use of reliable equipment?

1.8.3 Do master/Working Cell Banks and Seed Stocks have detailed records of:

1.8.3.1 history of cells including the number of generation doublings or passages of virus? Is there a maximum limit?

1.8.3.2 characterization according to the WHO TRS relevant to the product?

1.8.3.3 demonstration of purity?

1.8.3.4 manufacturing procedures?

1.8.3.5 appropriate storage and security with continuous monitoring of temperature, alarms and backup power supply?

1.8.3.6 inventory log?

1.8.3.7 adequately segregated storage to avoid mix-up or cross-contamination with other material?

1.8.3.8 routine monitoring of stability (viability/purity)?

Page 6 of 11

Audit Check List for Production Department

Yes No NA REMARKS

S. No DEPARTMENTS / SECTION CHECKPOINTS / CRITERIA

(To be filled by Auditor)

1.8.3.9 Demonstration of identity?

1.8.4 Does the Manufacturing Formulae adequately describe the complete production process?

1.8.5 Is the Manufacturing Formulae up-to-date and approved by QC/QA?

1.8.6 Is the Batch Production Record form an adequate representation of the Manufacturing Formulae?

1.9 ENVIRONMENTAL MONITORING

1.9.1 Are laminar flow hood and HVAC system which serve the sterile operation areas validated?

1.9.2 Is there an environmental monitoring program for the aseptic batching area, sterile filling room, sterile filling line and ancillary areas?

1.9.3 Is non-viable particulate sampling performed?

1.9.4 Are the results of environmental monitoring shared with production department?

1.10 ASEPTIC BATCHING AREA

1.10.1 Are the light fixtures and motors totally sealed and enclosed?

Page 7 of 11

Audit Check List for Production Department

Yes No NA REMARKS

S. No DEPARTMENTS / SECTION CHECKPOINTS / CRITERIA

(To be filled by Auditor)

1.10.2 Are the room air pressures, temperature, humidity and filter pressure drop automatically monitored, alarmed and recorded?

1.10.3 Do personnel sanitize their gloved hands?

1.10.4 Are equipment status labeled?

1.10.5 Are the results documented for the equipments calibrated?

1.10.6 Are sterilization/ clarification filters integrity tested and result documented?

1.10.7 What is the quality of water used for equipment cleaning?

1.10.8 Are tanks, hoses etc. sterilized prior to use and are these procedure user validated?

1.10.9 Are tanks labeled during processing indicating status of product?

1.10.10 Are aseptic operations performed using laminar flow hoods?

1.10.11 Are sampling and testing complete prior to or concurrent with the filling operation?

1.11 FILLING ROOMS

Page 8 of 11

Audit Check List for Production Department

Yes No NA REMARKS

S. No DEPARTMENTS / SECTION CHECKPOINTS / CRITERIA

(To be filled by Auditor)

1.11.1 Is the product contact equipment sterilized by hot air or steam?

Are room air pressures, temperature, humidity and filter pressure drop automatically monitored, alarmed and recorded?

1.11.3 What sanitizing agent is used to sanitize gloved hands? Is it sterilized by filtration?

- 1.11.4 Are aseptic operations performed using laminar flow hoods?
- 1.11.5 Is the sterile filter integrity tested before and/or after use?
- 1.11.6 Is the porosity and type of filter included with the batch records?
- 1.11.7 Are results of filter integrity test included with the batch records?
- 1.11.8 What is the quality of water used for equipment cleaning?
- 1.11.9 Is filling line use record kept?

2.0 FINISHED 'ACTIVE RAW MATERIAL' CONTROL

Do written procedures indicate how and who verifies that correct containers and 2.1 packages are used for finished ARM product during their finishing operation?
Page 9 of 11

Audit Check List for Production Department

Yes No NA REMARKS

S. No DEPARTMENTS / SECTION CHECKPOINTS / CRITERIA

(To be filled by Auditor)

Are written sampling and testing procedures and acceptance criteria available for 2.2 each product to ensure conformance to finished product specification?

3.0 Other Observations (Apart from the checklist)

3.1

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Page 10 of 11

Audit Check List for Production Department

Yes No NA REMARKS

S. No DEPARTMENTS / SECTION CHECKPOINTS / CRITERIA

(To be filled by Auditor)

3.10

3.11

Remarks:

—

Auditee's Sign/Date Auditor's Sign/Date

Each point of the checklist shall be filled as applicable and comments or remarks in addition shall be given under the remarks column in the checklist.

Note: Attach additional sheet if required for other observations.

Page 11 of 11

Audit Check List for Warehouse

Date of audit: Auditor: Annexure - 11

Department: Auditee:

Audit report No.:- Audit report no. assigned by (Sign/Date):

Yes No NA REMARKS

S. No CHECKPOINTS/CRITERIA

(To be filled by Auditor)

1.0 WAREHOUSING / DISTRIBUTION

1.1 Are materials stored under conditions as per the storage instructions on the label?

1.2 Is there adequate space for the orderly storage of material?

1.3 Is there a segregation of quarantined, approved & rejected materials?

Are there separate stores for raw material, packaging material, labels and finished

1.4 products?

Are materials stored on pallets, shelves, or racks, off the floor and off the walls in all

1.5

stores?

1.6 Are minimum and maximum temperatures in the store recorded?

1.7 Are air-conditioned and cold room facilities available?

Page 1 of 4

Audit Check List for Warehouse

Yes No NA REMARKS

S. No CHECKPOINTS/CRITERIA

(To be filled by Auditor)

1.8 Are flammable and hazardous chemicals stored in safe- conditions?

1.9 Are there no expired raw materials in the raw material store(s)?

1.10 Are materials dispensed according to prescribed SOP meeting GMP requirements?

After release by quality control, are raw materials and packaging materials released on

1.11

a FIFO/FEFO (First in First Out / First Expiry First Out) basis?

Do distribution records in the finished goods store enable specific batches to be

1.12

traced?

1.13 Is there updated list of approved vendors available with warehouse personnel?

1.14 Is the shipping validation done for all the products?

Is the list of approved transporters available with warehouse department and finished

1.15

products are dispatched through approved transporters only?

1.16 Is the data loggers sent with all shipments to monitor the temperature during transit?

1.17 Is the data obtained from data logger downloaded and reported to QA?

Page 2 of 4

Audit Check List for Warehouse

Yes No NA REMARKS

S. No CHECKPOINTS/CRITERIA

(To be filled by Auditor)

2.0 Other Observations (Apart from the checklist)

2.1

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Page 3 of 4

Audit Check List for Warehouse

Remarks:

Auditee's Sign/Date Auditor's Sign/Date

Each point of the checklist shall be filled as applicable and comments or remarks in addition shall be given under the remarks column in the checklist.

Note: Attach additional sheet if required for other observations.

Page 4 of 4

Audit Check List for Animal House

Annexure-12

Date of audit: Auditor:

Department: Auditee:

Audit report No.: Audit report no. assigned by (Sign/Date):

Yes No NA REMARKS

S. No. CHECKPOINTS/CRITERIA

(To be filled by Auditor)

1.0 ANIMAL HOUSE

1.1 Is there a list/record of all animals kept in animal house?

1.2 Are animals kept in quarantine before shifting to dedicated rooms?

1.3 Is the temperature/RH of quarantine area maintained?

Are necessary tests for animals performed before shifting to dedicated rooms? Are the
1.4
records maintained?

1.5 Is the animal bedding changed routinely? Are the records maintained?

1.6 Are animals tested prior to use? Are the records maintained?

1.7 Are the animals used for testing separated and identified from other animals?

Page 1 of 4
Audit Check List for Animal House
Yes No NA REMARKS
S. No. CHECKPOINTS/CRITERIA
(To be filled by Auditor)

1.8 Are animal cages autoclaved routinely? Are the records maintained?

1.9 Is animal feed tested in house? Are the records maintained?

1.10 Are the decadent animals kept separated? Are the records maintained?

1.11 Is the autopsy performed of dead animals? Are the records of animal autopsy kept?

1.12 Are animal tissues/organs archived (Wet and Dry)?

1.13 Are the animals checked routinely for health checkups? Are the records maintained?

1.14 Are the dead animals sent for incineration? Are the records maintained?

Is there SOP displayed at the entrance to the animal house? Is the record maintained
1.15
for every person visiting animal house?

1.16 Is there a proper pest control system in place? Are the records maintained?

2.0 Other Observations (Apart from the checklist)
Page 2 of 4
Audit Check List for Animal House
Yes No NA REMARKS
S. No. CHECKPOINTS/CRITERIA
(To be filled by Auditor)

2.1

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2.9

Page 3 of 4
Audit Check List for Animal House
Yes No NA REMARKS
S. No. CHECKPOINTS/CRITERIA
(To be filled by Auditor)

2.10

Remarks:

Auditee's Sign/Date Auditor's Sign/Date

Each point of the checklist shall be filled as applicable and comments or remarks in addition shall be given under the remarks column in the checklist.

Note: Attach additional sheet if required for other observations.

Page 4 of 4
Audit Check List for EOHS
Annexure-13
Date of audit: Auditor:

Department: Auditee:

Audit report No.:- Audit report no. assigned by (Sign/Date):

Yes No NA REMARKS
S. No. CHECKPOINTS/CRITERIA
(To be filled by Auditor)

1.0 Environment and Health

1.1 Are the bio-hazard products generated in plant identified?

1.2 Are all bio-hazard materials destroyed by incineration? Are records maintained?
Is the Effluent Treatment Plant checked regularly and required chemical dose given as
1.3 per requirement?

1.4 Is the BOD & COD of the treated water determined routinely? Are the records
maintained and values displayed on display board?

1.5 Is the non hazardous solid waste handed over to approve waste management
personnel?

1.6 Is the preventive maintenance schedule for ETP available? Does the Preventive
Maintenance performed as per the Schedule?

2.0 Safety

2.1 Is the list of fire hydrant available with EOHS department?

Page 1 of 4
Audit Check List for EOHS
Yes No NA REMARKS
S. No. CHECKPOINTS/CRITERIA
(To be filled by Auditor)

Is there is list of approved personnel available with contact details for handling of
2.2 emergency situations?
Is the personnel authorized for handling of emergency situations trained to handle the
2.3 situation?

2.4 Is the emergency exit checked routinely in all the departments for any obstacle? Are
the records maintained?

2.5 Is the contact number in case of emergency displayed in all critical areas including
areas with maximum personnel?

3.0 First Aid

3.1 Is there a list of approved personnel for giving first aid?

3.2 Are the personnel authorized for giving first aid trained for giving first aid?

3.3 Is there the list of items kept within first aid box?

Are the medicines kept within first aid box checked routinely for expiry date / other
3.4 physical observations?

Page 2 of 4
Audit Check List for EOHS
Yes No NA REMARKS
S. No. CHECKPOINTS/CRITERIA
(To be filled by Auditor)

Does all of the medicines kept in the first aid box are OTC category and not the drugs
3.5 as per Schedule H?

4.0 Other Observations (Apart from the checklist)

4.1

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4.6

4.7

Page 3 of 4
Audit Check List for EOHS
Remarks:

Auditee's Sign/Date Auditor's Sign/Date

Each point of the checklist shall be filled as applicable and comments or remarks in addition shall be given under the remarks column in the checklist.

Note: Attach additional sheet if required for other observations.

Page 4 of 4
Annexure-14

AUDIT REPORT

Audited Area: Reporting Date:
Audit Report No. Audit beginning date : Audit ending date:
Responsible person in audited area
Name Designation

Objective and Scope:

Reference Documents:

Distribution of report
Name Designation

Total Deficiencies :
Critical Deficiencies :
Major Deficiencies :
Minor Deficiencies :
Conclusion:

It is requested to _____ that, taking to consideration the deficiencies found, determine the necessary actions to be taken and submit the audit compliance report (SOP No.:- QA/024) to Quality Assurance Department on or before _____.
NAME OF AUDITORS SIGNATURE / DATE

Page 1 of 2
AUDIT REPORT

Audited Area: Reporting Date:
Audit Report No. Audit beginning date : Audit ending date:
S. No. Deficiency Classification

NAME OF AUDITORS SIGNATURE / DATE

Page 2 of 2
Annexure-15

SCHEDULE FOR UNANNOUNCED AUDIT

Year : Folio No.:
Department / Area
Jan. Feb. Mar. April May June July Aug. Sept. Oct. Nov. Dec.

Name Designation Signature Date

Written by:

Checked by:

Approved by:

Page 1 of 2
SCHEDULE FOR UNANNOUNCED AUDIT

DEVIATION FROM THE SCHEDULED AUDIT PROGRAM
Year : Folio :

Month Deviation number Justification for Deviation Signature

Page 2 of 2
Annexure-16

AUDIT COMPLIANCE REPORT

Department: Area:
Reference to audit report No.: Date of audit report:

To be filled by Auditee Department To be filled by QA

Sr. No. of
Immediate action (if any) and action Verification and Verified by
Audit QA Comments
taken on closure (Sign/Date)
observation

Prepared by/date: Checked by/date:
(Head/Designee auditee department) (QA)

Remarks:- _____

Authorized by/date:-

1.0	Objective		
	To lay down the procedure for performing Self Inspections.		
2.0	Scope		
	This Standard Operating Procedure is applicable to all facilities		
	pertaining to vaccine bulk manufacturing and testing including support		
	facilities of Company name.		
3.0	Responsibility		
	3.1	Head, Quality Assurance (QA) / his or her designee shall be	
	responsible for selection of audit team personnel from other		
	departments.		
	3.2	Designated QA personnel shall prepare the Audit program based on the	
	risk assessment.		
	3.3	Head, QA shall approve the Audit program.	
	3.4	Lead Auditor to execute the Audit as per Audit plan with support	
	from support Auditors.		
	3.5	Lead Auditor to prepare & issue the Audit Report to all concerned	
	departments.		
	3.6	Respective department HODs shall be responsible of Corrective Action	
	And Preventive Action (CAPA) for audit findings (as applicable).		
4.0	Abbreviations and Definitions		
	4.1	RM	Raw Materials
	4.2	PM	Packing Materials
	4.3	FG	Finished Goods
	4.4	HR	Human Resources
	4.5	EMP	Environmental Monitoring Program
Self –	Self – inspection is an examination and assessment of all		
inspection:-	processes and procedures with respect to quality system with		
	specific purpose of evaluating the compliance to established		
	quality systems and cGMP requirements as well as continually		
	improving the quality system.		
5.0	Procedure		
	5.1	General principles:	
	5.1.1	Self Inspections are performed:	
		5.1.1.1	To determine conformity with Quality System with
			respect to Good Manufacturing Practices.
		5.1.1.2	To ensure effective and on time corrective actions.
		5.1.1.3	To prevent delivery of non-conforming products.
		5.1.1.4	To address/check the deficiencies in the audited
		areas.	
	5.2	Scope of Audit	
	The scope of the audit extends depending upon the criticality of the		
	activity and events of nonconformance, deviations, change controls,		
	product complaints, product rejection and pre & post external audits		
	etc.		
	5.3	Self Inspections can be categorized into following:	
	5.3.1	Announced Audit:	
		Announced audit is planned and scheduled audit which is	
		performed or conducted on regular frequency by giving prior	
		intimation to respective department.	
	5.3.2	Unannounced Audit:	
		Unannounced audit is conducted as a surprise audit at any	
		time to check the effective implementation and compliance of	
		the quality system without prior intimation to the	
		department under audit.	
	5.3.3	Unscheduled Announced Audit:	
		Unscheduled announced audits are need based audits, planned	
		as the outcome of non-conformance such as complaints,	
		incidents, deviation, external audit observations etc.	
	5.4	Following is the list of areas (but not limited to) to be inspected	
	as a part of self-inspection:		
	5.4.1	Manufacturing, (Production, Packing, Process controls).	
	5.4.2	Warehouse (RM, PM), Sampling & Dispensing Areas and release	
		product(FG)	
	5.4.3	Quality Control (Chemical and Biological area).	
	5.4.4	Animal House (Breeding and Experimentation).	
	5.4.5	Quality Assurance.	
	5.4.6	Engineering.	
	5.4.7	HR & Administration.	
	5.4.8	EOHS (Environment Occupation Health and Safety).	
	5.5	The general aspects for compliance (but not limited to) to be taken	
	in consideration are:		
	5.5.1	Personnel and their training status.	
	5.5.2	Facilities.	
	5.5.3	Maintenance of buildings and equipment.	
	5.5.4	Storage of raw materials and final products.	
	5.5.5	Equipment.	
	5.5.6	Manufacturing and process controls.	
	5.5.7	Quality Control.	
	5.5.8	Documentation.	
	5.5.9	Health and Hygiene (Medical).	
	5.5.10	Validation and Revalidation Programs.	
	5.5.11	Calibration of Measuring Instruments.	
	5.5.12	Complaints & Product Recall.	
	5.5.13	General upkeep of the surrounding open areas.	
	5.5.14	Label Control.	
	5.5.15	Pest Control.	
	5.5.16	Shipping validations.	
	5.5.17	Changes made like pertaining to Facility/Approved document/	
		Equipment/ Process/ Validations/EMP.	

	5.6	Audit Program:	
	5.6.1	The designated QA personnel shall prepare the audit program	
		in the beginning of the year based upon risk analysis as per	
		Annexure 1 (Form No.). Assign a folio number to the audit	
		program as: XX/001,	
		Where XX is the last two digits of the year, / is a	
		separator and 001 is the first version of the audit program.	
		Risk analysis be carried out taking into account the below	
		mentioned factors but not limited:	
		Existing requirements for Regulatory registration.	
		Overall compliance status with respect to observations made	
		in External/Internal audit.	
		Consistency in the manufacturing process.	
		Number and significance of the quality defects (e.g.	
		complaints, recalls)	
		Changes with respect to the facility, equipment,	
		manufacturing process and key personnel.	
		Complexity of the process, product.	
		Each factor shall be assigned a number from 1 to 5 based on	
		the risk as per following parameters :-	
		Assessment of the risk	
		Risk Number	
		Very Slight	
		1	
		Slight	
		2	
		Minor	
		3	
		Moderate	
		4	
		Major	
		5	
		The numbers assigned to each factor shall be averaged to get	
		the final risk number for each block/department. The block	
		getting the highest risk number shall be audited first and	
		more than twice in the year. All the blocks/Departments	
		shall be audited month by month in descending order of risk	
		number; however all the departments shall be audited at	
		least twice in a year.	
		Prepare audit program based upon risk analysis as per	
		Annexure – 2. Mention the type of audit and the month during	
		which audit is to be carried. The second page is for writing	
		deviations from the program, if any. Distribute the approved	
		audit plan to all departments head.	
		To elaborate the Audit Program, consideration to the	
		following aspects is given:-	
		5.6.1.1	Quality System.
		5.6.1.2	Facilities & Equipment System.
		5.6.1.3	Materials System.
		5.6.1.4	Production System.
		5.6.1.5	Packaging & Labeling System.
		5.6.1.6	Laboratory Control System.
		5.6.2	Audit program shall be executed as per Form No. "Audit
		Program" (Annexure 2) and record the audit performance as	
		per Form No. "Record of Audit Performance" (Annexure 3).	
	5.7	The designated QA personnel informs in writing to concerned	
		departments to be audited about the date and scope of the audit at	
		least 15 days before its beginning for Announced Audits. The	
		duration of intimation can be lesser than 15 days in certain cases	
		like sudden announcement of external audit, critical noncompliance's	
		and product complaints etc.	
	5.8	Audits out of the program may be performed in special cases, as for	
		example; repetitive rejections, any incident, non-compliance with	
		GMP, before & after National Regulatory Authority audits/External	
		audits, etc. to assess the completeness. Such audits shall be	
		categorized as unscheduled announced audits.	
	5.9	Audit Team:	
	5.9.1	An audit team shall be formed by auditors selected from	
		cross functional departments e.g. Quality Assurance, Quality	
		Control, Production, Warehouse and Maintenance department	
		etc. to perform the audit.	
	5.9.2	Head, Quality Assurance department or his / her designee	
		shall select qualified and trained lead auditor and support	
		auditors from various departments/sections based on the	
		department to be audited. A minimum of 3 including the lead	
		auditor will be part of the team for each area to be	
		audited. Lead Auditor shall not be from the department being	
		audited. Selection of audit team shall be based upon their	
		competency, knowledge, expertise and experience in the area	
		to be audited.	
	5.9.3	A Lead auditor is one who leads the audit as per the scope	
		of the audit and guides the support auditors.	
	5.9.4	Support auditors are those who are the members of the audit	
		team.	
	5.10	Independence of Auditors:	
	5.10.1	Auditors should free from preferences or influences, which	

			might affect the objectivity of audits. In specific,	
			personnel routinely connected with the work in an area	
			cannot participate in audits performed to that area.	
		5.11	Criteria for selection & Responsibilities of Lead Auditor:	
		5.11.1	All the Lead auditors shall be qualified based on selection	
		criteria of Lead Auditor as per SOP and any qualified lead		
		auditor may play role as support auditor under guidance of		
		another qualified lead auditor based on requirement.		
		5.11.2	Responsibilities of Lead Auditor:	
		5.11.2.1	Has authority to make final decisions in the	
		management of the audit and audit findings.		
		5.11.2.2	To plan the audit, prepare work documents and give	
		instructions to the audit team.		
		5.11.2.3	To represent the audit team before the Head of the	
		audited area, as well as other levels of Company		
		Name.		
		5.11.2.4	To perform the audit according to the approved	
		plan.		
		5.11.2.5	To report immediately to the Head of Quality	
		Assurance Department in case of any difficulty that		
		might affect the performance of the audit.		
		5.11.2.6	To guide in preparation and finalization of the	
		final audit report.		
		5.11.2.7	To keep confidentiality of information.	
		5.12	Criteria for selection & Responsibilities of support auditors of	
		Audit Team:		
		5.12.1	All the support auditors shall be qualified based on	
		selection criteria of support auditor as per SOP. Support		
		auditor can be lead auditor only after if he/she qualify the		
		selection criteria of lead auditor as per SOP.		
		5.12.2	Responsibilities of Support Auditors:	
		5.12.2.1	To participate in audit planning and in the	
		preparation of work documents.		
		5.12.2.2	To perform assigned audit tasks in an objective	
		way.		
		5.12.2.3	To collect and analyze objective evidences that is	
		the base of audit results.		
		5.12.2.4	To report Leader Auditor of any difficulty found	
		while performing the audit.		
		5.12.2.5	To verify the effectiveness of the corrective	
		actions taken to solve deficiencies.		
		5.12.2.6	To keep confidentiality of information.	
		5.12.2.7	To keep custody of audit documentation.	
		5.12.2.8	To coordinate in the preparation of the final audit	
		report.		
		5.13	Responsibilities of Auditee:	
		5.13.1	To inform the area personnel, about the objectives and scope	
		of the audit.		
		5.13.2	To designate personnel who will accompany the auditors	
		during the performance of an audit.		
		5.13.3	To allow access of auditors to facilities, documentation,	
		etc.		
		5.13.4	To supply information and necessary resources to the	
		auditors for efficient auditing.		
		5.13.5	To co-operate with auditors so as to obtain audit	
		objectives.		
		5.13.6	To determine the causes of deficiencies found and to take	
		corrective actions to eliminate them.		
		5.13.7	To ensure that all area personnel give due respect and	
		support to the auditing team.		
		5.14	Performance of Announced Audit:	
		Announced audit shall be conducted as per approved Audit program by		
		following three steps as below:		
		5.14.1	Preparation of the Audit Plan.	
		5.14.2	Execution of audit as per audit plan.	
		5.14.3	Work documents preparation during the audit.	
		5.15	Preparation of the Audit Plan.	
		5.15.1	The audit plan shall be established in Form No. "Audit Plan"	
		(Annexure 4).		
		5.15.2	Audit planning typically consists of:	
		5.15.2.1	Opening discussion:	
			Opening Discussion shall include:	
			5.15.2.1.	Roles of the Auditor.
			1	
			5.15.2.1.	Introduction and apportionment of tasks
			2	to auditors.
			5.15.2.1.	To define the reason, purpose and scope
			3	of the inspection.
			5.15.2.1.	To evaluate the list of documents to be
			4	seen and area to be visited based on risk
			evaluation e.g. changes made to	
			facility/documents/process, any critical	
			observation during the last audit,	
			changes in the key personnel of the area,	
			any abandoned batches, any complaint from	
			customer etc.	
			5.15.2.2	Site Inspection (Key areas to be focused).
			5.15.2.3	Documentation to be checked.
			5.15.2.4	Concluding Discussion.
		5.15.3	Audit plan is prepared by Lead Auditor/designee of Lead	
		auditor, approved by Head of Quality Assurance and received		
		by Head/designee of the area being audited.		

	5.15.4	Audit plan shall be circulated to auditee department at	
		least 4 working days before the beginning of the	
		self-inspection.	
	5.16	Execution of audit as per audit plan	
	5.16.1	During the site inspection, the auditors shall follow the	
		flow of components from receipt, through the corresponding	
		manufacturing steps, to the packaging, to storage and	
		delivery. The quality control areas, critical system areas,	
		e.g. water, steam and air etc. and other auxiliary areas	
		(locks, change rooms etc.) must also be taken into	
		consideration.	
	5.16.2	The scope of the site inspection is dependent on the	
		objective and type of the inspection and can be adapted	
		during the inspection at any time.	
	5.16.3	Documentation Check : The documentation check can include	
		but not limited to the following:	
		5.16.3.1	Organization chart, job descriptions & its
		execution.	
		5.16.3.2	History of changes & Change control procedures.
		5.16.3.3	Evaluation of personnel health & Hygiene status.
		5.16.3.4	Evidence of training planning and execution.
		5.16.3.5	Manufacturing instructions and records.
		5.16.3.6	Test procedures and records.
		5.16.3.7	Evidence of calibration, qualification and
		validation.	
		5.16.3.8	Records of maintenance and repair.
		5.16.3.9	Records of environmental monitoring.
		5.16.3.1	Batch release documentation.
		0	
		5.16.3.1	Evidence for procurement of starting materials and
		1	packaging materials including their specifications.
		5.16.3.1	Procedure for deviations from specifications.
		2	
		5.16.3.1	Handling of OOS.
		3	
		5.16.3.1	Handling complaints.
		4	
		5.16.3.1	Records about recalls.
		5	
	5.16.4	Concluding Discussion.	
		All auditors involved in the audit shall meet before the	
		concluding discussion to coordinate results.	
	5.17	Work documents preparation during the audit.	
	5.17.1	The documents needed for auditing are:	
		5.17.1.1	Checklists
		The checklist shall used during audit in such a	
		way, as not to limit audit activities to the	
		checklist only.	
		A general checklist shall be considered for all the	
		departments as per form no. QA/024/F5-10 (Annexure	
		5).	
		Department Specific Checklist shall be used along	
		with general checklist for auditing concerned	
		department as established in Form No. QA/024/F6-10	
		"Audit Checklist for Quality Control" (Annexure 6),	
		- Form No. 0 "Audit Checklist for Quality	
		Assurance" (Annexure 7),	
		- Form No.QA/000/00"Audit Checklist for	
		Engineering" (Annexure 8),	
		- Form No. QA/000/00 "Audit Checklist for Human	
		Resource and Administration" (Annexure 9),	
		- Form No. QA/000/00 "Audit Checklist for	
		Production" (Annexure 10),	
		- Form No. QA/000/00 "Audit Checklist for	
		Warehouse" (Annexure 11),	
		- Form No. QA/000/00 "Audit Checklist for Animal	
		House" (Annexure 12) and	
		- Form No. QA/000/00 "Audit Checklist for EOHS"	
		(Annexure 13).	
	5.18	Final Report of the Audit:	
	5.18.1	Preparing the final report:	
	5.18.1.1	The lead auditor along with other auditors shall	
		summarize the observations/Recommendations in	
		wrap-up meeting with all the auditees along with	
		the Heads of the Audited area.	
	5.18.1.2	While demonstrating the deficiencies, theses should	
		be classified as Critical/ Major/ Minor according	
		to their severity as defined:-	
	5.18.1.2.	Critical: Non-conformance which have a	
		1	direct impact on the quality of the
		product.	
	5.18.1.2.	Major: Non-conformance which have an	
		2	in-direct impact on the quality of the
		product.	
	5.18.1.2.	Minor: Non-conformance which do not have	
		3	an impact on the quality of the product.
	5.18.1.2.	Recommendations: Are the observations	
		4	suggested for better understanding /
		clarity of any procedure or process,	
		however does not reflects any non –	
		conformance of the said procedure or	
		process.	

		5.18.1.3	It is vital in all cases that the auditee shall
			understand all conclusions and deficiencies
			submitted.
		5.18.1.4	The audit report must state clearly the
			deficiencies found/ recommendations as per Form No.
			QA/000/00 "Audit Report" (Annexure 14).
		5.18.1.5	The Lead Auditor shall prepare & issue the final
			audit report to all concerned departments within 10
			working days after completion of audit.
		5.18.1.6	Numbers of deficiencies observed during an audit
			shall be clearly stated in the conclusions of the
			audit report.
	5.18.2	Assigning Audit Report Number	
		5.19.1.5 5.18.2.1	
			For Announced Audit : WW/XX/YY/ZZ
			Where, WW : Shall define the type of Audit i.e.
			AN for Announced Audit.
			UN for Unannounced Audit.
			UA for Unscheduled Announced Audit
			"/" : Separator.
			XX : Last two digits of the year.
			YY : Serial Number of the Audit i.e. 01 for first audit,
			02 for second audit and so on.
			ZZ : Department Code i.e.
			VBI for Vaccine Block I
			VB II for Vaccine Block II
			VB III for Vaccine Block III
			VB IV for Vaccine Block IV.
			QC for Quality Control.
			QA for Quality Assurance.
			EN for Engineering.
			WH for Warehouse.
			AH for Animal House.
			ES for EOHS.
			HR for Human Resource.
			AD for Administration department.
	5.18.3	The final audit report shall be kept with Audits and	
		Inspection section of Quality Assurance department.	
	5.18.4	The final audit report (QA/000/00), filled checklist of the	
		audit (QA/000/00 to QA/ QA/000/00) and audit compliance plan	
		(QA/000/00) are confidential document and is not shown to	
		any external personnel or auditing/inspecting agency.	
	5.19	Performance of Unannounced Audit:	
		Unannounced audit shall be conducted as per approved schedule for	
		unannounced audit without prior intimation to auditee department. QA	
		shall prepare audit schedule as per "Schedule for unannounced audit"	
		– Annexure 15.	
	5.19.1	Head QA shall depute any internal Auditor(s) for execution	
		of unannounced audit as per schedule and no audit team	
		information is required for the same.	
	5.19.2	The Auditor(s) selection for unannounced audit should not be	
		from auditee department.	
	5.19.3	The auditor(s) shall plan and perform the audit by focusing	
		any particular area/section of auditee department. During	
		performance of un-announced audits, usage of audit	
		checklists (as mentioned under section 5.17.1.1) is not	
		applicable due to limited scope of the audit.	
	5.19.5	After Completion of unannounced audit the auditor(s) shall	
		discuss the observations/non-compliances discussed with	
		auditee and send the Audit Report as per Annexure – 14	
		(Audit Report) to Head of Auditee department and Head of QA	
		within ten working days. In case of delay in issue audit	
		report within ten days, a justification shall be given and	
		audit report shall be issued not later than next ten working	
		days.	
	5.20	Performance of Unscheduled Announced Audit:	
	5.20.1	Unscheduled announced audits are need based audits, planned	
		as the outcome of non-conformance such as complaints,	
		incidents, deviation, external audit observations etc.	
	5.20.2	Unscheduled announced audits shall be performed by following	
		the same procedure as described for Announced Audits.	
	5.21	Corrective Actions:	
	5.21.1	The concerned department Head/designee of the audited area	
		shall prepare an audit compliance report as per Annexure –	
		16 and submit it to QA within 10 (ten) working days after	
		receipt of the audit report.	
	5.21.2	For all audit observations classified as Critical,	
		investigation shall be performed as per SOP QA/000. In case	
		of Major observations investigation shall be performed on	
		case to case basis wherever required. In case of minor	
		observations investigations are not required.	
	5.21.3	Designated person of QA shall verify the compliance as per	
		Audit compliance report.	
	5.21.4	The concerned department Head/designee of the audited area	
		shall take necessary corrective and preventive actions as	
		per SOP for CAPA, SOP QA/025.	
	6.0	Forms and Records	
	6.1	Risk analysis for audit program - Annexure 1	
	6.2	Audit program - Annexure 2	

	6.3		Record of audit performance - Annexure 3	
	6.4		Audit plan - Annexure 4	
	6.5		General Audit checklist - Annexure 5	
	6.6		Audit Checklist for Quality Control - Annexure 6	
	6.7		Audit Checklist for Quality Assurance - Annexure 7	
	6.8		Audit Checklist for Engineering - Annexure 8	
	6.9		Audit Checklist for Human Resource and Administration - Annexure 9	
	6.10		Audit Checklist for Production department - Annexure 10	
	6.11		Audit Checklist for Warehouse - Annexure 11	
	6.12		Audit Checklist for Animal House - Annexure 12	
	6.13		Audit Checklist for EOHS - Annexure 13	
	6.14		Audit Report - Annexure 14	
	6.15		Schedule for unannounced audit - Annexure 15	
	6.16		Audit compliance report - Annexure 16	
	7.0		References:	
	7.1			
	8.0		Distribution	
	8.1			
	9.0		History	
	Date		Revision Number	
			Reason for Revision	

	Prepared By	Reviewed By	Approved by
Name:			
		Dept. Head	QA Head
Sign:			
Date:			
Effective Date:	Next Review Date:		

1.0	Objective	
	To lay down procedure for waste disposal management.	
2.0	Scope	
	This Standard Operating Procedure is applicable to Handling and Disposal	
	waste in PanEra Biotec Pvt Ltd.	
3.0	Responsibility	
	3.1	Head of the concern department or their designees shall be responsible
	for implementation of this SOP.	
	3.2	EOHS is responsible for disposal of waste whereas Admin is responsible
	for Handling and storage of waste.	
	3.3	Concerned department personnel shall be responsible for collection and
	transportation of waste at generation site.	
	3.4	Concerned department personnel shall be responsible for ensuring that
	the labels of empty containers are defaced before leaving their	
	department.	
	3.5	Concerned department personnel shall be responsible for collecting and
	transferring the waste to their earmarked storage location by Admin.	
	3.6	Administration department shall ensure the safe storage of waste in
	their earmarked storage location prior to its disposal to authorized	
	external party.	
	3.7	It is the responsibility of the Head of the production Department to
	bring to the notice of the Director (Operations & Projects) in case of	
	rejection of non-conforming intermediate product. He shall also bring	
	the same to the notice of Head QC, QA and EOHS Executive.	
	3.8	The Head, QC or his/her designee shall be responsible to bring the
	non-conformity to the notice of the Director (Operation & Project),	
	Head Production Department (if applicable) and Head QA & EOHS	
	Executive.	

4.0	Abbreviations and Definitions	
	ETP	: Effluent Treatment Plant.
	TERM	: Transport Emergency.
	SPCB	: State Pollution Control Board.
	PPE	: Personal Protective Equipment.
	EOHS	: Environment Occupational Health & Safety.
	Admin	: Administration
	HW	: Hazardous Waste.
	PCC	: Plain Cement Concrete
	Waste:	Any garbage, refuse, reject, sludge from a waste
	treatment plant, water supply treatment plant, or air	
	pollution control facility and other discarded material,	
	including solid, liquid, semisolid, or contained gaseous	
	material resulting from industrial, commercial, mining,	
	and agricultural operations, and from community	
	activities.	
	Hazardous waste:	Solid waste, or combination of solid wastes, which
	because of its quantity, concentration, or physical,	
	chemical, or infectious characteristics may - (A) cause,	
	or significantly contribute to an increase in mortality	
	or an increase in serious irreversible, or	
	incapacitating reversible illness or (B) pose a	
	substantial present or potential hazard to human health	
	or the environment when improperly treated, stored,	
	transported, or disposed of, or otherwise managed.	
	Biomedical waste:	Biomedical waste means any solid waste which is
	generated in the diagnosis, treatment, or immunization	
	of human beings or animals, in research pertaining	
	thereto, or in the production or testing of biological.	
	E-waste:	E-waste means waste electrical and electrical equipment,
	whole or in part or rejected from their manufacturing	
	and repair process, which in intended to be discarded.	

5.0	Procedure:	
	5.1	Various categories and type of waste procedure in our facilities.
	5.1.1	Categorization of the Hazardous waste as per Schedule :
		Sr. No
		Category of waste
		Type of Waste
		Schedule –1
		(HW Rule 2008)
		1.
		Hazardous Waste (Solid Waste)
		ETP Sludge
		34.3
		Incinerator Ash
		36.2
		Boiler Ash

			36.1
			Used Empty container
			33.3
			Herbal (Hexane Residue)
			36.4
			Oily Filters
			35.1
			Process Waste Powder
			28.1
			HVAC Filter
			Polystyrene Resins, Epoxy Silica Sand
			Fluorescent Tube / CFL's
			2.
			Acid Batteries
			Lead Acid Batteries/ Dry Acid Batteries
			3.
			Hazardous Waste
			(Liquid Waste)
			Used Oil/ Waste Oil
			5.1/5.2
			4.
			Biomedical waste
			Egg Waste
			Dead Animal
			Solid Culture waste coming out of Tubular centrifuges
			Sharp waste like needle, Syringe, blades
			Animal House waste, Bedding, faecal material, waste food ,Corn
			Cob
			Solid waste like Silicon tubing, Elisa plate, Tips, Rubber
			gloves, waste cotton/paper used in QC
			5.
			E-waste
			Electrical & Electronics components
			6.
			Solid waste
			Glass Wool
			Gutta Waste / paper waste / polythene waste

				Broken Glass
	5.1.2	Types of waste generated at different department		
			S. No.	
			Generator / Department / Location	
			Type of Waste	
			01.	
			Production	
			Solid: Powder, tablet, capsule, gelatin, container, packing	
			material, cassette, ultra-membrane, HVAC filters, oily Filters	
			paper, label, culture waste coming out of tubular centrifuges,	
			E-waste.	
			Semi-solid: Process paste, egg waste etc.	
			Liquid: Syrup, liquid process waste, waste water after	
			equipment washing, R.O. Rejected.	
			02.	
			Ware house	
			Solid: Expired raw material, paper, packing material, label,	
			E-waste, contaminated container, E-waste.	
			Semi-solid: N.A.	
			Liquid: N.A.	
			03.	
			Q.C.	
			Solid: Needle, syringe, blades, silicon tubing, Elisa plate,	
			Tips, Rubber gloves, waste cotton/ paper, E-waste, P.S.	
			dishes, , E-waste	
			Semi-solid: Culture medium.	
			Liquid: Waste water after equipment washing, expired reagent	
			solution.	
			04.	
			Animal House	
			Solid: Bedding (corn cob), dead animal, faecal material, waste	
			food, E-waste.	
			Semi-solid: N.A.	
			Liquid: N.A.	
			05.	
			Engineering	
			Solid: Paper, scrap, oil-rags, filters, resin, boiler chimney	
			shoot, CFL tubes, Acid/Dry batteries, E-waste, Epoxy Silica	
			Sand.	
			Semi-solid: Grease, oil sludge,	
			Liquid: Waste oil, used oil, waste water(high TDS)	
			06.	
			Office	
			Solid: Paper, packing waste, E-waste.	
			Semi-solid: N.A.	
			Liquid: N.A.	
			07.	
			Canteen	
			Solid: Waste food, packing waste.	
			Semi-solid: Waste food	
			Liquid: Waste food	
			08.	
			Toilet, hand, feet wash etc	
			Solid: Paper	
			Liquid: Sewage water, Effluent.	
			09.	
			Other	
			Solid: ETP Sludge, Incinerator ash	
			Semi-solid: N.A.	
			Liquid: Sewage water, Effluent.	
	5.1.3	Collection & Segregation:		
			5.1.3.1	Collection of different type of waste in segregated
			manner:	
			5.1.3.1.	Scrap: All the scrap material like Papers
			1	/Aluminum foil, Packaging material/ company
			related printed material, Gutta, Glass,	
			packing material Glass Bottle, Thermocol,	
			Glass wool, polythene generated from	
			different department should be placed at	
			specific location on temporary base/temporary	
			storages earmarked for same	
			5.1.3.1.	Biomedical waste: Biomedical waste like Dead
			2	Animal, Egg Waste, Animal House waste,
			Bedding, faecal material, Corn Curb, Organic	
			Waste, Solid Culture waste coming out of	
			Tubular Centrifuges, Elisa plate, Tips,	
			Rubber gloves, waste cotton/paper used in	

				QC, should be stored in yellow color polybags
				with biomedical hazard sign.
				5.1.3.1. Hazardous waste: The Hazardous waste like ETP
				3 Sludge, Incinerator ash, Boiler shoot should
				be packed in polybag with proper hazardous
				waste labeling and used oil/ waste oil should
				be stored in container with hazardous waste
				labeling. Used empty container should be
				collected and segregated at specific location
				earmarked in the facility.
				5.1.3.2 Sign of Segregation:
				5.1.3.2. The Sign of Labeling for different type of
				1 waste as follow:
				[pic] [pic]
				Biomedical waste
				Radioactive Waste
	5.2 Color coding and different type of waste:			
	Types of waste			
	Color code of polybag/ container			
	The paper, Gutta, Glass, packing material Glass Bottle			
	Grey			
	Solid waste like Silicon tubing, Elisa plate, Tips, Rubber gloves,			
	waste cotton/paper used in QC, Egg Waste, Dead Animal, Solid Culture			
	waste coming out of Tubular Centrifuges, Animal House waste, Bedding,			
	faecal material, Corn Cob, Organic Waste			
	Yellow with Biohazard Sign			
	[pic]			
	Sharp waste like needle, Syringe, blades.			
	Blue/White translucent with Biohazard Sign			
	Papers /Aluminium foil, Packaging material/ company related printed			
	material			
	Blue			
	Radioactive waste			
	Black with Radiation hazard sign			
	[pic]			
	5.3 Movement:			
	5.3.1 All the scrap material like Papers /Aluminum foil, Packaging			
	material/ company related printed material, Gutta, Glass,			
	packing material Glass Bottle, Thermocol, Glass wool,			
	polythene should be transported by Admin Dept. to dedicated			
	storage place.			
	5.3.2 Solid waste like Dead Animal, Egg Waste, Animal House waste,			
	Bedding, faecal material, Raw material, intermediate			
	product Corn Curb, Organic Waste, Solid Culture waste coming			
	out of Tubular Centrifuges, Elisa plate, Tips, Rubber			
	gloves, waste cotton/paper used in QC, waste should be			
	transported / producer to incineration area. Admin dept.			
	would provide transport facility upon request.			
	5.3.3 The Hazardous waste like ETP Sludge, Incinerator ash, Boiler			
	shoot should be transported to Hazardous storage place .Used			
	oil/ waste oil should be stored in container and transported			
	to storage place. Used empty container should be should be			
	transported to used container scrap yard. All the labels on			
	the container shall be defaced by crossing out with			
	permanent red color marker.			
	5.3.4 A request shall be filled by concerned department for			
	disposal of waste material generated in department as per			
	annexure- 1 in form no. ES/002/F1-00.			
	5.4 Storage of waste:			
	5.4.1 Hazardous waste must be in hazardous storage place for			
	period of ninety days only.			
	5.4.2 Biomedical waste must be stored only 48 hrs.			
	5.4.3 Egg waste must be stored only 24 hrs.			
	5.4.4 Scrap waste must be stored in Scrap yard.			
	5.4.5 Used container must be stored in used container scrap yard.			
	5.4.6 The storage facility must be such that it shall not allow			
	any leaching of the hazardous waste into the ground or spill			
	over into other areas or the storm water drains.			
	5.5 Treatment/disposal:			
	5.5.1 In-house Treatment:			
	5.5.1.1 Production, QC, QA should have all MSDS in their			
	possession for materials they are using such as			
	chemicals, reagents biological materials, cells,			
	antigens, cultures etc and inactivation should be			
	done as per respective MSDS/SOP.			
	5.5.1.2 The Biomedical Solid Culture waste coming out of			
	Tubular centrifuges, Animal House waste, Bedding,			
	faecal material ,Corn Cob, Egg Waste, Dead Animal,			
	powder, tablet, capsule, gelatin, container, packing			
	material, cassette, ultra-membrane, HVAC filters,			
	Joily Filters paper, label, culture waste coming out			

			of tubular centrifuges, packing material ,are
			disposed through incineration process. The waste
			water & sewage water are treated through ETP & STP
			respectively.
		5.5.2	To Contractor/authorized vendor:
		5.5.2.1	ETP Sludge, Incinerator ash, Boiler shoot & empty
			container are disposed to authorized vendor of SPCB.
		5.5.2.2	All printed packaging material to be disposed off
			should be defaced /shredded before disposal.
		5.5.2.3	The paper, Gutta, Glass, packing material after
			proper defacing, shredding is disposed to scrap
			dealers.
		5.5.3	Treatment and disposal of ETP Sludge/ Incineration ash/ Process
			waste/Boiler Shoot/Used oil/waste oil and Empty Container.
		5.5.3.1	This type of waste disposed through authorized party
			of State pollution Control Board.
		5.5.3.2	Used appropriate PPE like suitable PVC hand gloves,
			gum boots or safety shoes, general dust nose masks,
			and safety goggles while collecting and transporting
			hazardous waste.
		5.5.3.3	Clear boards indicate hazardous waste and quantity
			must be displayed.
		5.5.3.4	Administration department shall ensure the safe
			storage of empty containers in scrap yard before its
			disposal to authorized external party.
		5.5.3.5	EOHS department shall ensure the disposal of empty
			containers to the authorized external party.
		5.5.3.6	All HW should be packed in non-permeable material /
			leak proof containers or plastic bags.
		5.5.4	Regulatory requirement for ETP Sludge/ Incineration ash/
			Process waste/Boiler Shoot/Used oil/waste oil and Empty
			Container
		5.5.4.1	The Storage of ETP Sludge/Incineration ash in
			Hazardous waste store.
		5.5.4.2	The Disposal of ETP Sludge/Incineration ash should be
			within ninety Days.
		5.5.4.3	Authorization / Renewal of Authorization under
			sub-rule (3) of Rule 5 of the Hazardous Waste
			(Management, Handling and Trans-boundary Movement)
			Rules, 2008 for Collection / Reception / Treatment /
			Transport / Storage / Disposal of Hazardous waste
			from State Pollution Control Board.
		5.5.4.4	All Complies should be in accordance to State
			Pollution.
		5.5.5	Treatment and disposal of Dead Animal, Animal House Bedding/
			Corn Cob and Faecal Material
		5.5.5.1	For Microbiological, Biotechnological solid waste
			contaminated with biological fluids & Animal house
			waste use yellow colored plastic bags for collection.
		5.5.5.2	Secure the bags by properly closing the mouth with a
			string so that waste materials do not spill off even
			by accident.
		5.5.5.3	Label the poly bags properly with non-washable ink
			and dispose off by incineration.
		5.5.6	Regulatory requirement for Dead Animal, Animal House Bedding/
			Corn Cob and Faecal Material
		5.5.6.1	No untreated bio-medical waste shell be kept or
			stored beyond 48 hours.
		5.5.6.2	Use yellow colored plastic bags with Biohazard sign
			for collection.
		5.5.7	Procedure for Handling and disposal of Solid waste like silicon
			tubing, Elisa plate, Tips, Rubber gloves, waste cotton and
			Solid Culture waste coming out of Tubular centrifuges
		5.5.7.1	For solid waste like silicon tubing, rubber gloves,
			rubber stoppers and waste cotton article, pack the
			article in blue/ white translucent plastic bags
			securely by closing the month of the bag. Label them
			properly with non –washable ink. Autoclave at 121 OC
			for one hour. Dispose of by incineration.
		5.5.7.2	Solid waste coming out from tubular centrifuge is
			required to be securely collected in yellow colored
			poly bags and disposed off by incineration.
		5.5.8	Regulatory requirement for Solid waste like silicon tubing,
			Elisa plate, Tips, Rubber gloves, waste cotton/ paper etc and
			Solid Culture waste coming out of Tubular centrifuge
		5.5.8.1	No untreated bio-medical waste shell be kept or
			stored beyond 48 hours
		5.5.8.2	Use yellow colored plastic bags for collection.
		5.5.9	Treatment and disposal of Egg Waste :
		5.5.9.1	Transfer egg waste in poly-lined containers to
			incinerator room and dispose of by incineration.
		5.5.9.2	Dispose all packing materials off egg waste by
			incineration.
		5.5.9.3	Collect ash from the incinerator, after cooling to
			room temperature in Plastic Bag.
		5.5.10	Regulatory requirement for of Egg Waste:
		5.5.10.1	No egg waste shall be kept or stored beyond 24 hours.
		5.5.10.2	Use yellow colored plastic bags for collection.
		5.5.11	Treatment and disposal of Sharp waste like needle, Syringe,
			blades etc.
		5.5.11.1	For sharp waste like needle, syringes, scalpels,
			blades and glass etc disinfect with disinfectant,

			pack it in blue/ white translucent plastic bags/	
			puncture proof containers.	
			5.5.11.2	Autoclave the sharp waste like needle, syringes,
			scalpels, blades and glass etc at 121 OC.	
			5.5.11.3	Dispose of by deep burial after shredding.
			5.5.12	Regulatory requirement for Sharp waste like needle, Syringe,
			blades etc.	
			5.5.12.1	Pack it in blue/ white translucent plastic puncture
			proof containers	
			5.5.13	Treatment and disposal of Oily Filters:
			5.5.13.1	The oily filter placed in the identified location of
			storage oily filter.	
			5.5.13.2	Use required PPE's like Nose mask, Hand gloves and
			Safety Goggles during handling filter.	
			5.5.13.3	Dispose them through incineration.
			5.5.14	Treatment and disposal of Hexane Residue:
			5.5.14.1	The Hexane residue placed in the identified location
			of storage of Hexane residue.	
			5.5.14.2	Use required PPE's like Nose mask, Hand gloves
			(solvent Resistant) and Safety Goggles during	
			handling of Hexane Residue	
			5.5.14.3	Dispose them through incineration.
			5.5.15	Treatment and disposal of Lead Acid Batteries/ Dry Acid
			Batteries:	
			5.5.15.1	Lead Acid Batteries/ Dry Acid Batteries Contribute to
			heavy metals that potentially may leach slowly into	
			soil, groundwater or surface water.	
			5.5.15.2	Lead Acid Batteries/ Dry Acid Batteries placed in the
			identified location of storage of Lead Acid	
			Batteries/ Dry Acid Batteries.	
			5.5.15.3	Use required PPE's like Nose mask, Acid Resistance
			Hand gloves and Safety Goggles during handling of	
			Lead Acid Batteries/ Dry Acid Batteries.	
			5.5.15.4	These Lead Acid Batteries/ Dry Acid Batteries should
			be returned to supplier after use.	
			5.5.16	Treatment and disposal of Electrical & Electronics components:
			5.5.16.1	It Comes under E- Waste (electronic Waste) Category.
			5.5.16.2	Electrical & electronic component waste store in
			identified location of storage of electrical and	
			electronic component.	
			5.5.16.3	Electrical & electronic component waste should be
			disposed off to the SPCB authorized disposal sites.	
			5.5.17	Procedure for Handling and disposal of Thermocol, Glass Wool,
			Gutta waste, Broken Glass and Scrap:	
			5.5.17.1	Thermocol, Glass Wool, Gutta waste, Broken Glass and
			Scrap store in identified location of storage of	
			Scrap yard.	
			5.5.17.2	Thermocol, Glass Wool, Gutta waste, Broken Glass and
			Scrap dispose of through selling them to scrap	
			dealer.	
			5.5.18	Destruction of Product / Intermediate Product /Raw Material/
			Expired Material	
			5.5.18.1	Check all the relevant information with regard to the
			incoming material, labeling or packing material,	
			intermediate product, final ARM and final formulation	
			Bulk Vaccine that is meant for destruction. Verify	
			and be very clear that it needs destruction.	
			5.5.18.2	In case of final ARM, Intermediate Product and
			Formulated Bulk Vaccines the EOHS personnel shall	
			check Destruction order Form No.: QA/023/F1-04.	
			5.5.18.3	Request for destruction of products (Form No.:
			ES/002/F2-00) shall be filled, signed and forwarded	
			by the Head of the concerned area to EOHS Department.	
			5.5.18.4	The EOHS Department shall be responsible for
			destruction and disposal of product.	
			5.5.18.5	In Case vaccine filled vials, autoclave at 121 OC for
			30 minutes. Crush the vials and dispose by deep	
			burial.	
			5.5.18.6	In case of intermediate product, autoclave at 121 OC
			and dispose off same through drain or by incineration	
			whichever is applicable.	
			5.5.19	Treatment and disposal of Effluent and Sewage:
			5.5.19.1	Take all effluent from live area through biological
			drain pipe to the storage tank and Inactivate by	
			heating at 96 OC for 20 minutes. After cooling drain	
			the inactivated liquid through conventional drain to	
			the effluent treatment.	
			5.5.19.2	Drain all effluent from dead area through
			conventional drain system to effluent treatment plant	
			5.5.19.3	The entire waste water gathered from plant then
			collected at collection tank from collection tank	
			pump to Effluent Treatment plant/ sewage treatment	
			plant.	
			5.5.19.4	Treated waste can be used for Irrigation purposes.
			5.5.20	Regulatory requirement for Effluent and sewage.
			5.5.20.1	Consent to Establish/operate/renewal under
			section-25/26 of water (Prevention & control of	
			Pollution) act 1974 from State Pollution Control	
			Board.	
			5.5.20.2	All Complies should be in accordance to State
			Pollution Control Board as well Central Pollution	
			Control Board.	

[6.0]Forms and Records |
[6.1	Request for Disposal of waste materials generated in departments
	- Annexure 1
[6.2	Request for destruction of Products / Intermediate Product /Raw
	Material/ Expired Material
	- Annexure 2
[6.3	Log Book for destruction of Products / Intermediate Product /Raw
	Material/ Expired Material
	- Annexure 3
[6.4	Log book for material received for Incineration.
	- Annexure 4
[7.0]References:	
[7.1	Hazardous Waste (Management, Handling and Trans-boundary Movement)
	Rules, 2008 for Collection / Reception / Treatment / Transport /
	Storage / Disposal of Hazardous waste.
[7.2	Water (Prevention & Control of Pollution) Act, 1974.
[7.3	Bio- Medical Waste (Management and Handling) Rule, 1998.
[7.4	SOP QA/001 Standard Operating Procedure on Standard Operating
	Procedures.
[8.0]Distribution	
[8.1	Master Copy - Archived with Quality Assurance Department
[8.2	Controlled - Environment Occupational Health and Safety and
	Copy
[9.0]History	
[Date	Revision Number
	Rev. 00

Environment Occupation Health and Safety

[pic]

Registration of Pharmaceutical Product for General Sale
About the Service

This service enables the customers to submit applications to register simple pharmaceutical products with limited medicinal usage, which cannot be considered medicines and are intended for general sale. These include products such as dietary supplements, medical cosmetics and medical disinfectants.

Service Process

- Login to the MoHAP website or smart app using the UAE PASS
- Submit the request through the electronic service and complete the payment to meet all conditions
- The competent technical committees will deliberate the registration of products and recommendations will be submitted to the competent ministerial committee
- Declarations shall be addressed to the concerned companies stating the committee's decisions via the e-system
- The customer will follow up with the Pharmacological Analysis Section of at the Drug Department
- The company should complete the requirements and submit them online
- The relevant technical and ministerial committees will re-deliberate the registration of products that have been deferred in advance as soon as companies complete the requirements
- Issuing certificates of registration of products that have been approved for registration following fulfilling all conditions and requirements. Registration certificates are valid for five years effective the date of committee approval
- Issuing pharmaceutical certificates (for locally manufactured pharmaceutical products) valid for one year effective the date of issuance

Required Documents

- Certificate of pharmaceutical product in accordance with the WHO (World Health Organization) system or a certificate of free sale of the product issued by the competent authorities in the country of origin and certified by the Embassy of the United Arab Emirates. It should contain the following information: Product Brand Name: If the brand name required to be registered in UAE is different from the name in the country of origin, this shall be made clear in the certificate, besides stating the reason for that and making both names clear with an emphasis on their conformity in terms of formulation and other specificationsThe detailed formulation of the product contains active and inactive substances with their quantities and functions of inactive substancesName of the company entitled to marketing/ manufacturing/ manufacturing sites/ subcontract manufacturers along with their addresses
 - One samples of the product.
 - A valid registration certificate of the manufacturing company issued by MOHAP-UAE with intended production line.
 - Halal certificate issued by certified authorities and organizations.
 - A statement issued by the company confirming that the product to be registered is free of hormones, heavy metals, antibiotics, steroids, pig derivatives and any other natural or chemical substances that have a harmful impact on human beings biologically and behaviorally
 - If the product contains animal derived substances, the animal type and the part extracted from it should be mentioned with the percentage of alcohol used "if any and why it is used"
 - A copy of the certified contract signed between the marketing company and local agent indicating the products for which the agent will be responsible
 - A certified certificate from the competent authorities of the country of origin stating that the materials used in manufacturing the product is free from mad cow disease and its causes (if the product contains substances classified as potential substances for transmitting the disease) (Bovine Spongiform Encephalopathy (BSE)/ Transmissible Spongiform Encephalopathies (TSE) free certificate)
 - A copy of the outer and inner cover and the leaflet printed on letterhead paper stamped by the company and signed by the authorized person
 - Detailed Composition Certificate (active & inactive ingredients with their quantities).
 - Summary of Product Characteristics (SPC).
- Conditions & Requirements

1. Marketing Authorization Holder companies must be registered by MOHAP before they could register their products.
 2. The applicant must be a medical warehouse licensed by the Ministry of Health and have a valid license.
- FAQ's

Service completion duration

- 45 working days

Service fees

Service channels

Service locations

- Mohab Website
- MOHAP smart app

Support

- smartservicessupport@mohap.gov.ae
- drugreg.inquiries@mohap.gov.ae

Payment channels

- E- Payment

Target audience

- Medical warehouses
- Local pharmaceutical manufacturers
- Marketing officers

Resources

- Infographic - Registration of Pharmaceutical Product for General Sale - PDF 384KB
- Portal Manual_637716056042464487
- Login User Manual_637716056042643318
- Help Manual_637716056041058138

Department name

Drug

Sector name

Health Regulation

Main service

Medical, Pharmaceutical and Drug Licencing and Registration

Service Code

110-04-011-000

Service Classification

Transactional

Sub Service Type

Variation

Service Type

Government to Business

Related Services

- This service is not linked with other services

Service Bundle

This service is not linked with any service packages

Number of Users

43

Number of Transactions

336

Notes

Marketing Authorization Holder companies must be registered by MOHAP before they could register their products.

Sustainable Goals

Good Health And Well-Being

In the United Arab Emirates (UAE), the General Sale List (GSL) includes Dietary Supplements, Medicated Cosmetics, Antiseptic and Disinfectants and Miscellaneous products which contains pharmaceutical ingredients and/or a medical claim and cannot be classified as medicines.

The application can be made in English or Arabic. Marketing authorization for GSL products in the United Arab Emirates (UAE) can only be given to companies (Manufacturer) who are registered in the UAE ministry of health. Below is a list of the GSL product registration requirements followed by the requirements for registering a company in the UAE ministry of health. Applicants in the UAE should have Medical stores licensed by the Ministry of Health.

Registration of the GSL Product requires:

1. Application and document receiving form for the Registration of new GSL product (typed). Available from the Technical Affairs Section.
 2. Certificate of Pharmaceutical Product (CPP) as per the WHO Certification Scheme or Free Sale Certificate (FSC) for the product.
- To be issued by Competent Authorities in Country of Origin, it must be authenticated by the foreign affairs section of the UAE or any GCC embassy in the country of origin.

■The CPP must contain:

■ Trade Name of the Product.

If the Trade Name of the product intended for registration in UAE is different from that in the Country of Origin, difference should be explained clearly in the CPP/FSC and both Trade Names should be mentioned with confirmation of similarity in composition and other specifications.

■ Composition of the product in details (Active and inactive ingredients with their quantities).

■ Name of the Manufacturer/Marketing Authorization Holder/ Manufacturing Site(s)/ Contract Manufacturer(s) with address.

■ Shelf-life of the product with Storage conditions.

■ Declaration that the product is freely sold in country of Origin for less that 2 years.

2. Declaration that the product is newly sold in Canada for English and/or French language.
3. Package Insert for the product authenticated from Competent Authorities (if applicable).
4. Three (3) samples of the product in its final package.
5. Certificate of Analysis of the same batch of the samples submitted for registration.

The following information must be submitted on company letterhead with logo, signature and stamp:

6. Statement issued from the Principal Company showing that the product is devoid from hormones, heavy metals, antibiotics, steroids, derivatives of pork and any natural and chemical ingredients having harmful effects on human biological and behavioural functions. If the product contains an ingredient from animal source, the kind of animal and part extracted from it must be specified percentage of alcohol if any must be mentioned together with reasons thereof.

□ □ □ □ □ □ □ □ □ □ □ □ □ □

Adopted from MOH Circular No. 20 for 2001 issued by the Director of Drug Control Department

7. Halal certificate issued by recognizable organizations and authorities.

8 Authenticated contract between MAH & agent mentioning the products for which the agent is going to be responsible for.

9 The applicant should have a medical store licensed by the Ministry of Health. (as per the Circular No.
1
for year 2006)

10 valid company registration certificate issued by UAE / MOH drug control department

11 Submit BSE / TSE free certificate from competent authority in Country of origin (wherever applicable)

12 outer label, inner label and insert of the product on stamped company letterhead.

13 CD of artwork (outer, inner label and insert) of the product in JPEG format.

If the product marketing authorization holder (MAH) & manufacturer has not been registered in the UAE ministry of health, then this must be done before a product can be submitted for registration.

Documents required for GSL Company registration

Application is collected from Technical Affairs Section Secretary or downloaded from the MOH website, the forms are PART ONE (for MAH) & PART TWO (for manufacturing site), at the end of each form there is a list of required documents to be attached along with the form, applications are to be filled and stamped by the MAH & manufacturer, documents are to be attached along with each form.

N.B.: The GSL registration committee may request QCL analysis of the submitted products and the applicant will have to provide necessary samples and other analysis requirements requested by the drug control laboratory.

N.B:- All forms and complete copy of guidelines for company and product registration can be requested from the technical affairs section secretary, telephone 02- 6117325 or downloaded from the MOH website at the following link:

http://www.moh.gov.ae/moh_site/phar_med/moh_p_m.htm

UAE MOH Guidelines in Good Vigilance Practice (GVP) For Marketing Authorization Holders / Pharmaceutical Manufacturers In UAE

All pharmaceutical companies/ Marketing Authorization Holder (MAH) whose products are registered and marketed in UAE must have a system in place for documenting following Objectives:

Pharmacovigilance systems and their quality systems

Pharmacovigilance System Master File (PSMF)

Pharmacovigilance Inspections

Pharmacovigilance audits

Risk management systems

Management and reporting adverse reactions to medicinal

Products Periodic safety update reports (PSURs)

Post authorization safety studies

Signal management

Safety communication

Risk minimization measure

When submitting an application for new Pharmaceutical Product registration, the Applicant, should submit a description of the pharmacovigilance System and submit proof that the services of a Qualified Person Responsible for Pharmacovigilance (QPPV), hereafter referred to as the QPPV, are in place.

The MAH shall have permanently and continuously at its disposal an appropriately QPPV resident in UAE.

For multinational MAHs a local safety responsible (LSR) may be accepted Based in UAE. For local MAHs there should be a dedicated QPPV and he/she should be resident in UAE. The names and 24 hours contact details of the nominated QPPV and his alternate during absence should be submitted to PV section /Drug Department.

The QPPVs should have a minimum of bachelor degree in pharmacy or medicine, a basic training in epidemiology and biostatistics UAE MOH Guidelines in Good Vigilance Practice (GVP) For Marketing Authorization Holders / Pharmaceutical Manufacturers In UAE.

The QPPV shall be responsible for the establishment and maintenance of the marketing authorization holder's Pharmacovigilance System and therefore shall have sufficient authority to influence the performance of the quality system and the pharmacovigilance activities and to promote, maintain and improve compliance with the legal requirements. The applicant/MAH should provide the following requirements in order to get QPPV approval:

Letter of appointment from the company not the agent

Training certificate in PV

Experience certificate in PV

List of products covered by the company

ADR case reports submitted in UAE. (Local companies to report ADR cases of their products to the department within one year of the appointment)

SOP of the PV officer

Graduation certificate (minimum Bachelor Degree in Pharmacy or Medicine)

For the Multinational MAH/Applicant the following two documents are required to have

1. The PSMF (according to European Good Pharmacovigilance Practice) and/ or,
2. National pharmacovigilance system file (national PVSF) which describes the key elements of pharmacovigilance activities in the UAE. The content of National PVSF is as follow: Roles and Responsibilities of the Marketing Authorization Holder Monitoring of Compliance and Pharmacovigilance Inspections Qualified Person Responsible for Pharmacovigilance (QPPV) Pharmacovigilance Plan Organizational chart Quality Management System Requirements for Risk Management Systems: Clinical & Non-clinical Part of the Safety Specification Adverse Events/Adverse Reactions Identified and Potential Interactions including Food-Drug and Drug-Drug interactions Epidemiology The Risk Minimization Plan. Requirements for Expedited Reporting of Individual Case Safety Reports Requirements for Reporting in Special Situation Requirements for Periodic Safety Update Reports Training Documentation Overview of the parts and modules of the RMP The RMP is divided into several parts, with the safety specifications of the RMP organized into modules to increase flexibility. Part I Product(s) Overview Part II Safety Specification Module SI: Epidemiology of the indication(s) and target population(s) Module SII: Non-clinical part of the Safety Specification Module SIII: Clinical trial exposure Module SIV: Populations not studied in clinical trials Module SV: Post-Authorization Experience Module SVII: Identified and potential risks Module SVI: Additional requirements for the Safety Specification in UAE Module SVIII: Summary of the safety concerns Part III Pharmacovigilance Plan Part IV Plans for post-Authorization efficacy studies Part V Risk minimization measures (including evaluation of the effectiveness of risk minimization measures) Part VI Summary of the RMP Part VII Annexes Submission and updates for the RMP UAE MOH Guidelines in Good Vigilance Practice (GVP) For Marketing Authorization Holders / Pharmaceutical Manufacturers In UAE RMP for new molecules should be submitted with the registration application. RMP for new innovator, biological, and biosimilar should be submitted with the registration application. RMP for other registration status may be requested upon registration application.

Approvals of the URS:

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Department | Name | Signature | Date

BioBridge Healthcare Solutions Pvt. Ltd., India | BioBridge Healthcare Solutions Pvt. Ltd., India |

BioBridge Healthcare Solutions Pvt. Ltd., India | BioBridge Healthcare Solutions Pvt. Ltd., India

Prepared by: | | |

Checked/Reviewed by: | | |

Client | Client | Client | Client

Reviewed by: | | |

Reviewed by: | | |

Approved by: | | |

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1.0 | Objective | Objective

| This URS detail the requirement for Purified water generation system 500LPH Capacity | This URS detail the requirement for Purified water generation system 500LPH Capacity

2.0 | Scope | Scope

| This URS is applicable for the Purified water generation System 500LPH capacity used in the facility |

This URS is applicable for the Purified water generation System 500LPH capacity used in the facility

3.0 | Equipment description | Equipment description

| Purified water generation system 500LPH Capacity has to be designed for 365 working days. Working 8 hours in three shifts. It will be placed in the clean and closed utility area. | Purified water generation system 500LPH Capacity has to be designed for 365 working days. Working 8 hours in three shifts. It will be placed in the clean and closed utility area.

4.0 | Component details | Component details

| Quality Requirements for Purified water

It must ensure compliance with quality requirements

The quality of water will be according to current USP, EP and IP specifications.

Purified water USP 40 (current USP)

Conductivity $\leq 1.1 \mu\text{S}/\text{cm}$ (20° C)

TOC $\leq 0.5 \text{ mg}/\text{mL}$ (500 ppb)

Total Aerobic $\leq 100 \text{ cfu}/\text{mL}$

Purified water EP 9th (current EP)

Conductivity $\leq 4.3 \mu\text{S}/\text{cm}$ (20° C)

TOC $\leq 0.5 \text{ mg}/\text{mL}$ (500 ppb)

Total Aerobic $\leq 100 \text{ cfu}/\text{mL}$

Nitrates: $\leq 0.2 \text{ ppm}$

Heavy metals: $\leq 0.1 \text{ ppm}$

Traceability of all process information of pretreatment and osmosis plant.

Flexibility: All the information must be easy to process for qualified personnel.

Purified Water Generation

Design basis for Purified Water generation: Maximum Demand : 500 LPH

Purified water is produced from raw water from supply water network, based on the water analysis report provided by the client

Raw water from supply water network will be pre-treated to eliminate insoluble materials and calcium salts using filtration and ion exchange columns.

The equipment, piping and ducting must be designed to perform maintenance and cleaning activities in an easy way.

The equipment must be modular equipped with extension capacity.

After the EDI step of the pretreatment and osmosis skid the quality of Purified Water must be according to quality standards and requirements of monograph.

Pretreated water is treated through a two stage Reverse Osmosis system collected in a stainless-steel tank.

RO is equipped with a concentrate recovery membrane to get a water yield rate of at least 70%.

The pumps are able to vary their flow rate as a function of the tank level.

Pipes diameter, quality of materials and valves will be included in the P&ID.

All piping lines must be identified with product, direction flow and color codes (according to DIN 2403 standard).

All hydraulic elements must be labeled with their identifier tag attached to the element. P&ID must correspond with the real installation and the elements must be labeled accordingly.

Sampling points between all the operations must be defined on the design specification and installed for quality periodical monitoring.

Only after First step RO Membrane:

Pipe connections: Orbital welding wherever possible.

Welding must be performed by qualified welder.

Certificates of welding must be able.

Welding inspection: 100% of manual welds and 15% of the orbital welds.

All pipes must be treated before use. Installation of pretreatment and osmosis will be cleaned (caustic soda) and passivated (Nitric Acid).

Piping pressure leak test will be performed in Start-up

Temperature of operation in generation of water plant must be monitored.

Installed sensors must be easy to access and replace.

Conductivity sensor for each step of generation of water plant must be installed.

Critical parameters to be monitored:

Conductivity before the storage tank (at the end of the skid)

Temperature (normal operation)

The Level transmitter in the buffer tank should start and stop the RO plant for maximum level.

First step RO Membrane System should reduce Total dissolved solids below 70 .

Second step RO Membrane System should reduce Total dissolved solids below 5 .

EQUIPMENT AND MATERIALS OF PRETREATMENT AND OSMOSIS

EQUIPMENT AND MATERIALS

All metallic components in contact with purified water (piping, fittings...) must be stainless steel AISI 316L and end of surfaces must be polished Ra<0.5 µm. Supplier must provide the certificates of stainless steel (Ferrite content must be specified).

Valves and pumps must meet sanitary design, only after First step Reverse Osmosis Membrane.

Temperature and conductivity sensors must be installed.

Pressure, flow and level sensors must be installed.

Feed Water Tank

Feeding water tank xxxx L capacity, like PVC or similar. The Storage tank must be designed to prevent or minimize the development of biofilm, to minimize corrosion, to aid in the use of chemical sanitization of the tanks, and to safeguard mechanical integrity.

Pretreatment And Osmosis-Purified Water Generation

NaCl Dosage system is required to eliminate the bacteria contamination in the water and prevent any microbial growth.

A feed pump is required to feed the RO and the rest of the treatment with the proper pressure and flow.

Multimedia Filter is required to reduce the turbidity in the water by more than 99% giving a maximum particle size of 50 µm.

Before Reverse Osmosis Treatment, cartridge filter to prevent any particle ≥5 µm left by the multimedia filter to pass through the RO.

Before RO treatment Sodium Metabisulphite system for the reduction of free chlorine.

A redox control System is required to measure and display redox in water before the RO First step.

Before RO First a high-pressure pump.

Between first step and second step reverse osmosis membrane

Caustic soda (NaOH) with online pH measure system to ensure the feed water enters the 2nd step of RO with CO2 levels.

Electro-deionization module after the buffer tank should be able to reduce total dissolved solids and conductivity to below 1.1 micro-siemens (20 °C)

Control System of Pretreatment and Osmosis

Hardware:-

A PLC must be installed to control and monitor the system.

A touch screen should be installed as a human machine communication interface.

Software:-

In the Software the following aspects must be visualized, as a minimum:

- ✓ Set point values.
- ✓ Control panel setup screen.
- ✓ Alarm history.
- ✓ Manual operation most of automatic device.
- ✓ Automatic system activation.
- ✓ Washing mode automatic.
- ✓ Access and configuration of user accounts of the system.

The system must record and monitor critical parameters. | Quality Requirements for Purified water

It must ensure compliance with quality requirements

The quality of water will be according to current USP, EP and IP specifications.

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Valves and pumps must meet sanitary design, only after First step Reverse Osmosis Membrane.

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Pressure, flow and level sensors must be installed.

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Feeding water tank xxxx L capacity, like PVC or similar. The Storage tank must be designed to prevent or minimize the development of biofilm, to minimize corrosion, to aid in the use of chemical sanitization of the tanks, and to safeguard mechanical integrity.

Pretreatment And Osmosis-Purified Water Generation

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Electro-deionization module after the buffer tank should be able to reduce total dissolved solids and conductivity to below 1.1 micro-siemens (20 °C)

Control System of Pretreatment and Osmosis

Hardware:-

A PLC must be installed to control and monitor the system.

A touch screen should be installed as a human machine communication interface.

Software:-

In the Software the following aspects must be visualized, as a minimum:

- ✓ Set point values.
- ✓ Control panel setup screen.
- ✓ Alarm history.
- ✓ Manual operation most of automatic device.
- ✓ Automatic system activation.
- ✓ Washing mode automatic.
- ✓ Access and configuration of user accounts of the system.

The system must record and monitor critical parameters.

5.0 | Construction material | Construction material

| As per the GMP/GLP compliance. | As per the GMP/GLP compliance.

6.0 | Utility Requirements | Utility Requirements

| Energy Type | Specification

| Single Phase | 230V +10% / -6% 50Hz

| Three Phase / 4 Wire | 400V +10% / -6% 50Hz

| Control Voltage (Instruments) | 24V DC

| Control Voltage (Power circuits) | 230V +10% / -6% 50Hz

| Potable Water Supply:-

Potable water compliant with following specs:

- ✓ pH: 6,5 – 8,5
- ✓ Description: clear, colorless, odorless liquid
- ✓ Total Suspended Solids: NMT 200 ppm
- ✓ : NMT 500 cfu/ml
- ✓ Absence of Coli forms

Note: Conductivity of potable water from local supply up to 2.500 $\mu S/cm$ | Potable Water Supply:-

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- ✓ : NMT 500 cfu/ml
- ✓ Absence of Coli forms

Note: Conductivity of potable water from local supply up to 2.500 $\mu S/cm$

7.0 | Accessories | Accessories

| All accessories required for the normal operation of the equipment. | All accessories required for the normal operation of the equipment.

8.0 | Documents Required | Documents Required

| Complete documentation Package with following documents should be delivered with the system

Installation Qualification (IQ)
Operational Qualification (OQ)
Performance Qualification (PQ)
Operational manual (2 Sets)
Spare part list

A generic list of the documentation to be delivered by the supplier(s) before Design qualification DQ:

- ✓ Memory and design justifications.
- ✓ Electrical wiring/Schematic diagrams.
- ✓ Technical specifications.
- ✓ Installation drawings (layout, P&IDs).

The following is a generic list of the documentation to be delivery by the supplier(s) before Factory Acceptance Test (FAT):

- ✓ Materials and Equipment certifications.
- ✓ Welder's qualification.
- ✓ Passivation reports.
- ✓ Cleaning and flushing reports.
- ✓ Reports of pressure testing to check leaks in equipment or piping.
- ✓ Operation and maintenance manuals
- ✓ Updated backup copies of the software/configuration of programmable control systems (HMIs, PLCs, frequency drives

The operator and maintenance manuals must be in English. The manuals must be available paper and in electronic format. | Complete documentation Package with following documents should be delivered with the system

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- ✓ Reports of pressure testing to check leaks in equipment or piping.
- ✓ Operation and maintenance manuals
- ✓ Updated backup copies of the software/configuration of programmable control systems (HMIs, PLCs, frequency drives

The operator and maintenance manuals must be in English. The manuals must be available paper and in electronic format.

9.0 | Drawings | Drawings

| |

10.0 | Training | Training

| The vendor shall impart operation and maintenance for at least two personnel from user department and engineering /projects department and shall provide evaluation report. | The vendor shall impart operation and maintenance for at least two personnel from user department and engineering /projects department and shall provide evaluation report.

11.0 | Abbreviations | Abbreviations

| cGMP: current Good Manufacturing Practices

HW: Hardware

PC: Personal Computer

PW: Purified Water

P&ID: Piping and Diagram Instrumentation

EU: European Union

SW: Software

RO: Reverse Osmosis

TOC: Total Organic Carbon

EDI: electro-deionization

AISI: American Iron and Steel Institute

USP: United States Pharmacopeia

EP: European Pharmacopeia

IP: International Pharmacopeia (WHO) | cGMP: current Good Manufacturing Practices

HW: Hardware

PC: Personal Computer

PW: Purified Water

P&ID: Piping and Diagram Instrumentation

EU: European Union

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RO: Reverse Osmosis

TOC: Total Organic Carbon

EDI: electro-deionization

AISI: American Iron and Steel Institute

USP: United States Pharmacopeia

EP: European Pharmacopeia

IP: International Pharmacopeia (WHO)

12.0 | Other Requirements | Other Requirements

| - EudraLex - Volume 4 - Good Manufacturing Practice (GMP) guidelines.

- Code Federal Regulations Part 21. Current Good Manufacturing Practice for finished Pharmaceuticals.

- WHO good manufacturing practices for pharmaceutical products: main principles Annex2. Technical Report

Series 986, 2014 | - EudraLex - Volume 4 - Good Manufacturing Practice (GMP) guidelines.

- Code Federal Regulations Part 21. Current Good Manufacturing Practice for finished Pharmaceuticals.

- WHO good manufacturing practices for pharmaceutical products: main principles Annex2. Technical Report

Series 986, 2014

Approvals of the URS:

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Department | Name | Signature | Date

BioBridge Healthcare Solutions Pvt. Ltd., India | BioBridge Healthcare Solutions Pvt. Ltd., India |

BioBridge Healthcare Solutions Pvt. Ltd., India | BioBridge Healthcare Solutions Pvt. Ltd., India

Prepared by: | | |

Checked/Reviewed by: | | |

Client | Client | Client | Client

Reviewed by: | | |

Reviewed by: | | |

Approved by: | | |

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1.0 | Objective | Objective

| This URS detail the requirement for WFI generation system 700LPH and PSG system 300KgH Capacity | This

URS detail the requirement for WFI generation system 700LPH and PSG system 300KgH Capacity

2.0 | Scope | Scope

| This URS is applicable for the WFI generation System 700LPH and PSG 300KgH used in the facility | This

URS is applicable for the WFI generation System 700LPH and PSG 300KgH used in the facility

3.0 | Equipment description | Equipment description

| WFI generation system 700LPH Capacity + PSG 300KgH capacity has to be designed for 365 working days.

Working 8 hours in three shifts. It will be placed in the clean and closed utility area. | WFI generation

system 700LPH Capacity + PSG 300KgH capacity has to be designed for 365 working days. Working 8 hours in

three shifts. It will be placed in the clean and closed utility area.

4.0 | Component details | Component details

| Quality Requirements for Water For Injection and Pure Steam

It must be comply to the quality requirements

The quality of water will be according current USP, EP and IP specifications.

Water for injection USP & amendments thereof

Conductivity $\leq 1.1 \mu\text{S}/\text{cm}$ at 20°C

TOC $\leq 0.5 \text{ mg}/\text{mL}$ (500 ppb)

Endotoxins $\leq 0.25 \text{ EU}/\text{mL}$

Water for injection EP & amendments thereof

Conductivity $\leq 1.1 \mu\text{S}/\text{cm}$ (20°C)

TOC $\leq 0.5 \text{ mg}/\text{mL}$ (500 ppb)

Endotoxins $\leq 0.25 \text{ EU}/\text{mL}$

Total Aerobic $\leq 10 \text{ CFU}/100 \text{ mL}$

Nitrates $< 0.2 \text{ ppm}$

Pure Steam USP & amendments thereof

Conductivity $\leq 1.1 \mu\text{S}/\text{cm}$ (20°C)

TOC $\leq 0.5 \text{ mg}/\text{mL}$ (500 ppb)

Endotoxins $\leq 0.25 \text{ EU}/\text{mL}$

Supply steam must have a dryness value of no less than 0.95. Obtain no more than 3.5 ml of gases per 100

ml of condensate. Superheat to be less than 25°C when expanded to atmospheric pressure.

Traceability of all the process information of distiller plant

Flexibility: All the information must be easy to process for qualified personnel.

WATER FOR INJECTION AND PURE STEAM GENERATOR

Requirement:-

Water for Injection generation occurs from Purified Water.

Water for injection must be generated by Multi effect Distiller.

The quality of water for injection must be according quality standards and requirements of International monograph (IP, USP, Ph Eu and ISPE).

The equipment, piping and ducting must be designed to perform maintenance and cleaning activities in an easy way.

The equipment must be modular equipped with extension capacity.

Drain ability where necessary (pipe lines as well as equipment). Especially in between all the operations.

100% pipe line drainability for connection lines between generation plant and storage tank. Pipe slope minimum 1% of inclination to the drainable point.

Pipe connections: Orbital welding wherever possible.

Welding must be performed by qualified welder.

Certificates of welding must be able.

Valves and pumps must meet sanitary design.

All piping lines must be identified with product, direction flow and color codes (according DIN 2403 standard).

All hydraulic elements must be labeled with its identifier tag by a resistant plate attached to the element. P&ID must correspond with the real installation and elements must be labeled accordingly.

Sanitation at $\geq 100^\circ\text{C}$ in service mode and at every start.

Self-regulation of heating steam, feed- and cooling water.

EQUIPMENT AND MATERIALS OF DISTILLER

Distiller type three columns, steam heated, with increased capacity of column 1 for simultaneous production of WFI and Pure Steam. Columns designed for average velocity <1 m/sec.
Condenser coils with seamless tubes, sanitary connection and externally electro polished.
Built in 316L or equal according to GMP-lines. Mounted in stainless steel frame.
System to operate without moving parts and low sound level.
System must be with NCGR column.
Columns must be equipped with:
Sight glass
Steam temperature sensor, sanitary
Bottom flange.
Manual bottom drain valve
Internal surfaces of column and heat exchangers should be cleanable (caustic soda) and passivated (Nitric Acid) with $Ra < 0.8\mu m$.
Column heat Exchangers must withstand thermal stresses, high heat exchange efficiency and drainable.
Heating steam system with heating steam pressure reduction valve and ON/OFF valve, pneumatic. pressure sensor for heat steam regulation, pressure gauge and vacuum break.
Condensate returns system with float steam trap.
Cooling water system with cooling water proportional regulating and ON/OFF valve, Temperature sensor to cooling water regulation and pressure control water inlet.
Feed water inlet system with proportional regulating and ON/OFF valve, sanitary valve.
WFI Condensation system with tubes electro polished inside and outside.
A WFI sampling point and user points as per the requirements.

CONTROL SYSTEM OF DISTILLER

Hardware

A PLC must be installed for control and monitoring of the system.
A touch screen should be installed as a human machine communication interface.

Software

In the Software the following aspects must be visualized, as a minimum:

- ✓ Setpoint values.
- ✓ Screen for equipment calibration.
- ✓ Control panel setup screen.
- ✓ Alarm history.
- ✓ Manual operation of any automatic device.
- ✓ Automatic system activation.
- ✓ Powered from washing mode.
- ✓ Access and configuration of user accounts of the system.

Manual start & service from HMI panel
Auto WFI Start/Stop from tank float switches and / or
Auto PS Start/Stop on external signal.
Automatic shifting between the modes: WFI with Pure steam, Only WFI, Only Pure steam.
Operation modes: STOP & SERVICE /Manual .OP /Auto .Selected with the operation switch.

Stop alarms with text indication on the display:

A red alarm lamp will also light up.
HIGH CONDUCT WFI.
HIGH TEMP WFI.
HIGH PRESSURE COL1.
LOW PRESSURE COL1.
HIGH PRESSURE PURE STEAM
LOW PRESSURE PURE STEAM
ANALOG FAIL SIGNALS
OPERATION SWITCH FAIL
HIGH LEVEL COL1.
LOW LEVEL COL1.
LOW PRESSURE CW.
LOW PRESSURE FW.
LOW PRESSURE C.AIR
NO FW PUMP 1.
HIGH TEMP COL 3

The system must record and monitor critical parameters.
The system must generate reports of all processes, especially the sanitation process. | Quality Requirements for Water For Injection and Pure Steam

It must be comply to the quality requirements

The quality of water will be according current USP, EP and IP specifications.

Water for injection USP & amendments thereof
Conductivity $\leq 1.1\mu S/cm$ a $20^{\circ}C$
TOC $\leq 0.5 mg/mL$ (500 ppb)
Endotoxins $\leq 0.25 EU/mL$

Water for injection EP & amendments thereof
Conductivity $\leq 1.1\mu S/cm$ ($20^{\circ}C$)
TOC $\leq 0.5 mg/mL$ (500 ppb)
Endotoxins $\leq 0.25 EU/mL$
Total Aerobic $\leq 10 CFU/100 mL$
Nitrates $< 0.2 ppm$

Pure Steam USP & amendments thereof
Conductivity $\leq 1.1\mu S/cm$ ($20^{\circ}C$)
TOC $\leq 0.5 mg/mL$ (500 ppb)

Endotoxins ≤ 0.25 EU/mL

Supply steam must have a dryness value of no less than 0.95. Obtain no more than 3.5 ml of gases per 100 ml of condensate. Superheat to be less than 25°C when expanded to atmospheric pressure.

Traceability of all the process information of distiller plant

Flexibility: All the information must be easy to process for qualified personnel.

WATER FOR INJECTION AND PURE STEAM GENERATOR

Requirement:-

Water for Injection generation occurs from Purified Water.

Water for injection must be generated by Multi effect Distiller.

The quality of water for injection must be according quality standards and requirements of International monograph (IP, USP, Ph Eu and ISPE).

The equipment, piping and ducting must be designed to perform maintenance and cleaning activities in an easy way.

The equipment must be modular equipped with extension capacity.

Drain ability where necessary (pipe lines as well as equipment). Especially in between all the operations.

100% pipe line drainability for connection lines between generation plant and storage tank. Pipe slope minimum 1% of inclination to the drainable point.

Pipe connections: Orbital welding wherever possible.

Welding must be performed by qualified welder.

Certificates of welding must be able.

Valves and pumps must meet sanitary design.

All piping lines must be identified with product, direction flow and color codes (according DIN 2403 standard).

All hydraulic elements must be labeled with its identifier tag by a resistant plate attached to the element. P&ID must correspond with the real installation and elements must be labeled accordingly.

Sanitation at $\geq 100^\circ\text{C}$ in service mode and at every start.

Self-regulation of heating steam, feed- and cooling water.

EQUIPMENT AND MATERIALS OF DISTILLER

Distiller type three columns, steam heated, with increased capacity of column 1 for simultaneous production of WFI and Pure Steam. Columns designed for average velocity <1 m/sec.

Condenser coils with seamless tubes, sanitary connection and externally electro polished.

Built in 316L or equal according to GMP-lines. Mounted in stainless steel frame.

System to operate without moving parts and low sound level.

System must be with NCGR column.

Columns must be equipped with:

Sight glass

Steam temperature sensor, sanitary

Bottom flange.

Manual bottom drain valve

Internal surfaces of column and heat exchangers should be cleanable (caustic soda) and passivated (Nitric Acid) with $R_a < 0.8\mu\text{m}$.

Column heat Exchangers must withstand thermal stresses, high heat exchange efficiency and drainable.

Heating steam system with heating steam pressure reduction valve and ON/OFF valve, pneumatic. pressure sensor for heat steam regulation, pressure gauge and vacuum break.

Condensate returns system with float steam trap.

Cooling water system with cooling water proportional regulating and ON/OFF valve, Temperature sensor to cooling water regulation and pressure control water inlet.

Feed water inlet system with proportional regulating and ON/OFF valve, sanitary valve.

WFI Condensation system with tubes electro polished inside and outside.

A WFI sampling point and user points as per the requirements.

CONTROL SYSTEM OF DISTILLER

Hardware

A PLC must be installed for control and monitoring of the system.

A touch screen should be installed as a human machine communication interface.

Software

In the Software the following aspects must be visualized, as a minimum:

- ✓ Setpoint values.
- ✓ Screen for equipment calibration.
- ✓ Control panel setup screen.
- ✓ Alarm history.
- ✓ Manual operation of any automatic device.
- ✓ Automatic system activation.
- ✓ Powered from washing mode.
- ✓ Access and configuration of user accounts of the system.

Manual start & service from HMI panel

Auto WFI Start/Stop from tank float switches and / or

Auto PS Start/Stop on external signal.

Automatic shifting between the modes: WFI with Pure steam, Only WFI, Only Pure steam.

Operation modes: STOP & SERVICE /Manual .OP /Auto .Selected with the operation switch.

Stop alarms with text indication on the display:

A red alarm lamp will also light up.

HIGH CONDUCT WFI.

HIGH TEMP WFI.

HIGH PRESSURE COL1.

LOW PRESSURE COL1.

HIGH PRESSURE PURE STEAM

LOW PRESSURE PURE STEAM

ANALOG FAIL SIGNALS
OPERATION SWITCH FAIL
HIGH LEVEL COL1.
LOW LEVEL COL1.
LOW PRESSURE CW.
LOW PRESSURE FW.
LOW PRESSURE C.AIR
NO FW PUMP 1.
HIGH TEMP COL 3

The system must record and monitor critical parameters.

The system must generate reports of all processes, especially the sanitation process.

5.0 | Construction material | Construction material

| As per the GMP/GLP compliance. | As per the GMP/GLP compliance.

6.0 | Utility Requirements | Utility Requirements

| Energy Type | Specification

| Single Phase | 230V +10% / -6% 50Hz

| Three Phase / 4 Wire | 400V +10% / -6% 50Hz

| Control Voltage (Instruments) | 24V DC

| Control Voltage (Power circuits) | 230V +10% / -6% 50Hz

| Pneumatic Supply:-

Compressed Air :- 7 bar

Steam : 6 bar, Industrial steam | Pneumatic Supply:-

Compressed Air :- 7 bar

Steam : 6 bar, Industrial steam

7.0 | Accessories | Accessories

| All accessories required for the normal operation of the equipment. | All accessories required for the normal operation of the equipment.

8.0 | Documents Required | Documents Required

| Complete documentation Package with following documents should be delivered with the system

Installation Qualification (IQ)

Operational Qualification (OQ)

Performance Qualification (PQ)

Operational manual (2 Sets)

Spare part list

The following is a generic list of the documentation to be delivered by the supplier(s) before Design qualification DQ:

- ✓ Electrical wiring/Schematic diagrams.
- ✓ Technical specifications.
- ✓ Installation drawings (layout, P&IDs).

The following is a generic list of the documentation to be delivery by the supplier(s) before Factory Acceptance Test (FAT):

- ✓ Materials and Equipment certifications.
- ✓ Welder's qualification.
- ✓ Passivation reports.
- ✓ Cleaning and flushing reports.
- ✓ Reports of pressure testing to check leaks in equipment or piping.
- ✓ Operation and maintenance manuals
- ✓ Updated backup copies of the software/configuration of programmable control systems (HMIs, PLCs, frequency drives)

The operator and maintenance manuals must be in English. The manuals must be available in paper and in electronic format. | Complete documentation Package with following documents should be delivered with the system

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Operational manual (2 Sets)

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- ✓ Reports of pressure testing to check leaks in equipment or piping.
- ✓ Operation and maintenance manuals
- ✓ Updated backup copies of the software/configuration of programmable control systems (HMIs, PLCs, frequency drives)

The operator and maintenance manuals must be in English. The manuals must be available in paper and in electronic format.

9.0 | Drawings | Drawings

| P&ID, GA drawings, Electric, Isometric drawing | P&ID, GA drawings, Electric, Isometric drawing

10.0 | Training | Training

| The vendor shall impart successful training for operation and maintenance for at least two personnel from user department and engineering /projects department and shall provide evaluation report. | The

vendor shall impart successful training for operation and maintenance for at least two personnel from user department and engineering /projects department and shall provide evaluation report.

11.0 | Abbreviations | Abbreviations

| cGMP: current Good Manufacturing Practices

HW: Hardware

PC: Personal Computer

PW: Purified Water

P&ID: Piping and Diagram Instrumentation

EU: European Union

SW: Software

AISI: American Iron and Steel Institute

USP: United States Pharmacopeia

EP: European Pharmacopeia

IP: International Pharmacopeia (WHO)

WFI: Water for Injection

PS: Pure Steam

ISPE: International Society for Pharmaceutical Engineers

HMI: Human Machine Interface | cGMP: current Good Manufacturing Practices

HW: Hardware

PC: Personal Computer

PW: Purified Water

P&ID: Piping and Diagram Instrumentation

EU: European Union

SW: Software

AISI: American Iron and Steel Institute

USP: United States Pharmacopeia

EP: European Pharmacopeia

IP: International Pharmacopeia (WHO)

WFI: Water for Injection

PS: Pure Steam

ISPE: International Society for Pharmaceutical Engineers

HMI: Human Machine Interface

12.0 | Other Requirements | Other Requirements

| EudraLex - Volume 4 - Good Manufacturing Practice (GMP) guidelines.

Code Federal Regulations Part 21. Current Good Manufacturing Practice for finished Pharmaceuticals.

WHO good manufacturing practices for pharmaceutical products: main principles Annex2. Technical Report

Series 986, 2014 | EudraLex - Volume 4 - Good Manufacturing Practice (GMP) guidelines.

Code Federal Regulations Part 21. Current Good Manufacturing Practice for finished Pharmaceuticals.

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Approvals of the URS:

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Department | Name | Signature | Date

BioBridge Healthcare Solutions Pvt. Ltd., India | BioBridge Healthcare Solutions Pvt. Ltd., India |

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Prepared by: | | |

Checked/Reviewed by: | | |

Client | Client | Client | Client

Reviewed by: | | |

Reviewed by: | | |

Approved by: | | |

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8.0 | Documents Required | 4

9.0 | Drawings | 4

10.0 | Training | 4

11.0 | Abbreviations | 4

12.0 | Other Requirements | 4

1.0 | Objective | Objective

| This URS detail the requirement for WFI generation system 300LPH and PSG system 150KgH Capacity | This

URS detail the requirement for WFI generation system 300LPH and PSG system 150KgH Capacity

2.0 | Scope | Scope

| This URS is applicable for the WFI generation System 300LPH and PSG 150KgH used in the facility | This

URS is applicable for the WFI generation System 300LPH and PSG 150KgH used in the facility

3.0 | Equipment description | Equipment description

| WFI generation system 300LPH Capacity + PSG 150KgH capacity has to be designed for 365 working days.

Working 8 hours in three shifts. It will be placed in the clean and closed utility area. | WFI generation

system 300LPH Capacity + PSG 150KgH capacity has to be designed for 365 working days. Working 8 hours in

three shifts. It will be placed in the clean and closed utility area.

4.0 | Component details | Component details

| Quality Requirements for Water For Injection and Pure Steam

It must be the quality requirements

The quality of water will be according current USP, EP and IP specifications.

Water for injection USP

Conductivity $\leq 1,1\mu\text{S}/\text{cm}$ a 20°C

TOC $\leq 0,5 \text{ mg/mL}$ (500 ppb)

Endotoxins $\leq 0,25 \text{ EU/mL}$

Water for injection EP

Conductivity $\leq 1,1\mu\text{S}/\text{cm}$ (20°C)

TOC $\leq 0,5 \text{ mg/mL}$ (500 ppb)

Endotoxins $\leq 0,25 \text{ EU/mL}$

Total Aerobic $\leq 10 \text{ CFU/100 mL}$

Nitrates $< 0,2 \text{ ppm}$

Pure Steam USP

Conductivity $\leq 1,1\mu\text{S}/\text{cm}$ (20°C)

TOC $\leq 0,5 \text{ mg/mL}$ (500 ppb)

Endotoxins $\leq 0.25 \text{ EU/mL}$

Traceability of all the process information of distiller plant

Flexibility: All the information must be easy to process for qualified personnel.

WATER FOR INJECTION AND PURE STEAM GENERATOR

Requirement:-

Water for Injection generation occurs from Purified Water.

Water for injection must be generated by Multieffect Distiller.

The quality of water for injection must be according quality standards and requirements of International monograph (IP, USP, PhEu).

The equipment, piping and ducting must be designed to perform maintenance and cleaning activities in an easy way.

The equipment must be modular equipped with extension capacity.

Drain ability where necessary (pipe lines as well as equipment). Especially in between all the operations.

100% pipe line drainability for connection lines between generation plant and storage tank. Pipe slope

minimum 1% of inclination to the drainable point.

Pipe connections: Orbital welding wherever possible.

Welding must be performed by qualified welder.

Certificates of welding must be able.

Valves and pumps must meet sanitary design.

All piping lines must be identified with product, direction flow and color codes (according DIN 2403 standard).

All hydraulic elements must be labeled with its identifier tag by a resistant plate attached to the element. P&ID must correspond with the real installation and elements must be labeled accordingly.

Sanitation at 100°C in service mode and at every start.

Self-regulation of heating steam, feed- and cooling water.

EQUIPMENT AND MATERIALS OF DISTILLER

Distiller type three columns, steam heated, with increased capacity of column 1 for simultaneous production of WFI and Pure Steam. Columns designed for average velocity <1 m/sec.
 Condenser coils with seamless tubes, sanitary connection and externally electro polished.
 Built in 316L or equal according to GMP-lines. Mounted in stainless steel frame.
 System to operate without moving parts and low sound level.
 Columns must be equipped with:
 Sight glass
 Steam temperature sensor, sanitary
 Bottom flange.
 Manual bottom drain valve
 Internal surfaces of column and heat exchangers be (caustic soda) and passivated (Nitric Acid) with $Ra < 0,8\mu m$.
 Column heat Exchangers must withstand thermal stresses, high heat exchange efficiency and drainable.
 Heating steam system with heating steam pressure reduction valve and On/Off valve, pneumatic. ressure sensor heat steam regulation, pressure gauge and vacuum break.
 Condensate returns system with float steam trap.
 Cooling water system with cooling water proportional regulating and On/Off valve, Temperature sensor to cooling water regulation and pressure control water inlet.
 Feed water inlet system with proportional regulating and On/Off valve, sanitary.
 WFI Condensation system with tubes electro polished inside and outside.
 A WFI sampling point

CONTROL SYSTEM OF DISTILLER

Hardware

A PLC must be installed for control and monitoring of the system.
 A touch screen should be installed as a human machine communication interface.

Software

In the Software the following aspects must be visualized, as a minimum:

- ✓ Setpoint values.
- ✓ Screen for equipment calibration.
- ✓ Control panel setup screen.
- ✓ Alarm history.
- ✓ Manual operation of any automatic device.
- ✓ Automatic system activation.
- ✓ Powered from washing mode.
- ✓ Access and configuration of user accounts of the system.

Manual start & service from HMI panel
 Auto WFI Start/Stop from tank float switches and / or
 Auto PS Start/Stop on external signal.

Automatic shifting between the modes: WFI with Pure steam, Only WFI, Only Pure steam.

Operation modes: STOP & SERVICE /Manual .OP /Auto .Selected with the operation switch.

Stop alarms with text indication on the display:

A red alarm lamp will also light up.
 HIGH CONDUCT WFI.
 HIGH TEMP WFI.
 HIGH PRESSURE COL1.
 LOW PRESSURE COL1.
 HIGH PRESSURE PURE STEAM
 LOW PRESSURE PURE STEAM
 ANALOG FAIL SIGNALS
 OPERATION SWITCH FAIL
 HIGH LEVEL COL1.
 LOW LEVEL COL1.
 LOW PRESSURE CW.
 LOW PRESSURE FW.
 LOW PRESSURE C.AIR
 NO FW PUMP 1.
 HIGH TEMP COL 3

The system must record and monitor critical parameters.
 The system must generate reports of all processes, especially the sanitation process. | Quality Requirements for Water For Injection and Pure Steam

It must be the quality requirements

The quality of water will be according current USP, EP and IP specifications.

Water for injection USP
 Conductivity $\leq 1,1\mu S/cm$ a $20^{\circ}C$
 TOC $\leq 0,5$ mg/mL (500 ppb)
 Endotoxins $\leq 0,25$ EU/mL

Water for injection EP
 Conductivity $\leq 1,1\mu S/cm$ ($20^{\circ}C$)
 TOC $\leq 0,5$ mg/mL (500 ppb)
 Endotoxins $\leq 0,25$ EU/mL
 Total Aerobic ≤ 10 CFU/100 mL
 Nitrates $< 0,2$ ppm

Pure Steam USP
 Conductivity $\leq 1.1\mu S/cm$ ($20^{\circ}C$)
 TOC $\leq 0,5$ mg/mL (500 ppb)
 Endotoxins ≤ 0.25 EU/mL

Traceability of all the process information of distiller plant

Flexibility: All the information must be easy to process for qualified personnel.

WATER FOR INJECTION AND PURE STEAM GENERATOR

Requirement:-

Water for Injection generation occurs from Purified Water.

Water for injection must be generated by Multieffect Distiller.

The quality of water for injection must be according quality standards and requirements of International monograph (IP, USP, PhEu).

The equipment, piping and ducting must be designed to perform maintenance and cleaning activities in an easy way.

The equipment must be modular equipped with extension capacity.

Drain ability where necessary (pipe lines as well as equipment). Especially in between all the operations.

100% pipe line drainability for connection lines between generation plant and storage tank. Pipe slope minimum 1% of inclination to the drainable point.

Pipe connections: Orbital welding wherever possible.

Welding must be performed by qualified welder.

Certificates of welding must be able.

Valves and pumps must meet sanitary design.

All piping lines must be identified with product, direction flow and color codes (according DIN 2403 standard).

All hydraulic elements must be labeled with its identifier tag by a resistant plate attached to the element. P&ID must correspond with the real installation and elements must be labeled accordingly.

Sanitation at 100°C in service mode and at every start.

Self-regulation of heating steam, feed- and cooling water.

EQUIPMENT AND MATERIALS OF DISTILLER

Distiller type three columns, steam heated, with increased capacity of column 1 for simultaneous production of WFI and Pure Steam. Columns designed for average velocity <1 m/sec.

Condenser coils with seamless tubes, sanitary connection and externally electro polished.

Built in 316L or equal according to GMP-lines. Mounted in stainless steel frame.

System to operate without moving parts and low sound level.

Columns must be equipped with:

Sight glass

Steam temperature sensor, sanitary

Bottom flange.

Manual bottom drain valve

Internal surfaces of column and heat exchangers be (caustic soda) and passivated (Nitric Acid) with Ra < 0,8µm.

Column heat Exchangers must withstand thermal stresses, high heat exchange efficiency and drainable.

Heating steam system with heating steam pressure reduction valve and On/Off valve, pneumatic. ressure sensor heat steam regulation, pressure gauge and vacuum break.

Condensate returns system with float steam trap.

Cooling water system with cooling water proportional regulating and On/Off valve, Temperature sensor to cooling water regulation and pressure control water inlet.

Feed water inlet system with proportional regulating and On/Off valve, sanitary.

WFI Condensation system with tubes electro polished inside and outside.

A WFI sampling point

CONTROL SYSTEM OF DISTILLER

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Auto WFI Start/Stop from tank float switches and / or

Auto PS Start/Stop on external signal.

Automatic shifting between the modes: WFI with Pure steam, Only WFI, Only Pure steam.

Operation modes: STOP & SERVICE /Manual .OP /Auto .Selected with the operation switch.

Stop alarms with text indication on the display:

A red alarm lamp will also light up.

HIGH CONDUCT WFI.

HIGH TEMP WFI.

HIGH PRESSURE COL1.

LOW PRESSURE COL1.

HIGH PRESSURE PURE STEAM

LOW PRESSURE PURE STEAM

ANALOG FAIL SIGNALS

OPERATION SWITCH FAIL

HIGH LEVEL COL1.

LOW LEVEL COL1.

LOW PRESSURE CW.
LOW PRESSURE FW.
LOW PRESSURE C.AIR
NO FW PUMP 1.
HIGH TEMP COL 3

The system must record and monitor critical parameters.

The system must generate reports of all processes, especially the sanitation process.

5.0 | Construction material | Construction material

| As per the GMP/GLP compliance. | As per the GMP/GLP compliance.

6.0 | Utility Requirements | Utility Requirements

| Energy Type | Specification

| Single Phase | 230V +10% / -6% 50Hz

| Three Phase / 4 Wire | 400V +10% / -6% 50Hz

| Control Voltage (Instruments) | 24V DC

| Control Voltage (Power circuits) | 230V +10% / -6% 50Hz

| Pneumatic Supply:-

Compressed Air :- 7 bar

Steam : 6 bar, Industrial steam | Pneumatic Supply:-

Compressed Air :- 7 bar

Steam : 6 bar, Industrial steam

7.0 | Accessories | Accessories

| All accessories required for the normal operation of the equipment. | All accessories required for the normal operation of the equipment.

8.0 | Documents Required | Documents Required

| Complete documentation Package with following documents should be delivered with the system

Installation Qualification (IQ)

Operational Qualification (OQ)

Performance Qualification (PQ)

Operational manual (2 Sets)

Spare part list

The following is a generic list of the documentation to be delivered by the supplier(s) before Design qualification DQ:

- ✓ Electrical wiring/Schematic diagrams.
- ✓ Technical specifications.
- ✓ Installation drawings (layout, P&IDs).

The following is a generic list of the documentation to be delivery by the supplier(s) before Factory Acceptance Test (FAT):

- ✓ Materials and Equipment certifications.
- ✓ Welder's qualification.
- ✓ Passivation reports.
- ✓ Cleaning and flushing reports.
- ✓ Reports of pressure testing to check leaks in equipment or piping.
- ✓ Operation and maintenance manuals
- ✓ Updated backup copies of the software/configuration of programmable control systems (HMIs, PLCs, frequency drives)

The operator and maintenance manuals must be in English. The manuals must be available in paper and in electronic format. | Complete documentation Package with following documents should be delivered with the system

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- ✓ Operation and maintenance manuals
- ✓ Updated backup copies of the software/configuration of programmable control systems (HMIs, PLCs, frequency drives)

The operator and maintenance manuals must be in English. The manuals must be available in paper and in electronic format.

9.0 | Drawings | Drawings

| P&ID, GA drawings, Electric, Isometric drawing | P&ID, GA drawings, Electric, Isometric drawing

10.0 | Training | Training

| The vendor shall impart successful training for operation and maintenance for at least two personnel from user department and engineering /projects department and shall provide evaluation report. | The vendor shall impart successful training for operation and maintenance for at least two personnel from user department and engineering /projects department and shall provide evaluation report.

11.0 | Abbreviations | Abbreviations

| cGMP: current Good Manufacturing Practices

HW: Hardware

PC: Personal Computer

PW: Purified Water
 P&ID: Piping and Diagram Instrumentation
 EU: European Union
 SW: Software
 AISI: American Iron and Steel Institute
 USP: United States Pharmacopeia
 EP: European Pharmacopeia
 IP: International Pharmacopeia (WHO)
 WFI: Water for Injection
 PS: Pure Steam | cGMP: current Good Manufacturing Practices
 HW: Hardware
 PC: Personal Computer
 PW: Purified Water
 P&ID: Piping and Diagram Instrumentation
 EU: European Union
 SW: Software
 AISI: American Iron and Steel Institute
 USP: United States Pharmacopeia
 EP: European Pharmacopeia
 IP: International Pharmacopeia (WHO)
 WFI: Water for Injection
 PS: Pure Steam
 12.0 | Other Requirements | Other Requirements
 | - EudraLex - Volume 4 - Good Manufacturing Practice (GMP) guidelines.
 - Code Federal Regulations Part 21. Current Good Manufacturing Practice for finished Pharmaceuticals.
 - WHO good manufacturing practices for pharmaceutical products: main principles Annex2. Technical Report Series 986, 2014 | - EudraLex - Volume 4 - Good Manufacturing Practice (GMP) guidelines.
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Client | Client | Client | Client

Reviewed by: | | |

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Approved by: | | |

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11.0 | Abbreviations | 9

12.0 | Other Requirements | 10

1.0 | Objective | Objective

| This URS detail the requirement for Purified water storage and distribution system | This URS detail the requirement for Purified water storage and distribution system

2.0 | Scope | Scope

| This URS is applicable for the Purified water storage and distribution System used in the facility |

This URS is applicable for the Purified water storage and distribution System used in the facility

3.0 | Equipment description | Equipment description

| PW storage and distribution system has to be designed for 365 working days. Working 8 hours in three shifts. It will be placed in the clean and closed utility area. | PW storage and distribution system has to be designed for 365 working days. Working 8 hours in three shifts. It will be placed in the clean and closed utility area.

4.0 | Component details | Component details

| Purified water Storage and Distribution System

The tank is designed to have the minimum weld as possible.

The tank have level sensors to stop the entry of product, when there is maximum level, and to stop pumps, for minimum level.

The tank must keep the product stored at the same temperature as the loop.

The temperature of the tank is measured at the outlet.

There will be a venting system to keep the inside at atmospheric pressure. Using filters that prevent the entry of bacteria and nonviable particles into the tank.

Product entry in tank by spray balls. No clip. Clamp connection

All piping lines must be identified with product, direction flow and color codes (according to DIN 2403 standards).

All hydraulic elements must be labeled with its identifier tag attached to the element. P&ID must correspond with the real installation and elements must be labeled accordingly.

Both pipes and accessories will be provided with its quality certificate, identifying each of them individually. As far as possible, mixing different brand's materials will be avoided.

Pipe connections: Orbital welding wherever possible.

Welding must be performed by qualified welder.

Certificates of welding, must be available

Welding inspection: 100% of manual welds and 15% of the orbital welds.

Every time that a new clamp or a new connection is made in the purified water's distribution network, it will be tested at 150% working pressure. During 2 hours of such test , it has to resist hydrostatic pressure without any leaks

Pipe connections must be the minimum: only those which are completely necessary should be made similar.

Those surfaces which are in contact with the purified water system and with the distribution loop must have the slope necessary for a complete drainage of the installation. A maximum slope should be 1/100 or 1%.

Pipes diameters will be made according to the discharge.

Water will be maintained in constant movement to ensure a turbulent flow, with a Reynolds value equal or higher 10,000 in the lowest point. Water speed in event of lack of use should not be higher than 1 m/s

Instrumentation must be suitable for "in situ" sanitization by pure steam.

Loop's design must maintain purified water at room temperature

The Loop of purified water must be able to keep a minimum temperature of 121.1°C for 10 minutes to ensure a successful sanitization.

The temperature and time of sanitization must be adjustable by the personnel

Those loop sections located inside the clean rooms will be installed in a way that avoid creating hidden corners or non-sealed apertures, or surfaces that are difficult to clean.

Build and orientation of the valves will be made to ensure complete drainage of the system

Pick-up sampling valves will be located close to the tank returning and to the impulsion (after the pump).

Pick-up valves location must facilitate easy sampling and the complete drainage of the water.

Mechanical locks will be designed to prevent migration of lubricant fluids and particles towards the pump chamber. If refrigeration is made by water, the quality of this water must be equal to the quality of the water impelled and will be finally removed

The pump must be designed to allow the SIP with Pure Steam at 140°C. Also, a bypass is designed to allow the sterilization without interrupting the loop.

Pumps must be controlled by a frequency shifter in order to guarantee a suitable speed for the distribution loop'

Construction materials which are prone to maintaining and/or calibration (gaskets, valves, instruments, steam traps, etc.) must be easy-access and preferably in the technical area.

Those pipes which go through above the sterile area should be continuous, with no clamps or threads, since it can generate leak.

Pipes paths towards classified zones must be completely sealed in order to avoid particles entry.

All pipes must be treated before use. Installation of purified water generation will be cleaned (caustic

soda) and passivated (Nitric Acid).

Pipe's supports must be installed by dynamic supports in order to tolerate possible dilations.

The loop circuit and the pumping capacity should be designed for a circulation speed greater than 1 m/s and Reynolds number greater than 10,000 (measured at the end of the loop).

During recirculation without demand, the circulation speed could be dropped to about 0.5 to 0.8 m/s in order to avoid excessive increase of loop temperature.

Water must be permanently in recirculation. Turbulent flow in the re-circulating loop is required.

Recirculation return with spray ball at the top of storage tank to ensure that all inner surfaces are wet

Backflows must be avoided.

The following parameters must be monitored:

- Conductivity (return of the loop)

- Pressure and speed / flow of return of the loop

- Temperature of the loop (return of loop, during normal operation and sanitization)

Sampling points at the beginning of the loop and in the return of loop must be installed.

Equipment and Material

Water Tank

Material: Stainless-steel AISI 316L

Roughness: Ra ■ 0,5 µm

Volume: 3500 L.

Design Pressure: -1 to 3 bar

Design Temperature: 145 °C

Vent filter: Present

Pressure level transmitter: Present.

Plant loop connection: Present

(No Jacket / No Cladding required)

Manhole ø 500mm to access a tank.

The breathing of the tank is via 0,2 µm sub-micron hydrophobic vent filter in a stainless steel housing to reduce bio-burden and particles with clamp connection to the tank. Certificate of the filter quality and MOC must be provided by supplier.

The tank must have clamp connections for the installation of the necessary accessories. These accessories are, at a minimum:

- ✓ Rupture Disc

- ✓ Pressure transmitter

- ✓ Venting filter

- ✓ Spray ball / Loop return / PW inlet

Depressurizing mechanism, i.e. safety valve or rupture disc.

Water Storage tank must be constructed in stainless steel AISI 316L with an interior roughness Ra <0.8.

Centrifugal Sanitary Pump AISI 316L polished inside Ra<0,8 µm with Clamp Connection.

Instrumentation in contact with the product will have clamp connections, made of sanitary material (AISI 316L or similar) with Ra<0.8 µm.

Gaskets related to sanitary connections will be made of silicone/platinum

All pipes in contact with purified water must be stainless steel AISI 316L and surfaces finishes must be 180 grit or better, with a Ra<0.8 µm and whenever possible, electro-polished. Supplier must provide the certificates of stainless steel. Specifying Ferrite content,

Impulsion pumps will be made of Stainless steel, sanitary-type and centrifugal pump, designed to allow a complete drainage.

CONTROL SYSTEM STORAGE AND DISTRIBUTION FOR PURIFIED WATER

Hardware:-

A PLC must be installed for control and monitoring of the system.

A touch screen should be installed as a human machine communication interface.

Software:-

In the Software the following aspects must be visualized, as a minimum:

- ✓ Setpoint values.

- ✓ Control panel setup screen.

- ✓ Alarm history.

- ✓ Manual operation of any automatic device.

- ✓ Automatic system activation.

- ✓ Access and configuration of user accounts of the system.

The system must record and monitor critical parameters.

The system must generate reports of all processes, especially the sanitation process. | Purified water

Storage and Distribution System

The tank is designed to have the minimum weld as possible.

The tank have level sensors to stop the entry of product, when there is maximum level, and to stop pumps, for minimum level.

The tank must keep the product stored at the same temperature as the loop.

The temperature of the tank is measured at the outlet.

There will be a venting system to keep the inside at atmospheric pressure. Using filters that prevent the entry of bacteria and nonviable particles into the tank.

Product entry in tank by spray balls. No clip. Clamp connection

All piping lines must be identified with product, direction flow and color codes (according to DIN 2403 standards).

All hydraulic elements must be labeled with its identifier tag attached to the element. P&ID must correspond with the real installation and elements must be labeled accordingly.

Both pipes and accessories will be provided with its quality certificate, identifying each of them individually. As far as possible, mixing different brand's materials will be avoided.

Pipe connections: Orbital welding wherever possible.

Welding must be performed by qualified welder.

Certificates of welding, must be available

Welding inspection: 100% of manual welds and 15% of the orbital welds.

Every time that a new clamp or a new connection is made in the purified water's distribution network, it will be tested at 150% working pressure. During 2 hours of such test , it has to resist hydrostatic pressure without any leaks

Pipe connections must be the minimum: only those which are completely necessary should be made similar.

Those surfaces which are in contact with the purified water system and with the distribution loop must have the slope necessary for a complete drainage of the installation. A maximum slope should be 1/100 or 1%.

Pipes diameters will be made according to the discharge.

Water will be maintained in constant movement to ensure a turbulent flow, with a Reynolds value equal or

higher 10,000 in the lowest point. Water speed in event of lack of use should not be higher than 1 m/s
Instrumentation must be suitable for "in situ" sanitization by pure steam.
Loop's design must maintain purified water at room temperature
The Loop of purified water must be able to keep a minimum temperature of 121.1°C for 10 minutes to ensure a successful sanitization.
The temperature and time of sanitization must be adjustable by the personnel
Those loop sections located inside the clean rooms will be installed in a way that avoid creating hidden corners or non-sealed apertures, or surfaces that are difficult to clean.
Build and orientation of the valves will be made to ensure complete drainage of the system
Pick-up sampling valves will be located close to the tank returning and to the impulsion (after the pump).
Pick-up valves location must facilitate easy sampling and the complete drainage of the water.
Mechanical locks will be designed to prevent migration of lubricant fluids and particles towards the pump chamber. If refrigeration is made by water, the quality of this water must be equal to the quality of the water impelled and will be finally removed
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Equipment and Material

Water Tank

Material: Stainless-steel AISI 316L

Roughness: Ra ■ 0,5 µm

Volume: 3500 L.

Design Pressure: -1 to 3 bar

Design Temperature: 145 °C

Vent filter: Present

Pressure level transmitter: Present.

Plant loop connection: Present

(No Jacket / No Cladding required)

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The system must record and monitor critical parameters.

The system must generate reports of all processes, especially the sanitation process.

5.0 | Construction material | Construction material

| As per the GMP/GLP compliance. | As per the GMP/GLP compliance.

6.0 | Utility Requirements | Utility Requirements

| Energy Type | Specification

| Single Phase | 230V +10% / -6% 50Hz
 | Three Phase / 4 Wire | 400V +10% / -6% 50Hz
 | Control Voltage (Instruments) | 24V DC
 | Control Voltage (Power circuits) | 230V +10% / -6% 50Hz
 | Pneumatic Supply:-
 Compressed Air : 7 bar | Pneumatic Supply:-
 Compressed Air : 7 bar
 7.0 | Accessories | Accessories
 | Printer with all accessories required for the normal operation of the equipment. | Printer with all accessories required for the normal operation of the equipment.
 8.0 | Documents Required | Documents Required
 | Complete documentation Package with following documents should be delivered with the system
 Installation Qualification (IQ)
 Operational Qualification (OQ)
 Performance Qualification (PQ)
 Operational manual (2 Sets)
 Spare part list

The following is a generic list of the documentation to be delivered by the supplier(s) before Design qualification DQ:

- ✓ Technical Memory
- ✓ Electrical wiring/Schematic diagrams.
- ✓ Technical specifications.
- ✓ Installation drawings (layout, P&IDs).

The following is a generic list of the documentation to be delivery by the supplier(s) before Factory Acceptance Test (FAT):

- ✓ Materials and Equipment certifications.
- ✓ Welder's qualification.
- ✓ Passivation reports.
- ✓ Cleaning and flushing reports.
- ✓ Reports of pressure testing to check leaks in equipment or piping.
- ✓ Operation and maintenance manuals
- ✓ Updated backup copies of the software/configuration of programmable control systems (HMIs, PLCs, frequency drives

The operator and maintenance manuals must be in English. The manuals must be available in paper and in electronic format.

Training of personnel must be performed and documented by the supplier before the start up. | Complete documentation Package with following documents should be delivered with the system

Installation Qualification (IQ)
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The operator and maintenance manuals must be in English. The manuals must be available in paper and in electronic format.

Training of personnel must be performed and documented by the supplier before the start up.

9.0 | Drawings | Drawings
 | P&ID, GA drawings, Electric, Isometric drawing | P&ID, GA drawings, Electric, Isometric drawing
 10.0 | Training | Training

| The vendor shall impart successful training for operation and maintenance for at least two personnel from user department and engineering /projects department and shall provide evaluation report. | The vendor shall impart successful training for operation and maintenance for at least two personnel from user department and engineering /projects department and shall provide evaluation report.

11.0 | Abbreviations | Abbreviations
 | cGMP: current Good Manufacturing Practices

HW: Hardware
 PC: Personal Computer
 PW: Purified Water
 P&ID: Piping and Diagram Instrumentation
 EU: European Union
 SW: Software
 AISI: American Iron and Steel Institute
 USP: United States Pharmacopeia
 EP: European Pharmacopeia
 IP: International Pharmacopeia (WHO) | cGMP: current Good Manufacturing Practices
 HW: Hardware
 PC: Personal Computer
 PW: Purified Water
 P&ID: Piping and Diagram Instrumentation

EU: European Union

SW: Software

AISI: American Iron and Steel Institute

USP: United States Pharmacopeia

EP: European Pharmacopeia

IP: International Pharmacopeia (WHO)

12.0 | Other Requirements | Other Requirements

| EudraLex - Volume 4 - Good Manufacturing Practice (GMP) guidelines.

Code Federal Regulations Part 21. Current Good Manufacturing Practice for finished Pharmaceuticals.

WHO good manufacturing practices for pharmaceutical products: main principles Annex2. Technical Report

Series 986, 2014 | EudraLex - Volume 4 - Good Manufacturing Practice (GMP) guidelines.

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Series 986, 2014

Approvals of the URS:

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Department | Name | Signature | Date

BioBridge Healthcare Solutions Pvt. Ltd., India | BioBridge Healthcare Solutions Pvt. Ltd., India |

BioBridge Healthcare Solutions Pvt. Ltd., India | BioBridge Healthcare Solutions Pvt. Ltd., India

Prepared by: | | |

Checked/Reviewed by: | | |

Client | Client | Client | Client

Reviewed by: | | |

Reviewed by: | | |

Approved by: | | |

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6.0 | Utility Requirements | 4

7.0 | Accessories | 4

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11.0 | Abbreviations | 4

12.0 | Other Requirements | 4

1.0 | Objective | Objective

| This URS detail the requirement for Purified water storage and distribution system | This URS detail the requirement for Purified water storage and distribution system

2.0 | Scope | Scope

| This URS is applicable for the Purified water storage and distribution System used in the facility |

This URS is applicable for the Purified water storage and distribution System used in the facility

3.0 | Equipment description | Equipment description

| PW storage and distribution system has to be designed for 365 working days. Working 8 hours in three shifts. It will be placed in the clean and closed utility area. | PW storage and distribution system has to be designed for 365 working days. Working 8 hours in three shifts. It will be placed in the clean and closed utility area.

4.0 | Component details | Component details

| Quality Requirements for Purified water

It must ensure compliance with the quality requirements.

The quality of water will be according current USP, EP and IP specifications.

Traceability of all the process information of purified water distribution .

Flexibility: All the information must be easily to process for qualified personnel.

Purified water Storage and Distribution System

The tank is designed to have the minimum weld as possible.

The tank have level o stop the entry of product, when there is maximum level, and to stop pumps, for minimum level.

The tank must keep the product stored at the same temperature as the loop.

The temperature of the tank is measured at the outlet.

There will be a venting system to keep the inside at atmospheric pressure. Using filters that prevent the entry of bacteria and nonviable particles into the tank.

Product entry in tank by spray balls. No clip. Clamp connection

All piping lines must be identified with product, direction flow and color codes (according to DIN 2403 standards).

All hydraulic elements must be labeled with its identifier tag attached to the element. P&ID must correspond with the real installation and elements must be labeled accordingly.

Both pipes and accessories will be provided with its quality certificate, identifying each of them individually. As far as possible, mixing different brand's materials will be avoided.

Pipe connections: Orbital welding wherever possible.

Welding must be performed by qualified welder.

Certificates of welding, must be

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Every time that a new clamp or a new connection is made in the purified water's distribution network, it will be tested at 150% work pressure. During a 2 hours , it has to resist hydrostatic pressure without any leaks

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Those surfaces which are in contact with the purified water system and with the distribution loop must have the slope necessary for a complete drainage of the installation. A 1%

Pipes diameters will be made according to the discharge.

Water will be maintained in constant movement to ensure a turbulent flow, with a Reynolds value equal or higher 10,000 in the lowest point. Water speed in event of lack of use should not be higher than 1 m/s

Instrumentation must be suitable for "in situ" sanitization by pure steam.

Loop's design must maintain purified water at room temperature

The Loop of purified water must be able to keep a minimum temperature of 121.1°C for 10 minutes to ensure a successful sanitization.

The temperature and time of sanitization must be adjustable by the personnel

Those loop sections located inside the clean rooms will be installed in a way that avoid creating hidden corners or non-sealed apertures, or surfaces that are difficult to clean.

Build and orientation of the valves will be made to ensure complete drainage of the system

Pick-up sampling valves will be located close to the tank returning and to the impulsion (after the pump).

Pick-up valves location must facilitate the complete drainage of the

Mechanical locks will be designed to prevent migration of lubricant fluids and particles towards the pump chamber. If refrigeration is made by water, the quality of this water must be equal to the quality of the water impelled and will be finally removed

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- Temperature of the loop (return of loop, during normal operation and sanitization)

Sampling points at the beginning of the loop and in the return of loop must be installed.

Equipment and Material

Water Tank

Material: Stainless-steel AISI 316L

Roughness: Ra $\leq$ 0,5 μ m

Volume: 2000 L.

Design Pressure: -1 to 3 bar

Design Temperature: 145 °C

Vent filter: Present

Pressure level transmitter: Present.

Plant loop connection: Present

Manhole $\varnothing$ 500mm to access a tank.

The breathing of the tank is via 0,2 μ m sub-micron hydrophobic vent filter in a stainless steel housing to reduce bio-burden and particles with lamp connection to the tank. Certificate of the filter must be provided by supplier.

The tank must have lamp connections for the installation of the necessary accessories. These accessories are, at a minimum:

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5.0 | Construction material | Construction material

| As per the GMP/GLP compliance. | As per the GMP/GLP compliance.

6.0 | Utility Requirements | Utility Requirements

| Energy Type | Specification

| Single Phase | 230V +10% / -6% 50Hz

| Three Phase / 4 Wire | 400V +10% / -6% 50Hz

| Control Voltage (Instruments) | 24V DC

| Control Voltage (Power circuits) | 230V +10% / -6% 50Hz

| Pneumatic Supply:-

Compressed Air : 7 bar | Pneumatic Supply:-

Compressed Air : 7 bar

7.0 | Accessories | Accessories

| Printer with all accessories required for the normal operation of the equipment. | Printer with all accessories required for the normal operation of the equipment.

8.0 | Documents Required | Documents Required

| Complete documentation Package with following documents should be delivered with the system

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Spare part list

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- ✓ Technical Memory
- ✓ Electrical wiring/Schematic diagrams.
- ✓ Technical specifications.
- ✓ Installation drawings (layout, P&IDs).

The following is a generic list of the documentation to be delivery by the supplier(s) before Factory Acceptance Test (FAT):

- ✓ Materials and Equipment certifications.
- ✓ Welder's qualification.
- ✓ Passivation reports.
- ✓ Cleaning and flushing reports.
- ✓ Reports of pressure testing to check leaks in equipment or piping.
- ✓ Operation and maintenance manuals
- ✓ Updated backup copies of the software/configuration of programmable control systems (HMIs, PLCs, frequency drives

The operator and maintenance manuals must be in English. The manuals must be available in paper and in electronic format.

Training of personnel must be performed and documented by the supplier before the start up. | Complete documentation Package with following documents should be delivered with the system

Installation Qualification (IQ)

Operational Qualification (OQ)

Performance Qualification (PQ)

Operational manual (2 Sets)

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9.0 | Drawings | Drawings

| P&ID, GA drawings, Electric, Isometric drawing | P&ID, GA drawings, Electric, Isometric drawing

10.0 | Training | Training

| The vendor shall impart successful training for operation and maintenance for at least two personnel from user department and engineering /projects department and shall provide evaluation report. | The vendor shall impart successful training for operation and maintenance for at least two personnel from user department and engineering /projects department and shall provide evaluation report.

11.0 | Abbreviations | Abbreviations

| cGMP: current Good Manufacturing Practices

HW: Hardware

PC: Personal Computer

PW: Purified Water

P&ID: Piping and Diagram Instrumentation

EU: European Union

SW: Software

AISI: American Iron and Steel Institute

USP: United States Pharmacopeia

EP: European Pharmacopeia

IP: International Pharmacopeia (WHO) | cGMP: current Good Manufacturing Practices

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12.0 | Other Requirements | Other Requirements

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- Code Federal Regulations Part 21. Current Good Manufacturing Practice for finished Pharmaceuticals.

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Approvals of the URS:

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Department | Name | Signature | Date

BioBridge Healthcare Solutions Pvt. Ltd., India | BioBridge Healthcare Solutions Pvt. Ltd., India |

BioBridge Healthcare Solutions Pvt. Ltd., India | BioBridge Healthcare Solutions Pvt. Ltd., India

Prepared by: | | |

Checked/Reviewed by: | | |

Client | Client | Client | Client

Reviewed by: | | |

Reviewed by: | | |

Approved by: | | |

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1.0 | Objective | Objective

| This URS detail the requirement for WFI storage and distribution system | This URS detail the requirement for WFI storage and distribution system

2.0 | Scope | Scope

| This URS is applicable for the WFI storage and distribution System used in the facility | This URS is applicable for the WFI storage and distribution System used in the facility

3.0 | Equipment description | Equipment description

| WFI storage and distribution system has to be designed for 365 working days. Working 8 hours in three shifts. It will be placed in the clean and closed utility area. | WFI storage and distribution system has to be designed for 365 working days. Working 8 hours in three shifts. It will be placed in the clean and closed utility area.

4.0 | Component details | Component details

| WFI Storage and Distribution System

The tank is designed to have the minimum weld as possible.

The tank will have a level transmitter.

The tank must keep the product stored at the same temperature as the loop.

The temperature of the tank is measured at the outlet of it

There will be a venting system to keep the inside at atmospheric pressure. Using filters that prevent the entry of bacteria and nonviable particles into the tank.

All piping lines must be identified with product, direction flow and colour codes (according DIN 2403 standards).

All hydraulic elements must be labelled with its identifier tag attached to the element. P&ID must correspond with the real installation and elements must be labelled accordingly.

Both pipes and accessories will be provided with its quality certificate, identifying each of them individually. As far as possible, mixing different brand's materials will be avoided.

Pipe connections: Orbital welding wherever possible.

Welding must be performed by qualified welder.

Certificates of welding must be able.

Welding inspection: 100% of manual welds and 15% of the orbital welds.

Every time that a new clamp or a new connection is made in the water for injection's distribution network, it will be tested at 150% work ordinary pressure. During a 2 hours period of time, it has to resist hydrostatic pressure without any leaks

Pipe connections must be the minimum: only those which are completely necessary will be made. For details such as building, maintenance or inspection, it will be used accessories of sanitary design "Tri-Clamp" type or similar.

Those surfaces which are in contact with the water for injection system and with the distribution loop must have the slope necessary for a complete drainage of the installation. A minimum 1% slope will be considered.

Pipes diameters will be made according to the discharge brought.

Water will be maintained in constant movement to ensure a turbulent flow, with a Reynolds value equal or higher 10.000 in the lowest point. Water speed in event of lack of use should not be higher than 1 m/s

Instrumentation must be suitable for "in situ" sanitization by pure steam (max. Temp. 140°C).

Loop's design must maintain WFI in a temperature range of 80 – 90°C and must avoid lower or higher temperatures.

The Loop of water for injection must be able to keep a minimum temperature of 123°C for a one-hour period to ensure a successful sanitization.

Those loop sections located inside the clean rooms will be installed in a way that avoid creating hidden corners or non-sealed apertures, or surfaces that are difficult to clean.

Build and orientation of the valves will be made to ensure complete drainage of the system

Pick-up sampling valves will be located after the points of use and close to the tank returning and to the impulsion (after the pump).

Sampling valves location must facilitate the complete drainage of the installation while the valve is closed

Mechanical locks will be designed to prevent migration of lubricant fluids and particles towards de pump chamber. If refrigeration is made by water, the quality of this water must be equal to the quality of the water impelled and will be finally removed

Pumps must be designed to allow the SIP with Pure Steam (sterilization temperature of 140°C, P=2.5 bar).

Also, pumps have to allow sterilizing each one independently without interrupting the loop

Pumps must be controlled by a frequency shifter in order to guarantee a suitable speed for the distribution loop'

Heat exchangers must be internally made of stainless steel and it will have tri-clamp type connections.

Those parts which are in direct contact with purified water must be made of stainless steel AISI 316L and must have a maximum of 0.625µm Ra (25 µ-inch). A mean Ra of 0.5 µm is recommended.

Heat exchangers will be designed in order to prevent purified water contamination and corrosion, and also to allow the complete drainage and cleaning of it.

Heat exchangers must be made of double tube to avoid mixing cooling liquids with the water for injection.

Heat exchangers will be installed in technical areas in order to make its maintenance easier.

Construction materials which are prone to maintaining and/or calibration (gaskets, valves, instruments, steam traps, etc.) must be easy-access and preferably in the technical area.

Those pipes which go through above the sterile area should be continuous, with no clamps or threads, since it can generate leak.

Pipes paths towards classified zones must be completely sealed in order to avoid particles entry.

Those pipes which transport hot fluids will be installed with thermal insulation. The use of insulating with asbestos or chlorides contents is not allowed.

Those pipes and elements which contain low temperature fluids above production or office area will be completely lagged in order to avoid condensates

All pipes must be treated before use. Installation of purified water generation will be cleaned (caustic soda) and passivated (Nitric Acid).

The loop circuit and the pumping capacity should be designed for a circulation speed greater than 1 m/s and Reynolds number greater than 10,000 (measured at the end of the loop).

Water must be permanently in recirculation. Turbulent flow in the re-circulating loop is required.

Recirculation return with spray ball at the top of storage tank to ensure that all inner surfaces are wet

Backflows must be avoided.

The following parameters must be monitored:

- Conductivity (return of the loop)
- Pressure and speed / flow of return of the loop
- Temperature of the loop (return of loop, during normal operation and sanitization)

Sampling points at the beginning of the loop and in the return of loop must be installed.

EQUIPMENT AND MATERIALS

WFI STORAGE TANK

Material: Stainless-steel AISI 316L

Roughness: Ra $\leq$ 0.5 μ m

Volume: 4000 L.

Design Pressure: -1/3 bar

Design Temperature: 140 °C

Vent filter: Present

Pressure level transmitter: Present.

Plant loop connection: Present

(Partial Limpet jacket with steam & electrical heating and full cladding required)

WFI Storage tank must be constructed in stainless steel AISI 316L with an interior roughness Ra $<$ 0.5

Manhole $\varnothing$ 500mm to access a tank.

The breathing of the tank is via 0.2 μ m sub-micron hydrophobic vent filter in a stainless steel housing to reduce bio-burden and particles with clamp connection to the tank. Certificate of the filter quality and MOC must be provided by supplier.

The tank must have several clamp connections for the installation of the necessary accessories.

These accessories are, at a minimum:

✓ Rupture Disc

✓ Pressure transmitter

✓ Venting filter

✓ Spray ball / Loop return / WFI inlet

Depressurizing mechanism, i.e. safety valve or rupture disc.

Loop

All pipes in contact with water for injection must be stainless steel AISI 316L and surfaces finishes must be 180 grit or better, with a Ra $<$ 0.8 μ m and whenever possible, electro-polished. Supplier must provide the certificates of stainless steel. Ferrite content must be specified.

Centrifugal Sanitary Pump AISI 316L polished inside Ra $<$ 0.8 μ m with Clamp Connection.

Exchanger Double Tube in of stainless Steel.

Gaskets related to sanitary connections will be made of Silicone/platinum.

Impulsion pumps will be made of Stainless steel, sanitary-type and centrifugal pump, designed to allow a complete drainage.

CONTROL SYSTEM STORAGE AND DISTRIBUTION FOR WFI

Hardware:-

A PLC must be installed for control and monitoring of the system.

A touch screen should be installed as a human machine communication interface.

Software:-

In the Software the following aspects must be visualized, as a minimum:

✓ Setpoint values.

✓ Control panel setup screen.

✓ Alarm history.

✓ Manual operation of any automatic device.

✓ Automatic system activation.

✓ Access and configuration of user accounts of the system.

The system must record and monitor critical parameters.

The system must generate reports of all processes, especially the sanitation process. | WFI Storage and Distribution System

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Vent filter: Present

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Plant loop connection: Present

(Partial Limpet jacket with steam & electrical heating and full cladding required)

WFI Storage tank must be constructed in stainless steel AISI 316L with an interior roughness Ra <0.5
Manhole ø 500mm to access a tank.

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- ✓ Access and configuration of user accounts of the system.

The system must record and monitor critical parameters.

The system must generate reports of all processes, especially the sanitation process.

5.0 | Construction material | Construction material

| As per the GMP/GLP compliance. | As per the GMP/GLP compliance.

6.0 | Utility Requirements | Utility Requirements

| Energy Type | Specification

| Single Phase | 230V +10% / -6% 50Hz

| Three Phase / 4 Wire | 400V +10% / -6% 50Hz

| Control Voltage (Instruments) | 24V DC

| Control Voltage (Power circuits) | 230V +10% / -6% 50Hz

| |

7.0 | Accessories | Accessories

| All accessories required for the normal operation of the equipment. | All accessories required for the normal operation of the equipment.

8.0 | Documents Required | Documents Required

| Complete documentation Package with following documents should be delivered with the system

Installation Qualification (IQ)

Operational Qualification (OQ)

Performance Qualification (PQ)

Operational manual (2 Sets)

Spare part list

The following is a generic list of the documentation to be delivered by the supplier(s) before Design qualification DQ:

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- ✓ Reports of pressure testing to check leaks in equipment or piping.
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The operator and maintenance manuals must be in English. The manuals must be available in paper and in electronic format. | Complete documentation Package with following documents should be delivered with the system

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9.0 | Drawings | Drawings

| P&ID, GA drawings, Electric, Isometric drawing | P&ID, GA drawings, Electric, Isometric drawing

10.0 | Training | Training

| The vendor shall impart successful training for operation and maintenance for at least two personnel from user department and engineering /projects department and shall provide evaluation report. | The vendor shall impart successful training for operation and maintenance for at least two personnel from user department and engineering /projects department and shall provide evaluation report.

11.0 | Abbreviations | Abbreviations

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PC: Personal Computer

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1.0 | Objective | Objective

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4.0 | Component details | Component details

| Quality Requirements for WFI

It must ensure compliance with the quality requirements.

The quality of water will be according current USP, EP and IP specifications.

Water for Injection USP 40 (current USP)

Conductivity $\leq 1.1 \mu\text{S/cm}$ (20° C)

TOC $\leq 0.5 \text{ mg/mL}$ (500 ppb)

Endotoxins $\leq 0.25 \text{ EU/mL}$

Water for Injection EP 9th (current EP)

Conductivity $\leq 1.1 \mu\text{S/cm}$ (20° C)

TOC $\leq 0.5 \text{ mg/mL}$ (500 ppb)

Endotoxins $\leq 0.25 \text{ EU/mL}$

Nitrates: $\leq 0.2 \text{ ppm}$

Total Aerobic: $\leq 10 \text{ CFU} / 100\text{ml}$

Traceability of all the process information of WFI distribution plant.

Flexibility: All the information must be easily to process for qualified personnel.

WFI Storage and Distribution System

The tank is designed to have the minimum weld as possible.

The tank will have a level transmitter.

The tank must keep the product stored at the same temperature as the loop.

The temperature of the tank is measured at the outlet of it

There will be a venting system to keep the inside at atmospheric pressure. Using filters that prevent the entry of bacteria and nonviable particles into the tank.

Product entry in tank by spray balls. No clip. Clamp connection

All piping lines must be identified with product, direction flow and colour codes (according DIN 2403 standards).

All hydraulic elements must be labelled with its identifier tag attached to the element. P&ID must correspond with the real installation and elements must be labelled accordingly.

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Recirculation return with spray ball at the top of storage tank to ensure that all inner surfaces are wet

Backflows must be avoided.

The following parameters must be monitored:

- Conductivity (return of the loop)
- Pressure and speed / flow of return of the loop
- Temperature of the loop (return of loop, during normal operation and sanitization)

Sampling points at the beginning of the loop and in the return of loop must be installed.

EQUIPMENT AND MATERIALS

WATER TANK

Material: Stainless-steel AISI 316L

Roughness: Ra ■ 0.5 µm

Useful Volume: 1000 L.

Design Pressure: -1/3 bar

Design Temperature: 140 °C

Vent filter: Present

Pressure level transmitter: Present.

Plant loop connection: Present

Water Storage tank must be constructed in stainless steel AISI 316L with an interior roughness Ra <0.5

Manhole ø 500mm to access a tank.

The breathing of the tank is via 0.2 µm sub-micron hydrophobic vent filter in a stainless steel housing to reduce bio-burden and particles with Clamp Connection to the tank. Certificate of the filter must be provided by supplier.

The tank must have several Clamp Connections for the installation of the necessary accessories.

These accessories are, at a minimum:

- ✓ Rupture Disc
- ✓ Pressure transmitter
- ✓ Venting filter
- ✓ Spray ball / Loop return / WFI inlet

Depressurizing mechanism, i.e. safety valve or rupture disc.

Loop

All pipes in contact with water for injection must be stainless steel AISI 316L and surfaces finishes must be 180 grit or better, with a Ra<0.8 µm and whenever possible, electro-polished. Supplier must provide the certificates of stainless steel. Ferrite content must be specified.

Centrifugal Sanitary Pump AISI 316L polished inside Ra<0.8 µm with Clamp Connection.

Exchange Double Tube in of stainless Steel.

Gaskets related to sanitary connections will be made of Silicone/platinum.

Impulsion pumps will be made of Stainless steel, sanitary-type and centrifugal pump, designed to allow a complete drainage.

CONTROL SYSTEM STORAGE AND DISTRIBUTION FOR WFI

Hardware:-

A PLC must be installed for control and monitoring of the system.

A touch screen should be installed as a human machine communication interface.

Software:-

In the Software the following aspects must be visualized, as a minimum:

- ✓ Setpoint values.
- ✓ Control panel setup screen.
- ✓ Alarm history.
- ✓ Manual operation of any automatic device.
- ✓ Automatic system activation.
- ✓ Access and configuration of user accounts of the system.

The system must record and monitor critical parameters.

The system must generate reports of all processes, especially the sanitation process. | Quality Requirements for WFI

It must ensure compliance with the quality requirements.

The quality of water will be according current USP, EP and IP specifications.

Water for Injection USP 40 (current USP)

Conductivity $\leq 1.1 \mu\text{S/cm}$ (20° C)

TOC $\leq 0.5 \text{ mg/mL}$ (500 ppb)

Endotoxins $\leq 0.25 \text{ EU/mL}$

Water for Injection EP 9th (current EP)

Conductivity $\leq 1.1 \mu\text{S/cm}$ (20° C)

TOC $\leq 0.5 \text{ mg/mL}$ (500 ppb)

Endotoxins $\leq 0.25 \text{ EU/mL}$

Nitrates: $\leq 0.2 \text{ ppm}$

Total Aerobic: $\leq 10 \text{ CFU} / 100\text{ml}$

Traceability of all the process information of WFI distribution plant.

Flexibility: All the information must be easily to process for qualified personnel.

WFI Storage and Distribution System

The tank is designed to have the minimum weld as possible.

The tank will have a level transmitter.

The tank must keep the product stored at the same temperature as the loop.

The temperature of the tank is measured at the outlet of it

There will be a venting system to keep the inside at atmospheric pressure. Using filters that prevent the entry of bacteria and nonviable particles into the tank.

Product entry in tank by spray balls. No clip. Clamp connection

All piping lines must be identified with product, direction flow and colour codes (according DIN 2403 standards).

All hydraulic elements must be labelled with its identifier tag attached to the element. P&ID must correspond with the real installation and elements must be labelled accordingly.

Both pipes and accessories will be provided with its quality certificate, identifying each of them individually. As far as possible, mixing different brand's materials will be avoided.

Pipe connections: Orbital welding wherever possible.

Welding must be performed by qualified welder.

Certificates of welding must be able.

Welding inspection: 100% of manual welds and 15% of the orbital welds.

Every time that a new clamp or a new connection is made in the water for injection's distribution network, it will be tested at 150% work ordinary pressure. During a 2 hours period of time, it has to resist hydrostatic pressure without any leaks

Pipe connections must be the minimum: only those which are completely necessary will be made. For details such as building, maintenance or inspection, it will be used accessories of sanitary design "Tri-Clamp" type or similar.

Those surfaces which are in contact with the water for injection system and with the distribution loop must have the slope necessary for a complete drainage of the installation. A minimum 1% slope will be considered.

Pipes diameters will be made according to the discharge brought.

Water will be maintained in constant movement to ensure a turbulent flow, with a Reynolds value equal or higher 10.000 in the lowest point. Water speed in event of lack of use should not be higher than 1 m/s

Instrumentation must be suitable for "in situ" sanitization by pure steam (max. Temp. 140°C).

Loop's design must maintain WFI in a temperature range of 80 – 90°C and must avoid lower or higher temperatures.

The Loop of water for injection must be able to keep a minimum temperature of 123°C for a one-hour period to ensure a successful sanitization.

Those loop sections located inside the clean rooms will be installed in a way that avoid creating hidden corners or non-sealed apertures, or surfaces that are difficult to clean.

Build and orientation of the valves will be made to ensure complete drainage of the system

Pick-up sampling valves will be located after the points of use and close to the tank returning and to the impulsion (after the pump).

Sampling valves location must facilitate the complete drainage of the installation while the valve is closed

Mechanical locks will be designed to prevent migration of lubricant fluids and particles towards de pump chamber. If refrigeration is made by water, the quality of this water must be equal to the quality of the water impelled and will be finally removed

Pumps must be designed to allow the SIP with Pure Steam (sterilization temperature of 140°C, P=2.5 bar). Also, pumps have to allow sterilizing each one independently without interrupting the loop

Pumps must be controlled by a frequency shifter in order to guarantee a suitable speed for the distribution loop'

Heat exchangers must be internally made of stainless steel and it will have tri-clamp type connections.

Those parts which are in direct contact with purified water must be made of stainless steel AISI 316L and must have a maximum of 0.625µm Ra (25 µ-inch). A mean Ra of 0.5 µm is recommended.

Heat exchangers will be designed in order to prevent purified water contamination and corrosion, and also to allow the complete drainage and cleaning of it.

Heat exchangers must be made of double tube to avoid mixing cooling liquids with the water for injection.

Heat exchangers will be installed in technical areas in order to make its maintenance easier.

Construction materials which are prone to maintaining and/or calibration (gaskets, valves, instruments, steam traps, etc.) must be easy-access and preferably in the technical area.

Those pipes which go through above the sterile area should be continuous, with no clamps or threads, since it can generate leak.

Pipes paths towards classified zones must be completely sealed in order to avoid particles entry. Those pipes which transport hot fluids will be installed with thermal insulation. The use of insulating with asbestos or chlorides contents is not allowed. Those pipes and elements which contain low temperature fluids above production or office area will be completely lagged in order to avoid condensates. All pipes must be treated before use. Installation of purified water generation will be cleaned (caustic soda) and passivated (Nitric Acid). The loop circuit and the pumping capacity should be designed for a circulation speed greater than 1 m/s and Reynolds number greater than 10,000 (measured at the end of the loop). Water must be permanently in recirculation. Turbulent flow in the re-circulating loop is required. Recirculation return with spray ball at the top of storage tank to ensure that all inner surfaces are wet. Backflows must be avoided.

The following parameters must be monitored:

- Conductivity (return of the loop)
- Pressure and speed / flow of return of the loop
- Temperature of the loop (return of loop, during normal operation and sanitization)

Sampling points at the beginning of the loop and in the return of loop must be installed.

EQUIPMENT AND MATERIALS

WATER TANK

Material: Stainless-steel AISI 316L
Roughness: Ra ■ 0.5 µm
Useful Volume: 1000 L.
Design Pressure: -1/3 bar
Design Temperature: 140 °C
Vent filter: Present
Pressure level transmitter: Present.
Plant loop connection: Present
Water Storage tank must be constructed in stainless steel AISI 316L with an interior roughness Ra <0.5
Manhole ø 500mm to access a tank.
The breathing of the tank is via 0.2 µm sub-micron hydrophobic vent filter in a stainless steel housing to reduce bio-burden and particles with Clamp Connection to the tank. Certificate of the filter must be provided by supplier.
The tank must have several Clamp Connections for the installation of the necessary accessories.
These accessories are, at a minimum:
✓ Rupture Disc
✓ Pressure transmitter
✓ Venting filter
✓ Spray ball / Loop return / WFI inlet

Depressurizing mechanism, i.e. safety valve or rupture disc.

Loop

All pipes in contact with water for injection must be stainless steel AISI 316L and surfaces finishes must be 180 grit or better, with a Ra<0.8 µm and whenever possible, electro-polished. Supplier must provide the certificates of stainless steel. Ferrite content must be specified.
Centrifugal Sanitary Pump AISI 316L polished inside Ra<0.8 µm with Clamp Connection.
Exchanger Double Tube in of stainless Steel.
Gaskets related to sanitary connections will be made of Silicone/platinum.
Impulsion pumps will be made of Stainless steel, sanitary-type and centrifugal pump, designed to allow a complete drainage.

CONTROL SYSTEM STORAGE AND DISTRIBUTION FOR WFI

Hardware:-

A PLC must be installed for control and monitoring of the system.
A touch screen should be installed as a human machine communication interface.

Software:-

In the Software the following aspects must be visualized, as a minimum:

- ✓ Setpoint values.
- ✓ Control panel setup screen.
- ✓ Alarm history.
- ✓ Manual operation of any automatic device.
- ✓ Automatic system activation.
- ✓ Access and configuration of user accounts of the system.

The system must record and monitor critical parameters.

The system must generate reports of all processes, especially the sanitation process.

5.0 | Construction material | Construction material

| As per the GMP/GLP compliance. | As per the GMP/GLP compliance.

6.0 | Utility Requirements | Utility Requirements

| Energy Type | Specification

| Single Phase | 230V +10% / -6% 50Hz

| Three Phase / 4 Wire | 400V +10% / -6% 50Hz

| Control Voltage (Instruments) | 24V DC

| Control Voltage (Power circuits) | 230V +10% / -6% 50Hz

| |

7.0 | Accessories | Accessories

| All accessories required for the normal operation of the equipment. | All accessories required for the normal operation of the equipment.

8.0 | Documents Required | Documents Required

| Complete documentation Package with following documents should be delivered with the system

Installation Qualification (IQ)

Operational Qualification (OQ)

Performance Qualification (PQ)

Operational manual (2 Sets)

Spare part list

The following is a generic list of the documentation to be delivered by the supplier(s) before Design qualification DQ:

- ✓ Technical Memory
- ✓ Electrical wiring/Schematic diagrams.
- ✓ Technical specifications.
- ✓ Installation drawings (layout, P&IDs, Isometrics).

The following is a generic list of the documentation to be delivery by the supplier(s) before SAT:

- ✓ Materials and Equipment certifications.
- ✓ Welder's qualification.
- ✓ Passivation reports.
- ✓ Cleaning and flushing reports.
- ✓ Reports of pressure testing to check leaks in equipment or piping.
- ✓ Operation and maintenance manuals
- ✓ Updated backup copies of the software/configuration of programmable control systems (HMI's, PLCs, frequency drives)

The operator and maintenance manuals must be in English. The manuals must be available in paper and in electronic format. | Complete documentation Package with following documents should be delivered with the system

Installation Qualification (IQ)
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The operator and maintenance manuals must be in English. The manuals must be available in paper and in electronic format.

9.0 | Drawings | Drawings

| P&ID, GA drawings, Electric, Isometric drawing | P&ID, GA drawings, Electric, Isometric drawing

10.0 | Training | Training

| The vendor shall impart successful training for operation and maintenance for at least two personnel from user department and engineering /projects department and shall provide evaluation report. | The vendor shall impart successful training for operation and maintenance for at least two personnel from user department and engineering /projects department and shall provide evaluation report.

11.0 | Abbreviations | Abbreviations

| cGMP: current Good Manufacturing Practices

HW: Hardware

PC: Personal Computer

WFI: Water For Injection

P&ID: Piping and Diagram Instrumentation

EU: European Union

SW: Software

AISI: American Iron and Steel Institute

USP: United States Pharmacopeia

EP: European Pharmacopeia

IP: International Pharmacopeia (WHO) | cGMP: current Good Manufacturing Practices

HW: Hardware

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EU: European Union

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AISI: American Iron and Steel Institute

USP: United States Pharmacopeia

EP: European Pharmacopeia

IP: International Pharmacopeia (WHO)

12.0 | Other Requirements | Other Requirements

| - EudraLex - Volume 4 - Good Manufacturing Practice (GMP) guidelines.

- Code Federal Regulations Part 21. Current Good Manufacturing Practice for finished Pharmaceuticals.

- WHO good manufacturing practices for pharmaceutical products: main principles Annex2. Technical Report Series 986, 2014 | - EudraLex - Volume 4 - Good Manufacturing Practice (GMP) guidelines.

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Approvals of the URS:

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Department | Name | Signature | Date

BioBridge Healthcare Solutions Pvt. Ltd., India | BioBridge Healthcare Solutions Pvt. Ltd., India |

BioBridge Healthcare Solutions Pvt. Ltd., India | BioBridge Healthcare Solutions Pvt. Ltd., India

Prepared by: | | |

Checked/Reviewed by: | | |

Client | Client | Client | Client

Reviewed by: | | |

Reviewed by: | | |

Approved by: | | |

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8.0 | Documents Required | 4

9.0 | Drawings | 4

10.0 | Training | 4

11.0 | Abbreviations | 4

12.0 | Other Requirements | 4

1.0 | Objective | Objective

| This URS detail the requirement for Pure Steam distribution system | This URS detail the requirement for Pure Steam distribution system

2.0 | Scope | Scope

| This URS is applicable for the Pure Steam distribution System used in the facility | This URS is applicable for the Pure Steam distribution System used in the facility

3.0 | Equipment description | Equipment description

| Pure steam distribution system has to be designed for 365 working days. Working 8 hours in three shifts. It will be placed in the clean and closed utility area. | Pure steam distribution system has to be designed for 365 working days. Working 8 hours in three shifts. It will be placed in the clean and closed utility area.

4.0 | Component details | Component details

| Quality Requirements for Pure Steam

The quality of pure steam will be according current USP, EP and IP specifications.

Pure Steam USP 40 (current USP)

Conductivity $\leq 1.1 \mu\text{S/cm}$ (20° C)

TOC $\leq 0.5 \text{ mg/mL}$ (500 ppb)

Endotoxins $\leq 0.25 \text{ EU/mL}$

Pure Steam Distribution System

The condensate of the pure steam will not be recovered and will be lead towards the drainage greywater network

The finishing of metallic surface which are in direct contact with pure steam must be 180 grit or better, with a Ra lower or equal to $0.625\mu\text{m}$ (25 $\mu\text{-inch}$) and whenever possible, electropolished.

Both pipes and accessories will be provided with its quality certificate, identifying each of them individually. As far as possible, mixing different brand's materials will be avoided.

Pipe connections: Orbital welding wherever possible.

Welding must be performed by qualified welder.

Certificates of welding, must be able.

Welding inspection: 100% of manual welds and 15% of the orbital welds.

Every time that a new clamp or a new connection is made in the steam distribution network, it will be tested at 150% work ordinary pressure. During a 2 hours period of time, it has to resist hydrostatic pressure without any leaks

Pipe connections must be the minimum: only those which are completely necessary will be made.

For details such as building, maintenance or inspection, it will be used accessories of sanitary design "Tri-Clamp" type or similar.

Both distribution lines and points of use must have the slope necessary to allow successfully the drainage of the condensate towards the steam traps. Minimum slope of 1% must be considered. This slope must be according to the steam flow direction

The line will have steam traps before both low points, last lengths of the line, regulatory elements and points of use.

As far as possible, installation will be made with an only collector.

All pipes in contact with steam must be stainless steel AISI 316L and surfaces finishes must be 180 grit or better, with a Ra lower or equal of $0.625\mu\text{m}$ (25 $\mu\text{-inch}$) and whenever possible, electro-polished.

Supplier must provide the certificates of stainless steel. Ferrite content must be specified.

All piping lines must be identified with product, direction flow and color codes (according DIN 2403 standards).

All hydraulic elements must be labeled with its identifier tag attached to the element. P&ID must correspond with the real installation and elements must be labeled accordingly.

Both pipes and accessories will be provided with its quality certificate, identifying each of them individually. As far as possible, mixing different brand's materials will be avoided.

EQUIPMENT AND MATERIALS

Loop

Both pipes and auxiliary metallic materials must be made of stainless steel and those which are in contact with pure steam must be made of stainless steel under carbon, AISI 316L-type

Diaphragm Valve manual actuator, gaskets TEFLON/EPDM SS 316L Clamp Connection. Ra<0.5 must be installed.

Diagram Valve pneumatic actuator simple effect, gaskets TEFLON/EPDM. SS 316L .Clamp Connection. Ra<0.5 must be installed

The following points of use are needed in the loop: production tank 1, production tank 2, Filling Machine, WFI tank, PW tank.

Steam Trap. Clamp Connection must be installed.

CONTROL SYSTEM PURE STEAM DISTRIBUTION

Hardware:-

The control and monitoring of the system is done by the same PLC that rules the PW and WFI loops.

A touch screen should be installed as a human machine communication interface.

Software:-

In the Software the following aspects must be visualized, as a minimum:

- ✓ Setpoint values.
- ✓ Screen for Equipment Calibration.
- ✓ Control panel setup screen.
- ✓ Alarm history.
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- ✓ Manual operation of any automatic device.
- ✓ Automatic system activation.
- ✓ Access and configuration of user accounts of the system.

The system must record and monitor critical parameters.

5.0 | Construction material | Construction material

| As per the GMP/GLP compliance. | As per the GMP/GLP compliance.

6.0 | Utility Requirements | Utility Requirements

| Energy Type | Specification

| Single Phase | 230V +10% / -6% 50Hz

| Three Phase / 4 Wire | 400V +10% / -6% 50Hz

| Control Voltage (Instruments) | 24V DC

| Control Voltage (Power circuits) | 230V +10% / -6% 50Hz

||

7.0 | Accessories | Accessories

| All accessories required for the normal operation of the equipment. | All accessories required for the normal operation of the equipment.

8.0 | Documents Required | Documents Required

| Complete documentation Package with following documents should be delivered with the system

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Operational Qualification (OQ)
Performance Qualification (PQ)
Operational manual (2 Sets)
Spare part list

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 - ✓ Cleaning and flushing reports.
 - ✓ Reports of pressure testing to check leaks in equipment or piping.
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9.0 | Drawings | Drawings

| P&ID, GA drawings, Electric, Isometric drawing | P&ID, GA drawings, Electric, Isometric drawing

10.0 | Training | Training

| The vendor shall impart successful training for operation and maintenance for at least two personnel from user department and engineering /projects department and shall provide evaluation report. | The vendor shall impart successful training for operation and maintenance for at least two personnel from user department and engineering /projects department and shall provide evaluation report.

11.0 | Abbreviations | Abbreviations

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HW: Hardware

PC: Personal Computer

WFI: Water For Injection

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SW: Software

PS: Pure Steam

AISI: American Iron and Steel Institute

USP: United States Pharmacopeia

EP: European Pharmacopeia

IP: International Pharmacopeia (WHO) | cGMP: current Good Manufacturing Practices

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AISI: American Iron and Steel Institute

USP: United States Pharmacopeia

EP: European Pharmacopeia

IP: International Pharmacopeia (WHO)

12.0 | Other Requirements | Other Requirements

| - EudraLex - Volume 4 - Good Manufacturing Practice (GMP) guidelines.

- Code Federal Regulations Part 21. Current Good Manufacturing Practice for finished Pharmaceuticals.

- WHO good manufacturing practices for pharmaceutical products: main principles Annex2. Technical Report Series 986, 2014 | - EudraLex - Volume 4 - Good Manufacturing Practice (GMP) guidelines.
- Code Federal Regulations Part 21. Current Good Manufacturing Practice for finished Pharmaceuticals.
- WHO good manufacturing practices for pharmaceutical products: main principles Annex2. Technical Report Series 986, 2014

Approvals of the URS:

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Department | Name | Signature | Date

BioBridge Healthcare Solutions Pvt. Ltd., India | BioBridge Healthcare Solutions Pvt. Ltd., India |

BioBridge Healthcare Solutions Pvt. Ltd., India | BioBridge Healthcare Solutions Pvt. Ltd., India

Prepared by: | | |

Checked/Reviewed by: | | |

Client | Client | Client | Client

Reviewed by: | | |

Reviewed by: | | |

Approved by: | | |

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9.0 | Drawings | 4

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1.0 | Objective

| This URS detail the requirement analytical weighing balance used to weigh various reagents, chemicals, media that are used for preparation of Solutions in quality control department.

2.0 | Scope

| This URS is applicable mainly for the analytical weighing balance, to be installed in the quality control department.

3.0 | Equipment description

| The equipment should have the parameter, such as LCD/LED display, over load protection, printer, high accuracy of weighing.

4.0 | Component details

| Technical Specification:

LCD or LED

Overload protection

Applications: Formulation, Totaling, Dynamic Weighing, Piece Counting, Density, Percent Weighing, Check Weighing,

Maximum weight Load: 200gm

Minimum load: 10mg

Measuring scale can be changeable from mg to g and vice versa

Readability (g): 0.0001

Repeatability (g): 0.0001

Linearity (g): 0.0002

Settling time: 2.5 s

Weighing platform dimensions: Ø 90 mm

Safety glass /acrylic cover with sliding doors

Regulatory Compliance: ISO, GMP/GLP.

Operating temperature: 15°C to 30°C

Attached Printer.

5.0 | Construction material

| As per the GMP/GLP compliance.

6.0 | Utility Requirements

| As per recommended by manufacturer or supplier.

7.0 | Accessories

| Printer with all accessories required for the normal operation of the equipment.

8.0 | Documents Required

| Complete documentation Package with following documents should be delivered with the system

Installation Qualification (IQ)

Operational Qualification (OQ)

Performance Qualification (PQ)

Operational manual (2 Sets)

Spare part list

9.0 | Drawings

| NA

10.0 | Training

| The vendor shall impart training for operation and maintenance for at least two personnel from user department and engineering /projects department and shall provide evaluation report.

11.0 | Abbreviations

| QC: Quality control

gm: Gram

mg: Milligram

GLP: Good Laboratory Practice

GMP: Good Manufacturing Practice

12.0 | Other Requirements

| NA

Approvals of the URS:

TABLE OF CONTENTS

Department | Name | Signature | Date

BioBridge Healthcare Solutions Pvt. Ltd., India | BioBridge Healthcare Solutions Pvt. Ltd., India |

BioBridge Healthcare Solutions Pvt. Ltd., India | BioBridge Healthcare Solutions Pvt. Ltd., India

Prepared by: | | |

Checked/Reviewed by: | | |

Client | Client | Client | Client

Reviewed by: | | |

Reviewed by: | | |

Approved by: | | |

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1.0 | Objective | Objective

| This URS detail the requirement for peristaltic pump 200ml / min | This URS detail the requirement for peristaltic pump 200ml / min

2.0 | Scope | Scope

| This URS is applicable for the peristaltic pump 200ml/min used in the facility | This URS is applicable for the peristaltic pump 200ml/min used in the facility

3.0 | Equipment description | Equipment description

| Peristaltic pump used for the process activities in production /QC area | Peristaltic pump used for the process activities in production /QC area

4.0 | Component details | Component details

| Technical Specification

Peristaltic pump flow rate must be 200 ml/min

Head type : Spring loaded track – dual roller

It should be compatibility for tube Size : Minimum 1mm to Maximum 4mm (1.5mm wall thickness)

Bi-directional (clockwise & anti-clockwise) compatibility

Speed control with VFD

Start / Stop button | Technical Specification

Peristaltic pump flow rate must be 200 ml/min

Head type : Spring loaded track – dual roller

It should be compatibility for tube Size : Minimum 1mm to Maximum 4mm (1.5mm wall thickness)

Bi-directional (clockwise & anti-clockwise) compatibility

Speed control with VFD

Start / Stop button

5.0 | Construction material | Construction material

| As per the GMP/GLP compliance. | As per the GMP/GLP compliance.

6.0 | Utility Requirements | Utility Requirements

| Energy Type | Specification

| Single Phase | 220V, 50Hz

7.0 | Accessories | Accessories

| All accessories / Spare parts required for the normal operation of the equipment. | All accessories /

Spare parts required for the normal operation of the equipment.

8.0 | Documents Required | Documents Required

| Complete documentation Package with following documents should be delivered with the system

Installation Qualification (IQ)

Operational Qualification (OQ)

Performance Qualification (PQ)

Operational manual

Spare part list

Operating manual

Calibration certificates | Complete documentation Package with following documents should be delivered

with the system

Installation Qualification (IQ)

Operational Qualification (OQ)

Performance Qualification (PQ)

Operational manual

Spare part list

Operating manual

Calibration certificates

9.0 | Drawings | Drawings

| Not Applicable | Not Applicable

10.0 | Training | Training

| The vendor shall impart operation and maintenance training for at least two personnel from user

department and engineering /projects department and shall provide evaluation report. | The vendor shall

impart operation and maintenance training for at least two personnel from user department and engineering

/projects department and shall provide evaluation report.

11.0 | Abbreviations | Abbreviations

| cGMP: current Good Manufacturing Practices

RPM : revolutions per minute | cGMP: current Good Manufacturing Practices

RPM : revolutions per minute

12.0 | Other Requirements | Other Requirements

| Not Applicable | Not Applicable

Approvals of the URS:

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Department | Name | Signature | Date

BioBridge Healthcare Solutions Pvt. Ltd., India | BioBridge Healthcare Solutions Pvt. Ltd., India |

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9.0 | Drawings | 4

10.0 | Training | 4

11.0 | Abbreviations | 4

12.0 | Other Requirements | 4

1.0 | Objective

| This URS detail the requirement of GMP Lab Refrigerator (2-8°C), which will be used to store the solutions, chemicals and samples in the quality control Department.

2.0 | Scope

| This URS is applicable mainly for Lab Refrigerator (2-8°C), which will be installed in quality control department.

3.0 | Equipment description

| Lab Refrigerator (2-8°C) will be used to store the solutions, chemicals and samples. Lab Refrigerator (2-8°C) should have the LED digital display, Temperature controlling Safety control, and alarm system.

4.0 | Component details

| Technical Specification:

| Temperature working range: 2-8°C

Temperature Setting and display: Microprocessor Control with LED Digital display to 0.1°C

Temperature Fluctuation: ±0.5°C

Controls panel: Touch-pad icon-based controls & display

Alarm system: Audible and Visual alarms with battery back-up. With remote alarm contact

Alarm conditions: High & Low alarm, Door open, Power failure

Air Circulation: Forced air circulation fan

Defrost: Automatic cycle defrost

Indicating lamp: Fluorescent lamp

Capacity: 300 liters

Stainless steel outer body with 4 SS(304) Trays: 4

Full length inner glass door magnetic gasket.

Wheels for easy movement.

High quality PT-100 temperature sensor.

Validation port.

On line recording of real time temperature.

5.0 | Construction material

| SS 304 interior and exterior.

All parts to be designed and manufactured in line with the current standards.

6.0 | Utility Requirements

| 240VAC, 50Hz single phase or recommended by manufacturer or supplier.

7.0 | Accessories

| All accessories required for the normal operation of the equipment.

8.0 | Documents Required

| Complete documentation Package with following documents should be delivered with the system

Installation Qualification (IQ)

Operational Qualification (OQ)

Performance Qualification (PQ)

Operational manual (2 Sets)

Spare part list

Calibration certificate of sensor and temperature display.

Certificate of construction material.

9.0 | Drawings

| GA Drawing

10.0 | Training

| The vendor shall impart training for operation and maintenance for at least two personnel from user department and engineering /projects department and shall provide evaluation report.

11.0 | Abbreviations

| QC: Quality Control

GLP: Good Laboratory Practice

URS :User Requirement Specification

12.0 | Other Requirements

| NA

Approvals of the URS:

TABLE OF CONTENTS

Department | Name | Signature | Date

BioBridge Healthcare Solutions Pvt. Ltd., India | BioBridge Healthcare Solutions Pvt. Ltd., India |

BioBridge Healthcare Solutions Pvt. Ltd., India | BioBridge Healthcare Solutions Pvt. Ltd., India

Prepared by: | | |

Checked/Reviewed by: | | |

Client | Client | Client | Client

Reviewed by: | | |

Reviewed by: | | |

Approved by: | | |

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5.0 | Construction material | 4

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7.0 | Accessories | 4

8.0 | Documents Required | 4

9.0 | Drawings | 4

10.0 | Training | 4

11.0 | Abbreviations | 4

12.0 | Other Requirements | 4

1.0 | Objective

| This URS detail the requirement of pH and conductivity meter, which will be used to measure the pH and conductivity of the solution.

2.0 | Scope

| This URS is applicable mainly for the pH and conductivity meter, will be installed in different areas of manufacturing facilities where solution or reagents are prepared.

3.0 | Equipment description

| A pH meter is an analytical instrument that measures the acidity or basicity of a solution, a key parameter for many applications in various industries. It should provide intuitive, easy-to-use meters that accurately measure pH, as well as conductivity.

The pH and conductivity meter should have the multi-data screen that should display as well as print the pH, conductivity, Temperature, electrode status, calibrations, date and time of analysis.

4.0 | Component details

| Technical Specification:

| Parameter: pH and conductivity

Channel: Dual channel

pH measuring range: 1 to 14

pH resolution: 0.001, 0.01 and 0.1

pH accuracy (±): 0.002

pH calibration points: up to 5

Conductivity measurement range: 0.001µS/cm-500mS/cm

Conductivity resolution: 0.001 to 2

Conductivity accuracy (±): 1%

Conductivity calibration points: up to 5

Temperature Range: 0.0 to 100.0°C

Temperature Resolution: 0.1°C

Temperature accuracy: (±) 0.3°C

Temperature Normalization: 15.0 to 30°C

Temperature Compensation: ATC/MTC

Large, easy-to-read display

Printer

5.0 | Construction material

| All parts to be designed and manufactured in line with the current standards.

To resist laboratory working conditions

6.0 | Utility Requirements

| 9 V DC adapter, 1.3 A (100/240 VAC, SMPS) or recommended by manufacturer or supplier.

7.0 | Accessories

| Printer with all accessories required for the normal operation of the equipment.

8.0 | Documents Required

| Complete documentation Package with following documents should be delivered with the system

Installation Qualification (IQ)

Operational Qualification (OQ)

Operational manual (2 Sets)

Spare part list

9.0 | Drawings

| NA

10.0 | Training

| The vendor shall impart training for operation and maintenance for at least two personnel from user department and engineering /projects department and shall provide evaluation report.

11.0 | Abbreviations

| QC: Quality Control

GLP: Good Laboratory Practice

GMP: Good Manufacturing Practices

URS :User Requirement Specification

12.0 | Other Requirements

| NA

Approvals of the URS:

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BioBridge Healthcare Solutions Pvt. Ltd., India | BioBridge Healthcare Solutions Pvt. Ltd., India

Prepared by: | | |

Checked/Reviewed by: | | |

Client | Client | Client | Client

Reviewed by: | | |

Reviewed by: | | |

Approved by: | | |

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5.0 | Construction material | 4

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8.0 | Documents Required | 5

9.0 | Drawings | 6

10.0 | Training | 6

11.0 | Abbreviations | 6

12.0 | Other Requirements | 6

1.0 | Objective | Objective

| This URS detail the requirement for double door production autoclave | This URS detail the requirement for double door production autoclave

2.0 | Scope | Scope

| This URS is applicable for double door production autoclave used in the facility | This URS is applicable for double door production autoclave used in the facility

3.0 | Equipment description | Equipment description

| Double door production autoclave has to be designed for process activities in production area | Double door production autoclave has to be designed for process activities in production area

4.0 | Component details | Component details

| Technical and GMP features

This specification is intended to cover the minimum requirements to be met by the Vendor in the design, manufacture and supply of double door autoclave (dimensional/ functional) specification provided in this document are as a minimum and can be subjected to amendment after discussion with vendor and representative of Client for achieving higher degree of operational ease. | Technical and GMP features This specification is intended to cover the minimum requirements to be met by the Vendor in the design, manufacture and supply of double door autoclave (dimensional/ functional) specification provided in this document are as a minimum and can be subjected to amendment after discussion with vendor and representative of Client for achieving higher degree of operational ease.

| Technical Description:

Chamber Size: 400L

Door should be Sliding vertical and pneumatically operated

5" SIP able Vacuum break filter

Adjustable operating temperature up to 135 °C. Error $\pm$ 0.5 °C with empty chamber.

Programs required

Vacuum Leak Test

Bowie & Dick Process

Standard Cycle

H.P.H.V.

SIP of filter

Automatic F0 value calculation for each cycle.

PLC control with data recording facility for temperature and pressure.

Software should have three level access controls.

Indicator for Steam pressure in chamber and jacket. | Technical Description:

Chamber Size: 400L

Door should be Sliding vertical and pneumatically operated

5" SIP able Vacuum break filter

Adjustable operating temperature up to 135 °C. Error $\pm$ 0.5 °C with empty chamber.

Programs required

Vacuum Leak Test

Bowie & Dick Process

Standard Cycle

H.P.H.V.

SIP of filter

Automatic F0 value calculation for each cycle.

PLC control with data recording facility for temperature and pressure.

Software should have three level access controls.

Indicator for Steam pressure in chamber and jacket.

| Temperature control from the lowest temperature sensor point.

Temperature and pressure monitoring and recording during the sterilization process.

It must be compatible for standard and HPHV cycles

Audio visual alarm to be provided for :-

Cycle failure

Low and High temperature

Sterilization temperature not achieved

Steam generator water low level

Vacuum pump trip

Power failure

Door error

Door interlock

Simultaneous opening of both side doors.

Autoclave must have process-Lock to prevent opening of the door during the process.

Door obstruction safety.

Proper insulation should be provided for of all hot surfaces.

Validation port must be provided

Autoclave must be designed with inbuilt steam generator

Autoclave should be designed and built to comply with all requirements of EN 285, HTM,USFDA, EU, GMP, PIC/S etc. | Temperature control from the lowest temperature sensor point.
 Temperature and pressure monitoring and recording during the sterilization process.
 It must be compatible for standard and HPHV cycles
 Audio visual alarm to be provided for :-
 Cycle failure
 Low and High temperature
 Sterilization temperature not achieved
 Steam generator water low level
 Vacuum pump trip
 Power failure
 Door error
 Door interlock
 Simultaneous opening of both side doors.
 Autoclave must have process-Lock to prevent opening of the door during the process.
 Door obstruction safety.
 Proper insulation should be provided for of all hot surfaces.
 Validation port must be provided
 Autoclave must be designed with inbuilt steam generator

Autoclave should be designed and built to comply with all requirements of EN 285, HTM,USFDA, EU, GMP, PIC/S etc.

5.0 | Construction material | Construction material
 | As per the GMP/GLP compliance. | As per the GMP/GLP compliance.
 6.0 | Utility Requirements | Utility Requirements
 | Energy Type | Specification
 | Three Phase | 400V +10% / -6% 50Hz
 | Compressed Air | 6-7 bar
 7.0 | Accessories
 | All accessories / Spare parts as required for the normal operation of the equipment.
 Trolleys (2 Nos) for loading and unloading of material
 Pressure gauges and safety valve.
 Gasket for Doors
 8.0 | Documents Required
 | Complete documentation Package with following documents should be delivered with the system
 Installation Qualification (IQ)
 Operational Qualification (OQ)
 Performance Qualification (PQ)
 Operational manual (2 Sets)
 Spare part list
 Calibration Certificates

A generic list of the documentation to be delivered by the supplier(s) before Design qualification DQ:

- ✓ Memory and design justifications.
- ✓ Electrical wiring/Schematic diagrams.
- ✓ Technical specifications.
- ✓ Installation drawings (layout, P&IDs).

The following is a generic list of the documentation to be delivery by the supplier(s) before Factory Acceptance Test (FAT):

- ✓ Materials and Equipment certifications.
 - ✓ Welding qualification.
 - ✓ Reports of pressure testing to check leaks in equipment or piping.
 - ✓ Operation and maintenance manuals
 - ✓ Updated backup copies of the software/configuration of programmable control systems (HMIs, PLCs).
- The operator and maintenance manuals must be in English. The manuals must be available on paper and in electronic format.

9.0 | Drawings

| P&ID, GA drawings, Electric, Isometric drawing

10.0 | Training

| The vendor shall impart operation and maintenance training for at least two personnel from user department and engineering /projects department and shall provide evaluation report.

11.0 | Abbreviations

| cGMP: current Good Manufacturing Practices

WFI: Water for Injection

P&ID: Piping and Diagram Instrumentation

GA : General Arrangement

PLC: Programmable Logic Controller

HMI: Human Machine Interface

SCADA: Supervisory control and data acquisition

HTM: Health Technical Memorandum.

12.0 | Other Requirements

| EudraLex - Volume 4 - Good Manufacturing Practice (GMP) guidelines.

Code Federal Regulations Part 21. Current Good Manufacturing Practice for finished Pharmaceuticals.

WHO good manufacturing practices for pharmaceutical products: main principles Annex2. Technical Report Series 986, 2014

Health Technical Memorandum 01-01: Management and decontamination of surgical instruments (medical devices) used in acute care

Approvals of the URS:

TABLE OF CONTENTS

Department | Name | Signature | Date

BioBridge Healthcare Solutions Pvt. Ltd., India | BioBridge Healthcare Solutions Pvt. Ltd., India |

BioBridge Healthcare Solutions Pvt. Ltd., India | BioBridge Healthcare Solutions Pvt. Ltd., India

Prepared by: | Pradeep Yadav | |

Checked/Reviewed by: | | |

Client | Client | Client | Client

Reviewed by: | | |

Reviewed by: | | |

Approved by: | | |

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5.0 | Construction material | 4

6.0 | Utility Requirements | 4

7.0 | Accessories | 4

8.0 | Documents Required | 4-5

9.0 | Drawings | 5

10.0 | Training | 5

11.0 | Abbreviations | 5

12.0 | Other Requirements | 5

1.0 | Objective: | Objective: | Objective: | Objective: | Objective: | Objective:

| The objective of this document is to define and documents the User Requirement of dry heat sterilizer (DHS) | The objective of this document is to define and documents the User Requirement of dry heat sterilizer (DHS) | The objective of this document is to define and documents the User Requirement of dry heat sterilizer (DHS) | The objective of this document is to define and documents the User Requirement of dry heat sterilizer (DHS) | The objective of this document is to define and documents the User Requirement of dry heat sterilizer (DHS) | The objective of this document is to define and documents the User Requirement of dry heat sterilizer (DHS)

2.0 | Scope: | Scope: | Scope: | Scope: | Scope: | Scope:

| This document is applicable for the Dry Heat Sterilizer specified in this document.

The dry heat sterilizer includes advanced features such as programmable controls for temperature and time, safety alarms, and digital displays for monitoring the sterilization cycle. | This document is applicable for the Dry Heat Sterilizer specified in this document.

The dry heat sterilizer includes advanced features such as programmable controls for temperature and time, safety alarms, and digital displays for monitoring the sterilization cycle. | This document is applicable for the Dry Heat Sterilizer specified in this document.

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The dry heat sterilizer includes advanced features such as programmable controls for temperature and time, safety alarms, and digital displays for monitoring the sterilization cycle.

3.0 | Equipment description: | Equipment description: | Equipment description: | Equipment description: | Equipment description:

| Dry Heat Sterilizer, is used for the sterilization of Glass wares, Filter holders and other material compatible with dry heat sterilizer. | Dry Heat Sterilizer, is used for the sterilization of Glass wares, Filter holders and other material compatible with dry heat sterilizer. | Dry Heat Sterilizer, is used for the sterilization of Glass wares, Filter holders and other material compatible with dry heat sterilizer. | Dry Heat Sterilizer, is used for the sterilization of Glass wares, Filter holders and other material compatible with dry heat sterilizer. | Dry Heat Sterilizer, is used for the sterilization of Glass wares, Filter holders and other material compatible with dry heat sterilizer. | Dry Heat Sterilizer, is used for the sterilization of Glass wares, Filter holders and other material compatible with dry heat sterilizer.

4.0 | Component details: | Component details: | Component details: | Component details: | Component details: | Component details:

| The equipment shall be placed in a controlled environment with a temperature range of 50 to 320°C ± 5°C.

Chamber size (Dimension): 605 × 605 × 605 mm.

No. of Shelves: 3.

Dry Heat Sterilizer should have the built in HEPA filters, and should be DOP tested for efficiency of 99.97% down to 0.3-micron particle size.

The control panel should be housed with DTC, ON/OFF Switch, Indicating light, high/ low temp. Switch and is easily accessible. | The equipment shall be placed in a controlled environment with a temperature range of 50 to 320°C ± 5°C.

Chamber size (Dimension): 605 × 605 × 605 mm.

No. of Shelves: 3.

Dry Heat Sterilizer should have the built in HEPA filters, and should be DOP tested for efficiency of 99.97% down to 0.3-micron particle size.

The control panel should be housed with DTC, ON/OFF Switch, Indicating light, high/ low temp. Switch and is easily accessible. | The equipment shall be placed in a controlled environment with a temperature range of 50 to 320°C ± 5°C.

Chamber size (Dimension): 605 × 605 × 605 mm.

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Chamber size (Dimension): 605 × 605 × 605 mm.

No. of Shelves: 3.

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The control panel should be housed with DTC, ON/OFF Switch, Indicating light, high/ low temp. Switch and is easily accessible. | The equipment shall be placed in a controlled environment with a temperature range of 50 to 320°C ± 5°C.

Chamber size (Dimension): 605 × 605 × 605 mm.

No. of Shelves: 3.

Dry Heat Sterilizer should have the built in HEPA filters, and should be DOP tested for efficiency of 99.97% down to 0.3-micron particle size.

The control panel should be housed with DTC, ON/OFF Switch, Indicating light, high/ low temp. Switch and is easily accessible. | The equipment shall be placed in a controlled environment with a temperature range of 50 to 320°C ± 5°C.

Chamber size (Dimension): 605 × 605 × 605 mm.

No. of Shelves: 3.

Dry Heat Sterilizer should have the built in HEPA filters, and should be DOP tested for efficiency of 99.97% down to 0.3-micron particle size.

The control panel should be housed with DTC, ON/OFF Switch, Indicating light, high/ low temp. Switch and is easily accessible.

| Alarms & Safety features | Alarms & Safety features | Alarms & Safety features | Alarms & Safety features | Alarms & Safety features | Alarms & Safety features

|||||
|||||

5.0 | Construction material: | Construction material: | Construction material: | Construction material: | Construction material: | Construction material:

| Operating Temp: 250 degree with error +/-0.5 °C with empty chamber.

Chamber should be made up of Stainless Steel AISI 304 L with appropriate wall thickness.

Finish of inner surface Ra ≤ 0.8µm.

Jacket should be made up of Stainless Steel AISI 304 L with appropriate wall thickness.

Shelves should be Made of SS sheet / Aluminium Sheet, as per material of chamber with perforation.

The gap between the outer & middle wall should be filled with special grade insulation to prevent heat loss.

Heating Element should be of suitable capacity, and provided with air circulation path.

Fresh air should be sucked into the sterilizer through suction devices comprising of HEPA filter.

Validation ports for the placement of Temperature sensors. | Operating Temp: 250 degree with error +/-0.5 °C with empty chamber.

Chamber should be made up of Stainless Steel AISI 304 L with appropriate wall thickness.

Finish of inner surface Ra ≤ 0.8µm.

Jacket should be made up of Stainless Steel AISI 304 L with appropriate wall thickness.

Shelves should be Made of SS sheet / Aluminium Sheet, as per material of chamber with perforation.

The gap between the outer & middle wall should be filled with special grade insulation to prevent heat loss.

Heating Element should be of suitable capacity, and provided with air circulation path.

Fresh air should be sucked into the sterilizer through suction devices comprising of HEPA filter.

Validation ports for the placement of Temperature sensors. | Operating Temp: 250 degree with error +/-0.5 °C with empty chamber.

Chamber should be made up of Stainless Steel AISI 304 L with appropriate wall thickness.

Finish of inner surface Ra ≤ 0.8µm.

Jacket should be made up of Stainless Steel AISI 304 L with appropriate wall thickness.

Shelves should be Made of SS sheet / Aluminium Sheet, as per material of chamber with perforation.

The gap between the outer & middle wall should be filled with special grade insulation to prevent heat loss.

Heating Element should be of suitable capacity, and provided with air circulation path.

Fresh air should be sucked into the sterilizer through suction devices comprising of HEPA filter.

Validation ports for the placement of Temperature sensors. | Operating Temp: 250 degree with error +/-0.5 °C with empty chamber.

Chamber should be made up of Stainless Steel AISI 304 L with appropriate wall thickness.

Finish of inner surface Ra ≤ 0.8µm.

Jacket should be made up of Stainless Steel AISI 304 L with appropriate wall thickness.

Shelves should be Made of SS sheet / Aluminium Sheet, as per material of chamber with perforation.

The gap between the outer & middle wall should be filled with special grade insulation to prevent heat loss.

Heating Element should be of suitable capacity, and provided with air circulation path.

Fresh air should be sucked into the sterilizer through suction devices comprising of HEPA filter.

Validation ports for the placement of Temperature sensors. | Operating Temp: 250 degree with error +/-0.5 °C with empty chamber.

Chamber should be made up of Stainless Steel AISI 304 L with appropriate wall thickness.

Finish of inner surface Ra ≤ 0.8µm.

Jacket should be made up of Stainless Steel AISI 304 L with appropriate wall thickness.

Shelves should be Made of SS sheet / Aluminium Sheet, as per material of chamber with perforation.

The gap between the outer & middle wall should be filled with special grade insulation to prevent heat loss.

Heating Element should be of suitable capacity, and provided with air circulation path.

Fresh air should be sucked into the sterilizer through suction devices comprising of HEPA filter.

Validation ports for the placement of Temperature sensors. | Operating Temp: 250 degree with error +/-0.5 °C with empty chamber.

Chamber should be made up of Stainless Steel AISI 304 L with appropriate wall thickness.

Finish of inner surface Ra ≤ 0.8µm.

Jacket should be made up of Stainless Steel AISI 304 L with appropriate wall thickness.

Shelves should be Made of SS sheet / Aluminium Sheet, as per material of chamber with perforation.

The gap between the outer & middle wall should be filled with special grade insulation to prevent heat loss.

Heating Element should be of suitable capacity, and provided with air circulation path.

Fresh air should be sucked into the sterilizer through suction devices comprising of HEPA filter.

Validation ports for the placement of Temperature sensors.

6.0 | Utility Requirements: | Utility Requirements: | Utility Requirements: | Utility Requirements: |

Utility Requirements: | Utility Requirements:

| Power supply: 230V, 1 phase, 50 Hz, AC supply. | Power supply: 230V, 1 phase, 50 Hz, AC supply. | Power supply: 230V, 1 phase, 50 Hz, AC supply. | Power supply: 230V, 1 phase, 50 Hz, AC supply. | Power supply: 230V, 1 phase, 50 Hz, AC supply. | Power supply: 230V, 1 phase, 50 Hz, AC supply.

7.0 | Accessories: | Accessories: | Accessories: | Accessories: | Accessories: | Accessories:

| If applicable, Strip Chart Recorder, Spare parts with list. | If applicable, Strip Chart Recorder, Spare parts with list. | If applicable, Strip Chart Recorder, Spare parts with list. | If applicable, Strip Chart Recorder, Spare parts with list. | If applicable, Strip Chart Recorder, Spare parts with list. | If applicable, Strip Chart Recorder, Spare parts with list.

8.0 | Documents Required: | Documents Required: | Documents Required: | Documents Required: | Documents Required: | Documents Required:

Required: | Documents Required:

| Complete documentation Package with following documents should be delivered at least one month in advance.

DQ.

IQ.

OQ.

PQ.

Operational manual (2 Sets).

Maintenance manual (2 Sets).

Spare part list.

Calibration certificates. | Complete documentation Package with following documents should be delivered at least one month in advance.

DQ.

IQ.

OQ.

PQ.

Operational manual (2 Sets).

Maintenance manual (2 Sets).

Spare part list.

Calibration certificates. | Complete documentation Package with following documents should be delivered at least one month in advance.

DQ.

IQ.

OQ.

PQ.

Operational manual (2 Sets).

Maintenance manual (2 Sets).

Spare part list.

Calibration certificates. | Complete documentation Package with following documents should be delivered at least one month in advance.

DQ.

IQ.

OQ.

PQ.

Operational manual (2 Sets).

Maintenance manual (2 Sets).

Spare part list.

Calibration certificates. | Complete documentation Package with following documents should be delivered at least one month in advance.

DQ.

IQ.

OQ.

PQ.

Operational manual (2 Sets).

Maintenance manual (2 Sets).

Spare part list.

Calibration certificates. | Complete documentation Package with following documents should be delivered at least one month in advance.

DQ.

IQ.

OQ.

PQ.

Operational manual (2 Sets).

Maintenance manual (2 Sets).

Spare part list.

Calibration certificates.

9.0 | Drawings: | Drawings: | Drawings: | Drawings: | Drawings:

| GA. | GA. | GA. | GA. | GA. | GA.

10.0 | Training: | Training: | Training: | Training: | Training: | Training:

| The vendor shall impart training for operation and maintenance for at least two personnel from user department and engineering /projects department and shall provide evaluation report. | The vendor shall impart training for operation and maintenance for at least two personnel from user department and engineering /projects department and shall provide evaluation report. | The vendor shall impart training for operation and maintenance for at least two personnel from user department and engineering /projects department and shall provide evaluation report. | The vendor shall impart training for operation and maintenance for at least two personnel from user department and engineering /projects department and shall provide evaluation report. | The vendor shall impart training for operation and maintenance for at least two personnel from user department and engineering /projects department and shall provide evaluation report. | The vendor shall impart training for operation and maintenance for at least two personnel from user department and engineering /projects department and shall provide evaluation report. | The vendor shall impart training for operation and maintenance for at least two personnel from user department and engineering /projects department and shall provide evaluation report. | The vendor shall impart training for operation and maintenance for at least two personnel from user department and engineering /projects department and shall provide evaluation report. | The vendor shall impart training for operation and maintenance for at least two personnel from user department and engineering /projects department and shall provide evaluation report.

11.0 | Abbreviations: | Abbreviations: | Abbreviations: | Abbreviations: | Abbreviations: |

Abbreviations:

| URS: User Requirement Specification.

DQ: Design Qualification.

IQ: Installation Qualification.

OQ: Operational Qualification.

PQ: Performance Qualification

DHS: Dry Heat Sterilizer.

GA: General Arrangement.

NA: Not applicable. | URS: User Requirement Specification.

DQ: Design Qualification.

IQ: Installation Qualification.

OQ: Operational Qualification.

PQ: Performance Qualification

DHS: Dry Heat Sterilizer.

GA: General Arrangement.

NA: Not applicable. | URS: User Requirement Specification.

DQ: Design Qualification.

IQ: Installation Qualification.

OQ: Operational Qualification.

PQ: Performance Qualification

DHS: Dry Heat Sterilizer.
GA: General Arrangement.
NA: Not applicable. | URS: User Requirement Specification.
DQ: Design Qualification.
IQ: Installation Qualification.
OQ: Operational Qualification.
PQ: Performance Qualification
DHS: Dry Heat Sterilizer.
GA: General Arrangement.
NA: Not applicable. | URS: User Requirement Specification.
DQ: Design Qualification.
IQ: Installation Qualification.
OQ: Operational Qualification.
PQ: Performance Qualification
DHS: Dry Heat Sterilizer.
GA: General Arrangement.
NA: Not applicable. | URS: User Requirement Specification.
DQ: Design Qualification.
IQ: Installation Qualification.
OQ: Operational Qualification.
PQ: Performance Qualification
DHS: Dry Heat Sterilizer.
GA: General Arrangement.
NA: Not applicable.
12.0 | Other Requirements: | Other Requirements: | Other Requirements: | Other
Requirements: | Other Requirements:
| NA. | NA. | NA. | NA. | NA. | NA.

Approvals of the URS:

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Department | Name | Signature | Date

BioBridge Healthcare Solutions Pvt. Ltd., India | BioBridge Healthcare Solutions Pvt. Ltd., India |

BioBridge Healthcare Solutions Pvt. Ltd., India | BioBridge Healthcare Solutions Pvt. Ltd., India

Prepared by: | | |

Checked/Reviewed by: | | |

Client | Client | Client | Client

Reviewed by: | | |

Reviewed by: | | |

Approved by: | | |

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1.0 | Objective

| This URS detail the requirement of Laminar air flow, which will be used to separate room air from the working space air, and prevent airborne contaminants in the sterile activity.

2.0 | Scope

| This URS is applicable mainly for the Laminar air flow(Vertical) which will be installed in microbiology area of the quality control department.

3.0 | Equipment description

| The Laminar air flow should have the features such as high quality construction, quit operation ,ULPA/HEPA filter, Magnehelic gauge, energy efficient microprocessor control.

4.0 | Component details

| Technical Specification:

| Type: Vertical

Air Flow Velocity: Average of 0.3~0.5m/s

ULPA/HEPA filter: 99.995% efficiency at particle size between 0.1 to 0.3 µm

Pre filter: Polyester fiber, washable

Front Window: Tempered Glass, ,colorless and transparent

UV Lamp: 20W*1 Emission of 253.7 nanometers

Display: LCD display

Noise level: <65dB

Air flow sensor

Microprocessor control

Standards Compliance: Air quality: ISO 14644.1, Class 3, Worldwide, Filtration: EST-RP-CC001.3,

Worldwide, Electrical Safety: IEC61010-1, Worldwide

5.0 | Construction material

| Main body: 18-gauge electro-galvanized steel with white oven-baked epoxy-polyster powder-coated finish.

Work zone: stainless steel, grade 304, with 4B finish.

Side walls: stainless steel, grade 304, with 4B finish.

Sash window: Manual Sliding Sash: 5 mm (0.2") UV Resistant Tempered Glass with counter-balance system and ergonomic handle.

All parts to be designed and manufactured in line with the current standards.

MOC must be resistant to chemical decontamination.

6.0 | Utility Requirements

| Power Rating: 220-240 VAC, 50 / 60 Hz, 1 phase or as recommended by manufacturer or supplier.

7.0 | Accessories

| LED lamp, UV lamp x2, Base stand.

8.0 | Documents Required

| Complete documentation Package with following documents should be delivered with the system

Design Qualification (DQ)

Installation Qualification (IQ)

Operational Qualification (OQ)

Performance Qualification (PQ)

Operational manual (2 Sets)

Spare part list

Calibration certificates for measuring devices

MOC certificate

9.0 | Drawings

| GA Drawing

Electrical Drawing

10.0 | Training

| The vendor shall impart training for operation and maintenance for at least two personnel from user department and engineering /projects department and shall provide evaluation report.

11.0 | Abbreviations

| QC: Quality Control

GLP: Good Laboratory Practice

GMP: Good Manufacturing Practice

URS :User Requirement Specification

LAF: Laminar Air Flow

12.0 | Other Requirements

| NA

Approvals of the URS:

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BioBridge Healthcare Solutions Pvt. Ltd., India | BioBridge Healthcare Solutions Pvt. Ltd., India |

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5.0 | Construction material | 4

6.0 | Utility Requirements | 4

7.0 | Accessories | 4

8.0 | Documents Required | 4-5

9.0 | Drawings | 5

10.0 | Training | 5

11.0 | Abbreviations | 5

12.0 | Other Requirements | 5

1.0 | Objective

| This URS detail the requirement of Biosafety cabinet used to protect personnel, products and the environment from exposure to biohazards and cross contamination during routine microbiological procedures.

2.0 | Scope

| This URS is applicable to Biosafety Cabinet, to be installed in microbiology area of the quality control department.

3.0 | Equipment description

| The Biosafety cabinet should comply with one or more of the following international standards, and the manufacturer shall provide a certified copy of containment and performance tests as specified in the independent international standards for biological safety, electrical and other functional characteristics The BSC should have high quality construction in line with GXP expectations, NSF 49, EN 12469 certified , quit operation, high efficiency HEPA filter, energy efficient microprocessor control.

4.0 | Component details

| Technical Specification:

| Type: Class II, Type A2 Bio Safety Cabinet

Size: 4 feet

HEPA filter: 99.995% efficiency

Front Window: Tempered Glass, anti-UV, colorless and transparent

UV light with set timer, safer and high efficiency.

Display: Color LED screen, show the air velocity, time, pressure and alarm in real time.

Noise level: <65dB

Two anti-splash spare sockets and four service port in the cabinet.

Audio and visual alarm (Abnormal air velocity; Filter replacement; Front window at unsafe height etc.

Inflow Velocity: 0.53±0.025 m/s

Downflow Velocity: 0.33±0.025 m/s

Airflow System: 70% air recirculation, 30% air exhaust.

Certified to NSF 49, EN 12469

Airflow sensor to alert the user if airflow is insufficient for safety of the operator.

Light intensity:>1000 lux

5.0 | Construction material

| Main body: 18-gauge electro-galvanized steel with white oven-baked epoxy-polyester powder-coated finish.

Work zone: stainless steel, grade 304, with 4B finish.

Side walls: stainless steel, grade 304, with 4B finish.

Sash window: Manual Sliding Sash: 5 mm (0.2") UV Resistant Tempered Glass with counter-balance system and ergonomic handle.

All parts to be designed and manufactured in line with the current standards.

To resist laboratory working conditions

6.0 | Utility Requirements

| Gas connection with manual control valve.

Vacuum connection with manual control valve.

Changeable main HEPA filter from cabinet front, without having to dismount motors, power connections and ducts..

7.0 | Accessories

| LED lamp, UV lamp x2, Base stand.

8.0 | Documents Required

| Complete documentation Package with following documents should be delivered with the system

Design Qualification (DQ)

Installation Qualification (IQ)

Operational Qualification (OQ)

Performance Qualification (PQ)

Operational manual (2 Sets)

Spare part list

Calibration certificate

MOC certificate.

9.0 | Drawings

| GA Drawing

Electrical Drawing

10.0 | Training

| The vendor shall impart training for operation and maintenance for at least two personnel from user department and engineering /projects department and shall provide evaluation report.

11.0 | Abbreviations

| QC: Quality Control

GLP: Good Laboratory Practice

GMP: Good Manufacturing Practice

URS :User Requirement Specification

Approvals of the URS:

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7.0 | Accessories | 4

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9.0 | Drawings | 4

10.0 | Training | 4-5

11.0 | Abbreviations | 5

12.0 | Other Requirements | 5

1.0 | Objective

| This URS detail the requirement of Fume Hood (Table Top), which will be used to separate volumes of air, or prevent air contamination due to acid or alkali fumes in lab.

2.0 | Scope

| This URS is applicable for the Fume Hood, will be installed in chemical area of the quality control department.

3.0 | Equipment description

| The fume hood should have the features such as Durable and chemical-resistant construction, High performance centrifugal blower, Energy saving blowers and CE & ISO certified.

4.0 | Component details

| Technical Specification:

| Working Size in feet: 4' x 2' x 2'

Air Flow Velocity: > 0.5m/s

Door / Sash: Door vertical Folding Type

Noise level: <65dB

Glass Window: Two layer toughen glass, 5mm thickness, Motor Control, height adjustable.

Make-up Air: Air needed to replace the air exhausted from a room by the fume hood and other ventilation devices.

Face Velocity: The velocity of the air passing through the work opening of the fume hood measured in the plane of the sash. Measured in feet per minute (fpm), or liters per minute.

Illumination: Fluorescent Lamp.

Exhaust System: Motor Blower Assembly Covered with G.I Sheet.

Optional:

HEPA Filter

Caster wheels

SS 314 / SS 316 MOC

Temperature indicator

Face Velocity Meter

Airflow Indicator

Audio / Visual alarm

5.0 | Construction material

| Main body: CRCA 18G Epoxy Powder Coated.

Work zone: M.S Powder Coated Sheet Covered with P.P Sheet.

Stand: M.S Pipe (18G) Powder coated.

All parts to be designed and manufactured in line with the current standards.

6.0 | Utility Requirements

| Power Rating: 220-240 Volt or recommended by manufacturer or supplier.

7.0 | Accessories

| LED lamp, Base stand.

8.0 | Documents Required

| Complete documentation Package with following documents should be delivered with the system

Installation Qualification (IQ)

Operational Qualification (OQ)

Performance Qualification (PQ)

Operational manual (2 Sets)

Spare part list

Calibration certificate for measuring devices.

MOC certificate

9.0 | Drawings

| GA Drawing

Electrical Drawing

10.0 | Training

| The vendor shall impart training for operation and maintenance for at least two personnel from user department and engineering /projects department and shall provide evaluation report.

11.0 | Abbreviations

| QC: Quality control

GLP: Good Laboratory Practice

GMP: Good Manufacturing Practice

URS: User Requirement Specification

12.0 | Other Requirements

| NA

Approvals of the URS:

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BioBridge Healthcare Solutions Pvt. Ltd., India | BioBridge Healthcare Solutions Pvt. Ltd., India

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Checked/Reviewed by: | | |

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Reviewed by: | | |

Reviewed by: | | |

Approved by: | | |

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7.0 | Accessories | 4

8.0 | Documents Required | 4

9.0 | Drawings | 5

10.0 | Training | 5

11.0 | Abbreviations | 5

12.0 | Other Requirements | 5

1.0 | Objective | Objective

| This URS detail the requirement for Mobile recirculation CIP skid | This URS detail the requirement for Mobile recirculation CIP skid

2.0 | Scope | Scope

| This URS is applicable for Mobile recirculation CIP skid used in the facility | This URS is applicable for Mobile recirculation CIP skid used in the facility

3.0 | Equipment description | Equipment description

| Mobile recirculation CIP skid has to be designed to carry out CIP activities in production area |

Mobile recirculation CIP skid has to be designed to carry out CIP activities in production area

4.0 | Component details | Component details

| Technical and GMP features

This specification is intended to cover the minimum requirements to be met by the Vendor in the design, manufacture and supply of Mobile Recirculation CIP Skid (dimensional/ functional) specifications provided in this document are as a minimum and can be subjected to amendment after discussion with vendor and representative of Client for achieving higher degree of operational ease. | Technical and GMP features This specification is intended to cover the minimum requirements to be met by the Vendor in the design, manufacture and supply of Mobile Recirculation CIP Skid (dimensional/ functional) specifications provided in this document are as a minimum and can be subjected to amendment after discussion with vendor and representative of Client for achieving higher degree of operational ease.

| Technical Specification:

CIP Skid should be compatible to Integrate with tank systems

Outlet of CIP Skid connectable to the spray ball line of process tank / unit to be cleaned

Pump to re-circulate the sanitizing solution through spray ball, transfer lines, sample valve of the connected process unit to be cleaned

System must be designed with flow meter

CIP System program integration with respective vessels in the PLC

Conductivity sensor at inlet line.

Piping must be designed for self-draining with a continuous slope of $\geq 1\%$ and zero dead legs.

Automation and Control

PLC Control System

Operating panel (HMI) mounted on control panel | Technical Specification:

CIP Skid should be compatible to Integrate with tank systems

Outlet of CIP Skid connectable to the spray ball line of process tank / unit to be cleaned

Pump to re-circulate the sanitizing solution through spray ball, transfer lines, sample valve of the connected process unit to be cleaned

System must be designed with flow meter

CIP System program integration with respective vessels in the PLC

Conductivity sensor at inlet line.

Piping must be designed for self-draining with a continuous slope of $\geq 1\%$ and zero dead legs.

Automation and Control

PLC Control System

Operating panel (HMI) mounted on control panel

5.0 | Construction material | Construction material

| As per the GMP/GLP compliance. | As per the GMP/GLP compliance.

6.0 | Utility Requirements | Utility Requirements

| Energy Type | Specification

| Three Phase | 400V +10% / -6% 50Hz

7.0 | Accessories | Accessories

| All accessories / Spare parts as required for the normal operation of the equipment. | All accessories

/ Spare parts as required for the normal operation of the equipment.

8.0 | Documents Required | Documents Required

| Complete documentation Package with following documents should be delivered with the system

Installation Qualification (IQ)

Operational Qualification (OQ)

Performance Qualification (PQ)

Operational manual (2 Sets)

Spare part list

Calibration Certificates

A generic list of the documentation to be delivered by the supplier(s) before Design qualification DQ:

- ✓ Memory and design justifications.
- ✓ Electrical wiring/Schematic diagrams.
- ✓ Technical specifications.
- ✓ Installation drawings (layout, P&IDs).

The following is a generic list of the documentation to be delivery by the supplier(s) before Factory

Acceptance Test (FAT):

- ✓ Materials and Equipment certifications.
- ✓ Welding qualification.
- ✓ Reports of pressure testing to check leaks in equipment or piping. (Boroscopy reports)
- ✓ Operation and maintenance manuals
- ✓ Updated backup copies of the software/configuration of programmable control systems (HMIs, PLCs).

The operator and maintenance manuals must be in English. The manuals must be available on paper and in electronic format. | Complete documentation Package with following documents should be delivered with the system

Installation Qualification (IQ)
Operational Qualification (OQ)
Performance Qualification (PQ)
Operational manual (2 Sets)
Spare part list
Calibration Certificates

A generic list of the documentation to be delivered by the supplier(s) before Design qualification DQ:

- ✓ Memory and design justifications.
- ✓ Electrical wiring/Schematic diagrams.
- ✓ Technical specifications.
- ✓ Installation drawings (layout, P&IDs).

The following is a generic list of the documentation to be delivery by the supplier(s) before Factory Acceptance Test (FAT):

- ✓ Materials and Equipment certifications.
- ✓ Welding qualification.
- ✓ Reports of pressure testing to check leaks in equipment or piping. (Boroscopy reports)
- ✓ Operation and maintenance manuals
- ✓ Updated backup copies of the software/configuration of programmable control systems (HMIs, PLCs).

The operator and maintenance manuals must be in English. The manuals must be available on paper and in electronic format.

9.0 | Drawings | Drawings

| P&ID, GA drawings, Electric, Isometric drawing | P&ID, GA drawings, Electric, Isometric drawing

10.0 | Training | Training

| The vendor shall impart operation and maintenance training for at least two personnel from user department and engineering /projects department and shall provide evaluation report. | The vendor shall impart operation and maintenance training for at least two personnel from user department and engineering /projects department and shall provide evaluation report.

11.0 | Abbreviations | Abbreviations

| cGMP: Current Good Manufacturing Practices

P&ID: Piping and Diagram Instrumentation

GA: General Arrangement

PLC: Programmable Logic Controller

HMI: Human Machine Interface | cGMP: Current Good Manufacturing Practices

P&ID: Piping and Diagram Instrumentation

GA: General Arrangement

PLC: Programmable Logic Controller

HMI: Human Machine Interface

12.0 | Other Requirements | Other Requirements

| EudraLex - Volume 4 - Good Manufacturing Practice (GMP) guidelines.

Code Federal Regulations Part 21. Current Good Manufacturing Practice for finished Pharmaceuticals.

WHO good manufacturing practices for pharmaceutical products: main principles Annex2. Technical Report Series 986, 2014 | EudraLex - Volume 4 - Good Manufacturing Practice (GMP) guidelines.

Code Federal Regulations Part 21. Current Good Manufacturing Practice for finished Pharmaceuticals.

WHO good manufacturing practices for pharmaceutical products: main principles Annex2. Technical Report Series 986, 2014

Approvals of the URS:

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Department | Name | Signature | Date

BioBridge Healthcare Solutions Pvt. Ltd., India | BioBridge Healthcare Solutions Pvt. Ltd., India |

BioBridge Healthcare Solutions Pvt. Ltd., India | BioBridge Healthcare Solutions Pvt. Ltd., India

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Checked/Reviewed by: | | |

Client | Client | Client | Client

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Reviewed by: | | |

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5.0 | Construction material | 3

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7.0 | Accessories | 3

8.0 | Documents Required | 3

9.0 | Drawings | 4

10.0 | Training | 4

11.0 | Abbreviations | 4

12.0 | Other Requirements | 4

1.0 | Objective | Objective

| This URS detail the requirement for magnetic stirrer 5L | This URS detail the requirement for magnetic stirrer 5L

2.0 | Scope | Scope

| This URS is applicable for the magnetic stirrer 5L used in the manufacturing facility | This URS is applicable for the magnetic stirrer 5L used in the manufacturing facility

3.0 | Equipment description | Equipment description

| Magnetic stirrer 5L used for the process activities in production /QC area | Magnetic stirrer 5L used for the process activities in production /QC area

4.0 | Component details | Component details

| Speeds range 150 – 1500 rpm

Stainless Steel Hot Plate for temperatures up to 340°C

LCD for display of set & actual parameters

Monitoring of set & actual temperature / speed

temperature control with embedded sensors

Constant speed even with change of load / Voltage

Maintenance free brushless motor

Scratch proof & chemical resistance body

External temperature probe (PT-1000) for control of fluid temperature

Safety temperature control system

Settable safe temperature limit to avoid overheating

Residual Temperature warning when unit is switched off for operator safety | Speeds range 150 – 1500 rpm

Stainless Steel Hot Plate for temperatures up to 340°C

LCD for display of set & actual parameters

Monitoring of set & actual temperature / speed

temperature control with embedded sensors

Constant speed even with change of load / Voltage

Maintenance free brushless motor

Scratch proof & chemical resistance body

External temperature probe (PT-1000) for control of fluid temperature

Safety temperature control system

Settable safe temperature limit to avoid overheating

Residual Temperature warning when unit is switched off for operator safety

5.0 | Construction material | Construction material

| As per the GMP/GLP compliance. | As per the GMP/GLP compliance.

6.0 | Utility Requirements | Utility Requirements

| Energy Type | Specification

| Single Phase | 220V, 50Hz

7.0 | Accessories | Accessories

| All accessories / Spare parts required for the normal operation of the equipment. | All accessories /

Spare parts required for the normal operation of the equipment.

8.0 | Documents Required | Documents Required

| Complete documentation Package with following documents should be delivered with the system

Installation Qualification (IQ)

Operational Qualification (OQ)

Operational manual (2 Sets)

Spare part list

Operating manual

Calibration certificates | Complete documentation Package with following documents should be delivered with the system

Installation Qualification (IQ)

Operational Qualification (OQ)

Operational manual (2 Sets)

Spare part list

Operating manual

Calibration certificates

9.0 | Drawings | Drawings

| Not Applicable | Not Applicable

10.0 | Training | Training

| The vendor shall impart operation and maintenance training for at least two personnel from user department and engineering /projects department and shall provide evaluation report. | The vendor shall impart operation and maintenance training for at least two personnel from user department and engineering /projects department and shall provide evaluation report.

11.0 | Abbreviations | Abbreviations

| cGMP: current Good Manufacturing Practices

LCD: Liquid Crystal Display

WHO: World Health Organization
RPM : revolutions per minute | cGMP: current Good Manufacturing Practices
LCD: Liquid Crystal Display
WHO: World Health Organization
RPM : revolutions per minute
12.0 | Other Requirements | Other Requirements
| Not Applicable | Not Applicable

Approvals of the URS:

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Client | Client | Client | Client

Reviewed by: | | |

Reviewed by: | | |

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2.0 | Scope | 3

3.0 | Equipment description | 3

4.0 | Component details | 3

5.0 | Construction material | 3

6.0 | Utility Requirements | 3

7.0 | Accessories | 3

8.0 | Documents Required | 3

9.0 | Drawings | 4

10.0 | Training | 4

11.0 | Abbreviations | 4

12.0 | Other Requirements | 4

1.0 | Objective | Objective

| This URS detail the requirement for magnetic stirrer 20L | This URS detail the requirement for magnetic stirrer 20L

2.0 | Scope | Scope

| This URS is applicable for the magnetic stirrer 20L used in the facility | This URS is applicable for the magnetic stirrer 20L used in the facility

3.0 | Equipment description | Equipment description

| Magnetic stirrer 20L used for the process activities in production /QC area | Magnetic stirrer 20L used for the process activities in production /QC area

4.0 | Component details | Component details

| Speeds range 150 – 1500 rpm

LCD for display of set & actual parameters

Monitoring of set & actual speed

Constant speed even with change of load / Voltage

Maintenance free brushless motor

Scratch proof & chemical resistance body

Countdown timer | Speeds range 150 – 1500 rpm

LCD for display of set & actual parameters

Monitoring of set & actual speed

Constant speed even with change of load / Voltage

Maintenance free brushless motor

Scratch proof & chemical resistance body

Countdown timer

5.0 | Construction material | Construction material

| As per the GMP/GLP compliance. | As per the GMP/GLP compliance.

6.0 | Utility Requirements | Utility Requirements

| Energy Type | Specification

| Single Phase | 220V, 50Hz

7.0 | Accessories | Accessories

| All accessories / Spare parts required for the normal operation of the equipment. | All accessories /

Spare parts required for the normal operation of the equipment.

8.0 | Documents Required | Documents Required

| Complete documentation Package with following documents should be delivered with the system

Installation Qualification (IQ)

Operational Qualification (OQ)

Operational manual (2 Sets)

Spare part list

Operating manual

Calibration certificates | Complete documentation Package with following documents should be delivered with the system

Installation Qualification (IQ)

Operational Qualification (OQ)

Operational manual (2 Sets)

Spare part list

Operating manual

Calibration certificates

9.0 | Drawings | Drawings

| Not Applicable | Not Applicable

10.0 | Training | Training

| The vendor shall impart operation and maintenance training for at least two personnel from user department and engineering /projects department and shall provide evaluation report. | The vendor shall impart operation and maintenance training for at least two personnel from user department and engineering /projects department and shall provide evaluation report.

11.0 | Abbreviations | Abbreviations

| cGMP: current Good Manufacturing Practices

LCD: Liquid Crystal display

WHO: World Health Organization

RPM : revolutions per minute | cGMP: current Good Manufacturing Practices

LCD: Liquid Crystal display

WHO: World Health Organization

RPM : revolutions per minute

12.0 | Other Requirements | Other Requirements

| Not Applicable | Not Applicable

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2.0 | Scope | 3

3.0 | Equipment description | 3

4.0 | Component details | 3-6

5.0 | Construction material | 6

6.0 | Utility Requirements | 6

7.0 | Accessories | 6

8.0 | Documents Required | 6

9.0 | Drawings | 6

10.0 | Training | 7

11.0 | Abbreviations | 7

12.0 | Other Requirements | 7

1.0 | Objective

| This URS detail the requirement of Akta pure 150 purification system used to carrying out separation of bio-molecule in QC department.

2.0 | Scope

| This URS is applicable mainly for the Akta pure 150 purification system, will be installed in the quality Control department.

3.0 | Equipment description

| ÄKTA pure is intended for purification of bio-molecules, in particular proteins, for quality control purposes .

4.0 | Component details

| Technical Specification:

Control system UNICORN, version 7.6 or later

System pump

Pump type Piston pump, metering type

Flow rate Setting: 0.01 to 150 mL/min(up to 300 mL/min during column packing)

Flow rate specifications:

Accuracy: $\pm 1.5\%$

Precision: RSD < 0.5% (conditions: 1.0 to150 mL/min, < 3 MPa, 0.8 to 2 cP)

Pressure range :0 to 5 MPa

Viscosity range:0.35 to 5 cP

Sample pump

Pump type :Piston pump, metering type

Flow rate: 0.01 to 150 mL/min

Accuracy: $\pm 2\%$

Precision: RSD < 0.5% (conditions: 1.0 to150 mL/min, < 3 MPa, 0.8 to 3 cP)

Pressure range: 0 to 5 MPa

Viscosity range 0.7 to 10 cP

Mixer

Mixing principle Chamber with a magnetic stirrer

Mixer volume: 1.4 (mounted on system),5 mL (included with system), or 15 mL

Gradient flow rate range : 0.5 to 150 mL/min

Gradient composition and accuracy: $\pm 0.8\%$ (conditions: 5 to 95%)

2 to 150 mL/min, 0.2 to 2 MPa, 0.8 to 2 cP)

Valves

Type Rotary valves

Number of valves :Up to 12

Functions Standard: injection valve

Options: inlet selection, mixer by-pass, loop selection, column selection, pH, outlet etc.

Pressure sensors

Placement of sensors Standard: after system pump

Options: after sample pump, pre-column, post-column

Range: 0 to 20 MPa

Accuracy : ± 0.02 MPa or $\pm 2\%$, whichever is greater

Inlet valves

Inlet A :1, 2, or 7 inlets

Inlet B :1, 2, or 7 inlets

Sample inlet: 0, 1, or 7 inlets

Additional inlets Up to 16

UV monitors

Wavelength range:280 nm

Flow cells Standard: Optical path length 2 mm; illuminated cell, volume 2 μ L

Options: Optical path length 5 mm,illuminated cell, volume 6 ML.Optical path

length 0.4 mm; illuminated, cell volume 0.3 μ L

Resolution :0.001 mAU

Linearity: $\pm 5\%$ within 0–2 AU.

Drift : ≤ 0.2 mAU; AU/h,2 mm cell

Noise :< 0.1 mAU

Lamp operating time:> 10 000 h

Conductivity monitor

Conductivity reading range :0.01 mS/cm to 999.99 mS/cm

Accuracy ± 0.01 mS/cm or $\pm 2\%$, whichever is greater(within 0.3 to 300 mS/cm)

Operating pressure 0 to 5 MPa

Flow cell volume Standard: 22 μ L

Temperature monitor

Range :0°C to 99°C
 Temperature monitor accuracy: $\pm 1.5^{\circ}\text{C}$ within 4°C to 45°C
 pH monitor
 pH reading range :0 to 14
 Accuracy : ± 0.1 pH unit within pH 2 to 12
 Operating pressure: 0 to 0.5 MPa
 Flow cell volume: 129 μL
 Round fraction collector
 Number of F9-R :Up to 2 (two Round fraction collector F9-R or oneF9-R and one Flexible fraction collector, F9-C)
 Number of fractions: Up to 175 per fraction collector
 Vessels :175 (3 mL tubes),95 (8 or 15 mL tubes) and 40 (50 mL tubes)
 Fraction volumes: 0.1 to 50 mL
 Spillage-free mode: Drop Sync
 Delay volume: 473 μL (278 μL with optional, tubing kit, i.d. 0.5 mm)
 Outlet valves
 Number of outlets Valve V9-Os or V9H-Os: 3 (waste, fraction, collector, 1 outlet position)
 Valve V9-O or V9H-O: 12 (waste, fraction, collector, 10 outlet positions)
 Fraction volumes :2 mL to 10 000 mL
 Delay volume(UV – outlet valve): 296 μL (245 μL with optional tubing kit, i.d. 0.5 mm)
 Air sensors
 Number of sensors: Up to 7
 Placement of built-in sensors: Inlet valve V9-IA, Inlet valve V9-IB, Sample inlet valve V9-IS
 Placement of additional sensors: After the injection valve sensors, Before the system pumps, Before the sample pump
 Sensing principle: Ultrasonic
 5.0 | Construction material
 | All parts to be designed and manufactured in line with the current GMP standards. Inner chamber and external finishing powder coated.
 6.0 | Utility Requirements
 | Single Phase, Power Supply: 100–240 V, ~50–60 Hz or recommended by supplier and manufacturer.
 7.0 | Accessories
 | All accessories required for normal operation of equipment.
 8.0 | Documents Required
 | Operational manual (2 Sets)
 Installation Qualification
 Operational Qualification
 Performance Qualification
 Spare part list
 Maintenance manual (2Sets)
 9.0 | Drawings
 | NA
 10.0 | Training
 | The vendor shall impart training for operation and maintenance for at least two personnel from user department and engineering /projects department and shall provide evaluation report.
 11.0 | Abbreviations
 | QC: Quality control
 GLP: Good laboratory Practice
 GMP: Good manufacturing Practice
 URS :User requirement specification
 12.0 | Other Requirements
 | NA

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8.0 | Documents Required | 4

9.0 | Drawings | 4

10.0 | Training | 4

11.0 | Abbreviations | 4

12.0 | Other Requirements | 5

1.0 | Objective | Objective

| This URS detail the requirement for Filter Integrity Testing Machine | This URS detail the requirement for Filter Integrity Testing Machine

2.0 | Scope | Scope

| This URS is applicable for filter integrity testing machine used in the facility | This URS is applicable for filter integrity testing machine used in the facility

3.0 | Equipment description | Equipment description

| Filter integrity testing machine will be used in production / QC area | Filter integrity testing machine will be used in production / QC area

4.0 | Component details | Component details

| Technical Specification:

Operating range:

Maximum inlet pressure: 100 psi

Operating temperature :10-30°C

Relative humidity : 10-80%

Measuring ranges:

Test Pressure : 15-85 psi

Diffusion flow : 0.2-100ml/min.

Bubble point : 5-80 psi

Water intrusion rate :0.03-10ml/min

Accuracy:

Diffusion flow : ± 5% or ±0.1ml whichever is greater

Bubble point : ±0.7 psi

Water intrusion test : ± 5% or ± 0.02ml whichever is greater

Controls:

21CFR part 11 compliance for data security

Data storage

Wi-Fi connectivity for printer

Self-diagnosis for Internal and external Leakage | Technical Specification:

Operating range:

Maximum inlet pressure: 100 psi

Operating temperature :10-30°C

Relative humidity : 10-80%

Measuring ranges:

Test Pressure : 15-85 psi

Diffusion flow : 0.2-100ml/min.

Bubble point : 5-80 psi

Water intrusion rate :0.03-10ml/min

Accuracy:

Diffusion flow : ± 5% or ±0.1ml whichever is greater

Bubble point : ±0.7 psi

Water intrusion test : ± 5% or ± 0.02ml whichever is greater

Controls:

21CFR part 11 compliance for data security

Data storage

Wi-Fi connectivity for printer

Self-diagnosis for Internal and external Leakage

5.0 | Construction material | Construction material

| As per the GMP/GLP compliance. | As per the GMP/GLP compliance.

6.0 | Utility Requirements | Utility Requirements

| Energy Type | Specification

| Single Phase | 220V, 50Hz

| Compressed Air | 6-7 Bar

7.0 | Accessories | Accessories

| All accessories / Spare parts as required for the normal operation of the equipment.

Outlet tubing

SS connectors for Inlet / Outlet port

Wi-fi router

Laser printer with Wi-Fi connectivity

Power cord | All accessories / Spare parts as required for the normal operation of the equipment.

Outlet tubing

SS connectors for Inlet / Outlet port

Wi-fi router

Laser printer with Wi-Fi connectivity

Power cord

8.0 | Documents Required | Documents Required

| Complete documentation Package with following documents should be delivered with the system

Installation Qualification (IQ)

Operational Qualification (OQ)

Performance Qualification (PQ)

Operational manual

Calibration Certificates | Complete documentation Package with following documents should be delivered with the system

Installation Qualification (IQ)

Operational Qualification (OQ)

Performance Qualification (PQ)

Operational manual

Calibration Certificates

9.0 | Drawings | Drawings

| Not Applicable | Not Applicable

10.0 | Training | Training

| The vendor shall impart operation and maintenance training for at least two personnel from user department and engineering /projects department and shall provide evaluation report. | The vendor shall impart operation and maintenance training for at least two personnel from user department and engineering /projects department and shall provide evaluation report.

11.0 | Abbreviations | Abbreviations

| cGMP: current Good Manufacturing Practices

SS : Stainless steel

CFR: Code of Federal Regulations

Psi : Pounds per square inch

QC: Quality control | cGMP: current Good Manufacturing Practices

SS : Stainless steel

CFR: Code of Federal Regulations

Psi : Pounds per square inch

QC: Quality control

12.0 | Other Requirements | Other Requirements

| EudraLex - Volume 4 - Good Manufacturing Practice (GMP) guidelines.

Code Federal Regulations Part 21. Current Good Manufacturing Practice for finished Pharmaceuticals.

WHO good manufacturing practices for pharmaceutical products: main principles Annex2. Technical Report

Series 986, 2014 | EudraLex - Volume 4 - Good Manufacturing Practice (GMP) guidelines.

Code Federal Regulations Part 21. Current Good Manufacturing Practice for finished Pharmaceuticals.

WHO good manufacturing practices for pharmaceutical products: main principles Annex2. Technical Report

Series 986, 2014

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8.0 | Documents Required | 4

9.0 | Drawings | 4

10.0 | Training | 4-5

11.0 | Abbreviations | 5

12.0 | Other Requirements | 5

1.0 | Objective

| This URS detail the requirement of UV-VIS spectrophotometer, which will be used to measure the optical density of samples .

2.0 | Scope

| This URS is applicable for the UV-Vis spectrophotometer, will be installed in quality control department.

3.0 | Equipment description

| Ultraviolet-visible (UV-Vis) spectrophotometers use a light source to illuminate a sample with light across the UV to the visible wavelength range. The instruments then measure the light absorbed, transmitted, or reflected by the sample at each wavelength. should have the features high resolution, low stray light, high reproducibility, an ultra-fast scan function and easy-to-use interface on a color touch-screen display

4.0 | Component details

| Technical Specification:

| Wavelength range: 190-1100nm

Spectral bandwidth: 1nm (190-1100nm)

Wavelength repeatability: (± 0.025)

Lamp interchange wavelength: Automatic interchange linked to wavelength

Photometric system: Double beam optics

Light source: halogen lamp and deuterium lamp Built-in light source auto position adjustment.

Detector: Silicon photodiode

Environmental requirements: Temperature: 15°C to 35°C, Humidity: 35% to 80%

Output device: USB memory (optional) Extended memory (optional) Data files saved in text format.

PC compatibility. (optional)

Display: color touch screen

Photometric range: Absorbance: 0.3 to 3.0 Abs Transmittance: 0% to 200%

Photometric accuracy: $\leq 0.3\%$ T

Wavelength accuracy: ± 0.3 nm

Stray light: $\leq 0.05\%$ T

Scanning speed: 100 - 3000 nm/min.

UV-Vis Analyst software for more powerful data processing and expanded data collecting.

5.0 | Construction material

| All parts to be designed and manufactured in line with the current standards.

To resist laboratory working conditions

6.0 | Utility Requirements

| 110/220 V, 50/60 Hz or recommended by manufacturer or supplier.

7.0 | Accessories

| four optical glass cells, two quartz cells, dust cover, USB cable, UV-VIS Analyst software for PC control, cell holders for sample and reference, power cables and all accessories required for the normal operation of the equipment.

8.0 | Documents Required

| Complete documentation Package with following documents should be delivered with the system

Installation Qualification (IQ)

Operational Qualification (OQ)

Performance Qualification (PQ)

Operational manual (2 Sets)

Spare part list

Calibration certificates

9.0 | Drawings

| NA

10.0 | Training

| The vendor shall impart training for operation and maintenance for at least two personnel from user department and engineering /projects department and shall provide evaluation report.

11.0 | Abbreviations

| QC: Quality Control

GLP: Good Laboratory Practice

GMP: Good Manufacturing Practice

URS :User Requirement Specification

12.0 | Other Requirements

| NA

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6.0 | Utility Requirements | 8

7.0 | Accessories | 8

8.0 | Documents Required | 8

9.0 | Drawings | 9

10.0 | Training | 10

11.0 | Abbreviations | 10

12.0 | Other Requirements | 10

1.0 | Objective | Objective

| This URS detail the requirement for Powder containment station. | This URS detail the requirement for Powder containment station.

2.0 | Scope | Scope

| This URS is applicable for powder containment station used in the facility | This URS is applicable for powder containment station used in the facility

3.0 | Equipment description | Equipment description

| Dispensing booth required to carry out the weighing and dispensing activities in production area |

Dispensing booth required to carry out the weighing and dispensing activities in production area

4.0 | Component details | Component details

| Technical and GMP features

This specification is intended to cover the minimum requirements to be met by the Vendor in the design, manufacture and supply of dispensing booth (dimensional/ functional) specifications provided in this document are as a minimum and can be subjected to amendment after discussion with vendor and representative of Client for achieving higher degree of operational ease. | Technical and GMP features This specification is intended to cover the minimum requirements to be met by the Vendor in the design, manufacture and supply of dispensing booth (dimensional/ functional) specifications provided in this document are as a minimum and can be subjected to amendment after discussion with vendor and representative of Client for achieving higher degree of operational ease.

| Technical Specification

Automatic changeable frequency system to monitors the air velocity to ensure its stability in the work area.

Differential pressure gauge to real-time monitor the filters.

Uniform air flow design to protects the operators and prevent environmental pollution and cross contamination of products.

Alarm system for operation failure / fault alarms

Fitted with UV Lamp and hour meter

Curtains required

Port for validation activities

Fully automatic recording of parameters

Dimensions (W x D x H) : As per requirement

Clean Level : ISO 5 (Class 100), Class A

HEPA Filters : 99.99% efficiency at 0.3µm

Air Volume : Supply air Volume : ≤ 7500m³/h

Exhaust air volume : ≤ 2250m³/h ; Adjustable

Air Velocity : 0.35 – 0.65m/s, adjustable

Noise : ≤ 75db

Illumination : ≥ 800 lux | Technical Specification

Automatic changeable frequency system to monitors the air velocity to ensure its stability in the work area.

Differential pressure gauge to real-time monitor the filters.

Uniform air flow design to protects the operators and prevent environmental pollution and cross contamination of products.

Alarm system for operation failure / fault alarms

Fitted with UV Lamp and hour meter

Curtains required

Port for validation activities

Fully automatic recording of parameters

Dimensions (W x D x H) : As per requirement

Clean Level : ISO 5 (Class 100), Class A

HEPA Filters : 99.99% efficiency at 0.3µm

Air Volume : Supply air Volume : ≤ 7500m³/h

Exhaust air volume : ≤ 2250m³/h ; Adjustable

Air Velocity : 0.35 – 0.65m/s, adjustable

Noise : ≤ 75db

Illumination : ≥ 800 lux

5.0 | Construction material | Construction material

| As per the GMP/GLP compliance. | As per the GMP/GLP compliance.

6.0 | Utility Requirements | Utility Requirements

| Energy Type | Specification

| Three Phase | 400V +10%, 50Hz

7.0 | Accessories | Accessories

| All accessories / Spare parts as required for the normal operation of the equipment. | All accessories / Spare parts as required for the normal operation of the equipment.

8.0 | Documents Required | Documents Required

| Complete documentation Package with following documents should be delivered with the system

Design Qualification (DQ)

Installation Qualification (IQ)

Operational Qualification (OQ)

Performance Qualification (PQ)

Operational manual

Spare part list

Calibration Certificates

A generic list of the documentation to be delivered by the supplier(s) before Design qualification DQ:

- ✓ Memory and design justifications.
- ✓ Electrical wiring/Schematic diagrams.
- ✓ Technical specifications.
- ✓ Installation drawings (layout, P&IDs).

The following is a generic list of the documentation to be delivery by the supplier(s) before Factory Acceptance Test (FAT):

- ✓ Materials and Equipment certifications.
- ✓ Operation and maintenance manuals
- ✓ Updated backup copies of the software/configuration of programmable control systems (HMI, PLCs).

The operator and maintenance manuals must be in English. The manuals must be available on paper and in electronic format. | Complete documentation Package with following documents should be delivered with the system

Design Qualification (DQ)

Installation Qualification (IQ)

Operational Qualification (OQ)

Performance Qualification (PQ)

Operational manual

Spare part list

Calibration Certificates

A generic list of the documentation to be delivered by the supplier(s) before Design qualification DQ:

- ✓ Memory and design justifications.
- ✓ Electrical wiring/Schematic diagrams.
- ✓ Technical specifications.
- ✓ Installation drawings (layout, P&IDs).

The following is a generic list of the documentation to be delivery by the supplier(s) before Factory Acceptance Test (FAT):

- ✓ Materials and Equipment certifications.
- ✓ Operation and maintenance manuals
- ✓ Updated backup copies of the software/configuration of programmable control systems (HMI, PLCs).

The operator and maintenance manuals must be in English. The manuals must be available on paper and in electronic format.

9.0 | Drawings | Drawings

| P&ID, GA drawings & Electric drawing | P&ID, GA drawings & Electric drawing

10.0 | Training | Training

| The vendor shall impart operation and maintenance training to at least two personnel from user department and engineering /projects department and shall provide evaluation report. | The vendor shall impart operation and maintenance training to at least two personnel from user department and engineering /projects department and shall provide evaluation report.

11.0 | Abbreviations | Abbreviations

| cGMP: current Good Manufacturing Practices

P&ID: Piping and Diagram Instrumentation

GA : General Arrangement

PLC: Programmable Logic Controller

HMI: Human Machine Interface

V : Volt

SS: Stainless steel | cGMP: current Good Manufacturing Practices

P&ID: Piping and Diagram Instrumentation

GA : General Arrangement

PLC: Programmable Logic Controller

HMI: Human Machine Interface

V : Volt

SS: Stainless steel

12.0 | Other Requirements | Other Requirements

| EudraLex - Volume 4 - Good Manufacturing Practice (GMP) guidelines.

Code Federal Regulations Part 21. Current Good Manufacturing Practice for finished Pharmaceuticals.

WHO good manufacturing practices for pharmaceutical products: main principles Annex2. Technical Report Series 986, 2014 | EudraLex - Volume 4 - Good Manufacturing Practice (GMP) guidelines.

Code Federal Regulations Part 21. Current Good Manufacturing Practice for finished Pharmaceuticals.

WHO good manufacturing practices for pharmaceutical products: main principles Annex2. Technical Report Series 986, 2014

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5.0 | Construction material | 4

6.0 | Utility Requirements | 4

7.0 | Accessories | 4

8.0 | Documents Required | 4

9.0 | Drawings | 4

10.0 | Training | 4-5

11.0 | Abbreviations | 5

12.0 | Other Requirements | 5

1.0 | Objective

| This URS detail the requirement of Deep freezer (-20°C), which will be used to store the standard culture, chemicals and reference standard in the quality control Department.

2.0 | Scope

| This URS is applicable for Deep freezer (-20°C), which will be installed in quality control department.

3.0 | Equipment description

| Deep freezer (-20°C), which will be used to store the standard culture, chemicals and reference standard and any other material which is heat sensitive.

Deep freezer (-20°C) should have the LED digital display, Temperature controlling, Safety control, and alarm system.

4.0 | Component details

| Technical Specification:

| Temperature working range: -15°C to -25°C

Design configuration: Vertical / Horizontal

Outer cabinet: Powder coated MS

Inner chamber: Stainless steel 304

Door: Solid insulated door with lock & key

Insulation: Polyurethane Foam (PUF) insulation

Controller: Digital temperature controller with LED display

Alarm system: Audiovisual alarm for high / low set points

Sensor: NTC

Direct Cooling Type

Capacity: 500 liters

Shelves: SS wire mesh removable type

Full length inner glass door with magnetic gasket.

Mounting: Caster wheels with brakes.

CFC- free refrigerant.

On line recording system with software CFR 21 complied.

Optional:

Door open alarm

Data logger with USB port

Temperature chart recorder

Voltage stabilizer

5.0 | Construction material

| Outer surface: Powder coated MS, Inner chamber stainless steel 304.

All parts to be designed and manufactured in line with the current standards.

6.0 | Utility Requirements

| Power-220V/50HZ or recommended by manufacturer or supplier.

7.0 | Accessories

| All accessories required for the normal operation of the equipment.

8.0 | Documents Required

| Complete documentation Package with following documents should be delivered with the system

Installation Qualification (IQ)

Operational Qualification (OQ)

Performance Qualification (PQ)

Operational manual (2 Sets)

Spare part list

Calibration certificate of sensor and temperature display.

Certificate of construction material.

9.0 | Drawings

| GA Drawing

10.0 | Training

| The vendor shall impart training for operation and maintenance for at least two personnel from user department and engineering /projects department and shall provide evaluation report.

11.0 | Abbreviations

| QC: Quality Control

GLP: Good Laboratory Practice

GMP: Good Manufacturing Practice

URS :User Requirement Specification

12.0 | Other Requirements

| NA

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3.0 | Equipment description | 3

4.0 | Component details | 3-4

5.0 | Construction material | 4

6.0 | Utility Requirements | 4

7.0 | Accessories | 4

8.0 | Documents Required | 4

9.0 | Drawings | 4

10.0 | Training | 4

11.0 | Abbreviations | 4

12.0 | Other Requirements | 4

1.0 | Objective | Objective

| This URS detail the requirement for ultra centrifuge, continuous flow up to 35,000 rpm. | This URS detail the requirement for ultra centrifuge, continuous flow up to 35,000 rpm.

2.0 | Scope | Scope

| This document is applicable for the separation /purification of Rabies virus at pilot scale | This document is applicable for the separation /purification of Rabies virus at pilot scale

3.0 | Equipment description | Equipment description

| The ultra centrifuge is designed to develop protocols for the separation, fractionation and concentration of viral vectors, virus-like particles, viruses, viral and bacterial vaccines, macromolecules and cellular organelles. | The ultra centrifuge is designed to develop protocols for the separation, fractionation and concentration of viral vectors, virus-like particles, viruses, viral and bacterial vaccines, macromolecules and cellular organelles.

4.0 | Component details | Component details

| Component | Specification

| Speed range | Up to 35000 rpm

| Process Volume handling | 500ml to 10 L

| Rotor capacity | To match with the process volume

| Centrifugal force | $\geq 90,500 \times g$

| Drive | Electrical

| |

5.0 | Construction material | Construction material

| All parts which are in product contact to be designed and manufactured in line with the current standards. Rotar should made of material resistant to alcohols ,and non-particle shedding material. The internal finish | All parts which are in product contact to be designed and manufactured in line with the current standards. Rotar should made of material resistant to alcohols ,and non-particle shedding material. The internal finish

6.0 | Utility Requirements | Utility Requirements

| Power Supply: 380-400V 3-phase, 50/60 Hz. | Power Supply: 380-400V 3-phase, 50/60 Hz.

7.0 | Accessories | Accessories

| All accessories required for the normal operation of the system. It should also include all the system interconnections. | All accessories required for the normal operation of the system. It should also include all the system interconnections.

8.0 | Documents Required | Documents Required

| Complete documentation Package with following documents should be delivered with the system

Installation Qualification (IQ)

Operational Qualification (OQ)

Operational manual (2 Sets) | Complete documentation Package with following documents should be delivered with the system

Installation Qualification (IQ)

Operational Qualification (OQ)

Operational manual (2 Sets)

| Updated backup copies of the software/configuration of programmable control systems (HMIs, PLCs). |

Updated backup copies of the software/configuration of programmable control systems (HMIs, PLCs).

9.0 | Drawings | Drawings

| GA, P&ID, Electrical drawings | GA, P&ID, Electrical drawings

10.0 | Training | Training

| The vendor shall impart successful training for operation and maintenance to at least two personnel from user department and engineering /projects department and shall provide evaluation report. | The vendor shall impart successful training for operation and maintenance to at least two personnel from user department and engineering /projects department and shall provide evaluation report.

11.0 | Abbreviations | Abbreviations

| QC: Quality Control

GLP: Good Laboratory Practice

GMP: Good Manufacturing Practice

URS: User Requirement Specification | QC: Quality Control

GLP: Good Laboratory Practice

GMP: Good Manufacturing Practice

URS: User Requirement Specification

12.0 | Other Requirements | Other Requirements

| NA | NA

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8.0 | Documents Required | 10

9.0 | Drawings | 11

10.0 | Training | 11

11.0 | Abbreviations | 11

12.0 | Other Requirements | 11

1.0 | Objective | Objective | Objective | Objective

| This URS detail the requirement for TFF system | This URS detail the requirement for TFF system | This

URS detail the requirement for TFF system | This URS detail the requirement for TFF system

2.0 | Scope | Scope | Scope | Scope

| This URS is applicable for TFF system used in the facility | This URS is applicable for TFF system used

in the facility | This URS is applicable for TFF system used in the facility | This URS is applicable for

TFF system used in the facility

3.0 | Equipment description | Equipment description | Equipment description | Equipment description

| This system is required to carry out the purification of antigen or Biological materials in production.

| This system is required to carry out the purification of antigen or Biological materials in production.

| This system is required to carry out the purification of antigen or Biological materials in production.

| This system is required to carry out the purification of antigen or Biological materials in production.

4.0 | Component details | Component details | Component details | Component details

| Technical and GMP features

This specification is intended to cover the minimum requirements to be met by the Vendor in the design, manufacture and supply of TFF system (dimensional/ functional) specifications provided in this document are as a minimum and can be subjected to amendment after discussion with vendor and representative of Client for achieving higher degree of operational ease. | Technical and GMP features

This specification is intended to cover the minimum requirements to be met by the Vendor in the design, manufacture and supply of TFF system (dimensional/ functional) specifications provided in this document are as a minimum and can be subjected to amendment after discussion with vendor and representative of Client for achieving higher degree of operational ease. | Technical and GMP features

This specification is intended to cover the minimum requirements to be met by the Vendor in the design, manufacture and supply of TFF system (dimensional/ functional) specifications provided in this document are as a minimum and can be subjected to amendment after discussion with vendor and representative of Client for achieving higher degree of operational ease. | Technical and GMP features

This specification is intended to cover the minimum requirements to be met by the Vendor in the design, manufacture and supply of TFF system (dimensional/ functional) specifications provided in this document are as a minimum and can be subjected to amendment after discussion with vendor and representative of Client for achieving higher degree of operational ease.

| Technical Specification of TFF System: | Technical Specification of TFF System: | Technical

Specification of TFF System: | Technical Specification of TFF System:

| List of Major Instruments and Components required in TFF System:

Mechanical Components

Following components in the feed line:

Diaphragm pressure sensor

Level Switch

Retentate Line

Following components in retentate line:

Diaphragm pressure sensor

Diaphragm pressure control valve, pneumatically actuated

Flow meter

Permeate Line

Following components in the permeate line:

Diaphragm pressure sensor

Diaphragm pressure control valve, pneumatically actuated

Flow meter

Conductivity Sensor

pH Sensor

Recirculation Pump

Material : SS316L,

Surface : Ra U< U 0,6 µm e-polished

Performance : approx. 9 m³/hr @ 4 bar

Rotor type : Multilobe

Mechanical seal: Single-mechanical seal

SIC/SIC/EPDM

Static sealings : EPDM, FDA CFR 21, 177.2600

& USP Class VI

Addition Pump

Watson Marlow peristaltic pump

Performance: up to 900 LPH

Max. pressure: N/A

SIP: No

Process connection: Endless tubing

Power: 230V, 50Hz, 1 ph

CIP-Philosophy

Cross flow filtration skid should include the recirculation pump for an integrated CIP operation.

SIP Concept

The system and the associated piping must be sterilized in place by clean steam. SIP of the cross-flow system should be assisted by the cross-flow vessel.

All sterilization processes shall be executed from the PLC / IPC SCADA Control System. Sterilization temperature is monitored by temperature sensor at the cold spot.

The following process phases/steps will be applied for SIP of the system

SIP of the system without cassette holder

SIP of the system with cassette holder

Instrumentation and Controls

PLC Hardware (CPU)

PLC I/O Modules

Transmitters

Variable Frequency Drives

Emergency-stop buttons

Measurement

Pressure (Feed, Permeate, Retentate, Vessel)

Retentate flow rate

Permeate flow rate

Conductivity _ Permeate

pH Permeate

Control

Pressure (Feed, Permeate, Retentate, Vessel)

TMP

Flow control

Pump speed

Cabling

Cabling and pneumatic tubing should be routed via stainless steel protective pipes

Automation

The application software for the PLC and local HMIs 21 CFR Part 11 compliance should be provided with following measures:

WinCC standard functionality for access control

Audit trail

Additional SCADA functionality should be provided as below:

Storage of process data

Generation of batch reports

Audit trail function

User inputs

User messages

Process parameters

Calibration parameters of probes, totalizers and balances

Alarm parameters of process values and digital inputs

User Authorization

3 level password protection

Separate password for each operator

Activation / deactivation via administrator

Process Requirements:

The equipment should be designed such that 6.0 m²- 12m² of membrane can be used in processing.

The equipment should be designed to allow complete drainage and product removal after use.

Sampling Requirements

The equipment should be designed to permit sterile sampling of permeate, retentate, and drain during operation.

Technical Specification of Recirculation Vessel :

Vessel ports and connection as per listed below: | List of Major Instruments and Components required in TFF System:

Mechanical Components

Following components in the feed line:

Diaphragm pressure sensor

Level Switch

Retentate Line

Following components in retentate line:

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Diaphragm pressure control valve, pneumatically actuated

Flow meter

Permeate Line

Following components in the permeate line:

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Diaphragm pressure control valve, pneumatically actuated

Flow meter

Conductivity Sensor

pH Sensor

Recirculation Pump

Material : SS316L,

Surface : Ra U< U 0,6 µm e-polished

Performance : approx. 9 m³/hr @ 4 bar

Rotor type : Multilobe

Mechanical seal: Single-mechanical seal

SIC/SIC/EPDM

Static sealings : EPDM, FDA CFR 21, 177.2600

& USP Class VI

Addition Pump

Watson Marlow peristaltic pump

Performance: up to 900 LPH
Max. pressure: N/A
SIP: No
Process connection: Endless tubing
Power: 230V, 50Hz, 1 ph
CIP-Philosophy
Cross flow filtration skid should include the recirculation pump for an integrated CIP operation.

SIP Concept

The system and the associated piping must be sterilized in place by clean steam. SIP of the cross-flow system should be assisted by the cross-flow vessel.

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The following process phases/steps will be applied for SIP of the system

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SIP of the system with cassette holder

Instrumentation and Controls

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PLC I/O Modules

Transmitters

Variable Frequency Drives

Emergency-stop buttons

Measurement

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Retentate flow rate

Permeate flow rate

Conductivity _ Permeate

pH Permeate

Control

Pressure (Feed, Permeate, Retentate, Vessel)

TMP

Flow control

Pump speed

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Flow meter

Permeate Line

Following components in the permeate line:

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Diaphragm pressure control valve, pneumatically actuated

Flow meter

Conductivity Sensor

pH Sensor

Recirculation Pump

Material : SS316L,

Surface : Ra U< U 0,6 µm e-polished

Performance : approx. 9 m³/hr @ 4 bar

Rotor type : Multilobe

Mechanical seal: Single-mechanical seal

SIC/SIC/EPDM

Static sealings : EPDM, FDA CFR 21, 177.2600

& USP Class VI

Addition Pump

Watson Marlow peristaltic pump

Performance: up to 900 LPH

Max. pressure: N/A

SIP: No

Process connection: Endless tubing

Power: 230V, 50Hz, 1 ph

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SIP Concept

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SIP of the system with cassette holder

Instrumentation and Controls

PLC Hardware (CPU)

PLC I/O Modules

Transmitters

Variable Frequency Drives

Emergency-stop buttons

Measurement

Pressure (Feed, Permeate, Retentate, Vessel)

Retentate flow rate

Permeate flow rate

Conductivity _ Permeate

pH Permeate

Control

Pressure (Feed, Permeate, Retentate, Vessel)

TMP

Flow control

Pump speed

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Following components in retentate line:

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Diaphragm pressure control valve, pneumatically actuated

Flow meter

Permeate Line

Following components in the permeate line:

Diaphragm pressure sensor

Diaphragm pressure control valve, pneumatically actuated

Flow meter

Conductivity Sensor

pH Sensor

Recirculation Pump

Material : SS316L,
 Surface : Ra U< U 0,6 µm e-polished
 Performance : approx. 9 m³/hr @ 4 bar
 Rotor type : Multilobe
 Mechanical seal: Single-mechanical seal
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 Static sealings : EPDM, FDA CFR 21, 177.2600
 & USP Class VI
 Addition Pump
 Watson Marlow peristaltic pump
 Performance: up to 900 LPH
 Max. pressure: N/A
 SIP: No
 Process connection: Endless tubing
 Power: 230V, 50Hz, 1 ph
 CIP-Philosophy
 Cross flow filtration skid should include the recirculation pump for an integrated CIP operation.

SIP Concept

The system and the associated piping must be sterilized in place by clean steam. SIP of the cross-flow system should be assisted by the cross-flow vessel.
 All sterilization processes shall be executed from the PLC / IPC SCADA Control System. Sterilization temperature is monitored by temperature sensor at the cold spot.

The following process phases/steps will be applied for SIP of the system

SIP of the system without cassette holder
 SIP of the system with cassette holder

Instrumentation and Controls

PLC Hardware (CPU)
 PLC I/O Modules
 Transmitters
 Variable Frequency Drives
 Emergency-stop buttons
 Measurement
 Pressure (Feed, Permeate, Retentate, Vessel)
 Retentate flow rate
 Permeate flow rate
 Conductivity _ Permeate
 pH Permeate
 Control
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 TMP
 Flow control
 Pump speed

Cabling

Cabling and pneumatic tubing should be routed via stainless steel protective pipes

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The equipment should be designed to allow complete drainage and product removal after use.

Sampling Requirements

The equipment should be designed to permit sterile sampling of permeate, retentate, and drain during operation.

Technical Specification of Recirculation Vessel :

Vessel ports and connection as per listed below:

|||

5.0 | Construction material | Construction material | Construction material | Construction material

| As per the GMP/GLP compliance. | As per the GMP/GLP compliance. | As per the GMP/GLP compliance. | As per the GMP/GLP compliance.

| Item | Material | Surface Internal | Surface External

| Process vessel | 316 L | Ra < 0,8 µm, e-polished | Ra < 1,2 µm mill finished

| Process piping | 316 L | Ra < 0,8 µm, e-polished | Ra < 1,2 µm mill finished

| Utilities | 316 L | Ra < 0,8 µm, e-polished | Ra < 1,2 µm mill finished

6.0 | Utility Requirements | Utility Requirements | Utility Requirements | Utility Requirements

| Energy Type | Energy Type | Specification | Specification

| Three Phase | Three Phase | 400V +10%, 50Hz | 400V +10%, 50Hz

| Pure Steam | Pure Steam | 2.5 bar | 2.5 bar

| Instrument Air | Instrument Air | 6-7 bar | 6-7 bar

| Clean Air | Clean Air | 2.5 bar | 2.5 bar

| Chilled water | Chilled water | 2.5 bar | 2.5 bar

7.0 | Accessories | Accessories | Accessories | Accessories

| All accessories / Spare parts as required for the normal operation of the equipment. | All accessories / Spare parts as required for the normal operation of the equipment. | All accessories / Spare parts as required for the normal operation of the equipment. | All accessories / Spare parts as required for the normal operation of the equipment.

8.0 | Documents Required | Documents Required | Documents Required | Documents Required

| Complete documentation Package with following documents should be delivered with the system

Design Qualification (DQ)

Installation Qualification (IQ)

Operational Qualification (OQ)

Performance Qualification (PQ)

Operational manual

Spare part list

Calibration Certificates

A generic list of the documentation to be delivered by the supplier(s) before Design qualification DQ:

- ✓ Memory and design justifications.
- ✓ Electrical wiring/Schematic diagrams.
- ✓ Technical specifications.
- ✓ Installation drawings (layout, P&IDs).

The following is a generic list of the documentation to be delivery by the supplier(s) before Factory Acceptance Test (FAT):

- ✓ Materials and Equipment certifications.
- ✓ Welder's qualification.
- ✓ Reports of pressure testing to check leaks in equipment or piping.
- ✓ Operation and maintenance manuals
- ✓ Updated backup copies of the software/configuration of programmable control systems (HMIs, PLCs).

The operator and maintenance manuals must be in English. The manuals must be available on paper and in electronic format. | Complete documentation Package with following documents should be delivered with the system

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- ✓ Operation and maintenance manuals

✓ Updated backup copies of the software/configuration of programmable control systems (HMIs, PLCs).
The operator and maintenance manuals must be in English. The manuals must be available on paper and in electronic format.

9.0 | Drawings | Drawings | Drawings | Drawings

| P&ID, GA drawings, Electric, Isometric drawing | P&ID, GA drawings, Electric, Isometric drawing | P&ID, GA drawings, Electric, Isometric drawing | P&ID, GA drawings, Electric, Isometric drawing

10.0 | Training | Training | Training | Training

| The vendor shall impart operation and maintenance training for at least two personnel from user department and engineering /projects department and shall provide evaluation report. | The vendor shall impart operation and maintenance training for at least two personnel from user department and engineering /projects department and shall provide evaluation report. | The vendor shall impart operation and maintenance training for at least two personnel from user department and engineering /projects department and shall provide evaluation report. | The vendor shall impart operation and maintenance training for at least two personnel from user department and engineering /projects department and shall provide evaluation report.

11.0 | Abbreviations | Abbreviations | Abbreviations | Abbreviations

| cGMP: current Good Manufacturing Practices

P&ID: Piping and Diagram Instrumentation

GA : General Arrangement

PLC: Programmable Logic Controller

HMI: Human Machine Interface

V : Volt

RPM : Revolutions Per Minute

SS: Stainless steel

MOC: Material of Construction

TFF: Tangential Flow Filtration | cGMP: current Good Manufacturing Practices

P&ID: Piping and Diagram Instrumentation

GA : General Arrangement

PLC: Programmable Logic Controller

HMI: Human Machine Interface

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PLC: Programmable Logic Controller

HMI: Human Machine Interface

V : Volt

RPM : Revolutions Per Minute

SS: Stainless steel

MOC: Material of Construction

TFF: Tangential Flow Filtration

12.0 | Other Requirements | Other Requirements | Other Requirements | Other Requirements

| EudraLex - Volume 4 - Good Manufacturing Practice (GMP) guidelines.

Code Federal Regulations Part 21. Current Good Manufacturing Practice for finished Pharmaceuticals.

WHO good manufacturing practices for pharmaceutical products: main principles Annex2. Technical Report Series 986, 2014 | EudraLex - Volume 4 - Good Manufacturing Practice (GMP) guidelines.

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Approvals of the URS:

TABLE OF CONTENTS

Department | Name | Signature | Date

BioBridge Healthcare Solutions Pvt. Ltd., India | BioBridge Healthcare Solutions Pvt. Ltd., India |

BioBridge Healthcare Solutions Pvt. Ltd., India | BioBridge Healthcare Solutions Pvt. Ltd., India

Prepared by: | | |

Checked/Reviewed by: | | |

Client | Client | Client | Client

Reviewed by: | | |

Reviewed by: | | |

Approved by: | | |

S/No | Contents | Page No

1.0 | Objective | 3

2.0 | Scope | 3

3.0 | Equipment description | 3

4.0 | Component details | 3

5.0 | Construction material | 4

6.0 | Utility Requirements | 4

7.0 | Accessories | 4

8.0 | Documents Required | 4

9.0 | Drawings | 4

10.0 | Training | 4

11.0 | Abbreviations | 4

12.0 | Other Requirements | 4

1.0 | Objective | Objective

| This URS detail the requirement for hot air oven 250L capacity | This URS detail the requirement for hot air oven 250L capacity

2.0 | Scope | Scope

| This URS is applicable for hot air oven 250L capacity used in the facility | This URS is applicable for hot air oven 250L capacity used in the facility

3.0 | Equipment description | Equipment description

| Hot air oven will be used for depyrogenation of material used in production / QC area | Hot air oven will be used for depyrogenation of material used in production / QC area

4.0 | Component details | Component details

| Technical Specification:

Oven capacity: 250 Liter

Temp range: Ambient +30°C to 250°C.

Temperature accuracy: ±1°C

Digital PID temperature controller with timer, alarms and auto tuning

Temperature sensor: RTD PT100 sensor

Shelves: Chrome plated wire mesh cable trays (removable).

Door: Insulated solid door with spring latch.

Door gasket: High temperature silicone rubber gasket.

Automatic cut off of heater & blower when door opened.

Aero dynamic internal design for achieving horizontal air circulation.

Port hole for sensors & data monitoring

Vent port: fume vent port

Caster wheels with lock.

Safety: Over temperature thermostat.

Optional parameter:

Pass through (two side doors)

Digital timer 99 hours 99 minutes

Toughened glass window

Stainless steel exterior

Data logger with USB interface

HMI & PLC automation | Technical Specification:

Oven capacity: 250 Liter

Temp range: Ambient +30°C to 250°C.

Temperature accuracy: ±1°C

Digital PID temperature controller with timer, alarms and auto tuning

Temperature sensor: RTD PT100 sensor

Shelves: Chrome plated wire mesh cable trays (removable).

Door: Insulated solid door with spring latch.

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Automatic cut off of heater & blower when door opened.

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Port hole for sensors & data monitoring

Vent port: fume vent port

Caster wheels with lock.

Safety: Over temperature thermostat.

Optional parameter:

Pass through (two side doors)

Digital timer 99 hours 99 minutes

Toughened glass window

Stainless steel exterior

Data logger with USB interface

HMI & PLC automation

5.0 | Construction material | Construction material

| As per the GMP/GLP compliance. | As per the GMP/GLP compliance.

6.0 | Utility Requirements | Utility Requirements

| Energy Type | Specification

| Single Phase | 220V, 50Hz

7.0 | Accessories | Accessories

| All accessories / Spare parts as required for the normal operation of the equipment. | All accessories

/ Spare parts as required for the normal operation of the equipment.

8.0 | Documents Required | Documents Required

| Complete documentation Package with following documents should be delivered with the system

Installation Qualification (IQ)

Operational Qualification (OQ)

Performance Qualification (PQ)
Operational manual
Calibration Certificates | Complete documentation Package with following documents should be delivered with the system
Installation Qualification (IQ)
Operational Qualification (OQ)
Performance Qualification (PQ)
Operational manual
Calibration Certificates
9.0 | Drawings | Drawings
| Not Applicable | Not Applicable
10.0 | Training | Training
| The vendor shall impart operation and maintenance training for at least two personnel from user department and engineering /projects department and shall provide evaluation report. | The vendor shall impart operation and maintenance training for at least two personnel from user department and engineering /projects department and shall provide evaluation report.
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Checked/Reviewed by: | | |

Client | Client | Client | Client

Reviewed by: | | |

Reviewed by: | | |

Approved by: | | |

S/No | Contents | Page No

1.0 | Objective | 3

2.0 | Scope | 3

3.0 | Equipment description | 3

4.0 | Component details | 3

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6.0 | Utility Requirements | 3

7.0 | Accessories | 3

8.0 | Documents Required | 3

9.0 | Drawings | 4

10.0 | Training | 4

11.0 | Abbreviations | 4

12.0 | Other Requirements | 4

1.0 | Objective | Objective

| This URS detail the requirement for rotary sealer | This URS detail the requirement for rotary sealer

2.0 | Scope | Scope

| This URS is applicable for rotary sealer used in the facility | This URS is applicable for rotary sealer used in the facility

3.0 | Equipment description | Equipment description

| Rotary sealer is required to carry out the process activities in production / QC area | Rotary sealer is required to carry out the process activities in production / QC area

4.0 | Component details | Component details

| Technical Specification:-

LCD display

Sealing temp max 2200C

Sealing speed approx. 10m/min

Fix contact pressure

Seal distance from edge 0-35mm

Seal seam width b/w10-12mm

Sealing system ribbed

Automatic start and standby option

Compatible to sealing material

Sealable pouches and reels according to ISO EN 11607-1/EN 868-5 made of film and uncoated materials made of polyolefins according to EN 868-9 (e. g. Tyvek®1) | Technical Specification:-

LCD display

Sealing temp max 2200C

Sealing speed approx. 10m/min

Fix contact pressure

Seal distance from edge 0-35mm

Seal seam width b/w10-12mm

Sealing system ribbed

Automatic start and standby option

Compatible to sealing material

Sealable pouches and reels according to ISO EN 11607-1/EN 868-5 made of film and uncoated materials made of polyolefins according to EN 868-9 (e. g. Tyvek®1)

5.0 | Construction material | Construction material

| As per the GMP/GLP compliance. | As per the GMP/GLP compliance.

6.0 | Utility Requirements | Utility Requirements

| Energy Type | Specification

| Single Phase | 220V , 50Hz

7.0 | Accessories | Accessories

| All accessories / Spare parts as required for the normal operation of the equipment. | All accessories / Spare parts as required for the normal operation of the equipment.

8.0 | Documents Required | Documents Required

| Complete documentation Package with following documents should be delivered with the system

Installation Qualification (IQ)

Operational Qualification (OQ)

Operational manual

Spare part list

The operator and maintenance manuals must be in English. The manuals must be available on paper and in electronic format. | Complete documentation Package with following documents should be delivered with the system

Installation Qualification (IQ)

Operational Qualification (OQ)

Operational manual

Spare part list

The operator and maintenance manuals must be in English. The manuals must be available on paper and in electronic format.

9.0 | Drawings | Drawings

| Not Applicable | Not Applicable

10.0 | Training | Training

| The vendor shall impart operation and maintenance training to at least two personnel from user department and engineering /projects department and shall provide evaluation report. | The vendor shall impart operation and maintenance training to at least two personnel from user department and engineering /projects department and shall provide evaluation report.

11.0 | Abbreviations | Abbreviations

| cGMP: current Good Manufacturing Practices

LCD: Liquid Crystal Display

V : Volt | cGMP: current Good Manufacturing Practices

LCD: Liquid Crystal Display
V : Volt
12.0 | Other Requirements | Other Requirements
| Not Applicable | Not Applicable

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1.0 | Objective

| This URS detail the requirement of High-Quality Laboratory water generation system (10 L /hr) Milli -Q water in the quality control system.

2.0 | Scope

| This URS is applicable for the High-Quality Laboratory water generation system (10 L /hr.) Milli -Q water, will be installed in the quality control department.

3.0 | Equipment description

| Milli water system generate high quality water for lab work. water passes through a long-life dual wavelength UV lamp, which ensures organic molecule degradation by photooxidation. The Quantum® polishing cartridge then removes the remaining ionic and organic contaminants below trace levels. The

4.0 | Component details

| Technical Specification:

| Schedule-I: Water Purification System

The system should be designed to handle soft water having feed water Conductivity up to 2000 micro-S/cm, Free chlorine- 2 – 3 ppm & Fouling Index up to 20 or better.

There should be separate Pre-filtration unit with 5-micron filter to remove the particulate matter with inbuilt booster pump to maintain feed pressure of system should be provided from the same OEM.

The unit should have the following purification stages:

System should have pre-treatment with activated carbon, 0.5-micron filter and anti-scaling agent to enhance the life of downstream RO cartridge.

Reverse osmosis should be able to reject 95-99% of ions and organics having pre and post conductivity meter of RO cartridge.

RO should have capillary tube to minimize the RO wastewater by partial recirculation.

Self-regenerating electro-deionization module (EDI) to eliminate the requirement of frequent replacement of DI cartridge & Softener cartridge

EDI must have carbon bead at cathode to eliminate the requirement of any softener cartridge.

Water quality: Flow rate: 5 L/hr, Type 2 water (Ion's organics removal up to 99%, Resistivity: > 5-15 Mohm.cm, TOC < 30 ppb)

The Flow rate should be temperature independent resulting constant flow rate throughout the year due to Temperature feedback mechanism.

Type 2 water Reservoir minimum 50 Liters Tank with auto cut-off Tank level sensor.

System must have option to see the total throughput of the cartridge online.

Display: Water quality after every purification step (Feed water, RO feed water, RO product water, EDI water); Tank Level; Alarm and alert on water quality, No feed water alarm etc.

2nd stage system for Type 1 water analytical applications:

Ultrapure water machine producing water of: -

Resistivity: 18.2 MΩ.cm at 25 °C, TOC < 10 ppb (µg/L), Bacteria < 0.1 CFU/mL Endotoxin < 0.001 EU/mL, RNases < 0.01 ng/mL, DNases < 4 pg./µl

Automatic recirculation feature.

The polishing cartridge should contain mixture of activated carbon and mixed ion exchange beads to reduce Ionic and organic level,

System must have accurate online Resistivity (temperature compensated and nor compensated) for the dispensing water on system display.

System must have easy water delivery: Manual water delivery and Volumetric water delivery.

System should be delivered with an ultrafiltration final filter.

To guarantee compliance with minimum laboratory safety requirements, and to ensure that the water purification system meets internationally recognized safety norms, the Both the Type 1 & 2 water purification system shall be listed with Underwriters Laboratories (both UL and ULC), and will carry the CE mark, indicating compliance with EC Directives.

Both the polishing cartridge and final filter must be supplied with Quality certificate from the OEM.

Range: 4 ppb to 50 ppm

Precision: < 1% RSD

Accuracy: ± 2% or ± 0.5 ppb, whichever is greater

Display Readout: 3 significant digits

Sample Type: Autosampler or discrete grab sample

Calibration: Typically, stable for 12 months

Analysis Time: 2 minutes

Sample Temperature: 5–95 °C (41–203 °F)

Ambient Temperature: 5–40 °C (41–104 °F)

Instrument Sample Flow Rate: 0.5 mL/min

Outputs: USB device port (1), USB host ports (3); Ethernet

5.0 | Construction material

| NA

6.0 | Utility Requirements

| As recommended by manufacturer.

7.0 | Accessories

| All accessories required for the normal operation of the equipment.

8.0 | Documents Required

| Operational manual (2 Sets)

Installation Qualification (IQ)

Operational Qualification (OQ)

Performance Qualification (PQ)

Calibration certificate

Spare part list

9.0 | Drawings

| NA

10.0 | Training

| The vendor shall impart training for operation and maintenance for at least two personnel from user department and engineering /projects department and shall provide evaluation report.

11.0 | Abbreviations

| QC: Quality Control

GLP: Good Laboratory Practice

TOC: Total Organic Carbon

GMP: Good manufacturing Practice

URS: User requirement specification

12.0 | Other Requirements

| NA

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8.0 | Documents Required | 4

9.0 | Drawings | 4

10.0 | Training | 4

11.0 | Abbreviations | 4

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1.0 | Objective

| This URS detail the requirement of Ice flak machine used to producing ice flakes. The ice flakes are used to transport and store during testing of heat labile samples and material

2.0 | Scope

| This URS is applicable to the Ice flak machine, will be installed in the quality control department.

3.0 | Equipment description

| Laboratory ice flaking machines are known for their precise and controlled ice production. they are equipped with advanced technology to ensure consistent ice quality and uniform flake size. This makes them suitable for applications such as sample preservation, medical research, and temperature-controlled environments.

4.0 | Component details

| Technical Specification:

| Ice storage: 15kg

Capacity/24hrs:40kg

controlled by an auto microcomputer, making the machine easy to use and operate.

Water consumption (L/H) :1.6

Operating temperature:25±2°C

Ice type: small granular snow ice

Protection and display: Machine should be equipped with various protective features, including ice full display, lack of water detection, and too-cold protection. A fault warning display prompts protective shutdown when necessary. The machine should also have an automatic memory and recovery function.

Start/Stop switch

5.0 | Construction material

| All parts to be designed and manufactured in line with the current GMP standards. The outer body of the machine is made from high-quality stainless steel, ensuring durability and resistance to corrosion.

6.0 | Utility Requirements

| Power Supply: 50 Hz / 220 V or recommended by supplier and manufacturer.

7.0 | Accessories

| NA

8.0 | Documents Required

| Installation Qualification(IQ)

Operational Qualification(OQ)

Spare part list

Maintenance manual (2Sets)

9.0 | Drawings

| Electrical Circuit

10.0 | Training

| The vendor shall impart successful training for operation and maintenance for at least two personnel from user department and engineering /projects department and shall provide evaluation report.

11.0 | Abbreviations

| QC: Quality Control

GLP: Good Laboratory Practice

GMP: Good Manufacturing Practice

URS: User requirement specification

12.0 | Other Requirements

| NA

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8.0 | Documents Required | 4

9.0 | Drawings | 4

10.0 | Training | 4

11.0 | Abbreviations | 4

12.0 | Other Requirements | 4

1.0 | Objective

| This URS detail the requirement of TOC analyzer to used for measuring the concentration of organic carbon in WFI/Purified water samples.

2.0 | Scope

| This URS is applicable mainly for the TOC analyzer, will be installed in the quality control department.

3.0 | Equipment description

| Total Organic Carbon analyzer, which utilizes a catalytic oxidation combustion technique at high temperature (the temperature raises up to 720 °C), to convert organic carbon into CO₂. The CO₂ generated by oxidation is measured with a Non-dispersive Infra-Red (NDIR) sensor.

4.0 | Component details

| Technical Specification:

| Range: 4 ppb to 50 ppm

Precision: < 1% RSD

Accuracy: ± 2% or ± 0.5 ppb, whichever is greater

Display Readout: 3 significant digits

Sample Type: Autosampler or discrete grab sample

Calibration: Typically, stable for 12 months

Analysis Time: 2 minutes

Sample Temperature: 5–95 °C (41–203 °F)

Ambient Temperature: 5–40 °C (41–104 °F)

Instrument Sample Flow Rate: 0.5 mL/min

Outputs: USB device port (1), USB host ports (3); Ethernet

5.0 | Construction material

| NA

6.0 | Utility Requirements

| 100 – 240 V~, 50 – 60 Hz, 100 VA or recommended by manufacturer.

7.0 | Accessories

| All accessories required for the normal operation of the equipment.

Autosampler and data software (Optional)

8.0 | Documents Required

| Operational manual (2 Sets)

Installation Qualification (IQ)

Operational Qualification (OQ)

Performance Qualification (PQ)

Calibration certificate

9.0 | Drawings

| NA

10.0 | Training

| The vendor shall impart successful training for operation and maintenance for at least two personnel from user department and engineering /projects department and shall provide evaluation report.

11.0 | Abbreviations

| QC: Quality control

GLP: Good laboratory Practice

TOC: Total organic carbon

GMP: Good manufacturing Practice

URS: User requirement specification

12.0 | Other Requirements

| NA

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11.0 | Abbreviations | 4

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1.0 | Objective

| This URS detail the requirement of Colony Counter used to carrying out a microbial count in media petri plates which are exposed in clean areas for environmental monitoring.

2.0 | Scope

| This URS is applicable for the Colony Counter, will be placed in the quality control department.

3.0 | Equipment description

| Digital Colony Counter is used for quick and accurate counting of bacterial and mold colonies in petri dishes.

4.0 | Component details

| Technical Specification:

| Display: 3 Digit, 7-Seg. LED.

Max. Counts: 999

Dish Size: 110 mm

Audible Confirmation of Each Count.

Uniform Glare Free Illumination.

5.0 | Construction material

| NA

6.0 | Utility Requirements

| 230 V + 10% AC, 50 Hz, or as per r

7.0 | Accessories

| Marking Pen, Magnifying Lens, Lens Holder and Dust Cover.

8.0 | Documents Required

| Operational manual (2 Sets)

Installation Qualification

Operational Qualification

Performance Qualification

9.0 | Drawings

| NA

10.0 | Training

| The vendor shall impart training for operation and maintenance to at least two personnel from user department and engineering /projects department and shall provide evaluation report.

11.0 | Abbreviations

| QC: Quality Control

GLP: Good Laboratory Practice

GMP: Good Manufacturing Practice

URS :User Requirement Specification

12.0 | Other Requirements

| NA

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1.0 | Objective

| This URS detail the requirement of Tube heating block, which will be used to incubate the glass tubes for the endotoxin detection (LAL) test.

2.0 | Scope

| This URS is applicable to the Tube heating block which will be installed in the quality control department.

3.0 | Equipment description

| The tube heating block/dry bath are amongst the most important apparatus in laboratories. They are used to heat samples offering wide temperature ranges of temperature and are adaptable to different tube or flask sizes.

4.0 | Component details

| Technical Specification:

| Capacity: 50-60 tubes with the size10x75mm

Temperature range: 01 to 100°C

Temperature stability: ±01°C.

Temperature uniformity: ±01°C.

Digital display.

Heating Blocks: Aluminium, Anodized with 10mm Diameter Holes.

Timer :0 to 99 minutes

5.0 | Construction material

| Aluminum & Mild Steel/ All parts to be designed and manufactured in line with the current standards.

6.0 | Utility Requirements

| 240V/50HZ or as recommended by manufacturer or supplier.

7.0 | Accessories

| Accessories required for the normal operation of equipment.

8.0 | Documents Required

| Complete documentation Package with following documents should be delivered with the system

Installation Qualification (IQ)

Operational Qualification (OQ)

Performance qualification (PQ)

Calibration certificate for temperature sensors.

Thermal map for uniformity is temperature.

Operational manual (2 Sets)

9.0 | Drawings

| NA

10.0 | Training

| The vendor shall impart training for operation and maintenance for at least two personnel from user department and engineering /projects department and shall provide evaluation report.

11.0 | Abbreviations

| QC: Quality Control

GLP: Good Laboratory Practice

GMP: Good Manufacturing Practice

URS :User Requirement Specification

12.0 | Other Requirements

| NA

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6.0 | Utility Requirements | 3-4

7.0 | Accessories | 4

8.0 | Documents Required | 4

9.0 | Drawings | 4

10.0 | Training | 4

11.0 | Abbreviations | 4

12.0 | Other Requirements | 4

1.0 | Objective

| This URS detail the requirement of Air sampler, which will be used in controlled area. Where microbial contamination needs to be strictly monitored.

2.0 | Scope

| This URS is applicable for Air sampler, which will be used in controlled areas of production and quality control department.

3.0 | Equipment description

| This microbial air sampler works based on Andersen collision principle. Air is aspirated (at a fixed speed for variable times) through small holes in the instrument's head. The resulting laminar air flow is directed onto the surface of an agar plate. At the end of the sampling cycle, the plate is removed & incubated and the visible colonies are then counted for an assessment of the level of contamination in the controlled area

4.0 | Component details

| Technical Specification:

| Air flow :100 liter/min

Impactor nozzle number: 300

Air sample volume: 10~6000 liter

Air flow sensor: Automatically compensate

Data transfer: No/Sampler software

Applicable: Petri dish Φ 90mm

Error range: $\pm 2.5\%$

Cover material Silvery, bright blue aluminum alloy

Screen LCD

Battery: Rechargeable polymeric lithium battery

Charger 110~240V, 50~60HZ (Universal voltage lithium battery charger).

5.0 | Construction material

| All parts to be designed and manufactured in line with the current standards and to resist laboratory working conditions

6.0 | Utility Requirements

| Charger 110~240V, 50~60HZ (Universal voltage lithium battery charger) or recommended by manufacturer or supplier.

7.0 | Accessories

| All accessories required for the normal operation of the equipment.

8.0 | Documents Required

| Complete documentation Package with following documents should be delivered with the system

Installation Qualification (IQ)

Operational Qualification (OQ)

Operational manual (2 Sets)

9.0 | Drawings

| NA

10.0 | Training

| The vendor shall impart successful training for operation and maintenance for at least two personnel from user department and engineering /projects department and shall provide evaluation report.

11.0 | Abbreviations

| QC: Quality Control

GLP: Good Laboratory Practice

GMP: Good Manufacturing Practice

URS: User Requirement Specification

12.0 | Other Requirements

| NA

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8.0 | Documents Required | 4

9.0 | Drawings | 4

10.0 | Training | 4-5

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12.0 | Other Requirements | 5

1.0 | Objective

| This URS explain the details of equipment required to measure the cleanliness level of air inside the different clean areas especially in grade A/B by monitoring the particles/cubic ft of air.

2.0 | Scope

| This URS is applicable mainly for the potable particle counter, which will be used to monitor the air quality(particles,) during critical operations for example sterile filtration/filling operations.

3.0 | Equipment description

| Optical particle counters (OPC) or spectrometers employ a small sensing volume, of air by a laser source. Generally, commercial instruments handle concentrations of up to -10⁶ particles/liter. Beyond this concentration limit, sample dilution is usually used, which decreases the accuracy of the determination of the concentration. The range of particle diameter that single-particle instruments are capable of handling is -0.1 µm. The particle size distribution in single-particle counters is determined by comparing the detected pulse heights of the optical signals that correspond to the single particle flowthrough a small illuminated zone with a standard calibration curve, obtained from a set of uniform particles of known diameter.

For determination of the concentration, the flow rate (the volume of air tested) is also measured simultaneously. The different light detection geometries applied in different particle counters yield a wide range of instrument designs and constructions. In most airborne particle counters perpendicular scattering geometry is implement.

4.0 | Component details

| Technical Specification:

| Size Channels: 0.3 µm, 0.5 µm, 1.0 µm, 3.0 µm, 5.0 µm, 10.0 µm

Zero Count: 1 count or less in 5 minutes.

Coincidence Loss: Less than 5% at 400000 particles/ft

Flow Rate: 1.0 CFM

Counting Efficiency: 50 % for 0.3 µm

Light Source: Long Life Laser™ diode with 10-year Mean Time to Failure (MTTF)

Count display: LCD

Maximum Count: 9999999 display

Pump Type: Air vacuum, rated for continuous use

Sample /Hold Times: 1 second to 24 hours

Count Alarm: 1 to 9999999 counts

Count Cycles: Up to 100 while in automatic mode

Location label: 0 to 999, appears on printout

Data Storage: 5,000 samples, scrollable on Historical Data review

Screen.

Battery: Rechargeable Li

Operating Time (Battery): up to 6 hours

Battery Recharge Time: 7 hours minimum

5.0 | Construction material

| All parts to be designed and manufactured in line with the current standards. However, the external casing of the particle counter should me made of SS 304L

6.0 | Utility Requirements

| As recommended by manufacturer or supplier.

7.0 | Accessories

| All accessories required for the normal operation of the equipment.

8.0 | Documents Required

| Complete documentation Package with following documents should be delivered with the system

Installation Qualification (IQ)

Operational Qualification (OQ)

Performance Qualification (PQ)

Operational manual (2 Sets)

Certificate of calibration

9.0 | Drawings

| NA

10.0 | Training

| The vendor shall impart training for operation and maintenance for at least two personnel from user department and engineering /projects department and shall provide evaluation report.

11.0 | Abbreviations

| QC: Quality Control

GLP: Good Laboratory Practice

GMP: Good Manufacturing Practice

URS: User requirement specification

12.0 | Other Requirements

| NA

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8.0 | Documents Required | 4

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10.0 | Training | 4

11.0 | Abbreviations | 4-5

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1.0 | Objective

| This URS detail the requirement of incubator of variable volumes, which will be used to incubate the sterility sample/media and agar petri plates of culture in the quality control labs.

2.0 | Scope

| This URS is applicable mainly for incubator, which will be installed in quality control department.

3.0 | Equipment description

| Incubator will be used to incubate the sample/media etc.

Incubator should have the LED digital display, Temperature controller, Safety control, and alarm system.

4.0 | Component details

| Technical Specification:

| Temperature range: 10-60°C

Temperature Setting and display: Microprocessor Control with LED Digital

Temperature accuracy: ±0.5°C

Temperature uniformity: ±1°C

Controls panel: Touch-pad icon-based controls & display

Alarm system: Audible and Visual alarms with battery back-up. With remote alarm contact

Alarm conditions: High & Low alarm, Door open, Power failure

Air Circulation: Forced air circulation fan for uniform temperature

LED light for illumination and observation

Capacity: 500 liters

Stainless steel Trays

Full length inner glass door with magnetic gasket.

Wheels for easy movement.

High quality PT-100 temperature sensor.

Online temperature monitoring with MKT.

Automatic daily data backup.

Safety devices: Low and high temperature Cut-Off settable thermostats

5.0 | Construction material

| SS 304 interior and exterior either SS 304 of mild sheet.

6.0 | Utility Requirements

| 230 volt, 50Hz main supply or recommended by manufacturer or supplier.

7.0 | Accessories

| All accessories required for the normal operation of the equipment.

8.0 | Documents Required

| Complete documentation Package with following documents should be delivered with the system

Installation Qualification (IQ)

Operational Qualification (OQ)

Performance Qualification (PQ)

Operational manual (2 Sets)

Spare part list

Calibration certificate of sensor and temperature display.

Material test certificate.

9.0 | Drawings

| Process and Instrumentation Diagram (P & ID)

Instrument Listing

Control Schematics

Control Panel Assembly Drawing

Machine Assembly Drawings

10.0 | Training

| The vendor shall impart successful training for operation and maintenance for at least two personnel from user department and engineering /projects department and shall provide evaluation report.

11.0 | Abbreviations

| QC: Quality Control

GLP: Good laboratory Practice

URS: User requirement specification

12.0 | Other Requirements

| NA

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1.0 | Objective

| This URS detail the requirement of Vortex mixer, which will be used for mixing of solution and samples.

2.0 | Scope

| This URS is applicable for the Vortex mixer.

3.0 | Equipment description

| The Vortex mixer is used to mixing of solution and samples, should Support various test tube sizes, ,

Spill-proof ,speed electronic controler.

4.0 | Component details

| Technical Specification:

| Digital display of Speed and Time/manual speed controller

Continuous and Touch modes

Speed range: 200 to 3000 rpm

Quick set up to Max rpm:

Spill-proof electronic controls

5.0 | Construction material

| All parts to be designed and manufactured in line with the current standards.

To resist laboratory working conditions

6.0 | Utility Requirements

| 220 / 230 Volts 50 Hz or recommended by manufacturer or supplier.

7.0 | Accessories

| Optional Accessories:

Rubber Cup

Plate adapter with rubber disc

Plain adapter foam

Foam with 96X0.2 well

Foam with 60X0.5 well

Foam with 12mm well

Foam with different well

8.0 | Documents Required

| Complete documentation Package with following documents should be delivered with the system

Installation Qualification (IQ)

Operational Qualification (OQ)

Operational manual (2 Sets)

9.0 | Drawings

| NA

10.0 | Training

| NA

11.0 | Abbreviations

| QC: Quality Control

GLP: Good Laboratory Practice

GMP: Good Manufacturing Practice

URS: User requirement specification

12.0 | Other Requirements

| NA

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8.0 | Documents Required | 3-4

9.0 | Drawings | 4

10.0 | Training | 4

11.0 | Abbreviations | 4

12.0 | Other Requirements | 4

1.0 | Objective

| This URS detail the requirement of Vacuum Pump, which will be used for filtration in manufacturing and quality control departments.

2.0 | Scope

| This URS is applicable for the Vacuum Pump..

3.0 | Equipment description

| The Vacuum Pump is compact, lightweight and mobile oil – free piston.

4.0 | Component details

| Technical Specification:

| Maximum power :60W

Frequency: 50Hz

Maximum current: 0.3 A

Maximum vacuum: 100mbar

Maximum flow rate:17L/min

Motor rotation :1450 RPM

Noise level : <60dB

5.0 | Construction material

| All parts to be designed and manufactured in line with the current standards.

To resist laboratory working conditions

6.0 | Utility Requirements

| 220 V or recommended by manufacturer or supplier.

7.0 | Accessories

| Connectors, L shape connectors, Quick connectors, silencer, Vacuum regulator/filter with/without pressure gauge, foot pedal and tubing. OR All accessories required for the normal operation of the equipment.

8.0 | Documents Required

| Complete documentation Package with following documents should be delivered with the system

Operational manual (2 Sets).

Calibration certificate.

9.0 | Drawings

| NA

10.0 | Training

| NA

11.0 | Abbreviations

| QC: Quality Control

GLP: Good Laboratory Practice

GMP: Good Manufacturing Practice

URS :User Requirement Specification

12.0 | Other Requirements

| NA

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8.0 | Documents Required | 4

9.0 | Drawings | 4

10.0 | Training | 4-5

11.0 | Abbreviations | 5

12.0 | Other Requirements | 5

1.0 | Objective

| This URS detail the requirement of flame photometer, which will be used to test Sodium, Potassium, Calcium and Lithium elements.

2.0 | Scope

| This URS is applicable to the flame photometer, installed in quality control department.

3.0 | Equipment description

| Flame photometer is used for the determination of Sodium, Potassium, Calcium and Lithium. It uses the latest Microcontroller technology and advanced engineering techniques so as to give the reproducibility results. It has soft touch membrane key for ease of operations. the solution is aspirated through an atomizer, atomized sample and the fuel are mixed in the mixing chamber which is then sprayed as a very fine mist into the flame. The color of the flame is changed depending upon the concentration of elements present. Radiations from the flame passes through the sensing system and specific narrow band interference filter which permits only the characteristic radiation to pass to the photo-detector. The output of the photo-detector is then processed by the microcontroller and the final results are displayed on the digital display.

4.0 | Component details

| Technical Specification:

| Auto Gas Cutoff

Auto Flame Failure Detection

Four Measurements in Single Aspiration

Setup Storage Facility

Sample Data Storage Facility

Curve Calibration Programmability Using Maximum of 5 Standards

Multiple Calibration Curve Saving Facility

Data Processing Through Curve Fitting

Techniques, Least Square, Quadratic

Online Help Key

Direct Results in ppm.

Microprocessor Based with Printer Interface.

Gas Control: Adjustable With Knob

Display: Alphanumeric Backlit Display

Flame System: LPG and dry oil free air

Ignition System: Auto ignition

Atomizer: Axial flow type

Printer: Provision for attachment of dot matrix printer or LaserJet/ inkjet (non-windows) printers with Centronics parallel interface

PC & software.

5.0 | Construction material

| All parts to be designed and manufactured in line with the current standards.

6.0 | Utility Requirements

| 230 V $\pm$ 10% AC, 50 Hz or recommended by manufacturer or supplier.

7.0 | Accessories

| All accessories required for the normal operation of the equipment.

8.0 | Documents Required

| Complete documentation Package with following documents should be delivered with the system

Installation Qualification (IQ)

Operational Qualification (OQ)

Performance Qualification (PQ)

Operational manual (2 Sets)

9.0 | Drawings

| NA

10.0 | Training

| The vendor shall impart training for operation and maintenance for at least two personnel from user department and engineering /projects department and shall provide evaluation report.

11.0 | Abbreviations

| QC: Quality Control

GLP: Good Laboratory Practice

GMP: Good Manufacturing Practice

URS: User requirement specification

12.0 | Other Requirements

| NA

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9.0 | Drawings | 4

10.0 | Training | 4

11.0 | Abbreviations | 4

12.0 | Other Requirements | 4

1.0 | Objective

| This URS detail the requirement of Thermostatic water bath, which will be used to heat and/or keep samples and solutions of substances at a certain temperature.

2.0 | Scope

| This URS is applicable for the Thermostatic water bath, will be installed in quality control department.

3.0 | Equipment description

| The Water bath should have Digital LED display with high brightness Effective, prevention of steam leakage, Controlled high precision temperature.

4.0 | Component details

| Technical Specification:

| Capacity:20L

Temperature range: RT to 100°C

Temperature fluctuation: 0.5°C

Temperature accuracy: $\leq \pm 1^\circ\text{C}$

Digital temp. indicator-cum-controller

View glass on both sides

Light source to see the incubated tubes for reaction/ flocculation.

Attached water circulation device for achieving the uniformity in water temperature.

5.0 | Construction material

| All parts to be designed and manufactured in line with the current standards preferably SS 304L.

To resist laboratory working conditions

6.0 | Utility Requirements

| AC 110/220 V $\pm 10\%$, 50/60 Hz or recommended by manufacturer or supplier.

7.0 | Accessories

| All accessories required for the normal operation of the equipment.

8.0 | Documents Required

| Complete documentation Package with following documents should be delivered with the system

Installation Qualification (IQ)

Operational Qualification (OQ)

Operational manual (2 Sets)

9.0 | Drawings

| NA

10.0 | Training

| The vendor shall impart successful training for operation and maintenance for at least two personnel from user department and engineering /projects department and shall provide evaluation report.

11.0 | Abbreviations

| QC: Quality Control

GLP: Good Laboratory Practice

GMP: Good Manufacturing Practice

URS :User Requirement Specification

12.0 | Other Requirements

| NA

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1.0 | Objective

| This URS detail the requirement of Water Sonicator used to agitate the particles in the solutions

2.0 | Scope

| This URS is applicable for the Water sonicator, will be installed in the quality control department.

3.0 | Equipment description

| Sonication is defined as the process in which sound waves are used to agitate the particles in the solutions. These disruptions are used for mixing of the solutions, to increase the speed of dissolution of a solid into a liquid, and for the removal of dissolved gases from the liquids.

4.0 | Component details

| Technical Specification:

| Capacity: 6-10 Ltr.

Operating frequency: 40 ± 3 kHz

Temperature range: Ambient + 5°C to 65° C

Timer Range: 1 to 15 mins.

Stainless steel outer construction

Includes stainless steel tanks with lid, drain outlet and immersion basket.

Microcontroller based digital ultrasonic control circuit.

High quality transducers ensure high electro-acoustical efficiency and low heat generation.

Includes heater for thermostatic heating

Digital timer

Simple to use interface

Easy to read digital display

With non-slip rubber feet

5.0 | Construction material

| NA

6.0 | Utility Requirements

| 220 V AC ± 10%, 50-60 Hz or recommended by manufacturer or supplier.

7.0 | Accessories

| All accessories required for the normal operation of the equipment.

8.0 | Documents Required

| Complete documentation Package with following documents should be delivered with the system

Installation Qualification (IQ)

Operational Qualification (OQ)

Operational manual (2 Sets)

9.0 | Drawings

| NA

10.0 | Training

| NA

11.0 | Abbreviations

| QC: Quality Control

GLP: Good Laboratory Practice

GMP: Good Manufacturing Practice

URS :User Requirement Specification

12.0 | Other Requirements

| NA

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1.0 | Objective

| HPLC system is ensure the system can accurately separate, identify, quantify chemical components in a mixture and regulatory requirements that the system must meet.

2.0 | Scope

| This URS is applicable mainly for the HPLC system, will be placed in the quality control department.

3.0 | Equipment description

| HPLC system used for testing of in process and finished product of vaccine in the QC lab.

4.0 | Component details

| Flow rate setting range :0.0001 to ml/min

Flow rate precision: < 0.006% RSD to < 0.02% RSD

Maximum pressure: 44 mPa

Degassing unit: 5 lines (mobile phase 4 + Rinse solution 1)

Pumping method: Parallel type double plunger

Injection volume setting range: 0.1 to 100 µl

Injection volume accuracy: ±1%

Sample Vials capacity: approx. 300 (1mLVial)

Injection cycle time :15sec

Injection linearity :>0.9999

Sampler cooler: 5-45°C

Column size: 6 columns of 10cm and 3 columns of 30cm

Heating cooling method: Forced air circulation method

Wavelength range: 200-700nm

Wavelength reproducibility: < ±1 mm

5.0 | Construction material

| NA

6.0 | Utility Requirements

| As per manufacturer recommendation

7.0 | Accessories

| Solvent bottles with cap and accessory

Tool kit

Shim pack GIST C18 column

Vial, cap and septa

8.0 | Documents Required

| Operational manual (2 Sets)

Installation Qualification

Operational Qualification

Performance Qualification

9.0 | Drawings

| NA

10.0 | Training

| The vendor shall impart I training to operation and maintenance to at least two personnel from user department and engineering /projects department and shall provide evaluation report.

11.0 | Abbreviations

| QC: Quality Control

GLP: Good Laboratory Practice

GMP: Good Manufacturing Practice

HPLC: High-Performance Liquid Chromatography

12.0 | Other Requirements

| NA

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9.0 | Drawings | 4

10.0 | Training | 4

11.0 | Abbreviations | 4

12.0 | Other Requirements | 4

1.0 | Objective

| This URS detail the requirement of Volumetric Karl Fischer used in the chemical testing lab for water content testing.

2.0 | Scope

| This URS is applicable mainly for the KF which will be placed in the quality control labs.

3.0 | Equipment description

| This equipment is used to determine the water content of a sample. The fundamental principle behind it is based on the Bunsen Reaction between iodine and sulfur dioxide in an aqueous medium.

$\text{ROH} + \text{SO}_2 + \text{R}'\text{N} = [\text{R}'\text{NH}]\text{SO}_3\text{R} + \text{H}_2\text{O} + \text{I}_2 + 2\text{R}'\text{N} = 2[\text{R}'\text{NH}]\text{I} + [\text{R}'\text{NH}]\text{SO}_4\text{R}$

[alcohol] [base] [alkylsulfite salt] [water] [iodine] [hydroiodic acid salt] [alkylsulfate salt]

The alcohol reacts with sulfur dioxide (SO₂) and base to form an intermediate alkylsulfite salt, which is then oxidized by iodine to an alkylsulfate salt. This oxidation reaction consumes water. The reactive alcohol is typically methanol or 2-(2-Ethoxyethoxy)ethanol, also known as diethylene glycol monoethyl ether (DEGEE), or another suitable alcohol. Classic Karl Fisher reagents contained pyridine, a noxious carcinogen, as the base. The reagents most frequently used today are pyridine-free and contain imidazole or primary amines instead.

The volumetric titrator performs the following three key functions: 1) It dispenses KF titrating reagent containing iodine into the cell using the burette 2) It detects the endpoint of the titration using the double platinum pin indicator electrode 3) It calculates the end result based on the volume of KF reagent dispensed using the onboard microprocessor

4.0 | Component details

| Technical Specification:

| Ration Range :100 ppm to 100%

Titration Resolution :1 ppm to 0.0001%

Measurement Units: %, ppm, mg/g, µg/g, mg, µg, mg/mL, µg/mL, mg/pc, µg/pc

Sample Type: liquid or solid

Pre-Titration Conditioning: automatic

Background Drift Correction: automatic or user-selectable value

Endpoint Criteria: fixed mV persistence, relative drift stop or absolute drift stop

Dosing Type: dynamic with optional pre-dispensing

Result Statistics: mean, standard deviation

Dosing Accuracy: ±0.1% of full burette volume

Syringe:5 mL precision ground glass with PTFE plunger

Valve: motor-driven 3-way, PTFE liquid contact material

Tubing:PTFE with light block and thermal jacketing

Dispensing Tip glass, fixed position, anti-diffusing

Titration Vessel conical with operation volume between 50 and 150 mL

Solvent Handling System sealed system, integrated diaphragm air pump

Connection Type BNC

Polarization Current 1, 2, 5, 10, 15, 20, 30 or 40 uA

mV Range 2 mV to 1000 mV

mV Resolution 0.1 mV

mV Accuracy ±0.1%

Burette 5 mL burette

Dosing Pump Resolution 1/40000 of the burette volume (0.125 µL per dose) with 5 mL burette

Stirrer Type magnetic, optically regulated, digital stirrer

Stirrer Speed 200 - 2000 RPM

Stirrer Resolution 100 RPM

Titration Methods up to 100 standard and user defined

Reports-up to 100 complete titration reports and drift rate reports

Connectivity- USB

Built-in Magnetic Stirrer magnetic, optically regulated, digital stirrer; 200 to 2000 rpm; resolution 100 rpm

Display- 5.7" graphical color display with backlight

Keypad- polyester

Languages English

Power Supply 100-240 Vac, 50/60 Hz / 0.5 Amps

Environment Operating

Dimensions 315 x 205 x 375 mm (12.4 x 8.1 x 14.8")

Weight-Approximate 4.3 kg (9.5 lbs) .

5.0 | Construction material

| NA

6.0 | Utility Requirements

| NA

7.0 | Accessories

| All accessories required for the normal operation of the equipment.

8.0 | Documents Required

| Operational manual (2 Sets)

9.0 | Drawings

| NA

10.0 | Training

| The vendor shall impart training for operation and maintenance for at least two personnel from user department and engineering /projects department and shall provide evaluation report.

11.0 | Abbreviations

| QC: Quality Control

GLP: Good Laboratory Practice

GMP: Good Manufacturing Practice

URS: User requirement specification

12.0 | Other Requirements

| NA

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11.0 | Abbreviations | 4

12.0 | Other Requirements | 4

1.0 | Objective

| This URS detail the requirement of ELISA Microplate reader used to read the color reaction in microplate by various type of ELISA test.

2.0 | Scope

| This URS is applicable for the ELISA reader which will be installed in the quality control department.

3.0 | Equipment description

| The ELISA reader is a specialized spectrophotometer an apparatus for measuring the intensity of light throughout a part of the spectrum and used to measure the color reactions of antigens and antibodies in a solution at specific wave length of ligt spectrum.

4.0 | Component details

| Technical Specification:

| Wavelength range: 400-750 nm.

Photometric range: 0.000-3.500 OD

Linearity: $\pm 1.0\%$ from 0.000 to 2.000 OD, $\pm 2.0\%$ from 0.000 to 3.000 OD

Accuracy: $\pm 1.0\%$ or 0.010 from 0.000 to 3.000 OD at 490 nm

Precision: 1.0% or 0.005 OD.

Resolution: 0.001 OD

Filter wheel capacity: 8

Read time: 6 sec. at single wavelength, 10 sec. at dual wavelengths.

Factory options: Temperature control (5°C above ambient, up to 45°C, in 0.1°C increments), onboard graphical thermal printer.

5.0 | Construction material

| NA

6.0 | Utility Requirements

| As per manufacturer requirement.

7.0 | Accessories

| PC serial Cable, Printer, PC and software etc required for normal operation of equipment.

8.0 | Documents Required

| Operational manual (2 Sets)

Installation Qualification

Operational Qualification

Performance Qualification

Calibration certificate. For light filters

9.0 | Drawings

| NA

10.0 | Training

| The vendor shall impart training for operation and maintenance for at least two personnel from user department and engineering /projects department and shall provide evaluation report.

11.0 | Abbreviations

| QC: Quality Control

GLP: Good Laboratory Practice

ELISA: Enzyme Linked Immunosorbent Assay

URS: User Requirement Specification

12.0 | Other Requirements

| NA

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3.0 | Equipment description | 3

4.0 | Component details | 3-4

5.0 | Construction material | 4

6.0 | Utility Requirements | 4

7.0 | Accessories | 4

8.0 | Documents Required | 4-5

9.0 | Drawings | 5

10.0 | Training | 5

11.0 | Abbreviations | 5

12.0 | Other Requirements | 5

1.0 | Objective

| The objective of the document is to prepare URS for stability chambers used in the stability testing of various Biologics.

2.0 | Scope

| This document is applicable for stability chambers, which will be installed in the stability testing labs.

3.0 | Equipment description

| The stability chambers must be designed in such a manner that it allows precise, standard compliant simulation of all climatic conditions under a constant atmosphere for carrying out all long duration storage for stability tests according to the internationally valid EN standards and current pharmaceutical ICH guidelines Q1A & Q1B. The must deliver accurate and reliably temperature and must avoids any condensation inside the chamber. The equipment must be easy to clean. Machine should meet all cGMP and GEP standards, and comply CE safety and electrical standards. Machine needs to run continuously in fully automatic mode.

4.0 | Component details

| Technical Specification:

| Temperature range: 10-60°C

Set temperature: Knob

Capacity :200/500 L

Temperature Setting and display: Microprocessor Control with LED Digital

Temperature accuracy: ±0.5°C

Temperature uniformity: ±1°C

Controls panel: Touch-pad icon-based controls & display

Alarm system: Audible and Visual alarms with battery back-up. With remote alarm contact

Alarm conditions: High & Low alarm, Door open, Power failure

Air Circulation: Forced air circulation fan for uniform temperature

LED light for illumination and observation

Stainless steel Trays

Full length inner glass door magnetic gasket.

Wheels for easy movement.

High quality PT-100 temperature sensor.

Online temperature monitoring with MKT.

Automatic daily data backup.

Safety devices: Low and high temperature Cut-Off with thermostats

Door shall be provided with lock and key.

Safety devices including mobile messages. Internal lighting Water level sensor with audio/visual alarm

Proper Internal lighting.

Validation port.

5.0 | Construction material

| Interior made of stainless-steel SS 304

Exterior made of stainless steel SS304

Perforated stainless steel shelves (SS304)

Proper insulation with non-shedding material.

All parts to be designed and manufactured in line with the current standards.

6.0 | Utility Requirements

| 230 volt, 50Hz main supply or recommended by manufacturer or supplier.

7.0 | Accessories

| All accessories required for the normal operation of the equipment.

8.0 | Documents Required

| Complete documentation Package with following documents should be delivered with the system

Installation Qualification (IQ)

Operational Qualification (OQ)

Performance Qualification (PQ)

Operational manual (2 Sets)

Design Specification

Operator, Maintenance and Service Manuals

Spare part list

Calibration certificate of sensor and temperature display.

Material test certificate.

9.0 | Drawings

| Process and Instrumentation Diagram (P & ID)

Instrument Listing

Control Schematics

Control Panel Assembly Drawing

Machine Assembly Drawings

10.0 | Training

| The vendor shall impart training for operation and maintenance to at least two personnel from user department and engineering /projects department and shall provide evaluation report.

11.0 | Abbreviations

| QC: Quality Control

GLP: Good Laboratory Practice

URS: User Requirement Specification

12.0 | Other Requirements

| NA

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7.0 | Accessories | 4

8.0 | Documents Required | 5

9.0 | Drawings | 5

10.0 | Training | 5

11.0 | Abbreviations | 5

12.0 | Other Requirements | 5-6

1.0 | Objective

| The objective of this User Requirement Specifications (URS) is to define and document User Requirement Specification of cold room maintained at 2-8 degrees C.

2.0 | Scope

| This document is applicable for the cold room specified in this document.

3.0 | Equipment description

| This cold room shall be used for storing all the heat labile materials.

4.0 | Component details

| Furnish and install walk-in controlled environmental rooms (CER). Include all insulated walls, ceilings, floors, doors, hardware, refrigeration systems, mechanical systems, internal electrical systems, controls, gages, internal lighting, and other ancillary items required for a completely fabricated and operational CER.

Thermostat must be calibrated to ITS-90 (International Temperature Scale) and should be accurate to +/- 0.5°C;

There should be provision for temperature recording through following means- Digital Display (LED), graphically and on line recording in PC.

In case temperature is beyond the set limits there must be a provision of Alarm and Hooter.

Lighting must be ceiling mounted tungsten filament light fitted with external switch.

CFC Free refrigerant gas must be used;

All compressors to be fitted with hour meter to record the running duration for each unit.

5.0 | Construction material

| Wall and ceiling panels.

Insulated panels shall be minimum 4" thick, be of modular construction incorporating wall and ceiling panels should be of "woodless" type construction. Panels shall consist of insulation sandwiched between interior and exterior metal skin. Panel edges to have tongues and grooves that lock together assuring an air tight vapour proof joint. Construction shall allow disassembly for possible relocation or refurbishment at a later date.

All panel insulation shall be Class 1 "foamed-in-place" 2.2 lb density, polyurethane foam insulation expanded with "Ozone Safe" 245a and UL certified as having a flame spread rating of 25 or less and smoke density not greater than 450 per ASTM E84.

The sheet metal finish shall have Interior surface = 26 GA stucco embossed galvanized steel with a baked polyester white finish.

Exposed exterior surfaces = 26 GA stucco embossed galvanized steel with a baked polyester white finish.

Unexposed surfaces = 26 GA stucco embossed mill galvanized steel.

Door construction.

Door construction shall match the insulated panels. Each entrance to have a clear opening of 36" wide by 78" H. Doors shall be flush mounted, in-fitting and have a replaceable magnetic gasket on three sides and a neoprene adjustable gasket on all the sides of wall in which door fits in. Door hardware shall include three hinges, door closer and a latch/release. In addition, each door shall be equipped with a 14" x 14" heated viewing window. There should be a provision to lock the cold room

Floor panels.

Cold rooms not located on a concrete slab-on-grade will require sprayed-on insulation applied to the underside of the building deck below each cold room or insulated floor panels. If insulated floor panel are to be provided, the floor panels shall be capable of withstanding loads of up to 500 PSF. Wearing surface to be 18 GA type 304 #4 stainless steel over ½" non-wood underlayment. An ADA compliant transition ramp to be built into the insulated floor panel at each door.

6.0 | Utility Requirements

| The power supply can either be 220-240V single phase or 380-400V 3-phase, 50/60 Hz or recommended by manufacturer or supplier.

7.0 | Documents Required

| Complete documentation package with following documents should be delivered.

DQ

IQ

OQ

PQ

Operational manual (2 Sets)

Maintenance manual (2 Sets)

Spare part list

Calibration before installation.

8.0 | Drawings

| GA

Electrical drawings

9.0 | Training

| The vendor shall impart training for operation and maintenance for at least two personnel from user department and engineering /projects department and shall provide evaluation report.

10.0 | Abbreviations

| URS: User Requirement Specification

DQ: Design Qualification

IQ: Installation Qualification

OQ: Operational Qualification

PQ: Performance Qualification

GA: General Arrangement

NA: Not applicable

11.0 | Other Requirements

| NA

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BioBridge Healthcare Solutions Pvt. Ltd., India | BioBridge Healthcare Solutions Pvt. Ltd., India |
BioBridge Healthcare Solutions Pvt. Ltd., India | BioBridge Healthcare Solutions Pvt. Ltd., India
Prepared by: |||
Checked/Reviewed by: |||
Client | Client | Client | Client
Reviewed by: |||
Reviewed by: |||
Approved by: |||
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1.0 | Objective | Objective | Objective | Objective | Objective | Objective | Objective
| This URS detail the requirement for Lyophilizer 18 Sq.Mtr. Shelves Area with 300Kg Ice Condenser Capacity. | This URS detail the requirement for Lyophilizer 18 Sq.Mtr. Shelves Area with 300Kg Ice Condenser Capacity. | This URS detail the requirement for Lyophilizer 18 Sq.Mtr. Shelves Area with 300Kg Ice Condenser Capacity. | This URS detail the requirement for Lyophilizer 18 Sq.Mtr. Shelves Area with 300Kg Ice Condenser Capacity. | This URS detail the requirement for Lyophilizer 18 Sq.Mtr. Shelves Area with 300Kg Ice Condenser Capacity. | This URS detail the requirement for Lyophilizer 18 Sq.Mtr. Shelves Area with 300Kg Ice Condenser Capacity. | This URS detail the requirement for Lyophilizer 18 Sq.Mtr. Shelves Area with 300Kg Ice Condenser Capacity. | This URS detail the requirement for Lyophilizer 18 Sq.Mtr. Shelves Area with 300Kg Ice Condenser Capacity.
2.0 | Scope | Scope | Scope | Scope | Scope | Scope | Scope
| This document is applicable for the Lyophilizer specified in this document, will be installed with controlled environment, temperature range of 15 to 40 °C. | This document is applicable for the Lyophilizer specified in this document, will be installed with controlled environment, temperature range of 15 to 40 °C. | This document is applicable for the Lyophilizer specified in this document, will be installed with controlled environment, temperature range of 15 to 40 °C. | This document is applicable for the Lyophilizer specified in this document, will be installed with controlled environment, temperature range of 15 to 40 °C. | This document is applicable for the Lyophilizer specified in this document, will be installed with controlled environment, temperature range of 15 to 40 °C. | This document is applicable for the Lyophilizer specified in this document, will be installed with controlled environment, temperature range of 15 to 40 °C. | This document is applicable for the Lyophilizer specified in this document, will be installed with controlled environment, temperature range of 15 to 40 °C.
3.0 | Equipment description | Equipment description | Equipment description | Equipment description | Equipment description | Equipment description | Equipment description
| A Lyophilizer, commonly referred to as a freeze dryer, is a laboratory or industrial device used to remove moisture from substances through the process of sublimation. | A Lyophilizer, commonly referred to as a freeze dryer, is a laboratory or industrial device used to remove moisture from substances through the process of sublimation. | A Lyophilizer, commonly referred to as a freeze dryer, is a laboratory or industrial device used to remove moisture from substances through the process of sublimation. | A Lyophilizer, commonly referred to as a freeze dryer, is a laboratory or industrial device used to remove moisture from substances through the process of sublimation. | A Lyophilizer, commonly referred to as a freeze dryer, is a laboratory or industrial device used to remove moisture from substances through the process of sublimation. | A Lyophilizer, commonly referred to as a freeze dryer, is a laboratory or industrial device used to remove moisture from substances through the process of sublimation. | A Lyophilizer, commonly referred to as a freeze dryer, is a laboratory or industrial device used to remove moisture from substances through the process of sublimation.
4.0 | Component details | Component details | Component details | Component details | Component details | Component details | Component details
Component details | Component details | Component details
| Design Basis: | Design Basis: | Design Basis: | Design Basis: | Design Basis: | Design Basis: | Design Basis: | Design Basis:
| Product Name | Product Type | Product Type | Tray Size | Tray Size | Tray fill Volume | Batch Size | No of Trays/Batch
| NA | Liquid | Liquid | 10X9 inches | 10X9 inches | XXX ml | XX L | 264
| Capacities required:

Shelves : 18 Sq. Mtr
No. of working shelves : 12
Ice condenser capacity : 300 kg | Capacities required:

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No. of working shelves : 12
Ice condenser capacity : 300 kg | Capacities required:

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Ice condenser capacity : 300 kg | Capacities required:

Shelves : 18 Sq. Mtr
No. of working shelves : 12
Ice condenser capacity : 300 kg
| Process Requirements criteria: | Process Requirements criteria: | Process Requirements criteria: |
Process Requirements criteria: | Process Requirements criteria: | Process Requirements criteria: |
Process Requirements criteria: | Process Requirements criteria:
| Process control | Process control | Process control | Process control | Acceptance criteria |
Acceptance criteria | Acceptance criteria | Acceptance criteria
| Shelf Temperature ranges | Shelf Temperature ranges | Shelf Temperature ranges | Shelf Temperature
ranges | -60 to + 65 Deg C | -60 to + 65 Deg C | -60 to + 65 Deg C | -60 to + 65 Deg C
| Temperature control range | Temperature control range | Temperature control range | Temperature control
range | Across the shelves <+/- 2 Deg C
Within the shelves <+/- 2 Deg C | Across the shelves <+/- 2 Deg C
Within the shelves <+/- 2 Deg C | Across the shelves <+/- 2 Deg C
Within the shelves <+/- 2 Deg C | Across the shelves <+/- 2 Deg C
Within the shelves <+/- 2 Deg C
| Shelves Cooling rate (No Load) | Shelves Cooling rate (No Load) | Shelves Cooling rate (No Load) |
Shelves Cooling rate (No Load) | 20 Deg C to -40 Deg C in ≤ 30 Min | 20 Deg C to -40 Deg C in ≤ 30 Min |
20 Deg C to -40 Deg C in ≤ 30 Min | 20 Deg C to -40 Deg C in ≤ 30 Min
| Shelves Ultimate cooling (No Load) | Shelves Ultimate cooling (No Load) | Shelves Ultimate cooling (No
Load) | Shelves Ultimate cooling (No Load) | ≤ -60 Deg C | ≤ -60 Deg C | ≤ -60 Deg C | ≤ -60 Deg C
| Shelves heating rate (No Load) | Shelves heating rate (No Load) | Shelves heating rate (No Load) |
Shelves heating rate (No Load) | -40 Deg C to 20 Deg C ≤ 40 Min | -40 Deg C to 20 Deg C ≤ 40 Min | -40
Deg C to 20 Deg C ≤ 40 Min | -40 Deg C to 20 Deg C ≤ 40 Min
| Shelves Ultimate heating | Shelves Ultimate heating | Shelves Ultimate heating | Shelves Ultimate
heating | + 70 Deg | + 70 Deg | + 70 Deg | + 70 Deg
| Condenser cooling | Condenser cooling | Condenser cooling | Condenser cooling | < 20 Min | < 20 Min | <
20 Min | < 20 Min
| Condenser ultimate cooling | Condenser ultimate cooling | Condenser ultimate cooling | Condenser
ultimate cooling | ≤ - 80 Deg C | ≤ - 80 Deg C | ≤ - 80 Deg C | ≤ - 80 Deg C
| Ultimate vacuum reaching time | Ultimate vacuum reaching time | Ultimate vacuum reaching time |
Ultimate vacuum reaching time | < 30 Min | < 30 Min | < 30 Min | < 30 Min
| Ultimate vacuum | Ultimate vacuum | Ultimate vacuum | Ultimate vacuum | ≤ 0.001 Mbar Absolute | ≤ 0.001
Mbar Absolute | ≤ 0.001 Mbar Absolute | ≤ 0.001 Mbar Absolute
| Vacuum Control | Vacuum Control | Vacuum Control | Vacuum Control | 1000 Mbar to 0.001 Mbar Abs | 1000
Mbar to 0.001 Mbar Abs | 1000 Mbar to 0.001 Mbar Abs | 1000 Mbar to 0.001 Mbar Abs
| Process Requirements: | Process Requirements: | Process Requirements: | Process Requirements: | Process
Requirements: | Process Requirements: | Process Requirements: | Process Requirements:
| The Lyophilizer Shall be designed to allow the following operation to be undertaken automatically: |
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Lyophilizer Shall be designed to allow the following operation to be undertaken automatically:
| Process Requirement | Process Requirement | Process Requirement | Process Requirement | Type of
Operation | Type of Operation | Type of Operation | Type of Operation
| Loading / Unloading | Loading / Unloading | Loading / Unloading | Loading / Unloading | Automatic /
Semi- Automatic | Automatic / Semi- Automatic | Automatic / Semi- Automatic | Automatic / Semi- Automatic
| Lyophilization | Lyophilization | Lyophilization | Lyophilization | Automatic | Automatic | Automatic |
Automatic
| Shelves Moving mechanism | Shelves Moving mechanism | Shelves Moving mechanism | Shelves Moving
mechanism | Automatic / Hydraulic operation | Automatic / Hydraulic operation | Automatic / Hydraulic
operation | Automatic / Hydraulic operation
| Cleaning in Place (CIP) | Cleaning in Place (CIP) | Cleaning in Place (CIP) | Cleaning in Place (CIP) |
Automatic (With PW & WFI recirculation and rinses) | Automatic (With PW & WFI recirculation and rinses) |
Automatic (With PW & WFI recirculation and rinses) | Automatic (With PW & WFI recirculation and rinses)
| Sterilization in place (SIP) | Sterilization in place (SIP) | Sterilization in place (SIP) |
Sterilization in place (SIP) | Automatic / With Pure steam at 121 Deg C | Automatic / With Pure steam at
121 Deg C | Automatic / With Pure steam at 121 Deg C | Automatic / With Pure steam at 121 Deg C
| Filter Integrity Testing | Filter Integrity Testing | Filter Integrity Testing | Filter Integrity
Testing | Automatic | Automatic | Automatic | Automatic
| Vacuum Leak test | Vacuum Leak test | Vacuum Leak test | Vacuum Leak test | Automatic | Automatic |
Automatic | Automatic
| Defrosting Cycle | Defrosting Cycle | Defrosting Cycle | Defrosting Cycle | Automatic (With Pure steam)
| Automatic (With Pure steam) | Automatic (With Pure steam) | Automatic (With Pure steam)
| General Requirements:
All parts shall be made of non-oxidizing material as per FDA requirements
All internal corners and edges inside the chamber and condenser must have rounded and smooth finishing.
And accessible for easy cleaning
The Control cabinet shall be integrated on the machine frame. And the cabinets shall be equipped with
proper ventilators
Chamber shelves design shall allow minimum and maximum loading of the product and useable 90% out of
total shelves area available.
Remote access feature shall be available for providing technical support of the feature
Machine shall have feature to have control the speed increase / decrease the speed through frequency
control drive and shall be available to operate on HMI & SCADA
The machine shall be designed to be able to operate and maintain the temperature during the one
compressor failure

Redundancy Fluid pump shall be provided
Vacuum pump redundancy, shall be able to operate with one pump and maintain the set vacuum at any point of operation
All Hot surfaces shall be insulated with mineral wool and cladding with SS304 sheet.
The bellows used in the entire system shall be leak proof.

Chamber:

The chamber shall have the automatic Locking / Un-locking system
The Geometrical construction of the chamber shall be rectangular structure
The type of door support shall be operated with Hinged
Operating design pressure of the chamber shall be 2.0 Bar G
The slot door shall be installed at Loading /Unloading side and additional main door shall be installed at technical area for maintenance purpose and the door can be completely open-able up to 180 Deg direction.
Chamber shall have the jacket for cooling water circulation provision to do the jacket cooling after SIP and other heating operations.
The material of construction of the chamber shall be made SS316 material with surface finish Ra < 0.5 µs internally with mirror polish. And chamber body shall be insulated and clad with SS304 material
Chamber shall have the sight glass possible locations with light illumination

Shelves configuration:

Basic inter distance between the shelves : To be decided
Shelf dimensions (LxWxD) : To be decided
Total surface area : 18 Sq Mtr
Shelves heating rate : NLT 1.0 deg C/ Min
Shelves Minimum temperature : -65.0 Deg C
Shelves cooling source : Silicon Oil
Shelves moment for loading : Included
Shelves position control : Closed loop (Automatic)

The shelves shall allow the constant loading at fixed operating position/ height.
The shelves shall be made of SS316 L material and the surface finish shall be less than < 0.5 µm electro polished.
The shelves shall be designed to allowed minimum 2 formats for inter-distance for loading / Unloading
The Shelves loading should be from top to bottom direction and un-loading from bottom to top at constant height
Shelves shall be movable during CIP to ensure the 100% cleaning efficiency
Each shelf should have a temperature probe to measure the individual shelf temperature
The surface slope of the shelf maximum 0.5 mm/ Mtr

Refrigeration Compressors:

Total no. of compressors : To be defined
Compressors type : Screw type
Refrigerant : Non-CFC, Eco friendly

Vacuum System:

Vacuum pumps shall be dry pumps
The Pump capacity shall be design inline to the Chamber and condenser capacity
The ultimate vacuum shall be achieved with in 20 Min in ideal condition
The booster vacuum pump shall be dry type

Condenser:

Type of condenser : Close integrated
Number of condensers : 01 Nos
Design Pressure : 2.0 Bar G
Capacity of Ice thickness : To be defined
Minimum Temperature : >-75 Deg C

The construction of the condenser shall be cylindrical and should be suitable for vacuum applications during the lyophilization cycles
The condenser body and tubes shall be made of SS316 L and are mechanically polished and passivated
The ice condensing capacity shall be calculated on the basis of 24 Hr cycle
The suitable isolation valve shall be designed between the Chamber and condenser, and this shall be made up of SS316 L material.
The condenser cooling shall be done either with Gas/ Fluid
Sight glass with illumination shall be provided on body of the condenser for viewing purpose
The condenser shall be CIP and SIP able, and withstand with respect to the temperature and pressure

Functional Specification:

The Lyophilizer shall have interface with slot door with fixed loading/un-loading system
All Loading/ Unloading activities shall be from Grade A- Area
The Loading/ unloading system shall be designed as per the details and drawings provided by client
The loading / unloading operation shall be synchronized with lyophilizer operation

The system shall have the provision to back fill the equipment with sterile gases like Nitrogen/ Air and vacuum.
The vacuum control set point should be allowed from 1000 to 0.001 mbar absolute
SIP: The Sterilization cycle the temperature reference shall be measured at drain thermocouple, and all other temperature sensors shall be remains above the drain.
During SIP Automatic movement of shelves required from top to bottom to ensure the sterility of all components contained in the chamber
Vacuum pumps shall have the provision to provide the gas ballast to avoid the moisture and toxics impact during the process.

Integration of ISOLATOR for Loading/ Un-loading system:

The isolator unit shall be integrated with Lyophilizer as to meet the process requirements and current

good manufacturing practice

The design and integration of Isolator shall be in line to the Lyo loading and unloading chamber interface and dimensions

Tamper proof glass door shall be provided to allow a clear visibility into the entire space of the isolator.

The interior and all corners of the isolator must be easily accessible and cleanable when isolator is closed through glove ports

The surface in and outside side of the isolator shouldn't have any cracks, crevices and sharp edges. All corners and edges must be rounded

All HEPA Filters installed in the isolator must be accessible for maintenance and cleaning

Glare-free light, sealed edges of the lamps inside of isolator shall be provided

Calibration of all sensors must easily accessible and shall be removable for calibration and maintenance

Maximum expected operation of the equipment 24 Hrs./ Day

Technical Specification:

The temperature control maintained within the isolator in the range of 20 to 25 Deg C in all conditions.

Beginning of the batch

While continuous running of the batch

Fluctuation of outside environment shall be within tolerance

Relative Humidity control during production is maintained within the full humidity range of 2-5 % with aid of Nitrogen purge

HEPA filters are 99.99% efficient for removing particles greater than 0.3 Micron in size.

The room conditions of isolator installed is classified as grade B

Environment conditions surrounding room

Temp : NMT 26 Deg C

% RH : NMT 60%

The isolator unit is required to design to meet the cleanliness under the area to meet Grade 'A' requirement

The MOC of the isolator shall be compactable for VHP sterilization/ Nitrogen gas .and all product contact parts shall be made with SS316 L and surface finish will be $> 0.5 \mu$ RA.

Integrity test port shall be provided at upstream of HEPA Filters

Blower shall statically and dynamically balance to the supply air flow with sufficient capacity and static pressure

Viable/ non-viable particle count shall meet grade A requirements

The air velocity shall be 90 ± 20 FPM at 6 inches from the filter face

Initial and final pressure drop limits to be provided

During the VHP cycle, Gloves shall not be wrinkled or folded on surface

There shall be automatic leak test option available for isolator leak test and within the acceptance range; Leak test will be performed before start of the VHP cycle.

Interlocks/ Alarms:

Emergency Stop

Control power fault

Barrier open door

Glove port light curtain interrupted

Isolator contamination/ fault

Filter differential pressure alarm

Barrier high pressure alarm

Barrier low pressure alarm

Barrier differential pressure alarm

RH above 5% alarm

All machine surfaces must come in contact with H2O2, a minimum accessible area for H2O2 on the machine for example moving parts shall move slowly during VHP Cycle.

The whole decontamination cycle including aeration shall be completed within the specified operation time mentioned.

The isolator doors shall not open before the H2O2 concentration reaches to the neutral level.

Interface:

The Lyophilizer shall be provided with Siemens or Allen Bradley control logic platform. PLC, security, Recipe management process. Data logging shall be under taken Lyophilized dedicated SCADA.

Control system shall be accessible to authorized users only. It shall have individual login ID and Passwords to access the system.

All set parameters ranges are to be defined access shall be restricted as per user level

Alarm management shall be available

Programmed backup for PLC/HMI SCADA shall be provided

Generated data shall be saved by the software and storage path shall be defined through software only

Control system clock synchronization shall be available

11.8 Electronic records and signatures shall be available wherever applicable on the reports | General Requirements:

All parts shall be made of non-oxidizing material as per FDA requirements

All internal corners and edges inside the chamber and condenser must have rounded and smooth finishing. And accessible for easy cleaning

The Control cabinet shall be integrated on the machine frame. And the cabinets shall be equipped with proper ventilators

Chamber shelves design shall allow minimum and maximum loading of the product and useable 90% out of total shelves area available.

Remote access feature shall be available for providing technical support of the feature

Machine shall have feature to have control the speed increase / decrease the speed through frequency control drive and shall be available to operate on HMI & SCADA

The machine shall be designed to be able to operate and maintain the temperature during the one compressor failure

Redundancy Fluid pump shall be provided

Vacuum pump redundancy, shall be able to operate with one pump and maintain the set vacuum at any point of operation

All Hot surfaces shall be insulated with mineral wool and cladding with SS304 sheet.

The bellows used in the entire system shall be leak proof.

Chamber:

The chamber shall have the automatic Locking / Un-locking system
The Geometrical construction of the chamber shall be rectangular structure
The type of door support shall be operated with Hinged
Operating design pressure of the chamber shall be 2.0 Bar G
The slot door shall be installed at Loading /Unloading side and additional main door shall be installed at technical area for maintenance purpose and the door can be completely open-able up to 180 Deg direction.
Chamber shall have the jacket for cooling water circulation provision to do the jacket cooling after SIP and other heating operations.
The material of construction of the chamber shall be made SS316 material with surface finish Ra < 0.5 µm internally with mirror polish. And chamber body shall be insulated and clad with SS304 material
Chamber shall have the sight glass possible locations with light illumination

Shelves configuration:

Basic inter distance between the shelves : To be decided
Shelf dimensions (LxWxD) : To be decided
Total surface area : 18 Sq Mtr
Shelves heating rate : NLT 1.0 deg C/ Min
Shelves Minimum temperature : -65.0 Deg C
Shelves cooling source : Silicon Oil
Shelves moment for loading : Included
Shelves position control : Closed loop (Automatic)

The shelves shall allow the constant loading at fixed operating position/ height.
The shelves shall be made of SS316 L material and the surface finish shall be less than < 0.5 µm electro polished.
The shelves shall be designed to allowed minimum 2 formats for inter-distance for loading / Unloading
The Shelves loading should be from top to bottom direction and un-loading from bottom to top at constant height
Shelves shall be movable during CIP to ensure the 100% cleaning efficiency
Each shelf should have a temperature probe to measure the individual shelf temperature
The surface slope of the shelf maximum 0.5 mm/ Mtr

Refrigeration Compressors:

Total no. of compressors : To be defined
Compressors type : Screw type
Refrigerant : Non-CFC, Eco friendly

Vacuum System:

Vacuum pumps shall be dry pumps
The Pump capacity shall be design inline to the Chamber and condenser capacity
The ultimate vacuum shall be achieved with in 20 Min in ideal condition
The booster vacuum pump shall be dry type

Condenser:

Type of condenser : Close integrated
Number of condensers : 01 Nos
Design Pressure : 2.0 Bar G
Capacity of Ice thickness : To be defined
Minimum Temperature : >-75 Deg C

The construction of the condenser shall be cylindrical and should be suitable for vacuum applications during the lyophilization cycles
The condenser body and tubes shall be made of SS316 L and are mechanically polished and passivated
The ice condensing capacity shall be calculated on the basis of 24 Hr cycle
The suitable isolation valve shall be designed between the Chamber and condenser, and this shall be made up of SS316 L material.
The condenser cooling shall be done either with Gas/ Fluid
Sight glass with illumination shall be provided on body of the condenser for viewing purpose
The condenser shall be CIP and SIP able, and withstand with respect to the temperature and pressure

Functional Specification:

The Lyophilizer shall have interface with slot door with fixed loading/un-loading system
All Loading/ Unloading activities shall be from Grade A- Area
The Loading/ unloading system shall be designed as per the details and drawings provided by client
The loading / unloading operation shall be synchronized with lyophilizer operation

The system shall have the provision to back fill the equipment with sterile gases like Nitrogen/ Air and vacuum.
The vacuum control set point should be allowed from 1000 to 0.001 mbar absolute
SIP: The Sterilization cycle the temperature reference shall be measured at drain thermocouple, and all other temperature sensors shall be remains above the drain.
During SIP Automatic movement of shelves required from top to bottom to ensure the sterility of all components contained in the chamber
Vacuum pumps shall have the provision to provide the gas ballast to avoid the moisture and toxics impact during the process.

Integration of ISOLATOR for Loading/ Un-loading system:

The isolator unit shall be integrated with Lyophilizer as to meet the process requirements and current good manufacturing practice
The design and integration of Isolator shall be in line to the Lyo loading and unloading chamber interface and dimensions
Tamper proof glass door shall be provided to allow a clear visibility into the entire space of the isolator.
The interior and all corners of the isolator must be easily accessible and cleanable when isolator is closed through glove ports

The surface in and outside side of the isolator shouldn't have any cracks, crevices and sharp edges. All corners and edges must be rounded
All HEPA Filters installed in the isolator must be accessible for maintenance and cleaning
Glare-free light, sealed edges of the lamps inside of isolator shall be provided
Calibration of all sensors must easily accessible and shall be removable for calibration and maintenance
Maximum expected operation of the equipment 24 Hrs./ Day

Technical Specification:

The temperature control maintained within the isolator in the range of 20 to 25 Deg C in all conditions.
Beginning of the batch
While continuous running of the batch
Fluctuation of outside environment shall be within tolerance
Relative Humidity control during production is maintained within the full humidity range of 2-5 % with aid of Nitrogen purge
HEPA filters are 99.99% efficient for removing particles greater than 0.3 Micron in size.
The room conditions of isolator installed is classified as grade B
Environment conditions surrounding room
Temp : NMT 26 Deg C
% RH : NMT 60%
The isolator unit is required to design to meet the cleanliness under the area to meet Grade 'A' requirement
The MOC of the isolator shall be compactable for VHP sterilization/ Nitrogen gas .and all product contact parts shall be made with SS316 L and surface finish will be $> 0.5 \mu$ RA.
Integrity test port shall be provided at upstream of HEPA Filters
Blower shall statically and dynamically balance to the supply air flow with sufficient capacity and static pressure
Viable/ non-viable particle count shall meet grade A requirements
The air velocity shall be 90 ± 20 FPM at 6 inches from the filter face
Initial and final pressure drop limits to be provided
During the VHP cycle, Gloves shall not be wrinkled or folded on surface
There shall be automatic leak test option available for isolator leak test and within the acceptance range; Leak test will be performed before start of the VHP cycle.

Interlocks/ Alarms:

Emergency Stop
Control power fault
Barrier open door
Glove port light curtain interrupted
Isolator contamination/ fault
Filter differential pressure alarm
Barrier high pressure alarm
Barrier low pressure alarm
Barrier differential pressure alarm
RH above 5% alarm

All machine surfaces must come in contact with H₂O₂, a minimum accessible area for H₂O₂ on the machine for example moving parts shall move slowly during VHP Cycle.
The whole decontamination cycle including aeration shall be completed within the specified operation time mentioned.
The isolator doors shall not open before the H₂O₂ concentration reaches to the neutral level.

Interface:

The Lyophilizer shall be provided with Siemens or Allen Bradley control logic platform. PLC, security, Recipe management process. Data logging shall be under taken Lyophilized dedicated SCADA.
Control system shall be accessible to authorized users only. It shall have individual login ID and Passwords to access the system.
All set parameters ranges are to be defined access shall be restricted as per user level
Alarm management shall be available
Programmed backup for PLC/HMI SCADA shall be provided
Generated data shall be saved by the software and storage path shall be defined through software only
Control system clock synchronization shall be available
11.8 Electronic records and signatures shall be available wherever applicable on the reports | General Requirements:
All parts shall be made of non-oxidizing material as per FDA requirements
All internal corners and edges inside the chamber and condenser must have rounded and smooth finishing. And accessible for easy cleaning
The Control cabinet shall be integrated on the machine frame. And the cabinets shall be equipped with proper ventilators
Chamber shelves design shall allow minimum and maximum loading of the product and useable 90% out of total shelves area available.
Remote access feature shall be available for providing technical support of the feature
Machine shall have feature to have control the speed increase / decrease the speed through frequency control drive and shall be available to operate on HMI & SCADA
The machine shall be designed to be able to operate and maintain the temperature during the one compressor failure
Redundancy Fluid pump shall be provided
Vacuum pump redundancy, shall be able to operate with one pump and maintain the set vacuum at any point of operation
All Hot surfaces shall be insulated with mineral wool and cladding with SS304 sheet.
The bellows used in the entire system shall be leak proof.

Chamber:

The chamber shall have the automatic Locking / Un-locking system
The Geometrical construction of the chamber shall be rectangular structure
The type of door support shall be operated with Hinged
Operating design pressure of the chamber shall be 2.0 Bar G
The slot door shall be installed at Loading /Unloading side and additional main door shall be installed at technical area for maintenance purpose and the door can be completely open-able up to 180 Deg

direction.

Chamber shall have the jacket for cooling water circulation provision to do the jacket cooling after SIP and other heating operations.

The material of construction of the chamber shall be made SS316 material with surface finish $R_a < 0.5 \mu\text{s}$

internally with mirror polish. And chamber body shall be insulated and clad with SS304 material

Chamber shall have the sight glass possible locations with light illumination

Shelves configuration:

Basic inter distance between the shelves : To be decided

Shelf dimensions (LxWxD) : To be decided

Total surface area : 18 Sq Mtr

Shelves heating rate : NLT 1.0 deg C/ Min

Shelves Minimum temperature : -65.0 Deg C

Shelves cooling source : Silicon Oil

Shelves moment for loading : Included

Shelves position control : Closed loop (Automatic)

The shelves shall allow the constant loading at fixed operating position/ height.

The shelves shall be made of SS316 L material and the surface finish shall be less than $< 0.5 \mu\text{m}$ electro polished.

The shelves shall be designed to allowed minimum 2 formats for inter-distance for loading / Unloading

The Shelves loading should be from top to bottom direction and un-loading from bottom to top at constant height

Shelves shall be movable during CIP to ensure the 100% cleaning efficiency

Each shelf should have a temperature probe to measure the individual shelf temperature

The surface slope of the shelf maximum 0.5 mm/ Mtr

Refrigeration Compressors:

Total no. of compressors : To be defined

Compressors type : Screw type

Refrigerant : Non-CFC, Eco friendly

Vacuum System:

Vacuum pumps shall be dry pumps

The Pump capacity shall be design inline to the Chamber and condenser capacity

The ultimate vacuum shall be achieved with in 20 Min in ideal condition

The booster vacuum pump shall be dry type

Condenser:

Type of condenser : Close integrated

Number of condensers : 01 Nos

Design Pressure : 2.0 Bar G

Capacity of Ice thickness : To be defined

Minimum Temperature : $> -75 \text{ Deg C}$

The construction of the condenser shall be cylindrical and should be suitable for vacuum applications during the lyophilization cycles

The condenser body and tubes shall be made of SS316 L and are mechanically polished and passivated

The ice condensing capacity shall be calculated on the basis of 24 Hr cycle

The suitable isolation valve shall be designed between the Chamber and condenser, and this shall be made up of SS316 L material.

The condenser cooling shall be done either with Gas/ Fluid

Sight glass with illumination shall be provided on body of the condenser for viewing purpose

The condenser shall be CIP and SIP able, and withstand with respect to the temperature and pressure

Functional Specification:

The Lyophilizer shall have interface with slot door with fixed loading/un-loading system

All Loading/ Unloading activities shall be from Grade A- Area

The Loading/ unloading system shall be designed as per the details and drawings provided by client

The loading / unloading operation shall be synchronized with lyophilizer operation

The system shall have the provision to back fill the equipment with sterile gases like Nitrogen/ Air and vacuum.

The vacuum control set point should be allowed from 1000 to 0.001 mbar absolute

SIP: The Sterilization cycle the temperature reference shall be measured at drain thermocouple, and all other temperature sensors shall be remains above the drain.

During SIP Automatic movement of shelves required from top to bottom to ensure the sterility of all components contained in the chamber

Vacuum pumps shall have the provision to provide the gas ballast to avoid the moisture and toxics impact during the process.

Integration of ISOLATOR for Loading/ Un-loading system:

The isolator unit shall be integrated with Lyophilizer as to meet the process requirements and current good manufacturing practice

The design and integration of Isolator shall be in line to the Lyo loading and unloading chamber interface and dimensions

Tamper proof glass door shall be provided to allow a clear visibility into the entire space of the isolator.

The interior and all corners of the isolator must be easily accessible and cleanable when isolator is closed through glove ports

The surface in and outside side of the isolator shouldn't have any cracks, crevices and sharp edges. All corners and edges must be rounded

All HEPA Filters installed in the isolator must be accessible for maintenance and cleaning

Glare-free light, sealed edges of the lamps inside of isolator shall be provided

Calibration of all sensors must easily accessible and shall be removable for calibration and maintenance

Maximum expected operation of the equipment 24 Hrs./ Day

Technical Specification:

The temperature control maintained within the isolator in the range of 20 to 25 Deg C in all conditions.

Beginning of the batch

While continuous running of the batch

Fluctuation of outside environment shall be within tolerance

Relative Humidity control during production is maintained within the full humidity range of 2-5 % with aid of Nitrogen purge

HEPA filters are 99.99% efficient for removing particles greater than 0.3 Micron in size.

The room conditions of isolator installed is classified as grade B

Environment conditions surrounding room

Temp : NMT 26 Deg C

% RH : NMT 60%

The isolator unit is required to design to meet the cleanliness under the area to meet Grade 'A' requirement

The MOC of the isolator shall be compactable for VHP sterilization/ Nitrogen gas .and all product contact parts shall be made with SS316 L and surface finish will be $> 0.5 \mu$ RA.

Integrity test port shall be provided at upstream of HEPA Filters

Blower shall statically and dynamically balance to the supply air flow with sufficient capacity and static pressure

Viable/ non-viable particle count shall meet grade A requirements

The air velocity shall be 90 ± 20 FPM at 6 inches from the filter face

Initial and final pressure drop limits to be provided

During the VHP cycle, Gloves shall not be wrinkled or folded on surface

There shall be automatic leak test option available for isolator leak test and within the acceptance range; Leak test will be performed before start of the VHP cycle.

Interlocks/ Alarms:

Emergency Stop

Control power fault

Barrier open door

Glove port light curtain interrupted

Isolator contamination/ fault

Filter differential pressure alarm

Barrier high pressure alarm

Barrier low pressure alarm

Barrier differential pressure alarm

RH above 5% alarm

All machine surfaces must come in contact with H₂O₂, a minimum accessible area for H₂O₂ on the machine for example moving parts shall move slowly during VHP Cycle.

The whole decontamination cycle including aeration shall be completed within the specified operation time mentioned.

The isolator doors shall not open before the H₂O₂ concentration reaches to the neutral level.

Interface:

The Lyophilizer shall be provided with Siemens or Allen Bradley control logic platform. PLC, security, Recipe management process. Data logging shall be under taken Lyophilized dedicated SCADA.

Control system shall be accessible to authorized users only. It shall have individual login ID and Passwords to access the system.

All set parameters ranges are to be defined access shall be restricted as per user level

Alarm management shall be available

Programmed backup for PLC/HMI SCADA shall be provided

Generated data shall be saved by the software and storage path shall be defined through software only

Control system clock synchronization shall be available

11.8 Electronic records and signatures shall be available wherever applicable on the reports | General Requirements:

All parts shall be made of non-oxidizing material as per FDA requirements

All internal corners and edges inside the chamber and condenser must have rounded and smooth finishing.

And accessible for easy cleaning

The Control cabinet shall be integrated on the machine frame. And the cabinets shall be equipped with proper ventilators

Chamber shelves design shall allow minimum and maximum loading of the product and useable 90% out of total shelves area available.

Remote access feature shall be available for providing technical support of the feature

Machine shall have feature to have control the speed increase / decrease the speed through frequency control drive and shall be available to operate on HMI & SCADA

The machine shall be designed to be able to operate and maintain the temperature during the one compressor failure

Redundancy Fluid pump shall be provided

Vacuum pump redundancy, shall be able to operate with one pump and maintain the set vacuum at any point of operation

All Hot surfaces shall be insulated with mineral wool and cladding with SS304 sheet.

The bellows used in the entire system shall be leak proof.

Chamber:

The chamber shall have the automatic Locking / Un-locking system

The Geometrical construction of the chamber shall be rectangular structure

The type of door support shall be operated with Hinged

Operating design pressure of the chamber shall be 2.0 Bar G

The slot door shall be installed at Loading /Unloading side and additional main door shall be installed at technical area for maintenance purpose and the door can be completely open-able up to 180 Deg direction.

Chamber shall have the jacket for cooling water circulation provision to do the jacket cooling after SIP and other heating operations.

The material of constriction of the chamber shall be made SS316 material with surface finish $Ra < 0.5 \mu$ s internally with mirror polish. And chamber body shall be insulated and cladded with SS304 material

Chamber shall have the sight glass possible locations with light illumination

Shelves configuration:

Basic inter distance between the shelves : To be decided
Shelf dimensions (LxWxD) : To be decided
Total surface area : 18 Sq Mtr
Shelves heating rate : NLT 1.0 deg C/ Min
Shelves Minimum temperature : -65.0 Deg C
Shelves cooling source : Silicon Oil
Shelves moment for loading : Included
Shelves position control : Closed loop (Automatic)

The shelves shall allow the constant loading at fixed operating position/ height.
The shelves shall be made of SS316 L material and the surface finish shall be less than $< 0.5 \mu\text{m}$ electro polished.
The shelves shall be designed to allowed minimum 2 formats for inter-distance for loading / Unloading
The Shelves loading should be from top to bottom direction and un-loading from bottom to top at constant height
Shelves shall be movable during CIP to ensure the 100% cleaning efficiency
Each shelf should have a temperature probe to measure the individual shelf temperature
The surface slope of the shelf maximum 0.5 mm/ Mtr

Refrigeration Compressors:

Total no. of compressors : To be defined
Compressors type : Screw type
Refrigerant : Non-CFC, Eco friendly

Vacuum System:

Vacuum pumps shall be dry pumps
The Pump capacity shall be design inline to the Chamber and condenser capacity
The ultimate vacuum shall be achieved with in 20 Min in ideal condition
The booster vacuum pump shall be dry type

Condenser:

Type of condenser : Close integrated
Number of condensers : 01 Nos
Design Pressure : 2.0 Bar G
Capacity of Ice thickness : To be defined
Minimum Temperature : >-75 Deg C

The construction of the condenser shall be cylindrical and should be suitable for vacuum applications during the lyophilization cycles
The condenser body and tubes shall be made of SS316 L and are mechanically polished and passivated
The ice condensing capacity shall be calculated on the basis of 24 Hr cycle
The suitable isolation valve shall be designed between the Chamber and condenser, and this shall be made up of SS316 L material.
The condenser cooling shall be done either with Gas/ Fluid
Sight glass with illumination shall be provided on body of the condenser for viewing purpose
The condenser shall be CIP and SIP able, and withstand with respect to the temperature and pressure

Functional Specification:

The Lyophilizer shall have interface with slot door with fixed loading/un-loading system
All Loading/ Unloading activities shall be from Grade A- Area
The Loading/ unloading system shall be designed as per the details and drawings provided by client
The loading / unloading operation shall be synchronized with lyophilizer operation

The system shall have the provision to back fill the equipment with sterile gases like Nitrogen/ Air and vacuum.
The vacuum control set point should be allowed from 1000 to 0.001 mbar absolute
SIP: The Sterilization cycle the temperature reference shall be measured at drain thermocouple, and all other temperature sensors shall be remains above the drain.
During SIP Automatic movement of shelves required from top to bottom to ensure the sterility of all components contained in the chamber
Vacuum pumps shall have the provision to provide the gas ballast to avoid the moisture and toxics impact during the process.

Integration of ISOLATOR for Loading/ Un-loading system:

The isolator unit shall be integrated with Lyophilizer as to meet the process requirements and current good manufacturing practice
The design and integration of Isolator shall be in line to the Lyo loading and unloading chamber interface and dimensions
Tamper proof glass door shall be provided to allow a clear visibility into the entire space of the isolator.
The interior and all corners of the isolator must be easily accessible and cleanable when isolator is closed through glove ports
The surface in and outside side of the isolator shouldn't have any cracks, crevices and sharp edges. All corners and edges must be rounded
All HEPA Filters installed in the isolator must be accessible for maintenance and cleaning
Glare-free light, sealed edges of the lamps inside of isolator shall be provided
Calibration of all sensors must easily accessible and shall be removable for calibration and maintenance
Maximum expected operation of the equipment 24 Hrs./ Day

Technical Specification:

The temperature control maintained within the isolator in the range of 20 to 25 Deg C in all conditions.
Beginning of the batch
While continuous running of the batch
Fluctuation of outside environment shall be within tolerance
Relative Humidity control during production is maintained within the full humidity range of 2-5 % with

aid of Nitrogen purge

HEPA filters are 99.99% efficient for removing particles greater than 0.3 Micron in size.

The room conditions of isolator installed is classified as grade B

Environment conditions surrounding room

Temp : NMT 26 Deg C

% RH : NMT 60%

The isolator unit is required to design to meet the cleanliness under the area to meet Grade 'A' requirement

The MOC of the isolator shall be compactable for VHP sterilization/ Nitrogen gas .and all product contact parts shall be made with SS316 L and surface finish will be $> 0.5 \mu$ RA.

Integrity test port shall be provided at upstream of HEPA Filters

Blower shall statically and dynamically balance to the supply air flow with sufficient capacity and static pressure

Viable/ non-viable particle count shall meet grade A requirements

The air velocity shall be 90 ± 20 FPM at 6 inches from the filter face

Initial and final pressure drop limits to be provided

During the VHP cycle, Gloves shall not be wrinkled or folded on surface

There shall be automatic leak test option available for isolator leak test and within the acceptance range; Leak test will be performed before start of the VHP cycle.

Interlocks/ Alarms:

Emergency Stop

Control power fault

Barrier open door

Glove port light curtain interrupted

Isolator contamination/ fault

Filter differential pressure alarm

Barrier high pressure alarm

Barrier low pressure alarm

Barrier differential pressure alarm

RH above 5% alarm

All machine surfaces must come in contact with H₂O₂, a minimum accessible area for H₂O₂ on the machine for example moving parts shall move slowly during VHP Cycle.

The whole decontamination cycle including aeration shall be completed within the specified operation time mentioned.

The isolator doors shall not open before the H₂O₂ concentration reaches to the neutral level.

Interface:

The Lyophilizer shall be provided with Siemens or Allen Bradley control logic platform. PLC, security,

Recipe management process. Data logging shall be under taken Lyophilized dedicated SCADA.

Control system shall be accessible to authorized users only. It shall have individual login ID and

Passwords to access the system.

All set parameters ranges are to be defined access shall be restricted as per user level

Alarm management shall be available

Programmed backup for PLC/HMI SCADA shall be provided

Generated data shall be saved by the software and storage path shall be defined through software only

Control system clock synchronization shall be available

11.8 Electronic records and signatures shall be available wherever applicable on the reports | General

Requirements:

All parts shall be made of non-oxidizing material as per FDA requirements

All internal corners and edges inside the chamber and condenser must have rounded and smooth finishing.

And accessible for easy cleaning

The Control cabinet shall be integrated on the machine frame. And the cabinets shall be equipped with proper ventilators

Chamber shelves design shall allow minimum and maximum loading of the product and useable 90% out of total shelves area available.

Remote access feature shall be available for providing technical support of the feature

Machine shall have feature to have control the speed increase / decrease the speed through frequency

control drive and shall be available to operate on HMI & SCADA

The machine shall be designed to be able to operate and maintain the temperature during the one compressor failure

Redundancy Fluid pump shall be provided

Vacuum pump redundancy, shall be able to operate with one pump and maintain the set vacuum at any point of operation

All Hot surfaces shall be insulated with mineral wool and cladding with SS304 sheet.

The bellows used in the entire system shall be leak proof.

Chamber:

The chamber shall have the automatic Locking / Un-locking system

The Geometrical construction of the chamber shall be rectangular structure

The type of door support shall be operated with Hinged

Operating design pressure of the chamber shall be 2.0 Bar G

The slot door shall be installed at Loading /Unloading side and additional main door shall be installed at technical area for maintenance purpose and the door can be completely open-able up to 180 Deg

direction.

Chamber shall have the jacket for cooling water circulation provision to do the jacket cooling after SIP and other heating operations.

The material of constriction of the chamber shall be made SS316 material with surface finish $Ra < 0.5 \mu$ s internally with mirror polish. And chamber body shall be insulated and cladded with SS304 material

Chamber shall have the sight glass possible locations with light illumination

Shelves configuration:

Basic inter distance between the shelves : To be decided

Shelf dimensions (LxWxD) : To be decided

Total surface area : 18 Sq Mtr

Shelves heating rate : NLT 1.0 deg C/ Min

Shelves Minimum temperature : -65.0 Deg C

Shelves cooling source : Silicon Oil
Shelves moment for loading : Included
Shelves position control : Closed loop (Automatic)

The shelves shall allow the constant loading at fixed operating position/ height.
The shelves shall be made of SS316 L material and the surface finish shall be less than < 0.5 µm electro polished.
The shelves shall be designed to allowed minimum 2 formats for inter-distance for loading / Unloading
The Shelves loading should be from top to bottom direction and un-loading from bottom to top at constant height
Shelves shall be movable during CIP to ensure the 100% cleaning efficiency
Each shelf should have a temperature probe to measure the individual shelf temperature
The surface slope of the shelf maximum 0.5 mm/ Mtr

Refrigeration Compressors:

Total no. of compressors : To be defined
Compressors type : Screw type
Refrigerant : Non-CFC, Eco friendly

Vacuum System:

Vacuum pumps shall be dry pumps
The Pump capacity shall be design inline to the Chamber and condenser capacity
The ultimate vacuum shall be achieved with in 20 Min in ideal condition
The booster vacuum pump shall be dry type

Condenser:

Type of condenser : Close integrated
Number of condensers : 01 Nos
Design Pressure : 2.0 Bar G
Capacity of Ice thickness : To be defined
Minimum Temperature : >-75 Deg C

The construction of the condenser shall be cylindrical and should be suitable for vacuum applications during the lyophilization cycles
The condenser body and tubes shall be made of SS316 L and are mechanically polished and passivated
The ice condensing capacity shall be calculated on the basis of 24 Hr cycle
The suitable isolation valve shall be designed between the Chamber and condenser, and this shall be made up of SS316 L material.
The condenser cooling shall be done either with Gas/ Fluid
Sight glass with illumination shall be provided on body of the condenser for viewing purpose
The condenser shall be CIP and SIP able, and withstand with respect to the temperature and pressure

Functional Specification:

The Lyophilizer shall have interface with slot door with fixed loading/un-loading system
All Loading/ Unloading activities shall be from Grade A- Area
The Loading/ unloading system shall be designed as per the details and drawings provided by client
The loading / unloading operation shall be synchronized with lyophilizer operation

The system shall have the provision to back fill the equipment with sterile gases like Nitrogen/ Air and vacuum.
The vacuum control set point should be allowed from 1000 to 0.001 mbar absolute
SIP: The Sterilization cycle the temperature reference shall be measured at drain thermocouple, and all other temperature sensors shall be remains above the drain.
During SIP Automatic movement of shelves required from top to bottom to ensure the sterility of all components contained in the chamber
Vacuum pumps shall have the provision to provide the gas ballast to avoid the moisture and toxics impact during the process.

Integration of ISOLATOR for Loading/ Un-loading system:

The isolator unit shall be integrated with Lyophilizer as to meet the process requirements and current good manufacturing practice
The design and integration of Isolator shall be in line to the Lyo loading and unloading chamber interface and dimensions
Tamper proof glass door shall be provided to allow a clear visibility into the entire space of the isolator.
The interior and all corners of the isolator must be easily accessible and cleanable when isolator is closed through glove ports
The surface in and outside side of the isolator shouldn't have any cracks, crevices and sharp edges. All corners and edges must be rounded
All HEPA Filters installed in the isolator must be accessible for maintenance and cleaning
Glare-free light, sealed edges of the lamps inside of isolator shall be provided
Calibration of all sensors must easily accessible and shall be removable for calibration and maintenance
Maximum expected operation of the equipment 24 Hrs./ Day

Technical Specification:

The temperature control maintained within the isolator in the range of 20 to 25 Deg C in all conditions.
Beginning of the batch
While continuous running of the batch
Fluctuation of outside environment shall be within tolerance
Relative Humidity control during production is maintained within the full humidity range of 2-5 % with aid of Nitrogen purge
HEPA filters are 99.99% efficient for removing particles greater than 0.3 Micron in size.
The room conditions of isolator installed is classified as grade B
Environment conditions surrounding room
Temp : NMT 26 Deg C
% RH : NMT 60%
The isolator unit is required to design to meet the cleanliness under the area to meet Grade 'A'

requirement

The MOC of the isolator shall be compactable for VHP sterilization/ Nitrogen gas .and all product contact parts shall be made with SS316 L and surface finish will be $> 0.5 \mu$ RA.

Integrity test port shall be provided at upstream of HEPA Filters

Blower shall statically and dynamically balance to the supply air flow with sufficient capacity and static pressure

Viable/ non-viable particle count shall meet grade A requirements

The air velocity shall be 90 ± 20 FPM at 6 inches from the filter face

Initial and final pressure drop limits to be provided

During the VHP cycle, Gloves shall not be wrinkled or folded on surface

There shall be automatic leak test option available for isolator leak test and within the acceptance range; Leak test will be performed before start of the VHP cycle.

Interlocks/ Alarms:

Emergency Stop

Control power fault

Barrier open door

Glove port light curtain interrupted

Isolator contamination/ fault

Filter differential pressure alarm

Barrier high pressure alarm

Barrier low pressure alarm

Barrier differential pressure alarm

RH above 5% alarm

All machine surfaces must come in contact with H₂O₂, a minimum accessible area for H₂O₂ on the machine for example moving parts shall move slowly during VHP Cycle.

The whole decontamination cycle including aeration shall be completed within the specified operation time mentioned.

The isolator doors shall not open before the H₂O₂ concentration reaches to the neutral level.

Interface:

The Lyophilizer shall be provided with Siemens or Allen Bradley control logic platform. PLC, security,

Recipe management process. Data logging shall be under taken Lyophilized dedicated SCADA.

Control system shall be accessible to authorized users only. It shall have individual login ID and Passwords to access the system.

All set parameters ranges are to be defined access shall be restricted as per user level

Alarm management shall be available

Programmed backup for PLC/HMI SCADA shall be provided

Generated data shall be saved by the software and storage path shall be defined through software only

Control system clock synchronization shall be available

11.8 Electronic records and signatures shall be available wherever applicable on the reports | General Requirements:

All parts shall be made of non-oxidizing material as per FDA requirements

All internal corners and edges inside the chamber and condenser must have rounded and smooth finishing.

And accessible for easy cleaning

The Control cabinet shall be integrated on the machine frame. And the cabinets shall be equipped with proper ventilators

Chamber shelves design shall allow minimum and maximum loading of the product and useable 90% out of total shelves area available.

Remote access feature shall be available for providing technical support of the feature

Machine shall have feature to have control the speed increase / decrease the speed through frequency control drive and shall be available to operate on HMI & SCADA

The machine shall be designed to be able to operate and maintain the temperature during the one compressor failure

Redundancy Fluid pump shall be provided

Vacuum pump redundancy, shall be able to operate with one pump and maintain the set vacuum at any point of operation

All Hot surfaces shall be insulated with mineral wool and cladding with SS304 sheet.

The bellows used in the entire system shall be leak proof.

Chamber:

The chamber shall have the automatic Locking / Un-locking system

The Geometrical construction of the chamber shall be rectangular structure

The type of door support shall be operated with Hinged

Operating design pressure of the chamber shall be 2.0 Bar G

The slot door shall be installed at Loading /Unloading side and additional main door shall be installed at technical area for maintenance purpose and the door can be completely open-able up to 180 Deg direction.

Chamber shall have the jacket for cooling water circulation provision to do the jacket cooling after SIP and other heating operations.

The material of constriction of the chamber shall be made SS316 material with surface finish $Ra < 0.5 \mu$ s internally with mirror polish. And chamber body shall be insulated and cladded with SS304 material

Chamber shall have the sight glass possible locations with light illumination

Shelves configuration:

Basic inter distance between the shelves : To be decided

Shelf dimensions (LxWxD) : To be decided

Total surface area : 18 Sq Mtr

Shelves heating rate : NLT 1.0 deg C/ Min

Shelves Minimum temperature : -65.0 Deg C

Shelves cooling source : Silicon Oil

Shelves moment for loading : Included

Shelves position control : Closed loop (Automatic)

The shelves shall allow the constant loading at fixed operating position/ height.

The shelves shall be made of SS316 L material and the surface finish shall be less than $< 0.5 \mu$ m electro polished.

The shelves shall be designed to allowed minimum 2 formats for inter-distance for loading / Unloading
The Shelves loading should be from top to bottom direction and un-loading from bottom to top at constant height
Shelves shall be movable during CIP to ensure the 100% cleaning efficiency
Each shelf should have a temperature probe to measure the individual shelf temperature
The surface slope of the shelf maximum 0.5 mm/ Mtr

Refrigeration Compressors:

Total no. of compressors : To be defined
Compressors type : Screw type
Refrigerant : Non-CFC, Eco friendly

Vacuum System:

Vacuum pumps shall be dry pumps
The Pump capacity shall be design inline to the Chamber and condenser capacity
The ultimate vacuum shall be achieved with in 20 Min in ideal condition
The booster vacuum pump shall be dry type

Condenser:

Type of condenser : Close integrated
Number of condensers : 01 Nos
Design Pressure : 2.0 Bar G
Capacity of Ice thickness : To be defined
Minimum Temperature : >-75 Deg C

The construction of the condenser shall be cylindrical and should be suitable for vacuum applications during the lyophilization cycles
The condenser body and tubes shall be made of SS316 L and are mechanically polished and passivated
The ice condensing capacity shall be calculated on the basis of 24 Hr cycle
The suitable isolation valve shall be designed between the Chamber and condenser, and this shall be made up of SS316 L material.
The condenser cooling shall be done either with Gas/ Fluid
Sight glass with illumination shall be provided on body of the condenser for viewing purpose
The condenser shall be CIP and SIP able, and withstand with respect to the temperature and pressure

Functional Specification:

The Lyophilizer shall have interface with slot door with fixed loading/un-loading system
All Loading/ Unloading activities shall be from Grade A- Area
The Loading/ unloading system shall be designed as per the details and drawings provided by client
The loading / unloading operation shall be synchronized with lyophilizer operation

The system shall have the provision to back fill the equipment with sterile gases like Nitrogen/ Air and vacuum.
The vacuum control set point should be allowed from 1000 to 0.001 mbar absolute
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During SIP Automatic movement of shelves required from top to bottom to ensure the sterility of all components contained in the chamber
Vacuum pumps shall have the provision to provide the gas ballast to avoid the moisture and toxics impact during the process.

Integration of ISOLATOR for Loading/ Un-loading system:

The isolator unit shall be integrated with Lyophilizer as to meet the process requirements and current good manufacturing practice
The design and integration of Isolator shall be in line to the Lyo loading and unloading chamber interface and dimensions
Tamper proof glass door shall be provided to allow a clear visibility into the entire space of the isolator.
The interior and all corners of the isolator must be easily accessible and cleanable when isolator is closed through glove ports
The surface in and outside side of the isolator shouldn't have any cracks, crevices and sharp edges. All corners and edges must be rounded
All HEPA Filters installed in the isolator must be accessible for maintenance and cleaning
Glare-free light, sealed edges of the lamps inside of isolator shall be provided
Calibration of all sensors must easily accessible and shall be removable for calibration and maintenance
Maximum expected operation of the equipment 24 Hrs./ Day

Technical Specification:

The temperature control maintained within the isolator in the range of 20 to 25 Deg C in all conditions.
Beginning of the batch
While continuous running of the batch
Fluctuation of outside environment shall be within tolerance
Relative Humidity control during production is maintained within the full humidity range of 2-5 % with aid of Nitrogen purge
HEPA filters are 99.99% efficient for removing particles greater than 0.3 Micron in size.
The room conditions of isolator installed is classified as grade B
Environment conditions surrounding room
Temp : NMT 26 Deg C
% RH : NMT 60%
The isolator unit is required to design to meet the cleanliness under the area to meet Grade 'A' requirement
The MOC of the isolator shall be compactable for VHP sterilization/ Nitrogen gas .and all product contact parts shall be made with SS316 L and surface finish will be > 0.5 µ RA.
Integrity test port shall be provided at upstream of HEPA Filters
Blower shall statically and dynamically balance to the supply air flow with sufficient capacity and static pressure
Viable/ non-viable particle count shall meet grade A requirements

The air velocity shall be 90 ±20 FPM at 6 inches from the filter face
Initial and final pressure drop limits to be provided
During the VHP cycle, Gloves shall not be wrinkled or folded on surface
There shall be automatic leak test option available for isolator leak test and within the acceptance range; Leak test will be performed before start of the VHP cycle.

Interlocks/ Alarms:

Emergency Stop
Control power fault
Barrier open door
Glove port light curtain interrupted
Isolator contamination/ fault
Filter differential pressure alarm
Barrier high pressure alarm
Barrier low pressure alarm
Barrier differential pressure alarm
RH above 5% alarm

All machine surfaces must come in contact with H₂O₂, a minimum accessible area for H₂O₂ on the machine for example moving parts shall move slowly during VHP Cycle.
The whole decontamination cycle including aeration shall be completed within the specified operation time mentioned.
The isolator doors shall not open before the H₂O₂ concentration reaches to the neutral level.

Interface:

The Lyophilizer shall be provided with Siemens or Allen Bradley control logic platform. PLC, security, Recipe management process. Data logging shall be under taken Lyophilized dedicated SCADA. Control system shall be accessible to authorized users only. It shall have individual login ID and Passwords to access the system.
All set parameters ranges are to be defined access shall be restricted as per user level
Alarm management shall be available
Programmed backup for PLC/HMI SCADA shall be provided
Generated data shall be saved by the software and storage path shall be defined through software only
Control system clock synchronization shall be available
11.8 Electronic records and signatures shall be available wherever applicable on the reports | General Requirements:
All parts shall be made of non-oxidizing material as per FDA requirements
All internal corners and edges inside the chamber and condenser must have rounded and smooth finishing. And accessible for easy cleaning
The Control cabinet shall be integrated on the machine frame. And the cabinets shall be equipped with proper ventilators
Chamber shelves design shall allow minimum and maximum loading of the product and useable 90% out of total shelves area available.
Remote access feature shall be available for providing technical support of the feature
Machine shall have feature to have control the speed increase / decrease the speed through frequency control drive and shall be available to operate on HMI & SCADA
The machine shall be designed to be able to operate and maintain the temperature during the one compressor failure
Redundancy Fluid pump shall be provided
Vacuum pump redundancy, shall be able to operate with one pump and maintain the set vacuum at any point of operation
All Hot surfaces shall be insulated with mineral wool and cladding with SS304 sheet.
The bellows used in the entire system shall be leak proof.

Chamber:

The chamber shall have the automatic Locking / Un-locking system
The Geometrical construction of the chamber shall be rectangular structure
The type of door support shall be operated with Hinged
Operating design pressure of the chamber shall be 2.0 Bar G
The slot door shall be installed at Loading /Unloading side and additional main door shall be installed at technical area for maintenance purpose and the door can be completely open-able up to 180 Deg direction.
Chamber shall have the jacket for cooling water circulation provision to do the jacket cooling after SIP and other heating operations.
The material of construction of the chamber shall be made SS316 material with surface finish Ra < 0.5 µm internally with mirror polish. And chamber body shall be insulated and cladded with SS304 material
Chamber shall have the sight glass possible locations with light illumination

Shelves configuration:

Basic inter distance between the shelves : To be decided
Shelf dimensions (LxWxD) : To be decided
Total surface area : 18 Sq Mtr
Shelves heating rate : NLT 1.0 deg C/ Min
Shelves Minimum temperature : -65.0 Deg C
Shelves cooling source : Silicon Oil
Shelves moment for loading : Included
Shelves position control : Closed loop (Automatic)

The shelves shall allow the constant loading at fixed operating position/ height.
The shelves shall be made of SS316 L material and the surface finish shall be less than < 0.5 µm electro polished.
The shelves shall be designed to allowed minimum 2 formats for inter-distance for loading / Unloading
The Shelves loading should be from top to bottom direction and un-loading from bottom to top at constant height
Shelves shall be movable during CIP to ensure the 100% cleaning efficiency
Each shelf should have a temperature probe to measure the individual shelf temperature
The surface slope of the shelf maximum 0.5 mm/ Mtr

Refrigeration Compressors:

Total no. of compressors : To be defined
Compressors type : Screw type
Refrigerant : Non-CFC, Eco friendly

Vacuum System:

Vacuum pumps shall be dry pumps
The Pump capacity shall be design inline to the Chamber and condenser capacity
The ultimate vacuum shall be achieved with in 20 Min in ideal condition
The booster vacuum pump shall be dry type

Condenser:

Type of condenser : Close integrated
Number of condensers : 01 Nos
Design Pressure : 2.0 Bar G
Capacity of Ice thickness : To be defined
Minimum Temperature : >-75 Deg C

The construction of the condenser shall be cylindrical and should be suitable for vacuum applications during the lyophilization cycles
The condenser body and tubes shall be made of SS316 L and are mechanically polished and passivated
The ice condensing capacity shall be calculated on the basis of 24 Hr cycle
The suitable isolation valve shall be designed between the Chamber and condenser, and this shall be made up of SS316 L material.
The condenser cooling shall be done either with Gas/ Fluid
Sight glass with illumination shall be provided on body of the condenser for viewing purpose
The condenser shall be CIP and SIP able, and withstand with respect to the temperature and pressure

Functional Specification:

The Lyophilizer shall have interface with slot door with fixed loading/un-loading system
All Loading/ Unloading activities shall be from Grade A- Area
The Loading/ unloading system shall be designed as per the details and drawings provided by client
The loading / unloading operation shall be synchronized with lyophilizer operation

The system shall have the provision to back fill the equipment with sterile gases like Nitrogen/ Air and vacuum.
The vacuum control set point should be allowed from 1000 to 0.001 mbar absolute
SIP: The Sterilization cycle the temperature reference shall be measured at drain thermocouple, and all other temperature sensors shall be remains above the drain.
During SIP Automatic movement of shelves required from top to bottom to ensure the sterility of all components contained in the chamber
Vacuum pumps shall have the provision to provide the gas ballast to avoid the moisture and toxics impact during the process.

Integration of ISOLATOR for Loading/ Un-loading system:

The isolator unit shall be integrated with Lyophilizer as to meet the process requirements and current good manufacturing practice
The design and integration of Isolator shall be in line to the Lyo loading and unloading chamber interface and dimensions
Tamper proof glass door shall be provided to allow a clear visibility into the entire space of the isolator.
The interior and all corners of the isolator must be easily accessible and cleanable when isolator is closed through glove ports
The surface in and outside side of the isolator shouldn't have any cracks, crevices and sharp edges. All corners and edges must be rounded
All HEPA Filters installed in the isolator must be accessible for maintenance and cleaning
Glare-free light, sealed edges of the lamps inside of isolator shall be provided
Calibration of all sensors must easily accessible and shall be removable for calibration and maintenance
Maximum expected operation of the equipment 24 Hrs./ Day

Technical Specification:

The temperature control maintained within the isolator in the range of 20 to 25 Deg C in all conditions.
Beginning of the batch
While continuous running of the batch
Fluctuation of outside environment shall be within tolerance
Relative Humidity control during production is maintained within the full humidity range of 2-5 % with aid of Nitrogen purge
HEPA filters are 99.99% efficient for removing particles greater than 0.3 Micron in size.
The room conditions of isolator installed is classified as grade B
Environment conditions surrounding room
Temp : NMT 26 Deg C
% RH : NMT 60%
The isolator unit is required to design to meet the cleanliness under the area to meet Grade 'A' requirement
The MOC of the isolator shall be compactable for VHP sterilization/ Nitrogen gas .and all product contact parts shall be made with SS316 L and surface finish will be > 0.5 µ RA.
Integrity test port shall be provided at upstream of HEPA Filters
Blower shall statically and dynamically balance to the supply air flow with sufficient capacity and static pressure
Viable/ non-viable particle count shall meet grade A requirements
The air velocity shall be 90 ±20 FPM at 6 inches from the filter face
Initial and final pressure drop limits to be provided
During the VHP cycle, Gloves shall not be wrinkled or folded on surface
There shall be automatic leak test option available for isolator leak test and within the acceptance range; Leak test will be performed before start of the VHP cycle.

Interlocks/ Alarms:

Emergency Stop
Control power fault
Barrier open door
Glove port light curtain interrupted
Isolator contamination/ fault
Filter differential pressure alarm
Barrier high pressure alarm
Barrier low pressure alarm
Barrier differential pressure alarm
RH above 5% alarm

All machine surfaces must come in contact with H₂O₂, a minimum accessible area for H₂O₂ on the machine for example moving parts shall move slowly during VHP Cycle.

The whole decontamination cycle including aeration shall be completed within the specified operation time mentioned.

The isolator doors shall not open before the H₂O₂ concentration reaches to the neutral level.

Interface:

The Lyophilizer shall be provided with Siemens or Allen Bradley control logic platform. PLC, security, Recipe management process. Data logging shall be under taken Lyophilized dedicated SCADA. Control system shall be accessible to authorized users only. It shall have individual login ID and Passwords to access the system.

All set parameters ranges are to be defined access shall be restricted as per user level

Alarm management shall be available

Programmed backup for PLC/HMI SCADA shall be provided

Generated data shall be saved by the software and storage path shall be defined through software only

Control system clock synchronization shall be available

11.8 Electronic records and signatures shall be available wherever applicable on the reports | General Requirements:

All parts shall be made of non-oxidizing material as per FDA requirements

All internal corners and edges inside the chamber and condenser must have rounded and smooth finishing.

And accessible for easy cleaning

The Control cabinet shall be integrated on the machine frame. And the cabinets shall be equipped with proper ventilators

Chamber shelves design shall allow minimum and maximum loading of the product and useable 90% out of total shelves area available.

Remote access feature shall be available for providing technical support of the feature

Machine shall have feature to have control the speed increase / decrease the speed through frequency

control drive and shall be available to operate on HMI & SCADA

The machine shall be designed to be able to operate and maintain the temperature during the one compressor failure

Redundancy Fluid pump shall be provided

Vacuum pump redundancy, shall be able to operate with one pump and maintain the set vacuum at any point of operation

All Hot surfaces shall be insulated with mineral wool and cladding with SS304 sheet.

The bellows used in the entire system shall be leak proof.

Chamber:

The chamber shall have the automatic Locking / Un-locking system

The Geometrical construction of the chamber shall be rectangular structure

The type of door support shall be operated with Hinged

Operating design pressure of the chamber shall be 2.0 Bar G

The slot door shall be installed at Loading /Unloading side and additional main door shall be installed at technical area for maintenance purpose and the door can be completely open-able up to 180 Deg direction.

Chamber shall have the jacket for cooling water circulation provision to do the jacket cooling after SIP and other heating operations.

The material of constriction of the chamber shall be made SS316 material with surface finish Ra < 0.5 µm internally with mirror polish. And chamber body shall be insulated and cladded with SS304 material

Chamber shall have the sight glass possible locations with light illumination

Shelves configuration:

Basic inter distance between the shelves : To be decided

Shelf dimensions (LxWxD) : To be decided

Total surface area : 18 Sq Mtr

Shelves heating rate : NLT 1.0 deg C/ Min

Shelves Minimum temperature : -65.0 Deg C

Shelves cooling source : Silicon Oil

Shelves moment for loading : Included

Shelves position control : Closed loop (Automatic)

The shelves shall allow the constant loading at fixed operating position/ height.

The shelves shall be made of SS316 L material and the surface finish shall be less than < 0.5 µm electro polished.

The shelves shall be designed to allowed minimum 2 formats for inter-distance for loading / Unloading

The Shelves loading should be from top to bottom direction and un-loading from bottom to top at constant height

Shelves shall be movable during CIP to ensure the 100% cleaning efficiency

Each shelf should have a temperature probe to measure the individual shelf temperature

The surface slope of the shelf maximum 0.5 mm/ Mtr

Refrigeration Compressors:

Total no. of compressors : To be defined

Compressors type : Screw type

Refrigerant : Non-CFC, Eco friendly

Vacuum System:

Vacuum pumps shall be dry pumps
The Pump capacity shall be design inline to the Chamber and condenser capacity
The ultimate vacuum shall be achieved within 20 Min in ideal condition
The booster vacuum pump shall be dry type

Condenser:
Type of condenser : Close integrated
Number of condensers : 01 Nos
Design Pressure : 2.0 Bar G
Capacity of Ice thickness : To be defined
Minimum Temperature : >-75 Deg C

The construction of the condenser shall be cylindrical and should be suitable for vacuum applications during the lyophilization cycles
The condenser body and tubes shall be made of SS316 L and are mechanically polished and passivated
The ice condensing capacity shall be calculated on the basis of 24 Hr cycle
The suitable isolation valve shall be designed between the Chamber and condenser, and this shall be made up of SS316 L material.
The condenser cooling shall be done either with Gas/ Fluid
Sight glass with illumination shall be provided on body of the condenser for viewing purpose
The condenser shall be CIP and SIP able, and withstand with respect to the temperature and pressure

Functional Specification:

The Lyophilizer shall have interface with slot door with fixed loading/un-loading system
All Loading/ Unloading activities shall be from Grade A- Area
The Loading/ unloading system shall be designed as per the details and drawings provided by client
The loading / unloading operation shall be synchronized with lyophilizer operation

The system shall have the provision to back fill the equipment with sterile gases like Nitrogen/ Air and vacuum.
The vacuum control set point should be allowed from 1000 to 0.001 mbar absolute
SIP: The Sterilization cycle the temperature reference shall be measured at drain thermocouple, and all other temperature sensors shall be remains above the drain.
During SIP Automatic movement of shelves required from top to bottom to ensure the sterility of all components contained in the chamber
Vacuum pumps shall have the provision to provide the gas ballast to avoid the moisture and toxics impact during the process.

Integration of ISOLATOR for Loading/ Un-loading system:

The isolator unit shall be integrated with Lyophilizer as to meet the process requirements and current good manufacturing practice
The design and integration of Isolator shall be in line to the Lyo loading and unloading chamber interface and dimensions
Tamper proof glass door shall be provided to allow a clear visibility into the entire space of the isolator.
The interior and all corners of the isolator must be easily accessible and cleanable when isolator is closed through glove ports
The surface in and outside side of the isolator shouldn't have any cracks, crevices and sharp edges. All corners and edges must be rounded
All HEPA Filters installed in the isolator must be accessible for maintenance and cleaning
Glare-free light, sealed edges of the lamps inside of isolator shall be provided
Calibration of all sensors must easily accessible and shall be removable for calibration and maintenance
Maximum expected operation of the equipment 24 Hrs./ Day

Technical Specification:

The temperature control maintained within the isolator in the range of 20 to 25 Deg C in all conditions.
Beginning of the batch
While continuous running of the batch
Fluctuation of outside environment shall be within tolerance
Relative Humidity control during production is maintained within the full humidity range of 2-5 % with aid of Nitrogen purge
HEPA filters are 99.99% efficient for removing particles greater than 0.3 Micron in size.
The room conditions of isolator installed is classified as grade B
Environment conditions surrounding room
Temp : NMT 26 Deg C
% RH : NMT 60%
The isolator unit is required to design to meet the cleanliness under the area to meet Grade 'A' requirement
The MOC of the isolator shall be compactable for VHP sterilization/ Nitrogen gas .and all product contact parts shall be made with SS316 L and surface finish will be > 0.5 µ RA.
Integrity test port shall be provided at upstream of HEPA Filters
Blower shall statically and dynamically balance to the supply air flow with sufficient capacity and static pressure
Viable/ non-viable particle count shall meet grade A requirements
The air velocity shall be 90 ±20 FPM at 6 inches from the filter face
Initial and final pressure drop limits to be provided
During the VHP cycle, Gloves shall not be wrinkled or folded on surface
There shall be automatic leak test option available for isolator leak test and within the acceptance range; Leak test will be performed before start of the VHP cycle.

Interlocks/ Alarms:

Emergency Stop
Control power fault
Barrier open door
Glove port light curtain interrupted
Isolator contamination/ fault
Filter differential pressure alarm

Barrier high pressure alarm
Barrier low pressure alarm
Barrier differential pressure alarm
RH above 5% alarm

All machine surfaces must come in contact with H₂O₂, a minimum accessible area for H₂O₂ on the machine for example moving parts shall move slowly during VHP Cycle.

The whole decontamination cycle including aeration shall be completed within the specified operation time mentioned.

The isolator doors shall not open before the H₂O₂ concentration reaches to the neutral level.

Interface:

The Lyophilizer shall be provided with Siemens or Allen Bradley control logic platform. PLC, security, Recipe management process. Data logging shall be under taken Lyophilized dedicated SCADA. Control system shall be accessible to authorized users only. It shall have individual login ID and Passwords to access the system.

All set parameters ranges are to be defined access shall be restricted as per user level

Alarm management shall be available

Programmed backup for PLC/HMI SCADA shall be provided

Generated data shall be saved by the software and storage path shall be defined through software only

Control system clock synchronization shall be available

11.8 Electronic records and signatures shall be available wherever applicable on the reports

| General Requirements:

All parts shall be made of non-oxidizing material as per FDA requirements

All internal corners and edges inside the chamber and condenser must have rounded and smooth finishing.

And accessible for easy cleaning

The Control cabinet shall be integrated on the machine frame. And the cabinets shall be equipped with proper ventilators

Chamber shelves design shall allow minimum and maximum loading of the product and useable 90% out of total shelves area available.

Remote access feature shall be available for providing technical support of the feature

Machine shall have feature to have control the speed increase / decrease the speed through frequency

control drive and shall be available to operate on HMI & SCADA

The machine shall be designed to be able to operate and maintain the temperature during the one compressor failure

Redundancy Fluid pump shall be provided

Vacuum pump redundancy, shall be able to operate with one pump and maintain the set vacuum at any point of operation

All Hot surfaces shall be insulated with mineral wool and cladding with SS304 sheet.

The bellows used in the entire system shall be leak proof.

Chamber:

The chamber shall have the automatic Locking / Un-locking system

The Geometrical construction of the chamber shall be rectangular structure

The type of door support shall be operated with Hinged

Operating design pressure of the chamber shall be 2.0 Bar G

The slot door shall be installed at Loading /Unloading side and additional main door shall be installed at technical area for maintenance purpose and the door can be completely open-able up to 180 Deg direction.

Chamber shall have the jacket for cooling water circulation provision to do the jacket cooling after SIP and other heating operations.

The material of construction of the chamber shall be made SS316 material with surface finish Ra < 0.5 µm internally with mirror polish. And chamber body shall be insulated and cladded with SS304 material

Chamber shall have the sight glass possible locations with light illumination

Shelves configuration:

Basic inter distance between the shelves : To be decided

Shelf dimensions (LxWxD) : To be decided

Total surface area : 18 Sq Mtr

Shelves heating rate : NLT 1.0 deg C/ Min

Shelves Minimum temperature : -65.0 Deg C

Shelves cooling source : Silicon Oil

Shelves moment for loading : Included

Shelves position control : Closed loop (Automatic)

The shelves shall allow the constant loading at fixed operating position/ height.

The shelves shall be made of SS316 L material and the surface finish shall be less than < 0.5 µm electro polished.

The shelves shall be designed to allowed minimum 2 formats for inter-distance for loading / Unloading

The Shelves loading should be from top to bottom direction and un-loading from bottom to top at constant height

Shelves shall be movable during CIP to ensure the 100% cleaning efficiency

Each shelf should have a temperature probe to measure the individual shelf temperature

The surface slope of the shelf maximum 0.5 mm/ Mtr

Refrigeration Compressors:

Total no. of compressors : To be defined

Compressors type : Screw type

Refrigerant : Non-CFC, Eco friendly

Vacuum System:

Vacuum pumps shall be dry pumps

The Pump capacity shall be design inline to the Chamber and condenser capacity

The ultimate vacuum shall be achieved with in 20 Min in ideal condition

The booster vacuum pump shall be dry type

Condenser:

Type of condenser : Close integrated
Number of condensers : 01 Nos
Design Pressure : 2.0 Bar G
Capacity of Ice thickness : To be defined
Minimum Temperature : >-75 Deg C

The construction of the condenser shall be cylindrical and should be suitable for vacuum applications during the lyophilization cycles
The condenser body and tubes shall be made of SS316 L and are mechanically polished and passivated
The ice condensing capacity shall be calculated on the basis of 24 Hr cycle
The suitable isolation valve shall be designed between the Chamber and condenser, and this shall be made up of SS316 L material.
The condenser cooling shall be done either with Gas/ Fluid
Sight glass with illumination shall be provided on body of the condenser for viewing purpose
The condenser shall be CIP and SIP able, and withstand with respect to the temperature and pressure

Functional Specification:

The Lyophilizer shall have interface with slot door with fixed loading/un-loading system
All Loading/ Unloading activities shall be from Grade A- Area
The Loading/ unloading system shall be designed as per the details and drawings provided by client
The loading / unloading operation shall be synchronized with lyophilizer operation

The system shall have the provision to back fill the equipment with sterile gases like Nitrogen/ Air and vacuum.
The vacuum control set point should be allowed from 1000 to 0.001 mbar absolute
SIP: The Sterilization cycle the temperature reference shall be measured at drain thermocouple, and all other temperature sensors shall be remains above the drain.
During SIP Automatic movement of shelves required from top to bottom to ensure the sterility of all components contained in the chamber
Vacuum pumps shall have the provision to provide the gas ballast to avoid the moisture and toxics impact during the process.

Integration of ISOLATOR for Loading/ Un-loading system:

The isolator unit shall be integrated with Lyophilizer as to meet the process requirements and current good manufacturing practice
The design and integration of Isolator shall be in line to the Lyo loading and unloading chamber interface and dimensions
Tamper proof glass door shall be provided to allow a clear visibility into the entire space of the isolator.
The interior and all corners of the isolator must be easily accessible and cleanable when isolator is closed through glove ports
The surface in and outside side of the isolator shouldn't have any cracks, crevices and sharp edges. All corners and edges must be rounded
All HEPA Filters installed in the isolator must be accessible for maintenance and cleaning
Glare-free light, sealed edges of the lamps inside of isolator shall be provided
Calibration of all sensors must easily accessible and shall be removable for calibration and maintenance
Maximum expected operation of the equipment 24 Hrs./ Day

Technical Specification:

The temperature control maintained within the isolator in the range of 20 to 25 Deg C in all conditions.
Beginning of the batch
While continuous running of the batch
Fluctuation of outside environment shall be within tolerance
Relative Humidity control during production is maintained within the full humidity range of 2-5 % with aid of Nitrogen purge
HEPA filters are 99.99% efficient for removing particles greater than 0.3 Micron in size.
The room conditions of isolator installed is classified as grade B
Environment conditions surrounding room
Temp : NMT 26 Deg C
% RH : NMT 60%
The isolator unit is required to design to meet the cleanliness under the area to meet Grade 'A' requirement
The MOC of the isolator shall be compactable for VHP sterilization/ Nitrogen gas .and all product contact parts shall be made with SS316 L and surface finish will be > 0.5 µ RA.
Integrity test port shall be provided at upstream of HEPA Filters
Blower shall statically and dynamically balance to the supply air flow with sufficient capacity and static pressure
Viable/ non-viable particle count shall meet grade A requirements
The air velocity shall be 90 ±20 FPM at 6 inches from the filter face
Initial and final pressure drop limits to be provided
During the VHP cycle, Gloves shall not be wrinkled or folded on surface
There shall be automatic leak test option available for isolator leak test and within the acceptance range; Leak test will be performed before start of the VHP cycle.

Interlocks/ Alarms:

Emergency Stop
Control power fault
Barrier open door
Glove port light curtain interrupted
Isolator contamination/ fault
Filter differential pressure alarm
Barrier high pressure alarm
Barrier low pressure alarm
Barrier differential pressure alarm
RH above 5% alarm

All machine surfaces must come in contact with H2O2, a minimum accessible area for H2O2 on the machine for example moving parts shall move slowly during VHP Cycle.

The whole decontamination cycle including aeration shall be completed within the specified operation time mentioned.

The isolator doors shall not open before the H₂O₂ concentration reaches to the neutral level.

Interface:

The Lyophilizer shall be provided with Siemens or Allen Bradley control logic platform. PLC, security, Recipe management process. Data logging shall be under taken Lyophilized dedicated SCADA. Control system shall be accessible to authorized users only. It shall have individual login ID and Passwords to access the system.

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Programmed backup for PLC/HMI SCADA shall be provided

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And accessible for easy cleaning

The Control cabinet shall be integrated on the machine frame. And the cabinets shall be equipped with proper ventilators

Chamber shelves design shall allow minimum and maximum loading of the product and useable 90% out of total shelves area available.

Remote access feature shall be available for providing technical support of the feature

Machine shall have feature to have control the speed increase / decrease the speed through frequency

control drive and shall be available to operate on HMI & SCADA

The machine shall be designed to be able to operate and maintain the temperature during the one compressor failure

Redundancy Fluid pump shall be provided

Vacuum pump redundancy, shall be able to operate with one pump and maintain the set vacuum at any point of operation

All Hot surfaces shall be insulated with mineral wool and cladding with SS304 sheet.

The bellows used in the entire system shall be leak proof.

Chamber:

The chamber shall have the automatic Locking / Un-locking system

The Geometrical construction of the chamber shall be rectangular structure

The type of door support shall be operated with Hinged

Operating design pressure of the chamber shall be 2.0 Bar G

The slot door shall be installed at Loading /Unloading side and additional main door shall be installed at technical area for maintenance purpose and the door can be completely open-able up to 180 Deg direction.

Chamber shall have the jacket for cooling water circulation provision to do the jacket cooling after SIP and other heating operations.

The material of constriction of the chamber shall be made SS316 material with surface finish Ra < 0.5 µm

internally with mirror polish. And chamber body shall be insulated and cladded with SS304 material

Chamber shall have the sight glass possible locations with light illumination

Shelves configuration:

Basic inter distance between the shelves : To be decided

Shelf dimensions (LxWxD) : To be decided

Total surface area : 18 Sq Mtr

Shelves heating rate : NLT 1.0 deg C/ Min

Shelves Minimum temperature : -65.0 Deg C

Shelves cooling source : Silicon Oil

Shelves moment for loading : Included

Shelves position control : Closed loop (Automatic)

The shelves shall allow the constant loading at fixed operating position/ height.

The shelves shall be made of SS316 L material and the surface finish shall be less than < 0.5 µm electro polished.

The shelves shall be designed to allowed minimum 2 formats for inter-distance for loading / Unloading

The Shelves loading should be from top to bottom direction and un-loading from bottom to top at constant height

Shelves shall be movable during CIP to ensure the 100% cleaning efficiency

Each shelf should have a temperature probe to measure the individual shelf temperature

The surface slope of the shelf maximum 0.5 mm/ Mtr

Refrigeration Compressors:

Total no. of compressors : To be defined

Compressors type : Screw type

Refrigerant : Non-CFC, Eco friendly

Vacuum System:

Vacuum pumps shall be dry pumps

The Pump capacity shall be design inline to the Chamber and condenser capacity

The ultimate vacuum shall be achieved with in 20 Min in ideal condition

The booster vacuum pump shall be dry type

Condenser:

Type of condenser : Close integrated

Number of condensers : 01 Nos

Design Pressure : 2.0 Bar G

Capacity of Ice thickness : To be defined

Minimum Temperature : >-75 Deg C

The construction of the condenser shall be cylindrical and should be suitable for vacuum applications

during the lyophilization cycles

The condenser body and tubes shall be made of SS316 L and are mechanically polished and passivated

The ice condensing capacity shall be calculated on the basis of 24 Hr cycle

The suitable isolation valve shall be designed between the Chamber and condenser, and this shall be made up of SS316 L material.

The condenser cooling shall be done either with Gas/ Fluid

Sight glass with illumination shall be provided on body of the condenser for viewing purpose

The condenser shall be CIP and SIP able, and withstand with respect to the temperature and pressure

Functional Specification:

The Lyophilizer shall have interface with slot door with fixed loading/un-loading system

All Loading/ Unloading activities shall be from Grade A- Area

The Loading/ unloading system shall be designed as per the details and drawings provided by client

The loading / unloading operation shall be synchronized with lyophilizer operation

The system shall have the provision to back fill the equipment with sterile gases like Nitrogen/ Air and vacuum.

The vacuum control set point should be allowed from 1000 to 0.001 mbar absolute

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The surface in and outside side of the isolator shouldn't have any cracks, crevices and sharp edges. All corners and edges must be rounded

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Glare-free light, sealed edges of the lamps inside of isolator shall be provided

Calibration of all sensors must easily accessible and shall be removable for calibration and maintenance

Maximum expected operation of the equipment 24 Hrs./ Day

Technical Specification:

The temperature control maintained within the isolator in the range of 20 to 25 Deg C in all conditions.

Beginning of the batch

While continuous running of the batch

Fluctuation of outside environment shall be within tolerance

Relative Humidity control during production is maintained within the full humidity range of 2-5 % with aid of Nitrogen purge

HEPA filters are 99.99% efficient for removing particles greater than 0.3 Micron in size.

The room conditions of isolator installed is classified as grade B

Environment conditions surrounding room

Temp : NMT 26 Deg C

% RH : NMT 60%

The isolator unit is required to design to meet the cleanliness under the area to meet Grade 'A' requirement

The MOC of the isolator shall be compactable for VHP sterilization/ Nitrogen gas .and all product contact parts shall be made with SS316 L and surface finish will be > 0.5 μ RA.

Integrity test port shall be provided at upstream of HEPA Filters

Blower shall statically and dynamically balance to the supply air flow with sufficient capacity and static pressure

Viable/ non-viable particle count shall meet grade A requirements

The air velocity shall be 90 $\pm$ 20 FPM at 6 inches from the filter face

Initial and final pressure drop limits to be provided

During the VHP cycle, Gloves shall not be wrinkled or folded on surface

There shall be automatic leak test option available for isolator leak test and within the acceptance range; Leak test will be performed before start of the VHP cycle.

Interlocks/ Alarms:

Emergency Stop

Control power fault

Barrier open door

Glove port light curtain interrupted

Isolator contamination/ fault

Filter differential pressure alarm

Barrier high pressure alarm

Barrier low pressure alarm

Barrier differential pressure alarm

RH above 5% alarm

All machine surfaces must come in contact with H₂O₂, a minimum accessible area for H₂O₂ on the machine for example moving parts shall move slowly during VHP Cycle.

The whole decontamination cycle including aeration shall be completed within the specified operation time mentioned.

The isolator doors shall not open before the H₂O₂ concentration reaches to the neutral level.

Interface:

The Lyophilizer shall be provided with Siemens or Allen Bradley control logic platform. PLC, security,

Recipe management process. Data logging shall be under taken Lyophilized dedicated SCADA. Control system shall be accessible to authorized users only. It shall have individual login ID and Passwords to access the system.

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Chamber shall have the jacket for cooling water circulation provision to do the jacket cooling after SIP and other heating operations.

The material of constriction of the chamber shall be made SS316 material with surface finish Ra < 0.5 µm

internally with mirror polish. And chamber body shall be insulated and clad with SS304 material

Chamber shall have the sight glass possible locations with light illumination

Shelves configuration:

Basic inter distance between the shelves : To be decided

Shelf dimensions (LxWxD) : To be decided

Total surface area : 18 Sq Mtr

Shelves heating rate : NLT 1.0 deg C/ Min

Shelves Minimum temperature : -65.0 Deg C

Shelves cooling source : Silicon Oil

Shelves moment for loading : Included

Shelves position control : Closed loop (Automatic)

The shelves shall allow the constant loading at fixed operating position/ height.

The shelves shall be made of SS316 L material and the surface finish shall be less than < 0.5 µm electro polished.

The shelves shall be designed to allowed minimum 2 formats for inter-distance for loading / Unloading

The Shelves loading should be from top to bottom direction and un-loading from bottom to top at constant height

Shelves shall be movable during CIP to ensure the 100% cleaning efficiency

Each shelf should have a temperature probe to measure the individual shelf temperature

The surface slope of the shelf maximum 0.5 mm/ Mtr

Refrigeration Compressors:

Total no. of compressors : To be defined

Compressors type : Screw type

Refrigerant : Non-CFC, Eco friendly

Vacuum System:

Vacuum pumps shall be dry pumps

The Pump capacity shall be design inline to the Chamber and condenser capacity

The ultimate vacuum shall be achieved with in 20 Min in ideal condition

The booster vacuum pump shall be dry type

Condenser:

Type of condenser : Close integrated

Number of condensers : 01 Nos

Design Pressure : 2.0 Bar G

Capacity of Ice thickness : To be defined

Minimum Temperature : >-75 Deg C

The construction of the condenser shall be cylindrical and should be suitable for vacuum applications during the lyophilization cycles

The condenser body and tubes shall be made of SS316 L and are mechanically polished and passivated

The ice condensing capacity shall be calculated on the basis of 24 Hr cycle

The suitable isolation valve shall be designed between the Chamber and condenser, and this shall be made up of SS316 L material.

The condenser cooling shall be done either with Gas/ Fluid

Sight glass with illumination shall be provided on body of the condenser for viewing purpose

The condenser shall be CIP and SIP able, and withstand with respect to the temperature and pressure

Functional Specification:

The Lyophilizer shall have interface with slot door with fixed loading/un-loading system

All Loading/ Unloading activities shall be from Grade A- Area

The Loading/ unloading system shall be designed as per the details and drawings provided by client

The loading / unloading operation shall be synchronized with lyophilizer operation

The system shall have the provision to back fill the equipment with sterile gases like Nitrogen/ Air and vacuum.

The vacuum control set point should be allowed from 1000 to 0.001 mbar absolute

SIP: The Sterilization cycle the temperature reference shall be measured at drain thermocouple, and all other temperature sensors shall be remains above the drain.

During SIP Automatic movement of shelves required from top to bottom to ensure the sterility of all components contained in the chamber

Vacuum pumps shall have the provision to provide the gas ballast to avoid the moisture and toxics impact during the process.

Integration of ISOLATOR for Loading/ Un-loading system:

The isolator unit shall be integrated with Lyophilizer as to meet the process requirements and current good manufacturing practice

The design and integration of Isolator shall be in line to the Lyo loading and unloading chamber interface and dimensions

Tamper proof glass door shall be provided to allow a clear visibility into the entire space of the isolator.

The interior and all corners of the isolator must be easily accessible and cleanable when isolator is closed through glove ports

The surface in and outside side of the isolator shouldn't have any cracks, crevices and sharp edges. All corners and edges must be rounded

All HEPA Filters installed in the isolator must be accessible for maintenance and cleaning

Glare-free light, sealed edges of the lamps inside of isolator shall be provided

Calibration of all sensors must easily accessible and shall be removable for calibration and maintenance

Maximum expected operation of the equipment 24 Hrs./ Day

Technical Specification:

The temperature control maintained within the isolator in the range of 20 to 25 Deg C in all conditions.

Beginning of the batch

While continuous running of the batch

Fluctuation of outside environment shall be within tolerance

Relative Humidity control during production is maintained within the full humidity range of 2-5 % with aid of Nitrogen purge

HEPA filters are 99.99% efficient for removing particles greater than 0.3 Micron in size.

The room conditions of isolator installed is classified as grade B

Environment conditions surrounding room

Temp : NMT 26 Deg C

% RH : NMT 60%

The isolator unit is required to design to meet the cleanliness under the area to meet Grade 'A' requirement

The MOC of the isolator shall be compactable for VHP sterilization/ Nitrogen gas .and all product contact parts shall be made with SS316 L and surface finish will be $> 0.5 \mu\text{RA}$.

Integrity test port shall be provided at upstream of HEPA Filters

Blower shall statically and dynamically balance to the supply air flow with sufficient capacity and static pressure

Viable/ non-viable particle count shall meet grade A requirements

The air velocity shall be 90 ± 20 FPM at 6 inches from the filter face

Initial and final pressure drop limits to be provided

During the VHP cycle, Gloves shall not be wrinkled or folded on surface

There shall be automatic leak test option available for isolator leak test and within the acceptance range; Leak test will be performed before start of the VHP cycle.

Interlocks/ Alarms:

Emergency Stop

Control power fault

Barrier open door

Glove port light curtain interrupted

Isolator contamination/ fault

Filter differential pressure alarm

Barrier high pressure alarm

Barrier low pressure alarm

Barrier differential pressure alarm

RH above 5% alarm

All machine surfaces must come in contact with H₂O₂, a minimum accessible area for H₂O₂ on the machine for example moving parts shall move slowly during VHP Cycle.

The whole decontamination cycle including aeration shall be completed within the specified operation time mentioned.

The isolator doors shall not open before the H₂O₂ concentration reaches to the neutral level.

Interface:

The Lyophilizer shall be provided with Siemens or Allen Bradley control logic platform. PLC, security,

Recipe management process. Data logging shall be under taken Lyophilized dedicated SCADA.

Control system shall be accessible to authorized users only. It shall have individual login ID and Passwords to access the system.

All set parameters ranges are to be defined access shall be restricted as per user level

Alarm management shall be available

Programmed backup for PLC/HMI SCADA shall be provided

Generated data shall be saved by the software and storage path shall be defined through software only

Control system clock synchronization shall be available

11.8 Electronic records and signatures shall be available wherever applicable on the reports | General Requirements:

All parts shall be made of non-oxidizing material as per FDA requirements

All internal corners and edges inside the chamber and condenser must have rounded and smooth finishing.

And accessible for easy cleaning

The Control cabinet shall be integrated on the machine frame. And the cabinets shall be equipped with proper ventilators

Chamber shelves design shall allow minimum and maximum loading of the product and useable 90% out of total shelves area available.

Remote access feature shall be available for providing technical support of the feature

Machine shall have feature to have control the speed increase / decrease the speed through frequency

control drive and shall be available to operate on HMI & SCADA

The machine shall be designed to be able to operate and maintain the temperature during the one compressor failure

Redundancy Fluid pump shall be provided

Vacuum pump redundancy, shall be able to operate with one pump and maintain the set vacuum at any point of operation

All Hot surfaces shall be insulated with mineral wool and cladding with SS304 sheet.

The bellows used in the entire system shall be leak proof.

Chamber:

The chamber shall have the automatic Locking / Un-locking system

The Geometrical construction of the chamber shall be rectangular structure

The type of door support shall be operated with Hinged

Operating design pressure of the chamber shall be 2.0 Bar G

The slot door shall be installed at Loading /Unloading side and additional main door shall be installed at technical area for maintenance purpose and the door can be completely open-able up to 180 Deg direction.

Chamber shall have the jacket for cooling water circulation provision to do the jacket cooling after SIP and other heating operations.

The material of construction of the chamber shall be made SS316 material with surface finish Ra < 0.5 µm

internally with mirror polish. And chamber body shall be insulated and cladded with SS304 material

Chamber shall have the sight glass possible locations with light illumination

Shelves configuration:

Basic inter distance between the shelves : To be decided

Shelf dimensions (LxWxD) : To be decided

Total surface area : 18 Sq Mtr

Shelves heating rate : NLT 1.0 deg C/ Min

Shelves Minimum temperature : -65.0 Deg C

Shelves cooling source : Silicon Oil

Shelves moment for loading : Included

Shelves position control : Closed loop (Automatic)

The shelves shall allow the constant loading at fixed operating position/ height.

The shelves shall be made of SS316 L material and the surface finish shall be less than < 0.5 µm electro polished.

The shelves shall be designed to allowed minimum 2 formats for inter-distance for loading / Unloading

The Shelves loading should be from top to bottom direction and un-loading from bottom to top at constant height

Shelves shall be movable during CIP to ensure the 100% cleaning efficiency

Each shelf should have a temperature probe to measure the individual shelf temperature

The surface slope of the shelf maximum 0.5 mm/ Mtr

Refrigeration Compressors:

Total no. of compressors : To be defined

Compressors type : Screw type

Refrigerant : Non-CFC, Eco friendly

Vacuum System:

Vacuum pumps shall be dry pumps

The Pump capacity shall be design inline to the Chamber and condenser capacity

The ultimate vacuum shall be achieved within 20 Min in ideal condition

The booster vacuum pump shall be dry type

Condenser:

Type of condenser : Close integrated

Number of condensers : 01 Nos

Design Pressure : 2.0 Bar G

Capacity of Ice thickness : To be defined

Minimum Temperature : >-75 Deg C

The construction of the condenser shall be cylindrical and should be suitable for vacuum applications during the lyophilization cycles

The condenser body and tubes shall be made of SS316 L and are mechanically polished and passivated

The ice condensing capacity shall be calculated on the basis of 24 Hr cycle

The suitable isolation valve shall be designed between the Chamber and condenser, and this shall be made up of SS316 L material.

The condenser cooling shall be done either with Gas/ Fluid

Sight glass with illumination shall be provided on body of the condenser for viewing purpose

The condenser shall be CIP and SIP able, and withstand with respect to the temperature and pressure

Functional Specification:

The Lyophilizer shall have interface with slot door with fixed loading/un-loading system

All Loading/ Unloading activities shall be from Grade A- Area

The Loading/ unloading system shall be designed as per the details and drawings provided by client

The loading / unloading operation shall be synchronized with lyophilizer operation

The system shall have the provision to back fill the equipment with sterile gases like Nitrogen/ Air and vacuum.

The vacuum control set point should be allowed from 1000 to 0.001 mbar absolute

SIP: The Sterilization cycle the temperature reference shall be measured at drain thermocouple, and all other temperature sensors shall be remains above the drain.

During SIP Automatic movement of shelves required from top to bottom to ensure the sterility of all components contained in the chamber

Vacuum pumps shall have the provision to provide the gas ballast to avoid the moisture and toxics impact during the process.

Integration of ISOLATOR for Loading/ Un-loading system:

The isolator unit shall be integrated with Lyophilizer as to meet the process requirements and current good manufacturing practice

The design and integration of Isolator shall be in line to the Lyo loading and unloading chamber interface and dimensions

Tamper proof glass door shall be provided to allow a clear visibility into the entire space of the isolator.

The interior and all corners of the isolator must be easily accessible and cleanable when isolator is closed through glove ports

The surface in and outside side of the isolator shouldn't have any cracks, crevices and sharp edges. All corners and edges must be rounded

All HEPA Filters installed in the isolator must be accessible for maintenance and cleaning

Glare-free light, sealed edges of the lamps inside of isolator shall be provided

Calibration of all sensors must easily accessible and shall be removable for calibration and maintenance

Maximum expected operation of the equipment 24 Hrs./ Day

Technical Specification:

The temperature control maintained within the isolator in the range of 20 to 25 Deg C in all conditions.

Beginning of the batch

While continuous running of the batch

Fluctuation of outside environment shall be within tolerance

Relative Humidity control during production is maintained within the full humidity range of 2-5 % with aid of Nitrogen purge

HEPA filters are 99.99% efficient for removing particles greater than 0.3 Micron in size.

The room conditions of isolator installed is classified as grade B

Environment conditions surrounding room

Temp : NMT 26 Deg C

% RH : NMT 60%

The isolator unit is required to design to meet the cleanliness under the area to meet Grade 'A' requirement

The MOC of the isolator shall be compactable for VHP sterilization/ Nitrogen gas .and all product contact parts shall be made with SS316 L and surface finish will be $> 0.5 \mu$ RA.

Integrity test port shall be provided at upstream of HEPA Filters

Blower shall statically and dynamically balance to the supply air flow with sufficient capacity and static pressure

Viable/ non-viable particle count shall meet grade A requirements

The air velocity shall be 90 ± 20 FPM at 6 inches from the filter face

Initial and final pressure drop limits to be provided

During the VHP cycle, Gloves shall not be wrinkled or folded on surface

There shall be automatic leak test option available for isolator leak test and within the acceptance range; Leak test will be performed before start of the VHP cycle.

Interlocks/ Alarms:

Emergency Stop

Control power fault

Barrier open door

Glove port light curtain interrupted

Isolator contamination/ fault

Filter differential pressure alarm

Barrier high pressure alarm

Barrier low pressure alarm

Barrier differential pressure alarm

RH above 5% alarm

All machine surfaces must come in contact with H₂O₂, a minimum accessible area for H₂O₂ on the machine for example moving parts shall move slowly during VHP Cycle.

The whole decontamination cycle including aeration shall be completed within the specified operation time mentioned.

The isolator doors shall not open before the H₂O₂ concentration reaches to the neutral level.

Interface:

The Lyophilizer shall be provided with Siemens or Allen Bradley control logic platform. PLC, security,

Recipe management process. Data logging shall be under taken Lyophilized dedicated SCADA.

Control system shall be accessible to authorized users only. It shall have individual login ID and Passwords to access the system.

All set parameters ranges are to be defined access shall be restricted as per user level

Alarm management shall be available

Programmed backup for PLC/HMI SCADA shall be provided

Generated data shall be saved by the software and storage path shall be defined through software only

Control system clock synchronization shall be available

11.8 Electronic records and signatures shall be available wherever applicable on the reports | General Requirements:

All parts shall be made of non-oxidizing material as per FDA requirements

All internal corners and edges inside the chamber and condenser must have rounded and smooth finishing.

And accessible for easy cleaning

The Control cabinet shall be integrated on the machine frame. And the cabinets shall be equipped with

proper ventilators

Chamber shelves design shall allow minimum and maximum loading of the product and useable 90% out of total shelves area available.

Remote access feature shall be available for providing technical support of the feature

Machine shall have feature to have control the speed increase / decrease the speed through frequency

control drive and shall be available to operate on HMI & SCADA

The machine shall be designed to be able to operate and maintain the temperature during the one compressor failure

Redundancy Fluid pump shall be provided

Vacuum pump redundancy, shall be able to operate with one pump and maintain the set vacuum at any point of operation

All Hot surfaces shall be insulated with mineral wool and cladding with SS304 sheet.

The bellows used in the entire system shall be leak proof.

Chamber:

The chamber shall have the automatic Locking / Un-locking system

The Geometrical construction of the chamber shall be rectangular structure

The type of door support shall be operated with Hinged

Operating design pressure of the chamber shall be 2.0 Bar G

The slot door shall be installed at Loading /Unloading side and additional main door shall be installed at technical area for maintenance purpose and the door can be completely open-able up to 180 Deg direction.

Chamber shall have the jacket for cooling water circulation provision to do the jacket cooling after SIP and other heating operations.

The material of construction of the chamber shall be made SS316 material with surface finish Ra < 0.5 µm

internally with mirror polish. And chamber body shall be insulated and cladded with SS304 material

Chamber shall have the sight glass possible locations with light illumination

Shelves configuration:

Basic inter distance between the shelves : To be decided

Shelf dimensions (LxWxD) : To be decided

Total surface area : 18 Sq Mtr

Shelves heating rate : NLT 1.0 deg C/ Min

Shelves Minimum temperature : -65.0 Deg C

Shelves cooling source : Silicon Oil

Shelves moment for loading : Included

Shelves position control : Closed loop (Automatic)

The shelves shall allow the constant loading at fixed operating position/ height.

The shelves shall be made of SS316 L material and the surface finish shall be less than < 0.5 µm electro polished.

The shelves shall be designed to allowed minimum 2 formats for inter-distance for loading / Unloading

The Shelves loading should be from top to bottom direction and un-loading from bottom to top at constant height

Shelves shall be movable during CIP to ensure the 100% cleaning efficiency

Each shelf should have a temperature probe to measure the individual shelf temperature

The surface slope of the shelf maximum 0.5 mm/ Mtr

Refrigeration Compressors:

Total no. of compressors : To be defined

Compressors type : Screw type

Refrigerant : Non-CFC, Eco friendly

Vacuum System:

Vacuum pumps shall be dry pumps

The Pump capacity shall be design inline to the Chamber and condenser capacity

The ultimate vacuum shall be achieved with in 20 Min in ideal condition

The booster vacuum pump shall be dry type

Condenser:

Type of condenser : Close integrated

Number of condensers : 01 Nos

Design Pressure : 2.0 Bar G

Capacity of Ice thickness : To be defined

Minimum Temperature : >-75 Deg C

The construction of the condenser shall be cylindrical and should be suitable for vacuum applications during the lyophilization cycles

The condenser body and tubes shall be made of SS316 L and are mechanically polished and passivated

The ice condensing capacity shall be calculated on the basis of 24 Hr cycle

The suitable isolation valve shall be designed between the Chamber and condenser, and this shall be made up of SS316 L material.

The condenser cooling shall be done either with Gas/ Fluid

Sight glass with illumination shall be provided on body of the condenser for viewing purpose

The condenser shall be CIP and SIP able, and withstand with respect to the temperature and pressure

Functional Specification:

The Lyophilizer shall have interface with slot door with fixed loading/un-loading system

All Loading/ Unloading activities shall be from Grade A- Area

The Loading/ unloading system shall be designed as per the details and drawings provided by client

The loading / unloading operation shall be synchronized with lyophilizer operation

The system shall have the provision to back fill the equipment with sterile gases like Nitrogen/ Air and vacuum.

The vacuum control set point should be allowed from 1000 to 0.001 mbar absolute

SIP: The Sterilization cycle the temperature reference shall be measured at drain thermocouple, and all other temperature sensors shall be remains above the drain.

During SIP Automatic movement of shelves required from top to bottom to ensure the sterility of all components contained in the chamber
Vacuum pumps shall have the provision to provide the gas ballast to avoid the moisture and toxics impact during the process.

Integration of ISOLATOR for Loading/ Un-loading system:

The isolator unit shall be integrated with Lyophilizer as to meet the process requirements and current good manufacturing practice
The design and integration of Isolator shall be in line to the Lyo loading and unloading chamber interface and dimensions
Tamper proof glass door shall be provided to allow a clear visibility into the entire space of the isolator.
The interior and all corners of the isolator must be easily accessible and cleanable when isolator is closed through glove ports
The surface in and outside side of the isolator shouldn't have any cracks, crevices and sharp edges. All corners and edges must be rounded
All HEPA Filters installed in the isolator must be accessible for maintenance and cleaning
Glare-free light, sealed edges of the lamps inside of isolator shall be provided
Calibration of all sensors must easily accessible and shall be removable for calibration and maintenance
Maximum expected operation of the equipment 24 Hrs./ Day

Technical Specification:

The temperature control maintained within the isolator in the range of 20 to 25 Deg C in all conditions.
Beginning of the batch
While continuous running of the batch
Fluctuation of outside environment shall be within tolerance
Relative Humidity control during production is maintained within the full humidity range of 2-5 % with aid of Nitrogen purge
HEPA filters are 99.99% efficient for removing particles greater than 0.3 Micron in size.
The room conditions of isolator installed is classified as grade B
Environment conditions surrounding room
Temp : NMT 26 Deg C
% RH : NMT 60%
The isolator unit is required to design to meet the cleanliness under the area to meet Grade 'A' requirement
The MOC of the isolator shall be compactable for VHP sterilization/ Nitrogen gas .and all product contact parts shall be made with SS316 L and surface finish will be $> 0.5 \mu$ RA.
Integrity test port shall be provided at upstream of HEPA Filters
Blower shall statically and dynamically balance to the supply air flow with sufficient capacity and static pressure
Viable/ non-viable particle count shall meet grade A requirements
The air velocity shall be 90 ± 20 FPM at 6 inches from the filter face
Initial and final pressure drop limits to be provided
During the VHP cycle, Gloves shall not be wrinkled or folded on surface
There shall be automatic leak test option available for isolator leak test and within the acceptance range; Leak test will be performed before start of the VHP cycle.

Interlocks/ Alarms:

Emergency Stop
Control power fault
Barrier open door
Glove port light curtain interrupted
Isolator contamination/ fault
Filter differential pressure alarm
Barrier high pressure alarm
Barrier low pressure alarm
Barrier differential pressure alarm
RH above 5% alarm

All machine surfaces must come in contact with H₂O₂, a minimum accessible area for H₂O₂ on the machine for example moving parts shall move slowly during VHP Cycle.
The whole decontamination cycle including aeration shall be completed within the specified operation time mentioned.
The isolator doors shall not open before the H₂O₂ concentration reaches to the neutral level.

Interface:

The Lyophilizer shall be provided with Siemens or Allen Bradley control logic platform. PLC, security, Recipe management process. Data logging shall be under taken Lyophilized dedicated SCADA.
Control system shall be accessible to authorized users only. It shall have individual login ID and Passwords to access the system.
All set parameters ranges are to be defined access shall be restricted as per user level
Alarm management shall be available
Programmed backup for PLC/HMI SCADA shall be provided
Generated data shall be saved by the software and storage path shall be defined through software only
Control system clock synchronization shall be available
11.8 Electronic records and signatures shall be available wherever applicable on the reports | General Requirements:
All parts shall be made of non-oxidizing material as per FDA requirements
All internal corners and edges inside the chamber and condenser must have rounded and smooth finishing.
And accessible for easy cleaning
The Control cabinet shall be integrated on the machine frame. And the cabinets shall be equipped with proper ventilators
Chamber shelves design shall allow minimum and maximum loading of the product and useable 90% out of total shelves area available.
Remote access feature shall be available for providing technical support of the feature
Machine shall have feature to have control the speed increase / decrease the speed through frequency control drive and shall be available to operate on HMI & SCADA
The machine shall be designed to be able to operate and maintain the temperature during the one

compressor failure

Redundancy Fluid pump shall be provided

Vacuum pump redundancy, shall be able to operate with one pump and maintain the set vacuum at any point of operation

All Hot surfaces shall be insulated with mineral wool and cladding with SS304 sheet.

The bellows used in the entire system shall be leak proof.

Chamber:

The chamber shall have the automatic Locking / Un-locking system

The Geometrical construction of the chamber shall be rectangular structure

The type of door support shall be operated with Hinged

Operating design pressure of the chamber shall be 2.0 Bar G

The slot door shall be installed at Loading /Unloading side and additional main door shall be installed at technical area for maintenance purpose and the door can be completely open-able up to 180 Deg direction.

Chamber shall have the jacket for cooling water circulation provision to do the jacket cooling after SIP and other heating operations.

The material of construction of the chamber shall be made SS316 material with surface finish $R_a < 0.5 \mu\text{s}$ internally with mirror polish. And chamber body shall be insulated and cladded with SS304 material

Chamber shall have the sight glass possible locations with light illumination

Shelves configuration:

Basic inter distance between the shelves : To be decided

Shelf dimensions (LxWxD) : To be decided

Total surface area : 18 Sq Mtr

Shelves heating rate : NLT 1.0 deg C/ Min

Shelves Minimum temperature : -65.0 Deg C

Shelves cooling source : Silicon Oil

Shelves moment for loading : Included

Shelves position control : Closed loop (Automatic)

The shelves shall allow the constant loading at fixed operating position/ height.

The shelves shall be made of SS316 L material and the surface finish shall be less than $< 0.5 \mu\text{m}$ electro polished.

The shelves shall be designed to allowed minimum 2 formats for inter-distance for loading / Unloading

The Shelves loading should be from top to bottom direction and un-loading from bottom to top at constant height

Shelves shall be movable during CIP to ensure the 100% cleaning efficiency

Each shelf should have a temperature probe to measure the individual shelf temperature

The surface slope of the shelf maximum 0.5 mm/ Mtr

Refrigeration Compressors:

Total no. of compressors : To be defined

Compressors type : Screw type

Refrigerant : Non-CFC, Eco friendly

Vacuum System:

Vacuum pumps shall be dry pumps

The Pump capacity shall be design inline to the Chamber and condenser capacity

The ultimate vacuum shall be achieved with in 20 Min in ideal condition

The booster vacuum pump shall be dry type

Condenser:

Type of condenser : Close integrated

Number of condensers : 01 Nos

Design Pressure : 2.0 Bar G

Capacity of Ice thickness : To be defined

Minimum Temperature : $>-75 \text{ Deg C}$

The construction of the condenser shall be cylindrical and should be suitable for vacuum applications during the lyophilization cycles

The condenser body and tubes shall be made of SS316 L and are mechanically polished and passivated

The ice condensing capacity shall be calculated on the basis of 24 Hr cycle

The suitable isolation valve shall be designed between the Chamber and condenser, and this shall be made up of SS316 L material.

The condenser cooling shall be done either with Gas/ Fluid

Sight glass with illumination shall be provided on body of the condenser for viewing purpose

The condenser shall be CIP and SIP able, and withstand with respect to the temperature and pressure

Functional Specification:

The Lyophilizer shall have interface with slot door with fixed loading/un-loading system

All Loading/ Unloading activities shall be from Grade A- Area

The Loading/ unloading system shall be designed as per the details and drawings provided by client

The loading / unloading operation shall be synchronized with lyophilizer operation

The system shall have the provision to back fill the equipment with sterile gases like Nitrogen/ Air and vacuum.

The vacuum control set point should be allowed from 1000 to 0.001 mbar absolute

SIP: The Sterilization cycle the temperature reference shall be measured at drain thermocouple, and all other temperature sensors shall be remains above the drain.

During SIP Automatic movement of shelves required from top to bottom to ensure the sterility of all components contained in the chamber

Vacuum pumps shall have the provision to provide the gas ballast to avoid the moisture and toxics impact during the process.

Integration of ISOLATOR for Loading/ Un-loading system:

The isolator unit shall be integrated with Lyophilizer as to meet the process requirements and current good manufacturing practice
The design and integration of Isolator shall be in line to the Lyo loading and unloading chamber interface and dimensions
Tamper proof glass door shall be provided to allow a clear visibility into the entire space of the isolator.
The interior and all corners of the isolator must be easily accessible and cleanable when isolator is closed through glove ports
The surface in and outside side of the isolator shouldn't have any cracks, crevices and sharp edges. All corners and edges must be rounded
All HEPA Filters installed in the isolator must be accessible for maintenance and cleaning
Glare-free light, sealed edges of the lamps inside of isolator shall be provided
Calibration of all sensors must easily accessible and shall be removable for calibration and maintenance
Maximum expected operation of the equipment 24 Hrs./ Day

Technical Specification:

The temperature control maintained within the isolator in the range of 20 to 25 Deg C in all conditions.
Beginning of the batch
While continuous running of the batch
Fluctuation of outside environment shall be within tolerance
Relative Humidity control during production is maintained within the full humidity range of 2-5 % with aid of Nitrogen purge
HEPA filters are 99.99% efficient for removing particles greater than 0.3 Micron in size.
The room conditions of isolator installed is classified as grade B
Environment conditions surrounding room
Temp : NMT 26 Deg C
% RH : NMT 60%
The isolator unit is required to design to meet the cleanliness under the area to meet Grade 'A' requirement
The MOC of the isolator shall be compactable for VHP sterilization/ Nitrogen gas .and all product contact parts shall be made with SS316 L and surface finish will be > 0.5 µ RA.
Integrity test port shall be provided at upstream of HEPA Filters
Blower shall statically and dynamically balance to the supply air flow with sufficient capacity and static pressure
Viable/ non-viable particle count shall meet grade A requirements
The air velocity shall be 90 ±20 FPM at 6 inches from the filter face
Initial and final pressure drop limits to be provided
During the VHP cycle, Gloves shall not be wrinkled or folded on surface
There shall be automatic leak test option available for isolator leak test and within the acceptance range; Leak test will be performed before start of the VHP cycle.

Interlocks/ Alarms:

Emergency Stop
Control power fault
Barrier open door
Glove port light curtain interrupted
Isolator contamination/ fault
Filter differential pressure alarm
Barrier high pressure alarm
Barrier low pressure alarm
Barrier differential pressure alarm
RH above 5% alarm

All machine surfaces must come in contact with H2O2, a minimum accessible area for H2O2 on the machine for example moving parts shall move slowly during VHP Cycle.
The whole decontamination cycle including aeration shall be completed within the specified operation time mentioned.
The isolator doors shall not open before the H2O2 concentration reaches to the neutral level.

Interface:

The Lyophilizer shall be provided with Siemens or Allen Bradley control logic platform. PLC, security, Recipe management process. Data logging shall be under taken Lyophilized dedicated SCADA.
Control system shall be accessible to authorized users only. It shall have individual login ID and Passwords to access the system.
All set parameters ranges are to be defined access shall be restricted as per user level
Alarm management shall be available
Programmed backup for PLC/HMI SCADA shall be provided
Generated data shall be saved by the software and storage path shall be defined through software only
Control system clock synchronization shall be available
11.8 Electronic records and signatures shall be available wherever applicable on the reports | General Requirements:
All parts shall be made of non-oxidizing material as per FDA requirements
All internal corners and edges inside the chamber and condenser must have rounded and smooth finishing.
And accessible for easy cleaning
The Control cabinet shall be integrated on the machine frame. And the cabinets shall be equipped with proper ventilators
Chamber shelves design shall allow minimum and maximum loading of the product and useable 90% out of total shelves area available.
Remote access feature shall be available for providing technical support of the feature
Machine shall have feature to have control the speed increase / decrease the speed through frequency control drive and shall be available to operate on HMI & SCADA
The machine shall be designed to be able to operate and maintain the temperature during the one compressor failure
Redundancy Fluid pump shall be provided
Vacuum pump redundancy, shall be able to operate with one pump and maintain the set vacuum at any point of operation
All Hot surfaces shall be insulated with mineral wool and cladding with SS304 sheet.
The bellows used in the entire system shall be leak proof.

Chamber:

The chamber shall have the automatic Locking / Un-locking system

The Geometrical construction of the chamber shall be rectangular structure

The type of door support shall be operated with Hinged

Operating design pressure of the chamber shall be 2.0 Bar G

The slot door shall be installed at Loading /Unloading side and additional main door shall be installed at technical area for maintenance purpose and the door can be completely open-able up to 180 Deg direction.

Chamber shall have the jacket for cooling water circulation provision to do the jacket cooling after SIP and other heating operations.

The material of construction of the chamber shall be made SS316 material with surface finish Ra < 0.5 µs

internally with mirror polish. And chamber body shall be insulated and cladded with SS304 material

Chamber shall have the sight glass possible locations with light illumination

Shelves configuration:

Basic inter distance between the shelves : To be decided

Shelf dimensions (LxWxD) : To be decided

Total surface area : 18 Sq Mtr

Shelves heating rate : NLT 1.0 deg C/ Min

Shelves Minimum temperature : -65.0 Deg C

Shelves cooling source : Silicon Oil

Shelves moment for loading : Included

Shelves position control : Closed loop (Automatic)

The shelves shall allow the constant loading at fixed operating position/ height.

The shelves shall be made of SS316 L material and the surface finish shall be less than < 0.5 µm electro polished.

The shelves shall be designed to allowed minimum 2 formats for inter-distance for loading / Unloading

The Shelves loading should be from top to bottom direction and un-loading from bottom to top at constant height

Shelves shall be movable during CIP to ensure the 100% cleaning efficiency

Each shelf should have a temperature probe to measure the individual shelf temperature

The surface slope of the shelf maximum 0.5 mm/ Mtr

Refrigeration Compressors:

Total no. of compressors : To be defined

Compressors type : Screw type

Refrigerant : Non-CFC, Eco friendly

Vacuum System:

Vacuum pumps shall be dry pumps

The Pump capacity shall be design inline to the Chamber and condenser capacity

The ultimate vacuum shall be achieved with in 20 Min in ideal condition

The booster vacuum pump shall be dry type

Condenser:

Type of condenser : Close integrated

Number of condensers : 01 Nos

Design Pressure : 2.0 Bar G

Capacity of Ice thickness : To be defined

Minimum Temperature : >-75 Deg C

The construction of the condenser shall be cylindrical and should be suitable for vacuum applications during the lyophilization cycles

The condenser body and tubes shall be made of SS316 L and are mechanically polished and passivated

The ice condensing capacity shall be calculated on the basis of 24 Hr cycle

The suitable isolation valve shall be designed between the Chamber and condenser, and this shall be made up of SS316 L material.

The condenser cooling shall be done either with Gas/ Fluid

Sight glass with illumination shall be provided on body of the condenser for viewing purpose

The condenser shall be CIP and SIP able, and withstand with respect to the temperature and pressure

Functional Specification:

The Lyophilizer shall have interface with slot door with fixed loading/un-loading system

All Loading/ Unloading activities shall be from Grade A- Area

The Loading/ unloading system shall be designed as per the details and drawings provided by client

The loading / unloading operation shall be synchronized with lyophilizer operation

The system shall have the provision to back fill the equipment with sterile gases like Nitrogen/ Air and vacuum.

The vacuum control set point should be allowed from 1000 to 0.001 mbar absolute

SIP: The Sterilization cycle the temperature reference shall be measured at drain thermocouple, and all other temperature sensors shall be remains above the drain.

During SIP Automatic movement of shelves required from top to bottom to ensure the sterility of all components contained in the chamber

Vacuum pumps shall have the provision to provide the gas ballast to avoid the moisture and toxics impact during the process.

Integration of ISOLATOR for Loading/ Un-loading system:

The isolator unit shall be integrated with Lyophilizer as to meet the process requirements and current good manufacturing practice

The design and integration of Isolator shall be in line to the Lyo loading and unloading chamber interface and dimensions

Tamper proof glass door shall be provided to allow a clear visibility into the entire space of the isolator.

The interior and all corners of the isolator must be easily accessible and cleanable when isolator is

closed through glove ports

The surface in and outside side of the isolator shouldn't have any cracks, crevices and sharp edges. All corners and edges must be rounded

All HEPA Filters installed in the isolator must be accessible for maintenance and cleaning

Glare-free light, sealed edges of the lamps inside of isolator shall be provided

Calibration of all sensors must easily accessible and shall be removable for calibration and maintenance

Maximum expected operation of the equipment 24 Hrs./ Day

Technical Specification:

The temperature control maintained within the isolator in the range of 20 to 25 Deg C in all conditions.

Beginning of the batch

While continuous running of the batch

Fluctuation of outside environment shall be within tolerance

Relative Humidity control during production is maintained within the full humidity range of 2-5 % with aid of Nitrogen purge

HEPA filters are 99.99% efficient for removing particles greater than 0.3 Micron in size.

The room conditions of isolator installed is classified as grade B

Environment conditions surrounding room

Temp : NMT 26 Deg C

% RH : NMT 60%

The isolator unit is required to design to meet the cleanliness under the area to meet Grade 'A' requirement

The MOC of the isolator shall be compactable for VHP sterilization/ Nitrogen gas .and all product contact parts shall be made with SS316 L and surface finish will be $> 0.5 \mu$ RA.

Integrity test port shall be provided at upstream of HEPA Filters

Blower shall statically and dynamically balance to the supply air flow with sufficient capacity and static pressure

Viable/ non-viable particle count shall meet grade A requirements

The air velocity shall be 90 ± 20 FPM at 6 inches from the filter face

Initial and final pressure drop limits to be provided

During the VHP cycle, Gloves shall not be wrinkled or folded on surface

There shall be automatic leak test option available for isolator leak test and within the acceptance range; Leak test will be performed before start of the VHP cycle.

Interlocks/ Alarms:

Emergency Stop

Control power fault

Barrier open door

Glove port light curtain interrupted

Isolator contamination/ fault

Filter differential pressure alarm

Barrier high pressure alarm

Barrier low pressure alarm

Barrier differential pressure alarm

RH above 5% alarm

All machine surfaces must come in contact with H2O2, a minimum accessible area for H2O2 on the machine for example moving parts shall move slowly during VHP Cycle.

The whole decontamination cycle including aeration shall be completed within the specified operation time mentioned.

The isolator doors shall not open before the H2O2 concentration reaches to the neutral level.

Interface:

The Lyophilizer shall be provided with Siemens or Allen Bradley control logic platform. PLC, security,

Recipe management process. Data logging shall be under taken Lyophilized dedicated SCADA.

Control system shall be accessible to authorized users only. It shall have individual login ID and Passwords to access the system.

All set parameters ranges are to be defined access shall be restricted as per user level

Alarm management shall be available

Programmed backup for PLC/HMI SCADA shall be provided

Generated data shall be saved by the software and storage path shall be defined through software only

Control system clock synchronization shall be available

11.8 Electronic records and signatures shall be available wherever applicable on the reports | General Requirements:

All parts shall be made of non-oxidizing material as per FDA requirements

All internal corners and edges inside the chamber and condenser must have rounded and smooth finishing.

And accessible for easy cleaning

The Control cabinet shall be integrated on the machine frame. And the cabinets shall be equipped with proper ventilators

Chamber shelves design shall allow minimum and maximum loading of the product and useable 90% out of total shelves area available.

Remote access feature shall be available for providing technical support of the feature

Machine shall have feature to have control the speed increase / decrease the speed through frequency control drive and shall be available to operate on HMI & SCADA

The machine shall be designed to be able to operate and maintain the temperature during the one compressor failure

Redundancy Fluid pump shall be provided

Vacuum pump redundancy, shall be able to operate with one pump and maintain the set vacuum at any point of operation

All Hot surfaces shall be insulated with mineral wool and cladding with SS304 sheet.

The bellows used in the entire system shall be leak proof.

Chamber:

The chamber shall have the automatic Locking / Un-locking system

The Geometrical construction of the chamber shall be rectangular structure

The type of door support shall be operated with Hinged

Operating design pressure of the chamber shall be 2.0 Bar G

The slot door shall be installed at Loading /Unloading side and additional main door shall be installed

at technical area for maintenance purpose and the door can be completely open-able up to 180 Deg direction.

Chamber shall have the jacket for cooling water circulation provision to do the jacket cooling after SIP and other heating operations.

The material of construction of the chamber shall be made SS316 material with surface finish $Ra < 0.5 \mu s$ internally with mirror polish. And chamber body shall be insulated and cladded with SS304 material
Chamber shall have the sight glass possible locations with light illumination

Shelves configuration:

Basic inter distance between the shelves : To be decided

Shelf dimensions (LxWxD) : To be decided

Total surface area : 18 Sq Mtr

Shelves heating rate : NLT 1.0 deg C/ Min

Shelves Minimum temperature : -65.0 Deg C

Shelves cooling source : Silicon Oil

Shelves moment for loading : Included

Shelves position control : Closed loop (Automatic)

The shelves shall allow the constant loading at fixed operating position/ height.

The shelves shall be made of SS316 L material and the surface finish shall be less than $< 0.5 \mu m$ electro polished.

The shelves shall be designed to allowed minimum 2 formats for inter-distance for loading / Unloading

The Shelves loading should be from top to bottom direction and un-loading from bottom to top at constant height

Shelves shall be movable during CIP to ensure the 100% cleaning efficiency

Each shelf should have a temperature probe to measure the individual shelf temperature

The surface slope of the shelf maximum 0.5 mm/ Mtr

Refrigeration Compressors:

Total no. of compressors : To be defined

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Number of condensers : 01 Nos

Design Pressure : 2.0 Bar G

Capacity of Ice thickness : To be defined

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The condenser body and tubes shall be made of SS316 L and are mechanically polished and passivated

The ice condensing capacity shall be calculated on the basis of 24 Hr cycle

The suitable isolation valve shall be designed between the Chamber and condenser, and this shall be made up of SS316 L material.

The condenser cooling shall be done either with Gas/ Fluid

Sight glass with illumination shall be provided on body of the condenser for viewing purpose

The condenser shall be CIP and SIP able, and withstand with respect to the temperature and pressure

Functional Specification:

The Lyophilizer shall have interface with slot door with fixed loading/un-loading system

All Loading/ Unloading activities shall be from Grade A- Area

The Loading/ unloading system shall be designed as per the details and drawings provided by client

The loading / unloading operation shall be synchronized with lyophilizer operation

The system shall have the provision to back fill the equipment with sterile gases like Nitrogen/ Air and vacuum.

The vacuum control set point should be allowed from 1000 to 0.001 mbar absolute

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The surface in and outside side of the isolator shouldn't have any cracks, crevices and sharp edges. All corners and edges must be rounded

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Maximum expected operation of the equipment 24 Hrs./ Day

Technical Specification:

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While continuous running of the batch

Fluctuation of outside environment shall be within tolerance

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aid of Nitrogen purge

HEPA filters are 99.99% efficient for removing particles greater than 0.3 Micron in size.

The room conditions of isolator installed is classified as grade B

Environment conditions surrounding room

Temp : NMT 26 Deg C

% RH : NMT 60%

The isolator unit is required to design to meet the cleanliness under the area to meet Grade 'A' requirement

The MOC of the isolator shall be compactable for VHP sterilization/ Nitrogen gas .and all product contact parts shall be made with SS316 L and surface finish will be > 0.5 µ RA.

Integrity test port shall be provided at upstream of HEPA Filters

Blower shall statically and dynamically balance to the supply air flow with sufficient capacity and static pressure

Viable/ non-viable particle count shall meet grade A requirements

The air velocity shall be 90 ±20 FPM at 6 inches from the filter face

Initial and final pressure drop limits to be provided

During the VHP cycle, Gloves shall not be wrinkled or folded on surface

There shall be automatic leak test option available for isolator leak test and within the acceptance range; Leak test will be performed before start of the VHP cycle.

Interlocks/ Alarms:

Emergency Stop

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Recipe management process. Data logging shall be under taken Lyophilized dedicated SCADA.

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Passwords to access the system.

All set parameters ranges are to be defined access shall be restricted as per user level

Alarm management shall be available

Programmed backup for PLC/HMI SCADA shall be provided

Generated data shall be saved by the software and storage path shall be defined through software only

Control system clock synchronization shall be available

11.8 Electronic records and signatures shall be available wherever applicable on the reports

| Components / features | Components / features | Required Yes/ No | Required Yes/ No | Required Yes/ No

| Required Yes/ No | Description | Description

| Control Panel | Control Panel | Yes | Yes | Yes | Yes | The programmable logic controller in the control cabinet

The control system should monitor the

Machine functions

Controls the operational sequence of the machine | The programmable logic controller in the control cabinet

The control system should monitor the

Machine functions

Controls the operational sequence of the machine

| Indicators | Indicators | Yes | Yes | Yes | Yes | As applicable | As applicable

| Data Security | Data Security | Yes | Yes | Yes | Yes | As applicable | As applicable

| HMI/ PLC / SCADA Details | HMI/ PLC / SCADA Details | Yes | Yes | Yes | Yes | As applicable | As applicable

| User interface compatible to SCADA System | User interface compatible to SCADA System | Yes | Yes | Yes

| Yes | As applicable | As applicable

| Interface to other systems /Equipment / Instruments | Interface to other systems /Equipment / Instruments | Yes | Yes | Yes | Yes | As applicable | As applicable

| Data Storage Capacity | Data Storage Capacity | Yes | Yes | Yes | Yes | As applicable | As applicable

| Password Protection | Password Protection | Yes | Yes | Yes | Yes | As applicable | As applicable

| SCADA Features:

The SCADA shall allow the creation of batch reports, including the set points and limitations of all instruments.

It shall allow the editing of batch recipes

The SCADA shall be able to store and retrieve up to 999 Batch recipes

The generated batch report shall capture each and every action as audit trail

SCADA shall able to generate the all-auxiliary cycle's reports | SCADA Features:

The SCADA shall allow the creation of batch reports, including the set points and limitations of all instruments.

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The generated batch report shall capture each and every action as audit trail

SCADA shall able to generate the all-auxiliary cycle's reports

5.0 | Construction material | Construction material | Construction material | Construction material |

Construction material | Construction material | Construction material | Construction material

| Location | Location | Grade | Grade | Grade | Grade | Max Ra | Max Ra

| Chamber internal | Chamber internal | 316 L | 316 L | 316 L | 316 L | < 0.5 | < 0.5

| Chamber external

(Loading side) | Chamber external

(Loading side) | 316 L | 316 L | 316 L | 316 L | < 0.5 | < 0.5

| Chamber external

(Un-loading Side) | Chamber external

(Un-loading Side) | 316 L | 316 L | 316 L | 316 L | < 0.5 | < 0.5

| Chamber external top side | Chamber external top side | 316 L | 316 L | 316 L | 316 L | < 0.5 | < 0.5

| Chamber door loading side | Chamber door loading side | 316 L | 316 L | 316 L | 316 L | < 0.5 | < 0.5

| Chamber door un-load side | Chamber door un-load side | 316 L | 316 L | 316 L | 316 L | < 0.5 | < 0.5

| Condenser Internal | Condenser Internal | 316 L | 316 L | 316 L | 316 L | < 0.5 | < 0.5

| Condenser external | Condenser external | 316 L | 316 L | 316 L | 316 L | < 0.5 | < 0.5

| Condenser coil | Condenser coil | 316 L | 316 L | 316 L | 316 L | < 0.5 | < 0.5

| Chamber shelves | Chamber shelves | 316 L | 316 L | 316 L | 316 L | < 0.5 | < 0.5

| All SIP Piping's | All SIP Piping's | 316 L | 316 L | 316 L | 316 L | < 0.5 | < 0.5

| All CIP Piping's | All CIP Piping's | 316 L | 316 L | 316 L | 316 L | < 0.5 | < 0.5

| Internal piping's

(Chamber / condenser) | Internal piping's

(Chamber / condenser) | 316 L | 316 L | 316 L | 316 L | < 0.5 | < 0.5

| Vacuum Piping | Vacuum Piping | 316 L | 316 L | 316 L | 316 L | < 0.5 | < 0.5

| Silicon fluid Piping | Silicon fluid Piping | 316 L | 316 L | 316 L | 316 L | < 0.5 | < 0.5

| All product contact parts shall be made of SS316 L and non-contact parts SS304

All Gaskets used in the system. Shall be Silicon/ EPDM food grade and USFDA approved | All product

contact parts shall be made of SS316 L and non-contact parts SS304

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contact parts shall be made of SS316 L and non-contact parts SS304

All Gaskets used in the system. Shall be Silicon/ EPDM food grade and USFDA approved

6.0 | Utility Requirements | Utility Requirements | Utility Requirements | Utility Requirements | Utility

Requirements | Utility Requirements | Utility Requirements | Utility Requirements

| Electrical Supply : 415 V 3 phase, 50 Hz and 230 V 1 phase, 50Hz

Compressed air : 6 Bar

Nitrogen : As recommended | Electrical Supply : 415 V 3 phase, 50 Hz and 230 V 1 phase, 50Hz

Compressed air : 6 Bar

Nitrogen : As recommended | Electrical Supply : 415 V 3 phase, 50 Hz and 230 V 1 phase, 50Hz

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Design Qualification
Installation Qualification
Operational Qualification
Performance Qualification
Test and Calibration with traceability Certificates
Preventive Maintenance Procedures
Warranty Certificates
21 CFR Part 11 Compliant documents
PLC program verification report at vendor site | Operational manual (2 Sets)
Design Qualification
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PLC program verification report at vendor site | Operational manual (2 Sets)
9.0 | Drawings | Drawings | Drawings | Drawings | Drawings | Drawings | Drawings |
| General Arrangement Diagram
P & I Diagram
Electrical Wiring Diagram | General Arrangement Diagram
P & I Diagram
Electrical Wiring Diagram | General Arrangement Diagram
P & I Diagram
Electrical Wiring Diagram | General Arrangement Diagram
P & I Diagram
Electrical Wiring Diagram | General Arrangement Diagram
P & I Diagram
Electrical Wiring Diagram | General Arrangement Diagram
P & I Diagram
Electrical Wiring Diagram | General Arrangement Diagram
P & I Diagram
Electrical Wiring Diagram | General Arrangement Diagram
P & I Diagram
Electrical Wiring Diagram
10.0 | Training | Training | Training | Training | Training | Training | Training | Training |
| The vendor shall impart training for operation and maintenance for at least two personnel from user department and engineering /projects department and shall provide evaluation report. | The vendor shall impart training for operation and maintenance for at least two personnel from user department and engineering /projects department and shall provide evaluation report. | The vendor shall impart training for operation and maintenance for at least two personnel from user department and engineering /projects department and shall provide evaluation report. | The vendor shall impart training for operation and maintenance for at least two personnel from user department and engineering /projects department and shall provide evaluation report. | The vendor shall impart training for operation and maintenance for at least two personnel from user department and engineering /projects department and shall provide evaluation report. | The vendor shall impart training for operation and maintenance for at least two personnel from user department and engineering /projects department and shall provide evaluation report. | The vendor shall impart training for operation and maintenance for at least two personnel from user department and engineering /projects department and shall provide evaluation report. | The vendor shall impart training for operation and maintenance for at least two personnel from user department and engineering /projects department and shall provide evaluation report. | The vendor shall impart training for operation and maintenance for at least two personnel from user department and engineering /projects department and shall provide evaluation report. | The vendor shall impart training for operation and maintenance for at least two personnel from user department and engineering /projects department and shall provide evaluation report.
11.0 | Abbreviations | Abbreviations | Abbreviations | Abbreviations | Abbreviations | Abbreviations | Abbreviations |
Abbreviations | Abbreviations
| GMP: Good Manufacturing Practice
URS: User requirement specification

CFR: Code for federal regulation
FDA: Food and Drug Administration
HMI: Human – Machine Interface
Hz: Hertz
NEC: National Electrical Code
P&ID: Piping and Instrument Diagram
PLC: Programmable Logic Controller | GMP: Good Manufacturing Practice
URS: User requirement specification
CFR: Code for federal regulation
FDA: Food and Drug Administration
HMI: Human – Machine Interface
Hz: Hertz
NEC: National Electrical Code
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HMI: Human – Machine Interface
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NEC: National Electrical Code
P&ID: Piping and Instrument Diagram
PLC: Programmable Logic Controller
12.0 | Other Requirements | Other Requirements | Other Requirements | Other Requirements | Other
Requirements | Other Requirements | Other Requirements | Other Requirements
| NA | NA | NA | NA | NA | NA | NA | NA

Approvals of the URS:

TABLE OF CONTENTS

Department | Name | Signature | Date

BioBridge Healthcare Solutions Pvt. Ltd., India | BioBridge Healthcare Solutions Pvt. Ltd., India |

BioBridge Healthcare Solutions Pvt. Ltd., India | BioBridge Healthcare Solutions Pvt. Ltd., India

Prepared by: | | |

Checked/Reviewed by: | | |

Client | Client | Client | Client

Reviewed by: | | |

Reviewed by: | | |

Approved by: | | |

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10.0 | Training | 5

11.0 | Abbreviations | 5

12.0 | Other Requirements | 5

1.0 | Objective | Objective

| This URS detail the requirement for melting point apparatus. | This URS detail the requirement for melting point apparatus.

2.0 | Scope | Scope

| This document is applicable for the melting point apparatus which will be installed with controlled environment, temperature range of 15 to 40 °C. This is used test the melting point of certain chemicals in quality control department | This document is applicable for the melting point apparatus which will be installed with controlled environment, temperature range of 15 to 40 °C. This is used test the melting point of certain chemicals in quality control department

3.0 | Equipment description | Equipment description

| A melting point apparatus is a laboratory instrument used to determine the melting point of substances used to assess their purity and identity. | A melting point apparatus is a laboratory instrument used to determine the melting point of substances used to assess their purity and identity.

4.0 | Component details | Component details

| Functional requirements | Functional requirements

| Measurement Range | 25°C to 400°C or higher

| Accuracy | ±0.2°C or better in the temperature range of 25°C to 200°C

| Accuracy | ±0.5°C or better in the temperature range of 200°C to 400°C.

| Resolution | Temperature resolution of at least 0.1°C.

| Heating rate | Adjustable heating rate between 0.1°C/min to 10°C/min.

| Detection Method | Automatic detection of the melting point based on optical, thermal, or video recording technology.

| Detection Method | Capable of real-time monitoring with display of melting point transition.

| Sample Capacity | Able to accommodate at least 3-5 samples per run for parallel measurements.

| Control System | User-friendly touchscreen interface with intuitive operation.

| Control System | Multi-level user access control

| Control System | Storage of at least 100 programs and 500 results with metadata for cGMP compliance.

| Connectivity | Equipped with USB, Ethernet, or Wi-Fi connectivity for data transfer.

| Size and Dimensions | Compact design to optimize bench space in laboratory environments.

| Safety Features | Over-temperature protection to prevent equipment damage or operator injury.

| Safe sample loading and unloading mechanism to protect users from burns or exposure to chemicals

| Noise level | ≤ 60 dB(A).

| Heat dissipation | Should not affect the laboratory's ambient temperature significantly.

5.0 | Construction material | Construction material

| Materials in contact with the sample or the environment should be made of non-reactive, non-corrosive such as stainless steel or equivalent.

Smooth, crevice-free surfaces to facilitate easy cleaning and avoid contamination. | Materials in contact with the sample or the environment should be made of non-reactive, non-corrosive such as stainless steel or equivalent.

Smooth, crevice-free surfaces to facilitate easy cleaning and avoid contamination.

6.0 | Utility Requirements | Utility Requirements

| Power supply: 220-240V single phase, 50/60 Hz. | Power supply: 220-240V single phase, 50/60 Hz.

7.0 | Accessories | Accessories

| All accessories required for the normal operation of the system. | All accessories required for the normal operation of the system.

8.0 | Documents Required | Documents Required

| Complete documentation Package with following documents should be delivered with the system

Installation Qualification (IQ)

Operational Qualification (OQ)

Operational manual (2 Sets) | Complete documentation Package with following documents should be delivered with the system

Installation Qualification (IQ)

Operational Qualification (OQ)

Operational manual (2 Sets)

9.0 | Drawings | Drawings

| Electrical drawings | Electrical drawings

10.0 | Training | Training

| The vendor shall impart training for operation and maintenance for at least two personnel from user department and engineering /projects department and shall provide evaluation report. | The vendor shall impart training for operation and maintenance for at least two personnel from user department and engineering /projects department and shall provide evaluation report.

11.0 | Abbreviations | Abbreviations

| QC: Quality Control

GLP: Good Laboratory Practice

GMP: Good Manufacturing Practice

URS: User Requirement Specification | QC: Quality Control

GLP: Good Laboratory Practice
GMP: Good Manufacturing Practice
URS: User Requirement Specification
12.0 | Other Requirements | Other Requirements
| NA | NA

Approvals of the URS:

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Department | Name | Signature | Date

BioBridge Healthcare Solutions Pvt. Ltd., India | BioBridge Healthcare Solutions Pvt. Ltd., India |

BioBridge Healthcare Solutions Pvt. Ltd., India | BioBridge Healthcare Solutions Pvt. Ltd., India

Prepared by: | | |

Checked/Reviewed by: | | |

Client | Client | Client | Client

Reviewed by: | | |

Reviewed by: | | |

Approved by: | | |

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1.0 | Objective

| This URS detail the requirement of Power pack used to provides constant voltage or constant current to instruments used in electrophoresis

2.0 | Scope

| This URS is applicable mainly for the Power pack, which will be placed in the quality control department.

3.0 | Equipment description

| The Power Pack provides constant voltage or constant current to instruments used in electrophoresis. The power supply operates at the values specified for the constant parameter, to prevent damage to the electrophoresis cell.

4.0 | Component details

| Technical Specification:

| Voltage: Adjustable from 10 to 300 Volts, in 1-volt increments.

Current: Adjustable from 4 to 400 milliamperes (mA) in 1 mA increments.

Power: 75 watts (maximum).

Programmable constant voltage or constant current with automatic crossover.

Timer control from 0 to 999 minutes.

3-digit LED display.

Pause mode for editing running parameters.

Automatic detection of no-load conditions and rapid changes in resistance.

Power failure detection in timed modes allowing completion of run.

Operating Temperature 0–40°C.

Humidity 10–95%

5.0 | Construction material

| NA

6.0 | Utility Requirements

| NA

7.0 | Accessories

| All accessories required for the normal operation of the equipment.

8.0 | Documents Required

| Operational manual (2 Sets)

9.0 | Drawings

| NA

10.0 | Training

| The vendor shall impart training for operation and maintenance for at least two personnel from user department and engineering /projects department and shall provide evaluation report.

11.0 | Abbreviations

| QC: Quality Control

GLP: Good Laboratory Practice

GMP: Good Manufacturing Practice

URS :User Requirement Specification

12.0 | Other Requirements

| NA

Approvals of the URS:

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Note: In case of any specific requirement from the client the body of this document can be modified

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Prepared by: | | |

Checked/Reviewed by: | | |

Client | Client | Client | Client

Reviewed by: | | |

Reviewed by: | | |

Approved by: | | |

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8.0 | Documents Required |

9.0 | Drawings |

10.0 | Training |

11.0 | Abbreviations |

12.0 | Other Requirements |

1.0 | Objective | Objective | Objective

| The objective of the Vial Sealing Machine is to provide a reliable and efficient sealing of vaccine vials within the Fill Finish Facility | The objective of the Vial Sealing Machine is to provide a reliable and efficient sealing of vaccine vials within the Fill Finish Facility | The objective of the Vial Sealing Machine is to provide a reliable and efficient sealing of vaccine vials within the Fill Finish Facility

2.0 | Scope | Scope | Scope

| This specification is intended to cover the minimum requirements to be met by the vendor in the design, manufacture and supply of Vial sealing machine used in the Uganda facility | This specification is intended to cover the minimum requirements to be met by the vendor in the design, manufacture and supply of Vial sealing machine used in the Uganda facility | This specification is intended to cover the minimum requirements to be met by the vendor in the design, manufacture and supply of Vial sealing machine used in the Uganda facility

3.0 | Equipment description | Equipment description | Equipment description

| This specification is intended to cover the minimum requirements to be met by the Vendor in the design, manufacture and supply of Vial sealing machine provided in this document are at minimum and can be subjected to amendment after discussion with vendor and representative of Client for achieving higher degree of standards and operational ease

Technical and GMP features

6R Vial sealing machine should be able to place the flip off seals on the vials and tighten them properly to make it tamper proof.

This should include;

Infeed turntable type 01 No.

Automated sealing machine 01 No.

Spares | This specification is intended to cover the minimum requirements to be met by the Vendor in the design, manufacture and supply of Vial sealing machine provided in this document are at minimum and can be subjected to amendment after discussion with vendor and representative of Client for achieving higher degree of standards and operational ease

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6R Vial sealing machine should be able to place the flip off seals on the vials and tighten them properly to make it tamper proof.

This should include;

Infeed turntable type 01 No.

Automated sealing machine 01 No.

Spares

4.0 | Component details | Component details | Component details

| IN FEED TURN TABLE

Technical details:

In feed turntable should have SST finish, in feed pusher plate and connected to the sealing machine. Should have independent variable speed drive.

AUTOMATIC CLOSING MACHINE

Technical details:

This should be able to place the flip off seals on the vials and tighten them properly to make it tamper proof @ 220 vials/Min. (6R-Vial) It must meet the following requirements:

Aluminium plate type construction with stainless steel gladding,

Should have counters for vial number, running hour and output/min.
with variable drive motors
Suitable size parts consisting of infeed scroll, star wheel and guides.
Suitable automatic closing station for the mentioned seal.
For guarding purpose, it should have protection covers
It should have stainless steel conveying belt of.
No stopper- no seal control

Other features:

Should include switches for minimum and maximum accumulation at machine, at Infeed and at discharge, maintenance –free roller and ball bearings
Should have alarm for" tray full" | IN FEED TURN TABLE

Technical details:

In feed turntable should have SST finish, in feed pusher plate and connected to the sealing machine.
Should have independent variable speed drive.

AUTOMATIC CLOSING MACHINE

Technical details:

This should be able to place the flip off seals on the vials and tighten them properly to make it tamper proof @ 220 vials/Min. (6R-Vial) It must meet the following requirements:

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Should have alarm for" tray full"

| Instrumentation and automation

The system should have the necessary automation elements to ensure correct operation, as well as equipment and operator protection.

The following minimum requirements should be met: | Instrumentation and automation

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The system should have the necessary automation elements to ensure correct operation, as well as equipment and operator protection.

The following minimum requirements should be met:

| Interaction with other systems

The vial sealing machine control system must be able to communicate with a local PC-based monitoring system. The monitoring system will request the current values of all measured variables and will set the set points of all control points. The PC standard RS-232 is the preferred channel. If any other hardware is required, it must be supplied or fully specified (if it is commercially available)

Containment

The vial sealing machine should be designed so as to prevent the possibility generating and releasing aerosols into the environment without compromising the hygiene or integrity of the medium being filled and preferably within the enclosure of user friendly.

Alarms and signals

Low seal availability audiovisual signal | Interaction with other systems

The vial sealing machine control system must be able to communicate with a local PC-based monitoring system. The monitoring system will request the current values of all measured variables and will set the set points of all control points. The PC standard RS-232 is the preferred channel. If any other hardware is required, it must be supplied or fully specified (if it is commercially available)

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The vial sealing machine should be designed so as to prevent the possibility generating and releasing aerosols into the environment without compromising the hygiene or integrity of the medium being filled and preferably within the enclosure of user friendly.

Alarms and signals

Low seal availability audiovisual signal

| Other requirements and design characteristics

It should preferably include all the system interconnections.

It should include the required pumps: the pumps needed within the Vial washing machine to perform the required functions.

All the system must be capable of regular visual inspection and manual cleaning. The solution to this requirement in each case will depend on the size, geometry and placement of the machine and the proposal of the supplier.

The design and operation of the system and the degree of finish of its external surfaces must be compatible with the classification of the area in which they are to be installed and the chemicals commonly employed for cleaning and disinfecting in the biopharmaceutical industry.

The electrical system of the equipment shall comply with the requirements established by the following standard of the International Electrotechnical Commission: IEC 204 - Electrical equipment of industrial machines.

Part 1. General Requirements (This standard has been adopted and modified by the European Standard EN 60204-1:1992),

Part 2. Item designation and examples of drawings, diagrams, tables and instructions, and also with the applicable requirements of the attached document "Requirements for Engineered Equipment Wiring".

SWIVEL TYPE CONVEYER

Should be able to transfer filled vials to the capping machines infeed turntable. | Other requirements and design characteristics

It should preferably include all the system interconnections.

It should include the required pumps: the pumps needed within the Vial washing machine to perform the required functions.

All the system must be capable of regular visual inspection and manual cleaning. The solution to this requirement in each case will depend on the size, geometry and placement of the machine and the proposal of the supplier.

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SWIVEL TYPE CONVEYER

Should be able to transfer filled vials to the capping machines infeed turntable.

5.0 | Construction material | Construction material | Construction material

| Materials, welds and finish of parts in contact with the product

All metal components must be made of AISI 304 steel

The surface roughness of the metal parts of the vessels, interconnections and accessories must be Ra0.8m

All welds must comply with the FDA regulations (or the equivalent) for the biopharmaceutical industry. |

Materials, welds and finish of parts in contact with the product

All metal components must be made of AISI 304 steel

The surface roughness of the metal parts of the vessels, interconnections and accessories must be Ra0.8m

All welds must comply with the FDA regulations (or the equivalent) for the biopharmaceutical industry. |

Materials, welds and finish of parts in contact with the product

All metal components must be made of AISI 304 steel

The surface roughness of the metal parts of the vessels, interconnections and accessories must be Ra0.8m

All welds must comply with the FDA regulations (or the equivalent) for the biopharmaceutical industry.

6.0 | Utility Requirements | Utility Requirements | Utility Requirements

| Instrument Air | Instrument Air | 7.0 bar (g)

| Electrical Supply | Electrical Supply | 415 VAC, 3P, 50 Hz

| |

7.0 | Accessories | Accessories | Accessories

| Gaskets, SS clamps

The Vendors should provide as part of the quotation a list proposed spare parts for at least two years

operation | Gaskets, SS clamps

The Vendors should provide as part of the quotation a list proposed spare parts for at least two years

operation | Gaskets, SS clamps

The Vendors should provide as part of the quotation a list proposed spare parts for at least two years operation

8.0 | Documents Required | Documents Required | Documents Required

| Pre-Delivery Qualifications (PDQ)

Pre-Installation at the Vendor's Shop. The system or its parts as provided for in the scope of supply shall be pre-installed at the Vendor's shop prior to delivery to Customer's site. Installation will be completed and documented including mechanical parts as well as electrical connections of all parts to facilitate taking-over tests at Vendor's shop prior to delivery.

After the installation of the system at the Vendor's shop, Factory Acceptance Tests (FAT) shall be performed and accurately documented.

FAT tests shall be performed and documented according to a main protocol agreement between the Vendor and the Customer. At least two months before FAT test are scheduled to be performed; the documents where the results of such test will be recorded should be delivered to the Customer for approval.

General test to be performed during FAT:

Surface smoothness tests of metal surfaces.

Visual inspection for compliance with the cGMP, in accordance with the approved drawings.

Leakage Integrity test

CIP performance tests.

Alarms checkouts.

I/O Checks

Functional tests for all control loops.

Documents to be delivered by the Vendor

Two copies of the following documents must be provided, in the English language:

P&ID diagrams.

Operation Instruction Manual(s), including operating panel and safety operation warnings.

Maintenance Instruction Manual(s).

Assembly and Installation Manual(s).

Instrumentation and Calibration Manual(s).

Troubleshooting list.

Electrical and Wiring Manuals (including electrical drawing list, circuit diagrams, connection diagrams, cabinet layout, interconnection diagrams, etc.).

Component list.

Calibration certificate for each instrument installed in the equipment, including the description for re-calibration.

Certificate of materials used for constructing those parts in contact with the product stream(s).

Official approval document and manufacturing inspection protocol.

Welding procedures and visual inspection reports.

Mill test reports for those parts in contact with the product streams, which will be sanitary welded.

Qualification certificates of the technical personnel who have worked in welding and assembling of equipment shall be available on request.

Biological compatibility certificates of all non-metallic materials in contact with process streams (like diaphragms, gaskets, O-rings, seals, couplings, etc.).

Software and hardware validation documents for all computerized systems (microprocessor, PLC and computer-based systems).

Safety. Vendor should deliver to the Customer all legal documents relating to safety issues regulated by the National Official agencies where the equipment is built. The scope of these documents will be clearly defined during the contract negotiation.

Validation

All documents specified should be delivered at the Customer site at least two months before the installation of the equipment, in order to be used for the preparation of the Validation Protocols.

After installation of the equipment at Customer's site, complementary Installation and Operational Qualification (IQ, OQ) tests will be performed by the Customer and may be supervised (if it is of interest) by a member of Vendor's technical staff.

SPARES:

Vendor should provide as part of the tender a list of proposed spare parts for at least two years of operations. | Pre-Delivery Qualifications (PDQ)

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Component list.

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re-calibration.

Certificate of materials used for constructing those parts in contact with the product stream(s).

Official approval document and manufacturing inspection protocol.

Welding procedures and visual inspection reports.

Mill test reports for those parts in contact with the product streams, which will be sanitary welded.

Qualification certificates of the technical personnel who have worked in welding and assembling of equipment shall be available on request.

Biological compatibility certificates of all non-metallic materials in contact with process streams (like diaphragms, gaskets, O-rings, seals, couplings, etc.).

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Mill test reports for those parts in contact with the product streams, which will be sanitary welded.

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Software and hardware validation documents for all computerized systems (microprocessor, PLC and computer-based systems).

Safety. Vendor should deliver to the Customer all legal documents relating to safety issues regulated by the National Official agencies where the equipment is built. The scope of these documents will be clearly defined during the contract negotiation.

Validation

All documents specified should be delivered at the Customer site at least two months before the installation of the equipment, in order to be used for the preparation of the Validation Protocols.

After installation of the equipment at Customer's site, complementary Installation and Operational Qualification (IQ, OQ) tests will be performed by the Customer and may be supervised (if it is of interest) by a member of Vendor's technical staff.

SPARES:

Vendor should provide as part of the tender a list of proposed spare parts for at least two years of operations.

9.0 | Drawings | Drawings | Drawings

| P&ID, GA, Welding, electrical drawings | P&ID, GA, Welding, electrical drawings | P&ID, GA, Welding, electrical drawings

10.0 | Training | Training | Training

| Training of Customer's operators. The Vendor shall supply all available information for adequate equipment operation. To this end, the Customer's personnel will be trained by the Vendor's technical staff at the work place or at the Vendor's shop. | Training of Customer's operators. The Vendor shall supply all available information for adequate equipment operation. To this end, the Customer's personnel will be trained by the Vendor's technical staff at the work place or at the Vendor's shop. | Training of Customer's operators. The Vendor shall supply all available information for adequate equipment operation. To this end, the Customer's personnel will be trained by the Vendor's technical staff at the work place or at the Vendor's shop.

11.0 | Abbreviations | Abbreviations | Abbreviations
| URS | User Requirement Specification | User Requirement Specification
| P&ID | Piping and Instrumentation Diagram | Piping and Instrumentation Diagram
| G.A. | General Arrangement | General Arrangement
| No. | Number | Number
| Qty | Quantity | Quantity
| mm | Millimeter | Millimeter
| SS | Stainless Steel | Stainless Steel
| MOC | Material of Construction | Material of Construction
| DQ | Design Qualification | Design Qualification
| IQ | Installation Qualification | Installation Qualification
| OQ | Operational Qualification | Operational Qualification
| PQ | Performance Qualification | Performance Qualification
| FAT | Factory Acceptance test | Factory Acceptance test
| SAT | Site Acceptance test | Site Acceptance test
| PLC | Programmable Logical Control | Programmable Logical Control
| SCADA | Supervisory control and Data Acquisition | Supervisory control and Data Acquisition
| NA | Not Applicable | Not Applicable
| |
12.0 | Other Requirements | Other Requirements | Other Requirements
| NA | NA | NA

Approvals of the URS:

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Department | Name | Signature | Date

BioBridge Healthcare Solutions Pvt. Ltd., India | BioBridge Healthcare Solutions Pvt. Ltd., India |

BioBridge Healthcare Solutions Pvt. Ltd., India | BioBridge Healthcare Solutions Pvt. Ltd., India

Prepared by: | | |

Checked/Reviewed by: | | |

Client | Client | Client | Client

Reviewed by: | | |

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4.0 | Component details |

5.0 | Construction material |

6.0 | Utility Requirements |

7.0 | Accessories |

8.0 | Documents Required |

9.0 | Drawings |

10.0 | Training |

11.0 | Abbreviations |

12.0 | Other Requirements |

1.0 | Objective | Objective

| The objective of the Vial Filling Machine is to provide an accurate and efficient filling of vaccine vials in the Fill Finish Facility. It is designed to ensure precise dosage, minimizing the risk of contamination, and maintaining the sterility of the vaccine product throughout the filling process. | The objective of the Vial Filling Machine is to provide an accurate and efficient filling of vaccine vials in the Fill Finish Facility. It is designed to ensure precise dosage, minimizing the risk of contamination, and maintaining the sterility of the vaccine product throughout the filling process.

2.0 | Scope | Scope

| This specification is intended to cover the minimum requirements to be met by the vendor in the design, manufacture and supply of vial filling machine used in the Uganda facility | This specification is intended to cover the minimum requirements to be met by the vendor in the design, manufacture and supply of vial filling machine used in the Uganda facility

3.0 | Equipment description | Equipment description

| This specification is intended to cover the minimum requirements to be met by the Vendor in the design, manufacture and supply of Vial filling machine provided in this document are at minimum and can be subjected to amendment after discussion with vendor and representative of Client for achieving higher degree of standards and operational ease

Technical and GMP features

Vial filling machine 6R (180 Vials/ minute Multidose)

The vial filling machine will be used for the filling of liquid vaccines containing adjuvant (preferably Aluminum salts compounds) in the vials of 6R sizes and in volumes ranging from 5 ml to 6 ml. The presterilized tubular glass 6R vials shall be used in the process

Once filled, the preparation will be used parenterally in humans.

Filling Machine should include:

Fully automatic filling and stoppering machine 1 no.

Swivel conveyor 1 no.

Rejection Station 1 no.

Spares

Process description

Vial filling machine shall be used for the filling of vaccine to be used parenterally in humans. This machine shall be placed in class 100 (A) with class 1000 (A/B) as background.

The material to be filled shall be placed in a sterilized vessel and after making the necessary connections, filling shall be carried out.

On line stoppering of the vials will be done for applying the pre sterilized chloro butylated rubber stoppers for injectable products. In case of injectable, these shall be fixed in place by pressing.

In each case, after filling is concluded, the delivery needles will be subjected to a CIP process followed by SIP. | This specification is intended to cover the minimum requirements to be met by the Vendor in the design, manufacture and supply of Vial filling machine provided in this document are at minimum and can be subjected to amendment after discussion with vendor and representative of Client for achieving higher degree of standards and operational ease

Technical and GMP features

Vial filling machine 6R (180 Vials/ minute Multidose)

The vial filling machine will be used for the filling of liquid vaccines containing adjuvant (preferably Aluminum salts compounds) in the vials of 6R sizes and in volumes ranging from 5 ml to 6 ml. The presterilized tubular glass 6R vials shall be used in the process

Once filled, the preparation will be used parenterally in humans.

Filling Machine should include:

Fully automatic filling and stoppering machine 1 no.

Swivel conveyor 1 no.

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Spares

Process description

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connections, filling shall be carried out.

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4.0 | Component details | Component details

| Technological standards

VIAL FILLING MACHINE

Vial filling machine should include maintenance free roller and ball bearings, electrical installations, PLC, graphical touch operating panel, variable speed drive, central pump adjustment facility and designed to be used under laminar air flow.

The equipment should be able to fill and apply stoppers on vials up to a speed of 180 vials per minute for MD 6R vials.

Arrangement for filling of volumes ranging from 5 to 6 ml with high accuracy.

Provision of motor-controlled movement for pump stroke and rotation, needles stoppering turret and main drive.

Possibility of CIP and SIP for the vessel and all interconnections.

Volume based dosing facility

Possibility of having the break tank with stirring/magnetic stirring system for a homogenous solution delivery.

Break tank should be double jacketed and insulated with the provisions to keep the material at the required temperature.

Provisions for automatic insertion turret to apply rubber stoppers for injectable products on 6R vials.

Arrangement of hopper to feed the stoppers into the vibratory bowl.

VFL should have sanitary high-pressure safety devices to ensure product and system integrity, and meet established biosafety.

The closure feeding and sorting unit to be placed on the operator's side in order to be totally and easily accessible.

An on line non-destructive in process control of the filled volume based on weight for fast and easy control on the filled volumes.

Technical description

The treatment to the vials should be very gentle to avoid splashing of the product.

The operations should be very quiet.

Good accessibility to all machine components.

Good synchronization of movements of all working station

Fast and easy size changeover.

Vial filling lines should have a sanitary design that will ensure operation under aseptic conditions in line with the requirements established in the cGMP for the biopharmaceutical industry.

All contact parts must be CIP/SIP compatible.

Interconnections and access ports

The connection within the machine should preferably by flexible connection.

Pipes, fittings, traps, and other accessories for all system lines, within VFM limits (outlets and in- and outflow ports) shall be of sanitary design.

Connection to drainage for CIP/SIP must comply with the sanitary requirements established by cGMP.

Containment

The vial filling machine should be designed so as to prevent the possibility generating and releasing aerosols into the environment without compromising the hygiene or integrity of the medium being filled within the encloser with ease of operation.

CIP/SIP requirements (Optional)

The VFM should allow automatic CIP and SIP.

During the CIP cycle, the following agents may be used:

2% sodium hydroxide solution at 70°C

1% nitric acid solution at 70°C

c. VFM and all process lines should allow for sterilization with saturated steam, by a totally automatic procedure.

Instrumentation and automation

The system should have the necessary automation elements to ensure correct operation, as well as equipment and operator protection. The following minimum requirements should be met:

Indications:

Continuous Filling medium temperature indication

Records

Vials filled record (Cumulative)

Automatic sequences and operation modes

Standby

Filling (normal filling operation)

CIP of all lines and systems

SIP of all lines and systems

Sampling

Filling @ for 180 vials/min for 6R Vials

On line IPQC (statistical) preferably @2%

Interaction with other systems

The VFM control system must be able to communicate with a local PC-based monitoring system. The monitoring system will request the current values of all measured variables and will set the set points of all control points. The PC standard RS-232 is the preferred channel. If any other hardware is required, it must be supplied or fully specified (if it is commercially available)

Alarms and signals

Low vial availability audiovisual signal

Low stopper availability audiovisual signal

Operator actions

Activation/deactivation and modification of parameters for all sequences or operation modes

Establishment of set point, control loop mode (AUTOMATIC/MANUAL) and other parameters of control loops for all controlled variables

Manual activation/deactivation of all actuating elements, such as valves, pumps, etc. To this purpose, all these elements should be activated by the control system (AUT.) or manually by the operator (MAN.).

Other requirements and design characteristics

It should preferably include all the system interconnections.

It should include the required pumps: the pumps needed within the VFM to perform the required functions.

In the Break tank with agitation/stirring, the motor and electrical devices should be designed for working in normal operating conditions. Grade of protection IP 54 should be satisfactory.

All the system must be capable of regular visual inspection and manual cleaning. The solution to this requirement in each case will depend on the size, geometry and placement of the machine and the proposal of the supplier.

Corrosive or dangerous substances in the process streams: those specified for CIP

The design and operation of the system and the degree of finish of its external surfaces must be compatible with the classification of the area in which they are to be installed and the chemicals commonly employed for cleaning and disinfecting in the biopharmaceutical industry.

The electrical system of the equipment shall comply with the requirements established by the following standard of the International Electrotechnical Commission: IEC 204 - Electrical equipment of industrial machines. Part 1. General Requirements (This standard has been adopted and modified by the European Standard EN 60204-1:1992), Part 2. Item designation and examples of drawings, diagrams, tables and instructions, and also with the applicable requirements of the attached document "Requirements for Engineered Equipment Wiring".

SWIVEL TYPE CONVEYER

Should be able to transfer filled vials to the bung processing infeed turntable.

REJECTION STATION

Required for vials without stopper

This should have accumulation control | Technological standards

VIAL FILLING MACHINE

Vial filling machine should include maintenance free roller and ball bearings, electrical installations, PLC, graphical touch operating panel, variable speed drive, central pump adjustment facility and designed to be used under laminar air flow.

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All the system must be capable of regular visual inspection and manual cleaning. The solution to this requirement in each case will depend on the size, geometry and placement of the machine and the proposal of the supplier.
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The electrical system of the equipment shall comply with the requirements established by the following standard of the International Electrotechnical Commission: IEC 204 - Electrical equipment of industrial machines. Part 1. General Requirements (This standard has been adopted and modified by the European Standard EN 60204-1:1992), Part 2. Item designation and examples of drawings, diagrams, tables and instructions, and also with the applicable requirements of the attached document "Requirements for Engineered Equipment Wiring".

SWIVEL TYPE CONVEYER

Should be able to transfer filled vials to the bung processing infeed turntable.

REJECTION STATION

Required for vials without stopper
This should have accumulation control
5.0 | Construction material | Construction material
| All metal components that may come into contact with the product must be made of AISI 316 L steel.
Other components must be made of AISI 304 steel
All non-metal components that may come into contact with the product must be made of adequate material as provided for by FDA regulations (or the equivalent) for the biopharmaceutical industry.
The surface roughness of the metal parts that may come into contact with the product must be Ra 0.3m
All welds that may come into contact with the product must comply with the FDA regulations (or the equivalent) for the biopharmaceutical industry. | All metal components that may come into contact with the product must be made of AISI 316 L steel.
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The surface roughness of the metal parts that may come into contact with the product must be Ra 0.3m
All welds that may come into contact with the product must comply with the FDA regulations (or the equivalent) for the biopharmaceutical industry.
6.0 | Utility Requirements | Utility Requirements
| Pure steam
WFI water
Compressed air.
Chilled water
Power supply(415 volts, three phase,50Hz frequency) | Pure steam
WFI water
Compressed air.
Chilled water
Power supply(415 volts, three phase,50Hz frequency)
7.0 | Accessories | Accessories
| The Vendors should provide as part of the quotation a list proposed spare parts for at least two years operation | The Vendors should provide as part of the quotation a list proposed spare parts for at least two years operation
8.0 | Documents Required | Documents Required
| Pre-Delivery Qualifications (PDQ)
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After the installation of the system at the Vendor's shop, Factory Acceptance Tests (FAT) shall be performed and accurately documented.
FAT tests shall be performed and documented according to a main protocol agreement between the Vendor and the Customer. At least two months before FAT test are scheduled to be performed; the documents where the results of such test will be recorded should be delivered to the Customer for approval.
General test to be performed during FAT:
Surface smoothness tests of metal surfaces.
Visual inspection for compliance with the cGMP, in accordance with the approved drawings.
Leakage Integrity test
CIP performance tests.
Alarms checkouts.
I/O Checks
Functional tests for all control loops.
Documents to be delivered by the Vendor

Two copies of the following documents must be provided, in the English language:

P&ID diagrams.

Operation Instruction Manual(s), including operating panel and safety operation warnings.

Maintenance Instruction Manual(s).

Assembly and Installation Manual(s).

Instrumentation and Calibration Manual(s).

Troubleshooting list.

Electrical and Wiring Manuals (including electrical drawing list, circuit diagrams, connection diagrams, cabinet layout, interconnection diagrams, etc.).

Component list.

Calibration certificate for each instrument installed in the equipment, including the description for re-calibration.

Certificate of materials used for constructing those parts in contact with the product stream(s).

Official approval document and manufacturing inspection protocol.

Welding procedures and visual inspection reports.

Mill test reports for those parts in contact with the product streams, which will be sanitary welded.

Qualification certificates of the technical personnel who have worked in welding and assembling of equipment shall be available on request.

Biological compatibility certificates of all non-metallic materials in contact with process streams (like diaphragms, gaskets, O-rings, seals, couplings, etc.).

Passivation methods used.

Pressure tests reports or equivalent.

Software and hardware validation documents for all computerized systems (microprocessor, PLC and computer-based systems).

Safety. Vendor should deliver to the Customer all legal documents relating to safety issues regulated by the National Official agencies where the equipment is built. The scope of these documents will be clearly defined during the contract negotiation.

Validation

All documents specified should be delivered at the Customer site at least two months before the installation of the equipment, in order to be used for the preparation of the Validation Protocols.

After installation of the equipment at Customer's site, complementary Installation and Operational

Qualification (IQ, OQ) tests will be performed by the Customer and may be supervised (if it is of

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Official approval document and manufacturing inspection protocol.

Welding procedures and visual inspection reports.

Mill test reports for those parts in contact with the product streams, which will be sanitary welded.

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Software and hardware validation documents for all computerized systems (microprocessor, PLC and computer-based systems).

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Qualification (IQ, OQ) tests will be performed by the Customer and may be supervised (if it is of

interest) by a member of Vendor's technical staff.

9.0 | Drawings | Drawings

| P&ID, GA, Welding & electrical drawings | P&ID, GA, Welding & electrical drawings

10.0 | Training | Training

| Training of Customer's operators. The Vendor shall supply all available information for adequate equipment operation. To this end, the Customer's personnel will be trained by the Vendor's technical staff at the work place or at the Vendor's shop. | Training of Customer's operators. The Vendor shall

supply all available information for adequate equipment operation. To this end, the Customer's personnel will be trained by the Vendor's technical staff at the work place or at the Vendor's shop.

11.0 | Abbreviations | Abbreviations

| URS | User Requirement Specification

| P&ID | Piping and Instrumentation Diagram

| G.A. | General Arrangement

| No. | Number

| Qty | Quantity

| mm | Millimeter

| SS | Stainless Steel

| MOC | Material of Construction

| DQ | Design Qualification

| IQ | Installation Qualification

| OQ | Operational Qualification

| PQ | Performance Qualification

| FAT | Factory Acceptance test

| SAT | Site Acceptance test

| NA | Not Applicable

| PLC | Programmable Logical Control

| SCADA | Supervisory control and Data Acquisition

| VFM | Vial Filling Machine

| VFL | Vial Filling Line

12.0 | Other Requirements | Other Requirements

| NA | NA

Approvals of the URS:

TABLE OF CONTENTS

Note: In case of any specific requirement from the client the body of this document can be modified

Department | Name | Signature | Date

BioBridge Healthcare Solutions Pvt. Ltd., India | BioBridge Healthcare Solutions Pvt. Ltd., India |

BioBridge Healthcare Solutions Pvt. Ltd., India | BioBridge Healthcare Solutions Pvt. Ltd., India

Prepared by: | | |

Checked/Reviewed by: | | |

Client | Client | Client | Client

Reviewed by: | | |

Reviewed by: | | |

Approved by: | | |

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2.0 | Scope |

3.0 | Equipment description |

4.0 | Component details |

5.0 | Construction material |

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9.0 | Drawings |

10.0 | Training |

11.0 | Abbreviations |

12.0 | Other Requirements |

1.0 | Objective | Objective

| Vial sticker labelling machine will be used in the fill finish facility for labeling of vaccine vials.

| Vial sticker labelling machine will be used in the fill finish facility for labeling of vaccine vials.

2.0 | Scope | Scope

| This specification is intended to cover the minimum requirements to be met by the vendor in the design, manufacture and supply of labeling machine used in the Uganda facility | This specification is intended to cover the minimum requirements to be met by the vendor in the design, manufacture and supply of labeling machine used in the Uganda facility

3.0 | Equipment description | Equipment description

| This specification is intended to cover the minimum requirements to be met by the Vendor in the design, manufacture and supply of Vial Sticker Labelling Machine provided in this document are at minimum and can be subjected to amendment after discussion with vendor and representative of Client for achieving higher degree of standards and operational ease

Vial Sticker Labelling Machine

Capacity –250 Vial/ minute 6R capacity | This specification is intended to cover the minimum requirements to be met by the Vendor in the design, manufacture and supply of Vial Sticker Labelling Machine provided in this document are at minimum and can be subjected to amendment after discussion with vendor and representative of Client for achieving higher degree of standards and operational ease

Vial Sticker Labelling Machine

Capacity –250 Vial/ minute 6R capacity

4.0 | Component details | Component details

| Should have Output Speed: 250 Products/Minute (Speed depends on label length and products size)

It should be compatible to label 6R vials.

Should have Label Roll Empty - Machine Stop System with Alarm.

Should have Missing Label in Roll - Machine Stop System with Alarm.

Should have Special Label Sensor for sensing Transparent Film Labels which does not have any sensing mark on it.

Should have Option of Thermal Ink Jet Printer with various speed ranges available.

Toughened Glass or Acrylic Safety Cabinet for Machine.

Should have tower Light showing the machine status / operation.

Should have Color or Mono Touch Screen with PLC for machine operation and showing various error messages and information on display.

Should be Special Sensing system for checking missing label on product.

Should have be Low Compressed Air - Machine Stop with Alarm system for pneumatically operated coding system

Should have Vision System for checking Barcode, presence or absence of Coding or pharma code on labels.

Should have Online pneumatically operated product rejection system. | Should have Output Speed: 250

Products/Minute (Speed depends on label length and products size)

It should be compatible to label 6R vials.

Should have Label Roll Empty - Machine Stop System with Alarm.

Should have Missing Label in Roll - Machine Stop System with Alarm.

Should have Special Label Sensor for sensing Transparent Film Labels which does not have any sensing mark on it.

Should have Option of Thermal Ink Jet Printer with various speed ranges available.

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Should have Vision System for checking Barcode, presence or absence of Coding or pharma code on labels.

Should have Online pneumatically operated product rejection system.

5.0 | Construction material | Construction material

| All parts to be designed and manufactured in line with stainless steel 304 sheet to meet GMP norms. |

All parts to be designed and manufactured in line with stainless steel 304 sheet to meet GMP norms.

6.0 | Utility Requirements | Utility Requirements

| 415 v, three phase Power Supply | 415 v, three phase Power Supply

7.0 | Accessories | Accessories

| The Vendors should provide as part of the quotation a list proposed spare parts for at least two years operation | The Vendors should provide as part of the quotation a list proposed spare parts for at least two years operation

8.0 | Documents Required | Documents Required

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Certificate of Materials used for parts

Official approval document and manufacturing inspection protocol.

Welding procedures and visual inspection reports.

Qualification certificates of the technical personnel who has worked in welding and assembling of equipment shall be available on request.

Biological compatibility certificate of all non-metallic materials in contact with process streams (like diaphragms, gaskets, O-rings, seals, couplings, etc.).

Pressure test reports or equivalent.

Software and hardware validation documentation for all automated systems (microprocessor, PLC and computer-based systems).

Safety

The Vendor should provide the Customer with all legal documents related with safety issues regulated by the official agencies of the country where the equipment if built. The scope of these documents will be clearly defined during the negotiation of the contract.

Validation

1. All specified documents should be delivered at the Customer's site at least two months before the installation of the equipment so that they can be used in preparing the Validation Protocols.
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| URS | User Requirement Specification
| P&ID | Piping and Instrumentation Diagram
| G.A. | General Arrangement
| No. | Number
| Qty | Quantity
| mm | Millimeter
| SS | Stainless Steel
| MOC | Material of Construction
| DQ | Design Qualification
| IQ | Installation Qualification
| OQ | Operational Qualification
| PQ | Performance Qualification
| FAT | Factory Acceptance test
| SAT | Site Acceptance test
| NA | Not Applicable
| PLC | Programmable Logical Control
| SCADA | Supervisory control and Data Acquisition
12.0 | Other Requirements | Other Requirements
| NA | NA

1.0 |Objective |
|To lay down the procedure for determining the unitage (Lf/ml) of diphtheria|
|toxin/toxoid by limes flocculation test. |
2.0 |Scope |
|This Standard Operating Procedure is applicable to the diphtheria |
|toxin/toxoid samples. |
3.0 |Responsibility |
3.1	The Analyst shall be responsible for carrying out the procedure as
	established in this SOP.
3.2	Supervisor shall be responsible for verifying that the same has been
	carried out correctly.
3.3	Head QC or his designee shall be responsible for overall compliance
	of the SOP.
4.0	Abbreviations and Definitions
4.1	SOP
4.2	QA
4.3	QC
4.4	DATF
4.5	mL
4.6	°C
4.7	%
4.8	Lf/mL
4.9	w/v
5.0	Procedure:
5.1	Equipment, Materials and Reagents
	5.1.1
	5.1.2
	5.1.3
	5.1.4
	5.1.5
	5.1.6
	5.1.7
	5.1.8
	5.1.9
	5.1.10
	5.1.11
5.2	Preparation of solutions
	5.2.1
5.3	Preparation of sample
	5.3.1
	5.3.2
5.4	Test procedure
	5.4.1
	5.4.2
	5.4.3
	5.4.4

			second flocculation.	
			5.4.4.8	If flocculation occurs in the first tube increase the
			dilution fold or it occurs in the last tube decrease	
			the dilution fold of the sample.	
			5.4.4.9	Observe for first and second flocculation. The tube
			in which first flocculation occurs is the Lf value of	
			the sample. Record the time taken (Kf) for first and	
			second flocculation.	
			5.4.4.10	Multiply this by the dilution factor to get the Lf/ml
			of the original sample. Example:-	
			Tube No	
			5.4.4.12	If the first or the last tube flocculates most rapidly
			then the test has to be repeated in such a way that	
			the middle tube flocculates most rapidly.	
			5.4.4.13	After determining the broad range test perform narrow
			range test/ confirmatory test.	
	5.4.5	Narrow range test (Applicable for purified bulk)		
			For narrow range test keep 2 Lf difference between	
			subsequent tubes and perform the assay as described	
			above.	
	5.4.6	Confirmatory test		
			For confirmation, perform with 2 or 1 Lf difference	
			between the tubes as shown in Table above. Record the	
			results.	
	5.5	Calculation/Formula:		
	Limes floccules (Lf/mL)= DATF unit of tube which flocculate first X			
	dilution fold of sample.			
6.0	Forms and Records			
	6.1	Determination of Unitage (Lf/mL) of Diphtheria Toxin/Toxoid by		
	Flocculation Test.			
7.0	References			
	NA			

1.0	Objective		
	To lay down the procedure for detoxification test on Diphtheria toxin		
	after formalin treatment.		
2.0	Scope		
	This Standard Operating Procedure is applicable to all batches of		
	Diphtheria toxin after detoxification tested.		
3.0	Responsibility		
	3.1	The QC analyst shall be responsible for performing the described	
	procedure correctly.		
	3.2	Supervisor shall be responsible for verifying that the same has been	
	carried out correctly.		
	3.3	Head Animal House/Veterinarian/supervisor shall be responsible for	
	supply of quality animals and their maintenance.		
	3.4	Head QC or his designee shall be responsible for overall compliance	
	of the SOP.		
4.0	Abbreviations and Definitions		
	4.1	SOP	: Standard Operating Procedure
	4.2	QA	: Quality Assurance
	4.3	QC	: Quality Control
	4.4	mL	: milliliter
	4.5	ID	Intra Dermal
5.0	Procedure:		
	5.1	Equipment, Materials and Reagents	
	5.1.1	Weighing balance	
	5.1.2	Biosafety Cabinet	
	5.1.3	Disposable syringes with needle	
	5.1.4	Sterile glass vials/tubes	
	5.1.5	Water Bath (85°C)	
	5.1.6	Micropipette	
	5.2	Sample Preparation:	
	5.2.1	Take two glass vials and mark them as “H” and “UH”	
	respectively.		
	5.2.2	Aliquot 5 mL of the diphtheria toxoid in to these two glass	
	vials. This activity shall be carried out in biosafety		
	cabinet.		
	5.2.3	Set the water bath to 85°C. After temperature is attained,	
	incubate the tube marked as “H” (Heated) for 30 minutes at		
	85°C.		
	5.2.4	After 30 minutes, remove the “H” i.e., “Heated” sample from	
	the water bath and allow it to cool to room temperature. “H”		
	sample acts as a control sample.		
	5.2.5	The “UH” ie., “Unheated” sample is the test sample	
	5.3	Preparation of test animals:	
	5.3.1	Take one healthy Guinea Pigs weighing 250 to 350g.	
	5.3.2	Apply soap solution to the abdominal area of guinea pigs and	
	use wet cotton to produce lather for shaving.		
	5.3.3	Shave the hairs by using razor. Care should be taken to avoid	
	bruises over the skin. Clean the body by using cloth/cotton.		
	5.3.4	Use the animal for inoculation after body gets dried.	
	5.3.5	Mark Guinea pig with appropriate color scheme.	
	5.4	Animal Inoculation:	
	5.4.1	Inject intradermally 0.2 mL of the sample “UH” –“Unheated”	
	sample in the depilated skin of the guinea pig. Encircle the		
	site of inoculation using a permanent marker and mark it as		
	“UH”.		
	5.4.2	Inject intradermally 0.2 mL of the sample “H” – “Heated” near	
	the site of inoculation of unheated sample. Encircle the site		
	of inoculation using a permanent marker and mark it as “H”.		
	5.6	Observations:	
	5.6.1	Observe the injection site after 24 hours. Look for reactions	
	such as redness, inflammation and thickening of skin		
	5.6.2	Again observe injection site after 48 hours and look for any	
	increase or decrease in the reactions as compared to 24 hour		
	observations. Note any necrotic reactions at the injection		
	site.		
	5.6.3	If there is any inflammatory reaction, use a scale to measure	
	the redness and inflammation at the spot of injection.		
	5.7	Validity criteria	
	The test is valid if:		
	5.7.1	The control (Heated) sample shows no specific Diphtheria	
	erythema, necrosis and indurations.		
	5.7.2	If the test is invalid repeat the test.	
	5.8	Acceptance Criteria:	
	5.8.1	The sample complies with the test if no specific Diphtheria	
	erythema, necrosis and indurations are observed in case of		
	unheated samples.		
6.0	Forms and Records		
	Test for Detoxification of Diphtheria Toxin		
7.0	References		
	7.1	Indian Pharmacopoeia.	
	7.2	World Health Organization, Technical Report Series No. 800, 927	

[1.0]Objective: |
 ||To determine the organoleptic characteristics / physical appearance of |
 | active raw material, adjuvanted and formulated bulk of vaccines. |
 [2.0]Scope: |
 ||This standard operating procedure is applicable to the active raw |
 | material, adjuvanted and formulated bulk of vaccines |
 [3.0]Abbreviations and Definitions: |
 ||3.1 |ARM |: Active Raw Material |
 ||3.2 |NA |: Not applicable |
 [4.0]Procedure: |
	4.1	Equipment, Materials and Reagents	
		Not applicable	
	4.2	Organoleptic characteristics / physical appearance testing of Active	
		Raw Material.	
		4.2.1	The sample quantity of ARM required for the test is 10 ml in
			glass vial.
		4.2.2	Clean the work bench thoroughly with non lint shedding cloth
			before examining the sample.
		4.2.3	Clean the glass vial containing the test ARM with a lint
			free sterile cloth and keep on clean working bench.
		4.2.4	Carefully observe the ARM in glass vial for colour,
			turbidity and for the presence of suspended particles.
		4.2.5	The observations are made against white light.
		4.2.6	Check the characteristics of the sample by comparing the
			results with the established in house specifications for
			ARM.
	4.3	Organoleptic characteristics / physical appearance testing of	
	adjuvanted and formulated bulk of vaccine:		
		4.3.1	The sample quantity of vaccine required for the test is 10
			ml in glass vial.
		4.3.2	Clean the work bench thoroughly with non lint shedding cloth
			before examining the sample
		4.3.3	Clean the glass vial containing the test vaccine sample with
			a lint free sterile cloth and keep on clean working bench.
		4.3.4	Carefully observe the vaccine sample in vial for its colour,
			turbidity and for the presence of suspended particles. The
			observations are made with the naked eye. After observing
			the above characteristics, the vial is placed on the
			workbench undisturbed to observe if the gel layer separates
			from the clear liquid.
		4.3.5	Check the characteristics of the sample by comparing the
			results with the established in house specifications.
	4.4	Acceptance Criterion:	
		4.4.1	The ARM passes the organoleptic characteristics / physical
			appearance test, if the observations are as per the
			established specification.
		4.4.2	The sample passes, if the observations are as per the
			established in-house specifications.
	Note: Once the Organoleptic characteristics / Physical appearance of the		
sample have been determined, the same sample vial can be used to perform			
other test(s).			
[5.0]Forms and Records:			
	Fill the appropriate records		
[6.0]Reference:			
	NA		

1.0	Objective		
	To lay the procedure for specific toxicity test on bulk purified diphtheria		
	toxoid.		
2.0	Scope		
	This Standard Operating Procedure is applicable to all batches of bulk		
	purified diphtheria toxoid tested in quality control department.		
3.0	Responsibility		
	3.1	The QC analysts shall be responsible to carry out the described	
	procedure.		
	3.2	The supervisor shall be responsible for ensuring that the same is	
	carried out correctly.		
	3.3	Head Animal House/Veterinarian/supervisor shall be responsible for	
	supply of quality animals and their maintenance.		
	3.4	Head QC or his designee shall be responsible for overall compliance	
	of the SOP.		
4.0	Abbreviations and Definitions		
	4.1	SOP	: Standard Operating Procedure
	4.2	QC	: Quality Control
	4.3	mL	: Millilitre
	4.4	QA	: Quality Assurance
	4.5	AH	: Animal House
	4.6	W/V	: Weight / Volume
5.0	Procedure:		
	5.1	Equipment, Materials and Reagents	
		5.1.1	Weighing balance
		5.1.2	Biosafety cabinet
		5.1.3	Vortex mixer
		5.1.4	Micropipette
		5.1.5	Cotton
		5.1.6	Disposable syringes with needle
		5.1.7	Guinea pig
		5.1.8	Sterilized Glass Vials/ bottles
		5.1.9	Sterile micropipette tips
		5.1.10	70% Isopropyl alcohol
		5.1.11	Sodium chloride
		5.1.12	2- phenoxyethanol
	5.2	Preparation of solutions	
		5.2.1	Physiological saline (0.9% w/v) :
			Weigh 0.9 g of sodium chloride and dissolve in 60 ml of the
			purified water. Make up the volume up to 100 ml with purified
			water.
		5.2.2	50% w/v 2- phenoxyethanol Solution:
			Weigh 12.5 g of 2- phenoxyethanol and dissolve in 10 ml of the
			purified water. Make up the volume up to 25 ml with purified
			water.
	5.3	Diluent Preparation:	
		5.3.1	Prepare required quantity of diluent (Physiological Saline
			with 0.5% w/v 2- phenoxyethanol) by using Physiological saline
			and 50 % w/v 2- phenoxyethanol.
		5.3.2	For Example:
			Take 1.0 mL of 50% w/v 2- phenoxyethanol solution and transfer
			it in to a glass bottle containing 99.0 mL of 0.9 w/v
			Physiological Saline.
	5.4	Sample Preparation:	
		5.4.1	Prepare the biosafety cabinet for aseptic operation.
		5.4.2	Dilute the sample (of known Lf/mL) to contain 500 Lf/mL using
			the diluent Physiological Saline containing 2- phenoxyethanol
			(0.5% w/v)
	5.5	Preparation of test animals and inoculation:	
		5.5.1	Place request for Dunkin Hartley Guinea Pigs (250 – 350g) to
			the Head Animal House. After health status evaluation of
			animals by Veterinary Officer, receive the animals.
		5.5.2	Take 5 healthy Guinea Pigs weighing 250 to 350g, as per the
			set procedure
		5.5.3	Mark Guinea pig with appropriate color scheme.
		5.5.4	Inject 1.0 mL of diluted sample per guinea pig through
			subcutaneous route using disposable syringe and needle.
	5.6	Observation:	
		5.6.1	Observe the guinea pigs daily for symptoms of Diphtheria
			intoxication and weigh the animals at weekly interval for a
			period of 6 weeks (42 days).
		5.6.2	Animals that show symptoms like dullness, loss of appetite,
			loss of weight, sluggish movement, respiratory distress,
			abdominal respiration followed with death of the animal shall
			be examined by autopsy for symptoms of Diphtheria intoxication
			(Red adrenals).
		5.6.3	If death is not due to any of the symptoms above and there is
			no evidence of Red adrenals on autopsy then the death can be
			considered as Non Specific Death.
		5.6.4	Observe the health condition of the animals twice daily and
			record the observations.
	5.6.6	Percentage of survival is calculated by the following formula:	
			% Survival = A / B x 100
			A = Number of animal survive the test period.
			B = Number of animal injected with test sample.
	5.7	Acceptance Criteria:	
		The Sample complies with the test if:	
		5.7.1	No Guinea pig shows symptoms of specific Diphtheria
			intoxication or dies due to Diphtheria within 42 days or loses
			weight at the end of the test.

		5.7.2	At least 80 % of the animals survive the test period.
		5.7.3	If more than one animal dies from nonspecific causes, or loses
		weight repeat the test. If an animal dies or loses weight in	
		the second test the toxoid fails the test.	
6.0	Forms and Records		
	6.1	Test for Specific Toxicity on Bulk purified Diphtheria Toxoid	
7.0	References		
	7.1	Indian Pharmacopoeia.	
	7.2	World Health Organization, Technical Report Series No. 800, 927.	

[1. |Purpose: |
 |To lay down the procedure for quantifying the protein concentration by BCA|
 |method. |
 [2. |Applicable to: |
 |This Standard Operating Procedure is applicable to the samples of Active |
 |raw material of CRM, Active Raw Material of Pneumococcal protein conjugate|
 |vaccine in which protein concentration is to be determined. |
 [3. |Responsibility: |
 |3.1 |The analyst is responsible for executing the described procedure |
 |correctly. |
 |3.2 |Supervisor is responsible for verifying that the procedure is |
 |complied with and the results are checked. HOD shall be responsible |
 |for overall compliance of this SOP. |
 |3.3 |Departmental head shall ensure that the SOP is reviewed every two |
 |years or if required earlier. |
 [4. |Safety considerations: |
4.1	Wear laboratory coat while working in the laboratory.
4.2	Handle the quartz cuvette and glassware carefully.
4.3	Work in a fume hood while handling the acids.
[5.	Equipments:
5.1	UV-Visible Spectrophotometer
5.2	Analytical balance Sensitivity 0.01mg
5.3	Fume hood
5.4	Water bath (100 °C)
5.5	Vortex Mixer
5.6	Refrigerator
[6.	Materials and reagents:
6.1	Materials:
	6.1.1
	6.1.2
	6.1.3
	6.1.4
	6.1.5
	6.1.6
	6.1.7
	6.1.8
	6.1.9
	6.1.10
	6.1.11
6.2	Reagents:
	6.2.1
	6.2.2
	6.2.3
	6.2.4
[7.	Preparation of solutions:
7.1	Preparation of diluted Bovine Serum Albumin (BSA) standards: Each 1
ml ampule of 2.0 mg/ml (stock) albumin standard is used to prepare	
the curve as follows in microcentrifuge tube:	
Vial	Volume of
	Diluent
	A
	B
	C
	D
	E
	F
7.2	Preparation of BCA working Reagent (WR):
7.2.1	Use the following formula to determine total volume of WR
	required:
	sample) = Total volume of WR required
7.2.2	Prepare fresh working reagent by mixing 50 parts of BCA
	reagent A and 1 part of BCA reagent B.
	Note: when reagent B is initially added to reagent A,
	turbidity is observed that quickly disappears upon mixing to
	yield a clear-green Working Reagent.
[8.	Procedure:
8.1.	Pipette 0.1 ml of each standard, purified water (as standard blank),
and sample (diluted in case the concentration falls outside the	
calibration curve) into an appropriately labeled test tube in	
duplicates.	
8.2	Add 2.0 ml of Working reagent to each tube and vortex well.
8.3	Cover the tubes with Aluminium foil and incubate at 60°C in a water
bath for 30 minutes.	
8.4	Cool all the tubes to Room temperature.
8.5	Set the wavelength in spectrophotometer at 562 nm, Zero the
absorbance of standard blank and measure the absorbance of	
calibration curve and finally measure the absorbance of test	
samples.	
8.6	Note that all the readings of the assay tubes should be taken within
30 mins. of completion of the assay and maximum 10 samples should be	
analysed in one single assay.	
8.6	Calculations:
8.6.1	Proceed for calculations starting from the Calibration curve
	prepared The calculations are made according to the equation
	of minimum squares.

			Y = mX + b
			Where: m = slope
			b = intercept
			Y = absorbance
			X = concentration
			(Multiply by dilution factor)
		8.6.2	Use EXCEL programme for the preparation of calibration curve
		and Calculation of the results (Multiply by the dilution	
		factor).	
		8.7	Acceptance criteria:
		8.7.1	The coefficient of variation (CV) between the replicates of
		the standards should be $\leq 10\%$.	
		The Calibration curve is accepted if it is meeting the	
		parameters of Regression of ≥ 0.99 .	
9.	Abbreviations:		
	9.1	CV	: Coefficient of variation
	9.2	BCA	: Bicinchoninic Acid
	9.3	HOD	: Head of Department
	9.4	mg	: Milligram
	9.5	ml	: Millilitre
	9.6	μ l	: Micro litre
	9.7	AN	: Analytical
	9.8	PCV	: Pneumococcal conjugate vaccine
	9.9	RD	: Research and Development
	9.10	QA	: Quality Assurance

1.0 Purpose:

To lay down the procedure for the detection and identification of CRM197 protein by Western Blot method.

2.0 Scope:

This standard testing procedure is applicable to the purified protein samples of CRM197.

3.0 Responsibility:

3.1 Qualified Analysts shall be responsible for the preparation and execution of this STP.

3.2 Group leader shall supervise and check that the STP is followed properly.

3.3 Head, QC and Incharge, QA shall be responsible for the implementation and compliance of this STP.

4.0 Materials and Equipments:

4.1 Methanol

4.2 Glycine

4.3 Tris

4.4 Tween 20

4.5 NaCl

4.6 Skimmed Milk

4.7 Immobilon-P PVDF membrane

4.8 Mini Trans-Blot Electrophoretic Transfer Cell (BIORAD).

4.9 Gel who's SDS-PAGE has been run as per SOP No. BB/QC/CRM/001

4.10 Milli Q water.

4.11 Anti-Diphtheria Toxin Polyclonal Antibody, HRP conjugate (Ray Biotech, USA, Cat. No. DS-PB-00646).

4.12 DAB Tablets (Amresco, Biotechnology Grade, Cat. no. E733).

4.13 Powder free nitrile gloves.

4.14 Sodium dihydrogen orthophosphate, dihydrate, $\text{NaH}_2\text{PO}_4 \cdot 2\text{H}_2\text{O}$

4.15 Disodium hydrogen orthophosphate anhydrous, Na_2HPO_4

4.16 Plastic trough

4.17 Hydrogen peroxide

4.18 Gel Rocker

4.19 Densitometer (BIORAD).

4.20 Powerpack (Biorad, Model-Basic).

4.21 Weighing Balance (Mettler Toledo).

4.22 Single channel adjustable volume micropipette, 2-20 μl , 20-200 μl and 100-1000 μl .

4.23 Falcon tubes 15 ml

5.0 Definitions/ Abbreviations:

5.1 SM : Skimmed Milk

5.2 PBS : Phosphate Buffer Saline

5.3 PVDF : Poly Vinyl Di-fluoride

5.4 DAB : Diaminobenzidine

5.5 CRM : Cross Reacting Material

5.6 HRP : Horseradish peroxidase

5.7 NaCl : Sodium Chloride

5.8 Na_2HPO_4 : Disodium hydrogen Orthophosphate anhydrous

5.9 $\text{NaH}_2\text{PO}_4 \cdot 2\text{H}_2\text{O}$: Sodium dihydrogen orthophosphate dehydrate

5.10 H_2O_2 : Hydrogen Peroxide

6.0 Procedure:

6.1 Preparation of reagents:

6.1.1. Transfer buffer (pH 8.3): Dissolve 3.0285 g Tris base and 11.25 g of Glycine in 500 ml of milli Q water. To this, add 100 ml of methanol and make up the volume to 1000 ml with milli Q water. Store this buffer at 2-8°C. Transfer buffer can be re-used 4-5 times.

6.1.2. 0.05M Phosphate Buffer Saline (PBS), pH 7.2±0.2: Dissolve 8.5 of NaCl, 6.36g of $\text{NaH}_2\text{PO}_4 \cdot 2\text{H}_2\text{O}$ and 2.28g of Na_2HPO_4 in 900 ml milli Q water and make up to 1000 ml after checking pH. Adjust pH to 7.2± 0.2 with HCl.

6.1.3 Phosphate Buffer Saline with Tween 20 (PBST). To 1 litre of PBS 0.05M as prepared above, add 500 μl of tween 20 to it. Mix and can be stored at room temperature for one month.

6.1.4 Blocking buffer (3% skimmed milk in PBST): Dissolve 3gm of SM in PBST and make up the volume to 100ml. Make fresh every time.

6.1.5 Preparation of DAB substrate: Dissolve one DAB tablet in 10ml of milli-Q water. Just before use add 0.1% H_2O_2 . Prepare fresh during blot development and use within one hour.

6.2 Method:

6.2.1 Cut the Immobilon-P PVDF membrane of appropriate size and then immerse it in absolute methanol for 15 seconds and then into milli Q-water for two minutes. Equilibrate the membrane for at least 5 minutes in the transfer buffer.

6.2.2 Dip the sponges in Transfer Buffer.

6.2.3 Remove the gel from SDS-PAGE assembly and immerse it in

transfer buffer for 15 minutes to equilibrate.

6.2.4 Sandwich the gel & PVDF membrane in the gel cassette as follows:

6.2.5 Black cassette cover (Cathode)

Pre-wet sponge

Equilibrated gel

Pre-wet membrane

Pre-wet sponge

Red cassette cover (Anode)

6.2.6 Dip the gel cassette in the transfer buffer in the tank completely and cover the trans blot apparatus with the lid.

6.2.7 Blot the protein using the setting: 1.6V for 1.5 to 2 hours.

6.2.8 Remove the blot from the apparatus.

6.2.9 Immerse the protein transferred PVDF membrane in blocking buffer (i.e. 3% SMPBST on gel rocker for half hour and then keep the blot at 2-8oC for overnight).

6.2.10 Wash the membrane three times for 10 minutes each with PBST and one time with PBS on gel rocker.

6.2.11 Treat the blot with 1:200 dilution of anti-diphtheria toxin polyclonal antibody (50 µl of antibody in 10ml of PBST) for 2 hours on gel rocker.

6.2.12 Wash the membrane thrice for 10 minutes each with PBST and one time with PBS on gel rocker.

6.2.13 Add freshly prepared DAB substrate to the blot.

6.2.14 Observe for the appearance of CRM197 protein band along with two fragments A and B if any. Dry the blot by pressing between the layers of the filter paper.

6.3 Analysis of developed western blot:

6.3.1 Photograph and scan the developed blot in BIORAD densitometer and save for record. Make western blot report mentioning CRM197 protein sample, standard and marker details in Annexure 1.

6.3.2 Look for the presence of the intact CRM197 protein (major band, 59kD), two smaller more lighter bands as fragment A (21kD) and fragment B (37 kD) if any when compared with CRM 197 standard and maker. Detection of these bands also confirms the identity of CRM197 protein.

7.0 References:

7.1 Practical Biochemistry by Wilson & Walker.

7.2 Jian Qiaol, Karim Ghani, Manuel Caruso, Diphtheria toxin mutant CRM 197 is an inhibitor of protein synthesis that induces cellular toxicity. Toxicon 51 (2008) 473-477.

8.0 Annexures:

8.1 Format for "Detection of CRM197 by Western Blot" – Annexure 1.

S. No. | Tests | Acceptance limits

1 | Physical Appearance | Transparent Liquid of light yellow color, without suspended particles

2 | pH | 7.00 ± 0.20

3 | HBsAg (antigen) content*(ELISA) | 0.20 mg/ml

4 | Total Protein content (Lowry) | 0.22 mg/ml

5 | Protein content (OD @ 280 nm) | 0.30 mg/ml

6 | Specific Activity of HBsAg | 90.00%

7 | Antigen Purity (SDS-PAGE) | 95.00 % (Monomeric and dimer band)

8 | Purity by Chromatography | 95.00 % (Area under the Major peak)

9 | Host Cell Protein content | $< 100 \text{ ng} / \text{mg}$ of protein

10 | Host Cell DNA content | $< 100 \text{ pg} / 20 \text{ }\mu\text{g}$ of protein

11 | Carbohydrate Content | $< 10 \text{ }\mu\text{g} / 20 \text{ }\mu\text{g}$ of protein

12 | Cesium Content | $< 1 \text{ }\mu\text{g} / 20 \text{ }\mu\text{g}$ of protein

13 | Lipid Content | $20.00 \text{ }\mu\text{g} / 20 \text{ }\mu\text{g}$ HBsAg

14 | Potassium Thiocyanate Content | $1.00 \text{ }\mu\text{g} / 20 \text{ }\mu\text{g}$ HBsAg

15 | Endotoxin content | 30 EU / 20 μg HBsAg

16 | Sterility | There should be no evidence of microbial growth within 14 days of incubation.

* Test for antigen content also serves as test for identity. | * Test for antigen content also serves as test for identity. | * Test for antigen content also serves as test for identity.

1.0 |Objective |
 |To lay down the procedure for irreversibility test on bulk purified |
 |Diphtheria toxoid. |
 2.0 |Scope |
 |This Standard Operating Procedure is applicable to the batches of bulk |
 |purified Diphtheria toxoids. |
 3.0 |Responsibility |
 |3.1 |The QC analyst shall be responsible for performing the described |
 |procedure correctly. |
 |3.2 |Supervisor shall be responsible for verifying that the same has been |
 |carried out correctly. |
 |3.3 |Head Animal House/Veterinarian/supervisor shall be responsible for |
 |supply of quality animals and their maintenance. |
 |3.4 |Head QC or his designee shall be responsible for overall compliance |
 |of the SOP. |
 4.0 |Abbreviations and Definitions |
4.1	SOP	: Standard Operating Procedure
4.2	QA	: Quality Assurance
4.3	QC	: Quality control
4.4	AH	: Animal House
4.5	mL	: milliliter
5.0	Procedure:	
5.1	Equipment, Materials and Reagents	
5.1.1	Biosafety Cabinet	
5.1.2	Weighing balance	
5.1.3	Micropipette	
5.1.4	Vortex mixer	
5.1.5	Cotton	
5.1.6	Disposable syringes with needle	
5.1.7	Sterilized Glass Vials/ bottles	
5.1.8	Sterile micropipette tips	
5.1.9	70% Isopropyl alcohol	
5.1.10	Sodium chloride	
5.1.11	2- phenoxyethanol	
5.1.12	Incubator at 37 0 + 20 C	
5.1.13	Cold room at 2 – 8 0 C	
5.2	Preparation of solutions	
5.2.1	Physiological saline (0.9% w/v)	
Add 0.9 gm of weighed Sodium Chloride to 50ml of purified		
water, mix well and make up it to 100ml.		
5.2.2	50% w/v 2- phenoxyethanol Solution	
Add 50mg of weighed 2- phenoxyethanol to 50ml of purified		
water, mix well and make up it to 100ml.		
5.3	Diluent Preparation:	
Take 1.0 mL of 2- phenoxyethanol solution and transfer it in to a		
glass bottle containing 99.0 mL of Physiological Saline.		
5.4	Sample preparation and incubation:	
5.4.1	Prepare the biosafety cabinet for aseptic operation.	
5.4.2	Calculate the required quantity of bulk and diluent solution	
for the preparation of 100 mL sample containing 80 Lf/ml from		
its estimated Lf/mL.		
5.4.3	Dilute the sample (of known Lf/mL) to contain 80 Lf/mL using	
the diluent. Prepare 100 mL of diluted test sample. The		
dilution shall be done in two steps ensuring usage of at least		
1 mL of sample in the first step.		
5.4.4	Distribute the diluted sample in to two sterile vials labeled	
as "A" and "B", in two equal parts i.e. 50 mL in each bottle.		
5.4.5	Incubate Bottle "A" having 50 mL diluted sample at 37 + 20 C	
in incubator for 42 days (6 weeks).		
5.4.6	Expose the Bottle "B" having 50 mL of diluted sample at 2-80C	
in cold room for 42 Days (6 weeks).		
5.4.7	Label the each bottles appropriately before incubation.	
5.5	Preparation of test animals:	
5.5.1	Take one healthy Guinea Pigs weighing 250 to 350g.	
5.5.2	Apply soap solution to the abdominal area of guinea pigs and	
use wet cotton to produce lather for shaving.		
5.5.3	Shave the hairs by using razor. Care should be taken to avoid	
bruises over the skin. Clean the body by using cloth/cotton.		
5.5.4	Use the animal for inoculation after body gets dried.	
5.5.5	Mark Guinea pig with appropriate color scheme.	
5.6	Inoculation and observation:	
5.6.1	Inject 0.2 mL of the sample from Bottle "B" incubated at 2-8°C	
in the depilated skin of the flank region through intra-dermal		
route.		
5.6.2	Mark the site of inoculation by encircling using a permanent	
marker and label it as "B".		
5.6.3	Similarly inject 0.2 mL of the sample from Bottle "A"	
incubated at 37 + 20 C and mark the encircled area as "A".		
5.6.4	Inject similarly on the other side of the abdomen in the	
depilated skin of the animal and do the same procedures as		
mentioned above.		
5.6.5	Observe the injection site after 24 hours. Look for reactions	
like redness, inflammation and thickening of skin.		
5.6.6	Again observe injection site after 48 hours and look for any	
increase or decrease in the reactions as compared to 24 hour		
observations. Note any necrotic reactions at the injection		
site.		
5.6.7	If there is any inflammatory reaction, use a scale to measure	
the redness and inflammation at the spot of injection.		
5.6.8	Perform autopsy on dead animal if any during the test and	

		examine for symptoms of diphtheria intoxication (red adrenals)
		and record the observation.
5.7	Validity Criteria:	
	5.7.1	The test is valid if, the control sample (incubated at 2-8°C)
	shows no intradermal reaction i.e., redness, inflammation and	
	erythema.	
5.8	Acceptance criteria	
	5.8.1	The bulk purified Diphtheria toxoid passes the
	irreversibility test if there is no	
	Intradermal reaction i.e., redness, inflammation and	
	thickening of skin is observed in both the control sample and	
	the test sample.	
6.0	Forms and Records	
6.1	Irreversibility test on Bulk Purified Diphtheria Toxoid	
7.0	References	
7.1	Indian Pharmacopoeia, 2007.	
7.2	World Health Organization, Technical Report Series No. 800, 927	

[1.0]Objective: |
|To lay down the procedure for the detection of bacterial endotoxin in |
|samples by Gel clot method. |
[2.0]Scope |
|This Standard Operating Procedure is applicable to all Active Raw Material |
|and finish product samples of Hepatitis B vaccine samples. |
[3.0]Responsibility |
3.1	The analyst shall be responsible for carrying out the described
	procedure correctly.
3.2	The Supervisor of the laboratory shall be responsible for verifying
	that the procedure is complied with and for checking the results.
3.3	The departmental head/designee shall be responsible for overall
	compliance of the SOP.
3.4	Head QA/Designee shall be responsible for final approval and
	subsequent compliance of the SOP.
[4.0]Abbreviations and Definitions	
4.1	ARM
4.2	EU
4.3	LAL
4.4	LRW
4.5	MVD
4.6	OD
4.7	PPC
4.8	DQA
4.9	DQC
[5.0]Procedure:	
5.1	Equipments, Materials and Reagents
	5.1.1
	5.1.2
	5.1.3
	5.1.4
	5.1.5
	5.1.6
	5.1.7
	5.1.8
	5.1.9
	5.1.10
	5.1.11
	5.1.12
	5.1.13
5.2	Preparation of solutions:
	NA
5.3	Test Method:
	Note: Make sure that all the reagents are at room temperature before
	use.
	5.3.1
	5.3.2
	5.3.3

||| |5 |

||| |Note: Each dilution should be vortexed for at least 30 seconds |

||| |prior to proceeding to the next dilution. |

|| |5.3.4 |Preparation of samples: |

|| |5.3.4.|For this, calculate the Maximum Valid Dilution (MVD) of |

|| |1 |the sample in which endotoxin has to be detected. The |

|| |formula for calculating the MVD value is given below: |

|| |where: |

|| | = Endotoxin limit of sample |

|| | = Potency (concentration) of sample in 1 ml |

|| | = Sensitivity of LAL |

|| |Then the sample is diluted to its MVD value using LAL |

|| |Reagent water. |

|| |5.3.4.|Preparation of Product Positive Control (PPC): |

|| |2 |For this, dilute the samples according to MVD/2 value |

|| |with LAL Reagent Water. Then take 0.2 ml of this |

|| |dilution and add 0.2 ml of 4 λ standard endotoxin. |

|| |5.3.4.|Mark the reaction tubes for different dilutions of |

|| |3 |standard endotoxin, all product positive controls, |

|| |samples and negative controls in duplicates. |

|| |5.3.4.|Carefully transfer 0.1 ml of different dilutions of |

|| |4 |standard endotoxin, sample (appropriately diluted to MVD|

|| |value), PPCs and negative control into the appropriately|

|| |marked reaction tubes. |

|| |5.3.4.|Then add 0.1 ml of the reconstituted lysate to each tube|

|| |5 |beginning with the highest concentration of standard |

|| |endotoxin followed by negative control, test samples and|

|| |PPCs. |

|| |5.3.4.|Following the addition of lysate, incubate the tubes at |

|| |6 |37°C for 1 hour in a tube heating block. |

|| |Note: Assay tubes should not be disturbed prior to the |

|| |specified time of 1 hour. |

|| |5.3.4.|After 1 hour of incubation examine each reaction tube |

|| |7 |for gelation. |

|| |5.3.4.|A positive reaction is indicated by a firm gel that |

|| |8 |remains intact momentarily when tube is inverted at |

|| |180°. |

|| |5.3.5 |Results: |

|| |Gel Clot Method: The results are expressed as endpoint (the |

|| |endpoint dilution is the last dilution of endotoxin which still|

|| |yields a positive result). |

|| |Example for reporting results with Lysate of sensitivity 0.06 |

|| |EU/ml |

|| |Rep. |Endotoxin Dilutions (EU/ml)| | PPC |Endpoint|

|| | |LRW |Sample| EU/ml |

|| |

|| |

|| |6.0|References: |

| |Indian pharmacopeia |

1.0 |Objective |
 |To lay down the procedure for determining the Opacity (Cell concentration) |
 |of Bordetella Pertussis cell suspension by comparing with 5th International|
 |Reference Preparation. |
 2.0 |Scope |
 |This Standard Operating Procedure is applicable to all the samples of Cell |
 |bank, In process, ARM and other samples of Bordetella Pertussis used during|
 |analysis wherever opacity has to be determined. |
 3.0 |Responsibility |
 |3.1 |The Analyst shall be responsible for carrying out the procedure as |
 |established in this SOP. |
 |3.2 |Supervisor shall be responsible for verifying that the same has been |
 |carried out correctly. |
 |3.3 |Head QC or his designee shall be responsible for overall compliance |
 |of the SOP. |
 4.0 |Abbreviations and Definitions |
4.1	SOP	: Standard Operating Procedure
4.2	QA	: Quality Assurance
4.3	QC	: Quality control
4.4	I.O.U	: International Opacity Units
4.5	mL	: milliliter
5.0	Procedure:	
5.1	Equipment, Materials and Reagents	
5.1.1	5th International Reference Preparation of Opacity tube (10	
	IOU/ mL) (NIBSC code: 76/522)	
5.1.2	Auto pipettes	
5.1.3	Test tubes	
5.1.4	Test tube rack	
5.1.5	Vortex mixture	
5.1.6	Physiological saline (0.9% w/v)	
5.2	Preparation of solutions	
5.2.1	Physiological saline (0.9% w/v)	
	Add 0.9 gm of weighed Sodium Chloride to 50ml of purified	
	water, mix well and make up it to 100ml.	
5.3	Test procedure	
5.3.1	For fermenter cultures take neat sample and perform the test.(	
	As per Table 1)	
	S. No.	Volume of the
	sample	Physiological
		(mL)
		(mL)
	1	Neat sample
	2	5.0
	3	5.0
	4	5.0
	5	5.0
	6	5.0
	7	5.0
	Opacity (IOU/mL) = Final dilution which matches with the	
	opacity reference tube x 10	
	= 2.4 X 10 = 24 opacity units / mL	
5.3.2	In case of harvest culture and bulk dilute the sample with 10	
	fold difference and compare with the standard opacity tube. (	
	As per Table 2)	
	S.No.	Volume of the
	sample	Physiological
		(mL)
		(mL)
	1	1.0
	2	1.0
	3	1.0
	4	1.0
	5	1.0
	Details of further dilutions of the dilution showing turbidity	
	greater than opacity standard, Dilution fold selected for	
	further dilutions 30 Fold	
	S.No.	Volume of the
	sample	Physiological
		(mL)
		(mL)
	1	1.0
	2	1.0
	3	1.0
	4	1.0
	Opacity (IOU/mL) = Final dilution which matches with the	

		opacity reference	
		tube x 10	
		= 33 X 10 = 330 opacity units / mL	
	5.3.3	Select the highest dilution which is showing more turbidity	
		than the standard tube and perform further dilutions to obtain	
		the exact opacity units.	
	5.3.4	Compare the above dilutions with 5th International reference	
		standard opacity tube against a dark background or a written	
		page.	
	5.3.5	Compare the dilutions with standard tube until the opacity of	
		the sample matches and consider that particular dilution fold	
		multiplied by 10 IOU of the opacity standard tube as the	
		opacity of the sample.	
	5.3.6	Even though the final opacity of the sample is assigned the	
		successive dilution should be done and compare with the opacity	
		standard tube in order to prove the dilution is less turbid	
		than standard opacity tube and hence the previous dilution made	
		can be taken as the final dilution.	
	5.3.7	Record the results.	
	5.3.8	Calculation/ Formula:	
		5.3.8.1	Opacity (IOU/mL) = Final dilution which matches with
		the opacity	
		reference tube x 10	
		Note: 10 is the Opacity units of the International	
		reference Standard Opacity tube	
		IOU = International Opacity Units, 1 IOU = 1 x 10⁹	
		organisms/mL	
	5.3.8.2	For example:	
		For harvest sample at 30 fold the sample is matching	
		with the standard opacity tube the opacity of the	
		sample is considered as the following.	
		Opacity of the harvest sample = 30 x 10 = 300 IOU/ mL	
6.0	Forms and Records		
Determination of Opacity Units for B. pertussis Bacterial Suspension			
7.0	References		
WHO manual for the production and control of vaccines – Pertussis vaccine			
(BLG/UNDP/77.2 Rev 1)			

* PS: Polysaccharide

S. No | Test/Assay | Specifications

1 | Appearance | White Powder or Lump

2 | Identity | Should react with specific antibodies

3 | Sialic Acid Content | ≥ 56 % of dry weight of MenY Ps

4 | O acetyl Content | ≥ 0.3 mmol/gm of dry weight of MenY Ps

5 | KD Molecular Size Distribution | ≥ 80 % KD= 0.5

6 | Moisture content | To report results

7 | Protein Content | ≤ 5.0 % of dry weight of MenY Ps

8 | Nucleic acid content | ≤ 2.0 % of dry weight of MenY Ps

9 | Cell Wall polysaccharide content | ≤ 1.0 % of dry weight of MenY Ps

10 | Endotoxin content | < 25 IU/ μ g of MenY Ps

11 | Residual CTAB Content | ≤ 1.0 % of dry weight of MenY Ps

12 | Residual Formaldehyde | ≤ 1.0 % of dry weight of MenY Ps

13 | Residual DOC content | ≤ 1.0 % of dry weight of MenY Ps

- Specification of bulk Diphtheria Toxoid (BDS)
- S/No | Test/Assay | Acceptance Criteria
- 1 | Physical appearance | Clear, Yellowish brown solution without suspended particles
- 2 | Identity | To comply the test
- 3 | Antigen Content (Lf/ml) | Not less than 1000 Lf/ml
- 4 | Antigenic purity | Not less than 1500 Lf/ mg of protein nitrogen
- 5 | Sterility | To comply the test
- 6 | Specific Toxicity | To comply the test
- 7 | Irreversibility of toxoid | To comply the test
- 8 | Free formaldehyde | Not more than 0.2g/ L
- 9 | pH | 6.0 - 7.0

S. No | Test/Assay | Acceptance Criteria

1 | Physical appearance | White or almost white translucent suspension, settles into a clear liquid and gel phase on sedimentation.

2 | pH | 6.40 – 7.20

3 | In-vivo potency test of Hepatitis B surface antigen (In vivo potency and ELISA) | Upper confidence limit of relative potency should be > 1.0

4 | In-vitro potency* test of Hepatitis B surface antigen (ELISA method) | Upper confidence limit of relative potency should be > 1.0

5 | Adsorbed Antigen content | % adsorption of HBsAg > 80

6 | Aluminum content | Not more than 1.25mg/Single human dose

7 | Thiomersal Content (if applicable) | ≤ 0.1 mg/ml

8 | Endotoxin Content | ≤ 5 EU/Single human dose

9 | Abnormal toxicity | No signs of necrosis, illness or death should be observed in any of the animals during the 7 days incubation period

10 | Sterility | There should be no evidence of microbial growth within 14 days of incubation.

* Test for In vitro potency also serves as test for identity. | * Test for In vitro potency also serves as test for identity. | * Test for In vitro potency also serves as test for identity.

Date : 05-May-2016

CERTIFICATE OF ANALYSIS

Product: Bulk of Diphtheria Toxoid

Product code no.: B/DT 003

Appearance: Clear, Light yellowish brown solution without suspended particles

Form: Liquid

Buffer: Physiological saline

Manufacturing Date: April 2015

Expiry Date: Mar 2020

Antigen Content: 2000 Lf/ml

Antigenic Purity: 1800 Lf/mg of Protein Nitrogen

Sterility: Sterile Product

pH: 6.7

Specific Toxicity: Complied

Identity: Positive

Free formaldehyde: 0.06g/ L

Irreversibility of toxoid: Complied (No irreversibility observed)

Storage: 5±3°C (Do not freeze)

Note: To be used for research or in-vitro testing purpose only

For BioBridge Healthcare Solutions Pvt. Ltd.

Authorized Signatory

Address: SCO - 32, Sector - 33D, Chandigarh - 160020, India || Phone : +91-1724562168

Head Office: 13, Rachna Blossom, Jagadish Nagar, Aundh, Pune – 411007, (Mah), INDIA || Phone :

+91-9004349016, +91-9004349041

|| Website : www.biobridge.in

S. No | Test/Assay | Acceptance Criteria

1 | Physical appearance | Clear, Yellow to brownish solution without any suspended particles

2 | Identity | Positive

3 | Antigen Content (Lf/ml) | Not less than 1500 Lf/ml

4 | Antigenic purity | Not less than 1000 Lf/ mg of protein nitrogen

5 | Sterility | Sterile

6 | Specific Toxicity | To comply the test

7 | Irreversibility of toxoid | To comply the test

8 | Free formaldehyde | Not more than 0.2g/ L

9 | pH | 6.0 - 7.0

- Composition for 0.5 ml
- S.No | Tests/Assay | Acceptance Criteria
- 1 | Physical appearance | Whitish turbid, uniform suspension of Tetanus toxoid adsorbed on mineral carrier, free from extraneous particles/floccules. Settles into a clear liquid and gel phase on sedimentation.
- 2 | Identity | Positive
- 3 | Tetanus Toxoid | 7.5 Lf
- 4 | Adsorbed on Aluminum Phosphate (AlPO4) | ≥ 1.5 mg
- 5 | Sterility | Sterile
- 6 | Specific Toxicity | To comply the test
- 7 | Irreversibility of toxoid | To comply the test
- 8 | Free formaldehyde | Not more than 0.2g/ L
- 9 | pH | 6.4 - 7.2

1.0	Objective		
	To lay down the test for purity of Bordetella Pertussis cell bank by plate		
	streak method.		
2.0	Scope		
	This Standard Operating Procedure is applicable to Bordetella pertussis at		
	Cell bank,		
3.0	Responsibility		
	3.1	The Analyst shall be responsible for carrying out the procedure as	
	established in this SOP.		
	3.2	Supervisor shall be responsible for verifying that the same has been	
	carried out correctly.		
	3.3	Head QC or his designee shall be responsible for overall compliance	
	of the SOP.		
4.0	Abbreviations and Definitions		
	4.1	SOP	: Standard Operating Procedure
	4.2	QA	: Quality Assurance
	4.3	QC	: Quality control
	4.4	I.O.U	: International Opacity Units per Milliliter
	4.5	mL	: milliliter
	4.6	µL	: microliter
5.0	Procedure:		
	5.1	Equipment, Materials and Reagents	
		5.1.1	Micropipette
		5.1.2	Balance
		5.1.3	pH Meter
		5.1.4	Autoclave
		5.1.5	Biosafety Cabinet
		5.1.6	Incubator
		5.1.7	Bordet gangue agar plates
		5.1.8	Nutrient agar plates
	5.2	Preparation of solutions	
		5.2.1	Physiological saline (0.9% w/v)
			Add 0.9 gm of weighed Sodium Chloride to 50ml of purified
			water, mix well and make up it to 100ml.
	5.3	Test procedure	
		5.3.1	To check the purity of the cell bank, dilute the lyo cell bank
			in normal saline.
		5.3.2	Inoculate the Nutrient agar and Bordet- Gengou agar plates.
		5.3.3	Incubate the Nutrient agar and B.G agar plates at 35±1°C for
			48-72 hrs.
		5.3.4	Observe the growth and characteristic colonies.
	5.4	Validity criteria	
		5.4.1	Bordetella pertussis shows small colonies like mercury droplets
			and glistening on bordet gangue agar.and no growth should be
			appear on nutrient agar.
6.0	Forms and Records		
	Report for purity check of cell banks for B.Pertussis		
7.0	References		
	WHO manual for the production and control of vaccines – Pertussis Vaccine		
	(BLG/UNDP/77.2 Rev 1)		

1.0 |Objective: |
|To lay down the procedure for the Identification of albumin by SDS-PAGE. |
2.0 |Scope: |
|This standard testing procedure is applicable to the Identification of |
|albumin by SDS-PAGE. |
3.0 |Responsibility |
3.1	The QC analyst shall be responsible for executing the described
	procedure correctly.
3.2	The QC supervisor shall be responsible for verifying that the
	procedure is complied with and the results are checked.
3.3	The QC HOD shall be responsible for overall compliance of this SOP.
4.0	Abbreviations and Definitions:
4.1	Abbreviations:
	4.1.1
	4.1.2
	4.1.3
	4.1.4
	4.1.5
	4.1.6
	4.1.7
	4.1.8
	4.1.9
	4.1.1
	0
	4.1.1
	1
	4.1.1
	2
	4.1.1
	3
	4.1.1
	4
	4.1.1
	5
	4.1.1
	6
	4.1.1
	7
	4.1.1
	8
	4.1.1
	9
	4.1.2
	0
	4.1.2
	1
	4.1.2
	2
	4.1.2
	3
	4.1.2
	4
	4.1.2
	5
	4.1.2
	6
	4.1.2
	7
	4.1.2
	8
	4.1.2
	9
5.0	Procedure:
5.1	Preparation of solutions:
	5.1.1
	5.1.2
	5.1.3
	5.1.4
	5.1.5

	5.1.6	Gel running buffer (10X)
		Weigh 30.2 g Tris, 144 g Glycine and 10 g SDS & dissolve in
		800ml of Milli-Q water. Adjust the pH to 8.3 with HCl & make
		up the volume to 1000 ml.
	5.1.7	Sample Solubilising Buffer (6X)
		Mix 2.4 ml of 0.5M Tris (pH 6.8), 3 ml of Glycerol, 3 ml of
		20% SDS. 6 mg of Bromophenol blue. 1.6 ml of
		β-mercaptoethanol and make up the volume to 10 ml. Make
		aliquots of 1ml each and store at -20°C. For running samples
		in non reducing condition, do not add β-mercaptoethanol in
		sample solubilizing buffer.
	5.1.8	Coomassie Brilliant Blue stain:
		Weigh 1 g of Coomassie Brilliant Blue R250* and dissolve in
		100 ml of methanol, add 20 ml of glacial acetic acid and make
		up the volume to 200 ml with Milli-Q water Keep on magnetic
		stirrer for at least half an hour for proper dissolution.
	5.1.9	Destain Solution
		Mix 40 ml of methanol, 10 ml of glacial acetic acid and make
		up the volume to 100 ml with Milli-Q water. Prepare fresh at
		the time of destaining the gel.
	5.2	Casting of gel
	5.2.1	Clean the glass plates with detergent followed by washing
		with Milli-Q water and fix them in the casting frame.
	5.2.2	Prepare 12.5% Separating Gel mixture (7.5 ml) in a falcon
		tube consisting of following reagents:
		30% Acryl amide-3.125 ml
		1.5 M Tris (pH 8.8) — 1.875 ml
		20% SDS — 0.15 ml
		Milli-Q water — 2.275 ml
		10% APS- 75 µl
		TEMED- 5 µl
		Mix the contents gently by inverting the tube two-three times
	5.2.3	Immediately pour 12.5% Separating Gel mixture between the two
		glass plates leaving 4-5 cm space above for Stacking Gel and
		overlay it carefully with Milli-Q water.
	5.2.4	Prepare the Stacking Gel mixture (3.75 ml) in a tube
		consisting of following reagents:
		30% Acrylamide- 0.625 ml
		0.5 M Tris (pH 6.8) - 0.9375 ml
		20% SDS-0.075 ml
		Milli-Q water —2.0 ml
		10% APS-50 µl
		TEMED – 5 µl
	5.2.5	Pour the stacking gel mixture over the surface of the
		polymerized separating gel and immediately insert the 10-well
		comb of 0.75mm.
	5.2.6	Allow the gel to polymerize for 10 minutes.
	5.2.7	Dilute the 10X gel running buffer to 1 X by mixing 100 ml of
		the 10X stock solution in 900 ml of Milli-Q water.
	5.2.8	Remove the comb from the gel and fix the glass plate
		containing polymerized gel set on the gel cassette sandwich
		of the electrophoresis assembly.
	5.2.9	Add 1 X gel running buffer between and around the buffer tank
		of electrophoresis assembly.
	5.3	Standard and Sample Preparation :
	5.3.1	Take 0.05 mL of 10 mg/mL Reference solution/sample and
		diluted with 0.45 mL of purified water to obtain a
		concentration of 1 mg/mL.
	5.3.2	Take 5 µL of 1 mg/mL Reference solution/sample and added 8 µL
		of sample buffer (- β mercaptoethanol) and 27 µL of purified
		water to obtain a concentration of 5 µg.
	5.3.3	Loaded 20 µL of 5 µg Reference solution to obtain a
		concentration of 2.5 µg.
	5.4	Running SDS-PAGE gel:
		Connect the gel apparatus (the negative and positive electrodes)
		with the power pack and set the power pack at 300V for 45 min or
		until the dye front reaches at the bottom of the separating gel.
		Stop the power pack once dye front reaches at the bottom of the
		separating gel.
	5.5	Staining and de-staining of the gel:
	5.5.1	Once the gel run is over, remove the gel from the plate and
		place the gel in staining solution in a plastic tray for 20
		minutes with gentle agitation on gel rocker.
	5.5.2	De-stain the gel with de-staining solution to remove the
		extra stain till the gel background becomes clear and all the
		bands become clearly visible.
	5.6	Analysis of SDS-PAGE gel :
	5.6.1	After destaining, scan the gel in using densitometer and
		save for record
	6.	References:
	6.1	Bio-Rad Mini-Protean® 3 Cell Instruction Manual.
	6.2	Indian pharmacopeia

* Ps: Polysaccharide

S. No | Test/Assay | Specifications

1 | Physical appearance | Slightly turbid and yellowish liquid

2 | pH | 6.0 to 8.0

3 | Identity test | Serotype should be identified by reaction with type specific antibodies.

4 | Total Polysaccharide content

(Chemical Method) | Report Results

5 | Free polysaccharide content | $\leq 25\%$ of total polysaccharide

6 | Total Protein content | Report Results

7 | Free Carrier Protein content | $< 5\%$ of total protein content

8 | Polysaccharide to Protein Ratio | 0.30 to 3.00

9 | KD (Molecular size distribution) | $> 80\%$, KD = 0.50

10 | Endotoxin content | < 25 IU/ μ g of polysaccharide

11 | EDAC content | ≤ 10 ng / μ g of Polysaccharide

12 | Triethylamine content | ≤ 10 ng / μ g of Polysaccharide

13 | CDAP content | ≤ 20 ng / μ g of Polysaccharide

14 | Sterility | No growth should be observed during the 14 days incubation period

- S. No | Test/Assay | Acceptance Criteria
- 1 | Physical appearance | Clear, Yellow to dark brownish solution without any suspended particles
- 2 | Identity | Positive
- 3 | Antigen Content (Lf/ml) | Report Result
- 4 | Antigenic purity | Not less than 1500 Lf/ mg of protein nitrogen
- 5 | Total Protein content (Lowry) | Report Result
- 6 | Sterility | Sterile
- 7 | pH | 6.0 - 7.0

[1.0]Objective: |
| |To lay down the procedure for standardizing the diphtheria antitoxin for |
| |flocculation test. |
[2.0]Scope: |
| |This procedure is applicable to the diphtheria antitoxin concentrates which |
| |are to be standardized at 100 Lf/ml for flocculation test. |
[3.0]Responsibility: |
	3.1	The QC analyst shall be responsible for performing the described
	procedure correctly.	
	3.2	Supervisor shall be responsible for verifying that the same has been
	carried out correctly.	
	3.3	Head Animal House/Veterinarian/supervisor shall be responsible for
	supply of quality animals and their maintenance.	
	3.4	Head QC or his designee shall be responsible for overall compliance of
	the SOP.	
[4.0]Abbreviations and Definitions		
	4.1	DATF
	4.2	°C
	4.3	%
	4.4	IU/ml
	4.5	QA
	4.6	QC
	4.7	SOP
	4.8	NIBSC
[5.0]Procedures:		
	5.1	Equipment, Materials and Reagents
	5.1.1	Water bath with light source and temperature control for
	flocculation	
	5.1.2	Diphtheria antitoxin concentrate (candidate diphtheria
	antitoxin)	
	5.1.3	Test tubes
	5.1.4	Auto pipettes
	5.1.5	Flocculation tubes
	5.1.6	Flocculation tube rack
	5.1.7	Thermometer
	5.1.8	Stop watch
	5.1.9	Convex lens
	5.1.10	Test tube rack
	5.1.11	Standard Diphtheria toxoid: (NIBSC code: 07/216)
	5.1.12	Standard of Diphtheria Antitoxin for Flocculation (NIBSC code:
	63/007)	
	5.1.13	Sodium chloride
	5.2	Preparation of solutions:
	Weigh 0.9 g of sodium chloride and dissolve in 60 ml of the purified	
	water. Make up the volume up to 100 ml with purified water.	
	5.3	Dilute the diphtheria antitoxin concentrate of a given Lf value in
	normal saline to 100 Lf/ml (candidate diphtheria antitoxin).	
	5.4	Dilute standard diphtheria toxoid in normal saline to 50 Lf /ml.
	5.5	Determine the unitage of standard toxoid using the candidate diphtheria
	antitoxin at 100 Lf/ml by flocculation test.	
	5.6	If the candidate diphtheria antitoxin gives the unitage of standard
	toxoid more than 50 Lf/ml then make small additions of candidate	
	diphtheria antitoxin till it gives the unitage of standard toxoid 50	
	Lf/ml.	
	5.7	If the candidate antitoxin gives the unitage of standard toxoid less
	than 50 Lf/ml then make small additions of normal saline till it gives	
	the unitage of standard toxoid 50 Lf/ml.	
	5.8	Candidate diphtheria antitoxin when gives the unitage of standard
	toxoid as 50 Lf/ml is considered to be standardized for use as working	
	standard of diphtheria antitoxin for flocculation test and assigned the	
	value of 100 Lf/ml.	
[6.	Forms and Records	
	6.1	Form for standardization of diphtheria antitoxin for flocculation test
	(working standard)	
[7.	References	
	Indian Pharmacopoeia.	

Date : 05-May-2016
CERTIFICATE OF ANALYSIS
For BioBridge Healthcare Solutions Pvt. Ltd.
Authorized Signatory
Product: | Bulk of Diphtheria Toxoid
Product code no.: | B/DT 003
Appearance: | Clear, Light yellowish brown solution without suspended particles
Form: | Liquid
Buffer: | Physiological saline
Manufacturing Date: | April 2015
Expiry Date: | Mar 2020
Antigen Content: | 2000 Lf/ml
Antigenic Purity: | 1800 Lf/mg of Protein Nitrogen
Sterility: | Sterile Product
pH: | 6.7
Specific Toxicity: | Complied
Identity: | Positive
Free formaldehyde: | 0.06g/ L
Irreversibility of toxoid: | Complied (No irreversibility observed)
Storage: | 5±3°C (Do not freeze)
Note: To be used for research or in-vitro testing purpose only | Note: To be used for research or in-vitro testing purpose only

- * PS: Polysaccharide
- S. No | Test/Assay | Specifications
- 1 | Appearance | White Powder or Lump
- 2 | Identity | Should react with specific antibodies
- 3 | Sialic Acid Content | ≥ 80 % of dry weight of MenC Ps
- 4 | O acetyl Content | ≥ 1.5 mmol/gm of dry weight of MenC Ps
- 5 | KD Molecular Size Distribution | $\geq 75\%$ KD= 0.5
- 6 | Moisture content | To report results
- 7 | Protein Content | ≤ 1.0 % of dry weight of MenC Ps
- 8 | Nucleic acid content | ≤ 1.0 % of dry weight of MenC Ps
- 9 | Cell Wall polysaccharide content | ≤ 1.0 % of dry weight of MenC Ps
- 10 | Endotoxin content | < 25 IU/ μ g of MenC Ps
- 11 | Residual CTAB Content | ≤ 1.0 % of dry weight of MenC Ps
- 12 | Residual Formaldehyde | ≤ 1.0 % of dry weight of MenC Ps
- 13 | Residual DOC content | ≤ 1.0 % of dry weight of MenC Ps

[1.0]Objective: |
 |To lay down the procedure for the determination of free carrier protein|
 |in PRP-TT conjugate (bulk conjugate) of Haemophilus Influenzae type b |
 |conjugate vaccine. |
 [2.0]Scope: |
 |This Standard Operating Procedure is applicable for the determination |
 |of free carrier protein present in bulk conjugate of Haemophilus |
 |Influenzae type b conjugate vaccine. |
 [3.0]Abbreviations and Definitions: |
3.1	cm	: centimeter
3.2	g	: gram
3.3	HPLC	: High Performance Liquid Chromatography
3.4	id	: internal diameter
3.5	l	: length
3.6	L	: Litre
3.7	M	: Molar
3.8	mAu	: milli Absorbance unit
3.9	mg	: milligram
3.10	min	: minute
3.11	mL	: millilitre
3.12	mm	: millimeter
3.13	nm	: nanometer
3.14	No.	: Number
3.15	rpm	: rotation per minute
3.16	RSD	: Relative Standard Deviation
3.17	μL	: microliter
3.18	μm	: micrometer
3.19	μg	: microgram
3.20	SST	: System Suitability Test
3.21	System Suitability test : System suitability tests are an integral	
part of gas and liquid chromatographic methods. They are used to		
verify that the resolution and reproducibility of the		
chromatographic system are adequate for the analysis to be done.		
The tests are based on the concept that the equipment,		
electronics, analytical operations and samples to be analyzed		
constitute an integral system that can be evaluated as such.		
[4.0]Procedure		
4.1	Precautions to be taken:	
4.1.1	Always wear laboratory coat.	
4.1.2	Always use filtered solutions. Filtration of solution should	
	be done in order to eliminate the entry of possible	
	particles, if any, into the system as well as the column.	
4.1.3	Degas all solutions.	
4.1.4	Avoid entry of bubbles into the system as well as column.	
4.1.5	All the solution bottle openings should be well covered with	
	aluminum foil.	
4.1.6	Check the calibration status of all equipment and	
	instrument.	
4.2	Requirement:	
4.2.1	Equipment:	
	4.2.1.1	Calibrated HPLC System with ChemStation software.
	4.2.1.2	Ultrasonic Cleaner
	4.2.1.3	Vacuum Pump
4.2.2	Instrument:	
	4.2.2.1	pH Meter
	4.2.2.2	Magnetic stirrer
	4.2.2.3	Precision balance
	4.2.2.4	Analytical balance
4.2.3	Materials:	
	4.2.3.1	TSK gel G3000SWXL column {30cm(l), 7.8mm(i.d)}
	4.2.3.2	Laboratory Glassware
	4.2.3.3	Screw cap vials
	4.2.3.4	Solvac Filter Holder
	4.2.3.5	Nylon 6,6 membrane 0.20/0.22/0.45 μm membrane
	filter	
	4.2.3.6	Auto pipettes (10-100μL), (20-200(L), (100-1000(L)
	4.2.3.7	Tips for the Auto pipettes
4.2.4	Reagents:	
	4.2.4.1	Sodium Chloride (NaCl)
	4.2.4.2	Disodium hydrogen phosphate anhydrous (Na₂HPO₄)
	4.2.4.3	Sodium dihydrogen phosphate monohydrate
	4.2.4.4	Purified water
4.3	Preparation of solutions:	
	Prepare all the solutions as described below and Carry out the	
	labelling of the prepared solutions and assign the lot numbers.	
4.3.1	Isopropyl alcohol 70%	
	Take 700 mL of 2-propanol in a volumetric flask and make up	
	the volume to 1000 mL with Purified Water. This solution can	
	be stored at room temperature for 3 months.	
4.3.2	Solution I: [Na₂HPO₄ (0.1M), NaCl (0.1M)] For a volume of	
	1000 mL, weigh 14.19 g of Na₂HPO₄and 5.84 g of NaCl and	
	Dissolve in 700mL of purified water in a volumetric flask	
	and make up the volume to 1000mL. Prepare fresh before use.	
4.3.3	Solution II: (NaH₂PO₄.H₂O, 0.1 M) : For a volume of 500 mL,	
	weigh 6.85g of NaH₂PO₄.H₂O and dissolve in 400mL of	
	purified water in a volumetric flask and make up the volume	
	to 500mL. Prepare fresh before use.	
4.3.4	0.1M Sodium Phosphate buffer: (0.1 M Na₂HPO₄,0.1M NaCl	
	+0.1M NaH₂PO₄.H₂O, pH 7.0)	
	For a volume of 1000mL, take approximately 600mL of	

			Solution I and 400mL of Solution II. The pH of the solution
			should be 7.0 ± 0.2. Use the mobile phase within four days
			from the date of preparation if it remains clear and no
			turbidity or fibers observed in it.
		4.3.5	Column Preservation Solution: Prepare and use it.
		4.3.6	1 mg/mL of p-Aminobenzoic Acid Stock: For a Volume of 5 mL,
			weigh: 5 mg of p-Aminobenzoic acid (4-Aminobenzoic acid)
			and Dissolve in 5mL of purified water and make up the
			volume to 5mL.
		4.3.9	Working standard of p-Aminobenzoic Acid :Dilute 1 mg/mL of
			the stock solution to 1/10 with purified water.
			Note: All the solutions prepared for use are vacuum
			filtered using Nylon 6, 6 membrane 0.20 µm or 0.22 µm ,
			depending upon the nature of solution to be filtered. For
			organic solvents 0.45µm filters should be used (filtration
			not necessary if the solution is prepared in 0.2 µ filtered
			purified water).
		4.4	Preparing the HPLC System for run, Agilent Series:
		4.5	Run Conditions:
		4.5.1	Load method HIBTT FREE PROTEIN.M.
		4.5.2	Flow Rate: 0.75 mL/min
		4.5.3	Detection Wavelength: 280nm
		4.5.4	Run time: 20 minutes
		4.6	Equilibration of the system and the column:
			Equilibrate the column with mobile 0.1 M sodium phosphate buffer
			at 280nm until a stable baseline is observed on the online plot
			window.
		4.7	Sample Preparation and Application:
		4.7.1	Test sample and standards are loaded in screw cap vials and
			placed against the appropriate no. on the sample tray by
			clearly marking the screw cap vial.
		4.7.2	Load the required volume of sample as per the PRP
			concentration equivalent to 2.5 µg in the sample vial.
			Example: If a sample has a PRP concentration of 0.5 mg/mL,
			to calculate volume required for 2.5 µg of protein,
			0.5 mg (500 µg) ----1 mL
			2.5 µg-----X mL
			X mL = 2.5 µg x 1 mL
			500 µg
			X mL = 5 µL
			Therefore, 5 µL of volume is taken for sample having PRP
			concentration of 0.5 mg/mL.
		4.8	Running of sequence:
		4.8.1	Load sequence HIBTT FREE PROTEIN.S
		4.8.2	Open the sequence parameter option by clicking Sequence
			menu.
		4.8.3	Enter information about the data directory and operator
			name.
		4.8.4	In the shutdown option, click on the post-sequence option
			and select LAMPOFF/STANDBY.
		4.8.5	In sequence table option, specify the sample names, methods
			and vials to run in the sequence as per the following
			table:

S	No	Sample Code	Sample details	No. of	Injection
1	1	Blank	Eluent Blank	1	10
2	2	p-Aminobenzoic	System	6	1
3	3	Blank	Eluent Blank	1	10
4	4	Test Sample(s)	Sample	1	2.5µg of
					PRP

		5.8.6	Click on run sequence option in run control menu to start	
			the sequence.	
		5.8.7	Each sample run will be of 20 min. and after running all	
			the samples the sequence will automatically stop.	
		5.8.8	Inject the eluent blank after every 20 injections.	
			One extra vial with same SST sample can be used if 6	
			injections are not possible from single vial.	
		4.9	Calculations :	
			Area under curve for the standards/sample is analyzed by	
			Chemstation software.	
		4.9.1	Open the acquired chromatogram in the data analysis view in	
			the offline copy of the software.	
		4.9.2	Optimize integration with initial events in the signal	
			specific events table.	
		4.9.3	Enter the value of retention time of peak maxima in	
			integration with timed events option.	
		4.9.4	The percentage of free carrier protein in sample is	
			calculated by taking the percentage area of the peak maxima	
			corresponding to the lower molecular weight compared to the	
			peak maxima of higher molecular weight.	
		4.9.5	Calculations for System Suitability Test:	
			4.9.5.	Note the values of area under the curve and
			1	retention time for p-Aminobenzoic acid. (SST
			standard)	
			4.9.5.	Every time a sample is run with system suitability
			2	test, calculate the results for system suitability
			by creating a copy of Master EXCEL in the desired	
			folder and record the same.	
			4.9.5.	Attach the hard print of SST calculation along with
			3	chromatograms and the record form.

- | |4.10 |Acceptance Criteria : |
- | |4.10.1 |For System Suitability Test : |
- | | |4.10.1. |RSD calculated for Retention time should be <5.0%.|
- | | |1 |
- | | |4.10.1. |RSD calculated for Area should be <5.0%. |
- | | |2 |
- | |4.10.2 |Free protein for ARM of Haemophilus Influenzae type b |
- | | |conjugate vaccine Samples: |
- | | |4.10.2. |Free protein |< 1% |
- | | |1 |
- | |4.11 |Result Reporting: |
- | | |4.11.1 |Report the results of percentage of area of peak towards |
- | | | |higher Molecular weights (Conjugated) and percentage of |
- | | | |area of peak towards lower Molecular weights |
- | | | |(Unconjugated). |
- | | | |Note: Percentage of free carrier protein is the percentage|
- | | | |of area of peak towards lower molecular weights |
- | | | |(Unconjugated). |
- |5.0|Forms and Records: |
- | |Determination of free carrier protein in polyribosyl ribitol phosphate |
- | |(PRP) - tetanus toxoid conjugate by high performance size exclusion |
- | |chromatography |
- |6.0|References: |
- | |6.1 |United States Pharmacopeia 29. |
- | |6.2 |European pharmacopeia |

1.0	Objective		
	To lay down the procedure for identification of Bordetella pertussis		
	Agglutinogens 2 & 3 by slide agglutination method.		
2.0	Scope		
	This Standard Operating Procedure is applicable to all the samples of		
	Bordetella pertussis at Cell bank, In process and ARM stages.		
3.0	Responsibility		
	3.1	The Analyst shall be responsible for carrying out the procedure as	
	established in this SOP.		
	3.2	Supervisor shall be responsible for verifying that the same has been	
	carried out correctly.		
	3.3	Head QC or his designee shall be responsible for overall compliance	
	of the SOP.		
4.0	Abbreviations and Definitions		
	4.1	SOP	: Standard Operating Procedure
	4.2	QA	: Quality Assurance
	4.3	QC	: Quality control
	4.4	I.O.U	: International Opacity Units per Milliliter
	4.5	mL	: milliliter
	4.6	µL	: microliter
5.0	Procedure:		
	5.1	Equipment, Materials and Reagents	
		5.1.1	Micropipette
		5.1.2	Bordetella pertussis factor 2 antisera (NIBSC code: 06/124)
		5.1.3	Bordetella pertussis factor 3 antisera (NIBSC code: 06/128)
		5.1.4	5th International Reference Preparation of Opacity (NIBSC
		code: 76/522)	
		5.1.5	Physiological saline (0.9% w/v)
		5.1.6	Glass slides
		5.1.7	Vortex mixture
		5.1.8	Test tubes
	5.2	Preparation of solutions	
		5.2.1	Physiological saline (0.9% w/v)
		Add 0.9 gm of weighed Sodium Chloride to 50ml of purified	
		water, mix well and make up it to 100ml.	
	5.3	Test procedure	
		5.3.1	Dilute the sample to 10 IOU/mL with Physiological saline (0.9%
		w/v) from its initial opacity units.	
		5.3.2	Take clean grease free glass slides and number them as 1,2 &
		3(control)	
		5.3.3	Add 20µL of test sample suspension on slides1, 2 & 3.
		5.3.4	Add 20µL of Bordetella pertussis factor 2 and 3 antisera and
		Physiological saline (0.9% w/v) to slides 1, 2 &3 respectively.	
		5.3.5	Mix the sample with Bordetella pertussis factor antisera and
		Physiological saline (0.9% w/v) thoroughly. Rock the slides for	
		few minutes.	
		5.3.6	Observe for agglutination and record the observation
	5.4	Validity criteria	
		5.4.1	No agglutination should be observed on control slide 3
			(Physiological saline (0.9% w/v))
6.0	Forms and Records		
	Test for Pertussis Agglutinogens 2 and 3 by Agglutination Method		
7.0	References		
	WHO manual for the production and control of vaccines – Pertussis Vaccine		
	(BLG/UNDP/77.2 Rev 1)		

Date : 05-May-2016

CERTIFICATE OF ANALYSIS

Product: Bulk of Tetanus Toxoid

Product code no.: B/TT 002

Appearance: Clear, Yellow to brownish solution without any suspended particles

Form: Liquid

Buffer: Physiological Saline

Manufacturing Date: Feb 2015

Expiry Date: Jan 2020

Antigen Content: 2100 Lf/ml

Antigenic Purity: 1250 Lf/mg of Protein Nitrogen

Sterility: Sterile Product

pH: 6.6

Specific Toxicity: Complied

Identity: Positive

Free formaldehyde: 0.08g/ L

Irreversibility of toxoid: Complied (No irreversibility observed)

Storage: 5±3°C (Do not freeze)

Note: To be used for research or in-vitro testing purpose only

For BioBridge Healthcare Solutions Pvt. Ltd.

Authorized Signatory

Address: SCO - 32, Sector - 33D, Chandigarh - 160020, India II Phone : +91-1724562168

Head Office: 13, Rachna Blossom, Jagadish Nagar, Aundh, Pune – 411007, (Mah), INDIA II Phone :

+91-9004349016, +91-9004349041

II Website : www.biobridge.in

S. No. | Parameter | Specification

1 | Physical appearance | White lyophilized pellet which is resuspended using 0.9 % saline diluents.

2 | pH | 6.40 – 7.20

3 | Identity for Polysaccharide (PRP) | Line of precipitation should be observed on reaction against specific antibody

4 | Identity for Tetanus Toxoid | Line of precipitation should be observed on reaction against specific antibody

5 | Purified capsular polysaccharide – (PRP content) | 10 mcg ($\pm$ 20%) per single human dose of 0.5 ml (by chromatographic method)

6 | Tetanus Toxoid (carrier protein) | 19 to 30 mcg per single human dose

7 | Residual moisture | The average residual moisture content of all the vials analyzed should not be more than 2.5 %. And

None of the vial should have residual moisture content above 3%.

8 | Pyrogen content | Should pass the assay.

9 | Abnormal Toxicity | None of the animals should show any signs of illness or death during the 7 days of observation period.

10 | Sterility | No microbial growth should be observed during the 14 days of incubation period.

1.0	Objective		
	To lay down the procedure for determining the identity of diphtheria		
	component in vaccine.		
2.0	Scope		
	This Standard Operating Procedure is applicable to the diphtheria component		
	either single component or combined with other component by limes		
	flocculation test.		
3.0	Responsibility		
	3.1	The QC analyst shall be responsible for performing the described	
	procedure correctly.		
	3.2	Supervisor shall be responsible for verifying that the same has been	
	carried out correctly.		
	3.3	Head QC or his designee shall be responsible for overall compliance	
	of the SOP.		
4.0	Abbreviations and Definitions		
	4.1	SOP	: Standard Operating Procedure
	4.2	QA	: Quality Assurance
	4.3	QC	: Quality control
	4.4	DATF	: Diphtheria Antitoxin for Flocculation
	4.5	mL	: milliliter
	4.6	°C	: Degree Celsius
	4.7	Lf/ml	: Limes-flocculation per milliliter
5.0	Procedure:		
	5.1	Equipment, Materials and Reagents	
		5.1.1	Test sample
		5.1.2	Autopipettes
		5.1.3	Test tubes.
		5.1.4	Test tube rack
		5.1.5	Flocculation tubes.
		5.1.6	Thermometer
		5.1.7	Stop watch
		5.1.8	Convex lens
		5.1.9	Sterile 10ml & 50ml vials
		5.1.10	Physiological saline
		5.1.11	Working Standard of Diphtheria Antitoxin for Flocculation (100
			Lf/ml)
		5.1.12	Tri sodium citrate
	5.2	Test procedure	
		5.2.1	Take 10 ml vaccine sample in a centrifuge tube and add 1.0 gm
			of tri -Sodium citrate.
		5.2.2	Mix well and incubate the mixture at 35±20C for 16±2 hours.
		5.2.3	Centrifuge the sample at 4500 rpm for one hour.
		5.2.4	Collect the supernatant.
		5.2.5	Take 1 mL of the supernatant in flocculation tube, add to it
			0.40 mL of diphtheria antitoxin.
		5.2.6	Shake the mixture in tube thoroughly.
		5.2.7	Incubate the tubes at 50±20C in a see-through water bath.
			Ensure that 2/3rd of the flocculation tubes are immersed in
			water.
		5.2.8	Check for flocculation and note down the flocculation time.
	5.3	Acceptance criteria:	
		Flocculation against DATF confirms the presence of diphtheria	
		component in vaccines.	
6.0	Forms and Records		
	Identity Test for Diphtheria Vaccine		
7.0	References		
	7.1	World Health Organization, Technical Report Series No. 800, 927	
	7.2	Indian Pharmacopoeia.	

Test | Specifications

Organoleptic characteristics | A white to almost white suspension that separates in two phases on keeping.

Ph | 6.0 – 7.0

Specific Toxicity test for D & T component

(Invivo technique) | Animals should not show any ill health within six weeks of injection and at least 80% of the animals should survive the test period

Mouse Weight gain test for Inactivated w-B. pertussis component (In vivo technique) | Total weight of test animals in vaccine group on 3rd day should not be less than that on zero day and at the end of 7th day the average increase in weight in the vaccine group is not less than 60 % of that of in the control group

Potency test of Diphtheria Toxoid

(By challenge method) | ≥ 30 IU / SHD, the limit of the 95% confidence intervals of the estimated potency shall be within 50 to 200% of the estimated potency unless the lower limit of the 95% confidence interval of the estimated potency is greater than 30 IU /SHD. | ≥ 30 IU / SHD, the limit of the 95% confidence intervals of the estimated potency shall be within 50 to 200% of the estimated potency unless the lower limit of the 95% confidence interval of the estimated potency is greater than 30 IU /SHD.

Potency test of Tetanus Toxoid

(By challenge method) | ≥ 60 IU /SHD when test is carried out in mice. For three dilution assay, the limits of 95% confidence intervals of the estimate of potency shall be within 50 to 200% of the estimated potency unless the lower limit of the 95% confidence interval of the estimated potency is greater than 60 IU /SHD | ≥ 60 IU /SHD when test is carried out in mice. For three dilution assay, the limits of 95%

confidence intervals of the estimate of potency shall be within 50 to 200% of the estimated potency unless the lower limit of the 95% confidence interval of the estimated potency is greater than 60 IU /SHD

Potency test of Inactivated w-B. pertussis (By active challenge method) | ≥ 4.0 IU/ SHD (the lower fiducial limit (P= 0.95) of the estimated is not potency is less than the 2.0 IU).

Determination of PRP Content

(By HPAEC-PAD method) | Total PRP content = 16 to 24 μ g/ml

free PRP should be less than 20 % of total PRP content

In-vivo potency test of Hepatitis B surface antigen (In vivo potency and ELISA) | Upper confidence limit of relative potency should be > 1 .

In-vitro potency test of Hepatitis B surface antigen (ELISA method) | Upper confidence limit of relative potency should be > 1 .

Adsorption test of Diphtheria component | % adsorption of Diphtheria toxoid > 50

Adsorption test of Tetanus component | % adsorption of Tetanus toxoid > 30

Free formaldehyde content | Not more than 0.02% w/v (< 200 μ g/ml)

Aluminum content | ≤ 1.25 mg/ml.

Thiomersal content | ≤ 0.1 mg/ml

Abnormal toxicity test | No signs of sickness during observation period of 7 days

Extractable Volume | 0.60 ± 0.05 ml

Sterility | No microbial growth.

[1.0]Objective: |
	To lay down the procedure for the determination of protein concentration in
	Active Raw Material of Hepatitis B Vaccine based on the physical property
	of the proteins to absorb UV light at 280 nm. Since absorbance of this
	light and concentration of proteins are related, it is possible to
	calculate the concentration of protein with the help of extinction
	coefficient following Lambert – Beer's law.
[2.0]Scope:	
	This Standard Operating Procedure is applicable to the Active Raw Material
	(ARM) of Hepatitis-B vaccine (rDNA).
[3.0]Responsibility:	
	3.1
	executing the described procedure correctly.
	3.2
	that the procedure is followed correctly and for checking the results.
	3.3
	of the SOP. .
[4.0]Abbreviations and Definitions:	
	4.1
	4.2
	4.3
	4.4
	4.5
	4.6
	4.7
	4.8
[5.0] Procedure: Note: Make sure that all the reagents are at room temperature	
	before use.
	5.1
	5.2
	5.3
	5.4
	5.5
	5.7
	5.8
	5.9
[6.0]References:	
	NA

[1.0]Objective: |
|To detect and quantify mouse immunoglobulin (IgG) contaminants in samples by|
|using kit from Cygnus Technologies, Inc. |
[2.0]Scope: |
|This Standard Operating Procedure is applicable to the In process Samples |
|after immunoaffinity step and Active Raw Material (ARM) of Hepatitis-B |
|vaccine (rDNA). |
[3.0]Responsibility: |
|3.1 |The analyst of the Immunochemistry laboratory shall be responsible for |
|executing the described procedure correctly. |
|3.2 |The supervisor of the laboratory shall be responsible for verifying |
|that the procedure is followed correctly and for checking the results. |
|3.3 |The Departmental Head shall be responsible for the overall compliance |
|of the SOP and shall ensure that the SOP is reviewed every two years or|
|if required earlier. |
[4.0]Abbreviations and Definitions: |
[4.1] |
|ELISA |
: |
: Enzyme Linked Immunosorbent Assay |
| |
| |
| |
| |
[4.2] |
|IgG |
: |
: Immunoglobulin G |
| |
| |
| |
| |
[4.3] |
|ARM |
: |
: Active Raw Material |
| |
| |
| |
| |
[4.4] |
|PBS |
: |
: Phosphate buffered saline |
| |
| |
| |
| |
[4.5] |
|QC |
: |
: Quality Control |
| |
| |
| |
| |
[4.6] |
|QA |
: |
: Quality Assurance |
| |
| |
| |
| |
[4.7] |
|SOP |
: |
: Standard Operating Procedure |
| |
| |
| |
| |
[4.8] |
|PNPP |
: |
: p-nitrophenylphosphate |
| |
| |
| |
| |
[5.0] Procedure |
|Note: Make sure that all the reagents are at room temperature before use. |
|5.1 |Pipette 50µl of Standards, controls and samples into wells. |
|5.2 |Pipette 200µl of anti-mouse immunoglobulin: alkaline phosphatase into|
|each well. |
|5.3 |Cover and incubate on dancing shaker at room temperature for 2 hour. |
|5.4 |Aspirate the contents to waste. Blot and vigorously bang out residual|
|liquid over absorbent paper. Wash generously with diluted wash |
|solution by flooding the wells with solution by pipeting in 300µl. |
|Dump and bang again. Repeat for a total of 4 washes. Wipe off any |
|liquid from the bottom outside of the microtiter wells as any residue|
|can interfere in the reading step. |
5.5	Pipette 200µl of substrate provided in the kit.
5.6	Cover and incubate for the plate for 60 minutes. Do not shake.
5.7	Read absorbance at 405/490nm using a ELISA reader.
5.8	Follow the given pathway to save the Absorbance obtained from ELISA
reader:	
5.8.1	File→Save Data/Template as →My Computer →D: →Raw Data from

			Microplate Manager→IgG→ In-Process Batches/Regular Batches ARM	
			/Stability Batches→ File Name → Save → File→ Print	
			Note: File Name is given as Batch No. of ARM in case of Regular	
			Batches, In-Process stage in case of In-Process samples and	
			Batch No./Type of Stability/Interval in case of stability	
			batches.	
			5.9	Raw data processing and interpretation of results:
			5.9.1	Open the ELISA_3 program following the pathway given below:
			My Computer→C: →ELISA→ELISA_3	
			5.9.2	Enter the values of Absorbance obtained, in ELISA_3 software as
			follows: In the column entitled "Val (X)" type the different	
			concentrations of standard. In the columns entitled "Serie 1"	
			and "Serie 2" type the corresponding absorbance values of	
			replicates 1 and 2 for each concentration of the standard.	
			5.9.3	Press F5 and select the option "Reg/Tipo". Further select
			Parabolica".	
			5.9.4	Press F3 to perform the regression, followed by ESC to return
			to the previous screen.	
			5.9.5	Press F4 key. In the column entitled "Muestras" type the name
			or code of the sample. For each sample take a row. Under the	
			columns entitled "Abs1" and "Abs 2" type the absorbance values	
			of the replicates 1 and 2 of the sample. In the column entitled	
			"Dil" type the dilution of the sample. Do not try to fill the	
			last column entitled "Conc." as the estimated concentration of	
			each dilution of the sample will automatically appear. The	
			calculated results of the samples expressed in the program are	
			given in ng/ml.	
			5.9.6	Press F5 key and select the option "Files" using the arrow keys
			and press "Enter". Further select "Buscar" to open an already	
			existing file or "Salvar" to save a new file. The file name is	
			a eight-digit code consisting date, month and year (DDMMYYXX).	
			Here DDMMYY is Date, Month, Year and XX is the serial no. of	
			the file saved on that particular day. Press "ESC" to return to	
			the previous screen.	
			5.9.7	Press F5 key and select the option "Imprimir" using the arrow
			keys. Further select "Reporte" and press "enter" to print the	
			regression report. In order to print the sample results, select	
			"Muestras" and press "Enter". Press "ESC" key to return to the	
			previous screen. In order to exit the program, again press	
			"ESC" key. When a message appears on the screen, whether to	
			exit or not, press "S" to exit and "N" to continue.	
			5.9.8	The print out of report shall be authenticated by signing the
			document by the doer and checker and attached with the data	
			record form.	
			5.10	Reporting of results:
			5.10.1	For In process samples the value (calculated using ELISA_3
			program) shall be reported in term of ng/ml) as the final	
			results.	
			5.10.2	For ARM samples the results shall be reported in terms of
			percentage IgG contaminants and the calculations are done as	
			follows:	
			IgG contaminants in terms of nanograms per dose (Z)**	
			=[Concentration* of IgG contaminants (ng/ml) × Quantity of	
			HBsAg per dose (ng/dose)] / [Protein concentration (ng/ml)]	
			Where:	
			Protein concentration of the sample = X ng/ml (Calculated	
			using O.D. at 280 nm	
			Concentration* of IgG contaminants (ng/ml) = Y* ng/ml	
			(Calculated using ELISA_3 program).	
			Percentage IgG contaminants =	
			IgG contaminants in ng per dose (Z)** × 100	
			Concentration of HBsAg (ng/dose)	
			5.11	Validity criteria:
			5.11.1	The coefficient of regression must be greater than 0.96.
			5.11.2	The coefficient of variation between the replicates of same
			point of standard curve must be less than 10%.	
			6.0	References:
			Immunoenzymetric Assay for the Measurement of Total Mouse IgG	
			Immunoglobulin, Cygnus Technologies, inc.	

1.0 |Objective: |
 |To lay down the procedure for testing the potency of diphtheria toxoid. |
 2. 0|Scope: |
 |The Standard operating procedure is applicable to the batches of |
 |diphtheria toxoid vaccine as single component or combined with other |
 |components. |
 3.0 |Responsibility: |
3.1	The Analyst shall be responsible for carrying out the procedure as
	established in this SOP.
3.2	Supervisor shall be responsible for verifying that the same has been
	carried out correctly.
3.3	Head QC or his designee shall be responsible for overall compliance
	of the SOP.
4.0	Abbreviations and Definitions
4.1	ml
4.2	g
4.3	°C
4.4	%
4.5	QC
4.6	QA
4.7	AH
4.8	SOP
4.9	S.H.D
 5. |Procedure: |
 |5.1 |Equipment, Materials and Reagents | | |
 | |5.1.1 |Weighing balance |
 | |5.1.2 |Laminar air flow cabinet |
 | |5.1.3 |Autoclave |
 | |5.1.4 |Test sample |
 | |5.1.5 |Autopipettes |
 | |5.1.6 |Test tubes |
 | |5.1.7 |Test tube racks |
 | |5.1.8 |Glass vials, 10 ml |
 | |5.1.9 |Disposable syringes (2ml, 5 ml) |
 | |5.1.10 |Sodium Hydroxide |
 | |5.1.11 |Hydrochloric acid |
 | |5.1.12 |Sodium Chloride |
 | |5.1.13 |Peptone |
 | |5.1.14 |Purified Water |
 |5.2 |Preparation of solutions: |
 | |5.2.1 |Normal saline: |
 | | |Weigh 0.9 g of sodium chloride and dissolve in 60 ml of the |
 | | |purified water. Make up the volume up to 100 ml with |
 | | |purified water. |
 | |5.2.2 |Peptone water: |
 | | |Weigh 1 g of peptone and 0.5 g of sodium chloride. Dissolve |
 | | |in 60 ml of purified water. Make up the volume up to 100 ml |
 | | |with purified water. Sterilize by autoclaving at 121°C for |
 | | |20 minutes. |
 |5.3 |Selection of test animals: |
 | |5.3.1 |Select 116 guinea pigs weighing between 250 – 350 g at |
 | | |least one day before immunization. |
 | |5.3.2 |Preferably choose guinea pigs of the same sex. Otherwise |
 | | |equally distribute males and females in each group. |
 | |5.3.3 |Make six groups of 16 guinea pigs each and four groups of 4 |
 | | |guinea pigs each. |
 | |5.3.4 |Allocate three groups out of six to each of three dilutions |
 | | |of reference vaccine and other three groups to each of |
 | | |three dilutions of the test vaccine. |
 | |5.3.5 |Four groups of 4 guinea pigs each are to be used for LD₅₀ |
 | | |determination. Distribute all the guinea pigs in various |
 | | |groups by randomization. |
 | |5.3.6 |Mark each group of guinea pigs with appropriate color |
 | | |scheme. |
 |5.4 |Preparation of Vaccine dilution : |
 | |5.4.1 |Preparation of dilutions of reference vaccine: |
 | |5.4.2 |Dilute reference vaccine in normal saline so as to contain |
 | | |200 –300 IU/ml. Make further dilutions of the reference |
 | | |preparation as detailed below. |
 | | |5.4.2.1 |1/10 dilution: Add 1 ml of vaccine to 9 ml of |
 | | |normal saline and mix thoroughly. |
 | | |5.4.2.2 |1/50 dilution: Add 8 ml of 1/10 dilution of |
 | | |vaccine to 32 ml of normal saline and mix |
 | | |thoroughly. |
 | | |5.4.2.3 |1/100 dilution: Add 20 ml of 1/50 dilution of |
 | | |vaccine to 20 ml of normal saline and mix |
 | | |thoroughly. |
 | | |5.4.2.4 |1/200 dilution: Add 20 ml of 1/100 dilution of |
 | | |vaccine to 20 ml of normal saline and mix |
 | | |thoroughly. |
 | | |5.4.2.5 |Preparation of dilutions of test vaccine: Take |
 | | |test vaccine and dilute it following steps 5.4.2.1 |
 | | |to 5.4.2.4. |
 |5.5 |Immunization: |
 | |5.5.1 |Inject subcutaneously each of 16 guinea pigs in each group |
 | | |with 1.0 ml of appropriate dilution (1/50, 1/100 and 1/200) |
 | | |of test and reference vaccine allocated to that group. |
 | |5.5.2 |Keep the four groups of 5 guinea pigs each as uninoculated. |
 | |5.5.3 |Keep the immunized animals under observations for 28 days. |
 | | |The animals should show normal growth and good health |

			during the period of observation	
	5.6	Preparation of Challenge Diphtheria toxin:		
		5.6.1	Suitably dilute the stabilized diphtheria toxin in peptone	
			water in such a way that it contains approximately 100	
			LD50/ml.	
		5.6.2	1/25 dilution of challenge toxin: Add 1 ml of challenge	
			toxin prepared to 24 ml of peptone water. Mix it	
			thoroughly. (4LD50/ml)	
		5.6.3	1/50 dilution of challenge toxin: Add 5 ml of 1/25 dilution	
			of challenge toxin to 5 ml of peptone water. Mix it	
			thoroughly. (2 LD50/ml)	
		5.6.4	1/100 dilution of challenge toxin: Add 5 ml of 1/50	
			dilution of challenge toxin to 5 ml of peptone water. Mix	
			it thoroughly. (1 LD50/ml)	
		5.6.5	1/200 dilution of challenge toxin: Add 5 ml of 1/100	
			dilution of challenge toxin to 5 ml of peptone water. Mix	
			it thoroughly. (1/2 LD50/ml)	
		5.7	Inoculation of challenge toxin:	
			After twenty-eight days of immunization, inject all groups of guinea	
			pigs subcutaneously with 1.0 ml of the challenge toxin containing	
			approximately hundred times the 50% Lethal dose (100 LD50).	
		5.8	LD50 estimation:	
			Take four groups of 4 guinea pigs each (selected on the day of	
			immunization). Inject subcutaneously 1ml of challenge toxin	
			containing 4LD50, 2LD50, m, 1LD50 and ½ LD50 dose per ml respectively	
			in 4 guinea pigs each.	
		5.9	Observations:	
			Observe all the injected Guinea pigs daily and record the deaths	
		5.10	Calculations:	
			Count the number of surviving guinea pigs 5 days after injection of	
			the challenge toxin and calculate the potency of the vaccine being	
			examined relative to the potency of the standard preparation. This	
			may be done on the basis of the number of surviving animals in each	
			of the six groups of sixteen, using standard statistical methods. (By	
			probit analysis as per software programme evolved by F.R. Marsman	
			RIVM, Bilthoven, Netherland).	
		5.11	Validity criteria:	
			5.11.1	The slopes of the dose response curves of the vaccine under
			test and the standard are linear and parallel.	
			5.11.2	The 50% effective dose of each vaccine lies between the
			largest and the smallest immunizing doses.	
			5.11.3	Challenge toxin should be between 50-200% of lethal dose.
		5.12	Acceptance criteria:	
			5.12.1	The estimated potency is not less than 30 I.U. per single
			human dose.	
			5.12.2	The limit of 95% confidence intervals of the estimate of
			potency is within 50-200% unless the lower limit of the 95%	
			confidence interval of the estimated potency is greater	
			than 30 I.U. per single human dose.	
	6.0	Forms and Records		
		6.1	Raw data form for recording the details of potency test of diphtheria	
			toxoid by challenge method in Guinea pigs	
		6.2	Raw data form for recording the observations on Guinea pigs of	
			diphtheria potency test	
		6.3	Form for reporting the results of potency test of diphtheria toxoid	
			by challenge method in Guinea pigs	
	7.0	References		
		7.1	Indian Pharmacopoeia, 2007.	
		7.2	World Health Organization, Technical Report Series No. 800, 1990	

- Specification of bulk Tetanus Toxoid (BDS)
- S.No | Test/Assay | Acceptance Criteria
- 1 | Physical appearance | Clear, Yellow to brownish solution without any suspended particles
- 2 | Identity | Positive
- 3 | Antigen Content (Lf/ml) | Not less than 1500 Lf/ml
- 4 | Antigenic purity | Not less than 1000 Lf/ mg of protein nitrogen
- 5 | Sterility | Sterile
- 6 | Specific Toxicity | To comply the test
- 7 | Irreversibility of toxoid | To comply the test
- 8 | Free formaldehyde | Not more than 0.2g/ L
- 9 | pH | 6.0 - 7.0

* PS: Polysaccharide

S. No | Test/Assay | Specifications

1 | Appearance | White Powder or Lump

2 | Identity | Should react with specific antibodies

3 | Sialic Acid Content | ≥ 56 % of dry weight of MenW Ps

4 | O acetyl Content | ≥ 0.3 mmol/gm of dry weight of MenW Ps

5 | KD Molecular Size Distribution | ≥ 80 % KD= 0.5

6 | Moisture content | To report results

7 | Protein Content | ≤ 5.0 % of dry weight of MenW Ps

8 | Nucleic acid content | ≤ 2.0 % of dry weight of MenW Ps

9 | Cell Wall polysaccharide content | ≤ 1.0 % of dry weight of MenW Ps

10 | Endotoxin content | < 25 IU/ μ g of MenW Ps

11 | Residual CTAB Content | ≤ 1.0 % of dry weight of MenW Ps

12 | Residual Formaldehyde | ≤ 1.0 % of dry weight of MenW Ps

13 | Residual DOC content | ≤ 1.0 % of dry weight of MenW Ps

1.0 |Objective |
 |To lay down the procedure for preparation and standardization of diphtheria|
 |antitoxin (Working standard) for flocculation test of Diphtheria antigens. |
 2.0 |Scope |
 |This procedure is applicable for preparation and standardization of |
 |diphtheria antitoxin for flocculation test of Diphtheria antigens. |
 3.0 |Responsibility |
 |3.1 |The Analyst shall be responsible for carrying out the procedure as |
 |established in this SOP. |
 |3.2 |Supervisor shall be responsible for verifying that the same has been |
 |carried out correctly. |
 |3.3 |Head QC or his designee shall be responsible for overall compliance |
 |of the SOP. |
 4.0 |Abbreviations and Definitions |
4.1	SOP	: Standard Operating Procedure
4.2	QA	: Quality Assurance
4.3	QC	: Quality control
4.4	DATF	: Diphtheria Antitoxin for Flocculation
4.5	mL	: milliliter
4.6	°C	: Degree Celsius
4.7	%	: Percent
4.8	Lf/ml	: Limes-flocculation per milliliter
5.0	Procedure:	
5.1	Equipment, Materials and Reagents	
5.1.1	Diphtheria toxoid Standard (NIBSC code: 02/176)	
5.1.2	Micropipettes	
5.1.3	Micro tips	
5.1.4	Flocculation tube rack	
5.1.5	Flocculation tube	
5.1.6	Thermometer	
5.1.7	Stop watch	
5.1.8	Convex lens	
5.1.9	Sterile 10ml & 50ml vials	
5.1.10	Physiological saline (0.9% w/v)	
5.1.11	Thiomersal	
5.1.12	International Diphtheria Antitoxin for flocculation (NIBSC	
code: 63/007)		
5.2	Preparation of solutions	
5.2.1	Physiological saline (0.9% w/v)	
Add 0.9 gm of weighed Sodium Chloride to 50ml of purified		
water, mix well and make up it to 100ml.		
5.2.2	1% w/v Thiomersal Solution	
Add 1 gm of weighed Thiomersal to 50ml of purified water, mix		
well and make up it to 100ml.		
5.3	Preparation of Internal Reference Diphtheria toxoid (50Lf/mL)	
5.3.1	Obtain Diphtheria toxoid bulk purified from production	
facility/ external source which complies with the		
predetermined specifications.		
5.3.2	Dilute the toxoid to 50Lf/mL based on the previous estimated	
value.		
5.4	Standardization of Standard toxoid (50Lf/mL)	
5.4.1	Reconstitute the National/International reference standard of	
antitoxin for flocculation to 100Lf/mL based on the label		
claim with normal saline.		
5.4.2	Determine the unitage of standard toxoid using the	
National/International reference standard of antitoxin by		
flocculation test.		
5.4.3	The test is valid if the standard toxoid concentration is 50 ±	
1 Lf/mL.		
5.5	Preparation of Internal Reference Antitoxin	
5.5.1	Take the required quantity of antitoxin obtained from the	
reputed vendor in a suitable glass container		
5.5.2	Dilute the antitoxin to 100 Lf/mL with normal saline	
containing preservatives based on the label claim.		
5.6	Standardization of Antitoxin	
5.6.1	Determine the unitage of standard toxoid using the candidate	
diphtheria antitoxin at 100 Lf/ml by flocculation test.		
5.6.2	If the candidate diphtheria antitoxin gives the unitage of	
standard toxoid more than 50 Lf/ml then make small additions		
of candidate diphtheria antitoxin till it gives the unitage		
of standard toxoid 50 Lf/ml.		
5.6.3	If the candidate antitoxin gives the unitage of standard	
toxoid less than 50 Lf/ml then make small additions of normal		
saline till it gives the unitage of standard toxoid 50 Lf/ml.		
5.6.4	Candidate diphtheria antitoxin when gives the unitage of	
standard toxoid as 50 Lf/ml is considered to be standardized		
for use as working standard of diphtheria antitoxin for		
flocculation test and assigned the value of 100 Lf/ml.		
5.6.5	After standardization, sterile filter the antitoxin by using	
0.2μ filters and distributes it to 100 mL bottles.		
5.7	Monitoring of Internal Reference Antitoxin	
5.7.1	Check the Lf / mL concentration of Internal Reference	
antitoxin for flocculation once in three months with internal		
/ national / international standard toxoid.		
5.7.2	Discard the reference antitoxin if any drop is observed in Lf	
concentration and establish a new one.		
6.0	Forms and Records	
JNA		
7.0	References	
Determination of unitage (Lf/ml) of diphtheria toxin/toxoid by flocculation		

* DWB: Dry weight basis

S. No | Test/Assay | Specifications

1 | Moisture Content | To report results

2 | Identity | Positive

3 | KD (Molecular size distribution) | $PRP \geq 40\%$; $Kd \leq 0.3$

4 | Ribose Content | $> 32\%$ w/w (DWB)

5 | PRP content | > 0.73 mg/mg (DWB)

6 | Phosphorus Content | $6.8 - 9.0\%$ w/w (DWB)

7 | Protein Content | $\leq 1\%$ w/w (DWB)

8 | Nucleic acid Content | $\leq 1\%$ w/w (DWB)

9 | Endotoxin Content | < 25 IU/ μ g of PRP

1.0 |Objective |
 |To lay down the procedure for Specific toxicity test of pertussis component|
 |in pertussis bulk concentrate. |
 2.0 |Scope |
 |This Standard Operating Procedure is applicable to all the ARM samples and |
 |final lots of Pertussis vaccine. |
 3.0 |Responsibility |
 |3.1 |The QC analyst shall be responsible for performing the described |
 |procedure correctly. |
 |3.2 |Supervisor shall be responsible for verifying that the same has been |
 |carried out correctly. |
 |3.3 |Head Animal House/Veterinarian/supervisor shall be responsible for |
 |supply of quality animals and their maintenance. |
 |3.4 |Head QC or his designee shall be responsible for overall compliance |
 |of the SOP. |
 4.0 |Abbreviations and Definitions |
4.1	SOP	: Standard Operating Procedure
4.2	QA	: Quality Assurance
4.3	QC	: Quality control
4.4	OU	: Opacity Unit
4.5	AH	: Animal House
4.6	mL	: milliliter
4.7	%	: Percent
4.8	MWGT	: Mouse Weight Gain Test
5.0	Procedure:	
5.1	Equipment, Materials and Reagents	
5.1.1	Biosafety Cabinet	
5.1.2	Weighing balance	
5.1.3	Micropipette	
5.1.4	Vortex mixer	
5.1.5	Cotton	
5.1.6	Disposable syringes with needle	
5.1.7	Sterilized Glass Vials	
5.1.8	Sterile micropipette tips	
5.1.9	70% Isopropyl alcohol	
5.1.10	Sodium chloride	
5.1.11	2 phenoxyethanol	
5.2	Preparation of solutions	
5.2.1	Physiological saline (0.9% w/v)	
Add 0.9 gm of weighed Sodium Chloride to 50ml of purified		
water, mix well and make up it to 100ml.		
5.3	Place request for Swiss Albino mice (14-16g) to Head Animal House.	
After health status evaluation of animals by Veterinary officer		
receive the animals.		
5.4	Selection of animals for test:	
5.4.1	Select Swiss Albino mice (Test and Control) weighing 14 – 16	
g. Each group consisting of 10 animals. Color codes the		
animals.		
5.4.2	Take the group weight of the mice to be injected with test	
sample as well as with diluent (Control).		
5.4.3	The mice used for testing the test sample and the control	
group should be of the same sex. If both sexes are used, they		
should be equally distributed in both the groups.		
5.4.4	Animals should have free access to food and water for at least	
2 hours before injection and thereafter continuously		
throughout the duration of the test.		
5.4.5	The cages housing the mice shall bear labels.	
5.5	Preparation of diluent:	
5.5.1	For bulk containing 2 Phenoxyethanol as preservative:	
Add 5mg/mL of 2 phenoxyethanol in physiological saline. (0.5%)		
5.5.2	These solutions shall be prepared freshly for each test.	
5.6	Preparation of dilution for pertussis bulk:	
5.6.2	Based on the initial opacity, dilute the test sample with	
physiological saline containing preservative so that after		
dilution, bulk sample contains 40 OU/mL.		
5.6.3	Take 3 mL of diluted bulk and dilute it with 3 mL of diluent.	
Mix thoroughly with vortex mixer.		
5.7	Injection:	
5.7.1	Inject each mouse of the test group with 0.5 mL of the diluted	
Pertussis Bulk (20 OU/mL) intraperitoneally.		
5.7.2	Inject each mouse of the control group with 0.5 mL of the	
diluent intraperitoneally.		
5.8	Observations:	
5.8.1	Observe the health of each of the inoculated mice twice daily.	
5.8.2	Take the group weight of both control and test group at the	
end of 72 hours and at the end of 7 days, record the results		
5.9	Calculations:	
Percentage of mouse weight gain test is calculated for		
Pertussis bulk concentrate and test vaccine by the following		
formula:		
% MWGT = (B – A) / (D – C) x 100		
A = Group weight of mice injected with test vaccine on day – 0		
B = Group weight of mice injected with test vaccine on day – 7		
C = Group weight of mice injected with diluent(control) on day		
D = Group weight of mice injected		
with diluent(control) on day – 7		
5.10	Acceptance criteria:	
The test sample passes the test and is considered to be non		
toxic if		
5.10.1	At the end of 72 hours, total group weight of the	

				test animals is not less than that at the time of
			injection.	
			5.10.2	At the end of 7 days the average weight gain per
			mouse in test group mouse is not less than 60 % of	
			that of control group mice.	
			5.10.3	No deaths occur when 10 mice are used.
			5.11	Retest Criteria:
			Repeat the test if it fails to meet the acceptance criteria in the	
			1st test.	
			5.11.1	If repeat test (2nd test) fails, the sample fails the test.
			5.11.2	If the repeat tests (2nd test) passes then one additional test
			(3rd test) to be performed. If this test passes then sample	
			passes or otherwise sample fails the test.	
			6.0	Forms and Records
			Specific Toxicity Test of Whole Cell Pertussis Vaccine by Mouse Weight	
			Gain Method	
			7.0	References
			7.1	Indian Pharmacopoeia.
			7.2	World Health Organization, Technical Report Series No. 941.
			7.3	WHO manual for the production and control of vaccines – Pertussis
			Vaccine	
			(BLG/UNDP/77.2 Rev 1)	

Date : 27-May-2016
CERTIFICATE OF ANALYSIS
For BioBridge Healthcare Solutions Pvt. Ltd.
Authorized Signatory
Product: | Bulk of Hepatitis B surface Antigen (HBsAg)
Product code no.: | B/HepB 003
Appearance: | Colorless to light yellow color, clear liquid without any suspended particle
Form: | Liquid (Aqueous Solution)
Buffer: | Phosphates of Sodium and Potassium and Sodium Chloride
Manufacturing Date: | October 2015
Expiry Date: | September 2017
Antigen Content: | 1.03 mg/ml (as total protein by Lowry Method)
Purity: | 100% (by SDS), 99.98% (by HPLC)
Sterility: | Sterile Product
pH: | 7.08
Storage: | 5±3°C (Do not freeze)
Volume: | ~30ml x 1 Tube
Note: To be used for research or in-vitro testing purpose only | Note: To be used for research or in-vitro testing purpose only

* Ps : Polysaccharide

S.No | Test/Assay | Specifications

1 | Appearance | White Powder or Lump

2 | Identity | Should react with specific antibodies

3 | Phosphorous Content | ≥ 8.0 % of dry weight of MenA Ps

4 | O acetyl Content | ≥ 2 mmol/gm of dry weight of MenA Ps

5 | KD Molecular Size Distribution | $\geq 65\%$ KD= 0.5

6 | Moisture content | Report results

7 | Protein Content | ≤ 1.0 % of dry weight of MenA Ps

8 | Nucleic acid content | ≤ 1.0 % of dry weight of MenA Ps

9 | Cell Wall polysaccharide content | ≤ 1.0 % of dry weight of MenA Ps

10 | Endotoxin content | < 25 IU/ μ g of MenA Ps

11 | Residual CTAB Content | ≤ 1.0 % of dry weight of MenA Ps

12 | Residual Formaldehyde | ≤ 1.0 % of dry weight of MenA Ps

13 | Residual DOC content | ≤ 1.0 % of dry weight of MenA Ps

[1. |Objective: |
 ||To lay down the procedure to determine the sterility of products |
 ||purported to be sterile. |
 [2. |Scope: |
	This Standard Operating Procedure is applicable for all the In process,	
	ARM final bulk and final lot samples	
	2.1	For preparation, storage, growth promotion test and application of
	media, used in sterility testing.	
[3.	Abbreviations and Definitions:	
	3.1	°C
	3.2	IP
	3.3	ml
	3.4	USP
	3.5	EP
[4.	Procedure:	
	4.1	Equipments, Materials and Reagents
	4.1.1	Laminar airflow cabinet
	4.1.2	Incubators at 32.5 + 2.5°C and 22.5 + 2.5°C
	4.1.3	Sterile syringes
	4.1.4	Sterile pipettes
	4.1.5	Sterile gloves
	4.1.6	Sterile forceps
	4.1.7	McCartney bottles
	4.1.8	Purified water
	4.1.9	Tryptone Soya Broth
	4.1.10	Fluid Thioglycollate Medium
	4.1.11	Soya Bean Casein Digest media
	4.2	Preparation of Sterility Media:
	4.2.1	Fluid Thioglycollate Medium (FTM):
	4.2.1.1	Media for the test may be prepared as described
		below, or equivalent commercial media may be used
		provided that they comply with the growth promotion
		test.
	4.2.1.2	Prepare FTM media is prepared as per the produce
		mentioned below:
		Ingredient
		L-Cystine
		Agar
		Sodium chloride
		Glucose monohydrate/anhydrous
		Yeast extract (water-soluble)
		Pancreatic digest of casein
		Sodium thioglycollate or
		Thioglycollic acid
		Resazurin sodium solution (1 g/l
		of resazurin sodium), freshly
		prepared
		Water R
	4.2.1.3	Mix the L-cystine, agar, sodium chloride, glucose,
		water-soluble yeast extract and pancreatic digest of
		casein in 700 ml of purified water R and heat until
		the components are completely dissolved.
	4.2.1.4	Dissolve the sodium thioglycollate or thioglycollic
		acid in the solution and, if necessary, add sodium
		hydroxide (1 mol/l) so that, after sterilization,
		the solution will have a pH of 6.9 to 7.3.
	4.2.1.5	Add the resazurin sodium solution, mix and place the
		medium in suitable vessels which provide a ratio of
		surface to depth of medium such that not more than
		the upper half of the medium has undergone a colour
		change indicative of oxygen uptake at the end of the
		incubation period.
	4.2.1.6	Make up the volume of the media to 1000 ml with
		purified water and distribute the media in suitable
		bottles / containers
	4.2.1.7	Fill the media in sterile glass bottles / containers
		in a volume of 80 to 100ml. The volume of the media
		is validated such that the sample that needs to be
		analyzed does not get over diluted. If any
		antimicrobials or preservatives are present in the
		test sample should get sufficiently diluted so that
		their inhibitory effect is lost.
	4.2.1.8	Sterilize the bottles by autoclaving, after
		sterilization, store the media at a temperature
		between 2 °C and 25 °C. Do not use the medium for a
		longer storage period than has been validated.
	4.2.2	Soyabean Casein Digest Medium (SCDM):
	Media for the test may be prepared as described below, or	
	equivalent commercial media may be used provided that they	
	comply with the growth promotion test.	
	Prepare SCDM media is prepared as per the produce mentioned	
	below:	
	Ingredient	Content
	Pancreatic digest of casein	17.0 g
	Papaic digest of soya-bean	3.0 g
	meal	
	Sodium chloride	5.0 g
	Dipotassium hydrogen	2.5 g
	phosphate	
	Glucose monohydrate/anhydrous	2.5/2.3 g

			Water R	1000 ml
			4.2.2.1	Dissolve the solids in 700 ml of purified water R,
			warming slightly to dissolve the contents.	
			4.2.2.2	Cool the solution to room temperature and add sodium
			hydroxide (1 mol/l) VS, if necessary, so that after	
			sterilization the solution will have a pH of 7.1 to	
			7.5.	
			4.2.2.3	Make up the volume of the media to 1000 ml with
			purified water and distribute the media in suitable	
			bottles / containers to sterilize the media by	
			autoclaving.	
			4.2.2.4	Fill the media in sterile glass bottles / containers
			in a volume of 80 to 100ml. The volume of the media	
			is validated such that the sample that needs to be	
			analyzed does not get over diluted. If any	
			antimicrobials or preservatives are present in the	
			test sample should get sufficiently diluted so that	
			their inhibitory effect is lost.	
			4.2.2.5	Sterilize the bottles by autoclaving, after
			sterilization; store the media at a temperature	
			between 2 °C and 25 °C. Do not use the medium for a	
			longer storage period than has been validated.	
	4.3	Growth Promotion Test (GPT): Each batch of medium prepared either		
		from dehydrated medium or from ingredients has to be tested for		
		growth promotion test of aerobes, anaerobes and fungi.		
	4.3.1	GPT for Fluid Thioglycolate Media:		
			4.3.1.1	Prepare suspensions of Clostridium sporogenes,
			Pseudomonas aeruginosa and Staphylococcus aureus and	
			inoculate each suspension into separate containers	
			of FT media.	
			4.3.1.2	Inoculate each suspension as not more than 100 CFUs.
			4.3.1.3	Incubate the containers at 30–35 °C for next 3 days.
			4.3.1.4	The media is considered suitable for use if a
			clearly visible growth of the microorganisms occurs	
			at the end of incubation period.	
	4.3.2	GPT for Soyabean Casein Digest Medium:		
			4.3.2.1	Prepare suspensions of Aspergillus brasiliensis,
			Bacillus subtilis and Candida albicans. and	
			inoculate each suspension into separate containers	
			of SCD media.	
			4.3.2.2	Inoculate each suspension as not more than 100 CFUs.
			4.3.2.3	Incubate the containers at 20–25 °C for not more
			than 3 days in the case of bacteria and not more	
			than 5 days in the case of fungi.	
			4.3.2.4	The media is considered suitable for use if a
			clearly visible growth of the microorganisms occurs	
			at the end of incubation period.	
		Strains of the test microorganisms suitable for use in the Growth		
		promotion test:		
		Aerobic bacteria:		
			Staphylococcus aureus	ATCC 6538, CIP 4.83, NCTC
			10788, NCIMB 9518, NBRC 13276,	
			NBRC 13276	
			Bacillus subtilis	ATCC 6633, CIP 52.62, NCIMB
			8054, NBRC 3134	
			Pseudomonas aeruginosa	ATCC 9027, NCIMB 8626, CIP
			82.118, NBRC 13275	
			Anaerobic bacterium	
			Clostridium sporogenes	ATCC 19404, CIP 79.3, NCTC 532
			or ATCC 11437, NBRC 14293	
			Fungi	
			Candida albicans	ATCC 10231, IP 48.72, NCPF
			3179, NBRC 1594	
			Aspergillus brasiliensis	ATCC 16404, IP 1431.83, IMI
			149007, NBRC 9455	
	4.3.3	Execution of the test for liquid culture media:		
			4.3.3.1	Test each sterilized lot of the liquid media.
			Transfer 1 ml of inoculum in each test bottle of	
			dilution having 10-100 CFU/ml of each test	
			microorganism listed in the Table, and its previous	
			dilution.	
			4.3.3.2	Simultaneously transfer 0.1 ml of the selected
			dilution having 10-100 CFU/ml on Tryptone Soya Agar	
			Plate, which will act as control plate, with the	
			purpose of checking the viability and the presence	
			of contamination in the Inoculum.	
			4.3.3.3	Spread the inoculum with the help of spreader and
			thoroughly spread it to dry	
			4.3.3.4	Incubate the inoculated media bottles and Petri
			plates according to the requirement of test	
			microorganisms.	
			4.3.3.5	Count the cfu in the control plates, calculate the
			productivity ratio and % efficiency as described	
			below. Observe the turbidity in case of the	
			bottles and record the findings.	
			The Productivity Ratio is calculated in the following	
			way:	
			P.R = A x B	
			C x D	
			where: A: Average Count in the Test Plate	
			B: Dilution Factor of Test Plate	

				C: Average Count in the Control Plate				
				D: Dilution factor of the Control Plate				
				Calculate the % Efficiency as follows:				
				$\% \text{ Efficiency} = P.R \times 100$				
	4.3.5	Procedure for checking the Growth Promotion Properties of						
		Culture Medium/Media in case of Clost. sporogenes:						
		4.3.5.1	Take 48 hours growth of C. sporogenes in cooked meat					
			medium. Make serial dilution of the culture up to					
			10-8 dilution to determine the dilution containing					
			10 to 100 cfu.					
		4.3.5.2	Transfer 0.1 ml aliquot of the dilutions 10-5, 10-6,					
			10-7 and 10-8 in duplicate to Tryptone Soya Agar					
				with Glucose) to confirm the viability and number				
				of organisms in the dilution.				
		4.3.5.3	Place the plates in Anaerobic Gas Jar in an inverted					
			position.					
		4.3.5.4	Place an oxygen-absorbing pack inside the Gas jar					
			for maintaining an anaerobic environment and for the					
			detection of an anaerobic environment in the Gas jar					
				place an indicator tablet. Use the pack and				
				indicator tablet according to the supplier's				
				directions. Properly seal the Gas jar.				
		4.3.5.6	Incubate the Gas jar with plates at $36 \pm 2^\circ\text{C}$ for 48					
			to 72 hours.					
		4.3.5.7	Count the CFU in the plates and the growth should be					
			verified by Gram Staining for its purity and					
			determination of any contamination.					
		4.3.5.8	In case of Fluid Thioglycollate Medium, transfer 1					
			ml aliquot of the dilution having 10–100 CFU and the					
			preceding dilution in the respective bottles and					
			incubate at $33.0 \pm 2^\circ\text{C}$ for 48-72 hours					
		4.3.5.9	Observe for growth/turbidity at the completion of					
			the incubation period.					
	4.3.6	Interpretation of the results:						
		4.3.6.1	The test media are satisfactory if clear evidence of					
			growth or turbidity appears in the test media					
			container within specified period for bacteria and					
			fungal cultures as per specification. Follow the					
			table to interpret the results.					
				Sr.No	Observation	Test	Control	Result
					bottle	plate		
				1.	Growth	Positive	Positive	Passes
				2.	Growth	Negative	Positive	Fails
				3.	Growth	Negative	Negative	Repeat
					the test			
				4.	Negative	Negative	Positive	Passes
					control *			
		4.3.6.2	The growth should be verified by Gram Staining for					
			its purity and determination of any					
			contamination.					
	4.4	Control of the culture media:						
	4.4.1	Control of Fluid Thioglycollate Medium and Tryptone Soya						
		Broth (used in Sterility Testing):						
		4.4.2	Incubate the entire sterilized lot of the medium, Fluid					
			Thioglycollate at $33.0 \pm 2^\circ\text{C}$ and Tryptone Soya Broth at $22.5 \pm 2^\circ\text{C}$, for 48-72 h.					
		4.4.3	If 2 % or more of the batch is contaminated discard the					
			whole batch after decontamination otherwise re-incubate two					
			bottles of each media and observe for sterility for 14 days					
			along with the Sterility test.					
		4.4.4	However, the sterility test of the test media is considered					
			invalid if the growth promotion test fails.					
	4.5	Preliminary Operations for Sterility test:						
	Switch "ON" the laminar airflow cabinet for at least half an hour							
	before starting with the test.							
	4.5.1	Preparation of Material and equipment for sterility testing:						
		4.5.1.1	Label the previously prepared media bottles for					
			Fluid Thioglycollate Medium and Tryptone Soya Broth					
			Medium with the sample name and code.					
		4.5.1.2	Remove the first paper wrapping from all the					
			sterilized materials required in the test and place					
			them in the material transfer hatch of sterility					
			testing room from the sterilization and material					
			preparation laboratory.					
		4.5.1.3	Remove the second wrapping from the materials					
			whenever required, inside the laminar airflow					
			cabinet.					
	4.6	Procedure for performing the Sterility Test by membrane filtration						
	method:							
	4.6.1	Ensure that all three valves of the sterilized membrane						
			filtration assembly are in closed position.					
	4.6.2	Connect a sterilized filtration flask to the drain pipe of						
			the sterilized membrane filtration assembly.					
	4.6.3	Open the filtration funnels and place pre-sterilized 0.45						
			micron membrane filters with the help of sterile forceps.					
			Close the funnel lid.					
	4.6.4	Open the lid of the manifold of membrane filtration assembly						
			and add approximately 5 ml of sterile purified water/rinse					
			solution in all the funnels to wet the membrane filters.					
	4.6.5	Switch "ON" the vacuum Pump and open the valves of membrane						
			filtration assembly for drainage.					

	4.6.6	Close the valves and switch “OFF” the vacuum pump. Add the	
		required amount of sample. Now rinse with enough volume of	
		sterile rinse solution for ARM and Final Bulk of Vaccine.	
		Switch ON the vacuum pump and allow it to pass through the	
		membrane filter.	
	4.6.7	After filtration switch “OFF” the vacuum pump and open the	
		latch of the funnel. Aseptically remove the membrane(s) from	
		the holder, transfer the membrane in Fluid Thioglycollate	
		Medium and in Tryptone Soya Broth Medium	
	4.6.8	System suitability for membrane filtration method	
		After transferring the content of the container or	
		containers under test to the membrane, add an inoculum of a	
		small number of viable microorganisms (not more than 100	
		CFU) to the final portion of sterile diluent used to rinse	
		the filter.	
	4.7	Sterility by direct inoculation method	
	4.7.1	With sterile syringe(s) or pipette(s) withdraw the test	
		sample as per the specification for the test sample.	
	4.7.2	Dispense 1 ml aliquots of the sample in one bottle each of	
		Fluid Thioglycollate Medium and Tryptone Soya Broth	
		Medium/SCDM media.	
	4.7.3	System suitability for direct inoculation method	
		After transferring the contents of the container or	
		containers under test to the culture medium, add an inoculum	
		of a small number of viable microorganisms (not more than	
		100 CFU) to the medium.	
	4.7.4	In both system suitability tests, use the same	
		microorganisms as those described above under Growth	
		promotion test of aerobes, anaerobes and fungi.	
	4.7.5	Incubation conditions:	
		Incubate bottles containing Tryptone Soya Broth Medium at	
		22.5 + 2.5°C and those containing Fluid Thioglycollate	
		Medium at 32.5 + 2.5°C for 14 days.	
	4.7.6	Minimum quantity of sample to be used for each medium	
		Quantity per container	Minimum quantity to be used for
		each medium unless otherwise	
		Justified and authorized	
	Liquids		
		Less than 1 ml	The whole contents of each
		container	
		1-40 ml	Half the container of each
		container but not less than 1 ml	
		Greater than 40ml and	20ml
		not greater than 100ml	
		greater than 100ml	10 % of the content of the
		container but not less than 20ml	
		Antibiotic liquids	1ml
	Solids		
		Less than 50mg	The whole contents of each
		container	
		50 mg or more but not	Half the contents of each
		less than 300mg	container but not less than 50 mg
		300 mg -5 g	150mg
		Greater than 5g	500mg
	Final Lot (Vials, ampoules, PFS)		
	Batch Size	Sample quantity for sterility	
		assay	
		Not more than 100	10% or 4 containers whichever is
		containers	greater
		More than 100 but less	10 containers
		than 500 containers	
		More than 500 containers	2% or 10 containers whichever is
		less	
	4.8	Observations and Interpretation of the results:	
	4.8.1	During the incubation period and at its conclusion, examine	
		the media for evidence of microbial growth/turbidity.	
	4.8.2	When the material/sample being tested renders the medium	
		turbid so that the presence or absence of microbial growth	
		cannot be determined by visual examination, then transfer 1	
		ml of the medium to fresh bottles of the same medium at the	
		end of incubation period of 3 days.	
	4.8.3	Incubate the original and fresh bottles for a period of next	
		14 days.	
	4.8.4	If no evidence of microbial growth is found, the product	
		examined complies with the test for sterility.	
	4.8.5	If evidence of microbial growth is found the product	
		examined does not comply with test for sterility, unless it	
		can be clearly demonstrated that the test was invalid for	
		causes unrelated to the examined product.	
	4.8.6	If test is declared to be invalid, it is repeated using	
		sterility retention sample.	
	4.8.7	If no evidence of microbial growth is found in the sterility	
		repeat test, the product examined complies with the test for	
		sterility.	
	4.8.8	If microbial growth is found in the repeat test, the product	
		examined does not comply with the test for sterility.	
	4.9	Bacteriostasis and Fungistasis assay:	
	4.9.1	Before starting to analyze of a new product it should be	
		ensured that no inhibitory substances for bacterial or	
		fungal growth are present in it, therefore, it is necessary	
		to demonstrate that no inhibitory substances are present or	
		the test conditions are to be adjusted / modified to nullify	

			these effects. The bacteriostasis and fungistasis assay	
			shall be carried out one time.	
		4.9.2	Preparation of the inoculum for Bacteriostasis and	
			Fungistasis assay:	
			Take slants of strains, listed in Table 1 below, from	
			culture bank stored at 2-8°C. Streak them on fresh slants of	
			Tryptone Soya Agar and incubate at the temperature optimum	
			for that strain as mentioned in the respective dossier. To	
			the growth, add 9 ml of Peptone Water Saline and mix gently.	
			From this suspension make serial dilutions in Peptone Water	
			Saline for all organisms. Take Tryptone Soya Agar Petri	
			plates and spread dilutions 10⁻⁵ to 10⁻⁸. Incubate at the	
			optimum temperature for that microorganism as mentioned in	
			the respective dossier for the microorganism and enumerate	
			the Colony Forming Units at the end of the incubation	
			period. Select the dilution that has 10 to 100 CFU/ml	
		TABLE 1		
		Media	Microorganisms	
		Fluid	Bacillus subtilis	ATCC 6633, CIP 52.62, NCIMB 8054,
		Thioglycollate		NBRC 3134
		Medium		
		Clostridium sporogenes	ATCC 19404, CIP 79.3, NCTC 532 or	
		ATCC 11437, NBRC 14293		
		Pseudomonas aeruginosa	ATCC 9027, NCIMB 8626, CIP	
		82.118, NBRC 13275		
		Candida albicans	ATCC 10231, IP 48.72, NCPF 3179,	
		NBRC 1594		
		Staphylococcus aureus	ATCC 6538, CIP 4.83, NCTC 10788,	
		NCIMB 9518, NBRC 13276, NBRC		
		13276		
		Tryptone Soya	Bacillus subtilis	ATCC 6633, CIP 52.62, NCIMB 8054,
		Broth Medium		NBRC 3134
		Candida albicans	ATCC 10231, IP 48.72, NCPF 3179,	
		NBRC 1594		
		Aspergillus	ATCC 16404, IP 1431.83, IMI	
		brasiliensis	149007, NBRC 9455	
		4.9.3	Perform the test with the same volume of each medium as is	
			to be used in the sterility test and introduce in it the	
			same volume of sample whose sterility is to be checked as in	
			the main test. Also inoculate 1 ml aliquot of the dilution	
			containing 10 to 100 viable microorganisms/ml of each test	
			microorganism mentioned in Table 1. Incubate all bottles of	
			Fluid Thioglycollate Medium at 33.0 ± 2°C and that of	
			Tryptone Soya Broth at 22.5 + 2°C. Observe for appearance of	
			growth for 3 days in case of bacteria and 5 days in case	
			fungi.	
		4.9.4	Control: Take same volume of each medium as is to be used in	
			the sterility test. Inoculate it with 1 ml of the dilution	
			containing 10 to 100 viable microorganisms per ml of each	
			test microorganism mentioned in Table 1. Incubate all	
			bottles of Fluid Thioglycollate Medium at 33.0 + 2°C and	
			that of Tryptone Soya Broth at 22.5 + 2°C. Incubate for 3	
			days in case of bacteria and for 5 days for fungi.	
		4.9.5	Interpretation of results	
		4.9.5.1	If clearly visible growth of microorganisms is	
			obtained after the inoculation, visually comparable	
			to that in the control bottle, the test for	
			sterility can be carried out without further	
			modification in the amount of the sample and in the	
			medium to be used.	
		4.9.5.2	If clearly visible growth is not obtained in	
			presence of the sample under examination, visually	
			comparable to that in the control bottle without the	
			sample, the sample possesses Bacteriostasis and	
			Fungistasis activity. A neutralising agent may be	
			used adjusting the concentration of neutralizing	
			agent till the inhibitory effect of Bacteriostasis	
			and Fungistasis activity is completely neutralized.	
		4.9.5.3	If a neutralizing agent is not available modify the	
			amount of the sample and medium to be used in order	
			to eliminate Bacteriostasis and Fungistasis activity	
			and repeat the test.	
		4.9.5.4	When the sample exhibits antimicrobial activity in	
			the medium, increase the quantity of the medium so	
			that, the sample is sufficiently diluted to prevent	
			inhibition of growth	
		4.9.5.5	If no growth is present in both the control and test	
			bottles, repeat the test.	
		5.	References:	
		5.1	General Requirements for the Sterility of Biological Sub stances,	
		World Health Organization Technical Report Series No. 530, 1973.		
		5.2	United States Pharmacopeia, National Formulary.	
		5.3	Indian Pharmacopeia	
		5.4	European Pharmacopeia	

1. Purpose:

To analyze purified protein samples of CRM197 for the determination of percentage purities by Size Exclusion High Performance Liquid Chromatography using PDA detector.

2. Scope:

This standard testing procedure is applicable to the purified protein samples of CRM197.

3. Responsibility:

1. Qualified Analysts shall be responsible for the preparation and execution of this STP.

2. Group leader shall supervise and check that the STP is followed properly.

3. Head, R&D and Incharge, QA shall be responsible for the implementation and compliance of this STP.

4. Materials & Equipments:

1. Purified CRM1 97 Samples

2. HPLC system (Waters) with PDA detector (Waters)

3. Potassium dihydrogen phosphate anhydrous

4. Dipotassium hydrogen phosphate anhydrous

5. Sodium Chloride

6. Potassium Hydroxide

7. Sucrose

8. TSKgel G2000 SWxl column; 30 cm x 7.8 mm ID (Tosoh Bioscience, Cat. No. 8-08540)

9. TSKgel G2000 SWx1 Guard column; 4 cm x 6.0 mm ID (Tosoh Bioscience, Cat. No. 8-08543)

10. Milli-Q Water

11. HNC screw top vials 12 x 32 mm (Waters, Cat. No. 186000307C)

12. Glass filter assembly (Millipore, 1.5L)

13. Sonicator (Power Sonic 410, HWASHIN Technology Co.)

14. pH meter

15. Vacuum pump

16. 0.22 µ Syringe filter

17. 2.0 ml Disposable syringes

5. Definitions/ Abbreviations:

1. M : Molarity

2. mM : milli Mole

3. µg : Microgram

4. µl : Microliter

5. SE-HPLC : Size Exclusion High Performance Liquid Chromatography

6. PDA : Photo Diode Array

7. KOH : Potassium Hydroxide

6. Procedure:

1. Buffer preparation:

1. Preparation of 100 mM Potassium dihydrogen phosphate solution (1 litre):

To prepare 1000 ml solution of 100 mM Potassium dihydrogen phosphate solution, add 13.609 g of Potassium dihydrogen phosphate in 1000 ml of Milli-Q water and mix well.

2. Preparation of 100 mM Dipotassium hydrogen phosphate solution (500 ml):

To prepare 500ml solution of 100 mM Dipotassium hydrogen phosphate solution, add 8.709 g of Dipotassium hydrogen phosphate in 500 ml of Milli-Q water and mix well.

3. Preparation of 100 mM Potassium hydroxide solution (for pH adjustment):

To prepare 10 ml of 100 mM Potassium hydroxide solution, take 561.1 mg of Potassium hydroxide pellets and dissolve in 10 ml of Milli-Q water.

4. Preparation of 100 mM Potassium phosphate buffer, pH 6.8 (1 litre):

To prepare 1 liter 100 mM solution of Potassium Phosphate buffer (PH-6.8); add 490 ml of 100mM Dipotassium hydrogen phosphate solution and 510 ml of 100mM Potassium dihydrogen phosphate solution.

5. Preparation of 100 mM Potassium phosphate+ 100 mM NaCl working buffer(1 litre):

1. To prepare Potassium phosphate working buffer, add 5.844 g of NaCl in 1 liter of 100 mM Potassium Phosphate buffer, mix well and if required, adjust the pI-I to 6.8 using 1 M KOH solution.

2. Filter the working buffer with 0.22 µ filter and degas using the ultrasonication bath & vacuum pump. It can be stored at room temperature for one month.

2. Preparation of Sample Blank (10mM Potassium phosphate buffer with 5% sucrose):

1. To prepare 100 ml sample blank, take 5.1 ml 100 mM potassium dihydrogen phosphate and mix with 4.9 ml of 100 mN4 Dipotassium hydrogen phosphate solution.

2. Add 90 ml of Milli-Q water and adjust the pH to 6.8 using 1 M KOH solution, if required.

3. Add 5 g of sucrose to 100 ml of buffer as prepared in above mentioned step and filter the solution with 0.2 µ membranes.

(Note: Take Sample blank (10mM Potassium phosphate buffer with 5% sucrose) preferably from the purification group

in order to subtract any possible impurities carried by the sample buffer. Use the same sample1-.41-atk in diluting the concentrated protein samples for injection in HPLC. In case the buffer is not available, prepare the sample blank by the above mentioned method.)

3. Sample preparation for Purified CRM197:

1. If the concentration of the CRM 197 sample is more than 1mg/ml (as analysed by BCA Protein Assay kit), dilute the sample using sample blank to make the final concentration of 1mg/ml. In case protein concentration is less than 1mg/ml (as analysed by BCA Protein Assay kit method), use undiluted samples for HPLC.

2. Filter the sample with 0.22 µ syringe filter before injection.

4. HPLC Run conditions:

1. HPLC SEC Column used: TSKgel G2000 SWx1 column, 30 cm x 7.8 mm ID & TSKgel G2000 SWxi Guard column; 4 cm x 6.0 mm ID.

2. Mobile phase used : 100mM Potassium phosphate ± 100mM NaCl working buffer (pH-6.8)

3. Flow rate used: 0.6 ml/min.

4. Run Time: 30 min.

5. Injection volume: Sample Blank: 25 µl if sample concentration is 1mg/ml or 100 µl if sample concentration is < 1mg/ml.

6. Detector used : PDA

5. HPLC run procedure:

1. Equilibrate the column with 100 mM Potassium phosphate + 100mM NaCl working buffer (pH-6.8) for 1 hour at 200 to 300 nm wavelength of PDA detector.

2. After the PDA detector baseline is stabilized, inject Sample Blank (25 µl injection volume if the sample concentration is 1mg/ml or 100 µl if the sample concentration is less than 1 mg/ml).

3. Inject the purified CRM197 protein sample (25 µl injection volume if the sample concentration is 1 mg/ml or 100 µl if the sample concentration is less than 1 mg/ml).

(Note: Peak of the CRM 197 protein is obtained at 13.7 ± 0.5 minutes.)

6. Results Evaluation:

1. Compare the peak profile of purified CRM 197 protein sample and sample blank injected at same volume and calculate the percentage purity of the sample using the HPLC software.

2. To subtract the sample chromatogram with the sample blank in Waters HPLC, go to the project (Injection Select the Sample blank run file (Right click on the Sample blank file Select the Option Alter Sample (3 Type the text 'blank' in the column named Label 3 Save

3. Go to the Project (Methods (Select the Method set (eg.CRM SEC) (Derived Channel (select Instrument method CRM SEC (Fill the Channel name with the text '280nm' (Make sure, In the first only channel type Constant is 1X.

4. In the Option Channel PDA, select the option Single Wavelength (Type 280nm in the Box Wavelength (Second channel (select (-) for subtraction in the option Operation (Constant 1.0 (Channel 280nm (Click the box from injection labeled and type the text 'blank' in the box (PDA 3D Blank Subtraction (blank (Click the box save extended channel

5. For subtracting Right click on sample (Review (File (Open (Method Set (CRM SEC) (Open (See the chromatogram integrate and calibrates to reveal the peaks and see the calculated % Area for purity analysis.

Note: The absorbance of sucrose at UV wavelength 280 is not specific so even after subtracting the blank containing 5% sucrose from CRM 197 Sample's chromatogram, sometimes sucrose peak may not be completely subtracted. So the peak of sucrose need not to be integrated and should not be considered in determining purity of the samples. If blank subtraction option is not available in the HPLC software then for the determination of % purity of CRM 197 samples, integrate all peaks except the peak of sucrose (identified from the blank run) and determine the % Peak area of the CRM 197 peak, to determine the % purity.

7. References:

Product insert; Operating conditions and specification TSKgel(R) G2000 SWx1 products.

8. Annexures:

Format for "Analysis of Purified CRM 197 protein Samples for determination of percentage purity by Size Exclusion High Performance Liquid Chromatography (SE-HPLC)"

1.0	Objective:		
	To lay down the procedure for determination of pH.		
2.0	Scope:		
	This Standard Operating Procedure is applicable to all the In- process		
	and finished product samples.		
3.0	Abbreviations and Definitions:		
	Not applicable		
4.0	Procedure:		
	4.1	Equipment, Materials and Reagents	
		4.1.1	Tissue paper
		4.1.2	100 ml. vial
		4.1.3	100 ml beaker
		4.1.4	Wash bottle
		4.1.5	Purified water
	4.2	Preparation of solutions:	
		4.2.1	Buffer solution, pH =4.00
		4.2.2	Buffer solution, pH =7.00
		4.2.3	Buffer solution, pH=9.20
	4.3	pH Electrode Calibration:	
		4.3.1	Check that the equipment is correctly connected to electrical
		supply.	
		4.3.2	Switch on the instrument by pressing ON/OFF key on front
		panel.	
		Use a combination of two standard buffer for calibration	
		according to requirements i.e.:	
		Case 1- For solution with pH less than 7.00 uses standard	
		buffers of 4.00 & 7.00.	
		Case 2- For solution with pH more than 7.00 use standard	
		buffers of 7.00 & 9.20	
		(Note: The pH meter shall be set for the different buffers to	
		be used at the time of installation based on the future use.	
		This setting is changed only in case the range of pH readings	
		is different than preset.)	
		4.3.3	Clean pH electrode with purified water and blot dry.
		4.3.4	Put the standard buffers that are used in calibration in
		clean beakers.	
		4.3.5	Introduce pH electrode in first buffer according to case 1 or
		case 2 and wait for 2 seconds to stabilize the pH.	
		4.3.6	Remove pH electrode, rinse with purified water and blot dry
		and put in second standard buffer according to case 1 or case	
		2 and wait for 2 seconds to stabilize the pH.	
		4.3.7	Remove pH electrode, rinse with purified water and blot dry
		and put in potassium chloride solution.	
	4.4	pH Measurement:	
		4.4.1	Clean pH electrode with purified water and blot dry.
		4.4.2	Put the sample in clean beaker to check the pH.
		4.4.3	Introduce the pH electrode in beaker having sample solution
		4.4.4	Press [Read] key for pH measurement.
		4.4.5	Rinse, blot and store the electrode in a beaker having
		saturated KCl solution.	
5.	Reporting:		
	5.1	Record the information of operation and calibration in corresponding	
	log book		
6.	References:		
	NA		

1.0|Objective |
 |To lay down the procedure for the determination of Thiomersal concentration|
 |in Recombinant Hepatitis B vaccine. |
 2.0 |Scope |
 |This Standard Operating Procedure is applicable to the samples of Hepatitis|
 |B Vaccine that contains Thiomersal as a preservative, received in the |
 |Quality control department. |
 3.0 |Responsibility |
 |3.1 |The analyst shall be responsible for executing the described |
 |procedure correctly. |
 |3.2 |Supervisor shall be responsible for verifying that the procedure is |
 |complied with and the results are checked. |
 |3.3 |HOD shall be responsible for overall compliance of this SOP. |
 4.0 |Abbreviations and Definitions |
4.1	Abbreviations:		
	4.1.1	CV	:Coefficient of variation
	4.1.2	g	:Gram
	4.1.3	ID	:Identification
	4.1.4	mg	:Milligram
	4.1.5	mL	:Millilitre
	4.1.6	µL	:Micro litre
	4.1.7	µg	:Microgram
	4.1.8	nm	:Nanometer
	4.1.9	PN	:Part Number
	4.1.10	QA	:Quality Assurance
	4.1.11	QC	:Quality Control
	4.1.12	QCR	:Quality Control Record
	4.1.13	SD	:Standard Deviation
	4.1.14	SOP	:Standard Operating Procedure
	4.1.15	UV	:Ultra Violet
	4.1.16	HOD	:Head of Department
4.2	Definitions:		
		Not Applicable.	
5.0	Procedure:		
5.1	Preparation of Solutions:		
	5.1.1	0.002 % Dithizone Solution :	
		5.1.1.	Weigh 2 mg of Dithizone.
		1	
		5.1.1.	Transfer it to a 100 mL volumetric amber coloured
		2	flask, covered in such a way that no light should reach
		the solution.	
		5.1.1.	Make up the volume to 100 mL using acetone.
		3	
		5.1.1.	This solution should be stored in a refrigerator
		4	(temperature 2 - 8 °C) and used within 24 hours.
	5.1.2	0.01mg / mL Thiomersal :	
		5.1.2.	Weigh 50 mg of Thiomersal
		1	
		5.1.2.	Transfer it to a 100 mL volumetric flask and make up
		2	the volume to 100 mL with purified water (0.5 mg/mL).
		5.1.2.	Take 5 mL of the above solution (0.5 mg/mL) and
		3	transfer it to a 25 mL volumetric flask.
		5.1.2.	Make up the volume to 25 mL with purified water (0.1
		4	mg/mL).
		5.1.2.	Take 5 mL of the above solution (0.1 mg/mL) and
		5	transfer it to a 50 mL volumetric flask.
		5.1.2.	Make up the volume to 50 mL with purified water (0.01
		6	mg/mL).
		5.1.2.	The solution can be stored at room temperature for a
		7	maximum of 6 months.
	5.1.3	50 % Sodium Hydroxide Solution :	
		5.1.3.	Weigh 50 g of sodium hydroxide.
		1	
		5.1.3.	Transfer it to a 100 mL volumetric flask and dissolve
		2	in sufficient amount of purified water and finally make
		up the volume to 100 mL with purified water.	
		5.1.3.	This solution should be used within 6 months and stored
		3	at room temperature in plastic bottle.
5.2	Preparation of Calibration curve:		
	5.2.1	Prepare calibration curve (1-10 µg/mL) with Thiomersal	
		solution (0.01 mg/mL) in 10 labelled test tubes of 20 mL.	
		The solutions for making thiomersal calibration curve are	
		prepared as follows:	
		Final	0.01 mg/ml
		Concentration	Thiomersal
		(µg/ml)	Solution
			(ml)
		1	0.2
		2	0.4
		3	0.6
		4	0.8
		5	1.0
		6	1.2
		7	1.4
		8	1.6
		9	1.8
		10	2.0
		Note: The Calibration curve is accepted if it is meeting the	
		validated parameters of intercept, slope and correlation coefficient.	
5.3	Preparation of the samples:		

		5.3.1	Take 0.2 mL of the test sample and add it in 1.8 mL of
			purified water (1/10 dilution) in duplicate sets in 20 mL of
			labeled test tubes. However reference vaccine treated like
			test sample can be used in the assay as control, occasionally.
		Note: The dilution mentioned can vary depending upon the thiomersal	
		concentration. If there is any sample outside the range of the	
		calibration curve, carry out the calculation for more appropriate	
		dilution.	
		5.3.2	Treat 2 mL of purified water as Blank.
		5.4	Test:
		5.4.1	Add 2 mL of 0.002% dithizone solution to all the tubes of
			Calibration curve, Sample, Control and Blank and mix gently.
		5.4.2	Add 2 mL of acetone to all these test tubes and again mix
			gently using a vortex mixer.
		5.4.3	Now add 0.1 mL of 50% Sodium Hydroxide solution to each of
			these tubes and mix gently using a vortex mixer. The colour of
			the contents of each tube change to light pink.
		5.4.4	Set the wavelength in spectrophotometer at 538 nm. Zero the
			absorbance of blank and then measure the absorbance of
			calibration curve and test samples.
		5.4.5	The concentration of thiomersal in the test sample(s) is
			calculated by comparing results of its absorbance obtained
			against the plotted calibration curve of known thiomersal
			concentration.
		5.5	Calculations:
		5.5.1	Once the absorbance of the standards and the test samples have
			been taken, calculation shall be done using the excel sheet
			carrying formulas for the required calculation.
		5.5.2	Enter the absorbance for calculations starting from
			Calibration curve prepared between 1 and 10 g of Thiomersal
			and test samples. The calculations are made according to the
			equation of minimum squares.
		5.5.3	For calculating the Thiomersal concentration use the equation
			$Y = mX + b$.
			Where: m = slope
			b = intercept
			Y = absorbance
			X = concentration
		5.5.4	Record all the calculations in the data record form and also
			attach the hard copy of the excel sheet used for calculation
			along it.
		5.6	Acceptance criteria:
		5.6.1	The coefficient of variation (CV) between the replicates of
			the samples should be $\leq 5\%$.
		5.6.2	The value of the control should be within the acceptance
			limits.
		5.6.3	Thiomersal concentration in the sample should be within the
			specified limits.
		6.0	References
		6.1	A new method for Spectrophotometric Determination of Thiomersal in
			Biologicals: Shrivastaw, K.P and Sarbjit Singh. Biologicals (1995) 23,
			65-69.

1.0	Objective		
	To lay down the procedure for specific toxicity test on bulk purified		
	tetanus toxoid.		
2.0	Scope		
	Applicable to the batches of bulk purified tetanus toxoid.		
3.0	Responsibility:		
	3.1	The QC analyst shall be responsible for performing the described	
	procedure correctly.		
	3.2	Supervisor shall be responsible for verifying that the same has been	
	carried out correctly.		
	3.3	Head Animal House/Veterinarian/supervisor shall be responsible for	
	supply of quality animals and their maintenance.		
	3.4	Head QC or his designee shall be responsible for overall compliance	
	of the SOP.		
4.0	Abbreviations and Definitions		
	4.1	g	: Grams
	4.2	°C	: Degree Celsius
	4.3	%	: Percent
	4.4	IU/ml	: International unit per milliliter
	4.5	QA	: Quality Assurance
	4.6	SOP	: Standard Operating Procedure
	4.7	Lf/ml	: Limes-flocculation per milliliter
5.0	Procedure:		
	5.1	Equipment, Materials and Reagents	
		5.1.1	Weighing balance
		5.1.2	Glass Beakers (10ml, 50ml, 100ml, 500ml)
		5.1.3	Sodium Chloride
		5.1.4	Disposable syringes
	5.2	Preparation of Normal Saline:	
		Weigh 0.9 g of sodium chloride and dissolve in 60 ml of the purified	
	water. Make up the volume up to 100 ml with purified water. Adjust		
	the pH at 7 ± 0.1 . Sterilize by autoclaving at 121°C for 20		
	minutes.		
	5.3	Preparation of test animals:	
		5.3.1	Select five Guinea pigs each weighing between 250 -350 g
		5.3.2	Mark each Guinea pig with appropriate color scheme.
	5.4	Preparation of toxoid dilution:	
		Appropriately dilute the bulk purified tetanus toxoid in sterile	
	normal saline so that it contains 500 Lf per ml.		
	5.5	Injection:	
		Inject subcutaneously 1 ml of the toxoid diluted to 500 Lf/ml into	
	each of the five guinea pigs.		
	5.6	Observations:	
		Observe daily for three weeks all the injected Guinea pigs for any	
	symptoms of tetanus intoxication, Record the result.		
	5.7	Acceptance criteria	
		The bulk purified tetanus toxoid passes the test if no guinea pig	
	shows paralysis or any other signs of tetanus within 3 weeks of		
	injection and at least 80 % of the animals survive the test period.		
6.0	Forms and Records		
	Form for reporting the results of specific toxicity test on bulk purified		
	tetanus toxoid		
7.0	References:		
	WHO manual for the production and control of vaccines – Tetanus Toxoid		
	(BLG/UNDP/77.2 Rev 1)		

1.0 |Objective |
|To lay down the procedure for identification Bordetella Pertussis (wCP) |
|Agglutinin 1, 2 & 3 by slide agglutination method. |
2.0 |Scope |
|This Standard Operating Procedure is applicable to all the samples of |
|Bordetella pertussis(wCP) at Cell bank, In process and ARM stages. |
3.0 |Responsibility |
3.1	The Analyst shall be responsible for carrying out the procedure as
	established in this SOP.
3.2	Supervisor shall be responsible for verifying that the same has been
	carried out correctly.
3.3	Head QC or his designee shall be responsible for overall compliance
	of the SOP.
4.0	Abbreviations and Definitions
4.1	SOP
4.2	QA
4.3	QC
4.4	I.O.U
4.5	mL
4.6	µL
5.0	Procedure:
5.1	Equipment, Materials and Reagents
	5.1.1
	5.1.2
	5.1.3
	5.1.4
	5.1.5
	5.1.6
	5.1.7
	5.1.8
	5.1.9
5.2	Preparation of solutions
	5.2.1
5.3	Test procedure
	Identity test for solid media
	culture:
	5.3.1
	5.3.2
	5.3.3
	5.3.4
	5.3.5
	5.3.6
5.4	Identity test for Liquid media culture:
	5.4.1
	5.4.2
	5.4.3
	5.4.4
	5.4.5
	5.4.6
5.5	Validity criteria
	5.5.1
6.0	Forms and Records
	Not applicable
7.0	References
WHO manual for the production and control of vaccines – Pertussis Vaccine	
(BLG/UNDP/77.2 Rev 1)	

1.0	Objective							
	To lay down the procedure for the determination of Total Lipid content in							
	Active Raw Material (ARM) of recombinant Hepatitis B vaccine.							
2.0	Scope							
	This Standard Operating Procedure is applicable to all the samples of							
	Active Raw Material of recombinant Hepatitis B vaccine, , received in the							
	Quality control department.							
3.0	Responsibility							
	3.1	The analyst shall be responsible for executing the described						
		procedure correctly.						
	3.2	Supervisor shall be responsible for verifying that the procedure is						
		complied with and the results are checked.						
	3.3	HOD shall be responsible for overall compliance of this SOP.						
4.0	Abbreviations and Definitions							
	4.1	Abbreviations:						
		4.1.1	ARM	:Active Raw Material				
		4.1.2	CV	:Coefficient of variation				
		4.1.3	g	:Gram				
		4.1.4	ID	:Identification				
		4.1.5	mg	:Milligram				
		4.1.6	ml	:Millilitre				
		4.1.7	µl	:Micro litre				
		4.1.8	µg	:Microgram				
		4.1.9	nm	:Nanometer				
		4.1.10	RT	:Room Temperature				
		4.1.11	SD	:Standard Deviation				
		4.1.12	SOP	:Standard Operating Procedure				
	4.2	Definitions:						
		Not Applicable.						
5.0	Procedure:							
	5.1	Preparation of Solutions:						
		5.1.1	Extraction Mixture:					
			5.1.1.1	Prepare the extraction mixture with two volumes of				
				1	methanol and one volume of chloroform.			
		5.1.2	1mg/ml Cholesterol Solution:					
			5.1.2.1	Weigh 25 mg of Cholesterol and transfer it to a 25 ml				
				1	volumetric flask.			
			5.1.2.2	Make up the volume to 25 ml using Chloroform.				
				2				
			5.1.2.3	This solution must be prepared freshly.				
				3				
		5.1.3	13 mM Vanillin Solution :					
			5.1.3.1	Prepare 13 mM of Vanillin solution in 14 M of				
				1	Phosphoric acid.			
			5.1.3.2	Weigh 1.9779 g of Vanillin and dissolve it in 944 ml of				
				2	Phosphoric acid in 1 liter amber colour Volumetric			
				flask.				
			5.1.3.3	Make up the volume to 1000 ml with purified water and				
				3	mix with magnetic stirrer for twelve hours.			
			5.1.3.4	Protect from light and store this solution at Room				
				4	temperature and use within 6 months.			
	5.2	Preparation of Calibration curve:						
		5.2.1	Prepare calibration curve (50-250µg) with freshly prepared					
			cholesterol solution (1 mg/ml) in 5 labeled test tubes of 10					
			ml as follows:					
			Concentration	50	100	150	200	250
			(µg)					
			Cholesterol	50	100	150	200	250
			Solution 1					
			mg/ml (µl)					
			Evaporate the contents in respective test tubes to dryness by					
			purging nitrogen gas.					
		Note: The Calibration curve is accepted if it is meeting the						
		validated parameters of intercept, slope and correlation coefficient.						
	5.3	Test:						
		5.3.1	Take 0.1 ml of the Sample in a test tube (this volume may					
			vary, depending on the concentration of lipids in the sample).					
			However reference ARM treated like test sample can be used in					
			the assay as control, occasionally.					
		5.3.2	Add 2 ml of concentrated sulphuric acid slowly to the test					
			Sample, Control as well as the samples of calibration curve					
			containing dried cholesterol.					
		5.3.3	Mix thoroughly by shaking the tubes gently, cover the tubes					
			with cotton plug and put it in a boiling water bath for ten					
			minutes.					
		5.3.4	Allow the tubes to cool at room temperature					
		5.3.5	Pipette out 0.1 ml of the reaction mixture from each tube and					
			add to the new test tubes (perform the Sample, Control as well					
			as Calibration curve in duplicate).					
		5.3.6	Treat the blank in parallel using Concentrated sulphuric acid.					
		5.3.7	Add 2.5 ml of vanillin solution to all the tubes and keep them					
			at resting position for about 30 minutes.					
		5.3.8	Measure the absorbance at 530 nm against the Blank.					
		5.3.9	In case of samples having lipid content lower than the range					
			of standard curve (50-250 µg) then extraction must be done.					
	5.4	Extraction of Sample:						
		5.4.1	Place 1 ml of sample in a separating funnel and add 3.75 ml of					
			extraction mixture to it.					
		5.4.2	Shake the separating funnel vigorously					
		5.4.3	Add 1.25 ml of Chloroform to it and again shake vigorously.					

		5.4.4	Allow the funnel to stand still for some time to get two
			separate phases
		5.4.5	Collect the mixture settled at the bottom of funnel into a
			test tube.
		5.4.6	Evaporate and dry the collected sample by purging with
			nitrogen gas and re-dissolve it in 250 µl of chloroform.
		5.4.7	Again evaporate and dry the sample with nitrogen gas.
		5.5	Calculations:
		5.5.1	Once the absorbance of the standards and the test samples have
			been taken, calculation shall be done using the excel sheet
			carrying formulas for the required calculation.
		5.5.2	If the assay is performed with 100 µl of the sample, the
			result is to be multiplied by 10.5 (2.1/2 * 10) in order to
			correct the volume and in order to convert it to ml.
		5.5.3	If the assay is performed with 200 µl of the sample, the
			result is to be multiplied by 5.5 (2.2/2 * 5) in order to
			correct the volume and in order to convert it to ml.
		5.5.4	If extraction of sample is performed than the result is to be
			multiplied with 4 in order to convert it to ml.
		5.5.5	Enter the absorbance for calculations starting from
			Calibration curve prepared between 50 and 250 g of Cholesterol
			and test samples. The calculations are made according to the
			equation of minimum squares.
		5.5.6	$Y = mX + b$.
			Where: m = slope
			b = intercept
			Y = absorbance
			X = concentration
			Final lipid concentration is calculated by multiplying the
			value of X obtained from linear line equation with the
			conversion factor as per the volume of sample taken and the
			procedure used.
			$X_f = X * (\text{Conversion factor})$
		5.5.7	Taking into account the equivalence between the ARM volume and
			the antigen concentration, express the result per 20 g of
			protein. Enter the protein concentration estimated by OD at
			280nm in the excel sheet.
		5.5.8	If the Protein concentration of Active Raw Material of
			Hepatitis B Vaccine by Absorbance at 280 nm. is 1.40 mg/ml
			calculations are done as given below:
		5.5.9	$C = 20 \times X_f / 1.40 / 1000$
		5.5.10	Where:
			C = Carbohydrate Concentration per 20µg of Protein
			20 = Dose of Hepatitis B vaccine (20µg)
			Xf = Total concentration of Lipids calculated (µg/ml) after
			multiplying with the conversion factor.
			1.40mg/ml = Concentration of protein in the sample by
			Absorbance at 280 nm)
			1000 = Conversion factor
		5.5	Acceptance criteria:
		5.5.1	The coefficient of variation (CV) between the replicates of
			the samples $\leq 5\%$.
		5.5.2	The value of the control should be within the acceptance
			limits.
		5.5.3	Lipid content in the sample should be within the specified
			limits.
	6.0	References	
	6.1	Zollner, N. Y K. Kirsch, Z. exp. Med. 135 (1962), p. 545.	

1.0 |Purpose: |
 |To lay down a procedure for quantifying the protein concentration in HIB-TT|
 |conjugate samples received from the production process. |
 2.0 |Applicable to: |
 |This Standard Operating Procedure is applicable to all the samples of |
 |HIB-TT conjugate received from the production process. |
 3.0 |Equipment: |
 |3.1 |HIB-TT |: Haemophilus influenzae type b with tetanus toxoid |
 4.0 |Procedure: |
4.1	Materials and reagents:							
	4.1.1	Reagents						
	4.1.2	Copper Sulphate Pentahydrate						
	4.1.3	Sodium and potassium tartrate pentahydrate						
	4.1.4	Folin-Ciocalteu reagent 2N						
	4.1.5	Anhydrous Sodium Carbonate						
	4.1.6	Purified water						
	4.1.7	Lyophilized Bovine Serum Albumin						
	Materials:							
	4.1.9	20, 200, 1000 and 5000µl Automatic micropipettes, Pipette tips						
	4.1.10	5 or 10 ml Measuring cylinders						
	4.1.11	Thermo Resistant Test tubes (with a maximum capacity of 10 ml)						
	4.1.12	500 and 1000 ml Volumetric flasks (amber coloured)						
	4.1.13	25, 50 and 100 ml Beaker						
	4.1.14	3 ml Quartz cuvettes (1 cm width)						
	4.1.15	Tissue paper						
	4.1.16	Screw capped bottles (amber colored)						
4.2	Preparation of Solutions:							
	4.2.1	Alkaline Copper solution:						
		4.2.1.1	Dissolve 40 g of anhydrous sodium carbonate in 400 ml					
			1	of purified water.				
		4.2.1.2	Dissolve 0.6 g of sodium and potassium tartrate in 100					
			2	ml of purified water and add 0.46 g of copper sulphate				
			pentahydrate to it , shake untill it dissolves					
			completely.					
		4.2.1.3	Mix these two solutions gently. Transfer it to a 1000					
			3	ml volumetric flask and make up the volume to 1000 ml				
			with purified water. This solution should be stored in					
			an amber coloured bottle with a ground glass stopper at					
			2 - 8°C and should be used within 2 months .					
	4.2.2	1 N Folin-Ciocalteu reagent:						
		4.2.2.1	Dilute 2N Folin-Ciocalteu reagent to 1N. For this,					
			1	take 250 ml of the reagent and pour it in a 500 ml				
			amber flask. Make up the volume to 500 ml with purified					
			water. Store the solution in a screw cap amber flask at					
			2 to 8 °C, for a maximum of 6 months.					
	4.2.3	Lyophilized Bovine serum albumin reference material: 2 mg/vial						
		(for the calibration curves):						
		4.2.3.1	Prepare this solution according to the procedure					
			1	indicated in its documentation .The Reference material				
			of HIB-TT conjugate samples (as the control of the					
			technique) should be prepared according to the					
			procedure described earlier.					
4.3	Procedure:							
	4.3.1	Preparation of the Calibration Curve:						
		Reconstitute the reference material that is Bovine Serum						
		Albumin(BSA) in 2 ml of purified water						
		Now dilute the above material 10 times. This is done by taking						
		0.4 ml of BSA and adding to it 3.6 ml of purified water. A						
		final concentration of 100 µg/ml can be obtained .This is						
		further used in the preparation of low range calibration curve						
		(10-100 µg/ml).						
		Prepare the dilutions in test tubes, prepare a single dilution						
		for each concentration in a single tube, the volume should be						
		enough for the preparation of standard curve in duplicate.						
		The volumes taken for the preparation of Low Range Calibration						
		Curve are as follows:						
		Low range (10-100 µg/ml)						
		Water (ml)	0.9	0.8	0.6	0.4	0.2	-
		RM BSA 100 µg/ml (ml)	0.1	0.2	0.4	0.6	0.8	1.0
		Concentration (µg/ml)	10	20	40	60	80	100
		To obtain the high range calibration curve points, prepare the working						
		dilutions of Bovine Serum Albumin (reference material) at a						
		concentration of 1mg/ ml .						
		high range (100-500 µg/ml)						
		Water (ml)	0.9	0.8	0.7	0.6	0.5	
		RM BSA 1 mg/ml (ml)	0.1	0.2	0.3	0.4	0.5	
		Concentration (µg/ml)	100	200	300	400	500	
		Note:						
		Prepare the calibration curve whenever there is any change in the test						
		conditions (as when the new batches of reagents are prepared,						
		recalibration of the equipment, change in the analysts or new batch of						
		reference material etc.)						
		The test is accepted if the parameters of the intercept, the slope and						
		the correlation coefficient are found within the acceptance limits.						
		Record the results of the calibration curve.						
4.3.2	Treatment of samples with possible interference:							
		Each sample is treated according to the composition of elements that						
		may interfere in determination of protein concentration (Table 1). If						
		one or more interfering elements are present in the sample, carry out						
		the treatment, washing or acid precipitation as described below.						

	Table 1. Substances that interfere with Lowry test.			
	The numbers in parenthesis indicate the concentration limit that			
	should be present (according to the reports of the literature) (2).			
	Buffers			
	Tris (100 µmol/L), Phosphate (100 mmol/L), Tricine (100µmol/L), MOPS			
	(10 µmol/L), Hepes (1 µmol/L), Bicine (5 µmol/L), Citrate (1 mmol/L),			
	Glycylglycine (100 µmol/L).			
	Amino acids			
	Histidin (150 mmol/L), Tyrosine (150 mmol/L), Tryptophan (150 mmol/L),			
	Cysteine (40 µmol/L), Glycine (1 mmol/L), Glutamate (value not			
	described in the literature).			
	Carbohydrates			
	Saccharose (4 mmol/L), Monosaccharides (glucose, fructose, manose,			
	xilose, ramnose) (150 µmol/L), Glucosamine (5 mg/L), Glycerol (100			
	mL/L), Ethylenglycol (1 g/L), Ficoll, Polyvinylpyrrolidone.			
	Reductors			
	Phenol (except the nitrophenol), Dithiothreitol (20 µmol/L), Glutathion			
	(40 µmol/L), 2-Mercaptoethanol (700 µmol/L), Hydrazine (5 mg/L).			
	Ascorbic acid (100 µg/L).			
	Salts, metals, etc.			
	(NH₄)₂SO₄ (1.5 g/L), K⁺ (12 mmol/L), Hg²⁺ (15 µmol/L), Mn²⁺ (15			
	µmol/L), Co²⁺ (1 µmol/L), NaCl (0.7 mol/L), NaI (15 mmol/L)			
	Miscellaneous			
	EDTA (70 µmol/L), Haematin (15 µg/L), Triton X-100 (10 mg/mL),			
	oxidated lipids (10-200 µmol/L), Fatty acids (1 mmol/L), Ampholine,			
	Salicilate (20 µg/L), Penicillin (0.5 units/mL), Acetyl acetone (2			
	µmol/L).			
	4.3. Preparation of the samples and control:			
	3			
	4.3.3.1 Prepare the blank with water by substituting the volume			
	used for the sample.			
	4.3.3.2 For Preparing the control of HIB-TT conjugate samples,			
	dilute it 5 fold .			
	For example: Take 0.2 ml of HIB-TT conjugate samples and add 0.8 ml			
	of water to it. Prepare appropriate dilutions of the samples			
	accordingly. The dilutions are shown in the following table:			
	Sample	Dilution	Volume of the	Volume of water
		factor	sample (ml)	(ml)
	HIB-TT conjugate	1/5	0.200	0.800
	samples (ARM)			
	Note:			
	The dilutions mentioned can vary depending upon the protein			
	concentration. If there is any sample outside the range of the			
	calibration curve, carry out the calculation for more appropriate			
	dilution.			
	4.3. Coloured reaction:.			
	4			
	4.3.4.1 For the development of the colored reaction the solutions			
	should be mixed in the following proportion:			
	Sample	1.0 volume		
	Copper alkaline solution	5.0 volume		
	1N Folin reagent	0.5 volume		
	4.3. Test:			
	5			
	4.3.5.1 Switch on the thermostat bath and maintain it at a temperature			
	of 37 °C.			
	4.3.5.2 Take 0.4 ml of the blank , control and sample (already			
	diluted) and pour them in test tubes. (Perform the test			
	twice).			
	4.3.5.3 Add 2 ml of alkaline copper solution in each test tube.			
	4.3.5.4 Stir gently and incubate at 37 °C for 30 minutes.			
	4.3.5.5 Add 0.2 ml of 1 N Folin - Ciocalteau reagent. Stir immediately			
	after the addition of Folin until a homogeneous colour is			
	obtained.			
	4.3.5.6 Incubate at 37 °C for 20 minutes and read the absorbance at			
	730 nm in the spectrophotometer			
	4.3. Calculations:			
	6			
	For calculating the protein concentration use the equation $Y = mX + b$.			
	Where: m = slope			
	b = intercept			
	Y = absorbance			
	X = concentration			
	4.3.6.1 Use EXCEL programme for the preparation of calibration curve			
	and report the results (Multiply by the dilution factor.			
	Divide with 1000 to change µg/ml to mg/ml).			
	4.3. Acceptance criteria:			
	7			
	The variation coefficient for replicate of the samples, control of the			
	HIB-TT conjugate vaccine and Bovine Serum Albumin is (5 % .			
	The value of the HIB-TT conjugate controls is within the acceptance			
	limits according to the graph controlling it.			
	5.0 Reporting:			
	5.1 Calibration curve of Test method			
	6.0 References:			
	6.1 Lowry, O.H., J. Rosebrough, A.L.Farr and R.J. Randall,R.J. (1951) J.			
	Biol.Chem. 193, 265 - 275.			
	6.2 Text Book of General Chemistry,volume II. Collective of professors of			
	the University of the Havana. Ability of Pharmacy and Foods. Editorial			
	Town and Education, 1990.			

1.0	Objective		
	To lay down the procedure for determining the identity of tetanus component		
	in vaccine.		
2.0	Scope		
	This Standard Operating Procedure is applicable to the tetanus component		
	either single component or combined with other component by limes		
	floculation test.		
3.0	Responsibility		
	3.1	The QC analyst shall be responsible for performing the described	
		procedure correctly.	
	3.2	Supervisor shall be responsible for verifying that the same has been	
		carried out correctly.	
	3.3	Head QC or his designee shall be responsible for overall compliance	
		of the SOP.	
4.0	Abbreviations and Definitions		
	4.1	SOP	: Standard Operating Procedure
	4.2	QA	: Quality Assurance
	4.3	QC	: Quality control
	4.4	TATF	: Tetanus Antitoxin for Flocculation
	4.5	mL	: milliliter
	4.6	°C	: Degree Celsius
	4.7	%	: Percent
	4.8	Lf/ml	: Limes-flocculation per milliliter
5.0	Procedure:		
	5.1	Equipment, Materials and Reagents	
		5.1.1	Test sample
		5.1.2	Autopipettes
		5.1.3	Test tubes.
		5.1.4	Test tube rack
		5.1.5	Flocculation tubes.
		5.1.6	Thermometer
		5.1.7	Stop watch
		5.1.8	Convex lens
		5.1.9	Sterile 10ml & 50ml vials
		5.1.10	Physiological saline
		5.1.11	Working Standard of Tetanus Antitoxin for Flocculation (100
			Lf/ml)
		5.1.12	Tri sodium citrate
	5.2	Test procedure	
		5.2.1	Take 10 ml vaccine sample in a centrifuge tube and add 1.0 gm
			of tri -Sodium citrate.
		5.2.2	Mix well and incubate the mixture at 35±20C for 16±2 hours.
		5.2.3	Centrifuge the sample at 4500 rpm for one hour.
		5.2.4	Collect the supernatant.
		5.2.5	Take 1 mL of the supernatant in flocculation tube, add to it
			0.15 mL of tetanus antitoxin.
		5.2.6	Shake the mixture in tube thoroughly.
		5.2.7	Incubate the tubes at 50±20C in a see-through water bath.
			Ensure that 2/3rd of the flocculation tubes are immersed in
			water.
		5.2.8	Check for flocculation and note down the flocculation time.
	5.3	Acceptance criteria:	
		Flocculation against TATF confirms the presence of tetanus component	
		in vaccines.	
6.0	Forms and Records		
	6.1	Identity Test for Tetanus Vaccine	
7.0	References		
	7.1	WHO TRS 800,927	
	7.2	Indian Pharmacopoeia	

1.0 |Objective |
 |To lay down the procedure for the separation of proteins (Hepatitis B |
 |surface antigen) by High Performance Liquid Chromatography (HPLC) using Gel |
 |Filtration. |
 2.0 |Scope |
 |This Standard Operating Procedure is applicable to samples of Active Raw |
 |Material (ARM) of Hepatitis B surface antigen in the Quality Control |
 |Department. |
 3.0 |Responsibility |
3.1	The analyst shall be responsible for carrying out the described
	procedure correctly.
3.2	The Supervisor of the laboratory shall be responsible for verifying
	that the procedure is complied with and for checking the results.
3.3	The departmental head/designee shall be responsible for training of
	the staff and for overall compliance of the SOP.
3.4	Head QA/Designee shall be responsible for final approval and
	subsequent compliance of the SOP.
4.0	Abbreviations and Definitions
4.1	cm
4.2	g
4.3	HBsAg
4.4	HPLC
4.5	id
4.6	l
4.7	L
4.8	M
4.9	mAu
4.10	mg/mL
4.11	mL
4.12	mL/min
4.13	mm
4.14	N
4.15	nm
4.16	QA
4.17	QC
4.18	QCR
4.19	rpm
4.20	RSD
4.21	SOP
4.22	SST
4.23	µg
4.24	µL
4.25	µm
4.26	System
	Suitability
	test
5.0	Procedure
5.1	Precautions to be taken:
	5.1.1
	5.1.2
	5.1.3
	5.1.4
	5.1.5
	5.1.6
	5.1.7
5.2	Requirement:
	5.2.1
	5.2.2
	5.2.3

|| 5.2.4 |Reagents: |

||| 5.2.4.1 |Sodium Azide (NaN₃) |

||| 5.2.4.2 |Potassium Chloride (KCl) |

||| 5.2.4.3 |Sodium Chloride (NaCl) |

||| 5.2.4.4 |Disodium hydrogen phosphate anhydrous (Na₂HPO₄) |

||| 5.2.4.5 |Potassium dihydrogen phosphate (KH₂PO₄) |

||| 5.2.4.6 |Purified Water |

||| 5.2.4.7 |Hydrochloric Acid (HCl 37%) |

||| 5.2.4.8 |2- Propanol |

|| 5.3 |Preparation of solutions: |

|| 5.3.1 |Isopropyl alcohol 70% |

||| Take 700 mL of 2-propanol in a volumetric flask and make up the |

||| volume to 1000 mL with Purified Water. This solution can be |

||| stored at room temperature for 3 months. |

|| 5.3.2 |Running Buffer 10 X |

||| Na₂HPO₄ (79.0 mM), KH₂PO₄ (17.0 mM), KCl (27.0 mM) and NaCl |

||| (2.5 M) |

||| For a total volume of 500mL, weigh: |

||| Na₂HPO₄ |5.5 g |

||| KH₂PO₄ |1.2 g |

||| KCl |1 g |

||| NaCl |73 g |

||| Dissolve all the reagents in approximately 450 mL of purified |

||| water in a beaker. Adjust the pH to 7 with concentrated |

||| Hydrochloric Acid or 1N Sodium Hydroxide by continuous |

||| stirring. Make the volume to 500mL with purified water in a |

||| volumetric flask. Prepare fresh before use. |

|| 5.3.3 |Running Buffer 1 X |

||| For a volume of 2L, add 200mL of Running Buffer 10X in a |

||| volumetric flask and make the volume to 2 L with purified |

||| water. Use the mobile phase within four days from the date of |

||| preparation if it remains clear and no turbidity or fibers |

||| observed in it. |

|| 5.3.4 |Column Preservation Solution: |

||| 20% Ethanol solution: |

||| For a total volume of 500mL, take 100mL of absolute ethanol and |

||| make up the volume to 500mL with 0.2µ/0.22µ filtered purified |

||| water in a volumetric flask. Prepare fresh before use. |

||| Note: All the solutions prepared for use are vacuum filtered |

||| using Nylon 6, 6 membrane 0.20 µm or 0.22 µm , depending upon |

||| the nature of solution to be filtered. For organic solvents |

||| 0.45µm filters should be used (filtration not necessary if the |

||| solution is prepared in 0.2 µ filtered purified water). |

|| 5.4 |Preparing the HPLC System for run, Agilent Series: |

|| 5.5 |Run Conditions: |

|| 5.5.1 |Load method HBSAG.M. |

|| 5.5.2 |Flow Rate: 0.4 mL/min |

|| 5.5.3 |Detection Wavelength: 280nm |

|| 5.5.4 |Run time: 110 minutes |

|| 5.5.5 |Detector: DAD/VWD |

|| 5.6 |Equilibration of the system and the column: |

||| Equilibrate the system and the column with Running Buffer 1X at 280nm |

||| until a stable baseline is observed on the online plot window. |

|| 5.7 |Sample Preparation and Application: |

|| 5.7.1 |Centrifuge the standards/samples to be applied at 12000 rpm, |

||| for 3 to 5 minutes. |

|| 5.7.2 |Samples are loaded in screw cap vials and placed against the |

||| appropriate no. on the sample tray with proper labeling on the |

||| sample vials in coordination with the sequence loaded. |

|| 5.7.3 |Load the required volume of sample as per the protein |

||| concentration equivalent to 40µg in the sample vial. |

||| Example: If a sample has an antigen concentration of 0.5 mg/mL, |

||| to calculate volume required for 40µg of protein |

||| 0.5mg (500µg) ----1mL |

||| 40µg-----X mL |

||| X mL = 40µg x 1mL |

||| 500µg |

||| X mL = 80µL |

||| Therefore, 80µL of volume is taken for sample having protein |

||| concentration of 0.5 mg/mL. |

||| Note: Protein concentration is calculated as per SOP OD at 280 |

||| nm. |

|| 5.8 |Running of sequence: |

|| 5.8.1 |Load sequence HBSAG.S. |

|| 5.8.2 |Open the sequence parameter option by clicking Sequence menu. |

|| 5.8.3 |Enter information about the data directory and operator name. |

|| 5.8.4 |In the shutdown option, click on the post-sequence option and |

||| select LAMPOFF/STANDBY. |

|| 5.8.5 |In sequence table option, specify the sample names, methods and |

||| vials to run in the sequence as per the following table: |

||| S. No |Sample Code |Sample details |No. of |Injection |

||| | |Injections |Volume (µL) |

||| 1. |Blank |Eluent Blank |1 |10 |

||| 2. |In-house |System |6 |40 µg of |

||| |Reference |Suitability Test |Protein |

||| 3. |Blank |Eluent Blank |1 |10 |

||| 4. |Test |Test Sample |1 |40 µg of |

||| |Sample(s) |Protein |

|| 5.8.6 |Click on run sequence option in run control menu to start the |

||| sequence. |

|| 5.8.7 |Each sample run will be of 110 minutes and after running all |

||| the samples the sequence will automatically stop. |

		5.8.8	Inject the eluent blank after every 20 injections.	
			Note: One extra vial with same SST sample can be used if 6	
			injections are not possible from single vial.	
		5.9	Calculations :	
		5.9.1	Area under curve and retention time for the standards/sample is	
			analyzed by Chemstation software.	
		5.9.2	Calculations for System Suitability Test:	
			5.9.2.1	Note the values of area under the curve and retention
			time for reference (SST standard) and also in EXCEL	
			format for calculation of Relative standard deviation.	
			Path for Master EXCEL format is	
			My Computer → D drive → SST Master EXCEL (folder) →	
			SST Master EXCEL.	
			5.9.2.2	Every time a sample is run with system suitability
			test, calculate the results for system suitability by	
			creating a copy of Master EXCEL in the desired folder	
			and record.	
			5.9.2.3	Attach the hard print of SST calculation along with
			chromatograms and the record form.	
		5.10	Acceptance Criteria :	
		5.10.1	For System Suitability Test :	
			5.10.1.1	RSD calculated for Retention time should be <5.0%.
			5.10.1.2	RSD calculated for Area should be <5.0%.
		5.10.2	For ARM of Hepatitis-B vaccine :	
			5.10.2.1	Purity by Chromatography should be $\geq 95.00\%$ (Area
			under the major Peak of Curve).	
			5.10.2.2	The retention time of sample should be ± 1 minute to
			that of the average of the retention time of the ARM	
			Reference.	
	6.0	References:		
	6.1	United States Pharmacopeia 29.		
	6.2	European pharmacopoeia.		

1.0 |Objective: |
 |To lay down the procedure for the analysis of CRM197 protein by SDS-PAGE |
 |and quantification by gel densitometry. |
 2.0 |Scope: |
 |This standard testing procedure is applicable to the purified protein |
 |samples of CRM 197 |
 3.0 |Responsibility: |
3.1	The analyst is responsible for executing the described procedure
	correctly.
3.2	The supervisor of the laboratory shall be responsible for
	verifying that the procedure is carried out correctly and for
	checking the results.
3.3	The departmental Head shall be responsible for the overall
	compliance of the SOP.
4.0	Abbreviations and Definitions:
4.1	Abbreviations:
	4.1.1
	4.1.2
	4.1.3
	4.1.4
	4.1.5
	4.1.6
	4.1.7
	4.1.8
	4.1.9
	4.1.1
	0
	4.1.1
	1
	4.1.1
	2
	4.1.1
	3
	4.1.1
	4
	4.1.1
	5
	4.1.1
	6
	4.1.1
	7
	4.1.1
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4.2	Reagents:
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	4.2.5
	4.2.6
	4.2.7
	4.2.8
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	7	
	4.2.1	Coomassie Brilliant Blue
	8	
	4.2.1	Silver Nitrate
	9	
	4.2.2	Formaldehyde 37 %
	10	
	4.2.2	Citric acid
	11	
	4.2.2	Ammonia Solution 25 %
	12	
	4.2.2	Sodium hydroxide
	13	
	4.3	Equipments & materials:
	4.3.1	Water Bath
	4.3.2	pH Meter
	4.3.3	SDS-PAGE Assembly (Biorad, Mini-PROTEAN® 3 cell)
	4.3.4	Mini-PROTEAN 3 Spacer Plates with 0.75 mm spacers (Biorad,
		Cat. No. 1653310) & Mini- PROTEAN .3 Short Plates Biorad,
		Cat. No. 165-3308)
	4.3.5	Mini-PROTEAN 3 Casting Stand (Biorad, Cat. No. 165-3303) &
		Mini-PROTEAN 3 Casting Frame (Biorad, Cat. No. 165-3304)
	4.3.6	10-well combs 0.75 mm spacer (Biorad, Cat. No. 165-3354)
	4.3.7	Power pack (Biorad, Model-Basic)
	4.3.8	Gel staining box
	4.3.9	Sterile Microtips- 200µl and 1000 µl
	4.3.1	Sterile microcentrifuge tubes, 1.5 ml
	10	
	4.3.1	Refrigerator
	11	
	4.3.1	Single channel adjustable volume micropipette, 2-20µl, 20-200
	2 µl and 100-1000 µl.	
	4.3.1	Falcon tubes 15 ml
	13	
	4.3.1	Weighing Balance
	14	
	4.3.1	Densitometer (BIORAD)
	15	
	4.3.1	Gel Rocker
	16	
	4.3.1	CRM197 protein samples for analysis
	17	
	4.3.1	Aluminium foil
	18	
	5.0	Procedure:
	5.1	Preparation of solutions:
	5.1.1	30 % Acryl amide and 0.8% Bis-acrylamide solution:
		Weigh 29.2 g of acrylamide and 0.8 g of Bis-acrylamide and
		dissolve in 80ml Milli-Q water, make up the volume to 100 ml.
		Keep on stirring for at least 2 hours and finally filter the
		solution with 0.4511 µ filter and store at 2-8°C covered with
		aluminum foil.
	5.1.2	1.5 M Tris solution, (pH 8.8):
		Weigh 18.171 g of Tris and dissolve in 80ml of Milli-Q water.
		Adjust pH to 8.8 with concentrated Hydrochloric Acid (37%).
		Make up the volume to 100 ml and finally filter the solution
		with 0.4511 µ filter. Can be stored at room temperature for 3
		months.
	5.1.3	0.5 M Tris solution (pH 6.8)
		Weigh 6.057 g of Tris and dissolve in 80ml of Milli-Q water.
		Adjust pH to 8.8 with concentrated Hydrochloric Acid (37%).
		Make up the volume to 100 ml and finally filter the solution
		with 0.4511 µ filter. Can be stored at room temperature for 3
		months.
	5.1.4	20% SDS solution
		Weigh 20 g of SDS and dissolve in Milli-Q water, make up the
		volume to 100 ml. Can be stored at room temperature for 3
		months.
	5.1.5	10% Ammonium Persulphate solution
		Weigh 1 g of APS and dissolve in 9ml of Milli-Q water in
		5ml falcon tube, make up the volume to 10 ml. Make aliquots
		of 500µl and store at 20°C for 3 months.
	5.1.6	Gel running buffer (10X)
		Weigh 30.2 g Tris, 144 g Glycine and 10 g SDS & dissolve in
		800ml of Milli-Q water. Adjust the pH to 8.3 with HCl & make
		up the volume to 1000 ml.
	5.1.7	Sample Solubilising Buffer (6X)
		Mix 2.4 ml of 0.5M Tris (pH 6.8), 3 ml of Glycerol, 3 ml of
		20% SDS. 6 mg of Bromophenol blue. 1.6 ml of
		β-mercaptoethanol and make up the volume to 10 ml. Make
		aliquots of 1ml each and store at -20°C. For running samples
		in non reducing condition, do not add β-mercaptoethanol in
		sample solubilizing buffer.
	5.1.8	Coomassie Brilliant Blue stain:
		Weigh 1 g of Coomassie Brilliant Blue R250* and dissolve in
		100 ml of methanol, add 20 ml of glacial acetic acid and make
		up the volume to 200 ml with Milli-Q water Keep on magnetic
		stirrer for at least half an hour for proper dissolution.
	5.1.9	Destain Solution
		Mix 40 ml of methanol, 10 ml of glacial acetic acid and make
		up the volume to 100 ml with Milli-Q water. Prepare fresh at
		the time of destaining the gel.

	5.2	Casting of gel
	5.2.1	Clean the glass plates with detergent followed by washing
		with Milli-Q water and fix them in the casting frame.
	5.2.2	Prepare 12.5% Separating Gel mixture (7.5 ml) in a falcon
		tube consisting of following reagents:
		30% Acryl amide-3.125 ml
		1.5 M Tris (pH 8.8) — 1.875 ml
		20% SDS — 0.15 ml
		Milli-Q water — 2.275 ml
		10% APS- 75 µl
		TEMED- 5 µl
		Mix the contents gently by inverting the tube two-three times
	5.2.3	Immediately pour 12.5% Separating Gel mixture between the two
		glass plates leaving 4-5 cm space above for Stacking Gel and
		overlay it carefully with Milli-Q water.
	5.2.4	Prepare the Stacking Gel mixture (3.75 ml) in a tube
		consisting of following reagents:
		30% Acrylamide- 0.625 ml
		0.5 M Tris (pH 6.8) - 0.9375 ml
		20% SDS-0.075 ml
		Milli-Q water —2.0 ml
		10% APS-50 µl
		TEMED – 5 µl
	5.2.5	Pour the stacking gel mixture over the surface of the
		polymerized separating gel and immediately insert the 10-well
		comb of 0.75mm.
	5.2.6	Allow the gel to polymerize for 10 minutes.
	5.2.7	Dilute the 10X gel running buffer to 1 X by mixing 100 ml of
		the 10X stock solution in 900 ml of Milli-Q water.
	5.2.8	Remove the comb from the gel and fix the glass plate
		containing polymerized gel set on the gel cassette sandwich
		of the electrophoresis assembly.
	5.2.9	Add 1 X gel running buffer between and around the buffer tank
		of electrophoresis assembly.
	5.3	Sample Preparation For Upstream Samples:
	5.3.1	Prepare CRM197 standards from stock CRM197 standard
		concentration of 100 µg/ml in water to make four different
		concentrations (1.0µg, 0.5µg, 0.25µg and 0.125 µg per 12µl
		water) and mix with sample solubilizing buffer for reducing
		SDS PAGE.
	5.3.2	Take 50 µl of upstream samples and mix with 10µl of sample
		solubilizing buffer for reducing SDSPAGE. Boil standards and
		samples in water bath for 5 min or in microwave for 2 minutes
		at 100°C. Load 15µl each of prepared samples and standards in
		the gel.
	5.4	Preparation For Non reducing SDS PAGE samples:
		Adjust the concentration of purified CRM197 samples 2.0µg in 12 µl
		water by dilutions and mix with 3µl of sample solubilizing buffer.
		Load the samples in the gel along with CRM197 Standard (2.0µg) in
		one of the well of SDS PAGE gel.
	5.5	Preparation For reducing SDS PAGE samples:
	5.5.1	Adjust the concentration of purified CRM197 samples 2.0µg in
		12 µl water by dilutions and mix with 3 µl of sample
		solubilizing buffer. Boil these samples in water bath for 5
		min at 100°C.
	5.5.2	Load the samples in the gel along with CRM197 Standard (2.0
		µg) in one of the well of SDS PAGE gel.
	5.5.3	Note: Load the Molecular weight marker for determining the
		approx. molecular weight of CRM 197 protein as per the
		requirement.
	5.6	Running SDS-PAGE gel:
		Connect the gel apparatus (the negative and positive electrodes)
		with the power pack and set the power pack at 90V till the time dye
		front comes out of stacking gel. After that, turn the voltage to
		160V for further running of the samples. Stop the power pack once
		dye front reaches at the bottom of the separating gel.
	5.7	Staining and de-staining of the gel:
	5.7.1	Once the gel run is over, remove the gel from the plate and
		place the gel in staining solution in a plastic tray for 20
		minutes with gentle agitation on gel rocker.
	5.7.2	De-stain the gel with de-staining solution to remove the
		extra stain till the gel background becomes clear and all the
		bands become clearly visible.
	5.8	Analysis of SDS-PAGE gel (For Upstream samples)
	5.8.1	After destaining, scan the gel in BIORAD densitometer using
		Quantity one software and save for record
	5.8.2	For semi quantitative estimation of CRM 197 protein
		expression in upstream cell culture supernatant, open the
		saved image through the Quantity one software.
	5.8.3	Go to the volume menu and select the volume contour tool.
		Select the bands in volume contour/tool mode. For standard
		bands, double click on the band so that volume properties
		window appears. Select the standard option which leads to the
		concentration menu. Write the concentration of that
		particular band in given space. Click OK to save the data.
	5.8.4	After selecting the standard bands, select the unknown bands
		and mark them unknown such as U1, U2, U3, U4, U5 and U6.
	5.8.5	Double click on gel background and select the background in
		volume appear window to subtract the background as Bl. Again
		go to the volume and select the volume analysis report. Press
		done button to generate volume report.
	5.8.6	Print the volume report and calculate the unknown

			concentration by using following formula- Concentration X
			1000
			Volume of sample loaded in gel (in µl)
			This will give protein concentration in µg/ml.
		5.9	Analysis of SDS-PAGE gel (For Non reducing samples)
		5.9.1	After destaining, scan the gel in BIORAD densitometer using
			Quantity one software and save for record. For percentage
			purity analysis of purified CRM 197 protein, open the saved
			image through the Quantity one software.
		5.9.2	Go to the Lane menu, select the frame lanes and create no of
			lanes as per the samples loaded in the wells. Press OK to
			frame lanes.
		5.9.3	Go to the band menu and select the detect band option.
			Separate detect band window will open. Press OK to detect the
			bands in the lane.
		5.9.4	See carefully in the gel and make sure the all the bands in
			the lane are selected. If required, bands can be created or
			deleted through the options given in the band menu.
		5.9.5	Go to the band menu again and press the band attribute.
			Separate window will open. Select the relative quantity
			option and press OK. Percentage purity of each band in the
			lane will be displayed on the bands. Save or print the data
			for report preparation.
		5.10	Analysis of SDS-PAGE gel (For Reducing samples to calculate %
		Nicking)	
		5.10.	After destaining, scan the gel in BIORAD densitometer using
		1	Quantity one software and save for record.
		5.10.	For percentage nicking analysis of purified CRM 197 protein,
		2	open the saved image through the Quantity one software.
		5.10.	The electrophoretic separation of CRM197 protein in reduced
		3	conditions is being used to evaluate "nicking" of this
			protein in two fragments A and B. Look for the presence of
			three bands on the gel, the intact CRM197 protein (major
			band, 591kD), two smaller more lighter stained bands as
			fragment A (21kD)) and fragment B (37kD).
		5.10.	Go to the volume menu and select the volume contour or volume
		4	rectify tool to select all the bands present in the lanes.
		5.10.	Double click on gel background and select the background in
		5	volume appear window to subtract the background as BI
		5.10.	Go to the volume menu and press the volume analysis report
		6	option. Separate window will appear Make sure following
			options are selected:
			Name, Volume, Adj. volume, % volume, concentration, density,
			display image in report, print image on report, show overlays
			in image and global background subtraction.
		5.10.	Press done to generate the report. Print the report
		7	
		5.10.	Express the percent of nicked protein as percent of the
		8	summed areas (Adj. volume) for fragment A and B divided by
			the sum of all areas measured (fragment A+ fragment B+
			CRM197).
		6.0	References:
		6.1	Bio-Rad Mini-Protean® 3 Cell Instruction Manual.
		6.2	Jian Qiaol, Karim Ghani, Manuel Caruso. Diphtheria toxin mutant
		CRM197 is an inhibitor of protein synthesis that induces cellular	
		toxicity. Toxicon 51 (2008) 473-477.	

1.0	Objective							
	To lay down the procedure for Spectrophotometric determination of Potassium							
	Thiocyanate concentration in Active Raw Material (ARM) of recombinant							
	Hepatitis B vaccine.							
2.0	Scope							
	This Standard Operating Procedure is applicable to all the samples of							
	Active Raw Material of recombinant Hepatitis B surface antigen, , received							
	in the Quality control department.							
3.0	Responsibility							
	3.1	The analyst shall be responsible for executing the described						
	procedure correctly.							
	3.2	Supervisor shall be responsible for verifying that the procedure is						
	complied with and the results are checked.							
	3.3	HOD shall be responsible for overall compliance of this SOP.						
4.0	Abbreviations and Definitions							
	4.1	Abbreviations:						
	4.1.1	ARM	:Active Raw Material					
	4.1.2	CV	:Coefficient of variation					
	4.1.3	g	:Gram					
	4.1.4	ID	:Identification					
	4.1.5	mg	:Milligram					
	4.1.6	ml	:Millilitre					
	4.1.7	µl	:Micro litre					
	4.1.8	µg	:Microgram					
	4.1.9	nm	:Nanometer					
	4.1.10	SD	:Standard Deviation					
	4.1.11	SOP	:Standard Operating Procedure					
	4.2	Definitions:						
		Not Applicable.						
5.0	Procedure:							
	5.1	Preparation of Solutions:						
		5.1.1	37 % w/v Zinc Chloride solution :					
			5.1.1.1	Weigh 37 g of zinc chloride and transfer it to a 100 ml				
			1	volumetric flask.				
			5.1.1.1	Make up the volume to 100 ml with purified water.				
			2					
			5.1.1.1	Store at room temperature, the solution should be used				
			3	within 6months.				
		5.1.2	6 % w/v Ferric Chloride solution :					
			5.1.2.1	Weigh 3 g of ferric chloride and transfer it to a 50 ml				
			1	volumetric flask.				
			5.1.2.1	Make up the volume to 50 ml with concentrated				
			2	hydrochloric acid. Heat the solution slightly if not				
				dissolved properly.				
			5.1.2.1	Store at room temperature, the solution should be used				
			3	within 6 months.				
		5.1.3	1 mg/ml Potassium Thiocyanate Solution :					
			5.1.3.1	Weigh 50 mg of Potassium Thiocyanate and transfer it to				
			1	a 50 ml volumetric flask.				
			5.1.3.1	Make up the volume to 50 ml with Purified water.				
			2					
			5.1.3.1	This solution should be freshly prepared.				
			3					
	5.2	Preparation of Calibration curve:						
		5.2.1	Take 5 volumetric flasks of 25 ml each and label the details					
			with pen marker. Prepare calibration curve (5-25 µg/ml) by					
			adding 1mg/ml potassium thiocyanate to purified water as per					
			the following table:					
			Concentration (µg/ml)	5	10	15	20	25
			Potassium Thiocyanate	125	250	375	500	625
			(1mg/ml) taken (µl)					
			Volume Make up with	25	25	25	25	
			purified water in a					
			volumetric flask (ml)					
		Note: The Calibration curve is accepted if it is meeting the						
		validated parameters of intercept, slope and correlation coefficient.						
	5.3	Test:						
		5.3.1	Take 1ml from 5 to 25 µg/ml Potassium Thiocyanate solutions of					
			calibration curve, Test sample (ARM) and Blank (Purified water)					
			used as blank) in duplicates in separate 2 ml previously					
			labelled micro centrifuge tubes. However reference ARM treated					
			like test sample can be used in the assay as a control,					
			occasionally.					
		5.3.2	To these micro centrifuge tubes add 100 l of 37% Zinc Chloride					
			solution and 900 l of Purified water.					
		5.3.3	Allow the tubes to stand for 10 minutes and centrifuge at 3000					
			RPM for 20 minutes.					
		5.3.4	Take 1 ml of supernatant from each micro centrifuge tube and					
			transfer to the previously labeled fresh micro centrifuge					
			tubes after centrifugation.					
		5.3.5	Add 100 l of 6%w/v ferric chloride solution to each tube and					
			mix the contents using vortex mixer.					
		5.3.6	Allow these tubes to stand for five minutes.					
		5.3.7	Adjust the wavelength in spectrophotometer at 475 nm and					
			measure the absorbance.					
		Note: If absorbance of the sample falls outside the range of						
		calibration curve, dilute the sample appropriately so that it comes						
		under the range of calibration curve.						
	5.4	Calculations:						
		5.4.1	Once the absorbance of the standards and the test samples have					

			been taken, calculation shall be done using the excel sheet
			carrying formulas for the required calculation.
		5.4.2	Enter the absorbance for calculations starting from
			Calibration curve prepared between 5 and 25 g/ml of Potassium
			Thiocyanate solution and test samples. The calculations are
			made according to the equation of minimum squares.
		5.4.3	$Y = mX + b$.
			Where: m = slope
			b = intercept
			Y = absorbance
			X = concentration
			(multiply with the dilution factor)
		5.4.4	Taking into account the equivalence between the ARM volume and
			the antigen concentration, express the result per 20 g of
			protein. Enter the protein concentration estimated by OD at
			280nm in the excel sheet.
		5.4.5	If the Protein concentration of Active Raw Material of
			Hepatitis B Vaccine by Absorbance at 280 nm. is 1.40 mg/ml
			calculations are done as given below:
		5.4.6	$C = 20 \times X / 1.40 / 1000$
			Where:
			C = Potassium Thiocyanate concentration per 20 μ g of Protein
			20 = Dose of Hepatitis B vaccine (20 μ g)
			X = Total concentration of Potassium Thiocyanate calculated
			from the calibration curve (μ g/ml)
			1.40mg/ml = Concentration of protein in the sample by
			Absorbance at 280 nm)
			1000 = Conversion factor
		5.4.7	Record all the calculations in the data record form and also
			attach the hard copy of the excel sheet used for calculation
			along it.
		5.5	Acceptance criteria:
		5.5.1	The coefficient of variation (CV) between the replicates of
			the samples should be $\leq 5\%$.
		5.5.2	The value of the control should be within the acceptance
			limits.
		5.5.3	Potassium thiocyanate content in the sample should be within
			the specified limits.
		6.0	References
		6.1	LEVINE, V.E. (1961) Acids and Alkalis. Cap. 8. Toxicology Mechanisms
		and analytical methods. Vol. II.	

1.0 |Objective |
 |To lay down the procedure for separation of proteins by Sodium Dodecyl |
 |Sulphate Polyacrylamide Gel Electrophoresis (SDS-PAGE), for the detection |
 |of molecular aggregates, degradations and other contaminants, for staining |
 |of gel and evaluation of bands using Quantity one software. |
 2.0 |Scope |
 |This Standard operating procedure is applicable to carry out SDS-PAGE for |
 |proteins of samples of in-process, active raw material and final bulk |
 |vaccine in the Quality Control department. |
 3.0 |Responsibility |
 |3.1 |The analyst shall be responsible for carrying out the described |
 |procedure correctly. |
 |3.2 |The Supervisor of the laboratory shall be responsible for verifying |
 |that the procedure is complied with the SOP and for checking the |
 |results. |
 |3.3 |The departmental head/designee shall be responsible for training of |
 |the staff and for overall compliance of the SOP. |
 |3.4. |Head QA/Designee shall be responsible for final approval and |
 |subsequent compliance of the SOP. |
 4.0 |Abbreviations and Definitions |
4.1	APS	: Ammonium Per Sulphate
4.2	Bis	: Bis-acrylamide
4.3	BPB	: Bromophenol blue
4.4	CBB	: Coomassie Brilliant Blue
4.5	Cm	: centimeter
4.6	Dpi	: dots per inch
4.7	G	: gram
4.8	kDa	: Kilo Dalton

4.9	L	: Litre
4.10	M	: Molar
4.11	mA	: milli Ampere
4.12	Mg	: milligram
4.13	mL	: milliliter
4.14	Mm	: millimeter
4.15	OE	: Operation of equipment
4.16	PAGE	: Polyacrylamide Gel Electrophoresis
4.17	Rpm	: rotations per minute
4.18	SDS	: Sodium Dodecyl Sulphate
4.19	SOP	: Standard Operating Procedures
4.20	TEMED	: N',N', N',N'-Tetramethylethylenediamine
4.21	Tris	: Tris(hydroxymethyl)aminomethane
4.22	V	: Volt
4.23	°C	: Degree centigrade
4.24	µg	: microgram
4.25	µL	: microlitre

5.0 |Procedure |
 |5.1 |Precautions to be taken: |
 |5.1.1 |Handle Acrylamide and Bis-acrylamide carefully, with gloves,|
 |as these are neurotoxic. |
 |5.1.2 |Always wear laboratory coat, mask and personnel protective |
 |equipment. |
 |5.1.3 |Handle acids carefully under fume hood. |
 |5.1.4 |Acetic acid causes serious burns. So do not inhale vapours. |
 |In case of contact with the eyes, wash immediately with |
 |abundant water and get medical attention. |
 |5.1.5 |Methanol is toxic if inhaled or ingested and is also |
 |inflammable. Keep away from ignition sources. Avoid contact |
 |with the skin. Avoid inhalation. |
 |5.1.6 |Silver nitrate causes severe burns. In the event of contact |
 |with the eyes wash immediately with abundant water and get |
 |medical attention. |
 |5.1.7 |Ammonia solution causes irritation in eyes, skin and also in|
 |lungs on inhalation. In the event of contact with eyes, wash|
 |immediately with abundant water and get medical attention. |
 |5.1.8 |Never touch the gel with fingers, as the fingerprints get |
 |marked on the gel. |
5.2	Requirement:
5.2.1	Equipment:
5.2.1.1	Water Bath
5.2.1.2	Computer with Quantity one software and Image
Quant TL plus Software	
5.2.1.3	Minigel Twin Apparatus
5.2.1.4	Vertical Electrophoresis Unit
5.2.1.5	Laboratory Centrifuge
5.2.1.6	Tube Heating Block
5.2.1.7	Refrigerator
5.2.1.8	Deep Freezer (-20 ± 5°C)
5.2.1.9	Gel Documentation System
5.2.2	Instrument:
5.2.2.1	Precision Balance
5.2.2.2	Analytical Balance
5.2.2.3	Magnetic Stirrer
5.2.2.4	pH Meter
5.2.2.5	Power Supply
5.2.2.6	Gel Rocker
5.2.3	Material:
5.2.3.1	Autopipettes (20µl-200 µl,100 µl-1000 µl & 1000
µl-5000 µl) & Tips	
5.2.3.2	Eppendorfs (1.5ml and 2ml)

		5.2.3.3	Macro pipette controller	
		5.2.3.4	Whatman Filter Paper (No. 42)	
		5.2.3.5	Laboratory Glassware	
		5.2.3.6	Staining Trays	
	5.2.4	Reagents:		
		5.2.4.1	N',N', N',N'-Tetramethylethylenediamine (TEMED)	
		5.2.4.2	2-Butanol	
		5.2.4.3	2-Mercaptoethanol	
		5.2.4.4	Acrylamide	
		5.2.4.5	N,N'-Methylene-Bis-Acrylamide(Bis-acrylamide)	
		5.2.4.6	Ammonium Per Sulphate (APS)	
		5.2.4.7	Glycerol	
		5.2.4.8	Hydrochloric acid 37 % (Fuming)	
		5.2.4.9	Sodium Dodecyl Sulphate (SDS)	
		5.2.4.10	Bromophenol blue (BPB)	
		5.2.4.11	Tris(hydroxymethyl)aminomethane	
		5.2.4.12	Absolute alcohol	
		5.2.4.13	Glycine	
		5.2.4.14	Purified water	
		5.2.4.15	Methanol	
		5.2.4.16	Acetic Acid 100%	
		5.2.4.17	Charcoal activated	
		5.2.4.18	Coomassie Brilliant Blue	
		5.2.4.19	Silver Nitrate	
		5.2.4.20	Formaldehyde 37 %	
		5.2.4.21	Citric acid	
		5.2.4.22	Ammonia Solution 25 %	
		5.2.4.23	Sodium hydroxide	
	5.3	Preparation of solutions:		
	5.3.1	Monomeric Solution		
			(Acrylamide 30%; Bis-acrylamide 0.8%), For a total volume of	
			250mL, weigh:	
			Acrylamide	75.0g
			Bis-acrylamide	2.0g
			Dissolve in 170mL of purified water in a beaker. Make up the	
			volume to 250mL with purified water in a volumetric flask.	
			Keep on stirring for at least 2 hours and finally filter the	
			solution using whatman paper No.42. This solution can be	
			stored at 2-8°C in an amber colour reagent bottle for 2	
			months.	
	5.3.2	Separating Gel Buffer (1.5 M)		
			(Tris-HCl 1.5M; pH-8.8), For a volume of 250mL, weigh:	
			Tris	45.4g
			Dissolve in 170mL of purified water in a beaker. Adjust pH	
			to 8.8 with concentrated Hydrochloric Acid (37%). Make up	
			the volume to 250mL with purified water in a volumetric	
			flask and finally filter the solution, using whatman paper	
			No.42. This solution can be stored at 2-8°C for 6 months.	
	5.3.3	Stacking Gel Buffer (0.5 M)		
			(Tris-HCl 0.5 M; pH-6.8), For a volume of 50mL, weigh:	
			Tris	3.0g
			Dissolve in 20 mL purified water in a beaker. Adjust pH to	
			6.8 with concentrated Hydrochloric Acid (37%). Make up the	
			volume to 50mL with purified water in a volumetric flask and	
			finally filter the solution using whatman paper No.42. This	
			solution can be stored at 2-8°C for 6 months.	
	5.3.4	Stacking Gel Buffer (2.0 M)		
			(Tris-HCl 2M; pH-6.8), For a volume of 50mL, weigh:	
			Tris	12.11g
			Dissolve in 20mL of purified water in a beaker. Adjust pH to	
			6.8 with concentrated Hydrochloric acid (37%). Make up the	
			volume to 50mL with purified water in a volumetric flask and	
			finally filter the solution using whatman paper (No 42).	
			This solution can be stored at 2-8°C for 6 months.	
	5.3.5	Sodium Dodecyl Sulphate (SDS) – 10%		
			For a volume of 500mL, weigh:	
			SDS	50g
			Dissolve in 300 mL of purified water in a beaker. Make up	
			the volume to 500mL with purified water in a volumetric	
			flask. This solution can be stored at room temperature for	
			6 months.	
	5.3.6	Sodium Dodecyl Sulphate (SDS) – 20%		
			For a volume of 100mL, weigh:	
			SDS	20g
			Dissolve in 75mL of Purified water in a beaker. Make up the	
			volume to 100mL with Purified water in a volumetric flask.	
			This solution can be stored at room temperature for 6	
			months.	
	5.3.7	Ammonium per Sulphate (APS) – 10%		
			For a volume of 5mL, weigh:	
			APS	0.5g
			Dissolve in 4mL of purified water in a beaker. Make up the	
			volume to 5mL with purified water in a volumetric flask.	
			Divide in aliquots of 0.5mL in eppendorf. This can be stored	
			at -20±5°C for 2 months.	
	5.3.8	Bromophenol Blue (1%)		
			For a volume of 5mL, weigh:	
			Bromophenol blue	0.05g
			Dissolve in 4mL of purified water in a beaker. Make up the	
			volume to 5mL with purified water in a volumetric flask.	
			Divide in aliquots of 500µL in eppendorf. This can be stored	
			at -20±5°C for 6 months.	

		5.3.9	2X Reducer Buffer with dye	
		[[Tris(hydroxymethyl)aminomethane] 0.125M; Sodium Dodecyl		
		Sulphate 4%; Glycerol 20%; 2-Mercaptoethanol 10%;		
		Bromophenol blue 0.01%.]		
		For a volume of 10mL, add the following solutions:		
		Stacking Gel Buffer 2.0M	0.625mL	
		SDS 20%	2.0mL	
		Glycerol	2.0mL	
		2-Mercaptoethanol	1.0mL	
		Bromophenol Blue 1%	0.1mL	
		Make up the volume to 10mL with purified water in a		
		volumetric flask. Divide in aliquots of 500µL in eppendorf.		
		This can be stored at -20±5°C for 6 months.		
		5.3.10	5X Reducer Buffer with dye	
		[[Tris(hydroxymethyl) aminomethane] 0.3125M; Sodium Dodecyl		
		Sulphate 10%; Glycerol 50%; 2-Mercaptoethanol 25%;		
		Bromophenol blue 0.01%.]		
		For a volume of 10mL, add the following solutions in a		
		beaker, weigh SDS and dissolve it in that mixture:		
		Stacking Gel Buffer 2.0M	1.56mL	
		SDS	1.0g	
		Glycerol	5.0mL	
		2-Mercaptoethanol	2.5mL	
		Bromophenol Blue 1%	0.1mL	
		Make up the volume to 10mL with purified water in a		
		volumetric flask. Divide in aliquots of 500µL in eppendorf.		
		This can be stored at -20±5° C up to 6 months.		
		5.3.11	Running Buffer 10X	
		[[Tris (hydroxymethyl) aminomethane] 0.25M; Glycine 1.92M;		
		Sodium Dodecyl Sulphate 10%).]		
		For a volume of 1L, weigh:		
		Tris	30.2g	
		Glycine	144.0g	
		SDS	10.0g or SDS 10%	100mL
		Dissolve in 700mL of Purified water in a beaker. The pH of		
		the solution should be 8.3. Make up the volume to 1L with		
		purified water in a volumetric flask. This can be stored at		
		room temperature for 6 months.		
		5.3.12	Running Buffer, 1X	
		[[Tris-(hydroxymethyl) aminomethane 0.025M; Glycine 0.192M;		
		Sodium Dodecyl Sulphate 0.1%).]		
		For a volume of 1L, add 100mL of Running Buffer 10X, in a		
		volumetric flask and make up the volume to 1L with purified		
		water. This can be stored at room temperature up to 6		
		months.		
		5.3.13	2 - Butanol (50%)	
		For a volume of 100mL take:		
		2-Butanol	50mL	
		Make up the volume to 100mL with purified water in a		
		volumetric flask. Shake and allow to stand until the		
		appearance of two phases (upper phase: 2-Butanol saturated		
		with water, lower phase: water). Carefully collect the upper		
		phase for use. This solution can be stored at room		
		temperature up to 6 months.		
		5.3.14	Fixation Solution	
		(Methanol-50%; Acetic Acid-7%)		
		For a volume of 1L, take:		
		Methanol	500mL	
		Acetic Acid	70mL	
		Make up volume with purified water to 1L in a volumetric		
		flask. Store in amber colour bottle at room temperature. The		
		expiry date of this solution is 6 months, but it must not be		
		used more than 10 times.		
		5.3.15	Staining Solution	
		(Coomassie Brilliant Blue (CBB) G-250, 0.05%; Methanol 10%;		
		Acetic Acid 10%)		
		For a volume of 1L, in a beaker take:		
		Coomassie Brilliant Blue	0.5g	
		G-250		
		Methanol	100mL	
		Acetic Acid	100mL	
		Make up volume with purified water to 1L in a volumetric		
		flask. Stir it for 1 hour and filter it using Whatman Filter		
		Paper No.42. Store the staining solution in amber colour		
		bottle at room temperature. The expiry date of this solution		
		is 6 months, but it must not be used more than 10 times.		
		5.3.16	Destaining Solution	
		(Methanol 10%; Acetic Acid 10%)		
		For a volume of 1L, take:		
		Methanol	100mL	
		Acetic acid	100mL	
		Make up the volume with purified water to 1L in a volumetric		
		flask. Store in amber colour bottle at room temperature. The		
		expiry date of this solution is 6 months, but it must not be		
		used more than 10 times. The used Destaining solution is		
		filtered through activated charcoal for re-use.		
		5.3.17	Staining Solution II	
		(Coomassie Brilliant Blue (CBB) G-250, 0.2%; Methanol 40%;		
		Acetic Acid 10%)		
		For a volume of 500mL, in a beaker take:		
		Coomassie Brilliant Blue	1.0g	
		G-250		

			Methanol	200mL
			Acetic Acid	50mL
			Make up volume with purified water to 500mL in a volumetric	
			flask. Stir it for 1 hour and filter it using Whatman Filter	
			Paper No.42. Store in amber colour bottle at room	
			temperature. The expiry date of this solution is 6 months,	
			but it must not be used more than 5 times.	
			5.3.18	Destaining Solution II
			(Methanol 40%; Acetic Acid 10%)	
			For a volume of 500mL, take:	
			Methanol	200mL
			Acetic acid	50mL
			Make up the volume with purified water to 500mL in a	
			volumetric flask. Store in amber colour bottle at room	
			temperature. The expiry date of this solution is 6 months,	
			but it must not be used more than 5 times. The used	
			Destaining solution is filtered through activated charcoal	
			for re-use.	
			5.3.19	Sodium Hydroxide Solution (0.36%):
			For a volume of 1L, weigh:	
			Sodium Hydroxide	3.6g
			Dissolve in 700mL of purified water in a beaker. Make up	
			volume with purified water to 1L. This solution can be	
			stored in a glass bottle at room temperature for 6 months.	
			5.3.20	Citric Acid Solution (1%):
			For a volume of 10mL, weigh:	
			Citric Acid	0.1g
			Dissolve in 7mL of purified water in a beaker. Make up the	
			volume with purified water to 10mL in a volumetric flask and	
			pour aliquot of 1.35mL in eppendorf of 1.5mL. This solution	
			can be stored at -20 ±5 °C for 6 months.	
			5.3.21	Fixation and Stopping Solution:
			(Methanol-50%)	
			For a volume of 500mL, take:	
			Methanol	250mL
			Make up the volume to 500mL with purified water in a	
			volumetric flask.	
			Note: Prepare the solution prior to use.	
			5.3.22	Staining Solution:
			(AgNO₃ 0.4 %; NaOH 0.076%; NH₃ 0.38%)	
			For a volume of 100mL, take:	
			Silver nitrate	0.4g (dissolves in 4mL purified
			water)	
			Sodium hydroxide (0.36%)	21mL
			Ammonia (25%)	1.52mL
			Make up the volume to 100mL with purified water in a	
			volumetric flask.	
			For a volume of 250mL, take:	
			Silver nitrate	1g (dissolves in 10mL purified
			water)	
			Sodium hydroxide (0.36%)	52.5mL
			Ammonia (25%)	3.8mL
			Make up the volume to 250mL with purified water in a	
			volumetric flask.	
			Note: Prepare the solutions prior to use.	
			5.3.23	Developing Solution:
			(Formaldehyde 0.0185 %; Citric Acid 0.005%), For a volume of	
			250mL, take:	
			Formaldehyde (37%)	125μL
			Citric Acid (1%)	1.25mL
			Make up the volume to 250mL with purified water in a	
			volumetric flask.	
			Note: Always prepare this solution prior to use.	
			5.4	Preparation of Separating gel for Hoefer Series {140(w) x 240(l) x
			1.5(t)mm} :	
			5.4.1	Assembly of Electrophoresis Unit
			5.4.1.1	Take two of the cleaned and unchipped glass plates
			and two cleaned spacers. Lay one plate on a flat	
			surface and place the spacers along each edge.	
			Place the spacer mate in between for proper	
			alignment and lay the second glass plate on the	
			top. Tight the glass plates with the help of	
			clamps.	
			5.4.1.2	Place the blank silicone rubber gasket into the
			casting cradle with the foam side down.	
			5.4.1.3	Assemble the plates on the casting cradle.
			5.4.1.4	Check for leakage by pouring of Purified water
			between the assembled glass plates.	
			Note: In case of leakage apply thin film of gel	
			seal only the bottom corner surfaces created by	
			spacers and the plates.	
			5.4.1.5	Remove the water by tilting the chamber carefully.
			After that, dry the interior of chamber with	
			filter paper.	
			5.4.2	Prepare the gel of required percentage as given in Table-1
			as per Annexure 1.	
			5.4.2.1	APS 10% and TEMED should always be added just
			before casting the gel, without forming bubbles.	
			5.4.2.2	Slowly pour the above mixture of solutions,
			between the glass plates, using autopipette/glass	
			pipette with the help of a macro pipette	
			controller, up to a height approximately 5 cm	

			below the top. To check the polymerisation, leave	
			a small portion of the mixture in the beaker.	
			5.4.2.3	Add 100µl of 50% 2-Butanol immediately for
			complete polymerization of the gel. Leave for at	
			least 15 minutes to allow the polymerization of	
			the gel. When the gel has polymerized a brief	
			interface of liquid-gel appears.	
	5.5	Preparation of Separating gel for Minigel Apparatus (8.6 x 7.7 cm) :		
	5.5.1	Assemble the chamber of vertical minigel electrophoresis		
		equipment.		
	5.5.2	Take two of the thoroughly cleaned glass plates (one glass		
		plate with fixed spacers, and another notched glass plate).		
		Lay one plate with fixed spacers on a flat surface, and		
		place silicone rubber seal on it and lay the second notched		
		glass plate on the top. Fix the glass plate set on the		
		vertical chamber with clips.		
	5.5.3	Check for leakage by pouring of purified water between the		
		assembled glass plates.		
	5.5.4	Remove the water by tilting the chamber carefully. After		
		that, dry the interior of chamber with filter paper.		
	5.5.5	Prepare the gel of required percentage as given in Table-2		
		of Annexure 1.		
	5.5.6	APS 10% and TEMED should always be added just before casting		
		the gel, without forming bubbles.		
	5.5.7	Slowly pour the above mixture of solutions, between the		
		glass plates, using a 1mL autopipette. To check the		
		polymerization, leave a small portion of the mixture in the		
		beaker.		
	5.5.8	Add 50µl of 50% 2-Butanol immediately for complete		
		polymerization of the gel. Leave at least 10 minutes for		
		polymerization of the gel. When the gel has polymerized a		
		brief interface of liquid-gel appears.		
	5.6	Preparation of the Stacking gel for Hoefer Series (140 x 240 x 1. 5		
		mm) :		
	5.6.1	Remove 50% 2-Butanol from the surface of the gel, by tilting		
		the chamber carefully and rinsing with purified water. After		
		that, dry the interior of the chamber with filter paper.		
	5.6.2	Mix the solutions of the stacking gel according to the		
		quantities described in Table-1 of Annexure 1.		
	5.6.3	APS 10% and TEMED should always be added just before casting		
		the gel, without forming bubbles.		
	5.6.4	Pour the solution between the glass plates over the		
		separating gel and fill up to the top with the help of		
		autopipette/glass pipette with macro pipette controller.		
	5.6.5	Place the comb, between the glass plates in such a way that		
		there are no air bubbles under the teeth. It takes		
		approximately 30 minutes for the stacking gel to get		
		polymerized.		
	5.7	Preparation of the Stacking gel for Minigel Apparatus (8.6 x 7.7 cm)		
	:			
	5.7.1	Remove 50% 2-Butanol from the surface of the gel, by tilting		
		the chamber carefully and rinsing with purified water. After		
		that, dry the interior of the chamber with filter paper.		
	5.7.2	Mix the solutions of the stacking gel according to the		
		quantities described in Table-2 of Annexure 1.		
	5.7.3	APS 10% and TEMED should always be added just before casting		
		the gel, without forming bubbles.		
	5.7.4	Pour the solution between the glass plates over the		
		separating gel and fill up to the top with the help of		
		autopipette.		
	5.7.5	Place the comb, between the glass plates in such a way that		
		there are no air bubbles under the teeth. It takes		
		approximately 20 minutes for the stacking gel to get		
		polymerized.		
	5.7.6	After polymerization of the stacking gel, take out the comb		
		carefully without breaking the wells.		
	5.7.7	Clean the wells with purified water to remove the remains of		
		unpolymerised gel.		
	5.8	Preparation of the Samples: SDS is an anionic detergent, which		
		breaks the non-covalent intermolecular linkages in the proteins. SDS		
		binds strongly to proteins (1.4 g of SDS binds with one gram of		
		protein, forming negatively charged complex). The separation of the		
		proteins, in presence of SDS in polyacrylamide gel, depends		
		fundamentally on the molecular weight of proteins.		
	5.8.1	The description of the quantities, for loading of the		
		samples of various products with different protein		
		concentrations for both the apparatus, Hoefer Series and		
		Minigel are given in the Annexure 2.		
	5.8.2	Prepare the samples in eppendorf, taking equal volumes of		
		the sample and Reducer buffer 2X or 4 volumes of the sample		
		and 1 volume of Reducer Buffer 5X depending upon the protein		
		concentration.		
	5.8.3	In case of Hoefer Series the samples having protein		
		concentrations less than 0.5 mg/mL are to be treated with		
		Reducer Buffer 5X, so that the total volume to be applied in		
		the gel remains the same and protein samples having protein		
		concentration greater than 0.5mg/mL, use Reducer Buffer 2X		
		to treat the samples.		
		Maximum Loading Capacity = 180µL per well.		
	5.8.4	In case of Mini Gel Apparatus the samples having protein		
		concentrations between less than 0.6mg/mL are to be treated		
		with 5X reducer buffer so that the total volume to be		

			applied in the gel remains the same and samples having
			protein concentration greater than 0.6mg/mL, use Reducer
			Buffer 2X to treat the samples.
			Maximum Loading Capacity for Comb With 10 wells = 23µL per
			well.
			Maximum Loading Capacity for Comb With 5 wells = 60µL per
			well.
		5.8.5	Once all the samples and the reference material are prepared
			with the appropriate quantities of Reducer Buffer, heat the
			samples except molecular Weight marker at 75°C for 10
			minutes. Thereafter centrifuge all the samples at 8000rpm
			for 3 to 5 minutes. Then load the samples in the wells using
			an autopipette.
			Note: Samples for Anthrax are treated at 100°C for 1 minute
			in a water bath.
		5.9	Running of the Gel:
		5.9.1	Assemble the electrophoresis apparatus and fill both the
			reservoirs with 1X running buffer.
		5.9.2	Connect the electrophoresis chamber to the power supply,
			ensuring the correct polarity of the electrodes. The
			negative terminal from the power source is to be connected
			to the upper buffer reservoir and the positive terminal is
			to be connected to the lower buffer reservoir.
		5.9.3	Switch ON the Power Supply and set the current conditions as
			per the requirement for Hoefer Series; Current: 25mA whereas
			for Mini Gel Apparatus; Current: 12mA.
		5.9.4	Once sufficient dye front has traveled through the
			separating gel, press STOP and then switch off the power
			supply. Disassemble the chamber and remove the gel
			carefully.
		5.9.5	Separate the stacking gel and stain the separating gel.
		5.9.6	Wash the glass plates using detergent and tap water and then
			rinse with purified water. Finally clean with absolute
			ethanol. Dry the plates with lint free cloth or tissue
			paper.
		5.10	Staining of the gel run on Hoefer Series by Coomassie Brilliant Blue
		5.10.1	Place the gel in a tray containing 500mL of Fixation
			Solution for at least 30 minutes.
		5.10.2	Decant the Fixation Solution and add at least 500mL of
			Staining Solution.
		5.10.3	Leave it for overnight on a gel rocker at moderate speed.
		5.10.4	Decant the Staining Solution and add 500mL of Destaining
			Solution.
			Note: Mark on the label after each use of the Fixation,
			Staining and Destaining Solution. (See Table-1 of Annexure
			1).
		5.10.5	Destain the gel until a transparent background is achieved
			by changing the solution after every 30-45 minutes. Once the
			background of the gel is transparent remove the Destaining
			Solution, rinse the gel with purified water and scan the gel
			using GS-800 Calibrated Densitometer.
		5.11	Coomassie Staining of the gel run on Minigel Apparatus (8.6 x 7.7
			cm):
		5.11.1	Place the gel in a tray containing 250mL of Fixation
			Solution for at least 15 minutes.
		5.11.2	Decant the Fixation Solution and add at 250mL of Staining
			Solution II.
		5.11.3	Leave it for 2 hours on a gel rocker at moderate speed.
		5.11.4	Decant the Staining Solution II and add 250mL of Destaining
			Solution II.
			Note: Mark on the label after each use of the Fixation,
			Staining and Destaining Solution. (See Table-2 of Annexure
			1).
		5.11.5	Destain the gel until a transparent background is achieved
			by changing the solution after every 30minutes. Once the
			background of the gel is transparent remove the Destaining
			Solution, rinse the gel with purified water and scan the gel
			using GS-800 Calibrated Densitometer.
		5.12	Silver Nitrate Staining of the gel run on Hoefer Series (140 x 240 x
			1.5 mm):
		5.12.1	Introduce the gel in a tray, containing 250mL of fixation
			solution for minimum of 4 hours to overnight.
		5.12.2	Decant the Fixation Solution and rinse the gel with purified
			water. Carry out 3 serial rinses with 500mL of purified
			water, 5 to 7 minutes each.
		5.12.3	Decant the purified water and add 250mL Staining Solution,
			for 10 minutes with strong agitation on a gel rocker, so
			that the solution covers the surface of the gel. Perform
			this step in the dark, by covering the tray with aluminium
			foil.
		5.12.4	Remove the aluminium foil and decant the Staining Solution
			from the tray and rinse the gel with purified water. Carry
			out 3 serial rinses with 500mL purified water for a total
			time of 5 minutes, shaking slowly on a gel rocker.
		5.12.5	Add 250mL of the Developing Solution and place the tray on
			the gel rocker with moderate agitation. Until the appearance
			of band of 0.2(g of Reference Active Raw Material which is
			the Stopping Point for developing of the bands.
		5.12.6	Once the Stopping Point of enough intensity appears, decant
			the Developing Solution and rinse the gel with purified
			water and stop the developing reaction with addition of
			250mL of Stopping Solution, for 30 minutes.

A 10x4 grid of vertical bars representing a binary matrix. The bars are arranged in four columns. The first column has bars at rows 1, 3, 5, 7, 9. The second column has bars at rows 2, 4, 6, 8, 10. The third column has bars at rows 1, 3, 5, 7, 9. The fourth column has bars at rows 2, 4, 6, 8, 10. The bars are labeled 'B' at the top and bottom of the fourth column.

			on the "new standard" option in the dialog box and by
			entering the values of the molecular weight standard in the
			given table. Click the "type" button and then click on the
			lane, which contains molecular weight standards.
		5.14.13	Click "Band Attributes" then Band labels are displayed and
			select Molecular weight option to know the molecular weights
			of the bands in unknown lanes with respect to the standard
			lane.
		5.14.14	Similarly to find out the purity percentage of each band in
			a particular lane, click "band attributes" then Band labels
			are displayed and select Relative quantity to know the
			percentage of each band detected in a lane.
		5.14.15	With the relative quantity option, the quantity of a
			particular band in a lane is expressed as a percentage of
			either:
			(1) Percentage of lane.
			(2) Percentage of bands in lane.
			Select percentage of bands in lane by clicking Edit menu and
			select Preferences followed by Application.
		5.14.16	Click "Text Overlay Tools" in the Title bar and select Text
			Tools to name the samples loaded in various lanes of the
			gel. Save the gel with current analysis.
		5.14.17	To print the image, click "print image".
		5.14.18	The printed report is signed by the analyst who does the
			analysis and also checked by the other analyst.
6.0	References:		
	6.1	Gel Electrophoresis of Proteins: A Practical Approach. Ed. B. D.	
	Hames and D. Rick wood. IRL Press. 1981.		
	6.2	Hoefler Series operating manual and Minigel Twin operating manual.	
	6.3	Quantity one – Quantitation software user guide for version-4	
	Windows and Macintosh M/s BIO-RAD Laboratories (India) Pvt. Ltd.		
	6.4	Help option provided with each analysis option in the Quantity One	
	Imaging software.		

Annexure 1

QUALITY CONTROL DEPARTMENT |
 TABLES FOR PREPARATION OF SEPARATING AND STACKING GELS OF DIFFERENT PERCENTAGES |
 DEPENDING UPON THE SIZE OF THE GEL |

TABLE-1

(Hoefler Series)

Dimensions of the gel: 140 x 240 x 1.5 mm

| Separating gel | Stacking gel |

Name of the Solution | | |

| 7.5 % | 10 % | 12.5 % | 15 % | 18 % | 4 % |

Monomeric solution | 15.0 ml | 20.0 ml | 25.0 ml | 30.0 ml | 36.0 ml | 2.66 ml |

Separating Gel Buffer pH 8.8 | 14.0 ml | 15.0 ml | 15.0 ml | 15.0 ml | 15.0 ml | --- |

Stacking Gel Buffer (0.5 M) | --- | --- | --- | --- | 5.0 ml |

pH 6.8 | | | | |

Sodium Dodecyl Sulphate 10 % | 0.60 ml | 0.60 ml | 0.60 ml | 0.60 ml | 0.60 ml | 0.20 ml |

Purified water | 29.10 | 24.1 ml | 19.10 | 14.10 | 8.10 ml | 12.20 ml |

| ml | | ml | ml | | |

APS 10 % | 300 (l | 300 (l | 300(l | 300(l | 100(l |

TEMED | 40(l | 40(l | 40(l | 40(l | 40 (l | 20(l |

TABLE- 2

(Mini Gel)

Dimensions of the gel: 86 x 77 x 1.0 mm

| Name of the Solution | Separating gel | Stacking gel |

| 12.5 % | 15 % | 4 % |

Monomeric solution | 6.25 ml | 7.5 ml | 0.665 ml |

Separating Gel Buffer pH 8.8 | 3.75 ml | 3.5 ml | --- |

Stacking Gel Buffer (0.5 M) | --- | --- | 1.25 ml |

pH 6.8 | | | |

Sodium Dodecyl Sulphate 10 % | 150 µl | 150 µl | 50 ml |

Purified water | 4.77 ml | 3.52 ml | 3.05 ml |

APS 10 % | 75 (l | 75 (l | 25 (l |

TEMED | 10(l | 10(l | 5 (l |

Annexure 2

QUALITY CONTROL DEPARTMENT |
 QUANTITIES OF THE SAMPLES TO BE LOADED IN THE GEL |

1. HOEFER SERIES

1. 25µg of reduced protein should be applied for all the samples along with Reference Material.
2. 10µl of Molecular Weight Marker should be applied along with all ARM samples.
3. After running of the gels at a constant current of 25mA, they are stained with Coomassie Brilliant Blue G-250.

2. MINI GEL

1. 20µg of reduced protein should be applied for all the samples along with Reference Material.
2. 5µl of Molecular Weight Marker should be applied along with all ARM samples.
3. After running of the gels at a constant current of 12mA, they are stained with Coomassie Brilliant Blue G-250.

[1.0]Objective: |
 ||To lay down the procedure for calculating the Specific Activity of the |
 ||Active Raw Material of Recombinant Hepatitis B vaccine. |
 [2.0]Scope: |
 ||This Standard Operating Procedure is applicable to the Active Raw Material |
 ||(ARM) of Hepatitis-B vaccine (rDNA). |
 [3.0]Responsibility: |
	3.1	The analyst of the shall be responsible for executing the described
		procedure correctly.
	3.2	The supervisor of the laboratory shall be responsible for verifying
		that the procedure is followed correctly and for checking the results.
	3.3	The Departmental Head shall be responsible for the overall compliance
		of the SOP .
[4.0]Abbreviations and Definitions:		
	4.1	µg
	4.2	nm
	4.3	ARM
	4.4	QA
	4.5	QC
	4.6	SOP
[5.0] Procedure:		
	5.1	Note the results obtained of the test for the Determination of
		protein concentration by a specific method (example: ELISA Method)
		and the Determination of the concentration of total proteins by
		general method (example: Lowry's method).
	5.2	Calculations:
		5.2.1
		5.2.3
		5.2.4
		Note :The value of specific activity is expressed in percentage
	5.3	Acceptance limits:
		5.3.1
[6.0]References:		
	NA	

S. No | Test/Assay | Specifications

- 1 | Physical Appearance | Brownish to yellow liquid without suspended particles
- 2 | pH | 6.0 -7.4
- 3 | PRP Content | Not less than 0.5 mg/ml
- 4 | Protein content | Not less than 0.92 mg/ml
- 5 | PRP to protein ratio | 0.30-0.55
- 6 | Free Polysaccharide | < 20%
- 7 | Free Protein | < 1%
- 8 | KD (Molecular size distribution) | ≥ 85 % elution at KD < 0.5
- 9 | Identity | Positive
- 10 | Sterility | No microbial growth.
- 11 | EDAC Content | < 1.0 $\mu\text{g/ml}$
- 12 | CDAP Content | < 200 ng /dose (10 μg PRP)
- 13 | Endotoxin Content | <25 EU/ μg of PRP

1.0 |Objective: |
|To lay down the procedure for determination of pH. |
2.0 |Scope: |
|This Standard Operating Procedure is applicable to all the In- process |
|and finished product samples. |
3.0 |Abbreviations and Definitions: |
|Not applicable |
4.0 |Procedure: |
4.1	Equipment, Materials and Reagents	
	4.1.1	Tissue paper
	4.1.2	100 ml. vial
	4.1.3	100 ml beaker
	4.1.4	Wash bottle
	4.1.5	Purified water
4.2	Preparation of solutions:	
	4.2.1	Buffer solution, pH =4.00
	4.2.2	Buffer solution, pH =7.00
	4.2.3	Buffer solution, pH=9.20
4.3	pH Electrode Calibration:	
	4.3.1	Check that the equipment is correctly connected to electrical
	supply.	
	4.3.2	Switch on the instrument by pressing ON/OFF key on front
	panel.	
	Use a combination of two standard buffer for calibration	
	according to requirements i.e.:	
	Case 1- For solution with pH less than 7.00 uses standard	
	buffers of 4.00 & 7.00.	
	Case 2- For solution with pH more than 7.00 use standard	
	buffers of 7.00 & 9.20	
	Note: The pH meter shall be set for the different buffers to	
	be used at the time of installation based on the future use.	
	This setting is changed only in case the range of pH readings	
	is different than preset.)	
	4.3.3	Clean pH electrode with purified water and blot dry.
	4.3.4	Put the standard buffers that are used in calibration in
	clean beakers.	
	4.3.5	Introduce pH electrode in first buffer according to case 1 or
	case 2 and wait for 2 seconds to stabilize the pH.	
	4.3.6	Remove pH electrode, rinse with purified water and blot dry
	and put in second standard buffer according to case 1 or case	
	2 and wait for 2 seconds to stabilize the pH.	
	4.3.7	Remove pH electrode, rinse with purified water and blot dry
	and put in potassium chloride solution.	
4.4	pH Measurement:	
	4.4.1	Clean pH electrode with purified water and blot dry.
	4.4.2	Put the sample in clean beaker to check the pH.
	4.4.3	Introduce the pH electrode in beaker having sample solution
	4.4.4	Press [Read] key for pH measurement.
	4.4.5	Rinse, blot and store the electrode in a beaker having
	saturated KCl solution.	
5. |Reporting: |
|Record the information of operation and calibration in corresponding log |
|book |
6. |References: |
|pH meter instruction manual |

1.0 |Objective |
 | |To lay down Procedure of Nitrate reduction Test. |
 2.0 |Scope |
 | |This Standard Operating Procedure is applicable to identification of |
 | |Nitrate reduction organism for all samples. |
 3.0 |Responsibility |
	3.1	The Analyst shall be responsible for carrying out the procedure as
	established in this SOP.	
	3.2	Supervisor shall be responsible for verifying that the same has been
	carried out correctly.	
	3.3	Head QC or his designee shall be responsible for overall compliance
	of the SOP.	
4.0	Abbreviations and Definitions	
	4.1	SOP
	4.2	QA
	4.3	QC
	4.4	mL
	4.5	µL
5.0	Procedure:	
	5.1	Equipment, Materials and Reagents
	5.1.1	Autoclave
	5.1.4	Nitrate Reduction Kit (Cat no 73426,Milipore)
	5.1.5	Incubator
	5.1.6	Glass tubes, Durham tube
	5.1.7	Weighing balance
	5.1.8	pH meter
	5.1.9	Magnetic stirrer with hot plate
	5.2	Preparation of solutions
	5.2.1	Nitrate broth
		5.2.1.1
		5.2.1.2
		5.2.1.3
		5.2.1.4
		5.2.1.5
	5.2.2	Sulfanilic acid solution (Reagent A): Dissolve 8 g of
		sulfanilic acid in 1 litre 5N acetic acid. Store Reagent A at
		room temperature for up to 3 months, in dark.
	5.2.3	α-Naphthylamine solution (Reagent B): Dissolve 6 g of
		N,N-Dimethyl-1-naphthylamine in 1 litre 5N acetic acid. Store
		Reagent B at 2 to 8°C for up to 3 months, in dark.
	5.3	Test procedure
	5.3.1	Inoculate the nitrate broths with bacterial suspension.
	5.3.2	Incubate the tubes at the optimal temperature 30°C or 37°C for
		24 hours.
	5.3.3	After incubation look for N₂ gas first before adding reagents.
	5.3.4	After incubation, Put a 5 drops of reagent A and 5 drops of
		reagent B into the tube containing culture to be tested. Shake
		the tube well to mix reagents with medium.
	5.3.5	Observe for the reaction (color development).
	5.4	Interpretation of results:
	5.4.1	If the suspension turns pink-red color, the reaction is
	5.4.2	positive.
		If the suspension is colorless after the addition of reagents A
		and B, add a small amount ("sharp knife point") of zinc powder
		to the medium. Shake the tube vigorously and allow it to stand
		at room temperature for 10-15 min.
	5.4.3	If the medium remains colorless after the addition of Zn
		powder, the test result is positive.
	5.4.4	If the medium turns pink after the addition of Zn powder, the
		result is negative.
6.0	Forms and Records	
	Test for Citrate Utilization Test	
7.0	References	
	Society of American Bacteriologist, Pure Culture Study of Bacteria, 12 :	
	Leaflet 11, 8 (1944) 2. Ewing, Edwards and Ewings Identification of	
	Enterobacteriaceae, 4th ed., Elsevier Science Pub. Co., Inc., N.Y. (1986)	

Annexure -1
 Organisms (ATCC) Growth Nitrate
 reduction
 Acinetobacter calcoaceticus (19606) +++
 -
 Enterobacter aerogenes (13048) +++
 +
 Escherichia coli (25922) +++
 +
 Salmonella typhimurium (14028) +++
 +

[pic]

Date : 05-May-2016
CERTIFICATE OF ANALYSIS
For BioBridge Healthcare Solutions Pvt. Ltd.
Authorized Signatory
Product: | Bulk of Pertussis
Product code no.: | B/BP 002
Appearance: | Light creamish turbid suspension of inactivation harvests of Bordetella pertussis in saline.
Form: | Liquid
Buffer: | Phosphates of Sodium and Potassium and Sodium Chloride
Manufacturing Date: | OCT 2015
Expiry Date: | Mar 2018
Opacity: | ≥ 192 IOU / ml
Sterility: | Sterile Product
pH: | 7.2
Specific Toxicity: | Complied
Identity: | Positive (Presence of agglutinogens 1,2 and 3)
Potency
(at a concentration not more than 20 IOU/SHD, Adjuvanted) | Not less than 4 IU/dose
Lower confidence limit ($p=0.95$) : minimum 2 IU/dose
Storage: | $5\pm 3^{\circ}\text{C}$ (Do not freeze)
Note: To be used for research or in-vitro testing purpose only | Note: To be used for research or in-vitro testing purpose only

1.0 |Objective |
 |To lay down the procedure for preparation of Hepatitis B vaccine Active Raw |
 |Material (ARM) sample by desorption of Aluminium Hydroxide gel for SDS-PAGE.|
 2.0 |Scope |
 |This Standard Operating Procedure is applicable to all Active Raw Material |
 |samples adjuvanted to Aluminium Hydroxide gel. |
 3.0 |Responsibility |
3.1	The analyst shall be responsible for carrying out the described
	procedure correctly.
3.2	The Supervisor of the laboratory shall be responsible for verifying
	that the procedure is complied with and for checking the results.
3.3	The departmental head/designee shall be responsible for overall
	compliance of the SOP.
3.4	Head QA/Designee shall be responsible for final approval and
	compliance of the SOP.
4.0	Abbreviations and Definitions
4.1	ARM
4.2	BPB
4.3	g
4.4	M
4.5	mg
4.6	mL
4.7	QA
4.8	QC
4.9	rpm
4.10	SDS
4.11	SDS-PAGE
4.12	TEMED
4.13	°C
4.14	µg
4.15	µL
5.0	Procedure
5.1	Precautions to be taken:
	5.1.1
	5.1.2
	5.1.3
5.2	Requirement:
	5.2.1
	5.2.2
	5.2.3
5.3	Preparation of solutions:
	5.3.1
	5.3.2
	5.3.3
	5.3.4

||| |can be stored at 2-8°C for 6 months. |

|| |5.3.5 |Sodium Dodecyl Sulphate (SDS) – 20% |

|| | |For a volume of 100mL, weigh: |

|| | |SDS |20g |

|| | |Dissolve in 75mL of Purified water in a beaker. Make up the |

|| | |volume to 100mL with Purified water in a volumetric flask. This |

|| | |solution can be stored at room temperature for 6 months. |

|| |5.3.6 |Bromophenol Blue (1%) |

|| | |For a volume of 5mL, weigh: |

|| | |Bromophenol blue |0.05g |

|| | |Dissolve in 4mL of purified water in a beaker. Make up the |

|| | |volume to 5mL with purified water in a volumetric flask. Divide|

|| | |in aliquots of 500µL in eppendorf. This can be stored at -20±5°C|

|| | |for 6 months. |

|| |5.3.8 |Reducer 2X without dye: |

|| | |For a volume of 10 mL, add the following solutions: |

|| | |Stacking gel Buffer Tris HCl 2.0 M |0.625 mL |

|| | |SDS 20% |2.0 mL |

|| | |Glycerol |2.0 mL |

|| | |2 -Mercaptoethanol |1.0 mL |

|| | |Make up the volume to 10 mL with purified water. Divide in |

|| | |aliquots of 500µL in eppendorfs and freeze at – 20 ±2° C. This |

|| | |solution can be stored upto 6 months. |

|| |5.4 |Desorption of Active Raw Material (ARM) from Aluminium Hydroxide Gel: |

|| |5.4.1 |Homogenize the reference vaccine and the test vaccine vials to |

|| | |be analyzed and take 1mL of homogenized samples in different |

|| | |eppendorfs. |

|| |5.4.2 |Centrifuge the samples at 2000 rpm for 15 minutes, at room |

|| | |temperature. |

|| |5.4.3 |Discard the supernatant carefully without disturbing the pellet.|

|| |5.4.4 |Add 100 µL of Phosphate Buffer Solution to the pellet. |

|| |5.4.5 |Add 100 µL of Reducer 2X without dye to the pellet. |

|| |5.4.6 |Shake well to homogenize the sample mixture. |

|| |5.4.7 |Heat the eppendorfs containing sample mixture at 95°C for 3 |

|| | |minutes. |

|| |5.4.8 |Centrifuge the sample mixtures at 3000 rpm for 10 minutes, at |

|| | |room temperature. |

|| |5.5 |Preparation of Samples for loading in SDS PAGE: |

|| |5.6.1 |For gel running on Minigel, take 3 µL of supernatant from |

|| | |desorbed sample mixture in a fresh eppendorf and add 0.3 µL of |

|| | |1% Bromophenol Blue. 0.02(g of Reference Active Raw Material |

|| | |should also be loaded along with desorbed vaccine samples, which|

|| | |is heated at 75°C for 10 minutes, which acts the Stopping Point |

|| | |for silver nitrate staining. |

|| |5.6.2 |Homogenize and centrifuge all the samples at 8000 rpm for 3 to 5|

|| | |minutes. |

|| |5.6.3 |Load all desorbed vaccine sample that is 3.3 µL for Minigel. |

|| | |Load Reference Active Raw Material as per the above said protein|

|| | |concentration. |

|| |5.6.4 |Running of the Gel: |

|| | |As per QC Procedure For Separation Of Proteins By Sodium Dodecyl|

|| | |Sulphate Polyacrylamide Gel Electrophoresis (SDS-PAGE) |

|| |5.6.5 |Staining of the gel with Silver Nitrate: |

|| | |As per QC Procedure For Separation Of Proteins By Sodium Dodecyl|

|| | |Sulphate Polyacrylamide Gel Electrophoresis (SDS-PAGE) |

|| |5.6.6 |Analyzing the protein bands separated: |

|| | |Analyze the protein bands as per As per QC Procedure For |

|| | |Separation Of Proteins By Sodium Dodecyl Sulphate Polyacrylamide|

|| | |Gel Electrophoresis (SDS-PAGE). |

|| |6.0 |References: |

|| |6.1 |Gel Electrophoresis of Proteins: A Practical Approach. Ed. B. D. Hames|

|| | |and D. Rick wood. IRL Press. 1981. |

|| |6.2 |Quantity one – Quantitation software user guide for version-4 Windows |

|| | |and Macintosh M/s BIO-RAD Laboratories (India) Pvt. Ltd. |

|| |6.3 |Help option provided with each analysis option in the Quantity One |

|| | |Imaging software. |

1.0 |Objective |
|To lay down the procedure for determining the Maximal Toxic Value (MTV) of |
|tetanus toxin and also to determine the correct flocculation zone that has |
|be used to determine the Lf. |
2.0 |Scope |
|This Standard Operating Procedure is applicable to all batches of Teatnus |
|toxin. |
3.0 |Responsibility |
3.1	The QC analyst shall be responsible for performing the described
	procedure correctly.
3.2	Supervisor shall be responsible for verifying that the same has been
	carried out correctly.
3.3	Head Animal House/Veterinarian/supervisor shall be responsible for
	supply of quality animals and their maintenance.
3.4	Head QC or his designee shall be responsible for overall compliance
	of the SOP.
4.0	Abbreviations and Definitions
4.1	SOP
4.2	QA
4.3	QC
4.4	mL
4.5	SC
4.6	MTV
5.0	Procedure:
5.1	Equipment, Materials and Reagents
	5.1.1
	5.1.2
	5.1.3
	5.1.4
	5.1.5
	5.1.6
	5.1.7
	5.1.8
	5.1.9
5.2	Preparation of solutions
	5.2.1
5.3	Sample Preparation
	Maximal Toxic Value (MTV) is that value (expressed in terms of Lf/ml)
	where none of the mice show any symptoms of classical tetanus.
	5.3.1
	5.3.2
	5.3.3
	5.3.4
5.4	Preparation of test animals:
	5.4.1
	5.4.2
	5.4.3
5.5	Animal Inoculation:
	Inject subcutaneously 0.5 ml of each mixture into each of two mice.
5.6	Observations:
	5.6.1
	5.6.2
	5.6.3
6.0	Forms and Records
The following form shall be filled during the performance of the test and	
for the compilation of the results.	
6.1	Determination of Maximum Toxin Value (MTV) of Tetanus Toxin
7.0	References
WHO manual for the production and control of vaccines – Tetanus Toxoid	
(BLG/UNDP/77.2 Rev 1)	

1.0|Purpose: |
 |To lay down the procedure for the estimation of molecular size of PRP-TT |
 |conjugate in bulk conjugate of Haemophilus Influenzae type b conjugate |
 |vaccine. |
 2.0|Applicable to: |
 |This Standard Operating Procedure is applicable for evaluating the |
 |molecular size distribution and percentage elution of bulk conjugate of |
 |Haemophilus Influenzae type b conjugate vaccine. |
 3.0|Abbreviations and Definitions: |
3.1	CDAP	: 1-cyano-4-dimethylaminopyridinium tetrafluoroborate
3.2	cm	: centimeter
3.3	g	: gram
3.4	HIBTT	: Haemophilus Influenzae type b - Tetanus toxoid
3.5	HPLC	: High Performance Liquid Chromatography
3.6	i.d.	: Internal diameter
3.7	l	: length
3.8	M	: Molar
3.9	m	: meter
3.10	mAu	: milli Absorbance unit
3.11	mg	: milligram
3.12	min	: minute
3.13	mL	: Millilitre
3.14	mm	: milli meter
3.15	nm	: Nano meter
3.16	No.	: number
3.17	PRP-TT	: Polyribosyl Ribitol Phosphate - Tetanus Toxoid
3.18	RSD	: Relative Standard Deviation
3.19	SST	: System Suitability Test
3.20	°C	: Degree centigrade
3.21	µg	: microgram
3.22	µL	: microliter
3.23	µm	: micrometer
3.24	System	: System suitability tests are an integral part of gas and
	liquid chromatographic methods. They are used to verify	
	the resolution and reproducibility of the	
	chromatographic system are adequate for the analysis to be	
	done.	
	The tests are based on the concept that the equipment,	
	electronics, analytical operations and samples to be	
	analyzed constitute an integral system that can be	
	evaluated as such.	
 4. |Procedure: |
 |4.1 |Precautions to be taken: | | |
 | |4.1.1 |Always use filtered solutions. Filtration of solution |
 | |should be done in order to eliminate the entry of possible |
 | |particles, if any, into the system as well as the column. |
 | |4.1.2 |Degas all solutions. |
 | |4.1.3 |Avoid entry of bubbles into the system as well as column. |
 | |4.1.4 |All the solution bottle openings should be well covered |
 | |with aluminum foil. |
 | |4.1.5 |Check the calibration status of all equipment and |
 | |instrument. |
 |4.2 |Requirement: |
 | |4.2.1 |Equipment: |
 | | |4.2.1.1 |Calibrated HPLC System with Chem Station software. |
 | | |4.2.1.2 |Ultrasonic Cleaner |
 | | |4.2.1.3 |Vacuum Pump |
 | | |4.2.1.4 |Refrigerator |
 | |4.2.2 |Instrument: |
 | | |4.2.2.1 |pH Meter |
 | | |4.2.2.2 |Magnetic stirrer |
 | | |4.2.2.3 |Precision balance |
 | | |4.2.2.4 |Analytical balance |
 | |4.2.3 |Material: |
 | | |4.2.3.1 |TSK gel G5000PWxl column {30 cm (l), 7.8 mm (i.d)} |
 | | |4.2.3.2 |Laboratory Glassware |
 | | |4.2.3.3 |Screw cap vials |
 | | |4.2.3.4 |Solvac Filter Holder |
 | | |4.2.3.5 |Nylon 6,6 membrane 0.20/0.22/0.45 µm membrane filter |
 | | |4.2.3.6 |Autopipette (2-20µL), (20-200(L), (100-1000(L) |
 | | |4.2.3.7 |Tips for the Autopipette |
 | | |4.2.3.8 |Aluminium foil |
 | |4.2.4 |Reagents: |
 | | |4.2.4.1 |Sodium Chloride (NaCl) |
 | | |4.2.4.2 |Disodium hydrogen phosphate anhydrous (Na₂HPO₄) |
 | | |4.2.4.3 |Sodium dihydrogen phosphate monohydrate |
 | | |4.2.4.4 |Purified water |
 | | |4.2.4.5 |Sodium azide |
 | | |4.2.4.6 |Salmon DNA |
 | | |4.2.4.7 |2-Propanol |
 |4.3 |Preparation of solutions: |
 | |Prepare all the solutions and Carry out the labelling of the prepared |
 | |solutions and assign the lot numbers to the solutions. |
 | |4.3.1 |Isopropyl alcohol 70% |
 | | |Take 700 mL of 2-propanol in a volumetric flask and make up |
 | | |the volume upto 1000 mL with Purified Water. This solution can |
 | | |be stored at room temperature for 3 months. |
 | |4.3.2 |Solution I: [Na₂HPO₄ (0.1M), NaCl (0.1M)] |
 | | |For a volume of 1000 mL, weigh: |

||| |Na₂HPO₄ |14.19 g |
||| |NaCl |5.84 g |

		Dissolve in 700mL of purified water in a volumetric flask and
		make up the volume upto 1000mL. Prepare fresh solution before
		use.

		4.3.3	Solution II: NaH₂PO₄.H₂O, 0.1 M
		For a volume of 500 mL, weigh:	
		NaH₂P0₄.H₂O	6.85g
		Dissolve in 400mL of purified water in a volumetric flask and	
		make up the volume upto 500mL. Prepare fresh solution before	
		use.	

		4.3.4	0.1M Sodium Phosphate buffer
			(0.1 M Na₂HPO₄,0.1M NaCl +0.1M NaH₂PO₄.H₂O, pH 7.0)
			For a volume of 1000mL, take approximately 600mL of Solution I
			and 400mL of Solution II. The pH of the solution should be 7.0
			± 0.2. Use the mobile phase within four days from the date of
			preparation if it remains clear and no turbidity or fibers
			observed in it.

		4.3.5	Void Volume Marker (Salmon DNA):	
			For a volume of 5 mL, weigh:	
			Salmon DNA	5 mg
			Dissolve in purified water and make up volume upto 5 mL with	
			purified water in a volumetric flask. Store at -20 °C ± 5°C	
			for 1 month.	

		4.3.6	Total volume marker (Sodium Azide):	
			For a volume of 10 mL, weigh:	
			Sodium Azide	10 mg
			Dissolve in purified water and make up volume upto 10 mL with	
			purified water in a volumetric flask. Store at 2-8 °C for 1	
			month.	

		4.3.7	Column Preservation Solution:
			Note: All the solutions prepared for use are vacuum filtered
			using Nylon 6, 6 membrane 0.20 µm or 0.22 µm , depending upon
			the nature of solution to be filtered. For organic solvents
			0.45µm filters should be used.

|| |4.4 |Preparing the HPLC System for run, Agilent Series: |
|| |4.5 |Run Conditions: |

		4.5.1	Load method HIBTT.M.
		4.5.2	Flow Rate: 0.5 mL/min
		4.5.3	Injection Quantity : 2.5µg
		4.5.4	Detection Wavelength: 214nm
		4.5.5	Run time: 30 minutes

		4.6	Equilibration of the system and the column:
			Equilibrate the column with mobile 0.1 M sodium phosphate buffer at
			214nm until a stable baseline is observed on the online plot window.

|| |4.7 |Sample Preparation and Application: |

		4.7.1	Test sample and standards are loaded in screw cap vials and
			placed against the appropriate no. on the sample tray by
			clearly marking the screw cap vial.

		4.7.2	Load the required volume of sample as per the PRP
			concentration equivalent to 2.5 µg in the sample vial.
			Example: If a sample has a PRP concentration of 0.5 mg/mL, to
			calculate volume required for 2.5 µg of protein,
			0.5 mg (500 µg) ----1 mL
			2.5 µg-----X mL
			X mL = 2.5 µg x 1 mL
			500 µg
			X mL = 5 µL
			Therefore, 5 µL of volume is taken for sample having PRP
			concentration of 0.5 mg/mL.

|| |4.8 |Running of sequence: |

|| |4.8.1 |Load sequence HIBTT.S. |

|| |4.8.2 |Open the sequence parameter option by clicking Sequence menu. |

|| |4.8.3 |Enter information about the data directory and operator name. |

|| |4.8.4 |In the shutdown option, click on the post-sequence option and |
|| | |select LAMPOFF/STANDBY. |

|| |4.8.5 |In sequence table option, specify the sample names, methods |
|| | |and vials to run in the sequence as per the following table: |

S No	Sample Code	Sample details	No. of Injection
1	Blank	Eluent Blank	1
2	Sodium azide (Total	System	6
		volume marker)	Suitability
3	Blank	Eluent Blank	1
4	Salmon DNA	Void volume	1
5	Blank	Eluent Blank	1
6	Test Sample (s)	Sample	1

			Inject	volume (µL)			
				ons			
			1	Blank	Eluent Blank	1	10
			2	Sodium azide (Total	System	6	1
				volume marker)	Suitability		
			Test				
			3	Blank	Eluent Blank	1	10
			4	Salmon DNA	Void volume	1	10
				marker			
			5	Blank	Eluent Blank	1	10
			6	Test Sample (s)	Sample	1	2.5µg of PRP
			Click on run sequence option in run control menu to start				
			the sequence.				

|| |5.8.8|Each sample run will be of 30 min and after running all |
|| | |the samples the sequence will automatically stop. |

		4.9	Calculations :
			Area under curve for the standards/sample is analyzed by
			Chemstation software.

|| |4.9.1|The retention time of the void volume is determined with |
|| | |Salmon DNA. |

|| |4.9.2|The Retention time of the component (total volume) which |

				has full access to the pores of stationary phase is	
				determined with sodium azide.	
				4.9.3	Elution time of PRP-TT (Te) at KD = 0.50
				The calculations are made as per the following equation:	
				Te = KD (Tt - To) + To or Te = 0.50 (Tt -	
				To) + To	
				Where,	
				Te = Elution time of PRP-TT Bulk Conjugate at KD = 0.50	
				Tt = Retention time of sodium azide	
				To = Retention time of Salmon DNA	
				KD = Distribution Coefficient	
				Note: Take average RT of sodium azide as Tt from EXCEL	
				for calculation of SST.	
				4.9.4	Calculation of percentage of sample eluted before a given
				KD:	
				The percentage of sample eluted before a given KD is	
				calculated using ChemStation software.	
				4.9.4.1	Open the acquired chromatogram in the data
				analysis view in the offline copy of the	
				software.	
				4.9.4.2	Optimize integration with initial events in the
				signal specific events table.	
				4.9.4.3	Enter the value of Elution time (Te) calculated
				in step 5.9.3 in integration with timed events	
				option or with manual integration and select	
				split peak option.	
				4.9.4.4	The percentage of sample eluted at $KD \leq 0.5$ is
				calculated by adding the values of percentage of	
				area (Area %) obtained for bulk conjugate curve	
				till Te.	
				4.9.5	Calculations for System Suitability Test:
				4.9.5.1	Note the values of area under the curve and
				retention time for sodium azide (SST standard) in	
				Annexure 1 and also in EXCEL format for	
				calculation of Relative standard deviation.	
				4.9.5.2	Every time a sample is run with system
				suitability test, calculate the results for	
				system suitability.	
				4.9.5.3	Attach the hard print of SST calculation along
				with chromatograms and the record form.	
		4.10	Acceptance Criteria :		
		4.10.	For System Suitability Test :		
			1		
			4.10.1.	RSD calculated for Retention time should be <	
			1	5.0%.	
			4.10.1.	RSD calculated for Area should be < 5.0%.	
			2		
			4.10.	For ARM of Haemophilus Influenzae type b conjugate	
		2	vaccine Samples: KD (Molecular Size distribution): $\geq 85\%$,		
			KD ≤ 0.50		

5.0	Forms and Records:
5.1	Molecular sizing of polyribosyl ribitol phosphate (PRP) – Tetanus
	toxoid conjugate by high performance size exclusion chromatography:
6.0	References:
	United States Pharmacopeia 29.

[1. |Purpose: |
 |To lay down the procedure for quantifying the cesium chloride concentration. |
 [2. |Equipments: |
[2.1	Atomic absorption spectrophotometer
[2.2	Analytical balance Sensitivity 0.01mg
[2.3	Vortex Mixer
[3.	Materials and Reagents:
[3.1	Materials:
	[3.1.1
	[3.1.2
	[3.1.3
	[3.1.4
	[3.1.5
	[3.1.6
[3.2	Reagents:
	[3.2.1
	[3.2.2
	[3.2.3
	[3.2.4
	[3.2.5
	[3.2.6
[4.	Preparation of solutions:
[4.1	Cesium standard stock solution (1000ppm):
	[4.1.1
	[4.1.2
[4.2	Cesium working solution (50ppm)
	[4.2.1
	[4.2.2
[4.3	10% Potassium chloride (KCl) solution:
	[4.3.1
	[4.3.2
[4.4	Phosphate buffer solution:
	[4.4.1
	[4.4.2
	[4.4.3
[5.	Procedure:
[5.1	Preparation of Calibration Curve:
	The concentration of cesium chloride in the test sample (s) is
	calculated by comparing results of its absorbance obtained against
	the plotted calibration curve of known cesium concentration (0.5 –
	3.0ppm) Vs absorbance. The solutions for making cesium calibration
	curve (0.5 – 3.0ppm) are prepared as follows:
	S.No.
	1
	2
	3
	Prepare a blank solution for calibration curve by taking 2 ml of 10%
	KCl solution and make up the volume to 50 ml with purified water.
[5.2	Take 4.8 ml of phosphate buffer diluent and add 0.2 ml of 10% KCl
	solution. Treat this solution as sample blank.
[5.3	Take 4.8 ml of sample and add 0.2 ml of 10% KCl solution. Treat this
	solution as sample for taking absorbance.
[5.4	Conditions for Atomic absorption spectrophotometer:
	Flame
	Slit width
	Wavelength
	Calibration Type
	Integration time
	Replicates
	Read delay
	Technique
[5.5	The instrument aspirates the standards and samples to be tested and a
	standard curve is generated by the system by taking absorbance vs
	concentration.
	Cesium content (ppm) in sample is displayed directly.
[5.6	Calculations:
	Instrument reading (ppm) x total volume (5 ml)
	Cesium content (ppm) =

	Actual sample taken (4.8ml)
	Cesium content (ppm) x
	mol.wt. of CsCl
	Cesium chloride (ppm) =

	mol. wt. Of Cesium
[5.7	Acceptance criteria:

	5.7.1	The coefficient of variation (CV) between the replicates of
		the samples should be $\leq 5\%$.
	5.7.2	Cesium chloride content in the sample should be within the
		specified limits if specified.

[1. |Purpose: |
|To lay down the procedure for Qualitative determination of Hepatitis-B |
|surface antigen using monoclonal Hepatitis-B antibody. |
[2. |Applicable to: |
|This procedure is applicable to the Active Raw Material (ARM) of |
|Hepatitis-B vaccine (rDNA) |
[3. |Responsibility: |
3.1	The analyst of the Immunochemistry laboratory is responsible for
	executing the described procedure correctly.
3.2	The supervisor of the laboratory is responsible for verifying
	that the procedure is followed correctly and for checking the
	results.
3.3	The Departmental Head is responsible for the overall compliance
	of the SOP and shall ensure that the SOP is reviewed every two
	years or if required earlier.
[4.	Safety considerations:
4.1	Wear laboratory coat and gloves while working.
4.2	Check the polarity of electrodes before connecting the equipment.
4.3	Do not adjust the pH of the transfer buffer.
4.4	Handle acid and alkali carefully and in a fume hood.
[5.	Equipments:
5.1	Magnetic Stirrer
5.2	Gel Rocker
5.3	pH Meter
5.4	Trans-blot (Semi-dry transfer cell, BIORAD)
5.5	Analytical Balance
5.6	Precision Balance
5.7	PVC Thermal Sealer
5.8	Deep freezer (–20.0 + 5°C)
5.9	Refrigerator (2-8°C)
[6.	Materials and reagents:
6.1	Materials:
	6.1.1
	6.1.2
	6.1.3
	6.1.4
	6.1.5
	6.1.6
	6.1.7
6.2	Reagents:
	Name of the Reagent
	6.2.1
	6.2.2
	6.2.3
	6.2.4
	6.2.5
	6.2.6
	6.2.7
	6.2.8
	6.2.9
	6.2.10
	6.2.11
	6.2.12
	6.2.13
	6.2.14
	6.2.15
	6.2.16
	6.2.17
	6.2.18
	6.2.19
[7.	Preparation of solutions:
7.1	Phosphate Buffered Saline Solution 10 X (PBS 10X)
	NaCl 1.37M, KCl 0.0261 M, Na₂HPO₄ 0.0845 M, KH₂PO₄ 0.0169 M
	For a final volume of 1litre, weigh:
	Reagent
	Quantity
	NaCl
	80.0 g
	KCl
	2.0 g
	Na₂HPO₄ or
	Na₂HPO₄·7H₂O
	11.5 g or
	21.5 g
	KH₂PO₄
	2.4 g
	Dissolve the reagents in 750 ml of purified water, with magnetic
	stirring along with heat in order to make dissolution of reagents
	easier. Cool the solution to room temperature and adjust the pH
	to 7.2 with 40% NaOH. Make up the final volume to 1 litre with
	purified water. This solution can be stored at 2-8°C for 2
	months.

	7.2	Phosphate Buffered Saline Solution 1X (PBS 1X)	
		NaCl 0.137 M, KCl 0.00261 M, Na₂HPO₄ 0.00845 M, KH₂PO₄ 0.00169	
		M]	
		For a final volume of 1 litre take 100 ml of PBS 10X and add 750	
		ml of purified water. Mix well and make up the final volume with	
		purified water. This solution can be stored at 2-8°C for 1 month.	
	7.3	Semi-dry transfer buffer with SDS [48 mM Tris, 39 mM Glycine,	
		20 % Methanol and 1.3 mM SDS]	
		For a final volume of 1000 ml:	
		Reagent	
		Quantity	
		Tris	
		5.81 g	
		Glycine	
		2.93 g	
		SDS	
		0.0375 g	
		Methanol	
		200 ml	
		Dissolve the reagents, except methanol in 600 ml of purified	
		water and check that the pH is between 8.5 and 9.0, otherwise	
		discard the solution. Now add 200 ml methanol and make up the	
		final volume to 1litre with purified water. It can be stored at	
		room temperature for 1 week.	
		Note: Do not adjust the pH of the transfer buffer.	
	7.4	Washing Solution [0.1 % of Nonidet P-40 (v/v) in PBS-1X]	
		Using an autopipette, take 1000 µl of Nonidet P-40 and mix well	
		in 800 ml of PBS (1X). Make up the final volume to 1 litre using	
		PBS (1X). This solution can be stored at room temperature for 1	
		month.	
	7.5	Blocking Solution.	
		For a final volume of 100 ml:	
		Reagent	
		Quantity	
		Skimmed Milk powder	
		5.0 g	
		Tween-20	
		50 µl	
		Dissolve the skimmed milk powder in 80 ml of PBS (1X) and add	
		Tween-20. Mix thoroughly and dilute finally to 100 ml with PBS	
		(1X). This solution should be freshly prepared at the time of	
		use.	
	7.6	Substrate solution:	
		Dissolve 15 mg of 3,3'-Diaminobenzidine in 30 ml of PBS 1X and	
		just prior to use, add 15 µl H₂O₂ peroxide. This solution should	
		be freshly prepared.	
	7.7	NaOH (40%) solution:	
		Dissolve 40 g of NaOH in 80 ml of purified water and dilute	
		finally to 100 ml using purified water. This solution can be	
		stored at room temperature for two months.	
8.	Procedure:		
	8.1	Electrophoresis: Electrophoresis is carried out in the	
		Electrophoresis and Chromatography laboratory. The test sample	
		(ARM of Hepatitis-B vaccine (rDNA)) and a Reference ARM are	
		treated with reducer buffer and then applied on polyacrylamide	
		gel containing SDS. After the electrophoretic run is over, the	
		gel is handed over to Immunochemistry laboratory for further	
		processing.	
	8.2	Electro transfer (Semi-dry):	
		8.2.1	Cut a nitrocellulose membrane and two extra thick blotting
			sheets according to the size of the gel.
		8.2.2	Cut the upper left corner of the gel and the nitrocellulose
			membrane to identify the order in which samples were
			applied.
		8.2.3	Wet the nitrocellulose membrane, blotting sheets and the
			gel in the transfer buffer.
		8.2.4	Place one blotting sheet wetted in transfer buffer on a
			clean surface and place the nitrocellulose membrane on top
			of it. Now neatly place the gel above the nitrocellulose
			membrane, avoiding any air bubbles. Then place another
			blotting sheet wetted in transfer buffer, on top of the
			gel. This results in a gel and membrane sandwich between
			two blotting sheets.
			Note: The nitrocellulose membrane and the blotting sheets
			should be handled with gloves and there should be no air
			bubbles between the different layers of the gel-membrane
			sandwich.
		8.2.5	Lift the cover and the electrode plate of the transfer
			equipment carefully. Clean the electrode plate using a
			tissue paper soaked in purified water.
		8.2.6	Place the sandwich in the centre of the electrode plate
			such that the nitrocellulose membrane is on the lower side
			(anode) and the gel is on the upper side (cathode).

	8.2.7	The current is calculated by multiplying the area of the	
		gel by a factor (For a small gel this factor is 5.5 and for	
		bigger gels it is 3).	
	8.2.8	Place back the cathode plate and the safety cover.	
	8.2.9	Confirm the polarity of the electrodes.	
	8.2.10	Run the transfer for 30 minutes for both small and bigger	
		gels.	
	8.2.11	After the transfer is over, turn off the power supply,	
		disassemble the system and separate the sandwich carefully.	
	8.2.12	Again clean the electrode plate using a tissue paper soaked	
		in purified water.	
	8.2.13	Stain the gel. The absence of bands in the gel is an	
		indication of the complete transfer of proteins onto the	
		nitrocellulose membrane.	
	8.2.14	Proceed for the immunoidentification step.	
	8.3	Immunoidentification:	
	8.3.1	Blocking:	
		Put the nitrocellulose membrane in a PVC bag and seal from	
		three sides using a PVC thermal sealer. Add adequate volume	
		of blocking solution, so as to cover the entire membrane.	
		Remove air bubbles if any present and then seal the fourth	
		side of the PVC bag. Incubate for one hour at 37.0 + 2°C or	
		overnight at 2-8°C under gentle shaking conditions.	
		Note: After the blocking step, if required, the	
		nitrocellulose membrane can be kept between two blotting	
		sheets and stored in a sealed PVC bag for 24 hours at	
		2-8°C.	
	8.3.2	Treatment with Hepatitis B monoclonal antibody:	
	8.3.2.1	Dilute monoclonal antibody against Hepatitis-B	
		surface antigen, using blocking solution (without	
		Tween-20) as diluent to a most effective	
		concentration as determined by titration	
		(Annexure-1).	
	8.3.2.2	Put the nitrocellulose membrane in a PVC bag, add	
		adequate volume of antibody solution so as to cover	
		its entire surface and seal the PVC bag using a	
		thermal sealer.	
	8.3.2.3	Incubate for 1 hour 30 minutes at room temperature	
		for 16 hours at 2-8°C under gentle shaking	
		condition.	
	8.3.3	Washing:	
		Note: The washing is carried out under gentle agitation at	
		room temperature and the solution is discarded every time	
		after washing.	
	8.3.3.1	Wash the nitrocellulose membrane twice for 10	
		minutes with PBS (1X)	
	8.3.3.2	Wash the nitrocellulose membrane once for 10	
		minutes with Washing solution (1X).	
	8.3.3.3	Wash the nitrocellulose membrane twice for 10	
		minutes with PBS (1X).	
	8.3.4	Immunological visualization of results:	
	8.3.4.1	Protein-A peroxidase conjugate is used. Dilute the	
		conjugate in PBS (1X) as per its titration	
		(Annexure-2). The conjugate solution must be	
		prepared just before use.	
	8.3.4.2	Place the nitrocellulose membrane in the PVC bag	
		and add adequate volume of the conjugate solution	
		so as to cover the entire membrane and seal the	
		fourth side of the PVC bag using a thermal sealer.	
		Incubate for 1 hour at room temperature under	
		gentle conditions.	
	8.3.5	Washing:	
		Repeat the washing steps as mentioned in the step 8.3.3 of	
		the procedure.	
	8.3.6	Development of blot:	
	8.3.6.1	Submerge the nitrocellulose membrane in the	
		substrate solution and rock gently until the	
		bands become clearly visible but the background	
		remains clear.	
	8.3.6.2	Rinse the nitrocellulose membrane with purified	
		water and then dry it between two blotting	
		sheets.	
	8.3.7	Analysis of the results:	
		Compare the bands of the samples with that of the	
		Reference (an Active Raw Material previously	
		characterized).	
	9.	Abbreviations:	
	9.1	ARM	: Active Raw Material
	9.2	DAB	: 3,3'-Diaminobenzidine
	9.3	PBS	: Phosphate buffered Saline
	9.4	PVC	: Polyvinyl chloride
	9.5	SDS	: Sodium Dodecyl Sulphate
	9.6	SOP	: Standard Operating Procedure

1. Carry out SDS PAGE of Hepatitis-B surface antigen according to SOP QC/00/028.

In consecutive wells put 25 µg of extracted yeast proteins* as a negative control (NC) followed by 25µg of Positive Control (PC), which is an Active Raw Material (Reference Standard for Western for HBsAg).

Note: The extracted yeast proteins* are obtained by growing non-transformed, host strain of *Pichia pastoris* in fermentor, separating the cells, crushing in dynamill, precipitating, purifying and finally and

||
[pic]	PANERA BIOTEC PVT. LTD.	SOP: QC/00/014 Rev. 02	
	QUALITY CONTROL DEPARTMENT	Form No.:	Page No.: 3 of
		QCR/00/014/A	6
	LABORATORY: IMMUNOCHEMISTRY	Folio:	
	PROCEDURE FOR THE IMMUNOLOGICAL RECOGNITION OF HEPATITIS-B SURFACE		
	ANTIGEN PRESENT IN THE ACTIVE RAW MATERIAL (ARM) BY WESTERN BLOTTING		
PROCEDURE			
1.1 Size of the gel: Small () Big ()			
1.2 Acrylamide concentration in the gel: _____ %			
1.3 Initial time: _____ Final time: _____ Total Running time: _____			
1.4 Current applied (milli Amperes): _____ mA			
2. ELECTROTRANSFER (Semi-dry transfer)			
2.1 Wet the gel, nitrocellulose membrane and blotting sheets in Transfer buffer.			
2.2 Place the sandwich in the centre of the electrode plate such that the			
nitrocellulose membrane is on the lower side (anode) and the gel is on the upper			
side (cathode). Signature: _____			
2.3 Calculate the current to be applied according to the following equation:			
I = _____ • _____ • _____ = _____ mA			
2.4 Applied the current for 30 min.			
Start time: _____ End time: _____			
2.5 Check the transfer of proteins:			
Stain with Coomassie blue G-250 (According to SOP QC/00/029)			
Observations: Appearance of bands Yes [] No []			
Conclusion:			

[pic] |PANERA BIOTEC PVT. LTD. |SOP: QC/00/014 Rev. 02 |
	QUALITY CONTROL DEPARTMENT	Form No.:	Page No.: 4 of 6
		QCR/00/014/A	
	LABORATORY: IMMUNOCHEMISTRY	Folio:	
	PROCEDURE FOR THE IMMUNOLOGICAL RECOGNITION OF HEPATITIS-B SURFACE		
	ANTIGEN PRESENT IN THE ACTIVE RAW MATERIAL (ARM) BY WESTERN BLOTTING		
3 IMMUNOIDENTIFICATION			
3.1BLOCKING:			
Quantity of skimmed milk weighed: _____ Volume of Tween-20 taken: _____			
Final volume of blocking solution: _____ Signature: _____			
INCUBATION: Incubate the nitrocellulose membrane for 1 hour at 37.0 + 2°C []			
or Over night at 2-8°C [] (Tick which ever is applicable)			
Initial Time: _____ Final time: _____			
Signature : _____			
3.2 TREATMENT WITH MONOCLONAL HBs ANTIBODIES.			
3.2.1Dilution details of monoclonal HBs antibodies:			
Working dilution of antibody: _____ Volume of antibodies taken:			

Volume of diluent (blocking solution without Tween-20) taken: _____			
Final volume of antibodies solution: _____			
3.2.2 Incubation: 1 hour 30 min at room temperature [] or 16 h at 4°C []			
(Tick which ever is applicable)			
Initial time: _____ Final time: _____ Signature: _____			
3.3 WASHING:			
3.3.1 Wash twice with PBS (1X) for 10 minutes each. Yes [] No []			
3.3.2 Wash once with Washing solution (1X) for 10 minutes. Yes [] No []			
3.3.3 Wash twice with PBS (1X) for 10 minutes each. Yes [] No []			
Signature: _____			

[pic] |PANERA BIOTEC PVT. LTD. |SOP: QC/00/014 Rev. 02 |
	QUALITY CONTROL DEPARTMENT	Form No.:	Page No.: 5 of 6
		QCR/00/014/A	
	LABORATORY: IMMUNOCHEMISTRY	Folio:	
	PROCEDURE FOR THE IMMUNOLOGICAL RECOGNITION OF HEPATITIS-B SURFACE		
	ANTIGEN PRESENT IN THE ACTIVE RAW MATERIAL (ARM) BY WESTERN BLOTTING		
3.4 INCUBATION WITH CONJUGATE:			
Type of conjugate: Protein-A Peroxidase [] Other: [] Working dilution			
of conjugate: _____			
Volume of conjugate taken: _____ Volume of PBS (1X) taken:			

Incubation: 1 hour at room temperature. Yes [] No []			
Initial time: _____ Final time: _____			
Signature: _____			
3.5 WASHING:			
3.5.1 Wash twice with PBS (1X) for 10 minutes each. Yes [] No []			
3.5.2 Wash once with Washing solution (1X) for 10 minutes. Yes [] No []			
3.5.3 Wash twice with PBS (1X) for 10 minutes each. Yes [] No []			
Signature: _____			
3.6 COLOUR DEVELOPMENT:			
Quantity of 3,3'-Diaminobenzidine: _____ Volume of H2O2 added: _____			

[[pic] |PANERA BIOTEC PVT. LTD. |SOP: QC/00/014 Rev. 02 |
QUALITY CONTROL DEPARTMENT	Form No.:	Page No.: 6 of
	QCR/00/014/A	6
LABORATORY: IMMUNOCHEMISTRY	Folio:	
PROCEDURE FOR THE IMMUNOLOGICAL RECOGNITION OF HEPATITIS-B SURFACE		
ANTIGEN PRESENT IN THE ACTIVE RAW MATERIAL (ARM) BY WESTERN BLOTTING		
RESULTS		
Scanned image of the gel	Scanned image of the blot	

APPROVALS			
Name	Signature	Date	
Done by			
Checked by			
Approved by			
Annexure-4			

S. No.	Name of the Person	Signature	Date	Training imparted by
1	1	1	1	1
2	2	2	2	2
3	3	3	3	3
4	4	4	4	4
5	5	5	5	5
6	6	6	6	6
7	7	7	7	7
8	8	8	8	8
9	9	9	9	9
10	10	10	10	10
11	11	11	11	11
12	12	12	12	12
13	13	13	13	13
14	14	14	14	14
15	15	15	15	15
16	16	16	16	16
17	17	17	17	17
18	18	18	18	18
19	19	19	19	19
20	20	20	20	20
21	21	21	21	21
22	22	22	22	22
23	23	23	23	23
24	24	24	24	24
25	25	25	25	25
26	26	26	26	26
27	27	27	27	27
28	28	28	28	28
29	29	29	29	29
30	30	30	30	30
31	31	31	31	31
32	32	32	32	32
33	33	33	33	33
34	34	34	34	34
35	35	35	35	35
36	36	36	36	36
37	37	37	37	37
38	38	38	38	38
39	39	39	39	39
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41	41	41	41	41
42	42	42	42	42
43	43	43	43	43
44	44	44	44	44
45	45	45	45	45
46	46	46	46	46
47	47	47	47	47
48	48	48	48	48
49	49	49	49	49
50	50	50	50	50
51	51	51	51	51
52	52	52	52	52
53	53	53	53	53
54	54	54	54	54
55	55	55	55	55
56	56	56	56	56
57	57	57	57	57
58	58	58	58	58
59	59	59	59	59
60	60	60	60	60
61	61	61	61	61
62	62	62	62	62
63	63	63	63	63
64	64	64	64	64
65	65	65	65	65
66	66	66	66	66
67	67	67	67	67
68	68	68	68	68
69	69	69	69	69
70	70	70	70	70
71	71	71	71	71
72	72	72	72	72
73	73	73	73	73
74	74	74	74	74
75	75	75	75	75
76	76	76	76	76
77	77	77	77	77
78	78	78	78	78
79	79	79	79	79
80	80	80	80	80
81	81	81	81	81
82	82	82	82	82
83	83	83	83	83
84	84	84	84	84
85	85	85	85	85
86	86	86	86	86
87	87	87	87	

[illegible]

I - Current (mA).
L - Length of the gel (cm).
A - Breadth of the gel (cm).
C - Constant that can be:
C1 - for small gels = 5.5 mA/cm²
C2 - for big gels = 3 mA/cm²

[1.0]Objective |
 |To lay down the procedure for determining the Minimum Lethal Dose (MLD) of |
 |tetanus toxin. |
 [2.0]Scope |
 |This Standard Operating Procedure is applicable to all batches of Tetanus |
 |toxin. |
 [3.0]Responsibility |
3.1	The QC analyst shall be responsible for performing the described
	procedure correctly.
3.2	Supervisor shall be responsible for verifying that the same has been
	carried out correctly.
3.3	Head Animal House/Veterinarian/supervisor shall be responsible for
	supply of quality animals and their maintenance.
3.4	Head QC or his designee shall be responsible for overall compliance of
	the SOP.
[4.0]Abbreviations and Definitions	
4.1	SOP
4.2	QA
4.3	QC
4.4	mL
4.5	SC
4.6	MLD
[5.0]Procedure:	
5.1	Equipment, Materials and Reagents
	5.1.1
	5.1.2
	5.1.3
	5.1.4
	5.1.5
	5.1.7
	5.1.8
	5.1.9
	5.1.10
5.2	Preparation of solutions
	5.2.1
5.3	Sample Preparation
	Dilution of Clarified tetanus Toxin sample:
	5.3.1
	5.3.2
5.4	Preparation of test animals:
	5.4.1
	5.4.2
	5.4.3
5.5	Animal Inoculation:
	Inject subcutaneously 1.0 ml doses of each dilution from vial no.:
	6,7,8,9 and 10 in to the groups of 3 mice each.
5.6	Observations:
	5.6.1
	5.6.2
	5.6.3
	5.6.4
	5.6.5
[6.0]Forms and Records	
Determination of Minimum Lethal Dose (MLD) of Tetanus Toxin.	
 [7.0]References |

1. |Purpose: |
 |To lay down the procedure for the determination of 2-phenoxyethanol by High |
 |Performance Liquid Chromatography (HPLC). |
 2. |Applicable to: |
 |This Standard Operating Procedure is applicable to samples in which |
 |2-Phenoxyethanol is to be determined. |
 3. |Equipments: |
 |3.1 |Chromatographic column (Eclipse XDB-C18, 25cm (l) x 4.6 mm (i.d.) |
 |3.2 |HPLC-Agilent (1100 Series) |
 |3.3 |Chem Station Software |
 |3.4 |Laboratory Centrifuge |
 |3.5 |Refrigerator |
 |3.6 |pH Meter |
 |3.7 |Vacuum Pump |
 4. |Materials and Reagents: |
 |4.1 |Materials: | |
 | |4.1.1 |Screw cap vials (2.0ml) |
 | |4.1.2 |Solvac Filter Holder |
 | |4.1.3 |Nylon 6,6 membrane 0.45µm membrane filters |
 | |4.1.4 |Autopipettes (10-100(l), (20-200(l), (100-1000(l) |
 | |4.1.5 |Tips for the Autopipettes |
 | |4.1.6 |Laboratory Glassware |
 |4.2 |Reagents: |
 | |4.2.1 |Tetrahydrofuran |
 | |4.2.2 |Methanol |
 | |4.2.3 |Acetonitrile |
 | |4.2.4 |Purified Water |
 | |4.2.5 |2-Phenoxyethanol |
 | |4.2.6 |2-propanol |
 5. |Preparation of solutions: |
 |Prepare all the solutions as described below |
 |5.1 |Mobile phase | | | | | | |
 | |For 1 litre volume of mobile phase: |
 | |Tetrahydrofuran50 ml |
 | |Water600 ml |
 | |Methanol100 ml |
 | |Acetonitrile.....250 ml |
 | |This solution can be stored at room temperature for 1 month. |
 |5.2 |Isopropyl Alcohol 70% Solution |
 | |For storing and purging the HPLC system) |
 | |Take 700 ml of 2-propanol and make up the volume to 1000 ml with |
 | |Purified Water. |
 | |This solution can be stored at room temperature for 3 months. |
 | |Note: All the solutions prepared for use are vacuum filtered using |
 | |Nylon 6, 6 membrane 0.45 µm filters to eliminate the entry of possible |
 | |particles, if any into the system as well as the column. |
 |5.3 |100 mg/ml 2-Phenoxyethanol stock |
 | |5.3.1 |Take 9.09 ml 2-phenoxyethanol in a 10 ml volumetric flask and |
 | | |make up the volume to 10 ml with purified water. (1000 mg/ml) |
 | |5.3.2 |Take 1 ml of the above mentioned solution (step 7.3.1) and make |
 | | |up the volume to 10 ml in a 10ml volumetric flask. |
 |5.4 |Preparation of calibration standards (1-15 mg/ml): |
 | | |1.0mg/ml |2.5mg/ml|5mg/ml |10mg/ml |15mg/ml |
 | | |100 mg/ml Stock |0.05 |0.125 |0.250 |0.5 |0.750 |
 | | |sol.(ml) | | | |
 | | |Purified water (ml) |4.95 |4.875 |4.75 |4.5 |4.25 |
 6. |Procedure: |
 |6.1 |Preparing the HPLC System for run, Agilent 1100 Series: | | |
 | |6.1.1 |Mount the Chromatographic column on Agilent 1100 Series HPLC |
 | | |System. |
 | |6.1.2 |Establish On-Line connection of Agilent 1100 Series HPLC System |
 | | |with Chem Station online and offline software. |
 | |6.1.3 |Switch ON the Agilent 1100 Series HPLC System. |
 | |6.1.4 |Click on the Chem Station online software. |
 | |6.1.5 |Method and Run control mode which is the default window shall be |
 | | |displayed or it can be selected from the view menu. |
 | | |Note: Run Control, Sample injection and online data acquisition |
 | | |shall be done in Method and Run Control mode. |
 | |6.1.6 |Bring the pump flow to 0.0 ml/min and then click on the ON |
 | | |option on the Method and Run control mode window and activate |
 | | |the auto sampler, Quaternary pump and DAD Detector. |
 | |6.1.7 |Control of the instrument i.e pump (flow rate setting), |
 | | |autosampler (sample injection) and detector (UV-Vis wavelength) |
 | | |selection) is done through Instrument menu. |
 | |6.1.8 |Purging the system: |
 | | |6.1.8.1|Open the purge valve of pump (by turning it counter |
 | | | |clockwise) and set flow rate to 3-5ml/min |
 | | |6.1.8.2|Flush the vacuum degasser and all tubes with at least |
 | | | |30ml of Isopropyl alcohol 70%. |
 | |6.1.9 |After purging the pump set the flow at zero and mount the |
 | | |chromatographic column. |
 |6.2 |Equilibration of the column manually: |
 | |6.2.1 |Place the inlet tubing A in the reservoir for mobile phase. |
 | |6.2.2 |In Set up pump option in instrument menu, mention the desired |
 | | |flow rate of 1.5 ml/min and write 100% of Solvent A in the |
 | | |solvents field. |
 | |6.2.3 |In Set up DAD Signal option in instrument menu, enter the |
 | | |required wavelength of which is 280nm. |
 | |6.2.4 |Keep on running the column with mobile phase until a stable |
 | | |baseline at 280nm is obtained which can be viewed with the |

||| |online plot window in the Method and Run control mode. |

|6.3 |Preparation and application of the test sample and calibration |

|| |standards: |

|| |6.3.1 |Load 1 µl of each of the calibration standards (step 5.4) |

|| |6.3.2 |Centrifuge the test sample at 10000 rpm, for 3 to 5 minutes. |

|| |6.3.3 |Transfer the supernatant of the test samples into screw cap |

|| |vials placed against the appropriate no. on the sample tray. |

|| |6.3.4 |Load 1 µl of test sample. |

|6.4 |Running a sample manually. |

|| |6.4.1 |Fill the sample details in Sample Info option of Run Control |

|| |menu. |

|| |6.4.1. |Write the name of operator under Operator Name option. |

|| |1 |

|| |6.4.1. |In Data File field, click on prefix/counter and fill the |

|| |2 |prefix option alphabetically or alphanumerically. Fill |

|| |the counter option numerically. |

|| |6.4.1. |In subdirectory, specify name of the directory where the |

|| |3 |data file of the sample to be run should be stored. |

|| |6.4.1. |In sample parameters field, specify sample name, location|

|| |4 |of the sample vial and sample amount in the respective |

|| |options. |

|| |6.4.2 |Click on run method option in run control menu to start the run.|

|6.5 |Automatic injection of samples by running a sequence. |

|| |6.5.1 |In case of running samples automatically, load sequence |

|| |2-phenoxy.S. |

|| |6.5.1.1 |Open the sequence parameter option by clicking Sequence |

|| |menu. |

|| |6.5.1.2 |Enter information about the data directory and operator |

|| |name. |

|| |6.5.1.3 |Choose, According to run time check-time checklist in the|

|| |part of method to run option. |

|| |6.5.1.4 |In the shutdown option, click on the post-sequence option|

|| |and select LAMPOFF. |

|| |6.5.1.5 |In sequence table option, specify the sample names, |

|| |methods and vials to run in the sequence. |

|| |6.5.1.6 |Click on run sequence option in run control menu to start|

|| |the sequence. |

|6.6 |Finishing the Run |

|| |6.6.1 |Click on stop run option in the run control menu upon complete |

|| |elution of the injected sample as indicated by a stable UV |

|| |baseline or run will stop automatically if stop time (5min) is |

|| |specified. |

|| |6.6.2 |In case of running sequence, each sample run will be of 5 min |

|| |and after running all the samples the sequence will |

|| |automatically stop. |

|6.7 |Results of the sample are calculated from the calibration standards |

|| |using the chemstation software. |

|| |6.7.1 |Open the chromatogram corresponding to 1µg/µl calibration |

|| |standard in the offline copy of chemstation software in the Data|

|| |Analysis mode. |

|| |6.7.2 |From the calibration menu select calibration settings and write |

|| |µg/µl in the Amount units option. |

|| |6.7.3 |Select New Calibration Table from Calibration menu and enter |

|| |value1 in default amount option and value 1 in level option. |

|| |6.7.4 |Now open the chromatogram for 2.5µg/µl. Select add level option |

|| |from calibration menu and enter value 2.5 in default amount and |

|| |value 2 in level option. |

|| |6.7.5 |Similarly open chromatograms for 5.0µg/µl 10.0µg/µl and |

|| |15.0µg/µl and keep adding default amount and appropriate value |

|| |of level. |

|| |6.7.6 |Select calibration task icon and the calibration table with |

|| |required concentrations and levels can be viewed in the |

|| |calibration table view. A calibration curve is also displayed in|

|| |the calibration curve field with value of correlation |

|| |coefficient of the curve. |

|| |6.7.7 |Save the calibration method under appropriate name. |

|| |6.7.8 |Open the chromatogram of test sample and select the above |

|| |created calibration method for integration of the test sample. |

|| |6.7.9 |In the specify report option of Report menu, select ESTD in the |

|| |Calculate field. |

|| |6.7.10 |In the report layout view, create a new template or open an |

|| |existing template. |

|| |6.7.11 |Click on the show report preview with current template, method |

|| |and data file icon and view the result of the test sample in |

|| |terms of concentration (µg/µl). |

|| |6.7.12 |Click on the print icon to take print out of the chromatogram |

|| |with the results. |

|| |6.7.13 |The printed report is signed by the analyst who does the |

|| |analysis and also checked by the other analyst. |

|6.8 |Preservation and storage of the Column |

|| |When the column is not to be used for longer period then preservation |

|| |of the column is required. To preserve the column, pass acetonitrile |

|| |through the column at the flow rate of 1.5ml / minute. Dismount the |

|| |column and seal both ends of the column with the protective plugs |

|| |before keeping it at room temperature. |

|7. |Abbreviations: |

|| |7.1 |cm |centimeter |

|| |7.2 |g |gram |

|| |7.3 |HPLC | High Performance Liquid Chromatography |

|| |7.4 |id | internal diameter |

|| |7.5 |mAu | milli Absorbance unit |

|| |7.6 |µg | microgram |

7.7	mg/ml	milligram per milliliter
7.8	μl	microtitre
7.9	μm	micrometer
7.10	ml	milliliter
7.11	mm	millimeter
7.12	nm	nanometer
7.13	PN	Part Number
7.14	QC	Quality Control
7.15	rpm	rotation per minute

1.0	Objective		
	To lay down the procedure to check the viability of the Master/Working Cell		
	Bank by determining the number of viable cells used for the production of		
	recombinant Hepatitis B surface antigen.		
2.0	Scope		
	This Standard Operating Procedure is applicable to check the viability of		
	the Master/ Working Cell Bank by determining the number of viable cells in		
	the microbiology lab of Quality Control used for the production of		
	recombinant Hepatitis B surface antigen.		
3.0	Responsibility		
	3.1	Trained trainee scientific officer/ scientific officer shall be	
	responsible for carrying out the established procedure and reporting		
	the results on the same day of analysis after appropriate incubation		
	for on the next working day, if there is a holiday.		
	3.2	The senior scientific officer/ section head who is well versed with	
	the procedure shall be responsible for ensuring that the same is		
	carried out correctly.		
	3.3	The Departmental Head shall be responsible for overall compliance of	
	the SOP.		
4.0	Abbreviations and Definitions		
	4.1	CV	:Coefficient of variation
	4.2	CFU	:Colony forming unit
	4.3	°C	:Degree centigrade
	4.4	g	:Gram
	4.5	h	:Hours
	4.6	Kg/cm2	:Kilogram per centimetre square
	4.7	mg	:Milli gram
	4.8	µL	:Micro liter
	4.9	µm	:Micro meter
	4.10	ml	:Milli liter
	4.11	N	:Normal
	4.12	PN	:Part Number
	4.13	QA	:Quality Assurance
	4.14	QC	:Quality Control
	4.15	SOP	:Standard operating procedure
5.0	Procedure		
	5.1	Materials	
		5.1.1	General glassware
		5.1.2	Sterile tips
		5.1.3	Syringe Filters
	5.2	Reagents	
		Name	
		5.2.1	Purified Water
		5.2.2	NaCl
		5.2.3	Sabouraud Dextrose Agar
		5.2.4	Gram Reagents
		5.2.5	HCl
	5.3	Preparation of solutions:	
		5.3.1	Peptone Water Saline
		5.3.2	Solution of 1 N HCl
		5.3.3	Solution of NaOH 0.1 N
		5.3.4	Sabouraud Dextrose Agar (Yeast Culture medium):
		5.3.5	Gram Reagents
	5.4	Sampling	
		5.4.1	Carry out the sampling and Reception and storage of the sample
		shall be done.	
	5.5	Viability test of cell bank:	
		Note: If the testing for viability check of cell bank is not performed	
		on the day of receipt, the cell bank shall be kept in deep freezer at	
		-200C. The cell banks must be tested within 5 days of receipt of cell	
		bank.	
		5.5.1	Initial Control:
			For Initial analysis of new Master Cell Bank (MCB) or Working
			Cell Banks at least 6 vials shall be used.
		5.5.2	Take 0.5 ml of sample from each vial and add to a test tube
			containing 4.5 ml of peptone water saline; this is dilution
			10-1.
		5.5.3	Starting from the above dilution make serial dilutions up to
			10-8.
		5.5.4	For making serial dilutions, take 1 ml from the previous
			dilution and add it to the next tube containing 9 ml of
			peptone water/ saline solution and so on. To check viability
			of the master/ working cell bank, last three dilutions i.e.
			10-6, 10-7 and 10-8 shall be used for plating.
		5.5.5	Once the dilutions are over, the culture shall be inoculated
			on the Sabouraud Dextrose Agar plate by spread plate
			technique. At least 5 replicates for each dilution shall be
			performed.
		5.5.6	For spread plate, pipette out 0.1 ml of the dilutions on
			Sabouraud Dextrose Agar plates and using the sterile L-shaped
			rod and rotameter spread the culture on the surface of the
			agar.
		5.5.7	Incubate the plates in inverted position at 22.5 ± 2.5°C, for
			a period of 72 hours.
		5.5.8	After incubation for 72 hours, count the no. of colonies
			observed in the plates.
		5.5.9	All the colonies should show homogeneous growth with respect
			to morphology and appearance.
		5.5.10	All the plates which show CFU less than 30 or more than 300
			shall be rejected and not counted.

			Note: Significant no. of colony forming units (CFU) between 30	
			and 300.	
		5.5.11	If more than one type of colonies is observed, it indicates	
			contamination and in this case, repeat the test with same	
			number of samples.	
		5.5.12	Determine the total number of viable colony count for each	
			vial in the selected dilutions by counting the colony forming	
			units (CFU) from all the replicate.	
		5.5.13	Calculate the Mean number of viable cells, by dividing the	
			cumulative value of all replicates with 5.	
		5.5.14	This way 6 values will be obtained one for each vial, add	
			them and determine the average value of these by dividing	
			with 6 and multiply it by the inverse of the dilution for	
			which count was carried out. This value corresponds to CFU	
			per 0.1 ml in the bank. To convert into ml multiply by 10.	
		5.5.15	Determine the coefficient of variation for individual vial	
			(this value should not be more than 20%).	
		5.5.16	Evaluation of the homogeneity of the bank:	
			The cell bank is considered homogeneous if the dilution	
			selected for the viable cell counting is same for all the six,	
			sampled vials.	
		5.5.17	Acceptance Criteria for Initial Control:	
			5.5.17.1	For Yeasts the average CFU/ml should be more or equal
			to 10⁶.	
			5.7.17.2	The bank should be homogeneous.
		5.5.18	Follow up control:	
			For the follow up control 2 vials are to be checked.	
		5.5.19	The banks shall be analysed annually starting from the initial	
			control	
		5.5.20	Follow the steps 5.5.2 to 5.5.13 of the Initial Control, to	
			verify the follow up viability.	
		5.5.21	This way 2 values shall be obtained one for each vial, add	
			them and determine the average value of these by dividing by	
			2 and multiply it by the inverse of the dilution for which	
			count was carried out. This value corresponds to CFU per 0.1	
			ml in the bank. To convert into ml multiply by 10.	
		5.5.22	Determine the coefficient of variation for individual vial	
			(this value should not be more than 20%).	
		5.5.23	Interpretation of results:	
			An alert shall be given in writing when the viability of cell	
			bank falls to 10⁴ CFU/ml or if there is 2 log reductions from	
			the results of the viable count in the previous check of the	
			viability of the cell bank.	
		5.5.24	Reporting of the Result:	
			For both Initial control and Follow up control report the	
			results as CFU/ml.	
		5.6	Handling of results out of specification:	
		5.6.1	If the results are found out of specification, a team shall be	
			prepared from Quality Control and Quality Assurance and	
			investigations shall be carried out.	

1.0 |Objective: |
 | |Determination of phosphorus content of Meningococcal polysaccharide |
 | |serotype A and in turn to determine polysaccharide content. |
 2.0 |Scope: |
 | |This SOP is applicable to Men A polysaccharide, Men A polysaccharide in |
 | |conjugates (bulk or final lots) and stability samples. |
 3.0 |Responsibility: |
	3.1	The analyst of the quality control laboratory shall be responsible
	for executing the described procedure correctly.	
	3.2	The supervisor of the laboratory shall be responsible for verifying
	that the procedure is followed correctly and for checking the	
	results.	
	3.3	The Departmental Head shall be responsible for the overall
	compliance of the SOP and shall ensure that the SOP is reviewed	
	every two years or if required earlier.	
3.0	Abbreviations and Definitions:	
	3.1	SOP
	3.2	UV
	3.3	WFI
4.0	Procedure:	
	4.1	Materials, reagents & Equipments
	4.1.1	Laboratory vortex mixer
	4.1.2	Furnace
	4.1.3	UV-Vis Spectrophotometer
	4.1.4	Water bath, 37°C
	4.1.5	Glass test tubes (12 x 75 mm e.g. Borosil)
	4.1.6	Pipettes and tips (e.g. Eppendorf)
	4.1.7	Phosphorus standard, 20 µg/ml
	4.1.8	Sulfuric Acid
	4.1.9	Perchloric acid
	4.1.1	Hydrogen peroxide free from phosphorus
	0	
	4.1.1	Ascorbic acid sodium salt
	1	
	4.1.1	Ammonium molybdate tetra hydrate
	2	
	4.2	Preparation of Solution
	4.2.1	Reagent A (Sulphuric acid +Perchloric acid): Mix 2 volume of
	sulphuric acid and 1 volume perchloric acid. Prepare freshly	
	4.2.2	Reagent B (3 M sulphuric acid): Take 180 ml of Milli Q water
	in volumetric flask, to this adds 40.5 ml of conc. Sulphuric	
	acid slowly. Make up volume to 250 ml with WFI. Store at room	
	temperature. Expiry 6 months.	
	4.2.3	Reagent C (2.5% ammonium molybdate): Dissolve 5.0 g of
	ammonium molybdate in 200 ml of Milli Q water. Store at 4°C.	
	Expiry 6 month.	
	4.2.4	Reagent D (10 % ascorbic acid): Dissolve 10 g of ascorbic
	acid in 80 ml of Milli Q water. Make up final volume to 100	
	ml with Milli Q water. Store at 4°C expiry 1 month.	
	4.4.5	Reagent E: Mix 2 volume of milli Q water, 1 volume of reagent
	B, 1 volume of reagent C, 1 volume of reagent D. prepare	
	freshly before use.	
	4.3	Test Principle:
	This method requires a preliminary ashing step, in order to release	
	all organically bound phosphate. The conventional method involve	
	drying and burning of organic sample over Bunsen burner, in	
	presence of perchloric acid and sulphuric acid. A high temperature	
	Furnace is now used instead. The characteristic blue colour, which	
	is formed from reduction of phosphomolybdo complex by ascorbic	
	acid, is detected using a spectrophotometer at 825 nm. The method	
	is sensitive and gives reproducible results.	
	4.4	Preparation of Assay:
	4.4.1	Place series of standard solution from 20µg/ml Phosphorus
	standard solution on the day of measurement into 5 ml glass	
	tubes in duplicate according to the following table.	
	S.No.	Volume of 20µg/ml
		phosphorus standard
		(µl)
	Blank	0
	S 1	25
	S 2	50
	S 3	75
	S 4	100
	S 5	150
	S 6	200
	4.4.2	Add prediluted samples in duplicate, and make final volume to
	200 µl in 5 ml test tubes.	
	Note: The sample dilutions can vary depending upon the Sialic	
	acid concentration. If there is any sample outside the range	
	of the calibration curve, carry out the calculation for more	
	appropriate dilution.	
	In case of dry polysaccharide prepare a solution containing	
	about 1 mg/ml of polysaccharide in Lal reagent water.	
	4.4.3	Add 50µL reagent A to each tubes and vortex.
	4.4.4	Heat the tubes in a Furnace to 250°C for 4 1/2 hours. If
	charring occurs, add hydrogen peroxide and heat further.	
	4.4.5	Remove the tubes from the heating block. Cool down the tubes
	to room temperature. Add 200µL Milli Q H2O to each tube.	
	4.4.6	Add 1 ml Reagent E to each tube and vortex. Incubate in a
	37°C water bath for 2 hours.	

		4.4.7	Measure the absorbance of each tube at 825 nm with a
		spectrophotometer.	
		4.4.8	Note that all the readings of the assay tubes should be taken
		within 30 mins. of completion of the assay.	
		4.5	Calculations:
		4.5.1	Once the absorbance of the standards and the test samples
		have been taken, calculation shall be done using the excel	
		sheet carrying formulas for the required calculation.	
		4.5.2	Always copy the master excel sheet (sheet carrying formulas
		for the required calculation), never use the previously used	
		excel sheet for new calculations.	
		4.5.3	Copy the master excel sheet, keep it for use in respective
		folder and then fill in the required sample details (Batch	
		number), sample dilution (if applicable), name of analyst,	
		date, etc.	
		4.5.4	Calculate the phosphorous concentration of the samples using
		following formula: Phosphorous concentration	
		(μM/ml)= {(abs-intercept)/slope}*dilution factor	
		4.5.5	In case of dry polysaccharide, Phosphorous content (%) in
		polysaccharide on dry weight basis:	
			Phosphorous content in dry Polysacc. (μg/ml) =
			Phosphorous content in Polysacc. sol. (μg/ml) X 1 –
			{Moisture (%) / 100}
			Phosphorous content (%)
			= Phosphorous content in dry polysaccharide (μg/ml)
			_____ X
			100
			Conc. of polysaccharide solution (μg/ml)
		4.5.6	To calculate the pure polysaccharide content (mg) present in
		dry polysaccharide sample in of each serotype based on the	
		Sialic acid content obtained, following formula is applied:	
		Pure Polysaccharide content(mg) / Dry polysaccharide content	
		(mg)	
		= (Phosphorous content in dry Polysacc (mg) X Conversion	
		factor)	
		4.5.7	Calculate the polysaccharide concentration of the samples
		using following formula: Polysaccharide concentration=	
		Phosphorous concentration *11.11	
		4.6	Assay Specification:
		4.6.1	The linearity of the standard curve is NLT 0.99
		4.6.2	The absorbance value of test samples should fall inside the
		range of standard curve.	
		4.6.3	A coefficient of variation of less than 10% should be
		observed within the replicates.	
		5.0	Reference:
		5.1	Chen, P.S. Jr. Toribara T.Y. And Warner H. 1956. Micro
		determination of phosphorous. Anal. Chem. 28 1756-1758	
		5.2	WHO TRS 594: Requirement for Meningococcal polysaccharide vaccine.

Sr.No. | SOP TITLES

- 1 | Detection and quantification of mouse immunoglobulin (IgG) contaminants in samples by using kit from Cygnus Technologies, Inc.
- 2 | Determination of traces of yeast DNA in samples of active raw material of Hepatitis B vaccine (recombinant) by DOT- BLOT method
- 3 | Determination Of Cesium Chloride Content By Atomic Absorption Spectrophotometry
- 4 | Determination of Potassium Thiocyanate content in active raw material of recombinant Hepatitis b Vaccine
- 5 | Determination of Protein concentration by Lowry's method
- 6 | Determination Of Total Carbohydrate Content In Active Raw Material Of Recombinant Hepatitis B Vaccine
- 7 | Determination of total lipid content in active raw material of recombinant hepatitis b vaccine.
- 8 | Immunoenzymetric Assay for the measurement of Pichia pastoris Host cell proteins.
- 9 | Procedure for detection of surface antigen of the hepatitis b samples by enzyme immunoassay techniques.
- 10 | Procedure for preparation of active raw material Hepatitis b vaccine sample by desorption of aluminium hydroxide gel.
- 11 | Procedure for separation of proteins by Sodium Dodecyl Sulphate Polyacrylamide Gel Electrophoresis (SDS-PAGE)
- 12 | Procedure for testing of Bacterial Endotoxin by Lal Gel Clot method using reagents from associates of cape cod, inc. (Qualitative Lal test)
- 13 | Spectrophotometric determination of Thiomersal concentration in recombinant hepatitis b vaccine

1.0	Objective:		
	To lay down the procedure for detection of surface Antigen of the Hepatitis		
	B samples by enzyme immunoassay technique.		
2.0	Scope:		
	This Standard Operating Procedure is applicable to all the samples of Active		
	Raw Material of recombinant Hepatitis B surface antigen, received in the		
	Quality control department.		
3.0	Responsibility:		
	3.1	The analyst shall be responsible for executing the described	
	procedure correctly.		
	3.2	Supervisor shall be responsible for verifying that the procedure is	
	complied with and the results are checked.		
	3.3	HOD shall be responsible for overall compliance of this SOP.	
4.0	Abbreviations and Definitions		
	4.1	QC	:Quality Control
	4.2	QCR	:Quality Control Record
	4.3	SD	:Standard Deviation
	4.4	SOP	:Standard Operating Procedure
	4.5	ARM	:Active Raw Material
	4.6	CV	:Coefficient of variation
	4.7	g	:Gram
	4.8	ID	:Identification
	4.9	mg	:Milligram
	4.10	ml	:Millilitre
	4.11	µl	:Micro litre
	4.12	µg	:Microgram
	4.13	nm	:Nanometer

5.0	Procedure:			
	5.1	Reagents, Equipments and other materials		
		5.1.1	Reference standard	
		5.1.2	Test solution: HBsAg bulk protein	
		5.1.3	Sterile micro centrifuge tubes	
		5.1.4	Tube rack	
		5.1.5	Variable volume micropipettes and tips	
		5.1.6	Bead dispenser	
		5.1.7	Plate incubator (Commander from Abbott) or water bath or	
		Incubator oven		
		5.1.8	Quick wash (Abbott) or ELISA plate washer	
		5.1.9	Quantum II reader (Abbott) or ELISA plate reader or UV-Vis	
		Spectrophotometer		
		5.1.10	Computer with Parallel line assay software	
	5.2	Preparation of solutions		
		5.2.1	Phosphate buffered saline (PBS) pH 7.4	
			5.2.1.1	Dissolve 1.183 g Disodium hydrogen phosphate
				dihydrate, 0.19 g potassium dihydrogen phosphate
				and 8.0 g sodium chloride in about 800 ml of
				purified water.
			5.2.1.2	Adjust pH to 7.4 using dilute NaOH solution or
				dilute HCl.
			5.2.1.3	Adjust the final volume to 1000ml using a measuring
				cylinder.
		5.2.2	Diluent {PBS with 0.2 % w/v BSA}	
			5.2.2.1	Dissolve 0.2 g Bovine serum albumin in about 80 ml
				of PBS (pH 7.4) and make up the volume to 100 ml
				using PBS (pH 7.4).
			5.2.2.2	Auszyme monoclonal kit for HBsAg detection or any
				other validated kit (e.g. Monolisa Ag HBs Plus).
	5.3	Principle		
		Monoclonal or polyclonal antibodies coated on ELISA plates or on		
		beads capture HBsAg. After wash, second antibody conjugated to HRP is		
		added. A sandwich complex is formed wherein HBsAg is sandwiched		
		between the antibody and antibody HRP conjugate. After removing		
		excess conjugate substrate for HRP such as OPD or TMB is added. The		
		amount of color developed is directly proportional to content of		
		HBsAg in sample.		
	5.4	Dilution for test using Auszyme kit:		
		5.4.1	Dilute In House standard and test HBsAg (aq) bulk to 20	
			µg/ml of total protein (obtained from Lowry analysis) using	
			PBS with 0.2% BSA, in quadruplicate, as follows:	
		5.4.2	E. g., HBsAg protein concentration = 2.24 mg/ml,	
			100 µl bulk + 124 µl diluent → (1 mg/ml) -1	
			100 µl of 1 + 900 µl diluent → (100µg/ml) - 2	
			200 µl of 2 + 800 µl diluent → (20 µg/ml) -3	
		5.4.3	Dilute Standard further 1:1000 with diluent, in	
			quadruplicate, as follows:	
		5.4.4	100 µl of 20µg/ml Std + 900 µl diluent → (1:10 Dilution) -	
			1	
			100 µl of 1 + 900 µl diluent → (1:100 Dilution) - 2	
			100 µl of 2 + 900 µl diluent → (1:1000 Dilution) -3	
		5.4.5	Discard 1 and 2 dilutions after 3 have been made. Use 3	
			(1:1000 Dilution) for plating in the assay.	
		5.4.6	Dilute test sample of HBsAg further to 1:1000 dilution	
			using PBS with 0.2% BSA, in quadruplicate, as follows	
		5.4.7	100 µl of 20µg/ml sample + 900 µl diluent → (1:10 Dilution)	
			-1	
			100 µl of 1 + 900 µl diluent → (1:100 Dilution)-2	
			100 µl of 2 + 900 µl diluent → (1:1000 Dilution) -3	
		5.4.8	Discard 1 and 2 dilutions after dilution 3 has been made.	
			Use 3 (1:1000 Dilution) for plating in the assay.	

| | 5.5 | Assay |

| | 5.5.1 | Plate standard and test HBSAg sample in a total of 200 µl |

| | | in a manner that each plate will hold dilutions in |

| | | following manner. |

| | | For example: |

| | 5 µl of dilution 1:1000 + 195 µl diluent → 1: 40000 dilution |

| | 10 µl of dilution 1:1000 + 190 µl diluent → 1: 20000 dilution |

| | 20 µl of dilution 1:1000 + 180 µl diluent → 1: 10000 dilution |

| | 40 µl of dilution 1:1000 + 160 µl diluent → 1: 5000 dilution |

| | 80 µl of dilution 1:1000 + 120 µl diluent → 1: 2500 dilution |

| | (optional) |

| | 5.5.2 | Add 200 µl of kit positive control in two wells and 200 µl |

| | | of kit negative control in three wells. |

| | 5.5.3 | Add 50 µl of kit conjugate solution in each of the wells. |

| | 5.5.4 | Add one anti-HBs (mouse) monoclonal-coated bead provided in |

| | | the kit with the help of bead dispenser, to each well. |

| | 5.5.5 | Apply cover seal and gently tap the plate so as to remove |

| | | any air bubble trapped beneath the bead. |

| | 5.5.6 | Incubate the plate at 40°C ± 10°C for 3 Hrs ± 10 minutes in |

| | | commander incubator (Abbott) or water bath. |

| | 5.5.7 | During the last 5-10 minutes of incubation period, prepare |

| | | appropriate volume of OPD substrate solution depending on |

| | | number of beads used (1 tablet of OPD in 5 ml of OPD |

| | | diluent) considering a volume of 300 µl per bead. |

| | 5.5.8 | At the end of incubation period, remove the plates from |

| | | incubator and discard the cover seals. |

| | 5.5.9 | Wash the beads once with at least 12 ml of Milli-Q water |

| | | using Quick Wash (Abbott) or 3 times with penta wash II |

| | | (Abbott). |

| | 5.5.10 | Immediately transfer the beads to properly identified tubes |

| | | in boxes. |

| | 5.5.11 | Pipette 300 µl of freshly prepared OPD substrate solution |

| | | into two empty tubes (Blanks) and then into each tube |

| | | containing a bead. |

| | 5.5.12 | Cover the tubes (black cover plates provided with kit) and |

| | | incubate at room temperature for 30 ± 2 minutes. |

| | 5.5.13 | Stop the reaction by adding 1 ml of 1 N H₂SO₄ into each |

| | | tube. |

| | 5.5.14 | Blank spectrophotometer (e.g. Quantum II, Module A, Mode 0, |

| | | Abbott) with substrate blank at 492 nm. |

| | 5.5.15 | Measure absorbance of all the tubes at 492 nm within 2 |

| | | hours after addition of acid. |

| | 5.6 | Dilution for test using BIORAD Kit (Monolisa Ag HBs Plus): |

| | 5.6.1 | Dilute Standard HBSAg (aq) bulk & test sample of HBSAg to |

| | | 20 µg/ml of total protein (obtained from Lowry analysis) |

| | | using PBS with 0.2% BSA, in quadruplicate. |

| | | For example: HBSAg protein concentration = 2.29 mg/ml, |

| | | 100 µl bulk + 129 µl diluent -- (D) (1 mg/ml) |

| | | 100 µl D + 900 µl diluent -- (E) (100 µg/ml) |

| | | 200 µl E + 800 µl diluent -- (F) (20 µg/ml) |

| | 5.6.2 | Further dilute STANDARD & test sample of 20 µg/mL as |

| | | follows: |

| | | 100 µl (of 20 µg/mL) + 900 µl diluent -- 1:10 |

| | | 100 µl of 1:10 + 900 µl diluent -- 1:100 |

| | | 500 µl of 1:100 + 500 µl diluent -- 1:200 |

| | | 500 µl of 1:200 + 500 µl diluent -- 1:400 |

| | | 500 µl of 1:400 + 500 µl diluent -- 1:800 |

| | | 500 µl of 1:800 + 500 µl diluent -- 1:1600 |

| | | 500 µl of 1:1600 + 500 µl diluent -- 1:3200 |

| | | 500 µl of 1:3200 + 500 µl diluent -- 1:6400 |

| | | 500 µl of 1:6400 + 500 µl diluent -- 1:12800 |

| | | 500 µl of 1:12800 + 500 µl diluent -- 1:25600 |

| | | 500 µl of 1:25600 + 500 µl diluent -- 1:51200 |

| | | 500 µl of 1:51200 + 500 µl diluent -- 1:102400 |

| | | Use last four dilutions in quadruplicate for ELISA. |

| | 5.7 | Assay |

| | 5.7.1 | Take out the protective packing the support frame and the |

| | | necessary number of strip. Put the unused strip back in |

| | | their packing and close it. |

| | 5.7.2 | Add 100 µl of respective samples, 4 negative controls & 1 |

| | | positive control into respective wells as per plate layout. |

| | 5.7.3 | Add 50 µl of working conjugate solution in all wells. Shake |

| | | the plate gently. |

| | 5.7.4 | Cover the plate with adhesive tape and incubate at 37° C ± |

| | | 1°C for 120 ± 5 minutes |

| | 5.7.5 | Prepare Washing Solution and conjugate solution as per kit |

| | | manual. |

| | | Distribute the wells in the following order: |

| | | Wells A1, B1, C1 and D1: negative control |

| | | Well E1: Positive Control |

| | | Well F1 onwards for unknown samples |

| | 5.7.6 | After incubation is complete wash the plate five times with |

| | | washing buffer in ELISA plate washer. |

| | 5.7.7 | Prepare the enzymatic development solution as per the |

| | | instructions in the kit leaflet and dispense 100 µl of |

| | | development solution in each well. |

| | 5.7.8 | Incubate in dark for 30 minutes at room temperature. |

| | 5.7.9 | After completion of incubation add 100 µl stopping solution |

| | | in each well |

| | 5.7.10 | Read the plate at 450/620 nm, within the 30 minutes of |

| | | stopping reaction. |

| | 5.8 | Interpretation of results. |

		5.8.1	Validity of Kit
		5.8.1.1	Calculate average negative control absorbance value
		5.8.1.2	Calculate average positive control absorbance value
			for Auszyme kit only.
		5.8.1.3	Value of P-N should be ≥ 0.4 for the test to be
			valid for Auszyme kit. If not, technique or
			deterioration of reagents may be suspected and the
			run must be repeated.
		5.8.1.4	For Bio-Rad Kit average value of negative control
			should be lower than or equal to 0.080 O.D value
			and positive control should be over or equal to
			1.00 O.D value
		5.8.2	Calculation
		5.8.2.1	Parallel line assay software has to be used for
			calculation of potency of HBsAg.
		5.8.2.2	While feeding the data into the software assign a
			potency of 1.0 to the Standard.
		5.8.2.3	For the standard or test sample feed the
			quadruplicate absorbance for each dilution.
		5.8.2.4	If one of the values is significantly out of the
			linear fit it must be marked as an outlier and must
			not be used for calculation. Use the absorbance
			value of other three replicates of particular
			dilutions.
		5.8.2.5	Configure the software to consider log values for
			dose provided for ELISA.
		5.8.2.6	Results are reported in the form of the printout
			generated by parallel line assay
		5.8.3	Validity of results
		5.8.3.1	The analysis should not fail for parallelism and
			linearity, which is also printed on the report. If
			so, repeat the analysis.
		5.8.3.2	The percentage of lower and upper fiducial limits
			of the assay should lie between 80% to 125%
			respectively. If not, the assay and results are
			invalid and the test has to be repeated.
		5.8.4	Acceptance criteria
			The antigenic ratio should be ≥ 1.0
6.0	Reference		
6.1	Manual for BIORAD Kit (Monolisa Ag HBs Plus).		
6.2	Manual for Auszyme kit.		

1.0 |Objective |
 |To lay down the procedure for preparation and standardization of tetanus |
 |antitoxin (Working standard) for flocculation test of Tetanus antigens. |
 2.0 |Scope |
 |This procedure is applicable for preparation and standardization of tetanus|
 |antitoxin for flocculation test of Tetanus antigens. |
 3.0 |Responsibility |
3.1	The Analyst shall be responsible for carrying out the procedure as
	established in this SOP.
3.2	Supervisor shall be responsible for verifying that the same has been
	carried out correctly.
3.3	Head QC or his designee shall be responsible for overall compliance
	of the SOP.
4.0	Abbreviations and Definitions
4.1	SOP
4.2	QA
4.3	QC
4.4	TATF
4.5	mL
4.6	°C
4.7	%
4.8	Lf/ml
5.0	Procedure:
5.1	Equipment, Materials and Reagents
	5.1.1
	5.1.2
	5.1.3
	5.1.4
	5.1.5
	5.1.6
	5.1.7
	5.1.8
	5.1.9
	5.1.10
	5.1.11
	5.1.12
	5.1.13
5.2	Preparation of solutions
	5.2.1
	5.2.2
5.3	Preparation of Candidate Reference Tetanus toxoid (50Lf/mL)
	5.3.1
	5.3.2
5.4	Standardization of Standard toxoid (50Lf/mL)
	5.4.1
	5.4.2
	5.4.3
5.5	Preparation of Standard Reference Antitoxin
	5.5.1
	5.5.2
5.6	Standardization of Antitoxin
	5.6.1
	5.6.2
	5.6.3
	5.6.4
	5.6.5
5.7	Monitoring of Internal Reference Antitoxin
	5.7.1
	5.7.2
6.0	Forms and Records
Form for standardization of Tetanus antitoxin for flocculation test	
(working standard)	

1.0 |Objective |
 |To lay down the procedure for Estimation of Moisture Content in |
 |Meningitides polysaccharide sample by Karl Fischer titration method. |
 2.0 |Scope: |
 |This standard operation procedure is applicable to all the samples of |
 |Meningitides polysaccharide in which moisture content is to be determined. |
 3.0 |Responsibility |
3.1	The analyst is responsible for executing the described procedure
	correctly.
3.2	Supervisor is responsible for verifying that the procedure is carried
	out correctly and check the results.
3.3	HOD shall be responsible for overall compliance of this SOP and shall
	ensure that the SOP is reviewed every two years or if required
	earlier.
4.0	Abbreviations and Definitions
4.1	Abbreviations:
	4.1.1
	4.1.2
	4.1.3
	4.1.4
	4.1.5
	4.1.6
	4.1.7
	4.1.8
5.0	Procedure
5.1	Preparation of solutions: Not applicable
5.2	Testing procedure:
	Note: Operate the instrument as per the manual.
	5.2.1
	5.2.2
	5.2.3
5.3	Acceptance Criteria :
	The test sample should have the moisture content below the approved
	specifications
6.0	Forms and Records
Determination of Moisture content - Annexure 1	
7.0	Reference
7.1	Manual for operation of KF instrument
7.2	European Pharmacopoeia

1.0 |Objective |
 |To lay down the procedure for determining the unitage (Lf/ml) of tetanus |
 |toxin/toxoid by limes flocculation test. |
 2.0 |Scope |
 |This Standard Operating Procedure is applicable to the tetanus toxin/toxoid|
 |samples. |
 3.0 |Responsibility |
 |3.1 |The Analyst shall be responsible for carrying out the procedure as |
 |established in this SOP. |
 |3.2 |Supervisor shall be responsible for verifying that the same has been |
 |carried out correctly. |
 |3.3 |Head QC or his designee shall be responsible for overall compliance |
 |of the SOP. |
 4.0 |Abbreviations and Definitions |
4.1	SOP	: Standard Operating Procedure
4.2	QA	: Quality Assurance
4.3	QC	: Quality control
4.4	TATF	: Tetanus Antitoxin for Flocculation
4.5	mL	: milliliter
4.6	°C	: Degree Celsius
4.7	%	: Percent
4.8	Lf/mL	: Limes-flocculation per milliliter
4.9	w/v	: Weight/volume
5.0	Procedure:	
5.1	Equipment, Materials and Reagents	
	5.1.1	Auto pipettes
	5.1.2	Test tubes
	5.1.3	Test tube rack
	5.1.4	Flocculation tubes.
	5.1.5	Water bath
	5.1.6	Stop watch
	5.1.7	Convex lens
	5.1.8	Sterile 10ml & 50ml vials
	5.1.9	Physiological saline (0.9% w/v)
	5.1.10	Working Standard of Tetanus Antitoxin for Flocculation (100
		Lf/ml)
5.2	Preparation of solutions	
	5.2.1	Physiological saline (0.9% w/v)
		Add 0.9 gm of Sodium Chloride to 50ml of purified water, mix
		well and make up it to 100ml.
5.3	Preparation of sample	
	5.3.1	Based on the target concentration, dilute tetanus toxin/toxoid
		sample of which the Lf /ml is to be determined, in
		physiological saline (0.9% w/v) in such a way that it
		contains less than 100 Lf / ml.
	5.3.2	Dilute the sample with 0.9 % (w/v) physiological saline as
		prescribed in table - 1 prior to testing and mix thoroughly
		with the help of vortex mixer.
		Sample Description
		Crude toxin/Clarified Toxin
		Concentrated Toxin & Crude
		toxoid
		Purified toxoid
5.4	Test procedure	
	5.4.1	Set the temperature of the water bath at 50 + 2° C.
	5.4.2	Take 5–6 dry flocculation tubes of equal size, diameter and
		thickness.
	5.4.3	Label the flocculation tubes serially.
	5.4.4	Broad range test
		5.4.4.1
		5.4.4.2
		5.4.4.3
		5.4.4.4
		5.4.4.5
		5.4.4.6
		5.4.4.7

			second flocculation.	
			5.4.4.8	If flocculation occurs in the first tube increase the
			dilution fold or it occurs in the last tube decrease	
			the dilution fold of the sample.	
			5.4.4.9	Observe for first and second flocculation. The tube
			in which first flocculation occurs is the Lf value of	
			the sample. Record the time taken (Kf) for first and	
			second flocculation.	
			5.4.4.10	Multiply this by the dilution factor to get the Lf/ml
			of the original sample. Example:-	
			Tube No	
			5.4.4.12	If the first or the last tube flocculates most rapidly
			then the test has to be repeated in such a way that	
			the middle tube flocculates most rapidly.	
			5.4.4.13	After determining the broad range test perform narrow
			range test/ confirmatory test.	
			5.4.5	Narrow range test (Applicable for purified bulk)
			For narrow range test keep 2 Lf difference between subsequent	
			tubes and perform the assay as described above.	
			5.4.6	Confirmatory test
			For confirmation, perform with 2 or 1 Lf difference between the	
			tubes as shown in Table above and record the results.	
	5.5	Calculation/ Formula:		
	Limes floccules (Lf/mL) = TATF unit of tube which flocculate first X			
	dilution fold of sample.			
6.0	Forms and Records			
	6.1	Determination of Unitage (Lf/mL) of Tetanus Toxin/Toxoid by		
	Flocculation Test.			
7.0	References			
	WHO manual for the production and control of vaccines – Tetanus Toxoid			
	(BLG/UNDP/77.2 Rev 1)			

1.0 |Objective |
 |To lay down the procedure for estimation of residual formaldehyde content. |
 2.0 |Scope |
 |This Standard Operating Procedure is applicable to all the samples in which|
 |residual formaldehyde content is to be determined. |
 3.0 |Responsibility |
3.1	The analyst is responsible for executing the described procedure
	correctly.
3.2	Supervisor is responsible for verifying that the procedure is carried
	out correctly and check the results.
3.3	HOD shall be responsible for overall compliance of this SOP and shall
	ensure that the SOP is reviewed every two years or if required
	earlier.
4.0	Abbreviations and Definitions
4.1	Abbreviations:
	4.1.1
	4.1.2
	4.1.3
	4.1.4
	4.1.5
	4.1.6
	4.1.7
	4.1.8
	4.1.9
5.0	Procedure
5.1	Chemicals and reagents
	5.1.1
	5.1.2
	5.1.3
	5.1.4
5.2	Equipments, instruments and other materials
	5.2.1
	5.2.2
	5.2.3
	5.2.4
	5.2.5
	5.2.6
5.3	Preparation of solutions: Prepare all the solutions as described below
	and fill the form, label the solutions and assign lot numbers.
	5.3.1
	5.3.2
	5.3.3
	5.3.4
	5.3.5
	5.3.6
	5.3.7

||| |the supernatant collected. |

|| |5.3.8|Test Procedure: |

|| |5.3.8.|Prepare the sets of formaldehyde standards, check sample,|

|| |1 |in house control sample and test sample with buffer |

|| | |solution and acetylene acetone solution as mentioned in the|

|| | |table below: |

|| |Test |Vol. of Standard, Sample and |Vol. of |Vol. of |

|| |tube |Controls |Buffer sol. |acetyl |

|| |no. | |acetone sol. |

|| |1 |Purified water |-- |5.0 ml |

|| | |5.0 ml | | |

|| |2 |Std (10.0 µM/lit) |-- |5.0 ml |

|| | |5.0 ml | | |

|| |3 |Std (30.0 µM/lit) |-- |5.0 ml |

|| | |5.0 ml | | |

|| |4 |Std (50.0 µM/lit) |-- |5.0 ml |

|| | |5.0 ml | | |

|| |5 |Std (80.0 µM/lit) |-- |5.0 ml |

|| | |5.0 ml | | |

|| |6 |Std (100.0 µM/lit) |-- |5.0 ml |

|| | |5.0 ml | | |

|| |7 |Check sample (80 µmol/lit) |-- |5.0 ml |

|| | |5.0 ml | | |

|| |8 |In house control A |-- |5.0 ml |

|| | |5.0 ml | | |

|| |9 |In house control B |-- |5.0 ml |

|| | |5.0 ml | | |

|| |10 |In house control C |5.0 ml |-- |

|| | |5.0 ml | | |

|| |11 |Test sample A |-- |5.0 ml |

|| | |5.0 ml | | |

|| |12 |Test sample B |-- |5.0 ml |

|| | |5.0 ml | | |

|| |13 |Test sample C |5.0 ml |-- |

|| | |5.0 ml | | |

|| |5.3.8.|The test tube no. 10 and 13 are the color correction |

|| |2 |tubes of respective samples and hence acetyl acetone is |

|| | |not added to these tubes. |

|| |5.3.8.|After addition of all the components as mentioned above, |

|| |3 |place all these test tubes in a water bath at 37±1° C for|

|| | |50 minutes. |

|| |5.3.8.|On completion of incubation period, remove the test tubes|

|| |4 |and cool them under running water for 2 minutes. |

|| |5.3.8.|Read the absorbance of each tube within 30 mins. at 415 |

|| |5 |nm under a UV/VIS spectrophotometer. |

|| |5.3.8.|The standard tube no. 1 of purified water is considered |

|| |6 |to be as blank. |

|| |5.3.8.|The concentration of formaldehyde in the test sample(s) |

|| |7 |is calculated by comparing results of its absorbance |

|| | |obtained against the plotted calibration curve of known |

|| | |formaldehyde concentration (10 to 100 µM/lit) Vs |

|| | |absorbance difference at 415 nm |

|| |5.3.8.|Note that all the readings of the assay tubes should be |

|| |8 |taken within 30 mins. of completion of the assay and |

|| | |maximum 5 samples should be analysed in one single assay.|

|| |5.4 |Calculations: |

|| |5.4.1|Once the absorbance of the standards and the test samples have |

|| | |been taken, calculation shall be done using the excel sheet |

|| | |carrying formulas for the required calculation. |

|| |5.4.2|Always copy the master excel sheet (sheet carrying formulas for |

|| | |the required calculation), never use the previously used excel |

|| | |sheet for new calculations. |

|| |5.4.3|Copy the master excel sheet, keep it for use in respective |

|| | |folder and then fill in the required sample details (Batch |

|| | |number), sample dilution (if applicable), name of analyst, date,|

|| | |etc. |

|| |5.4.4|Proceed for calculations starting from the Calibration curve. |

|| | |The calculations are done according to the equation after |

|| | |subtractions blank OD. |

|| | |Y = mX + b. |

|| | |Where: m = slope |

|| | |b = intercept |

|| | |Y = absorbance |

|| | |X = concentration |

|| |5.4.5|To determine the formaldehyde concentration in mM/lit following |

|| | |formula is applied, |

|| | |Formaldehyde [Sample Conc. (µM/lit) – Color correction |

|| | |Conc. (µM/lit)] |

|| | |Concentration = |

|| | |

|| | |(mM/lit) 1000 X |

|| | |Dilution factor |

|| |5.4.6|Record all the calculations in the data record form and also |

|| | |attach the hard copy of the excel sheet used for calculation |

|| | |along it. |

|| |5.5 |Acceptance criteria: |

|| |5.5.1|Regression of the standard curve should not be less than 0.99. |

|| |6.4.2|The calculated check sample concentration of 80 µmol/lit should |

|| | |be within 10% of the true value. |

|| |6.4.3|The variation in OD values between the two replicates of same |

|| | |sample should not be more than 10%. |

|| |6.0 |Reference |

Date : 05-May-2016
CERTIFICATE OF ANALYSIS
For BioBridge Healthcare Solutions Pvt. Ltd.
Authorized Signatory
Product: | Bulk of Tetanus Toxoid
Product code no.: | B/TT 002
Appearance: | Clear, Yellow to brownish solution without any suspended particles
Form: | Liquid
Buffer: | Physiological Saline
Manufacturing Date: | Feb 2015
Expiry Date: | Jan 2020
Antigen Content: | 2100 Lf/ml
Antigenic Purity: | 1250 Lf/mg of Protein Nitrogen
Sterility: | Sterile Product
pH: | 6.6
Specific Toxicity: | Complied
Identity: | Positive
Free formaldehyde: | 0.08g/ L
Irreversibility of toxoid: | Complied (No irreversibility observed)
Storage: | 5±3°C (Do not freeze)
Note: To be used for research or in-vitro testing purpose only | Note: To be used for research or in-vitro testing purpose only

1.0 |Purpose: |
 |To lay down the procedure for the determination of CDAP |
 |(1-cyano-4-dimethylaminopyridinium tetrafluoroborate) by high performance |
 |liquid chromatography (HPLC). |
 2.0 |Applicable to: |
 |This Standard Operating Procedure is applicable for the analysis of the |
 |samples in which CDAP (1-cyano-4-dimethylaminopyridinium tetrafluoroborate)|
 |is to be determined. |
 3.0 |Abbreviations and Definitions |
3.1	CDAP	: 1-cyano-4-dimethylaminopyridinium tetrafluoroborate
3.2	cm	: centimetre
3.3	DNA	: Deoxyribonucleic acid
3.4	g	: Gram
3.5	HIBTT	: Haemophilus Influenzae type b - Tetanus toxoid
3.6	HPLC	: High Performance Liquid Chromatography
3.7	i.d	: Internal diameter
3.8	l	: length
3.9	m	: meter
3.10	mg	: milligram
3.11	mL	: Millilitre
3.12	mm	: millimetre
3.13	nm	: Nano meter
3.14	ng	: Nano gram
3.15	No.	: number
3.16	PRP-TT	: Polyribosyl Ribitol Phosphate - Tetanus Toxoid
3.17	RSD	: Relative Standard Deviation
3.18	SST	: System Suitability Test
3.19	°C	: Degree centigrade
3.20	µg	: microgram
3.21	µL	: Micro litre
3.22	µm	: Micro meter
3.23	System	: System suitability tests are an integral part of gas
	Suitability	and liquid chromatographic methods. They are used to
	Test	verify that the resolution and reproducibility of the
		chromatographic system are adequate for the analysis
		to be done.
	The tests are based on the concept that the equipment,	
	electronics, analytical operations and samples to be	
	analyzed constitute an integral system that can be	
	evaluated as such.	
4.0	Procedure	
4.1	Precautions to be taken:	
	4.1.1	Always use filtered solutions. Filtration of solution should
	be done in order to eliminate the entry of possible particles,	
	if any, into the system as well as the column.	
	4.1.2	Degas all solutions.
	4.1.3	Avoid entry of bubbles into the system as well as column.
	4.1.4	All the solution bottle openings should be well covered with
	aluminum foil.	
	4.1.5	Handle acids carefully and always manipulate in fume hood.
	4.1.6	Check the calibration status of all equipment and instrument.
4.2	Requirements	
	4.2.1	Equipment:
		4.2.1.1
		4.2.1.2
		4.2.1.3
		4.2.1.4
		4.2.1.5
	4.2.2	Instrument:
		4.2.2.1
		4.2.2.2
		4.2.2.3
		4.2.2.4
	4.2.3	Materials:
		4.2.3.1
		4.2.3.2
		4.2.3.3
		4.2.3.4
		4.2.3.5
		filters.
		4.2.3.6
		4.2.3.7
		4.2.3.8
	4.2.4	Reagents:
		4.2.4.1
		4.2.4.2
		4.2.4.3
		4.2.4.4
		4.2.4.5
		tetrafluoroborate)
		4.2.4.6
		4.2.4.7
4.3	Preparation of solutions:	
	Prepare all the solutions and carry out the labelling of the prepared	
	solutions and assign the lot numbers to the solutions.	
	4.3.1	Solution A
		(NaH₂PO₄. H₂O, 200mM)
		For a volume of 1 liter, weigh
		NaH₂PO₄. H₂O
		Adjust the pH to 3.0 with HCl 37%.

			Make up the solution to 1litre with purified water. Prepare			
			fresh before use.			
		4.3.2	Mobile phase for CDAP:			
			Take 700mL of methanol in a measuring cylinder and 300mL of			
			solution A to make the volume to 1000mL. Use the mobile phase			
			within four days from the date of preparation if it remains			
			clear and no turbidity or fibers observed in it.			
		4.3.3	Isopropyl Alcohol 70% Solution			
			(For storing and purging the HPLC system)			
			Take 700 mL of 2-propanol using a measuring cylinder and make			
			up the volume to 1000 mL with Purified Water in a volumetric			
			flask. This solution can be stored at room temperature for 3			
			months.			
		4.3.4	1 mg/mL of CDAP calibration standard stock:			
			Weigh 5mg of CDAP and make up the volume to 5mL with purified			
			water in a volumetric flask. This solution can be stored as			
			aliquots at -20°C for 2months.			
		4.3.5	Preparation of working calibration standard of 10µg/mL or			
			10ng/µL:			
			Take 10µL of 1 mg/mL of CDAP calibration standard stock and			
			add 990µL of purified water to make 1mL of solution in an			
			eppendorf or HPLC vial. Mix thoroughly. Prepare fresh before			
			use.			
			Note: The working calibration standard is prepared fresh			
			before use. All the solutions prepared for use are vacuum			
			filtered using Nylon 6, 6 membrane 0.20/0.22/0.45 µm filters			
			to eliminate the entry of possible particles, if any into the			
			system as well as the column. For organic solvents 0.45µm			
			filters should be used (filtration not necessary if the			
			solution is prepared in 0.2 µ filtered purified water).			
		4.4	Preparing the HPLC System for run:			
		4.5	Run Conditions:			
		4.5.1	Load method CDAP.M.			
		4.5.2	Flow Rate: 1.0 mL/min			
		4.5.3	Detection wavelength: 280 nm			
		4.5.4	Run time: 4 minutes			
		4.5.5	Mobile Phase: mobile phase for CDAP			
		4.6	Equilibration of the system and the column:			
		Equilibrate the system and column with mobile phase for CDAP until a				
		stable baseline at 280 nm is obtained which can be viewed with the				
		online plot window.				
		4.7	Sample Preparation and Application:			
		Test sample and standards are loaded in screw cap vials and placed				
		against the appropriate no. on the sample tray by clearly marking				
		the screw cap vial.				
		4.8	Running of sequence:			
		4.8.1	Load sequence CDAP.S.			
		4.8.2	Open the sequence parameter option by clicking Sequence menu.			
		4.8.3	Enter information about the data directory and operator name.			
		4.8.4	In the shutdown option, click on the post-sequence option and			
		select LAMPOFF/STANDBY.				
		4.8.5	In sequence table option, specify the sample names, methods			
		and vials to run in the sequence as per the following table:				
		S. No	Sample Code	Sample details	No. of	Injection
			injections	n volume		
			per Vial			
		1	Blank	Eluent Blank	1	10 µL
		2	Standard 3	System	6	0.5 µL
		(5ng/µL)	Suitability			
		Test standard				
		3	Blank	Eluent Blank	1	10 µL
		4	Standard 1	1ng/µL	1	0.1 µL
		Standard				
		5	Standard 2	3ng/µL	1	0.3 µL
		Standard				
		6	Standard 3	5ng/µL	1	0.5 µL
		Standard				
		7	Standard 4	10ng/µL	1	1.0 µL
		Standard				
		8	Standard 5	20ng/µL	1	2.0 µL
		Standard				
		9	Blank	Eluent Blank	1	10 µL
		10	Test Sample	Sample	1	1.0 µL
		(s)				
		4.8.6	Click on run sequence option in run control menu to start the			
		sequence.				
		4.8.7	Each sample run will be of 4 minutes and after running all			
		the samples the sequence will automatically stop.				
		4.8.8	Inject the eluent blank after every 20 injections.			
		Note: Here, eluent blank refers to mobile phase. One extra				
		vial with same SST sample can be used if 6 injections are not				
		possible from single vial.				
		4.9	Calculations:			
		4.9.1	Results of the sample are calculated from the calibration			
		standards using the chemstation software.				
		4.9.1.1	Open the chromatogram corresponding to 1ng/µL			
		calibration standard in the offline copy of				
		chemstation software in the Data Analysis mode.				
		4.9.1.2	From the calibration menu select calibration			
		settings and write ng/µL in the Amount unit's				
		option.				
		4.9.1.3	Select New Calibration Table from Calibration menu			

			and enter value1 in default amount option and value	
			1 in level option.	
			4.9.1.4	Now open the chromatogram for 3.0 ng/μL. Select add
			level option from calibration menu and enter value	
			3.0 in default amount and value 2 in level option.	
			4.9.1.5	Similarly open chromatograms for 5.0ng/μL,
			10.0ng/μL and 20.0ng/μL and keep adding default	
			amount and appropriate value of level.	
			4.9.1.6	Select calibration task icon and the calibration
			table with required concentrations and levels can	
			be viewed in the calibration table view. A	
			calibration curve is also displayed in the	
			calibration curve field with value of correlation	
			coefficient of the curve.	
			4.9.1.7	Save the calibration method under appropriate name.
			4.9.1.8	Open the chromatogram of test sample and select the
			above created calibration method for integration of	
			the test sample.	
			4.9.1.9	In the specify report option of Report menu, select
			ESTD in the Calculate field.	
			4.9.1.10	In the report layout view click on the show report
			preview with current template, method and data file	
			icon and view the result of the test sample in	
			terms of concentration (ng/μL).	
			4.9.1.11	Calculations for Standard curve and samples:
			The calculations are made according to the equation	
			of minimum squares.	
			$Y = mX + b.$	
			Where: m = slope	
			b = intercept	
			Y = absorbance	
			X = concentration	
		4.9.2	The calculations for determination of amount of CDAP per dose	
		(10μg) of ARM of Haemophilus Influenzae type b conjugate		
		vaccine is as follows:		
		Example: If a sample has a PRP concentration of 1.0 mg/mL and		
		the amount of CDAP present in the sample is 5ng/μL or 5μg/mL.		
		The amount of CDAP present per dose(10μg) is calculated as		
		follows:		
		1000μg -----5μg		
		10 μg(dose)----- X		
		$X = 5\mu\text{g} \times 10 \mu\text{g}$		
		1000 μg		
		= 0.05μg or		
		$X = 50 \text{ ng/dose}$		
		4.9.3	Calculations for System Suitability Test:	
		4.9.3.1	Note the values of area under the curve and	
		retention time for standard 3(SST standard) .		
		4.10	Acceptance Criteria System Suitability Test:	
		4.10.1	RSD calculated for Retention time of standard 3(5ng/μL)	
		should be <5.0%.		
		4.10.2	RSD calculated for Area of standard 3(5ng/μL) should be	
		<5.0%.		
		4.11	Result Reporting:	
		4.11.1	Open CDAP.FRP template and take print out of the report.	
		4.11.2	Print report: Click on print option to take a printout of a	
		ChemStation report using the current report template and data		
		from the currently loaded data file.		
		4.11.3	Report the results of quantity of CDAP present/dose (10μg).	
		4.11.4	The report generated can be viewed with report review icon.	
		4.11.5	The printed report is signed by the analyst who does the	
		analysis and also checked by the other analyst/supervisor.		
		5.0	Forms and Records:	
		5.1	Determination of CDAP by high performance liquid chromatography.	
		6.0	References:	
		6.1	European pharmacopoeia.	

[SHEET: MnPs QC]

QC Requirements for analysis of 16 batches of meningococcal polysaccharides (4 serotypes) | | | | |

S. No. | Equipments | Quantity | Reagents | Quantity | Consumables | Quantity |

1 | Analytical balance | 2 | PASTOREX MENINGITIS INDIVIDUAL LATEX TEST KIT, code 61608 | 5 | Disposable glass test tubes (approximately 12 100 mm) with disposable plastic caps are recommended. | 500 |
2 | Laboratory vortex mixer | 2 | Resorcinol reagent | 10 gm | Auto Pipettes tips (e.g. Eppendorf) | 5000 each |

3 | pH-Meter | 1 | Butanol | 2 lit | Volumetric flasks, 100–1000 mL. | 5 each |

4 | UV-Vis Spectrophotometer with Cuvettes | 1 | Butyl acetate | 5 lit | Measuring cylinders, 10–1000 mL. | 5 each |

5 | Water bath 100 deg. | 1 | N-acetyl neuraminic acid (NANA) | 2 gm | Glass test tubes (12 x 75 mm e.g. Borosil) | 200 |

6 | Furnace | 1 | Phosphorus standard, 20 µg/ml | 100 gm | Screw cap HPLC vials with/without inserts | 200 |

7 | Water bath, 37°C | 1 | Sulfuric Acid | 2 lit | Solvac Filter Holder | 20 |

8 | Calibrated HPLC system with RI detector | 1 | Perchloric acid | 2 lit | Nylon 6,6 membrane 0.2µm membrane filter | 20 |

9 | Precision balance | 1 | Hydrogen peroxide free from phosphorus | 1 lit | Syringe filters 0.2 µm | 500 |

10 | Vacuum Pump | 1 | Ascorbic acid sodium salt | 100 gm | Aluminum Foil | 20 |

11 | Refrigerator | 1 | Ammonium molybdate tetra hydrate | 25 gm | Volumetric flasks (10ml, 25ml & 50ml) | 5 each |

12 | Magnetic stirrer | 1 | Hydrochloric acid, 37% | 2 lit | Screw capped test tubes (30 ml) with stand | 100 |

13 | Ultrasonic Cleaner | 1 | Iron (III) chloride hex hydrate 399% | 100 gm | Glass Pipettes (5ml, 10ml & 25 ml) | 100 |

14 | Rocking and Tumbling Shaker | 1 | Sodium Hydroxide 399 % | 5000 gm | Eppendorf (1.5ml & 2ml) | 5000 |

15 | Sugar KS-804 (Shodex Column) | 1 | Sodium Acetate anhydrous ≥ 99.5 % | 50 gm | Gloves | 1000 |

16 | Sugar KS-805 (Shodex Column) | 1 | Hydroxylamine hydrochloride ≥ 99.0 % | 100 gm | Glass tubes 25 ml | 50 |

17 | Guard Column KS-G (Shodex) | 1 | Dextran | 1 gm | Depyrogenized glass dilution tubes | 100 |

18 | Auto pipettes with Tips (20-200µl, 100 to 1000 µl & 1000 to 5000 µl)r | 2 each size | Sodium azide | 50 gm | Depyrogenized glass reaction tubes | 100 |

19 | Vortex Mixer | 1 | Deuterium Oxide | 100 ml | Depyrogenized tips | 1000 |

20 | BDS Hypersil C18 silica column (250 x 4.6mm, Particle size:5µm) | 1 | Sodium Chloride | 5000 gm | Test tube rack | 10 |

21 | Laboratory Centrifuge | 1 | Sodium dihydrogen phosphate, dihydrate | 500 gm | | |

22 | Fume Hood | 1 | Disodium hydrogen phosphate, anhydrous | 500 gm | | |

23 | 96 well Maxisorp Nunc ELISA plates (Nunc Cat No. 439454) | 20 | Copper sulphate pentahydrate | 10 gm | | |

24 | Micro plate ELISA reader | 1 | Sodium tartarate | 10 gm | | |

25 | Incubator | 1 | Sodium hydroxide | 100 gm | | |

26 | Auto titrator | 1 | Sodium Carbonate | 100 gm | | |

27 | Multichannel auto pipette (50 to 300 µl) | 1 | Folin & Ciocalteu's phenol reagent | 500 ml | | |

28 | Tube heating block (37° C) | 1 | Bovine Serum Albumin | 100 gm | | |

29 | Deep Freezer (- 20.0 ± 5°C) | 1 | Deoxycholic acid (DCA) | 2 gm | | |

30 | Timer | 1 | Methanol | 2 lit | | |

31 | Thermogravimetric analyser to determine LOD | 1 | Acetonitrile | 2 lit | | |

32 | | | Sodium Acetate anhydrous | 20 gm | | |

33 | | | Orthophosphoric Acid | 100 ml | | |

34 | | | | |

35 | | | Ammonium acetate | 10 gm | | |

36 | | | acetic acid (glacial) | 100 ml | | |

37 | | | Acetyl acetone | 20 ml | | |

38 | | | Formaldehyde standard solution | 50 ml | | |

39 | | | Anti CWPS Polyclonal rabbit Antisera (Staten Serum Institute, Cat No. 3653) | 2 vials | | |

40 | | | Purified CWPS (Staten Serum Institute, Cat No. 3459) | 2 vials | | |

41 | | | Monoclonal antibodies against Phosphoryl Choline (Staten Serum Institute, Cat No. 87214) | 2 vials | | |

42 | | | Polyclonal Goat anti-mouse Immunoglobulin HRP conjugates (Dako Cat No. P0447) | 1 vial | | |

43 | | | Bovine Serum Albumin (Fraction V, Sigma Cat No. A9647) | 100 gm | | |

44 | | | Tween 20 | 100 ml | | |

45 | | | TMB Substrate (Sigma, Cat No. T4444) | 200 ml | | |

46 | | | Conc. Sulphuric acid | 1 lit | | |

47 | | | NaCl | 2000 gm | | |

48 | | | Na2HPO4 | 500 gm | | |

49 | | | NaH2PO4 | 500 gm | | |

50 | | | Dimidium Bromide/Disulphine blue indicator stock solution | 1 VIAL | | |

51 | | | Sodium dodecyl sulphate | 100 gm | | |

52 | | | Stock CTAB | 20 gm | | |

53 | | | Limulus Amoebocyte Lysate, Pyrotell | 10 Vials | | |

54 | | | Controlled Standard Endotoxin (CSE) | 2 vials | | |

55 | | | LAL reagent water (LRW) | 200 ml | | |

[SHEET: General QC]

| General QC testing equipment | | | Quantity | | |

| Water testing | | | pH meter and conductivity meter | 1 | | |

| | | TOC analysers | 1 | developmental batches can be taken without this | | |

| | | Microscope | 1 | | |

| | | Incubator 300L (37 deg. C) | 1 | | |

| | | Laminar flow cabinet vertical | 1 | | |

| | | Refrigerators (400L) | 1 | | |

| | | Steam and WFI sampler | 1 | developmental batches can be taken without this | | |

| | | Colony counter | 1 | | |

| Sterility testing | | | Isolator for sterility | 1 | developmental batches can be taken without this | | |

|

| | | Manifolds for sterility testing | 2 | | |

| | | Incubator 500L for 23 +/- 2 deg. C and 33 +/-2 deg. C respectively | 2 | | |

| Maintenance of reference cultures and cell banks | | Deep freezer (-20°C) (400L capacity) | 1 | | |

| | | Deep freezer (-80°C) (400L capacity) | 1 | | |

		Lyophilizer	1		
Stability Study		Stability Chambers (5±3° C, 25, 37 deg. C)	3		
		LAF Cabinet	1		
		Cold room (5±3° C)	1		
Media and solution preparation		Autoclave for sterilization	1		
		Autoclave for decontamination	1		
		High Quality Laboratory water preparation system (50 L /hr) (Milli Q water)	1	Not required, if	
equivalent water is available					
		Glassware washing machine	1		
		Hot air drying oven (200 L)	1		
		Media distribution system for plating	1		
		LAF Cabinet	1		
		pH meter and conductivity meter	1		
		Filter integrity tester	1		
		Ice Maker	1		
		Dry heat sterilizer	1		
Raw material testing		UV Vis Spectrophotometer	1		
		Table top centrifuges	1		
		Hot plate	1		
		Weighing Balance (0.01mg and 220 gm)	1		
		pH meter/conductivity meter	1		
		Vortex mixer	1		
		Heating plate magnetic stirrer (5-20 L volume)	1		
		Refrigerators (300L)	1		
		LAL test equipment	1		
		Osmolarity meter	1	developmental batches can be taken without this	
		Analytical balance	1		
		Fume hood	1		
		Oil Bath	1	developmental batches can be taken without this	
		Melting Point Apparatus	1		
		Magnetic stirrer	1		
		Vacuum Pump	1		
		High Temperature Muffle Furnace	1	developmental batches can be taken without this	
		Tube heating Block	1		
		Polarimeter	1	developmental batches can be taken without this	
Environment monitoring testing		Air sampler	1		
		Particle counter	1		
 [SHEET: Sheet3]

1.0 | Objective | Objective | Objective | Objective | Objective
| To lay down the procedure for preparation and standardization of tetanus antitoxin (Working standard) for flocculation test of Tetanus antigens. | To lay down the procedure for preparation and standardization of tetanus antitoxin (Working standard) for flocculation test of Tetanus antigens. | To lay down the procedure for preparation and standardization of tetanus antitoxin (Working standard) for flocculation test of Tetanus antigens. | To lay down the procedure for preparation and standardization of tetanus antitoxin (Working standard) for flocculation test of Tetanus antigens. | To lay down the procedure for preparation and standardization of tetanus antitoxin (Working standard) for flocculation test of Tetanus antigens.

2.0 | Scope | Scope | Scope | Scope | Scope
| This procedure is applicable for preparation and standardization of tetanus antitoxin for flocculation test of Tetanus antigens. | This procedure is applicable for preparation and standardization of tetanus antitoxin for flocculation test of Tetanus antigens. | This procedure is applicable for preparation and standardization of tetanus antitoxin for flocculation test of Tetanus antigens. | This procedure is applicable for preparation and standardization of tetanus antitoxin for flocculation test of Tetanus antigens. | This procedure is applicable for preparation and standardization of tetanus antitoxin for flocculation test of Tetanus antigens.

3.0 | Responsibility | Responsibility | Responsibility | Responsibility | Responsibility
| 3.1 | 3.1 | The Analyst shall be responsible for carrying out the procedure as established in this SOP. | The Analyst shall be responsible for carrying out the procedure as established in this SOP. | The Analyst shall be responsible for carrying out the procedure as established in this SOP.
| 3.2 | 3.2 | Supervisor shall be responsible for verifying that the same has been carried out correctly. | Supervisor shall be responsible for verifying that the same has been carried out correctly. | Supervisor shall be responsible for verifying that the same has been carried out correctly.
| 3.3 | 3.3 | Head QC or his designee shall be responsible for overall compliance of the SOP. | Head QC or his designee shall be responsible for overall compliance of the SOP. | Head QC or his designee shall be responsible for overall compliance of the SOP.

4.0 | Abbreviations and Definitions | Abbreviations and Definitions | Abbreviations and Definitions | Abbreviations and Definitions | Abbreviations and Definitions
| 4.1 | SOP | SOP | SOP | : Standard Operating Procedure
| 4.2 | QA | QA | QA | : Quality Assurance
| 4.3 | QC | QC | QC | : Quality control
| 4.4 | TATF | TATF | TATF | : Tetanus Antitoxin for Flocculation
| 4.5 | mL | mL | mL | : milliliter
| 4.6 | °C | °C | °C | : Degree Celsius
| 4.7 | % | % | % | : Percent
| 4.8 | Lf/ml | Lf/ml | Lf/ml | : Limes-flocculation per milliliter

5.0 | Procedure: | Procedure: | Procedure: | Procedure: | Procedure:
| 5.1 | 5.1 | Equipment, Materials and Reagents | Equipment, Materials and Reagents | Equipment, Materials and Reagents
| | 5.1.1 | Standard Tetanus toxoid (NIBSC code: 04/150) | Standard Tetanus toxoid (NIBSC code: 04/150)
| | 5.1.2 | Standard Tetanus Antitoxin for flocculation (NIBSC code: 66/021) | Standard Tetanus Antitoxin for flocculation (NIBSC code: 66/021)
| | 5.1.3 | Candidate Tetanus Antitoxin for flocculation | Candidate Tetanus Antitoxin for flocculation
| | 5.1.4 | Micropipettes | Micropipettes
| | 5.1.5 | Micro tips | Micro tips
| | 5.1.6 | Flocculation t tube rack | Flocculation t tube rack
| | 5.1.7 | Flocculation tube. | Flocculation tube.
| | 5.1.8 | Thermometer | Thermometer
| | 5.1.9 | Stop watch | Stop watch
| | 5.1.10 | Convex lens | Convex lens
| | 5.1.11 | Sterile 10ml & 50ml vials | Sterile 10ml & 50ml vials
| | 5.1.12 | Physiological saline (0.9% w/v) | Physiological saline (0.9% w/v)
| | 5.1.13 | Thiomersal | Thiomersal
| 5.2 | 5.2 | Preparation of solutions | Preparation of solutions | Preparation of solutions
| | 5.2.1 | Physiological saline (0.9% w/v) | Physiological saline (0.9% w/v)
| | | Add 0.9 gm of Sodium Chloride to 50ml of purified water, mix well and make up it to 100ml. | Add 0.9 gm of Sodium Chloride to 50ml of purified water, mix well and make up it to 100ml.
| | 5.2.2 | 1% w/v Thiomersal Solution | 1% w/v Thiomersal Solution
| | | Add 1 gm of Thiomersal to 50ml of purified water, mix well and make up it to 100ml. | Add 1 gm of Thiomersal to 50ml of purified water, mix well and make up it to 100ml.
| Standardization of In house Tetanus Antitoxin | Standardization of In house Tetanus Antitoxin | Standardization of In house Tetanus Antitoxin | Standardization of In house Tetanus Antitoxin | Standardization of In house Tetanus Antitoxin
| 5.3 | 5.3 | Preparation of Candidate Reference Tetanus toxoid (50Lf/mL) | Preparation of Candidate Reference Tetanus toxoid (50Lf/mL) | Preparation of Candidate Reference Tetanus toxoid (50Lf/mL)
| | 5.3.1 | 5.3.1 | Obtain Tetanus toxoid bulk purified from production facility/ external source which complies with the predetermined specifications.
| | 5.3.2 | 5.3.2 | Dilute the toxoid to 50Lf/mL based on the previous estimated value.
| 5.4 | 5.4 | Standardization of Standard toxoid (50Lf/mL) | Standardization of Standard toxoid (50Lf/mL) | Standardization of Standard toxoid (50Lf/mL)
| Standardization of Standard toxoid (50Lf/mL)
| | 5.4.1 | 5.4.1 | Reconstitute the National/International reference standard of antitoxin for flocculation to 100Lf/mL based on the label claim with physiological saline.
| | 5.4.2 | 5.4.2 | Determine the unitage of standard toxoid using the National/International reference standard of antitoxin by flocculation test following SOP.
| | 5.4.3 | 5.4.3 | The test is valid if the standard toxoid concentration is 50 ± 1 Lf/mL.
| 5.5 | 5.5 | Preparation of Standard Reference Antitoxin | Preparation of Standard Reference Antitoxin | Preparation of Standard Reference Antitoxin
| | 5.5.1 | 5.5.1 | Take the required quantity of antitoxin obtained from the reputed vendor in a suitable glass container
| | 5.5.2 | 5.5.2 | Dilute the antitoxin to 100 Lf/mL with physiological saline containing preservatives based on the label claim.
| 5.6 | 5.6 | Standardization of Antitoxin | Standardization of Antitoxin | Standardization of Antitoxin
| | 5.6.1 | 5.6.1 | Determine the unitage of standard toxoid using the candidate tetanus antitoxin at 100 Lf/ml by flocculation test.
| | 5.6.2 | 5.6.2 | If the candidate tetanus antitoxin gives the unitage of standard toxoid more than 50 Lf/ml then make small additions of candidate tetanus antitoxin till it gives the unitage of standard toxoid 50 Lf/ml.
| | 5.6.3 | 5.6.3 | If the candidate antitoxin gives the unitage of standard toxoid less than 50 Lf/ml then make small additions of normal saline till it gives the unitage of standard toxoid 50 Lf/ml.

||| 5.6.4 | 5.6.4 | Candidate tetanus antitoxin when gives the unitage of standard toxoid as 50 Lf/ml is considered to be standardized for use as working standard of tetanus antitoxin for flocculation test and assigned the value of 100 Lf/ml.

||| 5.6.5 | 5.6.5 | After standardization, sterile filter the antitoxin by using 0.2µ filters and distributes it to 100 mL bottles.

| 5.7 | 5.7 | Monitoring of Internal Reference Antitoxin | Monitoring of Internal Reference Antitoxin | Monitoring of Internal Reference Antitoxin

||| 5.7.1 | 5.7.1 | Check the Lf / mL concentration of Internal Reference antitoxin for flocculation once in three months with internal / national / international standard toxoid.

||| 5.7.2 | 5.7.2 | Discard the reference antitoxin if any drop is observed in Lf concentration and establish a new one.

| 5.8 | 5.8 | Conformation of International/national/in-house Tetanus Antitoxin Value in LF/ml | Conformation of International/national/in-house Tetanus Antitoxin Value in LF/ml | Conformation of International/national/in-house Tetanus Antitoxin Value in LF/ml

||| 5.8.1 | 5.8.1 | Reconstitute the National/International reference standard of tetanus toxoid to 50 Lf /ml with 0.9% normal saline.

||| 5.8.2 | 5.8.2 | Take the lyophilized ampoule of international tetanus antitoxin and reconstitute it to 150 IU/ml with 0.9 % normal saline.

||| 5.8.3 | 5.8.3 | Perform the flocculation test

||||| Set the temperature of the water bath at 50 + 2° C.

||||| Incubate the tubes at 50± 2°C, in a see-through water bath. Ensure that 2/3rd of the flocculation tubes are immersed in water.

||| Example: | Example: | Example:

|||||

||||| Observe the tube which flocculates most rapidly. This is tube with optimal concentration of antigen and antibody and gives the Lf /ml of sample.

||||| If the candidate antitoxin gives the unitage of standard toxoid more than 50 Lf/ml then make small additions of normal saline till it gives the unitage of standard toxoid 50 Lf/ml.

||||| Candidate tetanus antitoxin when gives the unitage of standard toxoid as 50 Lf/ml is considered to be standardized for use as working standard of tetanus antitoxin for flocculation test and assigned the value of 100 Lf/ml.

6.0 | Forms and Records | Forms and Records | Forms and Records | Forms and Records | Forms and Records

| Form for standardization of Tetanus antitoxin for flocculation test (working standard) | Form for standardization of Tetanus antitoxin for flocculation test (working standard) | Form for standardization of Tetanus antitoxin for flocculation test (working standard) | Form for standardization of Tetanus antitoxin for flocculation test (working standard) | Form for standardization of Tetanus antitoxin for flocculation test (working standard)

7.0 | References | References | References | References | References

| Indian Pharmacopeia | Indian Pharmacopeia | Indian Pharmacopeia | Indian Pharmacopeia | Indian Pharmacopeia

[1.0 |Purpose: |
 ||To lay down the procedure for quantifying nucleic acid by absorbance. |
 [2.0 |Applicable to: |
	This Standard Operating Procedure is applicable for the analysis of samples
	of PRP of Haemophilus Influenzae type b in which Nucleic acid concentration
	is to be determined
[3.0	Abbreviations:
	3.1
	3.2
	3.3
	3.4
	3.5
	3.6
	3.7
	3.8
[4.0	Procedure:
	4.1
	4.1.1
	4.1.2
	4.1.3
	4.1.4
	4.1.5
	4.2
	4.2.1
	4.2.2
	4.2.3
	4.2.4
	4.2.5
	4.2.6
	4.2.7
	4.2.8
	4.2.9
[5.0	Reporting:
	5.1
[6.0	Reference:
	Not applicable

1.0|Objective |
 |To lay down the procedure for testing the potency of tetanus toxoid by |
 |challenge method in Swiss albino mice. |
 2.0|Scope |
 |Applicable to the batches of tetanus toxoid as single component or combined |
 |with other components. |
 3.0|Responsibility: |
 |3.1 |The Analyst shall be responsible for carrying out the procedure as |
 |established in this SOP. |
 |3.2 |Supervisor shall be responsible for verifying that the same has been |
 |carried out correctly. |
 |3.3 |Head QC or his designee shall be responsible for overall compliance of|
 |the SOP. |
 4.0|Abbreviations and Definitions |
4.1	ml	: Milliliter
4.2	g	: Gram
4.3	°C	: Degree Celsius
4.4	%	: Percent
4.5	QC	: Quality Control
4.6	QA	: Quality Assurance
4.7	AH	: Animal House
4.8	SOP	: Standard Operating Procedure
5.0	Procedure:	
5.1	Equipment, Materials and Reagents	
	5.1.1	Standard Tetanus Toxoid Vaccine (NIBSC code: 08/218)
	5.1.2	Weighing balance
	5.1.3	Laminar air flow cabinet
	5.1.4	Autoclave
	5.1.5	Autopipettes
	5.1.6	Test tubes
	5.1.7	Test tube racks
	5.1.8	Glass vials, 10 ml
	5.1.9	Disposable syringes (2 ml, 5 ml)
	5.1.10	Peptone
	5.1.11	Sodium chloride
	5.1.12	Sodium Hydroxide
	5.1.13	Hydrochloric acid
5.2	Preparation of Normal saline:	
	Weigh 0.9 g of sodium chloride and dissolve in 60 ml of the purified	
	water. Make up the volume up to 100 ml with purified water. Adjust the	
	pH between 7.0 to 7.2 using 0.1 M sodium hydroxide/Hydrochloric acid	
	solution. Sterilize by autoclaving at 1210 C for 20 minutes.	
5.3	Peptone water:	
	Weigh 1 g of peptone and 0.5 g of sodium chloride. Dissolve in 60 ml of	
	purified water. Make up the volume up to 100 ml with purified water.	
	Adjust the pH between 7.0 to 7.2 using 0.1 M sodium	
	hydroxide/Hydrochloric acid solution. Sterilize by autoclaving at 1210C	
	for 20 minutes.	
5.4	Selection of test animals:	
	5.4.1	Select 116 Swiss albino mice each weighing between 13 to 17 g at
	least one day before immunization.	
	5.4.2	Preferably choose mice of the same sex. Otherwise distribute
	males and females equally in each group.	
	5.4.3	Make six groups of 16 mice each and four groups of 5 mice each.
	Allocate three groups out of six to each of three dilutions of	
	reference vaccine and other three groups to each of three	
	dilutions of the test vaccine. Four groups of 5 mice each are to	
	be used for LD50 determination.	
	5.4.4	Distribute all the mice in various groups by randomization.
	5.4.5	Mark each group of mice with appropriate color scheme.
5.5	Preparation of Vaccine dilution	
	5.5.1	Preparation of dilutions of reference vaccine:
	5.5.1.1	Suitable dilutions of the reference vaccine are selected
	1	in such a way so as that 80% to 90% protection is
	conferred in first dilution containing highest antigen	
	concentration, 40% to 60% protection is conferred in	
	middle dilution and 10% to 20% protection is conferred in	
	last dilution containing lowest antigen concentration.	
	5.5.1.1	Dilute reference vaccine in normal saline so as to
	2	contain 200 –300 IU/ml. Make further dilutions of the
	reference vaccine as detailed below	
	1/10 dilution: Add 0.5 ml of vaccine to 4.5 ml of normal	
	saline and mix thoroughly.	
	1/50 dilution: Add 4 ml of 1/10 dilution of vaccine	
	(section 8.2.1.1) to 16 ml of normal saline and mix	
	thoroughly	
	1/100 dilution: Add 10 ml of 1/50 dilution of vaccine	
	(section 8.2.1.2) to 10 ml of normal saline and mix	
	thoroughly.	
	1/200 dilution: Add 10 ml of 1/100 dilution of vaccine	
	(section 8.2.1.3) to 10 ml of normal saline and mix	
	thoroughly	
	5.5.2	Preparation of dilutions of test vaccine:
	Take test vaccine and dilute it following steps 8.2.1.1	
	to 8.2.1.4.	
5.6	Immunization:	
	5.6.1	Inject subcutaneously each of 16 Swiss mice in each group with
	0.5 ml of appropriate dilution (1/50, 1/100 and 1/200) of test	
	and reference vaccine allocated to that group.	

	5.6.2	Keep the four groups of 5 mice each as uninoculated.
	5.6.3	Keep the immunized animals under observations for 28 days. The
		animals should show normal growth and good health during the
		observation period.
	5.7	Preparation of Challenge tetanus toxin:
		Suitably dilute the stabilized tetanus toxin in peptone water in such a
		way so as to contain approximately 100 LD50/ml.
	5.7.1	1/10 dilution of challenge toxin: Add 1 ml of challenge toxin to
		9 ml of peptone water. Mix it thoroughly. (10 LD50/ml)
	5.7.2	1/50 dilution of challenge toxin: Add 3 ml of 1/10 dilution of
		challenge toxin to 12 ml of peptone water. Mix it thoroughly. (2
		LD50/ml)
	5.7.3	1/100 dilution of challenge toxin: Add 6 ml of 1/50 dilution of
		Challenge toxin to 6 ml of peptone water. Mix it thoroughly. (1
		LD50/ml)
	5.7.4	1/200 dilution of challenge toxin: Add 6 ml of 1/100 dilution of
		challenge toxin to 6 ml of peptone water. Mix it thoroughly.
		(1/2 LD50/ml)
	5.8	Inoculation of challenge toxin:
		After twenty-eight days of immunization, inject all groups of mice
		subcutaneously with 1.0 ml of the challenge dose containing
		approximately 100 LD50 per ml. Mice are challenged subcutaneously over
		the lumbar region of the spine where it produces distinctive hind limb
		paralysis which can be recognized at an early stage.
	5.9	LD50 estimation:
		Take four groups each consisting of 5 mice (selected on the day of
		immunization). Inject subcutaneously 1.0 ml of challenge dose
		containing approximately 100 LD50 and dilutions of challenge toxin
		containing 2 LD50, 1 LD50 and ½ LD50 dose per ml respectively in four
		groups of 5 mice each.
	5.10	Observations: Observe all the injected mice daily and record the deaths
	5.11	Calculations:
		Count the number of surviving mice 5 days after injection of the
		challenge toxin and calculate the potency of the vaccine being examined
		relative to the potency of the standard preparation. This may be done
		on the basis of the number of surviving animals in each of the six
		groups of sixteen, using standard statistical methods. (By probit
		analysis as per software programme evolved by F.R. Marsman RIVM,
		Bilthoven, Netherland).
	5.12	Validity criteria:
	5.12.1	The slopes of the dose – response curves of the vaccine under
		test and the standard are linear and parallel.
	5.12.2	The 50% effective dose of each vaccine lies between the
		largest and smallest immunizing doses.
	5.12.3	Challenge toxin was approximately 100 times the 50% lethal
		dose.
	5.13	Acceptance criteria:
	5.13.1	The estimated potency is not less than 40 I.U. per single
		human dose and not less than 60 I.U. per single human dose in
		case of combined vaccines containing whole cell pertussis.
	5.13.2	The limit of 95% confidence intervals of the estimate of
		potency is within 50-200% unless the lower limit of the 95%
		confidence interval of the estimated potency is not less than
		40 I.U. per single human dose and not less than 60 I.U. per
		single human dose in case of combined vaccines containing
		whole cell pertussis.
	6.0	Forms and Records
	6.1	Raw data form for recording the details of potency test of tetanus
		toxoid by challenge method in Swiss albino mice
	6.2	Raw data form for recording the observations on Swiss albino mice of
		tetanus potency test
	6.3	Form for reporting the results of potency test of tetanus toxoid by
		challenge method in Swiss albino mice
	7.0	References:
	7.1	World Health Organization, Technical Report Series No. 800, 1990
	7.2	Indian Pharmacopoeia, 1996 and amendments thereof.
	7.3	WHO manual for the production and control of vaccines – Tetanus Toxoid
		(BLG/UNDP/77.2 Rev 1)

2.0 | Scope | Scope | Scope | Scope | Scope | Scope | Scope | Scope | Scope

3.0 | Abbreviations and Definitions | Abbreviations and Definitions | Abbreviations and Definitions |

| 3.1 | cm | cm | : centimeter | : centimeter | : centimeter | : centimeter | : centimeter

3.3 | HPAEC-PAD | HPAEC-PAD : High Performance Anion Exchange Chromatography with Pulsed Amperometric

| 3.4 | ICS | ICS | : Ion Chromatography System | : Ion Chromatography System | : Ion Chromatography

| 3.5 | mL | mL | : milliliter | milliliter | : milliliter | : milliliter | : milliliter

3.7	psi	psi	: pressure per square inch	: pressure per square inch	: pressure per square inch
-----	-----	-----	----------------------------	----------------------------	----------------------------

| 3.8 | PRP-TT | PRP-TT | : Polvribosyl Ribitol Phosphate - Tetanus Toxoid | : Polvribosyl Ribitol

| 3.9 | rpm | rpm | : revolutions per minute | : revolutions per minute | : revolutions per minute | :

Deviation | : Relative Standard Deviation | : Relative Standard Deviation | : Relative Standard Deviation

Operating Procedures	Standard Operating Procedures	Standard Operating Procedures
3.12	SST	SST
	System Suitability Test	System Suitability Test

| 3.13 | °C | °C | : Degree centigrade | : Degree centigrade

| 3.14 | μg | μg | $\cdot \text{mic}$

3.15	µE	µE	: Micro litre	: Micro litre	: Micro litre	: Micro litre	: Micro litre
3.16	µm	µm	: Micro meter	: Micro meter	: Micro meter	: Micro meter	: Micro meter

3.17 | System Suitability Test | System Suitability Test | : System suitability tests are an integral part of gas and liquid chromatographic methods. They are used to verify that the resolution and

The tests are based on the concept that the equipment, electronics, analytical operations and samples to be analyzed constitute an integral system that can be evaluated as such. | : System suitability tests are an integral part of gas and liquid chromatographic methods. They are used to verify that the resolution

The tests are based on the concept that the equipment, electronics, analytical operations and samples to be analyzed constitute an integral system that can be evaluated as such. | : System suitability tests are an integral part of gas and liquid chromatographic methods. They are used to verify that the resolution

The tests are based on the concept that the equipment, electronics, analytical operations and samples to be analyzed constitute an integral system that can be evaluated as such. | : System suitability tests are an integral part of gas and liquid chromatographic methods. They are used to verify that the resolution

The tests are based on the concept that the equipment, electronics, analytical operations and samples to be analyzed constitute an integral system that can be evaluated as such. []: System suitability tests are an integral part of gas and liquid chromatographic methods. They are used to verify that the resolution

The tests are based on the concept that the equipment, electronics, analytical operations and samples to be analyzed constitute an integral system that can be evaluated as such.

with Chromeleon software.

|| | 4.2.2.3 | pH Meter | pH Meter | pH Meter | pH Meter | pH Meter | pH Meter

|| | 4.2.3 | Materials: | Materials: | Materials: | Materials: | Materials: | Materials: | Materials:

|| | 4.2.3.1 | CarboPac MA 1 column with CarboPac MA 1 pre-column guard | CarboPac MA 1 column with CarboPac MA 1 pre-column guard | CarboPac MA 1 column with CarboPac MA 1 pre-column guard | CarboPac MA 1 column with CarboPac MA 1 pre-column guard | CarboPac MA 1 column with CarboPac MA 1 pre-column guard | CarboPac MA 1 column with CarboPac MA 1 pre-column guard

|| | 4.2.3.2 | Laboratory Glassware | Laboratory Glassware | Laboratory Glassware | Laboratory Glassware | Laboratory Glassware | Laboratory Glassware

|| | 4.2.3.3 | Screw cap vials | Screw cap vials | Screw cap vials | Screw cap vials | Screw cap vials | Screw cap vials

|| | 4.2.3.4 | Solvac Filter Holder | Solvac Filter Holder | Solvac Filter Holder | Solvac Filter Holder | Solvac Filter Holder | Solvac Filter Holder

|| | 4.2.3.5 | Nylon 6, 6 membrane 0.20/0.22/.045 µm membrane filters. | Nylon 6, 6 membrane 0.20/0.22/.045 µm membrane filters. | Nylon 6, 6 membrane 0.20/0.22/.045 µm membrane filters. | Nylon 6, 6 membrane 0.20/0.22/.045 µm membrane filters. | Nylon 6, 6 membrane 0.20/0.22/.045 µm membrane filters. | Nylon 6, 6 membrane 0.20/0.22/.045 µm membrane filters.

|| | 4.2.3.6 | Autopipettes (2-20 L), (20-200 L), (100-1000 L) | Autopipettes (2-20 L), (20-200 L), (100-1000 L) | Autopipettes (2-20 L), (20-200 L), (100-1000 L) | Autopipettes (2-20 L), (20-200 L), (100-1000 L) | Autopipettes (2-20 L), (20-200 L), (100-1000 L) | Autopipettes (2-20 L), (20-200 L), (100-1000 L)

|| | 4.2.3.7 | Tips for the Autopipettes | Tips for the Autopipettes | Tips for the Autopipettes | Tips for the Autopipettes | Tips for the Autopipettes | Tips for the Autopipettes

|| | 4.2.3.8 | Syringe Filters (0.2µm porosity) | Syringe Filters (0.2µm porosity) | Syringe Filters (0.2µm porosity) | Syringe Filters (0.2µm porosity) | Syringe Filters (0.2µm porosity) | Syringe Filters (0.2µm porosity)

|| | 4.2.3.9 | 1.5 ml eppendorf tubes | 1.5 ml eppendorf tubes | 1.5 ml eppendorf tubes | 1.5 ml eppendorf tubes | 1.5 ml eppendorf tubes | 1.5 ml eppendorf tubes

|| | 4.2.4 | Reagents: | Reagents: | Reagents: | Reagents: | Reagents: | Reagents: | Reagents:

|| | 4.2.4.1 | Ribitol (Adonitol) | Ribitol (Adonitol) | Ribitol (Adonitol) | Ribitol (Adonitol) | Ribitol (Adonitol) | Ribitol (Adonitol)

|| | 4.2.4.2 | Sodium Hydroxide solution 50% GR or Sodium Hydroxide Pellets | Sodium Hydroxide solution 50% GR or Sodium Hydroxide Pellets | Sodium Hydroxide solution 50% GR or Sodium Hydroxide Pellets | Sodium Hydroxide solution 50% GR or Sodium Hydroxide Pellets | Sodium Hydroxide solution 50% GR or Sodium Hydroxide Pellets | Sodium Hydroxide solution 50% GR or Sodium Hydroxide Pellets

|| | 4.2.4.3 | Hydrochloric acid 37% (Fuming) | Hydrochloric acid 37% (Fuming) | Hydrochloric acid 37% (Fuming) | Hydrochloric acid 37% (Fuming) | Hydrochloric acid 37% (Fuming) | Hydrochloric acid 37% (Fuming)

|| | 4.2.4.4 | Deoxycholic acid, Sodium Salt (DOC) | Deoxycholic acid, Sodium Salt (DOC) | Deoxycholic acid, Sodium Salt (DOC) | Deoxycholic acid, Sodium Salt (DOC) | Deoxycholic acid, Sodium Salt (DOC) | Deoxycholic acid, Sodium Salt (DOC)

|| | 4.2.4.5 | Purified water (filtered through 0.20 µm filter) & vacuum degassed for at least 15 min.) | Purified water (filtered through 0.20 µm filter) & vacuum degassed for at least 15 min.) | Purified water (filtered through 0.20 µm filter) & vacuum degassed for at least 15 min.) | Purified water (filtered through 0.20 µm filter) & vacuum degassed for at least 15 min.) | Purified water (filtered through 0.20 µm filter) & vacuum degassed for at least 15 min.) | Purified water (filtered through 0.20 µm filter) & vacuum degassed for at least 15 min.)

|| | 4.2.4.6 | WFI (filtered through 0.20 µm filter) & vacuum degassed for at least 15 min.) | WFI (filtered through 0.20 µm filter) & vacuum degassed for at least 15 min.) | WFI (filtered through 0.20 µm filter) & vacuum degassed for at least 15 min.) | WFI (filtered through 0.20 µm filter) & vacuum degassed for at least 15 min.) | WFI (filtered through 0.20 µm filter) & vacuum degassed for at least 15 min.) | WFI (filtered through 0.20 µm filter) & vacuum degassed for at least 15 min.)

| 4.3 | Preparation of solutions:

Prepare all the solutions as described below. | Preparation of solutions:

Prepare all the solutions as described below. | Preparation of solutions:

Prepare all the solutions as described below. | Preparation of solutions:

Prepare all the solutions as described below. | Preparation of solutions:

Prepare all the solutions as described below. | Preparation of solutions:

Prepare all the solutions as described below. | Preparation of solutions:

Prepare all the solutions as described below.

|| | 4.3.1 | Preparation of Standards: | Preparation of Standards: | Preparation of Standards: | Preparation of Standards: | Preparation of Standards: | Preparation of Standards:

|| | 4.3.1.1 | Preparation of External Standard Ribitol Solution (150g/mL) – SE 1:

For a volume of 100 mL, weigh:

Ribitol. 15mg

Dissolve in 50mL of filtered purified water and make up the volume to 100mL. Filter the solution through 0.2µm porosity nylon filters. Store the solution in 1.5mL aliquots at -20°C±5°C for not more than six months. | Preparation of External Standard Ribitol Solution (150g/mL) – SE 1:

For a volume of 100 mL, weigh:

Ribitol. 15mg

Dissolve in 50mL of filtered purified water and make up the volume to 100mL. Filter the solution through 0.2µm porosity nylon filters. Store the solution in 1.5mL aliquots at -20°C±5°C for not more than six months. | Preparation of External Standard Ribitol Solution (150g/mL) – SE 1:

For a volume of 100 mL, weigh:

Ribitol. 15mg

Dissolve in 50mL of filtered purified water and make up the volume to 100mL. Filter the solution through 0.2µm porosity nylon filters. Store the solution in 1.5mL aliquots at -20°C±5°C for not more than six months. | Preparation of External Standard Ribitol Solution (150g/mL) – SE 1:

For a volume of 100 mL, weigh:

Ribitol. 15mg

Dissolve in 50mL of filtered purified water and make up the volume to 100mL. Filter the solution through 0.2µm porosity nylon filters. Store the solution in 1.5mL aliquots at -20°C±5°C for not more than six months. | Preparation of External Standard Ribitol Solution (150g/mL) – SE 1:

For a volume of 100 mL, weigh:

Ribitol. 15mg

Dissolve in 50mL of filtered purified water and make up the volume to 100mL. Filter the solution through 0.2µm porosity nylon filters. Store the solution in 1.5mL aliquots at -20°C±5°C for not more than six months. | Preparation of External Standard Ribitol Solution (150g/mL) – SE 1:

For a volume of 100 mL, weigh:

Ribitol. 15mg

Dissolve in 50mL of filtered purified water and make up the volume to 100mL. Filter the solution through 0.2µm porosity nylon filters. Store the solution in 1.5mL aliquots at -20°C±5°C for not more than six months.

|| 4.3.1.2 | Preparation of External Standard Ribitol solution (3.0µg/mL) - SE 2:

Take 1mL of solution SE 1 in step 4.3.1.1 and make up the volume to 50mL with filtered purified water in a volumetric flask. Store the solution in 1.5mL aliquots at -20°C±5°C for not more than three months. |

Preparation of External Standard Ribitol solution (3.0µg/mL) - SE 2:

Take 1mL of solution SE 1 in step 4.3.1.1 and make up the volume to 50mL with filtered purified water in a volumetric flask. Store the solution in 1.5mL aliquots at -20°C±5°C for not more than three months. |

Preparation of External Standard Ribitol solution (3.0µg/mL) - SE 2:

Take 1mL of solution SE 1 in step 4.3.1.1 and make up the volume to 50mL with filtered purified water in a volumetric flask. Store the solution in 1.5mL aliquots at -20°C±5°C for not more than three months. |

Preparation of External Standard Ribitol solution (3.0µg/mL) - SE 2:

Take 1mL of solution SE 1 in step 4.3.1.1 and make up the volume to 50mL with filtered purified water in a volumetric flask. Store the solution in 1.5mL aliquots at -20°C±5°C for not more than three months. |

Preparation of External Standard Ribitol solution (3.0µg/mL) - SE 2:

Take 1mL of solution SE 1 in step 4.3.1.1 and make up the volume to 50mL with filtered purified water in a volumetric flask. Store the solution in 1.5mL aliquots at -20°C±5°C for not more than three months. |

Preparation of External Standard Ribitol solution (3.0µg/mL) - SE 2:

Take 1mL of solution SE 1 in step 4.3.1.1 and make up the volume to 50mL with filtered purified water in a volumetric flask. Store the solution in 1.5mL aliquots at -20°C±5°C for not more than three months.

|| 4.3.2 | Preparation of Working Dilutions for External Standard solutions for analysis:

Prepare the working dilutions of the external standard for analysis as described below: | Preparation of

Working Dilutions for External Standard solutions for analysis:

Prepare the working dilutions of the external standard for analysis as described below: | Preparation of

Working Dilutions for External Standard solutions for analysis:

Prepare the working dilutions of the external standard for analysis as described below: | Preparation of

Working Dilutions for External Standard solutions for analysis:

Prepare the working dilutions of the external standard for analysis as described below: | Preparation of

Working Dilutions for External Standard solutions for analysis:

Prepare the working dilutions of the external standard for analysis as described below: | Preparation of

Working Dilutions for External Standard solutions for analysis:

Prepare the working dilutions of the external standard for analysis as described below: | Preparation of

Working Dilutions for External Standard solutions for analysis:

Prepare the working dilutions of the external standard for analysis as described below:

|| | Standard | Standard | Volume of SE 2 (µL) | Volume of Purified Water (µL) | Volume of Purified

Water (µL) | Final Concentration (µg/mL) | Final Concentration (µg/mL)

|| | Standard 1 | Standard 1 | 25 | 975 | 975 | 0.075 | 0.075

|| | Standard 2 | Standard 2 | 50 | 950 | 950 | 0.150 | 0.150

|| | Standard 3 | Standard 3 | 150 | 850 | 850 | 0.450 | 0.450

|| | Standard 4 | Standard 4 | 250 | 750 | 750 | 0.750 | 0.750

|| | Table 1 | Table 1 | Table 1 | Table 1 | Table 1 | Table 1 | Table 1

|| 4.3.3 | Preparation of the 1% w/v DOC at pH 6.8:

For 25mL solution, weigh:

DOC..... 250mg

The pH of the solution is around 8, and should be brought to 6.8 by drop wise addition of 1N HCl. Use magnetic stirring bar to avoid formation of lumps. Prepare this solution fresh prior to use. |

Preparation of the 1% w/v DOC at pH 6.8:

For 25mL solution, weigh:

DOC..... 250mg

The pH of the solution is around 8, and should be brought to 6.8 by drop wise addition of 1N HCl. Use magnetic stirring bar to avoid formation of lumps. Prepare this solution fresh prior to use. |

Preparation of the 1% w/v DOC at pH 6.8:

For 25mL solution, weigh:

DOC..... 250mg

The pH of the solution is around 8, and should be brought to 6.8 by drop wise addition of 1N HCl. Use magnetic stirring bar to avoid formation of lumps. Prepare this solution fresh prior to use. |

Preparation of the 1% w/v DOC at pH 6.8:

For 25mL solution, weigh:

DOC..... 250mg

The pH of the solution is around 8, and should be brought to 6.8 by drop wise addition of 1N HCl. Use magnetic stirring bar to avoid formation of lumps. Prepare this solution fresh prior to use. |

Preparation of the 1% w/v DOC at pH 6.8:

For 25mL solution, weigh:

DOC..... 250mg

The pH of the solution is around 8, and should be brought to 6.8 by drop wise addition of 1N HCl. Use magnetic stirring bar to avoid formation of lumps. Prepare this solution fresh prior to use. |

Preparation of the 1% w/v DOC at pH 6.8:

For 25mL solution, weigh:

DOC..... 250mg

The pH of the solution is around 8, and should be brought to 6.8 by drop wise addition of 1N HCl. Use magnetic stirring bar to avoid formation of lumps. Prepare this solution fresh prior to use. |

Preparation of the 1% w/v DOC at pH 6.8:

For 25mL solution, weigh:

DOC..... 250mg

The pH of the solution is around 8, and should be brought to 6.8 by drop wise addition of 1N HCl. Use magnetic stirring bar to avoid formation of lumps. Prepare this solution fresh prior to use.

|| 4.3.4 | Preparation of 6N HCl:

Take 25.5mL of HCl in 50mL volumetric flask and make up the volume to 50mL with filtered WFI. |

Preparation of 6N HCl:

Take 25.5mL of HCl in 50mL volumetric flask and make up the volume to 50mL with filtered WFI. |

Preparation of 6N HCl:

Take 25.5mL of HCl in 50mL volumetric flask and make up the volume to 50mL with filtered WFI. |

Preparation of 6N HCl:

Take 25.5mL of HCl in 50mL volumetric flask and make up the volume to 50mL with filtered WFI. |

Preparation of 6N HCl:

Take 25.5mL of HCl in 50mL volumetric flask and make up the volume to 50mL with filtered WFI. |

Preparation of 6N HCl:

Take 25.5mL of HCl in 50mL volumetric flask and make up the volume to 50mL with filtered WFI.

|| 4.3.5 | Preparation of 1M NaOH: | Preparation of 1M NaOH: | Preparation of 1M NaOH: | Preparation of

1M NaOH: | Preparation of 1M NaOH: | Preparation of 1M NaOH: | Preparation of 1M NaOH:

|| | 4.3.5.1 | Prepare 50% (w/v) stock solution of NaOH in filtered WFI. Keep it overnight without any disturbance or take readymade Sodium Hydroxide solution 50%. | Prepare 50% (w/v) stock solution of NaOH in filtered WFI. Keep it overnight without any disturbance or take readymade Sodium Hydroxide solution 50%. | Prepare 50% (w/v) stock solution of NaOH in filtered WFI. Keep it overnight without any disturbance or take readymade Sodium Hydroxide solution 50%. | Prepare 50% (w/v) stock solution of NaOH in filtered WFI. Keep it overnight without any disturbance or take readymade Sodium Hydroxide solution 50%. | Prepare 50% (w/v) stock solution of NaOH in filtered WFI. Keep it overnight without any disturbance or take readymade Sodium Hydroxide solution 50%. | Prepare 50% (w/v) stock solution of NaOH in filtered WFI. Keep it overnight without any disturbance or take readymade Sodium Hydroxide solution 50%.

|| | 4.3.5.2 | Take 52.3mL of 50% stock solution and make up the volume to 1000mL with filtered and vacuum degassed WFI. Shake well and keep. | Take 52.3mL of 50% stock solution and make up the volume to 1000mL with filtered and vacuum degassed WFI. Shake well and keep. | Take 52.3mL of 50% stock solution and make up the volume to 1000mL with filtered and vacuum degassed WFI. Shake well and keep. | Take 52.3mL of 50% stock solution and make up the volume to 1000mL with filtered and vacuum degassed WFI. Shake well and keep. | Take 52.3mL of 50% stock solution and make up the volume to 1000mL with filtered and vacuum degassed WFI. Shake well and keep. | Take 52.3mL of 50% stock solution and make up the volume to 1000mL with filtered and vacuum degassed WFI. Shake well and keep.

|| | 4.3.6 | Preparation of 70mM NaOH:

Take 7mL of 1M NaOH (as prepared in section 4.3.5.2) and make up the volume to 100mL using filtered and degassed purified water. | Preparation of 70mM NaOH:

Take 7mL of 1M NaOH (as prepared in section 4.3.5.2) and make up the volume to 100mL using filtered and degassed purified water. | Preparation of 70mM NaOH:

Take 7mL of 1M NaOH (as prepared in section 4.3.5.2) and make up the volume to 100mL using filtered and degassed purified water. | Preparation of 70mM NaOH:

Take 7mL of 1M NaOH (as prepared in section 4.3.5.2) and make up the volume to 100mL using filtered and degassed purified water. | Preparation of 70mM NaOH:

Take 7mL of 1M NaOH (as prepared in section 4.3.5.2) and make up the volume to 100mL using filtered and degassed purified water. | Preparation of 70mM NaOH:

Take 7mL of 1M NaOH (as prepared in section 4.3.5.2) and make up the volume to 100mL using filtered and degassed purified water. | Preparation of 70mM NaOH:

Take 7mL of 1M NaOH (as prepared in section 4.3.5.2) and make up the volume to 100mL using filtered and degassed purified water.

| 4.4 | Sample Preparation: | Sample Preparation: | Sample Preparation: | Sample Preparation: | Sample Preparation: | Sample Preparation: | Sample Preparation: | Sample Preparation:

|| | 4.4.1 | Preparation of blank sample:

Take 1mL of filtered purified water in a glass test tube and treat this sample for acid hydrolysis as described in section 4.4.5 | Preparation of blank sample:

Take 1mL of filtered purified water in a glass test tube and treat this sample for acid hydrolysis as described in section 4.4.5 | Preparation of blank sample:

Take 1mL of filtered purified water in a glass test tube and treat this sample for acid hydrolysis as described in section 4.4.5 | Preparation of blank sample:

Take 1mL of filtered purified water in a glass test tube and treat this sample for acid hydrolysis as described in section 4.4.5 | Preparation of blank sample:

Take 1mL of filtered purified water in a glass test tube and treat this sample for acid hydrolysis as described in section 4.4.5 | Preparation of blank sample:

Take 1mL of filtered purified water in a glass test tube and treat this sample for acid hydrolysis as described in section 4.4.5 | Preparation of blank sample:

Take 1mL of filtered purified water in a glass test tube and treat this sample for acid hydrolysis as described in section 4.4.5

|| | 4.4.2 | Preparation of sample for System Suitability Test: | Preparation of sample for System Suitability Test: | Preparation of sample for System Suitability Test: | Preparation of sample for System Suitability Test: | Preparation of sample for System Suitability Test: | Preparation of sample for System Suitability Test:

|| | 4.4.2.1 | Prepare external ribitol Standard 3 in duplicates as mentioned in Table1 in section 4.3.2 and as described in section 4.4.5. | Prepare external ribitol Standard 3 in duplicates as mentioned in Table1 in section 4.3.2 and as described in section 4.4.5. | Prepare external ribitol Standard 3 in duplicates as mentioned in Table1 in section 4.3.2 and as described in section 4.4.5. | Prepare external ribitol Standard 3 in duplicates as mentioned in Table1 in section 4.3.2 and as described in section 4.4.5. | Prepare external ribitol Standard 3 in duplicates as mentioned in Table1 in section 4.3.2 and as described in section 4.4.5. | Prepare external ribitol Standard 3 in duplicates as mentioned in Table1 in section 4.3.2 and as described in section 4.4.5.

|| | 4.4.2.2 | Analyze this sample for RSD for the area of the peak, tailing factor (Asymmetry) and theoretical plates as discussed in section 4.5.5. | Analyze this sample for RSD for the area of the peak, tailing factor (Asymmetry) and theoretical plates as discussed in section 4.5.5. | Analyze this sample for RSD for the area of the peak, tailing factor (Asymmetry) and theoretical plates as discussed in section 4.5.5. | Analyze this sample for RSD for the area of the peak, tailing factor (Asymmetry) and theoretical plates as discussed in section 4.5.5. | Analyze this sample for RSD for the area of the peak, tailing factor (Asymmetry) and theoretical plates as discussed in section 4.5.5. | Analyze this sample for RSD for the area of the peak, tailing factor (Asymmetry) and theoretical plates as discussed in section 4.5.5.

|| | 4.4.3 | Preparation of sample for total saccharide determination: | Preparation of sample for total saccharide determination: | Preparation of sample for total saccharide determination: | Preparation of sample for total saccharide determination: | Preparation of sample for total saccharide determination:

|| | 4.4.3.1 | Dilute HIBTT sample in filtered purified water to a final concentration of 20g/mL and take 1mL of the diluted sample in duplicates as T1 & T2.

Note: The dilution factor is entered in the sequence run for analysis of samples. | Dilute HIBTT sample in filtered purified water to a final concentration of 20g/mL and take 1mL of the diluted sample in duplicates as T1 & T2.

Note: The dilution factor is entered in the sequence run for analysis of samples. | Dilute HIBTT sample in filtered purified water to a final concentration of 20g/mL and take 1mL of the diluted sample in duplicates as T1 & T2.

Note: The dilution factor is entered in the sequence run for analysis of samples. | Dilute HIBTT sample in filtered purified water to a final concentration of 20g/mL and take 1mL of the diluted sample in duplicates as T1 & T2.

Note: The dilution factor is entered in the sequence run for analysis of samples. | Dilute HIBTT sample in filtered purified water to a final concentration of 20g/mL and take 1mL of the diluted sample in duplicates as T1 & T2.

Note: The dilution factor is entered in the sequence run for analysis of samples. | Dilute HIBTT sample

| | 4.4.5.8 | Analyze each sample with the method described below. | Analyze each sample with the method described below. | Analyze each sample with the method described below. | Analyze each sample with the method described below. | Analyze each sample with the method described below.

sequence. | Create the program, create a sequence and load the samples in the auto sampler as per the sequence. | Create the program, create a sequence and load the samples in the auto sampler as per the sequence. | Create the program, create a sequence and load the samples in the auto sampler as per the sequence.

|| | S No | Sample Code | Sample Code | Sample details | No. of injections/vial | No. of injections/vial
| No. of injections/vial

|| | 1 | EB | EB | Eluent Blank | 2 | 2 | 2

|| | 2 | B | B | Sample blank | 1 | 1 | 1

|| | 3 | Std 1

Std 2

Std 3

Std 4 | Std 1

Std 2

Std 3

Std 4 | Calibration Standards | 1 for each standard | 1 for each standard | 1 for each standard

|| | 4 | B | B | Sample blank | 1 | 1 | 1

|| | 5 | SST | SST | System Suitability Test | 5 | 5 | 5

|| | 6 | B | B | Sample blank | 1 | 1 | 1

|| | 7 | T1 & T2 | T1 & T2 | Total Saccharide | 1 for each T1 & T2 | 1 for each T1 & T2 | 1 for each T1 & T2

|| | 8 | F1 & F2 | F1 & F2 | Free Saccharide | 1 for each F1 & F2 | 1 for each F1 & F2 | 1 for each F1 & F2

|| | 9 | B | B | Sample blank | 1 | 1 | 1

|| | Table 2 | Table 2 | Table 2 | Table 2 | Table 2 | Table 2 | Table 2

|| | Note: One extra vial with same SST sample can be used if 5 injections are not possible from single vial. | Note: One extra vial with same SST sample can be used if 5 injections are not possible from single vial. | Note: One extra vial with same SST sample can be used if 5 injections are not possible from single vial. | Note: One extra vial with same SST sample can be used if 5 injections are not possible from single vial. | Note: One extra vial with same SST sample can be used if 5 injections are not possible from single vial. | Note: One extra vial with same SST sample can be used if 5 injections are not possible from single vial. | Note: One extra vial with same SST sample can be used if 5 injections are not possible from single vial.

|| | 4.5.2.3 | Start the batch after 'Ready check'. | Start the batch after 'Ready check'. | Start the batch after 'Ready check'. | Start the batch after 'Ready check'. | Start the batch after 'Ready check'.

|| | 4.5.2.4 | Inject eluent blank after every twenty injections and unknown blank after each set of samples. | Inject eluent blank after every twenty injections and unknown blank after each set of samples. | Inject eluent blank after every twenty injections and unknown blank after each set of samples. | Inject eluent blank after every twenty injections and unknown blank after each set of samples. | Inject eluent blank after every twenty injections and unknown blank after each set of samples.

|| | 4.5.2.5 | Record the calculated concentration in the analysis report. | Record the calculated concentration in the analysis report. | Record the calculated concentration in the analysis report. | Record the calculated concentration in the analysis report. | Record the calculated concentration in the analysis report.

|| | 4.5.3 | Storage and regeneration: | Storage and regeneration: | Storage and regeneration: | Storage and regeneration: | Storage and regeneration:

|| | 4.5.3.1 | At the end of each sequence, clean the system with filtered WFI without passing it through the guard and analytical column. | At the end of each sequence, clean the system with filtered WFI without passing it through the guard and analytical column. | At the end of each sequence, clean the system with filtered WFI without passing it through the guard and analytical column. | At the end of each sequence, clean the system with filtered WFI without passing it through the guard and analytical column. | At the end of each sequence, clean the system with filtered WFI without passing it through the guard and analytical column.

|| | 4.5.3.2 | Run the system with 1M NaOH of final concentration of 580mM after every two days, if the test is not to be performed. | Run the system with 1M NaOH of final concentration of 580mM after every two days, if the test is not to be performed. | Run the system with 1M NaOH of final concentration of 580mM after every two days, if the test is not to be performed. | Run the system with 1M NaOH of final concentration of 580mM after every two days, if the test is not to be performed. | Run the system with 1M NaOH of final concentration of 580mM after every two days, if the test is not to be performed.

|| | 4.5.4 | Calculations for Total Saccharides and Free saccharides: | Calculations for Total Saccharides and Free saccharides: | Calculations for Total Saccharides and Free saccharides: | Calculations for Total Saccharides and Free saccharides: | Calculations for Total Saccharides and Free saccharides:

|| | 4.5.4.1 | Total saccharide($\mu\text{g/mL}$) = $\{(T1+T2)/2\} \times 2.42$ | Total saccharide($\mu\text{g/mL}$) = $\{(T1+T2)/2\} \times 2.42$ | Total saccharide($\mu\text{g/mL}$) = $\{(T1+T2)/2\} \times 2.42$ | Total saccharide($\mu\text{g/mL}$) = $\{(T1+T2)/2\} \times 2.42$ | Total saccharide($\mu\text{g/mL}$) = $\{(T1+T2)/2\} \times 2.42$

|| | 4.5.4.2 | Free saccharide($\mu\text{g/mL}$) = $\{(F1+F2)/2\} \times 2.42$ | Free saccharide($\mu\text{g/mL}$) = $\{(F1+F2)/2\} \times 2.42$ | Free saccharide($\mu\text{g/mL}$) = $\{(F1+F2)/2\} \times 2.42$ | Free saccharide($\mu\text{g/mL}$) = $\{(F1+F2)/2\} \times 2.42$ | Free saccharide($\mu\text{g/mL}$) = $\{(F1+F2)/2\} \times 2.42$

|| | 4.5.4.3 | Percentage Free saccharide = Free saccharide (g/mL) / Total saccharide (g/mL) X 100 | Percentage Free saccharide = Free saccharide (g/mL) / Total saccharide (g/mL) X 100 | Percentage Free saccharide = Free saccharide (g/mL) / Total saccharide (g/mL) X 100 | Percentage Free saccharide = Free saccharide (g/mL) / Total saccharide (g/mL) X 100 | Percentage Free saccharide = Free saccharide (g/mL) / Total saccharide (g/mL) X 100

|| | 4.5.5 | Validity of Assay: | Validity of Assay: | Validity of Assay: | Validity of Assay: | Validity of Assay:

|| | 4.5.5.1 | The correlation coefficient of the standard curve must be more than or equal to 0.990. | The correlation coefficient of the standard curve must be more than or equal to 0.990. | The correlation coefficient of the standard curve must be more than or equal to 0.990. | The correlation coefficient of the standard curve must be more than or equal to 0.990. | The correlation coefficient of the standard curve must be more than or equal to 0.990.

|| | 4.5.5.2 | RSD of five areas should be $\leq 5.0\%$ | RSD of five areas should be $\leq 5.0\%$ | RSD of five areas should be $\leq 5.0\%$ | RSD of five areas should be $\leq 5.0\%$ | RSD of five areas should be $\leq 5.0\%$

|| | 4.5.5.3 | Theoretical plates should be ≥ 3000 | Theoretical plates should be ≥ 3000 | Theoretical plates should be ≥ 3000 | Theoretical plates should be ≥ 3000 | Theoretical plates should be ≥ 3000

|| | 4.5.5.4 | Tailing factor (asymmetry) should be ≤ 1.00 | Tailing factor (asymmetry) should be ≤ 1.00 | Tailing factor (asymmetry) should be ≤ 1.00 | Tailing factor (asymmetry) should be ≤ 1.00 | Tailing factor (asymmetry) should be ≤ 1.00

|| | 4.5.6 | Acceptance Criteria: | Acceptance Criteria: | Acceptance Criteria: | Acceptance Criteria: | Acceptance Criteria:

|| | 4.5.6.1 | % Free Saccharide: < 20% of Total Saccharide | % Free Saccharide: < 20% of Total Saccharide | % Free Saccharide: < 20% of Total Saccharide | % Free Saccharide: < 20% of Total Saccharide | % Free Saccharide: < 20% of Total Saccharide

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[1.0]Objective: |
	To lay down the procedure for determination of cell wall polysaccharide
	content in purified capsular polysaccharide of Neisseria meningitidis
	samples by ELISA technique.
[2.0]Scope:	
	This Standard Operating Procedure is applicable for determination of
	cell wall polysaccharide content in purified capsular polysaccharide of
	Neisseria meningitidis.
[3.0]Responsibility:	
	3.1
	executing the described procedure correctly.
	3.2
	that the procedure is carried out correctly and for checking the
	results.
	3.3
	compliance of the SOP.
[4.0]Abbreviations and Definitions:	
[4.1]	
QA	
: Quality Assurance	
.2	
QC	
: Quality Control	
[4.3]	
SOP	
: Standard Operating Procedure	
[4.4]	
PBS	
: Phosphate Buffer saline	
[4.5]	
CWPS	
: Cell Wall polysaccharide	
[4.6]	
ELISA	
: Enzyme linked Immuno sorbent Assay	
[4.7]	
HRP	
: Horse Raddish Peroxidase	
[4.8]	
TMB	
: 3,3',5,5' Tetra Methyl Benzidine	
[4.9]	
BSA	
: Bovine Serum Albumin	
[5.0]Procedure	
	5.1
	Wear hand gloves while performing the assay.
	5.2
	5.2.1
	Institute, Cat No. 3653)
	5.2.2
	5.2.3
	Serum Institute, Cat No. 87214)
	5.2.4
	Dako Cat No. P0447)
	5.2.5
	5.2.6
	5.2.7
	5.2.8
	5.2.9
	5.2.10
	5.2.11
	5.2.12
	5.2.13
	5.3
	5.3.1
	5.3.2
	5.3.3
	5.3.4
	5.4
	5.4.1
	standard stock solution add 10 ml of WFI in the 10 mg vial

			Pneumococcal Cell wall Polysaccharide Standard, keep the
			vial overnight at 2 to 8°C overnight with shaking. Prepare
			the aliquots as 1000 µl of 1 mg/ml, 100 µl of 1 mg/ml, and
			100 µl of 10 µg /ml. Store the standard aliquots at -70°C.
		5.4.2	Phosphate Buffered Saline (10X): Weigh 85.0 gm of NaCl, 22.8
			gm of Na2HPO4 and 63.6 gm of NaH2PO4 and dissolve in 800 ml
			of purified water. After complete dissolution of reagents,
			make up the total volume to 1000ml with purified water.
			Filter sterilizes the solution by passing through 0.22 µm
			filter.
		5.4.3	Phosphate Buffered Saline (1X): Add 100ml of Phosphate
			Buffered Saline (10X) to 900 ml sterile purified water.
		5.4.4	Dilution buffer (PBST): Add 50 µl (0.05% v/v) of Tween 20 to
			100 ml of PBS 1X solution.
		5.4.5	Blocking buffer (PBST with 1% BSA): Add 1 gm (1% w/v) of
			Bovine Serum Albumin (BSA) to 100 gm PBST solution. Do not
			agitate the solution for mixing.
		5.4.6	Carbonate / Bicarbonate buffer: Prepare 0.5 M stock solution
			of Na2CO3 and NaHCO3 by weighing 5.3 gm/100ml and 4.2
			gm/100ml respectively. Prepare the working stock of
			Carbonate / Bicarbonate buffer by adding 120 µl Na2CO3
			solution and 880 µl of NaHCO3 solution in 9 ml of purified
			water. pH of this buffer should be 9.6.
		5.4.7	Secondary antibody dilution: Calculate the total volume of
			working dilution of secondary antibody required for the
			assay. Prepare the working dilution of Polyclonal Goat
			anti-mouse Immunoglobulin HRP conjugate in PBST as 1:2000
			folds.
		5.4.8	Stop Solution: To prepare stop solution (1N H2SO4), add 2.8
			ml of H2SO4 to 97.2 ml of purified water.
		5.5	Test Method:
		5.5.1	Coating of wells:
		5.5.1.1	The ELISA plates are coated with anti CWPS
			polyclonal antibodies. Determine the number of wells
			to be coated. Prepare a 1: 32000 fold dilution of
			anti-CWPS polyclonal antibody dilution in carbonate
			/ bicarbonate coating buffer.
		5.5.1.2	Add 100 µl/ well of polyclonal antisera solution to
			each well. Incubate the plates at 2 to 8°C over
			night.
		5.5.2	Blocking of wells:
		5.5.2.1	On the next day, remove the coated plate from at 2
			to 8°C; discard the coating solution by flicking.
			Add 150 µl/ well of blocking buffer to each coated
			well.
		5.5.2.2	Incubate the plate at 37 ± 2°C for 30 minutes.
		5.5.3	Preparation of Polysaccharide Standard and Test sample
			dilutions:
		5.5.3.1	During the incubation for blocking of wells, prepare
			the dilutions for polysaccharide standards.
		5.5.3.2	Prepare 3 fold dilutions of polysaccharide standard
			in dilution buffer to get a concentrations of 1000
			ng/ml, 333.33 ng/ml, 111.11 ng/ml, 37.04 ng/ml,
			12.35 ng/ml, 4.12 ng/ml, 1.04 ng/ml and 0.46 ng/ml.
		5.5.3.3	Prepare 1: 10, 1: 100 and 1: 1000 fold dilution of
			test sample in dilution buffer.
		5.5.4	Washing of wells:
		5.5.4.1	After completion of 30 min. incubation of the
			blocked plate at 37°C, remove the plate from the
			incubator.
		5.5.4.2	Discard the blocking buffer from the wells by
			flicking. Wash the wells by adding 250 µl/ well of
			PBS 1X. After 30 sec. discard the PBS from wells by
			flicking and then gently tap the plate on to a
			blotting paper. Ensure that the washing solution is
			removed completely, do not allow the wells to dry.
			Repeat the procedure for three times.
		5.5.5	Addition of Standard, sample and controls to wells:
		5.5.5.1	Add 100 µl of all the eight dilutions of standard in
			column A and B of the plate in duplicate.
		5.5.5.2	Add 100 µl of each sample, in duplicate that is to
			be analysed in consecutive wells of column C.
		5.5.5.3	Add 100 µl of dilution buffer in duplicate; this
			will act as a negative control.
		5.5.5.4	After addition of all standards, samples and
			controls, seal the plate with plate sealer and
			incubate it in 37 ± 2°C for next 90 min.
		5.5.6	Addition of Primary antibody:
		5.5.6.1	Prepare 1: 1200 dilution of Monoclonal antibody for
			Phosphoryl Choline in dilution buffer as per the
			volume required.
		5.5.6.2	After completion of 90 min. of incubation period,
			remove the plate from incubator.
		5.5.6.3	Wash the wells for four times by PBS 1X as mentioned
			in the step no. 5.5.4
		5.5.6.4	Add 100 µl of 1: 1200 fold diluted Monoclonal
			antibody solution for Phosphoryl Choline to each
			well.
		5.5.6.5	After addition, seal the plate with plate sealer and
			incubate it in 37 ± 2°C for next 90 min.
		5.5.7	Addition of secondary antibody:
		5.5.7.1	Prepare 1:2000 dilution of Polyclonal Goat anti

			mouse Immunoglobulin / HRP conjugate solution in	
			dilution buffer as per the volume required.	
			5.5.7.2	After completion of 90 min. of incubation period,
			remove the plate from incubator.	
			5.5.7.3	Wash the wells for four times by PBS 1X as mentioned
			in the step no. 5.5.4.	
			5.5.7.4	Add 100 µl of 1: 2000 fold diluted secondary
			antibody solution to each well.	
			5.5.7.5	After addition, seal the plate with plate sealer and
			incubate it in 37 ± 2°C for next 60 min.	
			5.5.8	Addition of TMB Substrate:
			5.5.8.1	After completion of 60 min. of incubation period,
			remove the plate from incubator.	
			5.5.8.2	Wash the wells for four times by PBS 1X as mentioned
			in the step no. 5.5.4.	
			5.5.8.3	Add 100 µl of TMB Substrate, in dark to each well.
			5.5.8.4	After addition incubate the plate at room
			temperature for next 15 min. in dark.	
			5.5.9	Addition of Stop solution:
			After completion of 15 min. of incubation period, add 100 µl	
			of stop solution (1N H2SO4) to each well above the TMB	
			substrate to stop the reaction.	
			5.5.10	Reading of Plate:
			5.5.10.	After addition of stop solution, read the Optical
			1	density (OD) values of the wells on a Micro plate
			ELISA reader.	
			5.5.10.	Read the plate at a dual wavelength of 450 and 630
			2	nm. The readings should be taken after 5 mins but
			within 15 min from addition of stop solution.	
			5.5.11	Data analysis:
			5.5.11.1	Import the raw data from the software of ELISA
			reader into the window Excel file and get the	
			average OD values of the duplicate wells and	
			compile the data sheets comprising of standard	
			curve concentration and sample details along with	
			the average OD as per the plate design. For	
			standard curve, take concentration on X-axis and	
			its corresponding OD on Y-axis.	
			5.5.11.2	Prepare Standard curve using Standard CWPS ranging
			from 0.46 ng/ml to 1000 ng/ml concentration. The	
			Coefficient of correlation (R2) which should be ≥	
			0.98.	
			5.5.11.3	Once compiled data sheet is ready, use sigma plot
			software for finding concentration of CWPS in the	
			samples by extrapolating the standard curve values	
			i.e. concentration and its corresponding OD.	
			5.5.11.4	Open Sigma plot software, click on create new blank
			notebook and press ok.	
			5.5.11.5	Copy standard curve from compiled excel sheet onto
			the sigma plot note book and save the note book.	
			5.5.11.6	Click 'Pharmacology' and select 'standard curve
			analysis'. Window showing standard curve will	
			arrive. For curve, select 'Four parametric' options	
			and click for 'log X axis scale' and press OK.	
			Software will automatically process the data in the	
			form of three sheets below the notebook. Three	
			sheets will be named as	
			Data 2. Graph	
			3. Report	
			5.5.11.7	R2 for the standard curve will be determined from
			the report sheet.	
			5.5.11.8	Standard graph can be obtained from Graph sheet,
			where heading, X-axis and Y-axis information could	
			be incorporated by right clicking and opting 'Edit'	
			option.	
			5.5.11.9	Data sheet will give all the extrapolated values of
			concentration (X-axis) on column-6 and OD (Y-axis)	
			on column-7. Match OD of samples from excel	
			compiled sheets from column-7 of sigma plot and its	
			corresponding concentration from column-6.	
			Note: In case, the software gives two different	
			concentrations of CWPS for similar OD values of the	
			sample, higher dilutions should be considered for	
			this assay.	
			5.5.11.1	Similarly note the concentrations for all the
			0	samples and determine cell wall polysaccharide
			concentration by multiplying the dilution factor	
			with the concentration from the sigma plot.	
			5.5.12	Validity criteria:
			5.5.12.	Regression of standard curve (R2) should be ≥ 0.98.
			1	
			5.5.12.	Coefficient of Variation (%CV) for replicates should
			2	be < 15%.

1.0 |Objective |
 |To lay down the procedure for estimation of protein content in |
 |Meningococcal polysaccharide conjugate vaccine samples by Lowry's method. |
 2.0 |Scope |
 |This Standard Operating Procedure is applicable to all the samples of |
 |Meningococcal polysaccharide conjugate vaccine in which protein |
 |concentration is to be determined by Lowry's method. |
 3.0 |Responsibility |
3.1	The analyst is responsible for executing the described procedure
	correctly.
3.2	Supervisor is responsible for verifying that the procedure is carried
out correctly and check the results.	
3.3	HOD shall be responsible for overall compliance of this SOP and shall
ensure that the SOP is reviewed every two years or if required	
earlier.	
4.0	Abbreviations and Definitions
4.1	Abbreviations:
	4.1.1
	4.1.2
	4.1.3
	4.1.4
	4.1.5
	4.1.6
	4.1.7
	4.1.8
	4.1.9
	4.1.10
4.2	Reagents:
	4.2.1
	4.2.2
	4.2.3
	4.2.4
	4.2.5
	4.2.6
	4.2.7
4.3	Equipments and Material
	4.3.1
	4.3.2
	4.3.3
	4.3.4
	5000 µl)r
	4.3.5
	4.3.6
	4.3.7
5.0	Procedure
5.1	Preparation of solutions: Prepare all the solutions as described below
and fill the form, label the solutions and assign lot numbers.	
	5.1.1
	5.1.2
	5.1.3
	5.1.4
	5.1.5
	5.1.6
	5.1.7
	5.1.8
5.2	Preparation of Calibration Curve:
	5.2.1

			20	50	450
			40	100	400
			80	200	300
			160	400	100
	5.3	Test Procedure:			
	5.3.1	Take 500 µl of the standard dilutions from calibration curve,			
			Test Sample and Blank (0.1M sodium hydroxide) in duplicates in		
			separate test tubes. In case of dry polysaccharide prepare a		
			solution containing about 1-5 mg/ml of polysaccharide in Lal		
			reagent water and use it as test sample.		
			Note: The sample dilutions can vary depending upon the protein		
			concentration. If there is any sample outside the range of the		
			calibration curve, carry out the calculation for more		
			appropriate dilution.		
	5.3.2	Add. 2.5 ml of Cupric-tartaric solution to each of the standard			
			blank (0.1M Sodium hydroxide) and sample.		
	5.3.3	Vortex and incubate the test tubes in dark at room temperature			
			for 10 min.		
	5.3.4	Add 250 µl of 1 N Folin & Ciocalteu's phenol reagent and mix			
			well.		
	5.3.5	Incubate the test tube at room temperature for 30 minutes in			
			dark.		
	5.3.6	Observe for blue color which can vary with the concentration.			
	5.3.7	Take absorbance of the sample at 730nm against the blank using			
			spectrophotometer.		
	5.3.8	Note that all the readings of the assay tubes should be taken			
			within 30 mins. of completion of the assay and maximum 10		
			samples should be analysed in one single assay.		
	5.4	Calculations:			
	5.4.1	Once the absorbance of the standards and the test samples have			
			been taken, calculation shall be done using the excel sheet		
			carrying formulas for the required calculation.		
	5.4.2	Always copy the master excel sheet (sheet carrying formulas for			
			the required calculation), never use the previously used excel		
			sheet for new calculations.		
	5.4.3	Proceed for calculations starting from the Calibration curve.			
			The calculations are done according to the equation,		
			Y = mX + b.		
			Where: m = slope		
			b = intercept		
			Y = absorbance		
			X = concentration		
			(Multiplication by dilution factor)		
	5.4.4	Use EXCEL programme for the preparation of calibration curve			
			and Calculation of the results (Multiply by the dilution		
			factor. Divide with 1000 to change µg/ml to mg/ml).		
	5.4.5	In case of dry polysaccharide,			
			Protein (%) in polysaccharide on dry weight basis:		
			Protein in dry polysacc. (µg/ml) =		
			Protein in Polysacc. sol. (µg/ml) X 1 – {Moisture (%) / 100}		
			Protein in dry polysaccharide (µg/ml)		
			Protein (%) = _____		
			x 100		
			Concentration of polysaccharide solution (µg/ml)		
	5.5	Acceptance criteria:			
	5.5.1	The test is valid if Regression of the standard curve is ≥0.99			
			and the percentage of Coefficient of Variation is ≤ 15% within		
			the replicates.		
	5.5.2	Protein content in Purified Meningococcal polysaccharide and			
			conjugate should meet the laid specifications for each		
			serotype.		
	6.0	Forms and Records			
	Determination of Total Protein content - Annexure 1				
	7.0	Reference			
	7.1	Lowry, O.H., J. Rosebrough, A.L.Farr and R.J. Randall, R.J. (1951) J.			
			Biol.Chem. 193, 265 - 275.		
	7.2	European Pharmacopoeia, Volume 5, Protein in Polysaccharide Vaccines			
			2.5.16.		
	7.3	WHO TRS No. 977 Annexure 3: Recommendations to assure the quality,			
			safety and efficacy of pneumococcal conjugate vaccines.		

1.0 |Purpose: |
|To lay down the procedure for quantifying the protein concentration by |
|Lowry's method. |
2.0 |Applicable to: |
|This Standard Operating Procedure is applicable to all the samples in |
|which protein concentration is to be determined. |
3.0 |Responsibility: |
|The analyst is responsible for executing the described procedure |
|correctly. Supervisor is responsible for verifying that the procedure is |
|complied with and the results are checked. HOD shall be responsible for |
|overall compliance of this SOP. Departmental head shall ensure that the |
|SOP is reviewed every two years or if required earlier. |
4.0 |Abbreviations: |
4.1	CV	:Coefficient of variation
4.2	g	:Gram
4.3	mg	:Milligram
4.4	ml	:Millilitre
4.5	µl	:Micro litre
4.6	µg	:Microgram
4.7	nm	:Nanometer
4.8	QA	:Quality Assurance
4.9	QC	:Quality Control
4.10	QCR	:Quality Control Record
4.11	SD	:Standard Deviation
4.12	SOP	:Standard Operating Procedure
4.13	UV	:Ultra Violet
4.14	HOD	:Head of Department
5.0	Equipments:	
5.1	UV-Visible Spectrophotometer	
5.2	Analytical balance Sensitivity 0.01mg	
5.3	Fume hood	
5.4	Water bath (100 °C)	
5.5	Vortex Mixer	
5.6	Refrigerator	
6. |Materials and reagents: |
|6.1 |Materials: | |
| |6.1.1 | Autopipettes |
| |6.1.2 |5, 10, 25, 50 and 100 ml Volumetric Flasks |
| |6.1.3 |5 and 10 ml thermoresistant test tubes. |
| |6.1.4 |25, 50 and 100 ml Beaker |
| |6.1.5 |Quartz cuvettes (1 cm width) |
| |6.1.6 |Tissue paper |
| |6.1.7 |Autopipette tips |
|6.2 |Reagents: |
| |6.2.1 |Copper Sulphate Pentahydrate |
| |6.2.2 |Sodium potassium tartrate tetrahydrate |
| |6.2.3 |Folin-Ciocalteu reagent |
| |6.2.4 |Sodium Carbonate |
| |6.2.5 |Bovine serum albumin Fraction V |
7. |Preparation of solutions: |
|7.1 |Alkaline Copper Sulphate solution: | |
| |7.1.1 |Weigh 40 g of anhydrous sodium carbonate and dissolve it |
| | |in 400 ml of purified water. |
| |7.1.2 |Weigh 0.6 g of sodium and potassium tartrate and dissolve |
| | |in 100 ml of purified water and add 0.46 g of copper |
| | |sulphate pentahydrate to it, shake until it dissolves |
| | |completely. Mix the above two solutions gently. |
| |7.1.3 |Transfer it to a 1000 ml volumetric flask and make up the |
| | |volume to 1000 ml with purified water. |
| |7.1.4 |This solution should be stored in an amber coloured |
| | |bottle at 2 - 8°C and should be used within 2 months. |
|7.2 |1 N Folin-Ciocalteu Solution: |
| |7.2.1 |Dilute 2N Folin-Ciocalteu reagent to 1N. For this, take |
| | |250 ml of the 2N Folin-Ciocalteu reagent and pour it in |
| | |a 500 ml amber flask and make up the volume to 500 ml |
| | |with purified water. |
| |7.2.2 |Store the solution in a screw cap amber bottle at 2 to 8 |
| | |°C, for a maximum of 6 months. |
|7.3 |1 mg/ml Bovine serum albumin Solution: |
| |7.3.1 |Weigh 50 mg of Bovine Serum Albumin and transfer it to a |
| | |50 ml volumetric flask. |
| |7.3.2 |Dissolve and make up the volume to 50 ml with Purified |
| | |water. |
| |7.3.3 |Make aliquots of 1.5ml and maintain at –20°C, for a |
| | |maximum of 6 months. |
8. |Procedure: |
|8.1. |Preparation of Calibration Curve: | |
| |The concentration of Protein in the test sample(s) is calculated |
| |by comparing results of its absorbance obtained against the |
| |plotted calibration curve of known Protein concentration (10-100 |
| |µg/ml) Vs absorbance at 730 nm. The solutions for making Protein |
| |calibration curve (10-100 µg/ml) are prepared as follows: |
| |8.1.1 |Thaw the 1.5 ml vial of BSA by keeping at room |
| | |temperature. |
| |8.1.2 |Now dilute the above material 10 times. (This is done by |
| | |taking 1 ml of BSA and adding to it 9 ml of purified |
| | |water) A final concentration of 100µg/ml is obtained. |
| | |This dilution is used for the preparation of calibration |
| | |curve (10-100 µg/ml). |

			Concentration	10	20	40	60	80	100
			(µg/ml)						
			BSA100µg/ml (µl)	100	200	400	600	800	1000
			Volume of purified	900	800	600	400	200	-
			water added (µl)						
			Note: The Calibration curve is accepted if it is meeting						
			the acceptance criteria of correlation coefficient.						
		8.2	Test:						
		8.2.1	Dilute the sample to 1/10 dilution. (For preparing 1/10						
		dilution of sample take 100 µl of sample and add 900µl							
		purified water)							
		In case of dry polysaccharide prepare a solution							
		containing about 1-5 mg/ml of polysaccharide in purified							
		water and use it as test sample.							
		Note: The dilutions mentioned can vary depending upon the							
		protein concentration. If there is any sample outside the							
		range of the calibration curve, carry out the calculation							
		for more appropriate dilution.							
		8.2.2	Take 400 µl from 10 to 100 µg/ml BSA solutions from						
		calibration curve, Test Sample (already diluted) and							
		Blank (Purified water used as Blank) in duplicates in							
		separate test tubes.							
		8.2.3	To these test tubes add 2 ml of Alkaline copper sulphate						
		solution							
		8.2.4	Vortex the tubes and incubate in water bath at 37°C for						
		30 minutes.							
		8.2.5	Allow the tubes to cool and add 200 µl of 1N						
		Folin-Ciocalteu solution to all tubes.							
		8.2.6	Vortex the tubes and incubate in water bath at 37°C for						
		20 minutes.							
		8.2.7	Allow these tubes to cool at room temperature.						
		8.2.8	Measure the absorbance at 730 nm against the blank.						
		8.3	Calculations:						
		8.3.1	Proceed for calculations starting from the Calibration						
		curve prepared between 10 and 100 g/ml of BSA solution.							
		The calculations are done according to the equation of							
		minimum squares.							
		$Y = mX + b$.							
		Where: m = slope							
		b = intercept							
		Y = absorbance							
		X = concentration							
		(Multiply by dilution factor)							
		8.3.2	Use EXCEL programme for the preparation of calibration						
		curve and Calculation of the results (Multiply by the							
		dilution factor. Divide with 1000 to change µg/ml to							
		mg/ml).							
		8.4	Acceptance criteria:						
		8.4.1	The coefficient of variation (CV) between the replicates						
		of the samples should be $\leq 5\%$.							

1.0 |Objective: |
 |This method is used to determination of Sialic acid content in |
 |meningococcal polysaccharide of serotypes C, W135 and Y. |
 2.0 |Scope: |
 |This SOP is applicable to determination of Sialic acid content in |
 |meningococcal polysaccharide of serotypes C, W135 and Y samples. |
 3.0 |Responsibility: |
3.1	The analyst of the quality control laboratory shall be responsible
	for executing the described procedure correctly.
3.2	The supervisor of the laboratory shall be responsible for
	verifying that the procedure is followed correctly and for
	checking the results.
3.3	The Departmental Head shall be responsible for the overall
	compliance of the SOP and shall ensure that the SOP is reviewed
	every two years or if required earlier.
4.0	Abbreviations and Definitions:
4.1	QA
4.2	QC
4.3	SOP
5.0	Procedure:
5.1	Materials, reagents & Equipments
	5.5.1
	5.5.2
	5.5.3
	5.5.4
	5.5.5
	5.5.6
	5.5.7
	5.5.8
	5.5.9
	5.5.10
	5.5.11
	5.5.12
	5.5.13
	5.5.14
5.2	Preparation of solutions:
	5.2.1
	5.2.2
	5.2.3
	5.2.4
5.4	Assay Procedure:
	Place series of standard solution from 100µg/ml NANA standard
	solution on the day of measurement into 10 ml glass tubes in
	duplicate according to the following table:
	Working standard
	Volume (µl)
	0
	200
	400
	600
	800
	1000
	5.4.1
	5.4.2
	5.4.3
	5.4.4
	5.4.5
	5.4.6
	5.4.7

		5.4.8	Collect the upper phase carefully into other test tubes.
		Take care that the lower phase does not get mixed with the	
		upper phase.	
		5.4.9	Take absorbance of the sample at 580 nm against the blank
		using spectrophotometer.	
		5.4.10	Note that all the readings of the assay tubes should be
		taken within 30 mins. of completion of the assay, the test	
		tubes should be kept in a n ice tray during the readings	
		and maximum 10 samples should be analysed in one single	
		assay.	
		5.5	Calculations
		5.5.1	Once the absorbance of the standards and the test samples
		have been taken, calculation shall be done using the excel	
		sheet carrying formulas for the required calculation.	
		5.5.2	Always copy the master excel sheet (sheet carrying
		formulas for the required calculation), never use the	
		previously used excel sheet for new calculations.	
		5.5.3	Copy the master excel sheet, keep it for use in respective
		folder and then fill in the required sample details (Batch	
		number), sample dilution (if applicable), name of analyst,	
		date, etc.	
		5.5.4	Calculate the sialic acid concentration of the samples
		using following formula: Sialic acid concentration	
		(µg/ml)= {(abs-intercept)/slope}*dilution factor	
		5.5.5	In case of dry polysaccharide, Sialic acid (%) in
		polysaccharide on dry weight basis:	
			Sialic acid Conc. in dry Polysacc. (µg/ml) =
			Sialic acid in Polysacc. sol. (µg/ml) X 1 – {Moisture
			(%) / 100}
			Sialic acid in dry polysaccharide (µg/ml)
			Sialic acid (%) = _____
			X 100
			Conc. of polysaccharide solution (µg/ml)
		5.5.6	To calculate the pure polysaccharide content (mg) present
		in dry polysaccharide sample in of each serotype based on	
		the Sialic acid content obtained, following formula is	
		applied:	
		Pure Polysaccharide content(mg) / Dry polysaccharide	
		content (mg)	
		= (Sialic acid Conc. in dry Polysacc (mg) X Conversion	
		factor)	
		5.5.7	The sialic acid present in each serotype of C, W135 and Y
		is 80%. Therefore the polysaccharide content conversion	
		factor for each serotype shall be 1.25.	
		5.6	Assay specifications
		5.6.1	The linearity of the standard curve is NLT 0.99
		5.6.2	The absorbance value of test samples should fall inside
		the range of standard curve.	
		5.6.3	A coefficient of variation of less than 10% should be
		observed within the replicates.	
		6.0	Reference
		6.1	European pharmacopeia
		6.2	WHO TRS 594: Requirement for Meningococcal polysaccharide vaccine.

1.0 |Purpose: |
 |To lay down the procedure for the quantification of ribose concentration. |
 2.0 |Applicable to: |
 |This Standard Operating Procedure is applicable to all the samples of PRP |
 |and PRP-TT bulk conjugate of Haemophilus Influenzae type b in which ribose |
 |concentration is to be determined. |
 3.0 |Abbreviations |
3.1	Abbreviations:		
	3.1.1	CV	:Coefficient of variation
	3.1.2	g	:Gram
	3.1.3	mg	:Milligram
	3.1.4	mL	:Millilitre
	3.1.5	µl	:Micro litre
	3.1.6	µg	:Microgram
	3.1.7	nm	:Nanometer
	3.1.8	PRP	:Polyribosyl Ribitol Phosphate
	3.1.9	TT	:Tetanus Toxoid
	3.1.10	g/l	:Grams per Litre
	3.1.11	w/v	:Weight by Volume
4.0	Procedure:		
4.1	Preparation of Solutions:		
	Prepare all the solutions as described below and label the solutions.		
	4.1.1	0.5 g/l Ferric chloride solution:	
		4.1.1.1	Weigh 50 mg of Ferric Chloride hexahydrate and
			transfer it to 100 mL volumetric flask.
		4.1.1.2	Dissolve and make up the volume to 100 mL with
			concentrated hydrochloric acid.
		4.1.1.3	Store the solution in a screw cap amber bottle, for a
			maximum of 6 months at room temperature.
	4.1.2	100 g/l Orcinol solution:	
		4.1.2.1	Weigh 1gm of Orcinol monohydrate and transfer it to 10
			mL volumetric flask.
		4.1.2.2	Dissolve and make up the volume to 10 mL with absolute
			ethanol.
		4.1.2.3	Store the solution in a screw cap amber bottle at 2 to
			8 °C, for a maximum of 15 days.
	4.1.3	1 mg/ml Ribose Solution:	
		4.1.3.1	Weigh 25 mg of D-Ribose and transfer it to a 100 mL
			volumetric flask.
		4.1.3.2	Dissolve in purified water and make up the volume to
			100 mL with purified water.
		4.1.3.3	Prepare fresh prior to use.
	4.1.4	25 µg /mL Ribose Solution:	
		4.1.4.1	Take 1 mL of 1 mg/ml of Ribose solution and transfer
			it to a 10 mL volumetric flask.
		4.1.4.2	Make up the volume to 10 mL with purified water.
		4.1.4.3	Prepare fresh prior to use.
	4.1.5	10% w/v Cetyltrimethylammonium bromide (CTAB) Solution:	
		4.1.5.1	Weigh 1gm of Cetyltrimethylammonium bromide and
			transfer it to 10 mL volumetric flask.
		4.1.5.2	Dissolve and make up the volume to 10 mL with Purified
			water.
		4.1.5.3	Solution shall be prepared freshly.
	4.1.6	0.25 M Sodium hydroxide:	
		4.1.6.1	Weigh 1gm of Sodium hydroxide and dissolve in
			sufficient amount of purified water and make up the
			volume to 10 mL with Purified water.
		4.1.6.2	Store the solution in a screw cap bottle, for a
			maximum of 6 months at room temperature.
	4.2	Preparation of Calibration Curve:	
	4.2.1	Prepare calibration curve (2.5-25 µg) with freshly prepared	
		ribose solution (25 µg /ml) in previously labelled test tubes	
		of 20 mL.	
	4.2.2	Calibration Curve (2.5 - 25µg) is prepared in duplicates as	
		follow:	
		Ribose content (µg)	2.5
		25µg/ml ribose (µl)	100
		Purified water (µl)	900
	4.2.3	Make up the volume to 2 mL using purified water.	
		Note: The Calibration curve is accepted if it is meeting the	
		parameters of correlation coefficient ie. ≥ 0.990 .	
	4.3	Test:	
	4.3.1	Take sample volume such that the sample volumes used in test	
		contain 2.5 to 25 µg of ribose i.e. lies within the	
		calibration curve.	
		In case of dry polysaccharide prepare a solution containing	
		about 1-5 mg/mL of polysaccharide use it as test sample)	
	4.3.2	Make up the volume to 2 mL using purified water respectively.	
		Note: In case of fermentation samples, treat 1 mL of the sample with	
		50µl 10 % CTAB, vortex and keep for 10 minutes, centrifuge at 7000	
		RPM for 5 minutes. Discard the supernatant; add 1 mL purified water,	
		vortex and again centrifuge for 5 minutes at 7000 RPM. Discard the	
		supernatant and let the pellet air dry. Solubilise pellet with 1 mL	
		0.25 M NaOH. Treat this as sample for ribose estimation.	
	4.3.3	Treat 2 ml purified water as blank in thermoresistant test	
		tube.	
	4.3.4	Add 2 ml of 0.5 g/l Ferric Chloride solution to tubes of	
		calibration curve, Blank and sample.	
	4.3.5	Vortex and add 0.2 ml of 100 g/l Orcinol Solution to all test	
		tubes.	

	4.3.6	Close the tubes with aluminium foil and place the test tubes
		in boiling water bath for 20 minutes.
	4.3.7	Take out the tubes and cool in ice cold water.
	4.3.8	Allow the tubes to come at room temperature.
	4.3.9	Set the wavelength in spectrophotometer at 670 nm. Zero the
		absorbance of blank and then measure the absorbance of
		calibration curve and test samples.
	4.4	Calculations:
	4.4.1	Once the absorbance of the standards and the test samples
		have been taken, calculation shall be done using the excel
		sheet carrying formulas for the required calculation.
	4.4.2	Enter the absorbance for calculations starting from the
		Calibration curve prepared between 2.5 and 25 g of Ribose and
		test samples. The calculations are made according to the
		equation of minimum squares.
		$Y = mX + b$.
		Where: m = slope
		b = intercept
		Y = absorbance
		X = concentration
		(Multiply by dilution factor)
	4.4.3	Calculate the content of ribose in the test solution for each
		volume tested.
	4.4.4	Conversion factor for PRP:
		PRP content = Ribose content / 0.434
	4.4.5	In case of PRP,
		Ribose (%) in PRP on dry weight basis:
		Ribose content in dry PRP ($\mu\text{g/ml}$) =
		Ribose content in PRP($\mu\text{g/mL}$) X $1 - \{\text{Moisture} (\%) / 100\}$
		Ribose content in dry PRP ($\mu\text{g/mL}$)
		Ribose (%) = $\frac{\text{Ribose content in dry PRP } (\mu\text{g/mL})}{\text{Concentration of PRP solution } (\mu\text{g/mL})} \times 100$
		Concentration of PRP solution ($\mu\text{g/mL}$)
	4.5	Acceptance criteria:
	4.5.1	The coefficient of variation (CV) between the replicates of
		the samples should be $\leq 5\%$.
	4.5.2	Ribose content in the sample should be within the acceptance
		limits.
	5.0	Forms and Records:
		Determination Of Ribose Concentration By Orcinol Method .
	6.0	References:
		European pharmacopoeia.

[1.0 |Objective: |
| |To lay down the procedure for identification of Meningococcal |
| |Polysaccharide Antigens by agglutination method. |
[2.0 |Scope: |
| |This SOP is applicable to Meningococcal polysaccharide, Meningococcal |
| |conjugate bulk and final lot. |
[3.0 |Responsibility: |
	3.1	The analyst of the Immunochemistry laboratory shall be responsible
	for executing the described procedure correctly.	
	3.2	The supervisor of the laboratory shall be responsible for verifying
	that the procedure is followed correctly and for checking the	
	results.	
	3.3	The Departmental Head shall be responsible for the overall
	compliance of the SOP and shall ensure that the SOP is reviewed	
	every two years or if required earlier.	
[4.0	Abbreviations and Definitions:	
	4.1	QA
	4.2	QC
	4.3	SOP
[5.0	Procedure:	
	5.1	Material and Equipments
	5.1.1	PASTOREX MENINGITIS INDIVIDUAL LATEX TEST KIT, code 61608
	5.1.2	Pipettes and tips
	5.1.3	Laboratory vortex mixer
	5.2	Note:
	5.2.1	All reagents and the sample should be used at a temperature
	of between 18°C – 30°C.	
	5.2.2	Do not touch the reaction surface of the agglutination card.
	5.2.3	Shake the bottle of latex before use.
	5.2.4	Wipe the tip of the reagent dropper bottle in order to obtain
	a well calibrated drop	
	5.2.5	Hold the reagent bottle vertical to deposit drop.
	5.3	Principal of the Test
	Antigen contained in the specimen tested is identified using latex	
	particles coated with specific homologous antibodies. In the	
	presence of the homologous antigen, latex particles agglutinate. In	
	the absence of antigen, they remain in a homogenous suspension	
	5.4	Test Method
	5.4.1	Deposit two drop of latex reagent R6 on the black circle of
	the reaction slide.	
	5.4.2	Also deposit two drops of latex reagent R6 on the white
	circle of the reaction slide. This act as a negative control.	
	5.4.3	Added two drops of sample to the black circle and Diluent
	Solution R11 to the white circle.	
	5.4.4	Mix the sample and latex reagent with disposable mixing
	stick.	
	5.4.5	Observe the appearance of agglutination within 1 minute.
	5.5	Interpretation of results:
	5.5.1	Positive reaction is detected by the formation of clumps
	visible to the naked eye.	
	5.5.2	The negative control should be homogenous suspension, without
	clump	
[6.0	REFERENCES:	
	Manual of Pastorex Meningitis individual latex tests kit.	

[SHEET: Sheet1]
[SHEET: Sheet2]
[SHEET: Sheet3]

1.0 |Purpose: |
 |To lay down the procedure for the estimation of molecular size of PRP in |
 |purified polysaccharide of Haemophilus Influenzae type b conjugate vaccine. |
 2.0 |Applicable to: |
 |This Standard Operating Procedure is applicable for evaluating KD |
 |(molecular size distribution) of purified polysaccharide of Haemophilus |
 |Influenzae type b conjugate vaccine. |
 3.0 |Abbreviations and Definitions: |
3.1	cm	: centimetre
3.2	DNA	: Deoxyribonucleic acid
3.3	g	: Gram
3.4	HIBTT	: Haemophilus Influenzae type b - Tetanus toxoid
3.5	HPLC	: High Performance Liquid Chromatography
3.6	id	: Internal diameter
3.7	l	: length
3.8	m	: meter
3.9	mg	: milligram
3.10	mL	: Millilitre
3.11	mm	: milli meter
3.12	nm	: Nano meter
3.13	No.	: number
3.14	PRP	: Polyribosyl Ribitol Phosphate
3.15	RID	: Refractive index detector
3.16	RSD	: Relative Standard Deviation
3.17	SST	: System Suitability Test
3.18	°C	: Degree centigrade
3.19	µg	: microgram
3.20	µL	: Micro litre
3.21	µm	: Micro meter
3.22	System	: System suitability tests are an integral part of gas
Suitability	and liquid chromatographic methods. They are used to	
Test	verify that the resolution and reproducibility of the	
chromatographic system are adequate for the analysis to		
be done.		
The tests are based on the concept that the equipment,		
electronics, analytical operations and samples to be		
analyzed constitute an integral system that can be		
evaluated as such.		
4.0	Procedure:	
4.1	Precautions to be taken:	
4.1.1	Degas all solutions.	
4.1.2	Avoid entry of bubbles into the system as well as column.	
4.2	Requirements	
4.2.1	Equipment:	
4.2.1.1	Calibrated HPLC System with ChemStation software.	
1		
4.2.1.1	Refrigerator	
2		
4.2.1.1	Ultrasonic Cleaner	
3		
4.2.1.1	Vacuum Pump	
4		
4.2.2	Instrument:	
4.2.2.1	pH Meter	
1		
4.2.2.1	Analytical balance	
2		
4.2.2.1	Magnetic stirrer	
3		
4.2.2.1	Precision balance	
4		
4.2.3	Material:	
4.2.3.1	Two TSK gel G5000 PWxl columns {30 cm (l), 7.8 mm	
1	(i.d)} connected in series.	
4.2.3.1	Laboratory Glassware	
2		
4.2.3.1	Screw cap vials	
3		
4.2.3.1	Solvac Filter Holder	
4		
4.2.3.1	Nylon 6, 6 membrane 0.20/0.22/.045 µm membrane filters.	
5		
4.2.3.1	Auto pipette (2-20 (L), (20-200 (L), (100-1000 (L)	
6		
4.2.3.1	Tips for the Auto pipette	
7		
4.2.3.1	Eppendorf tubes	
8		
4.2.3.1	Aluminium foil	
9		
4.2.4	Reagents:	
4.2.4.1	Sodium Chloride (NaCl)	
1		
4.2.4.1	Sodium azide	
2		
4.2.4.1	Salmon DNA	
3		
4.2.4.1	Acetone	
4		
4.2.4.1	Purified water	

||| |5| |

|4.3 |Preparation of solutions: |

| |Prepare all the solutions as described below and label of the |

| |prepared solutions. |

| |4.3.1 |0.4% Sodium Chloride solution: |

| | |For a volume of 2litre, weigh: |

| | |NaCl |8.0 g |

| | |Dissolve in 700 mL of purified water in a beaker. Make up the |

| | |volume to 2 litres with purified water in a volumetric flask. |

| | |Use the mobile phase within four days from the date of |

| | |preparation if it remains clear and no turbidity or fibers |

| | |observed in it. |

| |4.3.2 |Void volume marker (Salmon DNA): |

| | |For a volume of 5 mL, weigh: |

| | |Salmon DNA |10 mg |

| | |Dissolve in 3mL of 0.2µm filtered purified water and make up |

| | |the volume to 5mL in a volumetric flask. Store at -20 °C ± 5 |

| | |°C for 1 month. |

| |4.3.3 |Total volume marker (Acetone): |

| | |Take 10µL of acetone and dissolve in 990µL of 0.2µm filtered |

| | |purified water in an eppendorf. Mix thoroughly. Prepare fresh |

| | |before use. |

| |4.3.4 |Column Preservation Solution: |

| | |Note: All the solutions prepared for use are vacuum filtered |

| | |using Nylon 6, 6 membrane 0.20/0.22/0.45 µm, depending upon |

| | |the nature of solution to be filtered (filtration not |

| | |necessary if the solution is prepared in 0.2µ filtered |

| | |purified water). For organic solvents filtration not |

| | |necessary if the solution is HPLC grade. |

|4.4 |Preparing the HPLC System for run. |

|4.5 |Run Conditions: |

| |4.5.1 |Load method HIBTT RI.M. |

| |4.5.2 |Flow Rate: 0.5 mL/min |

| |4.5.3 |Detector used: RID |

| |4.5.4 |Run time: 60 minutes |

| |4.5.5 |Mobile Phase: 0.4% sodium chloride solution |

|4.6 |Equilibration of the system and the column: |

| |Equilibrate the system and column with 0.4% sodium chloride solution |

| |unless a stable baseline for RID is obtained which can be viewed with |

| |the online plot window. |

|4.7 |Sample Preparation and Application: |

| |4.7.1 |Test sample and standards are loaded in screw cap vials and |

| | |placed against the appropriate no. on the sample tray by |

| | |clearly marking the screw cap vial. |

| |4.7.2 |Take 3 mg/mL solution of Purified polysaccharide and obtain |

| | |the value of PRP concentration. |

| |4.7.3 |Load the required volume of sample as per the PRP |

| | |concentration equivalent to 200µg in the sample vial. |

| | |Example: If a sample has a PRP concentration of 2mg/mL, to |

| | |calculate volume required for 200µg of polysaccharide, |

| | |2mg (2000µg) -----1mL |

| | |200µg-----X mL |

| | |

| | |X mL = 200µg x 1mL |

| | |2000µg |

| | |X mL = 0.1mL |

| | |Therefore, 100µL of volume is injected for sample having PRP |

| | |concentration of 2mg/mL. |

| | |Note: The application of the Calibration standards is carried |

| | |out every time whenever a test sample is applied. If several |

| | |samples are applied in one sequence then only one application |

| | |of the Calibration standards is enough. |

|4.8 |Running of sequence: |

| |4.8.1 |Load sequence PRP RI.S. |

| |4.8.2 |Open the sequence parameter option by clicking Sequence menu. |

| |4.8.3 |Enter information about the data directory and operator name. |

| |4.8.4 |In the shutdown option, click on the post-sequence option and |

| | |select LAMPOFF/STANDBY. |

| |4.8.5 |In sequence table option, specify the sample names, methods |

| | |and vials to run in the sequence as per the following table: |

| | |S |Sample Code |Sample details |No. of |Injection |

| | |No | | |injection |Volume |

| | | | |ns | |

| | |1 |Blank |Eluent Blank |1 |10 µL |

| | |2 |Acetone |System |6 |1 µL |

| | | | |(Total volume |Suitability Test| | |

| | | | |marker) | | |

| | |3 |Blank |Eluent Blank |1 |10 µL |

| | |4 |Salmon DNA |Void volume |1 |50 µL |

| | | | |marker | | |

| | |5 |Blank |Eluent Blank |1 |10 µL |

| | |6 |Test Sample |Sample |1 |200 µg of |

| | | |(s) | |PRP |

| |4.8.6 |Click on run sequence option in run control menu to start the |

| | |sequence. |

| |4.8.7 |Each sample run will be of 60 min and after running all the |

| | |samples the sequence will automatically stop. |

| |4.8.8 |Inject the eluent blank after every 20 injections. |

| | |Note: One extra vial with same SST sample can be used if 6 |

| | |injections are not possible from single vial. |

|4.9 |Calculations: |

| |4.9.1 |The retention time of the unretained compound is determined |

			with Salmon DNA.
		4.9.2	The Retention time of the component which has full access to
		the pores of stationary phase is determined with Acetone.	
		4.9.3	Proceed for calculating KD value.
		The calculations are made as per the following equation:	
		$KD = T_e - T_o$	
		$T_t - T_o$	
		Where,	
		T_e = Retention time of peak maxima of purified polysaccharide	
		sample	
		T_t = Retention time of Acetone	
		T_o = Retention time of Salmon DNA	
		Note: Take average RT of acetone as T_t from EXCEL for	
		calculation of SST.	
		4.9.4	Calculation of percentage of sample eluted before a given KD:
		The percentage of sample eluted before a given KD is	
		calculated using ChemStation software.	
		4.9.4.1	Open the acquired chromatogram in the data analysis
		view in the offline copy of the software.	
		4.9.4.2	Optimize integration with initial events in the
		signal specific events table.	
		4.9.4.3	Enter the value of Elution time (T_e) calculated in
		step 5.9.3 in integration with timed events option	
		or with manual integration and select split peak	
		option.	
		4.9.4.4	The percentage of sample eluted before a given KD
		is the value of percentage of area (Area %)	
		obtained for polysaccharide curve till the	
		retention time of peak maxima.	
		4.9.5	Calculations for System Suitability Test:
		4.9.5.1	Note the values of area under the curve and
		retention time for Acetone (SST standard).	
		4.9.5.2	Every time a sample is run with system suitability
		test, calculate the results for system suitability	
		and record the same.	
		4.9.5.3	Attach the hard print of SST calculation along with
		chromatograms and the record form.	
		4.10	Acceptance Criteria:
		4.10.1	For System Suitability Test:
		4.10.1.1	RSD calculated for RT of Acetone should be <5.0%.
		4.10.1.2	RSD calculated for area of Acetone should be <5.0%.
		4.10.2	For purified polysaccharide samples:
		KD (Molecular Size distribution): $\geq 40\%$, $KD \leq 0.3$	
		Reporting:	
		Molecular sizing of polyribosyl ribitol phosphate (PRP) by high performance	
		size exclusion chromatography.	
		6.0	Reference:
		European pharmacopoeia.	

* PS: Polysaccharide

S.No | Tests | Specifications

1 | Appearance | White powder or lump

2 | Serological Identity | Positive

3 | Methyl pentose content | $\geq 37\%$ of dry weight of PS

4 | Phosphorus content | 3.0-4.5 % of dry weight of PS

5 | Total nitrogen content | 0-1.0% of dry weight of PS

6 | Kav_g determination (SEC-HPLC with RI detector) | $Kav_g \leq 0.3$

7 | Protein content | ≤ 2.0 % of dry weight of PS

8 | Nucleic acid content | ≤ 2.0 % of dry weight of PS

9 | Cell wall polysaccharide content | ≤ 0.5 % of dry weight of PS

10 | Endotoxin content | < 0.5 IU/ μ g of PS

11 | Residual CTAB Content | ≤ 1.0 % of dry weight of PS

12 | Residual Formaldehyde content | ≤ 1.0 % of dry weight of PS

13 | Residual DOC Content | ≤ 1.0 % of dry weight of PS

14 | Moisture content | To report results

1.0	Objective		
	To lay down the procedure for analysis of Meningococcal polysaccharide		
	samples by High Performance Size Exclusion Chromatography.		
2.0	Scope		
	This Standard Operating Procedure is applicable for evaluating Kavg of		
	meningococcal polysaccharide samples at in process stage and lyophilized		
	purified polysaccharide of Meningococcal vaccine.		
3.0	Responsibility		
	3.1	The analyst shall be responsible for carrying out the described	
	procedure correctly.		
	3.2	The Supervisor of the laboratory shall be responsible for verifying	
	that the procedure is complied with and for checking the results.		
	3.3	The departmental head/designee shall be responsible for overall	
	compliance of the SOP.		
	3.4	Head QA/Designee shall be responsible for final approval and	
	subsequent compliance of the SOP.		
4.0	Abbreviations and Definitions		
	4.1	ATCC	: American Type Culture Collection
	4.2	D2O	: Deuterium oxide
	4.3	g	: Gram
	4.4	HOD	: Head of Department
	4.5	HPLC	: High Performance Liquid Chromatography
	4.6	M	: Molarity
	4.7	ml	: Milli litre
	4.8	nm	: Nano meter
	4.9	PBS	: Phosphate Buffer Saline
	4.10	PN	: Part Number
	4.11	QA	: Quality Assurance
	4.12	QC	: Quality Control
	4.13	RID	: Refractive Index Detector
	4.14	SOP	: Standard Operating Procedure
	4.15	Std.	: Standard
	4.16	°C	: Degree centigrade
	4.17	µg	: Microgram
	4.18	µl	: Micro litre
	4.19	µm	: Micro meter
5.0	Procedure		
	5.1	Precautions to be taken:	
		5.1.1	Always wear laboratory coat
		5.1.2	Always use solutions and samples filtered through 0.2 µm
			Membrane
		5.1.3	Degas all solutions
		5.1.4	Avoid entry of bubbles into the system as well as column
		5.1.5	All the solution bottle openings should be well covered with
			aluminum foil
		5.1.6	Check the calibration status of all equipment and instrument
		5.1.7	Handle acids carefully wearing gloves and always manipulate in
			fume hood.
	5.2	Equipment and Instruments	
		5.2.1	Calibrated HPLC system with RI detector
		5.2.2	Analytical balance
		5.2.3	Precision balance
		5.2.4	Vacuum Pump
		5.2.5	pH Meter
		5.2.6	Refrigerator
		5.2.7	Magnetic stirrer
		5.2.8	Ultrasonic Cleaner
		5.2.9	Rocking and Tumbling Shaker
	5.3	Material:	
		5.3.1	Sugar KS-804 (Shodex Column)
		5.3.2	Sugar KS-805 (Shodex Column)
		5.3.3	Guard Column KS-G (Shodex)
		5.3.4	Laboratory Glassware
		5.3.5	Screw cap HPLC vials with/without inserts
		5.3.6	Solvac Filter Holder
		5.3.7	Nylon 6,6 membrane 0.2µm membrane filter
		5.3.8	Auto pipettes (10-100 (l), (20-200 (l), (100-1000 (l)
		5.3.9	Tips for the Auto pipettes
		5.3.10	Syringe filters 0.2 µm
		5.3.11	Aluminum Foil
		5.3.12	Gloves
	5.4	Reagents:	
		5.4.1	Dextran
		5.4.2	Sodium azide
		5.4.3	Deuterium Oxide
		5.4.4	Sodium Chloride
		5.4.5	Purified water
		5.4.6	Sodium dihydrogen phosphate, dihydrate
		5.4.7	Disodium hydrogen phosphate, anhydrous
	5.5	Preparation of solutions: Prepare all the solutions as described	
		below. Carry out the labelling of the prepared solutions and assign	
		the lot numbers.	
		5.5.1	100mM Sodium dihydrogen phosphate, dihydrate, stock Solution:
			Take 15.602g of Sodium dihydrogen phosphate in 1L of purified
			water. Mix well and store the solution at room temperature for
			not more than one month.
		5.5.2	100mM Disodium hydrogen phosphate, anhydrous, stock Solution:
			Take 14.196g of Disodium hydrogen phosphate in 1L of purified
			water. Mix well and store the solution at room temperature for
			not more than one month.

		5.5.3	100mM Phosphate Buffer stock Solution:	
			Add 280 mL of 100mM Sodium dihydrogen phosphate, dihydrate and	
			720mL of 100 mM Disodium hydrogen phosphate, anhydrous.	
			Add 280mL of 100mM Sodium dihydrogen phosphate and 720mL of	
			100mM Disodium hydrogen phosphate. Store the solution at room	
			temperature for not more than one month.	
		5.5.4	10mM Phosphate buffer saline:	
			Add 100ml of 100 mM of phosphate buffer (pH-7.2) to 800 mL of	
			purified water and then add 8.5g of Sodium Chloride and adjust	
			the pH to 7.2 by adding 100mM Disodium hydrogen phosphate	
			solution. Make the final volume to 1000mL.	
			Filter the PBS solution with 0.2µm filter under vacuum and	
			degas the solution using the ultra-sonication bath.	
		5.5.5	Dextran (1 mg/mL) and Deuterium oxide (2.5 v/v) standard to	
			obtain void volume (Vo): Add 9.75mL of 10mM PBS slowly to 10mg	
			of Dextran in a clear glass bottle. Do not mix the solution.	
			Leave the glass bottle at 2-8°C for 2 hours Then add 0.25mL D2O	
			and gently shake the tube for 30 minutes on shaker.	
			Note: Prepare the standard solution fresh just prior to use	
			with the same 10mM PBS which is to be used during the run.	
		5.5.6	Sodium azide standard to obtain the total volume (Vt): Take	
			10mg of sodium azide, dissolve in 20mL of WFI/10mM Phosphate	
			Buffer Saline and then filter through 0.2µm syringe filter.	
			Note: Prepare this solution fresh just prior to use.	
		5.5.7	Purified polysaccharide sample preparation (Test sample):	
			Weigh the lyophilized purified sample and dissolve in 0.2µm	
			purified water to make 1mg/mL of sample. Keep it overnight at	
			2-8°C on shaker before use. Store at -20°C for not more than	
			one month.	
		5.5.8	Preservation Solution (Sodium azide 0.05%):	
			Weigh 1g of Sodium Azide and dissolve in approximately 1800 mL	
			of purified Water. Make up the volume to 2L with purified	
			water. This solution can be stored at room temperature for 2	
			months.	
			Note: All the solutions prepared for use are vacuum filtered	
			using Nylon 6,6 membrane 0.2µm filters to eliminate the entry	
			of possible particles into the system as well as the column.	
		5.6	HPLC method Run:	
		5.6.1	Mount Shodex guard column, Shodex 805 column and Shodex 804	
			column in series on calibrated HPLC system.	
		5.6.2	Place the inlet tubing A in the reservoir for 10 mM phosphate	
			buffer saline solution.	
		5.6.3	Establish On-Line connection of HPLC System with chromatography	
			software.	
		5.6.4	Switch ON the HPLC System.	
		5.6.5	Click on the chromatography software.	
		5.6.6	Purging the system:	
			5.6.6.1	Open the purge valve of pump (by turning it counter
				clockwise) and set flow rate to 3-5ml/min.
			5.6.6.2	Flush the vacuum degasser and all tubes with 10mM PBS
				for ten minutes.
			5.6.6.3	Set flow to at 0.0ml/min and close the purge valve. The
				flow shall be directed to the column.
		5.6.7	Run conditions:	
			5.6.7.1	Set the flow to 1.0ml/min
			5.6.7.2	Detector used: RID
			5.6.7.3	RID optical unit temperature : 30°C
			5.6.7.4	In case of running sample manually/sequence, run time
				is 30min.
		5.6.8	Purge reference cell of RID for 40 minutes after solvent	
			change. Further purging of reference cell may be required to	
			achieve stable baseline.	
		5.6.9	Keep on running the column with 10 mM phosphate buffer saline	
			solution unless a stable baseline for RID is obtained which can	
			be viewed on the chromatographic software.	
		5.7	Sample Preparation and injection to the system:	
		5.7.1	In process samples of meningococcal polysaccharide:	
			5.7.1.1	Filter the samples with 0.2 µm syringe filter in HPLC
				glass vials.
			5.7.1.2	Inject 50 µl of the sample on HPLC.
			Note: For the samples of this step if absorbance of the samples	
			in nRI Units observed is greater than 7000 nRI ,then the volume	
			of sample injected may be reduced accordingly and vice-versa.	
		5.7.5	Lyophilized purified polysaccharide:	
			5.7.5.1	Filter the samples with 0.2 µm syringe filter in HPLC
				glass vials.
			5.7.5.2	Inject 100 µl of the sample (1mg/ml) on HPLC.
		5.7.6	Inject 100µl of the Dextran standard for void volume	
			determination for final Lyophilized sample.	
		5.7.7	Inject 100µl of the sodium azide standard for total volume	
			determination for final Lyophilized sample.	
		5.8	Results Evaluation:	
		5.8.1	Proceed for calculating Kavg value of sample.	
			The calculations are made as per the following equation:	
			Kavg = Vr - Vo	
			Vt - Vo	
			Vr = Retention time of meningococcal polysaccharide sample.	
			Vt = Retention time of Sodium azide.	
			Vo = Retention time of Dextran(1st peak).	
		5.9	Reporting Results:	
		5.9.1	Open the report template showing retention time, area, height	
			of each sample.	

	5.9.2	Report the results of Kav_g with the retention time values of
		all the samples.
	5.9.3	Print report: Click this option to take a printout of the
		report generated for each sample, void volume marker and total
		volume marker.
	5.9.4	The printed report is signed by the analyst who does the
		analysis and also checked by the other analyst.
	5.10	Acceptance criteria:
	5.10.1	≥ 65 % KD= 0.5 for Polysaccharide of Meningococcal A serotype
	5.10.2	≥ 75 % KD= 0.5 for Polysaccharide of Meningococcal C serotype
	5.10.3	≥ 80 % KD= 0.5 for Polysaccharide of Meningococcal W135
		serotype
	5.10.4	≥ 80 % KD= 0.5 for Polysaccharide of Meningococcal Y serotype
6.0	Reference	
6.1	European pharmacopeia	
6.2	WHO TRS 594: Requirement for Meningococcal polysaccharide vaccine.	

1.0 |Purpose: |
 |To lay down the procedure for receipt and storage of samples at Quality |
 |Control department. |
 2.0 |Applicable to: |
 |This Standard Operating Procedure is applicable for receipt and storage of |
 |Retention samples, Stability Study samples, Reference Standards and under |
 |test samples from all the in process, Bulk, Formulated and Finished |
 |products, manufactured at Manufacturing dept. |
 3. |Responsibility: |
3.1	The analyst is responsible for executing the described procedure
	correctly.
3.2	Supervisor is responsible for verifying that the procedure is
	followed. HOD shall be responsible for overall compliance of this SOP.
3.3	Departmental head shall ensure that the SOP is reviewed every two
	years or if required earlier.
 4. |Safety consideration: |
 |4.1 |Wear laboratory coat and gloves while working. |
 |4.2 |Handle all the samples aseptically. |
 3.0 |Abbreviations: |
 |3.1 |FB: |Formulated Bulk |
 |3.2 |FP: |Finished Product |
 4.0 |Procedure: |
 |4.1 |Receipt of Samples: | | |
 | |4.1.1 |All the samples that are to be sent to the quality control |
 | | |department from the production facility shall be sent in |
 | | |carrier boxes in recommended storage temperatures. |
 | |4.1.2 |On receipt of the samples at QC dept, the analyst shall check |
 | | |the request form along with the sample label. |
 | |4.1.3 |The analyst shall physically verify the sample, the |
 | | |container's condition, the closure of the container and accept |
 | | |the samples only if all the parameters are meet. |
 | |4.1.4 |The QC analyst shall enter the details of the sample in the |
 | | |receipt note book and forward the samples to concerned |
 | | |sections within the QC or store it as per recommended. |
 | |4.1.5 |All the products, samples that are manufactured are to be |
 | | |stored at recommended storage conditions as mentioned in the |
 | | |table below. |
 | |4.1.6 |The containers that are used for the storage of samples should |
 | | |be clean and sterilized as per the approved sterilization |
 | | |procedures. |
 | |4.1.7 |The containers used for storage should be appropriately closed |
 | | |/ sealed by approved container closure systems. |
 | |4.1.8 |Label the containers with complete details mentioning; |
 | | |Product Name, Batch No., Date of Manufacturing, Quantity of |
 | | |sample, Storage Temperature and Signature of Stored by. |
 |4.2 |Stability Study sample: |
 | |4.2.1 |Label all the stability study vials as mentioned above also |
 | | |mention the type of stability study on the labels. |
 | |4.2.2 |The volume of each stability study vial, the duration of |
 | | |storage and the storage conditions are to be followed as |
 | | |mentioned in the respective stability study protocol. |
 |4.3 |Retention samples: |
 | |4.3.1 |The samples that are stored at quality control department as |
 | | |stand by samples are called as retention samples. |
 | |4.3.2 |The volume of each retention sample should be such that it |
 | | |should be sufficient to carry out of all assays, as per |
 | | |approved specification for 2 times. |
 | |4.3.3 |Store all the retention samples at recommended storage |
 | | |conditions after labeling it appropriately. |
 |4.4 |Reference Standards: |
 | |4.4.1 |The recommended storage conditions for reference standards is |
 | | |mentioned in the inserts or the documents provided by their |
 | | |suppliers. |
 | |4.4.2 |Follow all the instructions as per for the storage of reference |
 | | |standards. |
 |4.5 |Inprocess Samples: |
 | |Store all the in process samples at recommended storage conditions |
 | |until the analysis is completed. |
 |4.6 |Recommended storage conditions: |
 | |Following table mentions the storage conditions for all the samples. |
 | |Product |Sample |Storage Conditions |
 | |PolyRibosyl |Inprocess |(-25 °C ± 5) In screw capped |
 | |Ribitol Phosphate |glass bottles. |
 | |(PRP) | |
 | |Lyophilized PRP |(-25°C ± 5) glass vials with |
 | | |rubber bungs and flip off |
 | | |aluminum seals. |
 | |Hib-TT |Inprocess and Final |2 to 8°C In screw capped glass |
 | | |bottles. |
 | |Diphtheria toxoid |Inprocess and Bulk |2 to 8°C In screw capped glass |
 | | |bottles. |
 | |Tetanus toxoid |Inprocess and Bulk |2 to 8°C In screw capped glass |
 | | |bottles. |
 | |Pertussis whole |Inprocess and Bulk |2 to 8°C In screw capped glass |
 | | |cell |bottles. |
 | |Hepatitis B |Inprocess and Bulk |2 to 8°C In screw capped glass |
 | | |bottles. |
 | |Meningococcal |Inprocess |(-25 °C ± 5) In screw capped |
 | |Polysaccharide |glass bottles |
 | |Lyophilized |(-25 °C ± 5) glass vials with |

		polysaccharide rubber bungs and flip off	
			aluminum seals.
	Meningococcal Monovalent Bulk,	2 to 8°C In screw capped glass	
	Polysaccharide Pooled bulk,	bottles.	
	conjugated with Formulated Bulk,		
	TT Final Lot		
5.0 Reporting:			
Not applicable			

1.0 |Objective |
 |To lay down the procedure for the determination of total Carbohydrate |
 |content in Active Raw Material of Hepatitis B vaccine. |
 2.0 |Scope |
 |This Standard Operating Procedure is applicable for the analysis all the |
 |samples of Active Raw Material of Hepatitis B vaccine, received in the |
 |Quality control department. |
 3.0 |Responsibility |
 |3.1 |The analyst shall be responsible for executing the described |
 |procedure correctly. |
 |3.2 |Supervisor shall be responsible for verifying that the procedure is |
 |complied with and the results are checked. |
 |3.3 |HOD shall be responsible for overall compliance of this SOP. |
 4.0 |Abbreviations and Definitions |
4.1	Abbreviations:		
	4.1.1	ARM	:Active Raw Material
	4.1.2	CV	:Coefficient of variation
	4.1.3	g	:Gram
	4.1.4	mg	:Milligram
	4.1.5	mL	:Millilitre
	4.1.6	μL	:Micro litre
	4.1.7	μg	:Microgram
	4.1.8	nm	:Nanometer
	4.1.9	QC	:Quality Control
	4.1.10	QCR	:Quality Control Record
	4.1.11	SD	:Standard Deviation
	4.1.12	SOP	:Standard Operating Procedure
4.2	Definitions:		
		Not Applicable.	
5.0	Procedure:		
5.1	Preparation of Solutions:		
	5.1.1	1 mg/mL Anthrone solution :	
		5.1.1.	Weigh 100 mg of Anthrone and dissolve in sufficient
		1	quantity of conc. Sulphuric acid and make up to 100 mL
		with concentrated Sulphuric acid in a 100 mL volumetric	
		flask	
		5.1.1.	This solution must be prepared freshly.
		2	
	5.1.2	2 mg/mL Glucose solution :	
		5.1.2.	Weigh 100 mg of previously dried glucose and transfer
		1	it to a 50 mL volumetric flask and dissolve in
		sufficient quantity of purified water.	
		5.1.2.	Make up the volume to 50 mL using Purified water.
		2	
		5.1.2.	This solution must be prepared freshly.
		3	
5.2	Preparation of Calibration curve:		
	5.2.1	Take 5 volumetric flasks of 25 mL each and label the details	
		with pen marker. Prepare carbohydrate calibration curve	
		(20-100μg/mL) by adding 2mg/mL Glucose solution to purified	
		water as per the following table:	
		Concentration (μg/mL)	20
		Glucose Solution 2 mg/mL	250
		(μL)	
		Volume Make up with Purified	25
		water in a volumetric flask	
		(mL)	
		Note: The Calibration curve is accepted if it is meeting the	
		validated parameters of intercept, slope and correlation coefficient.	
5.3	Test:		
	5.3.1	Dilute the sample ½ using purified water as the diluent.	
		However reference ARM treated like test sample can be used in	
		the assay as control, occasionally. Use purified water as a	
		blank.	
	5.3.2	Take 1mL each from 20 to 100 μg/mL carbohydrate calibration	
		curve, previously diluted test sample, blank and control in	
		duplicates in labeled test tubes	
	5.3.3	Keep all the tubes in ice bath.	
	5.3.4	Add 3 mL of 1mg/mL Anthrone solution to each tube kept in ice	
		bath.	
	5.3.5	Take out the tubes from ice bath and transfer the tubes to	
		water bath. Heat all the tubes in boiling water for ten	
		minutes.	
	5.3.6	Take out the tubes from boiling water bath and allow it to	
		cool at room temperature.	
	5.3.7	Set the wavelength in spectrophotometer at 585 nm, Zero the	
		absorbance of blank and then measure the absorbance of	
		calibration curve and test samples.	
		Note: If absorbance of the sample falls outside the range of	
		calibration curve, dilute the sample appropriately so that it comes	
		under the range of calibration curve.	
5.4	Calculations:		
	5.4.1	Once the absorbance of the standards and the test samples have	
		been taken, calculation shall be done using the excel sheet	
		carrying formulas for the required calculation.	
	5.4.2	Enter the absorbance for calculations starting from the	
		Calibration curve prepared between 20 and 100 g/mL of	
		Glucose Solution and test samples. The calculations are made	
		according to the equation of minimum squares.	
	5.4.3	Y = mX + b.	

			Where: m = slope
			b = intercept
			Y = absorbance
			X = concentration
			(multiply with the dilution factor)
		5.4.4	Taking into account the equivalence between the ARM volume and
			the antigen concentration, express the result per 20 g of
			protein. Enter the protein concentration estimated by OD at
			280nm in the excel sheet.
			If the Protein concentration of Active Raw Material of
			Hepatitis B Vaccine by Absorbance at 280 nm. Is 1.40 mg/mL
			calculations are done as given below:
		5.4.5	C = $20 \times X / 1.40 / 1000$
		5.4.6	Where:
			C = Carbohydrate Concentration per 20µg of Protein
			20 =Dose of Hepatitis B vaccine (20µg)
			X =Total concentration of Carbohydrate calculated from the
			calibration curve (µg/mL)
			1.40mg/mL = Concentration of protein in the sample by
			Absorbance at 280 nm)
			1000 = Conversion factor
		5.5	Acceptance criteria:
		5.5.1	The coefficient of variation (CV) between the replicates of
			the samples should be $\leq 5\%$.
		5.5.2	The value of the control should be within the acceptance
			limits.
		5.5.3	Carbohydrate content in the sample should be within the
			specified limits.
		6.0	References
		6.1	Proteoglycans En: Carbohydrate analysis: a practical approach.
		Chaplin, M.F. y Kennedy, J.K. (eds.) IRL Press, Washington, USA pp:	
		135-136.	
		6.2	Biochem. J. 23, pp: 1824. Trevelyan W.E. y Harrison J.S. (1952).

1.0 | Objective: | Objective: | Objective: | Objective: | Objective:
 | Determination of Total Organic Carbon, pH and Conductivity of water samples. | Determination of Total Organic Carbon, pH and Conductivity of water samples. | Determination of Total Organic Carbon, pH and Conductivity of water samples. | Determination of Total Organic Carbon, pH and Conductivity of water samples. | Determination of Total Organic Carbon, pH and Conductivity of water samples.
 2.0 | Scope | Scope | Scope | Scope | Scope
 | This Standard Operating Procedure is applicable to all water samples received in Quality Control Department | This Standard Operating Procedure is applicable to all water samples received in Quality Control Department | This Standard Operating Procedure is applicable to all water samples received in Quality Control Department | This Standard Operating Procedure is applicable to all water samples received in Quality Control Department | This Standard Operating Procedure is applicable to all water samples received in Quality Control Department
 3.0 | Responsibility | Responsibility | Responsibility | Responsibility | Responsibility
 | 3.1 | The analyst shall be responsible for carrying out the described procedure correctly. | The analyst shall be responsible for carrying out the described procedure correctly. | The analyst shall be responsible for carrying out the described procedure correctly. | The analyst shall be responsible for carrying out the described procedure correctly. | The analyst shall be responsible for carrying out the described procedure correctly.
 | 3.2 | The Supervisor of the laboratory shall be responsible for verifying that the procedure is complied with and for checking the results. | The Supervisor of the laboratory shall be responsible for verifying that the procedure is complied with and for checking the results. | The Supervisor of the laboratory shall be responsible for verifying that the procedure is complied with and for checking the results. | The Supervisor of the laboratory shall be responsible for verifying that the procedure is complied with and for checking the results. | The Supervisor of the laboratory shall be responsible for verifying that the procedure is complied with and for checking the results.
 | 3.3 | The departmental head/designee shall be responsible for overall compliance of the SOP. | The departmental head/designee shall be responsible for overall compliance of the SOP. | The departmental head/designee shall be responsible for overall compliance of the SOP. | The departmental head/designee shall be responsible for overall compliance of the SOP. | The departmental head/designee shall be responsible for overall compliance of the SOP.
 | 3.4 | Head QA/Designee shall be responsible for final approval and subsequent compliance of the SOP. | Head QA/Designee shall be responsible for final approval and subsequent compliance of the SOP. | Head QA/Designee shall be responsible for final approval and subsequent compliance of the SOP. | Head QA/Designee shall be responsible for final approval and subsequent compliance of the SOP. | Head QA/Designee shall be responsible for final approval and subsequent compliance of the SOP.
 4.0 | Abbreviations and Definitions | Abbreviations and Definitions | Abbreviations and Definitions | Abbreviations and Definitions | Abbreviations and Definitions
 | 4.1 | QA | QA | QA | Quality Assurance
 | 4.2 | HOD | HOD | HOD | Head of Department
 | 4.3 | QC | QC | QC | Quality Control
 | 4.4 | SOP | SOP | SOP | Standard Operating Procedure
 | 4.5 | TOC | TOC | TOC | Total Organic Carbon
 | 4.6 | mL | mL | mL | Millilitre
 | 4.7 | RMS | RMS | RMS | Raw Material Specification
 | 4.8 | ppb | ppb | ppb | Parts per billion
 | 4.9 | S/cm | S/cm | S/cm | Micro Siemens per centimetre
 5.0 | Equipments, Materials and Reagents | Equipments, Materials and Reagents | Equipments, Materials and Reagents | Equipments, Materials and Reagents | Equipments, Materials and Reagents
 | 5.1 | 5.1 | Vortex Mixer | Vortex Mixer | Vortex Mixer | Vortex Mixer
 | 5.2 | 5.2 | pH and conductivity meter | pH and conductivity meter | pH and conductivity meter | pH and conductivity meter
 | 5.3 | 5.3 | Refrigerator (2-8°C) | Refrigerator (2-8°C) | Refrigerator (2-8°C) | Refrigerator (2-8°C)
 | 5.4 | 5.4 | TOC analyzer | TOC analyzer | TOC analyzer | TOC analyzer
 | 5.5 | 5.5 | pH buffers 4.01, 7.0 and 9.21 | pH buffers 4.01, 7.0 and 9.21 | pH buffers 4.01, 7.0 and 9.21 | pH buffers 4.01, 7.0 and 9.21
 | 5.6 | 5.6 | calibration standards i.e. 84µS/cm, 1413 µS/cm and 12.88mS/cm | calibration standards i.e. 84µS/cm, 1413 µS/cm and 12.88mS/cm | calibration standards i.e. 84µS/cm, 1413 µS/cm and 12.88mS/cm | calibration standards i.e. 84µS/cm, 1413 µS/cm and 12.88mS/cm
 6.0 | Preparation of solutions: | Preparation of solutions: | Preparation of solutions: | Preparation of solutions: | Preparation of solutions:
 | Not applicable | Not applicable | Not applicable | Not applicable | Not applicable | Not applicable
 7.0 | Procedure: | Procedure: | Procedure: | Procedure: | Procedure: | Procedure:
 | Note: Make sure that all the reagents are at room temperature before use. | Note: Make sure that all the reagents are at room temperature before use. | Note: Make sure that all the reagents are at room temperature before use. | Note: Make sure that all the reagents are at room temperature before use. | Note: Make sure that all the reagents are at room temperature before use.
 | 7.1 | 7.1 | Before carry out the analysis for pH,TOC and conductivity brought down the temperature of water samples to room temperature (20-25°C).. | Before carry out the analysis for pH,TOC and conductivity brought down the temperature of water samples to room temperature (20-25°C).. | Before carry out the analysis for pH,TOC and conductivity brought down the temperature of water samples to room temperature (20-25°C).. | Before carry out the analysis for pH,TOC and conductivity brought down the temperature of water samples to room temperature (20-25°C)..
 | 7.2 | 7.2 | Determination of Conductivity: | Determination of Conductivity: | Determination of Conductivity: | Determination of Conductivity:
 | | 7.2.1 | Calibrate the equipment as per manual. | Calibrate the equipment as per manual. | Calibrate the equipment as per manual. | Calibrate the equipment as per manual.
 | | 7.2.2 | Clean clean with WFI or purified water and blot dry. | Clean clean with WFI or purified water and blot dry. | Clean clean with WFI or purified water and blot dry. | Clean clean with WFI or purified water and blot dry.
 | | 7.2.3 | Put the sample in clean beaker to check the conductivity. | Put the sample in clean beaker to check the conductivity. | Put the sample in clean beaker to check the conductivity. | Put the sample in clean beaker to check the conductivity.
 | | 7.2.4 | Note: Rinse beaker with the sample before taking sample for checking conductivity. | Note: Rinse beaker with the sample before taking sample for checking conductivity. | Note: Rinse beaker with the sample before taking sample for checking conductivity. | Note: Rinse beaker with the sample before taking sample for checking conductivity.
 | | 7.2.5 | Put conductivity sensor in sample solution and press key [Read] to measure the conductivity .Wait till the value stabilizes and a stable reading is displayed depending on the end point format selected. | Put conductivity sensor in sample solution and press key [Read] to measure the conductivity .Wait till the value stabilizes and a stable reading is displayed depending on the end point format selected. | Put conductivity sensor in sample solution and press key [Read] to measure the conductivity .Wait till the value stabilizes and a stable reading is displayed depending on the end point format selected. | Put conductivity sensor in sample solution and press key [Read] to measure the conductivity .Wait till the value stabilizes and a stable reading is displayed depending on the end point format selected.
 | | 7.2.6 | Clean the cell with purified water & WFI and Blot dry. | Clean the cell with purified water & WFI and Blot dry. | Clean the cell with purified water & WFI and Blot dry. | Clean the cell with purified water & WFI and Blot dry.

| 7.3 | 7.3 | Determination of pH:. | Determination of pH:. | Determination of pH:. | Determination of pH:.

|| | 7.3.1 | Calibrate the equipment as per manual. | Calibrate the equipment as per manual. | Calibrate the equipment as per manual.

|| | 7.3.2 | Clean pH electrode with WFI or purified water and blot dry. | Clean pH electrode with WFI or purified water and blot dry. | Clean pH electrode with WFI or purified water and blot dry.

|| | 7.3.3 | Put the sample in clean beaker to check the pH. | Put the sample in clean beaker to check the pH. | Put the sample in clean beaker to check the pH.

|| | 7.3.4 | Introduce the pH electrode in beaker having sample solution | Introduce the pH electrode in beaker having sample solution | Introduce the pH electrode in beaker having sample solution

|| | 7.3.5 | Press [Read] key for pH measurement | Press [Read] key for pH measurement | Press [Read] key for pH measurement

|| | Rinse, blot and store the electrode in a beaker having saturated KCl solution. | Rinse, blot and store the electrode in a beaker having saturated KCl solution. | Rinse, blot and store the electrode in a beaker having saturated KCl solution.

| 7.4 | 7.4 | Determination of Total Organic Carbon | Determination of Total Organic Carbon | Determination of Total Organic Carbon | Determination of Total Organic Carbon | Determination of Total Organic Carbon

|| | 7.4.1 | Calibrate the system as per manual | Calibrate the system as per manual | Calibrate the system as per manual | Calibrate the system as per manual

|| | 7.4.2 | System suitability with 500 ppb sucrose and 500ppb | System suitability with 500 ppb sucrose and 500ppb | System suitability with 500 ppb sucrose and 500ppb | System suitability with 500 ppb sucrose and 500ppb

|| | 7.4.3 | On receipt of water samples, label the previously cleaned glass vials as per equipment requirement | On receipt of water samples, label the previously cleaned glass vials as per equipment requirement | On receipt of water samples, label the previously cleaned glass vials as per equipment requirement | On receipt of water samples, label the previously cleaned glass vials as per equipment requirement

|| | 7.4.3 | Rinse the glass vials twice with 10mL approximately of water sample. | Rinse the glass vials twice with 10mL approximately of water sample. | Rinse the glass vials twice with 10mL approximately of water sample. | Rinse the glass vials twice with 10mL approximately of water sample.

|| | 7.4.5 | Fill the glass vial with water sample up to the brim, then seal the glass vials. | Fill the glass vial with water sample up to the brim, then seal the glass vials. | Fill the glass vial with water sample up to the brim, then seal the glass vials. | Fill the glass vial with water sample up to the brim, then seal the glass vials.

|| | 7.4.6 | Introduce the sample with system for TOC reading. | Introduce the sample with system for TOC reading. | Introduce the sample with system for TOC reading. | Introduce the sample with system for TOC reading.

| 8.0 | 8.0 | Report the result to concern department | Report the result to concern department | Report the result to concern department | Report the result to concern department

- 2 phenoxy ethanol is missing in the spec.
- | |
- 1 | Physical appearance | White or almost white translucent suspension with mineral carrier, free from extraneous particles/floccules. Settles into a clear liquid and gel phase on sedimentation
- 2 | Preservative (Polysorbate) | 5mg/ml (Not less than 85% and Not greater than 115% of intended amount)
- 3 | Sterility | Sterile
- 4 | Identification | Positive when tested by immunochemical method
- 5 | Abnormal toxicity | All mice and Guinea pigs Should survive test with no sickness
- 6 | Aluminum content (Al3+) | 0.25mg/ml (acceptable range)
- 7 | Pyrogens | Inject each rabbit with 1 ml of dilution containing 2.5 µg of each polysaccharide.
- 8 | pH | 5.5 to 6.5
- 9 | Polysorbate 20 | 0.2mg/ml (acceptable range)
- 10 | Assay | Vaccine shall contain not less than 70% and not more than 130 % as per stated label of polysaccharide content.
- The confidence interval of assay shall be 80-120% of estimated content.
- 10 | Assay |
- 10 | Assay |

[1.0]Objective: |
|To lay down the procedure for the detection and quantification of |
|Hepatitis-B surface antigen using Bio-Rad Monolisa HBsAg ELISA kit. |
[2.0]Scope: |
|This Standard Operating Procedure is applicable to determine the antigen |
|content in all the In process and Active Raw Material for Hepatitis B |
|vaccine (rDNA) samples. |
[3.0]Responsibility: |
|3.1 |The analyst of the Quality control laboratory shall be responsible for |
|executing the described procedure correctly. |
|3.2 |The supervisor of the laboratory shall be responsible for verifying |
|that the procedure is followed correctly and for checking the results. |
|3.3 |The Departmental Head shall be responsible for the overall compliance |
|of the SOP and shall ensure that the SOP is reviewed every two years or |
|if required earlier. |
[4.0]Abbreviations and Definitions: |
[4.1 |
|ARM |
|: |
|Active Raw Material |
|Quality Assurance |
| |
| |
| |
| |
[4.2 |
|PBS |
|: |
|Phosphate buffered Saline |
|Quality Control |
| |
| |
| |
| |
[4.3 |
|QC |
|: |
|Quality Control |
|Quality control Record |
| |
| |
| |
| |
[4.4 |
|QA |
|: |
|Quality Assurance |
|Standard Operating Procedure |
| |
| |
| |
| |
[4.5 |
|OE |
|: |
|Operation of Equipment |
| |
| |
| |
| |
[4.6 |
|SOP |
|: |
|Standard Operating Procedure |
| |
| |
| |
| |
[5.0] Procedure |
|5.1 |Prepare the dilutions of the test sample and reference standard in |
|PBS (1X) with 1%BSA solution. |
|If ARM is to be analyzed, then prepare 4 dilutions of reference |
|standard, namely 10, 5, 2.5 and 1.25 ng/ml, with 3 independent |
|replicates each and for in process samples, 4 dilutions of reference |
|standard namely 10, 5, 2.5, 1.25, 0.65, 0.3125 and 0.156 ng/ml in |
|duplicate. |
|5.2 |If the sample is ARM, prepare 4 dilutions of the sample namely 10, 5, |
|2.5 and 1.25 ng/ml, with 3 independent replicates each while in case |
|of inprocess samples, prepare a working concentration of 5 ng/ml. |
|Protein concentration of ARM is estimated by lowry method, while for |
|in process samples, a pre-determined target concentration/absorbance |
|at 280 nm or results from Lowry can be used. |
|5.3 |Select the no. of ELISA strips that are required to conduct the |
|assay. Add 100µl of each dilution of Reference standard and the test |
|sample into the pre-coated ELISA wells, provided in the Bio-Rad |
|Monolisa ELISA kit. |
|5.4 |Add 100 µl of appropriate positive and negative controls also to the |
|pre determined ELISA wells. |
|5.5 |Prepare the conjugate solution as mentioned in the kit insert and add |
|50 µl of the conjugate solution into each well over the assay |
|controls, standard and sample solutions. |
|5.6 |Mix the conjugate solution properly by shaking and incubate the ELISA |
|plate at 37°C for next 90 mins. of duration. |
|5.7 |On completion of the incubation period, remove the plate from the |
|incubator and discard the contents in the wells. The wash solution |
|should be prepared before the completion of the incubation period. |

		Add 300 µl of wash solution and shake the plate for around 15 sec.		
		Discard the wash solution with a jerk and tap the plates on an		
		absorbent paper to remove all the residual was solution. Repeat the		
		process of washing for four more times (a total of five washes).		
		5.8	Freshly prepare the TMB substrate solution as per the instructions	
		given in the kit insert. Avoid direct exposure of solution to light.		
		On completion of washing steps, add 100 µl of TMB solution to each		
		well. And incubate the plate at room temperature in dark for 10 to 15		
		mins.		
		5.10	On completion of the incubation period add 100 µl of stop solution	
		into each well to stop the colour reaction.		
		5.11	Read the OD of wells at 450-620 nm on a ELISA plate reader.	
		5.12	Proceed to analyze the results. The analysis of results depends on	
		the type of sample.		
		5.12.1	In Process samples:	
		5.12.1.1	Proceed for calculations starting from the	
			Calibration curve prepared between 0.156 and 10 ng/ml	
			of Reference Standard solution.	
		5.12.1.2	$Y = mX + b$.	
			Where: m = slope	
			b = intercept	
			Y = absorbance	
			X = concentration	
			(Multiply by dilution factor)	
		5.12.1.3	Use EXCEL programme for the preparation of	
			calibration curve and Calculation of the results	
			(Multiply by the dilution factor. Divide with 1000 to	
			change ng/ml to µg/ml).	
		5.12.1.4	Validity criteria: For an assay to be valid	
			The coefficient of regression must be greater than	
			0.98.	
			The coefficient of variation between the replicates	
			of same point of standard curve must be less than	
			10%.	
		5.12.2	Active Raw Material:	
		5.12.2.1	The calculation of antigen content for ARM samples is	
			done by using Combistat software.	
		5.12.2.2	Entre the initial concentration of the Reference	
			standard, the dilution folds and the OD values	
			obtained for each replicate.	
		5.12.2.3	Enter the test sample dilution folds and the OD	
			values obtained for each replicate.	
		5.12.2.4	Pres on the calculate icon on the top of the screen	
			and the software shall directly give the antigen	
			content present in the sample.	
		5.12.2.5	Validity criteria of the assay:	
			The reference standard dilutions should pass in the	
			criteria for linearity.	
			The test sample dilutions should pass in the criteria	
			for linearity	
			The reference standard and the test sample should	
			pass the criteria for parallelism.	
		5.12.2.6	If all the above criteria are fulfilled, the assay is	
			valid otherwise reject it and repeat the assay. If	
			the assay is valid report the results.	
		6.0	References:	
		6.1	SOP QC/00/006 Determination of Protein concentration in Active Raw	
			Material of Recombinant Hepatitis-b Vaccine by Absorbance (280nm).	
		6.2	SOP QC/248 Determination of Protein concentration by Lowry's method.	
		7.0	Distribution:	
			Master Copy	Archived with Quality Assurance
			Department.	
			Controlled Copy 1	Quality Control Department.
			Subsequent Controlled Copies	Department(s) making requisition

1.0 |Objective: |
 |To lay down the procedure for quantifying Total and unconjugated (free) |
 |PRP concentration. |
 2.0 |Scope: |
 |This Standard Operating Procedure is applicable for the analysis of all |
 |the samples of PRP and PRP-TT bulk conjugate of Haemophilus Influenzae |
 |type b in which Total and unconjugated (free) PRP concentration is to be |
 |determined, received in the Quality control department. |
 3.0 |Abbreviations and Definitions |
3.1	Abbreviations:		
	3.1.1	CV	:Coefficient of variation
	3.1.2	g	:Gram
	3.1.3	mg	:Milligram
	3.1.4	mL	:Millilitre
	3.1.5	µL	:Micro litre
	3.1.6	µg	:Microgram
	3.1.7	nm	:Nanometer
	3.1.8	PRP	:Polyribosyl Ribitol Phosphate
	3.1.9	TT	:Tetanus Toxoid
	3.1.10	TNBS	: 2, 4, 6-Trinitrobenzenesulfonic acid
	3.1.11	DMF	: Dimethylformamide
3.2	Definitions:		
	Not Applicable.		
4.0	Procedure:		
4.1	0.5 g/l Ferric chloride solution:		
	4.1.1	Weigh 50 mg of Ferric Chloride hexahydrate and transfer it to	
		100 mL volumetric flask.	
	4.1.2	Dissolve and make up the volume to 100 mL with concentrated	
		hydrochloric acid.	
	4.1.3	Store the solution in screw cap amber bottle, for a maximum 6	
		months at room temperature.	
4.2	100 g/l Orcinol solution:		
	4.2.1	Weigh 1g Orcinol monohydrate and transfer it to 10 mL	
		volumetric flask.	
	4.2.2	Dissolve and make up the volume to 10 mL with absolute	
		ethanol.	
	4.2.3	Store the solution in a screw cap amber bottle at 2 to 8 °C,	
		for maximum of 15 days.	
4.3	0.1% w/v TNBS-10% DMF:		
	4.3.1	Take 1 mL of commercial TNBS solution (1%w/v in DMF) and	
		transfer it to a 10 mL volumetric flask.	
	4.3.2	Make up the volume to 10 mL with purified water.	
	4.3.3	Prepare fresh prior to use.	
4.4	0.25 g/1 Ribose Solution:		
	4.4.1	Weigh 25 mg of D-Ribose and transfer it to a 100 mL	
		volumetric flask.	
	4.4.2	Dissolve in purified water and make up the volume to 100 mL	
		with purified water.	
	4.4.3	Prepare fresh prior to use.	
4.5	25 µg /mL Ribose Solution:		
	4.5.1	Take 1 mL of 0.25 g/1 Ribose solution and transfer it to a 10	
		mL volumetric flask.	
	4.5.2	Make up the volume to 10 mL with purified water.	
	4.5.3	Prepare fresh prior to use.	
4.6	1% w/v Sodium deoxycholate (DOC):		
	4.6.1	Dissolve 250 mg sodium deoxycholate in 25 mL purified water	
		and make up the volume to 25 ml.	
	4.6.2	Adjust the pH to 6.8 by adding about 100 µL of 1M	
		hydrochloric acid drop wise.	
	4.6.3	The solution is prepared freshly.	
4.7	1 M Hydrochloric acid:		
	4.7.1	Take 85 mL of concentrated hydrochloric acid and make up the	
		volume to 1000 mL with purified water.	
	4.7.2	The solution should be stored at room temperature for maximum	
		6 months.	
4.8	Preparation of Calibration Curve:		
	4.8.1	Prepare calibration curve (2.5-25 µg) with freshly prepared	
		ribose solution (25 µg /mL) in previously labelled test	
		tubes of 20 ml.	
	4.8.2	Calibration Curve (2.5 - 25µg) is prepared in duplicate as	
		follow:	
		Ribose content	2.5
		(µg)	
		25 µg/ml Ribose	100
		(µl)	
		Purified water	900
		(µl)	
	4.8.3	Make up the volume to 2 mL using purified water.	
	Note: The Calibration curve is accepted if it is meeting the		
	parameters of correlation coefficient i.e. ≥ 0.990 .		
4.9	Preparation of sample for total PRP content:		
	4.9.1	Take sample volume such that the sample volumes used in test	
		contain 2.5 to 25 µg of ribose i.e. lies within the	
		calibration curve.	
	4.9.2	Make up the volume to 2 mL using purified water respectively.	
	Note: The sample volume can vary depending upon the ribose		
	concentration. If there is any sample outside the range of the		
	calibration curve, prepare an appropriate dilution of the sample.		
4.10	Unconjugated (Free) PRP:		

		4.10.1	Take 1 mL of sample (dilute if required) having about
		100-150 µg PRP in micro centrifuge tube in duplicate, add	
		100 µL of 1% DOC and mix it.	
		4.10.2	Put this tube in an ice bath for 30 minutes.
		4.10.3	Take out and add 50 µL 1M HCl and mix thoroughly.
		4.10.4	Centrifuge the mixture at 4°C for 15 minutes at 4000 RPM.
		4.10.5	Take 1 mL of supernatant in 20 mL labeled test tube.
		4.10.6	Make up the volume to 2 mL using purified water
		respectively.	
		4.10.7	The test is performed in duplicate for each sample.
		4.11	Treat 2 mL purified water as blank in thermoresistant test tube.
		4.12	Add 2 ml of 0.5 g/l Ferric Chloride solution to tubes of calibration
		curve, Blank and samples.	
		4.13	Vortex and add 0.2 mL of 100 g/l Orcinol Solution to all test tubes.
		4.14	Close the tubes with aluminium foil and place the test tubes in
		boiling water bath for 20 minutes.	
		4.15	Take out the tubes and cool in iced water.
		4.16	Allow the tubes to come at room temperature and set the wavelength
		in spectrophotometer at 670 nm. Zero the absorbance of blank and	
		then measure the absorbance of calibration curve and test samples.	
		4.17	Calculations:
		4.17.1	Once the absorbance of the standards and the test samples
		have been taken, calculation shall be done using the excel	
		sheet carrying formulas for the required calculation.	
		4.17.2	Always copy the master excel sheet (sheet carrying formulas
		for the required calculation), never use the previously	
		used excel sheet for new calculations.	
		4.17.3	Copy the master excel sheet, keep it for use in respective
		folder and then fill in the required sample details for	
		total & free sample treatments, sample dilution, name of	
		analyst, date, etc.	
		4.17.4	Enter the absorbance for calculations starting from the
		Calibration curve prepared between 2.5 and 25 g of Ribose	
		and test samples. The calculations are made according to	
		the equation of minimum squares.	
		Y = mX + b.	
		Where: m = slope	
		b = intercept	
		Y = absorbance	
		X = concentration	
		(Multiply by dilution factor)	
		4.17.5	Calculation of actual dilution for free PRP:
		Sample volume taken = 1000µL	
		Total volume = 1150µL (1000µL sample + 100µL 1% DOC + 50µL	
		1M HCl)	
		Correction Factor = 1150µL / 1000µL = 1.15	
		Actual dilution = dilution factor x 1.15	
		4.17.6	Calculate the content of ribose in the test solution for
		each volume tested and PRP content by the formula :	
		PRP content = Ribose content / 0.434	
		4.17.7	Record all the calculations in the data record form and
		also attach the hard copy of the excel sheet used for	
		calculation along it.	
		4.18	Acceptance criteria:
		The coefficient of variation (CV) between the replicates of the	
		samples should be ≤ 5%.	
5.0	Forms and Records:		
5.1	Determination of Total and Free PRP Concentration		
6.0	References:		
6.1	Biologicals (1998) 26, 33-38 – A simple and rapid method for		
		measuring unconjugated capsular polysaccharide (PRP) of Haemophilus	
		influenzae type b in PRP-TT conjugate vaccine.	
6.2	European pharmacopoeia.		

1. OBJECTIVE

This Standard Testing Procedure (STP) describes the test method

1.1 For the estimation of total protein nitrogen content by using semi-automatic Kjeldahl apparatus.

1.2 For the calculation of antigenic purity of purified Diphtheria toxoid bulk in terms of Lf units per mg of protein nitrogen.

2. SCOPE

This STP is applicable for

2.1 The estimation of total protein nitrogen content in in-process samples and purified bulks of Diphtheria toxoid and calculation of their antigenic purity.

2.2 The calculation of antigenic purity of purified Diphtheria toxoid bulk in terms of Lf units per mg of protein nitrogen.

3. RESPONSIBILITY

3.1 Analysts and supervisors of Quality Control Department are responsible for performing the test.

3.2 Quality Control In-Charge to ensure that this STP is followed.

4. EQUIPMENT/INSTRUMENTS

1. Semiautomatic Kjeldahl apparatus (GERHARDT)

2. Micro Pipettes

3. Centrifuge

4. Vortex mixer

5. Fume hood

4.6 Weighing balance

5. MATERIALS

1. Chemicals

1. Trichloro acetic acid (Grade-SQ. Make - Qualigens, Fischer scientific. Cat No.28445)

2. Potassium sulphate (Grade-. Make ExcelsaR - Qualigens, Fischer scientific. Cat No.13685)

3. Copper (II) sulphate (Grade-ExcelsaR. Make - Qualigens, Fischer scientific. Cat No.12515) (Make-Merck Cat NO.1.02790.0250)

4. Selenium dioxide (Grade-SQ. Make - Qualigens, Fischer scientific. Cat No.27243)

5. Nitrogen free concentrated Sulphuric acid (Grade-ExcelsaR. Make - Qualigens, Fischer scientific. Cat No.29997)

6. Ammonium sulphate (Grade-ExcelsaR. Make - Qualigens, Fischer scientific. Cat No.11535) and (Grade-SQ Make-Qualigens, Fischer scientific. Cat No.21495)

7. Glycine (Grade-SQ. Make - Qualigens, Fischer scientific. Cat No.24755)

(Grade-ExcelsaR. Make – Qualigens, Fischer scientific. Cat No.12835)

8. Sodium hydroxide (Grade-ExcelsaR. Make - Qualigens, Fischer scientific. Cat No.1589E)
(Grade-SQ. Make – Qualigens, Fischer scientific. Cat No.27815)

5.1.9 Hydrochloric acid (Grade-ExcelsaR. Make-Qualigens, Fischer scientific. Cat No.29507)
(Grade-SQ. Make-Qualigens, Fischer scientific. Cat No.29147)

5.1.10 Dextrose (Make-Merck, Cat No.61780905001730)

5.1.11 Methyl red (Make – LOBACHemie, Cat. No. 47012)

5.1.12 Bovine serum albumin (Make-Sigma Cat No.1000828615)

5.2 Reagents

5.2.1 70% Trichloroacetic acid

5.2.2 10 M Sodium hydroxide

5.2.3 1M Sodium hydroxide

5.2.4 0.01 M Sodium hydroxide

5.2.5 0.01M Hydrochloric acid

5.2.6 Water collected from WFI loop

5.2.7 Methyl red solution

5.3 Glassware and other materials

5.3.1 Conical flask – 250 mL

5.3.2 Centrifuge tubes (15.0 mL)

5.3.3 Digestion tubes

5.3.4 Sterile tips

5.3.5 Pestle and Mortar

6. ABBREVIATIONS

Abbreviation	Full form
STP	Standard Testing Procedure
RST	Record Sheet Template
SOP	Standard operating procedure
Lf	Limes Flocculation / Limits of
	flocculation
BSA	Bovine serum Albumin
PN2	Protein Nitrogen
RT	Room Temperature
DIP	Diphtheria

7. PROCEDURE

1. Preparation of digestion mixture

1. Weigh 100g of Potassium sulphate, 5 g of Copper (II) sulphate and 2.5 g of Selenium dioxide in a clean butter paper.

2. Pulverize the Copper (II) sulphate crystals in a Pestle and Mortar and transfer on to a clean butter paper.

3. Intermittently add above three chemicals in small quantities while mixing them in a Pestle and Mortar. Mix thoroughly.

4. After thorough mixing transfer the digestion mixture to an amber colored bottle and label with RST No.8610.

5. Store the digestion mixture at room temperature.

6. Use this mixture on or before one month from the date of preparation.

2. Preparation of Controls

1. For each assay prepare the standards freshly.

7.2.2 Weigh and dissolve 943 mg of Ammonium sulphate or 1072 mg of Glycine or 1214.3 mg of BSA in water collected from WFI loop and make up the volume to 100 mL.

Note: 1mL of each solution is equivalent to 2 mg of Total Protein Nitrogen.

7.3 Preparation of reagents

7.3.1 70% Trichloro acetic acid

7.3.1.1 Weigh 70 g of Trichloro acetic acid and dissolve in water collected from WFI loop and make up the volume to 100 mL.

7.3.1.2 Transfer the solution to an amber colored bottle and label the container with RST No.8610.

7.3.1.3 Store this reagent at room temperature and use it on or before one month from the date of preparation.

2. 10 M Sodium hydroxide solution

7.3.2.1 Weigh 400 g of Sodium hydroxide pellet and dissolve in sufficient quantity of water collected from WFI loop and make up the volume to 1000 mL.

7.3.2.2 Transfer the solution to a glass container and label with RST No.8610.

7.3.2.3 Store this reagent at room temperature and use it on or before one month from the date of preparation.

3. 0.01 M Sodium hydroxide solution

7.3.3.1 Weigh 400 mg of Sodium hydroxide pellet and dissolve in sufficient quantity of water collected from WFI loop and make up the volume to 1000 mL.

7.3.3.2 Standardize the solution to determine the exact molarity of solution.

7.3.3.3 Transfer the solution to a glass container and label with RST No.QC/8069.

7.3.3.4 Store this reagent at room temperature and use it on or before three months from the date of preparation.

4. 0.01 M Hydrochloric acid

7.3.4.1 Measure 0.85 mL of concentrated Hydrochloric acid and mix with sufficient quantity of water collected from WFI loop and make up the volume to 1000 mL.

7.3.4.2 Transfer the reagent to a glass bottle and label with RST No.QC/8069.

7.3.4.3 Store this reagent at room temperature and use it on or before three months from the date of preparation.

5. Preparation of Methyl red solution

7.3.5.1 Dissolve 50 mg of methyl red in a mixture containing 1.86 mL of 0.1M Sodium hydroxide and 50 mL of ethanol (96 %) and add sufficient quantity of water collected from WFI loop to make up to 100 mL.

1. Transfer the reagent to a glass bottle and label with RST No.8610.

2. Store this reagent at room temperature and use it on or before one year from date of preparation.

7.3.6 1M / 1N Sodium hydroxide

7.3.6.1 Weigh 40 g of Sodium hydroxide pellets and dissolve in sufficient quantity of water collected from WFI loop and make up the volume to 1000 mL.

7.3.6.2 Transfer the solution to a glass container and label with RST No.8610.

7.3.6.3 Store this reagent at room temperature and use it on or before one month from the date of preparation.

7.4 Preliminary checks

7.4.1 Safety

7.4.1.1 Follow appropriate dress code wherever applicable.

7.4.1.2 Wear gloves and safety goggles while performing the test.

7.4.1.3 Ensure that fume hood door is closed properly.

7.4.1.4 While carrying out the distillation process ensures that the door of the distillation unit is closed properly.

CAUTION

The chemicals and reagents used in this test are hazardous in nature. Avoid direct contact with any parts of the body. Perform the whole test with gloved hand. If any direct contact occurs wash the area with ample quantities of water and immediately contact the Physician for treatment.

7.4.2 Equipment and reagents

7.4.2.1 Ensure that the calibration or validation status of the equipment is within due date.

7.4.2.2 Ensure that the reagents are within expiry period.

7.5 Preparation of BSA and sample

7.5.1 Take 1.0 mL of BSA solution and 1.0 mL of sample in centrifuge tubes. Add 400µL of Trichloro acetic acid solution 70 % (w/v) and incubate at RT for 1 hour.

7.5.2 Centrifuge the incubated samples for 20 minutes at 6000 RPM and separate the precipitate and supernatant. Add 1.0 mL of 1M / 1 N NaOH and transfer the content to digestion flask.

7.5.3 Add an additional 400 µL of 70 % (w/v) Trichloro acetic acid to the supernatant to ensure the complete precipitation of protein.

7.5.4 If precipitation is observed centrifuge the treated supernatant for

20 minutes at 6000 RPM and discard the supernatant. Add 1.0 mL of 1M / 1 N NaOH and vortex the tube gently and transfer the content to digestion flask.

7.5.5 Rinse the tubes with minimum quantity of water collected from WFI loop and transfer the solution in the digestion tubes.

Note: BSA is used in each test as a control to ensure the activity of Trichloro acetic acid in precipitation of protein.

7.6 Digestion Process

7.6.1 Take 50.0 mg of Dextrose (blank) and 1.0 mL of freshly prepared Glycine or Ammonium Sulphate (equivalent to 2.0 mg of Protein Nitrogen) in duplicates in separate digestion tubes.

7.6.2 Add 4.0 g of digestion mixture and 5.0 mL of Sulphuric acid, allowing it to run down the sides of the tube to wash any adhering particles and mix the contents by swirling.

7.6.3 Place the digestion tubes (Kjeldatherm tubes) into the insert rack. Ensure that the insert rack is carrying all tubes in place for uniform 100% suction of fumes.

7.6.4 Place the insert rack onto lower support of the two – tier console.

7.6.5 Remove the drip tray and place the exhaust cassette on the insert rack containing the digestion tubes. Ensure that the exhaust cassette fits tightly to the insert rack. Switch on the suction device. Lower the combined insert rack and the exhaust cassette into the heating chamber of the equipment.

7.7 Operation of Digestion unit (Turbotherm)

7.7.1 Operate the fume hood as per SOP No. QC-047.

7.7.2 Programme and control the digestion unit via the key board of the control panel as per SOP No. QC-101.

7.7.3 Switch on the system.

7.7.4 Press programme button and enter into sub programmed h1 to set the conditions for heating step 1.

7.7.4.1 Set the heating capacity of the digestion unit as 100 % by using + or – key.

7.7.4.2 Bring the cursor to time mode by pressing the time button and set the time as 20 minutes by using + or – key.

7.7.5 Press programme button and enter into sub programme h2 to set the conditions for heating step 2.

7.7.5.1 Set the heating capacity of the digestion unit as 70 % by using + or – key to adjust the heating capacity.

7.7.5.2 Bring the cursor to time mode by pressing time button and set the time as 40 minutes by using + or – key.

7.7.6 Press the programme button and enter into sub programme h3 to set the conditions for heating step 3.

7.7.7 Set the heating capacity as 00 and time for digestion as 00 .

7.7.8 Enter the programme button to save the programme and come to main display (H2).

7.7.9 Press the run button to execute the saved programme.

7.7.10 After digestion, cool the digested sample before distillation.

7.8 Distillation

7.8.1 Prerequisites

7.8.1.1 Before transferring the digested material to distillation apparatus (Kjeldahl apparatus) wash the apparatus with steam for 7 minutes with 100 % steam capacity.

7.8.1.2 Ensure that the condensor is connected to water tap for condensation of distillate.

7.8.1.3 Ensure that the priming of distillation unit is done as per SOP No. QC-101.

7.9.2 Distillation process

7.9.2.1 Place an Erlenmeyer/conical flask into position containing 50.0 mL of 0.01M HCl with 0.2 mL of methyl red solution.

7.9.2.2 Place the steam inlet tube into the digested sample tube.

7.9.2.3 Place the quick clamping device and place the digestion tube

into position.

7.9.2.4 Check the tight fit of the tube against the Viton cone.

7.9.2.5 Close the protection door and switch the instrument on.

7.9.2.6 Programme the system as per SOP No. QC-101. Select the required steps given below and run the distillation unit.

Step 1: In programme 0, while programming skip the addition of boric acid by setting the time as 00 seconds as this addition is not required for this procedure.

Step 2: Set the time as 02 seconds to deliver approximately 25.0 mL of water (collected from WFI loop).

Step 3: Set the time as 03 seconds to deliver approximately 30.0 mL of 10M NaOH.

Step 4: Set the reaction time as 00 min and 00 seconds.

Step 5: Set the distillation time as 04 min 00 seconds.

Step 6: Set the steam power as 80 %.

Step 7: Set the sample suction power as 20 seconds.

7.9.2.7 After completion of distillation and evacuation of distilled sample from the tube. Remove the conical flask from the distillation apparatus and titrate.

7.9.2.8 Clean the system as per SOP No. QC-101.

7.10 Titration

7.10.1 Observe the distillate and if the Hydrochloric acid is neutralized by the ammonia during distillation process, as indicated by yellow color of the distillate. In this case dilute the sample to two fold and perform the test.

7.10.2 Titrate the distillate of blank, followed by controls and sample collected in collection flask against 0.01 M Sodium hydroxide using 0.2 mL of Methyl red solution as an indicator.

7.10.3 The difference between the first and second titration value should not exceed 0.5 mL.

7.10.4 Record the results in RST No. QC-F-0277.

7.11 Calculation

Note:

a. Each mL of 0.01 M Hydrochloric acid is equivalent to 0.1401 mg of Nitrogen.

7.11.1 Calculate the protein nitrogen content by using the following formula.

$$PN2 = \frac{(\text{Blank} - \text{Sample}) \times 0.1401 \times \text{Determined molarity of } 0.01 \text{ N NaOH} \times \text{Dilution fold}}{\text{Volume of the sample} \times 0.01 \text{ N HCl}}$$

= _____ mg PN2 / mL

7.11.2 Calculate the Antigenic Purity of purified toxoid by the following formula.

Lf content of toxoid (Lf / mL)

Protein Nitrogen / mL

= _____ Lf / mg PN2

7.11.3 Calculate the percent recovery of standard by using the following formula.

$$\% \text{ Recovery} = \frac{\text{Observed content of PN2}}{\text{Expected content of PN2 (2 mg)}} \times 100$$

Expected content of PN2 (2 mg)

7.11.4 Enter the results in RST No. QC-F-00277.

7.12 Validity

7.12.1 The difference between the first and second titration value should not exceed 0.5 mL.

7.12.2 The test is valid only when the recovery of the standard is 95 to 105%.

7.13 Acceptance criteria

The antigenic purity of Diphtheria toxoid for vaccine should be $\geq 1500 \text{Lf/mgP N2}$.

8. REFERENCE

8.1 In- house

8.2 Handling of out of specification results. SOP No. QC- 005.

8.3 Operation of fume hood SOP No. QC- 047.

8.4 Operation of Gerhardt Kjeldahl apparatus SOP No. QC- 101.

8.5 Estimation of total protein nitrogen content of Diphtheria toxoid by semi - automatic Kjeldahl apparatus RST No: QC-F-027700.

8.6 Preparation and standardization of volumetric solutions: SOP No. QC- 055.

8.7 Guidelines and numbering system for the preparation of reagents, buffers, indicators and volumetric solutions. SOP No. QC-054.

1.0 |Objective: |
 |This method is used to determine O-acetyl content in Meningococcal |
 |polysaccharide samples. |
 2.0 |Scope: |
 |This SOP is applicable to determination of O acetyl content in |
 |meningococcal polysaccharide of serotypes A, C, W135 and Y samples. |
 3.0 |Responsibility: |
3.1	The analyst of the Quality Control laboratory shall be responsible
	for executing the described procedure correctly.
3.2	The supervisor of the laboratory shall be responsible for
	verifying that the procedure is followed correctly and for
	checking the results.
3.3	The Departmental Head shall be responsible for the overall
	compliance of the SOP and shall ensure that the SOP is reviewed
	every two years or if required earlier.
4.0	Abbreviations and Definitions:
4.1	QA
4.2	QC
4.3	SOP
5.0	Procedure:
5.1	Materials, reagents & Equipments
	5.5.1
	5.5.2
	5.5.3
	5.5.4
	5.5.5
	5.5.6
	5.5.7
	5.5.8
	5.5.9
	5.5.10
	5.5.11
	5.5.12
	5.5.13
	5.5.14
	5.5.15
5.2	Preparation of solutions
	5.2.1
	5.2.2
	5.2.3
	5.2.4
	5.2.5
	5.2.6
	5.2.7
	5.2.8
5.3	Principle of the Test:
	O-acetyl group react with hydroxylamine in alkali to from
	hydroxamic acid. The hydroxamic acid formed is measured by the
	formation of a colored complex with in the acidic solution.
5.4	Preparation of the assay
	Place series of standard solution from 3mg/ml Accetylcholine
	standard solution on the day of measurement into 10 ml glass tubes
	in duplicate according to the following table:
	S.No.
	Blank
	S 1
	S 2
	S 3
	S 4
	S 5
	5.4.1
	5.4.2
	5.4.3

		5.4.4	Add 1 ml of Reagent F to bring pH to 1.2 ±0.2, and vortex.
			Add 1 ml of Reagent D and vortex.
		5.4.5	Read the absorbance of purple Brown colour developed in
			Assay at 540 nm promptly.
		5.4.6	Note that all the readings of the assay tubes should be
			taken within 30 mins. of completion of the assay.
		5.5	Calculations
		5.5.1	Once the absorbance of the standards and the test samples
			have been taken, calculation shall be done using the excel
			sheet carrying formulas for the required calculation.
		5.5.2	Always copy the master excel sheet (sheet carrying
			formulas for the required calculation), never use the
			previously used excel sheet for new calculations.
		5.5.3	Copy the master excel sheet, keep it for use in respective
			folder and then fill in the required sample details (Batch
			number), sample dilution (if applicable), name of analyst,
			date, etc.
		5.5.4	Read the average absorbance of the samples against the
			standard curve.
			O-acetyl concentration $\mu\text{mol/ ml} = \{(\text{abs-intercept})/\text{slope}\} =$
			X
		5.5.5	Calculate the O-Acetyl concentration of the samples
			considering pre-dilution factor.
			O-acetyl concentration $\mu\text{mol/ ml}$
			= (X) x Diln. Factor x 1000/Volume sample
		5.5.6	Enter all parameters in the MS Excel assay log worksheet.
		5.6	Assay specifications
		5.6.1	The linearity of the standard curve is NLT 0.99
		5.6.2	The absorbance value of test samples should fall inside
			the range of standard curve.
		5.6.3	A coefficient of variation of less than 10% should be
			observed within the replicates.
		6.0	Reference
		6.1	Hestrin S. 1949: The reaction of Accetylcholine and other
			carboxylic acid derivatives with Hydroxylamine and its analytical
			application. Journal of biological chemistry, 180: 249(1949).
		6.2	WHO TRS 594: Requirement for Meningococcal polysaccharide vaccine.

[1.0]Objective: |
|To lay down the procedure for Immunoenzymetric Assay for the measurement of |
|Pichia pastoris Host cell proteins. |
[2.0]Scope: |
|This Standard Operating Procedure is applicable to the Active Raw Material |
|(ARM) of Hepatitis-B vaccine (rDNA). |
[3.0]Responsibility: |
3.1	The analyst of the Immunochemistry laboratory shall be responsible for
	executing the described procedure correctly.
3.2	The supervisor of the laboratory shall be responsible for verifying
	that the procedure is followed correctly and for checking the results.
3.3	The Departmental Head shall be responsible for the overall compliance
	of the SOP and shall ensure that the SOP is reviewed every two years
	for if required earlier.
[4.0]Abbreviations and Definitions:	
4.1	ELISA
4.2	HCP
4.3	ARM
4.4	QA
4.5	QC
4.6	SOP
4.7	TMB
4.8	HRP
[5.0] Procedure: Note: Make sure that all the reagents are at room temperature	
before use.	
5.1	Pipette 25µl of Standards, controls and samples into wells. Prepare
	1:10 dilution of test ARM. (100µl of test ARM is taken and 900µl of
	PBS-1X is added to it).
5.2	Pipette 100µl of anti-P. pastoris: HRP into each well.
5.3	Cover and incubate on dancing shaker at room temperature for 1 hour.
5.4	Aspirate the contents to waste. Blot and vigorously bang out residual
	liquid over absorbent paper. Wash generously with diluted wash
	solution by flooding the wells with solution by pipeting in 300µl.
	Dump and bang again. Repeat for a total of 4 washes. Wipe off any
	liquid from the bottom outside of the microtiter wells as any residue
	can interfere in the reading step.
5.5	Pipette 100µl of TMB substrate provided in the kit into the wells.
	Take care to add TMB in dark.
5.6	Cover and incubate the plate for 30 minutes at room temperature. Do
	not shake.
5.7	Pipette 100µl of stopping solution provided in the kit.
5.8	Read absorbance at 450/655nm using a ELISA reader.
5.9	Follow the given pathway to save the Absorbance obtained from ELISA
	reader:
	5.9.1
5.10	Raw
	data processing and interpretation of results:
	5.10.1
	5.10.2
	5.10.3
	5.10.4
	5.10.5
	5.10.6
	5.10.7
5.11	Reporting of results:
	5.11.1

			Host cell protein (ng/dose)
			= HCP concentration (ng/ml)* X Quantity of HBsAg per dose
			(ng/dose)
			Protein concentration (ng/ml)
			Where:
			Protein concentration of the sample = X ng/ml (Calculated
			using O.D. at 280 nm
			*Concentration of HCP (ng/ml) is taken from ELISA_3 program.
	5.12	Validity criteria:	
		5.12.1	The coefficient of regression must be greater than 0.96.
		5.12.2	The coefficient of variation between the replicates of same
		point of standard curve must be less than 10%.	
		If the assay meets the validity criteria, then accept it	
		otherwise reject and repeat it.	
6.0	References:		
	Kit Manual for Immunoenzymetric Assay for the Determination of Pichia		
	pastoris Host Cell Proteins, Cygnus Technologies, Inc.		

1.0 |Objective: |
 |To lay down the procedure for quantifying the phosphorus concentration. |
 2.0 |Scope: |
 |This Standard Operating Procedure is applicable to all the samples of PRP |
 |of Haemophilus Influenzae type b in which phosphorous concentration is to |
 |be determined. |
 3.0 |Abbreviations |
3.1	Abbreviations:		
	3.1.1	CV	:Coefficient of variation
	3.1.2	g	:Gram
	3.1.3	mg	:Milligram
	3.1.4	mL	:Millilitre
	3.1.5	µl	:Micro litre
	3.1.6	µg	:Microgram
	3.1.7	nm	:Nanometer
	3.1.8	PN	:Part Number
	3.1.9	SD	:Standard Deviation
	3.1.10	PRP	:Polyribosyl Ribitol Phosphate
4.0	Procedure:		
4.1	Preparation of solutions:		
	4.1.1	100 µg/mL phosphorus solution:	
		4.1.1.	Weigh 0.2194 g of potassium dihydrogen phosphate and
		1	transfer it to 500 mL volumetric flask.
		4.1.1.	Dissolve and make up the volume to 500 mL with purified
		2	water.
		4.1.1.	This solution is stored at room temprature and used
		3	within 6 months.
	4.1.2	5 µg/mL Phosphorus solution:	
		4.1.2.	Take 5 mL of 100 µg/mL phosphorus solution and make up
		1	the volume to 100 mL with purified water. This solution
			is freshly prepared.
	4.1.3	25 mg/mL ammonium molybdate solution:	
		4.1.3.	Weigh 625 mg of ammonium molybdate and transfer it to
		1	25 mL volumetric flask.
		4.1.3.	Dissolve and make up the volume to 25 mL with purified
		2	water.
		4.1.3.	This solution is stored at room temperature and used
		3	within 6 months.
	4.1.4	100 mg/mL ascorbic acid solution:	
		4.1.4.	Weigh 2.5 g of ascorbic acid and transfer it to 25 mL
		1	volumetric flask.
		4.1.4.	Dissolve and make up the volume to 25 mL with purified
		2	water.
		4.1.4.	This solution is freshly prepared.
		3	
	4.1.5	3 M Sulphuric acid:	
		4.1.5.	Take approximately 500 mL of purified water in a 1
		1	litre volumetric flask. To this add 162 mL concentrated
			Sulphuric acid slowly.
		4.1.5.	Make up the volume to 1 litre with purified water.
		2	
		4.1.5.	This solution is stored at room temperature and used
		3	within 6 months.
	4.1.6	Ammonium molybdate reagent:	
		5.1.6.	Take 1 volume of 25g/l solution of ammonium molybdate,
		1	1 volume of 100 g/l solution of ascorbic acid, 1 volume
			of 3 M Sulphuric acid and 2 volumes of purified
			water. This solution is freshly prepared.
4.2	Preparation of Calibration Curve:		
	4.2.1	Prepare calibration curve (2.5-10 µg) with freshly prepared	
		5µg/mL phosphorus solution in previously labelled micro	
		centrifuge tube. Calibration curve is prepared in duplicates	
		as follow:	
		Phosphorus content (µg)	2.5
		5µg/mL phosphorus solution (µl)	500
		Note: The Calibration curve is accepted if it is meeting the	
		parameters of correlation coefficient i.e. ≥ 0.990	
4.3	Test:		
	4.3.1	Dilute the sample so that the sample volumes (1mL) used in	
		test contain about 6 µg of phosphorus i.e. lies within the	
		calibration curve.	
		(In case of dry polysaccharide prepare a solution containing	
		about 1-5 mg/mL of polysaccharide and use it as test sample)	
	4.3.2	Treat 2 mL purified water as blank in thermoresistant test	
		tube.	
	4.3.3	Add 0.2 mL of concentrated sulphuric acid to tubes of	
		calibration curve, Blank and sample.	
	4.3.4	Heat the tubes in oil bath or turbotherm at 120°C for 1 hour	
		and then at 160°C until white fumes appear (about 1 hour).	
	4.3.5	Add 0.1 mL of perchloric acid to all the tubes and heat the	
		tubes at 160°C until the solution is decolourised (about 90	
		minutes).	
	4.3.6	Cool at room temperature and add 4 mL of purified water and 4	
		mL of ammonium molybdate reagent.	
	4.3.7	Heat in water bath at 37°C for 90 minutes and cool.	
	4.3.8	Transfer the contents of each tube to 10 mL volumetric flasks	
		and make up the volume of each tube to 10 mL with purified	
		water.	
	4.3.9	Set the wavelength in spectrophotometer at 820 nm. Zero the	
		absorbance of blank and then measure the absorbance of	

		calibration curve and test samples. The blue colour is stable
		for several hours.
	4.4 Calculations:	
	4.4.1 Once the absorbance of the standards and the test samples	
	have been taken, calculation shall be done using the excel	
	sheet carrying formulas for the required calculation.	
	4.4.4 Enter the absorbance for calculations starting from the	
	Calibration curve prepared between 2.5 and 10 g of phosphorus	
	and test samples. The calculations are made according to the	
	equation of minimum squares.	
	$Y = mX + b$.	
	Where: m = slope	
	b = intercept	
	Y = absorbance	
	X = concentration	
	(Multiply by dilution factor)	
	4.4.5 Calculate the content of phosphorus (per mL) in the test	
	sample by multiplying with the dilution factor.	
	4.4.6 In case of PRP,	
	Phosphorous (%) in PRP on dry weight basis:	
	Phosphorous content in PRP($\mu\text{g/ml}$)	
	Phosphorous content in dry PRP =	

	$(\mu\text{g/mL}) \times 1 - \{\text{Moisture}\}$	
	$(\%) / 100$	
	Phosphorous content in dry PRP ($\mu\text{g/mL}$)	
	Phosphorous (%) = _____ x	
	100	
	Concentration of PRP solution ($\mu\text{g/mL}$)	
	4.5 Acceptance criteria:	
	The coefficient of variation (CV) between the replicates of the	
	samples should be $\leq 5\%$.	
	5.0 Forms and Records:	
	Determination of Phosphorous Concentration	
	6.0 Reference:	
	European pharmacopoeia.	

1.0 |Objective |
|To lay down the procedure for estimation of residual formaldehyde content. |
2.0 |Scope |
|This Standard Operating Procedure is applicable to all the samples in which|
|residual formaldehyde content is to be determined. |
3.0 |Responsibility |
3.1	The analyst is responsible for executing the described procedure
	correctly.
3.2	Supervisor is responsible for verifying that the procedure is carried
	out correctly and check the results.
3.3	HOD shall be responsible for overall compliance of this SOP and shall
	ensure that the SOP is reviewed every two years or if required
	earlier.
4.0	Abbreviations and Definitions
4.1	Abbreviations:
	4.1.1
	4.1.2
	4.1.3
	4.1.4
	4.1.5
	4.1.6
	4.1.7
	4.1.8
	4.1.9
5.0	Procedure
5.1	Preparation of solutions: Prepare all the solutions as described below
	and fill the form, label the solutions and assign lot numbers.
	5.1.1
	5.1.2
	5.1.3
	5.1.4
	5.1.5
5.2	Testing Procedure:
	5.2.1
	5.2.2
	5.2.3

1.0 |Objective |
 |To lay down the procedure for Estimation of the Nucleic acid content in |
 |Meningococcal polysaccharide samples. |
 2.0 |Scope |
 |This Standard Operating Procedure is applicable to all the samples of |
 |Meningococcal polysaccharide in which Nucleic acid is to be determined, |
 |which is received in the Quality control department. |
 3.0 |Responsibility |
3.1	The analyst is responsible for executing the described procedure
	correctly.
3.2	Supervisor is responsible for verifying that the procedure is carried
	out correctly and check the results.
3.3	HOD shall be responsible for overall compliance of this SOP and shall
	ensure that the SOP is reviewed every two years or if required
	earlier.
4.0	Abbreviations and Definitions
4.1	Abbreviations:
	4.1.1
	4.1.2
	4.1.3
	4.1.4
	4.1.5
	4.1.6
	4.1.7
	4.1.8
	4.1.9
	4.1.10
4.2	Reagents:
	4.2.1
4.3	Equipment
	4.3.1
5.0	Procedure
5.1	Preparation of solutions: Not Applicable
5.2	Testing procedure:
	5.2.1
5.3	Calculations:
	5.3.1
	50
	5.3.2
5.4	Validity of test:
	5.4.1
5.5	Acceptance criteria:
	5.5.1
	5.5.2
6.0	Forms and Records
Determination of Nucleic acid content - Annexure 1	
6.0	Reference
6.1	European Pharmacopoeia, volume 5, Nucleic Acid in polysaccharide
	vaccines, 2.5.17.
6.2	WHO TRS No. 977 Annexure 3: Recommendations to assure the quality,
	safety and efficacy of Meningococcal conjugate vaccines

| 1.0|Objective |
| |To lay down the procedure for abnormal toxicity test in vaccines. |
2.0 |Scope |
| |This Standard Operating Procedure is applicable to all the batches of |
| |vaccines. |
3.0 |Responsibility |
	3.1	The QC analyst shall be responsible for performing the described
	procedure correctly.	
	3.2	Supervisor shall be responsible for verifying that the same has been
	carried out correctly.	
	3.3	Head Animal House/Veterinarian/supervisor shall be responsible for
	supply of quality animals and their maintenance.	
	3.4	Head QC or his designee shall be responsible for overall compliance
	of the SOP.	
	3.5	Head QA or his designee shall be responsible for the verification of
	results.	
4.0	Abbreviations and Definitions	
	4.1	SOP
	4.2	QA
	4.3	QC
	4.4	AH
	4.5	G. pig
5.0	Procedure:	
	5.1	Equipment, Materials and Reagents
		5.1.1
		5.1.2
		5.1.3
		5.1.4
		5.1.5
		5.1.6
		5.1.7
	5.2	Place request for Dunken hartley Guinea pigs (250-350g) and Swiss
	mice (17-22g) to the Head Animal House. Receive the animals after	
	health status evaluation of animals by Veterinary officer.	
	5.3	Selection of animals for test:
		5.3.1
		pigs weighing 250-350 g.
		5.3.2
		of Swiss mice.
		5.3.3
		AH/005.
		5.3.4
		labels as per SOP AH/093.
	5.4	Injection:
		Note: Put the test number in Annexure 1 before preparation of
	dilutions as XXXX/YNNN	
		Where, XXXX is ABTX (Abnormal Toxicity Test),YY is the last two digit
	of the year e.g. 12 for year 2012 and NNN is consecutive test number	
	for the year YY.	
		5.4.1
		intraperitoneally. In case the vaccine contains pertussis as
		one of the components inject each mouse with half of the
		single humane dose of vaccine intraperitoneally.
		5.4.2
		intraperitoneally.
	5.5	Observations:
		5.5.1
		pigs twice daily for 7 days, remove and euthanize all the
		animals in moribund stage and mark it as dead. Record in daily
		observation sheet as per Annexure 2.
		5.5.2
		day and 7th day. Compile the report as per Annexure 1 in
		reference with Annexure 2.
		5.5.3
	5.6	Acceptance criteria:
		The sample passes the test, if none of animals dies or shows signs of
	ill health in seven days following the injection.	
	5.7	Retest Criteria:
		5.7.1
		health. The preparation passes the test if none of the animals
		in the repeat test dies or shows signs of ill health in seven
		days following the injection.
	5.8	Upon completion of the test and approval of the test results by Head
	QC.	

1.0 |Objective |
 |To lay down the procedure for testing the potency of whole cell pertussis |
 |vaccine by challenge method in Swiss albino mice. |
 2.0 |Scope |
 |This Standard Operating Procedure is applicable to all the batches of whole |
 |cell pertussis vaccine. |
 3.0 |Responsibility |
 |3.1 |The QC analysts shall be responsible to carry out the described |
 |procedure. |
 |3.2 |The supervisor shall be responsible for ensuring that the same is |
 |carried out correctly. |
 |3.3 |Head Animal House/Veterinarian/supervisor shall be responsible for |
 |supply of quality animals and their maintenance. |
 |3.4 |Head QC or his designee shall be responsible for overall compliance of |
 |the SOP. |
 4.0 |Abbreviations and Definitions |
4.1	SOP	: Standard Operating Procedure
4.2	QC	: Quality Control
4.3	mL	: Millilitre
4.4	g	: Gram
4.5	°C	: Degree centigrade
4.6	IU/mL	: International unit per milliliter
4.7	LD50	: 50% Lethal Dose
4.8	QA	: Quality Assurance
4.9	AH	: Animal House
4.10	EQP	: Equipment
4.11	I.O.U	: International Opacity Units
4.12	%	: Percent
4.13	QA	: Quality Assurance
4.14	AH	: Animal House
4.15	ED50	: 50% Effective Dose
4.16	S.H.D	: Single Human Dose
4.17	BG media	: Bordet Gengou media
4.18	NRS	: National Reference Standard
4.19	EDQM	: European Directorate for the Quality of Medicines &
HealthCare		
5.0	Procedure:	
5.1	Equipment, Materials and Reagents	
5.1.1	Weighing balance	
5.1.2	Microlab 501A dispenser system	
5.1.3	Incubator 35±2 °C	
5.1.4	Microscope	
5.1.5	Biosafety cabinet	
5.1.6	Micropipette	
5.1.7	Vortex mixer	
5.1.8	Reference Standard of Pertussis Vaccine (NIBSC code: 94/532)	
5.1.9	Test tubes (Size 18 x 150 mm, 20 x 200 mm)	
5.1.10	Micropipette tips	
5.1.11	Glass vials (5 mL, 10 mL, 100 mL)	
5.1.12	Disposable syringes with needle 22/23 gauge 1 inch.	
5.1.13	Test tube racks	
5.1.14	Hamilton syringes	
5.1.15	Hamilton 25 G, 5.1 mm needles	
5.1.16	Round bottom 500 mL capacity flask	
5.1.17	Glass beads	
5.1.18	Nylon cloth	
5.1.19	Inoculating loop	
5.1.20	International Opacity Standard (NIBSC code: 76/522)	
5.1.21	Cotton	
5.1.22	Pasteur pipettes	
5.1.23	70% Isopropyl alcohol	
5.1.24	Bordetella pertussis challenge strain ATCC 18323 (Freeze dried)	
5.1.26	Sodium Chloride	
5.1.27	Bordet Gengou Agar Base	
5.1.28	Casein acid hydrolysate	
5.1.29	Thermal box	
5.2	Preparation of solutions	
5.2.1	Physiological saline	
Add 0.9 gm of weighed Sodium Chloride to 50ml of purified water,		
mix well and make up it to 100ml.		
5.2.2	1% Casamino Acid Solution	
5.2.3	Bordet Gengou Medium	
5.3	Place request for Swiss Albino mice (13-16g) to the Head Animal House.	
After health status evaluation of animals by Veterinary officer, receive		
the animals.		
5.4	Selection of animals for test:	
5.4.1	Select 24 Swiss albino mice per dilution weighing 13-16 g.	
Difference in the weight of mice should not be more than 4 g.		
5.4.2	Preferably choose mice of the same sex; otherwise equally	
distribute males and females in each group.		
5.4.3	Distribute mice in three groups of 24 mice each per dilution for	
each batch of test sample and reference standard. Four groups of		
10 mice each are to be used for LD50 determination. All the mice		
in various groups should be distributed randomly.		
5.4.4	Mark each group of mice with appropriate color scheme. The cages	
housing the mice shall bear labels		
5.5	Preparation of reference dilution :	
Raise the request for Reference Standard required for testing and record		
its utilization. Take the vial of reference standard stored in		
temperature 5±3°C and place the reference standard in thermal box with		

		ice pack and transfer the reference material to room no. 15 of Animal				
		House Experimental for the preparation of dilutions in biosafety				
		cabinet.				
		5.5.1	Prepare the biosafety cabinet for aseptic operation.			
		5.5.2	Reconstitute the reference standard with normal saline so as to			
		contain 8 I.U. per mL.				
		5.5.3	Take 3 glass vials and label them as 1:4, 1:20, and 1:100.			
		Add 15 mL of physiological saline in to 1st vial and 16 mL				
		remaining 2nd, 3rd vials.				
		5.5.4	Add 5.0 mL of the reconstituted reference vaccine to the 1st vial,			
		mix properly with vortex mixer. (1/4 dilution)				
		5.5.5	Take 4 mL of 1/4 dilution and mix it thoroughly in 16 mL of			
		physiological Saline (2nd vial) with vortex				
		mixer (1/20 dilution).				
		5.5.6	Take 4 mL of 1/20 dilution and mix it thoroughly in 16 mL of			
		physiological Saline (3rd vial) with vortex mixer (1/100				
		dilution).				
		5.6	Preparation of test vaccine dilution:			
		5.6.1	Based on the Opacity units (IOU/mL) of the Pertussis Bulk samples,			
		formulate the sample to contain 32 IOU/mL with adjuvant as per				
		Annexure 4.				
		5.6.2	Take 3 vials and label them 1:4, 1:20 and 1:100			
		5.6.3	Add 15 mL of physiological saline in to 1st vial and 16mL each in			
		remaining 2nd and 3rd vial.				
		5.6.4	Add 5.0 mL of the formulated vaccine to the 1st vial, mix properly			
		with vortex mixer. (1/4 dilution)				
		5.6.5	Take 4 mL of 1/4 dilution and mix it thoroughly in 16 mL of			
		physiological Saline (2nd vial) with vortex				
		mixer (1/20 dilution).				
		5.6.6	Take 4 mL of 1/20 dilution and mix it thoroughly in 16 mL of			
		physiological Saline (3rd vial) with vortex mixer (1/100				
		dilution).				
		5.7	Immunization:			
		5.7.1	Inject each mouse in each group with 0.5 mL of appropriate			
		dilution of reference and vaccine intraperitoneally.				
		5.7.2	Keep the four groups of 10 mice each as uninoculated for LD50			
		(control group).				
		Keep the immunized animals under observations for 14 days. The				
		mortality, if any in the immunized mice before challenge should				
		not exceed 6% in each dilution of test as well as reference				
		vaccine.				
		5.7.3	Observe the animals daily twice (morning and evening) and record			
		the observation in Annexure 2 and 3				
		5.8	Preparation of Challenge culture:			
		5.8.1	Start preparing the challenge culture under aseptic conditions six			
		days prior to challenge. All the activities are to be carried out				
		in biosafety cabinet.				
		5.8.2	Reconstitute one ampoule of Bordetella pertussis challenge strain			
		(WC) 18323 in 6 – 7 drops of sterilized casamino acid solution.				
		5.8.3	Inoculate two B.G. plate with the reconstituted B.pertussis			
		challenge culture. Incubate the plate at 35 ± 2 0 C for				
		three days.(Revival)				
		5.8.4	On completion of incubation, check the growth and its purity			
		microscopically after Gram staining and subculture on two fresh				
		B.G. slopes/plates. Record the Grams staining observations.				
		Incubate the slopes/plates at 35 ± 2 0 C for 48 hours. (1st				
		subculture)				
		5.8.5	Check the purity of 48 hour growth microscopically after Gram			
		staining and make fresh subcultures on B. G. slopes/plates. Record				
		the Grams staining observations. Incubate the slopes/plates at 35				
		± 2 0 C for 24 hours. (2nd subculture)				
		5.8.6	Check the purity of 20- 24 hours old culture of B. pertussis			
		microscopically after Gram staining. Record the Grams staining				
		observations. This culture will be used for the challenge.				
		5.9	Preparation of challenge dose:			
		5.9.1	On the day of challenge, take the culture and wrap the slants/			
		plates of Bordet Gengou media in aluminium foil carefully and				
		keep it in thermal box. Take the thermal box to Animal house				
		experimentation, room no. 15 for preparation of dilutions in				
		biosafety cabinet.				
		5.9.2	Harvest the growth of B. pertussis culture obtained using curved			
		Pasteur pipette and suspends it into 5 mL Casamino acid in				
		sterile test tube with some glass beads.				
		5.9.3	Vortex the tube containing culture and filter the bacterial			
		suspension through sterile nylon cloth under strict aseptic				
		conditions to remove any clumps from the suspension.				
		5.9.4	Adjust the opacity of the suspension to 10 I.O.U. per mL by			
		comparing the opacity with International Opacity Reference				
		Standard visually.				
		5.9.5	From the preparation at (5.9.4), prepare the following dilutions			
		of challenge				
		Culture.				
		S.No.	No. of Organisms	Dilution	Challenge organism	Diluent in
				suspension in mL	mL	
		A	109/mL	10 Fold	1.0 (10 IOU/mL)	9.0
		B	108/mL	100 Fold	1.0 (A)	9.0
		C	107/mL	1000 Fold	2.0 (B)	18.0
		D	1,00,000/ 0.03 mL	3000 Fold	10.0 (C)	20.0
				(Approximately)		
				Challenge dose		
		E	10,000 / 0.03 mL	30000 Fold	1.0 (D)	9.0
		F	2,000 / 0.03 mL	1.5 × 105	2.0 (E)	8.0

			Fold						
		G	400 / 0.03 mL	7.5 × 10⁵	2.0 (F)	8.0			
			Fold						
		H	80 / 0.03 mL	37.5 × 10⁵	2.0 (G)	8.0			
			Fold						
		5.9.6	Vortex well each dilution and transfer each dilution to vial						
		individually.							
		5.10	Inoculation of challenge dose:						
		5.10.1	14-17 days after immunization, inject intracerebrally 0.03 mL						
		of the challenge dose(1,00,000 organisms/0.03mL) into all the							
		groups of immunized mice using Hamilton Microlab 501A dispenser							
		system or Hamilton syringe and 25 G, 5.1 mm needles.							
		5.10.2	Observe the animals twice daily for 14 days and record the						
		deaths. If any animal found in moribund stage euthanize the							
		animal and mark them as dead. Exclude the animals for							
		calculation dying within three days from the day of challenge.							
		5.11	LD50 Control:						
		5.11.1	Mark four groups of 10 mice each (selected on the day of						
		immunization with reference and test vaccine) appropriately.							
		5.11.2	Inject each of four groups with 0.03 mL of dilutions containing						
		10000, 2000, 400 and 80 organisms per 0.03 mL respectively.							
		5.11.3	Observe the animals twice daily for 14 days and record the						
		deaths on each day.							
		5.11.4	Exclude the deaths occurring within three days from the day of						
		challenge.							
		5.12	Percent Viable Count Determination:						
		5.12.1	Spread 0.1 mL of dilution containing 80 organisms per 0.03 mL						
		on two B.G. plates before challenge and on two plates after							
		completion of the challenge.							
		5.12.2	Incubate the plates at 35 ± 2 °C for 72 hours.						
		5.12.3	On completion of incubation period, count the number of						
		colonies on each plate and calculate the average.							
		5.12.4	Calculate the percent viable count by dividing the average no.						
		of colonies by 266 (total no. of organisms present in 0.1 mL of							
		dilution containing 80 organisms per 0.03 mL) and multiplying							
		with 100.							
		5.13	Observations:						
		Daily observe all the inoculated mice twice and record the sick and							
		deaths.							
		5.14	Calculations:						
		Count the number of surviving mice 14 days after the injection of the							
		challenge dose and calculate the potency of the test sample being							
		examined relative to the potency of the standard preparation on the							
		basis of the number of surviving mice in each of the three groups of							
		twenty-four, using standard statistical method (CombiStats Software,							
		EDQM). Determination of LD50 value by Reed and Muench formula.							
		Example : Suppose four groups of ten mice in each are injected with							
		doses expressed in number of organisms i.e. 10,000; 2000; 400; 80 (The							
		immunized mice are challenged with the virulent pertussis bacteria in a							
		dose containing 1,00,000 organisms)							
		Dose (No	Total	Observations for	Accumulated	Total	%		
		of	No of	14 days	observations	Mortality	Mortality		
		organisms)	animals			y	y		
			Deaths	Survival	Deaths	Survival			
				s	s				
		2000	10	9	1	17	1	17/18	94.44
		400	10	7	3	8	4	8/12	66.66
		80	10	1	9	1	13	1/14	7.14
		Proportional Distance (P.D.) = 50 – less than 50 x Log 5							
		More than 50 – less than 50							
		= (50 – 7.14)/ (66.66-7.14) x 0.69897							
		= 0.5033							
		Log LD50= Log of dose just below 50 % mortality + P.D.							
		= Log 80 + P. D							
		= 1.9030 + 0.5033= 2.406							
		LD50 = Antilog (2.406)							
		= 254 organisms							
		Total LD50 used = 100000/254 = 393							
		5.15	Validity of the test: The test is considered valid, if						
		5.15.1	It fulfills the criteria of Linearity.						
		5.15.2	It fulfills the criteria of Parallelism.						
		5.15.3	50 % protective doses lie between the largest and smallest						
		doses of the preparation.							
		5.15.4	The challenge dose contains 100 -1000 LD50.						
		5.15.5	The percent viable count ranges from 12.5% to 62.5%.						
		5.16	Criteria for acceptance: The sample passes the test, if:						
		The sample passes the test the requirement for potency, if the results							
		of the statistically valid assay so that the estimated potency of the							
		vaccine is not less than 4.0 IU per single human dose and the lower							
		fiducial limit (P=0.95) of the estimated potency is not less than 2.0							
		IU							
		5.17	Retest policy: Repeat the test if:						
		5.17.1	The test is invalid; it does not fulfill one or more validity						
		criteria.							
		5.17.2	The test is valid but the results are out of specification.						
		repeat the test once. The results of all valid tests are							
		combined to calculate the potency by the geometric mean							
		estimation and its lower fiducial limit.							
		5.17.3	The geometric mean of the valid tests is within the prescribed						
		limits, the vaccine batches pass the test. But in case the							
		geometric mean is still below the prescribed limit, the vaccine							
		batches fail the potency test.							

6.0	Forms and Records	
	Details of Potency Test for Pertussis Component by Challenge Method in	
	Combined Vaccines.	
7.0	References	
	7.1	Indian Pharmacopoeia.
	7.2	World Health Organization, Technical Report Series No. 800, 927, 941.
	7.3	WHO/VSQ/97.04 Manual of laboratory methods - for testing vaccines used
		in the WHO EPI

1.0	Objective			
	To lay down Procedure of Citrate Utilization Test.			
2.0	Scope			
	This Standard Operating Procedure is applicable to identification of			
	citrate utilization organism for all samples.			
3.0	Responsibility			
	3.1	The Analyst shall be responsible for carrying out the procedure as		
	established in this SOP.			
	3.2	Supervisor shall be responsible for verifying that the same has been		
	carried out correctly.			
	3.3	Head QC or his designee shall be responsible for overall compliance		
	of the SOP.			
4.0	Abbreviations and Definitions			
	4.1	SOP	: Standard Operating Procedure	
	4.2	QA	: Quality Assurance	
	4.3	QC	: Quality control	
	4.4	mL	: milliliter	
	4.5	µL	: microliter	
5.0	Procedure:			
	5.1	Equipment, Materials and Reagents		
		5.1.1	Autoclave	
		5.1.2	Positive control: Klebsiella pneumoniae ATCC 13883	
		5.1.3	Citrate Negative: Escherichia coli ATCC 25922	
		5.1.4	Simmons Citrate Agar (M099,Himedia)	
		5.1.5	Physiological saline (0.9% w/v)	
		5.1.6	Glass tubes for slant	
		5.1.7	Weighing balance	
		5.1.8	pH meter	
		5.1.9	Magnetic stirrer with hot plate	
	5.2	Preparation of solutions		
		5.2.1	Simmons Citrate Agar	
			5.2.1.1	Weigh the 28.24g Simmons Citrate Agar powder and
				dissolve in 700ml of Deionized water.
			5.2.1.2	Deionized water pH should between 6.5-7.0
			5.2.1.3	Gently heat, with mixing, to boiling until agar is
			dissolved.	
			5.2.1.4	Dispense the media 4.0 to 5.0 ml into 16-mm tubes.
			5.2.1.5	Autoclave at 121 degree C under 15 psi pressure for 15
			minutes	
			5.2.1.6	Cool in slanted position (long slant, shallow butt).
			5.2.1.7	Store the slant in refrigerator (2-8°C).
	5.3	Test procedure		
		5.3.1	Streak the slant with a light inoculum picked from the center	
			of a well-isolated colony and Parllely streak the negative and	
			positive control organism the slants.	
		5.3.2	Incubate the slants aerobically at 35 to 37°C for 4-7 days.	
		5.3.3	Observe a color in the slants.	
	5.4	Interpretation of results		
		5.4.1	Positive Reaction: Growth with color change from green to	
			intense blue in the slant.	
		5.4.2	Negative Reaction: No growth and No color change; Slant remains	
			green.	
			pic	
6.0	Forms and Records			
	Test for Citrate Utilization Test			
7.0	References			
	Clinical Microbiology Procedures Handbook 2nd Edition			

1.0 |Objective |
 |To lay down the procedure for Estimation of Moisture Content in |
 |Pneumococcal polysaccharide sample by Karl Fischer titration method. |
 2.0 |Scope: |
 |This standard operation procedure is applicable to all the samples of |
 |Pneumococcal polysaccharide in which moisture content is to be determined. |
 3.0 |Responsibility |
3.1	The analyst is responsible for executing the described procedure
	correctly.
3.2	Supervisor is responsible for verifying that the procedure is carried
	out correctly and check the results.
3.3	HOD shall be responsible for overall compliance of this SOP and shall
	ensure that the SOP is reviewed every two years or if required
	earlier.
4.0	Abbreviations and Definitions
4.1	Abbreviations:
	4.1.1
	4.1.2
	4.1.3
	4.1.4
	4.1.5
	4.1.6
	4.1.7
	4.1.8
5.0	Procedure
5.1	Preparation of solutions: Not applicable
5.2	Testing procedure:
	Note: Operate the instrument as per the manual.
	5.2.1
	5.2.2
	5.2.3
	5.3
	The test sample should have the moisture content below the approved
	specifications
6.0	Forms and Records
Determination of Moisture content - Annexure 1	
7.0	Reference
7.1	Manual for operation of KF instrument
7.2	European Pharmacopoeia

1.0|Objective: |
 |To lay down the procedure for the detection of bacterial endotoxin in |
 |samples by Gel clot method. |
 2.0|Scope |
 |This Standard Operating Procedure is applicable to all Active Raw Material |
 |and finish product samples of vaccine. |
 3.0|Responsibility |
3.1	The analyst shall be responsible for carrying out the described
	procedure correctly.
3.2	The Supervisor of the laboratory shall be responsible for verifying
	that the procedure is complied with and for checking the results.
3.3	The departmental head/designee shall be responsible for overall
	compliance of the SOP.
3.4	Head QA/Designee shall be responsible for final approval and
	subsequent compliance of the SOP.
4.0	Abbreviations and Definitions
4.1	ARM
4.2	EU
4.3	LAL
4.4	LRW
4.5	MVD
4.6	OD
4.7	PPC
4.8	DQA
4.9	DQC
5.0	Equipments, Materials and Reagents
5.1	Vortex Mixer
5.2	Tube heating block (37°C)
5.3	Refrigerator (2-8°C)
5.4	Deep Freezer (-20.0 + 5°C)
5.5	Depyrogenized glass dilution tubes
5.6	Depyrogenized glass reaction tubes
5.7	Depyrogenized tips
5.8	Autopipettes
5.9	Test tube rack
5.10	Timer
5.11	Limulus Amoebocyte Lysate, Pyrotell
5.12	Standard Endotoxin {Concentration as per certificate of analysis
	(when reconstituted with LAL reagent water)}
5.13	LAL reagent water (LRW)
6.0	Preparation of solutions:
Not applicable	
7.0	Procedure:
Note: Make sure that all the reagents are at room temperature before use.	
7.1	Clean the working shelf properly with 70% Isopropyl alcohol and keep
	all the materials and reagents.
7.2	Preparation of Limulus Amoebocyte Lysate:
7.2.1	This reagent must be reconstituted just prior to use.
7.2.2	Reconstitute the lyophilized lysate by adding LAL reagent
	water as per label directions. Swirl gently but thoroughly for
	at least 30 seconds.
	Note: Do not shake, as the contents will foam.
7.2.3	The reconstituted lysate can be stored for 24 hours at 2-8°C
	without loss of sensitivity or at -20°C for four weeks. Frozen
	liquid lysate should be thawed immediately before use and
	repeated thawing should be avoided.
7.3	Preparation of endotoxin standard.
7.3.1	Take an endotoxin vial. Reconstitute endotoxin control with
	LAL reagent water, as per label directions. The Concentration
	of standard endotoxin is as per the Certificate of Analysis
	(when reconstituted with LAL reagent water). For example, the
	concentration of endotoxin is 2500 EU/vial as per Certificate
	of Analysis. When reconstituted with 5 ml of LAL reagent water
	as per the label instructions, the concentration becomes
	2500EU/5ml or 500EU/ml
7.3.2	Vortex the vial vigorously for 1 minute, and then at 5-10
	minute intervals over a 30-60 minutes period at room
	temperature.
7.3.3	Dilute the reconstituted endotoxin with LAL reagent water to a
	concentration of 1 EU/ml.
7.3.4	Using 1 EU/ml endotoxin solution, prepare a serial two-fold
	dilution series in LAL Reagent water till two points lower than
	the endpoint of the lysate is reached.
	Dilution series of Standard endotoxin
	Tube
	number
	water
	1.
	2.
	3.
	4.
	5.
	6.
	Note: Each dilution should be vortex for at least 30 seconds
	prior to proceeding to the next dilution.
7.4	Preparation of samples:
For this, calculate the Maximum Valid Dilution (MVD) of the sample in	
which endotoxin has to be detected. The formula for calculating the MVD	
value is given below:	
	pic

	where:				
		[pic] = Endotoxin limit of sample			
		[pic] = Potency (concentration) of sample in 1 ml			
		[pic] = Sensitivity of LAL			
	Then the sample is diluted to its MVD value using LAL Reagent water.				
	7.5	Preparation of Product Positive Control (PPC):			
	For this, dilute the samples according to MVD/2 value with LAL Reagent				
	Water. Then take 0.2 ml of this dilution and add 0.2 ml of 4 λ standard				
	endotoxin.				
	7.6	Mark the reaction tubes for different dilutions of standard endotoxin,			
	all product positive controls, samples and negative controls in				
	duplicates.				
	7.7	Carefully transfer 0.1 ml of different dilutions of standard endotoxin,			
	sample (appropriately diluted to MVD value), PPCs and negative control				
	into the appropriately marked reaction tubes.				
	7.8	Then add 0.1 ml of the reconstituted lysate to each tube beginning with			
	the highest concentration of standard endotoxin followed by negative				
	control, test samples and PPCs.				
	7.9	Following the addition of lysate, incubate the tubes at 37°C for 1 hour			
	in a tube heating block.				
	Note: Assay tubes should not be disturbed prior to the specified time				
	of 1 hour.				
	7.10	After 1 hour of incubation examine each reaction tube for gelation.			
	7.11	A positive reaction is indicated by a firm gel that remains intact			
	momentarily when tube is inverted at 180°.				
	7.12	Results:			
	Gel Clot Method: The results are expressed as endpoint (the endpoint				
	dilution is the last dilution of endotoxin which still yields a				
	positive result).				
	Example for reporting results with Lysate of sensitivity 0.06 EU/ml.				
	Rep.	Endotoxin Dilutions (EU/ml)		PPC	Endpoint
		LRW	Sample		EU/ml

1.0|Objective: |
 |To lay down the procedure for the Determination of Yeast DNA in Samples of |
 |Active Raw Material of Hepatitis B vaccine (Recombinant) by PCR Method. |
 2.0|Scope: |
 |This Standard Operating Procedure is applicable to the Active Raw Material |
 |(ARM) of Hepatitis-B vaccine (rDNA). |
 3.0|Responsibility: |
3.1	The analyst shall be responsible for executing the described
	procedure correctly.
3.2	The supervisor of the laboratory shall be responsible for verifying
	that the procedure is followed correctly and for checking the results.
3.3	The Departmental Head shall be responsible for the overall compliance
	of the SOP .
4.0	Abbreviations and Definitions:
4.1	PCR
4.2	ARM
4.3	TBE
4.4	Sec
4.5	Min
4.6	SOP
4.7	DNA
4.8	UV
5.0	Procedure:
5.1	Safety considerations:.
	5.1.1
	5.1.2
	5.1.3
5.2	Equipments:
	5.2.1
	5.2.2
	5.2.3
	5.2.4
	5.2.4
5.3	Materials:
	5.3.1
	5.3.2
	5.3.3
	5.3.4
	5.3.5
	5.3.6
	5.3.7
	5.3.8
5.4	Reagents:
	5.4.1
	5.4.2
	5.4.3
	5.4.4
	5.4.5
	5.4.6
	5.4.7
	5.4.8
	5.4.9
	5.4.10
5.5	Preparation of solutions:
	5.5.1
	5.5.2
	5.5.3
5.6	Test procedure:
	5.6.1

			5.6.1.	Take 2 μ L from Eppendorf No.2 and repeat the procedure	
			3	as above until Eppendorf No.7.	
			5.6.1.	Thus serial dilutions of the Yeast DNA prepared are as	
			4	follows:	
				pic	
			5.6.1.	Thus making the concentration in the seven eppendorfs	
			5	as:	
				Eppendorf No.	
				1	
				2	
				3	
				4	
				5	
				6	
				Dilution	
				10-1	
				10-2	
				10-3	
				10-4	
				10-5	
				10-6	
				Conc. of DNA	
				700ng	
				70 ng	
				7 ng	
				0.7ng	
				0.07 ng	
				7pg	
		5.6.2	These dilutions shall be used as template for DNA		
			amplification by PCR as positive control. Purified water		
			shall be used as template in case of the negative control.		
		5.6.3	Prepare the PCR reaction as follows:		
			(for 9 rxns)		
			Reagents		
			Volume to be added		
			Taq polymerase		
			4.5 μ L		
			Yeast DNA F1 primer		
			4.5 μ L		
			Yeast DNA R1 primer		
			4.5 μ L		
			10 X PCR buffer		
			45 μ L		
			dNTPs		
			36 μ L		
			Purified water		
			357 μ L		
		5.6.4	Distribute the reaction mixture in 8 PCR tubes (45 μ L each)		
			and label them as 10-1, 10-2, 10-3, 10-4, 10-5,10-6 ,		
			Negative and Test		
		5.6.5	Add 5 μ L of each dilution prepared in 5.6.3 in the respective		
			PCR tube. Thus the concentration of DNA in the PCR reaction		
			comes out as:		
			10-1 : 175 ng		
			10-2 : 17.5 ng		
			10-3 : 1.75 ng		
			10-4 : 0.175 ng		
			10-5 : 17.5 ng		
			10-6 : 1.75 ng		
		5.6.6	Add 5 μ L of Purified water in the PCR tube labeled as		
			Negative.		
		5.6.7	Add 5 μ L of the ARM sample to be tested in PCR tube labeled		
			as Test.		
		5.6.8	Incubate the PCR tubes shall be incubated for 5 min at the		
			room temperature.		
		5.6.9	Switch ON the Mastercycler as per procedure.		
		5.6.10	Set the thermal cycler program as:		
			Cycles		
			Temperature/time		
			1 cycle		
			94°C for 2 min		
			40 cycles		
			94°C for 30 sec		
			54.3°C for 1 min		
			72°C for 45 sec		
		5.6.11	After the PCR reaction is completed take out the PCR tubes		

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| | |from Master cyclor. | |
| | |5.6.12 |Perform electrophoresis of all the samples to visualize the |
| | |bands. |
| | |5.6.13 |Prepare the samples for electrophoresis as follows: |
| | |Reagents |
| | |Volume to be added |
| | |
| | |Standard (10-1) |
| | |
| | |Amplified Test DNA from Mastercycler |
| | |10µL |
| | |
| | |6X loading Dye |
| | |
| | |3µL |
| | |
| | |Purified water |
| | |
| | |2 µL |
| | |
| | |Similarly prepare the samples for 10-2, 10-3, 10-4, 10 |
| | |-5,10-6 |
| | |
| | |5.6.14 |For Negative Control |
| | |Reagents |
| | |Volume to be added |
| | |
| | |Amplified Test DNA from Mastercycler |
| | |10µL |
| | |
| | |6X loading Dye |
| | |
| | |3µL |
| | |
| | |Purified water |
| | |
| | |2 µL |
| | |
| | |5.6.15 |For Test Sample |
| | |Reagents |
| | |Volume to be added |
| | |
| | |Amplified Test DNA from Mastercycler |
| | |10µL |
| | |
| | |6X loading Dye |
| | |
| | |3µL |
| | |
| | |Purified water |
| | |
| | |2 µL |
| | |
| | |
| | |
| | |
| | |
| | |
| | |
| | |Standard DNA Ladder 7µL |
| | |5.6.16 |Load each one of the above 8 samples in separate wells on an |
| | |agarose gel (2%). |
| | |5.6.17 |Perform electrophoresis of the samples |
| | |5.6.18 |Analyze the results by directly observing the gel on the U.V.|
| | |transilluminator and note the results by comparing the band |
| | |obtained with the corresponding bands in the DNA ladder. |
| | |6.0|References: |
| | |NA |
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1.0	Objective			
	To lay down the procedure for determination of contaminating yeast cell			
	DNA (host cell) in the active raw material (ARM) of Hepatitis B surface			
	antigen (HBsAg).			
2.0	Scope			
	This Standard Operating Procedure is applicable to ARM of Hepatitis B			
	surface antigen (HBsAg) produced in Pichia pastoris C-226.			
3.0	Responsibility			
	3.1	The analyst of the molecular biology laboratory shall be		
	responsible to follow the established procedure.			
	3.2	The Supervisor of the laboratory shall be responsible to ensure		
	that the procedure is followed correctly.			
	3.3	Head, Quality Control shall be responsible for overall compliance		
	of the SOP.			
4.0	Abbreviations and Definitions			
	4.1	ARM	: Active Raw Material	
	4.2	BSA	: Bovine Serum Albumin	
	4.3	Cm	: Centimeter	
	4.4	cpm	: Counts Per minute	
	4.5	dATP	: Deoxyadenosine tri phosphate	
	4.6	dCTP	: Deoxycytidine tri phosphate	
	4.7	dGTP	: Deoxyguanosine tri phosphate	
	4.8	DNA	: Deoxyribonucleic Acid	
	4.9	dTTP	: Deoxythymidine tri phosphate	
	4.10	oC	: Degree celcius	
	4.11	HBsAg	: Hepatitis B Surface Antigen	
	4.12	I.U.	: Internetal Unit	
	4.13	LGT	: Low Gelling Temperature	
	4.14	M	: Molar	
	4.15	mg	: Milli gram	
	4.16	(ci	: Micro curie	
	4.17	(g	: Micro gram	
	4.18	(L	: Microliter	
	4.19	Min	: Minute	
	4.20	NA	: Not Applicable	
	4.21	ng	: Nanogram	
	4.22	PBS	: phosphate Buffer Saline	
	4.23	pg	: Pico gram	
	4.24	PVC	: Poly Vinyl Cloride	
	4.25	QA	: Quality Assurance Department	
	4.26	QC	: Quality Control Department	
	4.27	rpm	: Revolution per minute	
	4.28	S.A.	: Specific Activity	
	4.29	SOP	: Standard Operating Procedure	
	4.30	TLD	: Thermo Luminiscent Dosimeter	
	4.31	tRNA	: Transfer Ribonucleic Acid	
	4.32	UV	: Ultra Violet	
	4.33	Vol	: Volume	
5.0	Procedure			
	5.1	Safety considerations:		
		5.1.1	All corrosive material should be handled carefully, in a	
		fume hood.		
		5.1.2	The personnel shall wear lab coat, surgical gloves,	
		covered shoes and protective face shield (acrylic) while		
		working in the Radioactivity Laboratory.		
		5.1.3	The personnel's safety shall be monitored using Thermo	
		Luminiscent dosimeter (TLD) cards.		
		5.1.4	All the waste material in the Radioactivity Laboratory	
		e.g. tips, Eppendorfs, gloves, plastic sheet, filter		
		paper, glassware, saran wrap etc shall be disposed in		
		acrylic waste bins.		
		5.1.5	No material/equipment/reagent shall be taken out from the	
		Radioactivity Laboratory.		
		5.1.6	Wear protective glasses while using the UV	
		transilluminator.		
		5.1.7	Agarose gels stained with ethidium bromide shall be	
		handled with gloves.		
	5.2	Equipments:		
		5.2.1	Radioactivity Laboratory	
			5.2.1.1	Fume hood
			5.2.1.2	Centrifuge
			5.2.1.3	Oven (42 ± 2°C)
			5.2.1.4	Water bath with thermostat
			5.2.1.5	Tube heating block
			5.2.1.6	β-Liquid Scintillation Counter
			5.2.1.7	Plastic bag sealer
			5.2.1.8	Radiation monitor
		5.2.2	Molecular Biology Laboratory	
			5.2.2.1	Centrifuge
			5.2.2.2	pH meter
			5.2.2.3	Magnetic stirrer
			5.2.2.4	Autoclave
			5.2.2.5	Dot-blot apparatus (Vacuum manifold)
			5.2.2.6	Refrigerator
			5.2.2.7	Ultra Low Temperature Freezer (-70±5°C)
			5.2.2.8	Deep Freezer (-20±2°C)
			5.2.2.9	UV Transilluminator
			5.2.2.10	UV / visible spectrophotometer
	5.3	Materials:		
		5.3.1	Autopipette (100-1000 (L)	

- || 5.3.2 |Autopipette (20-200 (L) |
- || 5.3.3 |Autopipette (2-20 (L) |
- || 5.3.4 |Eppendorfs (1.5 mL and 0.7 mL) and their rack |
- || 5.3.5 |Autopipette tips |
- || 5.3.6 |Laboratory glassware |
- || 5.3.7 |Gloves |
- || 5.3.8 |X- ray films and exposure cassette |
- || 5.3.9 |Glass wool |
- || 5.3.10 |Nylon Hybond N membrane |
- || 5.3.11 |Whatman filter paper |
- || 5.3.12 |Scintillation vials |
- || 5.3.13 |Saran wrap |
- || 5.3.14 |Plastic bags |
- || 5.4 |Reagents: |
- || 5.4.1 |Formamide |
- || 5.4.2 |DNA polymerase I (Klenow Large fragment) |
- || 5.4.3 |dCTP 10mM |
- || 5.4.4 |dGTP 10mM |
- || 5.4.5 |dTTP 10mM |
- || 5.4.6 |Restriction Endonuclease (EcoRI) |
- || 5.4.7 |Buffer EcoRI 10X |
- || 5.4.8 |Agarose Type II |
- || 5.4.9 |BSA fraction V (DNase free) |
- || 5.4.10 |(dATP32 10 μ Ci/ μ L |
- || 5.4.11 |pRSP3-EcoRI probe |
- || 5.4.12 |P. pastoris MP-36 Chromosomal DNA |
- || 5.4.13 |pRSP3 – Plasmid |
- || 5.5 |Preparation of solutions: |
- || 5.5.1 |PBS 1X |
- || 5.5.2 |SSC 20X |
- || 5.5.3 |SSC 10X |
- || 5.5.4 |SDS 20% |
- || 5.5.5 |Denaturation Solution |
- || 5.5.6 |Neutralizing Solution |
- || 5.5.7 |Denhardt's 50X |
- || 5.5.8 |Agarose 0.8 % |
- || 5.5.9 |TBE 1X |
- || 5.5.10 |Yeast-tRNA 10 mg/mL |
- || 5.5.11 |Calf thymus DNA (10 mg/mL) |
- || 5.5.12 |Nick Translation Buffer (10X) |
- || 5.5.13 |Sephadex G-25 |
- || 5.5.14 |Stop Solution |
- || 5.5.15 |Washing Solution No.1 |
- || 5.5.16 |Washing Solution No.2 |
- || 5.6 |Preparation of the standard curve of DNA obtained from Pichia |
- || |pastoris MP- 36: |
- || 5.6.1 |Isolate DNA from the pichia pastoris MP-36 by following |
- || |the procedure SOP No: QC/00/046 (Extraction Of High |
- || |Molecular Weight DNA (Chromosomal) from Yeast cells) |
- || 5.6.2 |Dilute the DNA to 250pg/ μ L with Milli Q water. |
- || 5.6.3 |In an Eppendorf take 18 μ L of PBS and 2 (L of yeast DNA |
- || |(250 pg/L).Vortex the solution and centrifuge at 6000 rpm|
- || |for 20 seconds. Label it as Eppendorf No. 1. |
- || 5.6.4 |Take 10 (L from Eppendorf No.1 (250 pg) and add to |
- || |Eppendorf No.2 having 10 (L of PBS (1X) so as to get a |
- || |final quantity of 250 pg in 20 (L in Eppendorf No.2. |
- || |Vortex the solution and centrifuge at 6000 rpm for 20 |
- || |seconds. |
- || |Take 10 (L from Eppendorf No.2 and repeat the procedure as|
- || |above until Eppendorf No.8 |
- || 5.6.5 |Discard 10 (L from Eppendorf No.8. |
- || |The final concentrations obtained in the eight Eppendorfs |
- || |are: |
- || |250 pg, 125 pg, 62.5 pg, 31.25 pg, 15.62 pg, 7.8 pg, 3.9 |
- || |pg and 1.95 pg. Label the Eppendorfs from 1 – 8. |
- || 1 2 3 |
- || 4 5 6 7 8 |
- || |
- || |
- || 250 pg 125 pg 62.5 pg 31.25 pg 15.62 pg 7.8 pg 3.9 pg 1.95 pg |
- || Eppendorf No. 1: 18 μ L of PBS 1X + 2 μ L of yeast DNA 250 pg/ μ L |
- || Eppendorf No. 2: 10 μ L of PBS 1X + 10 μ L from Eppendorf No.1 |
- || Eppendorf No. 3: 10 μ L of PBS 1X + 10 μ L from Eppendorf No.2 |
- || Eppendorf No. 4: 10 μ L of PBS 1X + 10 μ L from Eppendorf No.3 |
- || Eppendorf No. 5: 10 μ L of PBS 1X + 10 μ L from Eppendorf No.4 |
- || Eppendorf No. 6: 10 μ L of PBS 1X + 10 μ L from Eppendorf No.5 |
- || Eppendorf No. 7: 10 μ L of PBS 1X + 10 μ L from Eppendorf No.6 |
- || Eppendorf No. 8: 10 μ L of PBS 1X + 10 μ L from Eppendorf No.7 (discard |
- || 10 μ L) |
- || 5.6.6 |Add 90 (L of PBS (1X) to each Eppendorf so as to obtain a |
- || |final volume of 100 (L. |
- || 5.7 |Preparation of the test samples: |
- || |Taking into account that the dose to be evaluated is equivalent to |
- || |100 μ g of |
- || |protein, divide 100 by the protein concentration of the sample as |
- || |determined by UV absorption at 280. The quotient obtained, gives |
- || |the volume in μ L of the sample, which is to be evaluated. Once this|
- || |volume has been determined, the next step is to make a 10 times |
- || |dilution of the sample with PBS (1X). |
- || 5.8 |DNase free BSA (10 mg/mL) is used as negative control. Treat it, as|
- || |the test sample was treated above. The quotient in this case is |
- || |equal to 10 (L. To obtain a 10-fold dilution, make up the volume to|

[100 µL with 90 (L of PBS (1X).]
[5.9 [Wet the Nylon Hybond-N membrane in SSC (10X) for 5 minutes.]
[5.10 [Prepare the Dot-blot apparatus (Vacuum manifold) for applying the]
[samples and controls. For this, take the pre-wetted membrane and]
[place in the apparatus. Properly cover the wells that are not to be]
[used for loading any sample, with a plastic sheet. This shall be]
[done very precisely so that the wells are perfectly covered with]
[the sheet and there is no possibility of any leakage of vacuum.]
[5.11 [Denature the DNA of the standard, controls and test samples by]
[heating at 95°C]
[for 5 minutes in the tube-heating block. Apply the samples using an]
[auto pipette]
[in to the wells of the Dot-blot apparatus (Vacuum manifold). Create]
[vacuum in the apparatus with help of external vacuum pump. After]
[all the desired volume of the samples have been applied in the]
[required wells, switch off the vacuum pump and carefully dismount]
[the manifold system to remove the nylon membrane having samples on]
[it.]
[5.12 [Submerge the membrane (without agitation) in a container containing]
[denaturing solution for 5 minutes. Volume of the solution shall be]
[enough to completely cover the membrane.]
[5.13 [Transfer the membrane to a container containing neutralizing]
[solution and submerge it for 5 minutes. Do not agitate. Ensure]
[that the volume of the solution is sufficient to submerge the]
[membrane completely.]
[5.14 [Remove the membrane from neutralizing solution and place it on a]
[Whatman Paper No.1. Incubate the membrane in an Oven at 80°C for]
[10 min.]
[5.15 [Expose the membrane to UV light for 3 minutes ensuring that sample]
[side is facing UV Light.]
[5.16 [Prepare the pre-hybridization solution as follows:]
[For a volume of 2ml add:]
[Solutions]
[Volume to be added]
[SSC 20X]
[600 µL]
[Denhardt's reagent (50X)]
[200 µL]
[Formamide]
[1000 µL]
[SDS (20%)]
[50 µL]
[Calf thymus DNA or tRNA]
[20 µL]
[Sterile purified Water]
[130 µL]
[This solution is enough for membrane having dimensions of 8 cm X 2]
[cm. If the membrane is larger, prepare more volume of solution.]
[This solution must be prepared freshly, just prior to use to]
[prevent any of its components from precipitating.]
[5.17 [Put the membrane in a plastic bag and transfer 2 mL of]
[pre-hybridization solution, carefully remove all bubbles and seal]
[the bag with a PVC Bag Sealer. The pouch containing the membrane]
[and prehybridisation solution is then incubated at 42°C. The]
[incubation time can vary from 2 hours to 12 hours.]
[5.18 [Preparing the probe:]
[5.18.1 [Digest the plasmid pRSP3 (which acts as the probe) with]
[enzyme EcoRI at concentration 5 I.U. of enzyme per µg of]
[DNA. The procedure is as follows:]
[Mix the following in an Eppendorf:]
[Solutions]
[Volume to be added]
[plasmid pRSP3 (50ng)]
[2 µL]
[Buffer EcoRI (10X) in a way so that a concentration of 1X]
[remains]
[3 µL]
[5U of EcoRI]
[1 µL]
[Sterile purified water]
[25 µL]

			Incubate for 1 hour at 37°C.	
		5.18.2	Afterwards, check for the completion of digestion by	
		running agarose gel electrophoresis in 0.8% gel at 80V		
		using TBE (1X). Adjust the concentration of probe to		
		50ng/μL.		
		5.19	The procedure for Labeling the probe pRSP3 of concentration	
		50ng/μL with (-dATP32 (10μCi/μL) while the membrane is in the		
		pre-hybridization solution is as follows:		
		5.19.1	Add the following in an Eppendorf:	
		Solutions		
		Volume to be added		
				Probe (pRSP3 at 50 ng/μL)
		digested with EcoR I		
				1μL
				Nick translation buffer 10X
		3μL		
				dCTP 10 mM
		1μL		
				dGTP 10 mM
		1μL		
				dTTP 10 mM
		1μL		
				Klenow fragment 10U/μL(
		1 μL		
				(-dATP32 (10(Ci/μL)
		4 μL		
		Sterile purified water		
		18 μL		
		Final volume of the reaction		
		30μL		
		((If the concentration of the enzyme is more than 10U/μL,		
		then dilute with sterile purified water to a concentration		
		of 10U/μL)		
		5.20	Place the tube in a thermostat-regulated water bath at 37oC for 1	
		hour and then		
		add 70 μL of stopping solution to make up to 100 μL of final		
		volume.		
		5.21	Prepare Sephadex G-25 column:	
		5.21.1	Take one Eppendorf and with the help of a sterile needle	
		make a small hole at the base.		
		5.21.2	Put glass wool at the base and add 1 mL of Sephadex G-25.	
		5.21.3	Centrifuge at 6000 rpm for 30 seconds.	
		5.21.4	Wash the column with 200μL of sterile water and centrifuge	
		at 6000 rpm for 30 seconds.		
		5.21.5	Add 200μL of sterile water containing 2μL of tRNA of	
		concentration 10mg/mL or 2μL of calf thymus DNA (in this		
		case calf thymus DNA should be denatured at 95oC).		
		5.21.6	Centrifuge at 6000 rpm for 4 min (to saturate the column).	
		5.22	Pass 100μL of labeled probe through the Sephadex G-25 column.	
		Centrifuge at 6000 rpm for 4 min. Recover the elute and preserve at		
		-20oC if it is not to be used immediately.		
		Note: the labeled probe should not be stored for more than 24		
		hours.		
		5.23	Calculating the Specific Activity (S.A.) of the probe:	
		For this purpose, apply 1μL of the eluate on a disc of Whatman		
		filter paper No.1 (2 cm in diameter) and place it in a		
		scintillation vial. Determine the cpm (counts per minute) in a		
		(-liquid scintillation counter (model 1409 DSA).		
		After determining the cpm of the probe, calculate the specific		
		activity by means of the following formula:		
		S.A. = cpm incorporated(X total vol. of the probe (100μL)		
		μg of probe added (0.05μg) = cpm/μg of DNA		
		(cpm incorporated: the cpm value obtained from the Scintillation		
		counter.		
		Note: If the S.A. is less than 10 8 cpm/μg, which indicates low		
		sensitivity, it is not advisable to use the labeled probe.		
		Therefore it will be discarded in the acrylic bins provided for		
		this purpose in the lab and a new probe will be prepared.		
		5.24	Prepare the hybridization solution as follow:	
		Solutions		
		Volume to be added		
				SSC (20X)
		600μL		
				Formamide
		1.0 mL		
				SDS (20X)

- * PS: Polysaccharide
- S. No | Test/Assay | Specifications
- 1 | Appearance | White powder or lumps
- 2 | Serological Identity | Positive
- 3 | Uronic acid content | $\geq 45\%$ of dry weight of PS
- 4 | O-acetyl group | $\geq 1.8\%$ of dry weight of PS
- 5 | Phosphorus content | 0-1.5 % of dry weight of PS
- 6 | Total nitrogen content | 3.5-6.0% of dry weight of PS
- 7 | Kavg determination (SEC-HPLC with RI detector) | $K_{avg} \leq 0.3$
- 8 | Protein content | $\leq 2.0\%$ of dry weight of PS
- 9 | Nucleic acid content | $\leq 2.0\%$ of dry weight of PS
- 10 | Cell wall polysaccharide content | $\leq 0.5\%$ of dry weight of PS
- 11 | Endotoxin content | < 0.5 IU/ μ g of PS
- 12 | Residual CTAB Content | $\leq 1.0\%$ of dry weight of PS
- 13 | Residual Formaldehyde Content | $\leq 1.0\%$ of dry weight of PS
- 14 | Residual DOC content | $\leq 1.0\%$ of dry weight of PS
- 15 | Moisture content | To report results

* PS: Polysaccharide

S.No | Test/Assay | Specifications

1 | Appearance | White powder or lump

2 | Serological Identity | Positive

3 | Methyl pentose content | $\geq 13\%$ of dry weight of PS

4 | Hexosamine content | To report result

5 | O-Acetyl content | To report result

6 | Phosphorus content | 0-1.0 % of dry weight of PS

7 | Total nitrogen content | 1.5-4.0% of dry weight of PS

8 | Kav_g determination (SEC-HPLC with RI detector) | $Kav_g \leq 0.3$

9 | Protein content | $\leq 5.0\%$ of dry weight of PS

10 | Nucleic acid content | $\leq 2.0\%$ of dry weight of PS

11 | Cell wall polysaccharide content | $\leq 0.5\%$ of dry weight of PS

12 | Endotoxin content | < 0.5 IU/ μ g of PS

13 | Residual CTAB Content | $\leq 1.0\%$ of dry weight of PS

14 | Residual Formaldehyde Content | $\leq 1.0\%$ of dry weight of PS

15 | Residual DOC Content | $\leq 1.0\%$ of dry weight of PS

16 | Moisture content | To report results

[SHEET: MnPs QC]

QC Requirements for analysis of 10 batches of meningococcal polysaccharides (4 serotypes) | | | | |

S. No. | Equipments | Quantity | Reagents | Quantity | Consumables | Quantity |

1 | Analytical balance | 2 | PASTOREX MENINGITIS INDIVIDUAL LATEX TEST KIT, code 61608 | 5 | Disposable glass test tubes (approximately 12 100 mm) with disposable plastic caps are recommended. | 500 |
2 | Laboratory vortex mixer | 2 | Resorcinol reagent | 10 gm | Auto Pipettes tips (e.g. Eppendorf) | 5000 each |

3 | pH-Meter | 1 | Butanol | 2 lit | Volumetric flasks, 100–1000 mL. | 5 each |

4 | UV-Vis Spectrophotometer with Cuvettes | 1 | Butyl acetate | 5 lit | Measuring cylinders, 10–1000 mL. | 5 each |

5 | Water bath 100 deg. | 1 | N-acetyl neuraminic acid (NANA) | 2 gm | Glass test tubes (12 x 75 mm e.g. Borosil) | 200 |

6 | Furnace | 1 | Phosphorus standard, 20 µg/ml | 100 gm | Screw cap HPLC vials with/without inserts | 200 |

7 | Water bath, 37°C | 1 | Sulfuric Acid | 2 lit | Solvac Filter Holder | 20 |

8 | Calibrated HPLC system with RI detector | 1 | Perchloric acid | 2 lit | Nylon 6,6 membrane 0.2µm membrane filter | 20 |

9 | Precision balance | 1 | Hydrogen peroxide free from phosphorus | 1 lit | Syringe filters 0.2 µm | 500 |

10 | Vacuum Pump | 1 | Ascorbic acid sodium salt | 100 gm | Aluminum Foil | 20 |

11 | Refrigerator | 1 | Ammonium molybdate tetra hydrate | 25 gm | Volumetric flasks (10ml, 25ml & 50ml) | 5 each |

12 | Magnetic stirrer | 1 | Hydrochloric acid, 37% | 2 lit | Screw capped test tubes (30 ml) with stand | 100 |

13 | Ultrasonic Cleaner | 1 | Iron (III) chloride hex hydrate ³99% | 100 gm | Glass Pipettes (5ml, 10ml & 25 ml) | 100 |

14 | Rocking and Tumbling Shaker | 1 | Sodium Hydroxide ³99 % | 5000 gm | Eppendorf (1.5ml & 2ml) | 5000 |

15 | Sugar KS-804 (Shodex Column) | 1 | Sodium Acetate anhydrous ≥ 99.5 % | 50 gm | Gloves | 1000 |

16 | Sugar KS-805 (Shodex Column) | 1 | Hydroxylamine hydrochloride ≥ 99.0 % | 100 gm | Glass tubes 25 ml | 50 |

17 | Guard Column KS-G (Shodex) | 1 | Dextran | 1 gm | Depyrogenized glass dilution tubes | 100 |

18 | Auto pipettes with Tips (20-200µl, 100 to 1000 µl & 1000 to 5000 µl)r | 2 each size | Sodium azide | 50 gm | Depyrogenized glass reaction tubes | 100 |

19 | Vortex Mixer | 1 | Deuterium Oxide | 100 ml | Depyrogenized tips | 1000 |

20 | BDS Hypersil C18 silica column (250 x 4.6mm, Particle size:5µm) | 1 | Sodium Chloride | 5000 gm | Test tube rack | 10 |

21 | Laboratory Centrifuge | 1 | Sodium dihydrogen phosphate, dihydrate | 500 gm | | |

22 | Fume Hood | 1 | Disodium hydrogen phosphate, anhydrous | 500 gm | | |

23 | 96 well Maxisorp Nunc ELISA plates (Nunc Cat No. 439454) | 20 | Copper sulphate pentahydrate | 10 gm | | |

24 | Micro plate ELISA reader | 1 | Sodium tartarate | 10 gm | | |

25 | Incubator | 1 | Sodium hydroxide | 100 gm | | |

26 | Auto titrator | 1 | Sodium Carbonate | 100 gm | | |

27 | Multichannel auto pipette (50 to 300 µl) | 1 | Folin & Ciocalteu's phenol reagent | 500 ml | | |

28 | Tube heating block (37° C) | 1 | Bovine Serum Albumin | 100 gm | | |

29 | Deep Freezer (- 20.0 ± 5°C) | 1 | Deoxycholic acid (DCA) | 2 gm | | |

30 | Timer | 1 | Methanol | 2 lit | | |

31 | Thermogravimetric analyser to determine LOD | 1 | Acetonitrile | 2 lit | | |

32 | | | Sodium Acetate anhydrous | 20 gm | | |

33 | | | Orthophosphoric Acid | 100 ml | | |

34 | | | | |

35 | | | Ammonium acetate | 10 gm | | |

36 | | | acetic acid (glacial) | 100 ml | | |

37 | | | Acetyl acetone | 20 ml | | |

38 | | | Formaldehyde standard solution | 50 ml | | |

39 | | | Anti CWPS Polyclonal rabbit Antisera (Staten Serum Institute, Cat No. 3653) | 2 vials | | |

40 | | | Purified CWPS (Staten Serum Institute, Cat No. 3459) | 2 vials | | |

41 | | | Monoclonal antibodies against Phosphoryl Choline (Staten Serum Institute, Cat No. 87214) | 2 vials | | |

42 | | | Polyclonal Goat anti-mouse Immunoglobulin HRP conjugates (Dako Cat No. P0447) | 1 vial | | |

43 | | | Bovine Serum Albumin (Fraction V, Sigma Cat No. A9647) | 100 gm | | |

44 | | | Tween 20 | 100 ml | | |

45 | | | TMB Substrate (Sigma, Cat No. T4444) | 200 ml | | |

46 | | | Conc. Sulphuric acid | 1 lit | | |

47 | | | NaCl | 2000 gm | | |

48 | | | Na2HPO4 | 500 gm | | |

49 | | | NaH2PO4 | 500 gm | | |

50 | | | Dimidium Bromide/Disulphine blue indicator stock solution | 1 VIAL | | |

51 | | | Sodium dodecyl sulphate | 100 gm | | |

52 | | | Stock CTAB | 20 gm | | |

53 | | | Limulus Amoebocyte Lysate, Pyrotell | 10 Vials | | |

54 | | | Controlled Standard Endotoxin (CSE) | 2 vials | | |

55 | | | LAL reagent water (LRW) | 200 ml | | |

[SHEET: Sheet3]

1.0 |Objective: |
	To lay down the procedure for immunological identification of Purified
	Polysaccharide (PRP HIB) and Bulk Conjugate by immunodiffusion using
	polyclonal antibodies.
2.0	Scope:
	This Standard Operating Procedure is applicable to the Purified
	Polysaccharide (PRP HIB) and Bulk Conjugate of Haemophilus influenzae type
	b vaccine.
3.0	Abbreviations and Definitions:
	3.1
	3.2
	3.3
	3.4
	3.5
	3.6
	3.7
	3.8
	3.9
4.0	Procedure
	4.1
	4.2
	4.3
	4.4
	4.5
5.0	Reporting:
	Identification Of Purified Polysaccharide And Bulk Conjugate of Haemophilus
	influenzae Type B Conjugate Vaccine By Immunodiffusion Method.
6.0	References:
	6.1

* PS: Polysaccharide

S.No | Test/Assay | Specifications

1 | Appearance | White powder or lump

2 | Serological Identity | Positive

3 | Methyl pentose content | $\geq 20.0\%$ of dry weight of PS

4 | Hexosamine content | $\geq 12.0\%$ of dry weight of PS

5 | Phosphorus content | $3.0-7.0\%$ of dry weight of PS

6 | Total nitrogen content | $0.6-3.5\%$ of dry weight of PS

7 | Kav_g determination (SEC-HPLC with RI detector) | $Kav_g \leq 0.3$

8 | Protein content | $\leq 3.0\%$ of dry weight of PS

9 | Nucleic acid content | $\leq 2.0\%$ of dry weight of PS

10 | Cell wall polysaccharide content | $\leq 0.5\%$ of dry weight of PS

11 | Endotoxin content | < 0.5 IU/ μ g of PS

12 | Residual CTAB Content | $\leq 1.0\%$ of dry weight of PS

13 | Residual formaldehyde content | $\leq 1.0\%$ of dry weight of PS

14 | Residual DOC Content | $\leq 1.0\%$ of dry weight of PS

15 | Moisture content | To report results

1.0 | Objective | Objective | Objective

| To lay down the procedure for estimation of aluminum content by complexometry. | To lay down the procedure for estimation of aluminum content by complexometry. | To lay down the procedure for estimation of aluminum content by complexometry.

2.0 | Scope | Scope | Scope

| The standard operating procedure is applicable for estimation of aluminum content by complexometry. | The standard operating procedure is applicable for estimation of aluminum content by complexometry. | The standard operating procedure is applicable for estimation of aluminum content by complexometry.

3.0 | Responsibility | Responsibility | Responsibility

| 3.1 | The QC analyst shall be responsible to carry out the procedure as laid down in this SOP. | The QC analyst shall be responsible to carry out the procedure as laid down in this SOP.

| 3.2 | Supervisor shall be responsible for checking and compliance of instruction and action as mentioned in this SOP. | Supervisor shall be responsible for checking and compliance of instruction and action as mentioned in this SOP.

| 3.3 | The departmental head shall be responsible for overall compliance of this SOP. | The departmental head shall be responsible for overall compliance of this SOP.

4.0 | Abbreviations and Definitions | Abbreviations and Definitions | Abbreviations and Definitions

| 4.1 | μL | : Microliter

| 4.2 | mL | : Milliliter

| 4.3 | EDTA | : Ethylene diamine tetra acetic acid

| 4.4 | M | : Molarity

| 4.5 | % | : Percentage

| 4.6 | QC | : Quality Control

| 4.7 | SOP | : Standard Operating Procedure

5.0 | Procedure | Procedure | Procedure

5.1 | Equipment, Materials and Reagents | Equipment, Materials and Reagents

|| 5.1.1 | Vortex shaker

|| 5.1.2 | Fume hood

|| 5.1.3 | Hot Plate

|| 5.1.4 | Glass wares required are: Erlenmeyer flasks (250 mL & 500 mL), Pipettes (2 mL, 5 mL, 10 mL & 25 mL), Bottles (50 mL, 100 mL, 250 mL & 500 mL), Test tubes and Burettes.

|| 5.1.5 | Auto pipettes and their corresponding tips

|| 5.1.6 | Gloves

|| 5.1.7 | Conc. Sulphuric acid

|| 5.1.8 | Conc. Nitric acid

|| 5.1.9 | Methyl orange

|| 5.1.10 | EDTA solution

|| 5.1.11 | Copper Sulphate solution

|| 5.1.12 | 1-pyridyl-2-azonaphthol solution

|| 5.1.13 | 50% Sodium hydroxide solution

|| 5.1.14 | Glacial acetic acid

|| 5.1.15 | Sodium acetate

5.2 | Preparation of Solutions | Preparation of Solutions

|| Prepare all the solutions as described below. Carry out the labelling of the prepared solutions and assign the lot numbers. | Prepare all the solutions as described below. Carry out the labelling of the prepared solutions and assign the lot numbers.

|| 5.2.1 | 0.01M EDTA solution: - Take 1.47g EDTA in 500 mL of purified water. Mix well and store the solution at room temperature for not more than one month.

|| 5.2.2 | Sodium Acetate Solution: - Take 0.64 g of sodium acetate in 100 mL of purified water. Mix well and store the solution at room temperature for not more than one month.

|| 5.2.3 | Glacial acetic acid solution: - Take 1.5 mL of glacial acetic acid and made upto 100 mL with purified water.

|| 5.2.4 | Acetate buffer (pH 4.6 $\pm$ 2): - Take exactly 36.2 mL of sodium acetate solution into 250 mL flask and add 14.8 mL of glacial acetic acid solution, make the volume 100 ml using distilled water using purified water.

|| 5.2.5 | 0.005 M Copper Sulphate solutions: - Take 0.79g of Copper Sulphate in 1000 mL of purified water. Mix well and store the solution at room temperature for not more than one month.

5.3 | Sample preparation | Sample preparation

|| 5.3.1 | Dilute the sample and control (Reference Vaccine) to 1/10 dilution.

Note: For preparing the 1/10 dilution of the sample and control, take 1mL of sample / control and add 9 mL of Purified water.

|| 5.3.2 | Carry out the analysis of sample in triplicate and for control and blank in duplicate.

5.4 | Test Procedure | Test Procedure

|| 5.4.1 | Transfer 3 mL of purified water to 2 thermo resistant test tubes which are treated as Blank.

|| 5.4.2 | Transfer 3 mL of previously diluted sample to 3 thermo resistant test tubes.

|| 5.4.3 | Transfer 3 mL of previously diluted control to 2 thermo resistant test tubes.

|| 5.4.4 | To each tube add 1 mL of conc. Sulphuric acid and 125 μL of conc. Nitric Acid.

|| 5.4.5 | Place the tubes in a beaker containing boiling water and heat for ten minutes.

|| 5.4.6 | Take out the tubes and allow cooling at room temperature.

|| 5.4.7 | Transfer the content from each tube to separate Erlenmeyer flask and rinse the remaining material of the tube with 10 mL of purified water and add to the respective Erlenmeyer flask.

|| 5.4.8 | Add 2 drops of Methyl orange to each Erlenmeyer flask, pink colour will appear.

|| 5.4.9 | Add 50% Sodium hydroxide solution drop by drop until the colour of the contents of each Erlenmeyer flask changes from pink to light yellow.

|| 5.4.10 | Add 25 ml of purified water, 10 mL of 0.01M EDTA solution and 5 mL of the acetate buffer solution to each.

|| 5.4.11 | Boil the solution in the Erlenmeyer flask for about three minutes (Start the time count when the solution begins to boil). In the meantime fill 0.005 M Copper Sulphate solutions in a burette.

|| 5.4.12 | After heating is over remove the flask from the hot plate.

|| 5.4.13 | Add 1 mL of 1-pyridyl-2-azonaphthol solution to the Erlenmeyer flask and carry out the titration (while the solution is hot) with 0.005 M Copper Sulphate solution, until the colour of sample changes to purple - red colour, this colour is compared to the colour of Blank.

|| 5.4.14 | Note the volume of 0.005M Copper Sulphate solution used in the titration.

5.5 | Calculations | Calculations

|| 5.5.1 | Calculate V_b as the average volume of Copper Sulphate consumed in the titration of blank.

|| 5.5.2 | Calculate the volume of Copper Sulphate (0.005 M) consumed for titrating each replicate of Sample/Control. For this, subtract the average value of blank from each replicate of the sample/control independently. At the end, take mean of these values.

|| 5.5.3 | Concentration of the Al^{3+} ion:

$$C \text{ (mg/ml)} = (V_b - V_m)/A \times 0.1349 \times 10$$

Concentration of Aluminium Hydroxide:

Multiply with a factor of 4.52 to obtain the concentration of aluminium phosphate.

$C \text{ (mg/ml)} = (V_b - V_m)/A \times 0.1349 \times 10 \times 2.89$

| | 5.5.4 | Here:

V_b: Average volume of Copper Sulphate solution (0.005 M) consumed in the titration of blank.

V_m: Volume of Copper Sulphate solution (0.005M) consumed in the titration of each replicate of the sample.

A: Sample volume taken to perform the test.

0.1349: Mass of Aluminum equivalent to 1 ml Solution of Copper Sulphate (0.005 M).

4.52: Conversion factor of Aluminum ion to Aluminium Phosphate.

10: Dilution factor of the Sample.

| 5.6 | Acceptance criteria for the test | Acceptance criteria for the test

| | 5.6.1 | The Aluminium content in the control should be within the acceptable limits.

| | 5.6.2 | The Aluminium content in the sample should be within the specified limits.

6.0 | Reference: | Reference: | Reference:

| Indian Pharmacopeia | Indian Pharmacopeia | Indian Pharmacopeia

1. Objective: |
 |To lay down the procedure to determine the sterility of products |
 |purported to be sterile. |
 2. Scope: |
 |This Standard Operating Procedure is applicable for analysis all the In |
 |process, ARM final bulk and final lot samples |
 |2.1 For preparation, storage, growth promotion test and application of |
 |media, used in sterility testing. |
 |2.2 For analysis of samples by membrane filtration and direct |
 |inoculation technique. |
 |2.3 For analysis of test sample for presence of any inhibitory |
 |component by stasis test. |
 3. Abbreviations and Definitions: |
 |3.1 °C |: Degree centigrade |
 |3.2 IP |: Indian Pharmacopoeia |
 |3.3 ml |: Milli litre |
 |3.4 USP |: United State Pharmacopeia |
 |3.5 EP |: European Pharmacopoeia |
 4. Procedure: |
 |4.1 Equipments, Materials and Reagents |
 |4.1.1 Laminar airflow cabinet |
 |4.1.2 Incubators at 32.5 + 2.5°C and 22.5 + 2.5°C |
 |4.1.3 Sterile syringes |
 |4.1.4 Sterile pipettes |
 |4.1.5 Sterile gloves |
 |4.1.6 Sterile forceps |
 |4.1.7 McCartney bottles |
 |4.1.8 Purified water |
 |4.1.9 Tryptone Soya Broth |
 |4.1.10 Fluid Thioglycollate Medium |
 |4.1.11 Soya Bean Casein Digest media |
 |4.2 Preparation of Sterility Media: |
 |4.2.1 Fluid Thioglycollate Medium (FTM): |
 |4.2.1.1 Media for the test may be prepared as described |
 |below, or equivalent commercial media may be used |
 |provided that they comply with the growth promotion |
 |test. |
 |4.2.1.2 Prepare FTM media is prepared as per the produce |
 |mentioned below: |
 |Ingredient |Content |
 |L-Cystine |0.5 g |
 |Agar |0.75 g |
 |Sodium chloride |2.5 g |
 |Glucose monohydrate/anhydrous |5.5/5.0 g |
 |Yeast extract (water-soluble) |5.0 g |
 |Pancreatic digest of casein |15.0 g |
 |Sodium thioglycollate or |0.5 g |
 |Thioglycollic acid |0.3 ml |
 |Resazurin sodium solution (1 g/l |1.0 ml |
 |of resazurin sodium), freshly | |
 |prepared | |
 |Water R |1000 ml |
 |4.2.1.3 Mix the L-cystine, agar, sodium chloride, glucose, |
 |water-soluble yeast extract and pancreatic digest of |
 |casein in 700 ml of purified water R and heat until |
 |the components are completely dissolved. |
 |4.2.1.4 Dissolve the sodium thioglycollate or thioglycollic |
 |acid in the solution and, if necessary, add sodium |
 |hydroxide (1 mol/l) so that, after sterilization, |
 |the solution will have a pH of 6.9 to 7.3. |
 |4.2.1.5 Add the resazurin sodium solution, mix and place the |
 |medium in suitable vessels which provide a ratio of |
 |surface to depth of medium such that not more than |
 |the upper half of the medium has undergone a colour |
 |change indicative of oxygen uptake at the end of the |
 |incubation period. |
 |4.2.1.6 Make up the volume of the media to 1000 ml with |
 |purified water and distribute the media in suitable |
 |bottles / containers |
 |4.2.1.7 Fill the media in sterile glass bottles / containers |
 |in a volume of 80 to 100ml. The volume of the media |
 |is validated such that the sample that needs to be |
 |analyzed does not get over diluted. If any |
 |antimicrobials or preservatives are present in the |
 |test sample should get sufficiently diluted so that |
 |their inhibitory effect is lost. |
 |4.2.1.8 Sterilize the bottles by autoclaving, after |
 |sterilization, store the media at a temperature |
 |between 2 °C and 25 °C. Do not use the medium for a |
 |longer storage period than has been validated. |
 |4.2.2 Soyabean Casein Digest Medium (SCDM): |
 |Media for the test may be prepared as described below, or |
 |equivalent commercial media may be used provided that they |
 |comply with the growth promotion test. |
 |Prepare SCDM media is prepared as per the produce mentioned |
 |below: |
 |Ingredient |Content |
 |Pancreatic digest of casein |17.0 g |
 |Papain digest of soya-bean |3.0 g |
 |meal | |

		Sodium chloride	5.0 g
		Dipotassium hydrogen	2.5 g
		phosphate	
		Glucose monohydrate/anhydrous	2.5/2.3 g
		Water R	1000 ml
		4.2.2.1	Dissolve the solids in 700 ml of purified water R,
			warming slightly to dissolve the contents.
		4.2.2.2	Cool the solution to room temperature and add sodium
			hydroxide (1 mol/l) VS, if necessary, so that after
			sterilization the solution will have a pH of 7.1 to
			7.5.
		4.2.2.3	Make up the volume of the media to 1000 ml with
			purified water and distribute the media in suitable
			bottles / containers to sterilize the media by
			autoclaving.
		4.2.2.4	Fill the media in sterile glass bottles / containers
			in a volume of 80 to 100ml. The volume of the media
			is validated such that the sample that needs to be
			analyzed does not get over diluted. If any
			antimicrobials or preservatives are present in the
			test sample should get sufficiently diluted so that
			their inhibitory effect is lost.
		4.2.2.5	Sterilize the bottles by autoclaving, after
			sterilization; store the media at a temperature
			between 2 °C and 25 °C. Do not use the medium for a
			longer storage period than has been validated.
	4.3	Growth Promotion Test (GPT): Each batch of medium prepared either	
		from dehydrated medium or from ingredients has to be tested for	
		growth promotion test of aerobes, anaerobes and fungi.	
	4.3.1	GPT for Fluid Thioglycolate Media:	
	4.3.1.1	Prepare suspensions of Clostridium sporogenes,	
		Pseudomonas aeruginosa and Staphylococcus aureus and	
		inoculate each suspension into separate containers	
		of FT media.	
	4.3.1.2	Inoculate each suspension as not more than 100 CFUs.	
	4.3.1.3	Incubate the containers at 30–35 °C for next 3 days.	
	4.3.1.4	The media is considered suitable for use if a	
		clearly visible growth of the microorganisms occurs	
		at the end of incubation period.	
	4.3.2	GPT for Soyabean Casein Digest Medium:	
	4.3.2.1	Prepare suspensions of Aspergillus brasiliensis,	
		Bacillus subtilis and Candida albicans. and	
		inoculate each suspension into separate containers	
		of SCD media.	
	4.3.2.2	Inoculate each suspension as not more than 100 CFUs.	
	4.3.2.3	Incubate the containers at 20–25 °C for not more	
		than 3 days in the case of bacteria and not more	
		than 5 days in the case of fungi.	
	4.3.2.4	The media is considered suitable for use if a	
		clearly visible growth of the microorganisms occurs	
		at the end of incubation period.	
	Strains of the test microorganisms suitable for use in the Growth		
	promotion test:		
	Aerobic bacteria:		
	Staphylococcus aureus	ATCC 6538, CIP 4.83, NCTC	
		10788, NCIMB 9518, NBRC 13276,	
		NBRC 13276	
	Bacillus subtilis	ATCC 6633, CIP 52.62, NCIMB	
		8054, NBRC 3134	
	Pseudomonas aeruginosa	ATCC 9027, NCIMB 8626, CIP	
		82.118, NBRC 13275	
	Anaerobic bacterium		
	Clostridium sporogenes	ATCC 19404, CIP 79.3, NCTC 532	
		or ATCC 11437, NBRC 14293	
	Fungi		
	Candida albicans	ATCC 10231, IP 48.72, NCPF	
		3179, NBRC 1594	
	Aspergillus brasiliensis	ATCC 16404, IP 1431.83, IMI	
		149007, NBRC 9455	
	4.3.3	Execution of the test for liquid culture media:	
	4.3.3.1	Test each sterilized lot of the liquid media.	
		Transfer 1 ml of inoculum in each test bottle of	
		dilution having 10-100 CFU/ml of each test	
		microorganism listed in the Table, and its previous	
		dilution.	
	4.3.3.2	Simultaneously transfer 0.1 ml of the selected	
		dilution having 10-100 CFU/ml on Tryptone Soya Agar	
		Plate, which will act as control plate, with the	
		purpose of checking the viability and the presence	
		of contamination in the Inoculum.	
	4.3.3.3	Spread the inoculum with the help of spreader and	
		thoroughly spread it to dry	
	4.3.3.4	Incubate the inoculated media bottles and Petri	
		plates according to the requirement of test	
		microorganisms.	
	4.3.3.5	Count the cfu in the control plates, calculate the	
		productivity ratio and % efficiency as described	
		below. Observe the turbidity in case of the	
		bottles and record the findings.	
	The Productivity Ratio is calculated in the following		
	way:		

			P.R = A x B			
			C x D			
			where: A: Average Count in the Test Plate			
			B: Dilution Factor of Test Plate			
			C: Average Count in the Control Plate			
			D: Dilution factor of the Control Plate			
			Calculate the % Efficiency as follows:			
			% Efficiency = P.R x 100			
		4.3.5	Procedure for checking the Growth Promotion Properties of			
		Culture Medium/Media in case of Clost. sporogenes:				
		4.3.5.1	Take 48 hours growth of C. sporogenes in cooked meat			
		medium. Make serial dilution of the culture up to				
		10-8 dilution to determine the dilution containing				
		10 to 100 cfu.				
		4.3.5.2	Transfer 0.1 ml aliquot of the dilutions 10-5, 10-6,			
		10-7 and 10-8 in duplicate to Tryptone Soya Agar				
		(with Glucose) to confirm the viability and number				
		of organisms in the dilution.				
		4.3.5.3	Place the plates in Anaerobic Gas Jar in an inverted			
		position.				
		4.3.5.4	Place an oxygen-absorbing pack inside the Gas jar			
		for maintaining an anaerobic environment and for the				
		detection of an anaerobic environment in the Gas jar				
		place an indicator tablet. Use the pack and				
		indicator tablet according to the supplier's				
		directions. Properly seal the Gas jar.				
		4.3.5.6	Incubate the Gas jar with plates at 36 + 2 °C for 48			
		to 72 hours.				
		4.3.5.7	Count the CFU in the plates and the growth should be			
		verified by Gram Staining for its purity and				
		determination of any contamination.				
		4.3.5.8	In case of Fluid Thioglycollate Medium, transfer 1			
		ml aliquot of the dilution having 10–100 CFU and the				
		preceding dilution in the respective bottles and				
		incubate at 33.0 ± 2°C for 48-72 hours				
		4.3.5.9	Observe for growth/turbidity at the completion of			
		the incubation period.				
		4.3.6	Interpretation of the results:			
		4.3.6.1	The test media are satisfactory if clear evidence of			
		growth or turbidity appears in the test media				
		container within specified period for bacteria and				
		fungal cultures as per specification. Follow the				
		table to interpret the results.				
		Sr.No	Observation	Test	Control	Result
			bottle	plate		
		1.	Growth	Positive	Positive	Passes
		2.	Growth	Negative	Positive	Fails
		3.	Growth	Negative	Negative	Repeat
					the test	
		4.	Negative	Negative	Positive	Passes
			control *			
		4.3.6.2	The growth should be verified by Gram Staining for			
		its purity and determination of any				
		contamination.				
		4.4	Control of the culture media:			
		4.4.1	Control of Fluid Thioglycollate Medium and Tryptone Soya			
		Broth (used in Sterility Testing):				
		4.4.2	Incubate the entire sterilized lot of the medium, Fluid			
		Thioglycollate at 33.0 ± 2°C and Tryptone Soya Broth at 22.5				
		± 2°C, for 48-72 h.				
		4.4.3	If 2 % or more of the batch is contaminated discard the			
		whole batch after decontamination otherwise re-incubate two				
		bottles of each media and observe for sterility for 14 days				
		along with the Sterility test.				
		4.4.4	However, the sterility test of the test media is considered			
		invalid if the growth promotion test fails.				
		4.5	Preliminary Operations for Sterility test:			
		Switch "ON" the laminar airflow cabinet for at least half an hour				
		before starting with the test.				
		4.5.1	Preparation of Material and equipment for sterility testing:			
		4.5.1.1	Label the previously prepared media bottles for			
		Fluid Thioglycollate Medium and Tryptone Soya Broth				
		Medium with the sample name and code.				
		4.5.1.2	Remove the first paper wrapping from all the			
		sterilized materials required in the test and place				
		them in the material transfer hatch of sterility				
		testing room from the sterilization and material				
		preparation laboratory.				
		4.5.1.3	Remove the second wrapping from the materials			
		whenever required, inside the laminar airflow				
		cabinet.				
		4.6	Procedure for performing the Sterility Test by membrane filtration			
		method:				
		4.6.1	Ensure that all three valves of the sterilized membrane			
		filtration assembly are in closed position.				
		4.6.2	Connect a sterilized filtration flask to the drain pipe of			
		the sterilized membrane filtration assembly.				
		4.6.3	Open the filtration funnels and place pre-sterilized 0.45			
		micron membrane filters with the help of sterile forceps.				
		Close the funnel lid.				
		4.6.4	Open the lid of the manifold of membrane filtration assembly			

			and add approximately 5 ml of sterile purified water/rinse	
			solution in all the funnels to wet the membrane filters.	
		4.6.5	Switch "ON" the vacuum Pump and open the valves of membrane	
			filtration assembly for drainage.	
		4.6.6	Close the valves and switch "OFF" the vacuum pump. Add the	
			required amount of sample. Now rinse with enough volume of	
			sterile rinse solution for ARM and Final Bulk of Vaccine.	
			Switch ON the vacuum pump and allow it to pass through the	
			membrane filter.	
		4.6.7	After filtration switch "OFF" the vacuum pump and open the	
			latch of the funnel. Aseptically remove the membrane(s) from	
			the holder, transfer the membrane in Fluid Thioglycollate	
			Medium and in Tryptone Soya Broth Medium	
		4.6.8	System suitability for membrane filtration method	
			After transferring the content of the container or	
			containers under test to the membrane, add an inoculum of a	
			small number of viable microorganisms (not more than 100	
			CFU) to the final portion of sterile diluent used to rinse	
			the filter.	
		4.7	Sterility by direct inoculation method	
		4.7.1	With sterile syringe(s) or pipette(s) withdraw the test	
			sample as per the specification for the test sample.	
		4.7.2	Dispense 1 ml aliquots of the sample in one bottle each of	
			Fluid Thioglycollate Medium and Tryptone Soya Broth	
			Medium/SCDM media.	
		4.7.3	System suitability for direct inoculation method	
			After transferring the contents of the container or	
			containers under test to the culture medium, add an inoculum	
			of a small number of viable microorganisms (not more than	
			100 CFU) to the medium.	
		4.7.4	In both system suitability tests, use the same	
			microorganisms as those described above under Growth	
			promotion test of aerobes, anaerobes and fungi.	
		4.7.5	Incubation conditions:	
			Incubate bottles containing Tryptone Soya Broth Medium at	
			22.5 + 2.5°C and those containing Fluid Thioglycollate	
			Medium at 32.5 + 2.5°C for 14 days.	
		4.7.6	Minimum quantity of sample to be used for each medium	
		Quantity per container	Minimum quantity to be used for	
		each medium unless otherwise		
		justified and authorized		
		Liquids		
			Less than 1 ml	The whole contents of each
			container	
			1-40 ml	Half the container of each
			container but not less than 1 ml	
			Greater than 40ml and	20ml
			not greater than 100ml	
			greater than 100ml	10 % of the content of the
			container but not less than 20ml	
			Antibiotic liquids	1ml
		Solids		
			Less than 50mg	The whole contents of each
			container	
			50 mg or more but not	Half the contents of each
			less than 300mg	container but not less than 50 mg
			300 mg -5 g	150mg
			Greater than 5g	500mg
		Final Lot (Vials, ampoules, PFS)		
		Batch Size	Sample quantity for sterility	
			assay	
			Not more than 100	10% or 4 containers whichever is
			containers	greater
			More than 100 but less	10 containers
			than 500 containers	
			More than 500 containers	2% or 10 containers whichever is
			less	
		4.8	Observations and Interpretation of the results:	
		4.8.1	During the incubation period and at its conclusion, examine	
			the media for evidence of microbial growth/turbidity.	
		4.8.2	When the material/sample being tested renders the medium	
			turbid so that the presence or absence of microbial growth	
			cannot be determined by visual examination, then transfer 1	
			ml of the medium to fresh bottles of the same medium at the	
			end of incubation period of 3 days.	
		4.8.3	Incubate the original and fresh bottles for a period of next	
			14 days.	
		4.8.4	If no evidence of microbial growth is found, the product	
			examined complies with the test for sterility.	
		4.8.5	If evidence of microbial growth is found the product	
			examined does not comply with test for sterility, unless it	
			can be clearly demonstrated that the test was invalid for	
			causes unrelated to the examined product.	
		4.8.6	If test is declared to be invalid, it is repeated using	
			sterility retention sample.	
		4.8.7	If no evidence of microbial growth is found in the sterility	
			repeat test, the product examined complies with the test for	
			sterility.	
		4.8.8	If microbial growth is found in the repeat test, the product	
			examined does not comply with the test for sterility.	
		4.9	Bacteriostasis and Fungistasis assay:	
		4.9.1	Before starting to analyze of a new product it should be	

		ensured that no inhibitory substances for bacterial or	
		fungal growth are present in it, therefore, it is necessary	
		to demonstrate that no inhibitory substances are present or	
		the test conditions are to be adjusted / modified to nullify	
		these effects. The bacteriostasis and fungistasis assay	
		shall be carried out one time.	
	4.9.2	Preparation of the inoculum for Bacteriostasis and	
		Fungistasis assay:	
		Take slants of strains, listed in Table 1 below, from	
		culture bank stored at 2-8°C. Streak them on fresh slants of	
		Tryptone Soya Agar and incubate at the temperature optimum	
		for that strain as mentioned in the respective dossier. To	
		the growth, add 9 ml of Peptone Water Saline and mix gently.	
		From this suspension make serial dilutions in Peptone Water	
		Saline for all organisms. Take Tryptone Soya Agar Petri	
		plates and spread dilutions 10⁻⁵ to 10⁻⁸. Incubate at the	
		optimum temperature for that microorganism as mentioned in	
		the respective dossier for the microorganism and enumerate	
		the Colony Forming Units at the end of the incubation	
		period. Select the dilution that has 10 to 100 CFU/ml	
	TABLE 1		
	Media	Microorganisms	
	Fluid	Bacillus subtilis	ATCC 6633, CIP 52.62, NCIMB 8054,
	Thioglycollate		NBRC 3134
	Medium		
	Clostridium sporogenes	ATCC 19404, CIP 79.3, NCTC 532 or	
		ATCC 11437, NBRC 14293	
	Pseudomonas aeruginosa	ATCC 9027, NCIMB 8626, CIP	
		82.118, NBRC 13275	
	Candida albicans	ATCC 10231, IP 48.72, NCPF 3179,	
		NBRC 1594	
	Staphylococcus aureus	ATCC 6538, CIP 4.83, NCTC 10788,	
		NCIMB 9518, NBRC 13276, NBRC	
		13276	
	Tryptone Soya	Bacillus subtilis	ATCC 6633, CIP 52.62, NCIMB 8054,
	Broth Medium		NBRC 3134
	Candida albicans	ATCC 10231, IP 48.72, NCPF 3179,	
		NBRC 1594	
	Aspergillus	ATCC 16404, IP 1431.83, IMI	
	brasiliensis	149007, NBRC 9455	
	4.9.3	Perform the test with the same volume of each medium as is	
		to be used in the sterility test and introduce in it the	
		same volume of sample whose sterility is to be checked as in	
		the main test. Also inoculate 1 ml aliquot of the dilution	
		containing 10 to 100 viable microorganisms/ml of each test	
		microorganism mentioned in Table 1. Incubate all bottles of	
		Fluid Thioglycollate Medium at 33.0 ± 2°C and that of	
		Tryptone Soya Broth at 22.5 + 2°C. Observe for appearance of	
		growth for 3 days in case of bacteria and 5 days in case	
		fungi.	
	4.9.4	Control: Take same volume of each medium as is to be used in	
		the sterility test. Inoculate it with 1 ml of the dilution	
		containing 10 to 100 viable microorganisms per ml of each	
		test microorganism mentioned in Table 1. Incubate all	
		bottles of Fluid Thioglycollate Medium at 33.0 + 2°C and	
		that of Tryptone Soya Broth at 22.5 + 2°C. Incubate for 3	
		days in case of bacteria and for 5 days for fungi.	
	4.9.5	Interpretation of results	
	4.9.5.1	If clearly visible growth of microorganisms is	
		obtained after the inoculation, visually comparable	
		to that in the control bottle, the test for	
		sterility can be carried out without further	
		modification in the amount of the sample and in the	
		medium to be used.	
	4.9.5.2	If clearly visible growth is not obtained in	
		presence of the sample under examination, visually	
		comparable to that in the control bottle without the	
		sample, the sample possesses Bacteriostasis and	
		Fungistasis activity. A neutralising agent may be	
		used adjusting the concentration of neutralizing	
		agent till the inhibitory effect of Bacteriostasis	
		and Fungistasis activity is completely neutralized.	
	4.9.5.3	If a neutralizing agent is not available modify the	
		amount of the sample and medium to be used in order	
		to eliminate Bacteriostasis and Fungistasis activity	
		and repeat the test.	
	4.9.5.4	When the sample exhibits antimicrobial activity in	
		the medium, increase the quantity of the medium so	
		that, the sample is sufficiently diluted to prevent	
		inhibition of growth	
	4.9.5.5	If no growth is present in both the control and test	
		bottles, repeat the test.	
	5.	References:	
	5.1	General Requirements for the Sterility of Biological Sub stances,	
		World Health Organization Technical Report Series No. 530, 1973.	
	5.2	United States Pharmacopeia, National Formulary.	
	5.3	Indian Pharmacopeia	
	5.4	European Pharmacopeia	

- * PS: Polysaccharide
- S.No | Test/Assay | Specifications
- 1 | Appearance | White powder or lump
- 2 | Serological Identity | Positive
- 3 | Hexosamine content | $\geq 20\%$ of dry weight of PS
- 4 | Phosphorus content | $0-1.0\%$ of dry weight of PS
- 5 | Total nitrogen content | $1.5-4.0\%$ of dry weight of PS
- 6 | Kav_g determination (SEC-HPLC with RI detector) | $Kav_g \leq 0.3$
- 7 | Protein content | $\leq 5.0\%$ of dry weight of PS
- 8 | Nucleic acid content | $\leq 2.0\%$ of dry weight of PS
- 9 | Cell wall polysaccharide content | $\leq 0.5\%$ of dry weight of PS
- 10 | Endotoxin content | < 0.5 IU/ μ g of PS
- 11 | Residual CTAB Content | $\leq 1.0\%$ of dry weight of PS
- 12 | Residual Formaldehyde Content | $\leq 1.0\%$ of dry weight of PS
- 13 | Residual DOC Content | $\leq 1.0\%$ of dry weight of PS
- 14 | Moisture content | To report results

* PS: Polysaccharide

S.No | Tests/Assay | Specifications

1 | Appearance | White powder or lump

2 | Serological Identity | Positive

3 | Methyl pentose content | $\geq 20\%$ of dry weight of PS

4 | Hexosamine content | $\geq 12.5\%$ of dry weight of PS

5 | Phosphorus content | 3.0-5.5 % of dry weight of PS

6 | Total nitrogen content | 1.4-3.5% of dry weight of PS

7 | Kavg determination (SE-HPLC with RI detector) | $K_{avg} \leq 0.3$

8 | Protein content | $\leq 3.0\%$ of dry weight of PS

9 | Nucleic acid content | $\leq 2.0\%$ of dry weight of PS

10 | Cell wall polysaccharide content | $\leq 0.5\%$ of dry weight of PS

11 | Endotoxin content | < 0.5 IU/ μ g of PS

12 | Residual CTAB Content | $\leq 1.0\%$ of dry weight of PS

13 | Residual Formaldehyde Content | $\leq 1.0\%$ of dry weight of PS

14 | Residual DOC Content | $\leq 1.0\%$ of dry weight of PS

15 | Moisture content | To report results

* PS: Polysaccharide

S.No | Test/Assay | Specifications

1 | Appearance | White powder or lump

2 | Serological Identity | Positive

3 | Hexosamine Content | $\geq 13\%$ of dry weight of PS

4 | Uronic acid content | $\geq 15\%$ of dry weight of PS

5 | Total nitrogen content | 0.5-3.0% of dry weight of PS

6 | Phosphorus content | 0-1 % of dry weight of PS

7 | O-Acetyl content | To report result

9 | Kavg determination (SE-HPLC with RI detector) | $K_{avg} \leq 0.3$

10 | Protein content | $\leq 2.0\%$ of dry weight of PS

11 | Nucleic acid content | $\leq 2.0\%$ of dry weight of PS

12 | Cell wall polysaccharide content | $\leq 0.5\%$ of dry weight of PS

13 | Endotoxin content | < 0.5 IU/ μ g of PS

14 | Residual CTAB Content | $\leq 1.0\%$ of dry weight of PS

15 | Residual Formaldehyde Content | $\leq 1.0\%$ of dry weight of PS

16 | Residual DOC Content | $\leq 1.0\%$ of dry weight of PS

17 | Moisture content | To report result

1.0 |Objective |
 |To lay down the procedure for the determination of residual Cetyl Trimethyl|
 |Ammonium Bromide (CTAB) concentration in Meningococcal Purified |
 |Polysaccharide samples by Titration Method. |
 2.0 |Scope |
 |This Standard Operating Procedure is applicable to the final purified |
 |Polysaccharide samples. |
 3.0 |Responsibility |
3.1	The analyst is responsible for executing the described procedure
	correctly.
3.2	Supervisor is responsible for verifying that the procedure is carried
out correctly and check the results.	
3.3	HOD shall be responsible for overall compliance of this SOP and shall
ensure that the SOP is reviewed every two years or if required	
	earlier.
4.0	Abbreviations and Definitions
4.1	Abbreviations:
	4.1.1
	4.1.2
	4.1.3
	4.1.4
	4.1.5
	4.1.6
	4.1.7
	4.1.8
	4.1.9
	4.1.10
	4.1.11
	4.1.12
	4.1.13
	4.1.14
	4.1.15
5.0	Procedure
5.1	Preparation of solutions: Prepare all the solutions as described
	below and fill the form, label the solutions and assign lot numbers.
	5.1.1
	5.1.2
	5.1.3
	5.1.4
5.2	Testing procedure:
	5.2.1
	5.2.2
5.3	Calculations: For calculating the CTAB concentration, use the formula
	given below:
	5.3.1
5.4	Acceptance Criteria:
	The CTAB concentration in Meningococcal Purified Polysaccharide

		Samples of each serotype should be less than 1%.
6.0	Forms and Records	
	Estimation of CTAB concentration in Meningococcal Polysaccharide Samples	
	Annexure 1	
7.0	Reference	
	7.1	Standard Method ISO 2871-2-1990, "Surface Active Agents-
	Detergent-determination of Cationic-Active Matter content", Part 2:	
	Cationic –Active Matter of Low Molecular Mass (Between 200 and 500).	
	7.2	Ph.D. Thesis, "Treatment of Printing Ink Wastewater using High Liquid
	Carry over Mode Flootation", choon Jek ANG, Swinburne University of	
	Technology, Melbourne, Australia, August 2004.	

[1.0 |Purpose: |
 ||To lay down the procedure for the quantification of ribose concentration. |
 [2.0 |Applicable to: |
	This Standard Operating Procedure is applicable to all the samples of PRP
	and PRP-TT bulk conjugate of Haemophilus Influenzae type b in which ribose
	concentration is to be determined.
[3.0	Abbreviations
	3.1
[4.0	Procedure:
	4.1
[4.3	Test:
	4.3.1

	4.3.6	Close the tubes with aluminium foil and place the test
		tubes in boiling water bath for 20 minutes.
	4.3.7	Take out the tubes and cool in ice cold water.
	4.3.8	Allow the tubes to come at room temperature.
	4.3.9	Set the wavelength in spectrophotometer at 670 nm. Zero the
		absorbance of blank and then measure the absorbance of
		calibration curve and test samples.
	4.4	Calculations:
	4.4.1	Once the absorbance of the standards and the test samples
		have been taken, calculation shall be done using the excel
		sheet carrying formulas for the required calculation.
	4.4.2	Enter the absorbance for calculations starting from the
		Calibration curve prepared between 2.5 and 25 g of Ribose and
		test samples. The calculations are made according to the
		equation of minimum squares.
		$Y = mX + b$.
		Where: m = slope
		b = intercept
		Y = absorbance
		X = concentration
		(Multiply by dilution factor)
	4.4.3	Calculate the content of ribose in the test solution for each
		volume tested.
	4.4.4	Conversion factor for PRP:
		PRP content = Ribose content / 0.434
	4.4.5	In case of PRP,
		Ribose (%) in PRP on dry weight basis:
		Ribose content in dry PRP ($\mu\text{g/ml}$) =
		Ribose content in PRP($\mu\text{g/mL}$) $\times$ $1 - \{\text{Moisture} (\%) / 100\}$
		Ribose content in dry PRP ($\mu\text{g/mL}$)
		Ribose (%) = $\frac{\text{Ribose content in dry PRP } (\mu\text{g/mL})}{\text{Concentration of PRP solution } (\mu\text{g/mL})} \times 100$
		Concentration of PRP solution ($\mu\text{g/mL}$)
	4.5	Acceptance criteria:
	4.5.1	The coefficient of variation (CV) between the replicates of
		the samples should be $\leq 5\%$.
	4.5.2	Ribose content in the sample should be within the acceptance
		limits.
	5.0	Forms and Records:
		Determination Of Ribose Concentration By Orcinol Method .
	6.0	References:
		European pharmacopoeia.

1.0 |Objective |
 |To analyze purified and lyophilized polysaccharide samples of Neisseria |
 |meningitides for the determination of any residual Deoxycholic acid by |
 |reversed phase liquid chromatography with RI detector (HPLC). |
 2.0 |Scope |
 |This Standard Operating Procedure is applicable for purified and |
 |lyophilized polysaccharide samples of Neisseria meningitides in the QC. |
 3.0 |Responsibility |
 |3.1 |The analyst shall be responsible for carrying out the described |
 |procedure correctly. |
 |3.2 |The Supervisor of the laboratory shall be responsible for verifying |
 |that the procedure is complied with and for checking the results. |
 |3.3 |The departmental head/designee shall be responsible for training of |
 |the staff and for overall compliance of the SOP. |
 |3.4 |Head QA/Designee shall be responsible for final approval and |
 |subsequent compliance of the SOP. |
 4.0 |Abbreviations and Definitions |
4.1	cm	: centimeter
4.2	g	: gram
4.3	HPLC	: High Performance Liquid Chromatography
4.4	id	: internal diameter
4.5	l	: length
4.6	L	: Liter
4.7	M	: Molar
4.8	mAu	: milli Absorbance unit
4.9	mg	: milligram
4.10	min	: minute
4.11	mL	: milliliter
4.12	mm	: millimeter
4.13	nm	: nanometer
4.14	QA	: Quality Assurance
4.15	QC	: Quality Control
4.16	QCR	: Quality Control Record
4.17	rpm	: rotation per minute
4.18	r2	: Correlation Coefficient
4.19	RSD	: Relative Standard Deviation
4.20	SOP	: Standard Operating Procedures
4.21	SST	: System Suitability Test
4.22	°C	: Degree centigrade
4.23	µg	: microgram
4.24	µL	: microliter
4.25	µm	: micrometer
4.26	System Suitability test: System suitability tests are an integral	
part of gas and liquid chromatographic methods. They are used to		
verify that the resolution and reproducibility of the chromatographic		
system are adequate for the analysis to be done.		
The tests are based on the concept that the equipment, electronics,		
analytical operations and samples to be analyzed constitute an		
integral system that can be evaluated as such.		
5.0	Procedure	
5.1	Precautions to be taken:	
5.1.1	Always wear laboratory coat.	
5.1.2	Always use filtered solutions. Filtration of solution should be	
done in order to eliminate the entry of possible particles, if		
any, into the system as well as the column.		
5.1.3	Degas all solutions. Avoid entry of bubbles into the system as	
well as column.		
5.1.4	All the solution bottle openings should be well covered with	
aluminum foil.		
5.1.5	Check the calibration status of all equipment and instrument.	
5.1.6	Handle acids carefully wearing gloves and always manipulate in	
fume hood.		
5.2	Requirement:	
5.2.1	Equipment:	
5.2.1.1	Calibrated HPLC system with RI detector	
5.2.1.2	Laboratory Centrifuge	
5.2.1.3	Refrigerator	
5.2.1.4	Vacuum Pump	
5.2.1.5	Fume Hood	
5.2.2	Instrument:	
5.2.2.1	Analytical balance	
5.2.2.2	Precision balance	
5.2.2.3	pH Meter	
5.2.2.4	Magnetic stirrer	
5.2.3	Material:	
5.2.3.1	BDS Hypersil C18 silica column (250 x 4.6mm, Particle	
size:5µm)		
5.2.3.2	Screw cap vials	
5.2.3.3	Solvac Filter Holder	
5.2.3.4	Nylon 6,6 membrane 0.20 / 0.22 / 0.45 µm membrane	
filters		
5.2.3.5	Auto pipettes (20-200 (L), (100-1000 (L)	
5.2.3.6	Tips for the Auto pipettes	
5.2.3.7	Laboratory Glassware	
5.2.3.8	Eppendorf tubes	
5.2.3.9	Aluminium Foil	
5.2.3.10	Gloves	
5.2.4	Reagents:	
5.2.4.1	Deoxycholic acid (DCA)	
5.2.4.2	Methanol	

||| 5.2.4.3 |Acetonitrile |

||| 5.2.4.4 |Purified Water |

||| 5.2.4.5 |Sodium Acetate anhydrous |

||| 5.2.4.6 |Orthophosphoric Acid |

|| 5.3 |Preparation of solutions: |

|| |Prepare all the solutions as described below. Carry out the labelling|

|| |of the prepared solutions and assign the lot numbers to the |

|| |solutions. |

|| |5.3.1 |0.02M Sodium acetate Solution: To prepare 250mL of 0.02M sodium|

|| |acetate solution, weigh 410.2mg of sodium acetate anhydrous and|

|| |dissolve in 250mL of 0.2µm filtered purified water. |

|| |5.3.2 |1M Phosphoric Acid: To prepare 50mL of 1M phosphoric Acid, take|

|| |3.2mL of Orthophosphoric acid and dissolve in 46.8mL of 0.2µm |

|| |filtered purified water. |

|| |5.3.3 |Mobile phase for DCA |

|| |5.3.3.1 |To prepare 1 litre of mobile phase, add 375mL of |

|| |methanol and 375mL of acetonitrile to 250mL of 0.02M |

|| |sodium acetate solution. |

|| |5.3.3.2 |Adjust the pH to 4.3 using 1M Phosphoric acid. Filter |

|| |the mobile phase using 0.2µm nylon membrane filter and|

|| |degas the solution for 20 minutes using sonicator. The|

|| |mobile phase solution can be stored at room |

|| |temperature only for two days. |

|| |5.3.4 |3mg/mL DCA stock: |

|| |5.3.4.1 |Take 30mg of DCA in a 10mL volumetric flask and |

|| |dissolve in 7mL of mobile phase and make up the volume|

|| |to 10mL with mobile phase. |

|| |5.3.4.2 |Vortex the mixture until a clear solution is formed |

|| |and then filters it using 0.2µ inorganic membrane |

|| |filter. |

|| |5.3.5 |Working standard solutions: |

|| |5.3.5.1 |Prepare 2.0mL of 1200µg/mL standard solution by adding|

|| |800µL of 3mg/mL DCA stock in 1200µL of mobile phase. |

|| |5.3.5.2 |Make 5 concentrations of standard solutions (600µg/mL,|

|| |300µg/mL, 150µg/mL, 75µg/mL and 37.5µg/mL) as per |

|| |below table: |

S No	Standard	Volume of standard	Volume of Mobile
	Conc.	solution	Phase used
1	600µg/mL	750µL of 1200µg/mL	750µL
		solution	
2	300µg/mL	750µL of 600µg/mL	750µL
		solution	
3	150µg/mL	750µL of 300µg/mL	750µL
		solution	
4	75µg/mL	750µL of 150µg/mL	750µL
		solution	
5	37.5µg/mL	750µL of 75µg/mL	750µL
		solution	

|| |5.3.6 |Preparation of Lyophilized polysaccharide sample for HPLC |

|| |analysis: |

|| |5.3.7 |Prepare 1mg/ml of polysaccharide sample in 0.2µm filtered |

|| |purified water. Keep the sample at 2-8°C for overnight on gel |

|| |rocker. |

|| |

|| |5.3.8 |All the solutions prepared for use are vacuum filtered using |

|| |Nylon 6, 6 membrane 0.20 µm/0.22 µm, depending upon the nature |

|| |of solution to be filtered. For organic solvents 0.45µm filters|

|| |should be used (filtration not necessary if the solution is |

|| |prepared in 0.2 µ filtered purified water). |

|| 5.4 |Preparing the HPLC System for run: |

|| |5.4.1 |Flow Rate: 1.0 mL/min |

|| |5.4.2 |Detection using: RID at 30°C |

|| |5.4.3 |Run time: 18 minutes |

|| |5.4.4 |Load method : DCA.M |

|| |5.5 |Equilibration of the system and the column: |

|| |Equilibrate the column and the system at 1mL/min flow rate of mobile |

|| |phase until the RID baseline is stabilized. |

|| |5.6 |Transfer the blank, test sample(s) and standards into screw cap vials|

|| |and place against the appropriate no. on the sample tray. |

|| |5.7 |Running of sequence |

|| |5.8.1 |Load sequence DCA.S. |

|| |5.8.2 |Open the sequence parameter option by clicking Sequence menu. |

|| |5.8.3 |Enter information about the data directory and operator name. |

|| |5.8.4 |In the shutdown option, click on the post-sequence option and |

|| |select LAMPOFF/STANDBY. |

|| |5.8.5 |In sequence table option, specify the sample names, methods and|

|| |vials to run in the sequence as per the following table: |

S No	Sample Code	Sample details	No. of Injections
			Injections on
			Volume
1	Blank	Eluent Blank	1 20 µL
2	DCA Standard	3 System Suitability 6	20 µL
		test	
3	Blank	Eluent Blank	1 20 µL
4	DCA Standard	1 37.5 µg/mL	2 20 µL
		Standard	
5	DCA Standard	2 75 µg/mL Standard	2 20 µL
6	DCA Standard	3 150 µg/mL Standard	2 20 µL
7	DCA Standard	4 300 µg/mL Standard	2 20 µL
8	DCA Standard	5 600 µg/mL Standard	2 20 µL
9	Blank	Eluent Blank	1 20 µL
10	Test Sample (s)	Sample	2 20 µL

		5.8.6	Click on run sequence option in run control menu to start the
		sequence.	
		5.8.7	Each sample run will be of 18 min and after running all the
		samples the sequence will automatically stop.	
		5.8.8	Inject the eluent blank after every 20 injections of sample.
		Note: One extra vial with same SST sample can be used if 6	
		injections are not possible from single vial.	
		5.9	Calculations :
		5.9.1	Results of the sample are calculated from the calibration
		standards using the EXCEL worksheet.	
		5.9.1.1	Area under curve and retention time for the
		standards/sample is analyzed by Chemstation software.	
		5.9.1.2	Calculations for Standard curve and samples
		The calculations are made according to the equation of	
		minimum squares. $Y = mX + b$	
		Where: m = slope	
		b = intercept	
		Y = absorbance	
		X = concentration	
		5.9.1.3	Use EXCEL program for the preparation of calibration
		curve and calculation of the results.	
		5.9.1.4	Note the values of area under the curve and retention
		time for standards to prepare standard curve in	
		Annexure 1 and also in EXCEL format for calculation of	
		standard curve.	
		5.9.1.5	Prepare standard curve by creating a copy of Master
		EXCEL in the desired folder and record.	
		5.9.1.6	The amount of DCA present in the sample is calculated
		by using the formula	
		Concentration of DCA in sample (X) = $Y-b$	
		,	
		m	
		Where Y = peak area of DCA in test sample	
		b = intercept	
		m = slope	
		5.9.2	Calculations for System Suitability Test:
		5.9.2.1	Note the values of area under the curve and retention
		time for Standard 3 (SST standard) in Annexure 1 and	
		also in EXCEL format for calculation of Relative	
		standard deviation.	
		5.9.2.2	Every time a sample is run with system suitability
		test, calculate the results for system suitability.	
		5.10	Acceptance Criteria :
		5.10.1	For System Suitability Test :
		5.10.1.1	RSD calculated for Retention time of Standard 3 should
		be <5.0%.	
		5.10.1.2	RSD calculated for Area OF Standard 3 should be <5.0%.
		5.10.2	Parameters for Acceptance of the test:
		The Regression of the standard curve should be > 0.99.	
		5.10.3	The residual DCA content should be ≤1% of the dry weight of
		purified and lyophilized sample of meningitides polysaccharide.	
		5.11	Result Reporting :
		5.11.1	Using the File menu new reports can be templates, open existing
		templates, save completed reports, or print reports using the	
		current report template.	
		5.11.2	Open DCA.FRP template.
		5.11.3	The report generated can be viewed with report review icon.
		5.11.4	Print report: Click this option to take a printout of a
		Chemstation report using the current report template and data	
		from the currently loaded data file.	
		5.11.5	The printed report is signed by the analyst who does the
		analysis and checked by the other analyst/supervisor.	
		5.11.6	Report the results of DCA content of the sample(s).
		6.0	Forms and Records
		Determination of Total Protein content - Annexure 1	
		7.0	Reference
		7.1	European Pharmacopoeia, Volume 5, Protein in Polysaccharide Vaccines

S.No | Test/Assay | Specifications

- 1 | Appearance | White powder or lump
- 2 | Serological Identity | Positive
- 3 | Methyl pentose content | $\geq 15\%$ of dry weight of PS
- 4 | Phosphorus content | 2.5-5.0 % of dry weight of PS
- 5 | Total nitrogen content | 0-2.0% of dry weight of PS
- 6 | Kavg determination (SEC-HPLC with RI detector) | $K_{avg} \leq 0.3$
- 7 | Protein content | $\leq 3.0\%$ of dry weight of PS
- 8 | Nucleic acid content | $\leq 2.0\%$ of dry weight of PS
- 9 | Cell wall polysaccharide content | $\leq 0.5\%$ of dry weight of PS
- 10 | Endotoxin content | < 0.5 IU/ μ g of PS
- 11 | Residual CTAB Content | $\leq 1.0\%$ of dry weight of PS
- 12 | Residual formaldehyde content | $\leq 1.0\%$ of dry weight of PS
- 13 | Residual DOC Content | $\leq 1.0\%$ of dry weight of PS
- 14 | Moisture content | To report results

[1.0 |Objective: |
| |To lay down the procedure for the Identification of albumin and IVIG by |
| |SDS-PAGE. |
[2.0 |Scope: |
| |This standard testing procedure is applicable to the Identification of |
| |albumin and IVIG by SDS-PAGE. |
[3.0 |Responsibility |
	3.1	The QC analyst shall be responsible for executing the described
	procedure correctly.	
	3.2	The QC supervisor shall be responsible for verifying that the
	procedure is complied with and the results are checked.	
	3.3	The QC HOD shall be responsible for overall compliance of this SOP.
[4.0	Abbreviations and Definitions:	
	4.1	Abbreviations:
		4.1.1
		4.1.2
		4.1.3
		4.1.4
		4.1.5
		4.1.6
		4.1.7
		4.1.8
		4.1.9
		4.1.1
		0
		4.1.1
		1
		4.1.1
		2
		4.1.1
		3
		4.1.1
		4
		4.1.1
		5
		4.1.1
		6
		4.1.1
		7
		4.1.1
		8
		4.1.1
		9
		4.1.2
		0
		4.1.2
		1
		4.1.2
		2
		4.1.2
		3
		4.1.2
		4
		4.1.2
		5
		4.1.2
		6
		4.1.2
		7
		4.1.2
		8
		4.1.2
		9
[5.0	Procedure:	
	5.1	Preparation of solutions:
		5.1.1
		5.1.2
		5.1.3
		5.1.4
		5.1.5

			of 500µl and store at 20°C for 3 months.
		5.1.6	Gel running buffer (10X)
			Weigh 30.2 g Tris, 144 g Glycine and 10 g SDS & dissolve in
			800ml of Milli-Q water. Adjust the pH to 8.3 with HCl & make
			up the volume to 1000 ml.
		5.1.7	Sample Solubilising Buffer (6X)
			Mix 2.4 ml of 0.5M Tris (pH 6.8), 3 ml of Glycerol, 3 ml of
			20% SDS. 6 mg of Bromophenol blue. 1.6 ml of
			β-mercaptoethanol and make up the volume to 10 ml. Make
			aliquots of 1ml each and store at -20°C. For running samples
			in non reducing condition, do not add β-mercaptoethanol in
			sample solubilizing buffer.
		5.1.8	Coomassie Brilliant Blue stain:
			Weigh 1 g of Coomassie Brilliant Blue R250* and dissolve in
			100 ml of methanol, add 20 ml of glacial acetic acid and make
			up the volume to 200 ml with Milli-Q water Keep on magnetic
			stirrer for at least half an hour for proper dissolution.
		5.1.9	Destain Solution
			Mix 40 ml of methanol, 10 ml of glacial acetic acid and make
			up the volume to 100 ml with Milli-Q water. Prepare fresh at
			the time of destaining the gel.
		5.2	Casting of gel
		5.2.1	Clean the glass plates with detergent followed by washing
			with Milli-Q water and fix them in the casting frame.
		5.2.2	Prepare 12.5% Separating Gel mixture (7.5 ml) in a falcon
			tube consisting of following reagents:
			30% Acryl amide-3.125 ml
			1.5 M Tris (pH 8.8) — 1.875 ml
			20% SDS — 0.15 ml
			Milli-Q water — 2.275 ml
			10% APS- 75 µl
			TEMED- 5 µl
			Mix the contents gently by inverting the tube two-three times
		5.2.3	Immediately pour 12.5% Separating Gel mixture between the two
			glass plates leaving 4-5 cm space above for Stacking Gel and
			overlay it carefully with Milli-Q water.
		5.2.4	Prepare the Stacking Gel mixture (3.75 ml) in a tube
			consisting of following reagents:
			30% Acrylamide- 0.625 ml
			0.5 M Tris (pH 6.8) - 0.9375 ml
			20% SDS-0.075 ml
			Milli-Q water —2.0 ml
			10% APS-50 µl
			TEMED – 5 µl
		5.2.5	Pour the stacking gel mixture over the surface of the
			polymerized separating gel and immediately insert the 10-well
			comb of 0.75mm.
		5.2.6	Allow the gel to polymerize for 10 minutes.
		5.2.7	Dilute the 10X gel running buffer to 1 X by mixing 100 ml of
			the 10X stock solution in 900 ml of Milli-Q water.
		5.2.8	Remove the comb from the gel and fix the glass plate
			containing polymerized gel set on the gel cassette sandwich
			of the electrophoresis assembly.
		5.2.9	Add 1 X gel running buffer between and around the buffer tank
			of electrophoresis assembly.
		5.3	Standard and Sample Preparation :
		5.3.1	Take 0.05 mL of 10 mg/mL Reference solution/sample and
			diluted with 0.45 mL of purified water to obtain a
			concentration of 1 mg/mL.
		5.3.2	Take 5 µL of 1 mg/mL Reference solution/sample and added 8 µL
			of sample buffer (- β mercaptoethanol) and 27 µL of purified
			water to obtain a concentration of 5 µg.
		5.3.3	Loaded 20 µL of 5 µg Reference solution to obtain a
			concentration of 2.5 µg.
		5.4	Running SDS-PAGE gel:
			Connect the gel apparatus (the negative and positive electrodes)
			with the power pack and set the power pack at 300V for 45 min or
			until the dye front reaches at the bottom of the separating gel.
			Stop the power pack once dye front reaches at the bottom of the
			separating gel.
		5.5	Staining and de-staining of the gel:
		5.5.1	Once the gel run is over, remove the gel from the plate and
			place the gel in staining solution in a plastic tray for 20
			minutes with gentle agitation on gel rocker.
		5.5.2	De-stain the gel with de-staining solution to remove the
			extra stain till the gel background becomes clear and all the
			bands become clearly visible.
		5.6	Analysis of SDS-PAGE gel :
		5.6.1	After destaining, scan the gel in using densitometer and
			save for record
		6.	References:
		6.1	Bio-Rad Mini-Protean® 3 Cell Instruction Manual.
		6.2	Indian pharmacopeia

- 1 | Physical appearance | Slightly turbid and yellowish liquid
- 2 | pH | 6.0 to 8.0
- 3 | Identification of Polysaccharide
(Immunological assay) | Immunological reaction with serotype specific antibodies.
- 4 | Total Polysaccharide content
(Resorcinol Method) | Report Results
- 5 | Free polysaccharide content | $\leq 25\%$ of total polysaccharide
- 6 | Total Protein content | Report Results
- 7 | Free Carrier Protein content | $< 5\%$ of total protein content
- 8 | Polysaccharide to Protein Ratio | 0.30 to 3.00
- 9 | KD (Molecular size distribution) | $> 80\%$, KD = 0.50
- 10 | Endotoxin content | < 0.75 EU/ μ g of polysaccharide
- 11 | EDAC content | ≤ 10 ng / μ g of Polysaccharide
- 12 | Triethylamine content | ≤ 10 ng / μ g of Polysaccharide
- 13 | CDAP content | ≤ 20 ng / μ g of Polysaccharide
- 14 | Sterility | No growth should be observed during the 14 days incubation period

* PS: Polysaccharide

S.No | Test/Assay | Specifications

1 | Appearance | White powder or lump

2 | Serological Identity | Positive

3 | Methyl pentose content | $\geq 14\%$ of dry weight of PS

4 | O-Acetyl content | To report result

5 | Phosphorus content | 2.4-4.9 % of dry weight of PS

6 | Total nitrogen content | 0-1.0% of dry weight of PS

7 | Kavg determination (SE-HPLC with RI detector) | $K_{avg} \leq 0.3$

8 | Protein content | $\leq 3.0\%$ of dry weight of PS

9 | Nucleic acid content | $\leq 2.0\%$ of dry weight of PS

10 | Cell wall polysaccharide content | $\leq 0.5\%$ of dry weight of PS

11 | Endotoxin content | < 0.5 IU/ μ g of PS

12 | Residual CTAB Content | $\leq 1.0\%$ of dry weight of PS

13 | Residual Formaldehyde Content | $\leq 1.0\%$ of dry weight of PS

14 | Residual DOC Content | $\leq 1.0\%$ of dry weight of PS

15 | Moisture content | To report results

- * PS: Polysaccharide
- S. No | Test/Assay | Specifications
- 1 | Appearance | White powder or lumps
- 2 | Serological Identity | Positive
- 3 | Uronic acid content | $\geq 40\%$ of dry weight of PS
- 4 | Phosphorus content | 0-1.0 % of dry weight of PS
- 5 | Total nitrogen content | 0-1.0% of dry weight of PS
- 6 | Kavg determination (SEC-HPLC with RI detector) | $K_{avg} \leq 0.3$
- 7 | Protein content | ≤ 3.0 % of dry weight of PS
- 8 | Nucleic acid content | ≤ 2.0 % of dry weight of PS
- 9 | Cell wall polysaccharide content | ≤ 0.5 % of dry weight of PS
- 10 | Endotoxin content | < 0.5 IU/ μ g of PS
- 11 | Residual CTAB Content | ≤ 1.0 % of dry weight of PS
- 12 | Residual Formaldehyde Content | ≤ 1.0 % of dry weight of PS
- 13 | Residual DOC content | ≤ 1.0 % of dry weight of PS
- 14 | Moisture content | To report results

1.0 |Objective: |
 |To lay down a procedure to check the integrity of protein samples by |
 |distribution of molecular size using SEC HPLC. |
 2.0 |Scope: |
 |This Standard Operating Procedure is applicable to all the In- process and|
 |finished product samples. |
 3.0 |Responsibility |
3.1	The QC analyst shall be responsible for executing the described
	procedure correctly.
3.2	The QC supervisor shall be responsible for verifying that the
	procedure is complied with and the results are checked.
3.3	The QC HOD shall be responsible for overall compliance of this SOP.
4.0	Abbreviations and Definitions:
4.1	NaoH
4.2	g
4.3	mL
4.4	mg
4.5	M
4.6	min
4.7	µL
4.8	µg
4.9	°C
4.10	%
5.0	Procedure:
5.1	Equipment, Materials and Reagents
	5.1.1
	5.1.2
	5.1.3
	5.1.4
	5.1.5
	5.1.6
	5.1.7
	5.1.8
	5.1.9
	5.1.10
	5.1.11
5.2	Preparation of solutions:
	5.2.1
	5.2.2
5.3	Test method:
	5.3.1
	5.3.2
	5.3.3
	5.3.4

		the respective method file.
	5.3.5	After the first injection is completed, clean the column with
	mobile phase.	
		Flow rate: 0.5 mL/min
		Column volumes: 1
		Time: 30 min
	5.3.6	Second injection: Inject test solution by down loading the
		respective method file.
	5.3.7	After completing analysis, clean the column with mobile
	phase.	
		Flow rate: 0.5 mL/min
		Column volumes: 3
		Time: 1 Hour 30 min
	5.3.8	After completion of analysis, remove the column and purge the
	flow lines with purified water for 5 min.	
	5.3.9	Analyze the results using various integration icons as per
	the SOP	
	5.4	System suitability
	5.4.1	The reference solution injection is valid for 24 Hours.
	5.4.2	For checking System Suitability in between series of
	injections: Inject reference solution after every 6	
	injections of test solution.	
6.0	References:	
6.1	EP & IP	

1.0	Objective:			
	To lay down the Procedure for determination of human IgG antibodies against			
	Diphtheria Toxoid in plasma.			
2.0	Scope:			
	This Standard Operating Procedure is applicable to all the In- process and			
	finished product samples of IVIG.			
3.0	Responsibility			
	3.1	The QC analyst shall be responsible for executing the described		
		procedure correctly.		
	3.2	The QC supervisor shall be responsible for verifying that the		
		procedure is complied with and the results are checked.		
	3.3	The QC HOD shall be responsible for overall compliance of this SOP.		
4.0	Abbreviations and Definitions:			
	4.1	g	: Grams	
	4.2	mL	: Milliliter	
	4.3	mg	: Milligram	
	4.4	M	: Molarity	
	4.5	min	: Minute	
	4.6	μL	: Microliter	
	4.7	μg	: Microgram	
	4.8	°C	: Degree centigrade	
5.0	Procedure:			
	5.1	Equipment, Materials and Reagents		
		5.1.1	ELISA reader	
		5.1.2	Vortex mixer	
		5.1.3	Diphtheria Toxoid IgG Enzyme Immunoassay kit [IBL	
			International,REF:IB79219]	
		5.1.4	Purified water	
	5.2	Preparation of solutions:		
		5.2.1	Washing solution (1X)	
			5.2.1.1	S.No
				Volume of 10X washing solution (mL)
				Volume of Purified water (mL)
				Total Volume (mL)
				1
				10
				90
				100
	5.3	Test method:		
		5.3.1	Reference solution preparation:	
			5.3.1.1	Reference standard is Human Serum with the following
				concentrations of anti-diphtheria toxoid antibody: 1,
				0.5, 0.1, 0.01 and 0 IU/mL are part of kit content.
		5.3.2	Manufacturing pool plasma :	
			5.3.2.1	Perform 1:101 dilution fold for Manufacturing Pool
				Plasma sample.
			5.3.2.2	Required Dil. Folds
				Vol. of Pool plasma Sample (μL)
				Vol. of Specimen Diluent (μL)
				Total volume (μL)
				1:101
				5 μL of neat sample
				500
				505
		5.3.3	Test solution Preparation	
			5.3.3.1	Initially dilute the test sample 5F and 10F as
				mentioned below.
			5.3.3.2	Conc. (mg/mL)
				Required Dil. Folds
				Vol. of 5% Test Sample (μL)
				Vol. of Specimen Diluent (μL)
				Total volume
				(μL)
				50
				5F (10 mg/mL)
				20
				80
				100
				10F (5 mg/mL)
				50 μL of 5F diluted sample
				50
				100
			5.3.3.3	Then perform 1:100 dilutions for 5F and 10F diluted
				samples with sample diluent.
			5.3.3.4	Required further Dilution
				Volume of Test Sample (μL)
				Sample Diluent (μL)
				1:100
				5 μL of above 5F diluted sample
				495

				1:100
				5 µL of above 10F diluted sample
				495
		5.3.4	Internal Control:	
		5.3.4.1	Appropriate Internal control from In-House batches may	
				be considered or can use "CAL C" with 0.1 IU/mL as
				internal control. Consider the OD at 450 nm given in
				kit insert (acceptance range) as one of the control
				criteria and the second is activity obtained for 0.1
				IU/mL.
				Acceptance Criteria: The activity of internal control
				must be within 5% of the assigned value.
		5.3.6	Bring the kit to room temperature (18-25 °C) before performing	
		the test.		
		5.3.7	Before pipetting mix all reagents thoroughly by gentle tilting	
		for swinging. Vigorous shaking with formation of foam should be		
		avoided.		
		5.3.8	Place the required number of wells in the strip holder.	
		Position from well A1, filling columns from left to right		
		across the plate. When handling the plate squeeze the long		
		edges of the frame to prevent the wells falling out.		
		[Note: Return unused wells to the plastic bag immediately and		
		reseal tightly to minimize exposure to moisture. Store them dry		
		at 2-8 °C. Take care not to puncture or tear the foil bag.		
		WARNING: Exposure of wells to moisture or contamination by dust		
		or other particulate matter will result in antigen degradation,		
		leading to poor assay precision and potentially false results.		
		5.3.9	Dispense 100 µL of each calibrator, internal control and	
		diluted (1:100) samples into the appropriate wells of the plate		
		provided. Leave one well empty for the substrate blank.		
		[Note: samples should be added as quickly as possible to the		
		plate to minimize assay drift. Record the time for Incubation		
		after the addition of the last sample.		
		5.3.1	Cover plate with the enclosed foil and incubate at room	
		temperature for 60 minutes.		
		5.3.1	After incubation remove the plate and wash wells three times	
		with 300 µL wash buffer per well.		
		[Note: Avoid over dwelling of wells with wash buffer during wash		
		steps.		
		5.3.1	After the final wash invert the plate and tap the wells dry on	
		absorbent paper.		
		WARNING: The washing procedure is critical and requires special		
		attention. An improperly washed plate will give inaccurate		
		results with poor precision and high background.		
		5.3.1	Dispense 100 µL of conjugate into each well, blot the top of	
		the wells with tissue to remove any splashes. Leave one well		
		empty for the substrate blank.		
		[Note: To avoid contamination, never return the excess conjugant		
		to the reagent bottle.		
		5.3.1	Cover plate with the enclosed foil and incubate at room	
		temperature for 30 min.		
		5.3.1	After incubation remove the plate and wash wells three times	
		with 300 µL wash buffer per well.		
		5.3.1	Stop the reaction by dispensing 100 µL of Stop solution into	
		each well including substrate blank. This causes a change in		
		colour from Blue to yellow.		
		5.3.1	Read the optical density (OD) of each well at 450 nm on a micro	
		plate reader (using 620 nm as blank – optional), within 60 min		
		of stopping the reaction.		
		5.4	Calculations :	
		5.4.1	Plot the values using graph pad prism software using linear	
		concentrations on X-axis and absorbance values (O.D) values on		
		Y-axis. Multiply the extrapolated values obtained with the		
		dilution factor (initial 5 & 10 Folds only) to obtain final		
		concentration of antibodies to Diphtheria Toxoid Antigen in the		
		sample.		
		5.4.2	The sample dilution factor 1:100 must not be considered in the	
		calculation as it is already contained in the concentration		
		declaration of the standards.		
		5.4.3	In the same way calculate the concentration of Diphtheria	
		toxoid antibody in Manufacturing Pool Plasma sample and find		
		out the fold difference to determine the concentration of		
		Diphtheria antibodies in test sample to that of Manufacturing		
		Pool Plasma for fractionation.		
		5.4.4	Diphtheria Toxoid Antibodies in Drug Substance Sample:	
		Initial Conc. of Test sample (mg/mL)		
		Intermediate Dilutions		
		in folds		
		Activity		
		in IU/mL		
		Average		
		Activity in IU/L		
				50 mg/mL
				5
				Obtained activity * Dilution Folds
				Average activity in IU/mL * 1000
				10

		5.4.5	Diphtheria Toxoid antibodies in Manufacturing Pool Plasma for
		Fractionation:	
		Manufacturing Pooled Plasma For Fractionation	
		Activity in Intermediate dilution in IU/mL	
		Activity in IU/L	
		Neat	
		Obtained activity in neat sample from Calibration curve	
		Activity in IU/mL * 1000	
		5.4.6	Folds Difference
		Activity in Manufacturing	
		Pooled Plasma For Fractionation in IU/L	
		Activity in	
		Drug Substance	
		Sample in IU/L	
		Folds Difference	
		A	
		B	
		B/A	
		5.5	Method Validation Criteria:
		5.5.1	The calibrators must meet lot specific average OD criteria.
		5.5.2	The activity obtained for internal control is within 5% of
		expected value	
6.0	References:		
		6.1 Indian Pharmacopeia	

1.0 |Objective |
 |To lay down the procedure for the determination of residual Polysorbate 20 |
 |content as Oleic acid in pneumococcal polysaccharide vaccine. |
 2.0 |Scope |
 |This Standard Operating Procedure is applicable to the in pneumococcal |
 |polysaccharide vaccine samples. |
 3.0 |Responsibility |
3.1	The analyst is responsible for executing the described procedure
	correctly.
3.2	Supervisor is responsible for verifying that the procedure is carried
	out correctly and check the results.
3.3	HOD shall be responsible for overall compliance of this SOP and shall
	ensure that the SOP is reviewed every two years or if required
	earlier.
4.0	Abbreviations and Definitions
4.1	Abbreviations:
	4.1.1
	4.1.2
	4.1.3
	4.1.4
	4.1.5
	4.1.6
	4.1.7
	4.1.8
	4.1.9
	4.1.10
	4.1.11
	4.1.12
	4.1.13
	4.1.14
	4.1.15
	4.1.16
	4.1.17
4.2	Reagents:
	4.2.1
	4.2.2
	4.2.3
	4.2.4
	4.2.5
	4.2.6
4.3	Chromatographic system:
	4.3.1
	4.3.2
	4.3.3
	4.3.4
	4.3.5
	4.3.6
	4.3.7
5.0	Procedure
5.1	Preparation of solutions:
	Prepare all the solutions as described below and fill the form, label
	the solutions and assign lot numbers.
	5.1.1
	5.1.2
	5.1.3
	5.1.4
	5.1.5

		3	Parafilm.
		5.1.5.	Keep the solution for 12 hours at room temperature and
		4	then incubate it at 70oC on water bath for 6-8 hours.
	5.1.6	System suitability:	
		5.1.6.	Inject six replicates of Lauric acid standard in to the
		1	HPLC system, after filtering through 0.45µm membrane
			filter and record the chromatograms.
		5.1.6.	The system is not suitable until the following
		2	parameters are met:
		5.1.6.	The theoretical plates as determined on lauric acid peak
		3	are not less than 2500.
		5.1.6.	The relative standard deviation for six replicate
		4	injections, in terms of area determined on a lauric acid
			peak, is not more than 3.0%
	5.2	Test procedure:	
	5.2.1	Inject blank preparation (in single), Polysorbate 20	
		preparation and test preparation (in duplicate) after filtering	
		through 0.45µm membrane filter and record the chromatogram.	
	5.2.2	Compare the peak responses, in terms of area, in standard	
		preparation and test preparation.	
	5.3	Calculations:	
	5.3.1	Determination of Lauric acid content (mg/ml) in Polysorbate 20	
		(Input material):	
		Calculate the content of Lauric acid in Polysorbate 20 (input	
		material) using the below formula:	
		AT Ws 5	
		50 P	
		Lauric acid (mg/ml)= ----- X ----- X ----- X	

		AS 10 50 WT 100	
	5.3.2	Lauric Acid (in %) in Polysorbate 20 (input material):	
		Conc. Of Lauric Acid	
		(mg/ml)	
		Lauric Acid (in %) = -----	
		X 100	
		Conc. Of Polysorbate 20 (input) (mg/ml)	
	5.3.3	Conversion factor (to convert Lauric acid content to	
		Polysorbate 20):	
		100	
		Conversion factor= -----	
		Lauric acid (in %) in Polysorbate 20	
	5.3.4	Determination of Lauric acid content (mg/ml) in Test sample:	
		AT	
		WS 5 10 P	
		Lauric Acid content (mg/ml) = ----- X ----- X ----- X	
			----- X ----- X D
		AS 10 50 WS 100	
	5.3.5	Polysorbate 20 content (mg/mL) in test sample:	
		Polysorbate 20 (mg/mL) = Lauric acid content (mg/mL) X	
		Conversion factor	
	5.3.6	Assay of Polysorbate 20 (in %)	
		Polysorbate 20 content	
		in test (mg/mL)	
		Polysorbate 20 (in %) =	
			----- X 100
		Label claim (mg/mL)	
		Where,	
		AT = Average peak area of Lauric acid in test preparation.	
		As = Average peak area of Lauric acid in standard preparation.	
		Ws = Weight of the Lauric acid standard in the standard	
		preparation (in mg).	
		WT = Weight of the Polysorbate 20 (input material) taken (in	
		mg).	
		P = Potency of Lauric acid standard (in %).	
		D = Weight per mL of test sample (in g) calculated as per	
		IP2007.	
	6.0	Report	
	6.1	Report the content of Polysorbate 20/Tween 20 in %	

1.. |Purpose: |
 |To lay down the procedure for quantitative determination of |
 |Polysaccharides by HPAEC-PAD in Pneumococcal-CRM conjugate for serotype 4. |
 2. |Applicable to: |
 |This Standard Operating Procedure is applicable for quantitative |
 |determination of Polysaccharides by HPAEC-PAD of ARM of bulk conjugate of |
 |Pneumococcal CRM conjugate for serotype 4. |
 3. |Responsibility: |
3.1	Research scientist shall be responsible for executing the described
	procedure correctly.
3.2	Department Head shall be responsible for implementation of this SOP.
 4. |Safety considerations: |
 |4.1 |Always wear laboratory coat. |
 |4.2 |Always use solutions filtered through 0.22 µm membrane. |
 |4.3 |Degas all solutions. |
 |4.4 |Avoid entry of bubbles into the system as well as column. |
 |4.5 |Always store the column in NaOH solution. |
 |4.6 |Do not allow purified water to pass through the column. |
 |4.7 |Nitrogen gas pressure in four eluent bottles should always be |
 | |maintained around 5 psi. |
 |4.8 |Never apply potential to the electrode unless a stream of eluent or |
 | |water is flowing through the cell. |
 |4.9 |Do not allow the reference electrode of electrochemical detector to |
 | |dry out. |
 |4.10 |Always store the reference electrode in the storage cap filled with |
 | |saturated KCl solution when the cell is not in use. This prevents |
 | |the reference electrode membrane from drying out and damaging the |
 | |electrode. |
 5. |Equipments: |
 |2.1 |Dionex Chromatographic System (ICS-3000) |
 |2.2 |CarboPac PA 10 column with CarboPac PA 10 pre-column guard |
 |2.3 |Chromeleon software |
 |2.4 |Cooling Centrifuge |
 |2.5 |Deep-freezer (minus 20 0C ± 2 0C) |
 |2.6 |Oven |
 |2.7 |Vortex agitator |
 |2.8 |Nitrogen cylinder |
 |2.9 |Refrigerator |
 6. |Materials and reagents: |
 |6.1 |Materials: | |
 | |6.1.1 |Laboratory Glassware |
 | |6.1.2 |Screw cap vials |
 | |6.1.3 |Syringe Filters (0.2 (m) |
 | |6.1.4 |Autopipettes (10-100 (l), (20-200 (l), (100-1000 (l), |
 | | |(500-5000µl) |
 | |6.1.5 |Tips for the Autopipettes |
 |6.2 |Reagents: |
 | |6.2.1 |Galactose |
 | |6.2.2 |Sodium Hydroxide solution 50% GR |
 | |6.2.3 |Hydrochloric acid 37 %(Fuming) |
 | |6.2.4 |Deoxycholic acid, Sodium Salt (DOC) |
 | |6.2.5 |Purified water (filtered through 0.20 (m filter) & vacuum |
 | | |degassed for at least 15 min.) |
 7. |Preparation of solutions: |
 |Prepare all the solutions as described below and label the prepared |
 |solutions. |
 |7.1 |Preparation of Standards: | | | |
 | |7.1.1 |Preparation of External Standard Galactose Solution (100 (g |
 | | |(ml) |
 | | |For a volume of 100 ml, weigh: |
 | | |Galactose: 10 mg |
 | | |Dissolve in 50 ml of filtered purified water and make up the |
 | | |volume to 100 ml. Filter the solution through 0.2 (m porosity |
 | | |nylon filters. Store the solution in 1.5 ml aliquots at minus |
 | | |20 0C ± 2 0C for not more than six months. |
 |7.2 |Preparation of Working Dilutions for External Standard solutions for |
 | |analysis: |
 | |Prepare the working dilutions of the external standard for analysis |
 | |as described below: |
 | |Standard |Volume of |Volume of |Final Concentration |
 | | |Galactose |Purified Water |(µg/ml) |
 | | |standard (µl) |(µl) | |
 | |Standard 1 |50 |950 |0.25 |
 | |Standard 2 |100 |900 |0.50 |
 | |Standard 3 |150 |850 |0.75 |
 | |Standard 4 |200 |800 |1.00 |
 |7.3 |Preparation of 18mM NaOH: |
 | |Take 1.88 ml of 50% stock solution and make up the volume to 2000 ml |
 | |with filtered and vacuum degassed WFI. Shake well and keep. |
 |7.4 |Preparation of 200mM NaOH: |
 | |Take 21 ml of 50% stock solution and make up the volume to 2000 ml |
 | |with filtered and vacuum degassed WFI. Shake well and keep. |
 |7.5 |5% DOC: |
 | |Weigh and dissolve 250 mg of sodium deoxycholate in purified water |
 | |and make up the volume to 5 ml. Prepare the solution freshly. |
 8. |Procedure: |
 |8.1 |Sample Preparation: | |
 | |8.1.1 |Preparation of blank sample: |
 | | |Take 50µl of filtered purified water in a glass test tube and |
 | | |treat this sample for acid hydrolysis as described in section |

		8.1.5.	
	8.1.2	Preparation of Standards:	
		8.1.2.	Take 50 µl each of the galactose standard dilutions
		1	prepared in step 7.2 in glass test tubes. (The
			effective galactose amount will be 0.25, 0.50, 0.75 and
			1.0 µg).
		8.1.2.	Treat them for acid hydrolysis as described in section
		2	8.1.5.
	8.1.3	Preparation of sample for total Polysaccharide determination:	
		8.1.3.	Dilute sample in filtered purified water to a final
		1	concentration of 20-50 (g/ml and take 50 µl of the
			diluted sample.
			Note: The dilution factor is entered in the sequence
			run for analysis of samples.
		8.1.3.	Use this solution as the sample for acid hydrolysis as
		2	described in section 8.1.5.
	8.1.4	Preparation of sample for free Polysaccharide determination:	
		8.1.4.	Take 1 ml of sample (dilute if required) having about
		1	100-150 µg Polysaccharide in a microfuge tube, add 100
			µl of 5% DOC and mix it.
		8.1.4.	Put this tube in an ice bath for 30 minutes.
		2	
		8.1.4.	Take out and add 50 µl 1M HCl and mix thoroughly.
		3	
		8.1.4.	Centrifuge the mixture at 2-8°C for 15 minutes at 10000
		4	RPM.
		8.1.4.	Take 50µl supernatant volume in a test tube for free
		5	Polysaccharide estimation.
		8.1.4.	Note: The dilution factor is entered in the sequence
		6	run for analysis of samples.
		8.1.4.	Use this solution as the sample for acid hydrolysis as
		7	described in section 8.1.5.
	8.1.5	Acid Hydrolysis:	
		8.1.5.	Add 50 (l of Conc. HCl to each sample.
		1	
		8.1.5.	Seal each test tube with parafilm and then with
		2	aluminum foil and vortex.
		8.1.5.	Store the samples in dry hot air oven at 80 0C ± 5 0C
		3	for 2 hours.
		8.1.5.	After 2 hours, take out the samples from the oven and
		4	keep at room temperature for 15 minutes.
		8.1.5.	Dry the samples using Nitrogen gas flow.
		5	
		8.1.5.	Reconstitute the dried samples in 500 µl filtered
		6	purified water.
		8.1.5.	Analyze each sample with the method described below.
		7	
	8.2	Test Methodology:	
	8.2.1	Preparation and setting of equipment:	
		8.2.1.	Install the PA10 Guard and Analytical column.
		1	
		8.2.1.	Add 18mM NaOH (Eluent A) solution to eluent container
		2	and insert in the line of eluent A. Similarly, fill up
			second eluent container with 200mM NaOH (Eluent B) and
			insert in the line of eluent B.
		8.2.1.	Open the priming knob (one to three turns), set the
		3	gradient level of the eluents on the control panel for
			Eluent Line B = 100 % and switch on the priming through
			of the chromoleon server.
		8.2.1.	After the priming is over for Eluent B, set the
		4	gradient level of the eluents on the control panel for
			Eluent Line A = 100 % and switch on the priming
			through of the chromoleon server.
		8.2.1.	Repeat the steps 8.2.1.3 & 8.2.1.4 two more times for
		5	effective priming of both the eluents.
		8.2.1.	After the priming is over, close the priming knob and
		6	set the gradient level and flow rate of the eluents as
			follows:
			Eluent Line A (18mM NaOH) = 100% (0-20 minutes)
			Eluent Line B (200mM NaOH) = 100% (21 – 40 minutes)
			Flow rate= 1.0 ml/min
		8.2.1.	The waveform used for Polysaccharide estimation is the
		7	carbohydrate Quadruple potential.
	8.2.2	Conditioning of the system:	
		8.2.2.	At the start of each test, set the pump as described in
		1	8.2.1.6 and leave at the flow of 1.0 ml/min for at
			least 30 minutes to stabilize the baseline.
		8.2.2.	Create the program, create a sequence and load the
		2	samples in the auto sampler as per the sequence.
		8.2.2.	Start the batch after 'Ready check'.
		3	
		8.2.2.	Inject eluent blank after every twenty injections and
		4	unknown blank after each set of samples.
		8.2.2.	Record the calculated concentration in the analysis
		5	report.
	8.2.3	Calculations for Polysaccharide content:	
		8.2.3.	The Polysaccharide content can be calculated by
		1	dividing the galactose conc. obtained by the factor
			(0.215).
		8.2.3.	Derivation of Dilution factor :
		2	Theoretical volume sample volume taken = 1000µl
			Actual volume of sample taken=50µl

			Dilution Factor in this case = $1000\mu\text{l} / 50\mu\text{l} = 20$
			Dilution factor = 20
		8.2.3. Calculation of actual dilution for free Polysaccharide:	
		3 Sample volume taken = $1000\mu\text{l}$	
			Total volume = $1150\mu\text{l}$ ($1000\mu\text{l}$ sample + $100\mu\text{l}$ 5%DOC +
			$50\mu\text{l}$ 1M HCl)
			Correction Factor = $1150\mu\text{l} / 1000\mu\text{l} = 1.15$
			Actual dilution = dilution factor x 1.15 (as calculated
			in total PS)
			For Example: If sample is diluted 10 times
			calculation for the free Polysaccharide:
			$100\mu\text{l}$ sample + $900\mu\text{l}$ purified water + $100\mu\text{l}$ DOC + $50\mu\text{l}$
			1M HCl = $1150\mu\text{l}$
			Actual volume of sample taken = $50\mu\text{l}$
			Actual dilution = $20 \times 1.15 \times 10$ (sample dilution) = 230
		8.2.3. Note: Total volume of sample dilution prepared in case	
		4 of free is $1000\mu\text{l}$, dilution factor can be derived	
			according to the actual volume of sample taken for
			total Polysaccharide determination.
	8.2.4 Acceptance Criteria:		
		8.2.4. The test is valid if correlation coefficient of the	
		1 standard curve is ≥ 0.95	
		8.2.4. % Free Polysaccharide: < 25% of Total Polysaccharide	
		2	
9. Abbreviations:			
9.1	OC	: Degree centigrade	
9.2	g	: Gram	
9.3	ml	: Milli litre	
9.4	HPAEC-PAD	: High Performance Anion Exchange Chromatography with	
		Pulsed Amperometric Detection	
9.5	μl	: Micro litre	
9.6	AN	: Analytical	
9.7	PCV	: Pneumococcal conjugate vaccine	
9.8	RD	: Research And Development	
9.9	QA	: Quality Assurance	
9.10	HOD	: Head of Department	
9.11	WFI	: Water for Injection	

* PS: Polysaccharide

S.No | Test/Assay | Specifications

1 | Appearance | White powder or lump

2 | Serological Identity | Positive

3 | Uronic acid content | ≥ 12 % of dry weight of PS

4 | Hexosamine Content | $\geq 20\%$ of dry weight of PS

5 | Methyl Pentose | $\geq 25\%$ of dry weight of PS

6 | Phosphorus content | 2.0 % of dry weight of PS

7 | Total nitrogen content | 2.5-6.0% of dry weight of PS

8 | Kavg determination (SEC-HPLC with RI detector) | $K_{avg} \leq 0.3$

9 | Protein content | ≤ 7.5 % of dry weight of PS

10 | Nucleic acid content | ≤ 2.0 % of dry weight of PS

11 | Cell wall polysaccharide content | ≤ 0.5 % of dry weight of PS

12 | Endotoxin content | < 0.5 IU/ μ g of PS

13 | Residual CTAB Content | ≤ 1.0 % of dry weight of PS

14 | Residual Formaldehyde Content | ≤ 1.0 % of dry weight of PS

15 | Residual DOC Content | ≤ 1.0 % of dry weight of PS

16 | Moisture content | To report results

* PS: Polysaccharide

S.No | Test/Assay | Specifications

1 | Appearance | White powder or lump

2 | Serological Identity | Positive

3 | Methyl pentose content | $\geq 15\%$ of dry weight of PS

4 | Phosphorus content | 2.5-5.0 % of dry weight of PS

5 | Total nitrogen content | 0-2.0% of dry weight of PS

6 | Kav_g determination (SEC-HPLC with RI detector) | $Kav_g \leq 0.3$

7 | Protein content | $\leq 2.0\%$ of dry weight of PS

8 | Nucleic acid content | $\leq 2.0\%$ of dry weight of PS

9 | Cell wall polysaccharide content | $\leq 0.5\%$ of dry weight of PS

10 | Endotoxin content | < 0.5 IU/ μ g of PS

11 | Residual CTAB Content | $\leq 1.0\%$ of dry weight of PS

12 | Residual Formaldehyde Content | $\leq 1.0\%$ of dry weight of PS

13 | Residual DOC Content | $\leq 1.0\%$ of dry weight of PS

14 | Moisture content | To report results

1.0	Objective:			
	To lay down the Procedure for determination of IgG subclasses			
2.0	Scope:			
	This Standard Operating Procedure is applicable to all the In- process and			
	finished product samples of IVIG.			
3.0	Responsibility			
	3.1	The QC analyst shall be responsible for executing the described		
		procedure correctly.		
	3.2	The QC supervisor shall be responsible for verifying that the		
		procedure is complied with and the results are checked.		
	3.3	The QC HOD shall be responsible for overall compliance of this SOP.		
4.0	Abbreviations and Definitions:			
	4.1	g:	: Grams	
	4.2	mL	: Milliliter	
	4.3	mg	: Milligram	
	4.4	M	: Molarity	
	4.5	min	: Minute	
	4.6	µL	: Microliter	
	4.7	µg	: Microgram	
	4.8	°C	: Degree centigrade	
5.0	Procedure:			
	5.1	Equipment, Materials and Reagents		
		5.1.1	Multi scan Fc ELISA reader	
	5.2	Preparation of solutions:		
		5.2.1	Antibody coated microtitre plate strips	
			5.2.1.1	Each foil bag contains 2 plates of IgG Antibody-Coated
				Wells
			5.2.1.2	Each plate consists of 12 x 8 Well Strips (96 wells)
			5.2.1.3	Remove the number of strip-wells according to the
				experimental design
			5.2.1.4	Put the unused strips back into the bag
			5.2.1.5	Reseal the bag carefully and put it back into storage
				at 2 to 8°C to maintain plate integrity
			5.2.1.6	Storage and Validity: Once the vaccum sealed bag has
				been opened, carefully store the micro well strips at
				2-8 °C in the resealed bag
		5.2.2	Antibody	
			5.2.2.1	The kit contains 4 antibody vials each of 2.5 mL
				(liquid)
			5.2.2.2	Each vial consists of antibodies to 4 types of IgG
				subclasses.
				A vial of mAb Anti-Human IgG1
				A vial of mAb Anti-Human IgG2
				A vial of mAb Anti-Human IgG3
				A vial of mAb Anti-Human IgG4
			5.2.2.3	Storage and Validity: Store at 2-8 °C until expiration
				date
		5.2.3	Human Serum Control	
			5.2.3.1	The kit contains 2 lyophilized vials of human serum
				control contains 0.1% sodium azide
			5.2.3.2	Reconstitute the lyophilized control with 1.0 mL of
				diluent buffer provided in the kit
			5.2.3.3	Swirl or mix gently and allow to sit for 10 minutes to
				ensure complete reconstitution
			5.2.3.4	After reconstitution, the human serum control contains
				different IgG subtypes.
				Note: The different IgG subtypes concentration range
				for Human Serum Control vary for each lot
			5.2.3.5	Storage and Validity:
				Store the intact vials at 2-8 °C until expiration date.
				Use control within 1 hour of reconstitution
	5.2.4	Human IgG Subclass standard		
			5.2.4.1	The kit contains 2 lyophilized vials of standard
				contains 0.1% sodium azide
			5.2.4.2	Reconstitute each lyophilized standard vial with 1.0 mL
				of diluent buffer
			5.2.4.3	Swirl or mix gently and allow to sit for 10 minutes to
				ensure complete reconstitution
			5.2.4.4	When reconstituted in 1.0 mL using dilution buffer, the
				concentration of the standard is:
				13.72 µg/mL of IgG1,
				5.32 µg/mL of IgG2,
				1.34 µg/mL of IgG3, and
				0.76 µg/mL of IgG4
			5.2.4.5	Storage and Validity:
				Store the intact vials at 2-8 °C.
				Use control within 1 hour of reconstitution
	5.2.5	Secondary Antibody: Peroxidase Anti-Human IgG (50X Concentrate)		
			5.2.5.1	The kit contains 1 vial of 0.5 mL anti-human IgG
				peroxidase conjugate
			5.2.5.2	Dilute the secondary antibody in diluent buffer at a
				ratio of 1:50. For example, add 0.22 mL of conjugate to
				10.78 mL of diluent for each 96 well plate
				S.No
				Volume of 50X washing solution (mL)
				Volume of diluent buffer (mL)
				Total
				Volume (mL)
				1

0.22
10.78
11
5.2.5.3 Do not prepare more diluted antibody than required
5.2.5.4 Storage and Validity:
Store the intact vials at 2-8 °C until expiration date.
Discard any unused portion
5.2.6 Chromogen (TMB Solution)
5.2.6.1 The kit contains 1 vial of 25 mL TMB substrate solution
5.2.7 Stop Solution
5.2.7.1 The kit contains 1 vial of 25 mL stop solution
5.2.8 Diluent Buffer
5.2.8.1 The kit contains 1 vial of 135 mL diluent buffer
5.2.9 Wash Buffer Concentrate (25X)
5.2.9.1 The kit contains 1 bottle of 25X diluent buffer 100 mL
5.2.9.2 Dilute 1 volume of 25X wash buffer concentrate with 24
volumes of water.100 mL may be diluted to 2.5 L.
S.No
Volume of 25X washing solution(mL)
Volume of purified water (mL)
Total Volume(mL)
1
4
96
100
Storage & Validity: Store both the concentrate and the
working wash buffer in the refrigerator. Use the
diluted buffer within 14 days
5.2.1 Reference solution preparation
0
5.2.10. To generate a 6-point standard curve, make serial
1 dilutions of the standard using the diluent buffer
5.2.10. Take one lyophilized standard vial and with 1.0 mL of
2 diluent buffer to get 13.72 µg/mL of IgG1, 5.32 µg/mL
of IgG2, 1.34 µg/mL of IgG3, and 0.76 µg/mL of IgG4
5.2.10. Make five 2 fold serial dilutions of the standard to
3 get a series of concentrations for standard curve
construction
5.2.10. The concentrations of IgG subclasses in the diluted
4 series are as follows
Standard
Conc. in µg/mL
IgG1
IgG2
IgG3
IgG4
Dilution Fold
Neat
13.72
5.32
1.34
0.76
1:2
6.86
2.66
0.67
0.38
1:4
3.43
1.33
0.34
0.19
1:8
1.72
0.67
0.17
0.095
1:16
0.86
0.33
0.084
0.048
1:32
0.43
0.17
0.042
0.024

	5.2.1	Test solution preparation		
	1			
	5.2.11.	Test solution preparation for IgG1,IgG2,IgG3 & IgG4		
	1	subclass testing Dilute the sample 25F, 50F and 100F		
			dilutions as mentioned below:	
				Test sample Conc.
				Required Dilution Fold
				Volume of Test Sample (μL)
				Volume of Diluent(μL)
				50 mg/mL
				25F
				10 μL of 50 mg/mL test sample
				240
				50F
				125 μL of above (25 F diluted) sample
				125
				100F
				125 μL of above (50 F diluted) sample
				125
	5.2.11.	Then perform 1:100 dilution for 25F, 50F and 100F		
	2	diluted samples with sample diluent		
	5.2.11.	Add dilutions to the assay plate as mentioned in the		
	3	below table		
				Required further Dilution
				Volume of Test Sample (μL)
				Sample Diluent(μL)
				1:100
				5 μL of 25F diluted sample
				495
				1:100
				5 μL of 50F diluted sample
				495
				1:100
				5 μL of 100F diluted sample
				495
	5.3	Test method:		
	5.3.1	Bring the kit to room temperature (20-24°C)		
	5.3.2	Gently mix each kit component before use		
	5.3.3	Place the required number of wells in the strip holder.		
		Position from well A1, filling columns from left to right		
		across the plate. When handling the plate squeeze the long		
		edges of the frame to prevent the wells falling out.		
		Note: Return unused wells to the foil bag immediately with the		
		two desiccant pouches and reseal tigXXy to minimize exposure		
		to moisture. Take care not to puncture or tear the foil bag.		
		WARNING: Exposure of wells to moisture or contamination by dust		
		or other particulate matter will result in antigen degradation.		
		leading to poor assay precision and potentially false results		
	5.3.4	Add 50 μL of the appropriate human subclass specific antibody		
		(e.g., mAb Anti-Human IgG1, IgG2, IgG3 and IgG4 etc) to each		
		well in the strip.		
		Note: Take different coating antibodies for estimation of		
		different IgG subclasses		
		mAb Anti-Human IgG1 for IgG1 estimation		
		mAb Anti-Human IgG2 for IgG2 estimation		
		mAb Anti-Human IgG3 for IgG3 estimation		
		mAb Anti-Human IgG4 for IgG4 estimation		
	5.3.6	For the blank wells, add 50 μL of Diluent Buffer		
	5.3.7	Add 50 μL of diluted samples (8.3.12), standards (8.3.11), and		
		the ready-to-use Human Serum Control (8.3.3) to their		
		respective wells.		
		Note 1: Samples should be added as quickly as possible to the		
		plate to minimize assay drift.		
		Note 2: Different samples dilutions are to be added to the		
		assay plate for estimation of different IgG subclasses		
	5.3.8	Record the time for Incubation after the addition of the last		
		sample		
	5.3.9	Gently tap the plate on the side 10 times to mix		
	5.3.1	Incubate at room temperature for 30 min		
	0			
	5.3.1	After incubating for 30 min, wash the plate four times by		
	1	adding 300 μL of diluted wash buffer into each well. Soak for		
		10-15 seconds, then discard the wash buffer by inverting the		
		plate. Then remove excess wash buffer by inverting the plate		
		and tapping on absorbent paper.		
		Note: Avoid over dwelling of wells with wash buffer during wash		
		steps		
		WARNING: The washing procedure is critical and requires special		
		attention. An improperly washed plate will give inaccurate		
		results with poor precision and high back grounds.		
		Plate Layout for estimation of IgG1:		

			100
			1:100
			10000
			3
			IgG3
			50 mg/mL
			25
			1:100
			2500
			50
			1:100
			5000
			100
			1:100
			10000
			4
			IgG4
			50 mg/mL
			25
			1:100
			2500
			50
			1:100
			5000
			100
			1:100
			10000
			Note: Dilute samples producing signals greater than that of the
			highest standard in Standard curve with diluent Buffer and
			reanalyze. Multiply the concentration by the appropriate
			dilution factor
5.5	Validation Criteria:		
		5.5.1	The control values need to come within the range specified in
		the data sheet	
6.0	References:		
		6.1 Indian Pharmacopeia	

[1.0 |Objective: |
| |To lay down the procedure Identification of factor IX using Clotting |
| |Method. |
[2.0 |Scope: |
| |This Standard Operating Procedure is applicable to all the In- process and |
| |finished product samples. |
[3.0 |Responsibility |
	3.1	The QC analyst shall be responsible for executing the described
	procedure correctly.	
	3.2	The QC supervisor shall be responsible for verifying that the
	procedure is complied with and the results are checked.	
	3.3	The QC HOD shall be responsible for overall compliance of this SOP.
[4.0	Abbreviations and Definitions:	
	Not applicable	
[5.0	Procedure:	
	5.1	Equipment, Materials and Reagents
		5.1.1
		5.1.2
		5.1.3
		5.1.4
		5.1.5
		5.1.6
		— In house
		5.1.7
		5.1.8
		5.1.9
		Code: 14/148
	5.2	Preparation of solutions:
		5.2.1
		5.2.1.1.2
		5.2.2
		5.2.2.1
		5.2.2.1.4
		5.2.6
		5.2.6.1
		5.2.6.2
		5.2.6.3
		5.2.6.4
		5.2.6.5
		5.2.6.6
		5.2.6.7
[5.3	Test Method:	
	5.3.1	Predilution of Reference Standard (NIBSC) :
		5.3.1.1

				300	
				5.3.1.	Preparation of 0.5 and 0.25 IU/mL from 1 IU/mL IRS
				2	(Step-2):
				Required Activity of IRS (IU/mL)	
				Reconstituted IRS (μL)	
				Owren kollor buffer (μL)	
				Total volume	
				(μL)	
				0.5	
				150 μL of 1 IU/mL	
				150	
				300	
				0.25	
				150 μL of 0.5 IU/mL	
				150	
				300	
				5.3.2	Pre-dilution of working reference standard:
				5.3.2.	Working standard (WS) calibrated against WHO
				1	International Reference Standard
				Clotnine 600 IU I.P.:	
				Assigned Potency: 600 IU/vial i.e. ~120 IU/mL	
				5.3.2.	Reconstitute the entire vial with 5 mL of sterile water
				2	for injection and let the vial to stand at room
				temperature for 5 min to get a uniform solution.	
				5.3.2.	The activity of the WS after reconstitution with 5 mL of
				3	sterile water for injection is 120 IU/mL.
				5.3.2.	The assigned activity of the WS may vary Slightly from
				4	batch to batch.
				5.3.2.	Follow the dilutions according to the tables mentioned
				5	below.
				Pre-Dilution Step-1:	
				Dilution of 120 IU/mL to 10 IU/mL (12 Folds)	
				Required Activity of WS (IU/mL)	
				Reconstituted WS (μL)	
				Factor IX placebo (μL)	
				Total volume	
				(μL)	
				10 IU/mL	
				100 from 120 IU/mL	
				1100	
				1200	
				5.3.2.	Pre-Dilution Step-2:
				6	Dilution of 10 IU/mL to 1 IU/mL (10 Folds)
				Required Activity of WS (IU/mL)	
				Reconstituted WS (μL)	
				Factor IX Deficient (μL)	
				Total volume	
				(μL)	
				1 IU/mL	
				30 from 10 IU/mL	
				270	
				300	
				5.3.2.	Required Activity of WS (IU/mL)
				7	Reconstituted WS (μL)
				Owren kollor buffer (μL)	
				Total volume	
				(μL)	
				0.5	
				150 μL of 1 IU/mL	
				150	
				300	
				0.25	
				150 μL of 0.5 IU/mL	
				150	
				300	
				Preparation of 0.5 and 0.25 IU/mL from 1 IU/mL	
				5.3.3	Pre dilution of Test Sample:
				5.3.3.	Clotnine Drug Substance:
				1	
				5.3.3.	Prior to analysis bring the sample to room temperature.
				2	
				5.3.3.	Assigned Potency: 150 IU/mL
				3	Follow the dilutions according to the tables mentioned
				below.	
				5.3.3.	Pre-Dilution Step-1:
				4	Dilution of 150 IU/mL to 120 IU/mL (1.25 Folds)
				Required Activity of Test Sample (IU/mL)	
				Test Sample	
				(μL)	
				Factor IX Placebo (μL)	

				Total volume	
				(μL)	
				120 IU/mL	
				480	
				120	
				600	
				5.3.3.	Pre-Dilution Step-2:
				5	Dilution of 120 IU/mL to 10 IU/mL (12 Folds)
				Required Activity of Test Sample (IU/mL)	
				Test Sample	
				(μL)	
				Factor IX Placebo (μL)	
				Total volume	
				(μL)	
				10 IU/mL	
				100 from 120 IU/mL	
				1100	
				1200	
				5.3.3.	Pre-Dilution Step-3:
				6	Dilution of 10 IU/mL to 1 IU/mL (10 Folds)
				Required Activity of Test Sample (IU/mL)	
				Test Sample	
				(μL)	
				Factor IX Deficient (μL)	
				Total volume	
				(μL)	
				1 IU/mL	
				30 from 10 IU/mL	
				270	
				300	
				5.3.3.	Preparation of 0.5 and 0.25 IU/mL from 1 IU/mL :
				7	Required Activity of Test Sample (IU/mL)
				Test Sample (μL)	
				Owren koller buffer (μL)	
				Total volume	
				(μL)	
				0.5	
				150 μL of 1 IU/mL	
				150	
				300	
				0.25	
				150 μL of 0.5 IU/mL	
				150	
				300	
				5.3.4	Test procedure: Conduct the experiment in 2 steps.
				Pre assay testing to check placebo interference	
				Main experiment	
				5.3.4.	Pre assay testing to check placebo interference
				1	
				5.3.4.1.1	Check placebo interference before conducting
				the main experiment.	
				5.3.4.1.2	Switch on the Coagulometer 15 minutes prior to
				analysis.	
				5.3.4.1.3	Place all the reagents in appropriate slots of
				the Start – 4 Coagulometer for warming at 37°C	
				(5 minutes prior to analysis).	
				5.3.4.1.4	Place a Cuvette strip in the incubation area
				and pre warm at 37°C at least for 3 minutes.	
				Then dispense one magnetic ball to each	
				cuvette	
				5.3.4.1.5	Add 50 μL of Placebo of test sample under
				testing, 50 μL of Factor IX deficient plasma	
				and 50 μL of C.K. Prest to each cuvette.	
				5.3.4.1.6	Immediately start the timer corresponding to
				the incubation column for an incubation of 180	
				sec.	
				5.3.4.1.7	After 170 sec beep sound was produced.
				5.3.4.1.8	Then transfer the cuvette from the incubation
				chamber to the test column area.	
				5.3.4.1.9	Then add 50 μL of 0.025 M calcium chloride
				solution pre warmed at 37°C to each cuvette	
				with the help of Micro pipette, while adding	
				calcium chloride press timer to start the	
				measurement.	
				5.3.4.1.1	A clot should be formed within the established
				0	clotting time.
				5.3.4.1.1	If the placebo is in compliance with the
				1	established spec, proceed for the main
				experiment.	
				5.3.4.	Main experiment:
				2	
				5.3.4.2.1	Once the pre assay testing is clear, follow
				the procedure mentioned below to test the	

				sample activity.	
				5.3.4.2.2	Follow section 5.3.4.1.2 to 5.3.4.1.4 of
				current STP upto addition of samples/standard.	
				5.3.4.2.3	Add 50 μ L of reference standard/test sample,
				50 μ L of Factor IX deficient plasma and 50 μ L	
				of C.K. Prest to each cuvette.	
				5.3.4.2.4	Follow section 5.3.4.1.6 to 5.3.4.1.10 of
				current STP upto addition of 0.025 M calcium	
				chloride solution.	
				5.3.4.2.5	A clot will be formed in the reaction cuvette
				based on the activity of the sample.	
				5.3.4.2.6	Note down the clotting times which comes in
				the print out and interpret the result using	
				respective software's for data analysis.	
	5.4	Calculation and analysis:			
	5.4.1	The calculations of potency and analysis of the graphs should			
		be performed using both Combistat and Graph pad prism Software.			
6.0	References:				
	6.1 Indian Pharmacopeia				

Sr.no. | Standard details | Make | Code/Cat.No.

1. | 5th International Standard for Factor IX Concentrate , | NIBSC | 14/148
2. | WHO Reference standard (RS) (International Standard, Factor VIII Concentrate, | NIBSC | 07/350
3. | STA Deficient VIII Plasma | Diagnostica Stago | 00725
4. | FVIII-deficient plasma | Diagnostica Stago | 00621
5. | STA Deficient Factor II | Stago | 00745
6. | STA Deficient Factor VII | Stago | 00743
7. | STA Deficient Factor X | Stago | 00738
8. | Standard human plasma with known FVIII activity (e.g., 100%) | Stago | ORL17
9. | Factor IX Deficient Plasma | Stago | REF00724
10. | Working Standard (WS) - Human Normal Immunoglobulin for Intravenous use 5% I.P. (Immuglo 5% 100 mL)
calibrated against Sigma Protein Standard (200 mg/mL): | Sigma | P5369
11. | Platelet poor plasma | Merck | P2918
12. | 3rd International Standard for thrombin | NIBSC | 19/188
13. | 2nd International Standard for Fibrinogen concentrate | NIBSC | 09/242
14. | 20% Albumin Solution | Local | Any market product

- * PS: Polysaccharide
- S. No | Test/Assay | Specifications
- 1 | Appearance | White powder or lump
- 2 | Serological Identity | Positive
- 3 | Hexosamine Content | $\geq 40\%$ of dry weight of PS
- 4 | Phosphorus content | 0-1.5 % of dry weight of PS
- 5 | Total nitrogen content | 4.0-6.0% of dry weight of PS
- 6 | Kav_g determination (SEC-HPLC with RI detector) | $Kav_g \leq 0.3$
- 7 | Protein content | $\leq 3.0\%$ of dry weight of PS
- 8 | Nucleic acid content | $\leq 2.0\%$ of dry weight of PS
- 9 | Cell wall polysaccharide content | $\leq 0.5\%$ of dry weight of PS
- 10 | Endotoxin content | < 0.5 IU/ μ g of PS
- 11 | Residual CTAB Content | $\leq 1.0\%$ of dry weight of PS
- 12 | Residual Formaldehyde Content | $\leq 1.0\%$ of dry weight of PS
- 13 | Residual DOC Content | $\leq 1.0\%$ of dry weight of PS
- 14 | Moisture content | To report results

[1.0 |Objective: |
| |To lay down the procedure to determine the prekallikrein contaminants in |
| |plasma derived product. |
[2.0 |Scope: |
| |This Standard Operating Procedure is applicable to all the In- process and |
| |finished product samples. |
[3.0 |Responsibility |
	3.1	The QC analyst shall be responsible for executing the described
	procedure correctly.	
	3.2	The QC supervisor shall be responsible for verifying that the
	procedure is complied with and the results are checked.	
	3.3	The QC HOD shall be responsible for overall compliance of this SOP.
[4.0	Abbreviations and Definitions:	
	Not applicable	
[5.0	Procedure:	
	5.1	Equipment, Materials and Reagents
		5.1.1
		5.1.2
		5.1.3
		5.1.4
		5.1.5
		5.1.6
		5.1.7
		5.1.8
		5.1.9
		5.1.10
		5.1.11
	5.2	Preparation of solutions:
		NA
	5.3	Test method:
		5.3.1
		5.3.2
		5.3.3
		15
		____ µL (100 µL)
		11.25
		____ µL (75 µL) of 15 IU/mL
		____ µL (25 µL)
		7.5
		____ µL (66.6 µL) of 11.25 IU/mL
		____ µL (33.4 µL)
		5.625
		____ µL (75 µL) of 7.5 IU/mL
		____ µL (25 µL)
		3.75
		____ µL (66.6 µL) of 5.625 IU/mL
		____ µL (33.4 µL)
		1.88
		____ µL (50 µL) of 3.75 IU/mL
		____ µL (50 µL)
	5.3.4	Take neat sample for testing.
	5.3.5	Mark the microplate wells and add 50ul of prekallikerin
		Substrate to all blankk, standard and sample wells.
	5.3.6	Incubate the plate at 37±1°C for 2 minutes.
	5.3.7	Add 10ul of sample/standard in the respective wells.
	5.3.8	Incubate the plate at 37±1°C for 2-5 minutes.
	5.3.9	Measure the absorbance at 405nm in kinetic mode.
	5.3.10	Read the plate with 30 second interval for 5-10 minutes.
	5.3.11	Collect the data in the linear graph time vs absorbacne.
	5.3.12	Acceptance criteria for PKA should be less than 35IU/ml.
[6.0	References:	
	6.1 European Pharmacopeia	

Animal Experiments in Medicine

Animal houses, breeding of
experimental animals

József Kaszaki Ph.D
Institute of Surgical Research

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An experimental animal can serve as a model for a human being in all those functions in which there is a considerable degree of genetic similarity. The experiments can be acceptable for everybody only if the results are unquestionable and reproducible, i.e. exact. This can be the case only if the experimental (treatment) and control groups differ in only one aspect, i.e. the aspect under investigation. There are significant differences in sensitivity between species because of their differing genetic backgrounds. This is the reason why different species are appropriate or not for a given experiment.¹

An important requirement for laboratory animals (apart from price) is that they should be easy to keep. Today, rodents meet this demand best. Of all the mammals, mice and rats have been most frequently used in laboratories (approximately 90%), but in the last 20 years the proportion has decreased to 1/10. As compared with these species, the number of animals that can be kept as pets, such as guinea pigs, rabbits and dogs, is very low. The reasons for this are the increasing frequency of experiments without the use of animals, or the use of animals with particular characteristics (e.g. inbred, SPF, see below), and the rising prices.¹

1. Biology of the most frequently used mammals

1.1. Mouse

Seventy per cent of all animals used in biomedical activities are mice. More than one thousand stocks and strains of mice have been developed, as have hundreds of mutant stocks that are used as models of human diseases. In terms of genetics, the mouse is the most thoroughly characterized mammal. The life span is 1.5-3 years. Mice are very active animals. They are sensitive to changes in temperature, draughts and high humidity. The incisors are open-rooted and grow continuously. Mice will bite or "pinch" with their sharp incisors if mishandled. Mice should be fed a commercial pelleted mouse or rodent diet and water ad libitum. Mice are polyestrous animals and breed the year round; ovulation is spontaneous. A fertile postpartum estrus occurs 14-24 h following parturition. The duration of the estrous cycle is 4-5 days. The average gestation period is 20 days. The average litter size is 10. The pups are weaned at 3 weeks of age.

1.2. Rat

The rat is the second most commonly used animal in biomedical activities, surpassed only by its relative, the mouse. Various stocks, strains and mutants are available, but much fewer than for the mouse. Most laboratory rats are outbred albinos. Rats are less active and aggressive than mice. They are typical rodents, with continuously growing incisors. Rats should be fed a commercial pelleted rat or rodent diet and water ad libitum. The life span is 2.5-3.5 years. They are sensitive to draughts, a high level of ammonia, low humidity and noise. They are susceptible to respiratory diseases. Rats too are polyestrous animals and breed the year round; ovulation is spontaneous. The duration of the estrous cycle is 4-5 days. The average gestation period is 22 days. A fertile postpartum estrus occurs within 48 h of parturition. The average litter size is 6-12. The pups are weaned at 3 weeks of age.

1.3. Hamster

Hamsters are the third most commonly used research animal. Ninety per cent of them are Syrian hamsters. Hamsters have a tendency to bite if roughly or improperly handled, startled, injured or awakened. Females tend to be larger and more aggressive than males. The life span is 1.5-2 years. The incisors are open-rooted and grow continuously. The cheek pouches of the hamster extend to the scapulae and can be everted. Hamsters should be fed commercial pelleted hamster or rodent diet and water ad libitum. The flank or scent glands

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appear as dark patches on either flank. They are sebaceous glands that function in marking territory and mating behavior. The urethra of the female exists separately just ventral to the vulva. Breeding onset is at approximately 90 days. Hamsters are polyestrous animals and breed the year round; ovulation is spontaneous. The duration of the estrous cycle is 4 days. The average gestation period is 16 days. The average litter size is 5-9. Cannibalism is more common with hamsters than with other laboratory rodents. The young are weaned at 21 days.

1.4. Guinea pig

The most common stocks used in research are shorthairs. Guinea pigs panic easily in response to a new or frightening experience, scattering to the corner of the cage or freezing in place. Teeth-chattering of nervous guinea pigs can also be seen. The life span is 4-5 years. All teeth are open-rooted and grow continuously. The guinea pig has a large cecum. Guinea pigs have a dietary requirement for vitamin C, and should therefore be fed a commercial pelleted diet formulated specifically for guinea pigs. Guinea pigs are polyestrous and breed the year round; ovulation is spontaneous. The duration of the estrous cycle is 15-17 days and estrus itself lasts 24-48 h. The average gestation period is 65 days. The average litter size is 3-4. The young are precocious at birth 3-4; weaning occurs at 14-28 days.

1.5. Rabbit

The breeds of rabbit most commonly used in research are the New Zealand White and the Californian, which weigh 2-5 kg, i.e. they are large breeds. The life span is 5-6 years. All the teeth are open-rooted and grow continuously. The rabbit cannot vomit, like the rat and horse. The cecum is large, and terminates in the vermiform process or cecal appendix. This appendix contains a large amount of lymphoid tissue. Rabbits are coprophagous. They consume soft, moist fecal pellets produced at night (night stool). Rabbits should be fed a commercial pelleted rabbit diet; water should be provided ad libitum. Diets such as these are nutritionally complete and do not require supplementation. The highly vascularized ears are important in thermoregulation. The uterus is duplex with separate cervical and uterine openings. Breeding onset in the medium-sized breeds is at the age of 5-6 months. Rabbits are

induced ovulators with no regular estrus cycle. The vulva of the receptive doe is congested and purplish. The average gestation period is 32 days. Shortly prior to parturition (kindling), the doe builds a nest, pulling large amounts of hair from her mammary region and forelimbs. The average litter size is 7-8. Weaning occurs at 4-6 weeks.

1.6. Pig (potbellied pig, micropig, minipig)

Pigs are social animals and have a rigid dominance hierarchy. If pigs are group-housed, they will generally fight for the first 24-48 h to establish dominance, which is almost directly related to size. Pigs in general are friendly and docile, but will react severely to poor handling or a stressful environment. The life span of the potbellied pig (PBP) is probably 8-20 years with ~10-15 years typical. The PBP is sensitive to extremes of heat and cold and should be provided with a clean, dry, draught-free environment. Extended exposure to high temperatures combined with high humidity may be fatal to the PBP.

1.7. Cat

Cats are strict carnivores and predators. They have excellent vision and hearing, but their sense of smell is not as well developed as in the dog. Cats are generally solitary by nature, with strong territorial ties. At most they may form loose-knit social groups. Territories are marked by urine spraying and smearing surfaces with the scent glands located in the chin

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and in the head, in front of the ears. Cats are generally non social, but can adapt to group living. An ideal group is approximately 20 individuals, as this enables a hierarchy to form which tends to be relatively stable. Cats have certain physiological features that are more in common with those in humans than the laboratory rabbit or rodent; hence, they have been extensively used in behavioral and biomedical research, particularly in the neurological sciences. Bred cats can generally live for 10 to 13 years. Cats cannot survive on diets adequate for other carnivores, unlike dogs (e.g. cats require taurine). The female cat (or queen) is seasonally polyestrous, but a small percentage can breed continuously. Queens can be run as a harem-mating system. The only way to detect estrus is via the behavior changes (and the vaginal smear). Estrus behavior lasts for 4-6 days. The queen is an induced ovulator. The gestation period is 59-65 days. The average litter size is 4-6 kittens. The kittens are generally weaned at 6-7 weeks.

1.8. Dog

The domestic dog is a species with a large number of breeds rather than genetically defined strains. There are marked differences between the various breeds in size, appearance and life expectancy. Dogs are highly social, smart, intelligent animals. Both nonspecific and human social contacts are extremely important for their well-being. Their social isolation and solitary housing are therefore considered to be important stressors. They do best when housed either in pairs or in small compatible groups in an environment affording some level of complexity and choice. Daily contact with caretakers is important. The dog has been used as a model of many human conditions in areas such as cardiovascular research, pharmacology and toxicology. The average life span is 8-12 years. Dogs used for research or teaching may be purpose-bred. The typical laboratory dog is the beagle, but other breeds are also used in large numbers. Dogs are carnivores with monogastric digestive systems. They should be fed only complete and balanced diets. Many commercially available dog foods contain all essential nutrients in their required proportions. These foods are manufactured in dry, semi-moist and canned forms. Reproductive data: the age of puberty is 7-10 months. The length of the estrus cycle (ovarian cycle) is 21-28 days. In most breeds the interval between estrus periods is 6-7 months. Females (bitches) ovulate spontaneously. The average length of gestation (pregnancy) is 63 days. The average litter size is 3-8. The puppies are generally weaned at 6-7 weeks. 1, 4-6

2. Housing of experimental animals

2.1. General principles

Proper housing and good management of animal facilities are essential for the animal well-being, the quality of the research data, and the health and safety of the personnel. At a minimum, an animal must have sufficient space to turn around and to express normal postural adjustments, it must have ready access to food and water, and it must have enough clean-bedded or unobstructed area in which to move and rest. Animals should have opportunities to exhibit species-typical activity patterns. Whenever appropriate, social animals should be housed in pairs or groups, rather than individually, provided that such housing is not contraindicated by the protocol in question and does not pose an undue risk to the animals. Highly social and many other domesticated animals benefit from positive human interaction.

2.2. Physical facilities

The microenvironment of an animal is the physical environment immediately surrounding it (the primary enclosure), with its own specific factors (such as a cage or pen).

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The primary enclosure provides the limits of the animal's immediate environment. The macroenvironment is the physical environment of the secondary enclosure (such as an animal holding room, a barn, etc.). Basic types of establishments for animal keeping are breeding / supply establishments and user establishments. Important elements of the primary and secondary enclosures are as follows:

■ A cage is a permanently fixed or movable container that is enclosed by solid walls and, at least on one side, by bars or meshed wire, in which one or more animals are kept or transported; depending on the size of the container and the stocking density, the freedom of movement of the animals is relatively restricted.

■ A pen is an area enclosed by walls, bars or meshed wire, in which one or more animals are kept; depending on the size of the enclosure and the stocking density, the freedom of movement of the animals is usually less restricted than in a cage; pens must be designed for the well-being of the species; they must permit the satisfaction of certain ethological needs (for example, the need to climb, hide or shelter temporarily).

■ A run is an area enclosed by walls, fences and bars or meshed wire and frequently situated outside permanently fixed buildings, in which animals kept in cages or pens can move freely during certain periods of time.

■ An animal holding room is a room situated inside a building where animals are housed in cages or pens, either for breeding and stocking, or during the conductance of an experiment.

The majority of holding rooms are usually designed to house rodents. Such rooms may

also frequently be used to house larger species. Care should be taken not to house together species which are incompatible. Walls and floors should be smooth, impervious and have a non slippery, easily washable surface. Drains, if any, should be adequately covered. Such rooms should be provided with facilities for carrying out minor experiments and manipulations, where appropriate.

Service rooms are rooms for the storage of food, clean cages, bedding, cleaning and washing materials. The rooms for food and bedding must be cool and dry, and vermin- and insect-proof. The rooms for clean cages, instruments and other equipment must be adequately designed; the cleaning process must be arranged so as to separate the flow of clean and dry equipment in order to prevent the contamination of newly cleaned equipment. 1,2,4

2.3. Housing of the most frequently used species

Rodents, guinea pig, ferret

These animals are usually kept in plastic cages with solid bottom and walls, of different sizes (shoe-boxes), with roofs made of wire, and with bedding. Guinea pigs and ferrets may be housed in small pens, made of plastic, metal or concrete, enclosed by low walls, with open roof and with bedding (wood shavings). Ferrets and guinea pigs are social animals, and it is therefore best to keep them in pairs or in groups. They need a complex, dynamic environment. For guinea pigs, it is necessary to provide a shelter in their cage or pen, where they can seek safety.

Rabbit

Rabbits are usually kept individually in small cages with a solid or meshed-wire bottom. They are gregarious animals, and should therefore be housed in compatible groups in pens with bedding (indoor or indoor-outdoor).

Cat

Cats are kept in individual caging, or in group pens or cages. The housing of cats in cages should be strictly limited. They should be let out for exercising at least once a day, if this does

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not interfere with the experiment. In the group situation, the most essential criterion for acceptable housing is the provision of an adequate number of hides and escapes for cats. Cat pens should be equipped with dirt trays, an amply raised shelf room for resting, and objects suitable for climbing and claw-trimming. In a closed pen, many non convertible windows are desirable.

Dog, swine (miniature pig)

There is a great range of possibilities as concerns housing: it may be situated in indoor or outdoor pen areas, with open access to the outside run, or in an indoor environment with some sort of complexity built into it. Single housing may be necessary because of the demands of an experiment. Dogs can be given opportunities for activity by having access to a run or being moved into another area (such as a large cage or outdoor pen), which provides more space for movement.

Monkeys (non human primates)

Non human primates are social animals which, when deprived of companionship for an extended time, develop unmistakable signs of depression and frustration (as do dogs and pigs). They are physiologically and anatomically adapted to live in a complex, dynamic environment. Housing such animals in pairs offers a practicable alternative to group-housing. Raised resting surfaces or perches are also desirable for monkeys and dogs.

3. Husbandry

The person in charge of the establishment must ensure regular inspection of the animals and supervision of the accommodation by a veterinarian (or other competent person).

3.1. Food, drinking water and bedding

Feed

Animals should be fed palatable, non contaminated and nutritionally adequate food, daily or according to their particular requirements, unless the protocol in which they are being used requires otherwise. Nowadays, the feed is 'standard' mixed feed, pressed into pellets for almost all species (dry laboratory-animal diets). Packing, transport and storage must be such as to avoid contamination, deterioration or destruction. The major problems with feed may be an increase in the number of bacteria, rancidity, and the appearance of toxins caused by fungi. The feed distribution process may vary according to the species, but it must be such as to satisfy the physiological needs of the animal: for rodents *ad libitum* (as much they like), but for cats, dogs and pigs in portions. Feeders should be designed and placed so as to allow easy access to food and to minimize contamination.1-6

Drinking water

Animals should have access to potable, uncontaminated drinking water according to their particular requirements. (During transport, it is acceptable to provide water as part of a moist diet.) Watering devices should be checked daily to ensure their proper maintenance, cleanliness and operation. The methods commonly used in watering involve bottles, dishes and automatic systems. Bottles are often used for small animals such as rodents and rabbits, while dishes are customary for cats, dogs or pigs. When bottles are used, they must be made from translucent material in order to allow their contents to be monitored. All bottles and accessories must be taken to pieces, cleaned and sterilized at appropriate and regular periods. It is better to replace water bottles than to refill them, because of the potential for microbiologic cross-contamination.

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Bedding

Bedding must be dry, absorbent, non dusty, non toxic and free from infectious agents or vermin or any other form of contamination. Ordinarily, softwood shavings and chips are appropriate and used, made from non resinous wood. Special care must be taken to avoid sawdust or bedding material derived from wood which has been treated chemically. Bedding should be used in amounts sufficient to keep animals dry between cage changes.

Sanitation (cleaning)

Sanitation (the maintenance of conditions conducive to health) involves bedding changing (as appropriate), cleaning and disinfection. Cleaning removes excessive amounts of dirt and debris, and disinfection reduces or eliminates unacceptable concentrations of microorganisms. The standard of a facility depends very much on good hygiene. Clear instructions must be given for the changing of bedding in cages and pens, and adequate routines must be established for the cleaning, washing and decontamination of cages and

accessories, bottles and other equipment. All components of the animal facility, including animal rooms and support areas, should be cleaned regularly and disinfected as appropriate to the circumstances.

Contacts with the animals

The performance of an animal during an experiment depends very much on its confidence in man, something which has to be developed. It is therefore recommended that frequent contact should be maintained so that the animals become familiar with human presence and activity (adaptation). The staff must be sympathetic, gentle and firm when associating with the animals. 1-6

4. Factors in the macroenvironment

Temperature

The temperature must be measured continuously. The recommended (optimal) temperature range for mice, rats, guinea pigs and miniature pigs is 20–24 °C, while for dogs, cats and rabbits it is 15–21 °C. Some conditions might require an increased environmental temperature, such as postoperative recovery, the housing of some hairless rodents and the housing of neonates.

Humidity

Extreme variations in relative humidity (RH) have an adverse effect on the health and well-being of animals. The RH must ordinarily be maintained at 55–100%. Levels below 40% or above 70% for a prolonged period must be avoided.

Air ventilation

The purposes of ventilation are to supply adequate oxygen (fresh air); to remove thermal loads caused by animal respiration, lights and equipment; to dilute gaseous and particulate contaminants (e.g. ammonia or dust); to adjust the moisture content of the room air; and, where appropriate, to create static-pressure differentials between adjoining spaces. The air in the room should be changed at frequent intervals. A ventilation rate of 15–20 fresh-air changes per hour is normally adequate. The use of recycled air to ventilate animal rooms must be avoided. The ventilation system must be designed so as to avoid harmful draughts. The air used for ventilation should be heated in winter and cooled in summer. Frequent bedding changes and cage-cleaning, coupled with husbandry practices, such as a low animal

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density within the room, and lower environmental temperature and humidity, can also reduce the concentration of toxic or odor-causing gases in the animal room air.1-6

5. Microbiological environment (microorganisms and infections)

Infectious agents may affect animal populations in various ways. Some are pathogenic and may induce clinical signs with variable morbidity and mortality. However, most microorganisms induce no or only mild disease. Silent, asymptomatic infections (adventitious pathogens and opportunistic organisms) are often activated by experimental procedures (stress, immunosuppressant, etc.) or environmental influences (transportation, toxic gases, etc.). Frequently, certain strains of a given species are more sensitive to an infection, whereas the same agent may cause only milder or different symptoms in other strains, or the infection may be asymptomatic.

The types of organisms that can infect (experimental) animals include bacteria, protozoa, fungi, viruses, helminthes and arthropods. Many agents (even in the event of clinically silent infections) may exert impacts on the physiological parameters and hence on the results of animal experiments. It is obvious that experimental data obtained from diseased animals should, if at all, be used only with maximal caution. The use of laboratory animals that are free from unwanted microorganisms is an important prerequisite for the acquisition of reliable and reproducible results with a minimum of animals and is a significant contribution to animal welfare.1

5.1. Microbiological quality and classifications of experimental animals

The presence of unwanted microorganisms and the suitability of an animal population for a specific experiment can be demonstrated only by comprehensive health monitoring before and during experimentation. The quality of animals is most commonly characterized in terms of the microbiological (hygienic) status and of the system used in raising animals to ensure that a specific microbiological status is maintained. There are three major types of maintenance (hygienic levels):

■ **Isolator-maintained:** The animals are kept in a sterilizable chamber with a sterilized air supply, a mechanism for introducing sterilized materials, and a series of built-in gloves.

■ **Barrier-maintained:** The animals are bred and kept in a dedicated space, behind a barrier. For these facilities, personnel enter through a series of locks and are usually required to shower and use disinfected clothing. All equipment, supplies and conditioned air are sterilized or disinfected. Barrier facilities are designed to exclude organisms for which animals are the primary or preferred hosts, but will generally not exclude organisms for which humans are hosts. Barrier maintenance can also be achieved at the cage or rack level with equipment that can be sterilized or otherwise disinfected.

■ **Conventionally-maintained:** The animals are raised in areas that have no special impediments to the introduction of microorganisms. This method of maintaining animals cannot ensure the stability of the microbiological status, because unwanted organisms can be introduced at any time. Micro-isolation cages are generally used to protect animals in otherwise conventional rooms.

Various classifications have been developed to define the microbiological quality of laboratory animals. The most important classifications (hygienic categories) are as follows:

■ **Germ-free (GF)** animals that are derived by Cesarean section or embryo transfer and reared and maintained in an isolator with aseptic techniques.

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■ **Specified-pathogen-free (SPF)** animals that show no evidence (usually by serology, culturing or histopathology) of the presence of particular microorganisms. An animal can be classified as SPF if it is free of one or many pathogens. Commercial suppliers have coined various terms to indicate the SPF status. All terms are related to specified organisms from which the animals are stated to be free and for which they are regularly monitored.

■ **Virus-antibody-free (VAF)** animals that are free of antibodies to specified rodent

viruses. Animals might not be free of viruses other than those specified and might not be free of other microorganisms.

- Conventional animals in which the microbial burden is unknown, uncontrolled or both.
- Clean conventional or minimal disease (MD) animals that are maintained behind a low-security barrier and demonstrated to be free of selected pathogens. Use of this term should be avoided because of the lack of precision of its meaning.

5.2. Sources of infections / microbial contamination

The sources of microbial contamination include vermin, experimentally infected and spontaneously ill laboratory animals or their tissues or tumors, air, food, water, bedding, ancillary equipment and personnel. (The highest microbiological risk is man himself.) Good facility management practices and constant surveillance are necessary to minimize the introduction of unwanted microbes.

5.3. Zoonosis

Infectious diseases that may be transmitted from other animals, either wild or domestic, to humans, or from humans to animals, are called zoonoses (the latter is sometimes called reverse zoonosis). The list of potential zoonoses related to working with animals in research, teaching or testing is quite long, though in reality the risks are very low when the common small laboratory animal species are dealt with in the laboratory. There are a number of reasons for this low risk. Firstly, commercial suppliers of laboratory animals have done an excellent job of producing disease-free animals. Further, institutions have generally developed good occupational health and safety programs that include active veterinary monitoring and care programs. The risk of exposure to zoonotic diseases is greater for those who work with experimental animals from random sources (including cats, dogs and most livestock), and for field researchers studying wild animals in their habitat. Working with non human primates in the laboratory is a special case because of the many zoonotic concerns.¹⁻³

5.4. Animal quarantine and stabilization

Whenever possible, the health status of every animal should be ascertained before it is brought into the facility. Quarantine is the separation of a newly received animal from those already in the facility until the health of the new animal has been evaluated. Effective quarantine minimizes the introduction of disease agents into established colonies and prevents the possibility of zoonoses.

The quarantine period should be of sufficient duration to allow the expression of diseases present in the incubation stages. Some or all of the following should be achieved during the quarantine and stabilization period: diagnosis, control, prevention and treatment of diseases; physiological and nutritional stabilization; and grooming to include ectoparasite control (many zoonotic agents require an arthropod vector).

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The period of quarantine is in some cases laid down in the national animal health regulations. In others, it varies according to the circumstances and should be determined by a competent person, normally the veterinarian appointed by the establishment. Even when the animals are believed to be in sound health or, in the case of a higher microbiological status, after transportation (receiving), it is good husbandry for them to undergo a period of acclimatization (physiological, psychological, and nutritional stabilization) before being used in an experiment.

5.5. Symptoms, diagnosis, standard procedure for infected, sick animals

At the beginning of an illness, there are only general symptoms. It is usually too late to save the stock (group of animals) when specific signs are present. Clinical examination is difficult in some species (e.g. rodents). An animal that appears sick must be isolated immediately (in an SPF facility, it must be removed), and its surroundings must be disinfected. Apparently sick animal must be examined. In a conventional environment, isolated cases or the whole animal population can be treated.

6. Social environment and other environmental factors

The social environment usually involves physical contact and communications among members of the same species (conspecifics), although it can include non contact communication among individuals through visual, auditory and olfactory signals.

6.1. Space recommendations and group size

An animal's space needs are complex, and considerations of only the animal's body weight or surface area are insufficient. Space needs are given high priority in animal welfare regulations, where the minimal space needs and cage sizes are specified in detail. Some species benefit more from wall space (e.g. rodents), shelters (e.g. primates) or cage complexities (e.g. cats and primates) than from simple increases in floor space. Whenever appropriate, social animals should be housed in pairs or groups, rather than individually, provided that such housing is not contraindicated by the protocol in question and does not pose an undue risk to the animals. Space for group-housed animals should be based on the individual space needs, the behavior, the compatibility of the animals, the number of animals, and the goals of the housing situation.

6.2. Other factors

Effects between and within species (sound, olfactory and microbiological effects) should be considered. Because of these effects, all animal species should be kept (housed) in separated areas (rooms and barns). Noisy animals, such as dogs, pigs and nonhuman primates, should be housed away from quieter animals, such as rodents, rabbits and cats. Other effects involve light, color and the biological rhythm.^{1,2,3}

7. Transport conditions

All animals must be acquired lawfully. The use of purpose-bred research animals might be desirable and, for the most frequently used animals, is obligatory by the regulations. As a rule, vendors of purpose-bred animals regularly provide information that describes the genetic and pathogen status of their colonies or individual animals. All transportation is a stressful experience for animals and as a consequence, the physiological parameters may be seriously modified. For normalization of these modifications, rodents need at least 2 weeks, while animals with a highly developed nervous system (e.g. primates) need at least one

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month. All parts of the transportation process must be carried out according to the prevailing

animal health, transport and welfare regulations. All transportation of animals should be planned to minimize the transit time and the risk of zoonoses, to protect against environmental extremes, to avoid overcrowding, to provide food and water when indicated, and to protect against physical trauma. Animals must be in good health for transportation and it is the duty of the sender to ensure that they are so. Female animals which are likely to give birth during the transport must be excluded. Consignments of animals must be received and unpacked without avoidable delay. After inspection, the animals must be transferred to clean cages or pens and be supplied with feed and water as appropriate.1-3

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1. Kállai László: Laborállat könyv, Royal Press Hungary Kft, Budapest, 2003.
2. NRC ILAR Commission: Guide for the Care and Use of Laboratory Animals, Natl. Academy Press, Washington DC, 1996.
3. Olfert ED, Cross BM, McWilliam AA: Guide to the Care and Use of Experimental Animals I-II., Canadian Council on Animal Care, Ontario, 1993.
4. Tuffery AA: Laboratory Animals, J.Wiley, London, 1995.
5. NRC ILAR Commission: Laboratory Animal Management: Dogs, NAP, Washington DC, 1994.
6. NRC ILAR Commission: Laboratory Animal Management: Rodents, NAP, Washington DC, 1996.

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1.0 |Objective |
 |To lay down the procedure for Estimation of Methyl pentose content in |
 |Pneumococcal polysaccharide samples. |
 2.0 |Scope |
 |This Standard Operating Procedure is applicable to all the in process and |
 |polysaccharide samples of pneumococcal serotypes 6A, 6B, 7F, 18C, 19A, 19F |
 |and 23F in which the Methyl pentose content is to be determined. |
 3.0 |Responsibility |
3.1	The analyst is responsible for executing the described procedure
	correctly.
3.2	Supervisor is responsible for verifying that the procedure is carried
out correctly and check the results.	
3.3	HOD shall be responsible for overall compliance of this SOP and shall
ensure that the SOP is reviewed every two years or if required	
	earlier.
4.0	Abbreviations and Definitions
4.1	Abbreviations:
	4.1.1
	4.1.2
	4.1.3
	4.1.4
	4.1.5
	4.1.6
	4.1.7
	4.1.8
	4.1.9
4.2	Reagents:
	4.2.1
	4.2.2
	4.2.3
	4.2.4
4.3	Equipments and material
	4.3.1
	4.3.2
	4.3.3
	4.3.4
	4.3.5
	4.3.6
5.0	Procedure
5.1	Preparation of solutions: Prepare all the solutions as described below
and fill the form, label the solutions and assign lot numbers.	
	5.1.1
	5.1.2
	5.1.3
	5.1.4
5.2	Testing Procedure:
	5.2.1
	5.2.2

			outside the range of the calibration curve, carry out the	
			calculation for more appropriate dilution.	
			5.2.2.	Place the above test tubes in crushed ice and add 2.25 ml
			2	of cooled 15.7M Sulphuric acid drop-wise to each of the
			standard, blank and sample and mix well.	
			5.2.2.	Cool the test tubes to reach room temperature and
			3	incubate the tubes in water bath at 90°C for 7 min.
			5.2.2.	Allow the test tubes to cool down to room temperature for
			4	5 min and add 50µl of 0.2 M L- Cysteine hydrochloride
			solution and mix well.	
			5.2.2.	Incubate the test tube in dark at room temperature for
			5	two hours.
			5.2.2.	Observe for the appearance of greenish yellow color,
			6	which can vary with the concentration.
			5.2.2.	Measure the absorbance at the difference of 396nm and
			7	430nm against the blank using spectrophotometer.
			5.2.2.	Note that all the readings of the assay tubes should be
			8	taken within 30 mins. of completion of the assay and
			maximum 10 samples should be analysed in one single	
			assay.	
		5.3	Calculations:	
		5.3.1	Once the absorbance of the standards and the test samples have	
		been taken, calculation shall be done using the excel sheet		
		carrying formulas for the required calculation.		
		5.3.2	Always copy the master excel sheet (sheet carrying formulas for	
		the required calculation), never use the previously used excel		
		sheet for new calculations.		
		5.3.3	Copy the master excel sheet, keep it for use in respective	
		folder and then fill in the required sample details (Batch		
		number), sample dilution (if applicable), name of analyst, date,		
		etc.		
		5.3.4	Proceed for calculations starting from the Calibration curve.	
		The calculations are done according to the equation,		
		Y = mX + b.		
		Where: m = slope		
		b = intercept		
		Y = absorbance		
		X = concentration		
		(multiply with the dilution factor)		
		5.3.5	In case of dry polysaccharide,	
		Methyl pentose (%) in Polysaccharide on dry weight basis:		
		Methyl pentose in dry Polysaccharide. =		
		Methyl pentose in Polysaccharide. Sol. (µg/ml) X 1 – {Moisture		
		(%) / 100}		
		Methyl pentose in dry Polysaccharide (µg/ml)		
		Methyl pentose (%) = _____ x		
		100		
		Conc. of Polysaccharide solution (µg/ml)		
		5.3.6	To calculate the pure polysaccharide content (mg) present in dry	
		polysaccharide sample in of each serotype based on the methyl		
		pentose content obtained, following formula is applied:		
		Pure Polysaccharide content(mg) / Dry polysaccharide content		
		(mg)		
		= (Methyl pentose in dry polysacc (mg) X Conversion factor)		
		5.3.7	The conversion factor of each serotype is mentioned below	
		Serotype	Percentage of Methyl Pentose as	Conversion
		per WHO TRS 977	factor	
		6A	22.86 %	4.37
		6B	22.86 %	4.37
		7F	26.40 %	3.79
		18C	15.96 %	6.26
		19A	26.32 %	3.79
		19F	26.32 %	3.79
		23F	40.77 %	2.45
		5.3.6	Record all the calculations in the data record form and also	
		attach the hard copy of the excel sheet used for calculation		
		along it.		
		5.4	Acceptance criteria:	
		5.4.1	The test is valid if Regression of the standard curve is ≥0.99	
		and the percentage of Coefficient of Variation is ≤ 15% within		
		the replicates.		
		5.4.2	Methyl pentose content in Purified Pneumococcal polysaccharide	
		should meet the laid specifications for each serotype.		
		6.0	Forms and Records	
		Determination of Methyl Pentose content - Annexure 1		
		7.0	Reference	
		7.1	WHO TRS No. 977 Annexure 3: Recommendations to assure the quality,	
		safety and efficacy of pneumococcal conjugate vaccines.		
		7.2	Zacharibis Dische and Landrum B. Shettles (1948) "A specific color	
		reaction of methyl pentoses and a spectrophotometric micro method for		
		their estimation". J.Biol.Chem.175:595-603.		
		7.3	European Pharmacopoeia, volume 5, Methyl pentose in polysaccharide	
		vaccines, 2.5.21.		

[1.0 |Objective: |
| |To lay down the procedure to determine the thrombogenic (procoagulant) |
| |activity in IVIG. |
[2.0 |Scope: |
| |This Standard Operating Procedure is applicable to all the In- process and |
| |finished product samples of IVIG. |
[3.0 |Responsibility |
	3.1	The QC analyst shall be responsible for executing the described
	procedure correctly.	
	3.2	The QC supervisor shall be responsible for verifying that the
	procedure is complied with and the results are checked.	
	3.3	The QC HOD shall be responsible for overall compliance of this SOP.
[4.0	Abbreviations and Definitions:	
	Not applicable	
[5.0	Procedure:	
	5.1	Equipment, Materials and Reagents
		5.1.1
		5.1.2
		5.1.3
		5.1.4
		5.1.5
	5.2	Preparation of solutions:
		5.2.1
		5.2.2
		5.2.3
		5.2.4
	5.3	Test method:
		5.3.1
		5.3.2
		5.3.3
		5.3.4
		5.3.5
		5.3.6
		5.3.7
		5.3.8
		5.3.9

		samples into two different cuvettes.
	5.3.10	Then immediately transfer the cuvettes into the reaction chamber
		and immediately add 50 μL of 4.85 g/L calcium chloride into each
		cuvette with the help of micropipette.
	5.3.11	Note down the clotting time which comes in the print out and
		record the result
6.0	References:	
6.1 Indian Pharmacopeia		

[1.0 |Objective: |
| |To lay down the procedure for determine the human coagulation factor X |
| |activity in Factor IX. |
[2.0 |Scope: |
| |This Standard Operating Procedure is applicable to all the In- process and |
| |finished product samples. |
[3.0 |Responsibility |
	3.1	The QC analyst shall be responsible for executing the described
	procedure correctly.	
	3.2	The QC supervisor shall be responsible for verifying that the
	procedure is complied with and the results are checked.	
	3.3	The QC HOD shall be responsible for overall compliance of this SOP.
[4.0	Abbreviations and Definitions:	
	Not applicable	
[5.0	Procedure:	
	5.1	Equipment, Materials and Reagents
		5.1.1
		5.1.2
		5.1.3
		5.1.4
		5.1.5
	5.2	Preparation of solutions:
		5.2.1
		5.2.2
		5.2.3
[5.3	Pre-Dilution of reference standard and sample:	
	5.3.1	Reference solution preparation:
		Dilution fold
		Unicalibrator
		Owren koller buffer (µL)
		Total Volume (µL)

			1:10
			25 µL of neat Unicalibrator
			225
			250
			1:20
			125 µL of 1:10 dilution
			125
			250
			1:40
			125 µL of 1:20 dilution
			125
			250
			1:80
			125 µL of 1:40 dilution
			125
			250
			Perform serial dilutions of reference standard from 1:10
			dilution as per the below
			table.
		5.3.2	Test Solution preparation:
			Dilute the test sample 1:10 folds by taking 50 µL of Neat sample
			and 450 µL of Owren koller buffer.
		5.4	Test method:
		5.4.1	Switch on the Coagulometer 15 minutes prior to analysis
		5.4.2	Place all the reagents in appropriate slots of the Coagulometer
			for warming at 37°C (5 minutes prior to analysis).
		5.4.3	Place a Cuvette strip with required number of cuvettes in the
			incubation area and prewarm at 37°C at least for 3 minutes. Then
			dispense one magnetic ball to each cuvette.
		5.4.4	Add 50 µL of each of the dilution of reference standard, 50 µL
			of STA Deficient X into each cuvette.
		5.4.5	Immediately start the timer corresponding to the incubation
			column for an incubation of 60 sec.
		5.4.6	After 50 sec beep sound was produced. Then transfer the cuvette
			from the incubation chamber to the test column area.
		5.4.7	After 60 sec of incubation time, add 100 µL of Neoplastine CI
			Plus to each cuvette by pressing timer to start the measurement.
		5.4.8	Note down the clotting time in seconds.
		5.4.9	Then feed the Percentage of reference standard Factor X
			corresponding to its clotting time for each dilution in the
			instrument.
			Note: Instrument automatically plots the calibration curve.
		5.4.10	Then perform the test for the test samples as described above
			and note down the clotting time as well as Factor X activity
			which comes in the print out.
		5.5	Calculations:
		5.5.1	As calibration data is fed in Coagulometer, the instrument
			itself plots test sample clotting time on standard curve and
			calculates Factor X activity.
		5.5.2	Calculate % activity of Factor X to that of Factor IX using
			following formula
			[pic]
		6.0	References:
		6.1 Indian Pharmacopeia	

1.0	Objective:			
	To lay down the Procedure for assay for fibrinogen (Clottable protein).			
2.0	Scope:			
	This Standard Operating Procedure is applicable to all the In- process and			
	finished product samples of fibrinogen.			
3.0	Responsibility			
	3.1	The QC analyst shall be responsible for executing the described		
		procedure correctly.		
	3.2	The QC supervisor shall be responsible for verifying that the		
		procedure is complied with and the results are checked.		
	3.3	The QC HOD shall be responsible for overall compliance of this SOP.		
4.0	Abbreviations and Definitions:			
	4.1	g:	: Grams	
	4.2	mL	: Milliliter	
	4.3	mg	: Milligram	
	4.4	M	: Molarity	
	4.5	min	: Minute	
	4.6	μL	: Microliter	
	4.7	μg	: Microgram	
	4.8	°C	: Degree centigrade	
5.0	Procedure:			
	5.1	Equipment, Materials and Reagents		
		5.1.1	Desk Top Centrifuge (Eppendorf, 5430R)	
		5.1.2	K - 425 Speed digestion unit	
		5.1.3	K- 350 Distillation unit	
		5.1.4	Desk Top Centrifuge (Eppendorf, 5430R)	
		5.1.5	Buchi Scrubber B-414	
		5.1.6	Auto titrator (Mettler Toledo)	
		5.1.7	Sodium chloride	
		5.1.8	Sulphuric acid	
		5.1.9	Dipotassium Sulphate	
		5.1.1	Copper Sulphate	
		0		
		5.1.1	Hydrogen Peroxide	
		1		
		5.1.1	Sodium hydroxide	
		2		
		5.1.1	Boric acid	
		3		
		5.1.1	Methyl red	
		4		
		5.1.1	Methylene blue	
		5		
		5.1.1	Tris-Cl	
		6		
		5.1.1	Calcium Chloride	
		7		
		5.1.1	Human Thrombin preparation (500 IU/mL label claim vial)	
		8		
	5.2	Preparation of solutions:		
		5.2.1	Sodium chloride 9 g/L:	
			5.2.1.1	Dissolve 9 g of Sodium Chloride in 800 mL of Purified
				water and make up to 1000 mL with Purified water.
			5.2.1.2	Filter sterilize by using 0.45μ membrane filter and
				store at room temperature for 2 months.
		5.2.2	Tris–Cl (0.5 M) buffer pH 7.0 Stock solution:	
			5.2.2.1	Dissolve 60.5 g of Tris (hydroxyl methyl aminomethane)
				in 800 mL of Purified water.
			5.2.2.2	Adjust the pH to 7.0 with 1 M HCL.
			5.2.2.3	After pH adjustment make upto 1000 mL with Purified
				water.
			5.2.2.4	Filter sterilize by using 0.45μ membrane filter and
				store at room temperature for 2 months.
		5.2.3	0.5 M Calcium Chloride stock solution:	
			5.2.3.1	Dissolve 73.51 g of Calcium chloride in 1000 mL of
				Purified water.
			5.2.3.2	Filter sterilize by using 0.45μ membrane filter and
				store at room temperature for 2 months
		5.2.4	50 mM Tris Cl pH 7.0 containing 50 mM Calcium Chloride and 3	
			IU/mL Human thrombin:	
			5.2.4.1	S.No
				Stock Solution
				10 mL Volume (mL)
				1
				0.5 M Tris – Cl pH 7.0
				1
				2
				0.5 M Calcium Chloride
				1
				3
				Human Thrombin (500 IU/mL)
				0.06
				4
				Purified Water
				7.94

			5.2.4.2	Prepare the buffer freshly.
			NOTE: The above buffer preparation is based on 500 IU/mL of	
			Thrombin. If Thrombin activity is different calculate	
			accordingly. Also capture the thrombin details in ATDS.	
		5.2.5	Mixture of Dipotassium Sulphate and Copper sulphate in the	
			ratio of 10:1:	
			5.2.5.1	Mix 909 mg of Dipotassium Sulphate and 91 mg of Copper
			sulphate.	
		5.2.6	69 % Sulphuric acid:	
			5.2.6.1	Take 296 mL of purified water in a volumetric flask and
			make up the volume to 1000 mL by slowly adding 704 mL	
			of 98% H₂SO₄.Prepare the reagent freshly.	
		5.2.7	Boric acid – 4%	
			5.2.7.1	Take 40 g of Boric acid and dissolve in 500 mL of
			Purified water and adjust pH to 4.65 with 1 N NaOH and	
			make up the volume to 1000 mL with Purified water.	
			5.2.7.2	Filter sterilize by using 0.45µ membrane filter and
			store at room temperature for 2 months.	
		5.2.8	Methyl Red TS:	
			5.2.8.1	Dissolve 100 mg of Methyl red in 100 mL of alcohol, and
			filter if necessary.	
		5.2.9	Methylene Blue TS:	
			5.2.9.1	Dissolve 125 mg of Methylene blue in 100 mL of alcohol,
			and dilute with alcohol to 250 mL.	
		5.2.1	Methyl red- Methylene blue Indicator	
		0		
			5.2.10.	Add 10 mL of methyl red TS to 10 mL of Methylene blue
			1	TS, and mix.
			Note: Prepare the indicator freshly.	
		5.2.1	Sodium hydroxide Solution – 8 M	
		1		
			5.2.11.	Dissolve 1600 g of Sodium Hydroxide in 5000 mL of
			1	Purified water.
			NOTE :Filter the solution using 0.45 µm filter and store at	
			room temperature for 1 month.	
		5.2.1	Sulphuric acid 0.01 N	
		2		
			5.2.12.	Take 136 µL of 98% H₂SO₄ in Volumetric flask and make
			1	up the volume to 500 mL with Purified water.
			5.2.12.	Filter sterilize by using 0.45µ membrane filter and
			2	store at room temperature for 2 months. Prior to use,
			visually check Discard the solution if it is visually	
			unclear.	
		5.3	Test Method : This method consists of the following steps	
		Fibrin clot formation		
		Digestion, Distillation and Titration		
		5.3.1	Fibrin clot formation:	
			5.3.1.1	Mix 0.2 mL of the neat test sample/Reference standard
			with 2 mL of 50 mM Tris Cl pH 7.0 containing 50 mM	
			Calcium Chloride and 3 IU/mL Human Thrombin.	
			5.3.1.2	Incubate at 37 ° C for 20 min.
			5.3.1.3	Separate the Fibrin clot by centrifugation 5000 g (7200
			rpm) for 20 min.	
			5.3.1.4	Wash thoroughly with 9 g/L Sodium chloride R for two
			times.	
			5.3.1.5	Blank Solution :
			Add 2 mL of 9 g/L Sodium chloride (0.15 M) to the blank	
			digestion flask.	
		5.3.2	Digestion:	
			5.3.2.1	Set the required preheating temperature for speed
			digester (380°C) and let the system heat up for 10 to	
			15 minutes minimum.	
			5.3.2.2	Then transfer the Fibrin clot with 2 mL of Purified
			water to the digestion flask of the apparatus.	
			5.3.2.3	Add 1 g of a powdered mixture of potassium sulphate and
			cupric sulfate (10:1), and wash down any adhering	
			material from the neck of the flask with a fine jet of	
			water.	
			5.3.2.4	Add 7 mL of 69% sulphuric acid, allowing it to rinse
			down the wall of the flask, then while swirling the	
			flask, add 1 mL of 30 percent hydrogen peroxide	
			cautiously down the side of the flask. (Do not add	
			hydrogen peroxide during the digestion).	
			5.3.2.5	Heat the flask over a free flame or an electric heater
			for 90 min at 380 °C until the solution	
			has a clear blue colour and the sides of the flask are	
			free from carbonaceous material.	
		5.3.3	Distillation:	
			5.3.3.1	After cooling the flasks, add 80 mL of Purified water
			to the digested sample and to this add 30 mL of Sodium	
			Hydroxide solution 8 M by pressing reagent button	
			present in the distillation unit, in such manner as to	
			cause the solution, more alkaline. Close the apparatus,	
			and begin the distillation with steam immediately for 5	
			min.	
			5.3.3.2	Receive the distillate in 15 mL of 4% Boric acid
			solution to which 35 mL of Purified water has been	
			added, to which has been added 3 drops of methyl red –	
			methylene blue TS and check to cover the end of the	
			condensing tube in the Solution.	
			5.3.3.3	Remove the absorption flask; rinse the end of the
			condensing tube with a small quantity of water.	

		5.3.4	Titration:	
			5.3.4.1	Titrate the distillate with 0.01N sulphuric acid.
				Perform also a blank determination, and make any
				necessary correction.
			5.3.4.2	Each mL of 0.01N sulphuric acid is equal to 140.1 µg of
				nitrogen.
		5.3.5	Calculations:	
			5.3.5.1	Volume of 0.01 N Sulfuric acid consumed for blank = X
				mL
				Volume of 0.01 N Sulfuric acid consumed for Test sample
				= Y mL
				Final volume of 0.01 N Sulfuric acid consumed (Y-X) = Z
				mL
				Each mL of 0.01 N sulphuric acid VS is equivalent to
				140.1 µg of nitrogen.
				[pic]
				Nitrogen content (g) in 1.0 mL of sample (C) = A X B
				Nitrogen content (g) in 100.0 mL of sample or %Nitrogen
				(D) = C X100
				Clottable protein (mg/mL) = (D) X 6.0 X 10.0
				The suitable factor 6.0 is multiplied as per I.P. 2018.
6.0	References:			
	6.1 Indian Pharmacopeia			

 [pic]

1.0	Objective:			
	To lay down the Procedure for detection of Antibodies to HCV by Serology.			
2.0	Scope:			
	This Standard Operating Procedure is applicable to all the In- process and			
	finished product samples of IVIG.			
3.0	Responsibility			
	3.1	The QC analyst shall be responsible for executing the described		
	procedure correctly.			
	3.2	The QC supervisor shall be responsible for verifying that the		
	procedure is complied with and the results are checked.			
	3.3	The QC HOD shall be responsible for overall compliance of this SOP.		
4.0	Abbreviations and Definitions:			
	4.1	g: Grams		
	4.2	mL: Milliliter		
	4.3	mg: Milligram		
	4.4	M: Molarity		
	4.5	min: Minute		
	4.6	µL: Microliter		
	4.7	µg: Microgram		
	4.8	°C: Degree centigrade		
5.0	Procedure:			
	5.1	Equipment, Materials and Reagents		
		5.1.1	Multi scan Fc ELISA reader	
		5.1.2	Incubator	
		5.1.3	Laminar air flow unit	
	5.2	Preparation of solutions:		
		5.2.1	Preparation of Wash Buffer	
			5.2.1.1	Dilute 20 mL of 25X wash buffer to 1X concentrate with
			480 mL of water	
			Note: Wash buffer is stable for 2 months when stored at	
			2-80C	
			5.2.1.2	Check the buffer concentrate for the presence of salt
			crystals. If crystals are present in the solution,	
			resolubilize by warming at 370C until all crystals	
			dissolve	
		5.2.2	Preparation of 1X Working Conjugate	
			5.2.2.1	Dilute conjugate enzyme 1:100 in conjugate diluent. Do
			not store working conjugate	
			5.2.2.2	Prepare required amount of working conjugate solution
			from table given below.	
			pic	
			Note:	
			Do not store working conjugate. Prepare a fresh	
			dilution for each assay in a clean glass vessel.	
			In case any precipitate is found in conjugate	
			diluent/sample diluent, it should be allowed to settle	
			and the supernatant can be used for the test. The	
			precipitate does not interfere with the working of the	
			kit.	
		5.2.3	Preparation of working substrate solution	
			5.2.2.1	Mix TMB substrate and TMB diluent in 1:1 ratio to
			prepare working substrate	
			5.2.2.2	Prepare required amount of working substrate solution
			from table given below.	
			pic	
			Note: Do not store working substrate. Prepare a fresh	
			dilution for each assay in a clean plastic/glass	
			vessel. Mix solution thoroughly before use	
	5.3	Test method:		
		5.3.1	Switch on the laminar air flow unit	
		5.3.2	Wipe the laminar airflow unit with sterillium	
		5.3.3	Switch on the UV lamp for 5 to 10 min	
		5.3.4	Place the kit components in laminar air flow unit and bring the	
			kit components to room temperature (20 – 30 °C).	
			Note: Break off the required number of strips needed for the	
			assay and place in the well holder. Take the strip holder with	
			the required number of strips, taking into account that two	
			negative and two positive controls should be included in the	
			run while opening the fresh pack	
		5.3.6	Fit the strip holder with the required number of HCV Microlisa	
			strips	
		5.3.7	Add 100 µL of the sample diluent to one well as blank	
		5.3.8	Add 100 µL Negative Control, Positive Control & diluted test	
			sample to labelled wells. Negative control & Positive Control	
			is ready to use & hence no dilution is required	
		5.3.9	Carefully apply cover seal to the strips. Avoid spillage of the	
			contents	
		5.3.1	Then incubate at 37oC + 2oC for 30 mins + 2 minutes.	
		0	Note: While the samples are incubating, prepare working wash	
			solution and working conjugate	
		5.3.1	After the incubation is over, wash the wells 6 times with	
		1	working wash solution.	
			WASHING: Perform washing as mentioned below	
			5.3.11.	Incomplete washing will adversely affect the test
			1	outcome
			5.3.11.	Carefully aspirate the well contents completely into a
			2	waste container
			5.3.11.	Then fill the wells completely with wash buffer
			3	avoiding overflow of buffer from one well to another
			and allow to soak (approx. 30 seconds).	

		5.3.11.	Carefully aspirate completely and repeat the wash and
		4	soak procedure 4 additional times for a total of 5
			washes
		5.3.11.	Tap upside down on absorbent sheet till no droplets
		5	appear on the sheet, taking care not to dislodge the
			wells
	5.3.1	Add 100 µL of working conjugate solution in each well including	
	2	blank well	
	5.3.1	Carefully apply cover seal & incubate at 37°C ± 2°C for 30 mins	
	3	± 2 mins.	
		Note: While conjugate is incubating, prepare substrate solution	
		in last 5 minutes of incubation. Protect from light	
	5.3.1	Carefully aspirate and wash the wells after incubation as	
	4	described in section 5.3.11. Avoid spillage	
	5.3.1	Add 100 µL working substrate solution in each well including	
	5	blank well	
	5.3.1	Incubate at room temperature (20-30°C) in dark for 30 mins	
	6		
	5.3.1	Add 100 µL of stop solution	
	7		
	5.3.1	Read absorbance at 450 nm in ELISA reader. (Use of a reference	
	8	filter of 620 nm is preferred).	
		Note:	
		Care should be taken while performing the experiment in order	
		to avoid infection from plasma sample or positive control.	
		Wear nose mask and hand gloves throughout the experiment. Do	
		not reuse the gloves and masks; discard immediately after	
		completion of the experiment.	
		Discard the used tips in a beaker containing sodium	
		hypochlorite and close the beaker with aluminium foil.	
		Perform the experiment under laminar airflow unit	
	5.4	Calculations :	
	5.4.1	Cut-off value can be determined by using the following formula:	
		Cut-off Value = [pic]	
	5.5	Method Validation Criteria:	
	5.5.1	Blank acceptance criteria"	
		The OD value at 450/620 nm for Blank must be <0.100 in case of	
		differential filter being used. In case differential filter is	
		not available in the reader the blank value may go higher	
	5.5.2	Negative Control (NC) acceptance criteria:	
		The OD value at 450/620 nm for NC must be < 0.150. If it is not	
		so, the run is invalid and must be repeated.	
		Mean absorbance	
		[pic]	
	5.5.3	Positive Control (PC) acceptance criteria:	
		The OD value at 450/620 nm for PC or PCx must be >0.5. If it is	
		not so, the run is invalid and must be repeated.	
		Mean absorbance.	
		[pic]	
	6.0	References:	
	6.1	Indian Pharmacopeia	

1.0	Objective:		
	To lay down the procedure for osmolality test using osmometer method.		
2.0	Scope:		
	This Standard Operating Procedure is applicable to all the and finished		
	product samples.		
3.0	Responsibility		
	3.1	The QC analyst shall be responsible for executing the described	
	procedure correctly.		
	3.2	The QC supervisor shall be responsible for verifying that the	
	procedure is complied with and the results are checked.		
	3.3	The QC HOD shall be responsible for overall compliance of this SOP.	
4.0	Abbreviations and Definitions:		
	4.1	SOP	
	4.2	: Standard Operating Procedure	
	4.3		
	4.4	QA	
	4.5	: Quality Assurance	
		OE	
	: Operation of Equipment		
	mL		
	: milliliter		
	mOSm/Kg		
	: Milli Osmol per Kilogram of water		
5.0	Procedure:		
	5.1	Equipment, Materials and Reagents	
		5.1.1	Osmometer
		5.1.2	Sample tubes/tips
		5.1.3	Pipette
		5.1.4	Tissue paper
		5.1.5	Reference standard Clinitol™ 290mOSm/Kg
		5.1.6	Reference standard Clinitol™ 850mOSm/Kg
		5.1.7	Reference standard Clinitol™ 50mOSm/Kg
		5.1.8	Reference standard Clinitol™ 2000mOSm/Kg
	5.2	Preparation of solutions:	
		NA	
	5.3	Test Method:	
		5.3.1	Keep the osmometer in a clean dry level area.
		5.3.2	Plug in the power cord and switch on the electric supply.
		5.3.3	Turn the instrument on by pushing the rocker-style power
		switch on the instrument's back panel into the on position.	
		5.3.4	Run one minute self-diagnostic test by pressing the [yes]
		button. After completion of self-diagnostic test, the display	
		will read as "Osmometer Ready".	
		5.3.5	Run the Clinitol® 290 mOsm/kg H2O Reference Solution, before
		testing unknown samples. If the results meet the	
		specification i.e. 290 + 2 mOsm/kg H2O, the instrument is	
		considered ready to use. If the results do not meet the	
		specification then calibrate the instrument as per the	
		procedure given at step 5.3.4	
		5.3.6	For testing the sample, take the sampler and insert the
		sampler's tip into place ensuring that tip is straight and	
		firmly seated.	
		5.3.7	Depress the sampler's plunger and insert the sampler's tip at
		least ¼ inch (6 mm) below the surface of the fluid to be	
		tested.	
		5.3.8	Gently release the plunger to load a 20 µl sample. Visually
		inspect the sample to ensure that there are no large voids or	
		bubbles in the sample	
		5.3.9	Wipe the sides of the loaded sampler tip with a soft, no
		lint, non-ionic paper tissue to remove any clinging droplets	
		5.3.1	Quickly wipe the end of the sampler tip to remove any fluid
		0	protruding beyond the tip.
		5.3.1	Remove the chamber cleaner from the sample port and discard.
		1	
		5.3.1	Hold the sampler by barrel, insert the tip into the sample
		2	port then reset the sampler in the operating cradle.
		5.3.1	Start the test by pushing the operating cradle until it
		3	reaches a positive stop.
		5.3.1	Record the results which are presented in the format
		4	"Osmolality xxx mOsm".
		5.3.1	Pull back the operating cradle to a positive stop.
		5	
		5.3.1	Remove the sampler from the operating cradle and discard the
		6	used sampler tip.
		5.3.1	Insert a clean, dry chamber cleaner into the sample port and
		7	rotate it four or five times in both clockwise and
		counterclockwise direction. Withdraw the chamber cleaner and	
		insert the opposite end. Rotate the chamber cleaner in the	
		same manner and leave it in the sampler port until next test.	
		5.3.1	Calibrate the instrument as per operational manual
		8	
6.	References:		
	6.1 Indian Pharmacopeia		

1.0 |Objective: |
|To lay down the procedure Test for Identification of Human Albumin and |
|Human Normal Immunoglobulin for Intravenous use samples by using |
|Immuno-electrophoresis. |
2.0 |Scope: |
|This Standard Operating Procedure is applicable to all the In- process and |
|finished product samples. |
3.0 |Responsibility |
3.1	The QC analyst shall be responsible for executing the described
	procedure correctly.
3.2	The QC supervisor shall be responsible for verifying that the
	procedure is complied with and the results are checked.
3.3	The QC HOD shall be responsible for overall compliance of this SOP.
4.0	Abbreviations and Definitions:
4.1	TBE
4.2	g
4.3	mL
4.4	mg
4.5	M
4.6	min
4.7	µL
4.8	µg
4.9	°C
4.10	%
5.0	Procedure:
5.1	Equipment, Materials and Reagents
	5.1.1
	5.1.2
	5.1.3
	5.1.4
	5.1.5
	5.1.6
	5.1.7
	5.1.8
	5.1.9
	5.1.10
	5.1.11
	5.1.12
	5.1.13
	5.1.14
	5.1.15
	5.1.16
	5.1.17
	5.1.18
	5.1.19
5.2	Preparation of solutions:
	5.2.1
	5.2.2
	5.2.3

			S.No		
			Name of the Reagent		
			Quantity (100 mL)		
				1	
			Agarose		
				1 g	
				2	
			1X TBE buffer		
				100 mL	
			Warm for few minutes until the powder dissolves and the		
				liquid becomes semi-solid.	
			Prepare freshly at the time of experiment.		
		5.2.4	Anti Human Whole Serum Developed in Rabbit – 10 F dilution		
			Dilute 3 µL of neat Rabbit anti Human Whole Serum with 27 µL		
			of Electrophoresis buffer (10 F dilution).		
			Note: Prepare the reagent freshly.		
		5.2.5	Coomassie brilliant blue staining solution		
			5.2.5.	S.No	
				1	Name of the reagent
				Quantity(100 mL)	
				1	
				Coomassie brilliant blue R-250	
				0.1 g	
				2	
				Methanol	
				40 mL	
				3	
				Glacial acetic acid	
				10 mL	
			5.2.5.	Mix thoroughly and make up the final volume to 100 mL	
			2	with Purified water. This can be stored at room	
				temperature for future use.	
		5.2.6	Destaining solution		
			5.2.6.	S. No.	
				1	Name of the Reagent
				Quantity (100 mL)	
				1	
				Glacial acetic acid	
				10 mL	
				2	
				Methanol	
				40 mL	
				3	
				Purified water	
				50 mL	
			Note: Prepare the reagent freshly		
			This can be stored at room temperature for future use		
		5.2.7	Sodium chloride 0.15 M		
			5.2.7.	S. No.	
				1	Name of the Reagent
				Quantity (1000 mL)	
				1	
				Sodium chloride	
				9 g	
				2	
				Purified water	
				1000 mL	
			5.2.7.	Transfer the solution into a 1 liter clean and dry	
				2	reagent bottle.
		5.2.8	Reference standard (Normal Human Serum (Prepared In-house) :		
			5.2.8.	Prepare normal human serum from pooled donors of not	
				1	less than 100 or from a commercially available source.
				Aliquot 0.1 mL of sample in each micro centrifuge tube	
				and store frozen at -20°C for upto 12 months from the	
				date of preparation.	
			5.2.8.	Avoid repeated freeze or thaw cycles.	
				2	
		5.3	Method :		
		5.3.1	Reference solution preparation		
			5.3.1.	Prepare normal human serum by collecting 5 mL of blood	
				1	and incubate for 1 hour at room temperature.
			5.3.1.	Centrifuge at 12000 rpm for 15 min and collect the	
				2	supernatant containing serum.
			5.3.1.	Estimate the total protein concentration in serum by	
				3	Biuret method
			5.3.1.	Dilute normal human serum (reference solution) with 1X	

			4	Tris borate EDTA buffer pH 8.2 to a concentration of
				10 g/L.
		5.3.2	Test sample preparation:	
			5.3.2.	Test Sample Concentration(mg/ml)
			1	Required protein Concentration(mg/mL)
				Volume of Test sample
				(μL)
				Volume of
				1X TBE buffer pH 8.2
				(μL)
				Total volume
				(μL)
				From
				respective Concentration
				10
				100
				400
				500
		5.3.3	Place a clean and dry 7.5 x 2.5 cm glass plate on a leveled	
		surface.		
		5.3.4	Prepare 5 mL of 10 g/L solution of agarose for	
		electrophoresis (1% agarose solution) by melting in a water		
		bath for 30 to 40 minutes.		
		5.3.5	Pour the gel onto a micro-slide strip and wait until it gets	
		solidified.		
		5.3.6	Pour 5 mL of this solution onto a glass plate.	
		5.3.7	Leave the gel at room temperature for 15 minutes to	
		facilitate proper solidification.		
		5.3.8	Cut 2 wells 1.5 mm in diameter 0.5 cm and 0.5 cm from either	
		side of the plate and 1 cm from the cathode.		
		5.3.9	Using a suitable commercial device, dig two wells about 1.5	
		cm apart from the cathode and about 0.5 cm away from the		
		sides and at the same level perpendicular to the direction of		
		electrophoresis.		
		5.3.10	Cut a trough of 1.0 mm in between the wells such that the	
		distance between the trough and each well should be not more		
		than 3.1 mm.		
		5.3.11	Place 1 μL of reference solution into one well and an equal	
		volume of test solution into the other.		
		5.3.12	Place the micro-slide containing 1% agar gel between the two	
		buffer reservoirs of a suitable Immuno electrophoresis		
		apparatus.		
		5.3.13	Pour 125 mL of electrophoresis buffer in each of the buffer	
		reservoir such that the buffer overflows on the gel placed in		
		the middle of the electrophoresis chamber.		
		5.3.14	Put the cover on the electrophoresis chamber and wait 30 to	
		60 seconds before applying current. This allows the plate(s)		
		to equilibrate with the buffer.		
		5.3.15	Electrophores the samples for about 35 min applying 90 volts	
		current across the agar-gel strip.		
		5.3.16	Place in the central slot about 50 μL of the 10 F diluted	
		antiserum specific to human plasma proteins.		
		5.3.17	Place this micro-slide in a Petridish containing wet cotton	
		and incubate at 2 - 8°C for at least 24 hours.		
		5.3.18	Precipitin arcs appear in the gel for both sample and	
		standard after 24 hours of incubation at 2 - 8°C.		
		5.3.19	After incubation, place the gel slide in a 0.85% saline and	
		wash to stop the precipitin reaction.		
		5.3.20	After washing, put the gel slide in Coomassie brilliant blue	
		staining solution for 30 min.		
		5.3.21	De-stain the gel slide by keeping in a destaining solution	
		for overnight.		
		Note: Ensure that the gel slide is covered completely with the		
		De-staining solution.		
		5.3.22	Make sure that the De-staining is complete to visualize clear	
		and prominent arcs. If the De-staining is not complete, keep		
		the gel slide in fresh De-staining solution for 1 hour.		
		5.3.23	Two precipitin arcs appear in the gel. One arc corresponds to	
		normal human serum and another arc corresponds to test		
		sample.		
		5.3.24	To capture the image of the gel for documenting, switch on	
		the Epson scanner and then double click.		
		5.3.25	Double click on the Epson scan software from the PC connected	
		to it.		
		5.3.26	Epson scan window appears.	
		Set the following parameters on the Epson scan window		
		.		
		Window Name		
		Selection of specific window from the drop down list		
		.		
		Settings		
		Name		
		Current settings		
		.		
		Original		
		Document Type		
		Reflective		
		.		
		.		
		Document Source		

			Document table
			Auto Exposure Type
			Document
			Destination
			Image Type
			24 bit colour
			Resolution
			240 dpi
			Document size
			No modifications
		5.3.27	Place the gel to be scanned on the flat surface of the
		scanner.	
		5.3.28	After setting the above parameters, click on the preview
		button of the window to adjust the exact alignment and	
		preview image of the gel.	
		Note: This preview avails proper positioning of the gel before it is	
		finally scanned and documented.	
		5.3.30	Then we can notice preview scan in progress giving the
		picture of the gel in preview window.	
		5.3.31	Wait for few seconds until the preview of the gel appears.
		5.3.32	Once the image looks fine in the preview image, click on the
		scan button of Epson scan window for final gel image.	
		5.3.33	Then File save settings" window will open asking the format
		of the image in which gel has to be saved.	
		5.3.34	Select Image Format type as JPEG(*.jpg) and click on OK.
		5.3.35	Then Scanning will start. (Progress: scanning pop up window
		will open showing the time remaining for the scan).	
		5.3.36	After the scan is completed, the image of the gel in JPEG
		format will be directly saved into My Documents folder.	
		5.3.37	This will complete the gel scanning.
		5.3.38	Then open the JPEG file and click on Edit settings to adjust
		the brightness and contrast of the gel picture for the	
		precipitin arcs to appear clearly against a contrast	
		background.	
		5.3.39	Once we can observed the precipitin arcs clearly, label the
		file with the batch number and immediately take a print out.	
		5.4	Interpretation of results:
		When observed the scanned gel using Gel scanner instrument,	
		the agarose gel shows an arc for test solution that	
		corresponds to the respective arc depending on the product	
		tested.	
		6.	References:
		IP Monograph & In-House	

[1.0]Objective: |
 |To lay down the procedure for estimation of protein content by Lowry assay |
 |method. |
 [2.0]Scope: |
 |This Standard Operating Procedure is applicable to in process and finished |
 |product samples. |
 [3.0]Responsibility |
 |[3.1]The QC analyst shall be responsible for executing the described procedure |
 ||correctly. |
 |[3.2]The QC supervisor shall be responsible for verifying that the procedure is |
 ||complied with and the results are checked. |
 |[3.3]The QC HOD shall be responsible for overall compliance of this SOP. |
 [4.0]Abbreviations and Definitions: |
[4.1]TBE	: Tris Borate EDTA Buffer
[4.2]g	: Grams
[4.3]mL	: Milliliter
[4.4]mg	: Milligram
[4.5]M	: Molarity
[4.6]min	: Minute
[4.7]µL	: Microliter
[4.8]µg	: Microgram
[4.9]°C	: Degree centigrade
[4.10]%	: Percentage
[5.0]Procedure:	
[5.1]Equipment, Materials and Reagents	
	[5.1.1]UV/Visible-Spectrophotometer
	[5.1.2]Stop watch
	[5.1.3]Vortex mixer
	[5.1.4]Copper sulphate GR
	[5.1.5]Sodium tartrate GR
	[5.1.6]Anhydrous Sodium carbonate GR
	[5.1.7]Sodium Dodecyl Sulphate GR
	[5.1.8]Sodium Hydroxide GR
	[5.1.9]Bovine Serum Albumin (BSA): (Pierce, 23209)
	[5.1.1]Bovine Gamma Globulin (BGG): (Pierce, 23212)
	[0]
[5.2]Preparation of solutions:	
	[5.2.1]Preparation of 5% SDS solution
	[5.2.2]Sodium Hydroxide (0.8 M) solution
	[5.2.3]Preparation of Alkaline Copper Sulphate Reagent
	[5.2.3]Factor Reagent – Kit Contains 2 Vials of Factor reagent
	[5.2.4]Chromogenic Substrate – Kit Contains 1 Vial of Chromogenic reagent
[5.3]Method :	
	[5.3.1] Reconstitute of Reference standard (WHO Reference standard (RS)

				20	
				168	
				188	
				5.3.1.1	Step 2 Perform two 2-fold serial dilutions of RS from 1.0
				1	IU/mL to 0.25 IU/mL using dilution buffer.
				IRS Activity (IU/mL)	
				RS (μL)	
				Dilution Buffer (μL)	
				Total Volume (μL)	
				0.5	
				100 μL of 1.0 IU/mL (9.4 times diluted)	
				100	
				200	
				0.25	
				100 μL of 0.5 IU/mL	
				100	
				200	
				5.3.1.1	Internal Reference Standard (IRS) calibrated against WHO
				2	Reference Standard or Working Standard.
				5.3.1.1	Reconstitute IRS with 10 mL of sterile water for Injection
				3	and let the vial to stand at room temperature for 5 min to
				get a uniform solution.	
				5.3.1.1	If IRS assigned activity is 25 IU/mL
				4	
				5.3.1.1	Dilute the IRS 2.5 times with Factor VIII deficient plasma to
				5	get 10 IU/mL.
				5.3.1.1	Step 1 of Pre dilution: 2.5 times Dilution: Pre dilution from
				6	25 IU/mL to 10 IU/mL with F-VIII deficient plasma
				IRS Activity (IU/mL)	
				IRS Neat (μL)	
				F-VIII Deficient (μL)	
				Total Volume (μL)	
				10	
				40	
				60	
				100	
				Note: In case IRS activity is different from 25 IU/mL, dilute	
				accordingly to get a diluted preparation of 10 IU/mL	
				5.3.1.1	Step 2 of Pre dilution: Pre dilution from 10 IU/mL to 1 IU/mL
				7	with F-VIII deficient plasma (10 times Dilution)
				IRS Activity (IU/mL)	
				IRS from Step 1 (10 IU/mL) (μL)	
				F-VIII Deficient (μL)	
				Total Volume (μL)	
				1	
				30	
				270	
				300	
				5.3.1.1	Step 3 Perform two 2-fold serial dilutions of IRS from 1.0
				8	IU/mL to 0.25 IU/mL using dilution buffer.
				IRS Activity (IU/mL)	
				IRS (μL)	
				Dilution Buffer (μL)	
				Total Volume (μL)	
				0.5	
				100 μL of 1.0 IU/mL	
				100	
				200	
				0.25	
				100 μL of 0.5 IU/mL	
				100	
				200	
				5.3.2	Test Solution :
				5.3.2.1	Manufacturing pool for fractionation – Neat sample
				5.3.2.2	IPQC sample of Human Coagulation Factor VIII
				5.3.2.3	Pre dilution from 2 IU/mL to 1 IU/mL with F-VIII deficient
				plasma	
				Test Sample Expected activity (IU/mL)	
				Test Sample Neat (μL)	
				F-VIII deficient (μL)	
				Total Volume (μL)	
				1	
				50	
				50	
				100	

```

|||||
|||5.3.3|Drug Substance - Clotate 250 IU|
|||5.3.3.1|Step 1 of Pre dilution: 5 times Dilution|
||||Pre dilution from 50 IU/mL to 10 IU/mL with F-VIII deficient|
||||plasma|
||||Test Sample Expected activity (IU/mL)|
||||Neat Test Sample (μL)|
||||F-VIII deficient (μL)|
||||Total Volume (μL)|
|||||
||||10|
||||20|
||||80|
||||100|
|||||
|||5.3.3.2|Step 2 of Pre dilution: 10 times Dilution|
||||Pre dilution from 10 IU/mL to 1 IU/mL with F-VIII deficient|
||||plasma|
||||Test Sample Expected activity (IU/mL)|
||||Test Sample from Step 1 (10 IU/mL(μL)|
||||F-VIII deficient (μL)|
||||Total Volume (μL)|
|||||
||||1|
||||30|
||||270|
||||300|
|||||
|||5.3.3.3|Step 3: Perform two 2-fold serial dilutions of test sample|
||||from 1.0 IU/mL to 0.25 IU/mL using dilution buffer.|
||||Test Sample Expected activity (IU/mL)|
||||Test Sample (μL)|
||||Dilution Buffer (μL)|
||||Total Volume (μL)|
|||||
||||0.5|
||||100 μL of 1.0 IU/mL|
||||100|
||||200|
|||||
||||0.25|
||||100 μL of 0.5 IU/mL|
||||100|
||||200|
|||||
|||5.3.4|Drug Product - Clotate 250 IU/Vial/10 mL|
|||5.3.4.1|Assigned Potency after reconstitution with 10 mL of diluent:|
||||25 IU/mL|
||||Dilute the above test sample 2.5 times with dilution buffer|
||||to get 10 IU/mL.|
|||5.3.4.2|Step 1 of Pre dilution: 2.5 times Dilution|
||||Pre dilution from 25 IU/mL to 10 IU/mL with F-VIII deficient|
||||plasma|
||||Test Sample Expected activity (IU/mL)|
||||Neat Test Sample (μL)|
||||F-VIII deficient (μL)|
||||Total Volume (μL)|
|||||
||||10|
||||40|
||||60|
||||100|
|||||
|||5.3.4.3|Step 2 of Pre dilution: 10 times Dilution|
||||Pre dilution from 10 IU/mL to 1 IU/mL with F-VIII deficient|
||||plasma|
||||Test Sample Expected activity (IU/mL)|
||||Test Sample from Step 1 (10 IU/mL(μL)|
||||F-VIII deficient (μL)|
||||Total Volume (μL)|
|||||
||||1|
||||30|
||||270|
||||300|
|||||
|||5.3.4.4|Step 3: Perform two 2-fold serial dilutions of test sample|
||||from 1.0 IU/mL to 0.25 IU/mL using dilution buffer.|
||||Test Sample Expected activity (IU/mL)|
||||Test Sample (μL)|
||||Dilution Buffer (μL)|
||||Total Volume (μL)|
|||||
||||0.5|
||||100 μL of 1.0 IU/mL|
||||100|
||||200|
|||||
||||0.25|
||||100 μL of 0.5 IU/mL|
||||100|
||||200|

```


			50	
		5.3.8	Calculation and analysis	
		5.3.8.1	For calculations of potency using either graph pad prism or	
				Combistats software.
6.	References:			
	IP General Chapter 2.8.7			

[1.0 |Objective: |
 | |To lay down the Procedure for detection HBsAg by serology. |
 [2.0 |Scope: |
 | |This Standard Operating Procedure is applicable to all the In- process and |
 | |finished product samples of IVIG. |
 [3.0 |Responsibility |
	3.1	The QC analyst shall be responsible for executing the described
	procedure correctly.	
	3.2	The QC supervisor shall be responsible for verifying that the
	procedure is complied with and the results are checked.	
	3.3	The QC HOD shall be responsible for overall compliance of this SOP.
[4.0	Abbreviations and Definitions:	
	4.1	g:
	4.2	mL
	4.3	mg
	4.4	M
	4.5	min
	4.6	µL
	4.7	µg
	4.8	°C
[5.0	Procedure:	
	5.1	Equipment, Materials and Reagents
		5.1.1
		5.1.2
		5.1.3
	5.2	Preparation of solutions:
		5.2.1
		5.2.2
		5.2.3
[5.3	Test method:	
	5.3.1	Switch on the laminar air flow unit
	5.3.2	Wipe the laminar airflow unit with sterillium
	5.3.3	Switch on the UV lamp for 5 to 10 min
	5.3.4	Place the kit components in laminar air flow unit and bring the
		kit components to room temperature (20 – 30 °C).
	5.3.6	Leave blank well
	5.3.7	Add 100 µL Negative Control, Positive Control and Test sample
		to labelled wells
	5.3.8	Add 50 µL of working enzyme conjugate to each well except blank
		well. Gently shake the plate for 2-3 seconds to mix the sample
		and conjugate
	5.3.9	Carefully apply cover seal to the strips and incubate in an
		incubator at 37 oC ±10C for 60 ± 2 minutes.
		solution and working conjugate
	5.3.1	At the end of incubation period, take out the plate from
		incubator and wash with working wash buffer
	5.3.1	WASHING: Perform Washing as mentioned below
	1	
		5.3.11.
		1
		5.3.11.
		3
		5.3.11.
		4
		5.3.11.

		5	times on an absorbent tissue
	5.3.1	Tap dry the wells after washing and add 100 µL of working	
	2	substrate solution in all the wells including blank well	
	5.3.1	Cover the plate with an aluminium foil and incubate at room	
	3	temperature (20-25°C) for 30 minutes in dark	
	5.3.1	Stop the reaction by adding 100 µL of stop solution to each	
	4	well, mix gently	
	5.3.1	Read the absorbance of the wells at 450 nm in an ELISA Reader.	
	5	(Use of a reference filter of 620 nm is preferred)	
		Note:	
		Care should be taken while performing the experiment in order	
		to avoid infection from plasma sample or positive control.	
		Wear nose mask and hand gloves throughout the experiment. Do	
		not reuse the gloves and masks; discard immediately after	
		completion of the experiment.	
		Discard the used tips in a beaker containing sodium	
		hypochlorite and close the beaker with aluminium foil.	
		Perform the experiment under laminar airflow unit	
	5.4	Calculations :	
	5.4.1	Cut-off value can be determined by using the following formula:	
		Cut-off Value = [pic]	
	5.5	Method Validation Criteria:	
	5.5.1	Blank acceptance criteria"	
		The OD value at 450/620 nm for blank must be <0.100 in case of	
		differential filter being used. In case differential filter is	
		not available in the reader the blank value may go higher	
	5.5.2	Negative Control (NC) acceptance criteria:	
		The OD value at 450/620 nm for NC must be < 0.150. If it is not	
		so, the run is invalid and must be repeated.	
		Mean absorbance	
		[pic]	
	5.5.3	Positive Control (PC) acceptance criteria:	
		The OD value at 450/620 nm for PC or PCx must be >0.5. If it is	
		not so, the run is invalid and must be repeated.	
		Mean absorbance	
		[pic]	
	6.0	References:	
	6.1 Indian Pharmacopeia		

1.0 |Objective: |
 | |To lay down the procedure for determination of Heam content using |
 | |spectrophotometer method. |
 2.0 |Scope: |
 | |This Standard Operating Procedure is applicable to all the In- process |
 | |and finished product samples. |
 3.0 |Responsibility |
	3.1	The QC analyst shall be responsible for executing the described
	procedure correctly.	
	3.2	The QC supervisor shall be responsible for verifying that the
	procedure is complied with and the results are checked.	
	3.3	The QC HOD shall be responsible for overall compliance of this SOP.
4.0	Abbreviations and Definitions:	
	Not applicable	
5.0	Procedure:	
	5.1	Equipment, Materials and Reagents
		5.1.1
		5.1.2
		5.1.3
		5.1.4
		5.1.5
	5.2	Preparation of solutions:
		5.2.1
	5.3	Test Method:
		5.3.1
	5.3.2	Operate the UV/Visible Spectrophotometer as per operation
		SOP:
	5.3.3	Measure the absorbance of test solution at 403 nm using
		purified water as the compensation liquid three times and
		take the average value and record the result in the record
		data sheet.
 6. |References: |
 | |6.1 Indian Pharmacopeia |

1.0	Objective:		
	To lay down the procedure for determination of pH.		
2.0	Scope:		
	This Standard Operating Procedure is applicable to all the In- process		
	and finished product samples.		
3.0	Responsibility		
	3.1	The QC analyst shall be responsible for executing the described	
	procedure correctly.		
	3.2	The QC supervisor shall be responsible for verifying that the	
	procedure is complied with and the results are checked.		
	3.3	The QC HOD shall be responsible for overall compliance of this SOP.	
4.0	Abbreviations and Definitions:		
	Not applicable		
5.0	Procedure:		
	5.1	Equipment, Materials and Reagents	
		5.1.1	Tissue paper
		5.1.2	100 ml. vial
		5.1.3	100 ml beaker
		5.1.4	Wash bottle
		5.1.5	Purified water
	5.2	Preparation of solutions:	
		5.2.1	Buffer solution, pH =4.00
		5.2.2	Buffer solution, pH =7.00
		5.2.3	Buffer solution, pH=9.20
	5.3	pH Electrode Calibration:	
		5.3.1	Check that the equipment is correctly connected to electrical
		supply.	
		5.3.2	Switch on the instrument by pressing ON/OFF key on front
		panel.	
		Use a combination of two standard buffer for calibration	
		according to requirements i.e.:	
		Case 1- For solution with pH less than 7.00 uses standard	
		buffers of 4.00 & 7.00.	
		Case 2- For solution with pH more than 7.00 use standard	
		buffers of 7.00 & 9.20	
			(Note: The pH meter shall be set for the different buffers to
		be used at the time of installation based on the future use.	
		This setting is changed only in case the range of pH readings	
		is different than preset.)	
		5.3.3	Clean pH electrode with purified water and blot dry.
		5.3.4	Put the standard buffers that are used in calibration in
		clean beakers.	
		5.3.5	Introduce pH electrode in first buffer according to case 1 or
		case 2 and wait for 2 seconds to stabilize the pH.	
		5.3.6	Remove pH electrode, rinse with purified water and blot dry
		and put in second standard buffer according to case 1 or case	
		2 and wait for 2 seconds to stabilize the pH.	
		5.3.7	Remove pH electrode, rinse with purified water and blot dry
		and put in potassium chloride solution.	
	5.4	pH Measurement:	
		5.4.1	Clean pH electrode with purified water and blot dry.
		5.4.2	Put the sample in clean beaker to check the pH.
		5.4.3	Introduce the pH electrode in beaker having sample solution
		5.4.4	Press [Read] key for pH measurement.
		5.4.5	Rinse, blot and store the electrode in a beaker having
		saturated KCl solution.	
6.	References: NA		

[1.0 |Objective: |
| |To lay down the procedure for determine the human coagulation factor VII |
| |activity in Factor IX. |
[2.0 |Scope: |
| |This Standard Operating Procedure is applicable to all the In- process and |
| |finished product samples. |
[3.0 |Responsibility |
	3.1	The QC analyst shall be responsible for executing the described
	procedure correctly.	
	3.2	The QC supervisor shall be responsible for verifying that the
	procedure is complied with and the results are checked.	
	3.3	The QC HOD shall be responsible for overall compliance of this SOP.
[4.0	Abbreviations and Definitions:	
	Not applicable	
[5.0	Procedure:	
	5.1	Equipment, Materials and Reagents
		5.1.1
		5.1.2
		5.1.3
		5.1.4
		5.1.5
	5.2	Preparation of solutions:
		5.2.1
		5.2.2
		5.2.3
[5.3	Pre-Dilution of reference standard and sample:	
	5.3.1	Reference solution preparation:
		Dilution fold
		Unicalibrator
		Owren koller buffer (µL)
		Total Volume (µL)

[1.0 |Objective: |
 ||To lay down a procedure to determine the N-Acetyl Tryptophan using HPLC |
 | |method |
 [2.0 |Scope: |
 ||This Standard Operating Procedure is applicable to all the In- process and|
 | |finished product samples. |
 [3.0 |Responsibility |
	3.1	The QC analyst shall be responsible for executing the described
	procedure correctly.	
	3.2	The QC supervisor shall be responsible for verifying that the
	procedure is complied with and the results are checked.	
	3.3	The QC HOD shall be responsible for overall compliance of this SOP.
[4.0	Abbreviations and Definitions:	
	4.1	NaoH
	4.2	g
	4.3	mL
	4.4	mg
	4.5	M
	4.6	min
	4.7	µL
	4.8	µg
	4.9	°C
	4.10	%
[5.0	Procedure:	
	5.1	Equipment, Materials and Reagents
	5.1.1	HPLC
	5.1.2	HPLC Column
	5.1.3	Photo Diode Array detector
	5.1.4	UV detector
	5.1.5	Mobile phase filtration unit with 0.45 µm filter
	5.1.6	Sample filtration syringe filters 0.45 µm
	5.1.7	Column details
		Column Dimensions : 300 mm (l) x 8 mm (ID)
		Particle size : 5 µm
	5.1.8	Disodium hydrogen phosphate dihydrate (HPLC grade)
	5.1.9	Sodium dihydrogen phosphate monohydrate (HPLC grade)
	5.1.10	Sodium chloride GR
	5.1.11	Sodium azide GR
	5.2	Preparation of solutions:
	5.2.1	Reference Solution preparation (16 mM N -Acetyl Tryptophan):
		5.2.1.
		1
		and for complete dissolution of the reagent.
	5.2.2	Dilution of 16 mM reference standard :
		5.2.2.
		1
		treat the Reference standard similar to that of test
		solution. So dilute the Reference standard 20 F
	5.2.3	Test Solution preparation :
	5.2.3.	Dilute the preparation to be examined containing 16 mM
	1	N-Acetyl Tryptophan with 9 g/L solution of Sodium
		Chloride to a concentration of 10 g/L.
		Test Solution Conc.
		Dil. Folds
		Vol. of Ref Std (µL)
		Vol. of 9 g/L NaCl (µL)
		Total volume (µL)
	5.3	Test method:
	5.3.1	Follow the test procedure as per SOP for Operation of Liquid
		Chromatography.
	5.3.2	Initially purge the flow lines with purified water for 5 min
		and then with Mobile phase for 5 min.
	5.3.3	Equilibrate the column initially with mobile phase and auto
		zero if the base line is not straight:
		Flow rate: 0.5 mL/min
		Column volumes: 3
		Time: 1 Hour 30 min
	5.3.4	First injection: Inject reference solution by down loading
		the respective method file.

		5.3.5	After the first injection is completed, clean the column with
		mobile phase.	
		Flow rate: 0.5 mL/min	
		Column volumes: 1	
		Time: 30 min	
		5.3.6	Second injection: Inject test solution by down loading the
		respective method file.	
		5.3.7	After completing analysis, clean the column with mobile
		phase.	
		Flow rate: 0.5 mL/min	
		Column volumes: 3	
		Time: 1 Hour 30 min	
		5.3.8	After completion of analysis, remove the column and purge the
		flow lines with purified water for 5 min.	
		5.3.9	Analyze the results using various integration icons as per
		the SOP	
		5.4	Calculation:
		5.4.1	Compare the peak Area of N–Acetyl Tryptophan of test solution
		with that of Reference solution area and calculate in terms	
		of mmol/g of Protein (i.e. N-Acetyl Tryptophan).	
		5.4.2	The 20F Diluted 16 mM N-Acetyl Tryptophan standard equals to
		0.08 mmol/g of protein	
		Peak area of 20F diluted 16 mM concentration of N-Acetyl	
		Tryptophan standard = X	
		Peak area of 20F diluted N-Acetyl Tryptophan for sample = Y	
		[pic][pic][pic] mmol/g of protein	
6.0	References:		
6.1	EP & IP		

1.0 |Objective |
 |To lay down the procedure for estimation of Uronic acid content in |
 |Pneumococcal polysaccharide samples. |
 2.0 |Scope |
 |This Standard Operating Procedure is applicable to all the in process and |
 |polysaccharide samples of pneumococcal serotypes 1, 3, 5, 9V in which the |
 |Uronic acid content is to be determined. |
 3.0 |Responsibility |
3.1	The analyst is responsible for executing the described procedure
	correctly.
3.2	Supervisor is responsible for verifying that the procedure is carried
out correctly and check the results.	
3.3	HOD shall be responsible for overall compliance of this SOP and shall
ensure that the SOP is reviewed every two years or if required	
earlier.	
4.0	Abbreviations and Definitions
4.1	Abbreviations:
	4.1.1
	4.1.2
	4.1.3
	4.1.4
	4.1.5
	4.1.6
	4.1.7
	4.1.8
4.2	Reagents:
	4.2.1
	4.2.2
	4.2.3
	4.2.4
	4.2.5
	4.2.6
	4.2.7
4.3	Equipments and Material
	4.3.1
	4.3.2
	4.3.3
	4.3.4
	4.3.5
	4.3.6
	5000 µl)r
	4.3.7
	4.3.8
	4.3.9
5.0	Procedure
5.1	Preparation of solutions: Prepare all the solutions as described below
and fill the form, label the solutions and assign lot numbers.	
	5.1.1
	5.1.2
	5.1.3
	5.1.4
	5.1.5
5.2	Preparation of Calibration Curve:
	5.2.1
	5.2.2
	5.2.3
	5.0
	10.0
	20.0
	40.0
	80.0
	100.0
5.3	Sample preparation:
	5.3.1

			about 1 mg/ml of polysaccharide in Lal reagent water and use it		
			as test sample.		
			Note: The sample dilutions can vary depending upon the uronic		
			acid concentration. If there is any sample outside the range of		
			the calibration curve, carry out the calculation for more		
			appropriate dilution. Based on the expected uronic acid present		
			in the dry polysaccharide samples, the 1 mg/ml reconstituted		
			sample is further diluted as per the table below.		
			Serotype	Dilution fold	
			1	10 fold	
			3	10 fold	
			5	5 fold	
			9V	5 fold	
		5.3.2	In case on in process, take 1 ml of sample (dilute if required)		
			having about 100-150 µg Polysaccharide in eppendorf tube.		
		5.3.3	Add 100 µl of 5% DOC to the test sample and mix it, put this		
			tube in an ice bath for 30 minutes.		
		5.3.4	Take out the eppendorf and add 50 µl 1M HCl and mix thoroughly.		
			Centrifuge the mixture at 2-8°C for 15 minutes at 10000 RPM.		
			Collect the supernatant and use it as sample for uronic acid		
			estimation.		
		5.4	Test Procedure:		
		5.4.1	Take 500 µl of the standard dilutions from calibration curve,		
			Test Sample and Blank (Purified water) in duplicates in separate		
			test tubes.		
		5.4.2	Place the above test tubes in ice and add 2.5ml of cooled 9.94M		
			sodium tetraborate solution drop-wise to each tube of the		
			standard, blank and sample and mix well.		
		5.4.3	Cool the test tube to reach room temperature and incubate the		
			tubes in water bath at 80°C for 15 minutes.		
		5.4.4	Allow the test tube to cool down to room temperature for 5 min		
			and add 100 µl of 0.007M Carbazole solution to each tube.		
		5.4.5	Vortex the tubes and incubate in water bath at 80°C for 15		
			minutes.		
		5.4.6	Allow these tubes to cool at room temperature.		
		5.4.7	Observe for the appearance of purple pinkish color, which can		
			vary with the concentration.		
		5.4.8	Measure the absorbance at 530nm against the blank using		
			spectrophotometer.		
		5.4.9	Note that all the readings of the assay tubes should be taken		
			within 30 mins. of completion of the assay and maximum 10		
			samples should be analysed in one single assay.		
		5.5	Calculations:		
		5.5.1	Once the absorbance of the standards and the test samples have		
			been taken, calculation shall be done using the excel sheet		
			carrying formulas for the required calculation.		
		5.5.2	Always copy the master excel sheet (sheet carrying formulas for		
			the required calculation), never use the previously used excel		
			sheet for new calculations.		
		5.5.3	Copy the master excel sheet, keep it for use in respective		
			folder and then fill in the required sample details (Batch		
			number), sample dilution (if applicable), name of analyst, date,		
			etc.		
		5.5.4	Proceed for calculations starting from the Calibration curve.		
			The calculations are done according to the equation,		
			Y = mX + b, Where: m = slope		
			b = intercept		
			Y = absorbance		
			X = concentration		
			(Multiplication by dilution factor)		
		5.5.6	In case of dry polysaccharide,		
			Uronic acid (%) in polysaccharide on dry weight basis:		
			Uronic acid in dry polysacc. (µg/ml) =		
			Uronic acid in Polysacc. sol. (µg/ml) X 1 – {Moisture (%) / 100}		
			Uronic acid in dry polysaccharide (µg/ml)		
			Uronic acid (%) = _____		
			X 100		
			Conc. of polysaccharide solution (µg/ml)		
			To calculate the pure polysaccharide content (mg) present in dry		
			polysaccharide sample in of each serotype based on the Uronic		
			acid content obtained, following formula is applied:		
			Pure Polysaccharide content(mg) / Dry polysaccharide content		
			(mg)		
			= (Uronic acid in dry polysacc (mg) X Conversion factor)		
		5.5.7	The conversion factor of each serotype is mentioned below		
			Serotype	Percentage of Uronic acid as	Conversion factor
			per WHO TRS 977		
			1	55.17 %	1.81
			3	60.23 %	1.66
			5	23.59 %	4.24
			9V	22.33 %	4.47
		5.6	Acceptance criteria:		
		5.6.1	The test is valid if Regression of the standard curve is ≥ 0.99		
			and the percentage of CV is ≤ 15% within the replicates.		
		5.6.2	Uronic acid content in Purified Pneumococcal polysaccharide and		
			conjugate should meet the laid specifications for each serotype.		
		6.0	Forms and Records		
			Determination of Uronic Acid content - Annexure 1		
		7.0	Reference		
		7.1	WHO TRS No. 977 Annexure 3: Recommendations to assure the quality,		

	safety and efficacy of pneumococcal conjugate vaccines.	
	7.2	European Pharmacopoeia, Volume 5, 'Uronic acid in Polysaccharide
	Vaccine' 2.5.22.	

1.0 |Objective: |
 |To lay down a procedure to Procedure to determination of octanoic acid using |
 |HPLC method. |
 2.0 |Scope: |
 |This Standard Operating Procedure is applicable to all the In- process and |
 |finished product samples. |
 3.0 |Responsibility |
3.1	The QC analyst shall be responsible for executing the described
	procedure correctly.
3.2	The QC supervisor shall be responsible for verifying that the procedure
	is complied with and the results are checked.
3.3	The QC HOD shall be responsible for overall compliance of this SOP.
4.0	Abbreviations and Definitions:
4.1	NaoH
4.2	g
4.3	mL
4.4	mg
4.5	M
4.6	min
4.7	µL
4.8	µg
4.9	°C
4.10	%
5.0	Procedure:
5.1	Equipment, Materials and Reagents
	5.1.1
	5.1.2
	5.1.3
	5.1.4
	5.1.5
	5.1.6
	5.1.7
	5.1.8
	5.1.9
	5.1.10
	5.1.11
	5.1.12
	5.1.13
5.2	Preparation of solutions:
	5.2.1
5.3	Test procedure :
	5.3.1
	5.3.2
	5.3.2.

			Aspirate the supernatant from the pellet with the help of	
			a needle syringe and filter the sample using 0.45 µ	
			syringe filter and transfer into HPLC glass vial.	
				Note: No visible pellet is observed in the sample
				preparation of Reference solution.
		5.3.3	Test Sample Preparation:	
		5.3.3.	Take neat test sample.	
		1		
		5.3.3.	Dilute 0.4 mL of neat Test sample with 1.6 mL of Methanol	
		2	(HPLC grade) and vortex the solution for 3 minutes for	
			proper mixing. Then centrifuge the sample at 10,000 rpm	
			for 5 minutes. Aspirate the supernatant from the pellet	
			with the help of a needle syringe and filter the sample	
			using 0.45 µl syringe filter and transfer into HPLC glass	
			vial.	
			Note: A white precipitate is formed with the Test solution after	
			addition of Methanol. A visible pellet is observed after	
			centrifugation.	
		5.3.4	Analytical conditions	
			Wavelength : 214 nm	
			Flow rate : 0.8 mL/ min.	
			Run time : 12 min	
			Temperature : 40°C	
		5.3.5	Injection parameters	
			Injection volume : 50 µL	
		5.3.6	Method of Analysis:	
			5.3.6.1	Follow the Operation of Liquid Chromatography procedure
			as per SOP for Operation of Liquid Chromatography.	
			5.3.6.2	Initially purge the flow lines with purified water for 5
			min and then with mobile phase for 5 min.	
			5.3.6.3	Equilibrate the column initially with mobile phase and
			auto zero if the base line is not straight:	
			Flow rate: 0.8 mL/min	
			Column volumes: 3	
			Time: 40 min	
			5.3.6.4	First injection: Inject 50 µL of Reference Solution by
			down loading the respective method file.	
			Flow rate: 0.8 mL/min	
			Time: 12 min	
			5.3.6.5	Second injection: Inject 50 µL of Test solution by down
			loading the respective method file.	
			Flow rate: 0.8 mL/min	
			Time: 12 min	
			5.3.6.6	After completing analysis, clean the column with mobile
			phase	
			Flow rate: 0.8 mL/min	
			Column volumes: 3	
			Time: 15 min	
			5.3.6.7	After completion of analysis, remove the column and purge
			the flow lines with purified water for 5 min.	
			5.3.6.8	Analyze the results using various integration icons as
			per the SOP.	
		5.3.7	Calculation:	
			5.3.7.1	Compare the peak Area of Octanoic acid of test solution
			with that of Reference solution area and calculate in	
			terms of mmol/g of Protein (i.e. Octanoic acid) and	
			record the results in the analytical data sheet.	
			5.3.7.2	Octanoic acid 16 mM standard equals to 0.08 mmol/g of
			protein	
			Peak area of diluted 16 mM concentration of Octanoic acid	
			standard = X	
			Peak area of diluted Octanoic acid for Test sample = Y	
			[[pic]][pic][pic] mmol/g of protein	
6.0	References:			
6.1	EP & IP			

[1.0]Objective: |
	To lay down the procedure for determination of cell wall polysaccharide
	content in purified capsular polysaccharide of Streptococcus pneumoniae
	samples by ELISA technique.
[2.0]Scope:	
	This Standard Operating Procedure is applicable for determination of
	cell wall polysaccharide content in purified capsular polysaccharide of
	Streptococcus pneumoniae.
[3.0]Responsibility:	
	3.1
	executing the described procedure correctly.
	3.2
	that the procedure is carried out correctly and for checking the
	results.
	3.3
	compliance of the SOP.
[4.0]Abbreviations and Definitions:	
[4.1]	
QA	
: Quality Assurance	
.2	
QC	
: Quality Control	
[4.3]	
SOP	
: Standard Operating Procedure	
[4.4]	
PBS	
: Phosphate Buffer saline	
[4.5]	
CWPS	
: Cell Wall polysaccharide	
[4.6]	
ELISA	
: Enzyme linked Immuno sorbent Assay	
[4.7]	
HRP	
: Horse Raddish Peroxidase	
[4.8]	
TMB	
: 3,3',5,5' Tetra Methyl Benzidine	
[4.9]	
BSA	
: Bovine Serum Albumin	
[5.0]Procedure	
	5.1
	5.2
	5.3
	5.4

			Pneumococcal Cell wall Polysaccharide Standard, keep the
			vial overnight at 2 to 8°C overnight with shaking. Prepare
			the aliquots as 1000 µl of 1 mg/ml, 100 µl of 1 mg/ml, and
			100 µl of 10 µg /ml. Store the standard aliquots at -70°C.
		5.4.2	Phosphate Buffered Saline (10X): Weigh 85.0 gm of NaCl, 22.8
			gm of Na₂HPO₄ and 63.6 gm of NaH₂PO₄ and dissolve in 800 ml
			of purified water. After complete dissolution of reagents,
			make up the total volume to 1000ml with purified water.
			Filter sterilizes the solution by passing through 0.22 µm
			filter.
		5.4.3	Phosphate Buffered Saline (1X): Add 100ml of Phosphate
			Buffered Saline (10X) to 900 ml sterile purified water.
		5.4.4	Dilution buffer (PBST): Add 50 µl (0.05% v/v) of Tween 20 to
			100 ml of PBS 1X solution.
		5.4.5	Blocking buffer (PBST with 1% BSA): Add 1 gm (1% w/v) of
			Bovine Serum Albumin (BSA) to 100 gm PBST solution. Do not
			agitate the solution for mixing.
		5.4.6	Carbonate / Bicarbonate buffer: Prepare 0.5 M stock solution
			of Na₂CO₃ and NaHCO₃ by weighing 5.3 gm/100ml and 4.2
			gm/100ml respectively. Prepare the working stock of
			Carbonate / Bicarbonate buffer by adding 120 µl Na₂CO₃
			solution and 880 µl of NaHCO₃ solution in 9 ml of purified
			water. pH of this buffer should be 9.6.
		5.4.7	Secondary antibody dilution: Calculate the total volume of
			working dilution of secondary antibody required for the
			assay. Prepare the working dilution of Polyclonal Goat
			anti-mouse Immunoglobulin HRP conjugate in PBST as 1:2000
			folds.
		5.4.8	Stop Solution: To prepare stop solution (1N H₂SO₄), add 2.8
			ml of H₂SO₄ to 97.2 ml of purified water.
		5.5	Test Method:
		5.5.1	Coating of wells:
		5.5.1.1	The ELISA plates are coated with anti CWPS
			polyclonal antibodies. Determine the number of wells
			to be coated. Prepare a 1: 32000 fold dilution of
			anti-CWPS polyclonal antibody dilution in carbonate
			/ bicarbonate coating buffer.
		5.5.1.2	Add 100 µl/ well of polyclonal antisera solution to
			each well. Incubate the plates at 2 to 8°C over
			night.
		5.5.2	Blocking of wells:
		5.5.2.1	On the next day, remove the coated plate from at 2
			to 8°C; discard the coating solution by flicking.
			Add 150 µl/ well of blocking buffer to each coated
			well.
		5.5.2.2	Incubate the plate at 37 ± 2°C for 30 minutes.
		5.5.3	Preparation of Polysaccharide Standard and Test sample
			dilutions:
		5.5.3.1	During the incubation for blocking of wells, prepare
			the dilutions for polysaccharide standards.
		5.5.3.2	Prepare 3 fold dilutions of polysaccharide standard
			in dilution buffer to get a concentrations of 1000
			ng/ml, 333.33 ng/ml, 111.11 ng/ml, 37.04 ng/ml,
			12.35 ng/ml, 4.12 ng/ml, 1.04 ng/ml and 0.46 ng/ml.
		5.5.3.3	Prepare 1: 10, 1: 100 and 1: 1000 fold dilution of
			test sample in dilution buffer.
		5.5.4	Washing of wells:
		5.5.4.1	After completion of 30 min. incubation of the
			blocked plate at 37°C, remove the plate from the
			incubator.
		5.5.4.2	Discard the blocking buffer from the wells by
			flicking. Wash the wells by adding 250 µl/ well of
			PBS 1X. After 30 sec. discard the PBS from wells by
			flicking and then gently tap the plate on to a
			blotting paper. Ensure that the washing solution is
			removed completely, do not allow the wells to dry.
			Repeat the procedure for three times.
		5.5.5	Addition of Standard, sample and controls to wells:
		5.5.5.1	Add 100 µl of all the eight dilutions of standard in
			column A and B of the plate in duplicate.
		5.5.5.2	Add 100 µl of each sample, in duplicate that is to
			be analysed in consecutive wells of column C.
		5.5.5.3	Add 100 µl of dilution buffer in duplicate; this
			will act as a negative control.
		5.5.5.4	After addition of all standards, samples and
			controls, seal the plate with plate sealer and
			incubate it in 37 ± 2°C for next 90 min.
		5.5.6	Addition of Primary antibody:
		5.5.6.1	Prepare 1: 1200 dilution of Monoclonal antibody for
			Phosphoryl Choline in dilution buffer as per the
			volume required.
		5.5.6.2	After completion of 90 min. of incubation period,
			remove the plate from incubator.
		5.5.6.3	Wash the wells for four times by PBS 1X as mentioned
			in the step no. 5.5.4
		5.5.6.4	Add 100 µl of 1: 1200 fold diluted Monoclonal
			antibody solution for Phosphoryl Choline to each
			well.
		5.5.6.5	After addition, seal the plate with plate sealer and
			incubate it in 37 ± 2°C for next 90 min.
		5.5.7	Addition of secondary antibody:
		5.5.7.1	Prepare 1:2000 dilution of Polyclonal Goat anti

				mouse Immunoglobulin / HRP conjugate solution in	
				dilution buffer as per the volume required.	
				5.5.7.2	After completion of 90 min. of incubation period,
				remove the plate from incubator.	
				5.5.7.3	Wash the wells for four times by PBS 1X as mentioned
				in the step no. 5.5.4.	
				5.5.7.4	Add 100 µl of 1: 2000 fold diluted secondary
				antibody solution to each well.	
				5.5.7.5	After addition, seal the plate with plate sealer and
				incubate it in 37 ± 2°C for next 60 min.	
				5.5.8	Addition of TMB Substrate:
				5.5.8.1	After completion of 60 min. of incubation period,
				remove the plate from incubator.	
				5.5.8.2	Wash the wells for four times by PBS 1X as mentioned
				in the step no. 5.5.4.	
				5.5.8.3	Add 100 µl of TMB Substrate, in dark to each well.
				5.5.8.4	After addition incubate the plate at room
				temperature for next 15 min. in dark.	
				5.5.9	Addition of Stop solution:
				After completion of 15 min. of incubation period, add 100 µl	
				of stop solution (1N H2SO4) to each well above the TMB	
				substrate to stop the reaction.	
				5.5.10	Reading of Plate:
				5.5.10.	After addition of stop solution, read the Optical
				1	density (OD) values of the wells on a Micro plate
				ELISA reader.	
				5.5.10.	Read the plate at a dual wavelength of 450 and 630
				2	nm. The readings should be taken after 5 mins but
				within 15 min from addition of stop solution.	
				5.5.11	Data analysis:
				5.5.11.1	Import the raw data from the software of ELISA
				reader into the window Excel file and get the	
				average OD values of the duplicate wells and	
				compile the data sheets comprising of standard	
				curve concentration and sample details along with	
				the average OD as per the plate design. For	
				standard curve, take concentration on X-axis and	
				its corresponding OD on Y-axis.	
				5.5.11.2	Prepare Standard curve using Standard CWPS ranging
				from 0.46 ng/ml to 1000 ng/ml concentration. The	
				Coefficient of correlation (R2) which should be ≥	
				0.98.	
				5.5.11.3	Once compiled data sheet is ready, use sigma plot
				software for finding concentration of CWPS in the	
				samples by extrapolating the standard curve values	
				i.e. concentration and its corresponding OD.	
				5.5.11.4	Open Sigma plot software, click on create new blank
				notebook and press ok.	
				5.5.11.5	Copy standard curve from compiled excel sheet onto
				the sigma plot note book and save the note book.	
				5.5.11.6	Click 'Pharmacology' and select 'standard curve
				analysis'. Window showing standard curve will	
				arrive. For curve, select 'Four parametric' options	
				and click for 'log X axis scale' and press OK.	
				Software will automatically process the data in the	
				form of three sheets below the notebook. Three	
				sheets will be named as	
				Data 2. Graph	
				3. Report	
				5.5.11.7	R2 for the standard curve will be determined from
				the report sheet.	
				5.5.11.8	Standard graph can be obtained from Graph sheet,
				where heading, X-axis and Y-axis information could	
				be incorporated by right clicking and opting 'Edit'	
				option.	
				5.5.11.9	Data sheet will give all the extrapolated values of
				concentration (X-axis) on column-6 and OD (Y-axis)	
				on column-7. Match OD of samples from excel	
				compiled sheets from column-7 of sigma plot and its	
				corresponding concentration from column-6.	
				Note: In case, the software gives two different	
				concentrations of CWPS for similar OD values of the	
				sample, higher dilutions should be considered for	
				this assay.	
				5.5.11.1	Similarly note the concentrations for all the
				0	samples and determine cell wall polysaccharide
				concentration by multiplying the dilution factor	
				with the concentration from the sigma plot.	
				5.5.12	Validity criteria:
				5.5.12.	Regression of standard curve (R2) should be ≥ 0.98.
				1	
				5.5.12.	Coefficient of Variation (%CV) for replicates should
				2	be < 15%.

1.0	Objective:			
	To lay down the test procedure for Fc Function of Immunoglobulin			
2.0	Scope:			
	This Standard Operating Procedure is applicable to all the In- process and			
	finished product samples of IVIG.			
3.0	Responsibility			
	3.1	The QC analyst shall be responsible for executing the described		
		procedure correctly.		
	3.2	The QC supervisor shall be responsible for verifying that the		
		procedure is complied with and the results are checked.		
	3.3	The QC HOD shall be responsible for overall compliance of this SOP.		
4.0	Abbreviations and Definitions:			
	4.1	g:	: Grams	
	4.2	mL	: Milliliter	
	4.3	mg	: Milligram	
	4.4	M	: Molarity	
	4.5	min	: Minute	
	4.6	µL	: Microliter	
	4.7	µg	: Microgram	
	4.8	°C	: Degree centigrade	
5.0	Procedure:			
	5.1	Equipment, Materials and Reagents		
		5.1.1	Multi scan Fc ELISA reader	
		5.1.2	Incubator-Dryer	
		5.1.3	Desk Top Centrifuge	
	5.2	Preparation of solutions:		
		5.2.1	Stock Solutions	
			5.2.1.1	Magnesium and Calcium stock solution
			S.No	
			Reagents	
			Weight (g) / Volume (mL)	
			Rubella Antigen titration	
			Test for Fc Function	
			1	
			Calcium chloride	
			1.103 g (0.3 M)	
			0.11 g (0.0015 M)	
			2	
			Magnesium chloride	
			5.083 g (1 M)	
			0.508 g (0.005 M)	
			3	
			Purified water	
			25 mL	
			500 mL	
			Dissolve the above mentioned components in purified	
			water completely.	
			Note: Store at 4°C in glass containers upto 2 months	
			until the solution is visually clear	
		5.2.1.2	HEPES 5X buffer stock solution	
			S.No	
			Reagents	
			Weight (g) / Volume (mL)	
			1	
			Sodium chloride (0.71 M)	
			41.5 g	
			2	
			HEPES buffer (0.05 M)	
			11.9 g	
			3	
			Purified water	
			800 mL	
			Adjust to pH 7.3 using 1M Sodium hydroxide	
			4	
			Magnesium & Calcium stock solution (0.0025 M & 0.00075	
			M respectively)	
			2.5 mL	
			Dilute to 1000 mL with purified water	
			Note: Store at 4°C in transparent containers	
		5.2.1.3	Phosphate buffered saline (10X) pH 7.2	
			S.No	
			Reagents	
			Weight (g) / Volume (mL)	
			1	

				Anhydrous Disodium Hydrogen Phosphate (0.057 M)
				10.22 g
				2
				Anhydrous Sodium dihydrogen phosphate (0.024 M)
				3.36 g
				3
				Sodium chloride (1.5 M)
				87.66 g
				4
				Purified water
				750 mL
				Adjust to pH 7.2
				Dilute to 1000 mL with purified water
				Filter through 0.22 µm filter
				Note: Store at Room Temperature in glass containers
				upto 2 months until the solution is visually clear
				5.2.1.4Bovine Serum Albumin (5%)
				S.No
				Reagents
				Weight (g)/Volume (mL)
				1
				Bovine Serum
				lbumin
				1.25 g
				2
				Purified water
				25 mL
				Note: Distribute 25 mL of the solution into 1 mL
				aliquots and store at -20°C for further use
				5.2.2Working Solutions
				5.2.2.1Citrate solution pH 6.1
				S.No
				Reagents
				Weight (g) / Volume (mL)
				1
				Sodium citrate (0.027 M)
				8.0 g
				2
				Sodium chloride (0.071 M)
				4.2 g
				3
				Glucose (0.114 M)
				20.5 g
				4
				Purified water
				750 mL
				Adjust to pH 6.1 using a 100 g/L solution of citric
				acid
				Dilute to 1000 mL with Purified water
				Note: Store at 4°C upto 2 months. Use only if the
				solution is visually clear
				5.2.2.2Stabilized human blood
				Collect O group human red blood into Citrate solution
				(Anticoagulant solution).
				Note: Use 0.25 mL of citrate solution for 1 mL of
				blood. Store the stabilized blood at 4°C for not more
				than 3 weeks
				5.2.2.31X PBS Solution pH 7.2
				S.No
				Reagents
				Weight (g) / Volume (mL)
				1
				10X PBS pH 7.2
				10 mL
				2
				Purified water
				90 mL
				5.2.2.4Albumin PBS Solution : 0.15% BSA in 1X PBS (Phosphate
				buffer saline) containing 0.5 mM Magnesium Chloride and
				0.15 mM Calcium Chloride
				S.No

				Reagents
				Volume (mL)
				1
				10 X PBS stock
				10
				2
				Bovine Serum Albumin
				3
				3
				Magnesium and calcium chloride stock solution
				10
				4
				Purified water
				77
				5.2.2.5Tannic acid solution (0.01%)
				S.No
				Reagents
				Weight (g) / Volume (mL)
				1
				Tannic Acid
				10 mg
				2
				1X PBS pH 7.2
				100 mL
				Prepare immediately before use
				5.2.2.6Guinea pig complement
				Suitable dilution of Guinea pig complement available
				from a commercial source can be used. Follow the
				manufacturer's instructions for reconstitution of the
				complement
				5.2.2.7Rubella antigen (1:4 diluted)
				Suitable rubella antigen for
				haemagglutination-inhibition titre (HIT). Titre > 256
				HA units used.
				Note: Rubella Antigen available from a commercial
				source can be used
				Dilute Rubella antigen 1:4 (128 HA units) using Albumin
				PBS Solution.
				S.No
				Reagents
				Weight (g) / Volume (μL)
				1
				Rubella Antigen
				100
				2
				Albumin PBS Solution
				300
				Prepare immediately before use
				5.2.3Reference Solution preparation
				Ref Std Conc. in mg/mL
				Required Conc. in mg/mL
				Volume of Ref Std
				(μL)
				Volume of
				Albumin PBS Solution (μL)
				50
				30
				600
				400
				Adjust the final volume to 1.2 mL by adding 200 μL of Albumin
				PBS solution as mentioned below.
				Volume of Ref Std + Albumin PBS Solution
				(μL)
				Volume of
				Albumin PBS Solution (μL)
				Total volume
				(mL)
				1000
				200
				1.2
				5.2.4Test Solution preparation
				Test Sample Conc. in mg/mL
				Required Conc. in mg/mL
				Volume of Test Sample

			(μL)
			Volume of
			Albumin PBS Solution (μL)
			50
			30
			600
			400
			Adjust the final volume to 1.2 mL by adding 200 μL of Albumin
			PBS solution as mentioned below.
			Volume of Test Sample + Albumin PBS Solution(μL)
			Volume of
			Albumin PBS Solution (μL)
			Total volume
			(mL)
			1000
			200
			1.2
		5.2.5	Complement control preparation
		Albumin PBS Solution of 1.2 mL	
		5.3	Test method:
		5.3.1	Determination of titre of Rubella Antigen
		5.3.1.1	Rubella antigen received from a commercial source is
			initially titrated to find out the Haemagglutination
			–inhibition titre (HA titre)
			Note: The titre of the Rubella antigen should be
			greater than 256 HA units
		5.3.1.2	Prepare tanned RBC's by centrifuging 1 mL of O group
			Human red blood cells
		5.3.1.3	Wash the cells three times with 1X PBS pH 7.2
		5.3.1.4	Preparation of 2% V/V suspension of red blood cells
			S.No
			Reagents
			Volume (mL)
			1
			Washed human RBC
			0.1
			2
			1X PBS pH 7.2
			4.9
			Prepare immediately before use
		5.3.1.5	Preparation of Tannic acid solution
			S.No
			Reagents
			Volume (mL)
			1
			0.01% Tannic acid
			0.1
			2
			1X PBS pH 7.2
			7.4
			Prepare immediately before use
		5.3.1.6	Prepare tanned red blood cells by adding 3 mL of
			freshly prepared tannic acid solution and 3 mL of 2%
			v/v suspension of red blood cells.
			Freshly prepared Tannic acid solution
			3 mL
			2% V/V suspension of red blood cells
			3 mL
		5.3.1.7	Incubate tanned red blood cells at 37°C for 10 min
		5.3.1.8	Collect the cells by centrifugation at 2200 rpm for 10
			min. Discard the supernatant and wash the cells once
			with 1X PBS pH 7.2
		5.3.1.9	Preparation of 1% V/V Tanned red blood cells suspension
			S.No
			Reagents
			Volume (mL)
			1
			Tanned red blood cells
			0.05
			2
			1X PBS pH 7.2
			4.95
			Prepare immediately before use

| | | | 1:16 |

| | | | 1:32 |

| | | | | 1:64 |

| | | | 1:128

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|||||
|||1:512
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||| 1| |
||| 5[3.1.1]Initially add 50 µL of 1% Tanned RBC to wells labeled |
||| 2|from 1:2 to 1:1024 dilutions |
||| 5[3.1.1]Then add 10 µL of each dilution of the rubella antigen |
||| 3|prepared into respective wells |
||| 5[3.1.1]To controls, add 50 µL of 1% Tanned RBC and 10 µL of |
||| 4|Albumin HEPES buffer |
||| 5[3.1.1]Mix by shaking the plate and incubate for 1 hour at |
||| 5|37°C |
||| 5[3.1.1]Place the microtitre plate at an angle of 70° for 2 to |
||| 6|3 hrs at room temperature |
||| 5[3.1.1]Interpretation of results |
||| 7|Cell button ( ) indicates agglutination (Positive) and|
||| |Streaming of cells ( ) indicate non-agglutination |
||| |(Negative). |
||| |One HA unit is defined as the highest dilution of |
||| |antigen that gives complete haemagglutination of cells.|
||| |C1 |
||| |C2 |
||| |C3 |

```


			5.3.4.2	Mix well and incubate at 37°C for 60 min
			5.3.4.3	At the end of incubation, add 1.5 mL of Albumin PBS
			Solution and stir gently until homogenous	
			5.3.4.4	Collect the cells by centrifugation at 1000 g for 10
			min. Remove the supernatant and add approximately 3 mL	
			of Albumin PBS Solution. Repeat this operation up to 4	
			times in total, leaving a final volume of 300 µL	
		5.3.5	Complement-Initiated Haemolysis	
			5.3.5.1	To measure haemolysis, add 900 µL of Albumin PBS
			Solution warmed to 37°C to each test tube containing	
			300 µL of test/standard/complement treated cells and	
			re-suspend the cells carefully by repeated pipetting	
			(not fewer than 5 times)	
			5.3.5.2	The microtitre plate must be pre-warmed to 37°C before
			starting the test. Transfer 240 µL of each solution	
			into 4 microtitre plate wells	
			5.3.5.3	Incubate the microtitre plate at 37°C for 6 min
			5.3.5.4	Pre-warm diluted guinea pig complement for 30 sec at
			37°C.	
			Note: Use suitable dilution of guinea pig complement	
			5.3.5.5	To each microtitre plate well, add 60 µL of 1:50
			diluted guinea-pig complement	
			5.3.5.6	Mix for 10 s and immediately start recording the
			absorbance at 541 nm at 37°C measuring every 1 min	
			5.3.5.7	Stop the measurement if the absorbance as a function of
			time has clearly passed the inflexion point	
		5.4	Calculations :	
		5.4.1	For each well, determine the slope (S) of haemolysis curve at	
		the approximate inflexion point by segmenting the steepest		
		section in suitable time intervals Δt ($\Delta t=2$ min), and calculate		
		S between adjacent intersection points, expressed as ΔA per		
		minute		
		5.4.2	The largest value for S serves (S_{exp}). In addition, determine	
		the absorbance at the start of measurement (A_s) by		
		extrapolating the curve, which is almost linear and parallel to		
		the time axis within the first few minutes. Correct (S_{exp})		
		using the expression: $S' = S_{exp} / A_s$		
		5.4.3	Calculate the arithmetic mean of the values of S' for each	
		preparation(test & reference solution) .Calculate the index F_c		
		function (IF_c) from the expression.		
		[pic]		
		[pic]= Arithmetic mean of the corrected slope for the		
		preparation to be examined,		
		[pic]= Arithmetic mean of the corrected slope for the		
		reference preparation to be examined,		
		[pic]= Arithmetic mean of the corrected slope for the		
		complement control		
		5.4.4	Calculate the index of F_c function for the preparation to be	
		examined and record the result in the analytical test data		
		sheet.		
		6.0	References:	
		6.1 Indian Pharmacopeia		

1. |Purpose: |
 |To lay down the procedure for the estimation of molecular size of all |
 |serotypes of Pneumococcal CRM monovalent bulk conjugate of Pneumococcal |
 |conjugate vaccine. |
 2. |Applicable to: |
 |This Standard Operating Procedure is applicable for evaluating KD |
 |(molecular size distribution) of ARM of bulk conjugate of Pneumococcal CRM |
 |conjugate for all serotypes and for reporting of HPLC profile as obtained |
 |for inprocess Pneumococcal protein conjugate samples. |
 3. |Responsibility: |
 |The analyst is responsible for carrying out the described procedure |
 |correctly. Supervisor is responsible for ensuring that the procedure is |
 |carried out correctly and to check the results. |
 4. |Safety considerations: |
 |4.1 |Always wear laboratory coat. |
 |4.2 |Always use solutions and samples filtered through 0.2 µm membrane. |
 |4.3 |Degas all solutions. |
 |4.4 |Avoid entry of bubbles into the system as well as column. |
 |4.5 |All the solution bottle openings should be well covered with aluminum |
 || foil. |
 5. |Equipments: |
 |5.1 |TSKgelG5000PWxl column {30 cm (l), 7.8 mm (i.d)} |
 |5.2 |HPLC system with UV detector under routine calibration programme |
 |5.3 |HPLC software for evaluation of results |
 |5.4 |Analytical balance |
 |5.5 |Precision balance |
 |5.6 |Vacuum Pump |
 |5.7 |pH Meter |
 |5.8 |Refrigerator |
 |5.9 |Magnetic stirrer |
 |5.10 |Ultrasonic Cleaner |
 6. |Reagents: |
 |6.1 |Sodium Chloride (NaCl) |
 |6.2 |Sodium azide |
 |6.3 |Disodium hydrogen phosphate anhydrous (Na₂HPO₄) |
 |6.4 |Sodium dihydrogen phosphate monohydrate |
 |6.5 |Salmon DNA |
 |6.6 |Purified water |
 7. |Preparation of solutions: |
 |7.1 |Solution I: [Na₂HPO₄ (0.1M), NaCl (0.1M)] | |
 || |For a volume of 1000 ml, weigh 14.19 g of Na₂HPO₄ and 5.84 g of NaCl. |
 || |Dissolve in 700 ml of purified water. Make up the volume to 1000 ml |
 || |with purified water. |
 |7.2 |Solution II: (NaH₂PO₄. H₂O, 0.1 M) |
 || |For a volume of 500 ml, weigh 6.85 g of NaH₂PO₄. H₂O. Dissolve in 400 |
 || |ml of purified water. Make up the volume to 500 ml with purified |
 || |water. |
 |7.3 |0.1M Sodium Phosphate buffer: (0.1 M Na₂HPO₄, 0.1M NaCl +0.1M NaH₂PO₄ |
 || |.H₂O, pH 7.0) |
 || |For a volume of 1000 ml, take approximately 600 ml of Solution I and |
 || |400 ml of Solution II. The pH of the solution should be 7.0 ± 0.2. |
 || |This solution can be stored at room temperature for 1 month. |
 |7.4 |Void Volume Marker: |
 || |For a volume of 5 ml, weigh 5 mg of Salmon DNA. Dissolve in purified |
 || |water and make up volume to 5 ml with purified water. Prepare fresh |
 || |before use. |
 |7.5 |Total volume marker: For a volume of 10 ml, weigh 10 mg of Sodium |
 || |Azide. Dissolve in purified water and make up volume to 10 ml with |
 || |purified water. Prepare fresh before use. |
 |7.6 |Calibration Mixture: Take 1.6ml of void volume marker (step 7.4) and |
 || |400µl of total volume marker (step 7.5). Vortex for few seconds. |
 || |Prepare fresh before use. |
 |7.7 |Preservation Solution A: (Sodium azide 0.05%): Weigh 1 g of Sodium |
 || |Azide and dissolve in approximately 1800 ml of purified Water. Make up |
 || |the volume to 2.0 Litre with purified water. This solution can be |
 || |stored at room temperature for 2 months. |
 |7.8 |Preservation Solution B: (20% Ethanol): Take 200ml of ethanol and make |
 || |up the volume to 1.0 litre with purified water. This solution can be |
 || |stored at room temperature for 2 months. |
 || |Note: All the solutions prepared for use are vacuum filtered using |
 || |Nylon 6, 6 membrane 0.20 µm filters to eliminate the entry of possible |
 || |particles into the system as well as the column. |
 8. |Procedure: |
 |8.1 |Run conditions: | | |
 || |8.1.1 |Column: TSKgelG5000PWxl column{30 cm (l), 7.8 mm (i.d)} |
 || |8.1.2 |Buffer used: 0.1M sodium phosphate buffer |
 || |8.1.3 |Elution: Isocratic |
 || |8.1.4 |Flow rate: 0.5ml/min |
 || |Note: For in process samples of conjugation process, |
 || |the flow rate may be kept at 0.75ml/min. |
 || |8.1.5 |Run time: 35minutes or before if stable baseline is |
 || |obtained |
 || |8.1.6 |Sample Injection: |
 || |8.1.6.1 |Calibration of column: |
 || |1 |Calibration of the column is done to determine |
 || |void volume (retention time) and total volume |
 || | (retention time) every time when ever a test |
 || |sample is applied. |
 || |8.1.6.2 |After a stable baseline is obtained at 214nm, |
 || |2 |inject 5-20µl of calibration mixture. |

			8.1.6.	Injection of sample of Pneumococcal conjugate
			3	after sterile filtration:
			8.1.6.3.	Load 40-70µl of sample volume of
			1	pneumococcal polysaccharide conjugate.
			8.1.6.	Injection of sample of inprocess Pneumococcal
			4	conjugate samples.
			8.1.6.4.	For samples of Pneumococcal conjugate
			1	at 0hr, intermediate and final hour of
			conjugation, inject 10-20µl of sample	
			on the column.	
			8.1.6.4.	For samples received during
			2	concentration diafiltration cycles,
			inject 20-40µl of sample for retentate.	
			8.1.6.4.	For samples received before sterile
			3	filtration, inject 20-60µl of sample of
			purified pneumococcal conjugate.	
		8.1.7	Detector Used: UV detector at wavelength of 214nm.	
		8.2	Run procedure:	
		8.2.1	Equilibrate the column with 0.1M sodium phosphate	
		buffer for 2 hours and wait till a stable baseline is		
		obtained at 214nm.		
		8.2.2	For inprocess samples at various hours of conjugation	
		and concentration/diafiltration retentate, inject the		
		required volume of sample as mentioned in steps		
		8.1.6.4.		
		8.2.3	For samples of purified pneumococcal conjugate before	
		sterile filtration and after sterile filtration (Active		
		Raw Material) inject calibration standard and sample.		
		Note: If several production samples are applied in one		
		day then only one application of the Calibration		
		mixture is enough.		
		8.3	Column cleaning and storage:	
		8.3.1	Column storage: Pass 200ml of preservation solution A/	
		preservation solution B solution through the column for		
		long time storage.		
		8.3.2	Cleaning the column: If in case the peak shape observed	
		is not proper, then reverse the flow direction of		
		column and pass 200ml of purified water and then 200ml		
		of preservation solution B and then pass 200ml of		
		purified water with normal flow direction.		
		8.4	Chromatogram evaluation:	
		8.4.1	For samples received during conjugation hours:	
		8.4.1.1 For samples run with flow rate 0.5ml/min		
		8.4.1.1.	Open the acquired chromatogram.	
		1		
		8.4.1.1.	Optimize integration with various	
		2	integration parameters or manual	
		integration parameters by drawing		
		baseline under the curve obtained at		
		lower retention time (9-13minutes) and		
		curve obtained at little higher		
		retention time(~16-17minutes).		
		Note: Curve under the lower retention		
		time is the curve of conjugate.		
		8.4.1.1.	Baselines for peaks obtained at very	
		3	high retention times (20 minutes or	
		more) are not constructed.		
		8.4.1.1.	Acceptance Criteria: Area percentage	
		4	for curve under lower retention time	
		conjugated peak) with respect to		
		carrier protein peak should not be less		
		than 70% at 16-24 hrs. of conjugation		
		reaction.		
		8.4.1.	For samples run with flow rate: 0.75ml/min	
		2		
		8.4..2.1	Open the acquired chromatogram.	
		8.4.2.2	Optimize integration with various	
		integration parameters or manual		
		integration parameters by drawing		
		baseline under the curve obtained at		
		lower retention time (6-11minutes) and		
		curve obtained at little higher		
		retention time (~11.5-13minutes).		
		Note: Curve under the lower retention		
		time is the curve of conjugate.		
		8.4.2.3	Baselines for peaks obtained at very	
		high retention times (14minutes or		
		more) are not constructed.		
		8.4.2.4	Acceptance Criteria: Same as mentioned	
		in 8.4.1.1.4.		
		8.4.2	For samples received during concentration diafiltration	
		cycles:		
		8.4.2.	Flow rate: 0.5ml/min	
		1		
		8.4.2.1.	Open the acquired chromatogram.	
		1		
		8.4.2.1.	For a flow rate of 0.5ml/min optimize	
		2	integration with various integration	
		parameters or manual integration		
		parameters by drawing baseline under		
		the curve obtained at lower retention		
		time (9-13minutes) and curve obtained		

					at little higher retention	
					time(~16-17minutes).	
					8.4.2.1.	Also construct baseline under the
					3	curves obtained at higher retention
					times (20minutes or more).	
					8.4.2.1.	Acceptance Criteria: Area percentage
					4	for curve under lower retention time
					(conjugated peak) for final diafiltered	
					retentate should be $\geq 95\%$.	
					8.4.2.	Flow rate: 0.75ml/min
					2	
					8.4.2.2.	Open the acquired chromatogram.
					1	
					8.4.2.2.	For a flow rate of 0.75ml/min optimize
					2	integration with various integration
					parameters or manual integration	
					parameters by drawing baseline under	
					the curve obtained at lower retention	
					time (6-11minutes) and curve obtained	
					at little higher retention time	
					(~11.5-13minutes).	
					8.4.2.2.	Also construct baseline under the
					3	curves obtained at higher retention
					times (20minutes or more).	
					8.4.2.2.	Acceptance Criteria: Same as in
					4	8.4.2.1.4.
		8.4.3	For samples of purified conjugated material and ARM of			
		bulk conjugate:				
		Percentage of sample eluted at $KD=0.5$ is calculated for				
		these samples as per the below mentioned procedure:				
		8.4.3.	The retention time of the unretained compound			
		1	(To) is determined with Salmon DNA dissolved in			
		calibration mixture as the first peak in the				
		elution curve.				
		8.4.3.	The Retention time of the component which has			
		2	full access to the pores of stationary phase			
			(Tt) is determined with sodium azide dissolved			
		in calibration mixture as the second peak in the				
		elution curve or the peak of sodium azide if				
		applied separately.				
		8.4.3.	Calculations:			
		3	Elution time of Pneumococcal			
		polysaccharide-Protein (Te) at $KD = 0.5$				
		The calculations are made as per the following				
		equation:				
		$Te = KD(Tt - To) + To$ or $Te = 0.5$				
		$(Tt - To) + To$				
		Where,				
		Te= Elution time of PnPs -CRM Conjugate at KD				
		=0.5				
		Tt= Retention time of sodium azide				
		To= Retention time of Salmon DNA				
		KD= Distribution Coefficient				
		8.4.3.	Calculation of percentage of sample eluted at			
		4	KD = 0.5:			
		This is calculated using ChemStation software:				
		8.4.3.4.	Enter the value of Elution time (Te)			
		1	calculated in step 8.4.3.3 in			
		integration with timed events option or				
		with manual integration and select				
		split peak option.				
		8.4.3.4.	The percentage of sample eluted at $KD =$			
		2	0.5 is calculated by splitting the peak			
		at the Te and then taking the				
		percentage area for the first part of				
		the peak till Te.				
		8.4.3.4.	Acceptance Criteria: Percentage of			
		3	sample eluted at $KD = 0.5$ should be			
		$\geq 80\%$.				
		8.5	Reporting Results:			
		8.5.1	After optimizing integration, select Report Layout view			
		in the software.				
		8.5.2	For samples received during conjugation hours and			
		concentration/diafiltration samples, the print out of				
		integrated HPLC profile of sample is reported..				
		8.5.3	For samples received before and after sterile			
		filtration, the percentage of sample eluted at $KD = 0.5$				
		is reported. Also take printout of the sample and				
		calibration mixture with results.				
		9.	Abbreviations:			
		9.1	OC	: Degree centigrade		
		9.2	g	: Gram		
		9.3	ml	: Milli litre		
		9.4	μ l	: Micro litre		
		9.5	μ m	: Micro meter		
		9.6	nm	: Nano meter		
		9.7	AN	: Analytical		
		9.8	PCV	: Pneumococcal conjugate vaccine		
		9.9	ARM	: Active Raw Material		
		9.10	PnPs	: Pneumococcal polysaccharide		
		9.11	RD	: Research And Development		
		9.12	HOD	:Head of Department		

[1.0 |Objective: |
 ||To lay down the Procedure for molecular size by Liquid chromatography |
 [2.0 |Scope: |
 ||This Standard Operating Procedure is applicable to all the In- process and |
 ||finished product samples of IVIG. |
 [3.0 |Responsibility |
	3.1	The QC analyst shall be responsible for executing the described
		procedure correctly.
	3.2	The QC supervisor shall be responsible for verifying that the
		procedure is complied with and the results are checked.
	3.3	The QC HOD shall be responsible for overall compliance of this SOP.
[4.0	Abbreviations and Definitions:	
	4.1	g:
	4.2	mL
	4.3	mg
	4.4	M
	4.5	min
	4.6	µL
	4.7	µg
	4.8	°C
[5.0	Procedure:	
	5.1	Equipment, Materials and Reagents
		5.1.1
		5.1.2
		5.1.3
		5.1.4
		5.1.5
		5.1.6
		5.1.7
		5.1.8
		5.1.9
		5.1.1
		0
		5.1.1
		1
	5.2	Preparation of solutions:
		5.2.1
		5.2.2
[5.3	Test method:	
	5.3.1	Follow the test procedure as per SOP for Operation of Liquid
		Chromatography.
		5.3.2
		5.3.3
		5.3.4
		5.3.5
		5.3.6

		5.3.7	After completing analysis, clean the column with mobile phase.
			Flow rate: 0.5 mL/min
			Column volumes: 3
			Time: 1 Hour 30 min
		5.3.8	After completion of analysis, remove the column and purge the
			flow lines with purified water for 5 min.
		5.3.9	Analyze the results using various integration icons as per the
			SOP.
		5.4	System suitability:
		5.4.1	The reference solution injection is valid for 24 Hours.
		5.4.2	For checking System Suitability in between series of
			injections: Inject reference solution after every 6 injections
			of test solution.
		5.5	Method Validation Criteria:
		5.5.1	In the chromatogram obtained with reference solution, the
			principal peak corresponds to the IgG monomer and there is a
			peak corresponding to the dimer with a relative retention to
			the principal peak of about 0.85. Identify the peaks in the
			chromatogram obtained with the test solution by comparison with
			the chromatogram obtained with the reference solution; any peak
			with a retention time shorter than that of the dimer
			corresponds to polymers and aggregates
		5.5.2	Relative Retention time: For the monomer and for the dimer, the
			retention time relative to the corresponding peak in the
			chromatogram obtained with the reference solution is 1 ± 0.02 .
			Peak area: The sum of peak areas of the monomer and the dimer
			represent not less than 90 percent of the total area of the
			chromatogram and the sum of peak areas of polymers and
			aggregates represents not more than 3 percent of the total area
			of the chromatogram
		6.0	References:
		6.1 Indian Pharmacopeia	

S. No.	Parameter	Specification
1	Appearance	Clear transparent liquid
2	pH	6.8+0.2
3	Identity by Western Blot	Band of CRM should be observed on reacting with Specific antibodies
4	Total protein content	Report results
5	Purity by HPLC	> 90% of total protein
6	Nicking by SDS-PAGE	< 10% (In house specs)
7	Endotoxin content	<1EU /µg of protein
8	Sterility	No growth should be observed during the 14 days incubation period

1.0 |Objective |
 |To lay down the procedure for the determination of CDAP |
 |(1-cyano-4-dimethylaminopyridinium tetrafluoroborate) by High Performance |
 |Liquid Chromatography (HPLC). |
 2.0 |Scope |
 |This Standard Operating Procedure is applicable for the determination of |
 |CDAP (1-cyano-4-dimethylaminopyridinium tetrafluoroborate) by High |
 |Performance Liquid Chromatography (HPLC) in the QC. |
 3.0 |Responsibility |
3.1	The analyst shall be responsible for carrying out the described
	procedure correctly.
3.2	The Supervisor of the laboratory shall be responsible for verifying
	that the procedure is complied with and for checking the results.
3.3	The departmental head/designee shall be responsible for training of
	the staff and for overall compliance of the SOP.
3.4	Head QA/Designee shall be responsible for final approval and
	subsequent compliance of the SOP.
4.0	Abbreviations and Definitions
4.1	CDAP
4.2	Cm
4.3	DNA
4.4	G
4.5	HPLC
4.6	Id
4.7	L
4.8	L
4.9	M
4.10	mAu
4.11	Mg
4.12	Min
4.13	mL
4.14	mm
4.15	nm
4.16	QA
4.17	QC
4.18	QCR
4.19	rpm
4.20	R2
4.21	RSD
4.22	SOP
4.23	SST
4.24	°C
4.25	µg
4.26	µL
4.27	µm
4.28	System
	Suitability
	test
5.0	Procedure
5.1	Precautions to be taken:
	5.1.1
	5.1.2
	5.1.3
	5.1.4
	5.1.5
	5.1.6
	5.1.7
5.2	Requirement:
	5.2.1
	5.2.2
	5.2.3

||| |0 | |

|| |5.2.4 |Reagents: |

|| | |5.2.4.1|Sodium dihydrogen phosphate monohydrate |

|| | |5.2.4.2|Methanol |

|| | |5.2.4.3|Acetonitrile |

|| | |5.2.4.4|Purified Water |

|| | |5.2.4.5|CDAP (1-cyano-4-dimethylaminopyridinium |

|| | | |tetrafluoroborate) |

|| | |5.2.4.6|2-Propanol |

|| | |5.2.4.7|Hydrochloric acid 37 % |

|| |5.3 |Preparation of solutions: |

|| |Prepare all the solutions as described below. Carry out the labelling|

|| |of the prepared solutions and assign the lot numbers to the |

|| |solutions. |

|| |5.3.1 |Solution A (NaH₂PO₄. H₂O, 200mM): For a volume of 1 liter, |

|| | |weigh and add 27.6g NaH₂PO₄, H₂O. |

|| | |

|| | |Adjust the pH to 3.0 with HCl (37%). Make up the solution to |

|| | |1litre with purified water. Prepare fresh before use. |

|| |5.3.2 |Mobile phase for CDAP: |

|| | |Take 700mL of methanol in a measuring cylinder and 300mL of |

|| | |solution A to make the volume to 1000mL. Use the mobile phase |

|| | |within four days from the date of preparation if it remains |

|| | |clear and no turbidity or fibers observed in it. |

|| |5.3.3 |Isopropyl Alcohol 70% Solution: (For storing and purging the |

|| | |HPLC system) |

|| | |Take 700 mL of 2-propanol using a measuring cylinder and make |

|| | |up the volume to 1000 mL with Purified Water in a volumetric |

|| | |flask. This solution can be stored at room temperature for 3 |

|| | |months. |

|| |5.3.4 |1 mg/mL of CDAP calibration standard stock: |

|| | |Weigh 5mg of CDAP and make up the volume to 5mL with purified |

|| | |water in a volumetric flask. This solution can be stored as |

|| | |aliquots at -20°C for 2months. |

|| |5.3.5 |Preparation of working calibration standard of 10µg/mL or |

|| | |10ng/µL: |

|| | |Take 10µL of 1 mg/mL of CDAP calibration standard stock and add|

|| | |990µL of purified water to make 1mL of solution in an eppendorf|

|| | |for HPLC vial. Mix thoroughly. Prepare fresh before use. |

|| | |Note: The working calibration standard is prepared fresh before|

|| | |use. All the solutions prepared for use are vacuum filtered |

|| | |using Nylon 6, 6 membrane 0.20/0.22/0.45 µm filters to |

|| | |eliminate the entry of possible particles, if any into the |

|| | |system as well as the column. For organic solvents 0.45µm |

|| | |filters should be used (filtration not necessary if the |

|| | |solution is prepared in 0.2 µ filtered purified water). |

|| |5.3.6 |Preparation of Dilutions for standard Curve: |

|| | |Standard |Standard Conc. |Working Std. volume |Diluent (Mobile |

|| | | |(10µg/mL or 10ng/µL)|phase) volume |

|| | |1 (1 ng) |100 ng/mL |10 µL |990 µL |

|| | |2 (3 ng) |300 ng/ml |30 µL |970 µL |

|| | |3 (5 ng) |500 ng/ml |100 µL |1900 µL |

|| | |4 (10 ng)|1µg/ml |100 µL |900 µL |

|| | |5 (20 ng)|2µg/ml |200 µL |800 µL |

|| |5.4 |Preparing the HPLC System for run, Agilent Series: |

|| |5.5 |Run Conditions: |

|| |5.5.1 |Load method CDAP.M |

|| |5.5.2 |Flow Rate: 1.0 mL/min |

|| |5.5.3 |Detection wavelength : 280nm |

|| |5.5.4 |Run time: 4 minutes |

|| |5.5.5 |Mobile phase: mobile phase for CDAP |

|| |5.6 |Equilibration of the system and the column: |

|| | |Equilibrate the system and column with mobile phase for CDAP until a |

|| | |stable baseline at 280 nm is obtained which can be viewed with the |

|| | |online plot window. |

|| |5.7 |Sample Preparation and loading: |

|| |5.7.1|Based on the polysaccharide content obtained, dilute the sample|

|| | |to get a final working concentration of 100 µg/ml. |

|| |5.7.2|The injection volume for each standard, test sample and blank |

|| | |is of 10 µL, therefore load 20 µL of test sample, standards and|

|| | |blank in screw cap vials and place against the appropriate no. |

|| | |on the sample tray by clearly marking the screw cap vial. The |

|| | |sample of 10 µL which is loaded in the system contains 1 µg of |

|| | |polysaccharide. |

|| |5.8 |Running of sequence: |

|| |5.8.1 |Load sequence CDAP.S. |

|| |5.8.2 |Open the sequence parameter option by clicking Sequence menu. |

|| |5.8.3 |Enter information about the data directory and operator name. |

|| |5.8.4 |In the shutdown option, click on the post-sequence option and |

|| | |select LAMPOFF/STANDBY. |

|| |5.8.5 |In sequence table option, specify the sample names, methods and|

|| | |vials to run in the sequence as per the following table: |

|| | |S. No |Sample Code |Sample details |No. of injections |

|| | | |per Vial |

|| | |1 |Blank |Eluent Blank |1 X 10 µL |

|| | |2 |Standard 3 |System Suitability |6 X 10 µL |

|| | | |(5ng/µL) |test standard | |

|| | |3 |Blank |Eluent Blank |1 X 10 µL |

|| | |4 |Standard 1 |1ng Standard |1 X 10 µL |

|| | |5 |Standard 2 |3ng Standard |1 X 10 µL |

|| | |6 |Standard 3 |5ng Standard |1 X 10 µL |

|| | |7 |Standard 4 |10ng Standard |1 X 10 µL |

		8	Standard 5	20ng Standard	1 X 10 µL
		9	Blank	Eluent Blank	1 X 10 µL
		10	Test Sample (s)	Sample	1 X 10 µL
	5.8.6	Click on run sequence option in run control menu to start the			
		sequence.			
	5.8.7	Each sample run will be of 4 minutes and after running all the			
		samples the sequence will automatically stop.			
	5.8.8	Inject the eluent blank after every 20 injections.			
			Note: Here, eluent blank refers to mobile phase. One extra vial		
		with same SST sample can be used if 6 injections are not			
		possible from single vial.			
	5.9	Calculations :			
	5.9.1	Results of the sample are calculated from the calibration			
		standards using the chemstation software.			
		5.9.1.1	Open the chromatogram corresponding to 1ng calibration		
		standard in the offline copy of chemstation software in			
		the Data Analysis mode.			
		5.9.1.2	From the calibration menu select calibration settings		
		and write ng/µL in the Amount unit's option.			
		5.9.1.3	Select New Calibration Table from Calibration menu and		
		enter value1 in default amount option and value 1 in			
		level option.			
		5.9.1.4	Now open the chromatogram for 3.0 ng. Select add level		
		option from calibration menu and enter value 3.0 in			
		default amount and value 2 in level option.			
		5.9.1.5	Similarly open chromatograms for 5.0 ng, 10.0 ng and		
		20.0 ng and keep adding default amount and appropriate			
		value of level.			
		5.9.1.6	Select calibration task icon and the calibration table		
		with required concentrations and levels can be viewed			
		in the calibration table view. A calibration curve is			
		also displayed in the calibration curve field with			
		value of correlation coefficient of the curve.			
		5.9.1.7	Save the calibration method under appropriate name.		
		5.9.1.8	Open the chromatogram of test sample and select the		
		above created calibration method for integration of			
		the test sample.			
		5.9.1.9	In the specify report option of Report menu, select		
		ESTD in the Calculate field.			
		5.9.1.10	In the report layout view click on the show report		
		preview with current template, method and data file			
		icon and view the result of the test sample in terms			
		of concentration ng/µL or µg/mL.			
		5.9.1.11	Calculations for Standard curve and samples:		
		The calculations are made according to the equation of			
		minimum squares.			
		$Y = mX + b$.			
		Where: m = slope			
		b = intercept			
		Y = area of sample peak			
		X = concentration of sample (ng)			
	5.9.2	The calculations for determination of amount of CDAP per µg of			
	polysaccharide conjugate is as:				
		As the sample is diluted to a concentration of 1 µg/ 10 µL,			
		Therefore the CDAP content obtained for the sample shall be			
		considered in terms of ng/µg of polysaccharide.			
	5.9.3	Calculations for System Suitability Test:			
	5.9.2.1	Note the values of area under the curve and retention			
		time for standard 3(SST standard) in Annexure 1 and			
		also in EXCEL format for calculation of Relative			
		standard deviation.			
		Path for Master EXCEL format is			
		My Computer → D drive → SST Master EXCEL (folder) →			
		SST Master EXCEL QC/01/050 (File).			
	5.9.2.2	Every time a sample is run with system suitability			
		test, calculate the results for system suitability by			
		creating a copy of Master EXCEL in the desired folder			
		and record the same in Annexure 1.			
	5.9.2.3	Attach the hard print of SST calculation along with			
		chromatograms and the record form.			
	5.10	Acceptance Criteria :			
	5.10.1	RSD calculated for Retention time of standard 3(5ng) should be			
		< 5.0%.			
	5.10.2	RSD calculated for Area of standard 3(5ng) should be < 5.0%.			
	5.11	Result Reporting :			
	5.11.1	Open CDAP.FRP template and take print out of the report.			
	5.11.2	Print report: Click on print option to take a printout of a			
		ChemStation report using the current report template and data			
		from the currently loaded data file.			
	5.11.3	Report the results of quantity of CDAP present/µg.			
	5.11.4	The report generated can be viewed with report review icon.			
	5.11.5	The printed report is signed by the analyst who does the			
		analysis and also checked by the other analyst/supervisor.			
	5.11.6	The chromatogram and results for the sample and system			
		suitability test runs are attached along with the Annexure 1.			
	5.12	Preservation and storage of Column			
	5.12.1	Flush the column with 50mL of 100% Acetonitrile or 100			
		%Methanol at 1mL/min.			
	5.12.2	For long time storage, pass 100 mL of 100% Acetonitrile through			
		the column and screw the caps in both ends of the column.			

1.0	Objective:			
	To lay down the Procedure for detection anti-HBsAg antibodies using			
	Monolisa™ Anti HBsPLUS ELISA kit.			
2.0	Scope:			
	This Standard Operating Procedure is applicable to all the In- process and			
	finished product samples of IVIG.			
3.0	Responsibility			
	3.1	The QC analyst shall be responsible for executing the described		
		procedure correctly.		
	3.2	The QC supervisor shall be responsible for verifying that the		
		procedure is complied with and the results are checked.		
	3.3	The QC HOD shall be responsible for overall compliance of this SOP.		
4.0	Abbreviations and Definitions:			
	4.1	g	: Grams	
	4.2	mL	: Milliliter	
	4.3	mg	: Milligram	
	4.4	M	: Molarity	
	4.5	min	: Minute	
	4.6	µL	: Microliter	
	4.7	µg	: Microgram	
	4.8	°C	: Degree centigrade	
5.0	Procedure:			
	5.1	Equipment, Materials and Reagents		
		5.1.1	ELISA reader	
		5.1.2	Vortex mixer	
		5.1.3	Monolisa Anti-HBs Plus Kit [BioRad, 72566]	
		5.1.4	Purified water	
	5.2	Preparation of solutions:		
		5.2.1	Washing solution (1X)	
			5.2.1.1	S.No
				Volume of 20X washing solution (mL)
				Volume of Purified water (mL)
				Total Volume (mL)
				1
				5
				95
				100
			5.2.1.2	The diluted washing solution is stable for 2 weeks at 2
			-30 °C.	
	5.2.2	Substrate Buffer:		
		5.2.2.1	Dilute chromogen 1:11 using substrate buffer.	
		5.2.2.2	S.No	
			Volume of Neat Chromogen (µL)	
			Volume of Substrate buffer (mL)	
			Total Volume (mL)	
				1
				100
				1
				1.1
	5.3	Test method:		
		5.3.1	Reference solution preparation:	
			5.3.1.1	Reference standard is Anti-HBs antibodies of human with
				10 IU/L, 100 IU/L, 400 IU/L and 1000 IU/L Calibrator)
				in a preservative, are part of kit content.
			5.3.1.2	S.No
				Ref. Std Concentrations (IU/L)
				Amount of Calibrator (µL)
				Specimen Diluent (µL)
				1
				10
			75 µL of 10 IU/L	
			25	
				2
				*50
			5 µL of 1000 IU/L	
			95	
				3
			100	
			75 µL of 100 IU/L	
			25	
				4
				*200
			20 µL of 1000 IU/L	
			80	
				5
			400	
			75 µL of 400 IU/L	
			25	
				6
			1000	

				75 μ L of 1000 IU/L
				25
		5.3.2	Manufacturing pool plasma :	
		5.3.2.1	Required Dil. Folds	
				Vol. of Pool plasma Sample (μ L)
				Vol. of Specimen Diluent (μ L)
				Total volume (μ L)
				Neat
				NA
				NA
				NA
				2
				50 μ L of neat plasma sample
				50
				100
		5.3.3	Test solution Preparation	
		5.3.3.1	Conc. (mg/mL)	
				Required Dil. Folds
				Vol. of 5% Test Sample (μ L)
				Vol. of Specimen Diluent (μ L)
				Total volume
				(μ L)
				50
				5F (10 mg/mL)
				20
				80
				100
				10F (5 mg/mL)
				10
				90
				100
		5.3.4	Bring all reagents to room temperature (20-25°C) before	
		beginning the assay procedure.		
		5.3.6	Prepare conjugate working solution, working diluted substrate	
		solution and diluted washing solution.		
		5.3.7	Remove the microplate frame and strips from their protective	
		bag.		
		5.3.8	Add 25 μ L of Specimen diluent to each well first, followed by	
		75 μ L of Test sample or Calibrator or Control within 15		
		minutes. Then mix gently by a minimum of two aspirations to		
		avoid foaming.		
		Note: After adding the sample, the diluent will change from		
		purple to a blue colour.		
		5.3.9	Add directly the samples to the plate without prior washing of	
		the plate.		
		5.3.1	Cover the wells with adhesive film by pressing over the whole	
		0 surface to ensure tightness.		
		5.3.1	Incubate the plate for 60 $\pm$ 5 min at 37 $\pm$ 2°C.	
	1			
		5.3.1	Remove the adhesive film. Aspirate the contents of all wells	
		2 into a liquid waste container and add immediately a minimum of		
		300 μ L of washing solution to each well. Soak each well for 30		
		to 60 seconds between each wash cycle. Aspirate again. Repeat		
		the washing step 4 times (minimum of 5 washes).		
		Note: Blot the plate by turning it upside down on absorbent		
		paper.		
		5.3.1	Add quickly 100 μ L of the working diluted substrate solution to	
		3 each well. Allow the reaction to develop in the dark for 30 min		
		$\pm$ 5 min at room temperature.		
		Note: Do not use adhesive film during this incubation.		
		5.3.1	Add 100 μ L of stopping solution. Homogenize the reaction	
		4 mixture.		
		5.3.1	Carefully wipe the plate bottom. At least 4 min after stopping	
		5 solution addition and within 30 minutes of stopping the		
		reaction.		
		5.3.1	Read the optical density at 450/620 nm and 405/620 nm using a	
		6 plate reader.		
		5.4	Calculations :	
		5.4.1	Read the optical density at 450/620 nm using a plate reader.	
		5.4.2	For Absorbance values $\geq$ measured absorbance of 400 IU/L	
		calibrator, read the OD at 405/620 nm.		
		5.4.3	Plot the A450 of four calibrators C0, C1, C2 and C3 versus	
		their assigned concentrations, using a polynomial (quadratic)		
		regression.		
		Note:		
		The A450 of the 1000 IU/L Calibrator cannot be used in this		
		graph, as that absorbance value should be outside the range of		
		the spectrophotometer, hence the necessity of a second graph.		
		Samples with measured absorbance values less than 400 IU/L		
		calibrator are interpreted with the graph obtained with the		
		A450 of the four calibrators.		
		The A405 of 400 IU/L and 1000 IU/L calibrators are graphed		
		versus their assigned concentrations using point to point. A		
		straight line is drawn through the points.		

		5.4.4	Then the Anti-HBs concentration (IU/L) for each sample is read
		at the intersection of the respective absorbance values.	
		5.4.5	The A405 curve is used to determine the concentrations of
		samples whose are greater than 400 IU/L and less than or equal	
		to 1000 IU/L.	
		Note:	
		Samples with anti-HBs concentrations greater than 1000 IU/L can	
		be diluted using diluted washing solution and re-assayed.	
		5.5	Method Validation Criteria:
		5.5.1	The mean absorbance of the 10 IU/L calibrator (C1) is the
		cut-off value for the assay.	
		5.5.2	For Anti-HBs Negative Control, the measured absorbance value
		must be greater than 0.000 and less than or equal to 0.070.	
		5.5.3	For Positive Control, the measured absorbance value must be
		greater than 0.400.	
		Note: For Negative control and Positive control, if any of the	
		above criteria is not met, the assay is invalid and must be	
		repeated.	
		5.5.4	For 10 IU/L Calibrator, the measured absorbance value must be
		greater than or equal to 0.050 and less than or equal to 0.200.	
		Note: If one of the two ODs of 10 IU/L is outside the	
		acceptable range, the assay is invalid and must be repeated.	
		5.5.5	Each measured absorbance values must be greater than or equal
		to 1.5 the OD of the absorbance value of the Negative control.	
		6.0	References:
		6.1 Indian Pharmacopeia	

1.0 |Objective: |
 | |To lay down a procedure for estimating protein content in test samples by |
 | |Biuret method. |
 2.0 |Scope: |
 | |This Standard Operating Procedure is applicable to all the In- process and|
 | |finished product samples. |
 3.0 |Responsibility |
	3.1	The QC analyst shall be responsible for executing the described
		procedure correctly.
	3.2	The QC supervisor shall be responsible for verifying that the
		procedure is complied with and the results are checked.
	3.3	The QC HOD shall be responsible for overall compliance of this SOP.
4.0	Abbreviations and Definitions:	
	4.1	NaoH
	4.2	g
	4.3	mL
	4.4	mg
	4.5	M
	4.6	min
	4.7	µL
	4.8	µg
	4.9	°C
	4.10	%
5.0	Procedure:	
	5.1	Equipment, Materials and Reagents
		5.1.1
		5.1.2
		5.1.3
		5.1.4
		5.1.5
		5.1.6
		5.1.7
		5.1.8
		5.1.9
	5.2	Preparation of solutions:
		5.2.1
	5.2.2	Preparation of 6% NaOH Solution (1.5 N)
		5.2.2.
	5.2.3	Preparation of 9 g/L NaCl Solution (0.15 M)
		5.2.3.
	5.3	Method :
		5.3.1
	5.3.2	Add 2 mL of 9 g/L NaCl for blank tube.
	5.3.3	Add 2 mL of 6% NaOH (1.5 N) to all the tubes.
	5.3.4	Add 0.8 mL of Biuret reagent to each of these tubes.
		Note: After the addition of Biuret reagent, vortex the test tubes
		using vortex mixer for proper mixing.
	5.3.6	Note the timing of addition of Biuret reagent in the first
		tube.
	5.3.7	Allow all the tubes to stand at room temperature for 15
		minutes.
	5.3.8	After 15 minutes, determine the absorbance at 545 nm using
		the blank as compensation liquid.
	5.4	Interpretation of results:
 6. |References: |
 | |In-House |
 | |

1.0 |Objective |
 |To analyze purified and lyophilized polysaccharide samples of Streptococcus |
 |pneumoniae for the determination of any residual Deoxycholic acid by |
 |reversed phase liquid chromatography with RI detector (HPLC). |
 2.0 |Scope |
 |This Standard Operating Procedure is applicable for purified and |
 |lyophilized polysaccharide samples of Streptococcus pneumoniae in the QC. |
 3.0 |Responsibility |
3.1	The analyst shall be responsible for carrying out the described
	procedure correctly.
3.2	The Supervisor of the laboratory shall be responsible for verifying
	that the procedure is complied with and for checking the results.
3.3	The departmental head/designee shall be responsible for training of
	the staff and for overall compliance of the SOP.
3.4	Head QA/Designee shall be responsible for final approval and
	subsequent compliance of the SOP.
4.0	Abbreviations and Definitions
4.1	cm
4.2	g
4.3	HPLC
4.4	id
4.5	l
4.6	L
4.7	M
4.8	mAu
4.9	mg
4.10	min
4.11	mL
4.12	mm
4.13	nm
4.14	QA
4.15	QC
4.16	QCR
4.17	rpm
4.18	r2
4.19	RSD
4.20	SOP
4.21	SST
4.22	°C
4.23	µg
4.24	µL
4.25	µm
4.26	System Suitability test: System suitability tests are an integral
	part of gas and liquid chromatographic methods. They are used to
	verify that the resolution and reproducibility of the chromatographic
	system are adequate for the analysis to be done.
	The tests are based on the concept that the equipment, electronics,
	analytical operations and samples to be analyzed constitute an
	integral system that can be evaluated as such.
5.0	Procedure
5.1	Precautions to be taken:
	5.1.1
	5.1.2
	5.1.3
	5.1.4
	5.1.5
	5.1.6
5.2	Requirement:
	5.2.1
	5.2.2
	5.2.3
	5.2.4

||| 5.2.4.4 |Purified Water |

||| 5.2.4.5 |Sodium Acetate anhydrous |

||| 5.2.4.6 |Orthophosphoric Acid |

|| 5.3 |Preparation of solutions: |

|| |Prepare all the solutions as described below. Carry out the labelling|

|| |of the prepared solutions and assign the lot numbers to the |

|| |solutions. |

|| 5.3.1 |0.02M Sodium acetate Solution: To prepare 250mL of 0.02M sodium|

|| |acetate solution, weigh 410.2mg of sodium acetate anhydrous and|

|| |dissolve in 250mL of 0.2µm filtered purified water. |

|| 5.3.2 |1M Phosphoric Acid: To prepare 50mL of 1M phosphoric Acid, take|

|| |3.2mL of Orthophosphoric acid and dissolve in 46.8mL of 0.2µm |

|| |filtered purified water. |

|| 5.3.3 |Mobile phase for DCA |

|| |5.3.3.1 |To prepare 1 litre of mobile phase, add 375mL of |

|| |methanol and 375mL of acetonitrile to 250mL of 0.02M |

|| |sodium acetate solution. |

|| |5.3.3.2 |Adjust the pH to 4.3 using 1M Phosphoric acid. Filter |

|| |the mobile phase using 0.2µm nylon membrane filter and|

|| |degas the solution for 20 minutes using sonicator. The|

|| |mobile phase solution can be stored at room |

|| |temperature only for two days. |

|| 5.3.4 |3mg/mL DCA stock: |

|| |5.3.4.1 |Take 30mg of DCA in a 10mL volumetric flask and |

|| |dissolve in 7mL of mobile phase and make up the volume|

|| |to 10mL with mobile phase. |

|| |5.3.4.2 |Vortex the mixture until a clear solution is formed |

|| |and then filters it using 0.2µ inorganic membrane |

|| |filter. |

|| 5.3.5 |Working standard solutions: |

|| |5.3.5.1 |Prepare 2.0mL of 1200µg/mL standard solution by adding|

|| |800µL of 3mg/mL DCA stock in 1200µL of mobile phase. |

|| |5.3.5.2 |Make 5 concentrations of standard solutions (600µg/mL,|

|| |300µg/mL, 150µg/mL, 75µg/mL and 37.5µg/mL) as per |

|| |below table: |

S No	Standard	Volume of standard	Volume of Mobile
1	600µg/mL	750µL of 1200µg/mL	750µL
2	300µg/mL	750µL of 600µg/mL	750µL
3	150µg/mL	750µL of 300µg/mL	750µL
4	75µg/mL	750µL of 150µg/mL	750µL
5	37.5µg/mL	750µL of 75µg/mL	750µL

|| 5.3.6 |Preparation of Lyophilized polysaccharide sample for HPLC |

|| |analysis: |

|| 5.3.7 |Prepare 1mg/ml of polysaccharide sample in 0.2µm filtered |

|| |purified water. Keep the sample at 2-8°C for overnight on gel |

|| |rocker. |

|| |

|| 5.3.8 |All the solutions prepared for use are vacuum filtered using |

|| |Nylon 6, 6 membrane 0.20 µm/0.22 µm, depending upon the nature |

|| |of solution to be filtered. For organic solvents 0.45µm filters|

|| |should be used (filtration not necessary if the solution is |

|| |prepared in 0.2 µ filtered purified water). |

|| 5.4 |Preparing the HPLC System for run: |

|| 5.4.1 |Flow Rate: 1.0 mL/min |

|| 5.4.2 |Detection using: RID at 30°C |

|| 5.4.3 |Run time: 18 minutes |

|| 5.4.4 |Load method : DCA.M |

|| 5.5 |Equilibration of the system and the column: |

|| |Equilibrate the column and the system at 1mL/min flow rate of mobile |

|| |phase until the RID baseline is stabilized. |

|| 5.6 |Transfer the blank, test sample(s) and standards into screw cap vials|

|| |and place against the appropriate no. on the sample tray. |

|| 5.7 |Running of sequence |

|| 5.8.1 |Load sequence DCA.S. |

|| 5.8.2 |Open the sequence parameter option by clicking Sequence menu. |

|| 5.8.3 |Enter information about the data directory and operator name. |

|| 5.8.4 |In the shutdown option, click on the post-sequence option and |

|| |select LAMPOFF/STANDBY. |

|| 5.8.5 |In sequence table option, specify the sample names, methods and|

|| |vials to run in the sequence as per the following table: |

S No	Sample Code	Sample details	No. of Injections
1	Blank	Eluent Blank	1
2	DCA Standard	3	System Suitability
3	DCA Standard	1	37.5 µg/mL
4	DCA Standard	2	150 µg/mL
5	DCA Standard	2	75 µg/mL
6	DCA Standard	3	150 µg/mL
7	DCA Standard	4	300 µg/mL
8	DCA Standard	5	600 µg/mL
9	Blank	Eluent Blank	1
10	Test Sample (s)	Sample	2

|| 5.8.6 |Click on run sequence option in run control menu to start the |

|||sequence. |

||5.8.7|Each sample run will be of 18 min and after running all the |

|||samples the sequence will automatically stop. |

||5.8.8|Inject the eluent blank after every 20 injections of sample. |

|||Note: One extra vial with same SST sample can be used if 6 |

|||injections are not possible from single vial. |

||5.9|Calculations : |

||5.9.1|Results of the sample are calculated from the calibration |

|||standards using the EXCEL worksheet. |

||5.9.1.1|Area under curve and retention time for the |

|||standards/sample is analyzed by Chemstation software. |

||5.9.1.2|Calculations for Standard curve and samples |

|||The calculations are made according to the equation of |

|||minimum squares. $Y = mX + b$ |

|||Where: m = slope |

|||b = intercept |

|||Y = absorbance |

|||X = concentration |

||5.9.1.3|Use EXCEL program for the preparation of calibration |

|||curve and calculation of the results. |

||5.9.1.4|Note the values of area under the curve and retention |

|||time for standards to prepare standard curve in |

|||Annexure 1 and also in EXCEL format for calculation of |

|||standard curve. |

||5.9.1.5|Prepare standard curve by creating a copy of Master |

|||EXCEL in the desired folder and record. |

||5.9.1.6|The amount of DCA present in the sample is calculated |

|||by using the formula |

|||Concentration of DCA in sample (X) = $\frac{Y-b}{m}$ |

|||Where Y = peak area of DCA in test sample |

|||b = intercept |

|||m = slope |

||5.9.2|Calculations for System Suitability Test: |

||5.9.2.1|Note the values of area under the curve and retention |

|||time for Standard 3 (SST standard) in Annexure 1 and |

|||also in EXCEL format for calculation of Relative |

|||standard deviation. |

||5.9.2.2|Every time a sample is run with system suitability |

|||test, calculate the results for system suitability. |

||5.10|Acceptance Criteria : |

||5.10.1|For System Suitability Test : |

||5.10.1.1|RSD calculated for Retention time of Standard 3 should|

|||be <5.0%. |

||5.10.1.2|RSD calculated for Area OF Standard 3 should be <5.0%.|

||5.10.2|Parameters for Acceptance of the test: |

||The Regression of the standard curve should be > 0.99. |

||5.10.3|The residual DCA content should be ≤1% of the dry weight of |

|||purified and lyophilized sample of Pneumococcal polysaccharide.|

||5.11|Result Reporting : |

||5.11.1|Using the File menu new reports can be templates, open existing|

|||templates, save completed reports, or print reports using the |

|||current report template. |

||5.11.2|Open DCA.FRP template. |

||5.11.3|The report generated can be viewed with report review icon. |

||5.11.4|Print report: Click this option to take a printout of a |

|||Chemstation report using the current report template and data |

|||from the currently loaded data file. |

||5.11.5|The printed report is signed by the analyst who does the |

|||analysis and checked by the other analyst/supervisor. |

||5.11.6|Report the results of DCA content of the sample(s). |

||6.0|Forms and Records |

||Determination of Total Protein content - Annexure 1 |

||7.0|Reference |

||7.2|European Pharmacopoeia, Volume 5, Protein in Polysaccharide Vaccines |

||7.3|WHO TRS No. 977 Annexure 3: Recommendations to assure the quality, |

|||safety and efficacy of pneumococcal conjugate vaccines. |

1.0 |Objective: |
 |To lay down the Procedure for detection of Antibodies to HIV 1&2 by |
 |Serology. |
 2.0 |Scope: |
 |This Standard Operating Procedure is applicable to all the In- process and |
 |finished product samples of IVIG. |
 3.0 |Responsibility |
3.1	The QC analyst shall be responsible for executing the described
	procedure correctly.
3.2	The QC supervisor shall be responsible for verifying that the
	procedure is complied with and the results are checked.
3.3	The QC HOD shall be responsible for overall compliance of this SOP.
4.0	Abbreviations and Definitions:
4.1	g
4.2	mL
4.3	mg
4.4	M
4.5	min
4.6	μL
4.7	μg
4.8	°C
5.0	Procedure:
5.1	Equipment, Materials and Reagents
	5.1.1
	5.1.2
	5.1.3
5.2	Preparation of solutions:
	5.2.1
	5.2.2
	5.2.3
5.3	Test method:
	5.3.1
	5.3.2
	5.3.3
	5.3.4
	5.3.6
	5.3.7
	5.3.8
	5.3.9
	5.3.1
	5.3.1
	1

		3	the wash and soak procedure 4 additional times for a
			total of 5 washes. Avoid spillage
		5.3.11.	Tap upside down on absorbent sheet till no droplets
		4	appear on the sheet, taking care not to dislodge the
			wells
		5.3.11.	Immediately discard the tissue in a beaker that
		5	contains sodium hypochlorite and close the beaker with
			aluminium foil
	5.3.1	Add 100 µL of working conjugate solution in each well including	
	2	blank well	
	5.3.1	Carefully apply cover seal to the strips and cover with	
	3	Aluminium foil	
	5.3.1	Incubate at 37°C ± 2°C for 30 min ± 2 min	
	4		
	5.3.1	Carefully aspirate the well contents and wash as describe in	
	5	5.3.11	
	5.3.1	Add 100 µL of working substrate solution in each well including	
	6	blank well	
	5.3.1	Incubate at room temperature (20 - 30°C) for 30 min in dark.	
	7		
	5.3.1	Add 100 µL of stop solution	
	8		
	5.3.1	Read absorbance at 450 nm within 30 minutes in ELISA reader.	
	9	(Use of a reference filter of 620 nm is preferred)	
	5.4	Calculations :	
	5.4.1	Take average of the ODs of Negative control and Positive	
		Control separately	
	5.4.2	[pic]	
		[pic]	
	5.4.3	[pic]	
		[pic]	
	5.4.4	Then add the average values of both positive and negative	
		control and divide by 6 which give the cut off value.	
		[pic]	
	5.5	Method Validation Criteria:	
	5.5.1	Blank acceptance criteria"	
		The OD value at 450/620 nm for blank must be <0.100 in case of	
		differential filter being used. In case differential filter is	
		not available in the reader the blank value may go higher	
	5.5.2	Negative Control (NC) acceptance criteria:	
		The OD value at 450/620 nm for NC must be < 0.150. If it is not	
		so, the run is invalid and must be repeated.	
	5.5.3	Positive Control (PC) acceptance criteria:	
		5.5.3.1	The OD value at 450/620 nm for PC must be > 0.50
		5.5.3.2	Determine the mean value for PC triplicates, if one of
		three positive control values is outside of these	
		limits; recalculate mean PC based upon the two	
		acceptable positive control values.	
		5.5.3.3	Repeat the assay if two of the three positive control
		values are outside the limits, as the assay is invalid	
		Note:	
		Care should be taken while performing the experiment in	
		order to avoid infection from plasma sample or positive	
		control.	
		Wear nose mask and hand gloves throughout the	
		experiment. Do not reuse the gloves and masks; discard	
		immediately after completion of the experiment.	
		Discard the used tips in a beaker containing sodium	
		hypochlorite and close the beaker with aluminium foil.	
		Perform the experiment under laminar airflow unit.	
		Once the assay has started, complete the procedure	
		without interruption. All the reagents should be	
		dispensed in the centre of the well and the tip of the	
		pipette should not touch the wall of the microwell	
	6.0	References:	
	6.1	Indian Pharmacopeia	

1.0 |Objective |
 |To lay down the procedure for Estimation of the Nucleic acid content in |
 |Pneumococcal polysaccharide samples. |
 2.0 |Scope |
 |This Standard Operating Procedure is applicable to all the samples of |
 |Pneumococcal polysaccharide in which Nucleic acid is to be determined, |
 |which is received in the Quality control department. |
 3.0 |Responsibility |
3.1	The analyst is responsible for executing the described procedure
	correctly.
3.2	Supervisor is responsible for verifying that the procedure is carried
out correctly and check the results.	
3.3	HOD shall be responsible for overall compliance of this SOP and shall
ensure that the SOP is reviewed every two years or if required	
	earlier.
4.0	Abbreviations and Definitions
4.1	Abbreviations:
	4.1.1
	4.1.2
	4.1.3
	4.1.4
	4.1.5
	4.1.6
	4.1.7
	4.1.8
	4.1.9
	4.1.10
4.2	Reagents:
	4.2.1
4.3	Equipment
	4.3.1
5.0	Procedure
5.1	Preparation of solutions: Not Applicable
5.2	Testing procedure:
	5.2.1
5.3	Calculations:
	5.3.1
	50
	5.3.2
5.4	Validity of test:
	5.4.1
5.5	Acceptance criteria:
	5.5.1
	should meet the laid specifications for each serotype.
	5.5.2
	samples should be ≤ 5%.
6.0	Forms and Records
Determination of Nucleic acid content - Annexure 1	
6.0	Reference
6.1	European Pharmacopoeia, volume 5, Nucleic Acid in polysaccharide
	vaccines, 2.5.17.
6.2	WHO TRS No. 977 Annexure 3: Recommendations to assure the quality,
	safety and efficacy of pneumococcal conjugate vaccines

1.0	Objective			
	To lay down the procedure for identification of serotype specific purified			
	pneumococcal capsular polysaccharide by direct ELISA technique.			
2.0	Scope:			
	This Standard Operating Procedure is applicable for identification of			
	purified and lyophilized samples of pneumococcal capsular polysaccharide of			
	Streptococcus pneumoniae Type 1, 3, 4, 5, 6A, 6B, 7F, 9V, 14, 18C, 19A, 19F			
	and 23F by direct ELISA technique.			
3.0	Responsibility:			
	3.1	The analyst is responsible for executing the described procedure		
	correctly.			
	3.2	The supervisor of the laboratory shall be responsible for verifying		
	that the procedure is carried out correctly and for checking the			
	results.			
	3.3	The departmental Head shall be responsible for the overall compliance		
	of the SOP.			
4.0	Abbreviations and Definitions:			
	4.1	QA	: Quality Assurance	
	4.2	QC	: Quality Control	
	4.3	SOP	: Standard Operating Procedure	
	4.4	PBS	: Phosphate Buffer saline	
	4.5	ELISA	: Enzyme Linked Immunosorbent assay	
	4.6	CWPS	: Cell Wall Polysaccharide	
	4.7	BSA	: Bovine Serum Albumin	
	4.8	TMB	: Tetra Methyl Benzidine	
	4.9	ATCC	: American Type culture collection	
	4.10	HRP	: Horse reddish peroxidase	
	4.11	WFI	: Water for Injection	
	4.12	PBS	: Phosphate buffer saline	
5.0	Procedure			
	5.1	General Instructions:		
		Wear hand gloves while performing the assay.		
	5.2	Equipments and Materials:		
		5.2.1	96 well Maxisorp Nunc ELISA plates (Nunc Cat No. 439454)	
		5.2.2	Micro plate ELISA reader	
		5.2.3	Micro tips (2-20µl, 20-200µl, 100-1000µl and 50 -300µl)	
		5.2.4	Micro centrifuge tubes (1.5 ml)	
		5.2.5	Centrifuge tubes – 15 ml and 50 ml	
		5.2.6	Lint free paper towel	
		5.2.7	Vortex mixer	
	5.3	Reagents:		
		5.3.1	Pneumococcal rabbit antiserum for respective serotypes from	
			Staten Serum Institute (SSI)	
			Serotype	Cat No.
			Type 1	16744
			Type 3	16746
			Type 4	16747
			Type 5	16748
			Type 6A	16923
			Type 6B	16924
			Type 7F	16925
			Type 9V	16937
			Type 14	16753
			Type 18C	16958
			Type 19A	16963
			Type 19F	16962
			Type 23F	16968
		5.3.2	Polyclonal anti-rabbit IgG-HRP conjugated antibody (DAKO Cat	
			No. P0448).	
		5.3.3	Polyclonal anti-mouse IgG-HRP conjugated antibody (DAKO Cat No.	
			P0447).	
		5.3.4	Pneumococcal purified capsular polysaccharide antigen from ATCC	
			as a standard	
			for respective serotypes	
		Pneumococcal Serotype Antigen	ATCC Cat No.	
			Type 1	15-X
			Type 3	17-X
			Type 4	18-X
			Type 5	57-X
			Type 6A	14-X
			Type 6B	58-X
			Type 7F	27-X
			Type 9V	59-X
			Type 14	23-X
			Type 18C	28-X
			Type 19A	64-X
			Type 19F	24-X
			Type 23F	25-X
			Type 22F	528-X
		5.3.5	Purified Cell Wall Polysaccharide (CWPS) Staten Serum Institute	
			(SSI) Cat No.: 3459.	
		5.3.6	Bovine Serum Albumin (BSA)	
		5.3.7	Tween 20	
		5.3.8	Tetra Methyl Benzidine (TMB) substrate	
		5.3.9	Concentrated Sulphuric acid	
		5.3.10	Sodium chloride; NaCl	
		5.3.11	Sodium dihydrogen phosphate hepta hydrate; NaH2PO4·7H2O	
		5.3.12	Potassium Di hydrogen phosphate; KH2PO4	
		5.3.13	Potassium chloride; KCl	
		5.3.14	Milli- Q water	

	5.4	Preparation of solutions:	
		Prepare the solutions as described below. Label the solutions and	
		assign lot numbers.	
	5.4.1	Preparation of Standard Pneumococcal antigen: For preparation	
		of standard stock solution of pneumococcal antigens, add 2 ml	
		of WFI in 2 mg vial of standard and keep it overnight at 2-8°C	
		with shaking. Make aliquots of 100 µl. Store the aliquots at	
		-70°C for further use.	
	5.4.2	Phosphate buffer saline (pH 7.2 – 7.4) (10X)	
		5.4.2.1	Weigh 80.0 g NaCl, 20.61 g NaH₂PO₄·7 H₂O, 3.14 g of
		KH₂PO₄ and 1.60 g KCl, dissolve in 800ml of Milli Q	
		water.	
		5.4.2.2	Keep the beaker on a magnetic stirrer to dissolve the
		contents.	
		5.4.2.3	Make up the volume to 1L with milli Q water and
		sterilize the solution by filtering it through 0.22 µm	
		filter and store the solution at room temperature.	
	5.4.3	Wash buffer and coating buffer (1X PBS): Add 900 ml of sterile	
		Milli Q water to 100 ml of 10X PBS. Store the solution at room	
		temperature.	
	5.4.4	Wash / diluent and Antibody buffer (PBST): Add 50 µl of Tween	
		20 to 100 ml of 1X PBS solution (PBS with 0.05 % Tween 20). Mix	
		the solution thoroughly so that Tween 20 gets uniformly	
		dissolved in PBS.	
	5.4.5	Blocking buffer: (5% BSA in PBST): Add 5 g of BSA in 100 ml	
		PBST solution to give a final concentration of 5% BSA in PBST.	
	5.4.6	Secondary antibody: Working concentration of secondary antibody	
		is 1:2000 dilutions, prepared in diluent. The solution should	
		be freshly prepared just before the use.	
	5.4.7	Stop solution (1N H₂SO₄): Add 2.8 ml of Conc. H₂SO₄ to 97.2 ml	
		of Milli Q water to make up the volume to 100ml.	
	5.5	Procedure for ELISA:	
	5.5.1	Coating of plate:	
		5.5.1.1	Coat wells of ELISA plate with 100 µl/well of ATCC
		standard capsular polysaccharide antigen (1000 ng/ml of	
		dry weight) of the serotype under test as a positive	
		control on 26 wells as A1 to A7, B1 to B7, E1 to E6 and	
		F1 to F6. The dilutions of standard are to be prepared	
		in 1X PBS buffer.	
		5.5.1.2	Coat wells of ELISA plate with 100 µl/well of purified
		pneumococcal polysaccharide test sample (1000 ng/ml dry	
		weight) of serotype under test. Coat 26 wells as C1 to	
		C7, D1 to D7, G1 to G6 and H1 to H6. The dilutions of	
		standard are to be prepared in 1X PBS buffer.	
		5.5.1.3	Keep 4 wells uncoated as negative control (Blank) i.e.
		E7 to H7. Incubate the plates overnight at 37 ± 2°C.	
		as shown plate below.	
			1
			S1 is purified pneumococcal polysaccharide of Type 1
			serotype.
			NC is negative control (Uncoated wells)
	5.5.1.4	This pattern of coating of ATCC standard and test	
		sample shall be followed for each individual serotype	
		to be analyzed.	
	5.5.2	Preparation of Primary antibody:	
		Prepare adsorption solution: For studying cross reactivity of	
		type specific antisera, (specificity parameter). Prepare	
		adsorption solution by adding 22F polysaccharide (ATCC) and	
		purified CWPS (SSI) at 5µg/ml concentration to the antibody	
		diluent.	
		On next day prepare the dilutions of antibodies in adsorption	
		buffer solution in centrifuge tubes as per the dilution fold	
		mentioned below:	
		Serotype	Antibody dilution
		Type 1	1:60000
		Type 3	1:50000
		Type 4	1:50000
		Type 5	1:40000
		Type 6A	1:50000
		Type 6B	1:50000
		Type 7F	1:60000
		Type 9V	1:25000
		Type 14	1:50000
		Type 18C	1:60000
		Type 19A	1:40000
		Type 19F	1:40000
		Type 23F	1:15000
		Incubate the diluted primary antibody solution at 37 ± 2°C for	
		2 hrs.	
	5.5.3	Blocking of wells:	
		5.5.3.1	On next day remove the plate from incubator, discard
		the coating solution by a flick and add 300 µl/well of	
		blocking solution.	
		5.5.3.2	Incubate the plate at 37 ± 2°C for 1 hr. This step
		should be initiated after one hour of antibody	
		preparation.	
	5.5.4	1st Washing:	
		5.5.4.1	Discard the blocking solution after completion of one
		hour of incubation period by flicking.	
		5.5.4.2	Add 300 µl of PBST to each well and discard the buffer

			by flicking. Tap the plate on a lint free paper towel.	
			Ensure that the buffer is removed completely.	
			5.5.4.3	Repeat the procedure for two more times and then one
			time with 300 µl of PBS.	
		5.5.5	Addition of primary antibody:	
			5.5.5.1	The primary antibody dilutions for all the 13 serotype
			are prepared one hour before addition of blocking	
			buffer as mentioned in section 5.5.2.	
			5.5.5.2	Add 100 µl/well of primary antibody of all the 13
			serotypes as per pattern mentioned below.	
			Addition of Primary Antibody against Serotype	
			1	
		5.5.6	2nd washing:	
			5.5.6.1	Discard the primary antibody solution after completion
			of 90 mins.of incubation period by flicking.	
			5.5.6.2	Add 300 µl of PBST to each well and discard the buffer
			by flicking. Tap the plate on a lint free paper towel.	
			Ensure that the buffer is removed completely.	
			5.5.6.3	Repeat the procedure for three more times and then one
			time with 300 µl of PBS.	
		5.5.7	Addition of secondary antibody:	
			5.5.7.1	Prepare the recommended dilutions of secondary
			antibody.	
			5.5.7.2	For serotype 6B prepare 1:2000 dilution of anti-mouse
			IgG-HRP conjugated antibody. Prepare the dilutions in	
			adsorption solution.	
			5.5.7.3	For all other serotypes prepare 1:2000 dilution of
			anti-rabbit IgG-HRP conjugated antibody. Prepare the	
			dilutions in adsorption solution.	
			5.5.7.4	Add 100 µl of secondary antibody solution to
			respective wells and incubate the plate for 60 minutes	
			at 37 ± 2°C.	
		5.5.8	3rd washing:	
			5.5.8.1	Discard the secondary antibody solution after
			completion of 60 mins. of incubation period by	
			flicking.	
			5.5.8.2	Add 300 µl of PBST to each well and discard the buffer
			by flicking.	
			Tap the plate on a lint free paper towel. Ensure that	
			the buffer is removed completely.	
			5.5.8.3	Repeat the procedure for three more times and then one
			time with 300 µl of PBS.	
		5.5.9	Addition of TMB substrate:	
			5.5.9.1	Add 100 µl TMB to each well by a multichannel pipette.
			Take care that the TMB solution is not exposed to	
			direct light during addition.	
			5.5.9.2	Cover the plate and incubate at room temperature in
			dark preferably in a drover for next 15 mins.	
		5.5.10	Addition of Stop Solution:	
			5.5.10.1	On completion of 15 minutes incubation period of TMB.
			Carefully remove the plate out of drover. Take care	
			that the plate is not exposed to direct light during	
			addition.	
			5.5.10.2	Add 100 µl of stop solution over the TMB solution to
			each well in the dark.	
		5.5.11	Reading of plate and Data analysis:	
			5.5.11.1	Read the absorbance of plate at 450 and 630 nm
			wavelength on a ELISA reader. Calculate delta OD of	
			the readings.	
			5.5.11.2	Take care to take the readings after 5 mins and within
			15 mins. of addition of stop solution.	
			5.5.11.3	Get the average ODs of duplicate wells and compile the
			data sheets comprising of sample and ATCC standard.	
		5.5.12	Acceptance Criteria	
			5.5.12.1	OD values of Positive control and test sample should
			not be less than 0.8	
			5.5.12.2	OD values of Negative control should not be more than
			0.3	
6.0	Reference			
WHO TRS No. 977 Annexure 3: Recommendations to assure the quality, safety				
and efficacy of pneumococcal conjugate vaccines.				

1.0 |Purpose: |
 |To lay down the procedure for the estimation of residual Cyanogen Bromide |
 |(CNBr) by Spectrophotometry in Polyribosylribitol phosphate (PRP) |
 |Oligosaccharide conjugated to CRM. |
 2.0 |Applicable to: |
 |This Standard Operating Procedure is applicable for the analysis of the |
 |samples in which Cyanogen Bromide (CNBr) is to be determined. |
 3.0 |Abbreviations and Definitions |
3.1	CNBr	: Cyanogen Bromide
3.2	PRP	: Polyribosylribitol phosphate
3.3	CRM	: Cross Reacting Material
3.4	°C	: Degree centigrade
3.5	µg	: microgram
3.6	µL	: Micro litre
3.7	µm	: Micro meter
4.0	Procedure	
4.1	Precautions to be taken:	
4.1.1	Work in a fume hood.	
4.1.2	Wear gloves and safety goggles while handling DMBA.	
4.2	Requirements	
4.2.1	Equipment:	
4.2.1.1	Spectrophotometer	
4.2.1.2	Eppendorf tubes (1.5mL)	
4.2.1.3	Micropipette.	
4.2.1.4	Glass Cuvettes.	
4.2.1.5	Vortex.	
4.2.3	Materials:	
4.2.3.1	Autopipettes (10-100 (L), (20-200 (L), (100-1000 (L)	
4.2.3.2	Tips for the Autopipettes.	
4.2.3.3	Aluminium foil.	
4.2.3.4	0.22 µ Millipore syringe filters	
4.2.4	Reagents:	
4.2.4.1	Sodium Chloride (NaCl)	
4.2.4.2	Potassium Dihydrogen Orthophosphate (KH₂PO₄)	
4.2.4.3	Disodium Hydrogen Orthophosphate Dihydrate	
(Na₂HPO₄·2H₂O)		
4.2.4.4	Potassium Chloride (KCl)	
4.2.4.5	Sodium Hydroxide (NaOH)	
4.2.4.6	Dimethyl barbituric acid (DMBA)	
4.2.4.7	Cyanogen Bromide (CNBr) standard	
4.3	Preparation of solutions:	
Prepare all the solutions and carry out the labelling of the prepared		
solutions and assign the lot numbers to the solutions.		
4.3.1	Phosphate Buffer Saline (PBS), pH 7.4 :	
Dissolve the following in 800mL WFI,		
80g Sodium Chloride (NaCl)		
2.91g Potassium Dihydrogen Orthophosphate (KH₂PO₄),		
12.55g Disodium Hydrogen Orthophosphate Dihydrate (Na₂HPO₄		
2H₂O),		
1.6g Potassium Chloride (KCl).		
Adjust the pH to 7.4 with 10M Sodium Hydroxide (NaOH) and make		
up to 1000mL. Filter the buffer through a 0.22 µ Millipore		
filter.		
4.3.2	Dimethyl barbituric acid (DMBA) Reagent:	
Dissolve 0.19g of DMBA in 8mL Pyridine and 2mL WFI (use glass		
tube and glass pipette) use immediately.		
4.3.3	CNBr Stock Solution (1mg/mL):	
Dissolve 50mg of CNBr in 50mL PBS (freshly prepared).		
4.3.4	CNBr Working Solution (20µg/mL):	
Mix 200 µL CNBr Stock Solution with 9.8mL of PBS (freshly		
prepared).		
4.3.5	Preparation of standard dilutions: Prepare all standards	
dilutions from CNBr working solution in duplicate as		
mentioned below:		
CNBr Conc. (µg/mL)	CNBr Working	PBS (µL)
Standard (µL)		
Blank	0	1000
0.4	20	980
0.8	40	960
1.6	80	920
2.4	120	880
4.0	200	800
4.3.6	Spiking of samples with CNBr standard:	
Sample (µL)	CNBr Working	PBS (µL)
Standard (µL)		
200	0	800
200	80	720
4.3.7	Preparation of Test sample:	
4.3.7.1	Analyze all samples (with or without spike), the blank	
and standard in duplicate. Spike all samples where		
absorbance of CNBr has to be determined.		
4.3.7.2	Pipette out 200 µL of sample into each eppendorf	
tubes.		
4.3.7.4	Add 80 µL of CNBr working standard solution for spiked	
samples.		
4.3.7.5	Add 800 µL PBS into each tubes and mix (720 µL for	
spiked samples).		
4.4	Assay Procedure	
4.4.1	Add 100 µL of DMBA reagent to all tubes and mix well.	
4.4.2	Incubate for 45 minutes at room temperature.	

	4.4.3	Read the absorbance at 595 nm within 30 minutes.
	4.4.4	Auto zero is determined using a polystyrene cuvette filled
		with Blank.
	4.4.5	Calculate the concentration of CNBr in µg/mL and enter the
		results
	4.5	Calculations:
	4.5.1	Sample conc. in µg/mL = A mean sample OD – Intercept X
		Dilution fold
		Slope
	4.5.2	% Spike Recovery = Spike µg/mL – Sample µg/mL X 100
		1.6
	4.6	Acceptance Criteria:
	4.6.1	Spike Recovery should be between 70 – 140%.
	5.0	References:
	5.1	Wilchek, M., Miron, T., and kohn, J (1981) A Highly sensitive
		colorimetric method for the determination of carbodiimides. Anal,
		Biotchem. 114, 419-421.

1.0 |Objective: |
 |To lay down the procedure for potency assay using Chromogenic Method. |
 2.0 |Scope: |
 |This Standard Operating Procedure is applicable to estimation of Factor |
 |VIII activity in Human Coagulation Factor VIII, Manufacturing pool for |
 |fractionation samples by using Chromogenic Assay. |
 3.0 |Responsibility |
3.1	The QC analyst shall be responsible for executing the described
	procedure correctly.
3.2	The QC supervisor shall be responsible for verifying that the
	procedure is complied with and the results are checked.
3.3	The QC HOD shall be responsible for overall compliance of this SOP.
4.0	Abbreviations and Definitions:
4.1	TBE
4.2	g
4.3	mL
4.4	mg
4.5	M
4.6	min
4.7	µL
4.8	µg
4.9	°C
4.10	%
5.0	Procedure:
5.1	Equipment, Materials and Reagents
	5.1.1
	5.1.2
	5.1.3
	5.1.4
	5.1.5
	5.1.6
	5.1.7
	5.1.8
5.2	Preparation of solutions:
	5.2.1
	5.2.2
	5.2.3
	5.2.4
5.3	Method :
	5.3.1

			5.3.1.1	Pre dilution from 9.4 IU/mL to 1 IU/mL with F-VIII
			0	deficient plasma (9.4 Folds dilution)
				IRS activity (IU/mL)
				Neat RS (μL)
				F-VIII deficient (μL)
				Total Volume (μL)
				1
				20
				168
				188
			5.3.1.1	Step 2 Perform two 2-fold serial dilutions of RS
			1	from 1.0 IU/mL to 0.25 IU/mL using dilution buffer.
				IRS Activity (IU/mL)
				IRS (μL)
				Dilution Buffer (μL)
				Total Volume (μL)
				0.5
				100 μL of 1.0 IU/mL (9.4 times diluted)
				100
				200
				0.25
				100 μL of 0.5 IU/mL
				100
				200
			5.3.1.1	Internal Reference Standard (IRS) calibrated against
			2	WHO Reference Standard or Working Standard.
			5.3.1.1	Reconstitute IRS with 10 mL of sterile water for
			3	Injection and let the vial to stand at room
				temperature for 5 min to get a uniform solution.
			5.3.1.1	If IRS assigned activity is 25 IU/mL
			4	
			5.3.1.1	Dilute the IRS 2.5 times with Factor VIII deficient
			5	plasma to get 10 IU/mL.
			5.3.1.1	Step 1 of Pre dilution: 2.5 times Dilution: Pre
			6	dilution from 25 IU/mL to 10 IU/mL with F-VIII
				deficient plasma
				IRS Activity (IU/mL)
				IRS Neat (μL)
				F-VIII Deficient (μL)
				Total Volume (μL)
				10
				40
				60
				100
				Note: In case IRS activity is different from 25
				IU/mL, dilute accordingly to get a diluted
				preparation of 10 IU/mL
			5.3.1.1	Step 2 of Pre dilution: Pre dilution from 10 IU/mL
			7	to 1 IU/mL with F-VIII deficient plasma (10 times
				Dilution)
				IRS Activity (IU/mL)
				IRS from Step 1 (10 IU/mL) (μL)
				F-VIII Deficient (μL)
				Total Volume (μL)
				1
				30
				270
				300
			5.3.1.1	Step 3 Perform two 2-fold serial dilutions of IRS
			8	from 1.0 IU/mL to 0.25 IU/mL using dilution buffer.
				IRS Activity (IU/mL)
				IRS (μL)
				Dilution Buffer (μL)
				Total Volume (μL)
				0.5
				100 μL of 1.0 IU/mL
				100
				200
				0.25
				100 μL of 0.5 IU/mL
				100
				200
			5.3.2	Test Solution :
			5.3.2.1	Manufacturing pool for fractionation – Neat sample
			5.3.2.2	IPQC sample of Human Coagulation Factor VIII

```

||| |5.3.2.3|Pre dilution from 2 IU/mL to 1 IU/mL with F-VIII |
|||||deficient plasma |
|||||Test Sample Expected activity (IU/mL) |
|||||Test Sample Neat (μL) |
|||||F-VIII deficient (μL) |
|||||Total Volume (μL) |
|||||
|||||1 |
|||||50 |
|||||50 |
|||||100 |
|||||
|| |5.3.3 |Drug Substance - Clotate 250 IU |
|| |5.3.3.1|Step 1 of Pre dilution: 5 times Dilution |
||| |Pre dilution from 50 IU/mL to 10 IU/mL with F-VIII |
|||||deficient plasma |
|||||Test Sample Expected activity (IU/mL) |
|||||Neat Test Sample (μL) |
|||||F-VIII deficient (μL) |
|||||Total Volume (μL) |
|||||
|||||10 |
|||||20 |
|||||80 |
|||||100 |
|||||
|| |5.3.3.2|Step 2 of Pre dilution: 10 times Dilution |
||| |Pre dilution from 10 IU/mL to 1 IU/mL with F-VIII |
|||||deficient plasma |
|||||Test Sample Expected activity (IU/mL) |
|||||Test Sample from Step 1 (10 IU/mL(μL) |
|||||F-VIII deficient (μL) |
|||||Total Volume (μL) |
|||||
|||||1 |
|||||30 |
|||||270 |
|||||300 |
|||||
|| |5.3.3.3|Step 3: Perform two 2-fold serial dilutions of test |
||| |sample from 1.0 IU/mL to 0.25 IU/mL using dilution |
||| |buffer. |
|||||Test Sample Expected activity (IU/mL) |
|||||Test Sample (μL) |
|||||Dilution Buffer (μL) |
|||||Total Volume (μL) |
|||||
|||||0.5 |
||| |100 μL of 1.0 IU/mL |
|||||100 |
|||||200 |
|||||
|||||0.25 |
||| |100 μL of 0.5 IU/mL |
|||||100 |
|||||200 |
|||||
|| |5.3.4 |Drug Product - Clotate 250 IU/Vial/10 mL |
|| |5.3.4.1|Assigned Potency after reconstitution with 10 mL of |
|||||diluent: 25 IU/mL |
|||||Dilute the above test sample 2.5 times with dilution|
|||||buffer to get 10 IU/mL. |
|| |5.3.4.2|Step 1 of Pre dilution: 2.5 times Dilution |
||| |Pre dilution from 25 IU/mL to 10 IU/mL with F-VIII |
|||||deficient plasma |
|||||Test Sample Expected activity (IU/mL) |
|||||Neat Test Sample (μL) |
|||||F-VIII deficient (μL) |
|||||Total Volume (μL) |
|||||
|||||10 |
|||||40 |
|||||60 |
|||||100 |
|||||
|| |5.3.4.3|Step 2 of Pre dilution: 10 times Dilution |
||| |Pre dilution from 10 IU/mL to 1 IU/mL with F-VIII |
|||||deficient plasma |
|||||Test Sample Expected activity (IU/mL) |
|||||Test Sample from Step 1 (10 IU/mL(μL) |
|||||F-VIII deficient (μL) |
|||||Total Volume (μL) |
|||||
|||||1 |
|||||30 |
|||||270 |
|||||300 |
|||||
|| |5.3.4.4|Step 3: Perform two 2-fold serial dilutions of test |
||| |sample from 1.0 IU/mL to 0.25 IU/mL using dilution |
||| |buffer. |
|||||Test Sample Expected activity (IU/mL) |

```

Test Sample (μL)
Dilution Buffer (μL)
Total Volume (μL)
0.5
100 μL of 1.0 IU/mL
100
200
0.25
100 μL of 0.5 IU/mL
100
200
5.3.5 Assay dilutions: Assay dilutions of Factor VIII WHO / IRS /
Test Sample.
Activity (IU/mL)
Standard/Test Sample taken (μL)
Working buffer (μL)
1.42
25 μL of 1 IU/mL
1400
1.0
25 μL of 1 IU/mL
2000
0.5
25 μL of 0.5 IU/mL
2000
0.25
25 μL of 0.25 IU/mL
2000
5.3.6 Test Procedure:
5.3.6.1 Add 50 μL of Assay diluted samples / Controls /
Standards and one well for Buffer working solution as
blank to the wells of the microtitre plate.
5.3.6.2 Pre-warm the Factor reagent and Substrate reagent for
3 mins 30 secs using an incubator set at 37°C.
5.3.6.3 Simultaneously incubate the microtitre plate
containing the samples at 37°C for 3 mins 30 secs
inside the ELISA reader chamber set at 37°C.
5.3.6.4 Add 50 μL of 37°C pre-warmed Factor reagent.
5.3.6.5 Incubate at 37°C for 2 min inside the ELISA reader
chamber
5.3.6.6 Add 50 μL of 37°C pre-warmed Substrate.
5.3.6.7 Incubate at 37°C for 2 min inside the ELISA reader
chamber
5.3.6.8 Add 50 μL of 20% Acetic acid to all the wells
5.3.6.9 Read absorbance at 405 nm using the ELISA reader
5.3.7 The above described key steps starting from dispensing of
reagents in microtitre plate is detailed in the table below.
S.No
Ref.Std/ Test Sample activity (IU/mL)
Volume of diluted Ref.Std / Test Sample (μL)
Incubate at 37 °C for 3 Min. 30 Sec
Pre warmed Factor Reagent (μL)
Incubate at 37 °C for 2 Min
Pre warmed Substrate (μL)
Incubate at 37 °C for 2 Min
20% Acetic Acid - (μL)
OD 405 nm
1
1.42
50
50
50
50
50
2
1.0
50
50
50
50
50
3
0.5
50

			50
			50
			50
			4
			0.25
			50
			50
			50
			50
			50
			50
			50
		5.3.8	Calculation and analysis
		5.3.8.1	For calculations of potency using either graph pad
			prism or Combistats software.
6.	References:		
	IP General Chapter 2.8.7		

1.0	Objective:		
	Test for Description provides guidance to determine compliance with		
	requirements of individual monograph specifications.		
2.0	Scope:		
	This Standard Operating Procedure is applicable to all the In- process		
	and finished product samples.		
3.0	Responsibility		
	3.1	The QC analyst shall be responsible for executing the described	
	procedure correctly.		
	3.2	The QC supervisor shall be responsible for verifying that the	
	procedure is complied with and the results are checked.		
	3.3	The QC HOD shall be responsible for overall compliance of this SOP.	
4.0	Abbreviations and Definitions:		
	Not applicable		
5.0	Procedure:		
	5.1	Principle	
		5.1.1	Description is a general requirement given for information in
		most of the monographs. The property is not standard for	
		purity even though it may directly assist in the preliminary	
		evaluation.	
	5.2	Color:	
		5.2.1	Liquid Injectables:
			The color of the liquid biopharmaceutical product may be
		checked by taking the sample in a clean stoppered test	
		tube/vial. White color may be checked against a black	
		background and other colors may be checked against white	
		background.	
		5.2.2	Lyophilized Vials:
			Take out the contents from a vial on a white butter paper and
		observe the nature and color of the powder. Reconstitute the	
		injection in few vials by injecting a quantity of water for	
		injection as specified for the product in the pharmacopoeia	
		or in the directions for use. Observe and record the	
		solubility, color and nature of solution or suspension.	
6.	References:		
	In-house		

1.0 |Objective |
 |To lay down the procedure for determination of residual Ethyl dimethyl |
 |Aminopropyl carbodiimide (EDAC). |
 2.0 |Scope |
 |This Standard Operating Procedure is applicable to all the samples, in |
 |which the residual Ethyl dimethyl Aminopropyl carbodiimide (EDAC) content |
 |is to be determined, which is received in the QC. |
 3.0 |Responsibility |
3.1	The analyst shall be responsible for carrying out the described
	procedure correctly.
3.2	The Supervisor of the laboratory shall be responsible for verifying
	that the procedure is complied with and for checking the results.
3.3	The departmental head/designee shall be responsible for overall
	compliance of the SOP.
3.4	Head QA/Designee shall be responsible for final approval and
	subsequent compliance of the SOP.
4.0	Abbreviations and Definitions
4.1	g
4.2	HOD
4.3	M
4.4	ml
4.5	nm
4.6	PBS
4.7	QA
4.8	QC
4.9	SOP
4.10	Std.
4.11	µg
4.12	µl
4.13	µm
4.14	EDC
4.15	MES
4.16	DMBA
5.0	Procedure
5.1	Precautions to be taken:
	5.1.1
	5.1.2
	5.1.3
	5.1.4
5.2	Requirement:
	5.2.1
	5.2.2
5.3	Preparation of solutions:
	Prepare all the solutions as described below. Carry out the labelling
	of the prepared solutions and assign the lot numbers.
	5.3.1
	5.3.2
	5.3.3
	5.3.4

			Note: Use glass tube and glass pipettes for preparation.		
		5.3.5	Phosphate Buffer (0.2M NaCl, 10mM Phosphate Buffer pH 6.0):		
			Take 1.17 g NaCl, 0.054 g Na₂HPO₄·12H₂O and 0.133 g		
			NaH₂PO₄·2H₂O. Dissolve in 80 mL of MQW, adjust the pH with HCL		
			or NaOH and make up to 100mL. Filter through 0.22 µm filter.		
			This solution can be stored up to one month at room		
			temperature.		
		5.4	Test Procedure:		
		5.4.1	Preparation of Test Sample:		
			Dilute the test sample with phosphate buffer to 100 µg/mL of		
			polysaccharide based on its initial concentration and use 100µL		
			for the analysis.		
		5.4.2	Preparation of blank and positive control (1 µg/mL of EDC)		
			Prepare blank and positive control as described in the below		
			table.		
			EDC conc. (µg/mL)	EDC working stnd.	MES (µL)
			(µL)		
			Blank	0	100
			Positive control (1	5	95
			µg/mL)		
		5.4.3	Pipette 100 µL of the blank, the positive control and test		
			sample in duplicate into separately labeled 1.5 mL Microfuge		
			tubes.		
		5.4.4	Add 900µL DMBA reagent to all tubes using a glass pipette.		
		5.4.5	Close the tubes and vortex. Incubate them for 45 minutes at		
			room temperature.		
		5.4.6	Measure the absorbance at 595 nm (within 30 minutes after		
			incubation).		
		5.4.7	Record the results in record sheets		
		5.5	Acceptance criteria:		
			As per the specification the residual EDAC content should be less		
			than < 10 ng / µg Polysaccharides. Therefore the acceptance criterion		
			of the assay is "The absorbance value of the test sample containing		
			100 µg/mL of Polysaccharide should be less than the positive control		
			value (1 µg/mL of EDAC)".		
		6.0	Reference		
			Wilchek, M., Miron, T., and Kohn, J. (1981) A Highly sensitive		
			colorimetric method for the determination of carbodiimides. Anal.		
			Biochem.114, 419-421.		
			Giles, M.A., Hudson, A.Q., and Borders, C. L. Jr. (1990) Stability of		
			water-soluble carbodiimides in aqueous solution. Anal. Biochem. 184,		
			224-248.		

1.0 |Objective |
 |To lay down the procedure for quantifying adipic acid dihydrazide. |
 2.0 |Scope |
 |This Standard Operating Procedure is applicable for the analysis of all the |
 |samples of PRP-CRM bulk conjugate of Haemophilus Influenzae type b in which |
 |ADH concentration is to be determined.. |
 3.0 |Abbreviations and Definitions |
3.1	Abbreviations:		
	3.1.1	ADH	:Adipic acid dihydrazide
	3.1.2	CV	:Coefficient of variation
	3.1.3	g	:Gram
	3.1.4	mg	:Milligram
	3.1.5	mL	:Millilitre
	3.1.6	μL	:Micro litre
	3.1.7	μg	:Microgram
	3.1.8	nm	:Nanometer
	3.1.9	QA	:Quality Assurance
	3.1.10	QC	:Quality Control
	3.1.11	QCR	:Quality Control Record
	3.1.12	SD	:Standard Deviation
	3.1.13	SOP	:Standard Operating Procedure
	3.1.14	PRP	:Polyribosyl Ribitol Phosphate
	3.1.15	TT	:Tetanus Toxoid
	3.1.16	w/v	:Weight by Volume
3.2	Definitions:		
	Not Applicable.		
4.0	Procedure:		
4.1	Preparation of solutions:		
	Prepare all the solutions as described below and fill the form No.		
	QCR/166/A. Label the solutions and assign lot numbers as per SOP		
	QC/166.		
	4.1.1	6 M Sodium hydroxide:	
		4.1.1.1	Weigh 24 g sodium hydroxide and transfer it to 100 mL
			volumetric flask.
		4.1.1.2	Dissolve and make up the volume to 100 mL with purified
			water
		4.1.1.3	Store the solution in a screw cap bottle, for a maximum
			of 6 months at room temperature.
	4.1.2	0.2 M borate buffer,pH 8.7:	
		4.1.2.1	Weigh 6.2 g of boric acid and transfer it to 500 mL
			volumetric flask.
			Dissolve in 450 mL purified water and adjust to pH 8.7
			with 6 M sodium hydroxide.
		4.1.2.2	Make up the volume to 500 mL with purified water.
		4.1.2.3	Store the solution in a screw cap amber bottle at room
			temperature, for a maximum of 6 months.
	4.1.3	0.1% w/v TNBS-10% DMF:	
		4.1.3.1	Take 1 mL of commercial TNBS solution (1% w/v in DMF)
			and transfer it to a 10 mL volumetric flask.
		4.1.3.2	Make up the volume to 10 mL with purified water.
		4.1.3.3	Prepare fresh prior to use.
	4.1.4	10% w/v Sodium dodecyl sulphate (SDS):	
		4.1.4.1	Weigh 10 g of SDS and transfer it to 100 mL volumetric
			flask.
		4.1.4.2	Dissolve in purified water and make up the volume to
			100 mL.
		4.1.4.3	Store the solution in a screw cap amber bottle at room
			temperature, for a maximum of 6 months.
	4.1.5	1M Hydrochloric acid:	
		4.1.5.1	Slowly add 42 mL concentrated hydrochloric acid to 300
			mL purified water in a 500 mL volumetric flask.
		4.1.5.2	Make up the volume to 500 mL with purified water.
		4.1.5.3	Store the solution in a screw cap amber bottle at room
			temperature, for maximum of 6 months.
	4.1.6	1 mg/mL Adipic acid dihydrazide solution:	
		4.1.6.1	Take 25 mg ADH and transfer it to 25 mL volumetric
			flask.
		4.1.6.2	Dissolve in purified water and make up the final volume
			to 25 mL with purified water.
		4.1.6.3	Prepare fresh prior to use.
	4.1.7	10 μg/mL Adipic acid dihydrazide solution:	
		4.1.7.1	Take 1 mL of 1 mg/mL Adipic acid dihydrazide solution
			and transfer it to a 100 mL volumetric flask.
		4.1.7.2	Make up the volume to 100 mL with purified water.
		4.1.7.3	Prepare fresh prior to use.
	4.2	Set the water bath to 37°C.	
4.3	Preparation of Calibration Curve:		
	4.3.1	Prepare calibration curve (1–10μg/mL) with freshly prepared	
		ADH solution (10μg /mL) in duplicate in previously labelled	
		test tubes as per following table:	
		ADH (μg/ml)	
		1	
		2	
		4	
		6	
		8	
		10	
		10 μg/ml ADH(μL)	
		100	

			200
			400
			600
			800
			1000
			Purified water (µL)
			900
			800
			600
			400
			200
			-
		Note: The Calibration curve is accepted if it is meeting the	
		parameters of correlation coefficient i.e. ≥ 0.990 .	
		4.4	Test procedure:
		4.4.1	Take 1 ml of sample in duplicate in 10 mL thermoresistant test
		tubes after centrifugation.	
		Note: The sample dilution can vary depending upon the ADH	
		concentration. If there is any sample outside the range of the	
		calibration curve, dilute the sample appropriately.	
		4.4.2	Treat 1000 µL purified water as blank in thermoresistant test
		tube.	
		4.4.3	Add 500µL of 0.2 M borate buffer to tubes of calibration
		curve, blank and sample.	
		4.4.4	Vortex and add 500 µL of 0.1% w/v TNBS-10% DMF solution to all
		test tubes and again vortex gently.	
		4.4.5	Incubate for 30 minutes at 37°C in water bath.
		4.4.6	Allow the tubes to cool at room temperature, add 500µL 10% w/v
		SDS to each tube and vortex.	
		4.4.7	Add 1 mL of 1 M hydrochloric acid to each tube and vortex
		gently.	
		4.4.8	Allow the tubes stand at room temperature for 5 minutes and
		set the wavelength in spectrophotometer at 320 nm, as per	
		procedure described in SOP OE/058, OE/274. Zero the absorbance	
		of blank and then measure the absorbance of calibration curve	
		and test samples.	
		Note: The colour is stable for only a limited time, and absorbance	
		measurements should be made as soon as possible after colour	
		formation.	
		4.5	Calculations:
		4.5.1	Once the absorbance of the standards and the test samples have
		been taken, calculation shall be done using the excel sheet	
		carrying formulas for the required calculation.	
		4.5.2	Always copy the master excel sheet (sheet carrying formulas
		for the required calculation), never use the previously used	
		excel sheet for new calculations.	
		Master excel sheet can be assessed by following the below	
		mentioned path-	
		MY COMPUTER→ D DRIVE → MASTER EXCEL SHEETS → Master excel	
		sheet QC.01.19- Rev.04.	
		4.5.3	Copy the master excel sheet, keep it for use in respective
		folder and then fill in the required sample details, sample	
		dilution, name of analyst, date, etc.	
		4.5.4	Enter the absorbance for calculations starting from the
		Calibration curve prepared between 1 and 10 µg/mL of ADH and	
		test samples. The calculations are made according to the	
		equation of minimum squares.	
		$Y = mX + b$.	
		Where: m = slope	
		b = intercept	
		Y = absorbance	
		X = concentration	
		(Multiply by dilution factor)	
		4.5.5	Record all the calculations in the data record form and also
		attach the hard copy of the excel sheet used for calculation	
		along it.	
		4.6	Acceptance criteria:
		4.6.1	The coefficient of variation (CV) between the replicates of
		the samples should be $\leq 5\%$.	
		4.6.2	The ADH content in the sample should be within the specified
		limits.	
		Note: In case of any deviation follow the SOP QA/019 and in case some	
		parameters are not met, then proceed as per SOP QA/059.	
		5.0	References
		5.1	SOP OE/058 Operation of UV-Visible Spectrophotometer
		5.2	SOP QC/049 Procedure for Operation , Verification and Calibration of
		Autopipette	
		5.3	SOP OE/054 Procedure for Operation of Analytical Balance
		5.4	SOP OE/144 Operation and Calibration of pH and Conductivity Meter

1.0 |Objective: |
 |To lay down a procedure for determination of degree of activation for |
 |Meningococcal polysaccharide samples |
 2.0 |Scope: |
 |This SOP covers the procedure involved in testing of native and |
 |activated Meningococcal polysaccharide before conjugation |
 3.0 |Responsibility: |
 |3.1 |The analyst of the Immunochemistry laboratory shall be |
 |responsible for executing the described procedure correctly. |
 |3.2 |The supervisor of the laboratory shall be responsible for |
 |verifying that the procedure is followed correctly and for |
 |checking the results. |
 |3.3 |The Departmental Head shall be responsible for the overall |
 |compliance of the SOP and shall ensure that the SOP is reviewed |
 |every two years or if required earlier. |
 4.0 |Abbreviations and Definitions: |
4.1	QA	:Quality Assurance
4.2	QC	:Quality Control
4.3	SOP	:Standard Operating Procedure
5.0	Procedure:	
5.1	Materials, reagents & Equipments:	
5.1.1	BCA Assay reagent A: solution-containing Na₂CO₃, NaHCO₃,	
BCA detection reagent and sodium tart rate, in 0.2 M NaOH		
5.1.2	BCA Assay reagent B: 4% CuSO₄ (Sigma nr. C 2284 or Pierce	
nr. 23224)		
5.1.3	Milli Q water	
5.1.4	Glucose	
5.1.5	Glass test tubes (approximately 12 x 75 mm) with	
disposable plastic caps		
5.1.6	Pipettes and tips	
5.1.7	Analytical balance	
5.1.8	Laboratory vortex mixer	
5.1.9	Spectrophotometer	
5.1.10	Water bath with water circulation	
5.1.11	Cuvettes: 1 cm path length. Disposable polystyrene	
Cuvettes are recommended		
5.2	Preparation Of Solutions:	
5.2.1	Reagent A (commercial solution containing Na₂CO₃, NaHCO₃,	
BCA detection reagent, and sodium tartrate, in 0.2 M		
[NaOH]: total volume is 1000 mL. Store at RT (expiry 1		
year).		
5.2.2	Reagent B (commercial 4 % CuSO₄ solution): total volume is	
[25 mL. Store at RT (expiry 1 year).		
5.2.3	Reagent C: mix 1 mL of reagent B with 50 mL of reagent A.	
Prepare fresh		
5.2.4	Standard solution (0.4mM): Dissolve 0.0720 g of Glucose in	
[1000 ml MQW. Aliquot in small portion (e.g. 2.0 ml), and		
[store at ~-20°C (expiry 1 years).		
6.0	Procedure:	
6.1	Principle: Polysaccharides containing substituted or unsubstituted	
glycols in residues such as Glycerol, ribitol, arabinitol,		
furanosyl galactose, O-acetyl groups and sialic acid react with		
sodium periodate, and form an aldehyde, which further under goes		
redox reaction with oxidizing agents such as Cu⁺⁺ in presence of		
BCA and gives purple colour which is detected at 565 nm.		
6.2	Preparation for assay Standards:	
Prepare a series of standard solutions from 0.4 mM glucose		
standard solution into suitable glass tubes in duplicate according		
to the following table.		
Glucose	Volume of 0.4mM	Volume of MQW
Standard	glucose standard	(μl)
(μl)		
Blank	0	400
S1	25	375
S2	50	350
S3	100	300
S4	200	200
S5	400	0
6.2.1	Dilute the samples to desired concentration to fit in	
curve.		
6.2.2	Take appropriate aliquot of samples into the 5 ml glass	
tubes in duplicate and complete volume to 400 μl with MQW.		
6.2.3	Add 1 ml of BCA reagent to all the tubes and vortex	
gently. Cap the tubes.		
6.2.4	Incubate for 60 minutes at 60°C in a water bath.	
6.2.5	Cool the tubes to RT and read the absorbance at 550 nm	
6.3	Calculation:	
6.3.1	Data analysis can be done manually (calculator) or by	
[using computer and appropriate software (e.g. MS Excel).		
6.3.2	Construct the standard curve and calculate the aldehyde	
[concentration (mM) of the samples from their absorbance		
[and the standard curve (one mole glucose contains one mole		
aldehyde).		
[Aldehyde conc. (mM) = {(Mean absorbance –		
[Intercept)/Slope} = X		
6.3.3	Calculate the aldehyde concentration in the samples in	
[mM/ml taking into account the volume sample solution used		
[and dilution factor using formula.		
[Aldehyde conc. (mM/ml) = (X) x Diln.Factor x 1000/Volume		
[sample = Y		

	6.3.4	Calculate the MAPS concentration in the sample in mM using	
			formula.
			MAPS conc. (mM) = $m/340 \times 1000 = Z$
			Where m is MAPS concentration in mg/ml measured by
			Phosphorous assay, Z in mM assuming 340 Da for molecular
			weight of monomeric repeating unit of MAPS.
	6.3.5	Calculate the degree of activation in the sample by using	
			formula.
			Degree of activation (DA) = Y/Z
			Where Y is aldehyde concentration in mM measured by the
			BCA assay.
	6.4	Acceptance Criteria:	
		The correlation of calibration curve must be > 0.990	
	7.0	Reporting:	
	7.1	Record all the results in the data sheet.	
	7.2	Attach the print out of the spectrophotometer and the applicable	
		calculation sheet using MS Excel or any other software.	
	7.3	The report is submitted to QC in-charge for verification.	
	7.4	Report to QC in-charge in case of any problems such as out of	
		linearity and proceed as per instructions.	
	8.0	References:	
	8.1	SOP for Phosphorus contents of Meningococcal Polysaccharide	
	8.2	SOP for UV Spectrophotometer.	

1.0 |Objective: |
 |To lay down the procedure for determination of free polysaccharide |
 |content in Meningococcal polysaccharide A (PsA). |
 2.0 |Scope: |
 |This SOP is applicable to Men A polysaccharide in conjugates (bulk & |
 |final lots) and stability samples |
 3.0 |Responsibility: |
 |3.1 |The analyst of the Immunochemistry laboratory shall be |
 |responsible for executing the described procedure correctly. |
 |3.2 |The supervisor of the laboratory shall be responsible for |
 |verifying that the procedure is followed correctly and for |
 |checking the results. |
 |3.3 |The Departmental Head shall be responsible for the overall |
 |compliance of the SOP and shall ensure that the SOP is reviewed |
 |every two years or if required earlier. |
 4.0 |Abbreviations and Definitions: |
4.1	QA	Quality Assurance
4.2	QC	Quality Control
4.3	SOP	Standard Operating Procedure
5.0	Procedure:	
5.1	Materials, reagents & Equipments:	
5.1.1	Milli Q Water	
5.1.2	Laboratory vortex mixer (e.g. Spinex)	
5.1.3	Furnace (Barnstead thermolyne 6000 furnace)	
5.1.4	Spectrophotometer	
5.1.5	Water bath, 37°C	
5.1.6	Sonicator	
5.1.7	Centrifuge	
5.1.8	Glass test tubes (12 x 75 mm	
5.1.9	Cuvettes (1 cm path length)	
5.1.10	Pipettes and tips (e.g. Eppendorf)	
5.1.11	Phosphorus standard, 20µg/ml: KH₂PO₄ MW 137.66, ≥ 99.0%	
p.a. (e.g. Sigma P3869)		
5.1.12	Sulphuric Acid H₂SO₄ M.W.98.08 ≥ 99.0% p.a. (Fluka 84727)	
5.1.13	Perchloric acid HClO₄ M.W. 100.48 ≥ 70.0% p.a. (Aldrich	
24,425-2)		
5.1.14	Deoxycholic acid sodium salt C₂₄H₃₉O₄Na M.W. 414.2 >97%	
p.a. (Sigma D-6750) Harmful.		
5.1.15	Ascorbic acid sodium salt: C₆H₇NaO₆ MW 198.11 p.a.	
(e.g. Fluka 11140) Light and air sensitive.		
5.2	Preparation Of Solutions:	
5.2.1	Reagent A (Sulphuric acid +Perchloric acid): 2 volumes of	
Sulphuric acid and 1 volumes of Perchloric acid. Prepare		
fresh.		
5.2.2	Reagent B (1N Hydrochloric acid): Slowly add 21.5 ml	
concentrated HCl to 200 ml Milli Q water to final volume		
of 250 ml. Stored at room temperature (expiry 6 months).		
5.2.3	Reagent C (1 % DOC): Dissolve 250 mg of DOC in 25 ml of	
Milli Q water. Adjust pH to 6.8 with reagent B. prepare		
fresh.		
5.2.4	Reagent D (3 M Sulphuric acid): Take 180 ml of Milli Q	
water in volumetric flask, to this adds 40.5 ml of conc.		
Sulphuric acid slowly. Make up volume to 250 ml with Milli		
Q water. Store at room temperature. (Expiry 6 months).		
5.2.5	Reagent E (2.5% Ammonium Molybdate): Dissolve 5.0 g of	
ammonium molybdate in 200 ml of Milli Q water. Store at		
4°C. Expiry 6 month		
5.2.6	Reagent F (10 % Ascorbic acid): Dissolve 10 g of ascorbic	
acid in 80 ml of milli Q water. Make up final volume to		
100 ml with milli Q water. Store at 4°C (expiry 1 month).		
5.2.7	Reagent G: Mix 2 volume of Milli Q water, 1 volume of	
reagent D, 1 volume of reagent E, 1 volume of reagent F.		
prepare fresh.		
5.2.8	Standard solution: (2.0µg/ml): Dilute 10ml of 20µg/ml	
phosphorus standard (Sigma) with 90 ml of MQW. Aliquot in		
eppendorf tubes in small portion and store at ~-20°C		
(expiry 1 years).		
6.0	Procedure:	
6.1	Principle: The polysaccharide-protein conjugate is precipitated by	
using Deoxycholic acid (DOC) and 1N HCl. After centrifugation the		
unbound free polysaccharide remains in supernatant that is removed		
and tested by phosphorus assay.		
Phosphorus assay requires a preliminary ashing step, in order to		
release all organically bound phosphate. The conventional method		
involve drying and burning of organic sample over Bunsen burner,		
in presence of perchloric acid and Sulphuric acid. A high		
temperature furnace is now used instead. The characteristic blue		
color, which is formed from reduction of phosphomolybdo complex by		
ascorbic acid, is detected using a spectrophotometer at 825 nm.		
The method is sensitive and gives reproducible results.		
6.2	Preparation for assay Standards:	
Add standard solutions into glass test tubes in duplicate		
according to the following table.		
S.No.	Volume of	Volume of
standard (µl)	MQW (µl)	(µg /ml)
Blank	0	200
S 1	25	175
S 2	50	150
S 3	75	125
S 4	100	100

		S 5	150	50	1.5	0.01089
		S 6	200	0	2.0	0.01452
		6.3	DOC Precipitation method (sample preparation):			
		6.3.1	The total Meningococcal A polysaccharide content must be			
		determine separately (phosphorus assay.				
		6.3.2	Dilute the sample approximately 100 µg/ml with MQW into			
		conical microfuge tubes.				
		6.3.3	Add 100 µl of reagent C and vortex the sample.			
		6.3.4	Place the sample in refrigerator at 2°C- 8°C for 30 min.			
		6.3.5	Add 50 µl of reagent B and vortex. Centrifuge the sample			
		for 15 min at 6000Xg at 4°C.				
		6.3.6	Remove carefully the supernatant into another eppendorf			
		tube and mark it as supernatant.				
		6.4	Take appropriate aliquot of supernatant in duplicate in glass test			
		tubes				
		6.5	Add 75µL reagent A to each tubes and vortex			
		6.6	Ash the tubes in a furnace at 250 °C for 4hrs ± 15 minutes			
		6.7	Remove the tubes from the furnace. Cool down the tubes to room			
		temperature. Set the water bath to 37°C.				
		6.8	Add 1 ml Reagent G to each tube and vortex.			
		6.9	Cap the tubes with plastic caps and incubate in a 37°C water bath			
		for 90 ± 5 min.				
		6.10	Measure the absorbance of each tube at 825 nm with a			
		spectrophotometer.				
		6.11	Calculation:			
		6.11.1	Data analysis can be done manually (calculator) or by			
		using computer and appropriate software (e.g. MS Excel).				
		6.11.2	Construct the standard curve with the quantity of			
		phosphorous (q, in µg/ml) plotted versus the avg.				
		absorbance (A). The standard should fit a first degree				
		(i.e. linear) equation:				
		$y = mx + b$				
		6.11.3	Read the average absorbance of the samples against the			
		standard curve.				
		6.11.4	Calculate the phosphorous concentration of the samples			
		using following formula				
		Phosphorous concentration (µg/ml)=				
		$\frac{((\text{abs-intercept})/\text{slope}) \times \text{dilution factor}}{11.11}$				
		6.11.5	Calculate the polysaccharide concentration of the samples			
		using following formula				
		Polysaccharide concentration= Phosphorous concentration				
		$\times 11.11$				
		6.12	Acceptance Criteria:			
		The correlation of calibration curve must be > 0.990				
		7.0	Reporting:			
		7.1	Record all the results in the data sheet.			
		7.2	Attach the print out of the spectrophotometer and the applicable			
		calculation sheet using MS Excel or any other software.				
		7.3	The report is submitted to QC in-charge for verification.			
		7.4	Report to QC in-charge in case of any problems such as out of			
		linearity and proceed as per instructions.				
		8.0	References:			
		8.1	Chen, P.S. Jr. Toribara T.Y. And Warner H. 1956. Micro			
		determination of phosphorous. Anal. Chem. 28 1756-1758				
		8.2	SOP for UV Spectrophotometer.			

8. ANNEXURES:

|Annexure |Title |Format No. |Total Page No.|
 |No. |
 |1 |Analytical work sheet |035 5555/F1 |3 |
 |2 |SOP based questions |035 5555/F2 |1 |
 |

1.0	Objective:			
	To lay down the procedure for identification of thrombin.			
2.0	Scope:			
	This Standard Operating Procedure is applicable to all the In- process and			
	finished product samples.			
3.0	Responsibility			
	3.1	The QC analyst shall be responsible for executing the described		
		procedure correctly.		
	3.2	The QC supervisor shall be responsible for verifying that the		
		procedure is complied with and the results are checked.		
	3.3	The QC HOD shall be responsible for overall compliance of this SOP.		
4.0	Abbreviations and Definitions:			
	Not applicable			
5.0	Procedure:			
	5.1	Equipment, Materials and Reagents		
		5.1.1	Coagulometer	
		5.1.2	Imidazole	
		5.1.3	Bovine Serum Albumin	
		5.1.4	Calcium Chloride	
		5.1.5	Fibrinogen Concentrate	
	5.2	Preparation of solutions:		
		5.2.1	0.2 M Imidazole buffer (4X), pH 7.4:	
			5.2.1.1	Dissolve 3.4 g of Imidazole in 200 mL of Purified water
			1	and adjust the pH to 7.4 with 1M HCl and make up to 250
			mL with Milli- Q water.	
			5.2.1.1	Filter sterilize by using 0.45µ membrane filter and
			2	store at room temperature for 2 months. Prior to use,
			visually check Discard the buffer if it is visually	
			unclear.	
		5.2.2	0.5 M Calcium Chloride	
			5.2.2.1	Dissolve 73.51 g of Calcium chloride in 1000 mL of
			1	Purified water.
			5.2.2.1	Filter sterilize by using 0.45µ membrane filter and
			2	store at room temperature for 2 Months. And discard the
			buffer if it visually unclear.	
		5.2.3	10% BSA	
			5.2.3.1	Dissolve 5 g of BSA in 50 mL of Purified water and
			1	aliquot 1mL each in eppendorf tubes and store the
			aliquots at -20°C.	
		5.2.4	Assay buffer (0.05 M Imidazole buffer pH 7.4 containing 1% BSA)	
		S.No		
		Stock Solution		
		100 mL Volume (mL)		
			1	
			0.2 M Imidazole buffer pH 7.4 (4X)	
		25		
			2	
		0.5 M Calcium chloride		
		10		
		3		
		10 % BSA		
		10		
		4		
		Purified water		
	55			
		Note: Prepare the Assay buffer freshly.		
	5.2.5	Fibrinogen concentrate 1 mg/mL		
		5.2.5.1	Reconstitute Fibrinogen concentrate vial with 1 mL of	
		1	sterile water for injection,	
		5.2.5.1	Stated Clottable protein: 75 – 115 mg/mL	
		2		
		5.2.5.1	Assigned Clottable protein: 95 mg/mL	
		3		
		5.2.5.1	Dilute the Fibrinogen Concentrate to 1 mg/mL from 95	
		4	mg/mL with Assay buffer: Step:1	
			Fibrinogen mg/mL	
			Fibrinogen IRS-neat (µL)	
			Assay buffer (µL)	
			Total volume (µL)	
			10	
			50	
			425	
			475	
		5.2.5.1	Dilution from 10 mg/mL (from Step 1) to 1 mg/mL using	
		4	assay buffer.	
			Fibrinogen mg/mL	
			Fibrinogen	
			10 mg/mL from Step 1 (µL)	
			Assay buffer (µL)	
			Total volume (µL)	
			1	

			150	
			1350	
			1500	
			Note: In case the assigned clottable protein is more/less than	
		95 mg/mL, re-calculate the volume required to achieve 10 mg/mL		
		and record the same.		
	5.2.6	Human Thrombin Reference standard:		
		5.2.6.	WHO 3rd International Standard for Thrombin 19/188,	
		1	Activity: 90 IU/mL	
		5.2.6.	Reconstitute reference standard with 1 mL of specified	
		2	diluent as per data sheet. (WHO Standard with 1 mL of	
		Sterile water for Injection and IRS or Innovator product		
		with 1 mL of 40 mM CaCl₂).		
	5.3	Test Method:		
	5.3.1	Predilution of Reference Standard :		
		5.3.1.	Pre dilution from 90 IU/mL to 2.5 IU/mL with Assay	
		1	buffer.	
				IRS -Thrombin IU/mL
				Dilution Folds
				IRS-Thrombin from Step 1 (90 IU/mL) (μL)
				Assay Buffer (μL)
				Total volume (μL)
			10	
			09	
			50	
			400	
			450	
			5	
			18	
			200 of 10 IU	
			200	
			400	
			2.5	
			36	
			200 of 5 IU	
			200	
			400	
	5.3.2	Pre dilution of Thrombin Test Sample (DS 250 IU)		
		5.3.2.	Pre dilution from 250 IU/mL to 2.5 IU/mL with Assay	
		1	buffer.	
				IRS -Thrombin IU/mL
				Dilution Folds
				IRS-Thrombin from Step 1 (90 IU/mL) (μL)
				Assay Buffer (μL)
				Total volume (μL)
			100	
			2.5	
			200	
			300	
			500	
			10	
			25	
			50 of 100 IU	
			450	
			500	
			5	
			50	
			200 of 10 IU	
			200	
			400	
			2.5	
			100	
			200 of 5 IU	
			200	
			400	
	5.3.3	Place Cuvette strips in the incubation area for pre warming at		
		37°C for 2-3 minutes.		
	5.3.4	Then dispense one magnetic ball to each cuvette.		
	5.3.5	In Pre warmed cuvettes (37°C), Dispense 50 μL of Thrombin		
		reference / Test solution preparations (10 IU/mL, 5 IU/mL and		
		2.5 IU/mL).		
			Note: Perform the reaction for the Reference Standard as first	
		set and then the test sample as a second set.		
	5.3.6	Follow the procedure outlined in the table below:		
		Along with the diluted Test samples pre-warm at 37°C the		
		Fibrinogen Soln (1 mg/mL) and do not exceed 5 min in any case.		
		S.No		
		Thrombin WHO / IRS /Test Sample (IU/mL)		
		WHO / IRS / Test Sample – Vol. in (μL)		

		Incubate at 37 °C for 120 Sec
		Pre-warmed Fibrinogen (1 mg/mL) – (μL)
		1
		10
		50
		150
		2
		5
		50
		150
		3
		2.5
		50
		150
	5.3.7	Start the timer corresponding to the incubation column for an
		incubation of 120 sec.
	5.3.8	After 110 sec beep sound was produced.
	5.3.9	Then transfer the cuvette from the incubation chamber to the
		test column area.
	5.3.1	Dispense 150 μL of fibrinogen solution (1 mg/mL of Clottable
	0	protein) and measure the clotting time.
	5.3.1	Record the clotting time.
	1	
	5.4	Calculation and analysis:
	5.4.1	The calculations of potency and analysis of the graphs should
		be performed using both Combistat and Graph pad prism Software.
	5.4.2	Combistat analysis- Input sample and standard ODs with proper
		dilutions folds and assigned potency value in Thrombin assay
		template and follow the settings below
	5.4.3	Settings:
		Model : Parallel lines
		Design: Completely randomized
		Transformation: $y' = \log(y)$
		Variance : Observed residuals
		In order to get the slope ratios for sample with standard need
		to include equivalence testing.
	5.4.4	Graph pad prism analysis- Analyse both standard and sample for
		linear regression and determine slope and liner regression
		values. Calculate the slope ratio for slope of test
		Sample/slope of standard.
	6.0	References:
	6.1 Indian Pharmacopeia	

1. |Purpose: |
 |To lay down the procedure for the estimation of molecular size of |
 |Pneumococcal protein conjugate (bulk conjugate) of Pneumococcal protein |
 |conjugate vaccine. |
 2. |Applicable to: |
 |This Standard Operating Procedure is applicable for the determination of |
 |free carrier protein present in bulk conjugate of Pneumococcal protein |
 |conjugate for all serotypes. |
 3. |Responsibility: |
 |The analyst is responsible for carrying out the described procedure |
 |correctly. Supervisor is responsible for ensuring that the procedure is |
 |carried out correctly and to check the results. |
 4. |Safety considerations: |
 |Always use gloves and laboratory coat. |
 5. |Equipments: |
 |5.1 |TSKgelG3000SWXL column {30cm(l), 7.8mm(i.d)} |
 |5.2 |HPLC system with UV detector under routine calibration |
 |5.3 |Analytical balance |
 |5.4 |Precision balance |
 |5.5 |Vacuum Pump |
 |5.6 |pH Meter |
 |5.7 |Refrigerator |
 |5.8 |Magnetic stirrer |
 |5.9 |Ultrasonic Cleaner |
 6. |Reagents: |
 |6.1 |Sodium azide |
 |6.2 |Sodium Chloride (NaCl) |
 |6.3 |Disodium hydrogen phosphate anhydrous (Na₂HPO₄) |
 |6.4 |Sodium dihydrogen phosphate monohydrate |
 |6.5 |Absolute ethanol |
 |6.6 |Purified water |
 7. |Preparation of solutions: |
 |7.1 |Solution I |
 | |[[Na₂HPO₄ (0.1M;), NaCl (0.1M)]] |
 | |For a volume of 1000 ml, weigh: |
 | |Na₂HPO₄ 14.19 g |
 | |NaCl 5.84 g |
 | |Dissolve in 700 ml of purified water. |
 | |Make up the volume to 1000 ml with purified water. |
 |7.2 |Solution II |
 | |(NaH₂PO₄. H₂O, 0.1 M) |
 | |For a volume of 500 ml, weigh: |
 | |NaH₂PO₄. H₂O. 6.85g |
 | |Dissolve in 400 ml of purified water. |
 | |Make up the volume to 500 ml with purified water. |
 |7.3 |0.1M Sodium Phosphate buffer |
 | |(0.1 M Na₂HPO₄, 0.1M NaCl +0.1M NaH₂PO₄ .H₂O, pH 7.0) |
 | |For a volume of 1000 ml, take approximately 600 ml of Solution I and |
 | |400ml of Solution II. The pH of the solution should be 7.0 ± 0.2. |
 | |This solution can be stored at room temperature for 1 month. |
 |7.4 |Preservation Solution: |
 | |(Sodium azide 0.05%) |
 | |Weigh 1 g of Sodium Azide and dissolve in approximately 1800 ml of |
 | |purified Water. Make up the volume to 2.0 Litre with purified water. |
 | |This solution can be stored at room temperature for 2 months. |
 |7.5 |20% ethanol solution: |
 | |Take 200ml of absolute ethanol in a measuring cylinder and make up |
 | |the volume to 1 litre with purified water. |
 | |Note: All the solutions prepared for use are vacuum filtered using |
 | |Nylon 6, 6 membrane 0.22 µm filters to eliminate the entry of |
 | |possible particles into the system as well as the column. |
 8. |Procedure: |
 |8.1 |Run condition: | | |
 | |8.1.1 |Column: TSK gel SW3000xl column |
 | |8.1.2 |Buffer used: 0.1M sodium phosphate buffer |
 | |8.1.3 |Elution: Isocratic |
 | |8.1.4 |Flow rate: 0.75ml/min |
 | |8.1.5 |Run time: 30minutes |
 | |8.1.6 |Injection volume: 40-70µl |
 | |8.1.7 |Detector Used: UV detector at wavelength of 280nm |
 |8.2 |Run procedure: |
 | |8.2.1 |Equilibrate the column with 0.1M sodium phosphate buffer for |
 | |1 hour. |
 | |8.2.2 |Sample application: |
 | | |8.2.2.1 |Sample of purified conjugated material is injected to |
 | | |the column after a stable baseline at 280 nm is |
 | | |obtained. |
 |8.3 |Column cleaning and storage: |
 | |8.3.1 |Column storage: Pass 200 ml of preservation solution through |
 | |the column for long time storage. |
 | |8.3.2 |Cleaning the column: If peak shape observed is not proper, |
 | |then reverse the flow direction of column and pass 200 ml of |
 | |purified water and then 200 ml of 20% ethanol solution and |
 | |then pass 200ml of purified water with normal flow direction. |
 |8.4 |Chromatogram evaluation: |
 | |8.4.1 |Open the acquired chromatogram. |
 | |8.4.2 |Optimize integration with various integration parameters. |
 | |8.4.3 |The percentage of free carrier protein in sample is |
 | |calculated by taking the percentage area of the peak maxima |
 | |corresponding to the lower molecular weight compared to the |

			peak maxima of higher molecular weight.
		8.4.4	Acceptance Criteria:
		The percentage of Free Carrier Protein should be < 5%.	
8.5	Reporting Results		
	8.5.1	After optimizing integration, select Report Layout view in	
		the software.	
	8.5.2	Report the results of percentage of area of peak towards	
		higher Molecular weights (Conjugated) and percentage of area	
		of peak towards lower Molecular weights (Unconjugated).	
		Note: Percentage of free carrier protein is the percentage of	
		area of peak towards lower molecular weights (Unconjugated).	
	8.5.3	Take a printout of a report with the results.	
9.	Abbreviations:		
9.1	g	: Gram	
9.2	ml	: Milli litre	
9.3	μl	: Micro litre	
9.4	μm	: Micro meter	
9.5	nm	: Nano meter	
9.6	AN	: Analytical	
9.7	PCV	: Pneumococcal conjugate vaccine	

1.0 |Objective |
 |To lay down the procedure for analysis of Pneumococcal polysaccharide |
 |samples by High Performance Size Exclusion Chromatography. |
 2.0 |Scope |
 |This Standard Operating Procedure is applicable for evaluating Kavg of |
 |polysaccharide standard; polysaccharide samples at in process stage and |
 |lyophilized purified polysaccharide of pneumococcal vaccine. |
 3.0 |Responsibility |
 |3.1 |The analyst shall be responsible for carrying out the described |
 |procedure correctly. |
 |3.2 |The Supervisor of the laboratory shall be responsible for verifying |
 |that the procedure is complied with and for checking the results. |
 |3.3 |The departmental head/designee shall be responsible for overall |
 |compliance of the SOP. |
 |3.4 |Head QA/Designee shall be responsible for final approval and |
 |subsequent compliance of the SOP. |
 4.0 |Abbreviations and Definitions |
4.1	ATCC	: American Type Culture Collection
4.2	D₂O	: Deuterium oxide
4.3	g	: Gram
4.4	HOD	: Head of Department
4.5	HPLC	: High Performance Liquid Chromatography
4.6	M	: Molarity
4.7	ml	: Milli litre
4.8	nm	: Nano meter
4.9	PBS	: Phosphate Buffer Saline
4.10	PN	: Part Number
4.11	QA	: Quality Assurance
4.12	QC	: Quality Control
4.13	RID	: Refractive Index Detector
4.14	SOP	: Standard Operating Procedure
4.15	Std.	: Standard
4.16	°C	: Degree centigrade
4.17	µg	: Microgram
4.18	µl	: Micro litre
4.19	µm	: Micro meter
5.0	Procedure	
5.1	Precautions to be taken:	
5.1.1	Always wear laboratory coat	
5.1.2	Always use solutions and samples filtered through 0.2 µm	
membrane		
5.1.3	Degas all solutions	
5.1.4	Avoid entry of bubbles into the system as well as column	
5.1.5	All the solution bottle openings should be well covered with	
aluminum foil		
5.1.6	Check the calibration status of all equipment and instrument	
5.1.7	Handle acids carefully wearing gloves and always manipulate in	
fume hood		
5.2	Equipment and Instruments	
5.2.1	Calibrated HPLC system with RI detector	
5.2.2	Analytical balance	
5.2.3	Precision balance	
5.2.4	Vacuum Pump	
5.2.5	pH Meter	
5.2.6	Refrigerator	
5.2.7	Magnetic stirrer	
5.2.8	Ultrasonic Cleaner	
5.2.9	Rocking and Tumbling Shaker	
5.3	Material:	
5.3.1	Sugar KS-804 (Shodex Column)	
5.3.2	Sugar KS-805 (Shodex Column)	
5.3.3	Guard Column KS-G (Shodex)	
5.3.4	Laboratory Glassware	
5.3.5	Screw cap HPLC vials with/without inserts	
5.3.6	Solvac Filter Holder	
5.3.7	Nylon 6,6 membrane 0.2µm membrane filter	
5.3.8	Auto pipettes (10-100 (l), (20-200 (l), (100-1000 (l)	
5.3.9	Tips for the Auto pipettes	
5.3.10	Syringe filters 0.2 µm	
5.3.11	Aluminium Foil	
5.3.12	Gloves	
5.4	Reagents:	
5.4.1	Dextran	
5.4.2	Sodium azide	
5.4.3	Deuterium Oxide	
5.4.4	Sodium Chloride	
5.4.5	Purified water	
5.4.6	Sodium dihydrogen phosphate, dihydrate	
5.4.7	Disodium hydrogen phosphate, anhydrous	
5.5	Preparation of solutions: Prepare all the solutions as described	
below. Carry out the labelling of the prepared solutions and assign		
the lot numbers.		
5.5.1	100mM Sodium dihydrogen phosphate stock Solution:	
Take 15.602g of Sodium dihydrogen phosphate in 1L of purified		
water. Mix well and store the solution at room temperature for		
not more than one month.		
5.5.2	100mM Disodium hydrogen phosphate stock Solution:	
Take 14.196g of Disodium hydrogen phosphate in 1L of purified		
water. Mix well and store the solution at room temperature for		
not more than one month.		

		5.5.3	100mM Phosphate Buffer stock Solution:
		Add 280 mL of 100mM Sodium dihydrogen phosphate and 720mL of	
		100 mM Disodium hydrogen phosphate.	
		Add 280mL of 100mM Sodium dihydrogen phosphate and 720mL of	
		100mM Disodium hydrogen phosphate. Store the solution at room	
		temperature for not more than one month.	
		5.5.4	10mM Phosphate buffer saline:
		Add 100ml of 100 mM of phosphate buffer (pH-7.2) to 800 mL of	
		purified water and then add 8.5g of Sodium Chloride and adjust	
		the pH to 7.2 by adding 100mM Disodium hydrogen phosphate	
		solution. Make the final volume to 1000mL.	
		Filter the PBS solution with 0.2µm filter under vacuum and	
		degas the solution using the ultra-sonication bath.	
		5.5.5	Dextran (1 mg/mL) and Deuterium oxide (2.5 v/v) standard to
		obtain void volume (Vo): Add 9.75mL of 10mM PBS slowly to 10mg	
		of Dextran in a clear glass bottle. Do not mix the solution.	
		Leave the glass bottle at 2-8°C for 2 hours Then add 0.25mL D2O	
		and gently shake the tube for 30 minutes on shaker.	
		Note: Prepare the standard solution fresh just prior to use	
		with the same 10mM PBS which is to be used during the run.	
		5.5.6	Sodium azide standard to obtain the total volume (Vt) : Take
		10mg of sodium azide, dissolve in 20mL of WFI/10mM Phosphate	
		Buffer Saline and then filter through 0.2µm syringe filter.	
		Note: Prepare this solution fresh just prior to use.	
		5.5.7	Pneumococcal polysaccharide standard (2mg/mL):
		Take 2mg of ATCC standard and add 2ml of 0.2µm filtered	
		purified water and keep it overnight at 2-8°C on shaker. Take	
		the 1 mg/mL of the standard and aliquot in 120µL. Store at	
		-20°C for not more than 1 month.	
		5.5.8	Purified polysaccharide sample preparation (Test sample):
		Weigh the lyophilized purified sample and dissolve in 0.2µm	
		purified water to make 1mg/mL of sample. Keep it overnight at	
		2-8°C on shaker before use. Store at -20°C for not more than	
		one month.	
		5.5.9	Preservation Solution (Sodium azide 0.05%):
		Weigh 1g of Sodium Azide and dissolve in approximately 1800 mL	
		of purified Water. Make up the volume to 2L with purified	
		water. This solution can be stored at room temperature for 2	
		months.	
		Note: All the solutions prepared for use are vacuum filtered	
		using Nylon 6,6 membrane 0.2µm filters to eliminate the entry	
		of possible particles into the system as well as the column.	
		5.6	HPLC method Run:
		5.6.1	Mount Shodex guard column, Shodex 805 column and Shodex 804
		column in series on calibrated HPLC system.	
		5.6.2	Place the inlet tubing A in the reservoir for 10 mM phosphate
		buffer saline solution.	
		5.6.3	Establish On-Line connection of HPLC System with chromatography
		software.	
		5.6.4	Switch ON the HPLC System.
		5.6.5	Click on the chromatography software.
		5.6.6	Purging the system:
		5.6.6.1	Open the purge valve of pump (by turning it counter
		clockwise) and set flow rate to 3-5ml/min.	
		5.6.6.2	Flush the vacuum degasser and all tubes with 10mM PBS
		for ten minutes.	
		5.6.6.3	Set flow to at 0.0ml/min and close the purge valve. The
		flow shall be directed to the column.	
		5.6.7	Run conditions:
		5.6.7.1	Set the flow to 1.0ml/min
		5.6.7.2	Detector used: RID
		5.6.7.3	RID optical unit temperature : 30°C
		5.6.7.4	In case of running sample manually/sequence, run time
		is 30min.	
		5.6.8	Purge reference cell of RID for 40 minutes after solvent
		change. Further purging of reference cell may be required to	
		achieve stable baseline.	
		5.6.9	Keep on running the column with 10 mM phosphate buffer saline
		solution unless a stable baseline for RID is obtained which can	
		be viewed on the chromatographic software.	
		5.7	Sample Preparation and injection to the system:
		5.7.1	Diafiltration after 3-step ethanol precipitation samples:
		5.7.1.1	Filter the samples with 0.2 µm syringe filter in HPLC
		glass vials.	
		5.7.1.2	Inject 50 µl of the sample on HPLC.
		5.7.2	HA chromatography samples:
		5.7.2.1	Filter the samples with 0.2 µm syringe filter in HPLC
		glass vials.	
		5.7.2.2	Inject 50 µl of each sample on HPLC.
		5.7.3	HA pool sample:
		5.7.3.1	Filter the samples with 0.2µm syringe filter in HPLC
		glass vials.	
		5.7.3.2	Inject 50 µl of each sample on HPLC.
		5.7.4	After final diafiltration samples:
		5.7.4.1	Filter the samples with 0.2 µm syringe filter in HPLC
		glass vials.	
		5.7.4.2	Load 50 µl of the sample on HPLC.
		Note: For the samples of step 5.5.1 to 5.5.4 if absorbance of	
		the samples in nRI Units observed is greater than 7000 nRI	
		, then the volume of sample injected may be reduced accordingly	
		and vice-versa.	
		5.7.5	Lyophilized purified polysaccharide:

		5.7.5.1	Filter the samples with 0.2 µm syringe filter in HPLC
			glass vials.
		5.7.5.2	Inject 100 µl of the sample (1mg/ml) on HPLC.
	5.7.6	Inject 100µl of the respective ATCC Standard (1mg/ml)	
		pertaining to that particular serotype for all the above	
		samples.	
	5.7.7	Inject 100µl of the Dextran standard for void volume	
		determination for final Lyophilized sample.	
	5.7.8	Inject 100µl of the sodium azide standard for total volume	
		determination for final Lyophilized sample.	
		Note: For all the samples received on same day, only one injection of	
		ATCC polysaccharide standard is required.	
	5.8	Results Evaluation:	
	5.8.1	Compare the retention time of polysaccharide standards and	
		lyophilized sample.	
	5.8.2	Proceed for calculating Kavg value of standard and sample.	
		The calculations are made as per the following equation:	
		$K_{avg} = V_r - V_o$	
		$V_t - V_o$	
		Where,	
		V_r = Retention time of pneumococcal polysaccharide sample and	
		standard polysaccharide	
		V_t = Retention time of Sodium azide	
		V_o = Retention time of Dextran(1st peak)	
		Note: In case of 7F and 5 polysaccharide retention time of	
		first peak obtained is considered for Kavg calculation.	
	5.9	Reporting Results:	
	5.9.1	Open PNEUMOCOCCAL report template showing retention time, area,	
		height of each sample.	
	5.9.2	Report the results of Kavg with the retention time values of	
		all the samples.	
	5.9.3	Print report: Click this option to take a printout of the	
		report generated for each sample, void volume marker, total	
		volume marker and polysaccharide standard.	
	5.9.4	The printed report is signed by the analyst who does the	
		analysis and also checked by the other analyst.	
		Note: In case of HA chromatography samples, the printout	
		results of samples are attached to Annexure 1.	
	5.10	Acceptance criteria: The Kavg of the sample should be ≤ 0.30 .	

1.0 |Objective |
 |To lay down the procedure for the determination of residual Cetyl Trimethyl|
 |Ammonium Bromide (CTAB) concentration in Pneumococcal Purified |
 |Polysaccharide samples by Titration Method. |
 2.0 |Scope |
 |This Standard Operating Procedure is applicable to the final purified |
 |Polysaccharide samples. |
 3.0 |Responsibility |
3.1	The analyst is responsible for executing the described procedure
	correctly.
3.2	Supervisor is responsible for verifying that the procedure is carried
out correctly and check the results.	
3.3	HOD shall be responsible for overall compliance of this SOP and shall
ensure that the SOP is reviewed every two years or if required	
	earlier.
4.0	Abbreviations and Definitions
4.1	Abbreviations:
	4.1.1
	4.1.2
	4.1.3
	4.1.4
	4.1.5
	4.1.6
	4.1.7
	4.1.8
	4.1.9
	4.1.10
	4.1.11
	4.1.12
	4.1.13
	4.1.14
	4.1.15
5.0	Procedure
5.1	Preparation of solutions: Prepare all the solutions as described
	below and fill the form, label the solutions and assign lot numbers.
	5.1.1
	5.1.2
	5.1.3
	5.1.4
5.2	Testing procedure:
	5.2.1
	5.2.2
5.3	Calculations: For calculating the CTAB concentration, use the formula
	given below:
	5.3.1
5.4	Acceptance Criteria:
	The CTAB concentration in Pneumococcal Purified Polysaccharide

		Samples of each serotype should be less than 1%.
6.0	Forms and Records	
	Estimation of CTAB concentration in Pneumococcal Polysaccharide Samples	
	Annexure 1	
7.0	Reference	
	7.1	Standard Method ISO 2871-2-1990, "Surface Active Agents-
	Detergent-determination of Cationic-Active Matter content", Part 2:	
	Cationic –Active Matter of Low Molecular Mass (Between 200 and 500).	
	7.2	Ph.D. Thesis, "Treatment of Printing Ink Wastewater using High Liquid
	Carry over Mode Flootation", choon Jek ANG, Swinburne University of	
	Technology, Melbourne, Australia, August 2004.	

1.0 |Objective |
 |To lay down the procedure for Estimation of the O-acetyl Content in |
 |Pneumococcal polysaccharide samples. |
 2.0 |Scope |
 |This Standard Operating Procedure is applicable to final polysaccharide |
 |samples of pneumococcal serotypes in which the O-acetyl Content is to be |
 |determined. |
 3.0 |Responsibility |
3.1	The analyst is responsible for executing the described procedure
	correctly.
3.2	Supervisor is responsible for verifying that the procedure is carried
out correctly and check the results.	
3.3	HOD shall be responsible for overall compliance of this SOP and shall
ensure that the SOP is reviewed every two years or if required	
	earlier.
4.0	Abbreviations and Definitions
4.1	Abbreviations:
	4.1.1
	4.1.2
	4.1.3
	4.1.4
	4.1.5
	4.1.6
	4.1.7
	4.1.8
	4.1.9
4.2	Reagents
	4.2.1
	4.2.2
	4.2.3
	4.2.4
	4.2.5
	4.2.6
4.3	Equipments and Material
	4.3.1
	4.3.2
	4.3.3
	4.3.4
	4.3.5
	4.3.6
5.0	Procedure
5.1	Preparation of solutions: Prepare all the solutions as described below
and fill the form, label the solutions and assign lot numbers.	
5.1.1	4 M Hydrochloric acid:
	5.1.2
	5.1.3
	5.1.4
	5.1.5
	5.1.6
	5.1.7
	5.1.8

||| /ml): |

||| 5.1.8.1|Take 200 µl of 15 g/1 Acetylcholine chloride stock |

||| |solution and diluting to 10 ml with Purified water. |

||| 5.1.8.2|This solution prepared freshly. |

|| 5.2 |Testing procedure: |

|| 5.2.1 |Preparation of Calibration Curve: |

||| |From the working standard solution (600µg/ml), make the |

||| |following concentrations of the standard for the preparation of |

||| |a calibration curve. |

||| |

||| |

||| 5.2.1.1|Calibration curve for acetylcholine chloride (15-600 |

||| |µg/ml): |

||| |Acetylcholine |Working Stock- |Purified Water |

||| |chloride conc. |600 µg/ml |(µl) |

||| |(µg/ml) |(µl) |

||| |15 |25 |975 |

||| |30 |50 |950 |

||| |75 |125 |875 |

||| |120 |200 |800 |

||| |150 |250 |750 |

||| |360 |600 |400 |

||| |600 |1000 |0 |

|| 5.2.2 |Test Procedure: |

||| 5.2.2.1|Label the two sets of test tubes as "correction tubes" |

||| |and "reaction tubes". Correction tubes are used to |

||| |exclude the absorption due to non-specific binding in |

||| |sample. There is no need of the same for standards. |

||| 5.2.2.2|Introduce 1000µl of each of the concentration of the |

||| |standards, blank and sample in test tubes (in |

||| |duplicates). Sample shall be duplicated in two sets of |

||| |test tubes (correction as well as reaction) as stated |

||| |above in step 5.2.2.1 |

||| |In case of dry polysaccharide prepare a solution |

||| |containing about 1-5 mg/ml of polysaccharide in Lal |

||| |reagent water and use it as test sample. |

||| |Note: The sample dilutions can vary depending upon the |

||| |O-Acetyl concentration. If there is any sample outside |

||| |the range of the calibration curve, carry out the |

||| |calculation for more appropriate dilution. |

|| 5.2.3 |Correction tubes: |

||| 5.2.3.1|Add 0.2ml of 4M Hydrochloric acid to each of the |

||| |correction tubes, Vortex the tubes. |

||| 5.2.3.2|Add 0.4 ml of alkaline hydroxylamine solution to each |

||| |correction tubes, Vortex the tubes. |

||| 5.2.3.3|Allow the reaction to proceed exactly for 2 minutes at |

||| |room temperature, and follow the test procedure. |

|| 5.2.4 |Test procedure (Reaction tubes , Standard curve and Blank) |

||| 5.2.4.1|Add 0.4 ml of alkaline hydroxylamine to each of the |

||| |reaction tubes, standard curve and blank, Mix well. |

||| 5.2.4.2|Allow the reaction to proceed exactly for 2 minutes at |

||| |room temperature |

||| 5.2.4.3|Add 0.2 ml of 4M Hydrochloric acid to each of the |

||| |reaction tubes, standard curve and blank .Vortex the |

||| |tubes. |

||| 5.2.4.4|Add 0.2 ml of 0.74M Ferric chloride solution to all the |

||| |tubes (Blank, Reaction and correction, standard curve) |

||| |and again vortex for through mixing |

||| 5.2.4.5|Observed for the appearance for light yellow colour for |

||| |lower concentrations and yellowish brown colour for |

||| |higher concentration of acetylcholine or O-acetyl |

||| |groups. |

||| 5.2.4.6|Measure the absorbance at 540nm against the blank using |

||| |spectrophotometer. |

||| 5.2.4.7|Note that all the readings of the assay tubes should be |

||| |taken within 30 mins. of completion of the assay and |

||| |maximum 5 samples should be analysed in one single |

||| |assay. |

|| 5.3 |Calculations: |

|| 5.3.1 |Once the absorbance of the standards and the test samples have |

|| |been taken, calculation shall be done using the excel sheet |

|| |carrying formulas for the required calculation. |

|| 5.3.2 |Always copy the master excel sheet (sheet carrying formulas |

|| |for the required calculation), never use the previously used |

|| |excel sheet for new calculations. |

|| 5.3.3 |Copy the master excel sheet, keep it for use in respective |

|| |folder and then fill in the required sample details (Batch |

|| |number), sample dilution (if applicable), name of analyst, |

|| |date, etc. |

|| 5.3.4 |Proceed for calculations starting from the Calibration curve. |

|| |The calculations are done according to the equation |

|| |Y = mX + b Here: m = slope |

|| |b = intercept |

|| |Y = absorbance |

|| |X = concentration |

|| ||(Multiplication by dilution factor) |

|| 5.3.5 |Use EXCEL programme for the preparation of calibration curve |

|| |and Calculation of the results (Multiply by the dilution |

|| |factor. Divide with 1000 to change µg/ml to mg/ml). |

|| 5.3.6 |Subtract the concentration of the correction tube from the |

|| |corresponding reaction tube to estimate the accurate value of |

|| |acetylcholine chloride. |

	5.3.7	Divide the results obtained from the step above by 4.22 in
		order to get the quality of O-acetyl groups in the sample as
		the standard used is of acetylcholine chloride and 1 mole of
		acetylcholine chloride (181.7g) is equivalent to 1 mole of
		O-acetyl group (43.05g).
	5.3.8	In case of dry polysaccharide,
		O-acetyl (%) in polysaccharide on dry weight basis:
		O-acetyl Conc. in dry Polysacc. (µg/ml)
		= O-acetyl in Polysacc. sol. (µg/ml) X 1 – {Moisture (%) /
		{100}
		O-acetyl in dry polysaccharide (µg/ml)
		O-acetyl (%) = _____ X 100
		Conc. of polysaccharide solution (µg/ml)
	5.4	Acceptance criteria:
	5.4.1	The test is valid if Regression of the standard curve is ≥0.99
		and the percentage of Coefficient of Variation is ≤ 15% within
		the replicates.
	5.4.2	O-acetyl content in Purified Pneumococcal polysaccharide
		should meet the laid specifications.
	6.0	Forms and Records
		Determination of O-acetyl content - Annexure 1
	7.0	Reference
	7.1	WHO TRS No. 977 Annexure 3: Recommendations to assure the quality,
		safety and efficacy of pneumococcal conjugate vaccines.
	7.2	European Pharmacopoeia, Volume 5, 'O-acetyl content in Polysaccharide
		Vaccine' 2.5.19.

1.0 |Objective: |
 |To lay down the procedure Identification of factor IX using Clotting |
 | |Method. |
 2.0 |Scope: |
 |This Standard Operating Procedure is applicable to all the In- process and |
 |finished product samples. |
 3.0 |Responsibility |
3.1	The QC analyst shall be responsible for executing the described
	procedure correctly.
3.2	The QC supervisor shall be responsible for verifying that the
	procedure is complied with and the results are checked.
3.3	The QC HOD shall be responsible for overall compliance of this SOP.
4.0	Abbreviations and Definitions:
Not applicable	
5.0	Procedure:
5.1	Equipment, Materials and Reagents
	5.1.1
	5.1.2
	5.1.3
	5.1.4
	5.1.5
	5.1.6
	5.1.7
	5.1.8
	5.1.9
5.2	Preparation of solutions:
	5.2.1
	5.2.2
	5.2.6
5.3	Test Method:
	5.3.1

				300	
				5.3.1.	Preparation of 0.5 and 0.25 IU/mL from 1 IU/mL IRS
				2	(Step-2):
				Required Activity of IRS (IU/mL)	
				Reconstituted IRS (μL)	
				Owren kollor buffer (μL)	
				Total volume	
				(μL)	
				0.5	
				150 μL of 1 IU/mL	
				150	
				300	
				0.25	
				150 μL of 0.5 IU/mL	
				150	
				300	
				5.3.2	Pre-dilution of working reference standard:
				5.3.2.	Working standard (WS) calibrated against WHO
				1	International Reference Standard
				Clotnine 600 IU I.P.:	
				Assigned Potency: 600 IU/vial i.e. ~120 IU/mL	
				5.3.2.	Reconstitute the entire vial with 5 mL of sterile water
				2	for injection and let the vial to stand at room
				temperature for 5 min to get a uniform solution.	
				5.3.2.	The activity of the WS after reconstitution with 5 mL of
				3	sterile water for injection is 120 IU/mL.
				5.3.2.	The assigned activity of the WS may vary Slightly from
				4	batch to batch.
				5.3.2.	Follow the dilutions according to the tables mentioned
				5	below.
				Pre-Dilution Step-1:	
				Dilution of 120 IU/mL to 10 IU/mL (12 Folds)	
				Required Activity of WS (IU/mL)	
				Reconstituted WS (μL)	
				Factor IX placebo (μL)	
				Total volume	
				(μL)	
				10 IU/mL	
				100 from 120 IU/mL	
				1100	
				1200	
				5.3.2.	Pre-Dilution Step-2:
				6	Dilution of 10 IU/mL to 1 IU/mL (10 Folds)
				Required Activity of WS (IU/mL)	
				Reconstituted WS (μL)	
				Factor IX Deficient (μL)	
				Total volume	
				(μL)	
				1 IU/mL	
				30 from 10 IU/mL	
				270	
				300	
				5.3.2.	Required Activity of WS (IU/mL)
				7	Reconstituted WS (μL)
				Owren kollor buffer (μL)	
				Total volume	
				(μL)	
				0.5	
				150 μL of 1 IU/mL	
				150	
				300	
				0.25	
				150 μL of 0.5 IU/mL	
				150	
				300	
				Preparation of 0.5 and 0.25 IU/mL from 1 IU/mL	
				5.3.3	Pre dilution of Test Sample:
				5.3.3.	Clotnine Drug Substance:
				1	
				5.3.3.	Prior to analysis bring the sample to room temperature.
				2	
				5.3.3.	Assigned Potency: 150 IU/mL
				3	Follow the dilutions according to the tables mentioned
				below.	
				5.3.3.	Pre-Dilution Step-1:
				4	Dilution of 150 IU/mL to 120 IU/mL (1.25 Folds)
				Required Activity of Test Sample (IU/mL)	
				Test Sample	
				(μL)	
				Factor IX Placebo (μL)	

				Total volume	
				(μL)	
				120 IU/mL	
				480	
				120	
				600	
				5.3.3.	Pre-Dilution Step-2:
				5	Dilution of 120 IU/mL to 10 IU/mL (12 Folds)
				Required Activity of Test Sample (IU/mL)	
				Test Sample	
				(μL)	
				Factor IX Placebo (μL)	
				Total volume	
				(μL)	
				10 IU/mL	
				100 from 120 IU/mL	
				1100	
				1200	
				5.3.3.	Pre-Dilution Step-3:
				6	Dilution of 10 IU/mL to 1 IU/mL (10 Folds)
				Required Activity of Test Sample (IU/mL)	
				Test Sample	
				(μL)	
				Factor IX Deficient (μL)	
				Total volume	
				(μL)	
				1 IU/mL	
				30 from 10 IU/mL	
				270	
				300	
				5.3.3.	Preparation of 0.5 and 0.25 IU/mL from 1 IU/mL :
				7	Required Activity of Test Sample (IU/mL)
				Test Sample (μL)	
				Owren koller buffer (μL)	
				Total volume	
				(μL)	
				0.5	
				150 μL of 1 IU/mL	
				150	
				300	
				0.25	
				150 μL of 0.5 IU/mL	
				150	
				300	
				5.3.4	Test procedure: Conduct the experiment in 2 steps.
				Pre assay testing to check placebo interference	
				Main experiment	
				5.3.4.	Pre assay testing to check placebo interference
				1	
				5.3.4.1.1	Check placebo interference before conducting
				the main experiment.	
				5.3.4.1.2	Switch on the Coagulometer 15 minutes prior to
				analysis.	
				5.3.4.1.3	Place all the reagents in appropriate slots of
				the Start – 4 Coagulometer for warming at 37°C	
				(5 minutes prior to analysis).	
				5.3.4.1.4	Place a Cuvette strip in the incubation area
				and pre warm at 37°C at least for 3 minutes.	
				Then dispense one magnetic ball to each	
				cuvette	
				5.3.4.1.5	Add 50 μL of Placebo of test sample under
				testing, 50 μL of Factor IX deficient plasma	
				and 50 μL of C.K. Prest to each cuvette.	
				5.3.4.1.6	Immediately start the timer corresponding to
				the incubation column for an incubation of 180	
				sec.	
				5.3.4.1.7	After 170 sec beep sound was produced.
				5.3.4.1.8	Then transfer the cuvette from the incubation
				chamber to the test column area.	
				5.3.4.1.9	Then add 50 μL of 0.025 M calcium chloride
				solution pre warmed at 37°C to each cuvette	
				with the help of Micro pipette, while adding	
				calcium chloride press timer to start the	
				measurement.	
				5.3.4.1.1	A clot should be formed within the established
				0	clotting time.
				5.3.4.1.1	If the placebo is in compliance with the
				1	established spec, proceed for the main
				experiment.	
				5.3.4.	Main experiment:
				2	
				5.3.4.2.1	Once the pre assay testing is clear, follow
				the procedure mentioned below to test the	

				sample activity.	
				5.3.4.2.2	Follow section 5.3.4.1.2 to 5.3.4.1.4 of
				current STP upto addition of samples/standard.	
				5.3.4.2.3	Add 50 µL of reference standard/test sample,
				50 µL of Factor IX deficient plasma and 50 µL	
				of C.K. Prest to each cuvette.	
				5.3.4.2.4	Follow section 5.3.4.1.6 to 5.3.4.1.10 of
				current STP upto addition of 0.025 M calcium	
				chloride solution.	
				5.3.4.2.5	A clot will be formed in the reaction cuvette
				based on the activity of the sample.	
				5.3.4.2.6	Note down the clotting times which comes in
				the print out and interpret the result using	
				respective software's for data analysis.	
	5.4	Calculation and analysis:			
	5.4.1	The calculations of potency and analysis of the graphs should			
		be performed using both Combistat and Graph pad prism Software.			
	5.5	Specific activity calculation:			
	5.5.1	Specific activity =			
		Factor IX			
		Total protein mg/ml			
	6.0	References:			
	6.1 Indian Pharmacopeia				

1.0 |Objective |
 |To lay down the procedure for determination of Triethylamine (TEA) content |
 |in Pneumococcal Polysaccharide samples by Gas Chromatography. |
 2.0 |Scope |
 |This Standard Operating Procedure is applicable to all the samples, in |
 |which the Triethylamine (TEA) content is to be determined, which is |
 |received in the QC. |
 3.0 |Responsibility |
 |3.1 |The analyst shall be responsible for carrying out the described |
 |procedure correctly. |
 |3.2 |The Supervisor of the laboratory shall be responsible for verifying |
 |that the procedure is complied with and for checking the results. |
 |3.3 |The departmental head/designee shall be responsible for overall |
 |compliance of the SOP. |
 |3.4 |Head QA/Designee shall be responsible for final approval and |
 |subsequent compliance of the SOP. |
 4.0 |Abbreviations and Definitions |
4.1	g	: Gram
4.2	HOD	: Head of Department
4.3	M	: Molarity
4.4	ml	: Milli litre
4.5	nm	: Nano meter
4.6	PBS	: Phosphate Buffer Saline
4.7	QA	: Quality Assurance
4.8	QC	: Quality Control
4.9	SOP	: Standard Operating Procedure
4.10	Std.	: Standard
4.11	µg	: Microgram
4.12	µl	: Micro litre
4.13	µm	: Micro meter
4.14	TEA	: Triethyl amine
5.0	Procedure	
5.1	Precautions to be taken:	
5.1.1	Always wear laboratory coat.	
5.1.2	Always use solutions and samples filtered through 0.2 µm	
membrane.		
5.1.3	Check the calibration status of all equipment and instrument.	
5.1.4	Handle acids carefully wearing gloves and always manipulate in	
fume hood.		
5.2	Requirement:	
5.2.1	Equipment and Instruments:	
5.2.1.1	Gas Chromatography system with head space	
5.2.1.2	Micropipettes	
5.2.2	Material:	
5.2.2.1	Triethyl amine (Make: Fluka, Catalogue No. 90340)	
5.2.2.2	Sodium chloride (NaCl) (Make: Qualigens, Cat No: 15915	
or equivalent grade)		
5.2.2.3	Disodium hydrogen phosphate dodecahydrate	
(Na₂HPO₄·12H₂O) (Make: Sigma Aldrich, Cat No. 71650 or		
equivalent grade)		
5.2.2.4	Laboratory Glassware	
5.2.2.5	Sodium dihydrogen phosphate dehydrate (NaH₂PO₄·2H₂O)	
(Make: Merck, Cat No. 1.06342.0250 or equivalent grade)		
5.2.2.6	Water for Injection (WFI)	
5.2.2.7	Hydrogen gas (IOLAR)	
5.2.2.8	Nitrogen gas (IOLAR)	
5.2.2.9	Helium gas (IONAR)	
5.2.2.1	Zero gas (IOLAR)	
10		
5.2.2.1	Separation Column, e.g. OPTIMA–624–LB–1.80 µm –30 m x	
1	0.32 mm (Make: Macherey-Nagel, Catalogue No. 726 786.30)	
5.2.2.1	1.5 mL vials	
2		
5.2.2.1	Volumetric flasks- 100 mL	
3		
5.2.2.1	Glass pipettes	
4		
5.2.2.1	Head space vials and septa	
5		
5.2.2.1	Buffer (0.2 M NaCl, 10mM Phosphate Buffer pH 6.0)	
6		
5.3	Preparation of solutions:	
Prepare all the solutions as described below. Carry out the labelling		
of the prepared solutions and assign the lot numbers.		
5.3.1	0.2 M NaCl, 10mM Phosphate Buffer pH 6.0:	
Take 1.17 g NaCl, 0.054 g Na₂HPO₄·12H₂O and 0.133 g		
Na₂H₂PO₄·2H₂O. Dissolve in 80 mL of MQW, adjust the pH with HCL		
or NaOH and make up to 100mL. Filter through 0.22 µm filter.		
This solution can be stored up to one month at room		
temperature.		
5.3.2	TEA stock Standard (10,000 µg/mL):	
Take 1.370 mL of neat TEA and make up to 100 mL with 0.2 M		
NaCl, 10mM Phosphate Buffer in a volumetric flask. Prepare		
fresh on the day of use.		
5.3.3	Positive control (50µg/mL of TEA standards)	
Add 15µL of TEA stock to 3mL of 0.2 M NaCl, 10mM Phosphate		
Buffer.		
5.3.4	Test sample preparation:	
5.3.5	Dilute the test sample with 0.2 M NaCl, 10mM Phosphate Buffer	
to 10mg/ml of pneumococcal polysaccharide based on its initial		

||| |concentration. |

|| |5.4 |Test Procedure: |

|| |5.4.1 |Take 3 mL of 0.2 M NaCl, 10mM Phosphate Buffer as a blank, 3 mL |

|| |of positive control and 3mL of diluted test sample into |

|| |respectively labeled head space vials, and place these vials in |

|| |the head space carousel. |

|| |5.4.2 |Create a batch file in GC solution software and wait for GC |

|| |ready command and click start button. |

|| |5.4.3 |Then create a schedule file in HT3 TEKlink software for head |

|| |spacer. |

|| |5.4.7 |Chromatography conditions: |

|| |Set the chromatography conditions as described below: |

|| |5.4.7.1 |Carrier Gas |Helium |

|| |5.4.7.2 |Makeup Gas |Nitrogen (20 mL/minute) |

|| |5.4.7.3 |Hydrogen Gas |40mL/minute |

|| |5.4.7.4 |Zero Air |300mL/minute |

|| |5.4.7.5 |Flow (Pressure) |30kPa |

|| |5.4.7.6 |Split ratio |1:10 |

|| |5.4.7.7 |Furnance Temperature |5 minute at 40OC and then |

|| || |increase to 180 OC at a rate |

|| || |of 10 OC per minute and hold |

|| || |for 5 minutes at 180OC. |

|| |5.4.7.8 |Detector Temperature |250 OC |

|| |5.4.7.9 |Injection volume |3.0 mL |

|| |5.4.7.10 |Run Time |24 minutes |

|| |5.4.8 |Conditioning of the column |

|| |Perform the column conditioning using the following settings |

|| |5.4.8.1 |Injector temperature |200 OC |

|| |5.4.8.2 |Column temperature |220OC |

|| |5.4.8.3 |Detector temperature |280 OC |

|| |5.4.8.4 |Column Flow |0.5mL/min |

|| |5.4.9 |Head space system conditioning |

|| |Set the head spacer conditions as described below |

|| |5.4.9.1 |Constant Heat Time |On |

|| |5.4.9.2 |GC Cycle Time |35 minute |

|| |5.4.9.3 |Platen Temperature |155OC |

|| |5.4.9.4 |Value oven temperature |160 OC |

|| |5.4.9.5 |Transfer line |165 OC |

|| || |Temperature | |

|| |5.4.9.6 |Sample equilibration |30 minutes |

|| |5.4.9.7 |Pressurize |15 PSIG |

|| |5.4.9.8 |Loop fill Pressure |10 PSIG |

|| |5.4.9.9 |Injection Time |3.00 minutes |

|| |5.4.9.1 |Loop fill Time |4.00 minutes |

|| || |0 | |

|| |5.4.9.1 |Pressurize equilibration |0.20 minutes |

|| || |1 |time | |

|| |5.4.9.1 |Pressurize Time |2 minutes |

|| || |2 | | |

|| |5.4.10 |Communicating Head space system with Gas chromatography |

|| |5.4.10. |Wait for ready command in GC and then click start |

|| || |1 |button which in- turn gives stand by indication on GC |

|| || |software. |

|| |5.4.10. |Now click start button in head space system schedule oh |

|| || |2 |HT3 TEKlink software. |

|| |5.4.10. |Take the printout of the chromatograms and attach to |

|| || |3 |the prescribed annexure. |

|| |5.4.10. |Record the details in record sheets. |

|| || |4 | |

|| |5.5 |Acceptance criteria: |

|| |As per the specification the residual TEA content should be "less |

|| |than 10 ng/µg of polysaccharide". Therefore the acceptance criterion |

|| |of the assay is "The TEA peak areas of the test sample containing 10 |

|| |mg/mL of Polysaccharide should be less than the positive control peak |

|| |area values (50 µg/mL of TEA)" |

|| |5.6 |Reference |

|| |British pharmacopoeia. |

|| |Head space system Manual - TEKMAR TELEDYNE HT3 |

1. |Purpose: |
 |To lay down the procedure for quantifying Total and unconjugated (free) |
 |Polysaccharide concentration. |
 2. |Scope: |
 |This Standard Operating Procedure is applicable to all the samples of |
 |Polysaccharide-CRM bulk conjugate of Pneumococcal conjugate vaccine in |
 |which total and unconjugated (free) Polysaccharide concentration is to be |
 |determined. |
 3. |Responsibility: |
 |3.1 |The analyst is responsible for executing the described procedure |
 | |correctly. |
 |3.2 |Supervisor is responsible for verifying that the procedure is carried|
 | |out correctly and check the results. |
 |3.3 |HOD shall be responsible for overall compliance of this SOP and shall|
 | |ensure that the SOP is reviewed every two years or if required |
 | |earlier. |
 4. |Safety considerations: |
 |4.1 |Wear laboratory coat while working in the laboratory. |
 |4.2 |Handle the quartz cuvettes and glassware carefully. |
 |4.3 |Work in a fume hood while handling the acids. |
 5. |Equipments: |
 |5.1 |UV-Visible Spectrophotometer |
 |5.2 |Analytical balance Sensitivity 0.01mg |
 |5.3 |Fume hood |
 |5.4 |Water bath (100 °C) |
 |5.5 |Vortex Mixer |
 |5.6 |Refrigerator |
 |5.7 |Laboratory Centrifuge |
 6. |Materials and reagents: |
 |6.1 |Materials: | |
 | |6.1.1 |20, 200, 1000 and 5000µl Autopipettes |
 | |6.1.2 |Volumetric Flasks |
 | |6.1.3 |10 and 20 ml thermoresistant test tubes. |
 | |6.1.4 |25, 50 and 100 ml Beaker |
 | |6.1.5 |Quartz cuvettes |
 | |6.1.6 |Autopipette tips |
 | |6.1.7 |Eppendorf |
 |6.2 |Reagents: |
 | |6.2.1 |Resorcinol |
 | |6.2.2 |Sulphuric acid |
 | |6.2.3 |ATCC Standard of respective Polysaccharide |
 | |6.2.4 |Sodium Deoxycholate |
 | |6.2.5 |Purified water |
 7. |Preparation of solutions: |
 |7.1 |6.0 mg/ml Resorcinol solution: | |
 | |7.1.1 |Weigh 60 mg of Resorcinol and transfer it to 10 ml volumetric |
 | | |flask. Dissolve and make up the volume to 10 ml with Purified |
 | | |water. |
 | |7.1.2 |The solution is prepared freshly. |
 |7.2 |75% Sulphuric acid: |
 | |7.2.1 |Take 250 ml of purified water in a 1000 ml volumetric flask |
 | | |and add conc. H₂SO₄ slowly and make up the volume to 1000 |
 | | |ml. |
 | |7.2.2 |The solution should be stored at room temperature for a |
 | | |maximum of 6 months. |
 |7.3 |ATCC polysaccharide (Stock solution): |
 | |7.3.1 |Prepare a 2 mg/ml solution of ATCC standard of the respective |
 | | |polysaccharide in purified water. |
 | |7.3.2 |Using the 2 mg/ml solution prepared above, prepare 100µg/ml |
 | | |solution and aliquot it into 1.5 ml vials. |
 | |7.3.3 |Store the solution at -70°C for 1 week. |
 |7.5 |5% DOC solution: |
 | |7.5.1 |Weigh and dissolve 250 mg of sodium deoxycholate in purified |
 | | |water and make up the volume to 5 ml. |
 | |7.5.2 |The solution is prepared freshly. |
 |7.6 |1 M HCl: |
 | |7.6.1 |Take 85 ml of concentrated HCl and make up the volume to 1000 |
 | | |ml with purified water. |
 | |7.6.2 |The solution should be stored at room temperature for a maximum|
 | | |of 6 months. |
 8. |Procedure: |
 |8.1. |Preparation of Calibration Curve: | | | | | | |
 | |8.1.1 |Calibration Curve (10 - 50µg/ml) is prepared in duplicates as |
 | | |follow: |
 | | |Polysaccharide content |10 |20 |30 |40 |50 |
 | | | (µg/ml) | | | | |
 | | |Stock Sol. 100 µg/ml (µl) |40 |80 |120 |160 |200 |
 | | |Purified water (µl) |360 |320 |280 |240 |200 |
 |8.2 |Total Polysaccharide: |
 | |8.2.1 |Take 400µl sample in a test tube. |
 | | |Note: If there is any sample outside the range of the |
 | | |calibration curve, prepare an appropriate dilution of the |
 | | |sample. |
 |8.3 |Unconjugated (Free) Polysaccharide: |
 | |8.3.1 |Take 1 ml of sample (dilute if required) having about 100-150 |
 | | |µg Polysaccharide in a microfuge tube, add 100 µl of 5% DOC and|
 | | |mix it. |
 | |8.3.2 |Put this tube in an ice bath for 30 minutes. |
 | |8.3.3 |Take out and add 50 µl 1M HCl and mix thoroughly. |
 | |8.3.4 |Centrifuge the mixture at 2-8°C for 15 minutes at 10000 RPM. |

		8.3.5	Take 400µl supernatant volume in a test tube for free
			Polysaccharide estimation.
		8.3.6	The test is performed in duplicate for each sample.
		8.4	Treat 400 µl purified water as blank in thermoresistant test tube.
		8.5	Add 400 µl of Resorcinol solution to tubes of calibration curve,
		Blank and samples.	
		8.6	Vortex and add 2 ml of 75% H₂SO₄ to all test tubes and vortex well.
		8.7	Close the tubes with aluminium foil and place the test tubes in water
		bath at 90°C for 30 minutes.	
		8.8	Take out the tubes and cool in iced water and place in dark for 30
		minutes.	
		8.9	Allow to come at room temperature and measure the absorbance of each
		at 430 nm against the blank.	
		8.10	Note that all the readings of the assay tubes should be taken within
		30 mins. of completion of the assay and maximum 5 samples should be	
		analysed in one single assay.	
		8.11	Calculations:
		8.11.1	The calculations are made according to the equation of minimum
		squares.	
		$Y = mX + b$.	
		Where: m = slope	
		b = intercept	
		Y = absorbance	
		X = concentration	
		(Multiply by dilution factor)	
		8.11.2	Use EXCEL programme for the preparation of calibration curve
		and Calculation of the results.	
		8.11.3	Calculation of actual dilution for free Polysaccharide:
		Sample volume taken = 1000µl	
		Total volume = 1150µl (1000µl sample + 100µl 5%DOC + 50µl 1M	
		HCl)	
		Correction Factor = 1150µl / 1000µl = 1.15	
		Actual dilution = dilution factor x 1.15	
		8.12	Acceptance criteria:
		8.12.1	The test is valid if Regression of the standard curve is ≥ 0.99
		8.12.2	The coefficient of variation (CV) between the replicates of
		the samples should be $\leq 15\%$.	
		8.12.3	In case of In process (16-24 hour stage of conjugation
		reaction) sample:	
		% Free Saccharide: < 30-35% of Total Saccharide.	
		In case of final purified and ARM:	
		% Free Saccharide: < 25% of Total Saccharide.	
9.	Abbreviations:		
9.1	CV	:Coefficient of variation	
9.2	g	:Gram	
9.3	mg	:Milligram	
9.4	ml	:Millilitre	
9.5	µl	:Micro litre	
9.6	µg	:Microgram	
9.7	nm	:Nanometer	
9.8	PCV	: Pneumococcal conjugate vaccine	
9.9	HOD	:Head of Department	
9.10	QA	: Quality Assurance	

1.0 |Objective |
 |To lay down the procedure for determination of residual Ethyl Dimethyl |
 |Aminopropyl carbodiimide (EDAC). |
 2.0 |Scope |
 |This Standard Operating Procedure is applicable to all the samples, in |
 |which the residual Ethyl Dimethyl Aminopropyl carbodiimide (EDAC) content |
 |is to be determined. |
 3.0 |Abbreviations and Definitions |
3.1	PBS	: Phosphate Buffer Saline
3.2	Std.	: Standard
3.3	µg	: Microgram
3.4	µl	: Micro litre
3.5	µm	: Micro meter
3.6	EDAC	: Ethyl Dimethyl Aminopropyl Carbodiimide
3.7	MES	: Morpholino-ethanesulfonic acid
3.8	DMBA	: Dimethyl Barbituric acid
4.0	Procedure	
4.1	Precautions to be taken:	
	4.1.1	Always wear laboratory coat.
	4.1.2	Always use solutions and samples filtered through 0.2 µm
		membrane.
	4.1.3	Check the calibration status of all equipment and instrument.
	4.1.4	Handle acids carefully wearing gloves and always manipulate in
		fume hood.
4.2	Requirement:	
	4.2.1	Equipment and Instruments:
		4.2.1.
		1
		4.2.1.
		2
		4.2.1.
		3
		4.2.1.
		4
		4.2.1.
		5
	4.2.2	Material:
		4.2.2.
		1
		4.2.2.
		2
		4.2.2.
		3
		4.2.2.
		4
		4.2.2.
		5
		4.2.2.
		6
		4.2.2.
		7
4.3	Preparation of solutions:	
	Prepare all the solutions as described below. Carry out the labelling	
	of the prepared solutions and assign the lot numbers.	
	4.3.1	50 mM MES (Morpholino-ethanesulfonic acid):
		Take 1.074 g of MES and dissolve it in 60 mL of WFI and adjust
		the pH to 7.0 using 1M NaOH, and make up to 100 mL with WFI.
		Store at 2° to 8°C up to 1 year.
		Note: The Calculation is based on the purity (99%) as per the
		label claim)
	4.3.2	EDC Standard stock solution (1mg/mL):
		Take 100mg EDC and dissolve it in 100 mL of 50mM MES. Prepare
		fresh on the day of use.
	4.3.3	EDC working Standard (20µg/mL):
		Take 200µL EDC stock solution and dissolve it in 9.8 mL of 50mM
		MES. Prepare fresh on the day of use.
	4.3.4	Dimethyl Barbituric Acid (DMBA) reagent:
		Take 0.5 g of DMBA. Dissolve it in solution containing 8 mL
		pyridine and 2 mL water for injection. The UV absorbance of
		this solution at 500nm should be less than 0.2, higher
		absorbance indicates poor pyridine quality. Use this solution
		immediately.
		Note: Use glass tube and glass pipettes for preparation.
	4.3.5	Phosphate Buffer (0.2M NaCl, 10mM Phosphate Buffer pH 6.0):
		Take 1.17 g NaCl, 0.054 g Na₂HPO₄.12H₂O and 0.133 g
		NaH₂PO₄.2H₂O. Dissolve in 80 mL of MQW, adjust the pH with HCL
		for NaOH and make up to 100mL. Filter through 0.22 µm filter.
		This solution can be stored up to one month at room
		temperature.
4.4	Test Procedure:	
	4.4.1	Preparation of Test Sample:
		Dilute the test sample with phosphate buffer to 100 µg/mL of
		polysaccharide based on its initial concentration and use 100µL
		for the analysis.
	4.4.2	Preparation of blank and positive control (1 µg/mL of EDC)
		Prepare blank and positive control as described in the below
		table.
		EDC conc. (µg/mL)
		(µL)
		Blank

		Positive control (1	5	95
		µg/mL)		
	4.4.3	Pipette 100 µL of the blank, the positive control and test		
		sample in duplicate into separately labeled 1.5 mL Microfuge		
		tubes.		
	4.4.4	Add 900µL DMBA reagent to all tubes using a glass pipette.		
	4.4.5	Close the tubes and vortex. Incubate them for 45 minutes at		
		room temperature.		
	4.4.6	Measure the absorbance at 595 nm (within 30 minutes after		
		incubation).		
	4.4.7	Record the results in record sheets		
	4.5	Acceptance criteria:		
		As per the specification the residual EDAC content should be less		
		than < 10 ng / µg Polysaccharides. Therefore the acceptance criterion		
		of the assay is "The absorbance value of the test sample containing		
		100 µg/mL of Polysaccharide should be less than the positive control		
		value (1 µg/mL of EDAC)".		
5.0	Reference			
	5.1	Wilchek, M., Miron, T., and Kohn, J. (1981) A Highly sensitive		
		colorimetric method for the determination of carbodiimides. Anal.		
		Biochem.114, 419-421.		
	5.2	Giles, M.A., Hudson, A.Q., and Borders, C. L. Jr. (1990) Stability of		
		water-soluble carbodiimides in aqueous solution. Anal. Biochem. 184,		
		224-248.		

1.0 |Objective |
 |To lay down the procedure for estimation of residual formaldehyde content. |
 2.0 |Scope |
 |This Standard Operating Procedure is applicable to all the samples in which|
 |residual formaldehyde content is to be determined. |
 3.0 |Responsibility |
3.1	The analyst is responsible for executing the described procedure
	correctly.
3.2	Supervisor is responsible for verifying that the procedure is carried
	out correctly and check the results.
3.3	HOD shall be responsible for overall compliance of this SOP and shall
	ensure that the SOP is reviewed every two years or if required
	earlier.
4.0	Abbreviations and Definitions
4.1	Abbreviations:
	4.1.1
	4.1.2
	4.1.3
	4.1.4
	4.1.5
	4.1.6
	4.1.7
	4.1.8
	4.1.9
5.0	Procedure
5.1	Preparation of solutions: Prepare all the solutions as described below
	and fill the form, label the solutions and assign lot numbers.
	5.1.1
	5.1.2
	5.1.3
	5.1.4
	5.1.5
5.2	Testing Procedure:
	5.2.1
	5.2.2
	5.2.3

			2	Std (10.0 µM/lit)	--	5.0 ml
				5.0 ml		
			3	Std (30.0 µM/lit)	--	5.0 ml
				5.0 ml		
			4	Std (50.0 µM/lit)	--	5.0 ml
				5.0 ml		
			5	Std (80.0 µM/lit)	--	5.0 ml
				5.0 ml		
			6	Std (100.0 µM/lit)	--	5.0 ml
				5.0 ml		
			7	Check sample (80 µmol/lit)	--	5.0 ml
				5.0 ml		
			8	In house control A	--	5.0 ml
				5.0 ml		
			9	In house control B	--	5.0 ml
				5.0 ml		
			10	In house control C	5.0 ml	--
				5.0 ml		
			11	Test sample A	--	5.0 ml
				5.0 ml		
			12	Test sample B	--	5.0 ml
				5.0 ml		
			13	Test sample C	5.0 ml	--
				5.0 ml		
			5.2.3.	The test tube no. 10 and 13 are the color correction		
			2	tubes of respective samples and hence acetyl acetone is		
				not added to these tubes.		
			5.2.3.	After addition of all the components as mentioned above,		
			3	place all these test tubes in a water bath at 37±1° C for		
				50 minutes.		
			5.2.3.	On completion of incubation period, remove the test tubes		
			4	and cool them under running water for 2 minutes.		
			5.2.3.	Read the absorbance of each tube within 30 mins. at 415		
			5	nm under a UV/VIS spectrophotometer.		
			5.2.3.	The standard tube no. 1 of purified water is considered		
			6	to be as blank.		
			5.2.3.	The concentration of formaldehyde in the test sample(s)		
			7	is calculated by comparing results of its absorbance		
				obtained against the plotted calibration curve of known		
				formaldehyde concentration (10 to 100 µM/lit) Vs		
				absorbance difference at 415 nm		
			5.2.3.	Note that all the readings of the assay tubes should be		
			8	taken within 30 mins. of completion of the assay and		
				maximum 5 samples should be analysed in one single assay.		
		5.3	Calculations:			
		5.3.1	Once the absorbance of the standards and the test samples have			
				been taken, calculation shall be done using the excel sheet		
				carrying formulas for the required calculation.		
		5.3.2	Always copy the master excel sheet (sheet carrying formulas for			
				the required calculation), never use the previously used excel		
				sheet for new calculations.		
		5.3.3	Copy the master excel sheet, keep it for use in respective			
				folder and then fill in the required sample details (Batch		
				number), sample dilution (if applicable), name of analyst, date,		
				etc.		
		5.3.4	Proceed for calculations starting from the Calibration curve.			
				The calculations are done according to the equation after		
				subtractions blank OD.		
				Y = mX + b.		
				Where: m = slope		
				b = intercept		
				Y = absorbance		
				X = concentration		
		5.3.5	To determine the formaldehyde concentration in mM/lit following			
				formula is applied,		
				Formaldehyde [Sample Conc. (µM/lit) – Color correction		
				Conc. (µM/lit)]		
				Concentration =		

				(mM/lit) 1000 X		
				Dilution factor		
		5.3.6	Record all the calculations in the data record form and also			
				attach the hard copy of the excel sheet used for calculation		
				along it.		
		5.4	Acceptance criteria:			
		5.4.1	Regression of the standard curve should not be less than 0.99.			
		5.4.2	The calculated check sample concentration of 80 µmol/lit should			
				be within 10% of the true value.		
		5.4.3	The variation in OD values between the two replicates of same			
				sample should not be more the 10%.		
		6.0	Reference			
		6.1	European Pharmacopoeia, volume 5.Free formaldehyde monograph 2.4.18			

[SHEET: Regents]

Regents and chemical required for QC testing of pneumococcal Vaccine | | | | |

1 | Pneumococcal rabbit antiserum for respective serotypes from Staten Serum Institute (SSI) | | | | |

2 | Polyclonal anti-rabbit IgG-HRP conjugated antibody (DAKO Cat No. P0448). | | | | |

3 | Polyclonal anti-mouse IgG-HRP conjugated antibody (DAKO Cat No. P0447).. | | | | |

4 | Pneumococcal purified capsular polysaccharide antigen from ATCC as a standard for respective serotypes . | | | | |

5 | Bovine Serum Albumin (BSA) | | | | |

6 | Tween 20 | | | | |

7 | Tetra Methyl Benzidine (TMB) substrate | | | | |

8 | Concentrated Sulphuric acid | | | | |

9 | Sodium chloride; NaCl | | | | |

10 | Sodium dihydrogen phosphate hepta hydrate; NaH₂PO₄·7H₂O | | | | |

11 | Potassium Di hydrogen phosphate; KH₂PO₄ | | | | |

12 | Potassium chloride; KCl | | | | |

13 | Milli- Q water | | | | |

14 | Anti CWPS Polyclonal rabbit Antisera (Staten Serum Institute, Cat No. 3653) | | | | |

15 | Purified CWPS (Staten Serum Institute, Cat No. 3459) | | | | |

16 | Monoclonal antibodies against Phosphoryl Choline (Staten Serum Institute, Cat No. 87214) | | | | |

17 | Polyclonal Goat anti-mouse Immunoglobulin HRP conjugates (Dako Cat No. P0447) | | | | |

18 | Bovine Serum Albumin (Fraction V, Sigma Cat No. A9647) | | | | |

19 | Dextran | | | | |

20 | Sodium azide | | | | |

21 | Deuterium Oxide | | | | |

22 | Sodium dihydrogen phosphate, dihydrate | | | | |

23 | Disodium hydrogen phosphate, anhydrous | | | | |

24 | Hydrochloric Acid | | | | |

25 | Thymolphthalein | | | | |

26 | Sodium Hydroxide | | | | |

27 | Sodium carbonate | | | | |

28 | Acetyl acetone | | | | |

29 | Dimethylaminobenzaldehyde | | | | |

30 | Glucosamine hydrochloride | | | | |

31 | Mannosamine hydrochloride | | | | |

32 | Absolute Ethanol | | | | |

33 | LAL Reagent Water | | | | |

34 | L-Cysteine hydrochloride | | | | |

35 | Rhamnose | | | | |

36 | Sodium tetraborate | | | | |

37 | Carbazole | | | | |

38 | Glucuronic acid | | | | |

39 | Sodium deoxycholate | | | | |

40 | Hydroxylamine hydrochloride | | | | |

41 | Ferric chloride | | | | |

42 | Acetylcholine chloride | | | | |

43 | Ammonium heptamolybdate tetrahydrate | | | | |

44 | Ascorbic acid | | | | |

45 | Perchloric acid | | | | |

46 | Folin & Ciocalteu's phenol reagent | | | | |

47 | Copper sulphate pentahydrate | | | | |

48 | Sodium tartarate | | | | |

49 | Potassium hydroxide AR grade | | | | |

50 | Ethanol absolute AR grade 99.9% | | | | |

51 | Acetonitrile HPLC grade | | | | |

52 | Orthophosphoric acid AR grade | | | | |

53 | Potassium dihydrogen orthophosphate AR grade | | | | |

54 | formaldehyde standard | | | | |

55 | Deoxycholic acid (DCA) | | | | |

56 | Methanol | | | | |

57 | Acetonitrile | | | | |

58 | Sodium Acetate anhydrous | | | | |

59 | Orthophosphoric Acid | | | | |

60 | Triethyl amine (Make: Fluka, Catalogue No. 90340) | | | | |

61 | Disodium hydrogen phosphate dodecahydrate (Na₂HPO₄·12H₂O) (Make: Sigma Aldrich, Cat No. 71650 or equivalent grade) | | | | |

62 | Resorcinol | | | | |

63 | ATCC Standard of respective Polysaccharide | | | | |

64 | Sodium Deoxycholate | | | | |

65 | Galactose | | | | |

66 | Deoxycholic acid, Sodium Salt (DOC) | | | | |

67 | Sodium azide | | | | |

68 | Salmon DNA | | | | |

69 | CDAP (1-cyano-4-dimethylaminopyridinium tetrafluoroborate) | | | | |

70 | 2-Propanol | | | | |

71 | Hydrochloric acid 37 % | | | | |

72 | Morpholino-ethanesulfonic acid (MES) | | | | |

73 | Ethyl Dimethyl Aminopropyl Carbodiimide | | | | |

74 | Dimethyl Barbituric acid (DMBA) | | | | |

75 | Pyridine | | | | |

76 | Sodium phosphate dibasic dodecahydrate | | | | |

| Sucrose | | | | |

[SHEET: consumables]

List of Pneumococcal Consumables | | | | |

1 | 96 well Maxisorp Nunc ELISA plates (Nunc Cat No. 439454) | | | | |

2 | Micro tips (2-20µl, 20-200µl, 100-1000µl and 50 -300µl) | | | | |

3 | Micro centrifuge tubes (1.5 ml) | | | | |

4 | Centrifuge tubes – 15 ml and 50 ml | | | | |

5 | Lint free paper towel | | | | |

6 | Multichannel auto pipette (50 to 300 µl) | | | | |

7 | Auto pipettes (2 to 20µl, 20 to 200 µl and 100 to 1000µl) | | | | |

8 | Sugar KS-804 (Shodex Column) | | | |
 9 | Sugar KS-805 (Shodex Column) | | | |
 10 | Guard Column KS-G (Shodex) | | | |
 11 | Screw cap HPLC vials with/without inserts | | | |
 12 | Solvac Filter Holder | | | |
 13 | Nylon 6,6 membrane 0.2µm membrane filter | | | |
 14 | Syringe filters 0.2 µm | | | |
 15 | Aluminium Foil | | | |
 16 | Gloves | | | |
 17 | Screw capped test tubes (30 ml) with stand | | | |
 18 | Volumetric Flasks (5ml, 10ml, 50ml & 100ml) | | | |
 19 | Glass pipettes (5ml, 10ml & 25 ml) | | | |
 20 | Screw capped test tubes (5 ml) with stand | | | |
 21 | Volumetric flasks, Amber colored (10ml) | | | |
 22 | BDS Hypersil C18 silica column (250 x 4.6mm, Particle size:5µm | | | |
 23 | Water symmetry CIS,(4.6x 250) mm,5µm. | | | |
 24 | Hydrogen gas (IOLAR) | | | |
 25 | Nitrogen gas (IOLAR) | | | |
 26 | Helium gas (IONAR) | | | |
 27 | Zero gas (IOLAR) | | | |
 28 | Separation Column, e.g. OPTIMA–624–LB–1.80 µm –30 m x 0.32 mm | | | |
 29 | Head space vials and septa | | | |
 30 | Quartz cuvettes | | | |
 31 | 10 and 20 ml thermoresistant test tubes. | | | |
 32 | 25, 50 and 100 ml Beaker | | | |
 | TSKgel G2000 SWxl column; 30 cm x 7.8 mm ID | | | |
 | TSKgel G2000 SWxl column; 30 cm x 7.8 mm ID | | | |
 [SHEET: Sheet3]

1.0 |Objective |
 |To lay down the procedure for Estimation of Phosphorous content in |
 |Pneumococcal polysaccharide samples by Chen method. |
 2.0 |Scope |
 |This Standard Operating Procedure is applicable to all polysaccharide (13 |
 |serotypes) samples, in which phosphorous content is to be determined, which|
 |is received in the Quality control department. |
 3.0 |Responsibility |
3.1	The analyst is responsible for executing the described procedure
	correctly.
3.2	Supervisor is responsible for verifying that the procedure is carried
out correctly and check the results.	
3.3	HOD shall be responsible for overall compliance of this SOP and shall
ensure that the SOP is reviewed every two years or if required	
earlier.	
4.0	Abbreviations and Definitions
4.1	Abbreviations:
	4.1.1
	4.1.2
	4.1.3
	4.1.4
	4.1.5
	4.1.6
	4.1.7
	4.1.8
	4.1.9
4.2	Reagents:
	4.2.1
	4.2.2
	4.2.3
	4.2.4
	4.2.5
	4.2.6
4.3	Equipments and Material
	4.3.1
	4.3.2
	4.3.3
	4.3.4
	4.3.5
	4.3.6
	4.3.7
	4.3.8
5.0	Procedure
5.1	Preparation of solutions: Prepare all the solutions as described below
and fill the form, label the solutions and assign lot numbers.	
	Note: All the below mentioned solutions are prepared freshly prior to
use otherwise mentioned.	
5.1.1	Stock solution of Phosphorous (100µg/ml):
5.1.2	Working Standard solution (32µg/ml):
5.1.3	Ammonium heptamolybdate tetrahydrate solution (2.5% w/v):
5.1.4	6N Sulphuric acid :
5.1.5	10% w/v Ascorbic acid :
5.1.6	Ammonium Molybdate reagent:
5.1.7	Digestion Mixture :
5.2	Preparation of Calibration Curve:
5.2.1	The concentration of Phosphorus in the test sample(s) is
	calculated by comparing results of its absorbance obtained
	against the plotted calibration curve of known Phosphorus
	concentration (2-32 µg) Vs absorbance at 820 nm. The Phosphorus
	calibration curve (2-32 µg) is prepared as follows:

||| |32 |1000 |4.0 |

||5.3 |Sample preparation: |

|| |5.3.1 |In case of dry polysaccharide prepare a solution containing |

||| |about 1-5 mg/ml of polysaccharide in Lal reagent water and use |

||| |it as test sample. |

|| |5.3.2 |Take 1ml volume of the sample in duplicate in tubes. |

|| |5.3.3 |Add 500µl of Digestion mixture to each sample tube and vortex |

||| |well. |

|| |5.3.4 |Now heat the sample test tubes directly on flame with |

||| |intermittent shaking till the sample become clear and fumes |

||| |stop. |

|| |5.3.5 |Allow the tubes to cool down to room temperature. |

|| |5.3.6 |Wash the sample tubes five times vigorously with 5ml each of |

||| |purified water and transfer to 25ml volumetric flasks. |

|| |5.3.7 |Make up the volume with purified water to make any necessary |

||| |correction. |

|| |5.3.8 |Procedure for Untreated sample preparation: Take 1ml volume of |

||| |the sample and make it 25ml with purified water |

||5.4 |Test Procedure: |

|| |5.4.1 |Take 5ml of the standard dilutions from calibration curve, Test|

||| |Sample, Untreated sample and Blank (Purified water) in |

||| |duplicates in separate test tubes. |

|| |5.4.2 |Add 5ml of Ammonium molybdate reagent in all the test tubes. |

|| |5.4.3 |Vortex the test tubes and incubated at 37°C ± 2°C (dry heat) |

||| |for two hours. |

|| |5.4.4 |After incubation keep the tubes at 2-8 °C for 5 minutes. |

|| |5.4.5 |Observe for bluish color at the end of reaction in treated |

||| |sample and standards. |

|| |5.4.6 |Take absorbance of the sample at 820nm against the blank using |

||| |spectrophotometer. |

|| |5.4.7 |Note that all the readings of the assay tubes should be taken |

||| |within 30 mins. of completion of the assay and maximum 5 |

||| |samples should be analysed in one single assay. |

||5.5 |Calculations: |

|| |5.5.1 |Once the absorbance of the standards and the test samples have |

||| |been taken, calculation shall be done using the excel sheet |

||| |carrying formulas for the required calculation. |

|| |5.5.2 |Always copy the master excel sheet (sheet carrying formulas for|

||| |the required calculation), never use the previously used excel |

||| |sheet for new calculations. |

|| |5.5.3 |Copy the master excel sheet, keep it for use in respective |

||| |folder and then fill in the required sample details (Batch |

||| |number), sample dilution (if applicable), name of analyst, |

||| |date, etc. |

|| |5.5.4 |Proceed for calculations starting from the Calibration curve. |

||| |The calculations are done according to the equation. |

||| | $Y = mX + b$. |

||| |Where: m = slope |

||| |b = intercept |

||| |Y = absorbance |

||| |X = concentration |

|| |5.5.5 |Concentration of Phosphorus = |

||| |(Conc. of treated sample – Conc. of untreated sample) x |

||| |Dilution factor. |

|| |5.5.6 |Use EXCEL programme for the preparation of calibration curve |

||| |and Calculation of the results. |

|| |5.5.7 |In case of dry polysaccharide, |

||| |Phosphorus (%) in polysaccharide on dry weight basis: |

|||| |

|||| |Phosphorus in dry polysaccharide. (µg/ml) |

|||| | = Phosphorus in Polysacc. sol. (µg/ml) X 1 – {Moisture (%)|

|||| | / 100} |

|||| |

|||| |

|||| |

|||| |Phosphorus in dry polysaccharide (µg/ml) |

|||| |Phosphorus (%) = _____ x |

|||| |100 |

|||| |Conc. of polysaccharide solution (µg/ml) |

||5.6 |Acceptance criteria: |

|| |5.6.1 |The test is valid if Regression of the standard curve is ≥ 0.99|

||| |and the percentage of Coefficient of Variation is ≤ 15% within |

||| |the replicates. |

|| |5.6.2 |Phosphorous content in Purified Pneumococcal polysaccharide |

||| |should meet the laid specifications for each serotype. |

||6.0 |Forms and Records |

|| |Determination of Phosphorus content - Annexure 1 |

||7.0 |Reference |

|| |7.1 |WHO TRS No. 977 Annexure 3: Recommendations to assure the quality, |

||| |safety and efficacy of pneumococcal conjugate vaccines. |

|| |7.2 |European Pharmacopoeia, Volume 5, Phosphorus in Polysaccharide |

||| |Vaccines 2.5.18. |

|| |7.3 |Chen, P.S., T.Y. Toribara and H. Warner. Determination of phosphorus, |

|| |Analytical chemistry, 28:1756-8(1956). |

1.0 |Objective: |
 |To lay down the procedure for determine the human coagulation factor II |
 |activity in Factor IX. |
 2.0 |Scope: |
 |This Standard Operating Procedure is applicable to all the In- process and |
 |finished product samples. |
 3.0 |Responsibility |
 |3.1 |The QC analyst shall be responsible for executing the described |
 |procedure correctly. |
 |3.2 |The QC supervisor shall be responsible for verifying that the |
 |procedure is complied with and the results are checked. |
 |3.3 |The QC HOD shall be responsible for overall compliance of this SOP. |
 4.0 |Abbreviations and Definitions: |
 |Not applicable |
 5.0 |Procedure: |
5.1	Equipment, Materials and Reagents		
	5.1.1	Coagulometer	
	5.1.2	STA Deficient II	
	5.1.3	Neoplastine CI Plus	
	5.1.4	Owren koller buffer	
	5.1.5	Unicalibrator	
5.2	Preparation of solutions:		
	5.2.1	STA Deficient II:	
		5.2.1.	Freeze dried mixture of human serum and adsorbed bovine
		1	plasma, the latter providing factor V and fibrinogen to
		the test system.	
		5.2.1.	Preparation: Reconstitute each vial with 1 mL of purified
		2	water and allow to stand at room temperature (18-25 ° C)
		for 30 min. Then swirl the vial gently to obtain a	
		homogenous solution.	
		5.2.1.	Storage: The reagent in intact vial is stable until the
		3	expiration date indicated on the box label, when stored
		at 2-8° C.	
		5.2.1.	Stability after reconstitution: 8 hours at 37° C.
		4	
	5.2.2	Neoplastine CI Plus:	
		5.2.2.	Reagent 1: Lyophilized thromboplastin prepared from fresh
		1	rabbit cerebral tissues. The ISI (International sensitive
		index) value of Neoplastine CI plus is determined against	
		a secondary standard of the RBT [Rabbit brain	
		thromboplastin].	
		Note: See assay value insert provided in the box for the exact	
		ISI value of the lot being used.	
		5.2.2.	Once reconstituted, the Neoplastine CI Plus exhibits a
		2	low turbidity and does not sediment. The Neoplastine CI
		Plus reagent contains a specific heparin inhibitor. Any	
		prolongation of the prothrombin time is, therefore,	
		related to a real deficiency of factor II, V, VII, X and	
		/ or Fibrinogen.	
		5.2.2.	Reagent 2: Aqueous solvent containing calcium, 2 mL per
		3	vial.
		5.2.2.	Preparation of Neoplastin C1 plus reagent from reagent-1
		4	and reagent-2 : Transfer the entire contents of one vial
		of reagent 2 (R2) into one vial of Reagent 1 (R1) of the	
		same kit. Allow the reconstituted reagent to stand at	
		room temperature (18-25° C) for 30 min. Then swirl the	
		Neoplastin C1 plus reagent vial gently to obtain a	
		homogenous suspension.	
		5.2.2.	Storage: The reagents in intact vials are stable until
		5	the expiration date indicated on the box label, when
		stored at 2-8° C.Once reconstituted, the reagent 1	
		remains stable for 8 hours at 37° C.	
		48 hours at 20±5° C.	
		8 days at 2-8° C.	
		Do not freeze	
	5.2.3	Reference standard (Unicalibrator):	
		5.2.3.	The levels of factor II is determined against the
		1	secondary International standard.
		5.2.3.	Unicalibrator is a plasma intended for use as a
		2	calibration plasma for the assays of the following
		coagulation parameters: Prothrombin time (if to be	
		expressed as %), fibrinogen (Clauss method), factors II,	
		IV, VII, VIII, IX, X, XI, XII, anti-thrombin III (AT III),	
		protein C, protein S, plasminogen and antiplasmin.	
		5.2.3.	Preparation: Reconstitute each vial with exactly 1 mL of
		3	purified water. Allow the reconstituted material to stand
		at room temperature (18-25°C) for 30 min. Then swirl the	
		vial gently before use.	
		5.2.3.	Storage: The reagents in intact vials are stable until
		4	the expiration date indicated on the box label when
		stored at 2-8° C.	
		Once reconstituted; it remains stable for 4 hours at	
		20±5° C.	
		Do not freeze.	
5.3	Pre-Dilution of reference standard and sample:		
	5.3.1	Reference solution preparation:	
		Dilution fold	
		Unicalibrator	
		Owren koller buffer (µL)	
		Total Volume (µL)	

				1:10	
				25 µL of neat Unicalibrator	
				225	
				250	
				1:20	
				125 µL of 1:10 dilution	
				125	
				250	
				1:40	
				125 µL of 1:20 dilution	
				125	
				250	
				1:80	
				125 µL of 1:40 dilution	
				125	
				250	
				Perform serial dilutions of reference standard from 1:10	
				dilution as per the below	
				table.	
				5.3.2	Test Solution preparation:
				Dilute the test sample 1:10 folds by taking 50 µL of Neat sample	
				and 450 µL of Owren koller buffer.	
				5.4	Test method:
				5.4.1	Switch on the Coagulometer 15 minutes prior to analysis
				5.4.2	Place all the reagents in appropriate slots of the Coagulometer
				for warming at 37°C (5 minutes prior to analysis).	
				5.4.3	Place a Cuvette strip with required number of cuvettes in the
				incubation area and prewarm at 37°C at least for 3 minutes. Then	
				dispense one magnetic ball to each cuvette.	
				5.4.4	Add 50 µL of each of the dilution of reference standard, 50 µL
				of STA Deficient II into each cuvette.	
				5.4.5	Immediately start the timer corresponding to the incubation
				column for an incubation of 60 sec.	
				5.4.6	After 50 sec beep sound was produced. Then transfer the cuvette
				from the incubation chamber to the test column area.	
				5.4.7	After 60 sec of incubation time, add 100 µL of Neoplastine CI
				Plus to each cuvette by pressing timer to start the measurement.	
				5.4.8	Note down the clotting time in seconds.
				5.4.9	Then feed the Percentage of reference standard Factor II
				corresponding to its clotting time for each dilution in the	
				instrument.	
				Note: Instrument automatically plots the calibration curve.	
				5.4.10	Then perform the test for the test samples as described above
				and note down the clotting time as well as Factor II activity	
				which comes in the print out.	
				5.5	Calculations:
				5.5.1	As calibration data is fed in Coagulometer, the instrument
				itself plots test sample clotting time on standard curve and	
				calculates Factor II activity.	
				5.5.2	Calculate % activity of Factor II to that of Factor IX using
				following formula	
				[pic]	
				6.0	References:
				6.1 Indian Pharmacopeia	

1.0	Objective			
	To lay down the procedure for determination of conjugate antigen content in			
	Pneumococcal conjugate vaccine.			
2.0	Scope:			
	This Standard Operating Procedure is applicable to testing conjugate antigen			
	content of 13-valent pneumococcal conjugate vaccine.			
3.0	Responsibility:			
	3.1	The analyst is responsible for executing the described procedure		
	correctly.			
	3.2	The supervisor of the laboratory shall be responsible for verifying		
	that the procedure is carried out correctly and for checking the			
	results.			
	3.3	The departmental Head shall be responsible for the overall compliance		
	of the SOP.			
4.0	Abbreviations and Definitions:			
	4.1	QA	: Quality Assurance	
	4.2	QC	: Quality Control	
	4.3	SOP	: Standard Operating Procedure	
	4.4	PBS	: Phosphate Buffer saline	
	4.5	ELISA	: Enzyme Linked Immunosorbent assay	
	4.6	CWPS	: Cell Wall Polysaccharide	
	4.7	BSA	: Bovine Serum Albumin	
	4.8	TMB	: Tetra Methyl Benzidine	
	4.9	ATCC	: American Type culture collection	
	4.10	HRP	: Horse reddish peroxidase	
	4.11	WFI	: Water for Injection	
	4.12	PBS	: Phosphate buffer saline	
5.0	Procedure			
	5.1	General Instructions:		
		Wear hand gloves while performing the assay.		
	5.2	Equipments and Materials:		
		5.2.1	96 well Maxisorp Nunc ELISA plates (Nunc Cat No. 439454)	
		5.2.2	Micro plate ELISA reader	
		5.2.3	Micro tips (2-20µl, 20-200µl, 100-1000µl and 50 -300µl)	
		5.2.4	Micro centrifuge tubes (1.5 ml)	
		5.2.5	Centrifuge tubes – 15 ml and 50 ml	
		5.2.6	Lint free paper towel	
		5.2.7	Vortex mixer	
	5.3	Reagents:		
		5.3.1	Pneumococcal rabbit antiserum for respective serotypes from	
			Staten Serum Institute (SSI)	
			Serotype	Cat No.
			Type 1	16744
			Type 3	16746
			Type 4	16747
			Type 5	16748
			Type 6A	16923
			Type 6B	16924
			Type 7F	16925
			Type 9V	16937
			Type 14	16753
			Type 18C	16958
			Type 19A	16963
			Type 19F	16962
			Type 23F	16968
		5.3.2	Polyclonal anti-rabbit IgG-HRP conjugated antibody (DAKO Cat	
			No. P0448).	
		5.3.3	Polyclonal anti-mouse IgG-HRP conjugated antibody (DAKO Cat No.	
			P0447).	
		5.3.4	Pneumococcal purified capsular polysaccharide antigen from ATCC	
			as a standard	
			for respective serotypes	
			Pneumococcal Serotype Antigen	ATCC Cat No.
			Type 1	15-X
			Type 3	17-X
			Type 4	18-X
			Type 5	57-X
			Type 6A	14-X
			Type 6B	58-X
			Type 7F	27-X
			Type 9V	59-X
			Type 14	23-X
			Type 18C	28-X
			Type 19A	64-X
			Type 19F	24-X
			Type 23F	25-X
			Type 22F	528-X
		5.3.5	Purified Cell Wall Polysaccharide (CWPS) Staten Serum Institute	
			(SSI) Cat No.: 3459.	
		5.3.6	Bovine Serum Albumin (BSA)	
		5.3.7	Tween 20	
		5.3.8	Tetra Methyl Benzidine (TMB) substrate	
		5.3.9	Concentrated Sulphuric acid	
		5.3.10	Sodium chloride; NaCl	
		5.3.11	Sodium dihydrogen phosphate hepta hydrate; NaH₂PO₄·7H₂O	
		5.3.12	Potassium Di hydrogen phosphate; KH₂PO₄	
		5.3.13	Potassium chloride; KCl	
		5.3.14	Milli- Q water	
	5.4	Preparation of solutions:		
		Prepare the solutions as described below. Label the solutions and		

|| assign lot numbers. |

|| 5.4.1 |Preparation of Standard Pneumococcal antigen: For preparation |

|| |of standard stock solution of pneumococcal antigens, add 2 ml |

|| |of WFI in 2 mg vial of standard and keep it overnight at 2-8°C |

|| |with shaking. Make aliquots of 100 µl. Store the aliquots at |

|| |-70°C for further use. |

||||

|| 5.4.2 |Phosphate buffer saline (pH 7.2 – 7.4) (10X) |

|| |5.4.2.1|Weigh 80.0 g NaCl, 20.61 g NaH₂PO₄·7 H₂O, 3.14 g of |

|| |KH₂PO₄ and 1.60 g KCl, dissolve in 800ml of Milli Q |

|| |water. |

|| |5.4.2.2|Keep the beaker on a magnetic stirrer to dissolve the |

|| |contents. |

|| |5.4.2.3|Make up the volume to 1L with milli Q water and |

|| |sterilize the solution by filtering it through 0.22 µm |

|| |filter and store the solution at room temperature. |

|| 5.4.3 |Wash buffer and coating buffer (1X PBS): Add 900 ml of sterile |

|| |Milli Q water to 100 ml of 10X PBS. Store the solution at room |

|| |temperature. |

||||

|| 5.4.4 |Wash / diluent and Antibody buffer (PBST): Add 50 µl of Tween |

|| |20 to 100 ml of 1X PBS solution (PBS with 0.05 % Tween 20). Mix|

|| |the solution thoroughly so that Tween 20 gets uniformly |

|| |dissolved in PBS. |

|| 5.4.5 |Blocking buffer: (5% BSA in PBST): Add 5 g of BSA in 100 ml |

|| |PBST solution to give a final concentration of 5% BSA in PBST. |

|| 5.4.6 |Secondary antibody: Working concentration of secondary antibody|

|| |is 1:2000 dilutions, prepared in diluent. The solution should |

|| |be freshly prepared just before the use. |

|| 5.4.7 |Stop solution (1N H₂SO₄): Add 2.8 ml of Conc. H₂SO₄ to 97.2 ml |

|| |of Milli Q water to make up the volume to 100ml. |

|| 5.5 | Test Procedure |

|| 5.5.1 |Desorption treatment of test sample |

|| |5.5.1.1|Take 10 vials of 13-valent pneumococcal conjugate |

|| |vaccine and put them into 15ml centrifuge tube. Add |

|| |200µl of 0.6mo1/L sodium hydroxide solution into 4000µl|

|| |of above samples, and quickly add 200µl of 0.2mol/l |

|| |citric acid solution within 10s, mix well. The |

|| |desorption solution shall be completely clear and |

|| |transparent. |

|| |5.5.1.2|Take 4500µl of 13-valent free polysaccharide reference |

|| |substance and put it into 15ml centrifuge tube, add |

|| |200µl of 0.6mol/L sodium hydroxide solution into 4000µl|

|| |above samples, and quickly add 200µl of 0.2mol/l citric|

|| |acid solution within 10s, mix well. The desorption |

|| |solution shall be completely clear and transparent. |

|| |5.5.1.3|it is applicable to the content determination of |

|| |conjugate antigen except for type 3 in 13-valent |

|| |pneumococcal conjugate vaccine. |

|| |Free polysaccharide separation of samples: |

|| |Take 2000µl of D and Y respectively and put each into |

|| |50ml centrifuge tube, add 500µl of 5mol/L sodium |

|| |chloride solution, mix well, shake in ice water bath |

|| |at 150rpm for 30min. |

|| |5.5.1.4 |Add 10ml of absolute ethanol into each centrifuge |

|| |tube, mix well, and shake for 30min at 150rpm in ice |

|| |water bath. |

|| |5.5.1.5 |Carefully remove all the supernatant, add 2ml of |

|| |purified water into the centrifuge tube, and mix well.|

|| 5.5.3 |Determination of polysaccharide content: |

|| |5.5.3.1|Take 300µl of solution D, add 900µl of diluent 1 |

|| |(DIL1), mix well to get 4 times dilute solution, |

|| |determine the polysaccharide content on the IMMAGE |

|| |800, input 4.4 (i.e. 4×1.1) of off-line dilution |

|| |multiple. If the result is beyond the detection range,|

|| |adjust the dilute multiple to A, and input A×1.1 of |

|| |off-line dilution multiple. |

|| |5.5.3.2|Take 300µl of solution Y, add 600µl of diluent 1 |

|| |(DIL1), mix well to obtain 3 times dilute solution, |

|| |determine the polysaccharide content (except type 6B) |

|| |on the IMMAGE 800, input 3.3 (i.e. 3×1.1) of off-line |

|| |dilution multiple. Take 200µl of solution Y, add |

|| |1000µl of diluent 1 (DIL1), mix well to obtain 6 times|

|| |dilute solution, measure the polysaccharide content |

|| |(type 6B) on the IMMAGE 800, input 6.6 (i.e. 6×1.1) of|

|| |off-line dilution multiple. |

|| |Take 400µl of solution DS, add 800µl of diluent 1 |

|| |(DIL1), mix well to obtain 3 times dilute solution, |

|| |determine polysaccharide content on the IMMAGE 800, |

|| |input 3.3 (i.e. 3×1.1) for off-line dilution ratio. If|

|| |the result is beyond the detection range, adjust the |

|| |dilution ratio to A, and input A×1.1 for off-line |

|| |dilution multiple. |

|| 5.5.4 |1st Washing: |

|| |5.5.4.1|Discard the blocking solution after completion of one |

|| |hour of incubation period by flicking. |

|| |5.5.4.2|Add 300 µl of PBST to each well and discard the buffer|

|| |by flicking. Tap the plate on a lint free paper towel.|

|| |Ensure that the buffer is removed completely. |

|| |5.5.4.3|Repeat the procedure for two more times and then one |

|| |time with 300 µl of PBS. |

|| 5.5.5 |Addition of primary antibody: |

			5.5.5.1	The primary antibody dilutions for all the 13 serotype
			are prepared one hour before addition of blocking	
			buffer as mentioned in section 5.5.2.	
			5.5.5.2	Add 100 µl/well of primary antibody of all the 13
			serotypes as per pattern mentioned below.	
			Addition of Primary Antibody against Serotype	
			1	
		5.5.6	2nd washing:	
		5.5.6.1	Discard the primary antibody solution after completion	
		of 90 mins.of incubation period by flicking.		
		5.5.6.2	Add 300 µl of PBST to each well and discard the buffer	
		by flicking. Tap the plate on a lint free paper towel.		
		Ensure that the buffer is removed completely.		
		5.5.6.3	Repeat the procedure for three more times and then one	
		time with 300 µl of PBS.		
		5.5.7	Addition of secondary antibody:	
		5.5.7.1	Prepare the recommended dilutions of secondary	
		antibody.		
		5.5.7.2	For serotype 6B prepare 1:2000 dilution of anti-mouse	
		IgG-HRP conjugated antibody. Prepare the dilutions in		
		adsorption solution.		
		5.5.7.3	For all other serotypes prepare 1:2000 dilution of	
		anti-rabbit IgG-HRP conjugated antibody. Prepare the		
		dilutions in adsorption solution.		
		5.5.7.4	Add 100 µl of secondary antibody solution to	
		respective wells and incubate the plate for 60 minutes		
		at 37 ± 2°C.		
		5.5.8	3rd washing:	
		5.5.8.1	Discard the secondary antibody solution after	
		completion of 60 mins. of incubation period by		
		flicking.		
		5.5.8.2	Add 300 µl of PBST to each well and discard the buffer	
		by flicking.		
		Tap the plate on a lint free paper towel. Ensure that		
		the buffer is removed completely.		
		5.5.8.3	Repeat the procedure for three more times and then one	
		time with 300 µl of PBS.		
		5.5.9	Addition of TMB substrate:	
		5.5.9.1	Add 100 µl TMB to each well by a multichannel pipette.	
		Take care that the TMB solution is not exposed to		
		direct light during addition.		
		5.5.9.2	Cover the plate and incubate at room temperature in	
		dark preferably in a drover for next 15 mins.		
		5.5.10	Addition of Stop Solution:	
		5.5.10.1	On completion of 15 minutes incubation period of TMB.	
		Carefully remove the plate out of drover. Take care		
		that the plate is not exposed to direct light during		
		addition.		
		5.5.10.2	Add 100 µl of stop solution over the TMB solution to	
		each well in the dark.		
		5.5.11	Reading of plate and Data analysis:	
		5.5.11.1	Read the absorbance of plate at 450 and 630 nm	
		wavelength on a ELISA reader. Calculate delta OD of		
		the readings.		
		5.5.11.2	Take care to take the readings after 5 mins and within	
		15 mins. of addition of stop solution.		
		5.5.11.3	Get the average ODs of duplicate wells and compile the	
		data sheets comprising of sample and ATCC standard.		
		5.5.12	Acceptance Criteria	
		5.5.12.1	OD values of Positive control and test sample should	
		not be less than 0.8		
		5.5.12.2	OD values of Negative control should not be more than	
		0.3		
6.0	Reference			
WHO TRS No. 977 Annexure 3: Recommendations to assure the quality, safety				
and efficacy of pneumococcal conjugate vaccines.				

1.0 |Objective: |
 |To lay down the procedure for quantifying Total and unconjugated (free) |
 |PRP concentration. |
 2.0 |Scope: |
 |This Standard Operating Procedure is applicable for the analysis of all |
 |the samples of PRP and PRP-CRM bulk conjugate of Haemophilus Influenzae |
 |type b in which Total and unconjugated (free) PRP concentration is to be |
 |determined. |
 3.0 |Abbreviations and Definitions |
3.1	Abbreviations:		
	3.1.1	CV	:Coefficient of variation
	3.1.2	g	:Gram
	3.1.3	mg	:Milligram
	3.1.4	mL	:Millilitre
	3.1.5	µL	:Micro litre
	3.1.6	µg	:Microgram
	3.1.7	nm	:Nanometer
	3.1.8	PRP	:Polyribosyl Ribitol Phosphate
	3.1.9	TT	:Tetanus Toxoid
	3.1.10	TNBS	: 2, 4, 6-Trinitrobenzenesulfonic acid
	3.1.11	DMF	: Dimethylformamide
3.2	Definitions:		
	Not Applicable.		
4.0	Procedure:		
4.1	0.5 g/l Ferric chloride solution:		
	4.1.1	Weigh 50 mg of Ferric Chloride hexahydrate and transfer it to	
		100 mL volumetric flask.	
	4.1.2	Dissolve and make up the volume to 100 mL with concentrated	
		hydrochloric acid.	
	4.1.3	Store the solution in screw cap amber bottle, for a maximum 6	
		months at room temperature.	
4.2	100 g/l Orcinol solution:		
	4.2.1	Weigh 1g Orcinol monohydrate and transfer it to 10 mL	
		volumetric flask.	
	4.2.2	Dissolve and make up the volume to 10 mL with absolute	
		ethanol.	
	4.2.3	Store the solution in a screw cap amber bottle at 2 to 8 °C,	
		for maximum of 15 days.	
4.3	0.1% w/v TNBS-10% DMF:		
	4.3.1	Take 1 mL of commercial TNBS solution (1%w/v in DMF) and	
		transfer it to a 10 mL volumetric flask.	
	4.3.2	Make up the volume to 10 mL with purified water.	
	4.3.3	Prepare fresh prior to use.	
4.4	0.25 g/1 Ribose Solution:		
	4.4.1	Weigh 25 mg of D-Ribose and transfer it to a 100 mL	
		volumetric flask.	
	4.4.2	Dissolve in purified water and make up the volume to 100 mL	
		with purified water.	
	4.4.3	Prepare fresh prior to use.	
4.5	25 µg /mL Ribose Solution:		
	4.5.1	Take 1 mL of 0.25 g/1 Ribose solution and transfer it to a 10	
		mL volumetric flask.	
	4.5.2	Make up the volume to 10 mL with purified water.	
	4.5.3	Prepare fresh prior to use.	
4.6	1% w/v Sodium deoxycholate (DOC):		
	4.6.1	Dissolve 250 mg sodium deoxycholate in 25 mL purified water	
		and make up the volume to 25 mL.	
	4.6.2	Adjust the pH to 6.8 by adding about 100 µL of 1M	
		hydrochloric acid drop wise.	
	4.6.3	The solution is prepared freshly.	
4.7	1 M Hydrochloric acid:		
	4.7.1	Take 85 mL of concentrated hydrochloric acid and make up the	
		volume to 1000 mL with purified water.	
	4.7.2	The solution should be stored at room temperature for maximum	
		6 months.	
4.8	Preparation of Calibration Curve:		
	4.8.1	Prepare calibration curve (2.5-25 µg) with freshly prepared	
		ribose solution (25 µg /mL) in previously labelled test	
		tubes of 20 mL.	
	4.8.2	Calibration Curve (2.5 - 25µg) is prepared in duplicate as	
		follow:	
		Ribose content	2.5
		(µg)	
		25 µg/mL Ribose	100
		(µl)	
		Purified water	900
		(µl)	
	4.8.3	Make up the volume to 2 mL using purified water.	
	Note: The Calibration curve is accepted if it is meeting the		
	parameters of correlation coefficient i.e. ≥ 0.990 .		
4.9	Preparation of sample for total PRP content:		
	4.9.1	Take sample volume such that the sample volumes used in test	
		contain 2.5 to 25 µg of ribose i.e. lies within the	
		calibration curve.	
	4.9.2	Make up the volume to 2 mL using purified water respectively.	
	Note: The sample volume can vary depending upon the ribose		
	concentration. If there is any sample outside the range of the		
	calibration curve, prepare an appropriate dilution of the sample.		
4.10	Unconjugated (Free) PRP:		
	4.10.1	Take 1 mL of sample (dilute if required) having about	

		100-150 µg PRP in micro centrifuge tube in duplicate, add
		100 µL of 1% DOC and mix it.
	4.10.2	Put this tube in an ice bath for 30 minutes.
	4.10.3	Take out and add 50 µL 1M HCl and mix thoroughly.
	4.10.4	Centrifuge the mixture at 4°C for 15 minutes at 4000 RPM.
	4.10.5	Take 1 mL of supernatant in 20 mL labeled test tube.
	4.10.6	Make up the volume to 2 mL using purified water
		respectively.
	4.10.7	The test is performed in duplicate for each sample.
	4.11	Treat 2 mL purified water as blank in thermoresistant test tube.
	4.12	Add 2 ml of 0.5 g/l Ferric Chloride solution to tubes of calibration
		curve, Blank and samples.
	4.13	Vortex and add 0.2 mL of 100 g/l Orcinol Solution to all test tubes.
	4.14	Close the tubes with aluminium foil and place the test tubes in
		boiling water bath for 20 minutes.
	4.15	Take out the tubes and cool in iced water.
	4.16	Allow the tubes to come at room temperature and set the wavelength
		in spectrophotometer at 670 nm. Zero the absorbance of blank and
		then measure the absorbance of calibration curve and test samples.
	4.17	Calculations:
	4.17.1	Once the absorbance of the standards and the test samples
		have been taken, calculation shall be done using the excel
		sheet carrying formulas for the required calculation.
	4.17.2	Always copy the master excel sheet (sheet carrying formulas)
		for the required calculation), never use the previously
		used excel sheet for new calculations.
	4.17.3	Copy the master excel sheet, keep it for use in respective
		folder and then fill in the required sample details for
		total & free sample treatments, sample dilution, name of
		analyst, date, etc.
	4.17.4	Enter the absorbance for calculations starting from the
		Calibration curve prepared between 2.5 and 25 g of Ribose
		and test samples. The calculations are made according to
		the equation of minimum squares.
		$Y = mX + b$.
		Where: m = slope
		b = intercept
		Y = absorbance
		X = concentration
		(Multiply by dilution factor)
	4.17.5	Calculation of actual dilution for free PRP:
		Sample volume taken = 1000µL
		Total volume = 1150µL (1000µL sample + 100µL 1% DOC + 50µL
		1M HCl)
		Correction Factor = $1150\mu\text{L} / 1000\mu\text{L} = 1.15$
		Actual dilution = dilution factor x 1.15
	4.17.6	Calculate the content of ribose in the test solution for
		each volume tested and PRP content by the formula :
		PRP content = Ribose content / 0.434
	4.17.7	Record all the calculations in the data record form and
		also attach the hard copy of the excel sheet used for
		calculation along it.
	4.18	Acceptance criteria:
		The coefficient of variation (CV) between the replicates of the
		samples should be $\leq 5\%$.
		The free PRP Oligosaccharide content should not be more than 25% of
		total oligosaccharide content.
	5.0	References:
	5.1	Biologicals (1998) 26, 33-38 – A simple and rapid method for
		measuring unconjugated capsular polysaccharide (PRP) of Haemophilus
		influenzae type b in PRP-TT conjugate vaccine.
	5.2	European pharmacopoeia.

1.0 |Objective |
 |To lay down the procedure for determination of Hexosamine content in |
 |Pneumococcal Polysaccharide samples. |
 2.0 |Scope |
 |This Standard Operating Procedure is applicable to all the in process and |
 |polysaccharide samples of pneumococcal serotypes 4, 5, 9V, 14, 19A and 19F |
 |in which the Hexosamine content is to be determined. |
 3.0 |Responsibility |
 |3.1 |The analyst is responsible for executing the described procedure |
 |correctly. |
 |3.2 |Supervisor is responsible for verifying that the procedure is carried |
 |out correctly and check the results. |
 |3.3 |HOD shall be responsible for overall compliance of this SOP and shall |
 |ensure that the SOP is reviewed every two years or if required |
 |earlier. |
 4.0 |Abbreviations and Definitions |
4.1	Abbreviations:	
4.1.1	mg	:milligram
4.1.2	g	:gram
4.1.3	µg	:microgram
4.1.4	L	:Litre
4.1.5	µl	:microliter
4.1.6	CV	:Coefficient of Variation
4.1.7	QA	:Quality Assurance
4.1.8	QC	:Quality Control
4.1.9	HOD	:Head of Department
4.1.10	DMAB	:Dimethylaminobenzaldehyde
4.2	Reagents:	
4.2.1	Hydrochloric Acid	
4.2.2	Thymolphthalein	
4.2.3	Sodium Hydroxide	
4.2.4	Sodium carbonate	
4.2.5	Acetyl acetone	
4.2.6	Dimethylaminobenzaldehyde	
4.2.7	Glucosamine hydrochloride	
4.2.8	Mannosamine hydrochloride	
4.2.9	Absolute Ethanol	
4.2.10	LAL Reagent Water	
4.3	Equipment and materials	
4.3.1	UV/VIS Spectrophotometer	
4.3.2	Water Bath	
4.3.3	Vortex	
4.3.4	Screw capped test tubes (30 ml) with stand	
4.3.5	Volumetric Flasks (5ml, 10ml, 50ml & 100ml)	
4.3.6	Glass pipettes (5ml, 10ml & 25 ml)	
4.3.7	Auto pipettes with Tips (20-200µl, 100 to 1000 µl & 1000 to	
	5000 µl)	
4.3.8	Centrifuge Tubes (15ml & 50ml)	
5.0	Procedure	
5.1	Preparation of solutions:	
Prepare all the solutions as described below and fill the form, label		
the solutions and assign lot numbers.		
5.1.1	8M Hydrochloric acid:	
5.1.1.1	Take 69.0 ml of concentrated Hydrochloric acid in 100ml	
	volumetric flask.	
5.1.1.2	Make up the volume with purified water.	
5.1.1.3	The solution can be stored at Room temperature for a	
	maximum of 6 months.	
5.1.2	0.5% Thymolphthalein solution:	
5.1.2.1	Weigh 25mg of Thymolphthalein in a 5ml volumetric flask.	
5.1.2.2	Dissolve it in absolute ethanol and make up the volume	
	with the same solvent.	
5.1.2.3	The solution is prepared freshly prior to use.	
5.1.3	5N Sodium Hydroxide solution:	
5.1.3.1	Weigh 20g of sodium hydroxide in a 100ml volumetric	
	flask.	
5.1.3.2	Dissolve in purified water and make up the volume with	
	the same solvent.	
5.1.3.3	The solution can be stored at Room temperature for a	
	maximum of 6 months.	
5.1.4	1M Hydrochloric acid:	
5.1.4.1	Take 8.6 ml of hydrochloric acid in a 100ml volumetric	
	flask.	
5.1.4.2	Mix with purified water and make up the volume with the	
	same solvent.	
5.1.4.3	The solution can be stored at Room temperature for a	
	maximum of 6 months.	
5.1.5	5.3% Sodium carbonate solution:	
5.1.5.1	Weigh 5.3gm of sodium carbonate in a 100ml volumetric	
	flask.	
5.1.5.2	Dissolve in purified water and make up the volume with	
	the same solvent.	
5.1.5.3	The solution can be stored at Room temperature for a	
	maximum of 6 months.	
5.1.6	2.0% Acetyl acetone reagent:	
5.1.6.1	Take 1ml of acetyl acetone in a 50ml volumetric flask.	
5.1.6.2	Mix with 5.3% Sodium carbonate solution and make up the	
	volume with the same solvent.	
5.1.6.3	The solution is prepared freshly prior to use.	
5.1.7	2.67% Dimethylaminobenzaldehyde solution (DMAB):	

			5.1.7.1	Weigh 800mg of dimethylaminobenzaldehyde in a 50ml			
				centrifuge tube.			
			5.1.7.2	Dissolve in 15ml of absolute ethanol and 15ml of 8M			
				hydrochloric acid.			
			5.1.7.3	The solution is prepared freshly prior to use.			
		5.1.8	1000 µg/ml Glucosamine working solution:				
			5.1.8.1	Weigh 12mg of Glucosamine hydrochloride in a 10 ml			
				Volumetric flask.			
			5.1.8.2	Dissolve in purified water and make up the volume with			
				the same solvent.			
			5.1.8.3	It should be prepared freshly before use.			
		5.1.9	1000 µg/ml Mannosamine working solution:				
			5.1.9.1	Weigh 12mg of Mannosamine hydrochloride in a 10ml			
				Volumetric flask.			
			5.1.9.2	Dissolve in purified water and make up the volume with			
				the same solvent.			
			5.1.9.3	It should be prepared freshly before use.			
		5.1.10	2000 µg/ml Mannosamine working solution:				
			5.1.10.	Weigh 12mg of Mannosamine hydrochloride in a 5ml			
				1	Volumetric flask.		
			5.1.10.	Dissolve in purified water and make up the volume with			
				2	the same solvent.		
			5.1.10.	It should be prepared freshly before use.			
				3			
			Note: In case of Pneumococcal serotypes 5 and 14, Glucosamine				
				hydrochloride is used as a standard working solution, while for			
			Pneumococcal serotypes 4, 19F, 19A and 9V Mannosamine				
				hydrochloride is used as a standard working solution.			
		5.2	Preparation of Calibration Curve:				
		5.2.1	The concentration of Hexosamine in the test sample(s) is				
				calculated by comparing results of its absorbance difference			
				obtained against the plotted calibration curve of known			
				hexosamine concentration Vs absorbance at 530 nm. The solutions			
				for making hexosamine calibration curve are prepared as follows:			
			5.2.1.1	Calibration curve for Glucosamine (50-1000 µg/ml):			
				Glucosamine Conc.	Glucosamine working	Purified Water	
				(µg/ml)	Stock (1000µg/ml)	(µl)	
				(µl)			
				50	25	475	
				100	50	450	
				200	100	400	
				400	200	300	
				800	400	100	
				1000	500	0	
			5.2.1.2	Calibration curve for Mannosamine (50-2000µg /ml):			
				Mannosamine	Mannosamine	Mannosamine	Purified Water
				Conc. (µg/ml)	working Stock	working Stock	(µl)
				(1000µg/ml)	(2000µg/ml)	(µl)	
				(µl)			
				50	25	-	475
				100	50	-	450
				200	100	-	400
				400	200	-	300
				800	400	-	100
				1000	500	-	0
				2000	-	500	0
		5.3	Test Procedure:				
		5.3.1	Take 500 µl of the standard dilutions from calibration curve,				
				Test Sample and Blank (Purified water) in duplicates in			
				separate Graduated test tubes. In case of dry polysaccharide			
				prepare a solution containing about 1-5 mg/ml of polysaccharide			
				in Lal reagent water and use it as test sample.			
				Note: The sample dilutions can vary depending upon the			
				hexosamine concentration. If there is any sample outside the			
				range of the calibration curve, carry out the calculation for			
				more appropriate dilution.			
		5.3.2	Add 500µl of 8M HCl drop wise to each tube containing standard				
				dilutions, Test Sample and Blank (Purified water).			
		5.3.3	Vortex well, cap and incubate in water bath at 90°C for 1 hour.				
		5.3.4	Allow the tubes to cool at room temperature.				
		5.3.5	Add 25µl of 0.5% Thymolphthalein solution as indicator to each				
				tube, vortex well.			
		5.3.6	Titrate each tube with 5N sodium hydroxide solution until blue				
				colour is obtained.			
		5.3.7	Add 1M HCl solution drop wise to each tube and shake properly				
				until sample turns colourless.			
		5.3.8	Make up the volume in each graduated tube to 5ml with purified				
				water, mix well.			
		5.3.9	Take 500µl of the above solutions from each tube into separate				
				fresh graduated test tubes.			
		5.3.10	Add 500µl of 2% Acetyl acetone solution to each tube and vortex				
				well.			
		5.3.11	Cap the tubes and incubate in water bath at 90°C for 45				
				minutes.			
		5.3.12	Allow the tubes to cool at room temperature.				
		5.3.13	Add 1.25ml of absolute ethanol to each tube, mix well.				
		5.3.14	Add 500µl of 2.67% Dimethylaminobenzaldehyde solution to each				
				tube and vortex well.			
		5.3.15	Make up the volume to 5ml with Absolute ethanol in graduated				
				tubes, mix well.			

	5.3.16	Cap the tubes and incubate at room temperature for 90 minutes			
		in dark place.			
	5.3.17	Vortex each tube, observe for the appearance of pink colour.			
	5.3.18	Take absorbance of the sample at 530nm against the blank using			
		spectrophotometer.			
	5.3.19	Note that all the readings of the assay tubes should be taken			
		within 30 mins. of completion of the assay and maximum 10			
		samples should be analysed in one single assay.			
	5.4	Calculations:			
	5.4.1	Once the absorbance of the standards and the test samples have			
		been taken, calculation shall be done using the excel sheet			
		carrying formulas for the required calculation.			
	5.4.2	Always copy the master excel sheet (sheet carrying formulas for			
		the required calculation), never use the previously used excel			
		sheet for new calculations.			
	5.4.3	Copy the master excel sheet, keep it for use in respective			
		folder and then fill in the required sample details (Batch			
		number), sample dilution (if applicable), name of analyst, date,			
		etc.			
	5.4.4	Proceed for calculations starting from the Calibration curve.			
		The calculations are done according to the equation,			
		$Y = mX + b$.			
		Where: m = slope			
		b = intercept			
		Y = absorbance			
		X = concentration			
		(multiply with the dilution factor)			
	5.4.5	In case of dry polysaccharide, Hexosamine (%) in polysaccharide			
		on dry weight basis:			
			Hexosamine Conc. in dry Polysacc. (µg/ml) =		
			Hexosamine in Polysacc. sol. (µg/ml) X 1 – {Moisture (%) /		
			100}		
			Hexosamine in dry polysaccharide (µg/ml)		
			Hexosamine (%) = _____		
			X 100		
			Conc. of polysaccharide solution (µg/ml)		
	5.4.6	To calculate the pure polysaccharide content (mg) present in dry			
		polysaccharide sample in of each serotype based on the			
		hexosamine content obtained, following formula is applied:			
		Pure Polysaccharide content(mg) / Dry polysaccharide content			
		(mg)			
		= (Hexosamine Conc. in dry Polysacc (mg) X Conversion factor)			
	5.4.7	The conversion factor of each serotype is mentioned below			
		Serotype	Hexosamine	Percentage as per WHO	Conversion
			TR5 977	factor	
		4	Mannosamine	71.84 %	1.39
		5	Glucosamine	44.14 %	2.26
		9V	Mannosamine	20.89 %	4.79
		14	Glucosamine	29.44 %	3.39
		19A	Mannosamine	32.98 %	3.03
		19F	Mannosamine	32.98 %	3.03
	5.5	Acceptance criteria:			
	5.5.1	The test is valid if Regression of the standard curve is ≥ 0.99			
		and the percentage of Coefficient of Variation is $\leq 15\%$ within			
		the replicates.			
	5.5.2	Hexosamine content in Purified Pneumococcal polysaccharide			
		should meet the laid specifications for each serotype.			
	6.0	Forms and Records			
		Determination of Hexosamine content - Annexure 1.			
	7.0	Reference			
	7.1	WHO TRS No. 977 Annexure 3: Recommendations to assure the quality,			
		safety and efficacy of pneumococcal conjugate vaccines.			
	7.2	European Pharmacopoeia, volume 5, Hexosamine in polysaccharide			
		vaccine, 2.5.20.			
	7.3	Jin-Ho Jang, Hui Ching Hia, Michihiko Ike, Chiyo Inoue, Masanori			
		Fujita & Toshiomi Yoshida (2004)," Acid hydrolysis and quantitative			
		determination of total Hexosamines of an exopolysaccharide produced			
		by Citrobacter sp."			
	7.4	Surinder Kumar and P.M.T Hansen (1972), New reaction mixture of			
		spectrophotometric determination of N-acetyl Hexosamines.			

[1.0 |Purpose: |
 | |To lay down the procedure for quantifying the protein concentration by BCA|
 | |method. |
 [2.0 |Applicable to: |
	This Standard Operating Procedure is applicable to the samples of Active
	raw material of CRM, Active Raw Material of PRP- CRM conjugate vaccine in
	which protein concentration is to be determined.
[3.0	Abbreviations:
	3.1
	3.2
	3.3
	3.4
	3.5
	3.6
	3.7
	3.8
	3.9
[4.0	Procedure:
	4.1
	4.2
	4.3
	4.4
	4.5
	4.6

			X = concentration
			(Multiply by dilution factor)
		4.6.9	Use EXCEL programme for the preparation of calibration curve
			and Calculation of the results (Multiply by the dilution
			factor).
	4.7	Acceptance criteria:	
		4.7.1	The coefficient of variation (CV) between the replicates of
			the standards should be $\leq 10\%$.
		4.7.2	The Calibration curve is accepted if it is meeting the
			parameters of Regression of ≥ 0.99 .
5.0	References:		
	5.1	BCA TM Protein Assay Kit Insert	

1.0	Objective:		
	To lay down the procedure for immunological identification of Purified		
	Polysaccharide (PRP HIB) and Bulk Conjugate by immunodiffusion using		
	polyclonal antibodies.		
2.0	Scope:		
	This Standard Operating Procedure is applicable to the Purified		
	Polysaccharide (PRP HIB) and Bulk Conjugate of Haemophilus influenzae type		
	b vaccine.		
3.0	Abbreviations and Definitions:		
	3.1	PRP-CRM	: Polyribosyl Ribitol Phosphate conjugated with CRM
	3.2	HIB	: Haemophilus Influenzae Type-B
	3.3	mL	: Milliliter
	3.4	ARM	: Active Raw Material
	3.5	PRP	: Polyribosyl ribitol phosphate
	3.6	CRM	: Cross Reacting Material
	3.7	°C	: Degree Celsius
	3.8	%	: Percentage
	3.9	µl	: Micro-litre
4.0	Procedure		
	4.1	Reagents:	
		4.1.1	Anti PRP antisera (NIBSC: Code No. 09/222, Alpha Diagnostics
			Cat No. PRPB11-S)
		4.1.2	Anti CRM antisera (Alpha Diagnostics Cat No.940-220-DMG)
		4.1.3	Agarose powder (Electrophoresis grade)
		4.1.4	Physiological saline
		4.1.5	Microscopy slides
	4.2	Preparation of Agarose gel (0.7%):	
		4.2.1	Take 0.7% Agarose gel in Physiological saline and boil it for
			dissolution.
		4.2.2	Allow it to cool and when the temperature is around 50°C
			(bearable to touch at back of the hand) and pour 5 ml of it
			onto each microscopy glass slide.
		4.2.3	Allow the agarose to solidify and cut wells in the gel using a
			puncher at two equidistant points for Purified polysaccharide
			and three equidistant points for Bulk Conjugate.
	4.3	Load 20 µl samples of PRP HIB in one-well and PRP antisera in the	
		other. For Bulk Conjugate, load the sample of ARM HIB CRM in the	
		central well and PRP antisera in one of the surrounding wells and CRM	
		antitoxin in the other.	
	4.4	Allow the sample to be absorbed after the sample have dried place the	
		slides in a box with lid on top containing pre-wetted filter paper	
		sheets and incubate in an incubator at 37± 2°C for 48 hours and look	
		for the line of identity.	
	4.5	Scan and preserve the gel images.	
	4.6	Interpretation of Results: If line of identity formed, the product	
		passes identity test	
5.0	References:		
5.1	NA		

1.0|Objective: |
 |To lay down the procedure for the determination of free carrier protein|
 |in PRP-CRM conjugate (bulk conjugate) of Haemophilus Influenzae type b |
 |conjugate vaccine. |
 2.0|Scope: |
 |This Standard Operating Procedure is applicable for the determination |
 |of free carrier protein present in bulk conjugate of Haemophilus |
 |Influenzae type b conjugate vaccine. |
 3.0|Abbreviations and Definitions: |
3.1	HPLC	: High Performance Liquid Chromatography
3.2	Id	: internal diameter
3.5	mAu	: milli Absorbance unit
3.6	Mg	: milligram
3.7	RSD	: Relative Standard Deviation
3.8	µL	: microliter
3.9	µm	: micrometer
3.10	µg	: microgram
3.11	SST	: System Suitability Test
3.12	System Suitability test: System suitability tests are an integral	
part of gas and liquid chromatographic methods. They are used to		
verify that the resolution and reproducibility of the		
chromatographic system are adequate for the analysis to be done.		
The tests are based on the concept that the equipment,		
electronics, analytical operations and samples to be analyzed		
constitute an integral system that can be evaluated as such.		
4.0	Procedure	
4.1	Precautions to be taken:	
4.1.1	Always wear laboratory coat.	
4.1.2	Always use filtered solutions. Filtration of solution should	
be done in order to eliminate the entry of possible		
particles, if any, into the system as well as the column.		
4.1.3	Degas all solutions.	
4.1.4	Avoid entry of bubbles into the system as well as column.	
4.1.5	All the solution bottle openings should be well covered with	
aluminum foil.		
4.1.6	Check the calibration status of all equipment and	
instrument.		
4.2	Requirement:	
4.2.1	Equipment:	
4.2.1.1	Calibrated HPLC System with ChemStation software.	
4.2.1.2	Ultrasonic Cleaner	
4.2.1.3	Vacuum Pump	
4.2.2	Instrument:	
4.2.2.1	pH Meter	
4.2.2.2	Magnetic stirrer	
4.2.2.3	Precision balance	
4.2.2.4	Analytical balance	
4.2.3	Materials:	
4.2.3.1	TSK gel G3000SWXL column {30cm(l), 7.8mm(i.d)}	
4.2.3.2	Laboratory Glassware	
4.2.3.3	Screw cap vials	
4.2.3.4	Solvac Filter Holder	
4.2.3.5	Nylon 6,6 membrane 0.20/0.22/0.45 µm membrane	
filter		
4.2.3.6	Auto pipettes (10-100µL), (20-200(L), (100-1000(L)	
4.2.3.7	Tips for the Auto pipettes	
4.2.4	Reagents:	
4.2.4.1	Sodium Chloride (NaCl)	
4.2.4.2	Disodium hydrogen phosphate anhydrous (Na₂HPO₄)	
4.2.4.3	Sodium dihydrogen phosphate monohydrate	
4.2.4.4	Purified water	
4.3	Preparation of solutions:	
Prepare all the solutions as described below and Carry out the		
labelling of the prepared solutions and assign the lot numbers.		
4.3.1	Isopropyl alcohol 70%	
Take 700 mL of 2-propanol in a volumetric flask and make up		
the volume to 1000 mL with Purified Water. This solution can		
be stored at room temperature for 3 months.		
4.3.2	Solution I: [Na₂HPO₄ (0.1M), NaCl (0.1M)] For a volume of	
1000 mL, weigh 14.19 g of Na₂HPO₄ and 5.84 g of NaCl and		
Dissolve in 700mL of purified water in a volumetric flask		
and make up the volume to 1000mL. Prepare fresh before use.		
4.3.3	Solution II: (NaH₂PO₄.H₂O, 0.1 M) : For a volume of 500 mL,	
weigh 6.85g of NaH₂PO₄.H₂O and dissolve in 400mL of		
purified water in a volumetric flask and make up the volume		
to 500mL. Prepare fresh before use.		
4.3.4	0.1M Sodium Phosphate buffer: (0.1 M Na₂HPO₄, 0.1M NaCl	
+0.1M NaH₂PO₄.H₂O, pH 7.0)		
For a volume of 1000mL, take approximately 600mL of		
Solution I and 400mL of Solution II. The pH of the solution		
should be 7.0 ± 0.2. Use the mobile phase within four days		
from the date of preparation if it remains clear and no		
turbidity or fibers observed in it.		
4.3.5	Column Preservation Solution: Prepare and use it.	
4.3.6	1 mg/mL of p-Aminobenzoic Acid Stock: For a Volume of 5 mL,	
weigh: 5 mg of p-Aminobenzoic acid (4-Aminobenzoic acid)		
and Dissolve in 5mL of purified water and make up the		
volume to 5mL.		
4.3.9	Working standard of p-Aminobenzoic Acid: Dilute 1 mg/mL of	
the stock solution to 1/10 with purified water.		

			Note: All the solutions prepared for use are vacuum				
			filtered using Nylon 6, 6 membrane 0.20 µm or 0.22 µm ,				
			depending upon the nature of solution to be filtered. For				
			organic solvents 0.45µm filters should be used (filtration				
			not necessary if the solution is prepared in 0.2 µ filtered				
			purified water).				
	4.4	Preparing the HPLC System for run, Agilent Series:					
	4.5	Run Conditions:					
	4.5.1	Load method					
	4.5.2	Flow Rate: 0.75 mL/min					
	4.5.3	Detection Wavelength: 280nm					
	4.5.4	Run time: 20 minutes					
	4.6	Equilibration of the system and the column:					
			Equilibrate the column with mobile 0.1 M sodium phosphate buffer				
			at 280nm until a stable baseline is observed on the online plot				
			window.				
	4.7	Sample Preparation and Application:					
	4.7.1	Test sample and standards are loaded in screw cap vials and					
			placed against the appropriate no. on the sample tray by				
			clearly marking the screw cap vial.				
	4.7.2	Load the required volume of sample as per the PRP					
			concentration equivalent to 2.5 µg in the sample vial.				
			Example: If a sample has a PRP concentration of 0.5 mg/mL,				
			to calculate volume required for 2.5 µg of protein,				
			0.5 mg (500 µg) ----1 mL				
			2.5 µg-----X mL				
			X mL = 2.5 µg x 1 mL				
			500 µg				
			X mL = 5 µL				
			Therefore, 5 µL of volume is taken for sample having PRP				
			concentration of 0.5 mg/mL.				
	4.8	Running of sequence:					
	4.8.1	Load sequence					
	4.8.2	Open the sequence parameter option by clicking Sequence					
			menu.				
	4.8.3	Enter information about the data directory and operator					
			name.				
	4.8.4	In the shutdown option, click on the post-sequence option					
			and select LAMPOFF/STANDBY.				
	4.8.5	In sequence table option, specify the sample names, methods					
			and vials to run in the sequence as per the following				
			table:				
			S No	Sample Code	Sample details	No. of	Injection
					injections	Volume(µL)	
			1	Blank	Eluent Blank	1	10
			2	p-Aminobenzoic	System	6	1
			acid	Suitability Test			
			3	Blank	Eluent Blank	1	10
			4	Test Sample(s)	Sample	1	2.5µg of
					PRP		
	5.8.6	Click on run sequence option in run control menu to start					
			the sequence.				
	5.8.7	Each sample run will be of 20 min. and after running all					
			the samples the sequence will automatically stop.				
	5.8.8	Inject the eluent blank after every 20 injections.					
			One extra vial with same SST sample can be used if 6				
			injections are not possible from single vial.				
	4.9	Calculations :					
			Area under curve for the standards/sample is analyzed by				
			Chemstation software.				
	4.9.1	Open the acquired chromatogram in the data analysis view in					
			the offline copy of the software.				
	4.9.2	Optimize integration with initial events in the signal					
			specific events table.				
	4.9.3	Enter the value of retention time of peak maxima in					
			integration with timed events option.				
	4.9.4	The percentage of free carrier protein in sample is					
			calculated by taking the percentage area of the peak maxima				
			corresponding to the lower molecular weight compared to the				
			peak maxima of higher molecular weight.				
	4.9.5	Calculations for System Suitability Test:					
			4.9.5.1	Note the values of area under the curve and			
			1	retention time for p-Aminobenzoic acid. (SST			
			standard)				
			4.9.5.2	Every time a sample is run with system suitability			
			2	test, calculate the results for system suitability			
				by creating a copy of Master EXCEL in the desired			
				folder and record the same.			
			4.9.5.3	Attach the hard print of SST calculation along with			
			3	chromatograms and the record form.			
	4.10	Acceptance Criteria :					
	4.10.1	For System Suitability Test :					
			4.10.1.1	RSD calculated for Retention time should be <5.0%.			
			1				
			4.10.1.2	RSD calculated for Area should be <5.0%.			
			2				
	4.10.2	Free protein for ARM of Haemophilus Influenzae type b					
			conjugate vaccine Samples:				
			4.10.2.1	Free protein should be less than 2%.			
			1				
	4.11	Result Reporting:					
	4.11.1	Report the results of percentage of area of peak towards					

			higher Molecular weights (Conjugated) and percentage of
			area of peak towards lower Molecular weights
			(Unconjugated).
			Note: Percentage of free carrier protein is the percentage
			of area of peak towards lower molecular weights
			(Unconjugated).
5.0	References:		
5.1	United States Pharmacopeia.		
5.2	European pharmacopeia.		

1.0 |Objective |
 |To lay down the procedure for estimation of protein content in Pneumococcal|
 |polysaccharide conjugate vaccine samples by Lowry's method. |
 2.0 |Scope |
 |This Standard Operating Procedure is applicable to all the samples of |
 |Pneumococcal polysaccharide conjugate vaccine in which protein |
 |concentration is to be determined by Lowry's method. |
 3.0 |Responsibility |
3.1	The analyst is responsible for executing the described procedure
	correctly.
3.2	Supervisor is responsible for verifying that the procedure is carried
out correctly and check the results.	
3.3	HOD shall be responsible for overall compliance of this SOP and shall
ensure that the SOP is reviewed every two years or if required	
earlier.	
4.0	Abbreviations and Definitions
4.1	Abbreviations:
	4.1.1
	4.1.2
	4.1.3
	4.1.4
	4.1.5
	4.1.6
	4.1.7
	4.1.8
	4.1.9
	4.1.10
4.2	Reagents:
	4.2.1
	4.2.2
	4.2.3
	4.2.4
	4.2.5
	4.2.6
	4.2.7
4.3	Equipments and Material
	4.3.1
	4.3.2
	4.3.3
	4.3.4
	5000 µl)r
	4.3.5
	4.3.6
	4.3.7
5.0	Procedure
5.1	Preparation of solutions: Prepare all the solutions as described below
and fill the form, label the solutions and assign lot numbers.	
	5.1.1
	5.1.2
	5.1.3
	5.1.4
	5.1.5
	5.1.6
	5.1.7
	5.1.8
5.2	Preparation of Calibration Curve:
	5.2.1

			20	50	450
			40	100	400
			80	200	300
			160	400	100
		5.3	Test Procedure:		
		5.3.1	Take 500 µl of the standard dilutions from calibration curve,		
		Test Sample and Blank (0.1M sodium hydroxide) in duplicates in			
		separate test tubes. In case of dry polysaccharide prepare a			
		solution containing about 1-5 mg/ml of polysaccharide in Lal			
		reagent water and use it as test sample.			
		Note: The sample dilutions can vary depending upon the protein			
		concentration. If there is any sample outside the range of the			
		calibration curve, carry out the calculation for more			
		appropriate dilution.			
		5.3.2	Add. 2.5 ml of Cupric-tartaric solution to each of the standard		
		blank (0.1M Sodium hydroxide) and sample.			
		5.3.3	Vortex and incubate the test tubes in dark at room temperature		
		for 10 min.			
		5.3.4	Add 250 µl of 1 N Folin & Ciocalteu's phenol reagent and mix		
		well.			
		5.3.5	Incubate the test tube at room temperature for 30 minutes in		
		dark.			
		5.3.6	Observe for blue color which can vary with the concentration.		
		5.3.7	Take absorbance of the sample at 730nm against the blank using		
		spectrophotometer.			
		5.3.8	Note that all the readings of the assay tubes should be taken		
		within 30 mins. of completion of the assay and maximum 10			
		samples should be analysed in one single assay.			
		5.4	Calculations:		
		5.4.1	Once the absorbance of the standards and the test samples have		
		been taken, calculation shall be done using the excel sheet			
		carrying formulas for the required calculation.			
		5.4.2	Always copy the master excel sheet (sheet carrying formulas for		
		the required calculation), never use the previously used excel			
		sheet for new calculations.			
		5.4.3	Proceed for calculations starting from the Calibration curve.		
		The calculations are done according to the equation,			
		Y = mX + b.			
		Where: m = slope			
		b = intercept			
		Y = absorbance			
		X = concentration			
		(Multiplication by dilution factor)			
		5.4.4	Use EXCEL programme for the preparation of calibration curve		
		and Calculation of the results (Multiply by the dilution			
		factor. Divide with 1000 to change µg/ml to mg/ml).			
		5.4.5	In case of dry polysaccharide,		
		Protein (%) in polysaccharide on dry weight basis:			
		Protein in dry polysacc. (µg/ml) =			
		Protein in Polysacc. sol. (µg/ml) X 1 – {Moisture (%) / 100}			
		Protein in dry polysaccharide (µg/ml)			
		Protein (%) = _____			
		x 100			
		Concentration of polysaccharide solution (µg/ml)			
		5.5	Acceptance criteria:		
		5.5.1	The test is valid if Regression of the standard curve is ≥0.99		
		and the percentage of Coefficient of Variation is ≤ 15% within			
		the replicates.			
		5.5.2	Protein content in Purified Pneumococcal polysaccharide and		
		conjugate should meet the laid specifications for each			
		serotype.			
		6.0	Forms and Records		
		Determination of Total Protein content - Annexure 1			
		7.0	Reference		
		7.1	Lowry, O.H., J. Rosebrough, A.L.Farr and R.J. Randall, R.J. (1951) J.		
		Biol.Chem. 193, 265 - 275.			
		7.2	European Pharmacopoeia, Volume 5, Protein in Polysaccharide Vaccines		
		2.5.16.			
		7.3	WHO TRS No. 977 Annexure 3: Recommendations to assure the quality,		
		safety and efficacy of pneumococcal conjugate vaccines.			

1.0|Purpose: |
 |To lay down the procedure for the estimation of molecular size of PRP-CRM |
 |conjugate in bulk conjugate of Haemophilus Influenzae type b conjugate |
 |vaccine. |
 2.0|Applicable to: |
 |This Standard Operating Procedure is applicable for evaluating the |
 |molecular size distribution and percentage elution of bulk conjugate of |
 |Haemophilus Influenzae type b conjugate vaccine. |
 3.0|Abbreviations and Definitions: |
3.1	HPLC	: High Performance Liquid Chromatography
3.2	i.d.	: Internal diameter
3.3	l	: length
3.4	M	: Molar
3.5	m	: meter
3.6	mAu	: milli Absorbance unit
3.7	PRP-CRM	: Polyrribosyl Ribitol Phosphate – Cross Reacting Material
3.8	RSD	: Relative Standard Deviation
3.9	SST	: System Suitability Test
3.10	°C	: Degree centigrade
3.11	µg	: microgram
3.12	µL	: microliter
3.13	µm	: micrometer
3.14	System	: System suitability tests are an integral part of gas and
	Suitability	liquid chromatographic methods. They are used to verify
	ty test	that the resolution and reproducibility of the
		chromatographic system are adequate for the analysis to be
		done.
		The tests are based on the concept that the equipment,
		electronics, analytical operations and samples to be
		analyzed constitute an integral system that can be
		evaluated as such.
4.0	Procedure:	
4.1	Precautions to be taken:	
4.1.1	Always use filtered solutions. Filtration of solution	
		should be done in order to eliminate the entry of possible
		particles, if any, into the system as well as the column.
4.1.2	Degas all solutions.	
4.1.3	Avoid entry of bubbles into the system as well as column.	
4.1.4	All the solution bottle openings should be well covered	
		with aluminum foil.
4.1.5	Check the calibration status of all equipment and	
		instrument.
4.2	Requirement:	
4.2.1	Equipment:	
4.2.1.1	Calibrated HPLC System with Chem Station software.	
4.2.1.2	Ultrasonic Cleaner	
4.2.1.3	Vacuum Pump	
4.2.1.4	Refrigerator	
4.2.2	Instrument:	
4.2.2.1	pH Meter	
4.2.2.2	Magnetic stirrer	
4.2.2.3	Precision balance	
4.2.2.4	Analytical balance	
4.2.3	Material:	
4.2.3.1	TSK gel G5000PWxl column {30 cm (l), 7.8 mm (i.d)}	
4.2.3.2	Laboratory Glassware	
4.2.3.3	Screw cap vials	
4.2.3.4	Solvac Filter Holder	
4.2.3.5	Nylon 6,6 membrane 0.20/0.22/0.45 µm membrane filter	
4.2.3.6	Autopipette (2-20µL), (20-200(L), (100-1000(L)	
4.2.3.7	Tips for the Autopipette	
4.2.3.8	Aluminium foil	
4.2.4	Reagents:	
4.2.4.1	Sodium Chloride (NaCl)	
4.2.4.2	Disodium hydrogen phosphate anhydrous (Na₂HPO₄)	
4.2.4.3	Sodium dihydrogen phosphate monohydrate	
4.2.4.4	Purified water	
4.2.4.5	Sodium azide	
4.2.4.6	Salmon DNA	
4.2.4.7	2-Propanol	
4.3	Preparation of solutions:	
Prepare all the solutions and Carry out the labelling of the prepared		
solutions and assign the lot numbers to the solutions.		
4.3.1	Isopropyl alcohol 70%	
Take 700 mL of 2-propanol in a volumetric flask and make up		
the volume upto 1000 mL with Purified Water. This solution can		
be stored at room temperature for 3 months.		
4.3.2	Solution I: [Na₂HPO₄ (0.1M), NaCl (0.1M)]	
For a volume of 1000 mL, weigh:		
Na₂HPO₄	14.19 g	
NaCl	5.84 g	

	Dissolve in 700mL of purified water in a volumetric flask and
	make up the volume upto 1000mL. Prepare fresh solution before
	use.
4.3.3	Solution II: NaH₂PO₄.H₂O, 0.1 M
For a volume of 500 mL, weigh:	
NaH₂PO₄.H₂O	6.85g
Dissolve in 400mL of purified water in a volumetric flask and	
	make up the volume upto 500mL. Prepare fresh solution before

```

||| |Use. | | | | | | | |
||| |4.3.4 |0.1M Sodium Phosphate buffer |
||| |(0.1 M Na2HPO4,0.1M NaCl +0.1M NaH2PO4.H2O, pH 7.0) |
||| |For a volume of 1000mL, take approximately 600mL of Solution II |
||| |and 400mL of Solution II. The pH of the solution should be 7.0|
||| |± 0.2. Use the mobile phase within four days from the date of |
||| |preparation if it remains clear and no turbidity or fibers |
||| |observed in it. |
||| |4.3.5 |Void Volume Marker (Salmon DNA): |
||| |For a volume of 5 mL, weigh: |
||| |Salmon DNA |5 mg |
||| |Dissolve in purified water and make up volume upto 5 mL with |
||| |purified water in a volumetric flask. Store at -20 °C ± 5°C |
||| |for 1 month. |
||| |4.3.6 |Total volume marker (Sodium Azide): |
||| |For a volume of 10 mL, weigh: |
||| |Sodium Azide |10 mg |
||| |Dissolve in purified water and make up volume upto 10 mL with |
||| |purified water in a volumetric flask. Store at 2-8 °C for 1 |
||| |month. |
||| |4.3.7 |Column Preservation Solution: |
||| |Note: All the solutions prepared for use are vacuum filtered |
||| |using Nylon 6, 6 membrane 0.20 µm or 0.22 µm , depending upon |
||| |the nature of solution to be filtered. For organic solvents |
||| |0.45µm filters should be used. |
||| |4.4 |Preparing the HPLC System for run, Agilent Series: |
||| |4.5 |Run Conditions: |
||| |4.5.1 |Load method |
||| |4.5.2 |Flow Rate: 0.5 mL/min |
||| |4.5.3 |Injection Quantity : 2.5µg |
||| |4.5.4 |Detection Wavelength: 214nm |
||| |4.5.5 |Run time: 30 minutes |
||| |4.6 |Equilibration of the system and the column: |
||| |Equilibrate the column with mobile 0.1 M sodium phosphate buffer at |
||| |214nm until a stable baseline is observed on the online plot window. |
||| |4.7 |Sample Preparation and Application: |
||| |4.7.1 |Test sample and standards are loaded in screw cap vials and |
||| |placed against the appropriate no. on the sample tray by |
||| |clearly marking the screw cap vial. |
||| |4.7.2 |Load the required volume of sample as per the PRP |
||| |concentration equivalent to 2.5 µg in the sample vial. |
||| |Example: If a sample has a PRP concentration of 0.5 mg/mL, to |
||| |calculate volume required for 2.5 µg of protein, |
||| |0.5 mg (500 µg) ----1 mL |
||| |2.5 µg-----X mL |
||| |X mL = 2.5 µg x 1 mL |
||| |500 µg |
||| |X mL = 5 µL |
||| |Therefore, 5 µL of volume is taken for sample having PRP |
||| |concentration of 0.5 mg/mL. |
||| |4.8 |Running of sequence: |
||| |4.8.1 |Load sequence |
||| |4.8.2 |Open the sequence parameter option by clicking Sequence menu. |
||| |4.8.3 |Enter information about the data directory and operator name. |
||| |4.8.4 |In the shutdown option, click on the post-sequence option and |
||| |select LAMPOFF/STANDBY. |
||| |4.8.5 |In sequence table option, specify the sample names, methods |
||| |and vials to run in the sequence as per the following table: |
||| |S No|Sample Code |Sample details|No. of |Injection |
||| ||| |inject|volume (µL) |
||| ||| |Ions |
||| ||| |1 |Blank |Eluent Blank |1 |10 |
||| ||| |2 |Sodium azide (Total |System |6 |1 |
||| ||| |volume marker) |Suitability | | |
||| ||| |Test | | |
||| ||| |3 |Blank |Eluent Blank |1 |10 |
||| ||| |4 |Salmon DNA |Void volume |1 |10 |
||| ||| |marker | | |
||| ||| |5 |Blank |Eluent Blank |1 |10 |
||| ||| |6 |Test Sample (s) |Sample |1 |2.5µg of PRP |
||| ||| |Click on run sequence option in run control menu to start|
||| ||| |the sequence. |
||| ||| |5.8.8|Each sample run will be of 30 min and after running all |
||| ||| |the samples the sequence will automatically stop. |
||| ||| |4.9 |Calculations : |
||| ||| |Area under curve for the standards/sample is analyzed by |
||| ||| |Chemstation software. |
||| ||| |4.9.1|The retention time of the void volume is determined with |
||| ||| |Salmon DNA. |
||| ||| |4.9.2|The Retention time of the component (total volume) which |
||| ||| |has full access to the pores of stationary phase is |
||| ||| |determined with sodium azide. |
||| ||| |4.9.3|Elution time of PRP-TT (Te) at KD = 0.50 |
||| ||| |The calculations are made as per the following equation: |
||| ||| |Te = KD (Tt - To) + To or Te = 0.50 (Tt - |
||| ||| |To) + To |
||| ||| |Where, |
||| ||| |Te = Elution time of PRP-TT Bulk Conjugate at KD = 0.50 |
||| ||| |Tt = Retention time of sodium azide |
||| ||| |To = Retention time of Salmon DNA |
||| ||| |KD = Distribution Coefficient |
||| ||| |Note: Take average RT of sodium azide as Tt from EXCEL |

```

				for calculation of SST.	
				4.9.4	Calculation of percentage of sample eluted before a given
				KD:	
				The percentage of sample eluted before a given KD is	
				calculated using ChemStation software.	
				4.9.4.1	Open the acquired chromatogram in the data
				analysis view in the offline copy of the	
				software.	
				4.9.4.2	Optimize integration with initial events in the
				signal specific events table.	
				4.9.4.3	Enter the value of Elution time (Te) calculated
				in step 5.9.3 in integration with timed events	
				option or with manual integration and select	
				split peak option.	
				4.9.4.4	The percentage of sample eluted at $KD \leq 0.3$ to
				0.6 is calculated by adding the values of	
				percentage of area (Area %) obtained for bulk	
				conjugate curve till Te.	
				4.9.5	Calculations for System Suitability Test:
				4.9.5.1	Note the values of area under the curve and
				retention time for sodium azide (SST standard) in	
				Annexure 1 and also in EXCEL format for	
				calculation of Relative standard deviation.	
				4.9.5.2	Every time a sample is run with system
				suitability test, calculate the results for	
				system suitability.	
				4.9.5.3	Attach the hard print of SST calculation along
				with chromatograms and the record form.	
		4.10	Acceptance Criteria :		
		4.10.	For System Suitability Test :		
			1		
			4.10.1.	RSD calculated for Retention time should be <	
			1	5.0%.	
			4.10.1.	RSD calculated for Area should be < 5.0%.	
			2		
			4.10.	For ARM of Haemophilus Influenzae type b conjugate	
		2	vaccine Samples: KD (Molecular Size distribution): $\geq 50\%$,		
			KD $\leq 0.3 - 0.6$		
5.0	References:				
European Pharmacopeia.					

H: the conjugate antigen content of test samples, %; P: the test validity, which shall be 70-130%.

Method two: it is applicable for the content determination of type 3 conjugate antigen in 13-valent pneumococcal conjugate vaccine.

free polysaccharide separation of samples

Take 1000μl of solution D and Y and put into 2ml centrifuge tubes respectively, add 100μl of 1% sodium deoxycholate solution, mix well, and place at 2-8℃ for 30min

Add 50μl of 1mol/L hydrochloric acid solution into each centrifuge tube, mix well, and place at 2-8℃ for 30min.

Centrifuge the tubes at 8000g, 5min for 30min.

Carefully take 800μl of supernatant using a pipette respectively, which is recorded as DS and YS.

Determination of polysaccharide content

Take 300μl of solution D, add 900μl of diluent 1 (DIL1), mix well to obtain 4 times dilute solution, determine the polysaccharide content on the IMMAGE 800, input 4.4 (i.e. 4×1.1) of off-line dilution multiple. If the result is beyond the inspection range, adjust the dilution multiple to A, and input A×1.1 of off-line dilution multiple.

Take 300μl of solution Y, add 60μl of diluent 1 (DIL1), mix well to obtain 3 times dilute solution, determine the polysaccharide content on the IMMAGE 800, input 3.3 (i.e. 3×1.1) of off-line dilution multiple.

Take 400μl of DS solution, add 800μl of diluent 1 (DIL1), mix well to obtain 3 times dilute solution, and determine polysaccharide content on the IMMAGE 800. If the result is beyond the detection range, adjust the dilution multiple. The final result is calculated manually by the tester (i.e. test result × dilute multiple).

Take solution YS and directly put it on the IMMAGE 800 to determine the polysaccharide content. The final result is calculated manually by the tester (i.e. test result × dilution multiple).

Result calculation

Calculate the conjugate antigen content and test validity as following formula:

$$H = (1 - \frac{\text{Pneumococcal polysaccharide Standard substance}}{\text{Anti-serum of pneumococcal polysaccharide}}) \times 100\%$$

In the formula,

H =

$$\frac{\text{Conjugate antigen content determination SOP} \times \text{Version} \times \text{Page}}{\text{Conjugate antigen content determination SOP} \times \text{Confidential files No copy} \times \text{Confidential files No copy}} \times 100\%$$

H: the conjugate antigen content of test samples, %; P: the test validity, which shall be 70-130%.

Flow chart

N/A

Matters need attention

In the process of free polysaccharide separating from 13-valent pneumococcal polysaccharide vaccine samples(except type 3), the supernatant in the centrifuge tube should be directly poured out after centrifuging at 6000g, 5min for 90min, and the supernatant in the centrifuge tube should be completely sucked by inverted on the absorbent paper.

When removing supernatant from the centrifuge tube, pay special attention that the pipette tip is not allowed to touch the tube wall, and it is not allowed to take precipitate or floating substance.

Reference or standard basis

The production and test procedure for 13-valent pneumococcal conjugate vaccine

Revise history

| Conjugate antigen content determination SOP | Code | WSYX-QC-SOP-06-09-0003 | WSYX-QC-SOP-06-09-0003 | WSYX-QC-SOP-06-09-0003

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No. | Equipment name | Equipment model | Manufacturer | Notes

1 | Specific protein Analysis System | IMMAGE 800 | BECKMAN COULTER | Or equivalent equipment

2 | Centrifuge | CR21G/Multi fuge X1R | HITACHI/Thermo Scientific | Or equivalent equipment

3 | Electronic balance | BS124S/BSA 124S | Sartorius | Or equivalent equipment

No. | Reagents Name | Grade | Notes

1 | Diluent 1 | — | /

2 | Buffer 1 | — | /

3 | Wash solution | — | /

4 | Pneumococcal polysaccharide Standard substance | — | /

5 | Anti-serum of pneumococcal polysaccharide | — | /

| Conjugate antigen content determination SOP | Code | WSYX-QC-SOP-06-09-0003 | WSYX-QC-SOP-06-09-0003 | WSYX-QC-SOP-06-09-0003

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| Conjugate antigen content determination SOP | Confidential files No copy | Confidential files No copy | Confidential files No copy | Confidential files No copy

6 | Sodium hydroxide | AR | /

7 | Citric acid | AR | /

8 | Sodium deoxycholate | — | /

9 | Hydrochloric acid | AR | /

10 | Sodium Chloride | AR | /

11 | Absolute ethyl alcohol | AR | /

No. | Material name | No. | Material name

1 | Sample rack | 2 | Sample Cups

3 | Kit | 4 | 0.6ml/2ml/15ml/50ml Centrifuge Tube

5 | 100-1000ul Pipette | 6 | 20-200ul Pipette

7 | 500-5000ul Pipette | 8 | 5000ul/1000ul/200ul Pipette tip

| Conjugate antigen content determination SOP | Code | WSYX-QC-SOP-06-09-0003 | WSYX-QC-SOP-06-09-0003 | WSYX-QC-SOP-06-09-0003

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| Conjugate antigen content determination SOP | Code | WSYX-QC-SOP-06-09-0003 | WSYX-QC-SOP-06-09-0003 | WSYX-QC-SOP-06-09-0003

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WSYX-QC-SOP-06-09-0003
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Confidential files No copy | Confidential files No copy
| Conjugate antigen content determination SOP | Code | WSYX-QC-SOP-06-09-0003 | WSYX-QC-SOP-06-09-0003 |
WSYX-QC-SOP-06-09-0003
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Confidential files No copy | Confidential files No copy
| Conjugate antigen content determination SOP | Code | WSYX-QC-SOP-06-09-0003 | WSYX-QC-SOP-06-09-0003 |
WSYX-QC-SOP-06-09-0003
| Conjugate antigen content determination SOP | Version | 02 | Page | 7/7
| Conjugate antigen content determination SOP | Confidential files No copy | Confidential files No copy |
Confidential files No copy | Confidential files No copy
Version | Time in
force | Revision
Dept. | Revision basis, reason and content | Notes
00 | 13 Feb.
2019 | QC
Laboratory | Complete the content |
01 | 20 Jun.
2019 | QC
Laboratory | Complete the content and adjust the
sequence of operation steps |

[illegible]

3.0 | Abbreviations and Definitions | Abbreviations and Definitions | Abbreviations and Definitions |
Abbreviations and Definitions | Abbreviations and Definitions | Abbreviations and Definitions |
Abbreviations and Definitions | Abbreviations and Definitions | Abbreviations and Definitions |
Abbreviations and Definitions | Abbreviations and Definitions

| 3.1 | Cm | Cm | Cm | : centimetre | : centimetre | : centimetre | : centimetre | : centimetre | : centimetre

| 3.2 | G | G | G | : Gram | : Gram | : Gram | : Gram | : Gram | : Gram

[3.3] HPAEC-PAD | HPAEC-PAD | HPAEC-PAD | : High Performance Anion Exchange Chromatography with Pulsed Amperometric Detection | : High Performance Anion Exchange Chromatography with Pulsed Amperometric Detection | : High Performance Anion Exchange Chromatography with Pulsed Amperometric Detection | : High Performance Anion Exchange Chromatography with Pulsed Amperometric Detection | : High Performance Anion Exchange Chromatography with Pulsed Amperometric Detection

| 3.4 | ICS | ICS | ICS | : Ion Chromatography System | : Ion Chromatography System | : Ion
Chromatography System | : Ion Chromatography System | : Ion Chromatography System | : Ion Chromatography
System

| 3.5 | ml | ml | ml | : millilitre | : millilitre | : millilitre | : millilitre | : millilitre | :
millilitre

| 3.6 | Nm | Nm | Nm | : Nano meter | : Nano meter | : Nano meter | : Nano meter | : Nano meter | : Nano meter

| 3.7 | Psi | Psi | Psi | : pressure per square inch | : pressure per square inch | : pressure per square inch | : pressure per square inch | : pressure per square inch | : pressure per square inch

[3.8 | PRP-CRM | PRP-CRM | PRP-CRM] : Polyribosyl Ribitol Phosphate – Cross Reacting material | : Polyribosyl Ribitol Phosphate – Cross Reacting material | : Polyribosyl Ribitol Phosphate – Cross Reacting material | : Polyribosyl Ribitol Phosphate – Cross Reacting material | : Polyribosyl Ribitol Phosphate – Cross Reacting material

3.9 Rpm Rpm Rpm	: revolutions per minute	: revolutions per minute	: revolutions per minute
	: revolutions per minute	: revolutions per minute	: revolutions per minute

3.10 | RSD | RSD | RSD | : Relative Standard Deviation | : Relative Standard Deviation | : Relative Standard Deviation | : Relative Standard Deviation | : Relative Standard Deviation | : Relative Standard Deviation

| 3.11 | SOP | SOP | SOP | : Standard Operating Procedures | : Standard Operating Procedures | : Standard Operating Procedures | : Standard Operating Procedures | : Standard Operating Procedures | : Standard Operating Procedures

| 3.12 | SST | SST | SST | : System Suitability Test | : System Suitability Test | : System Suitability Test | : System Suitability Test | : System Suitability Test | : System Suitability Test

Test | : System Suitability Test | : System Suitability Test | : System Suitability Test
3.13 | °C | °C | °C | : Degree centigrade | : Degree centigrade | : Degree centigrade | : Degree centigrade | : Degree centigrade | : Degree centigrade

| 3.14 | μg | μg | μg | : microgram | : microgram | : microgram | : microgram | : microgram | : microgram

3.15 | μL | μL | μL | : Micro litre | : Micro litre | : Micro litre | : Micro litre | : Micro litre | :
Micro litre

Micro meter
3.16 | μm | μm | : Micro meter | : Micro meter | : Micro meter | : Micro meter | : Micro meter | :
Micro meter

| 3.17 | System Suitability Test | System Suitability Test | System Suitability Test | : System

The tests are based on the concept that the equipment, electronics, analytical operations and samples to be analyzed constitute an integral system that can be evaluated as such. [1] : System suitability tests are an integral part of gas and liquid chromatographic methods. They are used to verify that the resolution and reproducibility of the chromatographic system are adequate for the analysis to be done.

The tests are based on the concept that the equipment, electronics, analytical operations and samples to be analyzed constitute an integral system that can be evaluated as such. [1] : System suitability tests are an integral part of gas and liquid chromatographic methods. They are used to verify that the resolution and reproducibility of the chromatographic system are adequate for the analysis to be done.

The tests are based on the concept that the equipment, electronics, analytical operations and samples to be analyzed constitute an integral system that can be evaluated as such. 1. System suitability tests are an integral part of gas and liquid chromatographic methods. They are used to verify that the resolution and reproducibility of the chromatographic system are adequate for the analysis to be done.

[illegible]

| 4.1.1 | Always wear laboratory coat. | Always wear laboratory coat. | Always wear laboratory coat. |
Always wear laboratory coat. | Always wear laboratory coat. | Always wear laboratory coat. | Always wear
laboratory coat. | Always wear laboratory coat. | Always wear laboratory coat.

| 4.1.3 | Degas all solutions. | Degas all solutions. | Degas all solutions. | Degas all solutions. |
Degas all solutions. | Degas all solutions. | Degas all solutions. | Degas all solutions. | Degas al
solutions.

| | 4.1.5 | Always store the column in NaOH solution. | Always store the column in NaOH solution. |
Always store the column in NaOH solution. | Always store the column in NaOH solution. | Always store the
column in NaOH solution. | Always store the column in NaOH solution. | Always store the column in NaOH
solution. | Always store the column in NaOH solution. | Always store the column in NaOH solution.

[illegible][illegible]

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| | 4.4.5.5 | 4.4.5.5 | After 2 hours, take out the samples from the oven and keep at room temperature for 15 minutes. | After 2 hours, take out the samples from the oven and keep at room temperature for 15 minutes. | After 2 hours, take out the samples from the oven and keep at room temperature for 15 minutes. | After 2 hours, take out the samples from the oven and keep at room temperature for 15 minutes. |

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flow rate of the eluents as follows:

Eluent Line A (1M NaOH) = 58%

Eluent Line B (WFI) = 42%

Flow rate= 0.4 mL/min | After the priming is over, close the priming knob and set the gradient level and

flow rate of the eluents as follows:

Eluent Line A (1M NaOH) = 58%

Eluent Line B (WFI) = 42%

Flow rate= 0.4 mL/min | After the priming is over, close the priming knob and set the gradient level and

flow rate of the eluents as follows:

Eluent Line A (1M NaOH) = 58%

Eluent Line B (WFI) = 42%

Flow rate= 0.4 mL/min | After the priming is over, close the priming knob and set the gradient level and

flow rate of the eluents as follows:

Eluent Line A (1M NaOH) = 58%

Eluent Line B (WFI) = 42%

Flow rate= 0.4 mL/min | After the priming is over, close the priming knob and set the gradient level and

flow rate of the eluents as follows:

Eluent Line A (1M NaOH) = 58%

Eluent Line B (WFI) = 42%

Flow rate= 0.4 mL/min

|| 4.5.2 | Conditioning of the system: | Conditioning of the system: | Conditioning of the system: |

Conditioning of the system: | Conditioning of the system: | Conditioning of the system: | Conditioning of the system: | Conditioning of the system: | Conditioning of the system: | Conditioning of the system:

|| | 4.5.2.1 | 4.5.2.1 | At the start of each test, set the pump as described in 4.5.1.6 and leave at the flow of 0.4 ml/min for at least 40 minutes to stabilize the baseline. | At the start of each test, set the pump as described in 4.5.1.6 and leave at the flow of 0.4 ml/min for at least 40 minutes to stabilize the baseline. | At the start of each test, set the pump as described in 4.5.1.6 and leave at the flow of 0.4 ml/min for at least 40 minutes to stabilize the baseline. | At the start of each test, set the pump as described in 4.5.1.6 and leave at the flow of 0.4 ml/min for at least 40 minutes to stabilize the baseline. | At the start of each test, set the pump as described in 4.5.1.6 and leave at the flow of 0.4 ml/min for at least 40 minutes to stabilize the baseline. | At the start of each test, set the pump as described in 4.5.1.6 and leave at the flow of 0.4 ml/min for at least 40 minutes to stabilize the baseline. | At the start of each test, set the pump as described in 4.5.1.6 and leave at the flow of 0.4 ml/min for at least 40 minutes to stabilize the baseline.

|| | 4.5.2.2 | 4.5.2.2 | Create the program, create a sequence and load the samples in the auto sampler as per the sequence. Run each sample for duration of 40 mins. | Create the program, create a sequence and load the samples in the auto sampler as per the sequence. Run each sample for duration of 40 mins. | Create the program, create a sequence and load the samples in the auto sampler as per the sequence. Run each sample for duration of 40 mins. | Create the program, create a sequence and load the samples in the auto sampler as per the sequence. Run each sample for duration of 40 mins. | Create the program, create a sequence and load the samples in the auto sampler as per the sequence. Run each sample for duration of 40 mins. | Create the program, create a sequence and load the samples in the auto sampler as per the sequence. Run each sample for duration of 40 mins. | Create the program, create a sequence and load the samples in the auto sampler as per the sequence. Run each sample for duration of 40 mins.

|| | S No | Sample Code | Sample Code | Sample Code | Sample Code | No. of injections | No. of injections | Volume of injection | Volume of injection

|| | 1 | EB | EB | EB | Eluent Blank | 2 | 2 | 50 µl | 50 µl

|| | 2 | B | B | B | Sample blank | 1 | 1 | 50 µl | 50 µl

|| | 3 | Std 1 (75 ng/ml) | Std 1 (75 ng/ml) | Std 1 (75 ng/ml) | Calibration Standards | 1 | 1 | 50 µl

|| | 3 | Std 2 (150 ng/ml) | Std 2 (150 ng/ml) | Std 2 (150 ng/ml) | Calibration Standards | 1 | 1 | 50 µl

|| | 3 | Std 3 (450 ng/ml) | Std 3 (450 ng/ml) | Std 3 (450 ng/ml) | Calibration Standards | 1 | 1 | 50 µl

|| | 3 | Std 4 (750 ng/ml) | Std 4 (750 ng/ml) | Std 4 (750 ng/ml) | Calibration Standards | 1 | 1 | 50 µl

|| | 4 | B | B | B | Sample blank | 1 | 1 | 50 µl | 50 µl

|| | 5 | SST (450 ng/ml) | SST (450 ng/ml) | SST (450 ng/ml) | System Suitability Test | 5 | 5 | 50 µl | 50 µl

|| | 6 | B | B | B | Sample blank | 1 | 1 | 50 µl | 50 µl

|| | 7 | T1 & T2 | T1 & T2 | T1 & T2 | Total oligosaccharide | 1 for each | 1 for each | 50 µl | 50 µl

|| | 8 | F1 & F2 | F1 & F2 | F1 & F2 | Free oligosaccharide | 1 for each | 1 for each | 50 µl | 50 µl

|| | 9 | B | B | B | Sample blank | 1 | 1 | 50 µl | 50 µl

|| | Note: One extra vial with same SST sample can be used if 5 injections are not possible from single vial. | Note: One extra vial with same SST sample can be used if 5 injections are not possible from single vial. | Note: One extra vial with same SST sample can be used if 5 injections are not possible from single vial. | Note: One extra vial with same SST sample can be used if 5 injections are not possible from single vial. | Note: One extra vial with same SST sample can be used if 5 injections are not possible from single vial. | Note: One extra vial with same SST sample can be used if 5 injections are not possible from single vial. | Note: One extra vial with same SST sample can be used if 5 injections are not possible from single vial. | Note: One extra vial with same SST sample can be used if 5 injections are not possible from single vial. | Note: One extra vial with same SST sample can be used if 5 injections are not possible from single vial.

|| | 4.5.2.3 | 4.5.2.3 | Start the batch after 'Ready check'. | Start the batch after 'Ready check'. | Start the batch after 'Ready check'. | Start the batch after 'Ready check'. | Start the batch after 'Ready check'.

|| | 4.5.2.4 | 4.5.2.4 | Inject eluent blank after every twenty injections and unknown blank after each set of samples. | Inject eluent blank after every twenty injections and unknown blank after each set of samples. | Inject eluent blank after every twenty injections and unknown blank after each set of samples. | Inject eluent blank after every twenty injections and unknown blank after each set of samples. | Inject eluent blank after every twenty injections and unknown blank after each set of samples. | Inject eluent blank after every twenty injections and unknown blank after each set of samples. | Inject eluent blank after every twenty injections and unknown blank after each set of samples.

|| | 4.5.2.5 | 4.5.2.5 | Record the calculated concentration in the analysis report. | Record the calculated concentration in the analysis report. | Record the calculated concentration in the analysis report. | Record the calculated concentration in the analysis report. | Record the calculated concentration in the analysis report.

|| 4.5.3 | Storage and regeneration: | Storage and regeneration: | Storage and regeneration: | Storage and regeneration: | Storage and regeneration: | Storage and regeneration: | Storage and regeneration:

|| | 4.5.3.1 | 4.5.3.1 | At the end of each sequence, clean the system with filtered WFI without passing it through the guard and analytical column. | At the end of each sequence, clean the system with filtered WFI without passing it through the guard and analytical column. | At the end of each sequence, clean the system with filtered WFI without passing it through the guard and analytical column. | At the end of each sequence, clean the system with filtered WFI without passing it through the guard and analytical column.

[illegible]

| | 4.5.4.1 | 4.5.4.1 | Once the observations of the standards and the test samples have been taken,
calculation shall be done using the excel sheet carrying formulas for the required calculation. | Once
the observations of the standards and the test samples have been taken, calculation shall be done using
the excel sheet carrying formulas for the required calculation. | Once the observations of the standards
and the test samples have been taken, calculation shall be done using the excel sheet carrying formulas
for the required calculation. | Once the observations of the standards and the test samples have been
taken, calculation shall be done using the excel sheet carrying formulas for the required calculation. |
Once the observations of the standards and the test samples have been taken, calculation shall be done
using the excel sheet carrying formulas for the required calculation. | Once the observations of the
standards and the test samples have been taken, calculation shall be done using the excel sheet carrying
formulas for the required calculation.

[illegible]

|| 4.5.4.5 | 4.5.4.5 | Calculate the Ribitol concentration in the test sample by using following formula:

$b = \text{intercept}$

[illegible][illegible]

|| 4.5.5.3 | 4.5.5.3 | Theoretical plates should be ≥ 3000 | Theoretical plates should be ≥ 3000 |
Theoretical plates should be ≥ 3000 | Theoretical plates should be ≥ 3000 | Theoretical plates should be
 ≥ 3000 | Theoretical plates should be ≥ 3000

4.5.6	Acceptance Criteria:	Acceptance Criteria:	Acceptance Criteria:	Acceptance Criteria:
	Acceptance Criteria:	Acceptance Criteria:	Acceptance Criteria:	Acceptance Criteria:

5.0 | References: | References: | References: | References: | References: | References: | References:
References: | References: | References: | References:

[illegible][illegible][illegible]

1.0 |Purpose: |
 |To lay down the procedure for the determination of CDAP |
 |(1-cyano-4-dimethylaminopyridinium tetrafluoroborate) by high performance |
 |liquid chromatography (HPLC). |
 2.0 |Applicable to: |
 |This Standard Operating Procedure is applicable for the analysis of the |
 |samples in which CDAP (1-cyano-4-dimethylaminopyridinium tetrafluoroborate)|
 |is to be determined. |
 3.0 |Abbreviations and Definitions |
3.1	CDAP	: 1-cyano-4-dimethylaminopyridinium tetrafluoroborate
3.2	HPLC	: High Performance Liquid Chromatography
3.3	i.d	: Internal diameter
3.4	l	: length
3.5	RSD	: Relative Standard Deviation
3.6	SST	: System Suitability Test
3.7	°C	: Degree centigrade
3.8	µg	: microgram
3.9	µL	: Micro litre
3.10	µm	: Micro meter
3.11	System	: System suitability tests are an integral part of gas
	Suitability	and liquid chromatographic methods. They are used to
	Test	verify that the resolution and reproducibility of the
		chromatographic system are adequate for the analysis
		to be done.
		The tests are based on the concept that the equipment,
		electronics, analytical operations and samples to be
		analyzed constitute an integral system that can be
		evaluated as such.
4.0	Procedure	
4.1	Precautions to be taken:	
	4.1.1	Always use filtered solutions. Filtration of solution should
		be done in order to eliminate the entry of possible particles,
		if any, into the system as well as the column.
	4.1.2	Degas all solutions.
	4.1.3	Avoid entry of bubbles into the system as well as column.
	4.1.4	All the solution bottle openings should be well covered with
		aluminum foil.
	4.1.5	Handle acids carefully and always manipulate in fume hood.
	4.1.6	Check the calibration status of all equipment and instrument.
4.2	Requirements	
	4.2.1	Equipment:
		4.2.1.1
		4.2.1.2
		4.2.1.3
		4.2.1.4
		4.2.1.5
	4.2.2	Instrument:
		4.2.2.1
		4.2.2.2
		4.2.2.3
		4.2.2.4
	4.2.3	Materials:
		4.2.3.1
		4.2.3.2
		4.2.3.3
		4.2.3.4
		4.2.3.5
		4.2.3.6
		4.2.3.7
		4.2.3.8
	4.2.4	Reagents:
		4.2.4.1
		4.2.4.2
		4.2.4.3
		4.2.4.4
		4.2.4.5
		4.2.4.6
		4.2.4.7
4.3	Preparation of solutions:	
	Prepare all the solutions and carry out the labelling of the prepared	
	solutions and assign the lot numbers to the solutions.	
	4.3.1	Solution A
	4.3.2	Mobile phase for CDAP:
	4.3.3	Isopropyl Alcohol 70% Solution

			months.			
		4.3.4	1 mg/mL of CDAP calibration standard stock:			
		Weigh 5mg of CDAP and make up the volume to 5mL with purified				
		water in a volumetric flask. This solution can be stored as				
		aliquots at -20°C for 2months.				
		4.3.5	Preparation of working calibration standard of 10µg/mL or			
		10ng/µL:				
		Take 10µL of 1 mg/mL of CDAP calibration standard stock and				
		add 990µL of purified water to make 1mL of solution in an				
		leppendorf or HPLC vial. Mix thoroughly. Prepare fresh before				
		use.				
		Note: The working calibration standard is prepared fresh				
		before use. All the solutions prepared for use are vacuum				
		filtered using Nylon 6, 6 membrane 0.20/0.22/0.45 µm filters				
		to eliminate the entry of possible particles, if any into the				
		system as well as the column. For organic solvents 0.45µm				
		filters should be used (filtration not necessary if the				
		solution is prepared in 0.2 µ filtered purified water).				
		4.4	Preparing the HPLC System for run:			
		4.5	Run Conditions:			
		4.5.1	Load method			
		4.5.2	Flow Rate: 1.0 mL/min			
		4.5.3	Detection wavelength: 280 nm			
		4.5.4	Run time: 4 minutes			
		4.5.5	Mobile Phase: mobile phase for CDAP			
		4.6	Equilibration of the system and the column:			
		Equilibrate the system and column with mobile phase for CDAP until a				
		stable baseline at 280 nm is obtained which can be viewed with the				
		online plot window.				
		4.7	Sample Preparation and Application:			
		Test sample and standards are loaded in screw cap vials and placed				
		against the appropriate no. on the sample tray by clearly marking				
		the screw cap vial.				
		4.8	Running of sequence:			
		4.8.1	Load sequence.			
		4.8.2	Open the sequence parameter option by clicking Sequence menu.			
		4.8.3	Enter information about the data directory and operator name.			
		4.8.4	In the shutdown option, click on the post-sequence option and			
		select LAMPOFF/STANDBY.				
		4.8.5	In sequence table option, specify the sample names, methods			
		and vials to run in the sequence as per the following table:				
		S. No	Sample Code	Sample details	No. of	Injection
			injections	n volume		
			per Vial			
		1	Blank	Eluent Blank	1	10 µL
		2	Standard 3	System	6	0.5 µL
		(5ng/µL)	Suitability			
		test standard				
		3	Blank	Eluent Blank	1	10 µL
		4	Standard 1	1ng/µL	1	0.1 µL
		Standard				
		5	Standard 2	3ng/µL	1	0.3 µL
		Standard				
		6	Standard 3	5ng/µL	1	0.5 µL
		Standard				
		7	Standard 4	10ng/µL	1	1.0 µL
		Standard				
		8	Standard 5	20ng/µL	1	2.0 µL
		Standard				
		9	Blank	Eluent Blank	1	10 µL
		10	Test Sample	Sample	1	1.0 µL
		(s)				
		4.8.6	Click on run sequence option in run control menu to start the			
		sequence.				
		4.8.7	Each sample run will be of 4 minutes and after running all			
		the samples the sequence will automatically stop.				
		4.8.8	Inject the eluent blank after every 20 injections.			
		Note: Here, eluent blank refers to mobile phase. One extra				
		vial with same SST sample can be used if 6 injections are not				
		possible from single vial.				
		4.9	Calculations:			
		4.9.1	Results of the sample are calculated from the calibration			
		standards using the chemstation software.				
		4.9.1.1	Open the chromatogram corresponding to 1ng/µL			
		calibration standard in the offline copy of				
		chemstation software in the Data Analysis mode.				
		4.9.1.2	From the calibration menu select calibration			
		settings and write ng/µL in the Amount unit's				
		option.				
		4.9.1.3	Select New Calibration Table from Calibration menu			
		and enter value1 in default amount option and value				
		1 in level option.				
		4.9.1.4	Now open the chromatogram for 3.0 ng/µL. Select add			
		level option from calibration menu and enter value				
		3.0 in default amount and value 2 in level option.				
		4.9.1.5	Similarly open chromatograms for 5.0ng/µL,			
		10.0ng/µL and 20.0ng/µL and keep adding default				
		amount and appropriate value of level.				
		4.9.1.6	Select calibration task icon and the calibration			
		table with required concentrations and levels can				
		be viewed in the calibration table view. A				
		calibration curve is also displayed in the				

			calibration curve field with value of correlation
			coefficient of the curve.
		4.9.1.7	Save the calibration method under appropriate name.
		4.9.1.8	Open the chromatogram of test sample and select the
			above created calibration method for integration of
			the test sample.
		4.9.1.9	In the specify report option of Report menu, select
			ESTD in the Calculate field.
		4.9.1.10	In the report layout view click on the show report
			preview with current template, method and data file
			icon and view the result of the test sample in
			terms of concentration (ng/μL).
		4.9.1.11	Calculations for Standard curve and samples:
			The calculations are made according to the equation
			of minimum squares.
			$Y = mX + b$.
			Where: m = slope
			b = intercept
			Y = absorbance
			X = concentration
		4.9.2	The calculations for determination of amount of CDAP per dose
			(10μg) of ARM of Haemophilus Influenzae type b conjugate
			vaccine is as follows:
			Example: If a sample has a PRP concentration of 1.0 mg/mL and
			the amount of CDAP present in the sample is 5ng/μL or 5μg/mL.
			The amount of CDAP present per dose(10μg) is calculated as
			follows:
			1000μg -----5μg
			10 μg(dose)----- X
			$X = 5\mu\text{g} \times 10 \mu\text{g}$
			1000 μg
			= 0.05μg or
			$X = 50 \text{ ng/dose}$
		4.9.3	Calculations for System Suitability Test:
		4.9.3.1	Note the values of area under the curve and
			retention time for standard 3(SST standard) .
		4.10	Acceptance Criteria System Suitability Test:
		4.10.1	RSD calculated for Retention time of standard 3(5ng/μL)
			should be <5.0%.
		4.10.2	RSD calculated for Area of standard 3(5ng/μL) should be
			<5.0%.
		4.11	Result Reporting:
		4.11.1	Open the template and take print out of the report.
		4.11.2	Print report: Click on print option to take a printout of a
			ChemStation report using the current report template and data
			from the currently loaded data file.
		4.11.3	Report the results of quantity of CDAP present/dose (10μg).
		4.11.4	The report generated can be viewed with report review icon.
		4.11.5	The printed report is signed by the analyst who does the
			analysis and also checked by the other analyst/supervisor.
		5.0	References:
		5.1	European pharmacopoeia.

[1.0 |Objective |
 ||To lay down the procedure for determination of Triethylamine (TEA) content |
 ||in Polysaccharide Conjugate samples by Gas Chromatography. |
 [2.0 |Scope |
 ||This Standard Operating Procedure is applicable to all the samples, in |
 ||which the Triethylamine (TEA) content is to be determined. |
 [3.0 |Abbreviations and Definitions |
	3.1	PBS	: Phosphate Buffer Saline
	3.2	Std.	: Standard
	3.3	µg	: Microgram
	3.4	µl	: Micro litre
	3.5	µm	: Micro meter
	3.6	TEA	: Triethyl amine
[4.0	Procedure		
	4.1	Precautions to be taken:	
	4.1.1	Always wear laboratory coat.	
	4.1.2	Always use solutions and samples filtered through 0.2 µm	
		membrane.	
	4.1.3	Check the calibration status of all equipment and instrument.	
	4.1.4	Handle acids carefully wearing gloves and always manipulate in	
		fume hood.	
	4.2	Requirement:	
	4.2.1	Equipment and Instruments:	
	4.2.1.1	Gas Chromatography system with head space	
	4.2.1.2	Micropipettes	
	4.2.2	Material:	
	4.2.2.1	Triethyl amine (Make: Fluka, Catalogue No. 90340)	
	4.2.2.2	Sodium chloride (NaCl) (Make: Qualigens, Cat No: 15915)	
	4.2.2.3	Disodium hydrogen phosphate dodecahydrate	
		(Na₂HPO₄·12H₂O) (Make: Sigma Aldrich,Cat No.71650)	
	4.2.2.4	Laboratory Glassware	
	4.2.2.5	Sodium dihydrogen phosphate dehydrate (NaH₂PO₄·2H₂O)	
		(Make: Merck, Cat No.1.06342.0250)	
	4.2.2.6	Water for Injection (WFI)	
	4.2.2.7	Hydrogen gas (IOLAR)	
	4.2.2.8	Nitrogen gas (IOLAR)	
	4.2.2.9	Helium gas (IONAR)	
	4.2.2.1	Zero gas (IOLAR)	
		0	
	4.2.2.1	Separation Column, e.g. OPTIMA–624–LB–1.80 µm –30 m x	
		1	0.32 mm (Make: Macherey-Nagel, Catalogue No.726 786.30)
	4.2.2.1	1.5 mL vials	
		2	
	4.2.2.1	Volumetric flasks- 100 mL	
		3	
	4.2.2.1	Glass pipettes	
		4	
	4.2.2.1	Head space vials and septa	
		5	
	4.2.2.1	Buffer (0.2 M NaCl, 10mM Phosphate Buffer pH 6.0)	
		6	
[4.3	Preparation of solutions:		
	Prepare all the solutions as described below. Carry out the labelling		
	of the prepared solutions and assign the lot numbers.		
	4.3.1	0.2 M NaCl, 10mM Phosphate Buffer pH 6.0:	
		Take 1.17 g NaCl, 0.054 g Na₂HPO₄·12H₂O and 0.133 g	
		NaH₂PO₄·2H₂O. Dissolve in 80 mL of MQW, adjust the pH with HCL	
		or NaOH and make up to 100mL. Filter through 0.22 µm filter.	
		This solution can be stored up to one month at room	
		temperature.	
	4.3.2	TEA stock Standard (10,000 µg/mL):	
		Take 1.370 mL of neat TEA and make up to 100 mL with 0.2 M	
		NaCl, 10mM Phosphate Buffer in a volumetric flask. Prepare	
		fresh on the day of use.	
	4.3.3	Positive control (50µg/mL of TEA standards)	
		Add 15µL of TEA stock to 3mL of 0.2 M NaCl, 10mM Phosphate	
		Buffer.	
	4.3.4	Test sample preparation:	
	4.3.5	Dilute the test sample with 0.2 M NaCl, 10mM Phosphate Buffer	
		to 10mg/ml of pneumococcal polysaccharide based on its initial	
		concentration.	
[4.4	Test Procedure:		
	4.4.1	Take 3 mL of 0.2 M NaCl, 10mM Phosphate Buffer as a blank, 3 mL	
		of positive control and 3mL of diluted test sample into	
		respectively labeled head space vials, and place these vials in	
		the head space carousel.	
	4.4.2	Create a batch file in GC solution software and wait for GC	
		ready command and click start button.	
	4.4.3	Then create a schedule file in HT3 TEKlink software for head	
		spacer.	
	4.4.4	Chromatography conditions:	
		Set the chromatography conditions as described below:	
	4.4.4.1	Carrier Gas	Helium
	4.4.4.2	Makeup Gas	Nitrogen (20 mL/minute)
	4.4.4.3	Hydrogen Gas	40mL/minute
	4.4.4.4	Zero Air	300mL/minute
	4.4.4.5	Flow (Pressure)	30kPa
	4.4.4.6	Split ratio	1:10
	4.4.4.7	Furnance Temperature	5 minute at 40OC and then
			increase to 180 OC at a rate

				of 10 OC per minute and hold		
				for 5 minutes at 180OC.		
			4.4.4.8	Detector Temperature	250 OC	
			4.4.4.9	Injection volume	3.0 mL	
			4.4.4.10	Run Time	24 minutes	
		4.4.5	Conditioning of the column			
				Perform the column conditioning using the following settings		
			4.4.5.1	Injector temperature	200 OC	
			4.4.5.2	Column temperature	220OC	
			4.4.5.3	Detector temperature	280 OC	
			4.4.5.4	Column Flow	0.5mL/min	
		4.4.6	Head space system conditioning			
				Set the head spacer conditions as described below		
			4.4.6.1	Constant Heat Time	On	
			4.4.6.2	GC Cycle Time	35 minute	
			4.4.6.3	Platen Temperature	155OC	
			4.4.6.4	Value oven temperature	160 OC	
			4.4.6.5	Transfer line	165 OC	
				Temperature		
			4.4.6.6	Sample equilibration	30 minutes	
			4.4.6.7	Pressurize	15 PSIG	
			4.4.6.8	Loop fill Pressure	10 PSIG	
			4.4.6.9	Injection Time	3.00 minutes	
			4.4.6.1	Loop fill Time	4.00 minutes	
				0		
			4.4.6.1	Pressurize equilibration	0.20 minutes	
				1	time	
			4.4.6.1	Pressurize Time	2 minutes	
				2		
		4.4.7	Communicating Head space system with Gas chromatography			
				4.4.7.1	Wait for ready command in GC and then click start	
				button which in- turn gives stand by indication on GC		
				software.		
				4.4.7.2	Now click start button in head space system schedule oh	
				HT3 TEKlink software.		
				4.4.7.3	Take the printout of the chromatograms and attach to	
				the prescribed annexure.		
			4.4.7.4	Record the details in record sheets.		
		4.5	Acceptance criteria:			
		As per the specification the residual TEA content should be "less				
		than 10 ng/µg of polysaccharide". Therefore the acceptance criterion				
		of the assay is "The TEA peak areas of the test sample containing 10				
		µg/mL of Polysaccharide should be less than the positive control peak				
		area values (50 µg/mL of TEA)"				
		5.0	Reference			
		5.1	British pharmacopoeia.			
		5.2	Head space system Manual - TEKMAR TELEDYNE HT3			

3.13 | °C | °C | : Degree centigrade | : Degree centigrade | : Degree centigrade | : Degree centigrade
| : Degree centigrade | : Degree centigrade | : Degree centigrade | : Degree centigrade

| 3.15 | μL | μL | : Micro litre | : Micro litre | : Micro litre | : Micro litre | : Micro litre | :
Micro litre | : Micro litre | : Micro litre

3.17 | System Suitability Test | System Suitability Test : System suitability tests are an integral part of gas and liquid chromatographic methods. They are used to verify that the resolution and reproducibility of the chromatographic system are adequate for the analysis to be done.

The tests are based on the concept that the equipment, electronics, analytical operations and samples to be analyzed constitute an integral system that can be evaluated as such. 1. System suitability tests are an integral part of gas and liquid chromatographic methods. They are used to verify that the resolution and reproducibility of the chromatographic system are adequate for the analysis to be done.

The tests are based on the concept that the equipment, electronics, analytical operations and samples to be analyzed constitute an integral system that can be evaluated as such. I : System suitability tests are an integral part of gas and liquid chromatographic methods. They are used to verify that the resolution and reproducibility of the chromatographic system are adequate for the analysis to be done.

The tests are based on the concept that the equipment, electronics, analytical operations and samples to be analyzed constitute an integral system that can be evaluated as such. I : System suitability tests are an integral part of gas and liquid chromatographic methods. They are used to verify that the resolution and reproducibility of the chromatographic system are adequate for the analysis to be done.

The tests are based on the concept that the equipment, electronics, analytical operations and samples to be analyzed constitute an integral system that can be evaluated as such.

| 4.1 | Safety considerations/Precautions to be taken: | Safety considerations/Precautions to be taken: |
Safety considerations/Precautions to be taken: | Safety considerations/Precautions to be taken: | Safety
considerations/Precautions to be taken: | Safety considerations/Precautions to be taken: | Safety
considerations/Precautions to be taken: | Safety considerations/Precautions to be taken: | Safety
considerations/Precautions to be taken: | Safety considerations/Precautions to be taken: | Safety
considerations/Precautions to be taken:

| | 4.1.2 | Always use solutions filtered through 0.22 µm membrane. | Always use solutions filtered
through 0.22 µm membrane. | Always use solutions filtered through 0.22 µm membrane. | Always use
solutions filtered through 0.22 µm membrane. | Always use solutions filtered through 0.22 µm membrane. |
Always use solutions filtered through 0.22 µm membrane. | Always use solutions filtered through 0.22 µm
membrane. | Always use solutions filtered through 0.22 µm membrane. | Always use solutions filtered
through 0.22 µm membrane. | Always use solutions filtered through 0.22 µm membrane.

| | 4.1.4 | Avoid entry of bubbles into the system as well as column. | Avoid entry of bubbles into the
system as well as column. | Avoid entry of bubbles into the system as well as column. | Avoid entry of
bubbles into the system as well as column. | Avoid entry of bubbles into the system as well as column.
Avoid entry of bubbles into the system as well as column. | Avoid entry of bubbles into the system as
well as column. | Avoid entry of bubbles into the system as well as column. | Avoid entry of bubbles into
the system as well as column. | Avoid entry of bubbles into the system as well as column.

[illegible]

| 4.1.8 | Never apply potential to the electrode unless a stream of eluent or water is flowing through the cell. | Never apply potential to the electrode unless a stream of eluent or water is flowing through the cell. | Never apply potential to the electrode unless a stream of eluent or water is flowing through the cell.

[illegible]

[illegible][illegible]

| | | 4.2.3.2 | Screw cap vials | Screw cap vials | Screw cap vials | Screw cap vials | Screw cap vials |
Screw cap vials | Screw cap vials | Screw cap vials | Screw cap vials

[illegible][illegible]

| | | 4.2.3.8 | 1.5 ml eppendorf tubes | 1.5 ml eppendorf tubes | 1.5 ml eppendorf tubes | 1.5 ml
eppendorf tubes | 1.5 ml eppendorf tubes | 1.5 ml eppendorf tubes | 1.5 ml eppendorf tubes | 1.5 ml
eppendorf tubes | 1.5 ml eppendorf tubes

|| 4.2.4.1 | Glacial Acetic acid | Glacial Acetic acid | Glacial Acetic acid | Glacial Acetic acid |
Glacial Acetic acid | Glacial Acetic acid | Glacial Acetic acid | Glacial Acetic acid | Glacial Acetic
acid

	4.2.4.3	Sodium Hydroxide	Sodium Hydroxide	Sodium Hydroxide	Sodium Hydroxide	Sodium Hydroxide
Hydroxide		Sodium Hydroxide	Sodium Hydroxide	Sodium Hydroxide	Sodium Hydroxide	Sodium Hydroxide

[illegible][illegible][illegible]

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|| || || || || || || || || || || ||

[illegible]

| 4.5.5 | Calculations: | Calculations: | Calculations: | Calculations: | Calculations: | Calculations:
| Calculations: | Calculations: | Calculations:

[illegible]

|| | 4.5.5.4 | Enter the absorbance for calculations starting from the Calibration curve prepared between Pullulan standards and test samples. The calculations are made according to the equation of minimum squares. Plot a standard calibration curve as molecular weight versus retention time. | Enter the absorbance for calculations starting from the Calibration curve prepared between Pullulan standards and test samples. The calculations are made according to the equation of minimum squares. Plot a standard calibration curve as molecular weight versus retention time. | Enter the absorbance for calculations starting from the Calibration curve prepared between Pullulan standards and test samples. The calculations are made according to the equation of minimum squares. Plot a standard calibration curve as molecular weight versus retention time. | Enter the absorbance for calculations starting from the Calibration curve prepared between Pullulan standards and test samples. The calculations are made

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Reagent Details:

Formula & Calculations:

Formula: $M_1 \times V_1 = M_2 \times V_2$.

Calculation: To prepare 1 L of 1M HCl from 12M HCl (or 37% concentrated HCl) stock solution calculate quantity required of stock solution as below:

12M HCl stock solution required quantity= $1 \times 1 / 12 = 83.3\text{mL}$

Preparation Method:

To prepare 1M HCl, 1L solution from 12M HCl stock solution formula is $M_1(\text{Molarity of stock}) \times V_1(\text{volume of stock}) = M_2(\text{Molarity of working}) \times V_2(\text{volume of working})$.

Take 83.3mL of 12M HCl solution and make a 1 L solution.

Preparation for different volumes:

To prepare _____ of 1M HCl

Take 83.3mL x _____ L = _____ mL of 12M HCl

Make HCl solution to _____ L solution with WFI.

Check and record the details:

Preparation Details:

Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry

Date

|||

Final volume prepared | Weighed Quantity | Storage Condition:

||

Date of Preparation | Prepared By | Checked By

||

1) Sewage & Effluent Treatment: Clearly mark the physical location, dimensions, and layout of the Effluent Treatment Plant (ETP) / Sewage Treatment Plant (STP). Provide structural calculations proving adequate capacity for processing waste.
 Response: Physical location already marked in site layout.
 2) Organic Waste Converter (OWC): Designate a dedicated, ventilated space for a mechanical OWC, stating its processing capacity in kg/day based on your expected workforce.
 Response: Plasma: 20.4MT/year or 68T/day, LVP: 7.9MT/year or 26T/day
 3) Process Flow & Water Balance Chart: Submit a detailed quantitative water balance diagram showing domestic and industrial consumption, recycling rates, and a commitment to Zero Liquid Discharge (ZLD).
 Response:
 Plasma:
 LVP:
 4) Hazardous Waste Management Plan: Since ethanol-based fractionation is utilized, submit storage, handling, and disposal plans for hazardous/chemical solvents
 Response:
 Plasma:
 Note: Solid Biowaste will be treated with IPA and after decontamination it will be handed over to 3rd party.
 LVP:

Sr.No.	Item	Litres (L / year)	Cubic metres (m ³ / year)	Notes
A. Water Inputs				
1	Domestic water (506 ppl × 100 L/p/d)	1,84,69,000	18,469.00	Toilets, canteen, housekeeping
	WFI (process)	62,05,470	6,205.47	WFI for formulation, CIP, rinses
	Other process water (25% of WFI)	15,51,368	1,551.37	CIP makeup, utility washes
	Extra process water (10% of plasma vol)	41,400	41.4	Misc. process losses, small uses
2	Process Water	77,98,238	7,798	
	Total Water Input	2,62,67,238	26,267.24	A = sum of above
B. Water Outputs / Wastewater				
1	Domestic sewage → STP (80% of domestic)	1,47,75,200	14,775.20	Black/gray water to STP
2	Process wastewater → ETP (99% of process)	77,20,256	7,720.26	Process drains, WFI rejects, CIP effluent
	Total Wastewater Generated	2,24,95,456	22,495.46	B = sum of above
C. Balance / Losses / Product incorporation				
1	Unaccounted losses / product incorporation / evaporation (A – B)	37,71,782	3,771.78	~14.36% of input — includes water in products, steam losses, sampling, evaporation, bleed, equipment hold-up
Sr.No. Item Litres (L / year) Cubic metres (m ³ / year) Notes				
A. Water Inputs				
1	Domestic water (400 ppl × 100 L/p/d)	1,45,00,201	14,500.20	Toilets, canteen, housekeeping
	WFI (process)	2,42,40,000	24,240	WFI for LVP, CIP, rinses
	Other process water (25% of WFI)	60,60,000	6,060	CIP makeup, utility washes
	Extra process water (10% of vol)	6,06,000	606	Misc. process losses, small uses
2	Process Water	3,09,06,000	30,906	
	Total Water Input	4,54,06,201	45,406.20	A = sum of above
B. Water Outputs / Wastewater				
1	Domestic sewage → STP (80% of domestic)	1,16,00,160	11,600.16	Black/gray water to STP
2	Process wastewater → ETP (99% of process)	3,05,96,940	30,596.94	Process drains, WFI rejects, CIP effluent
	Total Wastewater Generated	4,21,97,100	42,197.10	B = sum of above
C. Balance / Losses / Product incorporation				
1	Unaccounted losses / product incorporation / evaporation (A – B)	32,09,101	3,209	~7.06% of input — includes water in products, steam losses, sampling, evaporation, bleed, equipment hold-up
Output Category Output Stream Estimated Quantity per Day Estimated Quantity per Year Remarks				
Treated Water Recovery Treated Water for Reuse / Discharge ~3,500 – 6,800 L ~840,000 – 1,632,000 L				
Recovered after primary and secondary treatment				
Reject Water / Losses RO Reject / Evaporation / Handling Losses ~500 – 1,600 L ~120,000 – 384,000 L				
Depends on treatment efficiency and recovery system				
Sludge Generation ETP Sludge / Solid Waste ~20 – 50 kg ~4,800 – 12,000 kg To be disposed through authorized waste handler				
Final Discharge Treated Effluent As per discharge norms As per regulatory limits Suitable for reuse/discharge after compliance testing				
Sr. No. Process / Source Type of Hazardous Waste Approx. Quantity / Year Proposed Method				
1	Solvent recovery & viral inactivation (TnBP + detergent process)	Spent solvent mixture (Tri-n-butyl phosphate + Triton X-100 / Polysorbate 80)	1.0 – 1.5 KL / year	ETP
2	Filtration & purification units	Used filter aids / depth filter media contaminated with protein or solvent	0.8 – 1.0 MT / year	3rd party
3	Activated carbon beds (viral removal, deodorization)	Spent activated carbon with adsorbed organic residues	0.5 – 0.7 MT / year	3rd party
4	Chromatography / TFF cleaning (CIP)	Residues of buffers, cleaning agents, NaOH / HCl neutralization sludge	2.0 – 3.0 MT / year (dry solids)	ETP
5	ETP operation	Sludge containing proteinaceous organic matter & trace solvents	5.0 – 6.0 MT / year	3rd party
6	ETP / RO reject water	High-TDS / COD effluent (liquid)	≈ 900 – 1,000 m ³ / year	ETP
7	Resin & membrane replacement	Spent chromatography resin, UF/DF membranes, PVDF filters	0.5 – 1.0 MT / year	3rd party
8	Laboratory & QC activities	Contaminated lab glassware, pipette tips, chemical bottles	200 – 250 kg / year	Scrap 3rd party
9	Oil from DG sets, compressors, hydraulic systems	Used lubricating oil	50 – 100 L / year	ETP
10	DG set maintenance	Oil-soaked cotton waste, filters	50 – 100 kg / year	Scrap 3rd party
11	Production packaging & vial washing	Contaminated packaging (plastic, foil, paper)	0.3 – 0.5 MT / year	Scrap 3rd party
12	Spill & floor washings	Absorbent material, mopping waste, PPE, gloves	0.5 MT / year	3rd party
Hazardous Waste API Residue / Chemical Sludge 50 – 200 kg 12000 – 48000 kg High-temperature incineration (>1,200°C). 3rd Party as proposed method				

Reagent Details:

Formula:

Tri Sodium Citrate (M.Wt): 258.07.

Citric acid (M.Wt): 192.124

Preparation of PF-RR-03 stock buffer:

Preparation of 0.2M Sodium Citrate pH 7.0.

Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).

Calculation: To prepare 1 L of 0.2M Citric Acid (MW = 192.124 g/mol)

Weight of solute(g)=0.2×192.124×1=38.424g.

Preparation Method:

Weigh 38.424g of Citric Acid and dissolve in water to make a 1 L solution.

Preparation for differential volumes:

To prepare _____ of 0.2M Citric Acid (MW = 38.424g/mol).

Weight of solute(g)=0.2×38.424 x _____ L= _____ g.

Weigh _____ gm of Citric Acid and dissolve in water to make a _____ L solution.

Calculations: To prepare the 0.2M Tris Sodium Citrate.

Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).

Calculation: To prepare 1 L of 0.2M Tris Sodium Citrate (MW = 258.07 g/mol)

Weight of solute(g)=0.2×258.07 x1=51.614g

Preparation Method:

Weigh 51.614g of Tris Sodium Citrate and dissolve in water to make a 1 L solution.

Preparation for different volumes:

To prepare _____ of 0.2M Tris Sodium Citrate (MW = 258.07 g/mol).

Weight of solute(g)=0.2×258.07x _____ L= _____ g.

Weigh _____ gm of Tris Sodium Citrate and dissolve in water to make a _____ L solution.

Procedure for Preparation of PF-RR-03 stock

To prepare a pH 7.0 solution, 0.2 M Tris Sodium Citrate solution is titrated with 0.2 M citric acid so that the pH attains pH 7.0 ±0.1.

For Example, to prepare a 1 L solution, take 0.2 M Tri Sodium Citrate and adjust the pH with the 0.2 M Citric acid to get the pH 7.0 ±0.1 and make up the final volume to 1L.

Check and record the details:

Preparation Details:

Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry

Date

Tri Sodium Citrate | | | |

Citric acid | | | |

Final volume prepared | pH (7.0 ± 0.1) | Conductivity (23± 2ms/cm) | Storage Condition | Reagent Record

| | | RT | PF-RR-03

Date of Preparation | Prepared By | Checked By

| |

Reagent Details:

Formula:

Sodium Citrate (M.Wt): 258.07.

Citric acid (M.Wt): 192.124

Sodium Chloride (M.Wt): 58.44

Calcium Chloride (M.Wt): 110.98

0.1% Tween 80

Preparation of FE-RR-010 buffer with Stock FE-RR-09.

Preparation of 20mM sodium citrate pH 6.7 + 0.13M sodium chloride + 2.55mM calcium chloride + 0.01% tween 80.

Stock Buffer (FE-RR-03): 0.2M Sodium Citrate pH 7.0

Formula:

$$C1 \times V1 = C2 \times V2$$

C1 = concentration of stock

V1 = volume of stock you need to take

C2 = desired (working) concentration

V2 = final total volume of working solution

Applying the formula to prepare the working buffer:

$$C1 \times V1 = C2 \times V2$$

C1 = 0.2M Sodium Citrate pH 7.0

V1 = volume of stock you need to take _____ L.

C2 = desired (working) concentration 20mM Sodium Citrate, pH 7.0

V2 = final total volume of working solution _____ L.

$$V1 = (C2 \times V2) / C1$$

$$C2 = 0.02M ; V2 = 1.0 L ; C1 = 0.2M$$

$$V1 = (0.02 \times 1.0) / 0.2$$

$$V1 = 0.1L$$

Take 0.1 L (FE-RR-03) Stock buffer solution and dilute with water, and make up to 1L.

Calculations: To prepare the 0.13M Sodium Chloride.

Formula: Weight of solute (g) = Molarity (M) × Molecular Weight (g/mol) × Volume (L).

Calculation: To prepare 1 L of 0.13M Sodium Chloride (MW = 58.44 g/mol)

$$\text{Weight of solute (g)} = 0.13 \times 58.44 \times 1 = 7.59g.$$

Preparation Method:

Weigh 7.59g of Sodium Chloride and dissolve in water to make a 1 L solution.

Preparation for different volumes:

To prepare _____ L of 0.13M Sodium Chloride (MW = 58.44 g/mol).

$$\text{Weight of solute (g)} = 0.13 \times 58.44 \times \text{L} = \text{g}.$$

Weigh _____ gm of Sodium Chloride and dissolve in water to make a _____ L solution.

Calculations: To prepare the 2.55mM Calcium Chloride.

Formula: Weight of solute (g) = Molarity (M) × Molecular Weight (g/mol) × Volume (L).

Calculation: To prepare 1 L of 2.55 mM Calcium Chloride (MW = 110.98g/mol)

$$\text{Weight of solute (g)} = 0.0025 \times 110.98 \times 1 = 0.277g.$$

Preparation Method:

Weigh 0.277g of Calcium Chloride and dissolve in water to make a 1 L solution.

Preparation for different volumes:

To prepare _____ L of 0.0025mM Calcium Chloride (MW = 110.98 g/mol).

$$\text{Weight of solute (g)} = 0.0025 \times 110.98 \times \text{L} = \text{g}.$$

Weigh _____ gm of Calcium Chloride and dissolve in water to make a _____ L solution.

Procedure

Calculation: to prepare the 0.01% Tween 80:

For 1000 mL of 0.01% solution:

$$0.01\% = 0.01g \text{ per } 100ml \text{ \& } 1000 \text{ mL} = 100 \text{ mg}$$

Since Tween 80 is a liquid, you can weigh it directly or use its density (~1.06 g/mL) to measure volume:

$$\text{Approx. } 100 \mu\text{L of Tween 80} = 0.1 \text{ g}$$

Add Tween 80 to a beaker or flask containing ~800 mL of water.

Mix thoroughly:

Use a magnetic stirrer or vortex until fully dissolved. Tween 80 is viscous, so mixing may take a few minutes.

Adjust volume:

After complete dissolution, bring the final volume up to 1000 mL with water.

Procedure for Preparation of FE-RR-10 buffer with the stock FE-RR-09.

To prepare a 1 L solution, take 0.2M Tri Sodium Citrate pH 7.0. To this, add the 0.13 M sodium chloride

and dissolve it and make up the volume with water to 900ml, and then to this add 0.0025 M Calcium

chloride, to this add the 0.01% Tween 80 then mix the solution till it dissolves and check the pH; if the

pH is less than 6.7 then adjust the pH with 1 M sodium hydroxide to get the pH 6.7 ± 0.1 and make up the

final volume to 1L.

Check and record the details:

Preparation Details:

Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry Date

Sodium Citrate | | | |

Citric Acid | | | |

Sodium Chloride | | | |

Calcium Chloride | | | |

Tween 80 | | | |

Stock buffer (FE-RR-09) taken volume(L) | pH (6.7 ± 0.1) | Conductivity (17 ± 1.0ms/cm) | Storage

Condition | Final volume prepared

|| RT |

Date of Preparation | Prepared By | Checked By

| |

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Sample Details :

Collect the concentrated sample from Bulk Protocol step No:7.3.4. record the details.

PRA-NF-005 FILTRATION

Filter the concentrated sample through a sterile PRA-SF-001 filter (pre-filter for PRA-NF-005 filtration) at 0.5 bar pressure under LAFU.

Check and record the following details:

Filtrate Volume: _____ L,

O.D. at 280 nm: _____, O.D. at 600 nm: _____.

Pass the pre-filtered sample through the PRA-NF-005 filter at 1.0 bar pressure.

Check and record the following details:

Filtrate Volume: _____ L,

O.D. at 280 nm: _____, O.D. at 600 nm: _____.

PRA-SF-001 FILTRATION

Filter the PRA-NF-005 filtered sample through a sterile PRA-SF-001 filter at 0.5 bar pressure under LAFU, and collect the filtrate into sterilised 50 ml polypropylene bottles.

Check and record the following details:

O. D. at 280 nm: _____, O.D. at 600 nm: _____.

Collect and send 10 ml of the PRA-SF-001 filtered sample to QC for in-process QC analysis. Record the QC Analysis report.

Store the bottles containing PRA-SF-001 filtrate at -20 °C until further use.

Details | Details

Volume (L) |

O.D. at 280 nm: |

Activity (IU/ml) |

Protein Conc. (mg/ml) |

Total protein (mg) |

Details | Details

Volume (L) |

Activity (IU/ml) |

Protein Conc. (mg/ml) |

Total protein (mg) |

Reagent Details:

Formula:

Sodium Citrate (M.Wt): 258.07.

Citric acid (M.Wt): 192.124

Preparation of FN-RR-08 stock buffer:

Preparation of 0.2M Sodium Citrate pH 7.4

Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).

Calculation: To prepare 1 L of 0.2M Citric Acid (MW = 192.124 g/mol)

Weight of solute(g)=0.2×192.124×1=38.42g.

Preparation Method:

Weigh 38.42 g of Citric Acid and dissolve in water to make a 1 L solution.

Preparation for required volumes:

To prepare _____ of 0.2M Citric Acid (MW = 192.124g/mol).

Weight of solute(g)=0.2×192.124x _____L= _____g.

Weigh _____gm of Citric Acid and dissolve in water to make a _____ L solution.

Calculations: To prepare the 0.2M Tris Sodium Citrate.

Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).

Calculation: To prepare 1 L of 0.2M Tris Sodium Citrate (MW = 258.07 g/mol)

Weight of solute(g)=0.2×258.07 x1=51.61g

Preparation Method:

Weigh 51.61g of Tris Sodium Citrate and dissolve in water to make a 1 L solution.

Preparation for different volumes:

To prepare _____ of 0.2M Tris Sodium Citrate (MW = 258.07 g/mol).

Weight of solute(g)=0.2×258.07x _____L= _____g.

Weigh _____gm of Tris Sodium Citrate and dissolve in water to make a _____ L solution.

Procedure for Preparation of FN-RR-08 stock

To prepare a pH 7.4 solution, 0.2 M Tris Sodium Citrate solution is titrated with 0.2 M citric acid so that the pH attains pH 7.4.

For Example, to prepare a 1 L solution, take 0.2 M Tri Sodium Citrate and adjust the pH with the 0.2 M Citric acid to get the pH 7.4 and make up the final volume to 1 L.

Check and record the details:

Preparation Details:

Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry

Date

Sodium Citrate | | | |

Citric Acid | | | |

0.2 M Tris Sodium Citrate Volume taken | 0.2 M Citric acid volume used to titrate | pH (7.4±0.1) |

Conductivity (2.8±0.2ms/cm) | Storage Condition | Final volume prepared

| | | | RT |

Date of Preparation | Prepared By | Checked By

| |

Query: In Utility RDS; transformer load looks high?

Response:

Query: In Utility RDS summary sheet; Dual fuel boiler (3000Kg/Hr) and normal boiler (1000 Kg/H) is mentioned but capacity for both is mentioned different, so what capacity is exactly required?

Response:

Query: Black steam is considered for PSG calculation?

Response:

Query: Common load like electric bulb load not considered in RDS as it is mentioned as 0KW

Response: We have considered all equipment load and Lux value is mentioned for each area, please considered electric bulb etc load on standard basis.

Query: For CNC area temperature not given?

Response: We have given temperate range where it is mandatory to control; however for CNC area RT can be considered.

Query: Differential Pressure difference between adjacent areas are always 5 Pascal ,may be it should be min.10 Pascal ,PI check

Response: We have considered DP basis on standard requirements.

Query: Where you use common AHU and where you use individual AHU

Response: AHU zoning layout is pending and in process.

Query: AHU type - CHW OR Dx system

Response:

Query: ACPH of Area is not mentioned in RDS.

Response: To be calculated

Query: AHU Zoning with AHU No.

Response: AHU zoning layout is pending and in process.

[SHEET: Organic waste Plasma]

Sr. No. | Source / Process Area | Type of Non-Hazardous Waste | Approx. Quantity Kg / Year | Solid (Kg/day) | Liquid (L/day) | Category | ETP/3rd party
 1 | ETP / STP treatment | Biological sludge (after dewatering) | 6237850 | 17090 | | Organic | 3rd party
 2 | Process area & lab washing | Treated water (ETP + RO permeate) | 27052340 L/year | | 74116 | Organic | Domestic use
 3 | Packaging line | Empty glass vials (non-contaminated rejects) (~50,000 pcs) | 1000 | 1250 | | Inorganic | Scrap 3rd party
 4 | Packaging & labeling | Paperboard cartons, labels, paper scrap | 2000 | 60 | | Organic | Scrap 3rd party
 5 | Procurement / stores area | Corrugated boxes, stretch film, plastic wrapping | 1500 | 25 | | Inorganic | Scrap 3rd party
 6 | Maintenance workshop | Metal scrap (iron, steel, aluminum parts) | 1200 | 1 | | Inorganic | Scrap 3rd party
 7 | Utility & HVAC area | Spent filters (air filters, bag filters, prefilters) | 500 | 1 | | Inorganic | 3rd party
 8 | Admin & office areas | Office paper waste, packaging waste | 1500 | 25 | | Organic | Scrap 3rd party
 9 | Canteen / pantry | Organic kitchen waste | 47000 | 128.76712328767124 | | Organic | Organic converter
 10 | General housekeeping | Sweepings, mopping dust, non-contaminated PPE | 800 | 2.191780821917808 | | Mixed | 3rd party
 11 | Cooling tower & utilities | Blowdown & scale sludge (non-toxic) | 1500 | 4.109589041095891 | | Organic | 3rd party
 12 | Electrical systems | Cable cuttings, wire ends, plastic conduits | 300 | 0.821917808219178 | | Inorganic | Scrap 3rd party
 13 | Gardening & green area | Dry leaves, grass clippings (~10 acre considering 33% green area) | 244000 | 668.4931506849315 | | Organic | Organic converter
 14 | Water purification & softener | Spent ion-exchange resins (non-toxic) | 200 | 0.547945205479452 | | Inorganic | 3rd party
 15 | Misc. activities | Empty HDPE / PET bottles (non-contaminated) | 200 | 0.547945205479452 | | Inorganic | 3rd party
		Domestic use	74.116	KL/Day	
		Organic converter	0.7972602739726027	T/day	
		3rd party	17.09839726027397	T/day	
		Scrap 3rd party	1.3618219178082194	T/day	
[SHEET: Organic waste LVP]					
Sr. No.	Source / Process Area	Type of Non-Hazardous Waste	Approx. Quantity Kg / Year	Solid (Kg/day)	Liquid (L/day)
1	ETP / STP treatment	Biological sludge (after dewatering)	4188375	11475	
2	Process area & lab washing	Treated water (ETP + RO permeate)	14699645 L/year		40273
3	Packaging line	Empty glass vials (non-contaminated rejects) (~50,000 pcs)	2200	6.027397260273973	
4	Packaging & labeling	Paperboard cartons, labels, paper scrap	1800000	4931.506849315068	
5	Procurement / stores area	Corrugated boxes, stretch film, plastic wrapping	4572000	12526.027397260274	
6	Maintenance workshop	Metal scrap (iron, steel, aluminum parts)	7200	19.726027397260275	
7	Utility & HVAC area	Spent filters (air filters, bag filters, prefilters)	1200	3.287671232876712	
8	Admin & office areas	Office paper waste, packaging waste	500000	1369.86301369863	
9	Canteen / pantry	Organic kitchen waste	36500	100	
10	General housekeeping	Sweepings, mopping dust, non-contaminated PPE	500000	1369.86301369863	
11	Cooling tower & utilities	Blowdown & scale sludge (non-toxic)	27000	73.97260273972603	
12	Electrical systems	Cable cuttings, wire ends, plastic conduits	600	1.643835616438356	
13	Water purification & softener	Spent ion-exchange resins (non-toxic)	43000	117.8082191780822	
14	Misc. activities	Empty HDPE / PET bottles (non-contaminated)	1300000	3561.6438356164385	
		Domestic use	40.273	KL/Day	
		Organic converter	0.1	T/day	
		3rd party	16.601575342465754	T/day	
		Scrap 3rd party	18.854794520547944	T/day	

Reagent Details:
Formula & Calculations:
Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).
Calculation: To prepare 1 L of 1N Sodium Hydroxide (MW = 40.0 g/mol)
Weight of solute(g)=1×40 = 40.00g.
Preparation Method:
Weigh 40.00 g of Sodium Hydroxide and dissolve in water to make a 1 L solution.
Preparation for different volumes:
To prepare _____ of 1N Sodium Hydroxide (MW = 40.0 g/mol).
Weight of solute(g)=1×40.0 x _____L= _____gm
Weigh _____gm of Sodium Hydroxide and dissolve in water to make a _____ L solution.
Check and record the details:
Preparation Details:
Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry
Date
Sodium Hydroxide | | | |
Final volume prepared | Weighed Quantity | pH (13.0±1.0) | Conductivity (180±1.0 ms/cm) | Storage
Condition:
| | | RT
Date of Preparation | Prepared By | Checked By
| |

Reagent Details:
Formula & Calculations:
Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).
Calculation: To prepare 1 L of 0.2M Citric Acid (MW = 192.124 g/mol)
Weight of solute(g)=0.2×192.124x1=38.424g.
Preparation Method:
Weigh 38.424g of Citric Acid and dissolve in water to make a 1 L solution.
Preparation for different volumes:
To prepare _____ of 0.2M Citric Acid (MW = 192.124g/mol).
Weight of solute(g)=0.2×192.124 x _____L= _____g.
Weigh _____gm of Citric Acid and dissolve in water to make a _____ L solution.
Check and record the details:
Preparation Details:
Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry
Date
Citric Acid | | | |
Final volume prepared | Weighed Quantity | pH (1.9±0.1) | Conductivity (4.0±0.5 ms/cm) | Storage
Condition | Reagent Record
| | | | FN-RR-07
Date of Preparation | Prepared By | Checked By
| |

ETP inlet parameter details required.

Response:

pH: Alkaline 10-12

Conductivity: 80-100 mS/cm

MITL Maharashtra Industrial township in place queries meeting

Domestic water consumption 170LPD or 100 LDP?

Raw water consumption is required

Per day calculation is required

Adjust per day calculations with recovery

Need to show etp/stp within premises

How much water to be reused for gardening after etp

Reagent Details:
Formula & Calculations:
Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).
Calculation: To prepare 1 L of 0.2M Sodium citrate monohydrate (MW = 294.1g/mol)
Weight of solute(g)=1×38.42=38.42g.
Preparation Method:
Weigh 38.42 g of Sodium citrate monohydrate and dissolve in water to make a 1 L solution.
Preparation for different volumes:
To prepare _____ of 0.2M Sodium citrate monohydrate (MW = 192.12g/mol).
Weight of solute(g)=1×38.42 _____L= _____gm
Weigh _____gm of Sodium citrate monohydrate and dissolve in water to make a _____ L solution.
Adjust the pH with sodium citrate or citric acid to 6.9
Check and record the details:
Preparation Details:
Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry
Date
| | | | |
Final volume prepared | Weighed Quantity | pH(7.0±0.1) | Conductivity (23±2ms/cm) | Storage Condition:
| | | | RT
Date of Preparation | Prepared By | Checked By
| |

Date- 18.05.2026

List of Documents

S/No	Title
1	ETP, STP water calculation details for Plasma and LVP plant
2	RDS for Plasma
3	Utility RDS for Plasma

Contact Details:

- 1. Dr. Ganesh Kumraj: ganesh@biobridge.in

Technical Information # BB-LN/IND/001

PROCESS INSTRUCTIONS AND DATA CAPTURE POINTS –HUMAN COAGULATION FACTOR VIII

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CHROMATOGRAPHY-12E

Take the following Fraction-1 from Chromatography-1F run of Plasma fractionation step 4.6.3.

Fraction 1: Take the Fraction 1 and load it on the Chromatography-FE-I column as Run-1. Collect the eluate and store at 2-8°C for processing of F-VIII.

Check and record the following details of Fraction 1:

Under stirring conditions, add Tween 80 to a final concentration of 0.1% to the above collected fraction volumes.

LOAD PREPARATION

Assemble PA-PO-007 filter and record the following details:

No. of filter(s) used: _____.

Equilibrate the filter by passing 0.5 L FE-RR-18 at 0.5 bar pressure.

Volume of FE-RR-18 used: _____ L.

Start filtration of the peak-1 fraction 1 at a pressure below 1.0 bar.

Total filtration time: _____.

After completion of the filtration, pass 0.5 L FE-RR-18 to flush out the protein solution.

Volume of FE-RR-18 used: _____ L.

Record the Chromatography-I load details:

Collect 10 mL of the Fraction 1 sample and send the sample to QC for in-process QC analysis. Record the obtained results as per the QC report:

pH: _____, Conductivity: _____ mS/cm.

Switch on the chromatography system. When the chromatography system is ready, open the UNICORN software.

Connect the column to the chromatography system and check the column connections to the system. Check for instrument air connection at 6 bar to the system.

Column Cleaning and Wash

Pass 2 CVs of WFI through the column at 32.7ml/min till the conductivity has dropped below 1.0 mS/cm.

Pass 2 CVs of FE-RR-15 through the column at 32.7ml/min .

Pass 2 CVs of FE-RR-16 through the column at 32.7ml/min .

Pass 2 CVs of WFI through the column at 32.7ml/min .

pH drop

Pass the column with upto 2 CVs of FE-RR-17 (50% gradient of FE-RR-02 against WFI) at a flow rate of 32.7ml/min till the pH has dropped down to 3.50 ± 0.50 . Record the following details:

Column Equilibration

Equilibrate the column with 3 CVs of FE-RR-18 (10% gradient of FE-RR-04 against WFI) at a flow rate of 32.7ml/min .

Loading and Elution

Before loading, Auto Zero the absorbance (A280nm) on the chromatography system and switch on the record button and enter the run details and date.

i) Load the fraction -1 sample as Run-1 at a flow rate of 32.7ml/min .

Pass 3 CVs of Wash-I buffer FE-RR-18 (10% gradient of FE-RR-04 against WFI) at a flow rate of 32.7ml/min .

Pass 3 CVs of Wash-II buffer FE-RR-19 (10% gradient of FE-RR-05 against WFI) at a flow rate of 32.7ml/min .

Pass 3 CVs of elution buffer (FE-RR-06) through the column at 32.7ml/min .

Start collecting the eluate when the absorbance rises from 100 mAU and collect until absorbance drops down to 100 mAU. Record the following details:

Eluate from Run. Check and record the following details.

O.D280/600 : _____. Final Volume: _____L.

Protein concentration: _____. Total protein: _____.

Regeneration

Pass 2 CVs of FE-RR-21(25% gradient of FE-RR-13 against WFI) through the column in reverse flow at 32.7ml/min and hold the column for 30 min.

Storage

Pass 2 CVs of FE-RR-14 through the column at a flow rate of 32.7ml/min and store the column in FE-RR-14.

Volume of FE-RR-14 used: _____ L.

SOLVENT / DETERGENT (S/D) TREATMENT FOR VIRAL INACTIVATION

Take the 1000 ml of plasma equivalent of the Elution sample from step 1.11.7 as Run-1 and add 50 mL/L of FE-RR-07 slowly to the sample under gentle stirring conditions.

Add 15 mL/L of FE-RR-23 slowly to the sample and incubate the sample at RT under gentle stirring conditions for 4 Hours.

CONDUCTIVITY ADJUSTMENT

3.1 After completion of incubation, adjust the conductivity of the Run-1 S/D treated sample to 14 – 17 mS/cm with FE-RR-08.

Collect 10 mL of the conductivity-adjusted sample and send the sample to QC for in-process QC analysis.

Record the obtained results as per the QC report.

pH: _____, Conductivity: _____ mS/cm.

CHROMATOGRAPHY-13E

Switch on the chromatography system. When the chromatography system is ready, open the UNICORN software.

Connect the column to the chromatography system and check the column connections to the system. Check for instrument air connection at 6 bar to the system.

Column Cleaning and Wash

Pass 2 CVs of WFI through the column at 16.40ml/min.

Pass 2 CVs of FE-RR-15 through the column at 16.40ml/min.

Pass 2 CVs of FE-RR-16 through the column at 16.40ml/min.

Pass 5CVs of WFI through the column at 16.40ml/min

Column Equilibration

Equilibrate the column with 3 CVs of FE-RR-20 (10% gradient of FE-RR-03 against WFI) at a flow rate of 16.40ml/min.

Loading and Elution

Before loading, Auto Zero the absorbance (A280) on the chromatography system and switch on the record button and enter the run details and date.

Load the S/D treated eluate obtained from step 3.1 at a flow rate of 16.40ml/min.

Note: NMT 176 mL of plasma per mL of resin.

After loading is completed, pass 15 CVs of FE-RR-20 (10% gradient of FE-RR-03) at a flow rate of 16.40ml/min

Pass 3 CVs of elution buffer (FE-RR-09) through the column at 16.40ml/min

Start collecting the eluate when the absorbance rises from 10 mAU and collect until absorbance drops down to 10 mAU.

Record the volume of eluate: _____ L

Regeneration

Pass 2 CVs of FE-RR-21 (25% gradient of FE-RR-13 against WFI) through the column in reverse flow at 16.40ml/min and hold the column for 30 minutes.

Storage

Pass 2 CVs of FE-RR-14 through the column at a flow rate of 16.40ml/min and store the column in FE-RR-14.

Volume of FE-RR-14 used: _____ L.

CONCENTRATION OF CHROMATOGRAPHY-13E ELUATE

Take the chromatography-13E pooled eluate sample from step 4.5.6 and record the following details:

pH: _____, Conductivity: _____ mS/cm.

O.D280/600 : _____, Volume: _____.

NWP calculation before concentration

Record the following details of the TFF cassettes connected in series horizontally to the TFF system:

No. of cassettes used: _____, Cassette ID Nos.: _____.

Wash the cassettes with WFI till conductivity reaches to < 0.1mS/cm and calculate NWP before concentration at a set TMP of ≤ 1.0 bar.

Volume of WFI used: _____ L.

Check and record the details as below:

Concentration of Eluate sample

After NWP calculation, flush the cassettes with FE-RR-09 till conductivity reaches to 52 to 53 mS/cm.

Volume of FE-RR-09 used: _____ L.

Concentrate the eluate sample to 10 folds at a set TMP of ≤ 1.0 bar.

Check and record the details as below:

Unload the concentrated sample and record the following details:

pH: _____, Conductivity: _____ mS/cm.

O.D280/600 : _____, Volume: _____.

Regeneration

Flush the cassettes with WFI and then with FE-RR-25 and carry out recirculation for 30 to 45 min.

Volume of FE-RR-25 used: _____ L.

Post NWP calculation

Repeat the step as 5.2.2 and record the details as below. Check if NWP is ≥ 120 LMH/Bar. If not, carry out the regeneration procedure again.

Storage

Flush the cassettes with FE-RR-14 and finally store the cassettes in FE-RR-14 at RT

Volume of FE-RR-14 used: _____ L.

CHROMATOGRAPHY-14E

Switch on the chromatography system. When the chromatography system is ready, open the UNICORN software.

Connect the column to the chromatography system and check the column connections to the system. Check for instrument air connection at 6 bar to the system.

Record the following details:

Column Height: _____ cm, Column Volume: _____ L,

HETP value: _____, Asymmetry: _____.

Column Cleaning and Wash

Pass 2 CVs of FE-RR-22(10% gradient of FE-RR-13 against WFI) through the column at a flow rate of 15 LPH and hold the column for 30 minutes.

Volume of FE-RR-22 used: _____ L.

Pass upto 5 CVs of WFI through the column at 15 LPH till the conductivity has dropped below 1.0 mS/cm.

Volume of WFI used: _____ L, Conductivity: _____ mS/cm

Pass 2 CVs of FE-RR-16 through the column at a flow rate of 15 LPH.

Volume of FE-RR-16 used: _____ L.

Pass upto 5 CVs of WFI through the column at 15 LPH till the conductivity has dropped below 0.1 mS/cm.

Volume of WFI used: _____ L, Conductivity: _____ mS/cm.

Column Equilibration

Equilibrate the column with 3 CVs of FE-RR-11 at a flow rate of 15 LPH.

Check and record the details

pH: _____, Conductivity: _____ mS/cm,

Volume of FE-RR-11 used: _____ L.

Loading and Elution

Before loading, Auto Zero the absorbance (A280) on the chromatography system and switch on the record button and enter the run details and date.

Note: Loading should not exceed 12.5% of column volume

(NMT 0.75 L per injection).

Load the concentrated chromatography-13E eluate obtained from step 5.3.4 at a flow rate of 15 LPH.

Check and record the details below.

After loading is completed, pass 2 CVs of FE-RR-11 at a flow rate of 15 LPH.

Start collecting the eluate when the absorbance rises from 5 mAU and collect until absorbance drops down to 5 mAU.

After conductivity peak comes down repeat steps 6.6.2 to 6.6.5 for Inj 2 and record the results in the above table.

After conductivity peak comes down repeat steps 6.6.2 to 6.6.5 for Inj 3 and record the results in the above table.

After conductivity peak comes down repeat steps 6.6.2 to 6.6.5 for Inj 4 and record the results in the above table.

Pool the eluates from injection1 to 4 from step 6.6.5.

Final Volume of Pooled eluate : _____ L.

Collect 12 mL of the sample and send the sample to QC for in-process QC analysis and record the obtained results as per the QC report.

pH: _____, Conductivity: _____ mS/cm.

Activity (IU/mL) _____ :

Regeneration

Pass 2 CVs of FE-RR-22(10% gradient of FE-RR-13 against WFI) through the column in reverse flow at 15 LPH.

Volume of FE-RR-22 used: _____ L.

Storage

Pass 2 CVs of FE-RR-14 through the column at a flow rate of 15 LPH and store in FE-RR-14.

Volume of FE-RR-14 used: _____ L.

STABILIZER ADDITION

Take the pooled chromatography-14E eluate volume from step 6.6.6 and add 80 mL/L of FE-RR-10.

Volume of FE-RR-10 added: _____ L.

Final Volume of the Eluate: _____ L.

Add 10 mL of Human Serum Albumin (HSA-20%) to the above sample and record the following details:

Volume of 20% HSA added: _____.

OD280/600: _____, Final Volume: _____ L,

pH: _____, Conductivity: _____ mS/cm.

If the pH of the pooled eluate is above 6.90, adjust to 6.70 - 6.80 with FE-RR-24.

pH: _____, Volume of FE-RR-24 used: _____ L.

CONCENTRATION OF CHROMATOGRAPHY-14E ELUATE

Take the pooled eluate sample from step 7.2 and carry out the concentration of the sample.

NWP calculation before concentration

Record the following details of the TFF cassettes connected in series horizontally to the TFF system:

No. of cassettes used: _____, Cassette ID Nos.: _____.

Wash the cassettes with WFI till conductivity reaches below 0.1mS/cm and calculate NWP before concentration at a set TMP of ≤ 1.0 bar.

Volume of WFI used: _____ L.

Check and record the details as below:

Concentration of Eluate sample

After NWP calculation, flush the cassettes with FE-RR-11 till the conditions of the buffer are attained.

Volume of FE-RR-11 used: _____ L.

Concentrate the eluate sample to the activity of > 40 IU/mL, at a set TMP of ≤ 1.0 bar.

Check and record the details as below:

Unload the concentrated sample and record the following details:

pH: _____, Conductivity: _____ mS/cm.

O.D.280/600: _____. Final Volume: _____.

Regeneration

Flush the cassettes with WFI and then with FE-RR-25 at 40 - 45°C and carry out recirculation for 30 to 45 min.

Volume of FE-RR-25 used: _____ L.

Post NWP calculation

Repeat the step as 8.2.2 and record the details as below. Check if NWP is ≥ 120 LMH/Bar. If not, carry out the regeneration procedure again.

Storage

Flush the cassettes with FE-RR-14 and finally store the cassettes in FE-RR-14.

Volume of FE-RR-14 used: _____ L.

FORMULATION

After concentration, add L-Arginine Hydrochloride and Sucrose to a final concentration of 100mM and 1% respectively.

Quantity of L-Arginine Hydrochloride added: _____

Quantity of Sucrose added: _____

Add and adjust the concentration of HSA-20% to a final concentration of 10 mg/mL of the concentrated sample volume

Volume of HSA-20% added before concentration in step 7.2 : _____.

Volume of HSA-20% added after concentration: _____.

Check and record the following details.

pH: _____, Conductivity: _____ mS/cm.

O.D280/600: _____. Final Volume: _____.

Conditions | Inj-1 | Inj-2 | Inj-3 | Inj-4 | Inj-5 | Inj-6 | Inj-7 | Inj-8

Fraction 1 vol. in L | | | | | | |

Visual Appearance | | | | | | |

Conditions | Inj-1 | Inj-2 | Inj-3 | Inj-4 | Inj-5 | Inj-6 | Inj-7 | Inj-8

Tween 80 added for Fraction-1 (mL). | | | | | | |

Details | Fraction 1

pH |

Conductivity (mS/cm) |

O.D280/600 |

Volume(L) |

Visual Appearance |

Details | Run-1

Volume of WFI used (L) |

Details | Run-1

Volume of FE-RR-15 used (L) |

Details | Run-1

Volume of FE-RR-16 used (L) |

Details | Run-1

Volume of WFI used (L) |

Details | Run-1

Volume of FE-RR-17 used (L) |

Details | Run-1

Volume of FE-RR-18 used (L) |

Details | Run-1

Load volume(L) |

Details | Run-1

Volume of FE-RR-18 used (L) |

Details | Run-1

Volume of FE-RR-19 used (L) |

Details | Run-1

Volume of FE-RR-06 used (L) |

Details | Run-1

pH |

Conductivity (mS/cm) |

O.D280/600 |

Details | Run-1

Volume of FE-RR-21 used (L) |

Details | Run-1

Volume of Elution sample(L) |

Volume of FE-RR-07 used (L) |

Details | Run-1
Volume of FE-RR-23 used (L) |
Final Volume (L) |
Incubation Time |
Details | Run-1
pH |
Conductivity (mS/cm) |
Volume of FE-RR-08 used (L) |
Final Volume (L) |
Details | Run-1
Volume of WFI used (L) |
Details | Run-1
Volume of FE-RR-15 used (L) |
Details | Run-1
Volume of FE-RR-16 used (L) |
Details | Run-1
Volume of WFI used (L) |
Details | Run-1
Volume of FE-RR-20 used (L) |
Details | Run-1
Volume of FE-RR-20 used (L) |
Details | Run-1
Volume of FE-RR-09 used (L) |
Details | Run-1
pH |
Conductivity (mS/cm) |
OD280/600 |
Volume of Eluate (L) |
Details | Run-1
Volume of FE-RR-21 used (L) |
PF (bar) | PR (bar) | PP (bar) | Flow (L/hr) | TMP (bar) | Flux (LMH) | NWP (LMH/bar)
| | | | |
Permeate Vol. (L) | PF (bar) | PR
(bar) | PP (bar) | Flow (L/hr) | TMP (bar) | Flux (LMH) | Permeate O.D280
PF (bar)	PR (bar)	PP (bar)	Flow (L/hr)
(bar)	Flux (LMH)	NWP (LMH/bar)	
Details	Inj 1	Inj 2	Inj 3
Volume of Load (L)			
Details	Inj 1	Inj 2	Inj 3
Volume of FE-RR-11 used (L)			
Details	Inj 1	Inj 2	Inj 3
Volume of eluate (L)			
pH			
Conductivity (mS/cm)			
OD280/600			
PF (bar)	PR (bar)	PP (bar)	Flow (L/hr)
Permeate Vol. (L)	PF (bar)	PR	
(bar)	PP (bar)	Flow (L/hr)	TMP (bar)
PF (bar)	PR (bar)	PP (bar)	Flow (L/hr)
(bar)	Flux (LMH)	NWP (LMH/bar)	

Reagent Details:
Preparation of FE-RR-08 buffer with Stock FE-RR-03.
Formula & Calculations:
Formula: $M_1 \times V_1 = M_2 \times V_2$.
Calculation: To prepare 1 L of 2mM sodium citrate from 0.2M sodium citrate stock solution calculate quantity required of stock solution as below:
 $0.2\text{M sodium citrate stock solution required quantity} = 0.002 \times 1 / 0.2 = 0.01\text{L}$
Preparation Method:
To prepare 2mM sodium citrate 1L solution from 0.2M sodium citrate stock solution formula is M_1 (Molarity of stock) $\times V_1$ (volume of stock) = M_2 (Molarity of working) $\times V_2$ (volume of working).
Take 0.01L of 0.2M sodium citrate solution and make a 1 L solution.
Preparation for different volumes:
To prepare _____ of 2mM sodium citrate
Take 0.01L $\times$ _____ L = _____ L of 0.2M sodium citrate
Make sodium hydroxide solution to _____ L solution.
Adjust the pH with citric acid or sodium citrate to pH 7.0
Check and record the details:
Preparation Details:
Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry
Date
| | | |
Final volume prepared | Weighed Quantity | pH (7.0) | Conductivity (0.6±0.1ms/cm) | Storage Condition:
| | | | RT
Date of Preparation | Prepared By | Checked By (QC/Analyst)
| |

Reagent Details:
Formula & Calculations:
Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).
Calculation: To prepare 1 L of 0.4M Sodium Acetate (MW = 80.03 g/mol)
Weight of solute(g)=0.4×80.03 = 32.01g.
Preparation Method:
Weigh 32.01g of Sodium Acetate and dissolve in water to make a 1 L solution.
Preparation for different volumes:
To prepare _____ of 0.4M Sodium Acetate (MW = 80.03 g/mol).
Weight of solute(g)=0.4×80.03 x _____L= _____gm
Weigh _____gm of Sodium Acetate and dissolve in water to make a _____ L solution.
Check and record the details:
Preparation Details:
Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry
Date
0.4M Sodium Acetate | | | |
Final volume prepared | Weighed Quantity | pH(8.7 ±0.2) | Conductivity (4.7 ±0.2ms/cm) | Storage
Condition:
| | | RT
Date of Preparation | Prepared By | Checked By
| |

Reagent Details:

Formula & Calculations:

Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).

Calculation: To prepare 1 L of 0.5M sodium chlorolride (MW = 58.44g/mol)

Weight of solute(g)=1×29.22=29.22g.

Preparation Method:

Weigh 29.22g of sodium chloride

To prepare 20mM Tris 1L solution from 1M Tris stock solution formula is $M_1(\text{Molarity of stock}) \times V_1$

(volume of stock)= $M_2(\text{Molarity of working}) \times V_2(\text{volume of working})$.

Take 20mL of 1M tris solution and add weighed amount of sodium chloride as required and dissolve in water to make a 1 L solution.

Preparation for different volumes:

To prepare _____ of 0.5M NaCl(MW = 58.44g/mol).

Weight of solute(g)=29.22 _____ L= _____ gm

Weigh _____ gm of NaCl.

To prepare _____ of 20mM Tris

Take 20mL x _____ L= _____ mL of 1M Tris

Dissolve pre weighed NaCl in 20mM Tris solution to make a _____ L solution.

Adjust the pH to 8 using concentrated HCl/NaOH solution

Check and record the details:

Preparation Details:

Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry

Date

||||

Final volume prepared | Weighed Quantity | pH (7.9±1.0) | Conductivity (49.5±1.0ms/cm) | Storage

Condition:

|||| RT

Date of Preparation | Prepared By | Checked By

||

Reagent Details:

Formula & Calculations:

Formula: Use the formula to calculate the molarity:

Molarity (M) = (Per cent by Weight / 100) * (Density in g/mL) * 1000 / Molar Mass.

Example: Molarity of Glacial Acetic Acid

Molar Mass of CH₃COOH: ~60.05 g/mol

Density: ~1.05 g/mL

Purity: 100%

Acetic Acid (MW = 60.05 g/mol).

Calculation:

Molarity = (100 / 100) * 1.05 g/mL * 1000 mL/L / 60.05 g/mol ≈ 17.5 M.

Calculation: To prepare 1 L of 0.4 M.

Applying the formula to prepare the working buffer:

C₁V₁=C₂V₂

C₁ = 17.5 M acetic acid.

V₁ = volume of stock needed to take.

C₂ = desired (working) concentration 0.4 M acetic acid.

V₂ = final total volume of working solution for one liter.

V₁ = (C₂ x V₂)/C₁.

C₂=0.4 M; V₂= 1.0 L; C₁=17.5 M

V₁ = (0.4 x 1.0)/17.5

V₁ = 0.0228 L

Take 0.0228 L of the Stock solution of acetic acid and add it to a volumetric flask, making up the volume to 1 L.

To prepare various volumes:

V₁ = volume of stock needed to take _____ L.

C₂ = desired (working) concentration 0.4M acetic acid.

V₂ = final total volume of working solution _____ L.

V₁ = (C₂ _____ x V₂ _____) /17.5.

V₁= _____ L.

Take V₁ _____ L of the Stock solution of acetic acid and add it to a volumetric flask, making up the final volume _____ L.

Check and record the details:

Preparation Details:

Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry

Date

||||

Final volume prepared | Measured Quantity | pH (2.5±0.1) | Conductivity (1.5±0.2ms/cm) | Storage

Condition | Reagent Record

|||| RT |

Date of Preparation | Prepared By | Checked By

||

Reagent Details:
Formula & Calculations:
Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).
Calculation: To prepare 1 L of 0.6M Sodium Acetate (MW = 80.03 g/mol)
Weight of solute(g)=0.6×80.03 = 48.018g.
Preparation Method:
Weigh 48.01g of Sodium Acetate and dissolve in water to make a 1 L solution.
Preparation for different volumes:
To prepare _____ of 0.6M Sodium Acetate (MW = 80.03 g/mol).
Weight of solute(g)=0.6×80.03 x _____L= _____gm
Weigh _____gm of Sodium Acetate and dissolve in water to make a _____ L solution.
Check and record the details:
Preparation Details:
Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry
Date
0.4M Sodium Acetate | | | |
Final volume prepared | Weighed Quantity | pH (8.4±0.5) | Conductivity (70±1.0ms/cm) | Storage Condition:
| | | | RT
Date of Preparation | Prepared By | Checked By
| |

[SHEET: Plasma]

List of Raw materials for Plasma Project | | |

S.No | Chemicals Name | UOM | Qty per year

1 | Sodium Citrate (dihydrate)(gm) | Kg | 115801.875
2 | Citric Acid (gm) | Kg | 99078.75000000001
3 | Sodium Chlorid (gm) | Kg | 598522.5
4 | Sodium Hydroxide (gm) | Kg | 919363.1250000001
5 | Calcium Chlorid (gm) | Kg | 5452.499999999999
6 | Sodium Acetate (anhydrous)(gm) | Kg | 200362.50000000003
7 | Acetic Acid (ml) | L | 47013.75
8 | Glycine (gm) | Kg | 35812.5
9 | TRIS (gm) | Kg | 17812.5
10 | 20% HAS (gm) | Kg | 566.25
11 | Sodium Caprylate (gm) | Kg | 2437.5
12 | Sodium N- Acetyl tryptophan | Kg | 2437.5
13 | Glycerol(ml) | L | 28125
14 | HCL(ml) | L | 56250
15 | L- Arginine (gm) | Kg | 326.62499999999994
16 | Sucroce(gm) | Kg | 187.5
17 | Tween80(ml) | L | 384.375
18 | TNBP(ml) | L | 37687.49999999999
19 | Ammonium Sulphate (gm) | Kg | 82500
20 | Triton X 100 (ml) | L | 37500
21 | Histidine (gm) | Kg | 187.5
22 | Nicotinamide(mg) | Kg | 75
23 | Maltose (gm) | Kg | 3750
24 | Iso Propanal Alcohol (ml) | L | 75000
25 | Sodium Hypo Chloride (ml) | L | 15000
26 | Sepharose 4 FF | L | 565.1999999999999
27 | DEAE sepharose FF | L | 1143.1170000000002
28 | Q Sepharose FF | L | 602.88
29 | CM Sepharose FF | L | 459.696
30 | Heparin Sepharose 6 FF | L | 19.0755
31 | Superdex 200PG | L | 50.868
32 | Q Sepharose high performance | L | 141.29999999999998
33 | SP Sepharose high performance | L | 19.075499999999998
34 | Superdex G25 | Kg | 38.150999999999996
35 | Boric acid GR | Kg | 8.75
36 | Agarose DNase, RNase free | Kg | 1.75
37 | Anti human serum antibody produced in Rabbit | L | 0.0175
38 | Coomassie brilliant blue R-250 | Kg | 0.21875
39 | Disodium hydrogen phosphate dihydrate (HPLC grade) | Kg | 4.375
40 | Copper sulphate GR | Kg | 4.375
41 | Prekallikrein Activator Assay Kit | Kit | 70
42 | Prekallikrein Activator Assay Ig Kit | Kit | 70
43 | Working Standard (WS) - Human Normal Immunoglobulin for Intravenous use 5% I.P. (Immuglo 5% 100 mL)
calibrated against Sigma Protein Standard (200 mg/mL): | vial | 5.833333333333333
44 | Platelet poor plasma | L | 0.35000000000000003
45 | Monolisa Anti-HBs Plus Kit | kit | 70
46 | Diptheria Toxoid IgG Enzyme Immunoassay kit | kit | 70
47 | Human IgG subclass profile Kit | kit | 70
48 | Anhydrous Sodium dihydrogen phosphate GR | Kg | 8.75
49 | Magnesium chloride GR | kg | 1.4583333333333333
50 | D (+) glucose GR | kg | 4.375
51 | Tannic acid GR | Kg | 0.875
52 | HEPES buffer AR | L | 0.875
53 | Guinea-pig complement | Vial | 2.9166666666666665
54 | Rubella antigen | Vial | 5.833333333333333
55 | COAMATIC Factor VIII kit | kit | 52.5
56 | WHO Reference standard (RS) (International Standard, Factor VIII Concentrate, | kit | 4.375
57 | STA Deficient VIII Plasma | kit | 8.75
58 | FVIII-deficient plasma | pack | 35
59 | CK presto kit | kit | 4.375
60 | (5th International Standard for Factor IX Concentrate) | vial | 5.833333333333333
61 | Standard human plasma with known FVIII activity (e.g., 100%) | kit | 4.375
62 | Factor IX Deficient Plasma | kit | 8.75
63 | Owren koller buffer - Ready to use. | pack | 4.375
64 | STA Deficient Factor II | pack | 4.375
65 | STA Deficient Factor VII | pack | 5.833333333333333
66 | STA Deficient Factor X | pack | 5.833333333333333
67 | Neoplastine CI Plus | pack | 5.833333333333333
68 | Unicalibrator | pack | 8.75
69 | Sodium tartrate GR | Kg | 4.375
70 | Folin-Ciocalteu's phenol reagent | L | 0.5833333333333334
71 | Bovine Serum Albumin (BSA) | vial | 8.75
72 | Bovine Gamma Globulin (BGG) | vial | 8.75
73 | Sulphuric acid | L | 8.75
74 | Dipotassium Sulphate | kg | 4.375
75 | Hydrogen Peroxide | L | 1.4583333333333333
76 | Methyl red | kg | 0.14583333333333334
77 | Methylene blue | Kg | 0.04375000000000004
78 | Tris-Cl buffer | L | 4.375
79 | Human Thrombin preparation (500 IU/mL label claim vial) | vial | 8.75
80 | Imidazole | Kg | 1.4583333333333333
81 | Fibrinogen Concentrate | Vial | 5.833333333333333
82 | Hbs Ag - HEPA ELISA (4th generation) | Kit | 52.5
83 | HCV ELISA (3rd generation) | Kit | 52.5
84 | HIV ELISA (4th generation) | Kit | 52.5
85 | Human IgA ELISA Development Kit (ALP) for 6 plates | Kit | 17.5
86 | Haemolysin [Antiserum against sheep red blood cells prepared in rabbits | vial | 8.75

87 | HPLC grade methanol | L | 21.875
 88 | Sodium Hydroxide pellets GR | Kg | 8.75
 89 | Trisodium citrate, anhydrous, 99%, | Kg | 4.375
 90 | Sodium carbonate anhydrous for | Kg | 14.583333333333334
 91 | Potassium Sodium Tartarate Tetra Hydrate | Kg | 4.375
 92 | Anhydrous Disodium Hydrogen Phosphate | Kg | 4.375
 93 | Anhydrous Sodium dihydrogen phosphate | Kg | 4.375
 94 | Bovine Serum Albumin heat shock fraction, protease free, essentially globulin free | Kg | 0.05833333333333334

[SHEET: LVP]

List of Raw materials for LVP Project | |

S. No. | Raw material | UOM | Qty Yearly

1 | Ciprofloxacin | Kg | 15000
 2 | Fluconazole | Kg | 7500
 3 | Levofloxacin | Kg | 18750
 4 | Linezolid | Kg | 7500
 5 | Mannitol | Kg | 2437500
 6 | Metronidazole | Kg | 18750
 7 | Moxifloxacin | Kg | 15000
 8 | Ofloxacin | Kg | 7500
 9 | Ornidazole | Kg | 18750
 10 | Paracetamol | Kg | 56250
 11 | Sodium Chloride | Kg | 435000
 12 | Dextrose | Kg | 2625000
 13 | Linezolid | Kg | 7500
 14 | Mannitol | Kg | 1125
 15 | Moxifloxacin | Kg | 15000
 16 | Tinidazole | Kg | 7500
 17 | Sodium lactate | Kg | 36000
 18 | Dextrose | Kg | 5062500

[SHEET: Reponse]		
S. No.	Query	Response
1	Confirmation of machine list with make, model, size, and weight.	Attached excel for make, size and weight
2	Final machine layout and required working clearances.	Layout attached, working clearance should be 2 meters
3	Floor-to-floor height confirmation as per machinery and HVAC requirements.	As per machinery 3.5 meters, HVAC requirement 6 meters
4	Verification of internal room sizes as per operational needs.	Mentioned in layout
5	Confirmation of surrounding space for vehicular movement and loading/unloading access.	
6	Utility requirements (water inlet/outlet, drainage, power load, and waste disposal).	Water inlet = 87557 L/day, Water outlet = 74984 L/day, Working load 4143.37 KVA and stand by load 1068.24 KVA, waste = 74984 L/day, ETP/STP: 900-1000 m3/year

[SHEET: Room sizes]
			Ok			
			New addition			
			Removed or to be relooked			
Room name and size for Plasma facility (Lifenity)						
S. No.	Room Name	Old Room Size		New Room Size		Comments
		L (m)	B (m)	L (m)	B (m)	
1	Reception	4.05	2.8	7	12	Change, Ok
		4.53	8.15			
2	Space	4.05	2.8	4.72	1.8	Change, Ok
3	Waiting room	5.3	4	4.67	5	
4	Gents Toilet	5.9	5.2			Removed
5	Gents Locker	5.47	5.3	6	5	Change, Ok
6	Visitor locker room	4.57	5.3	5.9	5	Change, Ok
7	Female locker room	4.57	5.3	6	5	Change, Ok
8	Ladies toilet room	5.9	5.2			Removed
9	Document room	2.5	5.3	4	5	Change, Ok
10	Office 3			4.7	5	New
11	Office 4			4.7	5	New
12	Office 1	4.63	5	4.3	5.1	Change, Ok
13	Office 2	4.63	5.3	4.3	5.1	Change, Ok
14	Janitor Room	3.42	2.37	2	5	Change, Ok
15	Corridor	33.4	2.8	36.7	1.8	Change, Ok
16	AL Male	4.77	5	4.7	5.1	Change, Ok
17	Primary Male Change	3.8	5	4	5.1	Change, Ok
18	Male Change - 1	3	5	3.3	5.1	Change, Ok
19	FeMale Change - 1	3.1	5	3.4	5.1	Change, Ok
20	Primary FeMale Change	3.1	5	3.19	5.1	Change, Ok
21	AL Female	3.64	5	3.64	5.1	Change, Ok
22	AL visitor	3.05	5	3.05	5.1	Change, Ok
23	Primary Visitor Change	2.7	5	2	5.1	Change, Ok
24	Visitor Change - 1	2.6	5	2.55	5.1	Change, Ok
25	Corridor	2.5	5	1.8	5.1	Change, Ok
26	Lift corridor	13.2	1.56			Removed, Not mentioned
27	Entry corridor	47.8	2	45.8	2.85	Change, Ok
28	Cold Room	4.4	8.1	4.4	7.09	Change, need to relook as cold room is shortened, may be increased by taking space from corridor mentioned at S. No. 27 for movement of DS in tank/vessel
29	Entry AL - 1	2.22	3.4	2.1	3.4	Change, Ok
30	Gown	3.47	3.4	2.47	3.4	Change, Ok
31	Entry AL - 2	2.2	3.4	2.2	3.4	Change, Ok
32	Exit AL - 1	2.2	3.4	2.2	3.4	Change, Ok
33	De Gown	3.47	3.4	2.47	3.4	Change, Ok
34	Exit AL - 2	2.22	3.4	2.1	3.4	Change, Ok
35A	Corridor	8.3	1.8	7.29	1.8	Space can be increased by taking space from corridor mentioned at S. No. 35B (2.62 meter) for movement of tanks
35B	Corridor	2.57	66	71.4	2.62	Change, Ok
36	Chemical day store	8.1	4	6.98	3.04	Room size may be increased by taking space from corridor mentioned at S. No. 35A for movement of tanks
37	Buffer preparation	4.29	8.1	4	7.09	Room size may be increased by taking space from corridor mentioned at S. No. 27 for movement of tanks
38	Tank Storage - 2	6.8	8.1	6.74	7.09	Room size may be increased by taking space from corridor mentioned at S. No. 27 for movement of tanks
39	CIP/SIP	5.6	8.1	5.6	7.09	Room size may be increased by taking space from corridor mentioned at S. No. 27 for movement of tanks
40	Waste in	2.5	2.4	2.5	2.34	Change, Ok
41	AL	4.13	2.4	4.13	2.34	Change, Ok
42	AL	4.01	2.4	3.4	2.34	Change, Ok
43	Sterilization	4.01	5.62	3.401	4.67	Room size may be increased by taking space from corridor mentioned at S. No. 27 for movement of tanks
44	Cool zone	2.76	8.1	2.75	7.09	Room size may be increased by taking space from corridor mentioned at S. No. 27 for movement of tanks
45	Decontamination	3.52	5.62	3.52	4.67	Change, Ok
46	Washing	3.11	5.62	3.11	4.67	Change, Ok
47	Corridor	37	2.8	36	2.85	Change, Ok
48	Bulk filtration	6.9	6	6.54	7.29	Change, Ok
49	Stage 6 I/ 11 F	14.7	16.7	14.7	18	Change, Ok
50	Stage 9 N	5.1	14.3	5.1	15.6	Change, Ok
51	Stage 10 T/13 E	8.1	12.2	7.48	13.5	Change, Ok
		6.25	2.05	5.63	2.05	Change, Ok
52	Corridor	2.57	67.9	2.62	74.2	Change, Ok
53	Column packing and Skid	6.9	10.6	6.54	10.6	Change, Ok
54	Entry AL - 1	2.22	4.33	2.1	4.76	Change, Ok
55	Gown	3.47	4.32	2.47	4.76	Change, Ok
56	Entry AL - 2	4.2	4.32	2.2	4.76	Change, Ok
57	Exit AL - 1	2.2	3.89	2.2	4.76	Change, Ok
58	De Gown	3.47	3.89	2.47	4.76	Change, Ok
59	Exit AL - 2	2.22	3.89	2.1	4.76	Change, Ok
60	Corridor	1.8	49.6	1.8	55.7	Space can be increased by taking space from corridor mentioned at S. No. 35B (2.62 meter) for movement of tanks
61	Cool zone	8.1	5.53	6.98	5.53	Change, Ok
62	Sterilization	6.5	5.85	5.38	6.85	Change, Ok
63	AL	1.5	5.73	1.5	6.88	Change, Ok
64	AL	1.5	4.42	1.5	4.42	Change, Ok
65	Washing	6.5	5	5.38	3	Change, Ok
66	Decontamination	6.5	5	5.38	3	Change, Ok
67	Waste Out	1.5	3.54	1.5	3.54	Change, Ok
68	CIP/SIP	8.1	5	6.98	5	Space may be increased as previously
69	Tank storage	8.1	6.58	6.98	8.58	Change, Ok
70	Buffer preparation	8.1	6	6.98	6	Space may be increased as previously
71	Plasma day store	8.1	8.77	6.98	10.7	Change, Ok

72	Janitor Room	5.2	1.8	5.4	1.8	Change, Ok
73	Stage 8N	6.17	8.68	5.84	11.2	Change, Ok
74	Stage 2 F	15.8	8.68	15.8	11.2	Change, Ok
75	Stage 3 F	12.8	8.68	12.1	11.2	Change, Ok
76	Corridor	34.1	2.8	34.1	1.8	This corridor width shall be 2.8 for movement of tanks
77	Plasma thawing	8	3.22	7.64	3.61	Change, Ok
78	Stage 1 F/ 12 E	11.1	8.21	11.1	10.6	Change, Ok
79	Corridor	2	28.8	2	32.2	Space may be increased for movement of tanks
80	Stage 4 F	13.6	13.5	13.6	15.8	Change, Ok
81	Bag cutting and pooling	8	7.8	7.64	9.8	Change, Ok
82	Stage 5 I A	11.1	18	11.1	19	Change, Ok
83	Stage 5 I B	8	7.2	7.64	7.2	Change, Ok
84	Stage 7 A	13.6	15.1	13.6	16.1	Change, Ok
85	Column packing and Skid	8	7.86	7.64	8.86	Change, Ok
86	Corridor	50.5	1.8	43.3	1.8	Change, Ok
87	Plasma storage room	5		15		Change, Ok (First floor layout required after revision)
88	Corridor	2.57	37	2.62	15.2	Change, Ok
89	Corridor	47.9	1.8	43.2	1.8	Change, Ok
90	Corridor	29.5	2.58	28.1	1.8	Change, Ok
91	Corridor	1.95	11.3	1.8	12	Change, Ok
92	Entry corridor	39.8	6.84	48.9	5	Change, Ok
93	Plasma receipt	2.4	4.95	2.4	4.95	No Change, Ok
94	Serology testing	3	4.95	3	4.95	No Change, Ok
95	Reject room	2.95	4.95	2.26	4.95	Change, Ok
96	Plasma quarantine	5	2.95	5	2.72	Change, Ok
	11.5	3.3	11.5	4.3	Change, Ok	
97	AL-1	1.87	1.4	1.87	1.4	No Change, Ok
98	Change room	1.87	1.4	1.87	1.4	No Change, Ok
99	Plasma sampling	1.78	2.9	1.76	2.9	Change, Ok
	3.68	2.05	3.68	2.05	No Change, Ok	
100	Janitor Room	1.45	2.9	1.45	2.72	Change, Ok
101	Washing area	2.3	2.9	2.3	2.72	Change, Ok
102	Bag Bar coding	2.6	2.9	2.6	2.72	Change, Ok
103	RM quarantine	6.21	5.77	5.85	6.55	Space constrain for 3 months stock approx 45 batches
104	Approved RM	4.63	4	3.65	4.49	Space constrain for 3 months stock approx 45 batches
	9.72	7.22	8.65	7.54	Space constrain for 3 months stock approx 45 batches	
105	AL -1	1.5	1.93	1.5	2.18	Change, Ok
106	Change room	1.5	1.93	1.5	2.18	Change, Ok
107	Dispensing room	3.48	3.95	3.4	4.44	Change, Ok
108	AL -1	2	1.21	2	1.21	Change, Ok
109	Change room	2	2.08	2	1.8	Change, Ok
110	RM sampling	1.07	3.39	3.16	2.31	Change, Ok
	3.16	2.03	1.07	3.11	Change, Ok	
111	Dedusting	3	2.6	2.64	2.6	Change, Ok
112	RM receipt	3	2.77	2.64	2.77	Change, Ok
113	Store Office	2.23	6.25	2.23	6.5	Change, Ok
114	FG dispatch area	13.9	3.84	10.64	6.49	Change, Ok
	9.22	2.45	4.7	4.08	Change, Ok	
	4.75	2.32	3	3.55	Change, Ok	
115	Reject room	2.3	2.35	2.3	2.35	No Change, Ok
116	Recall room	2.3	2.35	2.3	2.35	No Change, Ok
117	Store change room	4.7	2.37	4.7	2.37	No Change, Ok
	1.51	1.23			Removed	
118	Cold room 4	4.45	7.58	4.44	7.34	Change, Ok
119	Cold room 3	4.45	7.58	4.12	7.34	Change, Ok
120	Cold room 2	4.45	7.58	4.45	7.34	Change, Ok
121	Cold room 1	4.45	7.58	4.62	7.34	Change, Ok
122	Ice pack	2.91	7.58	2.49	7.34	Change, Ok
	Additional room at 2-8Deg (Cold room) may be required to store various products at intermediate stage					

Reagent Details:

Formula:

Glycine (M.Wt): 75.07 g/mol
 Calcium Chloride (M.Wt): 110.98 g/mol
 PEG 200 : (M.Wt): 200 g/mol
 0.01% Tween 80

Preparation of FE-RR-012 buffer.

Preparation of reagent :25 mM Glycine +0.52 mM PEG + 5 mM Calcium Chloride + 0.0025%Tween 80 pH 7.0

Calculations: To prepare the 5mM Calcium Chloride.

Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).

Calculation: To prepare 1 L of 5 mM Calcium Chloride (MW = 110.98g/mol)

Weight of solute(g)=0.005×110.98 x1=0.554g.

Preparation Method:

Weigh 0.554g of Calcium Chloride and dissolve in water to make a 1 L solution.

Preparation for different volumes:

To prepare _____ L of 0.005mM Calcium Chloride (MW =110.98 g/mol).

Weight of solute(g)=0.005 ×110.98x _____ L= _____ g.

Weigh _____ gm of Calcium Chloride and dissolve in water to make a _____ L solution.

Calculations: To prepare the 0.52mM PEG 200.

Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).

Calculation: To prepare 1 L of 0.52 mM PEG (MW = 200g/mol)

Weight of solute(g)=0.00052×200 x1=0.104g.

Preparation Method:

Weigh 0.104g of PEG and dissolve it in water to make a 1 L solution.

Preparation for different volumes:

To prepare _____ L of 0.00052mM PEG (MW =200g/mol).

Weight of solute(g)=0.00052 ×200x _____ L= _____ g. Weigh _____ gm of PEG and

dissolve in water to make a _____ L solution.

Procedure

Calculation: to prepare the 0.0025% Tween 80:

For 1000 mL of 0.0025% solution:

0.0025% =0.0025g per 100ml & 1000 mL=25 mg

Since Tween 80 is a liquid, you can weigh it directly or use its density

(~1.06

g/mL) to measure volume:

Approx. 25 µL of Tween 80 = 0.025 g

Add Tween 80 to a beaker or flask containing ~800 mL of water.

Mix thoroughly:

Use a magnetic stirrer or vortex until fully dissolved. Tween 80 is viscous,

so mixing may take a few minutes.

Calculations: To prepare the 25mM Glycine.

Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol)

×Volume (L).

Calculation: To prepare 1 L of 0.025M Glycine (MW = 75.07g/mol)

Weight of solute(g)=0.025×75.07 x1=1.876g.

Preparation Method:

Weigh 1.876g of Glycine and dissolve in water to make a 1 L solution.

Preparation for different volumes:

To prepare _____ L of 0.025M Glycine (MW =75.07g/mol).

Weight of solute(g)=0.025 ×75.07x _____ L= _____ g.

Weigh _____ gm of Glycine and dissolve in water to make a _____ L solution.

Adjust volume:

After complete dissolution, bring the final volume up to 1000 mL with water.

Procedure for Preparation of FE-RR-12 buffer.

To prepare a 1 L solution, take this add 5mM Calcium chloride, and then to this add 25mM Glycine and to

this add 0.52mM PEG and to this add the 0.0025% Tween 80 then mix the solution till it dissolves and

check the pH; if the pH is less than 7.0 then adjust the pH with 1 M sodium hydroxide to get the pH 7.0

±0.1 and make up the final volume to 1L.

Check and record the details:

Preparation Details:

Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry

Date

Glycine | | | |

PEG | | | |

Calcium Chloride | | | |

Tween 80 | | | |

Water taken volume(L) | pH (7.0 ± 0.1) | Conductivity (3.0±0.5ms/cm) | Storage Condition | Final volume

prepared

| | | RT |

Date of Preparation | Prepared By | Checked By

| |

[SHEET: Lifenity Layout comments GF]
			Ok				
			New addition				
			Removed or to be relooked with respect to room size				
Room name and size for Plasma facility (Lifenity)							
S. No.	Room Name	Old Room Size		New Room Size		Comments	
		L (m)	B (m)	L (m)	B (m)		
1	Reception	4.05	2.8	7	12	Change, Ok	
		4.53	8.15				
2	Space	4.05	2.8	4.72	1.8	Change, Ok	
3	Waiting room	5.3	4	4.67	5		
4	Gents Toilet	5.9	5.2			Removed	
5	Gents Locker	5.47	5.3	6	5	Change, Ok	
6	Visitor locker room	4.57	5.3	5.9	5	Change, Ok	
7	Female locker room	4.57	5.3	6	5	Change, Ok	
8	Ladies toilet room	5.9	5.2			Removed	
9	Document room	2.5	5.3	4	5	Change, Ok	
10	Office 3				4.7	5	New
11	Office 4				4.7	5	New
12	Office 1	4.63	5	4.3	5.1	Change, Ok	
13	Office 2	4.63	5.3	4.3	5.1	Change, Ok	
14	Janitor Room	3.42	2.37	2	5	Change, Ok	
15	Corridor	33.4	2.8	36.7	1.8	Change, Ok	
16	AL Male	4.77	5	4.7	5.1	Change, Ok	
17	Primary Male Change	3.8	5	4	5.1	Change, Ok	
18	Male Change - 1	3	5	3.3	5.1	Change, Ok	
19	FeMale Change - 1	3.1	5	3.4	5.1	Change, Ok	
20	Primary FeMale Change	3.1	5	3.19	5.1	Change, Ok	
21	AL Female	3.64	5	3.64	5.1	Change, Ok	
22	AL visitor	3.05	5	3.05	5.1	Change, Ok	
23	Primary Visitor Change	2.7	5	2	5.1	Change, Ok	
24	Visitor Change - 1	2.6	5	2.55	5.1	Change, Ok	
25	Corridor	2.5	5	1.8	5.1	Change, Ok	
26	Lift corridor	13.2	1.56			Removed, Not mentioned	
27	Entry corridor	47.8	2	45.8	2.85	Change, Ok	
28	Cold Room	4.4	8.1	4.4	7.09	Change, need to relook as cold room is shortened, may be increased by taking space from corridor mentioned at S. No. 27 for movement of DS in tank/vessel	
29	Entry AL - 1	2.22	3.4	2.1	3.4	Change, Ok	
30	Gown	3.47	3.4	2.47	3.4	Change, Ok	
31	Entry AL - 2	2.2	3.4	2.2	3.4	Change, Ok	
32	Exit AL - 1	2.2	3.4	2.2	3.4	Change, Ok	
33	De Gown	3.47	3.4	2.47	3.4	Change, Ok	
34	Exit AL - 2	2.22	3.4	2.1	3.4	Change, Ok	
35A	Corridor	8.3	1.8	7.29	1.8	Space can be increased by taking space from corridor mentioned at S. No. 35B (2.62 meter) for movement of tanks	
35B	Corridor	2.57	66	71.4	2.62	Change, Ok	
36	Chemical day store	8.1	4	6.98	3.04	Room size may be increased by taking space from corridor mentioned at S. No. 35A for movement of tanks	
37	Buffer preparation	4.29	8.1	4	7.09	Room size may be increased by taking space from corridor mentioned at S. No. 27 for movement of tanks	
38	Tank Storage - 2	6.8	8.1	6.74	7.09	Room size may be increased by taking space from corridor mentioned at S. No. 27 for movement of tanks	
39	CIP/SIP	5.6	8.1	5.6	7.09	Room size may be increased by taking space from corridor mentioned at S. No. 27 for movement of tanks	
40	Waste in	2.5	2.4	2.5	2.34	Change, Ok	
41	AL	4.13	2.4	4.13	2.34	Change, Ok	
42	AL	4.01	2.4	3.4	2.34	Change, Ok	
43	Sterilization	4.01	5.62	3.401	4.67	Room size may be increased by taking space from corridor mentioned at S. No. 27 for movement of tanks	
44	Cool zone	2.76	8.1	2.75	7.09	Room size may be increased by taking space from corridor mentioned at S. No. 27 for movement of tanks	
45	Decontamination	3.52	5.62	3.52	4.67	Change, Ok	
46	Washing	3.11	5.62	3.11	4.67	Change, Ok	
47	Corridor	37	2.8	36	2.85	Change, Ok	
48	Bulk filtration	6.9	6	6.54	7.29	Change, Ok	
49	Stage 6 I/ 11 F	14.7	16.7	14.7	18	Change, Ok	
50	Stage 9 N	5.1	14.3	5.1	15.6	Change, Ok	
51	Stage 10 T/13 E	8.1	12.2	7.48	13.5	Change, Ok	
		6.25	2.05	5.63	2.05	Change, Ok	
52	Corridor	2.57	67.9	2.62	74.2	Change, Ok	
53	Column packing and Skid	6.9	10.6	6.54	10.6	Change, Ok	
54	Entry AL - 1	2.22	4.33	2.1	4.76	Change, Ok	
55	Gown	3.47	4.32	2.47	4.76	Change, Ok	
56	Entry AL - 2	4.2	4.32	2.2	4.76	Change, Ok	
57	Exit AL - 1	2.2	3.89	2.2	4.76	Change, Ok	
58	De Gown	3.47	3.89	2.47	4.76	Change, Ok	
59	Exit AL - 2	2.22	3.89	2.1	4.76	Change, Ok	
60	Corridor	1.8	49.6	1.8	55.7	Space can be increased by taking space from corridor mentioned at S. No. 35B (2.62 meter) for movement of tanks	
61	Cool zone	8.1	5.53	6.98	5.53	Change, Ok	
62	Sterilization	6.5	5.85	5.38	6.85	Change, Ok	
63	AL	1.5	5.73	1.5	6.88	Change, Ok	
64	AL	1.5	4.42	1.5	4.42	Change, Ok	
65	Washing	6.5	5	5.38	3	Change, Ok	
66	Decontamination	6.5	5	5.38	3	Change, Ok	
67	Waste Out	1.5	3.54	1.5	3.54	Change, Ok	
68	CIP/SIP	8.1	5	6.98	5	Space may be increased as previously	
69	Tank storage	8.1	6.58	6.98	8.58	Change, Ok	
70	Buffer preparation	8.1	6	6.98	6	Space may be increased as previously	
71	Plasma day store	8.1	8.77	6.98	10.7	Change, Ok	

72	Janitor Room	5.2	1.8	5.4	1.8	Change, Ok
73	Stage 8N	6.17	8.68	5.84	11.2	Change, Ok
74	Stage 2 F	15.8	8.68	15.8	11.2	Change, Ok
75	Stage 3 F	12.8	8.68	12.1	11.2	Change, Ok
76	Corridor	34.1	2.8	34.1	1.8	This corridor width shall be 2.8 for movement of tanks
77	Plasma thawing	8	3.22	7.64	3.61	Change, Ok
78	Stage 1 F/ 12 E	11.1	8.21	11.1	10.6	Change, Ok
79	Corridor	2	28.8	2	32.2	Space may be increased for movement of tanks
80	Stage 4 F	13.6	13.5	13.6	15.8	Change, Ok
81	Bag cutting and pooling	8	7.8	7.64	9.8	Change, Ok
82	Stage 5 I A	11.1	18	11.1	19	Change, Ok
83	Stage 5 I B	8	7.2	7.64	7.2	Change, Ok
84	Stage 7 A	13.6	15.1	13.6	16.1	Change, Ok
85	Column packing and Skid	8	7.86	7.64	8.86	Change, Ok
86	Corridor	50.5	1.8	43.3	1.8	Change, Ok
87	Plasma storage room	5		15		Change, Ok (First floor layout required after revision)
88	Corridor	2.57	37	2.62	15.2	Change, Ok
89	Corridor	47.9	1.8	43.2	1.8	Change, Ok
90	Corridor	29.5	2.58	28.1	1.8	Change, Ok
91	Corridor	1.95	11.3	1.8	12	Change, Ok
92	Entry corridor	39.8	6.84	48.9	5	Change, Ok
93	Plasma receipt	2.4	4.95	2.4	4.95	No Change, Ok
94	Serology testing	3	4.95	3	4.95	No Change, Ok
95	Reject room	2.95	4.95	2.26	4.95	Change, Ok
96	Plasma quarantine	5	2.95	5	2.72	Change, Ok
	11.5	3.3	11.5	4.3	Change, Ok	
97	AL-1	1.87	1.4	1.87	1.4	No Change, Ok
98	Change room	1.87	1.4	1.87	1.4	No Change, Ok
99	Plasma sampling	1.78	2.9	1.76	2.9	Change, Ok
	3.68	2.05	3.68	2.05	No Change, Ok	
100	Janitor Room	1.45	2.9	1.45	2.72	Change, Ok
101	Washing area	2.3	2.9	2.3	2.72	Change, Ok
102	Bag Bar coding	2.6	2.9	2.6	2.72	Change, Ok
103	RM quarantine	6.21	5.77	5.85	6.55	Space constrain for 3 months stock approx 45 batches
104	Approved RM	4.63	4	3.65	4.49	Space constrain for 3 months stock approx 45 batches
	9.72	7.22	8.65	7.54	Space constrain for 3 months stock approx 45 batches	
105	AL -1	1.5	1.93	1.5	2.18	Change, Ok
106	Change room	1.5	1.93	1.5	2.18	Change, Ok
107	Dispensing room	3.48	3.95	3.4	4.44	Change, Ok
108	AL -1	2	1.21	2	1.21	Change, Ok
109	Change room	2	2.08	2	1.8	Change, Ok
110	RM sampling	1.07	3.39	3.16	2.31	Change, Ok
	3.16	2.03	1.07	3.11	Change, Ok	
111	Dedusting	3	2.6	2.64	2.6	Change, Ok
112	RM receipt	3	2.77	2.64	2.77	Change, Ok
113	Store Office	2.23	6.25	2.23	6.5	Change, Ok
114	FG dispatch area	13.9	3.84	10.64	6.49	Change, Ok
	9.22	2.45	4.7	4.08	Change, Ok	
	4.75	2.32	3	3.55	Change, Ok	
115	Reject room	2.3	2.35	2.3	2.35	No Change, Ok
116	Recall room	2.3	2.35	2.3	2.35	No Change, Ok
117	Store change room	4.7	2.37	4.7	2.37	No Change, Ok
	1.51	1.23			Removed	
118	Cold room 4	4.45	7.58	4.44	7.34	Change, Ok
119	Cold room 3	4.45	7.58	4.12	7.34	Change, Ok
120	Cold room 2	4.45	7.58	4.45	7.34	Change, Ok
121	Cold room 1	4.45	7.58	4.62	7.34	Change, Ok
122	Ice pack	2.91	7.58	2.49	7.34	Change, Ok
	Additional room at 2-8Deg (Cold room) may be required to store various products at intermediate stage					

Date- 30.05.2026

List of Documents

- [S/No |Title |
[1 |List of raw materials for LVP |
[2 |List of raw materials for Plasma |

Contact Details:

1. Dr. Ganesh Kumraj: ganesh@biobridge.in

Technical Information # BB-LN/IND/003

Reagent Details:

Formula:

Sodium Citrate (M.Wt): 258.07.

Citric acid (M.Wt): 192.124

Sodium Chloride (M.Wt): 58.44

Calcium Chloride (M.Wt): 110.98

0.1% Tween 80

Preparation of FE-RR-05 buffer with Stock FE-RR-03.

Preparation of 20mM sodium citrate pH 7.0 + 0.18M sodium chloride +2.6mM calcium chloride + 0.1% tween 80.

Stock Buffer (FE-RR-03):0.2M Sodium Citrate pH 7.0

Formula:

$$C1 \times V1 = C2 \times V2$$

C1 = concentration of stock

V1 = volume of stock you need to take

C2 = desired (working) concentration

V2 = final total volume of working solution

Applying the formula to prepare the working buffer:

$$C1 \times V1 = C2 \times V2$$

C1 = 0.2M Sodium Citrate pH 7.0

V1 = volume of stock you need to take _____ L.

C2 = desired (working) concentration 20mM Sodium Citrate, pH 7.0

V2 = final total volume of working solution _____ L.

$$V1 = (C2 \times V2) / C1$$

$$C2 = 0.02M ; V2 = 1.0 L ; C1 = 0.2M$$

$$V1 = (0.02 \times 1.0) / 0.2$$

$$V1 = 0.1L$$

Take 0.1 L (FE-RR-03) Stock buffer solution and dilute with water, and make up to 1L.

Calculations: To prepare the 0.18M Sodium Chloride.

Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).

Calculation: To prepare 1 L of 0.18M Sodium Chloride (MW = 58.44 g/mol)

$$\text{Weight of solute(g)} = 0.18 \times 58.44 \times 1 = 10.52g.$$

Preparation Method:

Weigh 10.52 g of Sodium Chloride and dissolve in water to make a 1 L solution.

Preparation for different volumes:

To prepare _____ L of 0.18M Sodium Chloride (MW = 58.44 g/mol).

$$\text{Weight of solute(g)} = 0.18 \times 58.44 \times \text{L} = \text{g}.$$

Weigh _____ gm of Sodium Chloride and dissolve in water to make a _____ L solution.

Calculations: To prepare the 2.6mM Calcium Chloride.

Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).

Calculation: To prepare 1 L of 2.6 mM Calcium Chloride (MW = 110.98g/mol)

$$\text{Weight of solute(g)} = 0.0026 \times 110.98 \times 1 = 0.288g.$$

Preparation Method:

Weigh 0.288g of Calcium Chloride and dissolve in water to make a 1 L solution.

Preparation for different volumes:

To prepare _____ L of 0.0026mM Calcium Chloride (MW =110.98 g/mol).

$$\text{Weight of solute(g)} = 0.0026 \times 110.98 \times \text{L} = \text{g}.$$

Weigh _____ gm of Calcium Chloride and dissolve in water to make a _____ L solution.

Procedure

Calculation: to prepare the 0.1% Tween 80:

For 1000 mL of 0.1% solution:

$$0.1\% = 0.1g \text{ per } 1000 \text{ mL} = 1000 \text{ mg}$$

Since Tween 80 is a liquid, you can weigh it directly or use its density (~1.06 g/mL) to measure volume:

$$\text{Approx. } 100 \mu\text{L of Tween 80} = 1.0 \text{ g}$$

Add Tween 80 to a beaker or flask containing ~800 mL of water.

Mix thoroughly:

Use a magnetic stirrer or vortex until fully dissolved. Tween 80 is viscous, so mixing may take a few minutes.

Adjust volume:

After complete dissolution, bring the final volume up to 1000 mL with water.

Procedure for Preparation of FE-RR-05 buffer with the stock FE-RR-03.

To prepare a 1 L solution, take 0.2M Tri Sodium Citrate pH 7.0. To this, add the 0.18M sodium chloride

and dissolve it and make up the volume with water to 900ml, and then to this add 0.026 M Calcium

chloride, to this add the 0.1% Tween 80 then mix the solution till it dissolves and check the pH; if the

pH is less than 7.0 then adjust the pH with 1 M sodium hydroxide to get the pH 7.0 ±0.1 and make up the

final volume to 1L.

Check and record the details:

Preparation Details:

Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry

Date

Sodium Citrate | | | |

Citric Acid | | | |

Sodium Chloride | | | |

Calcium Chloride | | | |

Tween 80 | | | |

Stock buffer (FE-RR-03) taken volume(L) | pH (7.0 ± 0.1) | Conductivity (20 ±2.0ms/cm) | Storage

Condition | Final volume prepared

| | | RT |

Date of Preparation | Prepared By | Checked By

| |

Application for Consent/ Authorisation

Sir,

I/We hereby apply for*

1. Consent to Establish/Operate/Renewal of consent under section 25 and 26 of the Water (Prevention & Control of Pollution) Act, 1974 as amended.
2. Consent to Establish/Operate/Renewal of consent under Section 21 of the Air (Prevention and Control of Pollution) Act, 1981, as amended.
3. Authorization/renewal of authorization under Hazardous and Other Wastes (Management and Transboundary Movement) Rules, 2016 in connection with my/our/existing/proposed/alterd/ additional manufacturing/processing activity from the premises as per the details given below.

Consent Information

UAN No: Application submitted on:
MPCB-CONSENT-0000274336 15-01-2026

Industry Information

Consent To: IIN No.: Submit to:
Establish (New) SRO - Chhatrapati
Sambhajanagar I

Type of institution: Industry Type: Category: Scale:
Industry R58 Pharmaceuticals Red M.S.I

Revised Industry Categorization as per CPCB Guidelines 2025

New Industry Type: Sector: New Category:
123.1 PHARMACEUTICAL INDUSTRY - Pharmaceuticals Industrial Sector Red
manufacturing PHARMACEUTICAL INDUSTRY123.1
PHARMACEUTICAL INDUSTRY - Pharmaceuticals manufacturing
PHARMACEUTICAL INDUSTRY - Pharmaceuticals manufacturing

Location of Name of MIDC:
industry/activity/etc:
MIDC MITL AURIC SHENDRA

EC Req'd.
No

Whether construction-buildup area is more than 20,000 No
sq.mtr.(Existing Expansion Unit)

General Information

1. Name, designation, office address with Telephone/Fax numbers, e-mail of the Applicant
Occupier/Industry/Institution / Local Body.

Name Address
MR. SANJAY UPADHYAY SECTOR 13, PLOT NO- 8A, AT AURIC SHENDRA,

Designation Taluka
DIRECTOR CHH. SAMBAJINAGAR

Area District
AURIC SHENDRA Chatrapati Sambhaji Nagar

Telephone Fax
7276066666 00

Email Pan Number
slandge@lifenitywellness.com AARCS3317A

2. (a) Name and location of the industrial unit/premises for which the application is made (Give revenue
Survey Number/Plot number
name of Taluka and District, also telephone and fax number)

Industry name
M/S: LIFENITY HEALTH LIMITED

Location of Unit Survey number/Plot Number
AT AURIC SHENDRA, SECTOR 13, PLOT NO- 8A,

Taluka District
CHH. SAMBAJINAGAR Chatrapati Sambhaji Nagar

(b) Details of the planning permission obtained from the local body/Town and Country Planning
authority/Metropolitan Development
authority/ designated Authority.

Planning permission Planning Authority
MITL MITL

Name of the local body under whose jurisdiction the unit is located and Name of the licence issuing authority

Name of Local Body Name of the licence issuing authority
AURIC SHENDRA MPCB

Sanction plan/ Approved layout Plan/Commencement
Certificate

3. Names, addresses with Telephone and Fax Number of Managing Director / Managing Partner and officer responsible for matters connected with pollution control and/or Hazardous waste disposal.

Name of Managing Director Telephone number
MR. SANJAY UPADHYAY 7276066666

Fax number Officer responsible for day to day business
00 MR. SANJAY UPADHYAY

4. (a.) Are you registered Industrial unit ? Yes

Registration number Date of registration
27AARCS3317A1ZH Apr 2, 2012

5. Gross capital investment of the unit without depreciation till the date of application (Cost of building, land, plant and machinery). (To be supported by an affidavit/undertaking on Rs.20/- stamp paper, annual report or certificate from a Chartered Accountant for proposed unit(s), give estimated figure)

Gross capital (in Lakh) * Verified * Terms * Consent Fee
29759.31 CA Certificate 1 595186.00

6. If the site is located near sea-shore/river bank/other water bodies/Highway, Indicate the crow fly distance and the name of the water body, if any.

Distance From Distance(Km) * Name
SH/NH 2.00 Aurangabad-Jalna Road

River 0.00 --NA--
Human Habitation 5.00

Religious Place 0.00 --NA--

Historical Place 0.00 --NA--

Creek/Sea 0.00 --NA--

6b. Enter Latitude and Longitude details of site

Latitude Longitude
00 00

7. Does the location satisfy the Requirements Under relevant Central/State Govt. Notification such as Coastal Regulation Zone. Notification on Ecologically Fragile Area, Industrial Location policy, etc. If so, give details.

Location Approved Industry Sensitive Area If Yes, Name Of Area Industry Location with Area Reference to CRZ
AURIC SHENDRA Yes No NA A1

8. If the site is situated in notified industrial estate,

Details

(a) Whether effluent collection, No PROPOSED treatment and disposal system has been provided by the authority.

(b) Will the applicant utilize the No WILL BE UTILIZED system, if provided.

(c) If not provided, details of proposed NA arrangement.

9.

(a) Total plot area (in square meter) (b) Built up area and (in square meter) (c) Area available for the use of treated sewage/ trade effluent for gardening/irrigation. (in square meter)
81920.50 47701.03 8194.31

10. Month and year of commissioning of the Unit.

2026-11-25

11. Number of workers and office staff

Workers staff Hrs. of shift Weekly office
450 50 08 SUNDAY

12.

(a) Do you have a residential No NA colony Within the premises in respect of Which the present application is Made ?
(b) If yes, please state population staying Number of person staying Water consumption Sewage generation Whether is STP provided?
00 00 00 No

(c) Indicate its location and distance with reference to plant site.
Number of person staying Water consumption
NA 00
13. List of products and by-products Manufactured in tonnes/month, KI/month or numbers/month with their types i.e.Dyes, drugs etc.
(Give figures corresponding to maximum installed production capacity

Products Name and Quantity

Product	UOM	Product Existing	Consented	Proposed	Total	Remarks
OTHERS	--NA--	0				

Pharmaceutical	No/M	ALBUMIN (GM)	00 00	13500	13500	VIALS PER s formulations MONTH
----------------	------	--------------	-------	-------	-------	--------------------------------

Pharmaceutical	No/M	HUMAN IGG	00 00	15000	15000	VIALS PER s formulations (GM) MONTH
----------------	------	-----------	-------	-------	-------	-------------------------------------

Pharmaceutical	No/M	FACTOR VIII IN	00 00	750	750	VIALS PER s formulations IU MONTH
----------------	------	----------------	-------	-----	-----	-----------------------------------

Pharmaceutical	No/M	FACTOR IX IN	00 00	2250	2250	VIALS PER s formulations IU MONTH
----------------	------	--------------	-------	------	------	-----------------------------------

Pharmaceutical	No/M	FIBRINOGEN	00 00	1200	1200	VIALS PER s formulations (GM) MONTH
----------------	------	------------	-------	------	------	-------------------------------------

Pharmaceutical	No/M	THROMBIN IN	00 00	21000	21000	VIALS PER s formulations IU MONTH
----------------	------	-------------	-------	-------	-------	-----------------------------------

Products Name and Quantity

Product	Name	UOM	Quantity	Remarks
NA	--NA--	00	NA	

14. List of raw materials and process chemicals with annual consumption corresponding to above stated production figures, in tonnes/month or kl/month or numbers/month.

Name of Raw Material	UOM	Quantity	Hazardous	Hazardous	Remarks
Waste Chemicals					
COPY ENCLOSED	KL/A	00	No	No	NA

15. Description of process of manufacture for each of the products showing input, output, quality and quantity of solid, liquid and gaseous wastes, if any from each unit process.

ENCLOSED

Part B : Waste Water aspects

16. Water consumption for different uses (m3/day)

Purpose	Consumption	Effluent Treatment	Remarks	Disposal	Remarks
Generation					
Domestic	Purpose 51.30	41.04	STP	NA	On Land for NA Gardening

Water gets Polluted	43.3	34.64	Primary + NA	On Land for NA & Pollutants are Secondary + Gardening Biodegradable Tertiary
---------------------	------	-------	--------------	--

Water gets	00 00	--NA--	NA	--NA--	NA
Polluted,Pollutants are not Biodegradable & Toxic					

Industrial	00 00	--NA--	NA	--NA--	NA
Cooling,spraying in mine pits or boiler feed					

Others NA
17. Source of water supply, Name of authority granting permission if applicable and quantity permitted.

Source of water supply	Name of MIDC	Name of authority granting	Qauntity permitted
MIDC	MITL AURIC SHENDRA	MITL AURIC SHENDRA	100

18. Quantity of waste water (effluent) generated (m3/day)

Domestic Boiler Blowdown Industrial Cooling water blowdown
41.04 00 00 00

Process DM Plants/Softening Washing Tail race discharge from
34.64 00 00 00

19. Water budget calculations accounting for difference between water consumption and effluent generated.

00

20. Present treatment of sewage/canteen effluent (Give sizes/capacities of treatment units).

Capacity of STP (m³/day)
60

Treatment unit Size (mxm) Retention time (hr)
STP WILL BE PROVIDED 00 00

21. Present treatment of trade effluent (Give sizes/capacities of treatment units) (A schematic diagram of the treatment scheme with inlet/outlet characteristics of each unit operation/process is to be provided. Include details of residue Management system (ETP sludges)

Capacity of ETP (m³/day)
50

Treatment unit Size (mxm) Retention time (hr)
STP WILL BE PROVIDED 00 00

22.

(i) Are sewage and trade effluents mixed together? No

If yes, state at which stage-Whether before, intermittently or after treatment. NA

23. Capacity of treated effluent sump, Guard Pond if any.

Capacity of treated effluent sump (m³) NA

Effluent sump/Guard pond details No NA

If yes, state at which stage-Whether No NA
before, intermittently or after
treatment.

24. Mode of disposal of treated effluent With respective quantity, m³/day

(i) into stream/river (name of NA (ii) into creek/estuary (name NA
river) of Creek/estuary)

(iii) into sea NA (iv) into drain/sewer (owner NA
of sewer)

(v) On land for irrigation on 00 (vi) Connected to CETP 00
owned land/ase land. Specify
cropped area.

(vii) Quantity of treated 00 Mode of disposal types Recycle
effluent reused/ recycled,
m³/day Provide a location
map of disposal
arrangement indicating the
outlet(s) for sampling.
Treated effluent reused /
recycled (m³/day)
Mode of disposal types other
(if any)

25. (a) Quality of untreated/treated effluents (Specify pH and concentration of SS, BOD,COD and specific
pollutants relevant to the
industry. TDS to be reported for disposal on land or into stream/river.

Untreated Effluent

pH 00

SS (mg/l) 00

BOD (mg/l) 00

COD (mg/l) 00

TDS (mg/l) 00

Specific pollutant if Name Value
any
1 NA NA

Treated Effluent

pH 00

SS (mg/l) 00

BOD (mg/l) 00

COD (mg/l) 00

TDS (mg/l) 00

Specific pollutant if Name Value
any
1 NA NA

(b) Enclose a copy of the latest report of analysis from the laboratory approved by State Board/
Committee/Central Board/Central
Government in the Ministry of Environment expected characteristics of the untreated/treated effluent

NA

26. Fuel consumption

Fuel Type UOM Fuel Consumption TPD/LKD Calorific value
HSD Lit/Day 83 00

Ash content Sulphur content Quantity Other (specify)
00 00 1 00

Fuel Type UOM Fuel Consumption TPD/LKD Calorific value
LPG Kg/Day 1.38 00

Ash content Sulphur content Quantity Other (specify)
00 00 1 00

Fuel Type UOM Fuel Consumption TPD/LKD Calorific value
Electricity MW 11.11 00

Ash content Sulphur content Quantity Other (specify)
00 00 1 00

Fuel Type UOM Fuel Consumption TPD/LKD Calorific value
CNG m3/day 416.6 00

Ash content Sulphur content Quantity Other (specify)
00 00 1 CNG/PNG

27. (a) Details of stack (process & fuel stacks: D. G.)

(a) Stack number(s) (b) Stack attached to (c) Capacity (d) Fuel Type
01 BOILER 0.5 PNG/CNG

(e) Fuel quantity (Kg/hr.) (f) Material of construction (g) Shape (h) Height, m (above ground
(round/rectangular) level)
34.71 MS ROUND 12

(i) Diameter/Size, in meters (j) Gas quantity, Nm3/hr. (k) Gas temperature (l) Exit gas velocity,
m/sec.
1.2 00 00 00

(m) Control equipment (n) Nature of pollutants (o) Emissions control system (p) In case of D.G. Set power
preceding the stack likely to present in stack provided generation capacity in KVA
gases such as Cl2, Nox, Sox
TPM etc.
SCRUBBER WILL BE PROVIDED SO2 SCRUBBER WILL BE PROVIDED 500 KVA

(a) Stack number(s) (b) Stack attached to (c) Capacity (d) Fuel Type
02 SD EXHAUST 00 NA

(e) Fuel quantity (Kg/hr.) (f) Material of construction (g) Shape (h) Height, m (above ground
(round/rectangular) level)
00 MS ROUND 10

(i) Diameter/Size, in meters (j) Gas quantity, Nm3/hr. (k) Gas temperature (l) Exit gas velocity,
m/sec.
1.2 00 00 00

(m) Control equipment (n) Nature of pollutants (o) Emissions control system (p) In case of D.G. Set power
preceding the stack likely to present in stack provided generation capacity in KVA
gases such as Cl2, Nox, Sox
TPM etc.
CARBON FILTERS FOLLOWED BY VOC CARBON FILTERS FOLLOWED BY 500 KVA
PACTBED WET SCRUBBER, WILL PACTBED WET SCRUBBER, WILL
BE PROVIDED BE PROVIDED

(a) Stack number(s) (b) Stack attached to (c) Capacity (d) Fuel Type
03 CLEAN ROOM NA NA

(e) Fuel quantity (Kg/hr.) (f) Material of construction (g) Shape (h) Height, m (above ground
(round/rectangular) level)
00 MS ROUND 03

(i) Diameter/Size, in meters (j) Gas quantity, Nm3/hr. (k) Gas temperature (l) Exit gas velocity,
m/sec.
1.5 00 00 00

(m) Control equipment (n) Nature of pollutants (o) Emissions control system (p) In case of D.G. Set power

preceding the stack likely to present in stack provided generation capacity in KVA
gases such as Cl2, Nox, Sox
TPM etc.
HEPA FILTERS WILL BE TPM HEPA FILTERS WILL BE NA
PROVIDED PROVIDED

(a) Stack number(s) (b) Stack attached to (c) Capacity (d) Fuel Type
04 LABORATORY 00 00

(e) Fuel quantity (Kg/hr.) (f) Material of construction (g) Shape (h) Height, m (above ground
(round/rectangular) level)
00 MS ROUND 03

(i) Diameter/Size, in meters (j) Gas quantity, Nm3/hr. (k) Gas temperature ■ (l) Exit gas velocity,
m/sec.
1.5 00 00 00

(m) Control equipment (n) Nature of pollutants (o) Emissions control system (p) In case of D.G. Set power
preceding the stack likely to present in stack provided generation capacity in KVA
gases such as Cl2, Nox, Sox
TPM etc.
SCRUBBER WILL BE PROVIDED GASES SCRUBBER WILL BE PROVIDED NA

(a) Stack number(s) (b) Stack attached to (c) Capacity (d) Fuel Type
05 DG SET 500 KVA X 02 HSD

(e) Fuel quantity (Kg/hr.) (f) Material of construction (g) Shape (h) Height, m (above ground
(round/rectangular) level)
07 MS ROUND 05

(i) Diameter/Size, in meters (j) Gas quantity, Nm3/hr. (k) Gas temperature ■ (l) Exit gas velocity,
m/sec.
1.2 00 00 00

(m) Control equipment (n) Nature of pollutants (o) Emissions control system (p) In case of D.G. Set power
preceding the stack likely to present in stack provided generation capacity in KVA
gases such as Cl2, Nox, Sox
TPM etc.
ACOUSTIC ENCLOSURES WILL BE TPM, SO2 ACOUSTIC ENCLOSURES WILL BE NA
PROVIDED PROVIDED

27. (B) Whether any release of odoriferous compounds such as Mercaptans, Phorate etc. Are coming out from
any storages or process
house.

NA

28. Do you have adequate facility for collection of samples of emissions in the form of port holes,
platform, ladder/etc. As per Central
Board Publication "Emission regulations Part-III" (December, 1985)

Poart hole No Details WILL BE PROVIDED

Platform No Details WILL BE PROVIDED

Ladder No Details WILL BE PROVIDED

29. Quality of treated ■ue gas emissions and process emissions. Quantity of treated ■ue gas emissions
and process emissions.

Sr. Stack attached to Parameter Concentration mg/Nm3 ■ow (Nm3/hr)
No

1 BOILER SO2 00 00

2 SD EXHAUST VOC 00 00

3 CLEAN ROOM TPM 00 00

4 LABORATORY GASES 00 00

5 DG SET SO2 TPM 00 00

(Specify concentration of criteria pollutants and industry/process-speci■c pollutants stack-wise. Enclose
a copy of the latest report of analysis from the laboratory approved by State Board/Central Board/
Central Government in the Ministry of Environment & Forests. For proposed unit furnish expected
characteristics of the emissions..

NA

Part - D: Hazardous Waste aspect
30 Information about Hazardous Waste Management as de■ned in Hazardous Waste (Management & Handling)
Rules, 1989 as amended in Jan.,2000.
Type/Category of Waste as per

Waste (Annually) Schedule I
Cat No Type Qty UOM
4.4 4.4 Organic residue from processes 0.5 MT/A

Max Method of collection Method of reception Method of storage
MANUAL DRUMS SEPARATE STORAGE

Method of transport Method of treatment Method of disposal
AUTHORIZED TRANSPORTER COPROCESSOR/REPROCESSOR/CHW COPROCESSOR/REPROCESSOR/CHW
TSDf TSDf

Cat No Type Qty UOM
7.2 7.2 Process residues 1.5 KL/A

Max Method of collection Method of reception Method of storage
MANUAL DRUMS SEPARATE STORAGE

Method of transport Method of treatment Method of disposal
AUTHORIZED TRANSPORTER COPROCESSOR/REPROCESSOR/CHW COPROCESSOR/REPROCESSOR/CHW
TSDf TSDf

Cat No Type Qty UOM
8.2 8.2 Sludge and filter cakes 05 MT/A

Max Method of collection Method of reception Method of storage
MANUAL DRUMS SEPARATE STORAGE

Method of transport Method of treatment Method of disposal
AUTHORIZED TRANSPORTER COPROCESSOR/REPROCESSOR/CHW COPROCESSOR/REPROCESSOR/CHW
TSDf TSDf

Cat No Type Qty UOM
5.1 5.1 Used or spent oil 50 Lit/Day

Max Method of collection Method of reception Method of storage
MANUAL DRUMS SEPARATE STORAGE

Method of transport Method of treatment Method of disposal
AUTHORIZED TRANSPORTER COPROCESSOR/REPROCESSOR/CHW COPROCESSOR/REPROCESSOR/CHW
TSDf TSDf

Cat No Type Qty UOM
35.3 35.3 Chemical sludge from waste 05 MT/A
water treatment

Max Method of collection Method of reception Method of storage
MANUAL DRUMS NASEPARATE STORAGE

Method of transport Method of treatment Method of disposal
NAAUTHORIZED TRANSPORTER COPROCESSOR/REPROCESSOR/CHW COPROCESSOR/REPROCESSOR/CHW
TSDf TSDf

Cat No Type Qty UOM
28.3 28.3 Spent carbon 0.5 MT/A

Max Method of collection Method of reception Method of storage
MANUAL DRUMS SEPARATE STORAGE

Method of transport Method of treatment Method of disposal
AUTHORIZED TRANSPORTER COPROCESSOR/REPROCESSOR/CHW COPROCESSOR/REPROCESSOR/CHW
TSDf TSDf

Waste (Annually) Schedule II
31. Details about use of hazardous waste

Name of hazardous Quantity used/month Party from whom purchased Party to whom sold
waste/Spent chemical
NA 00 NA NA

32.

a. Details about technical capability and equipments available with the applicant to handle the Hazardous
Waste
NA

b. Characteristics of hazardous waste(s) Specify concentration of relevant pollutants. Enclose a copy of
the latest report
of analysis from the laboratory approved by State Board/Central Board/Central Govt. in the ministry of
Environment &
Forests. For proposed units furnish expected characteristics
NA

33.

Copy of format of manifest/record Keeping practiced by the applicant.
NA

34.

Details of self-monitoring (source and environment system)
NA

35.

Are you using any imported hazardous waste. If yes, give details.
NA

36.

Copy of actual user Registration/certificate obtained from State Pollution Control Board/Ministry of Environment & Forests, Government of India, for use of hazardous waste.
NA

37.

Present treatment of hazardous waste, if any (give type and capacity of treatment units)
NA

38. Quantity of hazardous waste disposal

(i) Within factory
00

(ii) Outside the factory (specify location and enclose copies of agreement.)
00

(iii) Through sale (enclosed documentary proof and copies of agreement.)
00

(iv) Outside state/Union Territory, if yes particulars of (1 & 3) above.
00

(v) Other (Specify)
00

Part - E: Additional information

39.

a. Do you have any proposals to upgrade the present system for treatment and disposal of effluent/emissions and/or hazardous waste.
NA

b. If yes, give the details with time- schedule for the implementation and approximate expenditure to be incurred on it.
NA

40.

Capital and recurring (O & M) expenditure on various aspect of environment protection such as effluent, emission, hazardous waste, solid waste, tree- plantation, monitoring, data acquisition etc. (give figures separately for items implemented/to be implemented).
NA

41.

To which of the pollution control equipment, separate meters for recording consumption of electric energy are installed ?
NA

42.

Which of the pollution control items are connected to D.G. Set (captive power source) to ensure their running in the event of normal power failure
NA

43. Nature, quantity and method of disposal of non- hazardous solid waste generated separately from the process of manufacture and waste treatment.
(Give details of area/capacity available in applicant's land)

Type Quantity UOM Treatment Disposal Other Details
EMPTY GLASS VIALS 01 MT/A NA BY SALE TO AUTHORIZED NA
RECYCLER

PAPERBOARD CARTONS, 02 MT/A NA BY SALE TO AUTHORIZED NA
LABELS, PAPER SCRAP VENDOR

CORRUGATED BOXES, 01 MT/A NA BY SALE TO AUTHORIZED NA
STRETCH FILMS, PLASTIC VENDOR
WRAPPING

METAL SCRAP (ALUMINIUM, 1.2 MT/A NA BY SALE TO AUTHORIZED NA
IRON, STEEL) RECYCLER

CABLE CUTTINGS, WIRE 0.2 MT/A NA BY SALE TO AUTHORIZED NA
ENDS, PLASTIC CONDUITS VENDOR

EMPTY NON CONTAMINATED 0.2 MT/A NA BY SALE TO AUTHORIZED NA
HDPE/ PET BOTTLES RECYCLER

44. Hazardous Chemicals – Give details of Chemicals and quantities handled and Stored.

(i) Is the unit a Major Accident Hazard unit as per Mfg.Storage Import Hazardous Chemicals Rules ?
NA

(ii) Is the unit an isolated storage as defined under the MSIHC Rules ?

NA

(iii) Indicate status of compliance of Rules 5,7,10,11,12,13 and 18 of the MSIHC Rules.

NA

(iv) Has approval of site been obtained from the concerned authority?

NA

(v) Has the unit prepared an on-site Emergency Plan? Is it updated ?

NA

(vi) Has information on imports of Chemicals been provided to the concerned authority?

NA

(vii) Does the unit possess a policy under the PLI Act?

NA

45. Brief details of tree plantation/green belt development within applicant's premises (in hectors)

Open Space Availability Plantation Done On Number of Trees Planted

8194 Square meter 3000 Square meter(37 %) 20

46.

Information of schemes for waste Minimization, resource recovery and recycling - implemented and to be implemented, separately.

NA

47.

(a) The applicant shall indicate whether Industry comes under Public Hearing, if so, the relevant documents such as EIA, EMP, Risk Analysis etc. shall be submitted, if so, the relevant documents enclosed shall be indicated accordingly.

NA

(b) Any other additional information that the applicants desires to give

NA

(c) Whether Environmental Statement submitted ? If submitted, give date of submission.

NA

48.

I/We further declare that the information furnished above is correct to the best of my/our knowledge.

49.

I/We hereby submit that in case of any change from what is stated in this application in respect of raw materials, products, process of manufacture and treatment and/or disposal of effluent, emission, hazardous wastes etc. In quality and quantity; a fresh application for Consent/Authorization shall be made and until the grant of fresh Consent/Authorization no change shall be made.

50.

I/We undertake to furnish any other information within one month of its being called by the Board

Yours faithfully

Signature : MR. SANJAY UPADHYAY

Name : MR. SANJAY UPADHYAY

Designation : DIRECTOR

Additional Information

Air Pollution

Sr No. Air Pollution Source Pollutants APCS Provided Remark

1 BOILER SO₂ SCRUBBER WILL BE NA PROVIDED

2 SD EXHAUST VOC CARBON FILTERS FOLLOWED NA BY PACTBED WET SCRUBBER WILL BE PROVIDED

3 CLEAN ROOM TPM HEPA FILTERS WILL BE NA PROVIDED

4 LABORATORY GASES SCRUBBER WILL BE NA PROVIDED

5 DG SET SO₂, TPM ACOUSTIC ENCLOSURE WILL NA BE PROVIDED

Separate EM Provided No Other Emission Sources NA

Measures Proposed NA Foul Smell Coming Out No

Air Sampling Facility Details WILL BE PROVIDED

D.G. Set Details

Description Capacity(KVA) Remarks
DG SET 500 WILL BE USE IN POWER FAILURE

DG SET 500 WILL BE USE IN POWER FAILURE

Hazardous Waste Generation

Hazardous Waste Quantity UOM Treatment Disposal Other Details
7.2 Process residues 1.5 KL/A NA CHWTSDF/CO NA
PROCESSOR/RE
PROCESSOR

8.2 Sludge and ■iter 0.5 MT/A NA CHWTSDF/CO NA
cakes PROCESSOR/RE
PROCESSOR

28.3 Spent carbon 0.5 MT/A NA CHWTSDF/CO NA
PROCESSOR/RE
PROCESSOR

35.3 Chemical sludge 05 MT/A NA CHWTSDF/CO NA
from waste water PROCESSOR/RE
treatment PROCESSOR

4.4 Organic residue from 0.5 MT/A NA CHWTSDF/CO NA
processes PROCESSOR/RE
PROCESSOR

5.1 Used or spent oil 50 Lit/Day NA BY SALE AUTHORIZED NA
RECYCLER

CHWTSDF Details

Member of CHWTSDF CHWTSDF Name Remarks

Cess Details

Cess Applicable Cess Paid If Yes, UpTo
No No Jan 1 1900 12:00:00:000AM

Legal Actions

Legal Legal Record Of Company Legal Action Details Remarks
Action
Taken
No

No
Bank Guarantee Applicable:
Annexure

PROCESS INSTRUCTIONS AND DATA CAPTURE POINTS –FIBRINOGEN

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PRECIPITATION OF FRACTION-2 FROM CHROMATOGRAPHY-1F.

Collect 8 injections, Fraction-2 from Chromatography-1F.

Check and record the following details:

Total Volume of Fraction-2 _____ L.

Visual Appearance: _____, O.D280/600: _____.

To the above collected Fraction-2 and add FG-RR-03 to raise the conductivity to 100 ± 10 mS/cm.

Volume of FG-RR-03 used: _____ L.

Adjust the pH to 4.00 ± 0.2 with FG-RR-01 and record the details.

pH: _____, Conductivity: _____ mS/cm.

O.D280/600: _____, Volume of FG-RR-01 used: _____ L.

Under stirring conditions, incubate the above sample for 3 hours at RT.

Incubation time: _____; O.D280/600: _____.

Collection of Slurry from Precipitated Sample

Reinsert the Falcon with buffer FG-RR-06 and pour the sample into the Falcon, and place the Falcon in the high-speed Centrifuge and operate it at a speed of 9100 rpm.

Speed: _____ rpm,

Collect the slurry into a Falcon and record the volume of slurry collected.

Volume of Slurry collected: _____ L.

Add 12.5% of the volume of FG-RR-04 to a final volume of the slurry collected from step 1.6.2.

Volume of FG-RR-04 used: _____ L.

Adjust pH of the suspended slurry with FG-RR-15 to 7.5 ± 0.1

Final pH: _____, Cond.: _____ mS/cm.

Volume of FG-RR-15 used: _____ L

Resuspend the pH-adjusted slurry with FG-RR-06 in the ratio of 1:2 volumes.

Volume of FG-RR-06 used: _____ L.

Under stirring conditions, incubate the sample for 1 hour at RT.

Incubation time: _____

Filtration

Pass the resuspended slurry through PA-PO-008 filter and record the results below

Note: Divide the obtained value of O.D280 by the extinction coefficient of 1.51 to obtain protein concentration in mg/mL.

CONCENTRATION AND DIAFILTRATION

Take the filtered sample from step 1.7.1 for concentration.

2.2 NWP calculation before concentration

Record the following details of the TFF cassettes connected in series horizontally to the TFF system:

No. of cassettes used: _____, Cassette ID Nos.: _____.

Wash the cassettes with WFI and calculate NWP before concentration at a set TMP of ≤ 1.0 bar.

Volume of WFI used: _____ L.

Check and record the details as below:

Concentration of filtrate sample

After NWP calculation, flush the cassettes with FG-RR-10.

Volume of FG-RR-10 used: _____ L.

Concentrate the filtrate sample to 1.5-fold to 2-fold at a set TMP of ≤ 1.0 bar.

Final Volume: _____ L.

Check and record the details as below

Diafiltration

Collect the concentrated sample from step 2.3.2. and diafilter the

sample with 5 DF volumes using FG-RR-10.

Volume of FG-RR-10 used (L): _____

Unload the diafiltered sample. Check and record the following details

Regeneration

After unloading the concentrated sample, flush the cassettes with ambient WFI and then with FG-RR-15 at 40 to 45°C . Recirculate the cassettes with FG-RR-15 at 40°C to 45°C for 30 - 45 min.

Volume of FG-RR-15 used (L): _____.

Post-NWP calculation

Repeat the step as in 2.2.2 and record the details as below. Check if NWP is ≥ 120 LMH/Bar. If not, carry out the regeneration procedure again.

Storage

Flush the cassettes with FG-RR-14 and finally store the cassettes in FG-RR-14.

Remove the cassettes from the system, immerse in FG-RR-14 and store at $2-8^\circ\text{C}$.

Volume of FG-RR-14 used: _____ L.

SOLVENT / DETERGENT (S/D) TREATMENT FOR VIRAL INACTIVATION

Note: The amount of FG-RR-07 (20% stock solution) to be added to a final concentration of 1% (v/v) and FG-RR-08 (20% stock solution) to be added to a final concentration of 0.3% (v/v) is based on the sample volume.

Take the diafiltered sample from step 2.4.2 equivalent to 1 gram of protein/L/L of resin for Run and keep on the magnetic stirrer for thorough mixing, and store the remaining eluate at 2 to 8°C .

FG-RR-07 is added slowly to the diafiltered sample under gentle stirring conditions.

Then add FG-RR-08 slowly to the sample under gentle stirring conditions.

Incubate the sample at RT for 4 hours under stirring conditions and record the details below.

Repeat the steps from 3.1 to 3.3 for Run 2 and record the details in the above table.

CHROMATOGRAPHY-11FB

Switch on the chromatography system. When the chromatography system is ready, open the UNICORN software.

Connect the column to the chromatography system and check the column connections to the system. Check for instrument air connection at 6 bar to the system.

Record the following column details

Column Height: 20 cm, Column Volume: 393 ml,

HETP value: _____, Asymmetry: _____.

Column Wash

Pass 2 CVs of WFI through the column at 39.30 ml/min till the conductivity has dropped. Volume of WFI used

; _____ L.

pH Drop
 Pass 1.5 CVs of FG-RR-09 through the column at 39.30 ml/min.
 Volume of FG-RR-09 used: _____ L.
 Column Equilibration
 Equilibrate the column with 3 to 4 CVs of FG-RR-10 at a flow rate of 39.30 ml/min.
 Volume of FG-RR-10 used: _____ L.
 Loading and Elution
 Before loading, Auto Zero the absorbance (A280) on the chromatography system and switch on the record button, and enter the run details and date.
 Load the S/D treated sample obtained from step 3.4 at a flow rate of 39.30 ml/min. Volume of Load: _____.
 Note: Loading should be NMT 1 mg of protein / mL of resin.
 After loading is completed, pass 9 CVs of FG-RR-10 at a flow rate of 39.30 ml/min. Volume of FG-RR-10 used: _____.
 Pass 2 to 3 CVs of FG-RR-11 through the column at 39.30 ml/min.
 Volume of FG-RR-11 used: _____.
 Start collecting the eluate when the absorbance rises from 0.05 AU and collect until absorbance drops to 0.05 AU.
 Check and record the following details:
 Note: Divide the obtained value of O.D280 by the extinction coefficient of 1.51 to obtain protein concentration in mg/mL.
 Regeneration
 Pass 2 CVs of FG-RR-15 (25% gradient of FG-RR-13 against WFI) through The column in reverse flow at 39.30 ml/min.
 Volume of FG-RR-15 (L) used: _____
 Repeat steps 4.4 to 4.8 for Run2 and record the results in the above table.
 Storage
 Pass 2 CVs of FG-RR-14 through the column at a flow rate of 39.30 ml/min and store the column in FG-RR-14. Volume of FG-RR-14 used: _____ L
 Disconnect the column and store at 2-8 °C.
 CONCENTRATION AND DIAFILTRATION OF CHROMATOGRAPHY-11FB ELUATE
 The chromatography-11FB eluates from step 4.7.6 and record the following details.
 Note: Divide the obtained value of O.D280 by the extinction coefficient of 1.51 to obtain protein concentration in mg/mL.
 NWP calculation before concentration
 Record the following details of the TFF cassettes connected in series horizontally to the TFF system:
 No. of cassettes used: _____, Cassette ID Nos.: _____.
 Wash the cassettes with WFI and calculate NWP before concentration at a set TMP of ≤ 1.0 bar.
 Volume of WFI used: _____ L.
 Check and record the details as below:
 Concentration of Eluate sample
 After NWP calculation, flush the cassettes with FG-RR-11.
 Volume of FG-RR-11 used: _____ L.
 Concentrate the eluate sample to 12 to 16 mg/mL, at a set TMP of ≤ 1.0 bar.
 Collect and send 2 mL of the sample for in-process QC analysis.
 Record the obtained results as per the QC report:
 Protein Conc. (mg/mL): _____ .BET: _____.
 Concentrate the above sample from step 5.3.2 further to 35 to 50 mg/mL, at a set TMP of ≤ 1.0 bar.
 Check and record the details as below:
 Unload the concentrated sample and record the following details.
 Note: Divide the obtained value of O.D280 by the extinction coefficient of 1.51 to obtain protein concentration in mg/mL.
 Regeneration
 Flush the cassettes with ambient WFI and then with FG-RR-15, and carry out recirculation for 30 to 45 min.
 Volume of FG-RR-15 used: _____ L.
 Finally, wash the cassettes with ambient WFI.
 Volume of WFI used : _____ L
 Post-NWP calculation
 Repeat step 5.2.2 and record the details as below. Check if NWP is ≥ 120 LMH/bar if not, carry out the regeneration procedure again.
 Storage
 Flush the cassettes with FG-RR-14 and finally store the cassettes in FG-RR-14.
 Remove the cassettes from the system, immerse in FG-RR-14 and store at 2-8°C.
 Volume of FG-RR-14 used: _____ L.
 DS SAMPLE DETAILS:
 Record the QC Analysis report.
 Storage of the DS sample:
 Label the sample and store the DS sample in the -20°C freezer.
 Details | Details
 No. of Filters used |
 pH |
 Conductivity (mS/cm) |
 OD280/600 |
 Protein Conc. (mg/mL) |
 Volume (L) |
 Total protein (gms) |
 PF (bar) | PR
 (bar) | PP (bar) | Flow (L/hr) | TMP (bar) | Flux (LMH) | NWP (LMH/bar)
 |||||
 Permeate Vol. (L) | PF(bar) | PR
 (bar) | PP (bar) | Flow (L/hr) | TMP (bar) | Flux (LMH) | Permeate O.D280
 Details | Details
 pH |

Conductivity (mS/cm) |
OD280/600 |
Protein Conc. (mg/mL) |
Volume (L) |
Total protein (gms) |
% Recovery |
PF (bar) | PR (bar) | PP (bar) | Flow (L/Hr) | TMP
(bar) | Flux (LMH) | NWP (LMH/bar)
| | | | |
Details | Vol. Of
FG-RR-07 Added | Vol. Of
FG-RR-08 Added | Incubation Time | Incubation Time
Details | Vol. Of
FG-RR-07 Added | Vol. Of
FG-RR-08 Added | Start Time & Date | End Time & Date
Run 1 | | | |
Run 2 | | | |
Details | Eluate (L) | pH | Conductivity (mS/cm) | O.D280/600 | Protein Content (mg/mL)
Run 1 | | | | |
Run 2 | | | | |
Details | Details
pH |
Conductivity (mS/cm) |
OD280/600 |
Protein Conc. (mg/mL) |
Volume (L) |
Total protein (gms) |
PF(bar) | PR (bar) | PP(bar) | Flow (L/hr) | TMP (bar) | Flux (LMH) | NWP (LMH/bar)
| | | | |
Permeate Vol. (L) | PF (bar) | PR
(bar) | PP (bar) | Flow (L/hr) | TMP (bar) | Flux (LMH) | Permeate O.D280
Details	Details		
pH			
Conductivity (mS/cm)			
O.D280/600			
Protein Conc. (mg/mL)			
Volume (L)			
Total protein (gms)			
% Recovery			
PF (bar)	PR (bar)	PP (bar)	Flow (L/hr)
(bar)	Flux (LMH)	NWP (LMH/bar)	
Details | Details
pH |
Conductivity (mS/cm) |
OD280/600 |
Protein Conc. (mg/mL) |
Volume (L) |
Total protein (gms) |

Reagent Details:
Formula & Calculations:
Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).
Calculation: To prepare 1 L of 0.1N Sodium Chloride (MW = 58.44 g/mol)
Weight of solute(g)=0.1×58.44 = 5.844g.
Preparation Method:
Weigh 5.844 g of Sodium Chloride and dissolve in water to make a 1 L solution.
Preparation for different volumes:
To prepare _____ of 0.1N Sodium Chloride (MW = 58.44 g/mol).
Weight of solute(g)=0.1×58.44 x _____L= _____gm
Weigh _____gm of Sodium Chloride and dissolve in water to make a _____ L solution.
Check and record the details:
Preparation Details:
Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry
Date
Sodium Chloride | | | |
Final volume prepared | Weighed Quantity | pH (7.0±0.2) | Conductivity (10.0±2.0 ms/cm) | Storage
Condition:
| | | RT
Date of Preparation | Prepared By | Checked By
| |

Reagent Details:
Formula & Calculations:
Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).
Calculation: To prepare 1 L of 0.5M sodium hydroxide (MW = 40g/mol)
Weight of solute(g)=1×20=20g.
Preparation Method:
Weigh 20 g of Tris base and dissolve in water to make a 1 L solution.
Preparation for different volumes:
To prepare _____ of 0.5M sodium hydroxide (MW = 40g/mol).
Weight of solute(g)=1×20x _____L= _____gm
Weigh _____gm of sodium hydroxide and dissolve in water to make a _____ L solution.
Check and record the details:
Preparation Details:
Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry
Date
| | | | |
Final volume prepared | Weighed Quantity | Storage Condition:
| |
Date of Preparation | Prepared By | Checked By
| |

Reagent Details:
Formula & Calculations:
Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).
Calculation: To prepare 1 L of 0.12 M Sodium Chloride (MW = 58.44 g/mol)
Weight of solute(g)=0.12×58.44 = 7.012g.
Preparation Method:
Weigh 7.012 g of Sodium Chloride and dissolve in water to make a 1 L solution.
Preparation for different volumes:
To prepare _____ of 0.12 M Sodium Chloride (MW = 58.44 g/mol).
Weight of solute(g)=0.12×58.44 x _____L= _____gm
Weigh _____gm of Sodium Chloride and dissolve in water to make a _____ L solution.
Check and record the details:
Preparation Details:
Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry
Date
Sodium Chloride | | | |
Final volume prepared | Weighed Quantity | pH(7.0±0.1) | Conductivity (12.0± 1.0 ms/cm) | Storage
Condition:
| | | RT
Date of Preparation | Prepared By | Checked By
| |

HUMAN COAGULATION FACTOR-IX PROCESS

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9 PRA-SF-001 FILTRATION 15

1 CONCENTRATION OF ELUATE FROM CHROMATOGRAPHY-2F OF PLASMA FRACTIONATION

Take the eluate sample from Chromatography-2F of Plasma fractionation step 5.10 of Plasma Fractionation Stages.

Check and record the details:

Dilute the eluate with FN-RR-02 to obtain conductivity of 10 – 11 mS/cm.

NWP calculation before concentration

Record the following details of the TFF cassettes connected in series horizontally to the TFF system:

No. of cassettes used: _____,

Cassette ID Nos.: _____.

Wash the cassettes with 10 L of ambient WFI and calculate NWP before concentration at a set TMP of ≤ 1.0 bar.

Volume of ambient WFI used for Run: _____ L,

Check and record the details as below:

Concentration of Eluate sample

After NWP calculation, flush the cassettes with FN-RR-02.

Volume of FN-RR-02 used for Run: _____ L.

Concentrate the eluate sample to 7 to 9 folds based on plasma equivalent volume loaded on Chr.-PF-2F column; at a set TMP of ≤ 1.0 bar and Pump Speed of Run : _____ %.

Check and record the details as below:

Check and record the retentate details as below:

OD280/600: Run: _____,

pH: Run: _____,

Conductivity: Run : _____ mS/cm.

Regeneration

After unloading the concentrated sample, flush the cassettes with 10 L of ambient

WFI. Volume of WFI used for flush: Run : _____ L,

Then flush the cassettes with 80 L of FN-RR-01 at 40 to 45°C.

Volume of FN-RR-01 used: Run : _____ L, Temp. of FN-RR-01: Run: _____ °C.

Recirculate the cassettes with 10 L of FN-RR-01 at 40°C to 45°C for 30 - 45 min.

Volume of FN-RR-01 used: Run : _____ L,

Temp. of FN-RR-01: Run : _____ °C.

Finally wash the cassettes with 10 L of ambient WFI.

Volume of ambient WFI used: Run 1: _____ L.

Post NWP calculation

Repeat the step as in 1.4.2 and record the details as below. Check if NWP is ≥ 120 LMH/Bar. If not, carry out the regeneration procedure again.

Storage

Wash the cassettes with 5 L of FN-RR-17 and finally remove the cassettes and store in FN-RR-17. Volume of FN-RR-17 used: _____ L.

FILTRATION

2.1 Filter the Run concentrated sample through PRA-PO-003 filter. Collect the filtrate into a sterile vessel.

Check and record the following details:

Filtrate volume: Run : _____ L.

pH: Run : _____,

Conductivity: Run : _____ mS/cm,

OD280/600: Run : _____.

VIRUS INACTIVATION BY S/D TREATMENT

Note: The volume of FN-RR-19 (20% stock solution) and FN-RR-20 (20% stock solution) to be added to a final concentration of 1.05% (v/v) and 0.35% (v/v) respectively is based on the volume of FIX solution.

FN-RR-19 is added slowly to the run wise filtered sample under gentle stirring condition and continue stirring till the solution is homogenous.

Volume of FN-RR-19 used: Run : _____ L,

FN-RR-20 is added slowly to the sample under gentle stirring condition and continue stirring till the solution is homogenous.

Volume of FN-RR-20 used: Run: _____ L.

Incubate the sample under stirring condition at $25 \pm 1^\circ\text{C}$ for 4 hours.

Incubation temperature: Run : _____ °C.

Incubation time: Run : _____.

CHROMATOGRAPHY – 8N

Connect the column to the chromatography system and check the column connections to the system. Open the software.

Record the following details:

Column height: 15 cm, Column Volume: 79.60 L,

HETP value: _____, Asymmetry: _____.

Column Wash

Wash the column with up to 2 CVs of ambient WFI at a flow rate of 8.84ml/min till the conductivity drops below 1.0 ± 0.1 mS/cm. Volume of WFI used:

Run : _____ L,

pH drop

Wash the column with upto 2 CVs of FN-RR-05 at a flow rate of 8.84ml/min till pH has dropped down to 6.50 ± 0.20 . Volume of FN-RR-05 used: Run: _____ Final pH of the column: _____,

Column Equilibration

Equilibrate the column with upto 3 CVs of FN-RR-09 at a flow rate of 8.84ml/min .

Check and record the final pH: Run: _____,

Conductivity: Run : _____ mS/cm.

Volume of FN-RR-09 used: Run: _____ L

Sample preparation for loading

Check and record the conditions of virus inactivated sample from step 3.3:

pH: Run: _____.

Conductivity: Run : _____ mS/cm.

O.D. at 600 nm: Run: _____.

Volume: Run: _____ L.

Adjust the pH to 7.40 ± 0.10 with FN-RR-18.

Check and record the conditions below:

Volume of FN-RR-18 used: Run : _____ L.

O.D. at 600 nm: Run: _____.

Total Volume: Run : _____ L

pH: Run : _____.

Conductivity: Run : _____ mS/cm

ml of plasma load / ml of resin: _____. (Limit: ≤ 90 ml/ml).

Loading

Before loading, switch on the 'Record On' button and enter the run details and date. Auto Zero the absorbance at 280 nm on the chromatography system.

Load the sample at a flow rate of 8.84ml/min .

Wash-I

After loading is completed, wash the column with upto 2 CVs of FN-RR-09 at a flow rate of 8.84ml/min .

Record the following details:

Volume of FN-RR-09 used: Run : _____ L.

Wash II

After the A280 has reached the base line, pass upto 2 CVs of FN-RR-10 at a flow rate of 8.84ml/min .

Volume of FN-RR-10 used: Run : _____ L.

Elution

After the A280 has reached the base line, pass upto 2 CVs of FN-RR-11 at a flow rate of 8.84ml/min .

Collect eluate when the A280 reaches 10 mAU and collect the total peak till A280 drops down to 10 mAU.

Check and record the following eluate conditions:

OD280/600: Run : _____.

pH: Run : _____.

Cond.: Run : _____ mS/cm.

Volume of FN-RR-11 used: Run: _____ L.

Eluate vol.: Run : _____ L.

Protein conc.: Run : _____ mg/ml,

Total protein: Run : _____ g.

Cleaning

Clean the column by passing upto 2.0 CVs of FN-RR-12 at a flow rate of 8.84ml/min . Volume of FN-RR-12 used: Run : _____ L.

Regeneration

Wash the column by passing upto 2.0 CVs of ambient WFI at a flow rate of 8.84ml/min till the conductivity drops below 1.0 ± 0.1 mS/cm. Volume of WFI used: Run : _____ L.

Regenerate the column by passing upto 2.0 CVs of FN-RR-13 at a flow rate of 8.84ml/min and hold for 30 – 45 minutes.

Volume of FN-RR-13 used: Run : _____ L.

4.13.3 Wash the column by passing upto 2.0 CVs of ambient WFI at a flow rate of 8.84ml/min till the conductivity drops below 1.0 ± 0.1 mS/cm.

Volume of WFI used: Run : _____ L.

Storage

Store the column (at room temp.) by passing 2.0 CVs of FN-RR-14 at a flow rate of 8.84ml/min . Volume of FN-RR-14 used: _____ L.

Enter the process details as below:

CONCENTRATION OF ELUATE FROM CHROMATOGRAPHY-8N

Check and record the following details:

Eluate vol.: Run : _____ L.

pH: Run : _____.

Cond.: Run : _____ mS/cm.

OD280/600: Run: _____.

NWP calculation before concentration

Record the following details of the TFF cassettes connected in series horizontally to the TFF system:

No. of cassettes used: _____.

Cassette ID Nos.: _____.

Wash the cassettes with WFI and calculate NWP before concentration at a set TMP of ≤ 1.0 bar.

Volume of WFI used: Run : _____ L.

Check and record the details as below:

Concentration of Eluate sample

After NWP calculation, flush the cassettes with 10 L of FN-RR-11 of Chromatography-FN-I. Volume of FN-RR-11 used: Run : _____ L.

Concentrate the eluate sample to ≤ 10 mg/ml protein concentration at a set TMP of ≤ 1.0 bar. Flush the cassettes with FN-RR-11.

Check and record the details as below:

Unload the concentrated sample after obtaining the required protein concentration. Check and record the following details:

Volume of Retentate: Run: _____ L.

O.D.280/600: Run : _____.

Note: The time for concentration of Chromatography-FN-I Eluate and loading on Chromatography-FN-II should not exceed 6 hrs.

Regeneration

Flush the cassettes with 10 L of WFI and then with 10 L FN-RR-01 at $40 - 45^\circ\text{C}$ and carry out recirculation for 30 to 45 min.

Volume of FN-RR-01 used: Run : _____ L.

Post NWP calculation

Repeat the step as 5.1.1 and record the details as below. Check if NWP is ≥ 120 LMH/Bar. If not, carry out the regeneration procedure again.

Storage

Flush the cassettes with 5 L of FN-RR-17 and finally remove the cassettes and store in FN-RR-17 at $2 - 8^\circ\text{C}$. Volume of FN-RR-17 used: _____ L.

CHROMATOGRAPHY – 9N

Connect the column to the AKTA process chromatography system and check the column connections to the system. Check for instrument air connection at 6 bar to the system. Open the UNICORN software.

Record the following column details:

Column height: _____ cm, Column Volume: _____ L,

HETP value: _____, Asymmetry: _____.

Column Wash

Wash the column with upto 2.0 CVs of ambient WFI at a flow rate of 0.8 ml/min till the conductivity has dropped below 1.0 ± 0.1 mS/cm.

Volume of WFI used: _____ L, Conductivity: _____ mS/cm

Column Equilibration (With Formulation buffer)

Equilibrate the column with 2 - 3 column volumes of FN-RR-15 at a flow rate of 0.8 ml/min .

Check and record the final pH: _____,

Conductivity: _____ mS/cm,

Volume of FN-RR-15 used: _____ L.

Load Details

Take the pooled concentrated Chromatography-FN-I eluate as a load for Chromatography-FN-II.

Check and record the conditions as below:

pH: _____, Conductivity: _____ mS/cm,

O.D. at 280 nm for 10x diluted (with FN-RR-15) sample: _____,

O.D. at 600 nm: _____, Total Volume: _____ L,

Each injection Load Volume: _____ ml,

Load %: _____, (Limit: ≤ 2.0 %)

Number of injections = Total Volume / Each injection Load Volume = _____.

Loading and Elution

Before loading, switch on the 'Record On' button and enter the run details and date. Auto Zero the absorbance at 280 nm on the chromatography system.

Load 200 – 376 ml of the load volume for each injection at a flow rate of 0.8 ml/min .

After each injection, pass FN-RR-15 and collect Peak 3 observed at 50 - 60 % CV. Start peak collection when the A280 rises from 10 mAU, and stop collection when the A280 reaches to 10 mAU.

Check and record the details of the pooled peak 3 as below:

O.D.280: _____, O.D.600: _____,

pH: _____, Conductivity: _____ mS/cm,

Total volume of the Peak 3: _____ L,

Protein conc.: _____ mg/ml.

Note: pH and Conductivity of the pooled peak 3 should be in the range of 7.4 ± 0.2 and 17.0 ± 1.0 mS/cm respectively.

Collect and send 7.5 ml of the pooled peak 3 sample to QC for in-process QC analysis. Record the obtained result: Activity: _____ IU/ml.

Regeneration

Regenerate the column by passing up to 2.0 CVs of FN-RR-16 (10% gradient of FN-RR-18 against ambient WFI) at a flow rate of 0.8 ml/min and hold the column for 20 - 30 mins. Volume of FN-RR-16 used: _____ L.

Storage

Store the column (at room temp.) by passing upto 2.0 CVs of FN-RR-17 at a flow rate of 0.8 ml/min .

Volume of FN-RR-17 used: _____ L.

Enter the process details as below:

CONCENTRATION OF CHROMATOGRAPHY-9N ELUATE

Take the Chromatography eluate sample from step 6.6 and check and record the following details:

Eluate Volume: _____ L, pH: _____,

Conductivity: _____ mS/cm, OD280/600: _____.

NWP calculation before concentration

Record the following details of the TFF cassettes connected in series horizontally to the TFF system:

No. of cassettes used: _____,

Cassette ID Nos.: _____.

Wash the cassettes with 10 L of WFI and calculate NWP before concentration at a set TMP of ≤ 1.0 bar.

Volume of WFI used: _____ L.

Check and record the details as below:

Concentration of Eluate sample

After NWP calculation, flush the cassettes with 10 L of FN-RR-15.

Volume of FN-RR-15 used: _____ L.

Concentrate the eluate sample to the required activity based on the QC result of pooled peak 3 sample, at a set TMP of ≤ 1.0 bar; to achieve a final activity of ≥ 125 IU/ml. Flush the cassettes with FN-RR-15.

Check and record the details as below:

Unload the concentrated sample and record the following details:

Volume of Retentate: _____ L, O.D.280/600: _____.

Regeneration

Flush the cassettes with 10 L of WFI and then with 10 L of FN-RR-01 at 40 - 45°C and carry out recirculation for 30 to 45 min.

Volume of FN-RR-01 used: _____ L.

Post NWP calculation

Repeat the step as 7.2.2 and record the details as below. Check if NWP is ≥ 120 LMH/Bar. If not, carry out the regeneration procedure again.

Storage

Flush the cassettes with 5 L of FN-RR-17 and finally store the cassettes in FN-RR-17 at room temp. Volume of FN-RR-17 used: _____ L.

PRA-NF-005 FILTRATION

Filter the concentrated sample through sterile PRA-SF-001 filter (pre-filter for PRA-NF-005 filtration) at 0.5 bar pressure under LAFU.

Check and record the following details:

Filtrate Volume: _____ L,

O.D. at 280 nm: _____, O.D. at 600 nm: _____.

Pass the pre filtered sample through PRA-NF-005 filter at 1.0 bar pressure.

Check and record the following details:

Filtrate Volume: _____ L,

O.D. at 280 nm: _____, O.D. at 600 nm: _____.

PRA-SF-001 FILTRATION

Filter the PRA-NF-005 filtered sample through a sterile PRA-SF-001 filter at 0.5 bar pressure under LAFU; and collect the filtrate into sterilised 50 ml polypropylene bottles.

Check and record the following details:

Filtrate Volume: _____ L,

O.D. at 280 nm: _____, O.D. at 600 nm: _____.

Collect and send 10 ml of the PRA-SF-001 filtered sample to QC for in-process QC analysis. Store the bottles containing PRA-SF-001 filtrate at -25°C till further usage.

Details | Run

Eluate Volume (L) |

OD280/600 |

Protein Conc. (mg/ml) |

pH |

Conductivity (mS/cm) |
Details | Run
Volume of FN-RR-02 used (L) |
Run No. | PF (bar) | PR
(bar) | PP (bar) | Flow (L/hr) | TMP (bar) | Flux (LMH) | NWP
(LMH/bar)
1. | | | | | | | |
Run No. | Permeate Vol. (L) | PF (bar) | PR
(bar) | PP (bar) | Flow (L/hr) | TMP (bar) | Flux (LMH)
1 | | | | | | | |
1 | | | | | | | |
1 | | | | | | | |
Details | Details | Before Conc. | After Conc.
Run | Volume (L) | |
Run | Protein Conc. (mg/ml) | |
Run | Total protein (g) | |
Recovery (%) | Recovery (%) | |
PF (bar) | PR (bar) | PP (bar) | Flow (L/hr) | TMP
(bar) | Flux (LMH) | NWP (LMH/bar)
| | | | | |
Details | Load | Load | Eluate
Details | Run | Run | Run
Total volume (L) | | |
OD280/600 | | |
Protein conc. (mg/ml) | | |
Total protein (g) | | |
Details | PF (bar) | PR (bar) | PP (bar) | Flow (mL/min) | TMP
(bar) | Flux (LMH) | NWP (LMH/bar)
Run | | | | | | | |
Details | Permeate Vol. (L) | PF (bar) | PR
(bar) | PP (bar) | Flow (L/hr) | TMP (bar) | Flux (LMH) | Permeate O.D.280
Run | | | | | | | |
Run | | | | | | | |
Run | | | | | | | |
Run | | | | | | | |
Run | | | | | | | |
Run | | | | | | | |
Details | Run | Run
Details | Before Conc. | After Conc.
Volume (L) | |
Protein Conc. (mg/ml) | |
Total protein (g) | |
% Recovery | |
PF (bar) | PR (bar) | PP (bar) | Flow (mL/min) | TMP
(bar) | Flux (LMH) | NWP (LMH/bar)
| | | | | |
Details | Load | Eluate
Total volume (L) | |
Protein conc. (mg/ml) | |
Total protein (g) | |
PF (bar) | PR (bar) | PP (bar) | Flow (mL/min) | TMP (bar) | Flux (LMH) | NWP (LMH/bar)
| | | | | |
Permeate Vol. (L) | PF (bar) | PR
(bar) | PP (bar) | Flow (mL/min) | TMP (bar) | Flux (LMH) | Permeate O.D.280
| | | | | |
| | | | | |
| | | | | |
Details | Before Conc. | After Conc.
Volume (L) | |
Activity (IU/ml) | |
Protein Conc. (mg/ml) | |
Total protein (mg) | |
% Recovery | |
PF (bar) | PR (bar) | PP (bar) | Flow (mL/min) | TMP
(bar) | Flux (LMH) | NWP (LMH/bar)
| | | | | |

WATER BALANCE DIAGRAM

1. System Boundaries

Water-using blocks:

Plasma fractionation & purification

CIP/SIP systems

Utilities (boilers, cooling towers, HVAC)

Formulation & filling

QC laboratory

Domestic usage (staff)

Effluent treatment + ZLD system

2. Estimate Water Consumption

For biologics/plasma plants, water intensity is typically:

10–20 L water per L plasma processed (efficient facility)

Total annual water demand:

150,000 L plasma × 12 = 1,800,000 L/year (1,800 m³/year)

≈ 6,000 L/day

3. Break Down Water Usage

A. Process & CIP (Major Consumer ~50%)

Chromatography buffers

Equipment cleaning

Viral inactivation cleaning

~3,000 L/day

B. Utilities (~30%)

Cooling tower makeup

Boiler feed (WFI generation losses)

HVAC humidification

~1,800 L/day

C. Domestic (~5%)

Staff (say 40–50 people) ~600 L/day

D. QC + Misc (~5%)

Lab washing, testing ~250 L/day

4. Wastewater Generation

Not all input becomes effluent (some losses like evaporation, product incorporation)

Typical:

80–85% becomes wastewater

Effluent: ~5,000 L/day

5. ZLD Strategy

Typical ZLD system:

Segregation

High TDS (CIP, buffers)

Low TDS (cooling tower blowdown, domestic)

Treatment Train

ETP → UF → RO → MEE → ATFD

6. Recycling Assumptions (Practical & Achievable)

Recycled water: ~4,500 L/day recovered

~500 L/day losses (solid waste)

7. Final Water Balance (Daily)

INPUT

Fresh water: 6,000 L/day

OUTPUT (After ZLD)

Recycled: 4,500 L/day (75%)

Losses: 1,500 L/day

Evaporation: ~1,000 L

Sludge/salts: ~500 L

Liquid discharge: ZERO

Stream | Recovery

RO permeate | 70%

MEE condensate | 20%

Losses (salts, sludge) | 10%

ETP and STP water calculation for Plasma Plant (Lifenity)
ETP and STP water calculation for LVP Plant (Lifenity)
Calculation for ETP and STP | Calculation for ETP and STP | Calculation for ETP and STP
Parameter | Capacity | UOM
Plasma processing per year | 1,50,000 | L
Number of days | 365 | Days
Batch size | 1000 | L
Number of batches per year | 150 | Nos
Raw water required per batch | 2,50,000 | L
Number of days per batch | 5 | Days
Raw water required per batch per day | 50,000 | L per batch per day
Raw water required annually | 3,75,00,000 | L per Year
Raw water required per day | 1,02,740 | L per day
Product | Output | UOM
Albumin | 100 | L per batch
IVIG | 150 | L per batch
FIX | 2 | L per batch
FVIII | 0.2 | L per batch
Thrombin | 2 | L per batch
Fibrinogen | 0.5 | L per batch
Total output | 254.7 | L per batch
Product Output per day | 50.94 | L per day per batch
QC | 250 | L per day
Waste water | 1,02,939 | L per day
Evaporation Losses | 20,588 | 20 % of D
Input for ETP | 82,351 | L per day
Number of persons | 506 | No.
Water required per person per day | 35 | L/P/D
Input for STP | 17,710 | L per day
Calculation for ZLD | Calculation for ZLD | Calculation for ZLD | Calculation for ZLD
Parameter | Recovery % | Quantity | UOM
RO permeate (I) | 70% | 57,646 | L per day
MEE condensate (II) | 20% | 16,470 | L per day
Losses (salts, sludge) (III) | 10% | 8,235 | L per day
Final water balance | Final water balance | Final water balance
Input Raw water (D) | 1,02,740 | L per day
Output after ZLD | Output after ZLD | Output after ZLD
Recycled | 74,116 | L per day
Losses | 8,235 | L per day
Recycled | 72 | %
% loss (salts, sludge) | 8 | %
Liquid discharge | 0 | L per day
Calculation for ETP and STP | Calculation for ETP and STP | Calculation for ETP and STP
Parameter | Capacity | UOM
Capacity per year | 3,60,00,000 | Units
Assume number of working days | 365 | Days
Unit Volume | 0.5 | L
Production per day | 49,315 | L per day
CIP or rinse | 15,000 | L per day
Total process water (WFI , Pure steam and PW) requirement | 64,315 | L per day
Raw water required per day | 1,05,000 | L per day
Raw water required annually | 3,83,25,000 | L per Year
Raw water required per day | 1,05,000 | L per day
Product | Output | UOM
Product | 49,315 | L per day
Product Output per day | 49,315 | L per day
QC | 250 | L per day
Waste water | 55,935 | L per day
Evaporation Losses | 11,187 | 20 % of D
Input for ETP | 44,748 | L per day
Number of persons | 400 | No.
Water required per person per day | 35 | L/P/D
Input for STP | 14,000 | L per day
Calculation for ZLD | Calculation for ZLD | Calculation for ZLD
Parameter | Recovery % | Quantity | UOM
RO permeate (I) | 70% | 31,324 | L per day
MEE condensate (II) | 20% | 8,950 | L per day
Losses (salts, sludge) (III) | 10% | 4,475 | L per day
Final water balance | Final water balance | Final water balance
Input Raw water | 1,05,000 | L per day
Output after ZLD | Output after ZLD | Output after ZLD
Recycled | 40,273 | L per day
Losses (| 4,475 | L per day
Recycled (%) | 38 | %
% loss (salts, sludge) | 4 | %
Liquid discharge | 0 | L per day

Reagent Details:

Formula:

Sodium Citrate (M.Wt): 258.07.

Citric Acid (M.Wt): 192.124.

Calcium Chloride (M.Wt): 110.98

Preparation of FN-RR-09 working buffer from Stock buffer (FN-RR-08):

Calculations: For the Preparation of 20mM Sodium Citrate pH 7.4 + 2.5mM Calcium Chloride.

Stock Buffer (FN-RR-08):0.2M Sodium Citrate pH 7.4.

Formula:.

$C1 \times V1 = C2 \times V2$

C1 = concentration of stock

V1 = volume of stock you need to take

C2 = desired (working) concentration

V2 = final total volume of working solution

Applying the formula to prepare the working buffer:

$C1 \times V1 = C2 \times V2$

C1 = 0.2M Sodium Citrate pH 7.4.

V1 = volume of stock you need to take _____ L.

C2 = desired (working) concentration 20mM Sodium Citrate, pH 7.4

V2 = final total volume of working solution _____ L.

$V1 = (C2 \times V2) / C1$.

$C2 = 0.02M$; $V2 = 1.0 L$; $C1 = 0.2M$

$V1 = (0.02 \times 1.0) / 0.2$

$V1 = 0.1L$

Take 0.1 L (FN-RR-08) Stock buffer solution and dilute it with water to make up 1 L.

Calculations: 2.5mM Calcium Chloride.

Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).

Calculation: To prepare 1 L of 2.5mM Calcium Chloride (MW = 110.98 g/mol)

Weight of solute(g)=0.0025×110.98×1=0.277g.

Preparation Method:

Weigh 0.277g of Calcium Chloride and dissolve in water to make a 1 L solution.

Preparation for different volumes:

To prepare _____ L of 0.0025 M Calcium Chloride (MW = 110.98 g/mol).

Weight of solute(g)=0.0025 ×110.98× _____ L= _____ g.

Weigh _____ gm of Calcium Chloride and dissolve in water to make a _____ L solution.

Preparation of working buffer (FN-RR-09):

Take 0.1 L (FN-RR-08) Stock buffer solution and add 0.0025 M Calcium Chloride (0.277 gm). Dissolve it,

and then make up the final volume to 1 L with water.

Check and record the details:

Preparation Details:

Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry

Date

Sodium Citrate | | | |

Citric Acid | | | |

Calcium Chloride | | | |

Stock buffer (FN-RR-08) taken volume(L) | Calcium Chloride added(gm) | pH (7.4 ± 0.1) | Conductivity

(4.85 ± 0.1ms/cm) | Storage Condition | Final volume prepared (L)

| | | RT |

Date of Preparation | Prepared By | Checked By

| |

Reagent Details:

Formula:

Sodium Acetate(M.Wt): 82.03g/mol.

Acetic Acid (M.Wt): 60.05g/ mol

Preparation of 0.6 M Sodium Acetate pH 5.35 buffer

Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).

Calculation: To prepare 1 L of 0.6 M Sodium Acetate (MW = 82.03g/mol).

Weight of solute(g)=0.6×82.03=49.21

Preparation Method:

Weigh 49.21 g of Sodium Acetate and dissolve in water to make a 1 L solution.

Preparation for required volumes:

To prepare _____ of 0.6 M Sodium Acetate (MW = 82.03g/mol).

Weight of solute(g)=0.6 ×82.03x _____L= _____g.

Weigh _____gm of Sodium Acetate and dissolve in water to make a _____ L solution.

Calculations: To prepare the 0.6 M Acetic Acid.

Formula & Calculations:

Formula: Use the formula to calculate the molarity:

Molarity (M) = (Per cent by Weight / 100) * (Density in g/mL) * 1000 / Molar Mass.

Example: Molarity of Glacial Acetic Acid

Molar Mass of CH₃COOH: ~60.05 g/mol

Density: ~1.05 g/mL

Purity: 100%

Acetic Acid (MW = 60.05 g/mol).

Calculation:

Molarity = (100 / 100) * 1.05 g/mL * 1000 mL/L / 60.05 g/mol ≈ 17.5 M.

Calculation: To prepare 1 L of 1.0 M.

Applying the formula to prepare the working buffer:

C₁V₁=C₂V₂

C₁ = 17.5 M acetic acid.

V₁ = volume of stock needed to take.

C₂ = desired (working) concentration 0.6 M acetic acid.

V₂ = final total volume of working solution for one litre.

V₁ = (C₂ x V₂)/C₁.

C₂=0.6 M ; V₂= 1.0 L; C₁=17.5 M

V₁ = (0.6 x 1.0)/17.5

V₁ = 0.034 L

Take 0.034 L of the Stock solution of acetic acid and add it to a volumetric flask, making up the volume to 1 L.

To prepare various volumes:

V₁ = volume of stock needed to take _____L.

C₂ = desired (working) concentration 0.6 M acetic acid.

V₂ = final total volume of working solution _____L.

V₁ = (C₂ x V₂) /17.5.

V₁= _____L.

Take V₁ _____L of the Stock solution of acetic acid and add it to a volumetric flask, making up the final volume _____L.

Procedure for Preparation of 0.6 M Sodium Acetate pH 5.35.

To prepare a pH 5.35 solution, 0.6 M Sodium Acetate solution is titrated with 0.6 M Acetic acid so that the pH attains pH 5.35.

Preparation Details:

Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry

Date

Sodium Acetate | | | |

Acetic Acid | | | |

Final volume prepared | Measured Quantity | pH (5.35±0.1) | Conductivity (35±1ms/cm) | Storage Condition

| Reagent Record

| | | |

Date of Preparation | Prepared By | Checked By

| |

Reagent Details:
Formula & Calculations:
Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).
Calculation: To prepare 1 L of 0.4N N-Acetyl-DL-Tryptophan (MW = 246.26 g/mol)
Weight of solute(g)=0.4×246.26= 9.85g.
Preparation Method:
Weigh 9.85 g of N-Acetyl-DL-Tryptophan and dissolve in water to make a 1 L solution.
Preparation for different volumes:
To prepare _____ of 0.4 N N-Acetyl-DL-Tryptophan (MW = 246.26g/mol).
Weight of solute(g)=0.4×246.26× _____ L= _____ gm
Weigh _____ gm of N-Acetyl-DL-Tryptophan and dissolve in 1.0 N NaOH (1mg/ml of 1.0 N NaOH) and
make up with the water to _____ L solution.
Check and record the details:
Preparation Details:
Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry
Date
N-Acetyl-DL-Tryptophan | | | |
Final volume prepared | Weighed Quantity | pH | Conductivity | Storage Condition:
| | | |
Date of Preparation | Prepared By | Checked By
| |

Plasma
LVP
Calculation for ETP and STP | Calculation for ETP and STP | Calculation for ETP and STP
Parameter | Capacity | UOM
Plasma processing per year | 1,50,000 | L
Number of days | 365 | Days
Batch size | 1000 | L
Number of batches per year | 150 | Nos
Raw water required per batch | 2,50,000 | L
Number of days per batch | 5 | Days
Raw water required per batch per day | 50,000 | L per batch per day
Raw water required annually | 3,75,00,000 | L per Year
Raw water required per day (A) | 1,02,740 | L per day
Product | Output | UOM
Albumin | 100 | L per batch
IVIG | 150 | L per batch
FIX | 2 | L per batch
FVIII | 0.2 | L per batch
Thrombin | 2 | L per batch
Fibrinogen | 0.5 | L per batch
Total output | 254.7 | L per batch
| |
Product Output per day (B) | 50.94 | L per day per batch
QC (C) | 250 | L per day
Waste water (D=A-B+C) | 1,02,939 | L per day
Evaporation Losses (E) | 20,588 | 20 % of D
Input for ETP (F=D-E) | 82,351 | L per day
Number of persons | 506 | No.
Water required per person per day | 35 | L/P/D
Input for STP | 17,710 | L per day
Calculation for ZLD | Calculation for ZLD | Calculation for ZLD | Calculation for ZLD
Parameter | Recovery % | Quantity | UOM
RO permeate (I) | 70% | 57,646 | L per day
MEE condensate (II) | 20% | 16,470 | L per day
Losses (salts, sludge) (III) | 10% | 8,235 | L per day
| | |
Final water balance | | |
Input Raw water (D) | 1,02,740 | L per day |
Output after ZLD | | |
Recycled (G=I+II) | 74,116 | L per day |
Losses (III) | 8,235 | L per day |
Recycled (%) | 72 | % |
% loss (salts, sludge) | 8 | % |
Liquid discharge (D-G-E-III-B) | 0 | L per day |
Calculation for ETP and STP | Calculation for ETP and STP | Calculation for ETP and STP
Parameter | Capacity | UOM
Capacity per year | 3,60,00,000 | Units
Assume number of working days | 365 | Days
Unit Volume | 0.5 | L
Production per day | 49,315 | L per day
CIP or rinse | 15,000 | L per day
Total process water (WFI , Pure steam and PW) requirement | 64,315 | L per day
Raw water required per day | 1,05,000 | L per day
Raw water required annually | 3,83,25,000 | L per Year
Raw water required per day (A) | 1,05,000 | L per day
Product | Output | UOM
Product | 49,315 | L per day
Product Output per day (B) | 49,315 | L per day
QC (C) | 250 | L per day
Waste water (D=A-B+C) | 55,935 | L per day
Evaporation Losses (E) | 11,187 | 20 % of D
Input for ETP (F=D-E) | 44,748 | L per day
Number of persons | 400 | No.
Water required per person per day | 35 | L/P/D
Input for STP | 14,000 | L per day
Calculation for ZLD | Calculation for ZLD | Calculation for ZLD | Calculation for ZLD
Parameter | Recovery % | Quantity | UOM
RO permeate (I) | 70% | 31,324 | L per day
MEE condensate (II) | 20% | 8,950 | L per day
Losses (salts, sludge) (III) | 10% | 4,475 | L per day
| | |
Final water balance | | |
Input Raw water (D) | 1,05,000 | L per day |
Output after ZLD | | |
Recycled (G=I+II) | 40,273 | L per day |
Losses (III) | 4,475 | L per day |
Recycled (%) | 38 | % |
% loss (salts, sludge) | 4 | % |
Liquid discharge (D-G-E-III-B) | 0 | L per day |

Reagent Details:
Formula & Calculations:
Formula: $M_1 \times V_1 = M_2 \times V_2$.
Calculation: To prepare 1 L of 0.1M Tris from 1M tris stock solution calculate quantity required of stock solution as below:
 $1M \text{ tris stock solution required quantity} = 0.1 \times 1 / 1 = 0.1L$
Preparation Method:
To prepare 0.1M Tris 1L solution from 1M Tris stock solution formula is $M_1(\text{Molarity of stock}) \times V_1(\text{volume of stock}) = M_2(\text{Molarity of working}) \times V_2(\text{volume of working})$.
Take 0.1L of 1M tris solution and make a 1 L solution.
Preparation for different volumes:
To prepare _____ of 0.1M Tris
Take 0.1L x _____ L = _____ L of 1M Tris
Make Tris solution to _____ L solution.
Check and record the details:
Preparation Details:
Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry Date
|||
Final volume prepared | Weighed Quantity | pH (7.8±0.2) | Conductivity (4.0±1.0 ms/cm) | Storage Condition:
||| RT
Date of Preparation | Prepared By | Checked By
||

MAHARASHTRA POLLUTION CONTROL BOARD
Tel: 24010706/24010437 Kalpataru Point, 2nd, 3rd
Fax: 24023516 and 4th Floor, Opp. Cine
Website: <http://mpcb.gov.in> Planet Cinema, Near Sion
Email: cac-cell@mpcb.gov.in Circle, Sion (E),
Mumbai-400022

RED/L.S.I (R14) Date: 05/03/2025

No:- Format1.0/CC/UAN No.MPCB-
CONSENT-0000157866/CR/2503000490

To,
M/s. Aurangabad Industrial Township Ltd., (CETP),
(M/s. Delhi Mumbai Industrial Corridor Development
Corporation Ltd.,) .
Plot No. 1, Sector 9, AURIC, Shendra,
Your Service is Our Duty
Aurangabad.

Sub: Grant renewal of consent to operate under the provisions of Water
(P & CP) Act, 1974, Air (P & CP) Act, 1981 and Bio-Medical Waste
Management Rules, 2016 as amended and Hazardous Waste (M &
TM) Rules, 2016.

Ref: 1. Renewal of Consent to Operate granted vide No- Format1.0/CC/UAN
No.MPCB-CONSENT-0000126745/CR/2211000292 Date: 04/11/2022 valid
up to 31/12/2022
2. Minutes of 14th Consent Committee Meeting (Agenda C) of 2024-2025
held on 17/01/2025

Your application No.MPCB-CONSENT-0000157866 Dated 31.12.2022
For: grant of Consent to Operate under Section 26 of the Water (Prevention & Control of
Pollution) Act, 1974 & under Section 21 of the Air (Prevention & Control of Pollution) Act,
1981 and Authorization under Rule 6 and Rule 18(7) of the Hazardous & Other Wastes
(Management & Transboundary Movement) Rules 2016 is considered and the consent is
hereby granted subject to the following terms and conditions and as detailed in the
schedule I, II, III & IV annexed to this order:

1. The Renewal of consent to Operate is granted for period upto 31.12.2026
2. The capital investment of the project is Rs.60.3823 Crs. (As per C.A
Certificate submitted by industry Rs. 75.07 Crs (Previous Rs. 60.38 Crs))
3. Consent is valid for:
Sr Maximum
Treatment Facility UOM
No Quantity
Common Effluent Treatment Plant – The daily quantity of
13 MLD
industrial effluent to be treated shall not exceed

Common Effluent Treatment Plant for Mega Industrial Park in Shendra, District Aurangabad named as
Aurangabad Industrial City (Auric), Shendra, under the Delhi Mumbai Industrial Corridor (DMIC) Page 1 of
12
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4. Conditions under Water (P&CP), 1974 Act for discharge of effluent:

Sr Permitted
Description Standards to Disposal Path
No (in CMD)
The treated trade effluent
shall be 100% utilized for
As per Schedule-
1. Trade effluent 3000 industrial purposes,
I
flushing, HVAC make up,
gardening purposes, etc.
2. Domestic effluent 0 As per Schedule- Not Applicable
I

5. Conditions under Air (P & CP) Act, 1981 for air emissions:

Sr Stack Description of stack / Number of Standards to be
No. No. source Stack achieved

- 1 S-1 DG Set-125 KVA 01 As per Schedule -II
- 2 S-2 DG Set-320 KVA 01 As per Schedule -II
- 3 S-3 DG Set-500 KVA 01 As per Schedule -II
- 4 S-4 DG Set-720 KVA 01 As per Schedule -II

6. Non-Hazardous Wastes:

Sr No Type of Waste Quantity UoM Treatment Disposal
1 NA 0 --NA-- 0 NA

7. Conditions under Hazardous & Other Wastes (M & T M) Rules 2016 for
Collection, Segregation, Storage, Transportation, Treatment and Disposal of
hazardous waste:

Sr No Category No./ Type Quantity UoM Treatment Disposal
35.3 Chemical sludge from waste

12 MT/Day Landfill CHWTSDF
water treatment

8. The Board reserves the right to review, amend, suspend, revoke this consent and the
same shall be binding on the industry.

9. This consent should not be construed as exemption from obtaining necessary NOC/
permission from any other Government authorities.

10. PP shall comply with the conditions stipulated in EC vide dated 18.06.2015 & its
amendment dated 26.02.2016.

11. PP shall ensure 100% recycle of treated effluent in the member industrial units & shall
not discharge outside the premise in any case.

12. PP shall study feasibility of connection of effluent generating industries located at Shendra MIDC to CETP by pipe conveying systems.

Common Effluent Treatment Plant for Mega Industrial Park in Shendra, District Aurangabad named as Aurangabad Industrial City (Auric), Shendra, under the Delhi Mumbai Industrial Corridor (DMIC) Page 2 of 12

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13. The applicant shall make an application for renewal of consent 60 days prior to date of expiry of the consent.

14. The industry shall create an Environment Cell by appointing an Environmental Engineer OR Expert for looking after day-to-day activities related to Environment OR Pollution control.

This consent is issued on the basis of information/documents submitted by the Applicant/Project Proponent, if it has been observed that the information submitted by the Applicant/Project Proponent is false, misleading or fraudulent, the Board reserves its right to revoke the consent & further legal action will be initiated against the Applicant/Project Proponent.

Received Consent fee of -

Sr.No Amount(Rs.) Transaction/DR.No. Date Transaction Type

1 125000.00 MPCB-DR-25566 15/04/2024 NEFT

2 347603.00 MPCB-DR-32419 05/03/2025 RTGS

Copy to:

1. Regional Officer, MPCB, Chhatrapati Sambhajnagar and Sub-Regional Officer, MPCB, Chhatrapati Sambhajnagar I

- They are directed to ensure the compliance of the consent conditions.

2. Chief Accounts Officer, MPCB, Sion, Mumbai

3. CAC Desk - for record & website updating purposes.

Common Effluent Treatment Plant for Mega Industrial Park in Shendra, District Aurangabad named as Aurangabad Industrial City (Auric), Shendra, under the Delhi Mumbai Industrial Corridor (DMIC) Page 3 of 12

Project./CR/UAN No.MPCB-CONSENT-0000157866/Indus-Id.4709 (05-03-2025 10:16:58 pm) /QMS.PO6_F02/00

SCHEDULE-I

Terms & conditions for compliance of Water Pollution Control:

1. A] As per your application, you have provided Common Effluent Treatment Plant of capacity 3 MLD to treat the effluent of 3.0 MLD out of proposed CETP capacity 12 MLD consisting of receiving chamber, coarse screen, wet well, inlet chamber, fine screen, vortex grit chamber, equalization tank, flash mixing tank, flocculation tank, primary clarifier, ultra-fine screen, pre-anoxic tank, aeration tank, post-anoxic tank, membrane tank, chlorine contact tank, treated effluent outfall chamber, etc.

B] The Consent is valid for collection, storage and treatment of Industrial and Domestic Effluent conforming to the inlet standards specified here under:-

Limiting concentration not to exceed in

Sr.No Parameters

mg/l, except for pH

(1) pH 5.5 to 9.0

(2) Total Dissolved Solids 2100

(3) Suspended solids 600

(4) COD 1000

(5) BOD (3 days 27 °C) 700

(6) Total Nitrogen 80

(7) Total phosphorus 10

(8) Chlorides 600

(9) Sulphate 1000

(10) Total oil (free + emulsified) 350

(11) (a) Free Oil 100

(12) (b) Emulsified oil 250

(13) Arsenic (as As) 0.2

(14) Mercury (as Hg) 0.01

(15) Cadmium 1.0

(16) Selenium (as Se) 0.1

(17) Iron 5.0

(18) Lead 1.0

(19) Copper 3.0

(20) Zinc 15.0

(21) Nickel 3.0

(22) Chromium 1.0

(23) Hexavalent Chromium 0.1

(24) Cyanide 0.1

(25) Colour 150 Platinum Cobalt Unit

Common Effluent Treatment Plant for Mega Industrial Park in Shendra, District Aurangabad named as Aurangabad Industrial City (Auric), Shendra, under the Delhi Mumbai Industrial Corridor (DMIC) Page 4 of 12

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When the CETP is complying with discharge standards then, In case of SSI Unit, BOD of Maximum of 700 mg/l and COD of maximum 1000 mg/l will be allowed to inlet of CETP.

All other specific parameters including colour shall be complied before disposal to CETP. In case of other, primary and secondary treatment is required to meet consented standards before disposal to CETP.

i) All Large & Medium Scale Units irrespective of the quantity of effluent will have to achieve the standards as prescribed in the letter of Consent issued to them individually under the Water (P & CP) Act 1974, Air (P & CP) Act 1981, Hazardous Waste (M&H) Amendment thereto before discharging the effluent into CETP

C] The applicant shall operate the Common Effluent Treatment Plant (CETP) to treat the trade effluent so as to achieve the following standards prescribed by the

Board of under EP Act, 1986 & rules made there under from time to time, whichever is stringent. PP shall achieve the following standards from Sr. No. 01 to 06 for recycling back to member industry & from Sr. No. 07 to 31 for discharge on land for gardening.

Common Effluent Treatment Plant for Mega Industrial Park in Shendra, District Aurangabad named as Aurangabad Industrial City (Auric), Shendra, under the Delhi Mumbai Industrial Corridor (DMIC) Page 5 of 12

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Sr.No Parameters Standards (mg/l)

- 1 pH 6.0 to 9.0
- 2 BOD 3 Days 27 OC 5.0
- 3 COD 30
- 4 Total suspended Solids (TSS) 100
- 5 Fixed Dissolved Solids (FDS) 750
- 6 Oil & Grease 0.01
- 7 pH 6.0 to 9.0
- 8 BOD 3 Days 27 OC 30
- 9 COD 250
- 10 Ammoniacal Nitrogen 50
- 11 Nitrate-Nitrogen 10
- 12 Total Kjeldahl Nitrogen 50
- 13 Chlorides 1000
- 14 Sulphate (as SO₄) 1000
- 15 Fluoride 2
- 16 Sulphides (as S) 2
- 17 Phenolic Compound (as C₆H₅OH) 1
- 18 Total Residual Chlorine 1
- 19 Zinc 5
- 20 Iron 3
- 21 Copper 3
- 22 Trivalent Chromium 1.1
- 23 Manganese 2
- 24 Nickel 3
- 25 Arsenic 0.2
- 26 Cyanide (as CN) 0.2
- 27 Vanadium 0.2
- 28 Lead 0.1
- 29 Hexavalent Chromium 0.1
- 30 Selenium 0.05
- 31 Cadmium 0.05
- 32 Mercury 0.01
- 33 Phosphate as P 5.0
- 34 Colour 150 Platinum Cobalt Unit

Common Effluent Treatment Plant for Mega Industrial Park in Shendra, District Aurangabad named as Aurangabad Industrial City (Auric), Shendra, under the Delhi Mumbai Industrial Corridor (DMIC) Page 6 of 12

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Note:-

- i. All efforts should be made to remove colour and unpleasant odour as per as possible.
- ii. If the CETP is not able to achieve the outlet parameters, then all the members and the said Society would be individually and jointly responsible and liable for legal actions under the provisions of sections 47 of the Water (Prevention & Control of Pollution) Act, 1974.

D] The heavy metals if any shall be treated by providing separate treatment facility & the standard prescribed for metals shall meet at the outlet of metal treatment facility.

E] i] The treated trade effluent shall be 100% utilized for industrial purpose, flushing, HVAC make up, gardening etc. to the tune of 3.0 MLD generated in 1st phase as per your proposal. ii] In no case effluent shall find its way outside the designated area at any time. PP shall provide flow meter to ensure recycling of treated effluent & shall maintain the record with data logging system. iii] You shall demark the recycling effluent pipeline by colour coding for identification purpose & also shall demark the storage tanks used for the same purpose. iv] Online monitoring system shall be provided at the outlet of CETP for critical parameters such as BOD, COD, Chromium, O & G, Phenolic compound, TSS, flow & pH.

2. A] Generation - Nil

B] Treatment - NA

C] Disposal - NA

3. The Board reserves its rights to review plans, specifications or other data relating to plant setup for the treatment of waterworks for the purification thereof & the system for the disposal of sewage or trade effluent or in connection with the grant of any consent conditions. The Applicant shall obtain prior consent of the Board to take steps to establish the unit or establish any treatment and disposal system or an extension or addition thereto.

4. The industry shall ensure replacement of pollution control system or its parts after expiry of its expected life as defined by manufacturer so as to ensure the compliance of standards and safety of the operation thereof.

5. The Applicant shall comply with the provisions of the Water (Prevention & Control of Pollution) Act, 1974 and as amended, by installing water meters and other provisions as contained in the said act:

Water consumption quantity

Sr. No. Purpose for water consumed

(CMD)

Industrial Cooling, spraying in mine pits

1. 0.00

or boiler feed

2. Domestic purpose 2.50
Processing whereby water gets polluted
3. 0.00
& pollutants are easily biodegradable
Processing whereby water gets polluted
4. & pollutants are not easily biodegradable and are toxic 0.00
5. Gardening 0

Common Effluent Treatment Plant for Mega Industrial Park in Shendra, District Aurangabad named as Aurangabad Industrial City (Auric), Shendra, under the Delhi Mumbai Industrial Corridor (DMIC) Page 7 of 12

Project./CR/UAN No.MPCB-CONSENT-0000157866/Indus-Id.4709 (05-03-2025 10:16:58 pm) /QMS.PO6_F02/00

6. The Applicant shall provide Specific Water Pollution control system as per the conditions of EP Act, 1986 and rule made there under from time to time/ Environmental Clearance/ CREP guidelines.

SCHEDULE-II

Terms & conditions for compliance of Air Pollution Control:

1. As per your application, you have provided the Air pollution control (APC) system and erected following stack (s) to observe the following fuel pattern:

APC System Stack Type Sulphur

Stack

Source provided/pro Height(in of Content(in Pollutant Standard No.

posed mtr) Fuel %)

DG Set HSD

Acoustic 12.6

S-1 (125 3.00 26.25 1.0 SO2

Enclosure Kg/Day

KVA) Kg/Hr

DG Set HSD

Acoustic 32.25

S-2 (320 3.50 67.20 1.0 SO2

Enclosure Kg/Day

KVA) Kg/Hr

DG set HSD

Acoustic 50.4

S-3 (500 5.00 105 1.0 SO2

Enclosure Kg/Day

KVA) Kg/Hr

DG set HSD

Acoustic 75.6

S-4 (720 5.50 157.5 1.0 SO2

Enclosure Kg/Day

KVA) Kg/Hr

PP shall provide backup arrangement for proper supply towards smooth & continuous operation of CETP.

2. The Applicant shall provide Specific Air Pollution control equipments as per the conditions of EP Act, 1986 and rule made there under from time to time/ Environmental Clearance / CREP guidelines.

3. The Applicant shall obtain necessary prior permission for providing additional control equipment with necessary specifications and operation thereof or alteration or replacement/alteration well before its life come to an end or erection of new pollution control equipment.

4. The Board reserves its rights to vary all or any of the condition in the consent, if due to any technological improvement or otherwise such variation (including the change of any control equipment, other in whole or in part is necessary).

Common Effluent Treatment Plant for Mega Industrial Park in Shendra, District Aurangabad named as Aurangabad Industrial City (Auric), Shendra, under the Delhi Mumbai Industrial Corridor (DMIC) Page 8 of 12

Project./CR/UAN No.MPCB-CONSENT-0000157866/Indus-Id.4709 (05-03-2025 10:16:58 pm) /QMS.PO6_F02/00

SCHEDULE-III

Details of Bank Guarantees:

Consent

Sr.

(C2E/ Amt of BG Submission Compliance Validity

No Purpose of BG

C2O Imposed Period Period Date

.

/C2R)

C2R (Part- Extend existing Towards O & M

1 Rs. 5.0 Lakh 31.12.2026 30.04.2027

I) BG of CETP

Towards

C2R (Part- Extend existing compliance of

2 Rs. 5.0 Lakh 31.12.2026 30.04.2027

I) BG Consent

Conditions

**Existing BG obtained for above purpose if any, may be extended for period of validity as above.

If the above Bank Guarantee is not submitted within stipulated period, then 12% interest will be levied as a penalty as per circular dtd 29/02/2024 No.

BO/MPCB/AS(T)/Circular/B-240229FTS0122

BG Forfeiture History

Amount of Amount of Reason of

Consent Submission Purpose

Srno. BG BG BG

(C2E/C2O/C2R) Period of BG

imposed Forfeiture Forfeiture

NA

BG Return details

Srno. Consent (C2E/C2O/C2R) BG imposed Purpose of BG Amount of BG Returned
NA

Common Effluent Treatment Plant for Mega Industrial Park in Shendra, District Aurangabad named as Aurangabad Industrial City (Auric), Shendra, under the Delhi Mumbai Industrial Corridor (DMIC) Page 9 of 12

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General Conditions:

1. The Energy source for lighting purpose shall preferably be LED based
2. The PP shall harvest rainwater from roof tops of the buildings and storm water drains to recharge the ground water and utilize the same for different industrial applications within the plant
3. Conditions for D.G. Set
 - a) Noise from the D.G. Set should be controlled by providing an acoustic enclosure or by treating the room acoustically.
 - b) Industry should provide acoustic enclosure for control of noise. The acoustic enclosure/ acoustic treatment of the room should be designed for minimum 25 dB (A) insertion loss or for meeting the ambient noise standards, whichever is on higher side. A suitable exhaust muffler with insertion loss of 25 dB (A) shall also be provided. The measurement of insertion loss will be done at different points at 0.5 meters from acoustic enclosure/room and then average.
 - c) Industry should make efforts to bring down noise level due to DG set, outside industrial premises, within ambient noise requirements by proper siting and control measures.
 - d) Installation of DG Set must be strictly in compliance with recommendations of DG Set manufacturer.
 - e) A proper routine and preventive maintenance procedure for DG set should be set and followed in consultation with the DG manufacturer which would help to prevent noise levels of DG set from deteriorating with use.
 - f) D.G. Set shall be operated only in case of power failure.
 - g) The applicant should not cause any nuisance in the surrounding area due to operation of D.G. Set.
 - h) The applicant shall comply with the notification of MoEFCC, India on Environment (Protection) second Amendment Rules vide GSR 371(E) dated 17.05.2002 and its amendments regarding noise limit for generator sets run with diesel.
4. The applicant shall maintain good housekeeping.
5. The non-hazardous solid waste arising in the factory premises, sweepings, etc. be disposed of scientifically so as not to cause any nuisance / pollution. The applicant shall take necessary permissions from civic authorities for disposal of solid waste.
6. The applicant shall not change or alter the quantity, quality, the rate of discharge, temperature or the mode of the effluent/emissions or hazardous wastes or control equipments provided for without previous written permission of the Board. The industry will not carry out any activity, for which this consent has not been granted/without prior consent of the Board.
7. The industry shall ensure that fugitive emissions from the activity are controlled so as to maintain clean and safe environment in and around the factory premises.
8. The industry shall submit quarterly statement in respect of industries obligation towards consent and pollution control compliance's duly supported with documentary evidences (format can be downloaded from MPCB official site).
9. The industry shall submit official e-mail address and any change will be duly informed to the MPCB.
10. The industry shall achieve the National Ambient Air Quality standards prescribed vide Government of India, Notification No. B-29016/20/90/PCI-L dated. 18.11.2009 as amended.

Common Effluent Treatment Plant for Mega Industrial Park in Shendra, District Aurangabad named as Aurangabad Industrial City (Auric), Shendra, under the Delhi Mumbai Industrial Corridor (DMIC) Page 10 of 12

Project./CR/UAN No.MPCB-CONSENT-0000157866/Indus-Id.4709 (05-03-2025 10:16:58 pm) /QMS.PO6_F02/00

11. The Board reserves its rights to review plans, specifications or other data relating to plant setup for the treatment of waterworks for the purification thereof & the system for the disposal of sewage or trade effluent or in connection with the grant of any consent conditions. The Applicant shall obtain prior consent of the Board to take steps to establish the unit or establish any treatment and disposal system or an extension or addition thereto.
12. The industry shall ensure replacement of pollution control system or its parts after expiry of its expected life as defined by manufacturer so as to ensure the compliance of standards and safety of the operation thereof.
13. The PP shall provide personal protection equipment as per norms of Factory Act
14. Industry should monitor effluent quality, stack emissions and ambient air quality monthly/quarterly.
15. Whenever due to any accident or other unforeseen act or even, such emissions occur or is apprehended to occur in excess of standards laid down, such information shall be forthwith Reported to Board, concerned Police Station, Office of Directorate of Health Services, Department of Explosives, Inspectorate of Factories and Local Body. In case of failure of pollution control equipments, the production process connected to it shall be stopped.
16. The applicant shall provide an alternate electric power source sufficient to operate all pollution control facilities installed to maintain compliance with the terms and conditions of the consent. In the absence, the applicant shall stop, reduce or otherwise, control production to abide by terms and conditions of this consent.
17. The industry shall recycle/reprocess/reuse/recover Hazardous Waste as per the provision contain in the Hazardous and Other Wastes (M & TM) Rules 2016, which can be recycled /processed /reused /recovered and only waste which has to be incinerated shall go to incineration and waste which can be used for land filling and cannot be recycled/reprocessed etc. should go for that purpose, in order to reduce load on incineration and landfill site/environment.
18. An inspection book shall be opened and made available to the Board's officers during their visit to the applicant.
19. Industry shall strictly comply with the Water (P&CP) Act, 1974, Air (P&CP) Act, 1981 and Environmental Protection Act, 1986 and industry specific standard under EP Rules 1986 which are available on MPCB website (www.mpcb.gov.in).
20. Separate drainage system shall be provided for collection of trade and sewage

e■uents. Terminal manholes shall be provided at the end of the collection system with arrangement for measuring the ■ow. No e■uent shall be admitted in the pipes/sewers downstream of the terminal manholes. No e■uent shall ■nd its way other than in designed and provided collection system.

21. Neither storm water nor discharge from other premises shall be allowed to mix with the e■uents from the factory.

22. The industry should not cause any nuisance in surrounding area.

23. The industry shall take adequate measures for control of noise levels from its own sources within the premises so as to maintain ambient air quality standard in respect of noise to less than 75 dB (A) during day time and 70 dB (A) during night time. Day time is reckoned in between 6 a.m. and 10 p.m. and night time is reckoned between 10 p.m. and 6 a.m.

Common E■uent Treatment Plant for Mega Industrial Park in Shendra, District Aurangabad named as Aurangabad Industrial City (Auric), Shendra, under the Delhi Mumbai Industrial Corridor (DMIC) Page 11 of 12

Project./CR/UAN No.MPCB-CONSENT-0000157866/Indus-Id.4709 (05-03-2025 10:16:58 pm) /QMS.PO6_F02/00

24. The industry shall create the Environmental Cell by appointing an Environmental Engineer, Chemist and Agriculture expert for looking after day to day activities related to Environment and irrigation ■eld where treated e■uent is used for irrigation.

25. The applicant shall provide ports in the chimney(s) and facilities such as ladder, platform etc. for monitoring the air emissions and the same shall be open for inspection to/and for use of the Board's Sta■. The chimney(s) vents attached to various sources of emission shall be designated by numbers such as S-1, S-2, etc. and these shall be painted/ displayed to facilitate identi■cation.

26. The industry should comply with the Hazardous and Other Wastes (M & TM) Rules, 2016 and submit the Annual Returns as per Rule 6(5) & 20(2) of Hazardous and Other Wastes (M & TM) Rules, 2016 for the preceding year April to March in Form-IV by 30th June of every year.

27. The applicant shall install a separate meter showing the consumption of energy for operation of domestic and industrial e■uent treatment plants and air pollution control system. A register showing consumption of chemicals used for treatment shall be maintained.

28. The applicant shall bring minimum 33% of the available open land under green coverage/ plantation. The applicant shall submit a yearly statement by 30th September every year on available open plot area, number of trees surviving as on 31st March of the year and number of trees planted by September end.

29. The Board reserves its rights to review plans, speci■ications or other data relating to plant setup for the treatment of waterworks for the puri■ication thereof & the system for the disposal of sewage or trade e■uent or in connection with the grant of any consent conditions.

30. The ■rm shall submit to this o■ce, the 30th day of September every year, the Environment Statement Report for the ■nancial year ending 31st March in the prescribed FORM-V as per the provisions of Rule 14 of the Environment (Protection) (second Amendment) Rules, 1992.

31. The Applicant shall obtain necessary prior permission for providing additional control equipment with necessary speci■ications and operation thereof or alteration or replacement/alteration well before its life come to an end or erection of new pollution control equipment.

32. The Board reserves its rights to vary all or any of the condition in the consent, if due to any technological improvement or otherwise such variation (including the change of any control equipment, other in whole or in part is necessary).

33. The applicant shall provide facility for collection of environmental samples and samples of trade and sewage e■uents, air emissions and hazardous waste to the Board sta■ at the terminal or designated points and shall pay to the Board for the services rendered in this behalf.

This certi■icate is digitally & electronically signed.

Common E■uent Treatment Plant for Mega Industrial Park in Shendra, District Aurangabad named as Aurangabad Industrial City (Auric), Shendra, under the Delhi Mumbai Industrial Corridor (DMIC) Page 12 of 12

Project./CR/UAN No.MPCB-CONSENT-0000157866/Indus-Id.4709 (05-03-2025 10:16:58 pm) /QMS.PO6_F02/00

Reagent Details:
Formula & Calculations:
Formula: Use the formula to calculate the molarity:
 $Molarity (M) = (Per\ cent\ by\ Weight / 100) * (Density\ in\ g/mL) * 1000 / Molar\ Mass.$
Example: Molarity of Glacial Acetic Acid
Molar Mass of CH₃COOH: ~60.05 g/mol
Density: ~1.05 g/mL
Purity: 100%
Acetic Acid (MW = 60.05 g/mol).
Calculation:
 $Molarity = (100 / 100) * 1.05\ g/mL * 1000\ mL/L / 60.05\ g/mol \approx 17.5\ M.$
Calculation: To prepare 1 L of 0.1M.
Applying the formula to prepare the working buffer:
 $C1 \cdot V1 = C2 \cdot V2$
 $C1 = 17.5\ M$ acetic acid.
 $V1$ = volume of stock needed to take.
 $C2$ = desired (working) concentration 0.1M acetic acid.
 $V2$ = final total volume of working solution for one liter.
 $V1 = (C2 \times V2) / C1.$
 $C2=0.1\ M; V2= 1.0\ L; C1=17.5\ M$
 $V1 = (0.1 \times 1.0) / 17.5$
 $V1 =$ Take 0.0057 L of the Stock solution of acetic acid and add it to a volumetric flask, making up the volume to 1 L.
To prepare various volumes:
 $V1$ = volume of stock needed to take _____ L.
 $C2$ = desired (working) concentration 0.1M acetic acid.
 $V2$ = final total volume of working solution _____ L.
 $V1 = (C2 \times V2) / 17.5.$
 $V1=$ _____ L.
Take $V1$ _____ L of the Stock solution of acetic acid and add it to a volumetric flask, making up the final volume _____ L.
Check and record the details:
Preparation Details:
Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry Date
| | | |
Final volume prepared | Measured Quantity | pH (2.8±0.2) | Conductivity (1.8±0.1ms/cm) | Storage Condition | Reagent Record
| | | RT |
Date of Preparation | Prepared By | Checked By
| |

Reagent Details:
Formula & Calculations:
Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).
Calculation: To prepare 1 L of 1M Tris pH 7.80 (MW = 121.14g/mol)
Weight of solute(g)=1×121.14=121.14g.
Preparation Method:
Weigh 121.14 g of Tris base and dissolve in water to make a 1 L solution.
Preparation for different volumes:
To prepare _____ of 1M Tris base(MW = 121.14g/mol).
Weight of solute(g)=1×121.14x _____L= _____gm
Weigh _____gm of Tris base and dissolve in water to make a _____ L solution.
Check and record the details:
Adjust the pH upto 7.80 using concentrated HCl and record final pH and conductivity.
Preparation Details:
Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry
Date
| | | |
Final volume prepared | Weighed Quantity | pH(7.8±0.2) | Conductivity (56±2 mS/cm) | Storage Condition:
| | | RT
Date of Preparation | Prepared By | Checked By
| |

Technical Information # BB-LN/IND/003

Date- 30.05.2026
List of Documents

S/No Title
1 List of raw materials for LVP
2 List of raw materials for Plasma

Contact Details:

1. Dr. Ganesh Kumraj: ganesh@biobridge.in

www.biobridge.in Confidential Page 1 of 1

BioBridge Healthcare Solutions Private Limited
Address: # 40, 4th Floor, Block B, Chandigarh City Center (CCC), Zirakpur - 140603, India

Query: In Utility RDS; transformer load looks high?

Response: jameel to ask Rao

Query: In Utility RDS summary sheet; Dual fuel boiler (3000Kg/Hr) and normal boiler (1000 Kg/H) is mentioned but capacity for both is mentioned different, so what capacity is exactly required?

Response: jameel to ask Rao

Query: Black steam is considered for PSG calculation?

Response: Yes

Query: Common load like electric bulb load not considered in RDS as it is mentioned as 0KW

Response: We have considered all equipment load and Lux value is mentioned for each area, please considered electric bulb etc load on standard basis. Per sq feet yogesh

Query: For CNC area temperature not given?

Response: RDS revised

Query: Differential Pressure difference between adjacent areas are always 5 Pascal ,may be it should be min.10 Pascal ,PI check

Response: RDS revised

Query: Where you use common AHU and where you use individual AHU

Response: AHU zoning layout is pending and in process.

Query: AHU type - CHW OR Dx system

Response: CHW

Query: ACPH of Area is not mentioned in RDS.

Response: RDS revised

Query: AHU Zoning with AHU No.

Response: AHU zoning layout is pending and in process.

Reagent Details:

Formula:

Sodium Chloride (M.Wt):58.44

Glycine(M.Wt): 75.07.

Glycerol (M.Wt):92.094 g/mol

Calcium Chloride (M.Wt): 110.98

0.1% Tween 80

Preparation of FE-RR-06 buffer.

Preparation of 0.94M Sodium chloride +0.1M Glycine +12% Glycerol + 2.6mM Calcium Chloride +0.1% Tween-80,

pH- 6.0

Calculations: To prepare the 0.94M Sodium Chloride.

Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).

Calculation: To prepare 1 L of 0.94M Sodium Chloride (MW = 58.44 g/mol)

Weight of solute(g)=0.94×58.44×1=54.93g.

Preparation Method:

Weigh 54.93 g of Sodium Chloride and dissolve in water to make a 1 L solution.

Preparation for different volumes:

To prepare _____ L of 0.94M Sodium Chloride (MW = 58.44 g/mol).

Weight of solute(g)=0.94×58.44 x _____ L= _____ g.

Weigh _____ gm of Sodium Chloride and dissolve in water to make a _____ L solution.

Calculations: To prepare the 2.6mM Calcium Chloride.

Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).

Calculation: To prepare 1 L of 2.6 mM Calcium Chloride (MW = 110.98g/mol)

Weight of solute(g)=0.0026×110.98 x1=0.288g.

Preparation Method:

Weigh 0.288g of Calcium Chloride and dissolve in water to make a 1 L solution.

Preparation for different volumes:

To prepare _____ L of 0.0026mM Calcium Chloride (MW =110.98 g/mol).

Weight of solute(g)=0.0026 ×110.98x _____ L= _____ g.

Weigh _____ gm of Calcium Chloride and dissolve in water to make a _____ L solution.

Calculations: To prepare the 0.1mM Glycine.

Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).

Calculation: To prepare 1 L of 0.1M Glycine (MW = 75.07g/mol)

Weight of solute(g)=0.1×75.07 x1=7.507g.

Preparation Method:

Weigh 7.507g of Glycine and dissolve in water to make a 1 L solution.

Preparation for different volumes:

To prepare _____ L of 0.1M Glycine (MW =75.07 g/mol).

Weight of solute(g)=0.1 ×75.07x _____ L= _____ g.

Weigh _____ gm of Glycine and dissolve in water to make a _____ L solution.

Procedure

Calculation: to prepare the 12% Glycerol:

For 1000 mL of 12% solution:

12% =120g per 1000 mL=1200 mg

Since glycerol is a liquid, you can weigh it directly or use its density (~

1.261 g/mL) to measure volume:

Approx. 100 µL of Glycerol = 1.0 g

Add Glycerol to a beaker or flask containing ~800 mL of water.

Mix thoroughly:

Use a magnetic stirrer or vortex until fully dissolved. Glycerol is viscous, so mixing may take a few minutes.

Adjust volume:

After complete dissolution, bring the final volume up to 1000 mL with water.

Procedure

Calculation: to prepare the 0.1% Tween 80:

For 1000 mL of 0.1% solution:

0.1% =0.1g per 1000 mL=1000 mg

Since Tween 80 is a liquid, you can weigh it directly or use its density

(~1.06 g/mL) to measure volume:

Approx. 100 µL of Tween 80 = 1.0 g

Add Tween 80 to a beaker or flask containing ~800 mL of water.

Mix thoroughly:

Use a magnetic stirrer or vortex until fully dissolved. Tween 80 is viscous, so mixing may take a few minutes.

Adjust volume:

After complete dissolution, bring the final volume up to 1000 mL with water.

Procedure for Preparation of FE-RR-06 buffer.

To prepare a 1 L solution, To this, add the 0.94M sodium chloride and dissolve it and make up the volume

with water to 800ml, and then to this add 0.1M Glycine, then to this add 0.0026 M Calcium chloride, add

the 12% Glycerol then mix the solution till it dissolves, to this add the 0.1% Tween 80 then mix the

solution till it dissolves and check the pH; if the pH is less than 6.0 and need to adjust pH using HCl

or NaOH after dissolving all components and make up the final volume to 1L.

Check and record the details:

Preparation Details:

Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry

Date

Sodium Chloride | | | |

Glycine | | | |

Glycerol | | | |

Calcium Chloride | | | |

Tween 80 | | | |

Final volume prepared | pH (6.0 ± 0.1) | Conductivity (100 ±5.0ms/cm) | Storage Condition

| | | RT

Date of Preparation | Prepared By | Checked By

| |

[SHEET: LVP]

||||| Calculation for ETP and STP ||||| Calculation for ZLD |||
 | Water generation ||| | Parameter | Capacity | UOM || Parameter | Receovery % | Quantity | UOM
 | Raw water (100%) | 105000 | L || Capacity per year | 36000000 | Units || RO permeate (I) | 0.7 |
 31323.561643835616 | L per day
 | Soft water (95%) | 99750 | L || Assume number of working days | 365 | Days || MEE condensate (II) |
 0.2 | 8949.589041095891 | L per day
 | Purified Water (65% of Soft water) | 64837.5 | L || Unit Volume | 0.5 | L || Losses (salts, sludge)
 (III) | 0.1 | 4474.7945205479455 | L per day
 | Water For Injection (90% of PW) | 58353.75 | L || Production per day | 49315.068493150684 | L per day
				CIP or rinse	15000	L per day		Final water balance		
				Total process water (WFI , Pure steam and PW) requirement	64315.068493150684	L per day				
Input Raw water (D)	105000	L per day								
				Raw water required per day	105000	L per day		Output after ZLD		
				Raw water required annually	38325000	L per Year		Recycled (G=I+II)	40273.150684931505	
L per day										
				Raw water required per day (A)	105000	L per day		Losses (III)	4474.7945205479455	L
per day										
									Recycled (%)	38.35538160469667
				Product	Output	UOM		% loss (salts, sludge)	4.261709067188519	%
				Product	49315.068493150684	L per day		Liquid discharge (D-G-E-III-B)	0	L per day
				Product Output per day (B)	49315.068493150684	L per day				
				QC (C)	250	L per day				
				Waste water (D=A-B+C)	55934.931506849316	L per day				
				Evaporation Losses (E)	11186.986301369863	20 % of D				
				Input for ETP (F=D-E)	44747.94520547945	L per day				
				Number of persons	400	No.				
				Water required per person per day	35	L/P/D				
				Input for STP	14000	L per day				
[SHEET: Plasma]										
Water generation					Calculation for ETP and STP					Calculation for ZLD
Raw water (100%)	250000	L		Parameter	Capacity	UOM		Parameter	Receovery %	Quantity
UOM										
Soft water (95%)	237500	L		Plasma processing per year	150000	L		RO permeate (I)	0.7	
57645.72017534246	L per day									
Purified Water (65% of Soft water)	154375	L		Number of days	365	Days		MEE condensate (II)		
0.2	16470.20576438356	L per day								
Water For Injection (90% of PW)	138937.5	L		Batch size	1000	L		Losses (salts, sludge)		
(III)	0.1	8235.10288219178	L per day							
				Number of batches per year	150	Nos				
				Raw water required per batch	250000	L		Final water balance		
				Number of days per batch	5	Days		Input Raw water (D)	102739.72602739726	L per day
				Raw water required per batch per day	50000	L per batch per day		Output after ZLD		
				Raw water required annually	37500000	L per Year		Recycled (G=I+II)	74115.92593972602	
L per day										
				Raw water required per day (A)	102739.72602739726	L per day		Losses (III)		
8235.10288219178	L per day									
				Product	Output	UOM		Recycled (%)	72.13950124799999	%
				Albumin	100	L per batch		% loss (salts, sludge)	8.015500138666667	%
				IVIG	150	L per batch		Liquid discharge (D-G-E-III-B)	1.3244516594568267e-11	L per
day										
				FIX	2	L per batch				
				FVIII	0.2	L per batch				
				Thrombin	2	L per batch				
				Fibrinogen	0.5	L per batch				
				Total output	254.7	L per batch				
				Product Output per day (B)	50.94	L per day per batch				
				QC (C)	250	L per day				
				Waste water (D=A-B+C)	102938.78602739726	L per day				
				Evaporation Losses (E)	20587.757205479455	20 % of D				
				Input for ETP (F=D-E)	82351.0288219178	L per day				
				Number of persons	506	No.				
				Water required per person per day	35	L/P/D				
				Input for STP	17710	L per day				

PROPOSED METHOD FOR EFFLUENT TREATMENT FOR LIFENITY HEALTH LIMITED,
AURIC, AURANGABAD

This project of Wastewater Treatment is based on most conventional method of treatment "Activated Sludge Process". The Activated Sludge Process is first developed by Ardern and Lockett (1914) in England. In this proposed process of treatment, the wastewater undergoes Collection Tank, Bar Screen, Oil & Grease separation, Equalization, De-ammonification, Settlement, Aerobic Digestion, Clarification, Settlement, Disinfection followed by Filtration.

The treated water can be used for Green Belt.

DESIGN BASIS

275 KLD ETP PROCESS FLOW SHEET

M/s. LIFENITY HEALTH Ltd., Aurangabad, India.

BAR SCREEN

PROCESS DESCRIPTION

This project of Wastewater Treatment is based on the most conventional method of treatment, "Activated Sludge Process". The Activated Sludge Process is first developed by Ardern and Lockett (1914) in England. Wastewater is continuously exposed and subjected to biological degradation carried out by the microbial floc suspended in the aeration tank, into which oxygen is introduced by mechanical means. The wastewater from the aeration tank is allowed to settle, and a portion of the sludge is recycled to the tank itself for an effective microbial population.

In this proposed process of treatment, the wastewater undergoes Collection Tank, Bar Screen, Oil & Grease separation, De-ammonification, Equalization, Settlement, Aerobic Digestion, Clarification, Settlement, Disinfection, followed by Filtration. Sludge will pass onto the Sludge Beds. The treated water can be used for the Green Belt.

The process steps in brief are given below.

Primary Treatment,

Secondary Treatment,

Tertiary Treatment.

PRIMARY TREATMENT,

BAR SCREEN:

The effluents from various points from the Production and Utilities will flow by gravity into the Collection tank or sump through Bar Screen Chamber, where 1 No. SS Bar Screen is provided. The screen provided will arrest suspended trashy matter such as paper, cotton waste etc.

OIL & GREASE SEPARATION:

The effluents having Oil & Grease type material will be removed by mechanical equipment manually for further treatment.

EQUALIZATION TANK:

The pH of the wastewater is neutralized in this process. pH Correction chemical will be added in this process at justified dosage and uniform mixing. So that the wastewater is neutralized and pass onto further treatment process.

DEAMONIFICATION TANK:

The de-ammonification will be done in this by providing aeration and chemical dosing will be added in this process at justified dosage and uniform mixing. So that the wastewater is pass onto clarifier further treatment process.

PRIMARY SETTLING TANK:

In this process, the suspended Solids in the wastewater will be removed with help of coagulant agents. The Coagulants will be added at justified dosage and uniform mixing for settlement of Suspended Solids and enhancement treatment process.

SECONDARY TREATMENT:

AEROBIC DIGESTION TANKS:

The wastewater is passed into Aeration Tanks where an active mass of microorganisms is maintained with sufficient aeration. The biomass converts the organic matter present in the wastewater into cellular material. The Organic as well as Digestible Inorganic load of Wastewater will be decreased in this aerobic stabilization process.

TERTIARY TREATMENT.

CLARIFICATION TANK:

In this process, the suspended Solids in the wastewater will be removed with help of coagulant agents. The Coagulants will be added at justified dosage and uniform mixing for settlement of Suspended Solids and enhancement treatment process.

SECONDARY SETTLING CUM FILTER FEED TANK:

Suspended solids are settled at the bottom of the tank and in this disinfectant, will be used to avoid health hazard. The wastewater will be pumped to Filtration units for further treatment.

FILTRATION:

Final traces of solids are removed by filtration through Multi Grade Filter & Adsorption on Activated Carbon Filter. The treated water can be used for Green Belt.

SLUDGE DRYING BEDS:

The partial sludge settled in Settling Tanks, Clarifier and from the MGF & ACF will dry in the Sludge Drying Beds. In order to maintain sufficient cell concentration a part of the sludge settled in the Clarifiers is recycled as biomass. The dried solids from sludge systems will be send to PCB authorized Solid Disposal Facility as per PCB guidelines. The water from sludge beds is send collection for further treatment process.

10

TREATED WATER TANK-2

7.5X15.0X3.8 Mt-427.5 KL

TD-5X9.4X4 Mt

3.22

PLAN SHOWING THE PROPOSED CONSTRUCTION 160 KLD ETP AND 10 KLD STP [EFFLUENT AND SEWAGE TREATMENT PLANTS.

.

SPECIFICATIONS

FOUNDATIONS : RCC FOOTINGS BASEMENT : CRS IN CM SUPER STRUCTURE : B.K. IN CM

ROOF, BEAM, COLUMNS : R.C.C 1:2:4

FLOORING : MARBLE STONE

FINISHING : PLASTERING IN CM

JOINERY : MT WOOD

JOINERY

MD - 1.10M X 2.10M W - 1.5M X 1.2M D1 - 0.90M X 2.10M W1 - 1.2M X 1.2M D2 - 0.75M X 2.10M W2 - 1.2M X 0.76M

V - 0.6M X 0.3M

REFERENCE

PROPOSED

AREAS SQ.YDS SQ.MTS

PROPOSED FOR 275 KLD ETP
SIZE OF ETP TANK-1 : 00.00M X 00.00M X0.00M SIZE OF ETP TANK-2 : 00.0M X 00.00M X0.00M
SCALE : 1:100 & 1:200
PROJECT SYNOPSIS | PROJECT SYNOPSIS | PROJECT SYNOPSIS
Quantity | Units | Values
Design Capacity of Plant | M3/Day | 275
Normalized Hours of Operation | Hrs | 24 Hrs
Quality | --- | ---
Inlet BOD for Design | ppm | < 1,100.
Inlet COD for Design | ppm | < 2,250.
Inlet TSS for Design | ppm | < 200.
Inlet pH for Design | ppm | 10 - 12.
Outlet BOD | ppm | < 100
Outlet COD | ppm | < 250
Outlet TSS | ppm | < 50
Outlet pH | --- | 6.5 - 8.5

Reagent Details:
Preparation of AL-RR-20 working buffer from Stock buffer (AL-RR-18):
Calculations: For the Preparation of 0.01M Sodium Hydroxide.
Stock Buffer (AL-RR-18):2.0M Sodium Hydroxide,
Formula:
 $C1 \times V1 = C2 \times V2$
C1 = concentration of stock
V1 = volume of stock you need to take
C2 = desired (working) concentration
V2 = final total volume of working solution
Applying the formula to prepare the working buffer:
 $C1 \times V1 = C2 \times V2$
C1 = 2.0 M Sodium Hydroxide.
V1 = volume of stock you need to take _____ L.
C2 = desired (working) concentration 0.01M Sodium Hydroxide.
V2 = final total volume of working solution _____ L.
 $V1 = (C2 \times V2) / C1$
C2=0.01M; V2= 1.0 L; C1=2.0M
 $V1 = (0.01 \times 1.0) / 2.0$
V1 = 0.005L
Take 0.005 L (AL-RR-18) Stock buffer solution and dilute it with water, making up to 1 L.
Check and record the details:
Preparation Details:
Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry
Date
Sodium Hydroxide | | | |
Stock buffer (AL-RR-18) taken volume(L) | pH (12.0 ±1) | Conductivity (2.0 ±1.0 ms/cm) | Storage
Condition | Final volume prepared
| | RT |
Date of Preparation | Prepared By | Checked By
| |

Reagent Details:
Formula & Calculations:
Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).
Calculation: To prepare 1 L of 0.2N Sodium Hydroxide (MW = 40.0 g/mol)
Weight of solute(g)=0.2×40 = 8g.
Preparation Method:
Weigh 8 g of Sodium Hydroxide and dissolve in water to make a 1 L solution.
Preparation for different volumes:
To prepare _____ of 0.2N Sodium Hydroxide (MW = 40.0 g/mol).
Weight of solute(g)=0.2×40.0 x _____L= _____gm
Weigh _____gm of Sodium Hydroxide and dissolve in water to make a _____ L solution.
Check and record the details:
Preparation Details:
Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry
Date
Sodium Hydroxide | | | |
Final volume prepared | Weighed Quantity | pH (13.0±0.2) | Conductivity (43.0±1.0ms/cm) | Storage
Condition:
| | | RT
Date of Preparation | Prepared By | Checked By
| |

Reagent Details:
Formula & Calculations:
Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).
Calculation: To prepare 1 L of 0.5N Sodium Hydroxide (MW = 40.0 g/mol)
Weight of solute(g)=0.5×40 = 20.00g.
Preparation Method:
Weigh 20.00 g of Sodium Hydroxide and dissolve in water to make a 1 L solution.
Preparation for different volumes:
To prepare _____ of 0.5N Sodium Hydroxide (MW = 40.0 g/mol).
Weight of solute(g)=0.5×40.0 x _____L= _____gm
Weigh _____gm of Sodium Hydroxide and dissolve in water to make a _____ L solution.
Check and record the details:
Preparation Details:
Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry
Date
Sodium Hydroxide | | | |
Quantity of NaOH (FN-RR-01) taken (gm) | pH (13.5±2.0) | Conductivity (95±5.0ms/cm) | Storage Condition
| Final volume prepared
| | RT |
Date of Preparation | Prepared By | Checked By
| |

[SHEET: Organic waste Plasma]

Sr. No. | Source / Process Area | Type of Non-Hazardous Waste | Approx. Quantity / Year | Solid | Liquid
 | Category | ETP/3rd party
 1 | ETP / STP treatment | Biological sludge (after dewatering) | 8 – 10 MT/year (wet) | 10 | | Organic |
 3rd party
 2 | Process area & lab washing | Treated water (ETP + RO permeate) | ~12,000 m³/year | | 12000000 |
 Organic | Domestic use
 3 | Packaging line | Empty glass vials (non-contaminated rejects) | 0.5 – 1.0 MT/year (~50,000 pcs) | 1 |
 | Inorganic | Scrap 3rd party
 4 | Packaging & labeling | Paperboard cartons, labels, paper scrap | 1.5 – 2.0 MT/year | 2 | | Organic |
 Scrap 3rd party
 5 | Procurement / stores area | Corrugated boxes, stretch film, plastic wrapping | 1.0 – 1.5 MT/year |
 1.5 | | Inorganic | Scrap 3rd party
 6 | Maintenance workshop | Metal scrap (iron, steel, aluminum parts) | 0.8 – 1.2 MT/year | 12 | |
 Inorganic | Scrap 3rd party
 7 | Utility & HVAC area | Spent filters (air filters, bag filters, prefilters) | 0.3 – 0.5 MT/year | 0.5
 | | Inorganic | 3rd party
 8 | Admin & office areas | Office paper waste, packaging waste | 1.0 – 1.5 MT/year | 1.5 | | Organic |
 Scrap 3rd party
 9 | Canteen / pantry | Organic kitchen waste | 1.5 – 2.0 MT/year | 2 | | Organic | Organic converter
 10 | General housekeeping | Sweepings, mopping dust, non-contaminated PPE | 0.5 – 0.8 MT/year | 0.8 | |
 Mixed | 3rd party
 11 | Cooling tower & utilities | Blowdown & scale sludge (non-toxic) | 1.0 – 1.5 MT/year | 1.5 | |
 Organic | 3rd party
 12 | Electrical systems | Cable cuttings, wire ends, plastic conduits | 0.2 – 0.3 MT/year | 0.3 | |
 Inorganic | Scrap 3rd party
 13 | Gardening & green area | Dry leaves, grass clippings | 0.3 – 0.5 MT/year | 0.5 | | Organic | Organic
 converter
 14 | Water purification & softener | Spent ion-exchange resins (non-toxic) | 0.2 MT/year | 0.2 | |
 Inorganic | 3rd party
 15 | Misc. activities | Empty HDPE / PET bottles (non-contaminated) | 0.1 – 0.2 MT/year | 0.2 | |
 Inorganic | 3rd party
		Domestic use	32876.71232876712	L/Day	
		Organic converter	6.8493150684931505	T/day	
		3rd party	56.71232876712329	T/day	
		Scrap 3rd party	50.136986301369866	T/day	

Reagent Details:

Formula & Calculations:

Formula: Use the formula to calculate the molarity:

Molarity (M) = (Per cent by Weight / 100) * (Density in g/mL) * 1000 / Molar Mass.

Example: Molarity of Glacial Acetic Acid

Molar Mass of CH₃COOH: ~60.05 g/mol

Density: ~1.05 g/mL

Purity: 100%

Acetic Acid (MW = 60.05 g/mol).

Calculation:

Molarity = (100 / 100) * 1.05 g/mL * 1000 mL/L / 60.05 g/mol ≈ 17.5 M.

Calculation: To prepare 1 L of 0.6 M.

Applying the formula to prepare the working buffer:

$C_1 V_1 = C_2 V_2$

C_1 = 17.5 M acetic acid.

V_1 = volume of stock needed to take.

C_2 = desired (working) concentration 0.6 M acetic acid.

V_2 = final total volume of working solution for one liter.

$V_1 = (C_2 \times V_2) / C_1$.

$C_2 = 0.6$ M; $V_2 = 1.0$ L; $C_1 = 17.5$ M

$V_1 = (0.6 \times 1.0) / 17.5$

$V_1 = 0.034$ L

Take 0.034 L of the Stock solution of acetic acid and add it to a volumetric flask, making up the volume to 1 L.

To prepare various volumes:

V_1 = volume of stock needed to take _____ L.

C_2 = desired (working) concentration 0.6M acetic acid.

V_2 = final total volume of working solution _____ L.

$V_1 = (C_2 \times V_2) / 17.5$.

$V_1 =$ _____ L.

Take V_1 _____ L of the Stock solution of acetic acid and add it to a volumetric flask, making up the final volume _____ L.

Check and record the details:

Preparation Details:

Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry

Date

||||

Final volume prepared | Measured Quantity | pH (2.4±0.1) | Conductivity (30.0±5.0 ms/cm) | Storage

Condition | Reagent Record

|||| RT |

Date of Preparation | Prepared By | Checked By

||

[SHEET: Sheet1]

S. No.	TIF number	Details
1	Lifenity_TIF_001	ETP, STP water calculation details for Plasma and LVP plant
2		RDS for Plasma
3		Utility RDS for Plasma
4	Lifenity_TIF_002	Organic waste calculation and parameters for ETP
5	Lifenity_TIF_003	List of raw materials for LVP
6		List of raw materials for Plasma
7		
8		
9		
10		
11		

[SHEET: Sheet2]

[SHEET: Sheet3]

Reagent Details:
Formula & Calculations:
Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).
Calculation: To prepare 1 L of 0.1N Sodium Hydroxide (MW = 40.0 g/mol)
Weight of solute(g)=0.1×40 = 4g.
Preparation Method:
Weigh 4 g of Sodium Hydroxide and dissolve in water to make a 1 L solution.
Preparation for different volumes:
To prepare _____ of 0.1N Sodium Hydroxide (MW = 40.0 g/mol).
Weight of solute(g)=0.1×40.0 x _____L= _____gm
Weigh _____gm of Sodium Hydroxide and dissolve in water to make a _____ L solution.
Check and record the details:
Preparation Details:
Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry
Date
Sodium Hydroxide | | | |
Final volume prepared | Weighed Quantity | pH (13.0 ± 0.5) | Conductivity (22.0 ± 1.0ms/cm) | Storage
Condition:
| | | RT
Date of Preparation | Prepared By | Checked By
| |

Reagent Details:
Calculations: To prepare 1 L of 0.35% TNBP.
Decide on the total final volume to prepare (e.g., 1000 mL).
Calculate 0.35% of the total final volume to determine the amount of TNBP needed:
"TNBP required"="Total Volume"×0.35/100.
For 1000 mL final volume: mL TNBP.
Preparation Method:
Measure the required volume for one litre and take it in a beaker, and dilute it with water to make a 1 L solution.
Preparation for different volumes:
To prepare _____ of 0.35% TNBP.
TNBP required"="Total Volume"×0.35/100.
For _____ mL final volume: mL TNBP.
Measure the required volume of one liter and dissolve it in water to make a 1 L solution.
Check and record the details:
Preparation Details:
Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry
Date
TNBP | | | |
Final volume prepared | 0.35% TNBP Volume (ml) | pH (6.8 ± 2.0) | Conductivity (10 µs/cm) | Storage
Condition
| | | |
Date of Preparation | Prepared By | Checked By
| |

Engineering-grade ZLD system for PDMP plant
(150,000 L/year plasma processing).

1. Design Basis

Plant Throughput

Plasma processed: 150,000 L/year

Operating days: 300 days/year

1000 L/batch plasma

Water & Effluent

Total water consumption: 1,13,551 L/day (113.551 m³/day)

Effluent generation: 51,988 L/day (51.988 m³/day)

2. Effluent Segregation (Critical for Proper Sizing)

Low TDS → direct RO

High TDS → RO + MEE

3. Overall ZLD Philosophy

Effluent → Equalization → ETP → UF → RO → (Permeate reuse)

↓

RO Reject → MEE → ATFD → Solids

4. RO System Sizing

Design Assumptions

Feed: 5 m³/day

Recovery: 70%

Operating hours: 20 hr/day

Flow Calculations

Feed flow rate:

51.988 m³/day ÷ 20 hr = 2.5994 m³/hr

Permeate: 70% of 51.988 = 36.3916 m³/day

Reject:

15.5964 m³/day

RO Capacity Selection

Add design margin (25–30%):

■ Installed RO capacity: 0.35 m³/hr (~350 L/hr)

RO System Components

Multi-grade filter (MGF)

Activated carbon filter (ACF)

UF membrane

High-pressure RO skid

5. MEE (Multiple Effect Evaporator) Sizing

Feed to MEE

RO reject: 15.5964 m³/day

Assume concentration to 20–25% solids

Evaporation Load

Water to be evaporated: ~1.2 m³/day (80% evaporation)

Hourly evaporation: 1.2 ÷ 20 hr = 0.06 m³/hr (60 L/hr)

MEE Design Capacity

MEE capacity: 75–100 L/hr evaporation

Number of Effects

Typically: 2 or 3 effects

Steam Requirement

Typical:

1 kg steam evaporates ~0.8–1 kg water (2-effect)

Steam required:

~75 kg/hr steam

6. ATFD (Agitated Thin Film Dryer) Sizing

Feed to ATFD

Concentrated slurry from MEE:

■ ~0.3 m³/day (~300 L/day)

Solid Load

Assume:

20% solids

Solids:

~60 kg/day

ATFD Capacity

Operating hours: 10–12 hr/day

Required: 60 kg ÷ 12 hr = 5 kg/hr

With margin: ATFD capacity: 8–10 kg/hr solids handling

7. Final Water Recovery Summary

8. Utilities Requirement

Power

Total: ~25–35 kW

Steam

MEE: ~75 kg/hr

ATFD: ~20 kg/hr

Total: ~100 kg/hr steam

9. Equipment Sizing Summary (DPR Ready Table)

Stream Type | Source | Flow (m³/day) | TDS

Low TDS | Cooling tower blowdown, domestic | 2.0 | <2,000 ppm

High TDS | CIP, buffers, process streams | 3.0 | 5,000–15,000 ppm

Stage | Output

RO permeate | 3.5 m³/day

MEE condensate | 1.0 m³/day

Total recovery | 4.5 m³/day

Losses (salts/sludge) | 0.5 m³/day

Liquid discharge | ZERO

Equipment | Load

RO system | 5–7 kW

MEE | 10–15 kW

ATFD | 8–12 kW

Pumps & auxiliaries | 5 kW

Unit | Capacity

Equalization Tank | 10 m³

ETP | 5 m³/day
UF System | 0.3 m³/hr
RO System | 0.35 m³/hr
MEE | 100 L/hr evaporation
ATFD | 10 kg/hr solids

[SHEET: Index]

Sr.No. | Document List | Details provide

- 1 | MIDC Sanction Plan. | Revision suggested , awaited from Architect@ ddgroup
- 2 | MIDC Completion/Occupancy Certificate. | Possession Letter 6/8/2024
- 3 | Udyam Aadhar | NA
- 4 | Incorporation Certificate. | incorporation!A1
- 5 | Total Project Cost. (Esteemated) | 3065586099
- 6 | a) Built-up area | 40000 Sq Meter
- 6 | b) Area proposed for green belt | 27000 Sq Meter
- 6 | c) Tree plantation details. | Revision suggested , awaited from Architect@ ddgroup
- 7 | Proposed layout plan with area Statement. | Revision suggested , awaited from Architect@ ddgroup
- 8 | a) List of products | List of Products!A1
- 8 | b) Raw Material Balance Sheet. | Raw Material!A1
- 9 | a) List of HW waste will be generated from the process. | Hazardous Waste!A1
- 9 | b) List of Non-Hazardous waste will be generated from the process. | Non Haz Waste!A1
- 10 | Water Balance, Domestic Consumption & Process Consumption and Generation of Waste Water. | Water!A1
- 11 | Name of the Fuel will be used in the process: HSD/ Electricity/CNG. | Fuel Consmp!A1
- 12 | Details of any heat treatment or reaction tank, if any. | Heat treat_Reaction_Tank!A1
- 13 | Details of Equipment/Machinery with the Name | Equipments!A1
- 14 | Types and sources of Pollution and control measures (Pollution Control equipment). | Pol control meas!A1
- 15 | DG Set details. | DG Set !A1

[SHEET: incorporation]

||||| Back to Index

[SHEET: Possession Let]

||||| Back to Index

[SHEET: DG Set]

Sr.No. | Parameter | Description / Specification |

- 1 | Make / Model | Cummins / Kirloskar / Mahindra (example – confirm at procurement) | Back to Index
- 2 | Type | Silent Type, Auto-start, Water-cooled, Turbocharged Diesel Generator |
- 3 | Rating / Capacity | 2 × 1300 kVA each (standby operation, load sharing optional) |
- 4 | Fuel Used | High-Speed Diesel (HSD) – BS-VI grade, low sulphur (<0.05%) |
- 5 | Rated Power Output | ~400 kW per DG set (0.8 PF) |
- 6 | Total Installed Capacity | 2,600 kVA (2 sets × 1300 kVA) |
- 7 | Operating Hours / Year | ~800–1,000 hours per year (emergency + load testing only) |
- 8 | Fuel Consumption | ~55–60 L/hour @ full load |
- 9 | Estimated Annual HSD Consumption | ≈ 27,000 L / year (≈ 27 KL / year) |
- 10 | Storage Capacity (HSD Tank) | 10 KL aboveground double-walled MS tank with vent, dyke, and spill tray as per PESO norms |
- 11 | Cooling System | Water-cooled, with radiator and jacket water heater |
- 12 | Lube Oil System | Forced lubrication; oil drain point connected to collection tray |
- 13 | Starting System | 24V battery start with automatic mains failure (AMF) panel |
- 14 | Exhaust System | Silencer with flexible coupling; insulated exhaust manifold |
- 15 | Emission Parameters (at rated load) | • PM ≤ 0.3 g/kWh • NO_x + HC ≤ 7.5 g/kWh • CO ≤ 3.5 g/kWh (As per CPCB II norms) |
- 16 | Stack Details | Mild steel stack with flanged joints; height as per CPCB formula: $H = h + 0.2\sqrt{KVA} = 3 + 0.2\sqrt{1300} \approx 10.2$ m above roof level per set | 36.05551275463989
- 17 | Stack Quantity | 2 Nos (individual stack for each DG set) | 7.211102550927979
- 18 | Noise Level at 1 m distance | ≤ 75 dB(A) – within CPCB norm | 10.21110255092798
- 19 | Noise Control Measures | • Acoustic Enclosure (≤ 75 dB(A)) • Vibration isolators under skid base • Greenbelt & acoustic barriers along DG yard boundary |
- 20 | Location | Utility block (away from production & office area) |
- 21 | Ventilation System | Forced air intake & exhaust with acoustic louvers |
- 22 | Emission Control | • Stack emission monitoring (PM, NO_x, SO_x) – quarterly • Use of low-sulphur HSD |
- 23 | Lube Oil Handling | Drained to closed HDPE drum; sent to authorized recycler (HW 5.1) |
- 24 | Spill Control | Impervious floor, containment dyke, absorbent pads available |
- 25 | Safety & Fire Protection | Fire extinguishers (CO₂ + dry chemical) at DG yard; earthing pit per IS 3043 |
- 26 | Authority Approval | DG sets to comply with CPCB-II emission & noise certification; HSD tank under PESO license |

Sr.No. | Parameter | Standard / Limit | Plant DG Result / Provision

- 1 | Emissions (CPCB-II) | PM ≤ 0.3 g/kWh NO_x+HC ≤ 7.5 g/kWh CO ≤ 3.5 g/kWh | Cummins CPCB-II compliant engines
- 2 | Stack Height | $H = h + 0.2\sqrt{KVA}$ | 6.5 m above roof per set
- 3 | Noise | ≤ 75 dB(A) at 1 m | Acoustic enclosure fitted
- 4 | Fuel Type | Low-sulphur HSD | BS-VI HSD used
- 5 | Used Oil Disposal | To authorized recycler | As per HW Rule 5.1
- 6 | DG Area Containment | Impervious floor + bund | Provided

| Back to Index |

[SHEET: Pol control meas]

Sr.No. | Type | Source | Pollutants Likely | Control Measures / Equipment | Remarks |

- 1 | AIR POLLUTION | a. DG Set (2 × 500 kVA, standby) | PM, SO_x, NO_x, CO (from diesel combustion) | • Acoustic enclosure to control noise • Adequate stack height as per CPCB formula ($H = h + 0.2\sqrt{KVA}$) ~6.5 m above roof • Low-sulphur HSD (<0.05%) fuel • Regular maintenance & emission monitoring | Operates only during power outage | | Back to Index
- 2 | AIR POLLUTION | b. PNG / CNG-fired Boiler (500 kg steam/hr) | NO_x, trace CO | • Use of low-NO_x burner • Stack height 12 m above roof • Online temperature & pressure monitoring • Periodic stack emission testing | PNG is clean fuel, minimal SO_x |
- 3 | AIR POLLUTION | c. HVAC systems / AHUs | Dust, microbial aerosols | • HEPA filters (0.3 µm) at terminal outlets • Pre-filters (10 µm) at return air • Periodic filter replacement as per SOP | Ensures cleanroom Class 10,000–100,000 compliance |
- 4 | AIR POLLUTION | d. Solvent & detergent handling (TnBP, Triton X-100) | VOC vapours | • Closed mixing vessels, nitrogen blanketing • Solvent vapor condenser + activated-carbon adsorption unit • Explosion-proof zone classification (ATEX) | Emission < 50 mg/Nm³ VOC ensured |
- 5 | AIR POLLUTION | e. Laboratory fume hoods / exhausts | Trace acidic or organic vapours | • Fume hoods with local scrubbers or HEPA filters • Ducted exhaust at safe height | Minimal emissions, intermittent source |

Sr.No. | Type | Source | Type of Wastewater | Pollutants Likely | Treatment / Control Equipment |

- 1 | WATER POLLUTION | a. Process & WFI drains | Process effluent containing protein residues, buffers, mild solvents, detergents | pH 6–8, COD 1,000–2,000 mg/L, BOD 400–800 mg/L, low TDS | Effluent Treatment

Plant (ETP): • Collection & equalization tank (15 m³) • pH correction & neutralization tank • Primary clarifier • Biological MBR / MBBR reactor • Tertiary filtration (UF + RO) • RO permeate reused for gardening / utilities • RO reject → MEE + ATFD | |

2 | WATER POLLUTION | b. Solvent-detergent streams (TnBP phase) | High COD, solvent traces | Trace VOCs, COD > 10,000 mg/L | Dedicated solvent holding tank → evaporation or incineration via TSDF | |

3 | WATER POLLUTION | c. Domestic sewage (staff 506) | Black / grey water | BOD ~250 mg/L | STP (40 m³/day): MBBR + UF + chlorination | |

4 | WATER POLLUTION | d. Cooling tower blow-down, softener reject | High TDS, hardness | TDS 2,000–3,000 mg/L | Collected in neutralization tank → mixed with RO reject → MEE | |

Sr.No. | Type | Source | Type of Waste | Category | Control / Disposal Method | |

1 | SOLID / HAZARDOUS WASTE POLLUTION | Process residues, filter aids, spent carbon, ETP sludge | Solid / semi-solid | HW 28.2 / 35.3 | Sent to Authorized TSDF (CHWTSDF) for landfill/incineration | |

2 | SOLID / HAZARDOUS WASTE POLLUTION | Spent solvent mix (TnBP + detergent) | Liquid | HW 28.1 | Sent to Solvent recovery or TSDF | |

3 | SOLID / HAZARDOUS WASTE POLLUTION | Used oil from DG set | Liquid | HW 5.1 | Sold to MPCB-authorized recycler | |

4 | SOLID / HAZARDOUS WASTE POLLUTION | Biomedical waste (lab disposables, plasma bags) | Bio-infectious | BMW Cat. 4 / 5 | Sent to Authorized CBWTF | |

5 | SOLID / HAZARDOUS WASTE POLLUTION | Non-hazardous recyclable waste | Paper, plastic, metal, glass | Non-hazardous | Segregation → authorized recyclers | |

6 | SOLID / HAZARDOUS WASTE POLLUTION | Organic waste (canteen, sludge) | Biodegradable | Non-hazardous | Composting / manure | |

Sr.No. | Type | Source | Noise Level (dB-A) | Control Measures | | |

1 | NOISE POLLUTION | DG sets | 75–80 | Acoustic enclosure, antivibration mounts, green belt | | |

2 | NOISE POLLUTION | Compressors, pumps, blowers | 70–75 | Foundation damping pads, acoustic barrier | | |

3 | NOISE POLLUTION | HVAC & AHU | 60–65 | Acoustic lining, flexible connectors | | |

4 | NOISE POLLUTION | General utilities | < 55 | Periodic maintenance, ear protection for workers | | |

Sr.No. | Type | Equipment / Facility | Capacity / Function | Purpose | | |

1 | SOLID WASTE MANAGEMENT | ETP | 70 m³/day (MBBR + UF + RO + MEE) | Treat & recycle process effluent | | |

2 | SOLID WASTE MANAGEMENT | STP | 40 m³/day (MBBR + UF) | Treat domestic sewage | | |

3 | SOLID WASTE MANAGEMENT | Air pollution stacks | Boiler – 12 m; DG – 6.5 m above roof | Emission dispersion | | |

4 | SOLID WASTE MANAGEMENT | Activated carbon filter | 1 × 500 kg | VOC adsorption from S-D exhaust | | |

5 | SOLID WASTE MANAGEMENT | Scrubber (if acid used) | 1 × 3,000 m³/hr | Fume removal from buffer prep / lab exhaust | | |

6 | SOLID WASTE MANAGEMENT | Noise enclosure | DG & compressors | Noise reduction | | |

7 | SOLID WASTE MANAGEMENT | Greenbelt | ≥ 33 % of plot | Dust, noise & micro-climate control | | |

| Back to Index | | | | |

[SHEET: Equipments]

Sr.No. | Category | Equipment Name | Quantity | Estimated Amount | | Back to Index

1 | ADMIN EQUIPMENT | Office Cabinets for Documents | 20 | 240000 | |

2 | ADMIN EQUIPMENT | Computers / Laptops | 50 | 2500000 | |

3 | ADMIN EQUIPMENT | Office Furniture | NA | 1200000 | |

4 | ADMIN EQUIPMENT | Printer/scanner / xerox machine | 1 | 500000 | |

5 | ADMIN EQUIPMENT | Water Dispenser | 2 | 40000 | |

6 | ADMIN EQUIPMENT | Projector for conference / meeting room | 2 | 600000 | |

7 | CANTEEN EQUIPMENT | Refrigerator | 2 | 300000 | |

8 | CANTEEN EQUIPMENT | SS furniture (serving table etc) | NA | 400000 | |

9 | CANTEEN EQUIPMENT | SS Water cooler dispenser | 1 | 50000 | |

10 | CANTEEN EQUIPMENT | Microwaves | 2 | 60000 | |

11 | CANTEEN EQUIPMENT | Kitchenware | na | 500000 | |

12 | CANTEEN EQUIPMENT | Bain marie | 1 | 50000 | |

13 | Clean Utilities | Soft water generation 10000/hour and storage system | 1 | 40000000 | |

14 | Clean Utilities | Purified water generation 5000L/ hour storage tank 10KL and distribution system | 1 | | |

15 | Clean Utilities | Water for injection generation 1000L / hour, Storage tank 10KL and distribution system | 1 | | |

16 | Clean Utilities | Pure steam generator 1500Kg / hour and distribution system | 1 | | |

17 | Black Utilities | Plant steam generation 2000Kg/Hour and distribution System | 2 | 7000000 | |

18 | Black Utilities | Compressed air generation and distribution system | 2 | 7500000 | |

19 | Black Utilities | PASSENGER LIFT X 2 NOS | 2 | 7000000 | |

20 | Black Utilities | MATERIAL LIFT X 3 NOS | 3 | 6000000 | |

21 | PLASMA STORAGE EQUIPMENT | Analytical weighing Balance | 1 | 170000 | |

22 | PLASMA STORAGE EQUIPMENT | RLAf | 3 | 1200000 | |

23 | PLASMA STORAGE EQUIPMENT | Precision Balance | 1 | 144000 | |

24 | PLASMA STORAGE EQUIPMENT | Commercial Balance | 1 | 120000 | |

25 | PLASMA STORAGE EQUIPMENT | Refrigerator 200L | 2 | 218000 | |

26 | PLASMA STORAGE EQUIPMENT | Deep freezer -20°C (1000L) | 50 | 22500000 | |

27 | PLASMA STORAGE EQUIPMENT | Air curtain | 10 | 100000 | |

28 | PLASMA STORAGE EQUIPMENT | PVC pallets | 50 | 300000 | |

29 | PLASMA STORAGE EQUIPMENT | Storage racks SS | 25 | 700000 | |

30 | PLASMA STORAGE EQUIPMENT | Sampling rods | 2 | 10000 | |

31 | PLASMA STORAGE EQUIPMENT | Liquid sampler | 2 | 20000 | |

32 | PLASMA STORAGE EQUIPMENT | Dedusting tunnel | 1 | 1250000 | |

33 | PLASMA STORAGE EQUIPMENT | DPB | 4 | 600000 | |

34 | PLASMA STORAGE EQUIPMENT | LAF cabinet | 1 | 450000 | |

35 | PLASMA STORAGE EQUIPMENT | Computer / Laptops | 4 | 120000 | |

36 | PLASMA STORAGE EQUIPMENT | Office Furniture | NA | 200000 | |

37 | PLASMA STORAGE EQUIPMENT | Office Cabinets | 10 | 120000 | |

38 | BULK PLASMA MANUFACTURING EQUIPMENT | Chromatography AKTA-1 (471 L/hr (Pipe MOC SS) 1F | 2 | 34200000 | |

39 | BULK PLASMA MANUFACTURING EQUIPMENT | Chromatography Column Dia 1000mm x Height 500mm 1F | 4 | 89224400 | |

40 | BULK PLASMA MANUFACTURING EQUIPMENT | Chromatography AKTA-1 (785 L/hr pipe MOC SS) 2F | 2 | 40062840 | |

41 | BULK PLASMA MANUFACTURING EQUIPMENT | Chromatography Column Dia 1000mm x Height 500mm 2F | 2 | 44612200 | |

42 | BULK PLASMA MANUFACTURING EQUIPMENT | Chromatography AKTA-1 (1178 L/hr pipe MOC PP) 3F | 2 | 40062840 | |

43 | BULK PLASMA MANUFACTURING EQUIPMENT | Chromatography Column Dia 1000mm x Height 500mm 3F | 4 |

89224400 ||
44 | BULK PLASMA MANUFACTURING EQUIPMENT | Chromatography AKTA-1 (864 L/hr pipe MOC PP) 4l | 2 | 40062840
||
45 | BULK PLASMA MANUFACTURING EQUIPMENT | Chromatography Column Dia 1000mm x Height 500mm 4l | 2 |
44612200 ||
46 | BULK PLASMA MANUFACTURING EQUIPMENT | Chromatography AKTA-1 (864 L/hr pipe MOC PP) 5l | 1 | 20031420
||
47 | BULK PLASMA MANUFACTURING EQUIPMENT | Chromatography Column Dia 1000mm x Height 500mm 5l | 1 |
22306100 ||
48 | BULK PLASMA MANUFACTURING EQUIPMENT | Chromatography AKTA-1 (667 L/hr pipe MOC PP) 6l | 1 | 17100000
||
49 | BULK PLASMA MANUFACTURING EQUIPMENT | Chromatography Column Dia 1000mm x Height 500mm 6l | 1 |
22306100 ||
50 | BULK PLASMA MANUFACTURING EQUIPMENT | Chromatography AKTA-1 (785 L/hr pipe MOC PP) 7A | 1 | 20031420
||
51 | BULK PLASMA MANUFACTURING EQUIPMENT | Chromatography Column Dia 1000mm x Height 500mm 7A | 1 |
22306100 ||
52 | BULK PLASMA MANUFACTURING EQUIPMENT | Chromatography AKTA-1 (30 L/hr pipe MOC PP) 8N | 1 | 14657140
||
53 | BULK PLASMA MANUFACTURING EQUIPMENT | Chromatography Column Dia 300mm x Height 500mm 8N | 1 |
4896570 ||
54 | BULK PLASMA MANUFACTURING EQUIPMENT | Chromatography AKTA-1 (30 L/hr pipe MOC PP) 9N | 1 | 14657140
||
55 | BULK PLASMA MANUFACTURING EQUIPMENT | Chromatography Column Dia 300mm x Height 750mm 9N | 1 |
4896570 ||
56 | BULK PLASMA MANUFACTURING EQUIPMENT | Chromatography AKTA-1(85 L/hr pipe MOC PP) 10T | 1 | 12500000
||
57 | BULK PLASMA MANUFACTURING EQUIPMENT | Chromatography Column Dia 600mm x Height 500mm 10T | 1 |
14244570 ||
58 | BULK PLASMA MANUFACTURING EQUIPMENT | Chromatography AKTA-1 (340 L/hr pipe MOC PP) 11F | 1 |
30000000 ||
59 | BULK PLASMA MANUFACTURING EQUIPMENT | Chromatography Column Dia 300mm x Height 500mm 11F | 1 |
4896570 ||
60 | BULK PLASMA MANUFACTURING EQUIPMENT | Chromatography AKTA-1 (500 L/hr pipe MOC PP) 12E | 1 |
17100000 ||
61 | BULK PLASMA MANUFACTURING EQUIPMENT | Chromatography Column Dia 1000 mm x Height 500mm 12E | 1 |
22306100 ||
62 | BULK PLASMA MANUFACTURING EQUIPMENT | Chromatography AKTA-1(160 L/hr pipe MOC PP) 13E | 1 | 14657140
||
63 | BULK PLASMA MANUFACTURING EQUIPMENT | Chromatography Column Dia 450mm x Height 500mm 13E | 1 |
12018850 ||
64 | BULK PLASMA MANUFACTURING EQUIPMENT | TFF System 50 sqm | 3 | 12500000 ||
65 | BULK PLASMA MANUFACTURING EQUIPMENT | Cassette 50 sqm | 3 | 4000000 ||
66 | BULK PLASMA MANUFACTURING EQUIPMENT | TFF System 20 sqm | 1 | 9460000 ||
67 | BULK PLASMA MANUFACTURING EQUIPMENT | Cassette 20 sqm | 2 | 3010000 ||
68 | BULK PLASMA MANUFACTURING EQUIPMENT | TFF System 5 sqm | 4 | 14620000 ||
69 | BULK PLASMA MANUFACTURING EQUIPMENT | Cassette 5 sqm | 4 | 1376000 ||
70 | BULK PLASMA MANUFACTURING EQUIPMENT | Continuous Centrifuge 1000L/hr | 2 | 30000000 ||
71 | BULK PLASMA MANUFACTURING EQUIPMENT | Process Vessels 300 L | 4 | 8842500 ||
72 | BULK PLASMA MANUFACTURING EQUIPMENT | Process Vessels 100 L | 3 | 6226875 ||
73 | BULK PLASMA MANUFACTURING EQUIPMENT | Process Vessels 150 L | 1 | 2075625 ||
74 | BULK PLASMA MANUFACTURING EQUIPMENT | Process Vessels 250 L | 4 | 8842500 ||
75 | BULK PLASMA MANUFACTURING EQUIPMENT | Process Vessels 500 L | 4 | 10732500 ||
76 | BULK PLASMA MANUFACTURING EQUIPMENT | Process Vessels 1000 L | 6 | 19650600 ||
77 | BULK PLASMA MANUFACTURING EQUIPMENT | Process Vessels 1500 L | 29 | 100830825 ||
78 | BULK PLASMA MANUFACTURING EQUIPMENT | Process Vessels 2000 L | 11 | 42701175 ||
79 | BULK PLASMA MANUFACTURING EQUIPMENT | Buffer vessels 500L | 6 | 16098750 ||
80 | BULK PLASMA MANUFACTURING EQUIPMENT | Buffer vessels 1000L | 9 | 29475900 ||
81 | BULK PLASMA MANUFACTURING EQUIPMENT | Buffer vessels 1500L | 6 | 23291550 ||
82 | BULK PLASMA MANUFACTURING EQUIPMENT | Buffer vessels 2500L | 1 | 30000000 ||
83 | BULK PLASMA MANUFACTURING EQUIPMENT | Depth filtration system | 5 | 5212500 ||
84 | BULK PLASMA MANUFACTURING EQUIPMENT | Nano filter holder | 2 | 1000000 ||
85 | BULK PLASMA MANUFACTURING EQUIPMENT | Cold room 2-8°C | 4 | 3000000 ||
86 | BULK PLASMA MANUFACTURING EQUIPMENT | Rectangular tank | 2 | 860000 ||
87 | BULK PLASMA MANUFACTURING EQUIPMENT | Buffer preparation tank | 4 | 3440000 ||
88 | BULK PLASMA MANUFACTURING EQUIPMENT | Dispensing Booth | 2 | 800000 ||
89 | BULK PLASMA MANUFACTURING EQUIPMENT | Analytical Balance | 2 | 340000 ||
90 | BULK PLASMA MANUFACTURING EQUIPMENT | Precision Balance | 2 | 300000 ||
91 | BULK PLASMA MANUFACTURING EQUIPMENT | Commercial Balance | 2 | 240000 ||
92 | BULK PLASMA MANUFACTURING EQUIPMENT | PVC bag Sealer | 2 | 10000 ||
93 | BULK PLASMA MANUFACTURING EQUIPMENT | Rotary Bag Sealer | 2 | 600000 ||
94 | BULK PLASMA MANUFACTURING EQUIPMENT | pH/ Conductivity meter | 4 | 600000 ||
95 | BULK PLASMA MANUFACTURING EQUIPMENT | Magnetic Stirrer | 4 | 300000 ||
96 | BULK PLASMA MANUFACTURING EQUIPMENT | Peristaltic pump | 4 | 400000 ||
97 | BULK PLASMA MANUFACTURING EQUIPMENT | Filter Integrity machine | 2 | 4000000 ||
98 | BULK PLASMA MANUFACTURING EQUIPMENT | Fume Hood | 2 | 900000 ||
99 | BULK PLASMA MANUFACTURING EQUIPMENT | Autoclave 1000L | 4 | 8000000 ||
100 | BULK PLASMA MANUFACTURING EQUIPMENT | DHS 1000L | 2 | 4000000 ||
101 | BULK PLASMA MANUFACTURING EQUIPMENT | Ceiling LAF | 6 | 2400000 ||
102 | BULK PLASMA MANUFACTURING EQUIPMENT | Drying Oven | 4 | 2000000 ||
103 | BULK PLASMA MANUFACTURING EQUIPMENT | Acid tank 1000L (Pre and Post Virus) | 2 | 3870000 ||
104 | BULK PLASMA MANUFACTURING EQUIPMENT | Base tank 1000L (Pre and Post Virus) | 2 | 3870000 ||
105 | BULK PLASMA MANUFACTURING EQUIPMENT | Central CIP system | 4 | 4300000 ||
106 | BULK PLASMA MANUFACTURING EQUIPMENT | Refrigerator 200L | 2 | 300000 ||
107 | BULK PLASMA MANUFACTURING EQUIPMENT | Laminar air flow cabinet | 2 | 870000 ||
108 | BULK PLASMA MANUFACTURING EQUIPMENT | Dynamic pass box | 8 | 960000 ||
109 | BULK PLASMA MANUFACTURING EQUIPMENT | Tumble Dryer 25 Kg | 2 | 4000000 ||
110 | BULK PLASMA MANUFACTURING EQUIPMENT | Barrier washer Extractor 25Kg | 2 | 6000000 ||
111 | BULK PLASMA MANUFACTURING EQUIPMENT | SS Furniture (Table, chair, racks, Ladders etc) | NA |
8000000 ||
112 | BULK PLASMA MANUFACTURING EQUIPMENT | Lockers | NA | 400000 ||
113 | BULK PLASMA MANUFACTURING EQUIPMENT | Laptop / Computers | 20 | 1000000 ||

114 | BULK PLASMA MANUFACTURING EQUIPMENT | Office furniture / Cabinets | NA | 600000 ||
 115 | FILL FINISH EQUIPMENT | Formulation Vessel 100L | 1 | 2500000 ||
 116 | FILL FINISH EQUIPMENT | Ceiling LAF | 2 | 800000 ||
 117 | FILL FINISH EQUIPMENT | Laminar air flow cabinet | 1 | 435000 ||
 118 | FILL FINISH EQUIPMENT | Continuous Particle Counter | 1 | 3000000 ||
 119 | FILL FINISH EQUIPMENT | Dynamic pass box | 7 | 840000 ||
 120 | FILL FINISH EQUIPMENT | Pass Box static | 1 | 38000 ||
 121 | FILL FINISH EQUIPMENT | Vial washing machine | 1 | 4500000 ||
 122 | FILL FINISH EQUIPMENT | Sterilizing tunnel | 1 | 4500000 ||
 123 | FILL FINISH EQUIPMENT | Vial filling & Sealing line 120 vials /min | 1 | 7500000 ||
 124 | FILL FINISH EQUIPMENT | Lyophilizer (1sq meter) | 1 | 5500000 ||
 125 | FILL FINISH EQUIPMENT | Labelling machine | 1 | 9,00,000.00 ||
 126 | FILL FINISH EQUIPMENT | Double door autoclave 1000L | 1 | 2000000 ||
 127 | FILL FINISH EQUIPMENT | Cold room | 1 | 800000 ||
 128 | FILL FINISH EQUIPMENT | SS furniture (Table, Chair, racks, cabinets etc) | 1 | 200000 ||
 129 | QC & R&D EQUIPMENT | Akta pure -150 (For R&D) | 2 | 4000000 ||
 130 | QC & R&D EQUIPMENT | Columns XK 50/100 (For R & D) | 1 | 700000 ||
 131 | QC & R&D EQUIPMENT | Columns XK 50/60 (For R&D) | 1 | 700000 ||
 132 | QC & R&D EQUIPMENT | Column XK 50/30 (For R & D) | 3 | 1400000 ||
 133 | QC & R&D EQUIPMENT | HPLC-UV | 1 | 3400000 ||
 134 | QC & R&D EQUIPMENT | HPLC Columns and accessories | 1 | 1000000 ||
 135 | QC & R&D EQUIPMENT | Electrophoresis (Minigel) with trans blot | 2 | 600000 ||
 136 | QC & R&D EQUIPMENT | Gel Doc system densitometer | 1 | 500000 ||
 137 | QC & R&D EQUIPMENT | Power packs | 2 | 200000 ||
 138 | QC & R&D EQUIPMENT | Endotoxin Detection system (Tube heating block) | 1 | 200000 ||
 139 | QC & R&D EQUIPMENT | Micropipettes- variable volume range of 2-20µL, 20-200µL, 100-1000µL, 500-5000µL and multichannel 30-300µL | 2 set each | 300000 ||
 140 | QC & R&D EQUIPMENT | Lab Refrigerator 2-8 ° C (SS) with data control | 4 | 450000 ||
 141 | QC & R&D EQUIPMENT | Flame photometer | 1 | 450000 ||
 142 | QC & R&D EQUIPMENT | Weighing Balance Analytical | 2 | 350000 ||
 143 | QC & R&D EQUIPMENT | UV-VIS spectrophotometer Double Beam with data acquisition. | 1 | 900000 ||
 144 | QC & R&D EQUIPMENT | Volumetric KF | 1 | 700000 ||
 145 | QC & R&D EQUIPMENT | Moisture Analyser | 1 | 800000 ||
 146 | QC & R&D EQUIPMENT | Stability Chambers (5±3° C, 25, 37 deg. C) 200L (1set) | 3 | 1400000 ||
 147 | QC & R&D EQUIPMENT | Muffle Furnance | 1 | 80000 ||
 148 | QC & R&D EQUIPMENT | pH and conductivity meter with printer | 2 | 260000 ||
 149 | QC & R&D EQUIPMENT | Vaccum Oven | 1 | 120000 ||
 150 | QC & R&D EQUIPMENT | Melting point apparatus | 1 | 90000 ||
 151 | QC & R&D EQUIPMENT | Incubator Sterility Test 500L | 2 | 1440000 ||
 152 | QC & R&D EQUIPMENT | Incubators for Normal Media plates Incubation 500L | 1 | 300000 ||
 153 | QC & R&D EQUIPMENT | Incubator for EMP and water samples 500L | 2 | 600000 ||
 154 | QC & R&D EQUIPMENT | Hot Air Oven | 1 | 160000 ||
 155 | QC & R&D EQUIPMENT | Thermostatic water bath (10-20 L) | 2 | 200000 ||
 156 | QC & R&D EQUIPMENT | Vaccum pump | 2 | 120000 ||
 157 | QC & R&D EQUIPMENT | Desiccator Glass | 1 | 4000 ||
 158 | QC & R&D EQUIPMENT | High Quality Laboratory water generation system (10 L /hr) Milli -Q water | 1 | 1450000 ||
 159 | QC & R&D EQUIPMENT | Sonicator | 1 | 180000 ||
 160 | QC & R&D EQUIPMENT | Vortex Shaker | 6 | 210000 ||
 161 | QC & R&D EQUIPMENT | Magnetic Stirrer with hot plate | 4 | 400000 ||
 162 | QC & R&D EQUIPMENT | LAF Vertical | 2 | 600000 ||
 163 | QC & R&D EQUIPMENT | Biosafety Cabinet | 2 | 1400000 ||
 164 | QC & R&D EQUIPMENT | Autoclave Double door | 1 | 1250000 ||
 165 | QC & R&D EQUIPMENT | Dry Heat Steriliser | 1 | 750000 ||
 166 | QC & R&D EQUIPMENT | Fume Hood (Table Top) | 1 | 450000 ||
 167 | QC & R&D EQUIPMENT | Deep freezer (-20 Degrees C) with Data acquisition | 1 | 400000 ||
 168 | QC & R&D EQUIPMENT | Air sampler | 1 | 300000 ||
 169 | QC & R&D EQUIPMENT | Particle counter | 1 | 2000000 ||
 170 | QC & R&D EQUIPMENT | Decontamination autoclave | 1 | 1000000 ||
 171 | QC & R&D EQUIPMENT | Microscope Binocular | 1 | 100000 ||
 172 | QC & R&D EQUIPMENT | Colony Counter | 1 | 150000 ||
 173 | QC & R&D EQUIPMENT | Microplate reader ELISA | 1 | 600000 ||
 174 | QC & R&D EQUIPMENT | ELISA Auto Analyzers (Roche / Abbott)* | 2 | 1000000 ||
 175 | QC & R&D EQUIPMENT | Microplate washer | 1 | 350000 ||
 176 | QC & R&D EQUIPMENT | TOC analyser (Grab Samples) | 1 | 2500000 ||
 177 | QC & R&D EQUIPMENT | Timers/ Stop watch | 5 | 10000 ||
 178 | QC & R&D EQUIPMENT | Ice flaking machine | 1 | 250000 ||
 179 | QC & R&D EQUIPMENT | Standard Weight Box | 1 Set | 1000000 ||
 180 | QC & R&D EQUIPMENT | Printer/scanner / xerox machine | 1 | 500000 ||
 181 | QC & R&D EQUIPMENT | Centrifuge table top | 1 | 450000 ||
 182 | QC & R&D EQUIPMENT | Size exclusion | L | 150000 ||
 183 | QC & R&D EQUIPMENT | Anion | L | 150000 ||
 184 | QC & R&D EQUIPMENT | Cation | L | 150000 ||
 185 | QC & R&D EQUIPMENT | Affinity | L | 150000 ||
 186 | QC & R&D EQUIPMENT | SS furniture racks | 25 | 250000 ||
 187 | QC & R&D EQUIPMENT | Computers / Laptops | 10 | 500000 ||
 188 | QC & R&D EQUIPMENT | Office Furniture | NA | 500000 ||
 189 | QC & R&D EQUIPMENT | Office Cabinets for Documents | 10 | 120000 ||
 190 | QA EQUIPMENT | Anemometer | 1 | 15000 ||
 191 | QA EQUIPMENT | Lux meter | 1 | 12000 ||
 192 | QA EQUIPMENT | Datalogger 32 channel | 1 | 400000 ||
 193 | QA EQUIPMENT | Paper shredder | 1 | 10000 ||
 194 | QA EQUIPMENT | Laminator | 1 | 10000 ||
 195 | QA EQUIPMENT | Printer/scanner / xerox machine | 1 | 500000 ||
 196 | QA EQUIPMENT | Computers / Laptops | 10 | 500000 ||
 197 | QA EQUIPMENT | Numbering Machine | 5 | 25000 ||
 198 | QA EQUIPMENT | Spiral Binding Machine | 1 | 15000 ||
 199 | QA EQUIPMENT | A3 +A4 Lamination Machine | 1 | 25000 ||
 200 | QA EQUIPMENT | Bi-Focal Glass | 1 | 80000 ||
 201 | QA EQUIPMENT | Tachometer | 2 | 6000 ||
 202 | QA EQUIPMENT | Clamp Meter/ Multi meter | 5 | 10000 ||
 203 | QA EQUIPMENT | Level Indicator | 2 | 8000 ||

204 | QA EQUIPMENT | Measuring TAPE | 2 | 1500 | |
205 | QA EQUIPMENT | Stop Watch | 5 | 50000 | |
206 | QA EQUIPMENT | Temperature recorders 24 Nos. (data Loggers) | 20 | 200000 | |
207 | QA EQUIPMENT | Office Cabinets for Documents | 20 | 240000 | |
208 | QA EQUIPMENT | Projector for conference / meeting room | 1 | 300000 | |
209 | QA EQUIPMENT | Office Furniture | NA | 700000 | |
210 | FINISHED GOODS STORAGE | Air curtain | 10 | 100000 | |
211 | FINISHED GOODS STORAGE | PVC pallets | 50 | 300000 | |
212 | FINISHED GOODS STORAGE | Storage racks SS | 50 | 1500000 | |
213 | FINISHED GOODS STORAGE | Hydraulic trolley | 1 | 250000 | |
214 | ETP | Biowaste reception tank 15000L | 1 | 4000000 | |
215 | ETP | Kill tank 1000L | 2 | | |
216 | ETP | Bar Screen | NA | | |
217 | ETP | Collection tank | NA | | |
218 | ETP | Equalization tank | NA | | |
219 | ETP | Neutralization tank | NA | | |
220 | ETP | Chemical coagulation | NA | | |
221 | ETP | Biological oxidation tank | NA | | |
222 | ETP | Filtration System | NA | | |
[SHEET: Heat treat_Reaction_Tank]
Sr.No. | Vessel | Working vol (L) | No. of units recommended
1 | Plasma pooling tank (refrigerated) | 2000 | 1 primary + 1 standby (or reserve)
2 | Caprylic acid reaction tank (jacketed) | 2000 | 1 (or 2 for staggered batches)
3 | S-D mixing tank (jacketed, sealed) | 2000 | 1 (with dedicated solvent bund)
4 | Clarification / settling tank or DAF feed | 2500 | 1
5 | Pasteurizer (container) | capacity to hold per-batch vials (900) | 1 tunnel / basket line sized to throughput
6 | Buffer / formulation tanks (jacketed) | 500–1,500 | 2–3 (for different buffers)
7 | Waste / solvent hold tank (bunded) | 1,000–2,000 | 1
| Back to Index | |
[SHEET: Fuel Consmpr]
Sr. No. | Fuel Type | Purpose / Area of Use | Approx. Annual Consumption | Unit
1 | Electricity | Cleanroom HVAC, process equipment (centrifuges, lyophilizers, pumps, lighting, DG auxiliaries, WFI generation, chillers, lab & office) | 3.5 – 4.0 million kWh / year | kWh
2 | HSD (High-Speed Diesel) | Stand-by DG sets (2 × 500 kVA) for emergency power | ≈ 25 – 30 KL / year | KL (≈ 25,000 – 30,000 L)**
3 | CNG / PNG | Boiler / hot-water generation for WFI plant, sterilization steam, HVAC re-heat coils | ≈ 150,000 – 180,000 Sm³ / year | Standard m³ (Sm³)
4 | LPG (optional small cylinders) | Laboratory instruments / flame sterilizers | ≤ 500 kg / year | kg
| Back to Index | | |
[SHEET: Water]
Sr.No. | Item | Litres (L / year) | Cubic metres (m³ / year) | Notes
| A. Water Inputs | | |
1 | Domestic water (506 ppl × 100 L/p/d) | 18469000 | 18469 | Toilets, canteen, housekeeping
| WFI (process) | 6205470 | 6205.47 | WFI for formulation, CIP, rinses
| Other process water (25% of WFI) | 1551368 | 1551.37 | CIP makeup, utility washes
| Extra process water (10% of plasma vol) | 41400 | 41.4 | Misc. process losses, small uses
2 | Process Water | 7798238 | 7798.24 |
| Total Water Input | 26267238 | 26267.24 | A = sum of above
| B. Water Outputs / Wastewater | | |
1 | Domestic sewage → STP (80% of domestic) | 14775200 | 14775.2 | Black/gray water to STP
2 | Process wastewater → ETP (99% of process) | 7720256 | 7720.26 | Process drains, WFI rejects, CIP effluent
| Total Wastewater Generated | 22495456 | 22495.46 | B = sum of above
| C. Balance / Losses / Product incorporation | | |
1 | Unaccounted losses / product incorporation / evaporation (A – B) | 3771782 | 3771.78 | ~14.36% of input — includes water in products, steam losses, sampling, evaporation, bleed, equipment hold-up
| Back to Index | | |
[SHEET: Non Haz Waste]
Sr. No. | Source / Process Area | Type of Non-Hazardous Waste | Approx. Quantity / Year
1 | ETP / STP treatment | Biological sludge (after dewatering) | 8 – 10 MT/year (wet)
2 | Process area & lab washing | Treated water (ETP + RO permeate) | ~12,000 m³/year
3 | Packaging line | Empty glass vials (non-contaminated rejects) | 0.5 – 1.0 MT/year (~50,000 pcs)
4 | Packaging & labeling | Paperboard cartons, labels, paper scrap | 1.5 – 2.0 MT/year
5 | Procurement / stores area | Corrugated boxes, stretch film, plastic wrapping | 1.0 – 1.5 MT/year
6 | Maintenance workshop | Metal scrap (iron, steel, aluminum parts) | 0.8 – 1.2 MT/year
7 | Utility & HVAC area | Spent filters (air filters, bag filters, prefilters) | 0.3 – 0.5 MT/year
8 | Admin & office areas | Office paper waste, packaging waste | 1.0 – 1.5 MT/year
9 | Canteen / pantry | Organic kitchen waste | 1.5 – 2.0 MT/year
10 | General housekeeping | Sweepings, mopping dust, non-contaminated PPE | 0.5 – 0.8 MT/year
11 | Cooling tower & utilities | Blowdown & scale sludge (non-toxic) | 1.0 – 1.5 MT/year
12 | Electrical systems | Cable cuttings, wire ends, plastic conduits | 0.2 – 0.3 MT/year
13 | Gardening & green area | Dry leaves, grass clippings | 0.3 – 0.5 MT/year
14 | Water purification & softener | Spent ion-exchange resins (non-toxic) | 0.2 MT/year
15 | Misc. activities | Empty HDPE / PET bottles (non-contaminated) | 0.1 – 0.2 MT/year
| Back to Index | | |
[SHEET: Raw Material]
Sr.No. | Item | Annual requirement (rounded) | Unit | Notes |
1 | Source human plasma | 414000 | L | Primary raw material (cold chain required) | Back to Index
2 | Water for Injection (WFI) | 6205000 | L | Assumed 15,000 L WFI / 1,000 L plasma |
3 | Sodium chloride (NaCl) | 10350 | kg | 25 kg / 1,000 L plasma × 413.7 |
4 | Sodium citrate / acetate | 2070 | kg | 5 kg / 1,000 L |
5 | Glycine | 4140 | kg | 10 kg / 1,000 L |
6 | Caprylic (Octanoic) acid | 3310 | kg | 8 kg / 1,000 L |
7 | Triton X-100 / Polysorbate-80 | 2070 | L | 5 L / 1,000 L |
8 | Tri(n-butyl) phosphate (TnBP) | 1241 | L | 3 L / 1,000 L |
9 | PBS / Tris buffer components | 8275 | kg | 20 kg / 1,000 L |
10 | Acids / NaOH (dilute) | 82740 | L | 200 L / 1,000 L (process neutralization) |
11 | Activated carbon (GAC) | 20685 | kg | 50 kg / 1,000 L |
12 | Filter aids (depth filter media) | 41370 | kg | 100 kg / 1,000 L |
13 | 0.2 µm sterile filters (cartridges) | 8300 | units | 20 units / 1,000 L |
14 | PVDF / sterile filter cartridges | 4140 | units | 10 units / 1,000 L |

15 | Chromatography resins (total across steps) | 20685 | kg | 50 kg / 1,000 L |
 16 | TFF membranes / UF-DF modules | 4140 | modules | 10 modules / 1,000 L |
 17 | Albumin stabilizer (Na-octanoate, etc.) | 4965 | kg | 12 kg / 1,000 L |
 18 | Fibrin sealant reagents (thrombin, substrates) | 2070 | kg | 5 kg / 1,000 L |
 19 | Preservatives (if used) | 4140 | kg | 10 kg / 1,000 L |
 20 | Nitrogen gas (compressed) | 206850 | m³ | 500 m³ / 1,000 L |
 21 | Sterile glass vials (all sizes) | 1037500 | units | Total vials + 5% spare (see note below) |
 22 | Rubber stoppers | 1037500 | units | Match vials (one per vial) |
 23 | Flip-off seals | 1037500 | units | Match vials |
 24 | Labels | 1040000 | units | one per vial + spares |
 25 | Cartons (≈10 vials/carton) | 98810 | cartons | Round up to supplier pack sizes |
 26 | Laboratory consumables (pipette tips, plates, etc.) | Estimate 4,000 | packs | Itemised procurement per lab list |
 27 | Spare filters / membranes (replacement spares) | +20% of above | — | |
 28 | Cold-chain storage (freezer/refrigeration units) | qty per plant | — | |
 | | Back to Index | | |

[SHEET: List of Products]

Product | Protein availability (gm) | Protein Recovery | Conc. / vial | No. of vials /batch | Total
(Considering

15 Batches / month) | Per Annum

Albumin (gm) | 40000 | 20000 | 20 GM | 900 | 13500 | 162000

Human IgG (gm) | 12500 | 6250 | 5 GM | 1000 | 15000 | 180000

Factor VIII in IU | 1000000 | 12500 IU | 250 IU | 50 | 750 | 9000

Factor IX in IU | 1000000 | 90000 IU | 600 IU | 150 | 2250 | 27000

Fibrinogen (gm) | 3000 | 1500 | 70 mg | 80 | 1200 | 14400

Thrombin in IU | 900000 | 90000 | 1000 IU | 1400 | 21000 | 252000

Based on vessel size, the batch size assumed is 1000 liters, on every alternate date, a new batch can be initiated, therefore 15 batches shall be processed per month (one batch/ day). | | | | |

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[SHEET: Hazardous Waste]

Sr. No. | Process / Source | Type of Hazardous Waste | Approx. Quantity / Year

1 | Solvent recovery & viral inactivation (TnBP + detergent process) | Spent solvent mixture (Tri-n-butyl phosphate + Triton X-100 / Polysorbate 80) | 1.0 – 1.5 KL / year

2 | Filtration & purification units | Used filter aids / depth filter media contaminated with protein or solvent | 0.8 – 1.0 MT / year

3 | Activated carbon beds (viral removal, deodorization) | Spent activated carbon with adsorbed organic residues | 0.5 – 0.7 MT / year

4 | Chromatography / TFF cleaning (CIP) | Residues of buffers, cleaning agents, NaOH / HCl neutralization sludge | 2.0 – 3.0 MT / year (dry solids)

5 | ETP operation | Sludge containing proteinaceous organic matter & trace solvents | 5.0 – 6.0 MT / year

6 | ETP / RO reject water | High-TDS / COD effluent (liquid) | ≈ 900 – 1,000 m³ / year

7 | Resin & membrane replacement | Spent chromatography resin, UF/DF membranes, PVDF filters | 0.5 – 1.0 MT / year

8 | Laboratory & QC activities | Contaminated lab glassware, pipette tips, chemical bottles | 200 – 250 kg / year

9 | Oil from DG sets, compressors, hydraulic systems | Used lubricating oil | 50 – 100 L / year

10 | DG set maintenance | Oil-soaked cotton waste, filters | 50 – 100 kg / year

11 | Production packaging & vial washing | Contaminated packaging (plastic, foil, paper) | 0.3 – 0.5 MT / year

12 | Spill & floor washings | Absorbent material, mopping waste, PPE, gloves | 0.5 MT / year

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Reagent Details:

Formula:

Sodium Citrate (M.Wt): 258.07.

Citric Acid (M.Wt): 192.124

Sodium Chloride(M.Wt):58.44

Preparation of FN-RR-12 working buffer from Stock buffer (FN-RR-08):

Calculations: For the Preparation of 20mM Sodium Citrate pH 7.4 + 2 M Sodium Chloride.

Stock Buffer (FN-RR-08):0.2M Sodium Citrate pH 7.4.

Formula:.

$C1 \times V1 = C2 \times V2$

C1 = concentration of stock

V1 = volume of stock you need to take

C2 = desired (working) concentration

V2 = final total volume of working solution

Applying the formula to prepare the working buffer:

$C1 \times V1 = C2 \times V2$

C1 = 0.2M Sodium Citrate pH 7.4.

V1 = volume of stock you need to take _____ L.

C2 = desired (working) concentration 20mM Sodium Citrate, pH 7.4

V2 = final total volume of working solution _____ L.

$V1 = (C2 \times V2) / C1$.

$C2 = 0.02M$; $V2 = 1.0 L$; $C1 = 0.2M$

$V1 = (0.02 \times 1.0) / 0.2$

$V1 = 0.1L$

Take 0.1 L (FN-RR-08) Stock buffer solution and dilute it with water to make up 1 L.

Calculations: 2M Sodium Chloride.

Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).

Calculation: To prepare 1 L of 2 M Sodium Chloride (MW = 58.44 g/mol)

Weight of solute(g)=2×58.44 x1=116.88 g.

Preparation Method:

Weigh 116.88 g of Sodium Chloride and dissolve in water to make a 1 L solution.

Preparation for different volumes:

To prepare _____ L of 2 M Sodium Chloride (MW = 58.44 g/mol).

Weight of solute(g)=2×58.44× _____ L= _____ g.

Weigh _____ gm of Sodium Chloride and dissolve in water to make a _____ L solution.

Preparation of working buffer (FN-RR-12):

Take 0.1 L (FN-RR-08) Stock buffer solution and add 2 M Sodium Chloride (116.88 gm). Dissolve it, and

then make up the final volume to 1 L with water.

Check and record the details:

Preparation Details:

Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry

Date

Sodium Citrate | | | |

Citric Acid | | | |

Sodium Chloride | | | |

Stock buffer (FN-RR-08) taken volume(L) | Sodium Chloride added(gm) | pH (7.4 ± 0.1) | Conductivity (175

± 0.5 ms/cm) | Storage Condition | Final volume prepared (L)

| | | |

Date of Preparation | Prepared By | Checked By (QC/Analyst)

| |

Reagent Details:

Formula & Calculations:

Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).

Calculation: To prepare 1 L of 0.2M sodium chlorolride (MW = 58.44g/mol)

Weight of solute(g)=1×11.688=11.688g.

Preparation Method:

Weigh 11.688 g of sodium chloride

To prepare 20mM Tris 1L solution from 1M Tris stock solution formula is $M_1(\text{Molarity of stock}) \times V_1$

(volume of stock)= $M_2(\text{Molarity of working}) \times V_2(\text{volume of working})$.

Take 20mL of 1M tris solution and add weighed amount of sodium chloride as required and dissolve in water to make a 1 L solution.

Preparation for different volumes:

To prepare _____ of 0.2M NaCl(MW = 58.44g/mol).

Weight of solute(g)=11.688x _____ L= _____ gm

Weigh _____ gm of NaCl.

To prepare _____ of 20mM Tris

Take 20mL x _____ L= _____ mL of 1M Tris

Dissolve pre weighed NaCl in 20mM Tris solution to make a _____ L solution.

Adjust the pH to 8 using concentrated HCl/NaOH solution

Check and record the details:

Preparation Details:

Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry

Date

||||

Final volume prepared | Weighed Quantity | pH (7.9±1.0) | Conductivity (19.5±0.5ms/cm) | Storage

Condition:

|||| RT

Date of Preparation | Prepared By | Checked By

||

1) Sewage & Effluent Treatment: Clearly mark the physical location, dimensions, and layout of the Effluent Treatment Plant (ETP) / Sewage Treatment Plant (STP). Provide structural calculations proving adequate capacity for processing waste.
Response: Physical location already marked in site layout.

2) Organic Waste Converter (OWC): Designate a dedicated, ventilated space for a mechanical OWC, stating its processing capacity in kg/day based on your expected workforce.
Response: Plasma: 20.4MT/year or 68T/day, LVP: 7.9MT/year or 26T/day

3) Process Flow & Water Balance Chart: Submit a detailed quantitative water balance diagram showing domestic and industrial consumption, recycling rates, and a commitment to Zero Liquid Discharge (ZLD).
Response:
Plasma:
LVP:
4)Hazardous Waste Management Plan: Since ethanol-based fractionation is utilized, submit storage, handling, and disposal plans for hazardous/chemical solvents
Response:
Plasma:
LVP:

Sr.No.	Item	Litres (L / year)	Cubic metres (m ³ / year)	Notes
A. Water Inputs				
1	Domestic water (506 ppl × 100 L/p/d)	1,84,69,000	18,469.00	Toilets, canteen, housekeeping
	WFI (process)	62,05,470	6,205.47	WFI for formulation, CIP, rinses
	Other process water (25% of WFI)	15,51,368	1,551.37	CIP makeup, utility washes
	Extra process water (10% of plasma vol)	41,400	41.4	Misc. process losses, small uses
2	Process Water	77,98,238	7,798	
	Total Water Input	2,62,67,238	26,267.24	A = sum of above
B. Water Outputs / Wastewater				
1	Domestic sewage → STP (80% of domestic)	1,47,75,200	14,775.20	Black/gray water to STP
2	Process wastewater → ETP (99% of process)	77,20,256	7,720.26	Process drains, WFI rejects, CIP effluent
	Total Wastewater Generated	2,24,95,456	22,495.46	B = sum of above
C. Balance / Losses / Product incorporation				
1	Unaccounted losses / product incorporation / evaporation (A – B)	37,71,782	3,771.78	~14.36% of input — includes water in products, steam losses, sampling, evaporation, bleed, equipment hold-up
	Category Type of Water/Steam	Estimated Quantity per day	Estimated Quantity per year	Purpose
	Product Vehicle Water for Injection (WFI)	~62,200 – 62,500 L	~14928000 – 15000000 L	Main solvent; added q.s. to reach final batch volume.
	Equipment Cleaning Purified Water (PW)	5,000 – 8,000 L	1200000 – 1920000 L	Initial rinses for mixing tanks, piping, and filling lines (CIP).
	Final Rinse Water for Injection (WFI)	2,000 – 3,000 L	480000 – 720000 L	Final rinse of product-contact surfaces to ensure a pyrogenicity.
	Sterilization Pure Steam (Clean Steam)	~1,500 – 2,500 kg	~360000 – 600000 kg	Used in autoclaves for terminal sterilization and SIP of lines.
	Utility Heating Industrial Steam	~5,000 – 7,000 kg	~1200000 – 1680000 Kg	Indirect heating for jacketed vessels and WFI generation.
	Cooling Process Water	10,000 L	2400000+ L	Non-contact cooling for jacketed vessels after sterilization.
	Wastewater Cleaning-in-Place (CIP) Effluent	5,000 – 8,000 L	1200000 – 1920000 L	Treatment in an Effluent Treatment Plant (ETP)
	Output Category Output Stream	Estimated Quantity per Day	Estimated Quantity per Year	Remarks
	Treated Water Recovery Treated Water for Reuse / Discharge	~3,500 – 6,800 L	~840,000 – 1,632,000 L	
	Recovered after primary and secondary treatment			
	Reject Water / Losses RO Reject / Evaporation / Handling Losses	~500 – 1,600 L	~120,000 – 384,000 L	
	Depends on treatment efficiency and recovery system			
	Sludge Generation ETP Sludge / Solid Waste	~20 – 50 kg	~4,800 – 12,000 kg	To be disposed through authorized waste handler
	Final Discharge Treated Effluent	As per discharge norms As per regulatory limits	Suitable for reuse/discharge after compliance testing	
	Sr. No. Process / Source	Type of Hazardous Waste	Approx. Quantity / Year	
1	Solvent recovery & viral inactivation (TnBP + detergent process)	Spent solvent mixture (Tri-n-butyl phosphate + Triton X-100 / Polysorbate 80)	1.0 – 1.5 KL / year	
2	Filtration & purification units	Used filter aids / depth filter media contaminated with protein or solvent	0.8 – 1.0 MT / year	
3	Activated carbon beds (viral removal, deodorization)	Spent activated carbon with adsorbed organic residues	0.5 – 0.7 MT / year	
4	Chromatography / TFF cleaning (CIP)	Residues of buffers, cleaning agents, NaOH / HCl neutralization sludge	2.0 – 3.0 MT / year (dry solids)	
5	ETP operation	Sludge containing proteinaceous organic matter & trace solvents	5.0 – 6.0 MT / year	
6	ETP / RO reject water	High-TDS / COD effluent (liquid)	≈ 900 – 1,000 m ³ / year	
7	Resin & membrane replacement	Spent chromatography resin, UF/DF membranes, PVDF filters	0.5 – 1.0 MT / year	
8	Laboratory & QC activities	Contaminated lab glassware, pipette tips, chemical bottles	200 – 250 kg / year	
9	Oil from DG sets, compressors, hydraulic systems	Used lubricating oil	50 – 100 L / year	
10	DG set maintenance	Oil-soaked cotton waste, filters	50 – 100 kg / year	
11	Production packaging & vial washing	Contaminated packaging (plastic, foil, paper)	0.3 – 0.5 MT / year	
12	Spill & floor washings	Absorbent material, mopping waste, PPE, gloves	0.5 MT / year	
	Hazardous Waste API Residue / Chemical Sludge	50 – 200 kg	12000 – 48000 kg	High-temperature incineration (>1,200°C).

Reagent Details:
Formula & Calculations:
Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).
Calculation: To prepare 1 L of 1.0N Sodium Hydroxide (MW = 40.0 g/mol)
Weight of solute(g)=1.0 ×40 = 40g.
Preparation Method:
Weigh 40 g of Sodium Hydroxide and dissolve in water to make a 1 L solution.
Preparation for different volumes:
To prepare _____ of 1.0N Sodium Hydroxide (MW = 40.0 g/mol).
Weight of solute(g)=1.0×40.0 x _____L= _____gm
Weigh _____gm of Sodium Hydroxide and dissolve in water to make a _____ L solution.
Check and record the details:
Preparation Details:
Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry
Date
Sodium Hydroxide | | | |
Final volume prepared | Weighed Quantity | pH (13.5±0.5) | Conductivity (85±5.0 ms/cm) | Storage
Condition:
| | | RT
Date of Preparation | Prepared By | Checked By
| |

Reagent Details:

Formula:

Sodium Citrate (M.Wt): 258.07.

Citric acid (M.Wt): 192.124

Preparation of FN-RR-05 stock buffer:

Preparation of 0.1M Sodium Citrate pH 6.5

Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).

Calculation: To prepare 1 L of 0.1M Citric Acid (MW = 192.124 g/mol)

Weight of solute(g)=0.1×192.124×1=19.21g.

Preparation Method:

Weigh 19.21g of Citric Acid and dissolve in water to make a 1 L solution.

Preparation for required volumes:

To prepare _____ of 0.1M Citric Acid (MW = 192.124g/mol).

Weight of solute(g)=0.1×192.124× _____ L= _____ g.

Weigh _____ gm of Citric Acid and dissolve in water to make a _____ L solution.

Calculations: To prepare the 0.1M Tris Sodium Citrate.

Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).

Calculation: To prepare 1 L of 0.1M Tris Sodium Citrate (MW = 258.07 g/mol)

Weight of solute(g)=0.1×258.07 ×1=25.80g

Preparation Method:

Weigh 25.80g of Tris Sodium Citrate and dissolve in water to make a 1 L solution.

Preparation for different volumes:

To prepare _____ of 0.1M Tris Sodium Citrate (MW = 258.07 g/mol).

Weight of solute(g)=0.1×258.07× _____ L= _____ g.

Weigh _____ gm of Tris Sodium Citrate and dissolve in water to make a _____ L solution.

Procedure for Preparation of FN-RR-05 stock

To prepare a pH 6.5 solution, 0.1 M Tris Sodium Citrate solution is titrated with 0.1 M citric acid so that the pH attains pH 6.5.

For Example, to prepare a 1 L solution, take 0.1 M Tri Sodium Citrate and adjust the pH with the 0.1 M Citric acid to get the pH 6.5 and make up the final volume to 1 L.

Check and record the details:

Preparation Details:

Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry

Date

Sodium Citrate | | | |

Citric Acid | | | |

0.1 M Tris Sodium Citrate Volume taken | 0.1 M Citric acid volume used to titrate | pH (6.3±0.2) |

Conductivity (2.9±0.1 ms/cm) | Storage Condition | Final volume prepared

| | | | RT |

Date of Preparation | Prepared By | Checked By (QC/Analyst)

| |

Reagent Details:
Formula & Calculations:
Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).
Calculation: To prepare 1 L of 0.2M Calcium chloride (MW = 110.98/mol)
Weight of solute(g)=1×22.196=22.196g.
Preparation Method:
Weigh 22.196 g of calcium chloride and dissolve in water to make a 1 L solution.
Preparation for different volumes:
To prepare _____ of 0.2M calcium chloride (MW = 110.98g/mol).
Weight of solute(g)=1×22.196x _____L= _____gm
Weigh _____gm of calcium chloride and dissolve in water to make a _____ L solution.
Check and record the details:
Preparation Details:
Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry
Date
| | | |
Final volume prepared | Weighed Quantity | pH (6.5±0.5) | Conductivity (40±0.5 ms/cm) | Storage
Condition:
| | | RT
Date of Preparation | Prepared By | Checked By
| |

A. ETP and STP water calculation for Plasma Plant (Lifenity)

Calculation for ETP and STP

Parameter Capacity UOM

Plasma processing per year 1,50,000 L

Number of days 365 Days

Batch size 1000 L

Number of batches per year 150 Nos

Raw water required per batch 2,50,000 L

Number of days per batch 5 Days

Raw water required per batch per day 50,000 L per batch per day

Raw water required annually 3,75,00,000 L per Year

Raw water required per day 1,02,740 L per day

Product Output UOM

Albumin 100 L per batch

IVIG 150 L per batch

FIX 2 L per batch

FVIII 0.2 L per batch

Thrombin 2 L per batch

Fibrinogen 0.5 L per batch

Total output 254.7 L per batch

Product Output per day 50.94 L per day per batch

QC 250 L per day

Waste water 1,02,939 L per day

Evaporation Losses 20,588 20 % of D

Input for ETP 82,351 L per day

Number of persons 506 No.

Water required per person per day 35 L/P/D

Input for STP 17,710 L per day

Calculation for ZLD

Parameter Recovery % Quantity UOM

RO permeate (I) 70% 57,646 L per day

MEE condensate (II) 20% 16,470 L per day

Losses (salts, sludge) (III) 10% 8,235 L per day

Page 1 of 3

Final water balance

Input Raw water (D) 1,02,740 L per day

Output after ZLD

Recycled 74,116 L per day

Losses 8,235 L per day

Recycled 72 %

% loss (salts, sludge) 8 %

Liquid discharge 0 L per day

Page 2 of 3

B. ETP and STP water calculation for LVP Plant (Lifenity)

Calculation for ETP and STP

Parameter Capacity UOM

Capacity per year 3,60,00,000 Units

Assume number of working days 365 Days

Unit Volume 0.5 L

Production per day 49,315 L per day

CIP or rinse 15,000 L per day

Total process water (WFI , Pure steam and PW) requirement 64,315 L per day

Raw water required per day 1,05,000 L per day

Raw water required annually 3,83,25,000 L per Year

Raw water required per day 1,05,000 L per day

Product Output UOM

Product 49,315 L per day

Product Output per day 49,315 L per day

QC 250 L per day

Waste water 55,935 L per day

Evaporation Losses 11,187 20 % of D

Input for ETP 44,748 L per day

Number of persons 400 No.

Water required per person per day 35 L/P/D

Input for STP 14,000 L per day

Calculation for ZLD

Parameter Recovery % Quantity UOM

RO permeate (I) 70% 31,324 L per day

MEE condensate (II) 20% 8,950 L per day

Losses (salts, sludge) (III) 10% 4,475 L per day

Final water balance

Input Raw water 1,05,000 L per day

Output after ZLD

Recycled 40,273 L per day

Losses (4,475 L per day

Recycled (%) 38 %

% loss (salts, sludge) 4 %

Liquid discharge 0 L per day

Page 3 of 3

PROCESS INSTRUCTIONS AND DATA CAPTURE POINTS –THROMBIN

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2 CONCENTRATION OF CHROMATOGRAPHY-10T ELUATE 6

3 PA-SF-001FILTRATION 8

CHROMATOGRAPHY-10T

Note: Activation/Load preparation to be carried out every time for 300ml equivalent plasma.

Take the Wash-II peak from Chromatography-8N Run of Human coagulation factor IX and store at 2-8°C until it is used for processing of Thrombin.

Check and record the following details in the table below.

Note: Estimate the protein content (mg/mL) by using the extinction coefficient 1.83 by using the formula $O.D280/1.83$, and multiply the obtained value by the total volume to calculate total protein.

ACTIVATION STEP

Note: Take a 500 mL plasma equivalent sample each time for activation.

To the collected wash-II sample, add 125 mL/ml of TH-RR-07 and incubate it for 5Hours $\pm$ 10 Mins at RT (20-25°C).

LOAD PREPARATION

After incubation, filter the sample through a mesh to remove clots formed during incubation and followed by a PA-PO-007 filter.

No. of filters used: _____.

Lower the conductivity of the above sample to 1.9 – 2.4 mS/cm by adding TH-RR-06 buffer to the sample under constant stirring.

Check and record the Chromatography - 10T load details:

Note: Estimate the protein content (mg/mL) by using the extinction coefficient 1.83 by using the formula $O.D. 280 / 1.83$.

Switch on the chromatography system. When the chromatography system is ready, open the UNICORN software.

Connect the column to the chromatography system and check the Column connections to the system. Check for instrument air connection at 6 bar to the system.

Record the following details:

Column height: 20 cm, Column volume:393ml,

HETP value: _____, Asymmetry: _____.

Column Wash

Pass 2 CVs of WFI through the column at 39.3 ml/min till the conductivity has dropped below 1.0 mS/cm.

Volume of WFI used: _____.

pH drop

Pass up to 2 CVs of TH-RR-05 through the column at a flow rate of 39.3 ml/min till the pH has dropped to 7.5 ± 0.5 . Record the following details:

Column Equilibration

Equilibrate the column with 3 CVs of TH-RR-02 at a flow rate of 39.3 ml/min.

Loading and Elution

Before loading, Auto Zero the absorbance (O.D280) on the chromatography system and switch on the record button and enter the run details and date.

Load the sample from step 1.4.3 at a flow rate of 39.3 ml/min.

Loading should be done in equivalent of 300L plasma on chromatography-TH-1column (NMT 65mL of plasma/mL of resin).

Pass 3 CVs of TH-RR-02at a flow rate of 39.3 ml/min to wash the column.

Pass up to 7 CVs of TH-RR-03through the column at a flow rate of 39.3 ml/min to elute the bound protein.

Start collecting the eluate when the absorbance rises to 100mAU and collect until the absorbance drops down to 100mAU. Record the following details of the eluate.

Note: Protein concentration is estimated by dividing the O.D. 280 value by the extinction coefficient 1.83.

Stabiliser Addition

Add 10 mL of Human Serum Albumin (HSA-20%) to each eluate and record the following details.

Cleaning

Pass 2 CVs of cleaning buffer TH-RR-04 at a flow rate of 39.3 ml/min.

Regeneration

Pass 2 CVs of TH-RR-09 through the column in reverse flow at 39.3 ml/min and hold the column for 30min in TH-RR-09 buffer.

Repeat steps 1.7 to 1.13 for Run 2 and record the results in the above tables.

Storage

Pass 2 CVs of TH-RR-10 through the column at a flow rate of 39.3 ml/min to store the column in TH-RR-10.

Volume of TH-RR-10 used: _____ L.

CONCENTRATION OF CHROMATOGRAPHY-10T ELUATE

Take the Chromatography - 10T Run 1 eluate sample from step 1.11 and carry out the concentration of the sample.

NWP calculation before concentration

Record the following details of the TFF cassettes connected in series horizontally to the TFF system.

No. of cassettes used: _____,

Cassette ID Nos.: _____.

Wash the cassettes with WFI till conductivity reaches below 0.1mS/cm and calculate NWP before concentration at a set TMP of ≤ 1.0 bar.

Volume of WFI used: _____ L.

Check and record the details as below:

Concentration of Eluate sample

After NWP calculation, flush the cassettes with TH-RR-03 till the conditions of the buffer are reached.

Volume of TH-RR-03 used: _____ L.

Irrespective of the eluate volume, concentrate the sample to 400mL $\pm$ 50 mL, at a set TMP of ≤ 1.0 bar.

Check and record the details as below

Repeat the steps from 2.3 to 2.6.3 for Run 2 and record the details in the above tables.

Pool the concentrated eluates from step 2.6.3.

Volume of sample: _____ (mL).

Note: Store concentrated eluate from Run 1 at 2-8°C until it is pooled with Run 2 concentrated eluate.

Diafilter the pooled concentrated sample with 5 dia volumes of TH-RR-08 at a set TMP of ≤ 1.0 bar and concentrate the sample to 500 $\pm$ 100 mL.

Volume of TH-RR-08 used: _____.

pH: _____, Conductivity: _____ mS/cm.

O.D280/600: _____, Final volume (mL): _____.

Collect 2 mL of sample and send to QC for in-process QC analysis.

Record the obtained results as per the QC report :

Activity: _____, BET: _____.

Add and adjust Human Serum Albumin (HSA-20%) to a final concentration of 25 mg/mL to the diafiltered sample from step 2.6.6

Volume of diafiltered sample: _____ mL.

Volume of HSA-20% added before concentration in step 1.11.1: _____.

Volume of HSA-20% added after diafiltration: _____.

Regeneration

Flush the cassette with WFI till conductivity reaches below 0.1mS/cm and then with TH-RR-09at 40 - 45°C and recirculate TH-RR-09 for 30 to 45 min at a set TMP of ≤ 1.0 bar.

Volume of TH-RR-09 used: _____ L.

Post-NWP calculation

Repeat the step as in 2.2 and record the details of NWP in the table below. If NWP is <120 LMH/Bar, repeat the cleaning procedure.

Storage

Flush the cassettes with 2L of TH-RR-10 and finally store the cassettes inTH-RR-10. Remove the cassettes from the system, immerse in TH-RR-10 and store at 2-8°C.

Volume of TH-RR-10 used: _____ L.

PA-SF-001FILTRATION

Assemble the PA-SF-001 filter with sterilised inlet and outlet silicone tubing.

Check and record the following filter details:

No. of filter(s) used: _____.

Equilibrate the filter by passing 0.5L TH-RR-08.

Volume of TH-RR-08 passed: _____ L.

Filter the sample through a filter using a peristaltic pump under LAFU.

Collect 10 mL sample and send to QC for Drug Substance testing.

Check and record the following details:

Store the above sample at - 20°C from step 3.6 in a 500 mL Nalgene bottle till it is released for filling.

Sample Label ID: _____.

Details | Run

Volume of Wash-II peak |

pH |

Conductivity |

Visual Appearance |

O.D280/600 |

Total Protein (gms) |

Hold time of wash-II peak sample from Chr-8N run. |

Details | Run 1 | Run 2

Volume of TH-RR-07 added (L) ||

Incubation time ||

Visual Appearance after incubation ||

Details | Run 1 | Run 2

Volume of TH-RR-06 used(L) ||

Details | Run 1 | Run 2

Total Volume (L) ||

O.D280/600 ||

pH ||

Conductivity (mS/cm) ||

Protein Conc. (mg/mL) ||

Details

Details | Run 1 | Run 2

Volume of TH-RR-05 used (L) ||

Final pH of the column ||

Details | Run 1 | Run 2

Final pH of the column ||

Conductivity (mS/cm) ||

Volume of TH-RR-02 used (L) ||

Details | Run 1 | Run 2

Load Volume (L) ||

Details | Run 1 | Run 2

Volume of TH-RR-02 used (L) ||

Details | Run 1 | Run 2

Volume of TH-RR-03 used (L) ||

Details | Run 1 | Run 2

Volume of Eluate (L) ||

pH ||

Conductivity (mS/cm) ||

O.D280/600 ||

Protein Concentration (mg/mL) ||

Total Protein (gms) ||

% Recovery ||

Details | Eluate 1 | Eluate 2

Volume of 20% HSA added (mL) ||

Details | Run 1 | Run 2

Volume of TH-RR-04 used (L) ||

Details | Run 1 | Run 2

Volume of TH-RR-09 used (L) ||

PF(bar) | PR (bar) | PP(bar) | Flow (L/hr) | TMP (bar) | Flux (LMH) | NWP (LMH/bar)

|||||

Details | Permeate Vol. (L) | PF (bar) | PR (bar) | PP (bar) | Flow (L/hr) | TMP (bar) | Flux (LMH) | Permeate O.D280

Run-1 |||||||

Run-1 |||||||

Run-1 |||||||

Run-2 |||||||

Run-2 |||||||

Run-2 |||||||

Details | Run 1 | Run 2

Volume of Retentate (mL) ||

O.D280/600 ||

PF (bar) | PR (bar) | PP (bar) | Flow (L/hr) | TMP (bar) | Flux (LMH) | NWP (LMH/bar)

|||||

Details | Before Filtration | After Filtration
Volume (L) | |
O.D280/600 | |
pH | |
Conductivity (mS/cm) | |
Activity (IU/mL) as per QC report. | |

Reagent Details:

Formula:

Sodium Citrate (M.Wt): 258.07.

Citric Acid (M.Wt): 192.124

Sodium Chloride(M.Wt):58.44

Preparation of FN-RR-15 working buffer from Stock buffer (FN-RR-08):

Calculations: For the Preparation of 11.4mM Sodium Citrate pH 7.4 + 145m M Sodium Chloride.

Stock Buffer (FN-RR-08):0.2M Sodium Citrate pH 7.4.

Formula:.

$C1 \times V1 = C2 \times V2$

C1 = concentration of stock

V1 = volume of stock you need to take

C2 = desired (working) concentration

V2 = final total volume of working solution

Applying the formula to prepare the working buffer:

$C1 \times V1 = C2 \times V2$

C1 = 0.2M Sodium Citrate pH 7.4.

V1 = volume of stock you need to take _____ L.

C2 = desired (working) concentration 0.114mM Sodium Citrate, pH 7.4

V2 = final total volume of working solution _____ L.

$V1 = (C2 \times V2) / C1$.

$C2 = 0.0114M$; $V2 = 1.0 L$; $C1 = 0.2M$

$V1 = (0.0114 \times 1.0) / 0.2$

$V1 = 0.057L$

Take 0.057 L (FN-RR-08) Stock buffer solution and dilute it with water to make up 1 L.

Calculations: 0.145M Sodium Chloride.

Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).

Calculation: To prepare 1 L of 0.145 M Sodium Chloride (MW = 58.44 g/mol)

Weight of solute(g)=0.145×58.44 ×1=8.473 g.

Preparation Method:

Weigh 8.473 g of Sodium Chloride and dissolve in water to make a 1 L solution.

Preparation for different volumes:

To prepare _____ L of 0.145 M Sodium Chloride (MW = 58.44 g/mol).

Weight of solute(g)=0.145×58.44× _____ L= _____ g.

Weigh _____ gm of Sodium Chloride and dissolve in water to make a _____ L solution.

Preparation of working buffer (FN-RR-15):

Take 0.057 L (FN-RR-08) Stock buffer solution and add 0.145M Sodium Chloride (8.473 gm). Dissolve it, and

then make up the final volume to 1 L with water.

Check and record the details:

Preparation Details:

Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry

Date

Sodium Citrate | | | |

Citric Acid | | | |

Sodium Chloride | | | |

Stock buffer (FN-RR-08) taken volume(L) | Sodium Chloride added(gm) | pH (7.4± 0.1) | Conductivity (16.0

± 1ms/cm) | Storage Condition | Final volume prepared (L)

| | | RT |

Date of Preparation | Prepared By | Checked By

| |

Reagent Details:

Formula & Calculations:

Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).

Calculation: To prepare 1 L of 0.1M Tri-Sodium Citrate (MW = 258.06g/mol)

Weight of solute(g)=0.1×258.06 x 1=25.806g.

Preparation Method:

Weigh 25.806 g of Tri-sodium citrate and dissolve in water to make a 1 L solution.

Preparation for required volumes:

To prepare _____ of 0.1M Tri -Sodium Citrate (MW = 258.06g/mol).

Weight of solute(g)=0.1 x 258.06 x _____L= _____gm

Weigh _____gm of Tri-sodium citrate and dissolve in water to make a _____ L

solution.

Check and record the details:

Preparation Details:

Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry

Date

Tri-Sodium Citrate | | | |

Final volume prepared | Weighed Quantity | pH (8.9 ± 0.1) | Conductivity (23 ± 0.2 ms/cm) | Storage

Condition:

| | | | RT

Date of Preparation | Prepared By | Checked By

| |

SHAFT

3.30X2.45
DISPATCH BAY RECEIVING BAY RECEIVING BAY DISPATCH BAY
FIREMAN

LIFT
LIFT
2.80X2.60
5.00 M WIDE CORRIDOR
62.5X5.00
UP

LADIES TOILET

2.94 M WIDE CORRIDOR
3.30 x 4.35

WC WC WC
1.10X 1.10X 1.10X
1.70 1.70 1.70

6.77 M WIDE CORRIDOR
3600.000

2000.000
WC WC WC
1.10X 1.10X 1.10X
1.70 1.70 1.70

GENTS TOILET
3.30 x 4.35

SHAFT SHAFT

LIFT LIFT
2.10X 2.10X

3.24 M WIDE PASSAGE
2.00 2.00
FD
UP
FD
DIALYZER FD
UP
FD

48.54 x 91.34

3300.000

2.80 M WIDE PASSAGE

LADIES TOILET
3.30 x 4.35

WC WC WC
1.10X 1.10X 1.10X
1.70 1.70 1.70

WC WC WC
1.10X 1.10X 1.10X
1.70 1.70 1.70

GENTS TOILET
3.30 x 4.35
6.77 M WIDE CORRIDOR

6.77 M WIDE CORRIDOR
6.77X212.59

6.77X212.59
GENTS VISITOR FEMALE DOCUMENT JANITOR
SHAFT OFFICE 3 OFFICE 4 ROOM SHAFT
WAITING ROOM LOCKER LOCKER LOCKER ROOM
LIFT
4.00x5.00 4.70x5.00 4.70x5.00 2.00x5.00
LIFT 4.67x5.00 ROOM ROOM ROOM LIFT
5.90x5.00 6.00x5.00
2.10X 6.00x5.00 2.10X

3.24 M WIDE PASSAGE
2.00 2.00
FD FD FD FD
6D

RECEPTION 4.72x1.80 CORRIDOR
UP 36.7x1.80 UP
7.00x12.0

PRIMARY VISITOR
AIR LOCK AIR LOCK VISITOR OFFICE 1 OFFICE 2
FEMALE PRIMARY FEMALE CHANGE 1
AIR LOCK PRIMARY MALE MALE CHANGE 1 FEMALE 1.80x5.15 4.30x5.10 4.30x5.10
CHANGE 1 CHANGE VISITOR CHANGE 2.55x5.10
MALE CHANGE 3.30x5.10 3.40x5.10 3.19x5.10 3.64x5.10 3.05x5.10 2.70x5.10
4.70x5.10 4.00x5.10
8D

6D

DE
CORRIDOR
45.8x2.85

DPB
DPB DPB WASTE OUT
LAF DHS
3 WAY
6D

6D

DECONTAMINATION WASHING STERILIZATION
AUTOCLAVE

CHEMICAL DAY STORE
LAF

6.98X3.04 BUFFER TANK STORAGE-2 CIP SIP COLD ROOM 3.52x4.67 3.11x4.67 3.401x4.67
5.60x7.09
6D

PREPARATION 6.74x7.09 4.40x7.09
COOL ONE
4.00x7.09 2.75x7.09
DPB DPB
AL GOWN AL 7.29x1.80
2.10X3.40 2.47X3.40 2.20X3.40 WASTE AL AL
4.13x2.34 3.40x2.34
GS+HH

GS+HH

IN
LENAP LORTNOC LENAP LORTNOC
2.50x2.34
6D 6D 6D

AL DE GOWN AL
DE
2.10X3.40 2.47X3.40 2.20X3.40 CORRIDOR
36.0x2.85

D6 D6 D3
LAF
D6 D6
DPB AKTA-1 TFF
NANO
FILTER MH
JANITOR
1.80x2.00
LADIES TOILET
AL GOWN AL 1000 L
3.30 x 4.35
2.10X4.76 2.47X4.76 2.20X4.76 BULK FILTERATION
LAF

6.54X7.29 500 L
WC WC WC
6.77 M WIDE CORRIDOR

1.10X 1.10X 1.10X
2.80 M WIDE PASSAGE

MH

1.70 1.70 1.70
LAF

STAGE 9 N
MH

DE 500 L MH

50 L
WC WC WC AL GOWN AL LAF SL
MH

5.10X15.6 CHROMO SL
STAGE 13 E

1.10X 1.10X 1.10X 2.10X4.76 2.47X4.76 2.20X4.76 MH

COLUMN 250 L
1.70 1.70 1.70 600 L 600 L STAGE 10 T MH

500 L 7.48X13.5
STAGE 11 F
MH

SL
MH

600 L MH

200 L 5.63X2.05 250 L
AKTA-1

GENTS TOILET
STAGE 6 I
MH

3.30 x 4.35

COLUMN PACKING 14.7X18.0 250 L MH

250 L
MH

2000 L MH

AND SKIDS
MH

200 L

6.54X10.6 150 L 250 L
COOL ONE
MH MH

MH
1000 L MH
100 L MH

6.98x5.53 500 L
150 L MH

300 L
MH

MH

100 L MH

1000 L
MH

100 L
6770.000
150 L 300 L
LAF
DPB LAF
MH

MH

CHROMO MH

AKTA-1
100 L COLUMN

DHS
AUTOCLAVE
D6

500 L 500 L CHROMO
250 L 250 L 150 L 100 L COLUMN
6D

CHROMO MH
MH MH

AL MH MH
MH MH
MH

MH
COLUMN
CORRIDOR
2.62x71.4
1.50x6.88
STERILIZATION CORRIDOR
2000 L

D6

5.38x6.85
CLAF

2500 L 2500 L 2.62X74.2
CORRIDOR
6D

1.80X55.7
PRE - VIRUS CONTROLLED CORRIDOR

1000 L
TFF

MH

COLUMN PACKING
DPB
AND SKIDS MH

WASHING 100 L
7.64X8.86
AKTA-1

AL 5.38x3.00 STAGE 7A
AKTA-1

1.50x4.42 500 L
13.6X16.1 MH

STAGE 5 I A
D6

DPB LADIES TOILET
WASTE IN MH

500 L 3.30 x 4.35
250 L
11.1X19.0
TFF

MH

300 L
MH
2.00X32.2

WC WC WC
DECONTAMINATION 500 L 1.10X 1.10X 1.10X
250 L 1.70 1.70 1.70
OUT 5.38x5.00 MH
MH
MH

WASTE
D6

1.50x3.54 500 L

DPB STAGE 5 I B 1500 L 250 L
MH
MH

100 L
7.64X7.20 MH

250 L 250 L 250 L
SL
600 L 500 L WC WC WC
100 L 100 L
MH

MH MH MH 100 L MH
MH
1.10X 1.10X 1.10X
100 L MH MH MH

1.70 1.70 1.70

CIP SIP
MH

1500 L
SL
MH MH

MH MH MH

6.98x5.00 1000 L 100 L
MH
MH

500 L 500 L 1000 L 1000 L 1000 L
MH

GENTS TOILET
3.30 x 4.35
MH
6D

MH

1500 L
50 L 1500 L
GS+HH

GS+HH

SL
LENAP LORTNOC LENAP LORTNOC
2500 L
MH

1500 L

MH
250 L 250 L 250 L CHROMO
250 L
DPB MH MH MH COLUMN MH

WASTE OUT
6D

BAG CUTTING
POOLING MH
MH MH

MH

MH MH
STAGE 4 F
7.64X9.80 150 L 500 L
13.6X15.8
6D

500 L
SL 1000 L
TANK STORAGE 1000 L 1500 L 1500 L

6.98x8.58
D6

MH

STAGE 12 E
MH

250 L
MH

STAGE 1 F 500 L
CHROMO
COLUMN
11.1X10.6 250 L
300 L 300 L MH
TFF
D6

AKTA-1

6D
MH MH

CLAF

2500 L
6D

THAWING PLASMA
2000 L 2000 L
7.64X3.61
AKTA-1

MH

BUFFER DPB
500 L
250 L 250 L 300 L 300 L MH MH

PREPARATION MH
CHROMO
COLUMN
MH MH MH
MH MH

CHROMO
COLUMN
6.98x6.00 D6 D6 D6
6D

DE

SHAFT CORRIDOR SHAFT
DPB 34.1X1.80
LIFT D6 MH MH D6 D6 LIFT
2.10X 2.10X
MH MH MH MH

CHROMO MH
MH
MH MH MH MH

COLUMN
100 L 100 L 100 L 100 L MH

100 L 100 L CENTRIFUGE
6D

2.00 200 L 250 L 300 L 300 L 500 L 500 L 2.00
1500 L
3.24 M WIDE PASSAGE

3.24 M WIDE PASSAGE

FD FD FD FD
UP
PLASMA DAY STAGE 3 F UP
STAGE 2 F 12.1X11.2
STORE STAGE 8 N
15.8X11.2
5.84X11.2
6.98x10.7
AKTA-1

AKTA-1

UP
CHROMO MH

COLUMN

JANITOR SL
5.40x1.80 600 L 2000 L
1500 L 1500 L 1500 L 1500 L

300 L 300 L MH

DPB
MH MH MH
MH

100 L 100 L 100 L CHROMO CHROMO
MH
TFF MH

COLUMN COLUMN AKTA-1
MH MH MH

EMER.GLASS EMER.GLASS
CORRIDOR
43.3x1.80
6.77 M WIDE CORRIDOR

PLASMA STORAGE PLASMA STORAGE PLASMA STORAGE
~ 1000 L ~ 1000 L PLASMA STORAGE
~ 1000 L ~ 1000 L
~ 70 KL GV ~ 70 KL GV ~ 70 KL GV
9.59x11.5m 9.02x11.5m ~ 70 KL GV
9.35x11.5m 9.05x11.5m
CORRIDOR PLASMA STORAGE CORRIDOR
2.62x15.2 ~ 1000 L 2.62x15.2
~ 70 KL GV
5.98x11.5m

LADIES TOILET
3.30 x 4.35

D3 D3 D3 D3 D3
WC WC WC
CORRIDOR
2.80 M WIDE PASSAGE

1.10X 1.10X 1.10X
1.70 1.70 1.70 43.2x1.80
D6
CORRIDOR

28.1X1.80

D3 D6
WC WC WC DPB RLAF
D3

AIR LOCK-1
1.10X 1.10X 1.10X COLD ROOM -3 COLD ROOM -2 COLD ROOM -1 ICE PACK
1.70 1.70 1.70
1.50 X 2.18 RM QUARANTINE BAG BAR CODING, WASHING
JANITOR
PLASMA COLD ROOM -4
AREA QUARANTINE
MINI POOLS 4.62X7.34 2.49X7.34
6D

2.60X2.72
2.30X2.72
1.45X2.72
4.44X7.34 4.12X7.34 4.45X7.34
DISPENSING ROOM
3.40 X 4.44
3D D3 D3 D3

GENTS TOILET
3.30 x 4.35 CHANGE ROOM
1.50 X 2.18 5.85X6.55
3D

5.00X2.72
11.5X4.30
CORRIDOR
1.80X12.0
D3
3D

SD SD SD SD

DPB
6D

D3
APPROVED RM AIR LOCK-1
2.00X1.21
DPB D3 3D D3 D6 STORE 10.4X6.49
3.65X4.49 2.64X2.60 AIR LOCK-1 DPB
1.87X1.40 OFFICE 4.70X4.08
8.65X7.54 3D
2.23X6.50
CHANGE ROOM
2.00X1.80 3D
3.00X3.55
CHANGE ROOM

DE-DUSTING
1.87X1.40
REJECT SEROLOGY PLASMA
RECEIPT
D3

D3 TUNNEL
PLASMA ROOM TESTING 3D D3
STORE CHANGE
D3 SAMPLING
1.76X 2.90
2.26X4.95 3.00X 4.95 2.40X 4.95 ROOM
RM 4.70X2.37
SAMPLING 3.68X 2.05 RECALLED REJECT
RM FG DISCPATCH
3.16X2.31
RECEIPT GOODS ROOM FG IN
1.07X3.11 2.30X2.35 2.30X 2.35
2.64X2.77
RLAF RLAF 6D D6 D6
D3 D3

D6 D3

5.00 M WIDE CORRIDOR
3.30X2.45

6D

D6

48.9X5.00
LIFT

DISPATCH BAY RECEIVING BAY RECEIVING BAY DISPATCH BAY
40' CONTAINER YARD
CONTAINER
CONTAINER

CONTAINER

CONTAINER
12.19x2.44x2.59
12.19x2.44x2.59

12.19x2.44x2.59

12.19x2.44x2.59

GROUND FLOOR CLIENT :- LIFENITY-LARGE VOLUME PARENTERALS

DESIGN &
PROJECT BY :-BIOBRIDGE HEALTHCARE SOLUTION PVT.LTD.
PREPARED A.K
ADDRESS :-13, Rachna Blossom, Jagdishnagar,
Aundh, Pune -411007, Maharastra, India. CHECKED G.K.

DATE 01-04-2026
TITLE- CONCEPTUAL LAYOUT-GF REVISION R0
SCALE 1:1

SIZE A0 DWG.NO 01

[SHEET: Reponse]		
S. No.	Query	Response
1	Pls share the following Layout Plan , Manpower resources required, Height of the Plasma Ground Floor , 1st Floor and LVP 1st Floor height required considering machine height, HVAC and access requirements.	
2	Confirmation of machine list with make, model, size, and weight. Attached excel for make, size and weight, model can be confirmed after confirmation of vendor	
3	Final machine layout and required working clearances. Layout attached, working clearance should be 2 meters	
4	Floor-to-floor height confirmation as per machinery and HVAC requirements. As per machinery 3.5 meters, HVAC requirement 6 meters	
5	Verification of internal room sizes as per operational needs. As per layout	
6	Confirmation of surrounding space for vehicular movement and loading/unloading access.	
7	Utility requirements (water inlet/outlet, drainage, power load, and waste disposal). Water inlet = 87557 L/day, Water outlet = 74984 L/day, Working load 4143.37 KVA and stand by load 1068.24 KVA, waste = 74984 L/day, ETP/STP: 900-1000 m3/year	
8	Manpower resources required 508 as per DPR submitted earlier	

Reagent Details:
Formula & Calculations:
Formula: Use the formula to calculate the molarity:
 $Molarity (M) = (Per\ cent\ by\ Weight / 100) * (Density\ in\ g/mL) * 1000 / Molar\ Mass.$
Example: Molarity of Glacial Acetic Acid
Molar Mass of CH_3COOH : ~60.05 g/mol
Density: ~1.05 g/mL
Purity: 100%
Acetic Acid (MW = 60.05 g/mol).
Calculation:
 $Molarity = (100 / 100) * 1.05\ g/mL * 1000\ mL/L / 60.05\ g/mol \approx 17.5\ M.$
Calculation: To prepare 1 L of 1.0 M.
Applying the formula to prepare the working buffer:
 $C1 \cdot V1 = C2 \cdot V2$
 $C1 = 17.5\ M$ acetic acid.
 $V1$ = volume of stock needed to take.
 $C2$ = desired (working) concentration 1.0 M acetic acid.
 $V2$ = final total volume of working solution for one litre.
 $V1 = (C2 \times V2) / C1.$
 $C2=1.0\ M ; V2= 1.0\ L; C1=17.5\ M$
 $V1 = (1.0 \times 1.0) / 17.5$
 $V1 = 0.057\ L$
Take 0.057 L of the Stock solution of acetic acid and add it to a volumetric flask, making up the volume to 1 L.
To prepare various volumes:
 $V1$ = volume of stock needed to take _____ L.
 $C2$ = desired (working) concentration 1.0 M acetic acid.
 $V2$ = final total volume of working solution _____ L.
 $V1 = (C2 \times V2) / 17.5.$
 $V1=$ _____ L.
Take $V1$ _____ L of the Stock solution of acetic acid and add it to a volumetric flask, making up the final volume _____ L.
Preparation Details:
Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry Date
| | | |
Final volume prepared | Measured Quantity | pH (2.4±0.1) | Conductivity (40.0±5.0 ms/cm) | Storage Condition | Reagent Record
| | | RT |
Date of Preparation | Prepared By | Checked By
| |

[SHEET: Equipment capacities]
Equipment | Capacities to be installed |
| Plasma | LVP
Boiler | 3000 Kg/H | 5000 Kg/H
Chiller | 300 TR | 300 TR
Process chiller | 100 TR | 100 TR
Air compressor | 230 CFM |
Soft water generation | 20000 LPH generation, 50KL storage | 20000 LPH generation, 50KL storage
Purified water RO water generation and distribution,Storage | PW-12000 LPH generation , 20KL storage |
PW-15000 LPH generation, 30KL storage
Water for Injection generation and distribution,storage | 3500 LPH generation, 20KL storage | (5000 LPH
generation, 20KL storage)*2
Pure steam generation | 1000 Kg/H | 1000 Kg/H
Nitrogen | 50 M3 |
Co2 | 20 M3 | NA
Electrical transformer requirement and distribution and UPS power | 1400 KVA * 2 | 1000 KVA * 2
DG | 1300 KVA * 2 | 1000 KVA * 2
Hot water generation pump and distribution | 100000 Kcal/Hr |

[SHEET: GF_RDS]
ROOM DATA SHEET FOR LIFENITY |||||
|| Room Details ||||| Fire Detection & Protection System |||||
Room No | Room Name | Length (mtr) | Width(mtr) | Height(mtr) | Area (M²) | Volume | Temp.(oC) | RH(%) |
HVAC System | HVAC Type | Occupancy | Grade | Differential P | Fire Alarm System | Fire Sprinkler System
| Fire Extinguisher System | GPS Clock | LAN Port | Telephone Port | Door Interlock | Eye Washer | Body
Washer | Floor Finish | Electrical Lux level
| Reception Room | 8.3 | 12 | 3 | 99.60000000000001 | 298.8 | NA | NA | AC | NA | NA | NA | NA | Y | Y |
NA | NA | Y | Y | NA | NA | NA | Vitrified Tiles | 250
| Waiting Room | 4.67 | 5 | 3 | 23.35 | 70.05000000000001 | NA | NA | AC | NA | NA | NA | NA | Y | Y | NA
| NA | NA | NA | NA | NA | NA | Vitrified Tiles | 250
| Passage | 4.72 | 1.8 | 3 | 8.496 | 25.488 | NA | NA | NA | NA | NA | NA | NA | Y | Y | NA | Y | Y | NA
| NA | NA | NA | Vitrified Tiles | 250
| Corridor | 40.4 | 1.8 | 3 | 72.72 | 218.16 | NA | NA | NA | NA | NA | NA | NA | Y | Y | NA | Y | Y | NA
| NA | NA | NA | Vitrified Tiles | 250
| Gents Toilet Room | 9.94 | 4.9 | 3 | 48.706 | 146.118 | NA | NA | NA | Exhaust | NA | NA | NA | Y | NA
| NA | NA | NA | NA | NA | NA | NA | Vitrified Tiles | 250
| Gents locker Room | 6 | 5 | 3 | 30 | 90 | NA | NA | NA | Exhaust | NA | NA | NA | Y | NA | NA | NA | NA
| NA | NA | NA | NA | Vitrified Tiles | 250
| Visitor Locker Room | 5.9 | 5 | 3 | 29.5 | 88.5 | NA | NA | NA | Exhaust | NA | NA | NA | Y | NA | Y |
NA | NA | NA | NA | NA | NA | Vitrified Tiles | 250
| Female Locker Room | 6 | 5 | 3 | 30 | 90 | NA | NA | NA | Exhaust | NA | NA | NA | Y | NA | Y | NA | NA
| NA | NA | NA | NA | Vitrified Tiles | 350
| Ladies Toilet | 6.3 | 4.9 | 3 | 30.87 | 92.61 | NA | NA | NA | Exhaust | NA | NA | NA | Y | NA | Y | NA
| NA | NA | NA | NA | Vitrified Tiles | 250
| Female Change Room-1 | 3.4 | 5.1 | 3 | 17.34 | 52.019999999999996 | NA | NA | AHU | RECIRCULATION | 2 |
CNC | NA | Y | NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| Primary Female change Room | 3.19 | 5.1 | 3 | 16.269 | 48.806999999999995 | 22.5±2.5 °C | NMT 60% | AHU
| RECIRCULATION | 2 | D | 15 | Y | NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| Air lock Female | 3.64 | 5.1 | 3 | 18.564 | 55.692 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | 2 |
D | 20 | Y | NA | Y | NA | NA | NA | Y | NA | NA | Epoxy | 250
| Visitor Change Room | 2.75 | 5.1 | 3 | 14.024999999999999 | 42.074999999999996 | NA | NA | AHU |
RECIRCULATION | NA | CNC | NA | Y | NA | Y | NA | NA | NA | Y | NA | NA | Epoxy | 250
| Primary Visitor Change Room | 3.7 | 5.1 | 3 | 18.87 | 56.61 | 22.5±2.5 °C | NMT 60% | AHU |
RECIRCULATION | NA | D | 15 | Y | NA | Y | NA | NA | NA | Y | NA | NA | Epoxy | 250
| Air Lock Visitor | 3.05 | 5.1 | 3 | 15.554999999999998 | 46.664999999999999 | 22.5±2.5 °C | NMT 60% |
AHU | RECIRCULATION | NA | D | 20 | Y | NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| Male Change-1 | 3.3 | 5.1 | 3 | 16.83 | 50.489999999999995 | NA | NA | AHU | RECIRCULATION | 2 | CNC |
NA | Y | NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| Primary Male Change Room | 4 | 5.1 | 3 | 20.4 | 61.199999999999996 | 22.5±2.5 °C | NMT 60% | AHU |
RECIRCULATION | 2 | D | 15 | Y | NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| Air Lock Male | 4.7 | 5.1 | 3 | 23.97 | 71.91 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | 2 | D |
20 | Y | NA | NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| Corridor | 49 | 2.25 | 3 | 110.25 | 330.75 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | 2 | D | 25 |
Y | NA | NA | Y | Y | NA | NA | NA | NA | Epoxy | 250
| Cold Room | 5.87 | 7.5 | 3 | 44.025 | 132.075 | 2-8 C | NA | Cold room unit | NA | NA | NA | NA | Y |
NA | NA | NA | Y | Hooter | NA | NA | NA | Epoxy | 250
| Corridor | 74.2 | 1.8 | 3 | 133.56 | 400.68 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | NA | CNC |
NA | Y | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 250
| Passage | 5.15 | 1.8 | 3 | 9.270000000000001 | 27.810000000000002 | 22.5±2.5 °C | NMT 60% | AHU |
RECIRCULATION | 2 | D | 30 | Y | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 250
| Janitor Room | 3.41 | 5 | 3 | 17.05 | 51.150000000000006 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION
| NA | D | 25 | Y | NA | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 250
| Office-1 | 3.69 | 5.1 | 3 | 18.819 | 56.456999999999994 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION |
2 | D | 25 | Y | NA | NA | NA | Y | NA | NA | NA | NA | NA | Epoxy | 300
| Office-2 | 3.66 | 5.1 | 3 | 18.666 | 55.998000000000005 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION |
2 | D | 25 | Y | NA | NA | NA | Y | NA | NA | NA | NA | NA | Epoxy | 300
| Office-3 | 3.69 | 5.1 | 3 | 18.819 | 56.456999999999994 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION |
2 | D | 25 | Y | NA | NA | NA | Y | Y | NA | NA | NA | NA | Epoxy | 300
| Corridor | 71.9 | 2.57 | 3 | 184.783000000000002 | 554.349 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION
| 2 | D | 25 | Y | NA | NA | Y | Y | NA | NA | NA | NA | NA | Epoxy | 250
| Air Lock | 2.1 | 3.4 | 3 | 7.14 | 21.419999999999998 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | 2
| D | 30 | Y | NA | NA | NA | NA | NA | Y | NA | NA | NA | NA | Epoxy | 250
| Gown Room | 3.34 | 3.4 | 3 | 11.356 | 34.068 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | 2 | C | 45
| Y | NA | NA | NA | NA | NA | Y | NA | NA | NA | Epoxy | 250
| Air lock | 3.34 | 3.4 | 3 | 11.356 | 34.068 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | 2 | C | 50
| Y | NA | NA | NA | NA | NA | Y | NA | NA | NA | Epoxy | 250
| Air Lock | 3.34 | 3.4 | 3 | 11.356 | 34.068 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | 2 | C | 50
| Y | NA | NA | NA | NA | NA | Y | NA | NA | NA | Epoxy | 250
| De- Gown room | 3.34 | 3.4 | 3 | 11.356 | 34.068 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | 2 | C
| 45 | Y | NA | NA | NA | NA | NA | Y | NA | NA | NA | Epoxy | 250
| Air Lock | 2.1 | 3.4 | 3 | 7.14 | 21.419999999999998 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | 2
| D | 30 | Y | NA | NA | NA | NA | NA | Y | NA | NA | NA | NA | Epoxy | 250
| Chemical day store | 11.4 | 3.64 | 3 | 41.496 | 124.488 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION |
1 | C | 50 | Y | NA | NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 350
| 2.33 | 2.14 | 3 | 4.9862 | 14.9586 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | 1 | C | 50 | Y |
NA | NA | NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 350
| Corridor | 39.8 | 3.04 | 3 | 120.991999999999999 | 362.976 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION
| 2 | C | 60 | Y | NA | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 250
| Buffer preparation | 4 | 7.5 | 3 | 30 | 90 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | 4 | C | 50 |
Y | NA | NA | NA | NA | NA | Y | NA | Y | NA | NA | Epoxy | 350
| Tank storage-2 | 6.74 | 7.5 | 3 | 50.550000000000004 | 151.65 | 22.5±2.5 °C | NMT 60% | AHU |
RECIRCULATION | 1 | C | 55 | Y | NA | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 350
| CIP/SIP room | 5.6 | 7.5 | 3 | 42 | 126 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | 2 | C | 60 | Y
| NA | NA | NA | NA | Y | NA | NA | NA | NA | Epoxy | 350
| Waste In | 3.65 | 2 | 3 | 7.3 | 21.9 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | 1 | D | 30 | Y |
NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 350
| Air Lock | 4.13 | 2 | 3 | 8.26 | 24.78 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | 2 | D | 30 | Y |
NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 350
| Decontamination | 4.67 | 5.43 | 3 | 25.358099999999997 | 76.0743 | 22.5±2.5 °C | NMT 60% | AHU |

RECIRCULATION | 2 | D | 35 | Y | NA | NA | NA | NA | NA | NA | NA | Epoxy | 250
| Washing | 3.11 | 5.43 | 3 | 16.8873 | 50.6619 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | 2 | D | 35 | Y | NA | NA | NA | NA | NA | NA | NA | Epoxy | 250
| Air Lock | 3.6 | 2 | 3 | 7.2 | 21.6 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | 2 | D | 35 | Y | NA | NA | NA | NA | NA | NA | NA | Epoxy | 250
| Sterilization | 3.6 | 5.43 | 3 | 19.548 | 58.64399999999999 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | 2 | D | 40 | Y | NA | NA | NA | Y | Y | NA | NA | NA | Epoxy | 350
| Cool Zone | 3.25 | 7.5 | 3 | 24.375 | 73.125 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | 1 | C | 60 | Y | NA | NA | Y | Y | Y | NA | NA | NA | Epoxy | 350
| Stage 10 T / Stage 13 E | 7.18 | 13.5 | 3 | 96.92999999999999 | 290.78999999999996 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | 2 | C | 50 | Y | NA | NA | NA | Y | Y | NA | NA | NA | Epoxy | 500
| | 5.33 | 2.05 | 3 | 10.926499999999999 | 32.7795 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | 2 | C | 50 | Y | NA | NA | NA | Y | Y | NA | NA | NA | Epoxy | 500
| Stage 9 N | 7.25 | 15.6 | 3 | 113.1 | 339.29999999999995 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | 2 | C | 50 | Y | NA | NA | NA | Y | Y | NA | NA | NA | Epoxy | 500
| Stage 6 (I/ Stage 11 F) | 15 | 18 | 3 | 270 | 810 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | 2 | C | 50 | Y | NA | NA | NA | Y | Y | NA | NA | NA | Epoxy | 500
| Column packing and skids | 7.64 | 8.86 | 3 | 67.6904 | 203.07119999999998 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | 2 | C | 50 | Y | NA | NA | NA | Y | Y | NA | NA | NA | Epoxy | 250
| Waste Out | 1.5 | 3.54 | 3 | 5.3100000000000005 | 15.930000000000001 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | 2 | D | 30 | Y | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 250
| Air lock | 1.5 | 4.42 | 3 | 6.63 | 19.89 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | 2 | D | 30 | Y | NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| Decontamination | 7.4 | 5 | 3 | 37 | 111 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | 2 | D | 35 | Y | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 250
| Washing | 7.4 | 3 | 3 | 22.200000000000003 | 66.60000000000001 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | 2 | D | 35 | Y | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 250
| Air Lock | 1.5 | 6.85 | 3 | 10.274999999999999 | 30.824999999999996 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | 2 | D | 35 | Y | NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| Sterilization | 7.4 | 6.85 | 3 | 50.69 | 152.07 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | 2 | D | 40 | Y | NA | NA | NA | Y | Y | NA | NA | NA | Epoxy | 350
| Air lock | 2.1 | 4.76 | 3 | 9.996 | 29.988 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | 2 | D | 30 | Y | NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| Gown room | 3.34 | 4.76 | 3 | 15.898399999999999 | 47.6952 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | 2 | C | 45 | Y | NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| Air lock | 3.34 | 4.76 | 3 | 15.898399999999999 | 47.6952 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | 2 | C | 50 | Y | NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| Corridor | 3.5 | 55.7 | 3 | 194.95000000000002 | 584.85 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | 2 | C | 55 | Y | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 250
| Air lock | 3.34 | 4.76 | 3 | 15.898399999999999 | 47.6952 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | 2 | C | 50 | Y | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| De-gown | 3.34 | 4.76 | 3 | 15.898399999999999 | 47.6952 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | 2 | C | 45 | Y | NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| Air lock | 2.1 | 4.76 | 3 | 9.996 | 29.988 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | 2 | D | 30 | Y | NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| CIP/SIP | 9 | 5 | 3 | 45 | 135 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | 2 | C | 60 | Y | NA | NA | NA | NA | Y | NA | NA | NA | Epoxy | 350
| Cool Zone | 9 | 5.53 | 3 | 49.77 | 149.31 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | 1 | C | 60 | Y | NA | NA | Y | Y | Y | NA | NA | NA | Epoxy | 350
| Buffer Preparation | 9 | 6 | 3 | 54 | 162 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | 2 | C | 60 | Y | NA | NA | NA | NA | Y | NA | NA | NA | Epoxy | 350
| Tank storage | 9 | 8.58 | 3 | 77.22 | 231.66 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | 2 | C | 65 | Y | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 350
| Plasma Day Store | 9 | 10.7 | 3 | 96.3 | 288.9 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | 1 | C | 50 | Y | NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 350
| | 2.66 | 2.29 | 3 | 6.0914 | 18.2742 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | 1 | C | 50 | Y | NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 350
| Janitor Room | 3.5 | 5.4 | 3 | 18.900000000000002 | 56.7 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | NA | C | 50 | Y | NA | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 250
| Corridor | 36.2 | 2.5 | 3 | 90.5 | 271.5 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | 2 | C | 65 | Y | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 350
| Plasma Thawing | 7.64 | 3.61 | 3 | 27.580399999999997 | 82.74119999999999 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | 2 | C | 55 | Y | NA | NA | NA | NA | Y | NA | NA | NA | Epoxy | 350
| Bag cutting & Pooling | 7.64 | 9.8 | 3 | 74.872 | 224.61599999999999 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | 2 | C | 55 | Y | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 350
| Stage 1 (F) / Stage 12 € | 10 | 10.6 | 3 | 106 | 318 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | 2 | C | 55 | Y | NA | NA | Y | Y | Y | NA | NA | NA | Epoxy | 350
| Stage 2 (F) | 17.7 | 10.5 | 3 | 185.85 | 557.55 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | 2 | C | 55 | Y | NA | NA | NA | Y | Y | NA | NA | NA | Epoxy | 350
| Stage 3 (F) | 12.4 | 10.5 | 3 | 130.20000000000002 | 390.6 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | 2 | C | 55 | Y | NA | NA | NA | Y | Y | NA | NA | NA | Epoxy | 350
| Buffer systems | 9 | 6 | 3 | 54 | 162 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | 2 | C | 60 | Y | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 350
| Column packing and Skid | 7.64 | 8.86 | 3 | 67.6904 | 203.07119999999998 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | 2 | C | 60 | Y | NA | NA | NA | Y | Y | NA | NA | NA | Epoxy | 350
| Pre-Virus Controlled Corridor | 3.5 | 32.1 | 3 | 112.35000000000001 | 337.05 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | 2 | C | 55 | Y | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 350
| Stage 8 N | 5.84 | 10.5 | 3 | 61.32 | 183.96 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | 2 | C | 45 | Y | NA | NA | NA | Y | Y | NA | NA | NA | Epoxy | 500
| Stage 5 (I) | 10 | 19 | 3 | 190 | 570 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | 2 | C | 45 | Y | NA | NA | NA | Y | Y | NA | NA | NA | Epoxy | 500
| Stage 7A | 14.7 | 16.1 | 3 | 236.67000000000002 | 710.01 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | 2 | C | 45 | Y | NA | NA | NA | Y | Y | NA | NA | NA | Epoxy | 500
| Stage 4 (F) | 14.7 | 15.8 | 3 | 232.26 | 696.78 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | 2 | C | 45 | Y | NA | NA | NA | Y | Y | NA | NA | NA | Epoxy | 500
| Corridor | 53.9 | 3.17 | 3 | 170.863 | 512.5889999999999 | NA | NA | AHU | RECIRCULATION | 2 | CNC | NA | Y | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 250
| RM Receipt De-Dusting Loading side | 3.54 | 2.77 | 3 | 9.8058 | 29.4174 | NA | NA | AHU | RECIRCULATION | NA | CNC | NA | Y | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 250
| De-Dusting unloading side | 3.54 | 2.6 | 3 | 9.204 | 27.612000000000002 | NA | NA | AHU | RECIRCULATION | 1 | CNC | NA | Y | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 250
| RM quarantine | 6.85 | 6.55 | 3 | 44.8675 | 134.6025 | NA | NA | AHU | RECIRCULATION | 2 | CNC | NA | Y

| NA | NA | NA | NA | Y | NA | NA | NA | Epoxy | 250
| Air lock-1 | 2.13 | 1.21 | 3 | 2.5772999999999997 | 7.7318999999999996 | NA | NA | AHU | RECIRCULATION
| 2 | CNC | NA | Y | NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| Change Room | 2.13 | 1.8 | 3 | 3.834 | 11.502 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | 2 | D |
15 | Y | NA | NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 350
| RM Sampling | 3.26 | 2.31 | 3 | 7.5306 | 22.5918 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | 2 | D |
20 | Y | NA | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 500
| 1.07 | 3.11 | 3 | 3.3277 | 9.9831 | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA |
NA | NA | NA | Epoxy | NA
| Approved RM | 10.1 | 7.54 | 3 | 76.154 | 228.462 | NA | NA | AHU | RECIRCULATION | 1 | CNC | NA | Y |
NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 350
| 5.15 | 4.49 | 3 | 23.123500000000003 | 69.3705 | NA | NA | AHU | RECIRCULATION | NA | NA | NA | NA | NA |
NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | NA
| Air lock -1 | 1.5 | 2.18 | 3 | 3.2700000000000005 | 9.810000000000002 | NA | NA | AHU | RECIRCULATION |
2 | CNC | NA | Y | NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| Change Room | 1.5 | 2.18 | 3 | 3.2700000000000005 | 9.810000000000002 | 22.5±2.5 °C | NMT 60% | AHU |
RECIRCULATION | 2 | D | 15 | Y | NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| Dispensing Room | 3.4 | 4.44 | 3 | 15.096 | 45.288 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | 2 |
D | 20 | Y | NA | NA | Y | Y | Y | NA | Y | NA | Epoxy | 500
| Plasma receipt room | 2.55 | 4.95 | 3 | 12.622499999999999 | 37.867499999999999 | NA | NA | AHU |
RECIRCULATION | 1 | CNC | NA | Y | NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| Plasma quarantine room | 5.15 | 2.72 | 3 | 14.008000000000003 | 42.02400000000001 | NA | NA | AHU |
RECIRCULATION | 2 | CNC | NA | Y | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 250
| 11.6 | 4.3 | 3 | 49.87999999999995 | 149.64 | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA |
NA | NA | NA | NA | NA | NA | Epoxy | NA
| Serology Testing | 3 | 4.95 | 3 | 14.850000000000001 | 44.550000000000004 | NA | NA | AHU |
RECIRCULATION | 2 | CNC | NA | Y | NA | NA | NA | NA | Y | NA | NA | NA | Epoxy | 350
| Air lock-1 | 1.87 | 1.4 | 3 | 2.618 | 7.853999999999999 | NA | NA | AHU | RECIRCULATION | 2 | CNC | NA |
Y | NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| Change Room | 1.87 | 1.4 | 3 | 2.618 | 7.853999999999999 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION
| 2 | D | 15 | Y | NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| Plasma Sampling | 1.76 | 2.9 | 3 | 5.104 | 15.312000000000001 | 22.5±2.5 °C | NMT 60% | AHU |
RECIRCULATION | 2 | D | 20 | Y | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 500
| 3.68 | 2.05 | 3 | 7.544 | 22.631999999999998 | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA |
NA | NA | NA | NA | NA | NA | Epoxy | NA
| Reject Room | 2.26 | 4.95 | 3 | 11.187 | 33.561 | NA | NA | AHU | RECIRCULATION | NA | CNC | NA | Y |
NA | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 250
| Bag Bar Coding Mini Pools | 2.6 | 2.67 | 3 | 6.942 | 20.826 | NA | NA | AHU | RECIRCULATION | 1 | CNC |
NA | Y | NA | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 250
| Washing area | 2.3 | 2.67 | 3 | 6.140999999999999 | 18.423 | NA | NA | AHU | RECIRCULATION | 1 | CNC |
NA | Y | NA | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 250
| Janitor Room | 1.45 | 2.67 | 3 | 3.8714999999999997 | 11.6145 | NA | NA | AHU | RECIRCULATION | NA |
CNC | NA | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 250
| Corridor | 28 | 1.8 | 3 | 50.4 | 151.2 | NA | NA | AHU | RECIRCULATION | 2 | CNC | NA | Y | NA | NA |
NA | NA | Y | NA | NA | Y | Epoxy | 250
| Corridor | 1.8 | 12 | 3 | 21.6 | 64.80000000000001 | NA | NA | AHU | RECIRCULATION | 2 | CNC | NA | Y |
NA | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 250
| Plasma Storage -1000L | 9.1 | 15.2 | 3 | 138.32 | 414.96 | NA | NA | AHU | RECIRCULATION | NA | CNC |
NA | Y | NA | NA | NA | Y | NA | NA | NA | NA | NA | Epoxy | 350
| Plasma Storage -1000L | 10.5 | 15.2 | 3 | 159.6 | 478.79999999999995 | NA | NA | AHU | RECIRCULATION |
NA | CNC | NA | Y | NA | NA | NA | Y | NA | NA | NA | NA | Epoxy | 350
| Plasma Storage -1000L | 9.36 | 15.2 | 3 | 142.272 | 426.816 | NA | NA | AHU | RECIRCULATION | NA | CNC |
NA | Y | NA | NA | NA | Y | NA | NA | NA | NA | NA | Epoxy | 350
| Plasma Storage -1000L | 10.3 | 15.2 | 3 | 156.56 | 469.68 | NA | NA | AHU | RECIRCULATION | NA | CNC |
NA | Y | NA | NA | NA | Y | NA | NA | NA | NA | NA | Epoxy | 350
| Plasma Storage -1000L | 9.1 | 15.2 | 3 | 138.32 | 414.96 | NA | NA | AHU | RECIRCULATION | NA | CNC |
NA | Y | NA | NA | NA | Y | NA | NA | NA | NA | NA | Epoxy | 350
| Store change Room | 2.95 | 2.88 | 3 | 8.496 | 25.488 | NA | NA | AHU | RECIRCULATION | 2 | CNC | NA | Y |
NA | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 250
| Store office | 2.23 | 6.5 | 3 | 14.495 | 43.485 | NA | NA | AHU | RECIRCULATION | 2 | CNC | NA | Y | NA |
NA | NA | Y | Y | NA | NA | NA | Epoxy | 350
| FG dispatch/FG In | 12.9 | 6.49 | 3 | 83.721 | 251.163 | NA | NA | AHU | RECIRCULATION | 2 | CNC | NA |
Y | NA | NA | NA | NA | NA | NA | NA | NA | Kota Stone | 350
| 4.7 | 4.09 | 3 | 19.223 | 57.669 | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA |
NA | NA | NA | NA | NA | NA
| 3 | 3.6 | 3 | 10.8 | 32.400000000000006 | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA |
NA | NA | NA | NA | NA | NA
| Recalled goods | 2.3 | 2.35 | 3 | 5.404999999999999 | 16.214999999999996 | NA | NA | AHU |
RECIRCULATION | NA | CNC | NA | Y | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | Kota Stone | 350
| reject room | 2.3 | 2.35 | 3 | 5.404999999999999 | 16.214999999999996 | NA | NA | AHU | RECIRCULATION |
NA | CNC | NA | Y | NA | NA | NA | NA | NA | NA | NA | NA | Kota Stone | 250
| Ice Pack | 2.5 | 7.34 | 3 | 18.35 | 55.050000000000004 | NA | NA | AHU | RECIRCULATION | NA | CNC | NA |
Y | NA | NA | NA | NA | NA | NA | NA | NA | Kota Stone | 250
| Cold room-1 | 5.04 | 7.34 | 3 | 36.9936 | 110.9808 | 2-8 °C | NA | Cold room unit | NA | NA | NA | NA |
NA | NA | NA | NA | Y | Hooter | NA | NA | NA | Epoxy | 350
| Cold room-2 | 5.02 | 7.34 | 3 | 36.846799999999995 | 110.54039999999998 | 2-8 °C | NA | Cold room unit
| NA | NA | NA | NA | NA | NA | NA | NA | Y | Hooter | NA | NA | NA | Epoxy | 350
| Cold room-3 | 5 | 7.34 | 3 | 36.7 | 110.10000000000001 | 2-8 °C | NA | Cold room unit | NA | NA | NA |
NA | NA | NA | NA | NA | Y | Hooter | NA | NA | NA | Epoxy | 350
| Cold room-4 | 5.04 | 7.34 | 3 | 36.9936 | 110.9808 | 2-8 °C | NA | Cold room unit | NA | NA | NA | NA |
NA | NA | NA | NA | Y | Hooter | NA | NA | NA | Epoxy | 350
[SHEET: FF_Fill & Finish_RDS]
| Room Details | Fire Detection & Protection System |
Room No | Room Name | Length (mtr) | Width(mtr) | Height | Area (M²) | Volume | Temp.(oC) | RH(%) | HVAC
System | HVAC Type | Occupancy | Grade | Differential P | Fire Alarm System | Fire Sprinkler System |
Fire Extinguisher System | GPS Clock | LAN Port | Telephone Port | Door Interlock | Eye Washer | Body
Washer | Floor Finish | Electrical Lux level
| Corridor | 25.2 | 2.85 | 3 | 71.82 | 215.45999999999998 | NA | NA | AHU | RECIRCULATION | NA | CNC | NA |
Y | Y | Y | Y | Y | NA | NA | NA | NA | NA | Epoxy | 250
| Bulk Staging cold Room | 21 | 8.07 | 3 | 169.47 | 508.40999999999997 | NA | NA | AHU | RECIRCULATION |
NA | CNC | NA | Y | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 350
| Corridor | 25.5 | 1.8 | 3 | 45.9 | 137.7 | NA | NA | AHU | RECIRCULATION | NA | CNC | NA | Y | NA | Y |

NA | NA | NA | NA | NA | NA | Epoxy | 250
| De-cartoning room | 3.89 | 3.32 | 3 | 12.9148 | 38.7444 | NA | NA | AHU | RECIRCULATION | 5 | CNC | NA
| Y | NA | NA | NA | NA | Y | NA | NA | NA | Epoxy | 250
| Pal-1 | 2.27 | 1.64 | 3 | 3.7228 | 11.1684 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | 2 | D | 15 |
Y | NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| Pal-1 | 2.27 | 1.64 | 3 | 3.7228 | 11.1684 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | 2 | D | 20 |
Y | NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| Vial washing | 6.42 | 9.65 | 3 | 61.953 | 185.859 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | 1 | D
| 25 | Y | NA | NA | NA | NA | Y | NA | NA | NA | Epoxy | 350
| Janitor Room | 2.32 | 1 | 3 | 2.32 | 6.959999999999999 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION |
NA | D | 20 | Y | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 250
| Cold room | 4.19 | 3.27 | 3 | 13.701300000000002 | 41.1039 | 2-8 °C | NA | Cold room Unit | NA | NA |
NA | NA | Y | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | NA
| 2.26 | 2.53 | 3 | 5.717799999999999 | 17.153399999999998 | 2-8 °C | NA | Cold room Unit | NA | NA |
NA | NA | Y | NA | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | NA
| Primary change room male | 5.31 | 5.81 | 3 | 30.851099999999995 | 92.55329999999998 | 22.5±2.5 °C | NMT
60% | AHU | RECIRCULATION | 2 | D | 15 | Y | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| Primary change room female | 4.43 | 3.93 | 3 | 17.4099 | 52.2297 | 22.5±2.5 °C | NMT 60% | AHU |
RECIRCULATION | 2 | D | 15 | Y | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| 3.63 | 1.92 | 3 | 6.9696 | 20.9088 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | 2 | D | 15 | Y |
NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| Primary change room visitor | 4.51 | 5.78 | 3 | 26.0678 | 78.203399999999999 | 22.5±2.5 °C | NMT 60% |
AHU | RECIRCULATION | 2 | D | 15 | Y | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| Mal entry | 2.85 | 5.78 | 3 | 16.473000000000003 | 49.419000000000001 | 22.5±2.5 °C | NMT 60% | AHU |
RECIRCULATION | NA | NA | 10 | Y | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| Corridor | 16.7 | 1.8 | 3 | 30.06 | 90.179999999999999 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | 2
| D | 20 | Y | NA | Y | NA | NA | NA | NA | NA | NA | Epoxy | 250
| Corridor | 34.1 | 1.8 | 3 | 61.38 | 184.140000000000001 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION |
2 | D | 25 | Y | NA | Y | Y | Y | NA | NA | NA | NA | Epoxy | 250
| PAL | 2.84 | 1.45 | 3 | 4.117999999999999 | 12.354 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | 2 |
D | 30 | Y | NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| MAL | 2.84 | 1.45 | 3 | 4.117999999999999 | 12.354 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | NA |
D | 30 | Y | NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| Material prep. & wash | 8.15 | 3.07 | 3 | 25.0205 | 75.0615 | 22.5±2.5 °C | NMT 60% | AHU |
RECIRCULATION | 2 | D | 35 | Y | NA | NA | NA | Y | Y | Y | NA | Y | NA | Epoxy | 350
| 5.26 | 1.64 | 3 | 8.626399999999999 | 25.879199999999997 | 22.5±2.5 °C | NA | NA | NA | | NA | NA |
NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA |
| 2.21 | 1.46 | 3 | 3.2266 | 9.6798 | 22.5±2.5 °C | NA | NA | NA | | NA | NA | NA | NA | NA | NA | NA | NA |
NA | NA | NA | NA | NA | NA | NA |
| Disinfectant | 8.41 | 2.62 | 3 | 22.034200000000002 | 66.102600000000001 | 22.5±2.5 °C | NMT 60% | AHU |
RECIRCULATION | | D | 40 | Y | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 350
| MAL | 2.04 | 1.47 | 3 | 2.9988 | 8.99640000000000001 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | NA |
D | 40 | Y | NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| PAL | 2.04 | 2.29 | 3 | 4.6716 | 14.0148 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | 2 | D | 40 | Y
| NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| Unloading & packing area | 2.94 | 5.13 | 3 | 15.0822 | 45.2466 | 22.5±2.5 °C | NMT 60% | AHU |
RECIRCULATION | 1 | D | 45 | Y | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 350
| PAL | 1.41 | 1.64 | 3 | 2.3124 | 6.937199999999999 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | 2 |
D | 30 | Y | NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| Mal | 1.45 | 1.64 | 3 | 2.3779999999999997 | 7.133999999999999 | 22.5±2.5 °C | NMT 60% | AHU |
RECIRCULATION | NA | D | 30 | Y | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| Laundry area | 5.1 | 1.57 | 3 | 8.007 | 24.021 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | 1 | D |
35 | Y | NA | NA | NA | NA | Y | NA | NA | NA | Epoxy | 250
| 2.98 | 3.88 | 3 | 11.5624 | 34.687200000000004 | 22.5±2.5 °C | NA | NA | NA | NA | NA | NA | NA | NA |
NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA |
| PAL-1 | 4.54 | 1.87 | 3 | 8.4898 | 25.4694 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | 2 | D | 30 |
Y | NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| PAL-2 | 4.4 | 1.87 | 3 | 8.228000000000002 | 24.684000000000005 | 22.5±2.5 °C | NMT 60% | AHU |
RECIRCULATION | | C | 45 | Y | NA | NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| PAL-3 | 4.23 | 3.82 | 3 | 16.1586 | 48.4758 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | | B | 60 |
Y | NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| PAL-4 | 3.2 | 3.82 | 3 | 12.224 | 36.672 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | | B | 65 | Y |
NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| Exit-1 | 4.4 | 1.87 | 3 | 8.228000000000002 | 24.684000000000005 | 22.5±2.5 °C | NMT 60% | AHU |
RECIRCULATION | 2 | B | 65 | Y | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| Exit-2 | 4.54 | 1.87 | 3 | 8.4898 | 25.4694 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | | C | 50 |
Y | NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| Sterile Corridor | 15.5 | 1.8 | 3 | 27.900000000000002 | 83.7 | 22.5±2.5 °C | NMT 60% | AHU |
RECIRCULATION | 2 | B | 70 | Y | NA | NA | Y | Y | Y | NA | NA | NA | NA | Epoxy | 300
| 8.49 | 1.8 | 3 | 15.282 | 45.846000000000004 | 22.5±2.5 °C | NA | NA | NA | NA | NA | NA | NA | NA |
NA | NA | NA | NA | NA | NA | NA | NA | NA |
| Formulation | 8.41 | 3.59 | 3 | 30.1919 | 90.5757 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | 2 | B
| 75 | Y | NA | NA | Y | Y | Y | NA | NA | NA | Epoxy | 350
| Clean JC | 1.88 | 2.48 | 3 | 4.6624 | 13.9872 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | NA | | |
Y | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 250
| Unload | 8.41 | 3.4 | 3 | 28.594 | 85.782000000000001 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | 1
| B | 75 | Y | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 350
| Fill Finish | 6.42 | 8.3 | 3 | 53.286 | 159.858 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | 2 | B |
75 | Y | NA | NA | Y | Y | Y | NA | NA | NA | NA | Epoxy | 500
| Capping | 8.29 | 4.04 | 3 | 33.4916 | 100.474799999999999 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION
| 1 | B | 80 | Y | NA | NA | Y | Y | Y | NA | NA | NA | Epoxy | 500
| Corridor | 1.8 | 117.9 | 3 | 212.220000000000003 | 636.66000000000001 | 22.5±2.5 °C | NMT 60% | AHU |
RECIRCULATION | 2 | D | 25 | Y | NA | Y | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 250
| Lyo loading | 8.29 | 3.55 | 3 | 29.429499999999994 | 88.28849999999998 | 22.5±2.5 °C | NMT 60% | AHU |
RECIRCULATION | 2 | D | 30 | Y | NA | NA | Y | Y | Y | NA | NA | NA | Epoxy | 300
| Lyo service area | 10.9 | 3.55 | 3 | 38.695 | 116.085000000000001 | 22.5±2.5 °C | NMT 60% | AHU |
RECIRCULATION | 2 | D | 30 | Y | NA | NA | NA | NA | Y | NA | NA | NA | Epoxy | 300
| PPM Store | 15.2 | 12 | 3 | 182.399999999999998 | 547.199999999999999 | NA | NA | AHU | Exhaust | NA | CNC
| NA | Y | Y | NA | NA | NA | Y | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA |
| 11.8 | 3.03 | 3 | 35.754 | 107.262 | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA |
NA | NA | NA | NA | NA |
| Approved Rubber bung Store | 3.36 | 2.97 | 3 | 9.9792 | 29.937600000000003 | NA | NA | AHU | Exhaust |

NA | CNC | NA | Y | Y | Y | NA | NA | NA | NA | NA | NA | Epoxy | 350
| Corridor | 2.8 | 27.9 | 3 | 78.11999999999999 | 234.35999999999996 | NA | NA | AHU | Exhaust | NA | CNC
| NA | Y | Y | Y | NA | NA | NA | NA | NA | NA | Epoxy | 250
| | 49 | 2.59 | 3 | 126.91 | 380.73 | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA
| NA | NA | NA | NA
| Packing Change Female | 4.14 | 3.08 | 3 | 12.751199999999999 | 38.2536 | 22.5±2.5 °C | NA | AHU |
RECIRCULATION | 2 | CNC | NA | Y | NA | NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| Packing Change Male | 4.14 | 4.23 | 3 | 17.5122 | 52.5366 | 22.5±2.5 °C | NA | AHU | RECIRCULATION | 2
| CNC | NA | Y | NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| Packing Change Visitor | 4.41 | 3.08 | 3 | 13.5828 | 40.748400000000004 | 22.5±2.5 °C | NA | AHU |
RECIRCULATION | 2 | CNC | NA | Y | NA | NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| Corridor | 11.8 | 1.8 | 3 | 21.240000000000002 | 63.720000000000006 | 22.5±2.5 °C | NA | AHU |
RECIRCULATION | 2 | CNC | NA | Y | NA | Y | NA | NA | NA | NA | NA | NA | NA | Epoxy | 250
| Packing office | 4.14 | 4.5 | 3 | 18.63 | 55.89 | 22.5±2.5 °C | NA | AHU | RECIRCULATION | 2 | CNC | NA
| Y | NA | NA | NA | Y | Y | NA | NA | NA | NA | Epoxy | 500
| Visual Inspection Room | 9.79 | 3.87 | 3 | 37.887299999999996 | 113.66189999999999 | 22.5±2.5 °C | NA |
AHU | RECIRCULATION | 2 | CNC | NA | Y | NA | NA | NA | Y | Y | NA | NA | NA | Epoxy | 350
| | 2.88 | 1.14 | 3 | 3.2831999999999995 | 9.849599999999999 | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA
| NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA
| Buffer Room | 2.83 | 2.68 | 3 | 7.5844000000000005 | 22.7532 | 22.5±2.5 °C | NA | AHU | RECIRCULATION |
2 | CNC | NA | Y | NA | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 350
| Labelling And Packing | 12.6 | 5.4 | 3 | 68.04 | 204.12 | 22.5±2.5 °C | NA | AHU | RECIRCULATION | 10 |
CNC | NA | Y | NA | NA | NA | Y | Y | NA | NA | NA | Epoxy | 500
| Finished Good Staging | 12.6 | 5.67 | 3 | 71.442 | 214.32599999999996 | 22.5±2.5 °C | NA | AHU |
RECIRCULATION | 5 | CNC | NA | Y | NA | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 350
| | 3.14 | 1.84 | 3 | 5.7776000000000005 | 17.332800000000002 | 22.5±2.5 °C | NA | AHU | RECIRCULATION |
5 | CNC | NA | Y | NA | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 350
| Reject room | 3.58 | 2.98 | 3 | 10.6684 | 32.0052 | NA | NA | AHU | Exhaust | NA | CNC | NA | Y | Y |
NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 250
| SPM Receipt | 3.59 | 2.98 | 3 | 10.6982 | 32.0946 | NA | NA | AHU | Exhaust | NA | CNC | NA | Y | Y |
NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 250
| SPM Store | 14.9 | 9.32 | 3 | 138.868 | 416.604 | NA | NA | AHU | Exhaust | NA | CNC | NA | Y | Y | NA
| NA | NA | Y | NA | NA | NA | Epoxy | 350
| | 7.22 | 2.69 | 3 | 19.421799999999998 | 58.265399999999999 | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA
| NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA
| Labeling Store | 7.68 | 5.67 | 3 | 43.5456 | 130.6368 | NA | NA | AHU | Exhaust | NA | CNC | NA | Y | Y |
NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 350
| Control Room | 5.55 | 3.55 | 3 | 19.702499999999997 | 59.107499999999999 | 22.5±2.5 °C | NMT 60% | AHU |
RECIRCULATION | 2 | D | 30 | Y | NA | NA | NA | NA | Y | NA | NA | NA | Epoxy | 300
| Plasma Storage -1000L | 12.2 | 17.1 | 3 | 208.62 | 625.86 | NA | NA | AHU | RECIRCULATION | NA | CNC |
NA | Y | NA | NA | NA | Y | NA | NA | NA | NA | Epoxy | 350
| Plasma Storage -1000L | 12.1 | 17.1 | 3 | 206.910000000000003 | 620.73 | NA | NA | AHU | RECIRCULATION |
NA | CNC | NA | Y | NA | NA | NA | Y | NA | NA | NA | NA | Epoxy | 350
| Plasma Storage -1000L | 12.1 | 17.1 | 3 | 206.910000000000003 | 620.73 | NA | NA | AHU | RECIRCULATION |
NA | CNC | NA | Y | NA | NA | NA | Y | NA | NA | NA | NA | Epoxy | 350
| Plasma Storage -1000L | 12.1 | 17.1 | 3 | 206.910000000000003 | 620.73 | NA | NA | AHU | RECIRCULATION |
NA | CNC | NA | Y | NA | NA | NA | Y | NA | NA | NA | NA | Epoxy | 350
| Plasma Storage -1000L | 12.1 | 17.1 | 3 | 206.910000000000003 | 620.73 | NA | NA | AHU | RECIRCULATION |
NA | CNC | NA | Y | NA | NA | NA | Y | NA | NA | NA | NA | Epoxy | 350
| Corridor | 49 | 2.59 | 3 | 126.91 | 380.73 | NA | NA | AHU | RECIRCULATION | NA | CNC | NA | Y | NA |
NA | NA | Y | NA | NA | NA | NA | Epoxy | 350
[SHEET: FF_QC & QA_RDS]
| | Room Details | | | | | | | | Fire Detection & Protection System | | | | | | | |
Room No | Room Name | Length (mtr) | Width(mtr) | Height | Area (M²) | Volume | Temp.(oC) | RH(%) | HVAC
System | HVAC Type | Occupancy | Grade | Differential P | Fire Alarm System | Fire Sprinkler System |
Fire Extinguisher System | GPS Clock | LAN Port | Telephone Port | Door Interlock | Eye Washer | Body
Washer | Floor Finish | Electrical Lux level
| Corridor | 48.5 | 2.9 | 3 | 140.65 | 421.95000000000005 | NA | NA | AHU | RECIRCULATION | NA | CNC | NA
| Y | NA | Y | NA | NA | NA | NA | NA | NA | NA | Vitrified Tiles | 250
| Corridor | 2.62 | 120.8 | 3 | 316.496 | 949.48799999999999 | NA | NA | AHU | RECIRCULATION | NA | CNC |
NA | Y | NA | Y | NA | NA | NA | NA | NA | NA | Vitrified Tiles | 250
| Gents change room | 4.73 | 3.42 | 3 | 16.1766 | 48.5298 | NA | NA | AHU | RECIRCULATION | NA | CNC | NA
| Y | NA | NA | NA | NA | NA | NA | NA | NA | NA | Vitrified Tiles | 250
| Ladies change room | 4.73 | 3.42 | 3 | 16.1766 | 48.5298 | NA | NA | AHU | RECIRCULATION | NA | CNC |
NA | Y | NA | NA | NA | NA | NA | NA | NA | NA | Vitrified Tiles | 250
| Corridor | 22.5 | 1.8 | 3 | 40.5 | 121.5 | NA | NA | AHU | RECIRCULATION | NA | CNC | NA | Y | NA | Y |
NA | NA | NA | NA | NA | NA | NA | Vitrified Tiles | 250
| QC Sample Receipt | 3.11 | 3.93 | 3 | 12.2223 | 36.6669 | NA | NA | AHU | RECIRCULATION | NA | CNC | NA
| Y | NA | NA | NA | NA | NA | NA | NA | NA | NA | Vitrified Tiles | 350
| Janitor | 1.5 | 3.93 | 3 | 5.8950000000000005 | 17.685000000000002 | NA | NA | AHU | RECIRCULATION | NA
| CNC | NA | Y | NA | NA | NA | NA | NA | NA | NA | Vitrified Tiles | 350
| QC sample preparation | 4.73 | 5.42 | 3 | 25.6366 | 76.9098 | NA | NA | AHU | RECIRCULATION | NA | CNC
| NA | Y | NA | NA | NA | NA | NA | NA | NA | NA | Vitrified Tiles | 350
| Document storage | 4.73 | 6.62 | 3 | 31.312600000000003 | 93.937800000000001 | NA | NA | AHU |
RECIRCULATION | NA | CNC | NA | Y | NA | NA | NA | NA | NA | NA | Vitrified Tiles | 500
| QC manager | 4.73 | 5.91 | 3 | 27.954300000000003 | 83.862900000000001 | NA | NA | AHU | RECIRCULATION |
NA | CNC | NA | Y | NA | NA | NA | Y | Y | NA | NA | NA | Vitrified Tiles | 350
| QC staff documentation | 4.03 | 6.62 | 3 | 26.678600000000003 | 80.035800000000001 | NA | NA | AHU |
RECIRCULATION | NA | CNC | NA | Y | NA | NA | NA | Y | Y | NA | NA | NA | Vitrified Tiles | 500
| Chemical/ Glass ware Storage | 7.71 | 6.62 | 3 | 51.0402 | 153.1206 | NA | NA | AHU | RECIRCULATION |
NA | CNC | NA | Y | NA | NA | NA | NA | NA | NA | Y | NA | Vitrified Tiles | 350
| PM testing room | 9.94 | 6.64 | 3 | 66.0016 | 198.0048 | NA | NA | AHU | RECIRCULATION | NA | CNC | NA
| Y | NA | NA | NA | NA | Y | NA | NA | NA | NA | Vitrified Tiles | 350
| Corridor | 10.1 | 1.8 | 3 | 18.18 | 54.54 | NA | NA | AHU | RECIRCULATION | NA | CNC | NA | Y | NA | NA
| NA | NA | NA | NA | NA | NA | NA | Vitrified Tiles | 350
| Instrument room 1 HPLC | 9.94 | 7.04 | 3 | 69.9776 | 209.9328 | NA | NA | AHU | RECIRCULATION | NA |
CNC | NA | Y | NA | NA | NA | NA | NA | NA | NA | NA | Vitrified Tiles | 350
| Retention sample room | 9.94 | 3.31 | 3 | 32.9014 | 98.704200000000001 | NA | NA | AHU | RECIRCULATION |
NA | CNC | NA | Y | NA | NA | NA | NA | NA | NA | NA | NA | Vitrified Tiles | 350
| RM control sample | 9.94 | 3.31 | 3 | 32.9014 | 98.704200000000001 | NA | NA | AHU | RECIRCULATION | NA
| CNC | NA | Y | NA | NA | NA | NA | NA | NA | NA | NA | Vitrified Tiles | 350
| Corridor | 1.8 | 29.2 | 3 | 52.56 | 157.68 | NA | NA | AHU | RECIRCULATION | NA | CNC | NA | Y | NA |

NA | NA | NA | NA | NA | NA | NA | Vitrified Tiles | 350
| Conference | 9.31 | 6.07 | 3 | 56.511700000000005 | 169.5351 | NA | NA | AHU | RECIRCULATION | NA | CNC
| NA | Y | NA | NA | NA | NA | NA | NA | NA | Vitrified Tiles | 500
| Wet lab | 9.31 | 4.85 | 3 | 45.1535 | 135.4605 | NA | NA | AHU | RECIRCULATION | NA | CNC | NA | Y | NA
| NA | Y | Y | Y | NA | Y | NA | Vitrified Tiles | 350
| Balance room | 4.82 | 3.37 | 3 | 16.2434 | 48.7302 | NA | NA | AHU | RECIRCULATION | NA | CNC | NA | Y
| NA | NA | NA | NA | NA | NA | NA | NA | NA | Vitrified Tiles | 350
| GeneralFTIR | 4.38 | 3.37 | 3 | 14.7606 | 44.281800000000004 | NA | NA | AHU | RECIRCULATION | NA | CNC
| NA | Y | NA | NA | NA | NA | NA | NA | NA | Vitrified Tiles | 350
| Instrument room-II | 15.3 | 5.43 | 3 | 83.079 | 249.23699999999997 | NA | NA | AHU | RECIRCULATION | NA
| CNC | NA | Y | NA | NA | Y | Y | Y | NA | Y | NA | Vitrified Tiles | 500
| Reference standard | 5.87 | 2.98 | 3 | 17.4926 | 52.4778 | NA | NA | AHU | RECIRCULATION | NA | CNC |
NA | Y | NA | NA | Y | Y | Y | NA | Y | NA | Vitrified Tiles | 500
| GC room | 4.83 | 2.98 | 3 | 14.3934 | 43.1802 | NA | NA | AHU | RECIRCULATION | NA | CNC | NA | Y | NA
| NA | Y | Y | Y | NA | Y | NA | Vitrified Tiles | 500
| TOC/ LPC | 4.44 | 2.98 | 3 | 13.231200000000001 | 39.6936 | NA | NA | AHU | RECIRCULATION | NA | CNC |
NA | Y | NA | NA | NA | Y | NA | NA | NA | NA | Vitrified Tiles | 350
| Stability room | 15.3 | 6.04 | 3 | 92.412 | 277.236 | NA | NA | AHU | RECIRCULATION | NA | CNC | NA | Y
| NA | NA | NA | Y | NA | NA | NA | NA | NA | Vitrified Tiles | 350
| Hot Room | 5.9 | 5.46 | 3 | 32.214 | 96.642 | NA | NA | AHU | RECIRCULATION | NA | CNC | NA | Y | NA |
NA | NA | NA | NA | NA | NA | NA | NA | NA | Vitrified Tiles | 350
| Wash area | 5.9 | 2.76 | 3 | 16.284 | 48.852 | NA | NA | AHU | RECIRCULATION | NA | CNC | NA | Y | NA |
NA | NA | NA | NA | NA | NA | NA | NA | NA | Vitrified Tiles | 350
| Canteen | 12.4 | 10.6 | 3 | 131.44 | 394.32 | NA | NA | AHU | RECIRCULATION | NA | CNC | NA | Y | NA |
NA | NA | NA | NA | NA | NA | NA | NA | NA | Vitrified Tiles | 350
| 5.98 | 6.12 | 3 | 36.5976 | 109.7928 | NA | NA | AHU | RECIRCULATION | NA | CNC | NA | Y | NA | NA |
NA | NA | NA | NA | NA | NA | NA | Vitrified Tiles | 350
| 2.35 | 8.03 | 3 | 18.8705 | 56.6115 | NA | NA | AHU | RECIRCULATION | NA | CNC | NA | Y | NA | NA |
NA | NA | NA | NA | NA | NA | NA | Vitrified Tiles | 350
| Training Hall | 14.8 | 19 | 3 | 281.2 | 843.5999999999999 | NA | NA | AHU | RECIRCULATION | NA | CNC |
NA | Y | NA | NA | NA | NA | Y | Y | Y | NA | NA | NA | Vitrified Tiles | 500
| Corridor | 1.8 | 117.9 | 3 | 212.22000000000003 | 636.6600000000001 | NA | NA | AHU | RECIRCULATION |
NA | CNC | NA | Y | NA | NA | NA | NA | NA | NA | NA | NA | Vitrified Tiles | 350
| Ladies change room | 4.78 | 3.12 | 3 | 14.9136 | 44.7408 | NA | NA | AHU | RECIRCULATION | NA | CNC |
NA | Y | NA | NA | NA | NA | NA | NA | NA | NA | NA | Vitrified Tiles | 250
| Gents change room | 4.78 | 2.83 | 3 | 13.527400000000002 | 40.58220000000001 | NA | NA | AHU |
RECIRCULATION | NA | CNC | NA | Y | NA | NA | NA | NA | NA | NA | NA | Vitrified Tiles | 250
| Corridor | 2.5 | 8.31 | 3 | 20.775000000000002 | 62.325 | NA | NA | AHU | RECIRCULATION | NA | CNC | NA
| Y | NA | Y | Y | Y | NA | NA | NA | NA | Vitrified Tiles | 300250
| BET room | 4.78 | 2.83 | 3 | 13.527400000000002 | 40.58220000000001 | NA | NA | AHU | RECIRCULATION |
NA | CNC | NA | Y | NA | NA | NA | NA | NA | NA | NA | Vitrified Tiles | 350
| Media Store | 4.78 | 3.54 | 3 | 16.921200000000002 | 50.76360000000001 | NA | NA | AHU | RECIRCULATION
| NA | CNC | NA | Y | NA | NA | NA | NA | NA | NA | NA | NA | Vitrified Tiles | 350
| Air Lock | 2.5 | 4.24 | 3 | 10.600000000000001 | 31.800000000000004 | 22.5±2.5 °C | NMT 60% | AHU |
RECIRCULATION | 2 | D | 15 | Y | NA | NA | Y | Y | Y | NA | Y | NA | NA | Epoxy | 250
| Media Prep./ Storage room | 7.38 | 3.88 | 3 | 28.6344 | 85.9032 | 22.5±2.5 °C | NMT 60% | AHU |
RECIRCULATION | 2 | D | 20 | Y | NA | NA | NA | NA | Y | NA | NA | NA | Epoxy | 350
| Preparation and Sterility area | 8.11 | 5.91 | 3 | 47.930099999999996 | 143.7903 | 22.5±2.5 °C | NMT
60% | AHU | RECIRCULATION | 2 | D | 20 | Y | NA | NA | Y | Y | Y | NA | NA | NA | Epoxy | 350
| CH-1 | 4.81 | 2.02 | 3 | 9.716199999999999 | 29.148599999999995 | 22.5±2.5 °C | NMT 60% | AHU |
RECIRCULATION | 2 | D | 15 | Y | NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| CH-2 | 2.73 | 2.02 | 3 | 5.5146 | 16.543799999999997 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | |
C | 30 | Y | NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| CH-3 | 2.02 | 2.02 | 3 | 4.0804 | 12.2412 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | | B | 45 | Y
| NA | NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| Air Lock | 5.36 | 2.02 | 3 | 10.827200000000001 | 32.4816 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION
| 2 | B | 40 | Y | NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| Exit-1 | 3.49 | 2.02 | 3 | 7.0498 | 21.1494 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | | C | 25 |
Y | NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| Exit-2 | 2.63 | 2.02 | 3 | 5.3126 | 15.9378 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | | D | 10 |
Y | NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| Sterile Passage | 3.53 | 2.02 | 3 | 7.130599999999999 | 21.391799999999996 | 22.5±2.5 °C | NMT 60% |
AHU | RECIRCULATION | 2 | B | 50 | Y | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 250
| Sterility room | 4.65 | 3.78 | 3 | 17.577 | 52.73100000000001 | 22.5±2.5 °C | NMT 60% | AHU |
RECIRCULATION | 2 | B | 60 | Y | NA | NA | Y | Y | Y | NA | NA | NA | Epoxy | 500
| Cooling zone | 3.38 | 3.78 | 3 | 12.776399999999999 | 38.3292 | 22.5±2.5 °C | NMT 60% | AHU |
RECIRCULATION | 1 | B | 55 | Y | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 250
| 1.83 | 1.91 | 3 | 3.4953 | 10.485899999999999 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | 1 | B |
55 | Y | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 250
| Wash area | 5.221 | 2.02 | 3 | 10.54642 | 31.63926 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | 1 |
D | 25 | Y | NA | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 250
| Air Lock-1 | 5.36 | 2.02 | 3 | 10.827200000000001 | 32.4816 | 22.5±2.5 °C | NMT 60% | AHU |
RECIRCULATION | 2 | D | 15 | Y | NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| CH-1 | 6.04 | 2.02 | 3 | 12.200800000000001 | 36.6024 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | |
D | 25 | Y | NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| MLT | 4.65 | 3.77 | 3 | 17.5305 | 52.591499999999996 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | 2
| D | 15 | Y | NA | NA | NA | NA | Y | NA | NA | NA | Epoxy | 350
| Corridor | 1.8 | 8.5 | 3 | 15.3 | 45.900000000000006 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | 2
| D | 25 | Y | NA | NA | NA | NA | NA | NA | Y | NA | NA | NA | Epoxy | 350
| Micro Office | 5.17 | 4.24 | 3 | 21.9208 | 65.7624 | NA | NA | AHU | RECIRCULATION | NA | CNC | NA | Y
| NA | NA | NA | Y | Y | NA | NA | NA | Vitrified Tiles | 350
| Micro documentation room | 3.49 | 4.24 | 3 | 14.797600000000001 | 44.3928 | NA | NA | AHU |
RECIRCULATION | NA | CNC | NA | Y | NA | NA | NA | NA | NA | Y | Y | NA | NA | NA | Vitrified Tiles | 350
| Micro sample receipt | 2.63 | 4.24 | 3 | 11.1512 | 33.453599999999994 | NA | NA | AHU | RECIRCULATION |
NA | CNC | NA | Y | NA | NA | NA | Y | Y | NA | NA | NA | Vitrified Tiles | 350
| Change Room | 2.39 | 2.86 | 3 | 6.8354 | 20.5062 | 22.5±2.5 °C | NMT 60% | AHU | RECIRCULATION | 2 | D
| 15 | Y | NA | NA | NA | NA | NA | Y | NA | NA | NA | Epoxy | 250
| Exit | 2.15 | 2.86 | 3 | 6.148999999999999 | 18.446999999999996 | 22.5±2.5 °C | NMT 60% | AHU |
RECIRCULATION | | D | 25 | Y | NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| Culture Room | 4.65 | 3.78 | 3 | 17.577 | 52.73100000000001 | 22.5±2.5 °C | NMT 60% | AHU |
RECIRCULATION | 2 | D | 15 | Y | NA | NA | NA | NA | Y | NA | NA | NA | Epoxy | 350
| Scrap out | 5.49 | 2.86 | 3 | 15.7014 | 47.1042 | NA | NA | AHU | RECIRCULATION | NA | CNC | NA | Y |

NA | NA | NA | Y | Y | NA | NA | NA | Vitrifified Tiles | 350
| Air Lock | 2.66 | 2.86 | 3 | 7.6076 | 22.8228 | NA | NA | AHU | RECIRCULATION | NA | CNC | NA | Y | NA
| NA | NA | Y | Y | NA | NA | NA | Vitrifified Tiles | 350
| Corridor | 2.8 | 7.77 | 3 | 21.755999999999997 | 65.267999999999999 | NA | NA | AHU | RECIRCULATION | NA
| CNC | NA | Y | NA | Y | NA | NA | NA | NA | NA | NA | Vitrifified Tiles | 250
| | 3.76 | 3.78 | 3 | 14.212799999999998 | 42.638399999999999 | NA | NA | AHU | RECIRCULATION | NA | CNC |
NA | Y | NA | NA | NA | NA | NA | NA | NA | NA | NA | Vitrifified Tiles | NA
| Media fill incubation-1 | 5.33 | 3.78 | 3 | 20.147399999999998 | 60.442199999999999 | NA | NA | AHU |
RECIRCULATION | NA | CNC | NA | Y | NA | NA | NA | NA | NA | NA | NA | Vitrifified Tiles | 350
| Media fill incubation-2 | 5.33 | 3.77 | 3 | 20.0941 | 60.282300000000006 | NA | NA | AHU |
RECIRCULATION | NA | CNC | NA | Y | NA | NA | NA | NA | NA | NA | NA | Epoxy | 350
| Media fill incubation-3 | 4.36 | 3.78 | 3 | 16.480800000000002 | 49.442400000000006 | NA | NA | AHU |
RECIRCULATION | NA | CNC | NA | Y | NA | NA | NA | NA | NA | NA | NA | Epoxy | 350
| Media Distribution | 7.34 | 2.02 | 3 | 14.8268 | 44.4804 | NA | NA | AHU | RECIRCULATION | NA | CNC |
NA | Y | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 350
| Decontamination | 5.54 | 2.02 | 3 | 11.1908 | 33.5724 | NA | NA | AHU | RECIRCULATION | NA | CNC | NA |
Y | NA | NA | NA | NA | NA | NA | NA | NA | Vitrifified Tiles | 350
| Corridor | 34.1 | 1.8 | 3 | 61.38 | 184.14000000000001 | NA | NA | AHU | RECIRCULATION | NA | CNC | NA
| Y | NA | Y | Y | Y | NA | NA | NA | Y | Vitrifified Tiles | 250
| QA Office | 18 | 10.8 | 3 | 194.4 | 583.2 | NA | NA | AHU | RECIRCULATION | NA | CNC | NA | Y | NA | NA
| NA | Y | Y | NA | NA | NA | Vitrifified Tiles | 350
| Storage | 4.13 | 4.19 | 3 | 17.3047 | 51.914100000000005 | NA | NA | AHU | RECIRCULATION | NA | CNC |
NA | Y | NA | NA | NA | Y | Y | NA | NA | NA | Vitrifified Tiles | 350
| Cabin 1 | 3.39 | 4.14 | 3 | 14.0346 | 42.1038 | NA | NA | AHU | RECIRCULATION | NA | CNC | NA | Y | NA
| NA | NA | Y | Y | NA | NA | NA | Vitrifified Tiles | 350
| Cabin 2 | 4.05 | 4.14 | 3 | 16.767 | 50.301 | NA | NA | AHU | RECIRCULATION | NA | CNC | NA | Y | NA |
NA | NA | Y | Y | NA | NA | NA | Vitrifified Tiles | 350
| Cabin 3 | 3.43 | 4.14 | 3 | 14.200199999999999 | 42.6006 | NA | NA | AHU | RECIRCULATION | NA | CNC |
NA | Y | NA | NA | NA | Y | Y | NA | NA | NA | Vitrifified Tiles | 350
| Cabin 4 | 2.88 | 4.14 | 3 | 11.923199999999998 | 35.7696 | NA | NA | AHU | RECIRCULATION | NA | CNC |
NA | Y | NA | NA | NA | Y | Y | NA | NA | NA | Vitrifified Tiles | 350
| Audit | 2.85 | 5.81 | 3 | 16.5585 | 49.6755 | NA | NA | AHU | RECIRCULATION | NA | CNC | NA | Y | NA |
NA | NA | Y | Y | NA | NA | NA | Vitrifified Tiles | 350
| QA manager | 2.85 | 4.06 | 3 | 11.571 | 34.713 | NA | NA | AHU | RECIRCULATION | NA | CNC | NA | Y | NA
| NA | NA | Y | Y | NA | NA | NA | Vitrifified Tiles | 350
| QA Head | 2.85 | 4.47 | 3 | 12.7395 | 38.2185 | NA | NA | AHU | RECIRCULATION | NA | CNC | NA | Y | NA
| NA | NA | Y | Y | NA | NA | NA | Vitrifified Tiles | 350
| QA document | 21 | 9.82 | 3 | 206.22 | 618.66 | NA | NA | AHU | RECIRCULATION | NA | CNC | NA | Y | NA
| NA | NA | Y | Y | NA | NA | NA | Vitrifified Tiles | 350
| Corridor | 16.7 | 1.8 | 3 | 30.06 | 90.179999999999999 | NA | NA | AHU | RECIRCULATION | NA | CNC | NA |
Y | NA | NA | NA | NA | NA | NA | NA | NA | Vitrifified Tiles | NA
| R&D Entrance | 7.48 | 2.9 | 3 | 21.692 | 65.076 | NA | NA | AHU | RECIRCULATION | NA | CNC | NA | Y |
NA | NA | NA | NA | NA | NA | NA | NA | NA | Vitrifified Tiles | 250
| R&D Office | 7.31 | 2.9 | 3 | 21.198999999999998 | 63.596999999999994 | NA | NA | AHU | RECIRCULATION |
NA | CNC | NA | Y | NA | NA | NA | Y | Y | NA | NA | NA | Vitrifified Tiles | 350
| Corridor | 14.8 | 1.8 | 3 | 26.64 | 79.92 | NA | NA | AHU | RECIRCULATION | NA | CNC | NA | Y | NA | Y
| NA | NA | NA | NA | NA | NA | NA | Vitrifified Tiles | 250
| Corridor | 13.3 | 1.8 | 3 | 23.94 | 71.82000000000001 | NA | NA | AHU | RECIRCULATION | NA | CNC | NA |
Y | NA | NA | NA | Y | Y | NA | NA | NA | NA | Vitrifified Tiles | 350
| R&D Lab-1 | 6.47 | 6.61 | 3 | 42.7667 | 128.3001 | NA | NA | AHU | RECIRCULATION | NA | CNC | NA | Y |
NA | NA | NA | Y | Y | NA | NA | NA | NA | Vitrifified Tiles | 350
| R&D Lab-2 | 6.47 | 6.61 | 3 | 42.7667 | 128.3001 | NA | NA | AHU | RECIRCULATION | NA | CNC | NA | Y |
NA | NA | NA | Y | Y | NA | NA | NA | NA | Vitrifified Tiles | 350
| R&D Lab-3 | 6.47 | 6.61 | 3 | 42.7667 | 128.3001 | NA | NA | AHU | RECIRCULATION | NA | CNC | NA | Y |
NA | NA | NA | Y | Y | NA | NA | NA | NA | Vitrifified Tiles | 350
| R&D Lab-4 | 6.47 | 6.61 | 3 | 42.7667 | 128.3001 | NA | NA | AHU | RECIRCULATION | NA | CNC | NA | Y |
NA | NA | NA | Y | Y | NA | NA | NA | NA | Vitrifified Tiles | 350

PLASMA THAWING, CUTTING, POOLING AND FILTRATION
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- 1. ISSUANCE OF PLASMA BAGS AND THAWING OF PLASMA BAGS 3
- 2. CUTTING AND POOLING OF PLASMA BAGS 3
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ISSUANCE OF QC TESTED PLASMA BAGS:

THAWING OF PLASMA BAGS

Collect purified water (PW) up to 80% of the volume of the thawing tank and ensure that the temperature is < 30 0C.

Collect the QC test released plasma bags from -25 0C freezer, disinfect with 70% IPA and place them in the thawing tank, lot-wise.

Drain the purified water from the tank when the water temperature drops below 20■C. Refill the tank with fresh, purified water.

Repeat until the plasma bags are completely thawed, take out the plasma bags from the tank and place them in plastic crates.

Transfer these plastic crates with plasma bags from the plasma thawing room to the plasma cutting and pooling room.

CUTTING AND POOLING OF PLASMA BAGS

The plasma bags are cut manually using SS scissors under the LAF unit, and collected in a sterilized glass bottle.

After completion, check and record the following details:

PLASMA FILTRATION THROUGH PP FILTERS 5µm + 1.2 µm FILTERS.

Filter the pooled plasma through sterilised PP filters connected serially.

Note: Collect a 65 ml sample for in-process QC analysis.

Check and record the following details:

Plasma Bags Details | Plasma Bags Details

Total number of plasma bags received |

Total volume of plasma bags (kg) |

Total plasma volume considering the density of 1.05. Volume(L) = plasma volume in kg/density. |

Plasma Bags Details | Plasma Bags Details

Total number of plasma bags received |

Total number of Rejects due to clots & RBCs |

Total number of Rejects due to leaks |

Total number of plasma bags pooled |

Final pooled plasma volume (L) |

Details |

Plasma volume after filtration (L) |

OD at 600 nm |

pH |

Conductivity (mS/cm) |

[SHEET: Sheet1]

Sr.No. | Action Point | Owner | Target Date | BB Response 12.05.26

1 | Sewage & Effluent Treatment: Clearly mark the physical location, dimensions, and layout of the Effluent Treatment Plant (ETP) / Sewage Treatment Plant (STP). Provide structural calculations proving adequate capacity for plasma processing waste. | LVP , Plasma : Mr. Budhish
Dialyzer : Mr. Vinod , Mr. Laxminarayan, Mr. Abhijit | 2026-05-13 00:00:00 | Physical location already marked in site layout.

2 | Organic Waste Converter (OWC): Designate a dedicated, ventilated space for a mechanical OWC, stating its processing capacity in kg/day based on your expected workforce. | LVP , Plasma : Mr. Budhish
Dialyzer : Mr. Vinod , Mr. Laxminarayan, Mr. Abhijit | 2026-05-13 00:00:00 | See attached word file

3 | Green Belt Compliance: Highlight the mandatory 33% Green Belt area (27,033.6 Sqm of your 81,920 Sqm plot) on the master layout plan with a clear index. | Mr. Dashrath Dhupal | 2026-05-13 00:00:00 | NA

4 | Peripheral Driveway: Ensure the peripheral driveway around the 44,190 Sqm industrial shed shows a minimum clear width of 6.00 meters to comply with MIDC Fire Protection Norms. | Mr. Dashrath Dhupal | 2026-05-13 00:00:00 | NA

5 | Process Flow & Water Balance Chart: Submit a detailed quantitative water balance diagram showing domestic and industrial consumption, recycling rates, and a commitment to Zero Liquid Discharge (ZLD). | LVP , Plasma : Mr. Budhish

Dialyzer : Mr. Vinod , Mr. Laxminarayan, Mr. Abhijit | 2026-05-13 00:00:00 | See attached word file for procecss flow water calculations.

6 | Hazardous Waste Management Plan: Since ethanol-based fractionation is utilized, submit storage, handling, and disposal plans for hazardous/chemical solvents. | LVP , Plasma : Mr. Budhish
Dialyzer : Mr. Vinod , Mr. Laxminarayan, Mr. Abhijit | 2026-05-13 00:00:00 | See attached word file

7 | MPCB Status: Submit a copy of the submitted application or acknowledgment for Consent to Establish (CTE) from the Maharashtra Pollution Control Board (MPCB). | Mr.Kiran ECOSIA | 2026-05-13 00:00:00 | NA

8 | Legal Undertaking: Submit a draft copy of the notarized ■500 Stamp Paper Affidavit confirming that no physical construction or site preparation will commence post-IOD until the final EC from SEIAA is obtained and uploaded to MITL. | Ashok | 2026-05-13 00:00:00 | NA

SHAFT

3.30X2.45
DISPATCH BAY RECEIVING BAY RECEIVING BAY DISPATCH BAY
FIREMAN

LIFT
LIFT
2.80X2.60
5.00 M WIDE CORRIDOR
62.5X5.00
UP

LADIES TOILET

2.94 M WIDE CORRIDOR
3.30 x 4.35

WC WC WC
1.10X 1.10X 1.10X
1.70 1.70 1.70

6.77 M WIDE CORRIDOR
3600.000

2000.000
WC WC WC
1.10X 1.10X 1.10X
1.70 1.70 1.70

GENTS TOILET
3.30 x 4.35

SHAFT SHAFT

LIFT LIFT
2.10X 2.10X

3.24 M WIDE PASSAGE
2.00 2.00
FD
UP
FD
DIALYZER FD
UP
FD

48.54 x 91.34

3300.000

2.80 M WIDE PASSAGE

LADIES TOILET
3.30 x 4.35

WC WC WC
1.10X 1.10X 1.10X
1.70 1.70 1.70

WC WC WC
1.10X 1.10X 1.10X
1.70 1.70 1.70

GENTS TOILET
3.30 x 4.35
6.77 M WIDE CORRIDOR

6.77 M WIDE CORRIDOR
6.77X212.59

6.77X212.59
GENTS VISITOR FEMALE DOCUMENT JANITOR
SHAFT OFFICE 3 OFFICE 4 ROOM SHAFT
WAITING ROOM LOCKER LOCKER LOCKER ROOM
LIFT
4.00x5.00 4.70x5.00 4.70x5.00 2.00x5.00
LIFT 4.67x5.00 ROOM ROOM ROOM LIFT
5.90x5.00 6.00x5.00
2.10X 6.00x5.00 2.10X

3.24 M WIDE PASSAGE
2.00 2.00
FD FD FD FD
6D

RECEPTION 4.72x1.80 CORRIDOR
UP 36.7x1.80 UP
7.00x12.0

PRIMARY VISITOR
AIR LOCK AIR LOCK VISITOR OFFICE 1 OFFICE 2
FEMALE PRIMARY FEMALE CHANGE 1
AIR LOCK PRIMARY MALE MALE CHANGE 1 FEMALE 1.80x5.15 4.30x5.10 4.30x5.10
CHANGE 1 CHANGE VISITOR CHANGE 2.55x5.10
MALE CHANGE 3.30x5.10 3.40x5.10 3.19x5.10 3.64x5.10 3.05x5.10 2.70x5.10
4.70x5.10 4.00x5.10
8D

6D

This space can be adjusted to increase
DE
CORRIDOR
chemical day store room 45.8x2.85

DPB
DPB DPB WASTE OUT
LAF DHS
3 WAY
6D

6D

DECONTAMINATION WASHING STERILIZATION
AUTOCLAVE

CHEMICAL DAY STORE
LAF

6.98X3.04 BUFFER TANK STORAGE-2 CIP SIP COLD ROOM 3.52x4.67 3.11x4.67 3.401x4.67
5.60x7.09
6D

PREPARATION 6.74x7.09 4.40x7.09
COOL ONE
4.00x7.09 2.75x7.09
DPB DPB
AL GOWN AL 7.29x1.80
2.10X3.40 2.47X3.40 2.20X3.40 WASTE AL AL
4.13x2.34 3.40x2.34
GS+HH

GS+HH

IN
LENAP LORTNOC LENAP LORTNOC
2.50x2.34
6D 6D 6D

AL DE GOWN AL
DE

2.10X3.40 2.47X3.40 2.20X3.40 CORRIDOR
36.0x2.85

D6 D6 D3
LAF
D6 D6
DPB AKTA-1 TFF
NANO
FILTER MH
JANITOR
1.80x2.00
LADIES TOILET
AL GOWN AL 1000 L
3.30 x 4.35
2.10X4.76 2.47X4.76 2.20X4.76 BULK FILTRATION
LAF

6.54X7.29 500 L
WC WC WC
6.77 M WIDE CORRIDOR

1.10X 1.10X 1.10X
2.80 M WIDE PASSAGE

MH

1.70 1.70 1.70
LAF

STAGE 9 N
MH

DE 500 L MH

50 L
WC WC WC AL GOWN AL LAF SL
MH

5.10X15.6 CHROMO SL

STAGE 13 E
1.10X 1.10X 1.10X 2.10X4.76 2.47X4.76 2.20X4.76 MH

COLUMN 250 L
1.70 1.70 1.70 600 L 600 L STAGE 10 T MH

500 L 7.48X13.5
STAGE 11 F
MH

SL
MH

600 L MH

200 L 5.63X2.05 250 L
AKTA-1

GENTS TOILET
STAGE 6 I
MH

3.30 x 4.35

COLUMN PACKING 14.7X18.0 250 L MH

250 L
MH

2000 L MH

AND SKIDS
MH

200 L

6.54X10.6 150 L 250 L
COOL ONE
MH MH

MH
1000 L MH
100 L MH

6.98x5.53 500 L
150 L MH

300 L
MH

MH

100 L MH

1000 L
MH

100 L
6770.000
150 L 300 L
LAF
DPB LAF
MH

MH

CHROMO MH

AKTA-1
100 L COLUMN

DHS
AUTOCLAVE
D6

500 L 500 L CHROMO
250 L 250 L 150 L 100 L COLUMN
6D

CHROMO MH
MH MH

AL MH MH
MH MH
MH

MH
COLUMN
CORRIDOR
2.62x71.4
1.50x6.88
STERILIZATION CORRIDOR

2000 L
D6

5.38x6.85
CLAF

2500 L 2500 L 2.62X74.2
CORRIDOR
6D

1.80X55.7
PRE - VIRUS CONTROLLED CORRIDOR

1000 L
TFF

MH

COLUMN PACKING
DPB
AND SKIDS MH

WASHING 100 L
7.64X8.86
AKTA-1

AL 5.38x3.00 STAGE 7A
AKTA-1

1.50x4.42 500 L
13.6X16.1 MH

STAGE 5 I A
D6

DPB LADIES TOILET
WASTE IN MH

500 L 3.30 x 4.35
250 L
11.1X19.0
TFF

MH

300 L
MH
2.00X32.2

WC WC WC
DECONTAMINATION 500 L 1.10X 1.10X 1.10X
250 L 1.70 1.70 1.70
OUT 5.38x5.00 MH
MH
MH

WASTE
D6

1.50x3.54 500 L

DPB STAGE 5 I B 1500 L 250 L
MH
MH

100 L
7.64X7.20 MH

250 L 250 L 250 L
SL
600 L 500 L WC WC WC
100 L 100 L
MH

MH MH MH 100 L MH
MH
1.10X 1.10X 1.10X
100 L MH MH MH

1.70 1.70 1.70

CIP SIP
MH

1500 L
SL
MH MH

MH MH MH

6.98x5.00 1000 L 100 L
MH

MH
500 L 500 L 1000 L 1000 L 1000 L
MH

GENTS TOILET
3.30 x 4.35
MH
6D

MH

1500 L
50 L 1500 L
GS+HH

GS+HH

SL
LENAP LORTNOC LENAP LORTNOC
2500 L
MH

1500 L

MH
250 L 250 L 250 L CHROMO
250 L
DPB MH MH MH COLUMN MH

WASTE OUT
6D

BAG CUTTING
POOLING MH
MH MH

MH

MH MH
STAGE 4 F
7.64X9.80 150 L 500 L
13.6X15.8
6D

500 L
SL 1000 L
TANK STORAGE 1000 L 1500 L 1500 L

6.98x8.58
D6

MH

STAGE 12 E
MH

250 L
MH

STAGE 1 F 500 L
CHROMO
COLUMN
11.1X10.6 250 L
300 L 300 L MH
TFF
D6

AKTA-1

6D
MH MH

CLAF

2500 L
6D

THAWING PLASMA
2000 L 2000 L
7.64X3.61
AKTA-1

MH

BUFFER DPB
500 L
250 L 250 L 300 L 300 L 300 L MH MH

PREPARATION MH
CHROMO
COLUMN
MH MH MH

MH MH

CHROMO
COLUMN
6.98x6.00 D6 D6 D6
6D

DE

SHAFT CORRIDOR SHAFT
DPB 34.1X1.80
LIFT D6 MH MH D6 D6 LIFT
2.10X 2.10X
MH MH MH MH

CHROMO MH
MH
MH MH MH MH

COLUMN
100 L 100 L 100 L 100 L MH

100 L 100 L CENTRIFUGE
6D

2.00 200 L 250 L 300 L 300 L 500 L 500 L 2.00
1500 L
3.24 M WIDE PASSAGE

3.24 M WIDE PASSAGE

FD FD FD FD
UP
PLASMA DAY STAGE 3 F UP
STAGE 2 F 12.1X11.2
STORE STAGE 8 N
15.8X11.2
5.84X11.2
6.98x10.7
AKTA-1

AKTA-1

UP
CHROMO MH

COLUMN

JANITOR SL
5.40x1.80 600 L 2000 L
1500 L 1500 L 1500 L 1500 L

300 L 300 L MH

DPB
MH MH MH
MH

100 L 100 L 100 L CHROMO CHROMO
MH
TFF MH

COLUMN COLUMN AKTA-1
MH MH MH

EMER.GLASS EMER.GLASS
CORRIDOR
43.3x1.80
6.77 M WIDE CORRIDOR

PLASMA STORAGE PLASMA STORAGE PLASMA STORAGE
~ 1000 L ~ 1000 L PLASMA STORAGE
~ 1000 L ~ 1000 L
~ 70 KL GV ~ 70 KL GV ~ 70 KL GV
9.59x11.5m 9.02x11.5m ~ 70 KL GV
9.35x11.5m 9.05x11.5m
CORRIDOR PLASMA STORAGE CORRIDOR
2.62x15.2 ~ 1000 L 2.62x15.2
~ 70 KL GV
5.98x11.5m

LADIES TOILET
3.30 x 4.35

D3 D3 D3 D3 D3
WC WC WC
CORRIDOR
2.80 M WIDE PASSAGE

1.10X 1.10X 1.10X
1.70 1.70 1.70 43.2x1.80
D6

CORRIDOR
28.1X1.80

D3 D6
WC WC WC DPB RLAF
D3

AIR LOCK-1
1.10X 1.10X 1.10X COLD ROOM -3 COLD ROOM -2 COLD ROOM -1 ICE PACK
1.70 1.70 1.70
1.50 X 2.18 RM QUARANTINE BAG BAR CODING, WASHING
JANITOR
PLASMA COLD ROOM -4
AREA QUARANTINE
MINI POOLS 4.62X7.34 2.49X7.34
6D

2.60X2.72
2.30X2.72
1.45X2.72
4.44X7.34 4.12X7.34 4.45X7.34
DISPENCING ROOM
3.40 X 4.44
3D D3 D3 D3

GENTS TOILET
3.30 x 4.35 CHANGE ROOM
1.50 X 2.18 5.85X6.55
3D

5.00X2.72
11.5X4.30
CORRIDOR
1.80X12.0
D3
3D

SD SD SD SD

DPB
6D

D3
APPROVED RM AIR LOCK-1
2.00X1.21
DPB D3 3D D3 D6 STORE 10.4X6.49
3.65X4.49 2.64X2.60 AIR LOCK-1 DPB
1.87X1.40 OFFICE 4.70X4.08
8.65X7.54 3D
2.23X6.50
CHANGE ROOM
2.00X1.80 3D
3.00X3.55
CHANGE ROOM

DE-DUSTING
1.87X1.40
REJECT SEROLOGY PLASMA
RECEIPT
D3

D3 TUNNEL
PLASMA ROOM TESTING 3D D3
STORE CHANGE
D3 SAMPLING
1.76X 2.90
2.26X4.95 3.00X 4.95 2.40X 4.95 ROOM
RM 4.70X2.37
SAMPLING 3.68X 2.05 RECALLED REJECT
RM FG DISCPATCH
3.16X2.31
RECEIPT GOODS ROOM FG IN
1.07X3.11 2.30X2.35 2.30X 2.35
2.64X2.77
RLAF RLAF 6D D6 D6
D3 D3

D6 D3

5.00 M WIDE CORRIDOR
3.30X2.45

6D

D6

48.9X5.00
LIFT

DISPATCH BAY RECEIVING BAY RECEIVING BAY DISPATCH BAY
40' CONTAINER YARD
CONTAINER

CONTAINER

CONTAINER

CONTAINER
12.19x2.44x2.59
12.19x2.44x2.59

12.19x2.44x2.59

12.19x2.44x2.59

GROUND FLOOR CLIENT :- LIFENITY-LARGE VOLUME PARENTERALS

DESIGN &
PROJECT BY :-BIOBRIDGE HEALTHCARE SOLUTION PVT.LTD.
PREPARED A.K
ADDRESS :-13, Rachna Blossom, Jagdishnagar,
Aundh, Pune -411007, Maharastra, India. CHECKED G.K.

DATE 01-04-2026
TITLE- CONCEPTUAL LAYOUT-GF REVISION R0
SCALE 1:1

SIZE A0 DWG.NO 01

Reagent Details:
Preparation of AL-RR-23 working buffer from Stock buffer (AL-RR-09):
Calculations: For the Preparation of 150 mM Sodium Acetate pH 5.2.
Stock Buffer (AL-RR-09):0.6M Sodium acetate, pH 5.35.
Formula:
 $C1 \times V1 = C2 \times V2$
C1 = concentration of stock
V1 = volume of stock you need to take
C2 = desired (working) concentration
V2 = final total volume of working solution
Applying the formula to prepare the working buffer:
 $C1 \times V1 = C2 \times V2$
C1 = 0.2M Sodium Acetate pH 5.35.
V1 = volume of stock you need to take _____ L.
C2 = desired (working) concentration 150mM Sodium Acetate pH 5.2
V2 = final total volume of working solution _____ L.
 $V1 = (C2 \times V2) / C1$
C2=0.15M ; V2= 1.0 L; C1=0.6M
 $V1 = (0.15 \times 1.0) / 0.6$
V1 = 0.25L
Take 0.25 L (AL-RR-09) Stock buffer solution, dilute with water, and make up to 1 L.
Check and record the details:
Preparation Details:
Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry
Date
Sodium Acetate | | | |
Acetic Acid | | | |
Stock buffer (AL-RR-09) taken volume(L) | pH(5.2±0.1) | Conductivity (1.4±0.1ms/cm) | Storage Condition
| Final volume prepared
| | RT |
Date of Preparation | Prepared By | Checked By
| |

1. Size of dispatch and receiving bay:
FG Dispatch room: 9.42 x 6.25 meter, 4.70 x 3.85 meter, 4.55 x 2.32 meter,
Plasma receiving bay: 10 x 6.84 meter
RM receiving bay: 10 x 6.84 meter
SPM receiving bay: 10 x 6.84 meter
FG dispatch corridor: 10 x 6.84 meter
2. Loading/unloading dock height: 3 meter
3. Clear floor height requirement: 3 meter
4. Slab load capacity (kg/sq.m):
5. Point load of heavy machinery:
6. Crane requirement (capacity & hook height):
Heavy load items for first floor are: Vial washing machine gross weight is 4000 Kg, Autoclave gross weight is 2500 Kg
Filling machine 2200 Kg
Lyophilizer machine 7500 Kg
7. Machine foundation requirement: No machine will be placed on foundation it will be rested on flaps
8. Total electrical load (kW): Working load 4143.37 KVA and stand by load 1068.24 KVA
9. Power backup requirement (DG/UPS): 1000 KVA (2 sets x 500 KVA)
10. Water inlet size and pressure:
11. Daily water consumption (liters/day): Water consumption (including process water): 87557 L/day
12. Process water requirement (RO/DM): purified water, WFI
13. Waste water generation per day: 74984 L/day
14. ETP/STP requirement: 900-1000 m3/year
15. Drainage line size:
16. Harmful chemicals used (if any): 1.0-4.5 KL/ year (spent solvent, triton x100 and polysorbate)
17. Chemical storage room requirement:
Approved RM area is 9.72x7.22 meter and 4.63 x 4.00 meter
RM quarantine area is 6.21 x 5.77 meter
18. Biomedical waste disposal system: Biomedical waste disposal shall be done by 3rd party
19. Thermal insulation requirement: Modular panels as per GMP design will be used
Roof/wall insulation requirement: Modular panels as per GMP design will be used
20. Epoxy or anti-static flooring requirement: Epoxy shall be done
21. Male/Female toilet requirement (as per staff):
Gents toilet size is 5.90 x 5.20 and Female toilet size is 5.90 x 5.20
22. Future expansion provision upper floors

Reagent Details:
Formula & Calculations:
Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).
Calculation: To prepare 1 L of 2N Sodium Hydroxide (MW = 40.0 g/mol)
Weight of solute(g)=2×40 = 80.00g.
Preparation Method:
Weigh 80.00 g of Sodium Hydroxide and dissolve in water to make a 1 L solution.
Preparation for different volumes:
To prepare _____ of 2N Sodium Hydroxide (MW = 40.0 g/mol).
Weight of solute(g)=2×40.0 x _____L= _____gm
Weigh _____gm of Sodium Hydroxide and dissolve in water to make a _____ L solution.
Check and record the details:
Preparation Details:
Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry
Date
Sodium Hydroxide | | | |
Final volume prepared | Weighed Quantity | pH (13.0±1.0) | Conductivity (405.0±5.0ms/cm) | Storage
Condition:
| | | RT
Date of Preparation | Prepared By | Checked By
| |

TABLE OF CONTENTS

CONCENTRATED SAMPLE.....2

FORMULATION.....2

QC ANALYSIS REPORT.....3

CONCENTRATED SAMPLE:

1.1 Concentrate the sample and record the following details:

pH: _____, Conductivity: _____ mS/cm.

O.D.280/600: _____.Final Volume: _____.

FORMULATION

After concentration, add 100mM L-Arginine Hydrochloride and 1% Sucrose to a final concentration respectively.

Quantity of L-Arginine Hydrochloride added: _____

Quantity of Sucrose added: _____

Add and adjust the concentration of HSA-20% to a final concentration of 10 mg/mL of the concentrated sample volume

Volume of HSA-20% added before concentration from BMR step 7.2 : _____.

Volume of HSA-20% added after concentration: _____.

Check and record the following details.

pH: _____, Conductivity: _____ mS/cm.

O.D280/600: _____. Final Volume: _____.

Record the QC Analysis report.

Store the bottles containing PRA-SF-001 filtrate at -20 °C until further use.

Details | Details

Volume (L) |

Activity (IU/ml) |

Protein Conc. (mg/ml) |

Total protein (mg) |

List of Raw materials for LVP Project

S. No. Raw material UOM Qty Yearly

- 1 Ciprofloxacin Kg 15,000
- 2 Fluconazole Kg 7,500
- 3 Levofloxacin Kg 18,750
- 4 Linezolid Kg 7,500
- 5 Mannitol Kg 24,37,500
- 6 Metronidazole Kg 18,750
- 7 Moxifloxacin Kg 15,000
- 8 Ofloxacin Kg 7,500
- 9 Ornidazole Kg 18,750
- 10 Paracetamol Kg 56,250
- 11 Sodium Chloride Kg 4,35,000
- 12 Dextrose Kg 26,25,000
- 13 Linezolid Kg 7,500
- 14 Mannitol Kg 1,125
- 15 Moxifloxacin Kg 15,000
- 16 Tinidazole Kg 7,500
- 17 Sodium lactate Kg 36,000
- 18 Dextrose Kg 50,62,500

FOR OFFICE USE

LIFT
(O/H)TANK(2) MACHINE
ROOM

1200.0000

TERRACE

LIFT
6100.0000

6100.0000
23400.0000
BUILDING L.V.P.

8500.0000
BUILDING DIALIZER
T-0.30m.
R-0.15m.

FRONT ELEVATION
SECTION A-A SCALE 1:100
SCALE 1:100

SHAFT SHAFT SHAFT

3.30X2.45

3.30X2.45

3.30X2.45
DISPATCH BAY RECEIVING BAY RECEIVING BAY DISPATCH BAY
FIREMAN FIREMAN FIREMAN

LIFT

LIFT

LIFT
LIFT LIFT LIFT
2.80X2.60 2.80X2.60 2.80X2.60
5.00 M WIDE CORRIDOR

UP

PLAN OF PROPOSED FACTORY

AD
BUILDINGON PLOT NO 8A, SECTOR 13,

DE RO
LADIES TOILET LADIES TOILET LADIES TOILET

DMIC, AURIC , SHENDRA,

2.94 M WIDE CORRIDOR

2.94 M WIDE CORRIDOR

2.94 M WIDE CORRIDOR
WI
3.30 x 4.35 3.30 x 4.35 3.30 x 4.35

WC WC WC WC WC WC WC WC WC
1.10X 1.10X 1.10X 1.10X 1.10X 1.10X 1.10X 1.10X 1.10X
1.70 1.70 1.70 1.70 1.70 1.70 1.70 1.70 1.70

6.77 M WIDE CORRIDOR

6.77 M WIDE CORRIDOR

6.77 M WIDE CORRIDOR
60.00 M.

3600.000

3600.000

3600.000
CHATTRAPATI SAMBHAJI NAGAR

2000.000

2000.000

2000.000
WC WC WC WC WC WC WC WC WC
1.10X 1.10X 1.10X 1.10X 1.10X 1.10X 1.10X 1.10X 1.10X

1.70 1.70 1.70 1.70 1.70 1.70 1.70 1.70 1.70

GENTS TOILET GENTS TOILET GENTS TOILET
3.30 x 4.35 3.30 x 4.35 3.30 x 4.35

FOR M/S LIFENITY LIMITED
ROAD
WIDE
60.00 M.

AD
60.
00

RO
M.
W
ID
PLOT NO. 1B

E
E

ID
SHAFT SHAFT SHAFT SHAFT SHAFT SHAFT

RO

60.00 M.
LIFT LIFT LIFT LIFT LIFT LIFT

W
AD 2.10X 2.10X 2.10X 2.10X 2.10X 2.10X

3.24 M WIDE PASSAGE

3.24 M WIDE PASSAGE

3.24 M WIDE PASSAGE
2.00 2.00 2.00 2.00 2.00 2.00

DIALYZER LVP WORKSHOP

M.
FD FD FD FD FD FD FD FD
UP UP UP UP UP UP

00
48.54 x 91.34 48.54 x 96.57 48.54 x 96.57

30.
AREA STATEMENT
WIDE PROPOSED SITE AREA IN

3300.000
3300.000 3300.000

NO PARTICULARS
SQM.

ROAD
60
.00M

.
W 1 AREA OF PLOT NO 56 SECTOR 13, DMIC, SHENDRA 81920.00
ID
PLOT NO. 1C E
RO
AD AS PER OWNERSHIP DOCUMENTS 81920.00
OAD
45.00 M. W I D E R

2.80 M WIDE PASSAGE

2.80 M WIDE PASSAGE

2.80 M WIDE PASSAGE
LADIES TOILET LADIES TOILET LADIES TOILET
3.30 x 4.35 3.30 x 4.35 3.30 x 4.35

45.00 M. W I D E R
OA D WC
1.10X
1.70
WC
1.10X
1.70
WC
1.10X
1.70
WC
1.10X

1.70
WC
1.10X
1.70
WC
1.10X
1.70
WC
1.10X
1.70
WC
1.10X
1.70
WC
1.10X
1.70
AS PER MEASUREMENT SHEET 81920.00

30.00 M. W I D E R O A D
AS PER SITE
WC WC WC WC WC WC WC WC WC

81920.00
1.10X 1.10X 1.10X 1.10X 1.10X 1.10X 1.10X 1.10X 1.10X
1.70 1.70 1.70 1.70 1.70 1.70 1.70 1.70 1.70

GENTS TOILET GENTS TOILET GENTS TOILET

GATE-2
3.30 x 4.35 3.30 x 4.35 3.30 x 4.35

53285.2997

2 DEDUCTIONS FOR
ENTRY
a. PROPOSED DP ROAD 0.00
46971.8561

6.77 M WIDE CORRIDOR

6.77 M WIDE CORRIDOR

6.77 M WIDE CORRIDOR

6.77 M WIDE CORRIDOR

6.77 M WIDE CORRIDOR

6.77 M WIDE CORRIDOR
SECURITY

b. ANY RESERVATIONS 0.00
CABIN-2

TOTAL (a+b)
LOCATION PLAN 0.00

DRIVER'S 3 BALANCE AREA OF PLOT (1-2) (A) 81920.00
SCALE :- NTS
ROOM 4 AMENITY SPACE IF APPLICABLE 5% 0.00
SHAFT SHAFT SHAFT SHAFT SHAFT SHAFT

LIFT
2.10X
LIFT
2.10X
LIFT
2.10X
LIFT
2.10X
LIFT
2.10X
LIFT
2.10X
a. Required 0.00

3.24 M WIDE PASSAGE

3.24 M WIDE PASSAGE

3.24 M WIDE PASSAGE
2.00 2.00 2.00 2.00 2.00 2.00
FD FD FD FD FD FD FD FD FD
UP UP UP UP UP UP

b. Adjustment of 2(b) if any 0.00
c. Balanced proposed 0.00

5 NET PLOT AREA (3-4(c))) PRO RATA 81920.00
OPEN SPACE-1 6 RECREATIONAL OPEN SPACE IF APPLICABLE 10%
4024.28 LADIES TOILET
3.30 x 4.35

LADIES TOILET
3.30 x 4.35
LADIES TOILET
3.30 x 4.35 a. Required
WC WC WC WC WC WC WC WC WC

6.77 M WIDE CORRIDOR

6.77 M WIDE CORRIDOR

6.77 M WIDE CORRIDOR
1.10X 1.10X 1.10X 1.10X 1.10X 1.10X 1.10X 1.10X 1.10X

2.80 M WIDE PASSAGE

2.80 M WIDE PASSAGE

2.80 M WIDE PASSAGE
1.70 1.70 1.70 1.70 1.70 1.70 1.70 1.70 1.70

b. Proposed

12000.000
WC WC WC WC WC WC WC WC WC
1.10X 1.10X 1.10X 1.10X 1.10X 1.10X 1.10X 1.10X 1.10X
1.70 1.70 1.70 1.70 1.70 1.70 1.70 1.70 1.70

GENTS TOILET GENTS TOILET GENTS TOILET
3.30 x 4.35 3.30 x 4.35 3.30 x 4.35

7 INTERNAL ROAD AREA 0.00

12000.000
6770
6770 6770 8 PLOTABLE AREA IF APPLICABLE (as per sanctioned layout) 81920.00

9 B/UP area with reference to basic fsi as per front road width(1.0) 81920.00

60
INTERNAL ROAD 12.00 M WIDE
.00 LADIES TOILET

WC
3.30 x 4.35

1.10X
1.70
WC
1.10X
1.70
WC
1.10X
1.70
LADIES TOILET

WC
3.30 x 4.35

1.10X
1.70
WC
1.10X
1.70
WC
1.10X
1.70
LADIES TOILET

WC
3.30 x 4.35

1.10X
1.70
WC
1.10X
1.70
WC
1.10X
1.70
LADIES TOILET

10 ADDITIONAL FSI ON PAYMENT OF PREMIUM
a. Maximum permissible premium FSI / TOD Zone. 2.00
M. PLASMA EXTRACTION
48.54 x 115.59
WC
1.10X
1.70
WC
1.10X
1.70
WC
1.10X
1.70

PLASMA EXTRACTION

48.54 x 120.82

WC

1.10X

1.70

WC

1.10X

1.70

WC

1.10X

1.70

WORKSHOP

48.54 x 120.82

WC

1.10X

1.70

WC

1.10X

1.70

WC

1.10X

1.70

b. Proposed FSI of payment of Premium 0.00

115590

GENTS TOILET GENTS TOILET GENTS TOILET

3.30 x 4.35 3.30 x 4.35 3.30 x 4.35

W

ID

7224

11 IN SITU TDR / FSI LOADING

I

N

E a. In situ area against Statutory Raod (2 xSr No 2(a)) if any 0.00

7.500 12.000 12.000 12.000 12.000 7.500

SHAFT

3.30X2.45

T

DISPATCH BAY RECEIVING BAY RECEIVING BAY DISPATCH BAY

FIREMAN

LIFT

LIFT

2.80X2.60

E

R

5.00 M WIDE CORRIDOR

62.5X5.00

12 ADDITIONAL FSI AREA UNDER CHAPTER NO.6 0.000

R

UP

O

SHAFT SHAFT SHAFT SHAFT SHAFT SHAFT

LIFT LIFT LIFT LIFT LIFT LIFT

N

2.10X 2.10X 2.10X 2.10X 2.10X 2.10X

2.00 2.00 2.00 2.00 2.00 2.00

AD 13 TOTAL ENTITLEMENT OF FSI IN THE PROPOSAL

3.24 M WIDE PASSAGE

3.24 M WIDE PASSAGE

3.24 M WIDE PASSAGE

3.24 M WIDE PASSAGE

3.24 M WIDE PASSAGE

3.24 M WIDE PASSAGE

FD FD FD FD FD FD FD FD

0.000

A UP UP UP UP UP UP

L UP UP

a. 9 +10(b) +11(a) or 9+12 which ever is applicable

0.00

LADIES TOILET

Permissible

2.94 M WIDE CORRIDOR

3.30 x 4.35

WC
1.10X
1.70
WC
1.10X
1.70
WC
1.10X
1.70
R a. 9 +10(b) +11(a) or 9+ 12 which ever is applicable
6.77 M WIDE CORRIDOR

3.600

O
2.000
3600.000

Proposed 0.00
2000.000

WC WC WC
1.10X 1.10X 1.10X

A
1.70 1.70 1.70

GENTS TOILET b. Permissible Ancillary area FSI up to 80% with payment
D
3.30 x 4.35

0.00
315211.3172

PERMISSIBLE
1

6.77 M WIDE CORRIDOR

6.77 M WIDE CORRIDOR

6.77 M WIDE CORRIDOR
b. Proposed Ancillary area FSI up to 60% or 80% with payment
2
.0 LADIES TOILET
3.30 x 4.35
LADIES TOILET
3.30 x 4.35
LADIES TOILET
3.30 x 4.35
c.Total Entitlement permissible
0 WC WC WC WC WC WC WC WC WC
(a+b) 81920.00

2.80 M WIDE PASSAGE

2.80 M WIDE PASSAGE

2.80 M WIDE PASSAGE
1.10X 1.10X 1.10X 1.10X 1.10X 1.10X 1.10X 1.10X 1.10X
1.70 1.70 1.70 1.70 1.70 1.70 1.70 1.70 1.70

M WC WC WC WC WC WC WC WC WC c.Total Entitlement proposed
W 81920.00
1.10X 1.10X 1.10X 1.10X 1.10X 1.10X 1.10X 1.10X 1.10X
1.70 1.70 1.70 1.70 1.70 1.70 1.70 1.70 1.70

SHAFT SHAFT
GENTS TOILET GENTS TOILET GENTS TOILET
(a+b)
ID
3.30 x 4.35 3.30 x 4.35 3.30 x 4.35
LIFT LIFT
2.10X 2.10X
Maximum utilization limit of FSI (Building Potential) permissible as per
3.24 M WIDE PASSAGE

2.00 2.00
FD FD
DIALYZER FD FD

14 81920.00
UP

48.54 x 91.34
UP

E road width ((as per Regulation No. 5.5 of Chapter - 5)) x 1.6 or 1.8)

15 TOTAL Built up area in proposal (excluding area as per sr. no 17b)
3.3003300.000

SE
C
CA UR 5.00 M WIDE CORRIDOR
a. Existing Built -up Area.

3.30X2.45

3.30X2.45

3.30X2.45
0.000

LIFT

LIFT

LIFT
BI ITY
DISPATCH BAY RECEIVING BAY RECEIVING BAY DISPATCH BAY

N- b. Proposed Built- up area (as per P line) 40968.18
1 GA
EN TE- WORKSHOP BUILDING GROUND FLOOR PLAN WORKSHOP BUILDING FIRST FLOOR PLAN WORKSHOP BUILDING SECOND
FLOOR PLAN
c. Total (a+b) 40968.18
TR 1 SCALE 1:100 SCALE 1:100 SCALE 1:100

Y
2.80 M WIDE PASSAGE

LADIES TOILET
3.30 x 4.35

WC
1.10X
1.70
WC
1.10X
1.70
WC
1.10X
1.70
16 FSI CONSUMED (15/13) (should not be more than serial 0.50
WC WC WC
No.14 above.)
1.10X 1.10X 1.10X
1.70 1.70 1.70

GENTS TOILET
3.30 x 4.35

CERTIFICATE OF AREA :-
19500 Certified that the plot under reference was surveyed by me on
330840.1374

63000
02/06/2025 and the Dimensions of sides etc.of plot stated on plan area

R I D M.
AREA CALCULATION
6.77 M WIDE CORRIDOR
6.77 M WIDE CORRIDOR

6.77 M WIDE CORRIDOR

as measured on site and the area so worked out tallies with the stated in
6.77X212.59

6.77X212.59

documents of ownership/ DMIC/Land records Department/City Survey

W .00
O E
Records
Ar. Dashrath Dhumal

AD
30
Owners Decleration :-
SHAFT
WAITING ROOM
GENTS
LOCKER
VISITOR
LOCKER
FEMALE
LOCKER
DOCUMENT

ROOM
4.00x5.00
OFFICE 3
4.70x5.00
OFFICE 4
4.70x5.00
JANITOR
ROOM
2.00x5.00 LIFT
SHAFT 3600 I/We undersigned hereby confirmed that I/We would abide by plans
R O A D 1 2 . 0 0 M W I D E

R O A D 1 2 . 0 0 M W I D E

LIFT 4.67x5.00 ROOM ROOM ROOM LIFT
6.00x5.00

approved by Authority/ Collector. I / We would execute the structure as
5.90x5.00
2.10X 2.10X
BLOCK A = 63.00 X 218.05 = 13737.15
6.00x5.00
GREEN BELT 6 M WIDE

3.24 M WIDE PASSAGE

2.00 2.00

2000
FD FD FD FD
6D

RECEPTION CORRIDOR

per the approved plans.Also I/We would execute the work under
UP 4.72x1.80 36.7x1.80 UP
7.00x12.0

supervision of proper technical person so as to ensure the quality and
PRIMARY VISITOR
AIR LOCK AIR LOCK VISITOR OFFICE 1 OFFICE 2
FEMALE PRIMARY FEMALE CHANGE 1
AIR LOCK PRIMARY MALE MALE CHANGE 1 FEMALE VISITOR 1.80x5.15 4.30x5.10 4.30x5.10
CHANGE 1 CHANGE CHANGE 2.55x5.10
MALE CHANGE 3.30x5.10

DEDUCTIONS STAIRCASE SHAFT
3.40x5.10 3.19x5.10 3.64x5.10 3.05x5.10 2.70x5.10
4.70x5.10 4.00x5.10

safety at the work site.
8D

SHAFT 1 = 1.40 X 3.30 = 4.62
6D

DE

CORRIDOR
45.8x2.85

DPB Mr. Sarang Landage.
R O A D 1 2 . 0 0 M W I D E

DPB DPB WASTE OUT
LAF DHS
3 WAY
6D

6D

DECONTAMINATION WASHING STERILIZATION

6 SHAFTS = 4.62 X 6 = 27.72
AUTOCLAVE

CHEMICAL DAY STORE
LAF

6.98X3.04 BUFFER TANK STORAGE-2 CIP / SIP COLD ROOM 3.52x4.67 3.11x4.67 3.401x4.67
5.60x7.09
6D

PREPARATION 6.74x7.09 4.40x7.09
COOL ZONE

A/L
2.10X3.40
GOWN
2.47X3.40
A/L
2.20X3.40

7.29x1.80
4.00x7.09
DPB

WASTE A/L A/L
DPB
2.75x7.09

Architects Name and Signature
4.13x2.34 3.40x2.34
GS+HH

GS+HH

IN

DEDUCTIONS TOILET SHAFT
LENAP LORTNOC LENAP LORTNOC
2.50x2.34
6D 6D 6D

A/L DE GOWN A/L

OPEN SPACE-2 3300
DE

1400

1400
2.10X3.40 2.47X3.40 2.20X3.40 CORRIDOR
36.0x2.85 3300
DPB
D6 D6
AKTA-1 TFF LAF
D6 D6 D3

SHAFT 1 = 3.60 X 2.00 = 7.20 Ar. Dashrath Dhumal
4178.44 sq.m.
NANO
FILTER MH
JANITOR
1.80x2.00
LADIES TOILET
A/L GOWN A/L 1000 L
3.30 x 4.35
BULK FILTRATION
Drn. By. Checked By Registration no of Architect
2.10X4.76 2.47X4.76 2.20X4.76
LAF

WC WC WC
6.54X7.29 500 L

5 SHAFTS = 7.20 X 5 = 36.00 Job No. Drg. No. Scale
6.77 M WIDE CORRIDOR

1.10X 1.10X 1.10X
2.80 M WIDE PASSAGE

MH

1.70 1.70 1.70
LAF

WC
1.10X
1.70
WC
1.10X
1.70
WC
1.10X
1.70
A/L
2.10X4.76
DE
GOWN
2.47X4.76
A/L
2.20X4.76
LAF MH
SL
600 L
500 L
MH
STAGE 9 N
5.10X15.6 CHROMO
COLUMN
SL
600 L
MH

STAGE 13 E
STAGE 10 T/
50 L

250 L
MH
MH

001 1:100 Ar. Dashrath Dhumal CA/2005/37017
500 L 7.48X13.5
STAGE 11 F
MH

SL
MH

600 L MH

200 L 5.63X2.05 250 L
AKTA-1

GENTS TOILET
STAGE 6 (I)
MH

DEDUCTIONS LIFT SHAFT
3.30 x 4.35
14.7X18.0

PARKING STATEMENT
250 L

C
COLUMN PACKING MH

2000 L MH
MH

250 L

AND SKIDS
MH

200 L

6.54X10.6 150 L 250 L
COOL ZONE
MH MH

MH
1000 L MH
100 L MH

6.98x5.53 500 L

LIFT 1 = 3.30 X 2.45 = 8.085
150 L MH

300 L
MH

100 L

a GARAGE PERMISSIBLE GARAGE PROPOSED NIL
MH
MH

1000 L
MH

100 L
6.770
6770.000

150 L 300 L
LAF
DPB LAF
MH

MH

CHROMO MH

AKTA-1
100 L COLUMN

DHS
AUTOCLAVE

2 LIFTS = 8.085 X 2 = 16.17 PARKING PARKING
D6

500 L 500 L CHROMO
250 L 250 L 150 L 100 L COLUMN
26191.6800
6D

CHROMO
MH MH

b TYPE
MH

COLUMN
A/L MH MH
MH MH
MH

MH

CORRIDOR
2.62x71.4
1.50x6.88
STERILIZATION CORRIDOR

REQUIRED PROPOSED
2000 L
D6

5.38x6.85
CLAF

2500 L 2500 L 2.62X74.2
CORRIDOR
6D

1.80X55.7
PRE - VIRUS CONTROLLED CORRIDOR

LIFT 2 = 2.00 X 2.10 = 4.20
1000 L
TFF

CAR
MH

COLUMN PACKING

2000
DPB
AND SKIDS MH

WASHING 100 L
7.64X8.86
AKTA-1

5.38x3.00 STAGE 7A
6 LIFTS = 4.20 X 6 = 25.20
A/L
AKTA-1

1.50x4.42 500 L
I N T E R N A L

I N T E R N A L

13.6X16.1 MH

STAGE 5 (I) A
SCOOTER
D6

DPB LADIES TOILET
WASTE IN MH

500 L 3.30 x 4.35
250 L
11.1X19.0
TFF

MH

300 L
MH
2.00X32.2

OUT
DECONTAMINATION
5.38x5.00 MH
250 L
MH
500 L
MH
WC

1.10X
1.70
WC
1.10X
1.70
WC
1.10X
1.70
3600 8192 SQ.M. 8192 SQ.M.
CYCLE
WASTE
D6

500 L

PLASMA EXTRACTION
1.50x3.54
STAGE 5(I)B
TOTAL B/UP AREA = 13737.15
1500 L 250 L
DPB
MH

MH

100 L
7.64X7.20 MH

250 L 250 L 250 L
SL
600 L 500 L
WC WC WC
100 L 100 L
MH

48.54 x 115.59
MH MH MH 100 L MH
MH
1.10X 1.10X 1.10X
100 L MH MH MH

1.70 1.70 1.70

CIP / SIP
VISITORS
MH

1500 L
SL
MH MH

60
MH MH MH

6.98x5.00 1000 L 100 L

6 SHAFTS = 4.62 X 6 = 27.72
500 L GENTS TOILET
MH

MH
MH

500 L 1000 L 1000 L 1000 L
3.30 x 4.35
MH
6D

MH

1500 L
50 L 1500 L
GS+HH

GS+HH

SL
2500 L

LOADING UNLOADING SPACES
LENAP LORTNOC LENAP LORTNOC

MH

D
1500 L

W .00 M
250 L 250 L 250 L 250 L

5 SHAFTS = 7.20 X 5 = 36.00
MH

CHROMO
DPB MH MH MH COLUMN MH

WASTE OUT
6D

BAG CUTTING
POOLING MH
MH MH

MH

MH MH
STAGE 4 (F)
7.64X9.80 150 L 500 L
13.6X15.8
6D

500 L
SL 1000 L
TANK STORAGE
a LOADING UNLOADING REQUIRED
1000 L 1500 L 1500 L

6.98x8.58
2 LIFTS = 8.085 X 2 = 16.17 6
D6

MH

STAGE 12 (E)
MH

250 L

R IDE .
MH

STAGE 1 (F)/ 500 L
CHROMO
COLUMN
11.1X10.6
300 L 300 L
250 L
MH

6 LIFTS = 4.20 X 6 = 25.20 b TOTAL LOADING UNLOADING PROVIDED 6
TFF
D6

AKTA-1

6D
MH MH

CLAF

2500 L
6D

THAWING PLASMA
2000 L 2000 L
7.64X3.61
AKTA-1

MH

BUFFER DPB
500 L
250 L 250 L 300 L 300 L 300 L MH MH
I N T E R N A L

PREPARATION CHROMO

O
COLUMN CHROMO
MH MH MH MH
MH MH

COLUMN
6.98x6.00

TOTAL B/UP AREA = 13632.06
D6 D6 D6
6D

DE

CORRIDOR

LEGENDS
SHAFT SHAFT

DPB 34.1X1.80
LIFT LIFT

AD
D6 MH MH D6 D6

2.10X 2.10X
MH MH MH MH

CHROMO MH MH
MH
MH
MH MH

COLUMN
100 L 100 L 100 L 100 L MH

100 L 100 L CENTRIFUGE
6D

2.00 200 L 250 L 300 L 300 L 500 L 500 L 2.00

PLOT AREA = 81920.00
1500 L
3.24 M WIDE PASSAGE

3.24 M WIDE PASSAGE

FD FD FD FD
UP
PLASMA DAY STAGE 3 (F) UP
STAGE 2 (F) 12.1X11.2
STORE STAGE 8 N
15.8X11.2
5.84X11.2
PLOT BOUNDRY SHOWN IN

1400
6.98x10.7
AKTA-1

AKTA-1

AMENITY SPACE
D

UP
3300
JANITOR
5.40x1.80
CHROMO
COLUMN

SL
600 L
MH

2000 L
B/UP AREA SHOWN IN

1400
1500 L 1500 L 1500 L

3300
1500 L
300 L

4125.33 sq.m.
A

300 L MH

DPB
MH MH
MH MH

100 L 100 L 100 L CHROMO CHROMO
MH
TFF MH

COLUMN COLUMN AKTA-1

WATER LINE SHOWN IN
MH MH MH

EMER.GLASS EMER.GLASS
CORRIDOR
43.3x1.80
O

FORM OF STATEMENT DRAINAGE LINE SHOWN IN
79701.6798

218050
R

AREA CALCULATIONS
6.77 M WIDE CORRIDOR

PLASMA STORAGE PLASMA STORAGE PLASMA STORAGE
~ 1000 L ~ 1000 L PLASMA STORAGE
~ 1000 L ~ 1000 L
~ 70 KL GV ~ 70 KL GV ~ 70 KL GV
9.59x11.5m 9.02x11.5m ~ 70 KL GV
9.35x11.5m 9.05x11.5m
CORRIDOR PLASMA STORAGE CORRIDOR
2.62x15.2 ~ 1000 L 2.62x15.2
~ 70 KL GV
5.98x11.5m BLOCK NO BLOCK NAME FLOOR SIZE AREA
SCHEDULE OF DOORS AND WINDOWS
E

LADIES TOILET 1 BUILDING-01 WORKSHOP GROUND 13632.06
3.30 x 4.35

WC WC WC
D3 D3 D3

CORRIDOR
D3 D3

FIRST 13632.06 TYPE SIZE DISCRIPTION
2.80 M WIDE PASSAGE

1.10X 1.10X 1.10X
43.2x1.80
ID

1.70 1.70 1.70

D6
CORRIDOR
28.1X1.80

DPB
D3 D6 SECOND 13632.06 RS 6000 X 6000 M.S ROLLING SHUTTER
WC WC WC RLAF
D3

AIR LOCK-1
1.10X 1.10X 1.10X COLD ROOM -3 COLD ROOM -2 COLD ROOM -1 ICE PACK
1.70 1.70 1.70
1.50 X 2.18 RM QUARANTINE BAG BAR CODING, WASHING
JANITOR
PLASMA COLD ROOM -4
AREA QUARANTINE
MINI POOLS 4.62X7.34 2.49X7.34
6D

2.60X2.72
2.30X2.72
1.45X2.72
4.44X7.34 4.12X7.34 4.45X7.34
DISPENCING ROOM
3.40 X 4.44
3D D3 D3 D3
TOTAL OF BUILT-UP AREA 40896.18 D1 750 X 2100 R.C.C. FRAME FLUSH DOOR
W

GENTS TOILET
3.30 x 4.35 CHANGE ROOM
1.50 X 2.18 5.85X6.55
3D

D4 1000 X 2100
5.00X2.72
11.5X4.30
CORRIDOR
1.80X12.0 D3 2 DRIVERS ROOM GROUND 72.00 R.C.C. FRAME FLUSH DOOR
3D

SD SD SD SD

DPB
6D

N
D3
APPROVED RM AIR LOCK-1 DPB
STORE 10.4X6.49
FD 1000 X 2100 FIRE DOOR
2.00X1.21 D3 3D D3 D6
3.65X4.49 2.64X2.60 AIR LOCK-1 DPB
1.87X1.40 OFFICE 4.70X4.08

8.65X7.54 3D
2.23X6.50
CHANGE ROOM
3.00X3.55
40968.18
2.00X1.80 3D

TOTAL B/UP AREA OF ALL BUILDINGS
M.

CHANGE ROOM

DE-DUSTING
1.87X1.40
REJECT SEROLOGY PLASMA
RECEIPT
D3

TESTING
ROOM
W1 3000 X 1500 ALLU. SLIDING WINDOW
TUNNEL 3D D3

2000
D3 PLASMA

3600
D3 STORE CHANGE
SAMPLING
1.76X 2.90
2.26X4.95 3.00X 4.95 2.40X 4.95 ROOM
RM 4.70X2.37
SAMPLING 3.68X 2.05 RECALLED REJECT
RM FG DISCPATCH
3.16X2.31
RECEIPT GOODS ROOM FG IN
1.07X3.11 2.30X2.35 2.30X 2.35

F S I CONSUMED
2.64X2.77
RLAF RLAF 6D D6 D6
D3

0.50
D3

D6 D3

V 600 X 900 M.S. LOUVERED VENTILATOR
.00

5.00 M WIDE CORRIDOR
3.30X2.45

6D

D6

48.9X5.00
LIFT

DISPATCH BAY RECEIVING BAY RECEIVING BAY DISPATCH BAY

PLAN OF PROPOSED FACTORY
45
40' CONTAINER YARD

7224

BUILDINGON PLOT NO 8A, SECTOR 13,
CONTAINER
CONTAINER

CONTAINER

CONTAINER
12.19x2.44x2.59
12.19x2.44x2.59

12.19x2.44x2.59

12.19x2.44x2.59

DMIC, AURIC , SHENDRA,
INTERNAL ROAD 12 . 0 0 M W I D E
CHATRAPATI SAMBHAJI NAGAR
WORKSHOP BUILDING GROUND FLOOR PLAN

GREEN BELT 6 M WIDE

A
899.7116

4500.0000

4500.0000

3300
231187.1690

Security Cabin
4.00 x 5.00
899.1372

12000.0000

1400

1400
3300
10449.9866

A
14000.0000

UP
A
14000.0000

FOR M/S LIFENITY LIMITED
5999.5206

A
5454.7337

VISITORS
4.00 x 6.74

AREA KEY PLAN
(Driver's Room)

AREA KEY PLAN
12000.0000

Pump Room

TOILET
2.40 x 1.50 A W.C. A
10449.9867

W.B. 1.20 x
1.20 BATH
1.20 x
2.55
TYPICAL PLAN-1,2&3. AREA KEY PLAN
(Security Cabin)
A Security Cabin DRIVER'S ROOM
8.84 x 4.54
5999.5207

A PUMP ROOM A
5454.7660

LOCKER'S RM.
10.00 X 5.00

45.00 M. W I D E R O A D
2.55 x 1.84

UP
UP

PLAN 1996.9025
2997.1267

1514.1437

1198.8496

1514.1437

999.0415

229.7794

(Driver's Room)
10449.9870
599.4248

PARAPET LVL. PLAN (Pump Room)
SLAB LVL.
599.4248

Parapet LVL.
PARAPET LVL.

1598.4661

Slab LVL.

SIGN OF OWNER
SLAB LVL. LINTEL LVL.
RCC Chajja
3996.1654
1598.4661

1498.5620

Lintel LVL.
2397.6992

LINTEL LVL. SILL LVL.
3996.1654

899.1372

PLINTH LVL.
1498.5620

Sill LVL.
G.LVL. G.LVL.
2397.6992

SILL LVL.
449.5686

FRONT ELEVATION
Plinth LVL.
899.1372

PLINTH LVL. (Driver's Room)
G. LVL. G. LVL.
G.LVL. G.LVL.
449.5686

FRONT ELEVATION
LEFT SIDE ELEVATION
(Security Cabin)

PARAPET LVL.
TERRACE TERRACE Parapet LVL.
599.4248

SLAB LVL.
Terrace Slab LVL.
1598.4662
899.7115

LINTEL LVL. RCC Chajja
Lintel LVL.
3996.1654
1498.5620

3996.1654
2397.6992

3496.6447

SILL LVL.
1498.5620

2000
3600
2397.6992

2397.6992

Sill LVL.
DRIVER'S ROOM
899.1372

Security Cabin VISITORS TOILET W.C. BATH
PLINTH LVL. SUNK SUNK
G.LVL. PUMP ROOM
899.1372

SUNK
G.LVL. Plinth LVL.
G. LVL. G. LVL.
449.5686

SECTION- AA
SECTION- AA SECTION- AA
(Driver's Room)
(Security Cabin)

SIGN OF ARCHITECT

LAYOUT PLAN
CREATION

JOB No. DRG NO. SCALE DATE DRN BY CHK. BY

01 1 :100

Reagent Details:

Formula:

Sodium Acetate (M.Wt): 82.03g/mol.

Acetic acid (M.Wt): 60.05 g/mol

Glycine (M.Wt): 75.07 g/mol

Sodium Chloride (M.Wt):58.44 g/mol

Calcium Chloride (M.Wt): 110.98 g/mol

0.01% Tween 80

Preparation of FE-RR-010 buffer with Stock FE-RR-09.

Preparation of reagent :30mM sodium acetate ph 6.7 +50mM glycine +0.2Msodium chloride +5 mM calcium chloride+0.01% tween 80, pH-7.0.

Stock Buffer (FE-RR-09):0.2M Sodium Citrate pH 6.9.

Formula:.

$$C1 \times V1 = C2 \times V2$$

C1 = concentration of stock

V1 = volume of stock you need to take

C2 = desired (working) concentration

V2 = final total volume of working solution

Applying the formula to prepare the working buffer:

$$C1 \times V1 = C2 \times V2$$

C1 = 0.2M Sodium Citrate pH 6.9

V1 = volume of stock you need to take _____ L.

C2 = desired (working) concentration 0.03M Sodium Citrate, pH 7.0

V2 = final total volume of working solution _____ L.

$$V1 = (C2 \times V2) / C1$$

C2=0.03M ; V2= 1.0 L; C1=0.2M

$$V1 = (0.03 \times 1.0) / 0.2$$

$$V1 = 0.15L$$

Take 0.15 L (FE-RR-09) Stock buffer solution and dilute with water, and make up to 1L.

Calculations: To prepare the 0.2M Sodium Chloride.

Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).

Calculation: To prepare 1 L of 0.2M Sodium Chloride (MW = 58.44 g/mol)

$$\text{Weight of solute(g)} = 0.2 \times 58.44 \times 1 = 11.688g$$

Preparation Method:

Weigh 11.688g of Sodium Chloride and dissolve in water to make a 1 L solution.

Preparation for different volumes:

To prepare _____ L of 0.2M Sodium Chloride (MW = 58.44 g/mol).

$$\text{Weight of solute(g)} = 0.2 \times 58.44 \times \text{L} = \text{g}$$

Weigh _____ gm of Sodium Chloride and dissolve in water to make a _____ L solution.

Calculations: To prepare the 5mM Calcium Chloride.

Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).

Calculation: To prepare 1 L of 5 mM Calcium Chloride (MW = 110.98g/mol)

$$\text{Weight of solute(g)} = 0.005 \times 110.98 \times 1 = 0.554g$$

Preparation Method:

Weigh 0.554g of Calcium Chloride and dissolve in water to make a 1 L solution.

Preparation for different volumes:

To prepare _____ L of 0.005mM Calcium Chloride (MW =110.98 g/mol).

$$\text{Weight of solute(g)} = 0.005 \times 110.98 \times \text{L} = \text{g}$$

Weigh _____ gm of Calcium Chloride and dissolve in water to make a _____ L solution.

Procedure

Calculation: to prepare the 0.01% Tween 80:

For 1000 mL of 0.01% solution:

$$0.01\% = 0.01g \text{ per } 100ml \text{ \& } 1000 \text{ mL} = 100 \text{ mg}$$

Since Tween 80 is a liquid, you can weigh it directly or use its density (~1.06 g/mL) to measure volume:

$$\text{Approx. } 100 \mu\text{L of Tween 80} = 0.1 \text{ g}$$

Add Tween 80 to a beaker or flask containing ~800 mL of water.

Mix thoroughly:

Use a magnetic stirrer or vortex until fully dissolved. Tween 80 is viscous, so mixing may take a few minutes.

Calculations: To prepare the 50mM Glycine.

Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).

Calculation: To prepare 1 L of 0.05M Glycine (MW = 75.07g/mol)

$$\text{Weight of solute(g)} = 0.05 \times 75.07 \times 1 = 3.753g$$

Preparation Method:

Weigh 3.753g of Glycine and dissolve in water to make a 1 L solution.

Preparation for different volumes:

To prepare _____ L of 0.05M Glycine (MW =75.07g/mol).

$$\text{Weight of solute(g)} = 0.05 \times 75.07 \times \text{L} = \text{g}$$

Weigh _____ gm of Glycine and dissolve in water to make a _____ L solution.

Adjust volume:

After complete dissolution, bring the final volume up to 1000 mL with water.

Procedure for Preparation of FE-RR-11 buffer with the stock FE-RR-09.

To prepare a 1 L solution, take 0.2M Tri Sodium Citrate pH 6.9. To this, add the 0.2 M sodium chloride and dissolve it and make up the volume with water to 900ml, and then to this add 5mM Calcium chloride, and then to this add 50mM Glycine and to this add the 0.01% Tween 80 then mix the solution till it dissolves and check the pH; if the pH is less than 7.0 then adjust the pH with 1 M sodium hydroxide to get the pH 7.0 ±0.1 and make up the final volume to 1L.

Check and record the details:

Preparation Details:

Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry

Date

Sodium Acetate | | | |

Acetic Acid | | | |

Glycine | | | |

Sodium Chloride | | | |

Calcium Chloride | | | |

Tween 80 | | | |

Stock buffer (FE-RR-09) taken volume(L) | pH (7.0 ± 0.1) | Conductivity (24±1.0ms/cm) | Storage

Condition | Final volume prepared

| | RT |

List of Raw materials for Plasma Project

S.No Chemicals Name UOM Qty per year

- 1 Sodium Citrate (dihydrate)(gm) Kg 115801.88
- 2 Citric Acid (gm) Kg 99078.75
- 3 Sodium Chlorid (gm) Kg 598522.50
- 4 Sodium Hydroxide (gm) Kg 919363.13
- 5 Calcium Chlorid (gm) Kg 5452.50
- 6 Sodium Acetate (anhydrous)(gm) Kg 200362.50
- 7 Acetic Acid (ml) L 47013.75
- 8 Glycine (gm) Kg 35812.50
- 9 TRIS (gm) Kg 17812.50
- 10 20% HAS (gm) Kg 566.25
- 11 Sodium Caprylate (gm) Kg 2437.50
- 12 Sodium N- Acetyl tryptophan Kg 2437.50
- 13 Glycerol(ml) L 28125.00
- 14 HCL(ml) L 56250.00
- 15 L- Arginine (gm) Kg 326.63
- 16 Sucroce(gm) Kg 187.50
- 17 Tween80(ml) L 384.38
- 18 TNBP(ml) L 37687.50
- 19 Ammonium Sulphate (gm) Kg 82500.00
- 20 Triton X 100 (ml) L 37500.00
- 21 Histidine (gm) Kg 187.50
- 22 Nicotinamide(mg) Kg 75.00
- 23 Maltose (gm) Kg 3750.00
- 24 Iso Propanal Alcohol (ml) L 75000.00
- 25 Sodium Hypo Chloride (ml) L 15000.00
- 26 Sepharose 4 FF L 565.20
- 27 DEAE sepharose FF L 1143.12
- 28 Q Sepharose FF L 602.88
- 29 CM Sepharose FF L 459.70
- 30 Heparin Sepharose 6 FF L 19.08
- 31 Superdex 200PG L 50.87
- 32 Q Sepharose high performance L 141.30
- 33 SP Sepharose high performance L 19.08
- 34 Superdex G25 Kg 38.15
- 35 Boric acid GR Kg 8.75
- 36 Agarose DNase, RNase free Kg 1.75
- 37 Anti human serum antibody produced in Rabbit L 0.02
- 38 Coomassie brilliant blue R-250 Kg 0.22
- 39 Disodium hydrogen phosphate dihydrate (HPLC grade) Kg 4.38
- 40 Copper sulphate GR Kg 4.38
- 41 Prekallikrein Activator Assay Kit Kit 70.00
- 42 Prekallikrein Activator Assay Ig Kit Kit 70.00
- Working Standard (WS) - Human Normal Immunoglobulin vial 5.83
- 43 for Intravenous use 5% I.P. (Immuglo 5% 100 mL) calibrated against Sigma Protein Standard (200 mg/mL):
- 44 Platelet poor plasma L 0.35
- 45 Monolisa Anti-HBs Plus Kit kit 70.00
- 46 Diptheria Toxoid IgG Enzyme Immunoassay kit kit 70.00
- 47 Human IgG subclass profile Kit kit 70.00
- 48 Anhydrous Sodium dihydrogen phosphate GR Kg 8.75
- 49 Magnesium chloride GR kg 1.46
- 50 D (+) glucose GR kg 4.38
- 51 Tannic acid GR Kg 0.88
- 52 HEPES buffer AR L 0.88
- 53 Guinea-pig complement Vial 2.92
- 54 Rubella antigen Vial 5.83
- 55 COAMATIC Factor VIII kit kit 52.50
- WHO Reference standard (RS) (International Standard, kit 4.38
- 56
- Factor VIII Concentrate,
- 57 STA Deficient VIII Plasma kit 8.75
- 58 FVIII-deficient plasma pack 35.00
- 59 CK presto kit kit 4.38
- 60 (5th International Standard for Factor IX Concentrate) vial 5.83
- Standard human plasma with known FVIII activity (e.g., kit 4.38
- 61
- 100%)
- 62 Factor IX Deficient Plasma kit 8.75
- 63 Owren koller buffer - Ready to use. pack 4.38
- 64 STA Deficient Factor II pack 4.38
- 65 STA Deficient Factor VII pack 5.83
- 66 STA Deficient Factor X pack 5.83
- 67 Neoplastine Cl Plus pack 5.83
- 68 Unicalibrator pack 8.75
- 69 Sodium tartrate GR Kg 4.38
- 70 Folin-Ciocalteu's phenol reagent L 0.58
- 71 Bovine Serum Albumin (BSA) vial 8.75
- 72 Bovine Gamma Globulin (BGG) vial 8.75
- 73 Sulphuric acid L 8.75
- 74 Dipotassium Sulphate kg 4.38
- 75 Hydrogen Peroxide L 1.46
- 76 Methyl red kg 0.15
- 77 Methylene blue Kg 0.04
- 78 Tris-Cl buffer L 4.38
- Human Thrombin preparation (500 IU/mL label claim vial) vial 8.75
- 79
- 80 Imidazole Kg 1.46
- 81 Fibrinogen Concentrate Vial 5.83

82 Hbs Ag - HEPA ELISA (4th generation) Kit 52.50
83 HCV ELISA (3rd generation) Kit 52.50
84 HIV ELISA (4th generation) Kit 52.50
85 Human IgA ELISA Development Kit (ALP) for 6 plates Kit 17.50
Haemolysin [Antiserum against sheep vial 8.75
86
red blood cells prepared in rabbits
87 HPLC grade methanol L 21.88
88 Sodium Hydroxide pellets GR Kg 8.75
89 Trisodium citrate, anhydrous, 99%, Kg 4.38
90 Sodium carbonate anhydrous for Kg 14.58
91 Potassium Sodium Tartarate Tetra Hydrate Kg 4.38
92 Anhydrous Disodium Hydrogen Phosphate Kg 4.38
93 Anhydrous Sodium dihydrogen phosphate Kg 4.38
Bovine Serum Albumin heat shock fraction, protease free, Kg 0.06
94
essentially globulin free

Reagent Details:

Formula & Calculations:

Formula: $M_1 \times V_1 = M_2 \times V_2$.

Calculation: To prepare 1 L of 2mM Tris from 1M tris stock solution calculate quantity required of stock solution as below:

1M tris stock solution required quantity= $2 \times 1 / 1000 = 2 \text{ mL}$

Preparation Method:

To prepare 0.1M Tris 1L solution from 1M Tris stock solution formula is $M_1 (\text{Molarity of stock}) \times V_1 (\text{volume of stock}) = M_2 (\text{Molarity of working}) \times V_2 (\text{volume of working})$.

Take 2mL of 1M tris solution and make a 1 L solution.

Preparation for different volumes:

To prepare _____ of 2mM Tris

Take 2mL x _____ L = _____ mL of 1M Tris

Make Tris solution to _____ L solution.

Check and record the details:

Preparation Details:

Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry

Date

||| |

Final volume prepared | Weighed Quantity | pH(7.9±0.1) | Conductivity (0.1 ms/cm) | Storage Condition:

||| | RT

Date of Preparation | Prepared By | Checked By

||

Reagent Details:

Formula:

Sodium Citrate (M.Wt): 258.07.

Citric Acid (M.Wt): 192.124

Calcium Chloride(M.Wt): 110.98

Sodium Chloride(M.Wt):58.44

Preparation of FN-RR-10 working buffer from Stock buffer (FN-RR-08):

Calculations: For the Preparation of 20mM Sodium Citrate pH 7.4 + 2.5mM Calcium Chloride.

Stock Buffer (FN-RR-08):0.2M Sodium Citrate pH 7.4.

Formula:

$$C1V1=C2V2$$

C1 = concentration of stock

V1 = volume of stock you need to take

C2 = desired (working) concentration

V2 = final total volume of working solution

Applying the formula to prepare the working buffer:

$$C1V1=C2V2$$

C1 = 0.2M Sodium Citrate pH 7.4.

V1 = volume of stock you need to take _____ L.

C2 = desired (working) concentration 20mM Sodium Citrate, pH 7.4

V2 = final total volume of working solution _____ L.

$$V1 = (C2 \times V2)/C1.$$

C2=0.02M ; V2= 1.0 L; C1=0.2M

$$V1 = (0.02 \times 1.0)/0.2$$

$$V1 = 0.1L$$

Take 0.1 L (FN-RR-08) Stock buffer solution and dilute it with water to make up 1 L.

Calculations: 2.5mM Calcium Chloride.

Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).

Calculation: To prepare 1 L of 2.5mM Calcium Chloride (MW = 110.98 g/mol)

$$\text{Weight of solute(g)}=0.0025 \times 110.98 \times 1=0.277g.$$

Preparation Method:

Weigh 0.277g of Calcium Chloride and dissolve in water to make a 1 L solution.

Preparation for different volumes:

To prepare _____ L of 0.0025 M Calcium Chloride (MW = 110.98 g/mol).

$$\text{Weight of solute(g)}=0.0025 \times 110.98 \times \text{_____} L= \text{_____} g.$$

Weigh _____ gm of Calcium Chloride and dissolve in water to make a _____ L solution.

Calculations: 200mM Sodium Chloride.

Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).

Calculation: To prepare 1 L of 200mM Sodium Chloride (MW = 58.44 g/mol)

$$\text{Weight of solute(g)}=0.2 \times 58.44 \times 1=11.688g.$$

Preparation Method:

Weigh 11.688 g of Sodium Chloride and dissolve in water to make a 1 L solution.

Preparation for different volumes:

To prepare _____ L of 0.2 M Sodium Chloride (MW = 58.44 g/mol).

$$\text{Weight of solute(g)}=0.2 \times 58.44 \times \text{_____} L= \text{_____} g.$$

Weigh _____ gm of Sodium Chloride and dissolve in water to make a _____ L solution.

Preparation of working buffer (FN-RR-10):

Take 0.1 L (FN-RR-08) Stock buffer solution, add 0.0025 M Sodium Chloride (0.277 gm) and add 200mM Sodium

Chloride (11.688gm). Dissolve it, and then make up the final volume to 1 L with water.

Check and record the details:

Preparation Details:

Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry

Date

Sodium Citrate | | | |

Citric Acid | | | |

Calcium Chloride | | | |

Sodium Chloride | | | |

Stock buffer (FN-RR-08) taken volume(L) | Calcium Chloride added (gm) | Sodium Chloride added(gm) | pH

(7.4±0.1) | Conductivity (23.0 ± 0.2 ms/cm) | Storage Condition | Final volume prepared (L)

| | | | RT |

Date of Preparation | Prepared By | Checked By (QC/Analyst)

| |

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Formulation

Check the QC result for Bulk from step 11.3.3:

Protein Concentration _____, pH: _____,

Conductivity: _____ uS/cm, Total Volume: _____ L,

Total Protein: _____ g.

Formulation with 10% (W/V) Maltose:

Finally, the sample concentration should be 5%.

Folds to be diluted to the total volume of the Bulk sample.

A= Protein Concentration of Bulk sample _____/50 mg.

A= _____.

Final Total volume to be made up after formulation =A x Bulk total volume.

B=A _____ x _____.

B= _____ L.

Difference between = final volume - bulk volume.

C= B _____ - _____.

C= _____ L.

Calculation of the 10 % maltose.

D=Final volume (B) x 10% (w/v) Maltose.

D=B _____ x 10% _____.

D= _____ kg.

Dissolving the maltose:

Add the calculated amount of the maltose slowly to the Bulk Sample under stirring (150 rpm) for 15 to 20 min.

Note: Check with 10% of maltose dissolved in the 5 ml of water, and the rise of the water level and check the difference between the final rise volume and mine's initial volume. The difference in the volume is considered a standard rise value for the formulated bulk.

Standard rise in the formulated bulk volume:

= rise in volume – dissolved in 5 ml.

E= _____.

Rise in volume after the addition of Maltose to the Bulk Sample.

F= D (kg) _____ + Bulk volume _____ x E _____.

F= _____ L.

Calculating the Dilution of the formulated bulk sample with the dilution buffer to achieve the final 5% concentration.

G = (B) final total volume -(F) Rise volume after maltose addition.

G = _____.

Formulate Bulk Details:

Check and record the following details:

OD at 280 nm: _____, OD at 600 nm: _____,

pH: _____, Conductivity: _____ uS/cm,

Protein Concentration _____.

Total Volume: _____ L, Total Protein: _____ g.

Collect 25 ml of sample and send to QC for Formulated Bulk-API analysis. Record the obtained results as per the QC report:

Protein Concentration _____, pH: _____.

Conductivity: _____ uS/cm,

Calculate and record the following details according to the QC result:

pH: _____, Conductivity: _____ uS/cm,

Protein Concentration _____.

Total Volume: _____ L, Total Protein: _____ g.

Adjusting the Protein concentration. If it is more than 5.5% then dilute the sample with the dilution buffer.

Final protein concentration should be 5.0. % accepted criteria is $\pm 0.5\%$.

Do the sterile filtration with a 0.2 μ m filter.

Submit the final sample for the QC testing as a drug substance.

Record the Drug substance QC result.

pH: _____, Conductivity: _____ uS/cm,

Protein Concentration _____.

Total Volume: _____ L, Total Protein: _____ g.

Storage of DS sample.

The DS sample is stored in the 2 to 8 °C.

Reagent Details:

Formula:

Sodium Citrate (M.Wt): 258.07.

Citric Acid (M.Wt): 192.124

Calcium Chloride(M.Wt): 110.98

Sodium Chloride(M.Wt):58.44

Preparation of FN-RR-11 working buffer from Stock buffer (FN-RR-08):

Calculations: For the Preparation of 20mM Sodium Citrate pH 7.4 + 2.5mM Calcium Chloride + 370 mM Sodium Chloride.

Stock Buffer (FN-RR-08):0.2M Sodium Citrate pH 7.4.

Formula:.

$C1 \times V1 = C2 \times V2$

C1 = concentration of stock

V1 = volume of stock you need to take

C2 = desired (working) concentration

V2 = final total volume of working solution

Applying the formula to prepare the working buffer:

$C1 \times V1 = C2 \times V2$

C1 = 0.2M Sodium Citrate pH 7.4.

V1 = volume of stock you need to take _____ L.

C2 = desired (working) concentration 20mM Sodium Citrate, pH 7.4

V2 = final total volume of working solution _____ L.

$V1 = (C2 \times V2) / C1$.

C2=0.02M ; V2= 1.0 L; C1=0.2M

$V1 = (0.02 \times 1.0) / 0.2$

V1 = 0.1L

Take 0.1 L (FN-RR-08) Stock buffer solution and dilute it with water to make up 1 L.

Calculations: 2.5mM Calcium Chloride.

Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol)×Volume (L).

Calculation: To prepare 1 L of 2.5mM Calcium Chloride (MW = 110.98 g/mol)

Weight of solute(g)=0.0025×110.98×1=0.277g.

Preparation Method:

Weigh 0.277g of Calcium Chloride and dissolve in water to make a 1 L solution.

Preparation for different volumes:

To prepare _____ L of 0.0025 M Calcium Chloride (MW = 110.98 g/mol).

Weight of solute(g)=0.0025 ×110.98× _____ L= _____ g.

Weigh _____ gm of Calcium Chloride and dissolve in water to make a _____ L solution.

Calculations: 370 mM Sodium Chloride.

Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).

Calculation: To prepare 1 L of 370 mM Sodium Chloride (MW = 58.44 g/mol)

Weight of solute(g)=0.37×58.44 ×1=21.622g.

Preparation Method:

Weigh 21.622 g of Sodium Chloride and dissolve in water to make a 1 L solution.

Preparation for different volumes:

To prepare _____ L of 0.37 M Sodium Chloride (MW = 58.44 g/mol).

Weight of solute(g)=0.37 ×58.44× _____ L= _____ g.

Weigh _____ gm of Sodium Chloride and dissolve in water to make a _____ L solution.

Preparation of working buffer (FN-RR-11):

Take 0.1 L (FN-RR-08) Stock buffer solution, add 0.0025 M Calcium Chloride (0.277 gm) and add 370mM

Sodium Chloride (21.622gm). Dissolve it, and then make up the final volume to 1 L with water.

Check and record the details:

Preparation Details:

Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry

Date

Sodium Citrate | | | |

Citric Acid | | | |

Calcium Chloride | | | |

Sodium Chloride | | | |

Stock buffer (FN-RR-08) taken volume(L) | Calcium Chloride added (gm) | Sodium Chloride added(gm) | pH

(7.4 ± 0.1) | Conductivity (37.0 ± 0.5ms/cm) | Storage Condition | Final volume prepared (L)

| | | | RT |

Date of Preparation | Prepared By | Checked By (QC/Analyst)

| |

Reagent Details:

Formula & Calculations:

Formula: $M_1 \times V_1 = M_2 \times V_2$.

Calculation: To prepare 1 L of 0.01M sodium hydroxide from 0.5M sodium hydroxide stock solution calculate quantity required of stock solution as below:

0.5M sodium hydroxide stock solution required quantity= $0.01 \times 1 / 0.5 = 0.02$ L

Preparation Method:

To prepare 0.01M sodium hydroxide 1L solution from 0.5M sodium hydroxide stock solution formula is

$M_1 (\text{Molarity of stock}) \times V_1 (\text{volume of stock}) = M_2 (\text{Molarity of working}) \times V_2 (\text{volume of working})$.

Take 0.02L of 0.5M sodium hydroxide solution and make a 1 L solution.

Preparation for different volumes:

To prepare _____ of 0.01M sodium hydroxide

Take 0.02L x _____ L = _____ L of 0.5M sodium hydroxide

Make sodium hydroxide solution to _____ L solution.

Check and record the details:

Preparation Details:

Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry

Date

|||

Final volume prepared | Weighed Quantity | Storage Condition:

||

Date of Preparation | Prepared By | Checked By

||

Reagent Details:
Formula & Calculations:
Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).
Calculation: To prepare 1 L of 0.1M citric acid (MW = 192.124g/mol)
Weight of solute(g)=1×19.212=19.212g.
Preparation Method:
Weigh 19.212 g of citric acid and dissolve in water to make a 1 L solution.
Preparation for different volumes:
To prepare _____ of 0.1M citric acid (MW = 192.124g/mol).
Weight of solute(g)=1×19.212x _____L= _____gm
Weigh _____gm of citric acid base and dissolve in water to make a _____ L solution.
Check and record the details:
Preparation Details:
Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry
Date
|||||
Final volume prepared | Weighed Quantity | pH (2.0±0.2) | Conductivity (27±2.0ms/cm) | Storage Condition:
|||| RT
Date of Preparation | Prepared By | Checked By
||

Reagent Details:

Formula & Calculations:

Formula: $M_1 \times V_1 = M_2 \times V_2$.

Calculation: To prepare 1 L of 0.01M sodium hydroxide from 1M sodium hydroxide stock solution calculate quantity required of stock solution as below:

0.5M sodium hydroxide stock solution required quantity= $0.01 \times 1 / 1 = 0.1L$

Preparation Method:

To prepare 0.01M sodium hydroxide 1L solution from 1M sodium hydroxide stock solution formula is

$M_1 (\text{Molarity of stock}) \times V_1 (\text{volume of stock}) = M_2 (\text{Molarity of working}) \times V_2 (\text{volume of working})$.

Take 0.1L of 1M sodium hydroxide solution and make a 1 L solution.

Preparation for different volumes:

To prepare _____ of 0.01M sodium hydroxide

Take 0.1L x _____ L = _____ L of 1M sodium hydroxide

Make sodium hydroxide solution to _____ L solution.

Check and record the details:

Preparation Details:

Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry

Date

|||

Final volume prepared | Weighed Quantity | Storage Condition:

||

Date of Preparation | Prepared By | Checked By

||

WATER BALANCE DIAGRAM – LVP plant

1. Design Basis (BFS Technology)

Production

36 million units/year

500 mL each

18 million L/year

Daily Production

300 days/year

30,000 L/day product

2. Water Consumption Benchmark

BFS plants are more efficient:

Typical range: 5–8 L water / L product

Total Water Requirement

30,000 L/day × 6

180,000 L/day (180 m³/day)

3. Revised Water Consumption Breakdown

A. Industrial (~80%)

1. WFI Generation

Includes distillation losses

90 m³/day

2. BFS Machine Cooling + Process

Mould cooling, machine cleaning

30 m³/day

3. CIP/SIP Systems

Tanks, pipelines, filters

20 m³/day

4. Utilities

Boiler + cooling tower

25 m³/day

B. Domestic (~20%)

15 m³/day

4. Wastewater Generation

~80% becomes effluent

~145 m³/day

5. ZLD-Based Segregation

6. Recycling & Losses

Final Recovery

Recycled water: ~125 m³/day (~69%)

Losses:

Evaporation: ~12 m³/day

Sludge/salts: ~8 m³/day

Liquid discharge = ZERO

7. Key KPIs

Stream | Flow | Type

Low TDS | 70 m³/day | Cooling + domestic

High TDS | 75 m³/day | WFI reject + CIP

Parameter | Value

RO recovery | 70%

MEE condensate | 20%

Losses | 10%

Parameter | Value

Water intensity | 6 L per L product

Reduction vs glass system | ~40% lower

Recycling rate | ~69–70%

ZLD compliance | 100%

[SHEET: Organic waste Plasma]

Sr. No. | Source / Process Area | Type of Non-Hazardous Waste | Approx. Quantity Kg / Year | Solid (Kg/day) | Liquid (L/day) | Category | ETP/3rd party
1 | ETP / STP treatment | Biological sludge (after dewatering) | 6237850 | 17090 | | Organic | 3rd party
2 | Process area & lab washing | Treated water (ETP + RO permeate) | 27052340 L/year | | 74116 | Organic | Domestic use
3 | Packaging line | Empty glass vials (non-contaminated rejects) (~50,000 pcs) | 1000 | 2.73972602739726 | | Inorganic | Scrap 3rd party
4 | Packaging & labeling | Paperboard cartons, labels, paper scrap | 2000 | 5.47945205479452 | | Organic | Scrap 3rd party
5 | Procurement / stores area | Corrugated boxes, stretch film, plastic wrapping | 1500 | 4.109589041095891 | | Inorganic | Scrap 3rd party
6 | Maintenance workshop | Metal scrap (iron, steel, aluminum parts) | 1200 | 1 | | Inorganic | Scrap 3rd party
7 | Utility & HVAC area | Spent filters (air filters, bag filters, prefilters) | 500 | 1 | | Inorganic | 3rd party
8 | Admin & office areas | Office paper waste, packaging waste | 1500 | 25 | | Organic | Scrap 3rd party
9 | Canteen / pantry | Organic kitchen waste | 47000 | 128.76712328767124 | | Organic | Organic converter
10 | General housekeeping | Sweepings, mopping dust, non-contaminated PPE | 800 | 2.191780821917808 | | Mixed | 3rd party
11 | Cooling tower & utilities | Blowdown & scale sludge (non-toxic) | 1500 | 4.109589041095891 | | Organic | 3rd party
12 | Electrical systems | Cable cuttings, wire ends, plastic conduits | 300 | 0.821917808219178 | | Inorganic | Scrap 3rd party
13 | Gardening & green area | Dry leaves, grass clippings (~10 acre considering 33% green area) | 73200 | 200.54794520547946 | | Organic | Organic converter
14 | Water purification & softener | Spent ion-exchange resins (non-toxic) | 200 | 0.547945205479452 | | Inorganic | 3rd party
15 | Misc. activities | Empty HDPE / PET bottles (non-contaminated) | 200 | 0.547945205479452 | | Inorganic | 3rd party
	Domestic use	74.116	KL/Day	
	Organic converter	0.32931506849315073	T/day	
	3rd party	17.09839726027397	T/day	
	Scrap 3rd party	0.03915068493150684	T/day	

[SHEET: Organic waste LVP]

Sr. No. | Source / Process Area | Type of Non-Hazardous Waste | Approx. Quantity Kg / Year | Solid (Kg/day) | Liquid (L/day) | Category | ETP/3rd party
1 | ETP / STP treatment | Biological sludge (after dewatering) | 4188375 | 11475 | | Organic | 3rd party
2 | Process area & lab washing | Treated water (ETP + RO permeate) | 14699645 L/year | | 40273 | Organic | Domestic use
3 | Packaging line | Empty glass vials (non-contaminated rejects) (~50,000 pcs) | 2200 | 6.027397260273973 | | Inorganic | Scrap 3rd party
4 | Packaging & labeling | Paperboard cartons, labels, paper scrap | 36500 | 100 | | Organic | Scrap 3rd party
5 | Procurement / stores area | Corrugated boxes, stretch film, plastic wrapping | 18250 | 50 | | Inorganic | Scrap 3rd party
6 | Maintenance workshop | Metal scrap (iron, steel, aluminum parts) | 7200 | 19.726027397260275 | | Inorganic | Scrap 3rd party
7 | Utility & HVAC area | Spent filters (air filters, bag filters, prefilters) | 1200 | 3.287671232876712 | | Inorganic | 3rd party
8 | Admin & office areas | Office paper waste, packaging waste | 5000 | 13.698630136986301 | | Organic | Scrap 3rd party
9 | Canteen / pantry | Organic kitchen waste | 36500 | 100 | | Organic | Organic converter
10 | General housekeeping | Sweepings, mopping dust, non-contaminated PPE | 20000 | 54.794520547945204 | | Mixed | 3rd party
11 | Cooling tower & utilities | Blowdown & scale sludge (non-toxic) | 27000 | 73.97260273972603 | | Organic | 3rd party
12 | Electrical systems | Cable cuttings, wire ends, plastic conduits | 600 | 1.643835616438356 | | Inorganic | Scrap 3rd party
13 | Water purification & softener | Spent ion-exchange resins (non-toxic) | 20000 | 54.794520547945204 | | Inorganic | 3rd party
14 | Misc. activities | Empty HDPE / PET bottles (non-contaminated) | 19000 | 52.054794520547944 | | Inorganic | 3rd party
	Domestic use	40.273	KL/Day	
	Organic converter	0.1	T/day	
	3rd party	11.71390410958904	T/day	
	Scrap 3rd party	0.19109589041095892	T/day	

PROCESS INSTRUCTIONS FOR (IMMUNOGLOBULIN)

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CONCENTRATION OF FLOW THROUGH FROM CHROMATOGRAPHY–4I RUN OF PLASMA FRACTIONATION

Take the flow through sample from the two runs of Chromatography-4I run of Plasma fractionation step

10.8. (Refer to PROCESS INSTRUCTIONS of Plasma Fractionation)

Check and record the following details of the runs of Chromatography-4I run of Plasma fractionation step

10.8. (Refer to PROCESS INSTRUCTIONS of Plasma Fractionation):

NWP calculation before concentration

Record the following details of the TFF cassettes connected to the TFF system:

No. of cassettes used: _____,

Cassette ID Nos.: _____.

Wash the cassettes with 5 L of PW and calculate NWP before concentration at a set TMP of ≤ 1.0 bar.

Volume of PW used: _____ L.

Check and record the details as below:

Concentration of Flow through Sample

After NWP calculation, flush the cassettes with 5 L of equilibration buffer PF-RR-22 of Chromatography-4I of Plasma fractionation step 10.5. (Refer to PROCESS INSTRUCTIONS of Plasma Fractionation)

Volume of PF-RR-22 used: _____ L.

Based on the initial volume of the flow through the sample, concentrate the sample to the initial batch volume at a set TMP of ≤ 1.0 bar.

Check and record the details as below:

Unload the concentrated sample based on initial batch volume; by flushing 5 L of PF-RR-22. Check and record the following details:

Regeneration

After unloading the concentrated sample, flush the cassettes with 5 L of PW and then with 5 L of IG-RR-23. Recirculate the cassettes with IG-RR-23 at 40°C to 45°C for 30 - 45 min.

Finally, wash the cassettes with 5 L of PW.

Repeat steps 1.4 to 1.5 for Run 2 and record the results in the above tables.

Post-NWP calculation

Repeat the step as in 1.3.2 and record the details as below. Check if NWP is ≥ 120 LMH/Bar. If not, carry out the regeneration procedure again.

Storage

Wash the cassettes with 5 L of IG-RR-24 and finally store the cassettes in IG-RR-24 at room temp. Volume of IG-RR-24 used: _____ L.

CHROMATOGRAPHY 5I

Check the column connections to the AKTA process chromatography system. Check for instrument air connection at 6 bar to the system. Open the UNICORN software.

Record the following column details:

Column height: _____ cm, Column Volume: _____ L,

HETP value: _____, Asymmetry: _____.

Column Wash

Wash the column with 1-2 CVs of PW at a flow rate of 36 ml/min till the conductivity drops below 1.0 mS/cm.

pH drop

Pass up to 2 CVs of IG-RR-02 (10% gradient of IG-RR-09 against PW) at a flow rate of 36 ml/min till the pH has dropped to 3.50 ± 0.50 . Record the following details:

Column Equilibration

Equilibrate the column with 2 CVs of IG-RR-06 (10% gradient of IG-RR-05 against PW) at a flow rate of 36 ml/min .

Sample preparation for loading

Take the Concentrated Flow Through sample from step 1.4 as load.

Check and record the conditions as below:

If the pH is below 5.15, adjust it to 5.20 ± 0.05 with IG-RR-01.

Bring down the conductivity to 1200 – 1400 μ S/cm with IG-RR-25.

Check and record the following conditions:

Filtration of the load sample

Filter the Load sample using a sterilised PRA-SF-003 filter based on sample clarity.

Record the following details of the final load sample:

Loading

Before loading, switch on the 'Record On' button and enter the run details and date. Auto Zero the absorbance at 280 nm on the chromatography system.

Load the sample at a flow rate of 36 ml/min (Limit: ≤ 35 mg of protein/ml of resin)

Collect the Flow through sample when the A280 rises from 50 mAU.

Wash

After loading is completed, wash the column with 1 - 2 CVs of IG-RR-06 at a flow rate of 36 ml/min, and stop collecting the flow when the A280 reaches baseline. (50 - 100 mAU).

Check and record the flow through the sample details as below:

pH adjustment for flow-through sample

Check and adjust the pH of the flow-through sample to 4.00 ± 0.10 with IG-RR-09.

Record the following conditions:

Cleaning

Clean the column by passing 2 column volumes of IG-RR-08 (25% gradient of IG-RR-07 against PW) at a flow rate of 36 ml/min and hold the column for 20 - 30 min.

Regeneration

Regenerate the column by passing 2 column volumes of IG-RR-23 (25% gradient of IG-RR-22 against PW) at a flow rate of 36 ml/min and hold the column for 30 min.

Repeat steps 2.2 to 2.12 for Run 2 and record the results in the above tables.

Storage

Store the column by passing 2 CVs of IG-RR-24 at a flow rate of 36 ml/min. Volume of IG-RR-24 used: _____ L.

Record the volume of flow through after pH adjustment: Run 1: _____ L; Run 2: _____ L

CONCENTRATION OF FT FROM CHROMATOGRAPHY 5I

Take the Run-1 flow through sample after pH adjustment from step 2.10 of Chromatography-IG-5I to do concentration.

NWP calculation before concentration

Record the following details of the TFF cassettes connected to the TFF system:

No. of cassettes used: _____,

Cassette ID Nos.: _____.

Wash the cassettes with 5 L of PW and calculate NWP before concentration at a set TMP of ≤ 1.0 bar.

Volume of PW used: _____ L.

Check and record the details as below:

Concentration of Flow through Sample

After NWP calculation, flush the cassettes with 5 L of IG-RR-06 of Chromatography-IG-5I.

Volume of IG-RR-06 used: Run 1: _____ L, Run 2: _____ L.

Concentrate the FT sample to 59 - 63 mg/ml of protein concentration at a set TMP of ≤ 1.0 bar.

Check and record the details as below:

Unload the concentrated sample when the desired protein concentration is reached; by flushing 10 L of IG-RR-06. Check and record the following details:

Regeneration

Flush the cassettes with 5 L of PW and then with 5 L of IG-RR-23 at 40 - 45°C and carry out recirculation for 30 to 45 min. Flush the cassettes with 30 L of PW again.

Repeat the steps 3.0 to 3.4 for Run 2 and record the results as above.

Post NWP calculation

Repeat the step as 3.2.2 and record the details as below. Check if NWP is ≥ 120 LMH/Bar. If not, carry out the regeneration procedure again.

Storage

Flush the cassettes with 5 L of IG-RR-24 and finally store the cassettes in IG-RR-24 at room temp. Volume of IG-RR-24 used: _____ L.

CONDUCTIVITY ADJUSTMENT

Take the concentrated FT sample run wise from step 3.2.4 and adjust the conductivity to 2.0 mS/cm (Limit: 2.0 ± 0.2 mS/cm) with IG-RR-10.

Record the following conditions:

Repeat the steps 4.1 to 4.2 for Run 2 and record the results as above.

Pool 25 ml of the sample and send it to QC for in-process QC analysis. Record the obtained results as per the QC report: pH: _____,

Cond.: _____ mS/cm and Protein Conc. (mg/ml): _____.

VIRUS INACTIVATION BY LOW pH TREATMENT

Take the conductivity adjusted sample run-wise from step 4.0 and incubate the sample at $\text{pH } 4.00 \pm 0.10$ at $37.0 \pm 0.5^\circ\text{C}$ for 12 hrs. in a process vessel with water circulation in jacket.

Record the incubation details:

Check and record the sample details after incubation:

OD at 280 nm for 100x diluted sample: Run 1: _____,

Run 2: _____. (Dilution buffer is IG-RR-06)

OD at 600 nm: Run 1: _____, Run 2: _____.

Repeat the steps 5.1 to 5.3 for Run 2 and record the results as above.

VIRUS INACTIVATION BY S/D TREATMENT

Note: The amount of IG-RR-12 (20% stock solution) and IG-RR-13 (20% stock solution) to be added to a final concentration of 1% (w/w) is based on the weight of IgG solution.

IG-RR-12 is added slowly to the run wise low pH treated sample for 15 - 20 mins. under gentle stirring condition and continue stirring till the solution is homogenous.

Then add IG-RR-13 slowly to the sample for 15 - 20 mins. under gentle stirring condition and continue stirring till the solution is homogenous.

Check and record the following details:

Incubate the sample at $30 \pm 3^\circ\text{C}$ for 6 hrs. under stirring conditions in a process vessel with water circulation in the jacket.

Check and record OD600 after 10 min. of incubation:

OD600: Run 1: _____, Run 2: _____.

Check and record the sample details after incubation:

Total incubation time: Run 1: _____, Run 2: _____.

OD600 after incubation: Run 1: _____, Run 2: _____.

Repeat the steps 6.1 to 6.6 for Run 2 and record the results as above.

FILTRATION

Filter the S/D treated virus inactivated sample run wise using sterilized PRA-PO-008 and PRA-PO-005 filters connected serially.

Check and record the following details:

Volume after filtration: Run 1: _____ L, Run 2: _____ L.

OD at 280 nm for 10x diluted sample: Run 1: _____, Run 2: _____. (Dilution buffer is IG-RR-25)

OD at 600 nm: Run 1: _____, Run 2: _____.

Repeat the steps 7.1 to 7.2 for Run 2 and record the results as above.

CHROMATOGRAPHY 6I

Check the column connections to the AKTA process chromatography system. Check for instrument air connection at 6 bar to the system. Open the UNICORN software.

Record the following column details:

Column height: _____ cm, Column Volume: _____ L,

HETP value: _____, Asymmetry: _____.

Column Wash

Wash the column with upto 2 CVs of PW at a flow rate of 27.8 ml/min till the conductivity has dropped below 0.50 mS/cm.

pH Drop

Wash the column with up to 2 CVs of IG-RR-02 (10% gradient of IG-RR-09 against PW) at a flow rate of 27.8 ml/min till the pH drops to 4.50 ± 0.10 .

Column Equilibration

Equilibrate the column with 6 CVs of IG-RR-15 (10% gradient of IG-RR-14 against PW) at a flow rate of 27.8 ml/min.

Check and record the following results:

Sample details

Take the filtered sample from step 7.0 (run wise) and record the following conditions:

Loading

Before loading, switch on the 'Record On' button and enter the run details and date. Auto Zero the absorbance at 280 nm on the chromatography system.

Load the sample at a flow rate of 27.8 ml/min and start collecting the flow through sample.

Wash-I

After loading is completed, wash the column with 4 CVs of IG-RR-15 (10% gradient of IG-RR-14 against PW) at a flow rate of 27.8 ml/min and collect the wash in the same flow-through vessel until A280 reaches baseline (50 – 100 mAU).

Wash II

After the UV absorbance has reached the baseline, pass 6 CVs of IG-RR-17 (25% gradient of IG-RR-16 against PW) at a flow rate of 27.8 ml/min in order to remove the virus inactivation chemicals.

Elution

After the A280 reaches the base line, pass 4 – 5 CVs of IG-RR-19 (25% gradient of IG-RR-18 against PW). Collect the peak when the A280 rises and collect the total peak till the A280 reaches base line (50 – 100 mAU).

Check and record the following conditions:

pH adjustment for eluate

Adjust the pH of the eluate to 4.00 ± 0.10 with IG-RR-20.

Check and record the following conditions:

Regeneration

Regenerate the column by passing 2 CVs of IG-RR-23 (25% gradient of IG-RR-22 against PW) at a flow rate of 27.8 ml/min and hold the column for 30 - 60 mins.

Repeat steps 8.2 to 8.11 for Run 2 and record the results as above.

Storage

Store the column (at room temp.) by passing 2 CVs of IG-RR-24 at a flow rate of 27.8 ml/min.

Volume of IG-RR-24 used: _____ L.

Enter the process details as below:

ADJUSTMENT OF PROTEIN CONCENTRATION

Take the CHROMATOGRAPHY 5I pooled eluate sample and check the protein concentration. Record the following details:

O.D. at 280 nm for 10x diluted sample: _____ (Dilution buffer is IG-RR-26),

Protein conc.: _____ mg/ml, Pooled Eluate volume: _____ L.

Note: Divide OD280 by the extinction coefficient of 1.40 to obtain protein concentration. in mg/ml.

If protein concentration is more than 11 mg/ml, dilute the pooled eluate sample with IG-RR-26 to a final protein concentration of 8 - 11 mg/ml. Record the following details:

Volume of IG-RR-26 used: _____ L,

O.D. at 280 nm: _____, Protein conc.: _____ mg/ml, Final Eluate volume: _____ L.

PRA-DE-005 & PRA-NF-003 FILTRATION

Flush 5 L of PW through PRA-DE-005 and equilibrate with 5 L of IG-RR-26 buffer. Volume of PW used: _____ L,

Volume of IG-RR-26 used: _____ L.

Take the concentration-adjusted sample from step 9.2 and filter the sample through the PRA-DE-005 filter.

Check and record the following details:

OD at 280 nm: _____, OD at 600 nm: _____,

Protein conc.: _____ mg/ml, Final Volume after filtration: _____ L.

Flush 5 L of PW through PRA-NF-003 and equilibrate with 5 L of IG-RR-26 buffer. Volume of PW used: _____ L.

Volume of IG-RR-26 used: _____ L.

Take the PRA-DE-005 filtrate from step 10.2 and filter the sample through the PRA-NF-003 filter.

Check and record the following details:

Final Volume after filtration: _____ L,

pH: _____, Conductivity: _____ mS/cm,

OD at 280 nm: _____, OD at 600 nm: _____,

Protein conc.: _____ mg/ml, Total Protein: _____ g.

CONCENTRATION AND DIAFILTRATION

NWP calculation before concentration

Record the following details of the TFF cassettes connected to the TFF system:

No. of cassettes used: _____,

Cassette ID Nos.: _____.

Wash the cassettes with 5 L of ambient WFI and calculate NWP before concentration at a set TMP of ≤ 1.0 bar.

Volume of ambient WFI used: _____ L.

Check and record the details as below:

Concentration of filtrate sample

After NWP calculation, flush the cassettes with 5 L of IG-RR-26.

Volume of IG-RR-26 used: _____ L.

Based on the initial protein concentration of the filtrate sample, concentrate the sample to 5.0 - 5.5% at a set TMP of ≤ 1.0 bar.

Check and record the following details:

Note: Divide the obtained value of OD280 with extinction coefficient of 1.40 to obtain protein concentration in mg/ml.

Diafiltration

Diafilter the sample from step 11.2 with IG-RR-28 to reach a final conductivity of permeate of 900 – 1300 uS/cm.

Unload the concentrated sample when the required protein concentration of 5.5 - 6.5% is reached; by flushing 5 L of IG-RR-26. Check and record the following details:

Volume of Retentate: _____ L, OD280/600: _____.

Collect 25 ml of sample and send to QC for Bulk-API analysis. Record the obtained results as per the QC report: pH: _____.

Check and record the following details:

OD at 280 nm: _____, OD at 600 nm: _____,

pH: _____, Conductivity: _____ uS/cm,

Total Volume: _____ L, Total Protein: _____ g.

Regeneration

Flush the cassettes with 5 L of ambient WFI and then with IG-RR-23 at 40 - 45°C and carry out recirculation for 30 to 60 min.

Volume of IG-RR-23 used: _____ L.

Post NWP calculation

Repeat the step as 11.1.2 and record the details as below. Check if NWP is ≥ 120 LMH/Bar. If not, carry out the regeneration procedure again.

Storage

Flush the cassettes with 5 L of IG-RR-24 and finally store the cassettes in IG-RR-24 at room temp.

Volume of IG-RR-24 used: _____ L.

Enter the process details as below:

Formulation

Check the QC result for Bulk from step 11.3.3:

Protein Concentration _____, pH: _____,

Conductivity: _____ uS/cm, Total Volume: _____ L,

Total Protein: _____ g.

Formulation with 10% (W/V) Maltose:

Finally, the sample concentration should be 5%.

Folds to be diluted to the total volume of the Bulk sample.

A= Protein Concentration of Bulk sample _____/50 mg.

A= _____.

Final Total volume to be made up after formulation =A x Bulk total volume.

B=A _____ x _____.

B= _____ L.

Difference between = final volume - bulk volume.

C= B _____ - _____.

C= _____ L.

Calculation of the 10 % maltose.

D=Final volume (B) x 10% (w/v) Maltose.

D=B _____ x 10% _____.

D= _____ kg.

Dissolving the maltose:

Add the calculated amount of the maltose slowly to the Bulk Sample under stirring (150 rpm) for 15 to 20 min.

Note: Check with 10% of maltose dissolved in the 5 ml of water, and the rise of the water level and check the difference between the final rise volume and mine's initial volume. The difference in the volume is considered a standard rise value for the formulated bulk.

Standard rise in the formulated bulk volume:

= rise in volume – dissolved in 5 ml.

E= _____.

Rise in volume after the addition of Maltose to the Bulk Sample.

F= D (kg) _____ + Bulk volume _____ x E _____.

F= _____ L.

Calculating the Dilution of the formulated bulk sample with the dilution buffer to achieve the final 5% concentration.

G = (B) final total volume -(F) Rise volume after maltose addition.

G = _____.

Formulate Bulk Details:

Check and record the following details:

OD at 280 nm: _____, OD at 600 nm: _____,

pH: _____, Conductivity: _____ uS/cm,

Protein Concentration _____.

Total Volume: _____ L, Total Protein: _____ g.

Collect 25 ml of sample and send to QC for Formulated Bulk-API analysis. Record the obtained results as per the QC report:

Protein Concentration _____, pH: _____.

Conductivity: _____ uS/cm,

Calculate and record the following details according to the QC result:

pH: _____, Conductivity: _____ uS/cm,

Protein Concentration _____.

Total Volume: _____ L, Total Protein: _____ g.

Adjusting the Protein concentration. If it is more than 5.5% then dilute the sample with the dilution buffer.

Final protein concentration should be 5.0. % accepted criteria is $\pm 0.5\%$.

Do the sterile filtration with a 0.2 μ m filter.

Submit the final sample for the QC testing as a drug substance.

Record the Drug substance QC result.

pH: _____, Conductivity: _____ uS/cm,

Protein Concentration _____.

Total Volume: _____ L, Total Protein: _____ g.

Storage of DS sample.

DS sample is stored in the 2 to 8 °C.

Details / Run | 1 | 2

Total Volume (L) | |

Total Protein (g) | |

PF (bar) | PR

(bar) | PP (bar) | Flow (L/hr) | TMP (bar) | Flux (LMH) | NWP

(LMH/bar)

| | | | |

Details | Permeate Vol. (L) | PF (bar) | PR

(bar) | PP (bar) | Flow (L/hr) | TMP (bar) | Flux (LMH) | Permeate OD280

Run 1 | | | | | | |

Run 1 | | | | | | |

Run 1 | | | | | | |

Run2 | | | | | | |

Run2 | | | | | | |

Run2 | | | | | | |

Run2 | | | | | | |

Details | Run 1 | Run 2

Volume of retentate (L) | |

OD280 | |

OD600 | |

Protein Conc. (mg/ml) | |

Total protein (g) | |

% Recovery | |

Details | Run 1 | Run 2

Volume of IG-RR-23 used (L) | |

Details | Run 1 | Run 2

Volume of PW used (L) | |

PF (bar) | PR

(bar) | PP (bar) | Flow (L/hr) | TMP (bar) | Flux (LMH) | NWP

(LMH/bar)

|||||
 Details | Run 1 | Run 2
 Volume of PW used (L) ||
 Details | Run 1 | Run 2
 Volume of IG-RR-02 used (L) ||
 Final pH of the column ||
 Details | Run 1 | Run 2
 Final pH ||
 Final Conductivity ($\mu\text{S}/\text{cm}$) ||
 Volume of IG-RR-06 used (L) ||
 Details | Run 1 | Run 2
 pH ||
 Conductivity ($\mu\text{S}/\text{cm}$) ||
 Volume (L) ||
 Details | Run 1 | Run 2
 pH ||
 Conductivity ($\mu\text{S}/\text{cm}$) ||
 Volume of IG-RR-01 used (L) ||
 Details | Run 1 | Run 2
 pH ||
 Conductivity ($\mu\text{S}/\text{cm}$) ||
 Volume of IG-RR-25 used (L) ||
 Total Volume (L) ||
 Details | Run 1 | Run 2
 Volume of Load (L) ||
 OD280 ||
 Protein conc. (mg/ml) ||
 Total protein (g) ||
 Details | Run 1 | Run 2
 Volume of IG-RR-06 used (L) ||
 OD280 ||
 OD600 ||
 pH ||
 Conductivity (mS/cm) ||
 Flow Through Volume (L) ||
 Protein conc. (mg/ml) ||
 Total protein (g) ||
 % Recovery ||
 Details | Run 1 | Run 2
 Volume of IG-RR-09 used (L) ||
 pH ||
 Conductivity (mS/cm) ||
 Final Volume (L) ||
 Details | Run 1 | Run 2
 Volume of IG-RR-08 used (L) ||
 Details | Run 1 | Run 2
 Volume of IG-RR-23 used (L) ||
 Details | Permeate Vol. (L) | PF (bar) | PR
 (bar) | PP (bar) | Flow (L/hr) | TMP (bar) | Flux (LMH)
 Run 1 |||||
 Run 1 |||||
 Run 1 |||||
 Run 2 |||||
 Run 2 |||||
 Run 2 |||||
 PF (bar) | PR (bar) | PP (bar) | Flow (L/hr) | TMP
 (bar) | Flux (LMH) | NWP (LMH/bar)
 |||||
 Details | Run 1 | Run 1 | Run 2 | Run 2
 Details | Before Conc. | After Conc. | Before Conc. | After Conc.
 Volume (L) ||||
 OD280/600 ||||
 Protein Conc. (mg/ml) ||||
 Total protein (g) ||||
 % Recovery ||||
 Details | Run 1 | Run 2
 Volume of PW used (L) ||
 Volume of IG-RR-23 used (L) ||
 PF (bar) | PR (bar) | PP (bar) | Flow (L/hr) | TMP
 (bar) | Flux (LMH) | NWP (LMH/bar)
 |||||
 Details | Run 1 | Run 2
 Volume of IG-RR-10 used (L) ||
 OD600 ||
 pH ||
 Conductivity (mS/cm) ||
 Final Volume of the sample 00)(L) ||
 Details | Run 1 | Run 2
 Temp. at Start time ($^{\circ}\text{C}$) ||
 Temp. at End time ($^{\circ}\text{C}$) ||
 Total incubation time (hrs.) ||
 Details | Details | Run 1 | Run 1 | Run 2
 Volume of IG-RR-12 added (L) | Volume of IG-RR-12 added (L) |||
 Volume of IG-RR-13 added (L) | Volume of IG-RR-13 added (L) |||
 Volume of IG-RR-25 added to adjust protein conc. to 50 mg/ml (L) | Volume of IG-RR-25 added to adjust
 protein conc. to 50 mg/ml (L) ||
 pH | pH ||
 Final volume of the sample 00)(L) | Final volume of the sample 00)(L) ||
 Details | Run 1 | Run 2
 Volume of PW used (L) ||
 Details | Run 1 | Run 2
 Volume of IG-RR-02 used (L) ||

Details | Run 1 | Run 2
pH ||
Conductivity (mS/cm) ||
Volume of IG-RR-15 used (L) ||
Details | Run 1 | Run 2
Final Volume (L) ||
pH ||
OD600 ||
Total Load protein (g) ||
mg of Protein/ml of resin (mg/ml)
(Limit: ≤ 80 mg/ml) ||
Details | Run 1 | Run 2
Volume of IG-RR-15 used (L) ||
Details | Run 1 | Run 2
Volume of IG-RR-17 used (L) ||
Details | Run 1 | Run 2
Volume of IG-RR-19 used (L) ||
pH ||
Conductivity (mS/cm) ||
Volume of Eluate (L) ||
Details | Run 1 | Run 2
Volume of IG-RR-20 used (L) ||
OD280 ||
Protein Conc. (mg/ml) ||
OD600 ||
pH ||
Conductivity (mS/cm) ||
Volume of the Eluate (L) ||
Details | Run 1 | Run 2
Volume of IG-RR-23 used (L) ||
Run No. | Details | Load | Eluate
1. | Total Volume (L) ||
1. | Total protein (g) ||
1. | % Recovery ||
2. | Total Volume (L) ||
2. | Total protein (g) ||
2. | % Recovery ||
PF
(bar) | PR
(bar) | PP
(bar) | Flow (L/hr) | TMP (bar) | Flux (LMH) | NWP (LMH/bar)
|||||
Permeate Vol. (L) | PF (bar) | PR
(bar) | PP (bar) | Flow (L/hr) | TMP (bar) | Flux (LMH) | Permeate OD280
|||||||
|||||||
|||||||
Details | Before Conc. | After Conc.
Volume (L) ||
Protein Conc. (mg/ml) ||
Total protein (g) ||
% Recovery ||
No. of Dia Filtration | Vol. of IG-RR-28 added (L) | Conductivity (mS/cm) of permeate | Permeate OD280
|||
|||
|||
|||
|||
|||
|||
PF (bar) | PR (bar) | PP (bar) | Flow (L/hr) | TMP
(bar) | Flux (LMH) | NWP (LMH/bar)
|||||
Details | After Conc. | After Diafiltr. & Conc.
Volume (L) ||
Protein Conc. (mg/ml) ||
Total protein (g) ||
% Recovery ||

[SHEET: Equipment capacities]
Equipment | Capacities to be installed |
| Plasma | LVP
Boiler | 3000 Kg/H | 5000 Kg/H
Chiller | 300 TR | 250 TR
Process chiller | 100 TR | 100 TR
Air compressor | 230 CFM | 250 CFM
Soft water generation | 20000 LPH generation, 50KL storage | 20000 LPH generation, 50KL storage
Purified water RO water generation and distribution,Storage | PW-12000 LPH generation , 20KL storage |
PW-15000 LPH generation, 30KL storage
Water for Injection generation and distribution,storage | 3500 LPH generation, 20KL storage | (5000 LPH
generation, 20KL storage)*2
Pure steam generation | 1000 Kg/H | 1000 Kg/H
Nitrogen | 50 M3 | 50 M3
Co2 | 20 M3 | NA
Electrical transformer requirement and distribution and UPS power | 1400 KVA * 2 | 1000 KVA * 2
DG | 1300 KVA * 2 | 1000 KVA * 2
Hot water generation pump and distribution | NA | NA

Reagent Details:
Preparation of PF-RR-26 working buffer from Stock buffer (PF-RR-25):
Calculations: For the Preparation of 25mM Sodium acetate pH 5.2.
Stock Buffer (PF-RR-25):0.1M Sodium acetate, pH 5.35 + 4.0 M Sodium Chloride.
Formula:
 $C1 \times V1 = C2 \times V2$
C1 = concentration of stock
V1 = volume of stock you need to take
C2 = desired (working) concentration
V2 = final total volume of working solution
Applying the formula to prepare the working buffer:
 $C1 \times V1 = C2 \times V2$
C1 = 0.1M Sodium Acetate pH 5.35.
V1 = volume of stock you need to take _____ L.
C2 = desired (working) concentration 25mM Sodium Acetate pH 5.2
V2 = final total volume of working solution _____ L.
 $V1 = (C2 \times V2) / C1$
C2=0.025M; V2= 1.0 L; C1=0.1M
 $V1 = (0.025 \times 1.0) / 0.1$
V1 = 0.25L
Take 0.25 L (PF-RR-25) Stock buffer solution and dilute it with water to make up a final volume of 1 L.
Check and record the details:
Preparation Details:
Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry
Date
Sodium Acetate | | | |
Acetic Acid | | | |
Sodium Chloride | | | |
Stock buffer (PF-RR-25) taken volume(L) | pH (5.2±0.1) | Conductivity (88.0 ±0.2ms/cm) | Storage
Condition | Final volume prepared
| | RT |
Date of Preparation | Prepared By | Checked By
| |

Reagent Details:

Formula & Calculations:

Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).

Calculation: To prepare 1 L of 1M sodium hydroxide (MW = 40g/mol)

Weight of solute(g)=1×40=40g.

Preparation Method:

Weigh 40 g of Sodium hydroxide and dissolve in water to make a 1 L solution.

Preparation for different volumes:

To prepare _____ of 1M sodium hydroxide (MW = 40g/mol).

Weight of solute(g)=1×40× _____ L= _____ gm

Weigh _____ gm of sodium hydroxide and dissolve in water to make a _____ L solution.

Check and record the details:

Preparation Details:

Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry

Date

|||||

Final volume prepared | Weighed Quantity | Storage Condition:

||

Date of Preparation | Prepared By | Checked By

||

Reagent Details:

Formula & Calculations:

Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).

Calculation: To prepare 1 L of 0.2M sodium hydroxide (MW = 40g/mol)

Weight of solute(g)=1×8=8g.

Preparation Method:

Weigh 8 g of Sodium hydroxide and dissolve in water to make a 1 L solution.

Preparation for different volumes:

To prepare _____ of 0.2M sodium hydroxide (MW = 40g/mol).

Weight of solute(g)=1×8× _____ L= _____ gm

Weigh _____ gm of sodium hydroxide and dissolve in water to make a _____ L solution.

Check and record the details:

Preparation Details:

Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry

Date

|||||

Final volume prepared | Weighed Quantity | Storage Condition

||

Date of Preparation | Prepared By | Checked By

||

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Record The Concentrated Sample Details:

Formulation:

Preparation Of Stabiliser (FG-RR-12):

60mM Histidine + 3.5mg/ml Nicotinamide + 0.05% Tween 80 + 8mg/ml Human Serum Albumin pH7.2

Preparation 60mM L-Histidine Molecular weight: (155.15 g/mol).

To prepare _____ of 60mM L-Histidine (MW = 155.15g/mol).

Weight of solute(g)=9.31x _____ L= _____ gm

Weigh _____ gm of L-Histidine.

Prepared according to the final protein concentration requirements, as defined in the dilution scope.

Preparation of 3.5mg/ml Nicotinamide.

After adjusting the protein concentration, the final protein concentration volume _____ ml, and calculating the 3.5mg/ml Nicotinamide to the sample.

Final sample volume: _____ ml X 3.5mg/ml of Nicotinamide.

Added Nicotinamide _____ mg/ml.

Addition of 0.05% Tween 80 .

After adjusting the protein concentration, the final protein concentration volume _____ ml and measure the volume to be add 0.05% Tween 80.

Final sample volume: _____ ml X 0.05% Tween 80.

Added Tween 80 volume _____ ml.

Addition of 8mg/ml Human Serum Albumin.

After adjusting the protein concentration, the final volume _____ ml is calculated for the 8mg/ml Human Serum Albumin required, and it is taken from the 20% Albumin sample.

Final sample volume: _____ ml X 8mg/ml Human Serum Albumin.

Added 20% Human Serum Albumin volume _____ ml.

pH Adjustment:

After the addition of the stabilisers check the pH _____.

Adjust the pH to pH 7.2 with the 0.5M NaOH/ HCL Added volume _____.

Send 2 mL of the sample for in-process QC analysis.

Record the QC Analysis report.

Storage of the DS sample:

Label the sample and store the DS sample in the -20°C freezer.

Details | Details

pH |

Conductivity (mS/cm) |

O.D280/600 |

Protein Conc. (mg/mL) |

Volume (L) |

Total protein (gms) |

% Recovery |

Details | Details

pH |

Conductivity (mS/cm) |

O.D280/600 |

Protein Conc. (mg/mL) |

Volume (L) |

Total protein (gms) |

Reagent Details:
Formula & Calculations:
Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).
Calculation: To prepare 1 L of 50 mM Sodium Acetate (MW = 80.03 g/mol)
Weight of solute(g)=0.05×80.03 = 4g.
Preparation Method:
Weigh 4 g of Sodium Acetate and dissolve in water to make a 1 L solution.
Preparation for different volumes:
To prepare _____ of 0.05mM Sodium Acetate (MW = 80.03 g/mol).
Weight of solute(g)=0.05×80.03 x _____ L= _____ gm
Weigh _____ gm of Sodium Acetate and dissolve in water to make a _____ L solution.
Preparation of 20%Ethanol:
For per cent v/v:
Calculation:
Volume of ethanol needed = (% desired / 100) × final solution volume.
Examples (digit-by-digit arithmetic)
To make 1000 mL of 20% (v/v) ethanol
Ethanol: (20 / 100) × 1000 mL = 0.20 × 1000 mL = 200.0 mL
Water: 1000 mL – 200.0 mL = 800.0 mL
Procedure for Preparation of 50mM Sodium Acetate + 20% Ethanol.
To prepare a 50 mM Sodium Acetate solution, 4gm added to the water and completely dissolved and to this the 200 mL of 20% Ethanol was added and made up to 1 L.
Check and record the details:
Preparation Details:
Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry
Date
Sodium Acetate | | | |
Ethanol | | | |
Final volume prepared | pH (8.9 ± 0.1) | Conductivity (5 .0± 1.0 ms/cm) | Storage Condition:
| | | RT
Date of Preparation | Prepared By | Checked By (QC/Analyst)
| |

ALBUMIN PROCESS

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CHROMATOGRAPHY 7A

Check the XK column connections to the chromatography system. Check for instrument air connection at 6 bar to the system. Open the software.

Record the following column details :

Column height: 20 cm, Column Volume: 393ml,

HETP value: _____, Asymmetry: _____.

Column Wash

Wash the column with upto 2 CVs of PW at a maximum flow rate of 40ml/min till the conductivity drops below 1.0 mS/cm.

pH Drop

Pass up to 2 CVs of AL-RR-22 (10% gradient of AL-RR-02 against PW) at a maximum flow rate of 40ml/min till the pH drops to 4.50 ± 0.10 .

Column Equilibration

Equilibrate the column with up to 6 CVs of AL-RR-06 (12.5% gradient of AL-RR-05 against PW) at a flow rate of 40ml/min.

Check and record the details:

Sample preparation for loading

Take Eluate from step 10.9 of Plasma fractionation step as load for CHROMATOGRAPHY 7A. (Refer PROCESS

INSTRUCTIONS of Plasma Fractionation)

Check and record the conditions as below:

Adjust the pH to 4.50 (Limit: 4.50 ± 0.10) with AL-RR-02.

Record the following conditions:

Loading and Washing

Before loading, switch on the 'Record On' button and enter the run details and date. Auto Zero the absorbance at 280 nm on the chromatography system.

Load the sample based on ≤ 60 mg of protein per ml of resin at a flow rate of 40ml/min

After loading is completed, wash the column with up to 2 CVs of AL-RR-06 (12.5% gradient of AL-RR-05 against PW).

Elution

After the column wash, pass 2 - 3 CVs of AL-RR-23 (25% gradient of AL-RR-09 against PW) through the column.

Start collecting Eluate when the A280 rises to 100 mAU and collect till A280 drops to 400 mAU.

Cleaning

Clean the column by passing upto 2 CVs of AL-RR-11 (25% gradient of AL-RR-10 against PW) in up flow direction at a flow rate of 40ml/min .

Regeneration

Regenerate the column by passing up to 2 CVs of AL-RR-19 (25% gradient of AL-RR-18 against PW) in the up-flow direction at a flow rate of 40ml/min and hold the column for 30 min.

Repeat the steps 1.3 to 1.10 for Run 2 and record the results as above.

Check and record the following conditions of the pooled eluate sample:

Storage

Store the column (at room temp.) by passing upto 2 CVs of AL-RR-20 at a flow rate of 40ml/min .

Volume of AL-RR-20 used: _____ L.

CONCENTRATIONS OF ELUATE FROM CHROMATOGRAPHY 7A

Take the eluate from the above 2 runs from step 1.8 of Chromatography-I.

pH adjustment

Adjust the pH to 5.20 (Limit: 5.20 ± 0.10) with AL-RR-12 under stirring condition, run-wise.

NWP calculation before concentration

Record the following details of the TFF cassettes connected in series horizontally to the TFF system:

No. of cassettes used: _____,

Cassette ID Nos.: _____.

Wash the cassettes with 20L of PW and calculate NWP at a set TMP of ≤ 1.0 bar. Volume of PW used: _____ L.

Check and record the details as below:

Concentration

After NWP calculation, flush the cassettes with 30 L of AL-RR-23 of CHROMATOGRAPHY 7A.

Volume of AL-RR-23 used: Run 1: _____ L, Run 2: _____ L.

Based on the initial protein concentration of eluate from step no. 1.8.2, concentrate the pooled eluate sample to 60 – 70 mg/ml at a set TMP of ≤ 1.0 bar and Pump Speed: Run 1: _____, Run 2: _____.

Check and record the details as below:

Unload the concentrated sample when the desired protein concentration is reached by flushing 30 L of AL-RR-23. Check and record the following details as below:

Note: Divide the obtained value of OD280 by the extinction coefficient of 0.56 to obtain protein concentration in mg/ml.

Regeneration

Flush the cassettes with 10 L of PW and then with 10 L of AL-RR-19 at 40 - 45°C and carry out recirculation for 30 to 45 min.

Volume of AL-RR-19 used: Run 1: _____ L, Run 2: _____ L.

Flush the cassettes with 25 L of PW.

Volume of PW used: Run 1: _____ L, Run 2: _____ L.

Repeat steps 2.4 to 2.5 for Run 2 and record the results as above.

Post NWP calculation

Repeat the step as 2.3.2 and record the details as below. Check if NWP is ≥ 120 LMH/Bar. If not, carry out the regeneration procedure again.

Storage

Flush the cassettes with 50 L of AL-RR-21 and store the cassettes in AL-RR-21 at room temp.

Volume of AL-RR-21 used: _____ L.

HEAT TREATMENT

Take the concentrated eluate from step 2.4. Check and record pH: _____.

If the pH is not 5.20 ± 0.10 , then adjust to 5.20 ± 0.10 with AL-RR-12.

Volume of AL-RR-12 used: _____ L.

Check and record the following details:

Final pH: _____, Cond.: _____ mS/cm,

OD280/600: _____, Final volume of sample: _____ L,

Protein Conc. (mg/ml): _____, Total protein (g): _____.

Incubation

Incubate the pH-adjusted sample at 57°C (Limit: $57 \pm 0.5^\circ\text{C}$) for up to 6 hrs. in a process vessel with hot water circulation in the jacket.

Temp. achieved: _____. Temp. achieved at: _____.

Record the temperature and stirrer speed at regular intervals:

Record the details after heat treatment. OD600: _____.

After incubation at 57°C, shift the incubation temperature to $< 13^\circ\text{C}$ by circulating chilled water through the jacket of process vessel.

Temp. achieved: _____. Temp. achieved at: _____.

Separation of Supernatant from Heat-treated sample

Take the Heat treatment sample in a batch mode centrifuge bottle and spin at a speed of 10000 rpm.

Collect the supernatant into a sterilized bottle, and discard the pellet in a centrifuge bottle.

Check and record the following details:

Collect the supernatant into a sterilized glass bottle.

Check and record the following details of supernatant:

OD280/600: _____, Protein conc.: _____ mg/ml,

pH: _____, Cond.: _____ mS/cm,

Final Volume: _____ L, Total protein (g): _____.

Concentration of Heat-treated sample

Pre-NWP calculation

Record the following details of the TFF cassettes connected in series horizontally to the TFF system: No. of cassettes used: _____,

Cassette ID Nos.: _____.

Wash the cassettes with 15 L of PW and calculate NWP before concentration at a set TMP of ≤ 1.0 bar.

Volume of PW used: _____ L.

Check and record the details as below:

Concentration

After NWP calculation, flush the cassettes with 30 L of AL-RR-23 of CHROMATOGRAPHY 7A.

Volume of AL-RR-23 used: _____ L.

Based on the initial protein conc., concentrate the sample to 120 to 130 mg/ml at a set TMP of ≤ 1.0 bar and Pump speed: _____ %.

Check and record the details as below:

Unload the concentrated sample when the desired protein concentration is reached; by flushing 30 L of AL-RR-23. Check and record the following details: Volume of AL-RR-23 used: _____ L.

Note: Divide the obtained value of OD280 with extinction coefficient of 0.56 to obtain protein concentration in mg/ml.

Regeneration

Flush the cassettes with 10 L of PW and then with 10 L of AL-RR-19 at $40 - 45^\circ\text{C}$ and carry out recirculation for 30 to 45 min.

Volume of AL-RR-19 used: _____ L.

Flush the cassettes with 15 L of PW.

Volume of PW used: _____ L.

Post NWP calculation

Repeat the step as 4.2.2 and record the details as below. Check if NWP is ≥ 120 LMH/Bar. If not, carry out the regeneration procedure again.

Storage

Flush the cassettes with AL-RR-21 and finally store the cassettes at room temp. in

AL-RR-21 Volume of AL-RR-21 used: _____ L.

5.0 PRA-SF-003 Filtration

Filter the above concentrated sample using PRA-SF-003 filter.

Check and record the following details after filtration:

OD600: _____, Final Volume: _____ L.

6.0 Virus inactivation by AL-RR-14 and low pH treatment

6.1 AL-RR-14 addition

Add AL-RR-14 to a final concentration of 32 mM.

Volume of AL-RR-14 used: _____ L.

pH and protein concentration adjustment

After AL-RR-14 addition, adjust the pH of the sample to 4.50 (Limit: 4.50 ± 0.10) with AL-RR-02; and at a final albumin concentration of 100 ± 05 mg/ml.

Record the details as below:

Volume of AL-RR-02 used: _____ L, Final Volume: _____ L,

pH: _____, Cond.: _____ mS/cm,

OD280/600: _____, Protein Conc. (mg/ml): _____.

Collect and send 15 ml of the pH and concentration adjusted sample for in-process QC analysis. Record the obtained results as per the QC report: pH: _____, Protein Conc. (mg/ml): _____.

Viral inactivation by low pH incubation

Incubate the pH adjusted sample at 30°C (Limit: $30 \pm 2.0^\circ\text{C}$) for upto 10 hrs. in a process vessel with water circulation in jacket

Shift the temperature to 5°C (Limit: $5 \pm 3^\circ\text{C}$) by giving chilled water circulation to the jacket of the process vessel.

Record the temperature and stirrer speed at regular intervals as below:

Record the details after incubation:

OD600: _____.

Filtration of viral inactivated sample through PRA-DE-002 filter

Check and record the following details after filtration:

OD600: _____, pH: _____, Cond.: _____ mS/cm,

Filtrate Volume: _____ L.

Filtration of PRA-DE-002 filtrate through PRA-SF-003 filter

Check and record the following details after filtration:

OD600: _____, pH: _____, Cond.: _____ mS/cm,

Filtrate Volume: _____ L.

DIAFILTRATION AND CONCENTRATION

USING TFF SYSTEM

pH adjustment before diafiltration

After PRA-SF-003 filtration, adjust the pH of the sample to 7.00 (Limit: 7.00 ± 0.30) with AL-RR-12.

Record the details as below after pH adjustment:

Volume of AL-RR-12 used: _____ L.

OD280/600: _____, Protein Conc. (mg/ml): _____,

pH: _____, Cond.: _____ mS/cm,

Volume: _____ L and Total Protein (g): _____.

Pre NWP calculation

Record the following details of the TFF cassettes connected in series horizontally to the TFF system:

No. of cassettes used: _____, Cassette ID Nos.: _____.

Wash the cassettes with 15 L of PW and calculate NWP before concentration at a set TMP of ≤ 1.0 bar.

Volume of PW used: _____ L.

Check and record the details as below:

Diafiltration

Take the pH adjusted sample for diafiltration.

Diafilter the sample with 3 dia volumes of AL-RR-15 at a set TMP of ≤ 1.15 bar.

Check and record the following details after diafiltration.

Concentration

Concentrate the diafiltered sample to 190 – 210 mg/ml at a set TMP of ≤ 1.15 bar. Pump Speed: _____ %.

Check and record the following details:

Unload the concentrated sample when the desired protein concentration is reached; by flushing 10 L of AL-RR-15. Check and record the following details:

pH: _____, Cond.: _____ mS/cm,

OD280/600: _____, Volume of AL-RR-15 used: _____ L.

Regeneration

Flush the cassettes with 10 L of PW and then with 10 L of AL-RR-19 and carry out recirculation for 30 to 45 min. Volume of AL-RR-19 used: _____ L.

Post NWP calculation

Repeat the step as 7.2.2 and record the details as below. Check if NWP is ≥ 120 LMH/Bar. If not, carry out the regeneration procedure again.

Storage

Flush the cassettes with 5 L of AL-RR-21 and finally store the cassettes at room temp. in AL-RR-21.

Volume of AL-RR-21 used: _____ L.

PRA-SF-003 FILTRATION

Filter the above c concentrated sample using sterilized PRA-SF-003 filter.

Check and record the following details after filtration:

OD280/600: _____, Protein Conc. (mg/ml): _____,

pH: _____, Cond.: _____ mS/cm,

Final Volume: _____ L, Total Protein (g): _____.

PASTEURIZATION

Incubate the PRA-SF-003 filtered sample at 60°C (Limit: 60 ± 1.0 °C) for 5 $\pm$ 1 hrs. in a process vessel with hot water circulation in the jacket.

Record the temperature and stirrer speed at regular intervals as below:

After 4 to 6 hours of incubation, adjust the pH of the sample to 5.20 (Limit: 5.20 ± 0.10) with AL-RR-02.

Record the following details:

Volume of AL-RR-02 used: _____ L. Final pH: _____.

Incubate the pH adjusted sample at 60°C (Limit: 60 ± 1.0 °C) for another 5 $\pm$ 1 hrs. in the same process vessel with hot water circulation in jacket.

Record the temperature and stirrer speed at every 3 hrs. interval as below:

Record the details after completion of pasteurization:

OD600: _____.

Cool the pasteurized sample to 37.0°C by giving chilled water circulation to the jacket of the process vessel.

PRA-DE-004 FILTRATION

Filter the pasteurized sample using PRA-DE-004 filter.

Check and record the following details after filtration:

OD280/600: _____, pH: _____, Cond.: _____ mS/cm,

Final Volume: _____ L, and Total protein (g): _____.

pH ADJUSTMENT

Adjust the pH of the sample to 6.80 (Limit: 6.80 ± 0.10) with AL-RR-12. Volume of

AL-RR-12 used: _____ L.

Record the details as below:

OD280/600: _____, Protein Conc. (mg/ml): _____,

pH: _____, Cond.: _____ mS/cm,

Final Volume: _____ L, and Total protein (g): _____.

DIAFILTRATION AND CONCENTRATION OF pH ADJUSTED

SAMPLE.

Pre NWP calculation

Record the following details of the TFF cassettes connected in series horizontally to the TFF system: No. of cassettes used: _____,

Cassette ID Nos.: _____.

Wash the cassettes with 10 L of WFI and calculate NWP before concentration at a set TMP of ≤ 1.0 bar.

Volume of WFI used: _____ L.

Check and record the details as below:

Diafiltration

Take the pH adjusted sample for diafiltration.

Diafilter the sample with 2 dia volumes of AL-RR-24 at a TMP of ≤ 1.0 bar.

Check and record the following details after diafiltration.

Volume of AL-RR-24 used: _____ L.

Concentration

Based on the initial protein conc., concentrate the sample to 220 to 230 mg/ml at a set TMP of ≤ 1.0 bar and Pump speed: _____ %; to obtain Bulk-API.

Check and record the details as below:

Unload the concentrated sample when the desired protein concentration is reached; by flushing 10 L of AL-RR-24. Check and record the following details:

Note: Divide the obtained value of OD280 with extinction coefficient of 0.56 to obtain protein concentration in mg/ml.

Regeneration

Flush the cassettes with 10 L of WFI and then with 100 L of AL-RR-19 at 40 - 45°C and carry out recirculation for 30 to 45 min.

Flush the cassettes with 10 L of ambient WFI.

Volume of ambient WFI used: _____ L.

Post NWP calculation

Repeat the step as 12.1.2 and record the details as below. Check if NWP is ≥ 120 LMH/Bar. If not, repeat the regeneration procedure.

Storage

Flush the cassettes with 5 L of AL-RR-21 and finally store the cassettes at room temp. in AL-RR-21.

Volume of AL-RR-21 used: _____ L.

PRA-SF-003 FILTRATION

Filter the above concentrated sample using sterile PRA-SF-003 filter under LAFU.

Check and record the following details after filtration:

OD280/600: _____, Protein Conc. (mg/ml): _____,

pH: _____, Cond.: _____ mS/cm,

Filtrate Volume: _____ L, Total Protein (g): _____.

Collect and send 25 ml of the sample after PRA-SF-003 filtration to QC for Bulk- API analysis. Record the obtained results as per the QC report: pH: _____.

Formulation

Check and record the QC results:

Protein Conc. (mg/ml): _____,

pH: _____, Cond.: _____ mS/cm,

Filtrate Volume: _____ L, Total Protein (g): _____.

Calculation of the protein concentration for 20%.

Required Dilution fold calculation for bulk sample(A) = QC estimated protein concentration $\div$ Required protein concentration(200mg).

A= _____.

Depending on the No. of fold to be diluted to the bulk sample to get the 20% final sample concentration.

Final Dilution of bulk sample volume (B) = Initial bulk volume _____ x A _____.

Final dilution volume(B)= _____.

PREPARATION OF THE FORMULATION BUFFERS:

Formulation with AL-RR-16 & AL-RR-17 buffers should be prepared based on the total protein & concentration of both buffers should be 16 mM per gram of protein AL-RR-16 required is (0.0133).

Preparation of 0.15 M AL-RR-16(C):

Molecular weight =166.2.

C = 0.0133x Total protein _____ gm.

C= _____ gm.

Required 0.15M=1/0.15.

= 6.666-fold dilution.

0.15 M=Volume (1M) x 6.666 folds.

From a 1M solution, dissolve 6.666-fold to get a 0.15M solution.

AL-RR-16 Required volume of 0.15 M = _____ L.

Dissolution of AL-RR-16 buffer.

Dissolution of the Sodium Caprylate with hot WFI, and after dissolving the salt and make up the volume with ambient WFI to attain the 0.15M.

Preparation of 0.4 M AL-RR-17(D):

Molecular weight = 246.26.

D = 0.0197x Total protein _____ gm.

D= _____ gm.

Required 0.4M=1/0.4.= 2.5-fold dilution.

0.4 M=Volume (1M) x 2.5 folds.

From a 1M solution, dissolve 2.5-fold to get a 0.4M solution.

AL-RR-17 Required volume of 0.4 M = _____ L.

Dissolution of AL-RR-17 buffer.

Dissolution of the Sodium N-Acetyl Tryptophan with 1M NaOH solution (25mg salt will be dissolved in 1ml 1M NaOH solution after dissolving the salt, then make up with ambient WFI.

Addition of the formulation buffer:

Add the C & D formulation buffer to the bulk sample while stirring at 150 rpm.

Note: Formulated sample protein concentration should be checked against the formulated buffer.

Check the sample details:

OD280/600: _____, Protein Conc. (mg/ml): _____,

pH: _____, Cond.: _____ mS/cm,

Filtrate Volume: _____ L, Total Protein (g): _____.

Protein concentration criteria are 200mg/ml $\pm$ 10 mg. If the concentration is higher side to be adjusted with dilution buffer AL-RR-24.

AL-RR-24 added volume _____ L.

Check the sample details:

OD280/600: _____, Protein Conc. (mg/ml): _____,

pH: _____, Cond.: _____ mS/cm,

Filtrate Volume: _____ L, Total Protein (g): _____.

Sterile filtration:

Filter the sample with PRA-SF-003 filter.

Collect and send 25 ml of the sample after PRA-SF-003 filtration to QC for Drug substance-API analysis.

Record the obtained results as per the QC report:

Protein Conc. (mg/ml): _____, pH: _____, Cond.: _____ mS/c Filtrate Volume:

_____, L, Total Protein (g): _____.

Details | Run 1 | Run 2

Volume of PW used (L) | |

Details | Run 1 | Run 2

Volume of AL-RR-02 used (L) | |

Details | Run 1 | Run 2

Final pH | |

Final Conductivity (mS/cm) | |

Volume of AL-RR-06 used (L) | |

Details | Result

pH |

Conductivity (mS/cm) |

OD280/600 |

Final Volume (L) |

Details | Result

Final pH |

Volume of AL-RR-02 used (L) |

OD280/600 |

Final Volume (L) |

Protein conc. (mg/ml) |

Total protein (g) |
Details | Run 1 | Run 2
Load Volume (L) ||
mg of protein loaded/ml of resin | |
Details | Run 1 | Run 2
Volume of AL-RR-06 used (L) | |
Details | Run 1 | Run 2
Volume of AL-RR-23 used (L) | |
Details | Run 1 | Run 2
Volume of AL-RR-11 used (L) | |
Details | Run 1 | Run 2
Volume of AL-RR-19 used (L) | |
Details | Result
Pooled Eluate Volume (L) |
pH |
Conductivity (mS/cm) |
OD280/600 |
Protein conc. (mg/ml) |
Total protein (g) |
% Recovery |
Details | Run 1 | Run 2
Final pH | |
Conductivity (mS/cm) | |
Volume of AL-RR-12 used (L) | |
Final Eluate Volume (L) | |
PF (bar) | PR (bar) | PP (bar) | Flow (L/hr) | TMP
(bar) | Flux (LMH) | NWP (LMH/bar)
| | | | | |
Run No. | Permeate Vol. (L) | PF (bar) | PR
(bar) | PP (bar) | Flow (L/hr) | TMP (bar) | Flux (LMH)
1. | | | | | | | |
1. | | | | | | | |
1. | | | | | | | |
2. | | | | | | | |
2. | | | | | | | |
2. | | | | | | | |
Details | Run 1 | Run 2
Volume (L) | |
OD280/600 | |
Protein Conc. (mg/ml) | |
Total protein (g) | |
PF (bar) | PR (bar) | PP (bar) | Flow (L/hr) | TMP
(bar) | Flux (LMH) | NWP (LMH/bar)
| | | | | |
Time | Temperature (■C) | Stirring Speed (rpm)
| |
| |
| |
PF (bar) | PR (bar) | PP (bar) | Flow (L/hr) | TMP (bar) | Flux (LMH) | NWP (LMH/bar)
| | | | | |
Permeate Vol.(L) | PF (bar) | PR
(bar) | PP (bar) | Flow (L/hr) | TMP (bar) | Flux (LMH)
| | | | | |
| | | | | |
| | | | | |
| | | | | |
| | | | | |
Details | Before Conc. | After Conc.
Volume (L) | |
OD280/600 | |
Protein Conc. (mg/ml) | |
Total protein (g) | |
% Recovery | |
PF (bar) | PR (bar) | PP (bar) | Flow (L/hr) | TMP
(bar) | Flux (LMH) | NWP (LMH/bar)
| | | | | |
Date | Time | Temperature (■C) | Stirring Speed (rpm)
| | |
| | |
| | |
| | |
PF (bar) | PR (bar) | PP (bar) | Flow (L/hr) | TMP (bar) | Flux (LMH) | NWP (LMH/bar)
| | | | | |
No. of Dia Filtr. | Permeate Vol.(L) | PF (bar) | PR
(bar) | PP (bar) | Flow (L/hr) | TMP (bar) | Flux (LMH)
| | | | | |
| | | | | |
| | | | | |
Permeate Vol. (L) | PF (bar) | PR
(bar) | PP (bar) | Flow (L/hr) | TMP (bar) | Flux (LMH)
| | | | | |
| | | | | |
| | | | | |
Details | Before Conc. | After Conc.
Volume (L) | |
Protein Conc. (mg/ml) | |
Total protein (g) | |
% Recovery | |
PF (bar) | PR (bar) | PP (bar) | Flow (L/hr) | TMP
(bar) | Flux (LMH) | NWP (LMH/bar)
| | | | | |
Date | Time | Temperature (■C) | Stirring Speed (rpm)
| | |

Date	Time
PF (bar)	PR (bar)
No. of Diafiltr.	Permeate Vol.(L)
(bar)	PP (bar)
Permeate Vol.(L)	PF (bar)
(bar)	PP (bar)
Details	Before Conc.
Volume (L)	
OD280/600	
Protein Conc. (mg/ml)	
Total protein (g)	
% Recovery	
PF (bar)	PR (bar)
(bar)	Flux (LMH)

[SHEET: Comments]

S. No. | Comment

- 1 | Emergency door may be shows
- 2 | DPB may be provided in sterility room
- 3 | Autoclave room may be shifted towards PAL - 4 entry and entry to formulation area may be shifted to autoclave room
- 4 | Quarantine cold room may be provided in Secondary packing area

To,
Sub- Regional Officer,
Maharashtra Pollution Control Board
Jog Center, 2nd Floor,
Mumbai Pune Road,
Wakdewadi, Pune 411003
Subject-Detailed Proposal of Pollution Control System
We M/s. (Entity Name with Full Address)here by submit that we have proposed to provide pollution control
system as follows.
Thank You
M/s.

Sr. No.	Source	Capacity	APC System Proposed	Stack Height (m)	Type of Fuel
1	DG Set-1				
2	Paint Booth				
3	Boiler (Working)				
4	Boiler (Stand By)				

Reagent Details:

Formula & Calculations:

Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).

Calculation: To prepare 1 L of 0.5M sodium hydroxide (MW = 40g/mol)

Weight of solute(g)=1×20=20g.

Preparation Method:

Weigh 20 g of Sodium hydroxide and dissolve in water to make a 1 L solution.

Preparation for different volumes:

To prepare _____ of 0.5M sodium hydroxide (MW = 40g/mol).

Weight of solute(g)=1×20x _____L= _____gm

Weigh _____gm of sodium hydroxide and dissolve in water to make a _____ L solution.

Check and record the details:

Preparation Details:

Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry

Date

|||||

Final volume prepared | Weighed Quantity | Storage Condition:

||

Date of Preparation | Prepared By | Checked By

||

Reagent Details:

Formula & Calculations:

Formula: Use the formula to calculate the molarity:

Molarity (M) = (Per cent by Weight / 100) * (Density in g/mL) * 1000 / Molar Mass.

Example: Molarity of Glacial Acetic Acid

Molar Mass of CH₃COOH: ~60.05 g/mol

Density: ~1.05 g/mL

Purity: 100%

Acetic Acid (MW = 60.05 g/mol).

Calculation:

Molarity = (100 / 100) * 1.05 g/mL * 1000 mL/L / 60.05 g/mol ≈ 17.5 M.

Calculation: To prepare 1 L of 1.0 M.

Applying the formula to prepare the working buffer:

$C_1 V_1 = C_2 V_2$

C_1 = 17.5 M acetic acid.

V_1 = volume of stock needed to take.

C_2 = desired (working) concentration 1.0 M acetic acid.

V_2 = final total volume of working solution for one litre.

$V_1 = (C_2 \times V_2) / C_1$.

$C_2 = 1.0$ M ; $V_2 = 1.0$ L; $C_1 = 17.5$ M

$V_1 = (1.0 \times 1.0) / 17.5$

$V_1 = 0.057$ L

Take 0.057 L of the Stock solution of acetic acid and add it to a volumetric flask, making up the volume to 1 L.

To prepare various volumes:

V_1 = volume of stock needed to take _____ L.

C_2 = desired (working) concentration 1.0 M acetic acid.

V_2 = final total volume of working solution _____ L.

$V_1 = (C_2 \times V_2) / 17.5$.

$V_1 =$ _____ L.

Take V_1 _____ L of the Stock solution of acetic acid and add it to a volumetric flask, making up the final volume _____ L.

Preparation Details:

Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry

Date

|||

Final volume prepared | Measured Quantity | pH

(2.3 ± 0.1) | Conductivity

(1.9 ± 0.1) | Storage Condition | Reagent Record

||| RT |

Date of Preparation | Prepared By | Checked By

||

Reagent Details:
Formula & Calculations:
Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).
Calculation: To prepare 1 L of 7mM sodium chroloride (MW = 58.44g/mol)
Weight of solute(g)=1×0.409=0.409■g.
Preparation Method:
Weigh 0.409 g of sodium chloride
To prepare 20mM Tris 1L solution from 1M Tris stock solution formula is M1(Molarity of stock) x V1
(volume of stock)= M2 (Molarity of working) xV2 (volume of working).
Take 20mL of 1M tris solution and add weighed amount of sodium chloride as required and dissolve in water
to make a 1 L solution.
Preparation for different volumes:
To prepare _____ of 7mM NaCl(MW = 58.44g/mol).
Weight of solute(g)=0.409x _____ L= _____ gm
Weigh _____ gm of NaCl.
To prepare _____ of 20mM Tris
Take 20mL x _____ L= _____ mL of 1M Tris
Dissolve pre weighed NaCl in 20mM Tris solution to make a _____ L solution.
Adjust the pH to 8 using concentrated HCl/NaOH solution
Check and record the details:
Preparation Details:
Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry
Date
| | | |
Final volume prepared | Weighed Quantity | pH (7.9±1.0) | Conductivity (1.2±0.5 ms/cm) | Storage
Condition:
| | | | RT
Date of Preparation | Prepared By | Checked By
| |

Reagent Details:
Preparation of AL-RR-19 working buffer from Stock buffer (AL-RR-18):
Calculations: For the Preparation of 0.5M Sodium Hydroxide.
Stock Buffer (AL-RR-18):2.0M Sodium Hydroxide,
Formula:
 $C1 \times V1 = C2 \times V2$
C1 = concentration of stock
V1 = volume of stock you need to take
C2 = desired (working) concentration
V2 = final total volume of working solution
Applying the formula to prepare the working buffer:
 $C1 \times V1 = C2 \times V2$
C1 = 2.0 M Sodium Hydroxide.
V1 = volume of stock you need to take _____ L.
C2 = desired (working) concentration 0.5M Sodium Hydroxide.
V2 = final total volume of working solution _____ L.
 $V1 = (C2 \times V2) / C1$
C2=0.5M; V2= 1.0 L; C1=2.0M
 $V1 = (0.5 \times 1.0) / 2.0$
V1 = 0.25L
Take 0.25 L (AL-RR-18) Stock buffer solution and dilute with water, and make up to 1L.
Check and record the details:
Preparation Details:
Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry
Date
Sodium Hydroxide | | | |
Stock buffer (AL-RR-18) taken volume(L) | pH (13.3 ± 0.2) | Conductivity (95 ±5.0 ms/cm) | Storage
Condition | Final volume prepared
| | RT |
Date of Preparation | Prepared By | Checked By
| |

Technical Information # BB-LN/IND/001

Date- 18.05.2026

List of Documents

S/No Title

1 ETP, STP water calculation details for Plasma and LVP plant

2 RDS for Plasma

3 Utility RDS for Plasma

Contact Details:

1. Dr. Ganesh Kumraj: ganesh@biobridge.in

www.biobridge.in Confidential Page 1 of 1

BioBridge Healthcare Solutions Private Limited

Address: # 40, 4th Floor, Block B, Chandigarh City Center (CCC), Zirakpur - 140603, India

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Formulation

Check and record the QC results:

Protein Conc. (mg/ml): _____,

pH: _____, Cond.: _____ mS/cm,

Filtrate Volume: _____ L, Total Protein (g): _____.

Calculation of the protein concentration for 20%.

Required Dilution fold calculation for bulk sample(A) = QC estimated protein concentration ÷ Required protein concentration(200mg).

A=_____.

Depending on the No. of fold to be diluted to the bulk sample to get the 20% final sample concentration.

Final Dilution of bulk sample volume (B) = Initial bulk volume _____ x A _____.

Final dilution volume(B)=_____.

Preparation Of the Formulation Buffers

Formulation with AL-RR-16 & AL-RR-17 buffers should be prepared based on the total protein & concentration of both buffers should be 16 mM per gram of protein. The amount of AL-RR-16 required is (0.0133).

Preparation of 0.15 M AL-RR-16(C):

Molecular weight =166.2.

C = 0.0133x Total protein _____ gm.

C=_____ gm.

Required 0.15M=1/0.15 = 6.666-fold dilution.

0.15 M=Volume (1M) x 6.666 folds.

From a 1M solution, dissolve 6.666-fold to get a 0.15M solution.

AL-RR-16 Required volume of 0.15 M = _____ L.

Dissolution of AL-RR-16 buffer.

Dissolution of the Sodium Caprylate with hot WFI, and after dissolving the salt, make up the volume with ambient WFI to attain the 0.15M.

Preparation of 0.4 M AL-RR-17(D):

Molecular weight = 246.26.

D = 0.0197x Total protein _____ gm.

D=_____ gm.

Required 0.4M=1/0.4.= 2.5-fold dilution.

0.4 M=Volume (1M) x 2.5 folds.

From a 1M solution, dissolve 2.5-fold to get a 0.4M solution.

AL-RR-17 Required volume of 0.4 M = _____ L.

Dissolution of AL-RR-17 buffer.

Dissolution of the Sodium N-Acetyl Tryptophan with 1M NaOH solution (25mg salt will be dissolved in 1ml 1M NaOH solution after dissolving the salt, then make up with ambient WFI.

Addition of the formulation buffer:

Add the C & D formulation buffer to the bulk sample while stirring at 150 rpm.

Note: Formulated sample protein concentration should be checked against the formulated buffer.

Check the sample details:

OD280/600: _____, Protein Conc. (mg/ml): _____,

pH: _____, Cond.: _____ mS/cm,

Filtrate Volume: _____ L, Total Protein (g): _____.

Protein concentration criteria are 200mg/ml ± 10 mg. If the concentration is higher side to be adjusted with dilution buffer AL-RR-24.

AL-RR-24 added volume _____ L.

Check the sample details:

OD280/600: _____, Protein Conc. (mg/ml): _____,

pH: _____, Cond.: _____ mS/cm,

Filtrate Volume: _____ L, Total Protein (g): _____.

Sterile filtration:

Filter the sample with the PRA-SF-003 filter.

Collect and send 25 ml of the sample after PRA-SF-003 filtration to QC for Drug substance-API analysis.

Record the obtained results as per the QC report:

Protein Conc. (mg/ml): _____, pH: _____, Cond: _____ mS/c Filtrate Volume:

_____ L, Total Protein (g): _____.

DS sample stored in 2 to 8 °Cold room.

[SHEET: Process Flow]
SPINNING PROCESS FLOW CHART ||||| MOLDING PROCESS FLOW CHART |||||
FINAL ASSEMBLY PROCESS FLOW CHART |||||

Reagent Details:
Preparation of AL-RR-21 working buffer from Stock buffer (AL-RR-18):
Calculations: For the Preparation of 0.1M Sodium Hydroxide.
Stock Buffer (AL-RR-18):2.0M Sodium Hydroxide,
Formula:
 $C1 \times V1 = C2 \times V2$
C1 = concentration of stock
V1 = volume of stock you need to take
C2 = desired (working) concentration
V2 = final total volume of working solution
Applying the formula to prepare the working buffer:
 $C1 \times V1 = C2 \times V2$
C1 = 2.0 M Sodium Hydroxide.
V1 = volume of stock you need to take _____ L.
C2 = desired (working) concentration 0.1M Sodium Hydroxide.
V2 = final total volume of working solution _____ L.
 $V1 = (C2 \times V2) / C1$
C2=0.1M; V2= 1.0 L; C1=2.0M
 $V1 = (0.1 \times 1.0) / 2.0$
V1 = 0.05L
Take 0.05 L (AL-RR-18) Stock buffer solution and dilute it with water, making up to 1 L.
Check and record the details:
Preparation Details:
Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry
Date
Sodium Hydroxide | | | |
Stock buffer (AL-RR-18) taken volume(L) | pH (13.0 ± 1.0) | Conductivity (22.0 ± 1.0 ms/cm) | Storage
Condition | Final volume prepared
| | RT |
Date of Preparation | Prepared By | Checked By
| |

TANK LAYOUT GF
40' CONTAINER YARD

CONTAINER

CONTAINER
12.19x2.44x2.59

CONTAINER
CONTAINER
12.19x2.44x2.59

12.19x2.44x2.59

12.19x2.44x2.59
1 2 3 4 5 6

SPM
FG PLASMA RM RECEIVING
DISCPATCH RECEIVING RECEIVING BAY
BAY BAY

A

CORRIDOR
40.1X6.84

D3 6D

D6 D6 3D D3 D6
RLAF RM RLAF
REJECT RECALLED PLASMA RECEIPT
RM
SAMPLING
FG DISCPATCH FG IN ROOM GOODS 3.00X2.77 SAMPLING
STORE CHANGE 3.68X 2.05
2.30X 2.35
ROOM 2.30X2.35
PLASMA SEROLOGY REJECT 1.76X 2.90 3.16X2.03
4.50X3.88 STORE 1.07X3.39

RECEIPT ROOM D3 D3

D3
D3 OFFICE TESTING DE-DUSTING
D3
2.40X 4.95 2.95X4.95 TUNNEL
2.23X6.25 3.00X 4.95 CHANGE ROOM
1.87X1.40 CHANGE ROOM
9.42X6.25 2.00X2.08

B 4.70X3.85
D3

3.00X2.60
D3
4.55X2.32 CORRIDOR
AIR LOCK-1
1.87X1.40
1.95X11.3 D6 D3 D3 DPB D3 AIR LOCK-1
2.00X1.21
DPB D3

D6
APPROVED RM
11.5X3.30 9.72X7.22
5.00X2.95
3D
SD SD SD SD
6.21X5.77 4.63X4.00

3D
DPB

3D
CHANGE ROOM
3D 3D D3
1.50 X 1.93
D3
DISPENCING ROOM
3.48X 3.95
ICE PACK COLD ROOM -1 COLD ROOM -2 COLD ROOM -3 COLD ROOM -4 JANITOR WASHING BAG BAR CODING,
MINI POOLS RM QUARANTINE
1.45X2.90 AREA

6D
PLASMA
2.91X7.58 4.45X7.58 4.45X7.58 4.45X7.58 4.45X7.58 QUARANTINE 2.30X2.90 2.60X2.90
AIR LOCK-1
1.50X 1.93

D3
RLAF
D6 D3 DPB

CORRIDOR
29.5 X 2.58

C D6

CORRIDOR
D3 D3 47.9x1.80 D3 D3 D3
699

PLASMA STORAGE PLASMA STORAGE PLASMA STORAGE PLASMA STORAGE PLASMA STORAGE
~ 1000 L ~ 1000 L ~ 1000 L ~ 1000 L ~ 1000 L
~ 70 KL GV ~ 70 KL GV ~ 70 KL GV ~ 70 KL GV ~ 70 KL GV
8.15x11.5m 9.95x11.5m 9.95x11.5m 9.95x11.5m 7.10x11.5m

D 800 800 800 800 850
----- 9M WIDE ROAD -----

CORRIDOR
D3 D3
45.3x1.80 D3 D3 D3
1125

800 800 800 800

PLASMA STORAGE 800 PLASMA STORAGE PLASMA STORAGE PLASMA STORAGE PLASMA STORAGE
~ 1000 L ~ 1000 L ~ 1000 L ~ 1000 L ~ 1000 L
~ 70 KL GV ~ 70 KL GV ~ 70 KL GV ~ 70 KL GV ~ 70 KL GV
8.15x9.62m 9.95x9.62m 9.95x9.62m 9.95x9.62m 7.10x9.62m

12M WIDE ROAD
E CORRIDOR
2.57x35.1
CORRIDOR
2.57x37.0

CORRIDOR
D3 D3
45.3x1.80 D3 D3 D3

800 800 800 800

F PLASMA STORAGE
~ 1000 L
~ 70 KL GV
PLASMA STORAGE
~ 1000 L
~ 70 KL GV
PLASMA STORAGE
~ 1000 L
~ 70 KL GV
PLASMA STORAGE
~ 1000 L
~ 70 KL GV
PLASMA STORAGE
~ 1000 L
~ 70 KL GV
8.15x9.62m 9.95x9.62m 9.95x9.62m 9.95x9.62m 7.10x9.62m

CORRIDOR
EMER.GLASS 50.5x1.80 EMER.GLASS
MH MH MH

CHROMO CHROMO
TFF
MH

AKTA-1" 100 L 100 L 100 L
MH

COLUMN COLUMN
DPB
G
MH MH

300 L
MH MH
MH

300 L
1500 L
JANITOR
1500 L 1500 L 1500 L SL
2000 L 5.20x1.80
600 L
CHROMO
MH
COLUMN

STAGE 8 N
6.17X8.68
STAGE 2 (F) PLASMA DAY
AKTA-1"

STAGE 3 (F) D4

STORE
AKTA-1"

12.8X8.68 15.8X8.68
8.10x8.77
1500 L
CENTRIFUGE
500 L 500 L 300 L 250 L 200 L
300 L
100 L 100 L MH 100 L 100 L CHROMO 100 L 100 L
MH MH MH MH

COLUMN
6D

MH
MH

6D 6D 6D
MH MH MH MH

MH MH
DPB

CORRIDOR
35.1X2.80
DE

6D

H D6 D6 D6
BUFFER
DPB
CHROMO
COLUMN
MH MH
MH MH MH MH
CHROMO
COLUMN
MH

PREPARATION
AKTA-1"

MH MH

300 L 250 L 250 L 250 L
MH
300 L 300 L 500 L
PLASMA 8.10x6.00
THAWING
6D

2000 L 2000 L
8.00X3.22
1000 L
SL

MH

2500 L STAGE 1 (F)/
AKTA-1"

STAGE 12 (E) D6
TFF

D6

CLAF
11.1X8.21
MH MH

MH

300 L 300 L
250 L
CHROMO
D6

COLUMN
500 L STAGE 4 (F)
1500 L 1500 L BAG CUTTING & TANK STORAGE
13.6X13.5
MH

1000 L
6D

500 L 500 L
150 L
POOLING 8.10x6.58
8.00X7.80
MH MH
MH

MH MH
MH
6D

MH MH MH MH

MH

I
MH

250 L 250 L 250 L 250 L
1500 L 1500 L WASTE OUT
SL CONTROL PANEL CONTROL PANEL

2500 L
1500 L DPB
PRE - VIRUS CONTROLLED CORRIDOR

50 L
GS+HH

GS+HH

CHROMO
COLUMN
6D

MH

MH

MH

MH
1000 L 1000 L 1000 L
500 L 500 L
100 L
MH

SL
CIP / SIP
1000 L
8.10x5.00
MH MH MH

MH MH

1500 L MH

MH
MH

100 L
MH

100 L
MH

100 L
MH

MH MH MH

MH
100 L
STAGE 5 (I) B
250 L 250 L 250 L
SL
500 L
8.00X7.20
MH

600 L
MH
STAGE 5 (I) A 100 L
DPB
MH

250 L 1500 L 11.1X18.0 WASTE
2.00X28.8

500 L OUT
D6

DECONTAMINATION 1.50x3.54
STAGE 7A
MH
MH

6.50x5.00
MH

250 L
500 L
13.6X15.1
4D
TFF

MH

MH

WASTE IN

J 250 L
4D

500 L
DPB
D6

D4

MH

300 L
COLUMN PACKING
AKTA-1"

D4

500 L
WASHING A/L
MH

AND SKIDS
AKTA-1"

1.50x4.42
100 L
8.00X7.86 6.50x3.00
MH

DPB
4D
TFF

MH
D4
6D

1000 L
STERILIZATION
CLAF

A/L
6.50x5.85
D6

2500 L 2500 L 1.50x5.73
2000 L
CHROMO MH MH
MH

MH
MH MH

COLUMN
6D

100 L 250 L 250 L
150 L
MH MH MH

500 L 500 L
D6

CHROMO
COLUMN AUTOCLAVE

CORRIDOR DHS
CORRIDOR

K 2.57X67.9
300 L
MH
AKTA-1" CHROMO
COLUMN
100 L
MH

150 L
MH
STAGE 6 (I) LAF LAF DPB
STAGE 11 F
1000 L

100 L MH

MH

100 L MH

14.7X16.7
COOL ZONE
MH

300 L 150 L
COLUMN PACKING
MH
1000 L

MH
100 L
500 L MH
MH

AND SKIDS 8.10x5.53
250 L MH
MH 150 L 6.90X10.6
200 L
2000 L

STAGE 10 T/
MH
MH
D4

MH
1.80X49.6
250 L MH

250 L

STAGE 13 E 200 L
AKTA-1"

MH
600 L

4D

4D

8.10X12.2
MH
SL

250 L MH

MH
DE
500 L
MH
6.25X2.05 STAGE 9 N A/L
2.20X3.89
GOWN
3.47X3.89
A/L
600 L

250 L SL CHROMO 2.22X3.89
SL

600 L COLUMN
5.10X14.3 MH

LAF
MH
D4

50 L 500 L
4D

MH

MH

BULK FILTRATION
LAF

L MH
6.90X6.00
D4
LAF

500 L
D4

1000 L A/L GOWN A/L
2.20X4.32 3.47X4.32 2.22X4.32
JANITOR MH

NANO
1.80x2.00
4D

4D

AKTA-1"
D3 D6 D6 LAF
FILTER TFF
D6 D6 DPB
4D

4D

CORRIDOR
DE CORRIDOR
37.0x2.80 A/L A/L
GOWN 2.57x66.0
DE

2.20X3.40 3.47X3.40 2.22X3.40
4D
D4

D4 4D D4
D6 D6 D6
D4

A/L A/L WASTE
D4
D4

4.01x2.40 4.13x2.40 IN
2.50x2.40 CONTROL PANEL CONTROL PANEL

A/L GOWN A/L
2.20X3.40 4.00X3.40 2.22X3.40
DPB

4D 4D D4
DPB

GS+HH

GS+HH

M 8.30x1.80
4D

4D

STERILIZATION
4.01x5.62 BUFFER
AUTOCLAVE

COLD ROOM CIP / SIP TANK STORAGE-2 PREPARATION
6D
LAF

WASHING DECONTAMINATION
3.11x5.62 3.52x5.62 4.40x8.10 5.60x8.10 6.80x8.10 4.29x8.10 CHEMICAL DAY STORE
8.10X4.00
COOL ZONE
2.76x8.10
6D

6D

DPB
DPB

WASTE OUT
DHS

LAF

DPB

3 WAY
D4

CORRIDOR
47.8x2.00
DE

6D

D4 D4 D4

D4

FEMALE
MALE PRIMARY MALE AIR LOCK
PRIMARY AIR LOCK AIR LOCK PRIMARY FEMALE
VISITOR
VISITOR VISITOR FEMALE CHANGE CHANGE 1 CHANGE 1 CHANGE MALE
OFFICE 1 OFFICE 2 2.50x5.00 CHANGE 1
3.64x5.00 3.10x5.00 3.10x5.00 3.00x5.00 3.80x5.00 4.77x5.00
2.60x5.00 CHANGE 3.05x5.00
4.83x5.00 4.83x5.00 2.70x5.00
N
D4

D4

4D

4D

D4

D4

D4 D4 D4 D4 D4

RECEPTION
8D

CORRIDOR 8.00x6.10
34.9x2.80 4.05x2.80
6D

JANITOR
4D 4D 4D 4D 4D 4D 4D 4D
ROOM
D1

3.50x1.65 WAITING ROOM
5.30x4.00
D1 D1 D1
GENTS TOILET
DOCUMENT FEMALE VISITOR GENTS 5.90x5.20
OFFICE 3 LADIES TOILET
ROOM LOCKER LOCKER LOCKER
LIFT 3.13x5.30 3.13x5.30 5.90x5.20
ROOM ROOM ROOM
1D 1D 1D 1D 1D 1D 1D 1D 1D 1D
4.57x5.30 4.57x5.30 4.57x5.30

O

-----6M WIDE ROAD-----

Reagent Details:
Formula & Calculations:
Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).
Calculation: To prepare 1 L of 0.2M Citric Acid (MW = 192.12g/mol)
Weight of solute(g)=1×38.424=38.424g.
Preparation Method:
Weigh 38.424 g of Citric acid and dissolve in water to make a 1 L solution.
Preparation for different volumes:
To prepare _____ of 0.2M Citric acid (MW = 192.12g/mol).
Weight of solute(g)=1×38.424 _____ L= _____ gm
Weigh _____ gm of Citric acid and dissolve in water to make a _____ L solution.
Check and record the details:
Preparation Details:
Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry
Date
| | | |
Final volume prepared | Weighed Quantity | pH 1.9±0.1 | Conductivity (4.7±0.2ms/cm) | Storage Condition:
| | | | RT
Date of Preparation | Prepared By | Checked By
| |

Reagent Details:
Preparation of PF-RR-30 working buffer from Stock buffer (PF-RR-27):
Calculations: For the Preparation of 0.01M Sodium Hydroxide.
Stock Buffer (PF-RR-27):2.0M Sodium Hydroxide,
Formula:
 $C1 \times V1 = C2 \times V2$
C1 = concentration of stock
V1 = volume of stock you need to take
C2 = desired (working) concentration
V2 = final total volume of working solution
Applying the formula to prepare the working buffer:
 $C1 \times V1 = C2 \times V2$
C1 = 2.0 M Sodium Hydroxide.
V1 = volume of stock you need to take _____ L.
C2 = desired (working) concentration 0.01M Sodium Hydroxide.
V2 = final total volume of working solution _____ L.
 $V1 = (C2 \times V2) / C1$
C2=0.01M; V2= 1.0 L; C1=2.0M
 $V1 = (0.01 \times 1.0) / 2.0$
V1 = 0.005L
Take 0.005 L (PF-RR-30) Stock buffer solution and dilute it with water, making up to 1 L.
Check and record the details:
Preparation Details:
Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry
Date
Sodium Hydroxide | | | |
Stock buffer (PF-RR-27) taken volume(L) | pH (12.0 ±0.5) | Conductivity (62.0 ±3.0ms/cm) | Storage
Condition | Final volume prepared
| | |
Date of Preparation | Prepared By | Checked By
| |

Reagent Details:
Calculations: To prepare 1 L of 1.05% Tween 80.
Decide on the total final volume to prepare (e.g., 1000 mL).
Calculate 1.05% of the total final volume to determine the amount of Tween-80 needed:
"Tween-80 required"="Total Volume"×1.05/100.
For 1000 mL final volume: mL Tween-80.
Preparation Method:
Measure the required volume of one litre and dissolve it in water to make a 1 L solution.
Preparation for different volumes:
To prepare _____ of 1.05% Tween 80.
Tween-80 required"="Total Volume"×1.05/100.
For _____ mL final volume: mL Tween-80.
Measure the required volume of one liter and dissolve it in water to make a 1 L solution.
Check and record the details:
Preparation Details:
Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry
Date
Tween 80 | | | |
Final volume prepared | 1.05% Tween 80 Volume taken | pH (6.0±1.0) | Conductivity (100µs/cm) | Storage
Condition
| | | | RT
Date of Preparation | Prepared By | Checked By (QC/Analyst)
| |

Reagent Details:

Formula:

Sodium Acetate(M.Wt): 82.03g/mol.

Acetic Acid (M.Wt): 60.05g/ mol

Sodium Chloride:58.44g/mol

Preparation of 0.1 M Sodium Acetate pH 5.35 buffer

Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).

Calculation: To prepare 1 L of 0.1 M Sodium Acetate (MW = 82.03g/mol).

Weight of solute(g)=0.1×82.03× 1.0=8.203g.

Preparation Method:

Weigh 8.203 g of Sodium Acetate and dissolve in water to make a 1 L solution.

Preparation for required volumes:

To prepare _____ of 0.1M Sodium Acetate (MW = 82.03g/mol).

Weight of solute(g)=0.1×82.03× _____ L= _____ g.

Weigh _____ gm of Sodium Acetate and dissolve in water to make a _____ L solution.

Calculations: To prepare the 0.1 M Acetic Acid.

Formula & Calculations:

Formula: Use the formula to calculate the molarity:

Molarity (M) = (Per cent by Weight / 100) * (Density in g/mL) * 1000 / Molar Mass.

Example: Molarity of Glacial Acetic Acid

Molar Mass of CH₃COOH: ~60.05 g/mol

Density: ~1.05 g/mL

Purity: 100%

Acetic Acid (MW = 60.05 g/mol).

Calculation:

Molarity = (100 / 100) * 1.05 g/mL * 1000 mL/L / 60.05 g/mol ≈ 17.5 M.

Calculation: To prepare 1 L of 0.1 M acetic acid

Applying the formula to prepare the working buffer:

C₁V₁=C₂V₂

C₁ = 17.5 M acetic acid.

V₁ = volume of stock needed to take.

C₂ = desired (working) concentration 0.1 M acetic acid.

V₂ = final total volume of working solution for one litre.

V₁ = (C₂ × V₂)/C₁.

C₂=0.1 M ; V₂= 1.0 L; C₁=17.5 M

V₁ = (0.1 × 1.0)/17.5

V₁ = 0.0057 L

Take 0.0057 L of the Stock solution of acetic acid and add it to a volumetric flask, making up the volume to 1 L.

To prepare various volumes:

V₁ = volume of stock needed to take _____ L.

C₂ = desired (working) concentration 0.1 M acetic acid.

V₂ = final total volume of working solution _____ L.

V₁ = (C₂ _____ × V₂ _____) /17.5.

V₁= _____ L.

Take V₁ _____ L of the Stock solution of acetic acid and add it to a volumetric flask, making up the final volume _____ L.

Preparation of 4.0 M Sodium Chloride.

Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).

Calculation: To prepare 1 L of 4 M Sodium Chloride (MW = 58.44g/mol).

Weight of solute(g)=4×58.44 × 1.0= 233.76g.

Preparation Method:

Weigh 233.76g of Sodium Chloride and dissolve in water to make a 1 L solution.

Preparation for required volumes:

To prepare _____ of 4 M Sodium Chloride (MW = 58.44g/mol).

Weight of solute(g)=0.1×58.44× _____ L= _____ g.

Weigh _____ gm of Sodium Chloride and dissolve in water to make a _____ L solution.

Procedure for Preparation of 0.1 M Sodium Acetate + 4 M Sodium Chloride pH 5.35

To prepare a pH 5.35 solution, 0.1 M Sodium Acetate solution was added to this 4M Sodium Chloride was added and completely dissolved and then titrated with 0.1 M Acetic acid so that the pH attains pH 5.35 and made up to 1 L.

Check and record the details:

Preparation Details:

Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry

Date

Sodium Acetate | | | |

Acetic Acid | | | |

Final volume prepared | Measured Quantity | pH (5.35± 0.1) | Conductivity (9.0± 1ms/cm) | Storage

Condition | Reagent Record

| | | |

Stock buffer (PF-RR-23) taken volume(L) | pH (5.35± 0.1) | Conductivity (340 ± 10 ms/cm) | Storage

Condition | Final volume prepared

| | | RT |

Date of Preparation | Prepared By | Checked By

| |

Reagent Details:
Formula & Calculations:
Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).
Calculation: To prepare 1 L of 0.1M Citric Acid (MW = 192.124 g/mol)
Weight of solute(g)=0.1×192.124x1=19.21g.
Preparation Method:
Weigh 19.21g of Citric Acid and dissolve in water to make a 1 L solution.
Preparation for different volumes:
To prepare _____ of 0.1M Citric Acid (MW = 192.124g/mol).
Weight of solute(g)=0.1×192.124 x _____L= _____g.
Weigh _____gm of Citric Acid and dissolve in water to make a _____ L solution.
Check and record the details:
Preparation Details:
Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry
Date
Citric Acid | | | |
Final volume prepared | Weighed Quantity | pH (2.1± 0.1) | Conductivity (3.0± 0.5ms/cm) | Storage
Condition
| | | RT
Date of Preparation | Prepared By | Checked By
| |

ACPH=volumetric air flow rate 8 60/volume of area
60-150
20-60

Reagent Details:
Formula & Calculations:
Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).
Calculation: To prepare 1 L of 0.15MSodium Caprylate (MW = 166.19 g/mol)
Weight of solute(g)=0.15×166.19 = 24.92■g.
Preparation Method:
Weigh 24.92 g of Sodium Caprylate and dissolve in water to make a 1 L solution.
Preparation for different volumes:
To prepare _____ of 0.15M Sodium Caprylate (MW = 166.19 g/mol).
Weight of solute(g)=0.15×166.19 x _____L= _____gm
Weigh _____gm of Sodium Caprylate and dissolve in water to make a _____ L solution.
Check and record the details:
Preparation Details:
Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry
Date
Sodium Caprylate | | | |
Final volume prepared | Weighed Quantity | pH | Conductivity | Storage Condition:
| | |
Date of Preparation | Prepared By | Checked By
| |

Reagent Details:
Formula & Calculations:
Formula: Use the formula to calculate the molarity:
 $Molarity (M) = (Per\ cent\ by\ Weight / 100) * (Density\ in\ g/mL) * 1000 / Molar\ Mass.$
Example: Molarity of Glacial Acetic Acid
Molar Mass of CH_3COOH : ~60.05 g/mol
Density: ~1.05 g/mL
Purity: 100%
Acetic Acid (MW = 60.05 g/mol).
Calculation:
 $Molarity = (100 / 100) * 1.05\ g/mL * 1000\ mL/L / 60.05\ g/mol \approx 17.5\ M.$
Calculation: To prepare 1 L of 1.0 M.
Applying the formula to prepare the working buffer:
 $C_1 V_1 = C_2 V_2$
 $C_1 = 17.5\ M$ acetic acid.
 V_1 = volume of stock needed to take.
 C_2 = desired (working) concentration 1.0 M acetic acid.
 V_2 = final total volume of working solution for one litre.
 $V_1 = (C_2 \times V_2) / C_1.$
 $C_2 = 1.0\ M ; V_2 = 1.0\ L ; C_1 = 17.5\ M$
 $V_1 = (1.0 \times 1.0) / 17.5$
 $V_1 = 0.057\ L$
Take 0.057 L of the Stock solution of acetic acid and add it to a volumetric flask, making up the volume to 1 L.
To prepare various volumes:
 V_1 = volume of stock needed to take _____ L.
 C_2 = desired (working) concentration 1.0 M acetic acid.
 V_2 = final total volume of working solution _____ L.
 $V_1 = (C_2 \times V_2) / 17.5.$
 $V_1 =$ _____ L.
Take V_1 _____ L of the Stock solution of acetic acid and add it to a volumetric flask, making up the final volume _____ L.
Preparation Details:
Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry Date
|||
Final volume prepared | Measured Quantity | pH (2.36 ±0.2) | Conductivity (0.36±0.2ms/cm) | Storage Condition | Reagent Record
||| RT |
Date of Preparation | Prepared By | Checked By
||

[SHEET: Warehousing (GF)]

Sr. No. | Equipment Tag | Working Volume in Litres | Room/Stage (As per Layout) | Type | Warehouse | Fixed / Mobile | Gross Weight in Kg (Empty- Tentative) | Full weight in Kg | Dimension (L mm x W mmx H mm) | Make

1 | De-Dusting tunnel | NA | RM Receipt Room | De-Dusting Machine | Warehouse | Fixed | 200 | NA | 2200 L X 1200 W X 2400 H | Pridetech
2 | RLAF | NA | RM Sampling Room | Sampling Booth | Warehouse | Fixed | 150 | NA | 2500 L x 2000 Wx 1000 H | Clean air, Fabtech
3 | DBP | NA | RM Sampling Room | Equipment | Warehouse | Fixed | 250 | NA | 1200 L x 1000 W x 2100 H | Clean air, Fabtech
4 | DPB | NA | Approved RM | Equipment | Warehouse | Fixed | 250 | NA | 1200 L x 1000 W x 2100 H | Clean air, Fabtech
5 | DPB | NA | Dispensing Room | Equipment | Warehouse | Fixed | 250 | NA | 1200 L x 1000 W x 2100 H | Clean air, Fabtech
6 | RLAF | NA | Dispensing Room | Dispensing Booth | Warehouse | Fixed | 150 | NA | 2500 L x 2000 Wx 1000 H | Clean air, Fabtech
7 | DPB | NA | Plasma Sampling | Equipment | Warehouse | Fixed | 250 | NA | 1200 L x 1000 W x 2100 H | Clean air, Fabtech
8 | RLAF | NA | Plasma Sampling | Plasma Sampling Booth | Warehouse | Fixed | 150 | NA | 2500 L x 2000 Wx 1000 H | Clean air, Fabtech
9 | DPB | NA | Reject Room | Equipment | Warehouse | Fixed | 250 | NA | 1200 L x 1000 W x 2100 H | Clean air, Fabtech
10 | freezer-1000L | 1000 L | Plasma Storage Area | Freezer | Warehouse | Fixed | 425 | 1625 | 1400 L X 950 W X 2100 H | Newtronic lifecare, REMI
11 | freezer-1000L | 1000 L | Plasma Storage Area | Freezer | Warehouse | Fixed | 425 | 1625 | 1400 L X 950 W X 2100 H | Newtronic lifecare, REMI
12 | freezer-1000L | 1000 L | Plasma Storage Area | Freezer | Warehouse | Fixed | 425 | 1625 | 1400 L X 950 W X 2100 H | Newtronic lifecare, REMI
13 | freezer-1000L | 1000 L | Plasma Storage Area | Freezer | Warehouse | Fixed | 425 | 1625 | 1400 L X 950 W X 2100 H | Newtronic lifecare, REMI
14 | freezer-1000L | 1000 L | Plasma Storage Area | Freezer | Warehouse | Fixed | 425 | 1625 | 1400 L X 950 W X 2100 H | Newtronic lifecare, REMI

[SHEET: Bulk MFG (GF)]

Sr. No. | Vessel Tag | Working Volume in Litres | Room/Stage (As per Layout) | Type | Pre Viral/Post Viral | Fixed / Mobile | Empty (Kg) | Full load (Kg) | Dimension (mm x mmx mm) | Make

1 | TW 100 | 500 | THAWING | Process Vessel | Pre-Viral | Fixed | 450 | 950 | 2200L X 2200 W X 2800H | Praj and Aaruni
2 | DPB | NA | THAWING | Equipment | Pre-Viral | Fixed | 50 | NA | 1200 L x 1000 W x 2100 H | Clean air, Fabtech
3 | PV 101.1 | 300 | CUTTING | Process Vessel | Pre-Viral | Fixed | 390 | 690 | 2100L X 2000 X 2700H | Praj and Aaruni
4 | PV 101.2 | 300 | CUTTING | Process Vessel | Pre-Viral | Fixed | 390 | 690 | 2100L X 2000 X 2700H | Praj and Aaruni
6 | PV 102.1 | 300 | 1F/12E | Process Vessel | Pre-Viral | Fixed | 390 | 690 | 2100L X 2000 X 2700H | Praj and Aaruni
7 | PV 102.2 | 300 | 1F/12E | Process Vessel | Pre-Viral | Fixed | 390 | 690 | 2100L X 2000 X 2700H | Praj and Aaruni
8 | PV 103.1 | 1500 | 1F/12E | Process Vessel | Pre-Viral | Fixed | 900 | 2400 | 2100L X 2000 X 2700H | Praj and Aaruni
9 | PV 103.2 | 1500 | 1F/12E | Process Vessel | Pre-Viral | Fixed | 900 | 2400 | 3200L X 3500 W X 4200H | Praj and Aaruni
10 | PV 103.3 | 500 | 1F/12E | Process Vessel | Pre-Viral | Fixed | 450 | 950 | 3200L X 3500 W X 4200H | Praj and Aaruni
11 | PV 103.4 | 500 | 1F/12E | Process Vessel | Pre-Viral | Fixed | 450 | 950 | 2200L X 2200 W X 2800H | Praj and Aaruni
12 | PV 103.5 | 500 | 1F/12E | Process Vessel | Pre-Viral | Fixed | 450 | 950 | 2200L X 2200 W X 2800H | Praj and Aaruni
13 | SPARE-1 | 250 | 1F/12E | Process Vessel | Pre-Viral | Fixed | 340 | 590 | 2200L X 2200 W X 2800H | Praj and Aaruni
14 | PF-RR-03 | 1000 | 1F/12E | Buffer | Pre-Viral | Fixed | 650 | 1650 | 2100L X 2000 X 2600H | Praj and Aaruni
15 | PF-RR-27 | 250 | 1F/12E | Buffer | Pre-Viral | Fixed | 340 | 590 | 2700L X 2700W X 3200H | Praj and Aaruni
16 | PV 128 | 300 | 1F/12E | Process Vessel | Pre-Viral | Fixed | 390 | 690 | 2100L X 2000 X 2600H | Praj and Aaruni
17 | FE-RR-02 | 150 | 1F/12E | Buffer | Pre-Viral | Mobile | 250 | 430 | 2100L X 2000 X 2700H | Praj and Aaruni
18 | FE-RR-06 | 250 | 1F/12E | Buffer | Pre-Viral | Fixed | 340 | 590 | 1900L X 2000W X 2800H | Praj and Aaruni
19 | FE-RR-06 | 250 | 1F/12E | Buffer | Pre-Viral | Fixed | 340 | 590 | 2100L X 2000 X 2600H | Praj and Aaruni
| FE-RR-14 | 250 | 1F/12E | Buffer | Pre-Viral | Mobile | 340 | 590 | 2100L X 2000 X 2600H | Praj and Aaruni
20 | Chromatography Column- 1000 X 500 | NA | 1F/12E | Equipment | Pre-Viral | Fixed | 400 | 600 | 1500L X 1500 W X 2700H | Truking, Cytiva, Aaruni
21 | Akta- 500 L/ Hr | NA | 1F/12E | Equipment | Pre-Viral | Fixed | 1200 | NA | 2000L X 2000 X 2000H | Truking, Cytiva, Aaruni
22 | Chromatography Column-1000 X 500 | NA | 1F/12E | Equipment | Pre-Viral | Fixed | 400 | 600 | 1500L X 1500 W X 2700H | Truking, Cytiva, Aaruni
23 | Mobile CIP | NA | 1F/12E | Equipment | Pre-Viral | Mobile | 150 | 200 | 2000L X 2000 X 2000H | Praj & DDE
24 | PV 104.1 | 1500 | 2F | Process Vessel | Pre-Viral | Fixed | 900 | 2400 | 3200L X 3500 W X 4200H | Praj and Aaruni
25 | PV 104.2 | 1500 | 2F | Process Vessel | Pre-Viral | Fixed | 900 | 2400 | 3200L X 3500 W X 4200H | Praj and Aaruni
26 | PV 105.1 | 1500 | 2F | Process Vessel | Pre-Viral | Fixed | 900 | 2400 | 3200L X 3500 W X 4200H | Praj and Aaruni
27 | PV 105.2 | 1500 | 2F | Process Vessel | Pre-Viral | Fixed | 900 | 2400 | 3200L X 3500 W X 4200H | Praj and Aaruni
28 | PV 105.3 | 300 | 2F | Process Vessel | Pre-Viral | Fixed | 390 | 690 | 2100L X 2000 X 2700H | Praj

and Aaruni
29 | PF–RR-35 | 200 | 2F | Buffer | Pre-Viral | Mobile | 300 | 520 | 2100L X 2000 X 2600H | Praj and Aaruni
30 | PF–RR-31 | 100 | 2F | Buffer | Pre-Viral | Mobile | 220 | 370 | 1800L X 1900W X 2800H | Praj and Aaruni
31 | PF–RR-11 | 100 | 2F | Buffer | Pre-Viral | Mobile | 220 | 370 | 1800L X 1900W X 2800H | Praj and Aaruni
32 | PF–RR-12 | 250 | 2F | Buffer | Pre-Viral | Fixed | 340 | 590 | 2100L X 2000 X 2600H | Praj and Aaruni
33 | PF–RR-32 | 100 | 2F | Buffer | Pre-Viral | Mobile | 220 | 370 | 1800L X 1900W X 2800H | Praj and Aaruni
34 | PF–RR-27 | 100 | 2F | Buffer | Pre-Viral | Mobile | 220 | 370 | 1800L X 1900W X 2800H | Praj and Aaruni
| PF–RR-28 | 100 | 2F | Buffer | Pre-Viral | Mobile | 220 | 370 | 1800L X 1900W X 2800H | Praj and Aaruni
35 | PF–RR-30 | NALGENE | 2F | Bottle | Pre-Viral | NA | 50 | NA | 500L X 500 W x 1000 H | Praj and Aaruni
36 | TFF | NA | 2F | Equipment | Pre-Viral | Fixed | 3000 | 5000 | 2500L X 2000 W X 2700H | Praj, Cytiva, Sartorius, Aaruni
37 | Chromatography Column-1000 X 500 | NA | 2F | Equipment | Pre-Viral | Fixed | 400 | 600 | 1500L X 1500 W X 2700H | Truking, Cytiva, Aaruni
38 | Akta-785 L/ hr | NA | 2F | Equipment | Pre-Viral | Fixed | 2000 | 4000 | 2000L X 2000 X 2000H | Truking, Cytiva, Aaruni
39 | Mobile CIP | NA | 2F | Equipment | Pre-Viral | Mobile | 150 | 200 | 2000L X 2000 X 2000H | Praj & DDE
40 | PV 106 | 1500 | 3F | Process Vessel | Pre-Viral | Fixed | 900 | 2400 | 3200L X 3500 W X 4200H | Praj and Aaruni
41 | PV 107 | 2000 | 3F | Process Vessel | Pre-Viral | Fixed | 1200 | 3200 | 3600L X 3600W X 4300H | Praj and Aaruni
42 | PV 103.6 | 500 | 3F | Process Vessel | Pre-Viral | Fixed | 450 | 950 | 2200L X 2200 W X 2800H | Praj and Aaruni
43 | PV 128 | 500 | 3F | Process Vessel | Pre-Viral | Fixed | 450 | 950 | 2200L X 2200 W X 2800H | Praj and Aaruni
44 | PV 129 | 300 | 3F | Process Vessel | Pre-Viral | Fixed | 390 | 690 | 2100L X 2000 X 2700H | Praj and Aaruni
45 | PV 119.1 | 300 | 3F | Process Vessel | Pre-Viral | Fixed | 390 | 690 | 2100L X 2000 X 2700H | Praj and Aaruni
46 | PV 130 | 100 | 3F | Process Vessel | Pre-Viral | Mobile | 220 | 370 | 1800L X 1900W X 2800H | Praj and Aaruni
47 | Centrifuge | NA | 3F | Equipment | Pre-Viral | Fixed | 800 | 900 | 2000L x 1500 W x 2000H | Alfa Laval
48 | Chromatography Column | NA | 3F | Equipment | Pre-Viral | Fixed | 400 | 600 | 1500L X 1500 W X 2700H | Truking, Cytiva, Aaruni
49 | Chromatography Column-1000 X 500 | NA | 3F | Equipment | Pre-Viral | Fixed | 400 | 600 | 1500L X 1500 W X 2700H | Truking, Cytiva, Aaruni
50 | Akta-1178 L/Hr | NA | 3F | Equipment | Pre-Viral | Fixed | 3000 | NA | 2000L X 2000 X 2000H | Truking, Cytiva, Aaruni
51 | Mobile CIP | NA | 3F | Equipment | Pre-Viral | Mobile | 150 | 200 | 2000L X 2000 X 2000H | Praj & DDE
52 | PV 108 | 1000 | 4I | Process Vessel | Pre-Viral | Fixed | 650 | 1650 | 2700L X 2700W X 3200H | Praj and Aaruni
53 | PV 109.1 | 2000 | 4I | Process Vessel | Pre-Viral | Fixed | 1200 | 3200 | 3600L X 3600W X 4300H | Praj and Aaruni
54 | PV 109.2 | 2000 | 4I | Process Vessel | Pre-Viral | Fixed | 1200 | 3200 | 3600L X 3600W X 4300H | Praj and Aaruni
55 | PV 109.3 | 1000 | 4I | Process Vessel | Pre-Viral | Fixed | 650 | 1650 | 2700L X 2700W X 3200H | Praj and Aaruni
56 | PV 110.1 | 2500 | 4I | Process Vessel | Pre-Viral | Fixed | 1500 | 4000 | 3800L X 3800W X 4300H | Praj and Aaruni
57 | PV 110.2 | 2500 | 4I | Process Vessel | Pre-Viral | Fixed | 1500 | 4000 | 3800L X 3800W X 4300H | Praj and Aaruni
58 | PF-RR-18 | 250 | 4I | Buffer | Pre-Viral | Fixed | 340 | 590 | 2100L X 2000 X 2600H | Praj and Aaruni
59 | PF-RR-21 | 1000 | 4I | Buffer | Pre-Viral | Fixed | 650 | 1650 | 2700L X 2700W X 3200H | Praj and Aaruni
60 | PF-RR-21 | NALGENE | 4I | Bottle | Pre-Viral | NA | 50 | NA | 500L X 500 W x 1000 H | Praj and Aaruni
61 | PF-RR-23 | 500 | 4I | Buffer | Pre-Viral | Fixed | 450 | 950 | 2200L X 2200 W X 2800H | Praj and Aaruni
62 | PF-RR-25 | 500 | 4I | Buffer | Pre-Viral | Fixed | 450 | 950 | 2200L X 2200 W X 2800H | Praj and Aaruni
63 | PF-RR-27 | 500 | 4I | Buffer | Pre-Viral | Fixed | 450 | 950 | 2200L X 2200 W X 2800H | Praj and Aaruni
64 | Akta-3454 L/Hr | NA | 4I | Equipment | Pre-Viral | Fixed | 3000 | NA | 2000L X 2000 X 2000H | Truking, Cytiva, Aaruni
65 | Chromatography Column-2000 X 500 | NA | 4I | Equipment | Pre-Viral | Fixed | 3500 | 5000 | 2200L x 3200 W x 3000H | Truking, Cytiva, Aaruni
66 | TFF | NA | 4I | Equipment | Pre-Viral | Fixed | 2000 | NA | 2500L X 2000 W X 2700H | Praj, Cytiva, Sartorius, Aaruni
67 | Mobile CIP | NA | 4I | Equipment | Pre-Viral | Mobile | 150 | 200 | 2000L X 2000 X 2000H | Praj & DDE
68 | PV 110 | 1500 | 5I | Process Vessel | Pre-Viral | Fixed | 900 | 2400 | 3200L X 3500 W X 4200H | Praj and Aaruni
69 | PV 111 | 1500 | 5I | Process Vessel | Pre-Viral | Fixed | 900 | 2400 | 3200L X 3500 W X 4200H | Praj and Aaruni
70 | PV 112.1 | 1500 | 5I | Process Vessel | Pre-Viral | Fixed | 900 | 2400 | 3200L X 3500 W X 4200H | Praj and Aaruni
71 | PV 112.2 | 1500 | 5I | Process Vessel | Pre-Viral | Fixed | 900 | 2400 | 3200L X 3500 W X 4200H | Praj and Aaruni
72 | IG-RR-09 | 100 | 5I | Buffer | Pre-Viral | Mobile | 220 | 370 | 1800L X 1900W X 2800H | Praj and Aaruni
73 | IG-RR-05 | 300 | 5I | Buffer | Pre-Viral | Fixed | 390 | 690 | 2100L X 2000 X 2700H | Praj and Aaruni

74 | IG-RR-07 | 250 | 5l | Buffer | Pre-Viral | Fixed | 340 | 590 | 2100L X 2000 X 2600H | Praj and Aaruni

75 | IG-RR-22 | 250 | 5l | Buffer | Pre-Viral | Fixed | 340 | 590 | 2100L X 2000 X 2600H | Praj and Aaruni

76 | IG-RR-23 | 50 | 5l | Buffer | Pre-Viral | Mobile | 70 | 150 | 1800L X 1800W X 2800H | Praj and Aaruni

77 | PV 113 | 250 | 5l | Process Vessel | Pre-Viral | Fixed | 340 | 590 | 2100L X 2000 X 2600H | Praj and Aaruni

78 | PV113.1 | 250 | 5l | Process Vessel | Pre-Viral | Fixed | 340 | 590 | 2100L X 2000 X 2600H | Praj and Aaruni

79 | IG-RR-06 | NALGENE | 5l | Bottle | Pre-Viral | NA | 50 | NA | 500L X 500 W x 1000 H | Praj and Aaruni

80 | IG-RR-23 | 100 | 5l | Buffer | Pre-Viral | Mobile | 220 | 370 | 1800L X 1900W X 2800H | Praj and Aaruni

81 | IG-RR-24 | 100 | 5l | Buffer | Pre-Viral | Mobile | 220 | 370 | 1800L X 1900W X 2800H | Praj and Aaruni

82 | Akta-2211 L/ Hr | NA | 5l | Equipment | Pre-Viral | Fixed | 3000 | NA | 2000L X 2000 X 2000H | Truking, Cytiva, Aaruni

83 | Chromatography Column-1600X 500 | NA | 5l | Equipment | Pre-Viral | Fixed | 2500 | 5000 | 2200L x 3200 W x 3000H | Truking, Cytiva, Aaruni

84 | TFF | NA | 5l | Equipment | Pre-Viral | Fixed | 2000 | NA | 2500L X 2000 W X 2700H | Praj, Cytiva, Sartorius, Aaruni

85 | Depth filters | NA | 5l | Equipment | Pre-Viral | Fixed | 150 | NA | 500L X 500 W x 1000 H | Aaruni, Millipore, 3M

86 | Mobile CIP | NA | 5l | Equipment | Pre-Viral | Mobile | 150 | 200 | 2000L X 2000 X 2000H | Praj & DDE

87 | PV 118.1 | 2500 | 7A | Process Vessel | Pre-Viral | Fixed | 1500 | 4000 | 3800L X 3800W X 4300H | Praj and Aaruni

88 | PV 118.2 | 2500 | 7A | Process Vessel | Pre-Viral | Fixed | 1500 | 4000 | 3800L X 3800W X 4300H | Praj and Aaruni

89 | PV 119 | 2000 | 7A | Process Vessel | Pre-Viral | Fixed | 1200 | 3200 | 3600L X 3600W X 4300H | Praj and Aaruni

90 | PV 120 | 1000 | 7A | Process Vessel | Pre-Viral | Fixed | 650 | 1650 | 2700L X 2700W X 3200H | Praj and Aaruni

91 | AL-RR-02 | 100 | 7A | Buffer | Pre-Viral | Mobile | 220 | 370 | 1800L X 1900W X 2800H | Praj and Aaruni

92 | AL-RR-05 | 500 | 7A | Buffer | Pre-Viral | Fixed | 450 | 950 | 2200L X 2200 W X 2800H | Praj and Aaruni

93 | AL-RR-05 | 500 | 7A | Buffer | Pre-Viral | Fixed | 450 | 950 | 2200L X 2200 W X 2800H | Praj and Aaruni

94 | AL-RR-09 | 500 | 7A | Buffer | Pre-Viral | Fixed | 450 | 950 | 2200L X 2200 W X 2800H | Praj and Aaruni

95 | AL-RR-11 | 250 | 7A | Buffer | Pre-Viral | Fixed | 340 | 590 | 2100L X 2000 X 2600H | Praj and Aaruni

96 | AL-RR-19 | 250 | 7A | Buffer | Pre-Viral | Fixed | 340 | 590 | 2100L X 2000 X 2600H | Praj and Aaruni

97 | AL-RR-19 | 250 | 7A | Buffer | Pre-Viral | Fixed | 340 | 590 | 2100L X 2000 X 2600H | Praj and Aaruni

98 | PV 120 | 500 | 7A | Process Vessel | Pre-Viral | Fixed | 450 | 950 | 2200L X 2200 W X 2800H | Praj and Aaruni

99 | PV 121 | 500 | 7A | Process Vessel | Pre-Viral | Fixed | 450 | 950 | 2200L X 2200 W X 2800H | Praj and Aaruni

100 | PV 122 | 250 | 7A | Process Vessel | Pre-Viral | Fixed | 340 | 590 | 2100L X 2000 X 2600H | Praj and Aaruni

101 | PV 122.S | 250 | 7A | Process Vessel | Pre-Viral | Fixed | 340 | 590 | 2100L X 2000 X 2600H | Praj and Aaruni

102 | PV 123 | 100 | 7A | Process Vessel | Pre-Viral | Mobile | 220 | 370 | 1800L X 1900W X 2800H | Praj and Aaruni

103 | PV 124 | 100 | 7A | Process Vessel | Pre-Viral | Mobile | 220 | 370 | 1800L X 1900W X 2800H | Praj and Aaruni

104 | AL-RR-23 | 250 | 7A | Buffer | Pre-Viral | Fixed | 340 | 590 | 2100L X 2000 X 2600H | Praj and Aaruni

105 | AL-RR-19 | 100 | 7A | Buffer | Pre-Viral | Mobile | 220 | 370 | 1800L X 1900W X 2800H | Praj and Aaruni

106 | AL-RR-21 | NALGENE | 7A | Bottle | Pre-Viral | NA | 50 | NA | 500L X 500 W x 1000 H | Praj and Aaruni

107 | Akta- 1356 L/ Hr | NA | 7A | Equipment | Pre-Viral | Fixed | 2000 | NA | 2200L x 3200 W x 3000H | Truking, Cytiva, Aaruni

108 | Chromatography Column-1200 X 500 | NA | 7A | Equipment | Pre-Viral | Fixed | 3000 | 5000 | 2200L x 3200 W x 3000H | Truking, Cytiva, Aaruni

109 | TFF | NA | 7A | Equipment | Pre-Viral | Fixed | 2000 | NA | 2500L X 2000 W X 2700H | Praj, Cytiva, Sartorius, Aaruni

110 | Mobile CIP | NA | 7A | Equipment | Pre-Viral | Mobile | 150 | 200 | 2000L X 2000 X 2000H | Praj & DDE

111 | PV 125 | 100 | 8N | Process Vessel | Pre-Viral | Mobile | 220 | 370 | 1800L X 1900W X 2800H | Praj and Aaruni

112 | PV 126 | 100 | 8N | Process Vessel | Pre-Viral | Mobile | 220 | 370 | 1800L X 1900W X 2800H | Praj and Aaruni

113 | PV 125.1 | 100 | 8N | Process Vessel | Pre-Viral | Mobile | 220 | 370 | 1800L X 1900W X 2800H | Praj and Aaruni

114 | FN-RR-02 | 300 | 8N | Buffer | Pre-Viral | Fixed | 390 | 690 | 2100L X 2000 X 2700H | Praj and Aaruni

115 | PV 127 | 100 | 8N | Process Vessel | Pre-Viral | Mobile | 220 | 370 | 1800L X 1900W X 2800H | Praj and Aaruni

116 | PV 127.1 | 100 | 8N | Process Vessel | Pre-Viral | Mobile | 220 | 370 | 1800L X 1900W X 2800H | Praj and Aaruni

117 | Chromatography Column-300 X 500 | NA | 8N | Equipment | Pre-Viral | Fixed | 70 | 150 | 500L X 500 W x 1000 H | Truking, Cytiva, Aaruni

118 | Akta- 70 L/ Hr | NA | 8N | Equipment | Pre-Viral | Fixed | 1200 | NA | 2200L x 3200 W x 3000H | Truking, Cytiva, Aaruni

119 | Mobile CIP | NA | 8N | Equipment | Pre-Viral | Mobile | 150 | 200 | 2000L X 2000 X 2000H | Praj & DDE

120 | Centralised CIP - 2 tank system | TBD | CIP/SIP Room | Equipment | Pre-Viral | Fixed | 1000 | NA | 2200L x 3200 W x 3000H | Praj and Aaruni

119 | Decontamination Autoclave 300L | TBD | Decontamination Room | Equipment | Pre-Viral | Fixed | 300 | NA | 500L x 500 W x 1200 H | Pharma Lab, Indu IonPure

120 | Dynamic Pass Box (DPB) | NA | Decontamination Room | Cleanroom Equipment | Pre-Viral | Fixed | 250 | NA | 1200 L x 1000 W x 2100 H | Clean air, Fabtech

121 | Dynamic Pass Box (DPB) | NA | Decontamination Room | Cleanroom Equipment | Pre-Viral | Fixed | 250 | NA | 1200 L x 1000 W x 2100 H | Clean air, Fabtech

122 | Dynamic Pass Box (DPB) | NA | Sterilization Room | Cleanroom Equipment | Pre-Viral | Fixed | 250 | NA | 1200 L x 1000 W x 2100 H | Clean air, Fabtech

123 | Autoclave Double Door 1000L | TBD | Sterilization Room | Equipment | Pre-Viral | Fixed | 2500 | NA | 1200 L x 1000 W x 2100 H | Pharma Lab, Indu IonPure

126 | Ceiling LAF | NA | Cooling zone | Equipment | Pre-Viral | Fixed Under Ceiling | 100 | NA | 2500 L x 2000 W x 1000 H | Clean air, Fabtech

127 | Dynamic Pass Box (DPB) | NA | Corridor | Equipment | Pre-Viral | Fixed | 250 | NA | 1200 L x 1000 W x 2100 H | Clean air, Fabtech

128 | PV 114 | 250 | 6I/11F | Process Vessel | Post Viral | Fixed | 340 | 590 | 2100L X 2000 X 2600H | Praj and Aaruni

129 | PV 115 | 1000 | 6I/11F | Process Vessel | Post Viral | Fixed | 650 | 1650 | 2700L X 2700W X 3200H | Praj and Aaruni

130 | PV 116 | 1000 | 6I/11F | Process Vessel | Post Viral | Fixed | 650 | 1650 | 2700L X 2700W X 3200H | Praj and Aaruni

131 | IG-RR-09 | 100 | 6I/11F | Buffer | Post Viral | Mobile | 220 | 370 | 1800L X 1900W X 2800H | Praj and Aaruni

132 | IG-RR-14 | 500 | 6I/11F | Buffer | Post Viral | Fixed | 450 | 950 | 2200L X 2200 W X 2800H | Praj and Aaruni

133 | IG-RR-16 | 500 | 6I/11F | Buffer | Post Viral | Fixed | 450 | 950 | 2200L X 2200 W X 2800H | Praj and Aaruni

134 | IG-RR-18 | 500 | 6I/11F | Buffer | Post Viral | Fixed | 450 | 950 | 2200L X 2200 W X 2800H | Praj and Aaruni

135 | IG-RR-22 | 250 | 6I/11F | Buffer | Post Viral | Fixed | 340 | 590 | 2100L X 2000 X 2600H | Praj and Aaruni

136 | PV 117 | 150 | 6I/11F | Process Vessel | Post Viral | Mobile | 250 | 430 | 1900L X 2000W X 2800H | Praj and Aaruni

137 | IG-RR-26 | 150 | 6I/11F | Buffer | Post Viral | Mobile | 250 | 430 | 1900L X 2000W X 2800H | Praj and Aaruni

138 | IG-RR-28 | 500 | 6I/11F | Buffer | Post Viral | Fixed | 450 | 950 | 2200L X 2200 W X 2800H | Praj and Aaruni

139 | IG-RR-23 | 150 | 6I/11F | Buffer | Post Viral | Mobile | 250 | 430 | 1900L X 2000W X 2800H | Praj and Aaruni

140 | IG-RR-24 | 150 | 6I/11F | Buffer | Post Viral | Mobile | 250 | 430 | 1900L X 2000W X 2800H | Praj and Aaruni

141 | FG-RR-09 | 1000 | 6I/11F | Buffer | Post Viral | Fixed | 650 | 1650 | 2700L X 2700W X 3200H | Praj and Aaruni

142 | FG-RR-10 | 2000 | 6I/11F | Buffer | Post Viral | Fixed | 1200 | 3200 | 3600L X 3600W X 4300H | Praj and Aaruni

143 | FG-RR-11 | 500 | 6I/11F | Buffer | Post Viral | Fixed | 450 | 950 | 2200L X 2200 W X 2800H | Praj and Aaruni

144 | FG-RR-13 | 100 | 6I/11F | Buffer | Post Viral | Mobile | 220 | 370 | 2100L X 2000 X 2600H | Praj and Aaruni

144 | FG-RR-14 | 250 | 6I/11F | Buffer | Post Viral | Fixed | 340 | 590 | 2100L X 2000 X 2600H | Praj and Aaruni

145 | Akta-1" -667 L/Hr | NA | 6I/11F | Equipment | Post Viral | Fixed | 2000 | NA | 2200L x 3200 W x 3000H | TruKing, Cytiva, Aaruni

146 | Chromatography Column 1000X 500 | NA | 6I/11F | Equipment | Post Viral | Fixed | 400 | 600 | 1500L X 1500 W X 2700H | TruKing, Cytiva, Aaruni

147 | TFF | NA | 6I/11F | Equipment | Post Viral | Fixed | 2000 | NA | 2500L X 2000 W X 2700H | Praj, Cytiva, Sartorius, Aaruni

148 | Nano Filter | NA | 6I/11F | Single use | Post Viral | Fixed | 20 | NA | 500L x 500 W x 1200 H | Planova

149 | Mobile CIP | NA | 6I/11F | Equipment | Post Viral | Mobile | 150 | 200 | 2000L X 2000 X 2000H | Praj & DDE

150 | FN-RR-15 | 500 | 9N | Buffer | Post Viral | Fixed | 450 | 950 | 2200L X 2200 W X 2800H | Praj and Aaruni

151 | FN-RR-18 | 100 | 9N | Buffer | Post Viral | Mobile | 220 | 370 | 1800L X 1900W X 2800H | Praj and Aaruni

152 | FN-RR-17 | 100 | 9N | Buffer | Post Viral | Mobile | 220 | 370 | 1800L X 1900W X 2800H | Praj and Aaruni

153 | Akta-1" -30 L/Hr | NA | 9N | Equipment | Post Viral | Fixed | 200 | NA | 2200L x 1800 W x 2000H | TruKing, Cytiva, Aaruni

154 | Chromatography Column-300 X 750 | NA | 9N | Equipment | Post Viral | Fixed | 80 | 200 | 500L x 500 W x 2000 H | TruKing, Cytiva, Aaruni

155 | Mobile CIP | NA | 9N | Equipment | Post Viral | Mobile | 150 | 200 | 2000L X 2000 X 1800H | Praj & DDE

156 | PV 131 | 300 | 10T/13E | Process Vessel | Post Viral | Fixed | 390 | 690 | 2100L X 2000 X 2700H | Praj and Aaruni

157 | PV 131.1 | 200 | 10T/13E | Process Vessel | Post Viral | Mobile | 300 | 520 | 2100L X 2000 X 2600H | Praj and Aaruni

158 | PV 132 | 250 | 10T/13E | Process Vessel | Post Viral | Fixed | 340 | 590 | 2100L X 2000 X 2600H | Praj and Aaruni

159 | TH-RR-05 | 50 | 10T/13E | Buffer | Post Viral | Mobile | 70 | 120 | 1800L X 1800W X 2800H | Praj and Aaruni

160 | TH-RR-02 | 100 | 10T/13E | Buffer | Post Viral | Mobile | 220 | 370 | 1800L X 1900W X 2800H | Praj and Aaruni

161 | TH-RR-02 | 100 | 10T/13E | Buffer | Post Viral | Mobile | 220 | 370 | 1800L X 1900W X 2800H | Praj and Aaruni

162 | TH-RR-03 | 200 | 10T/13E | Buffer | Post Viral | Mobile | 300 | 520 | 2100L X 2000 X 2600H | Praj and Aaruni

163 | NALGENE | 50 | 10T/13E | Bottle | Post Viral | NA | 20 | NA | 500L x 500 W x 1200 H | Praj and Aaruni

164 | PV 132 | 250 | 10T/13E | Process Vessel | Post Viral | Fixed | 340 | 590 | 2100L X 2000 X 2600H | Praj and Aaruni

165 | PV 128 | 300 | 10T/13E | Process Vessel | Post Viral | Fixed | 390 | 690 | 2100L X 2000 X 2700H | Praj and Aaruni

166 | FE-RR-09 | 250 | 10T/13E | Buffer | Post Viral | Fixed | 340 | 590 | 2100L X 2000 X 2600H | Praj and Aaruni

167 | FE-RR-09 | 250 | 10T/13E | Buffer | Post Viral | Fixed | 340 | 590 | 2100L X 2000 X 2600H | Praj and Aaruni

168 | Akta-1" -160 L/Hr | NA | 10T/13E | Equipment | Post Viral | Fixed | 200 | NA | 2200L x 1800 W x 2000H | Truking, Cytiva, Aaruni

169 | Chromatography Column-450 X 500 | NA | 10T/13E | Equipment | Post Viral | Fixed | 100 | 250 | 500L x 500 W x 2000 H | Truking, Cytiva, Aaruni

170 | Mobile CIP | NA | 10T/13E | Equipment | Post Viral | Mobile | 150 | 200 | 2000L X 2000 X 2000H | Praj & DDE

171 | Centralised CIP - 2 tank system | TBD | CIP/SIP Room | Equipment | Post Viral | Fixed | 1000 | NA | 2200L x 3200 W x 3000H | Praj and Aaruni

173 | Decontamination Autoclave 300L | TBD | Decontamination Room | Equipment | Post Viral | Fixed | 300 | NA | 500L x 500 W x 1200 H | Pharma Lab, Indu IonPure

174 | Dynamic Pass Box (DPB) | NA | Decontamination Room | Cleanroom Equipment | Post Viral | Fixed | 250 | NA | 1200 L x 1000 W x 2100 H | Clean air, Fabtech

175 | Dynamic Pass Box (DPB) | NA | Decontamination Room | Cleanroom Equipment | Post Viral | Fixed | 250 | NA | 1200 L x 1000 W x 2100 H | Clean air, Fabtech

176 | Dynamic Pass Box (DPB) | NA | Sterilization Room | Cleanroom Equipment | Post Viral | Fixed | 250 | NA | 1200 L x 1000 W x 2100 H | Clean air, Fabtech

177 | Sterilization Autoclave 1000L | TBD | Sterilization Room | Equipment | Post Viral | Fixed | 2500 | NA | 1200 L x 1000 W x 2100 H | Pharma Lab, Indu IonPure

178 | Ceiling LAF | NA | Cooling zone | Equipment | Post Viral | Fixed Under Ceiling | 100 | NA | 2500 L x 2000 W x 1000 H | Clean air, Fabtech

179 | Buffer Preparation | 1500 | Buffer Preparation Room | Process Vessel | Post Viral | TBD | 900 | 2400 | 3200L X 3500 W X 4200H | Praj and Aaruni

180 | Buffer Preparation | 1000 | Buffer Preparation Room | Process Vessel | Post Viral | TBD | 650 | 1650 | 2700L X 2700W X 3200H | Praj and Aaruni

181 | Static Pass box | NA | Buffer Preparation Room | Equipment | Post Viral | NA | 50 | NA | 1200 L x 1000 W x 2100 H | Clean air, Fabtech

182 | Static Pass box | NA | Chemical Day Store | Equipment | Post Viral | NA | 50 | NA | 1200 L x 1000 W x 2100 H | Clean air, Fabtech

[SHEET: Fill Finish(F)]

Sr. No. | Vessel Tag | Room/Stage (As per Layout) | Type | Fill Finish | Fixed / Mobile | Gross Weight in Kg | Full Weight in Kg | Dimension (L mm x W mm x H mm) | Make |

1 | DPB | De-Carbonning Room | Equipment | Plasma | Fixed | 250 | NA | 1200 L x 1000 W x 2100 H | Clean air, Fabtech |

2 | Vial washing | Vial Washing Room | Vial Washing Machine | Plasma | Fixed | 2800 | NA | 3500 L X 1800 W X 2100 H | NKP, Harikrushna |

3 | DPB | Vial Washing Room | Equipment | Plasma | Fixed | 250 | NA | 1200 L x 1000 W x 2100 H | Clean air, Fabtech |

4 | Autoclave Double Door 1000L | Material Prep & Wash | Equipment | Plasma | Fixed | 2500 | NA | 1200 L x 1000 W x 2100 H | Pharma Lab, Indu IonPure |

5 | DPB | Material Prep & Wash | Equipment | Plasma | Fixed | 250 | NA | 1200 L x 1000 W x 2100 H | Clean air, Fabtech |

6 | Decont Autoclave 300L | Decontamination | Equipment | Plasma | Fixed | 300 | NA | 1200 L x 800 W x 1400 H | Pharma Lab, Indu IonPure |

7 | Washing machine 25Kg | Garment Wash | Washing Machine | Plasma | Mobile | 80 | NA | 1200 L x 1300 W x 1700 H | IFB |

8 | Dryer 25Kg | Garment Wash | Drying machine | Plasma | Mobile | 80 | NA | 1800 L x 1500 W x 2600 H | IFB |

9 | Ceiling LAF | Unload Area | Equipment | Plasma | Fixed under Ceiling | 100 | NA | 2500 L x 2000 W x 1000 H | Clean air, Fabtech |

10 | Empty/ Filled Vessel- 500L | Formulation | Formulation Vessel | Plasma | Fixed | 400 | 900 | 2200L X 2200 W X 2800H | Praj and Aaruni |

11 | Empty/ Filled Vessel- 500L | Formulation | Formulation Vessel | Plasma | Fixed | 400 | 900 | 2200L X 2200 W X 2800H | Praj and Aaruni |

12 | Ceiling LAF | Formulation | Equipment | Plasma | Fixed under Ceiling | 100 | NA | 2500 L x 2000 W x 1000 H | Clean air, Fabtech |

13 | Filling Machine | Filling Area | Filling machine | Plasma | Fixed | 2200 | NA | 3200 L X 1700 W X 2300 H | NKP, Harikrushna |

14 | Ceiling LAF | Filling Area | Equipment | Plasma | Fixed under Ceiling | 100 | NA | 2500 L x 2000 W x 1000 H | Clean air, Fabtech |

15 | Ceiling LAF | Capping Room | Equipment | Plasma | Fixed under Ceiling | 100 | NA | 2500 L x 2000 W x 1000 H | Clean air, Fabtech |

16 | Capping Machine | Capping Room | Capping Machine | Plasma | Fixed | 600 | NA | 2200 L X 1500 W X 2000 H | NKP, Harikrushna |

17 | Turn Table | Capping Room | SS Turn Table | Plasma | Fixed | 150 | NA | 1200 L X 1300 W X 900 H | NKP, Harikrushna |

18 | Swing Conveyor | Capping Room | Swing Conveyor With Vial Collection Tray | Plasma | Fixed | 220 | NA | 2500 L X 1200 W X 900 H | NKP, Harikrushna |

19 | Lyo Machine 3.3msq | Lyo Loading Side | Lyo machine | Plasma | Fixed | 7500 | NA | 4200 L X 2200 W X 2400 H | LSI |

20 | DPB | Buffer Room | Equipment | Plasma | Fixed | 250 | NA | 1200 L x 1000 W x 2100 H | Clean air, Fabtech |

21 | Visual Inspection Booth | Visual Inspection Room | Visual Inspection Booth | Plasma | Fixed | 250 | NA | 3000 L X 1800 W X 2400 H | NKP, Harikrushna |

22 | Labelling machine | Secondary Packing Area | Labeling Machine | Plasma | Fixed | 600 | NA | 3000 L X 1800 W X 2000 H | NKP, Harikrushna |

23 | Cartoning machine | Secondary Packing Area | Cartoning Machine | Plasma | Fixed | 200 | NA | 6000 L X 2000 W X 2100 H | NKP, Harikrushna |

24 | Bulk | Bulk staging | 3D Bag/Tote | Plasma | Mobile | 100 | 1500 | 1000 L x 800 x 1000 H | Sartorius |

15 | freezer-1000L | 1000 L | Plasma Storage Area | Freezer | Warehouse | Fixed | 425 | 1625 | 1400 L X 950 W X 2100 H | Newtronic lifecare, REMI

16 | freezer-1000L | 1000 L | Plasma Storage Area | Freezer | Warehouse | Fixed | 425 | 1625 | 1400 L X 950 W X 2100 H | Newtronic lifecare, REMI

17 | freezer-1000L | 1000 L | Plasma Storage Area | Freezer | Warehouse | Fixed | 425 | 1625 | 1400 L X 950 W X 2100 H | Newtronic lifecare, REMI

18 | freezer-1000L | 1000 L | Plasma Storage Area | Freezer | Warehouse | Fixed | 425 | 1625 | 1400 L X

950 W X 2100 H | Newtronic lifecare, REMI
19 | freezer-1000L | 1000 L | Plasma Storage Area | Freezer | Warehouse | Fixed | 425 | 1625 | 1400 L X
950 W X 2100 H | Newtronic lifecare, REMI
20 | freezer-1000L | 1000 L | Plasma Storage Area | Freezer | Warehouse | Fixed | 425 | 1625 | 1400 L X
950 W X 2100 H | Newtronic lifecare, REMI
21 | freezer-1000L | 1000 L | Plasma Storage Area | Freezer | Warehouse | Fixed | 425 | 1625 | 1400 L X
950 W X 2100 H | Newtronic lifecare, REMI
22 | freezer-1000L | 1000 L | Plasma Storage Area | Freezer | Warehouse | Fixed | 425 | 1625 | 1400 L X
950 W X 2100 H | Newtronic lifecare, REMI
23 | freezer-1000L | 1000 L | Plasma Storage Area | Freezer | Warehouse | Fixed | 425 | 1625 | 1400 L X
950 W X 2100 H | Newtronic lifecare, REMI
24 | freezer-1000L | 1000 L | Plasma Storage Area | Freezer | Warehouse | Fixed | 425 | 1625 | 1400 L X
950 W X 2100 H | Newtronic lifecare, REMI
| | | | | Empty | Full | | |
24 | PW Generation(MGF, Softener, Soft Water Storage tank, RO and RO water storage tank) | Water System & Engineering Store | Water System & Utilities | Utilities equipment | Fixed | 500 | NA | | Praj, CN water
|
25 | PW storage 20,000L | Water System & Engineering Store | Water System & Utilities | Utilities equipment | Fixed | 2500 | 22500 | | Praj, CN water |
26 | WFI Generation and storage 20000L | Water System & Engineering Store | Water System & Utilities | Utilities equipment | Fixed | 1100 | 21100 | | Praj, CN water |
27 | PSG System 1000Kg | Water System & Engineering Store | Water System & Utilities | Utilities equipment | Fixed | 760 | NA | | Praj, CN water |
28 | Compressed Air System 300 CFM | Water System & Engineering Store | Water System & Utilities | Utilities equipment | Fixed | 120 | 125 | | Atlas copco |
29 | AHUs | Service Area | HVAC | Utilities equipment | Fixed | NA | NA | | Clean air, Fabtech |
30 | Electrical Control Panel | Service Area | Electrical | Utilities equipment | Fixed | NA | NA | | Clean air, Fabtech |

Reagent Details:
Formula & Calculations:
Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).
Calculation: To prepare 1 L of 2mM Tri-sodium citrate (MW = 258.06g/mol)
Weight of solute(g)=0.002×258.06=0.516g.
Preparation Method:
Weigh 0.516 g of Tri-sodium citrate and dissolve in water to make a 1 L solution.
Preparation for different volumes:
To prepare _____ of 2mM Tri-sodium citrate (MW = 258.06g/mol).
Weight of solute(g)=0.002×258.06x _____L= _____gm
Weigh _____gm of Tri-sodium citrate and dissolve in water to make a _____ L solution.
Check and record the details
Preparation Details:
Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry Date
Tri-sodium citrate | | | |
Final volume prepared | Weighed Quantity | pH(7.0±0.1) | Conductivity(0.650 ± 0.1 µs/cm) | Storage Condition:
| | | RT
Date of Preparation | Prepared By | Checked By (QC/Analyst)
| |

Reagent Details:

Formula:

Sodium Acetate(M.Wt): 82.03g/mol.

Acetic Acid (M.Wt): 60.05g/ mol

Sodium Chloride:58.44g/mol

Preparation of 0.2 M Sodium Acetate pH 5.35 buffer

Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).

Calculation: To prepare 1 L of 0.2 M Sodium Acetate (MW = 82.03g/mol).

Weight of solute(g)=0.2×82.03x 1.0=16.406 g.

Preparation Method:

Weigh 16.406 g of Sodium Acetate and dissolve in water to make a 1 L solution.

Preparation for required volumes:

To prepare _____ of 0.2M Sodium Acetate (MW = 82.03g/mol).

Weight of solute(g)=0.2×82.03x _____ L= _____ g.

Weigh _____ gm of Sodium Acetate and dissolve in water to make a _____ L solution.

Calculations: To prepare the 0.2 M Acetic Acid.

Formula & Calculations:

Formula: Use the formula to calculate the molarity:

Molarity (M) = (Per cent by Weight / 100) * (Density in g/mL) * 1000 / Molar Mass.

Example: Molarity of Glacial Acetic Acid

Molar Mass of CH₃COOH: ~60.05 g/mol

Density: ~1.05 g/mL

Purity: 100%

Acetic Acid (MW = 60.05 g/mol).

Calculation:

Molarity = (100 / 100) * 1.05 g/mL * 1000 mL/L / 60.05 g/mol ≈ 17.5 M.

Calculation: To prepare 1 L of 0.2 M acetic acid

Applying the formula to prepare the working buffer:

C₁V₁=C₂V₂

C₁ = 17.5 M acetic acid.

V₁ = volume of stock needed to take.

C₂ = desired (working) concentration 0.2 M acetic acid.

V₂ = final total volume of working solution for one litre.

V₁ = (C₂ × V₂)/C₁.

C₂=0.2 M ; V₂= 1.0 L; C₁=17.5 M

V₁ = (0.2 × 1.0)/17.5

V₁ = 0.011 L

Take 0.011 L of the Stock solution of acetic acid and add it to a volumetric flask, making up the volume to 1 L.

To prepare various volumes:

V₁ = volume of stock needed to take _____ L.

C₂ = desired (working) concentration 0.2 M acetic acid.

V₂ = final total volume of working solution _____ L.

V₁ = (C₂ _____ x V₂ _____) /17.5.

V₁= _____ L.

Take V₁ _____ L of the Stock solution of acetic acid and add it to a volumetric flask, making up the final volume _____ L.

Preparation of 4.0 M Sodium Chloride.

Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).

Calculation: To prepare 1 L of 4 M Sodium Chloride (MW = 58.44g/mol).

Weight of solute(g)=4×58.44 x 1.0= 233.76g.

Preparation Method:

Weigh 233.76g of Sodium Chloride and dissolve in water to make a 1 L solution.

Preparation for required volumes:

To prepare _____ of 4 M Sodium Chloride (MW = 58.44g/mol).

Weight of solute(g)=0.2×58.44x _____ L= _____ g.

Weigh _____ gm of Sodium Chloride and dissolve in water to make a _____ L solution.

Procedure for Preparation of 0.2 M Sodium Acetate + 4 M Sodium Chloride pH 5.35

To prepare a pH 5.35 solution, 0.2 M Sodium Acetate solution was added, to then 4M Sodium Chloride was added and completely dissolved and then titrated with 0.2 M Acetic acid so that the pH attains pH 5.35 and made up to 1 L.

Check and record the details:

Preparation Details:

Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry

Date

Sodium Acetate | | | |

Acetic Acid | | | |

Final volume prepared | Measured Quantity | pH | Conductivity | Storage Condition | Reagent Record

| | | |

Stock buffer (AL-RR-10) taken volume(L) | pH

(5.35 ±0.1) | Conductivity

(180±5ms/cm) | Storage Condition | Final volume prepared

| | | RT |

Date of Preparation | Prepared By | Checked By

| |

Reagent Details:

Formula:

Sodium Citrate (M.Wt): 258.07.

Citric acid (M.Wt): 192.124

Sodium Chloride (M.Wt): 58.44

Calcium Chloride (M.Wt): 110.98

0.1% Tween 80

Preparation of FE-RR-04 buffer with Stock FE-RR-03.

Preparation of 20mM sodium citrate pH 7.0 + 1.3M sodium chloride + 2.6mM calcium chloride + 0.1% tween 80.

Stock Buffer (FE-RR-03): 0.2M Sodium Citrate pH 7.0

Formula:

$$C1 \times V1 = C2 \times V2$$

C1 = concentration of stock

V1 = volume of stock you need to take

C2 = desired (working) concentration

V2 = final total volume of working solution

Applying the formula to prepare the working buffer:

$$C1 \times V1 = C2 \times V2$$

C1 = 0.2M Sodium Citrate pH 7.0

V1 = volume of stock you need to take _____ L.

C2 = desired (working) concentration 20mM Sodium Citrate, pH 7.0

V2 = final total volume of working solution _____ L.

$$V1 = (C2 \times V2) / C1$$

C2 = 0.02M ; V2 = 1.0 L ; C1 = 0.2M

$$V1 = (0.02 \times 1.0) / 0.2$$

V1 = 0.1L

Take 0.1 L (FE-RR-03) Stock buffer solution and dilute with water, and make up to 1L.

Calculations: To prepare the 1.3M Sodium Chloride.

Formula: Weight of solute (g) = Molarity (M) × Molecular Weight (g/mol) × Volume (L).

Calculation: To prepare 1 L of 1.3M Sodium Chloride (MW = 58.44 g/mol)

$$\text{Weight of solute (g)} = 1.3 \times 58.44 \times 1 = 75.972\text{g}$$

Preparation Method:

Weigh 75.972g of Sodium Chloride and dissolve in water to make a 1 L solution.

Preparation for different volumes:

To prepare _____ L of 1.3M Sodium Chloride (MW = 58.44 g/mol).

$$\text{Weight of solute (g)} = 1.3 \times 58.44 \times \text{L} = \text{g}$$

Weigh _____ gm of Sodium Chloride and dissolve in water to make a _____ L solution.

Calculations: To prepare the 2.6mM Calcium Chloride.

Formula: Weight of solute (g) = Molarity (M) × Molecular Weight (g/mol) × Volume (L).

Calculation: To prepare 1 L of 2.6 mM Calcium Chloride (MW = 110.98g/mol)

$$\text{Weight of solute (g)} = 0.0026 \times 110.98 \times 1 = 0.288\text{g}$$

Preparation Method:

Weigh 0.288g of Calcium Chloride and dissolve in water to make a 1 L solution.

Preparation for different volumes:

To prepare _____ L of 0.0026mM Calcium Chloride (MW = 110.98 g/mol).

$$\text{Weight of solute (g)} = 0.0026 \times 110.98 \times \text{L} = \text{g}$$

Weigh _____ gm of Calcium Chloride and dissolve in water to make a _____ L solution.

Procedure

Calculation: to prepare the 0.1% Tween 80:

For 1000 mL of 0.1% solution:

$$0.1\% = 0.1\text{g per } 1000\text{ mL} = 1000\text{ mg}$$

Since Tween 80 is a liquid, you can weigh it directly or use its density (~1.06 g/mL) to measure volume:

$$\text{Approx. } 100\text{ }\mu\text{L of Tween 80} = 1.0\text{ g}$$

Add Tween 80 to a beaker or flask containing ~800 mL of water.

Mix thoroughly:

Use a magnetic stirrer or vortex until fully dissolved. Tween 80 is viscous, so mixing may take a few minutes.

Adjust volume:

After complete dissolution, bring the final volume up to 1000 mL with water.

Procedure for Preparation of FE-RR-04 buffer with the stock FE-RR-03.

To prepare a 1 L solution, take 0.2M Tri Sodium Citrate pH 7.0. To this, add the 1.3 M sodium chloride

and dissolve it and make up the volume with water to 900ml, and then to this add 0.026 M Calcium

chloride, to this add the 0.1% Tween 80 then mix the solution till it dissolves and check the pH; if the

pH is less than 7.0 then adjust the pH with 1 M sodium hydroxide to get the pH 7.0 ± 0.1 and make up the

final volume to 1L.

Check and record the details:

Preparation Details:

Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry

Date

Sodium Citrate | | | |

Citric Acid | | | |

Sodium Chloride | | | |

Calcium Chloride | | | |

Tween 80 | | | |

Stock buffer (PF-RR-31) taken volume(L) | pH (7.0 ± 0.1) | Conductivity (135 ± 5.0ms/cm) | Storage

Condition | Final volume prepared

| | | RT |

Date of Preparation | Prepared By | Checked By

| |

ZLD system for LVP plant
(180 m³/day water demand, 145 m³/day effluent)

1. Design Basis (Confirmed)
Effluent Load
Total effluent: 145 m³/day
Operating hours: 20 hr/day
Hourly flow = 145 / 20 = 7.25 m³/hr

2. Effluent Segregation (Critical for Cost Optimization)

3. Overall ZLD Scheme
Equalization → ETP → UF → RO → Permeate reuse
↓
RO Reject → MEE → ATFD → solids

4. RO SYSTEM SIZING

4.1 Feed to RO
Total after ETP/UF: 145 m³/day
Hourly: 7.25 m³/hr

4.2 Design Assumptions
Recovery: 70%
Flux: 18–22 LMH (industrial pharma)
Operating: 20 hr/day
Safety margin: 25%

4.3 Flow Calculations
Permeate:
70% of 145 = 101.5 m³/day
Reject:
43.5 m³/day

4.4 Design Capacity
Required:
7.25 m³/hr
With margin:
~9 m³/hr RO system

4.5 Membrane Area
Assume flux = 20 LMH
Area:
9,000 L/hr ÷ 20 = 450 m² membrane area

4.6 RO Configuration
2-stage RO (70% recovery)
Skid size: ~9 m³/hr
Pumps: ~12–15 kW

5. MEE (MULTIPLE EFFECT EVAPORATOR) SIZING

5.1 Feed to MEE
RO reject: 43.5 m³/day
Hourly: 2.2 m³/hr

5.2 Concentration Target
From ~5–8% solids → 25–30% slurry

5.3 Evaporation Load
Water to evaporate (~80%):
~35 m³/day
Hourly evaporation:
35 ÷ 20 = 1.75 m³/hr

5.4 MEE Capacity
Add margin:
Design MEE capacity: 2.0 m³/hr evaporation

5.5 Number of Effects
Recommended: 3-effect MEE
Reason:
Better steam economy
Lower OPEX

5.6 Steam Requirement
Typical:
1 kg steam evaporates ~1.8–2.2 kg water (3-effect)
Steam needed:
1,750 kg/hr ÷ 2 ≈ 875 kg/hr steam
Design:
~900–1,000 kg/hr steam

5.7 Power Requirement
Pumps + vacuum system: 25–35 kW

6. ATFD (AGITATED THIN FILM DRYER) SIZING

6.1 Feed to ATFD
Concentrated slurry: ~8.5 m³/day

6.2 Solids Load
Assume:
25% solids
Solids:
~2,100 kg/day

6.3 Hourly Load
Operating: 16 hr/day
Solids/hr:
2100 ÷ 16 ≈ 130 kg/hr

6.4 ATFD Capacity
With margin:
Design capacity: 150–180 kg/hr solids

6.5 Steam Requirement
~1.2–1.5 kg steam/kg evaporation
~200–250 kg/hr steam

6.6 Power
Motor + scraper: 15–20 kW

7. FINAL WATER BALANCE (ZLD PERFORMANCE)

8. EQUIPMENT SIZING SUMMARY (DPR TABLE)

9. UTILITIES REQUIREMENT

Power
Total: ~80–100 kW
Steam
Total: ~1.2 TPH boiler capacity
Stream | Flow (m³/day) | TDS | Treatment Path
Low TDS | 70 | <2,000 ppm | ETP → RO
High TDS | 75 | 5,000–15,000 ppm | ETP → RO → MEE → ATFD
Stage | Flow (m³/day)
RO permeate | 101.5
MEE condensate | 32–35
Total recovery | ~135 m³/day
Losses (sludge + evaporation) | ~10 m³/day
Liquid discharge | ZERO
Unit | Capacity
Equalization Tank | 200 m³
ETP | 150 m³/day
UF System | 10 m³/hr
RO System | 9 m³/hr
MEE | 2.0 m³/hr evaporation
ATFD | 150–180 kg/hr solids
System | Load
RO | 15 kW
MEE | 30 kW
ATFD | 20 kW
Pumps/ETP | 20 kW
System | Steam
MEE | 900–1,000 kg/hr
ATFD | 200–250 kg/hr

Reagent Details:
Formula & Calculations:
Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).
Calculation: To prepare 1 L of 0.01N Sodium Hydroxide (MW = 40.0 g/mol)
Weight of solute(g)=0.01×40 = 0.4g.
Preparation Method:
Weigh 0.4 g of Sodium Hydroxide and dissolve in water to make a 1 L solution.
Preparation for different volumes:
To prepare _____ of 0.01N Sodium Hydroxide (MW = 40.0 g/mol).
Weight of solute(g)=0.01×40.0 x _____L= _____gm
Weigh _____gm of Sodium Hydroxide and dissolve in water to make a _____ L solution.
Check and record the details:
Preparation Details:
Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry
Date
Sodium Hydroxide | | | |
Final volume prepared | Weighed Quantity | pH (12.5 ±0.5) | Conductivity (2.5±0.5 ms/cm) | Storage
Condition:
| | | RT
Date of Preparation | Prepared By | Checked By
| |

Reagent Details:

Formula:

Sodium Acetate(M.Wt): 82.03g/mol.

Acetic Acid (M.Wt): 60.05g/ mol

Preparation of 0.4 M Sodium Acetate pH 4.6 buffer

Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).

Calculation: To prepare 1 L of 0.4 M Sodium Acetate (MW = 82.03g/mol).

Weight of solute(g)=0.4×82.03× 1.0=32.81g.

Preparation Method:

Weigh 32.81 g of Sodium Acetate and dissolve in water to make a 1 L solution.

Preparation for required volumes:

To prepare _____ of 0.4 M Sodium Acetate (MW = 82.03g/mol).

Weight of solute(g)=0.4×82.03× _____ L= _____ g.

Weigh _____ gm of Sodium Acetate and dissolve in water to make a _____ L solution.

Calculations: To prepare the 0.4 M Acetic Acid.

Formula & Calculations:

Formula: Use the formula to calculate the molarity:

Molarity (M) = (Per cent by Weight / 100) * (Density in g/mL) * 1000 / Molar Mass.

Example: Molarity of Glacial Acetic Acid

Molar Mass of CH₃COOH: ~60.05 g/mol

Density: ~1.05 g/mL

Purity: 100%

Acetic Acid (MW = 60.05 g/mol).

Calculation:

Molarity = (100 / 100) * 1.05 g/mL * 1000 mL/L / 60.05 g/mol ≈ 17.5 M.

Calculation: To prepare 1 L of 0.4 M.

Applying the formula to prepare the working buffer:

C₁V₁=C₂V₂

C₁ = 17.5 M acetic acid.

V₁ = volume of stock needed to take.

C₂ = desired (working) concentration 0.4 M acetic acid.

V₂ = final total volume of working solution for one litre.

V₁ = (C₂ x V₂)/C₁.

C₂=0.4 M ; V₂= 1.0 L; C₁=17.5 M

V₁ = (0.4 x 1.0)/17.5

V₁ = 0.022 L

Take 0.022 L of the Stock solution of acetic acid and add it to a volumetric flask, making up the volume to 1 L.

To prepare various volumes:

V₁ = volume of stock needed to take _____ L.

C₂ = desired (working) concentration 0.4 M acetic acid.

V₂ = final total volume of working solution _____ L.

V₁ = (C₂ x V₂) / 17.5.

V₁= _____ L.

Take V₁ _____ L of the Stock solution of acetic acid and add it to a volumetric flask, making up the final volume _____ L.

Procedure for Preparation of 0.4 M Sodium Acetate pH 4.6.

To prepare a pH 4.6 solution, 0.4 M Sodium Acetate solution is titrated with 0.4 M Acetic acid so that the pH attains pH 4.6.

Check and record the details:

Preparation Details:

Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry Date

Sodium Acetate | | | |

Acetic Acid | | | |

Final volume prepared | Measured Quantity | pH (4.6± 0.1) | Conductivity (24.0 ± 0.1ms/cm) | Storage Condition | Reagent Record

| | | RT |

Date of Preparation | Prepared By | Checked By

| |

Reagent Details:
Preparation of PF-RR-28 working buffer from Stock buffer (PF-RR-27):
Calculations: For the Preparation of 0.5M Sodium Hydroxide.
Stock Buffer (PF-RR-27):2.0M Sodium Hydroxide,
Formula:
 $C1 \times V1 = C2 \times V2$
C1 = concentration of stock
V1 = volume of stock you need to take
C2 = desired (working) concentration
V2 = final total volume of working solution
Applying the formula to prepare the working buffer:
 $C1 \times V1 = C2 \times V2$
C1 = 2.0 M Sodium Hydroxide.
V1 = volume of stock you need to take _____ L.
C2 = desired (working) concentration 0.5M Sodium Hydroxide.
V2 = final total volume of working solution _____ L.
 $V1 = (C2 \times V2) / C1$.
C2=0.5M; V2= 1.0 L; C1=2.0M
 $V1 = (0.5 \times 1.0) / 2.0$
V1 = 0.25L
Take 0.25 L (PF-RR-27) Stock buffer solution and dilute with water, and make up to 1L.
Check and record the details:
Preparation Details:
Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry
Date
Sodium Hydroxide | | | |
Stock buffer (PF-RR-27) taken volume(L) | pH (13.0± 1.0) | Conductivity (250.0± 3.0ms/cm) | Storage
Condition | Final volume prepared
| | RT |
Date of Preparation | Prepared By | Checked By
| |

Reagent Details:
Preparation of AL-RR-11 working buffer from Stock buffer (AL-RR-10):
Calculations: For the Preparation of 50mM Sodium acetate pH 5.2.
Stock Buffer (AL-RR-10):0.2M Sodium acetate, pH 5.35 + 4.0 M Sodium Chloride.
Formula:
 $C1 \times V1 = C2 \times V2$
C1 = concentration of stock
V1 = volume of stock you need to take
C2 = desired (working) concentration
V2 = final total volume of working solution
Applying the formula to prepare the working buffer:
 $C1 \times V1 = C2 \times V2$
C1 = 0.2M Sodium Acetate pH 5.35.
V1 = volume of stock you need to take _____ L.
C2 = desired (working) concentration 50mM Sodium Acetate pH 5.2
V2 = final total volume of working solution _____ L.
 $V1 = (C2 \times V2) / C1$
C2=0.05M; V2= 1.0 L; C1=0.2M
 $V1 = (0.05 \times 1.0) / 0.1$
V1 = 0.5L
Take 0.5 L (AL-RR-10) Stock buffer solution and dilute it with water to make up a final volume of 1 L.
Check and record the details:
Preparation Details:
Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry
Date
Sodium Acetate | | | |
Acetic Acid | | | |
Sodium Chloride | | | |
Stock buffer (AL-RR-10) taken volume(L) | pH (5.2±0.1) | Conductivity (100 ±5ms/cm) | Storage Condition
| Final volume prepared
| | RT |
Date of Preparation | Prepared By | Checked By
| |

Reagent Details:
Formula & Calculations:
Formula: Use the formula to calculate the molarity:
Molarity (M) = (Per cent by Weight / 100) * (Density in g/mL) * 1000 / Molar Mass.
Example: Molarity of Glacial Acetic Acid
Molar Mass of CH₃COOH: ~60.05 g/mol
Density: ~1.05 g/mL
Purity: 100%
Acetic Acid (MW = 60.05 g/mol).
Calculation:
Molarity = (100 / 100) * 1.05 g/mL * 1000 mL/L / 60.05 g/mol ≈ 17.5 M.
Calculation: To prepare 1 L of 0.1M.
Applying the formula to prepare the working buffer:
C₁V₁=C₂V₂
C₁ = 17.5 M acetic acid.
V₁ = volume of stock needed to take.
C₂ = desired (working) concentration 0.1M acetic acid.
V₂ = final total volume of working solution for one liter.
V₁ = (C₂ x V₂)/C₁.
C₂=0.1 M; V₂= 1.0 L; C₁=17.5 M
V₁ = (0.1 x 1.0)/17,5
V₁ = Take 0.0057 L of the Stock solution of acetic acid and add it to a volumetric flask, making up the volume to 1 L.
To prepare various volumes:
V₁ = volume of stock needed to take _____ L.
C₂ = desired (working) concentration 0.1M acetic acid.
V₂ = final total volume of working solution _____ L.
V₁ = (C₂ _____ x V₂ _____) /17.5.
V₁= _____ L.
Take V₁ _____ L of the Stock solution of acetic acid and add it to a volumetric flask, making up the final volume _____ L.
Check and record the details:
Preparation Details:
Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry Date
| | | |
Final volume prepared | Measured Quantity | pH (2.3±0.3) | Conductivity (4.3±0.3ms/cm) | Storage Condition | Reagent Record
| | | RT |
Date of Preparation | Prepared By | Checked By
| |

W W W W W W W W W

FIREMAN

FOLDING DOOR
CORRIDOR
Goods LIFT 53.5x2.77 LIFT
3.50X3.00 D1
6D
2.70X2.50
4D

D3
CH R3 CH R2
LOBBY
3.00X11.77 6.34 x 5.05 P.M RECEIVING LAF 3.72x2.60 3.72x2.60 LOBBY
3.00X11.77
FIRE STAIR
CORRIDOR FINISHED GOODS ROOM AREA up
SAMPLING

3D
OPEN TO 2.57x9.00 OPEN TO ALL
5.00x5.00
FIRST 24.6 x 8.95 ROOM D3
FLOORS
FLOOR 7.51x7.60

D3
(I.V.D.)
ONLY D4
EXIT CH R1
D1 D1
3.80x2.60 3.80x2.60
DPB

3D

3D
DE DUSTING

4D
MATERIAL

DIALYZER RECEIVING
2.94 M WIDE CORRIDOR

AREA
5.05x6.60 W
2.50 x 13.0
WC WC WC

BELOW

D4
1.10X 1.10X 1.10X
1.70 1.70 1.70

6.77 M WIDE CORRIDOR
REJECT
RECEIVING D1
19.8x30.3 AREA
WC WC WC 5.05x4.94
1.10X 1.10X 1.10X
1.70 1.70 1.70
W

D4
W W

FINISHED GOODS ROOM
LOBBY 24.6 x 10.9
6.77X11.77 QUARENTAINE RM STORE
D1

STAIRCASE
12.9x29.5
W
AND 7.80x11.6 W
LIFT ONLY
FOR
FIRST FL0OR
(IVD)
6D
PACKING MATERIAL STORE
18.6 x 15.9
11.3 x 27.9
Packing table

Packing table

3.40x3.46 3.40x3.46
PP GRANUAL SECOND FLOOR
AUTO CLAVE RM DISPENSING
ROOM
PRODUCTION
OFFICE & IPQA
TOOL ROOM
4.25x7.00
HOPPER LOADER
ROOM
3D
(ADMIN BLOCK)
4.05x7.00 2.92x7.00
ROOM
D3
4.11x7.00
CH R3 EXIT
XX

13.2x11.5 XX

XX

3.40x3.46 3.40x3.46

SPB LAF
4D 4D 4D
D3 D3
EMERGENCY EXIT
3D

CORRIDOR
D4

29.2x2.79
D4 PW U2
WFI U2

3D 4D 4D
12.3x6.07 DPB
D4

1.25x4.25 COOLING ZONE
4D

7.16x4.54 5.15X 2.00
LOBBY 9.03x4.54 3 WAY
12.47X6.77 6.92x1.52 SPB
AUTO CLAVE

13.7x6.06 Pass. LIFT
2.30X2.00
5.15X3.99
3D

PRODUCTION DPBEURO CAP
IPQA OFFICE AIR LOCK
OFFICE 1.76X1.20
STORE 3.04 x 1.44
3D
3.37x1.44 2.71x1.443D 3D
DPB D3

CORRIDOR
53.5x3.00

3D 3D 3D

ELECTRIC
D3

D3
D3

3.00X11.7 ROOMW
3.50X6.00
MALE PRIMARY PRIMARY
Goods LIFT CONFERENCE
HALL
KITCHEN
2.00X6.40
OFFICE 04 OFFICE 03 OFFICE 02 OFFICE 01 AIR LOCK MALE MALE RECEPTION
FEMALE
AIR LOCK FEMALE FEMALE VISITOR
PRIMARY
VISITOR
VISITOR
3.00X6.40 3.00X6.40 3.00X6.40 2.50X6.40 3.50X6.40 CHANGE CHANGE CHANGE
3.50X3.00 4.80X6.40 3.50X6.40 3.50X6.40 5.00X6.40 4.00X6.40
CHANGE
3.00X6.40

CHANGE
3.00X6.40
AIR LOCK
3.00X6.40 CHANGE
3.00X6.40
3.00X6.40

LOADING / UNLOADING
3D

3D
3D

D6
LIFT ONLY D3 D3 D3 D3 D3 D3 D3 D3
W
FOR
SERVER
SECOND FLOOR ROOM
CORRIDOR 3.50X6.57
7D

D7

53.5x3.27

W W W W W W W W W W

FIRST FLOOR PLAN BUILDING -A CLIENT :- LIFENITY- HEALTH LIMITED (DIALYZER, LVP & PLASMA)
ADDRESS :-Auric city, chhatrapati sambhaji nagar

SCALE 1:100

DESIGN &
PROJECT BY :-BIOBRIDGE HEALTHCARE SOLUTION PVT.LTD.
PREPARED A.K
ADDRESS :-13, Rachna Blossom, Jagdishnagar,
Aundh, Pune -411007, Maharastra, India. CHECKED G.K.

DATE 18-06-2026
TITLE- CONCEPTUAL LAYOUT-FF-A REVISION R0
SCALE 1:1

SIZE A0 DWG.NO 02

TABLE OF CONTENTS

POOLED CONCENTRATED SAMPLE 2

QC FOR IN-PROCESS2

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QC ANALYSIS REPORT.....3

Pooled Concentrated Sample:

Diafilter the pooled concentrated sample with 5 dia volumes of TH-RR-08 at a set TMP of ≤ 1.0 bar and concentrate the sample to 500 ± 100 mL.

Volume of TH-RR-08 used: _____.

pH: _____, Conductivity: _____ mS/cm.

O.D280/600: _____, Final volume (mL): _____.

Collect 2 mL of sample and send to QC for in-process QC analysis.

Record the obtained results as per the QC report :

Activity: _____, BET: _____.

Add and adjust Human Serum Albumin (HSA-20%) to a final concentration of 25 mg/mL to the diafiltered sample from BMR step 2.6.6

Volume of diafiltered sample: _____ mL.

Volume of HSA-20% added before concentration from BMR step 1.11.1: _____.

Volume of HSA-20% added after diafiltration: _____.

PA-SF-001FILTRATION

Assemble the PA-SF-001 filter with sterilised inlet and outlet silicone tubing.

Check and record the following filter details:

No. of filter(s) used: _____.

Equilibrate the filter by passing 0.5L TH-RR-08.

Volume of TH-RR-08 passed: _____ L.

Filter the sample through a filter using a peristaltic pump under LAFU.

Collect 10 mL sample and send to QC for Drug Substance testing.

Check and record the following details:

Store the above sample at -20°C from step 3.6 in a 500 mL Nalgene bottle till it is released for filling.

Sample Label ID: _____.

Details | Before Filtration | After Filtration

Volume (L) | |

O.D280/600 | |

pH | |

Conductivity (mS/cm) | |

Activity (IU/mL) as per QC report. | |

Reagent Details:
Formula & Calculations:
Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).
Calculation: To prepare 1 L of 0.2M Sodium Acetate (MW = 80.03 g/mol)
Weight of solute(g)=0.2×80.03 = 16.00g.
Preparation Method:
Weigh 16.00 g of Sodium Acetate and dissolve in water to make a 1 L solution.
Preparation for different volumes:
To prepare _____ of 0.2M Sodium Acetate (MW = 80.03 g/mol).
Weight of solute(g)=0.2×80.03 x _____L= _____gm
Weigh _____gm of Sodium Acetate and dissolve in water to make a _____ L solution.
Check and record the details:
Preparation Details:
Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry
Date
| | | |
Final volume prepared | Weighed Quantity | pH (8.9±0.1) | Conductivity (17±0.1ms/cm) | Storage Condition:
| | | | RT
Date of Preparation | Prepared By | Checked By
| |

[SHEET: Page -1]

| Check- sheet for Application for Consent To Establish (CTE) / Authorisation |||||

|

| Industry Unit & Plant Details | Plant Address along with Pin code |||||

| Industry Information |||||

| Consent to | Date of Commissioning | UAN No.: | Submit to : |||||

| Renewal Type | Validity of existing content |||||

| Industrial Type | Category | Scale |||||

| Location of industry/activity/etc. | Name of MIDC |||||

| Whether the project requires clearance under the Environment (Protection) Act 1986 (Environmental Clearance) |||||

| Whether construction-built-up area is more than 20,000 sq.mtr.(Existing Expansion Unit) |||||

|||||

| Part A: General |||||

1 | 1. Name, designation, office address with : Telephone/Fax numbers, e-mail of the Applicant

Occupier/Industry/Institution / Local Body. || Address |||||

| Designation || Taluka |||||

| Area || District |||||

| Telephone || Fax |||||

| Email || PAN number |||||

2 | Name and location of the industrial unit/premises for which the application is made (Give revenue Survey Number/Plot number name of Taluka and District, also telephone and fax number) |||||

|||||

| Industry name || Survey number/Plot Number |||||

| Telephone | Fax |||||

b | Details of the planning permission obtained from the local body /town and Country planning authority

/ metropolitan development authority designated authority Planning Permission |||||

| Planning Authority |||||

| Name of the local body under whose jurisdiction the units is located and name of the lincses issuing authority |||||

| Name of Local Body |||||

| Name of the lincses issuing authority |||||

3 | Names, addresses with Telephone and Fax Number of Managing Director / Managing Partner and officer

responsible for matters connected with pollution control and/or Hazardous waste disposal |||||

|||||

| Name of Managing Director | Telephone number | Fax number |||||

| Officer responsible for day to day business |||||

4 | 4. (a.) Are you registered Industrial/HCE unit ? | (b.) If yes, give the number and date of

registration | Registration number | Date of registration |||||

5 | Gross capital investment of the unit without depreciation till the date of application (Cost of building, land, plant and machinery). (To be supported by an affidavit/undertaking on Rs.20/- stamp paper, annual report or certificate from a Chartered Accountant for proposed unit(s), give estimated figure) |||||

| Gross capital (in Lakh) | Verified | Terms | Consent fee |||||

6a | If the site is located near sea-shore/river bank/other water bodies/Highway, Indicate the crow fly

distance and the name of the water body, if any |||||

| Distance From | Distance(Km) | Name |||||

| SH/NH |||| 0 |||||

| River |||||

| Human Habitation |||||

| Religious Place |||||

| Historical Place |||||

| Creek/Sea |||||

6b | Enter Latitude and Longitude details of site (In degrees) |||||

| Latitude | Longitude |||||

7 | Does the location satisfy the requirement under relevant central/state govt. notification such as coastal regulation zone . Notification on ecologically fragile area industrial location policy , etc. if so, give details |||||

| Location | Approved Industrial Area | Sensitive Area | If YES, Name of Area | Industrial Location with Reference to CRZ |||||

8 | If the site is situated in notification industrial estate |||||

(a) | (a.) Whether effluent collection , treatment and disposal system has been provided by the authority

|||||

(b) | (b) Will the application utilize the system if provided |||||

© | (c) if not provided details of proposed arrangement |||||

9 (a) | Total plot area (in square meter) |||||

(b) | * (b) Built up area and (in square meter) |||||

© | (c) Area available for the use of treated sewage/ trade effluent for gardening/irrigation. (in square meter) |||||

10 | Month and year of commissioning of the Unit. |||||

11 | Number of Worker and office staff |||||

| Workers | Staff | Hrs. of Shift | Weekly off |||||

12 (a) | Do you have a residential colony within the premises in respect of which the present application is made? |||||

13 | List of products and by-products Manufactured in tonnes/month, KI/month or numbers/month with their types i.e. Dyes, drugs etc.

(Give figures corresponding to maximum installed production capacity |||||

| Products Name and Quantity |||||

| Products Name and Quantity |||||

| Product Name | * UOM | * Product Name | * Existing | * Consented | * Proposed Revision | * Total |

Remarks |||||

| BY Products Name and Quantity |||||

| By product Name | * UOM | Quantity | Remark |||||

14 | List of raw materials and process chemicals with annual consumption corresponding to above stated production figures, in tonnes/month or kl/month or numbers/month |||||

| Raw Material With Quantity |||||

| Name of Raw Material | * UOM | Quantity | Hazardous Waste(Y/N) | * Hazardous Chemicals(Y/N) |||||

|||||

15 | Decryption of process of manufacture for each of the products input, output, quality and quantity of solid, liquid and gaseous waste , if from each units |||||

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43 | Hazardous chemical - give details of chemicals and quantities handle and stored. |||||
(i) is the units a major accident hazard units as per mfg. storage import hazardous chemical rule?				
(ii) is the unit an isolated storage as defined under the MSIHC rules ?				
(iii) indicate status of compliance of Rule 5,7,10,11,12,13,& 18 of MSIHC rules?				
(iv) Has approval if site been obtained from the concerned authority ?				
(vi) has information on imports of chemicals been provided to the concerned authority?				
(vii) does the unit process a policy under the PLI act?				
44	Brief details of tree plantation / green belt development within applicants premises			
Tree-plantation				
Open Space Availability	Square meter	Plantation Done on	Square meter	
Number of Tree Planted				
 45 | Information of schemes for waste minimization , resource recycling - implemented and to be implemented severalty |||||
 46. (a) | The applicant shall indicate whether industry comes under public hearing if so the relevant document such as EIA, EMP, risk analysis etc. shall be submitted , if so the relevant documents enclosed shall be indicated accordingly |||||
 (b) | any other additional information that the application desires to give |||||
 © | whether environmental statement submitted ? If submitted , give date of submission |||||
 ||||| |
 | Bank Guarantee |||||
 | Bank Guarantee applicable |||||
 47 | I/we undertake to furnish any other information within one month of its being called by the board |
 |||||
 | Signature | Name | Designation |||||
 [SHEET: Page-2]
 | Water Pollution Qty in CMD ||||| | | |
 | CETP Member (PLEASE TICK) || CETP Name || CETP Capacity | Alloted Capacity | Industry Capacity (CMD) ||
 | NO | YES |||||
 | Crep Unit (PLEASE TICK) || Provision Of Alternate Electric Supply (PLEASE TICK) || Environment Cell || Treated Effluent Sump Capacity ||
 | NO | YES || NO | YES | NO | YES ||
 | Air Pollution A.P.C.S (Air Pollution Control System) |||||
 | * Air Pollution Source || Pollutants | APCS Provided | Remark |||
 | Separate Electricity Meter Provided (PLEASE TICK) || Other Emission Sources |||||
 | NO | YES |||||
 | Foul Smell Coming Out (PLEASE TICK) || Air Sampling Facility Details |||||
 | NO | YES |||||
 | Hazardous Waste Generation (Annual Returns Details | Inventory Data Details | Form IV | Details) ||
 |||||
 | Hazardous Waste Name | * Quantity | * UOM | Treatment | Disposal | * Other Details |||
 | Member of CHWTSDf (PLEASE TICK) || CHWTSDf Name || * Remarks |||
 | NO | YES |||||
 | Cess Details (PLEASE TICK) |||||
 | Cess Applicable || Cess Paid |||||
 | NO | YES | NO | YES |||||
 | If Yes, UpTo |||||
 | Legal Actions[If Any] |||||
 | Legal Action Taken (PLEASE TICK) |||||
 | NO | YES |||||
 | Legal Record Of Company |||||
 | Legal Action Details |||||
 | Remarks |||||
 | Personal Details |||||
 | Name | Place | Date | Designation |||||
 [SHEET: Sheet2]
 Industry Type || Scale || Location of industry/activity/etc || Consent to || Verified || Product Name || Select Domestic purpose Treatment ||| DISPOSAL || Source of Water Supply || Select Fuel type || Select Hazardous Waste || Select Waste Name | TERMS
 Local Body || L.S.I || MIDC || Establish || Annual Report || Aatta-Chakkies || NA || NA || NA || MIDC || NA || NA || NA || 1
 G64 Railway Statios(Waste water Generation<10KLD) || M.S.I || Local Body || Operate || CA Certificate || Agrobases || STP || Primary || Recycle || Local Body || Bagasse || 1.1 Furnance or reactor residue and debris || A1 Arsenic | 2
 G65 Gaushalas || S.S.I || Private Industry estate || Renewal || Balance Sheet || Aluminium Smelter | Septic Tank || Primary+Secondary || CETP || Well/Bore Well water || Briquettes || 1.2 Tarry residue and still bottoms from distillation || A2 Barium | 3
 G24 Heat treatment with any of the new technology like ultrasound probe, induction hardening, ionization beam, gas carburizing etc. ||||| Undertaking || Apparel making || Septic Tank & Soak Pit || Primary+Tertiary || Local Bodies || Tankers || Coal || 1.3 Oily sludge emulsion || A3 Cadmium | 4
 G40 Rolling mill (gas fired) and cold rolling mill ||||| Asbestos and asbestos based Industry ||| Primary+Secondary+Tertiary || Shared ETP || Irrigation Department || Diesel || 1.4 Organic residues || A4(2) Chromium and/or Chromium (iii) compounds | 5
 G01 Aluminium utensils from aluminium circles ||||| Assembly of Air Coolers & Conditioners ||| OTHERS || River Disposal || Other || Electricity || 1.5 Residues from alkali wash of fuels | A5 Lead |
 G2 Ayurvedic and homeopathic medicines(without boiler)\ ||||| Assembly of bicycles, baby carraige ||||| Create Disposal ||| Ethanol || 1.6 Spent Catalyst and molecular sieves || A6 Manganese |
 G3 Bakery/confectionery/sweets products (with production capacity <1 tpd(with gas or electrical oven) || |||| 3 ||||| On Land for gardening ||| Furnace Oil || 1.7 Oil from waste water treatment || A7 Mercury |
 G4 Bi-axially oriented PP film along with metalizing operations ||||| Automobile Manufacturing ||||| Others ||| GAS || 2.1 Drill cutting excluding those from water based mud | A8 Selenium |

G5 Biomass briquettes (sun drying without using toxic hazardous wastes) ||||| Automobile Servicing & repair stations ||||| HSD || 2.2 Sludge containing Oil || A9 Silver |

G6 Blending of melamine resins & different powder, additives by physical mixing ||||| Ayurvedic ||||| Husk || 2.3 Drilling mud containing oil || A10 Ammonia |

G7 Brass and bell metal utensils manufacturing from circles(dry mechanical operation without re-rolling facility) ||||| Bakery products ,biscuits ||||| Kerosin || 3.1 Cargo residue, washing water and sludge containing oil || A11 Cyanide |

G8 Candy ||||| Bamboo & Cane Products(only dry operations) ||||| LDO || 3.2 Cargo residue and sludge containing chemicals || A12 Nitrate(as nitrate-nitrogen) |

G9 Cardboard or corrugated box and paper products (excluding paper or pulp manufacturing and without using boilers) ||||| Basic Drugs ||||| LPG || 3.3. Sludge and filters contaminated with oil || A13 Sulphide(as H₂ S) |

G10 Carpentry & wooden furniture manufacturing (excluding saw mill) with the help of electrical (motorized) machines such as electrical wood planner, steel saw cutting circular blade, etc. ||||| Block Making for printing ||||| LSHS || 3.4 Ballast water containing oil from ships || A14 1,1-Dichloroethylene |

G11 Cement products (without using asbestos/ boiler/ steam curing) like pipe, pillar, jafri, well ring, block/tiles etc.(should be done in closed covered shed to control fugitive emissions) ||||| Bone Mill ||||| Natural Gas || 4.1 Oil Sludge or emulsion || A15 1,2-Dichloroethane |

G12 Ceramic colour manufacturing by mixing & blending only(not using boiler and wastewater recycling process) ||||| Bricks ||||| Petrol || 4.2 Spent Catalyst || A16 1,4-Dichlorobenzene |

G13 Chilling plant, cold storage and ice mnaking ||||| Candles ||||| PNG || 4.3 Slop Oil || A17 2,4,5-Trichlorophenol |

G14 Coke briquetting(sun drying) ||||| Cardboard or corrugated Box And paper products ||||| Wood || 4.4 Organic Residue from processes || A18 2,4,6- trichlorophenol |

G15 Cotton spinning and weaving (small scale) ||||| Carpentry(excluding saw mill) ||||| CNG || 4.5 Spent Clay containing Oil || A19 2,4-Dinitroene |

G16 Dal Mills ||||| Carpet Weaving ||||| 5.1 Used or spent oil || A20 Benzene |

G17 Decoration of ceramic cups and plates by electric furnace ||||| Casting ||||| 5.2 Wastes or residues containing oil || A21 Benzo (a) Pyrene |

G18 Digital printing on PVC Clothes ||||| Cement ||||| 5.3 Waste cutting oils || A22 Bromodichloromethane |

G19 Facility of handling, storage and transportation of food grains in bulk ||||| Ceramics ||||| 6.1 Sludge and filter press cake arising out of production of Zinc Sulphate and other Zinc Compound || A23 Bromoform |

G20 Flour mills(dry process) ||||| Chemical, petrochemical& Electrochemical ||||| 6.2 Zinc fines or dust or ash or skimmings in dispersible form || A24 Carbon tetrachloride |

G21 Glass, ceramic, earthen potteries, tile and tile manufacturing using electrical kiln or not involving fossil fuel kiln ||||| Chilling plants and cold storages ||||| 6.3 Other residues from processing of zinc ash or skimmings || A25 Chlorobenzene |

G22 Glue from starch (physical; mixing) with gas / electrically operated oven / boiler ||||| Chlorates, perchlorates & peroxides ||||| 6.4 Flue gas dust and other particulates |

A26 Chloroform |

G23 Gold and silver smithy (purification with acid smelting operation and sulphuric acid polishing operation)(using less or equal to 1 litre of sulphuric acid/ nitric acid per month) ||||| Chlorine,fluorine& their compounds ||||| 7.1 Flue gas dust from roasting || A27 Cresol (ortho+ meta+ para) |

G25 Insulation and other coated papers (excluding paper or pipe manufacturing) ||||| Copper Smelter ||||| 7.2 Process residues || A28 Dibromochloromethane |

G26 Leather foot wear and leather products (excluding tanning and hide processing except cottage scale | ||||| Cosmetic & Herbal ||||| 7.3 Arsenic-bearing sludge || A29 Hexachlorobenzene |

G27 Lubricating oil, graeses or petroleum based products (only blending at normal temperature) ||||| Cotton and woolen hosiery ||||| 7.4 Non-ferrous metal bearing sludge and residue || A30 Hexachlorobutadiene |

G28 Manufacturing of pasted venners using gas fired boiler or thermic fluid heater and by sun drying || ||||| Cotton spinning and weaving ||||| 7.5 Sludge from scrubbers || A31 Hexachloroethane |

G29 Oil mill Ghani and extraction (no hydrogenation / refining) ||||| Dal Mills ||||| ||||| 8.1 Spent electrolytic solutions || A32 Methyl ethyl ketone |

G30 Packing materials manufacturing from non asbestos fibre, vegetable fibre yarn ||||| Distillery including Fermentation Industry ||||| 8.2 Sludge and filter cakes || A33 Napthalene |

G31 Phenyl/toilet cleaner formulation and bottling ||||| Dry Coal Processing/mines Processing ||||| 8.3 Flue gas dust and other particulates || A34 Nitrobenzene |

G32 Polythene and plastic processed products manufacturing (virgin plastic) ||||| Dyes and Dye intermediaries ||||| 9.1 Lead bearing residues || A35 Pentachlorophenol |

G33 Polutry, Hatchery and Piggery ||||| Electronic and Electrical goods ||||| || 9.2 Lead ash or particulate from the flue gas || A36 Pyridine |

G34 Power looms (without dye and bleaching) ||||| Electronics equipment(Assembly) ||||| ||||| 9.3 Acid from used batteries || A37 tetrachloroethylene |

G35 Puffed rice (muri) (using gas or electrical heating system) ||||| Electroplating ||||| ||||| 10.1 Residues containing cadmium and arsenic || A38 Trichloroethylene |

G36 Pulverization of bamboo and scrap wood ||||| Explosive Including detonators etc. ||||| ||||| 11.1 Sludges from off-gas treatment || A39 Vinyl Chloride |

G37 Ready mix cement concrete ||||| Fermentation Industry Including manufacture of yeast, beer etc. ||||| 11.2 Cathode residues including pot lining wastes || A40 2,4,5-TP (Silvex) |

G38 Reprocessing of waste cotton ||||| Fertiliser(Basic) (excluding formulation) ||||| ||||| 11.3 Tar containing wastes || A41 2,4-Dichlorophenoxyacetic acid |

G39 Rice mill (Rice hullers only) ||||| Fire Crackers ||||| 11.4 Flue gas dust and other particulates || A42 Alachlor |

G 41 Rubber goods industry (with gas opertaed baby boiler) ||||| Fish Processing ||||| ||||| 11.5 Drosses and waste from treatment of salt sludge || A43 Alpha HCH |

G42 Saw mills ||||| Flour Mills (excluding Domestic AattaChakki) ||||| 11.6 Used anode butts || A44 Atrazine |

G43 Soap manufacturing (hand made without staem boiling/ boiler) ||||| Food additives, nutrients and flavours ||||| 11.7 Vanadium sludge from alumina refineries || A45 Beta HCH |

G44 Spice grinding (upto 20 HP motor) ||||| Food Including fruits & vegetable Processing |

||||| 12.1 Acidic and alkaline residue || A46 Butachlor |
G45 spice grinding (>20 hp motor) ||| | | | | | | | | Footware(rubber) ||| | | | | | | | | 12.2 Spent acid and alkali || A47 Chlordane |
G46 Steel furniture without spray painting ||| | | | | | | | | Forging ||| | | | | | | | | 12.3 Spent bath and sludge containing sulphide, cyanide and toxic metals || A48 Chlorpyrifos |
G47 Stepping and processing of grains ||| | | | | | | | | Fountain Pens ||| | | | | | | | | 12.4 Sludge from bath containing organic solvents || A49 Delta HCH |
G48 Tyres and tube retreating (without boiler) ||| | | | | | | | | Fragrances and industrial perfumes ||| | | | | | | | | 12.5 Phosphate sludge || A50 Endosulfan(alpha+beta+ sulphate) |
G49 Chilling plant and ice making without using ammonia ||| | | | | | | | | Furniture (Wooden & Steel) |
||||| 12.6 Sludge from staining bath || A51 Endrin |
G50 CO2 recovery ||| | | | | | | | | Garments stitching 7 Tailoring ||| | | | | | | | | 12.7 Copper etching residues || A52 Ethion |
G51 Distilled water (without boiler) with electricity as source of heat ||| | | | | | | | | Glass and Fibreglass Production & Processing ||| | | | | | | | | 12.8 Plating metal sludge || A53 Heptachlor (& its Epoxide) |
G52 Hotels(up to 20 rooms and without boilers) ||| | | | | | | | | Glue & gelatin ||| | | | | | | | |
13.1 Spent pickling liquor || A54 Isoproturon |
G53 Manufacturing of optical lenses (using electrical furnace) ||| | | | | | | | | Gold & Silver thread zari work ||| | | | | | | | | 13.2 Sludge from acid recovery unit || A55 Lindane |
G54 Mineralized water ||| | | | | | | | | Gold and Silver smithy ||| | | | | | | | | 13.3 Benzol acid sludge || A56 Malathion |
G55 Tamarind powder manufacturing ||| | | | | | | | | Groundnut decorticating(dry) ||| | | | | | | | |
| 13.4 Decanter tank tar sludge || A57 Methoxychlor |
G56 Cutting, sizing and polishing of marble stone ||| | | | | | | | | Handloom Weaving ||| | | | | | | | |
|| 13.5 Tar storage tank residue || A58 Methyl Parathion |
G57 Emery powder (fine dust of sand) manufacturing ||| | | | | | | | | Hotels & restaurants ||| | | | | | | | |
|||| 13.6 Residues from coke oven by product plant || A59 Monocrotophos |
G58 Flyash export, transport & disposal facilities ||| | | | | | | | | Hydrocarbons ||| | | | | | | | |
| 14.1 Cyanide, nitrate, or nitrite- containing sludge || A60 Phorate |
G59 Mineral stack yard / Railway siding ||| | | | | | | | | Hydrocyanic acid & its derivatives ||| | | | | | | | |
|||| 14.2 Spent hardening salt || A61 Toxaphene |
G60 Oil and gas transportation pipeline ||| | | | | | | | | Ice cream or ice making ||| | | | | | | | |
| 15.1 Asbestos -containing residues || A62 Antimony |
G61 Seasoning of wood in steam heated chamber ||| | | | | | | | | Incineration Plant ||| | | | | | | | |
|| 15.2 Discarded asbestos || A63 Beryllium |
G62 Synthetic detergent formulation ||| | | | | | | | | Industrial Carbon including electrodes and graphic blocks ||| | | | | | | | | 15.3 Dust or particulates from exhaust gas treatment || A64 Chromium (VI) |
|
G63 Tea processing (with boiler) ||| | | | | | | | | Industrial or Inorganic Gases ||| | | | | | | | |
| 16.1 Mercury bearing sludge generated from mercury cell process || A65 Cobalt |
01 Dismantling of rolling stocks (wagons/ coaches) ||| | | | | | | | | Industry or Process involving electroplating operations ||| | | | | | | | | 16.2 Residue or sludge and filter cakes || A66 Copper |
|
02 Bakery and confectionery units with production capacity > 1 TPD.(with ovens/ furnaces) ||| | | | | | | | |
|| Industry or process involving foundry operations ||| | | | | | | | | 16.3 Brine sludge || A67 Molybdenum |
03 Chanachur and laddoo from puffed and beaten rice (muri and shira) using husk fired oven ||| | | | | | | | |
|| Ink ||| | | | | | | | | 17.1 Process acidic residue, filter cake , dust || A68 Nickel |
04 Coated electrode manufacturing ||| | | | | | | | | Instant tea/coffee,coffee processing ||| | | | | | | | |
|||| 17.2 Spent catalyst || A69 Thallium |
05 Compact disc computer floppy and cassette manufacturing / Reel manufacturing ||| | | | | | | | |
Insulation and other coated papers ||| | | | | | | | | 18.1 Spent catalyst || A70 Vanadium |
06 Flakes from rejected PET bottle ||| | | | | | | | | Iron & Steel ||| | | | | | | | | 18.2 Carbon residue || A71 Zinc |
07 Food and food processing including fruits and vegetable processing ||| | | | | | | | | ITES & ITES Activity ||| | | | | | | | | 18.3 Sludge or residue containing arsenic || A72 Fluoride |
08 Jute processing without dyeing ||| | | | | | | | | Jobbing & Machining ||| | | | | | | | | 18.4 Chromium sludge from water cooling tower || A73 Aldrin |
09 Manufacturing of silica gel ||| | | | | | | | | Khandsari Sugar ||| | | | | | | | | 19.1 Residue or sludge containing phenol || A74 Dichlorodiphenyltrichloroethane(DDT), Dichlorophenyldichloroethylene(DDE), Dichlorodiphenyldichloroethane(DDD) |
010 Manufacturing of tooth powder, toothpaste, talcum powder and other cosmetic items ||| | | | | | | | |
Laboratory-wares ||| | | | | | | | | 19.2 Spent catalyst || A75 Dieldrin |
011 Printing or etching of glass sheet using hydrofluoric acid ||| | | | | | | | | Laboratory chemicals (distillation & Purification Process) ||| | | | | | | | | 20.1 Contaminated aromatic, aliphatic or naphthenic solvent may or may not be fit for reuse || A76 Kepone |
012 Silk screen printing, sari printing by wooden blocks ||| | | | | | | | | Lead processing & battery reconditioning & Manufacturing ||| | | | | | | | | 20.2 Spent solvents || A77 Mirex |
013 Synthetic detergents and soaps(excluding formulation) ||| | | | | | | | | Leather footwear and leather excluding tanning ||| | | | | | | | | 20.3 Distillation residue || A78 Polychlorinated biphenyls |
015 Cotton spinning and weaving (medium and large scale) ||| | | | | | | | | Lime Manufacturing ||| | | | | | | | |
|||| 20.4 Process Sludge || A79 Dioxin(2,3,7,8-TCDD) |
016 Almirah, Grill Manufacturing (Dry Mechanical Process) ||| | | | | | | | | Lubricating Oils Greases or petroleum bases products ||| | | | | | | | | 21.1 Process Wastes, residues and sludges || B1 Asbestos |
017 Aluminium & copper extraction from scrap oil fired furnace (dry process only) ||| | | | | | | | |
Malated food ||| | | | | | | | | 21.2 Spent Solvent || B2 Total Petroleum Hydrocarbons(TPH)(C5-C36) |
018 Automobile servicing, repairing and painting (excluding only oil dispensing) ||| | | | | | | | |
Manufacture of Mirror from sheet glass and photoframing ||| | | | | | | | | 22.1 Spent Catalyst ||
C1 Flammable |
019 Ayurvedic and homeopathic medicine ||| | | | | | | | | Manufacture of steel trunks and suitcases ||| | | | | | | | |
|||| 22.2 Process residues || C2 Corrosive |
020 Brickfields (excluding fly ash brick manufacturing using lime process)||| | | | | | | | | Medical & Surgical Instruments ||| | | | | | | | | 23.1 Waste or residues(not made with vegetable or animal materials) || C3 Reactive or explosive |
021 Building and construction project more than 20,000 sq. m built up are ||| | | | | | | | | Metal ||| | | | | | | | |
|||| 23.2 Spent solvents || C4 Toxic |
022 Ceramics and Refractories ||| | | | | | | | | Milk processing and dairy products(integrated project) ||| | | | | | | | |
|||| 24.1 Chemical residues || C5 Substances or wastes liable to spontaneous combustion |

023 Coal Washeries ||||| Mineralised Water ||||| 25.1 Chemical residues |
| C6 Substances or Wastes which in contact with water emit flammable gases |
024 Dairy and dairy products (small scale) ||||| Minerals ||||| 25.2
Residues from wood alkali bath || C5 Oxidizing |
025 DG Set of capacity > 1 MVA but < 5 MVA ||||| Mining ||||| 26.1
Process waste sludge /residues containing acid, toxic metals , organic compounds || C8 Organic Peroxides
|
026 dry coal processing, mineral processing, industries involving ore sintering, pelletisating, grinding
& pulverization ||||| Musical Instruments manufacturing ||||| 26.2 Dust
from air filtration system || C9 Poisons(acute) |
027 Fementation industry including manufacture of yeast, beer, distillation of alcohol (Extra Neutral
Alcohol) ||||| Non-alcoholic beverages (soft drinks) ||||| 26.3 Spent
acid || C10 Infectious substances |
028 Ferrous and Non- ferrous metal extraction involving different furnances through melting , refining,
re-processing, casting and alloy-making ||||| NPK Fertilisers/Granulation |||||
|| 26.4 Spent solvent || C11 Liberation of toxic gases in contact with air or water |
029 Fertilizer (granulation/ formulation/ blending only) ||||| Oil ginning/expelling(no
hydrogenation/refining) ||||| 26.5 Spent Catalyst || C12 Eco-Toxic |
030 Fish feed, polutry feed and cattle feed ||||| Oil Refinary(Mineral Oil or Petro
Refineries) ||||| 27.1 Process residue || C13 Capable |
031 Fish processing and packing (excluding chilling of fishes) ||||| Optical Frames ||||
||||| 28.1 Process Residue and wastes |||
032 Forging of ferrous and non-ferrous metals (using oil and gas fired furnaces) |||||
Organic nutrients ||||| 28.2 Spent catalyst |||
033 Formulation/ pelletization of camphor tablets, naphthalene balls from camphor/ naphthalene powders |
||||| OTHERS ||||| 28.3 Spent Carbon |||
034 Glass ceramics, earthen potteries and tile manufacturing using oil and gas fired kilns, coating on
glasses using cerium fluorides and magnesium fluoride etc. ||||| Paint(by mixing process
only) ||||| 28.4 Off specification products |||
035 Gravure printing, digital printing on flex, vinyl ||||| Paints & Varnishes(excluding
Blending & Mixing) ||||| 28.5 Date-expied products |||
037 Hot mix plants ||||| Paper Pins & U-Clips ||||| 28.6 Spent organic
solvent |||
038 Hotels (<3 star) or hotels having >20 rooms and less than 100 rooms |||||
Pesticides(technical)(excluding Formulation) ||||| 29.1 Process waste or residues ||
|
039 Ice Cream ||||| Pesticides/Insecti/fungicides/Hebicides ||||| 29.2
Sludge containing residual pesticides |||
042 Industry or processes involving foundry operations ||||| Petrochemicals |||||
||| 29.3 Date expired and off specification pesticides |||
043 Lime manufacturing (using lime klin) ||||| Petroleum Products involving storage,
transfer or processing ||||| 29.4 Spent solvents |||
044 Liquid floor cleaner, black phenyl, liquid soap, glycerol mono sterate manufacturing |||||
|| Pharmaceuticals formulations ||||| 29.5 Spent catalyst |||
045 Manufacturing of glass ||||| Pharmaceuticals (excluding formulations) |||||
||| 29.6 Spent acid |||
046 Manufacturing of iodized salt from crude/ raw salt ||||| Phosphaphate rock processing
plants ||||| 30.1 Chromium bearing residue and sludge |||
047 Manufacturing of mirror from sheet glass ||||| Photographic films & Chemicals ||||
||||| 31.1 Process residue and wastes |||
048 Manufacturing of mosquito repellent coil ||||| Pigments and Intermediaries |||||
||||| 31.2 Spent etching chemicals and solvents |||
049 Manufacturing of Starch/Sago ||||| Plastic processed goods ||||| 32.1
Spent chemicals |||
050 Mechanized laundry using oil fired boiler ||||| Polethene, plastic & PVC goods through
extrusion/moulding ||||| 32.3 Process sludge containing adsorbable |||
051 Modular wooden furniture from particle board , MDF<swan timber etc, Ceiling tiles/ partition board
from saw dust, wood chips etc., and other agricultural waste using synthetic adhesive resin, wooden box
making (with boiler) ||||| Potable Alcohol (IMFL) by blending ||||| 33.1
Empty barrels/ containers/liners contaminated with hazardous chemicals /wastes |||
052 New highway construction project ||||| Power Generating plants (excluding D.G Sets) ||
||||| 33.2 Contaminated cotton rags or other cleaning materials |||
053 Non-alcoholic beverage(soft drink) & bottling of alcohol/ non alcoholic products |||||
Power looms/handloom(without dyeing & bleaching) ||||| 34.1 Chemical-containing residue
arising from decontamination. |||
054 paint blending and mixing (Ball mill) ||||| Printing Press ||||| 35.1
Exhaust air or gas cleaning residue |||
055 Paints and varnishes (mixing and blending) ||||| Process involving chlorinated
hydrocarbon ||||| 35.2 Spent ion exchange resin containing toxic metals |||
056 ply-board manufacturing (including Veneer and laminate) with oil fired boiler/ thermic fluid heater
(without resin plant) ||||| Process involving Metal Treatment ||||| 35.3
Chemical sludge from waste water treatment |||
057 Potable alcohol(IMFL) by blending, bottling of alcohol products ||||| Pulp and Paper |
||||| 35.4 Oil and grease skimming |||
058 Printing ink manufacturing ||||| Pulping and fermenting of coffeese beans |||||
||||| 35.5 Chromium sludge from cooling water |||
059 Printing press ||||| Radio Assembling ||||| 36.1 Any process or
distillation residue |||
060 Reprocessing of waste plastic including PVC ||||| Refractory |||||
36.2 Spent carbon or filter medium |||
062 Spray painting, paint baking, paint shipping ||||| Rice mullors |||||
37.1 Sludge from wet scrubbers |||
063 Steel and steel products using various furnaces like blast furnace/ open hearth furnace/ submerged
arc furnace/ basic oxygen furnace / hot rolling reheated furnace ||||| RMC |||||
||| 37.2 Ash from incinerator and due gas cleaning residue |||
064 Stone crushers ||||| Rope(Cotton & Plastic) ||||| 37.3 Concentration
or evaporation residues |||
065 Surgical and medical products including prophylactics and latex ||||| Rubber Goods
Industry ||||| 38.1 Process residues |||
066 Tephlon based products ||||| Scientific, mathematical Instruments |||||
| 38.2 Spent acid |||
068 Tobacco products including cigarettes and tobacco/ opium processes ||||| Shoelace

Manufacturing ||||||||| Other Hazardous Waste |||

069 Transformer repairing/ manufacturing (dry process only) ||||||||| Slaughter house & meat Processing Units ||||||||| Micro Biological Waste |||

070 Tyres and tubes vulcanization/hot retreating ||||||||| Sports goods |||||||||

|||

071 vegetable oil manufacturing including solvent extraction and refinery / hydrogenated oils ||||||

||| Steel and Steel Products including coke plants |||||||||

072 Wire drawing and wire netting ||||||||| Stepping & Processing of Grains |||||||||

|||

073 Dry cell battery (excluding manufacturing of electrodes) and assembling & charging of acid lead battery on micro scale ||||||||| Stone Crushers |||||||||

074 Pharmaceutical formulation and R & D purpose (For sustained release/ extended release of drugs only and not for commercial purpose) ||||||||| Sugar (excluding Khandsari) |||||||||

|||

075 Synthetic resins ||||||||| Supari (Betelnut) and masala griding |||||||||

|

076 Synthetic rubber excluding molding ||||||||| Surgical & Medical Products not involving effluent/emmission |||||||||

077 Cashew nut processing ||||||||| Surgical and medical products involving prophylatics & latex |||||||||

078 Coffee seed processing ||||||||| Synthetic detergent & Soap |||||||||

079 Parboiled Rice Mills ||||||||| Synthetic Resins |||||||||

080 Foam Manufacturing ||||||||| Synthetic Rubber |||||||||

083 Producer gas plant using conventional up drift coal gasification (linked to rolling mills glass and ceramic industry refectories for dedicated fuel supply) ||||||||| Synthetis Fibres(rayon, tyre cord, polyester, filayarn) |||||||||

014 Thermometer manufacturing ||||||||| Tailoring & Garment Making |||||||||

|

036 Heat treatment using oil fired furnance (without cyaniding) ||||||||| Tanneries ||||||

|||||||

040 Industries engaged in recycling/ reprocessing/ recovery/ reuse of Hazardous waste under schedule iv of HW (M.H & TBM) rules, 2008- items namely-B148 Paint and ink sludges /residues |||||||||

Thermal Power Plants |||||||||

041 Industries engaged in recycling/ reprocessing/ recovery/ reuse of Hazardous waste under schedule iv of HW (M>H& TBM) rules, 2008- Items namely- Brass Dross, Copper Dross, Copper Oxide Mill Scale, Copper reverts, Cake & residues, Waste Waste Copper and copper alloys in dispersible form, Slags from copper processing for further processing or refining, Insulated Copper Wire, Scrap/Copper with PVC Sheathing including ISRI- code Material namely "Druid", jelly filled copper cables, Zinc Dross- Hot dip Galvanizers SLAB, Zinc Dross-Bottom Dross, Zinc ash/Skimming arising from galvanizing and die casting operations, Zinc ash/Skimming/other zinc bearing wastes arising from smelting and refining , Zinc ash and residues including zinc alloy residues in dispersible form ||||||||| Tobacco Products(Cigarettes & tobacco processing) |||||||||

061 Rolling mill (oil or coal fired) and cold rolling mill ||||||||| Tyres & Tubes (excluding vulcanisation/ Retreating/moulding) |||||||||

067 Thermocol manufacturing(with boiler) ||||||||| Tyres and Tubes vulcanisation, vulcanisation, retreading |||||||||

081 Industries enagagged in recycling / reprocessing / recovery/ reuse of Hazardous Waste under schedule iv of HW (M.H & TBM) rules, 2008- Items namely- Used Oil- As per specifications prescribed from time to time ||||||||| Vegetable oils including solvent extracted oils, hydrog. Oils |||||||||

|||||||

082 Industries engaged in recycling / reprocessing / recovery/ reuse of Hazardous Waste under schedule iv of HW(M.H& TBM) rules, 2008- Items namely- Waste Oil----As per specifications prescribed from time to time ||||||||| Wire Drawing (Cold Process) and bailing straps |||||||||

Resturant activity having over all waste water generated upto 10 KLD ||||||||| Yarn & Textile processing (Scouring, bleaching, dyeing, etc.) |||||||||

084 railway Stations (waste water Generation >= 10 KLD, but <100 KLD) ||||||||| Zinc Smelter ||

|||||||

085 Scrapping Centers (for End of Life if Vehicles and other scraps such as plant and machineries, structural materail, railway coaches and wagons etc.) a. Collection, De-Pollution, Dismantling Centers and Shredding Centres. B. Collection, De-Pollution , Dismantling Centers and Shredding Centre. C. Shredding Centres (Can include white goods*/other scrap Also) |||||||||

|||

086 Compressed /Refined Bio-Gas Production from Bio-degradables Wates |||||||||

|||||||

087 Dairy farms |||||||||

Assaying & hallmarking centers |||||||||

R61 Dead Animal incineration plant (Carcass disposal plant) |||||||||

||

R89 (i.e. Sterilization process by using Gamma Rays using CoBALT-60). |||||||||

|||||||

R61 Railway Stations (Water Water Generation >= 100 KLD) |||||||||

Pyrolysis oil from plastic waste |||||||||

Municipal corpotaion stp |||||||||

Restaurant activity having over all waste water generated more than 10KLD |||||||||

|||||||

R3 Industries engaged in recycling / reprocessing/ recovery/ reuse of Hazardous Waste under schedule iv of HW (m.H& TBM) rules, 2008- Items namely - Spent cleared metal catalyst containing copper, Spent cleared metal catalyst containing Zinc |||||||||

R10 Industries engaged in recycling / reprocessing / recovery / reuse of Hazardous Waste under schedule iv of HW (M, H & TBM) rules, 2008- Items namely -Spent catalyst containing nickel, cadmium, Zinc, Copper, arsenic, vanadium and cobalt, |||||||||

R15 Industries engaged in recycling / reprocessing / recovery / reuse of Hazardous Waste under schedule iv of HW (M, H & TBM) rules, 2008- Items namely - Dismantlers Recycling Plants-- Components of waste electrical and electronic assemblies comprising accumulators and other batteries included on list A, mercury-switches, actived glass cullets from cathode-ray tubes and other activated glass and PCB-Capacitors or any other component contaminated with Schedule 2 constituents (e.g. cadmium, mercury, lead, polychlorinated biphenyl) to an extent that they exhibit hazard characteristics indicated in part C of this Schedule |||||||||

R18 Pulp & Paper (waste paper based without bleaching process to manufacture Kraft paper) ||||||||

|||||||

R32 Industries engaged in recycling /reprocessing/recovery/reuse of Hazardous Waste under Schedule iv of HW (M, H & TBM) |||||||||

R33 Industries engaged in recycling / reprocessing/ recovery / reuse of Hazardous Waste under Schedule iv of HW (M, H & TBM) rules, 2008 -Items Namely -Integrated Recycling Plants--Components of waste electrical and electronic assemblies comprising accumulators and other batteries included on list A, mercury- switches, activated glass cullets from cathode-ray tubes and other activated glass and PCB-Capacitors, or any other component contaminated with Schedule 2 constituents(e.g, cadmium, mercury, lead, polychlorinated biphenyl) to an extent that they exhibit hazard characteristics indicated in part C of this Schedule. |||

R54 Pulp & Paper (waste paper based units with bleaching process to manufacture writing & printing paper) |||

R59 Pulp 7 Paper (Large-Agro + wood), Small Pulp & Paper (agro based-wheat straw/ rice husk) |||

R1 Isolated storage of hazardous chemicals (as per schedule of manufacturing , storage of hazardous chemicals rules, 1989 as amended) |||

R2 Automobile Manufacturing (Integrated facilities) |||

R4 Manufacturing of lubricating oils, grease and petroleum based Products |||

R5 DG Set of capacity> 5 MVA |||

R6 Industrial carbon including electrodes and graphite blocks, activated carbon, carbon black |||

R7 Lead acid battery manufacturing (excluding assembling and charging of lead- acid battery in micro scale) |||

R8 Phosphate rock processing plant |||

R9 Power generation plant [except Wind and solar renewable power plants of all capacities and Mini Hydel power plant of capacity <25MW] |||

R11 processes involving chlorinated hydrocarbons |||

R12 Sugar(excluding Khandsari) |||

R13 Fibre glass production and processing (excluding moulding) |||

R14 Fire Crackers manufacturing and bulk storage facilities |||

R16 Milk processes and dairy products(integrated project) |||

R17 Phosphorus and its compounds |||

R19 Coke making, liquefaction, coal tar distillation or fuel gas making |||

R20 Manufacturing of explosives, detonators, fuses including management and handling activities |||

R21 Manufacturing of paints varnishes, pigments and intermediate (excluding blending /mixing) |||

R22 Organic Chemicals manufacturing |||

R23 Airports and Commercial Air Strips |||

R24 Asbestos and asbestos based industries |||

R25 Basic chemicals and electro chemicals and its derivatives including manufacturing of acids |||

R26 Cement |||

R27 Chlorates, per-chlorates & peroxides |||

R28 Chlorine, fluorine, bromine, iodine and their compounds |||

R29 Dyes and Dye- Intermediates |||

R31 Hotels having overall waste-water generation @ 100 KLD and more |||

R34 Manufacturing of glue and gelatin |||

R35 Mining and ore beneficiation |||

R36 Nuclear power plant |||

R37 Pesticides (technical) (excluding formulation) |||

R38 Photographic film and its chemicals |||

R39 Railway locomotive workshop/ Integrated road transport workshop/ Authorized service centre |||

R40 Yarn/ Textile processing involving any effluent/ emission generating processes including bleaching, dyeing, printing and colouring |||

R41 Chlor Alkali |||

R42 Ship Breaking Industries |||

R43 Oil and gas extraction including CBM(offshore & on-shore extraction through drilling wells) |||

R44 Industry or process involving metal surface treatment or process such as pickling/ electroplating/paint stripping/heat treatment using cyanide bath/phosphating or finishing and anodizing / enamellings/ galvanizing |||

R45 Tanneries |||

R46 Ports and harbour, jetties and dredging operations |||

R47 Synthetic fibres including rayon, tyre cord, polyester filament yarn |||

R48 Thermal Power Plants |||

R49 Slaughter house (as per notification S.O.270 E dated (26.03.2001) and meat processing industries , bone mill , processing of animal horn, hoofs and other body parts |||

R50 Aluminium Smelter |||

R51 Copper Smelter |||

R52 Fertilizer(basic) (excluding formulation) |||

R53 Iron & Steel 9involving processing from ore/ integrated steel plants) and or Sponge Iron units |||

R55 Zinc Smelter |||

R56 Oil refinery (mineral Oil or Petro refineries) |||

R57 Petrochemicals Manufacturing (including Processing or Emulsions of oil and water) |||

R58 Pharmaceuticals |||

R60 Distillery(molasses/ grain / yeast based) |||

Other |||

Reagent Details:
Formula & Calculations:
Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).
Calculation: To prepare 1 L of 0.2M Tri Sodium Citrate (MW = 258.06g/mol)
Weight of solute(g)=0.2×258.06=51.66g.
Preparation Method:
Weigh 51.66 g of Tri-sodium citrate and dissolve in water to make a 1 L solution.
Preparation for different volumes:
To prepare _____ of 0.2M Tri Sodium Citrate (MW = 258.06g/mol).
Weight of solute(g)=0.2×258.06x _____L= _____gm
Weigh _____gm of Tri-sodium citrate and dissolve in water to make a _____ L solution.
Check and record the details:
Preparation Details:
Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry Date
Tri Sodium Citrate | | | |
Final volume prepared | Weighed Quantity | pH (8.4±0.2) | Conductivity (35±0.5 ms/cm) | Storage Condition:
| | | RT
Date of Preparation | Prepared By | Checked By
| |

PLASMA
LIFT
(O/H)TANK(2) MACHINE
ROOM

1200.0000

TERRACE

LIFT
6650.0000

6100.0000

6650.0000

6100.0000

23650.0000

BUILDING PLASMA

9150.0000

8500.0000

BUILDING PLASMA
T-0.30m.

1199.9996
R-0.15m.

SECTION A-A
SCALE 1:100

SHAFT SHAFT

3.30X2.45

3.30X2.45
CORRIDOR
FIREMAN FIREMAN 49.1x2.56

LIFT

LIFT
LIFT LIFT
2.80X2.60 2.80X2.60 4D
D3
4D 4D

D3
CH R2 CH R1
3.71x2.60 3.79x2.60 REJECT RECEIVING
MATERIAL RECEIVING
P.M RECEIVING 4.51x5.05 D3 D3
AREA AREA

AREA 6.86x5.00 4.94x5.00
5.00x5.00 CH R3 EXIT
3.71x2.60 3.79x2.60 DE DUSTING
D4 D4 D4
D3 D3

SAMPLING
ROOM
2.94 M WIDE CORRIDOR

2.94 M WIDE CORRIDOR
7.59x8.46

LADIES TOILET LADIES TOILET

DPB
3.30 4.35 3.30 4.35 FINISHED GOODS ROOM
19.8x30.3 LAF
WC WC WC WC WC WC
1.10X 1.10X 1.10X 1.10X 1.10X 1.10X
1.70 1.70 1.70 1.70 1.70 1.70

6.77 M WIDE CORRIDOR

6.77 M WIDE CORRIDOR
WC WC WC WC WC WC
1.10X 1.10X 1.10X 1.10X 1.10X 1.10X
1.70 1.70 1.70 1.70 1.70 1.70

GENTS TOILET GENTS TOILET PACKING MATERIAL STORE
3.30 4.35 3.30 4.35
9.56 x 25.3

11.9 x 15.7

QUARENTAINE RM STORE
19.5x32.2
11.8x8.85

4D

400
400

730
730

400
400

P

P
P

P

L
2
L
2
SHAFT SHAFT SHAFT SHAFT

/
/
LIFT LIFT LIFT LIFT

L
1
L
1

V
3.24 M WIDE PASSAGE

3.24 M WIDE PASSAGE
3.24 M WIDE PASSAGE

3.24 M WIDE PASSAGE
V
2.10X 2.10X 2.10X 2.10X

3.24 M WIDE PASSAGE

3.24 M WIDE PASSAGE
2.00 2.00 2.00 2.00

UP
FD
DIALYZER FD
UP UP
FD 17.3x21.0 FD
UP

53.54 x 96.57 CORRIDOR
80.7x2.57 CORRIDOR
80.7x1.80

D4

047
047
4D APPROVED RM RAW MATERIAL STORAGE
24.2x14.0
7.30x9.75
COOL ZONE
17.3x8.86

4D
GENERAL STORE
STATIONARY
LADIES TOILET & ENG STORE LADIES TOILET
3.30 4.35 4D 3.30 4.35
7.25x5.97
1100 TROLLEY WIDTH 1100 TROLLEY WIDTH
1100 TROLLEY WIDTH

3D
7.40x2.00

1100

1100
CH 2 CH 1
1200 1200

DPB 2.56x1.92 3.20x1.92

3D
1200

WC WC WC WC WC WC
1.10X 1.10X 1.10X
DPB 1.10X 1.10X 1.10X
1.70 1.70 1.70 3D 1.70 1.70 1.70

3D

3D
MI ING BLENDING AREA

1100
XX
XX
13.1x8.69 CH R2 CH R1
6.45x2.88
XX

3.40x3.46 3.40x3.46
PP GRANUAL

WC
1.10X
WC
1.10X
WC
1.10X
AUTO CLAVE RM DISPENSING
ROOM
PRODUCTION
OFFICE & IPQA
TOOL ROOM
4.25x7.00
HOPPER LOADER
ROOM
3D
WC
1.10X
WC
1.10X
WC
1.10X
4.05x7.00 2.92x7.00
ROOM
D3
1.70 1.70 1.70 4.11x7.00 1.70 1.70 1.70

CHAMBER LG. 8400

CHAMBER LG. 8400
PITT LENGTH 9650

PITT LENGTH 9650

CHAMBER LG. 8400
PITT LENGTH 9650
CH R3 EXIT
13.2x11.5 3.40x3.46 3.40x3.46

PITT LENGTH 9650
XX
XX

XX

GENTS TOILET GENTS TOILET
3.30 4.35 3.30 4.35
DIA. 2250 DIA. 2250
DIA. 2250
SPB LAF
4D 4D 4D
D3 D3

EMERGENCY EXIT
3D
700

700

700
CORRIDOR

D4
1326 1326
1326

29.2x2.79

D4
DPB
PW U2
WFI U2

3D 4D 4D
12.3x6.07
1100

1100
3500 PIT WIDTH 3500 PIT WIDTH 3500 PIT WIDTH

D4
1.25x4.25 COOLING ZONE
4D

1100 TROLLEY WIDTH 1100 TROLLEY WIDTH 1100 TROLLEY WIDTH
7.16x4.54 5.15X 2.00

COLLECTED AREA 9.03x4.54 3 WAY
SPB
FILLED BOTTLE 6.92x1.52 AUTO CLAVE

13.7x6.06
5.15X3.99

3D
PRODUCTION DPBEURO CAP
IPQA OFFICE AIR LOCK
1.76X1.20
OFFICE STORE 3.04 x 1.44
3.37x1.44 2.71x1.44 DPB
6.77 M WIDE CORRIDOR

6.77 M WIDE CORRIDOR

6.77 M WIDE CORRIDOR

6.77 M WIDE CORRIDOR
D3
3D 3D 3D

CORRIDOR
53.5x1.80
D4 D4
D4
D1

1D

D1

1D

KITCHEN OFFICE
7.30x3.19 5.56x3.19 OFFICE
PRIMARY FEMALE
6.64x3.19
1D

D1

1D

PRIMARY MALE PRIMARY VISITOR
WASHING CHANGE ROOM
CHANGE ROOM CHANGE ROOM
D1

6.50x5.18 2.50x5.18 6.50x5.18
3D D3 3D 6.32x5.18
1D

D1

1D
D1

WAITING ROOM
CORRIDOR
1D

D1

1D

8.46x2.49 5.04x2.49
D4 D4 3D D4
3D
D1

CONFERENCE 3D 3D

1D

D1

1D

HALL
LADIES TOILET F- REST ROOM
GENTS TOILET 6.00x8.10 M- REST ROOM CORRIDOR
5.80x11.3 4.39x2.96 12.5x2.96 4.93x2.99
1D

D1

1D

5.80x11.3
5.47x3.11

4D

4D
RECEPTION
1D

D1

1D

13.5x5.55

4D
D4
M- WASH ROOM
1D

D1

1D

F- WASH ROOM
8.20x3.09 8.20x3.06
D7 D7
D1 D1 D1

Employees
CORRIDOR Entry
53.5x2.50
7D

D7

3.24 M WIDE PASSAGE
3.24 M WIDE PASSAGE

D1 D1 D1 D1 D1 D1 D1 D1 D1 D1

GENTS VISITOR FEMALE CORRIDOR
SHAFT GENTS TOILET LADIES TOILET ANITOR SHAFT SHAFT 48.5 2.90 SHAFT
WAITING ROOM LOCKER LOCKER LOCKER LIFT LIFT
9.94 4.90 6.30 4.90 ROOM
LIFT 4.67 5.00 ROOM ROOM ROOM 3.41 5.00 LIFT LIFT LIFT
3.24 M WIDE PASSAGE

D6
5.90 5.00 6.00 5.00
3.24 M WIDE PASSAGE

2.10X 6.00 5.00 1D 1D 1D 1D 1D 1D 1D 1D 1D
2.10X 2.10X 2.10X
D3

2.00 D1 D1
2.00 2.00 2.00
FD FD FD FD FD DOCUMENT STORE CHEMICAL /GLASS
FD
6D

C STAFF
RECEPTION 4.72 1.80
CORRIDOR 4.73 6.62 DOCUMENTATION WARE STORAGE
CONFERENCE
9.31 6.07 CANTEEN
UP 40.4 1.80 UP UP 4.03 6.62 7.71 6.62
12.4 10.6 UP
8.30 12.0
5.98 6.12
2.35 8.03
D3
PRIMARY VISITOR
AIR LOCK OFFICE 1 OFFICE 2 OFFICE 3
FEMALE PRIMARY FEMALE AIR LOCK VISITOR CHANGE 1 D3

AIR LOCK PRIMARY MALE MALE CHANGE 1 1.80 5.15 3.69 5.10 3.66 5.10 3.69 5.10
FEMALE VISITOR
D3

CHANGE 1 CHANGE CHANGE 2.75 5.10
MALE CHANGE 3.30 5.10 3.40 5.10 3.19 5.10 3.64 5.10 3.05 5.10 3.70 5.10
4.70 5.10 4.00 5.10

WASH AREA
WET LAB D3 5.90 2.76
8D

D3 9.31 4.85

C MANEGER
PM TESTING ROOM
6D

4.73 5.91 9.94 6.64

D6
CORRIDOR D1
D3
DE

49.0 2.25
DPB
DPB DPB WASTE OUT
LAF DHS HOT ROOM/
CHEMICAL DAY STORE 3 WAY BALANCE GENERALFTIR FUME HOOD
W ROOM 5.90 5.46
6D

6D

11.4 3.64 DECONTAMINATION WASHING STERILIZATION D3 D3
4.82 3.37
4.38 3.37
W
AUTOCLAVE

2.33 2.14 3.11 5.43 3.60 5.43
4.67 5.43
LAF

BUFFER TANK STORAGE-2 CIP / SIP COLD ROOM 1.80 M WIDE CORRIDOR
D3
D3
10.1 1.80
PREPARATION 6.74 7.50 5.60 7.50 5.87 7.50 COOL ZONE
C SAMPLE PREP.
4.73 5.42
4.00 7.50 3.25 7.50
DPB DPB CORRIDOR
A/L GOWN A/L 6D 22.5 1.80
2.10 3.40 3.34 3.40 3.34 3.40 D1 INSTRUMENT ROOM II
WASTE IN A/L A/L 15.3 5.43
GS+HH

GS+HH

LENAP LORTNOC LENAP LORTNOC
3.65 2.00 4.13 2.00 3.60 2.00 W INSTRUMENT ROOM I
CORRIDOR
W 6D 6D 6D DPB HPLC
1.80 29.2
9.94 7.04
D1
TRANING HALL
DE

A/L DE GOWN A/L C SAMPLE
RECEIPT
ANITOR
D3 D3
14.8 19.0
1.50 3.93
2.10 3.40 3.34 3.40 3.34 3.40 CORRIDOR 3.11 3.93
39.8 3.04 REFERENCE TOC/LPC ROOM
D3 GC ROOM
D4 STANDARD 4.44 2.98
4.83 2.98
5.87 2.98
D6 D6 D6 D6 D3
DPB AKTA-1" TFF
NANO
LAF D3
FILTER MH

ANITOR
1.80 2.00 GENTS CH ROOM
LADIES TOILET A/L LADIES TOILET 4.73 3.42 RETENTION

3.30 4.35
A/L GOWN 1000 L 3.30 4.35 SAMPLES ROOM
3.34 4.76
COB

3.34 4.76 D1
2.10 4.76 BULK FILTERATION W (COLD ROOM 2 TO 8 C)
LAF

9.94 3.31
6.54 7.29 500 L D3 D3
WC WC WC WC WC WC
6.77 M WIDE CORRIDOR

6.77 M WIDE CORRIDOR

1.10X 1.10X 1.10X 1.10X 1.10X 1.10X
2.94 M WIDE PASSAGE

2.94 M WIDE PASSAGE

1.70 1.70 1.70 MH
SL MH

1.70 1.70 1.70
600 L
STABILITY ROOM

V1 V1 V1 LADIES CH ROOM 15.3 6.04
500 L 4.73 3.42
LAF

MH SL RM CONTROL SAMPLE
9.94 3.31
COB

600 L MH
D1
V1 V1 V1 MH

DE MH

50 L
A/L D3 D3 D1
WC WC WC A/L GOWN LAF 500 L STAGE 13 E WC WC WC
D6

1.10X 1.10X 1.10X 3.34 4.76
MH

2000 L CHROMO SL 1.10X 1.10X 1.10X
2.10 4.76 3.34 4.76 COLUMN 250 L
1.70 1.70 1.70 MH 600 L STAGE 10 T/ MH
1.70 1.70 1.70
D3 D3
WASH AREA DPB AC
D3
MEDIA DIST.
W
STAGE 9 N 7.18 13.5 5.221 2.02 DECONTAMINATION ROOM 7.34 2.02
STAGE 11 F
MH

5.54 2.02
3D

D3

1000 L
5.33 2.05 250 L MEDIA PREP. PREPARATION & STERILIZATION
MH

250 L
7.25 15.6 200 L GRADE D 8.11 5.91
AKTA-1"

GENTS TOILET GENTS TOILET D3
STAGE 6 (I) 7.38 3.88
MH

MH

GRADE D
DPB

3.30 4.35 3.30 4.35 D3
IC-1 DPB
COLUMN PACKING MH

1000 L 15.0 18.0 MH

250 L GRADE B

DPB

INCUBATOR STERILITY
AND SKIDS
MH

200 L DPB
FOR MEDIA TESTING ROOM IC-1
4.65 3.78 GRADE CNC IC-4
6.54 10.6 250 L D3
INCUBATOR
COOL ZONE
MH
MH

150 L GRADE B 5.33 3.78
100 L MH

COOL ZONE IC-2
3.38 3.78 R&D LAB-4
9.00 5.53
D3
MH

500 L
D3 IC-3 R&D LAB -3
1.83 1.91 CORRIDOR
3D

MEDIA 6.47 6.61 6.47 6.61
CHROMO GRADE D OBSERVATION
13.3 1.80
COLUMN
MH

300 L STORE A/L GRADE B
150 L MH

2.50 4.24
100 L MH

4.78 3.54
D4 D4
CORRIDOR D3
MH

100 L GRADE B GRADE B 3.53 2.02 DPB
GRADE C
CH-I CH-II CH-III 1.72 1.86
300 L 4.81 2.02 2.73 2.02 2.02 2.02
LAF 150 L
W DPB
MH

LAF D4 GRADE CNC
MH

100 L
CHROMO
COLUMN AKTA-1"
MH

W D4
D4 D4 DPB GRADE C
MLT ROOM
IC-1
INCUBATOR IC-4
D4 4.65 3.77 5.33 3.77 2.80 7.77
AUTOCLAVE DHS
BET ROOM IC-2
GRADE C GRADE B
GRADE CNC GRADE D IC-3
E IT-II
D6

4.78 2.83 A/L E IT-I D3
500 L 500 L CHROMO
5.36 2.02 2.63 2.02 OBSERVATION
150 L 100 L COLUMN D4 D4 3.49 2.02
250 L 250 L
6D

MH MH
D3
R&D LAB -2 R&D LAB -1
MH

A/L GRADE C FRIDGE
CORRIDOR
MH MH MH

MH MH MH

CORRIDOR CORRIDOR 6.47 6.61 6.47 6.61
2.57 71.9 1.50 6.85 CORRIDOR BSC
GRADE CNC 1.80 8.50
3.50 55.7 1.80 74.2 GRADE D GRADE C GRADE CNC
STERILIZATION 2000 L CORRIDOR A/L CH-I
D6

INCUBATION ROOM
COB

2500 L GRADE D
CLAF

D4 5.36 2.02 6.04 2.02
2500 L 2.50 8.31 4.36 3.78
7.40 6.85
D4 D4 3.76 3.78
CULTURE ROOM
GENTS CH ROOM 4.65 3.78
6D

GRADE CNC D3 D3
4.78 2.83 DPB
D3

D3 D3
W W
D3

3D

DPB
PRE - VIRUS CONTROLLED CORRIDOR

1000 L DPB D3

GRADE CNC
AKTA-1"

GRADE CNC D4 D4
TFF

MH
GRADE CNC DPB
MICRO OFFICE MICRO MICRO CORRIDOR
COLUMN PACKING SAMPLE RECEIPT 14.8 1.80
COB

5.17 4.24 DOCUMENT STORE GRADE D
2.63 4.24 GRADE D SCRAP OUT
3.49 4.24 E IT GRADE CNC
LADIES CH ROOM CH ROOM 5.49 2.86
DPB
AND SKIDS 2.15 2.86 A/L D3 D3
MH

GRADE CNC 2.39 2.86
2.66 2.86
4.78 3.12
WASHING 7.64 8.86
100 L D3 D3
D3 D3
2.94 M WIDE PASSAGE

D3
A/L
7.40 3.00 R&D OFFICE
6.77 M WIDE CORRIDOR
AKTA-1"

1.50 4.42
STAGE 7A 500 L 7.31 2.90 R&D ENTRANCE
6.77 M WIDE CORRIDOR

CORRIDOR 7.48 2.90
34.1 1.80
D3

MH

STAGE 5 (I) A 14.7 16.1
D6

DPB LADIES TOILET LADIES TOILET
WASTE IN 500 L 3.30 4.35 D4 D4
3.30 4.35
2.94 M WIDE PASSAGE

MH

250 L

10.0 19.0 CORRIDOR
TFF

W MH

16.7 1.80
3.50 32.1

300 L
MH
STORAGE

CABIN 1
WC WC WC CABIN 4 CABIN 3 CABIN 2 WC WC WC
3.39 4.14
DECONTAMINATION 250 L
500 L 1.10X
1.70
1.10X
1.70
1.10X
1.70
2.88 4.14 3.43 4.14 4.05 4.14 D4 D4 D4 D4
1.10X
1.70
1.10X
1.70
1.10X
1.70
7.40 5.00
MH

OUT MH
MH

4.13 4.19 PAL-1
WASTE V1 V1 AUDIT MAL ENTRY PRIMARY CHANGE PRIMARY CHANGE 2.27 1.64
D6

V1 VISITOR FEMALE COLD ROOM
1.50 3.54 500 L 2.85 5.81 2.85 5.78
4.51 5.78 4.43 3.93

DPB STAGE 5(I)B 1500 L 250 L
MH
MH

V1 V1 V1
D4 D4 D4 D4 3.63 1.92
PRIMARY CHANGE
4.19 3.27
2.26 2.53
PAL-2
D4 DE-CARTONING
3.89 3.32
DPB

MALE
7.64 7.20 100 L
MH

250 L 250 L 250 L
SL
600 L 500 L WC WC WC
5.31 5.81
2.27 1.64
WC WC WC
D4

MH

MH
MH MH 100 L 100 L 100 L MH
MH

1.10X 1.10X 1.10X
D4
1.10X 1.10X 1.10X
100 L MH MH MH

1.70 1.70 1.70 D4 D4 D4
DPB 1.70 1.70 1.70
CORRIDOR CLEAN C

CIP / SIP
MH

W 1500 L 2.62 120.8 1.88 2.48
SL MH MH
CORRIDOR

9.00 5.00

MH MH MH

1000 L 100 L 9.00 1.80
D4

DPB
D4

D4

CORRIDOR
MH

500 L MH GENTS TOILET GENTS TOILET
500 L 1.80 117.9
MH
6D

1000 L 1000 L 1000 L 3.30 4.35
A OFFICE 3.30 4.35
MH

1500 L
MH

QA MANGER
50 L 1500 L
GS+HH

GS+HH

SL
LENAP LORTNOC LENAP LORTNOC

MH
2500 L 18.0 10.8 2.85 4.06
MATERIAL PREP WASH
8.15 3.07
AC
1.00 2.40 UNLOAD
8.41 3.40
D4

1500 L 5.26 1.64
D4

2.21 1.46
250 L 250 L 250 L 250 L
DRYING ZONE

CHROMO MH

DPB DPB
VIAL WASHING
6D

COLUMN MH MH MH MH 1.00 1.00
WASTE OUT
6.42 9.65
BAG CUTTING &
D6

D4

D4

PAL
STERILIZING ZONE

CIP S

STAGE 4 (F) 2.84 1.45
CIP S
CIP STATION
D4

POOLING MH
MH MH

MH
QA HEAD
DISINFECTANT
STERILE CORRIDOR
15.5 1.80

W
MH MH

2.85 4.47
7.64 9.80 14.7 15.8 8.41 2.62
D3 D3
150 L

6D

500 L 500 L
COOLING ZONE
MAL
D4

D4

W SL 1000 L
MAL
MAL PAL
TANK STORAGE 2.84 1.45
D4

1000 L 1500 L 1500 L 1.45 1.64 1.41 1.64
STABLIZING ZONE

PASS THROUGH

9.00 8.58
D6

D4

MH

D3 D3
D3

250 L STAGE 12 (E) MH

D6
PAL
2.04 2.29
D3

D4

TUMBLE
MH

STAGE 1 (F)/ 500 L DRYER FORMULATION
8.41 3.59
CHROMO
1.50 2.00 PACKING AREA
COLUMN
10.0 10.6 250 L
D3

D4

MAL
CONVEYOR
SWING

300 L 300 L MH
2.04 1.47 DOUBLE
DOOR
TFF
D6

AKTA-1"

MH MH

CLAF D3 WASHER
W 6D UNLOADING AREA
36" TURNTABLE
NEW ADD

W A DOCUMENT
LAUNDRY AREA 1.50 2.00
5.10 1.57 2.94 5.13 8.49 1.80
2500 L 2.98 3.88

21.0 9.82
& LOADING TRAY
6D

VIAL COLLECTION

THAWING PLASMA CHROMO
COLUMN 2000 L 2000 L D2

FILL FINISH
7.64 3.61
D4
AKTA-1"

MH

6.42 8.30
D4

DPB
500 L 250 L 300 L 300
300LL E IT 2 E IT 1
BUFFER 250 L MH MH

4.54 1.87
D4

4.40 1.87
MH MH MH

CHROMO
MH MH MH

PREPARATION COLUMN
CORRIDOR
PAL 3
4.23 3.82
PAL 4
3.20 3.82
9.00 6.00 D6 D6 D6
2.85 25.2 PAL 1 PAL 2
4.54 1.87 4.40 1.87
D4

D4

D4

D4

CORRIDOR
6D

DE

SHAFT SHAFT SHAFT SHAFT
DPB 36.2 2.50 D4

LIFT LIFT LIFT LIFT
3.24 M WIDE PASSAGE

3.24 M WIDE PASSAGE

2.10X D6 D6 D6
2.10X 2.10X 2.10X
6D

MH MH

2.00 2.00 2.00 LYO SERVICE AREA
2.00
MH MH MH MH

CHROMO MH MH

CONTROL ROOM
MH MH MH
MH
LYO LOAD
D4

COLUMN
100 L 100 L 100 L 100 L MH

100 L 100 L CENTRIFUGE 5.55 3.55 10.9 3.55 8.29 3.55
200 L 250 L
D8
3.24 M WIDE PASSAGE

3.24 M WIDE PASSAGE

3.24 M WIDE PASSAGE

300 L 300 L 500 L 500 L
FD FD FD FD FD D8 FD
AUTOMATIC FOUR HEAD ALUMINUM CAP
SEALING MACHINE, MODEL-AEWCS-120

1500 L
UP UP UP D4
UP
PLASMA DAY STAGE 3 (F)
STAGE 2 (F)
SWING
CONVEYOR

DPB
STORE 12.4 10.5 BULK STAGING COLD ROOM

VIAL COLLECTION

STAGE 8 N

9.00 10.7 17.7 10.5 SECONDARY PACKING AREA

16.6 4.04 CAPPING

TRAY

2.66 2.29

5.84 10.5

21.0 8.07 8.29 4.04

AKTA-1"

AKTA-1"

DPB

D8

UP DPB

CORRIDOR

ANITOR

3D

CHROMO 25.5 1.80 C

3.50 5.40

MH

COLUMN

2.32 1.00

SL

600 L 1500 L 2000 L

1500 L 1500 L DPB

D4

1500 L

UNDERTEST

D4

300 L 300 L MH

MH

BUFFER ROOM

MH MH

VISUAL &

MH

100 L 100 L CHROMO CHROMO PACKING CHANGE

100 L TFF COLUMN COLUMN AKTA-1" 2.83 2.68

DPB

MH

MH

INSPECTION FEMALE

4.14 3.08

MH MH MH

9.79 3.87

D4

EMER.GLASS 2.88 1.14

SAMPLING

D4

W D8

SPM STORE

D4

W PPM STORE 11.8 1.80

15.2 12.0 14.9 9.32

D4

PACKING CHANGE

11.8 3.03 7.22 2.69

MALE

4.14 4.23

UARANTINE

LABELING & PACKING

12.6 5.40

D4

PB

D4

PACKING

D6

PLASMA STORAGE PLASMA STORAGE PLASMA STORAGE
PLASMA STORAGE 1000 L 1000 L 1000 L
PLASMA STORAGE W VISITOR
W 1000 L 1000 L CORRIDOR 4.41 3.08
6.77 M WIDE CORRIDOR

D4
70 KL GV 70 KL GV 70 KL GV
70 KL GV 70 KL GV 1.80 15.2 D8
D4

10.5 15.2 9.36 15.2 10.3 15.2
D8

9.10 15.2 9.52 15.2
D4

D8 D8
D4 PACKING OFFICE
LABELING STORE 4.14 4.50
FINISHED GOODS STAGING
7.68 5.67
12.6 5.67
6.77 M WIDE CORRIDOR

3.14 1.84
APPROVED RUBBER
SPM RECEIPT RE ECT
BUNG STORE
3.36 2.97 3.59 2.98 3.58 2.98
D8 D8 D8
LADIES TOILET LADIES TOILET
3.30 4.35 3.30 4.35
W
WC WC WC WC WC WC
2.94 M WIDE PASSAGE

1.10X 1.10X 1.10X 1.10X 1.10X 1.10X
1.70 1.70 1.70 1.70 1.70 1.70

V1 V1 V1 D3 D3 D3 D3 D3

CORRIDOR
2.94 M WIDE PASSAGE

CORRIDOR
49.1 1.80 2.80 27.9
V1 V1 V1
D6
WC WC WC CORRIDOR WC WC WC
1.10X 1.10X 1.10X 1.10X 1.10X 1.10X
1.70 1.70 1.70
28.0 1.80 1.70 1.70 1.70

D3 D6 W
DPB RLAF
D3

AIR LOCK-1
GENTS TOILET 1.50 2.18 RM UARANTINE WASHING PLASMA COLD ROOM -4 COLD ROOM -3 COLD ROOM -2 COLD ROOM -1
ICE PACK GENTS TOILET
BAG BAR CODING ANITOR
AREA UARANTINE
3.30 4.35 MINI POOLS 2.50 7.34 3.30 4.35
6D

2.30 2.67
1.45 2.67 5.04 7.34 5.02 7.34 5.02 7.34 5.04 7.34
2.60 2.67
DISPENCING ROOM
3.40 4.44
3D D3 D3
D3
PLASMA STORAGE
1000 L
70 KL GV PLASMA STORAGE PLASMA STORAGE PLASMA STORAGE
CHANGE ROOM
12.2 17.1 1000 L 1000 L 1000 L
1.50 2.18 6.85 6.55 70 KL GV 70 KL GV 70 KL GV
3D

12.1 17.1 12.1 17.1 12.1 17.1
5.15 2.72
CORRIDOR
11.6 4.30 1.80 12.0
D3
3D

SD SD SD SD

DPB
6D W

W D3
DPB
AIR LOCK-1
2.13 1.21
D3
DPB
3D D3 D6 STORE 12.9 6.49
APPROVED RM 3.54 2.60
AIR LOCK-1
1.87 1.40 OFFICE 4.70 4.09
3D
10.1 7.54 2.23 6.50
5.15 4.49 CHANGE ROOM 3D
3.00 3.60
2.13 1.80 CHANGE ROOM
1.87 1.40
RE ECT SEROLOGY PLASMA
DE-DUSTING
TESTING RECEIPT
D3

D3 TUNNEL
PLASMA ROOM 3D D3
STORE CHANGE
D3
SAMPLING 2.26 4.95 3.00 4.95 2.55 4.95 ROOM
RM 1.76 2.90 2.95 2.88
SAMPLING 3.68 2.05 RECALLED REJECT
3.26 2.31
RM
GOODS ROOM FG DISCPATCH
1.07 3.11
RECEIPT
2.30 2.35 2.30 2.35
FG IN
3.54 2.77
RLAF RLAF
6D D3 D3 D6 D6

D6 D3

CORRIDOR
3.30X2.45

3.30X2.45

3.30X2.45

3.30X2.45

D3 D3 D3 D3

53.9 3.17
6D

D6

D6

D6
LIFT

LIFT

LIFT

LIFT

CORRIDOR
49.0 2.59
DISPATCH BAY RECEIVING BAY RECEIVING BAY DISPATCH BAY
40' CONTAINER YARD

CLIENT :- LIFENITY-LARGE VOLUME PARENTERALS

DESIGN &
CONTAINER

CONTAINER

CONTAINER

CONTAINER

PROJECT BY :- BIOBRIDGE HEALTHCARE SOLUTION PVT.LTD.
12.19 2.44 2.59

12.19 2.44 2.59

12.19 2.44 2.59

12.19 2.44 2.59

PREPARED A.K
ADDRESS :-13 R B
A P -411007 M I .

FIRST FLOOR
CHECKED G.K.

DATE 18-04-2026
TITLE- CONCEPTUAL LAYOUT-FF REVISION R0
SCALE 1:1

SIZE A0 DWG.NO 01

CLIENT :- LIFENITY-LARGE VOLUME PARENTERALS

DESIGN &
PROJECT BY :-BIOBRIDGE HEALTHCARE SOLUTION PVT.LTD.
PREPARED A.K
ADDRESS :-13 R B
A P -411007 M I .

GROUND FLOOR
CHECKED G.K.

DATE 18-04-2026
TITLE- CONCEPTUAL LAYOUT-GF REVISION R0
SCALE 1:1

SIZE A0 DWG.NO 01

CONCEPTUAL LAYOUT GF
40' CONTAINER YARD

CONTAINER

CONTAINER
12.19x2.44x2.59

CONTAINER

CONTAINER
12.19x2.44x2.59

12.19x2.44x2.59
12.19x2.44x2.59
1 2 3 4 5 6

SPM
FG PLASMA RM RECEIVING
DISCPATCH RECEIVING RECEIVING BAY
BAY BAY

A

CORRIDOR
40.1X6.84

D3 6D

D6 D6 3D D3 D6
RLAF RM RLAF
REJECT RECALLED PLASMA RECEIPT
RM
SAMPLING
FG DISCPATCH FG IN ROOM GOODS 3.00X2.77 SAMPLING
STORE CHANGE 3.68X 2.05
2.30X 2.35
ROOM 2.30X2.35
PLASMA SEROLOGY REJECT 1.76X 2.90 3.16X2.03
4.50X3.88 STORE 1.07X3.39

RECEIPT ROOM D3 D3

D3
D3 OFFICE TESTING DE-DUSTING
D3
2.40X 4.95 2.95X4.95 TUNNEL
2.23X6.25 3.00X 4.95 CHANGE ROOM
1.87X1.40 CHANGE ROOM
9.42X6.25 2.00X2.08

B 4.70X3.85
D3

3.00X2.60
D3
4.55X2.32 CORRIDOR
AIR LOCK-1
1.87X1.40
1.95X11.3 D6 D3 D3 DPB D3 AIR LOCK-1
2.00X1.21
DPB D3

D6
APPROVED RM
11.5X3.30 9.72X7.22
5.00X2.95
3D
SD SD SD SD
6.21X5.77 4.63X4.00

3D
DPB

3D
CHANGE ROOM
3D 3D D3
1.50 X 1.93
D3
DISPENCING ROOM
3.48X 3.95
ICE PACK COLD ROOM -1 COLD ROOM -2 COLD ROOM -3 COLD ROOM -4 JANITOR WASHING BAG BAR CODING,
MINI POOLS RM QUARANTINE
1.45X2.90 AREA

6D
PLASMA
2.91X7.58 4.45X7.58 4.45X7.58 4.45X7.58 4.45X7.58 QUARANTINE 2.30X2.90 2.60X2.90
AIR LOCK-1
1.50X 1.93

D3
RLAF
D6 D3 DPB

CORRIDOR
29.5 X 2.58

C D6

CORRIDOR
D3 D3 47.9x1.80 D3 D3 D3
699

PLASMA STORAGE PLASMA STORAGE PLASMA STORAGE PLASMA STORAGE PLASMA STORAGE
~ 1000 L ~ 1000 L ~ 1000 L ~ 1000 L ~ 1000 L
~ 70 KL GV ~ 70 KL GV ~ 70 KL GV ~ 70 KL GV ~ 70 KL GV
8.15x11.5m 9.95x11.5m 9.95x11.5m 9.95x11.5m 7.10x11.5m

D 800 800 800 800 850
----- 9M WIDE ROAD -----

CORRIDOR
D3 D3
45.3x1.80 D3 D3 D3
1125

800 800 800 800

PLASMA STORAGE 800 PLASMA STORAGE PLASMA STORAGE PLASMA STORAGE PLASMA STORAGE
~ 1000 L ~ 1000 L ~ 1000 L ~ 1000 L ~ 1000 L
~ 70 KL GV ~ 70 KL GV ~ 70 KL GV ~ 70 KL GV ~ 70 KL GV
8.15x9.62m 9.95x9.62m 9.95x9.62m 9.95x9.62m 7.10x9.62m

12M WIDE ROAD
E CORRIDOR
2.57x37.0

CORRIDOR
2.57x35.1

CORRIDOR
D3 D3
45.3x1.80 D3 D3 D3

800 800 800 800

F PLASMA STORAGE
~ 1000 L
~ 70 KL GV
PLASMA STORAGE
~ 1000 L
~ 70 KL GV
PLASMA STORAGE
~ 1000 L
~ 70 KL GV
PLASMA STORAGE
~ 1000 L
~ 70 KL GV
PLASMA STORAGE
~ 1000 L
~ 70 KL GV
8.15x9.62m 9.95x9.62m 9.95x9.62m 9.95x9.62m 7.10x9.62m

CORRIDOR
EMER.GLASS 50.5x1.80 EMER.GLASS

DPB
G JANITOR
5.20x1.80

STAGE 8 N
6.17X8.68
STAGE 3 (F) STAGE 2 (F) D4
PLASMA DAY
12.8X8.68 15.8X8.68 STORE
8.10x8.77
6D

6D 6D 6D
DPB

CORRIDOR
35.1X2.80
DE

6D

H D6 D6 D6
BUFFER
DPB

PREPARATION
PLASMA 8.10x6.00
THAWING
6D

8.00X3.22
STAGE 1 (F)/
STAGE 12 (E) D6
D6

CLAF
11.1X8.21
D6

STAGE 4 (F)
BAG CUTTING & TANK STORAGE
13.6X13.5
6D

POOLING 8.10x6.58
8.00X7.80
6D

I WASTE OUT
CONTROL PANEL CONTROL PANEL

DPB
PRE - VIRUS CONTROLLED CORRIDOR

GS+HH

GS+HH
6D

CIP / SIP
8.10x5.00
STAGE 5(I)B
STAGE 5 (I) A 8.00X7.20
DPB
11.1X18.0 WASTE
2.00X28.8

OUT
D6

DECONTAMINATION 1.50x3.54
STAGE 7A 6.50x5.00
13.6X15.1
4D

WASTE IN

J
4D

DPB
D6

D4

COLUMN PACKING
D4

AND SKIDS WASHING A/L
1.50x4.42
8.00X7.86 6.50x3.00
DPB
4D
D4
6D

STERILIZATION
CLAF

A/L
6.50x5.85
D6

1.50x5.73
6D

CORRIDOR
2.57X67.9
D6

AUTOCLAVE

CORRIDOR DHS

K STAGE 6 (I) LAF LAF DPB
STAGE 11 F
14.7X16.7

COLUMN PACKING COOL ZONE
AND SKIDS 8.10x5.53
6.90X10.6

STAGE 10 T/
D4

1.80X49.6

STAGE 13 E
4D

4D

8.10X12.2 DE
6.25X2.05 STAGE 9 N A/L
2.20X3.89
GOWN
3.47X3.89
A/L
2.22X3.89
5.10X14.3
LAF
D4

4D

BULK FILTRATION
LAF

L 6.90X6.00
D4
LAF

D4

A/L GOWN A/L
2.20X4.32 3.47X4.32 2.22X4.32
JANITOR
1.80x2.00
4D

4D

D3 D6 D6 LAF D6 D6 DPB
4D

4D

CORRIDOR
DE CORRIDOR
37.0x2.80 A/L A/L
GOWN 2.57x66.0
DE

2.20X3.40 3.47X3.40 2.22X3.40
4D
D4

D4 4D D4
D6 D6 D6
D4

A/L A/L WASTE
D4
D4

4.01x2.40 4.13x2.40 IN
2.50x2.40 CONTROL PANEL CONTROL PANEL

A/L GOWN A/L
2.20X3.40 4.00X3.40 2.22X3.40
DPB

4D 4D D4
DPB

GS+HH

GS+HH

M 8.30x1.80
4D

4D

STERILIZATION
4.01x5.62 BUFFER
AUTOCLAVE

COLD ROOM CIP / SIP TANK STORAGE-2 PREPARATION
6D

LAF

WASHING DECONTAMINATION
3.11x5.62 3.52x5.62 4.40x8.10 5.60x8.10 6.80x8.10 4.29x8.10 CHEMICAL DAY STORE
8.10x4.00
COOL ZONE
2.76x8.10
6D

6D

DPB
DPB

WASTE OUT
DHS

LAF

DPB

3 WAY
D4

CORRIDOR
47.8x2.00
DE

6D

D4 D4 D4

D4

FEMALE
MALE PRIMARY MALE AIR LOCK
PRIMARY AIR LOCK AIR LOCK PRIMARY FEMALE
VISITOR
VISITOR VISITOR FEMALE CHANGE CHANGE 1 CHANGE 1 CHANGE MALE
OFFICE 1 OFFICE 2 2.50x5.00 CHANGE 1
3.64x5.00 3.10x5.00 3.10x5.00 3.00x5.00 3.80x5.00 4.77x5.00
2.60x5.00 CHANGE 3.05x5.00
4.83x5.00 4.83x5.00 2.70x5.00
N
D4

D4

4D

4D

D4

D4

D4 D4 D4 D4 D4

RECEPTION
8D

CORRIDOR 8.00x6.10
34.9x2.80 4.05x2.80
6D

JANITOR
4D 4D 4D 4D 4D 4D D4 4D
ROOM
D1

3.50x1.65 WAITING ROOM
5.30x4.00
D1 D1 D1
GENTS TOILET
DOCUMENT FEMALE VISITOR GENTS 5.90x5.20
OFFICE 3 LADIES TOILET
ROOM LOCKER LOCKER LOCKER
LIFT 3.13x5.30 3.13x5.30 5.90x5.20
ROOM ROOM ROOM
1D 1D 1D 1D 1D 1D 1D 1D 1D
4.57x5.30 4.57x5.30 4.57x5.30

O

-----6M WIDE ROAD-----

Reagent Details:

Formula & Calculations:

Formula: Use the formula to calculate the molarity:

Molarity (M) = (Per cent by Weight / 100) * (Density in g/mL) * 1000 / Molar Mass.

Example: Molarity of Glacial Acetic Acid

Molar Mass of CH₃COOH: ~60.05 g/mol

Density: ~1.05 g/mL

Purity: 100%

Acetic Acid (MW = 60.05 g/mol).

Calculation:

Molarity = (100 / 100) * 1.05 g/mL * 1000 mL/L / 60.05 g/mol ≈ 17.5 M.

Calculation: To prepare 1 L of 2M.

Applying the formula to prepare the working buffer:

$C_1 V_1 = C_2 V_2$

C_1 = 17.5 M acetic acid.

V_1 = volume of stock needed to take.

C_2 = desired (working) concentration 2 M acetic acid.

V_2 = final total volume of working solution for one litre.

$V_1 = (C_2 \times V_2) / C_1$.

$C_2 = 2$ M; $V_2 = 1.0$ L; $C_1 = 17.5$ M

$V_1 = (2 \times 1.0) / 17.5$

$V_1 = 0.11$ L

Take 0.11 L of the Stock solution of acetic acid and add it to a volumetric flask, making up the volume to 1 L.

To prepare various volumes:

V_1 = volume of stock needed to take _____ L.

C_2 = desired (working) concentration 2 M acetic acid.

V_2 = final total volume of working solution _____ L.

$V_1 = (C_2 \times V_2) / 17.5$.

$V_1 =$ _____ L.

Take V_1 _____ L of the Stock solution of acetic acid and add it to a volumetric flask, making up the final volume _____ L.

Check and record the details:

Preparation Details:

Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry

Date

||||

Final volume prepared | Measured Quantity | pH (1.24±0.1) | Conductivity (1.1±0.1 ms/cm) | Storage

Condition | Reagent Record

|||| RT |

Date of Preparation | Prepared By | Checked By

||

Reagent Details:
Preparation of PF-RR-14 buffer :
Calculations: For the Preparation of 20mM Sodium Citrate pH 7.0 + 0.3 M Sodium Chloride.
Stock Buffer (PF-RR-12):40mM Sodium Citrate + 0.6M Sodium Chloride.
Formula:.
 $C1 \times V1 = C2 \times V2$
C1 = concentration of stock
V1 = volume of stock you need to take
C2 = desired (working) concentration
V2 = final total volume of working solution
Applying the formula to prepare the working buffer:
 $C1 \times V1 = C2 \times V2$
C1 = 0.4M Sodium Citrate, pH 7.0.
V1 = volume of stock you need to take _____ L.
C2 = desired (working) concentration 20mM Sodium Citrate, pH 7.0.
V2 = final total volume of working solution _____ L.
 $V1 = (C2 \times V2) / C1$.
C2=0.02M ; V2= 1.0 L; C1=0.4M
 $V1 = (0.02 \times 1.0) / 0.4$
V1 = 0.05L
Take 0.05 L (PF-RR-12) Stock buffer solution .
Preparation of Stock buffer (PF-RR-12):
Take 0.05 L (PF-RR-12) Stock buffer solution and then make up the volume to 1 L with water.
Check and record the details:
Preparation Details:
Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry
Date
Sodium Citrate | | | |
Citric Acid | | | |
Sodium Chloride | | | |
Stock buffer (PF-RR-12) taken volume(L) | pH (7.0± 0.1) | Conductivity (30 ±0.1ms/cm) | Storage
Condition | Final volume prepared
| | | RT |
Date of Preparation | Prepared By | Checked By
| |

Reagent Details:

Preparation of PF-RR-11 Stock buffer :

Calculations: For the Preparation of 40mM Sodium Citrate pH 7.0 + 0.32M Sodium Chloride.

Stock Buffer (PF-RR-09):0.4M Sodium Citrate pH 7.0.

Formula:

$$C1 \times V1 = C2 \times V2$$

C1 = concentration of stock

V1 = volume of stock you need to take

C2 = desired (working) concentration

V2 = final total volume of working solution

Applying the formula to prepare the working buffer:

$$C1 \times V1 = C2 \times V2$$

C1 = 0.4M Sodium Citrate pH 7.0 .

V1 = volume of stock you need to take _____ L.

C2 = desired (working) concentration 40mM Sodium Citrate, pH 7.0.

V2 = final total volume of working solution _____ L.

$$V1 = (C2 \times V2) / C1$$

$$C2 = 0.04M ; V2 = 1.0 L ; C1 = 0.4M$$

$$V1 = (0.04 \times 1.0) / 0.4$$

$$V1 = 0.1L$$

Take 0.1 L (PF-RR-09) Stock buffer solution and add 0.32M sodium Chloride.

Calculations: To prepare the 0.32M Sodium Chloride.

Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).

Calculation: To prepare 1 L of 0.32M Sodium Chloride (MW = 58.44 g/mol)

$$\text{Weight of solute(g)} = 0.32 \times 58.44 \times 1 = 18.7g.$$

Preparation Method:

Weigh 18.7 g of Sodium Chloride and dissolve in water to make a 1 L solution.

Preparation for different volumes:

To prepare _____ L of 0.32 M Sodium Chloride (MW = 58.44 g/mol).

$$\text{Weight of solute(g)} = 0.32 \times 58.44 \times \text{_____} L = \text{_____} g.$$

Weigh _____ gm of Sodium Chloride and dissolve in water to make a _____ L solution.

Preparation of Stock buffer (PF-RR-11):

Take 0.1 L (PF-RR-09) Stock buffer solution and add 0.32 M sodium chloride. Dissolve it, and then make up the volume to 1 L with water.

Check and record the details:

Preparation Details:

Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry

Date

Sodium Citrate | | | |

Citric Acid | | | |

Sodium Chloride | | | |

Stock buffer (PF-RR-09) taken volume(L) | pH (7.0 ±0.1) | Conductivity (31±0.2) | Storage Condition |

Final Buffer volume prepared

| | | RT |

Date of Preparation | Prepared By | Checked By

| |

CHROMATOGRAPHY -1F ,2F & 4F

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CHROMATOGRAPHY-1F

Check the column connections to the chromatography system. Check for instrument air connection at 6 bars to the system. Open the software.

Record the following details for column:

Column height: 60 cm, Column Volume: 1.178 L

HETP value: _____, Asymmetry: _____.

Column wash

Wash the columns with 1 - 2 CVs of PW at a flow rate of 23.3ml/min till the conductivity has dropped below 1.0 mS/cm.

Record the volume of PW used: _____ L.

Column Equilibration

Equilibrate the columns with 2 CVs of PF-RR-04 (10% gradient of PF-RR-03 against PW) at a flow rate of 23.3ml/min.

Check and record the final pH: _____.

Conductivity: _____ mS/cm,

Volume of PF-RR-04 consumed: _____ L.

Loading & Elution

Before loading, switch on the 'Record On' button and enter the run details and date. Auto Zero the absorbance at 280 nm on the chromatography system.

Load the filtered plasma (250 ml) for first injection, and after loading is completed, pass PF-RR-04 (10% gradient of PF-RR-03 against PW) at a flow rate of 16.7 ml/min.

Collect fractions 1 & 2 at a flow rate of 16.7 ml/min, and for fraction 3 collection, increase the flow rate to 23.3ml/min (Column pressure should be ≤ 2.5 bar). Collect the three fractions separately.

Note: Fraction 1 should be further sent for FE processing and Fraction 2 for FG processing.

Repeat steps 1.5.2 and 1.5.3 for Inj. 2.

Repeat steps 1.5.2 and 1.5.3 for Inj. 3.

Repeat steps 1.5.2 and 1.5.3 for Inj. 4.

Record the details of four injections: Frac-1,2&3.

Regeneration

Regenerate the column with 2 CV of PF-RR-28 (25% gradient of PF-RR-27 against PW) in up flow direction at a flow rate of 23.3ml/min and hold the column for 30 – 60 minutes. Volume of PF-RR-28 used: _____ L.

Storage

Pass 2 CVs of PF-RR-30 through the column at a flow rate of 23.3ml/min and finally store the column in PF-RR-30 at room temp.

Volume of PF-RR-30 used: _____ L.

CHROMATOGRAPHY-2F

Check the column connections to the chromatography system. Check for instrument air connection at 6 bar to the system. Open the UNICORN software.

Record the following column details:

Column height: 20 cm, Column Volume: 0.393 L,

HETP value: _____, Asymmetry: _____.

Column Wash

Wash the column with upto 2 CVs of PW at a flow rate of 40 ml/min till the conductivity has dropped below 1.0 mS/cm.

pH Drop

Wash the column with upto 3 CVs of PF-RR-06 (25% gradient of PF-RR-35 against PW) to drop the pH to 4.50 (4.50 ± 0.50) at a flow rate of 40 ml/min.

Column Equilibration

Equilibrate the column with up to 8 CVs of PF-RR-10 (10% gradient of PF-RR-31 against PW) at a flow rate of 40 ml/min to reach a final pH of 7.00 (7.00 ± 0.20).

Sample preparation for loading

Take fraction 3 (pooled 4 injections) from Chromatography-1F (step 1.5.7) as a load for Chromatography-1 and record the following conditions:

If pH is not 7.00 ± 0.20 , then adjust the pH to 7.00 ± 0.20 with PF-RR-05 under stirring condition.

Record the following conditions:

ml of plasma loaded / ml of resin: _____. (Limit: ≤ 9.0 ml/ml)

Loading

Before loading, switch on the 'Record On' button and enter the run details and date. Auto Zero the absorbance at 280 nm on the chromatography system.

Load the sample at a flow rate of 33.3 ml/min.

Collect the flow through sample when the A280 rises from 100 mAU.

Wash I

After loading is completed, wash the column with up to 2 CVs of PF-RR-10 (10% gradient of PF-RR-31 against PW) at a flow rate of 33.3 ml/min and stop collecting the flow through when the A280 drops down to 100 mAU. Volume of PF-RR-10 used:

Wash II

After Wash I, pass 3.5 CVs of PF-RR-29 (50% gradient of PF-RR-11 against PW) at a flow rate of 33.3 ml/min and do not collect the peak as it contains most of the impurities.

Elution

After the A280 has reached the baseline, elute the sample by passing up to 3 CVs of PF-RR-14 (50% gradient of PF-RR-12 against PW) at a flow rate of 33.3 ml/min.

Note: Elution buffer (PF-RR-14) conductivity should be in the range of 33 ± 1.5 mS/cm.

Collect the eluate when the A280 rises from 100 mAU and collect the total peak until the A280 drops down to 100 mAU.

Check and record the following conditions of eluate sample:

Note: Send the eluate fraction for further purification of FN.

Cleaning

Clean the column by passing up to 2 CVs of PF-RR-13 (25% gradient of PF-RR-32 against PW) in upflow direction at a flow rate of 33.3 ml/min.

Regeneration

Regenerate the column by passing 2 CVs of PF-RR-28 (25% gradient of PF-RR-27 against PW) in up flow

direction at a flow rate of 33.3 ml/min and hold the column for 30 – 60 minutes.

Storage

Store the column (at room temp.) by passing 2 CVs of PF-RR-30 at a flow rate of 33.3 ml/min. Volume of PF-RR-30 used: _____ L.

Enter the process details as below:

CONCENTRATION AND DIAFILTRATION OF CHROMATOGRAPHY-2F FT USING TFF SYSTEM

Take the FT from Chromatography-2F step (Step 2.8.1).

Pre-NWP calculation

Record the following details of the TFF cassettes connected in series horizontally to the TFF system:

No. of cassettes used: _____, Cassette ID Nos.: _____,

Wash the cassettes with 10 L of PW to remove storage solution and calculate NWP before concentration at a set TMP of ≤ 1.0 bar.

Check and record the details as below:

Concentration of FT

After NWP calculation, wash the cassettes with 5 L of PF-RR-10 at ambient temperature to equilibrate the cassettes. Based on the initial volume, concentrate the sample to the batch volume at a set TMP of ≤ 1.0 bar and Pump Speed of 25 to 32%. Volume of PF-RR-10 used: Run : _____ L.

After collecting the retentate, flush the cassettes and system with 1.0 L of PF-RR-15 to collect the holdup volume and pool into the collected retentate. Volume of PF-RR-15 passed: Run : _____ L.

Check and record the following details:

Record the retentate details:

Diafiltration

Dia filter the concentrated sample from step 3.3 with PF-RR-15 to reach a final conductivity of permeate of 2000 – 2500 $\mu\text{S}/\text{cm}$.

Check and record the following details of the Dia filtered sample:

Cassette Regeneration

After the diafiltration step, wash the cassettes with 5 L of PF-RR-15 and then with 15 L of PW.

Volume of PF-RR-15 used: Run : _____ L, Flush the cassettes with 10 L of PF-RR-28 at 40 - 45°C and carry out recirculation for 30 min – 45 min.

After recirculation, wash the cassettes with 20 L of PW.

Post NWP calculation

Repeat the step from 3.2.2 and record the details below. Check if NWP is ≥ 120 LMH/Bar. If not, perform the regeneration procedure again.

Storage

Flush the cassettes with 10 L of PF-RR-30. After flushing, store the cassettes in PF-RR-30 at room temp.

Volume of PF-RR-30 used: _____ L.

DILUTION & pH ADJUSTMENT

Take the Dia filtered sample from step 3.4 and split it into two runs.

Make up the volume to _____ ml (run-wise) with PF-RR-15 under stirring conditions. Ensure that conductivity is NMT 1100 $\mu\text{S}/\text{cm}$.

Adjust the pH of the sample to 5.20 (run-wise) (Limit: 5.20 ± 0.10) with PF-RR-16 under stirring conditions.

Collect 20 ml of sample from Run and send to QC for in-process QC analysis. Record the obtained results:

pH: _____, Cond. (mS/cm): _____.

Record the details below:

EUGLOBULIN PRECIPITATION

Incubate the sample after pH adjustment in the sterilized 10 L glass bottle for 6 – 16 hrs. at 2 to 8°C under stirring conditions.

Check and record the following details after incubation:

SEPARATION OF PRECIPITATE

Take Euglobulin precipitate sample in a batch mode centrifuge bottle and spin at a speed of 10000 rpm.

Collect the supernatant into a sterilized bottle and pellet in a centrifuge bottle is discarded.

Check and record the following details:

CHROMATOGRAPHY-4I

Check the column connections to the chromatography system. Check for instrument air connection at 6 bar to the system. Open the UNICORN software.

Record the following column details:

Column height: 25 cm, Column Volume: 491 ml ,

HETP value: _____, Asymmetry: _____.

Column Wash

Wash the column with upto 2 CVs of PW at a flow rate of 36.7ml/min till the conductivity has dropped below 1.0 mS/cm. Record the following details:

pH drop

Wash the column with upto 2 CVs of PF-RR-36 (10% gradient of PF-RR-18 against PW) to drop the pH to 4.25 ± 0.25 at a flow rate of 36.7ml/min. Record the following details:

Column Equilibration

Equilibrate the column with upto 4 CVs of PF-RR-22(10% gradient of PF-RR-21 against PW) at a flow rate of 36.7ml/min, to reach a final pH of 5.20 (5.20 ± 0.10).

After equilibration, check and record the details below:

Sample preparation for loading

Take the supernatant sample from step 6.2 (run-wise) after Euglobulin precipitate separation step and divide it into two runs.

Adjust the conductivity of the sample (run-wise) to 1.4 ± 0.1 mS/cm with PF-RR-17.

Check and record the following conditions:

ml of plasma loaded/ml of resin: _____. (Limit: ≤ 1.12 ml/ml).

Loading

Before loading, switch on the 'Record On' button and enter the run details and date. Auto Zero the absorbance at 280 nm on the chromatography system.

Load the sample at a flow rate of 36.7ml/min.

Start collecting the flow through sample when the A280 rises from 100 mAU.

Wash-I

After loading is completed, wash the column with up to 2 CVs of PF-RR-22 (10% gradient of PF-RR-21 against PW) at a flow rate of 36.7ml/min; and collect the flow through till the A280 drops down to 100 mAU.

Check and record the details as below:

Elution

After the A280 has reached the baseline, pass up to 3 CVs of PF-RR-24 (12.5% gradient of PF-RR-23 against PW) at a flow rate of 36.7ml/min. Record the details:

Collect the eluate when the A280 rises from 200 mAU and collect the total peak until the A280 drops down to 200 mAU.

Check and record the following conditions:

Cleaning
Clean the column by passing 2 CVs of PF-RR-26 (25% gradient of PF-RR-25 against PW) in up-flow direction at a flow rate of 36.7ml/min.

Regeneration
Regenerate the column by passing 2 CVs of PF-RR-28 (25% gradient of PF-RR-27 against PW) in up flow direction at a flow rate of 36.7ml/min.

Repeat the steps 7.3 to 7.11 for Run 2. Record the results as above.

Storage
Store the column by passing 2 CVs of PF-RR-30 at a flow rate of 36.7ml/min. Volume of PF-RR-30 used: _____ L.

Pool Flow Through sample after completion of 2 runs for further processing of IG.

Final Volume of Flow Through: _____ L.

Start sending the Eluate sample after completion of 1runs for further processing of AL. Final Volume of Eluate: _____ L.

Details | Inj. 1 | Inj. 2 | Inj. 3 | Inj. 4

Volume of Load (L) | | | |

Volume of PF-RR-04 used (L) | | | |

Inj. No. | Sample | OD280/OD600 | pH | Cond. (mS/cm) | Total volume in L

1 to 4 | Frac. 1 | | | |

1 to 4 | Frac. 2 | | | |

1 to 4 | Frac. 3 | | | |

Details | Run

Volume of PW used (L) |

Details | Run

Volume of PF-RR-06 used (L) |

Details | Run

Final pH |

Final Conductivity (mS/cm) |

Volume of PF-RR-10 used (L) |

Details | Run

pH |

Conductivity (mS/cm) |

Volume of Load (L) |

Details | Run

Volume of PF-RR-05 used (L) |

Final pH |

Final Conductivity (mS/cm) |

OD280/600 |

Protein Conc. (mg/ml) |

Total Volume (L) |

Details | Run

Volume of PF-RR-10 used (L) |

OD280/600 |

Protein Conc. (mg/ml) |

Volume of Flow Through (L) |

Details | Run

Volume of PF-RR-29 used (L) |

Conductivity (mS/cm) |

Details | Run

Volume of PF-RR-14 used (L) |

Conductivity (mS/cm) |

Details | Run

OD280/600 |

pH |

Conductivity (mS/cm) |

Volume of Eluate (L) |

Details | Run

Volume of PF-RR-13 used (L) |

Details | Run

Volume of PF-RR-28 used (L) |

Run No. | Details | Load | Flow Through | Eluate

1. | Total volume (L) | | |

1. | Total protein (g) | | |

1. | Product | NA | IG & AL | FN

1. | Total protein (g) | | |

1. | Product | NA | IG & AL | FN

PF (bar) | PR (bar) | PP (bar) | Flow (L/hr) | TMP (bar) | Flux (LMH) | NWP

| | | | |

Permeate Vol. (L) | PF (bar) | PR(bar) | PP (bar) | Flow (L/hr) | TMP (bar) | Flux (LMH)

| | | | |

| | | | |

| | | | |

| | | | |

| | | | |

| | | | |

| | | | |

| | | | |

Details | Run

pH |

Conductivity (mS/cm) |

No. of Dia Filtration | Vol. of PF-RR-15 added (L) | Conductivity(μS/cm) of permeate

| |

| |

| |

| |

| |

| |

| |

Details | Run

OD280/600 |

pH |

Conductivity (μS/cm) |

Total Volume (L) |

PF (bar) | PR (bar) | PP (bar) | Flow (L/hr) | TMP

(bar) | Flux (LMH) | NWP
 |||||
 Details | Run
 pH |
 Cond. (μS/cm) |
 OD600 |
 Volume of PF-RR-15 used (L) |
 Volume of PF-RR-16 used (L) |
 Total volume of the sample (L) |
 Details | Euglobulin precipitation
 Incubation temperature (■C) |
 OD600 |
 Incubation Start Time & Date |
 Incubation End Time & Date |
 Total Incubation Hrs. |
 Details | Run 1 | Run 2
 pH ||
 Cond. (μS/cm) ||
 OD600 ||
 Separation Start Time & Date ||
 Separation End Time & Date ||
 Supernatant Volume (L) ||
 Details | Run 1 | Run 2
 PW used (L) ||
 Details | Run 1 | Run 2
 PF-RR-36 used (L) ||
 Final pH ||
 Details | Run 1 | Run 2
 pH ||
 Conductivity (mS/cm) ||
 PF-RR-22 used (L) ||
 Details | Run 1 | Run 2
 PF-RR-17 used (L) ||
 Final Volume (L) ||
 pH ||
 Conductivity (μS/cm) ||
 OD600 ||
 OD280 ||
 Protein Conc. (mg/ml) ||
 Total protein (g) ||
 Details | Run 1 | Run 2
 PF-RR-22 used (L) ||
 OD280 ||
 OD600 ||
 pH ||
 Conductivity (mS/cm) ||
 Volume of FT (L) ||
 Protein Conc. (mg/ml) ||
 Total protein (g) ||
 Recovery (%) ||
 Details | Run 1 | Run 2
 PF-RR-24 used (L) ||
 Details | Run 1 | Run 2
 OD280 ||
 OD600 ||
 pH ||
 Conductivity (mS/cm) ||
 Volume of Eluate (L) ||
 Protein Conc. (mg/ml) ||
 Total protein (g) ||
 Recovery (%) ||
 Details | Run 1 | Run 2
 PF-RR-26 used (L) ||
 Details | Run 1 | Run 2
 PF-RR-28 used (L) ||

Reagent Details:

Formula & Calculations:

Formula: Use the formula to calculate the molarity:

Molarity (M) = (Per cent by Weight / 100) * (Density in g/mL) * 1000 / Molar Mass.

Example: Molarity of Glacial Acetic Acid

Molar Mass of CH₃COOH: ~60.05 g/mol

Density: ~1.05 g/mL

Purity: 100%

Acetic Acid (MW = 60.05 g/mol).

Calculation:

Molarity = (100 / 100) * 1.05 g/mL * 1000 mL/L / 60.05 g/mol ≈ 17.5 M.

Calculation: To prepare 1 L of 0.2M.

Applying the formula to prepare the working buffer:

C₁V₁=C₂V₂

C₁ = 17.5 M acetic acid.

V₁ = volume of stock needed to take.

C₂ = desired (working) concentration 0.2 M acetic acid.

V₂ = final total volume of working solution for one litre.

V₁ = (C₂ x V₂)/C₁.

C₂=0.2 M; V₂= 1.0 L; C₁=17.5 M

V₁ = (0.2 x 1.0)/17,5

V₁ = 0.011 L

Take 0.011 L of the Stock solution of acetic acid and add it to a volumetric flask, making up the volume to 1 L.

To prepare various volumes:

V₁ = volume of stock needed to take _____ L.

C₂ = desired (working) concentration 0.2 M acetic acid.

V₂ = final total volume of working solution _____ L.

V₁ = (C₂ _____ x V₂ _____) /17.5.

V₁= _____ L.

Take V₁ _____ L of the Stock solution of acetic acid and add it to a volumetric flask, making up the final volume _____ L.

Check and record the details:

Preparation Details:

Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry

Date

||||

Final volume prepared | Measured Quantity | pH (2.8±0.1) | Conductivity (1.2± 0.1ms/cm) | Storage

Condition | Reagent Record

|||| RT |

Date of Preparation | Prepared By | Checked By

||

Reagent Details:
Formula & Calculations:
Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).
Calculation: To prepare 1 L of 0.1M Citric Acid (MW = 192.12g/mol)
Weight of solute(g)=1×19.212=19.212g.
Preparation Method:
Weigh 19.212 g of Citric acid and dissolve in water to make a 1 L solution.
Preparation for different volumes:
To prepare _____ of 0.1M Citric acid (MW = 192.12g/mol).
Weight of solute(g)=1×19.212 _____ L= _____ gm
Weigh _____ gm of Citric acid and dissolve in water to make a _____ L solution.
Check and record the details:
Preparation Details:
Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry
Date
| | | |
Final volume prepared | Weighed Quantity | pH (2.1±0.1) | Conductivity (27±3.0 ms/cm) | Storage
Condition:
| | | RT
Date of Preparation | Prepared By | Checked By
| |

SHAFT

3.30X2.45
FIREMAN

LIFT
LIFT
2.80X2.60

2.94 M WIDE CORRIDOR
LADIES TOILET
3.30 4.35

WC WC WC
1.10X 1.10X 1.10X
1.70 1.70 1.70

6.77 M WIDE CORRIDOR
WC WC WC
1.10X 1.10X 1.10X
1.70 1.70 1.70

GENTS TOILET
3.30 4.35

SHAFT SHAFT

LIFT LIFT

3.24 M WIDE PASSAGE

3.24 M WIDE PASSAGE
2.10X 2.10X

3.24 M WIDE PASSAGE
2.00 2.00

UP
FD
DIALYZER FD
UP

53.54 x 96.57

LADIES TOILET
3.30 4.35

WC WC WC
1.10X 1.10X 1.10X
1.70 1.70 1.70

WC WC WC
1.10X 1.10X 1.10X
1.70 1.70 1.70

GENTS TOILET
3.30 4.35
6.77 M WIDE CORRIDOR

6.77 M WIDE CORRIDOR
3.24 M WIDE PASSAGE
D1 D1 D1 D1 D1 D1 D1 D1 D1 D1

GENTS VISITOR FEMALE
SHAFT GENTS TOILET LADIES TOILET ANITOR SHAFT
WAITING ROOM LOCKER LOCKER LOCKER LIFT
9.94 4.90 6.30 4.90 ROOM
LIFT 4.67 5.00 ROOM ROOM ROOM 3.41 5.00 LIFT
5.90 5.00 6.00 5.00
3.24 M WIDE PASSAGE

2.10X 6.00 5.00
1D 1D 1D 1D 1D 1D 1D 1D 1D 1D
2.10X
2.00 D1 D1
2.00
FD FD FD FD
6D

RECEPTION CORRIDOR
UP 4.72 1.80 40.4 1.80 UP
8.30 12.0

PRIMARY VISITOR
AIR LOCK OFFICE 1 OFFICE 2 OFFICE 3
FEMALE PRIMARY FEMALE AIR LOCK VISITOR CHANGE 1
AIR LOCK PRIMARY MALE MALE CHANGE 1 FEMALE 1.80 5.15 3.69 5.10 3.66 5.10 3.69 5.10
CHANGE 1 CHANGE VISITOR CHANGE 2.75 5.10
MALE CHANGE 3.30 5.10 3.40 5.10 3.19 5.10 3.64 5.10 3.05 5.10 3.70 5.10
4.70 5.10 4.00 5.10

8D

6D

CORRIDOR
DE

49.0 2.25
DPB
DPB DPB WASTE OUT
LAF DHS
CHEMICAL DAY STORE 3 WAY
6D

6D

W 11.4 3.64 DECONTAMINATION WASHING STERILIZATION
3.60 5.43 W
AUTOCLAVE

2.33 2.14 3.11 5.43
4.67 5.43
LAF

BUFFER TANK STORAGE-2 CIP / SIP COLD ROOM
PREPARATION 6.74 7.50 5.60 7.50 5.87 7.50 COOL ZONE
4.00 7.50 3.25 7.50
DPB DPB
A/L GOWN A/L 6D
2.10 3.40 3.34 3.40 3.34 3.40
WASTE IN A/L A/L
GS+HH

GS+HH

LENAP LORTNOC LENAP LORTNOC
3.65 2.00 4.13 2.00 3.60 2.00 W
W 6D 6D 6D
DE

A/L DE GOWN A/L
2.10 3.40 3.34 3.40 3.34 3.40 CORRIDOR
39.8 3.04

D6 D6 D3
LAF
D6 D6
DPB AKTA-1" TFF
NANO
FILTER MH
ANITOR
1.80 2.00
LADIES TOILET A/L
A/L GOWN 1000 L
3.30 4.35 3.34 4.76
2.10 4.76 3.34 4.76 BULK FILTRATION W
LAF

6.54 7.29 500 L
WC WC WC
6.77 M WIDE CORRIDOR

1.10X 1.10X 1.10X
2.94 M WIDE PASSAGE

1.70 1.70 1.70 MH
SL MH

600 L

V1 V1 V1
500 L
LAF

MH SL
600 L MH

V1 V1 V1 MH

DE MH

50 L
A/L
WC WC WC A/L GOWN LAF MH

2000 L 500 L
SL
STAGE 13 E
1.10X 1.10X 1.10X 2.10 4.76 3.34 4.76 3.34 4.76 CHROMO
COLUMN 250 L
1.70 1.70 1.70 MH 600 L STAGE 10 T/ MH

W
STAGE 9 N 7.18 13.5
STAGE 11 F
MH

1000 L 250 L 5.33 2.05 250 L
MH

7.25 15.6 200 L
AKTA-1"

GENTS TOILET
STAGE 6 (I)
MH

MH

3.30 4.35

COLUMN PACKING MH

1000 L 15.0 18.0 MH

250 L

AND SKIDS
MH

200 L

6.54 10.6 250 L
COOL ZONE
MH MH

150 L 100 L MH

9.00 5.53 CHROMO
MH

500 L
COLUMN
MH

300 L
150 L MH

100 L MH

100 L
MH

300 L
LAF 150 L
W DPB LAF
MH

MH

100 L
CHROMO
COLUMN AKTA-1"
MH

W
AUTOCLAVE
DHS
D6

500 L 500 L CHROMO
250 L 250 L 150 L 100 L COLUMN
6D

MH
MH MH

CORRIDOR A/L MH MH
MH MH
MH

MH

CORRIDOR CORRIDOR
2.57 71.9 1.50 6.85
3.50 55.7 1.80 74.2
STERILIZATION 2000 L
D6

CLAF

2500 L 2500 L
7.40 6.85
6D

W W
PRE - VIRUS CONTROLLED CORRIDOR

1000 L
AKTA-1"

TFF

MH

COLUMN PACKING
DPB
AND SKIDS MH

WASHING 7.64 8.86
100 L
2.94 M WIDE PASSAGE

A/L
7.40 3.00
AKTA-1"

1.50 4.42
STAGE 7A 500 L
6.77 M WIDE CORRIDOR

MH

STAGE 5 (I) A 14.7 16.1
D6

DPB LADIES TOILET
WASTE IN MH 500 L 3.30 4.35
250 L
10.0 19.0
TFF

W MH
3.50 32.1

300 L
MH

WC WC WC
DECONTAMINATION 250 L
500 L 1.10X
1.70
1.10X
1.70
1.10X
1.70
7.40 5.00
MH

OUT MH
MH

WASTE V1 V1
D6

500 L
V1
1.50 3.54
DPB STAGE 5(I)B 1500 L 250 L
MH
MH

V1 V1 V1
7.64 7.20 100 L
MH

250 L 250 L 250 L
SL
600 L 500 L
WC WC WC
MH

MH
MH MH 100 L 100 L 100 L MH
MH

1.10X 1.10X 1.10X
100 L 1.70 1.70 1.70
MH MH MH

CIP / SIP
MH

W 1500 L
SL MH MH

9.00 5.00
MH MH MH

1000 L 100 L
MH
MH

500 L MH GENTS TOILET
500 L
6D

MH

1000 L 1000 L 1000 L 3.30 4.35
MH

1500 L
50 L 1500 L
GS+HH

GS+HH

SL
LENAP LORTNOC LENAP LORTNOC

MH
2500 L
1500 L

CHROMO MH
250 L 250 L 250 L 250 L
DPB COLUMN
6D

MH MH MH MH

WASTE OUT

BAG CUTTING &
POOLING MH
MH MH

MH
STAGE 4 (F)
W
MH MH

7.64 9.80 150 L 500 L 14.7 15.8
6D

500 L
W SL 1000 L
TANK STORAGE 1000 L 1500 L 1500 L

9.00 8.58
D6

MH

STAGE 12 (E)
MH

250 L
MH

STAGE 1 (F)/ 500 L
CHROMO
COLUMN
10.0 10.6 250 L
300 L 300 L MH
TFF
D6

AKTA-1"

MH MH

CLAF
W 6D
W
2500 L
6D

THAWING PLASMA CHROMO
COLUMN 2000 L 2000 L
7.64 3.61
AKTA-1"

MH

DPB
500 L 250 L 250 L 300 L 300

300L L
BUFFER MH MH

CHROMO
MH MH MH MH
MHMH

PREPARATION D6 D6 D6
COLUMN

9.00 6.00

SHAFT CORRIDOR SHAFT
6D

DE

DPB 36.2 2.50
LIFT LIFT
3.24 M WIDE PASSAGE

2.10X D6 D6 D6
2.10X
6D

MH MH

2.00 2.00
MH MH MH MH

CHROMO MH
MH
MH MH MH MH

COLUMN
100 L 100 L 100 L 100 L MH

100 L 100 L CENTRIFUGE
200 L 250 L
3.24 M WIDE PASSAGE

3.24 M WIDE PASSAGE

300 L 300 L 500 L 500 L
FD FD FD FD
1500 L
UP UP
PLASMA DAY STAGE 3 (F)
STORE STAGE 2 (F) 12.4 10.5
STAGE 8 N
9.00 10.7
5.84 10.5
17.7 10.5
2.66 2.29
AKTA-1"

AKTA-1"

UP
ANITOR
3D

CHROMO MH

3.50 5.40 COLUMN

SL
600 L 1500 L 2000 L
1500 L 1500 L 1500 L

300 L 300 L MH MH MH
MH
MH

100 L 100 L 100 L CHROMO CHROMO
DPB MH
TFF MH

COLUMN COLUMN AKTA-1"
MH MH MH

EMER.GLASS

W
W

PLASMA STORAGE PLASMA STORAGE PLASMA STORAGE
PLASMA STORAGE 1000 L 1000 L 1000 L
PLASMA STORAGE W
W 1000 L
70 KL GV
70 KL GV 70 KL GV 70 KL GV

1000 L
70 KL GV
CORRIDOR
1.80 15.2
10.5 15.2 9.36 15.2 10.3 15.2
9.10 15.2 9.52 15.2
CORRIDOR
2.57 17.0
6.77 M WIDE CORRIDOR

LADIES TOILET
3.30 4.35
W
WC WC WC
1.10X 1.10X 1.10X
1.70 1.70 1.70

V1 V1 V1 D3 D3 D3 D3 D3

CORRIDOR
2.94 M WIDE PASSAGE

49.1 1.80
V1 V1 V1
D6
WC WC WC CORRIDOR
1.10X 1.10X 1.10X
1.70 1.70 1.70
28.0 1.80

D3 D6 W
DPB RLAF
D3

AIR LOCK-1
GENTS TOILET 1.50 2.18 RM UARANTINE WASHING PLASMA COLD ROOM -4 COLD ROOM -3 COLD ROOM -2 COLD ROOM -1
ICE PACK
BAG BAR CODING ANITOR
AREA UARANTINE
3.30 4.35 MINI POOLS
5.04 7.34 5.02 7.34 5.04 7.34 2.50 7.34
6D

2.60 2.67
2.30 2.67
1.45 2.67 5.02 7.34
DISPENCING ROOM
3.40 4.44
3D D3 D3 D3

CHANGE ROOM
1.50 2.18 6.85 6.55
3D

5.15 2.72
CORRIDOR
11.6 4.30 1.80 12.0
D3
3D

SD SD SD SD

DPB W
6D

W D3
DPB
AIR LOCK-1
2.13 1.21
D3
DPB
3D D3 D6 STORE 12.9 6.49
AIR LOCK-1
APPROVED RM 3.54 2.60
1.87 1.40 OFFICE 4.70 4.09
3D
10.1 7.54 2.23 6.50
5.15 4.49 CHANGE ROOM 3D
3.00 3.60
2.13 1.80 CHANGE ROOM
1.87 1.40
RE ECT SEROLOGY PLASMA
DE-DUSTING
RECEIPT
D3

D3 TUNNEL
PLASMA ROOM TESTING 3D D3
STORE CHANGE
D3 SAMPLING
1.76 2.90
2.26 4.95 3.00 4.95 2.55 4.95 ROOM

RM 2.95 2.88
SAMPLING 3.68 2.05 RECALLED REJECT
RM FG DISCPATCH
3.26 2.31
RECEIPT GOODS ROOM FG IN
1.07 3.11 2.30 2.35 2.30 2.35
3.54 2.77
RLAF RLAF 6D D6 D6
D3 D3

D6 D3

CORRIDOR
3.30X2.45

3.30X2.45

53.9 3.17
6D

D6
LIFT

LIFT

DISPATCH BAY RECEIVING BAY RECEIVING BAY DISPATCH BAY
40' CONTAINER YARD
CONTAINER
CONTAINER

CONTAINER

CONTAINER
12.19 2.44 2.59
12.19 2.44 2.59

12.19 2.44 2.59

12.19 2.44 2.59

CLIENT :- LIFENITY-LARGE VOLUME PARENTERALS

DESIGN &
PROJECT BY :-BIOBRIDGE HEALTHCARE SOLUTION PVT.LTD.
PREPARED A.K
ADDRESS :-13 R B
A P -411007 M I .

GROUND FLOOR
CHECKED G.K.

DATE 18-04-2026
TITLE- CONCEPTUAL LAYOUT-GF REVISION R0
SCALE 1:1

SIZE A0 DWG.NO 01

Reagent Details:
Preparation of AL-RR-06 working buffer from Stock buffer (AL-RR-05):
Calculations: For the Preparation of 50mM Sodium acetate pH 4.5.
Stock Buffer (AL-RR-05):0.4M Sodium acetate, pH 4.6.
Formula:
 $C1 \times V1 = C2 \times V2$
C1 = concentration of stock
V1 = volume of stock you need to take
C2 = desired (working) concentration
V2 = final total volume of working solution
Applying the formula to prepare the working buffer:
 $C1 \times V1 = C2 \times V2$
C1 = 0.4M Sodium Acetate pH 4.6.
V1 = volume of stock you need to take _____ L.
C2 = desired (working) concentration 50mM Sodium Acetate pH 4.5
V2 = final total volume of working solution _____ L.
 $V1 = (C2 \times V2) / C1$.
C2=0.05M ; V2= 1.0 L; C1=0.4M
 $V1 = (0.05 \times 1.0) / 0.4$
V1 = 0.125L
Take 0.125 L (AL-RR-05) Stock buffer solution, dilute with water, and make up to 1 L.
Check and record the details:
Preparation Details:
Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry
Date
Sodium Acetate | | | |
Acetic Acid | | | |
Stock buffer (AL-RR-05) taken volume(L) | pH
(4.5±0.1) | Conductivity (2.0±0.2ms/cm) | Storage Condition | Final volume prepared
| | | RT |
Date of Preparation | Prepared By | Checked By
| |

Reagent Details:
Preparation of PF-RR-06 working buffer from Stock buffer (PF-RR-05):
Calculations: For the Preparation of 50mM Citric Acid.
Stock Buffer taken from (PF-RR-05):1.0 M Citric Acid.
Dilution Formula:
 $C1 \times V1 = C2 \times V2$
C1 = concentration of stock
V1 = volume of stock you need to take
C2 = desired (working) concentration
V2 = final total volume of working solution
Applying the formula to prepare the working buffer:
 $C1 \times V1 = C2 \times V2$
C1 = Concentration of Stock Citric Acid.
V1 = volume of stock you need to take _____ L.
C2 = desired (working) concentration 50mM Citric Acid.
V2 = final total volume of working solution _____ L.
 $V1 = (C2 \times V2) / C1$
C2=0.05M ; V2= 1.0 L; C1=1.0M
 $V1 = (0.05 \times 1.0) / 1.0$
V1 = 0.05L
Take 0.05 L (PF-RR-05) Stock buffer solution and dilute with water, and make up to 1L.
Check and record the details:
Preparation Details:
Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry
Date
| | | |
Final volume prepared | pH (2 to 3) | Conductivity (4 to 8 ms/cm) | Storage Condition(RT) | Reagent
Record
| | | PF-RR-06
Date of Preparation | Prepared By | Checked By
| |

[SHEET: GF_UTILITY_RDS]									
PROJECT LIFENITY FOR UTILITY RDS.									
Raw power GMP Drain Purified Water,									
(PW) Water For Injection,									
(WFI) Puer Steam,									
(PS) Soft Water,									
(SW) Hot Water,									
(HW) Chilled Water, (CW) Gases Compressed Air,									
CA Black Steam									
(BS)									
Sr.									
No. Room No. Room Name Equipment Name and									
Capacity QTY Electrical Load (KW) Three Phase Total Load (KW) Single Phase Electrical Load (KW)									
Single Phase Total Load (KW) UPS Load (KW) Floor Drain Condensate Drain Process Drain POU									
Liter Per Hour. Liter Per Day POU Liter Per Hour. Liter Per Day Temp. POU kg/hr Temp.									
POU Liter Per Hour. Liter Per Day Temp. POU POU Hydrogen Nitrogen POU BAR POU kg/hr									
Temp.									
1 Reception Room NA 0 0 0 0 0 0 Y NA NA NA NA NA NA NA NA NA NA NA									
NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA									
2 Waiting Room NA 0 0 0 0 0 0 Y NA NA NA NA NA NA NA NA NA NA NA									
NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA									
3 Passage NA 0 0 0 0 0 0 Y NA NA NA NA NA NA NA NA NA NA NA									
NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA									
4 Gents Toilet Room NA 0 0 0 0 0 0 Y NA NA NA NA NA NA NA NA NA NA NA									
NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA									
5 Gents locker Room NA 0 0 0 0 0 0 0 NA NA NA NA NA NA NA NA NA NA NA									
NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA									
6 Visitor Locker Room NA 0 0 0 0 0 0 0 NA NA NA NA NA NA NA NA NA NA NA									
NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA									
7 Female Locker Room NA 0 0 0 0 0 0 0 NA NA NA NA NA NA NA NA NA NA NA									
NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA									
8 Ladies Toilet NA 0 0 0 0 0 0 Y NA NA NA NA NA NA NA NA NA NA NA									
NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA									
9 Female Change Room-1 NA 0 0 0 0 0 0 0 NA NA NA NA NA NA NA NA NA NA NA									
NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA									
10 Primary Female change Room NA 0 0 0 0 0 0 0 NA NA NA NA NA NA NA NA NA NA									
NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA									
11 Air lock Female NA 0 0 0 0 0 0 0 NA NA NA NA NA NA NA NA NA NA NA									
NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA									
12 Visitor Change Room NA 0 0 0 0 0 0 0 NA NA NA NA NA NA NA NA NA NA NA									
NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA									
13 Primary Visitor Change Room NA 0 0 0 0 0 0 0 NA NA NA NA NA NA NA NA NA NA									
NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA									
14 Air Lock Visitor NA 0 0 0 0 0 0 0 NA NA NA NA NA NA NA NA NA NA NA									
NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA									
15 Male Change-1 NA 0 0 0 0 0 0 0 NA NA NA NA NA NA NA NA NA NA NA									
NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA									
16 Primary Male Change Room NA 0 0 0 0 0 0 0 NA NA NA NA NA NA NA NA NA NA									
NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA									
17 Air Lock Male NA 0 0 0 0 0 0 0 NA NA NA NA NA NA NA NA NA NA NA									
NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA									
18 Corridor NA 0 0 0 0 0 0 Y NA NA NA NA NA NA NA NA NA NA NA									
NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA									
19 Cold Room NA 1 5.5 5.5 0 0 0 0 NA NA NA NA NA NA NA NA NA NA NA									
NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA									
20 Corridor NA 0 0 0 0 0 0 Y NA NA NA NA NA NA NA NA NA NA NA									
NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA									
21 Passage NA 0 0 0 0 0 0 0 NA NA NA NA NA NA NA NA NA NA NA									
NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA									
22 Janitor Room NA 0 0 0 0 0 0 0 Y NA NA NA NA NA NA NA NA NA NA NA									
NA NA Y NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA									
23 Office-1 Desktop 2 0 0 0.07 0.14 0.14 NA NA NA NA NA NA NA NA NA NA NA									
NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA									
24 Office-2 Desktop 2 0 0 0.07 0.14 0.14 NA NA NA NA NA NA NA NA NA NA NA									
NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA									
25 Office-3 Desktop 2 0 0 0.07 0.14 0.14 NA NA NA NA NA NA NA NA NA NA NA									
NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA									
26 Corridor NA 0 0 0 0 0 0 Y NA NA NA NA NA NA NA NA NA NA NA									
NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA									
27 Air Lock NA 0 0 0 0 0 0 0 NA NA NA NA NA NA NA NA NA NA NA									
NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA									
28 Gown Room NA 0 0 0 0 0 0 0 NA NA NA NA NA NA NA NA NA NA NA									
NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA									
29 Air lock NA 0 0 0 0 0 0 0 NA NA NA NA NA NA NA NA NA NA NA									
NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA									
30 Air Lock NA 0 0 0 0 0 0 0 NA NA NA NA NA NA NA NA NA NA NA									
NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA									
31 De- Gown room NA 0 0 0 0 0 0 0 NA NA NA NA NA NA NA NA NA NA NA									
NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA									
32 Air Lock NA 0 0 0 0 0 0 0 NA NA NA NA NA NA NA NA NA NA NA									
NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA									
33 Corridor NA 0 0 0 0 0 0 Y NA NA NA NA NA NA NA NA NA NA NA									
NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA									
34 Chemical day store DPB 1 0 0 0.5 0.5 0 NA NA NA NA NA NA NA NA NA NA									
NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA									
35 Corridor NA 0 0 0 0 0 0 Y									
36 Buffer preparation Tank-1500 L 1 1.2 1.2 0.1 0.1 Y Y NA 1 1000 5000									
1 1500 800 80 1 270 121 NA NA NA NA NA 1 NA NA 1 06-08 bar 1 270									
110									
37 Tank-1000 L 1 1 1 0.1 0.1 0.1 NA NA NA NA 1000 3000 NA 1500 500									

BioBridge Master Knowledge — Page 3000

[illegible]

BioBridge Master Knowledge — Page 3002

BioBridge Master Knowledge — Page 3003

BioBridge Master Knowledge — Page 3004

[illegible]

BioBridge Master Knowledge — Page 3006

BioBridge Master Knowledge — Page 3007

| NA | NA | NA | NA | NA | Y | NA | NA | NA | NA | NA | Y | Y | NA | NA | NA | NA | NA |
||| HPLC Columns and accessories | 1 | 0 | 0 | 0 | 0 | 0 | NA | NA | NA | NA | NA | NA | NA | NA | NA |
NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA |
||| Electrophoresis (Minigel) with trans blot | 1 | 0 | 0 | 0.1 | 0.1 | 0 | NA | NA | NA | NA | NA | NA | NA |
NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA |
||| Gel Doc system densitometer | 1 | 0 | 0 | 0.1 | 0.1 | 0 | NA | NA | NA | NA | NA | NA | NA | NA | NA |
NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA |
||| Power packs | 1 | 0 | 0 | 0.1 | 0.1 | 0 | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA |
NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA |
||| Microplate reader ELISA | 1 | 0 | 0 | 0.1 | 0.1 | 0 | NA | NA | NA | NA | NA | NA | NA | NA | NA |
NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA |
||| Centrifuge table top | 1 | 0 | 0 | 0.2 | 0.2 | 0 | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA |
NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA |
||| Timer and stop watch | 1 | 0 | 0 | 0 | 0 | 0 | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA |
NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA |
||| Microplate washer | 1 | 0 | 0 | 0.1 | 0.1 | 0 | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA |
NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA |
||| Gas chromatography | 1 | 0 | 0 | 0.7 | 0.7 | 0.7 | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA |
NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA |
||| Atomic absorption Spectrophotometer | 1 | 0 | 0 | 0.2 | 0.2 | 0 | NA | NA | NA | NA | NA | NA | NA | NA |
NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA |
||| Kjeldhal apparatus | 1 | 0 | 0 | 0.1 | 0.1 | 0 | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA |
NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA |
||| Zone electrophoresis | 1 | 0 | 0 | 0.1 | 0.1 | 0 | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA |
NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA |
||| pH and conductivity meter | 1 | 0 | 0 | 0.1 | 0.1 | 0 | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA |
NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA |
||| Magnetic Stirrer with hot plate | 1 | 0 | 0 | 0.25 | 0.25 | 0 | NA | NA | NA | NA | NA | NA | NA | NA |
NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA |
||| Lab Refrigerator 2-8 ° C (SS) with data control | 1 | 0 | 0 | 0.2 | 0.2 | 0.1 | NA | NA | NA | NA |
NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA |
NA | NA |
47 || TOC/ LPC | Toc analyzer | 1 | 0 | 0 | 0.5 | 0.5 | 0 | Y | NA | NA | NA | NA | NA | NA | NA | NA |
NA | NA | NA | NA | NA | Y | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA |
||| Liquid particle counter | 1 | 0 | 0 | 0.1 | 0.1 | 0 | NA | NA | NA | NA | NA | NA | NA | NA | NA |
NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA |
||| LAF | 1 | 0 | 0 | 0.5 | 0.5 | 0 | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA |
NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA |
48 || Chemical/ Glass ware Storage | 0 | 0 | 0 | 0 | 0 | 0 | NA | NA | NA | NA | NA | NA | NA | NA |
NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA |
49 || Corridor | NA | 0 | 0 | 0 | 0 | 0 | 0 | Y | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA |
NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA |
50 || R&D Entrance | 0 | 0 | 0 | 0 | 0 | 0 | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA |
NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA |
51 || R&D Office | Desktop | 3 | 0 | 0 | 0.07 | 0.21000000000000002 | 0.21 | NA | NA | NA | NA | NA | NA |
NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA |
52 || Passage | 0 | 0 | 0 | 0 | 0 | 0 | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA |
NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA |
53 || R&D Lab-1 | 0 | 0 | 0 | 0 | 0 | 0 | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA |
NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA |
54 || R&D Lab-2 | 0 | 0 | 0 | 0 | 0 | 0 | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA |
NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA |
55 || R&D Lab-3 | 0 | 0 | 0 | 0 | 0 | 0 | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA |
NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA |
56 || R&D Lab-4 | 0 | 0 | 0 | 0 | 0 | 0 | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA |
NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA |
TOTAL |||| 89 | 5 | 5 | 42.930000000000004 | 44.640000000000004 | 3.9500000000000006 | 0 | 0 | 0 | 0 | 0
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 110 | 1200 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0
[SHEET: FINAL SHEET]
PROJECT LIFENITY FOR UTILITY RDS. ||||||||||||||||||||||||||||||||
||||| Electrical ||||| GMP Drain ||| Purified Water,
(PW) ||| Water For Injection,
(WFI) |||| Pure Steam,
(PS) ||| Soft Water,
(SW) |||| Hot Water,
(HW) | Chilled Water, (CW) | Gases || Compressed Air,
CA || Black Steam
(BS) ||
Sr.
No. | NAME ||| QTY | Electrical Load (KW) Three Phase | Three Phase Total Load (KW) | Single Pase
Electrical Load (KW) | Total Load (KW) | UPS Load (KW) | Floor Drain | Condensate Drain | Process Drain |
POU | Liter Per Hour. | Liter Per Day | POU | Liter Per Hour. | Liter Per Day | Temp. | POU | kg/hr |
Temp. | POU | Liter Per Hour. | Liter Per Day | Temp. | POU | POU | Hydrogen | Nitrogen | POU | BAR | POU
| kg/hr | Temp.
A | Production Block ||||||||||||||||||||||||||||
1 | GF DS UTILITY RDS ||| 128 | 130.50000000000003 | 130.50000000000003 | 31.030000000000022 |
31.310000000000027 | 9.019999999999985 | 0 | 0 | 0 | 32 | 55500 | 195600 | 37 | 108850 | 53150 | 5740 |
20 | 10230 | 6905 | 0 | 300 | 600 | 0 | 0 | 18 | 0 | 0 | 39 | 0 | 18 | 10130 | 6182.2
2 | FF_Fill & Finish_UTILITY_RDS ||| 34 | 36.5 | 36.5 | 17.4 | 17.4 | 0.8999999999999999 | 0 | 0 | 0
| 3 | 400 | 550 | 2 | 600 | 200 | 0 | 1 | 0 | 242 | 0 | 0 | 0 | 0 | 0 | 0 | 3 | 0 | 1 | 190 | 110
4 | FF_QC & QA_RDS |||| 89 | 5 | 5 | 42.930000000000004 | 42.930000000000004 | 44.640000000000004 |
3.9500000000000006 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 110 | 1200 | 0 | 0 | 0 | 0 |
0 | 1 | 0 | 0 | 0
| TOTAL |||| 172.00000000000003 | 172.00000000000003 | 91.36000000000007 | 263.3600000000001 |
54.56000000000003 | 3.9500000000000006 | 0 | 0 | 0 | 55900 | 196150 | 109450 | 53350 | 5740 | 10230 ||
||||||||| 10320 |
| Diversity 60% | 0.6 ||||| 158.01600000000008 | 32.73600000000002 |||| 33540 | 117690 ||
65670 | 32010 || 6138 ||||| 6192 |
Proposed Capacity (Round off)					200																			
Location of Following Utilities at Service Floor	Capacity	Quantity	KW																					
Brine Chiller(-10C/-15C) Air Cooled	50TR	2	100				200																	

|||||
 | Purified Water Pump | 5000 M3/hr | 1 | 40 |||| 40 |||||
 ||
WFI (MCDP) or Vapour Thermo compression	3000 M3/hr	1	10				10				
PSG Pure Steam Generator	500 Kg/Hr	2	5				10				
Hot Water Generator Pump	100000 Kcal/Hr	2	5				10				
Nitrogen	50 M3	1	50				50				
Co2	20 M3	1	20				20				
AHU Load	10 KW	55	5				275				
Brine Circulation Pump	As per Chiller Requirement	2	10				20				
Vaccume Pump		2	15				30				
Scrubber		4	10				40				
Hydrogen		1	2				2				
B	Admin Block										
Admin Block Lighting and Domestic Power Point					125						
Admin Block HVAC					150						
Admin Block Equipments for Canteen & Conference Training					20						
C	Utility Block / Boiler House / Pump House / Fire / ETP										
Utility Block / Boiler House Lighting					10						
Chiller (7/12 C)Water Cooled (HVAC)	300 TR	2	200				400				
Process Chiller (-5C /- 10C)Brine Water Cooled	100TR	2	150				300				
Cooling Tower Process Water	200000 K/Cal	2	15				30				
Cooling Tower for AHU Chiller	300TR	2	15				30				
Pre Treatment of Raw Water including Softning Plant Pump	10 M3 / hr	2	15				30				
Compressed Air with receiver and air dryer and filters	600 CFM at 8 Bar	2	150				300				
Compressed Air with receiver and air dryer and filters	300 CFM at 8 Bar	2	75				150				
Steam Boiler Dual Fuel	3000 Kg/hr	1	15				15				
Steam Boiler	1000 Kg/hr	1	7.5				7.5				
Hot Water Generator Pump	100000 Kcal/Hr	2	5				10				
Domestic Pumps	2000 Ltr/hr	2	7.5				15				
ETP	50000 Ltrs	1	15				15				
STP Pump	20000 Ltrs	1	5				5				
Fire Pumps (Electrical + Deseal + Jocky)	1 Set	1	75				75				
Nitrogen	50 M3	1	50				50				
Co2	20 M3	1	20				20				
Chilled Water Pumps (HVAC)	As per Chiller Requirement	3	15				45				
Chilled Water Pumps (Process)	As per Chiller Requirement	3	10				30				
Condenser/ Cooling Water Pump (HVAC)	As per Chiller Requirement	3	20				60				
Condenser/ Cooling Water Pump (Process)	As per Chiller Requirement	2	10				20				
Fuel Transfer Pump		2	1				2				
D	Security Office Lighting and HVAC					10					
E	Scrap Yard Lighting					5					
F	Street Light					20					
G	Misc. Load					125					
GRAND TOTAL OF CONNECTED LOAD				KW				3044.86	KW		
TAKING OVERALL DIVERSITY					0.6						
GRAND TOTAL OF DIVERSIFIED LOAD				KW				1826.916	KW		
CONSIDERING PF 0.85 (i.e. LOAD REQUIREMENT IN KVA)				KVA				2149.3129411764708	KVA		
CONSIDERING 80% CONTINUOUS LOADING ON TRANSFORMER				KVA				2686.641176470588	KVA		
Recommended Transformer Rating (At 3Ø, 380 Volt)				KVA				2 Nos. x 1400	KVA		
CONSIDERING PF 0.80 (i.e. LOAD REQUIREMENT IN KVA)				KVA				2283.645	KVA		
CONSIDERING 90% LOADING ON DG				KVA				2537.383333333333	KVA		
RECOMMENDED DG RATING				KVA				2Nos. x 1300	KVA		

[SHEET: LVP]

||| Calculation for ETP and STP ||| Calculation for ZLD |||

Water generation ||| Parameter | Capacity | UOM || Parameter | Receovery % | Quantity | UOM

Raw water | 105000 | L || Capacity per year | 36000000 | Units || RO permeate (I) | 0.7 |

31323.561643835616 | L per day

Soft water | 99750 | L || Number of days | 365 | Days || MEE condensate (II) | 0.2 | 8949.589041095891

| L per day

PW | 64837.5 | L || Unit quantity | 0.5 | L || Losses (salts, sludge) (III) | 0.1 | 4474.7945205479455

| L per day

WFI | 58353.75 | L || Production per day | 49315.068493150684 | L per day || || |

||| CIP or rinse | 15000 | L per day || Final water balance || |

||| Total process water requirement | 64315.068493150684 | L per day || Input Raw water (D) | 105000

| L per day |

||| Raw water required per day | 105000 | L per day || Output after ZLD || |

||| Raw water required annually | 38325000 | L per Year || Recycled (G=I+II) | 40273.150684931505 |

L per day |

||| Raw water required per day (A) | 105000 | L per day || Losses (III) | 4474.7945205479455 | L per day |

||| Product | Output | UOM || Recycled (%) | 38.35538160469667 | % |

||| Product | 49315.068493150684 | L per day || % loss (salts, sludge) | 4.261709067188519 | % |

||| Product Output per day (B) | 49315.068493150684 | L per day || Liquid discharge (D-G-E-III-B+C)

| 0 | L per day |

||| QC (C) | 250 | L per day || || |

||| Waste water (D=A-B+C) | 55934.931506849316 | L per day || || |

||| Evaporation Losses (E) | 11186.986301369863 | 20 % of D || || |

||| Input for ETP (F=D-E) | 44747.94520547945 | L per day || || |

||| Number of persons | 400 | No. || || |

||| Water required per person per day | 35 | L/P/D || || |

||| Input for STP | 14000 | L per day || || |

[SHEET: Plasma]

||| Calculation for ETP and STP ||| Calculation for ZLD |||

Water generation ||| Parameter | Capacity | UOM || Parameter | Receovery % | Quantity | UOM

Raw water | 250000 | L || Plasma processing per year | 150000 | L || RO permeate (I) | 0.7 |

57645.72017534246 | L per day

Soft water | 237500 | L || Number of days | 365 | Days || MEE condensate (II) | 0.2 | 16470.20576438356

| L per day

PW | 154375 | L || Batch size | 1000 | L || Losses (salts, sludge) (III) | 0.1 | 8235.10288219178 | L

per day

WFI | 138937.5 | L || Number of batches per year | 150 | Nos || || |

||| Raw water required per batch | 250000 | L || Final water balance || |

||| Number of days per batch | 5 | Days || Input Raw water (D) | 102739.72602739726 | L per day |

||| Raw water required per batch per day | 50000 | L per batch per day || Output after ZLD || |

||| Raw water required annually | 37500000 | L per Year || Recycled (G=I+II) | 74115.92593972602 | L

per day |

||| Raw water required per day (A) | 102739.72602739726 | L per day || Losses (III) |

8235.10288219178 | L per day |

||| Product | Output | UOM || Recycled (%) | 72.13950124799999 | % |

||| Albumin | 100 | L per batch || % loss (salts, sludge) | 8.015500138666667 | % |

||| IVIG | 150 | L per batch || Liquid discharge (D-G-E-III-B+C) | 1.3244516594568267e-11 | L per

day |

||| FIX | 2 | L per batch || || |

||| FVIII | 0.2 | L per batch || || |

||| Thrombin | 2 | L per batch || || |

||| Fibrinogen | 0.5 | L per batch || || |

||| Total output | 254.7 | L per batch || || |

||| Product Output per day (B) | 50.94 | L per day per batch || || |

||| QC (C) | 250 | L per day || || |

||| Waste water (D=A-B+C) | 102938.78602739726 | L per day || || |

||| Evaporation Losses (E) | 20587.757205479455 | 20 % of D || || |

||| Input for ETP (F=D-E) | 82351.0288219178 | L per day || || |

||| Number of persons | 506 | No. || || |

||| Water required per person per day | 35 | L/P/D || || |

||| Input for STP | 17710 | L per day || || |

[D726] Budhish-20260806T040948Z-1-001(1)/Budhish/Lifenity/Buffer Annexure/PF Annexure/Annexure-003 PF-RR-03 preparation of buffer solution.

Reagent Details:

Formula:

Sodium Citrate (M.Wt): 258.07.

Citric acid (M.Wt): 192.124

Sodium Chloride (M.Wt): 58.44

Calcium Chloride (M.Wt): 110.98

Preparation of PF-RR-03 stock buffer:

Preparation of 0.2M Sodium Citrate pH 7.0 + 1.2M Sodium Chloride + 26mM Calcium Chloride.

Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).

Calculation: To prepare 1 L of 0.2M Citric Acid (MW = 192.124 g/mol)

Weight of solute(g)=0.2×192.124×1=38.424g.

Preparation Method:

Weigh 38.424g of Citric Acid and dissolve in water to make a 1 L solution.

Preparation for differential volumes:

To prepare _____ of 0.2M Citric Acid (MW = 38.424g/mol).

Weight of solute(g)=0.2×38.424 x _____ L= _____g.

Weigh _____ gm of Citric Acid and dissolve in water to make a _____ L solution.

Calculations: To prepare the 0.2M Tris Sodium Citrate.

Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).

Calculation: To prepare 1 L of 0.2M Tris Sodium Citrate (MW = 258.07 g/mol)

Weight of solute(g)=0.2×258.07 x1=51.614g

Preparation Method:

Weigh 51.614g of Tris Sodium Citrate and dissolve in water to make a 1 L solution.

Preparation for different volumes:

To prepare _____ of 0.2M Tris Sodium Citrate (MW = 258.07 g/mol).

Weight of solute(g)=0.2×258.07x _____ L= _____g.

Weigh _____ gm of Tris Sodium Citrate and dissolve in water to make a _____ L solution.

Calculations: To prepare the 1.2M Sodium Chloride.

Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).

Calculation: To prepare 1 L of 1.2M Sodium Chloride (MW = 58.44 g/mol)

Weight of solute(g)=1.2×58.44x1=70.128g.

Preparation Method:

Weigh 70.128g of Sodium Chloride and dissolve in water to make a 1 L solution.

Preparation for different volumes:

To prepare _____ L of 1.2M Sodium Chloride (MW = 58.44 g/mol).

Weight of solute(g)=1.2×58.44 x _____ L= _____g.

Weigh _____ gm of Sodium Chloride and dissolve in water to make a _____ L solution.

Calculations: To prepare the 26mM Calcium Chloride.

Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).

Calculation: To prepare 1 L of 26 mM Calcium Chloride (MW = 110.98g/mol)

Weight of solute(g)=0.026×110.98 x1=2.88g.

Preparation Method:

Weigh 2.88g of Calcium Chloride and dissolve in water to make a 1 L solution.

Preparation for different volumes:

To prepare _____ L of 0.026mM Calcium Chloride (MW =110.98 g/mol).

Weight of solute(g)=0.026 x110.98x _____ L= _____g.

Weigh _____ gm of Calcium Chloride and dissolve in water to make a _____ L solution.

Procedure for Preparation of PF-RR-03 stock

To prepare a pH 7.0 solution, 0.2 M Tris Sodium Citrate solution is titrated with 0.2 M citric acid so that the pH attains pH 7.0 ±0.1.

For Example, to prepare a 1 L solution, take 0.2 M Tri Sodium Citrate and adjust the pH with the 0.2 M Citric acid to pH 7.0. To this, add the 1.2 M sodium chloride and dissolve it and make up the volume with water to 900ml, and then to this add 0.026 M Calcium chloride, then mix the solution till the salt dissolves and check the pH; if the pH is less than 7.0 then adjust the pH with 1 M sodium hydroxide to get the pH 7.0 ±0.1 and make up the final volume to 1L.

Check and record the details:

Preparation Details:

Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry Date

Sodium Citrate | | | |

Sodium Chloride | | | |

Calcium Chloride | | | |

Final volume prepared | pH (7.0 ± 0.1) | Conductivity (130 to 150 ms/cm) | Storage Condition | Reagent

Record

| | | RT | PF-RR-03

Date of Preparation | Prepared By | Checked By

| |

ROAD

FOR OFFICE USE

1.200

T-300 MM

TERRACE

WIDE

R-150 MM

5.100

FORM OF STATEMENT

5.100

AREA CALCULATIONS

60.00 M.

5.100

5.100

5.100

BLOCK NO BLOCK NAME FLOOR SIZE AREA

1 BUILDING-01 ADMIN GROUND 4033.57

FIRST 3478.10

8.553

8.550

8.550

SECOND 4033.57

ROAD

T-0.30m.

R-0.15m.

WIDE

D

60.00 M.

T-0.30m.

R-0.15m.

LABORATORY 1.200

A

2 BUILDING-02 GROUND 13673.472

60

O

FIRST .00 SECTION A-A

ELEVATION BUILDING 2

R

3 13673.472 M.

E

W

PLOT NO. 1B BUILDING-2

ID

4 SECOND 7946.736 ID SECTION BB

ER

W

TOTAL 46838.92 O RIGHT SIDE

M.

5 AD

.00

TOTAL B/UP AREA OF ALL FLOORS+ EXTRA HEIGHT 46838.92

30

PROPOSED SITE

CANOPY ABOVE CANOPY BELOW

60 SHAFT

RECEIVING BAY DISPATCH BAY DISPATCH BAY DISPATCH BAY

SHAFT SHAFT SHAFT SHAFT SHAFT

.00 FIREMAN

LIFT

2.80X2.60

LIFT

3.30X2.45

FIREMAN

LIFT

2.80X2.60

LIFT
3.30X2.45
FIREMAN
LIFT
2.80X2.60
LIFT
3.30X2.45
M.
W CORRIDOR
LIFT
2.10x
LIFT
2.10x
LIFT
2.10x
ID
PLOT NO. 1C
UP 1.90 UP 1.90 1.90
UP

E DISPATCH / RECEIVING BAY LIFT LIFT LIFT
R 2.10X 2.10X 2.10X
O 2.00 2.00 2.00

WIDEROAD
A
D 703 sq.mt.
45.00 M.

OAD
45.00 M. W I D E R
30.00 M. W I D E R O A D

FORM OF STATEMENT AREA CALCULATION
PLAN OF PROPOSED FACTORY
SECURITY CABIN 1,2&3 189.00 -- -- 189.00
BUILDINGON PLOT NO 8A, SECTOR 13,
DRIVER ROOM 60.00 --- -- 60.00
DMIC, AURIC , SHENDRA,
PUMP ROOM 57.11 -- -- 57.11
CHATRAPATI SAMBHAJI NAGAR
FOR M/S LIFENITY LIMITED
LOCATION PLAN OCCUPANT LOAD AS PER TABLE NO 8E
CLAUSE NO. 10.6.3 CDCPR WATER REQUIREMENT
NOS.

TOTAL B/UP AREA = 47494.288 SQ.M. DIALYZER DIALYZER TERRACE
400NOS.

SHAFT SHAFT SHAFT

FIREMAN FIREMAN FIREMAN
OCCUPANT LOAD CALATION OCCUPANT LOAD = 47494.288/10 = 4749 NOS. SHAFT SHAFT SHAFT
NO OF PERSONS

LIFT LIFT LIFT

SCALE :- NTS OCCUPANT LOAD actual = 400 NOS. 59.60 x 83.80 2.80X2.60
59.60 x 95.80 2.80X2.60
59.60 x 95.80 2.80X2.60
STAFF --100

FD FD FD FD FD FD FD FD FD

BUILT UP AREA 47494.288 SQMT NO ACTIVITY WATER IN LITS. UP UP UP UP UP UP
OCCUPANTS

AREA PER PERSON 10 SQMT
1 FOR W.C. 30 X 400 = 12000
UP
CARPET AREA 4994 .48 SQ.MT. CARPET AREA 5709.68 SQ.MT.
2 FOR INDUSTRIAL 100000 LITS/DAY UP

4749.42
OFFICE

NOS TOTAL /DAY 112000 LITR/DAY

60.00 M.
OFOF

AS PER ACTUAL OCCUPANTS IN THE BUDILING FOR DOMESTIC = 25000 LTR.
NO.

FOR INDUSTRIAL = 200000 LTR.
NO.

NO. OF WORKERS - 300 NOS.
TERRACE AREA STATEMENT
WIDE
RCC SLAB

SANITATION REQUIREMENT (TABLE NO. 10D) G.L. G.L.
BELOW
2.5MX2.5MX3M BAFFLE WALL 5MX2.5MX3M DESILTING AREA IN
RAIN WATER HARVESTING OCCUPANT LOAD 47494 / 10 4749.40 NO PARTICULARS
CHAMBER SQM.
PIT
ROAD
PCC BED
TOTAL NO. OF PERSON 4749 SECTION AT CC
U.G. WATER TANK
FIXTURES OFFICES / VISITORS WORKERS 1 AREA OF PLOT NO 56 SECTOR 13, DMIC, SHENDRA 81920.00
100 300
50 50 200 100
AS PER OWNERSHIP DOCUMENTS 81920.00
MALE FEMALE MALE FEMALE
25X25MM 112000 LTR. 112000 LTR.
1.2MX2.5MX1.5M 1 FOR 25 1 FOR 15 1 FOR 20 1 FOR 15
AS PER MEASUREMENT SHEET
MANHOLE
600 MM x 600
MM
BAR SCREEN REQUIRED WC
after 2.5% thereafter 3% thereafter 2.5% thereafter 4% 81920.00
GREASE TRAP 1+1 = 2 1+2=3 1+5=6 1+4=5 U.G. WATER TANK PLAN
1 up to10
thereafter 2.5%
-
1 up to10
thereafter 2.5%
- AS PER SITE 81920.00
53.285 X
MANHOLE

X
REQUIRED URINALS
1+1 = 2 1+5 = 6
2 DEDUCTIONS FOR
REQUIRED

46.9 PROVIDED
SHAFT

FIREMAN
LIFT SHAFT
FIREMAN
LIFT
SHAFT

SHAFT
SHAFT

FIREMAN
LIFT SHAFT a. PROPOSED DP ROAD 0.00
72
UP 2.80X2.60 2.80X2.60 2.80X2.60

NO. OF W.C. FOR STAFF (MALE) 2 2 FD FD FD FD FD FD FD FD FD

NO. OF URINALS FOR STAFF 2 2 SEPTIC TANK REQUIREMENT UP UP UP UP UP UP

NO. OF W.C. FOR STAFF (FEMALE) 3 3 TOTAL B/UP AREA = 47494.288 SQ.M. b. ANY RESERVATIONS 0.00
OCCUPANT LOAD = 47494/10 = 4749 NOS. UP
NO. OF W.C. FOR WORKERS (MALE) 6 6
OCCUPANT LOAD actual = 400 NOS.
CAPACITY 3.00X 3.00 X 1.50 X 3 nos= 40500 LITS

RWHP FLOOR PLAN - NO. OF URINALS FOR WORKERS 6 6 TOTAL (a+b) 0.00
GATE-2 NO. OF W.C. FOR WORKERS
(FEMALE)
5 5 NO ACTIVITY
1 FOR W.C. & URINALS
WATER IN LITS.
30 X 400 = 12000
SCALE - 1:100
ENTRY 3 BALANCE AREA OF PLOT (1-2) (A) 81920.00
SECURITY

TOTAL /DAY 12000 LITS/DAY
CABIN-2

DETENTION 12000 X 2 = 24000

DRIVER'S PARKING REQUIREMENTS CLAUSE NO 7.20 MIDC CDCPR
PERIOD 2 DAYS

SLUDGE SETTLEMENT / PERSON = 30 LIT/YR.
4 AMENITY SPACE IF APPLICABLE 5% 0.00
AS PER TABLE 7C AND 7D FOR 47494.288 SQM. MULTIPLICATION FACTOR 0.8
ROOM SEPTIC TANK
TYPE OF VEHICLE PARKING SIZE REQUIRED PARKING REQUIRED REQUIRED PROVIDED
CLEANING PERIOD = 1 YEAR
TOTAL SLUDGE = 30 X 400 = 12000 LITS

SHAFT

LIFT
3.30X2.45
SHAFT

LIFT
3.30X2.45
LIFT
SHAFT

3.30X2.45
a. Required 0.00
CAR 2.5M X 5.0M 1 FOR EVERY 200SQM 237.47 0.80 190 TANK CAPACITY = 24000 +12000 = 36000 LITS LIFT LIFT
LIFT
SCOOTER / MOTOR CYCLE 2.0 SQ.M. (1.0M X 2.0M) 2 FOR EVERY 200SQM 474.94 0.80 380 100 2.10x 2.10x 2.10x
1.90 UP 1.90 1.90
CYCLE
TRUCK
VENT PIPE
LIFT
2.10X
2.00
LIFT
2.10X
2.00
LIFT
2.10X
2.00
b. Adjustment of 2(b) if any 0.00
3.75M X 10.0M 1 FOR EVERY 1000 SQM 47 47
BRICK WALL
In case of plots pf 1ha and above minimum parking space shall be 10% of plot area BRICK BAT

For plot area 81920 sq.m. parking should be 8192 sq.m.

SPACE
OPEN SPACE-1 PCC BED SEPTIC TANK SECTION AT DD
c. Balanced proposed 0.00
FOR
ETP 4399.62 sq.m. RAIN WATER HARVESTING TANK CAPACITY 5 NET PLOT AREA (3-4(c))) PRO RATA
AVERAGE ANNUAL RAINFALL=100MM (0.1M/HOUR) 81920.00
RAIN WATER PLOT AREA= 81920 SQ.M
HARVESTING VOLUMETRIC RAINFALL= 81920 X 0.1= 8192 CUM/HOUR
FIRE TANK SEPTIC TANK PLAN
SPACE
200K
PIT = 136.53 CUM/MINUTE
MAIN GATE
6 RECREATIONAL OPEN SPACE IF APPLICABLE 10%
RAIN WATER TANK CAPACITY PROPOSED= 5x 5 X 3 X 2= 150 CUM SCALE 1:50 BOUNDARY WALL DETAIL

FOR RAIN WATER TANK CAPACITY PROPOSED= 150000 LITER
STP
RAIN WATER
PIPE
FRONT ELEVATION COMPOUND WALL & GATE
A A A A a. Required
SHAFT SHAFT SHAFT
DOMESTIC FIRE TANK MURUM
GL GL
A FIREMAN A FIREMAN A FIREMAN A
TANK FILLING
SHAFT LIFT SHAFT LIFT SHAFT LIFT
50K 200K SAND
FILLING 2.80X2.60 2.80X2.60 2.80X2.60

PUMP SECURITY RUBBLE
FILLING

WATER HARVESTING SECTION
(SCALE - 1:50)
FD
UP
FD FD
UP
FD
UP
FD FD
UP
FD
UP
FD FD
UP
FD
b. Proposed
ROOM CABIN-3
RAINWATER PIT CAPACITY 5.00 X 5.00 X3 x 2 nos = 150 K LTS
UP
UP

7 INTERNAL ROAD AREA 0.00

PROCESS TANK
200K
8 PLOTABLE AREA IF APPLICABLE (as per sanctioned layout) 81920.00

9 B/UP area with reference to basic fsi as per front road width(1.0) 81920.00

60 10 ADDITIONAL FSI ON PAYMENT OF PREMIUM
IRNOTAEDR9N. A
0 0L M W I D E .0
0 PLASMA EXTRACTION PLASMA EXTRACTION LVP
a. Maximum permissible premium FSI / TOD Zone. 2.00

M 59.60 x 120.60 59.60 x 132.60 59.60 x 132.60 b. Proposed FSI of payment of Premium 0.00
SHAFT

FIREMAN
LIFT
2.80X2.60
RECEIVING BAY
CANOPY ABOVE
DISPATCH BAY DISPATCH BAY DISPATCH BAY
SHAFT

LIFT
3.30X2.45
.W CARPET AREA 7187 .00 SQ.MT. CARPET AREA 7902.96 SQ.MT. CARPET AREA 7902.96 SQ.MT.

11 IN SITU TDR / FSI LOADING
ID
LIFT
UP CORRIDOR 2.10x
1.90 UP

DISPATCH / RECEIVING BAY LIFT
2.10X
2.00 a. In situ area against Statutory Raod (2 xSr No 2(a)) if any 0.00
703 sq.mt.
IRN
OT E TERRACE WITH
WATERPROOFING
12 ADDITIONAL FSI AREA UNDER CHAPTER NO.6 0.000
AE R
O
SHAFT SHAFT SHAFT

DR FD
SHAFT

FD FD
FIREMAN
LIFT
2.80X2.60
FD
SHAFT

FD FD
FIREMAN
LIFT
2.80X2.60
FD
SHAFT

FD FD
FIREMAN
LIFT
2.80X2.60
FD
SHAFT

13 TOTAL ENTITLEMENT OF FSI IN THE PROPOSAL 0.000
10
N
A
UP UP UP UP UP UP

a. 9 +10(b) +11(a) or 9+12 which ever is applicable
6
NO A UP

Permissible 0.00
D
UP

21
S
SC L 31 a. 9 +10(b) +11(a) or 9+ 12 which ever is applicable
2 OO Proposed 0.00
NO
S
SC
TE
R
5 .2 b. Permissible Ancillary area FSI up to 80% with payment
0.00

10
6
NO
OO 1 1
PERMISSIBLE
b. Proposed Ancillary area FSI up to 60% or 80% with payment
TE
S R
SC
DIALYZER (G+1) SHAFT

OO c.Total Entitlement permissible
SHAFT
FIREMAN
LIFT

(a+b) 81920.00
FD FD
59.60 x 83.80 FD
2.80X2.60
FD
TE
UP UP

CARPET AREA 4994 .48 SQ.MT. R c.Total Entitlement proposed
UP
(a+b) 81920.00
Car Parking

UP

Maximum utilization limit of FSI (Building Potential) permissible as per
14 road width ((as per Regulation No. 5.5 of Chapter - 5)) x 1.6 or 1.8) 81920.00

SE CORRIDOR
15 TOTAL Built up area in proposal (excluding area as per sr. no 17b)
LIFT LIFT LIFT

CA CU
3.30X2.45 3.30X2.45 3.30X2.45
DISPATCH / RECEIVING BAY
41

UP
LIFT
2.10x
703 sq.mt.
LIFT
2.10x
LIFT
2.10x
a. Existing Built -up Area. 0.000
BI RIT
1.90 1.90 1.90
LIFT LIFT LIFT
2.10X 2.10X 2.10X
.6

2.00 2.00 2.00

B
N- Y GA SHAFT RECEIVING BAY DISPATCH BAY DISPATCH BAY DISPATCH BAY SHAFT SHAFT

b. Proposed Built- up area (as per P line) 46838.92
1
19

EN TE CANOPY ABOVE CANOPY BELOW
UP

FD

TR -1 GROUND FLOOR PLAN FIRST FLOOR PLAN SECOND FLOOR PLAN TERRACE FLOOR c. Total (a+b) 46838.92
UP

Y
FD

PLAN
(BUILDING 2) (BUILDING 2) (BUILDING 2)
LOBBY (BUILDING 2)
27 X 4.77
16 FSI CONSUMED (15/13) (should not be more than serial 0.571
No.14 above.)
330.840

CERTIFICATE OF AREA :-
AUDITORIUM
27 X 14.77
Certified that the plot under reference was surveyed by me on
.
R ID M

SHAFT

FIREMAN
02/06/2025 and the Dimensions of sides etc.of plot stated on plan area
LIFT SHAFT

36.575
2.80X2.60

as measured on site and the area so worked out tallies with the stated in
UP
FD FD FD FD FD FD FD FD

UP UP
W .00

A E

UP UP
documents of ownership/ DMIC/Land records Department/City Survey
UP

Records
D

Ar. Dashrath Dhumal
30

Owners Declaration :-
O

I/We undersigned hereby confirmed that I/We would abide by plans
SHAFT
approved by Authority/ Collector. I / We would execute the structure as
ELEVATION BUILDING 1
LIFT
3.30X2.45

LIFT
per the approved plans.Also I/We would execute the work under
2.10x
1.90 UP
ADMIN AREA
supervision of proper technical person so as to ensure the quality and
LIFT
2.10X
27.04X50.53
FRONT SIDE
2.00
safety at the work site.
LVP or Plasma
0 0L M W I D E

LIFT
2.45X3.30 LIFT
DUCT
4.20X1.03

LIFT
Mr. Sarang Landage.
2.00X2.10 2.00X2.10
LIFT MACHINE
OHWT ROOM

Architects Name and Signature
17.840 TERRACE 1.200 TERRACE 1.200

OPEN SPACE-2
5.100

5.100

5.100

5.100

BUILDING - 1 ADMIN BUILDING - 1 ADMIN Ar. Dashrath Dhumal
3793.43 sq.m. LADIES TOI. LADIES TOI.

Job No. Drg. No. Scale Dm. By. Checked By Registration no of Architect
5.100

FD FD FD FD
5.100

5.100

5.100

D4 D4

122.100 BUILDING - 1 ADMIN
SHAFT
10.200

CA/2005/37017
LIFT WELL

LIFT WELL

LIFT WELL

SHAFT
FIREMAN
LIFT
2.80X2.60
FD
UP

D1

WC
D1

WC
D1

WC
D1

WC WC
D1
UP
LADIES TOI. LADIES TOI.
001 1:100 Ar. Dashrath Dhumal
5.100

FD FD FD 1.05X 1.05X 1.05X 1.05X
IRNOTAEDR9N. A

5.100

5.100

5.100

1.70 1.70 1.70 1.70 1.70
UP UP
V V V V V

AUDITORIUM CANTEEN
DUCT
5.90X1.43
V V V V V

UP
WC WC WC WC WC T-300 MM
C PARKING STATEMENT
26.192

1.05X 1.05X 1.05X 1.05X 1.05X
R-150 MM LADIES TOI.
UP
1.70 1.70 1.70 1.70 1.70 LADIES TOI.
D1 D1 D1 D1 D1 T-300 MM
R-150 MM LIFT LIFT LIFT
PIT PIT PIT
a GARAGE PERMISSIBLE GARAGE PROPSD NIL
SECTION B-B
IRNOTAEDR1N2A. 0L 0 M W I D E

D4
D4

PARKING PARKING
b TYPE
REQUIRED PROPOSED
BUILDING-1
CAR

SCOOTER
8192 SQ.M. 8192 SQ.M.
CYCLE
B B B B
PLASMA EXTRACTION (G+2)
Car Parking

VISITORS
60
UP

FD

59.60 x 120.60 CANTEEN
27.04 X 59.00
IRNOTAEDR1N2A. 0L 0 M W I D E

CARPET AREA 7187 .00 SQ.MT. D LOADING UNLOADING SPACES
UP

UP

UP

W .00
FD

FD

FD

FD

LOBBY
LOBBY 27 X 4.77
27 X 4.77
a LOADING UNLOADING REQUIRED 6

R I D M.
FD FD FD FD

b TOTAL LOADING UNLOADING PROVIDED 6
DOUBLE HEIGHT
UP UP AUDITORIUM

O E AUDITORIUM
LEGENDS
A
27 X 14.77
D

SHAFT

FIREMAN

D
SHAFT

PLOT BOUNDRY SHOWN IN
LIFT
2.80X2.60
FD FD FD FD
A

AMENITY SPACE
UP UP
FD FD FD FD FD FD FD FD

UP
B/UP AREA SHOWN IN
4164.17 sq.m.
UP
O

UP UP

WATER LINE SHOWN IN
02

FD FD
DOOR
R

DRAINAGE LINE SHOWN IN
.7

DOOR
79

FLUSH
E

FLUSH

LIFT
SCHEDULE OF DOORS AND WINDOWS
FRAME
ID

2.10X2.00
COMPRESSOR
4.20X1.03
DUCT

PANEL ROOM

ALLU. SLIDING WINDOW

LIFT
18.28 X 10.62 2.10X2.00
TYPE SIZE DISCRIPTION
FRAME

LIFT
3.30X2.45
R.C.C.

RS 6000 X 6000 M.S ROLLING SHUTTER

124.700
FD

.W

FIRE DOOR

R.C.C.

ADMIN AREA ADMIN AREA ADMIN AREA
TERRACE WITH WATER PROOFING
D1 750 X 2100
27.04X50.53 27.04X50.53 27.04X70.54
FD

UP

B D4 1000 X 2100

GROND FLOOR PLAN FD 1000 X 2100
M

DUCT DUCT DUCT
4.20X1.03 4.20X1.03 4.20X1.03

LIFT LIFT LIFT
2.45X3.30 LIFT LIFT 2.45X3.30 LIFT LIFT 2.45X3.30 LIFT LIFT

(BUILDING 1) 2.00X2.10 2.00X2.10 2.00X2.10 2.00X2.10 2.00X2.10 2.00X2.10
W1 3000 X 1500
0

A
V 600 X 900 M.S. LOUVERED VENTILATOR
0.900

LIFT CORRIDOR 4.500
4.500
.0

3.30X2.45

LIFT
DISPATCH / RECEIVING BAY Security Cabin
4.00 x 5.00
2.10x
UP
1.90 703 sq.mt. 0.899

PLAN OF PROPOSED FACTORY
45

LIFT
12.000
2.10X
10.450
2.00
14.000

RECEIVING BAY DISPATCH BAY DISPATCH BAY DISPATCH BAY
14.000

SHAFT UP
A
5.000

A FD FD FD FD FD FD FD
FD

BUILDINGON PLOT NO 8A, SECTOR 13,
5.455

A D4 D4 D4 D4 D4 D4
VISITORS

CANOPY ABOVE 4.00 x 6.74

AREA KEY PLAN
(Driver's Room) UP UP

AREA KEY PLAN
12.000 Pump Room
D1 D1 D1 D1 D1 D1 D1 D1 D1 D1 D1 D1 D1 D1 D1

DMIC, AURIC , SHENDRA,
21.572

WC WC WC WC WC WC WC WC WC WC WC WC WC WC WC WC
TOILET 1.05X 1.05X 1.05X 1.05X 1.05X 1.05X 1.05X 1.05X 1.05X 1.05X 1.05X 1.05X 1.05X 1.05X 1.05X 1.05X
W.B. 2.40 x 1.50 A W.C.
1.20 x
A 10.450 1.70 1.70 1.70 1.70 1.70 1.70 FD 1.70 1.70 1.70 1.70 1.70 FD 1.70 1.70 1.70 1.70 1.70
1.20 BATH
1.20 x V V V V V V V V V V V V V V V V
2.55 DUCT DUCT DUCT DUCT
TYPICAL PLAN-1,2&3.
A AREA KEY PLAN
V 5.90X1.43 5.90X1.43 5.90X1.43 5.90X1.43 5.90X1.43
5.000

V V V V V V V V V V V V V V V V
(Security Cabin) Security Cabin DRIVER'S ROOM
8.84 x 4.54
WC WC WC WC WC WC WC WC WC WC WC WC WC WC WC WC
1.05X 1.05X 1.05X 1.05X 1.05X 1.05X 1.05X 1.05X 1.05X 1.05X 1.05X 1.05X 1.05X 1.05X 1.05X
A A
5.455

PUMP ROOM 1.70 1.70 1.70 1.70 1.70 1.70 1.70 1.70 1.70 1.70 1.70 1.70 1.70 1.70 1.70 1.70
LOCKER'S RM.
10.00 X 5.00

CHATRAPATI SAMBHAJI NAGAR
2.55 x 1.84 D1 D1 D1 D1 D1 D1 D1 D1 D1 D1 D1 D1 D1 D1 D1

IRNOTAEDR1N2A. 0L 0 M W I D E PLAN
(Driver's Room)
UP

PARAPET LVL.
1.997 2.997 1.514
10.450
PLAN (Pump Room)
1.199
UP

1.514 0.999 0.230 D4
D4
D4
D4
D4
D4
1.598 0.599

SLAB LVL.

Parapet LVL.
PARAPET LVL.
1.598 0.599

Slab LVL.
SLAB LVL. LINTEL LVL.
3.996

RCC Chajja
1.499
2.398

Lintel LVL.
LINTEL LVL. 0.899
SILL LVL.
3.996

1.499

PLINTH LVL.
0.450
2.398

Sill LVL.
0.899 G.LVL. G.LVL.
SILL LVL.
FRONT ELEVATION Plinth LVL.
PLINTH LVL. (Driver's Room)
0.450

G. LVL. G. LVL.
G.LVL. G.LVL.
FRONT ELEVATION
LEFT SIDE ELEVATION
(Security Cabin)

PARAPET LVL.
1.598 0.599

TERRACE TERRACE Parapet LVL.
SLAB LVL.
Terrace Slab LVL.

0.900
LINTEL LVL. RCC Chajja
3.996

Lintel LVL.
1.499

3.996
2.398

3.497

1.499

0.899 SILL LVL.
2.398

2.398

Sill LVL.
DRIVER'S ROOM W.C. BATH

231.187
Security Cabin VISITORS TOILET
0.899

PLINTH LVL. SUNK
SUNK
SUNK
G.LVL. G.LVL. PUMP ROOM
Plinth LVL.
0.450

G. LVL. G. LVL.
SECTION- AA
SECTION- AA SECTION- AA
(Driver's Room)
(Security Cabin)

CANTEEN
27.04 X 59.00

FOR M/S LIFENITY LIMITED
FD FD FD FD FD FD FD FD

UP UP
ADMIN AREA ADMIN AREA

45.00 M. W I D E R O A D
27.04X69.77 27.04X69.77
FD FD

SIGN OF OWNER

SIGN OF ARCHITECT
LIFT LIFT LIFT
2.10X2.00 2.10X2.00 2.10X2.00
COMPRESSOR
4.20X1.03

4.20X1.03

4.20X1.03

N
DUCT

DUCT

DUCT

PANEL ROOM
LIFT LIFT LIFT
18.28 X 10.62 2.10X2.00 2.10X2.00 2.10X2.00

LIFT LIFT LIFT
3.30X2.45 3.30X2.45 3.30X2.45

LAYOUT PLAN
FD
FD

FD

FD

FD

UP

B B B B

GROND FLOOR PLAN FIRST FLOOR PLAN SECOND FLOOR PLAN TERRACE FLOOR PLAN
SCALE :- 1:100 (BUILDING 1) (BUILDING 1) (BUILDING 1) (BUILDING 1)
N CREATION

WE
JOB No. DRG NO. SCALE DATE DRN BY CHK. BY

S 01 1 :100

[SHEET: Index]

Sr.No. | Document List | Details provide

- 1 | MIDC Sanction Plan. |
- 2 | MIDC Completion/Occupancy Certificate. | Possession Letter 6/8/2024
- 3 | Udyam Aadhar | NA
- 4 | Incorporation Certificate. | incorporation!A1
- 5 | Total Project Cost. (Esteemated) | 3065586099
- 6 | a) Built-up area | 40000 Sq Meter
- 7 | b) Area proposed for green belt | 27000 Sq Meter
- 8 | c) Tree plantation details. |
- 9 | Proposed layout plan with area Statement. |
- 10 | a) List of products | List of Products!A1
- 11 | b) Raw Material Balance Sheet. | Raw Material!A1
- 12 | a) List of HW waste will be generated from the process. | Hazardous Waste!A1
- 13 | b) List of Non-Hazardous waste will be generated from the process. | Non Haz Waste!A1
- 14 | Water Balance, Domestic Consumption & Process Consumption and Generation of Waste Water. | Water!A1
- 15 | Name of the Fuel will be used in the process: HSD/ Electricity/CNG. | Fuel Consmp!A1
- 16 | Details of any heat treatment or reaction tank, if any. | Heat treat_Reaction_Tank!A1
- 17 | Details of Equipment/Machinery with the Name | Equipments!A1
- 18 | Types and sources of Pollution and control measures (Pollution Control equipment). | Pol control meas!A1
- 19 | DG Set details. | DG Set !A1

[SHEET: incorporation]

||||| Back to Index

[SHEET: Possession Let]

||||| Back to Index

[SHEET: DG Set]

Sr.No. | Parameter | Description / Specification |

- 1 | Make / Model | Cummins / Kirloskar / Mahindra (example – confirm at procurement) | Back to Index
- 2 | Type | Silent Type, Auto-start, Water-cooled, Turbocharged Diesel Generator |
- 3 | Rating / Capacity | 2 × 1300 kVA each (standby operation, load sharing optional) |
- 4 | Fuel Used | High-Speed Diesel (HSD) – BS-VI grade, low sulphur (<0.05%) |
- 5 | Rated Power Output | ~400 kW per DG set (0.8 PF) |
- 6 | Total Installed Capacity | 2,600 kVA (2 sets × 1300 kVA) |
- 7 | Operating Hours / Year | ~800–1,000 hours per year (emergency + load testing only) |
- 8 | Fuel Consumption | ~55–60 L/hour @ full load |
- 9 | Estimated Annual HSD Consumption | ≈ 27,000 L / year (≈ 27 KL / year) |
- 10 | Storage Capacity (HSD Tank) | 10 KL aboveground double-walled MS tank with vent, dyke, and spill tray as per PESO norms |
- 11 | Cooling System | Water-cooled, with radiator and jacket water heater |
- 12 | Lube Oil System | Forced lubrication; oil drain point connected to collection tray |
- 13 | Starting System | 24V battery start with automatic mains failure (AMF) panel |
- 14 | Exhaust System | Silencer with flexible coupling; insulated exhaust manifold |
- 15 | Emission Parameters (at rated load) | • PM ≤ 0.3 g/kWh • NO_x + HC ≤ 7.5 g/kWh • CO ≤ 3.5 g/kWh (As per CPCB II norms) |
- 16 | Stack Details | Mild steel stack with flanged joints; height as per CPCB formula: $H = h + 0.2\sqrt{KVA} = 3 + 0.2\sqrt{1300} \approx 10.2$ m above roof level per set | 36.05551275463989
- 17 | Stack Quantity | 2 Nos (individual stack for each DG set) | 7.211102550927979
- 18 | Noise Level at 1 m distance | ≤ 75 dB(A) – within CPCB norm | 10.21110255092798
- 19 | Noise Control Measures | • Acoustic Enclosure (≤ 75 dB(A)) • Vibration isolators under skid base • Greenbelt & acoustic barriers along DG yard boundary |
- 20 | Location | Utility block (away from production & office area) |
- 21 | Ventilation System | Forced air intake & exhaust with acoustic louvers |
- 22 | Emission Control | • Stack emission monitoring (PM, NO_x, SO_x) – quarterly • Use of low-sulphur HSD |
- 23 | Lube Oil Handling | Drained to closed HDPE drum; sent to authorized recycler (HW 5.1) |
- 24 | Spill Control | Impervious floor, containment dyke, absorbent pads available |
- 25 | Safety & Fire Protection | Fire extinguishers (CO₂ + dry chemical) at DG yard; earthing pit per IS 3043 |
- 26 | Authority Approval | DG sets to comply with CPCB-II emission & noise certification; HSD tank under PESO license |

Sr.No. | Parameter | Standard / Limit | Plant DG Result / Provision

- 1 | Emissions (CPCB-II) | PM ≤ 0.3 g/kWh NO_x+HC ≤ 7.5 g/kWh CO ≤ 3.5 g/kWh | Cummins CPCB-II compliant engines
 - 2 | Stack Height | $H = h + 0.2\sqrt{KVA}$ | 6.5 m above roof per set
 - 3 | Noise | ≤ 75 dB(A) at 1 m | Acoustic enclosure fitted
 - 4 | Fuel Type | Low-sulphur HSD | BS-VI HSD used
 - 5 | Used Oil Disposal | To authorized recycler | As per HW Rule 5.1
 - 6 | DG Area Containment | Impervious floor + bund | Provided
- | Back to Index | |

[SHEET: Pol control meas]

Sr.No. | Type | Source | Pollutants Likely | Control Measures / Equipment | Remarks | |

- 1 | AIR POLLUTION | a. DG Set (2 × 500 kVA, standby) | PM, SO_x, NO_x, CO (from diesel combustion) | • Acoustic enclosure to control noise • Adequate stack height as per CPCB formula ($H = h + 0.2\sqrt{KVA}$) ~6.5 m above roof • Low-sulphur HSD (<0.05%) fuel • Regular maintenance & emission monitoring | Operates only during power outage | | Back to Index
- 2 | AIR POLLUTION | b. PNG / CNG-fired Boiler (500 kg steam/hr) | NO_x, trace CO | • Use of low-NO_x burner • Stack height 12 m above roof • Online temperature & pressure monitoring • Periodic stack emission testing | PNG is clean fuel, minimal SO_x | |
- 3 | AIR POLLUTION | c. HVAC systems / AHUs | Dust, microbial aerosols | • HEPA filters (0.3 μm) at terminal outlets • Pre-filters (10 μm) at return air • Periodic filter replacement as per SOP | Ensures cleanroom Class 10,000–100,000 compliance | |
- 4 | AIR POLLUTION | d. Solvent & detergent handling (TnBP, Triton X-100) | VOC vapours | • Closed mixing vessels, nitrogen blanketing • Solvent vapor condenser + activated-carbon adsorption unit • Explosion-proof zone classification (ATEX) | Emission < 50 mg/Nm³ VOC ensured | |
- 5 | AIR POLLUTION | e. Laboratory fume hoods / exhausts | Trace acidic or organic vapours | • Fume hoods with local scrubbers or HEPA filters • Ducted exhaust at safe height | Minimal emissions, intermittent source | |

Sr.No. | Type | Source | Type of Wastewater | Pollutants Likely | Treatment / Control Equipment | |

- 1 | WATER POLLUTION | a. Process & WFI drains | Process effluent containing protein residues, buffers, mild solvents, detergents | pH 6–8, COD 1,000–2,000 mg/L, BOD 400–800 mg/L, low TDS | Effluent Treatment

Plant (ETP): • Collection & equalization tank (15 m³) • pH correction & neutralization tank • Primary clarifier • Biological MBR / MBBR reactor • Tertiary filtration (UF + RO) • RO permeate reused for gardening / utilities • RO reject → MEE + ATFD | |

2 | WATER POLLUTION | b. Solvent-detergent streams (TnBP phase) | High COD, solvent traces | Trace VOCs, COD > 10,000 mg/L | Dedicated solvent holding tank → evaporation or incineration via TSDF | |

3 | WATER POLLUTION | c. Domestic sewage (staff 506) | Black / grey water | BOD ~250 mg/L | STP (40 m³/day): MBBR + UF + chlorination | |

4 | WATER POLLUTION | d. Cooling tower blow-down, softener reject | High TDS, hardness | TDS 2,000–3,000 mg/L | Collected in neutralization tank → mixed with RO reject → MEE | |

Sr.No. | Type | Source | Type of Waste | Category | Control / Disposal Method | |

1 | SOLID / HAZARDOUS WASTE POLLUTION | Process residues, filter aids, spent carbon, ETP sludge | Solid / semi-solid | HW 28.2 / 35.3 | Sent to Authorized TSDF (CHWTSDF) for landfill/incineration | |

2 | SOLID / HAZARDOUS WASTE POLLUTION | Spent solvent mix (TnBP + detergent) | Liquid | HW 28.1 | Sent to Solvent recovery or TSDF | |

3 | SOLID / HAZARDOUS WASTE POLLUTION | Used oil from DG set | Liquid | HW 5.1 | Sold to MPCB-authorized recycler | |

4 | SOLID / HAZARDOUS WASTE POLLUTION | Biomedical waste (lab disposables, plasma bags) | Bio-infectious | BMW Cat. 4 / 5 | Sent to Authorized CBWTF | |

5 | SOLID / HAZARDOUS WASTE POLLUTION | Non-hazardous recyclable waste | Paper, plastic, metal, glass | Non-hazardous | Segregation → authorized recyclers | |

6 | SOLID / HAZARDOUS WASTE POLLUTION | Organic waste (canteen, sludge) | Biodegradable | Non-hazardous | Composting / manure | |

Sr.No. | Type | Source | Noise Level (dB-A) | Control Measures | | |

1 | NOISE POLLUTION | DG sets | 75–80 | Acoustic enclosure, antivibration mounts, green belt | | |

2 | NOISE POLLUTION | Compressors, pumps, blowers | 70–75 | Foundation damping pads, acoustic barrier | | |

3 | NOISE POLLUTION | HVAC & AHU | 60–65 | Acoustic lining, flexible connectors | | |

4 | NOISE POLLUTION | General utilities | < 55 | Periodic maintenance, ear protection for workers | | |

Sr.No. | Type | Equipment / Facility | Capacity / Function | Purpose | | |

1 | SOLID WASTE MANAGEMENT | ETP | 70 m³/day (MBBR + UF + RO + MEE) | Treat & recycle process effluent | | |

2 | SOLID WASTE MANAGEMENT | STP | 40 m³/day (MBBR + UF) | Treat domestic sewage | | |

3 | SOLID WASTE MANAGEMENT | Air pollution stacks | Boiler – 12 m; DG – 6.5 m above roof | Emission dispersion | | |

4 | SOLID WASTE MANAGEMENT | Activated carbon filter | 1 × 500 kg | VOC adsorption from S-D exhaust | | |

5 | SOLID WASTE MANAGEMENT | Scrubber (if acid used) | 1 × 3,000 m³/hr | Fume removal from buffer prep / lab exhaust | | |

6 | SOLID WASTE MANAGEMENT | Noise enclosure | DG & compressors | Noise reduction | | |

7 | SOLID WASTE MANAGEMENT | Greenbelt | ≥ 33 % of plot | Dust, noise & micro-climate control | | |

| Back to Index | | | | |

[SHEET: Equipments]

Sr.No. | Category | Equipment Name | Quantity | Estimated Amount | | Back to Index

1 | ADMIN EQUIPMENT | Office Cabinets for Documents | 20 | 240000 | |

2 | ADMIN EQUIPMENT | Computers / Laptops | 50 | 2500000 | |

3 | ADMIN EQUIPMENT | Office Furniture | NA | 1200000 | |

4 | ADMIN EQUIPMENT | Printer/scanner / xerox machine | 1 | 500000 | |

5 | ADMIN EQUIPMENT | Water Dispenser | 2 | 40000 | |

6 | ADMIN EQUIPMENT | Projector for conference / meeting room | 2 | 600000 | |

7 | CANTEEN EQUIPMENT | Refrigerator | 2 | 300000 | |

8 | CANTEEN EQUIPMENT | SS furniture (serving table etc) | NA | 400000 | |

9 | CANTEEN EQUIPMENT | SS Water cooler dispenser | 1 | 50000 | |

10 | CANTEEN EQUIPMENT | Microwaves | 2 | 60000 | |

11 | CANTEEN EQUIPMENT | Kitchenware | na | 500000 | |

12 | CANTEEN EQUIPMENT | Bain marie | 1 | 50000 | |

13 | Clean Utilities | Soft water generation 10000/hour and storage system | 1 | 40000000 | |

14 | Clean Utilities | Purified water generation 5000L/ hour storage tank 10KL and distribution system | 1 | | |

15 | Clean Utilities | Water for injection generation 1000L / hour, Storage tank 10KL and distribution system | 1 | | |

16 | Clean Utilities | Pure steam generator 1500Kg / hour and distribution system | 1 | | |

17 | Black Utilities | Plant steam generation 2000Kg/Hour and distribution System | 2 | 7000000 | |

18 | Black Utilities | Compressed air generation and distribution system | 2 | 7500000 | |

19 | Black Utilities | PASSENGER LIFT X 2 NOS | 2 | 7000000 | |

20 | Black Utilities | MATERIAL LIFT X 3 NOS | 3 | 6000000 | |

21 | PLASMA STORAGE EQUIPMENT | Analytical weighing Balance | 1 | 170000 | |

22 | PLASMA STORAGE EQUIPMENT | RLAf | 3 | 1200000 | |

23 | PLASMA STORAGE EQUIPMENT | Precision Balance | 1 | 144000 | |

24 | PLASMA STORAGE EQUIPMENT | Commercial Balance | 1 | 120000 | |

25 | PLASMA STORAGE EQUIPMENT | Refrigerator 200L | 2 | 218000 | |

26 | PLASMA STORAGE EQUIPMENT | Deep freezer -20°C (1000L) | 50 | 22500000 | |

27 | PLASMA STORAGE EQUIPMENT | Air curtain | 10 | 100000 | |

28 | PLASMA STORAGE EQUIPMENT | PVC pallets | 50 | 300000 | |

29 | PLASMA STORAGE EQUIPMENT | Storage racks SS | 25 | 700000 | |

30 | PLASMA STORAGE EQUIPMENT | Sampling rods | 2 | 10000 | |

31 | PLASMA STORAGE EQUIPMENT | Liquid sampler | 2 | 20000 | |

32 | PLASMA STORAGE EQUIPMENT | Dedusting tunnel | 1 | 1250000 | |

33 | PLASMA STORAGE EQUIPMENT | DPB | 4 | 600000 | |

34 | PLASMA STORAGE EQUIPMENT | LAF cabinet | 1 | 450000 | |

35 | PLASMA STORAGE EQUIPMENT | Computer / Laptops | 4 | 120000 | |

36 | PLASMA STORAGE EQUIPMENT | Office Furniture | NA | 200000 | |

37 | PLASMA STORAGE EQUIPMENT | Office Cabinets | 10 | 120000 | |

38 | BULK PLASMA MANUFACTURING EQUIPMENT | Chromatography AKTA-1 (471 L/hr (Pipe MOC SS) 1F | 2 | 34200000 | |

39 | BULK PLASMA MANUFACTURING EQUIPMENT | Chromatography Column Dia 1000mm x Height 500mm 1F | 4 | 89224400 | |

40 | BULK PLASMA MANUFACTURING EQUIPMENT | Chromatography AKTA-1 (785 L/hr pipe MOC SS) 2F | 2 | 40062840 | |

41 | BULK PLASMA MANUFACTURING EQUIPMENT | Chromatography Column Dia 1000mm x Height 500mm 2F | 2 | 44612200 | |

42 | BULK PLASMA MANUFACTURING EQUIPMENT | Chromatography AKTA-1 (1178 L/hr pipe MOC PP) 3F | 2 | 40062840 | |

43 | BULK PLASMA MANUFACTURING EQUIPMENT | Chromatography Column Dia 1000mm x Height 500mm 3F | 4 |

89224400 ||
44 | BULK PLASMA MANUFACTURING EQUIPMENT | Chromatography AKTA-1 (864 L/hr pipe MOC PP) 4l | 2 | 40062840
||
45 | BULK PLASMA MANUFACTURING EQUIPMENT | Chromatography Column Dia 1000mm x Height 500mm 4l | 2 |
44612200 ||
46 | BULK PLASMA MANUFACTURING EQUIPMENT | Chromatography AKTA-1 (864 L/hr pipe MOC PP) 5l | 1 | 20031420
||
47 | BULK PLASMA MANUFACTURING EQUIPMENT | Chromatography Column Dia 1000mm x Height 500mm 5l | 1 |
22306100 ||
48 | BULK PLASMA MANUFACTURING EQUIPMENT | Chromatography AKTA-1 (667 L/hr pipe MOC PP) 6l | 1 | 17100000
||
49 | BULK PLASMA MANUFACTURING EQUIPMENT | Chromatography Column Dia 1000mm x Height 500mm 6l | 1 |
22306100 ||
50 | BULK PLASMA MANUFACTURING EQUIPMENT | Chromatography AKTA-1 (785 L/hr pipe MOC PP) 7A | 1 | 20031420
||
51 | BULK PLASMA MANUFACTURING EQUIPMENT | Chromatography Column Dia 1000mm x Height 500mm 7A | 1 |
22306100 ||
52 | BULK PLASMA MANUFACTURING EQUIPMENT | Chromatography AKTA-1 (30 L/hr pipe MOC PP) 8N | 1 | 14657140
||
53 | BULK PLASMA MANUFACTURING EQUIPMENT | Chromatography Column Dia 300mm x Height 500mm 8N | 1 |
4896570 ||
54 | BULK PLASMA MANUFACTURING EQUIPMENT | Chromatography AKTA-1 (30 L/hr pipe MOC PP) 9N | 1 | 14657140
||
55 | BULK PLASMA MANUFACTURING EQUIPMENT | Chromatography Column Dia 300mm x Height 750mm 9N | 1 |
4896570 ||
56 | BULK PLASMA MANUFACTURING EQUIPMENT | Chromatography AKTA-1(85 L/hr pipe MOC PP) 10T | 1 | 12500000
||
57 | BULK PLASMA MANUFACTURING EQUIPMENT | Chromatography Column Dia 600mm x Height 500mm 10T | 1 |
14244570 ||
58 | BULK PLASMA MANUFACTURING EQUIPMENT | Chromatography AKTA-1 (340 L/hr pipe MOC PP) 11F | 1 |
30000000 ||
59 | BULK PLASMA MANUFACTURING EQUIPMENT | Chromatography Column Dia 300mm x Height 500mm 11F | 1 |
4896570 ||
60 | BULK PLASMA MANUFACTURING EQUIPMENT | Chromatography AKTA-1 (500 L/hr pipe MOC PP) 12E | 1 |
17100000 ||
61 | BULK PLASMA MANUFACTURING EQUIPMENT | Chromatography Column Dia 1000 mm x Height 500mm 12E | 1 |
22306100 ||
62 | BULK PLASMA MANUFACTURING EQUIPMENT | Chromatography AKTA-1(160 L/hr pipe MOC PP) 13E | 1 | 14657140
||
63 | BULK PLASMA MANUFACTURING EQUIPMENT | Chromatography Column Dia 450mm x Height 500mm 13E | 1 |
12018850 ||
64 | BULK PLASMA MANUFACTURING EQUIPMENT | TFF System 50 sqm | 3 | 12500000 ||
65 | BULK PLASMA MANUFACTURING EQUIPMENT | Cassette 50 sqm | 3 | 4000000 ||
66 | BULK PLASMA MANUFACTURING EQUIPMENT | TFF System 20 sqm | 1 | 9460000 ||
67 | BULK PLASMA MANUFACTURING EQUIPMENT | Cassette 20 sqm | 2 | 3010000 ||
68 | BULK PLASMA MANUFACTURING EQUIPMENT | TFF System 5 sqm | 4 | 14620000 ||
69 | BULK PLASMA MANUFACTURING EQUIPMENT | Cassette 5 sqm | 4 | 1376000 ||
70 | BULK PLASMA MANUFACTURING EQUIPMENT | Continuous Centrifuge 1000L/hr | 2 | 30000000 ||
71 | BULK PLASMA MANUFACTURING EQUIPMENT | Process Vessels 300 L | 4 | 8842500 ||
72 | BULK PLASMA MANUFACTURING EQUIPMENT | Process Vessels 100 L | 3 | 6226875 ||
73 | BULK PLASMA MANUFACTURING EQUIPMENT | Process Vessels 150 L | 1 | 2075625 ||
74 | BULK PLASMA MANUFACTURING EQUIPMENT | Process Vessels 250 L | 4 | 8842500 ||
75 | BULK PLASMA MANUFACTURING EQUIPMENT | Process Vessels 500 L | 4 | 10732500 ||
76 | BULK PLASMA MANUFACTURING EQUIPMENT | Process Vessels 1000 L | 6 | 19650600 ||
77 | BULK PLASMA MANUFACTURING EQUIPMENT | Process Vessels 1500 L | 29 | 100830825 ||
78 | BULK PLASMA MANUFACTURING EQUIPMENT | Process Vessels 2000 L | 11 | 42701175 ||
79 | BULK PLASMA MANUFACTURING EQUIPMENT | Buffer vessels 500L | 6 | 16098750 ||
80 | BULK PLASMA MANUFACTURING EQUIPMENT | Buffer vessels 1000L | 9 | 29475900 ||
81 | BULK PLASMA MANUFACTURING EQUIPMENT | Buffer vessels 1500L | 6 | 23291550 ||
82 | BULK PLASMA MANUFACTURING EQUIPMENT | Buffer vessels 2500L | 1 | 30000000 ||
83 | BULK PLASMA MANUFACTURING EQUIPMENT | Depth filtration system | 5 | 5212500 ||
84 | BULK PLASMA MANUFACTURING EQUIPMENT | Nano filter holder | 2 | 1000000 ||
85 | BULK PLASMA MANUFACTURING EQUIPMENT | Cold room 2-8°C | 4 | 3000000 ||
86 | BULK PLASMA MANUFACTURING EQUIPMENT | Rectangular tank | 2 | 860000 ||
87 | BULK PLASMA MANUFACTURING EQUIPMENT | Buffer preparation tank | 4 | 3440000 ||
88 | BULK PLASMA MANUFACTURING EQUIPMENT | Dispensing Booth | 2 | 800000 ||
89 | BULK PLASMA MANUFACTURING EQUIPMENT | Analytical Balance | 2 | 340000 ||
90 | BULK PLASMA MANUFACTURING EQUIPMENT | Precision Balance | 2 | 300000 ||
91 | BULK PLASMA MANUFACTURING EQUIPMENT | Commercial Balance | 2 | 240000 ||
92 | BULK PLASMA MANUFACTURING EQUIPMENT | PVC bag Sealer | 2 | 10000 ||
93 | BULK PLASMA MANUFACTURING EQUIPMENT | Rotary Bag Sealer | 2 | 600000 ||
94 | BULK PLASMA MANUFACTURING EQUIPMENT | pH/ Conductivity meter | 4 | 600000 ||
95 | BULK PLASMA MANUFACTURING EQUIPMENT | Magnetic Stirrer | 4 | 300000 ||
96 | BULK PLASMA MANUFACTURING EQUIPMENT | Peristaltic pump | 4 | 400000 ||
97 | BULK PLASMA MANUFACTURING EQUIPMENT | Filter Integrity machine | 2 | 4000000 ||
98 | BULK PLASMA MANUFACTURING EQUIPMENT | Fume Hood | 2 | 900000 ||
99 | BULK PLASMA MANUFACTURING EQUIPMENT | Autoclave 1000L | 4 | 8000000 ||
100 | BULK PLASMA MANUFACTURING EQUIPMENT | DHS 1000L | 2 | 4000000 ||
101 | BULK PLASMA MANUFACTURING EQUIPMENT | Ceiling LAF | 6 | 2400000 ||
102 | BULK PLASMA MANUFACTURING EQUIPMENT | Drying Oven | 4 | 2000000 ||
103 | BULK PLASMA MANUFACTURING EQUIPMENT | Acid tank 1000L (Pre and Post Virus) | 2 | 3870000 ||
104 | BULK PLASMA MANUFACTURING EQUIPMENT | Base tank 1000L (Pre and Post Virus) | 2 | 3870000 ||
105 | BULK PLASMA MANUFACTURING EQUIPMENT | Central CIP system | 4 | 4300000 ||
106 | BULK PLASMA MANUFACTURING EQUIPMENT | Refrigerator 200L | 2 | 300000 ||
107 | BULK PLASMA MANUFACTURING EQUIPMENT | Laminar air flow cabinet | 2 | 870000 ||
108 | BULK PLASMA MANUFACTURING EQUIPMENT | Dynamic pass box | 8 | 960000 ||
109 | BULK PLASMA MANUFACTURING EQUIPMENT | Tumble Dryer 25 Kg | 2 | 4000000 ||
110 | BULK PLASMA MANUFACTURING EQUIPMENT | Barrier washer Extractor 25Kg | 2 | 6000000 ||
111 | BULK PLASMA MANUFACTURING EQUIPMENT | SS Furniture (Table, chair, racks, Ladders etc) | NA |
8000000 ||
112 | BULK PLASMA MANUFACTURING EQUIPMENT | Lockers | NA | 400000 ||
113 | BULK PLASMA MANUFACTURING EQUIPMENT | Laptop / Computers | 20 | 1000000 ||

114 | BULK PLASMA MANUFACTURING EQUIPMENT | Office furniture / Cabinets | NA | 600000 ||
115 | FILL FINISH EQUIPMENT | Formulation Vessel 100L | 1 | 2500000 ||
116 | FILL FINISH EQUIPMENT | Ceiling LAF | 2 | 800000 ||
117 | FILL FINISH EQUIPMENT | Laminar air flow cabinet | 1 | 435000 ||
118 | FILL FINISH EQUIPMENT | Continuous Particle Counter | 1 | 3000000 ||
119 | FILL FINISH EQUIPMENT | Dynamic pass box | 7 | 840000 ||
120 | FILL FINISH EQUIPMENT | Pass Box static | 1 | 38000 ||
121 | FILL FINISH EQUIPMENT | Vial washing machine | 1 | 4500000 ||
122 | FILL FINISH EQUIPMENT | Sterilizing tunnel | 1 | 4500000 ||
123 | FILL FINISH EQUIPMENT | Vial filling & Sealing line 120 vials /min | 1 | 7500000 ||
124 | FILL FINISH EQUIPMENT | Lyophilizer (1sq meter) | 1 | 5500000 ||
125 | FILL FINISH EQUIPMENT | Labelling machine | 1 | 9,00,000.00 ||
126 | FILL FINISH EQUIPMENT | Double door autoclave 1000L | 1 | 2000000 ||
127 | FILL FINISH EQUIPMENT | Cold room | 1 | 800000 ||
128 | FILL FINISH EQUIPMENT | SS furniture (Table, Chair, racks, cabinets etc) | 1 | 200000 ||
129 | QC & R&D EQUIPMENT | Akta pure -150 (For R&D) | 2 | 4000000 ||
130 | QC & R&D EQUIPMENT | Columns XK 50/100 (For R & D) | 1 | 700000 ||
131 | QC & R&D EQUIPMENT | Columns XK 50/60 (For R&D) | 1 | 700000 ||
132 | QC & R&D EQUIPMENT | Column XK 50/30 (For R & D) | 3 | 1400000 ||
133 | QC & R&D EQUIPMENT | HPLC-UV | 1 | 3400000 ||
134 | QC & R&D EQUIPMENT | HPLC Columns and accessories | 1 | 1000000 ||
135 | QC & R&D EQUIPMENT | Electrophoresis (Minigel) with trans blot | 2 | 600000 ||
136 | QC & R&D EQUIPMENT | Gel Doc system densitometer | 1 | 500000 ||
137 | QC & R&D EQUIPMENT | Power packs | 2 | 200000 ||
138 | QC & R&D EQUIPMENT | Endotoxin Detection system (Tube heating block) | 1 | 200000 ||
139 | QC & R&D EQUIPMENT | Micropipettes- variable volume range of 2-20µL, 20-200µL, 100-1000µL, 500-5000µL and multichannel 30-300µL | 2 set each | 300000 ||
140 | QC & R&D EQUIPMENT | Lab Refrigerator 2-8 ° C (SS) with data control | 4 | 450000 ||
141 | QC & R&D EQUIPMENT | Flame photometer | 1 | 450000 ||
142 | QC & R&D EQUIPMENT | Weighing Balance Analytical | 2 | 350000 ||
143 | QC & R&D EQUIPMENT | UV-VIS spectrophotometer Double Beam with data acquisition. | 1 | 900000 ||
144 | QC & R&D EQUIPMENT | Volumetric KF | 1 | 700000 ||
145 | QC & R&D EQUIPMENT | Moisture Analyser | 1 | 800000 ||
146 | QC & R&D EQUIPMENT | Stability Chambers (5±3° C, 25, 37 deg. C) 200L (1set) | 3 | 1400000 ||
147 | QC & R&D EQUIPMENT | Muffle Furnance | 1 | 80000 ||
148 | QC & R&D EQUIPMENT | pH and conductivity meter with printer | 2 | 260000 ||
149 | QC & R&D EQUIPMENT | Vaccum Oven | 1 | 120000 ||
150 | QC & R&D EQUIPMENT | Melting point apparatus | 1 | 90000 ||
151 | QC & R&D EQUIPMENT | Incubator Sterility Test 500L | 2 | 1440000 ||
152 | QC & R&D EQUIPMENT | Incubators for Normal Media plates Incubation 500L | 1 | 300000 ||
153 | QC & R&D EQUIPMENT | Incubator for EMP and water samples 500L | 2 | 600000 ||
154 | QC & R&D EQUIPMENT | Hot Air Oven | 1 | 160000 ||
155 | QC & R&D EQUIPMENT | Thermostatic water bath (10-20 L) | 2 | 200000 ||
156 | QC & R&D EQUIPMENT | Vaccum pump | 2 | 120000 ||
157 | QC & R&D EQUIPMENT | Desiccator Glass | 1 | 4000 ||
158 | QC & R&D EQUIPMENT | High Quality Laboratory water generation system (10 L /hr) Milli -Q water | 1 | 1450000 ||
159 | QC & R&D EQUIPMENT | Sonicator | 1 | 180000 ||
160 | QC & R&D EQUIPMENT | Vortex Shaker | 6 | 210000 ||
161 | QC & R&D EQUIPMENT | Magnetic Stirrer with hot plate | 4 | 400000 ||
162 | QC & R&D EQUIPMENT | LAF Vertical | 2 | 600000 ||
163 | QC & R&D EQUIPMENT | Biosafety Cabinet | 2 | 1400000 ||
164 | QC & R&D EQUIPMENT | Autoclave Double door | 1 | 1250000 ||
165 | QC & R&D EQUIPMENT | Dry Heat Steriliser | 1 | 750000 ||
166 | QC & R&D EQUIPMENT | Fume Hood (Table Top) | 1 | 450000 ||
167 | QC & R&D EQUIPMENT | Deep freezer (-20 Degrees C) with Data acquisition | 1 | 400000 ||
168 | QC & R&D EQUIPMENT | Air sampler | 1 | 300000 ||
169 | QC & R&D EQUIPMENT | Particle counter | 1 | 2000000 ||
170 | QC & R&D EQUIPMENT | Decontamination autoclave | 1 | 1000000 ||
171 | QC & R&D EQUIPMENT | Microscope Binocular | 1 | 100000 ||
172 | QC & R&D EQUIPMENT | Colony Counter | 1 | 150000 ||
173 | QC & R&D EQUIPMENT | Microplate reader ELISA | 1 | 600000 ||
174 | QC & R&D EQUIPMENT | ELISA Auto Analyzers (Roche / Abbott)* | 2 | 1000000 ||
175 | QC & R&D EQUIPMENT | Microplate washer | 1 | 350000 ||
176 | QC & R&D EQUIPMENT | TOC analyser (Grab Samples) | 1 | 2500000 ||
177 | QC & R&D EQUIPMENT | Timers/ Stop watch | 5 | 10000 ||
178 | QC & R&D EQUIPMENT | Ice flaking machine | 1 | 250000 ||
179 | QC & R&D EQUIPMENT | Standard Weight Box | 1 Set | 1000000 ||
180 | QC & R&D EQUIPMENT | Printer/scanner / xerox machine | 1 | 500000 ||
181 | QC & R&D EQUIPMENT | Centrifuge table top | 1 | 450000 ||
182 | QC & R&D EQUIPMENT | Size exclusion | L | 150000 ||
183 | QC & R&D EQUIPMENT | Anion | L | 150000 ||
184 | QC & R&D EQUIPMENT | Cation | L | 150000 ||
185 | QC & R&D EQUIPMENT | Affinity | L | 150000 ||
186 | QC & R&D EQUIPMENT | SS furniture racks | 25 | 250000 ||
187 | QC & R&D EQUIPMENT | Computers / Laptops | 10 | 500000 ||
188 | QC & R&D EQUIPMENT | Office Furniture | NA | 500000 ||
189 | QC & R&D EQUIPMENT | Office Cabinets for Documents | 10 | 120000 ||
190 | QA EQUIPMENT | Anemometer | 1 | 15000 ||
191 | QA EQUIPMENT | Lux meter | 1 | 12000 ||
192 | QA EQUIPMENT | Datalogger 32 channel | 1 | 400000 ||
193 | QA EQUIPMENT | Paper shredder | 1 | 10000 ||
194 | QA EQUIPMENT | Laminator | 1 | 10000 ||
195 | QA EQUIPMENT | Printer/scanner / xerox machine | 1 | 500000 ||
196 | QA EQUIPMENT | Computers / Laptops | 10 | 500000 ||
197 | QA EQUIPMENT | Numbering Machine | 5 | 25000 ||
198 | QA EQUIPMENT | Spiral Binding Machine | 1 | 15000 ||
199 | QA EQUIPMENT | A3 +A4 Lamination Machine | 1 | 25000 ||
200 | QA EQUIPMENT | Bi-Focal Glass | 1 | 80000 ||
201 | QA EQUIPMENT | Tachometer | 2 | 6000 ||
202 | QA EQUIPMENT | Clamp Meter/ Multi meter | 5 | 10000 ||
203 | QA EQUIPMENT | Level Indicator | 2 | 8000 ||

204 | QA EQUIPMENT | Measuring TAPE | 2 | 1500 | |
205 | QA EQUIPMENT | Stop Watch | 5 | 50000 | |
206 | QA EQUIPMENT | Temperature recorders 24 Nos. (data Loggers) | 20 | 200000 | |
207 | QA EQUIPMENT | Office Cabinets for Documents | 20 | 240000 | |
208 | QA EQUIPMENT | Projector for conference / meeting room | 1 | 300000 | |
209 | QA EQUIPMENT | Office Furniture | NA | 700000 | |
210 | FINISHED GOODS STORAGE | Air curtain | 10 | 100000 | |
211 | FINISHED GOODS STORAGE | PVC pallets | 50 | 300000 | |
212 | FINISHED GOODS STORAGE | Storage racks SS | 50 | 1500000 | |
213 | FINISHED GOODS STORAGE | Hydraulic trolley | 1 | 250000 | |
214 | ETP | Biowaste reception tank 15000L | 1 | 4000000 | |
215 | ETP | Kill tank 1000L | 2 | | |
216 | ETP | Bar Screen | NA | | |
217 | ETP | Collection tank | NA | | |
218 | ETP | Equalization tank | NA | | |
219 | ETP | Neutralization tank | NA | | |
220 | ETP | Chemical coagulation | NA | | |
221 | ETP | Biological oxidation tank | NA | | |
222 | ETP | Filtration System | NA | | |
[SHEET: Heat treat_Reaction_Tank]
Sr.No. | Vessel | Working vol (L) | No. of units recommended
1 | Plasma pooling tank (refrigerated) | 2000 | 1 primary + 1 standby (or reserve)
2 | Caprylic acid reaction tank (jacketed) | 2000 | 1 (or 2 for staggered batches)
3 | S-D mixing tank (jacketed, sealed) | 2000 | 1 (with dedicated solvent bund)
4 | Clarification / settling tank or DAF feed | 2500 | 1
5 | Pasteurizer (container) | capacity to hold per-batch vials (900) | 1 tunnel / basket line sized to throughput
6 | Buffer / formulation tanks (jacketed) | 500–1,500 | 2–3 (for different buffers)
7 | Waste / solvent hold tank (bunded) | 1,000–2,000 | 1
| Back to Index | |
[SHEET: Fuel Consmpt]
Sr. No. | Fuel Type | Purpose / Area of Use | Approx. Annual Consumption | Unit
1 | Electricity | Cleanroom HVAC, process equipment (centrifuges, lyophilizers, pumps, lighting, DG auxiliaries, WFI generation, chillers, lab & office) | 3.5 – 4.0 million kWh / year | kWh
2 | HSD (High-Speed Diesel) | Stand-by DG sets (2 × 500 kVA) for emergency power | ≈ 25 – 30 KL / year | KL (≈ 25,000 – 30,000 L)**
3 | CNG / PNG | Boiler / hot-water generation for WFI plant, sterilization steam, HVAC re-heat coils | ≈ 150,000 – 180,000 Sm³ / year | Standard m³ (Sm³)
4 | LPG (optional small cylinders) | Laboratory instruments / flame sterilizers | ≤ 500 kg / year | kg
| Back to Index | | |
[SHEET: Water]
Sr.No. | Item | Litres (L / year) | Cubic metres (m³ / year) | Notes
| A. Water Inputs | | |
1 | Domestic water (506 ppl × 100 L/p/d) | 18469000 | 18469 | Toilets, canteen, housekeeping
| WFI (process) | 6205470 | 6205.47 | WFI for formulation, CIP, rinses
| Other process water (25% of WFI) | 1551368 | 1551.37 | CIP makeup, utility washes
| Extra process water (10% of plasma vol) | 41400 | 41.4 | Misc. process losses, small uses
2 | Process Water | 7798238 | 7798.24 |
| Total Water Input | 34065476 | 34065.48 | A = sum of above
| B. Water Outputs / Wastewater | | |
1 | Domestic sewage → STP (80% of domestic) | 14775200 | 14775.2 | Black/gray water to STP
2 | Process wastewater → ETP (99% of process) | 15440511.24 | 15440.51124 | Process drains, WFI rejects, CIP effluent
| Total Wastewater Generated | 30215711.240000002 | 30215.71124 | B = sum of above
| C. Balance / Losses / Product incorporation | | |
1 | Unaccounted losses / product incorporation / evaporation (A – B) | 3849764.759999998 | 3849.7687600000027 | ~11.3% of input — includes water in products, steam losses, sampling, evaporation, bleed, equipment hold-up
| Back to Index | | |
[SHEET: Raw Material]
Sr.No. | Item | Annual requirement (rounded) | Unit | Notes |
1 | Source human plasma | 414000 | L | Primary raw material (cold chain required) | Back to Index
2 | Water for Injection (WFI) | 6205000 | L | Assumed 15,000 L WFI / 1,000 L plasma |
3 | Sodium chloride (NaCl) | 10350 | kg | 25 kg / 1,000 L plasma × 413.7 |
4 | Sodium citrate / acetate | 2070 | kg | 5 kg / 1,000 L |
5 | Glycine | 4140 | kg | 10 kg / 1,000 L |
6 | Caprylic (Octanoic) acid | 3310 | kg | 8 kg / 1,000 L |
7 | Triton X-100 / Polysorbate-80 | 2070 | L | 5 L / 1,000 L |
8 | Tri(n-butyl) phosphate (TnBP) | 1241 | L | 3 L / 1,000 L |
9 | PBS / Tris buffer components | 8275 | kg | 20 kg / 1,000 L |
10 | Acids / NaOH (dilute) | 82740 | L | 200 L / 1,000 L (process neutralization) |
11 | Activated carbon (GAC) | 20685 | kg | 50 kg / 1,000 L |
12 | Filter aids (depth filter media) | 41370 | kg | 100 kg / 1,000 L |
13 | 0.2 µm sterile filters (cartridges) | 8300 | units | 20 units / 1,000 L |
14 | PVDF / sterile filter cartridges | 4140 | units | 10 units / 1,000 L |
15 | Chromatography resins (total across steps) | 20685 | kg | 50 kg / 1,000 L |
16 | TFF membranes / UF-DF modules | 4140 | modules | 10 modules / 1,000 L |
17 | Albumin stabilizer (Na-octanoate, etc.) | 4965 | kg | 12 kg / 1,000 L |
18 | Fibrin sealant reagents (thrombin, substrates) | 2070 | kg | 5 kg / 1,000 L |
19 | Preservatives (if used) | 4140 | kg | 10 kg / 1,000 L |
20 | Nitrogen gas (compressed) | 206850 | m³ | 500 m³ / 1,000 L |
21 | Sterile glass vials (all sizes) | 1037500 | units | Total vials + 5% spare (see note below) |
22 | Rubber stoppers | 1037500 | units | Match vials (one per vial) |
23 | Flip-off seals | 1037500 | units | Match vials |
24 | Labels | 1040000 | units | one per vial + spares |
25 | Cartons (≈10 vials/carton) | 98810 | cartons | Round up to supplier pack sizes |
26 | Laboratory consumables (pipette tips, plates, etc.) | Estimate 4,000 | packs | Itemised procurement per lab list |
27 | Spare filters / membranes (replacement spares) | +20% of above | — | |
28 | Cold-chain storage (freezer/refrigeration units) | qty per plant | — | |
| Back to Index | | |
[SHEET: List of Products]

Product | Protein availability (gm) | Protein Recovery | Conc. / vial | No. of vials /batch | Total
(Considering
15 Batches / month) | Per Annum
Albumin (gm) | 40000 | 20000 | 20 GM | 900 | 13500 | 162000
Human IgG (gm) | 12500 | 6250 | 5 GM | 1000 | 15000 | 180000
Factor VIII in IU | 1000000 | 12500 IU | 250 IU | 50 | 750 | 9000
Factor IX in IU | 1000000 | 90000 IU | 600 IU | 150 | 2250 | 27000
Fibrinogen (gm) | 3000 | 1500 | 70 mg | 80 | 1200 | 14400
Thrombin in IU | 900000 | 90000 | 1000 IU | 1400 | 21000 | 252000
Based on vessel size, the batch size assumed is 1000 liters, on every alternate date, a new batch can be initiated, therefore 15 batches shall be processed per month (one batch/ day). |||||
Back to Index |||||
[SHEET: Non Haz Waste]
Sr. No. | Source / Process Area | Type of Non-Hazardous Waste | Approx. Quantity / Year | Solid | Liquid
| Category | ETP/3rd party
1 | ETP / STP treatment | Biological sludge (after dewatering) | 8 – 10 MT/year (wet) | 10 | | Organic |
3rd party
2 | Process area & lab washing | Treated water (ETP + RO permeate) | ~12,000 m³/year | | 12000000 |
Organic | Domestic use
3 | Packaging line | Empty glass vials (non-contaminated rejects) | 0.5 – 1.0 MT/year (~50,000 pcs) | 1 |
| Inorganic | Scrap 3rd party
4 | Packaging & labeling | Paperboard cartons, labels, paper scrap | 1.5 – 2.0 MT/year | 2 | | Organic |
Scrap 3rd party
5 | Procurement / stores area | Corrugated boxes, stretch film, plastic wrapping | 1.0 – 1.5 MT/year |
1.5 | | Inorganic | Scrap 3rd party
6 | Maintenance workshop | Metal scrap (iron, steel, aluminum parts) | 0.8 – 1.2 MT/year | 12 | |
Inorganic | Scrap 3rd party
7 | Utility & HVAC area | Spent filters (air filters, bag filters, prefilters) | 0.3 – 0.5 MT/year | 0.5
| | Inorganic | 3rd party
8 | Admin & office areas | Office paper waste, packaging waste | 1.0 – 1.5 MT/year | 1.5 | | Organic |
Scrap 3rd party
9 | Canteen / pantry | Organic kitchen waste | 1.5 – 2.0 MT/year | 2 | | Organic | Organic converter
10 | General housekeeping | Sweepings, mopping dust, non-contaminated PPE | 0.5 – 0.8 MT/year | 0.8 | |
Mixed | 3rd party
11 | Cooling tower & utilities | Blowdown & scale sludge (non-toxic) | 1.0 – 1.5 MT/year | 1.5 | |
Organic | 3rd party
12 | Electrical systems | Cable cuttings, wire ends, plastic conduits | 0.2 – 0.3 MT/year | 0.3 | |
Inorganic | Scrap 3rd party
13 | Gardening & green area | Dry leaves, grass clippings | 0.3 – 0.5 MT/year | 0.5 | | Organic | Organic
converter
14 | Water purification & softener | Spent ion-exchange resins (non-toxic) | 0.2 MT/year | 0.2 | |
Inorganic | 3rd party
15 | Misc. activities | Empty HDPE / PET bottles (non-contaminated) | 0.1 – 0.2 MT/year | 0.2 | |
Inorganic | 3rd party
Back to Index		Domestic use	32876.71232876712	L/Day	
		Organic converter	6.8493150684931505	T/day	
		3rd party	56.71232876712329	T/day	
		Scrap 3rd party	50.136986301369866	T/day	
[SHEET: Hazardous Waste]					
Sr. No.	Process / Source	Type of Hazardous Waste	Approx. Quantity / Year	Category	
1	Solvent recovery & viral inactivation (TnBP + detergent process)	Spent solvent mixture (Tri-n-butyl			
phosphate + Triton X-100 / Polysorbate 80)	1.0 – 1.5 KL / year	Inorganic	ETP		
2	Filtration & purification units	Used filter aids / depth filter media contaminated with protein or			
solvent	0.8 – 1.0 MT / year	mixed	3rd party		
3	Activated carbon beds (viral removal, deodorization)	Spent activated carbon with adsorbed organic			
residues	0.5 – 0.7 MT / year	organic	3rd party		
4	Chromatography / TFF cleaning (CIP)	Residues of buffers, cleaning agents, NaOH / HCl neutralization			
sludge	2.0 – 3.0 MT / year (dry solids)	Inorganic	ETP		
5	ETP operation	Sludge containing proteinaceous organic matter & trace solvents	5.0 – 6.0 MT / year		
organic	3rd party				
6	ETP / RO reject water	High-TDS / COD effluent (liquid)	~ 900 – 1,000 m³ / year	Inorganic	ETP
7	Resin & membrane replacement	Spent chromatography resin, UF/DF membranes, PVDF filters	0.5 – 1.0		
MT / year	Inorganic	3rd party			
8	Laboratory & QC activities	Contaminated lab glassware, pipette tips, chemical bottles	200 – 250		
kg / year	Inorganic	Scrap 3rd party			
9	Oil from DG sets, compressors, hydraulic systems	Used lubricating oil	50 – 100 L / year	organic	
ETP					
10	DG set maintenance	Oil-soaked cotton waste, filters	50 – 100 kg / year	Inorganic	Scrap 3rd
party					
11	Production packaging & vial washing	Contaminated packaging (plastic, foil, paper)	0.3 – 0.5 MT /		
year	Inorganic	Scrap 3rd party			
12	Spill & floor washings	Absorbent material, mopping waste, PPE, gloves	0.5 MT / year	Inorganic	
3rd party					
Back to Index					

Reagent Details:

Preparation of PF-RR-12 Stock buffer :

Calculations: For the Preparation of 40mM Sodium Citrate pH 7.0 + 0.6M Sodium Chloride.

Stock Buffer (PF-RR-09):0.4M Sodium Citrate pH 7.0.

Formula:

$$C1 \times V1 = C2 \times V2$$

C1 = concentration of stock

V1 = volume of stock you need to take

C2 = desired (working) concentration

V2 = final total volume of working solution

Applying the formula to prepare the working buffer:

$$C1 \times V1 = C2 \times V2$$

C1 = 0.4M Sodium Citrate, pH 7.0.

V1 = volume of stock you need to take _____ L.

C2 = desired (working) concentration 40mM Sodium Citrate, pH 7.0.

V2 = final total volume of working solution _____ L.

$$V1 = (C2 \times V2) / C1$$

$$C2 = 0.04M ; V2 = 1.0 L ; C1 = 0.4M$$

$$V1 = (0.04 \times 1.0) / 0.4$$

$$V1 = 0.1L$$

Take 0.1 L (PF-RR-09) Stock buffer solution and add 0.6M sodium Chloride.

Calculations: To prepare the 0.6M Sodium Chloride.

Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).

Calculation: To prepare 1 L of 0.6M Sodium Chloride (MW = 58.44 g/mol)

$$\text{Weight of solute(g)} = 0.6 \times 58.44 \times 1 = 35.06 \text{ g.}$$

Preparation Method:

Weigh 35.06 g of Sodium Chloride and dissolve in water to make a 1 L solution.

Preparation for different volumes:

To prepare _____ L of 0.6 M Sodium Chloride (MW = 58.44 g/mol).

$$\text{Weight of solute(g)} = 0.6 \times 58.44 \times \text{_____ L} = \text{_____ g.}$$

Weight _____ gm of Sodium Chloride and dissolve in water to make a _____ L solution.

Preparation of Stock buffer (PF-RR-12):

Take 0.1 L (PF-RR-09) Stock buffer solution and add 0.6M sodium Chloride and dissolve it, and make up the volume with water to one litre.

Check and record the details:

Preparation Details:

Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry

Date

Sodium Citrate | | | |

Citric Acid | | | |

Sodium Chloride | | | |

Stock buffer (PF-RR-09) taken volume(L) | pH (7.0 ± 0.1) | Conductivity (58 ± 0.2) | Storage Condition

| Final volume prepared

| | | |

Date of Preparation | Prepared By | Checked By

| |

1
FOR OFFICE USE 2

water tank

inlet pipe
ates
1.200 solar pl
TERRACE
TERRACE FLOOR

6.750
6.000
SECOND FLOOR

6.750
6.000
24.000
FIRST FLOOR
BUILDING L.V.P.

PLAN OF PROPOSED FACTORY

9.300
8.500
BUILDING ON PLOT NO 8A, SECTOR 13,
GROUND FLOOR
BUILDING DIALIZER
1.200
DMIC, AURIC , SHENDRA,
T-0.30m.
R-0.15m.

SECTION A-A
CHATTRAPATI SAMBHAJI NAGAR
SECTION AA
FOR M/S LIFENITY HEALTH LIMITED
FRONT ELEVATION
front elevation

SCALE 1:100 SCALE 1:100

AREA STATEMENT
AREA IN
NO PARTICULARS
SQM.

1 AREA OF PLOT NO 56 SECTOR 13, DMIC, SHENDRA 81920.00

AS PER OWNERSHIP DOCUMENTS 81920.00
RS1 SHAFT SHAFT SHAFT
RS1 RS1

3.30X2.45

3.30X2.45

3.30X2.45

3.30X2.45
DISPATCH BAY RS1 RS1 RS1 RS1
RECEIVING BAY RECEIVING BAY DISPATCH BAY
FIREMAN FIREMAN FIREMAN FIREMAN

LIFT

LIFT

LIFT

LIFT
LIFT
2.80X2.60
LIFT
2.80X2.60
LIFT
2.80X2.60
LIFT
2.80X2.60 AS PER MEASUREMENT SHEET 81920.00
5.00 M WIDE CORRIDOR
D1

UP
D1 D1 D1
AS PER SITE 81920.00

2 DEDUCTIONS FOR

W W
a. PROPOSED DP ROAD 0.00

b. ANY RESERVATIONS 0.00
TOTAL (a+b)

6.77 M WIDE CORRIDOR
0.00
2.94 M WIDE CORRIDOR

2.94 M WIDE CORRIDOR

2.94 M WIDE CORRIDOR
W
WC WC WC WC WC WC WC WC WC
1.10X
1.70
1.10X
1.70
1.10X
1.70
1.10X
1.70
1.10X
1.70
1.10X
1.70
1.10X
1.70
1.10X
1.70
1.10X
1.70
1.10X
1.70
1.10X
1.70
1.10X
1.70
3 BALANCE AREA OF PLOT (1-2) (A) 81920.00

6.77 M WIDE CORRIDOR

6.77 M WIDE CORRIDOR
V1 V1 V1
ventilation shaft D1 D1 ventilation shaft ventilation shaft
V1 V1 V1

WC
1.10X
WC
1.10X
WC
1.10X
WC
1.10X
WC
1.10X
WC
1.10X
WC
1.10X
WC
1.10X
WC
1.10X
WC
1.10X
WC
1.10X
WC
1.10X
4 AMENITY SPACE IF APPLICABLE 5% 0.00
1.70 1.70 1.70 1.70 1.70 1.70 1.70 1.70 1.70
W
a. Required 4096.00
b. Adjustment of 2(b) if any 0.00
c. Balanced proposed 4125.33
W W

5 NET PLOT AREA (3-4(c))) PRO RATA 81920.00

6 RECREATIONAL OPEN SPACE IF APPLICABLE 10%
W W

a. Required 8192.00

ventilation shaft
ventilation shaft
ventilation shaft
ventilation shaft
ventilation shaft
ventilation shaft
b. Proposed 8202.72
SHAFT SHAFT SHAFT SHAFT SHAFT SHAFT SHAFT SHAFT

LIFT LIFT LIFT LIFT LIFT LIFT LIFT LIFT
2.10X 2.10X 2.10X 2.10X 2.10X 2.10X 2.10X 2.10X
7 INTERNAL ROAD AREA 0.00

3.24 M WIDE PASSAGE

3.24 M WIDE PASSAGE
FD
2.00
FD

DIALYZER FD

2.00

FD

2.00

FD

LVP FD

2.00 2.00

FD

WORKSHOP 2.00

FD

2.00

FD FD

2.00

UP

D1 D1

UP UP UP UP UP

8 PLOTABLE AREA IF APPLICABLE (as per sanctioned layout) 81920.00

53.54 x 91.34 53.54 x 96.57 53.54 x 96.57

9 B/UP area with reference to basic fsi as per front road width(1.0) 81920.00

10 ADDITIONAL FSI ON PAYMENT OF PREMIUM

a. Maximum permissible premium FSI / TOD Zone. 2.00

W

b. Proposed FSI of payment of Premium 0.00

W

11 IN SITU TDR / FSI LOADING

W W a. In situ area against Statutory Raod (2 xSr No 2(a)) if any 0.00

12 ADDITIONAL FSI AREA UNDER CHAPTER NO.6 0.000

13 TOTAL ENTITLEMENT OF FSI IN THE PROPOSAL 0.000

2.80 M WIDE PASSAGE

2.80 M WIDE PASSAGE

2.80 M WIDE PASSAGE

W

WC

1.10X

WC

1.10X

WC

1.10X

WC

1.10X

WC

1.10X

WC

1.10X

WC

1.10X

WC

1.10X

WC

1.10X

a. 9 +10(b) +11(a) or 9+12 which ever is applicable

1.70 1.70 1.70

V1

1.70 1.70 1.70 1.70 1.70 1.70

Permissible 0.00

V1 V1

D1

D1

ventilation shaft

V1 V1 V1

ventilation shaft ventilation shaft

a. 9 +10(b) +11(a) or 9+ 12 which ever is applicable

WC

1.10X

WC

1.10X

WC

1.10X

WC

1.10X

WC

1.10X

WC

1.10X

WC

1.10X

WC

1.10X

WC

1.10X

Proposed 0.00

1.70 1.70 1.70 1.70 1.70 1.70 1.70 1.70 1.70

W
b. Permissible Ancillary area FSI up to 80% with payment
PERMISSIBLE 0.00
b. Proposed Ancillary area FSI up to 60% or 80% with payment

c.Total Entitlement permissible
W (a+b) 81920.00
W
c.Total Entitlement proposed
(a+b) 81920.00
Maximum utilization limit of FSI (Building Potential) permissible as per
14 road width ((as per Regulation No. 5.5 of Chapter - 5)) x 1.6 or 1.8) 81920.00
6.77 M WIDE CORRIDOR

6.77 M WIDE CORRIDOR

6.77 M WIDE CORRIDOR

6.77 M WIDE CORRIDOR

6.77 M WIDE CORRIDOR
W W

15 TOTAL Built up area in proposal (excluding area as per sr. no 17b)

a. Existing Built -up Area. 0.000

W W
b. Proposed Built- up area (as per P line) 44238.93

c. Total (a+b) 44238.93

16 FSI CONSUMED (15/13) (should not be more than serial 0.54
W W
No.14 above.)

CERTIFICATE OF AREA :-
ventilation shaft ventilation shaft ventilation shaft
ventilation shaft
SHAFT SHAFT
ventilation shaft
SHAFT SHAFT
ventilation shaft
SHAFT SHAFT SHAFT SHAFT Certified that the plot under reference was surveyed by me on
LIFT LIFT LIFT LIFT LIFT LIFT LIFT LIFT
2.10X 2.10X 2.10X 2.10X 2.10X 2.10X 2.10X 03/03/2026 and the Dimensions of sides etc.of plot stated
on plan area

3.24 M WIDE PASSAGE
2.00 2.00 2.00 2.00 2.00 2.00 2.00 2.00
FD FD FD FD FD FD FD FD FD as measured on site and the area so worked out tallies with the stated in
UP
D1 D1
UP UP UP UP UP
documents of ownership/ DMIC/Land records Department/City Survey
Records
Ar. Dashrath Dhumal

Owners Declaration :-
I/We undersigned hereby confirmed that I/We would abide by plans
W
W
approved by Authority/ Collector. I / We would execute the structure as
per the approved plans.Also I/We would execute the work under
supervision of proper technical person so as to ensure the quality and
safety at the work site.
W
W
Lifenity Health Limited.

Architects Name and Signature

W
Ar. Dashrath Dhumal
Job No. Drg. No. Scale Drn. By. Checked By Registration no of Architect
WC WC WC WC WC WC WC WC WC
6.77 M WIDE CORRIDOR

6.77 M WIDE CORRIDOR

6.77 M WIDE CORRIDOR
1.10X 1.10X 1.10X 1.10X 1.10X 1.10X 1.10X 1.10X
2.80 M WIDE PASSAGE

2.80 M WIDE PASSAGE

2.80 M WIDE PASSAGE
1.70 1.70 1.70 1.70 1.70 1.70 1.70 1.70

V1 V1 V1

ventilation shaft D1 ventilation shaft ventilation shaft 001 1:100 Ar. Dashrath Dhumal CA/2005/37017
V1 V1 V1

WC WC WC WC WC WC WC WC WC
1.10X 1.10X 1.10X 1.10X 1.10X 1.10X 1.10X 1.10X 1.10X
1.70 1.70 1.70
W
1.70 1.70 1.70 1.70 1.70 1.70
C PARKING STATEMENT
GARAGE PERMISSIBLE GARAGE PROPOSED NIL
PARKING PARKING
b TYPE
REQUIRED PROPOSED
CAR
W W

SCOOTER
8192 SQ.M. 8195.53 SQ.M.
CYCLE

W
VISITORS
W

D LOADING UNLOADING SPACES
a LOADING UNLOADING REQUIRED 44
b TOTAL LOADING UNLOADING PROVIDED 44
W
WC WC WC WC WC WC WC WC WC
1.10X 1.10X 1.10X 1.10X 1.10X 1.10X 1.10X 1.10X 1.10X
1.70 1.70 1.70 1.70 1.70 1.70 1.70 1.70 1.70
LEGENDS
WORKSHOP TERRACE
V1 V1 V1

PLASMA EXTRACTION D1

WC
ventilation shaft

V1 V1

WC
V1

WC
PLASMA EXTRACTION ventilation shaft

WC WC WC WC
ventilation shaft

WC WC
PLOT BOUNDARY SHOWN IN
B/UP AREA SHOWN IN
53.54 x 115.59 53.54 x 120.82 53.54 x 120.82
1.10X 1.10X 1.10X 1.10X 1.10X 1.10X 1.10X 1.10X 1.10X
1.70 1.70 1.70 1.70 1.70 1.70 1.70 1.70 1.70

W

WATER LINE SHOWN IN
GREY WATER LINE SHOWN IN
SEWAGE LINE SHOWN IN
EFFLUNT LINE
W
W

SCHEDULE OF DOORS AND WINDOWS
TYPE SIZE DISCRPTION

W W
RS1 2500 X 6000 M.S ROLLING SHUTTER

D1 1800 X 2400 FLUSH DOOR

D2 900 X 2400 FLUSH DOOR

ventilation shaft ventilation shaft ventilation shaft ventilation shaft ventilation shaft ventilation
shaft
D3 750 X 2400 FLUSH DOOR
SHAFT SHAFT SHAFT SHAFT SHAFT SHAFT SHAFT SHAFT

LIFT LIFT LIFT LIFT LIFT LIFT LIFT LIFT FD 1000 X 2400 FIRE DOOR
2.10X 2.10X 2.10X 2.10X 2.10X 2.10X 2.10X 2.10X
2.00 2.00 2.00 2.00 2.00 2.00 2.00 2.00 W 2500 X 2500 ALLU. SLIDING WINDOW

3.24 M WIDE PASSAGE
3.24 M WIDE PASSAGE

3.24 M WIDE PASSAGE

3.24 M WIDE PASSAGE

FD FD FD FD FD FD FD FD FD
UP UP UP UP UP UP
V 600 X 900 ALU LOUVERED VENTILATOR
D1 D1

PLAN OF PROPOSED FACTORY
UP
BUILDING ON PLOT NO 8A, SECTOR 13,
DMIC, AURIC , SHENDRA,
W
W
CHATRAPATI SAMBAJI NAGAR

W
W
6.77 M WIDE CORRIDOR

6.77 M WIDE CORRIDOR

6.77 M WIDE CORRIDOR
W
WC WC WC WC WC WC WC WC WC
FOR M/S LIFENITY HEALTH LIMITED
2.80 M WIDE PASSAGE

2.80 M WIDE PASSAGE

2.80 M WIDE PASSAGE

1.10X 1.10X 1.10X 1.10X 1.10X 1.10X 1.10X 1.10X 1.10X
1.70 1.70 1.70 1.70 1.70 1.70 1.70 1.70 1.70

V1 V1 V1
ventilation shaft D1 D1 ventilation shaft ventilation shaft
V1 V1 V1

WC WC WC WC WC WC WC WC WC
1.10X 1.10X 1.10X 1.10X 1.10X 1.10X 1.10X 1.10X 1.10X
1.70 1.70 1.70 1.70 1.70 1.70 1.70 1.70 1.70

W

SIGN OF OWNER

W
W

SOLAR WATER HEATING

D1 D1 D1 D1
W

5.00 M WIDE CORRIDOR
3.30X2.45

3.30X2.45

3.30X2.45

3.30X2.45
SIGN OF ARCHITECT
LIFT

LIFT

LIFT

LIFT
RS1 DISPATCH BAY RS1 RS1 RECEIVING BAY RS1 RS1 RS1 DISPATCH BAY RS1
RS1 RECEIVING BAY

WORKSHOP BUILDING GROUND FLOOR PLAN WORKSHOP BUILDING FIRST FLOOR PLAN WORKSHOP BUILDING SECOND FLOOR
PLAN WORKSHOP BUILDING TERRACE FLOOR PLAN
SCALE 1:100 SCALE 1:100 SCALE 1:100 SCALE 1:100
GROUND FLOOR PLAN FIRST FLOOR PLAN SECOND FLOOR PLAN TERRACE FLOOR PLAN

CREATION

JOB No. DRG NO. SCALE DATE DRN BY CHK. BY

01 1 :100

Reagent Details:
Formula & Calculations:
Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).
Calculation: To prepare 1 L of 0.2M Citric Acid (MW = 192.124 g/mol)
Weight of solute(g)=0.2×192.124x1=38.424g.
Preparation Method:
Weigh 38.424g of Citric Acid and dissolve in water to make a 1 L solution.
Preparation for different volumes:
To prepare _____ volume of 0.2M Citric Acid (MW = 192.124g/mol).
Weight of solute(g)=0.2×192.124 x _____ L= _____ g.
Weigh _____ gm of Citric Acid and dissolve in water to make a _____ L solution.
Check and record the details:
Preparation Details:
Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry
Date
Citric Acid | | | |
Final volume prepared | Weighed Quantity | pH
(7.0 to 8.5) | Conductivity
(1.0 to 3.0ms/cm) | Storage Condition | Reagent Record
| | | | PF-RR-02
Date of Preparation | Prepared By | Checked By
| |

Reagent Details:
Preparation of IG-RR-08 working buffer from Stock buffer (IG-RR-07):
Calculations: For the Preparation of 25mM Sodium acetate pH 5.2 +1.0M Sodium Chloride.
Stock Buffer (PF-RR-07):0.1M Sodium acetate, pH 5.35 + 4.0 M Sodium Chloride.
Formula:
 $C1 \times V1 = C2 \times V2$
C1 = concentration of stock
V1 = volume of stock you need to take
C2 = desired (working) concentration
V2 = final total volume of working solution
Applying the formula to prepare the working buffer:
 $C1 \times V1 = C2 \times V2$
C1 = 0.1M Sodium Acetate pH 5.35.
V1 = volume of stock you need to take _____ L.
C2 = desired (working) concentration 25mM Sodium Acetate pH 5.2
V2 = final total volume of working solution _____ L.
 $V1 = (C2 \times V2) / C1$
 $C2 = 0.025M$; $V2 = 1.0\text{ L}$; $C1 = 0.1M$
 $V1 = (0.025 \times 1.0) / 0.1$
 $V1 = 0.25L$
Take 0.25 L (PF-RR-07) Stock buffer solution and dilute it with water to make up a final volume of 1 L.
Check and record the details:
Preparation Details:
Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry
Date
Sodium Acetate | | | |
Acetic Acid | | | |
Sodium Chloride | | | |
Stock buffer (PF-RR-07) taken volume(L) | pH(5.2 ± 0.1) | Conductivity (130 ± 0.1 ms/cm) | Storage
Condition | Final volume prepared
| | RT |
Date of Preparation | Prepared By | Checked By (QC/Analyst)
| |

Reagent Details:
Preparation of PF-RR-035 working buffer from Stock buffer (PF-RR-05):
Calculations: For the Preparation of 0.2M Citric Acid.
Stock Buffer taken from (PF-RR-05):1.0 M Citric Acid.
Dilution Formula:
 $C1 \times V1 = C2 \times V2$
C1 = concentration of stock
V1 = volume of stock you need to take
C2 = desired (working) concentration
V2 = final total volume of working solution
Applying the formula to prepare the working buffer:
 $C1 \times V1 = C2 \times V2$
C1 = Concentration of Stock Citric Acid.
V1 = volume of stock you need to take _____ L.
C2 = desired (working) concentration 0.2M Citric Acid.
V2 = final total volume of working solution _____ L.
 $V1 = (C2 \times V2) / C1$
C2=0.2M ; V2= 1.0 L; C1=1.0M
 $V1 = (0.2 \times 1.0) / 1.0$
V1 = 0.2L
Take 0.2 L (PF-RR-05) Stock buffer solution and dilute with water, and make up to 1L.
Check and record the details:
Preparation Details:
Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry
Date
| | | |
Final volume prepared | pH (1.8±0.2) | Conductivity (10.0±0.5ms/cm) | Storage Condition | Reagent Record
| | | RT | PF-RR-035
Date of Preparation | Prepared By | Checked By
| |

Reagent Details:
Preparation of PF-RR-29 working buffer from Stock buffer (PF-RR-11):
Calculations: For the Preparation of 20mM Sodium Citrate pH 7.0 + 0.16M Sodium Chloride.
Stock Buffer (PF-RR-11):0.04M Sodium Citrate pH 7.0 + 0.32M Sodium Chloride
Formula:.
 $C1 \times V1 = C2 \times V2$
C1 = concentration of stock
V1 = volume of stock you need to take
C2 = desired (working) concentration
V2 = final total volume of working solution
Applying the formula to prepare the working buffer:
 $C1 \times V1 = C2 \times V2$
C1 = 0.04M Sodium Citrate pH 7.0 + 0.32M Sodium Chloride.
V1 = volume of stock you need to take _____ L.
C2 = desired (working) concentration 20mM Sodium Citrate, pH 7.0 + 0.16M Sodium Chloride.
V2 = final total volume of working solution _____ L.
 $V1 = (C2 \times V2) / C1$.
C2=0.02M ; V2= 1.0 L; C1=0.04M
 $V1 = (0.02 \times 1.0) / 0.04$
V1 = 0.5L
Take 0.5 L (PF-RR-11) Stock buffer solution and dilute with water, and make up to 1L.
Check and record the details:
Preparation Details:
Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry
Date
Sodium Citrate | | | |
Citric Acid | | | |
Sodium Chloride | | | |
Stock buffer (PF-RR-11) taken volume(L) | pH (7.0±0.1) | Conductivity (20±0.1ms/cm) | Storage Condition
| Final volume prepared
| | RT |
Date of Preparation | Prepared By | Checked By
| |

Reagent Details:

Formula:

Sodium Acetate(M.Wt): 82.03g/mol.

Acetic Acid (M.Wt): 60.05g/ mol

Preparation of 1.0 M Sodium Acetate pH 5.2 buffer

Formula:

Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol)×Volume (L).

Calculation: To prepare 1 L of 1.0 M Sodium Acetate (MW = 82.03g/mol).

Weight of solute(g)=1.0×82.03× 1.0=82.03g.

Preparation Method:

Weigh 82.03g of Sodium Acetate and dissolve in water to make a 1 L solution.

Preparation for required volumes:

To prepare _____ of 1.0 M Sodium Acetate (MW = 82.03g/mol).

Weight of solute(g)=1.0×82.03× _____ L= _____ g.

Weigh _____ gm of Sodium Acetate and dissolve in water to make a _____ L solution.

Calculations: To prepare the 1.0 M Acetic Acid.

Formula & Calculations:

Formula: Use the formula to calculate the molarity:

Molarity (M) = (Per cent by Weight / 100) * (Density in g/mL) * 1000 / Molar Mass.

Example: Molarity of Glacial Acetic Acid

Molar Mass of CH₃COOH: ~60.05 g/mol

Density: ~1.05 g/mL

Purity: 100%

Acetic Acid (MW = 60.05 g/mol).

Calculation:

Molarity = (100 / 100) * 1.05 g/mL * 1000 mL/L / 60.05 g/mol ≈ 17.5 M.

Calculation: To prepare 1 L of 1.0 M.

Applying the formula to prepare the working buffer:

C₁V₁=C₂V₂

C₁ = 17.5 M acetic acid.

V₁ = volume of stock needed to take.

C₂ = desired (working) concentration 1.0 M acetic acid.

V₂ = final total volume of working solution for one litre.

V₁ = (C₂ × V₂)/C₁.

C₂=1.0 M ; V₂= 1.0 L; C₁=17.5 M

V₁ = (1.0 × 1.0)/17.5

V₁ = 0.057 L

Take 0.057 L of the Stock solution of acetic acid and add it to a volumetric flask, making up the volume to 1 L.

To prepare various volumes:

V₁ = volume of stock needed to take _____ L.

C₂ = desired (working) concentration 1.0 M acetic acid.

V₂ = final total volume of working solution _____ L.

V₁ = (C₂ _____ × V₂ _____) /17.5.

V₁= _____ L.

Take V₁ _____ L of the Stock solution of acetic acid and add it to a volumetric flask, making up the final volume _____ L.

Procedure for Preparation of 1.0 M Sodium Acetate pH 5.2.

To prepare a pH 5.2 solution, 1.0 M Sodium Acetate solution is titrated with 1.0 M Acetic acid so that the pH attains pH 5.2.

Preparation Details:

Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry Date

Sodium Acetate | | | |

Acetic Acid | | | |

Final volume prepared | Measured Quantity | pH (5.2 ± 0.1) | Conductivity (85 ±0.5ms/cm) | Storage Condition | Reagent Record

| | | |

Date of Preparation | Prepared By | Checked By

| |

[SHEET: Room sizes]

Room name and size for Plasma facility (Lifenity)			
S. No.	Room Name	Existing Room Size	Comments
L (m) B (m)			
1	Buffer preparation	8.1 6	Ok
2	Tank Storage	8.1 6.58	Ok
3	CIP/SIP	8.1 5	Ok
4	Plasma day store	8.1 8.77	Ok
5	Stage 8N	6.17 8.68	Ok
6	Stage 2 F	15.8 8.68	Ok
7	Stage 3 F	12.8 8.68	Ok
8	Plasma thawing	8 3.22	Ok
9	Stage 1 F/ 12 E	11.1 8.21	Ok
10	Stage 4 F	13.6 13.5	Ok
11	Bag cutting and pooling	8 7.8	Ok
12	Stage 5 I A	11.1 18	Ok
13	Stage 5 I B	8 7.2	Ok
14	Stage 7 A	13.6 15.1	Ok
15	Column packing and Skid	8 7.86	Ok
16	Column packing and Skid	6.9 10.6	Ok
17	Bulk filtration	6.9 6	Ok
18	Stage 6 I/ 11 F	14.7 16.7	Ok
19	Stage 9 N	5.1 14.3	Ok
20	Stage 10 T/13 E	6.25 2.05	Ok
21		8.1 12.2	Ok
22	Tank storage - 2	6.8 8.1	Ok
23	Buffer preparation	4.29 8.1	Ok
24	CIP/SIP	5.6 8.1	Ok
25	Chemical day store	8.1 4	Room size may be increased by including space from corridor and from the adjacent change rooms.

Reagent Details:
Preparation of PF-RR-24 working buffer from Stock buffer (PF-RR-23):
Calculations: For the Preparation of 50mM Sodium acetate pH 4.5.
Stock Buffer (PF-RR-23):0.4M Sodium acetate, pH 4.6.
Formula:
 $C1 \times V1 = C2 \times V2$
C1 = concentration of stock
V1 = volume of stock you need to take
C2 = desired (working) concentration
V2 = final total volume of working solution
Applying the formula to prepare the working buffer:
 $C1 \times V1 = C2 \times V2$
C1 = 0.4M Sodium Acetate pH 4.6.
V1 = volume of stock you need to take _____ L.
C2 = desired (working) concentration 50mM Sodium Acetate pH 4.5
V2 = final total volume of working solution _____ L.
 $V1 = (C2 \times V2) / C1$.
C2=0.05M ; V2= 1.0 L; C1=0.4M
 $V1 = (0.05 \times 1.0) / 0.4$
V1 = 0.125L
Take 0.125 L (PF-RR-23) Stock buffer solution and dilute with water, and make up to 1L.
Check and record the details:
Preparation Details:
Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry
Date
Sodium Acetate | | | |
Acetic Acid | | | |
Stock buffer (PF-RR-23) taken volume(L) | pH
(4.5±0.1) | Conductivity (2.0±0.2ms/cm) | Storage Condition | Final volume prepared
| | | RT |
Date of Preparation | Prepared By | Checked By
| |

Reagent Details:
Formula & Calculations:
Formula: Use the formula to calculate the molarity:
 $Molarity (M) = (Per\ cent\ by\ Weight / 100) * (Density\ in\ g/mL) * 1000 / Molar\ Mass.$
Example: Molarity of Glacial Acetic Acid
Molar Mass of CH_3COOH : ~60.05 g/mol
Density: ~1.05 g/mL
Purity: 100%
Acetic Acid (MW = 60.05 g/mol).
Calculation:
 $Molarity = (100 / 100) * 1.05\ g/mL * 1000\ mL/L / 60.05\ g/mol \approx 17.5\ M.$
Calculation: To prepare 1 L of 1.0 M.
Applying the formula to prepare the working buffer:
 $C1 \cdot V1 = C2 \cdot V2$
 $C1 = 17.5\ M$ acetic acid.
 $V1$ = volume of stock needed to take.
 $C2$ = desired (working) concentration 1.0 M acetic acid.
 $V2$ = final total volume of working solution for one litre.
 $V1 = (C2 \times V2) / C1.$
 $C2 = 1.0\ M ; V2 = 1.0\ L ; C1 = 17.5\ M$
 $V1 = (1.0 \times 1.0) / 17.5$
 $V1 = 0.057\ L$
Take 0.057 L of the Stock solution of acetic acid and add it to a volumetric flask, making up the volume to 1 L.
To prepare various volumes:
 $V1$ = volume of stock needed to take _____ L.
 $C2$ = desired (working) concentration 1.0 M acetic acid.
 $V2$ = final total volume of working solution _____ L.
 $V1 = (C2 \times V2) / 17.5.$
 $V1 =$ _____ L.
Take $V1$ _____ L of the Stock solution of acetic acid and add it to a volumetric flask, making up the final volume _____ L.
Check and record the details:
Preparation Details:
Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry Date
|||
Final volume prepared | Measured Quantity | pH (2.7) | Conductivity 1.7ms/cm | Storage Condition | Reagent Record
|||
Date of Preparation | Prepared By | Checked By
||

Reagent Details:
Preparation of IG-RR-24 working buffer from Stock buffer (IG-RR-22):
Calculations: For the Preparation of 0.01M Sodium Hydroxide.
Stock Buffer (IG-RR-22):2.0M Sodium Hydroxide,
Formula:
 $C1 \times V1 = C2 \times V2$
C1 = concentration of stock
V1 = volume of stock you need to take
C2 = desired (working) concentration
V2 = final total volume of working solution
Applying the formula to prepare the working buffer:
 $C1 \times V1 = C2 \times V2$
C1 = 2.0 M Sodium Hydroxide.
V1 = volume of stock you need to take _____ L.
C2 = desired (working) concentration 0.01M Sodium Hydroxide.
V2 = final total volume of working solution _____ L.
 $V1 = (C2 \times V2) / C1$
C2=0.01M; V2= 1.0 L; C1=2.0M
 $V1 = (0.01 \times 1.0) / 2.0$
V1 = 0.005L
Take 0.005 L (IG-RR-22) Stock buffer solution and dilute with water, and make up to 1L.
Check and record the details:
Preparation Details:
Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry
Date
Sodium Hydroxide | | | |
Stock buffer (IG-RR-22) taken volume(L) | pH (12.0±0.5) | Conductivity (20.0±0.5) | Storage Condition |
Final volume prepared
| | RT |
Date of Preparation | Prepared By | Checked By
| |

A. Organic waste calculation for Plasma

Sr. Source / Process Approx. Quantity Kg Solid Liquid

Type of Non-Hazardous Waste Category ETP/3rd party

No. Area / Year (Kg/day) (L/day)

1 ETP / STP treatment Biological sludge (after dewatering) 6237850 17090 Organic 3rd party

Process area & lab

2 Treated water (ETP + RO permeate)

washing 27052340 L/year 74116 Organic Domestic use

Empty glass vials (non-contaminated

3 Packaging line 1000 Scrap 3rd party

rejects) (~50,000 pcs) 2.7 Inorganic

Packaging &

4 Paperboard cartons, labels, paper scrap 2000 Scrap 3rd party

labeling 5.5 Organic

Procurement / stores Corrugated boxes, stretch film, plastic

5 1500 Scrap 3rd party

area wrapping 4.1 Inorganic

Maintenance Metal scrap (iron, steel, aluminum

6 1200 Scrap 3rd party

workshop parts) 1 Inorganic

Utility & HVAC Spent filters (air filters, bag filters,

7 500 3rd party

area prefilters) 1 Inorganic

Admin & office

8 Office paper waste, packaging waste 1500 Scrap 3rd party

areas 25 Organic

9 Canteen / pantry Organic kitchen waste 47000 128.77 Organic Organic converter

General Sweepings, mopping dust, non-

10 800 3rd party

housekeeping contaminated PPE 2.19 Mixed

Cooling tower &

11 Blowdown & scale sludge (non-toxic) 1500 3rd party

utilities 4.11 Organic

Cable cuttings, wire ends, plastic

12 Electrical systems 300 Scrap 3rd party

conduits 0.82 Inorganic

Gardening & green Dry leaves, grass clippings (~10 acre

13 73200 200.55 Organic Organic converter

area considering 33% green area)

Water purification &

14 Spent ion-exchange resins (non-toxic) 200 3rd party

softener 0.55 Inorganic

Empty HDPE / PET bottles (non-

15 Misc. activities 200 3rd party

contaminated) 0.55 Inorganic

Page 1 of 5

ETP treated water

consumption 74.12 KL/Day

Organic converter T/day

0.33

Disposal by 3rd party

17.10 T/day

rd

Scrap to 3 party

0.04 T/day

Page 2 of 5

B. Organic waste calculation for LVP Plant

Sr. Approx. Quantity Solid Liquid ETP/3rd

Source / Process Area Type of Non-Hazardous Waste Category

No. Kg / Year (Kg/day) (L/day) party

1 ETP / STP treatment Biological sludge (after dewatering) 4188375 11475 Organic 3rd party

2 Process area & lab washing Treated water (ETP + RO permeate) 14699645 L/year 40273 Organic Domestic use

Empty glass vials (non-contaminated

3 Packaging line 2200 Scrap 3rd party

rejects) (~50,000 pcs) 6.03 Inorganic

Paperboard cartons, labels, paper

4 Packaging & labeling 36500 Scrap 3rd party

scrap 100 Organic

Corrugated boxes, stretch film,

5 Procurement / stores area 18250 Scrap 3rd party

plastic wrapping 50 Inorganic

Metal scrap (iron, steel, aluminum

6 Maintenance workshop 7200 Scrap 3rd party

parts) 19.73 Inorganic

Spent filters (air filters, bag filters,

7 Utility & HVAC area 1200 3rd party

prefilters) 3.29 Inorganic

8 Admin & office areas Office paper waste, packaging waste 5000 13.70 Organic Scrap 3rd party

Organic

9 Canteen / pantry Organic kitchen waste 36500

100.00 Organic converter

Sweepings, mopping dust, non-

10 General housekeeping 20000 3rd party

contaminated PPE 54.79 Mixed

Blowdown & scale sludge (non-

11 Cooling tower & utilities 27000 3rd party

toxic) 73.97 Organic

Cable cuttings, wire ends, plastic

12 Electrical systems 600 Scrap 3rd party
conduits 1.64 Inorganic
Spent ion-exchange resins (non-
13 Water purification & softener 20000 3rd party
toxic) 54.79 Inorganic
Empty HDPE / PET bottles (non-
14 Misc. activities 19000 3rd party
contaminated) 52.05 Inorganic

Page 3 of 5
ETP treated water
consumption 40.273 KL/Day
Organic converter 0.1 T/day
Disposal by 3rd party 11.71 T/day
Scrap to 3rd party 0.19 T/day

Page 4 of 5
C. List of parameters for ETP
LIST OF FEED PARAMETERS TO BE CONSIDER FOR DESIGN

Sr. No. Parameter Value Unit

- 1. pH 10-12 NA
- 2. Fee Oil <60 mg/L
- 3. TDS 2000-2500 mg/L
- 4. TSS <500 mg/L
- 5. COD <2250 mg/L
- 6. BOD <1100 mg/L

LIST OF OUTLET PARAMETERS

Sr. No. Parameter Value Unit

- 1. pH 6.5-8.5 NA
- 2. Fee Oil <20 mg/L
- 3. TDS <1200 mg/L
- 4. TSS <300 mg/L
- 5. COD <500 mg/L
- 6. BOD <450 mg/L

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[SHEET: Warehousing (GF)]

Sr. No. | Equipment Tag | Working Volume in Litres | Room/Stage (As per Layout) | Type | Warehouse | Fixed / Mobile | Gross Weight in Kg (Empty- Tentative) | Full weight in Kg | Dimension (L mm x W mmx H mm) | Make

1 | De-Dusting tunnel | NA | RM Receipt Room | De-Dusting Machine | Warehouse | Fixed | 200 | NA | 2200 L X 1200 W X 2400 H | Pridetech
2 | RLAF | NA | RM Sampling Room | Sampling Booth | Warehouse | Fixed | 150 | NA | 2500 L x 2000 Wx 1000 H | Clean air, Fabtech
3 | DPB | NA | RM Sampling Room | Equipment | Warehouse | Fixed | 250 | NA | 1200 L x 1000 W x 2100 H | Clean air, Fabtech
4 | DPB | NA | Approved RM | Equipment | Warehouse | Fixed | 250 | NA | 1200 L x 1000 W x 2100 H | Clean air, Fabtech
5 | DPB | NA | Dispensing Room | Equipment | Warehouse | Fixed | 250 | NA | 1200 L x 1000 W x 2100 H | Clean air, Fabtech
6 | RLAF | NA | Dispensing Room | Dispensing Booth | Warehouse | Fixed | 150 | NA | 2500 L x 2000 Wx 1000 H | Clean air, Fabtech
7 | DPB | NA | Plasma Sampling | Equipment | Warehouse | Fixed | 250 | NA | 1200 L x 1000 W x 2100 H | Clean air, Fabtech
8 | RLAF | NA | Plasma Sampling | Plasma Sampling Booth | Warehouse | Fixed | 150 | NA | 2500 L x 2000 Wx 1000 H | Clean air, Fabtech
9 | DPB | NA | Reject Room | Equipment | Warehouse | Fixed | 250 | NA | 1200 L x 1000 W x 2100 H | Clean air, Fabtech
10 | freezer-1000L | 1000 L | Plasma Storage Area | Freezer | Warehouse | Fixed | 425 | 1625 | 1400 L X 950 W X 2100 H | Newtronic lifecare, REMI
11 | freezer-1000L | 1000 L | Plasma Storage Area | Freezer | Warehouse | Fixed | 425 | 1625 | 1400 L X 950 W X 2100 H | Newtronic lifecare, REMI
12 | freezer-1000L | 1000 L | Plasma Storage Area | Freezer | Warehouse | Fixed | 425 | 1625 | 1400 L X 950 W X 2100 H | Newtronic lifecare, REMI
13 | freezer-1000L | 1000 L | Plasma Storage Area | Freezer | Warehouse | Fixed | 425 | 1625 | 1400 L X 950 W X 2100 H | Newtronic lifecare, REMI
14 | freezer-1000L | 1000 L | Plasma Storage Area | Freezer | Warehouse | Fixed | 425 | 1625 | 1400 L X 950 W X 2100 H | Newtronic lifecare, REMI

[SHEET: Bulk MFG (GF)]

Sr. No. | Vessel Tag | Working Volume in Litres | Room/Stage (As per Layout) | Type | Pre Viral/Post Viral | Fixed / Mobile | Empty (Kg) | Full load (Kg) | Dimension (mm x mmx mm) | Make

1 | TW 100 | 500 | THAWING | Process Vessel | Pre-Viral | Fixed | 450 | 950 | 2200L X 2200 W X 2800H | Praj and Aaruni
2 | DPB | NA | THAWING | Equipment | Pre-Viral | Fixed | 50 | NA | 1200 L x 1000 W x 2100 H | Clean air, Fabtech
3 | PV 101.1 | 300 | CUTTING | Process Vessel | Pre-Viral | Fixed | 390 | 690 | 2100L X 2000 X 2700H | Praj and Aaruni
4 | DPB | NA | CUTTING | Equipment | Pre-Viral | Fixed | 50 | NA | 1200 L x 1000 W x 2100 H | Clean air, Fabtech
5 | Ceiling LAF | NA | CUTTING | Equipment | Pre-Viral | Fixed Under Ceiling | 100 | NA | 2500 L x 2000 Wx 1000 H | Clean air, Fabtech
6 | PV 101.2 | 300 | CUTTING | Process Vessel | Pre-Viral | Fixed | 390 | 690 | 2100L X 2000 X 2700H | Praj and Aaruni
7 | PV 102.1 | 300 | 1F/12E | Process Vessel | Pre-Viral | Fixed | 390 | 690 | 2100L X 2000 X 2700H | Praj and Aaruni
8 | PV 102.2 | 300 | 1F/12E | Process Vessel | Pre-Viral | Fixed | 390 | 690 | 2100L X 2000 X 2700H | Praj and Aaruni
9 | PV 103.1 | 1500 | 1F/12E | Process Vessel | Pre-Viral | Fixed | 900 | 2400 | 2100L X 2000 X 2700H | Praj and Aaruni
10 | PV 103.2 | 1500 | 1F/12E | Process Vessel | Pre-Viral | Fixed | 900 | 2400 | 3200L X 3500 W X 4200H | Praj and Aaruni
11 | PV 103.3 | 500 | 1F/12E | Process Vessel | Pre-Viral | Fixed | 450 | 950 | 3200L X 3500 W X 4200H | Praj and Aaruni
12 | PV 103.4 | 500 | 1F/12E | Process Vessel | Pre-Viral | Fixed | 450 | 950 | 2200L X 2200 W X 2800H | Praj and Aaruni
13 | PV 103.5 | 500 | 1F/12E | Process Vessel | Pre-Viral | Fixed | 450 | 950 | 2200L X 2200 W X 2800H | Praj and Aaruni
14 | SPARE-1 | 250 | 1F/12E | Process Vessel | Pre-Viral | Fixed | 340 | 590 | 2200L X 2200 W X 2800H | Praj and Aaruni
15 | PF-RR-03 | 1000 | 1F/12E | Buffer | Pre-Viral | Fixed | 650 | 1650 | 2100L X 2000 X 2600H | Praj and Aaruni
16 | PF-RR-27 | 250 | 1F/12E | Buffer | Pre-Viral | Fixed | 340 | 590 | 2700L X 2700W X 3200H | Praj and Aaruni
17 | PV 128 | 300 | 1F/12E | Process Vessel | Pre-Viral | Fixed | 390 | 690 | 2100L X 2000 X 2600H | Praj and Aaruni
18 | FE-RR-02 | 150 | 1F/12E | Buffer | Pre-Viral | Mobile | 250 | 430 | 2100L X 2000 X 2700H | Praj and Aaruni
19 | FE-RR-06 | 250 | 1F/12E | Buffer | Pre-Viral | Fixed | 340 | 590 | 1900L X 2000W X 2800H | Praj and Aaruni
20 | FE-RR-06 | 250 | 1F/12E | Buffer | Pre-Viral | Fixed | 340 | 590 | 2100L X 2000 X 2600H | Praj and Aaruni
21 | FE-RR-14 | 250 | 1F/12E | Buffer | Pre-Viral | Mobile | 340 | 590 | 2100L X 2000 X 2600H | Praj and Aaruni
22 | Chromatography Column- 1000 X 500 | NA | 1F/12E | Equipment | Pre-Viral | Fixed | 400 | 600 | 1500L X 1500 W X 2700H | Truking, Cytiva, Aaruni
23 | Akta- 500 L/ Hr | NA | 1F/12E | Equipment | Pre-Viral | Fixed | 1200 | NA | 2000L X 2000 X 2000H | Truking, Cytiva, Aaruni
24 | Chromatography Column-1000 X 500 | NA | 1F/12E | Equipment | Pre-Viral | Fixed | 400 | 600 | 1500L X 1500 W X 2700H | Truking, Cytiva, Aaruni
25 | Mobile CIP | NA | 1F/12E | Equipment | Pre-Viral | Mobile | 150 | 200 | 2000L X 2000 X 2000H | Praj & DDE
26 | PV 104.1 | 1500 | 2F | Process Vessel | Pre-Viral | Fixed | 900 | 2400 | 3200L X 3500 W X 4200H | Praj and Aaruni
27 | PV 104.2 | 1500 | 2F | Process Vessel | Pre-Viral | Fixed | 900 | 2400 | 3200L X 3500 W X 4200H | Praj and Aaruni
28 | PV 105.1 | 1500 | 2F | Process Vessel | Pre-Viral | Fixed | 900 | 2400 | 3200L X 3500 W X 4200H |

Praj and Aaruni
29 | PV 105.2 | 1500 | 2F | Process Vessel | Pre-Viral | Fixed | 900 | 2400 | 3200L X 3500 W X 4200H |
Praj and Aaruni
30 | PV 105.3 | 300 | 2F | Process Vessel | Pre-Viral | Fixed | 390 | 690 | 2100L X 2000 X 2700H | Praj
and Aaruni
30 | PF–RR-35 | 200 | 2F | Buffer | Pre-Viral | Mobile | 300 | 520 | 2100L X 2000 X 2600H | Praj and
Aaruni
31 | PF–RR-31 | 100 | 2F | Buffer | Pre-Viral | Mobile | 220 | 370 | 1800L X 1900W X 2800H | Praj and
Aaruni
32 | PF–RR-11 | 100 | 2F | Buffer | Pre-Viral | Mobile | 220 | 370 | 1800L X 1900W X 2800H | Praj and
Aaruni
33 | PF–RR-12 | 250 | 2F | Buffer | Pre-Viral | Fixed | 340 | 590 | 2100L X 2000 X 2600H | Praj and
Aaruni
34 | PF–RR-32 | 100 | 2F | Buffer | Pre-Viral | Mobile | 220 | 370 | 1800L X 1900W X 2800H | Praj and
Aaruni
34 | PF–RR-27 | 100 | 2F | Buffer | Pre-Viral | Mobile | 220 | 370 | 1800L X 1900W X 2800H | Praj and
Aaruni
35 | PF–RR-28 | 100 | 2F | Buffer | Pre-Viral | Mobile | 220 | 370 | 1800L X 1900W X 2800H | Praj and
Aaruni
36 | PF–RR-30 | NALGENE | 2F | Bottle | Pre-Viral | NA | 50 | NA | 500L X 500 W x 1000 H | Praj and
Aaruni
37 | TFF | NA | 2F | Equipment | Pre-Viral | Fixed | 3000 | 5000 | 2500L X 2000 W X 2700H | Praj, Cytiva,
Sartorius, Aaruni
38 | Chromatography Column-1000 X 500 | NA | 2F | Equipment | Pre-Viral | Fixed | 400 | 600 | 1500L X
1500 W X 2700H | Truking, Cytiva, Aaruni
38 | Akta-785 L/ hr | NA | 2F | Equipment | Pre-Viral | Fixed | 2000 | 4000 | 2000L X 2000 X 2000H |
Truking, Cytiva, Aaruni
39 | Mobile CIP | NA | 2F | Equipment | Pre-Viral | Mobile | 150 | 200 | 2000L X 2000 X 2000H | Praj &
DDE
40 | PV 106 | 1500 | 3F | Process Vessel | Pre-Viral | Fixed | 900 | 2400 | 3200L X 3500 W X 4200H | Praj
and Aaruni
41 | PV 107 | 2000 | 3F | Process Vessel | Pre-Viral | Fixed | 1200 | 3200 | 3600L X 3600W X 4300H | Praj
and Aaruni
42 | PV 103.6 | 500 | 3F | Process Vessel | Pre-Viral | Fixed | 450 | 950 | 2200L X 2200 W X 2800H | Praj
and Aaruni
43 | PV 128 | 500 | 3F | Process Vessel | Pre-Viral | Fixed | 450 | 950 | 2200L X 2200 W X 2800H | Praj
and Aaruni
44 | PV 129 | 300 | 3F | Process Vessel | Pre-Viral | Fixed | 390 | 690 | 2100L X 2000 X 2700H | Praj and
Aaruni
45 | PV 119.1 | 300 | 3F | Process Vessel | Pre-Viral | Fixed | 390 | 690 | 2100L X 2000 X 2700H | Praj
and Aaruni
46 | PV 130 | 100 | 3F | Process Vessel | Pre-Viral | Mobile | 220 | 370 | 1800L X 1900W X 2800H | Praj
and Aaruni
47 | Centrifuge | NA | 3F | Equipment | Pre-Viral | Fixed | 800 | 900 | 2000L x 1500 W x 2000H | Alfa
Laval
48 | Chromatography Column | NA | 3F | Equipment | Pre-Viral | Fixed | 400 | 600 | 1500L X 1500 W X 2700H
| Truking, Cytiva, Aaruni
49 | Chromatography Column-1000 X 500 | NA | 3F | Equipment | Pre-Viral | Fixed | 400 | 600 | 1500L X
1500 W X 2700H | Truking, Cytiva, Aaruni
50 | Akta-1178 L/Hr | NA | 3F | Equipment | Pre-Viral | Fixed | 3000 | NA | 2000L X 2000 X 2000H |
Truking, Cytiva, Aaruni
51 | Mobile CIP | NA | 3F | Equipment | Pre-Viral | Mobile | 150 | 200 | 2000L X 2000 X 2000H | Praj &
DDE
52 | PV 108 | 1000 | 4I | Process Vessel | Pre-Viral | Fixed | 650 | 1650 | 2700L X 2700W X 3200H | Praj
and Aaruni
53 | PV 109.1 | 2000 | 4I | Process Vessel | Pre-Viral | Fixed | 1200 | 3200 | 3600L X 3600W X 4300H |
Praj and Aaruni
54 | PV 109.2 | 2000 | 4I | Process Vessel | Pre-Viral | Fixed | 1200 | 3200 | 3600L X 3600W X 4300H |
Praj and Aaruni
55 | PV 109.3 | 1000 | 4I | Process Vessel | Pre-Viral | Fixed | 650 | 1650 | 2700L X 2700W X 3200H |
Praj and Aaruni
56 | PV 110.1 | 2500 | 4I | Process Vessel | Pre-Viral | Fixed | 1500 | 4000 | 3800L X 3800W X 4300H |
Praj and Aaruni
57 | PV 110.2 | 2500 | 4I | Process Vessel | Pre-Viral | Fixed | 1500 | 4000 | 3800L X 3800W X 4300H |
Praj and Aaruni
58 | PF-RR-18 | 250 | 4I | Buffer | Pre-Viral | Fixed | 340 | 590 | 2100L X 2000 X 2600H | Praj and
Aaruni
59 | PF-RR-21 | 1000 | 4I | Buffer | Pre-Viral | Fixed | 650 | 1650 | 2700L X 2700W X 3200H | Praj and
Aaruni
60 | PF-RR-21 | NALGENE | 4I | Bottle | Pre-Viral | NA | 50 | NA | 500L X 500 W x 1000 H | Praj and
Aaruni
61 | PF-RR-23 | 500 | 4I | Buffer | Pre-Viral | Fixed | 450 | 950 | 2200L X 2200 W X 2800H | Praj and
Aaruni
62 | PF-RR-25 | 500 | 4I | Buffer | Pre-Viral | Fixed | 450 | 950 | 2200L X 2200 W X 2800H | Praj and
Aaruni
63 | PF-RR-27 | 500 | 4I | Buffer | Pre-Viral | Fixed | 450 | 950 | 2200L X 2200 W X 2800H | Praj and
Aaruni
64 | Akta-3454 L/Hr | NA | 4I | Equipment | Pre-Viral | Fixed | 3000 | NA | 2000L X 2000 X 2000H |
Truking, Cytiva, Aaruni
65 | Chromatography Column-2000 X 500 | NA | 4I | Equipment | Pre-Viral | Fixed | 3500 | 5000 | 2200L x
3200 W x 3000H | Truking, Cytiva, Aaruni
66 | TFF | NA | 4I | Equipment | Pre-Viral | Fixed | 2000 | NA | 2500L X 2000 W X 2700H | Praj, Cytiva,
Sartorius, Aaruni
67 | Mobile CIP | NA | 4I | Equipment | Pre-Viral | Mobile | 150 | 200 | 2000L X 2000 X 2000H | Praj &
DDE
68 | PV 110 | 1500 | 5I | Process Vessel | Pre-Viral | Fixed | 900 | 2400 | 3200L X 3500 W X 4200H | Praj
and Aaruni
69 | PV 111 | 1500 | 5I | Process Vessel | Pre-Viral | Fixed | 900 | 2400 | 3200L X 3500 W X 4200H | Praj
and Aaruni
70 | PV 112.1 | 1500 | 5I | Process Vessel | Pre-Viral | Fixed | 900 | 2400 | 3200L X 3500 W X 4200H |
Praj and Aaruni
71 | PV 112.2 | 1500 | 5I | Process Vessel | Pre-Viral | Fixed | 900 | 2400 | 3200L X 3500 W X 4200H |

Praj and Aaruni
72 | IG-RR-09 | 100 | 5l | Buffer | Pre-Viral | Mobile | 220 | 370 | 1800L X 1900W X 2800H | Praj and Aaruni
73 | IG-RR-05 | 300 | 5l | Buffer | Pre-Viral | Fixed | 390 | 690 | 2100L X 2000 X 2700H | Praj and Aaruni
74 | IG-RR-07 | 250 | 5l | Buffer | Pre-Viral | Fixed | 340 | 590 | 2100L X 2000 X 2600H | Praj and Aaruni
75 | IG-RR-22 | 250 | 5l | Buffer | Pre-Viral | Fixed | 340 | 590 | 2100L X 2000 X 2600H | Praj and Aaruni
76 | IG-RR-23 | 50 | 5l | Buffer | Pre-Viral | Mobile | 70 | 150 | 1800L X 1800W X 2800H | Praj and Aaruni
77 | PV 113 | 250 | 5l | Process Vessel | Pre-Viral | Fixed | 340 | 590 | 2100L X 2000 X 2600H | Praj and Aaruni
78 | PV113.1 | 250 | 5l | Process Vessel | Pre-Viral | Fixed | 340 | 590 | 2100L X 2000 X 2600H | Praj and Aaruni
79 | IG-RR-06 | NALGENE | 5l | Bottle | Pre-Viral | NA | 50 | NA | 500L X 500 W x 1000 H | Praj and Aaruni
80 | IG-RR-23 | 100 | 5l | Buffer | Pre-Viral | Mobile | 220 | 370 | 1800L X 1900W X 2800H | Praj and Aaruni
81 | IG-RR-24 | 100 | 5l | Buffer | Pre-Viral | Mobile | 220 | 370 | 1800L X 1900W X 2800H | Praj and Aaruni
82 | Akta-2211 L/ Hr | NA | 5l | Equipment | Pre-Viral | Fixed | 3000 | NA | 2000L X 2000 X 2000H | Truking, Cytiva, Aaruni
83 | Chromatography Column-1600X 500 | NA | 5l | Equipment | Pre-Viral | Fixed | 2500 | 5000 | 2200L x 3200 W x 3000H | Truking, Cytiva, Aaruni
84 | TFF | NA | 5l | Equipment | Pre-Viral | Fixed | 2000 | NA | 2500L X 2000 W X 2700H | Praj, Cytiva, Sartorius, Aaruni
85 | Depth filters | NA | 5l | Equipment | Pre-Viral | Fixed | 150 | NA | 500L X 500 W x 1000 H | Aaruni, Millipore, 3M
86 | Mobile CIP | NA | 5l | Equipment | Pre-Viral | Mobile | 150 | 200 | 2000L X 2000 X 2000H | Praj & DDE
87 | PV 118.1 | 2500 | 7A | Process Vessel | Pre-Viral | Fixed | 1500 | 4000 | 3800L X 3800W X 4300H | Praj and Aaruni
88 | PV 118.2 | 2500 | 7A | Process Vessel | Pre-Viral | Fixed | 1500 | 4000 | 3800L X 3800W X 4300H | Praj and Aaruni
89 | PV 119 | 2000 | 7A | Process Vessel | Pre-Viral | Fixed | 1200 | 3200 | 3600L X 3600W X 4300H | Praj and Aaruni
90 | PV 120 | 1000 | 7A | Process Vessel | Pre-Viral | Fixed | 650 | 1650 | 2700L X 2700W X 3200H | Praj and Aaruni
91 | AL-RR-02 | 100 | 7A | Buffer | Pre-Viral | Mobile | 220 | 370 | 1800L X 1900W X 2800H | Praj and Aaruni
92 | AL-RR-05 | 500 | 7A | Buffer | Pre-Viral | Fixed | 450 | 950 | 2200L X 2200 W X 2800H | Praj and Aaruni
93 | AL-RR-05 | 500 | 7A | Buffer | Pre-Viral | Fixed | 450 | 950 | 2200L X 2200 W X 2800H | Praj and Aaruni
94 | AL-RR-09 | 500 | 7A | Buffer | Pre-Viral | Fixed | 450 | 950 | 2200L X 2200 W X 2800H | Praj and Aaruni
95 | AL-RR-11 | 250 | 7A | Buffer | Pre-Viral | Fixed | 340 | 590 | 2100L X 2000 X 2600H | Praj and Aaruni
96 | AL-RR-19 | 250 | 7A | Buffer | Pre-Viral | Fixed | 340 | 590 | 2100L X 2000 X 2600H | Praj and Aaruni
97 | AL-RR-19 | 250 | 7A | Buffer | Pre-Viral | Fixed | 340 | 590 | 2100L X 2000 X 2600H | Praj and Aaruni
98 | PV 120 | 500 | 7A | Process Vessel | Pre-Viral | Fixed | 450 | 950 | 2200L X 2200 W X 2800H | Praj and Aaruni
99 | PV 121 | 500 | 7A | Process Vessel | Pre-Viral | Fixed | 450 | 950 | 2200L X 2200 W X 2800H | Praj and Aaruni
100 | PV 122 | 250 | 7A | Process Vessel | Pre-Viral | Fixed | 340 | 590 | 2100L X 2000 X 2600H | Praj and Aaruni
101 | PV 122.S | 250 | 7A | Process Vessel | Pre-Viral | Fixed | 340 | 590 | 2100L X 2000 X 2600H | Praj and Aaruni
102 | PV 123 | 100 | 7A | Process Vessel | Pre-Viral | Mobile | 220 | 370 | 1800L X 1900W X 2800H | Praj and Aaruni
103 | PV 124 | 100 | 7A | Process Vessel | Pre-Viral | Mobile | 220 | 370 | 1800L X 1900W X 2800H | Praj and Aaruni
104 | AL-RR-23 | 250 | 7A | Buffer | Pre-Viral | Fixed | 340 | 590 | 2100L X 2000 X 2600H | Praj and Aaruni
105 | AL-RR-19 | 100 | 7A | Buffer | Pre-Viral | Mobile | 220 | 370 | 1800L X 1900W X 2800H | Praj and Aaruni
106 | AL-RR-21 | NALGENE | 7A | Bottle | Pre-Viral | NA | 50 | NA | 500L X 500 W x 1000 H | Praj and Aaruni
107 | Akta- 1356 L/ Hr | NA | 7A | Equipment | Pre-Viral | Fixed | 2000 | NA | 2200L x 3200 W x 3000H | Truking, Cytiva, Aaruni
108 | Chromatography Column-1200 X 500 | NA | 7A | Equipment | Pre-Viral | Fixed | 3000 | 5000 | 2200L x 3200 W x 3000H | Truking, Cytiva, Aaruni
109 | TFF | NA | 7A | Equipment | Pre-Viral | Fixed | 2000 | NA | 2500L X 2000 W X 2700H | Praj, Cytiva, Sartorius, Aaruni
110 | Mobile CIP | NA | 7A | Equipment | Pre-Viral | Mobile | 150 | 200 | 2000L X 2000 X 2000H | Praj & DDE
111 | PV 125 | 100 | 8N | Process Vessel | Pre-Viral | Mobile | 220 | 370 | 1800L X 1900W X 2800H | Praj and Aaruni
112 | PV 126 | 100 | 8N | Process Vessel | Pre-Viral | Mobile | 220 | 370 | 1800L X 1900W X 2800H | Praj and Aaruni
113 | PV 125.1 | 100 | 8N | Process Vessel | Pre-Viral | Mobile | 220 | 370 | 1800L X 1900W X 2800H | Praj and Aaruni
114 | FN-RR-02 | 300 | 8N | Buffer | Pre-Viral | Fixed | 390 | 690 | 2100L X 2000 X 2700H | Praj and Aaruni
115 | PV 127 | 100 | 8N | Process Vessel | Pre-Viral | Mobile | 220 | 370 | 1800L X 1900W X 2800H | Praj and Aaruni
116 | PV 127.1 | 100 | 8N | Process Vessel | Pre-Viral | Mobile | 220 | 370 | 1800L X 1900W X 2800H | Praj and Aaruni
117 | Chromatography Column-300 X 500 | NA | 8N | Equipment | Pre-Viral | Fixed | 70 | 150 | 500L X 500 W

x 1000 H | Truking, Cytiva, Aaruni
118 | Akta- 70 L/ Hr | NA | 8N | Equipment | Pre-Viral | Fixed | 1200 | NA | 2200L x 3200 W x 3000H | Truking, Cytiva, Aaruni
119 | Mobile CIP | NA | 8N | Equipment | Pre-Viral | Mobile | 150 | 200 | 2000L X 2000 X 2000H | Praj & DDE
120 | Centralised CIP - 2 tank system | TBD | CIP/SIP Room | Equipment | Pre-Viral | Fixed | 1000 | NA | 2200L x 3200 W x 3000H | Praj and Aaruni
121 | Decontamination Autoclave 300L | TBD | Decontamination Room | Equipment | Pre-Viral | Fixed | 300 | NA | 500L x 500 W x 1200 H | Pharma Lab, Indu IonPure
122 | Dynamic Pass Box (DPB) | NA | Decontamination Room | Cleanroom Equipment | Pre-Viral | Fixed | 250 | NA | 1200 L x 1000 W x 2100 H | Clean air, Fabtech
123 | Dynamic Pass Box (DPB) | NA | Decontamination Room | Cleanroom Equipment | Pre-Viral | Fixed | 250 | NA | 1200 L x 1000 W x 2100 H | Clean air, Fabtech
124 | Dynamic Pass Box (DPB) | NA | Sterilization Room | Cleanroom Equipment | Pre-Viral | Fixed | 250 | NA | 1200 L x 1000 W x 2100 H | Clean air, Fabtech
125 | Autoclave Double Door 1000L | TBD | Sterilization Room | Equipment | Pre-Viral | Fixed | 2500 | NA | 1200 L x 1000 W x 2100 H | Pharma Lab, Indu IonPure
126 | Ceiling LAF | NA | Cooling zone | Equipment | Pre-Viral | Fixed Under Ceiling | 100 | NA | 2500 L x 2000 W x 1000 H | Clean air, Fabtech
127 | Dry heat Sterilizer (DHS) (1000 L) | NA | Cooling zone | Equipment | Pre-Viral | Fixed | 250 | NA | 1000 L x 1100 W x 1000 H | Black Gold Engineers, NBS, Drc Engineering
128 | Ceiling LAF | NA | Cooling zone | Equipment | Pre-Viral | Fixed Under Ceiling | 100 | NA | 2500 L x 2000 W x 1000 H | Clean air, Fabtech
129 | Dynamic Pass Box (DPB) | NA | Corridor | Equipment | Pre-Viral | Fixed | 250 | NA | 1200 L x 1000 W x 2100 H | Clean air, Fabtech
130 | PV 114 | 250 | 6I/11F | Process Vessel | Post Viral | Fixed | 340 | 590 | 2100L X 2000 X 2600H | Praj and Aaruni
131 | PV 115 | 1000 | 6I/11F | Process Vessel | Post Viral | Fixed | 650 | 1650 | 2700L X 2700W X 3200H | Praj and Aaruni
132 | PV 116 | 1000 | 6I/11F | Process Vessel | Post Viral | Fixed | 650 | 1650 | 2700L X 2700W X 3200H | Praj and Aaruni
133 | IG-RR-09 | 100 | 6I/11F | Buffer | Post Viral | Mobile | 220 | 370 | 1800L X 1900W X 2800H | Praj and Aaruni
134 | IG-RR-14 | 500 | 6I/11F | Buffer | Post Viral | Fixed | 450 | 950 | 2200L X 2200 W X 2800H | Praj and Aaruni
135 | IG-RR-16 | 500 | 6I/11F | Buffer | Post Viral | Fixed | 450 | 950 | 2200L X 2200 W X 2800H | Praj and Aaruni
136 | IG-RR-18 | 500 | 6I/11F | Buffer | Post Viral | Fixed | 450 | 950 | 2200L X 2200 W X 2800H | Praj and Aaruni
137 | IG-RR-22 | 250 | 6I/11F | Buffer | Post Viral | Fixed | 340 | 590 | 2100L X 2000 X 2600H | Praj and Aaruni
138 | PV 117 | 150 | 6I/11F | Process Vessel | Post Viral | Mobile | 250 | 430 | 1900L X 2000W X 2800H | Praj and Aaruni
139 | IG-RR-26 | 150 | 6I/11F | Buffer | Post Viral | Mobile | 250 | 430 | 1900L X 2000W X 2800H | Praj and Aaruni
140 | IG-RR-28 | 500 | 6I/11F | Buffer | Post Viral | Fixed | 450 | 950 | 2200L X 2200 W X 2800H | Praj and Aaruni
141 | IG-RR-23 | 150 | 6I/11F | Buffer | Post Viral | Mobile | 250 | 430 | 1900L X 2000W X 2800H | Praj and Aaruni
142 | IG-RR-24 | 150 | 6I/11F | Buffer | Post Viral | Mobile | 250 | 430 | 1900L X 2000W X 2800H | Praj and Aaruni
143 | FG-RR-09 | 1000 | 6I/11F | Buffer | Post Viral | Fixed | 650 | 1650 | 2700L X 2700W X 3200H | Praj and Aaruni
144 | FG-RR-10 | 2000 | 6I/11F | Buffer | Post Viral | Fixed | 1200 | 3200 | 3600L X 3600W X 4300H | Praj and Aaruni
145 | FG-RR-11 | 500 | 6I/11F | Buffer | Post Viral | Fixed | 450 | 950 | 2200L X 2200 W X 2800H | Praj and Aaruni
146 | FG-RR-13 | 100 | 6I/11F | Buffer | Post Viral | Mobile | 220 | 370 | 2100L X 2000 X 2600H | Praj and Aaruni
147 | FG-RR-14 | 250 | 6I/11F | Buffer | Post Viral | Fixed | 340 | 590 | 2100L X 2000 X 2600H | Praj and Aaruni
148 | Akta-1" -667 L/Hr | NA | 6I/11F | Equipment | Post Viral | Fixed | 2000 | NA | 2200L x 3200 W x 3000H | Truking, Cytiva, Aaruni
149 | Chromatography Column 1000X 500 | NA | 6I/11F | Equipment | Post Viral | Fixed | 400 | 600 | 1500L X 1500 W X 2700H | Truking, Cytiva, Aaruni
150 | TFF | NA | 6I/11F | Equipment | Post Viral | Fixed | 2000 | NA | 2500L X 2000 W X 2700H | Praj, Cytiva, Sartorius, Aaruni
151 | Nano Filter | NA | 6I/11F | Single use | Post Viral | Fixed | 20 | NA | 500L x 500 W x 1200 H | Planova
152 | Mobile CIP | NA | 6I/11F | Equipment | Post Viral | Mobile | 150 | 200 | 2000L X 2000 X 2000H | Praj & DDE
153 | FN-RR-15 | 500 | 9N | Buffer | Post Viral | Fixed | 450 | 950 | 2200L X 2200 W X 2800H | Praj and Aaruni
154 | FN-RR-18 | 100 | 9N | Buffer | Post Viral | Mobile | 220 | 370 | 1800L X 1900W X 2800H | Praj and Aaruni
155 | FN-RR-17 | 100 | 9N | Buffer | Post Viral | Mobile | 220 | 370 | 1800L X 1900W X 2800H | Praj and Aaruni
156 | Akta-1" -30 L/Hr | NA | 9N | Equipment | Post Viral | Fixed | 200 | NA | 2200L x 1800 W x 2000H | Truking, Cytiva, Aaruni
157 | Chromatography Column-300 X 750 | NA | 9N | Equipment | Post Viral | Fixed | 80 | 200 | 500L x 500 W x 2000 H | Truking, Cytiva, Aaruni
158 | Ceiling LAF | NA | 9N | Equipment | Post Viral | Fixed Under Ceiling | 100 | NA | 2500 L x 2000 W x 1000 H | Clean air, Fabtech
159 | Mobile CIP | NA | 9N | Equipment | Post Viral | Mobile | 150 | 200 | 2000L X 2000 X 1800H | Praj & DDE
160 | PV 131 | 300 | 10T/13E | Process Vessel | Post Viral | Fixed | 390 | 690 | 2100L X 2000 X 2700H | Praj and Aaruni
161 | PV 131.1 | 200 | 10T/13E | Process Vessel | Post Viral | Mobile | 300 | 520 | 2100L X 2000 X 2600H | Praj and Aaruni
162 | PV 132 | 250 | 10T/13E | Process Vessel | Post Viral | Fixed | 340 | 590 | 2100L X 2000 X 2600H | Praj and Aaruni
163 | TH-RR-05 | 50 | 10T/13E | Buffer | Post Viral | Mobile | 70 | 120 | 1800L X 1800W X 2800H | Praj

and Aaruni
164 | TH-RR-02 | 100 | 10T/13E | Buffer | Post Viral | Mobile | 220 | 370 | 1800L X 1900W X 2800H | Praj and Aaruni
165 | TH-RR-02 | 100 | 10T/13E | Buffer | Post Viral | Mobile | 220 | 370 | 1800L X 1900W X 2800H | Praj and Aaruni
166 | TH-RR-03 | 200 | 10T/13E | Buffer | Post Viral | Mobile | 300 | 520 | 2100L X 2000 X 2600H | Praj and Aaruni
167 | NALGENE | 50 | 10T/13E | Bottle | Post Viral | NA | 20 | NA | 500L x 500 W x 1200 H | Praj and Aaruni
168 | PV 132 | 250 | 10T/13E | Process Vessel | Post Viral | Fixed | 340 | 590 | 2100L X 2000 X 2600H | Praj and Aaruni
169 | PV 128 | 300 | 10T/13E | Process Vessel | Post Viral | Fixed | 390 | 690 | 2100L X 2000 X 2700H | Praj and Aaruni
170 | FE-RR-09 | 250 | 10T/13E | Buffer | Post Viral | Fixed | 340 | 590 | 2100L X 2000 X 2600H | Praj and Aaruni
171 | FE-RR-09 | 250 | 10T/13E | Buffer | Post Viral | Fixed | 340 | 590 | 2100L X 2000 X 2600H | Praj and Aaruni
172 | Akta-1" -160 L/Hr | NA | 10T/13E | Equipment | Post Viral | Fixed | 200 | NA | 2200L x 1800 W x 2000H | Truking, Cytiva, Aaruni
173 | Chromatography Column-450 X 500 | NA | 10T/13E | Equipment | Post Viral | Fixed | 100 | 250 | 500L x 500 W x 2000 H | Truking, Cytiva, Aaruni
174 | Ceiling LAF | NA | 10T/13E | Equipment | Post Viral | Fixed Under Ceiling | 100 | NA | 2500 L x 2000 Wx 1000 H | Clean air, Fabtech
175 | Mobile CIP | NA | 10T/13E | Equipment | Post Viral | Mobile | 150 | 200 | 2000L X 2000 X 2000H | Praj & DDE
176 | Centralised CIP - 2 tank system | TBD | CIP/SIP Room | Equipment | Post Viral | Fixed | 1000 | NA | 2200L x 3200 W x 3000H | Praj and Aaruni
173 | Decontamination Autoclave 300L | TBD | Decontamination Room | Equipment | Post Viral | Fixed | 300 | NA | 500L x 500 W x 1200 H | Pharma Lab, Indu IonPure
174 | Dynamic Pass Box (DPB) | NA | Decontamination Room | Cleanroom Equipment | Post Viral | Fixed | 250 | NA | 1200 L x 1000 W x 2100 H | Clean air, Fabtech
175 | Dynamic Pass Box (DPB) | NA | Decontamination Room | Cleanroom Equipment | Post Viral | Fixed | 250 | NA | 1200 L x 1000 W x 2100 H | Clean air, Fabtech
176 | Dynamic Pass Box (DPB) | NA | Sterilization Room | Cleanroom Equipment | Post Viral | Fixed | 250 | NA | 1200 L x 1000 W x 2100 H | Clean air, Fabtech
177 | Sterilization Autoclave 1000L | TBD | Sterilization Room | Equipment | Post Viral | Fixed | 2500 | NA | 1200 L x 1000 W x 2100 H | Pharma Lab, Indu IonPure
178 | Ceiling LAF | NA | Cooling zone | Equipment | Post Viral | Fixed Under Ceiling | 100 | NA | 2500 L x 2000 Wx 1000 H | Clean air, Fabtech
179 | Dry heat Sterilizer (DHS) (1000 L) | NA | Cooling zone | Equipment | Post Viral | Fixed | 250 | NA | 1000 L x 1100 W x 1000 H | Black Gold Engineers, NBS, Drc Engineering
180 | Buffer Preparation | 1500 | Buffer Preparation Room | Process Vessel | Post Viral | TBD | 900 | 2400 | 3200L X 3500 W X 4200H | Praj and Aaruni
181 | Buffer Preparation | 1000 | Buffer Preparation Room | Process Vessel | Post Viral | TBD | 650 | 1650 | 2700L X 2700W X 3200H | Praj and Aaruni
182 | Static Pass box | NA | Buffer Preparation Room | Equipment | Post Viral | NA | 50 | NA | 1200 L x 1000 W x 2100 H | Clean air, Fabtech
183 | Static Pass box | NA | Chemical Day Store | Equipment | Post Viral | NA | 50 | NA | 1200 L x 1000 W x 2100 H | Clean air, Fabtech
[SHEET: Fill Finish(FF)]
Sr. No. | Vessel Tag | Room/Stage (As per Layout) | Type | Fill Finish | Fixed / Mobile | Gross Weight in Kg | Full Weight in Kg | Dimension (L mm x W mmx H mm) | Make
1 | DPB | De-Contonning Room | Equipment | Plasma | Fixed | 250 | 250 | 1200 L x 1000 W x 2100 H | Clean air, Fabtech
2 | Vial washing | Vial Washing Room | Vial Washing Machine | Plasma | Fixed | 2800 | 2800 | 3500 L X 1800 W X 2100 H | NKP, Harikrushna
3 | DPB | Vial Washing Room | Equipment | Plasma | Fixed | 250 | 250 | 1200 L x 1000 W x 2100 H | Clean air, Fabtech
4 | Autoclave Double Door 1000L | Material Prep & Wash | Equipment | Plasma | Fixed | 2500 | 2500 | 1200 L x 1000 W x 2100 H | Pharma Lab, Indu IonPure
5 | DPB | Material Prep & Wash | Equipment | Plasma | Fixed | 250 | 250 | 1200 L x 1000 W x 2100 H | Clean air, Fabtech
6 | Decont Autoclave 300L | Decontamination | Equipment | Plasma | Fixed | 300 | 300 | 1200 L x 800 W x 1400 H | Pharma Lab, Indu IonPure
7 | Washing machine 25Kg | Garment Wash | Washing Machine | Plasma | Mobile | 80 | 80 | 1200 L x 1300 W x 1700 H | IFB
8 | Dryer 25Kg | Garment Wash | Drying machine | Plasma | Mobile | 80 | 80 | 1800 L x 1500 W x 2600 H | IFB
9 | Ceiling LAF | Unload Area | Equipment | Plasma | Fixed under Ceiling | 100 | 100 | 2500 L x 2000 Wx 1000 H | Clean air, Fabtech
10 | Empty/ Filled Vessel- 500L | Formulation | Formulation Vessel | Plasma | Fixed | 400 | 900 | 2200L X 2200 W X 2800H | Praj and Aaruni
11 | Empty/ Filled Vessel- 500L | Formulation | Formulation Vessel | Plasma | Fixed | 400 | 900 | 2200L X 2200 W X 2800H | Praj and Aaruni
12 | Ceiling LAF | Formulation | Equipment | Plasma | Fixed under Ceiling | 100 | 100 | 2500 L x 2000 Wx 1000 H | Clean air, Fabtech
13 | Filling Machine | Filling Area | Filling machine | Plasma | Fixed | 2200 | 2200 | 3200 L X 1700 W X 2300 H | NKP, Harikrushna
14 | Ceiling LAF | Filling Area | Equipment | Plasma | Fixed under Ceiling | 100 | 100 | 2500 L x 2000 Wx 1000 H | Clean air, Fabtech
15 | Ceiling LAF | Capping Room | Equipment | Plasma | Fixed under Ceiling | 100 | 100 | 2500 L x 2000 Wx 1000 H | Clean air, Fabtech
16 | Capping Machine | Capping Room | Capping Machine | Plasma | Fixed | 600 | 600 | 2200 L X 1500 W X 2000 H | NKP, Harikrushna
17 | Turn Table | Capping Room | SS Turn Table | Plasma | Fixed | 150 | 150 | 1200 L X 1300 W X 900 H | NKP, Harikrushna
18 | Swing Conveyor | Capping Room | Swing Conveyor With Vial Collection Tray | Plasma | Fixed | 220 | 220 | 2500 L X 1200 W X 900 H | NKP, Harikrushna
19 | Lyo Machine 3.3msq | Lyo Loading Side | Lyo machine | Plasma | Fixed | 7500 | 7500 | 4200 L X 2200 W X 2400 H | LSI
20 | DPB | Buffer Room | Equipment | Plasma | Fixed | 250 | 250 | 1200 L x 1000 W x 2100 H | Clean air, Fabtech

21 | Visual Inspection Booth | Visual Inspection Room | Visual Inspection Booth | Plasma | Fixed | 250 | 250 | 3000 L X 1800 W X 2400 H | NKP, Harikrushna

22 | Labelling machine | Secondary Packing Area | Labeling Machine | Plasma | Fixed | 600 | 600 | 3000 L X 1800 W X 2000 H | NKP, Harikrushna

23 | Cartoning machine | Secondary Packing Area | Cartoning Machine | Plasma | Fixed | 200 | 200 | 6000 L X 2000 W X 2100 H | NKP, Harikrushna

24 | Bulk | Bulk staging | 3D Bag/Tote | Plasma | Mobile | 100 | 1500 | 1000 L x 800 x 1000 H | Sartorius

25 | freezer-1000L | 1000 L | Plasma Storage Area | Freezer | Warehouse | 425 | 1625 | 1400 L X 950 W X 2100 H | Newtronic lifecare, REMI

26 | freezer-1000L | 1000 L | Plasma Storage Area | Freezer | Warehouse | 425 | 1625 | 1400 L X 950 W X 2100 H | Newtronic lifecare, REMI

27 | freezer-1000L | 1000 L | Plasma Storage Area | Freezer | Warehouse | 425 | 1625 | 1400 L X 950 W X 2100 H | Newtronic lifecare, REMI

28 | freezer-1000L | 1000 L | Plasma Storage Area | Freezer | Warehouse | 425 | 1625 | 1400 L X 950 W X 2100 H | Newtronic lifecare, REMI

[SHEET: Utilities (SF)]

Sr. No. | Vessel Tag | Room/Stage (As per Layout) | Type | Fill Finish | Fixed / Mobile | Gross Weight in Kg | Full Weight in Kg | Make

| | | | | Empty | Full |

1 | PW Generation(MGF, Softener, Soft Water Storage tank, RO and RO water storage tank) | Water System & Engineering Store | Water System & Utilities | Utilities equipment | Fixed | 500 | 500 | Praj, CN water

2 | PW storage 20,000L | Water System & Engineering Store | Water System & Utilities | Utilities equipment | Fixed | 2500 | 22500 | Praj, CN water

3 | WFI Generation and storage 20000L | Water System & Engineering Store | Water System & Utilities | Utilities equipment | Fixed | 1100 | 21100 | Praj, CN water

4 | PSG System 1000Kg | Water System & Engineering Store | Water System & Utilities | Utilities equipment | Fixed | 760 | 760 | Praj, CN water

5 | Compressed Air System 300 CFM | Water System & Engineering Store | Water System & Utilities | Utilities equipment | Fixed | 120 | 125 | Atlas copco

6 | AHUs | Service Area | HVAC | Utilities equipment | Fixed | NA | NA | Clean air, Fabtech

7 | Electrical Control Panel | Service Area | Electrical | Utilities equipment | Fixed | NA | NA | Clean air, Fabtech

Reagent Details:
Formula & Calculations:
Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).
Calculation: To prepare 1 L of 0.4M Sodium Acetate (MW = 80.03 g/mol)
Weight of solute(g)=0.4×80.03 = 32.01g.
Preparation Method:
Weigh 32.01g of Sodium Acetate and dissolve in water to make a 1 L solution.
Preparation for different volumes:
To prepare _____ of 0.4M Sodium Acetate (MW = 80.03 g/mol).
Weight of solute(g)=0.4×80.03 x _____L= _____gm
Weigh _____gm of Sodium Acetate and dissolve in water to make a _____ L solution.
Check and record the details:
Preparation Details:
Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry
Date
0.4M Sodium Acetate | | | |
Final volume prepared | Weighed Quantity | pH (7.0±0.2) | Conductivity (340.0±10.0 ms/cm) | Storage
Condition:
| | | RT
Date of Preparation | Prepared By | Checked By
| |

[SHEET: Warehousing (GF)]

Sr. No. | Equipment Tag | Working Volume in Litres | Room/Stage (As per Layout) | Type | Warehouse | Fixed / Mobile | Gross Weight in Kg (Empty- Tentative) | Dimension (L mm x W mmx H mm) | Make
1 | De-Dusting tunnel | NA | RM Receipt Room | De-Dusting Machine | Warehouse | Fixed | 200 | 2200 L X 1200 W X 2400 H | Pridetech
2 | RLAF | NA | RM Sampling Room | Sampling Booth | Warehouse | Fixed | 150 | 2500 L x 2000 Wx 1000 H | Clean air, Fabtech
3 | DBP | NA | RM Sampling Room | Equipment | Warehouse | Fixed | 180 | 1200 L x 1000 W x 2100 H | Clean air, Fabtech
4 | DPB | NA | Approved RM | Equipment | Warehouse | Fixed | 180 | 1200 L x 1000 W x 2100 H | Clean air, Fabtech
5 | DPB | NA | Dispensing Room | Equipment | Warehouse | Fixed | 180 | 1200 L x 1000 W x 2100 H | Clean air, Fabtech
6 | RLAF | NA | Dispensing Room | Dispensing Booth | Warehouse | Fixed | 150 | 2500 L x 2000 Wx 1000 H | Clean air, Fabtech
7 | DPB | NA | Plasma Sampling | Equipment | Warehouse | Fixed | 180 | 1200 L x 1000 W x 2100 H | Clean air, Fabtech
8 | RLAF | NA | Plasma Sampling | Plasma Sampling Booth | Warehouse | Fixed | 150 | 2500 L x 2000 Wx 1000 H | Clean air, Fabtech
9 | DPB | NA | Reject Room | Equipment | Warehouse | Fixed | 180 | 1200 L x 1000 W x 2100 H | Clean air, Fabtech
10 | freezer-1000L | 1000 L | Plasma Storage Area | Freezer | Warehouse | Fixed | 425 | 1400 L X 950 W X 2100 H | Newtronic lifecare, REMI
11 | freezer-1000L | 1000 L | Plasma Storage Area | Freezer | Warehouse | Fixed | 425 | 1400 L X 950 W X 2100 H | Newtronic lifecare, REMI
12 | freezer-1000L | 1000 L | Plasma Storage Area | Freezer | Warehouse | Fixed | 425 | 1400 L X 950 W X 2100 H | Newtronic lifecare, REMI
13 | freezer-1000L | 1000 L | Plasma Storage Area | Freezer | Warehouse | Fixed | 425 | 1400 L X 950 W X 2100 H | Newtronic lifecare, REMI
14 | freezer-1000L | 1000 L | Plasma Storage Area | Freezer | Warehouse | Fixed | 425 | 1400 L X 950 W X 2100 H | Newtronic lifecare, REMI
15 | freezer-1000L | 1000 L | Plasma Storage Area | Freezer | Warehouse | Fixed | 425 | 1400 L X 950 W X 2100 H | Newtronic lifecare, REMI
16 | freezer-1000L | 1000 L | Plasma Storage Area | Freezer | Warehouse | Fixed | 425 | 1400 L X 950 W X 2100 H | Newtronic lifecare, REMI
17 | freezer-1000L | 1000 L | Plasma Storage Area | Freezer | Warehouse | Fixed | 425 | 1400 L X 950 W X 2100 H | Newtronic lifecare, REMI
18 | freezer-1000L | 1000 L | Plasma Storage Area | Freezer | Warehouse | Fixed | 425 | 1400 L X 950 W X 2100 H | Newtronic lifecare, REMI
19 | freezer-1000L | 1000 L | Plasma Storage Area | Freezer | Warehouse | Fixed | 425 | 1400 L X 950 W X 2100 H | Newtronic lifecare, REMI
20 | freezer-1000L | 1000 L | Plasma Storage Area | Freezer | Warehouse | Fixed | 425 | 1400 L X 950 W X 2100 H | Newtronic lifecare, REMI
21 | freezer-1000L | 1000 L | Plasma Storage Area | Freezer | Warehouse | Fixed | 425 | 1400 L X 950 W X 2100 H | Newtronic lifecare, REMI
22 | freezer-1000L | 1000 L | Plasma Storage Area | Freezer | Warehouse | Fixed | 425 | 1400 L X 950 W X 2100 H | Newtronic lifecare, REMI
23 | freezer-1000L | 1000 L | Plasma Storage Area | Freezer | Warehouse | Fixed | 425 | 1400 L X 950 W X 2100 H | Newtronic lifecare, REMI
24 | freezer-1000L | 1000 L | Plasma Storage Area | Freezer | Warehouse | Fixed | 425 | 1400 L X 950 W X 2100 H | Newtronic lifecare, REMI

[SHEET: Bulk MFG (GF)]

Sr. No. | Vessel Tag | Working Volume in Litres | Room/Stage (As per Layout) | Type | Pre Viral/Post Viral | Fixed / Mobile | Empty (Kg) | Full load (Kg) | Emergency weight | Dimension (mm x mmx mm) | Make
1 | TW 100 | 500 | THAWING | Process Vessel | Pre-Viral | Fixed | 450 | 950 | | 2200L X 2200 W X 2800H | Praj and Aaruni
2 | DPB | NA | THAWING | Equipment | Pre-Viral | Fixed | 50 | - | | 1200 L x 1000 W x 2100 H | Clean air, Fabtech
3 | PV 101.1 | 300 | CUTTING | Process Vessel | Pre-Viral | Fixed | 390 | 690 | | 2100L X 2000 X 2700H | Praj and Aaruni
4 | PV 101.2 | 300 | CUTTING | Process Vessel | Pre-Viral | Fixed | 390 | 690 | | 2100L X 2000 X 2700H | Praj and Aaruni
6 | PV 102.1 | 300 | 1F/12E | Process Vessel | Pre-Viral | Fixed | 390 | 690 | | 2100L X 2000 X 2700H | Praj and Aaruni
7 | PV 102.2 | 300 | 1F/12E | Process Vessel | Pre-Viral | Fixed | 390 | 690 | | 2100L X 2000 X 2700H | Praj and Aaruni
8 | PV 103.1 | 1500 | 1F/12E | Process Vessel | Pre-Viral | Fixed | 900 | 2400 | | 2100L X 2000 X 2700H | Praj and Aaruni
9 | PV 103.2 | 1500 | 1F/12E | Process Vessel | Pre-Viral | Fixed | 900 | 2400 | | 3200L X 3500 W X 4200H | Praj and Aaruni
10 | PV 103.3 | 500 | 1F/12E | Process Vessel | Pre-Viral | Fixed | 450 | 950 | | 3200L X 3500 W X 4200H | Praj and Aaruni
11 | PV 103.4 | 500 | 1F/12E | Process Vessel | Pre-Viral | Fixed | 450 | 950 | | 2200L X 2200 W X 2800H | Praj and Aaruni
12 | PV 103.5 | 500 | 1F/12E | Process Vessel | Pre-Viral | Fixed | 450 | 950 | | 2200L X 2200 W X 2800H | Praj and Aaruni
13 | SPARE-1 | 250 | 1F/12E | Process Vessel | Pre-Viral | Fixed | 340 | 590 | | 2200L X 2200 W X 2800H | Praj and Aaruni
14 | PF-RR-03 | 1000 | 1F/12E | Buffer | Pre-Viral | Fixed | 650 | 1650 | | 2100L X 2000 X 2600H | Praj and Aaruni
15 | PF-RR-27 | 250 | 1F/12E | Buffer | Pre-Viral | Fixed | 340 | 590 | | 2700L X 2700W X 3200H | Praj and Aaruni
16 | PV 128 | 300 | 1F/12E | Process Vessel | Pre-Viral | Fixed | 390 | 690 | | 2100L X 2000 X 2600H | Praj and Aaruni
17 | FE-RR-02 | 150 | 1F/12E | Buffer | Pre-Viral | Mobile | 250 | 430 | | 2100L X 2000 X 2700H | Praj and Aaruni
18 | FE-RR-06 | 250 | 1F/12E | Buffer | Pre-Viral | Fixed | 340 | 590 | | 1900L X 2000W X 2800H | Praj and Aaruni
19 | FE-RR-06 | 250 | 1F/12E | Buffer | Pre-Viral | Fixed | 340 | 590 | | 2100L X 2000 X 2600H | Praj and Aaruni

| FE-RR-14 | 250 | 1F/12E | Buffer | Pre-Viral | Mobile | 340 | 590 | | 2100L X 2000 X 2600H | Praj and Aaruni
20 | Chromatography Column- 1000 X 500 | NA | 1F/12E | Equipment | Pre-Viral | Fixed | 400 | 600 | | 1500L X 1500 W X 2700H | Truking, Cytiva, Aaruni
21 | Akta- 500 L/ Hr | NA | 1F/12E | Equipment | Pre-Viral | Fixed | 1200 | - | | 2000L X 2000 X 2000H | Truking, Cytiva, Aaruni
22 | Chromatography Column-1000 X 500 | NA | 1F/12E | Equipment | Pre-Viral | Fixed | 400 | 600 | | 1500L X 1500 W X 2700H | Truking, Cytiva, Aaruni
23 | Mobile CIP | NA | 1F/12E | Equipment | Pre-Viral | Mobile | 150 | 200 | | 2000L X 2000 X 2000H | Praj & DDE
24 | PV 104.1 | 1500 | 2F | Process Vessel | Pre-Viral | Fixed | 900 | 2400 | | 3200L X 3500 W X 4200H | Praj and Aaruni
25 | PV 104.2 | 1500 | 2F | Process Vessel | Pre-Viral | Fixed | 900 | 2400 | | 3200L X 3500 W X 4200H | Praj and Aaruni
26 | PV 105.1 | 1500 | 2F | Process Vessel | Pre-Viral | Fixed | 900 | 2400 | | 3200L X 3500 W X 4200H | Praj and Aaruni
27 | PV 105.2 | 1500 | 2F | Process Vessel | Pre-Viral | Fixed | 900 | 2400 | | 3200L X 3500 W X 4200H | Praj and Aaruni
28 | PV 105.3 | 300 | 2F | Process Vessel | Pre-Viral | Fixed | 390 | 690 | | 2100L X 2000 X 2700H | Praj and Aaruni
29 | PF-RR-35 | 200 | 2F | Buffer | Pre-Viral | Mobile | 300 | 520 | | 2100L X 2000 X 2600H | Praj and Aaruni
30 | PF-RR-31 | 100 | 2F | Buffer | Pre-Viral | Mobile | 220 | 370 | | 1800L X 1900W X 2800H | Praj and Aaruni
31 | PF-RR-11 | 100 | 2F | Buffer | Pre-Viral | Mobile | 220 | 370 | | 1800L X 1900W X 2800H | Praj and Aaruni
32 | PF-RR-12 | 250 | 2F | Buffer | Pre-Viral | Fixed | 340 | 590 | | 2100L X 2000 X 2600H | Praj and Aaruni
33 | PF-RR-32 | 100 | 2F | Buffer | Pre-Viral | Mobile | 220 | 370 | | 1800L X 1900W X 2800H | Praj and Aaruni
34 | PF-RR-27 | 100 | 2F | Buffer | Pre-Viral | Mobile | 220 | 370 | | 1800L X 1900W X 2800H | Praj and Aaruni
| PF-RR-28 | 100 | 2F | Buffer | Pre-Viral | Mobile | 220 | 370 | | 1800L X 1900W X 2800H | Praj and Aaruni
35 | PF-RR-30 | NALGENE | 2F | Bottle | Pre-Viral | NA | 50 | - | | 500L X 500 W x 1000 H | Praj and Aaruni
36 | TFF | NA | 2F | Equipment | Pre-Viral | Fixed | 3000 | 5000 | | 2500L X 2000 W X 2700H | Praj, Cytiva, Sartorius, Aaruni
37 | Chromatography Column-1000 X 500 | NA | 2F | Equipment | Pre-Viral | Fixed | 400 | 600 | | 1500L X 1500 W X 2700H | Truking, Cytiva, Aaruni
38 | Akta-785 L/ hr | NA | 2F | Equipment | Pre-Viral | Fixed | 2000 | 4000 | | 2000L X 2000 X 2000H | Truking, Cytiva, Aaruni
39 | Mobile CIP | NA | 2F | Equipment | Pre-Viral | Mobile | 150 | 200 | | 2000L X 2000 X 2000H | Praj & DDE
40 | PV 106 | 1500 | 3F | Process Vessel | Pre-Viral | Fixed | 900 | 2400 | | 3200L X 3500 W X 4200H | Praj and Aaruni
41 | PV 107 | 2000 | 3F | Process Vessel | Pre-Viral | Fixed | 1200 | 3200 | | 3600L X 3600W X 4300H | Praj and Aaruni
42 | PV 103.6 | 500 | 3F | Process Vessel | Pre-Viral | Fixed | 450 | 950 | | 2200L X 2200 W X 2800H | Praj and Aaruni
43 | PV 128 | 500 | 3F | Process Vessel | Pre-Viral | Fixed | 450 | 950 | | 2200L X 2200 W X 2800H | Praj and Aaruni
44 | PV 129 | 300 | 3F | Process Vessel | Pre-Viral | Fixed | 390 | 690 | | 2100L X 2000 X 2700H | Praj and Aaruni
45 | PV 119.1 | 300 | 3F | Process Vessel | Pre-Viral | Fixed | 390 | 690 | | 2100L X 2000 X 2700H | Praj and Aaruni
46 | PV 130 | 100 | 3F | Process Vessel | Pre-Viral | Mobile | 220 | 370 | | 1800L X 1900W X 2800H | Praj and Aaruni
47 | Centrifuge | NA | 3F | Equipment | Pre-Viral | Fixed | 800 | 900 | | 2000L x 1500 W x 2000H | Alfa Laval
48 | Chromatography Column | NA | 3F | Equipment | Pre-Viral | Fixed | 400 | 600 | | 1500L X 1500 W X 2700H | Truking, Cytiva, Aaruni
49 | Chromatography Column-1000 X 500 | NA | 3F | Equipment | Pre-Viral | Fixed | 400 | 600 | | 1500L X 1500 W X 2700H | Truking, Cytiva, Aaruni
50 | Akta-1178 L/Hr | NA | 3F | Equipment | Pre-Viral | Fixed | 3000 | - | | 2000L X 2000 X 2000H | Truking, Cytiva, Aaruni
51 | Mobile CIP | NA | 3F | Equipment | Pre-Viral | Mobile | 150 | 200 | | 2000L X 2000 X 2000H | Praj & DDE
52 | PV 108 | 1000 | 4I | Process Vessel | Pre-Viral | Fixed | 650 | 1650 | | 2700L X 2700W X 3200H | Praj and Aaruni
53 | PV 109.1 | 2000 | 4I | Process Vessel | Pre-Viral | Fixed | 1200 | 3200 | | 3600L X 3600W X 4300H | Praj and Aaruni
54 | PV 109.2 | 2000 | 4I | Process Vessel | Pre-Viral | Fixed | 1200 | 3200 | | 3600L X 3600W X 4300H | Praj and Aaruni
55 | PV 109.3 | 1000 | 4I | Process Vessel | Pre-Viral | Fixed | 650 | 1650 | | 2700L X 2700W X 3200H | Praj and Aaruni
56 | PV 110.1 | 2500 | 4I | Process Vessel | Pre-Viral | Fixed | 1500 | 4000 | | 3800L X 3800W X 4300H | Praj and Aaruni
57 | PV 110.2 | 2500 | 4I | Process Vessel | Pre-Viral | Fixed | 1500 | 4000 | | 3800L X 3800W X 4300H | Praj and Aaruni
58 | PF-RR-18 | 250 | 4I | Buffer | Pre-Viral | Fixed | 340 | 590 | | 2100L X 2000 X 2600H | Praj and Aaruni
59 | PF-RR-21 | 1000 | 4I | Buffer | Pre-Viral | Fixed | 650 | 1650 | | 2700L X 2700W X 3200H | Praj and Aaruni
60 | PF-RR-21 | NALGENE | 4I | Bottle | Pre-Viral | NA | 50 | - | | 500L X 500 W x 1000 H | Praj and Aaruni
61 | PF-RR-23 | 500 | 4I | Buffer | Pre-Viral | Fixed | 450 | 950 | | 2200L X 2200 W X 2800H | Praj and Aaruni
62 | PF-RR-25 | 500 | 4I | Buffer | Pre-Viral | Fixed | 450 | 950 | | 2200L X 2200 W X 2800H | Praj and Aaruni
63 | PF-RR-27 | 500 | 4I | Buffer | Pre-Viral | Fixed | 450 | 950 | | 2200L X 2200 W X 2800H | Praj and Aaruni

64 | Akta-3454 L/Hr | NA | 4l | Equipment | Pre-Viral | Fixed | 3000 | - | | 2000L X 2000 X 2000H | Truking, Cytiva, Aaruni

65 | Chromatography Column-2000 X 500 | NA | 4l | Equipment | Pre-Viral | Fixed | 3500 | 5000 | | 2200L x 3200 W x 3000H | Truking, Cytiva, Aaruni

66 | TFF | NA | 4l | Equipment | Pre-Viral | Fixed | 2000 | - | | 2500L X 2000 W X 2700H | Praj, Cytiva, Sartorius, Aaruni

67 | Mobile CIP | NA | 4l | Equipment | Pre-Viral | Mobile | 150 | 200 | | 2000L X 2000 X 2000H | Praj & DDE

68 | PV 110 | 1500 | 5l | Process Vessel | Pre-Viral | Fixed | 900 | 2400 | | 3200L X 3500 W X 4200H | Praj and Aaruni

69 | PV 111 | 1500 | 5l | Process Vessel | Pre-Viral | Fixed | 900 | 2400 | | 3200L X 3500 W X 4200H | Praj and Aaruni

70 | PV 112.1 | 1500 | 5l | Process Vessel | Pre-Viral | Fixed | 900 | 2400 | | 3200L X 3500 W X 4200H | Praj and Aaruni

71 | PV 112.2 | 1500 | 5l | Process Vessel | Pre-Viral | Fixed | 900 | 2400 | | 3200L X 3500 W X 4200H | Praj and Aaruni

72 | IG-RR-09 | 100 | 5l | Buffer | Pre-Viral | Mobile | 220 | 370 | | 1800L X 1900W X 2800H | Praj and Aaruni

73 | IG-RR-05 | 300 | 5l | Buffer | Pre-Viral | Fixed | 390 | 690 | | 2100L X 2000 X 2700H | Praj and Aaruni

74 | IG-RR-07 | 250 | 5l | Buffer | Pre-Viral | Fixed | 340 | 590 | | 2100L X 2000 X 2600H | Praj and Aaruni

75 | IG-RR-22 | 250 | 5l | Buffer | Pre-Viral | Fixed | 340 | 590 | | 2100L X 2000 X 2600H | Praj and Aaruni

76 | IG-RR-23 | 50 | 5l | Buffer | Pre-Viral | Mobile | 70 | 150 | | 1800L X 1800W X 2800H | Praj and Aaruni

77 | PV 113 | 250 | 5l | Process Vessel | Pre-Viral | Fixed | 340 | 590 | | 2100L X 2000 X 2600H | Praj and Aaruni

78 | PV113.1 | 250 | 5l | Process Vessel | Pre-Viral | Fixed | 340 | 590 | | 2100L X 2000 X 2600H | Praj and Aaruni

79 | IG-RR-06 | NALGENE | 5l | Bottle | Pre-Viral | NA | 50 | - | | 500L X 500 W x 1000 H | Praj and Aaruni

80 | IG-RR-23 | 100 | 5l | Buffer | Pre-Viral | Mobile | 220 | 370 | | 1800L X 1900W X 2800H | Praj and Aaruni

81 | IG-RR-24 | 100 | 5l | Buffer | Pre-Viral | Mobile | 220 | 370 | | 1800L X 1900W X 2800H | Praj and Aaruni

82 | Akta-2211 L/ Hr | NA | 5l | Equipment | Pre-Viral | Fixed | 3000 | - | | 2000L X 2000 X 2000H | Truking, Cytiva, Aaruni

83 | Chromatography Column-1600X 500 | NA | 5l | Equipment | Pre-Viral | Fixed | 2500 | 5000 | | 2200L x 3200 W x 3000H | Truking, Cytiva, Aaruni

84 | TFF | NA | 5l | Equipment | Pre-Viral | Fixed | 2000 | - | | 2500L X 2000 W X 2700H | Praj, Cytiva, Sartorius, Aaruni

85 | Depth filters | NA | 5l | Equipment | Pre-Viral | Fixed | 150 | - | | 500L X 500 W x 1000 H | Aaruni, Millipore, 3M

86 | Mobile CIP | NA | 5l | Equipment | Pre-Viral | Mobile | 150 | 200 | | 2000L X 2000 X 2000H | Praj & DDE

87 | PV 118.1 | 2500 | 7A | Process Vessel | Pre-Viral | Fixed | 1500 | 4000 | | 3800L X 3800W X 4300H | Praj and Aaruni

88 | PV 118.2 | 2500 | 7A | Process Vessel | Pre-Viral | Fixed | 1500 | 4000 | | 3800L X 3800W X 4300H | Praj and Aaruni

89 | PV 119 | 2000 | 7A | Process Vessel | Pre-Viral | Fixed | 1200 | 3200 | | 3600L X 3600W X 4300H | Praj and Aaruni

90 | PV 120 | 1000 | 7A | Process Vessel | Pre-Viral | Fixed | 650 | 1650 | | 2700L X 2700W X 3200H | Praj and Aaruni

91 | AL-RR-02 | 100 | 7A | Buffer | Pre-Viral | Mobile | 220 | 370 | | 1800L X 1900W X 2800H | Praj and Aaruni

92 | AL-RR-05 | 500 | 7A | Buffer | Pre-Viral | Fixed | 450 | 950 | | 2200L X 2200 W X 2800H | Praj and Aaruni

93 | AL-RR-05 | 500 | 7A | Buffer | Pre-Viral | Fixed | 450 | 950 | | 2200L X 2200 W X 2800H | Praj and Aaruni

94 | AL-RR-09 | 500 | 7A | Buffer | Pre-Viral | Fixed | 450 | 950 | | 2200L X 2200 W X 2800H | Praj and Aaruni

95 | AL-RR-11 | 250 | 7A | Buffer | Pre-Viral | Fixed | 340 | 590 | | 2100L X 2000 X 2600H | Praj and Aaruni

96 | AL-RR-19 | 250 | 7A | Buffer | Pre-Viral | Fixed | 340 | 590 | | 2100L X 2000 X 2600H | Praj and Aaruni

97 | AL-RR-19 | 250 | 7A | Buffer | Pre-Viral | Fixed | 340 | 590 | | 2100L X 2000 X 2600H | Praj and Aaruni

98 | PV 120 | 500 | 7A | Process Vessel | Pre-Viral | Fixed | 450 | 950 | | 2200L X 2200 W X 2800H | Praj and Aaruni

99 | PV 121 | 500 | 7A | Process Vessel | Pre-Viral | Fixed | 450 | 950 | | 2200L X 2200 W X 2800H | Praj and Aaruni

100 | PV 122 | 250 | 7A | Process Vessel | Pre-Viral | Fixed | 340 | 590 | | 2100L X 2000 X 2600H | Praj and Aaruni

101 | PV 122.S | 250 | 7A | Process Vessel | Pre-Viral | Fixed | 340 | 590 | | 2100L X 2000 X 2600H | Praj and Aaruni

102 | PV 123 | 100 | 7A | Process Vessel | Pre-Viral | Mobile | 220 | 370 | | 1800L X 1900W X 2800H | Praj and Aaruni

103 | PV 124 | 100 | 7A | Process Vessel | Pre-Viral | Mobile | 220 | 370 | | 1800L X 1900W X 2800H | Praj and Aaruni

104 | AL-RR-23 | 250 | 7A | Buffer | Pre-Viral | Fixed | 340 | 590 | | 2100L X 2000 X 2600H | Praj and Aaruni

105 | AL-RR-19 | 100 | 7A | Buffer | Pre-Viral | Mobile | 220 | 370 | | 1800L X 1900W X 2800H | Praj and Aaruni

106 | AL-RR-21 | NALGENE | 7A | Bottle | Pre-Viral | NA | 50 | - | | 500L X 500 W x 1000 H | Praj and Aaruni

107 | Akta- 1356 L/ Hr | NA | 7A | Equipment | Pre-Viral | Fixed | 2000 | - | | 2200L x 3200 W x 3000H | Truking, Cytiva, Aaruni

108 | Chromatography Column-1200 X 500 | NA | 7A | Equipment | Pre-Viral | Fixed | 3000 | 5000 | | 2200L x 3200 W x 3000H | Truking, Cytiva, Aaruni

109 | TFF | NA | 7A | Equipment | Pre-Viral | Fixed | 2000 | - | | 2500L X 2000 W X 2700H | Praj, Cytiva, Sartorius, Aaruni

110 | Mobile CIP | NA | 7A | Equipment | Pre-Viral | Mobile | 150 | 200 | | 2000L X 2000 X 2000H | Praj & DDE

111 | PV 125 | 100 | 8N | Process Vessel | Pre-Viral | Mobile | 220 | 370 | | 1800L X 1900W X 2800H | Praj and Aaruni

112 | PV 126 | 100 | 8N | Process Vessel | Pre-Viral | Mobile | 220 | 370 | | 1800L X 1900W X 2800H | Praj and Aaruni

113 | PV 125.1 | 100 | 8N | Process Vessel | Pre-Viral | Mobile | 220 | 370 | | 1800L X 1900W X 2800H | Praj and Aaruni

114 | FN-RR-02 | 300 | 8N | Buffer | Pre-Viral | Fixed | 390 | 690 | | 2100L X 2000 X 2700H | Praj and Aaruni

115 | PV 127 | 100 | 8N | Process Vessel | Pre-Viral | Mobile | 220 | 370 | | 1800L X 1900W X 2800H | Praj and Aaruni

116 | PV 127.1 | 100 | 8N | Process Vessel | Pre-Viral | Mobile | 220 | 370 | | 1800L X 1900W X 2800H | Praj and Aaruni

117 | Chromatography Column-300 X 500 | NA | 8N | Equipment | Pre-Viral | Fixed | 70 | 150 | | 500L X 500 W x 1000 H | Truking, Cytiva, Aaruni

118 | Akta- 70 L/ Hr | NA | 8N | Equipment | Pre-Viral | Fixed | 1200 | - | | 2200L x 3200 W x 3000H | Truking, Cytiva, Aaruni

119 | Mobile CIP | NA | 8N | Equipment | Pre-Viral | Mobile | 150 | 200 | | 2000L X 2000 X 2000H | Praj & DDE

120 | Centralised CIP - 2 tank system | TBD | CIP/SIP Room | Equipment | Pre-Viral | Fixed | 1000 | - | | 2200L x 3200 W x 3000H | Praj and Aaruni

119 | Decontamination Autoclave | TBD | Decontamination Room | Equipment | Pre-Viral | Fixed | 300 | - | | 500L x 500 W x 1200 H | Pharma Lab, Indu IonPure

120 | Dynamic Pass Box (DPB) | NA | Decontamination Room | Cleanroom Equipment | Pre-Viral | Fixed | 50 | - | | 1200 L x 1000 W x 2100 H | Clean air, Fabtech

121 | Dynamic Pass Box (DPB) | NA | Decontamination Room | Cleanroom Equipment | Pre-Viral | Fixed | 50 | - | | 1200 L x 1000 W x 2100 H | Clean air, Fabtech

122 | Dynamic Pass Box (DPB) | NA | Sterilization Room | Cleanroom Equipment | Pre-Viral | Fixed | 50 | - | | 1200 L x 1000 W x 2100 H | Clean air, Fabtech

123 | Sterilization Autoclave | TBD | Sterilization Room | Equipment | Pre-Viral | Fixed | 1600 | - | | 1200 L x 1000 W x 2100 H | Pharma Lab, Indu IonPure

126 | Ceiling LAF | NA | Cooling zone | Equipment | Pre-Viral | Fixed Under Ceiling | 150 | - | | 2500 L x 2000 W x 1000 H | Clean air, Fabtech

127 | Dynamic Pass Box (DPB) | NA | Corridor | Equipment | Pre-Viral | Fixed | 50 | | | 1200 L x 1000 W x 2100 H | Clean air, Fabtech

128 | PV 114 | 250 | 6I/11F | Process Vessel | Post Viral | Fixed | 340 | 590 | | 2100L X 2000 X 2600H | Praj and Aaruni

129 | PV 115 | 1000 | 6I/11F | Process Vessel | Post Viral | Fixed | 650 | 1650 | | 2700L X 2700W X 3200H | Praj and Aaruni

130 | PV 116 | 1000 | 6I/11F | Process Vessel | Post Viral | Fixed | 650 | 1650 | | 2700L X 2700W X 3200H | Praj and Aaruni

131 | IG-RR-09 | 100 | 6I/11F | Buffer | Post Viral | Mobile | 220 | 370 | | 1800L X 1900W X 2800H | Praj and Aaruni

132 | IG-RR-14 | 500 | 6I/11F | Buffer | Post Viral | Fixed | 450 | 950 | | 2200L X 2200 W X 2800H | Praj and Aaruni

133 | IG-RR-16 | 500 | 6I/11F | Buffer | Post Viral | Fixed | 450 | 950 | | 2200L X 2200 W X 2800H | Praj and Aaruni

134 | IG-RR-18 | 500 | 6I/11F | Buffer | Post Viral | Fixed | 450 | 950 | | 2200L X 2200 W X 2800H | Praj and Aaruni

135 | IG-RR-22 | 250 | 6I/11F | Buffer | Post Viral | Fixed | 340 | 590 | | 2100L X 2000 X 2600H | Praj and Aaruni

136 | PV 117 | 150 | 6I/11F | Process Vessel | Post Viral | Mobile | 250 | 430 | | 1900L X 2000W X 2800H | Praj and Aaruni

137 | IG-RR-26 | 150 | 6I/11F | Buffer | Post Viral | Mobile | 250 | 430 | | 1900L X 2000W X 2800H | Praj and Aaruni

138 | IG-RR-28 | 500 | 6I/11F | Buffer | Post Viral | Fixed | 450 | 950 | | 2200L X 2200 W X 2800H | Praj and Aaruni

139 | IG-RR-23 | 150 | 6I/11F | Buffer | Post Viral | Mobile | 250 | 430 | | 1900L X 2000W X 2800H | Praj and Aaruni

140 | IG-RR-24 | 150 | 6I/11F | Buffer | Post Viral | Mobile | 250 | 430 | | 1900L X 2000W X 2800H | Praj and Aaruni

141 | FG-RR-09 | 1000 | 6I/11F | Buffer | Post Viral | Fixed | 650 | 1650 | | 2700L X 2700W X 3200H | Praj and Aaruni

142 | FG-RR-10 | 2000 | 6I/11F | Buffer | Post Viral | Fixed | 1200 | 3200 | | 3600L X 3600W X 4300H | Praj and Aaruni

143 | FG-RR-11 | 500 | 6I/11F | Buffer | Post Viral | Fixed | 450 | 950 | | 2200L X 2200 W X 2800H | Praj and Aaruni

| FG-RR-13 | 100 | 6I/11F | Buffer | Post Viral | Mobile | 220 | 370 | | 2100L X 2000 X 2600H | Praj and Aaruni

144 | FG-RR-14 | 250 | 6I/11F | Buffer | Post Viral | Fixed | 340 | 590 | | 2100L X 2000 X 2600H | Praj and Aaruni

145 | Akta-1" -667 L/Hr | NA | 6I/11F | Equipment | Post Viral | Fixed | 2000 | - | | 2200L x 3200 W x 3000H | Truking, Cytiva, Aaruni

146 | Chromatography Column 1000X 500 | NA | 6I/11F | Equipment | Post Viral | Fixed | 400 | 600 | | 1500L X 1500 W X 2700H | Truking, Cytiva, Aaruni

147 | TFF | NA | 6I/11F | Equipment | Post Viral | Fixed | 2000 | - | | 2500L X 2000 W X 2700H | Praj, Cytiva, Sartorius, Aaruni

148 | Nano Filter | NA | 6I/11F | Single use | Post Viral | Fixed | 20 | - | | 500L x 500 W x 1200 H | Planova

149 | Mobile CIP | NA | 6I/11F | Equipment | Post Viral | Mobile | 150 | 200 | | 2000L X 2000 X 2000H | Praj & DDE

150 | FN-RR-15 | 500 | 9N | Buffer | Post Viral | Fixed | 450 | 950 | | 2200L X 2200 W X 2800H | Praj and Aaruni

151 | FN-RR-18 | 100 | 9N | Buffer | Post Viral | Mobile | 220 | 370 | | 1800L X 1900W X 2800H | Praj and Aaruni

152 | FN-RR-17 | 100 | 9N | Buffer | Post Viral | Mobile | 220 | 370 | | 1800L X 1900W X 2800H | Praj and Aaruni

153 | Akta-1" -30 L/Hr | NA | 9N | Equipment | Post Viral | Fixed | 200 | - | | 2200L x 1800 W x 2000H | Truking, Cytiva, Aaruni

154 | Chromatography Column-300 X 750 | NA | 9N | Equipment | Post Viral | Fixed | 80 | 200 | | 500L x 500 W x 2000 H | Truking, Cytiva, Aaruni

155 | Mobile CIP | NA | 9N | Equipment | Post Viral | Mobile | 150 | 200 | | 2000L X 2000 X 1800H | Praj & DDE

156 | PV 131 | 300 | 10T/13E | Process Vessel | Post Viral | Fixed | 390 | 690 | | 2100L X 2000 X 2700H | Praj and Aaruni

157 | PV 131.1 | 200 | 10T/13E | Process Vessel | Post Viral | Mobile | 300 | 520 | | 2100L X 2000 X 2600H | Praj and Aaruni

158 | PV 132 | 250 | 10T/13E | Process Vessel | Post Viral | Fixed | 340 | 590 | | 2100L X 2000 X 2600H | Praj and Aaruni

159 | TH-RR-05 | 50 | 10T/13E | Buffer | Post Viral | Mobile | 280 | - | | 1800L X 1800W X 2800H | Praj and Aaruni

160 | TH-RR-02 | 100 | 10T/13E | Buffer | Post Viral | Mobile | 220 | 370 | | 1800L X 1900W X 2800H | Praj and Aaruni

161 | TH-RR-02 | 100 | 10T/13E | Buffer | Post Viral | Mobile | 220 | 370 | | 1800L X 1900W X 2800H | Praj and Aaruni

162 | TH-RR-03 | 200 | 10T/13E | Buffer | Post Viral | Mobile | 300 | 520 | | 2100L X 2000 X 2600H | Praj and Aaruni

163 | NALGENE | 50 | 10T/13E | Bottle | Post Viral | NA | 280 | | | 500L x 500 W x 1200 H | Praj and Aaruni

164 | PV 132 | 250 | 10T/13E | Process Vessel | Post Viral | Fixed | 340 | 590 | | 2100L X 2000 X 2600H | Praj and Aaruni

165 | PV 128 | 300 | 10T/13E | Process Vessel | Post Viral | Fixed | 390 | 690 | | 2100L X 2000 X 2700H | Praj and Aaruni

166 | FE-RR-09 | 250 | 10T/13E | Buffer | Post Viral | Fixed | 340 | 590 | | 2100L X 2000 X 2600H | Praj and Aaruni

167 | FE-RR-09 | 250 | 10T/13E | Buffer | Post Viral | Fixed | 340 | 590 | | 2100L X 2000 X 2600H | Praj and Aaruni

168 | Akta-1" -160 L/Hr | NA | 10T/13E | Equipment | Post Viral | Fixed | 200 | - | | 2200L x 1800 W x 2000H | Truking, Cytiva, Aaruni

169 | Chromatography Column-450 X 500 | NA | 10T/13E | Equipment | Post Viral | Fixed | 100 | 250 | | 500L x 500 W x 2000 H | Truking, Cytiva, Aaruni

170 | Mobile CIP | NA | 10T/13E | Equipment | Post Viral | Mobile | 150 | 200 | | 2000L X 2000 X 2000H | Praj & DDE

171 | Centralised CIP - 2 tank system | TBD | CIP/SIP Room | Equipment | Post Viral | Fixed | 1000 | - | | 2200L x 3200 W x 3000H | Praj and Aaruni

173 | Decontamination Autoclave | TBD | Decontamination Room | Equipment | Post Viral | Fixed | 300 | - | | 500L x 500 W x 1200 H | Pharma Lab, Indu IonPure

174 | Dynamic Pass Box (DPB) | NA | Decontamination Room | Cleanroom Equipment | Post Viral | Fixed | 50 | - | | 1200 L x 1000 W x 2100 H | Clean air, Fabtech

175 | Dynamic Pass Box (DPB) | NA | Decontamination Room | Cleanroom Equipment | Post Viral | Fixed | 50 | - | | 1200 L x 1000 W x 2100 H | Clean air, Fabtech

176 | Dynamic Pass Box (DPB) | NA | Sterilization Room | Cleanroom Equipment | Post Viral | Fixed | 50 | - | | 1200 L x 1000 W x 2100 H | Clean air, Fabtech

177 | Sterilization Autoclave | TBD | Sterilization Room | Equipment | Post Viral | Fixed | 1600 | - | | 1200 L x 1000 W x 2100 H | Pharma Lab, Indu IonPure

178 | Ceiling LAF | NA | Cooling zone | Equipment | Post Viral | Fixed Under Ceiling | 150 | - | | 2500 L x 2000 W x 1000 H | Clean air, Fabtech

179 | Buffer Preparation | 1500 | Buffer Preparation Room | Process Vessel | Post Viral | TBD | 900 | 2400 | | 3200L X 3500 W X 4200H | Praj and Aaruni

180 | Buffer Preparation | 1000 | Buffer Preparation Room | Process Vessel | Post Viral | TBD | 650 | 1650 | | 2700L X 2700W X 3200H | Praj and Aaruni

181 | Pass box | NA | Buffer Preparation Room | Equipment | Post Viral | NA | 50 | - | | 1200 L x 1000 W x 2100 H | Clean air, Fabtech

182 | Pass box | NA | Chemical Day Store | Equipment | Post Viral | NA | 50 | - | | 1200 L x 1000 W x 2100 H | Clean air, Fabtech

[SHEET: Fill Finish(FF)]

Sr. No. | Vessel Tag | Room/Stage (As per Layout) | Type | Fill Finish | Fixed / Mobile | Gross Weight in Kg (Empty Vessel) | Dimension (L mm x W mmx H mm) | Make

1 | DPB | De-Contamination Room | Equipment | Plasma | Fixed | 50 | 1200 L x 1000 W x 2100 H | Clean air, Fabtech

2 | Vial washing | Vial Washing Room | Vial Washing Machine | Plasma | Fixed | 4000 | 3500 L X 1800 W X 2100 H | NKP, Harikrushna

3 | DPB | Vial Washing Room | Equipment | Plasma | Fixed | 50 | 1200 L x 1000 W x 2100 H | Clean air, Fabtech

4 | Autoclave | Material Prep & Wash | Equipment | Plasma | Fixed | 2500 | 1200 L x 1000 W x 2100 H | Pharma Lab, Indu IonPure

5 | DPB | Material Prep & Wash | Equipment | Plasma | Fixed | 50 | 1200 L x 1000 W x 2100 H | Clean air, Fabtech

6 | Autoclave | Decontamination | Equipment | Plasma | Fixed | 300 | 1200 L x 1000 W x 2100 H | Pharma Lab, Indu IonPure

7 | Washing machine | Garment Wash | Washing Machine | Plasma | Mobile | 80 | 1200 L x 1300 W x 1700 H | IFB

8 | Dryer | Garment Wash | Drying machine | Plasma | Mobile | 80 | 1800 L x 1500 W x 2600 H | IFB

9 | Ceiling LAF | Unload Area | Equipment | Plasma | Fixed under Ceiling | NA | 2500 L x 2000 W x 1000 H | Clean air, Fabtech

10 | Vessel | Formulation | Formulation Vessel | Plasma | Fixed | 500 | 2200L X 2200 W X 2800H | Praj and Aaruni

11 | Vessel | Formulation | Formulation Vessel | Plasma | Fixed | 500 | 2200L X 2200 W X 2800H | Praj and Aaruni

12 | Ceiling LAF | Formulation | Equipment | Plasma | Fixed under Ceiling | NA | 2500 L x 2000 W x 1000 H | Clean air, Fabtech

13 | Filling Machine | Filling Area | Filling machine | Plasma | Fixed | 2200 | 3200 L X 1700 W X 2300 H | NKP, Harikrushna

14 | Ceiling LAF | Filling Area | Equipment | Plasma | Fixed under Ceiling | NA | 2500 L x 2000 W x 1000 H | Clean air, Fabtech

15 | Ceiling LAF | Capping Room | Equipment | Plasma | Fixed under Ceiling | NA | 2500 L x 2000 W x 1000 H | Clean air, Fabtech

16 | Capping Machine | Capping Room | Capping Machine | Plasma | Fixed | 600 | 2200 L X 1500 W X 2000 H | NKP, Harikrushna

17 | Turn Table | Capping Room | SS Turn Table | Plasma | Fixed | 300 | 1200 L X 1300 W X 900 H | NKP, Harikrushna

18 | Swing Conveyor | Capping Room | Swing Conveyor With Vial Collection Tray | Plasma | Fixed | 350 | 2500 L X 1200 W X 900 H | NKP, Harikrushna

19 | Lyo Machine | Lyo Loading Side | Lyp machine | Plasma | Fixed | 7500 | 4200 L X 2200 W X 2400 H | LSI

20 | DPB | Buffer Room | Equipment | Plasma | Fixed | 50 | 1200 L × 1000 W × 2100 H | Clean air, Fabtech

21 | Visual Inspection Booth | Visual Inspection Room | Visual Inspection Booth | Plasma | Fixed | 425 | 3000 L X 1800 W X 2400 H | NKP, Harikrushna

22 | Labelling machine | Secondary Packing Area | Labeling Machine | Plasma | Fixed | 600 | 3000 L X 1800 W X 2000 H | NKP, Harikrushna

23 | Cartonng machine | Secondary Packing Area | Cartonng Machine | Plasma | Fixed | 200 | 6000 L X 2000 W X 2100 H | NKP, Harikrushna

24 | PW Generation(MGF, Softener, Soft Water Storage tank, RO and RO water storage tank) | Water System & Engineering Store | Water System & Utilities | Utilities equipment | Fixed | 500 | | Praj, CN water

25 | PW storage 15,000L | Water System & Engineering Store | Water System & Utilities | Utilities equipment | Fixed | 17500 | | Praj, CN water

26 | WFI Generation and storage 5000L | Water System & Engineering Store | Water System & Utilities | Utilities equipment | Fixed | 6500 | | Praj, CN water

27 | PSG System | Water System & Engineering Store | Water System & Utilities | Utilities equipment | Fixed | 350 | | Praj, CN water

28 | Compressed Air System 300 CFM | Water System & Engineering Store | Water System & Utilities | Utilities equipment | Fixed | 750 | | Atlas copco

29 | AHUs | Service Area | HVAC | Utilities equipment | Fixed | | | Clean air, Fabtech

30 | Electrical Control Panel | Service Area | Electrical | Utilities equipment | Fixed | | | Clean air, Fabtech

Reagent Details:

Formula:

Sodium Acetate(M.Wt): 82.03g/mol.

Acetic Acid (M.Wt): 60.05g/ mol

Preparation of 0.2 M Sodium Acetate pH 4.2 buffer

Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol)

×Volume (L).

Calculation: To prepare 1 L of 0.2 M Sodium Acetate (MW = 82.03g/mol).

Weight of solute(g)=0.2×82.03× 1.0=16.406 g.

Preparation Method:

Weigh 16.406 g of Sodium Acetate and dissolve in water to make a 1 L solution.

Preparation for required volumes:

To prepare _____ of 0.2 M Sodium Acetate (MW = 82.03g/mol).

Weight of solute(g)=0.2×82.03× _____ L= _____ g.

Weigh _____ gm of Sodium Acetate and dissolve in water to make a _____ L solution.

Calculations: To prepare the 0.2 M Acetic Acid.

Formula & Calculations:

Formula: Use the formula to calculate the molarity:

Molarity (M) = (Per cent by Weight / 100) * (Density in g/mL) * 1000 / Molar Mass.

Example: Molarity of Glacial Acetic Acid

Molar Mass of CH₃COOH: ~60.05 g/mol

Density: ~1.05 g/mL

Purity: 100%

Acetic Acid (MW = 60.05 g/mol).

Calculation:

Molarity = (100 / 100) * 1.05 g/mL * 1000 mL/L / 60.05 g/mol ≈ 17.5 M.

Calculation: To prepare 1 L of 0.2 M.

Applying the formula to prepare the working buffer:

C₁V₁=C₂V₂

C₁ = 17.5 M acetic acid.

V₁ = volume of stock needed to take.

C₂ = desired (working) concentration 0.2 M acetic acid.

V₂ = final total volume of working solution for one litre.

V₁ = (C₂ × V₂)/C₁.

C₂=0.2 M ; V₂= 1.0 L; C₁=17.5 M

V₁ = (0.2 × 1.0)/17.5

V₁ = 0.011 L

Take 0.011 L of the Stock solution of acetic acid and add it to a volumetric flask, making up the volume to 1 L.

To prepare various volumes:

V₁ = volume of stock needed to take _____ L.

C₂ = desired (working) concentration 0.2M acetic acid.

V₂ = final total volume of working solution _____ L.

V₁ = (C₂ _____ × V₂ _____) /17.5.

V₁= _____ L.

Take V₁ _____ L of the Stock solution of acetic acid and add it to a volumetric flask, making up the final volume _____ L.

Procedure for Preparation of 0.2 M Sodium Acetate pH 4.2.

To prepare a pH 4.2 solution, 0.2 M Sodium Acetate solution is titrated with 0.2 M Acetic acid so that the pH attains pH 4.2.

Preparation Details:

Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry Date

Sodium Acetate | | | |

Acetic Acid | | | |

Final volume prepared | Measured Quantity | pH (4.2 ± 0.1) | Conductivity (15 ± 0.2ms/cm) | Storage Condition | Reagent Record

| | | RT |

Date of Preparation | Prepared By | Checked By

| |

[SHEET: Warehousing (GF)]

Sr. No. | Equipment Tag | Working Volume in Litres | Room/Stage (As per Layout) | Type | Warehouse | Fixed / Mobile | Gross Weight in Kg (Empty- Tentative) | Dimension (L mm x W mmx H mm) | Make
1 | De-Dusting tunnel | NA | RM Receipt Room | De-Dusting Machine | Warehouse | Fixed | 200 | 2200 L X 1200 W X 2400 H | Pridetech
2 | RLAF | NA | RM Sampling Room | Sampling Booth | Warehouse | Fixed | 150 | 2500 L x 2000 Wx 1000 H | Clean air, Fabtech
3 | DBP | NA | RM Sampling Room | Equipment | Warehouse | Fixed | 180 | 1200 L x 1000 W x 2100 H | Clean air, Fabtech
4 | DPB | NA | Approved RM | Equipment | Warehouse | Fixed | 180 | 1200 L x 1000 W x 2100 H | Clean air, Fabtech
5 | DPB | NA | Dispensing Room | Equipment | Warehouse | Fixed | 180 | 1200 L x 1000 W x 2100 H | Clean air, Fabtech
6 | RLAF | NA | Dispensing Room | Dispensing Booth | Warehouse | Fixed | 150 | 2500 L x 2000 Wx 1000 H | Clean air, Fabtech
7 | DPB | NA | Plasma Sampling | Equipment | Warehouse | Fixed | 180 | 1200 L x 1000 W x 2100 H | Clean air, Fabtech
8 | RLAF | NA | Plasma Sampling | Plasma Sampling Booth | Warehouse | Fixed | 150 | 2500 L x 2000 Wx 1000 H | Clean air, Fabtech
9 | DPB | NA | Reject Room | Equipment | Warehouse | Fixed | 180 | 1200 L x 1000 W x 2100 H | Clean air, Fabtech
10 | freezer-1000L | 1000 L | Plasma Storage Area | Freezer | Warehouse | Fixed | 425 | 1400 L X 950 W X 2100 H | Newtronic lifecare, REMI
11 | freezer-1000L | 1000 L | Plasma Storage Area | Freezer | Warehouse | Fixed | 425 | 1400 L X 950 W X 2100 H | Newtronic lifecare, REMI
12 | freezer-1000L | 1000 L | Plasma Storage Area | Freezer | Warehouse | Fixed | 425 | 1400 L X 950 W X 2100 H | Newtronic lifecare, REMI
13 | freezer-1000L | 1000 L | Plasma Storage Area | Freezer | Warehouse | Fixed | 425 | 1400 L X 950 W X 2100 H | Newtronic lifecare, REMI
14 | freezer-1000L | 1000 L | Plasma Storage Area | Freezer | Warehouse | Fixed | 425 | 1400 L X 950 W X 2100 H | Newtronic lifecare, REMI
15 | freezer-1000L | 1000 L | Plasma Storage Area | Freezer | Warehouse | Fixed | 425 | 1400 L X 950 W X 2100 H | Newtronic lifecare, REMI
16 | freezer-1000L | 1000 L | Plasma Storage Area | Freezer | Warehouse | Fixed | 425 | 1400 L X 950 W X 2100 H | Newtronic lifecare, REMI
17 | freezer-1000L | 1000 L | Plasma Storage Area | Freezer | Warehouse | Fixed | 425 | 1400 L X 950 W X 2100 H | Newtronic lifecare, REMI
18 | freezer-1000L | 1000 L | Plasma Storage Area | Freezer | Warehouse | Fixed | 425 | 1400 L X 950 W X 2100 H | Newtronic lifecare, REMI
19 | freezer-1000L | 1000 L | Plasma Storage Area | Freezer | Warehouse | Fixed | 425 | 1400 L X 950 W X 2100 H | Newtronic lifecare, REMI
20 | freezer-1000L | 1000 L | Plasma Storage Area | Freezer | Warehouse | Fixed | 425 | 1400 L X 950 W X 2100 H | Newtronic lifecare, REMI
21 | freezer-1000L | 1000 L | Plasma Storage Area | Freezer | Warehouse | Fixed | 425 | 1400 L X 950 W X 2100 H | Newtronic lifecare, REMI
22 | freezer-1000L | 1000 L | Plasma Storage Area | Freezer | Warehouse | Fixed | 425 | 1400 L X 950 W X 2100 H | Newtronic lifecare, REMI
23 | freezer-1000L | 1000 L | Plasma Storage Area | Freezer | Warehouse | Fixed | 425 | 1400 L X 950 W X 2100 H | Newtronic lifecare, REMI
24 | freezer-1000L | 1000 L | Plasma Storage Area | Freezer | Warehouse | Fixed | 425 | 1400 L X 950 W X 2100 H | Newtronic lifecare, REMI

[SHEET: Bulk MFG (GF)]

Sr. No. | Vessel Tag | Working Volume in Litres | Room/Stage (As per Layout) | Type | Pre Viral/Post Viral | Fixed / Mobile | Empty (Kg) | Full load (Kg) | Dimension (mm x mmx mm) | Make
1 | TW 100 | 500 | THAWING | Process Vessel | Pre-Viral | Fixed | 450 | 950 | 2200L X 2200 W X 2800H | Praj and Aaruni
2 | DPB | NA | THAWING | Equipment | Pre-Viral | Fixed | 50 | - | 1200 L x 1000 W x 2100 H | Clean air, Fabtech
3 | PV 101.1 | 300 | CUTTING | Process Vessel | Pre-Viral | Fixed | 390 | 690 | 2100L X 2000 X 2700H | Praj and Aaruni
4 | PV 101.2 | 300 | CUTTING | Process Vessel | Pre-Viral | Fixed | 390 | 690 | 2100L X 2000 X 2700H | Praj and Aaruni
6 | PV 102.1 | 300 | 1F/12E | Process Vessel | Pre-Viral | Fixed | 390 | 690 | 2100L X 2000 X 2700H | Praj and Aaruni
7 | PV 102.2 | 300 | 1F/12E | Process Vessel | Pre-Viral | Fixed | 390 | 690 | 2100L X 2000 X 2700H | Praj and Aaruni
8 | PV 103.1 | 1500 | 1F/12E | Process Vessel | Pre-Viral | Fixed | 900 | 2400 | 2100L X 2000 X 2700H | Praj and Aaruni
9 | PV 103.2 | 1500 | 1F/12E | Process Vessel | Pre-Viral | Fixed | 900 | 2400 | 3200L X 3500 W X 4200H | Praj and Aaruni
10 | PV 103.3 | 500 | 1F/12E | Process Vessel | Pre-Viral | Fixed | 450 | 950 | 3200L X 3500 W X 4200H | Praj and Aaruni
11 | PV 103.4 | 500 | 1F/12E | Process Vessel | Pre-Viral | Fixed | 450 | 950 | 2200L X 2200 W X 2800H | Praj and Aaruni
12 | PV 103.5 | 500 | 1F/12E | Process Vessel | Pre-Viral | Fixed | 450 | 950 | 2200L X 2200 W X 2800H | Praj and Aaruni
13 | SPARE-1 | 250 | 1F/12E | Process Vessel | Pre-Viral | Fixed | 340 | 590 | 2200L X 2200 W X 2800H | Praj and Aaruni
14 | PF-RR-03 | 1000 | 1F/12E | Buffer | Pre-Viral | Fixed | 650 | 1650 | 2100L X 2000 X 2600H | Praj and Aaruni
15 | PF-RR-27 | 250 | 1F/12E | Buffer | Pre-Viral | Fixed | 340 | 590 | 2700L X 2700W X 3200H | Praj and Aaruni
16 | PV 128 | 300 | 1F/12E | Process Vessel | Pre-Viral | Fixed | 390 | 690 | 2100L X 2000 X 2600H | Praj and Aaruni
17 | FE-RR-02 | 150 | 1F/12E | Buffer | Pre-Viral | Mobile | 250 | 430 | 2100L X 2000 X 2700H | Praj and Aaruni
18 | FE-RR-06 | 250 | 1F/12E | Buffer | Pre-Viral | Fixed | 340 | 590 | 1900L X 2000W X 2800H | Praj and Aaruni
19 | FE-RR-06 | 250 | 1F/12E | Buffer | Pre-Viral | Fixed | 340 | 590 | 2100L X 2000 X 2600H | Praj and Aaruni

| FE-RR-14 | 250 | 1F/12E | Buffer | Pre-Viral | Mobile | 340 | 590 | 2100L X 2000 X 2600H | Praj and Aaruni

20 | Chromatography Column- 1000 X 500 | NA | 1F/12E | Equipment | Pre-Viral | Fixed | 400 | 600 | 1500L X 1500 W X 2700H | Truking, Cytiva, Aaruni

21 | Akta- 500 L/ Hr | NA | 1F/12E | Equipment | Pre-Viral | Fixed | 1200 | - | 2000L X 2000 X 2000H | Truking, Cytiva, Aaruni

22 | Chromatography Column-1000 X 500 | NA | 1F/12E | Equipment | Pre-Viral | Fixed | 400 | 600 | 1500L X 1500 W X 2700H | Truking, Cytiva, Aaruni

23 | Mobile CIP | NA | 1F/12E | Equipment | Pre-Viral | Mobile | 150 | 200 | 2000L X 2000 X 2000H | Praj & DDE

24 | PV 104.1 | 1500 | 2F | Process Vessel | Pre-Viral | Fixed | 900 | 2400 | 3200L X 3500 W X 4200H | Praj and Aaruni

25 | PV 104.2 | 1500 | 2F | Process Vessel | Pre-Viral | Fixed | 900 | 2400 | 3200L X 3500 W X 4200H | Praj and Aaruni

26 | PV 105.1 | 1500 | 2F | Process Vessel | Pre-Viral | Fixed | 900 | 2400 | 3200L X 3500 W X 4200H | Praj and Aaruni

27 | PV 105.2 | 1500 | 2F | Process Vessel | Pre-Viral | Fixed | 900 | 2400 | 3200L X 3500 W X 4200H | Praj and Aaruni

28 | PV 105.3 | 300 | 2F | Process Vessel | Pre-Viral | Fixed | 390 | 690 | 2100L X 2000 X 2700H | Praj and Aaruni

29 | PF-RR-35 | 200 | 2F | Buffer | Pre-Viral | Mobile | 300 | 520 | 2100L X 2000 X 2600H | Praj and Aaruni

30 | PF-RR-31 | 100 | 2F | Buffer | Pre-Viral | Mobile | 220 | 370 | 1800L X 1900W X 2800H | Praj and Aaruni

31 | PF-RR-11 | 100 | 2F | Buffer | Pre-Viral | Mobile | 220 | 370 | 1800L X 1900W X 2800H | Praj and Aaruni

32 | PF-RR-12 | 250 | 2F | Buffer | Pre-Viral | Fixed | 340 | 590 | 2100L X 2000 X 2600H | Praj and Aaruni

33 | PF-RR-32 | 100 | 2F | Buffer | Pre-Viral | Mobile | 220 | 370 | 1800L X 1900W X 2800H | Praj and Aaruni

34 | PF-RR-27 | 100 | 2F | Buffer | Pre-Viral | Mobile | 220 | 370 | 1800L X 1900W X 2800H | Praj and Aaruni

| PF-RR-28 | 100 | 2F | Buffer | Pre-Viral | Mobile | 220 | 370 | 1800L X 1900W X 2800H | Praj and Aaruni

35 | PF-RR-30 | NALGENE | 2F | Bottle | Pre-Viral | NA | 50 | - | 500L X 500 W x 1000 H | Praj and Aaruni

36 | TFF | NA | 2F | Equipment | Pre-Viral | Fixed | 3000 | 5000 | 2500L X 2000 W X 2700H | Praj, Cytiva, Sartorius, Aaruni

37 | Chromatography Column-1000 X 500 | NA | 2F | Equipment | Pre-Viral | Fixed | 400 | 600 | 1500L X 1500 W X 2700H | Truking, Cytiva, Aaruni

38 | Akta-785 L/ hr | NA | 2F | Equipment | Pre-Viral | Fixed | 2000 | 4000 | 2000L X 2000 X 2000H | Truking, Cytiva, Aaruni

39 | Mobile CIP | NA | 2F | Equipment | Pre-Viral | Mobile | 150 | 200 | 2000L X 2000 X 2000H | Praj & DDE

40 | PV 106 | 1500 | 3F | Process Vessel | Pre-Viral | Fixed | 900 | 2400 | 3200L X 3500 W X 4200H | Praj and Aaruni

41 | PV 107 | 2000 | 3F | Process Vessel | Pre-Viral | Fixed | 1200 | 3200 | 3600L X 3600W X 4300H | Praj and Aaruni

42 | PV 103.6 | 500 | 3F | Process Vessel | Pre-Viral | Fixed | 450 | 950 | 2200L X 2200 W X 2800H | Praj and Aaruni

43 | PV 128 | 500 | 3F | Process Vessel | Pre-Viral | Fixed | 450 | 950 | 2200L X 2200 W X 2800H | Praj and Aaruni

44 | PV 129 | 300 | 3F | Process Vessel | Pre-Viral | Fixed | 390 | 690 | 2100L X 2000 X 2700H | Praj and Aaruni

45 | PV 119.1 | 300 | 3F | Process Vessel | Pre-Viral | Fixed | 390 | 690 | 2100L X 2000 X 2700H | Praj and Aaruni

46 | PV 130 | 100 | 3F | Process Vessel | Pre-Viral | Mobile | 220 | 370 | 1800L X 1900W X 2800H | Praj and Aaruni

47 | Centrifuge | NA | 3F | Equipment | Pre-Viral | Fixed | 800 | 900 | 2000L x 1500 W x 2000H | Alfa Laval

48 | Chromatography Column | NA | 3F | Equipment | Pre-Viral | Fixed | 400 | 600 | 1500L X 1500 W X 2700H | Truking, Cytiva, Aaruni

49 | Chromatography Column-1000 X 500 | NA | 3F | Equipment | Pre-Viral | Fixed | 400 | 600 | 1500L X 1500 W X 2700H | Truking, Cytiva, Aaruni

50 | Akta-1178 L/Hr | NA | 3F | Equipment | Pre-Viral | Fixed | 3000 | - | 2000L X 2000 X 2000H | Truking, Cytiva, Aaruni

51 | Mobile CIP | NA | 3F | Equipment | Pre-Viral | Mobile | 150 | 200 | 2000L X 2000 X 2000H | Praj & DDE

52 | PV 108 | 1000 | 4I | Process Vessel | Pre-Viral | Fixed | 650 | 1650 | 2700L X 2700W X 3200H | Praj and Aaruni

53 | PV 109.1 | 2000 | 4I | Process Vessel | Pre-Viral | Fixed | 1200 | 3200 | 3600L X 3600W X 4300H | Praj and Aaruni

54 | PV 109.2 | 2000 | 4I | Process Vessel | Pre-Viral | Fixed | 1200 | 3200 | 3600L X 3600W X 4300H | Praj and Aaruni

55 | PV 109.3 | 1000 | 4I | Process Vessel | Pre-Viral | Fixed | 650 | 1650 | 2700L X 2700W X 3200H | Praj and Aaruni

56 | PV 110.1 | 2500 | 4I | Process Vessel | Pre-Viral | Fixed | 1500 | 4000 | 3800L X 3800W X 4300H | Praj and Aaruni

57 | PV 110.2 | 2500 | 4I | Process Vessel | Pre-Viral | Fixed | 1500 | 4000 | 3800L X 3800W X 4300H | Praj and Aaruni

58 | PF-RR-18 | 250 | 4I | Buffer | Pre-Viral | Fixed | 340 | 590 | 2100L X 2000 X 2600H | Praj and Aaruni

59 | PF-RR-21 | 1000 | 4I | Buffer | Pre-Viral | Fixed | 650 | 1650 | 2700L X 2700W X 3200H | Praj and Aaruni

60 | PF-RR-21 | NALGENE | 4I | Bottle | Pre-Viral | NA | 50 | - | 500L X 500 W x 1000 H | Praj and Aaruni

61 | PF-RR-23 | 500 | 4I | Buffer | Pre-Viral | Fixed | 450 | 950 | 2200L X 2200 W X 2800H | Praj and Aaruni

62 | PF-RR-25 | 500 | 4I | Buffer | Pre-Viral | Fixed | 450 | 950 | 2200L X 2200 W X 2800H | Praj and Aaruni

63 | PF-RR-27 | 500 | 4I | Buffer | Pre-Viral | Fixed | 450 | 950 | 2200L X 2200 W X 2800H | Praj and Aaruni

64 | Akta-3454 L/Hr | NA | 4I | Equipment | Pre-Viral | Fixed | 3000 | - | 2000L X 2000 X 2000H | Truking, Cytiva, Aaruni

65 | Chromatography Column-2000 X 500 | NA | 4I | Equipment | Pre-Viral | Fixed | 3500 | 5000 | 2200L x

3200 W x 3000H | TruKing, Cytiva, Aaruni

66 | TFF | NA | 4l | Equipment | Pre-Viral | Fixed | 2000 | - | 2500L X 2000 W X 2700H | Praj, Cytiva, Sartorius, Aaruni

67 | Mobile CIP | NA | 4l | Equipment | Pre-Viral | Mobile | 150 | 200 | 2000L X 2000 X 2000H | Praj & DDE

68 | PV 110 | 1500 | 5l | Process Vessel | Pre-Viral | Fixed | 900 | 2400 | 3200L X 3500 W X 4200H | Praj and Aaruni

69 | PV 111 | 1500 | 5l | Process Vessel | Pre-Viral | Fixed | 900 | 2400 | 3200L X 3500 W X 4200H | Praj and Aaruni

70 | PV 112.1 | 1500 | 5l | Process Vessel | Pre-Viral | Fixed | 900 | 2400 | 3200L X 3500 W X 4200H | Praj and Aaruni

71 | PV 112.2 | 1500 | 5l | Process Vessel | Pre-Viral | Fixed | 900 | 2400 | 3200L X 3500 W X 4200H | Praj and Aaruni

72 | IG-RR-09 | 100 | 5l | Buffer | Pre-Viral | Mobile | 220 | 370 | 1800L X 1900W X 2800H | Praj and Aaruni

73 | IG-RR-05 | 300 | 5l | Buffer | Pre-Viral | Fixed | 390 | 690 | 2100L X 2000 X 2700H | Praj and Aaruni

74 | IG-RR-07 | 250 | 5l | Buffer | Pre-Viral | Fixed | 340 | 590 | 2100L X 2000 X 2600H | Praj and Aaruni

75 | IG-RR-22 | 250 | 5l | Buffer | Pre-Viral | Fixed | 340 | 590 | 2100L X 2000 X 2600H | Praj and Aaruni

76 | IG-RR-23 | 50 | 5l | Buffer | Pre-Viral | Mobile | 70 | 150 | 1800L X 1800W X 2800H | Praj and Aaruni

77 | PV 113 | 250 | 5l | Process Vessel | Pre-Viral | Fixed | 340 | 590 | 2100L X 2000 X 2600H | Praj and Aaruni

78 | PV113.1 | 250 | 5l | Process Vessel | Pre-Viral | Fixed | 340 | 590 | 2100L X 2000 X 2600H | Praj and Aaruni

79 | IG-RR-06 | NALGENE | 5l | Bottle | Pre-Viral | NA | 50 | - | 500L X 500 W x 1000 H | Praj and Aaruni

80 | IG-RR-23 | 100 | 5l | Buffer | Pre-Viral | Mobile | 220 | 370 | 1800L X 1900W X 2800H | Praj and Aaruni

81 | IG-RR-24 | 100 | 5l | Buffer | Pre-Viral | Mobile | 220 | 370 | 1800L X 1900W X 2800H | Praj and Aaruni

82 | Akta-2211 L/ Hr | NA | 5l | Equipment | Pre-Viral | Fixed | 3000 | - | 2000L X 2000 X 2000H | TruKing, Cytiva, Aaruni

83 | Chromatography Column-1600X 500 | NA | 5l | Equipment | Pre-Viral | Fixed | 2500 | 5000 | 2200L x 3200 W x 3000H | TruKing, Cytiva, Aaruni

84 | TFF | NA | 5l | Equipment | Pre-Viral | Fixed | 2000 | - | 2500L X 2000 W X 2700H | Praj, Cytiva, Sartorius, Aaruni

85 | Depth filters | NA | 5l | Equipment | Pre-Viral | Fixed | 150 | - | 500L X 500 W x 1000 H | Aaruni, Millipore, 3M

86 | Mobile CIP | NA | 5l | Equipment | Pre-Viral | Mobile | 150 | 200 | 2000L X 2000 X 2000H | Praj & DDE

87 | PV 118.1 | 2500 | 7A | Process Vessel | Pre-Viral | Fixed | 1500 | 4000 | 3800L X 3800W X 4300H | Praj and Aaruni

88 | PV 118.2 | 2500 | 7A | Process Vessel | Pre-Viral | Fixed | 1500 | 4000 | 3800L X 3800W X 4300H | Praj and Aaruni

89 | PV 119 | 2000 | 7A | Process Vessel | Pre-Viral | Fixed | 1200 | 3200 | 3600L X 3600W X 4300H | Praj and Aaruni

90 | PV 120 | 1000 | 7A | Process Vessel | Pre-Viral | Fixed | 650 | 1650 | 2700L X 2700W X 3200H | Praj and Aaruni

91 | AL-RR-02 | 100 | 7A | Buffer | Pre-Viral | Mobile | 220 | 370 | 1800L X 1900W X 2800H | Praj and Aaruni

92 | AL-RR-05 | 500 | 7A | Buffer | Pre-Viral | Fixed | 450 | 950 | 2200L X 2200 W X 2800H | Praj and Aaruni

93 | AL-RR-05 | 500 | 7A | Buffer | Pre-Viral | Fixed | 450 | 950 | 2200L X 2200 W X 2800H | Praj and Aaruni

94 | AL-RR-09 | 500 | 7A | Buffer | Pre-Viral | Fixed | 450 | 950 | 2200L X 2200 W X 2800H | Praj and Aaruni

95 | AL-RR-11 | 250 | 7A | Buffer | Pre-Viral | Fixed | 340 | 590 | 2100L X 2000 X 2600H | Praj and Aaruni

96 | AL-RR-19 | 250 | 7A | Buffer | Pre-Viral | Fixed | 340 | 590 | 2100L X 2000 X 2600H | Praj and Aaruni

97 | AL-RR-19 | 250 | 7A | Buffer | Pre-Viral | Fixed | 340 | 590 | 2100L X 2000 X 2600H | Praj and Aaruni

98 | PV 120 | 500 | 7A | Process Vessel | Pre-Viral | Fixed | 450 | 950 | 2200L X 2200 W X 2800H | Praj and Aaruni

99 | PV 121 | 500 | 7A | Process Vessel | Pre-Viral | Fixed | 450 | 950 | 2200L X 2200 W X 2800H | Praj and Aaruni

100 | PV 122 | 250 | 7A | Process Vessel | Pre-Viral | Fixed | 340 | 590 | 2100L X 2000 X 2600H | Praj and Aaruni

101 | PV 122.S | 250 | 7A | Process Vessel | Pre-Viral | Fixed | 340 | 590 | 2100L X 2000 X 2600H | Praj and Aaruni

102 | PV 123 | 100 | 7A | Process Vessel | Pre-Viral | Mobile | 220 | 370 | 1800L X 1900W X 2800H | Praj and Aaruni

103 | PV 124 | 100 | 7A | Process Vessel | Pre-Viral | Mobile | 220 | 370 | 1800L X 1900W X 2800H | Praj and Aaruni

104 | AL-RR-23 | 250 | 7A | Buffer | Pre-Viral | Fixed | 340 | 590 | 2100L X 2000 X 2600H | Praj and Aaruni

105 | AL-RR-19 | 100 | 7A | Buffer | Pre-Viral | Mobile | 220 | 370 | 1800L X 1900W X 2800H | Praj and Aaruni

106 | AL-RR-21 | NALGENE | 7A | Bottle | Pre-Viral | NA | 50 | - | 500L X 500 W x 1000 H | Praj and Aaruni

107 | Akta- 1356 L/ Hr | NA | 7A | Equipment | Pre-Viral | Fixed | 2000 | - | 2200L x 3200 W x 3000H | TruKing, Cytiva, Aaruni

108 | Chromatography Column-1200 X 500 | NA | 7A | Equipment | Pre-Viral | Fixed | 3000 | 5000 | 2200L x 3200 W x 3000H | TruKing, Cytiva, Aaruni

109 | TFF | NA | 7A | Equipment | Pre-Viral | Fixed | 2000 | - | 2500L X 2000 W X 2700H | Praj, Cytiva, Sartorius, Aaruni

110 | Mobile CIP | NA | 7A | Equipment | Pre-Viral | Mobile | 150 | 200 | 2000L X 2000 X 2000H | Praj & DDE

111 | PV 125 | 100 | 8N | Process Vessel | Pre-Viral | Mobile | 220 | 370 | 1800L X 1900W X 2800H | Praj and Aaruni

112 | PV 126 | 100 | 8N | Process Vessel | Pre-Viral | Mobile | 220 | 370 | 1800L X 1900W X 2800H | Praj and Aaruni

113 | PV 125.1 | 100 | 8N | Process Vessel | Pre-Viral | Mobile | 220 | 370 | 1800L X 1900W X 2800H | Praj and Aaruni

114 | FN-RR-02 | 300 | 8N | Buffer | Pre-Viral | Fixed | 390 | 690 | 2100L X 2000 X 2700H | Praj and Aaruni

115 | PV 127 | 100 | 8N | Process Vessel | Pre-Viral | Mobile | 220 | 370 | 1800L X 1900W X 2800H | Praj and Aaruni

116 | PV 127.1 | 100 | 8N | Process Vessel | Pre-Viral | Mobile | 220 | 370 | 1800L X 1900W X 2800H | Praj and Aaruni

117 | Chromatography Column-300 X 500 | NA | 8N | Equipment | Pre-Viral | Fixed | 70 | 150 | 500L X 500 W x 1000 H | Truking, Cytiva, Aaruni

118 | Akta- 70 L/ Hr | NA | 8N | Equipment | Pre-Viral | Fixed | 1200 | - | 2200L x 3200 W x 3000H | Truking, Cytiva, Aaruni

119 | Mobile CIP | NA | 8N | Equipment | Pre-Viral | Mobile | 150 | 200 | 2000L X 2000 X 2000H | Praj & DDE

120 | Centralised CIP - 2 tank system | TBD | CIP/SIP Room | Equipment | Pre-Viral | Fixed | 1000 | - | 2200L x 3200 W x 3000H | Praj and Aaruni

119 | Decontamination Autoclave | TBD | Decontamination Room | Equipment | Pre-Viral | Fixed | 300 | - | 500L x 500 W x 1200 H | Pharma Lab, Indu IonPure

120 | Dynamic Pass Box (DPB) | NA | Decontamination Room | Cleanroom Equipment | Pre-Viral | Fixed | 50 | - | 1200 L x 1000 W x 2100 H | Clean air, Fabtech

121 | Dynamic Pass Box (DPB) | NA | Decontamination Room | Cleanroom Equipment | Pre-Viral | Fixed | 50 | - | 1200 L x 1000 W x 2100 H | Clean air, Fabtech

122 | Dynamic Pass Box (DPB) | NA | Sterilization Room | Cleanroom Equipment | Pre-Viral | Fixed | 50 | - | 1200 L x 1000 W x 2100 H | Clean air, Fabtech

123 | Sterilization Autoclave | TBD | Sterilization Room | Equipment | Pre-Viral | Fixed | 1600 | - | 1200 L x 1000 W x 2100 H | Pharma Lab, Indu IonPure

126 | Ceiling LAF | NA | Cooling zone | Equipment | Pre-Viral | Fixed Under Ceiling | 150 | - | 2500 L x 2000 W x 1000 H | Clean air, Fabtech

127 | Dynamic Pass Box (DPB) | NA | Corridor | Equipment | Pre-Viral | Fixed | 50 | | 1200 L x 1000 W x 2100 H | Clean air, Fabtech

128 | PV 114 | 250 | 6I/11F | Process Vessel | Post Viral | Fixed | 340 | 590 | 2100L X 2000 X 2600H | Praj and Aaruni

129 | PV 115 | 1000 | 6I/11F | Process Vessel | Post Viral | Fixed | 650 | 1650 | 2700L X 2700W X 3200H | Praj and Aaruni

130 | PV 116 | 1000 | 6I/11F | Process Vessel | Post Viral | Fixed | 650 | 1650 | 2700L X 2700W X 3200H | Praj and Aaruni

131 | IG-RR-09 | 100 | 6I/11F | Buffer | Post Viral | Mobile | 220 | 370 | 1800L X 1900W X 2800H | Praj and Aaruni

132 | IG-RR-14 | 500 | 6I/11F | Buffer | Post Viral | Fixed | 450 | 950 | 2200L X 2200 W X 2800H | Praj and Aaruni

133 | IG-RR-16 | 500 | 6I/11F | Buffer | Post Viral | Fixed | 450 | 950 | 2200L X 2200 W X 2800H | Praj and Aaruni

134 | IG-RR-18 | 500 | 6I/11F | Buffer | Post Viral | Fixed | 450 | 950 | 2200L X 2200 W X 2800H | Praj and Aaruni

135 | IG-RR-22 | 250 | 6I/11F | Buffer | Post Viral | Fixed | 340 | 590 | 2100L X 2000 X 2600H | Praj and Aaruni

136 | PV 117 | 150 | 6I/11F | Process Vessel | Post Viral | Mobile | 250 | 430 | 1900L X 2000W X 2800H | Praj and Aaruni

137 | IG-RR-26 | 150 | 6I/11F | Buffer | Post Viral | Mobile | 250 | 430 | 1900L X 2000W X 2800H | Praj and Aaruni

138 | IG-RR-28 | 500 | 6I/11F | Buffer | Post Viral | Fixed | 450 | 950 | 2200L X 2200 W X 2800H | Praj and Aaruni

139 | IG-RR-23 | 150 | 6I/11F | Buffer | Post Viral | Mobile | 250 | 430 | 1900L X 2000W X 2800H | Praj and Aaruni

140 | IG-RR-24 | 150 | 6I/11F | Buffer | Post Viral | Mobile | 250 | 430 | 1900L X 2000W X 2800H | Praj and Aaruni

141 | FG-RR-09 | 1000 | 6I/11F | Buffer | Post Viral | Fixed | 650 | 1650 | 2700L X 2700W X 3200H | Praj and Aaruni

142 | FG-RR-10 | 2000 | 6I/11F | Buffer | Post Viral | Fixed | 1200 | 3200 | 3600L X 3600W X 4300H | Praj and Aaruni

143 | FG-RR-11 | 500 | 6I/11F | Buffer | Post Viral | Fixed | 450 | 950 | 2200L X 2200 W X 2800H | Praj and Aaruni

143 | FG-RR-13 | 100 | 6I/11F | Buffer | Post Viral | Mobile | 220 | 370 | 2100L X 2000 X 2600H | Praj and Aaruni

144 | FG-RR-14 | 250 | 6I/11F | Buffer | Post Viral | Fixed | 340 | 590 | 2100L X 2000 X 2600H | Praj and Aaruni

145 | Akta-1" -667 L/Hr | NA | 6I/11F | Equipment | Post Viral | Fixed | 2000 | - | 2200L x 3200 W x 3000H | Truking, Cytiva, Aaruni

146 | Chromatography Column 1000X 500 | NA | 6I/11F | Equipment | Post Viral | Fixed | 400 | 600 | 1500L X 1500 W X 2700H | Truking, Cytiva, Aaruni

147 | TFF | NA | 6I/11F | Equipment | Post Viral | Fixed | 2000 | - | 2500L X 2000 W X 2700H | Praj, Cytiva, Sartorius, Aaruni

148 | Nano Filter | NA | 6I/11F | Single use | Post Viral | Fixed | 20 | - | 500L x 500 W x 1200 H | Planova

149 | Mobile CIP | NA | 6I/11F | Equipment | Post Viral | Mobile | 150 | 200 | 2000L X 2000 X 2000H | Praj & DDE

150 | FN-RR-15 | 500 | 9N | Buffer | Post Viral | Fixed | 450 | 950 | 2200L X 2200 W X 2800H | Praj and Aaruni

151 | FN-RR-18 | 100 | 9N | Buffer | Post Viral | Mobile | 220 | 370 | 1800L X 1900W X 2800H | Praj and Aaruni

152 | FN-RR-17 | 100 | 9N | Buffer | Post Viral | Mobile | 220 | 370 | 1800L X 1900W X 2800H | Praj and Aaruni

153 | Akta-1" -30 L/Hr | NA | 9N | Equipment | Post Viral | Fixed | 200 | - | 2200L x 1800 W x 2000H | Truking, Cytiva, Aaruni

154 | Chromatography Column-300 X 750 | NA | 9N | Equipment | Post Viral | Fixed | 80 | 200 | 500L x 500 W x 2000 H | Truking, Cytiva, Aaruni

155 | Mobile CIP | NA | 9N | Equipment | Post Viral | Mobile | 150 | 200 | 2000L X 2000 X 1800H | Praj & DDE

156 | PV 131 | 300 | 10T/13E | Process Vessel | Post Viral | Fixed | 390 | 690 | 2100L X 2000 X 2700H | Praj and Aaruni

157 | PV 131.1 | 200 | 10T/13E | Process Vessel | Post Viral | Mobile | 300 | 520 | 2100L X 2000 X 2600H | Praj and Aaruni

158 | PV 132 | 250 | 10T/13E | Process Vessel | Post Viral | Fixed | 340 | 590 | 2100L X 2000 X 2600H | Praj and Aaruni

159 | TH-RR-05 | 50 | 10T/13E | Buffer | Post Viral | Mobile | 280 | - | 1800L X 1800W X 2800H | Praj and Aaruni

160 | TH-RR-02 | 100 | 10T/13E | Buffer | Post Viral | Mobile | 220 | 370 | 1800L X 1900W X 2800H | Praj and Aaruni

161 | TH-RR-02 | 100 | 10T/13E | Buffer | Post Viral | Mobile | 220 | 370 | 1800L X 1900W X 2800H | Praj and Aaruni

162 | TH-RR-03 | 200 | 10T/13E | Buffer | Post Viral | Mobile | 300 | 520 | 2100L X 2000 X 2600H | Praj and Aaruni

163 | NALGENE | 50 | 10T/13E | Bottle | Post Viral | NA | 280 | | 500L x 500 W x 1200 H | Praj and Aaruni

164 | PV 132 | 250 | 10T/13E | Process Vessel | Post Viral | Fixed | 340 | 590 | 2100L X 2000 X 2600H | Praj and Aaruni

165 | PV 128 | 300 | 10T/13E | Process Vessel | Post Viral | Fixed | 390 | 690 | 2100L X 2000 X 2700H | Praj and Aaruni

166 | FE-RR-09 | 250 | 10T/13E | Buffer | Post Viral | Fixed | 340 | 590 | 2100L X 2000 X 2600H | Praj and Aaruni

167 | FE-RR-09 | 250 | 10T/13E | Buffer | Post Viral | Fixed | 340 | 590 | 2100L X 2000 X 2600H | Praj and Aaruni

168 | Akta-1" -160 L/Hr | NA | 10T/13E | Equipment | Post Viral | Fixed | 200 | - | 2200L x 1800 W x 2000H | Truking, Cytiva, Aaruni

169 | Chromatography Column-450 X 500 | NA | 10T/13E | Equipment | Post Viral | Fixed | 100 | 250 | 500L x 500 W x 2000 H | Truking, Cytiva, Aaruni

170 | Mobile CIP | NA | 10T/13E | Equipment | Post Viral | Mobile | 150 | 200 | 2000L X 2000 X 2000H | Praj & DDE

171 | Centralised CIP - 2 tank system | TBD | CIP/SIP Room | Equipment | Post Viral | Fixed | 1000 | - | 2200L x 3200 W x 3000H | Praj and Aaruni

173 | Decontamination Autoclave | TBD | Decontamination Room | Equipment | Post Viral | Fixed | 300 | - | 500L x 500 W x 1200 H | Pharma Lab, Indu IonPure

174 | Dynamic Pass Box (DPB) | NA | Decontamination Room | Cleanroom Equipment | Post Viral | Fixed | 50 | - | 1200 L x 1000 W x 2100 H | Clean air, Fabtech

175 | Dynamic Pass Box (DPB) | NA | Decontamination Room | Cleanroom Equipment | Post Viral | Fixed | 50 | - | 1200 L x 1000 W x 2100 H | Clean air, Fabtech

176 | Dynamic Pass Box (DPB) | NA | Sterilization Room | Cleanroom Equipment | Post Viral | Fixed | 50 | - | 1200 L x 1000 W x 2100 H | Clean air, Fabtech

177 | Sterilization Autoclave | TBD | Sterilization Room | Equipment | Post Viral | Fixed | 1600 | - | 1200 L x 1000 W x 2100 H | Pharma Lab, Indu IonPure

178 | Ceiling LAF | NA | Cooling zone | Equipment | Post Viral | Fixed Under Ceiling | 150 | - | 2500 L x 2000 W x 1000 H | Clean air, Fabtech

179 | Buffer Preparation | 1500 | Buffer Preparation Room | Process Vessel | Post Viral | TBD | 900 | 2400 | 3200L X 3500 W X 4200H | Praj and Aaruni

180 | Buffer Preparation | 1000 | Buffer Preparation Room | Process Vessel | Post Viral | TBD | 650 | 1650 | 2700L X 2700W X 3200H | Praj and Aaruni

181 | Pass box | NA | Buffer Preparation Room | Equipment | Post Viral | NA | 50 | - | 1200 L x 1000 W x 2100 H | Clean air, Fabtech

182 | Pass box | NA | Chemical Day Store | Equipment | Post Viral | NA | 50 | - | 1200 L x 1000 W x 2100 H | Clean air, Fabtech

[SHEET: Fill Finish(FF)]

Sr. No. | Vessel Tag | Room/Stage (As per Layout) | Type | Fill Finish | Fixed / Mobile | Gross Weight in Kg (Empty Vessel) | Dimension (L mm x W mm x H mm) | Make

1 | DPB | De-Contanning Room | Equipment | Plasma | Fixed | 50 | 1200 L x 1000 W x 2100 H | Clean air, Fabtech

2 | Vial washing | Vial Washing Room | Vial Washing Machine | Plasma | Fixed | 4000 | 3500 L X 1800 W X 2100 H | NKP, Harikrushna

3 | DPB | Vial Washing Room | Equipment | Plasma | Fixed | 50 | 1200 L x 1000 W x 2100 H | Clean air, Fabtech

4 | Autoclave | Material Prep & Wash | Equipment | Plasma | Fixed | 2500 | 1200 L x 1000 W x 2100 H | Pharma Lab, Indu IonPure

5 | DPB | Material Prep & Wash | Equipment | Plasma | Fixed | 50 | 1200 L x 1000 W x 2100 H | Clean air, Fabtech

6 | Autoclave | Decontamination | Equipment | Plasma | Fixed | 300 | 1200 L x 1000 W x 2100 H | Pharma Lab, Indu IonPure

7 | Washing machine | Garment Wash | Washing Machine | Plasma | Mobile | 80 | 1200 L x 1300 W x 1700 H | IFB

8 | Dryer | Garment Wash | Drying machine | Plasma | Mobile | 80 | 1800 L x 1500 W x 2600 H | IFB

9 | Ceiling LAF | Unload Area | Equipment | Plasma | Fixed under Ceiling | NA | 2500 L x 2000 W x 1000 H | Clean air, Fabtech

10 | Vessel | Formulation | Formulation Vessel | Plasma | Fixed | 500 | 2200L X 2200 W X 2800H | Praj and Aaruni

11 | Vessel | Formulation | Formulation Vessel | Plasma | Fixed | 500 | 2200L X 2200 W X 2800H | Praj and Aaruni

12 | Ceiling LAF | Formulation | Equipment | Plasma | Fixed under Ceiling | NA | 2500 L x 2000 W x 1000 H | Clean air, Fabtech

13 | Filling Machine | Filling Area | Filling machine | Plasma | Fixed | 2200 | 3200 L X 1700 W X 2300 H | NKP, Harikrushna

14 | Ceiling LAF | Filling Area | Equipment | Plasma | Fixed under Ceiling | NA | 2500 L x 2000 W x 1000 H | Clean air, Fabtech

15 | Ceiling LAF | Capping Room | Equipment | Plasma | Fixed under Ceiling | NA | 2500 L x 2000 W x 1000 H | Clean air, Fabtech

16 | Capping Machine | Capping Room | Capping Machine | Plasma | Fixed | 600 | 2200 L X 1500 W X 2000 H | NKP, Harikrushna

17 | Turn Table | Capping Room | SS Turn Table | Plasma | Fixed | 300 | 1200 L X 1300 W X 900 H | NKP, Harikrushna

18 | Swing Conveyor | Capping Room | Swing Conveyor With Vial Collection Tray | Plasma | Fixed | 350 | 2500 L X 1200 W X 900 H | NKP, Harikrushna

19 | Lyo Machine | Lyo Loading Side | Lyp machine | Plasma | Fixed | 7500 | 4200 L X 2200 W X 2400 H | LSI

20 | DPB | Buffer Room | Equipment | Plasma | Fixed | 50 | 1200 L x 1000 W x 2100 H | Clean air, Fabtech

21 | Visual Inspection Booth | Visual Inspection Room | Visual Inspection Booth | Plasma | Fixed | 425 | 3000 L X 1800 W X 2400 H | NKP, Harikrushna

22 | Labelling machine | Secondary Packing Area | Labeling Machine | Plasma | Fixed | 600 | 3000 L X 1800 W X 2000 H | NKP, Harikrushna
23 | Cartoning machine | Secondary Packing Area | Cartoning Machine | Plasma | Fixed | 200 | 6000 L X 2000 W X 2100 H | NKP, Harikrushna
24 | PW Generation(MGF, Softener, Soft Water Storage tank, RO and RO water storage tank) | Water System & Engineering Store | Water System & Utilities | Utilities equipment | Fixed | 500 | | Praj, CN water
25 | PW storage 15,000L | Water System & Engineering Store | Water System & Utilities | Utilities equipment | Fixed | 17500 | | Praj, CN water
26 | WFI Generation and storage 5000L | Water System & Engineering Store | Water System & Utilities | Utilities equipment | Fixed | 6500 | | Praj, CN water
27 | PSG System | Water System & Engineering Store | Water System & Utilities | Utilities equipment | Fixed | 350 | | Praj, CN water
28 | Compressed Air System 300 CFM | Water System & Engineering Store | Water System & Utilities | Utilities equipment | Fixed | 750 | | Atlas copco
29 | AHUs | Service Area | HVAC | Utilities equipment | Fixed | | | Clean air, Fabtech
30 | Electrical Control Panel | Service Area | Electrical | Utilities equipment | Fixed | | | Clean air, Fabtech

Reagent Details:
Preparation of PF-RR-10 working buffer from Stock buffer (PF-RR-31):
Calculations: For the Preparation of 20mM Sodium Citrate pH 7.0 + 0.06M Sodium Chloride.
Stock Buffer (PF-RR-31):0.2M Sodium Citrate pH 7.0 + 0.6M Sodium Chloride
Formula:.
 $C1 \times V1 = C2 \times V2$
C1 = concentration of stock
V1 = volume of stock you need to take
C2 = desired (working) concentration
V2 = final total volume of working solution
Applying the formula to prepare the working buffer:
 $C1 \times V1 = C2 \times V2$
C1 = 0.2M Sodium Citrate pH 7.0 + 0.6M Sodium Chloride.
V1 = volume of stock you need to take _____ L.
C2 = desired (working) concentration 20mM Sodium Citrate, pH 7.0 + 0.06M Sodium Chloride.
V2 = final total volume of working solution _____ L.
 $V1 = (C2 \times V2) / C1$.
C2=0.02M ; V2= 1.0 L; C1=0.2M
 $V1 = (0.02 \times 1.0) / 0.2$
V1 = 0.1L
Take 0.1 L (PF-RR-31) Stock buffer solution and dilute with water, and make up to 1L.
Check and record the details:
Preparation Details:
Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry
Date
Sodium Citrate | | | |
Citric Acid | | | |
Sodium Chloride | | | |
Stock buffer (PF-RR-31) taken volume(L) | pH (7.0 ± 0.1) | Conductivity (10± 0.5ms/cm) | Storage
Condition | Final volume prepared
| | |
Date of Preparation | Prepared By | Checked By
| |

51000
G
FD FD FD FD
UP UP

UP UP

F

E

D

SHAFT SHAFT

72860 FD FD FD FD
UP UP

UP UP

C

B

A

UP UP

UP UP
FD FD FD FD
O
FD FD D1 D1 D1 D1 D1 D1 D1 D1 D1 D1 FD FD
GENTS VISITOR
UP FEMALE LADIES TOILET DOCUMENT OFFICE 2 UP
WAITING ROOM LOCKER LOCKER LOCKER ROOM 4.63x5.30
5.30x4.00 GENTS TOILET ROOM 5.90x5.20 2.50x5.30
ROOM ROOM
5.90x5.20 D1 D1 D1
4.57x5.30 4.57x5.30 4.57x5.30
UP UP
D4 D4 D4 D4 D4 D4 D4 D4

13.2x1.56
6D

4.05x2.80 CORRIDOR
33.4x2.80

4D D4 4D D4

JANITOR
D4

D4

4D

4D

D4

D4

D1
ROOM
PRIMARY VISITOR
3.42x2.37 N
RECEPTION FEMALE PRIMARY FEMALE AIR LOCK AIR LOCK VISITOR CHANGE 1
OFFICE 1
PRIMARY MALE MALE CHANGE 1 4.63x5.00
4.05x2.80 CHANGE 3.00x5.00
CHANGE 1 CHANGE FEMALE VISITOR CHANGE 2.60x5.00
3.10x5.00 3.10x5.00 3.64x5.00 3.05x5.00 2.70x5.00
4.53x8.15 AIR LOCK
3.80x5.00 2.50x5.00
MALE
4.77x5.00
LIFT
8D

D4

D4 4D D4

6D

DE
CORRIDOR
47.8x2.00
D4

DPB
DPB DPB WASTE OUT
LAF DHS
This place can 3 WAY
6D

6D

CHEMICAL DAY STORE
be taken for
chemical day
8.10X4.00
room and door COLD ROOM DECONTAMINATION WASHING STERILIZATION

AUTOCLAVE

LAF
may be BUFFER TANK STORAGE-2 CIP / SIP 4.40x8.10
3.52x5.62 3.11x5.62 4.01x5.62
5.60x8.10
6D

6.80x8.10
shifted to corridor
PREPARATION COOL ZONE
4.29x8.10 2.76x8.10 M
4D

4D

DPB D4 D4
DPB
GS+HH

GS+HH

8.30x1.80
D4
A/L GOWN A/L
2.22X3.40 3.47X3.40 2.20X3.40 LENAP LORTNOC LENAP LORTNOC

WASTE A/L A/L
4.13x2.40 4.01x2.40
D4

IN
D4

2.50x2.40
D4

6D 6D 6D D4 D4 D4
D4
4D

CORRIDOR
2.57x66.0 A/L DE GOWN A/L
DE
2.22X3.40 3.47X3.40 2.20X3.40 CORRIDOR
37.0x2.80
4D

4D

D3
LAF
D6 D6 D6 D6
DPB AKTA-1" TFF
4D

4D

NANO
FILTER MH

JANITOR
1.80x2.00
A/L GOWN A/L 1000 L
2.22X4.32 3.47X4.32 2.20X4.32
BULK FILTRATION
LAF

6.90X6.00
D4

500 L
D4

L
LAF

MH

STAGE 9 N
MH

500 L
4D

50 L
MH

STAGE 13 E
D4

LAF SL
MH

5.10X14.3 CHROMO SL
250 L
MH

COLUMN
DE 600 L 600 L
A/L
2.22X3.89
GOWN A/L
500 L
STAGE 10 T/ MH

3.47X3.89 2.20X3.89
STAGE 11 F
MH

213630 MH SL
600 L 6.25X2.05 250 L
4D

4D

200 L
MH
AKTA-1"

STAGE 6 (I) 8.10X12.2
MH

COLUMN PACKING 2000 L 14.7X16.7 250 L MH

250 L
D4

MH

MH

AND SKIDS
MH

CORRIDOR
200 L
1.80X49.6 6.90X10.6 250 L
COOL ZONE 150 L
MH MH

MH
1000 L MH
100 L MH

8.10x5.53 150 L
500 L
MH

300 L
CORRIDOR
MH
2.57X67.9
MH

100 L MH

1000 L
MH

100 L
150 L 300 L
LAF K
DPB LAF
MH

MH

CHROMO MH

AKTA-1"
100 L COLUMN

DHS
AUTOCLAVE CHROMO
D6

COLUMN
500 L 500 L MH
MH MH

250 L 250 L 150 L 100 L
6D

CHROMO
SHAFT A/L MH MH
MH MH
MH

MH
COLUMN SHAFT
CLAF

1.50x5.73
STERILIZATION 2000 L
2500 L
2500 L
D6

FD FD 6.50x5.85 FD FD
PRE - VIRUS CONTROLLED CORRIDOR
6D

1000 L
UP UP
D4

TFF

MH

D4
COLUMN PACKING
DPB
UP AND SKIDS MH

UP
WASHING 100 L
8.00X7.86
AKTA-1"

A/L 6.50x3.00 STAGE 7A
AKTA-1"

MH

1.50x4.42 500 L
D4

300 L 13.6X15.1
D4

MH

STAGE 5 (I) A
D6

DPB J
WASTE IN
4D

500 L
250 L
11.1X18.0
TFF

MH

MH
2.00X28.8

DECONTAMINATION
4D

500 L
250 L
OUT 6.50x5.00 MH
MH
MH

WASTE
D6

1.50x3.54 500 L

DPB STAGE 5(I)B 1500 L 250 L
MH
MH

100 L
8.00X7.20 MH

250 L 250 L 250 L
SL
600 L 500 L
100 L 100 L
MH

MH MH MH 100 L MH
MH

100 L MH MH MH

MH

1500 L
CIP / SIP SL
1000 L 100 L
MH MH

MH MH MH

8.10x5.00
MH

500 L 500 L 1000 L 1000 L 1000 L
MH

MH

MH
6D

MH

CHROMO
50 L COLUMN
GS+HH

GS+HH

DPB 2500 L
LENAP LORTNOC LENAP LORTNOC
WASTE OUT 1500 L
1500 L 1500 L
SL I
MH
250 L 250 L 250 L 250 L
MH

MH MH MH MH
6D

BAG CUTTING &
TANK STORAGE POOLING
MH
MH MH

MH

MH MH
STAGE 4 (F)
150 L 500 L 500 L
8.00X7.80
6D

1000 L
8.10x6.58 1500 L 1500 L 13.6X13.5 MH

500 L
D6

CHROMO
COLUMN

250 L
300 L 300 L STAGE 12 (E) MH

STAGE 1 (F)/
TFF
D6

AKTA-1"

6D
MH MH

CLAF

11.1X8.21
1000 L

2500 L
SL

MH

THAWING PLASMA
2000 L 2000 L
8.00X3.22
6D

BUFFER
AKTA-1"

MH

500 L 250 L 300 L 300 L 300 L
250 L 250 L
PREPARATION
MH MH

CHROMO CHROMO
COLUMN
MH MH MH MH

COLUMN
MH MH MH

DPB D6 D6 D6

8.10x6.00 H
140770
6D

DE

DPB CORRIDOR
35.1X2.80
D6 MH MH D6 D6
6D

MH MH MH MH

CHROMO MH
MH
MH MH MH MH

COLUMN
100 L 100 L 100 L 100 L MH

100 L 100 L CENTRIFUGE
200 L 250 L 300 L 300 L 500 L 500 L
1500 L

PLASMA DAY STAGE 3 (F)
AKTA-1"

STORE 4D
STAGE 2 (F) 12.8X8.68
AKTA-1"

STAGE 8 N
8.10x8.77 6.17X8.68
15.8X8.68
CHROMO MH

JANITOR COLUMN

5.20x1.80 SL
600 L 1500 L 2000 L
1500 L 1500 L 1500 L
300 L
G
DPB 300 L MH MH MH
MH
MH

100 L 100 L 100 L CHROMO CHROMO
MH
TFF MH

COLUMN COLUMN AKTA-1"
MH MH MH

EMER.GLASS EMER.GLASS

CORRIDOR
50.5x1.80

PLASMA STORAGE PLASMA STORAGE PLASMA STORAGE PLASMA STORAGE PLASMA STORAGE
~ 1000 L ~ 1000 L ~ 1000 L ~ 1000 L ~ 1000 L
~ 70 KL GV ~ 70 KL GV ~ 70 KL GV ~ 70 KL GV ~ 70 KL GV
7.10x9.62m 9.95x9.62m 9.95x9.62m 9.95x9.62m 8.15x9.62m

F
800 800 800 800

SHAFT SHAFT
D3 D3 D3 D3 D3

CORRIDOR
45.3x1.80
FD FD FD FD
UP UP

UP UP

CORRIDOR
2.57x37.0 CORRIDOR
2.57x35.1

PLASMA STORAGE PLASMA STORAGE PLASMA STORAGE PLASMA STORAGE PLASMA STORAGE
~ 1000 L
~ 70 KL GV
~ 1000 L
~ 70 KL GV
~ 1000 L
~ 70 KL GV
~ 1000 L
~ 70 KL GV
~ 1000 L
~ 70 KL GV
E
7.10x9.62m 9.95x9.62m 9.95x9.62m 9.95x9.62m 8.15x9.62m

800

800 800 800 800
1125

D3 D3 D3 D3 D3

CORRIDOR
45.3x1.80

D
850 800 800 800 800

PLASMA STORAGE PLASMA STORAGE PLASMA STORAGE PLASMA STORAGE PLASMA STORAGE
~ 1000 L ~ 1000 L ~ 1000 L ~ 1000 L ~ 1000 L
~ 70 KL GV ~ 70 KL GV ~ 70 KL GV ~ 70 KL GV ~ 70 KL GV
7.10x11.5m 9.95x11.5m 9.95x11.5m 9.95x11.5m 8.15x11.5m
699

D3 D3 D3 D3 D3

CORRIDOR
47.9x1.80
D6
C
29.5 X 2.58
CORRIDOR

D3 D6 COLD ROOM -4 COLD ROOM -3 COLD ROOM -2 COLD ROOM -1 ICE PACK
DPB RLAF
4.45X7.58 2.91X7.58
D3

AIR LOCK-1 4.45X7.58 4.45X7.58 4.45X7.58
1.50X 1.93 RM QUARANTINE WASHING PLASMA
BAG BAR CODING, JANITOR
AREA QUARANTINE
MINI POOLS
6D

1.45X2.90
2.30X2.90
2.60X2.90
DISPENCING ROOM
3D
3.48X 3.95
CHANGE ROOM D3 D3 D3
1.50 X 1.93
3D

DPB 6.21X5.77
3D

5.00X2.95 D3
11.5X3.30 SD SD SD SD

6D

D3
CORRIDOR
APPROVED RM AIR LOCK-1 DPB 1.95X11.3
4.63X4.00
2.00X1.21 D3
AIR LOCK-1 DPB
3D D3 D6 STORE 13.9X3.84
3.00X2.60
9.72X7.22 1.87X1.40 OFFICE 9.22X2.45
3D
CHANGE ROOM
2.00X2.08 3D
2.23X6.25
4.75X2.32
B
CHANGE ROOM
1.87X1.40 STORE CHANGE
DE-DUSTING REJECT SEROLOGY PLASMA ROOM
TUNNEL 4.70X2.37
ROOM TESTING RECEIPT 3D D3 1.51X1.23
D3

PLASMA
D3 D3 SAMPLING
1.76X 2.90
2.95X4.95 3.00X 4.95 2.40X 4.95
RM
SAMPLING 3.68X 2.05 RECALLED REJECT
RM FG DISCPATCH
1.07X3.39
RECEIPT GOODS ROOM FG IN
3.16X2.03 2.30X2.35 2.30X 2.35
3.00X2.77
RLAF RLAF 6D D6 D6
D3 D3

D6 D3

UP UP
CORRIDOR
39.8X6.84
UP UP
FD FD FD FD
A
SPM RM PLASMA
RECEIVING FG
RECEIVING RECEIVING
BAY DISCPATCH
BAY BAY

CANOPY ABOVE
40' CONTAINER YARD
CONTAINER

CONTAINER

CONTAINER

CONTAINER
12.19x2.44x2.59
12.19x2.44x2.59

6 5 4 3 2 1
12.19x2.44x2.59

12.19x2.44x2.59

CLIENT :- LIFENITY-LARGE VOLUME PARENTERALS

DESIGN &
PROJECT BY :-BIOBRIDGE HEALTHCARE SOLUTION PVT.LTD.
PREPARED

GROUND FLOOR
A.K
ADDRESS :-13, Rachna Blossom, Jagdishnagar,
Aundh, Pune -411007, Maharastra, India. CHECKED G.K.

DATE 19-03-2026
TITLE- CONCEPTUAL LAYOUT-GF REVISION R0
SCALE 1:1

SIZE A0 DWG.NO 02

6D UP

FIRE EXIT
CORRIDOR
50.3x1.97

D1
D1 D1 D1 D1

2000
LADIES TOILET D6 D1
7.77x2.74
LIFT

FIRE EXIT
1.20x7.17
GENTS LOCKER LOBBY
GENTS TOILET ROOM
7.77x2.74
OFFICE 1 VISITOR PRIMARY VISITOR VISITOR 4.17x2.70 1.70x3.92 6.77x11.77
2.65x5.15 AIR LOCK FEMALE CHANGE PRIMARY MALE
D1 D1 D1 D1
2.90x5.15
CHANGE
2.50x5.15
CHANGE
2.50x5.15
FEMALE AIR LOCK PRIMARY FEMALE
3.50x5.15 RECEPTION MALE CHANGE
CHANGE
MALE AIR LOCK OPEN TO ALL

D1
4.36x5.15 CHANGE 3.94x5.15 4.50x5.15
4.00x5.15
7.95x5.15
4.00x5.15
2.20x2.13 FLOORS

D1
FEMALE LOCKER
JANITOR
UP DRINK
ROOM
4.17x2.40 ROOM FIREMAN
Goods LIFT
WATER
3.90x1.22
LIFT
3.50X3.00
1D
2.70X2.50
CORRIDOR
50.9x1.80
D6
DPB
DPB DPB WASTE OUT
LAF DHS
CHEMICAL DAY STORE 3 WAY FIRE STAIR
LOBBY

6D

6D

DE
11.4X3.64 DECONTAMINATION WASHING STERILIZATION up
3.00X11.30 3.60x5.43

AUTOCLAVE
2.33X2.14 3.11x5.43
4.67x5.43

LAF
OPEN TO
FIRST BUFFER TANK STORAGE-2 CIP SIP COLD ROOM UP
UP FLOOR PREPARATION 6.74x7.50 5.60x7.50 5.87x7.50 COOL ONE
4.00x7.50 3.25x7.50
DPB DPB
AL GOWN AL 6D
2.10X3.40 3.34X3.40 3.34X3.40
WASTE IN AL AL

GS+HH

GS+HH
LENAP LORTNOC LENAP LORTNOC
3.65x2.00 4.13x2.00 3.60x2.00
6D 6D 6D

DE
AL DE GOWN AL
2.10X3.40 3.34X3.40 3.34X3.40 CORRIDOR
39.8x3.04
FIRE CONTROL
D3 ROOM UP
LAF
D6 D6 D6 D6
DPB AKTA-1 TFF
NANO 3.50X6.54
FILTER MH
JANITOR
1.80x2.00
AL GOWN AL
1000 L
3.34X4.76
2.10X4.76 3.34X4.76 BULK FILTERATION

LAF
6.54X7.29 500 L
12000.000

MH
SL MH

600 L

500 L

LAF
MH SL
600 L MH

MH
FIRE PUMP
DE MH

50 L ROOM
AL
AL GOWN LAF MH

2000 L 500 L
SL
STAGE 13 E 3.50X5.00
2.10X4.76 3.34X4.76 3.34X4.76 CHROMO
COLUMN 250 L
MH 600 L STAGE 10 T MH
UP
STAGE 9 N 7.18X13.5
STAGE 11 F
MH

7.25X15.6 200 L 5.33X2.05 250 L

AKTA-1
STAGE 6 I
MH

COLUMN PACKING 15.0X18.0 250 L
MH

250 L
1000 L
AND SKIDS
MH

200 L
MH

MH
2.80 M WIDE PASSAGE

LADIES TOILET 6.54X10.6 250 L
COOL ONE
MH MH

3.30 x 4.35 150 L 100 L MH

1000 L
9.00x5.53
MH

500 L

6.77 M WIDE CORRIDOR
MH

MH

300 L
WC WC WC 150 L MH

1.10X 1.10X 1.10X CHROMO 100 L MH

COLUMN
100 L
MH

1.70 1.70 1.70
300 L
LAF 150 L
DPB LAF
MH

CHROMO MH

AKTA-1
MH

100 L COLUMN

AUTOCLAVE
DHS

D6
WC WC WC 500 L 500 L CHROMO
250 L 250 L 150 L 100 L COLUMN

6D
1.10X 1.10X 1.10X MH
MH MH

1.70 1.70 1.70 CORRIDOR AL MH MH
MH MH
MH

MH

CORRIDOR CORRIDOR
2.57x71.7 1.50x6.85
3.50X55.7 1.80X71.9
STERILIZATION 2000 L

D6

CLAF
2500 L 2500 L
GENTS TOILET 7.40x6.85

6D
3.30 x 4.35

PRE - VIRUS CONTROLLED CORRIDOR
1000 L

TFF
MH

UP COLUMN PACKING
DPB
AND SKIDS MH

UP
AKTA-1

WASHING 7.64X8.86
100 L
AL
7.40x3.00

AKTA-1
1.50x4.42
STAGE 7A 500 L
MH

STAGE 5 I A 14.7X16.1
D6

DPB
WASTE IN MH 500 L
250 L
10.0X19.0
TFF

LOBBY MH

3.50X32.1
300 L
MH

6.77X11.77
DECONTAMINATION 250 L
500 L

6.00
7.40x5.00
MH

OUT MH
MH

WASTE
D6

1.50x3.54 500 L

DPB STAGE 5 I B 1500 L 250 L
MH
MH

7.64X7.20 100 L
MH

250 L 250 L 250 L
SL
600 L 500 L
100 L 100 L
6D

100 L
MH

MH
MH MH MH
MH

100 L
MH MH MH

CIP SIP
MH

1500 L
SL
UP
MH MH

MH MH MH

9.00x5.00 1000 L
MH

500 L 500 L
MH

MH

1000 L 1000 L 1000 L
UP
100 L 1500 L
1500 L
GS+HH

GS+HH

MH SL
LENAP LORTNOC LENAP LORTNOC

MH
2500 L
MH

1500 L LADIES TOILET
50 L
Pass. LIFT CHROMO MH
250 L 250 L 250 L 250 L 3.30 x 4.35
2.30X2.00 DPB COLUMN
6D

MH MH MH MH

WASTE OUT

BAG CUTTING WC WC WC

POOLING MH
MH MH

MH

MH MH
STAGE 4 F 1.10X
1.70
1.10X
1.70
1.10X

1.70

7.64X9.80 150 L 500 L 14.7X15.8
6D

500 L
SL 1000 L
TANK STORAGE 1000 L 1500 L 1500 L

9.00x8.58
D6

MH

WC WC WC
STAGE 12 E
MH

250 L 1.10X 1.10X 1.10X
MH

STAGE 1 F 1.70 1.70 1.70
CHROMO
500 L
COLUMN
10.0X10.6 250 L
300 L 300 L MH

GENTS TOILET
TFF

3.30 x 4.35
D6

AKTA-1

6D
MH MH

CLAF

2500 L

THAWING PLASMA 2000 L 2000 L
UP 7.64X3.61
AKTA-1

MH

DPB
500 L 250 L 250 L 300 L 300
300L L
BUFFER
6D

MH MH

CHROMO
MH
COLUMN MH MH MH
MHMH

CHROMO
PREPARATION D6 D6 D6
COLUMN

9.00x6.00
CORRIDOR
6D

DE
DPB 36.2X2.50
3.24 M WIDE PASSAGE

D6 D6 D6
6D

MH MH

MH MH MH MH

CHROMO MH
MH
MH MH MH MH

COLUMN
100 L 100 L 100 L 100 L MH

100 L 100 L CENTRIFUGE
200 L 250 L 300 L 300 L 500 L 500 L
1500 L

PLASMA DAY STAGE 3 F
STORE STAGE 2 F 12.4X10.5
STAGE 8 N
9.00x10.7
5.84X10.5
17.7X10.5
2.66x2.29
AKTA-1

AKTA-1

UP

JANITOR CHROMO MH
UP
3.50x5.40 COLUMN
3D

SL
600 L 1500 L 2000 L
LADIES TOILET 1500 L 1500 L 1500 L
3.30 x 4.35 300 L
300 L MH MH MH
MH
MH

100 L 100 L 100 L CHROMO CHROMO
DPB MH
TFF MH

COLUMN COLUMN AKTA-1
MH MH MH

WC WC WC
1.10X 1.10X 1.10X EMER.GLASS

1.70 1.70 1.70
LOBBY
6.77X11.00
6.00

PLASMA STORAGE
WC WC WC ~ 1000 L STAIRCASE
1.10X 1.10X 1.10X ~ 70 KL GV
15.9x13.2m
AND
1.70 1.70 1.70
PLASMA STORAGE
PLASMA STORAGE
~ 1000 L PLASMA STORAGE LIFT ONLY
~ 1000 L CORRIDOR
~ 70 KL GV
~ 70 KL GV ~ 1000 L
~ 70 KL GV 1.80x15.2 FOR
11.9x13.2m
9.10x13.2m
GENTS TOILET CORRIDOR
11.9x13.2m SECOND FL0OR
13200

3.30 x 4.35 2.57x17.0 (L.V.P.)

UP

UP

W

D3 D3 D3 D3 D3

CORRIDOR
49.1x1.80
D6
LOBBY CORRIDOR
12.47X6.77 28.0X1.80

D3 D6
Pass. LIFT W
DPB RLAF 2.30X2.00
D3

AIR LOCK-1
1.50 X 2.18 RM QUARANTINE BAG BAR CODING, WASHING PLASMA COLD ROOM -4 COLD ROOM -3 COLD ROOM -2 COLD ROOM
-1 ICE PACK
AREA JANITOR QUARANTINE
MINI POOLS
5.04X7.34 5.02X7.34 5.04X7.34 2.50X7.34
6D

2.60X2.67
2.30X2.67
1.45X2.67 5.02X7.34

DISPENCING ROOM
3.40 X 4.44
3D D3 D3
UP
D3

CHANGE ROOM
1.50 X 2.18 6.85X6.55
3D

5.15X2.72
CORRIDOR
11.6X4.30 1.80X12.0
D3
3D

SD SD SD SD

DPB
6D
ELECTRIC
D3 ROOM
AIR LOCK-1
2.13X1.21
DPB
D3
DPB
3D D3 D6 STORE 12.9X6.49 3.50X6.00
APPROVED RM 3.54X2.60
AIR LOCK-1
1.87X1.40 OFFICE 4.70X4.09
3D
LOBBY 10.1X7.54 2.23X6.50
5.15X4.49 3.00X3.60
3.00X11.53 CHANGE ROOM
2.13X1.80 CHANGE ROOM
3D

OPEN TO DE-DUSTING
1.87X1.40
REJECT SEROLOGY PLASMA
RECEIPT
D3

SECOND D3 TUNNEL
PLASMA ROOM TESTING 3D D3
STORE CHANGE
D3 SAMPLING 2.26X4.95 3.00X 4.95 2.55X 4.95 ROOM
FLOOR RM 1.76X 2.90
REJECT 2.95X2.88
SAMPLING
RM
3.68X 2.05 RECALLED
3.26X2.31 GOODS ROOM FG DISCPATCH
1.07X3.11 RECEIPT FG IN
3.54X2.77 2.30X2.35 2.30X 2.35
RLAF RLAF 6D D6 D6
D3 D3

D6 D3
SERVER
Goods LIFT CORRIDOR ROOM
3.50X3.00 53.9X3.17
6D

D6

3.50X6.57
DISPATCH BAY RECEIVING BAY RECEIVING BAY DISPATCH BAY

RS3
40' CONTAINER YARD
CONTAINER
CONTAINER

CONTAINER

CONTAINER
12.19x2.44x2.59
12.19x2.44x2.59

12.19x2.44x2.59

12.19x2.44x2.59

CLIENT :- LIFENITY-LARGE VOLUME PARENTERALS

DESIGN &
PROJECT BY :-BIOBRIDGE HEALTHCARE SOLUTION PVT.LTD.
PREPARED A.K
ADDRESS :-13, Rachna Blossom, Jagdishnagar,
Aundh, Pune -411007, Maharastra, India.

GROUND FLOOR
CHECKED G.K.

DATE 05-06-2026
TITLE- CONCEPTUAL LAYOUT-GF REVISION R0
SCALE 1:1

SIZE A0 DWG.NO 01

Reagent Details:

Preparation of PF-RR-31 Stock buffer :

Calculations: For the Preparation of 0.2M Sodium Citrate pH 7.0 + 0.6M Sodium Chloride.

Stock Buffer (PF-RR-09):0.4M Sodium Citrate pH 7.0.

Formula:

$$C1 \times V1 = C2 \times V2$$

C1 = concentration of stock

V1 = volume of stock you need to take

C2 = desired (working) concentration

V2 = final total volume of working solution

Applying the formula to prepare the working buffer:

$$C1 \times V1 = C2 \times V2$$

C1 = 0.4M Sodium Citrate pH 7.0 .

V1 = volume of stock you need to take _____ L.

C2 = desired (working) concentration 0.2M Sodium Citrate, pH 7.0.

V2 = final total volume of working solution _____ L.

$$V1 = (C2 \times V2) / C1$$

C2=0.2M; V2= 1.0 L; C1=0.4M

$$V1 = (0.2 \times 1.0) / 0.4$$

$$V1 = 0.5L$$

Take 0.5 L (PF-RR-09) Stock buffer solution and add 0.6M sodium Chloride.

Calculations: To prepare the 0.6M Sodium Chloride.

Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).

Calculation: To prepare 1 L of 0.6M Sodium Chloride (MW = 58.44 g/mol)

$$\text{Weight of solute(g)} = 0.6 \times 58.44 \times 1 = 35.06g$$

Preparation Method:

Weigh 35.06 g of Sodium Chloride and dissolve in water to make a 1 L solution.

Preparation for different volumes:

To prepare _____ L of 0.6 M Sodium Chloride (MW = 58.44 g/mol).

$$\text{Weight of solute(g)} = 0.6 \times 58.44 \times \text{_____ L} = \text{_____ g}$$

Weigh _____ gm of Sodium Chloride and dissolve in water to make a _____ L solution.

Preparation of Stock buffer (PF-RR-31):

Take 0.5 L (PF-RR-09) Stock buffer solution and add 0.6 M sodium chloride. Dissolve it, and then make up the volume to 1 L with water.

Check and record the details:

Preparation Details:

Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry

Date

Sodium Citrate | | | |

Citric Acid | | | |

Sodium Chloride | | | |

Stock buffer (PF-RR-09) taken volume(L) | pH (7.0 ± 0.1) | Conductivity (63.0 ±0.2ms/cm) | Storage

Condition | Final volume prepared

| | | RT |

Date of Preparation | Prepared By | Checked By

| |

Reagent Details:
Formula & Calculations:
Formula: Use the formula to calculate the molarity:
 $Molarity (M) = (Per\ cent\ by\ Weight / 100) * (Density\ in\ g/mL) * 1000 / Molar\ Mass.$
Example: Molarity of Glacial Acetic Acid
Molar Mass of CH_3COOH : ~60.05 g/mol
Density: ~1.05 g/mL
Purity: 100%
Acetic Acid (MW = 60.05 g/mol).
Calculation:
 $Molarity = (100 / 100) * 1.05\ g/mL * 1000\ mL/L / 60.05\ g/mol \approx 17.5\ M.$
Calculation: To prepare 1 L of 0.4 M.
Applying the formula to prepare the working buffer:
 $C1 \cdot V1 = C2 \cdot V2$
 $C1 = 17.5\ M$ acetic acid.
 $V1$ = volume of stock needed to take.
 $C2$ = desired (working) concentration 0.4 M acetic acid.
 $V2$ = final total volume of working solution for one liter.
 $V1 = (C2 \times V2) / C1.$
 $C2=0.4\ M; V2= 1.0\ L; C1=17.5\ M$
 $V1 = (0.4 \times 1.0) / 17.5$
 $V1 = 0.0228\ L$
Take 0.0228 L of the Stock solution of acetic acid and add it to a volumetric flask, making up the volume to 1 L.
To prepare various volumes:
 $V1$ = volume of stock needed to take _____ L.
 $C2$ = desired (working) concentration 0.4M acetic acid.
 $V2$ = final total volume of working solution _____ L.
 $V1 = (C2 \times V2) / 17.5.$
 $V1=$ _____ L.
Take $V1$ _____ L of the Stock solution of acetic acid and add it to a volumetric flask, making up the final volume _____ L.
Check and record the details:
Preparation Details:
Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry Date
|||
Final volume prepared | Measured Quantity | pH (2.6±0.1) | Conductivity (1.6±0.1ms/cm) | Storage Condition | Reagent Record
||| RT |
Date of Preparation | Prepared By | Checked By
||

1
FOR OFFICE USE 2

AD
BLOCK A = 68.00 X 218.05 = 14827.40

WI DE RO
DEDUCTIONS STAIRCASE SHAFT

60.00 M.
SHAFT 1 = 1.40 X 3.30 = 4.62
6 SHAFTS = 4.62 X 6 = 27.72

DEDUCTIONS TOILET SHAFT
SHAFT 1 = 3.60 X 2.00 = 7.20 ROAD
WIDE
60.00 M.

AD
60.
5 SHAFTS = 7.20 X 5 = 36.00 00

RO
M.
W
ID
PLOT NO. 1B

E
E

ID
RO

W
AD
DEDUCTIONS LIFT SHAFT

M.
00
30.
LIFT 1 = 3.30 X 2.45 = 8.085
2 LIFTS = 8.085 X 2 = 16.17 PROPOSED SITE
LIFT 2 = 2.00 X 2.10 = 4.20 60
.00
M.
6 LIFTS = 4.20 X 6 = 25.20 W
ID
PLOT NO. 1C E
RO
GATE-2 53.285 TOTAL B/UP AREA = 14827.40 45. 00 M. W I D E R O
AD
AD

ENTRY 6 SHAFTS = 4.62 X 6 = 27.72 OAD
45.00 M. W I D E R
46.9 5 SHAFTS = 7.20 X 5 = 36.00 PLAN OF PROPOSED FACTORY
7

30.00 M. W I D E R O A D
2 2 LIFTS = 8.085 X 2 = 16.17 BUILDING ON PLOT NO 8A, SECTOR 13,
6 LIFTS = 4.20 X 6 = 25.20 DMIC, AURIC , SHENDRA,
WATCHMAN ROOM

SECURITY
CABIN-2

DRIVER'S
ROOM
TOTAL B/UP AREA = 14722.31 CHATTRAPATI SAMBHAJI NAGAR
PLOT AREA = 81920.00 FOR M/S LIFENITY HEALTH LIMITED
VCP VCP VCP VCP VCP VCP VCP VCP VCP VCP VCP VCP VCP VCP VCP VCP

LOCATION PLAN
SCALE :- NTS
VCP VCP VCP VCP VCP VCP VCP VCP VCP VCP VCP VCP VCP VCP VCP VCP

OPEN SPACE-1 PUMP ROOM
PUMP FORM OF STATEMENT
ROOM

VCP VCP VCP VCP VCP VCP VCP VCP VCP VCP VCP VCP VCP VCP VCP VCP 4024.28
AREA CALCULATIONS AREA STATEMENT
BLOCK NO BLOCK NAME FLOOR SIZE AREA AREA IN
NO PARTICULARS
BUILDING-01 WORKSHOP GROUND 14722.31
SQM.
SPACE 1

PARKING 1 FOR

STP PROCESS
FIRST 14722.31 1 AREA OF PLOT NO 56 SECTOR 13, DMIC, SHENDRA 81920.00
GRAY WATER LINE

TANK
SECOND 14722.31
AS PER OWNERSHIP DOCUMENTS
VCP VCP VCP VCP VCP VCP
2159.69
VCP VCP VCP VCP VCP VCP VCP VCP VCP VCP
TOTAL OF BUILT-UP AREA 44166.93
OCCUPANT LOAD AS PER TABLE NO 8E
CLAUSE NO. 10.6.3 CDCPR
81920.00
SPACE
FOR
EFFLUENT TREATMENT PLANT
WATER LINE

2 DRIVERS ROOM GROUND 72.00 OCCUPANT LOAD CALATION
AS PER MEASUREMENT SHEET 81920.00
ETP
44238.93 SQMT
BUILT UP AREA
AS PER SITE 81920.00
SC

SC

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SC

FIRE TANK TOTAL B/UP AREA OF ALL BUILDINGS 44238.93 MAIN GATE
SCALE 1:50 BOUNDARY WALL DETAIL
10
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AREA PER PERSON SQMT

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F S I CONSUMED 2 DEDUCTIONS FOR
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0.54 FRONT ELEVATION COMPOUND WALL & GATE
NO OF PERSONS 4424 NOS

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11

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22

22

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EFFLUNT LINE

SEV

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30

a. PROPOSED DP ROAD 0.00

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RAIN WATER NO. OF OCCUPANTS - 900 NOS.

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HARVESTING
RAIN WATER HARVESTING
NO. OF OFFICE STAFF - 200 NOS.

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PORTABLE WATER TANKS NO. OF WORKERS - 700 NOS.

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PIT b. ANY RESERVATIONS 0.00

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TOTAL NO. OF PERSON 4424

NO ACTIVITY WATER IN LITS.

200 700

100 100 450 250 a. Required 4096 00

b. Adjustment of 2(b) if any 0.00

33

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TOTAL CAPACITY = 3,00,000 LITS

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after 2.5% thereafter 3% thereafter 2.5% thereafter 4%

ID REQUIRED WC

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c. Balanced proposed

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thereafter 2.5% thereafter 2.5%
5 NET PLOT AREA (3-4(c))) PRO RATA 81920.00
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REQUIRED URINALS

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SECTION AT CC
6 RECREATIONAL OPEN SPACE IF APPLICABLE 10%
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U.G. WATER TANK

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PORTABLE CAPACITY 10 X 6 X 2.50 = 150000 LITS
a. Required 8192.00
N PORTABLE CAPACITY 10 X 6 X 2.50 = 150000 LITS
REQUIRED PROVIDED

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A NO. OF W.C. FOR STAFF (MALE) 3 3

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NO. OF URINALS FOR STAFF 3 3 b. Proposed 8202.72

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NO. OF W.C. FOR WORKERS (MALE) 12 12
7 INTERNAL ROAD AREA 0.00
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NO. OF W.C. FOR WORKERS (MALE) 12 12
7 INTERNAL ROAD AREA 0.00

7 INTERNAL ROAD AREA 0.00

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NO. OF URINALS FOR WORKERS 12 12
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NO. OF W.C. FOR WORKERS 10 10 8 PLOTABLE AREA IF APPLICABLE (as per sanctioned layout)
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PROPOSED 5. U.G. WATER TANK PLAN (FEMALE)
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SEWAGE LINE

10 ADDITIONAL FSI ON PAYMENT OF PREMIUM
2 SEPTIC TANK REQUIREMENT
CP-L CP-L CP-L CP-L CP-L CP-L CP-L CP-L CP-L CP-L CP-L

6035.84
CP-L
CP-L CP-L CP-L CP-L

35 .0 TOTAL B/UP AREA = 44238.93 SQ.M.
OCCUPANT LOAD = 44238.93 /10 = 4424 NOS.
a. Maximum permissible premium FSI / TOD Zone. 2.00
PARKING REQUIREMENTS CLAUSE NO 7.20 MIDC CDCPR b. Proposed FSI of payment of Premium
CP-L
CP-L CP-L CP-L CP-L CP-L CP-L CP-L CP-L CP-L CP-L CP-L CP-L CP-L CP-L CP-L CP-L CP-L CP-L 0 AS PER TABLE 7C
AND 7D FOR 44238.93 SQM. MULTIPLICATION FACTOR 0.8
OCCUPANT LOAD actual = 900 NOS.
0.00
38 M TYPE OF VEHICLE PARKING SIZE REQUIRED PARKING REQUIRED REQUIRED REQUIRED PROVIDED NO ACTIVITY WATER
IN LITS.

W CAR 2.5M X 5.0M 1 FOR EVERY 200SQM 221.19 0.80 177
1 FOR W.C. & URINALS 30 X 900 = 27000
11 IN SITU TDR / FSI LOADING
GRAY WATER LINE

SCOOTER / MOTOR CYCLE 2.0 SQ.M. (1.0M X 2.0M) 442.38 0.80 553
ID CYCLE
TRUCK
2 FOR EVERY 200SQM
8195.53Sq.m
TOTAL /DAY 27000 LITS/DAY
a. In situ area against Statutory Raod (2 xSr No 2(a)) if any 0.00
3.75M X 10.0M 1 FOR EVERY 1000 SQM 44 44 DETENTION 27000 X 2 = 54000
CP-L
CP-L CP-L CP-L CP-L CP-L CP-L CP-L CP-L CP-L CP-L CP-L CP-L CP-L CP-L CP-L CP-L CP-L CP-L CP-L CP-L

42 E In case of plots pf 1ha and above minimum parking space shall be 10% of plot area
For plot area 81920 sq.m. parking should be 8192 sq.m.
PERIOD 2 DAYS

SLUDGE SETTLEMENT / PERSON = 30 LIT/YR. 12 ADDITIONAL FSI AREA UNDER CHAPTER NO.6 0.000
CP-L
CP-L
CP-L CP-L CP-L
SE CLEANING PERIOD = 1 YEAR
TOTAL SLUDGE = 30 X 900 = 27000 LITS 13 TOTAL ENTITLEMENT OF FSI IN THE PROPOSAL 0.000
45 C
CA UR
TANK CAPACITY = 54000 +27000 = 81000 LITS
CAPACITY 5.00X 3.00 X 2.00 X 3 nos= 90000 LITS a. 9 +10(b) +11(a) or 9+12 which ever is applicable
RAIN WATER HARVESTING TANK CAPACITY
BI ITY AVERAGE ANNUAL RAINFALL=100MM (0.1M/HOUR) VENT PIPE
Permissible 0.00
N- PLOT AREA= 81920 SQ.M BRICK WALL
BRICK BAT a. 9 +10(b) +11(a) or 9+ 12 which ever is applicable
1 GA VOLUMETRIC RAINFALL= 81920 X 0.1= 8192 CUM/HOUR
= 136.53 CUM/MINUTE PCC BED SEPTIC TANK SECTION AT DD Proposed 0.00
49 EN TE-
CP-L
CP-L
CP-L CP-L CP-L CP-L CP-L CP-L CP-L
RAIN WATER TANK CAPACITY PROPOSED= 5x 5 X 3 X 2= 150 CUM
RAIN WATER TANK CAPACITY PROPOSED= 150000 LITER
b. Permissible Ancillary area FSI up to 80% with payment
0.00
TR 1 PERMISSIBLE
b. Proposed Ancillary area FSI up to 60% or 80% with payment
Y
RAIN WATER
PIPE

GL GL
MURUM
FILLING
SAND
FILLING

RUBBLE
FILLING
SEPTIC TANK PLAN c.Total Entitlement permissible
WATER HARVESTING SECTION
(SCALE - 1:50)
(a+b) 81920.00
RAINWATER PIT CAPACITY 5.00 X 5.00 X3 x 2 nos = 150 K LTS

c.Total Entitlement proposed
81920.00
330.840

(a+b)

Maximum utilization limit of FSI (Building Potential) permissible as per
14 road width ((as per Regulation No. 5.5 of Chapter - 5)) x 1.6 or 1.8) 81920.00

15 TOTAL Built up area in proposal (excluding area as per sr. no 17b)

R I D M.
a. Existing Built -up Area. 0.000
b. Proposed Built- up area (as per P line) 44238.93

W .00
O E
CROSS DRAIN
c. Total (a+b) 44238.93

AD
30
16 FSI CONSUMED (15/13) (should not be more than serial 0.54
No.14 above.)

CERTIFICATE OF AREA :-
R O A D 1 2 . 0 0 M W I D E

R O A D 1 2 . 0 0 M W I D E
GREEN BELT 6 M WIDE

Certified that the plot under reference was surveyed by me on
03/03/2026 and the Dimensions of sides etc.of plot stated on plan area
as measured on site and the area so worked out tallies with the stated in
documents of ownership/ DMIC/Land records Department/City Survey
A-1 (WORKSHOP)

Records
Ar. Dashrath Dhumal

R O A D 1 2 . 0 0 M W I D E
Owners Declaration :-
I/We undersigned hereby confirmed that I/We would abide by plans
OPEN SPACE-2 approved by Authority/ Collector. I / We would execute the structure as
FUTURE 4178.44 sq.m. per the approved plans.Also I/We would execute the work under
supervision of proper technical person so as to ensure the quality and
EXPANSION safety at the work site.
Lifenity Health Limited.

Architects Name and Signature

26.192
Ar. Dashrath Dhumal
Job No. Drg. No. Scale Drn. By. Checked By Registration no of Architect
001 1:100 Ar. Dashrath Dhumal CA/2005/37017

C PARKING STATEMENT
I N T E R N A L

I N T E R N A L

GARAGE PERMISSIBLE GARAGE PROPOSED NIL
PARKING PARKING
b TYPE
REQUIRED PROPOSED

60 CAR

W .00 M
SCOOTER
8192 SQ.M. 8195.53 SQ.M.
CYCLE

R I D E .
VISITORS
D LOADING UNLOADING SPACES
I N T E R N A L

O a LOADING UNLOADING REQUIRED 44

AD b TOTAL LOADING UNLOADING PROVIDED 44
SewageLine

LEGENDS
AMENITY SPACE

D
PLOT BOUNDARY SHOWN IN
B/UP AREA SHOWN IN
4125.33 sq.m.

A
2
WATER LINE SHOWN IN

70

O

GREY WATER LINE SHOWN IN
SEWAGE LINE SHOWN IN

.

R
79
EFFLUNT LINE

E
SCHEDULE OF DOORS AND WINDOWS

ID
TYPE SIZE DISCRIPTION

RS1 2500 X 6000 M.S ROLLING SHUTTER
SECURITY CABIN DRIVERS CABIN PUMP ROOM
D1 1800 X 2400 FLUSH DOOR

W
A

0.900
4.500
4.500
12.000 10.450 D2 900 X 2400 FLUSH DOOR
C3

D3 750 X 2400 FLUSH DOOR

5.455
A A
Security Cabin

6.000
4.00 x 5.00

0.899
FD 1000 X 2400 FIRE DOOR

M.
C4
AREA KEY PLAN
ELECTRICMETER ROOM
METER ROOM
ALLU. SLIDING WINDOW
Pump Room

W 2500 X 2500

14.000
AREA KEY PLAN

14.000
UP (Driver's Room)
A 10.450
12.000

VISITORS
V 600 X 900 ALU LOUVERED VENTILATOR

.00
4.00 x 6.74
A W.C.
1.20 x
A
1.20 BATH
1.20 x

5.455
A A

PLAN OF PROPOSED FACTORY
2.55 PUMP ROOM
10.00 X 5.00
DRIVER'S ROOM

6.000
8.84 x 4.54

LOCKER'S RM.
TOILET

BUILDING ON PLOT NO 8A, SECTOR 13,
2.40 x 1.50
W.B.
UP
45
2.55 x 1.84
EFFLUNT LINE TYPICAL PLAN-1,2&3. AREA KEY PLAN 1.997 2.997 1.514 1.199 1.514 0.999 0.230
(Security Cabin)
A Security Cabin 10.450

UP
PLAN (Pump Room)

N
PLAN Parapet LVL.

DMIC, AURIC , SHENDRA,
ELECTRIC SUBSTATION (Driver's Room)
Slab LVL.
PARAPET LVL.

1.598 0.599
PARAPET LVL.
SLAB LVL.

1.598 0.599
RCC Chajja

N
SLAB LVL.
Lintel LVL.

LINTEL LVL.

3.996
LINTEL LVL.

1.499
3.996
Sill LVL.

CHATTRAPATI SAMBHAJI NAGAR

2.398
1.499
0.899SILL LVL.

2.398
Plinth LVL.
0.899
SILL LVL. G. LVL. G. LVL.
PLINTH LVL.

0.450
PLINTH LVL. G.LVL. G.LVL.
FRONT ELEVATION

0.450
G.LVL. G.LVL. FRONT ELEVATION
(Driver's Room)
LEFT SIDE ELEVATION
(Security Cabin)

I N T E R N A L R O A D 1 2 . 0 0 M W I D E
Parapet LVL.
PARAPET LVL.
Terrace

1.598 0.599
Slab LVL.
TERRACE SLAB LVL.

RCC Chajja
TERRACE
0.900 Lintel LVL.

3.996
LINTEL LVL.

3.497

1.499
3.996
1.499

2.398

2.398
2.398
Sill LVL.
0.899

0.899
SILL LVL.
PUMP ROOM
Plinth LVL.

0.450
Security Cabin VISITORS TOILET PLINTH LVL. G. LVL. G. LVL.
SUNK
G.LVL.
DRIVER'S ROOM W.C. BATH SECTION- AA

SUNK SUNK
G.LVL.
SECTION- AA
(Security Cabin)
SECTION- AA
(Driver's Room)

SECURITY CABIN AREA CALCULATION DRIVER'S ROOM AREA CALCULATION PUMP ROOM AREA CALCULATION

GREEN BELT 6 M WIDE
Sr. No. L W AREA NOS TOTAL AREA Sr. No. L W AREA NOS TOTAL AREA Sr. No. L W AREA NOS TOTAL AREA

A 4.50 14.00 63.00 2 126.00 B 6.00 12.00 72.00 1 72.00 A 5.455 10.45 57.00 1 57.00
TOTAL B/P AREA 126.00 Sq,m, TOTAL B/P AREA 72.00 Sq,m, TOTAL B/P AREA 57.00 Sq,m,

FOR M/S LIFENITY HEALTH LIMITED
231.187 SECURITY CABIN 1,2 189.00 -- -- 189.00

PROPOSED PARKING POLYLINE 2.5MX2.5MX3M BAFFLE WALL 5MX2.5MX3M DESILTING
Parking
PARKING
Nos
913
Area
0.00 DRIVER ROOM 72.00 -- -- 72.00
RAIN WATER HARVESTING CHAMBER SIGN OF OWNER
PIT
PUMP ROOM 57.00 -- -- 57.00

45.00 M. W I D E R O A D MANHOLE
600 MM x 600
1.2MX2.5MX1.5M
25X25MM
BAR SCREEN SIGN OF ARCHITECT
MM

GREASE TRAP
AREA CALCULATION
MANHOLE

X X
CREATION

LAYOUT PLAN RWHP FLOOR PLAN -
SCALE - 1:100
JOB No.

01
DRG NO. SCALE

1 :100
DATE DRN BY CHK. BY

Reagent Details:
Preparation of IG-RR-17 working buffer from Stock buffer (IG-RR-16):
Calculations: For the Preparation of 0.2M Glycine pH 7.0.
Stock Buffer (IG-RR-16):0.8M Glycine pH 7.0.
Formula:
 $C1 \times V1 = C2 \times V2$
C1 = concentration of stock
V1 = volume of stock you need to take
C2 = desired (working) concentration
V2 = final total volume of working solution
Applying the formula to prepare the working buffer:
 $C1 \times V1 = C2 \times V2$
C1 = 0.8M Glycine pH 7.0.
V1 = volume of stock you need to take _____ L.
C2 = desired (working) concentration 0.2 M Glycine pH 7.0
V2 = final total volume of working solution _____ L.
 $V1 = (C2 \times V2) / C1$
C2=0.2M ; V2= 1.0 L; C1=0.8M
 $V1 = (0.2 \times 1.0) / 0.8$
V1 = 0.25L
Take 0.25 L (IG-RR-16) Stock buffer solution and dilute with water, and make up to 1L.
Check and record the details:
Preparation Details:
Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry
Date
Glycine ||||
NaOH ||||
Stock buffer (IG-RR-16) taken volume(L) | pH (7.0±0.1) | Conductivity (1.0 ±0.1ms/cm) | Storage
Condition | Final volume prepared
|| RT |
Date of Preparation | Prepared By | Checked By
||

Reagent Details:
Formula & Calculations:
Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).
Calculation: To prepare 1 L of 4N Sodium Chloride (MW = 58.44 g/mol)
Weight of solute(g)=4×58.44 = 233.76■g.
Preparation Method:
Weigh 233.76g of Sodium Chloride and dissolve in water to make a 1 L solution.
Preparation for different volumes:
To prepare _____ of 4N Sodium Chloride (MW = 58.44 g/mol).
Weight of solute(g)=4×58.44 x _____L= _____gm
Weigh _____gm of Sodium Chloride and dissolve in water to make a _____ L solution.
Check and record the details:
Preparation Details:
Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry
Date
Sodium Chloride | | | |
Final volume prepared | Weighed Quantity | pH (7.0 ±0.1) | Conductivity (340 ±10.0 ms/cm) | Storage
Condition:
| | |
Date of Preparation | Prepared By | Checked By
| |

Reagent Details:
Formula & Calculations:
Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).
Calculation: To prepare 1 L of 1N Sodium Hydroxide (MW = 40.0 g/mol)
Weight of solute(g)=1×40 = 40.00g.
Preparation Method:
Weigh 40.00 g of Sodium Hydroxide and dissolve in water to make a 1 L solution.
Preparation for different volumes:
To prepare _____ of 1N Sodium Hydroxide (MW = 40.0 g/mol).
Weight of solute(g)=1×40.0 x _____ L= _____ gm
Weigh _____ gm of Sodium Hydroxide and dissolve in water to make a _____ L solution.
Check and record the details:
Preparation Details:
Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry
Date
Sodium Hydroxide | | | |
Final volume prepared | Weighed Quantity | pH (13.5 ± 0.5) | Conductivity (295 ± 5.0ms/cm) | Storage
Condition:
| | | RT
Date of Preparation | Prepared By | Checked By
| |

CTE excel sheet-application for consent to establish (CTE) displayed.
List of raw materials page no. 115 to be prepared

Reagent Details:

Formula:

Sodium Acetate(M.Wt): 82.03g/mol.

Acetic Acid (M.Wt): 60.05g/ mol

Sodium Chloride:58.44g/mol

Preparation of 0.1 M Sodium Acetate pH 5.35 buffer

Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).

Calculation: To prepare 1 L of 0.1 M Sodium Acetate (MW = 82.03g/mol).

Weight of solute(g)=0.1×82.03× 1.0=8.203g.

Preparation Method:

Weigh 8.203 g of Sodium Acetate and dissolve in water to make a 1 L solution.

Preparation for required volumes:

To prepare _____ of 0.1M Sodium Acetate (MW = 82.03g/mol).

Weight of solute(g)=0.1×82.03× _____ L= _____ g.

Weigh _____ gm of Sodium Acetate and dissolve in water to make a _____ L solution.

Calculations: To prepare the 0.1 M Acetic Acid.

Formula & Calculations:

Formula: Use the formula to calculate the molarity:

Molarity (M) = (Per cent by Weight / 100) * (Density in g/mL) * 1000 / Molar Mass.

Example: Molarity of Glacial Acetic Acid

Molar Mass of CH₃COOH: ~60.05 g/mol

Density: ~1.05 g/mL

Purity: 100%

Acetic Acid (MW = 60.05 g/mol).

Calculation:

Molarity = (100 / 100) * 1.05 g/mL * 1000 mL/L / 60.05 g/mol ≈ 17.5 M.

Calculation: To prepare 1 L of 0.1 M acetic acid

Applying the formula to prepare the working buffer:

C₁V₁=C₂V₂

C₁ = 17.5 M acetic acid.

V₁ = volume of stock needed to take.

C₂ = desired (working) concentration 0.1 M acetic acid.

V₂ = final total volume of working solution for one litre.

V₁ = (C₂ × V₂)/C₁.

C₂=0.1 M ; V₂= 1.0 L; C₁=17.5 M

V₁ = (0.1 × 1.0)/17.5

V₁ = 0.0057 L

Take 0.0057 L of the Stock solution of acetic acid and add it to a volumetric flask, making up the volume to 1 L.

To prepare various volumes:

V₁ = volume of stock needed to take _____ L.

C₂ = desired (working) concentration 0.1 M acetic acid.

V₂ = final total volume of working solution _____ L.

V₁ = (C₂ _____ × V₂ _____) /17.5.

V₁= _____ L.

Take V₁ _____ L of the Stock solution of acetic acid and add it to a volumetric flask, making up the final volume _____ L.

Preparation of 4.0 M Sodium Chloride.

Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).

Calculation: To prepare 1 L of 4 M Sodium Chloride (MW = 58.44g/mol).

Weight of solute(g)=4×58.44 × 1.0= 233.76g.

Preparation Method:

Weigh 233.76g of Sodium Chloride and dissolve in water to make a 1 L solution.

Preparation for required volumes:

To prepare _____ of 4 M Sodium Chloride (MW = 58.44g/mol).

Weight of solute(g)=0.1×58.44× _____ L= _____ g.

Weigh _____ gm of Sodium Chloride and dissolve in water to make a _____ L solution.

Procedure for Preparation of 0.1 M Sodium Acetate + 4 M Sodium Chloride pH 5.35

To prepare a pH 5.35 solution, 0.1 M Sodium Acetate solution was added to this 4M Sodium Chloride was added and completely dissolved and then titrated with 0.1 M Acetic acid so that the pH attains pH 5.35 and made up to 1 L.

Check and record the details:

Preparation Details:

Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry

Date

Sodium Acetate | | | |

Acetic Acid | | | |

Stock buffer taken volume(L) | pH (5.35 ± 0.2) | Conductivity (220.0± 5.0ms/cm) | Storage Condition |

Final volume prepared

| | |

Date of Preparation | Prepared By | Checked By

| |

Reagent Details:
Formula & Calculations:
Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).
Calculation: To prepare 1 L of 1.0 M Citric Acid (MW = 192.124 g/mol)
Weight of solute(g)=1×192.124x1=192.124 g.
Preparation Method:
Weigh 192.124 g of Citric Acid and dissolve in water to make a 1 L solution.
Preparation for different volumes:
To prepare _____ of 1.0 M Citric Acid (MW = 192.124g/mol).
Weight of solute(g)=1.0 ×192.124 x _____ L= _____g.
Weigh _____gm of Citric Acid and dissolve in water to make a _____ L solution.
Check and record the details:
Preparation Details:
Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry
Date
Citric Acid | | | |
Final volume prepared | Weighed Quantity | pH (2 to 3) | Conductivity (4.5 to 5.0 ms/cm) | Storage
Condition | Reagent Record
| | | RT | PF-RR-05
Date of Preparation | Prepared By | Checked By
| |

Reagent Details:
Preparation of IG-RR-23 working buffer from Stock buffer (IG-RR-22):
Calculations: For the Preparation of 0.5M Sodium Hydroxide.
Stock Buffer (IG-RR-22):2.0M Sodium Hydroxide,
Formula:
 $C1 \times V1 = C2 \times V2$
C1 = concentration of stock
V1 = volume of stock you need to take
C2 = desired (working) concentration
V2 = final total volume of working solution
Applying the formula to prepare the working buffer:
 $C1 \times V1 = C2 \times V2$
C1 = 2.0 M Sodium Hydroxide.
V1 = volume of stock you need to take _____ L.
C2 = desired (working) concentration 0.5M Sodium Hydroxide.
V2 = final total volume of working solution _____ L.
 $V1 = (C2 \times V2) / C1$.
C2=0.5M; V2= 1.0 L; C1=2.0M
 $V1 = (0.5 \times 1.0) / 2.0$
V1 = 0.25L
Take 0.25 L (IG-RR-22) Stock buffer solution and dilute with water, and make up to 1L.
Check and record the details:
Preparation Details:
Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry
Date
Sodium Hydroxide | | | |
Stock buffer (IG-RR-22) taken volume(L) | pH (13.5±0.3) | Conductivity (240±15 ms/cm) | Storage
Condition | Final volume prepared
| | RT |
Date of Preparation | Prepared By | Checked By (QC/Analyst)
| |

Reagent Details:
Preparation of PF-RR-04 working buffer from Stock buffer (PF-RR-03):
Calculations: For the Preparation of 20mM Sodium Citrate pH 7.0 + 0.12M Sodium Chloride + 2.6 mM Calcium Chloride.
Stock Buffer (PF-RR-03):0.2M Sodium Citrate pH 7.0 + 1.2M Sodium Chloride + 26mM Calcium Chloride.
Formula:
 $C1 \times V1 = C2 \times V2$
C1 = concentration of stock
V1 = volume of stock you need to take
C2 = desired (working) concentration
V2 = final total volume of working solution
Applying the formula to prepare the working buffer:
 $C1 \times V1 = C2 \times V2$
C1 = 0.2M Sodium Citrate pH 7.0 + 1.2M Sodium Chloride + 26mM Calcium Chloride.
V1 = volume of stock you need to take _____L.
C2 = desired (working) concentration 20mM Sodium Citrate, pH 7.0 + 0.12M Sodium Chloride + 2.6 mM Calcium Chloride.
V2 = final total volume of working solution _____L.
 $V1 = (C2 \times V2) / C1$
C2=0.02M; V2= 1.0 L; C1=0.2M
 $V1 = (0.02 \times 1.0) / 0.2$
V1 = 0.1L
Take 0.1 L (PF-RR-03) Stock buffer solution and dilute with water, and make up to 1L.
Check and record the details:
Preparation Details:
Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry
Date
Sodium Citrate | | | |
Citric Acid | | | |
Sodium Chloride | | | |
Calcium Chloride | | | |
Final volume prepared | pH (7.0 ±0.1) | Conductivity (17± 1ms/cm) | Storage Condition | Reagent Record
| | | RT | PF-RR-03
Date of Preparation | Prepared By | Checked By
| |

Reagent Details:
Formula & Calculations:
Formula: Use the formula to calculate the molarity:
 $Molarity (M) = (Per\ cent\ by\ Weight / 100) * (Density\ in\ g/mL) * 1000 / Molar\ Mass.$
Example: Molarity of Glacial Acetic Acid
Molar Mass of CH_3COOH : ~60.05 g/mol
Density: ~1.05 g/mL
Purity: 100%
Acetic Acid (MW = 60.05 g/mol).
Calculation:
 $Molarity = (100 / 100) * 1.05\ g/mL * 1000\ mL/L / 60.05\ g/mol \approx 17.5\ M.$
Calculation: To prepare 1 L of 0.1M.
Applying the formula to prepare the working buffer:
 $C_1 V_1 = C_2 V_2$
 $C_1 = 17.5\ M$ acetic acid.
 V_1 = volume of stock needed to take.
 C_2 = desired (working) concentration 0.1M acetic acid.
 V_2 = final total volume of working solution for one liter.
 $V_1 = (C_2 \times V_2) / C_1.$
 $C_2 = 0.1\ M; V_2 = 1.0\ L; C_1 = 17.5\ M$
 $V_1 = (0.1 \times 1.0) / 17.5$
 $V_1 =$ Take 0.0057 L of the Stock solution of acetic acid and add it to a volumetric flask, making up the volume to 1 L.
To prepare various volumes:
 V_1 = volume of stock needed to take _____ L.
 C_2 = desired (working) concentration 0.1M acetic acid.
 V_2 = final total volume of working solution _____ L.
 $V_1 = (C_2 \times V_2) / 17.5.$
 $V_1 =$ _____ L.
Take V_1 _____ L of the Stock solution of acetic acid and add it to a volumetric flask, making up the final volume _____ L.
Preparation Details:
Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry
Date
|||
Final volume prepared | Measured Quantity | pH (2.9) | Conductivity (250 to 300 $\mu S/cm$) | Storage
Condition | Reagent Record
|||
Date of Preparation | Prepared By | Checked By
||

Reagent Details:

Preparation of PF-RR-13 Stock buffer :

Calculations: For the Preparation of 20mM Sodium Citrate pH 7.0 + 1.0 M Sodium Chloride.

Stock Buffer (PF-RR-09):0.4M Sodium Citrate pH 7.0.

Formula:

$$C1 \times V1 = C2 \times V2$$

C1 = concentration of stock

V1 = volume of stock you need to take

C2 = desired (working) concentration

V2 = final total volume of working solution

Applying the formula to prepare the working buffer:

$$C1 \times V1 = C2 \times V2$$

C1 = 0.4M Sodium Citrate, pH 7.0.

V1 = volume of stock you need to take _____ L.

C2 = desired (working) concentration 20mM Sodium Citrate, pH 7.0.

V2 = final total volume of working solution _____ L.

$$V1 = (C2 \times V2) / C1$$

$$C2 = 0.02M ; V2 = 1.0 L ; C1 = 0.4M$$

$$V1 = (0.02 \times 1.0) / 0.4$$

$$V1 = 0.05L$$

Take 0.05 L (PF-RR-09) Stock buffer solution and add 1.0M sodium Chloride.

Calculations: To prepare the 1 M Sodium Chloride.

Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).

Calculation: To prepare 1 L of 1 M Sodium Chloride (MW = 58.44 g/mol)

$$\text{Weight of solute(g)} = 1 \times 58.44 \times 1 = 58.44 \text{ g.}$$

Preparation Method:

Weigh 58.44 g of Sodium Chloride and dissolve in water to make a 1 L solution.

Preparation for different volumes:

To prepare _____ L of 1.0 M Sodium Chloride (MW = 58.44 g/mol).

$$\text{Weight of solute(g)} = 1.0 \times 58.44 \times \text{_____} = \text{_____} \text{ g.}$$

Weight _____ gm of Sodium Chloride and dissolve in water to make a _____ L solution.

Preparation of Stock buffer (PF-RR-13):

Take 0.05 L (PF-RR-09) Stock buffer solution and add 1.0 M sodium chloride. Dissolve it, and then make up the volume to 1 L with water.

Check and record the details:

Preparation Details:

Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry

Date

Sodium Citrate | | | |

Citric Acid | | | |

Sodium Chloride | | | |

Stock buffer (PF-RR-09) taken volume(L) | pH (7.0±0.1) | Conductivity (88 ±0.2) | Storage Condition |

Final volume prepared

| | | RT |

Date of Preparation | Prepared By | Checked By

| |

6D UP

FIRE EXIT

D1
D1 D1 D1 D1

D6 D1
LIFT

FIRE EXIT
LOBBY
6.77X11.77
D1 D1 D1 D1 RECEPTION OPEN TO ALL

D1
FLOORS

D1
6D

UP FIREMAN
Goods LIFT D6
LIFT
3.50X3.00
1D
2.70X2.50

D6
DPB
DPB DPB WASTE OUT
LAF DHS
CHEMICAL DAY STORE 3 WAY FIRE STAIR
LOBBY

6D

6D

DE
DECONTAMINATION WASHING STERILIZATION up
3.00X11.30

AUTOCLAVE

LAF
OPEN TO COLD ROOM
FIRST BUFFER TANK STORAGE-2 CIP / SIP UP
UP FLOOR PREPARATION COOL ZONE

DPB DPB
A/L GOWN A/L 6D

WASTE IN A/L A/L

GS+HH

GS+HH
LENAP LORTNOC LENAP LORTNOC

6D 6D 6D

DE
A/L DE GOWN A/L
CORRIDOR

FIRE CONTROL
D3 ROOM UP
LAF
D6 D6 D6 D6
DPB AKTA-1" TFF
NANO 3.50X6.54
FILTER MH

A/L GOWN A/L
1000 L
BULK FILTRATION

LAF
500 L

MH
SL MH

600 L

500 L

LAF
MH SL

600 L MH

MH
FIRE PUMP
DE MH

50 L ROOM
A/L
A/L GOWN LAF MH

2000 L 500 L
CHROMO SL
STAGE 13 E 3.50X5.00
COLUMN 250 L
MH 600 L STAGE 10 T/ MH
UP

STAGE 11 F STAGE 9 N MH

250 L
200 L

AKTA-1"
STAGE 6 (I)
MH

COLUMN PACKING 250 L
MH

250 L
1000 L
AND SKIDS
MH

200 L
MH

MH

2.80 M WIDE PASSAGE
250 L
COOL ZONE
MH MH

150 L 100 L MH

MH

1000 L 500 L

6.77 M WIDE CORRIDOR
MH

MH

300 L
WC WC WC 150 L MH

1.10X 1.10X 1.10X CHROMO 100 L MH

COLUMN
100 L
MH

1.70 1.70 1.70
300 L
LAF 150 L
DPB LAF
MH

CHROMO MH

AKTA-1"
MH

100 L COLUMN

AUTOCLAVE
DHS

D6
WC WC WC 500 L 500 L CHROMO
250 L 250 L 150 L 100 L COLUMN

6D
1.10X 1.10X 1.10X MH
MH MH

1.70 1.70 1.70 CORRIDOR A/L MH MH
MH MH
MH

MH

CORRIDOR CORRIDOR

STERILIZATION 2000 L

D6

CLAF
2500 L 2500 L

6D

PRE - VIRUS CONTROLLED CORRIDOR
1000 L

TFF
MH

UP COLUMN PACKING
DPB
AND SKIDS MH

UP

AKTA-1"
WASHING 100 L
A/L

AKTA-1"
STAGE 7A 500 L
MH

STAGE 5 (I) A

D6
DPB
WASTE IN MH 500 L
250 L

TFF
LOBBY 300 L
MH
MH

6.77X11.77
DECONTAMINATION 250 L
500 L
MH

OUT MH
MH

WASTE

D6
500 L

DPB STAGE 5(I)B 1500 L 250 L
MH
MH

100 L SL
MH

250 L 250 L 250 L 600 L 500 L
100 L 100 L

6D
100 L
MH

MH
MH MH MH
MH

100 L
MH MH MH

CIP / SIP
MH

1500 L
SL
UP
MH MH

MH MH MH

1000 L
MH

500 L 500 L
MH

MH

1000 L 1000 L 1000 L
UP
100 L 1500 L
1500 L

GS+HH

GS+HH
MH SL
LENAP LORTNOC LENAP LORTNOC

MH
2500 L
MH

1500 L
50 L
Pass. LIFT CHROMO MH
250 L 250 L 250 L 250 L
2.30X2.00 DPB COLUMN

6D
MH MH MH MH

WASTE OUT

BAG CUTTING & WC WC WC

POOLING MH
MH MH

MH

MH MH
STAGE 4 (F) 1.10X
1.70
1.10X
1.70
1.10X
1.70

150 L 500 L

6D
500 L
SL 1000 L
TANK STORAGE 1000 L 1500 L 1500 L

D6
MH

WC WC WC
STAGE 12 (E)
MH

250 L 1.10X 1.10X 1.10X
MH

STAGE 1 (F)/ 1.70 1.70 1.70
CHROMO
500 L
COLUMN

250 L
300 L 300 L MH

TFF
D6

AKTA-1"
6D
MH MH

CLAF

2500 L

THAWING PLASMA 2000 L 2000 L
UP

AKTA-1"
MH

DPB
500 L 250 L 250 L 300 L 300
300L L
BUFFER

6D
MH MH

CHROMO
MH
COLUMN MH MH MH
MHMH

CHROMO
PREPARATION D6 D6 D6
COLUMN

CORRIDOR

6D

DE
DPB
3.24 M WIDE PASSAGE

D6 D6 D6

6D
MH MH

MH MH MH MH

CHROMO MH
MH
MH MH MH MH

COLUMN
100 L 100 L 100 L 100 L MH

100 L 100 L CENTRIFUGE
200 L 250 L 300 L 300 L 500 L 500 L
1500 L

PLASMA DAY STAGE 3 (F)
STORE STAGE 2 (F)
STAGE 8 N

AKTA-1"

AKTA-1"
UP

CHROMO
COLUMN
MH
UP
3D

SL
600 L 1500 L 2000 L
1500 L 1500 L 1500 L

300 L 300 L MH MH MH
MH
MH

100 L 100 L 100 L CHROMO CHROMO
DPB MH
TFF MH

COLUMN COLUMN AKTA-1"
MH MH MH

WC WC WC
1.10X 1.10X 1.10X EMER.GLASS

1.70 1.70 1.70
LOBBY
6.77X11.00

WC WC WC
STAIRCASE
1.10X 1.10X 1.10X AND
1.70 1.70 1.70
LIFT ONLY
CORRIDOR
FOR
CORRIDOR
SECOND FLOOR
(L.V.P.)

UP

UP

W

D3 D3 D3 D3 D3

CORRIDOR

D6
LOBBY CORRIDOR
12.47X6.77
D3 D6
Pass. LIFT W
DPB
2.30X2.00
D3

6D

3D D3 D3
UP
D3
3D

CORRIDOR

D3
3D

SD SD SD SD

DPB
6D
ELECTRIC
D3
ROOM
DPB
D3
DPB
3D D3 D6 STORE 3.50X6.00
OFFICE
3D
LOBBY
3.00X11.53 3D

OPEN TO

D3
SECOND D3
D3
3D D3

FLOOR REJECT
RECALLED
GOODS ROOM FG DISCPATCH
FG IN
6D D3 D3 D6 D6

D6 D3
SERVER
Goods LIFT CORRIDOR
ROOM
3.50X3.00
6D

D6
3.50X6.57
DISPATCH BAY RECEIVING BAY RECEIVING BAY DISPATCH BAY

RS3
40' CONTAINER YARD

CLIENT :- LIFENITY-LARGE VOLUME PARENTERALS

DESIGN &
PROJECT BY :-BIOBRIDGE HEALTHCARE SOLUTION PVT.LTD.
PREPARED A.K
ADDRESS :-13 R B
A P -411007 M I .

GROUND FLOOR
CHECKED G.K.

DATE 05-06-2026
TITLE- CONCEPTUAL LAYOUT-GF REVISION R0
SCALE 1:1

SIZE A0 DWG.NO 01

Organic waste calculation for Plasma
Organic waste calculation for LVP Plant
List of parameters for ETP
LIST OF FEED PARAMETERS TO BE CONSIDER FOR DESIGN
LIST OF OUTLET PARAMETERS
Sr. No. | Source / Process Area | Type of Non-Hazardous Waste | Approx. Quantity Kg / Year | Solid (Kg/day) | Liquid (L/day) | Category | ETP/3rd party
1 | ETP / STP treatment | Biological sludge (after dewatering) | 6237850 | 17090 | | Organic | 3rd party
2 | Process area & lab washing | Treated water (ETP + RO permeate) | 27052340 L/year | | 74116 | Organic | Domestic use
3 | Packaging line | Empty glass vials (non-contaminated rejects) (~50,000 pcs) | 1000 | 2.7 | | Inorganic | Scrap 3rd party
4 | Packaging & labeling | Paperboard cartons, labels, paper scrap | 2000 | 5.5 | | Organic | Scrap 3rd party
5 | Procurement / stores area | Corrugated boxes, stretch film, plastic wrapping | 1500 | 4.1 | | Inorganic | Scrap 3rd party
6 | Maintenance workshop | Metal scrap (iron, steel, aluminum parts) | 1200 | 1 | | Inorganic | Scrap 3rd party
7 | Utility & HVAC area | Spent filters (air filters, bag filters, prefilters) | 500 | 1 | | Inorganic | 3rd party
8 | Admin & office areas | Office paper waste, packaging waste | 1500 | 25 | | Organic | Scrap 3rd party
9 | Canteen / pantry | Organic kitchen waste | 47000 | 128.77 | | Organic | Organic converter
10 | General housekeeping | Sweepings, mopping dust, non-contaminated PPE | 800 | 2.19 | | Mixed | 3rd party
11 | Cooling tower & utilities | Blowdown & scale sludge (non-toxic) | 1500 | 4.11 | | Organic | 3rd party
12 | Electrical systems | Cable cuttings, wire ends, plastic conduits | 300 | 0.82 | | Inorganic | Scrap 3rd party
13 | Gardening & green area | Dry leaves, grass clippings (~10 acre considering 33% green area) | 73200 | 200.55 | | Organic | Organic converter
14 | Water purification & softener | Spent ion-exchange resins (non-toxic) | 200 | 0.55 | | Inorganic | 3rd party
15 | Misc. activities | Empty HDPE / PET bottles (non-contaminated) | 200 | 0.55 | | Inorganic | 3rd party
| | | | |
ETP treated water consumption | 74.12 | KL/Day |
Organic converter | 0.33 | T/day |
Disposal by 3rd party | 17.10 | T/day |
Scrap to 3rd party | 0.04 | T/day |
Sr. No. | Source / Process Area | Type of Non-Hazardous Waste | Approx. Quantity Kg / Year | Solid (Kg/day) | Liquid (L/day) | Category | ETP/3rd party
1 | ETP / STP treatment | Biological sludge (after dewatering) | 4188375 | 11475 | | Organic | 3rd party
2 | Process area & lab washing | Treated water (ETP + RO permeate) | 14699645 L/year | | 40273 | Organic | Domestic use
3 | Packaging line | Empty glass vials (non-contaminated rejects) (~50,000 pcs) | 2200 | 6.03 | | Inorganic | Scrap 3rd party
4 | Packaging & labeling | Paperboard cartons, labels, paper scrap | 36500 | 100 | | Organic | Scrap 3rd party
5 | Procurement / stores area | Corrugated boxes, stretch film, plastic wrapping | 18250 | 50 | | Inorganic | Scrap 3rd party
6 | Maintenance workshop | Metal scrap (iron, steel, aluminum parts) | 7200 | 19.73 | | Inorganic | Scrap 3rd party
7 | Utility & HVAC area | Spent filters (air filters, bag filters, prefilters) | 1200 | 3.29 | | Inorganic | 3rd party
8 | Admin & office areas | Office paper waste, packaging waste | 5000 | 13.70 | | Organic | Scrap 3rd party
9 | Canteen / pantry | Organic kitchen waste | 36500 | 100.00 | | Organic | Organic converter
10 | General housekeeping | Sweepings, mopping dust, non-contaminated PPE | 20000 | 54.79 | | Mixed | 3rd party
11 | Cooling tower & utilities | Blowdown & scale sludge (non-toxic) | 27000 | 73.97 | | Organic | 3rd party
12 | Electrical systems | Cable cuttings, wire ends, plastic conduits | 600 | 1.64 | | Inorganic | Scrap 3rd party
13 | Water purification & softener | Spent ion-exchange resins (non-toxic) | 20000 | 54.79 | | Inorganic | 3rd party
14 | Misc. activities | Empty HDPE / PET bottles (non-contaminated) | 19000 | 52.05 | | Inorganic | 3rd party
ETP treated water consumption | 40.273 | KL/Day |
Organic converter | 0.1 | T/day |
Disposal by 3rd party | 11.71 | T/day |
Scrap to 3rd party | 0.19 | T/day |
Sr. No. | Parameter | Value | Unit
| pH | 10-12 | NA
| Fee Oil | <60 | mg/L
| TDS | 2000-2500 | mg/L
| TSS | <500 | mg/L
| COD | <2250 | mg/L
| BOD | <1100 | mg/L
Sr. No. | Parameter | Value | Unit
| pH | 6.5-8.5 | NA
| Fee Oil | <20 | mg/L
| TDS | <1200 | mg/L
| TSS | <300 | mg/L
| COD | <500 | mg/L
| BOD | <450 | mg/L

Reagent Details:
Formula & Calculations:
Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).
Calculation: To prepare 1 L of 0.4M Tri-Sodium Citrate (MW = 258.06g/mol)
Weight of solute(g)=0.4×258.06 x 1=103.224g.
Preparation Method:
Weigh 103.224g of Tri-sodium citrate and dissolve in water to make a 1 L solution.
Preparation for required volumes:
To prepare _____ of 0.4M Tri -Sodium Citrate (MW = 258.06g/mol).
Weight of solute(g)=0.4 x 258.06 x _____L= _____gm
Weigh _____gm of Tri-sodium citrate and dissolve in water to make a _____ L solution.
Check and record the details:
Preparation Details:
Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry Date
| | | |
Final volume prepared | Weighed Quantity | pH (8 to 9) | Conductivity (3 to 5 mS/cm) | Storage Condition:
| | |
Date of Preparation | Prepared By | Checked By
| |

Reagent Details:
Formula:
Glycine : (M.Wt): 75.07g/mol.
Sodium Hydroxide : (M.Wt): 40 g/ mol
Preparation of 0.8 M Glycine pH 7.0 buffer
Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).
Calculation: To prepare 1 L of 0.8 M Glycine (MW = 75.07g/mol).
Weight of solute(g)=0.8×75.07.0x 1.0=60.056 g.
Preparation Method:
Weigh 60.056 g of Glycine and dissolve it in water to make a 1 L solution.
Preparation for required volumes:
To prepare _____ of 0.8 M Glycine (MW = 75.07g/mol).
Weight of solute(g)=0.8×75.07█x _____ L= _____g.
Weigh _____gm of Glycine and dissolve in water to make a _____ L solution.
Calculations: To prepare the 1 M Sodium Hydroxide.
Formula & Calculations:
Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).
Calculation: To prepare 1 L of 1N Sodium Hydroxide (MW = 40.0 g/mol)
Weight of solute(g)=1×40 = 40.00█g.
Preparation Method:
Weigh 40.00 g of Sodium Hydroxide and dissolve in water to make a 1 L solution.
Preparation for different volumes:
To prepare _____ of 1N Sodium Hydroxide (MW = 40.0 g/mol).
Weight of solute(g)=1×40.0 x _____ L= _____gm
Weigh _____gm of Sodium Hydroxide and dissolve in water to make a _____ L solution.
Procedure for Preparation of 0.8 M Glycine pH 7.0.
To prepare a pH 7.0 solution, 0.8 M Glycine solution is pH adjusted with 1 M sodium Hydroxide so that the pH attains 7.0.
Check and record the details:
Preparation Details:
Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry
Date
Glycine | | | |
NaOH | | | |
Final volume prepared | Measured Quantity | pH (7.0 ± 0.5) | Conductivity (40 ± 10 ms/cm) | Storage
Condition | Reagent Record
| | | RT |
Date of Preparation | Prepared By | Checked By
| |

Reagent Details:
Formula & Calculations:
Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).
Calculation: To prepare 1 L of 0.2M Sodium citrate monohydrate (MW = 294.1g/mol)
Weight of solute(g)=1×38.42=38.42g.
Preparation Method:
Weigh 38.42 g of Sodium citrate monohydrate and dissolve in water to make a 1 L solution.
Preparation for different volumes:
To prepare _____ of 0.2M Sodium citrate monohydrate (MW = 192.12g/mol).
Weight of solute(g)=1×38.42 _____L= _____gm
Weigh _____gm of Sodium citrate monohydrate and dissolve in water to make a _____ L
solution.
Check and record the details:
Preparation Details:
Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry
Date
| | | |
Final volume prepared | Weighed Quantity | pH (9.5±0.3) | Conductivity (33±0.3ms/cm) | Storage Condition:
| | | RT
Date of Preparation | Prepared By | Checked By
| |

Reagent Details:

Formula & Calculations:

Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).

Calculation: To prepare 1 L of 2N Sodium Hydroxide (MW = 40.0 g/mol)

Weight of solute(g)=2×40 = 80.00g.

Preparation Method:

Weigh 80.00 g of Sodium Hydroxide and dissolve in water to make a 1 L solution.

Preparation for different volumes:

To prepare _____ of 2N Sodium Hydroxide (MW = 40.0 g/mol).

Weight of solute(g)=2×40.0 x _____ L= _____ gm

Weigh _____ gm of Sodium Hydroxide and dissolve in water to make a _____ L solution.

Check and record the details:

Preparation Details:

Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry

Date

Sodium Hydroxide | | | |

Final volume prepared | Weighed Quantity | pH (14.0) | Conductivity (480 ± 10ms/cm) | Storage Condition:

| | | | RT

Date of Preparation | Prepared By | Checked By

| |

Reagent Details:
Formula & Calculations:
Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).
Calculation: To prepare 1 L of 2N Sodium Hydroxide (MW = 40.0 g/mol)
Weight of solute(g)=2×40 = 80.00g.
Preparation Method:
Weigh 80.00 g of Sodium Hydroxide and dissolve in water to make a 1 L solution.
Preparation for different volumes:
To prepare _____ of 2N Sodium Hydroxide (MW = 40.0 g/mol).
Weight of solute(g)=2×40.0 x _____L= _____gm
Weigh _____gm of Sodium Hydroxide and dissolve in water to make a _____ L solution.
Check and record the details:
Preparation Details:
Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry
Date
Sodium Hydroxide | | | |
Final volume prepared | Weighed Quantity | pH (13.5±0.5) | Conductivity (300±1.0 ms/cm) | Storage
Condition:
| | | RT
Date of Preparation | Prepared By | Checked By
| |

1 2 3 4 5 6

A
CORRIDOR
43.8x2.77

D8 D8 D8
APPROVED RUBBER
BUNG STORE
3.50 X 3.30
REJECT SPM RECEIPT
3.00 X 3.30 3.30 X 4.00 CORRIDOR
2.00x6.15
FINISHED GOODS STAGING
13.15 X 6.10
LABELING STORE D8 D8 4D
8.0 X 6.10

CORRIDOR
6.98x2.00

8D
D4 D8

6D
PACKING OFFICE

B 3.00X2.55

D4
LABELING & PACKING
PB
10.6x5.89

PPM STORE
PACKING 13.9x11.8
VISITOR

QUARANTINE
4.19x3.35
SPM STORE

D4
3.00X2.89

D4

D4
15.1x11.7
2.05X11.7 7.05x2.80
D8

D4
PACKING CHANGE
MALE

SAMPLING
3.0 X 3.0

D4
VISUAL &
INSPECTION

D4
7.60X5.84
3.00X3.00

D4
PACKING CHANGE

UNDERTTEST
FEMALE BUFFER ROOM
3.00 X 3.19 2.95X2.79 CORRIDOR
2.80x27.9

D4
C

D4
DPB

JC

1D
1.50X1.00 CORRIDOR
DPB
24.3X2.00
DPB

DPB

BULK STAGING
20.5x6.00

D4
1.00X1.00
CAPPING
5.83X3.98

VIAL COLLECTION
TRAY
AL LYO LOAD LYO SERVICE AREA
2.59X3.98 5.58X3.98 8.06X3.98
CONVEYOR
SWING

AUTOMATIC FOUR HEAD ALUMINUM CAP
SEALING MACHINE, MODEL-AEWCS-120

4D
4D

D4
D4

D4

D4

D4

D4
PAL 2 PAL 1
4.50X2.50 4.53X2.50

FILL FINISH
PAL 4 PAL 3
5.03X9.16
2.50X5.08 2.50X5.08 CORRIDOR
2.00X22.9

D

D4
EXIT 1 EXIT 2
VIAL COLLECTION
& LOADING TRAY

4.50X2.50 4.53X2.50
36" TURNTABLE
NEW ADD

D4
4D
2D
QA DOCUMENT
CONVEYOR
SWING

STERILE CORRIDOR
DOUBLE LAUNDRY AREA
2.81X16.1
6.24X1.80
UNLOADING AREA
3.07X4.76 DOOR
WASHER
4.88X1.60
3.57X3.16 D4
20.5x10.6
1.50X2.00
MAL
1.23X1.53
4D

4D
FORMULATION PACKING AREA TUMBLE
6.16X2.88 DRYER
4D

1.50X2.00 PAL
1.23X1.50
STABLIZING ZONE

4D

4D
D3 3D
COOLING ZONE

6D
PASS THROUGH
MAL
PAL MAL 2.53X1.50

4D

4D
1.50X1.50 1.47X1.47
STERILIZING ZONE

D4
DISINFECTANT 3D D3
6.16X3.08
PAL QA HEAD
4D

D4

2.53X1.50
2.50x4.23

4D

4D
DRYING ZONE

E DPB
1.00X1.00

6D
4D

MATERIAL PREP WASH
VIAL WASHING 6.98X3.55
UNLOAD AC
5.03X10.0 1.00X2.50 5.45X1.58
6.16X3.55
2.35X1.55

D4
QA MANGER
2.50x4.23
QA OFFICE
4D

DPB
17.9x12.1
4D

D4
CORRIDOR
9.11X2.00
CLEAN JC
2.54X3.00 D4 D4 D4
DPB D4

D4
PAL-2
1.80X1.74
PRIMARY CHANGE PRIMARY CHANGE PRIMARY CHANGE MAL ENTRY
D4 MALE FEMALE VISITOR
COLD ROOM 2.00X4.97
DE-CARTONING 4.50X7.05 4.50X4.97 4.50X4.97
3.59X4.00
2.97X4.46
PAL-1 AUDIT
1.00X3.05
1.80X2.67 2.50x6.70

D4 D4 D4 D4

F DPB

STORAGE
D4 D4 D4 D4 D4 D4 D4

CABIN 1 CABIN 2 CABIN 3 CABIN 4
CORRIDOR 4.00x3.13 4.00x3.13 4.00x3.13 3.05x3.13
24.3X2.40
PRINTER
D6

G

CORRIDOR
2.77x121.4

SERVICE AREA
44.9x23.4
34.8x11.0

H

CORRIDOR
2.77x125.8

D6

I WATER SYSTEM &
ENGINEERING STORE
10.0x11.0
D6

CORRIDOR
3D

R&D ENTRANCE R&D OFFICE 32.7x2.00
6.04x2.27 6.04x2.27

D3
D3 D3 D3 D3
D3 D3
J CORRIDOR
GRADE CNC
A/L
GRADE D
EXIT
GRADE D
CH ROOM
LADIES CH ROOM
GRADE CNC
COB

2.10x2.68 SCRAP OUT 2.19x2.68 2.23x2.68 4.98x2.95
12.1x2.22 5.93x2.68 GRADE CNC GRADE CNC GRADE CNC
MICRO MICRO
DPB MICRO OFFICE
D4 D4 SAMPLE RECEIPT DOCUMENT STORE 3.62x4.11
D3 3.23x4.11 2.81x4.11
D3

3D

DPB DPB D3 D3
DPB D3 D3
GENTS CH ROOM
R&D LAB -1 R&D LAB -2 GRADE CNC GRADE D GRADE CNC
CULTURE ROOM D4 D4 GRADE D 4.98x2.95
COB

4.85x7.00 5.19x7.00 INCUBATION ROOM GRADE C D4 GRADE CNC
8.15x3.93 4.54x3.93 CH-I A/L
5.95x2.10 3.82x2.10 CORRIDOR
1.68x8.09
GRADE C 2.80x8.35
BSC CORRIDOR
FRIDGE 1.69x8.55
D3
D4 GRADE D D4
OBSERVATION GRADE B GRADE C BET ROOM
D3 EXIT-II
A/L
EXIT-I 3.62x2.10 GRADE CNC
IC-3 3.23x2.10 2.81x2.10 4.98x2.95
IC-2
GRADE CNC GRADE C D4
D3
IC-4 INCUBATOR MLT ROOM D4 D4
6.35x3.93 IC-1 4.54x3.93 DPB D4
GRADE B GRADE B
CH-II GRADE C
GRADE B CH-III
2.10x2.10 2.84x2.10 CH-I GRADE D
DPB CORRIDOR 2.41x2.10
D3 D4 D4 A/L

K OBSERVATION D3
3.98x2.10
2.10x1.51
2.80x4.42 MEDIA
STORE
4.98x3.69
R&D LAB -3 D3 GRADE B
R&D LAB-4 IC-3 COOL ZONE D3
5.19x7.00 3.52x3.93
GRADE CNC IC-2 GRADE B
4.85x7.00 2.21x1.99
INCUBATOR STERILITY D3
IC-4 6.35x3.93
TESTING ROOM
IC-1 INCUBATOR DPB
4.54x3.93
DPB

FOR MEDIA

D3 DPB IC-1
DPB

D3 MEDIA PREP.
 PREPARATION & STERILIZATION GRADE D
 5.84x6.14 7.88x4.04
 DECONTAMINATION ROOM
 3D

D3

MEDIA DIST. WASH AREA GRADE D
 7.22x2.10 5.47x2.10 5.74x2.10
 D3 AC DPB D3 D3
 6D

D1 D3 LADIES CH ROOM D3
 4.92x2.53
 RM CONTROL SAMPLE
 COB

D1 7.08x7.3.45

STABILITY ROOM
 14.6x6.29 D3 GENTS CH ROOM D3
 L 4.92x2.53
 COB

RETENTION
 SAMPLES ROOM
 D1 (COLD ROOM 2 TO 8 °C)
 7.08x3.45 D4 D3
 JANITOR QC SAMPLE

TRANING HALL D3 1.56x2.53
 RECEIPT
 3.25x2.53

12.1x19.8 TOC/LPC ROOM GC ROOM REFERENCE DPB
 5.03x3.10 STANDARD
 4.20x3.10 CORRIDOR
 5.18x3.10
 2.85x14.5

QC SAMPLE PREP.
 D3 D3 4.92x3.23
 D1 INSTRUMENT ROOM I
 HPLC
 7.08x7.32
 D3
 INSTRUMENT ROOM II
 14.6x5.65
 QC MANEGER
 D1 4.92x3.16

M D3
 CORRIDOR
 2.80x27.9
 1.80 M WIDE CORRIDOR
 D3 10.0x1.80
 D3
 D3

HOT ROOM/ BALANCE
 D3
 FUME HOOD GENERAL FTIR DOCUMENT STORE
 ROOM
 6D

5.72x5.27 4.56x3.51 4.10x3.51 4.92x5.63

PM TESTING ROOM
 4.90x5.23
 D3

D3 D1
 QC STAFF
 DOCUMENTATION
 4.72x11.4
 WET LAB
 WASH AREA
 D3 8.77x5.05
 5.72x3.29 D3

N CORRIDOR
 5.53x2.00
 2.00x2.30
 D3 CHEMICAL /GLASS
 WARE STORAGE
 4.90x6.09
 CORRIDOR
 CONFERENCE 4.95x3.97
 8.77x3.75
 CANTEEN

12.9x6.83
2.54x4.01
CORRIDOR 12.1x4.14
8.07x2.00
4.19x2.00 CORRIDOR
29.6x2.12

1D 1D
6D GENTS TOILET
LADIES TOILET
9.60x1.15 8.05x1.15

1D 1D 1D 1D 1D 1D 1D 1D 1D 1D 1D

CORRIDOR
12.8x2.00 1.60x2.37
W.C. W.C. W.C. W.C. W.C. W.C. W.C. W.C. W.C. W.C.
4.73x3.67 1.30x2.37
1.30x2.37 1.30x2.37 1.30x2.37 1.20x2.37 1.30x2.37 1.30x2.37 1.30x2.37 1.30x2.37 1.30x2.37 1.30x2.37
1.30x2.37

O

FIRST FLOOR

Reagent Details:

Formula:

Sodium Acetate(M.Wt): 82.03g/mol.

Acetic Acid (M.Wt): 60.05g/ mol

Preparation of 1.0 M Sodium Acetate pH 5.8 buffer

Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).

Calculation: To prepare 1 L of 1.0 M Sodium Acetate (MW = 82.03g/mol).

Weight of solute(g)=1.0×82.03× 1.0=82.03g.

Preparation Method:

Weigh 82.03g of Sodium Acetate and dissolve in water to make a 1 L solution.

Preparation for required volumes:

To prepare _____ of 1.0 M Sodium Acetate (MW = 82.03g/mol).

Weight of solute(g)=1.0×82.03× _____ L= _____ g.

Weigh _____ gm of Sodium Acetate and dissolve in water to make a _____ L solution.

Calculations: To prepare the 1.0 M Acetic Acid.

Formula & Calculations:

Formula: Use the formula to calculate the molarity:

Molarity (M) = (Per cent by Weight / 100) * (Density in g/mL) * 1000 / Molar Mass.

Example: Molarity of Glacial Acetic Acid

Molar Mass of CH₃COOH: ~60.05 g/mol

Density: ~1.05 g/mL

Purity: 100%

Acetic Acid (MW = 60.05 g/mol).

Calculation:

Molarity = (100 / 100) * 1.05 g/mL * 1000 mL/L / 60.05 g/mol ≈ 17.5 M.

Calculation: To prepare 1 L of 1.0 M.

Applying the formula to prepare the working buffer:

C₁V₁=C₂V₂

C₁ = 17.5 M acetic acid.

V₁ = volume of stock needed to take.

C₂ = desired (working) concentration 1.0 M acetic acid.

V₂ = final total volume of working solution for one litre.

V₁ = (C₂ × V₂)/C₁.

C₂=1.0 M ; V₂= 1.0 L; C₁=17.5 M

V₁ = (1.0 × 1.0)/17.5

V₁ = 0.057 L

Take 0.057 L of the Stock solution of acetic acid and add it to a volumetric flask, making up the volume to 1 L.

To prepare various volumes:

V₁ = volume of stock needed to take _____ L.

C₂ = desired (working) concentration 1.0 M acetic acid.

V₂ = final total volume of working solution _____ L.

V₁ = (C₂ × V₂) /17.5.

V₁= _____ L.

Take V₁ _____ L of the Stock solution of acetic acid and add it to a volumetric flask, making up the final volume _____ L.

Procedure for Preparation of 1.0 M Sodium Acetate pH 5.8.

To prepare a pH 5.8 solution, 1.0 M Sodium Acetate solution is titrated with 1.0 M Acetic acid so that the pH attains pH 5.8.

Preparation Details:

Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry

Date

Sodium Acetate | | | |

Acetic Acid | | | |

Final volume prepared | Measured Quantity | pH (5.8±0.1) | Conductivity (8.5±0.5ms/cm) | Storage

Condition | Reagent Record

| | | |

Date of Preparation | Prepared By | Checked By

| |

Reagent Details:
Preparation of PF-RR-22 working buffer from Stock buffer (PF-RR-21):
Calculations: For the Preparation of 20mM Sodium acetate pH 5.2.
Stock Buffer (PF-RR-21):0.2M Sodium acetate pH 5.35.
Formula:
 $C1 \times V1 = C2 \times V2$
C1 = concentration of stock
V1 = volume of stock you need to take
C2 = desired (working) concentration
V2 = final total volume of working solution
Applying the formula to prepare the working buffer:
 $C1 \times V1 = C2 \times V2$
C1 = 0.2M Sodium Acetate pH 5.35.
V1 = volume of stock you need to take _____ L.
C2 = desired (working) concentration 20mM Sodium Acetate pH 5.2
V2 = final total volume of working solution _____ L.
 $V1 = (C2 \times V2) / C1$.
C2=0.02M ; V2= 1.0 L; C1=0.2M
 $V1 = (0.02 \times 1.0) / 0.2$
V1 = 0.1L
Take 0.1 L (PF-RR-21) Stock buffer solution and dilute with water, and make up to 1L.
Check and record the details:
Preparation Details:
Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry
Date
Sodium Acetate | | | |
Acetic Acid | | | |
Stock buffer (PF-RR-21) taken volume(L) | pH(5.2±0.1) | Conductivity (1.4±0.1ms/cm) | Storage Condition
| Final volume prepared
| | | RT |
Date of Preparation | Prepared By | Checked By
| |

Reagent Details:

Formula:

Sodium Citrate (M.Wt): 258.07.

Citric acid (M.Wt): 192.124

Preparation of PF-RR-09 stock buffer:

Preparation of 0.4M Sodium Citrate pH 7.0

Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).

Calculation: To prepare 1 L of 0.4M Citric Acid (MW = 192.124 g/mol)

Weight of solute(g)=0.4×192.124×1=76.85g.

Preparation Method:

Weigh 76.85g of Citric Acid and dissolve in water to make a 1 L solution.

Preparation for required volumes:

To prepare _____ of 0.4M Citric Acid (MW = 192.124g/mol).

Weight of solute(g)=0.4×76.85x _____ L= _____ g.

Weigh _____ gm of Citric Acid and dissolve in water to make a _____ L solution.

Calculations: To prepare the 0.4M Tris Sodium Citrate.

Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).

Calculation: To prepare 1 L of 0.4M Tris Sodium Citrate (MW = 258.07 g/mol)

Weight of solute(g)=0.4×258.07 x1=103.228g

Preparation Method:

Weigh 103.228g of Tris Sodium Citrate and dissolve in water to make a 1 L solution.

Preparation for different volumes:

To prepare _____ of 0.4M Tris Sodium Citrate (MW = 258.07 g/mol).

Weight of solute(g)=0.4×258.07x _____ L= _____ g.

Weigh _____ gm of Tris Sodium Citrate and dissolve in water to make a _____ L solution.

Procedure for Preparation of PF-RR-09 stock

To prepare a pH 7.0 solution, 0.4 M Tris Sodium Citrate solution is titrated with 0.4 M citric acid so that the pH attains pH 7.0.

For Example, to prepare a 1 L solution, take 0.4 M Tri Sodium Citrate and adjust the pH with the 0.4 M Citric acid to get the pH 7.0 and make up the final volume to 1 L.

Check and record the details:

Preparation Details:

Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry

Date

Sodium Citrate | | | |

Citric Acid | | | |

0.4 M Tris Sodium Citrate Volume taken | 0.4 M Citric acid volume used to titrate | pH (7.0 ± 0.1) |

Conductivity (40±0.2ms/cm) | Storage Condition | Final volume prepared

| | | |

Date of Preparation | Prepared By | Checked By

| |

Reagent Details:
Formula & Calculations:
Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).
Calculation: To prepare 1 L of 30mM sodium chlorolride (MW = 58.44g/mol)
Weight of solute(g)=1×1.753=1.753g.
Preparation Method:
Weigh 1.753 g of sodium chloride
To prepare 25mM Tris 1L solution from 1M Tris stock solution formula is M1(Molarity of stock) x V1
(volume of stock)= M2 (Molarity of working) xV2 (volume of working).
Take 25mL of 1M tris solution and add weighed amount of sodium chloride as required and dissolve in water
to make a 1 L solution.
Preparation for different volumes:
To prepare _____ of 30mM NaCl (MW = 58.44g/mol).
Weight of solute(g)=1.753x _____ L= _____ gm
Weigh _____ gm of NaCl.
To prepare _____ of 25mM Tris
Take 25mL x _____ L= _____ mL of 1M Tris
Dissolve pre weighed NaCl in 25mM Tris solution to make a _____ L solution.
Adjust the pH to 7.5 using concentrated HCl/NaOH solution
Check and record the details:
Preparation Details:
Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry
Date
| | | |
Final volume prepared | Weighed Quantity | pH (7.5±0.2) | Conductivity (0.6±0.2 ms/cm) | Storage
Condition:
| | | | RT
Date of Preparation | Prepared By | Checked By
| |

FOR OFFICE USE
4
4

2.5MX2.5MX3M BAFFLE WALL 5MX2.5MX3M DESILTING B
RAIN WATER HARVESTING CHAMBER 12.57

2.50 1.50 7.65

PIT MBBR

2.10
WD-3.30M
MH
NATURAL
VENTILATION
ANOXIC
WD-3.50M
CUTOUT PLAN OF PROPOSED FACTORY
1.50 2.42 5.00 BUILDING ON PLOT NO 8A, SECTOR 13,
A MH
A DMIC, AURIC , SHENDRA,

2.10

2.10
SLUDGE FILTER SETTLER
WD-2.90M SWD-2.10M
COLLECTION
WD-3.40M
WD-3.50M
CHATRAPATI SAMBHAJI NAGAR

8.17
MH MH
MH FOR M/S LIFENITY HEALTH LIMITED
6.15 3.00

25X25MM
MH
OGT

1.2MX2.5MX1.5M
BAR SCREEN
PLANT ROOM

3.50

3.50
MANHOLE TREATED
600 MM x 600
MM
OGT WD-3.50M

GREASE TRAP BS MH PSF ACF MH
ENTRANCE

LAYOUT AREA STATEMENT
MANHOLE
B NO PARTICULARS
AREA IN
SQM.

1 AREA OF PLOT 81920.00

AS PER OWNERSHIP DOCUMENTS 81920.00
NATURAL

3.00
VENTILATION
CUTOUT AS PER MEASUREMENT SHEET 81920.00
TOP OF STP TANK
LVL: (+1.2M) MH MH MH MH MH
AS PER SITE 81920.00
ABOVE GROUND
PLANT ROOM
GROUND LEVEL ANOXIC SLUDGE 2 DEDUCTIONS FOR
LVL: (±0.0M) SETTLER WD-3.50M WD-3.50M

2.30
COLLECTION SLUDGE FILTER
WD-3.40M SWD-2.10M a. PROPOSED DP ROAD
WD-3.50M WD-2.90M 3.50 0.00

2.10

4.00
1.75
3.50

3.50
3.40
b. ANY RESERVATIONS 0.00

2.90
TOTAL (a+b) 0.00

1.30
1.00

1.00
3 BALANCE AREA OF PLOT (1-2) (A) 81920.00

0.30
BOTTOM OF STP
TANK LVL: (-2.8M)
4 AMENITY SPACE IF APPLICABLE 5% 0.00
2.50 1.50 2.42 5.00 2.10 2.10
4.44 a. Required 0.00
12.11

RWHP FLOOR PLAN - SECTION A-A SECTION B-B b. Adjustment of 2(b) if any

c. Balanced proposed
0.00

0.00

S.T.P. DETAILS 5 NET PLOT AREA (3-4(c))) PRO RATA 81920.00

SCALE - NTS 6 RECREATIONAL OPEN SPACE IF APPLICABLE 10%

a. Required 8192.00

b. Proposed 8210.48

7 INTERNAL ROAD AREA 0.00

8 PLOTABLE AREA IF APPLICABLE (as per sanctioned layout) 0.00

9 B/UP area with reference to basic fsi as per
front road width(1.0) 81920.00

10 ADDITIONAL FSI ON PAYMENT OF PREMIUM
a. Maximum permissible premium FSI / TOD Zone. 1,63,840.00
b. Proposed FSI of payment of Premium

E WASTE
CURING RACK CURING RACK
100 CRATES 100 CRATES 11 IN SITU TDR / FSI LOADING
CLIENT
SCOPE

E a. In situ area against Statutory Raod (2 xSr No 2(a)) if any 0.00
SERVICE WATER INLET

DRY WASTE
ProEco
SCOPE

POLY ACID ALKALI PAC POLY
BATTERY LIMITS
HYPO WATER REQUIREMENT 12 ADDITIONAL FSI AREA UNDER CHAPTER NO.6 0.000
DOSING
UNIT
DOSING
UNIT
DOSING
UNIT
DOSING
UNIT
DOSING
UNIT
DOSING
UNIT
PIPE CODING DOSING
UNIT 13 TOTAL ENTITLEMENT OF FSI IN THE PROPOSAL 0.000
EFFLUENT LINE A
PLAZMA & LVP INLET
TOTAL B/UP AREA = 42467.00 SQ.M. a. 9 +10(b) +11(a) or 9+12 which ever is applicable
Permissible 81920.00
DU-106 DU-101 DU-102 DU-103 DU-104 SLUDGE LINE
B
UTILITY INLET DU-105 OCCUPANT LOAD = 42467.00/10 = 4246 NOS. a. 9 +10(b) +11(a) or 9+ 12 which ever is
applicable
C
TREATED LINE DIALYZER INLET 42467.00
OCCUPANT LOAD actual = 1400 NOS. Proposed

BIO WASTE
D EXCESS STP TREATED WATER INLET (IF REQUIRED
BACKWASH LINE b. Permissible Ancillary area FSI up to 80% with payment
E
65536.00

SLUDGE FILTRATE LINE SERVICE WATER INLET
NO ACTIVITY WATER IN LITS. PERMISSIBLE
b. Proposed Ancillary area FSI up to 60% or 80% with payment
F
CURING RACK

ENTRANCE
AB-101 SERVICE WATER LINE DRIED SLUDGE DISPOSAL OUTLET
OWC MACHINE 0.00
DOSING LINE
G
TREATED OUTLET 1 FOR W.C. 45 X 1400 = 63000 80-120 KG/HR 100 CRATES c.Total Entitlement permissible
(a+b) 108003.00
AB-102
2 FOR INDUSTRIAL 500000 LITS/DAY c.Total Entitlement proposed
(a+b) 42467.00
Maximum utilization limit of FSI (Building Potential) permissible as
TOTAL /DAY 563000 LITR/DAY 14 per road width ((as per Regulation No. 5.5 of Chapter - 5)) x 1.6 or 1.8
) 108003.00
AIR LINE
ORGANIC WEST CONVERTER
15 TOTAL Built up area in proposal (excluding area as per sr. no 17b)

AIR GRID
TOTAL CAPACITY = 5,63,000 LITS a. Existing Built -up Area. 0.000
MBBR MEDIA
PMP-101
CLIENT
SCOPE
b. Proposed Built- up area (as per P line) 42467.00
A
PLAZMA & LVP INLET c. Total (a+b) 42467.00
RCC SLAB
ProEco

FLOW: 127.0 KLD
SCOPE

PMP-102 16 FSI CONSUMED (15/13) (should not be more than serial 0.52
PH: 10 - 12 FM-101 FL-101 AIR DIFFUSERS G.L. G.L. No.14 above.)
BOD: < 1100 mg/lit
COD: < 2250 mg/lit PST-101 MBBR-101 SST-101
EQT-101
TSS: <500 mg/lit
TDS: 2000 - 2500 mg/lit
O&G: < 60 mg/lit CERTIFICATE OF AREA :-
Certified that the plot under reference was surveyed by me on
PMP-104

PMP-106
PMP-103

PMP-105

03/03/2026 and the Dimensions of sides etc.of plot stated on plan area
CLIENT
PMP-109 OCCUPANT LOAD AS PER TABLE NO 8E as measured on site and the area so worked out tallies with the
stated in
SCOPE
documents of ownership/ DMIC/Land records Department/City Survey
F
VOLUTE PRESS
CLAUSE NO. 10.6.3 CDCPR Records
SOLID TO DISPOSAL
PANEL
Ar. Dashrath Dhumal
ProEco
SCOPE

PMP-110 PCC BED
SHT-101 CLIENT
SCOPE
SECTION AT CC OCCUPANT LOAD CALATION Owners Declaration :-
B
To EQT OR SHT UTILITY INLET U.G. WATER TANK I/We undersigned hereby confirmed that I/We would abide by
plans
approved by Authority/ Collector. I / We would execute the structure as
FLOW: 15.0 KLD
TDS: 1000 mg/lit
ProEco
SCOPE

BUILT UP AREA 42467.00 SQMT per the approved plans.Also I/We would execute the work under
supervision of proper technical person so as to ensure the quality and
CLIENT CLIENT
safety at the work site.
10
SCOPE SCOPE

EXCESS STP TREATED
D
DIALYZER INLET

C AREA PER PERSON SQMT LIFENITY HEALTH LIMITED.
ProEco FLOW: 31.0 KLD ProEco
Architects Name and Signature
4247
EXCESS STP TREATED WATER SCOPE PH: 7.5 SCOPE
BOD: 15 mg/lit

NOS
IF REQUIRED FOR DILUTION

NO OF PERSONS
COD: 40 mg/lit
TSS: 20 mg/lit
TDS: 405 mg/lit
O&G: 5 mg/lit

Ar. Dashrath Dhumal
PMP-111 PMP-107
CLIENT
SCOPE
Job No. Drg. No. Scale Drn. By. Checked By Registration no of Architect
G
001 1:100 Ar. Dashrath Dhumal CA/2005/37017
TREATED OUTLET TO CETP AS PER ACTUAL OCCUPANTS IN THE BUDILING
ProEco
SCOPE

PMP-112
A.C.F-101 P.S.F-101 PMP-108
NO. OF OCCUPANTS - 1400 NOS.
C PARKING STATEMENT
PH: 5.5 - 9.0
BOD: < 700 mg/lit
COD: < 1000 mg/lit TWT-101 FFT-101
U.G. WATER TANK PLAN NO. OF OFFICE STAFF - 200 NOS.
a GARAGE PERMISSIBLE
PARKING
GARAGE PROPOSED NIL
PARKING
TSS: < 600 mg/lit b TYPE
REQUIRED PROPOSED
TDS: < 2100 mg/lit
WATER CAPACITY 10 X 6 X 3.00 = 180000 X4 =720000 LITS NO. OF WORKERS - 1200 NOS.
CAR

SCOOTER
8192 SQ.M. 8192.96 SQ.M.
CYCLE
SANITATION REQUIREMENT (TABLE NO. 10D) VISITORS
D LOADING UNLOADING SPACES
OCCUPANT LOAD 42467.00 / 10 4247
E T P DETAILS SCALE :- NTS a LOADING UNLOADING REQUIRED
b TOTAL LOADING UNLOADING PROVIDED
44
44

TOTAL NO. OF PERSON 4247 LEGENDS
PLOT BOUNDRY SHOWN IN
B/UP AREA SHOWN IN
FIXTURES OFFICES / VISITORS WORKERS
WATER LINE SHOWN IN
GREY WATER LINE SHOWN IN
200 1200 SEWAGE LINE SHOWN IN
EFFLUNT LINE
100 100 800 400
SCHEDULE OF DOORS AND WINDOWS
TYPE SIZE DISCRPTION

Treated Water Tank : 100m3 × 1 MALE FEMALE MALE FEMALE RS1 2500 X 6000 M.S ROLLING SHUTTER

D1 1800 X 2400 FLUSH DOOR

D2 900 X 2400 FLUSH DOOR

Temporary Tank : 5m3 × 1 1 FOR 25 1 FOR 15 1 FOR 20 1 FOR 15
D3 750 X 2400 FLUSH DOOR

Scubber FD 1000 X 2400 FIRE DOOR

after 2.5% thereafter 3% thereafter 2.5% thereafter 4% W 2500 X 2500 ALLU. SLIDING WINDOW

REQUIRED WC V 600 X 900 ALU LOUVERED VENTILATOR
Tank Area for Waster water and 60% DMAC

1+2 = 3 1+3=4 1+19.50=21 1+15.40=16
PLAN OF PROPOSED FACTORY
Purification Unit BUILDING ON PLOT NO 8A, SECTOR 13,
Tank Area for Treated Water 1 up to10 1 up to10
12.0m × 8.0m × H30m DMIC, AURIC , SHENDRA,
(out of our scope) CROSS DRAIN thereafter 2.5% - thereafter 2.5% -
12.0m × 8.0m × H15m REQUIRED URINALS CHATTRAPATI SAMBHAJI NAGAR
Raw Material Tank : 100m3 × 2

60% DMAC Tank : 100m³ × 1

1+2 = 3 1+19.75 = 21 FOR M/S LIFENITY HEALTH LIMITED
OFF NO 301, S NO 5, PLOT
NO 1, BLDG A 1 , Navkar Avenue, NDA Pashan
Tank Area for DMAC
5.0m × 24.0m × H15m

TOTAL W.C. REQUIRED = 3+4+21+16= 44 road, BHAVDHAN, PUNE 411021
13.0m × 20.0m × H10m * Concentration Unit
* 12.0m × 8.0m × H20m RAIN WATER HARVESTING TANK CAPACITY Phone no: 7276066666
AVERAGE ANNUAL RAINFALL=100MM (0.1M/HOUR) Email : slandge@lifenitywellness.com
DMAC Product Tank : 14m³ × 2 TOTAL URINALS REQUIRED = 3+21= 24
PLOT AREA= 81920 SQ.M
DMAC Feed Tank : 30m³ × 2
VOLUMETRIC RAINFALL= 81920 X 0.1= 8192 CUM/HOUR
DMAC & PVP Tank : 10m³ × 1
= 136.53 CUM/MINUTE
DMAC Return Tank : 10m³ × 1 Dryer SIGN OF DIRECTOR
RAIN WATER TANK CAPACITY PROPOSED= 5x 5 X 3 X 2= 150 CUM
DMAC&HB Tank : 1m³ × 1
Concentration Unit RAIN WATER TANK CAPACITY PROPOSED= 150000 LITER REQUIRED PROVIDED : ARCHITECT
12.0m × 8.0m × H20m NO. OF W.C. FOR STAFF (MALE) 3 30 Vastushilp Architects,
#03,Alok Nagar, Beed bypass Road,
RAIN WATER
PIPE
NO. OF URINALS FOR STAFF 3 30 Chh. Sambahi Nagar 431001
Ar. Dashrath K. Dhumal.
NO. OF W.C. FOR STAFF (FEMALE) 4 30
GL GL
MURUM
FILLING
SAND
Phone: 9850207562 Email: darchitect.dashrath@gmail.com
FILLING

NO. OF W.C. FOR WORKERS (MALE)
Recovery plant RUBBLE
FILLING

WATER HARVESTING SECTION
21 60

(SCALE - 1:50) NO. OF URINALS FOR WORKERS 21 60
RAINWATER PIT CAPACITY 5.00 X 5.00 X3 x 2 nos = 150 K LTS

NO. OF W.C. FOR WORKERS 16 60
(FEMALE)
JOB No. DRG NO. SCALE DATE DRN BY CHK. BY

01 1 :100

Reagent Details:

Formula:

Sodium Acetate(M.Wt): 82.03g/mol.

Acetic Acid (M.Wt): 60.05g/ mol

Preparation of 0.2 M Sodium Acetate pH 5.35 buffer

Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).

Calculation: To prepare 1 L of 0.2 M Sodium Acetate (MW = 82.03g/mol).

Weight of solute(g)=0.2×82.03× 1.0=16.40g.

Preparation Method:

Weigh 16.40g of Sodium Acetate and dissolve in water to make a 1 L solution.

Preparation for required volumes:

To prepare _____ of 0.2 M Sodium Acetate (MW = 82.03g/mol).

Weight of solute(g)=0.2×82.03× _____ L= _____ g.

Weigh _____ gm of Sodium Acetate and dissolve in water to make a _____ L solution.

Calculations: To prepare the 0.2 M Acetic Acid.

Formula & Calculations:

Formula: Use the formula to calculate the molarity:

Molarity (M) = (Per cent by Weight / 100) * (Density in g/mL) * 1000 / Molar Mass.

Example: Molarity of Glacial Acetic Acid

Molar Mass of CH₃COOH: ~60.05 g/mol

Density: ~1.05 g/mL

Purity: 100%

Acetic Acid (MW = 60.05 g/mol).

Calculation:

Molarity = (100 / 100) * 1.05 g/mL * 1000 mL/L / 60.05 g/mol ≈ 17.5 M.

Calculation: To prepare 1 L of 1.0 M.

Applying the formula to prepare the working buffer:

C₁V₁=C₂V₂

C₁ = 17.5 M acetic acid.

V₁ = volume of stock needed to take.

C₂ = desired (working) concentration 0.2 M acetic acid.

V₂ = final total volume of working solution for one litre.

V₁ = (C₂ x V₂)/C₁.

C₂=0.2 M ; V₂= 1.0 L; C₁=17.5 M

V₁ = (0.2 x 1.0)/17.5

V₁ = 0.011 L

Take 0.011 L of the Stock solution of acetic acid and add it to a volumetric flask, making up the volume to 1 L.

To prepare various volumes:

V₁ = volume of stock needed to take _____ L.

C₂ = desired (working) concentration 0.2 M acetic acid.

V₂ = final total volume of working solution _____ L.

V₁ = (C₂ x V₂) /17.5.

V₁= _____ L.

Take V₁ _____ L of the Stock solution of acetic acid and add it to a volumetric flask, making up the final volume _____ L.

Procedure for Preparation of 0.2 M Sodium Acetate pH 5.35.

To prepare a pH 5.35 solution, 0.2 M Sodium Acetate solution is titrated with 0.2 M Acetic acid so that the pH attains pH 5.35.

Preparation Details:

Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry Date

Sodium Acetate | | | |

Acetic Acid | | | |

Final volume prepared | Measured Quantity | pH (5.35±0.1) | Conductivity (11.5±0.1ms/cm) | Storage Condition | Reagent Record

| | | | RT |

Date of Preparation | Prepared By | Checked By

| |

Reagent Details:

Formula & Calculations:

Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).

Calculation: To prepare 1 L of 0.2M Sodium Citrate (MW = 258.06g/mol)

Weight of solute(g)=0.2×258.06=51.66g.

Preparation Method:

Weigh 51.66 g of Tri-sodium citrate and dissolve in water to make a 1 L solution.

Preparation for different volumes:

To prepare _____ volume of 0.2M Sodium Citrate (MW = 258.06g/mol).

Weight of solute(g)=0.2×258.06x _____L= _____gm

Weigh _____gm of Tri-sodium citrate and dissolve in water to make a _____ L solution.

Check and record the details:

Preparation Details:

Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry Date

||| |

Final volume prepared | Weighed Quantity | pH (8.0 to 9.0) | Conductivity (1.5 to 2.5 ms/cm) | Storage

Condition:

||| | RT

Date of Preparation | Prepared By | Checked By

||

Date- 19.05.2026

List of Documents

S/No	Title
1	Organic waste calculation and parameters for ETP

Contact Details:

1. Dr. Ganesh Kumraj: ganesh@biobridge.in

Technical Information # BB-LN/IND/002

LIFT
2.70X2.50
Goods LIFT
3.50X3.00
LIFT
2.70X2.50
Goods LIFT
3.50X3.00
LIFT
2.70X2.50 B B AS PER MEASUREMENT SHEET 81920.00

FIRE STAIR FIRE STAIR FIRE STAIR
AS PER SITE 81920.00
up up up

2 DEDUCTIONS FOR
UP

a. PROPOSED DP ROAD 0.00

b. ANY RESERVATIONS 0.00
TOTAL (a+b) 0.00

W

W

W
B B B B
BALANCE AREA OF PLOT (1-2) (A)
MEZZANINE FLOOR

24.77 X 12.54 C C W 3 81920.00

4 AMENITY SPACE IF APPLICABLE 5% 0.00

W

W

W
MEZZANINE FLOOR MEZZANINE FLOOR MEZZANINE FLOOR
H.V.A.C. UNITS
6.77 X 29.04
H.V.A.C. UNITS
6.77 X 29.04
H.V.A.C. UNITS
6.77 X 29.04
a. Required 0.00
b. Adjustment of 2(b) if any 0.00
W
c. Balanced proposed 0.00

W

W
W

W

W
MEZZANINE FLOOR MEZZANINE FLOOR MEZZANINE FLOOR
H.V.A.C. UNITS H.V.A.C. UNITS H.V.A.C. UNITS
6.77 X 16.27 6.77 X 16.27 6.77 X 16.27
5 NET PLOT AREA (3-4(c))) PRO RATA 81920.00
Y
ADMIN / WORKSHOP
W

W

W
W

W

W
6 RECREATIONAL OPEN SPACE IF APPLICABLE 10%
BELOW 8192.00
a. Required

E E b. Proposed 8210.48
Pass. LIFT
2.30X2.00
Pass. LIFT
2.30X2.00
Pass. LIFT
2.30X2.00 D X D 7 INTERNAL ROAD AREA 0.00
DIALYZER L.V.P.
C 8 PLOTABLE AREA IF APPLICABLE (as per sanctioned layout) 0.00
BELOW BELOW
9 B/UP area with reference to basic fsi as per

front road width(1.0) 81920.00

10 ADDITIONAL FSI ON PAYMENT OF PREMIUM
a. Maximum permissible premium FSI / TOD Zone. 1,63,840.00
b. Proposed FSI of payment of Premium
W

W

W
W

W

W
11 IN SITU TDR / FSI LOADING
W

W

W
W

W

W
MEZZANINE FLOOR MEZZANINE FLOOR MEZZANINE FLOOR MEZZANINE FLOOR MEZZANINE FLOOR MEZZANINE FLOOR
H.V.A.C. UNITS H.V.A.C. UNITS H.V.A.C. UNITS H.V.A.C. UNITS H.V.A.C. UNITS H.V.A.C. UNITS
6.77 X 17.04 6.77 X 29.04 6.77 X 17.04 6.77 X 29.04 6.77 X 17.04 6.77 X 29.04
a. In situ area against Statutory Raod (2 xSr No 2(a)) if any 0.00

12 ADDITIONAL FSI AREA UNDER CHAPTER NO.6 0.000

13 TOTAL ENTITLEMENT OF FSI IN THE PROPOSAL 0.000
W

W

W
a. 9 +10(b) +11(a) or 9+12 which ever is applicable
81920.00
C C Permissible
a. 9 +10(b) +11(a) or 9+ 12 which ever is applicable
W

W

W
Proposed 42467.00
b. Permissible Ancillary area FSI up to 80% with payment
65536.00
PERMISSIBLE
b. Proposed Ancillary area FSI up to 60% or 80% with payment 0.00
c.Total Entitlement permissible
(a+b) 108003.00
c.Total Entitlement proposed
(a+b) 42467.00
A A
Maximum utilization limit of FSI (Building Potential) permissible as
14 108003.00
Pass. LIFT
2.30X2.00
Pass. LIFT
2.30X2.00
Pass. LIFT
2.30X2.00 D per road width ((as per Regulation No. 5.5 of Chapter - 5)) x 1.6 or 1.8)

E 15 TOTAL Built up area in proposal (excluding area as per sr. no 17b)

a. Existing Built -up Area. 0.000
b. Proposed Built- up area (as per P line) 42467.00
W

W

W
Goods LIFT
3.50X3.00 MEZZANINE FLOOR
H.V.A.C. UNITS
Goods LIFT
3.50X3.00 MEZZANINE FLOOR
H.V.A.C. UNITS
Goods LIFT
3.50X3.00 MEZZANINE FLOOR
H.V.A.C. UNITS
B B c. Total (a+b) 42467.00
6.77 X 11.80 6.77 X 11.80 6.77 X 11.80
W

W

W

1.10X
1.70
1.10X
1.70 CAR
6.77 M WIDE CORRIDOR

6.77 M WIDE CORRIDOR
height 11 m
D1 D1 D1 D1 D1
SCOOTER
WC
1.10X
WC
1.10X
WC
1.10X
WC
1.10X
WC
1.10X
WC
1.10X
WC
1.10X
WC
1.10X
WC
1.10X
WC
1.10X 8192 SQ.M. 8192.96 SQ.M.
1.70 1.70 1.70
FIRE PUMP
1.70 1.70 1.70
W
1.70 1.70 1.70
CYCLE
ROOM
3.50X5.00
W

UP
VISITORS

UP UP D LOADING UNLOADING SPACES
C
PHARMACEUTICAL PARTITION

PHARMACEUTICAL PARTITION

PHARMACEUTICAL PARTITION
PHARMACEUTICAL PARTITION

a LOADING UNLOADING REQUIRED 44
W W W W
b TOTAL LOADING UNLOADING PROVIDED 44
LOBBY LOBBY LOBBY
6.77X11.77 6.77X11.77 6.77X11.77

LEGENDS
LOBBY D1 D1
6.77X11.77
STAIRCASE STAIRCASE STAIRCASE
AND AND AND
LIFT ONLY W LIFT ONLY W W LIFT ONLY W
FOR
FIRST FL0OR
FOR
FIRST FL0OR
FOR
FIRST FL0OR
PLOT BOUNDRY SHOWN IN
(IVD) (IVD) (IVD)

UP UP
B/UP AREA SHOWN IN
WATER LINE SHOWN IN
Pass. LIFT
2.30X2.00
Pass. LIFT
2.30X2.00
BUILDING A
Pass. LIFT
2.30X2.00
Pass. LIFT
2.30X2.00
BUILDING A GREY WATER LINE SHOWN IN
BUILDING A WC WC WC WC WC WC BUILDING A WC WC WC
SEWAGE LINE SHOWN IN
W

TERRACE
1.10X 1.10X 1.10X 1.10X 1.10X 1.10X 1.10X 1.10X

DIALYZER

1.70 1.70 1.70
L.V.P. 1.70 1.70 1.70
ADMIN / WORKSHOP
1.70 1.70 1.70

EFFLUNT LINE
53.54 x 86.11
D1
D1
53.54 x 86.11 D1
D1 53.54 x 86.11
D1

WC WC WC WC WC WC WC WC WC
1.10X 1.10X 1.10X 1.10X 1.10X 1.10X 1.10X 1.10X 1.10X
1.70 1.70 1.70 1.70 1.70 1.70 1.70 1.70 1.70 SCHEDULE OF DOORS AND WINDOWS
TYPE SIZE DISCRPTION

RS1 2500 X 6000 M.S ROLLING SHUTTER

D1 1800 X 2400 FLUSH DOOR
UP
D2 900 X 2400 FLUSH DOOR

D3 750 X 2400 FLUSH DOOR

W W W W W FD 1000 X 2400 FIRE DOOR

W 2500 X 2500 ALLU. SLIDING WINDOW
LOBBY V 600 X 900 ALU LOUVERED VENTILATOR
D1 6.77X11.77 D1 D1

W W
W W W

UP
PLAN OF PROPOSED FACTORY
UP BUILDING ON PLOT NO 8A, SECTOR 13,
2.80 M WIDE PASSAGE

2.80 M WIDE PASSAGE

2.80 M WIDE PASSAGE

WC WC WC WC WC WC WC WC WC
DMIC, AURIC , SHENDRA,
1.10X 1.10X 1.10X 1.10X 1.10X 1.10X 1.10X 1.10X 1.10X

CHATTRAPATI SAMBHAJI NAGAR
1.70 1.70 1.70 1.70 1.70 1.70 1.70 1.70 1.70
LOBBY
6.77X11.00
LOBBY
6.77X11.00
LOBBY
6.77X11.00 D
WC
1.10X
1.70
WC
1.10X
1.70
WC
1.10X
1.70
STAIRCASE
AND
LIFT ONLY
WC
1.10X
1.70
WC
1.10X
1.70
WC
1.10X
1.70
STAIRCASE
AND
LIFT ONLY
WC
1.10X
1.70
WC
1.10X
1.70
WC
1.10X
1.70
STAIRCASE
AND
LIFT ONLY

01 1 :100

Reagent Details:
Formula & Calculations:
Formula: $M1 \times V1 = M2 \times V2$.
Calculation: To prepare 1 L of 0.01M sodium hydroxide from 0.5M sodium hydroxide stock solution calculate quantity required of stock solution as below:
0.5M sodium hydroxide stock solution required quantity= $0.01 \times 1 / 0.5 = 0.02$ L
Preparation Method:
To prepare 0.01M sodium hydroxide 1L solution from 0.5M sodium hydroxide stock solution formula is $M1(\text{Molarity of stock}) \times V1(\text{volume of stock}) = M2(\text{Molarity of working}) \times V2(\text{volume of working})$.
Take 0.02L of 0.5M sodium hydroxide solution and make a 1 L solution.
Preparation for different volumes:
To prepare _____ of 0.01M sodium hydroxide
Take 0.02L x _____ L = _____ L of 0.5M sodium hydroxide
Make sodium hydroxide solution to _____ L solution.
Check and record the details:
Preparation Details:
Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry
Date
|||
Final volume prepared | Weighed Quantity | Storage Condition:
||
Date of Preparation | Prepared By | Checked By (QC/Analyst)
||

Reagent Details:

Formula & Calculations:

Formula: Use the formula to calculate the molarity:

Molarity (M) = (Per cent by Weight / 100) * (Density in g/mL) * 1000 / Molar Mass.

Example: Molarity of Hydrochloric Acid

Molar Mass of HCL: ~36.46 g/mol

Density: ~1.18 g/mL

Purity: 37%

HCL Acid (Molarity = 12.0M).

Calculation:

Molarity = (37 / 100) * 1.18 g/mL * 1000 mL/L / 36.46 g/mol ≈ 12M.

Calculation: To prepare 1 L of 1.0 M.

Applying the formula to prepare the working buffer:

$C1 \cdot V1 = C2 \cdot V2$

$C1 = 12$ M HCL acid.

$V1$ = volume of stock needed to take.

$C2$ = desired (working) concentration 1.0 M HCL acid.

$V2$ = final total volume of working solution for one litre.

$V1 = (C2 \times V2) / C1$.

$C2 = 1.0$ M ; $V2 = 1.0$ L; $C1 = 12$ M

$V1 = (1.0 \times 1.0) / 12$.

$V1 = 0.0833$ L

Take 0.0833 L of the Stock solution of HCL acid and add it to a volumetric flask, making up the volume to 1 L.

To prepare various volumes:

$V1$ = volume of stock needed to take _____ L.

$C2$ = desired (working) concentration 1.0 M HCL acid.

$V2$ = final total volume of working solution _____ L.

$V1 = (C2 \times V2) / 12$.

$V1 =$ _____ L.

Take $V1$ _____ L of the Stock solution of HCL acid and add it to a volumetric flask, making up the final volume _____ L.

Preparation Details:

Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry

Date

HYROCHLORIC ACID | | | |

Final volume prepared | Measured Quantity | pH | Conductivity | Storage Condition | Reagent Record

| | | |

Date of Preparation | Prepared By | Checked By (QC/Analyst)

| |

Goods LIFT
3.50X3.00
LIFT
2.70X2.50
Goods LIFT
3.50X3.00
LIFT
2.70X2.50 B B AS PER MEASUREMENT SHEET 81920.00

FIRE STAIR FIRE STAIR FIRE STAIR
AS PER SITE 81920.00
up up up

2 DEDUCTIONS FOR
UP

a. PROPOSED DP ROAD 0.00

b. ANY RESERVATIONS 0.00
TOTAL (a+b) 0.00

W

W

W
B B B B
BALANCE AREA OF PLOT (1-2) (A)
MEZZANINE FLOOR

24.77 X 12.54 C C W 3 81920.00

4 AMENITY SPACE IF APPLICABLE 5% 0.00

W

W

W
MEZZANINE FLOOR MEZZANINE FLOOR MEZZANINE FLOOR
H.V.A.C. UNITS
6.77 X 29.04
H.V.A.C. UNITS
6.77 X 29.04
H.V.A.C. UNITS
6.77 X 29.04
a. Required 0.00
b. Adjustment of 2(b) if any 0.00
W
c. Balanced proposed 0.00

W

W
W

W

W
MEZZANINE FLOOR MEZZANINE FLOOR MEZZANINE FLOOR
H.V.A.C. UNITS H.V.A.C. UNITS H.V.A.C. UNITS
6.77 X 16.27 6.77 X 16.27 6.77 X 16.27
5 NET PLOT AREA (3-4(c))) PRO RATA 81920.00
Y
ADMIN / WORKSHOP
W

W

W
W

W

W
6 RECREATIONAL OPEN SPACE IF APPLICABLE 10%
BELOW 8192.00
a. Required

E E b. Proposed 8210.48
Pass. LIFT
2.30X2.00
Pass. LIFT
2.30X2.00
Pass. LIFT
2.30X2.00 D X D 7 INTERNAL ROAD AREA 0.00
DIALYZER L.V.P.
C 8 PLOTABLE AREA IF APPLICABLE (as per sanctioned layout) 0.00
BELOW BELOW
9 B/UP area with reference to basic fsi as per
front road width(1.0) 81920.00

10 ADDITIONAL FSI ON PAYMENT OF PREMIUM

a. Maximum permissible premium FSI / TOD Zone. 1,63,840.00

b. Proposed FSI of payment of Premium

W

W

W

W

W

W

11 IN SITU TDR / FSI LOADING

W

W

W

W

W

W

MEZZANINE FLOOR MEZZANINE FLOOR MEZZANINE FLOOR MEZZANINE FLOOR MEZZANINE FLOOR MEZZANINE FLOOR

H.V.A.C. UNITS H.V.A.C. UNITS H.V.A.C. UNITS H.V.A.C. UNITS H.V.A.C. UNITS H.V.A.C. UNITS H.V.A.C. UNITS

6.77 X 17.04 6.77 X 29.04 6.77 X 17.04 6.77 X 29.04 6.77 X 17.04 6.77 X 29.04 6.77 X 29.04

a. In situ area against Statutory Raod (2 xSr No 2(a)) if any 0.00

12 ADDITIONAL FSI AREA UNDER CHAPTER NO.6 0.000

13 TOTAL ENTITLEMENT OF FSI IN THE PROPOSAL 0.000

W

W

W

a. 9 +10(b) +11(a) or 9+12 which ever is applicable

81920.00

C C Permissible

a. 9 +10(b) +11(a) or 9+ 12 which ever is applicable

W

W

W

Proposed 42467.00

b. Permissible Ancillary area FSI up to 80% with payment

65536.00

PERMISSIBLE

b. Proposed Ancillary area FSI up to 60% or 80% with payment 0.00

c.Total Entitlement permissible

(a+b) 108003.00

c.Total Entitlement proposed

(a+b) 42467.00

A A

Maximum utilization limit of FSI (Building Potential) permissible as

14 108003.00

Pass. LIFT

2.30X2.00

Pass. LIFT

2.30X2.00

Pass. LIFT

2.30X2.00 D per road width ((as per Regulation No. 5.5 of Chapter - 5)) x 1.6 or 1.8)

E 15 TOTAL Built up area in proposal (excluding area as per sr. no 17b)

a. Existing Built -up Area. 0.000

b. Proposed Built- up area (as per P line) 42467.00

W

W

W

Goods LIFT

3.50X3.00 MEZZANINE FLOOR

H.V.A.C. UNITS

Goods LIFT

3.50X3.00 MEZZANINE FLOOR

H.V.A.C. UNITS

Goods LIFT

3.50X3.00 MEZZANINE FLOOR

H.V.A.C. UNITS

B B c. Total (a+b) 42467.00

6.77 X 11.80 6.77 X 11.80 6.77 X 11.80

W

W

W

16 FSI CONSUMED (15/13) (should not be more than serial 0.52

No.14 above.)

RS3 W

FOLDING DOOR

VV

2.94 M WIDE CORRIDOR

1.70 BELOW WC

1.10X
1.70
WC
1.10X
1.70
WC
1.10X
1.70 CAR
6.77 M WIDE CORRIDOR

6.77 M WIDE CORRIDOR
D1 D1 D1 D1 D1
SCOOTER
WC
1.10X
WC
1.10X
WC
1.10X
WC
1.10X
WC
1.10X
WC
1.10X
WC
1.10X
WC
1.10X 8192 SQ.M. 8192.96 SQ.M.
1.70 1.70 1.70
FIRE PUMP
1.70 1.70 1.70
W
1.70 1.70 1.70
CYCLE
ROOM
3.50X5.00
W

UP
VISITORS

UP UP D LOADING UNLOADING SPACES
C
PHARMACEUTICAL PARTITION

PHARMACEUTICAL PARTITION

PHARMACEUTICAL PARTITION
PHARMACEUTICAL PARTITION

a LOADING UNLOADING REQUIRED 44
W W W W
b TOTAL LOADING UNLOADING PROVIDED 44
LOBBY LOBBY LOBBY
6.77X11.77 6.77X11.77 6.77X11.77

LEGENDS
LOBBY D1 D1
6.77X11.77
STAIRCASE STAIRCASE STAIRCASE
AND AND AND
LIFT ONLY W LIFT ONLY W W LIFT ONLY W
FOR
FIRST FL0OR
FOR
FIRST FL0OR
FOR
FIRST FL0OR
PLOT BOUNDRY SHOWN IN
(IVD) (IVD) (IVD)

UP UP
B/UP AREA SHOWN IN
WATER LINE SHOWN IN
Pass. LIFT
2.30X2.00
Pass. LIFT
2.30X2.00
BUILDING A
Pass. LIFT
2.30X2.00
Pass. LIFT
2.30X2.00
BUILDING A GREY WATER LINE SHOWN IN
BUILDING A WC WC WC WC WC WC BUILDING A WC WC WC
SEWAGE LINE SHOWN IN
W

TERRACE

1.10X 1.10X 1.10X 1.10X 1.10X 1.10X 1.10X 1.10X 1.10X

DIALYZER
1.70 1.70 1.70
L.V.P. 1.70 1.70 1.70
ADMIN / WORKSHOP
1.70 1.70 1.70

EFFLUNT LINE
53.54 x 86.11
D1
D1
53.54 x 86.11 D1
D1 53.54 x 86.11
D1

WC WC WC WC WC WC WC WC WC
1.10X 1.10X 1.10X 1.10X 1.10X 1.10X 1.10X 1.10X 1.10X
1.70 1.70 1.70 1.70 1.70 1.70 1.70 1.70 1.70 SCHEDULE OF DOORS AND WINDOWS
TYPE SIZE DISCRPTION

RS1 2500 X 6000 M.S ROLLING SHUTTER

D1 1800 X 2400 FLUSH DOOR
UP
D2 900 X 2400 FLUSH DOOR

D3 750 X 2400 FLUSH DOOR

W W W W W FD 1000 X 2400 FIRE DOOR

W 2500 X 2500 ALLU. SLIDING WINDOW
LOBBY V 600 X 900 ALU LOUVERED VENTILATOR
D1 6.77X11.77 D1 D1

W W
W W W

UP
PLAN OF PROPOSED FACTORY
UP BUILDING ON PLOT NO 8A, SECTOR 13,
2.80 M WIDE PASSAGE

2.80 M WIDE PASSAGE

2.80 M WIDE PASSAGE

WC WC WC WC WC WC WC WC WC
DMIC, AURIC , SHENDRA,
1.10X 1.10X 1.10X 1.10X 1.10X 1.10X 1.10X 1.10X 1.10X

CHATTRAPATI SAMBHAJI NAGAR
1.70 1.70 1.70 1.70 1.70 1.70 1.70 1.70 1.70
LOBBY
6.77X11.00
LOBBY
6.77X11.00
LOBBY
6.77X11.00 D
WC
1.10X
1.70
WC
1.10X
1.70
WC
1.10X
1.70
STAIRCASE
AND
LIFT ONLY
WC
1.10X
1.70
WC
1.10X
1.70
WC
1.10X
1.70
STAIRCASE
AND
LIFT ONLY
WC
1.10X
1.70
WC
1.10X
1.70
WC
1.10X
1.70

01 1 :100

Reagent Details:
Formula & Calculations:
Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).
Calculation: To prepare 1 L of 25mM Sodium citrate monohydrate (MW = 294.1g/mol)
Weight of solute(g)=1×7.28=7.28g.
Preparation Method:
Weigh 7.28 g of Sodium citrate monohydrate and dissolve in water to make a 1 L solution.
Preparation for different volumes:
To prepare _____ of 25mM Sodium citrate monohydrate (MW = 192.12g/mol).
Weight of solute(g)=1×7.28 _____ L= _____ gm
Weigh _____ gm of Sodium citrate monohydrate and dissolve in water to make a _____ L solution.
Adjust the pH to 7.5 with sodium hydroxide or hydrochloric acid.
Check and record the details:
Preparation Details:
Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry
Date
| | | | |
Final volume prepared | Weighed Quantity | pH(7.4 ±0.1) | Conductivity (4.5±0.5mS/cm) | Storage Condition:
| | | | RT
Date of Preparation | Prepared By | Checked By
| |

Reagent Details:
Formula & Calculations:
Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).
Calculation: To prepare 1 L of 0.4M Citric Acid (MW = 192.124 g/mol)
Weight of solute(g)=0.4×192.124x1=76.85g.
Preparation Method:
Weigh 76.85g of Citric Acid and dissolve in water to make a 1 L solution.
Preparation for different volumes:
To prepare _____ of 0.4M Citric Acid (MW = 192.124g/mol).
Weight of solute(g)=0.4×192.124 x _____L= _____g.
Weigh _____gm of Citric Acid and dissolve in water to make a _____ L solution.
Check and record the details:
Preparation Details:
Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry
Date
| | | |
Final volume prepared | Weighed Quantity | pH (1.6 to 2.1) | Conductivity (34.5 mS/cm) | Storage
Condition
| | | |
Date of Preparation | Prepared By | Checked By
| |

Reagent Details:

Formula & Calculations:

Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).

Calculation: To prepare 1 L of 3M Ammonium sulphate (MW = 132.14g/mol)

Weight of solute(g)=1×132.14=396.42g.

Preparation Method:

Weigh 396.42 g of Ammonium sulphate and dissolve in water to make a 1 L solution.

Preparation for different volumes:

To prepare _____ of 3M Ammonium sulphate (MW = 132.14g/mol).

Weight of solute(g)=1×396.42 _____ L= _____ gm

Weigh _____ gm of Ammonium sulphate and dissolve in water to make a _____ L solution.

Adjust the pH to desired level with ammonia solution/HCL

Check and record the details:

Preparation Details:

Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry

Date

||| |

Final volume prepared | Weighed Quantity | pH (4.5±0.2) | Conductivity (450±10.0ms/cm) | Storage

Condition:

||| | RT

Date of Preparation | Prepared By | Checked By

||

W W W W W W W W W

FIREMAN

FOLDING DOOR
CORRIDOR
Goods LIFT 53.5x2.77 LIFT
3.50X3.00 D1
6D
2.70X2.50
4D

D3
CH R3 CH R2
LOBBY
3.00X11.77 6.34 x 5.05 P.M RECEIVING LAF 3.72x2.60 3.72x2.60 LOBBY
3.00X11.77
FIRE STAIR
CORRIDOR FINISHED GOODS ROOM AREA up
SAMPLING

3D
OPEN TO 2.57x9.00 OPEN TO ALL
5.00x5.00
FIRST 24.6 x 8.95 ROOM D3
FLOORS
FLOOR 7.51x7.60

D3
(I.V.D.)
ONLY D4
EXIT CH R1
D1 D1
3.80x2.60 3.80x2.60
DPB

3D

3D
DE DUSTING

4D
MATERIAL

DIALYZER RECEIVING

2.94 M WIDE CORRIDOR
AREA
5.05x6.60 W
2.50 x 13.0
WC WC WC

BELOW

D4
1.10X 1.10X 1.10X
1.70 1.70 1.70

6.77 M WIDE CORRIDOR
REJECT
RECEIVING D1
19.8x30.3 AREA
WC WC WC 5.05x4.94
1.10X 1.10X 1.10X
1.70 1.70 1.70
W

D4
W W

FINISHED GOODS ROOM
LOBBY 24.6 x 10.9
6.77X11.77 QUARENTAINE RM STORE
D1

STAIRCASE
12.9x29.5
W
AND 7.80x11.6 W
LIFT ONLY
FOR
FIRST FLOOR
(IVD)
6D
PACKING MATERIAL STORE
18.6 x 15.9
11.3 x 27.9

Packing table

Packing table
Pass. LIFT

Packing table

Packing table
2.30X2.00
WC WC WC

Labeling
Labeling
1.10X 1.10X 1.10X

2
2
1.70 1.70 1.70

Vertical label m/c
Vertical label m/c

Labeling
Labeling

1
1
D1

conveyor
conveyor
D1

17.3x21.1 WC
1.10X
WC
1.10X
WC
1.10X
1.70 1.70 1.70

CORRIDOR
59.0x2.57 CORRIDOR
1.80x81.0

DPB
W D4
W

047
047
D1

D6
W APPROVED RM RAW MATERIAL STORAGE W
24.2x14.0
7.30x9.75

COOL ZONE
2.80 M WIDE PASSAGE

4D
17.3x8.86
WC WC WC
1.10X 1.10X 1.10X GENERAL STORE
4D

1.70 1.70 1.70
STATIONARY
& ENG STORE LOBBY
7.25x5.97 6.77X11.00

3D
7.40x2.00
CH 2 CH 1
DPB 2.56x1.92 3.20x1.92

3D
DPB
WC WC WC 3D
STAIRCASE
1.10X 1.10X 1.10X AND

3D

3D
1.70 1.70 1.70
LIFT ONLY
13.1x8.69 CH R2 CH R1
XX
XX

FOR
6.45x2.88

XX

3.40x3.46 3.40x3.46
PP GRANUAL SECOND FLOOR
AUTO CLAVE RM DISPENSING
ROOM
PRODUCTION
OFFICE & IPQA
TOOL ROOM
4.25x7.00
HOPPER LOADER
ROOM
3D
(ADMIN BLOCK)
4.05x7.00 2.92x7.00
ROOM
D3
4.11x7.00
CH R3 EXIT

XX

13.2x11.5 XXX

XX

3.40x3.46 3.40x3.46

SPB LAF

Emergency Exit
4D 4D 4D
D3 D3

EMERGENCY EXIT
3D
CORRIDOR
D4

29.2x2.79
D4 PW U2
WFI U2

3D 4D 4D
12.3x6.07 DPB

D4
1.25x4.25 COOLING ZONE
4D

7.16x4.54 5.15X 2.00
LOBBY 9.03x4.54 3 WAY
12.47X6.77 6.92x1.52 SPB
AUTO CLAVE

13.7x6.06 Pass. LIFT
2.30X2.00
5.15X3.99

3D
PRODUCTION DPBEURO CAP

3D
IPQA OFFICE AIR LOCK
1.76X1.20
OFFICE STORE 3.04 x 1.44
3D
3.37x1.44 2.71x1.443D 3D
DPB D3

CORRIDOR
53.5x3.00
DPB

3D 3D 3D

ELECTRIC

D3

D3
D3

3.00X11.7 ROOMW
3.50X6.00
MALE PRIMARY PRIMARY
Goods LIFT CONFERENCE
HALL
KITCHEN
2.00X6.40
OFFICE 04 OFFICE 03 OFFICE 02 OFFICE 01 AIR LOCK MALE MALE RECEPTION
FEMALE
AIR LOCK FEMALE FEMALE VISITOR

PRIMARY
VISITOR
VISITOR
3.00X6.40 3.00X6.40 3.00X6.40 2.50X6.40 3.50X6.40 CHANGE CHANGE CHANGE
3.50X3.00 4.80X6.40 3.50X6.40 3.50X6.40 5.00X6.40 4.00X6.40
CHANGE
3.00X6.40
CHANGE
3.00X6.40
AIR LOCK
3.00X6.40 CHANGE
3.00X6.40
3.00X6.40

LOADING / UNLOADING
3D

3D
3D

D6
LIFT ONLY D3 D3 D3 D3 D3 D3 D3 D3
W
FOR
SERVER
SECOND FLOOR ROOM
CORRIDOR 3.50X6.57
7D

D7
53.5x3.27

W W W W W W W W W W

FIRST FLOOR PLAN BUILDING -A CLIENT :- LIFENITY- HEALTH LIMITED (DIALYZER, LVP & PLASMA)
ADDRESS :-Auric city, chhatrapati sambhaji nagar

SCALE 1:100

DESIGN &
PROJECT BY :-BIOBRIDGE HEALTHCARE SOLUTION PVT.LTD.
PREPARED A.K
ADDRESS :-13, Rachna Blossom, Jagdishnagar,
Aundh, Pune -411007, Maharastra, India. CHECKED G.K.

DATE 18-06-2026
TITLE- CONCEPTUAL LAYOUT-FF-A REVISION R0
SCALE 1:1

SIZE A0 DWG.NO 02

Reagent Details:
Formula & Calculations:
Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).
Calculation: To prepare 1 L of 5 mM Sodium acetate (MW = 82.0343g/mol)
Weight of solute(g)=0.005×82.0343=0.41017g.
Preparation Method:
Weigh 82.034 g of sodium Acetate and dissolve in water to make a 1 L solution.
Preparation for different volumes:
To prepare _____ of 0.005 M Sodium Acetate (MW = 82.034 g/mol).
Weight of solute(g)=0.005×82.034 _____L= _____gm
Weigh _____gm of Sodium Acetate and dissolve in water to make a _____ L solution.
Check and record the details:
Preparation Details:
Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry
Date
|||||
Final volume prepared | Weighed Quantity | pH (8.0±0.5) | Conductivity (430± 50 µs/cm) | Storage
Condition:
||| RT
Date of Preparation | Prepared By | Checked By
||

FOR OFFICE USE
3
4
AREA CALCULATION BUILDING B AREA CALCULATION BUILDING B
AREA CALCULATION BUILDING B MEZZANINE WEST SIDE MEZZANINE EAST SIDE

BLOCK L = 68.00 X 114.00 = 7752.00 BLOCK M = 7.23 X 114.00 = 824.22 BLOCK N = 7.23 X 114.00 = 824.22
water tank
BLOCK M+BLOCK N = 824.22 + 824.22 = 1648.44 SQ.M.
inlet pipe
DEDUCTIONS
ates
1.200 solar pl
TERRACE BLOCK B = 3.50 X 3.00 X 2 = 21.00 DEDUCTIONS DEDUCTIONS
BLOCK C = 3.60 X 2.00 X 3 = 21.60 BLOCK B = 3.50 X 3.00 X 2 = 21.00
BLOCK D = 2.30 X 2.00 X 2 = 9.20 BLOCK C = 3.60 X 2.00 X 2 = 14.40 BLOCK C = 3.60 X 2.00 X 1 = 7.20

7.000
BLOCK E = 3.50 X 1.97 X 2 = 13.79 BLOCK D = 2.30 X 2.00 X 1 = 4.60 BLOCK D = 2.30 X 2.00 X 1 = 4.60
BLOCK F = 2.70 X 2.50 X 1 = 6.75 BLOCK E = 3.50 X 1.97 X 1 = 6.895 BLOCK E = 3.50 X 1.97 X 1 = 6.895

15.200
BLOCK F = 2.70 X 2.50 X 1 = 6.75

LIFT WELL
TOTAL DEDUCTIONS = 72.34 sq.m.

7.000
DEDUCTIONS = 46.895 sq.m. DEDUCTIONS = 25.445 sq.m. PLAN OF PROPOSED FACTORY
TOTAL B/UP AREA G.F. = 7679.66
1.200 TOTAL B/UP AREA F.F. = 7679.66 TOTAL DEDUCTIONS BLOCK Y+ BLOCK Z = 46.895 + 25.445 = 72.34 SQ.M.
BUILDING ON PLOT NO 8A, SECTOR 13,
TOTAL B/UP AREA S.F. = 7679.66 GROUND FLOOR MEZZANINE = 1648.44 - 72.34 = 1576.10 SQ.M. DMIC, AURIC ,
SHENDRA,
1.500
FIRST FLOOR MEZZANINE = 648.44 - 72.34 = 1576.10 SQ.M.
BUILDING B - SECTION C-C
CHATRAPATI SAMBHAJI NAGAR
BUILDING B - FRONT ELEVATION
FOR M/S LIFENITY HEALTH LIMITED

W W W W W W W W W W

Goods LIFT
3.50X3.00
FIREMAN
LIFT
2.70X2.50
Goods LIFT
3.50X3.00
FIREMAN
LIFT
2.70X2.50
B F B F
FIRE STAIR FIRE STAIR
up up

AREA STATEMENT
AREA IN
NO PARTICULARS
SQM.

1 AREA OF PLOT 81920.00

AS PER OWNERSHIP DOCUMENTS 81920.00
W

W
AS PER MEASUREMENT SHEET 81920.00
MEZZANINE FLOOR MEZZANINE FLOOR
H.V.A.C. UNITS
6.77 X 17.04
H.V.A.C. UNITS
6.77 X 17.04
AS PER SITE 81920.00
W

W
2 DEDUCTIONS FOR

a. PROPOSED DP ROAD 0.00

BUILDING B AREA CALCULATION Plot Area 81920.00 b. ANY RESERVATIONS 0.00
TOTAL (a+b)

W

W
0.00
BLOCK NO

C C BLOCK NAME FLOOR AREA
3 BALANCE AREA OF PLOT (1-2) (A) 81920.00
PLASMA EXTRACTION BUILDING-B
1

W

W
PLASMA EXTRACTION
BELOW
BELOW MEZZANINE FLOOR
H.V.A.C. UNITS
MEZZANINE FLOOR
H.V.A.C. UNITS GROUND 7679.66
4 AMENITY SPACE IF APPLICABLE 5% 0.00
6.77 X 41.04 6.77 X 41.04

MEZZANINE 1576.10 a. Required 0.00
FIRST 7679.66 b. Adjustment of 2(b) if any 0.00
1576.10

W

W
MEZZANINE FLOOR MEZZANINE FLOOR
MEZZANINE c. Balanced proposed
W

W
H.V.A.C. UNITS H.V.A.C. UNITS 0.00
6.77 X 17.04 6.77 X 17.04
M TOTAL B/UP AREA BUILDING B 18511.52 NET PLOT AREA (3-4(c))) PRO RATA
W
N 5 81920.00

W
W

W
6 RECREATIONAL OPEN SPACE IF APPLICABLE 10%

a. Required 8192.00
E E
b. Proposed 8210.48
Pass. LIFT
2.30X2.00
Pass. LIFT
2.30X2.00 D L D
7 INTERNAL ROAD AREA 0.00
C C
8 PLOTABLE AREA IF APPLICABLE (as per sanctioned layout) 0.00

9 B/UP area with reference to basic fsi as per
front road width(1.0) 81920.00

10 ADDITIONAL FSI ON PAYMENT OF PREMIUM
F.S.I. STATEMENT Plot Area 81920.00 a. Maximum permissible premium FSI / TOD Zone. 1,63,840.00
b. Proposed FSI of payment of Premium
W

W
W

W

MEZZANINE FLOOR MEZZANINE FLOOR
BLOCK NO BLOCK NAME FLOOR SIZE AREA
MEZZANINE FLOOR MEZZANINE FLOOR
H.V.A.C. UNITS
H.V.A.C. UNITS
H.V.A.C. UNITS
H.V.A.C. UNITS
11 IN SITU TDR / FSI LOADING
W

W
6.77 X 29.04 6.77 X 29.04
W

W

6.77 X 17.04 6.77 X 17.04
1 BUILDING-A 23883.83
a. In situ area against Statutory Raod (2 xSr No 2(a)) if any 0.00
2 BUILDING-B 18511.52
12 ADDITIONAL FSI AREA UNDER CHAPTER NO.6 0.000
3 DRIVERS ROOM GROUND 72.00
13 TOTAL ENTITLEMENT OF FSI IN THE PROPOSAL 0.000
TOTAL 42467.35
W

W

a. 9 +10(b) +11(a) or 9+12 which ever is applicable
Permissible 81920.00
C C SAY TOTAL B/UP AREA OF ALL BUILDINGS 42467.00 a. 9 +10(b) +11(a) or 9+ 12 which ever is applicable
Proposed 42467.00
W

W

b. Permissible Ancillary area FSI up to 80% with payment
F S I CONSUMED 0.52 65536.00
PERMISSIBLE
b. Proposed Ancillary area FSI up to 60% or 80% with payment 0.00
c.Total Entitlement permissible
W (a+b) 108003.00
W

c.Total Entitlement proposed
MEZZANINE FLOOR MEZZANINE FLOOR (a+b) 42467.00
H.V.A.C. UNITS H.V.A.C. UNITS
6.77 X 17.04 6.77 X 17.04
Maximum utilization limit of FSI (Building Potential) permissible as
14 108003.00
W

per road width ((as per Regulation No. 5.5 of Chapter - 5)) x 1.6 or 1.8)
Pass. LIFT
2.30X2.00
W Pass. LIFT
2.30X2.00
W
D D
15 TOTAL Built up area in proposal (excluding area as per sr. no 17b)
E E
a. Existing Built -up Area. 0.000
b. Proposed Built- up area (as per P line) 42467.00
MEZZANINE FLOOR
H.V.A.C. UNITS c. Total (a+b) 42467.00
6.77 X 11.77

16 FSI CONSUMED (15/13) (should not be more than serial 0.52
Goods LIFT Goods LIFT
No.14 above.)
3.50X3.00 3.50X3.00
B B
RS3 RS2 RS1 RS1 RS1 RS2 RS3 W W W W W W W W W
CERTIFICATE OF AREA :-
MAIN GATE
SCALE 1:50 BOUNDARY WALL DETAIL
Certified that the plot under reference was surveyed by me on
GROUND FLOOR, FIRST FLOOR, AREA CALCULATIONS 03/03/2026 and the Dimensions of sides etc.of plot stated on
plan area
GROUND FLOOR MEZZANINE PLAN - BUILDING B FIRST FLOOR MEZZANINE PLAN - BUILDING B as measured on site and
the area so worked out tallies with the stated in
MEZZANINE FLOORS AREA CALCULATIONS documents of ownership/ DMIC/Land records Department/City Survey
SCALE 1:100 SCALE 1:100 FRONT ELEVATION COMPOUND WALL & GATE Records
Ar. Dashrath Dhumal

RS2 RS2
UP

FIRE EXIT
Owners Declaration :-
LOADING / UNLOADING I/We undersigned hereby confirmed that I/We would abide by plans
FIRE EXIT

D1 5.00 M WIDE CORRIDOR LOBBY D1 5.00 M WIDE CORRIDOR LOBBY
6.77X11.77 3.00X11.77
OPEN TO ALL OPEN TO ALL approved by Authority/ Collector. I / We would execute the structure as
FLOORS FLOORS
per the approved plans.Also I/We would execute the work under
UP
Goods LIFT
FIREMAN
LIFT Goods LIFT
FIREMAN
LIFT Goods LIFT
FIREMAN
LIFT
supervision of proper technical person so as to ensure the quality and
3.50X3.00 2.70X2.50 3.50X3.00 2.70X2.50 3.50X3.00 2.70X2.50
safety at the work site.
FIRE STAIR FIRE STAIR SECURITY CABIN DRIVERS CABIN PUMP ROOM LIFENITY HEALTH LIMITED.
LOBBY LOBBY
up up up
3.00X11.30 3.00X11.30
D1 D1
OPEN TO
FIRST UP
OPEN TO
FIRST
A Architects Name and Signature

0.900

UP FLOOR FLOOR
4.500
12.000 10.450
4.500

Ar. Dashrath Dhumal

5.455
A A
Drn. By. Checked By Registration no of Architect
Security Cabin

Job No. Drg. No. Scale

6.000
4.00 x 5.00

0.899

AREA KEY PLAN
001 1:100 Ar. Dashrath Dhumal CA/2005/37017
FIRE CONTROL Pump Room

14.000
AREA KEY PLAN
ROOM UP

14.000
UP (Driver's Room)
3.50X6.54
W W

VISITORS
A
12.000
10.450
C PARKING STATEMENT
4.00 x 6.74
A W.C.
1.20 x
A a GARAGE PERMISSIBLE GARAGE PROPSD NIL
D1 1.20 BATH
D1 D1 1.20 x

5.455
A A
2.55 PUMP ROOM
PARKING PARKING
DRIVER'S ROOM
10.00 X 5.00
b TYPE
REQUIRED

6.000
8.84 x 4.54
PROPOSED
FIRE PUMP W W LOCKER'S RM.
TOILET
ROOM
3.50X5.00
W.B. 2.40 x 1.50
UP
CAR
UP 2.55 x 1.84

TYPICAL PLAN-1,2&3. 1.997 2.997 1.514 1.199 1.514 0.999 0.230
(Security Cabin)
A AREA KEY PLAN
Security Cabin 10.450
UP
PLAN (Pump Room) SCOOTER
PLAN
(Driver's Room)
Parapet LVL.
8192 SQ.M. 8192.96 SQ.M.
PARAPET LVL. Slab LVL. CYCLE

1.598 0.599
PARAPET LVL.
2.80 M WIDE PASSAGE

2.80 M WIDE PASSAGE

SLAB LVL.

1.598 0.599
RCC Chajja
SLAB LVL.
Lintel LVL.
VISITORS
6.77 M WIDE CORRIDOR

6.77 M WIDE CORRIDOR

W LINTEL LVL.
WC WC WC WC WC WC

3.996
1.10X 1.10X 1.10X 1.10X 1.10X 1.10X LINTEL LVL.

LOADING UNLOADING SPACES

1.499
3.996
1.70 1.70 1.70 1.70 1.70 1.70 Sill LVL.

D

2.398
1.499
0.899SILL LVL.

2.398
Plinth LVL.
0.899
SILL LVL. G. LVL. G. LVL.
D1 D1 PLINTH LVL.

0.450
FRONT ELEVATION
PLINTH LVL. G.LVL. G.LVL.
a LOADING UNLOADING REQUIRED

0.450
WC WC WC WC WC WC
G.LVL. G.LVL. FRONT ELEVATION
(Driver's Room)
44
1.10X 1.10X 1.10X 1.10X 1.10X 1.10X LEFT SIDE ELEVATION
1.70 1.70 1.70 1.70 1.70 1.70 (Security Cabin)
W
PARAPET LVL. Terrace
Parapet LVL. b TOTAL LOADING UNLOADING PROVIDED 44

1.598 0.599
Slab LVL.
TERRACE SLAB LVL.

RCC Chajja
TERRACE

LEGENDS
0.900 Lintel LVL.

3.996
LINTEL LVL.

3.497

1.499
3.996
1.499

2.398

2.398
2.398
Sill LVL.
UP 0.899

0.899
SILL LVL.
PUMP ROOM
Plinth LVL.
PLOT BOUNDRY SHOWN IN

0.450
UP Security Cabin VISITORS TOILET PLINTH LVL. G. LVL. G. LVL.
SUNK
G.LVL.
DRIVER'S ROOM W.C. BATH SECTION- AA
SUNK

B/UP AREA SHOWN IN
SUNK
G.LVL.
SECTION- AA
W W (Security Cabin)
SECTION- AA

LOBBY
BUILDING B LOBBY
BUILDING B SECURITY CABIN AREA CALCULATION
(Driver's Room)

1.10X
1.70
AND
LIFT ONLY
FOR
1.10X
1.70
1.10X
1.70
1.10X
1.70
AND
LIFT ONLY
FOR
W Email : slandge@lifenitywellness.com
SECOND FLOOR SECOND FLOOR
(L.V.P.) (L.V.P.)

UP

UP

W W
W

D1
SIGN OF DIRECTOR

N
LOBBY C LOBBY
12.47X6.77
Pass. LIFT W
W
12.47X6.77
Pass. LIFT Pass. LIFT
: ARCHITECT
2.30X2.00 2.30X2.00 2.30X2.00
Vastushilp Architects,
UP
#03,Alok Nagar, Beed bypass Road,
Chh. Sambahi Nagar 431001
C
ELECTRIC
ROOM W
Ar. Dashrath K. Dhumal.
SOLAR WATER HEATING
W
3.50X6.00
Phone: 9850207562 Email: darchitect.dashrath@gmail.com
E
LOBBY LOBBY
3.00X11.53 3.00X11.53
OPEN TO OPEN TO ALL
SECOND FOLDING DOOR 6 M FLOORS FOLDING DOOR 6 M PHARMACEUTICAL
PARTITION FOLDING DOOR 6 M
FLOOR

W
SERVER
Goods LIFT 5.00 M WIDE CORRIDOR Goods LIFT 5.00 M WIDE CORRIDOR Goods LIFT

S
ROOM
3.50X3.00 LOADING / UNLOADING 3.50X6.57 3.50X3.00 3.50X3.00
LOADING / UNLOADING

RS3 RS2 RS1 RS1 RS1 RS1 RS2

FIRST FLOOR PLAN - BUILDING B TERRACE FLOOR PLAN - BUILDING B
GROUND FLOOR PLAN - BUILDING B SCALE 1:100 SCALE 1:100 JOB No. DRG NO. SCALE DATE DRN BY CHK. BY

SCALE 1:100 01 1 :100

Reagent Details:
Preparation of IG-RR-06 working buffer from Stock buffer (IG-RR-05):
Calculations: For the Preparation of 20mM Sodium acetate pH 5.7.
Stock Buffer (IG-RR-05):0.2M Sodium acetate pH 5.8.
Formula:

$C1 \times V1 = C2 \times V2$
C1 = concentration of stock
V1 = volume of stock you need to take
C2 = desired (working) concentration
V2 = final total volume of working solution
Applying the formula to prepare the working buffer:
 $C1 \times V1 = C2 \times V2$
C1 = 0.2M Sodium Acetate pH 5.8.
V1 = volume of stock you need to take _____ L.
C2 = desired (working) concentration 20mM Sodium Acetate pH 5.7
V2 = final total volume of working solution _____ L.
 $V1 = (C2 \times V2) / C1$
C2=0.02M ; V2= 1.0 L; C1=0.2M
 $V1 = (0.02 \times 1.0) / 0.2$
V1 = 0.1L
Take 0.1 L (IG-RR-05) Stock buffer solution and dilute with water, and make up to 1L.
Check and record the details:

Preparation Details:
Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry
Date
Sodium Acetate | | | |
Acetic Acid | | | |
Stock buffer (IG-RR-05) taken volume(L) | pH (5.7 ± 0.1) | Conductivity (1.7± 0.5ms/cm) | Storage
Condition | Final volume prepared
| | | |
Date of Preparation | Prepared By | Checked By
| |

Technical Information # BB-LN/IND/002

Date- 19.05.2026

List of Documents

S/No Title

1 Organic waste calculation and parameters for ETP

Contact Details:

1. Dr. Ganesh Kumraj: ganesh@biobridge.in

www.biobridge.in Confidential Page 1 of 1

BioBridge Healthcare Solutions Private Limited

Address: # 40, 4th Floor, Block B, Chandigarh City Center (CCC), Zirakpur - 140603, India

FOR OFFICE USE

1
4 4
PARKING REQUIREMENTS CLAUSE NO 7.20 MIDC CDCPR
8
3 AS PER TABLE 7C AND 7D FOR 42467.00 SQM. MULTIPLICATION FACTOR 0.8
2 TYPE OF VEHICLE PARKING SIZE REQUIRED PARKING REQUIRED REQUIRED PROVIDED
CAR 2.5M X 5.0M 1 FOR EVERY 200SQM 212.33 0.80 170
SCOOTER / MOTOR CYCLE 2.0 SQ.M. (1.0M X 2.0M) 2 FOR EVERY 200SQM 106.16 0.80 85
8192.96 Sq.m
CYCLE
TRUCK 3.75M X 10.0M 1 FOR EVERY 1000 SQM 42 0.80 34
In case of plots pf 1ha and above minimum parking space shall be 10% of plot area
For plot area 81920 sq.m. parking should be 8192 sq.m.

1
424.41 PL-1
PLOT AREA CALCULATION
PLAN OF PROPOSED FACTORY
Sr. No. L W Nos. AREA SQ.M. BUILDING ON PLOT NO 8A, SECTOR 13,
93
6 .59 1 209.66 93.59 0.5 1 9811.04 SQ.M. DMIC, AURIC , SHENDRA,
7 1 2 209.66 89.87 0.5 1 9421.07 SQ.M.
CHATTRAPATI SAMBHAJI NAGAR
3 267.25 93.00 0.5 1 12427.13 SQ.M.
4 318.65 106.78 0.5 1 17012.72 SQ.M.
FOR M/S LIFENITY HEALTH LIMITED
5 360.94 87.04 0.5 1 15708.11 SQ.M.
89
.87 6 360.94 25.50 0.5 1 4601.99 SQ.M.
2 7 355.87 47.40 0.5 1 8434.12 SQ.M.
PL 1 POLYLINE 424.41 SQ.M.
GATE-2 PL 2 POLYLINE 4079.92 SQ.M.
SECURITY
CABIN-2 5 81920.50 SQ.M.

DRIVER'S ELECTRIC OPEN SPACE 1
ROOM SUB STATION
AREA STATEMENT

6
AREA IN
NO PARTICULARS

9.6
SQM.
GENERATOR

20
1 AREA OF PLOT 81920.00

93.00
SET
AS PER OWNERSHIP DOCUMENTS 81920.00
3
LOADING OPEN SPACE 2 AS PER MEASUREMENT SHEET 81920.00
/UNLOADING AS PER SITE 81920.00
20 X 50
OPEN SPACE 1 2 DEDUCTIONS FOR

2878.52 OPEN SPACE AREA CALCULATION a. PROPOSED DP ROAD 0.00

BOILERS Sr. No. L W Nos. AREA SQ.M. b. ANY RESERVATIONS 0.00
TRAILER TRUCK

TRAILER TRUCK

TRAILER TRUCK

1 45.00 56.32 1 1 2534.4 SQ.M. TOTAL (a+b) 0.00
2 35.71 12.64 0.50 1 225.68 SQ.M. BALANCE AREA OF PLOT (1-2) (A)
3 81920.00
3 9.29 12.64 1 1 117.42 SQ.M.
4 9.29 0.22 0.5 1 1.0219 SQ.M. 267.25 4 AMENITY SPACE IF APPLICABLE 5% 0.00

87.0
a. Required 0.00
5 87.85 25.25 1 1 2218.21 SQ.M.
6 15.49 59.05 1 1 914.68 SQ.M.
5 b. Adjustment of 2(b) if any 0.00

4
60 7 72.36 59.05 0.5 1 2136.42 SQ.M. c. Balanced proposed 0.00
.00 8 POLYLINE 62.65 SQ.M.
5 NET PLOT AREA (3-4(c))) PRO RATA 81920.00
I N T E R N A L R O A D 1 2 . 0 0 M W I D E
Recovery plant
M .
8210.48 SQ.M.
6 RECREATIONAL OPEN SPACE IF APPLICABLE 10%

W

25.5

a. Required 8192.00

70 X 35 6

RWH

ID .65 b. Proposed 8210.48

0

4 8

10

31

E

6.7

7 INTERNAL ROAD AREA 0.00

8

R 8 PLOTABLE AREA IF APPLICABLE (as per sanctioned layout) 0.00

O 9 B/UP area with reference to basic fsi as per
front road width(1.0) 81920.00

AD .94

10 ADDITIONAL FSI ON PAYMENT OF PREMIUM

360 a. Maximum permissible premium FSI / TOD Zone. 1,63,840.00

47.40

b. Proposed FSI of payment of Premium

I

N 7

T

water

E GR 3 55.87 11 IN SITU TDR / FSI LOADING

R EE a. In situ area against Statutory Raod (2 xSr No 2(a)) if any 0.00
tanks N N

A BE 12 ADDITIONAL FSI AREA UNDER CHAPTER NO.6 0.000

L LT3 13 TOTAL ENTITLEMENT OF FSI IN THE PROPOSAL 0.000

(3

R 87 a. 9 +10(b) +11(a) or 9+12 which ever is applicable

GREEN BELT 1 2 . 00 M WIDE

O 9.9 Permissible 81920.00

A 3s 4079.92 PL-2 a. 9 +10(b) +11(a) or 9+ 12 which ever is applicable

D q.m Proposed 42467.00

PROPOSED PUMP .) b. Permissible Ancillary area FSI up to 80% with payment

65536.00

ROOM 9 PERMISSIBLE

BUILDING A .0 b. Proposed Ancillary area FSI up to 60% or 80% with payment

W

AT 0

PLOT AREA TRANGULATION LAYOUT 0.00

c.Total Entitlement permissible

ER M PARKING AREA CALCULATION 108003.00

LI W SCALE :- 1:2000 SQ.M.

(a+b)

c.Total Entitlement proposed

NE GA Sr. No. L W Nos. AREA

42467.00

ID TE 1 53.62 95.31 1 1 5110.52 SQ.M.

(a+b)

Maximum utilization limit of FSI (Building Potential) permissible as

E -1 2 34.74 42.94 0.50 1 745.86 SQ.M. 14 per road width ((as per Regulation No. 5.5 of Chapter - 5))

x 1.6 or 1.8) 108003.00

3 26.07 34.74 1 1 839.71 SQ.M.

OPEN SPACE 2 15 TOTAL Built up area in proposal (excluding area as per sr. no 17b)

5331.96 4 2.53 26.07 0.5 1 32.97 SQ.M.

a. Existing Built -up Area. 0.000

D

5 32.21 26.31 0.5 1 423.72 SQ.M.

330.840

SE

A

RO

C 6 42.25 34.51 0.5 1 914.68 SQ.M. b. Proposed Built- up area (as per P line) 42467.00

CA UR

R O A D 1 2 . 0 0 M W I D E

SQ.M.

E

7 11.37 16.31 1 1 185.44
BI ITY

ID
125.72
c. Total (a+b) 42467.00
R O A D 9 . 0 0 M W I D E

SQ.M.
N- 8 POLYLINE

W
1 CP

R I D M.
M
8192.96 FSI CONSUMED (15/13) (should not be more than serial
-L

CP
SQ.M. 16 0.52

12
-L

CP
-L
No.14 above.)
CP

W .00
-L

O E
CP
-L

8 CERTIFICATE OF AREA :-
CP
-L

AD
CP
-L

30
CP
-L
Certified that the plot under reference was surveyed by me on
CP
-L
03/03/2026 and the Dimensions of sides etc.of plot stated on plan area
CP
-L

CP
-L
as measured on site and the area so worked out tallies with the stated in
CP documents of ownership/ DMIC/Land records Department/City Survey

SC

SC

CS
-L

SC

SC

CS
Records
CP

6
-L

SC

SC

CS
7 Ar. Dashrath Dhumal

SC

SC

CS
CP
-L

SC

SC

CS
CP
-L

SC

SC

CS
SC

SC

CS
CP
-L

SC

SC

CS
CP

SC

SC

CS
-L

Owners Decleration :-

SC

SC

CS
CP
-L

SC

SC

CS
CP
-L

SC

SC

CS
CP

CP
-L

I/We undersigned hereby confirmed that I/We would abide by plans

SC

SC

CS
-L
CP
-L

SC

SC

CS
CP
-L

approved by Authority/ Collector. I / We would execute the structure as

SC

SC

CS
CP CP
-L -L

SC

SC

CS
CP

per the approved plans.Also I/We would execute the work under

-L
CP CP

SC

SC

CS
-L -L
CP
-L

SC

SC

CS
CP CP
-L -L

supervision of proper technical person so as to ensure the quality and

SC

SC

CS
CP
-L
CP CP

SC

SC

CS
-L -L
CP

safety at the work site.

SC

SC

CS
-L
CP CP
-L -L

SC

SC

CS
CP
-L

SC

SC

CS
CP CP
-L -L

LIFENITY HEALTH LIMITED.

SC

SC

CS
CP
-L
CP CP
-L

SC

SC

CS
-L

INTERNAL

INTERNAL

CP
-L

SC

SC

CS
CP CP
-L -L

Architects Name and Signature

GRAY WATER LINE
SC

SC

CS
CP
-L

ROAD 9.00 M WIDE
CP CP

SC

SC

CS
-L -L
CP

SC

SC

CS
-L
CP CP
-L -L

SC

SC

CS
CP
-L

SC

SC

CS
CP
-L
CP

Ar. Dashrath Dhumal

SC

SC

CS
-L

CP-L

CP-L
CP
-L

5
SC

SC

CS

CP-L
CP
-L

SC

SC

CS
Drn. By. Checked By Registration no of Architect
CP

Job No. Drg. No. Scale
-L

SC

SC

CS

CP-L

CP-L
CP

CP-L
SC

SC

CS
-L
SC

SC

CS
CP

001 1:100 Ar. Dashrath Dhumal CA/2005/37017

CP-L

CP-L
-L
SC

SC

CS

CP-L
SC

SC

CS
SC

SC

CS

CP-L

CP-L
C PARKING STATEMENT

CP-L
SC

SC

CS
SC

SC

CS 60

CP-L

CP-L
SC

SC

CS

CP-L
a GARAGE PERMISSIBLE GARAGE PROPOSED NIL
SC

SC

CS
SC

SC

CS

CP-L

CP-L
PARKING AREA

W .00 M

CP-L
PARKING PARKING
SC

SC

CS

b TYPE
SC

SC

CS

CP-L

CP-L
8192.96 sq.m. REQUIRED PROPOSED
SC

SC

CS

CP-L
SC

SC

CS

-L
CP
SC

SC

CS

CP-L

CP-L
CAR

CP-L
-L
SC

SC

CS

R IDE .

CP
SC

SC

CS

-L
CP
CP-L

CP-L

-L
SC

SC

CS

CP
3

CP-L
SCOOTER
FUTURE EXPANSION AREA

SC

SC

CS

-L
CP
8192 SQ.M. 8192.96 SQ.M.

-L
GREEN BELT 2 (6889 sq.m.)
SC

SC

CS

CP-L

CP-L

CP

CP-L
SC

SC

CS

-L
CYCLE

CP
O

-L
87.85 x 128.27
SC

SC

CS

CP
PROPOSED
CP-L

CP-L
1
SC

SC

CS

-L
CP
-L
CP
VISITORS
SC

SC

CS

CP
AD

-L
-L
SC

SC

CS

CP-L

CP-L

CP
BUILDING B (11261 sq.m.)

-L
CP
SC

SC

CS

CP
LOADING UNLOADING SPACES

-L
D

-L
SC

SC

CS

4

CP
CP-L

CP-L

-L
CP
SC

SC

CS

CP
-L
-L
SC

SC

CS

CP
-L
CP
a LOADING UNLOADING REQUIRED
SC

SC

CS

CP-L

CP-L

CP
44

-L
-L
SC

SC

CS
SEWAGE LINE

CP
-L
CP
SC

SC

CS

CP
CP-L

CP-L

-L
-L
b TOTAL LOADING UNLOADING PROVIDED

CP
SC

SC

CS

44

-L
CP
SC

SC

CS

CP
-L
-L
CP
SEWAGE LINE

SC

SC

CS

-L
CP
SC

SC

CS

CP
-L
-L
LEGENDS

CP
INTERNAL
SC

SC

CS

-L
CP
SC

SC

CS

CP
-L
-L
CP
SC

SC

CS

-L
CP

CP
SC

SC

CS

-L
-L
PLOT BOUNDRY SHOWN IN

CP
SC

SC

CS

-L
CP

CP
2
SC

SC

CS

-L

-L
CP
SC

SC

CS

B/UP AREA SHOWN IN

CP
SC

SC

CS

-L
-L
CP
SC

SC

CS

CP
WATER LINE SHOWN IN

-L

D
CP
-L
GREY WATER LINE SHOWN IN

CP
-L
ETP LINE

STP LINE

SEWAGE LINE SHOWN IN

A
CP
-L
EFFLUNT LINE

CP
-L

O
CP
-L
SCHEDULE OF DOORS AND WINDOWS
CP
-L

R
TYPE SIZE DISCRIPTION
GREEN BELT AREA CALCULATION
CP
-L

Plot Area 81920.00
GRAY WATER LINE

GRAY WATER LINE

RS1 2500 X 6000 M.S ROLLING SHUTTER
CP
-L

CP-L CP-L

D1 1800 X 2400 FLUSH DOOR

E
BLOCK NO AREA D2 900 X 2400 FLUSH DOOR

D3 750 X 2400 FLUSH DOOR

ID
FD 1000 X 2400 FIRE DOOR
1 GREEN BELT 1 8087.00
W 2500 X 2500 ALLU. SLIDING WINDOW

GREEN BELT 2 6889.00 V 600 X 900 ALU LOUVERED VENTILATOR
OWC SPACE FOR STP

W

2

3 GREEN BELT 3 3879.93
15 X 12 X 4.50 OPEN SPACE 8210.48

4
PARKING AREA CALCULATION PLAN OF PROPOSED FACTORY
M.
TOTAL GREEN BELT PROPOSED 27066.41

TOTAL GREEN BELT REQUIRED 33% 27033.60
BUILDING ON PLOT NO 8A, SECTOR 13,
.00

SPACE FOR ETP DMIC, AURIC , SHENDRA,
16 X 12 X 5.50 CHATTRAPATI SAMBHAJI NAGAR
45

RWH SEWAGE LINE ETP LINE F.S.I. STATEMENT Plot Area 81920.00
FOR M/S LIFENITY HEALTH LIMITED

AD
GRAY WATER LINE
GRAY WATER LINE

OFF NO 301, S NO 5, PLOT

IDE RO
BLOCK NO BLOCK NAME FLOOR SIZE AREA
NO 1, BLDG A 1 , Navkar Avenue, NDA Pashan
I N T E R N A L R O A D 9 . 0 0 M W I D E 1 BUILDING-A 23883.83

W
road, BHAVDHAN, PUNE 411021

60.00 M.
2 BUILDING-B 18511.52
Phone no: 7276066666
3 Email : slandge@lifenitywellness.com
GREEN BELT 1 (8087 sq.m.) DRIVERS ROOM GROUND 72.00

TOTAL 42467.35
ROAD
WIDE
60.00 M.
SAY TOTAL B/UP AREA OF ALL BUILDINGS 42467.00

AD
60.
0

RO
OM
. W
ID
PLOT NO. 1B

E
F S I CONSUMED E

ID
0.52 RO

W
AD

M.
00
30.
SIGN OF DIRECTOR
PROPOSED SITE
: ARCHITECT
60
.00
M. Vastushilp Architects,

N
W
ID
PLOT NO. 1C E
RO
#03,Alok Nagar, Beed bypass Road,
AD
45.00 M. W I D E R
OAD Chh. Sambahi Nagar 431001
45.00 M. W I D E R O A D 45.00 M. W I D E R
OAD Ar. Dashrath K. Dhumal.
Phone: 9850207562 Email: darchitect.dashrath@gmail.com
W

30.00 M. W I D E R O A D
E
S

LAYOUT PLAN LOCATION PLAN SCALE :- NTS
JOB No. DRG NO. SCALE DATE DRN BY CHK. BY

01 1 :100

Reagent Details:
Preparation of IG-RR-25 working buffer from Stock buffer (IG-RR-05):
Calculations: For the Preparation of 0.002M Sodium Acetate pH 5.7.
Stock Buffer (IG-RR-05):0.2M Sodium acetate, pH 5.8.
Formula:

$C1 \times V1 = C2 \times V2$
C1 = concentration of stock
V1 = volume of stock you need to take
C2 = desired (working) concentration
V2 = final total volume of working solution
Applying the formula to prepare the working buffer:
 $C1 \times V1 = C2 \times V2$
C1 = 0.2M Sodium Acetate pH 5.8.
V1 = volume of stock you need to take _____ L.
C2 = desired (working) concentration 0.002M Sodium Acetate pH 5.7
V2 = final total volume of working solution _____ L.
 $V1 = (C2 \times V2) / C1$
C2=0.02M ; V2= 1.0 L; C1=0.2M
 $V1 = (0.02 \times 1.0) / 0.2$
V1 = 0.1L
Take 0.1 L (IG-RR-05) Stock buffer solution and dilute with water, and make up to 1L.
Check and record the details:

Preparation Details:
Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry
Date
Sodium Acetate | | | |
Acetic Acid | | | |
Stock buffer (IG-RR-05) taken volume(L) | pH (5.7± 0.1) | Conductivity (0.25± 0.1 ms/cm) | Storage
Condition | Final volume prepared
| | | RT |
Date of Preparation | Prepared By | Checked By
| |

Reagent Details:

Formula:

Sodium Acetate(M.Wt): 82.03g/mol.

Acetic Acid (M.Wt): 60.05g/ mol

Preparation of 0.2 M Sodium Acetate pH 5.8 buffer

Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol)

×Volume (L).

Calculation: To prepare 1 L of 0.2 M Sodium Acetate (MW = 82.03g/mol).

Weight of solute(g)=0.2×82.03× 1.0=16.40g.

Preparation Method:

Weigh 16.40g of Sodium Acetate and dissolve in water to make a 1 L solution.

Preparation for required volumes:

To prepare _____ of 0.2 M Sodium Acetate (MW = 82.03g/mol).

Weight of solute(g)=0.2×82.03× _____ L= _____g.

Weigh _____ gm of Sodium Acetate and dissolve in water to make a _____ L solution.

Calculations: To prepare the 0.2 M Acetic Acid.

Formula & Calculations:

Formula: Use the formula to calculate the molarity:

Molarity (M) = (Per cent by Weight / 100) * (Density in g/mL) * 1000 / Molar Mass.

Example: Molarity of Glacial Acetic Acid

Molar Mass of CH₃COOH: ~60.05 g/mol

Density: ~1.05 g/mL

Purity: 100%

Acetic Acid (MW = 60.05 g/mol).

Calculation:

Molarity = (100 / 100) * 1.05 g/mL * 1000 mL/L / 60.05 g/mol ≈ 17.5 M.

Calculation: To prepare 1 L of 1.0 M.

Applying the formula to prepare the working buffer:

C₁V₁=C₂V₂

C₁ = 17.5 M acetic acid.

V₁ = volume of stock needed to take.

C₂ = desired (working) concentration 0.2 M acetic acid.

V₂ = final total volume of working solution for one litre.

V₁ = (C₂ × V₂)/C₁.

C₂=0.2 M ; V₂= 1.0 L; C₁=17.5 M

V₁ = (0.2 × 1.0)/17.5

V₁ = 0.011 L

Take 0.011 L of the Stock solution of acetic acid and add it to a volumetric flask, making up the volume to 1 L.

To prepare various volumes:

V₁ = volume of stock needed to take _____ L.

C₂ = desired (working) concentration 0.2 M acetic acid.

V₂ = final total volume of working solution _____ L.

V₁ = (C₂ _____ × V₂ _____) /17.5.

V₁= _____ L.

Take V₁ _____ L of the Stock solution of acetic acid and add it to a volumetric flask, making up the final volume _____ L.

Procedure for Preparation of 0.2 M Sodium Acetate pH 5.8.

To prepare a pH 5.8 solution, 0.2 M Sodium Acetate solution is titrated with 0.2 M Acetic acid so that the pH attains pH 5.8.

Check and record the details:

Preparation Details:

Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry

Date

Sodium Acetate | | | |

Acetic Acid | | | |

Final volume prepared | Measured Quantity | pH (5.8 ± 0. 1) | Conductivity (13± 1.0 ms/cm) | Storage

Condition | Reagent Record

| | | |

Date of Preparation | Prepared By | Checked By

| |

Reagent Details:

Formula & Calculations:

Formula: $M_1 \times V_1 = M_2 \times V_2$.

Calculation: To prepare 1 L of 0.5M sodium hydroxide from 2M sodium hydroxide stock solution calculate quantity required of stock solution as below:

0.5M sodium hydroxide stock solution required quantity= $0.5 \times 1/2 = 0.25\text{L}$

Preparation Method:

To prepare 0.5M sodium hydroxide 1L solution from 2M sodium hydroxide stock solution formula is

$M_1(\text{Molarity of stock}) \times V_1(\text{volume of stock}) = M_2(\text{Molarity of working}) \times V_2(\text{volume of working})$.

Take 0.25L of 2M sodium hydroxide solution and make a 1 L solution.

Preparation for different volumes:

To prepare _____ of 0.5M sodium hydroxide

Take 0.25L x _____ L = _____ L of 2M sodium hydroxide

Make sodium hydroxide solution to _____ L solution.

Check and record the details:

Preparation Details:

Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry

Date

|||

Final volume prepared | Weighed Quantity | Storage Condition:

||

Date of Preparation | Prepared By | Checked By (QC/Analyst)

||

Reagent Details:
Preparation of IG-RR-15 working buffer from Stock buffer (IG-RR-14):
Calculations: For the Preparation of 20mM Sodium acetate pH 4.0.
Stock Buffer (IG-RR-14):0.2M Sodium acetate pH 4.0.
Formula:
 $C1 \times V1 = C2 \times V2$
C1 = concentration of stock
V1 = volume of stock you need to take
C2 = desired (working) concentration
V2 = final total volume of working solution
Applying the formula to prepare the working buffer:
 $C1 \times V1 = C2 \times V2$
C1 = 0.2M Sodium Acetate pH 4.0.
V1 = volume of stock you need to take _____ L.
C2 = desired (working) concentration 20mM Sodium Acetate pH 4.0
V2 = final total volume of working solution _____ L.
 $V1 = (C2 \times V2) / C1$.
C2=0.02M ; V2= 1.0 L; C1=0.2M
 $V1 = (0.02 \times 1.0) / 0.2$
V1 = 0.1L
Take 0.1 L (IG-RR-14) Stock buffer solution and dilute with water, and make up to 1L.
Check and record the details:
Preparation Details:
Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry
Date
Sodium Acetate | | | |
Acetic Acid | | | |
Stock buffer (IG-RR-14) taken volume(L) | pH (4.0±0.1) | Conductivity (2.0±0.1ms/cm) | Storage Condition
| Final volume prepared
| | | RT |
Date of Preparation | Prepared By | Checked By
| |

Reagent Details:
Formula & Calculations:
Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).
Calculation: To prepare 1 L of 0.2M Sodium Acetate (MW = 80.03 g/mol)
Weight of solute(g)=0.2×80.03 = 16.00g.
Preparation Method:
Weigh 16.00 g of Sodium Acetate and dissolve in water to make a 1 L solution.
Preparation for different volumes:
To prepare _____ of 0.2M Sodium Acetate (MW = 80.03 g/mol).
Weight of solute(g)=0.2×80.03 x _____L= _____gm
Weigh _____gm of Sodium Acetate and dissolve in water to make a _____ L solution.
Check and record the details:
Preparation Details:
Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry
Date
| | | |
Final volume prepared | Weighed Quantity | pH (8.8±0.2) | Conductivity (18.0±2.0ms/cm) | Storage
Condition:
| | | |
Date of Preparation | Prepared By | Checked By
| |

Reagent Details:

Formula:

Sodium Acetate(M.Wt): 82.03g/mol.

Acetic Acid (M.Wt): 60.05g/ mol

Preparation of 1.0 M Sodium Acetate pH 4.0 buffer

Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).

Calculation: To prepare 1 L of 1.0 M Sodium Acetate (MW = 82.03g/mol).

Weight of solute(g)=1.0×82.03× 1.0=82.03g.

Preparation Method:

Weigh 82.03g of Sodium Acetate and dissolve in water to make a 1 L solution.

Preparation for required volumes:

To prepare _____ of 1.0 M Sodium Acetate (MW = 82.03g/mol).

Weight of solute(g)=1.0×82.03× _____ L= _____ g.

Weigh _____ gm of Sodium Acetate and dissolve in water to make a _____ L solution.

Calculations: To prepare the 1.0 M Acetic Acid.

Formula & Calculations:

Formula: Use the formula to calculate the molarity:

Molarity (M) = (Per cent by Weight / 100) * (Density in g/mL) * 1000 / Molar Mass.

Example: Molarity of Glacial Acetic Acid

Molar Mass of CH₃COOH: ~60.05 g/mol

Density: ~1.05 g/mL

Purity: 100%

Acetic Acid (MW = 60.05 g/mol).

Calculation:

Molarity = (100 / 100) * 1.05 g/mL * 1000 mL/L / 60.05 g/mol ≈ 17.5 M.

Calculation: To prepare 1 L of 1.0 M.

Applying the formula to prepare the working buffer:

C₁V₁=C₂V₂

C₁ = 17.5 M acetic acid.

V₁ = volume of stock needed to take.

C₂ = desired (working) concentration 1.0 M acetic acid.

V₂ = final total volume of working solution for one litre.

V₁ = (C₂ x V₂)/C₁.

C₂=1.0 M ; V₂= 1.0 L; C₁=17.5 M

V₁ = (1.0 x 1.0)/17.5

V₁ = 0.057 L

Take 0.057 L of the Stock solution of acetic acid and add it to a volumetric flask, making up the volume to 1 L.

To prepare various volumes:

V₁ = volume of stock needed to take _____ L.

C₂ = desired (working) concentration 1.0 M acetic acid.

V₂ = final total volume of working solution _____ L.

V₁ = (C₂ x V₂) /17.5.

V₁= _____ L.

Take V₁ _____ L of the Stock solution of acetic acid and add it to a volumetric flask, making up the final volume _____ L.

Procedure for Preparation of 1.0 M Sodium Acetate pH 4.0.

To prepare a pH 4.0 solution, 1.0 M Sodium Acetate solution is titrated with 1.0 M Acetic acid so that the pH attains pH 4.0.

Check and record the details:

Preparation Details:

Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry

Date

Sodium Acetate | | | |

Acetic Acid | | | |

Final volume prepared | Measured Quantity | pH (4.0±0.1) | Conductivity (45.0±2.0ms/cm) | Storage

Condition | Reagent Record

| | | | RT |

Date of Preparation | Prepared By | Checked By

| |

Reagent Details:
Preparation of IG-RR-19 working buffer from Stock buffer (IG-RR-18):
Calculations: For the Preparation of 0.2M Glycine pH 9.0 + 0.15 M Sodium Chloride.
Stock Buffer (IG-RR-18):0.8M Glycine pH 9.0 + 0.6M Sodium Chloride.
Formula:
 $C1 \times V1 = C2 \times V2$
C1 = concentration of stock
V1 = volume of stock you need to take
C2 = desired (working) concentration
V2 = final total volume of working solution
Applying the formula to prepare the working buffer:
 $C1 \times V1 = C2 \times V2$
C1 = 0.8M Glycine pH 9.0 + 0.6 M Sodium Chloride.
V1 = volume of stock you need to take _____ L.
C2 = desired (working) concentration 0.2 M Glycine, pH 9.0 + 0.15 M Sodium Chloride.
V2 = final total volume of working solution _____ L.
 $V1 = (C2 \times V2) / C1$
 $C2 = 0.2M$; $V2 = 1.0 L$; $C1 = 0.8M$
 $V1 = (0.2 \times 1.0) / 0.8$
 $V1 = 0.25L$
Take 0.25 L (IG-RR-18) Stock buffer solution and dilute with water, and make up to 1L.
Check and record the details:
Preparation Details:
Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry
Date
Glycine | | | |
Sodium Hydroxide | | | |
Sodium Chloride | | | |
Stock buffer (IG-RR-18) taken volume(L) | pH (9.0±0.1) | Conductivity (1.8±0.2 ms/cm) | Storage
Condition | Final volume prepared
| | RT |
Date of Preparation | Prepared By | Checked By
| |

SHAFT

3.30X2.45
CORRIDOR
FIREMAN 49.1x2.56

LIFT
LIFT
2.80X2.60 4D
D3
4D 4D

D3
CH R2 CH R1
3.71x2.60 3.79x2.60
MATERIAL RECEIVING REJECT RECEIVING
P.M RECEIVING 4.51x5.05 D3 D3
AREA AREA

AREA 6.86x5.00 4.94x5.00
5.00x5.00 CH R3 EXIT
3.71x2.60 3.79x2.60 DE DUSTING
D4 D4 D4
D3 D3

SAMPLING
ROOM

2.94 M WIDE CORRIDOR
7.59x8.46

LADIES TOILET

DPB
3.30 x 4.35 FINISHED GOODS ROOM
19.8x30.3 LAF
WC WC WC
1.10X 1.10X 1.10X
1.70 1.70 1.70

6.77 M WIDE CORRIDOR
WC WC WC
1.10X 1.10X 1.10X
1.70 1.70 1.70

GENTS TOILET PACKING MATERIAL STORE
3.30 x 4.35
9.56 x 25.3
11.9 x 15.7

QUARENTAINE RM STORE
19.5x32.2
11.8x8.85

4D

400
400

730
730

400
400

Packing table

Packing table
Packing table

Packing table

Labeling
Labeling

2
2
SHAFT SHAFT

Vertical label m/c
Vertical label m/c

Labeling
Labeling
LIFT LIFT

1
1
3.24 M WIDE PASSAGE

3.24 M WIDE PASSAGE
conveyor
2.10X 2.10X

conveyor

3.24 M WIDE PASSAGE
2.00 2.00
FD 17.3x21.0 FD
UP UP

CORRIDOR
80.7x2.57 CORRIDOR
80.7x1.80

D4

047
047
4D APPROVED RM RAW MATERIAL STORAGE
24.2x14.0
7.30x9.75
COOL ZONE
17.3x8.86

4D
GENERAL STORE
STATIONARY
& ENG STORE LADIES TOILET
4D 3.30 x 4.35
7.25x5.97
1100 TROLLEY WIDTH 1100 TROLLEY WIDTH
1100 TROLLEY WIDTH

3D
7.40x2.00
1100

1100
CH 2 CH 1
1200 1200

DPB 2.56x1.92 3.20x1.92

3D
1200

WC WC WC
DPB 1.10X 1.10X 1.10X
3D 1.70 1.70 1.70

3D

3D
MIXING BLENDING AREA

1100
XX
XX
13.1x8.69 CH R2 CH R1
6.45x2.88
XX

3.40x3.46 3.40x3.46
PP GRANUAL
AUTO CLAVE RM DISPENSING
ROOM
PRODUCTION
OFFICE & IPQA
TOOL ROOM
4.25x7.00
HOPPER LOADER
ROOM
3D
WC
1.10X
WC
1.10X
WC
1.10X
4.05x7.00 2.92x7.00
ROOM
D3
4.11x7.00 1.70 1.70 1.70
CHAMBER LG. 8400

CHAMBER LG. 8400
PITT LENGTH 9650

PITT LENGTH 9650

CHAMBER LG. 8400

CH R3 EXIT
13.2x11.5 3.40x3.46 3.40x3.46
PITT LENGTH 9650

XX

EMERGENCY EXIT
3D
700

700

D4
1326 1326
1326

29.2x2.79
D4 PW U2
WFI U2
DPB
3D 4D 4D
12.3x6.07
1100

3500 PIT WIDTH 3500 PIT WIDTH
3500 PIT WIDTH

D4
1.25x4.25 COOLING ZONE
4D

1100 TROLLEY WIDTH 1100 TROLLEY WIDTH
1100 TROLLEY WIDTH
7 16x4 54 5 15X 2 00

COLLECTED AREA 9.03x4.54 3 WAY
SPB
FILLED BOTTLE 6.92x1.52 AUTO CLAVE

13.7x6.06
5.15X3.99

3D
PRODUCTION DPBEURO CAP
IPQA OFFICE AIR LOCK
OFFICE 1.76X1.20
STORE 3.04 x 1.44
3.37x1.44 2.71x1.443D DPB
6.77 M WIDE CORRIDOR

6.77 M WIDE CORRIDOR
D3
3D 3D

CORRIDOR
53.5x1.80
D4 D4
D4
D1

1D

D1

1D

KITCHEN OFFICE
7.30x3.19 5.56x3.19 OFFICE
PRIMARY FEMALE
1D

D1

1D

6.64x3.19 PRIMARY MALE PRIMARY VISITOR
WASHING CHANGE ROOM
CHANGE ROOM CHANGE ROOM
D1

6.50x5.18 2.50x5.18 6.50x5.18
6.32x5.18
1D

D1

1D

3D D3 3D
D1

WAITING ROOM
CORRIDOR
1D

D1

1D

8.46x2.49 5.04x2.49 D4 D4 3D D4
3D
D1

CONFERENCE 3D 3D
1D

D1

1D

HALL
LADIES TOILET F- REST ROOM
GENTS TOILET 6.00x8.10 M- REST ROOM CORRIDOR
5.80x11.3 4.39x2.96 12.5x2.96 4.93x2.99
1D

D1

1D

5.80x11.3
5.47x3.11
4D

4D
RECEPTION
1D

D1

1D

13.5x5.55

4D
D4
M- WASH ROOM
1D

D1

1D

F- WASH ROOM
8.20x3.09 8.20x3.06
D7 D7
D1 D1 D1

Employees
CORRIDOR Entry
53.5x2.50
7D

D7

3.24 M WIDE PASSAGE
CORRIDOR
SHAFT 48.5x2.90 SHAFT
LIFT
LIFT LIFT
3.24 M WIDE PASSAGE

D6

2.10X 2.10X

D3

2.00 2.00
FD DOCUMENT STORE QC STAFF CHEMICAL /GLASS FD
CONFERENCE
UP
4.73x6.62 DOCUMENTATION WARE STORAGE
7.71x6.62
9.31x6.07 CANTEEN UP
4.03x6.62 12.4x10.6
5.98x6.12
2.35x8.03
D3
D3
D3

WASH AREA
WET LAB D3 5.90x2.76
D3 9.31x4.85

QC MANEGER
PM TESTING ROOM
4.73x5.91 9.94x6.64
D6

D1 D3

HOT ROOM/
BALANCE GENERALFTIR FUME HOOD
D3 D3 ROOM 4.38x3.37 5.90x5.46
4.82x3.37

D3
1.80 M WIDE CORRIDOR
D3
QC SAMPLE PREP. 10.1x1.80
4.73x5.42

CORRIDOR
22.5x1.80
D1 INSTRUMENT ROOM II
15.3x5.43
CORRIDOR
INSTRUMENT ROOM I
1.80x29.2
DPB HPLC
9.94x7.04
D1
D3 TRANING HALL
QC SAMPLE JANITOR
D3
RECEIPT
1.50x3.93
14.8x19.0
3.11x3.93

REFERENCE TOC/LPC ROOM
D3 GC ROOM
D4 STANDARD 4.44x2.98
5.87x2.98 4.83x2.98

D3
GENTS CH ROOM
LADIES TOILET 4.73x3.42 RETENTION
3.30 x 4.35 SAMPLES ROOM
COB

(COLD ROOM 2 TO 8 °C)
D1
9.94x3.31
D3 D3
WC WC WC
6.77 M WIDE CORRIDOR

1.10X 1.10X 1.10X
2.94 M WIDE PASSAGE

1.70 1.70 1.70
STABILITY ROOM
LADIES CH ROOM 15.3x6.04
4.73x3.42
RM CONTROL SAMPLE
9.94x3.31
COB

D1

D3 D3 D1
WC WC WC
D6

1.10X 1.10X 1.10X

1.70 1.70 1.70
D3 D3 D3
AC
WASH AREA DPB MEDIA DIST.
5.221x2.02 DECONTAMINATION ROOM 7.34x2.02
5.54x2.02
3D

D3

MEDIA PREP. PREPARATION & STERILIZATION
GRADE D 8.11x5.91
GENTS TOILET 7.38x3.88
D3
GRADE D
DPB

3.30 x 4.35 D3
IC-1 DPB
GRADE B
DPB

INCUBATOR STERILITY
DPB
FOR MEDIA TESTING ROOM IC-1
D3 4.65x3.78 GRADE CNC IC-4
INCUBATOR
GRADE B 5.33x3.78
COOL ZONE IC-2
D3 3.38x3.78 D3 IC-3 R&D LAB -3 R&D LAB-4
1.83x1.91
3D

MEDIA 6.47x6.61 CORRIDOR 6.47x6.61
GRADE D OBSERVATION 13.3x1.80
STORE A/L GRADE B
4.78x3.54 2.50x4.24 D4 D4
CORRIDOR D3
GRADE C GRADE B GRADE B 3.53x2.02 DPB
CH-II CH-III 1.72x1.86
CH-I 2.02x2.02
4.81x2.02 2.73x2.02
D4 GRADE C GRADE CNC
DPB IC-1
D4
D4 D4
MLT ROOM INCUBATOR IC-4
D4 4.65x3.77 5.33x3.77 2.80x7.77
BET ROOM IC-2
GRADE C GRADE B
GRADE CNC GRADE D IC-3
A/L EXIT-II D3
4.78x2.83 EXIT-I
5.36x2.02 2.63x2.02 OBSERVATION
D4 D4 3.49x2.02
D3
GRADE C FRIDGE
R&D LAB -2 R&D LAB -1
CORRIDOR 6.47x6.61 6.47x6.61
BSC
GRADE CNC 1.80x8.50
GRADE D GRADE C GRADE CNC
CORRIDOR A/L CH-I INCUBATION ROOM
COB

2.50x8.31
D4 5.36x2.02 D4 6.04x2.02 GRADE D 4.36x3.78
D4
CULTURE ROOM 3.76x3.78
GENTS CH ROOM 4.65x3.78
GRADE CNC D3 D3
4.78x2.83 DPB
D3

D3 D3
D3

3D

DPB DPB D3

GRADE CNC GRADE CNC GRADE CNC D4 D4
MICRO DPB CORRIDOR
MICRO OFFICE MICRO
SAMPLE RECEIPT 14.8x1.80
COB

5.17x4.24 DOCUMENT STORE GRADE D
2.63x4.24 GRADE D SCRAP OUT
3.49x4.24 EXIT GRADE CNC
LADIES CH ROOM CH ROOM 5.49x2.86
2.39x2.86 2.15x2.86 A/L D3 D3

GRADE CNC 2.66x2.86
D3 4.78x3.12 D3
D3 D3
D3
R&D OFFICE
6.77 M WIDE CORRIDOR

7.31x2.90 R&D ENTRANCE
CORRIDOR 7.48x2.90
34.1x1.80
D3

LADIES TOILET
D4 D4 D4
3.30 x 4.35
2.94 M WIDE PASSAGE

CORRIDOR
16.7X1.80
STORAGE

CABIN 1
CABIN 4 CABIN 3 CABIN 2 WC WC WC
3.39x4.14
2.88x4.14 3.43x4.14 4.05x4.14 D4 D4 D4 D4 1.10X 1.10X 1.10X
1.70 1.70 1.70
4.13x4.19 PAL-1
PRIMARY CHANGE PRIMARY CHANGE
AUDIT MAL ENTRY
FEMALE
2.27X1.64
2.85X5.78 VISITOR COLD ROOM
2.85x5.81 4.43X3.93
4.51X5.78 4.19X3.27 D4 DE-CARTONING DPB
3.63X1.92 2.26X2.53 3.89X3.32
D4 D4 D4 D4
PRIMARY CHANGE
MALE PAL-2
2.27X1.64
5.31X5.81
WC WC WC
D4

D4
1.10X 1.10X 1.10X
DPB 1.70 1.70 1.70
CORRIDOR
D4 D4 D4
CLEAN JC
2.62x120.8 1.88X2.48
CORRIDOR
9.00X1.80
D4

DPB
D4
D4

CORRIDOR
GENTS TOILET
1.80x117.9

QA OFFICE QA MANGER
3.30 x 4.35

18.0x10.8 2.85x4.06
MATERIAL PREP WASH
8.15X3.07
AC
1.00X2.40 UNLOAD
8.41X3.40
D4

5.26X1.64
D4

2.21X1.46
DRYING ZONE

DPB
1.00X1.00 VIAL WASHING
6.42X9.65
D6

D4

D4

PAL
STERILIZING ZONE

2.84X1.45

D4

STERILE CORRIDOR
QA HEAD 15.5X1.80
DISINFECTANT
2.85x4.47 D3 D3 8.41X2.62
COOLING ZONE
D4
D4

MAL
MAL PAL
2.84X1.45
D4

1.45X1.64 1.41X1.64 PASS THROUGH
STABLIZING ZONE
D4

D3 D3
D3

PAL
D6
2.04X2.29
D3

D4

TUMBLE
DRYER FORMULATION
PACKING AREA 8.41X3.59
1.50X2.00
D3

D4

MAL
CONVEYOR
SWING

2.04X1.47 DOUBLE
DOOR
D3 WASHER
UNLOADING AREA
36" TURNTABLE
NEW ADD

LAUNDRY AREA 1.50X2.00
QA DOCUMENT 5.10X1.57
2.98X3.88
2.94X5.13 8.49X1.80

21.0x9.82
& LOADING TRAY
VIAL COLLECTION

D2

FILL FINISH
D4

6.42X8.30
D4

EXIT 2 EXIT 1
4.54X1.87 4.40X1.87
D4

PAL 3 PAL 4
4.23X3.82 3.20X3.82
CORRIDOR
2.85X25.2 PAL 1 PAL 2
4.54X1.87 4.40X1.87
D4

D4

D4

D4

SHAFT SHAFT
D4

LIFT LIFT
3.24 M WIDE PASSAGE

2.10X 2.10X
2.00 CONTROL ROOM
LYO SERVICE AREA
LYO LOAD 2.00

D4

5.55X3.55 10.9X3.55 8.29X3.55
D8
3.24 M WIDE PASSAGE

FD FD
AUTOMATIC FOUR HEAD ALUMINUM CAP

D8
SEALING MACHINE, MODEL-AEWCS-120

UP D4
UP
SWING
CONVEYOR

BULK STAGING
VIAL COLLECTION

SECONDARY PACKING AREA
16.6X4.04 CAPPING
TRAY

21.0x8.07 8.29X4.04
DPB

D8

DPB
CORRIDOR
25.5X1.80 JC
DPB 2.32X1.00

DPB
D4
D4
UNDERTEST

VISUAL & BUFFER ROOM PACKING CHANGE
INSPECTION 2.83X2.68 FEMALE
9.79X3.87 4.14 X 3.08
D4

2.88X1.14
D4
SAMPLING

D8
D4

PPM STORE SPM STORE 11.8X1.80
15.2x12.0 14.9x9.32
D4

7.22x2.69 PACKING CHANGE
11.8x3.03 MALE
4.14 X 4.23
QUARANTINE

LABELING & PACKING
12.6x5.40
D4

PB
D4

PACKING
D6

VISITOR
4.41 X 3.08
6.77 M WIDE CORRIDOR

D4
D8
D4
D8

D4

D8 D8
D4 PACKING OFFICE
LABELING STORE 4.14X4.50
FINISHED GOODS STAGING
7.68 X 5.67
12.6X 5.67
3.14 X 1.84
APPROVED RUBBER
SPM RECEIPT REJECT
BUNG STORE
3.36 X 2.97 3.59 X 2.98 3.58 X 2.98

D8 D8 D8
LADIES TOILET
3.30 x 4.35

WC WC WC
2.94 M WIDE PASSAGE

1.10X 1.10X 1.10X
1.70 1.70 1.70

CORRIDOR
2.80x27.9

WC WC WC
1.10X 1.10X 1.10X
1.70 1.70 1.70

GENTS TOILET
3.30 x 4.35

PLASMA STORAGE
~ 1000 L
~ 70 KL GV PLASMA STORAGE PLASMA STORAGE PLASMA STORAGE
12.2x17.1m ~ 1000 L ~ 1000 L ~ 1000 L
~ 70 KL GV ~ 70 KL GV ~ 70 KL GV
12.1x17.1m 12.1x17.1m 12.1x17.1m
3.30X2.45

3.30X2.45

D3 D3 D3 D3
D6

D6
LIFT

LIFT

CORRIDOR
49.0x2.59

CLIENT :- LIFENITY-LARGE VOLUME PARENTERALS

DESIGN &
PROJECT BY :-BIOBRIDGE HEALTHCARE SOLUTION PVT.LTD.
PREPARED A.K
ADDRESS :-13, Rachna Blossom, Jagdishnagar,
Aundh, Pune -411007, Maharastra, India.

FIRST FLOOR
CHECKED G.K.

DATE 18-04-2026
TITLE- CONCEPTUAL LAYOUT-FF REVISION R0
SCALE 1:1

SIZE A0 DWG.NO 01

Reagent Details:
Formula & Calculations:
Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).
Calculation: To prepare 1 L of 0.18M sodium chloride (MW = 58.44g/mol)
Weight of solute(g)=1×10.519=10.519g.
Preparation Method:
Weigh 10.519g of sodium chloride
To prepare 25mM Tris 1L solution from 1M Tris stock solution formula is M1(Molarity of stock) x V1
(volume of stock)= M2 (Molarity of working) xV2 (volume of working).
Take 25mL of 1M tris solution and add weighed amount of sodium chloride as required and dissolve in water
to make a 1 L solution.
Preparation for different volumes:
To prepare _____ of 0.18M NaCl (MW = 58.44g/mol).
Weight of solute(g)=10.519x _____ L= _____ gm
Weigh _____ gm of NaCl.
To prepare _____ of 25mM Tris
Take 25mL x _____ L= _____ mL of 1M Tris
Dissolve pre weighed NaCl in 25mM Tris solution to make a _____ L solution.
Adjust the pH to 7.5 using concentrated HCl/NaOH solution
Check and record the details:
Preparation Details:
Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry
Date
| | | |
Final volume prepared | Weighed Quantity | pH(7.5±0.1) | Conductivity (18.0±2.0 ms/cm) | Storage
Condition:
| | | | RT
Date of Preparation | Prepared By | Checked By
| |

Reagent Details:

Formula:

Sodium Acetate(M.Wt): 82.03g/mol.

Acetic Acid (M.Wt): 60.05g/ mol

Preparation of 0.4 M Sodium Acetate pH 4.6 buffer

Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).

Calculation: To prepare 1 L of 0.4 M Sodium Acetate (MW = 82.03g/mol).

Weight of solute(g)=0.4×82.03× 1.0=32.81g.

Preparation Method:

Weigh 32.81 g of Sodium Acetate and dissolve in water to make a 1 L solution.

Preparation for required volumes:

To prepare _____ of 0.4 M Sodium Acetate (MW = 82.03g/mol).

Weight of solute(g)=0.4×82.03× _____ L= _____ g.

Weigh _____ gm of Sodium Acetate and dissolve in water to make a _____ L solution.

Calculations: To prepare the 0.4 M Acetic Acid.

Formula & Calculations:

Formula: Use the formula to calculate the molarity:

Molarity (M) = (Per cent by Weight / 100) * (Density in g/mL) * 1000 / Molar Mass.

Example: Molarity of Glacial Acetic Acid

Molar Mass of CH₃COOH: ~60.05 g/mol

Density: ~1.05 g/mL

Purity: 100%

Acetic Acid (MW = 60.05 g/mol).

Calculation:

Molarity = (100 / 100) * 1.05 g/mL * 1000 mL/L / 60.05 g/mol ≈ 17.5 M.

Calculation: To prepare 1 L of 0.4 M.

Applying the formula to prepare the working buffer:

C₁V₁=C₂V₂

C₁ = 17.5 M acetic acid.

V₁ = volume of stock needed to take.

C₂ = desired (working) concentration 0.4 M acetic acid.

V₂ = final total volume of working solution for one litre.

V₁ = (C₂ x V₂)/C₁.

C₂=0.4 M ; V₂= 1.0 L; C₁=17.5 M

V₁ = (0.4 x 1.0)/17.5

V₁ = 0.022 L

Take 0.022 L of the Stock solution of acetic acid and add it to a volumetric flask, making up the volume to 1 L.

To prepare various volumes:

V₁ = volume of stock needed to take _____ L.

C₂ = desired (working) concentration 0.4 M acetic acid.

V₂ = final total volume of working solution _____ L.

V₁ = (C₂ _____ x V₂ _____) /17.5.

V₁= _____ L.

Take V₁ _____ L of the Stock solution of acetic acid and add it to a volumetric flask, making up the final volume _____ L.

Procedure for Preparation of 0.4 M Sodium Acetate pH 4.6.

To prepare a pH 4.6 solution, 0.4 M Sodium Acetate solution is titrated with 0.4 M Acetic acid so that the pH attains pH 4.6.

Check and record the details:

Preparation Details:

Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry

Date

Sodium Acetate | | | |

Acetic Acid | | | |

Final volume prepared | Measured Quantity | pH (4.6± 0.1) | Conductivity (13.7±0.1ms/cm) | Storage

Condition | Reagent Record

| | | | RT |

Date of Preparation | Prepared By | Checked By

| |

Reagent Details:

Formula & Calculations:

Formula: Use the formula to calculate the molarity:

Molarity (M) = (Per cent by Weight / 100) * (Density in g/mL) * 1000 / Molar Mass.

Example: Molarity of Glacial Acetic Acid

Molar Mass of CH₃COOH: ~60.05 g/mol

Density: ~1.05 g/mL

Purity: 100%

Acetic Acid (MW = 60.05 g/mol).

Calculation:

Molarity = (100 / 100) * 1.05 g/mL * 1000 mL/L / 60.05 g/mol ≈ 17.5 M.

Calculation: To prepare 1 L of 0.2M.

Applying the formula to prepare the working buffer:

$C_1 V_1 = C_2 V_2$

C_1 = 17.5 M acetic acid.

V_1 = volume of stock needed to take.

C_2 = desired (working) concentration 0.2 M acetic acid.

V_2 = final total volume of working solution for one litre.

$V_1 = (C_2 \times V_2) / C_1$.

$C_2 = 0.2$ M; $V_2 = 1.0$ L; $C_1 = 17.5$ M

$V_1 = (0.2 \times 1.0) / 17.5$

$V_1 = 0.011$ L

Take 0.011 L of the Stock solution of acetic acid and add it to a volumetric flask, making up the volume to 1 L.

To prepare various volumes:

V_1 = volume of stock needed to take _____ L.

C_2 = desired (working) concentration 0.2 M acetic acid.

V_2 = final total volume of working solution _____ L.

$V_1 = (C_2 \times V_2) / 17.5$.

$V_1 =$ _____ L.

Take V_1 _____ L of the Stock solution of acetic acid and add it to a volumetric flask, making up the final volume _____ L.

Check and record the details:

Preparation Details:

Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry

Date

|||||

Final volume prepared | Measured Quantity | pH (2.8) | Conductivity (6 to 7ms/cm) | Storage Condition |

Reagent Record

|||||

Date of Preparation | Prepared By | Checked By

||

6D UP

FIRE EXIT

D6 D1
LIFT

FIRE EXIT
LOBBY
6.77X11.77
RECEPTION OPEN TO ALL
FLOORS

D1
6D

UP FIREMAN
DPB
Goods LIFT D6
LIFT
3.50X3.00
1D
2.70X2.50

D6

DPB DPB WASTE IN A/L A/L DHS
CHEMICAL DAY STORE 3 WAY FIRE STAIR
LOBBY

6D

6D

DE
up
3.00X11.30

LAF
DPB DPB
OPEN TO
FIRST BUFFER TANK STORAGE-2 CIP / SIP COLD ROOM LAF UP
UP FLOOR PREPARATION DECONTAMINATION WASHING STERILIZATION COOL ZONE

AUTOCLAVE
A/L GOWN A/L 6D

GS+HH

GS+HH
DPB
LENAP LORTNOC LENAP LORTNOC

6D 6D 6D
DPB
WASTE OUT

DE
A/L DE GOWN A/L
CORRIDOR

FIRE CONTROL
D6 D6 D6 D6 D3 ROOM UP
DPB AKTA-1" TFF
NANO
LAF 3.50X6.54
FILTER MH

A/L GOWN A/L
1000 L
BULK FILTRATION

LAF
500 L

MH
SL MH

600 L

500 L

LAF
MH SL
600 L MH

FIRE PUMP

MH
DE MH

50 L ROOM
A/L
A/L GOWN LAF MH

2000 L 500 L
CHROMO SL
STAGE 13 E 3.50X5.00
COLUMN 250 L
MH 600 L STAGE 10 T/ MH
UP

STAGE 11 F STAGE 9 N MH

250 L
200 L

AKTA-1"
STAGE 6 (I)
MH

COLUMN PACKING 250 L
MH

250 L
1000 L
AND SKIDS
MH

200 L
MH

MH

2.80 M WIDE PASSAGE
250 L
COOL ZONE
MH MH

150 L 100 L MH

MH

1000 L 500 L

6.77 M WIDE CORRIDOR
MH

MH

300 L
WC WC WC 150 L MH

1.10X 1.10X 1.10X CHROMO 100 L MH

COLUMN
100 L
MH

1.70 1.70 1.70
300 L
LAF 150 L
DPB LAF
MH

CHROMO MH

AKTA-1"
MH

100 L COLUMN

AUTOCLAVE
DHS

D6
WC WC WC 500 L 500 L CHROMO
250 L 250 L 150 L 100 L COLUMN

6D
1.10X 1.10X 1.10X MH
MH MH

1.70 1.70 1.70 CORRIDOR A/L MH MH
MH MH
MH

MH

CORRIDOR CORRIDOR

STERILIZATION 2000 L

D6

CLAF
2500 L 2500 L

6D

PRE - VIRUS CONTROLLED CORRIDOR
1000 L

TFF
MH

UP COLUMN PACKING
DPB
AND SKIDS MH

UP

AKTA-1"
WASHING 100 L
A/L

AKTA-1"
STAGE 7A 500 L
MH

STAGE 5 (I) A

D6
DPB
WASTE IN MH 500 L
250 L

TFF
LOBBY 300 L
MH
MH

Emergency Exit
6.77X11.77
DECONTAMINATION 250 L
500 L
MH

OUT MH
MH

WASTE

D6
500 L

DPB STAGE 5(I)B 1500 L 250 L
MH
MH

100 L SL
MH

250 L 250 L 250 L 600 L 500 L
100 L 100 L

6D
100 L
MH

MH
MH MH MH
MH

100 L
MH MH MH

CIP / SIP
MH

1500 L
SL
UP
MH MH

MH MH MH

1000 L
MH

500 L 500 L
MH

MH

1000 L 1000 L 1000 L
UP
100 L 1500 L
1500 L

GS+HH

GS+HH
MH SL
LENAP LORTNOC LENAP LORTNOC

MH
2500 L
MH

1500 L
50 L
Pass. LIFT CHROMO MH
250 L 250 L 250 L 250 L
2.30X2.00 DPB COLUMN

6D
MH MH MH MH

WASTE OUT

BAG CUTTING & WC WC WC

POOLING MH
MH MH

MH

MH MH
STAGE 4 (F) 1.10X
1.70
1.10X
1.70
1.10X
1.70

150 L 500 L

6D
500 L
SL 1000 L
TANK STORAGE 1000 L 1500 L 1500 L

D6
MH

WC WC WC
STAGE 12 (E)
MH

250 L 1.10X 1.10X 1.10X
MH

STAGE 1 (F)/ 1.70 1.70 1.70
CHROMO
500 L
COLUMN

250 L
300 L 300 L MH

TFF
D6

AKTA-1"
6D
MH MH

CLAF

2500 L

THAWING PLASMA 2000 L 2000 L
UP

AKTA-1"
MH

DPB
500 L 250 L 250 L 300 L 300
300L L
BUFFER

6D
MH MH

CHROMO
MH
COLUMN MH MH MH
MH MH

CHROMO
PREPARATION D6 D6 D6
COLUMN

CORRIDOR

6D

DE
DPB
3.24 M WIDE PASSAGE

D6 D6 D6

6D
MH MH

MH MH MH MH

CHROMO MH
MH
MH MH MH MH

COLUMN
100 L 100 L 100 L 100 L MH

100 L 100 L CENTRIFUGE
200 L 250 L 300 L 300 L 500 L 500 L
1500 L

PLASMA DAY STAGE 3 (F)
STORE STAGE 2 (F)
STAGE 8 N

AKTA-1"

AKTA-1"
UP

CHROMO
COLUMN
MH
UP
3D

SL
600 L 1500 L 2000 L
1500 L 1500 L 1500 L

300 L 300 L MH MH MH
MH
MH

100 L 100 L 100 L CHROMO CHROMO
DPB MH
TFF MH

COLUMN COLUMN AKTA-1"
MH MH MH

WC WC WC
1.10X 1.10X 1.10X EMER.GLASS

1.70 1.70 1.70
LOBBY
6.77X11.00

WC WC WC
STAIRCASE
1.10X 1.10X 1.10X AND
1.70 1.70 1.70
LIFT ONLY
CORRIDOR
FOR
CORRIDOR
SECOND FLOOR
(L.V.P.)

UP

UP

W

D6 D6 D6 D6 D6

CORRIDOR

D6
LOBBY CORRIDOR
12.47X6.77
D3 D6
Pass. LIFT W
DPB
2.30X2.00
D3

6D

3D D3 D3
UP
D3
3D

CORRIDOR

D3
3D

SD SD SD SD

DPB
6D
ELECTRIC
D3 ROOM
DPB D3
DPB
3D D3 D6 STORE 3.50X6.00
OFFICE
3D
LOBBY
3.00X11.53 3D

OPEN TO
SECOND D3 3D D3

D3
D3 REJECT
FLOOR ROOM
RECALLED
GOODS FG DISCPATCH
FG IN
PB 6D D3 D3 PB D6 D6

D6 D3
SERVER
Goods LIFT CORRIDOR
ROOM
3.50X3.00
6D

D6
3.50X6.57
DISPATCH BAY RECEIVING BAY RECEIVING BAY DISPATCH BAY

RS3
40' CONTAINER YARD

CLIENT :- LIFENITY- HEALTH LIMITED (DIALYZER, LVP & PLASMA)
ADDRESS :-Auric city, chhatrapati sambhaji nagar

DESIGN &
PROJECT BY :-BIOBRIDGE HEALTHCARE SOLUTION PVT.LTD.
PREPARED A.K
ADDRESS :-13 R B
A P -411007 M I .

GROUND FLOOR
CHECKED G.K.

DATE 18-06-2026
TITLE- CONCEPTUAL LAYOUT-GF REVISION R0
SCALE 1:1

SIZE A0 DWG.NO 01

Reagent Details:
Formula & Calculations:
Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).
Calculation: To prepare 1 L of 0.2M Sodium citrate monohydrate (MW = 294.1g/mol)
Weight of solute(g)=1×38.42=38.42g.
Preparation Method:
Weigh 38.42 g of Sodium citrate monohydrate and dissolve in water to make a 1 L solution.
Preparation for different volumes:
To prepare _____ of 0.2M Sodium citrate monohydrate (MW = 192.12g/mol).
Weight of solute(g)=1×38.42 _____L= _____gm
Weigh _____gm of Sodium citrate monohydrate and dissolve in water to make a _____ L solution.
Adjust the pH to 7.5 with sodium hydroxide or hydrochloric acid.
Check and record the details:
Preparation Details:
Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry
Date
| | | | |
Final volume prepared | Weighed Quantity | pH (7.3±0.2) | Conductivity (33±0.2ms/cm) | Storage Condition:
| | | | RT
Date of Preparation | Prepared By | Checked By
| |

Reagent Details:
Preparation of IG-RR-11 working buffer from Stock buffer (IG-RR-10):
Calculations: For the Preparation of 20mM Sodium acetate pH 4.0.
Stock Buffer (IG-RR-10):1.0M Sodium acetate, pH 4.0.
Formula:
 $C1 \times V1 = C2 \times V2$
C1 = concentration of stock
V1 = volume of stock you need to take
C2 = desired (working) concentration
V2 = final total volume of working solution
Applying the formula to prepare the working buffer:
 $C1 \times V1 = C2 \times V2$
C1 = 1.0 M Sodium Acetate pH 4.0.
V1 = volume of stock you need to take _____ L.
C2 = desired (working) concentration 20mM Sodium Acetate pH 4.0
V2 = final total volume of working solution _____ L.
 $V1 = (C2 \times V2) / C1$.
C2=0.02M ; V2= 1.0 L; C1=1.0M
 $V1 = (0.02 \times 1.0) / 1.0$
V1 = 0.02L
Take 0.02 L (IG-RR-10) Stock buffer solution and dilute it with water, making up to 1 L.
Check and record the details:
Preparation Details:
Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry
Date
Sodium Acetate | | | |
Acetic Acid | | | |
Stock buffer (IG-RR-10) taken volume(L) | pH(4.0 ± 0.1) | Conductivity (1.85 ± 0.2) | Storage Condition
| Final volume prepared
| | |
Date of Preparation | Prepared By | Checked By
| |

SHAFT

3.30X2.45
FIREMAN

LIFT
LIFT
2.80X2.60

LADIES TOILET

2.94 M WIDE CORRIDOR
3.30 x 4.35

WC WC WC
1.10X 1.10X 1.10X
1.70 1.70 1.70

6.77 M WIDE CORRIDOR
3600.000

2000.000
WC WC WC
1.10X 1.10X 1.10X
1.70 1.70 1.70

GENTS TOILET
3.30 x 4.35

SHAFT SHAFT

LIFT LIFT
2.10X 2.10X

3.24 M WIDE PASSAGE
2.00 2.00

UP
FD
LVP FD
UP

48.54 x 96.57

2.80 M WIDE PASSAGE

LADIES TOILET
3.30 x 4.35

WC WC WC
1.10X 1.10X 1.10X
1.70 1.70 1.70

WC WC WC
1.10X 1.10X 1.10X
1.70 1.70 1.70

GENTS TOILET
3.30 x 4.35
6.77 M WIDE CORRIDOR

6.77 M WIDE CORRIDOR
CORRIDOR
SHAFT 48.5x2.90 SHAFT
LIFT
LIFT D6 LIFT
2.10X 2.10X
D3

3.24 M WIDE PASSAGE
2.00 2.00
FD DOCUMENT STORE QC STAFF CHEMICAL /GLASS CANTEEN FD FD
CONFERENCE
4.73x6.62 DOCUMENTATION WARE STORAGE 9.31x6.07
UP 4.03x6.62 5.21x6.62 15.2x6.12 UP
11.9x4.57
2.30x3.38

D3
D3
D3

WET LAB WASH AREA
D3 9.31x4.85 D3 5.49x3.16

PM TESTING ROOM
7.44x6.65
QC MANEGER
4.73x5.91
D1 D3

HOT ROOM/
D6

BALANCE GENERALFTIR FUME HOOD
D3 D3 ROOM 4.38x3.37 5.49x5.06
4.82x3.37

D3
1.80 M WIDE CORRIDOR
D3
QC SAMPLE PREP. 7.66x1.80
4.73x5.42

CORRIDOR
22.5x1.80
D1 INSTRUMENT ROOM II TRAINING HALL
14.9x5.43 11.9x19.0
INSTRUMENT ROOM I CORRIDOR
1.80x29.2
DPB HPLC
7.44x6.87
D1
QC SAMPLE JANITOR
D3 D3
RECEIPT
3.12x3.93 1.50x3.93

REFERENCE TOC/LPC ROOM
D3 GC ROOM
D4 STANDARD 4.03x2.98
5.87x2.98 4.83x2.98

D3
GENTS CH ROOM
LADIES TOILET 4.73x3.42 RETENTION
3.30 x 4.35
COB

SAMPLES ROOM D1
(COLD ROOM 2 TO 8 °C)
7.44x3.31
D3 D3
WC WC WC
6.77 M WIDE CORRIDOR

1.10X 1.10X 1.10X
2.80 M WIDE PASSAGE

1.70 1.70 1.70
STABILITY ROOM
LADIES CH ROOM 14.9x6.04
4.73x3.43
RM CONTROL SAMPLE
COB

7.44x3.31
D1

D3 D3 D1
WC WC WC
D6

1.10X 1.10X 1.10X
1.70 1.70 1.70
D3 D3 D3
AC
DPB MEDIA DIST.
WASH AREA DECONTAMINATION ROOM 6.93x2.02
5.51x2.02 5.25x2.02
3D

D3

MEDIA PREP. PREPARATION & STERILIZATION
GRADE D 5.61x5.91
GENTS TOILET 7.38x3.88 GRADE D
D3
DPB

3.30 x 4.35 D3
IC-1 DPB
GRADE B
DPB

INCUBATOR STERILITY
DPB
FOR MEDIA TESTING ROOM IC-1
D3 4.36x3.78 GRADE CNC IC-4
INCUBATOR
GRADE B 6.09x3.78 R&D LAB-4
COOL ZONE IC-2

D3 3.38x3.78 5.05x6.61
D3 IC-3 R&D LAB -3
2.13x1.91 CORRIDOR
MEDIA D3 5.05x6.61
GRADE D OBSERVATION 13.3x1.80
STORE A/L GRADE B
4.78x3.54 2.50x4.24 D4 D4 CORRIDOR D3
GRADE C GRADE B GRADE B 3.82x2.02 DPB
CH-I CH-II CH-III 2.01x1.86
2.31x2.02 2.73x2.02 2.02x2.02
D4 GRADE C GRADE CNC
D4 D4 DPB IC-1
D3 MLT ROOM INCUBATOR IC-4
D4 4.36x3.77 6.09x3.77
BET ROOM GRADE C IC-2
GRADE D GRADE B
GRADE CNC EXIT-I EXIT-II IC-3
4.78x2.83 A/L D3
D4 3.47x2.02 D4 2.69x2.02 2.97x2.02 OBSERVATION
D3
FRIDGE R&D LAB -2 R&D LAB -1
GRADE C
CORRIDOR BSC 5.05x6.61 5.05x6.61
GRADE CNC GRADE D GRADE C 1.80x8.50 GRADE CNC
CORRIDOR A/L CH-I
COB

INCUBATION ROOM
2.50x8.31
D4
3.66x2.02 5.54x2.02 GRADE D
D4 D4 7.82x3.78
CULTURE ROOM
GENTS CH ROOM 4.36x3.78 1.61x7.77
GRADE CNC D3 D3
D3 4.78x2.83 D3 DPB
D3

3D

DPB DPB
D3
GRADE CNC GRADE CNC D4 D4
GRADE CNC MICRO DPB
MICRO
MICRO OFFICE SAMPLE RECEIPT CORRIDOR
COB

DOCUMENT STORE 2.92x4.24 GRADE D
3.47x4.24 2.69x4.24 GRADE D SCRAP OUT 11.9x1.80
CH ROOM EXIT GRADE CNC
LADIES CH ROOM 5.69x2.86
2.10x2.86 2.15x2.86 A/L D3 D3
GRADE CNC 2.02x2.86
D3 4.78x3.12 D3
D3 D3
D3
R&D OFFICE R&D ENTRANCE
CORRIDOR 5.89x2.90 6.00x2.90
31.2x1.80
D3

LADIES TOILET
3.30 x 4.35

WC WC WC
1.10X 1.10X 1.10X
1.70 1.70 1.70

WC WC WC
1.10X 1.10X 1.10X
1.70 1.70 1.70

CORRIDOR
2.62x115.3
CORRIDOR
PLASMA STORAGE PLASMA STORAGE PLASMA STORAGE PLASMA STORAGE GENTS TOILET
2.62x115.3
~ 1000 L ~ 1000 L ~ 1000 L ~ 1000 L 3.30 x 4.35
~ 70 KL GV ~ 70 KL GV ~ 70 KL GV ~ 70 KL GV
10.5x17.1m 11.5x17.1m 10.9x17.1m 10.1x17.1m

D3 D3 D3 D3
D6

CORRIDOR
43.1X1.80

PRINTER

CORRIDOR
24.2X1.80

STORAGE

CABIN 1
CABIN 4 CABIN 3 CABIN 2
3.39x4.14
2.88x4.14 3.43x4.14 4.05x4.14 D4 D4 D4 D4 D4 D4
DPB
PAL-1
2.58x4.19 2.62X1.64
COLD ROOM
4.19X3.27
AUDIT MAL ENTRY PRIMARY CHANGE PRIMARY CHANGE 2.26X2.53
D4 DE-CARTONING
3.12X3.32
SHAFT D4 D4 D4 D4 2.35x6.31 2.00X3.97 VISITOR FEMALE SHAFT
4.51X3.93 4.57X3.93 PRIMARY CHANGE
PAL-2
MALE
LIFT 2.67X5.81
2.62X1.64
LIFT
2.10X 2.10X
D4
DPB
D4

D4
D4 D4
CLEAN JC
2.00 1.88X2.48 2.00
CORRIDOR
3.24 M WIDE PASSAGE

3.24 M WIDE PASSAGE

9.16X1.80
D4

FD FD
D4

DPB
D4

UP QA OFFICE QA MANGER
UP

16.5x10.8 2.35x4.06
MATERIAL PREP WASH
8.15X3.07
AC
1.00X2.40

UNLOAD
D4

5.26X1.64 5.91X3.40
D4

2.21X1.46
DRYING ZONE

DPB
1.00X1.00 VIAL WASHING
6.00X9.65
D6

D4

D4

PAL
STERILIZING ZONE

2.84X1.45
D4

STERILE CORRIDOR
QA HEAD 15.5X1.80

2.35x4.47 D3 D3 DISINFECTANT
5.91X2.62
COOLING ZONE
D4
D4

MAL
MAL PAL
2.84X1.45
D4

1.45X1.64 1.41X1.64

STABLIZING ZONE

PASS THROUGH
D4

D3 D3
D3

D6 PAL
2.04X2.29
D3

D4

TUMBLE
DRYER FORMULATION
1.50X2.00 PACKING AREA
5.91X3.59
D3

D4

MAL
CONVEYOR
SWING

2.04X1.47 DOUBLE
DOOR
D3 WASHER
UNLOADING AREA
36" TURNTABLE
NEW ADD

LAUNDRY AREA 1.50X2.00
QA DOCUMENT 5.10X1.57
2.98X3.88
2.94X5.13 6.00X1.80

18.9x9.82
& LOADING TRAY
VIAL COLLECTION

D2
D4
D4

EXIT 2 EXIT 1
FILL FINISH
4.54X1.87 4.54X1.87
D4

6.00X8.30
PAL 3 PAL 4
CORRIDOR 2.50X3.82 2.50X3.82
2.00X25.2 PAL 1 PAL 2
4.54X1.87 4.54X1.87
D4

D4

D4

D4

6.77 M WIDE CORRIDOR

D4

LYO SERVICE AREA
CONTROL ROOM LYO LOAD
D4

5.55X3.55 9.52X3.55 6.83X3.55
D8

AUTOMATIC FOUR HEAD ALUMINUM CAP

D8
SEALING MACHINE, MODEL-AEWCS-120

D4

SWING
CONVEYOR

LADIES TOILET BULK STAGING
VIAL COLLECTION

SECONDARY PACKING AREA
3.30 x 4.35 15.1X4.04 CAPPING
TRAY

18.9x8.07 6.83X4.04

WC WC WC
DPB
2.80 M WIDE PASSAGE

1.10X 1.10X 1.10X D8
1.70 1.70 1.70
DPB
CORRIDOR
24.1X1.80 JC
DPB
D3

CORRIDOR 1.50X1.00
2.80x27.9

DPB
D4
UNDERTEST

WC WC WC
1.10X 1.10X 1.10X
D4

BUFFER ROOM PACKING CHANGE
1.70 1.70 1.70 VISUAL &
2.83X2.68
INSPECTION FEMALE
7.29X3.87 3.32 X 3.08
D4

2.88X1.14

GENTS TOILET
D4
SAMPLING

D8
3.30 x 4.35
SPM STORE
D4

10.8X1.80
PPM STORE 14.5x9.32 PACKING CHANGE
D4

6.77x2.69
13.1x12.0 MALE
9.78x3.03 3.32 X 3.05
QUARANTINE

D4

LABELING & PACKING

PB 10.1x5.40
D4

PACKING
VISITOR
3.32X3.08
D4
D6

D4
D8
D8

D4

D8 D8
D4
LABELING STORE PACKING OFFICE
FINISHED GOODS STAGING
7.68 X 5.67 3.32X5.67
10.1 X 5.67
4.12 X 1.84
APPROVED RUBBER
SPM RECEIPT REJECT
BUNG STORE
3.36 X 2.98 3.84 X 2.98 2.88 X 2.98
D8 D8 D8
3.30X2.45
LIFT

CORRIDOR
43.3x2.62

FIRST FLOOR CLIENT :- LIFENITY-LARGE VOLUME PARENTERALS

DESIGN &

PROJECT BY :-BIOBRIDGE HEALTHCARE SOLUTION PVT.LTD.
PREPARED A.K
ADDRESS :-13, Rachna Blossom, Jagdishnagar,
Aundh, Pune -411007, Maharastra, India. CHECKED G.K.

DATE 06-04-2026
TITLE- CONCEPTUAL LAYOUT-FF REVISION R0
SCALE 1:1

SIZE A0 DWG.NO 01

Reagent Details:
Formula & Calculations:
Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).
Calculation: To prepare 1 L of 0.2M Citric Acid (MW = 192.12g/mol)
Weight of solute(g)=1×38.424=38.424g.
Preparation Method:
Weigh 38.424 g of Citric acid and dissolve in water to make a 1 L solution.
Preparation for different volumes:
To prepare _____ of 0.2M Citric acid (MW = 192.12g/mol).
Weight of solute(g)=1×38.424 _____ L= _____ gm
Weigh _____ gm of Citric acid and dissolve in water to make a _____ L solution.
Check and record the details:
Preparation Details:
Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry
Date
| | | |
Final volume prepared | Weighed Quantity | pH (1.95± 0.2) | Conductivity (4.5± 0.5ms/cm) | Storage
Condition:
| | | RT
Date of Preparation | Prepared By | Checked By
| |

6D UP

FIRE EXIT

D6 D1
LIFT

FIRE EXIT
LOBBY
6.77X11.77
RECEPTION OPEN TO ALL
FLOORS

D1
6D

UP FIREMAN
Goods LIFT D6
LIFT
3.50X3.00
1D
2.70X2.50
DPB

D6

DPB DPB WASTE IN A/L A/L DHS
CHEMICAL DAY STORE 3 WAY FIRE STAIR
LOBBY

6D

6D

DE
up
3.00X11.30

LAF
DPB DPB
OPEN TO
FIRST BUFFER TANK STORAGE-2 CIP / SIP COLD ROOM LAF UP
UP FLOOR PREPARATION DECONTAMINATION WASHING STERILIZATION COOL ZONE

AUTOCLAVE
DPB
A/L GOWN A/L 6D

GS+HH

GS+HH
LENAP LORTNOC LENAP LORTNOC

6D 6D 6D DPB
WASTE OUT

DE
A/L DE GOWN A/L
CORRIDOR

FIRE CONTROL
D3 ROOM UP
LAF
D6 D6 D6 D6
DPB AKTA-1" TFF
NANO 3.50X6.54
FILTER MH

A/L GOWN A/L
1000 L
BULK FILTRATION

LAF
500 L

MH
SL MH

600 L

500 L

LAF
MH SL
600 L MH

MH
FIRE PUMP
DE MH

50 L ROOM
A/L
A/L GOWN LAF MH

2000 L 500 L
CHROMO SL
STAGE 13 E 3.50X5.00
COLUMN 250 L
MH 600 L STAGE 10 T/ MH
UP

STAGE 11 F STAGE 9 N MH

250 L
200 L

AKTA-1"
STAGE 6 (I)
MH

COLUMN PACKING 250 L
MH

250 L
1000 L
AND SKIDS
MH

200 L
MH

MH

2.80 M WIDE PASSAGE
250 L
COOL ZONE
MH MH

150 L 100 L MH

MH

1000 L

6.77 M WIDE CORRIDOR
MH

500 L
MH

300 L
WC WC WC 150 L MH

1.10X 1.10X 1.10X CHROMO 100 L MH

COLUMN
100 L
MH

1.70 1.70 1.70
300 L
LAF 150 L
DPB LAF
MH

CHROMO MH

AKTA-1"
MH

100 L COLUMN

AUTOCLAVE
DHS

D6
WC WC WC 500 L 500 L CHROMO
250 L 250 L 150 L 100 L COLUMN

6D
1.10X 1.10X 1.10X MH
MH MH

1.70 1.70 1.70 CORRIDOR A/L MH MH
MH MH
MH

MH

CORRIDOR CORRIDOR

STERILIZATION 2000 L

D6

CLAF
2500 L 2500 L

6D

PRE - VIRUS CONTROLLED CORRIDOR
1000 L

TFF
MH

UP COLUMN PACKING
DPB
AND SKIDS MH

UP

AKTA-1"
WASHING 100 L
A/L

AKTA-1"
STAGE 7A 500 L
MH

STAGE 5 (I) A

D6
DPB
WASTE IN MH 500 L
250 L

TFF
LOBBY 300 L
MH
MH

6.77X11.77
DECONTAMINATION 250 L
500 L
MH

OUT MH
MH

WASTE

D6
500 L

DPB STAGE 5(I)B 1500 L 250 L
MH
MH

100 L SL
MH

250 L 250 L 250 L 600 L 500 L
100 L 100 L

6D
100 L
MH

MH
MH MH MH
MH

100 L
MH MH MH

CIP / SIP
MH

1500 L
SL MH MH

UP
MH MH MH

1000 L
MH

500 L 500 L
MH

MH

1000 L 1000 L 1000 L

UP
100 L 1500 L
1500 L

GS+HH

GS+HH
MH SL
LENAP LORTNOC LENAP LORTNOC

MH
MH
2500 L
1500 L
50 L
Pass. LIFT CHROMO MH
250 L 250 L 250 L 250 L
2.30X2.00 DPB COLUMN

6D
MH MH MH MH

WASTE OUT

BAG CUTTING & WC WC WC

POOLING MH
MH MH

MH

MH MH
STAGE 4 (F) 1.10X
1.70
1.10X
1.70
1.10X
1.70

150 L 500 L

6D
500 L
SL 1000 L
TANK STORAGE 1000 L 1500 L 1500 L

D6
MH

WC WC WC
STAGE 12 (E)
MH

250 L 1.10X 1.10X 1.10X
MH

STAGE 1 (F)/ 1.70 1.70 1.70
CHROMO
500 L
COLUMN

250 L
300 L 300 L MH

TFF
D6

AKTA-1"
6D
MH MH

CLAF

2500 L

THAWING PLASMA 2000 L 2000 L
UP

AKTA-1"
MH

DPB
500 L 250 L 250 L 300 L 300
300L L
BUFFER

6D
MH MH

CHROMO
MH
COLUMN MH MH MH

MH MH

CHROMO
PREPARATION D6 D6 D6
COLUMN

CORRIDOR

6D

DE
DPB
3.24 M WIDE PASSAGE

D6 D6 D6

6D
MH MH

MH MH MH MH

CHROMO MH
MH
MH MH MH MH

COLUMN
100 L 100 L 100 L 100 L MH

100 L 100 L CENTRIFUGE
200 L 250 L 300 L 300 L 500 L 500 L
1500 L

PLASMA DAY STAGE 3 (F)
STORE STAGE 2 (F)
STAGE 8 N

AKTA-1"

AKTA-1"
UP

CHROMO
COLUMN
MH
UP
3D

SL
600 L 1500 L 2000 L
1500 L 1500 L 1500 L

300 L 300 L MH MH MH
MH
MH

100 L 100 L 100 L CHROMO CHROMO
DPB MH
TFF MH

COLUMN COLUMN AKTA-1"
MH MH MH

WC WC WC
1.10X 1.10X 1.10X EMER.GLASS

1.70 1.70 1.70
LOBBY
6.77X11.00

WC WC WC
STAIRCASE
1.10X 1.10X 1.10X AND
1.70 1.70 1.70
LIFT ONLY
CORRIDOR
FOR
CORRIDOR
SECOND FLOOR
(L.V.P.)

UP

UP

W

D6 D6 D6 D6 D6

CORRIDOR

D6
LOBBY CORRIDOR

12.47X6.77
D3 D6
Pass. LIFT W
DPB
2.30X2.00
D3

6D

3D D3 D3
UP
D3
3D

CORRIDOR

D3
3D

SD SD SD SD

DPB
6D
ELECTRIC
D3 ROOM
DPB
D3
DPB
3D D3 D6 STORE 3.50X6.00
OFFICE
3D
LOBBY
3.00X11.53 3D

OPEN TO

D3
SECOND D3
D3
3D D3

FLOOR REJECT
RECALLED
GOODS ROOM FG DISCPATCH
FG IN
6D D3 D3 D6 D6

D6 D3
SERVER
Goods LIFT CORRIDOR ROOM
3.50X3.00
6D

D6
3.50X6.57
DISPATCH BAY RECEIVING BAY RECEIVING BAY DISPATCH BAY

RS3
40' CONTAINER YARD

CLIENT :- LIFENITY- HEALTH LIMITED (DIALYZER, LVP & PLASMA)
ADDRESS :-Auric city, chhatrapati sambhaji nagar

DESIGN &
PROJECT BY :-BIOBRIDGE HEALTHCARE SOLUTION PVT.LTD.
PREPARED A.K
ADDRESS :-13 R B
A P -411007 M I .

GROUND FLOOR
CHECKED G.K.

DATE 18-06-2026
TITLE- CONCEPTUAL LAYOUT-GF REVISION R0
SCALE 1:1

SIZE A0 DWG.NO 01

Reagent Details:

Formula:

Glycine : (M.Wt): 75.07g/mol.

Sodium Hydroxide : (M.Wt): 40 g/ mol.

Sodium Chloride : (M.Wt): 58.44 g/ mol.

Preparation of 0.8 M Glycine pH 7.0 buffer

Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).

Calculation: To prepare 1 L of 0.8 M Glycine (MW = 75.07g/mol).

Weight of solute(g)=0.8×75.07×1.0=60.056 g.

Preparation Method:

Weigh 60.056 g of Glycine and dissolve it in water to make a 1 L solution.

Preparation for required volumes:

To prepare _____ of 0.8 M Glycine (MW = 75.07g/mol).

Weight of solute(g)=0.8×75.07× _____ L= _____ g.

Weigh _____ gm of Glycine and dissolve in water to make a _____ L solution.

Calculations: To prepare the 1 M Sodium Hydroxide.

Formula & Calculations:

Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).

Calculation: To prepare 1 L of 1N Sodium Hydroxide (MW = 40.0 g/mol)

Weight of solute(g)=1×40 = 40.00g.

Preparation Method:

Weigh 40.00 g of Sodium Hydroxide and dissolve in water to make a 1 L solution.

Preparation for different volumes:

To prepare _____ of 1N Sodium Hydroxide (MW = 40.0 g/mol).

Weight of solute(g)=1×40.0 x _____ L= _____ gm

Weigh _____ gm of Sodium Hydroxide and dissolve in water to make a _____ L solution.

Calculations: To prepare the 0.6 M Sodium Chloride.

Formula & Calculations:

Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).

Calculation: To prepare 1 L of 0.6 M Sodium Chloride (MW = 58.44 g/mol)

Weight of solute(g)=0.6 × 58.44 x 1 = 35.06g.

Preparation Method:

Weigh 35.06 g of Sodium Chloride and dissolve in water to make a 1 L solution.

Preparation for different volumes:

To prepare _____ of 0.6M Sodium Chloride (MW = 58.44 g/mol).

Weight of solute(g)=0.6×58.44 x _____ L= _____ gm

Weigh _____ gm of Sodium Chloride and dissolve in water to make a _____ L solution.

Procedure for Preparation of 0.8 M Glycine+ 0.6M Sodium Chloride

pH 9.0.

To prepare a pH 9.0 solution, 0.8 M Glycine solution is pH adjusted with 1 M sodium Hydroxide so that the pH attains 9.0 and 0.6M Sodium Chloride salt is added to the solution and dissolved to make up the final volume.

Preparation Details:

Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry Date

Glycine | | | | |

Sodium Hydroxide | | | | |

Sodium Chloride | | | | |

Final volume prepared | Measured Quantity | pH (9.0± 0.1) | Conductivity (7± 1 ms/cm) | Storage Condition

| Reagent Record

| | | | RT |

Date of Preparation | Prepared By | Checked By

| |

Reagent Details:

Formula:

Sodium Acetate(M.Wt): 82.03g/mol.

Acetic Acid (M.Wt): 60.05g/ mol

Preparation of 0.2 M Sodium Acetate pH 4.0 buffer

Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol)

×Volume (L).

Calculation: To prepare 1 L of 0.2 M Sodium Acetate (MW = 82.03g/mol).

Weight of solute(g)=0.2×82.03× 1.0=16.406 g.

Preparation Method:

Weigh 16.406 g of Sodium Acetate and dissolve in water to make a 1 L solution.

Preparation for required volumes:

To prepare _____ of 0.2 M Sodium Acetate (MW = 82.03g/mol).

Weight of solute(g)=0.2×82.03× _____ L= _____g.

Weigh _____ gm of Sodium Acetate and dissolve in water to make a _____ L solution.

Calculations: To prepare the 0.2 M Acetic Acid.

Formula & Calculations:

Formula: Use the formula to calculate the molarity:

Molarity (M) = (Per cent by Weight / 100) * (Density in g/mL) * 1000 / Molar Mass.

Example: Molarity of Glacial Acetic Acid

Molar Mass of CH₃COOH: ~60.05 g/mol

Density: ~1.05 g/mL

Purity: 100%

Acetic Acid (MW = 60.05 g/mol).

Calculation:

Molarity = (100 / 100) * 1.05 g/mL * 1000 mL/L / 60.05 g/mol ≈ 17.5 M.

Calculation: To prepare 1 L of 0.2 M.

Applying the formula to prepare the working buffer:

C₁V₁=C₂V₂

C₁ = 17.5 M acetic acid.

V₁ = volume of stock needed to take.

C₂ = desired (working) concentration 0.2 M acetic acid.

V₂ = final total volume of working solution for one litre.

V₁ = (C₂ × V₂)/C₁.

C₂=0.2 M ; V₂= 1.0 L; C₁=17.5 M

V₁ = (0.2 × 1.0)/17.5

V₁ = 0.011 L

Take 0.011 L of the Stock solution of acetic acid and add it to a volumetric flask, making up the volume

to 1 L.

To prepare various volumes:

V₁ = volume of stock needed to take _____ L.

C₂ = desired (working) concentration 0.2M acetic acid.

V₂ = final total volume of working solution _____ L.

V₁ = (C₂ _____ × V₂ _____) /17.5.

V₁= _____ L.

Take V₁ _____ L of the Stock solution of acetic acid and add it to a volumetric flask, making up

the final volume _____ L.

Procedure for Preparation of 0.2 M Sodium Acetate pH 4.0.

To prepare a pH 4.0 solution, 0.2 M Sodium Acetate solution is titrated with 0.2

M Acetic acid so that the pH attains pH 4.0.

Check and record the details:

Preparation Details:

Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry

Date

Sodium Acetate | | | |

Acetic Acid | | | |

Final volume prepared | Measured Quantity | pH (4.0±0.1) | Conductivity (11±1.0 ms/cm) | Storage

Condition | Reagent Record

| | | |

Date of Preparation | Prepared By | Checked By

| |

49.1x2.77

D1 D6
LOBBY
3.00X11.77

D3
OPEN TO ALL
FLOORS
DOCUMENT STORE QC STAFF CHEMICAL /GLASS CONFERENCE
4.73x6.62 DOCUMENTATION WARE STORAGE
7.71x6.62
9.31x6.07 CANTEEN
4.03x6.62 14.8x10.7
5.98x6.12 FIREMAN
Goods LIFT 2.35x8.03 LIFT
3.50X3.00 2.70X2.50
D3
D3

D3
FIRE STAIR
LOBBY WASH AREA
WET LAB D3 5.90x2.76 up
3.00X11.30 D3 9.31x4.85
D1
OPEN TO
FIRST QC MANEGER
FLOOR 4.73x5.91
PM TESTING ROOM
9.94x7.06

D6
D1 D3

HOT ROOM/
BALANCE GENERALFTIR FUME HOOD
D3 ROOM 4.38x3.76 5.90x5.85
D3 4.82x3.76

D3
1.80 M WIDE CORRIDOR
10.1x2.80 D3
QC SAMPLE PREP.
4.73x5.42

CORRIDOR
22.5x1.80
INSTRUMENT ROOM II
15.3x5.04
W W
DPB HPLC D1
9.94x5.63

INSTRUMENT ROOM I
D1
D3 TRANING HALL
QC SAMPLE JANITOR
D3
D1 RECEIPT
1.50x3.93
14.8x19.0 D1
3.11x3.93
CORRIDOR
1.80x29.2
REFERENCE TOC/LPC ROOM
D3 GC ROOM
D4 STANDARD 4.44x2.98
5.87x2.98 4.83x2.98

D3
W W
GENTS CH ROOM
4.73x3.42 RETENTION
SAMPLES ROOM
COB

(COLD ROOM 2 TO 8 °C)
D1
9.94x3.31
D3 D3

STABILITY ROOM
LADIES CH ROOM 15.3x6.04
4.73x3.42
RM CONTROL SAMPLE
9.94x3.31
COB

D1

2.80 M WIDE PASSAGE

D3 D3 D1

D6

6.77 M WIDE CORRIDOR
D3 D3 D3
DPB AC
MEDIA DIST.
W
WASH AREA
WC WC WC DECONTAMINATION ROOM
5.221x2.02 7.34x2.02
1.10X 1.10X 1.10X
5.54x2.02

3D

D3
1.70 1.70 1.70 MEDIA PREP. PREPARATION & STERILIZATION
GRADE D 8.11x5.91 D3
7.38x3.88
GRADE D
DPB D3
IC-1 DPB D1
GRADE B
DPB

INCUBATOR STERILITY
DPB
FOR MEDIA TESTING ROOM IC-1
D3 4.65x3.78 GRADE CNC IC-4
WC WC WC INCUBATOR
GRADE B 5.33x3.78
1.10X 1.10X 1.10X COOL ZONE IC-2
1.70 1.70 1.70
D3 3.38x3.78 D3 IC-3 R&D LAB -3 R&D LAB-4
1.83x1.91

3D
CORRIDOR
MEDIA GRADE D OBSERVATION
6.47x6.61
13.3x1.80
6.47x6.61 W
STORE A/L GRADE B
4.78x3.54 2.50x4.24 D4 D4
CORRIDOR D3
GRADE C GRADE B GRADE B 3.53x2.02 DPB
CH-II CH-III 1.72x1.86
CH-I 2.02x2.02
4.81x2.02 2.73x2.02
D4 GRADE C GRADE CNC
DPB IC-1
D4
D4 D4
MLT ROOM INCUBATOR IC-4
D4 5.33x4.91 2.80x8.90
4.65x4.91
BET ROOM IC-2
GRADE C GRADE B IC-3
GRADE CNC A/L EXIT-II
4.78x2.83 GRADE D EXIT-I
2.63x2.02
5.36x2.53 D4 3.49x2.02
D3 D4
GRADE C D3 R&D LAB -2 R&D LAB -1
CORRIDOR OBSERVATION 6.47x6.61 6.47x6.61
GRADE CNC 1.80x8.50
GRADE D GRADE C
CORRIDOR A/L CH-I
COB

D4 FRIDGE

3D
2.50x8.31 5.36x1.52 D4 6.04x1.52 D4 BSC 3.77x3.51
GENTS CH ROOM GRADE CNC
W GRADE CNC D3 D3 INCUBATION ROOM W
D3 4.78x2.83 D3 DPB 4.36x3.51
GRADE D

D3

3D
CULTURE ROOM
4.65x3.51
GRADE CNC GRADE CNC GRADE CNC
LOBBY MICRO OFFICE MICRO MICRO
SAMPLE RECEIPT
DPB DPB D3 CORRIDOR

14.8x1.80
COB

5.17x4.24
6.77X11.77 DOCUMENT STORE 2.63x4.24 D4 D4 SCRAP OUT DPB
D1
D1 LADIES CH ROOM
3.49x4.24
GRADE D 5.49x2.86 GRADE CNC
GRADE D A/L D3 D3
GRADE CNC EXIT 2.66x2.86
4.78x3.12 CH ROOM
D3 D3 2.39x2.01 2.15x2.01
D3 D3
D3
R&D OFFICE
7.31x2.90 R&D ENTRANCE
W CORRIDOR
34.1x1.80
7.48x2.90 W

D3
D4 D4 D4

CORRIDOR
CABIN 1 16.7X1.80
CABIN 4 CABIN 3 CABIN 2
STORAGE

3.39x4.14
2.88x4.14 3.43x4.14 4.05x4.14
D4 D4 D4 D4

4.13x4.19 PAL-1
PRIMARY CHANGE PRIMARY CHANGE
AUDIT FEMALE
2.27X1.64
MAL ENTRY VISITOR COLD ROOM
2.85x5.81 4.43X3.93
2.85X5.78 4.51X5.78 4.19X3.27 D4 DE-CARTONING DPB
3.63X1.92 2.26X2.53 3.89X3.32
D4 D4 D4 D4
PRIMARY CHANGE
MALE PAL-2
2.27X1.64
5.31X5.81
Pass. LIFT
D4

D4
2.30X2.00 DPB
CORRIDOR
D4 D4 D4
CLEAN JC CORRIDOR
2.62x100.6 1.88X2.48 1.80x100.6 WC WC WC
CORRIDOR 1.10X 1.10X 1.10X
9.00X1.80 1.70 1.70 1.70
D4

DPB

D4
D4

D1
QA OFFICE QA MANGER

18.0x10.8 2.85x4.06
MATERIAL PREP WASH
8.15X3.07
AC
1.00X2.40 UNLOAD
8.41X3.40
D4

5.26X1.64
WC WC WC
D4

2.21X1.46
1.10X 1.10X 1.10X

DRYING ZONE
DPB
VIAL WASHING 1.70 1.70 1.70
1.00X1.00
6.42X9.65
D6

D4

D4

PAL

STERILIZING ZONE
2.84X1.45
D4

STERILE CORRIDOR
QA HEAD 15.5X1.80
DISINFECTANT
2.85x4.47 D3 D3 8.41X2.62

COOLING ZONE
D4
D4

MAL
MAL PAL
2.84X1.45
D4

1.45X1.64 1.41X1.64 PASS THROUGH

STABLIZING ZONE
D4

D3 D3
D3

PAL
D6
2.04X2.29
D3

D4

TUMBLE
DRYER FORMULATION
PACKING AREA 8.41X3.59
1.50X2.00
D3

D4

MAL
CONVEYOR
SWING

2.04X1.47 DOUBLE
DOOR
W D3 WASHER W
UNLOADING AREA
36" TURNTABLE
NEW ADD

LAUNDRY AREA 1.50X2.00
QA DOCUMENT 5.10X1.57 2.94X5.13 8.49X1.80
3.24 M WIDE PASSAGE

2.98X3.88

21.0x9.82
& LOADING TRAY
VIAL COLLECTION

D2

FILL FINISH
D4

6.42X8.30
D4

D1 EXIT 2
4.54X1.87
EXIT 1
4.40X1.87
D1
D4

PAL 3 PAL 4
4.23X3.82 3.20X3.82
CORRIDOR
2.85X25.2 PAL 1 PAL 2
4.54X1.87 4.40X1.87
D4

D4

D4

D4

W W
D4

LYO SERVICE AREA
CONTROL ROOM LYO LOAD
D4

5.55X3.55 10.9X3.55 8.29X3.55
D8

AUTOMATIC FOUR HEAD ALUMINUM CAP

D8
SEALING MACHINE, MODEL-AEWCS-120

D4

SWING
CONVEYOR

BULK STAGING
VIAL COLLECTION

SECONDARY PACKING AREA
16.6X4.04 CAPPING
TRAY

21.0x8.07 8.29X4.04
W
WC WC WC
DPB

1.10X 1.10X 1.10X D8
1.70 1.70 1.70
DPB LOBBY
CORRIDOR
25.5X1.80 JC
6.77X11.00
DPB
D4

2.32X1.00

DPB
D4

STAIRCASE
D4
UNDERTEST

WC WC WC
1.10X 1.10X 1.10X
VISUAL & BUFFER ROOM
AND
1.70 1.70 1.70 PACKING CHANGE
INSPECTION 2.83X2.68 FEMALE LIFT ONLY W
9.79X3.87 4.14 X 3.08
FOR
D4

2.88X1.14
SECOND FLOOR
D4
SAMPLING

D8
(L.V.P.)
D4

PPM STORE SPM STORE 11.8X1.80
15.2x12.0 14.9x9.32
D4

7.22x2.69 PACKING CHANGE
11.8x2.89 MALE
4.14 X 4.23
QUARANTINE

LABELING & PACKING
12.6x5.40
D4

PB
D4

PACKING
W
D6

VISITOR
D4 4.41 X 3.08

D8
D4
D8

D4

D8 D8
D4 PACKING OFFICE
LABELING STORE 4.14X4.42
FINISHED GOODS STAGING
7.68 X 5.59
12.6 X 5.59
LOBBY 3.14 X 3.05
APPROVED RUBBER
SPM RECEIPT REJECT
12.47X6.77 BUNG STORE
3.58 X 2.89
W 3.36 X 2.89 3.59 X 2.89
D8 D8 D8 Pass. LIFT
2.30X2.00

CORRIDOR
53.5x1.80

D6 D6 D6 D6

ELECTRIC
ROOMW
CORRIDOR
3.04x11.7
3.50X6.00
LOBBY
3.00X11.53
OPEN TO ALL
FLOORS

W
SERVER
Goods LIFT ROOM
3.50X3.00 3.50X6.57

CLIENT :- LIFENITY- HEALTH LIMITED (DIALYZER, LVP & PLASMA)
ADDRESS :-Auric city, chhatrapati sambhaji nagar

FIRST FLOOR DESIGN &
PROJECT BY :-BIOBRIDGE HEALTHCARE SOLUTION PVT.LTD.

ADDRESS :-13, Rachna Blossom, Jagdishnagar,
PREPARED A.K

Aundh, Pune -411007, Maharastra, India. CHECKED G.K.

DATE 18-06-2026
TITLE- CONCEPTUAL LAYOUT-FF REVISION R0
SCALE 1:1

SIZE A0 DWG.NO 02

Reagent Details:
Preparation of IG-RR-28 working buffer from Stock buffer (IG-RR-27):
Calculations: For the Preparation of 15mM Sodium acetate pH 4.2.
Stock Buffer (IG-RR-27):0.2M Sodium acetate pH 4.2.
Formula:
 $C1 \times V1 = C2 \times V2$
C1 = concentration of stock
V1 = volume of stock you need to take
C2 = desired (working) concentration
V2 = final total volume of working solution
Applying the formula to prepare the working buffer:
 $C1 \times V1 = C2 \times V2$
C1 = 0.2M Sodium Acetate pH 4.2.
V1 = volume of stock you need to take _____ L.
C2 = desired (working) concentration 15mM Sodium Acetate pH 4.2
V2 = final total volume of working solution _____ L.
 $V1 = (C2 \times V2) / C1$
C2=0.015M ; V2= 1.0 L; C1=0.2M
 $V1 = (0.015 \times 1.0) / 0.2$
V1 = 0.75L
Take 0.75 L (IG-RR-27) Stock buffer solution and dilute with water, and make up to 1L.
Check and record the details:
Preparation Details:
Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry
Date
Sodium Acetate | | | |
Acetic Acid | | | |
Stock buffer (IG-RR-27) taken volume(L) | pH (4.2±0.1) | Conductivity (1.15±0.1ms/cm) | Storage
Condition | Final volume prepared
| | | RT |
Date of Preparation | Prepared By | Checked By
| |

1) Sewage & Effluent Treatment: Clearly mark the physical location, dimensions, and layout of the Effluent Treatment Plant (ETP) / Sewage Treatment Plant (STP). Provide structural calculations proving adequate capacity for processing waste.
Response: ETP and STP layout with required details is pending and shall be shared ASAP.

2) Organic Waste Converter (OWC): Designate a dedicated, ventilated space for a mechanical OWC, stating its processing capacity in kg/day based on your expected workforce.
Response: Plasma: 20.4MT/year or 68T/day, LVP: 7.9MT/year or 26T/day

3) Process Flow & Water Balance Chart: Submit a detailed quantitative water balance diagram showing domestic and industrial consumption, recycling rates, and a commitment to Zero Liquid Discharge (ZLD).
Response:
Plasma:
LVP:

4) Hazardous Waste Management Plan: Since ethanol-based fractionation is utilized, submit storage, handling, and disposal plans for hazardous/chemical solvents
Response:
Plasma:
LVP:

Sr.No. | Item | Litres (L / year) | Cubic metres (m³ / year) | Notes
| A. Water Inputs | | |
1 | Domestic water (506 ppl × 100 L/p/d) | 1,84,69,000 | 18,469.00 | Toilets, canteen, housekeeping
| WFI (process) | 62,05,470 | 6,205.47 | WFI for formulation, CIP, rinses
| Other process water (25% of WFI) | 15,51,368 | 1,551.37 | CIP makeup, utility washes
| Extra process water (10% of plasma vol) | 41,400 | 41.4 | Misc. process losses, small uses
2 | Process Water | 77,98,238 | 7,798 |
| Total Water Input | 2,62,67,238 | 26,267.24 | A = sum of above
| | |
| B. Water Outputs / Wastewater | | |
1 | Domestic sewage → STP (80% of domestic) | 1,47,75,200 | 14,775.20 | Black/gray water to STP
2 | Process wastewater → ETP (99% of process) | 77,20,256 | 7,720.26 | Process drains, WFI rejects, CIP effluent
| Total Wastewater Generated | 2,24,95,456 | 22,495.46 | B = sum of above
| | |
| C. Balance / Losses / Product incorporation | | |
1 | Unaccounted losses / product incorporation / evaporation (A – B) | 37,71,782 | 3,771.78 | ~14.36% of input — includes water in products, steam losses, sampling, evaporation, bleed, equipment hold-up
Category | Type of Water/Steam | Estimated Quantity per day | Estimated Quantity per year | Purpose
Product Vehicle | Water for Injection (WFI) | ~62,200 – 62,500 L | ~14928000 – 15000000 L | Main solvent; added q.s. to reach final batch volume.
Equipment Cleaning | Purified Water (PW) | 5,000 – 8,000 L | 1200000 – 1920000 L | Initial rinses for mixing tanks, piping, and filling lines (CIP).
Final Rinse | Water for Injection (WFI) | 2,000 – 3,000 L | 480000 – 720000 L | Final rinse of product-contact surfaces to ensure a pyrogenicity.
Sterilization | Pure Steam (Clean Steam) | ~1,500 – 2,500 kg | ~360000 – 600000 kg | Used in autoclaves for terminal sterilization and SIP of lines.
Utility Heating | Industrial Steam | ~5,000 – 7,000 kg | ~1200000 – 1680000 Kg | Indirect heating for jacketed vessels and WFI generation.
Cooling | Process Water | 10,000 L | 2400000+ L | Non-contact cooling for jacketed vessels after sterilization.
Wastewater | Cleaning-in-Place (CIP) Effluent | 5,000 – 8,000 L | 1200000 – 1920000 L | Treatment in an Effluent Treatment Plant (ETP)
Output Category | Output Stream | Estimated Quantity per Day | Estimated Quantity per Year | Remarks
Treated Water Recovery | Treated Water for Reuse / Discharge | ~3,500 – 6,800 L | ~840,000 – 1,632,000 L
| Recovered after primary and secondary treatment
Reject Water / Losses | RO Reject / Evaporation / Handling Losses | ~500 – 1,600 L | ~120,000 – 384,000 L
| Depends on treatment efficiency and recovery system
Sludge Generation | ETP Sludge / Solid Waste | ~20 – 50 kg | ~4,800 – 12,000 kg | To be disposed through authorized waste handler
Final Discharge | Treated Effluent | As per discharge norms | As per regulatory limits | Suitable for reuse/discharge after compliance testing
Sr. No. | Process / Source | Type of Hazardous Waste | Approx. Quantity / Year
1 | Solvent recovery & viral inactivation (TnBP + detergent process) | Spent solvent mixture (Tri-n-butyl phosphate + Triton X-100 / Polysorbate 80) | 1.0 – 1.5 KL / year
2 | Filtration & purification units | Used filter aids / depth filter media contaminated with protein or solvent | 0.8 – 1.0 MT / year
3 | Activated carbon beds (viral removal, deodorization) | Spent activated carbon with adsorbed organic residues | 0.5 – 0.7 MT / year
4 | Chromatography / TFF cleaning (CIP) | Residues of buffers, cleaning agents, NaOH / HCl neutralization sludge | 2.0 – 3.0 MT / year (dry solids)
5 | ETP operation | Sludge containing proteinaceous organic matter & trace solvents | 5.0 – 6.0 MT / year
6 | ETP / RO reject water | High-TDS / COD effluent (liquid) | ~ 900 – 1,000 m³ / year
7 | Resin & membrane replacement | Spent chromatography resin, UF/DF membranes, PVDF filters | 0.5 – 1.0 MT / year
8 | Laboratory & QC activities | Contaminated lab glassware, pipette tips, chemical bottles | 200 – 250 kg / year
9 | Oil from DG sets, compressors, hydraulic systems | Used lubricating oil | 50 – 100 L / year
10 | DG set maintenance | Oil-soaked cotton waste, filters | 50 – 100 kg / year
11 | Production packaging & vial washing | Contaminated packaging (plastic, foil, paper) | 0.3 – 0.5 MT / year
12 | Spill & floor washings | Absorbent material, mopping waste, PPE, gloves | 0.5 MT / year
Hazardous Waste | API Residue / Chemical Sludge | 50 – 200 kg | 12000 – 48000 kg | High-temperature incineration (>1,200°C).

Reagent Details:

Formula & Calculations:

Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).

Calculation: To prepare 1 L of 2M sodium hydroxide (MW = 40g/mol)

Weight of solute(g)=1×80=80g.

Preparation Method:

Weigh 80 g of Sodium hydroxide and dissolve in water to make a 1 L solution.

Preparation for different volumes:

To prepare _____ of 2M sodium hydroxide (MW = 80g/mol).

Weight of solute(g)=1×80x _____L= _____gm

Weigh _____gm of sodium hydroxide and dissolve in water to make a _____ L solution.

Check and record the details:

Preparation Details:

Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry

Date

|||||

Final volume prepared | Weighed Quantity | Storage Condition:

||

Date of Preparation | Prepared By | Checked By

||

Reagent Details:

Formula & Calculations:

Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).

Calculation: To prepare 1 L of 1M Citric Acid (MW = 192.12g/mol)

Weight of solute(g)=1×192.12=192.12g.

Preparation Method:

Weigh 192.12 g of Citric acid and dissolve in water to make a 1 L solution.

Preparation for different volumes:

To prepare _____ of 1M Citric acid (MW = 192.12g/mol).

Weight of solute(g)=1×192.12× _____ L= _____ gm

Weigh _____ gm of Citric acid and dissolve in water to make a _____ L solution.

Check and record the details:

Preparation Details:

Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry

Date

|||||

Final volume prepared | Weighed Quantity | pH (1.57) | Conductivity (10± 0.5 ms/cm) | Storage Condition:

|||| RT

Date of Preparation | Prepared By | Checked By

||

Reagent Details:

Formula:

Glycine : (M.Wt): 75.07g/mol.

HCL: 12M

Sodium Chloride (M.Wt): 58.44g/mol

Preparation of 0.2 M Glycine pH 4.0 buffer

Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).

Calculation: To prepare 1 L of 0.2 M Glycine (MW = 75.07g/mol).

Weight of solute(g)=0.2×75.07.0x 1.0=15.01 g.

Preparation Method:

Weigh 15.01 g of Glycine and dissolve it in water to make a 1 L solution.

Preparation for required volumes:

To prepare _____ of 0.2 M Glycine (MW = 75.07g/mol).

Weight of solute(g)=0.2×75.07.0x _____ L= _____ g.

Weigh _____ gm of Glycine and dissolve in water to make a _____ L solution.

Applying the formula to prepare the 1M HCL working buffer:

$C1 \times V1 = C2 \times V2$

$C1 = 12$ M HCL acid.

$V1$ = volume of stock needed to take.

$C2$ = desired (working) concentration 1.0 M HCL acid.

$V2$ = final total volume of working solution for one litre.

$V1 = (C2 \times V2)/C1$.

$C2=1.0$ M ; $V2= 1.0$ L; $C1=12$ M

$V1 = (1.0 \times 1.0)/12$.

$V1 = 0.0833$ L

Take 0.0833 L of the Stock solution of HCL acid and add it to a volumetric flask, making up the volume to 1 L.

To prepare various volumes:

$V1$ = volume of stock needed to take _____ L.

$C2$ = desired (working) concentration 1.0 M HCL acid.

$V2$ = final total volume of working solution _____ L.

$V1 = (C2 \times V2) / 12$.

$V1 =$ _____ L.

Take $V1$ _____ L of the Stock solution of HCL acid and add it to a volumetric flask, making up the final volume _____ L.

Calculations: To prepare the 0.15 M Sodium Chloride.

Formula & Calculations:

Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).

Calculation: To prepare 1 L of 0.15 M Sodium Chloride (MW = 58.44 g/mol)

Weight of solute(g)=0.15 × 58.44 × 1 = 8.766g.

Preparation Method:

Weigh 8.766 g of Sodium Chloride and dissolve in water to make a 1 L solution.

Procedure for Preparation of 0.2 M Glycine pH 4.0 + 0.15 M NaCl.

To prepare a pH 4.0 solution, 0.2 M Glycine solution is pH adjusted with 1 M HCL so that the pH attains 4.0, then add the 0.15 M sodium chloride, then make up the final volume.

Preparation Details:

Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry

Date

Glycine |||||

HCL |||||

|||||

Final volume prepared | Measured Quantity | pH | Conductivity | Storage Condition | Reagent Record

|||||

Final volume prepared | Measured Quantity | pH (4.0± 0.1) | Conductivity (17± 0.1ms/cm) | Storage Condition | Reagent Record

|||||

Date of Preparation | Prepared By | Checked By (QC/Analyst)

||

51000 I

H

G

F

E

80000

D

C

B

A

51000 O

W.C. W.C. W.C. W.C. W.C. W.C. W.C. W.C. W.C. W.C.

1.30x2.37 1.30x2.37 1.30x2.37 1.30x2.37 1.20x2.37 1.30x2.37 1.30x2.37 1.30x2.37 1.30x2.37 CORRIDOR

1.30x2.37 1.30x2.37 1.30x2.37 1.30x2.37

1.60x2.37

12.8x2.00

4.73x3.67

D1 D1 D1 D1 D1 D1 D1 D1 D1 D1 D1

LADIES TOILET GENTS TOILET

8.05x1.15 D6

9.60x1.15

D1 D1

CORRIDOR

29.6x2.12 CORRIDOR

8.07x2.00

4.19x2.00

CANTEEN

12.9x6.83

2.54x4.01

CONFERENCE

CORRIDOR 8.77x3.75 12.1x4.14

4.95x3.97

CHEMICAL /GLASS

WARE STORAGE D3

4.90x6.09 CORRIDOR

5.53x2.00

2.00x2.30 N

D3 WET LAB WASH AREA

8.77x5.05 D3 5.72x3.29

QC STAFF

DOCUMENTATION

4.72x11.4

D1 D3

D3

PM TESTING ROOM

4.90x5.23

HOT ROOM/

D6

DOCUMENT STORE BALANCE FUME HOOD

GENERALFTIR

4.92x5.63 ROOM 5.72x5.27

D3 4.10x3.51 4.56x3.51

D3

D3

1.80 M WIDE CORRIDOR

D3

10.0x1.80

CORRIDOR

D3 2.80x27.9

M

QC MANEGER D1

4.92x3.16

INSTRUMENT ROOM II

14.6x5.65

D3 INSTRUMENT ROOM I

HPLC

7.08x7.32 D1

QC SAMPLE PREP.

D3 D3

4.92x3.23

CORRIDOR TRANING HALL

2.85x14.5

REFERENCE

GC ROOM

TOC/LPC ROOM 12.1x19.8
STANDARD 4.20x3.10
5.18x3.10 5.03x3.10
DPB

QC SAMPLE JANITOR D3
RECEIPT 1.56x2.53
3.25x2.53
D3 D4 RETENTION
SAMPLES ROOM D1
(COLD ROOM 2 TO 8 °C)
7.08x3.45
COB

L
GENTS CH ROOM
D3 4.92x2.53 D3
STABILITY ROOM
14.6x6.29
COB

RM CONTROL SAMPLE
D1
7.08x7.3.45
LADIES CH ROOM
D3 4.92x2.53
D3 D1
D6

D3 D3 D3
AC
DPB MEDIA DIST.
WASH AREA DECONTAMINATION ROOM 7.22x2.10
D3

D3

PREPARATION & STERILIZATION 5.74x2.10 5.47x2.10
MEDIA PREP.
GRADE D 5.84x6.14
GRADE D
D3
7.88x4.04
DPB

D3
IC-1 DPB
GRADE B
DPB

INCUBATOR STERILITY
DPB
FOR MEDIA TESTING ROOM IC-1
D3 4.54x3.93 GRADE CNC IC-4
INCUBATOR
GRADE B 6.35x3.93 R&D LAB -3 R&D LAB-4
COOL ZONE IC-2
D3 3.52x3.93 5.19x7.00 4.85x7.00
D3 IC-3
2.21x1.99
MEDIA GRADE D
D3
OBSERVATION K
STORE A/L GRADE B
4.98x3.69 2.80x4.42 D4 D4 CORRIDOR D3
GRADE C GRADE B GRADE B 3.98x2.10 DPB
CH-I CH-II CH-III 2.10x1.51
2.41x2.10 2.84x2.10 2.10x2.10
D4 GRADE C GRADE CNC
D4 D4 DPB IC-1
D3 MLT ROOM INCUBATOR IC-4
D4 4.54x3.93 6.35x3.93
BET ROOM GRADE C IC-2
GRADE CNC GRADE D GRADE B IC-3
A/L EXIT-I D3
4.98x2.95 2.81x2.10 EXIT-II OBSERVATION
D4 3.62x2.10 D4 3.23x2.10
D3
FRIDGE R&D LAB -2 R&D LAB -1
GRADE C
CORRIDOR BSC 5.19x7.00 4.85x7.00
GRADE CNC GRADE D GRADE C 1.69x8.55 GRADE CNC
CORRIDOR A/L CH-I
COB

INCUBATION ROOM
2.80x8.35
D4
3.82x2.10 5.95x2.10 GRADE D
D4 D4 8.15x3.93
CULTURE ROOM
GENTS CH ROOM 4.54x3.93 1.68x8.09

GRADE CNC D3 D3
D3 4.98x2.95 D3 DPB
DPB DPB
D3
D3

D3
GRADE CNC GRADE CNC D4 D4
GRADE CNC MICRO DPB
MICRO
MICRO OFFICE SAMPLE RECEIPT
COB

DOCUMENT STORE GRADE D SCRAP OUT CORRIDOR
3.62x4.11 3.23x4.11 CH ROOM GRADE D GRADE CNC
2.81x4.11 EXIT 5.93x2.68 A/L 12.1x2.22

D3
LADIES CH ROOM
GRADE CNC
2.23x2.68 2.19x2.68 2.10x2.68
J
4.98x2.95 D3 D3 D3
D3 D3
D3

R&D OFFICE R&D ENTRANCE
CORRIDOR 6.04x2.27 6.04x2.27
32.7x2.00
D3
D6

WATER SYSTEM &
ENGINEERING STORE
10.0x11.0

I
D6

CORRIDOR
2.77x125.8

H
141002
SERVICE AREA
44.9x21.0
34.8x11.0
CORRIDOR
2.77x121.4

G
D6

PRINTER

CORRIDOR
STORAGE

CABIN 1 24.3X2.40
CABIN 4 CABIN 3 CABIN 2
4.00x4.44 4.00x4.44 4.00x4.44
3.05x4.44 D4 D4 D4 D4 D4 D4 D4
DPB
PAL-1
COLD ROOM 1.80X1.70
4.09X3.41
AUDIT MAL ENTRY PRIMARY CHANGE PRIMARY CHANGE
1.50X2.63 D4 DE-CARTONING
2.97X3.46
D4 D4 D4 D4 2.50x6.70 2.00X3.97 VISITOR FEMALE
4.27X3.97 4.76X3.97 PAL-2
1.80X1.70
F
PRIMARY CHANGE D4
MALE
DPB
D4

D4 D4 D4 4.00X6.05
CLEAN JC
2.54X2.58
CORRIDOR
D4

D4

DPB
D4

9.11X2.00

QA OFFICE QA MANGER

17.9x11.0 2.50x4.23
MATERIAL PREP WASH
6.98X3.55
AC
1.00X2.50

UNLOAD
D4

2.40X3.16 6.16X3.55
D4

3.10X1.29
DRYING ZONE

DPB
1.03X1.03 VIAL WASHING
5.03X10.0
D6

STERILIZING ZONE
D4

D4

PAL STERILE CORRIDOR
D4

QA HEAD 2.53X1.50 2.81X16.1
2.50x4.23 D3 D3
DISINFECTANT
COOLING ZONE
D4

MAL PAL 6.16X3.08
D4
D4

MAL
D4

1.47X1.47 1.50X1.50
STABLIZING ZONE

2.53X1.50

D3 D3
PASS THROUGH E
D4
D4

D6 PAL
1.23X1.50
FORMULATION
D4

TUMBLE
6.16X2.88
DRYER
1.50X2.00 PACKING AREA
D4

D4

MAL
1.23X1.53
CONVEYOR
SWING

D4 DOUBLE
DOOR
LAUNDRY AREA WASHER
UNLOADING AREA 6.24X1.80
36" TURNTABLE
NEW ADD

4.88X1.60 1.50X2.00
QA DOCUMENT 3.57X3.16 3.07X4.76

20.5x10.4
& LOADING TRAY
D2 VIAL COLLECTION
D4
D4

EXIT 2 EXIT 1
4.53X2.24 4.50X2.24
PAL 3 PAL 4
D4

2.50X4.57 FILL FINISH

2.50X4.57
5.03X8.65

CORRIDOR
2.00X22.9 PAL 1 PAL 2
4.53X2.24 4.50X2.24
D4

D4

D4

D4

D4 D
LYO SERVICE AREA
CONTROL ROOM LYO LOAD
D4

5.35X3.79 9.92X3.79 6.84X3.79
D8

AUTOMATIC FOUR HEAD ALUMINUM CAP
SEALING MACHINE, MODEL-AEWCS-120

D8

D4

SWING
CONVEYOR

BULK STAGING
VIAL COLLECTION

SECONDARY PACKING AREA
15.3X4.00 CAPPING
TRAY

20.5x8.41 6.84X4.00
DPB

D8

DPB

CORRIDOR
24.3X2.00 JC
DPB
D1

1.50X1.00

DPB
C
D4
UNDERTTEST

D4

CORRIDOR PACKING CHANGE
BUFFER ROOM
2.80x27.9 2.95X2.79 FEMALE
3.00 X 3.19
D4
D4
SAMPLING

VISUAL &
INSPECTION
D4

7.60X5.84
3.00X3.00 PACKING CHANGE
MALE
3.0 X 3.0
D4

D8
QUARANTINE

SPM STORE 2.05X11.7
D4

PPM STORE 15.1x11.7
D4

7.05x2.80
13.9x11.8
D4

4.19x3.35

PACKING
VISITOR
3.00X2.89
LABELING & PACKING

PB 10.6x5.89
D4

PACKING OFFICE
B
3.00X2.55
D6

D4
D8
D8

CORRIDOR
6.98x2.00

D8 D8
D4
LABELING STORE
FINISHED GOODS STAGING
8.0 X 6.10
13.15 X 6.10
CORRIDOR
2.00x6.15
APPROVED RUBBER SPM RECEIPT REJECT
BUNG STORE 3.00 X 3.30
3.50 X 3.30 3.30 X 4.00
D8 D8 D8

CORRIDOR
43.8x2.77

A

6 5 4 3 2 1

CLIENT :- LIFENITY-LARGE VOLUME PARENTERALS

DESIGN &
PROJECT BY :-BIOBRIDGE HEALTHCARE SOLUTION PVT.LTD.

FIRST FLOOR
PREPARED A.K
ADDRESS :-13, Rachna Blossom, Jagdishnagar,
Aundh, Pune -411007, Maharastra, India. CHECKED G.K.

DATE 04-03-2026
TITLE- CONCEPTUAL LAYOUT-FF REVISION R0
SCALE 1:1

SIZE A0 DWG.NO 02

QUESTIONNAIRE FOR ETP PLANT

[Company Information |
[Company Name | |
[Address | |
[Telephone Numbers | |
[Contact person with Designation | |
[Cell | |
[E-mail | |
[Requirement Details |
[Type Of Industry |Bio Pharma (Plasma), Injectable (LVP) |
[(Pharma/Chemical/Auto etc.) | |
[Process of Industry , if Pharma |API, Formulation |
[type | |
[(API / Formulation) | |
[Current Scheme (If any) | |
[(Please attached Process Flow | |
[Diagram) | |
[Source of Feed |Water systems & Manufacturing/injectable |
| |waste |
[Feed Rate, m3/day or KLD | |
[Feed Parameters to be consider |Please refer next page |
[for design | |
[(Please attached Test Report) | |
[Outlet Parameters to be required|Please refer next page |
[Name of Solvents & concentration|NA |
[present in Feed (If Any) | |
[Plant/Site Location | |
[Available Space for New Plant | |
[Established (Length X | |
[Width) | |
[Any other information you would like to mention: |
[This is composite project where multiple units are located (LVP, Plasma, |
[Dialyzer, QC, QA, Admin, etc.) There is dedicated ETP for complete plant. |
| |
| |
| |

LIST OF FEED PARAMETERS TO BE CONSIDER FOR DESIGN

[Sr. No.|Parameter |Value |Unit |
	pH	10-12	NA
	Oil & Grease	20-150	mg/L
	TDS	1000-8000	mg/L
	TSS	<200	ppm
	COD	<2250	ppm
	BOD	<1100	ppm

LIST OF OUTLET PARAMETERS REQUIRED

[Sr. No.|Parameter |Value |Unit |
	pH	6.5-8.5	NA
	Oil & Grease	<100	mg/L
	TDS	<500	mg/L
	TSS	<50	mg/L
	COD	<250	mg/L
	BOD	<100	mg/L

[SHEET: Index]

Sr.No. | Document List | Details provide

- 1 | MIDC Sanction Plan. |
- 2 | MIDC Completion/Occupancy Certificate. | Possession Letter 6/8/2024
- 3 | Udyam Aadhar | NA
- 4 | Incorporation Certificate. | incorporation!A1
- 5 | Total Project Cost. (Esteemated) | 3065586099
- 6 | a) Built-up area | 40000 Sq Meter
- b) Area proposed for green belt | 27000 Sq Meter
- c) Tree plantation details. |
- 7 | Proposed layout plan with area Statement. |
- 8 | a) List of products | List of Products!A1
- b) Raw Material Balance Sheet. | Raw Material!A1
- 9 | a) List of HW waste will be generated from the process. | Hazardous Waste!A1
- b) List of Non-Hazardous waste will be generated from the process. | Non Haz Waste!A1
- 10 | Water Balance, Domestic Consumption & Process Consumption and Generation of Waste Water. | Water!A1
- 11 | Name of the Fuel will be used in the process: HSD/ Electricity/CNG. | Fuel Consmp!A1
- 12 | Details of any heat treatment or reaction tank, if any. | Heat treat_Reaction_Tank!A1
- 13 | Details of Equipment/Machinery with the Name | Equipments!A1
- 14 | Types and sources of Pollution and control measures (Pollution Control equipment). | Pol control meas!A1
- 15 | DG Set details. | DG Set !A1

[SHEET: incorporation]

||||| Back to Index

[SHEET: Possession Let]

||||| Back to Index

[SHEET: DG Set]

Sr.No. | Parameter | Description / Specification |

- 1 | Make / Model | Cummins / Kirloskar / Mahindra (example – confirm at procurement) | Back to Index
- 2 | Type | Silent Type, Auto-start, Water-cooled, Turbocharged Diesel Generator |
- 3 | Rating / Capacity | 2 × 1300 kVA each (standby operation, load sharing optional) |
- 4 | Fuel Used | High-Speed Diesel (HSD) – BS-VI grade, low sulphur (<0.05%) |
- 5 | Rated Power Output | ~400 kW per DG set (0.8 PF) |
- 6 | Total Installed Capacity | 2,600 kVA (2 sets × 1300 kVA) |
- 7 | Operating Hours / Year | ~800–1,000 hours per year (emergency + load testing only) |
- 8 | Fuel Consumption | ~55–60 L/hour @ full load |
- 9 | Estimated Annual HSD Consumption | ≈ 27,000 L / year (≈ 27 KL / year) |
- 10 | Storage Capacity (HSD Tank) | 10 KL aboveground double-walled MS tank with vent, dyke, and spill tray as per PESO norms |
- 11 | Cooling System | Water-cooled, with radiator and jacket water heater |
- 12 | Lube Oil System | Forced lubrication; oil drain point connected to collection tray |
- 13 | Starting System | 24V battery start with automatic mains failure (AMF) panel |
- 14 | Exhaust System | Silencer with flexible coupling; insulated exhaust manifold |
- 15 | Emission Parameters (at rated load) | • PM ≤ 0.3 g/kWh • NO_x + HC ≤ 7.5 g/kWh • CO ≤ 3.5 g/kWh (As per CPCB II norms) |
- 16 | Stack Details | Mild steel stack with flanged joints; height as per CPCB formula: $H = h + 0.2\sqrt{KVA} = 3 + 0.2\sqrt{1300} \approx 10.2$ m above roof level per set | 36.05551275463989
- 17 | Stack Quantity | 2 Nos (individual stack for each DG set) | 7.211102550927979
- 18 | Noise Level at 1 m distance | ≤ 75 dB(A) – within CPCB norm | 10.21110255092798
- 19 | Noise Control Measures | • Acoustic Enclosure (≤ 75 dB(A)) • Vibration isolators under skid base • Greenbelt & acoustic barriers along DG yard boundary |
- 20 | Location | Utility block (away from production & office area) |
- 21 | Ventilation System | Forced air intake & exhaust with acoustic louvers |
- 22 | Emission Control | • Stack emission monitoring (PM, NO_x, SO_x) – quarterly • Use of low-sulphur HSD |
- 23 | Lube Oil Handling | Drained to closed HDPE drum; sent to authorized recycler (HW 5.1) |
- 24 | Spill Control | Impervious floor, containment dyke, absorbent pads available |
- 25 | Safety & Fire Protection | Fire extinguishers (CO₂ + dry chemical) at DG yard; earthing pit per IS 3043 |
- 26 | Authority Approval | DG sets to comply with CPCB-II emission & noise certification; HSD tank under PESO license |

Sr.No. | Parameter | Standard / Limit | Plant DG Result / Provision

- 1 | Emissions (CPCB-II) | PM ≤ 0.3 g/kWh NO_x+HC ≤ 7.5 g/kWh CO ≤ 3.5 g/kWh | Cummins CPCB-II compliant engines
- 2 | Stack Height | $H = h + 0.2\sqrt{KVA}$ | 6.5 m above roof per set
- 3 | Noise | ≤ 75 dB(A) at 1 m | Acoustic enclosure fitted
- 4 | Fuel Type | Low-sulphur HSD | BS-VI HSD used
- 5 | Used Oil Disposal | To authorized recycler | As per HW Rule 5.1
- 6 | DG Area Containment | Impervious floor + bund | Provided

| Back to Index |

[SHEET: Pol control meas]

Sr.No. | Type | Source | Pollutants Likely | Control Measures / Equipment | Remarks |

- 1 | AIR POLLUTION | a. DG Set (2 × 500 kVA, standby) | PM, SO_x, NO_x, CO (from diesel combustion) | • Acoustic enclosure to control noise • Adequate stack height as per CPCB formula ($H = h + 0.2\sqrt{KVA}$) ~6.5 m above roof • Low-sulphur HSD (<0.05%) fuel • Regular maintenance & emission monitoring | Operates only during power outage | | Back to Index
- 2 | AIR POLLUTION | b. PNG / CNG-fired Boiler (500 kg steam/hr) | NO_x, trace CO | • Use of low-NO_x burner • Stack height 12 m above roof • Online temperature & pressure monitoring • Periodic stack emission testing | PNG is clean fuel, minimal SO_x |
- 3 | AIR POLLUTION | c. HVAC systems / AHUs | Dust, microbial aerosols | • HEPA filters (0.3 µm) at terminal outlets • Pre-filters (10 µm) at return air • Periodic filter replacement as per SOP | Ensures cleanroom Class 10,000–100,000 compliance |
- 4 | AIR POLLUTION | d. Solvent & detergent handling (TnBP, Triton X-100) | VOC vapours | • Closed mixing vessels, nitrogen blanketing • Solvent vapor condenser + activated-carbon adsorption unit • Explosion-proof zone classification (ATEX) | Emission < 50 mg/Nm³ VOC ensured |
- 5 | AIR POLLUTION | e. Laboratory fume hoods / exhausts | Trace acidic or organic vapours | • Fume hoods with local scrubbers or HEPA filters • Ducted exhaust at safe height | Minimal emissions, intermittent source |

Sr.No. | Type | Source | Type of Wastewater | Pollutants Likely | Treatment / Control Equipment |

- 1 | WATER POLLUTION | a. Process & WFI drains | Process effluent containing protein residues, buffers, mild solvents, detergents | pH 6–8, COD 1,000–2,000 mg/L, BOD 400–800 mg/L, low TDS | Effluent Treatment

Plant (ETP): • Collection & equalization tank (15 m³) • pH correction & neutralization tank • Primary clarifier • Biological MBR / MBBR reactor • Tertiary filtration (UF + RO) • RO permeate reused for gardening / utilities • RO reject → MEE + ATFD | |

2 | WATER POLLUTION | b. Solvent-detergent streams (TnBP phase) | High COD, solvent traces | Trace VOCs, COD > 10,000 mg/L | Dedicated solvent holding tank → evaporation or incineration via TSDF | |

3 | WATER POLLUTION | c. Domestic sewage (staff 506) | Black / grey water | BOD ~250 mg/L | STP (40 m³/day): MBBR + UF + chlorination | |

4 | WATER POLLUTION | d. Cooling tower blow-down, softener reject | High TDS, hardness | TDS 2,000–3,000 mg/L | Collected in neutralization tank → mixed with RO reject → MEE | |

Sr.No. | Type | Source | Type of Waste | Category | Control / Disposal Method | |

1 | SOLID / HAZARDOUS WASTE POLLUTION | Process residues, filter aids, spent carbon, ETP sludge | Solid / semi-solid | HW 28.2 / 35.3 | Sent to Authorized TSDF (CHWTSDF) for landfill/incineration | |

2 | SOLID / HAZARDOUS WASTE POLLUTION | Spent solvent mix (TnBP + detergent) | Liquid | HW 28.1 | Sent to Solvent recovery or TSDF | |

3 | SOLID / HAZARDOUS WASTE POLLUTION | Used oil from DG set | Liquid | HW 5.1 | Sold to MPCB-authorized recycler | |

4 | SOLID / HAZARDOUS WASTE POLLUTION | Biomedical waste (lab disposables, plasma bags) | Bio-infectious | BMW Cat. 4 / 5 | Sent to Authorized CBWTF | |

5 | SOLID / HAZARDOUS WASTE POLLUTION | Non-hazardous recyclable waste | Paper, plastic, metal, glass | Non-hazardous | Segregation → authorized recyclers | |

6 | SOLID / HAZARDOUS WASTE POLLUTION | Organic waste (canteen, sludge) | Biodegradable | Non-hazardous | Composting / manure | |

Sr.No. | Type | Source | Noise Level (dB-A) | Control Measures | | |

1 | NOISE POLLUTION | DG sets | 75–80 | Acoustic enclosure, antivibration mounts, green belt | | |

2 | NOISE POLLUTION | Compressors, pumps, blowers | 70–75 | Foundation damping pads, acoustic barrier | | |

3 | NOISE POLLUTION | HVAC & AHU | 60–65 | Acoustic lining, flexible connectors | | |

4 | NOISE POLLUTION | General utilities | < 55 | Periodic maintenance, ear protection for workers | | |

Sr.No. | Type | Equipment / Facility | Capacity / Function | Purpose | | |

1 | SOLID WASTE MANAGEMENT | ETP | 70 m³/day (MBBR + UF + RO + MEE) | Treat & recycle process effluent | | |

2 | SOLID WASTE MANAGEMENT | STP | 40 m³/day (MBBR + UF) | Treat domestic sewage | | |

3 | SOLID WASTE MANAGEMENT | Air pollution stacks | Boiler – 12 m; DG – 6.5 m above roof | Emission dispersion | | |

4 | SOLID WASTE MANAGEMENT | Activated carbon filter | 1 × 500 kg | VOC adsorption from S-D exhaust | | |

5 | SOLID WASTE MANAGEMENT | Scrubber (if acid used) | 1 × 3,000 m³/hr | Fume removal from buffer prep / lab exhaust | | |

6 | SOLID WASTE MANAGEMENT | Noise enclosure | DG & compressors | Noise reduction | | |

7 | SOLID WASTE MANAGEMENT | Greenbelt | ≥ 33 % of plot | Dust, noise & micro-climate control | | |

| Back to Index | | | | |

[SHEET: Equipments]

Sr.No. | Category | Equipment Name | Quantity | Estimated Amount | | Back to Index

1 | ADMIN EQUIPMENT | Office Cabinets for Documents | 20 | 240000 | |

2 | ADMIN EQUIPMENT | Computers / Laptops | 50 | 2500000 | |

3 | ADMIN EQUIPMENT | Office Furniture | NA | 1200000 | |

4 | ADMIN EQUIPMENT | Printer/scanner / xerox machine | 1 | 500000 | |

5 | ADMIN EQUIPMENT | Water Dispenser | 2 | 40000 | |

6 | ADMIN EQUIPMENT | Projector for conference / meeting room | 2 | 600000 | |

7 | CANTEEN EQUIPMENT | Refrigerator | 2 | 300000 | |

8 | CANTEEN EQUIPMENT | SS furniture (serving table etc) | NA | 400000 | |

9 | CANTEEN EQUIPMENT | SS Water cooler dispenser | 1 | 50000 | |

10 | CANTEEN EQUIPMENT | Microwaves | 2 | 60000 | |

11 | CANTEEN EQUIPMENT | Kitchenware | na | 500000 | |

12 | CANTEEN EQUIPMENT | Bain marie | 1 | 50000 | |

13 | Clean Utilities | Soft water generation 10000/hour and storage system | 1 | 40000000 | |

14 | Clean Utilities | Purified water generation 5000L/ hour storage tank 10KL and distribution system | 1 | | |

15 | Clean Utilities | Water for injection generation 1000L / hour, Storage tank 10KL and distribution system | 1 | | |

16 | Clean Utilities | Pure steam generator 1500Kg / hour and distribution system | 1 | | |

17 | Black Utilities | Plant steam generation 2000Kg/Hour and distribution System | 2 | 7000000 | |

18 | Black Utilities | Compressed air generation and distribution system | 2 | 7500000 | |

19 | Black Utilities | PASSENGER LIFT X 2 NOS | 2 | 7000000 | |

20 | Black Utilities | MATERIAL LIFT X 3 NOS | 3 | 6000000 | |

21 | PLASMA STORAGE EQUIPMENT | Analytical weighing Balance | 1 | 170000 | |

22 | PLASMA STORAGE EQUIPMENT | RLAf | 3 | 1200000 | |

23 | PLASMA STORAGE EQUIPMENT | Precision Balance | 1 | 144000 | |

24 | PLASMA STORAGE EQUIPMENT | Commercial Balance | 1 | 120000 | |

25 | PLASMA STORAGE EQUIPMENT | Refrigerator 200L | 2 | 218000 | |

26 | PLASMA STORAGE EQUIPMENT | Deep freezer -20°C (1000L) | 50 | 22500000 | |

27 | PLASMA STORAGE EQUIPMENT | Air curtain | 10 | 100000 | |

28 | PLASMA STORAGE EQUIPMENT | PVC pallets | 50 | 300000 | |

29 | PLASMA STORAGE EQUIPMENT | Storage racks SS | 25 | 700000 | |

30 | PLASMA STORAGE EQUIPMENT | Sampling rods | 2 | 10000 | |

31 | PLASMA STORAGE EQUIPMENT | Liquid sampler | 2 | 20000 | |

32 | PLASMA STORAGE EQUIPMENT | Dedusting tunnel | 1 | 1250000 | |

33 | PLASMA STORAGE EQUIPMENT | DPB | 4 | 600000 | |

34 | PLASMA STORAGE EQUIPMENT | LAF cabinet | 1 | 450000 | |

35 | PLASMA STORAGE EQUIPMENT | Computer / Laptops | 4 | 120000 | |

36 | PLASMA STORAGE EQUIPMENT | Office Furniture | NA | 200000 | |

37 | PLASMA STORAGE EQUIPMENT | Office Cabinets | 10 | 120000 | |

38 | BULK PLASMA MANUFACTURING EQUIPMENT | Chromatography AKTA-1 (471 L/hr (Pipe MOC SS) 1F | 2 | 34200000 | |

39 | BULK PLASMA MANUFACTURING EQUIPMENT | Chromatography Column Dia 1000mm x Height 500mm 1F | 4 | 89224400 | |

40 | BULK PLASMA MANUFACTURING EQUIPMENT | Chromatography AKTA-1 (785 L/hr pipe MOC SS) 2F | 2 | 40062840 | |

41 | BULK PLASMA MANUFACTURING EQUIPMENT | Chromatography Column Dia 1000mm x Height 500mm 2F | 2 | 44612200 | |

42 | BULK PLASMA MANUFACTURING EQUIPMENT | Chromatography AKTA-1 (1178 L/hr pipe MOC PP) 3F | 2 | 40062840 | |

43 | BULK PLASMA MANUFACTURING EQUIPMENT | Chromatography Column Dia 1000mm x Height 500mm 3F | 4 |

89224400 ||
44 | BULK PLASMA MANUFACTURING EQUIPMENT | Chromatography AKTA-1 (864 L/hr pipe MOC PP) 4l | 2 | 40062840
||
45 | BULK PLASMA MANUFACTURING EQUIPMENT | Chromatography Column Dia 1000mm x Height 500mm 4l | 2 |
44612200 ||
46 | BULK PLASMA MANUFACTURING EQUIPMENT | Chromatography AKTA-1 (864 L/hr pipe MOC PP) 5l | 1 | 20031420
||
47 | BULK PLASMA MANUFACTURING EQUIPMENT | Chromatography Column Dia 1000mm x Height 500mm 5l | 1 |
22306100 ||
48 | BULK PLASMA MANUFACTURING EQUIPMENT | Chromatography AKTA-1 (667 L/hr pipe MOC PP) 6l | 1 | 17100000
||
49 | BULK PLASMA MANUFACTURING EQUIPMENT | Chromatography Column Dia 1000mm x Height 500mm 6l | 1 |
22306100 ||
50 | BULK PLASMA MANUFACTURING EQUIPMENT | Chromatography AKTA-1 (785 L/hr pipe MOC PP) 7A | 1 | 20031420
||
51 | BULK PLASMA MANUFACTURING EQUIPMENT | Chromatography Column Dia 1000mm x Height 500mm 7A | 1 |
22306100 ||
52 | BULK PLASMA MANUFACTURING EQUIPMENT | Chromatography AKTA-1 (30 L/hr pipe MOC PP) 8N | 1 | 14657140
||
53 | BULK PLASMA MANUFACTURING EQUIPMENT | Chromatography Column Dia 300mm x Height 500mm 8N | 1 |
4896570 ||
54 | BULK PLASMA MANUFACTURING EQUIPMENT | Chromatography AKTA-1 (30 L/hr pipe MOC PP) 9N | 1 | 14657140
||
55 | BULK PLASMA MANUFACTURING EQUIPMENT | Chromatography Column Dia 300mm x Height 750mm 9N | 1 |
4896570 ||
56 | BULK PLASMA MANUFACTURING EQUIPMENT | Chromatography AKTA-1(85 L/hr pipe MOC PP) 10T | 1 | 12500000
||
57 | BULK PLASMA MANUFACTURING EQUIPMENT | Chromatography Column Dia 600mm x Height 500mm 10T | 1 |
14244570 ||
58 | BULK PLASMA MANUFACTURING EQUIPMENT | Chromatography AKTA-1 (340 L/hr pipe MOC PP) 11F | 1 |
30000000 ||
59 | BULK PLASMA MANUFACTURING EQUIPMENT | Chromatography Column Dia 300mm x Height 500mm 11F | 1 |
4896570 ||
60 | BULK PLASMA MANUFACTURING EQUIPMENT | Chromatography AKTA-1 (500 L/hr pipe MOC PP) 12E | 1 |
17100000 ||
61 | BULK PLASMA MANUFACTURING EQUIPMENT | Chromatography Column Dia 1000 mm x Height 500mm 12E | 1 |
22306100 ||
62 | BULK PLASMA MANUFACTURING EQUIPMENT | Chromatography AKTA-1(160 L/hr pipe MOC PP) 13E | 1 | 14657140
||
63 | BULK PLASMA MANUFACTURING EQUIPMENT | Chromatography Column Dia 450mm x Height 500mm 13E | 1 |
12018850 ||
64 | BULK PLASMA MANUFACTURING EQUIPMENT | TFF System 50 sqm | 3 | 12500000 ||
65 | BULK PLASMA MANUFACTURING EQUIPMENT | Cassette 50 sqm | 3 | 4000000 ||
66 | BULK PLASMA MANUFACTURING EQUIPMENT | TFF System 20 sqm | 1 | 9460000 ||
67 | BULK PLASMA MANUFACTURING EQUIPMENT | Cassette 20 sqm | 2 | 3010000 ||
68 | BULK PLASMA MANUFACTURING EQUIPMENT | TFF System 5 sqm | 4 | 14620000 ||
69 | BULK PLASMA MANUFACTURING EQUIPMENT | Cassette 5 sqm | 4 | 1376000 ||
70 | BULK PLASMA MANUFACTURING EQUIPMENT | Continuous Centrifuge 1000L/hr | 2 | 30000000 ||
71 | BULK PLASMA MANUFACTURING EQUIPMENT | Process Vessels 300 L | 4 | 8842500 ||
72 | BULK PLASMA MANUFACTURING EQUIPMENT | Process Vessels 100 L | 3 | 6226875 ||
73 | BULK PLASMA MANUFACTURING EQUIPMENT | Process Vessels 150 L | 1 | 2075625 ||
74 | BULK PLASMA MANUFACTURING EQUIPMENT | Process Vessels 250 L | 4 | 8842500 ||
75 | BULK PLASMA MANUFACTURING EQUIPMENT | Process Vessels 500 L | 4 | 10732500 ||
76 | BULK PLASMA MANUFACTURING EQUIPMENT | Process Vessels 1000 L | 6 | 19650600 ||
77 | BULK PLASMA MANUFACTURING EQUIPMENT | Process Vessels 1500 L | 29 | 100830825 ||
78 | BULK PLASMA MANUFACTURING EQUIPMENT | Process Vessels 2000 L | 11 | 42701175 ||
79 | BULK PLASMA MANUFACTURING EQUIPMENT | Buffer vessels 500L | 6 | 16098750 ||
80 | BULK PLASMA MANUFACTURING EQUIPMENT | Buffer vessels 1000L | 9 | 29475900 ||
81 | BULK PLASMA MANUFACTURING EQUIPMENT | Buffer vessels 1500L | 6 | 23291550 ||
82 | BULK PLASMA MANUFACTURING EQUIPMENT | Buffer vessels 2500L | 1 | 30000000 ||
83 | BULK PLASMA MANUFACTURING EQUIPMENT | Depth filtration system | 5 | 5212500 ||
84 | BULK PLASMA MANUFACTURING EQUIPMENT | Nano filter holder | 2 | 1000000 ||
85 | BULK PLASMA MANUFACTURING EQUIPMENT | Cold room 2-8°C | 4 | 3000000 ||
86 | BULK PLASMA MANUFACTURING EQUIPMENT | Rectangular tank | 2 | 860000 ||
87 | BULK PLASMA MANUFACTURING EQUIPMENT | Buffer preparation tank | 4 | 3440000 ||
88 | BULK PLASMA MANUFACTURING EQUIPMENT | Dispensing Booth | 2 | 800000 ||
89 | BULK PLASMA MANUFACTURING EQUIPMENT | Analytical Balance | 2 | 340000 ||
90 | BULK PLASMA MANUFACTURING EQUIPMENT | Precision Balance | 2 | 300000 ||
91 | BULK PLASMA MANUFACTURING EQUIPMENT | Commercial Balance | 2 | 240000 ||
92 | BULK PLASMA MANUFACTURING EQUIPMENT | PVC bag Sealer | 2 | 10000 ||
93 | BULK PLASMA MANUFACTURING EQUIPMENT | Rotary Bag Sealer | 2 | 600000 ||
94 | BULK PLASMA MANUFACTURING EQUIPMENT | pH/ Conductivity meter | 4 | 600000 ||
95 | BULK PLASMA MANUFACTURING EQUIPMENT | Magnetic Stirrer | 4 | 300000 ||
96 | BULK PLASMA MANUFACTURING EQUIPMENT | Peristaltic pump | 4 | 400000 ||
97 | BULK PLASMA MANUFACTURING EQUIPMENT | Filter Integrity machine | 2 | 4000000 ||
98 | BULK PLASMA MANUFACTURING EQUIPMENT | Fume Hood | 2 | 900000 ||
99 | BULK PLASMA MANUFACTURING EQUIPMENT | Autoclave 1000L | 4 | 8000000 ||
100 | BULK PLASMA MANUFACTURING EQUIPMENT | DHS 1000L | 2 | 4000000 ||
101 | BULK PLASMA MANUFACTURING EQUIPMENT | Ceiling LAF | 6 | 2400000 ||
102 | BULK PLASMA MANUFACTURING EQUIPMENT | Drying Oven | 4 | 2000000 ||
103 | BULK PLASMA MANUFACTURING EQUIPMENT | Acid tank 1000L (Pre and Post Virus) | 2 | 3870000 ||
104 | BULK PLASMA MANUFACTURING EQUIPMENT | Base tank 1000L (Pre and Post Virus) | 2 | 3870000 ||
105 | BULK PLASMA MANUFACTURING EQUIPMENT | Central CIP system | 4 | 4300000 ||
106 | BULK PLASMA MANUFACTURING EQUIPMENT | Refrigerator 200L | 2 | 300000 ||
107 | BULK PLASMA MANUFACTURING EQUIPMENT | Laminar air flow cabinet | 2 | 870000 ||
108 | BULK PLASMA MANUFACTURING EQUIPMENT | Dynamic pass box | 8 | 960000 ||
109 | BULK PLASMA MANUFACTURING EQUIPMENT | Tumble Dryer 25 Kg | 2 | 4000000 ||
110 | BULK PLASMA MANUFACTURING EQUIPMENT | Barrier washer Extractor 25Kg | 2 | 6000000 ||
111 | BULK PLASMA MANUFACTURING EQUIPMENT | SS Furniture (Table, chair, racks, Ladders etc) | NA |
8000000 ||
112 | BULK PLASMA MANUFACTURING EQUIPMENT | Lockers | NA | 400000 ||
113 | BULK PLASMA MANUFACTURING EQUIPMENT | Laptop / Computers | 20 | 1000000 ||

114 | BULK PLASMA MANUFACTURING EQUIPMENT | Office furniture / Cabinets | NA | 600000 ||
 115 | FILL FINISH EQUIPMENT | Formulation Vessel 100L | 1 | 2500000 ||
 116 | FILL FINISH EQUIPMENT | Ceiling LAF | 2 | 800000 ||
 117 | FILL FINISH EQUIPMENT | Laminar air flow cabinet | 1 | 435000 ||
 118 | FILL FINISH EQUIPMENT | Continuous Particle Counter | 1 | 3000000 ||
 119 | FILL FINISH EQUIPMENT | Dynamic pass box | 7 | 840000 ||
 120 | FILL FINISH EQUIPMENT | Pass Box static | 1 | 38000 ||
 121 | FILL FINISH EQUIPMENT | Vial washing machine | 1 | 4500000 ||
 122 | FILL FINISH EQUIPMENT | Sterilizing tunnel | 1 | 4500000 ||
 123 | FILL FINISH EQUIPMENT | Vial filling & Sealing line 120 vials /min | 1 | 7500000 ||
 124 | FILL FINISH EQUIPMENT | Lyophilizer (1sq meter) | 1 | 5500000 ||
 125 | FILL FINISH EQUIPMENT | Labelling machine | 1 | 9,00,000.00 ||
 126 | FILL FINISH EQUIPMENT | Double door autoclave 1000L | 1 | 2000000 ||
 127 | FILL FINISH EQUIPMENT | Cold room | 1 | 800000 ||
 128 | FILL FINISH EQUIPMENT | SS furniture (Table, Chair, racks, cabinets etc) | 1 | 200000 ||
 129 | QC & R&D EQUIPMENT | Akta pure -150 (For R&D) | 2 | 4000000 ||
 130 | QC & R&D EQUIPMENT | Columns XK 50/100 (For R & D) | 1 | 700000 ||
 131 | QC & R&D EQUIPMENT | Columns XK 50/60 (For R&D) | 1 | 700000 ||
 132 | QC & R&D EQUIPMENT | Column XK 50/30 (For R & D) | 3 | 1400000 ||
 133 | QC & R&D EQUIPMENT | HPLC-UV | 1 | 3400000 ||
 134 | QC & R&D EQUIPMENT | HPLC Columns and accessories | 1 | 1000000 ||
 135 | QC & R&D EQUIPMENT | Electrophoresis (Minigel) with trans blot | 2 | 600000 ||
 136 | QC & R&D EQUIPMENT | Gel Doc system densitometer | 1 | 500000 ||
 137 | QC & R&D EQUIPMENT | Power packs | 2 | 200000 ||
 138 | QC & R&D EQUIPMENT | Endotoxin Detection system (Tube heating block) | 1 | 200000 ||
 139 | QC & R&D EQUIPMENT | Micropipettes- variable volume range of 2-20µL, 20-200µL, 100-1000µL, 500-5000µL and multichannel 30-300µL | 2 set each | 300000 ||
 140 | QC & R&D EQUIPMENT | Lab Refrigerator 2-8 ° C (SS) with data control | 4 | 450000 ||
 141 | QC & R&D EQUIPMENT | Flame photometer | 1 | 450000 ||
 142 | QC & R&D EQUIPMENT | Weighing Balance Analytical | 2 | 350000 ||
 143 | QC & R&D EQUIPMENT | UV-VIS spectrophotometer Double Beam with data acquisition. | 1 | 900000 ||
 144 | QC & R&D EQUIPMENT | Volumetric KF | 1 | 700000 ||
 145 | QC & R&D EQUIPMENT | Moisture Analyser | 1 | 800000 ||
 146 | QC & R&D EQUIPMENT | Stability Chambers (5±3° C, 25, 37 deg. C) 200L (1set) | 3 | 1400000 ||
 147 | QC & R&D EQUIPMENT | Muffle Furnance | 1 | 80000 ||
 148 | QC & R&D EQUIPMENT | pH and conductivity meter with printer | 2 | 260000 ||
 149 | QC & R&D EQUIPMENT | Vaccum Oven | 1 | 120000 ||
 150 | QC & R&D EQUIPMENT | Melting point apparatus | 1 | 90000 ||
 151 | QC & R&D EQUIPMENT | Incubator Sterility Test 500L | 2 | 1440000 ||
 152 | QC & R&D EQUIPMENT | Incubators for Normal Media plates Incubation 500L | 1 | 300000 ||
 153 | QC & R&D EQUIPMENT | Incubator for EMP and water samples 500L | 2 | 600000 ||
 154 | QC & R&D EQUIPMENT | Hot Air Oven | 1 | 160000 ||
 155 | QC & R&D EQUIPMENT | Thermostatic water bath (10-20 L) | 2 | 200000 ||
 156 | QC & R&D EQUIPMENT | Vaccum pump | 2 | 120000 ||
 157 | QC & R&D EQUIPMENT | Desiccator Glass | 1 | 4000 ||
 158 | QC & R&D EQUIPMENT | High Quality Laboratory water generation system (10 L /hr) Milli -Q water | 1 | 1450000 ||
 159 | QC & R&D EQUIPMENT | Sonicator | 1 | 180000 ||
 160 | QC & R&D EQUIPMENT | Vortex Shaker | 6 | 210000 ||
 161 | QC & R&D EQUIPMENT | Magnetic Stirrer with hot plate | 4 | 400000 ||
 162 | QC & R&D EQUIPMENT | LAF Vertical | 2 | 600000 ||
 163 | QC & R&D EQUIPMENT | Biosafety Cabinet | 2 | 1400000 ||
 164 | QC & R&D EQUIPMENT | Autoclave Double door | 1 | 1250000 ||
 165 | QC & R&D EQUIPMENT | Dry Heat Steriliser | 1 | 750000 ||
 166 | QC & R&D EQUIPMENT | Fume Hood (Table Top) | 1 | 450000 ||
 167 | QC & R&D EQUIPMENT | Deep freezer (-20 Degrees C) with Data acquisition | 1 | 400000 ||
 168 | QC & R&D EQUIPMENT | Air sampler | 1 | 300000 ||
 169 | QC & R&D EQUIPMENT | Particle counter | 1 | 2000000 ||
 170 | QC & R&D EQUIPMENT | Decontamination autoclave | 1 | 1000000 ||
 171 | QC & R&D EQUIPMENT | Microscope Binocular | 1 | 100000 ||
 172 | QC & R&D EQUIPMENT | Colony Counter | 1 | 150000 ||
 173 | QC & R&D EQUIPMENT | Microplate reader ELISA | 1 | 600000 ||
 174 | QC & R&D EQUIPMENT | ELISA Auto Analyzers (Roche / Abbott)* | 2 | 1000000 ||
 175 | QC & R&D EQUIPMENT | Microplate washer | 1 | 350000 ||
 176 | QC & R&D EQUIPMENT | TOC analyser (Grab Samples) | 1 | 2500000 ||
 177 | QC & R&D EQUIPMENT | Timers/ Stop watch | 5 | 10000 ||
 178 | QC & R&D EQUIPMENT | Ice flaking machine | 1 | 250000 ||
 179 | QC & R&D EQUIPMENT | Standard Weight Box | 1 Set | 1000000 ||
 180 | QC & R&D EQUIPMENT | Printer/scanner / xerox machine | 1 | 500000 ||
 181 | QC & R&D EQUIPMENT | Centrifuge table top | 1 | 450000 ||
 182 | QC & R&D EQUIPMENT | Size exclusion | L | 150000 ||
 183 | QC & R&D EQUIPMENT | Anion | L | 150000 ||
 184 | QC & R&D EQUIPMENT | Cation | L | 150000 ||
 185 | QC & R&D EQUIPMENT | Affinity | L | 150000 ||
 186 | QC & R&D EQUIPMENT | SS furniture racks | 25 | 250000 ||
 187 | QC & R&D EQUIPMENT | Computers / Laptops | 10 | 500000 ||
 188 | QC & R&D EQUIPMENT | Office Furniture | NA | 500000 ||
 189 | QC & R&D EQUIPMENT | Office Cabinets for Documents | 10 | 120000 ||
 190 | QA EQUIPMENT | Anemometer | 1 | 15000 ||
 191 | QA EQUIPMENT | Lux meter | 1 | 12000 ||
 192 | QA EQUIPMENT | Datalogger 32 channel | 1 | 400000 ||
 193 | QA EQUIPMENT | Paper shredder | 1 | 10000 ||
 194 | QA EQUIPMENT | Laminator | 1 | 10000 ||
 195 | QA EQUIPMENT | Printer/scanner / xerox machine | 1 | 500000 ||
 196 | QA EQUIPMENT | Computers / Laptops | 10 | 500000 ||
 197 | QA EQUIPMENT | Numbering Machine | 5 | 25000 ||
 198 | QA EQUIPMENT | Spiral Binding Machine | 1 | 15000 ||
 199 | QA EQUIPMENT | A3 +A4 Lamination Machine | 1 | 25000 ||
 200 | QA EQUIPMENT | Bi-Focal Glass | 1 | 80000 ||
 201 | QA EQUIPMENT | Tachometer | 2 | 6000 ||
 202 | QA EQUIPMENT | Clamp Meter/ Multi meter | 5 | 10000 ||
 203 | QA EQUIPMENT | Level Indicator | 2 | 8000 ||

204 | QA EQUIPMENT | Measuring TAPE | 2 | 1500 | |
205 | QA EQUIPMENT | Stop Watch | 5 | 50000 | |
206 | QA EQUIPMENT | Temperature recorders 24 Nos. (data Loggers) | 20 | 200000 | |
207 | QA EQUIPMENT | Office Cabinets for Documents | 20 | 240000 | |
208 | QA EQUIPMENT | Projector for conference / meeting room | 1 | 300000 | |
209 | QA EQUIPMENT | Office Furniture | NA | 700000 | |
210 | FINISHED GOODS STORAGE | Air curtain | 10 | 100000 | |
211 | FINISHED GOODS STORAGE | PVC pallets | 50 | 300000 | |
212 | FINISHED GOODS STORAGE | Storage racks SS | 50 | 1500000 | |
213 | FINISHED GOODS STORAGE | Hydraulic trolley | 1 | 250000 | |
214 | ETP | Biowaste reception tank 15000L | 1 | 4000000 | |
215 | ETP | Kill tank 1000L | 2 | | |
216 | ETP | Bar Screen | NA | | |
217 | ETP | Collection tank | NA | | |
218 | ETP | Equalization tank | NA | | |
219 | ETP | Neutralization tank | NA | | |
220 | ETP | Chemical coagulation | NA | | |
221 | ETP | Biological oxidation tank | NA | | |
222 | ETP | Filtration System | NA | | |
[SHEET: Heat treat_Reaction_Tank]
Sr.No. | Vessel | Working vol (L) | No. of units recommended
1 | Plasma pooling tank (refrigerated) | 2000 | 1 primary + 1 standby (or reserve)
2 | Caprylic acid reaction tank (jacketed) | 2000 | 1 (or 2 for staggered batches)
3 | S-D mixing tank (jacketed, sealed) | 2000 | 1 (with dedicated solvent bund)
4 | Clarification / settling tank or DAF feed | 2500 | 1
5 | Pasteurizer (container) | capacity to hold per-batch vials (900) | 1 tunnel / basket line sized to throughput
6 | Buffer / formulation tanks (jacketed) | 500–1,500 | 2–3 (for different buffers)
7 | Waste / solvent hold tank (bunded) | 1,000–2,000 | 1
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[SHEET: Fuel Consmpr]
Sr. No. | Fuel Type | Purpose / Area of Use | Approx. Annual Consumption | Unit
1 | Electricity | Cleanroom HVAC, process equipment (centrifuges, lyophilizers, pumps, lighting, DG auxiliaries, WFI generation, chillers, lab & office) | 3.5 – 4.0 million kWh / year | kWh
2 | HSD (High-Speed Diesel) | Stand-by DG sets (2 × 500 kVA) for emergency power | ≈ 25 – 30 KL / year | KL (≈ 25,000 – 30,000 L)**
3 | CNG / PNG | Boiler / hot-water generation for WFI plant, sterilization steam, HVAC re-heat coils | ≈ 150,000 – 180,000 Sm³ / year | Standard m³ (Sm³)
4 | LPG (optional small cylinders) | Laboratory instruments / flame sterilizers | ≤ 500 kg / year | kg
| Back to Index | | |
[SHEET: Water]
Sr.No. | Item | Litres (L / year) | Cubic metres (m³ / year) | Notes
| A. Water Inputs | | |
1 | Domestic water (506 ppl × 100 L/p/d) | 18469000 | 18469 | Toilets, canteen, housekeeping
| WFI (process) | 6205470 | 6205.47 | WFI for formulation, CIP, rinses
| Other process water (25% of WFI) | 1551368 | 1551.37 | CIP makeup, utility washes
| Extra process water (10% of plasma vol) | 41400 | 41.4 | Misc. process losses, small uses
2 | Process Water | 7798238 | 7798.24 |
| Total Water Input | 26267238 | 26267.24 | A = sum of above
| B. Water Outputs / Wastewater | | |
1 | Domestic sewage → STP (80% of domestic) | 14775200 | 14775.2 | Black/gray water to STP
2 | Process wastewater → ETP (99% of process) | 7720256 | 7720.26 | Process drains, WFI rejects, CIP effluent
| Total Wastewater Generated | 22495456 | 22495.46 | B = sum of above
| C. Balance / Losses / Product incorporation | | |
1 | Unaccounted losses / product incorporation / evaporation (A – B) | 3771782 | 3771.78 | ~14.36% of input — includes water in products, steam losses, sampling, evaporation, bleed, equipment hold-up
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[SHEET: Non Haz Waste]
Sr. No. | Source / Process Area | Type of Non-Hazardous Waste | Approx. Quantity / Year
1 | ETP / STP treatment | Biological sludge (after dewatering) | 8 – 10 MT/year (wet)
2 | Process area & lab washing | Treated water (ETP + RO permeate) | ~12,000 m³/year
3 | Packaging line | Empty glass vials (non-contaminated rejects) | 0.5 – 1.0 MT/year (~50,000 pcs)
4 | Packaging & labeling | Paperboard cartons, labels, paper scrap | 1.5 – 2.0 MT/year
5 | Procurement / stores area | Corrugated boxes, stretch film, plastic wrapping | 1.0 – 1.5 MT/year
6 | Maintenance workshop | Metal scrap (iron, steel, aluminum parts) | 0.8 – 1.2 MT/year
7 | Utility & HVAC area | Spent filters (air filters, bag filters, prefilters) | 0.3 – 0.5 MT/year
8 | Admin & office areas | Office paper waste, packaging waste | 1.0 – 1.5 MT/year
9 | Canteen / pantry | Organic kitchen waste | 1.5 – 2.0 MT/year
10 | General housekeeping | Sweepings, mopping dust, non-contaminated PPE | 0.5 – 0.8 MT/year
11 | Cooling tower & utilities | Blowdown & scale sludge (non-toxic) | 1.0 – 1.5 MT/year
12 | Electrical systems | Cable cuttings, wire ends, plastic conduits | 0.2 – 0.3 MT/year
13 | Gardening & green area | Dry leaves, grass clippings | 0.3 – 0.5 MT/year
14 | Water purification & softener | Spent ion-exchange resins (non-toxic) | 0.2 MT/year
15 | Misc. activities | Empty HDPE / PET bottles (non-contaminated) | 0.1 – 0.2 MT/year
| Back to Index | | |
[SHEET: Raw Material]
Sr.No. | Item | Annual requirement (rounded) | Unit | Notes |
1 | Source human plasma | 414000 | L | Primary raw material (cold chain required) | Back to Index
2 | Water for Injection (WFI) | 6205000 | L | Assumed 15,000 L WFI / 1,000 L plasma |
3 | Sodium chloride (NaCl) | 10350 | kg | 25 kg / 1,000 L plasma × 413.7 |
4 | Sodium citrate / acetate | 2070 | kg | 5 kg / 1,000 L |
5 | Glycine | 4140 | kg | 10 kg / 1,000 L |
6 | Caprylic (Octanoic) acid | 3310 | kg | 8 kg / 1,000 L |
7 | Triton X-100 / Polysorbate-80 | 2070 | L | 5 L / 1,000 L |
8 | Tri(n-butyl) phosphate (TnBP) | 1241 | L | 3 L / 1,000 L |
9 | PBS / Tris buffer components | 8275 | kg | 20 kg / 1,000 L |
10 | Acids / NaOH (dilute) | 82740 | L | 200 L / 1,000 L (process neutralization) |
11 | Activated carbon (GAC) | 20685 | kg | 50 kg / 1,000 L |
12 | Filter aids (depth filter media) | 41370 | kg | 100 kg / 1,000 L |
13 | 0.2 µm sterile filters (cartridges) | 8300 | units | 20 units / 1,000 L |
14 | PVDF / sterile filter cartridges | 4140 | units | 10 units / 1,000 L |

15 | Chromatography resins (total across steps) | 20685 | kg | 50 kg / 1,000 L |
 16 | TFF membranes / UF-DF modules | 4140 | modules | 10 modules / 1,000 L |
 17 | Albumin stabilizer (Na-octanoate, etc.) | 4965 | kg | 12 kg / 1,000 L |
 18 | Fibrin sealant reagents (thrombin, substrates) | 2070 | kg | 5 kg / 1,000 L |
 19 | Preservatives (if used) | 4140 | kg | 10 kg / 1,000 L |
 20 | Nitrogen gas (compressed) | 206850 | m³ | 500 m³ / 1,000 L |
 21 | Sterile glass vials (all sizes) | 1037500 | units | Total vials + 5% spare (see note below) |
 22 | Rubber stoppers | 1037500 | units | Match vials (one per vial) |
 23 | Flip-off seals | 1037500 | units | Match vials |
 24 | Labels | 1040000 | units | one per vial + spares |
 25 | Cartons (≈10 vials/carton) | 98810 | cartons | Round up to supplier pack sizes |
 26 | Laboratory consumables (pipette tips, plates, etc.) | Estimate 4,000 | packs | Itemised procurement per lab list |
 27 | Spare filters / membranes (replacement spares) | +20% of above | — | |
 28 | Cold-chain storage (freezer/refrigeration units) | qty per plant | — | |
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[SHEET: List of Products]

Product | Protein availability (gm) | Protein Recovery | Conc. / vial | No. of vials /batch | Total
 (Considering

15 Batches / month) | Per Annum

Albumin (gm) | 40000 | 20000 | 20 GM | 900 | 13500 | 162000

Human IgG (gm) | 12500 | 6250 | 5 GM | 1000 | 15000 | 180000

Factor VIII in IU | 1000000 | 12500 IU | 250 IU | 50 | 750 | 9000

Factor IX in IU | 1000000 | 90000 IU | 600 IU | 150 | 2250 | 27000

Fibrinogen (gm) | 3000 | 1500 | 70 mg | 80 | 1200 | 14400

Thrombin in IU | 900000 | 90000 | 1000 IU | 1400 | 21000 | 252000

Based on vessel size, the batch size assumed is 1000 liters, on every alternate date, a new batch can be initiated, therefore 15 batches shall be processed per month (one batch/ day). | | | | |

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[SHEET: Hazardous Waste]

Sr. No. | Process / Source | Type of Hazardous Waste | Approx. Quantity / Year

1 | Solvent recovery & viral inactivation (TnBP + detergent process) | Spent solvent mixture (Tri-n-butyl phosphate + Triton X-100 / Polysorbate 80) | 1.0 – 1.5 KL / year

2 | Filtration & purification units | Used filter aids / depth filter media contaminated with protein or solvent | 0.8 – 1.0 MT / year

3 | Activated carbon beds (viral removal, deodorization) | Spent activated carbon with adsorbed organic residues | 0.5 – 0.7 MT / year

4 | Chromatography / TFF cleaning (CIP) | Residues of buffers, cleaning agents, NaOH / HCl neutralization sludge | 2.0 – 3.0 MT / year (dry solids)

5 | ETP operation | Sludge containing proteinaceous organic matter & trace solvents | 5.0 – 6.0 MT / year

6 | ETP / RO reject water | High-TDS / COD effluent (liquid) | ≈ 900 – 1,000 m³ / year

7 | Resin & membrane replacement | Spent chromatography resin, UF/DF membranes, PVDF filters | 0.5 – 1.0 MT / year

8 | Laboratory & QC activities | Contaminated lab glassware, pipette tips, chemical bottles | 200 – 250 kg / year

9 | Oil from DG sets, compressors, hydraulic systems | Used lubricating oil | 50 – 100 L / year

10 | DG set maintenance | Oil-soaked cotton waste, filters | 50 – 100 kg / year

11 | Production packaging & vial washing | Contaminated packaging (plastic, foil, paper) | 0.3 – 0.5 MT / year

12 | Spill & floor washings | Absorbent material, mopping waste, PPE, gloves | 0.5 MT / year

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[SHEET: GF_RDS]
ROOM DATA SHEET FOR LIFENITY |||||
|| Room Details ||||| Fire Detection & Protection System |||||
Room No | Room Name | Length (mtr) | Width(mtr) | Height(mtr) | Area (M²) | Volume | Temp.(oC) | RH(%) |
HVAC System | HVAC Type | Occupancy | Grade | Differential P | Fire Alarm System | Fire Sprinkler System
| Fire Extinguisher System | GPS Clock | LAN Port | Telephone Port | Door Interlock | Eye Washer | Body
Washer | Floor Finish | Electrical Lux level
| Reception Room | 8.3 | 12 | 3 | 99.60000000000001 | 298.8 | NA | NA | AC | NA | NA | NA | NA | Y | Y |
NA | NA | Y | Y | NA | NA | NA | NA | Vitrified Tiles | 250
| Waiting Room | 4.67 | 5 | 3 | 23.35 | 70.05000000000001 | NA | NA | AC | NA | NA | NA | NA | Y | Y | NA
NA | NA | NA | NA | NA | NA | Vitrified Tiles | 250
| Passage | 4.72 | 1.8 | 3 | 8.496 | 25.488 | NA | NA | NA | NA | NA | NA | NA | Y | Y | NA | Y | Y | NA
NA | NA | NA | Vitrified Tiles | 250
| Corridor | 40.4 | 1.8 | 3 | 72.72 | 218.16 | NA | NA | NA | NA | NA | NA | NA | Y | Y | NA | Y | Y | NA
NA | NA | NA | Vitrified Tiles | 250
| Gents Toilet Room | 9.94 | 4.9 | 3 | 48.706 | 146.118 | NA | NA | NA | Exhaust | NA | NA | NA | Y | NA
NA | NA | NA | NA | NA | NA | NA | Vitrified Tiles | 250
| Gents locker Room | 6 | 5 | 3 | 30 | 90 | NA | NA | NA | Exhaust | NA | NA | NA | Y | NA | NA | NA | NA
NA | NA | NA | NA | Vitrified Tiles | 250
| Visitor Locker Room | 5.9 | 5 | 3 | 29.5 | 88.5 | NA | NA | NA | Exhaust | NA | NA | NA | Y | NA | Y |
NA | NA | NA | NA | NA | NA | Vitrified Tiles | 250
| Female Locker Room | 6 | 5 | 3 | 30 | 90 | NA | NA | NA | Exhaust | NA | NA | NA | Y | NA | Y | NA | NA
NA | NA | NA | NA | Vitrified Tiles | 350
| Ladies Toilet | 6.3 | 4.9 | 3 | 30.87 | 92.61 | NA | NA | NA | Exhaust | NA | NA | NA | Y | NA | Y | NA
NA | NA | NA | NA | NA | Vitrified Tiles | 250
| Female Change Room-1 | 3.4 | 5.1 | 3 | 17.34 | 52.019999999999996 | NA | NA | AHU | RECIRCULATION | 2 |
CNC | NA | Y | NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| Primary Female change Room | 3.19 | 5.1 | 3 | 16.269 | 48.806999999999995 | 23±4 °C | NMT 60% | AHU |
RECIRCULATION | 2 | D | 15 | Y | NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| Air lock Female | 3.64 | 5.1 | 3 | 18.564 | 55.692 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 2 | D |
20 | Y | NA | Y | NA | NA | NA | Y | NA | NA | Epoxy | 250
| Visitor Change Room | 2.75 | 5.1 | 3 | 14.024999999999999 | 42.074999999999996 | NA | NA | AHU |
RECIRCULATION | NA | CNC | NA | Y | NA | Y | NA | NA | NA | Y | NA | NA | Epoxy | 250
| Primary Visitor Change Room | 3.7 | 5.1 | 3 | 18.87 | 56.61 | 23±4 °C | NMT 60% | AHU | RECIRCULATION |
NA | D | 15 | Y | NA | Y | NA | NA | NA | Y | NA | NA | Epoxy | 250
| Air Lock Visitor | 3.05 | 5.1 | 3 | 15.554999999999998 | 46.664999999999999 | 23±4 °C | NMT 60% | AHU |
RECIRCULATION | NA | D | 20 | Y | NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| Male Change-1 | 3.3 | 5.1 | 3 | 16.83 | 50.489999999999995 | NA | NA | AHU | RECIRCULATION | 2 | CNC |
NA | Y | NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| Primary Male Change Room | 4 | 5.1 | 3 | 20.4 | 61.199999999999996 | 23±4 °C | NMT 60% | AHU |
RECIRCULATION | 2 | D | 15 | Y | NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| Air Lock Male | 4.7 | 5.1 | 3 | 23.97 | 71.91 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 2 | D | 20 |
Y | NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| Corridor | 49 | 2.25 | 3 | 110.25 | 330.75 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 2 | D | 25 | Y |
NA | NA | Y | Y | NA | NA | NA | NA | Epoxy | 250
| Cold Room | 5.87 | 7.5 | 3 | 44.025 | 132.075 | 2-8 C | NA | Cold room unit | NA | NA | NA | NA | Y |
NA | NA | NA | Y | Hooter | NA | NA | NA | Epoxy | 250
| Corridor | 74.2 | 1.8 | 3 | 133.56 | 400.68 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | NA | D | 30 | Y
NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 250
| Passage | 5.15 | 1.8 | 3 | 9.270000000000001 | 27.810000000000002 | 23±4 °C | NMT 60% | AHU |
RECIRCULATION | 2 | D | 30 | Y | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 250
| Janitor Room | 3.41 | 5 | 3 | 17.05 | 51.150000000000006 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | NA
D | 25 | Y | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 250
| Office-1 | 3.69 | 5.1 | 3 | 18.819 | 56.456999999999994 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 2 |
D | 25 | Y | NA | NA | NA | Y | NA | NA | NA | NA | Epoxy | 300
| Office-2 | 3.66 | 5.1 | 3 | 18.666 | 55.998000000000005 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 2 |
D | 25 | Y | NA | NA | NA | Y | NA | NA | NA | NA | Epoxy | 300
| Office-3 | 3.69 | 5.1 | 3 | 18.819 | 56.456999999999994 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 2 |
D | 25 | Y | NA | NA | NA | Y | Y | NA | NA | NA | Epoxy | 300
| Corridor | 71.9 | 2.57 | 3 | 184.783000000000002 | 554.349 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 2
D | 25 | Y | NA | NA | Y | Y | NA | NA | NA | NA | Epoxy | 250
| Air Lock | 2.1 | 3.4 | 3 | 7.14 | 21.419999999999998 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 2 | D
30 | Y | NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| Gown Room | 3.34 | 3.4 | 3 | 11.356 | 34.068 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 2 | C | 45 | Y
NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| Air lock | 3.34 | 3.4 | 3 | 11.356 | 34.068 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 2 | C | 50 | Y
NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| Air Lock | 3.34 | 3.4 | 3 | 11.356 | 34.068 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 2 | C | 50 | Y
NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| De- Gown room | 3.34 | 3.4 | 3 | 11.356 | 34.068 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 2 | C | 45
Y | NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| Air Lock | 2.1 | 3.4 | 3 | 7.14 | 21.419999999999998 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 2 | D
30 | Y | NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| Chemical day store | 11.4 | 3.64 | 3 | 41.496 | 124.488 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 1 |
C | 50 | Y | NA | NA | NA | NA | NA | NA | Y | NA | Epoxy | 350
| 2.33 | 2.14 | 3 | 4.9862 | 14.9586 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 1 | C | 50 | Y | NA |
NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 350
| Corridor | 39.8 | 3.04 | 3 | 120.99199999999999 | 362.976 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 2
C | 60 | Y | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 250
| Buffer preparation | 4 | 7.5 | 3 | 30 | 90 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 4 | C | 50 | Y |
NA | NA | NA | NA | Y | NA | Y | NA | NA | Epoxy | 350
| Tank storage-2 | 6.74 | 7.5 | 3 | 50.550000000000004 | 151.65 | 23±4 °C | NMT 60% | AHU | RECIRCULATION
1 | C | 55 | Y | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 350
| CIP/SIP room | 5.6 | 7.5 | 3 | 42 | 126 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 2 | C | 60 | Y | NA
NA | NA | NA | Y | NA | NA | NA | Epoxy | 350
| Waste In | 3.65 | 2 | 3 | 7.3 | 21.9 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 1 | D | 30 | Y | NA |
NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 350
| Air Lock | 4.13 | 2 | 3 | 8.26 | 24.78 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 2 | D | 30 | Y | NA
NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 350
| Decontamination | 4.67 | 5.43 | 3 | 25.358099999999997 | 76.0743 | 23±4 °C | NMT 60% | AHU |

RECIRCULATION | 2 | D | 35 | Y | NA | NA | NA | NA | NA | NA | NA | Epoxy | 250
| Washing | 3.11 | 5.43 | 3 | 16.8873 | 50.6619 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 2 | D | 35 | Y | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 250
| Air Lock | 3.6 | 2 | 3 | 7.2 | 21.6 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 2 | D | 35 | Y | NA | NA | NA | NA | NA | NA | NA | Epoxy | 250
| Sterilization | 3.6 | 5.43 | 3 | 19.548 | 58.64399999999999 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 2 | D | 40 | Y | NA | NA | NA | Y | Y | NA | NA | NA | Epoxy | 350
| Cool Zone | 3.25 | 7.5 | 3 | 24.375 | 73.125 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 1 | C | 60 | Y | NA | NA | Y | Y | Y | NA | NA | NA | Epoxy | 350
| Stage 10 T / Stage 13 E | 7.18 | 13.5 | 3 | 96.92999999999999 | 290.78999999999996 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 2 | C | 50 | Y | NA | NA | NA | NA | Y | Y | NA | NA | NA | Epoxy | 500
| | 5.33 | 2.05 | 3 | 10.926499999999999 | 32.7795 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 2 | C | 50 | Y | NA | NA | NA | Y | Y | NA | NA | NA | Epoxy | 500
| Stage 9 N | 7.25 | 15.6 | 3 | 113.1 | 339.29999999999995 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 2 | C | 50 | Y | NA | NA | NA | Y | Y | NA | NA | NA | Epoxy | 500
| Stage 6 (I/ Stage 11 F) | 15 | 18 | 3 | 270 | 810 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 2 | C | 50 | Y | NA | NA | NA | Y | Y | NA | NA | NA | Epoxy | 500
| Column packing and skids | 7.64 | 8.86 | 3 | 67.6904 | 203.07119999999998 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 2 | C | 50 | Y | NA | NA | NA | Y | Y | NA | NA | NA | Epoxy | 250
| Waste Out | 1.5 | 3.54 | 3 | 5.3100000000000005 | 15.930000000000001 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 2 | D | 30 | Y | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 250
| Air lock | 1.5 | 4.42 | 3 | 6.63 | 19.89 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 2 | D | 30 | Y | NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| Decontamination | 7.4 | 5 | 3 | 37 | 111 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 2 | D | 35 | Y | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 250
| Washing | 7.4 | 3 | 3 | 22.200000000000003 | 66.60000000000001 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 2 | D | 35 | Y | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 250
| Air Lock | 1.5 | 6.85 | 3 | 10.274999999999999 | 30.824999999999996 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 2 | D | 35 | Y | NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| Sterilization | 7.4 | 6.85 | 3 | 50.69 | 152.07 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 2 | D | 40 | Y | NA | NA | NA | Y | Y | NA | NA | NA | Epoxy | 350
| Air lock | 2.1 | 4.76 | 3 | 9.996 | 29.988 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 2 | D | 30 | Y | NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| Gown room | 3.34 | 4.76 | 3 | 15.898399999999999 | 47.6952 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 2 | C | 45 | Y | NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| Air lock | 3.34 | 4.76 | 3 | 15.898399999999999 | 47.6952 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 2 | C | 50 | Y | NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| Corridor | 3.5 | 55.7 | 3 | 194.95000000000002 | 584.85 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 2 | C | 55 | Y | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 250
| Air lock | 3.34 | 4.76 | 3 | 15.898399999999999 | 47.6952 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 2 | C | 50 | Y | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| De-gown | 3.34 | 4.76 | 3 | 15.898399999999999 | 47.6952 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 2 | C | 45 | Y | NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| Air lock | 2.1 | 4.76 | 3 | 9.996 | 29.988 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 2 | D | 30 | Y | NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| CIP/SIP | 9 | 5 | 3 | 45 | 135 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 2 | C | 60 | Y | NA | NA | NA | NA | Y | NA | NA | NA | Epoxy | 350
| Cool Zone | 9 | 5.53 | 3 | 49.77 | 149.31 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 1 | C | 60 | Y | NA | NA | Y | Y | Y | NA | NA | NA | Epoxy | 350
| Buffer Preparation | 9 | 6 | 3 | 54 | 162 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 2 | C | 60 | Y | NA | NA | NA | NA | Y | NA | NA | NA | Epoxy | 350
| Tank storage | 9 | 8.58 | 3 | 77.22 | 231.66 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 2 | C | 65 | Y | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 350
| Plasma Day Store | 9 | 10.7 | 3 | 96.3 | 288.9 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 1 | C | 50 | Y | NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 350
| | 2.66 | 2.29 | 3 | 6.0914 | 18.2742 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 1 | C | 50 | Y | NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 350
| Janitor Room | 3.5 | 5.4 | 3 | 18.900000000000002 | 56.7 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | NA | C | 50 | Y | NA | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 250
| Corridor | 36.2 | 2.5 | 3 | 90.5 | 271.5 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 2 | C | 65 | Y | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 350
| Plasma Thawing | 7.64 | 3.61 | 3 | 27.580399999999997 | 82.74119999999999 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 2 | C | 55 | Y | NA | NA | NA | NA | Y | NA | NA | NA | Epoxy | 350
| Bag cutting & Pooling | 7.64 | 9.8 | 3 | 74.872 | 224.61599999999999 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 2 | C | 55 | Y | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 350
| Stage 1 (F) / Stage 12 € | 10 | 10.6 | 3 | 106 | 318 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 2 | C | 55 | Y | NA | NA | Y | Y | Y | NA | NA | NA | Epoxy | 350
| Stage 2 (F) | 17.7 | 10.5 | 3 | 185.85 | 557.55 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 2 | C | 55 | Y | NA | NA | NA | Y | Y | NA | NA | NA | Epoxy | 350
| Stage 3 (F) | 12.4 | 10.5 | 3 | 130.20000000000002 | 390.6 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 2 | C | 55 | Y | NA | NA | NA | Y | Y | NA | NA | NA | Epoxy | 350
| Buffer systems | 9 | 6 | 3 | 54 | 162 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 2 | C | 60 | Y | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 350
| Column packing and Skid | 7.64 | 8.86 | 3 | 67.6904 | 203.07119999999998 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 2 | C | 60 | Y | NA | NA | NA | NA | Y | Y | NA | NA | NA | Epoxy | 350
| Pre-Virus Controlled Corridor | 3.5 | 32.1 | 3 | 112.35000000000001 | 337.05 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 2 | C | 55 | Y | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 350
| Stage 8 N | 5.84 | 10.5 | 3 | 61.32 | 183.96 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 2 | C | 45 | Y | NA | NA | NA | Y | Y | NA | NA | NA | Epoxy | 500
| Stage 5 (I) | 10 | 19 | 3 | 190 | 570 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 2 | C | 45 | Y | NA | NA | NA | Y | Y | NA | NA | NA | Epoxy | 500
| Stage 7A | 14.7 | 16.1 | 3 | 236.67000000000002 | 710.01 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 2 | C | 45 | Y | NA | NA | NA | Y | Y | NA | NA | NA | Epoxy | 500
| Stage 4 (F) | 14.7 | 15.8 | 3 | 232.26 | 696.78 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 2 | C | 45 | Y | NA | NA | NA | Y | Y | NA | NA | NA | Epoxy | 500
| Corridor | 53.9 | 3.17 | 3 | 170.863 | 512.58899999999999 | NA | NA | AHU | RECIRCULATION | 2 | CNC | NA | Y | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 250
| RM Receipt De-Dusting Loading side | 3.54 | 2.77 | 3 | 9.8058 | 29.4174 | NA | NA | AHU | RECIRCULATION | NA | CNC | NA | Y | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 250
| De-Dusting unloading side | 3.54 | 2.6 | 3 | 9.204 | 27.612000000000002 | NA | NA | AHU | RECIRCULATION | 1 | CNC | NA | Y | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 250
| RM quarantine | 6.85 | 6.55 | 3 | 44.8675 | 134.6025 | NA | NA | AHU | RECIRCULATION | 2 | CNC | NA | Y

| NA | NA | NA | NA | Y | NA | NA | NA | Epoxy | 250
| Air lock-1 | 2.13 | 1.21 | 3 | 2.5772999999999997 | 7.7318999999999996 | NA | NA | AHU | RECIRCULATION
| 2 | CNC | NA | Y | NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| Change Room | 2.13 | 1.8 | 3 | 3.834 | 11.502 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 2 | D | 15 |
Y | NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 350
| RM Sampling | 3.26 | 2.31 | 3 | 7.5306 | 22.5918 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 2 | D | 20
| Y | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 500
| | 1.07 | 3.11 | 3 | 3.3277 | 9.9831 | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA |
NA | NA | NA | Epoxy | NA
| Approved RM | 10.1 | 7.54 | 3 | 76.154 | 228.462 | NA | NA | AHU | RECIRCULATION | 1 | CNC | NA | Y |
NA | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 350
| | 5.15 | 4.49 | 3 | 23.123500000000003 | 69.3705 | NA | NA | AHU | RECIRCULATION | | NA | NA | NA | NA
| NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | NA
| Air lock -1 | 1.5 | 2.18 | 3 | 3.2700000000000005 | 9.810000000000002 | NA | NA | AHU | RECIRCULATION |
2 | CNC | NA | Y | NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| Change Room | 1.5 | 2.18 | 3 | 3.2700000000000005 | 9.810000000000002 | 23±4 °C | NMT 60% | AHU |
RECIRCULATION | 2 | D | 15 | Y | NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| Dispensing Room | 3.4 | 4.44 | 3 | 15.096 | 45.288 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 2 | D |
20 | Y | NA | NA | Y | Y | Y | NA | Y | NA | Epoxy | 500
| Plasma receipt room | 2.55 | 4.95 | 3 | 12.622499999999999 | 37.867499999999999 | NA | NA | AHU |
RECIRCULATION | 1 | CNC | NA | Y | NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| Plasma quarantine room | 5.15 | 2.72 | 3 | 14.008000000000003 | 42.02400000000001 | NA | NA | AHU |
RECIRCULATION | 2 | CNC | NA | Y | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 250
| | 11.6 | 4.3 | 3 | 49.87999999999995 | 149.64 | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA |
NA | NA | NA | NA | NA | Epoxy | NA
| Serology Testing | 3 | 4.95 | 3 | 14.850000000000001 | 44.550000000000004 | NA | NA | AHU |
RECIRCULATION | 2 | CNC | NA | Y | NA | NA | NA | NA | Y | NA | NA | NA | Epoxy | 350
| Air lock-1 | 1.87 | 1.4 | 3 | 2.618 | 7.853999999999999 | NA | NA | AHU | RECIRCULATION | 2 | CNC | NA
| Y | NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| Change Room | 1.87 | 1.4 | 3 | 2.618 | 7.853999999999999 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 2
| D | 15 | Y | NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| Plasma Sampling | 1.76 | 2.9 | 3 | 5.104 | 15.312000000000001 | 23±4 °C | NMT 60% | AHU | RECIRCULATION
| 2 | D | 20 | Y | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 500
| | 3.68 | 2.05 | 3 | 7.544 | 22.631999999999998 | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA |
NA | NA | NA | NA | NA | Epoxy | NA
| Reject Room | 2.26 | 4.95 | 3 | 11.187 | 33.561 | NA | NA | AHU | RECIRCULATION | NA | CNC | NA | Y |
NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 250
| Bag Bar Coding Mini Pools | 2.6 | 2.67 | 3 | 6.942 | 20.826 | NA | NA | AHU | RECIRCULATION | 1 | CNC |
NA | Y | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 250
| Washing area | 2.3 | 2.67 | 3 | 6.140999999999999 | 18.423 | NA | NA | AHU | RECIRCULATION | 1 | CNC |
NA | Y | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 250
| Janitor Room | 1.45 | 2.67 | 3 | 3.8714999999999997 | 11.6145 | NA | NA | AHU | RECIRCULATION | NA |
CNC | NA | | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 250
| Corridor | 28 | 1.8 | 3 | 50.4 | 151.2 | NA | NA | AHU | RECIRCULATION | 2 | CNC | NA | Y | NA | NA |
NA | NA | Y | NA | NA | Y | Epoxy | 250
| Corridor | 1.8 | 12 | 3 | 21.6 | 64.80000000000001 | NA | NA | AHU | RECIRCULATION | 2 | CNC | NA | Y |
NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 250
| Plasma Storage -1000L | 9.1 | 15.2 | 3 | 138.32 | 414.96 | NA | NA | AHU | RECIRCULATION | NA | CNC |
NA | Y | NA | NA | NA | NA | Y | NA | NA | NA | NA | NA | Epoxy | 350
| Plasma Storage -1000L | 10.5 | 15.2 | 3 | 159.6 | 478.79999999999995 | NA | NA | AHU | RECIRCULATION |
NA | CNC | NA | Y | NA | NA | NA | Y | NA | NA | NA | NA | Epoxy | 350
| Plasma Storage -1000L | 9.36 | 15.2 | 3 | 142.272 | 426.816 | NA | NA | AHU | RECIRCULATION | NA | CNC
| NA | Y | NA | NA | NA | Y | NA | NA | NA | NA | Epoxy | 350
| Plasma Storage -1000L | 10.3 | 15.2 | 3 | 156.56 | 469.68 | NA | NA | AHU | RECIRCULATION | NA | CNC |
NA | Y | NA | NA | NA | Y | NA | NA | NA | NA | Epoxy | 350
| Store change Room | 2.95 | 2.88 | 3 | 8.496 | 25.488 | NA | NA | AHU | RECIRCULATION | 2 | CNC | NA | Y
| NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 250
| Store office | 2.23 | 6.5 | 3 | 14.495 | 43.485 | NA | NA | AHU | RECIRCULATION | 2 | CNC | NA | Y | NA
| NA | NA | Y | Y | NA | NA | NA | NA | Epoxy | 350
| FG dispatch/FG In | 12.9 | 6.49 | 3 | 83.721 | 251.163 | NA | NA | AHU | RECIRCULATION | 2 | CNC | NA |
Y | NA | NA | NA | NA | NA | NA | NA | NA | Kota Stone | 350
| | 4.7 | 4.09 | 3 | 19.223 | 57.669 | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA |
NA | NA | NA | NA | NA
| | 3 | 3.6 | 3 | 10.8 | 32.400000000000006 | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA |
NA | NA | NA | NA | NA
| Recalled goods | 2.3 | 2.35 | 3 | 5.404999999999999 | 16.214999999999996 | NA | NA | AHU |
RECIRCULATION | NA | CNC | NA | Y | NA | NA | NA | NA | NA | NA | NA | NA | Kota Stone | 350
| reject room | 2.3 | 2.35 | 3 | 5.404999999999999 | 16.214999999999996 | NA | NA | AHU | RECIRCULATION |
NA | CNC | NA | Y | NA | NA | NA | NA | NA | NA | NA | NA | NA | Kota Stone | 250
| Ice Pack | 2.5 | 7.34 | 3 | 18.35 | 55.050000000000004 | NA | NA | AHU | RECIRCULATION | NA | CNC | NA
| Y | NA | NA | NA | NA | NA | NA | NA | NA | Kota Stone | 250
| Cold room-1 | 5.04 | 7.34 | 3 | 36.9936 | 110.9808 | 2-8 °C | NA | Cold room unit | NA | NA | NA | NA |
NA | NA | NA | NA | Y | Hooter | NA | NA | NA | Epoxy | 350
| Cold room-2 | 5.02 | 7.34 | 3 | 36.846799999999995 | 110.54039999999998 | 2-8 °C | NA | Cold room unit
| NA | NA | NA | NA | NA | NA | NA | NA | Y | Hooter | NA | NA | NA | Epoxy | 350
| Cold room-3 | 5 | 7.34 | 3 | 36.7 | 110.10000000000001 | 2-8 °C | NA | Cold room unit | NA | NA | NA |
NA | NA | NA | NA | NA | Y | Hooter | NA | NA | NA | Epoxy | 350
| Cold room-4 | 5.04 | 7.34 | 3 | 36.9936 | 110.9808 | 2-8 °C | NA | Cold room unit | NA | NA | NA | NA |
NA | NA | NA | NA | Y | Hooter | NA | NA | NA | Epoxy | 350
[SHEET: FF_Fill & Finish_RDS]
| | Room Details | | | | | | | | | Fire Detection & Protection System | | | | | | | | |
Room No | Room Name | Length (mtr) | Width(mtr) | Height | Area (M²) | Volume | Temp.(oC) | RH(%) | HVAC
System | HVAC Type | Occupancy | Grade | Differential P | Fire Alarm System | Fire Sprinkler System |
Fire Extinguisher System | GPS Clock | LAN Port | Telephone Port | Door Interlock | Eye Washer | Body
Washer | Floor Finish | Electrical Lux level
| Corridor | 25.2 | 2.85 | 3 | 71.82 | 215.45999999999998 | NA | NA | AHU | RECIRCULATION | NA | CNC | NA
| Y | NA | Y | Y | Y | NA | NA | NA | NA | Epoxy | 250
| Bulk Staging cold Room | 21 | 8.07 | 3 | 169.47 | 508.40999999999997 | NA | NA | AHU | RECIRCULATION |
NA | CNC | NA | Y | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 350
| Corridor | 25.5 | 1.8 | 3 | 45.9 | 137.7 | NA | NA | AHU | RECIRCULATION | NA | CNC | NA | Y | NA | Y |
NA | NA | NA | NA | NA | NA | NA | Epoxy | 250
| De-cartoning room | 3.89 | 3.32 | 3 | 12.9148 | 38.7444 | NA | NA | AHU | RECIRCULATION | 5 | CNC | NA

| Y | NA | NA | NA | NA | Y | NA | NA | NA | Epoxy | 250
| Pal-1 | 2.27 | 1.64 | 3 | 3.7228 | 11.1684 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 2 | D | 15 | Y |
NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| Pal-1 | 2.27 | 1.64 | 3 | 3.7228 | 11.1684 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 2 | D | 20 | Y |
NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| Vial washing | 6.42 | 9.65 | 3 | 61.953 | 185.859 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 1 | D |
25 | Y | NA | NA | NA | NA | Y | NA | NA | NA | Epoxy | 350
| Janitor Room | 2.32 | 1 | 3 | 2.32 | 6.959999999999999 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | NA |
D | 20 | Y | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 250
| Cold room | 4.19 | 3.27 | 3 | 13.701300000000002 | 41.1039 | 2-8 °C | NA | Cold room Unit | NA | NA |
NA | NA | Y | NA | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | NA
| | 2.26 | 2.53 | 3 | 5.717799999999999 | 17.153399999999998 | 2-8 °C | NA | Cold room Unit | NA | NA |
NA | NA | Y | NA | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | NA
| Primary change room male | 5.31 | 5.81 | 3 | 30.851099999999995 | 92.55329999999998 | 23±4 °C | NMT 60%
| AHU | RECIRCULATION | 2 | D | 15 | Y | NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| Primary change room female | 4.43 | 3.93 | 3 | 17.4099 | 52.2297 | 23±4 °C | NMT 60% | AHU |
RECIRCULATION | 2 | D | 15 | Y | NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| | 3.63 | 1.92 | 3 | 6.9696 | 20.9088 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 2 | D | 15 | Y | NA |
NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| Primary change room visitor | 4.51 | 5.78 | 3 | 26.0678 | 78.20339999999999 | 23±4 °C | NMT 60% | AHU |
RECIRCULATION | 2 | D | 15 | Y | NA | NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| Mal entry | 2.85 | 5.78 | 3 | 16.473000000000003 | 49.419000000000001 | 23±4 °C | NMT 60% | AHU |
RECIRCULATION | NA | NA | 10 | Y | NA | NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| Corridor | 16.7 | 1.8 | 3 | 30.06 | 90.17999999999999 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 2 | D |
20 | Y | NA | Y | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 250
| Corridor | 34.1 | 1.8 | 3 | 61.38 | 184.14000000000001 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 2 |
D | 25 | Y | NA | Y | Y | Y | NA | NA | NA | NA | NA | Epoxy | 250
| PAL | 2.84 | 1.45 | 3 | 4.117999999999999 | 12.354 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 2 | D |
30 | Y | NA | NA | NA | NA | NA | Y | NA | NA | NA | Epoxy | 250
| MAL | 2.84 | 1.45 | 3 | 4.117999999999999 | 12.354 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | NA | D |
30 | Y | NA | NA | NA | NA | NA | Y | NA | NA | NA | Epoxy | 250
| Material prep. & wash | 8.15 | 3.07 | 3 | 25.0205 | 75.0615 | 23±4 °C | NMT 60% | AHU | RECIRCULATION |
2 | D | 35 | Y | NA | NA | Y | Y | Y | NA | Y | NA | Y | NA | Epoxy | 350
| | 5.26 | 1.64 | 3 | 8.626399999999999 | 25.879199999999997 | 23±4 °C | NA | NA | NA | | NA | NA | NA |
NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA |
| | 2.21 | 1.46 | 3 | 3.2266 | 9.6798 | 23±4 °C | NA | NA | NA | | NA | NA | NA | NA | NA | NA | NA | NA | NA |
NA | NA | NA | NA | NA |
| Disinfectant | 8.41 | 2.62 | 3 | 22.034200000000002 | 66.102600000000001 | 23±4 °C | NMT 60% | AHU |
RECIRCULATION | | D | 40 | Y | NA | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 350
| MAL | 2.04 | 1.47 | 3 | 2.9988 | 8.9964000000000001 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | NA | D |
40 | Y | NA | NA | NA | NA | NA | Y | NA | NA | NA | Epoxy | 250
| PAL | 2.04 | 2.29 | 3 | 4.6716 | 14.0148 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 2 | D | 40 | Y |
NA | NA | NA | NA | NA | Y | NA | NA | NA | Epoxy | 250
| Unloading & packing area | 2.94 | 5.13 | 3 | 15.0822 | 45.2466 | 23±4 °C | NMT 60% | AHU |
RECIRCULATION | 1 | D | 45 | Y | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 350
| PAL | 1.41 | 1.64 | 3 | 2.3124 | 6.937199999999999 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 2 | D |
30 | Y | NA | NA | NA | NA | NA | Y | NA | NA | NA | Epoxy | 250
| MaL | 1.45 | 1.64 | 3 | 2.377999999999999 | 7.133999999999999 | 23±4 °C | NMT 60% | AHU |
RECIRCULATION | NA | D | 30 | Y | NA | NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| Laundry area | 5.1 | 1.57 | 3 | 8.007 | 24.021 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 1 | D | 35 |
Y | NA | NA | NA | NA | Y | NA | NA | NA | NA | Epoxy | 250
| | 2.98 | 3.88 | 3 | 11.5624 | 34.6872000000000004 | 23±4 °C | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA |
NA | NA | NA | NA | NA | NA | NA | NA | NA |
| PAL-1 | 4.54 | 1.87 | 3 | 8.4898 | 25.4694 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 2 | D | 30 | Y |
NA | NA | NA | NA | NA | NA | Y | NA | NA | NA | Epoxy | 250
| PAL-2 | 4.4 | 1.87 | 3 | 8.2280000000000002 | 24.6840000000000005 | 23±4 °C | NMT 60% | AHU |
RECIRCULATION | | C | 45 | Y | NA | NA | NA | NA | NA | Y | NA | NA | NA | Epoxy | 250
| PAL-3 | 4.23 | 3.82 | 3 | 16.1586 | 48.4758 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | | B | 60 | Y |
NA | NA | NA | NA | NA | NA | Y | NA | NA | NA | Epoxy | 250
| PAL-4 | 3.2 | 3.82 | 3 | 12.224 | 36.672 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | | B | 65 | Y | NA
| NA | NA | NA | NA | Y | NA | NA | NA | Epoxy | 250
| Exit-1 | 4.4 | 1.87 | 3 | 8.2280000000000002 | 24.6840000000000005 | 23±4 °C | NMT 60% | AHU |
RECIRCULATION | 2 | B | 65 | Y | NA | NA | NA | NA | NA | Y | NA | NA | NA | Epoxy | 250
| Exit-2 | 4.54 | 1.87 | 3 | 8.4898 | 25.4694 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | | C | 50 | Y |
NA | NA | NA | NA | NA | Y | NA | NA | NA | Epoxy | 250
| Sterile Corridor | 15.5 | 1.8 | 3 | 27.9000000000000002 | 83.7 | 23±4 °C | NMT 60% | AHU | RECIRCULATION
| 2 | B | 70 | Y | NA | NA | Y | Y | NA | NA | NA | NA | NA | Epoxy | 300
| | 8.49 | 1.8 | 3 | 15.282 | 45.8460000000000004 | 23±4 °C | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA |
NA | NA | NA | NA | NA | NA | NA | NA | NA |
| Formulation | 8.41 | 3.59 | 3 | 30.1919 | 90.5757 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 2 | B |
75 | Y | NA | NA | Y | Y | Y | NA | NA | NA | NA | Epoxy | 350
| Clean JC | 1.88 | 2.48 | 3 | 4.6624 | 13.9872 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | NA | | Y |
NA | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 250
| Unload | 8.41 | 3.4 | 3 | 28.594 | 85.782000000000001 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 1 | B
| 75 | Y | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 350
| Fill Finish | 6.42 | 8.3 | 3 | 53.286 | 159.858 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 2 | B | 75
| Y | NA | NA | Y | Y | Y | NA | NA | NA | NA | Epoxy | 500
| Capping | 8.29 | 4.04 | 3 | 33.4916 | 100.47479999999999 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 1
| B | 80 | Y | NA | NA | Y | Y | Y | NA | NA | NA | NA | NA | Epoxy | 500
| Corridor | 1.8 | 117.9 | 3 | 212.220000000000003 | 636.66000000000001 | 23±4 °C | NMT 60% | AHU |
RECIRCULATION | 2 | D | 25 | Y | NA | Y | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 250
| Lyo loading | 8.29 | 3.55 | 3 | 29.429499999999994 | 88.28849999999998 | 23±4 °C | NMT 60% | AHU |
RECIRCULATION | 2 | D | 30 | Y | NA | NA | Y | Y | Y | NA | NA | NA | NA | Epoxy | 300
| Lyo service area | 10.9 | 3.55 | 3 | 38.695 | 116.085000000000001 | 23±4 °C | NMT 60% | AHU |
RECIRCULATION | 2 | D | 30 | Y | NA | NA | NA | NA | Y | NA | NA | NA | NA | Epoxy | 300
| PPM Store | 15.2 | 12 | 3 | 182.39999999999999 | 547.19999999999999 | NA | NA | AHU | Exhaust | NA | CNC
| NA | Y | Y | NA | NA | NA | Y | NA | NA | NA | NA | Epoxy | 350
| | 11.8 | 3.03 | 3 | 35.754 | 107.262 | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA |
NA | NA | NA | NA |
| Approved Rubber bung Store | 3.36 | 2.97 | 3 | 9.9792 | 29.9376000000000003 | NA | NA | AHU | Exhaust |
NA | CNC | NA | Y | Y | Y | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 350
| Corridor | 2.8 | 27.9 | 3 | 78.11999999999999 | 234.35999999999996 | NA | NA | AHU | Exhaust | NA | CNC

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| NA | Y | Y | Y | NA | Y | NA | Vitrified Tiles | 350
| Balance room | 4.82 | 3.37 | 3 | 16.2434 | 48.7302 | NA | NA | AHU | RECIRCULATION | NA | CNC | NA | Y
| NA | NA | NA | NA | NA | NA | NA | NA | Vitrified Tiles | 350
| GeneralFTIR | 4.38 | 3.37 | 3 | 14.7606 | 44.281800000000004 | NA | NA | AHU | RECIRCULATION | NA | CNC
| NA | Y | NA | NA | NA | NA | NA | NA | NA | Vitrified Tiles | 350
| Instrument room-II | 15.3 | 5.43 | 3 | 83.079 | 249.23699999999997 | NA | NA | AHU | RECIRCULATION | NA
| CNC | NA | Y | NA | NA | Y | Y | Y | NA | Y | NA | Vitrified Tiles | 500
| Reference standard | 5.87 | 2.98 | 3 | 17.4926 | 52.4778 | NA | NA | AHU | RECIRCULATION | NA | CNC |
NA | Y | NA | NA | Y | Y | Y | NA | Y | NA | Vitrified Tiles | 500
| GC room | 4.83 | 2.98 | 3 | 14.3934 | 43.1802 | NA | NA | AHU | RECIRCULATION | NA | CNC | NA | Y | NA
| NA | Y | Y | Y | NA | Y | NA | Vitrified Tiles | 500
| TOC/ LPC | 4.44 | 2.98 | 3 | 13.231200000000001 | 39.6936 | NA | NA | AHU | RECIRCULATION | NA | CNC |
NA | Y | NA | NA | NA | Y | NA | NA | NA | NA | Vitrified Tiles | 350
| Stability room | 15.3 | 6.04 | 3 | 92.412 | 277.236 | NA | NA | AHU | RECIRCULATION | NA | CNC | NA | Y
| NA | NA | NA | Y | NA | NA | NA | NA | Vitrified Tiles | 350
| Hot Room | 5.9 | 5.46 | 3 | 32.214 | 96.642 | NA | NA | AHU | RECIRCULATION | NA | CNC | NA | Y | NA |
NA | NA | NA | NA | NA | NA | NA | NA | Vitrified Tiles | 350
| Wash area | 5.9 | 2.76 | 3 | 16.284 | 48.852 | NA | NA | AHU | RECIRCULATION | NA | CNC | NA | Y | NA |
NA | NA | NA | NA | NA | NA | NA | NA | Vitrified Tiles | 350
| Canteen | 12.4 | 10.6 | 3 | 131.44 | 394.32 | NA | NA | AHU | RECIRCULATION | NA | CNC | NA | Y | NA |
NA | NA | NA | NA | NA | NA | NA | NA | Vitrified Tiles | 350
| | 5.98 | 6.12 | 3 | 36.5976 | 109.7928 | NA | NA | AHU | RECIRCULATION | NA | CNC | NA | Y | NA | NA |
NA | NA | NA | NA | NA | NA | NA | NA | Vitrified Tiles | 350
| | 2.35 | 8.03 | 3 | 18.8705 | 56.6115 | NA | NA | AHU | RECIRCULATION | NA | CNC | NA | Y | NA | NA |
NA | NA | NA | NA | NA | NA | NA | NA | Vitrified Tiles | 350
| Training Hall | 14.8 | 19 | 3 | 281.2 | 843.5999999999999 | NA | NA | AHU | RECIRCULATION | NA | CNC |
NA | Y | NA | NA | NA | Y | Y | NA | NA | NA | NA | Vitrified Tiles | 500
| Corridor | 1.8 | 117.9 | 3 | 212.22000000000003 | 636.6600000000001 | NA | NA | AHU | RECIRCULATION |
NA | CNC | NA | Y | NA | NA | NA | NA | NA | NA | NA | NA | Vitrified Tiles | 350
| Ladies change room | 4.78 | 3.12 | 3 | 14.9136 | 44.7408 | NA | NA | AHU | RECIRCULATION | NA | CNC |
NA | Y | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | Vitrified Tiles | 250
| Gents change room | 4.78 | 2.83 | 3 | 13.527400000000002 | 40.58220000000001 | NA | NA | AHU |
RECIRCULATION | NA | CNC | NA | Y | NA | NA | NA | NA | NA | NA | NA | NA | Vitrified Tiles | 250
| Corridor | 2.5 | 8.31 | 3 | 20.775000000000002 | 62.325 | NA | NA | AHU | RECIRCULATION | NA | CNC | NA
| Y | NA | Y | Y | Y | NA | NA | NA | NA | NA | NA | Vitrified Tiles | 300250
| BET room | 4.78 | 2.83 | 3 | 13.527400000000002 | 40.58220000000001 | NA | NA | AHU | RECIRCULATION |
NA | CNC | NA | Y | NA | NA | NA | NA | NA | NA | NA | NA | Vitrified Tiles | 350
| Media Store | 4.78 | 3.54 | 3 | 16.921200000000002 | 50.76360000000001 | NA | NA | AHU | RECIRCULATION
| NA | CNC | NA | Y | NA | NA | NA | NA | NA | NA | NA | NA | NA | Vitrified Tiles | 350
| Air Lock | 2.5 | 4.24 | 3 | 10.600000000000001 | 31.800000000000004 | 23±4 °C | NMT 60% | AHU |
RECIRCULATION | 2 | D | 15 | Y | NA | NA | Y | Y | Y | NA | Y | NA | NA | Epoxy | 250
| Media Prep./ Storage room | 7.38 | 3.88 | 3 | 28.6344 | 85.9032 | 23±4 °C | NMT 60% | AHU |
RECIRCULATION | 2 | D | 20 | Y | NA | NA | NA | NA | Y | NA | NA | NA | Epoxy | 350
| Praparation and Sterility area | 8.11 | 5.91 | 3 | 47.930099999999996 | 143.7903 | 23±4 °C | NMT 60% |
AHU | RECIRCULATION | 2 | D | 20 | Y | NA | NA | Y | Y | Y | NA | NA | NA | Epoxy | 350
| CH-1 | 4.81 | 2.02 | 3 | 9.716199999999999 | 29.148599999999995 | 23±4 °C | NMT 60% | AHU |
RECIRCULATION | 2 | D | 15 | Y | NA | NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| CH-2 | 2.73 | 2.02 | 3 | 5.5146 | 16.543799999999997 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | | C |
30 | Y | NA | NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| CH-3 | 2.02 | 2.02 | 3 | 4.0804 | 12.2412 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | | B | 45 | Y | NA
| NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| Air Lock | 5.36 | 2.02 | 3 | 10.827200000000001 | 32.4816 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 2
| B | 40 | Y | NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| Exit-1 | 3.49 | 2.02 | 3 | 7.0498 | 21.1494 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | | C | 25 | Y | Y
NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| Exit-2 | 2.63 | 2.02 | 3 | 5.3126 | 15.9378 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | | D | 10 | Y | Y
NA | NA | NA | NA | NA | Y | NA | NA | Epoxy | 250
| Sterile Passage | 3.53 | 2.02 | 3 | 7.130599999999999 | 21.391799999999996 | 23±4 °C | NMT 60% | AHU |
RECIRCULATION | 2 | B | 50 | Y | NA | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 250
| Sterility room | 4.65 | 3.78 | 3 | 17.577 | 52.73100000000001 | 23±4 °C | NMT 60% | AHU | RECIRCULATION
| 2 | B | 60 | Y | NA | NA | Y | Y | Y | NA | NA | NA | Epoxy | 500
| Cooling zone | 3.38 | 3.78 | 3 | 12.776399999999999 | 38.3292 | 23±4 °C | NMT 60% | AHU | RECIRCULATION
| 1 | B | 55 | Y | NA | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 250
| | 1.83 | 1.91 | 3 | 3.4953 | 10.485899999999999 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 1 | B | 55
| Y | NA | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 250
| Wash area | 5.221 | 2.02 | 3 | 10.54642 | 31.63926 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 1 | D |
25 | Y | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 250
| Air Lock-1 | 5.36 | 2.02 | 3 | 10.827200000000001 | 32.4816 | 23±4 °C | NMT 60% | AHU | RECIRCULATION |
2 | D | 15 | Y | NA | NA | NA | NA | NA | NA | Y | NA | NA | NA | Epoxy | 250
| CH-1 | 6.04 | 2.02 | 3 | 12.200800000000001 | 36.6024 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | | D |
25 | Y | NA | NA | NA | NA | NA | Y | NA | NA | NA | Epoxy | 250
| MLT | 4.65 | 3.77 | 3 | 17.5305 | 52.591499999999996 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 2 | D
| 15 | Y | NA | NA | NA | NA | Y | NA | NA | NA | NA | Epoxy | 350
| Corridor | 1.8 | 8.5 | 3 | 15.3 | 45.900000000000006 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 2 | D
| 25 | Y | NA | NA | NA | NA | NA | Y | NA | NA | NA | Epoxy | 350
| Micro Office | 5.17 | 4.24 | 3 | 21.9208 | 65.7624 | NA | NA | AHU | RECIRCULATION | NA | CNC | NA | Y
| NA | NA | NA | Y | Y | NA | NA | NA | NA | Vitrified Tiles | 350
| Micro documentation room | 3.49 | 4.24 | 3 | 14.797600000000001 | 44.3928 | NA | NA | AHU |
RECIRCULATION | NA | CNC | NA | Y | NA | NA | NA | NA | Y | Y | NA | NA | NA | NA | Vitrified Tiles | 350
| Micro sample receipt | 2.63 | 4.24 | 3 | 11.1512 | 33.453599999999994 | NA | NA | AHU | RECIRCULATION |
NA | CNC | NA | Y | NA | NA | NA | Y | Y | NA | NA | NA | NA | Vitrified Tiles | 350
| Change Room | 2.39 | 2.86 | 3 | 6.8354 | 20.5062 | 23±4 °C | NMT 60% | AHU | RECIRCULATION | 2 | D | 15
| Y | NA | NA | NA | NA | NA | NA | Y | NA | NA | NA | Epoxy | 250
| Exit | 2.15 | 2.86 | 3 | 6.148999999999999 | 18.446999999999996 | 23±4 °C | NMT 60% | AHU |
RECIRCULATION | | D | 25 | Y | NA | NA | NA | NA | NA | Y | NA | NA | NA | Epoxy | 250
| Culture Room | 4.65 | 3.78 | 3 | 17.577 | 52.73100000000001 | 23±4 °C | NMT 60% | AHU | RECIRCULATION |
2 | D | 15 | Y | NA | NA | NA | NA | Y | NA | NA | NA | NA | Epoxy | 350
| Scrap out | 5.49 | 2.86 | 3 | 15.7014 | 47.1042 | NA | NA | AHU | RECIRCULATION | NA | CNC | NA | Y |
NA | NA | NA | Y | Y | NA | NA | NA | NA | Vitrified Tiles | 350
| Air Lock | 2.66 | 2.86 | 3 | 7.6076 | 22.8228 | NA | NA | AHU | RECIRCULATION | NA | CNC | NA | Y | NA
| NA | NA | Y | Y | NA | NA | NA | NA | Vitrified Tiles | 350
| Corridor | 2.8 | 7.77 | 3 | 21.755999999999997 | 65.267999999999999 | NA | NA | AHU | RECIRCULATION | NA

| CNC | NA | Y | NA | Y | NA | NA | NA | NA | NA | NA | Vitrified Tiles | 250
| | 3.76 | 3.78 | 3 | 14.212799999999998 | 42.638399999999999 | NA | NA | AHU | RECIRCULATION | NA | CNC |
NA | Y | NA | NA | NA | NA | NA | NA | NA | NA | Vitrified Tiles | NA
| Media fill incubation-1 | 5.33 | 3.78 | 3 | 20.147399999999998 | 60.442199999999999 | NA | NA | AHU |
RECIRCULATION | NA | CNC | NA | Y | NA | NA | NA | NA | NA | NA | NA | Vitrified Tiles | 350
| Media fill incubation-2 | 5.33 | 3.77 | 3 | 20.0941 | 60.2823000000000006 | NA | NA | AHU |
RECIRCULATION | NA | CNC | NA | Y | NA | NA | NA | NA | NA | NA | NA | Epoxy | 350
| Media fill incubation-3 | 4.36 | 3.78 | 3 | 16.4808000000000002 | 49.4424000000000006 | NA | NA | AHU |
RECIRCULATION | NA | CNC | NA | Y | NA | NA | NA | NA | NA | NA | NA | Epoxy | 350
| Media Distribution | 7.34 | 2.02 | 3 | 14.8268 | 44.4804 | NA | NA | AHU | RECIRCULATION | NA | CNC |
NA | Y | NA | NA | NA | NA | NA | NA | NA | NA | NA | Epoxy | 350
| Decontamination | 5.54 | 2.02 | 3 | 11.1908 | 33.5724 | NA | NA | AHU | RECIRCULATION | NA | CNC | NA |
Y | NA | NA | NA | NA | NA | NA | NA | NA | Vitrified Tiles | 350
| Corridor | 34.1 | 1.8 | 3 | 61.38 | 184.140000000000001 | NA | NA | AHU | RECIRCULATION | NA | CNC | NA
| Y | NA | Y | Y | Y | NA | NA | NA | Y | Vitrified Tiles | 250
| QA Office | 18 | 10.8 | 3 | 194.4 | 583.2 | NA | NA | AHU | RECIRCULATION | NA | CNC | NA | Y | NA | NA
| NA | Y | Y | NA | NA | NA | Vitrified Tiles | 350
| Storage | 4.13 | 4.19 | 3 | 17.3047 | 51.9141000000000005 | NA | NA | AHU | RECIRCULATION | NA | CNC |
NA | Y | NA | NA | NA | Y | Y | NA | NA | NA | Vitrified Tiles | 350
| Cabin 1 | 3.39 | 4.14 | 3 | 14.0346 | 42.1038 | NA | NA | AHU | RECIRCULATION | NA | CNC | NA | Y | NA
| NA | NA | Y | Y | NA | NA | NA | NA | Vitrified Tiles | 350
| Cabin 2 | 4.05 | 4.14 | 3 | 16.767 | 50.301 | NA | NA | AHU | RECIRCULATION | NA | CNC | NA | Y | NA |
NA | NA | Y | Y | NA | NA | NA | Vitrified Tiles | 350
| Cabin 3 | 3.43 | 4.14 | 3 | 14.200199999999999 | 42.6006 | NA | NA | AHU | RECIRCULATION | NA | CNC |
NA | Y | NA | NA | NA | Y | Y | NA | NA | NA | Vitrified Tiles | 350
| Cabin 4 | 2.88 | 4.14 | 3 | 11.923199999999998 | 35.7696 | NA | NA | AHU | RECIRCULATION | NA | CNC |
NA | Y | NA | NA | NA | Y | Y | NA | NA | NA | Vitrified Tiles | 350
| Audit | 2.85 | 5.81 | 3 | 16.5585 | 49.6755 | NA | NA | AHU | RECIRCULATION | NA | CNC | NA | Y | NA |
NA | NA | Y | Y | NA | NA | NA | Vitrified Tiles | 350
| QA manager | 2.85 | 4.06 | 3 | 11.571 | 34.713 | NA | NA | AHU | RECIRCULATION | NA | CNC | NA | Y | NA
| NA | NA | Y | Y | NA | NA | NA | Vitrified Tiles | 350
| QA Head | 2.85 | 4.47 | 3 | 12.7395 | 38.2185 | NA | NA | AHU | RECIRCULATION | NA | CNC | NA | Y | NA
| NA | NA | Y | Y | NA | NA | NA | Vitrified Tiles | 350
| QA document | 21 | 9.82 | 3 | 206.22 | 618.66 | NA | NA | AHU | RECIRCULATION | NA | CNC | NA | Y | NA
| NA | NA | Y | Y | NA | NA | NA | Vitrified Tiles | 350
| Corridor | 16.7 | 1.8 | 3 | 30.06 | 90.179999999999999 | NA | NA | AHU | RECIRCULATION | NA | CNC | NA |
Y | NA | NA | NA | NA | NA | NA | NA | NA | Vitrified Tiles | NA
| R&D Entrance | 7.48 | 2.9 | 3 | 21.692 | 65.076 | NA | NA | AHU | RECIRCULATION | NA | CNC | NA | Y |
NA | NA | NA | NA | NA | NA | NA | NA | Vitrified Tiles | 250
| R&D Office | 7.31 | 2.9 | 3 | 21.198999999999998 | 63.596999999999994 | NA | NA | AHU | RECIRCULATION |
NA | CNC | NA | Y | NA | NA | NA | Y | Y | NA | NA | NA | Vitrified Tiles | 350
| Corridor | 14.8 | 1.8 | 3 | 26.64 | 79.92 | NA | NA | AHU | RECIRCULATION | NA | CNC | NA | Y | NA | Y
| NA | NA | NA | NA | NA | NA | NA | Vitrified Tiles | 250
| Corridor | 13.3 | 1.8 | 3 | 23.94 | 71.820000000000001 | NA | NA | AHU | RECIRCULATION | NA | CNC | NA |
Y | NA | NA | NA | Y | Y | NA | NA | NA | Vitrified Tiles | 350
| R&D Lab-1 | 6.47 | 6.61 | 3 | 42.7667 | 128.3001 | NA | NA | AHU | RECIRCULATION | NA | CNC | NA | Y |
NA | NA | NA | Y | Y | NA | NA | NA | Vitrified Tiles | 350
| R&D Lab-2 | 6.47 | 6.61 | 3 | 42.7667 | 128.3001 | NA | NA | AHU | RECIRCULATION | NA | CNC | NA | Y |
NA | NA | NA | Y | Y | NA | NA | NA | Vitrified Tiles | 350
| R&D Lab-3 | 6.47 | 6.61 | 3 | 42.7667 | 128.3001 | NA | NA | AHU | RECIRCULATION | NA | CNC | NA | Y |
NA | NA | NA | Y | Y | NA | NA | NA | Vitrified Tiles | 350
| R&D Lab-4 | 6.47 | 6.61 | 3 | 42.7667 | 128.3001 | NA | NA | AHU | RECIRCULATION | NA | CNC | NA | Y |
NA | NA | NA | Y | Y | NA | NA | NA | Vitrified Tiles | 350

51000
I

H

G

F

E
80000

D

C

B

A

-----6M WIDE ROAD -----
51000
O
D1 D1 D1 D1 D1 D1 D1 D1 D1 D1
GENTS VISITOR FEMALE LADIES TOILET DOCUMENT OFFICE 3
WAITING ROOM LOCKER LOCKER LOCKER LOCKER ROOM 2.65x5.30
5.30x4.00 GENTS TOILET ROOM 5.90x5.20 2.50x5.30
ROOM ROOM
5.90x5.20 D1 D1 D1
4.57x5.30 4.57x5.30 4.57x5.30
JANITOR

D1
D4 D4 D4 D4 D4 D4 D4
D4
ROOM
3.50x1.65

RECEPTION
6D

4.05x2.80 CORRIDOR
34.9x2.80 LIFT
8.00x6.10
8D

4D D4 4D D4 D4
D4

D4

4D

4D

D4

D4

FEMALE PRIMARY FEMALE AIR LOCK AIR LOCK
PRIMARY
VISITOR
VISITOR
CHANGE 1
OFFICE 2 OFFICE 1
N
PRIMARY MALE MALE CHANGE 1 FEMALE VISITOR CHANGE 4.83x5.00 4.83x5.00
CHANGE 1 CHANGE 2.60x5.00
CHANGE 3.00x5.00 3.64x5.00
3.10x5.00 3.10x5.00 3.05x5.00 2.70x5.00
3.80x5.00 2.50x5.00
AIR LOCK
MALE
4.77x5.00

D4

D4 4D D4

6D

DE
CORRIDOR
47.8x2.00
D4
DPB
DPB DPB WASTE OUT
LAF DHS
3 WAY
6D

6D

CHEMICAL DAY STORE
8.10X4.00 COLD ROOM DECONTAMINATION WASHING STERILIZATION
AUTOCLAVE

LAF

BUFFER TANK STORAGE-2 CIP / SIP 4.40x8.10
3.52x5.62 3.11x5.62 4.01x5.62
5.60x8.10
6D

PREPARATION 6.80x8.10
COOL ZONE
4.29x8.10 2.76x8.10 M
4D

4D

DPB D4 D4
DPB
GS+HH

GS+HH

8.30x1.80
D4
A/L GOWN A/L
2.22X3.40 3.47X3.40 2.20X3.40 LENAP LORTNOC LENAP LORTNOC

WASTE A/L A/L
4.13x2.40 4.01x2.40
D4

IN
D4

2.50x2.40
D4

6D 6D 6D D4 D4 D4
D4
4D

CORRIDOR
A/L DE GOWN A/L
DE

2.57x66.0
2.22X3.40 3.47X3.40 2.20X3.40 CORRIDOR
37.0x2.80
4D

4D

D6 D6 D3
LAF
D6 D6
DPB AKTA-1" TFF
4D

4D

NANO
FILTER MH

JANITOR
1.80x2.00
A/L GOWN A/L 1000 L
2.22X4.32 3.47X4.32 2.20X4.32
BULK FILTRATION
LAF

6.90X6.00
D4

500 L
D4

L
LAF

MH

STAGE 9 N
MH

500 L
4D

50 L
MH

STAGE 13 E
D4

LAF SL
MH

5.10X14.3 CHROMO SL
250 L
MH

COLUMN
DE 600 L 600 L
A/L
2.22X3.89
GOWN A/L
500 L
STAGE 10 T/ MH

3.47X3.89 2.20X3.89
STAGE 11 F
MH

MH SL
600 L 6.25X2.05 250 L
4D

4D

200 L
MH
AKTA-1"

STAGE 6 (I) 8.10X12.2
MH

COLUMN PACKING 2000 L 14.7X16.7 250 L MH

250 L
D4

MH

MH

AND SKIDS
MH

CORRIDOR
200 L
1.80X49.6 6.90X10.6 250 L
COOL ZONE 150 L
MH MH

MH
1000 L MH
100 L MH

8.10x5.53 150 L
500 L
MH

300 L
CORRIDOR
MH
2.57X67.9
MH

100 L MH

1000 L
MH

100 L
150 L 300 L
LAF K
DPB LAF
MH

MH

CHROMO MH

AKTA-1"
100 L COLUMN

DHS
AUTOCLAVE CHROMO
D6

COLUMN
500 L 500 L MH
MH MH

250 L 250 L 150 L 100 L
6D

CHROMO
COLUMN
A/L MH MH
MH MH
MH

MH
CLAF

1.50x5.73
STERILIZATION 2000 L
2500 L
2500 L
D6

6.50x5.85
PRE - VIRUS CONTROLLED CORRIDOR
6D

1000 L
D4

TFF

MH

D4
COLUMN PACKING
DPB
AND SKIDS MH

WASHING 100 L
8.00X7.86
AKTA-1"

A/L 6.50x3.00 STAGE 7A
AKTA-1"

MH

1.50x4.42 500 L
D4

300 L 13.6X15.1
D4

MH

STAGE 5 (I) A
D6

DPB J
WASTE IN
4D

500 L
250 L
11.1X18.0
TFF

MH

MH
2.00X28.8

DECONTAMINATION
4D

500 L
250 L
OUT 6.50x5.00 MH
MH
MH

WASTE
D6

1.50x3.54 500 L

DPB STAGE 5(I)B 1500 L 250 L
MH
MH

100 L
8.00X7.20 MH

250 L 250 L 250 L
SL
600 L 500 L
100 L 100 L
MH

MH MH MH 100 L MH
MH

100 L MH MH MH

MH

1500 L
CIP / SIP SL
1000 L 100 L
MH MH

MH MH MH

8.10x5.00
MH

500 L 500 L 1000 L 1000 L 1000 L
MH

MH

MH
6D

MH

CHROMO
50 L COLUMN
GS+HH

GS+HH

DPB 2500 L
LENAP LORTNOC LENAP LORTNOC
WASTE OUT 1500 L
1500 L 1500 L
SL I
MH
250 L 250 L 250 L 250 L
MH

MH MH MH MH
6D

BAG CUTTING &
TANK STORAGE POOLING
MH
MH MH

MH

MH MH
STAGE 4 (F)
150 L 500 L 500 L
8.00X7.80
6D

1000 L
8.10x6.58 1500 L 1500 L 13.6X13.5 MH

500 L
D6

CHROMO
COLUMN

250 L
300 L 300 L STAGE 12 (E) MH

STAGE 1 (F)/
TFF
D6

AKTA-1"

6D
MH MH

CLAF

11.1X8.21
1000 L

2500 L

SL

MH

THAWING PLASMA
2000 L 2000 L
8.00X3.22
6D

BUFFER
AKTA-1"

MH

500 L 250 L 300 L 300 L 300 L
250 L 250 L
PREPARATION
MH MH

CHROMO CHROMO
COLUMN
MH MH MH MH

COLUMN
MH MH MH

DPB D6 D6 D6

8.10x6.00 H
141000
6D

DE

DPB CORRIDOR
35.1X2.80
D6 MH MH D6 D6
6D

MH MH MH MH

CHROMO MH
MH
MH MH MH MH

COLUMN
100 L 100 L 100 L 100 L MH

100 L 100 L CENTRIFUGE
200 L 250 L 300 L 300 L 500 L 500 L
1500 L

PLASMA DAY STAGE 3 (F)
AKTA-1"

STORE STAGE 2 (F)
----- 9M WIDE ROAD -----

4D
12.8X8.68
AKTA-1"

STAGE 8 N
8.10x8.77 6.17X8.68
15.8X8.68
CHROMO MH

JANITOR COLUMN

5.20x1.80 SL
600 L 1500 L 2000 L
1500 L 1500 L 1500 L
300 L
G
DPB 300 L MH MH MH
MH
MH

100 L 100 L 100 L CHROMO CHROMO
MH
TFF MH

COLUMN COLUMN AKTA-1"
MH MH MH

EMER.GLASS EMER.GLASS
CORRIDOR
50.5x1.80
12M WIDE ROAD

PLASMA STORAGE PLASMA STORAGE PLASMA STORAGE PLASMA STORAGE PLASMA STORAGE
~ 1000 L ~ 1000 L ~ 1000 L ~ 1000 L ~ 1000 L

~ 70 KL GV ~ 70 KL GV ~ 70 KL GV ~ 70 KL GV ~ 70 KL GV
7.10x9.62m 9.95x9.62m 9.95x9.62m 9.95x9.62m 8.15x9.62m

F
800 800 800 800

D3 D3 D3 D3 D3

CORRIDOR
45.3x1.80

CORRIDOR
2.57x37.0 CORRIDOR
2.57x35.1

PLASMA STORAGE PLASMA STORAGE PLASMA STORAGE PLASMA STORAGE PLASMA STORAGE
~ 1000 L
~ 70 KL GV
~ 1000 L
~ 70 KL GV
~ 1000 L
~ 70 KL GV
~ 1000 L
~ 70 KL GV
~ 1000 L
~ 70 KL GV
E

7.10x9.62m 9.95x9.62m 9.95x9.62m 9.95x9.62m 8.15x9.62m

800

800 800 800 800
1125

D3 D3 D3 D3 D3

CORRIDOR
45.3x1.80

D
850 800 800 800 800

PLASMA STORAGE PLASMA STORAGE PLASMA STORAGE PLASMA STORAGE PLASMA STORAGE
~ 1000 L ~ 1000 L ~ 1000 L ~ 1000 L ~ 1000 L
~ 70 KL GV ~ 70 KL GV ~ 70 KL GV ~ 70 KL GV ~ 70 KL GV
7.10x11.5m 9.95x11.5m 9.95x11.5m 9.95x11.5m 8.15x11.5m
699

D3 D3 D3 D3 D3

CORRIDOR
47.9x1.80
D6
C
29.5 X 2.58
CORRIDOR

D3 D6 COLD ROOM -4 COLD ROOM -3 COLD ROOM -2 COLD ROOM -1 ICE PACK
DPB RLAF
4.45X7.58 2.91X7.58
D3

AIR LOCK-1 4.45X7.58 4.45X7.58 4.45X7.58
1.50X 1.93 RM QUARANTINE WASHING PLASMA
BAG BAR CODING, JANITOR
AREA QUARANTINE
MINI POOLS
6D

1.45X2.90
2.30X2.90
2.60X2.90
DISPENCING ROOM
3D
3.48X 3.95
CHANGE ROOM D3 D3 D3
1.50 X 1.93
3D

DPB 6.21X5.77
3D

5.00X2.95 D3
11.5X3.30 SD SD SD SD

6D

D3
CORRIDOR
APPROVED RM AIR LOCK-1 DPB 1.95X11.3
4.63X4.00
2.00X1.21 D3

AIR LOCK-1 DPB
3D D3 D6 STORE 4.55X2.32
3.00X2.60
9.72X7.22 1.87X1.40 OFFICE 4.70X3.85
3D
CHANGE ROOM
2.00X2.08 3D
2.23X6.25
9.42X6.25
B
CHANGE ROOM
1.87X1.40
DE-DUSTING REJECT SEROLOGY PLASMA
TUNNEL STORE CHANGE
PLASMA ROOM TESTING RECEIPT 3D D3 ROOM
D3

D3 D3 SAMPLING
1.76X 2.90
2.95X4.95 3.00X 4.95 2.40X 4.95 4.50X3.88
RM
SAMPLING 3.68X 2.05 RECALLED REJECT
RM FG DISCPATCH
1.07X3.39
RECEIPT GOODS ROOM FG IN
3.16X2.03 2.30X2.35 2.30X 2.35
3.00X2.77
RLAF RLAF 6D D6 D6
D3 D3

D6 D3

CORRIDOR
40.1X6.84

A
SPM RM PLASMA
RECEIVING FG
RECEIVING RECEIVING
BAY DISCPATCH
BAY BAY
40' CONTAINER YARD
CONTAINER

CONTAINER

CONTAINER

CONTAINER
12.19x2.44x2.59

12.19x2.44x2.59

6 5 4 3 2 1
12.19x2.44x2.59

12.19x2.44x2.59

CLIENT :- LIFENITY-LARGE VOLUME PARENTERALS

DESIGN &
PROJECT BY :-BIOBRIDGE HEALTHCARE SOLUTION PVT.LTD.
PREPARED

GROUND FLOOR
A.K
ADDRESS :-13, Rachna Blossom, Jagdishnagar,
Aundh, Pune -411007, Maharastra, India. CHECKED G.K.

DATE 04-03-2026
TITLE- CONCEPTUAL LAYOUT-GF REVISION R0
SCALE 1:1

SIZE A0 DWG.NO 01

6D UP

FIRE EXIT

D1
D6 D1
LIFT

FIRE EXIT
LOBBY
6.77X11.77
RECEPTION OPEN TO ALL

D1
FLOORS

D1
6D

UP FIREMAN
Goods LIFT D6
LIFT
3.50X3.00
1D
2.70X2.50
DPB

D6

DPB DPB WASTE IN A/L A/L DHS
CHEMICAL DAY STORE 3 WAY FIRE STAIR
LOBBY

6D

6D

DE
up
3.00X11.30

LAF
DPB DPB
OPEN TO COLD ROOM
FIRST BUFFER TANK STORAGE-2 CIP / SIP LAF UP
UP FLOOR PREPARATION DECONTAMINATION WASHING STERILIZATION COOL ZONE

AUTOCLAVE
DPB
A/L GOWN A/L 6D

GS+HH

GS+HH
LENAP LORTNOC LENAP LORTNOC

6D 6D 6D
DPB
WASTE OUT

DE
A/L DE GOWN A/L
CORRIDOR

FIRE CONTROL
D3 ROOM UP
LAF
D6 D6 D6 D6
DPB AKTA-1" TFF
NANO 3.50X6.54
FILTER MH

A/L GOWN A/L
1000 L
BULK FILTRATION

LAF
500 L

MH
SL MH

600 L

500 L

LAF
MH SL
600 L MH

MH
FIRE PUMP
DE MH

50 L ROOM
A/L
A/L GOWN LAF MH

2000 L 500 L
CHROMO SL
STAGE 13 E 3.50X5.00
COLUMN 250 L
MH 600 L STAGE 10 T/ MH
UP

STAGE 11 F STAGE 9 N MH

250 L
200 L

AKTA-1"
STAGE 6 (I)
MH

COLUMN PACKING 250 L
MH

250 L
1000 L
AND SKIDS
MH

200 L
MH

MH

2.80 M WIDE PASSAGE
250 L
COOL ZONE
MH MH

150 L 100 L MH

MH

1000 L

6.77 M WIDE CORRIDOR
MH

500 L
MH

300 L
WC WC WC 150 L MH

1.10X 1.10X 1.10X CHROMO 100 L MH

COLUMN
100 L
MH

1.70 1.70 1.70
300 L
LAF 150 L
DPB LAF
MH

CHROMO MH

AKTA-1"
MH

100 L COLUMN

AUTOCLAVE
DHS

D6
WC WC WC 500 L 500 L CHROMO
250 L 250 L 150 L 100 L COLUMN

6D
1.10X 1.10X 1.10X MH
MH MH

1.70 1.70 1.70 CORRIDOR A/L MH MH
MH MH
MH

MH

CORRIDOR CORRIDOR

STERILIZATION 2000 L

D6

CLAF
2500 L 2500 L

6D

PRE - VIRUS CONTROLLED CORRIDOR
1000 L

TFF
MH

UP COLUMN PACKING
DPB
AND SKIDS MH

UP

AKTA-1"
WASHING 100 L
A/L

AKTA-1"
STAGE 7A 500 L
MH

STAGE 5 (I) A

D6
DPB
WASTE IN MH 500 L
250 L

TFF
LOBBY 300 L
MH
MH

6.77X11.77
DECONTAMINATION 250 L
500 L
MH

OUT MH
MH

WASTE

D6
500 L

DPB STAGE 5(I)B 1500 L 250 L
MH
MH

100 L SL
600 L 500 L
250 L 250 L
MH

250 L
100 L 100 L

6D
100 L
MH

MH
MH MH MH
MH

100 L
MH MH MH

CIP / SIP
MH

1500 L
SL MH MH

UP
MH MH MH

1000 L
MH

500 L 500 L
MH

1000 L 1000 L 1000 L
MH

UP
100 L 1500 L
1500 L

GS+HH

GS+HH
MH SL
LENAP LORTNOC LENAP LORTNOC

MH
2500 L

MH
1500 L
50 L
Pass. LIFT CHROMO MH
250 L 250 L 250 L 250 L
2.30X2.00 DPB COLUMN

6D
MH MH MH MH

WASTE OUT

BAG CUTTING & WC WC WC

POOLING MH
MH MH

MH

MH MH
STAGE 4 (F) 1.10X
1.70
1.10X
1.70
1.10X
1.70

150 L 500 L

6D
500 L
SL 1000 L
TANK STORAGE 1000 L 1500 L 1500 L

D6
MH

WC WC WC
STAGE 12 (E)
MH

250 L 1.10X 1.10X 1.10X
MH

STAGE 1 (F)/ 1.70 1.70 1.70
CHROMO
500 L
COLUMN

250 L
300 L 300 L MH

TFF
D6

AKTA-1"
6D
MH MH

CLAF

2500 L

THAWING PLASMA 2000 L 2000 L
UP

AKTA-1"
MH

DPB
500 L 250 L 250 L 300 L 300
300L L
BUFFER

6D
MH MH

CHROMO
MH
COLUMN MH MH MH
MHMH

CHROMO
PREPARATION D6 D6 D6
COLUMN

CORRIDOR

6D

DE
DPB
3.24 M WIDE PASSAGE

D6 D6 D6

6D
MH MH

MH MH MH MH

CHROMO MH
MH
MH MH MH MH

COLUMN
100 L 100 L 100 L 100 L MH

100 L 100 L CENTRIFUGE
200 L 250 L 300 L 300 L 500 L 500 L
1500 L

PLASMA DAY STAGE 3 (F)
STORE STAGE 2 (F)
STAGE 8 N

AKTA-1"

AKTA-1"
UP

CHROMO
COLUMN
MH
UP
3D

SL
600 L 1500 L 2000 L
1500 L 1500 L 1500 L

300 L 300 L MH MH MH
MH
MH

100 L 100 L 100 L CHROMO CHROMO
DPB MH
TFF MH

COLUMN COLUMN AKTA-1"
MH MH MH

WC WC WC
1.10X 1.10X 1.10X EMER.GLASS

1.70 1.70 1.70
LOBBY
6.77X11.00

WC WC WC
STAIRCASE
1.10X 1.10X 1.10X AND
1.70 1.70 1.70
LIFT ONLY
CORRIDOR
FOR
CORRIDOR
SECOND FLOOR
(L.V.P.)

UP

UP

W

D3 D3 D3 D3 D3

CORRIDOR

D6
LOBBY CORRIDOR
12.47X6.77
D3 D6
Pass. LIFT W
DPB
2.30X2.00
D3

6D

3D D3 D3
UP
D3
3D

CORRIDOR

D3
3D

SD SD SD SD

DPB
6D
ELECTRIC
D3 ROOM
DPB D3
DPB
3D D3 D6 STORE 3.50X6.00
OFFICE
3D
LOBBY
3.00X11.53 3D

OPEN TO

D3
SECOND D3
D3
3D D3

FLOOR REJECT
RECALLED
GOODS ROOM FG DISCPATCH
FG IN
6D D3 D3 D6 D6

D6 D3
SERVER
Goods LIFT CORRIDOR ROOM
3.50X3.00
6D

D6
3.50X6.57
DISPATCH BAY RECEIVING BAY RECEIVING BAY DISPATCH BAY

RS3
40' CONTAINER YARD

CLIENT :- LIFENITY-LARGE VOLUME PARENTERALS

DESIGN &
PROJECT BY :-BIOBRIDGE HEALTHCARE SOLUTION PVT.LTD.
PREPARED A.K
ADDRESS :-13 R B
A P -411007 M I .

GROUND FLOOR
CHECKED G.K.

DATE 05-06-2026
TITLE- CONCEPTUAL LAYOUT-GF REVISION R0
SCALE 1:1

SIZE A0 DWG.NO 01

Reagent Details:
Formula & Calculations:
Formula: Weight of solute (g)=Molarity (M)×Molecular Weight (g/mol) ×Volume (L).
Calculation: To prepare 1 L of 30mM sodium chlorolride (MW = 58.44g/mol)
Weight of solute(g)=1×1.753=1.753■g.
Preparation Method:
Weigh 1.753 g of sodium chloride
To prepare 25mM Tris 1L solution from 1M Tris stock solution formula is M1(Molarity of stock) x V1
(volume of stock)= M2 (Molarity of working) xV2 (volume of working).
Take 25mL of 1M tris solution and add weighed amount of sodium chloride as required and dissolve in water
to make a 1 L solution.
Preparation for different volumes:
To prepare _____ of 30mM NaCl (MW = 58.44g/mol).
Weight of solute(g)=1.753x _____ L= _____ gm
Weigh _____ gm of NaCl.
To prepare _____ of 25mM Tris
Take 25mL x _____ L= _____ mL of 1M Tris
Dissolve pre weighed NaCl in 25mM Tris solution to make a _____ L solution.
Adjust the pH to 2 using concentrated HCl/NaOH solution
Check and record the details:
Preparation Details:
Name of Reagent | Grade / Purity | Manufacturer / Supplier | Lot / Batch No | Storage Condition | Expiry
Date
| | | |
Final volume prepared | Weighed Quantity | pH (2.0) | Conductivity (10.0±2.0mS/cm) | Storage Condition:
| | | | RT
Date of Preparation | Prepared By | Checked By
| |

FOR OFFICE USE

AD
DE RO
1.200
T-300 MM
R-150 MM TERRACE

5.100
5.100
FORM OF STATEMENT

WI
AREA CALCULATIONS

60.00 M.

5.100

5.100
5.100
BLOCK NO BLOCK NAME FLOOR SIZE AREA

1 BUILDING-01 ADMIN GROUND 4033.57

FIRST 3478.10

8.553

8.550
8.550
SECOND 4033.57

T-0.30m.

ROAD
R-0.15m.

WIDE
60.00 M.
T-0.30m.
R-0.15m.

BUILDING-02 LABORATORY 1.200

AD
2 GROUND 13673.472
60.
00 SECTION A-A

RO
FIRST
3 13673.472

PLOT NO. 1B
M . W
ID
BUILDING-2
ELEVATION BUILDING 2

E
SECOND 7946.736
4 E

ID
SECTION BB

RO
RIGHT SIDE

W
5 TOTAL 46838.92 AD

M.
00
TOTAL B/UP AREA OF ALL FLOORS+ EXTRA HEIGHT 46838.92

30.
PROPOSED SITE
CANOPY ABOVE CANOPY BELOW
60 SHAFT
RECEIVING BAY DISPATCH BAY DISPATCH BAY DISPATCH BAY
SHAFT SHAFT SHAFT SHAFT SHAFT

.00 FIREMAN
LIFT LIFT
FIREMAN
LIFT LIFT
FIREMAN
LIFT LIFT

M. 2.80X2.60 3.30X2.45 2.80X2.60 3.30X2.45 2.80X2.60 3.30X2.45

W LIFT LIFT LIFT
ID CORRIDOR 2.10x 2.10x 2.10x

PLOT NO. 1C
UP 1.90 UP 1.90 1.90
E UP

RO DISPATCH / RECEIVING BAY LIFT
2.10X
2.00
LIFT
2.10X
2.00
LIFT
2.10X
2.00
AD
OA D 703 sq.mt.
45.00 M. W I D E R

OAD
45.00 M. W I D E R

PLAN OF PROPOSED FACTORY

30.00 M. W I D E R O A D
FORM OF STATEMENT AREA CALCULATION

SECURITY CABIN 1,2&3 189.00 -- -- 189.00
BUILDING ON PLOT NO 8A, SECTOR 13,
DRIVER ROOM 60.00 -- -- 60.00
DMIC, AURIC , SHENDRA,
PUMP ROOM 57.11 -- -- 57.11
CHATTRAPATI SAMBHAJI NAGAR
FOR M/S LIFENITY LIMITED
LOCATION PLAN OCCUPANT LOAD AS PER TABLE NO 8E
CLAUSE NO. 10.6.3 CDCPR WATER REQUIREMENT
TOTAL B/UP AREA = 47494.288 SQ.M.
DIALYZER SHAFT
DIALYZER SHAFT
TERRACE SHAFT

FIREMAN FIREMAN FIREMAN
OCCUPANT LOAD CALATION OCCUPANT LOAD = 47494.288/10 = 4749 NOS. SHAFT LIFT SHAFT LIFT SHAFT LIFT

SCALE :- NTS BUILT UP AREA 47494.288 SQMT
OCCUPANT LOAD actual = 400 NOS.
NO ACTIVITY WATER IN LITS.
FD
UP
FD
59.60 x 83.80 FD
2.80X2.60

UP
FD
UP
FD
59.60 x 95.80 FD
2.80X2.60

UP
FD
UP
FD
59.60 x 95.80 FD
2.80X2.60

UP
FD

AREA PER PERSON 10 SQMT
1 FOR W.C. 30 X 400 = 12000
UP
CARPET AREA 4994 .48 SQ.MT. CARPET AREA 5709.68 SQ.MT.
2 FOR INDUSTRIAL 100000 LITS/DAY UP

4749.42 NOS
NO OF PERSONS TOTAL /DAY 112000 LITR/DAY

60.00 M. AS PER ACTUAL OCCUPANTS IN THE BUDILING
NO. OF OCCUPANTS - 400 NOS.
NO. OF OFFICE STAFF - 100 NOS.
NO. OF WORKERS - 300 NOS.
FOR DOMESTIC = 25000 LTR.
FOR INDUSTRIAL = 200000 LTR.

TERRACE AREA STATEMENT
WIDE
RCC SLAB

SANITATION REQUIREMENT (TABLE NO. 10D) G.L. G.L.
BELOW
2.5MX2.5MX3M BAFFLE WALL 5MX2.5MX3M DESILTING AREA IN
RAIN WATER HARVESTING OCCUPANT LOAD 47494 / 10 4749.40 NO PARTICULARS
CHAMBER SQM.
PIT
ROAD
PCC BED
TOTAL NO. OF PERSON 4749 SECTION AT CC
U.G. WATER TANK
FIXTURES OFFICES / VISITORS WORKERS 1 AREA OF PLOT NO 56 SECTOR 13, DMIC, SHENDRA 81920.00
100 300
50 50 200 100
AS PER OWNERSHIP DOCUMENTS 81920.00
MALE FEMALE MALE FEMALE
25X25MM 112000 LTR. 112000 LTR.
1.2MX2.5MX1.5M 1 FOR 25 1 FOR 15 1 FOR 20 1 FOR 15
AS PER MEASUREMENT SHEET
MANHOLE
600 MM x 600
MM
BAR SCREEN REQUIRED WC
after 2.5% thereafter 3% thereafter 2.5% thereafter 4% 81920.00
GREASE TRAP 1+1 = 2 1+2=3 1+5=6 1+4=5 U.G. WATER TANK PLAN
1 up to10
thereafter 2.5% -
1 up to10
thereafter 2.5% - AS PER SITE 81920.00
53.285 REQUIRED URINALS
MANHOLE

1+1 = 2 1+5 = 6

X X 2 DEDUCTIONS FOR

46.9 REQUIRED PROVIDED
SHAFT

FIREMAN
LIFT SHAFT
FIREMAN
LIFT
SHAFT

SHAFT
SHAFT

FIREMAN
LIFT SHAFT a. PROPOSED DP ROAD 0.00
72
UP 2.80X2.60 2.80X2.60 2.80X2.60

NO. OF W.C. FOR STAFF (MALE) 2 2 FD FD FD FD FD FD FD FD FD FD

NO. OF URINALS FOR STAFF 2 2 SEPTIC TANK REQUIREMENT UP UP UP UP UP UP

NO. OF W.C. FOR STAFF (FEMALE) 3 3 TOTAL B/UP AREA = 47494.288 SQ.M. b. ANY RESERVATIONS 0.00
OCCUPANT LOAD = 47494/10 = 4749 NOS. UP
NO. OF W.C. FOR WORKERS (MALE) 6 6
OCCUPANT LOAD actual = 400 NOS.

RWHP FLOOR PLAN - NO. OF URINALS FOR WORKERS 6 6 TOTAL (a+b) 0.00
GATE-2 NO. OF W.C. FOR WORKERS
(FEMALE)
5 5 NO ACTIVITY
1 FOR W.C. & URINALS
WATER IN LITS.
30 X 400 = 12000
SCALE - 1:100
ENTRY TOTAL /DAY 12000 LITS/DAY
3 BALANCE AREA OF PLOT (1-2) (A) 81920.00
DETENTION 12000 X 2 = 24000
SECURITY

DRIVER'S PERIOD 2 DAYS
4 AMENITY SPACE IF APPLICABLE 5%
CABIN-2

PARKING REQUIREMENTS CLAUSE NO 7.20 MIDC CDCPR 0.00
SLUDGE SETTLEMENT / PERSON = 30 LIT/YR.
AS PER TABLE 7C AND 7D FOR 47494.288 SQM. MULTIPLICATION FACTOR 0.8
ROOM SEPTIC TANK
TYPE OF VEHICLE PARKING SIZE REQUIRED PARKING REQUIRED REQUIRED REQUIRED PROVIDED
CLEANING PERIOD = 1 YEAR
TOTAL SLUDGE = 30 X 400 = 12000 LITS
SHAFT

LIFT
3.30X2.45
SHAFT

LIFT
3.30X2.45
SHAFT

LIFT
3.30X2.45
a. Required 0.00
CAR 2.5M X 5.0M 1 FOR EVERY 200SQM 237.47 0.80 190 TANK CAPACITY = 24000 +12000 = 36000 LITS LIFT LIFT
LIFT
SCOOTER / MOTOR CYCLE 2.0 SQ.M. (1.0M X 2.0M) 2 FOR EVERY 200SQM 474.94 0.80 380 100 2.10x 2.10x 2.10x
CAPACITY 3.00X 3.00 X 1.50 X 3 nos= 40500 LITS 1.90 UP 1.90 1.90
CYCLE
TRUCK
VENT PIPE
LIFT
2.10X
2.00
LIFT
2.10X
2.00
LIFT
2.10X
2.00
b. Adjustment of 2(b) if any 0.00
3.75M X 10.0M 1 FOR EVERY 1000 SQM 47 47
BRICK WALL
BRICK BAT
In case of plots pf 1ha and above minimum parking space shall be 10% of plot area
For plot area 81920 sq.m. parking should be 8192 sq.m.
OPEN SPACE-1 PCC BED SEPTIC TANK SECTION AT DD
c. Balanced proposed 0.00
SPACE
FOR
ETP 4399.62 sq.m. RAIN WATER HARVESTING TANK CAPACITY 5 NET PLOT AREA (3-4(c))) PRO RATA
AVERAGE ANNUAL RAINFALL=100MM (0.1M/HOUR) 81920.00
RAIN WATER PLOT AREA= 81920 SQ.M
HARVESTING VOLUMETRIC RAINFALL= 81920 X 0.1= 8192 CUM/HOUR
FIRE TANK SEPTIC TANK PLAN
SPACE
200K
PIT = 136.53 CUM/MINUTE
MAIN GATE
6 RECREATIONAL OPEN SPACE IF APPLICABLE 10%
RAIN WATER TANK CAPACITY PROPOSED= 5x 5 X 3 X 2= 150 CUM SCALE 1:50 BOUNDARY WALL DETAIL

FOR RAIN WATER TANK CAPACITY PROPOSED= 150000 LITER
STP
RAIN WATER
PIPE
FRONT ELEVATION COMPOUND WALL & GATE
A A A A a. Required
SHAFT SHAFT SHAFT
DOMESTIC FIRE TANK MURUM
GL GL
A FIREMAN
A FIREMAN
A FIREMAN
A
TANK FILLING
SHAFT LIFT SHAFT LIFT SHAFT LIFT
50K 200K SAND
FILLING 2.80X2.60 2.80X2.60 2.80X2.60

PUMP SECURITY RUBBLE
FILLING

WATER HARVESTING SECTION
(SCALE - 1:50)
FD
UP
FD FD
UP
FD
UP
FD FD
UP
FD
UP
FD FD
UP
FD
b. Proposed
ROOM CABIN-3
RAINWATER PIT CAPACITY 5.00 X 5.00 X3 x 2 nos = 150 K LTS
UP
UP

7 INTERNAL ROAD AREA 0.00
PROCESS TANK
200K
8 PLOTABLE AREA IF APPLICABLE (as per sanctioned layout) 81920.00

9 B/UP area with reference to basic fsi as per front road width(1.0) 81920.00

60

I N T E R N A L R O A D 9.00 M WIDE .00 10 ADDITIONAL FSI ON PAYMENT OF PREMIUM

a. Maximum permissible premium FSI / TOD Zone. 2.00

CANOPY ABOVE

M. PLASMA EXTRACTION

59.60 x 120.60

CARPET AREA 7187 .00 SQ.MT.

PLASMA EXTRACTION

59.60 x 132.60

CARPET AREA 7902.96 SQ.MT.

LVP

59.60 x 132.60

CARPET AREA 7902.96 SQ.MT.

b. Proposed FSI of payment of Premium 0.00

SHAFT

FIREMAN

RECEIVING BAY DISPATCH BAY DISPATCH BAY DISPATCH BAY

SHAFT

W

ID

LIFT LIFT

2.80X2.60 3.30X2.45

LIFT 11 IN SITU TDR / FSI LOADING

UP CORRIDOR 2.10x

1.90 UP

DISPATCH / RECEIVING BAY LIFT

E a. In situ area against Statutory Raod (2 xSr No 2(a)) if any 0.00

2.10X

2.00

703 sq.mt. TERRACE WITH

I

N R WATERPROOFING

12 ADDITIONAL FSI AREA UNDER CHAPTER NO.6 0.000

T O SHAFT SHAFT SHAFT

AD 13 TOTAL ENTITLEMENT OF FSI IN THE PROPOSAL

FIREMAN FIREMAN FIREMAN

0.000

E

SHAFT LIFT SHAFT LIFT SHAFT LIFT

2.80X2.60 2.80X2.60 2.80X2.60

FD FD FD FD FD FD FD FD FD FD

10 UP UP UP UP UP UP UP

a. 9 +10(b) +11(a) or 9+12 which ever is applicable

6N

OS

R UP

Permissible 0.00

N

UP

SC a. 9 +10(b) +11(a) or 9+ 12 which ever is applicable

21

2 NO

OO

TE A 31 Proposed 0.00

SS R L 5. b. Permissible Ancillary area FSI up to 80% with payment

10

6N

CO

OT 21 PERMISSIBLE 0.00

OS ER R 1 b. Proposed Ancillary area FSI up to 60% or 80% with payment

SC

OO

O

SHAFT

DIALYZER (G+1)

59.60 x 83.80

SHAFT

FIREMAN

LIFT

2.80X2.60

TE

R

A c.Total Entitlement permissible

(a+b) 81920.00

FD FD

UP

CARPET AREA 4994 .48 SQ.MT.
FD
UP
FD

D c.Total Entitlement proposed
UP
UP (a+b) 81920.00
Maximum utilization limit of FSI (Building Potential) permissible as per
Car Parking

14 road width ((as per Regulation No. 5.5 of Chapter - 5)) x 1.6 or 1.8) 81920.00

15 TOTAL Built up area in proposal (excluding area as per sr. no 17b)
SE LIFT CORRIDOR LIFT LIFT

C 3.30X2.45
DISPATCH / RECEIVING BAY 3.30X2.45 3.30X2.45

CA UR
41

UP
LIFT
2.10x
1.90 703 sq.mt.
LIFT
2.10x
1.90
LIFT
2.10x
1.90
a. Existing Built -up Area. 0.000
BI ITY LIFT
2.10X
LIFT
2.10X
LIFT
2.10X
.6

N- GA
2.00
SHAFT RECEIVING BAY DISPATCH BAY DISPATCH BAY DISPATCH BAY
2.00
SHAFT
2.00
SHAFT

b. Proposed Built- up area (as per P line) 46838.92
1
19

EN TE-
B
CANOPY ABOVE CANOPY BELOW
UP

FD

TR 1 TERRACE FLOOR c. Total (a+b) 46838.92
Y GROUND FLOOR PLAN FIRST FLOOR PLAN SECOND FLOOR PLAN
UP
FD

PLAN
(BUILDING 2) (BUILDING 2) (BUILDING 2)
LOBBY (BUILDING 2)
27 X 4.77
16 FSI CONSUMED (15/13) (should not be more than serial 0.571
No.14 above.)
R O A D 9 . 0 0 M W I D E
330.840

CERTIFICATE OF AREA :-
AUDITORIUM
27 X 14.77
Certified that the plot under reference was surveyed by me on
SHAFT
02/06/2025 and the Dimensions of sides etc.of plot stated on plan area

R I D M.
FIREMAN
LIFT SHAFT

36.575
2.80X2.60

as measured on site and the area so worked out tallies with the stated in
UP

FD FD FD FD FD FD FD FD

UP UP

documents of ownership/ DMIC/Land records Department/City Survey

W .00
UP UP

O E
UP

Records
Ar. Dashrath Dhumal

AD
30
R O A D 1 2 . 0 0 M W I D E

Owners Declaration :-
I/We undersigned hereby confirmed that I/We would abide by plans
SHAFT
approved by Authority/ Collector. I / We would execute the structure as
ELEVATION BUILDING 1
LIFT
3.30X2.45

LIFT
per the approved plans.Also I/We would execute the work under
2.10x
1.90 UP
ADMIN AREA
supervision of proper technical person so as to ensure the quality and
LIFT
2.10X
27.04X50.53
FRONT SIDE
2.00
safety at the work site.
LIFT
DUCT
4.20X1.03
Mr. Sarang Landage.
R O A D 1 2 . 0 0 M W I D E

2.45X3.30 LIFT LIFT
2.00X2.10 2.00X2.10
LIFT MACHINE
ROOM
OHWT

Architects Name and Signature
17.840 TERRACE 1.200 TERRACE 1.200

OPEN SPACE-2

5.100

5.100

5.100

5.100
I N T E R N A L

BUILDING - 1 ADMIN BUILDING - 1 ADMIN Ar. Dashrath Dhumal
3793.43 sq.m. LADIES TOI. LADIES TOI.

Job No. Drg. No. Scale Drn. By. Checked By Registration no of Architect

5.100
FD FD FD FD

5.100

5.100

5.100
D4 D4

122.100 BUILDING - 1 ADMIN

10.200
SHAFT

LIFT WELL

LIFT WELL

LIFT WELL
SHAFT
FIREMAN

LIFT
2.80X2.60
FD
UP

D1 D1 D1 D1 D1
UP
LADIES TOI. LADIES TOI.
001 1:100 Ar. Dashrath Dhumal CA/2005/37017

5.100
FD FD FD WC WC WC WC WC
1.05X 1.05X 1.05X 1.05X 1.05X

5.100

5.100

5.100
1.70 1.70 1.70 1.70 1.70
UP UP
V V V V V

AUDITORIUM CANTEEN
DUCT
5.90X1.43
V V V V V

UP
WC WC WC WC WC T-300 MM
C PARKING STATEMENT
26.192
1.05X 1.05X 1.05X 1.05X 1.05X
R-150 MM LADIES TOI.
UP
1.70 1.70 1.70 1.70 1.70 LADIES TOI.
D1 D1 D1 D1 D1
T-300 MM
R-150 MM LIFT LIFT LIFT
PIT PIT PIT
a GARAGE PERMISSIBLE GARAGE PROPOSED NIL
SECTION B-B
D4
D4

PARKING PARKING
b TYPE
REQUIRED PROPOSED
BUILDING-1
CAR

SCOOTER
8192 SQ.M. 8192 SQ.M.
I N T E R N A L

CYCLE
B B B B
PLASMA EXTRACTION (G+2)
VISITORS

UP
60

FD
59.60 x 120.60 CANTEEN
Car Parking

27.04 X 59.00

CARPET AREA 7187 .00 SQ.MT. D LOADING UNLOADING SPACES
W .00 M

UP

UP

UP
FD

FD

FD
FD

LOBBY
LOBBY 27 X 4.77
27 X 4.77
a LOADING UNLOADING REQUIRED 6

R IDE .
FD FD FD FD

b TOTAL LOADING UNLOADING PROVIDED 6
DOUBLE HEIGHT
UP UP AUDITORIUM
I N T E R N A L

O LEGENDS
AD
AUDITORIUM
27 X 14.77

SHAFT

FIREMAN
SHAFT

PLOT BOUNDRY SHOWN IN
LIFT
2.80X2.60
FD FD FD FD

AMENITY SPACE
D

UP UP
FD FD FD FD FD FD FD FD

UP
B/UP AREA SHOWN IN
4164.17 sq.m.
UP
A

UP UP

WATER LINE SHOWN IN
02

FD FD
O

DRAINAGE LINE SHOWN IN
.7

R
79

LIFT
SCHEDULE OF DOORS AND WINDOWS
E

2.10X2.00
COMPRESSOR
4.20X1.03
DUCT

PANEL ROOM
LIFT
18.28 X 10.62 2.10X2.00
TYPE SIZE DISCRPTION
ID

LIFT
3.30X2.45
RS 6000 X 6000 M.S ROLLING SHUTTER

124.700
FD

ADMIN AREA ADMIN AREA ADMIN AREA D1 750 X 2100 R.C.C. FRAME FLUSH DOOR
W

27.04X50.53 27.04X50.53 27.04X70.54 TERRACE WITH WATER PROOFING
FD

UP

B D4 1000 X 2100 R.C.C. FRAME FLUSH DOOR

GROND FLOOR PLAN LIFT
DUCT
4.20X1.03

LIFT
DUCT
4.20X1.03

LIFT
DUCT
4.20X1.03
FD 1000 X 2100 FIRE DOOR
.

2.45X3.30 LIFT LIFT 2.45X3.30 LIFT LIFT 2.45X3.30 LIFT LIFT

(BUILDING 1) W1 3000 X 1500 ALLU. SLIDING WINDOW
OM

2.00X2.10 2.00X2.10 2.00X2.10 2.00X2.10 2.00X2.10 2.00X2.10

A
V 600 X 900 M.S. LOUVERED VENTILATOR

0.900
LIFT CORRIDOR 4.500
4.500

3.30X2.45
DISPATCH / RECEIVING BAY
.0

LIFT Security Cabin
4.00 x 5.00
2.10x
UP
1.90 703 sq.mt. 0.899

LIFT

PLAN OF PROPOSED FACTORY
45

12.000
2.10X
2.00 10.450

14.000

14.000
SHAFT RECEIVING BAY DISPATCH BAY DISPATCH BAY DISPATCH BAY UP
A

5.000
A FD FD FD FD FD FD FD
FD

BUILDINGON PLOT NO 8A, SECTOR 13,

5.455
A D4 D4 D4 D4 D4 D4
VISITORS

CANOPY ABOVE 4.00 x 6.74

AREA KEY PLAN
(Driver's Room) UP UP
AREA KEY PLAN
12.000 Pump Room
21.572

D1 D1 D1 D1 D1 D1 D1 D1 D1 D1 D1 D1 D1 D1 D1

W.B.

TYPICAL PLAN-1,2&3.
TOILET
2.40 x 1.50

A AREA KEY PLAN
A W.C.
1.20 x
1.20 BATH
1.20 x
2.55
A 10.450
WC
1.05X
1.70

V
WC
1.05X
1.70

V
WC
1.05X
1.70

V
WC
1.05X
1.70

DUCT

V

5.90X1.43
WC
1.05X
1.70

V
FD
WC
1.05X
1.70

V
WC
1.05X
1.70

V
WC
1.05X
1.70

V
DUCT
5.90X1.43
WC
1.05X
1.70

V
WC
1.05X
1.70

V
FD
WC
1.05X
1.70

V
WC
1.05X
1.70

V
WC
1.05X
1.70

V
DUCT
5.90X1.43
WC
1.05X
1.70

V
WC
1.05X
1.70

V
DUCT
5.90X1.43
DMIC, AURIC , SHENDRA,

5.000
V V V V V V V V V V V V V V V V
(Security Cabin) Security Cabin DRIVER'S ROOM
8.84 x 4.54
WC WC WC WC WC WC WC WC WC WC WC WC WC WC WC WC
1.05X 1.05X 1.05X 1.05X 1.05X 1.05X 1.05X 1.05X 1.05X 1.05X 1.05X 1.05X 1.05X 1.05X 1.05X 1.05X

5.455
A PUMP ROOM A 1.70 1.70 1.70 1.70 1.70 1.70 1.70 1.70 1.70 1.70 1.70 1.70 1.70 1.70 1.70 1.70
LOCKER'S RM.

CHATRAPATI SAMBHAJI NAGAR
2.55 x 1.84 10.00 X 5.00
D1 D1 D1 D1 D1 D1 D1 D1 D1 D1 D1 D1 D1 D1 D1

I N T E R N A L R O A D 1 2 . 0 0 M W I D E P L A N
(Driver's Room)
UP

PARAPET LVL.
1.997 2.997 1.514
10.450
PLAN (Pump Room)
1.199

UP

1.514 0.999 0.230 D4
D4
D4
D4
D4
D4

1.598 0.599
SLAB LVL.

Parapet LVL.
PARAPET LVL.
1.598 0.599

Slab LVL.
SLAB LVL. LINTEL LVL.

3.996
RCC Chajja

1.499
2.398
Lintel LVL.
LINTEL LVL. 0.899
SILL LVL.
3.996

1.499

PLINTH LVL.

0.450
2.398

Sill LVL.
0.899 G.LVL. G.LVL.
SILL LVL.
FRONT ELEVATION Plinth LVL.
PLINTH LVL. (Driver's Room)
0.450

G. LVL. G. LVL.
G.LVL. G.LVL.
FRONT ELEVATION
LEFT SIDE ELEVATION
(Security Cabin)

PARAPET LVL.
1.598 0.599

TERRACE TERRACE Parapet LVL.
SLAB LVL.
Terrace Slab LVL.

0.900
LINTEL LVL. RCC Chajja
3.996

Lintel LVL.
1.499

3.996
2.398

3.497

1.499
0.899 SILL LVL.

2.398

2.398
Sill LVL.
Security Cabin DRIVER'S ROOM W.C. BATH

231.187
VISITORS TOILET PLINTH LVL.

0.899
SUNK SUNK
SUNK
G.LVL. G.LVL. PUMP ROOM
Plinth LVL.

0.450
G. LVL. G. LVL.
SECTION- AA
SECTION- AA SECTION- AA
(Driver's Room)
(Security Cabin)

CANTEEN
27.04 X 59.00

FOR M/S LIFENITY LIMITED
FD FD FD FD FD FD FD FD

UP UP
ADMIN AREA ADMIN AREA

45.00 M. W I D E R O A D
27.04X69.77 27.04X69.77
FD FD

SIGN OF OWNER

SIGN OF ARCHITECT
LIFT LIFT LIFT
2.10X2.00 2.10X2.00 2.10X2.00
COMPRESSOR

4.20X1.03

4.20X1.03

4.20X1.03
N

DUCT

DUCT

DUCT
PANEL ROOM
LIFT LIFT LIFT
18.28 X 10.62 2.10X2.00 2.10X2.00 2.10X2.00

LIFT LIFT LIFT
3.30X2.45 3.30X2.45 3.30X2.45

LAYOUT PLAN

FD

FD

FD
FD

FD

UP
B B B B

GROND FLOOR PLAN FIRST FLOOR PLAN SECOND FLOOR PLAN TERRACE FLOOR PLAN
SCALE :- 1:100 N
(BUILDING 1) (BUILDING 1) (BUILDING 1) (BUILDING 1)
CREATION

W E JOB No. DRG NO. SCALE DATE DRN BY CHK. BY

S 01 1 :100

6D UP

FIRE EXIT

D1
D6 D1
LIFT

FIRE EXIT
LOBBY
6.77X11.77
RECEPTION OPEN TO ALL

D1
FLOORS

D1
6D

UP FIREMAN
Goods LIFT D6
LIFT
3.50X3.00
1D
2.70X2.50
DPB

D6

DPB DPB WASTE IN A/L A/L DHS
CHEMICAL DAY STORE 3 WAY FIRE STAIR
LOBBY

6D

6D

DE
up
3.00X11.30

LAF
DPB DPB
OPEN TO
FIRST BUFFER TANK STORAGE-2 CIP / SIP COLD ROOM LAF UP
UP FLOOR PREPARATION DECONTAMINATION WASHING STERILIZATION COOL ZONE

AUTOCLAVE
DPB
A/L GOWN A/L 6D

GS+HH

GS+HH
LENAP LORTNOC LENAP LORTNOC

6D 6D 6D DPB
WASTE OUT

DE
A/L DE GOWN A/L
CORRIDOR

FIRE CONTROL
D3 ROOM UP
LAF
D6 D6 D6 D6
DPB AKTA-1" TFF
NANO 3.50X6.54
FILTER MH

A/L GOWN A/L
1000 L
BULK FILTRATION

LAF
500 L

MH
SL MH

600 L

500 L

LAF
MH SL
600 L MH

MH

FIRE PUMP
DE MH

50 L ROOM
A/L
A/L GOWN LAF MH

2000 L 500 L
CHROMO SL
STAGE 13 E 3.50X5.00
COLUMN 250 L
MH 600 L STAGE 10 T/ MH
UP

STAGE 11 F STAGE 9 N MH

250 L
200 L

AKTA-1"
STAGE 6 (I)
MH

COLUMN PACKING 250 L
MH

250 L
1000 L
AND SKIDS
MH

200 L
MH

MH
2.80 M WIDE PASSAGE

250 L
COOL ZONE
MH MH

150 L 100 L MH

MH

1000 L 500 L

6.77 M WIDE CORRIDOR
MH

MH

300 L
WC WC WC 150 L MH

1.10X 1.10X 1.10X CHROMO 100 L MH

COLUMN
100 L
MH

1.70 1.70 1.70
300 L
LAF 150 L
DPB LAF
MH

CHROMO MH

AKTA-1"
MH

100 L COLUMN

AUTOCLAVE
DHS

D6
WC WC WC 500 L 500 L CHROMO
250 L 250 L 150 L 100 L COLUMN

6D
1.10X 1.10X 1.10X MH
MH MH

1.70 1.70 1.70 CORRIDOR A/L MH MH
MH MH
MH

MH

CORRIDOR CORRIDOR

STERILIZATION 2000 L

D6

CLAF
2500 L 2500 L

6D

PRE - VIRUS CONTROLLED CORRIDOR
1000 L

TFF
MH

UP COLUMN PACKING
DPB
AND SKIDS MH

UP
AKTA-1"

WASHING 100 L
A/L

AKTA-1"
STAGE 7A 500 L
MH

STAGE 5 (I) A
D6

DPB
WASTE IN MH 500 L
250 L
TFF

LOBBY 300 L
MH
MH

6.77X11.77
DECONTAMINATION 250 L
500 L
MH

OUT MH
MH

WASTE
D6

500 L

DPB STAGE 5(I)B 1500 L 250 L
MH
MH

100 L SL
MH

250 L 250 L 250 L 600 L 500 L
100 L 100 L
6D

100 L
MH

MH
MH MH MH
MH

100 L
MH MH MH

CIP / SIP
MH

1500 L
SL
UP
MH MH

MH MH MH

1000 L
MH

500 L 500 L
MH

MH

1000 L 1000 L 1000 L
UP
100 L 1500 L
1500 L
GS+HH

GS+HH

MH SL
LENAP LORTNOC LENAP LORTNOC

MH
2500 L
MH

1500 L
50 L
Pass. LIFT CHROMO MH
250 L 250 L 250 L 250 L
2.30X2.00 DPB COLUMN
6D

MH MH MH MH

WASTE OUT

BAG CUTTING & WC WC WC

POOLING MH
MH MH

MH

MH MH
STAGE 4 (F) 1.10X
1.70
1.10X
1.70
1.10X
1.70

150 L 500 L
6D

500 L
SL 1000 L
TANK STORAGE 1000 L 1500 L 1500 L
D6

MH

WC WC WC
STAGE 12 (E)
MH

250 L 1.10X 1.10X 1.10X
MH

STAGE 1 (F)/ 1.70 1.70 1.70
CHROMO
500 L
COLUMN

250 L
300 L 300 L MH
TFF
D6

AKTA-1"

6D
MH MH

CLAF

2500 L

THAWING PLASMA 2000 L 2000 L
UP
AKTA-1"

MH

DPB
500 L 250 L 250 L 300 L 300
300L L
BUFFER
6D

MH MH

CHROMO
MH
COLUMN MH MH MH
MHMH

CHROMO
PREPARATION D6 D6 D6
COLUMN

CORRIDOR
6D

DE
DPB
3.24 M WIDE PASSAGE

D6 D6 D6
6D

MH MH

MH MH MH MH

CHROMO MH
MH
MH MH MH MH

COLUMN
100 L 100 L 100 L 100 L MH

100 L 100 L CENTRIFUGE
200 L 250 L 300 L 300 L 500 L 500 L
1500 L

PLASMA DAY STAGE 3 (F)
STORE STAGE 2 (F)
STAGE 8 N
AKTA-1"

AKTA-1"

UP

CHROMO
COLUMN
MH
UP
3D

SL
600 L 1500 L 2000 L
1500 L 1500 L 1500 L

300 L 300 L MH MH MH
MH
MH

100 L 100 L 100 L CHROMO CHROMO
DPB MH
TFF MH

COLUMN COLUMN AKTA-1"
MH MH MH

WC WC WC
1.10X 1.10X 1.10X EMER.GLASS

1.70 1.70 1.70
LOBBY
6.77X11.00

WC WC WC
STAIRCASE
1.10X 1.10X 1.10X AND
1.70 1.70 1.70
LIFT ONLY
CORRIDOR
FOR
CORRIDOR
SECOND FLOOR
(L.V.P.)

UP

UP

W

D6 D6 D6 D6 D6

CORRIDOR

D6
LOBBY CORRIDOR
12.47X6.77
D3 D6
Pass. LIFT W
DPB
2.30X2.00
D3

6D

3D D3 D3
UP
D3
3D

CORRIDOR

D3
3D

SD SD SD SD

DPB
6D
ELECTRIC
D3 ROOM
DPB
D3
DPB
3D D3 D6 STORE 3.50X6.00
OFFICE
3D
LOBBY
3.00X11.53 3D

OPEN TO
D3

SECOND D3
D3
3D D3

FLOOR REJECT
RECALLED
GOODS ROOM FG DISCPATCH
FG IN
6D D3 D3 D6 D6

D6 D3
SERVER
Goods LIFT CORRIDOR ROOM
3.50X3.00
6D

D6

3.50X6.57
DISPATCH BAY RECEIVING BAY RECEIVING BAY DISPATCH BAY

RS3
40' CONTAINER YARD

CLIENT :- LIFENITY- HEALTH LIMITED (DIALYZER, LVP & PLASMA)
ADDRESS :-Auric city, chhatrapati sambhaji nagar

DESIGN &
PROJECT BY :-BIOBRIDGE HEALTHCARE SOLUTION PVT.LTD.
PREPARED A.K
ADDRESS :-13 R B
A P -411007 M I .

GROUND FLOOR
CHECKED G.K.

DATE 18-06-2026
TITLE- CONCEPTUAL LAYOUT-GF REVISION R0
SCALE 1:1

SIZE A0 DWG.NO 01

W W W W W W W W W

FIREMAN

FOLDING DOOR
CORRIDOR
Goods LIFT 53.5x2.77 LIFT
3.50X3.00 D1
6D
2.70X2.50
4D

D3
CH R3 CH R2
LOBBY
3.00X11.77 6.34 x 5.05 P.M RECEIVING LAF 3.72x2.60 3.72x2.60 LOBBY
3.00X11.77
FIRE STAIR
CORRIDOR FINISHED GOODS ROOM AREA up
SAMPLING

3D
OPEN TO 2.57x9.00 OPEN TO ALL
5.00x5.00
FIRST 24.6 x 8.95 ROOM D3
FLOORS
FLOOR 7.51x7.60

D3
(I.V.D.)
ONLY D4
EXIT CH R1
D1 D1
3.80x2.60 3.80x2.60
DPB

3D

3D
DE DUSTING

4D
MATERIAL

DIALYZER RECEIVING

2.94 M WIDE CORRIDOR
AREA
5.05x6.60 W
2.50 x 13.0
WC WC WC

BELOW

D4
1.10X 1.10X 1.10X
1.70 1.70 1.70

6.77 M WIDE CORRIDOR
REJECT
RECEIVING D1
19.8x30.3 AREA
WC WC WC 5.05x4.94
1.10X 1.10X 1.10X
1.70 1.70 1.70
W

D4
W W

FINISHED GOODS ROOM
LOBBY 24.6 x 10.9
6.77X11.77 QUARENTAINE RM STORE
D1

STAIRCASE
12.9x29.5
AND 7.80x11.6
W LIFT ONLY W
FOR
FIRST FL0OR
(IVD)
6D
PACKING MATERIAL STORE
18.6 x 15.9
11.3 x 27.9

Packing table

Packing table

3.40x3.46 3.40x3.46
PP GRANUAL SECOND FLOOR
AUTO CLAVE RM DISPENSING
ROOM
PRODUCTION
OFFICE & IPQA
TOOL ROOM
4.25x7.00
HOPPER LOADER
ROOM
3D
(ADMIN BLOCK)
4.05x7.00 2.92x7.00
ROOM
D3
4.11x7.00
CH R3 EXIT
XX

13.2x11.5 XX

XX

3.40x3.46 3.40x3.46

SPB LAF
4D 4D 4D
D3 D3

EMERGENCY EXIT
3D
CORRIDOR
D4

29.2x2.79
D4 PW U2
WFI U2

3D 4D 4D
12.3x6.07 DPB

D4
1.25x4.25 COOLING ZONE
4D

7.16x4.54 5.15X 2.00
LOBBY 9.03x4.54 3 WAY
12.47X6.77 6.92x1.52 SPB
AUTO CLAVE

13.7x6.06 Pass. LIFT
2.30X2.00
5.15X3.99

3D
PRODUCTION DPBEURO CAP
IPQA OFFICE AIR LOCK
OFFICE 1.76X1.20
STORE 3.04 x 1.44
3D
3.37x1.44 2.71x1.443D 3D
DPB D3

CORRIDOR
53.5x3.00

3D 3D 3D

ELECTRIC

D3

D3
D3

3.00X11.7 ROOMW
3.50X6.00
MALE PRIMARY PRIMARY
Goods LIFT CONFERENCE
HALL
KITCHEN
2.00X6.40
OFFICE 04 OFFICE 03 OFFICE 02 OFFICE 01 AIR LOCK MALE MALE RECEPTION
FEMALE
AIR LOCK FEMALE FEMALE VISITOR
PRIMARY
VISITOR
VISITOR
3.00X6.40 3.00X6.40 3.00X6.40 2.50X6.40 3.50X6.40 CHANGE CHANGE CHANGE
3.50X3.00 4.80X6.40 3.50X6.40 3.50X6.40 5.00X6.40 4.00X6.40
CHANGE

3.00X6.40
CHANGE
3.00X6.40
AIR LOCK
3.00X6.40 CHANGE
3.00X6.40
3.00X6.40

LOADING / UNLOADING
3D

3D
3D

D6
LIFT ONLY D3 D3 D3 D3 D3 D3 D3 D3
W
FOR
SERVER
SECOND FL00R ROOM
CORRIDOR 3.50X6.57
7D

D7
53.5x3.27

W W W W W W W W W

FIRST FLOOR PLAN BUILDING -A CLIENT :- LIFENITY- HEALTH LIMITED (DIALYZER, LVP & PLASMA)
ADDRESS :-Auric city, chhatrapati sambhaji nagar

SCALE 1:100

DESIGN &
PROJECT BY :-BIOBRIDGE HEALTHCARE SOLUTION PVT.LTD.
PREPARED A.K
ADDRESS :-13, Rachna Blossom, Jagdishnagar,
Aundh, Pune -411007, Maharastra, India. CHECKED G.K.

DATE 18-06-2026
TITLE- CONCEPTUAL LAYOUT-FF-A REVISION R0
SCALE 1:1

SIZE A0 DWG.NO 02

60.00 M.
WIDE
ROAD

GATE-2
ENTRY

60
INTERNAL ROAD 9.00 M WIDE .00
M.
W
20311 20075 19729 19622 ID
I
E
NR
51000
G
TO
FD

UP
UP
FD FD
UP
FD

UP
E
RAD
N
A
F
L

315211
R
O
A
E

D
D

SHAFT SHAFT

FD FD FD FD
UP UP

UP UP

C

GA
ENTE-
TR 1
B
Y
ROAD 9.00 M WIDE

A

UP UP
330840

UP UP

R I D M.
FD FD FD FD
UP
O
FD FD D1 D1 D1 D1 D1 D1 D1 D1 D1 D1 FD FD
GENTS VISITOR
UP FEMALE LADIES TOILET DOCUMENT OFFICE 2 UP
WAITING ROOM LOCKER LOCKER ROOM
5.30x4.00 LOCKER 5.90x5.20
4.63x5.30
GENTS TOILET ROOM ROOM 2.50x5.30
5.90x5.20 ROOM

W .00
D1 D1 D1
4.57x5.30 4.57x5.30 4.57x5.30
UP UP

O E
D4 D4 D4 D4 D4 D4 D4
13.2x1.56
6D

4.05x2.80 CORRIDOR

33.4x2.80

AD
4D D4 4D D4

30
JANITOR
D4

D4

4D

4D

D4

D4

D1

ROOM
PRIMARY VISITOR
3.42x2.37 N
RECEPTION FEMALE PRIMARY FEMALE AIR LOCK AIR LOCK VISITOR CHANGE 1
OFFICE 1
PRIMARY MALE MALE CHANGE 1 4.63x5.00
4.05x2.80 FEMALE VISITOR
R O A D 1 2 . 0 0 M W I D E

CHANGE 1 CHANGE CHANGE 2.60x5.00
CHANGE 3.00x5.00 3.64x5.00
3.10x5.00 3.10x5.00 3.05x5.00 2.70x5.00
4.53x8.15 AIR LOCK
3.80x5.00 2.50x5.00
MALE
4.77x5.00
LIFT
8D

D4

D4 4D D4
6D

DE

CORRIDOR
47.8x2.00
D4
DPB
DPB DPB WASTE OUT
LAF DHS
3 WAY
6D

6D

CHEMICAL DAY STORE
8.10x4.00 COLD ROOM DECONTAMINATION WASHING STERILIZATION
AUTOCLAVE

LAF

BUFFER TANK STORAGE-2 CIP / SIP 4.40x8.10
3.52x5.62 3.11x5.62 4.01x5.62
5.60x8.10
6D

PREPARATION 6.80x8.10
COOL ZONE
4.29x8.10 2.76x8.10 M
4D

4D

DPB D4 D4
DPB
GS+HH

GS+HH

8.30x1.80
D4
A/L GOWN A/L
2.22X3.40 3.47X3.40 2.20X3.40 LENAP LORTNOC LENAP LORTNOC

WASTE A/L A/L
4.13x2.40 4.01x2.40
D4

IN

D4

2.50x2.40
D4

6D 6D 6D D4 D4 D4
D4
4D

CORRIDOR
2.57x66.0 A/L DE GOWN A/L
DE

2.22X3.40 3.47X3.40 2.20X3.40 CORRIDOR
37.0x2.80
4D

4D

D6 D6 D3
LAF
D6 D6
DPB AKTA-1" TFF
4D

4D

NANO
FILTER MH

JANITOR
1.80x2.00
A/L GOWN A/L 1000 L
2.22X4.32 3.47X4.32 2.20X4.32
BULK FILTRATION
LAF

6.90X6.00
D4

500 L
D4

L
LAF

MH

STAGE 9 N
MH

500 L
4D

R O A D 1 2 . 0 0 M W I D E
50 L
MH

STAGE 13 E
D4

LAF SL
MH

5.10X14.3 CHROMO SL
250 L
MH

COLUMN
DE 600 L 600 L
A/L
2.22X3.89
GOWN A/L
500 L
STAGE 10 T/ MH

3.47X3.89 2.20X3.89
STAGE 11 F
MH

213630 MH SL
600 L 6.25X2.05 250 L
4D

4D

200 L
MH
AKTA-1"

STAGE 6 (I) 8.10X12.2
MH

COLUMN PACKING 2000 L 14.7X16.7 250 L MH

250 L
D4

MH

MH

AND SKIDS
MH

CORRIDOR
200 L
17840

1.80X49.6 6.90X10.6 250 L
COOL ZONE 150 L
MH MH

MH
1000 L MH
100 L MH

8.10x5.53 150 L
500 L
MH

300 L
CORRIDOR
MH
2.57X67.9
MH

100 L MH

1000 L
MH

100 L
I N T E R N A L

150 L 300 L
LAF K
DPB LAF
MH

MH

CHROMO MH

AKTA-1"
100 L COLUMN

DHS
AUTOCLAVE CHROMO
D6

COLUMN
500 L 500 L MH
MH MH

250 L 250 L 150 L 100 L
6D

CHROMO
SHAFT A/L MH MH
MH MH
MH

MH
COLUMN SHAFT
CLAF

1.50x5.73
STERILIZATION 2000 L
2500 L
2500 L
D6

FD FD 6.50x5.85 FD FD
PRE - VIRUS CONTROLLED CORRIDOR
6D

1000 L
UP UP
D4

TFF

MH

D4
COLUMN PACKING
DPB
UP AND SKIDS MH

UP
WASHING 100 L
8.00X7.86
AKTA-1"

A/L 6.50x3.00 STAGE 7A
AKTA-1"

MH

1.50x4.42 500 L
D4

300 L 13.6X15.1
D4

MH

STAGE 5 (I) A
D6

DPB J
WASTE IN
4D

500 L
250 L
11.1X18.0
TFF

MH

MH
2.00X28.8

DECONTAMINATION
4D

500 L
250 L
OUT 6.50x5.00 MH
MH
MH

WASTE
D6

1.50x3.54 500 L

DPB STAGE 5(I)B 1500 L 250 L
MH
MH

100 L
8.00X7.20 MH

250 L 250 L 250 L
SL
600 L 500 L
100 L 100 L
MH

MH MH MH 100 L MH
MH

100 L MH MH MH

MH

1500 L
CIP / SIP SL
1000 L 100 L
MH MH

MH MH MH

8.10x5.00
MH

500 L 500 L 1000 L 1000 L 1000 L
MH

MH

MH
6D

MH

26192
CHROMO
50 L COLUMN
GS+HH

GS+HH

DPB 2500 L
LENAP LORTNOC LENAP LORTNOC
WASTE OUT 1500 L
1500 L 1500 L
SL I
MH
250 L 250 L 250 L 250 L
MH

MH MH MH MH
6D

BAG CUTTING &
TANK STORAGE POOLING
MH
MH MH

MH

MH MH
STAGE 4 (F)
150 L 500 L 500 L
8.00X7.80
6D

1000 L
8.10x6.58 1500 L 1500 L 13.6X13.5 MH

500 L
D6

CHROMO
COLUMN

250 L
300 L 300 L STAGE 12 (E) MH

STAGE 1 (F)/
TFF
D6

AKTA-1"

6D
MH MH

CLAF

11.1X8.21
1000 L

2500 L
SL

MH

THAWING PLASMA
2000 L 2000 L
8.00X3.22
6D

BUFFER
AKTA-1"

MH

500 L 300 L 300 L
I N T E R N A L

250 L 250 L 250 L 300 L
PREPARATION
MH MH

CHROMO CHROMO
COLUMN
MH MH MH MH MH

COLUMN
MH MH

DPB D6 D6 D6

8.10x6.00 H
6D

DE

DPB CORRIDOR
35.1X2.80
D6 MH MH
D6 D6
6D

MH MH MH MH

CHROMO MH MH
MH MH MH MH

COLUMN
100 L 100 L 100 L 100 L MH

100 L 100 L CENTRIFUGE

60
200 L 250 L 300 L 300 L 500 L 500 L
1500 L

PLASMA DAY STAGE 3 (F)
AKTA-1"

STORE 4D
STAGE 2 (F) 12.8X8.68
AKTA-1"

STAGE 8 N
15.8X8.68

W .00 M
8.10x8.77 6.17X8.68
CHROMO MH

JANITOR COLUMN

5.20x1.80 SL
600 L 1500 L 2000 L
1500 L 1500 L 1500 L
300 L
G
DPB 300 L MH MH MH
MH
MH

100 L 100 L 100 L CHROMO CHROMO
MH
TFF MH

COLUMN COLUMN AKTA-1"
MH MH MH

EMER.GLASS EMER.GLASS
CORRIDOR
9000

I
50.5x1.80
7500

R DE .
12000
7500

I N T E R N A L
O
PLASMA STORAGE PLASMA STORAGE PLASMA STORAGE PLASMA STORAGE PLASMA STORAGE
~ 1000 L ~ 1000 L ~ 1000 L ~ 1000 L ~ 1000 L
~ 70 KL GV ~ 70 KL GV ~ 70 KL GV ~ 70 KL GV ~ 70 KL GV
7.10x9.62m 9.95x9.62m 9.95x9.62m 9.95x9.62m 8.15x9.62m

F

AD
800 800 800 800

SHAFT SHAFT
D3 D3 D3 D3 D3

CORRIDOR
45.3x1.80
FD FD FD FD
UP UP

UP UP

D

CORRIDOR
2.57x37.0 CORRIDOR
2.57x35.1

PLASMA STORAGE PLASMA STORAGE PLASMA STORAGE PLASMA STORAGE PLASMA STORAGE
~ 1000 L
~ 70 KL GV
~ 1000 L
~ 70 KL GV
~ 1000 L
~ 70 KL GV
~ 1000 L
~ 70 KL GV
~ 1000 L
~ 70 KL GV
E
7.10x9.62m 9.95x9.62m 9.95x9.62m 9.95x9.62m 8.15x9.62m

A
800

800 800 800 800
1125

O
D3 D3 D3 D3 D3

CORRIDOR
45.3x1.80

79702

R
D
850 800 800 800 800

PLASMA STORAGE PLASMA STORAGE PLASMA STORAGE PLASMA STORAGE PLASMA STORAGE
~ 1000 L ~ 1000 L ~ 1000 L ~ 1000 L ~ 1000 L

E
~ 70 KL GV ~ 70 KL GV ~ 70 KL GV ~ 70 KL GV ~ 70 KL GV
7.10x11.5m 9.95x11.5m 9.95x11.5m 9.95x11.5m 8.15x11.5m

ID
699

D3 D3 D3 D3 D3

CORRIDOR
47.9x1.80
D6
C
29.5 X 2.58
CORRIDOR

W
D3 D6 COLD ROOM -4 COLD ROOM -3 COLD ROOM -2 COLD ROOM -1 ICE PACK
DPB RLAF
4.45X7.58 2.91X7.58
D3

AIR LOCK-1 4.45X7.58 4.45X7.58 4.45X7.58
1.50X 1.93 RM QUARANTINE WASHING PLASMA
BAG BAR CODING, JANITOR
AREA QUARANTINE
MINI POOLS
6D

1.45X2.90
2.30X2.90
2.60X2.90
DISPENCING ROOM
3D
3.48X 3.95
CHANGE ROOM D3 D3 D3
1.50 X 1.93
3D

DPB 6.21X5.77
3D

5.00X2.95 D3
11.5X3.30 SD SD SD SD

6D
CORRIDOR
D3

M.
APPROVED RM AIR LOCK-1 DPB 1.95X11.3
4.63X4.00
2.00X1.21 D3

AIR LOCK-1 DPB
3D D3 D6 STORE 13.9X3.84
3.00X2.60
9.72X7.22 1.87X1.40 OFFICE 9.22X2.45
3D
CHANGE ROOM
2.00X2.08 3D
2.23X6.25
4.75X2.32
B
CHANGE ROOM
1.87X1.40 STORE CHANGE
DE-DUSTING REJECT SEROLOGY PLASMA ROOM
TUNNEL 4.70X2.37
ROOM TESTING RECEIPT 3D D3 1.51X1.23
D3

PLASMA
D3 D3 SAMPLING
1.76X 2.90
2.95X4.95 3.00X 4.95 2.40X 4.95
RM
SAMPLING 3.68X 2.05 RECALLED REJECT
RM FG DISCPATCH
1.07X3.39
RECEIPT GOODS ROOM FG IN
3.16X2.03 2.30X2.35 2.30X 2.35
3.00X2.77
RLAF RLAF

.00
6D D3 D3 D6 D6

D6 D3

UP UP
CORRIDOR
39.8X6.84

45
UP UP
FD FD FD FD
A
SPM RM PLASMA
RECEIVING FG
RECEIVING RECEIVING
BAY DISCPATCH
BAY BAY

3548 CANOPY ABOVE
40' CONTAINER YARD
CONTAINER

CONTAINER

CONTAINER

CONTAINER
12.19x2.44x2.59

12.19x2.44x2.59

6 5 4 3 2 1
12.19x2.44x2.59

12.19x2.44x2.59

17148
13600
INTERNAL ROAD 12.00 M WIDE

231187

CLIENT :- LIFENITY-LARGE VOLUME PARENTERALS
45.00 M. W I D E R O A D

DESIGN &
N

PROJECT BY :- BIOBRIDGE HEALTHCARE SOLUTION PVT.LTD.

LAYOUT PLAN ADDRESS :-13, Rachna Blossom, Jagdishnagar,
Aundh, Pune -411007, Maharastra, India.
PREPARED A.K

CHECKED G.K.
SCALE :- 1:100 DATE 19-03-2026
TITLE- PLOT LAYOUT-GF REVISION R0
SCALE 1:1

SIZE A0 DWG.NO 01

49.1x2.77

D1 D6
LOBBY
PB 3.00X11.77

D3
OPEN TO ALL
FLOORS
DOCUMENT STORE QC STAFF CHEMICAL /GLASS CONFERENCE
4.73x6.62 DOCUMENTATION WARE STORAGE
7.71x6.62
9.31x6.07 CANTEEN
4.03x6.62 14.8x10.7
5.98x6.12 FIREMAN
Goods LIFT 2.35x8.03 LIFT
3.50X3.00 2.70X2.50
D3
D3

D3
FIRE STAIR
LOBBY WASH AREA
WET LAB
D3 5.90x2.76 up
3.00X11.30 D3 9.31x4.85
D1
OPEN TO
FIRST QC MANEGER
PM TESTING ROOM
FLOOR 4.73x5.91 9.94x7.06

D6
D1 D3

HOT ROOM/
BALANCE GENERALFTIR FUME HOOD
D3 ROOM 4.38x3.76 5.90x5.85
D3 4.82x3.76

D3
1.80 M WIDE CORRIDOR
10.1x2.80 D3
QC SAMPLE PREP.
4.73x5.42

CORRIDOR
22.5x1.80
INSTRUMENT ROOM II
15.3x5.04
W W
DPB HPLC D1
9.94x5.63

INSTRUMENT ROOM I
D1
D3 TRANING HALL
QC SAMPLE JANITOR
D3
D1 RECEIPT
1.50x3.93
14.8x19.0 D1
3.11x3.93
CORRIDOR
1.80x29.2
REFERENCE TOC/LPC ROOM
D3 GC ROOM
D4 STANDARD 4.44x2.98
5.87x2.98 4.83x2.98

D3
W W
GENTS CH ROOM
4.73x3.42 RETENTION
SAMPLES ROOM

COB
(COLD ROOM 2 TO 8 °C)
D1
9.94x3.31
D3 D3

STABILITY ROOM
LADIES CH ROOM 15.3x6.04
4.73x3.42
RM CONTROL SAMPLE
9.94x3.31

COB
D1

2.80 M WIDE PASSAGE
D3 D3 D1
DPB

D6

6.77 M WIDE CORRIDOR
D3 D3 D3
DPB AC
MEDIA DIST.
W
WASH AREA
WC WC WC 5.221x2.02 DECONTAMINATION ROOM
1.10X 1.10X 1.10X 7.34x2.02
5.54x2.02

3D

D3
1.70 1.70 1.70 MEDIA PREP. PREPARATION & STERILIZATION
GRADE D
7.38x3.88 8.11x5.91 D3
GRADE D

DPB
D3
IC-1 DPB D1
GRADE B

DPB
INCUBATOR STERILITY
DPB
FOR MEDIA TESTING ROOM IC-1
D3 4.65x3.78 GRADE CNC IC-4
INCUBATOR
WC WC WC GRADE B
COOL ZONE 5.33x3.78
1.10X 1.10X 1.10X IC-2
1.70 1.70 1.70
D3 3.38x3.78 D3 IC-3 R&D LAB -3 R&D LAB-4
1.83x1.91

3D
CORRIDOR
MEDIA GRADE D OBSERVATION
6.47x6.61
13.3x1.80
6.47x6.61 W
STORE A/L GRADE B
4.78x3.54 2.50x4.24 D4 D4 CORRIDOR D3
GRADE C GRADE B GRADE B 3.53x2.02 DPB
CH-I CH-II CH-III 1.72x1.86

Emergency Exit
4.81x2.02 2.73x2.02 2.02x2.02
D4 GRADE C GRADE CNC
DPB IC-1
D4
D4 D4
MLT ROOM INCUBATOR IC-4
D4 5.33x4.91 2.80x8.90
4.65x4.91
BET ROOM IC-2
GRADE C GRADE B IC-3
GRADE CNC A/L EXIT-II
4.78x2.83 GRADE D EXIT-I
2.63x2.02
D3 5.36x2.53 D4 3.49x2.02
D4 R&D LAB -2 R&D LAB -1
GRADE C D3
CORRIDOR OBSERVATION 6.47x6.61 6.47x6.61
GRADE CNC 1.80x8.50
GRADE D GRADE C
CORRIDOR A/L CH-I

COB
D4 FRIDGE

3D
2.50x8.31 5.36x1.52 D4 6.04x1.52 BSC
D4 3.77x3.51
GENTS CH ROOM GRADE CNC
W GRADE CNC D3 D3 INCUBATION ROOM W
D3 4.78x2.83 D3 DPB 4.36x3.51
GRADE D

D3

3D
CULTURE ROOM
4.65x3.51

GRADE CNC GRADE CNC GRADE CNC
LOBBY MICRO OFFICE MICRO MICRO
SAMPLE RECEIPT
DPB DPB D3 CORRIDOR
14.8x1.80

COB
5.17x4.24
6.77x11.77 DOCUMENT STORE 2.63x4.24 D4 D4 SCRAP OUT DPB
D1
D1 LADIES CH ROOM
3.49x4.24
GRADE D 5.49x2.86 GRADE CNC
GRADE D A/L D3 D3
GRADE CNC EXIT 2.66x2.86
4.78x3.12 CH ROOM
D3 D3 2.39x2.01 2.15x2.01
D3 D3
D3
R&D OFFICE
7.31x2.90 R&D ENTRANCE
W CORRIDOR
34.1x1.80
7.48x2.90 W

D3
D4 D4 D4

CORRIDOR
CABIN 1 16.7X1.80
CABIN 4 CABIN 3 CABIN 2

STORAGE
3.39x4.14
2.88x4.14 3.43x4.14 4.05x4.14
D4 D4 D4 D4

4.13x4.19 PAL-1
PRIMARY CHANGE PRIMARY CHANGE
AUDIT FEMALE
2.27X1.64
MAL ENTRY VISITOR COLD ROOM
2.85x5.81 4.43X3.93
2.85X5.78 4.51X5.78 4.19X3.27 D4 DE-CARTONING DPB
3.63X1.92 2.26X2.53 3.89X3.32
D4 D4 D4 D4
PRIMARY CHANGE
MALE PAL-2
2.27X1.64
5.31X5.81
Pass. LIFT

D4
D4
2.30X2.00 DPB
CORRIDOR
D4 D4 D4
CLEAN JC CORRIDOR
2.62x100.6 1.88X2.48 1.80x100.6 WC WC WC
CORRIDOR 1.10X 1.10X 1.10X
9.00X1.80 1.70 1.70 1.70

D4
DPB

D4
D4
D1
QA OFFICE QA MANGER

18.0x10.8 2.85x4.06
MATERIAL PREP WASH
8.15X3.07
AC
1.00X2.40 UNLOAD
8.41X3.40

D4
5.26X1.64
WC WC WC

D4
2.21X1.46
1.10X 1.10X 1.10X

DRYING ZONE
DPB
VIAL WASHING 1.70 1.70 1.70
1.00X1.00
6.42X9.65

D6

D4

D4
PAL

STERILIZING ZONE
2.84X1.45

D4
STERILE CORRIDOR
QA HEAD 15.5X1.80
DISINFECTANT
2.85x4.47 D3 D3 8.41X2.62

COOLING ZONE
D4
D4
MAL
2.84X1.45 MAL PAL

D4
1.45X1.64 1.41X1.64 PASS THROUGH

STABLIZING ZONE
D4
D3 D3

D3
PAL
D6
2.04X2.29

D3

D4
TUMBLE
DRYER FORMULATION
PACKING AREA 8.41X3.59
1.50X2.00

D3

D4
MAL

CONVEYOR
SWING
2.04X1.47 DOUBLE
DOOR
W D3 WASHER W
UNLOADING AREA

36" TURNTABLE
NEW ADD
LAUNDRY AREA 1.50X2.00
QA DOCUMENT 5.10X1.57 2.94X5.13 8.49X1.80
3.24 M WIDE PASSAGE

2.98X3.88

21.0x9.82
& LOADING TRAY
VIAL COLLECTION

D2

FILL FINISH

D4
6.42X8.30

D4
D1 EXIT 2
4.54X1.87
EXIT 1
4.40X1.87
D1

D4
PAL 3 PAL 4
4.23X3.82 3.20X3.82
CORRIDOR
2.85X25.2 PAL 1 PAL 2
4.54X1.87 4.40X1.87

D4

D4

D4

D4	
W W	
D4	
LYO SERVICE AREA	
CONTROL ROOM LYO LOAD	
D4	
5.55X3.55 10.9X3.55 8.29X3.55	
D8	
AUTOMATIC FOUR HEAD ALUMINUM CAP	
D8	
SEALING MACHINE, MODEL-AEWCS-120	
D4	
SWING	
CONVEYOR	
BULK STAGING	
VIAL COLLECTION	
SECONDARY PACKING AREA	
16.6X4.04 CAPPING	
TRAY	
21.0x8.07 8.29X4.04	
W	
WC WC WC	
DPB	
1.10X 1.10X 1.10X D8	
1.70 1.70 1.70	
DPB LOBBY	
CORRIDOR	
25.5X1.80 JC	
6.77X11.00	
DPB	
D4	
2.32X1.00	
DPB	
D4	
STAIRCASE	
UNDERTEST	
D4	
WC WC WC	
1.10X 1.10X 1.10X	
VISUAL & BUFFER ROOM	
AND	
1.70 1.70 1.70 PACKING CHANGE	
INSPECTION 2.83X2.68 FEMALE LIFT ONLY W	
9.79X3.87 4.14 X 3.08	
FOR	
D4	
2.88X1.14	
SECOND FLOOR	
D4	
SAMPLING	
D8	
(L.V.P.)	
D4	
PPM STORE SPM STORE 11.8X1.80	
15.2x12.0 14.9x9.32	
D4	
7.22x2.69 PACKING CHANGE	
11.8x2.89 MALE	
4.14 X 4.23	
QUARANTINE	
LABELING & PACKING	
12.6x5.40	
D4	
PB	
D8	
D4	
PACKING	
W	
D6	

VISITOR
D4 4.41 X 3.08
D8

D4
D8

D4
D8 D8
D4 PACKING OFFICE
LABELING STORE 4.14X4.42
FINISHED GOODS STAGING
7.68 X 5.59
12.6 X 5.59
LOBBY 3.14 X 3.05
APPROVED RUBBER
SPM RECEIPT REJECT
12.47X6.77 BUNG STORE
W 3.36 X 2.89 3.59 X 2.89 3.58 X 2.89
D8 D8 D8 Pass. LIFT
2.30X2.00

CORRIDOR
53.5x1.80

D6 D6 D6 D6

ELECTRIC
ROOMW
CORRIDOR
3.04x11.7
3.50X6.00
LOBBY
3.00X11.53
OPEN TO ALL
FLOORS

W
SERVER
Goods LIFT ROOM
3.50X3.00 3.50X6.57

CLIENT :- LIFENITY- HEALTH LIMITED (DIALYZER, LVP & PLASMA)
ADDRESS :-Auric city, chhatrapati sambhaji nagar

FIRST FLOOR DESIGN &
PROJECT BY :-BIOBRIDGE HEALTHCARE SOLUTION PVT.LTD.

ADDRESS :-13, Rachna Blossom, Jagdishnagar,
PREPARED A.K

Aundh, Pune -411007, Maharastra, India. CHECKED G.K.

DATE 18-06-2026
TITLE- CONCEPTUAL LAYOUT-FF REVISION R0
SCALE 1:1

SIZE A0 DWG.NO 02

DRAWING NO: LIF_MP_A-1000-R0

585.5 KEY PLAN:

5.0
58
586.0
585.5
585.0

.5 GATE-2
584
ID. (RAW MATERIAL IN)

4.5
58
.0
584

DRIVER'S ROOM OPEN SPACE-1
4572.69 Sq.M.
SECURITY
CHILLER SPACE
CABIN-2 FOR
ETP

SPACE
COOLING TOWER
COOLING FOR
WATER TANKS SECURITY .0
120 TR
STP

TOWERS PUMP ROOM 584

583
.5 COOLING TOWER
120 TR
COOLING TOWER
240 TR FIRE TANK
150K
PROCESS TANK
100K
CABIN-3 GENERAL NOTES:

FUTURE
40' CONTAINER YARD CONTAINER YARD
GATE-3
(MATERIAL OUT)
----- 15M WIDE ROAD -----
1 2 3 4 5 6

A
MATERIAL MATERIAL

.5
LIFT LIFT

LIFT
LIFT
583

B

--
--
-7
.5M
LEGEND:
C

.0
WI
583 DE
RO
D
AD
--
--
----- 9M WIDE ROAD -----

-
----- 12M WIDE ROAD -----
BUILDING-3
E CANTEEN
12M WIDE ROAD

582.5 1510.67 Sq.m.

F

TEMPLE

G

GATE-1

H OFFICE ENTRY

.5

583

I

J

SECURITY

CABIN-1

0

0.

K

58

.0

584

BUILDING- 4

7191.00 Sq.m.

L

230000

M

.0

582

N

BUILDING- 6 PARKING

(FUTURE EXPANSION) BUILDING-2 BUILDING-1

13310.98 Sq.m. ADMIN R&D

LIFT

LIFT 11730.00 Sq.m.

O

(G+1)

174000

3783.64 Sq.m.

-----6M WIDE ROAD -----

A

OPEN SPACE-2 DESCRIPTION OF PROPOSAL

.0 3793.43 Sq.m.

582

BUILDING- 5

4080.00 Sq.m.

PROPOSED MASTER PLAN AT

AURIC CITY, CH. SAMBHAJI NAGAR.

B

AMENITY-5%

----- 7.5M WIDE ROAD -----

3837.25 Sq.m. 584.0

C

582

.0

D

CLIENT :

583.5

M/S LIFENITY HEALTH LIMITED

TITLE :

E

F

MASTER PLAN

.5

581 583.0

0.5 COMPRESSOR

58

G

PURPOSE:

DG

H 51000 58

0.0

76500 FOR TENDER

PANEL 582.5

ROOM

.5

580

I

TRANSFORMER

0 .0 METERING

58 0.5

58
KIOSK
582.0
-----12M WIDE ROAD ----- NORTH JOB.NO.
.5 GATE-4 DRG.NO.
581 LIF_MP_A-1000
(CONTINGENCY GATE)
SCALE AS WRITTEN
.0
582
DATE 09-04-2025
581.5

CHECKED BY DP
DRAWN BY MP
581.0
58
58

1.5
2.0

01 MASTER PLAN ESIGN
ESTINATION We Design Spaces to create better Destination

Office no. 205, Second Floor Shubham Heights, Near Max care hosp.,Pimple Saudagar,Pune-27
Contact Details -9881074567 e-mail - designdestination2015@gmail.com